Alexander Grothendieck

Les Dérivateurs

an integrated reader, with commentary

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Chapter I

Généralités sur les (pré)dérivateurs

p. –

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre I

Généralités sur les (pré)dérivateurs

p. –

Ce texte a été déchiffré et transcrit en $\mathcal{AMS}$-TEX par G. Maltsiniotis. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou les rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

Dérivateurs

1. Préliminaires sur les foncteurs images inverses des préfaisceaux sur des catégories d'indices, et leurs adjoints multiples.

A "catégorie de coefficients",
I "catégorie d'indices",
$A(I) = \underline{\mathsf{Hom}}(I^\circ, A)$ préfaisceaux sur I à valeurs dans A.

Si $f : I \to J$, on en déduit :

$$f^* : A(J) \to A(I) \quad ,$$

et on s'intéresse à l'existence des foncteurs adjoints successifs à gauche et à droite. Notation provisoire :

$$f^? \; f_! \; f^* \; f_* \; f^! \quad .$$

Voici les faits principaux :

Commentary · Exposition [ms. 1]

Variance and the two Kan extensions

The elementary presheaf calculation fixes the conventions for everything that follows. Throughout these pages an I-diagram has arrows in the direction of I^{op}. Thus a diagram on $0 \to 1$ is an arrow $F_1 \to F_0$. We retain this convention in the commentary. If $f : I \to J$, the adjunctions are

$$f_! \dashv f^* \dashv f_*.$$

They are adjunctions of ordinary categories at this stage. Their pointwise formulas, whenever the indicated limits exist, are

$$(f_*F)_j = \lim_{(f/j)^{\mathrm{op}}} F \circ \mathrm{pr}, \qquad (f_!F)_j = \operatorname*{colim}_{(j/f)^{\mathrm{op}}} F \circ \mathrm{pr}\,.$$

Here f/j has objects $(i, f(i) \to j)$, whereas j/f has objects $(i, j \to f(i))$. The opposite signs are essential: forgetting one exchanges the two formulas. These are the usual pointwise Kan formulas applied to f^{op}; see Riehl (n.d., §6.2).

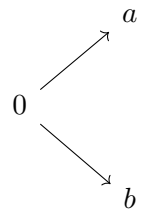
For completeness, a map $G \to f_*F$ consists, at j, of a compatible family of maps $G_j \to F_i$, one for every $f(i) \to j$. A map $f^*G \to F$ supplies this family by composing $G_j \to G_{f(i)} \to F_i$. Conversely, the members indexed by identities recover the original map. The two constructions are inverse and natural. The left adjunction follows by reversing all arrows. This proves the existence assertion without any adjoint functor theorem; for finite categories the comma categories are finite, so finite limits or colimits suffice.

The word "immersion" here denotes the inclusion of a full subcategory, with "open" meaning a sieve and "closed" a cosieve. A sieve is closed under taking the source of an arrow whose target belongs to it; a cosieve is closed under taking the target. In a poset these are, respectively, downward and upward closed subsets. This agrees with the conventions forced by the displayed examples. For a fully faithful f, the identity at i is terminal in $f/f(i)$ and initial in $f(i)/f$. The formulas therefore give $f^*f_*F \cong F \cong f^*f_!F$. Both Kan extensions are fully faithful. Calling f^* a localization expresses these adjunction identities, not a claim that restriction itself is fully faithful.

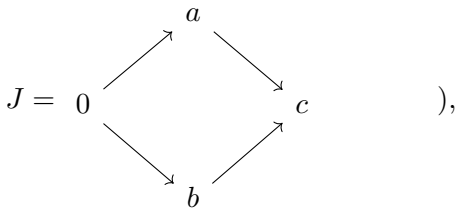
a) Sous réserve de l'existence des $\varprojlim$ (resp. $\varinjlim$) pertinentes, f_* (resp. $f_!$) existe. Par exemple s'il existe `[des]` $\varprojlim$ (resp. $\varinjlim$) *finies*, alors f_* (resp. $f_!$) existe si I, J sont des catégories finies.

b) Même si A est une catégorie abélienne, avec $\varprojlim$ et $\varinjlim$ (par exemple Ab), les *biadjoints* n'existent pas toujours. Par exemple, dans le cas particulier où $J = e$ est la catégorie ponctuelle,

et I l'ensemble ordonné



(dans le cas de $f^!$), ou l'ensemble ordonné opposé (dans le cas de $f^?$). Même si f est une immersion ouverte, $f^!$ n'existe pas forcément (par exemple elle n'existe pas si I est comme `[ci-]`dessus, et



et si f est une immersion fermée, $f^?$ n'existe pas forcément.

Commentary · Examples [ms. 1]

Why the next adjoint can fail

The first counterexample is already a calculation with abelian groups. On the fork $0 \to a, 0 \to b$, a presheaf is a cospan $A \to C \leftarrow B$, and its limit is $A \times_C B$. Consider the componentwise surjection of cospans

$$(\mathbb{Z} \xrightarrow{2} \mathbb{Z} \leftarrow 0) \longrightarrow (\mathbb{Z}/2 \xrightarrow{0} \mathbb{Z}/2 \leftarrow 0),$$

whose nonzero components are reduction modulo 2. The induced map on limits is $0 \to \mathbb{Z}/2$, which is not surjective. A functor having a right adjoint is a left adjoint and hence preserves cokernels; in this abelian diagram category it would preserve epimorphisms. The limit functor therefore has no right adjoint. This verifies precisely the failure of $f^!$ asserted in (b).

For the opposite fork the colimit is a pushout. The componentwise injection from the span $\mathbb{Z} \xrightarrow{2} \mathbb{Z}, \mathbb{Z} \to 0$, to the analogous span with $\mathbb{Q}$ in place of $\mathbb{Z}$ induces $\mathbb{Z}/2 \to 0$ on pushouts. It fails to preserve a monomorphism and cannot be a right adjoint. Thus its left adjoint $f^?$ does not exist. These examples use neither an absence of ordinary limits nor a failure of abelian structure.

The open-immersion example is the same obstruction localized at the new vertex c. For the inclusion of the fork into the diamond, $(f_*F)_c$ is the preceding pullback. Evaluation at c preserves colimits, so a right adjoint to f_* would force this pullback functor to preserve cokernels. The same surjection disproves that possibility. The dual argument treats a closed immersion and $f^?$. The point of the examples is that iterated adjunction is a property of a particular functor, not a consequence of the ambient categories having all limits and colimits.

c) Si f est une *immersion fermée*, et A est *ponctuée* (*i.e.* admet un objet à la fois initial et final), alors $f^!$ existe. Si f est une immersion ouverte, $f^?$ existe. (Toujours sous réserve de p. 2
l'existence dans A des limites pertinentes). Si f est une immersion, f_* et $f_!$ `[sont]` pleinement fidèles *i.e.* f^* `[est]` un foncteur de localisation.

d) Supposons seulement l'existence de l'objet final e dans A (plus au besoin certaines $\varprojlim$ finies). Si

[page 2]

$$i : Y \hookrightarrow S$$

est une immersion fermée, de complémentaire U

$$Y \xhookrightarrow{i} S \xhookleftarrow{j} U$$

de sorte que j soit une immersion ouverte, alors $A(Y)$ s'identifie par le foncteur i_* à la sous-catégorie pleine de $A(S)$ formée des $F \in \mathsf{Ob}\ A(S)$ tels que $F|U \simeq e_U$ (préfaisceau final). De même, si l'objet initial $\emptyset$ existe dans A, et les $\varinjlim$ pertinentes, alors $A(U)$ s'identifie par $j_!$ à la sous-catégorie pleine de $A(S)$ formée des F tels que $F|Y = \emptyset_Y$ (préfaisceau initial, prenant partout la valeur $\emptyset$).

Si A est même *ponctuée* ($e = \emptyset$), alors on a, pour tout $F \in \mathsf{Ob}\ A(S)$, un carré à la fois cartésien et cocartésien, fonctoriel en F

(1)
$$\begin{array}{ccccc} & & F & & \\ & \overset{\text{adj}}{\nearrow} & & \overset{\text{adj}}{\searrow} & \\ j_!j^*F & & & & i_*i^*F \\ & \searrow & & \nearrow & \\ & & e & & \end{array} \quad .$$

Dans le cas A abélienne, ce diagramme prend la forme d'une suite exacte

$$0 \longrightarrow j_!j^*F \longrightarrow F \longrightarrow i_*i^*F \longrightarrow 0 \quad .$$

e) Mêmes conditions pour i et j. On suppose que A a les $\varprojlim$ pertinentes, dont objet final. On a alors un carré cartésien fonctoriel en $F \in \mathsf{Ob}\ A(S)$ `[il faut supposer la catégorie` A `ponctuée]`

(2)
$$\begin{array}{ccc} F & \longleftarrow & i_*i^!F \\ {\scriptstyle\text{adj}}\downarrow & & \downarrow \\ j_*j^*F & \longleftarrow & e \end{array}$$

i.e. $i^!F$ s'identifie par le foncteur pleinement fidèle i_* au *noyau* de $F \to j_*j^*F$. Dans le cas abélien, cela prend la forme d'une suite exacte

$$0 \longrightarrow i_*i^!F \longrightarrow F \longrightarrow j_*j^*F \quad .$$

Commentary · Exposition [ms. 2]

Support, zero extension, and the exceptional inverse image

Let $U \subset S$ be a sieve and $Y = S \setminus U$ its complementary cosieve. A presheaf on Y extends by the terminal object on U under i_*. Indeed, Y/u is empty for $u \in U$, whereas restriction to Y recovers the original presheaf. Conversely, if $F|_U$ is terminal, the unit $F \to i_*i^*F$ is an isomorphism at every object. This proves the description of the essential image in (d). Dually, $j_!$ extends a presheaf on U by the initial object on Y.

Assume now that A has a zero object and the required limits. Form

$$K = F \times_{j_*j^*F} 0.$$

Its restriction to U is zero, so $K = i_*G$ for a unique G up to the canonical equivalence above. For $H \in A(Y)$, the maps $i_*H \to j_*j^*F$ are in bijection with maps $j^*i_*H = 0 \to j^*F$, hence form a singleton. The universal property of the pullback gives

$$\mathrm{Hom}(H, G) \cong \mathrm{Hom}(i_*H, K) \cong \mathrm{Hom}(i_*H, F).$$

Thus $G = i^!F$, proving both the extra adjunction and square (2). This argument explains the editors' pointedness correction: a terminal object alone supplies no canonical map from it to j_*j^*F. For $A = \mathsf{Set}$ and $S = [1]$, the closed inclusion of 1 has $i_*B = (B \to *)$. It does not preserve the initial object, so cannot have a right adjoint.

The dual construction uses

$$Q = F \amalg_{i_!i^*F} 0.$$

Now $Q|_Y = 0$, hence $Q = j_! j^? F$, and the same universal-property argument proves $j^? \dashv j_!$. The first square of (d) is bicartesian for a different, simpler reason: at each point it is a square with an identity on one side and a zero object on the other. Checking at points proves both universal properties. No stability hypothesis is needed for this particular square. In an abelian category its associated short exact sequence need not split as a sequence of diagrams, even though every component is split. For $S = [1]$ and $F = (\mathbb{Z} \xrightarrow{1} \mathbb{Z})$, the sequence is

$$0 \longrightarrow (0 \to \mathbb{Z}) \longrightarrow (\mathbb{Z} \xrightarrow{1} \mathbb{Z}) \longrightarrow (\mathbb{Z} \to 0) \longrightarrow 0.$$

A splitting would require the identity map on the first copy of $\mathbb{Z}$ to have zero composite with the displayed identity arrow, which is impossible.

Je me dispense d'écrire le carré cocartésien dual, quand A a les $\varinjlim$ pertinentes, y compris objet initial. Il semble qu'elle n'ait pas servi en pratique. Les foncteurs adjoints et biadjoints classiquement utiles sont inclus ci-dessous (où les "pas utiles" sont entre crochets) p. 3 [page 3]

$$[i_!]\ i^*\ i_*\ i^!$$
$$[j^?]\ j_!\ j^*\ j_*$$

i.e. les foncteurs $i_!$ et $j^?$ n'ont pas servi classiquement.

f) Cas des catégories $\mathbf{\Delta}_i$ (simplexes type, $i \in \mathbb{N}$), et des foncteurs entre ceux-ci (applications semi-simpliciales). Il semblerait que pour une telle f, si A est ponctuée, il y ait les *bi-adjoints* à la fois à gauche et à droite. De plus, pour $\mathbf{\Delta}_i$ et $\mathbf{\Delta}_j$ fixés quand on regarde toutes les $f : \mathbf{\Delta}_i \to \mathbf{\Delta}_j$ et toutes les $g : \mathbf{\Delta}_j \to \mathbf{\Delta}_i$, que les suites totales de foncteurs multi-adjoints pour chacun des f^*, g^* correspondants sont toutes les mêmes `[ce n'est pas vrai]`. Ça a l'air vrai du moins si $j = i + 1$ (et je l'ai vérifié pour $i = 0$, $i = 1$), on doit trouver une "suite d'adjonctions" de longueur $2i + 7 = (i + 1) + (i + 2) + 2 + 2$. De façon précise, la suite comporte quatre termes "extrêmes" (les deux premiers, les deux derniers) et les $(i + 1) + (i + 2)$ termes intermédiaires correspondant aux foncteurs de la forme f^* ou g^*, où $f : \mathbf{\Delta}_{i+1} \to \mathbf{\Delta}_i$ est une des $(i + 1)$ applications de dégénérescence, et $g : \mathbf{\Delta}_i \to \mathbf{\Delta}_{i+1}$ une des $i + 2$ applications bord `[ce n'est pas tout à fait vrai; il faut se limiter aux applications qui sont injectives ou surjectives]`.

Commentary · Examples [ms. 3] p. 4 [page 4]

The simplex adjunctions, with their endpoints

Write $C_n = A^{[n]^{\mathrm{op}}}$, so its objects are strings $X_n \to \cdots \to X_0$. Let $\delta^k : [n] \to [n+1]$ omit k, and let $\sigma^k : [n+1] \to [n]$ identify k and $k+1$. The inequalities in the finite ordinals give

$$\delta^{k+1} \dashv \sigma^k \dashv \delta^k.$$

Applying contravariant presheaves reverses both arrows and natural transformations; consequently the restrictions form the adjunction chain

$$(\delta^{n+1})^* \dashv (\sigma^n)^* \dashv (\delta^n)^* \dashv \cdots \dashv (\sigma^0)^* \dashv (\delta^0)^*.$$

One can also check each bijection of Hom sets by writing down the compatible component maps. There are $2n + 3$ terms here.

If A is pointed with finite limits and colimits, two more terms occur at either end. On the left, let $L : C_n \to C_{n+1}$ prepend zero at the high end: $L(X) = (0 \to X_n \to \cdots \to X_0)$. Then $L \dashv (\delta^{n+1})^*$. Define $Q : C_{n+1} \to C_n$ by

$$(QX)_r = X_r \amalg_{X_{n+1}} 0 \qquad (0 \le r \le n).$$

A map $X \to LY$ is exactly a compatible family $X_r \to Y_r$ killing X_{n+1}, so $Q \dashv L$. At the right end let $R(X) = (X_n \to \cdots \to X_0 \to 0)$. Then $(\delta^0)^* \dashv R$. Its right adjoint is the string $K(X)_r = X_{r+1} \times_{X_0} 0$, since a map $RY \to X$ is exactly a family of maps to those fibers. The full chain has $2n + 7$ terms, as suggested in the manuscript.

For $n = 0$ it is particularly transparent:

$$\mathrm{coker} \dashv (0 \to -) \dashv \mathrm{ev}_0 \dashv \mathrm{const} \dashv \mathrm{ev}_1 \dashv (- \to 0) \dashv \ker.$$

Here kernel and cokernel mean fiber and cofiber over the zero object, so abelian structure is unnecessary. This also proves the first conjecture in (f): every monotone map of finite ordinals factors into elementary faces and degeneracies, each restriction occurring far enough inside one of these chains to have two adjoints on both sides. Composing the corresponding adjunctions supplies $f^?$ and $f^!$.

The stronger statement about one common chain for *all* maps between two fixed ordinals is false. Already for [1] and itself, the identity restriction has only identity adjoints up to isomorphism, whereas the restriction along a constant map is not the identity on C_1, for example when $A = \mathsf{Ab}$. Between $[n]$ and $[n+1]$, the central chain displayed above contains exactly the injective faces and surjective degeneracies. Other monotone maps are composites, not additional entries in that chain. The editors' qualifications should therefore be retained; the nearby count is a correct statement about the specified chain, not a classification of every restriction functor.

2. Préliminaires heuristiques sur les dérivateurs.

On se donne une sous-catégorie pleine Dia (comme "diagrammes") de $\mathcal{C}at$. Ça peut être tout $\mathcal{C}at$, ou plus petit. Mais on supposera le plus souvent que Dia contient au moins les "catégories ordonnées" (correspondant aux ensembles ordonnés, *i.e.* telles que *a*) tout morphisme est un mono et *b*) tout iso est une identité `[ce n'est pas vrai : la catégorie réduite à une double flèche satisfait à ces conditions, mais n'est pas un ensemble ordonné]`). Par abus de langage, on identifie un ensemble ordonné avec la catégorie correspondante (et une catégorie au topos qu'elle définit?). On considère Dia comme une 2-*catégorie.* [1]

Commentary · Context [ms. 4]

From derived coefficients to diagram categories

The opening calculations make the geometric language of supports available before any homotopy theory is introduced. There is a documented reason for this starting point. In *Pursuing Stacks*, §69, Grothendieck traces his concern to the inadequacy of a single derived category for the trace and determinant formulas arising around SGA 5. The proposed replacement assigns a derived category to every diagram shape; filtered complexes and diagrams of triangles are instances of this larger construction. The demand that the same structure reappear at every shape already anticipates the shifted derivators of manuscript p. 18 (Grothendieck, 1983, §69, pp. 1–3 of the edited extract).

His letter to Thomason of 2 April 1991 returns to the same question: which intrinsic structure survives passage from models to categories of coefficients? That letter explicitly connects the renewed work to the 1983 notes and discusses both derivators and the role of small categories as homotopy models (Grothendieck, 1991a, pp. 2–4 of the edited letter). These are statements by Grothendieck about his program; they do not establish a retrospective priority claim for every later theorem. Heller's independently formulated homotopy theories already organized homotopy categories of diagrams, homotopy Kan extensions, and pointed operations in a related axiomatic framework (Heller, 1988, Chs. II–III).

The terminology on the next pages must be read provisionally. The superscript $\circ$ on Dia reverses both functors and transformations. A transformation $\alpha : f \Rightarrow g$ induces $\alpha^* : g^* \Rightarrow f^*$ because a presheaf sends $\alpha_i : f(i) \to g(i)$ to $F(g(i)) \to F(f(i))$. In modern notation the input 2-category would therefore be $\mathsf{Dia}^{\mathrm{coop}}$. A common modern convention uses covariant diagrams and writes $\mathscr{D} : \mathsf{Cat}^{\mathrm{op}} \to \mathsf{CAT}$, reversing functors but retaining the direction of 2-cells. The translation is $\mathscr{D}(I) = \mathbb{D}(I^{\mathrm{op}})$. A left derivator in this manuscript has right Kan extensions; this is the terminology of left exactness, not that of a left adjoint. The preliminary labels Der_1, Der_2, and Der_3 will also be reassigned when the systematic axioms begin.

The small correction about posets is substantive but completely local. A category is a poset precisely when it is thin and skeletal: each Hom set has at most one member, and isomorphic objects are equal. Having only monomorphisms and no nonidentity isomorphisms does not imply thinness. The parallel-pair category printed in the editors' note is a counterexample, since every composable arrow that could test cancellation is an identity. No later construction depends on the incorrect characterization.

Un *dérivateur* $\mathbb{D}$ (sur Dia, comme "*domaine*" du dérivateur) est un 2-foncteur contravariant de Dia dans $\mathcal{C}at$, satisfaisant certaines conditions. Donc

$$\mathbb{D} : \mathsf{Dia}^{\circ} \longrightarrow \mathcal{C}at \quad .$$

Si $I \in \mathsf{Ob}_0\mathsf{Dia}$ est une catégorie (ou un "type de diagramme") du domaine de $\mathbb{D}$, on doit donc avoir une catégorie $\mathbb{D}(I)$. Les objets de $\mathbb{D}(I)$ s'appelleront les *objets-coefficients* (ou *coefficients*) de type $\mathbb{D}$, au dessus de I. Pour

$$f : I \longrightarrow J$$

1. N.B. Il faut des conditions de stabilité pour Dia, au moins la stabilité par $I \times J$ et $I \amalg J$, en fait, nettement plus …

une 1-flèche de Dia, on doit avoir un foncteur

$$f^*_{\mathbb{D}} \text{ ou } f^* : \mathbb{D}(J) \longrightarrow \mathbb{D}(I) \quad ,$$

dit *"foncteur image inverse"*. Si F est un coefficient de type $\mathbb{D}$ sur J son "*image inverse*" est donc un coefficient de type $\mathbb{D}$ sur I. On suppose (pour simplifier!) que le 2-foncteur $\mathbb{D}$ est *strict*, donc qu'on a transitivité stricte

$$(gf)^* = f^* g^* \ , \qquad \text{si} \qquad I \xrightarrow{f} J \xrightarrow{g} K \qquad \text{dans Dia} \ .$$

De plus, si on a deux foncteurs $f, g : I \rightrightarrows J$, et un morphisme entre eux [page 5]

$$I \underset{g}{\overset{f}{\underset{\downarrow u}{\overset{\downarrow u}{\rightrightarrows}}}} J \quad ,$$

on se donne un morphisme $u^*_{\mathbb{D}}$ ou u^* entre les foncteurs g^* et f^*, d'où

$$\mathbb{D}(I) \underset{g^*}{\overset{f^*}{\underset{\uparrow u^*}{\overset{\uparrow u^*}{\leftrightarrows}}}} \mathbb{D}(J) \quad .$$

On doit encore avoir transitivité pour une composition de morphismes de foncteurs : p. 5

$$I \underset{h}{\overset{f}{\underset{g \downarrow v}{\overset{\downarrow u}{\rightrightarrows}}}} J$$

doit donner

$$(vu)^* = u^* v^*$$

donc

$$\mathbb{D}(I) \underset{h^*}{\overset{f^*}{\underset{g^* \uparrow v^*}{\overset{\uparrow u^*}{\leftrightarrows}}}} \mathbb{D}(J) \quad .$$

De plus, si f est une identité, f^* aussi

$$(\mathsf{id}_I)^* = \mathsf{id}_{\mathbb{D}(I)} \quad ,$$

et de même si u est une identité

$$(\mathsf{id}_f)^* = \mathsf{id}_{f^*} \quad .$$

J'indique tout de suite l'axiome le plus important sur un dérivateur (au-delà de la donnée [page 6]
d'un 2-contra foncteur strict) :

Der$_1$: Si $f : I \to J$ dans Dia, alors $f^*_{\mathbb{D}} = f^* : \mathbb{D}(J) \to \mathbb{D}(I)$ a un adjoint à droite et un adjoint à gauche.

Ces foncteurs seront notés (du moins provisoirement) $f^{\mathbb{D}}_*$ ou f_* , et $f^{\mathbb{D}}_!$ ou $f_!$, soit

$$f_! \, , \ f_* : \mathbb{D}(I) \rightrightarrows \mathbb{D}(J) \quad ,$$

de sorte qu'on trouve une suite de *trois* foncteurs adjoints entre $\mathbb{D}(I)$ et $\mathbb{D}(J)$

$$f_! \, , \ f^* , \ f_* \quad .$$

N.B. Pour la plupart des développements on n'aura pas besoin de l'existence à la fois des $f_!$ et des f_* . Un *dérivateur à gauche* ne suppose que les f_* (adjoints à droite des f^*), et [page 7]
l'existence de $f_!$ seulement quand f est une *immersion ouverte*. Dualement, un *dérivateur à* p. 6

droite ne suppose que les $f_!$ (adjoints à gauche des f^*), et f_* seulement si f est une *immersion fermée*. Ce sont sans doute les dérivateurs à gauche qui sont les plus importants.

Der$_2$: Si $f : I \to J$ est dans Dia, si c'est une immersion *fermée*, le biadjoint à droite $f^!_{\mathbb{D}}$ ou $f^!$ de f^* existe. Si c'est une immersion ouverte, le biadjoint à gauche $f^?$ existe. [2] Dans les deux cas, on veut que f^* soit un foncteur de localisation, ce qui équivaut à dire que f_* est pleinement fidèle, ou que $f_!$ est pleinement fidèle. [3]

N.B. Dans le cas d'un dérivateur à gauche, on suppose seulement l'existence de $f^!$ si f est une immersion fermée. Donc pour résumer, pour un tel dérivateur, on suppose l'existence de

$$\begin{cases} f_* \text{ quel que soit } f : I \to J \text{ dans Dia} \\ f_! \text{ si } f \text{ est une immersion ouverte} \\ f^! \text{ si } f \text{ est une immersion fermée (cas ponctué seulement).} \end{cases}$$

Dualement, pour un dérivateur à droite, on suppose l'existence de

$$\begin{cases} f_! \text{ quel que soit } f : I \to J \text{ dans Dia} \\ f_* \text{ si } f \text{ est une immersion fermée} \\ f^? \text{ si } f \text{ est une immersion ouverte (cas ponctué seulement).} \end{cases}$$

Je voudrais formuler des axiomes astucieux, pour le cas d'un couple

$$Y \overset{i}{\hookrightarrow} I \overset{j}{\hookleftarrow} U$$

d'une immersion ouverte j et d'une immersion fermée i dans Dia, complémentaires l'une de l'autre. D'abord une formulation si on suppose $\mathbb{D}$ "*ponctué*", *i.e.* les catégories $\mathbb{D}(I)$ ponctuées, et les foncteurs f^* compatibles avec les ponctuations. On veut "interpréter" le carré (1) page 2 :

[page 8]

Commentary · Exposition [ms. 7]

Coherent squares and what the axioms remember

The passage from a square in $\mathbb{D}(I)$ to an object of $\mathbb{D}(I \times [1] \times [1])$ is the first genuinely homotopical issue. The underlying-diagram functor discards part of the data. A commutative square in a homotopy category records equality of two homotopy classes; a coherent square also records the homotopy implementing that equality, together with whatever compatibility is required in a larger diagram. It is therefore the coherent object to which a homotopy pullback or pushout operation applies.

Put $\square = [1] \times [1]$ and let W be the full subcategory obtained by deleting $(1, 1)$. For presheaves, W^{op} is a cospan. A coherent square $X \in \mathbb{D}(I \times \square)$ is cartesian if it is in the essential image of right Kan extension along $I \times W \hookrightarrow I \times \square$. Equivalently, its unit comparison with that right Kan extension is an isomorphism. The opposite corner and left Kan extension define cocartesian squares. These definitions use the universal property of an adjoint inside the whole family of diagram categories, rather than an ordinary pullback in $\mathbb{D}(I)$. In the represented derivator of a complete category they reduce to ordinary cartesian and cocartesian squares.

The desire to lift diagrams does not say that every diagram in $\mathbb{D}(I)$ has a preferred lift. A frequently imposed additional axiom, strongness, asks that the underlying arrow functor $\mathbb{D}(I \times [1]) \to \mathbb{D}(I)^{[1]^{\mathrm{op}}}$ be full and essentially surjective. Even that says neither that the lift is unique nor that the functor is faithful. The special support squares have more structure: they arise from restrictions, adjunction units or counits, and zero extensions. Those operations give the coherent constructions that an arbitrary incoherent square lacks.

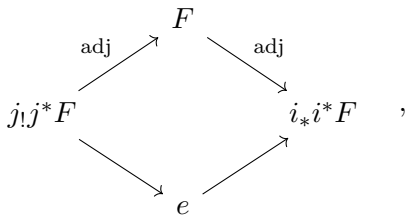
In the modern full derivator axioms, pointedness can be required only at $\mathbb{D}(e)$. It then propagates to all diagram categories, and the exceptional adjoints required here follow; this is proved in Groth

2. Il n'y a lieu d'exiger les biadjoints que pour un dérivateur ponctué. HOT n'y satisfait pas.

3. Il faut de plus caractériser l'image de f_* si f est une immersion fermée (les F tels que $g^*F \xrightarrow{\sim} e_{\mathbb{D}(U)}$) et celle de $f_!$ si c'est une immersion ouverte (les F tels que $g^*F \simeq \emptyset_{\mathbb{D}(Y)}$), où g est l'immersion complémentaire.

(2013, Proposition 3.2 and Corollary 3.8). Thus the manuscript's extra-adjoint requirement was a useful way to expose the intended geometry, but is not an additional independent condition on an already pointed derivator. This does not justify imposing exceptional adjoints on the unpointed derivator of spaces: the initial and terminal spaces differ. The qualification in footnote 2 is necessary.

(1)



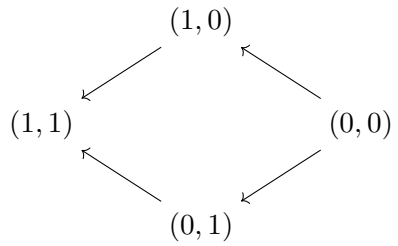
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Der′$_2$: Pour tous $I, J \in \mathsf{Dia}$, on a

$$\mathbb{D}(I \amalg J) \xrightarrow[(\alpha^*,\beta^*)]{\sim} \mathbb{D}(I) \times \mathbb{D}(J) \quad ,$$

où α, β sont les injections canoniques de I et J dans $I \amalg J$.

carré à la fois cartésien et cocartésien. On cherche donc un carré dans $\mathbb{D}(I)$, mais mieux que ça. Soit $J = \mathbf{\Delta}_1 \times \mathbf{\Delta}_1$ le carré standard p. 7



et considérons la catégorie $I \times J$. On a

$$\begin{aligned} J &\longrightarrow \underline{\mathsf{Hom}}(I, I \times J) \\ j &\longmapsto \big(i \longmapsto (i, j)\big) \end{aligned}$$

D'autre part, par $\mathbb{D}$

$$\underline{\mathsf{Hom}}(I, I \times J)^\circ \longrightarrow \underline{\mathsf{Hom}}(\mathbb{D}(I \times J), \mathbb{D}(I))$$

d'où un composé

$$J^\circ \longrightarrow \underline{\mathsf{Hom}}(\mathbb{D}(I \times J), \mathbb{D}(I))$$

ou encore

$$\mathbb{D}(I \times J) \longrightarrow \underline{\mathsf{Hom}}(J^\circ, \mathbb{D}(I)) \quad .$$

Ceci vaut pour tout $J \in \mathsf{Ob}\,\mathsf{Dia}$. On peut donc dire que la donnée d'un "coefficient" Φ sur une catégorie produit $I \times J$ donne naissance à un diagramme de type J° dans la catégorie $\mathbb{D}(I)$ des coefficients sur I. La philosophie des dérivateurs consiste en grande partie, inversement, chaque fois qu'on a un diagramme dans une catégorie de coefficients $\mathbb{D}(I)$ (sur un diagramme I), disons diagramme de type $K : K \to \mathbb{D}(I)$, de se demander s'il ne provient pas "canoniquement" d'un objet de $\mathbb{D}(I \times K^\circ)$ *i.e.* d'un coefficient sur $I \times K^\circ$. [page 9]

Ainsi pour le carré p. 8. On voudrait qu'il provienne de $\mathbb{D}(I \times J)$, où $J = \mathbf{\Delta}_1 \times \mathbf{\Delta}_1$, et ceci fonctoriellement, donc on désire un foncteur "canonique"

$$\mathbb{D}(I) \longrightarrow \mathbb{D}(I \times \mathbf{\Delta}_1 \times \mathbf{\Delta}_1)$$

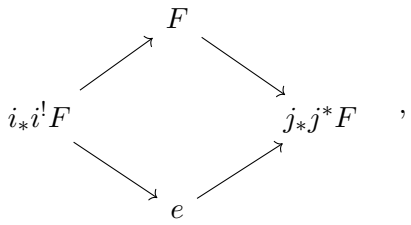
p. 8

qui en composant avec

$$\mathbb{D}(I \times \mathbf{\Delta}_1 \times \mathbf{\Delta}_1) \longrightarrow \underline{\mathsf{Hom}}((\mathbf{\Delta}_1 \times \mathbf{\Delta}_1)^\circ, \mathbb{D}(I))$$

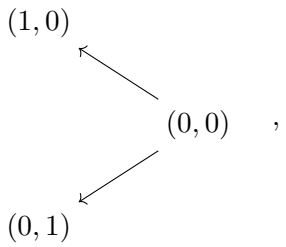
donne le foncteur décrit par le carré (1). De plus, il faut exprimer dans le contexte présent la condition "bicartésienne" du carré (1), dans le contexte de la page 7. Question analogue pour le carré (2) de la page 2 :

(2)



,

où il faut de plus interpréter la condition cartésienne. On va définir en fait un coefficient sur la sous-catégorie ouverte W suivante de $\mathbf{\Delta}_1 \times \mathbf{\Delta}_1$



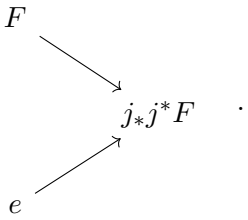
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plus exactement sur $W \times I$, et si α

[page 10]

$$\alpha : W \times I \longrightarrow \mathbf{\Delta}_1 \times \mathbf{\Delta}_1 \times I$$

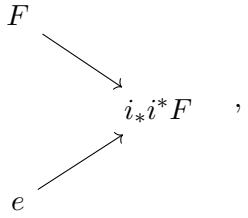
est l'inclusion canonique, on définira le carré "cartésien" comme l'image par α_* de ce coefficient. Ceci nous ramène donc au problème de "globaliser" d'abord le diagramme partiel



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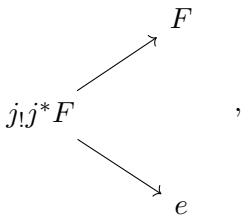
Et de même pour le carré (1) sauf qu'on est embarrassé si on choisit de le regarder surtout comme cartésien (ce qui suggère à procéder comme tantôt) ou comme <u>co</u>cartésien (ce qui suggère de procéder de façon duale). Mais les deux coefficients qu'on trouvera par ces deux procédés, produits soit de

p. 9



,

soit de



,

devraient être *canoniquement* isomorphes.
Si $\mathbb{D}$ est un dérivateur, la catégorie [page 11]

$$\mathcal{A} = A_{\mathbb{D}} = \mathbb{D}(e)$$

joue un rôle important. On l'appelle la *catégorie de base,* ou *la catégorie fondamentale*, du dérivateur, ces objets sont les *coefficients fondamentaux.* Pour I quelconque dans Dia, on a donc pour $F \in \mathbb{D}(I)$, coefficient sur I

$$I^\circ \longrightarrow \mathcal{A} \quad , \qquad i \longmapsto F_i$$

un diagramme de type I° dans $\mathcal{A}$, fonctoriellement, *i.e.*

$$\mathbb{D}(I) \longrightarrow \underline{\mathsf{Hom}}(I^\circ, \underset{\mathbb{D}(e)}{\mathcal{A}})$$

(*cf.* page 8).
Der$_3$: Pour tout $I \in \mathsf{Dia}$, ce foncteur est *conservatif,*[4] *i.e.* pour que

$$F \longrightarrow G$$

dans $\mathbb{D}(I)$ soit [un] isomorphisme, il faut et il suffit que pour tout $i \in I$, p. 10

$$F_i \longrightarrow G_i$$

le soit.
Corollaire. *Le foncteur canonique (pour I, $J \in \mathsf{Dia}$)*

$$\mathbb{D}(I \times J) \xrightarrow{\theta} \underline{\mathsf{Hom}}(J^\circ, \mathbb{D}(I))$$

est conservatif.

En effet, si F est dans $\mathbb{D}(I \times J)$, on a pour $(i, j) \in I \times J$

$$F_{i,j} \simeq \theta(F)(j)_i \quad .$$

Intuitivement, on regarde un dérivateur $\mathbb{D}$ comme "étant" la catégorie fondamentale $\mathcal{A}$, avec une "structure" supplémentaire, [à] savoir la donnée des "extensions" de $\mathcal{A}$ (comme catégorie [page 12] des coefficients ponctuels) aux diagrammes $I \in \mathsf{Dia}$, de façon contravariante en I. Un coefficient F dans $\mathbb{D}(I) \overset{\text{déf}}{=} \mathcal{A}_I$ est regardé comme "étant" le diagramme $(F_i)_{i \in I}$ correspondant d'objets fondamentaux, avec une donnée supplémentaire de "globalisation". Les F_i, ou "fibres" de F en les $i \in I$, sont les caractéristiques locaux, du coefficient "global sur I" F.

Commentary · Examples [ms. 11]
Conservative is not faithful

The conservativity axiom detects whether a *given coherent morphism* is invertible. It does not reconstruct that morphism from its components. A small derived example makes the distinction exact. In the abelian category of arrows $X_1 \to X_0$, put

$$P = (\mathbb{Z} \to 0), \qquad Q = (0 \to \mathbb{Z}), \qquad E = (\mathbb{Z} \xrightarrow{1} \mathbb{Z}).$$

The nonsplit sequence

$$0 \longrightarrow Q \longrightarrow E \longrightarrow P \longrightarrow 0$$

defines a nonzero element of $\mathrm{Ext}^1(P, Q)$, and hence a nonzero morphism $P \to Q[1]$ in the derived category of arrows. Its two evaluated components are $\mathbb{Z} \to 0[1]$ and $0 \to \mathbb{Z}[1]$, both zero. This is

4. N.B. Mais ce foncteur n'a aucune envie d'être *fidèle.*

the same nonsplitting calculated in the support example, now viewed as a morphism in a derived category. It exhibits exactly the information lost by

$$D(\mathsf{Ab}^{[1]^{\mathrm{op}}}) \longrightarrow D(\mathsf{Ab})^{[1]^{\mathrm{op}}}.$$

There is no contradiction with conservativity. The nonzero extension class is not a morphism all of whose components are isomorphisms.

The corollary for partial underlying diagrams is proved in the text, not left as a gap. If a coherent morphism over $I \times J$ becomes an isomorphism at every j in $\mathbb{D}(I)$, evaluating again at every i gives an isomorphism at each (i, j). The original conservativity axiom on $I \times J$ now applies. In particular, a shifted family retains the same test for invertibility. This short argument does not need strongness, full faithfulness, a model structure, or a choice of representatives.

Tous les dérivateurs connus sont obtenus de la façon standard suivante. On part d'une "catégorie de modèles"

$$\mathcal{M} \quad , \quad \Sigma$$

munie d'un ensemble de flèches, appelées "équivalences", et satisfaisant aux axiomes pertinents pour le passage à une catégorie des fractions

$$\mathcal{A} = \mathcal{M}\Sigma^{-1}$$

qui sera la "catégorie fondamentale". On définit pour tout I dans Dia

$$\begin{aligned} \mathcal{M}_I &= \underline{\mathsf{Hom}}(I^\circ, \mathcal{M}) \\ \Sigma_I &= \{u \in \mathsf{Fl}(\mathcal{M}_I) \mid \forall i \in I, u_i \in \Sigma\} \\ \mathcal{A}_I &= \mathcal{M}_I \Sigma_I^{-1} \end{aligned}$$

Il est clair que $\mathcal{A}_I \overset{\text{déf}}{=} \mathbb{D}_{\mathcal{M},\Sigma}(I)$ dépend de I de façon fonctorielle contravariante, et même de façon 2-fonctorielle. [page 13]

Bien entendu, pour avoir vraiment un dérivateur, il faut vérifier les axiomes. Le plus fort est l'existence des $f_!$, f_*. Elle est prouvée par Thomason lorsque $(\mathcal{M}, \Sigma)$ provient d'une "catégorie de modèles" au sens de Quillen. Ce n'est pas clair pour moi qu'alors $(\mathcal{M}_I, \Sigma_I)$ provienne également d'une catégorie de modèles (dont par exemple les fibrations et les cofibrations seraient définies à partir de ces données dans $\mathcal{M}$, comme tantôt pour les équivalences). p. 11

Quand on prend pour $(\mathcal{M}, \Sigma)$ une catégorie de modèles pour Hot (par exemple les ensembles semi-simpliciaux $\mathcal{M} = \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \mathcal{E}ns)$, d'où $\mathcal{M}_I \simeq \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ \times I^\circ, \mathcal{E}ns) \simeq \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \underbrace{\underline{\mathsf{Hom}}(I^\circ, \mathcal{E}ns)}_{\widehat{I}})$, catégorie des objets semi-simpliciaux dans le topos $\widehat{I}$, on trouve une description d'un dérivateur (non additif), qui jouera un rôle particulièrement important. Je le désigne par

$$\mathsf{HOT} \quad ,$$

sa catégorie des coefficients fondamentaux est Hot, la catégorie homotopique standard.

Commentary · Later developments [ms. 13]

Localizing diagrams and deriving their Kan extensions

For a relative category $(\mathcal{M}, W)$, objectwise weak equivalences are preserved by every restriction. Thus localization supplies the displayed prederivator, subject to the usual size convention ensuring that its Hom classes are sets. Naturality of the localized transformations follows from the universal property of localization. One may use a functorial localization construction to keep restriction strictly functorial. Arbitrary independent choices of localized categories instead give canonically equivalent data with coherence isomorphisms.

The difficult assertion is the existence and pointwise behavior of the derived Kan extensions. Grothendieck attributes the model-category case to Thomason here, without specifying a publication.

We have not identified a published Thomason theorem with exactly the generality of this attribution. A precise later source is Cisinski (2003, Theorem 6.11): a Quillen model category admitting small limits gives a left derivator in the presheaf convention. Applying the result to the opposite model category gives the dual assertion when small colimits exist. The proof also establishes local smallness of the diagram localizations. This settles the mathematical existence question without requiring an ordinary Quillen model structure on every diagram category.

For a combinatorial model category there is a familiar more direct route. The projective diagram model structure has objectwise fibrations and weak equivalences; the injective structure has objectwise cofibrations and weak equivalences. Their homotopy categories are the same localization, although their replacements differ. Projective replacements derive left Kan extension, while injective replacements derive right Kan extension. In general one cannot declare *both* fibrations and cofibrations objectwise and expect the lifting axioms. The passage from models to presheaves is treated in a more flexible formalism by Cisinski (2010, Theorem 6.22 and Corollary 6.24).

For the indicated model of spaces, the exponential law gives

$$(\mathsf{sSet})^{I^{\mathrm{op}}} \cong \mathsf{Set}^{\Delta^{\mathrm{op}} \times I^{\mathrm{op}}} \cong (\mathsf{Set}^{I^{\mathrm{op}}})^{\Delta^{\mathrm{op}}}.$$

The weak equivalences in this construction are evaluated at each i. They are not a newly imposed local weak-equivalence relation for an arbitrary Grothendieck topology. Also, "semi-simplicial" is historical terminology here: the displayed Δ includes degeneracies and gives what is now called a simplicial set. Replacing it silently by the category of injective order maps would change the model category under discussion.

La notion de "morphisme de Dérivateurs" est claire, c'est un morphisme entre deux 2-foncteurs stricts [page 14]

$$\mathsf{Dia}^{\circ} \longrightarrow \mathcal{C}at$$

Donc

$$u : \mathbb{D} \longrightarrow \mathbb{D}'$$

est défini par la donnée, pour tout $I \in \mathsf{Dia}$, de

$$u_I : \mathbb{D}(I) \longrightarrow \mathbb{D}'(I)$$

avec commutativité (stricte?) pour $f : I \to J$

$$\begin{array}{ccc} \mathbb{D}(J) & \xrightarrow{f^*_{\mathbb{D}}} & \mathbb{D}(I) \\ {\scriptstyle u_J}\downarrow & & \downarrow{\scriptstyle u_I} \\ \mathbb{D}'(J) & \xrightarrow{f^*_{\mathbb{D}'}} & \mathbb{D}'(I) \end{array} \quad . \tag{1}$$

On a

$$\begin{array}{ccc} \underline{\mathsf{Hom}}(I,J)^{\circ} & \xrightarrow{\mathbb{D}_{I,J}} & \underline{\mathsf{Hom}}(\mathbb{D}(J),\mathbb{D}(I)) \\ {\scriptstyle \mathbb{D}'_{I,J}}\downarrow & & \downarrow{\scriptstyle \phi \mapsto u_I \circ \phi} \\ \underline{\mathsf{Hom}}(\mathbb{D}'(J),\mathbb{D}'(I)) & \xrightarrow[\psi \mapsto \psi \circ u_J]{} & \underline{\mathsf{Hom}}(\mathbb{D}(J),\mathbb{D}'(I)) \end{array} \tag{2}$$

et la commutativité de (1) signifie la commutativité de (2) *pour les objets*. Mais bien sûr on la veut aussi pour les flèches, *i.e.* on veut la commutativité du diagramme (2). [5]

Sans doute on veut un peu plus sur les morphismes de dérivateurs - on veut une commutation aux f_* et (?) aux $f_!$ (cette fois seulement à isomorphisme canonique près) ... [6]

5. Donner l'exemple de morphisme de dérivateurs défini par un foncteur de catégories de modèles $(\mathcal{M}, \Sigma) \to (\mathcal{M}', \Sigma')$.

6. Commutation à f_* (resp. $f_!$) morphisme *exact à gauche* (resp. *exact à droite*). Les morphismes *exacts* sont ceux à la fois exacts à gauche et à droite.

Commentary · Exposition [ms. 14]

Morphisms and their comparison maps

A morphism between the entire families of coefficient categories must respect restriction, including its dependence on natural transformations of indexing functors. The square (2) is the condition that is lost if one checks only the individual functors in square (1). In modern treatments one normally allows coherent invertible comparison maps

$$\gamma_f : f^*_{\mathbb{D}'} u_J \xrightarrow{\sim} u_I f^*_{\mathbb{D}}$$

rather than demanding strict equality. The coherence conditions concern identity functors, composites, and natural transformations. Modifications between such pseudonatural morphisms provide the 2-cells. A functor of relative categories preserving weak equivalences gives an example by applying it at every component and then localizing; no preservation of Kan extensions follows merely from this construction.

Adjunction does give canonical comparison maps, with precisely the orientations on manuscript pp. 15–16. For the right Kan extension the map is the composite

$$\begin{aligned} u_J f_*^{\mathbb{D}} &\longrightarrow f_*^{\mathbb{D}'} f^*_{\mathbb{D}'} u_J f_*^{\mathbb{D}} \\ &\xrightarrow{f_*^{\mathbb{D}'} \gamma_f f_*^{\mathbb{D}}} f_*^{\mathbb{D}'} u_I f^*_{\mathbb{D}} f_*^{\mathbb{D}} \longrightarrow f_*^{\mathbb{D}'} u_I . \end{aligned}$$

The outer arrows are the unit in $\mathbb{D}'$ and the counit in $\mathbb{D}$. The left Kan comparison $f_!^{\mathbb{D}'} u_I \to u_J f_!^{\mathbb{D}}$ is the adjoint of

$$u_I \longrightarrow u_I f^*_{\mathbb{D}} f_!^{\mathbb{D}} \xrightarrow{\gamma_f^{-1}} f^*_{\mathbb{D}'} u_J f_!^{\mathbb{D}} .$$

An invertible restriction comparison need not have an invertible mate. Preservation of limits or colimits is the additional assertion that the corresponding mates are invertible. These definitions also explain which comparison is meant; an unspecified isomorphism of the two functors is not sufficient.

In the manuscript, left and right exactness at this point mean preservation of all the relevant right and left Kan extensions. Modern "exact morphism of stable derivators" usually refers to finite homotopy limits and colimits. Stability makes the two finite notions coincide; it does not equate preservation of arbitrary products and coproducts. For example, derived tensor product with $\mathbb{Q}$ is exact on $D(\mathbb{Z})$ and preserves coproducts. The map

$$\mathbb{Q} \otimes \prod_{n \geq 1} \mathbb{Z} \longrightarrow \prod_{n \geq 1} \mathbb{Q}$$

is not surjective: an element on the left has a common denominator, whereas $(1/n!)_{n \geq 1}$ does not. The footnote about triangulated categories is correct in its finite setting and should not be promoted to an assertion about arbitrary Kan extensions. For model categories, a right Quillen adjoint yields a derived morphism preserving homotopy limits, with the dual statement for a left Quillen adjoint; Cisinski (2003, Proposition 6.12 and its dual) provide the relevant construction.

p. 12
[page 15]

Par exemple, considérons le dérivateur

$$\mathsf{HOTAB}$$

dont la catégorie fondamentale est Hotab, la sous-catégorie pleine de la catégorie dérivée $\mathsf{Der}(\mathsf{Ab})$ de Ab, formée des complexes à cohomologie nulle en degré > 0. On peut décrire HOTAB par la catégorie de modèles $\underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \mathsf{Ab})$ des groupes abéliens semi-simpliciaux, avec comme équivalences les quasi-isomorphismes au sens habituel. Le "foncteur abélianisation" canonique

$$\mathsf{Hot} \longrightarrow \mathsf{Hotab}$$

se prolonge en un morphisme de dérivateurs

$$\mathsf{HOT} \longrightarrow \mathsf{HOTAB} \quad .$$

La question de sa compatibilité aux f_* ne semble pas tautologique. Elle impliquerait, par exemple, que si

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[j]{} & Y' \end{array}$$

est un diagramme cartésien dans $\mathcal{M} = \underline{\mathsf{Hom}}(\mathbf{\Delta}^\circ, \mathcal{E}ns)$ avec f une fibration (de Thom dans le cadre des espaces topologiques, avec f par exemple fibration localement triviale) alors le diagramme de Hotab

$$\begin{array}{ccc} \mathbb{R}\Gamma_!(X) & \longleftarrow & \mathbb{R}\Gamma_!(X') \\ \downarrow & & \downarrow \\ \mathbb{R}\Gamma_!(Y) & \longleftarrow & \mathbb{R}\Gamma_!(Y') \end{array}$$

est hot-cartésien[7]. Et de même pour une hot-somme amalgamée (carrée hot-cocartésien dans Hot).
Si c'est vrai, et pas connu de plus, ce serait quand même scandaleux! [page 16]

Commentary · Examples [ms. 15]

Abelianization preserves homotopy colimits

The model for HOTAB is simplicial abelian groups. Normalization identifies these with nonnegative chain complexes, or equivalently cochain complexes in nonpositive degrees; see The Stacks Project Authors (2026c, Theorem 14.24.3, Tag 019D). Its homotopy category is $D^{\leq 0}(\mathsf{Ab})$. This is an additive, pointed theory, but it is not stable: suspension need not be invertible inside the connective category. Consequently one must not infer preservation of homotopy limits from an additive appearance.

The morphism from spaces is induced by the degreewise free abelian group functor $\mathbb{Z}[-]$. It is left adjoint to the forgetful functor on simplicial objects. The latter preserves fibrations and trivial fibrations in the standard model structures, so this is a Quillen adjunction. Its derived left adjoint preserves homotopy colimits, including the homotopy pushouts in the last sentence of manuscript p. 15. That sentence therefore has a positive answer, although the preceding pullback assertion does not. Under normalization, this is the usual chain construction, with its coherent diagram version.

A finite discrete counterexample makes the failure of the pullback assertion explicit. Let $X = Y' = \{0, 1\}$ and $Y = *$, so $X' = X \times Y'$ has four points. All objects are Kan complexes and the maps to the point are fibrations. After abelianization the square has upper left and lower right terms $\mathbb{Z}^2$, lower left term $\mathbb{Z}$, and upper right term $\mathbb{Z}^4$. Each map $\mathbb{Z}^2 \to \mathbb{Z}$ is augmentation. Since these maps are surjective, their ordinary pullback computes their homotopy pullback in simplicial abelian groups, and it is

$$\{(a_0, a_1, b_0, b_1) \in \mathbb{Z}^4 : a_0 + a_1 = b_0 + b_1\} \cong \mathbb{Z}^3.$$

The comparison from the chains of X' sends the basis vector e_{ij} to (e_i, e_j). Its kernel is generated by

$$e_{00} - e_{01} - e_{10} + e_{11}.$$

It is therefore not an isomorphism. This proves the correction in footnote 7 while keeping separate the valid colimit statement. Already the terminal-object test would rule out continuity, since $\mathbb{Z}[*] = \mathbb{Z}$ and the terminal simplicial abelian group is zero, but the displayed example addresses exactly the nonempty square that Grothendieck is discussing.

En procédant comme pour les morphismes de changements de base pour les Rf_*, on trouve en tout cas un morphisme canonique, pour $f : I \to J$

$$u_J f_*^{\mathbb{D}} \longrightarrow f_*^{\mathbb{D}'} u_I$$

7. C'est faux déjà si Y est l'espace ponctuel : trivialement le $\mathbb{R}\Gamma_!$ ne transforme pas produits en sommes! Ce foncteur est sûrement [pas] "exact à gauche" (comme morphisme de dérivateurs).

et de même p. 13

$$f_!^{\mathbb{D}'} u_I \longrightarrow u_J f_!^{\mathbb{D}} \quad ,$$

et la question est de savoir si c'est des isomorphismes. Questions analogues pour les $f^!$, pour f immersion fermée (ou pour $f^?$, pour f immersion ouverte). Comme la notion de morphisme de dérivateurs devrait quand même "coller" à la notion bien connue de morphisme (triangulé) de catégories triangulées [8], donc être exact en un sens convenable, cela montre bien que l'on doit exiger la commutation d'un morphisme de dérivateurs aux f_* et aux $f_!$. [9] Je devrais demander à Thomason s'il a regardé, pour un morphisme

$$\mathcal{M} \longrightarrow \mathcal{M}'$$

entre catégories de modèles, qui soit disons *exact à gauche*, et "compatible" avec les équivalences, fibrations (et cofibrations (???)).

Notation : DER pour la 2-catégorie des *dérivateurs.*

Dualité. Si $\mathbb{D}$ est un dérivateur, on va définir un dérivateur *dual* $\mathbb{D}^\circ$ par [page 17]

$$\mathbb{D}^\circ(I) \overset{\text{déf}}{=} \mathbb{D}(I^\circ)^\circ \quad .$$

Les coefficients sur I de type $\mathbb{D}^\circ$ sont les coefficient sur la catégorie opposée I° à I de type $\mathbb{D}$, et en tant que catégorie c'est la catégorie *opposée* de celle des coefficients de type $\mathbb{D}$ sur I°. Pour définir la loi 2-fonctorielle, on note que pour I, J dans Dia on a

$$\begin{array}{ccc} \underline{\mathsf{Hom}}(I,J)^\circ & \xrightarrow{\mathbb{D}_{I,J}} & \underline{\mathsf{Hom}}(\mathbb{D}(J),\mathbb{D}(I)) \\ \wr\downarrow \text{dualité} & & \wr\downarrow \text{dualité} \\ \underline{\mathsf{Hom}}(I^\circ,J^\circ) & \xrightarrow{\theta} & \underline{\mathsf{Hom}}(\mathbb{D}(J)^\circ,\mathbb{D}(I)^\circ)^\circ \end{array} \quad ,$$

où θ est l'unique foncteur rendant le diagramme commutatif. Alors θ°est un foncteur

$$\underline{\mathsf{Hom}}(I^\circ,J^\circ)^\circ \longrightarrow \underline{\mathsf{Hom}}(\mathbb{D}(J)^\circ,\mathbb{D}(I)^\circ) \quad ,$$

c'est ça par définition $\mathbb{D}^\circ_{I,J}$ [plutôt $\mathbb{D}^\circ_{I^\circ,J^\circ}$].

Si $\mathbb{D}$ est défini par $(\mathcal{M},\Sigma)$, alors $\mathbb{D}^\circ$ par $(\mathcal{M}^\circ,\Sigma^\circ)$, puisque

$$(\mathcal{M}^\circ)_I \overset{\text{déf}}{=} \underline{\mathsf{Hom}}(I^\circ,\mathcal{M}^\circ) \simeq \underline{\mathsf{Hom}}(I,\mathcal{M})^\circ \simeq (\mathcal{M}_{I^\circ})^\circ$$

et $\Sigma^\circ{}_I$ est déduit de Σ_{I°, d'où p. 14

$$(\mathcal{M}^\circ)_I(\Sigma^\circ{}_I)^{-1} \simeq (\mathcal{M}_{I^\circ}\Sigma_{I^\circ}^{-1})^\circ = \mathbb{D}(I^\circ)^\circ \quad .$$

[page 18]

Commentary · Later developments [ms. 17]

What the whole diagram theory remembers

A modern comparison makes precise the extra information carried by the variable I. For an ∞-category $\mathcal{C}$, define, in the convention of this manuscript,

$$\mathbb{D}_{\mathcal{C}}(I) = h\,\mathrm{Fun}(N(I)^{\mathrm{op}},\mathcal{C}).$$

Here N is the nerve and h takes the homotopy category. One first forms coherent diagrams and then passes to homotopy; reversing these operations would give $\mathrm{Fun}(I^{\mathrm{op}},h\mathcal{C})$, which is generally different. Restriction and its natural transformations supply the prederivator structure. If $\mathcal{C}$ has all small

8. Non! Le cas des catégories triangulées est "misleading", car pour celles-ci l'exactitude à gauche et à droite revient au même.

9. N.B. On conçoit qu'un foncteur entre dérivateurs puisse être exact à gauche ou exact à droite ou les deux.

limits and colimits, its pointwise Kan extensions supply the two adjoints and the comma-square formula. Riehl and Verity (2017, §5.3, especially Proposition 5.3.9 and Remark 5.3.10) explain this passage using the homotopy 2-category of an ∞-cosmos. Their convention uses covariant diagrams; replacing I by I^{op} yields the formula above.

The formula also clarifies the philosophical remark on manuscript page 18. Passing to $\mathbb{D}(e) = h\mathcal{C}$ discards most of the indexing variable. Retaining the entire prederivator can retain much more. Arlin (2020, Theorem 19 and Corollary 20) proves a fully faithful embedding using strict transformations and the specified simplicial enrichment. With pseudonatural transformations, his Theorem 29 gives a bicategorically fully faithful embedding for *small* quasicategories when the domain contains all small categories. The size hypothesis, the domain, and the kind of transformation belong to the theorem; none can be silently omitted. In the model-category setting, Renaudin (2009, Theorems 3.3.2 and 3.4.4) obtains the corresponding local equivalence, and then a biequivalence with derivators of small presentation, from left proper combinatorial model categories localized at Quillen equivalences, with adjunctions as morphisms.

These results concern the entire diagram theory. They do not assert that every functor $h\mathcal{C} \to h\mathcal{E}$ admits a coherent lift, or that a lift is determined by its value at e. Nor does an objectwise description of a prederivator by localized diagram categories dispense with the proof of its Kan-extension axioms. The comparison is useful precisely because it keeps three questions separate: existence of coherent diagrams, existence of homotopy Kan extensions, and reconstruction of an enhanced category from all its diagram categories. Grothendieck's successive shifts $\mathbb{D}_I$ belong to this richer level of information.

Philosophie. Chaque fois qu'on a un foncteur

$$\mathcal{A} \xrightarrow{u_0} \mathcal{A}'$$

entre deux catégories qui peuvent s'interpréter naturellement comme les catégories fondamentales de deux dérivateurs $\mathbb{D}$, $\mathbb{D}'$, il se pose la question d'interpréter u_0 comme provenant d'un morphisme de dérivateurs

$$\mathbb{D} \xrightarrow{u} \mathbb{D}' \quad .$$

Celui-ci est, en pratique, toujours déterminé canoniquement, à isomorphisme canonique près. (Mais je doute pourtant que le foncteur restriction aux catégories fondamentales

$$\underline{\mathsf{Hom}}(\mathbb{D}, \mathbb{D}') \longrightarrow \underline{\mathsf{Hom}}(\mathcal{A}, \mathcal{A}')$$

soit pleinement fidèle …)

Commentary · Examples [ms. 18]

A negative answer to the evaluation question

For unrestricted morphisms of derivators, the doubt about full faithfulness can be answered within the elementary examples already at hand. Let $A = [1]$ and let $\mathscr{R}(I) = \mathrm{Fun}(I^{\mathrm{op}}, A)$. Since the two-element ordered set has all small meets and joins, this represented prederivator is a derivator: its Kan extensions are the ordinary pointwise ones. The represented construction is standard; compare Groth (2013, Examples 1.2(1) and 1.6).

For any prederivator $\mathscr{E}$ in our convention and any $X \in \mathscr{E}(A^{\mathrm{op}})$, define a morphism $U_X : \mathscr{R} \to \mathscr{E}$ as follows. An object $F \in \mathscr{R}(I)$ is a functor $F : I^{\mathrm{op}} \to A$, so $F^{\mathrm{op}} : I \to A^{\mathrm{op}}$, and we set

$$(U_X)_I(F) = (F^{\mathrm{op}})^* X.$$

If $F \Rightarrow G$ is a natural transformation, taking opposites gives $G^{\mathrm{op}} \Rightarrow F^{\mathrm{op}}$; the reversal of 2-cells by $\mathscr{E}$ gives the required map $(U_X)_I(F) \to (U_X)_I(G)$. The composition and restriction laws follow directly from those of $\mathscr{E}$. A map $m : X \to Y$ likewise induces a modification $U_m : U_X \Rightarrow U_Y$ with component $(F^{\mathrm{op}})^* m$ at F.

Now take $\mathscr{E}$ to be the unbounded derived derivator of abelian groups. Its category $\mathscr{E}(A^{\mathrm{op}})$ is the derived category of arrows $0 \to 1$. As in the preceding nonfaithfulness example, the extension

$$0 \to (0 \to \mathbb{Z}) \to (\mathbb{Z} \xrightarrow{1} \mathbb{Z}) \to (\mathbb{Z} \to 0) \to 0$$

yields a nonzero map $m : (\mathbb{Z} \to 0) \to (0 \to \mathbb{Z})[1]$ with both evaluated components zero. Therefore $(U_m)_e$ is the zero natural transformation between the base-category functors. Nevertheless U_m itself is nonzero: at $I = A^{\mathrm{op}}$ and $F = \mathrm{id}_A$, its value is exactly m. Evaluation at e is thus not faithful, and consequently not fully faithful.

This settles the question for the usual category of all pseudonatural morphisms, and already for the strictly natural morphisms just constructed. It does not assert the same counterexample for a subcategory in which every morphism must preserve all left or right Kan extensions. The manuscript is still negotiating that convention. The proof identifies the precise obstruction in the unrestricted case: an entire coherent arrow can carry a morphism which vanishes at each of its two endpoints.

On va appliquer cette philosophie aux foncteurs f^*, $f_!$, f_* (et éventuellement $f^!$, $f^?$), pour $f : I \to J$, donc pour les foncteurs entre $\mathbb{D}(I)$ et $\mathbb{D}(J)$. Cela nous amène donc à regarder $\mathbb{D}(I)$ et $\mathbb{D}(J)$ comme des catégories fondamentales de dérivateurs convenables.

Dérivateur induit par un dérivateur $\mathbb{D}$, sur une catégorie d'indices $I \in \mathsf{Dia}$. C'est un dérivateur $\mathbb{D}_I$ défini par

$$\mathbb{D}_I(J) = \mathbb{D}(I \times J) \quad .$$

Le foncteur associé à $\mathbb{D}_I$, pour deux catégories J, $J' \in \mathsf{Dia}$ [page 19]

$$\underline{\mathsf{Hom}}(J, J')^\circ \xrightarrow{(\mathbb{D}_I)_{J,J'}} \underline{\mathsf{Hom}}(\mathbb{D}_I(J'), \mathbb{D}_I(J)) = \underline{\mathsf{Hom}}(\mathbb{D}(I \times J'), \mathbb{D}(I \times J))$$

est le composé

$$\underline{\mathsf{Hom}}(J, J')^\circ \longrightarrow \underline{\mathsf{Hom}}(I \times J, I \times J')^\circ \xrightarrow{\mathbb{D}_{I\times J, I\times J'}} \underline{\mathsf{Hom}}(\mathbb{D}(I \times J'), \mathbb{D}(I \times J)) \quad ,$$

où la première flèche est le foncteur

$$f \longmapsto \mathsf{id}_I \times f \quad .$$

En d'autres termes, $\mathbb{D}_I$ est le composé

$$\begin{aligned} &\mathsf{Dia}^\circ \longrightarrow \mathsf{Dia}^\circ \xrightarrow{\mathbb{D}} \mathcal{C}at \\ &J \longmapsto I \times J \qquad . \end{aligned}$$

On a

$$f^*_{\mathbb{D}_I} = (\mathsf{id} \times f)^*_{\mathbb{D}} \quad .$$

Si les f_* (resp. les $f_!$) existent pour $\mathbb{D}$, ils existent pour $\mathbb{D}_I$. (De même pour toute espèce de multiadjoints.) Comme $J \mapsto I \times J$ transforme immersion fermée (resp. ouverte) en itou, l'axiome $\mathbf{Der}_2$ (plus précisément, chacun de ces ingrédients, *cf.* annotations marginales p. 7) est stable par passage de $\mathbb{D}$ à $\mathbb{D}_I$. De même pour $\mathbf{Der}_3$. (Pour chaque nouvel axiome qu'on dégagera, il faudra vérifier la stabilité.)

p. 15
[page 20]

Commentary · Exposition [ms. 19]

Duality and the change of coefficient category

The two opposites in $\mathbb{D}^{\mathrm{op}}(I) = \mathbb{D}(I^{\mathrm{op}})^{\mathrm{op}}$ have distinct roles. The first reverses the direction of the indexing arrows; the second reverses arrows between coefficient objects. Both are needed to obtain another prederivator with the same variance convention. In a model presentation this is the canonical identification

$$\mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M}^{\mathrm{op}}) \cong \mathrm{Fun}(I, \mathcal{M})^{\mathrm{op}},$$

followed by localization. Localization commutes with taking opposites by its universal property: a functor inverting the selected arrows on either side is exactly the opposite of such a functor on the other side. Thus the dual model statement is proved by the displayed argument. The incorrect indices in the definition of the dual action on Hom categories are already corrected by the editors.

The shift $\mathbb{D}_I(J) = \mathbb{D}(I \times J)$ changes which objects count as coefficients. An entire coherent I-diagram now becomes a single coefficient, and a coherent J-diagram of such coefficients is an object over $I \times J$. At the level of a relative category this is the elementary exponential law

$$(\mathcal{M}^{I^{\mathrm{op}}})^{J^{\mathrm{op}}} \cong \mathcal{M}^{(I\times J)^{\mathrm{op}}},$$

with exactly the same objectwise weak equivalences. This is why the shift is intrinsic to the proposed language: it permits an argument about one coefficient to be reapplied to a family with parameters.

The preliminary stability checks on p. 19 are valid for the axioms already written. Products preserve sieves and cosieves. Adjoints to $(\mathrm{id}_I \times f)^*$ are supplied by the corresponding adjoints in $\mathbb{D}$, and the conservativity argument uses evaluation at each (i, j). The full pointwise Kan-extension axiom still needs a separate verification. In the completed theory this is the shift theorem, Groth (2013, Theorem 1.25); it is not an unproved additional assumption about an arbitrary family.

Likewise the candidate components $(f \times \mathrm{id}_J)_!$ and $(f \times \mathrm{id}_J)_*$ on p. 20 do not constitute a morphism of derivators merely by being written down. One must show that their comparison maps for changes of J are invertible and satisfy coherence. The product base-change squares supply this, and the coherence follows from the calculus of mates. Restriction is strictly natural before choices of identifications, whereas adjoint extensions are naturally pseudonatural. The following pages are devoted to precisely these exchange maps. It is appropriate to call the present construction a sketch awaiting those checks, not an error in the definition of the shift.

Soit

$$f : I \longrightarrow I' \quad .$$

Considérons entre $\mathbb{D}(I)$, $\mathbb{D}(I')$ les foncteurs

$$\mathbb{D}(I) \;\underset{\xrightarrow[f_*]{}}{\overset{\xrightarrow{f_!}}{\xleftarrow{f^*}}}\; \mathbb{D}(I')$$

(sous réserve d'existence des f_* resp. $f_!$). Peut on les interpréter en termes de morphismes de dérivateurs

$$\mathbb{D}_I \;\underset{\xrightarrow[\overline{f_*}]{}}{\overset{\xrightarrow{\overline{f_!}}}{\xleftarrow{\overline{f^*}}}}\; \mathbb{D}_{I'} \qquad ?$$

Donc il faut définir, pour J dans Dia, des foncteurs correspondants entre $\mathbb{D}_I(J)$ et $\mathbb{D}_{I'}(J)$, *i.e.* entre $\mathbb{D}(I \times J)$ et $\mathbb{D}(I' \times J)$. Or on a bien

$$f \times \mathsf{id}_J : I \times J \longrightarrow I' \times J$$

et on posera (sous réserve d'existence des deux derniers)

$$\begin{cases} \overline{f^*}(J) = (f \times \mathsf{id}_J)^*_{\mathbb{D}} \\ \overline{f_!}(J) = (f \times \mathsf{id}_J)^{\mathbb{D}}_! \\ \overline{f_*}(J) = (f \times \mathsf{id}_J)^{\mathbb{D}}_* \quad . \end{cases}$$

Autrement dit, on peut considérer

[page 21]

$$\mathsf{Dia} \ni I \longmapsto \mathbb{D}_I \quad , \qquad \mathbb{D} \in \mathsf{DER} \quad ,$$

comme fonctoriel en I de trois façons (une contravariante, les deux autres covariantes), suivant qu'on utilise les f^*, les $f_!$ (s'ils existent) ou les f_* (s'ils existent). On désignera par

$$f^{\mathbb{D}}_! \quad , \qquad f^*_{\mathbb{D}} \quad , \qquad f^{\mathbb{D}}_*$$

indifféremment les foncteurs entre $\mathbb{D}(I)$ et $\mathbb{D}(I')$ ou (de façon plus fine) entre $\mathbb{D}_I$ et $\mathbb{D}_{I'}$, dont les précédents se déduisent en passant aux catégories fondamentales correspondantes. De même pour $f^!$ ou pour $f^?$ sous réserve d'existence. Par exemple, si on suppose que les $f^!$ existent pour $\mathbb{D}$, quel que soit f immersion fermée, alors on trouve pour un tel f un morphisme de dérivateurs

$$\mathbb{D}_I \xleftarrow{f^!_{\mathbb{D}}} \mathbb{D}_{I'} \quad .$$

Il s'impose de regarder si tous ces morphismes de dérivateurs sont exacts, *i.e.* si pour

$$g : J \longrightarrow J'$$

dans Dia, ils commutent à $g_!$, g_*.

Pour expliciter cette question, on est amenés à regarder le diagramme de catégories

p. 16
[page 22 vide]
[page 23]

$$\begin{array}{ccc} I \times J & \xrightarrow{G=\mathsf{id}_I \times g} & I \times J' \\ {\scriptstyle F=f\times \mathsf{id}_J}\downarrow & & \downarrow{\scriptstyle F'=f\times \mathsf{id}_{J'}} \\ I' \times J & \xrightarrow{G'=\mathsf{id}_{I'}\times g} & I' \times J' \end{array} \quad ,$$

qui est d'ailleurs un carré cartésien dans $\mathcal{C}at$. La commutativité de $f^*_{\mathbb{D}}$ aux g_* s'explicite alors par la commutativité dans

$$\begin{array}{ccc} \mathbb{D}(I \times J) & \xrightarrow{G_*} & \mathbb{D}(I \times J') \\ {\scriptstyle F^*}\uparrow & \overset{\text{iso?}}{\nwarrow} & \uparrow{\scriptstyle F'^*} \\ \mathbb{D}(I' \times J) & \xrightarrow{G'_*} & \mathbb{D}(I' \times J') \end{array} \quad .$$

Cette commutativité sera assurée par des axiomes ultérieures sur les dérivateurs. (cf. plus bas page 59). Ainsi, $f^*_{\mathbb{D}}$ est exact à droite `[plutôt à gauche]`, et dualement, il doit être aussi exact à gauche `[plutôt à droite]`. Donc $f^*_D : \mathbb{D}_{I'} \to \mathbb{D}_I$ est *exact*. Considérons $f^{\mathbb{D}}_*$, cette fois c'est la commutativité de

$$\begin{array}{ccc} \mathbb{D}(I \times J) & \xrightarrow{G_*} & \mathbb{D}(I \times J') \\ {\scriptstyle F_*}\downarrow & & \downarrow{\scriptstyle F'_*} \\ \mathbb{D}(I' \times J) & \xrightarrow{G'_*} & \mathbb{D}(I' \times J') \end{array} \quad ,$$

c'est trivial (à isomorphisme canonique près). Ainsi, on trouve que $f^{\mathbb{D}}_* : \mathbb{D}_I \to \mathbb{D}_{I'}$ est *exact à gauche*, et dualement $f^{\mathbb{D}}_!$ est *exact à droite.* Mais est-ce que $f^{\mathbb{D}}_*$ est aussi exact à *droite*? Cela signifie donc commutativité de [page 24]

$$\begin{array}{ccc} \mathbb{D}(I \times J) & \xrightarrow{G_!} & \mathbb{D}(I \times J') \\ {\scriptstyle F_*}\downarrow & & \downarrow{\scriptstyle F'_*} \\ \mathbb{D}(I' \times J) & \xrightarrow[G'_!]{} & \mathbb{D}(I' \times J') \end{array} \quad .$$

Commentary · Exposition [ms. 23]

Three different interchange questions

The extra variable in a shifted derivator is a parameter. An object of $\mathbb{D}(I \times J)$ is therefore a coherent I-family of coherent J-diagrams; it is not, in general, merely a functor from I° to $\mathbb{D}(J)$. Keeping the two variables coherent makes the question on these pages well posed. There are three operations to compare: restriction in one variable with a Kan extension in the other; two right, or two left, Kan extensions; and a right Kan extension with a left Kan extension. Only the last comparison introduces a new exactness condition.

Here is the pointwise mechanism for the first comparison. For a projection in the J-direction, the slice occurring in its right Kan formula at (i, j') has the form

$$(I/i) \times (g/j').$$

The first factor has terminal object $(i, 1_i)$. Since our diagrams are contravariant, taking cohomology in that factor amounts to evaluating at this terminal object. Consequently the value is the right Kan formula in the J-variable applied to the i-th fibre. The same calculation with i replaced by $f(i)$ gives the restriction comparison. Der 2 promotes these pointwise isomorphisms to an isomorphism of coherent diagrams. The dual argument treats left Kan extensions. This is the content of Groth (2013, Proposition 2.5), in the convention of covariant diagrams.

For the second comparison, no pointwise calculation is necessary. Restriction around the product square agrees in either order. The two composites of right adjoints are therefore right adjoint to the same functor, and uniqueness of an adjoint gives their canonical isomorphism. The left-adjoint assertion is identical. These are the derivator forms of Fubini's rule for limits and for colimits.

For the mixed comparison there is still a canonical map, in the direction

$$G'_! F_* \longrightarrow F'_* G_!,$$

but no formal argument makes it invertible. Its construction uses a unit, the already established restriction comparison, and an adjunction. The asymmetry of its direction is meaningful: it is the familiar comparison from a colimit of limits to a limit of colimits. The source's provisional word *exact* means preservation of all the Kan extensions available in the chosen domain. This is stronger than the finite exactness used in modern treatments of stable derivators. The distinction is needed before reading the subsequent appeal to triangulated categories.

On arrive bien à définir p. 17

$$G'_! F_* \xrightarrow{?} F'_* G_!$$

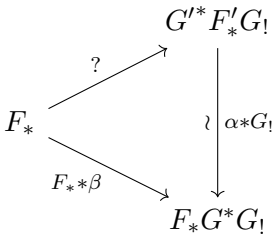
ou ce qui revient au même par adjonction

$$F_* \xrightarrow{?} G'^* F'_* G_!$$

en utilisant le morphisme de changement de base

$$G'^* F'_* \xrightarrow{\alpha}_{\sim} F_* G^*$$

qui est [un] *isomorphisme* (on l'a vu), d'où



où $\beta : \mathsf{id} \to G^* G_!$ est le morphisme d'adjonction, d'où le morphisme voulu. (On peut aussi chercher $F'^*(G'_! F_*) \to G_!$, et utiliser la commutation $G_! F^* \xrightarrow{\sim} F'^* G'_!$, *i.e.* l'exactitude à *droite* de $f^*_{\mathbb{D}}$.) [page 25]

Mais ce morphisme est-il un isomorphisme? Je pense que dans le cas additif ça peut être raisonnable, mais dans le cas non additif ça n'a aucune raison d'être vrai. On voit en tout cas que si pour tout $f : I \to I'$ dans Dia, $f_*^{\mathbb{D}} : \mathbb{D}_I \to \mathbb{D}_{I'}$ est (non seulement exact à gauche mais aussi) exact à droite, alors pour tout $g : J \to J'$ dans Dia, $g_!^{\mathbb{D}}$ est (non seulement exact à droite mais même) exacte à gauche.

Prenons le dérivateur des préfaisceaux d'ensembles sur des I variables, sa catégorie fondamentale est $\mathcal{E}ns$. C'est un dérivateur non ponctué. Considérons $I \to I'$ une immersion fermée non surjective, $f_* : \mathbb{D}(I) \to \mathbb{D}(I')$. Si $f_*^{\mathbb{D}}$ était exact à droite f_* devrait transformer objet initial de $\mathbb{D}(I)$ en objet initial de $\mathbb{D}(I')$, sauf erreur (*cf.* plus bas). Or il n'en est rien, puisque pour tout $F \in \mathsf{Im}\big(f_*^{\mathbb{D}} : \mathbb{D}(I) \to \mathbb{D}(I')\big)$ la restriction de F à $I' \setminus I$ est le faisceau final $e_U = (i \mapsto e)$, et non le faisceau initial $\emptyset_U = (i \mapsto \emptyset)$.

On va prouver ceci :

Proposition. *Si $\mathbb{D}$ est un dérivateur à gauche (à droite), alors les produits finis (sommes finies) existent dans les $\mathbb{D}(I)$. Tout morphisme $\varphi : \mathbb{D} \to \mathbb{D}'$ de dérivateurs qui est exact à gauche (à droite) commute aux dits produits (sommes), chaque I donnant $\varphi(I) : \mathbb{D}(I) \to \mathbb{D}'(I)$. En particulier, il transforme objet final (initial) en itou.*

p. 18
[page 26]

Commentary · Examples [ms. 25]

The empty shape and finite products

The simplest test for a claim about preservation of all homotopy colimits is the empty diagram. If $i : Y \hookrightarrow X$ is a proper closed inclusion of index categories, then in the represented derivator of sets i_* extends a presheaf by the terminal set outside Y. Thus $(i_*\varnothing_Y)_x = \{*\}$ for $x \notin Y$, whereas $(\varnothing_X)_x = \varnothing$. A right adjoint can preserve every limit and still fail this first colimit test.

The finite-product proposition uses a different feature of derivators: discrete coherent diagrams carry no hidden coherence. For a finite set S, Der 1 identifies

$$\mathbb{D}(I \times S) \simeq \prod_{s \in S} \mathbb{D}(I).$$

Restriction along $\pi : I \times S \to I$ becomes the diagonal. Its right adjoint π_* is therefore the ordinary product in the category $\mathbb{D}(I)$: for (F_s) and E the adjunction reads

$$\mathrm{Hom}(E, \pi_*(F_s)) \cong \prod_{s \in S} \mathrm{Hom}(E, F_s).$$

This proves the assertion including $S = \varnothing$, when the right-hand set is a singleton. The left-adjoint calculation gives coproducts. A morphism preserving the corresponding Kan extensions preserves these ordinary products or coproducts because it identifies their representing adjunctions. This argument explains exactly why the empty-diagram test is available. The phrase "adjoint à gauche" at the beginning of manuscript page 26 is an edition-preserved slip: for products it must be a right adjoint, as the same paragraph subsequently says.

Dire que les produits finis indexés par l'ensemble fini I existent dans la catégorie $\mathcal{A} = \mathbb{D}(e)$ (disons) équivaut à dire que le foncteur diagonal

$$\mathcal{A} \xrightarrow{\Delta} \mathcal{A}^I$$

admet un adjoint à gauche. Or si

$$f : I \to e$$

est la projection de I (considéré comme ensemble ordonné discret) sur l'ensemble ordonné ponctuel e, le foncteur Δ s'identifie à $f^* : \mathbb{D}(e) \to \mathbb{D}(I)$ (grâce à l'axiome $\mathbf{Der}_2$ sur le comportement du dérivateur pour une somme discrète de catégories d'indices). Donc l'existence du produit revient à celle de l'adjoint à droite f_* de f^*, OK pour dérivateur à gauche. Et un homomorphisme exact à gauche de dérivateurs à gauche, comme il commute aux f_*, commute aux produits finis. Itou pour les sommes finies, et les dérivateurs à droite.

L'exemple précédent utilise le fait que le dérivateur envisagé n'est pas ponctué. Mais même pour un dérivateur ponctué $\mathbb{D}$, un morphisme $f_*^{\mathbb{D}} : \mathbb{D}_I \to \mathbb{D}_{I'}$ n'est pas forcément exact à droite. Si on prend le dérivateur associé à une catégorie $\mathcal{M}$ quelconque (avec $\Sigma = \emptyset$),

$$\mathbb{D}_{\mathcal{M}}(I) = \underline{\mathsf{Hom}}(I^\circ, \mathcal{M}) \overset{\text{déf}}{=} \mathcal{M}(I)$$

alors le dérivateur induit

$$(\mathbb{D}_{\mathcal{M}})_I = \{J \mapsto \mathbb{D}_{\mathcal{M}}(I \times J) = \underline{\mathsf{Hom}}((I \times J)^\circ, \mathcal{M}) \simeq \underline{\mathsf{Hom}}(J^\circ, \underline{\mathsf{Hom}}(I^\circ, \mathcal{M}))\}$$

s'identifie à [page 27]

$$\mathbb{D}_{\underline{\mathsf{Hom}}(I^\circ, \mathcal{M})} = \mathbb{D}_{\mathcal{M}(I)}$$

i.e.

$$(\mathbb{D}_{\mathcal{M}})_I = \mathbb{D}_{\mathcal{M}(I)}$$

N.B. Ici $\mathcal{M}$ et $\mathbb{D}_{\mathcal{M}}$ se déterminent mutuellement et le foncteur

$$\mathcal{M} \longmapsto \mathbb{D}_{\mathcal{M}}$$

$$\mathcal{C}at \longrightarrow \mathsf{DER}$$

est 2-fidèle, *i.e.* si $\mathcal{M}$, $\mathcal{M}'$ sont deux catégories

$$\underline{\mathsf{Hom}}(\mathcal{M}, \mathcal{M}') \xrightarrow{\simeq} \mathsf{Hom}(\mathbb{D}_{\mathcal{M}}, \mathbb{D}_{\mathcal{M}'})$$

est une équivalence de catégories. D'autre part, on vérifie tout de suite que $u : \mathcal{M} \to \mathcal{M}'$ est exact à gauche (ou à droite) si et seulement si $\mathbb{D}_u : \mathbb{D}_{\mathcal{M}} \to \mathbb{D}_{\mathcal{M}'}$ l'est.

Ainsi, pour voir si un morphisme de dérivateurs de la forme

$$f_*^{\mathbb{D}} : \quad \begin{array}{ccc} (\mathbb{D}_{\mathcal{M}})_I & \longrightarrow & (\mathbb{D}_{\mathcal{M}})_{I'} \\ \| & & \| \\ \mathbb{D}_{\mathcal{M}(I)} & & \mathbb{D}_{\mathcal{M}(I')} \end{array}$$

est exact à *droite,* il revient au même de dire

que le foncteur pour les catégories fondamentales associées [p. 19] [page 28]

$$\mathcal{M}(I) \xrightarrow{f_*^{\mathcal{M}}} \mathcal{M}(I')$$

est exact à droite. Or ici c'est un foncteur image directe de préfaisceaux ordinaires, et ce foncteur n'a aucune envie d'être exact à *droite.* (Même si $\mathcal{M}$ est *abélienne*, d'ailleurs!) Prenons par exemple pour $f : I \hookrightarrow I'$ une immersion ouverte avec $I' = \mathbf{\Delta}_1 \times \mathbf{\Delta}_1$, $I = I' - \{c\}$, c objet final de I'

$$I' = \begin{array}{ccc} c & \longleftarrow & b \\ \uparrow & & \uparrow \\ a & \longleftarrow & 0 \end{array} \qquad I = \begin{array}{ccc} & & b \\ & & \uparrow \\ a & \longleftarrow & 0 \end{array}$$

Le foncteur f_* est celui qui associe à un diagramme

$$F = \left\{ \begin{array}{ccc} & & F_b \\ & & \downarrow \\ F_a & \longrightarrow & F_0 \end{array} \right\}$$

le carré cartésien complété

$$\overline{F} = \left\{ \begin{array}{ccc} F_a \times_{F_0} F_b & \longrightarrow & F_b \\ \downarrow & & \downarrow \beta \\ F_a & \xrightarrow{\alpha} & F_0 \end{array} \right\}$$

Or si $F \mapsto \overline{F}$ était exact à droite, $F \mapsto F_a \times_{F_0} F_b$ le serait aussi. Or $F_a \times_{F_0} F_b = K(F) = \mathsf{Ker}(F_a \times F_b \xrightarrow{\alpha pr_1 - \beta pr_2} F_0)$, en tant que foncteur additif en F, n'est pas exact à droite, par exemple il ne transforme pas nécessairement épimorphisme en épimorphisme. Prenons par exemple

$$F' = \left\{ \begin{array}{ccc} & & F_b \\ & & \downarrow \\ F_a & \longrightarrow & 0 \end{array} \right\}$$

et l'homomorphisme évident $F \to F'$ qui est l'identité sur F_a et sur F_b. C'est un épimorphisme et l'homomorphisme correspondant $\overline{F} \to \overline{F'}$ est l'inclusion [page 29]

$$F_a \times_{F_0} F_b \longrightarrow F_a \times F_b$$

qui n'est un épimorphisme que si $\alpha = \beta = 0$.

Commentary · Exposition [ms. 27]

What represented prederivators remember

For a category $\mathcal{M}$, the assignment $\mathbb{D}_{\mathcal{M}}(I) = \mathrm{Fun}(I^\circ, \mathcal{M})$ is always a prederivator. The existence of the limits or colimits demanded by the diagram domain is needed before calling it a derivator. Its shift is represented by another ordinary category because currying gives

$$\mathrm{Fun}((I \times J)^\circ, \mathcal{M}) \cong \mathrm{Fun}(J^\circ, \mathrm{Fun}(I^\circ, \mathcal{M})).$$

Both sides carry the same objects, maps and composition laws after currying; the identification commutes with restriction in J and hence identifies the prederivators, rather than only their separate values.

The fully faithful assertion about categories and represented prederivators can also be made explicit. A functor $u : \mathcal{M} \to \mathcal{M}'$ acts by postcomposition at every shape. Conversely, let $\Phi : \mathbb{D}_{\mathcal{M}} \to \mathbb{D}_{\mathcal{M}'}$ be a pseudonatural morphism and put $u = \Phi_e$. Its pseudonaturality at $i : e \to I$ identifies the i-th value of $\Phi_I(F)$ with $u(F_i)$. For an arrow $i \to j$, the corresponding natural transformation between these point functors makes these identifications commute with the presheaf arrow $F_j \to F_i$. They therefore form a natural isomorphism $\Phi_I(F) \cong uF$. Naturality in F and in changes of shape follows from the coherence conditions for Φ. A modification between two such morphisms is determined, by the same evaluation argument, by its component at e, and every natural transformation $u \Rightarrow v$ extends by postcomposition. This proves the asserted equivalence of hom-categories. With strict morphisms one uses strict naturality instead.

There is a useful qualification to the next assertion. Preservation of *all relevant* limits by u implies preservation of right Kan extensions because these are calculated pointwise by those limits. The converse follows by taking Kan extensions to the point. If the domain contains infinite categories, ordinary finite left exactness of u is not enough. The same qualification applies dually to colimits. Thus the represented example detects a genuine obstruction to the manuscript's strong exactness condition; it does not show that homotopical exactness has failed to remember an elementary property of $\mathcal{M}$.

Perplexité : On devrait pouvoir définir une notion de dérivateur (additif) "*triangulé*" (qui correspondrait à l'intuition des catégories triangulées; c'est un dérivateur satisfaisant à des axiomes assez forts), de sorte que pour les morphismes entre ceux-ci l'exactitude à gauche et l'exactitude à droite soient *équivalentes.* p. 20

"Théorème de changement de base" pour dérivateurs. Soit $\mathbb{D}$ dérivateur à gauche (existence de f_*) et considérons

$$f : I \longrightarrow J \qquad \text{dans } \mathsf{Dia} \quad ,$$

soit $j_0 \in \mathsf{Ob}\ J$, on se propose de "calculer" le foncteur

$$F \mapsto f_*(F)_{j_0} = \alpha_{j_0}^*(f_*(F)) \quad ,$$

où

$$\alpha_{j_0} : e \longrightarrow J$$

est l'inclusion de la catégorie ponctuelle e dans J, de valeur j_0.

Considérons le diagramme [page 30]

$$(1) \qquad \begin{array}{ccc} I/j_0 & \xrightarrow{q} & I \\ {\scriptstyle f'}\downarrow & \swarrow{\scriptstyle \gamma} & \downarrow{\scriptstyle f} \\ e & \xrightarrow[p=\alpha_{j_0}]{} & J \end{array} \quad ,$$

où I/j_0 est la catégorie $I \times_J J/j_0$ des couples

$$\{(i, \varphi) \mid i \in \mathsf{Ob}\ I,\ \varphi : f(i) \to j_0\}$$

et q est le foncteur $(i, \varphi) \mapsto i$. Ce diagramme n'est pas commutatif, mais il y a un foncteur `[plutôt morphisme de foncteurs]` canonique

$$fq \xrightarrow{\gamma} pf' \quad ,$$

où fq est le foncteur $(i, \varphi) \mapsto f(i)$ et pf' le foncteur constant $(i, \varphi) \mapsto j_0$ et où $\gamma(i, \varphi) : f(i) \to j_0$ est défini par $\gamma(i, \varphi) = \varphi$.

Commentary · Examples [ms. 29]

The pullback obstruction, and the finite stable repair

The abelian counterexample can be made numerical. Let the coefficient category be abelian groups and choose

$$F_a = F_0 = \mathbb{Z}, \qquad F_b = 0, \qquad \alpha = 1_{\mathbb{Z}}, \quad \beta = 0.$$

Then $K(F) = \mathbb{Z} \times_{\mathbb{Z}} 0 = 0$. Replace the central value F_0 by zero to obtain F'. The map $F \to F'$ is an epimorphism at each object of the indexing cospan, but $K(F') = \mathbb{Z}$, and the induced map of missing corners is $0 \to \mathbb{Z}$. Hence the right Kan extension fails to preserve epimorphisms and cannot be right exact even in the ordinary finite abelian sense. Every group and every map in this example is visible in the original three-object shape.

The ensuing "Perplexité" points toward stability, with a necessary change in the meaning of exactness. A pointed derivator is stable when its coherent cartesian and cocartesian squares coincide. Between stable derivators a morphism preserving zero objects and cartesian squares therefore preserves cocartesian squares as well, and conversely. This settles the finite version of the question; see Groth (2013, Definition 3.15 and Corollary 4.17). Additivity alone cannot imply stability. In a represented pointed derivator the pushout of $0 \leftarrow A \rightarrow 0$ is always zero, so its suspension functor is zero. If that derivator were stable, suspension would be an equivalence, forcing every object to be zero.

Even stability does not make arbitrary limits commute with arbitrary colimits. Work in the derivator of complexes over a nonzero field k, and regard vector spaces as complexes in degree zero. On the discrete shape $\mathbb{N} \times \mathbb{N}$ put

$$X_{nm} = \begin{cases} k, & n = m, \\ 0, & n \neq m. \end{cases}$$

Products and coproducts of vector spaces are exact, so here their derived values are their ordinary values. The mixed comparison is

$$\bigoplus_m \prod_n X_{nm} = k^{(\mathbb{N})} \longrightarrow \prod_n \bigoplus_m X_{nm} = k^{\mathbb{N}}.$$

It is the inclusion of finitely supported sequences; the sequence $(1, 1, 1, \ldots)$ is absent from its image. This disproves the unrestricted interpretation of the anticipated stable exactness theorem.

The precise positive result is stronger than merely commuting a finite product with a finite sum: in a stable derivator homotopy finite limits commute with arbitrary homotopy colimits, and dually. A homotopy finite category is equivalent to a finite skeletal category with no nonidentity endomorphisms; finiteness of the underlying set of arrows alone is insufficient. The equivalence of this commutation property with stability is proved in Groth and Shulman (2017, Theorem 2.14). It identifies the exact boundary between the true stable intuition and the stronger interchange asked for on these pages.

Plus généralement, considérons un diagramme dans Dia

(2)
$$\begin{array}{ccc} I' & \xrightarrow{q} & I \\ {\scriptstyle f'}\downarrow & \swarrow{\scriptstyle \gamma} & \downarrow{\scriptstyle f} \\ J' & \xrightarrow[p]{} & J \end{array} \quad ,$$

avec $\gamma : fq \to pf'$. On définit alors un *morphisme de changement de base pour les images directes* relatif à ce diagramme p. 21

$$p^* f_* \xrightarrow{c_\gamma} f'_* q^*$$

de deux façons, qui donnent le même résultat (histoire générale de foncteurs adjoints à expliciter une bonne fois, en termes d'un diagramme de foncteurs [page 31]

$$\begin{array}{ccc} X' & \xleftarrow{q^*} & X \\ {\scriptstyle f'^*}\uparrow & \nearrow & \uparrow{\scriptstyle f^*} \\ Y' & \xleftarrow[p^*]{} & Y \end{array}$$

en supposant que les foncteurs adjoints à gauche f'_*, f_* des deux foncteurs verticaux existent). Un cas intéressant est celui où

$$pf' = fq \quad , \qquad \gamma = \mathsf{id}_{fq}$$

et où on pose $c_\gamma = c$.

Ici

$$p^* f_* \xrightarrow{?} f'_* q^*$$

revient à

$$f_* \xrightarrow{?} p_* f'_* q^* \simeq \qquad \begin{array}{c} (pf')_* q^* \\ \uparrow{\scriptstyle \gamma_* * q^*} \\ (fq)_* q^* \simeq (f_* q_*) q^* = f_*(q_* q^*) \end{array} \quad .$$

Or on a

$$\mathsf{id} \xrightarrow{\alpha} q_* q^* \qquad \text{(adjonction)} \quad ,$$

d'où

$$f_* \xrightarrow{f_* * \alpha} f_*(q_* q^*)$$

et en composant avec $\gamma * q^*$, on trouve ce qu'on veut. Et on peut au lieu de ceci chercher

$$\begin{array}{ccc} f'^*(p^*f_*) & \xrightarrow{\ ?\ } & q^* \\ \| & & \\ (pf')^*f_* & & \\ \downarrow{\scriptstyle \gamma^**f_*} & & \\ (fq)^*f_* \simeq q^*(f^*f_*) & \xrightarrow{q^**\gamma'} & q^* \end{array} ,$$

où $\gamma' : f^*f_* \to \mathsf{id}$ est l'adjonction.

Der$_4$: Le morphisme de changement de base pour images directes f_* associé à un diagramme (1) est toujours un isomorphisme (dérivateurs à gauche). Condition duale pour dérivateurs à droite et le changement de base pour images directes $f_!$, pour le diagramme

p. 22 [page 32]

$$(1') \qquad \begin{array}{ccc} j_0\backslash I & \longrightarrow & I \\ \downarrow & \overset{\gamma}{\nearrow} & \downarrow{\scriptstyle f} \\ e & \xrightarrow[p=\alpha_{j_0}]{} & J \end{array}$$

associé à $f : I \to J$ dans Dia, et $j_0 \in \mathsf{Ob}\ J$.

Commentary · Context [ms. 31]

The base-change map as a mate

The requested general adjunction calculation belongs to the calculus of mates. Its two-categorical setting was already available in Kelly and Street (1974); a derivator-oriented account is Groth et al. (2014a, Appendix A). Here we spell out the map in the manuscript's variance, since reversing a single arrow changes the statement.

Suppose $\gamma : fq \Rightarrow pf'$ is given. Contravariance on 2-cells gives $\gamma^* : f'^*p^* \Rightarrow q^*f^*$. Write η' for the unit of $f'^* \dashv f'_*$ and ε for the counit of $f^* \dashv f_*$. The canonical transformation is the composite

$$\begin{aligned} p^*f_* &\xrightarrow{\eta'\, p^*f_*} f'_*f'^*p^*f_* \\ &\xrightarrow{f'_*\gamma^*f_*} f'_*q^*f^*f_* \xrightarrow{f'_*q^*\varepsilon} f'_*q^*. \end{aligned}$$

In particular, this construction needs only the adjoints f_* and f'_*; adjoints to p^* and q^* are convenient for the alternative calculation, but are not part of its minimal input.

The equality of the two calculations is an equality of natural transformations. Under the adjunction bijection

$$\mathrm{Nat}(p^*f_*, f'_*q^*) \cong \mathrm{Nat}(f'^*p^*f_*, q^*)$$

the displayed composite is sent to $q^*\varepsilon \circ (\gamma^*f_*)$, by the triangular identity for $f'^* \dashv f'_*$. In the alternative construction, transpose first across $p^* \dashv p_*$; replace $(pf')_*$ and $(fq)_*$ by the composites of right adjoints; then transpose back across $f'^* \dashv f'_*$. Naturality of the units moves them past γ^*, and the unit–counit pair for $q^* \dashv q_*$ cancels by its triangular identity. The resulting transformation is exactly $q^*\varepsilon \circ (\gamma^*f_*)$. Since the adjunction bijection is injective, the original two transformations agree.

A strictly commuting square supplies $\gamma = 1$, but does not supply invertibility of its mate. The role of Der 4 is to assert invertibility for the particular comma squares that compute fibres. More general base-change theorems will require conditions on the square. The words "adjoints à gauche" on this page are another source slip: the functors bearing a subscript $*$ in this calculation are right adjoints.

Der$_5$ [10] : Soit $f : I \to J$ une *hot-équivalence stricte, i.e.* $\forall j_0 \in \mathsf{Ob}\ J$, I/j_0 est contractile. Alors f^* est pleinement fidèle, *i.e.* le morphisme d'adjonction

$$\mathsf{id}_{\mathbb{D}(J)} \longrightarrow f_*f^*$$

10. Totalement canulé, devient raisonnable que pour coefficients constants sur F.

est un isomorphisme ($\mathbb{D}$ dérivateur à gauche). Dualement, si f est une équivalence co-stricte, *i.e.* $f^\circ : I^\circ \to J^\circ$ est une équivalence stricte, *i.e.* $\forall j_0 \in \mathsf{Ob}\ J$, $j_0\backslash I$ est contractile, alors (si $\mathbb{D}$ est un dérivateur à droite) le foncteur f^* est pleinement fidèle, *i.e.* le morphisme d'adjonction

$$f_! f^* \longrightarrow \mathsf{id}_{\mathbb{D}(J)}$$

est un isomorphisme.

N.B. Donc les foncteurs f_* (resp. $f_!$) sont des foncteurs de localisation.

N.B. Les pages 33 à 40 ont été suprimées, car canulées.

Je vais reprendre la terminologie et les axiomes relatifs aux dérivateurs. [page 41]

Commentary · Examples [ms. 32]

The cancelled axiom and the walking arrow

The cancellation is mathematically necessary and should remain visible. Let $Y = [1] = (0 \to 1)$ and let $f : e \to Y$ select 0. Each slice f/y has one object and one arrow, so f satisfies the stated strict hot-equivalence condition. In the represented derivator of abelian groups a coefficient on Y is a map $F_1 \to F_0$. The restriction f^* remembers F_0, and its right adjoint sends A to the constant map $A \xrightarrow{1} A$. Consequently the unit is

$$(F_1 \longrightarrow F_0) \longrightarrow (F_0 \xrightarrow{1} F_0),$$

whose component at 1 is the original structure map. With $F_1 = 0$ and $F_0 = \mathbb{Z}$ it is not invertible. Contractibility of the slice controls a constant coefficient system on that slice; it does not turn the restriction of an arbitrary F into such a system.

The failure can also be read directly as failure of full faithfulness. The distinct endomorphisms of the diagram $\mathbb{Z} \to 0$ obtained from the identity and zero on $\mathbb{Z}$ have the same restriction to the object 0. Thus f^* is not faithful. The dual example, using the final endpoint and left Kan extension, disproves the dual part of the cancelled axiom.

There is a valid constant-coefficient conclusion. If $F = p_Y^* M$, the pointwise right Kan formula gives

$$(f_* f^* p_Y^* M)_y \simeq \Gamma_{\mathbb{D}}(f/y, p_{f/y}^* M).$$

Under the valid contractibility theorem the right-hand side is M, and the isomorphism is precisely the component of the unit. Der 2 then shows that the unit on $p_Y^* M$ is an isomorphism. This is the repair anticipated in footnote 10. Full faithfulness of restriction on all coherent diagrams is a stronger condition, usually called a homotopical epimorphism. It must not be substituted for either orientation of cofinality without an additional argument.

Un *prédérivateur* est un 2-foncteur

$$\mathsf{Dia}^\circ \xrightarrow{\mathbb{D}} \mathcal{C}at \quad ,$$

où Dia est une 2-sous-catégorie pleine de $\mathcal{C}at$.[11] On suppose tout au moins que Dia contient les *ensembles ordonnés finis,* que Dia est stable par $\varprojlim$ finies, par sommes finies. Qu'avec toute catégorie C et toute sous-catégorie strictement pleine ouverte (resp. fermée) elle contienne la sous-catégorie complémentaire.[12]

On définit le prédérivateur $\mathbb{D}_{\mathcal{M},\Sigma}$ associé à un système p. 23 [page 42]

$$(\mathcal{M}, \Sigma) \qquad \mathcal{M} \text{ Catégorie}, \quad \Sigma \subset \mathsf{Fl}(\mathcal{M}) \quad .$$

Si $\Sigma = \emptyset$, on écrit $\mathbb{D}_{\mathcal{M}}$.

11. Dia *domaine* du dérivateur. Les objets sont les *catégories d'indices* pour $\mathbb{D}$. Les $\mathbb{D}(I)$ sont les catégories de coefficients pour $\mathbb{D}$, les objets de $\mathbb{D}(I)$ les "coeficients de type $\mathbb{D}$ sur I".

12. Notation f^* : image inverse des coefficients.

Commentary · Exposition [ms. 41]

Restarting the axioms without changing variance

The return to definitions is a revision of the theory, not an additional list to be imposed on top of the cancelled axioms. We keep $\mathbb{D}(I)$ for contravariant coherent diagrams. In fully explicit modern notation its source is $\mathsf{Dia}^{\mathrm{coop}}$, with both 1-cells and 2-cells reversed. Equivalently one can work with the covariant-diagram convention by setting $\mathbb{D}_{\mathrm{cov}}(A) = \mathbb{D}(A^\circ)$. This translation explains why modern formulas often use the opposite comma category; compare Groth et al. (2014a, Warning 2.2).

The diagram domain is part of the hypotheses. The operations used below require products, coproducts, opposites, and the relevant comma categories. The stated finite-limit and complement closures do not explicitly include all these operations. Taking all small categories in a fixed universe makes the present formulas unambiguous; when using a smaller domain, one must require the corresponding closure conditions before writing the dual prederivator or the pointwise Kan formulas. No assertion about a shape outside the domain follows formally from an axiom restricted to that domain.

For a relative category $(\mathcal{M}, \Sigma)$ the diagram localization exists as a prederivator once the chosen universes allow the required localizations. Saturating the weak equivalences does not by itself prove the derivator axioms. On the other hand, the shift identity does not demand a new homotopical theorem. The exponential law identifies $\mathcal{M}^{(I\times J)^\circ}$ with $(\mathcal{M}^{I^\circ})^{J^\circ}$, and identifies the two classes of pointwise weak equivalences before localization. Localizing therefore gives the claimed equivalence of shifted prederivators. Functoriality in relative functors and in natural transformations is descended through the same universal property. This is a useful place to separate what follows from localization alone from the homotopy Kan extension properties that require additional assumptions.

Les prédérivateurs forment une 2- catégorie PREDER, et

$$(\mathcal{M}, \Sigma) \longmapsto \mathbb{D}_{\mathcal{M},\Sigma}$$

est un 2-foncteur

$$\mathsf{PREMOD}\begin{pmatrix}\text{“Catégorie des catégories}\\ \text{de modèles” (faibles)}\end{pmatrix} \longrightarrow \mathsf{PREDER} \quad .$$

Pour $\mathbb{D}$ donné, on définit le dérivateur induit ($I \in \mathsf{Dia}$)

$$\mathbb{D}_I = \big(J \longmapsto \mathbb{D}(I \times J)\big) \quad ,$$

dépendant de I de façon contravariante, d'où un 2-foncteur

$$\begin{aligned}\mathsf{Dia}^\circ &\longrightarrow \mathsf{PREDER}\\ I &\longmapsto \mathbb{D}_I\end{aligned}$$

ou encore un bi-2-foncteur

$$\begin{aligned}\mathsf{PREDER}\times\mathsf{Dia}^\circ &\longrightarrow \mathsf{PREDER}\\ (\mathbb{D}, I) &\longmapsto \mathbb{D}_I \quad .\end{aligned}$$

Si $\mathbb{D}$ est défini par $(\mathcal{M}, \Sigma)$, alors $\mathbb{D}_I$ est défini par $(\mathcal{M}(I), \Sigma(I))$:

$$(\mathbb{D}_{\mathcal{M},\Sigma})_I \simeq \mathbb{D}_{\mathcal{M}(I),\Sigma(I)} \quad ,$$

où $\mathcal{M}(I) = \underline{\mathsf{Hom}}(I^\circ, \mathcal{M})$ (préfaisceaux sur I à valeurs dans $\mathcal{M}$), et $\Sigma(I) \subset \mathsf{Fl}\,\mathcal{M}(I)$ est déduit de Σ en regardant fibre par fibre (supposant maintenant Σ saturé). [page 43]

Pour tous axiomes qu'on dégagera pour certains prédérivateurs, il faudra vérifier qu'il sont stables par passage aux prédérivateurs induits $\mathbb{D}_I$, et aussi comment ils se comportent par dualité (passage de $\mathbb{D}$ à $\mathbb{D}^\circ$).

Définition du prédérivateur dual $\mathbb{D}^\circ$

$$\mathbb{D}^\circ(I) \overset{\text{déf}}{=} \mathbb{D}(I^\circ)^\circ \qquad\qquad \textit{etc.}$$

Der 1 (Premier axiome de localisation) : *a*) Si I, $J \in \mathsf{Dia}$, alors

$$\mathbb{D}(I \amalg J) \longrightarrow \mathbb{D}(I) \times \mathbb{D}(J)$$

est une équivalence de catégories.

b) $\mathbb{D}(\emptyset)$ est la catégorie finale.

Stable par passage aux prédérivateurs induits, et passage au dual.

Corollaire. *Soit* $(I_\alpha)_{\alpha \in A}$ [une] *famille finie de catégories de* Dia, $I = \coprod_{\alpha \in A} I_\alpha$. *Alors*

$$\mathbb{D}(I) \longrightarrow \prod_{\alpha \in A} \mathbb{D}(I_\alpha)$$

est une équivalence de catégories.

Der 2 (Deuxième axiome de localisation) : Pour tout $I \in \mathsf{Dia}$, le foncteur canonique p. 24

$$\mathbb{D}(I) \longrightarrow \underline{\mathsf{Hom}}(I^\circ, \mathcal{A}_\mathbb{D}) \quad ,$$

$\mathcal{A}_\mathbb{D} = \mathbb{D}(e)$ catégorie fondamentale pour le dérivateur $\mathbb{D}$, est conservatif.

Stable par induction, et par dualité.

[page 44]

[page 44]

Commentary · Exposition [ms. 43]

What the localization axioms do and do not say

Der 1 decomposes coefficients over disconnected indexing categories. By induction on the number of summands, its binary and empty cases give the displayed assertion for every finite family. It survives a shift because $I \times (J \amalg K)$ is canonically $(I \times J) \amalg (I \times K)$ and $I \times \varnothing = \varnothing$. It survives duality because taking opposites commutes with products of categories and with the corresponding indexing coproducts.

Der 2 is a detection statement for morphisms already present in $\mathbb{D}(I)$. If such a morphism becomes invertible at every object of I, it was invertible. This gives neither a lifting of every incoherent diagram nor a lifting of every map between incoherent diagrams. The later strongness axiom addresses a different issue and must not be read into this one. To check Der 2 after shifting by J, apply it first to the J-valued fibres and then to the pairs of objects of $I \times J$. The converse direction follows because every functor preserves isomorphisms. Duality preserves the same criterion.

The next axiom names f_*F relative cohomology. Its transitivity is an adjunction calculation: both g_*f_* and $(gf)_*$ are right adjoint to $f^*g^* = (gf)^*$. This gives a specified canonical comparison, compatible with further composition. The existence of finite products again follows by applying this axiom to finite discrete projections. A right adjoint f_* preserves them automatically. Preservation by restriction f^* in a one-sided derivator uses the independent-variable base-change assertion discussed at manuscript page 23; the final pointwise axiom supplies the missing justification. The corresponding homological statements follow by duality.

Un prédérivateur est appelé un *dérivateur à gauche* s'il satisfait **Der 1** et **Der 2**, et de plus les axiomes suivants (dont voici le premier).

Der 3 (g) (Existence des images directes) : Pour tout $f : I \to J$ dans Dia, le foncteur $f^* : \mathbb{D}(J) \to \mathbb{D}(I)$ admet un *adjoint à droite* f_*.

Stable par induction.

On appelle f_* le foncteur image directe (cohomologique) pour f. L'objet $f_*(F) \in \mathbb{D}(J)$ (pour $F \in \mathbb{D}(I)$) s'appelle aussi l'objet de "*cohomologie relative*" de I sur J (par f), à coefficients dans F. On a des isomorphismes canoniques

$$(gf)_* \simeq g_*f_* \quad .$$

Corollaire. *Les catégories* $\mathbb{D}(I)$ *ont des produits finis, les foncteurs* f^* *et* f_* *y commutent.*

Utilise **Der 1** pour $I = \emptyset$ et $I = e \amalg e$, et **Der 3** pour $I \xrightarrow{f} e$.

On a un axiome dual :
Der 3 (d) : existence des adjoints à gauche $f_!$ des f^*, le foncteur $f_!$ s'appelle "image directe homologique" par f, et on a une notion d'*homologie relative* de I sur J (pour f), à coefficients dans F.

Par la suite, la mention (g) ou (d) à coté de la désignation **Der** n d'un axiome, signifie qu'il s'applique aux dérivateurs à gauche, respectivement à droite. De même pour les énoncés, définitions *etc.*
[page 45] [page 45]

Der 4$_0$ (g) [13] : Soit $U \overset{j}{\hookrightarrow} X \overset{i}{\hookleftarrow} Y$ dans Dia, immersions ouverte et fermée complémentaires l'une de l'autre. Alors le morphisme d'adjonction dans $\mathbb{D}(Y)$

$$i^* i_*(F_Y) \longrightarrow F_Y$$

est un isomorphisme, donc

$$i_* : \mathbb{D}(Y) \longrightarrow \mathbb{D}(X)$$

est pleinement fidèle, et $i^* : \mathbb{D}(X) \to \mathbb{D}(Y)$ est un foncteur de localisation. L'image essentielle de i_* est formée des objets F de $\mathbb{D}(X)$ tels que $j^*(F)$ soit l'objet final de $\mathbb{D}(U)$.
Der 4$_0$ (d) : Avec les notations précédentes pour i, j, le morphisme d'adjonction dans $\mathbb{D}(U)$ p. 25

$$F_U \longrightarrow j^* j_! F_U$$

est un isomorphisme, - $j_!$ est pleinement fidèle, j^* un foncteur de localisation. L'image essentielle de $j_!$ est la sous-catégorie pleine de $\mathbb{D}(X)$ formée des F tels que $i^*(F)$ soit objet initial de $\mathbb{D}(Y)$.

Il y a les deux carrés de la page 8 et 9, mais pour interpréter la condition cartésienne ou cocartésienne c'est prématuré ici.
[page 46] [page 46]

Commentary · Exposition [ms. 45]

Extension by an empty fibre

The proposed additional axiom Der 4_0 is a consequence of the pointwise Kan formula. This is a useful calculation because it distinguishes the special behaviour of an immersion from the false full-faithfulness statement for arbitrary cofinal functors.

In the manuscript's convention a closed full inclusion $i : Y \hookrightarrow X$ is a cosieve: an arrow whose source lies in Y has target in Y. For $y \in Y$, the comma category $(i/i(y))$ has terminal object $(y, 1)$. Since the diagram is contravariant, its cohomology is the value at that terminal object. Thus $(i_*E)_{i(y)} \cong E_y$, and these are the components of the counit $i^* i_* E \to E$. Der 2 makes the counit invertible, proving full faithfulness of i_*.

For $x \notin Y$, the comma category (i/x) is empty: an arrow $i(y) \to x$ would contradict the cosieve condition. Hence $(i_*E)_x$ is terminal, the right Kan extension over an empty category. Conversely let $F \in \mathbb{D}(X)$ have terminal fibres on the complement. In the unit $F \to i_* i^* F$, the components on Y are isomorphisms by the preceding calculation, and the components off Y are maps between terminal objects. Der 2 makes the whole unit invertible, so F belongs to the essential image. This proves both assertions of Der $4_0(g)$.

For an open inclusion $j : U \hookrightarrow X$, which is a sieve, the left Kan formula uses (x/j). Outside U this is empty, and inside U its initial object becomes terminal after taking the opposite indexing category for the colimit. Therefore $j_!$ is fully faithful and extends by initial objects. This proves the dual assertion. The calculation is recorded in modern covariant notation in Groth (2013, Proposition 1.23). Only the empty and pointwise Kan formulas are doing work here; no separate extension-by-zero axiom is needed. Pointedness later identifies the terminal and initial extensions with extension by the same zero object.

13. Inutile comme axiome, va devenir un théorème (résulte des autres axiomes).

Axiomes de changement de base.
Der 4 (g) (Fibres d'une image directe) : Soient $f : X \to Y$ dans Dia, $y \in Y$, considérons le diagramme connu

$$\begin{array}{ccc} X/y & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ e & \xrightarrow[p=\alpha_y]{} & Y \end{array}$$

et le morphisme de changement de base correspondant

$$f_*(F)_y \overset{\text{déf}}{=} p^* f_*(F) \longrightarrow f'_*(q^*(F)) (\overset{\text{déf}}{=} \Gamma_{\mathbb{D}}(X/y, q^*(F))) \quad .$$

C'est toujours un isomorphisme.
[page 47]
Der 5 (g) (Axiome de cofinalité pour la cohomologie)[14] : Soit $\alpha : X' \to X$ dans Dia "cohomologiquement cofinal" *i.e.* les $x\backslash X'$ sont contractiles ($x \in X$) *i.e.* $\alpha^\circ : X'^\circ \to X^\circ$ est une hot-équivalence "localement sur X°". Alors $\forall F \in \mathbb{D}(X)$, l'homomorphisme [page 47]

$$\Gamma_{\mathbb{D}}(X, F) \longrightarrow \Gamma_{\mathbb{D}}(X', \alpha^* F)$$

est un isomorphisme (dans $\mathcal{A} = \mathbb{D}(e)$).
N.B. Si $f : X \to e$, on pose $\Gamma_{\mathbb{D}}(X, F) = f_*(F)$ (cohomologie "absolue" de X, à coefficients dans F).
Même chose[15] si α est "homologiquement cofinal", i.e. les X'/x sont contractiles [phrase barrée].
Remarque. On devrait pouvoir renforcer considérablement cet axiome, en exigeant que p. 26

$$\alpha_!(\varepsilon_{X'}) \xrightarrow{\sim} \varepsilon_X \qquad \text{dans} \quad \mathsf{HOT}(X) \quad ,$$

où $\varepsilon_{X'}$, ε_X [sont] les objets finaux de $\mathsf{HOT}(X')$, $\mathsf{HOT}(X)$, et où $\alpha_!(\varepsilon_{X'}) \to \varepsilon_X$ est l'homomorphisme d'adjonction, compte tenu que $\varepsilon_{X'} \simeq \alpha^*(\varepsilon_X)$. On dit que X' est *homotopiquement trivial sur* X. (Et si la condition duale est satisfaite dans $\mathsf{HOT}(X^\circ)$, on dit que X' est cohomotopiquement trivial sur X.) Cette condition n'est sans doute pas autoduale. Elle est vérifiée aussi bien si α est cohomologiquement cofinal, que s'il est homologiquement cofinal (hot-équivalence costricte ou stricte).
N.B. Pour $f : X \to e$, dire que f est cohomologiquement cofinal, homologiquement cofinal, ou que X est contractile, est équivalent.
Corollaire 1 (g) (Théorème d'asphéricité pour les coefficients constants en cohomologie). *Soit $f : X \to e$ dans* Dia*, X contractile, $M \in \mathbb{D}(e)$. Alors l'homomorphisme d'adjonction*

$$M \xrightarrow{\sim} f_* f^*(M) \quad (\overset{\text{déf}}{=} \Gamma(X, f^*(M)))$$

est un isomorphisme.
Corollaire 2 (g)[16] *Soit $f : X \to Y$* [une] *hot-équivalence stricte (i.e. $\forall y \in Y$, X/y contractile). Alors pour $F \in \mathsf{Ob}\,\mathbb{D}(Y)$*

$$F \overset{\text{adj}}{\xrightarrow{\sim}} f_* f^* F$$

[est un] *isomorphisme.* [Ce corollaire est barré, mais pas sa démonstration.]
[page 48]

[page 48]

14. Il est conséquence d'énoncés plus faibles.
15. Faux!
16. Faux, sauf pour coefficients constants.

Commentary · Present state [ms. 47]

Cofinality retains an orientation

The valid cohomological cofinality condition is contractibility of (x/α), whose objects are arrows $x \to \alpha(x')$. For contravariant coefficients it preserves the homotopy limit. Contractibility of (α/x) is the opposite condition and preserves the homotopy colimit. These conditions become equivalent for a map to the point, but not for a general map. The cofinality theorem is part of the established derivator calculus; in covariant notation see Groth et al. (2014a, Corollary 3.13 and Remark 3.14). The theorem's derivation for all small categories does not remove the need to check domain hypotheses in a restricted theory.

The walking-arrow example in the commentary at manuscript page 32 settles the crossed-out sentence. For the inclusion of 0 in $Y = [1]$, every (α/y) is a point, but $\Gamma(Y, F) = F_1$ while $\Gamma(e, \alpha^* F) = F_0$. The comparison is the given structure map $F_1 \to F_0$, which may be $0 \to \mathbb{Z}$. Homotopy invariance with arbitrary varying coefficients cannot ignore its direction.

The proposed condition involving the terminal object of HOT can be calculated completely. In the homotopy theory of presheaves of spaces, the pointwise left Kan formula yields

$$(\alpha_! \varepsilon_{X'})_x \simeq \operatorname*{hocolim}_{(x/\alpha)^\circ} * \simeq B(x/\alpha).$$

For the second identification, the simplicial replacement of the constant point has one simplex for each composable string of arrows in the indexing category; its realization is the nerve realization. Passing to the opposite category reverses these strings and gives the same homotopy type. The counit in the remark has component the augmentation $B(x/\alpha) \to *$. Der 2 therefore identifies its invertibility with weak contractibility of every (x/α). Thus this condition is a reformulation of the valid cohomological cofinality hypothesis, not a substantial strengthening of it.

It also refutes the assertion that either orientation suffices. For the same initial-endpoint inclusion, $(1/\alpha)$ is empty, so the component of $\alpha_! \varepsilon$ at 1 is the empty space and its map to the point is not a weak equivalence. The opposite-endpoint inclusion supplies the dual discrepancy. The remark's prediction that the condition is not self-dual is correct, while its claimed implication from both cofinality conditions is false.

The constant-coefficient corollary has no such problem. If $p : X \to e$ and NX is weakly contractible, the valid cohomological cofinality theorem applied to M gives $M \cong p_* p^* M$. This is exactly the unit, rather than an unspecified isomorphism. What fails in the following cancelled corollary is the replacement of a genuinely varying restriction of F by a constant coefficient system on a contractible slice.

On applique **Der 2**, **Der 4** (g), pour se ramener à regarder, pour $y_0 \in Y$, l'homomorphisme α du diagramme (commutatif!) suivant :

$$\begin{array}{ccc} \Gamma(\{y_0\}, F_{y_0}) = F_{y_0} & \xrightarrow{\quad\alpha\quad} & \Gamma_{\mathbb{D}}(X/y_0, f^*(F)) \\ \varphi\wr \uparrow & \nearrow_{\alpha'} & \\ \Gamma(Y/y_0, F) & & \end{array} ,$$

où α' est déduit du foncteur induit par f

$$X/y_0 \longrightarrow Y/y_0$$

et où φ est déduit du foncteur

$$\{y_0\} \longrightarrow Y/y_0 \qquad \text{(objet final)}\,.$$

Donc φ est un isomorphisme par **Der 5** (g) [17] (N.B. Si $z \in Z$, d'où $\{z\} \to Z$, ce foncteur est cohomologiquement cofinal si et seulement si z est objet final, homologiquement final `[plutôt cofinal]` si et seulement si z est objet initial). Donc la conclusion du corollaire équivaut à p. 27

17. *cf.* corollaire 3, page 50, qu'on devrait remonter ici.

dire que α' est un isomorphisme. Mais $X/y_0 \to Y/y_0$ est aussi homologiquement cofinal, car si $u : y_1 \to y_0$ est dans Y/y_0, on a un isomorphisme canonique

$$(X/y_0)/(y_1, u) \simeq X/y_1$$

qui est contractile. Donc par la deuxième partie de **Der 5** (g),[18] on gagne.
[page 49]

[page 49]

Commentary · Exposition [ms. 48]

Locating the failed inference in the surviving proof

Most of the surviving proof is correct. The category Y/y_0 has terminal object $(y_0, 1)$, and cohomology of a contravariant diagram therefore evaluates to F_{y_0}. The inclusion of that terminal object is cohomologically cofinal: its comma category over an object is the one-element set consisting of the unique map to the terminal object. More generally, for $z \in Z$ the comma category $(x/\{z\})$ is the discrete set $Z(x, z)$. Its nerve is contractible exactly when that set has one element. Hence the singleton inclusion is cohomologically cofinal precisely when z is terminal; the dual assertion characterizes initial objects.

The displayed slice identification is also exact. An object of $(X/y_0)/(y_1, u)$ consists of $x \in X$, a map $v : f(x) \to y_0$, and a map $w : f(x) \to y_1$ with $uw = v$. Forgetting v, which is determined by w, identifies this category with X/y_1, on morphisms as well as objects. This proves the *homological* cofinality of $X/y_0 \to Y/y_0$ under the stated hypothesis. It does not prove cohomological cofinality. The final sentence invokes exactly the opposite-orientation clause that was crossed out on manuscript page 47. No additional fault in the slice calculation is needed to account for the failure. The author and editors explicitly mark the false premise; a modern repair retains the calculation and changes the coefficient hypothesis or the cofinality hypothesis, as in the preceding commentary.

Corollaire 4 (g) (critère de changement de base pour images directes cohomologiques). *Soit*

$$\begin{array}{ccc} X' & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & \overset{\gamma}{\swarrow} & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[p]{} & Y \end{array}$$

dans Dia (*cf.* page 30), *d'où l'homomorphisme de changement de base*

$$p^* f_*(F) \longrightarrow f'_* q^*(F) \qquad (F \in \mathsf{Ob}\, \mathbb{D}(X)) \quad .$$

Pour que ce soit un isomorphisme pour tous F, il suffit que pour tout $y' \in \mathsf{Ob}\, Y'$, le foncteur induit par (p. 9)

$$X'/y' \longrightarrow X/p(y')$$

soit une hot-équivalence stricte (ou costricte);[19] *cela signifie aussi (dans le cas strict) que pour*

$$y'_0 \in \mathsf{Ob}\, Y' , \quad x_0 \in \mathsf{Ob}\, X , \quad u_0 : f(x_0) \longrightarrow p(y'_0)$$

la catégorie $K(y'_0, x_0, u_0)$ des objets de la forme

$$(x', v', u) \mid x' \in \mathsf{Ob}\, X', \quad v' : f'(x') \to y'_0, \quad u : q(x') \overset{\leftarrow\!-\!-}{\longrightarrow} x_0$$

rendant commutatif le carré dans Y'

(∗)

$$\begin{array}{ccc} f(q(x')) & \xrightarrow{\gamma(x')} & p(f'(x')) \\ {\scriptstyle f(u)}\downarrow\uparrow & \text{cas de hot-équivalence} \leftarrow \text{costricte} & \downarrow{\scriptstyle p(v')} \\ f(x_0) & \xrightarrow[u_0]{} & p(y'_0) \end{array}$$

18. (qui est fausse)
19. Faux pour "stricte", ne marche que pour "costricte".

est contractile.
N.B. Dans le cas de la hot équivalence costricte, c'est la même description de p. 28

$$K = (x_0, u_0)\backslash(X'/y'_0) \quad ,$$

sauf que la flèche u est en sens inverse $x_0 \to q(x')$ et itou dans le diagramme commutatif $(*)$.
On applique **Der 2**, **Der 4** (g) et corollaire 1 `[plutôt` **Der 5** (g)`]`.
`[page 50]` [page 50]

Commentary · Exposition [ms. 49]

The factorization category controls base change

The base-change criterion is a statement about the square, including its specified natural transformation. It does not assert that every cartesian square is exact, and it does not require the square to be cartesian. To see what it tests, fix $y' \in Y'$, put $y = p(y')$, and write

$$h_{y'} : X'/y' \longrightarrow X/y, \qquad (x', v : f'(x') \to y') \longmapsto (q(x'), p(v)\gamma_{x'}).$$

The pointwise formula for right Kan extension identifies the value of the Beck–Chevalley map at y' with restriction of cohomology along $h_{y'}$. Since the coefficients are contravariant, the appropriate cofinality test uses arrows *from* an object of X/y *to* the image of $h_{y'}$. This is precisely the orientation retained in footnote 19. The discarded orientation is the mistake already isolated in the commentary at manuscript p. 32, "The cancelled axiom and the walking arrow."

For clarity, the category attached to $x \in X$ and $\alpha : f(x) \to p(y')$ has objects

$$(x', u : x \to q(x'), v : f'(x') \to y'), \qquad p(v)\gamma_{x'} f(u) = \alpha.$$

A morphism to (x'', u', v') is an arrow $t : x' \to x''$ such that $q(t)u = u'$ and $v' f'(t) = v$. Thus this is the comma category $(x, \alpha)\backslash h_{y'}$, with all compatibility conditions retained. Weak contractibility of every such category implies that restriction along $h_{y'}$ preserves homotopy limits of the contravariant diagrams in question. Applying this at every y', and then using **Der 2**, proves the criterion. The words "dans Y'" before the displayed square are a slip in the edition: the square is in Y.

The resulting invariant is now called a *homotopy exact square*. In modern covariant notation its factorization categories are those of Groth et al. (2014a, Definition 3.7 and Theorem 3.8); passage to opposite indexing categories recovers exactly the arrows above. Their Theorem 3.16 also gives the converse when exactness is required for every derivator. The useful content of this later formulation is that the elementary categories of factorizations, rather than any particular presentation by model categories, determine the universal base-change property.

Corollaire 5 (g). *Soit $f : X \to Y$ pleinement fidèle dans* Dia*, et $F \in \mathbb{D}(X)$. Alors l'homomorphisme d'adjonction*

$$f^* f_*(F) \longrightarrow F$$

est un isomorphisme.

En efffet, il suffit de le vérifier fibre par fibre (**Der 2**). Soit $x \in X$, $y = f(x) \in Y$, alors

$$f^* f_*(F)_x \simeq f_*(F)_y \simeq \Gamma_{\mathbb{D}}(X/y, F)$$

or à cause de la pleine fidélité $X/y \simeq X/x$.

Corollaire 3 (remonter page 48).[20] *Soit $Z \in$ Dia, supposons que Z a un objet final e_Z. Soit $p : e \to Z$ le foncteur correspondant. Alors l'homomorphisme canonique (pour F dans $\mathbb{D}(Z)$)*

$$\begin{array}{ccc} Z & & \\ \uparrow p & \Big\downarrow f & \\ e & & \end{array} \qquad\qquad f_*(F) \longrightarrow p^*(F)$$

20. Va résulter de **Der 6** (g) plus fort ci-dessous.

est un isomorphisme. [21]

Cet homomorphisme est [un] cas particulier de celui-ci, pour

$$\begin{array}{ccc} Z' & \xrightarrow{\quad p \quad} & Z \\ & {\scriptstyle f'}\searrow \quad \swarrow{\scriptstyle f} & \\ & S & \end{array}$$

(en faisant $Z' = S = e$, $f' = \mathsf{id}_e$),

$$f_*(F) \longrightarrow f'_*(p^*(F)) \quad , \qquad (\, F \in \mathsf{Ob}\, \mathbb{D}(Z)\,) \quad ,$$

et c'est un cas particulier [22] des morphismes de changement de base pour

$$\begin{array}{ccc} Z' & \xrightarrow{\ p\ } & Z \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S & \xrightarrow{\ \mathsf{id}\ } & S \end{array} \quad .$$

p. 29
[page 51]

Commentary · Exposition [ms. 50]

Fully faithful direct image and extension by a terminal object

There are two separate assertions behind the next localization calculation. First, a fully faithful inclusion retains all coefficients on its image. If $f : X \to Y$ is fully faithful, the map

$$X/x \longrightarrow X/f(x), \qquad (x', a : x' \to x) \longmapsto (x', f(a)),$$

is an isomorphism of categories. The right member here is the comma category formed with f. Fullness and faithfulness give, respectively, existence and uniqueness of the inverse on objects and arrows. Since X/x has terminal object $(x, 1_x)$, the pointwise formula gives

$$(f^* f_* F)_x \simeq \Gamma_{\mathbb{D}}(X/x, F) \simeq F_x.$$

The map so obtained is the counit, not merely an unspecified isomorphism between its source and target. Conservativity therefore proves that f_* is fully faithful.

Second, for a cosieve $i : Y \hookrightarrow X$, right extension in the presheaf convention fills the complementary sieve U with terminal objects. If $u \in U$, there can be no arrow from an object of Y to u, so Y/u is empty. The pointwise formula consequently makes $(i_* F)_u$ terminal. Conversely, let $E \in \mathbb{D}(X)$ be terminal on U. On Y, the unit $E \to i_* i^* E$ is invertible by the first assertion. On U, it is a map between terminal objects and is again invertible. The two parts cover all objects of X, so **Der 2** finishes the proof. This argument uses no zero object. In a pointed theory the terminal values happen also to be initial, which explains the later terminology "extension by zero."

These computations also explain why the argument survives the loss of ordinary diagrammatic information. We do not recover an arbitrary coherent diagram from a list of its values. We construct a specific morphism, the unit or counit, and show that this morphism is invertible at every object. That is exactly what conservativity permits. The corresponding modern statement is Groth (2013, Proposition 1.23), with sieve and cosieve interchanged when changing the variance convention.

Remarque : On retrouve le fait que $i^* i_*(F_Y) \xrightarrow{\sim} F_Y$ est [un] isomorphisme, pour i une immersion fermée. De plus, avec les notations de **Der 4** (g), $j^*(i_*(F_Y))$ se calcule par le théorème de changement de base (corollaire 2 [plutôt 4] ci-dessus), qui s'applique dans toute situation commutative

$$\begin{array}{ccc} X' & \xrightarrow{\ q\ } & X \\ {\scriptstyle f'}\downarrow & \text{cart} & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{\ p\ } & Y \end{array}$$

21. Axiome de la catégorie d'indices à objet final (coefficient quelconque).
22. déjà envisagé dans cor. 2, p. 47

quand p est une immersion ouverte (et $\gamma = \mathsf{id}$) (alors $X'/y' \to X/p(y')$ est même un isomorphisme!), et on trouve, revenant aux notations de **Der 4** (g),

$$\begin{array}{ccc} \emptyset & \xrightarrow{q} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle i} \\ U & \xrightarrow[j]{} & X \end{array}$$

i.e. $j^*(i_*(F_Y)) \simeq f'_*(\varphi)$, où $\varphi = q^*(F_Y)$ est l'unique élément de la catégorie (équivalente à une catégorie ponctuelle, par **Der 1**) $\mathbb{D}(\emptyset)$. D'autre part, f'_* transforme objet final en objet final, d'où $j^*i_*(F_Y)$ est objet final.

Inversement, soit F dans $\mathbb{D}(X)$, tel que $j^*(F) = e_{\mathbb{D}(U)}$. Considérons

$$F \xrightarrow{u} i_*i^*F \qquad \text{(adjonction)}\,;$$

je dis que c'est un isomorphisme. En effet, par l'axiome de localisation **Der 2**, il suffit de prouver que $i^*(u)$ et $j^*(u)$ sont `[des]` isomorphismes. Pour $i^*(u)$ cela résulte du fait que i^* soit foncteur de localisation, et pour j^* du fait que $j^*(F)$ et $j^*(i_*i^*(F))$ sont des objets finaux dans $\mathbb{D}(U)$. [page 52]

p. 30
[page 53]

Commentary · Exposition [ms. 52]

Properness as a condition on all lifts

The geometric terminology begins with a computational requirement. A right direct image over y is initially computed on the comma category X/y. To compute it on the strict fibre X_y instead, the inclusion $X_y \to X/y$ must be cofinal in the orientation appropriate to contravariant coefficients. Properness asserts this simultaneously for every y.

There is a concrete way to express the requirement without derived functors. For $x \in X$ and $\alpha : f(x) \to y$, let $L_f(x,\alpha)$ be the category whose objects are arrows

$$v : x \longrightarrow x_1, \qquad f(v) = \alpha,$$

and whose morphisms from v to v' are arrows $t : x_1 \to x'_1$ with $tv = v'$ and $f(t) = 1_y$. This is a category of lifts of one fixed base arrow, with a fixed source upstairs. There is a canonical isomorphism

$$L_f(x,\alpha) \cong (x,\alpha)\backslash X_y.$$

Indeed, a morphism $(x,\alpha) \to (x_1, 1_y)$ in X/y is exactly such a lift v, and a morphism between two of them is exactly the triangle defining t. Thus f is proper precisely when every $L_f(x,\alpha)$ has weakly contractible nerve. In particular, a lift must exist, all choices must be connected, and their higher homotopy must vanish. Mere existence of lifts is insufficient.

For smoothness one fixes the *target* upstairs instead. Given $x \in X$ and $\beta : y \to f(x)$, the incoming lift category consists of $v : x_0 \to x$ over β and triangles over 1_y. Smoothness says that these categories are weakly contractible. The first display on manuscript p. 54 should consequently have target $y\backslash X$, not $y\backslash Y$; the latter is what the edition actually prints. The error is typographical, since the succeeding characterization by fibres of $X/x \to Y/f(x)$ has the correct meaning. In the modern literature this elementary equivalence is stated in Maltsiniotis (2020, Definition 3.2.1 and Proposition 3.2.3); Maltsiniotis (2005, Theorem 1.2) gives its formulation by base change.

The stability under strict pullback can now be proved without any cohomology. In a pullback $X' = X \times_Y Y'$, choose $x' = (x, y')$ and an arrow $\alpha' : y' \to z'$ in Y'. A lift of α' out of x' is exactly a lift of $p(\alpha')$ out of x, its second component already being prescribed. The same correspondence identifies every morphism between lifts. The lift categories are therefore isomorphic. Reversing all arrows proves the smooth case.

Several small consequences deserve to be read literally. A cosieve inclusion is proper because every allowed outgoing arrow has one lift; a sieve inclusion is smooth by the dual argument. If f is proper, nonemptiness of $L_f(x,\alpha)$ shows that every arrow out of an object of its image is itself in the image. Thus the strict image is a full cosieve. Finally, $X \to e$ is both proper and smooth: the lift categories are $x\backslash X$ and X/x, which have initial and terminal objects. Product projections inherit both properties by strict pullback.

Morphismes homotopiquement propres, homotopiquement lisses.

Soit $f : X \to Y$ [un] morphisme de catégories, considérons $y_0 \in \mathsf{Ob}\, Y$, et le foncteur canonique

$$X_{y_0} \xrightarrow{\varphi} X/y_0 \qquad \varphi(x) = (x, \mathsf{id}_{f(x)}) \tag{1}$$

(X_{y_0} fibre de X en y_0). Soit $(x_0, u_0) \in \mathsf{Ob}\, X/y_0$ donc $x_0 \in \mathsf{Ob}\, X$, $u_0 : f(x_0) \to y_0$. Considérons la catégorie coinduite par φ sur (x_0, u_0)

$$(x_0, u_0)\backslash(X_{y_0}) \quad ,$$

pour tester si φ est une hot-équivalence costricte *i.e.* homologiquement [plutôt cohomologiquement] cofinal. On va l'interpréter via le foncteur induit par f

$$x_0\backslash X \xrightarrow{\psi} f(x_0)\backslash Y$$

allant de la catégorie des spécialisations de x_0 dans X vers celle des spécialisations de $f(x_0)$ dans Y. On trouve alors un isomorphisme canonique de la catégorie coinduite $(x_0, u_0)\backslash X_{y_0}$ par X_{y_0} au dessus de l'objet (x_0, u_0) de X/y_0 et de la catégorie fibre $(x_0\backslash X)_{u_0}$ de $x_0\backslash X$ au dessus de $f(x_0)\backslash Y$ en u_0. Donc

Proposition. *Pour que les foncteurs $\varphi : X_{y_0} \to X/y_0$ ($y_0 \in \mathsf{Ob}\, Y$) soient des hot-équivalences costrictes, il faut et il suffit que les fibres des foncteurs $x_0\backslash X \to f(x_0)\backslash Y$ induits par f soient contractiles.*

On dit alors que f est *homotopiquement propre.* La notion duale s'exprime, soit par la condition que les foncteurs induits par f [page 54]

$$X_{y_0} \longrightarrow y_0\backslash Y$$

soient des hot-équivalences strictes *i.e.* qu'ils soient cohomologiquement [plutôt homologiquement] cofinaux, soit que les fibres des morphismes induits sur les "catégories localisées" (catégories des générisations)

$$X/x_0 \longrightarrow Y/f(x_0)$$

soient contractiles. (N.B. On utilise ici qu'une catégorie Z est contractile si et seulement si Z° l'est.) On dit alors que f est *homotopiquement lisse.*

Proposition. *La notion de foncteur propre (resp. lisse) est stable par changement de base, par composition. Les isomorphismes, et plus généralement, les immersions fermées sont propres. L'image d'un morphisme propre est une sous-catégorie fermée.* [23]

N.B. Le foncteur $X \to e$ est toujours propre et lisse, car les catégories $x_0\backslash X$ et X/x_0 sont contractiles (ayant objet initial ou objet final). Donc tout foncteur de projection $X \times Y \xrightarrow{pr_2} Y$ est à la fois propre et lisse.

DÉMONSTRATION. Stabilité par changement de base : p. 31 [page 55]

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \text{cart} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array} \quad ,$$

d'où pour $x' \in \mathsf{Ob}\, X'$

$$\begin{array}{ccc} X/x & \longleftarrow & X'/x' \\ \downarrow & & \downarrow \\ Y/y & \longleftarrow & Y'/y' \end{array} \quad ,$$

23. Attention, une équivalence de catégories n'est pas propre! Si $X' \hookrightarrow X$ [est] une inclusion d'une sous-catégorie de X, elle n'est propre que si c'est une immersion fermée.

où x, y', y sont les images de x' dans X, Y', Y. On vérifie que ce diagramme est *cartésien*, donc les fibres de $X'/x' \longrightarrow Y'/y'$ sont isomorphes à des fibres de $X/x \longrightarrow Y/y$, donc contractiles si celles-ci le sont.

Commentary · Examples [ms. 54]

Why an equivalence can fail properness

Let J be the groupoid with two objects $0, 1$ and exactly one morphism between each ordered pair, and let $i : e \to J$ select 0. The functor i is fully faithful and essentially surjective, hence an equivalence. Nevertheless, for the unique isomorphism $\alpha : 0 \to 1$, the outgoing lift category $L_i(*, \alpha)$ is empty. There is no object of the strict domain lying over 1. The incoming lift category for the inverse isomorphism is empty for the same reason, so i is neither proper nor smooth.

This failure is visible directly in base change. Pull i back along $j : e \to J$ selecting 1. The strict pullback is empty. For an ordinary set M, or a space M, the right Kan extension i_*M has value M at 1: the comma category $e/1$ has a single object. The right direct image from the empty pullback is terminal. Hence the comparison at 1 is

$$M \longrightarrow *,$$

which fails to be invertible, for example when M is a discrete two-point space. This example verifies the caution in footnote 23 and shows exactly where the strictness matters. Replacing the strict pullback by an equivalence-invariant pullback would change the problem under discussion.

For an equivalence $f : X \to Y$, the preceding argument gives a more precise criterion: f is proper if and only if it is surjective on objects, and the same criterion characterizes smoothness. Necessity follows because every object of Y is isomorphic to one in the image and properness supplies a lift of that isomorphism. For sufficiency, fix $\alpha : f(x) \to y$ and choose x_1 with $f(x_1) = y$. Full faithfulness gives a unique lift $x \to x_1$ of α. Any two lift objects have a unique morphism between them over 1_y, again by full faithfulness. Their lift category is therefore nonempty and equivalent to a point. The incoming version is identical. This is the criterion recorded in Maltsiniotis (2020, Remark 3.2.5).

Dans un cas $X \xrightarrow{f} Y$ on vérifie que pour un foncteur bilocalisé de f

$$X/x \xrightarrow{\varphi = f/x} Y/y \quad ,$$

tout “bilocalisé”

$$(X/x)/\xi \longrightarrow (Y/y)/\eta \qquad\qquad (\xi : x' \to x\,, \quad \eta : y' \to y)$$

de φ s’identifie encore à un bilocalisé de f

$$X/x' \longrightarrow Y/y' \quad .$$

Donc si f est lisse, les bilocalisés f/x ($x \in \mathsf{Ob}\, X$) le sont aussi. Notons d’autre part.

Proposition.[24] *Tout morphisme propre (ou dualement lisse) à fibres contractiles est une hot-équivalence.*

Cela provient du fait que pour un tel foncteur propre, si F est un préfaisceau, ou complexe de préfaisceaux abéliens sur X, la formation des $Rf_*(F)$ commute au passage aux fibres, et de même pour $R^1f_*(F)$ si F est un préfaisceau en groupes (pas même abéliens). [page 56]

Commentary · Exposition [ms. 55]

Contractible fibres and the missing steps in composition

The assertion about contractible fibres does not need a separate theorem detecting weak equivalences by abelian and nonabelian cohomology. Suppose first that f is proper and every X_y is weakly contractible. By definition $X_y \to X/y$ is coaspherical, hence a weak equivalence on nerves by Quillen’s Theorem A. Therefore every X/y is weakly contractible. A second application of

24. *cf.* amélioration dans cor. 2, page suivante.

Theorem A, now to f, proves that Nf is a weak equivalence. Opposite categories give the smooth case. This is a completion of the proof sketched here, using Quillen (1973, §1, Theorem A), rather than a correction of the theorem.

The same argument proves the stronger assertion on manuscript p. 56. Base change along $Y/y \to Y$ and $y\backslash Y \to Y$ preserves properness or smoothness and preserves the contractibility of strict fibres. Each base has contractible nerve, since it has a terminal or an initial object. The total categories are X/y and $y\backslash X$, respectively, and the fibre theorem makes each of them contractible. Thus f is both aspherical and coaspherical.

Here is also a full proof of the composition assertion. Let $f : X \to Y$ and $g : Y \to Z$ be proper, fix $x \in X$, and fix $\alpha : gf(x) \to z$. There is a functor

$$L_{gf}(x, \alpha) \longrightarrow L_g(f(x), \alpha), \qquad (v : x \to x_1) \longmapsto f(v).$$

The undercategory functor $x\backslash X \to f(x)\backslash Y$ is proper: after fixing an object $v : x \to x_1$, its outgoing lift categories identify with the corresponding outgoing lift categories of f at x_1. Its strict fibres are contractible by the definition of properness of f. The displayed functor is its strict pullback along $L_g(f(x), \alpha) \to f(x)\backslash Y$. It is consequently proper with contractible fibres, so the preceding theorem makes it a weak equivalence. Its target is contractible because g is proper. Its source is therefore contractible, and this is the required condition for gf. Duality proves composition for smooth functors. The argument is the elementary mechanism behind Maltsiniotis (2020, Propositions 3.2.6 and 3.2.10).

There is no cancellation theorem with the other two positions in a composite. The functors $[1] \to e$ and $e \to e$ are proper, but their factor $e \to [1]$ selecting 0 is not: the arrow $0 \to 1$ has no lift. The statement about sections in the text has precisely this content. It should not be read as a defect of the proof of composition.

Corollaire 1. *Si $f : X \to Y$ est cohomologiquement propre à fibres contractiles, alors Y est contractile si et seulement si X l'est.*

Grâce à ceci, on trouve que le composé de deux foncteurs propres (lisses) est propre (lisse). N.B. Contrairement à ce qui a lieu en topologie pour la notion de propreté, si gf est propre et g propre, f ne l'est pas nécessairement, *i.e.* un S-morphisme de S-catégories X, Y propres sur S n'est pas nécessairement propre. Ainsi une section de X/S, pour X propre sur S, n'est pas nécessairement propre, *i.e.* n'est pas nécessairement une immersion fermée. C'est déjà faux si $S = e$ (auquel cas $f : X \to e$ est toujours propre). p. 32

Corollaire 2. *Un morphisme propre ou lisse, à fibres contractiles est à la fois une hot-équivalence stricte et costricte.*

En effet, on applique le corollaire 1 à $X/y \to Y/y$ ou $y\backslash X \to y\backslash Y$.

Théorème (g) *Soit une situation de changement de base dans* Dia

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \text{cart} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array} \qquad , \qquad p^* f_*(F) \xrightarrow{c} f'_* q^*(F)$$

($F \in \mathbb{D}(X)$), $\mathbb{D}$ dérivateur exact à gauche. Si f est (homotopiquement) propre ou p homotopiquement lisse, alors c est un isomorphisme.

La question c'est si pour tout $y' \in Y'$, posant $y = p(y')$ et considérant le foncteur induit par q [page 57]

$$X/y \longleftarrow X'/y'$$

le morphisme correspondant

$$\Gamma_{\mathbb{D}}(X/y, F) \longrightarrow \Gamma_{\mathbb{D}}(X'/y', q^* F)$$

est [un] isomorphisme. Dans le cas f propre, on considère

$$\begin{array}{ccc} X_y & \xleftarrow{\sim} & X_{y'} \\ {\scriptstyle\sim}\ \text{costricte}\downarrow & & \downarrow{\scriptstyle\sim}\ \text{costricte} \\ X/y & \longleftarrow & X'/y' \end{array}$$

d'où la conclusion, puisque par **Der 5** (g), dans

$$\begin{array}{ccc} \Gamma(X_y) & \longrightarrow & \Gamma(X_{y'}) \\ \uparrow & & \uparrow \\ \Gamma(X/y) & \longrightarrow & \Gamma(X'/y') \end{array}$$

les flèches verticales et l'horizontale supérieure sont [des] isomorphismes, donc aussi l'horizontale inférieure.

Dans le cas p lisse, considérons du coté Y, Y' le carré de changement de base p. 33

$$\begin{array}{ccc} Y & \longleftarrow & Y' \\ \uparrow & & \uparrow \\ Y/y & \longleftarrow & Y'/y' \end{array}$$

d'où le diagramme contigu [page 58]

$$\begin{array}{ccccc} X & \longleftarrow & X' & & \\ & \nwarrow & \downarrow & \nwarrow & \\ & X/y & \xleftarrow[\kappa]{} & X'/y' & \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & & \\ & \nwarrow & & \nwarrow & \\ & Y/y & \xleftarrow[\pi]{} & Y'/y' & \end{array} ,$$

où les carrés latéraux (verticaux) sont cartésiens. On note que π est lisse et à fibres contractiles (car p est lisse) donc κ aussi. Donc κ est une hot-équivalence costricte, donc $\Gamma(X/y) \xrightarrow{\sim} \Gamma(X'/y')$, on gagne encore.

N.B. Sauf erreur, on a une réciproque du théorème, [à] savoir : Si $f : X \to Y$ est tel que le théorème de changement de base pour $f_*(F)$ est vrai pour *tout* dérivateur (il suffit sans doute de HOTAB et HOT) et pour tout $Y' \to Y$, alors f est homotopiquement propre. D'autre part, si $g : Y' \to Y$ est tel que le théorème de changement de base pour g est vrai pout tout $X \to Y$ et tout $F \in \mathbb{D}(X)$ ($\mathbb{D}$ dérivateur exact à droite quelconque), alors g est homotopiquement lisse.

Du côté homologique les foncteurs lisses f sont exactement ceux pour lesquels $f_!$ commute à tout changement de base. Les foncteurs propres g sont exactement ceux pour lesquels le changement de base par g commute à $f_!$, pour tout $f : X \to Y$ et tout $\mathbb{D}$ dérivateur exact à gauche.

Ainsi, les foncteurs propres $f : X \to Y$ se caractérisent de *quatre* façons remarquables [page 59]

a) f_* commute à tout changement de base ($\forall$ dérivateur $\mathbb{D}$ exact à gauche);

b) le changement de base f^* commute à tout $g_!$ ($g : Y' \to Y$, $\mathbb{D}$ dérivateur exact à droite);

c) $\forall y \in \mathsf{Ob}\, Y$, $X_y \to X/y$ est une hot-équivalence costricte *i.e.* est un foncteur homologiquement [plutôt cohomologiquement] cofinal;

d) $\forall x \in \mathsf{Ob}\, X$ le foncteur induit sur les colocalisés

$$x\backslash X \longrightarrow f(x)\backslash Y$$

Commentary · Present state [ms. 58]

The converse, and why spaces suffice

The conjectural converse is proved in the subsequent theory. Its exact scope is universal base change: for a fixed $f : X \to Y$, one considers every strict pullback square and every coefficient in the indicated derivator. The class of homotopy exact squares is already detected by the derivator of spaces; Groth et al. (2014a, Theorem 3.16) equate this test with exactness for all derivators and with weak contractibility of the factorization categories. Thus the proposed pair HOTAB and HOT can be reduced to HOT alone. The proper/smooth theory is developed systematically by Maltsiniotis (2020, §3.2). We spell out the small tests that detect the two converse assertions, because they explain why the conjecture has the stated strength.

Fix $y \in Y$ and form the fibre square, with $j : X_y \to X$ and $s : X_y \to e$. In the derivator of spaces its right comparison

$$y^* f_* \longrightarrow s_* j^*$$

is invertible if and only if its left-adjoint mate

$$j_! s^* \longrightarrow f^* y_!$$

is invertible. This is taking the adjoint of a transformation between right adjoints, so invertibility is preserved in both directions. Apply the left mate to the terminal space $*$. At $x \in X$ its target is the discrete space $\mathrm{Hom}_Y(f(x), y)$, because $y_! *$ is the representable presheaf $\mathrm{Hom}_Y(-, y)$. Its source is

$$(j_! *)_x \simeq \coprod_{\alpha : f(x) \to y} |NL_f(x, \alpha)|.$$

Indeed, the left pointwise formula is the homotopy colimit of the point over $x\backslash X_y$ (with the opposite convention making no difference to its realization), and this category is the disjoint union of the lift categories indexed by α. The mate collapses each summand to its indexing point. It is an equivalence precisely when every summand is weakly contractible. This proves properness from universal right base change, and even shows that the fibre squares alone suffice for this converse.

For the smooth converse, fix $p : Y' \to Y$ and assume that base change along p commutes with every right direct image in the derivator of spaces. Choose $f : e \to Y$ selecting any y_0, and evaluate at $y' \in Y'$. For every space M, the comparison becomes

$$\mathrm{Map}(S, M) \longrightarrow \mathrm{Map}(|N(Y'_{y_0}/y')|, M), \qquad S = \mathrm{Hom}_Y(y_0, p(y')),$$

where S is discrete. The map underlying this comparison sends an arrow $z \to y'$ with $p(z) = y_0$ to its image in Y. If the displayed map is an equivalence for every M, Yoneda in the homotopy category of spaces says that this underlying map to S is an equivalence. Its fibre over $\beta : y_0 \to p(y')$ is the nerve of the incoming lift category at y'. All these categories are therefore weakly contractible, which is smoothness.

The converse statements thus require no conjectural detection principle left over from these pages. There is, however, a notational qualification in the printed paragraph: a comparison involving f_* requires right Kan extensions, which the manuscript calls a derivator "à gauche." The subsequent sentence concerning $f_!$ requires left Kan extensions. Where the edition switches these adjectives, the adjunctions identify the intended assertion without ambiguity. Passing to the left-adjoint mate also explains condition (b) on manuscript p. 59: for a pullback square, properness of the right vertical functor controls

$$q_! f'^* \longrightarrow f^* p_!,$$

just as it controls $p^* f_* \to f'_* q^*$. These are two adjoint forms of one exactness statement.

est à fibres contractiles.

Autre situation standard où le changement de base marche (mais où les foncteurs qui interviennent n'ont aucune raison d'être propres ou lisses).

Soient

$$f : I \longrightarrow I' \quad , \qquad g : J \longrightarrow J'$$

dans Dia, d'où le diagramme (comparer page 23)

$$\begin{array}{ccc} X' = I \times J & \xrightarrow{G'} & I \times J' = X \\ {\scriptstyle F'}\downarrow & & \downarrow{\scriptstyle F} \\ Y' = I' \times J & \xrightarrow{G} & I' \times J' = Y \end{array} \quad .$$

Ce diagramme est tel que pour tout dérivateur à gauche

[page 60]

Commentary · Exposition [ms. 59]

Independent changes of the two variables

The product square illustrates exactness that does not arise from properness of either original functor. Its proof is worth expanding because it also checks the correction in brackets on manuscript p. 60. Fix

$$x_0 = (i_0, j_0'), \quad y_0' = (i_0', j_0), \quad \alpha = (u_0' : f(i_0) \to i_0',\ v_0' : j_0' \to g(j_0)).$$

An object of the factorization category is specified by $i \in I$, $j \in J$ and arrows

$$a : i_0 \to i, \quad r : f(i) \to i_0', \quad b : j_0' \to g(j), \quad s : j \to j_0$$

with $rf(a) = u_0'$ and $g(s)b = v_0'$. The first equation and the second are independent. The same separation holds for morphisms, giving the isomorphism

$$K \cong (i_0, u_0')\backslash(I/i_0') \ \times \ (j_0'\backslash J)/(j_0, v_0').$$

The first factor has initial object $((i_0, u_0'), 1)$; the second has terminal object $((j_0, v_0'), 1)$. Their nerves are therefore contractible, as is the product. This proves the desired base-change isomorphism. The product need not itself have an initial object, which is why the second part of the edition's bracketed correction is necessary as well as the corrected formula.

The result says that a homotopy Kan extension in one variable can be performed while retaining the other variable as a parameter. Restriction in that other variable commutes with the extension. It is one reason that shifted derivators $\mathbb{D}^J(I) = \mathbb{D}(I \times J)$ provide an efficient language for parametrized constructions. It does not say that an arbitrary homotopy limit commutes with an arbitrary homotopy colimit. The latter exchange is the stronger question raised earlier in the chapter and requires additional hypotheses.

on a le changement de base

$$G^* F_* \simeq F'_* G'^* \quad ,$$

la raison ici étant que les foncteurs induits

$$X'/y' \longrightarrow X/y$$

sont des hot-équivalences costrictes, *i.e.* que les catégories K (page 49) sont contractiles. En effet, si $x_0 = (i_0, j_0') \in \mathsf{Ob}X$, $y_0' = (i_0', j_0) \in \mathsf{Ob}Y'$, $u_0 = (u_0', v_0') : F(x_0) \to G(y_0')$, où $u_0' : f(i_0) \to i_0'$, $v_0' : j_0' \to g(j_0)$ sont donnés, on vérifie que la catégorie K cherchée est isomorphe simplement à $(i_0, u_0')\backslash(I/i_0') \times (j_0, v_0')\backslash(J/j_0')$ qui est contractile ayant un objet initial `[plutôt` $(i_0, u_0')\backslash(I/i_0') \times (j_0'\backslash J)/(j_0, v_0')$ `qui est contractile étant produit d'une catégorie ayant un objet initial et d'une catégorie ayant un objet final]`. Ouf!

Je m'aperçois d'un très gros canular. Je croyais me rappeler que pour $f : X \to Y$ qui est une hot-équivalence stricte, on avait pour tout $F \in \mathbb{D}(Y)$, $F \xrightarrow{\sim} f_* f^* F$ (par exemple pour $\mathbb{D} = \mathsf{Der}(Ab)$, ou $\mathbb{D} = \mathsf{Hot}$), *i.e.* f^* pleinement fidèle, d'où aussi que $\Gamma_{\mathbb{D}}(Y, F) \to \Gamma_{\mathbb{D}}(X, f^* F)$ est `[un]` isomorphisme. Mais ce n'est vrai que si F est un coefficient *constant* sur Y, *i.e.* de la forme $q^*(M)$ ($q : Y \to e$, $M \in \mathbb{D}(e) = \mathcal{A}_{\mathbb{D}}$). C'est faux, pour F général, déjà dans le cas de l'immersion $f : \{\varphi\} \to Y$, où φ est un objet initial. Alors, f est toujours une hot-équivalence stricte, mais l'homomorphisme $\Gamma(Y, F) \to \Gamma(X, f^* F)$ n'est en général pas un isomorphisme.

Commentary · Development (commentators') [ms. 60]

A cone construction exposing the homotopy obstruction

The lifting criterion allows an entire space of obstructions to be placed into a functor over the walking arrow. The criterion itself is established mathematics. We did not find the following explicit organization, including its universal ordinary base-change test, in the passages checked in Maltsiniotis (2005, §1 and §4) and Groth et al. (2014a, §3). We record the construction and its proof as a development of the present discussion, without a claim of priority.

Proposition (the cone test). [ours] All categories are small relative to the fixed indexing universe, and completeness below refers to its small limits. Let K be a small category and let $X = K^{\triangleleft}$ be obtained by freely adjoining an initial object o. Define

$$f : X \longrightarrow [1], \qquad f(o) = 0, \quad f(k) = 1 \quad (k \in K).$$

Then f is smooth and a weak equivalence on nerves. It is proper if and only if NK is weakly contractible. If K is nonempty and connected, every strict base-change square for f satisfies right Beck–Chevalley for ordinary diagrams in every complete category. If in addition NK is not weakly contractible, the fibre square at 1 fails right Beck–Chevalley for the derivator of spaces.

Proof. For an incoming arrow into o, the only relevant base arrow is 1_0, and its lift category is a point. For a target $k \in K$ and base arrow $0 \to 1$, the unique lift is $o \to k$, so again the lift category is a point. For base arrow 1_1, the incoming lift category is K/k, with terminal object 1_k. This proves smoothness. Both X and $[1]$ have initial objects; their nerves are contractible, and every functor between them is a weak equivalence on nerves.

The outgoing lift categories are equally explicit. At o over 1_0 the lift category is a point; at $k \in K$ over 1_1 it is $k\backslash K$, with initial object 1_k. The remaining lift category, at o over $0 \to 1$, is exactly K. Thus properness is equivalent to weak contractibility of NK.

For the assertion about ordinary base change, let $p : Y' \to [1]$ be arbitrary and let $X' = X \times_{[1]} Y'$. Fix $y' \in Y'$ and consider

$$h : X'/y' \longrightarrow X/p(y').$$

If $p(y') = 0$, its target is the one-object category consisting of o, and its source is Y'/y', since every arrow into y' must lie over 1_0. The source has a terminal object. If $p(y') = 1$, identify $X/1$ with X. For $k \in K$ there is an isomorphism

$$k\backslash h \cong (k\backslash K) \times (Y'_1/y'),$$

where Y'_1 is the full subcategory of objects sent to 1. Both factors are nonempty and connected: the first has an initial object, the second a terminal object. For the remaining object o, the comma category $o\backslash h$ is X'/y' itself. It contains the full subcategory $K \times \{y'\}$, which is connected. Every object $(a, z \to y')$ maps to this subcategory: if $a \in K$, use (a, y'); if $a = o$, choose any $k \in K$ and use $o \to k$. Hence $o\backslash h$ is nonempty and connected as well.

For completeness, this connectedness criterion preserves ordinary limits of contravariant diagrams for the following direct reason. Given a cone with vertex T over Fh, define its component at a by choosing $(b, a \to h(b))$ in $a\backslash h$ and composing $T \to F(h(b)) \to F(a)$. Morphisms in the comma category give equal composites, and connectedness makes the choice irrelevant. Applying this to an arrow $a \to a'$ proves compatibility of the constructed components. It is the unique cone over F inducing the given cone. Thus restriction induces a bijection on cones with every vertex T, proving the limit comparison isomorphism and consequently ordinary right base change.

Finally take $p : e \to [1]$ to select 1. For a constant space coefficient M, the comparison is

$$M \simeq \mathrm{Map}(|NX|, M) \longrightarrow \mathrm{Map}(|NK|, M).$$

If this were an equivalence for every M, the map $|NK| \to *$ would be an isomorphism in the homotopy category by Yoneda, contradicting the hypothesis. This proves the last assertion. □

A finite example makes the failure explicit. Take K to have four objects a_0, a_1, b_0, b_1, with the four nonidentity arrows $a_i \to b_j$ and no others. Its nerve has four vertices, four edges, and no nondegenerate higher simplices; its realization is a circle. The category X has five objects. All the

ordinary base-change comparisons in the proposition are invertible. In the derivator of spaces, take $M = S^1$. The comparison for the fibre at 1 is

$$S^1 \longrightarrow \mathrm{Map}(S^1, S^1),$$

sending a point to the constant map. The source has one connected component, whereas the target has components indexed by the integer degree. The map lands in degree zero and is not surjective on components. Here the distinction between connected lift categories and contractible lift categories accounts for the entire obstruction.

Notions de $\mathbb{D}$-équivalence et de $\mathbb{D}$-contractibilité.

On va donc reprendre les questions d'acyclicité et de changement de base.

Définition 1. Soit $\mathbb{D}$ un prédérivateur, et soit

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & e & \end{array}$$

dans Dia. On dit que f est une $\mathbb{D}$-*équivalence* si le foncteur

$$F \longmapsto f^*F \quad , \qquad \mathbb{D}(Y) \longrightarrow \mathbb{D}(X)$$

induit un foncteur pleinement fidèle sur la sous-catégorie de $\mathbb{D}(Y)$ formée des $\mathbb{D}$-coefficients *constants*, *i.e.* si pour tous $M, N \in \mathsf{Ob}\mathcal{A} = \mathbb{D}(e)$ ("coefficients absolus"), on a

$$\mathsf{Hom}(q^*M, q^*N) \xrightarrow{\sim} \mathsf{Hom}(p^*M, p^*N) \quad . \tag{1}$$

Supposons que p_* et q_* existent. On posera, pour un faisceau F sur X

$$H^*_{\mathbb{D}}(X, F) \overset{\text{déf}}{=} p_*(F) \qquad (\text{et de même } \ H^{\mathbb{D}}_*(X, F) = p_!(F)\ ).$$

N.B. Et si $F = p^*M$, $M \in \mathcal{A}$, on note simplement $H^*_{\mathbb{D}}(X, M)$, $H^{\mathbb{D}}_*(X, M)$.
L'homomorphisme (1) s'interprète par adjonction comme

$$\begin{array}{ccc} \mathsf{Hom}(M, q_*q^*N) & \longrightarrow & \mathsf{Hom}(M, p_*p^*N) \\ \| & & \| \\ \mathsf{Hom}(M, H^*_{\mathbb{D}}(Y, q^*N)) & \longrightarrow & \mathsf{Hom}(M, H^*_{\mathbb{D}}(X, p^*N)) \end{array}$$

déduit de l'homomorphisme

$$q_*q^*N \longrightarrow p_*p^*N \quad , \qquad i.e. \quad H^*_{\mathbb{D}}(Y, N) \longrightarrow H^*_{\mathbb{D}}(X, N) \quad . \tag{2}$$

Donc dire que (1) est un isomorphisme pour tous M, N signifie que (2) est un isomorphisme pour tout N. De même, si $p_!$, $q_!$ existent cela équivaut à dire que

$$p_!p^*N \longrightarrow q_!q^*N \quad , \qquad i.e. \quad H^{\mathbb{D}}_*(X, N) \longrightarrow H^{\mathbb{D}}_*(Y, N) \tag{2'}$$

est un isomorphisme dans $\mathcal{A}$ pour tout $N \in \mathcal{A}$.

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Commentary · Exposition [ms. 61]

Equivalence measured by constant coefficients

The definition of $\mathbb{D}$-equivalence deliberately retains less information than an equivalence $\mathbb{D}(Y) \simeq \mathbb{D}(X)$. It compares only the full subcategories formed by constant coefficients, while retaining every morphism between them in the coherent diagram categories. Such morphisms can contain cohomological information not visible in the values of the constant diagrams themselves.

Write $p : X \to e$ and $q : Y \to e$, with $p = qf$. If the required right adjoints exist, adjunction gives

$$\mathrm{Hom}_{\mathbb{D}(Y)}(q^*M, q^*N) \cong \mathrm{Hom}_{\mathbb{D}(e)}(M, q_*q^*N).$$

The analogous formula for X identifies the restriction map with the map represented by $q_*q^*N \to p_*p^*N$. It is bijective for every M exactly when this representing map is invertible. This is ordinary Yoneda in $\mathbb{D}(e)$; it does not require an enrichment. With left adjoints one instead writes

$$\mathrm{Hom}_{\mathbb{D}(Y)}(q^*M, q^*N) \cong \mathrm{Hom}_{\mathbb{D}(e)}(q_!q^*M, N)$$

and uses contravariant Yoneda while varying N. This proves both characterizations on manuscript p. 62, including the weaker existence hypothesis that only the representing objects for constant coefficients are needed.

The elementary formal properties also hold already for prederivators. A natural transformation $f \Rightarrow g$ induces the identity on the restrictions of constant coefficients, because its composite with $Y \to e$ is the identity transformation. The maps on their Hom sets therefore agree; a zigzag of transformations has the same consequence. For composable functors, the maps on these sets compose. Two-out-of-three for bijections proves two-out-of-three for $\mathbb{D}$-equivalences. Taking the terminal category as target then gives the characterization of $\mathbb{D}$-contractibility by full faithfulness of the constant-diagram functor.

For $\mathbb{D} = \mathsf{HOT}$ the representing objects have familiar meanings:

$$H^*_{\mathsf{HOT}}(X, M) \simeq \mathrm{Map}(|NX|, M), \qquad H_*^{\mathsf{HOT}}(X, M) \simeq |NX| \times M.$$

The first expression is the derived *mapping space*, considered as an object of the homotopy category. It is not the ordinary Hom set of that category: taking only its connected components would lose higher information. This is the intended reading of the printed $\mathrm{Hom}_{\mathcal{A}}$ in the example. The second formula follows because the homotopy colimit of a constant space is its tensor with the classifying space of the indexing category. These two constructions are adjoint: products with $|NX|$ are left adjoint to mapping spaces out of $|NX|$. Setting $M = *$ in the homological formula proves that a HOT-equivalence is precisely a weak equivalence of nerves; the converse follows by taking products with arbitrary M. The coefficient-sensitive definition thus recovers the ordinary homotopy type for this particular derivator, without asserting that every coefficient theory detects the same class of functors.

En résumé :

Proposition 1. *Si p_* et q_* existent, i.e. si $H^*_{\mathbb{D}}(X, \bullet)$ et $H^*_{\mathbb{D}}(Y, \bullet)$ existent (il suffit même que ça existe pour des coefficients constants), alors f est une $\mathbb{D}$-équivalence si et seulement si f induit un isomorphisme pour les $\mathbb{D}$-cohomologies à coefficients constants de X et Y. Si $p_!$ et $q_!$ existent, i.e. si $H^{\mathbb{D}}_*(X, \bullet)$ et $H^{\mathbb{D}}_*(Y, \bullet)$ existent, f est une $\mathbb{D}$-équivalence si et seulement si f induit un isomorphisme pour les $\mathbb{D}$-homologies à coefficients constants de X et Y.*

Remarque 1. Pour que f soit `[une]` $\mathbb{D}$-équivalence, il *suffit* que pour tout F *constant* sur Y le morphisme d'adjonction $F \to f_*f^*F$ soit `[un]` isomorphisme.

Remarque 2. Si f, $f' : X \to Y$ sont dans une même composante connexe de $\underline{\mathsf{Hom}}(X, Y)$, alors f est $\mathbb{D}$-équivalence $\Leftrightarrow$ f' $\mathbb{D}$-équivalence.

Proposition 2. *Les isomorphismes sont des $\mathbb{D}$-équivalences. Si $X \xrightarrow{f} Y \xrightarrow{g} Z$, si deux parmis f, g, gf sont des $\mathbb{D}$-équivalences, le troisième aussi.*

Définition 2. On dit que X dans Dia est $\mathbb{D}$-*asphérique* (ou $\mathbb{D}$-*contractile*, ou $\mathbb{D}$-*trivial*) si $X \xrightarrow{p} e$ est une $\mathbb{D}$-équivalence, *i.e.* si p^* est pleinement fidèle.

Si p_* existe, cela signifie donc que pour $M \in \mathsf{Ob}(\mathcal{A})$

$$M \xrightarrow[\sim]{\text{adj}} p_* p^* M \quad , \qquad i.e. \quad M \xrightarrow{\sim} H^*_{\mathbb{D}}(X, M) \quad ,$$

et si $p_!$ existe, cela signifie que pour tout M

$$p_! p^* M \xrightarrow[\sim]{\text{adj}} M \quad , \qquad i.e. \quad H^{\mathbb{D}}_*(X, M) \xrightarrow{\sim} M \quad .$$

Corollaire. *Si Y* (resp. X) *est contractile, alors $f : X \to Y$ est une $\mathbb{D}$-équivalence si et seulement si X* (resp. Y) *est contractile.*

Exemple. Prenons $\mathbb{D} = \mathsf{HOT}$. (On peut prendre Dia = catégorie des petites catégories). Alors, pour $M \in \mathcal{A} = \mathsf{Hot}$ et X petite catégorie, on a

$$H^*_{\mathsf{HOT}}(X, M) \simeq \mathsf{Hom}_{\mathcal{A}}(\mathsf{hot}(X), M) \quad ,$$

où

$$\mathsf{hot} : \mathcal{C}at \longrightarrow \mathcal{A} = \mathsf{Hot}$$

est le foncteur canonique qui associe à tout X son "type d'homotopie". On a

$$H^{\mathsf{HOT}}_*(X, M) = \mathsf{hot}(X) \times M \quad .$$

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Commentary · Examples [ms. 62]

How much constant coefficients can detect

The definition of $\mathbb{D}$-equivalence invites a small calculation which makes the dependence on the coefficient theory completely explicit. This calculation concerns represented *prederivators*; no existence of Kan extensions is required. We use $\pi_0 I$ for the set of equivalence classes of objects under zigzags of arrows, equivalently the connected components of NI.

Proposition (coefficient sensitivity of a represented theory). Let A be a locally small category and let $\mathbb{D}_A(I) = \mathrm{Fun}(I^{\mathrm{op}}, A)$. For a functor $f : X \to Y$, precisely the following alternatives occur.

(i) If some set $\mathrm{Hom}_A(M, N)$ contains two different elements, then f is a $\mathbb{D}_A$-equivalence if and only if $\pi_0 f : \pi_0 X \to \pi_0 Y$ is a bijection.

(ii) If every hom-set has at most one element and at least one hom-set is empty, then f is a $\mathbb{D}_A$-equivalence if and only if X is empty exactly when Y is empty.

(iii) If A is empty, or if every hom-set of the nonempty category A is a singleton, then every functor is a $\mathbb{D}_A$-equivalence. In the latter case A is equivalent to the terminal category.

Proof. A natural transformation between constant A-valued diagrams with values M and N consists of arrows $a_i : M \to N$. Naturality along any arrow of I says that the two corresponding arrows a_i are equal. Thus, canonically,

$$\mathrm{Hom}_{\mathbb{D}_A(I)}(M_I, N_I) \cong \mathrm{Hom}_A(M, N)^{\pi_0 I}.$$

Restriction along f is precomposition with $u = \pi_0 f$. If u is a bijection, all these restriction maps are bijections. Conversely, suppose a hom-set H contains distinct elements a, b. If u is not surjective, choose a component of Y outside its image. The constant a-valued function on $\pi_0 Y$ and the function which takes value b only at that component have the same restriction. If u is not injective, choose different components x_1, x_2 with the same image. The function on $\pi_0 X$ taking value b at x_2 and a elsewhere cannot be a restriction. Hence bijectivity on H-valued functions forces u to be a bijection, proving (i).

For (ii), every hom-set is either a singleton or the empty set. Singleton-valued functions never impose a condition. For an empty hom-set, its power by a set S is a singleton when S is empty and is empty otherwise. Therefore the induced map is bijective exactly when $\pi_0 X$ and $\pi_0 Y$ have the

same emptiness status. The hypothesis that an empty hom-set exists makes this condition necessary. For (iii), there are either no pairs (M, N) to test, or each tested set of transformations is a singleton. Finally, if all hom-sets in a nonempty A are singletons, the unique arrows between any two objects are inverse, since the only endomorphism is the identity. This proves the stated equivalence with the terminal category. □

For $A = \mathsf{Set}$, or $A = \mathsf{Ab}$, alternative (i) applies. For the two-element ordered category $0 < 1$, alternative (ii) applies: all functors between nonempty indexing categories are equivalences for this coefficient test. More strikingly, if G is a nontrivial group, the functor $BG \to e$ is a $\mathbb{D}_A$-equivalence for *every* represented prederivator. Both categories have one connected component, so the displayed formula proves the assertion without assumptions on A. To see directly why spaces distinguish it, write $B = |N(BG)|$ and take the constant coefficients $M = *$ and $N = B$. The tested map identifies with

$$[*, B] \longrightarrow [B, B].$$

Its image consists of the homotopy class of constant maps. The identity of B is not in this image: it induces the identity on the nontrivial fundamental group G, whereas a constant map induces the trivial homomorphism. The identification follows from the constant-coefficient homotopy-limit formula $\operatorname{holim}_{(BG)^{\mathrm{op}}} B \simeq \mathrm{Map}(B, B)$ explained in the preceding source discussion. Thus this functor is not a HOT-equivalence, and constant coefficients in an ordinary functor category and constant coefficients in the localized theory of spaces have quite different detecting power.

The hom-set calculation is the elementary naturality condition for constant functors; compare Riehl (n.d., §§1.4 and 1.7). The connected-component test in the first case is standard. We have written out the thin and empty cases as well, so that no nondegeneracy assumption on the coefficient category is hidden. This is an elementary worked consequence of naturality, not an additional axiom on a derivator.

On en déduit que $f : X \to Y$ est une HOT-équivalence (au sens de la définition 1) si et seulement si c'est une hot-équivalence (ou "équivalence faible") *i.e.* si et seulement si

$$\mathsf{hot}(f) : \mathsf{hot}(X) \xrightarrow{\sim} \mathsf{hot}(Y)$$

est un isomorphisme.

Proposition 2. *Soit $f : X \to Y$. Conditions équivalentes (si $\mathbb{D}$ `[est un]` dérivateur à gauche).* p. 37

a) *f est une $\mathbb{D}$-équivalence, et le reste après tout changement de base $Y' \to Y$ qui est un isomorphisme local (par exemple pour $Y/F \to Y$, $F \in \mathsf{Ob}\widehat{Y}$).*

b) *Pour tout $y \in \mathsf{Ob}Y$, le foncteur induit $X/y \to Y/y$ est une $\mathbb{D}$-équivalence.*

c) *Pour tout $y \in \mathsf{Ob}Y$, X/y est $\mathbb{D}$-contractile.*[25]

N.B. On dit que $g : Z \to Y$ est un *isomorphisme local* si pour tout $z \in \mathsf{Ob}Z$, posons $y = g(z)$, le morphisme induit sur les catégories localisées en z et y

$$Z/z \xrightarrow{\sim} Y/y$$

est un isomorphisme. Par exemple toute immersion ouverte, tout foncteur induit $Y/y \to Y$ (foncteur de localisation) sont des isomorphismes locaux.

Il en résulte a) ⇒ b) trivialement.

b) ⇔ c). Résulte du corollaire de la page précédente, si l'on sait que Y/y est $\mathbb{D}$-contractile[26]. Cela résultera de **Der 5** ci-dessous.

c) ⇒ a) "Passage du local au global". Comme l'hypothèse c) est stable par changement de base par immersion ouverte, il suffit de prouver que c) implique que f est une $\mathbb{D}$-équivalence, *i.e.* que pour tout $F = M_Y$ ($M \in \mathsf{Ob}\mathcal{A}$), coefficient constant sur Y, l'homomorphisme d'adjonction [page 64]

$$M_Y \longrightarrow f_* f^* M_Y = f_* M_X$$

25. Autres conditions équivalentes :
d) *Pour tout faisceau constant M_Y sur Y, on a $M_Y \xrightarrow{\sim} f_* f^* M_Y$.*
d′) *Comme* d) *avec faisceau localement constant.*

26. Cf. cor. 1 de **Der 5** (g)

est [un] isomorphisme. Il suffit de prouver que c'est un isomorphisme fibre par fibre (**Der 2**), donc que pour tout $y \in Y$,

$$M_y \longrightarrow f_*(M_X)_y$$

est un isomorphisme. Par **Der 4** (g), on a

$$f_*(M_X)_y \xrightarrow{\sim} \Gamma(X/y, M) \quad ,$$

et l'hypothèse que X/y est $\mathbb{D}$-contractile donne ce qu'on veut.

Proposition 2′. *Dualement, si $\mathbb{D}$ est un dérivateur à gauche* [plutôt à droite]*, conditions équivalentes sur $f : X \to Y$:*

a) *f $\mathbb{D}$-équivalence,* [et] *le reste par tout changement de base $g : Y' \to Y$ qui est un "isomorphisme colocal" (i.e. pour tout $y' \in \mathsf{Ob}Y'$, le foncteur induit sur les catégories "colocalisées" $y'\backslash Y' \xrightarrow{\sim} y\backslash Y$ (où $y = g(y')$)* [est un isomorphisme]*).*

b) *Pour tout $y \in Y$, le foncteur induit $y\backslash X \to y\backslash Y$ est une $\mathbb{D}$-équivalence.*

c) *Pour tout $y \in Y$, $y\backslash Y$ est $\mathbb{D}$-contractile.*

Pour le prouver, on doit utiliser que les $y\backslash Y$ sont $\mathbb{D}$-contractiles (cf. cor. 1 de **Der 5** (d)).

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[page 65]

Commentary · Exposition [ms. 64]

Local isomorphisms and local constancy

A local isomorphism in this text is a discrete fibration of categories. Indeed, the requirement that $Z/z \to Y/g(z)$ be an isomorphism says that every arrow $y_0 \to g(z)$ has exactly one lift with target z. Conversely, unique lifts give the required inverse on slice categories, including their morphisms. A category of elements of a presheaf has this property: lifting an arrow into an element means taking its restriction. The projection $Y/y \to Y$ is the representable case.

This description supplies a small omitted step in the implication (c)⇒(a). The text mentions stability under open immersions, while (a) quantifies over all local isomorphisms. If $g : Y' \to Y$ is a discrete fibration and $X' = X \times_Y Y'$, then for every $y' \in Y'$ there is a canonical isomorphism

$$X'/y' \cong X/g(y').$$

To recover the left-hand object from $(x, \alpha : f(x) \to g(y'))$, lift α uniquely with target y'. Its source y'_0 satisfies $g(y'_0) = f(x)$ and gives the required pair (x, y'_0). Unique lifting also identifies all arrows. Hence condition (c) is stable under every base change in (a), not merely the sieve inclusions mentioned in the proof. The remainder of the argument is complete: the unit on a constant coefficient, evaluated at y, is

$$M \longrightarrow \Gamma_{\mathbb{D}}(X/y, M),$$

which is invertible by $\mathbb{D}$-contractibility, and **Der 2** makes the unit invertible globally.

The locally constant strengthening in footnote 25 has a similarly short proof. Let $F \in \mathbb{D}(Y)$ have invertible structure maps. On Y/y, the natural transformation from the projection to the constant functor at y induces, in the presheaf convention, a morphism from the constant coefficient with value F_y to the restriction of F. Every component is invertible. By **Der 2** this is an isomorphism of coherent coefficients. Its pullback to X/y is therefore constant as well, so the preceding argument applies. Conversely, a statement for every locally constant coefficient includes every constant coefficient. This proves the equivalence claimed in the footnote, rather than requiring local constancy to be an additional axiom.

In Proposition $2'$ the displayed condition (c) must read $y\backslash X$, as duality and condition (b) require. The edition prints $y\backslash Y$. Read literally, that condition holds automatically because $y\backslash Y$ has an initial object, and cannot characterize f. For example, $\emptyset \to e$ fails to be an equivalence for the derivator represented by sets although the printed (c) holds. The following sentence about the contractibility of $y\backslash Y$ is correct: that is the *base* category used to pass between (b) and the repaired (c).

The closure assertions on manuscript p. 65 now have transparent proofs. Pullback preserves discrete fibrations. After a local base change over Z, a composite of strict $\mathbb{D}$-equivalences remains a composite of $\mathbb{D}$-equivalences. The same argument and two-out-of-three prove cancellation when f and gf are strict. For a categorical equivalence f, the pullback along a discrete fibration is again an equivalence: full faithfulness follows by uniquely lifting the arrow upstairs through f, and essential surjectivity follows by lifting an isomorphism with prescribed target through the discrete fibration. Any prederivator preserves categorical equivalences. The assertion about equivalences therefore needs no stronger hypothesis at this point.

Définition 3. Soit $\mathbb{D}$ un prédérivateur (peut-être même un dérivateur à gauche ou à droite). On dit que f est une $\mathbb{D}$-équivalence *stricte* si f satisfait la condition a) de `[la]` proposition 2. On dit que c'est une $\mathbb{D}$-équivalence *costricte* si f satisfait celle de `[la]` proposition $2'$; on dit aussi parfois que f est $\mathbb{D}$-cofinal. (La condition duale, *i.e.* $\mathbb{D}$-équivalence stricte, s'appelerait co-$\mathbb{D}$-cofinal!) [27]

Proposition 3. *Soit $X \xrightarrow{f} Y \xrightarrow{g} Z$. Si f, g sont des $\mathbb{D}$-équivalences strictes, alors gf aussi. Si gf et f sont des $\mathbb{D}$-équivalences strictes, alors g aussi. Si $\mathbb{D}$* `[est un]` *dérivateur à gauche ou à droite, toute équivalence* `[de catégories]` *est une $\mathbb{D}$-équivalence stricte.* (Car elle a un adjoint à droite.)

27. Meilleure terminologie : f est $\mathbb{D}$-*asphérique* ($\mathbb{D}$-équivalence stricte), $\mathbb{D}$-*coasphérique* ($\mathbb{D}$-équivalence costricte).

On utilise le fait que la notion d'isomorphisme local est stable par changement de base.

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Proposition 4. *Soient $\mathbb{D}$ un dérivateur à droite, $f : X \to Y$ un foncteur dans* Dia*, tel que pour $y \in Y$, $y\backslash X$ soit $\mathbb{D}$-contractile (donc f une $\mathbb{D}$-équivalence costricte, sous réserve que $\mathbb{D}$ soit un dérivateur à droite …). Alors pour tout F dans $\mathbb{D}(Y)$ (pas nécessairement à coefficients constant) on a*

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*(F))$$

est un isomorphisme. Réciproquement, si la condition est vraie pour tout F, alors les $y\backslash X$ sont $\mathbb{D}$-contractiles.

Démonstration. Il revient au même de dire que pour tout $M \in \mathsf{Ob}\mathcal{A} = \mathbb{D}(e)$, on a

$$\mathsf{Hom}(M, H^*_{\mathbb{D}}(Y, F)) \xrightarrow{\text{iso}} \mathsf{Hom}(M, H^*_{\mathbb{D}}(X, f^*(F))) \ .$$

Mais par adjonction, cette flèche s'interprète comme

$$(*) \qquad \mathsf{Hom}(M_Y, F) \longrightarrow \mathsf{Hom}(M_X, f^*F)$$

D'autre part, si $f_!$ existe, on a

$$\mathsf{Hom}(M_X, f^*F) = \mathsf{Hom}(f_!(M_X), F) \ ,$$

et l'homomorphisme $(*)$ est déduit de l'homomorphisme d'adjonction

$$(**) \qquad f_!(M_X) = f_!(f^*(M_Y)) \longrightarrow M_Y \ ,$$

en prenant les Hom dans F. Donc la condition équivaut à dire que cet homomorphisme est [un] isomorphisme, quel que soit le coefficient constant M_Y sur Y. Pour voir si c'est un isomorphisme, on regarde fibre par fibre (**Der 2**), or les fibres s'interprètent comme $H^{\mathbb{D}}_*(y\backslash X, M_X)$ (par **Der 4** (d)) et comme M.

Commentary · Exposition [ms. 66]

Cofinality and the final Yoneda argument

The final proposition explains why coasphericity controls arbitrary coefficients although $\mathbb{D}$-equivalence was defined using constant ones. Its proof varies the *target* coefficient while keeping the source constant. That change in the Yoneda argument is what strengthens the conclusion.

There is one existence qualification. A derivator "à droite" in the terminology of these pages supplies $f_!$, not the absolute right Kan extensions used to define $H^*_{\mathbb{D}}$. The cohomological statement consequently also requires the existence of p_* and q_*, where $p : X \to e$ and $q : Y \to e$. Without them the Hom-set formulation in the proof remains meaningful and gives the correct one-sided statement. We prove that formulation first.

Suppose left Kan extensions and their pointwise formulas exist. For $M \in \mathbb{D}(e)$, abbreviate p^*M and q^*M to M_X and M_Y. Adjunction identifies the map

$$\mathrm{Hom}_{\mathbb{D}(Y)}(M_Y, F) \longrightarrow \mathrm{Hom}_{\mathbb{D}(X)}(M_X, f^*F)$$

with precomposition by the counit

$$\varepsilon_M : f_!M_X \longrightarrow M_Y.$$

For fixed M, this is a bijection for every F if and only if ε_M is invertible, by Yoneda in $\mathbb{D}(Y)$. By conservativity and the left pointwise formula, that happens for every M if and only if

$$H^{\mathbb{D}}_*(y\backslash X, M) \longrightarrow M$$

is invertible for every y and M. By the definition and characterization of $\mathbb{D}$-contractibility, this is precisely the requirement that every $y\backslash X$ be $\mathbb{D}$-contractible. Thus the forward implication and the converse in the Hom-set statement are both proved using only the left Kan extensions.

Now assume in addition that p_* and q_* exist. Adjunction identifies the same Hom comparison with

$$\mathrm{Hom}_{\mathbb{D}(e)}(M, q_*F) \longrightarrow \mathrm{Hom}_{\mathbb{D}(e)}(M, p_*f^*F).$$

Varying M and applying Yoneda once more makes its bijectivity equivalent to invertibility of $q_*F \to p_*f^*F$. This proves the cohomological statement exactly as intended, with its existence hypothesis made explicit. No additional repair of the algebraic proof is needed. In modern terminology this is the cofinality criterion for the relevant homotopy limits, with the comma orientation dictated throughout by the presheaf convention.

Overview · I-a

Chapter I, first part: from presheaf operations to homotopical base change

These pages construct a language in which a homotopy theory can carry its own diagram categories. The point of departure is elementary: a functor of indexing categories gives restriction of presheaves, restriction has Kan-extension adjoints when the required limits exist, and special inclusions admit further adjoints when the coefficient category is pointed. What changes in passing to homotopy theory is the meaning of a diagram. A diagram in a homotopy category records only the images of objects and arrows after localization. Localizing a category of diagrams retains additional morphisms and coherence. Grothendieck proposes to organize all these localized categories together, so that a universal construction can be performed on a coherent diagram and remain compatible with further parameters.

The historical motivation is stated explicitly in *Pursuing Stacks*, §69: a single derived category does not provide an adequate setting for the trace and determinant identities encountered around SGA 5 (Grothendieck, 1983, §69). Categories of filtered complexes and diagrams of complexes supply more information, but they too must be organized into a structure stable under introducing new diagram shapes. The letter to Thomason of 2 April 1991 describes the renewed investigation of this intrinsic structure (Grothendieck, 1991a, pp. 2–4). Heller's homotopy theories form a closely related independent axiomatic development (Heller, 1988). The present opening chapter is exploratory: proposed axioms are changed, conclusions are withdrawn, and the final definitions are reached through computations that reveal which hypotheses are actually needed.

The convention that governs the calculations. We work in a fixed universe of small indexing categories; a restricted domain must contain every opposite, product, coproduct, and comma category used in a particular assertion. The manuscript uses presheaves,

$$\mathbb{D}_A(I) = \mathrm{Fun}(I^{\mathrm{op}}, A), \qquad f_! \dashv f^* \dashv f_*.$$

Thus an object over $[1] = (0 \to 1)$ has an underlying arrow $F_1 \to F_0$. Restriction reverses both functors and natural transformations, so the explicit modern source is $\mathsf{Dia}^{\mathrm{coop}}$. The manuscript calls a theory with right Kan extensions a derivator *à gauche*, because these are the operations of left exactness. A theory with left Kan extensions is called *à droite*. The systematic axioms starting at manuscript p. 41 replace the earlier provisional numbering. These changes of convention are substantive reading instructions: a remembered modern formula with the opposite variance can reverse the cofinality condition.

The ordinary pointwise formulas already display the relevant slices:

$$(f_*F)_y = \lim_{(f/y)^{\mathrm{op}}} F, \qquad (f_!F)_y = \operatorname*{colim}_{(y/f)^{\mathrm{op}}} F.$$

For derivators these are homotopy Kan formulas, expressed by invertibility of specified base-change maps. The map, not merely an isomorphism between its source and target, is part of the assertion. This explains the care devoted to adjunction units, counits, and the two equivalent constructions of the base-change mate.

Supports, coherent squares, and shifts. Open inclusions are sieves and closed inclusions are cosieves. In the presheaf convention a closed inclusion extends by terminal objects under i_*; an open inclusion extends by initial objects under $j_!$. Pointedness identifies the two values with zero and permits the exceptional adjoints $i^!$ and $j^?$. The commentary proves these assertions from their universal properties and works out the nonsplit short exact sequence of arrow diagrams underlying the support decomposition. It also gives the finite-ordinal adjunction chain, including its four exceptional endpoints, while retaining the edition's correction of the stronger claim about every monotone map.

The same nonsplit extension exhibits a nonzero morphism between coherent derived diagrams whose components are all zero. Consequently the underlying-diagram functor is conservative without being faithful. This is precisely the distinction the localization axiom requires. Homotopy cartesian squares are coherent squares obtained by right Kan extension from a missing-corner diagram; they need not be ordinary pullbacks in the base homotopy category. Strongness, when imposed, addresses lifting arrows and morphisms of arrows. It is a separate condition from conservativity.

The shift $\mathbb{D}_I(J) = \mathbb{D}(I \times J)$ turns an entire coherent I-diagram into one coefficient. Its elementary formal properties explain the repeated demand that each new axiom survive shifting. The completed shift theorem and product base change establish the proposed morphisms induced by f^*, $f_!$, and f_*. At manuscript p. 18 the question whether evaluation at the base category is fully faithful also has a concrete negative answer for unrestricted morphisms: the invisible extension class yields a nonzero modification between morphisms out of the represented derivator of [1] whose component at the point is zero. The example does not impose continuity on these morphisms.

For a Quillen model category, existence of the homotopy Kan extensions is a theorem rather than a formal property of localization. Cisinski supplies a precise general treatment (Cisinski, 2003, Theorem 6.11). In the ∞-categorical formulation one takes

$$\mathbb{D}_{\mathcal{C}}(I) = h\,\mathrm{Fun}(N(I)^{\mathrm{op}}, \mathcal{C});$$

the pointwise extensions of a complete and cocomplete ∞-category give the derivator operations (Riehl and Verity, 2017, §5.3). Reconstruction results for the whole prederivator retain their hypotheses on size and morphisms. Renaudin's comparison concerns combinatorial model theories and adjunctions (Renaudin, 2009, Theorems 3.3.2 and 3.4.4), while Arlin distinguishes strict transformations from pseudonatural ones (Arlin, 2020, Theorems 19 and 29). These are stronger invariants than the base category alone.

The exactness question has a finiteness boundary. The connective derived theory HOTAB is additive and pointed but is not stable. Its abelianization morphism from spaces preserves homotopy colimits. It fails to preserve homotopy pullbacks: for two discrete two-point spaces over a point, the comparison is a map from $\mathbb{Z}^4$ to $\mathbb{Z}^3$ with an explicitly displayed rank-one kernel. This separates the true pushout assertion from the false pullback assertion in the same passage.

Stability answers the later "Perplexité" after the scope of exactness is fixed. Between stable derivators, preservation of zero objects and cartesian squares is equivalent to preservation of zero objects and cocartesian squares (Groth, 2013, Corollary 4.17). Homotopy finite limits commute with homotopy colimits in the corresponding precise stable theorem (Groth and Shulman, 2017, Theorem 2.14). Arbitrary products and coproducts still need not commute. The diagonal array of copies of a field gives the comparison $k^{(\mathbb{N})} \to k^{\mathbb{N}}$, which omits every sequence with infinite support. The modern finite theorem therefore does not validate an unrestricted preservation claim for all Kan extensions.

Cofinality and universal base change. The decisive correction in the chapter concerns the coefficient system. Contractibility of every f/y does not make restriction fully faithful on all coefficients. For the inclusion of the initial endpoint into [1], every such comma category is a point, but the unit on $F_1 \to F_0$ contains the original structure map. The example $0 \to \mathbb{Z}$ disproves the proposed axiom. The surviving proof of the later cancelled corollary contains a valid slice identification; its last inference uses exactly the false, opposite-orientation cofinality clause. The audit identifies this inference without inventing an additional gap in the calculation.

For right direct image in the manuscript's convention, the correct cofinality test uses arrows from an object of the target to the image of the functor. Applied pointwise to a square,

it produces categories of factorizations of a fixed base arrow. Their weak contractibility is the universal homotopy-exactness criterion. Properness specializes it to the comparison between a strict fibre and a comma category; smoothness is the dual incoming-lift condition. The commentary completes the composition proof by identifying the relevant lift-category functor as a strict pullback and applying Quillen's Theorem A (Quillen, 1973, §1).

The conjectured converse is established: spaces alone detect universal homotopy exactness, so the proposed simultaneous use of HOT and HOTAB is unnecessary (Groth et al., 2014a, Theorem 3.16). Proper and smooth functors are treated systematically by Maltsiniotis (Maltsiniotis, 2005, Theorem 1.2). Strict pullbacks matter here. An equivalence of categories that misses an object can fail both properness and smoothness, as the two-object groupoid example shows.

The commentators' marked cone construction makes the strength of the homotopical test explicit. Adjoin an initial object to a small category K, send that object to 0, and send K to 1. The resulting functor to [1] is always smooth and a weak equivalence of nerves; it is proper exactly when NK is weakly contractible. If K is nonempty and connected, all its strict base-change squares pass the right Kan test for every complete ordinary coefficient category. A four-object crown gives $|NK| \simeq S^1$ and a failure for space coefficients, detected by the degree components of $\mathrm{Map}(S^1, S^1)$. The construction, including its universal ordinary-coefficient assertion, is proved in full and carries no claim of priority.

The final definition and its scope. The corrected notion of $\mathbb{D}$-equivalence compares every morphism between constant coherent coefficients. By Yoneda it is equivalent, when the representing objects exist, to invariance of constant-coefficient homology or cohomology. Its two-out-of-three property is therefore already formal at the prederivator level. The local versions distinguish discrete fibrations from discrete opfibrations and lead to asphericity and coasphericity. The omitted stability check under all local isomorphisms is supplied by an explicit isomorphism of comma categories. Locally constant coefficients are handled by a canonical morphism on a slice and conservativity.

The last proposition turns this constant-coefficient test into a criterion for arbitrary coefficients by varying the target in a Yoneda argument. Its Hom-set formulation needs left Kan extensions only; its cohomological formulation additionally needs the absolute right Kan extensions that define H^*. This is an existence qualification, not a failure of the argument. The questions treated here consequently have precise answers once variance, coefficient class, finiteness, and existence of adjoints are stated. The attribution to Thomason, which the text leaves unspecified, remains a documentary uncertainty.

Donc la flèche envisagée (∗∗) s'interprète comme ed. 39

[page 67]

$$H_*^{\mathbb{D}}(y\backslash X, M_X) \longrightarrow M$$

et c'est l'homomorphisme d'adjonction $p_!(p^*(M)) \to M$.

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & e & \end{array}$$

Donc c'est un isomorphisme pour tout M si et seulement si $y\backslash X$ est $\mathbb{D}$-contractile.
On va énoncer ce qu'on a prouvé.

Théorème 1. *Soit*

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & e & \end{array}$$

dans Dia. *On suppose que* $\mathbb{D}$ *est un prédérivateur satisfaisant* **Der 2**, *que* p_*, q_* *et* $f_!$ *existent, que* $f_!$ *satisfait à* **Der 4** (d) *pour le calcul des fibres. Conditions équivalentes :*

a) f *est une* $\mathbb{D}$*-équivalence costricte, i.e.* (prop. 2) *pour tout* $y \in \mathsf{Ob}\, Y$, $y\backslash X$ *est* $\mathbb{D}$*-contractile ;*

b) *pour tout* $F \in \mathsf{Ob}\, \mathbb{D}(Y)$, *l'homomorphisme suivant est* `[un]` *isomorphisme*

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*(F)) \ ;$$

c) *pour tout* $M \in \mathsf{Ob}(\mathcal{A} = \mathbb{D}(e))$, *l'homomorphisme d'adjonction est* `[un]` *isomorphisme*

$$f_!(M_X) = f_!(f^*(M_Y)) \xrightarrow{\sim} M_Y \ .$$

Commentary · Exposition [ms. 67]

The conclusion reached on manuscript page 67

Manuscript pages 60–66 are commented with the first part of the chapter; its blocks "Equivalence measured by constant coefficients", "How much constant coefficients can detect", "Local isomorphisms and local constancy", and "Cofinality and the final Yoneda argument" give the definitions and the argument now being completed. The distinction needed below is quite exact. A $\mathbb{D}$-equivalence preserves the morphisms between constant objects; a coaspherical functor preserves cohomology with *every* coefficient object on its target. Neither assertion, by itself, says that inverse image is fully faithful on the entire coefficient category.

For the theorem on manuscript page 67, the relevant test is the counit

$$f_! p^* M \longrightarrow q^* M, \qquad p = qf.$$

Applying $\mathrm{Hom}(-, F)$ and then the two adjunctions identifies this test with cohomological invariance for all F. Evaluation at y identifies it instead with the augmentation of homology of $(y\backslash f)$ with constant value M. These are two readings of the *same natural transformation*. This observation explains the appearance of homology in a theorem about cohomology; it also prevents a spurious gap from being introduced into an otherwise complete Yoneda argument. The theorem explicitly supplies the right adjoints to the terminal projections, whose existence was less explicit in the proposition on page 66.

Je vais renuméroter les axiomes sur le prédérivateurs.
Der 1, **Der 2** comme avant (p. 45).

Der 3 [28] : Soit X dans Dia, $p : X \to e$. Supposons X contractile. [29] Alors

$$p^* : M \longmapsto M_X \quad , \qquad p^* : \mathcal{A} = \mathbb{D}(e) \longrightarrow \mathbb{D}(X)$$

est pleinement fidèle.
Der 4 (g) : existence des f_*.
Der 4 (d) : existence des $f_!$.
Der 5 (g) Calcul des fibres de $f_*(F)$.
Der 5 (d) Calcul des fibres de $f_!(F)$.
L'ancien axiome **Der 5** (g) résulte de **Der 2**, **Der 3**, **Der 4** (g,d), **Der 5** (d), grâce au théorème précédent.

ed. 40
[page 68]

Commentary · Exposition [ms. 68]

Keeping the axiom numbers separate

Grothendieck is reorganising the assumptions after the failure diagnosed on manuscript page 60. The new numbering must be kept separate from both the preceding pages and present-day derivator terminology. Here **Der 3** asserts that a contractible indexing category has fully faithful constant-diagram functor. The two parts of **Der 4** assert existence of right and left Kan extensions; the two parts of **Der 5** give their pointwise comma-category formulas. In the now common numbering, existence is usually (Der3), and the pointwise formulas are (Der4); compare Groth (2013, Definition 1.5). The extra strongness axiom used to lift incoherent arrows is a different matter and is not being imposed by this renumbering.

Throughout these comments we retain the manuscript's presheaf convention: a coefficient object over X has underlying diagram $X^{\mathrm{op}} \to \mathbb{D}(e)$. Thus

$$(f_*F)_y \simeq H^*_{\mathbb{D}}(X/y, F), \qquad (f_!F)_y \simeq H^{\mathbb{D}}_*(y\backslash X, F).$$

A modern reference using covariant diagrams reverses the comma categories. The right adjoint is still the right adjoint; the difference comes from replacing every indexing category by its opposite.

The new contractibility axiom is not an assertion that the whole category $\mathbb{D}(X)$ collapses to $\mathbb{D}(e)$. It concerns only constant objects and their morphisms. For example, over the walking arrow a coefficient is a coherent arrow, and its two values need not be isomorphic even though the walking arrow has contractible nerve. What the axiom guarantees is that taking a constant object over this shape introduces no new endomorphisms. This is precisely the amount of invariance used when the lifting categories in the following theorem are contractible.

The comment that both homological and cohomological axioms seem indispensable should consequently be read as a description of this proof strategy, not as a proved independence theorem. Modern treatments may begin with a one-sided derivator and formulate the corresponding one-sided conclusions. The argument here deliberately uses the opposite adjoint to make a universal-coefficient test accessible to pointwise calculation.

Un aspect intéressant du théorème précédent, si on l'envisage comme énonçant l'équivalence de a) et b), c'est que cet énoncé ne fait pas appel à des notions homologiques, mais que sa démonstration y fait appel, sous une forme relativement forte (calcul des fibres de $f_!(F)$). Pour les théorèmes de changement de base pour la $\mathbb{D}$-cohomologie relative, *i.e.* pour les $f^{\mathbb{D}}_*$, on a besoin aussi constamment du calcul des fibres des $f_*(F)$. Donc il semble qu'on ne peut se passer d'aucun des axiomes précédents, y compris **Der 4** (d) et **Der 5** (d), même si on ne veut s'intéresser qu'au formalisme des f^*, f_* (aspect cohomologique). Par la suite, on va donc supposer ces axiomes satisfaits.

28. Axiome des catégories d'indices asphériques, *cf.* variantes **Der 3** a) b) c) page 97, et surtout pages 111-112.
29. N.B. Il suffit de supposer que X a un objet final. Mais ce n'est pas un axiome autodual.

Commentary · Exposition [ms. 68]

The category of lifts behind properness

There is a useful concrete form of the first two conditions of Theorem 2. Given $x \in X$ and $\alpha : f(x) \to y$, let $L_f(x,\alpha)$ have objects

$$h : x \longrightarrow z, \qquad f(z) = y, \qquad f(h) = \alpha.$$

A morphism from h to h' is a map $k : z \to z'$ with $f(k) = 1_y$ and $kh = h'$. These are strict lifting data; replacing equality by an unspecified isomorphism would change the condition. The theorem says that f is $\mathbb{D}$-proper precisely when every $L_f(x,\alpha)$ is $\mathbb{D}$-contractible.

Indeed, if $i_y : X_y \to X/y$ is the canonical inclusion, an object of X/y is exactly a pair (x,α) as above, and

$$((x,\alpha)\backslash i_y) \cong L_f(x,\alpha).$$

Condition (a), applied to the definition of coasphericity, is therefore condition (b). This also proves its stability under arbitrary strict base change. For a pullback $f' : X \times_Y Y' \to Y'$, the lift of an arrow $\alpha' : y'_0 \to y'_1$ from (x, y'_0) is the lift of its image in Y, together with the already fixed second coordinate y'_1. The resulting lifting categories are isomorphic.

The significance of these categories is stronger than the existence of lifts. An opfibration gives a preferred cocartesian lift, which is an initial object of $L_f(x,\alpha)$. Properness permits a whole category of choices, requiring only that constant coefficients cannot distinguish its homotopy from a point. In particular, requiring merely that each strict fibre X_y be contractible misses the lifting condition altogether.

To obtain the base-change statement, first replace Y by any $Y' \to Y$. At an object y', the right Kan extension is calculated over X'/y'. By coasphericity of $X'_{y'} \to X'/y'$ and Theorem 1, restriction induces

$$H^*_{\mathbb{D}}(X'/y', F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X'_{y'}, F|_{X'_{y'}})$$

for *every* coefficient object F on X'/y'. Consequently right direct image is calculated by the actual fibres. After a further base change, the actual fibre is unchanged. The two evaluations of the Beck–Chevalley comparison are therefore identified by their canonical restriction maps; **Der 2** makes the comparison an isomorphism.

Conversely, suppose the fibre formula holds after localisation $Y/y \to Y$. The category Y/y has terminal object 1_y, so evaluating the right direct image there identifies its global cohomology with that of X/y. The asserted fibre formula then states that restriction along i_y preserves cohomology for every object of $\mathbb{D}(X/y)$. Theorem 1 gives the coasphericity of i_y. This completes the equivalence of (a), (b), (c), (c′), and (c″), without substituting constant coefficients where the argument needs arbitrary ones.

Théorème 2. *Soit $f : X \to Y$ dans* Dia. *Conditions équivalentes :*

a) *Pour tout $y_0 \in \mathsf{Ob}\, Y$, le foncteur canonique*

$$X_{y_0} \xrightarrow{\varphi} X/y_0 \quad (\simeq X \times_Y Y/y_0)\ ,$$

où X_{y_0} désigne la fibre en y_0, est une $\mathbb{D}$-équivalence co-stricte.

b) *Pour tout $x_0 \in \mathsf{Ob}\, X$, le foncteur induit sur les catégories colocalisées en x_0 et en $y_0 = f(x_0)$*

$$x_0\backslash X \longrightarrow y_0\backslash Y$$

est à fibres $\mathbb{D}$-contractiles.

c) *La formation des f_*F (F dans $\mathbb{D}(X)$) commute à tout changement de base $Y' \to Y$, et ça reste vrai après tout changement de base – i.e. si on a $Y'' \to Y' \to Y$, d'où* [page 69]

$$\begin{array}{ccccc} X'' & \longrightarrow & X' & \longrightarrow & X \\ {\scriptstyle f''}\downarrow & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ Y'' & \xrightarrow[p']{} & Y' & \xrightarrow[p]{} & Y \end{array}$$

le "théorème de changement de base" est vrai pour f'_* et p'^*.

c′) *La formation des $f_*(F)$ ($F \in \mathsf{Ob}\,\mathbb{D}(X)$) commute au passage aux fibres et ceci reste vrai après tout changement de base $p : Y' \to Y$.*

c″) *Comme* c′) *avec $Y' \to Y$ un morphisme de localisation $Y/y \to Y$.*

d) *Si $Y'' \xrightarrow{p} Y' \to Y$, avec p coasphérique, alors $q : X'' = X \times_Y Y'' \to X' = X \times_Y Y'$ est coasphérique.*

d′) *Comme* d) *avec $Y'' = e$* (donc Y' est catégorie avec objet final e). ed. 41

d″) *Comme* d) *avec $Y'' = \{1\}$, $Y' = \mathbf{\Delta}_1$.*

.

Démonstration. a) ⇔ b) cf. proposition page 53. Montrons que ces conditions impliquent c). Comme elles sont stables par changement de base (*cf.* démonstration immédiate page 55) donc par **Der 2** on est ramené à prouver que $H^*_{\mathbb{D}}(X/y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Y_y, F)$ ($F \in \mathsf{Ob}\,\mathbb{D}(Y)$), ce qui résulte du théorème 1.[30] (Cf. démonstration du théorème page 56, qui marche avec des hypothèses plus générales.)

On a c) ⇔ c′) ⇒ c″) (l'équivalence c) ⇔ c′) par **Der 2**).

Prouvons c″) ⇒ a).

Considérons le diagramme à carrés cartésiens

[page 70]

$$\begin{array}{ccccc} X_{y_0} & \xrightarrow{q} & X/y_0 & \longrightarrow & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle f} \\ \{y_0\} & \longrightarrow & Y/y_0 & \longrightarrow & Y \end{array} \quad .$$

L'hypothèse c″) prouve que pour tout F dans $\mathbb{D}(X/y_0)$, on a $H^*_{\mathbb{D}}(X/y_0, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X_{y_0}, F)$. Par le théorème 1, ceci implique que f est un $\mathbb{D}$-équivalence co-stricte.

Commentary · Exposition [ms. 70]

Completing the tests in the proper base-change theorem

The three conditions labelled (d) are not proved in the displayed argument. They have a short verification which also explains why a single walking arrow is a sufficient test. Write a pullback square as

$$\begin{array}{ccc} X'' & \xrightarrow{q} & X' \\ {\scriptstyle f''}\downarrow & & \downarrow{\scriptstyle f'} \\ Y'' & \xrightarrow[p]{} & Y'. \end{array}$$

If f' is proper, the isomorphism $p^* f'_* \simeq f''_* q^*$ has, on passing to its left adjoint, the canonical isomorphism

$$q_! f''^* \simeq f'^* p_!.$$

Here we take the left adjoints of the two *composite functors*; no general assertion that every mate of every invertible cell is invertible is needed. If p is coaspherical, its counit is an isomorphism on constants by Theorem 1. Applying the displayed comparison to a constant object on Y'' identifies the counit $q_! M_{X''} \to M_{X'}$ with its pullback. Theorem 1 now makes q coaspherical. This proves (c)⇒(d).

Condition (d) implies its tests on inclusions of terminal objects, and these imply the test $\{1\} \to [1]$. Conversely, pull f back along the functor $[1] \to Y$ determined by $\alpha : f(x) \to y$. Apply (d″) to the terminal-object inclusion $\{1\} \to [1]$. In the resulting coaspherical functor $X_y \to X \times_Y [1]$, the comma category under $(x, 0)$ is exactly $L_f(x, \alpha)$. Thus (d″) implies (b). All the asserted conditions are now accounted for.

The parenthesis in (d′), "donc Y' est catégorie avec objet final e", should be understood as specifying the terminal-object tests. A $\mathbb{D}$-coaspherical functor from the point need not literally select a terminal object for every possible coefficient theory. For example, take the represented derivator of the complete lattice $\{0 < 1\}$. Every nonempty discrete category is $\mathbb{D}$-contractible, since its diagonal

30. Donc on utilise **Der 5** (d).

constant-diagram functor is fully faithful. The inclusion $e \to BC_2$ is then $\mathbb{D}$-coaspherical: its only comma category is the discrete two-element set C_2. The unique object of BC_2 is not terminal. Reading (d′) as the restricted terminal-object test avoids attributing this false converse to the proof.

Several other slips here are already in the edition. The proof on page 69 requires X_y, not Y_y, and a coefficient on X, not on Y. At the end of the argument on page 70 the coaspherical functor is the displayed $q : X_{y_0} \to X/y_0$, not f. Definition 4 refers to Theorem 2, not Theorem 1. These corrections repair the typing and references; they do not alter the theorem.

Corollaire 1. Énoncé dual du théorème, pour les foncteurs $X_{y_0} \to y_0\backslash X$, $X/x_0 \to Y/y_0$, et le changement de base des $j_!(F)$ …

Définition 4. Sous les conditions équivalentes du théorème 1, on dit que f est $\mathbb{D}$-*propre.* La condition duale s'appelle f $\mathbb{D}$-*lisse.*

Corollaire 2. *Soit $f : X \to Y$ dans* Dia. *Si f est $\mathbb{D}$-propre ou $\mathbb{D}$-lisse, et à fibres $\mathbb{D}$-contractiles, alors f est une $\mathbb{D}$-équivalence stricte et costricte. Mieux, f est une $\mathbb{D}$-équivalence* universelle *i.e. est une $\mathbb{D}$-équivalence et le reste après tout changement de base.*[31]

Il suffit de prouver que f est une $\mathbb{D}$-équivalence. Supposons par exemple f $\mathbb{D}$-propre. Il suffit ed. 42
de prouver que si Y est $\mathbb{D}$-contractile, alors X l'est (quitte à l'appliquer aux Y/y_0 après [page 71]
changement de base, pour montrer que les X/y_0 sont contractiles). Pour ceci, il suffit de prouver que f est une $\mathbb{D}$-équivalence, et on sait que ceci est le cas si pour tout F *constant* sur Y, l'homomorphisme $F \xrightarrow{\text{adj}} f_*f^*F$ est `[un]` isomorphisme. En fait, c'est vrai pour *tout* F.

Corollaire 3. *Si $f : X \to Y$ est $\mathbb{D}$-propre ou $\mathbb{D}$-lisse, à fibres $\mathbb{D}$-contractiles, alors $f^* : \mathbb{D}(Y) \to \mathbb{D}(X)$ est pleinement fidèle, i.e. les homomorphismes d'adjonction*

$$F \longrightarrow f_*f^*F \quad , \qquad f_!f^*F \longrightarrow F$$

dans $\mathbb{D}(Y)$ sont des isomorphismes.

Il suffit de le voir pour f propre, et cela résulte du calcul des fibres de f_*f^*F …

Proposition 5. *La notion de foncteur $\mathbb{D}$-propre* (resp. $\mathbb{D}$-*lisse*) *est stable par changement de base,* (trivial), *et par composition* (utiliser `[le]` corollaire 2). *Toute immersion fermée est $\mathbb{D}$-propre, toute immersion ouverte $\mathbb{D}$-lisse. Plus généralement, tout isomorphisme local est $\mathbb{D}$-lisse, tout co-isomorphisme local est $\mathbb{D}$-propre.*

Proposition 6. *Soient X, Y sur S, $f : X \to Y$ un S-morphisme. Supposons X, Y propres sur S, ou X et Y lisses sur S, et que pour tout $s \in \mathsf{Ob}\, S$, $f_s : X_s \to Y_s$ soit une $\mathbb{D}$-équivalence, alors f aussi.*[32]

Commentary · Examples [ms. 71]

Properness does not require constant fibres

Let $A : [1] \to \mathcal{C}at$ send 0 to the point and 1 to the discrete category $\{a, b\}$, with the transition selecting a. Its Grothendieck construction X has three objects u, a, b, one nonidentity arrow $u \to a$, and projection $f : X \to [1]$ sending u to 0 and a, b to 1. The map $u \to a$ is a cocartesian lift of $0 \to 1$.

31. Se généralise : **Corollaire 2′.** *Soit*

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

avec X, Y, $\mathbb{D}$-propres sur S. Si f induit une $\mathbb{D}$-équivalence sur les fibres c'est une $\mathbb{D}$-équivalence. Itou dans le cas lisse.

C'est la proposition 6 ci-dessous.

32. Démonstration. On prouvera même que sur la sous-catégorie des coefficients sur Y qui proviennent de $\mathbb{D}(Y)$ `[plutôt de` $\mathbb{D}(S)$`]`, $f^* : \mathbb{D}(Y) \to \mathbb{D}(X)$ induit un foncteur pleinement fidèle. Si f `[est]` propre, je l'interprète comme $q_*q^*F \xrightarrow{\sim} p_*p^*F$ pour tout F, et cela résulte du calcul de p_*, q_* par passage aux fibres (propre), et de l'hypothèse sur les f_s. OK.

The corresponding lift category is a point; the lift categories over identities have their identity lifts as initial objects. Thus f is proper for every derivator under consideration.

For a contravariant coefficient diagram F on X, the right direct image is

$$(f_*F)_0 \simeq F_u, \qquad (f_*F)_1 \simeq F_a \times F_b,$$

and its transition is the composite $F_a \times F_b \to F_a \to F_u$. In the derivator of complexes, these are derived products, which for a finite pair agree with the ordinary finite product. For constant value M, the transition is the projection $M \times M \to M$. Taking $M = \mathbb{Q}$ in degree zero proves that it is not locally constant. The map is therefore proper but not a cohomological fibration in the definition about to be introduced.

This example separates two uses of a fibre calculation. Properness identifies a direct image with the cohomology of the fibres. A fibration condition additionally requires that these cohomologies be transported by isomorphisms along the base. There is no contradiction in a proper map having fibres with different homotopy types.

By contrast, if a proper map has $\mathbb{D}$-contractible fibres, then for every $F \in \mathbb{D}(Y)$ the component at y of its unit is

$$F_y \longrightarrow H^*_{\mathbb{D}}(X_y, (F_y)_{X_y}),$$

which is an isomorphism. Pointwise detection proves that f^* is fully faithful. Since f^* has both adjoints, this is equivalent both to invertibility of the unit of $f^* \dashv f_*$ and to invertibility of the counit of $f_! \dashv f^*$. This proves Corollary 3 and its smooth counterpart. It also gives universal $\mathbb{D}$-equivalence: properness and the fibres are preserved by base change. The claims of strict and costrict equivalence follow by applying that universal assertion to both kinds of local-isomorphism base changes.

Théorème 3. *Soit*

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array}$$

[page 72]

cartésien dans Dia. *Alors le morphisme de changement de base*

$$p^* f_* \longrightarrow f'_* q^*$$

est un isomorphisme si f est $\mathbb{D}$-propre, ou p $\mathbb{D}$-lisse.
Le cas $\mathbb{D}$-propre déjà prouvé. Le cas $\mathbb{D}$-lisse : la démonstration p. 56 est OK.

Commentary · Context [ms. 72]

The geometric pattern and its categorical content

The vocabulary "proper" and "smooth" records a specific geometric analogy. In étale cohomology, proper direct image commutes with base change, and a smooth change of base gives a second base-change theorem under the appropriate torsion hypotheses. The comparison map has the same shape as the one displayed here. Maltsiniotis (2005, Introduction and Theorem 1.2) explicitly traces the categorical definitions to this pattern in *Pursuing Stacks* and identifies the corresponding SGA 4 exposés. His Theorems 4.13 and 5.4 develop the relation between lifting categories and base change in the framework of asphericity structures.

There is no implication here that a proper functor between categories is a proper morphism of spaces in the topological sense. The definition asks for a Beck–Chevalley isomorphism in a coefficient theory. The lifting categories supply its combinatorial content. The smooth half of Theorem 3 is obtained by applying that same calculation with the two adjunctions interchanged; it is not obtained merely by observing that the underlying square is a pullback. In the covariant-diagram convention, Groth (2013, Propositions 1.24 and 1.26) formulate the corresponding exactness of pullback squares for Grothendieck fibrations and opfibrations and of comma squares. Opposing all four indexing categories returns the convention of this manuscript.

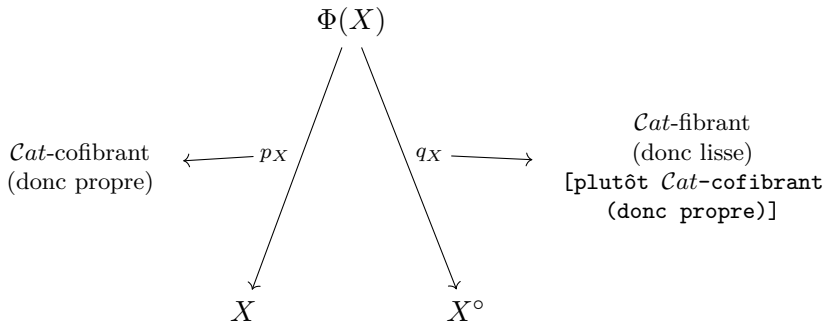
For Proposition 6 one needs only to compare q_*q^*M and p_*p^*M over the common base S. Their components are constant-coefficient cohomologies of Y_s and X_s. The hypothesis identifies those components, and **Der 2** finishes the proof. The footnote's stronger statement for coefficients pulled

back from S follows by the same calculation with M replaced pointwise by F_s, followed by adjunction. This argument is valid without requiring the comparison map $X \to Y$ itself to be proper.

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Comparaison de X et X°.

Soit X [une] catégorie, on définit une catégorie $\Phi(X)$ et des foncteurs



qui seront des hot-équivalences, et des $\mathbb{D}$-équivalences si $\mathbb{D}$ est un dérivateur dont le domaine contient X et $\Phi(X)$. On pose

$$\mathsf{Ob}\,\Phi(X) = \mathsf{Fl}(X) \quad ,$$

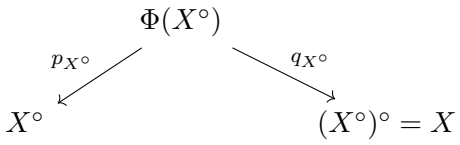
donc mêmes objets que $\mathcal{F}\ell(X) = \underline{\mathsf{Hom}}(\mathbf{\Delta}_1, X)$, mais les morphismes sont différents :

$$\begin{array}{ccc} Y_1 & \xrightarrow{u} & Y_0 \\ {\scriptstyle v_1}\uparrow & & \downarrow{\scriptstyle v_0} \\ Y_1' & \xrightarrow[u']{} & Y_0' \end{array}$$

[page 73]

un morphisme de u dans u' est un couple $v_0 : Y_0 \to Y_0'$, $v_1 : Y_1' \to Y_1$ tel que $v_0 u v_1 = u'$. Ainsi source et but $s(u)$ et $b(u)$ sont l'une contravariante, l'autre covariant en u. On prend pour q le foncteur source, pour p le foncteur but.

Si on applique ceci à X°, on trouve



On a $\mathsf{Ob}\,\Phi(X^\circ) = \mathsf{Fl}(X^\circ) \simeq \mathsf{Fl}(X)$

$$\begin{array}{ccc} Y_1 & \xrightarrow{u} & Y_0 \\ {\scriptstyle v_0}\uparrow & & \downarrow{\scriptstyle v_1} \\ Y_1' & \xrightarrow[u']{} & Y_0' \end{array} \qquad \begin{array}{l} \text{dans } X \\ \\ \text{dans } X \end{array}$$

et mêmes homomorphismes au sens de $\Phi(X)$ et au sens de $\Phi(X^\circ)$. Donc

$$\begin{cases} \Phi(X^\circ) = \Phi(X) \\ p_{X^\circ} = q_X \\ q_{X^\circ} = p_X \quad , \end{cases}$$

ed. 44

le passage de X à X° revient à échanger p_X et q_X. La fibre de $\Phi(X)$ en $x \in \mathsf{Ob}X$ est $\simeq (X/x)^\circ$, celle en $y^\circ \in \mathsf{Ob}X^\circ$ est $\simeq y\backslash X$

$$\begin{cases} p_X^{-1}(\{x\}) \simeq (X/x)^\circ \\ q_X^{-1}(\{y^\circ\}) \simeq y\backslash X \end{cases}$$

N.B. : p_X est W-lisse `[plutôt W-propre]` à fibres W-asphériques, donc dans W^u [W^u = `ensemble des morphismes qui sont dans` W `et le restent après tout changement de base`], q_X est W-propre à fibres W-asphériques, donc dans W^u.

Commentary · Exposition [ms. 73]

The twisted-arrow category and its two projections

The category $\Phi(X)$ is the twisted-arrow category, with the orientation explicitly fixed by the equation $u' = v_0 u v_1$. Keeping that equation in view avoids confusing it with the ordinary arrow category. An object $u : a \to b$ maps to b under p_X and to a, considered in X^{op}, under q_X. Composition is componentwise in the appropriate orders. Replacing u by u^{op} interchanges the two components of a twisted morphism and gives a canonical isomorphism

$$\Phi(X^{\mathrm{op}}) \cong \Phi(X),$$

not the opposite category on the right. The two projection formulas follow under this identification.

Both projections are opfibrations. For p_X, an arrow $\beta : b \to b'$ is lifted from u by the twisted arrow $u \to \beta u$, with source component 1_a and target component β. If a twisted arrow from u to $w : c \to d$ has target component $\gamma\beta$, it factors uniquely through this chosen lift, using target component γ and the same source component. This is the cocartesian universal property. For q_X, an arrow $a \to a'$ in X^{op} is an arrow $\alpha : a' \to a$ in X; its cocartesian lift is $u \to u\alpha$. The same factorisation argument applies.

The fibre of p_X over b is $(X/b)^{\mathrm{op}}$: its arrows are the reversed triangles between maps to b. Its object 1_b is initial. The fibre of q_X over a is $a\backslash X$, whose object 1_a is initial as well. Consequently both maps are proper with contractible fibres. The preceding corollaries prove directly that they are universal $\mathbb{D}$-equivalences and that inverse image along either is fully faithful. This gives a shorter proof of the assertion announced on page 74; it uses exactly the corrected "cofibrant, donc propre" label in the edition.

The displayed slice identities are also literal categorical isomorphisms, and not merely equivalences of homotopy types. An object of $\Phi(X)/x$ is a composable pair $a \xrightarrow{u} b \xrightarrow{h} x$. It determines the arrow $(a, hu) \to (b, h)$ in X/x, hence an object of $\Phi(X/x)$. A twisted morphism consists of $v_1 : a' \to a$, $v_0 : b \to b'$ with $u' = v_0 u v_1$ and $h' v_0 = h$. The remaining slice equation $h' u' = h u v_1$ then follows. These are precisely the morphisms in $\Phi(X/x)$. Reversing the construction is immediate. The second identity follows by applying this argument to X^{op} and interchanging the endpoints.

The last comma notation in the succeeding proof requires a correction. The strict-equivalence test for $q_Z : \Phi(Z) \to Z^{\mathrm{op}}$ uses $\Phi(Z)/z_0^{\mathrm{op}}$, whose additional arrow in Z is $z_0 \to s(u)$. It is this category that is isomorphic to $\Phi(z_0\backslash Z)$. The edition instead prints $z_0\backslash\Phi(Z)$; interpreted literally as a comma category under q_Z, that symbol would require $s(u) \to z_0$ in Z and reverse the intended arrow. With the intended comma category restored, the proof continues correctly. If Z has a terminal object, $z_0\backslash Z$ has both an initial and a terminal object. The unique arrow between these is terminal in its twisted-arrow category, exactly as Grothendieck observes.

On vérifie

$$\begin{cases} \Phi(X)/x \simeq \Phi(X/x) & x \in \mathsf{Ob}X \\ \Phi(X)/x^\circ \simeq \Phi(x\backslash X) & x^\circ \in \mathsf{Ob}X^\circ \end{cases}$$

[page 74]

Je dis que p_X, q_X sont des $\mathbb{D}$-*équivalences strictes* et *costrictes.* Par les formules précédentes, cela signifie que les $\Phi(X/x)$, $\Phi(x\backslash X)$ sont $\mathbb{D}$-contractiles. Donc que si Z a un objet final, $\Phi(Z)$ est $\mathbb{D}$-contractile. Mais considérons

$$\begin{array}{ccc} \Phi(Z) & & \\ & \searrow^{q_Z} & ; \\ & & Z^\circ \end{array}$$

il suffit de prouver que c'est une $\mathbb{D}$-équivalence stricte (puisque Z°, ayant un objet initial, est $\mathbb{D}$-contractil), donc que les $z_0\backslash\Phi(Z)$ sont $\mathbb{D}$-contractiles. On a par les formules précédentes

$$z_0\backslash\Phi(Z) \simeq \Phi(z_0\backslash Z) = \Phi(T) \quad ,$$

où cette fois T a un objet initial $\emptyset$ et un objet final e. Considérons alors l'objet

$$\emptyset \longrightarrow e$$

de $\Phi(T)$, c'est un objet final, car pour tout $x \to y$ dans $\Phi(T)$, il y a une unique flèche

$$\begin{array}{ccc} x & \longrightarrow & y \\ \uparrow & & \downarrow \\ \emptyset & \longrightarrow & e \end{array}$$

Commentary · Examples [ms. 74]

Intervals in a finite ordered set

For $X = [1]$, the twisted-arrow category has objects 1_0, $u : 0 \to 1$, and 1_1, with precisely the nonidentity arrows

$$1_0 \longrightarrow u \longleftarrow 1_1.$$

Thus u is terminal. Under the target projection, the fibre at 0 is the point 1_0, whereas the fibre at 1 is the arrow $1_1 \to u$. Under the source projection, the roles are reversed. Each fibre is contractible, although the projections are not isomorphisms of categories. Their nerves subdivide the interval into two edges.

For the chain $[n]$, objects are intervals (i, j) with $0 \le i \le j \le n$. There is exactly one morphism

$$(i, j) \longrightarrow (i', j') \quad \Longleftrightarrow \quad i' \le i \le j \le j'.$$

The largest interval $(0, n)$ is terminal. More generally this description works for comparable pairs in any poset, showing immediately that Φ preserves finite posets. This is the domain-closure observation at the end of manuscript page 75; it is not an assertion that every finite category is a finite poset.

The comparison with the opposite category is useful because it is natural. For a functor $f : X \to Y$, applying f to arrows and to both components gives $\Phi(f)$, and both endpoint squares commute. Since their vertical projections are $\mathbb{D}$-equivalences, two-out-of-three first gives

$$f \in W_{\mathbb{D}} \iff \Phi(f) \in W_{\mathbb{D}},$$

and then the corresponding equivalence with $f^{\mathrm{op}} \in W_{\mathbb{D}}$. One must have $X^{\mathrm{op}}, Y^{\mathrm{op}}, \Phi(X), \Phi(Y)$ in the domain. There is no claim that X and X^{op} are equivalent as ordinary categories. The natural span compares precisely the information detected by the chosen coefficient theory.

Pour résumer ed. 45

Proposition 1 : *p_X et q_X sont des* hot*-équivalences strictes,* i.e. Hot*-asphériques (donc aussi des $\mathbb{D}$-équivalences strictes si $\Phi(X)$ est dans le domaine de $\mathbb{D}$) à fibres les $(X/x)^\circ$ et $y\backslash X$ respectivement (donc à fibres contractiles).* [page 75]

Corollaire 1. *Pour que X soit $\mathbb{D}$-contractile, il faut et il suffit que X° le soit.*

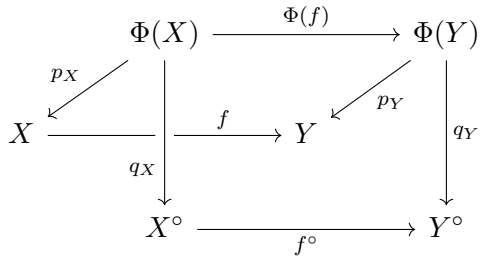
Plus généralement :

Corollaire 2. *Soit $f : X \to Y$. Pour que f soit une $\mathbb{D}$-équivalence, il faut et il suffit que f° le soit.*

On utilisera que le diagramme

$$\begin{array}{ccccc} & & \Phi(X) & & \\ & \swarrow^{p_X} & & \searrow^{q_X} & \\ X & & & & X^\circ \end{array}$$

est fonctoriel en X. On a donc



les deux carrés commutatifs. Comme p_X et p_Y sont des $\mathbb{D}$-équivalences, f est une $\mathbb{D}$-équivalence si et seulement si $\Phi(f)$ l'est, et de même pour f°. Ces raisonnements supposent que si X est dans le domaine de $\mathbb{D}$, $\Phi(X)$ aussi. C'est le cas si Dia est formé des catégories ordonnées finies.

Commentary · Later developments [ms. 75]

Twisted arrows after the opposite-category comparison

The construction has become a standard indexing device for homotopy ends and coends. Riehl (2014, Example 7.2.9 and the footnote in Section 7.8) describe the twisted-arrow category and the identification of its nerve with edgewise subdivision. This supplies a simplicial interpretation of the span $X \leftarrow \Phi(X) \to X^{\mathrm{op}}$. Its realization compares the two classifying spaces without choosing an equivalence of categories.

For a derivator written with the covariant convention, Ponto and Shulman (2016, Section 5) use the same category to define the homotopy coend of $A \in \mathbb{V}(X^{\mathrm{op}} \times X)$ by

$$\int^X A = (\pi_{\Phi(X)^{\mathrm{op}}})_!(t^{\mathrm{op}}, s^{\mathrm{op}})^* A.$$

The opposite on the indexing category is essential in this formula. Ends use the other variance. The passage from this formula to composition of homotopy profunctors explains why the two endpoint projections occur together so persistently in later derivator calculus. Grothendieck uses them here for a narrower purpose: to show that the weak-equivalence class obtained from constant coefficients is unchanged by taking opposites. The later coend formalism extends the utility of the construction; it is not an extra assumption in his proof.

ed. 46

[page 76]

$\mathbb{D}$-fibrations

Proposition 1. *Soit $f : X \to Y$ dans* Dia*, conditions équivalentes :*

a) *Pour tout $y \in Y$, l'inclusion $X_y \to X/y$ est une $\mathbb{D}$-équivalence, et pour toute flèche $y_1 \to y_0$ de Y, $X/y_1 \to X/y_0$ est une $\mathbb{D}$-équivalence.*

b) *Pour tout coefficient* constant *F sur Y, $f_* f^* F$ est localement constant,*[33] *et "se calcule fibre par fibre"* i.e. *l'homomorphisme canonique*

$$f_* f^*(F)_y \xrightarrow{\sim} H^*_{\mathbb{D}}(X_y, F|X_y)$$

`[est un isomorphisme]`.

Ces conditions sont équivalentes, et impliquent les conditions de (b) pour tout F *localement* constant sur Y.

La première condition dans (a) signifie que pour tout $y \in Y$, et tout $M \in \mathsf{Ob}\mathcal{A}$,

$$H^*_{\mathbb{D}}(X/y, M) \longrightarrow H^*_{\mathbb{D}}(X_y, M)$$

33. Par là on entend que pour tout $u : y \to y'$ dans Y, $G_{y'} \to G_y$ est `[un]` isomorphisme. Sans axiomes supplémentaires sur $\mathbb{D}$, il n'est pas clair que cela implique que les "restrictions" de G aux Y/y ou $y\backslash Y$, soient isomorphes à des coefficients constants. Voir cependant l'axiome **Der 3** a′), page 97, qui assure que c'est pareil.

est un isomorphisme. Or cette flèche s'interprète comme

$$f_*(M_X)_y \longrightarrow H^*_{\mathbb{D}}(X_y, M)$$

est un isomorphisme, *i.e.* que $f_*(M_X)$ se calcule fibre par fibre. Comme $M_X \simeq f^*(M_Y)$, c'est la deuxième condition (b). La deuxième condition dans (a), pour $y_1 \to y_0$ signifie que pour M dans $\mathcal{A}$,

$$H^*_{\mathbb{D}}(X/y_0, M) \longrightarrow H^*_{\mathbb{D}}(X/y_1, M)$$

est [un] isomorphisme, or cette flèche s'interprète comme

$$f_*(M_X)_{y_0} \longrightarrow f_*(M_X)_{y_1} \quad ,$$

que ce soit toujours un isomorphisme, c'est la première condition (b). Que (b) soit valable aussi pour F localement constant sur Y est immédiat, car c'est une condition de nature locale par rapport à Y.

Commentary · Exposition [ms. 76]

Local constancy and the first fibration condition

The new fibration condition retains two separate tests. First, the canonical inclusion of an actual fibre into its comma category must be a $\mathbb{D}$-equivalence. Second, the transition functors between comma categories must be $\mathbb{D}$-equivalences. The first identifies the values of f_*M_X with fibre cohomology. The second says that their transport maps are isomorphisms. Since $\mathbb{D}$-equivalence tests all absolute coefficients M, this is an equivalence of conditions, not an inference from the cohomology of a single distinguished coefficient.

The extension from constant to locally constant coefficients is valid with the usual two-dimensional functoriality and pointwise detection axioms. There is a direct proof of the local issue raised in footnote 33. If a category A has terminal object a, the natural transformation $1_A \to ap$ induces, in the manuscript's presheaf convention, a morphism

$$p^*a^*F \longrightarrow F.$$

For arrowwise locally constant F, every component is an isomorphism; **Der 2** makes the whole morphism invertible. If A has an initial object, the corresponding natural transformation gives the comparison in the other direction, with the same conclusion. Applying these observations to Y/y and $y\backslash Y$ proves that restrictions of a locally constant object are constant there. This only uses categories possessing the indicated distinguished object. The stronger assertion for *every* weakly contractible category is the separate issue addressed by the later variants of **Der 3**.

Now fix an arrow $y_1 \to y_0$. Pull a locally constant F back to Y/y_0, where it is isomorphic to the constant object with value F_{y_0}. Pulling this isomorphism further to X/y_0, X/y_1, and the two fibres turns the desired comparison into the one already established for constant coefficients. The isomorphism $F_{y_0} \to F_{y_1}$ supplies the change of value at the other endpoint. This proves the statement for locally constant F, including the compatibility of transport, rather than relying only on the phrase "de nature locale".

For model-category derivators, a systematic treatment is Cisinski (2009, Theorem 1.3 and Section 3): weak equivalences of indexing nerves induce equivalences on the homotopy categories of locally constant diagrams. The distinction between constant and locally constant objects remains necessary when the base has nontrivial fundamental group. It is exactly this distinction that will invalidate the proposed implication on manuscript pages 80–81.

La proposition duale fait appel aux foncteurs

$$y_0\backslash X \longrightarrow y_1\backslash X \qquad \text{et} \qquad X_y \longrightarrow y\backslash X \quad , \qquad \text{[page 77]}$$

et aux faisceaux $f_!f^*M$, pour M constant, ou localement constant sur Y.

Définition 1. Si les conditions équivalentes de la proposition 1 sont satisfaites, on dit ed. 47
que f est une $\mathbb{D}$-*fibration cohomologique* (ou $\mathbb{D}$-*fibration gauche*). Si les conditions duales sont satisfaites, on dit que c'est une $\mathbb{D}$-*fibration homologique* (ou $\mathbb{D}$-*fibration droite*).

N.B. Les $\mathbb{D}$-fibrations cohomologiques sont stables par tout changement de base qui est un isomorphisme local. Les $\mathbb{D}$-fibrations homologiques par tout changement de base qui est un co-isomorphisme local. Mais ce ne sont pas des notions stables par *tout* changement de base[34]. Mais la notion de $\mathbb{D}$-fibration cohomologique est stable par changement de base $\mathbb{D}$-lisse, et celle de $\mathbb{D}$-fibration homologique par changement de base $\mathbb{D}$-propre.

Commentary · Development (commentators') [ms. 77]

A two-object failure of arbitrary base change

The footnote asks for an example showing that the first fibration condition need not survive arbitrary base change. We can supply one before the more elaborate counterexamples later in the manuscript. We checked the treatments of smoothness and asphericity in Maltsiniotis (2005, Sections 1 and 4) and of locally constant diagrams in Cisinski (2009); we did not find this precise small example there. The construction below is an elementary verification for this commentary, without a claim of priority.

A non-universal cohomological fibration. [ours] Let J be the groupoid with two objects a, b and exactly one arrow between every ordered pair of objects. Let $f : [1] \to J$ send 0 to a, 1 to b, and the nonidentity arrow to the unique isomorphism $a \to b$. For the space derivator, f is both a cohomological and a homological fibration in Definition 1, but one strict base change of f has neither property.

Proof. Each strict fibre of f is a point. For either $y \in J$, an object of $[1]/y$ consists of $i \in [1]$ and an arrow $f(i) \to y$. The latter is unique. Projection therefore gives an isomorphism $[1]/y \cong [1]$. Under these isomorphisms every transition functor is the identity. The inclusion of either fibre is an endpoint inclusion $e \to [1]$, hence a weak equivalence of nerves. This proves the cohomological condition. The same argument with $y\backslash[1]$ proves the homological condition. More generally these tests succeed for any coefficient theory in which the point and the walking arrow have the usual constant-coefficient invariance.

Define $g : [1] \to J$ in the reverse order: $g(0) = b$, $g(1) = a$. Form the strict pullback

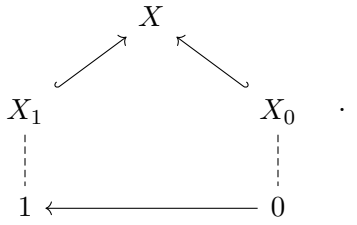
$$P = [1] \times_J [1] \xrightarrow{f'} [1],$$

with f on the first factor and g on the second. Its objects are exactly $(1, 0)$ and $(0, 1)$. There is no arrow from the first to the second, because there is no arrow $1 \to 0$ in the first factor. There is none in the reverse direction, because there is no arrow $1 \to 0$ in the second factor. Thus P is the discrete two-object category, with one object in each fibre of f'.

The comma category $P/1$ is discrete with two objects, whereas P_1 is a point. Hence $P_1 \to P/1$ is not a weak equivalence, and f' is not a cohomological fibration. Dually $0\backslash P$ is discrete with two objects while P_0 is a point, so the homological condition fails too. For complexes of rational vector spaces the same failure is visible in degree zero: the fibre comparison is $\mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$, a projection with nonzero kernel. □

Both [1] and J in the original pair of maps are contractible, and every original strict fibre is a point. The defect is the incompatible choice of direction for a strict lift of the base isomorphism. This example therefore isolates the need for a universal condition without invoking monodromy or a failure of constant coefficients to detect it. The later counterexamples address a different, stronger assertion: whether a universal fibration preserves all $\mathbb{D}$-equivalences under pullback.

Je vais essayer de construire un exemple où le changement de base canule, en posant $Y = \mathbf{\Delta}_1$. Se donner X sur $\mathbf{\Delta}_1$ revient à se donner une sous-catégorie ouverte X_0, et son complémentaire fermé X_1

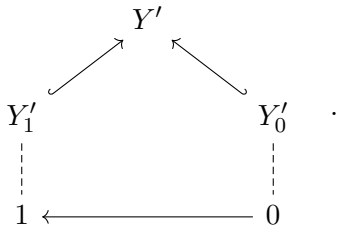


34. ? donner exemple!

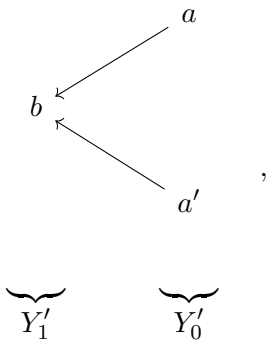
Dire que $X \to Y = \mathbf{\Delta}_1$ est une $\mathbb{D}$-fibration cohomologique signifie simplement que les restrictions [page 78]

$$X_0 \xrightarrow{j} X \quad , \qquad X_1 \xrightarrow{i} X$$

sont des $\mathbb{D}$-équivalences. Prenons un changement de base $Y' \to Y = \mathbf{\Delta}_1$, *i.e.* un Y' avec Y'_0, Y'_1



On voit que la condition Y' lisse sur $\mathbf{\Delta}_1$ signifie que pour tout $y'_1 \in \mathsf{Ob}Y'_1$, la catégorie Y'_0/y'_1 des objets y_0 de Y'_0 au dessus de y'_1 (*i.e.* munis de $y_0 \to y'_1$) est contractile. On voit que si $Y'_1 \neq \emptyset$ alors $Y'_0 \neq \emptyset$ – mais un exemple avec $Y'_0 = \emptyset$, où $Y'_1 = \emptyset$, n'est pas intéressant (car une projection $I \times J \to I$ est toujours une $\mathbb{D}$-fibration cohomologique). Prenons par exemple pour Y' ed. 48



somme amalgamée de deux copies de $Y = \mathbf{\Delta}_1$, donc $X' = X \times_Y Y'$ peut se voir comme somme amalgamée $X \amalg_{X_1} X$. Mais ça semble *pas* être un contre-exemple. Il faudrait que $X_1 \to X'$ ne soit *pas* une $\mathbb{D}$-équivalence.

Commentary · Examples [ms. 78]

Why the first doubled-arrow attempt does not work for spaces

The fork $a \to b \leftarrow a'$ is a useful test even though it does not provide the desired counterexample for the space derivator. A functor $X \to [1]$ decomposes X into a sieve X_0 and a cosieve X_1. Pullback along that fork is two copies of X identified along X_1:

$$X' = X \amalg_{X_1} X.$$

Every nonidentity chain in X' lies in one of the copies. Indeed, after it enters the common cosieve X_1, it cannot leave it to reach the other copy of X_0. Consequently the same pushout description holds degree by degree for nerves:

$$NX' = NX \amalg_{NX_1} NX.$$

The maps $NX_1 \to NX$ are inclusions of simplicial subsets. Their realizations are inclusions of CW subcomplexes. If $X \to [1]$ is a cohomological fibration for spaces, both $X_0 \to X$ and $X_1 \to X$ are weak homotopy equivalences. In particular the latter inclusion is a homotopy equivalence of CW complexes and a cofibration. Gluing a second copy along it preserves its homotopy type: the homotopy extension property, applied to an inverse and its homotopy, makes the inclusion of either copy in the pushout a homotopy equivalence. It follows that $|NX_1| \to |NX'|$ is a weak equivalence.

At a and a' the new fibre and comma category are copies of X_0. At b they are X_1 and X'; the preceding paragraph supplies their comparison. The transport from either of the first two comma categories to the last is a copy of $X_0 \to X \to X'$, hence a weak equivalence. Thus this particular base change is still a cohomological fibration. This establishes Grothendieck's expressed suspicion in the space case; it does not assert that arbitrary base changes work.

Proposition 2. *Soit $f : X \to Y$ dans* Dia. *Conditions équivalentes :*
a) *Pour tout $Y'' \xrightarrow{p} Y' \to Y$, d'où* [page 79]

$$\begin{array}{ccccc} X'' & \xrightarrow{q} & X' & \longrightarrow & X \\ \downarrow & \text{cart} & \downarrow & \text{cart} & \downarrow f \\ Y'' & \xrightarrow[p]{} & Y' & \longrightarrow & Y \end{array} ,$$

si p est une $\mathbb{D}$-équivalence, [alors] *q aussi.*[35]

b) *f est une $\mathbb{D}$-fibration cohomologique "universelle",* i.e. *est une fibration cohomologique, et le reste après tout changement de base.*

b′) *Condition duale de* (b) *: f est une $\mathbb{D}$-fibration homologique universelle.*[36]

a) ⇒ *b*) : Supposons *a*), soit $Y' \to Y$ changement de base, d'où

$$\begin{array}{ccc} X' & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \longrightarrow & Y \end{array} .$$

Pour prouver que f' est une $\mathbb{D}$-fibration cohomologique, on regarde au dessus de Y' les foncteurs de la forme $\{y'\} \to Y'/y'$ ($y' \in \mathsf{Ob}\, Y'$) et $Y'/y'_1 \to Y'/y'_0$ (si $u' : y'_1 \to y'_0$ flèche de Y'), et on voit que les flèches dans $\mathcal{C}at/Y'$ déduites de ceux-ci par le changement de base $X' \xrightarrow{f'} Y'$ restent des $\mathbb{D}$-équivalences ed. 49

$$\begin{array}{ccccccc} X'_1 & \xrightarrow{q} & X'_0 & \longrightarrow & X' & \longrightarrow & X \\ {\scriptstyle f'_1}\downarrow & \text{cart} & \downarrow{\scriptstyle f'_0} & \text{cart} & \downarrow{\scriptstyle f'} & \text{cart} & \downarrow{\scriptstyle f} \\ Y'_1 & \xrightarrow[p]{} & Y'_0 & \longrightarrow & Y' & \longrightarrow & Y \end{array} .$$ [page 80]

Comme p est une $\mathbb{D}$-équivalence, cela résulte bien de a).

b) ⇒ *a*) : Comme la condition *b*) est évidemment stable par changement de base, on est ramené à prouver que si f satisfait *b*), et si on a

(∗)
$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \text{cart} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array}$$

si p est une $\mathbb{D}$-équivalence, alors q aussi, *i.e.*

$$\begin{array}{ccc} H^*_{\mathbb{D}}(X, M) & \xrightarrow{\sim} & H^*_{\mathbb{D}}(X', M) \\ \wr\downarrow & & \wr\downarrow \\ H^*_{\mathbb{D}}(Y, f_*(M_X)) & & H^*_{\mathbb{D}}(Y', f'_*(M_{X'})) \end{array}$$

35. Il faut supposer Y', Y'' Hot-asphériques, cf. page 96, et supposer vérifié l'axiome Der 3 a') de la page 97.

36. b) ⇒ a) canulée, *cf.* annotation marginale,[37] *cf.* contre-exemple page 89. Mais ça marche pour les dérivateurs $\mathbb{D}_i$ ($i \in \mathbb{N}$) associés aux "localisateurs fondamentaux" W_i, *i.e.* à la théorie des i-types d'homotopie.

37. L'implication b) ⇒ a) pas prouvée, seulement sous forme plus faible :

a′) si Y'', Y' sont $\mathbb{D}$-asphériques et 1-connexes, alors $X'' \to X'$ est une $\mathbb{D}$-équivalence. (Il suffit même de le supposer pour Y', Y'' ayant des objets finaux (pour impliquer b) ou ayant des objets initiaux (pour impliquer b′).)

pour tout M dans $\mathcal{A} = \mathbb{D}(e)$. Mais $F = f_*(M_X)$ est localement constant et se calcule "fibre par fibre", de même pour $F' = f'_*(M_{X'})$.

L'équivalence $a) \Leftrightarrow b')$ se déduit de $a) \Leftrightarrow b)$ par dualité (la condition $a)$ étant autoduale), ou en reprenant le même raisonnement pour $a) \Rightarrow b')$, et pour $b') \Rightarrow a)$ aussi (en remplaçant la cohomologie par l'homologie). [page 81]

Commentary · Exposition [ms. 81]

The coefficient lost in the attempted proof

The proposition is an experiment whose unsuccessful part Grothendieck subsequently identifies himself. Its three conditions must therefore be read together with the marginal corrections. Condition (a) says that pullback along every base change of f preserves the entire class of $\mathbb{D}$-equivalences. Conditions (b) and (b′) say that constant coefficients have universally well-behaved cohomological or homological direct images. The first condition implies the other two. The converse requires information about coefficients which the first definition of a $\mathbb{D}$-equivalence does not provide.

Here is the precise point. In a pullback square with lower horizontal map $p : Y' \to Y$, universal fibrewise calculation gives an isomorphism

$$p^* f_* M_X \simeq f'_* M_{X'}.$$

To conclude that $q : X' \to X$ is a $\mathbb{D}$-equivalence, one must compare the cohomology of Y and Y' with coefficient $F = f_* M_X$. This coefficient is locally constant. It need not be constant. The assumption that p is a $\mathbb{D}$-equivalence compares cohomology only with constant coefficients. There is no applicable hypothesis at the last step of the displayed argument. This is the gap, and the examples on manuscript pages 87–95 make it an actual obstruction.

The surviving implication can be proved without this step. For an object y of a base category, the categories Y/y and $\{y\}$ have final objects. The functors between them, and the functors $Y/y_1 \to Y/y_0$ induced by arrows of Y, are $\mathbb{D}$-equivalences under the asphericity axiom. Pulling them back along f gives exactly the inclusions $X_y \to X/y$ and the transition functors between the comma categories. Condition (a), applied also after a preliminary base change, therefore implies (b). Using undercategories gives (b′). Nothing in this argument proves the converse.

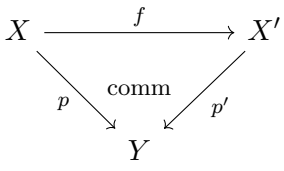
The correction in footnote 35, restricting the test categories to weakly contractible categories, is adequate once locally constant objects on such categories are constant. The qualification is essential: replacing weak contractibility by $\mathbb{D}$-acyclicity is insufficient. Nor does connectedness of a base remove monodromy. We shall return below to a two-object covering that makes this visible with set-valued coefficients alone. Thus the sentence in footnote 36 about all truncated homotopy theories cannot be retained under the usual interpretation of $\mathbb{D}_i$; its case $i = 0$ already fails.

Définition 2. Si les conditions équivalentes de la proposition 2 sont satisfaites, on dit que f est une $\mathbb{D}$-*fibration*.

N.B. C'est une notion *à la fois* locale et "colocale" sur Y (alors que fibration gauche est locale, fibration droite colocale).

Proposition 3. a) *La notion de $\mathbb{D}$-fibration est stable par changement de base et par composition.*[38] ed. 50

b) *Soient*



avec p, p' des $\mathbb{D}$-fibrations (il suffit que p, p' soient des $\mathbb{D}$-fibrations cohomologiques, ou que ce soient des $\mathbb{D}$-fibrations homologiques, ou que p, p' soient propres, ou p, p' lisses). Si f induit des $\mathbb{D}$-équivalences dans les fibres, alors f est une $\mathbb{D}$-équivalence.[39]

38. La stabilité par composition *n'est* pas claire.

39. Peut se préciser considérablement dans le cas où p, p' sont des $\mathbb{D}$-fibrations, *cf.* prop. 6, page 84.

Démonstration. *a*) est immédiat sous la condition *a*) (prop. 2) de la notion de $\mathbb{D}$-fibration. Pour *b*), supposons que f induise des $\mathbb{D}$-équivalences sur les fibres. On veux prouver que

$$H^*_{\mathbb{D}}(X, M) \longrightarrow H^*_{\mathbb{D}}(X', M)$$

est un isomorphisme, pour tout M dans $\mathcal{A}$, mais cette flèche s'identifie à

$$H^*_{\mathbb{D}}(Y, p_*(M_X)) \longrightarrow H^*_{\mathbb{D}}(Y, p'_*(M_{X'}))$$

déduite de

$$p_*(M_X) \longrightarrow p'_*(M_{X'}) \qquad (\text{NB } \ M_X \simeq f^* M_{X'})$$

et il suffit de prouver que cette flèche est un isomorphisme. Pour ceci, on regarde fibre par fibre, et on utilise le fait que "les images directes se calculent fibre par fibre". [page 82]

Commentary · Exposition [ms. 82]

Fibration terminology during the revision

The notation temporarily conceals two different classes of functors. If "$\mathbb{D}$-fibration" means the pullback-preservation condition (a), closure under composition is immediate: a $\mathbb{D}$-equivalence is first pulled back along one factor and then along the other. Base change stability is immediate for the same reason. If the word instead means universal cohomological fibrewise calculation, that proof is unavailable. In a composite, the intermediate coefficient is a locally constant direct image, and the available comparison controls only direct images of coefficients pulled back from the base. The reservation in footnote 38 is consequently a substantive correction to the argument, rather than a doubt about the formal stability of condition (a).

The fibrewise comparison theorem in Proposition 3(b) survives. Write $f : X \to X'$ over Y, with projections p and p'. The natural cohomological comparison is

$$p'_* M_{X'} \longrightarrow p_* M_X,$$

obtained by applying p'_* to the unit $M_{X'} \to f_* f^* M_{X'}$. If both direct images are computed on strict fibres, its component at y is the cohomological map induced by $f_y : X_y \to X'_y$. A $\mathbb{D}$-equivalence on each fibre therefore makes this map invertible by the detection axiom, and applying the direct image $Y \to e$ proves the assertion. The three displayed cohomological arrows in the printed proof point in the reverse direction; the theorem does not depend on that slip. The homological proof uses the counit and has the covariant direction.

The lemma on manuscript page 85 strengthens the sufficient criterion. The property that f_y is a $\mathbb{D}$-equivalence propagates across every arrow of Y, by applying two-out-of-three to the square of comma categories. It is therefore enough to test one object in each connected component. The proposed converse, from the total-space equivalence to fibre equivalences, fails because taking global cohomology can annihilate a nonzero local system. This does not contradict pointwise detection in $\mathbb{D}(Y)$: the functor of global sections is generally not conservative.

Question. Soit $f : X \to Y$ une $\mathbb{D}$-fibration, et soit $F \in \mathbb{D}(Y)$. Alors $f_* f^* F$ se calcule-t-il par passage aux fibres? Même question pour $f_! f^* F$. Pour la première propriété, concernant f_*, on montre que cela marche si F est de la forme $p_*(M_Y)$, où $p : Y' \to Y$ est $\mathbb{D}$-*propre,* et $M \in \mathsf{Ob}\,\mathcal{A}$. D'autre part, la sous-catégorie pleine C de $\mathbb{D}(Y)$ formée des F pour lesquels ça marche est (sauf erreur) "stable par $\mathbb{D}$-hot-$\varprojlim$" en le sens suivant : si on a une catégorie I avec objet final e formant une sous-catégorie fermée (*i.e.* $e \to x \Rightarrow x = e$), et si on a $\mathcal{F} \in \mathbb{D}(Y \times I)$, d'où $I^\circ \to \mathbb{D}(Y)$, $i \mapsto \mathcal{F}_i$ (diagramme squelettique), si les $\mathcal{F}_i$ pour $i \neq e$ sont dans C, et si $\mathcal{F}$ est dans l'image essentielle de j_*, où $j : Y \times (I \setminus \{e\}) \hookrightarrow Y \times I$ est l'inclusion, alors $\mathcal{F}_e$ est aussi dans C.

Commentary · Exposition [ms. 82]

Two useful classes of coefficients

The question about arbitrary coefficients should be separated from the assertion actually needed in the preceding proof. A universally cohomological fibration already calculates f_*f^*F on strict fibres whenever F is locally constant, under the local constancy axiom. Two further operations on coefficients explain the tentative observations here.

Suppose $p : Y' \to Y$ is $\mathbb{D}$-proper, $M \in \mathbb{D}(e)$, and $F = p_*M_{Y'}$. Form $X' = X \times_Y Y'$, with maps $q : X' \to X$ and $f' : X' \to Y'$. Proper base change gives

$$f^*p_*M_{Y'} \simeq q_*M_{X'}, \qquad f_*f^*p_*M_{Y'} \simeq p_*f'_*M_{X'}.$$

The fibre formula for p_* turns evaluation at y into cohomology over Y'_y. Universal fibrewise calculation for f' identifies the restriction of $f'_*M_{X'}$ to Y'_y with the direct image along $X_y \times Y'_y \to Y'_y$. The resulting object is $H^*_{\mathbb{D}}(X_y \times Y'_y, M)$. On the other side, $F_y = H^*_{\mathbb{D}}(Y'_y, M)$, and the product interchange formula identifies $H^*_{\mathbb{D}}(X_y, (F_y)_{X_y})$ with the same object. All identifications are the canonical mates, so this proves the desired comparison, not merely an accidental isomorphism between its endpoints. The printed $p_*(M_Y)$ must here be understood as $p_*(M_{Y'})$.

There is also a formal closure argument. Fix y. The two sides of the fibre comparison, as functors of $F \in \mathbb{D}(Y)$, preserve coherent homotopy limits: restrictions preserve limits computed in the diagram variable, and right Kan extensions compose. Their comparison is a natural transformation between these continuous functors. If it is invertible on a coherent diagram of coefficients, it is invertible on its homotopy limit, by applying the two functors and then the canonical limit comparison. In particular this proves the punctured-cone assertion in the text, provided the indicated diagram shapes belong to Dia. No claim about arbitrary incoherent diagrams in the category $\mathbb{D}(Y)$ is required. This distinction matters because an arbitrary diagram in a homotopy category need not have a chosen coherent lift.

On peut cependant considérer que si on veux que $f : X \to Y$ satisfasse à toutes les propriétés cohomologiques et homologiques qui s'attachent à la notion d'une "fibration" ("localement triviale"), on devrait mettre dans les conditions que f est *lisse*, et *propre*, pour que les changements de base par f commutent aux foncteurs g_*, $g_!$.

Commentary · Development (commentators') [ms. 82]

The fibre formula for coefficients from the base

The question on this page has an affirmative answer for the universal notion of fibration retained on manuscript pages 117–118. The restriction on the coefficient is essential to the argument: it may vary arbitrarily on the base, but the coefficient on the total category is its inverse image. We give the argument because it also answers the question about trivial fibrations on page 84. Throughout this block the domain contains the comma categories, categories of factorizations, and their pullbacks used below. The usual domain of all small categories has this property; so does the domain of finite posets when the categories under consideration are finite posets.

Proposition (fibre calculation for inverse images). [ours] Suppose that $\mathbb{D}$ takes admissible disjoint unions to products, has both homotopy Kan extensions and their pointwise formulas, and detects isomorphisms pointwise. All disjoint unions indexed by the hom-sets below are assumed admissible. Let $f : X \to Y$ be a weak cohomological $\mathbb{D}$-fibration after every base change. Then, for every $F \in \mathbb{D}(Y)$ and $y \in Y$, the canonical comparison

$$(f_*f^*F)_y \longrightarrow H^*_{\mathbb{D}}(X_y, (F_y)_{X_y})$$

is an isomorphism. These comparisons also imply base change for f_*f^*F along an arbitrary functor into Y. The dual assertion holds for $f_!f^*F$ when f is universally a weak homological fibration.

Fix y and write $H = X/y$, $S = X_y$, $v : H \to Y$ for $(x, f(x) \to y) \mapsto f(x)$, and $s : H \to e$, $t : S \to e$ for the projections. The two functors in the comparison are the right adjoints

$$T = s_*v^*, \qquad R = t_*t^*y^* : \mathbb{D}(Y) \longrightarrow \mathbb{D}(e).$$

The pointwise formula identifies T with evaluation of f_*f^* at y. Their left adjoints are respectively $v_!s^*$ and $y_!t_!t^*$. It therefore suffices to prove that the left mate of the comparison,

$$y_!t_!t^* \longrightarrow v_!s^*,$$

is invertible. We can test this on $M \in \mathbb{D}(e)$ and then at each object $z \in Y$.

For an arrow $\gamma : z \to y$, let E_γ be the category of factorizations

$$z \xrightarrow{\beta} c \xrightarrow{\alpha} y, \qquad \alpha\beta = \gamma.$$

A morphism from (c, β, α) to (c', β', α') is $h : c \to c'$ with $h\beta = \beta'$ and $\alpha' h = \alpha$. This category has initial object $(z, 1_z, \gamma)$ and terminal object $(y, \gamma, 1_y)$. Put

$$K_\gamma = X \times_Y E_\gamma,$$

where $E_\gamma \to Y$ remembers the middle object. Thus an object of K_γ is a factorization $z \to f(x) \to y$ of γ, and its morphisms are arrows of X making the two triangles commute.

The pullback $K_\gamma \to E_\gamma$ is a weak cohomological fibration by hypothesis. Its strict fibre over the terminal factorization is X_y. Since slicing over a terminal object changes nothing, the first clause in the definition of a weak cohomological fibration says precisely that

$$X_y \longrightarrow K_\gamma, \qquad x \longmapsto (x, \gamma, 1_y),$$

is a $\mathbb{D}$-equivalence. Consequently it induces an isomorphism on homology with every constant coefficient M. Here we use the equivalence of the homological and cohomological tests for a $\mathbb{D}$-equivalence established on manuscript pages 61–62; no assertion about arbitrary local systems is being substituted for it.

Now $z\backslash v$ is the disjoint union of the K_γ over the discrete set $Y(z, y)$. Indeed, its objects give the two arrows $z \to f(x) \to y$, and every morphism preserves their composite. The pointwise formula for left Kan extension and the coproduct axiom identify the component of the mate at z with

$$\coprod_{\gamma \in Y(z,y)} H^{\mathbb{D}}_*(X_y, M) \longrightarrow \coprod_{\gamma \in Y(z,y)} H^{\mathbb{D}}_*(K_\gamma, M).$$

Each summand is induced by the displayed inclusion of X_y and is an isomorphism. Hence the mate is an isomorphism by pointwise detection, and so is the original comparison between right adjoints. This proves the fibre formula. For a base change $a : Y' \to Y$, both sides of the base-change map at $y' \in Y'$ identify with the same cohomology of $X_{a(y')}$ with value $F_{a(y')}$. Naturality of the comparisons gives the base-change assertion. Applying the argument with the opposite indexing categories and opposite coefficient categories gives the homological assertion. □

The mechanism is the relative form of the factorization-category proof of homotopy exactness in Groth et al. (2014a, Theorem 3.8). The full-faithfulness consequence in the next block has an all-derivator formulation in the homotopical epimorphism criterion of Groth and Štovíček (2016, Theorem 6.10). We have checked these references, as well as Maltsiniotis (2005) and Cisinski (2009); we did not locate this precise answer to the page-82 question in those texts. The label marks our application and proof here, not a claim of priority. Notice what the proof does not require: neither properness of f nor local constancy of F on Y.

Signalons : ed. 51

Proposition 4. *Soit $f : X \to Y$* propre. *Pour que f soit une $\mathbb{D}$-fibration, il faut et il suffit que pour tout $y_1 \to y_0$ dans Y, le foncteur $X/y_1 \to X/y_0$ induit soit une $\mathbb{D}$-équivalence* (i.e. *que f soit une $\mathbb{D}$-fibration* cohomologique). *Dualement, si f est $\mathbb{D}$-lisse, pour que ce soit une $\mathbb{D}$-fibration, il faut et il suffit que pour tout $y_1 \to y_0$ dans Y, $y_0\backslash X \to y_1\backslash X$ soit une $\mathbb{D}$-équivalence.* [page 83]

On pourrait appeler $\mathbb{D}$-fibrations *parfaites* les foncteurs à la fois propres et lisses qui sont des $\mathbb{D}$-fibrations cohomologiques (ou ce qui revient au même des $\mathbb{D}$-fibrations homologiques).

Corollaire. *Supposons f $\mathcal{C}at$-cofibration (donc propre). Pour que f soit une $\mathbb{D}$-fibration, il faut et il suffit que les foncteurs $X_{y_0} \to X_{y_1}$ (pour $y_0 \to u_1$ dans Y) soient des $\mathbb{D}$-équivalences. Duallement, pour $\mathcal{C}at$-fibration (donc lisse).*

Proposition 5. *Soit $f : X \to Y$ dans* Dia. *Conditions équivalentes :*

a) *f est une $\mathbb{D}$-équivalence, et le reste par tout changement de base ;*

b) *f est une $\mathbb{D}$-fibration à fibres $\mathbb{D}$-contractiles.*

Démonstration. *a*) ⇒ *b*) : Que *a*) implique que f soit une $\mathbb{D}$-fibration est immédiat sur la définition de celle-ci (condition *a*) de la proposition 2), et que les fibres sont contractiles est tautologique.

b) ⇒ *a*) : Il suffit de montrer que *b*) implique que f est une $\mathbb{D}$-équivalence. Mais c'est un cas particulier de la proposition 3 *b*) (en y prenant $X' = Y$).

Commentary · Exposition [ms. 83]

Properness separates the two tests

For a proper functor the first half of the cohomological-fibration condition is already supplied by proper base change: the inclusion $X_y \to X/y$ is coaspherical, hence a $\mathbb{D}$-equivalence, and right Kan extensions calculate on strict fibres. What remains is local constancy of f_*M_X for every M. This remaining condition is stable under arbitrary change of base. Indeed properness identifies the new direct image with the restriction of the old one, and restriction preserves local constancy. This proves the criterion of Proposition 4 in the universal cohomological sense. Its identification with the revised bilateral fibration notion uses the local-constancy hypotheses explained at manuscript pages 97–100.

For a Grothendieck opfibration, a choice of cocartesian lifts gives transition functors $\alpha_! : X_{y_0} \to X_{y_1}$ for arrows $\alpha : y_0 \to y_1$. They are defined up to canonical natural isomorphism. The map on constant-coefficient cohomology between the two strict fibres is the map induced by $\alpha_!$, with the cohomological direction reversed. Consequently local constancy says precisely that every transition functor is a $\mathbb{D}$-equivalence. The criterion is independent of the chosen cleavage. The dual statement uses cartesian transport in a Grothendieck fibration.

The later adjective "trivial" is also justified without the false converse of Proposition 2. A universal $\mathbb{D}$-equivalence has $\mathbb{D}$-contractible strict fibres by pulling back to points. Its further pullbacks are again universal equivalences. It has the cohomological and homological fibration tests because the relevant maps form triangles whose other two sides are these universal equivalences. Conversely, a universal cohomological fibration with $\mathbb{D}$-contractible fibres has $M_Y \to f_*M_X$ invertible at every point, hence invertible in $\mathbb{D}(Y)$. Taking global cohomology proves that it is a $\mathbb{D}$-equivalence, and the argument repeats after every base change. Thus Proposition 5 survives the revision. The stronger full-faithfulness question on page 84 requires the separate factorization-category argument given in its commentary.

Quand ces conditions sont satisfaites, on s'attend à ce que le foncteur

$$f^* : \mathbb{D}(Y) \longrightarrow \mathbb{D}(X)$$

[page 84]

soit pleinement fidèle, *i.e.* que les morphisme d'adjonction

$$F \longrightarrow f_*f^*F \quad , \qquad f_!f^* \longrightarrow F$$

dans $\mathbb{D}(Y)$ soient des isomorphismes. (C'est vrai si f est $\mathbb{D}$-propre ou $\mathbb{D}$-lisse, cf. cor. 3, p.71.) Cela revient encore à dire que les f_*f^*F se calculent par passage aux fibres, ou encore itou pour les $f_!f^*F$. Cela nous ramène à la question de la page 82.

Commentary · Development (commentators') [ms. 84]

Universal equivalences give fully faithful restriction

The expectation in the preceding paragraph follows from the fibre formula just proved. It is stronger than the assertion that f preserves cohomology of constant objects, because it controls all morphisms between arbitrary coefficients on Y.

Corollary (the question about trivial fibrations). [ours] Under the domain and derivator hypotheses of the preceding development, suppose that every pullback of $f : X \to Y$ is a $\mathbb{D}$-equivalence. Then

$$f^* : \mathbb{D}(Y) \longrightarrow \mathbb{D}(X)$$

is fully faithful. Both the unit $F \to f_*f^*F$ and the counit $f_!f^*F \to F$ are isomorphisms, for every $F \in \mathbb{D}(Y)$.

To prove this, first pull back over a point $y \to Y$. The hypothesis says that X_y is $\mathbb{D}$-aspherical. Pulling back over Y/y likewise shows that X/y is $\mathbb{D}$-aspherical, since Y/y has a terminal object. The inclusion $X_y \to X/y$ and all transition functors between these comma categories are therefore $\mathbb{D}$-equivalences by two-out-of-three. The same argument works after any base change. Thus f is universally a weak cohomological fibration. At y, compose the adjunction unit with the comparison proved in the preceding block:

$$F_y \longrightarrow (f_* f^* F)_y \longrightarrow H^*_{\mathbb{D}}(X_y, (F_y)_{X_y}).$$

The composite is the constant-coefficient unit for $X_y \to e$, which is invertible by asphericity. The second arrow is invertible by the proposition. The first arrow is therefore invertible, and pointwise detection proves that $F \to f_* f^* F$ is an isomorphism. This is equivalent to full faithfulness of f^*. Since f^* also has the left adjoint $f_!$, full faithfulness makes the other displayed adjunction counit invertible as well. □

There is also a useful direct connection with the modern criterion. For each arrow $\gamma : z \to y$, the category E_γ of its factorizations has an initial and a terminal object. Universality makes $K_\gamma = X \times_Y E_\gamma$ aspherical. These are exactly the factorization categories controlling full faithfulness. In the special case of weak equivalences of nerves, Groth and Ševčík (2016, Theorem 6.10) therefore supplies the same conclusion simultaneously for every derivator. The argument above gives the coefficient-relative conclusion asked for in the manuscript and explains why universality, rather than the single global equivalence $X \to Y$, is the relevant hypothesis.

Définition 3. On dit alors que f est une $\mathbb{D}$-fibration *triviale*.

Proposition 6 (précise prop. 3 page 81). *Considérons un triangle commutatif dans* Dia

$$\begin{array}{ccc} X & \xrightarrow{\ f\ } & X' \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle p'} & \\ & S & \end{array} \quad .$$

a) *Supposons que p et p' soient des $\mathbb{D}$-fibrations gauches, ou que p et p' soient des $\mathbb{D}$-fibrations droites. Soit S_0 une partie de S qui rencontre toute composante connexe de S. Pour que f soit une $\mathbb{D}$-équivalence, il* suffit *que pour tout $s_0 \in S_0$, f induise une $\mathbb{D}$-équivalence $X_{s_0} \to Y_{s_0}$.* ed. 52

b) *Cette condition est aussi nécessaire*[40] *si p et p' sont des $\mathbb{D}$-fibrations. Donc dans ce cas, f est une $\mathbb{D}$-équivalence si et seulement si f induit des $\mathbb{D}$-équivalences fibre par fibre.*

Démonstration. a) Il suffit, en vertu de la proposition 3, de montrer que l'hypothèse implique que pour *tout* $s \in S$, f induit une $\mathbb{D}$-équivalence $f_s : X_s \to X'_s$. Cela résulte aussitôt du lemme : [page 85]

Lemme. *Si p et p' sont toutes deux soit des $\mathbb{D}$-fibrations gauches, soit des $\mathbb{D}$-fibrations droites, et si $s_1 \to s_0$ est une flèche dans S, alors f_{s_1} est une $\mathbb{D}$-équivalence si et seulement si f_{s_0} l'est. (Donc l'ensemble $S' \subset \mathsf{Ob}\, S$ des $s \in S$ tels que f_s soit une $\mathbb{D}$-équivalence est une réunion de composantes connexes de S.)*

Supposons par exemple que p, p' soient des $\mathbb{D}$-fibrations gauches. Considérons, pour $s \in S$, le diagramme

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & X'_s \\ \downarrow & & \downarrow \\ X/s & \xrightarrow[f/s]{} & X'/s \end{array} \quad .$$

40. Pas prouvé![41] Même si $X' = S$, ce n'est pas clair, même si $\mathbb{D} = \mathsf{HOTAB}$, f propre et lisse. Mais OK si S est $\mathbb{D}$-contractile.

41. Faux, *cf.* contre-exemple, p. 87.

Les flèches verticales sont des $\mathbb{D}$-équivalences par l'hypothèse sur p, p', donc f_s est une $\mathbb{D}$-équivalence si et seulement si f/s l'est. Considérons maintenant

$$\begin{array}{ccc} X/s_1 & \xrightarrow{f/s_1} & X'/s_1 \\ \downarrow & & \downarrow \\ X/s_0 & \xrightarrow[f/s_0]{} & X'/s_0 \end{array} \quad .$$

Ici encore, l'hypothèse sur p, p' implique que les flèches verticales sont des $\mathbb{D}$-équivalences, donc f_{s_1} est une $\mathbb{D}$-équivalence si et seulement si f_{s_0} l'est.

Si $Y' = S$, l'énoncé (non prouvé) dit que si $f : X \to S$ est une $\mathbb{D}$-fibration et une $\mathbb{D}$-équivalence, alors les fibres sont $\mathbb{D}$-contractiles. Ce n'est pas sûr même pour une projection $X = T \times S \to S$. Si $\eta = hot_0(S)$, $\xi = hot_0(T)$, la question se réduit à celle-ci : Si pour tout M dans $\mathcal{A}$, on a : l'homomorphisme [page 86]

$$\alpha_M : \xi \times_{\mathbb{D}} M \longrightarrow M$$

satisfait ed. 53

$$\eta \times_{\mathbb{D}} \alpha_M : \eta \times_{\mathbb{D}} (\xi \times_{\mathbb{D}} M) \xrightarrow{\sim} \eta \times_{\mathbb{D}} M$$

est un isomorphisme, alors α_M est lui même un isomorphisme.

Commentary · Development (commentators') [ms. 86]

Why a product cannot supply the counterexample

The product case singled out on manuscript page 86 has a short resolution that does not use an additive coefficient theory. We give it to isolate the role of twisting in the counterexamples which follow. It is a formal retract argument; we claim no priority for it. The locally constant-coefficient results of Cisinski (2009) address the general homotopical framework, but we did not find this particular paragraph of the manuscript treated there.

Cancellation for a nonempty product base. [ours] Assume the action of indexing homotopy types on $\mathbb{D}(e)$ is functorial and that S has an object. If the product projection $S \times T \to S$ is a $\mathbb{D}$-equivalence, then $T \to e$ is a $\mathbb{D}$-equivalence.

Proof. Choose $s : e \to S$ and write $p : S \to e$. Then $ps = 1_e$. For each coefficient M, let $\alpha_M : T \times_{\mathbb{D}} M \to M$ be the augmentation. Functoriality gives a retract diagram in the arrow category of $\mathbb{D}(e)$,

$$\alpha_M \longrightarrow S \times_{\mathbb{D}} \alpha_M \longrightarrow \alpha_M,$$

whose composite is the identity: the first map is induced by s, the second by p. The middle arrow is an isomorphism by the product hypothesis and Fubini for homology. A retract of an isomorphism is an isomorphism. Explicitly, if an arrow a is a retract of an invertible arrow b, with inclusions i_0, i_1 and retractions r_0, r_1, then $r_0 b^{-1} i_1$ is an inverse of a, as the two commutative squares show. Hence every α_M is invertible. This is the asserted $\mathbb{D}$-contractibility of T. □

The nonempty-base hypothesis cannot be omitted: $\varnothing \times T \to \varnothing$ is an isomorphism for every T. The argument establishes that the proposed product example cannot work; it does not establish the analogous assertion for a bundle with monodromy. The next example exploits exactly that distinction.

Voici un *contre-exemple*[42] que je donne d'abord dans le contexte des espaces topologiques, au lieu de $\mathcal{C}at$. Je vais construire une fibration localement triviale et non triviale [page 87]

$$\begin{array}{ccc} X & \supset & S^1 \\ {\scriptstyle f}\downarrow & & \\ S & = & S^1 \end{array} \quad ,$$

42. Voir un contre-exemple encore plus simple pages 93, 94. (Mais il a l'inconvenient de ne pas être à fibres connexes.)

de fibre S^1, telle que f soit une $\mathbb{D}$-équivalence pour tout $\mathbb{D} = \mathsf{Der}(k\text{-}\mathsf{Mod})$, k étant un anneau où 2 est inversible. Cela signifie que

$$H_*(X,k) \longrightarrow H_*(S,k)$$

est un isomorphisme modulo 2-torsion (noyau et conoyau sont annulés par une puissance de 2 - en fait par 2 ici), ou encore $H_*(X,k) \xrightarrow{\sim} H_*(S,k)$ si 2 inversible dans k. Mais la fibre S^1 n'est pas $\mathbb{D}$-contractile (sauf si $k = 0!$). [43]

La fibration est donnée à isomorphisme unique près, en se fixant un point $s \in S$, et en se donnant un automorphisme u de la fibre X_s ("monodromie"). (La classe d'isomorphie du fibré ne dépend que de la classe d'isotopie de u). Je prend pour u la symétrie (inversant l'orientation) $z \mapsto z^{-1}$ du cercle standard $\{z \in \mathbb{C} \mid |z| = 1\}$. Cette fibration "se réalise" par des ensembles ordonés $X \to S$, S^1 étant représenté (sur la base, comme sur les fibres) par

$$S \;=\; \begin{array}{ccccc} & & a & & \\ & \nearrow & & \nwarrow & \\ 0 & & & & 0' \\ & \searrow & & \swarrow & \\ & & b & & \end{array} \;=\; S' \amalg_R S'' \quad ,$$

où R est la catégorie discrète ayant deux éléments a, b. On définit X comme $X' \amalg_Y X''$, où $X' = X|S'$, $X'' = X|S''$, $Y = X|R$. [44]

On pose

ed. 54

[page 88]

$$X' = S' \times \bar{S} \qquad\qquad X'' = S'' \times \bar{S}$$

($\bar{S}$ copie conforme de S, cf ci-dessous) et il faut, pour recoller X' et X'', se donner un isomorphisme entre $X'|R \simeq R \times \bar{S} = \bar{S} \amalg \bar{S}$ et $X''|R \simeq R \times \bar{S} = \bar{S} \amalg \bar{S}$ qui soit un R-foncteur, *i.e.* qui provient de deux isomorphismes

$$\begin{aligned} \bar{S} &\xrightarrow{u_a} \bar{S} \\ \bar{S} &\xrightarrow{u_b} \bar{S} \qquad . \end{aligned}$$

On prend

$$\begin{aligned} u_a &= \mathsf{id}_{\bar{S}} \\ u_b &= \text{symétrie par rapport à l'axe } \bar{0}, \bar{0}' \text{ } \textit{i.e.} \\ &\qquad u_b \text{ laisse fixes } \bar{0}, \bar{0}' \text{ et intervertit } \bar{a} \text{ et } \bar{b}. \end{aligned}$$

(C'est plus clair si on désidentifie la base et la fibre, en prenant fibre

$$\bar{S} \;=\; \begin{array}{ccccc} & & \bar{a} & & \\ & \nearrow & & \nwarrow & \\ \bar{0} & & & & \bar{0}' \\ & \searrow & & \swarrow & \\ & & \bar{b} & & \end{array} \qquad).$$

43. Exemple beaucoup plus simple encore : on prend $X = S^1$, et $X \to S$ le revêtement à deux feuillets. Si on prend le revêtement à n feuillets, c'est aussi OK si k anneau où n est inversible.

44. N.B. S', S'', $R = S' \cap S''$ fermés dans S, X', X'', $Y = X' \cap X''$ fermés dans X.

Commentary · Examples [ms. 88]

The Klein bottle and its sign coefficient

Take a nonzero commutative ring k in which 2 is invertible. Let $r : S^1 \to S^1$ be complex conjugation and form its mapping torus

$$K = (S^1 \times [0,1])/((z,1) \sim (r(z),0)).$$

Projection to $[0,1]/(0 \sim 1)$ gives the Klein-bottle bundle $f : K \to S^1$. We use the cohomological convention in which $k[-1]$ has its copy of k in degree 1. Grothendieck's displayed $k[1]$ is explicitly annotated "en degré 1" and should be translated accordingly, not corrected as a transcription error.

A cell structure has one vertex, two edges a, b, and one two-cell with attaching word $aba^{-1}b$. Here a projects once around the base and b runs around the fibre. The cellular complex is

$$0 \longrightarrow k \xrightarrow{(0,2)} ka \oplus kb \xrightarrow{0} k \longrightarrow 0.$$

The map to the cellular complex of the base sends a to its generator and kills b and the two-cell. Its kernel is the two-term complex $k \xrightarrow{2} k$, which is contractible. Consequently the projection induces an equivalence with every constant derived k-module coefficient, including unbounded complexes: the displayed chain contraction survives applying the appropriate Hom complex. Integrally, one instead gets $H_1(K;\mathbb{Z}) = \mathbb{Z} \oplus \mathbb{Z}/2$, and the kernel of the map to $H_1(S^1;\mathbb{Z})$ is the indicated 2-torsion. The fibre still has $H_1(S^1;k) = k$ and is not $\mathbb{D}$-contractible.

The same computation in local-coefficient language explains why global cohomology loses the fibre. Let L be the rank-one local system on the base with monodromy -1. The fibre cohomology gives

$$Rf_*k_K \simeq k_{S^1} \oplus L[-1].$$

This splitting can be obtained from the cellular circle complex with its reflection action. Alternatively, the fixed point of the reflection gives a section and splits the degree-zero summand. The remaining summand is $L[-1]$. Global sections of L are computed by $k \xrightarrow{-2} k$, in degrees $0, 1$; the complex is zero in the derived category. Thus the extra summand is visible over each point and invisible after taking global sections.

The finite categorical model in the text has four objects in the base and four in each fibre. Choose $S' = \{0, a, b\}$ and $S'' = \{0', a, b\}$. Their intersection is the discrete category $\{a, b\}$. Gluing the products with the four-object circle by the identity over a and by reflection over b produces sixteen objects. Each arrow in the base transports the fibre by an isomorphism. The projection is therefore both a Grothendieck fibration and an opfibration, and remains so after every base change. Its nerve realizes the same monodromy bundle. This supplies the categorical counterexample required by the manuscript, without appealing only to a topological analogy. Cellular calculations and the general language of local coefficients are reviewed in Hatcher (2002, §§2.2, 3.H).

Pour voir que X est une $\mathbb{D}$-fibration sur S (quelque soit $\mathbb{D}$) prouvons qu'il est *propre*[45] sur S - on est ramenés à voir que $X' \to S'$ et $X'' \to S''$ sont propres, ce qui est trivial. Pour voir que c'est une $\mathbb{D}$-fibration, on est ramenés également au cas de $X' \to S'$ et $X'' \to S''$, où c'est aussi trivial. On a utilisé le :

Lemme. *Soient $f : X \to S$, $(S_i)_{i\in I}$ une famille de sous-catégories fermées recouvrant S. Pour tout i, soit $X_i = X \times_S S_i \xrightarrow{f_i} S_i$. Pour que f soit propre, il faut et il suffit que les f_i le soient. Pour que f soit une $\mathbb{D}$-fibration propre il faut et il suffit que les f_i le soient.*

Commentary · Exposition [ms. 88]

Why the closed-cover test is sufficient

The gluing lemma has a local proof which does not require a descent theorem for arbitrary closed covers. A closed full subcategory here is a cosieve: whenever it contains the source of an arrow it contains its target. In the lifting-category criterion for properness, fix $x \in X$ and an arrow

45. N.B. $f : X \to S$ est également *lisse*, car c'est un fibré trivial (un produit) au dessus de deux ouverts $S - \{a\}$, $S - \{b\}$ qui recouvrent S. Mieux : f est à la fois $\mathcal{C}at$-fibrant et $\mathcal{C}at$-cofibrant (car cette condition n'utilise que les propriétés des $X \times_S \mathbf{\Delta}_2 \to \mathbf{\Delta}_2$, pour $\mathbf{\Delta}_2 \to S$).

$f(x) \to s$ in the base. Choose a member S_i of the cover containing $f(x)$. Its cosieve property forces s and every intermediate object relevant to this lift into S_i. The entire category of lifts is therefore the same one computed in $X_i \to S_i$. Its $\mathbb{D}$-contractibility follows from properness there. Necessity follows by base change.

Once properness has been obtained globally, the fibration condition asks whether the fibre-cohomology comparison along each arrow is invertible. Choose S_i containing its source; the arrow and both endpoints lie in that subcategory. The local condition proves the comparison, and the pointwise criterion proves the global condition. This explains both assertions of the lemma. For the displayed four-object circle, the two closed subcategories S' and S'' contain respectively 0 and $0'$, together with both targets a, b, so they meet exactly these requirements. The dual open-cover argument explains smoothness.

Voici le petit calcul qui montre que $f : X \to S$ est une $\mathbb{D}$-équivalence. On fait le calcul en cohomologie. [46] On doit prouver que [page 89]

$$(*) \qquad H^*(S,k) \longrightarrow H^*(X,k)$$

est un isomorphisme (et de même pour tout coefficient constant M sur S). Or ed. 55

$$H^*(X,k) \overset{\text{def}}{=} \mathbb{R}\Gamma_X(k_X) = \mathbb{R}\Gamma_S(\mathbb{R}f_*(k_X)) \quad ,$$

et $(*)$ est induit par

$$i : k_S \longrightarrow \mathbb{R}f_*k_X \quad .$$

Or on a

$$\mathbb{R}f_*k_X \simeq \underbrace{k_S}_{=\mathsf{Im}(i)} \quad \oplus \quad \underbrace{k_S[1]}_{k_S \text{ en degré } 1} \quad \underline{\text{tordu}}$$

savoir "tordu" par l'action de monodromie sur $S = S^1$, laquelle est la symétrie $\lambda \mapsto -\lambda$. On a

$$\mathbb{R}\Gamma_S(k_S[1]^{\text{tordu}}) = H^1(\underbrace{\pi_1(S)}_{=\mathbb{Z}}, k_S^{\text{tordu}}) = k_{(\mathbb{Z})} = k/(u - \mathsf{id})k = k/2k = 0 \ ,$$

d'où le résultat.

Notons que si on regarde

$$X' = X \times_S X \xrightarrow{f'} S \quad ,$$

on a par Künneth relatif

$$\mathbb{R}f'_*(k_{X'}) \simeq \mathbb{R}f_*(k_X) \otimes^{\mathbb{L}} \mathbb{R}f_*(k_X) \simeq k_S \oplus k_S[1]^t \oplus k_S[1]^t \oplus \underbrace{k_S[1]^t \otimes k_S[1]^t}_{\simeq k_S[2] \text{ non tordu}} \ ,$$

d'où

$$H^*_{\mathbb{D}}(X',k) \simeq \mathbb{R}\Gamma_{X'}(k_{X'}) \simeq \mathbb{R}\Gamma_S(\mathbb{R}f'_*(k_{X'})) \simeq \mathbb{R}\Gamma_S(k_S) \oplus \mathbb{R}\Gamma_S(k_S)[2]$$

et

$$H^*_{\mathbb{D}}(S,k) \longrightarrow H^*_{\mathbb{D}}(X',k)$$

n'est *pas* un isomorphisme.

Ainsi $f : X \to S$ qui est une $\mathbb{D}$ fibration (propre et lisse) satisfait bien les conditions équivalentes $b), b')$ de la proposition 2 (page 79), mais pas la condition $a)$, puisque le changement de base [page 90]

$$\begin{array}{ccc} X' = X \times_S X & \xrightarrow{q=pr_1} & X \\ {\scriptstyle f'=pr_2}\downarrow & & \downarrow{\scriptstyle f} \\ S' = X & \xrightarrow[p=f]{} & S \end{array}$$

46. Mais dans le cas du revêtement à 2 feuillets de S, il n'y a pas même de calcul à faire!

est une $\mathbb{D}$-équivalence $S' \to S$, alors que $X' \to X$ ne l'est pas.

Commentary · Examples [ms. 90]

Self-pullback restores the missing cohomology

The self-pullback $K \times_{S^1} K$ is the mapping torus of $r \times r : T^2 \to T^2$. Its fibre has cohomology k in degrees 0 and 2, and k^2 in degree 1. The monodromy acts trivially in degrees $0, 2$ and as -1 on both degree-one generators. The degree-two action is $(-1)(-1) = 1$. The two sign summands again have acyclic global sections, whereas the degree-two summand does not. Hence

$$H^j(K \times_{S^1} K; k) = \begin{cases} k, & j = 0, 1, 2, 3, \\ 0, & \text{otherwise.} \end{cases}$$

The same conclusion follows directly from the mapping-torus complex: in each fibre degree its extra differential is $1 - (r \times r)^*$, which is either zero or multiplication by 2. This calculation avoids extension ambiguities and is valid over the stated ring.

The map $K \to S^1$ is a constant-coefficient equivalence, but its pullback $K \times_{S^1} K \to K$ cannot be one, because the source has the additional nonzero groups in degrees $2, 3$. Thus even a bundle which is both proper and smooth in the categorical sense does not preserve all constant-coefficient equivalences under pullback. Properness controls the base-change comparison for a direct-image functor; it does not assert that the base-change map is an equivalence for whichever weak-equivalence class one has chosen on the base.

The easier covering example in footnote 43 exhibits the same distinction. The degree-n map $S^1 \to S^1$ is a k-homology equivalence when n is invertible in k, since it is the identity in degree 0 and multiplication by n in degree 1. Its strict fibre has n points. Its self-pullback is a disjoint union of n circles, so the pulled-back projection fails already on H_0 for $n > 1$. This example also avoids any question about a derived splitting.

Voici un autre contre-exemple, sur le même principe mais un peu plus simple. Soit G un groupe fini (par exemple $G = \mathbb{Z}/2\mathbb{Z}$), faisons le opérer sur un objet Z de $\mathcal{C}at$, soit $S = [G] = B_G$ la catégorie réduite à un objet a, avec $G = \mathsf{End}(a)$. Soit E_G la catégorie (isomorphe à $(B_G)/a$)

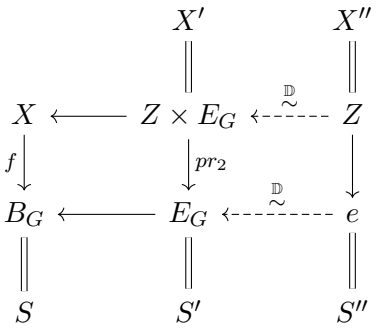
$$\mathsf{Ob}\ E_G = G \quad , \qquad \mathsf{Hom}_{E_G}(g, g') = \{h \in G \mid g = g'h\} \quad ,$$

composition évidente - alors E_G est un groupoïde trivial (1-connexe). Le goupe G y opère à gauche librement par translations, et $B_G = E_G/G$.

Si G opère à gauche sur Z, on peut alors l'utiliser pour tordre Z, par ed. 56

$$X = E_G \times_G Z = (E_G \times Z)/\text{opérations “diagonales” de G} \quad . \qquad \text{[page 91]}$$

Donc on a un diagramme cartésien



Comme le changement de base $S' = E_G = (B_G)/a \to B_G = S$ transforme X sur B_G en un produit (lequel est un $\mathbb{D}$-fibré propre et lisse sur S'), on voit que X sur B_G est un $\mathbb{D}$-fibré propre et lisse (car c'est là une condition locale sur B_G).

On suppose maintenant que G est fini d'ordre $n > 1$, et soit k un anneau où n est invesible, $\mathbb{D} = \mathsf{Der}(k - \mathsf{Mod})$. Alors $E_G \to B_G$ est une $\mathbb{D}$-équivalence, ou ce qui revient au même $e \to B_G$ l'est.[47] Pourtant, après changement de base soit par $S' = E_G$, soit par $S'' = e$, $X' \to X$ (ou

47. En fait, e, E_G, B_G sont $\mathbb{D}$-asphériques.

ce qui revient au même, $X'' \to X$) n'a guère de raison d'en être aussi. On a en cohomologie

$$\mathbb{R}\Gamma_X(k_X) \simeq \mathbb{R}\Gamma_{BG}(f_*(k_X)) \simeq \mathbb{R}\Gamma_G(G, H^*(Z,k))$$

donc (n étant inversible dans k) :

$$H^*(X,k) = \mathbb{R}\Gamma_X(k_X) \simeq H^*(Z,k)^G \longrightarrow H^*(Z,k) \quad , \qquad \text{[page 92]}$$

ce n'est pas en général un isomorphisme. On peut prendre par exemple $G = \mathbb{Z}/2\mathbb{Z}$, $Z \sim S^1$ comme avant, et G opérant par symétrie de S^1 comme précédemment.

Cet exemple est particulièrement convainquant, car c'est un exemple *avec S, S' $\mathbb{D}$-asphériques,* et $S' \to S$ un isomorphisme local (en fait, un revêtement principal de S). C'est aussi un exemple d'un $\mathbb{D}$-fibré lisse et propre $X \to S$, sur une base *asphérique* S, et tel que l'inclusion d'une fibre dans X

$$X_s \longrightarrow X$$

ne soit *pas* une $\mathbb{D}$-équivalence.

Le seul reproche qu'on puisse faire à cet exemple, en comparaison avec le précédent, c'est que les catégories X, S ne sont pas des ensembles ordonnés. Et S ne peut être un ensemble ordonné fini, car l'espace classifiant d'un groupe fini a de la cohomologie en toutes dimensions (sauf erreur).

Commentary · Examples [ms. 92]

Averaging over a finite group

The second counterexample places the same phenomenon over a base which is already $\mathbb{D}$-acyclic. Let G be finite of order invertible in k. The category E_G has one arrow between every ordered pair of objects of G; in particular each object is both initial and final. The quotient by left translation is BG. A G-category Z gives an associated bundle

$$X = (E_G \times Z)/G \longrightarrow BG.$$

An arrow in the base acts on the fibre by an isomorphism, so this is again a fibration and an opfibration with universally locally constant fibre transport. Pulling back to the unique object of BG recovers Z.

For a $k[G]$-module V, the averaging operator

$$e_G(v) = |G|^{-1} \sum_{g \in G} gv$$

is an idempotent with image V^G. More than a splitting, it proves exactness of invariants: an invariant element of a quotient can be lifted arbitrarily and its lift averaged. The same argument applies degreewise to complexes, so cohomology commutes with invariants. Thus

$$H^*(X;k) = H^*(Z;k)^G, \qquad H^*(BG;k) = k$$

with the second k in degree zero. The first formula is a formula for cohomology; a functorial complex-level statement uses derived sections of Z with their action, followed by invariants. It does not require choosing a splitting of an arbitrary complex into its cohomology groups.

Take $G = C_2$ and let Z be the four-object circle with reflection. Its cohomology is the trivial representation in degree zero and the sign representation in degree one. Therefore X and BG are $\mathbb{D}$-acyclic, whereas their common fibre Z is not. This disproves the qualification "OK si S est $\mathbb{D}$-contractile" in footnote 40 as well as the unqualified necessity assertion of Proposition 6(b). The replacement needed there is complete $\mathbb{D}$-asphericity, which controls locally constant coefficients, or ordinary weak contractibility under the appropriate axiom.

An even smaller choice is the discrete regular G-set $Z = G$. Then X is equivalent to the contractible category E_G, while its fibre has $|G|$ connected components. This is a wholly categorical example with finitely many objects, although its nerve is not finite-dimensional. A nontrivial finite group cannot have a finite-poset classifying space: if it did, the finite covering corresponding to its universal cover would be a finite contractible complex, so its Euler characteristic would be both 1 and $|G|$ times an integer. The phrase "cohomology in all dimensions" in the manuscript should be read as the obstruction to a bounded-dimensional classifying space, not as a claim of nonvanishing in every positive degree for every coefficient ring.

Mais on peut s'inspirer de cet exemple pour le modifier de façon qu'il se transpose à des ed. 57
ensembles ordonnés. Il suffit de réaliser, pour G donné : [page 93]

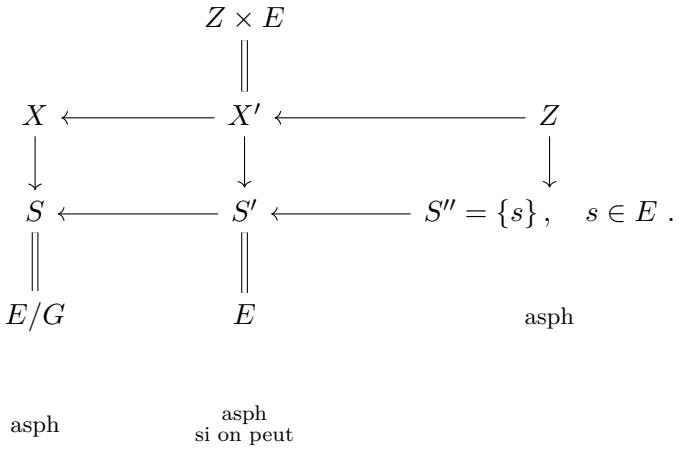
a) Une opération *libre* de G sur un objet E de $\mathcal{C}at$, telle que $H^*(E,k)^G \simeq k$, *i.e.* $B = E/G$ soit $\mathbb{D}$-asphérique.

b) Une opération de G sur un objet Z, telle que l'inclusion

$$H^*(Z,k)^G \longrightarrow H^*(Z,k)^G$$

ne soit pas [un] isomorphisme.

c) Choisir pour E, Z des ensembles ordonnés, et s'assurer que $B = E/G$ est également représenté par une catégorie ordonnée. Sauf erreur, il en sera de même pour $X = E \times Z/G$. On aura en tout les cas



Cela donne encore un exemple avec X $\mathbb{D}$-fibré propre et lisse, sur une base S $\mathbb{D}$-asphérique, tel que l'inclusion d'une fibre X_s dans X ne soit pas une $\mathbb{D}$-équivalence. On peut même se débrouiller pour que $H^*(Z,k)^G \simeq k$ ce qui implique que X est lui aussi asphérique (mais les fibres de $X \to S$ ne sont pas $\mathbb{D}$-asphériques). Cela donne donc un autre contre-exemple au b) [page 94]
de la prop. 6 (p. 84).

Voici la façon la plus économique, je crois, de réaliser la situation que j'ai dite. Topologiquement, je prend

$G = \{\pm 1\}$	
$E = S^2$	sphère euclidienne, sur laquelle $\{\pm 1\}$
	opère par antipodisme $x \mapsto -x$
$B = S^2/\{\pm 1\}$	plan projectif réel
$Z = S^1$	où G opère comme précédemment par la symétrie.

C'est donc un exemple très proche de celui du début (p. 87), où on pouvait aussi considérer la base ($B = S^1$) comme quotient d'un E ($= S^1$) par $G = \{\pm 1\}$ opérant librement (par l'antipodisme $z \mapsto -z$ dans $\mathbb{C} \simeq \mathbb{R}^2$). Mais dans cet exemple, la base n'était pas $\mathbb{D}$-asphérique. ed. 58

La façon la plus économique de réaliser la sphère S^2 par un ensemble ordonné est sans doute par sa décomposition "globulaire" par trois paires de cellules de dimension 0, 1, 2,

donnant l'ensemble ordonné

$$I \quad = \quad \begin{array}{ccc} S_2^- & & S_2^+ \\ \uparrow & \nwarrow\!\nearrow & \uparrow \\ S_1^- & & S_1^+ \\ \uparrow & \nwarrow\!\nearrow & \uparrow \\ S_0^- & & S_0^+ \end{array}$$

et l'antipodisme échange S_i^- et S_i^+, mais le quotient n'est pas ordonné, c'est la catégorie (importante pour la théorie des 2-catégories) :

[page 95]

$$\begin{array}{c} S_2 \\ {\scriptstyle p_1^-}\uparrow\uparrow{\scriptstyle p_1^+} \\ S_1 \\ {\scriptstyle p_0^-}\uparrow\uparrow{\scriptstyle p_0^+} \\ S_0 \end{array} \qquad \text{avec} \qquad \begin{cases} p_1^- p_0^- = p_1^- p_0^+ \\ p_1^+ p_0^- = p_1^+ p_0^+ \end{cases}$$

Mais prenons la subdivision barycentrique de la décomposition cellulaire précédente de S dont les sommets correspondent aux éléments de $\mathsf{Ob}\ I$, les simplexes aux parties totalement ordonnés de $\mathsf{Ob}\ I$. On trouve l'octaèdre, ayant 6 sommets, 12 arêtes, 8 faces, et G opère librement sur l'ensemble des simplexes, donc le quotient : 3 sommets, 6 arêtes, 4 faces, décomposition cellulaire du plan projectif. (Sa décomposition par trois droites.) On présente S^1 par l'ensemble ordonné (comme au début)

$$\begin{array}{ccc} S_1^- & & S_1^+ \\ \uparrow & \nwarrow\!\nearrow & \uparrow \\ S_0^- & & S_0^+ \end{array} \qquad \textit{i.e.} \qquad \begin{array}{ccc} & S_1^+ & \\ \nearrow & & \nwarrow \\ 0 = S_0^- & & S_0^+ = 0' \\ \searrow & & \swarrow \\ & S_1^- & \end{array}$$

et G y opère par symétrie échangeant S_1^- et S_1^+. Donc l'ensemble ordonné X aura $4 \times 13 = 52$ sommets et c'est modeste …

Commentary · Development (commentators') [ms. 95]

The finite poset and its residual top class

The passage to finite posets deserves a complete computation. The number 52 in the text is correct. The claimed acyclicity of the resulting total space is not. We give both verifications, retaining the printed construction and its error in the source. The transfer calculation used below is standard; see Hatcher (2002, §3.G). We did not find a published discussion of this particular 52-object calculation in the sources checked.

Footnote 42 describes the later example as having disconnected fibres, whereas the explicit choice $Z = S^1$ on manuscript page 94 has connected fibres. The covering alternative of footnote 43 and a discrete regular G-set instead give disconnected fibres; the intended reference in footnote 42 cannot be fixed unambiguously.

The corrected finite model. [ours] Let E be the face poset of the boundary of an octahedron. Its vertices are $\{\pm1, \pm2, \pm3\}$; a nonempty face is a subset containing at most one element of each antipodal pair. There are $6 + 12 + 8 = 26$ faces. The involution τ negates every vertex. Let Z be the four-object circle poset with two vertices v_0, v_1 below two edges e_0, e_1, and let τ fix the vertices and exchange the edges. Then $B = E/C_2$ and $X = (E \times Z)/C_2$ are posets with 13 and 52 objects. Their realizations satisfy

$$|NB| \simeq \mathbb{RP}^2, \qquad |NX| \simeq (S^2 \times S^1)/((x, z) \sim (-x, \bar{z})).$$

For $2 \in k^\times$, their cohomology is

$$H^*(B; k) = k, \qquad H^j(X; k) = \begin{cases} k, & j = 0, 3, \\ 0, & \text{otherwise.} \end{cases}$$

The fibre is a circle, so its inclusion into X is not a k-cohomology equivalence, but $X \to B$ is not a k-cohomology equivalence either.

Proof. If nonempty faces σ, ρ satisfy both $\sigma \subseteq \rho$ and $\sigma \subseteq -\rho$, then ρ contains an antipodal pair, which is impossible. Hence for fixed representatives there is at most one group element transporting a possible upper face to a face containing the lower one. Quotient incidence therefore gives at most one morphism between two orbit objects. Ranks prevent nontrivial cycles. The same argument applied to the first coordinate proves the assertion for $E \times Z$. Every chain in either quotient lifts uniquely once its first object is chosen, so the nerve of the quotient is the quotient of the nerve. The nerve of E is the barycentric subdivision of the octahedral sphere, and the nerve of Z is a circle. Realization commutes with these finite products and these free simplicial quotients. This proves the assertions about realization and object counts.

For a finite regular covering with group G, pullback identifies cochains on the quotient with invariant cochains upstairs: every simplex has $|G|$ lifts, permuted freely and transitively. Averaging makes invariants exact when $|G|$ is invertible. Consequently the quotient cohomology is the invariant part of the covering-space cohomology. The antipodal map on S^2 acts by -1 on H^2, whereas reflection on S^1 acts by -1 on H^1. In $H^*(S^2 \times S^1; k)$ the actions in degrees $0, 1, 2, 3$ are respectively $+1, -1, -1, +1$. This gives exactly the displayed answer. The product of the two anti-invariant generators is invariant and survives in degree 3. It is the term omitted by the proposed implication on manuscript pages 93–94. □

There is an earlier incidence slip on the same page. For the six-object globular poset, label the two orbits of arrows between consecutive levels by the product of the endpoint signs. Composition multiplies these signs. The orbit relations are consequently

$$p_1^- p_0^- = p_1^+ p_0^+, \qquad p_1^- p_0^+ = p_1^+ p_0^-.$$

The displayed relations in the edition instead hold p_1 fixed while changing p_0; they do not describe this antipodal quotient. This does not affect the face-poset construction just proved, which gives an unambiguous replacement.

The broader recipe (a)–(c) also needs care. If E is the discrete regular G-set, then E/G is a point and $X = (E \times Z)/G \cong Z$. Taking a nontrivial action on $H^*(Z; k)$ satisfies the proposed invariant-cohomology conditions, yet the fibre inclusion is the identity. Thus those conditions alone do not force a counterexample. For the selected connected $E = S^2$ the fibre inclusion does fail, as calculated above, but invariants of the separate factors do not determine invariants of their tensor product.

As an independent check, the accompanying exact rational chain computation gives simplex counts $(13, 36, 24)$ for NB, with boundary ranks $(12, 24)$, and $(52, 340, 576, 288)$ for NX, with boundary ranks $(51, 289, 287)$. The resulting Betti numbers are $(1, 0, 0)$ and $(1, 0, 0, 1)$. These finite calculations corroborate the proof; they are not used to infer the missing degree-three class.

Finalement, il semble qu'il ne reste plus rien de la proposition 2 (p.79) introduisant la notion de $\mathbb{D}$-fibration – pas même l'équivalence des deux conditions b) et b') ($\mathbb{D}$-fibration à gauche universelle, $\mathbb{D}$-fibration à droite universelle). On songe à leur relation à une condition a) modifiée ainsi (pour $f : X \to Y$) : ed. 59 [page 96]

a) Pour tout $Y'' \to Y' \to Y$, avec Y' et Y'' Hot-asphériques (et pas seulement $\mathbb{D}$-

asphériques) $X'' = X \times_Y Y'' \to X' = X \times_Y Y'$ est une $\mathbb{D}$-équivalence. (Quand on suppose que Y'' et Y' ont des éléments finaux, c'est la condition que f soit une $\mathbb{D}$-fibration à gauche universelle; quand on suppose que Y'' et Y' ont des éléments initiaux, c'est la condition $\mathbb{D}$-fibration à droite universelle.) [48]

On a évidemment $a) \Rightarrow b)$ et $a) \Rightarrow b')$ et on aimerait que ce soient des équivalences. Voici un axiome intéressant [49] sur $\mathbb{D}$, qui semble vérifié dans tous les cas connus, qui assure l'équivalence souhaitée. Je vais en même temps reformuler **Der 3** en trois volets (*cf.* p. 67).

Der 3 a) Si $Z \in$ Dia est Hot-asphérique, alors $p : Z \to e$ est une $\mathbb{D}$-équivalence, *i.e.* par exemple $p^* : \mathcal{A} = \mathbb{D}(e) \to \mathbb{D}(Z)$ est pleinement fidèle.

[page 97]

b) Soit $f : Z \to Z'$ une flèche dans Dia. Si f est une Hot-équivalence, c'est une $\mathbb{D}$-équivalence (*i.e.* $f^* : \mathbb{D}(Z') \to \mathbb{D}(Z)$ induit une équivalence entre les sous-catégories de Z, Z' [plutôt de $\mathbb{D}(Z)$, $\mathbb{D}(Z')$] formées des objets constants).

a') (Précise considérablement a). Si $Z \in$ Dia est Hot-asphérique, alors on a a), et l'image essentielle de $\mathbb{D}(e) = \mathcal{A}$ dans $\mathbb{D}(Z)$ est formée des objets localement constants de Z, *i.e.* des objets tels que le foncteur

$$Z^\circ \longrightarrow \mathcal{A} \qquad z \longmapsto F_z$$

soit localement constant (*i.e.* transforme toute flèche en isomorphisme) donc isomorphe à un foncteur constant (puisque Z est Hot-asphérique, donc 1-connexe).

Commentary · Later developments [ms. 97]

Locally constant objects as a homotopy invariant

The additional axiom sought here became a theorem for the usual derivator framework. Cisinski (2009, Corollary 3.8) proves that a weak equivalence of nerves induces an equivalence on locally constant objects for every derivator on small categories. The model-category statement is Theorem 1.3 of the same paper; Theorem 3.13 organizes locally constant objects into a derivator. Thus Der 3 b′ is available under those modern hypotheses, rather than remaining an unresolved extra condition. This does not retroactively prove the assertion from only the provisional axioms currently being tried in the manuscript.

The modern interpretation uses the fundamental ∞-groupoid $\Pi_\infty|NZ|$. For a coefficient ∞-category $\mathcal{C}$ presenting the homotopy theory, locally constant coherent diagrams are functors from this groupoid to $\mathcal{C}$, and the category denoted $\mathbb{D}(Z)_{\mathrm{lc}}$ is their homotopy category. If $|NZ|$ is contractible, evaluation gives the desired equivalence with $\mathbb{D}(e)$. Constancy of the underlying functor into $\mathbb{D}(e)$ alone is weaker: it forgets higher coherence. This is the same distinction emphasized in the commentary at manuscript p. 7 ("Coherent squares and what the axioms remember"), now applied to local systems.

Il faut bien noter que dans a'), il ne suffit pas de supposer que Z soit $\mathbb{D}$-asphérique, au lieu de Hot-asphérique. Dès que la $\mathbb{D}$-équivalence est strictement plus faible que Hot, il semble que a') devient toujours faux, en tant que condition sur $\mathbb{D}$, si on y suppose seulement que Z soit $\mathbb{D}$-asphérique. Par exemple, même si $\mathbb{D} = \mathbb{D}_i$ (théorie des i-types d'homotopie relative), la catégorie des objets localement constants sur Z s'exprime en termes du $(i+1)$-type d'homotopie, mais pas en termes de son i-type d'homotopie seul.

[page 98]

Commentary · Examples [ms. 98]

Sets already detect the missing monodromy

The shift from an i-type to an $(i+1)$-type has a concrete beginning. For the represented derivator $\mathbb{D}_0(A) = \mathsf{Set}^{A^{\mathrm{op}}}$, natural maps between constant diagrams on a nonempty connected category are simply maps of sets. Hence $e \to BG$ is a $\mathbb{D}_0$-equivalence for every group G. Locally constant set-valued diagrams on BG, however, are G-sets, and restriction to e forgets the action.

48. Il faut expliciter le raisonnement. Il suffit de tester même sur $Y'' \to Y'$, où Y'' est ponctuelle, et où Y' a un objet final (resp. initial).

49. *cf.* **Der 3** a') page suivante.

For a nontrivial G, this functor is not full: there are maps of sets from a one-point set to the regular G-set, but no equivariant such map. Thus a 0-type records connected components, while local systems of 0-types already record the fundamental group.

This also supplies a direct test of footnote 36. Take $G = C_2$ and the covering $f : E_G \to BG$. Its strict fibres are the discrete two-element set G. Since the covering is a discrete fibration and a discrete opfibration, every base change is a covering of the same kind. Right Kan extension of a constant set M has fibre M^G, and left Kan extension has fibre $G \times M$; arrows in the base act by permutations of G. Both resulting diagrams are locally constant and both Kan extensions calculate on strict fibres. Therefore conditions (b) and (b′) hold universally for $\mathbb{D}_0$.

Now pull back along $e \to BG$. The induced map of total categories is $G_{\mathrm{disc}} \to E_G$. It is not a $\mathbb{D}_0$-equivalence: for a two-element coefficient set M, the constant-coefficient limit on the connected category E_G is M, whereas on the discrete fibre it is M^2. The comparison is the diagonal map and is not bijective. Thus (b) does not imply (a), even for the derivator of 0-types. The correct positive assertion for ordinary weak homotopy equivalences uses their invariance of locally constant coefficients, as in the preceding block; it does not replace them by $\mathbb{D}_0$-equivalences. If Dia is restricted to finite posets, the connected double cover of the four-object circle gives the same counterexample. Its total poset has eight objects and its self-pullback is the disjoint union of two copies of that total poset. Both bases are connected, while the pulled-back total categories have different numbers of components.

Il semble bien que ce soit l'axiome **Der** 3 (a') qui assure l'existence d'une bonne notion ("bilatère") de $\mathbb{D}$-fibration, qui soit stable par changement de base, et qui se réduise soit à "$\mathbb{D}$-fibration à gauche universelle" soit à "$\mathbb{D}$-fibration à droite universelle" et donc qui soit en plus une condition vérifiable pratiquement. Dans le cas où f est propre ou lisse, le critère est particulièrement simple, puisque il se borne à la condition que les $f_* f^* a_Y$, ou les $f_! f^* a_Y$ (pour $a \in \mathcal{A}$), soient des objets localements constants sur Y (donc constants si Y est Hot-asphérique, en vertu de **Der** 3 (a'). ed. 60

Je vais également introduire une variante renforcée de **Der** 3 b) :

Der 3 b') Soit $Z' \to Z$ une Hot-équivalence dans Dia. Alors $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ induit une équivalence entre les catégories des coefficients localements constants sur Z et sur Z'.[50]

Dans certaines questions il suffit de savoir que le dit foncteur est pleinement fidèle. Cela implique que [page 99]

$$H^*_{\mathbb{D}}(Z, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Z', f^* F)$$

quand F est localement constant sur Z[51] (mais il n'est pas clair que cette condition implique toujours la pleine fidélité[52]) et de même pour l'homologie

$$H^{\mathbb{D}}_*(Z', f^* F) \longrightarrow H^{\mathbb{D}}_*(Z, F) \quad .$$

Moyennant cette condition, on a la :

Proposition : *On suppose* **Der** 3 b′) *satisfait. Soit $f : X \to Y$ dans* Dia. *Pour que f soit une $\mathbb{D}$-fibration, il faut et il suffit que pour $Y'' \to Y'$ sur Y qui soit une* Hot*-équivalence, $X'' = X \times_Y Y'' \to X' = X \times_Y Y'$ soit une $\mathbb{D}$-équivalence.*

À vrai dire, on peut énoncer quelque chose de raisonnable, sans supposer l'axiome **Der** 3 b'). On doit introduire la :

Définition : Soit $f : Z' \to Z$ dans Dia. On dit que f et une $\mathbb{D}$-équivalence 1-*complète* si le foncteur $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ est pleinement fidèle sur la sous-catégorie pleine de $\mathbb{D}(Z)$ formée

50. **Der** 3 b'_0) : le foncteur est fidèle
b'_1) : le foncteur est pleinement fidèle
b'_2) : le foncteur est une équivalence

51. On pourra dire que $f : Z' \to Z$ est une $\mathbb{D}$-équivalence 0-*complète à gauche,* et 0-*complète à droite* dans le cas homologique.

52. Mais c'est OK si on a des Hom internes dans $\mathbb{D}(Z)_{\mathrm{loc.const.}}$ satisfaisant

$$\mathsf{Hom}(F, G) \simeq \mathsf{Hom}(e_Z, \underline{\mathsf{Hom}}(F, G)) \quad ,$$

où e est l'élément final de $\mathcal{A}$, les Hom étant compatibles aux images inverses.

des coefficients localement constants. On dit que f est une $\mathbb{D}$-équivalence 2-*complète* si elle induit une équivalence entre les catégories des coefficients localement constants.

On dit que Z est *complètement* $\mathbb{D}$-*asphérique*, si $Z \to e$ est une $\mathbb{D}$-équivalence 2-complète.

Ainsi, l'axiome **Der** 3 a') peut se reformuler, en disant que si Z est Hot-asphérique, alors Z est complètement $\mathbb{D}$-asphérique. Et de même, **Der** 3 b') en disant que si $f : X \to Y$ dans Dia est une Hot-équivalence, alors f est une $\mathbb{D}$-équivalence 2-complète. Nous allons donner une variante affaiblie, de **Der** 3 a'), qui suffit pour donner une notion raisonnable de $\mathbb{D}$-fibration : [ed. 61 [page 100]]

Der 3 a'') Tout Z dans Dia qui a soit un objet final, soit un objet initial, est complètement $\mathbb{D}$-asphérique.

Définition. On dit que $f : X \to Y$ dans Dia est une $\mathbb{D}$-fibration, si pour

$$Y'' \longrightarrow Y' \longrightarrow Y$$

avec $Y'' \to Y'$ une $\mathbb{D}$-équivalence 1-complète (par exemple Y' et Y'' complètement $\mathbb{D}$-asphèriques), $X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$ est une $\mathbb{D}$-équivalence.

Cela équivaut à chacune des conditions b), b') de la prop. 2 (page 79), *i.e.* que f soit universellement une $\mathbb{D}$ fibration à gauche, ou universellement une $\mathbb{D}$-fibration à droite (grâce à **Der 3** b'' `[plutôt` a''`]`).

Définition 1. $f : X \to Y$ dans Dia est appelé $\mathbb{D}$-*équivalence cohomologique* (sous-entendu : *complète*) si pour tout $F \in \mathsf{Ob}\,\mathbb{D}(Y)$, F localement constant, on a [page 101]

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*(F)) \quad .$$

Notion duale $\mathbb{D}$-*équivalence homologique*

$$H^{\mathbb{D}}_*(X, f^*(F)) \xrightarrow{\sim} H^{\mathbb{D}}_*(Y, F) \quad .$$

Si Y est complètement $\mathbb{D}$-asphérique, alors les coefficients localement constants sur Y sont constants, et f est une $\mathbb{D}$-équivalence cohomologique (resp. homologique) si et seulement si c'est une $\mathbb{D}$-équivalence *i.e.* si X est $\mathbb{D}$-asphérique. [53]

Proposition 1 (Soritale). *Soient $X \xrightarrow{f} Y \xrightarrow{g} Z$. Si f et g sont des $\mathbb{D}$-équivalences cohomologiques, alors gf aussi. Si gf et f sont des $\mathbb{D}$-équivalences cohomologiques, alors g aussi. Si gf est une $\mathbb{D}$-équivalence cohomologique, et si g est une $\mathbb{D}$-équivalence 2-complète* (donc tout F localement constant sur Y provient d'un G localement constant sur Z), *alors f est une $\mathbb{D}$-équivalence cohomologique.*

Commentary · Exposition [ms. 101]

Complete equivalences repair the base change argument

The adjective "complete" records which coefficients are being tested. A 1-complete equivalence is fully faithful on the category of locally constant objects. A 2-complete equivalence is in addition essentially surjective there. These are levels of categorical information, not degrees of homology. In particular a 1-complete equivalence $p : Y' \to Y$ compares both homology and cohomology with every locally constant coefficient F on Y.

For cohomology, adjunction gives

$$\mathsf{Hom}_{\mathbb{D}(e)}(M, H^*_{\mathbb{D}}(Y, F)) = \mathsf{Hom}_{\mathbb{D}(Y)}(M_Y, F).$$

Both arguments on the right are locally constant. Full faithfulness identifies this group of morphisms with $\mathsf{Hom}_{\mathbb{D}(Y')}(M_{Y'}, p^*F)$, naturally in M. Yoneda gives the asserted cohomology isomorphism. Applying the same argument to morphisms from F to a constant object proves homological invariance. The internal-Hom qualification in footnote 52 is a converse criterion: if locally constant internal Homs exist and commute with pullback, cohomological invariance applied to $\underline{\mathsf{Hom}}(F, G)$ yields full faithfulness. Its unit must be the stated cartesian unit, or an explicitly chosen monoidal unit; the zero object of a stable theory cannot play that role merely by being terminal.

53. L'axiome **Der** 3 b'_1) implique que toute Hot-équivalence est une $\mathbb{D}$-équivalence cohomologique.

We can now complete the repaired proof. Suppose f is universally cohomological, and let p be 1-complete. For a constant coefficient M, put $F = f_*M_X$. The coefficient F is locally constant, and universal fibre calculation identifies p^*F with $f'_*M_{X'}$. The preceding Yoneda argument compares the global cohomology of these two objects and proves that $X' \to X$ is a $\mathbb{D}$-equivalence. This establishes the new pullback-preservation condition. Conversely, under Der 3 a$''$, every functor between categories having final objects is 2-complete, because both locally constant categories are equivalent to $\mathbb{D}(e)$ and the restriction agrees with the identity on constants. Testing the new condition on $\{y\} \to Y/y$ and on transitions between slices therefore gives universal cohomological fibre calculation. Categories with initial objects give the homological condition. These arguments prove both directions and explain why the weaker endpoint axiom suffices at this stage.

The cancellation statements on manuscript page 101 have the same logic. Composing two comparisons proves closure under composition; cancelling the comparison for f proves the implication from gf and f to g. To recover f from gf, one must first express an arbitrary locally constant coefficient on Y as the pullback of a coefficient on Z. The 2-complete hypothesis on g supplies exactly that essential-surjectivity step, and its full faithfulness supplies the comparison for g. Ordinary two-out-of-three would have omitted the coefficient which must descend.

Définition 2. 1°) $f : X \to Y$ dans Dia est appelé une $\mathbb{D}$-*fibration cohomologique complète,* je dis bien complète, si

a) Pour tout $y \in \mathsf{Ob}\, Y$, $X_y \to X/y$ est une $\mathbb{D}$-équivalence cohomologique (complète).

b) Pour tout $y' \to y$ dans Y, $X/y' \to X/y$ est une $\mathbb{D}$-équivalence cohomologique (complète).

c) Ces conditions restent satisfaites après tout changement de base $Y' \to Y$. ed. 62

2°) Notion duale de fibration homologique complète. [page 102]

Exemple. Si f est Hot-propre, les flèches des *a*) et *b*) sont des Hot-équivalences, donc en vertu de **Der 3** b'_1) (qu'il faut admettre), ce sont des $\mathbb{D}$-équivalences cohomologiques. Cela signifie que *a*) et *b*) sont valables, et *c*) aussi, car la notion de flèche Hot-propre est stable par changement de base.

Commentary · Exposition [ms. 102]

What the completion requires

The quantifier on the coefficients is the decisive change. A weak cohomological fibration is tested by the constant objects M_X. A complete cohomological fibration must work with every locally constant object of each category obtained by base change. Such an object need not descend from X, and need not become constant on X. The monodromy examples just discussed explain why that distinction cannot be suppressed. The role of completion is to make the coefficients produced by one direct image admissible for a second direct image.

We keep the variance of the manuscript: objects of $\mathbb{D}(X)$ behave as coherent contravariant diagrams, and the pointwise formula for a right direct image is

$$(f_*F)_y \simeq H^*_{\mathbb{D}}(X/y, F|_{X/y}).$$

Write $\mathcal{L}(X) \subseteq \mathbb{D}(X)$ for the full subcategory of locally constant objects. Local constancy means that every transition morphism is invertible; it does not mean that an isomorphism to a constant object has been chosen. For $r : A \to B$, the cohomological condition is the isomorphism

$$H^*_{\mathbb{D}}(B, L) \longrightarrow H^*_{\mathbb{D}}(A, r^*L) \qquad (L \in \mathcal{L}(B)).$$

The conditions on $X_y \to X/y$ therefore compare a true fibre with a comma category for every local system on the latter. The conditions on $X/y \to X/y'$ then control the transport of those cohomology objects along the base. Universal repetition of the first comparison is what makes the resulting fibre formula compatible with arbitrary base changes.

The example immediately preceding Proposition 2 contains a genuine slip: Hot-properness gives the fibre-to-comma comparison, but does not make the comma categories locally constant as y varies. Take the inclusion $\{1\} \hookrightarrow [1]$, where $[1] = \{0 \to 1\}$. This is a closed immersion and hence proper, but its comma categories at 0 and 1 are respectively empty and terminal. The transition $\varnothing \to e$ is not a weak homotopy equivalence. Thus the example becomes correct after adding the transport condition, or by assuming that f is both Hot-proper and a weak HOT-fibration. This is precisely the hypothesis separated out on manuscript page 107 and proved usable in Proposition 6 on page 118. The printed assertion is retained in the source and is audited as false as written.

Proposition 2. *Soient $f : X \to Y$ $\mathbb{D}$-fibration cohomologique complète, $F \in \mathsf{Ob}\,\mathbb{D}(X)$ localement constant sur X. Alors $f_*(F)$ est localement constant, sa formation commute à tout changement de base, et en particulier se calcule fibre par fibre.*[54]

Corollaire 1. *Soit un diagramme cartésien*

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array} \quad .$$

On suppose f $\mathbb{D}$-fibration cohomologique et p une $\mathbb{D}$-équivalence cohomologique. Alors q est une $\mathbb{D}$-équivalence cohomologique.

En effet, on a pour F localement constant sur X

$$H^*_{\mathbb{D}}(X,F) \simeq H^*_{\mathbb{D}}(Y,f_*F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Y',p^*f_*F) \simeq H^*_{\mathbb{D}}(Y',f'_*q^*F) \simeq H^*_{\mathbb{D}}(X',q^*F) \qquad \text{[page 103]}$$

qed.

Proposition 3. *La notion de $\mathbb{D}$-fibration cohomologique complète est stable par changement de base* (trivial) *et par composition.*

Pour la stabilité par composition, il suffit de noter le corollaire suivant de la proposition 2.

Corollaire 2 (de prop. 2). *Soit $f : X \to Y$. Pour que f soit une $\mathbb{D}$-fibration cohomologique complète, il faut et il suffit que pour tout $Y'' \xrightarrow{p} Y' \to Y$ sur Y, avec p une $\mathbb{D}$-équivalence cohomologique, la flèche correspondante $X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$ soit aussi une $\mathbb{D}$-équivalence cohomologique.*

C'est nécessaire par le corollaire 1, compte tenu que $f' : X' = X \times_Y Y' \to Y'$ est aussi une $\mathbb{D}$-fibration cohomologique. C'est suffisant, car il suffit de l'appliquer aux cas $\{y\} \to Y/\{y\}$, $Y/y \to Y/y'$, qui sont des $\mathbb{D}$-équivalences cohomologiques en vertu de **Der** 3 b'_1) (ou seulement **Der** 3 a')).

[page 104] ed. 63

Proposition 4. *On suppose que* **Der** 3 a′) *est satisfait* (*i.e.* tout Z dans Dia qui est Hot-asphérique, est complètement $\mathbb{D}$-asphérique). *Soit $f : X \to Y$ dans* Dia. *Les conditions suivantes sont équivalentes*

1°) *f est une $\mathbb{D}$-fibration cohomologique complète* (conditions *a*), *b*), *c*) de `[la]` définition 2, page 101).

2°) *Pour tout $F \in \mathsf{Ob}\,\mathbb{D}(X)$, localement constant, $f_*(F)$ est localement constant, se calcule fibre par fibre; et ça reste vrai après tout changement de base $Y' \to Y$ (avec F' localement constant sur $X' = X \times_Y Y'$). (Alors la formation de $f_*(F)$ commute à tout changement de base).*

3°) *Pour tout $Y'' \xrightarrow{p} Y'$ sur Y, si p est une $\mathbb{D}$-équivalence cohomologique, alors $X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$ aussi.*

3′) *Comme 3, en supposant que Y'' est la catégorie ponctuelle, et que Y' a un élément final : il faut que $X'' \to X'$ soit une $\mathbb{D}$-équivalence cohomologique.*

Commentary · Exposition [ms. 104]

The universal criterion, with the quantifiers restored

Proposition 4 is more than a convenient restatement: it supplies the missing reason that the strengthened notion composes. We give the argument, keeping the hypothesis that every weakly contractible indexing category has only constant local systems, as in **Der** 3 a'. The same argument will apply to Proposition 9, after the terminology is revised.

First assume the comma-category conditions of Definition 2, universally in the base. If $F \in \mathcal{L}(X)$, its restriction to X/y is locally constant. The comparison $X_y \to X/y$ gives

$$(f_*F)_y \simeq H^*_{\mathbb{D}}(X_y, F|_{X_y}).$$

54. Il y a une réciproque *cf.* prop. 3′.

For an arrow $y \to y'$, the functor $X/y \to X/y'$ is a cohomological equivalence, so the resulting transition $(f_*F)_{y'} \to (f_*F)_y$ is an isomorphism. Thus f_*F is locally constant. In a pullback square with base map $p : Y' \to Y$ and induced maps f', q, evaluate the base-change map

$$p^* f_* F \longrightarrow f'_* q^* F$$

at y'. Both sides identify with the cohomology of the same strict fibre $X_{p(y')}$ with the same restricted coefficient. The identification of the comparison with the identity is the naturality of the pointwise Kan-extension formula. Pointwise detection of isomorphisms proves base change.

Suppose now that this direct-image property holds after every base change. For a cohomological equivalence $p : Y'' \to Y'$, and $L \in \mathcal{L}(X')$, put $G = f'_* L$. The preceding property makes G locally constant and identifies p^*G with $f''_* q^* L$. The canonical comparison is consequently the composite

$$H^*_{\mathbb{D}}(X', L) \simeq H^*_{\mathbb{D}}(Y', G) \longrightarrow H^*_{\mathbb{D}}(Y'', p^*G) \simeq H^*_{\mathbb{D}}(X'', q^*L),$$

whose middle map is invertible. This proves the universal preservation criterion. Notice that it tests all L on X', rather than just coefficients pulled back from X.

It remains to recover the comma conditions from the apparently much smaller test in $3'$. Every functor $e \to B$ with B admitting a terminal object is a cohomological equivalence under a': all local systems on B are constant and B is $\mathbb{D}$-aspherical. Apply the test to the base change $Y/y \to Y$, selecting the terminal object 1_y. Its inverse image is X_y, and the total space is X/y. This gives the first comma comparison. For $u : y \to y'$, use $Y/y' \to Y$ and select its object u. We get a cohomological equivalence $X_y \to X/y'$. It factors as

$$X_y \longrightarrow X/y \longrightarrow X/y'.$$

The first map and the composite are cohomological equivalences. Testing a local system on X/y' in the two comparisons shows that the second map is also one. This is the valid cancellation direction from Proposition 1 on manuscript page 101; no unqualified two-out-of-three assertion for cohomological equivalences is being used. As the point test was required over every base change, all the comparisons are universal.

These arguments also justify the parenthesis in Proposition 9 that only the fibre comparison needs to be imposed universally. Applied to Y/y' and to its arbitrary object u, that one comparison already recovers transport. They explain why the converse concerns arbitrary locally constant coefficients on the pullbacks. Merely proving the fibre formula for restrictions of coefficients from the original X would leave a real gap.

Définition 3. On dit que $f : X \to Y$ dans Dia est une $\mathbb{D}$-*fibration complète* si c'est une $\mathbb{D}$-fibration cohomologique complète et une $\mathbb{D}$-fibration homologique complète. [page 105]

C'est stable par changement de base, par composition.

La définition 3 équivaut à ceci : pour tout $p : Y'' \to Y'$ sur Y, avec Y'' et Y' Hot-asphériques, le morphisme correspondant $X'' \to X'$ est à la fois une $\mathbb{D}$-équivalence cohomologique, et une $\mathbb{D}$-équivalence homologique. (Et il suffit même de le supposer pour $Y'' = e$ et pour Y' avec objet final ou objet initial.) On peut songer à renforcer encore cette condition en exigeant (sans regarder à gauche ni à droite) que $X'' \to X'$ soit une $\mathbb{D}$-équivalence 1-complète. Mais il n'est pas clair que cette notion renforcée soit stable par composition – à moins de la renforcer encore, en exigeant que si $Y'' \to Y'$ est une $\mathbb{D}$-équivalence 1-complète, alors $X'' \to X'$ l'est aussi. L'ennui avec cette définition renforcée, c'est que faute d'avoir une description cohomologique ou homologique de la notion de $\mathbb{D}$-équivalence 1-complète, je ne me sens pas capable de donner des exemples intéressants (non totalement triviaux) de $\mathbb{D}$-fibrations en ce sens très fort. Sauf les Hot-fibrations triviales (*i.e.* à fibres Hot-asphériques) – mais ce n'est pas drôle! [page 106]

Mais quand on a des produits $*$ associatifs et unitaires et des $\underline{\mathsf{Hom}}$ internes correspondants dans les $\mathbb{D}(X)_{\text{loc. const.}}$, compatibles avec `[les]` images inverses, donc on aura sur X si F et G sont localement constants dans $\mathbb{D}(X)$, et si ε_X désigne l'unité pour $*$ dans $\mathbb{D}(X)$,

$$\mathsf{Hom}(F, G) \simeq \mathsf{Hom}(\varepsilon_X * F, G) \simeq \mathsf{Hom}(\varepsilon_X, \underline{\mathsf{Hom}}(F, G)) \simeq \mathsf{Hom}_{\mathcal{A}}(\varepsilon_e, p_* \underline{\mathsf{Hom}}(F, G)) \ ,$$

car $\varepsilon_X = p^*(\varepsilon_e)$, où $p : X \to e$, donc dans ce cas une flèche $f : X \to Y$ dans Dia est une $\mathbb{D}$-équivalence 1-complète si (et seulement si) c'est une $\mathbb{D}$-équivalence cohomologique. Donc dans ce cas, si f est une $\mathbb{D}$-fibration cohomologique complète, c'est aussi une $\mathbb{D}$-fibration homologique complète, donc aussi une $\mathbb{D}$-fibration complète (définition 3). (Et même au sens soit-disant renforcé de tantôt : si $Y'' \to Y'$ sur Y est une $\mathbb{D}$-équivalence 1-complète ($\Leftrightarrow$ une $\mathbb{D}$-équivalence cohomologique!) alors $X'' \to X'$ aussi.

Commentary · Exposition [ms. 106]

What internal Hom does, and what it does not do

The internal-Hom argument is sound, provided its compatibility hypotheses are stated at the level where they are used. Suppose that each $\mathcal{L}(X)$ is a closed monoidal category, that its tensor unit is $\varepsilon_X = p_X^*\varepsilon_e$, and that restriction preserves the internal Hom between locally constant objects. For $F, G \in \mathcal{L}(Y)$, adjunction gives

$$\mathsf{Hom}_{\mathbb{D}(Y)}(F, G) \simeq \mathsf{Hom}_{\mathbb{D}(e)}\big(\varepsilon_e, p_{Y*}\underline{\mathsf{Hom}}_Y(F, G)\big).$$

If $f : X \to Y$ is a cohomological equivalence, apply its defining comparison to the locally constant object $\underline{\mathsf{Hom}}_Y(F, G)$. The assumed comparison

$$f^*\underline{\mathsf{Hom}}_Y(F, G) \simeq \underline{\mathsf{Hom}}_X(f^*F, f^*G)$$

then identifies the displayed Hom set with $\mathsf{Hom}_{\mathbb{D}(X)}(f^*F, f^*G)$. Thus f^* is fully faithful on $\mathcal{L}(Y)$: f is 1-complete.

Conversely, full faithfulness on local systems compares the morphisms from every constant object M_Y to $F \in \mathcal{L}(Y)$. By adjunction these are the morphisms from M to $p_{Y*}F$. Yoneda gives the cohomological comparison. Comparing morphisms from F to every M_Y, instead, gives the homological comparison. This last argument needs no tensor product. It proves the two outgoing implications from 1-completeness in the diagram on page 113, while the internal-Hom hypothesis supplies the converse only on the cohomological side.

There is no need to assume that the unit detects isomorphisms. In the forward direction the cohomological comparison is already an isomorphism before $\mathsf{Hom}(\varepsilon_e, -)$ is applied; in the reverse direction Yoneda uses every M. Nor does the argument identify 1-completeness with 2-completeness: essential surjectivity of local systems is an additional demand.

One must also retain the adjective "cohomological". On page 122 the printed sentence says that an unqualified $\mathbb{D}$-equivalence is already 1-complete in the presence of internal Homs. That stronger assertion is false. For example, with $\mathbb{D} = \mathsf{Der}(\mathbb{Q}\text{-}\mathsf{Mod})$ and $G = C_2$, the map $e \to BG$ is a constant-coefficient equivalence, but restriction sends the sign representation to a one-dimensional vector space. There are no G-equivariant maps from the trivial representation to the sign representation, whereas after restriction there are nonzero maps. The categories of local systems have the usual derived tensor and internal Hom. The wording on pages 106 and 113, and the diagram on page 123, supply the correct repair: cohomological equivalence, not constant-coefficient equivalence.

Il se pose la question de trouver des f qui soient des $\mathbb{D}$-fibrations cohomologiques, ou homologiques. Je ne sais si je serai capable d'en trouver, qui ne soient déjà des HOT-fibrations cohomologiques, ou homologiques (cela revient maintenant au même [55]), car pratiquement je ne sais guère dire qu'une flèche $Z' \to Z$ dans $\mathcal{C}at$ est une $\mathbb{D}$-équivalence cohomologique complète, que si cette flèche est une HOT-équivalence (et en admettant l'axiome **Der** 3 b'_1), ou sa variante affaiblie cohomologique). ed. 64 [page 107]

Commentary · Examples [ms. 107]

A complete fibration beyond ordinary homotopy

The request for examples that are not already HOT-fibrations has a simple answer with rational coefficients. Let $G = C_2$, let BG denote its one-object category, and take the Grothendieck opfibration

$$f : E = \int_{[1]} (BG \longrightarrow e) \longrightarrow [1].$$

55. Les HOT-équivalences cohomologiques = HOT-équivalences homologiques = HOT-équivalences sans plus.

Explicitly, E has objects a and t, with $\mathrm{End}(a) = G$, $\mathrm{End}(t) = \{1\}$, one arrow $a \to t$, and no arrow $t \to a$. The unique arrow to t absorbs every $g \in G$. The fibres are BG and e, and the transport is the collapse $r : BG \to e$.

For $\mathbb{D} = \mathsf{Der}(\mathbb{Q}\text{-}\mathsf{Mod})$, the functor r is a cohomological equivalence. All coefficients on its target are constant, and invariants for a finite group over $\mathbb{Q}$ are exact. Indeed, averaging $v \mapsto |G|^{-1} \sum_{g \in G} gv$ is a natural projection onto invariants, and averaging a lift proves that invariants preserve epimorphisms. Consequently their right derived functor on any complex is computed degreewise. On a trivial G-complex M, this gives $R\Gamma(BG, M) = M$.

More generally, an opfibration whose transport functors are cohomological equivalences is a complete cohomological fibration. Properness supplies the fibre formula for every coefficient. For a locally constant coefficient L on the total category and an arrow in the base, the cocartesian arrows give an isomorphism between L on the source fibre and the pullback of L on the target fibre. The cohomological equivalence of transport therefore makes the direct-image transition invertible. This argument survives an arbitrary pullback, since the new transport functors are the old ones indexed by the new arrows. Proposition 4, or Proposition 9 in the revised terminology, proves the assertion.

Applying this observation to f shows that it is a complete cohomological fibration. Derived tensor products and internal Homs on rational local systems satisfy the hypotheses explained at page 106, so it is a complete homological fibration as well. It is not a HOT-fibration: its transport $BG \to e$ is not a weak homotopy equivalence, since $\pi_1(BG) = C_2$. Thus stronger coefficient invariance is still relative to the chosen coefficients. Reversing the transport to $e \to BG$ destroys cohomological equivalence, and is exactly the feature exploited by the composition counterexample at manuscript page 118.

Je vais donc chercher les HOT-fibrations, *i.e.* les $f : X \to Y$ tels que pour tout $Y'' \to Y' \to Y$, avec Y'' et Y' asphériques (sous-entendu : pour HOT), $X'' \to X'$ est une Hot-équivalence. (Il suffit même de supposer Y'' réduit à un point, Y' avec objet final ...)

Supposons par exemple f propre (*i.e.* HOT-propre), donc la condition a) de la définition 2, p. 101 satisfaite et de même après tout changement de base. Suffit-il de vérifier de plus la condition b), *i.e.* que

$$X/y \longrightarrow X/y' \quad , \qquad \text{pour } y \to y' \text{ dans } Y \ ,$$

sont des Hot-équivalences? (J'entends en restant sur Y, sans supposer aussi d'avoir la même chose après changement de base). Mais cela reste-il vrai après tout changement de base?

La situation semble ici meilleur pour f lisse (au lieu de propre), en interprétant les conditions a) et b) sous forme homologique, comme signifiant que $f_!(e_X)$ est un objet localement constant de $\mathsf{HOT}(Y)$, qui se calcule fibre par fibre. Comme f lisse implique que $f_!$ commute à tout changement de base, cette hypothèse sur f (que f soit une HOT-fibration cohomologique, peut-être pas complète) est stable par changement de base, donc implique que c'est une HOT-fibration cohomologique complète. Mais cet argument utilise les propriétés générales des dérivateurs pour HOT, que je n'ai pas encore établies - surtout pas dans le domaine $\mathcal{C}at$ tout entier! Donc il est sain de me méfier. Je préfère tâter la difficulté en restant d'abord dans le cadre du formalisme cohomologique (non commutatif), dont j'ai une certaine maîtrise. Je suppose donc f propre à nouveau, et que les [page 108]

$$X_y \longrightarrow X/y \quad , \qquad X/y \longrightarrow X/y'$$

soient des HOT-équivalences, je veux prouver la même chose après tout changement de base, au moins que dans la situation [page 109]

$$\begin{array}{ccccc} X & \xleftarrow{q} & X' & \xleftarrow{q'} & X'' \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f''} \\ Y & \xleftarrow[p]{} & Y' & \xleftarrow[p']{} & Y'' \end{array}$$

si Y', Y'' sont asphériques, alors q' est une Hot-équivalence. Si je veux l'établir par critère cohomologique, je dois commencer par prendre sur X' un faisceau d'*ensembles* localement constant F, et prouver que $\Gamma(X', F) \xrightarrow{\sim} \Gamma(X'', q'^* F)$. Comme ce faisceau ne provient pas de X, je suis ennuyé pour appliquer l'hypothèse sur f, qui concerne à priori des faisceaux d'ensembles constants sur X, ou sur les X/y. Mais on devrait avoir ceci : ed. 65

Lemme. *Soit $f : X'' \to X'$. Alors f est une* HOT-*équivalence* i.e.

$$H^*_{\mathsf{HOT}}(X', a_{X'}) \xrightarrow{\sim} H^*_{\mathsf{HOT}}(X'', a_{X''})$$

est [un] *isomorphisme pour tout $a \in \mathsf{Hot} = \mathsf{HOT}(e)$, si et seulement si f est une* Hot-*équivalence,*[56] *et alors f est même une* HOT-*équivalence* 2-*complète* (j'ai besoin seulement de : HOT-équivalence cohomologique et homologique complète) i.e. *pour tout $\xi \in \mathbb{D}(X')$ localement constant sur X', on a*[57]

$$H^*_{\mathsf{HOT}}(X', \xi) \xrightarrow{\sim} H^*_{\mathsf{HOT}}(X'', q'^*(\xi)) \quad .$$

La première assertion doit résulter des formules [page 110]

$$(*) \qquad H^*_{\mathsf{HOT}}(Z, a_Z) \simeq \underline{\mathsf{Hom}}_{\mathsf{Hot}}(\mathsf{hot}(Z), a)$$

$$\text{(et aussi} \qquad H^{\mathsf{HOT}}_*(Z, a_Z) \simeq \mathsf{hot}(Z) \times a \qquad \text{)}$$

donc si

$$(**) \qquad \mathsf{hot}(X'') \xrightarrow{\sim} \mathsf{hot}(X')$$

est [un] isomorphisme, alors

$$H^*_{\mathsf{HOT}}(X', a_{X'}) \longrightarrow H^*_{\mathsf{HOT}}(X'', a_{X''})$$

est [un] isomorphisme, puisque c'est déduit de $(*)$ en prenant $\underline{\mathsf{Hom}}_{\mathsf{Hot}}(\,.\,, a)$. Inversement, si $(*)$ est toujours [un] isomorphisme, exprimant les deux membres [mot illisible] de $\underline{\mathsf{Hom}}_{\mathsf{Hot}}(\,.\,, a)$ et prenant les π_0, on trouve que

$$\mathsf{Hom}_{\mathsf{Hot}}(\mathsf{hot}(X'), a) \xrightarrow{\sim} \mathsf{Hom}_{\mathsf{Hot}}(\mathsf{hot}(X''), a)$$

est un isomorphisme pour tout a, donc $(**)$ est un isomorphisme.

Mais le fait que ça implique que f'' soit une HOT-équivalence cohomologique complète, *i.e.* que la relation $(*)$ reste vraie pour un $\xi \in \mathbb{D}(X')$ localement constant sur X', me paraît une propriété délicate du dérivateur HOT, sur laquelle il me faudra revenir.

Donc le lemme est loin d'être établi, faute d'avoir seulement construit le dérivateur HOT, sans même parler d'en établir les propriétés spéciales délicates! Mais je n'ai aucune raison de douter du lemme, il fait partie de la kyrielle des intuitions de base que je voudrais fonder solidement.

Commentary · Later developments [ms. 110]

The lemma for spaces

The lemma on pages 109–110 is proved by the modern theory of locally constant diagrams. We can separate its two assertions, which need different arguments. Put $BX = |NX|$, and let $\mathcal{S}$ denote the homotopy theory of spaces. In the homotopy derivator HOT, the constant-diagram formulas are

$$H^*_{\mathsf{HOT}}(X, M_X) \simeq \mathrm{Map}(BX, M), \qquad H^{\mathsf{HOT}}_*(X, M_X) \simeq BX \times M.$$

56. Attention ici [mot illisible] la distinction entre HOT-équivalence et Hot-équivalence.

57. Est-ce vrai aussi pour les dérivateurs $\mathbb{D}_i$?

These are derived mapping spaces and derived products, viewed in Hot. Thus a weak equivalence $BX \to BY$ induces both constant-coefficient comparisons. Conversely, if all the cohomological comparisons are isomorphisms, their sets of components give bijections

$$[BY, M] \longrightarrow [BX, M]$$

for every homotopy type M. Yoneda in Hot makes $BX \to BY$ an isomorphism. Homologically one can simply take $M = e$. This proves the identification of the manuscript's uppercase HOT-equivalences with lowercase Hot-equivalences, without conflating the homotopy derivator with its value at a point.

The local-coefficient assertion has another content: all coherence in a locally constant diagram must be transportable along a weak homotopy equivalence of the base. An invariant modern formulation identifies locally constant space-valued diagrams on X with space-valued functors on the homotopy groupoid $\Pi_\infty(BX)$. Restriction along an equivalence of these groupoids is an equivalence of functor categories, and remains so on passing to homotopy categories. Hence a Hot-equivalence is 2-complete for HOT. It preserves cohomology and homology with every local coefficient by the adjunction argument in the preceding block.

A precise derivator reference is Cisinski (2009, Theorem 1.3 and Corollary 3.8): a nerve weak equivalence induces an equivalence of categories of locally constant objects, for model categories and, in the latter result, for arbitrary Grothendieck derivators on small categories. The construction of those local systems as a derivator, with both adjoints to its inclusion into the diagram derivator, is Theorem 3.13. Thus the strongest b-axiom proposed below is available in the usual modern setting; it is not a currently unproved property of spaces.

The higher-categorical description is compatible with straightening and unstraightening (Lurie, 2009, Theorem 2.2.1.2); the homotopy limit of a space-valued local system is its space of sections (Lurie, 2009, Corollary 3.3.3.4). This gives a geometric reading of the two comparisons: transport the family, then compare its sections or its total homotopy colimit. It also indicates exactly what is absent from the ordinary diagram in the homotopy category: the coherence of transport around paths and higher simplices.

Footnote 57 requires a qualification. If its question asks whether a genuine Hot-equivalence induces 2-completeness for the derivators of truncated homotopy types, the answer is yes, by the same invariance theorem. If it asks whether a $\mathbb{D}_i$-equivalence alone has the same consequence, the answer is no already for $i = 0$. The example in "Sets already detect the missing monodromy" at manuscript page 98 has $e \to BC_2$ a constant-coefficient equivalence for sets, while restriction of C_2-sets is not full: a map from the one-point trivial action to the regular action exists after forgetting the action and does not exist before. Truncating the coefficients can lose information about the base that the category of local systems still retains.

Le lemme implique qu'une $\mathbb{D} = \mathsf{HOT}$-fibration (au sens de la définition 2 page 81, revue et corrigée à la page 96-97) est en fait déjà une HOT-fibration cohomologique et homologique complète. Or on a vu que si f est propre, f est une $\mathbb{D}$-fibration si et seulement si les conditions précédentes sur Y (sans changement de base) sont satisfaites. Et dualement pour f lisse. ed. 66 [page 111]

11.11.90

Je sens le besoin de revoir les notions de $\mathbb{D}$-fibrations en tous genres (p. 76 ff). D'abord les diverses variantes utiles de l'axiome **Der** 3 sur les dérivateurs :

Der 3 a) [58] Si $Z \in$ Dia est Hot-asphérique, il est $\mathbb{D}$-asphérique *i.e.* $\mathcal{A} = \mathbb{D}(e) \to \mathbb{D}(Z)$ est pleinement fidèle.

a') Comme a), avec en plus la caractérisation de l'image essentielle de $\mathcal{A}$ dans $\mathbb{D}(Z)$, comme formée des $\mathbb{D}$-coefficients localements constants.

Der 3 b) Si $f : Z' \to Z$ dans Dia est une Hot-équivalence, alors c'est une $\mathbb{D}$-équivalence, *i.e.* le foncteur $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ induit une équivalence entre les sous-catégories formées des coefficients constants (ou encore un foncteur pleinement fidèle $\mathbb{D}(Z)_{\text{loc. const.}} \longrightarrow \mathbb{D}(Z')$).

58. *cf.* aussi la série **Der 3** $c\ldots$, page 124 ff.

b'_1) Si f comme ci-dessus, alors f est une $\mathbb{D}$-équivalence 1-complète, *i.e.* le foncteur $f^* : \mathbb{D}(Z) \to \mathbb{D}(Z')$ induit un foncteur pleinement fidèle entre les sous-catégories pleines formées des objets localement constants. [page 112]

b'_2) Comme b'_1), mais avec $\mathbb{D}$-équivalence 2-complète au lieu de 1-complète, *i.e.* f^* induit une équivalence entre les sous-catégories formées des coefficients localement constants.

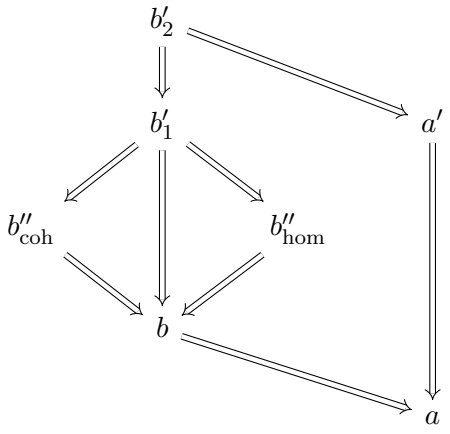
b''_{coh}) Si $f : Z' \to Z$ dans Dia est une Hot-équivalence, alors c'est une $\mathbb{D}$-équivalence cohomologique complète, *i.e.* pour tout $F \in \mathsf{Ob}\ \mathbb{D}(Z)$ qui est localement constant, on a

$$H^*_{\mathbb{D}}(Z, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Z', f^*F) \quad .$$

b''_{hom}) Si $f : Z' \to Z$ dans Dia est une Hot-équivalence, c'est une $\mathbb{D}$-équivalence homologique complète, *i.e.* pour tout $F \in \mathsf{Ob}\ \mathbb{D}(Z)$ qui est localement constant, on a [59]

$$H^{\mathbb{D}}_*(Z', f^*F) \xrightarrow{\sim} H^{\mathbb{D}}_*(Z, F) \quad .$$

On a le diagramme d'implications tautologiques pour les variantes de **Der** 3 : ed. 67



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Dans tous les cas à ma connaissance, les dérivateurs satisfont la condition la plus forte **Der** 3 b'_2.

[page 113]

Commentary · Exposition [ms. 112]

Two families of axioms with different antecedents

The *a*- and *b*-conditions should be read as assertions about the strength of the coefficient theory, not as successive definitions of the same phrase. Condition *a* says that a weakly contractible base does not create new morphisms between constant coefficients. Condition a' adds that it does not create new locally constant objects either. The *b*-conditions ask for analogous invariance under arbitrary weak homotopy equivalences, rather than just maps to a point.

Here is a direct check of the diagram. If f^* is an equivalence on local systems, it is fully faithful there. Full faithfulness gives cohomological and homological invariance by testing morphisms from and to all constant objects. Each of these implies invariance for constant coefficients. Applying the latter to $Z \to e$ proves *a*. Applying the equivalence on local systems to this same map proves a', because every object at e is constant. Finally a' contains *a* by definition. These are the arrows drawn in the source; the diagram does not claim that full faithfulness alone gives essential surjectivity.

There is a small but consequential inconsistency in the parenthesis defining *b* on page 111. Its first wording concerns constant coefficients, but the parenthesis writes $\mathbb{D}(Z)_{\mathrm{loc.const.}}$. Taken literally this is b'_1, the stronger condition introduced on the next line. For the displayed implication diagram and the earlier definition of a $\mathbb{D}$-equivalence, the subscript in that parenthesis must be read as "constant". We retain the edition's wording and record the distinction in the audit.

Modern homotopy invariance supplies b'_2 for ordinary Grothendieck derivators with all small categories as shapes. A restricted diagram domain still requires care: the statement must only

59. *cf.* variante ± spéciale pour HOT pages 124-125.

> quantify over shapes in that domain, and a proof that passes through simplicial resolutions or larger indexing categories cannot be imported without an extension hypothesis. This is the reason the manuscript continues to list these properties as separate assumptions. Its present calculation needs the explicitly stated hypothesis, not an unannounced change of domain.

Je rappelle aussi la :

Définition 1 : Soit $f : X \to Y$ dans Dia. On dit que f est une $\mathbb{D}$-*équivalence cohomologique* (sous-entendu : "*complète*") si pour tout $\mathbb{D}$-coefficient F localement constant sur Y,

$$H^*_{\mathbb{D}}(Y, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(X, f^*F) \quad ;$$

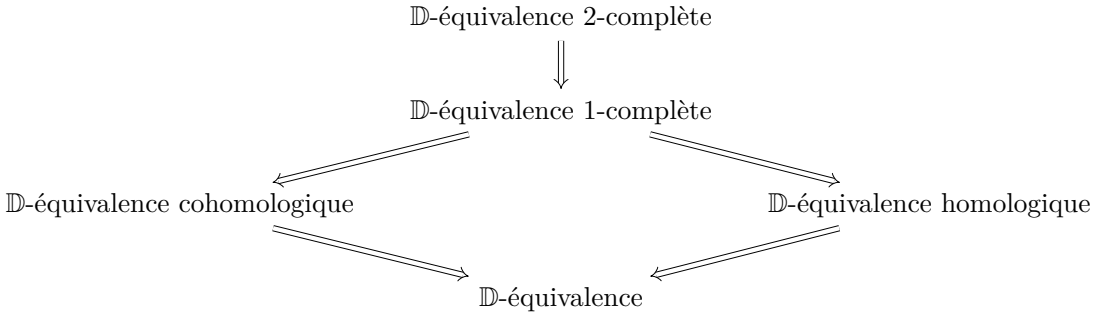
on dit que f est une $\mathbb{D}$-*équivalence homologique* ("*complète*") si pour tout F comme [ci-]dessus,

$$H^{\mathbb{D}}_*(X, f^*F) \xrightarrow{\sim} H^{\mathbb{D}}_*(Y, F) \quad .$$

On dit que f est une $\mathbb{D}$-*équivalence* 1-*complète* (resp. 2-*complète*) si le foncteur

$$f^* : \mathbb{D}(Y)_{\text{loc. const.}} \longrightarrow \mathbb{D}(X)_{\text{loc. const.}}$$

est pleinement fidèle (resp. une équivalence).

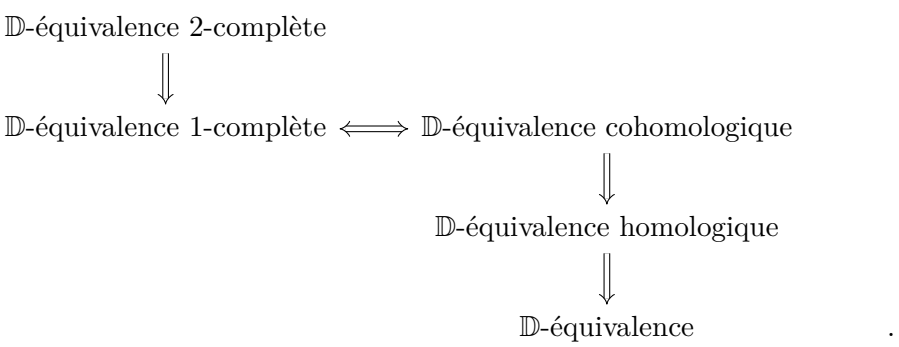


Si dans les sous-catégories $\mathbb{D}(X)_{\text{loc. const.}}$ il y a une loi $*$ (associative, unitaire, commutative) et des Hom internes associés, préservés par les images inverses, alors les $\mathbb{D}$-équivalences cohomologiques sont déjà 1-complètes et sont donc des $\mathbb{D}$-équivalences homologiques, donc dans ce cas particulier

ed. 68

[114: vide]

[page 115]



Dans $\mathbb{D} = \mathsf{HOT}$, toutes ces notions devraient être équivalentes, et équivalentes à la condition encore plus faible en apparence, que $f : X \to Y$ soit une Hot-équivalence. Mais je ne connais que HOT comme dérivateur où la $\mathbb{D}$-équivalence d'une flèche $f : X \to Y$ dans Dia implique déjà la $\mathbb{D}$-équivalence cohomologique (complète), ou la $\mathbb{D}$-équivalence homologique (complète), voire même la $\mathbb{D}$-équivalence 1-complète. (Je reviendrai là-dessus tantôt.)

Proposition 1. *Soit $f : X \to Y$ dans* Dia. *Conditions équivalentes :*

a) *Pour tout $y \in \mathsf{Ob}\, Y$, l'inclusion $X_y \to X/y$ est une $\mathbb{D}$-équivalence, et pour toute flèche $y' \to y$ dans Y, $X/y' \to X/y$ est une $\mathbb{D}$-équivalence.*

b) *Pour tout coefficient constant F sur Y, f_*f^*F est localement constant, et se calcule "fibre par fibre".*

(C'est la prop. 1, p. 76).

Proposition 1′. Dual.

Définition 2. Sous les conditions équivalentes de la proposition 1, on dit que f est une $\mathbb{D}$-*fibration cohomologique faible.* Sous les conditions duales, que f est une $\mathbb{D}$-*fibration homologique faible.* [page 116]

C'est ce que dans la définition 1, p. 77 j'appelais "$\mathbb{D}$-fibration cohomologique" ou "$\mathbb{D}$-fibration à gauche", et dualement. Maintenant la notion me paraît trop faible pour mériter ce nom, que je réserve pour le cas "complet".

Proposition 2. *Les $\mathbb{D}$-fibrations cohomologiques faibles sont stables par changements de base lisses, les $\mathbb{D}$-fibrations homologiques faibles stables par changements de base propres.*

Je devrais construire un exemple où f est une $\mathbb{D}$-fibration cohomologique faible, et ne l'est plus après changement de base.

Commentary · Development (commentators') [ms. 116]

A failure already among finite posets

The example at manuscript page 77 uses isomorphisms in the base to expose the failure of arbitrary base change. The same failure occurs for finite posets; it is therefore relevant even to the restricted domains repeatedly considered in these pages.

A four-object example. [ours] Let $Y = [2] = \{0 < 1 < 2\}$ and let X be the poset whose distinct nonidentity relations are

$$x < a, \qquad b < a, \qquad b < z.$$

Define $f : X \to Y$ by $f(x) = 0$, $f(a) = f(b) = 1$, and $f(z) = 2$. Then f is both a weak cohomological and a weak homological HOT-fibration. Its pullback along the long edge $[1] \to [2]$, selecting 0 and 2, is neither.

To prove the first assertion, the strict fibres are $\{x\}$, $\{b < a\}$, and $\{z\}$, all contractible. Since Y is a poset, the overcategories X/y are the subposets on the elements mapping to $\{0, \dots, y\}$. They are respectively

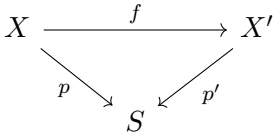
$$\{x\}, \qquad \{x \to a \leftarrow b\}, \qquad X.$$

Their nerves realize to a point, an interval subdivided into two edges, and an interval subdivided into three edges. They are all contractible. Every fibre-to-overcategory inclusion and every induced transition between these categories is consequently a weak homotopy equivalence. This is exactly the criterion for a weak cohomological fibration. Dually, the undercategories $y\backslash X$ are X, the poset $\{a \leftarrow b \to z\}$, and $\{z\}$; the same argument proves the weak homological condition.

In the pullback, only x and z remain. There is no morphism $x \to z$ in X: the connecting path in its nerve traverses $b \to a$ backwards. Thus the pullback category X' is the discrete two-point category, mapping to the two endpoints of $[1]$. The overcategory at the terminal endpoint is all of X', whereas that strict fibre is a point. Its inclusion is not an equivalence on components. The dual failure occurs at the initial endpoint. Even ordinary degree-zero cohomology detects the defect: for a nonzero coefficient field k, the relevant comparison is $k \times k \to k$, not an isomorphism.

This settles the request for an example on page 116 without any infinite category or nontrivial automorphism. The original weak conditions hold for every coefficient derivator in which weakly contractible shapes are completely aspherical; the failure is visible for both spaces and ordinary derived coefficients. The construction is supplied here with no priority claim. We checked the local-system and smoothness treatments cited in this commentary and did not locate this precise four-object example. In particular, the "weakly smooth" functors of Maltsiniotis (2005, Definition 4.1 and Proposition 4.3) have a different, comma-asphericity condition; their terminology must not be substituted for the weak cohomological condition used here.

Proposition 3. *Soit*



dans Dia*, avec p et p' des $\mathbb{D}$-fibrations cohomologiques faibles* (resp. *des $\mathbb{D}$-fibrations homologiques faibles*)*, soit $S_0 \subset S$ tel que $\pi_0(S_0) \to \pi_0(S)$ soit surjective. Pour que f soit une $\mathbb{D}$-équivalence, il suffit que les $f_s : X_s \to X'_s$, pour $s \in S_0$, le soient.*

Proposition 4. *Si $f : X \to Y$ dans* Dia *est $\mathbb{D}$-propre* (resp. *$\mathbb{D}$-lisse*)*, alors f est une $\mathbb{D}$-fibration cohomologique* (resp. *homologique*) *faible si et seulement si pour tout F constant sur Y, f_*f^*F* (resp. *$f_!f^*F$*) *est localement constant sur Y. Et s'il en est ainsi, pour tout $Y' \to Y$ dans* Dia*, $f' : X' = X \times_Y Y' \to Y'$ est encore une $\mathbb{D}$-fibration cohomologique* (resp. *homologique*) *faible.*

Immédiat sur le critère *b*) de la proposition 1.

Proposition 5. *Soit $f : X \to Y$ dans* Dia*. Considérons les conditions suivantes :*

a) *Pour tout $Y'' \to Y'$ sur Y dans* Dia*, si Y'' et Y' sont* Hot*-asphériques, alors* [page 117]

$$X'' = X \times_Y Y'' \longrightarrow X' = X \times_Y Y'$$

est une $\mathbb{D}$-équivalence.

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \underset{p}{\longleftarrow} & Y'' \end{array}$$

a′) *Comme* a)*, mais en supposant $p : Y'' \to Y'$ une* Hot*-équivalence (au lieu de supposer Y', Y''* Hot*-asphériques).*

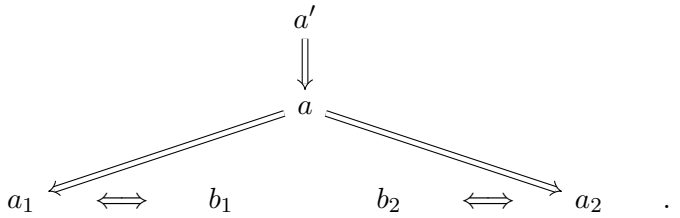
a$_1$) *Comme* a)*, mais en supposant (au lieu de l'asphéricité) que Y' a un objet final, et Y'' est la catégorie unité.*

a$_2$) *Comme* a)*, mais en supposant que Y' a un objet cofinal, et Y'' est la catégorie unité.*

b$_1$) *f est une $\mathbb{D}$-fibration cohomologique faible, et le reste après tout changement de base.*

b$_2$) *f est une $\mathbb{D}$-fibration homologique faible, et le reste après tout changement de base.*

On a alors le diagramme d'implications tautologiques ed. 70



Lorsque $\mathbb{D}$ satisfait à l'axiome **Der** 3 a′ (p. 111)*, alors toutes ces conditions, sauf* a′)*, sont équivalentes. Lorsque $\mathbb{D}$ satisfait à l'axiome* **Der** 3 b$'_1$ *(ou seulement* **Der** 3 b$''_{\text{coh}}$ ou b$''_{\text{hom}}$*), alors toutes les conditions (y inclus* a′*) sont équivalentes.*

Pour simplifier, on va par la suite supposer que l'axiome **Der** 3 *a*′) est satisfait.

Définition 3. Sous les conditions équivalentes a, a_1, b_1, a_2, b_2 de la proposition 4, on dit que f est une $\mathbb{D}$-*fibration*.

Commentary · Exposition [ms. 117]

Why universality identifies the two weak conditions

Proposition 5 organizes the earlier failed argument by changing its input. Under a', one may pull back between weakly contractible bases. The preceding example shows why this is an additional requirement; it is not implied by the fibre criterion on the original base.

Suppose first that the weak cohomological condition holds after every base change. Over a category B with a terminal object, the direct image of a constant coefficient is locally constant and hence constant by a'. Its cohomology over B is therefore its value at any selected object b. The pointwise formula identifies that value with the cohomology of the strict fibre. Hence every point inclusion into such a base produces a $\mathbb{D}$-equivalence on total categories. Conversely, these point tests applied over Y/y, and then at its objects $u : y \to y'$, give the fibre and transport comparisons, exactly as in the proof of Proposition 4 above, now with constant coefficients only. This proves $a_1 \Leftrightarrow b_1$.

More generally, take a pullback $p : B' \to B$ with both bases weakly contractible. Put $G = f_* M_X$ and $G' = f'_* M_{X'}$. They are locally constant. Their pointwise comparison is an isomorphism because both universal weak fibre formulas use the same strict fibre; thus $G' \simeq p^* G$. Since G is constant by a', the comparison of its cohomology over the two contractible bases is invertible. This proves $b_1 \Rightarrow a$. The dual proof gives $b_2 \Rightarrow a$, and the remaining implications are immediate. In a_2, the printed "objet cofinal" must carry the dual, initial-object meaning required by this argument.

If the stronger b''_{coh} is assumed, the same calculation works with an arbitrary Hot-equivalence p: G need not become constant, because p preserves cohomology of every locally constant coefficient. That proves a'. The homological version is dual. Thus the two different uses of the prime on a' must be kept apart: **Der** 3 a' is an axiom about coefficients on contractible categories, whereas condition a' of Proposition 5 is a universal pullback test using all weak homotopy equivalences.

Proposition 6. *Supposons que $f : X \to Y$ dans* Dia *soit $\mathbb{D}$-propre* (resp. $\mathbb{D}$-*lisse*). *Conditions équivalentes :* [page 118]

a) *f est une $\mathbb{D}$-fibration.*

b) *f est une $\mathbb{D}$-fibration cohomologique* (resp. *homologique*) *faible.*

c) *Pour tout $y' \to y$ dans Y, $X/y' \to X/y$* (resp. $y\backslash X \to y'\backslash X$) *est une $\mathbb{D}$-équivalence.*

d) *Pour tout F constant sur Y, $f_* f^* F$* (resp. $f_! f^* F$) *est localement constant sur Y.*

d′) *Pour tout F localement constant sur Y, $f_* f^* F$* (resp. $f_! f^* F$) *est localement constant sur Y.*

Cela résulte formellement de Prop. 4 et Prop. 5.

Corollaire. *Supposons $f : X \to Y$ $\mathcal{C}at$-fibrant* (resp. $\mathcal{C}at$-*cofibrant*). *Pour que f soit une $\mathbb{D}$-fibration, il faut et il suffit que pour tout $u : y \to y'$ dans Y, $u^* : X_{y'} \to X_y$* (resp. $u_* : X_y \to X_{y'}$) *soit une $\mathbb{D}$-équivalence.*

Il est immédiat que la notion de $\mathbb{D}$-fibration est stable par changement de base, mais il est douteux qu'elle le soit par composition.[60] (Ici encore, j'ai omis de faire un contre-exemple, avec des dérivateurs $\mathbb{D} = \mathsf{Der}(k\text{-}\mathsf{Mod})$ disons.) Autre lacune de la notion : il n'est pas clair, si f est une $\mathbb{D}$-fibration et F un $\mathbb{D}$-coefficient sur Y, si

$$f_* f^* F \qquad \text{et} \qquad f_! f^* F$$

se calculent "fibre par fibre", comme on s'attendrait dans le cas d'un "fibré". ed. 71

Par contre, voici un résultat sympa :

Commentary · Development (commentators') [ms. 118]

Two fibrations whose composite is not a fibration

The doubt about composition can be settled by a small categorical example. It uses the same loss of monodromy information as the preceding counterexamples, but places that loss in a transition functor. We work with $\mathbb{D} = \mathsf{Der}(\mathbb{Q}\text{-}\mathsf{Mod})$ on all small categories and with the final definition of $\mathbb{D}$-fibration on page 117, which includes universality under base change.

Proposition (failure of composition). [ours] There are $\mathbb{D}$-fibrations $p : X \to Y$ and $q : Y \to [1]$ such that qp is not even a weak cohomological $\mathbb{D}$-fibration. The first map can be chosen to be a covering functor, hence both proper and smooth; the second is a Grothendieck opfibration.

60. N.B. Si les $\mathbb{D}$-équivalences sont déjà des Hot-équivalences, alors la notion coïncide avec celle de Hot-fibration, et elle est stable par composition. Également, si les $\mathbb{D}$-équivalences sont au sens des $\mathbb{D}$-équivalences cohomologiques, ou homologiques, alors la notion coïncide avec celle des $\mathbb{D}$-fibrations cohomologiques (resp. homologiques) ci-dessous, et est également stable par composition.

Let $G = C_2 = \{1, \sigma\}$. Define Y to have two objects a, b, with

$$Y(a,a) = \{1_a\}, \quad Y(b,b) = G, \quad Y(a,b) = G, \quad Y(b,a) = \varnothing.$$

Composition on $Y(b,b)$ is multiplication, and postcomposition of the arrow $g : a \to b$ by $h \in G$ is the arrow hg. Send a to 0 and b to 1. This defines $q : Y \to [1]$. It is the Grothendieck construction of the functor $e \to BG$. More explicitly, any arrow $g : a \to b$ is opcartesian: for every arrow $k : a \to b$, the unique factorization through g is multiplication by kg^{-1}. Its endpoint fibres are e and BG, and its transition functor is $e \to BG$.

This transition is a $\mathbb{D}$-equivalence. For a complex M of rational vector spaces with trivial G-action, averaging by $(1+\sigma)/2$ makes invariants exact, and coinvariants are exact for the same reason. Thus

$$\mathbb{R}\Gamma(G, M) = M, \qquad \mathbb{Q} \otimes^{\mathbb{L}}_{\mathbb{Q}[G]} M = M.$$

These identities are natural for arbitrary, including unbounded, complexes: the trivial module is projective over $\mathbb{Q}[G]$, and the averaging contraction is degreewise compatible with the differential. The opfibration criterion on page 118 now makes q a $\mathbb{D}$-fibration, with all base changes covered by that criterion.

Let Z be the discrete category on the set G. Define a functor $L : Y \to \mathcal{C}at$ by $L(a) = L(b) = Z$, and let every arrow labeled g act by left translation $z \mapsto gz$. Put $X = \int_Y L$ and let p be the projection. All its transport functors are isomorphisms. Every arrow of Y therefore has a unique lift with either prescribed source or prescribed target. In particular p is both a discrete fibration and a discrete opfibration, and the transport criterion makes it a $\mathbb{D}$-fibration (indeed a HOT-fibration).

The composite qp is an opfibration whose fibre at 0 is the discrete two-point category Z. Its fibre at 1 is the action groupoid $G \ltimes G$ for the left regular action. There is exactly one arrow between any ordered pair of objects of this groupoid, so it is equivalent to the terminal category. The transition functor is the inclusion

$$Z_{\mathrm{disc}} \longrightarrow G \ltimes G.$$

On cohomology in degree zero it induces

$$\mathbb{Q} \longrightarrow \mathbb{Q}^2, \qquad r \longmapsto (r, r),$$

which is not an isomorphism. Since qp is proper, the criterion of Proposition 6 identifies this failure with failure to be a weak cohomological fibration. Equivalently, $(qp)_*\mathbb{Q}_X$ has values $\mathbb{Q}^2$ at 0 and $\mathbb{Q}$ at 1, and its transition is the displayed diagonal. This completes the construction and the proof.□

The example explains why the positive answer concerning f_*f^*F does not force closure under composition. The coefficient $p_*\mathbb{Q}_X$ on Y is the permutation local system attached to the regular G-set. It is not pulled back from a coefficient on [1]: every such pullback has trivial G-action at b. The constant-coefficient test for q cannot control this local system. The stronger cohomological fibrations introduced on page 121 impose exactly the missing condition. Our literature check covered the smooth-functor formalism of Maltsiniotis (2005) and the locally constant coefficient theory of Cisinski (2009); we did not find this particular composition counterexample there. It is an elementary development of the mechanism already present in Grothendieck's finite-group example.

Proposition 7. *Soit $f : X \to Y$ dans* Dia. *Conditions équivalentes :*

a) *f est une $\mathbb{D}$-équivalence "universelle",* i.e. *pour tout $Y' \to Y$ dans* Dia, *$f' : X' = X \times_Y Y' \to Y'$ est une $\mathbb{D}$-équivalence.* [page 119]

b) *f est une $\mathbb{D}$-fibration à fibres $\mathbb{D}$-asphériques.*

Définition 4. Sous les conditions équivalentes de la proposition 7, on dit que f est une $\mathbb{D}$-fibration *triviale*.

Mais attention, f peut-être une $\mathbb{D}$-fibration et une $\mathbb{D}$-équivalence, sans que la $\mathbb{D}$-fibration soit triviale *i.e.* sans que les fibres soient $\mathbb{D}$-asphériques, cf. p. 87 et suivantes. Et également (devrait remonter) même si $\mathbb{D}$ satisfait **Der** 3 b_2 (la plus forte des variantes de **Der** 3), donc quand <u>toutes</u> les conditions de la proposition 5, y inclus a'), sont équivalentes, celles-ci (*i.e.* que f soit une $\mathbb{D}$-fibration) n'impliquent pourtant pas, en général, que dans la condition a') on puisse se borner à $p : Y'' \to Y'$ une $\mathbb{D}$-équivalence (au lieu d'une Hot-équivalence). En

d'autres termes, si on a un diagramme cartésien dans Dia

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array} \quad ,$$

avec f une $\mathbb{D}$-fibration, et p une $\mathbb{D}$-équivalence, il ne s'ensuit pas forcément que q soit une $\mathbb{D}$-équivalence. (Mais ce sera pourtant le cas, "bien sûr", si $\mathbb{D} = \mathsf{HOT}$.)

Par contre, on a :

Proposition 8. *Soit $f : X \to Y$ dans* Dia*. Conditions équivalentes :* [page 120]

a) *f est une $\mathbb{D}$-fibration.*

b) *Pour tout $p : Y'' \to Y'$ sur Y, si p est une $\mathbb{D}$-équivalence* 1*-complète* (déf. 1, p. 113)*, alors $q : X'' \to X'$ est une $\mathbb{D}$-équivalence.*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow{q} & X'' \\ {\scriptstyle f}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f''} \\ Y & \longleftarrow & Y' & \xleftarrow[p]{} & Y'' \end{array}$$

Corollaire. *Supposons que dans* Dia*, toute $\mathbb{D}$-équivalence soit* 1*-complète.* (N.B. Je ne connais que le dérivateur HOT pour satisfaire à cette condition, qui paraît quasiment caractéristique de HOT.) *Alors $f : X \to Y$ est une $\mathbb{D}$-fibration si et seulement si pour tout $p : Y'' \to Y'$ sur Y, si p est une $\mathbb{D}$-équivalence, alors $q : X'' \to X'$ aussi. Dans ce cas exceptionnel parmi les dérivateurs, la notion de $\mathbb{D}$-fibration est stable par composition* (ce qui est tautologique sur le critère que je viens de donner.)

Commentary · Exposition [ms. 120]

Universal equivalence and the 1-complete test

Two different universal assertions occur here. A universal $\mathbb{D}$-equivalence is a functor whose every base change is a $\mathbb{D}$-equivalence. A $\mathbb{D}$-fibration preserves certain equivalences in diagrams over its base; it need not itself be a $\mathbb{D}$-equivalence. Proposition 7 identifies their intersection with the condition that all strict fibres be $\mathbb{D}$-aspherical. For the forward implication, base change to each point gives aspherical fibres, while the two-out-of-three property applied to the two universally equivalent projections gives the fibration criterion. For the reverse implication, the universal weak fibre formula makes f_*M_X a constant object M_Y: its unit is an isomorphism at every y because X_y is $\mathbb{D}$-aspherical. Taking global cohomology proves the equivalence, and the same argument applies after base change.

This result does not permit replacement of "universal" by "ordinary". The rational monodromy examples already show that a fibration which is a $\mathbb{D}$-equivalence can have a fibre that is not $\mathbb{D}$-aspherical. The issue is not a failure of the fibre formula, but a loss of information upon taking global cohomology of a nonconstant local system.

Proposition 8 repairs the pullback assertion by strengthening the equivalence in the base. If $p : Y'' \to Y'$ is 1-complete, it preserves cohomology of every local system. For a constant M, universality of the weak fibration condition makes $f'_*M_{X'}$ locally constant and identifies its pullback with $f''_*M_{X''}$. Taking cohomology proves that $X'' \to X'$ is a $\mathbb{D}$-equivalence. Conversely, apply the preservation assertion to $e \to B$ with B weakly contractible. Under a', restriction on local systems is an equivalence and hence 1-complete, so the point tests of Proposition 5 follow.

There is a useful distinction between the result here and the stronger definition below. Proposition 8 asks that pulling back a 1-complete equivalence produce a constant-coefficient equivalence. It does not assert that the pullback is itself 1-complete. Demanding that stronger conclusion supplies a composition-stable class by definition, but it is an extra condition unless the internal-Hom argument identifies the relevant equivalence classes.

On va renforcer la notion de $\mathbb{D}$-fibration, pour la rendre plus souple, et notamment stable par composition.

Proposition 9. *Soit $f : X \to Y$ dans* Dia. *Conditions équivalentes :*

a) *Pour tout $y \in Y$, $X_y \to X/y$ est une $\mathbb{D}$-équivalence cohomologique* (déf. 1, p. 113), *et de même pour $X/y \to X/y'$ pour toute flèche $y \to y'$ dans Y. Et ces conditions sur f restent valables après tout changement de base* (N.B. il suffit que la première de ces conditions le reste).

b) *pour tout $F \in \mathsf{Ob}\ \mathbb{D}(X)$ localement constant, f_*F est localement constant sur Y et se calcule "fibre par fibre" – et itou après tout changement de base.* [page 121]

c) *Pour tout $p : Y'' \to Y'$ sur Y, si p est une $\mathbb{D}$-équivalence cohomologique, alors il en est de même de $q : X'' \to X'$.*

c′) *Comme* c, *mais en supposant comme hypothèse sur p que Y'' soit la catégorie finale, et Y' ait un objet final.*

Démonstration. Dans l'ordre (± tautologique)

$$\mathrm{a} \Longrightarrow \mathrm{b} \Longrightarrow \mathrm{c} \Longrightarrow \mathrm{c}' \Longrightarrow \mathrm{a} \quad .$$

Définition 5. Sous les conditions équivalentes de la proposition 9, on dit que f est une $\mathbb{D}$-*fibration cohomologique*. Sous les conditions duales, on dit que f est une $\mathbb{D}$-*fibration homologique*. On dit que f est une $\mathbb{D}$-*fibration forte*, si c'est à la fois une $\mathbb{D}$-fibration cohomologique et une $\mathbb{D}$-fibration homologique.

Ces notions renforcent la notion de $\mathbb{D}$-fibration. Elles ont l'avantage de plus d'être stables par composition, non seulement par changement de base.

Commentary · Exposition [ms. 121]

The reason the strong notion composes

We may now prove the composition assertion without an appeal to intuition about bundles. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be cohomological fibrations in the revised sense. For $L \in \mathcal{L}(X)$, the object f_*L is locally constant on Y, and then g_*f_*L is locally constant on Z. The canonical composition isomorphism identifies it with $(gf)_*L$. This is the coefficient condition that was missing for the earlier notion.

For the universal base-change assertion, pull back the composite along any $Z' \to Z$, and factor the resulting rectangle into the two cartesian squares for g and f. The comparison for g, applied to the locally constant object f_*L, is invertible. The comparison for f, applied to L, is invertible. Pasting gives the comparison for gf. The same reasoning works for every locally constant coefficient on the new total category, because the hypotheses on f and g were universal. Specializing to a point in Z gives the fibre formula. Alternatively, condition c of Proposition 9 proves composition by applying two successive pullbacks to a cohomological equivalence. The two arguments express the same closure property at the levels of coefficients and indexing categories.

The dual reasoning applies to homological fibrations, and hence to strong fibrations. It does not prove that every $\mathbb{D}$-fibration composes: a weak fibration controls direct images of constant coefficients, whereas the first direct image in a composite is usually only locally constant. The explicit counterexample at page 118 realizes exactly this obstruction. The questions about f_*f^*F and $f_!f^*F$, for an arbitrary coefficient on the base, are different again; the factorization-category argument at page 82 proves the required fibre formulas under the stated domain hypotheses. There is no conflict, because a locally constant object produced on the intermediate category need not be pulled back from the final base.

For the proposed 1-complete and 2-complete fibrations of page 122, the phrase "not stable under composition (as far as I know)" supplies no proof. In the compatible closed monoidal situation, the 1-complete variant reduces to the cohomological one and therefore composes. The 2-complete variant nevertheless fails to compose, even for the represented set-valued derivator: the development "The 2-complete variant also fails to compose" constructs two functors between finite categories from a normalized C_2-valued cocycle on a triangulated two-sphere. Their composite fails a permitted contractible-base test because restriction gives only trivial C_2-actions. This restriction is fully faithful but not essentially surjective. An internal Hom controls morphisms between objects already

descended; it does not supply descent of every local system. The counterexample thus settles the 2-complete question without extending its negative conclusion to the 1-complete class.

La notion de $\mathbb{D}$-fibration forte a été choisie pour coiffer les deux notions duales de $\mathbb{D}$-fibration cohomologique et homologique (à défaut d'une meilleur notion) tout en restant stable par composition. On pourrait exiger que pour tout [page 122]

$$Y'' \longrightarrow Y'$$

sur Y, avec Y', Y'' Hot-asphériques, $q : X'' \to X'$ soit une $\mathbb{D}$-équivalence 1-complète (voire même 2-complète), appeler ça les $\mathbb{D}$-*fibrations* 1-*complètes* (resp. 2-*complètes*). Ça renforce de façon assez élégante la notion de $\mathbb{D}$-fibration forte, mais a le grand tort de ne pas être stable par composition (que je sache!), sauf dans le cas particulier où on a des $\underline{\mathsf{Hom}}$ internes sympas (cf. bas de la page 113), auquel cas la notion de $\mathbb{D}$-équivalence implique déjà celle de $\mathbb{D}$-équivalence 1-complète, donc aussi celle de $\mathbb{D}$-équivalence homologique. Donc dans ce cas ed. 73

[page 123]

$$\begin{array}{ccc} \mathbb{D}\text{-fibration cohomologique} & \Longleftrightarrow & \mathbb{D}\text{-fibration 1-complète} \\ \Big\Downarrow & & \\ \mathbb{D}\text{-fibration homologique} & , & \end{array}$$

et il n'y a pas à chercher plus loin que les $\mathbb{D}$-fibrations cohomologiques.

Commentary · Development (commentators') [ms. 122]

The 2-complete variant also fails to compose

The qualification "que je sache" is not a proof of failure. Nevertheless, the proposed 2-complete class really need not be closed under composition, even for a represented derivator with all small categories as shapes. We give a categorical counterexample, including the coherence responsible for the failure. The existence of the local-systems derivator in Cisinski (2009, Theorem 3.13) does not supply the missing composition property. We checked the local-systems treatment just cited and did not find this particular construction there. The construction below is a verification given here, without a claim of priority.

Proposition. [ours] Let $\mathbb{D}_{\mathrm{set}}(A) = \mathrm{Fun}(A^{\mathrm{op}}, \mathsf{Set})$. There are functors

$$X \xrightarrow{p} Y \xrightarrow{q} [1]$$

which are both 2-complete fibrations in the precise sense proposed on manuscript page 122, while qp is not one. The categories X and Y are finite, and the functor p is even a HOT-fibration. Thus the failure persists in the usual domain of all small categories, with cartesian closed coefficients.

Write $\Pi_1(A)$ for the fundamental groupoid of $|NA|$, or equivalently the groupoid obtained by inverting all arrows of A. A locally constant set-valued presheaf factors through this groupoid. Consequently

$$\mathcal{L}_{\mathrm{set}}(A) \simeq \mathrm{Fun}(\Pi_1(A)^{\mathrm{op}}, \mathsf{Set}).$$

In particular, an equivalence of fundamental groupoids induces a 2-complete equivalence. We will verify this condition for the two factors, and exhibit an object which cannot descend for the composite.

Let B be the poset of nonempty proper subsets of $\{0,1,2,3\}$, ordered by inclusion. Its nerve is the barycentric subdivision of the boundary of a tetrahedron, so $|NB|$ is a two-sphere. Fix the maximal chain

$$\sigma = (\{0\} < \{0,1\} < \{0,1,2\}).$$

Write $C_2 = \mathbb{Z}/2\mathbb{Z}$ additively. For composable arrows $u : b \to b'$ and $v : b' \to b''$, set $c(u,v) = 1$ when they constitute this chain, and set $c(u,v) = 0$ otherwise. In particular it is zero if either arrow is an identity. This is a normalized C_2-valued 2-cocycle: there is no chain of three nonidentity arrows in B, and the cocycle identity with an identity arrow cancels term by term.

Define E_c to have the same objects as B, with one arrow (u, n) for each arrow u of B and label $n \in C_2$. Put

$$(v, n) \circ (u, m) = (vu, m + n + c(u, v)), \qquad 1_b = (1_b, 0).$$

The cocycle identity is precisely associativity. The projection $E_c \to B$ is both a Grothendieck fibration and an opfibration. Indeed, after fixing a lift of one base arrow, any required factorization is obtained by uniquely solving for the C_2-label of the other lift. Its fibre at each object is the one-object groupoid BC_2, and its transport functors act identically on C_2. The coherence is nontrivial, although the individual transport functors are identities.

We need the fundamental group of E_c, not the full homotopy type of its nerve. Put $t_b = (1_b, 1)$ and $h_{bb'} = (b \leq b', 0)$. Its groupoid completion has these generators and the relations

$$t_b^2 = 1, \qquad t_{b'} h_{bb'} = h_{bb'} t_b, \qquad h_{b'b''} h_{bb'} = h_{bb''} t_b^{c(b,b',b'')}.$$

Choose a spanning tree of the one-skeleton of NB and use it to identify all objects. The first relations identify the t_b with a single central generator t, with $t^2 = 1$. Every triangle other than σ gives the ordinary, untwisted edge relation. These relations kill all remaining edge generators, because the complex obtained from the triangulated sphere by removing the interior of σ is a disk, with the same one-skeleton. The remaining triangle relation then says $1 = t$. This proves that E_c is connected and that $\Pi_1(E_c)$ is equivalent to the terminal groupoid.

Now choose $b_0 = \{0\}$ and let

$$q : Y = \int_{[1]} (e \xrightarrow{b_0} B) \longrightarrow [1]$$

be the Grothendieck construction. Concretely, Y consists of B and one new object a, with

$$Y(a, b) = B(b_0, b), \quad Y(b, a) = \varnothing, \quad Y(a, a) = \{1_a\}.$$

The functor q sends a to 0 and B to 1. There is a retraction $r : Y \to B$ sending a to b_0. If $i : B \hookrightarrow Y$ is the inclusion, then $ri = 1$ and the arrow $a \to b_0$ gives a natural transformation $1_Y \to ir$. Finally define

$$X = Y \times_B E_c, \qquad p : X \longrightarrow Y.$$

The retraction and its natural transformation lift to $X \to E_c$, using the arrow of C_2-label zero over $a \to b_0$. Naturality follows from the normalization of c, since this arrow maps to 1_{b_0} under r. Hence $|NX| \simeq |NE_c|$, and X is simply connected. The strict fibre of qp at 0 is BC_2.

We now check the universal quantifiers in the definition. First, p is a pullback of the opfibration $E_c \to B$; all its transports are equivalences of groupoids. The opfibration criterion already established on page 118 therefore makes p a HOT-fibration, after every base change. A functor between weakly contractible bases is a nerve weak equivalence. Pullback along p consequently gives a nerve weak equivalence, and hence a 2-complete equivalence for sets.

There is also a direct verification of this last assertion. For any weakly contractible category $A \to Y$, the pulled-back cocycle on A is a coboundary, since $H^2(|NA|; C_2) = 0$. Writing it as δd and replacing the label n of an arrow u by $n + d(u)$ identifies the pulled-back category with $A \times BC_2$. Choose d normalized, so that this identification preserves the vertical C_2-loops. A comparison induced by a functor between two such bases preserves their vertical generator, and hence induces an isomorphism of their fundamental groups, both equal to C_2. Both total categories are connected. Thus their fundamental groupoids are equivalent, as required.

For q, we prove a stronger assertion. For every $A \to [1]$, with no contractibility assumption, the projection

$$q_A : Y_A = Y \times_{[1]} A \longrightarrow A$$

induces an equivalence of fundamental groupoids. Its strict fibres are e or B, and it has a section selecting a over 0 and b_0 over 1. The transport functors preserve these selected objects. Within each fibre there is a unique morphism in its fundamental groupoid from the selected object to any given object, because the fibre is connected and simply connected. These fibre paths give a natural isomorphism, in $\Pi_1(Y_A)$, from the section followed by the projection to the identity. For naturality, factor an arrow into its opcartesian lift and a vertical arrow; transport of the chosen path and composition with the vertical arrow give the same path in the target fibre, by uniqueness. The section is therefore a quasi-inverse to $\Pi_1(q_A)$.

If $A' \to A$ is a functor over [1] with both bases weakly contractible, the two projections just considered and the equivalence $\Pi_1(A') \to \Pi_1(A)$ show that $Y_{A'} \to Y_A$ is a 2-complete equivalence. This proves exactly the page-122 condition for q.

For the composite, use the permitted test $\{0\} \to [1]$. Both bases are contractible, but the upstairs comparison is

$$BC_2 = X_0 \longrightarrow X.$$

Since X is connected and simply connected, all its locally constant set-valued presheaves are constant. Restriction to BC_2 therefore has only trivial C_2-actions in its essential image. A two-element set on which $1 \in C_2$ exchanges the two elements is a locally constant coefficient on BC_2 which does not extend to X. Restriction is not essentially surjective, so the comparison is not 2-complete. This proves the proposition. □

The failure is precisely in descent of objects. In this final test the functor from sets to C_2-sets with trivial action is fully faithful; the obstruction does not concern morphisms between objects already descended. Thus the example is consistent with closure of the 1-complete class under the internal-Hom hypotheses, and explains why that argument cannot settle the 2-complete variant. Theorem 3.13 of Cisinski constructs coherent diagrams of local systems for a fixed indexing category; it does not assert that the restricted pullback tests defining the proposed class preserve essential surjectivity under a composite.

On pourrait songer, il est vrai, à définir (dans le cas général) une notion plus forte, en exigeant que pour tout $p : Y'' \to Y'$ sur Y qui soit une $\mathbb{D}$-équivalence 1-complète, $q : X'' \to X'$ soit aussi une $\mathbb{D}$-équivalence 1-complète. Cette fois-ci, la notion est stable par composition, mais elle me paraît néanmoins inutilisable (en dehors du cas qu'on vient de dire, où elle coïncide avec la notion de $\mathbb{D}$-fibration cohomologique, ou encore de $\mathbb{D}$-fibration forte), car n'étant pas en mesure de décrire les $\mathbb{D}$-équivalences 1-complètes, je suis aussi incapable de donner des exemples non triviaux de $\mathbb{D}$-fibrations "strictement 1-complètes".

Pour vérifier qu'une $f : X \to Y$ dans Dia est une $\mathbb{D}$-fibration cohomologique, il faut savoir vérifier que des flèches $q : Z' \to Z$ dans Dia sont des $\mathbb{D}$-équivalences cohomologiques, donc que $H^*_{\mathbb{D}}(Z, F) \xrightarrow{\sim} H^*_{\mathbb{D}}(Z', q^*(F))$ est [un] isomorphisme pour tout $F \in \mathsf{Ob}\ \mathbb{D}(Z)$ qui est localement constant sur Z. Or je ne sais vérifier un énoncé de cette nature (en dehors du cas où je sais d'avance que Z est complètement $\mathbb{D}$-asphérique, *i.e.* que les coefficients localement constants sur Z sont en fait constant) que si q est déjà une Hot-équivalence. C'est dire que je ne sais exhiber des $\mathbb{D}$-fibrations cohomologiques, que quand ce sont déjà des HOT-fibrations cohomologiques. Je veux, dans la suite, donner des critères utilisables pour des HOT-fibrations. Mais on va axiomatiser ces développements, en gardant un $\mathbb{D}$ général, sur lequel je vais faire une hypothèse si draconienne que pratiquement, je ne connais que le dérivateur HOT qui y satisfait. Il s'agit de c_1 ci dessous. [page 124]

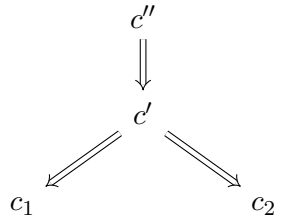
Der 3 c_1) Toute $\mathbb{D}$-équivalence est une $\mathbb{D}$-équivalence cohomologique.

c_2) Toute $\mathbb{D}$-équivalence est une $\mathbb{D}$-équivalence homologique.

c') Toute $\mathbb{D}$-équivalence est 1-complète.

c'') Toute $\mathbb{D}$-équivalence est 2-complète. [page 125]

N.B. On a



tautologiquement. Je ne connais que HOT pour satisfaire l'ensemble des deux plus faibles conditions c_1, c_2. Et en fait, sous réserve d'écrire la théorie du dérivateur HOT, il doit être vrai qu'elle satisfait même la forme la plus forte c''. (Et les HOT-équivalences doivent être simplement les Hot-équivalences.) (Cf. le "lemme" p. 109.) ed. 74

La proposition suivante est une tautologie.

Proposition 10. *Supposons que* $\mathbb{D}$ *satisfasse la condition* **Der** 3 c_1 (p. 124), *par exemple* $\mathbb{D} = \mathsf{HOT}$. *Alors la notion de* $\mathbb{D}$*-fibration coïncide avec celle de* $\mathbb{D}$*-fibration cohomologique. Elle est donc stable par composition. Pour que* $f : X \to Y$ *soit une* $\mathbb{D}$*-fibration, il faut et il suffit que pour* $p : Y'' \to Y'$ *sur* Y*, si* p *est une* $\mathbb{D}$*-équivalence, il en est de même de* $q : X'' \to X'$.

Corollaire 1. *Si* $f : X \to Y$ *dans* Dia *est* $\mathbb{D}$*-propre, alors* f *est une* $\mathbb{D}$*-fibration cohomologique si et seulement si pour toute flèche* $y' \to y$ *dans* Y*,* $X/y' \to X/y$ *est une* $\mathbb{D}$*-équivalence* (i.e. *si et seulement si* f *est une* $\mathbb{D}$*-fibration cohomologique faible*).

Appliquer le critère c) de `[1a]` prop. 6.

Corollaire 2. *Supposons que* $\mathbb{D}$ *satisfasse* **Der** 3 (c_1, c_2) (par exemple $\mathbb{D} = \mathsf{HOT}$). *Alors pour une flèche* $f : X \to Y$ *dans* Dia*, il revient au même que* f *soit une* $\mathbb{D}$*-fibration, une* $\mathbb{D}$*-fibration cohomologique, ou une* $\mathbb{D}$*-fibration homologique. Si* f *est propre* (resp. *lisse*)*, il revient au même de dire que* f *est une* $\mathbb{D}$*-fibration cohomologique* (resp. *homologique*)*, faible.* [page 126]

Commentary · Present state [ms. 126]

The closing axioms and the theorem they isolate

The c-conditions have an antecedent different from that of the b-conditions. They start with a $\mathbb{D}$-equivalence, which may be much weaker than a weak homotopy equivalence of nerves. Thus the modern validity of b'_2 does not make the c-conditions automatic. Rational coefficients and the map $e \to BC_2$ already separate them: all the homotopy-invariance axioms hold, while invariance for constant coefficients does not imply invariance for the sign local system.

For spaces, the argument at page 110 proves all of c_1, c_2, c', c''. Propositions 5 and 9 therefore describe one class, and the criterion says that its members preserve weak homotopy equivalences under arbitrary pullback over their base. This is the intrinsic categorical fibration property being isolated here. It should not be identified with the class of fibrations in every model structure on categories, or with an assertion that the realization of the displayed functor is a literal Serre fibration. The invariant content is the comparison with homotopy fibres and the stability of the appropriate pullback squares.

The transition criterion has a classical source. In the dual form of Quillen (1973, §1, Theorem B and its corollary), weak equivalences between the comma categories along every arrow identify these comma categories with homotopy fibres of the induced map of classifying spaces. For an opfibration, transport functors between strict fibres provide the corresponding test. Properness then gives the strict-fibre-to-comma comparison required in the manuscript. The resulting criterion is the content of Corollary 1 here; its dual gives Corollary 2. These statements were not awaiting a new homotopy group calculation: they were awaiting a coefficient formalism in which the calculation could be made functorial and self-dual.

The scope of the later theorems is worth stating exactly. The ordinary homotopy derivator exists on all small categories (Cisinski, 2003, Theorem 6.11), and its locally constant objects have the invariance needed in the lemma. Within that setting, the question on pages 107–111 and the final conditional assertions are settled affirmatively. The weaker notions for other coefficient derivators retain the distinctions exposed by the examples in these pages. The manuscript's remark that HOT is the only example known to him is autobiographical, not a uniqueness theorem: for instance a product of HOT with another derivator satisfying b'_2 still detects nerve weak equivalences and satisfies c''.

The last statement follows directly. A map is an equivalence for the product derivator exactly when it is one for both factors. The HOT factor forces it to be a nerve weak equivalence. The b'_2 property of each factor then makes restriction an equivalence on their local systems, and local systems in a product are computed factorwise. This proves c'' for the product without identifying the coefficient category with spaces. What the axiom singles out is the strength of detection and transport, not a unique presentation of the theory.

On voit donc que dans $\mathcal{C}at$, on doit avoir une seule notion raisonnable de HOT-fibration, qui coïncidera avec celle de HOT-fibration cohomologique ou homologique. Mais pour que cette notion soit dès à présent disponible, il faudrait que le dérivateur HOT soit déjà construit sur le domaine $\mathsf{Dia} = \mathcal{C}at$ tout entier. Or ce n'est pas chose faite encore (et il n'a donc pas pu

être question de prouver les propriétés de HOT dont il vient d'être question, disons la plus forte **Der** 3 c'').

Néanmoins, je voudrais développer de façon directe ("élémentaire") la notion de Hot-fibration, sans me référer à la théorie du dérivateur HOT, et aux développements théoriques précédents.

Commentary · Context [ms. 126]

The later letter and the meaning of a fibration

There is a useful chronological comparison at this stopping point. In his letter to Thomason of 2 April 1991, Section 1, Grothendieck starts with a category equipped with a class W of weak equivalences and defines its W-fibrations by requiring pullback to carry W-equivalences to W-equivalences (Grothendieck, 1991a). Closure under composition then follows by composing pullback functors. This is the strong universal preservation condition proposed on manuscript page 79, and it becomes the final condition in this chapter when the supplementary assumptions on $\mathbb{D}$ identify the appropriate equivalence classes.

The comparison explains why the word "fibration" must always be accompanied by its test class. A W-fibration defined by preservation of W has a formal closure property. A $\mathbb{D}$-fibration defined by preservation of weak equivalences of nerves only as $\mathbb{D}$-equivalences has two different test classes, so that formal argument does not apply. The composition example on page 118 makes the difference explicit. In the case $\mathbb{D} = \mathsf{HOT}$, the classes agree and the difficulty disappears.

The earlier motivation in Section 69 of *Pursuing Stacks* was to retain coherent diagram categories and the two Kan extensions as part of the coefficient theory (Grothendieck, 1983). The present pages show why that enrichment matters even before stability or triangulated categories enter: the coefficient produced by one direct image can have nontrivial transport, and the next direct image must retain it. The mathematical work of these pages is the progressive adjustment of the admissible coefficients and the admissible base changes until this operation can be iterated. It should be read with those successive adjustments intact.

Overview · I-b

Coefficients, monodromy, and the search for a usable fibration.

These pages of Chapter I begin with a correction to an overly strong use of cofinality and end with a programme for a theory of fibrations in the category of small categories. Their continuity lies in a single question: which coefficients must a comparison preserve if that comparison is to remain useful after another operation? Preservation of constant coefficients is enough to define a class of weak equivalences. It is not, in general, enough to carry out base change or to compose fibrations. The passage from constants to locally constant coherent coefficients is therefore the principal mathematical movement of the text.

These are pages 34–74 of the edition, manuscript pages 60–126. The text follows the editors' electronic edition (Grothendieck, 1990–1991a). Manuscript pages 60–66 also close the first part of the chapter, where their commentary, claims and readings are given. Printed and manuscript numbers are kept distinct throughout. No assertion in the commentary silently corrects a mathematical mistake in the French source: genuine transcription errors are corrected, whereas errors already present in the edition remain visible and are explained where they affect the argument.

We use the manuscript's convention that an object of $\mathbb{D}(X)$ has underlying diagram $X^{\mathrm{op}} \to \mathbb{D}(e)$. Thus the value of a right Kan extension f_*F at y is calculated on X/y, and the value of the left Kan extension $f_!F$ is calculated on $y\backslash X$. The symbols $H^*_{\mathbb{D}}$ and $H^{\mathbb{D}}_*$ denote objects in the coefficient category, not merely lists of cohomology or homology groups. This convention is responsible for several apparently reversed statements when one compares the manuscript with modern covariant-diagram treatments; the variance distinction is explained explicitly in Groth et al. (2014a, Warning 2.2).

From constant equivalences to base change.

For $f : X \to Y$, with terminal projections p and q, a $\mathbb{D}$-equivalence means that restriction induces bijections

$$\mathrm{Hom}_{\mathbb{D}(Y)}(q^*M, q^*N) \longrightarrow \mathrm{Hom}_{\mathbb{D}(X)}(p^*M, p^*N)$$

for all $M, N \in \mathbb{D}(e)$. Adjunction and Yoneda identify this with invariance of constant-coefficient homology or cohomology whenever the relevant Kan extensions exist. It does not say that f^* is fully faithful on all of $\mathbb{D}(Y)$. The distinction is already decisive for an endpoint inclusion in the walking arrow.

Theorem 1 on manuscript page 67 gives the stronger cofinality test actually needed: cohomological invariance for every coefficient on Y is equivalent to asphericity of the comma categories $y\backslash X$. The proof uses the counit $f_!p^*M \to q^*M$. One adjunction reads it as a cohomology comparison; pointwise evaluation reads it as constant homology of a comma category. This explains the apparent use of homology inside a cohomological theorem without adding a missing hypothesis to a valid argument.

The proper and smooth base-change results then have a precise local content. Properness requires the categories of lifts of every arrow $f(x) \to y$ starting at x to be $\mathbb{D}$-aspherical. The requirement involves both objects and their morphisms, rather than mere nonemptiness of the set of lifts. A Grothendieck opfibration satisfies it because an opcartesian lift is an initial object of the corresponding lift category. The dual account gives smoothness. The commentary supplies the omitted implications among the endpoint, lifting, and base-change tests in Theorem 2, while preserving the edition's mistyped objects and cross-references as source readings.

This vocabulary retains the geometry of proper and smooth base change, but defines a property of functors relative to a coefficient theory. The later theory of asphericity structures provides an appropriate comparison, rather than an identification with topological proper maps (Maltsiniotis, 2005, Theorems 1.2, 4.13, and 5.4). The intervening construction $\Phi(X)$, now called the twisted-arrow category, gives natural comparisons between X and X^{op}. Both

endpoint projections are opfibrations with contractible fibres. Their universal equivalence proves that taking opposites preserves the equivalence class detected by $\mathbb{D}$. This is a comparison of homotopy information, not an equivalence of the original categories.

What the counterexamples distinguish.

From manuscript page 76 onward, the word "fibration" changes meaning as the argument develops. Its first version asks that the actual fibre and its comma-category model have the same constant-coefficient theory and that transport between comma categories preserve that theory. Requiring this after every base change is an additional condition. We exhibit both a two-object example and an example involving only finite posets where the first condition holds but fails after a strict base change.

The more substantial failure concerns the proposed implication that a universal fibration must preserve every $\mathbb{D}$-equivalence of bases. For rational complexes, a base can be cohomologically indistinguishable from a point while still carrying nontrivial local systems. Grothendieck's reflected-circle bundle over the circle makes the point concretely. The degree-one cohomology of its fibres is a sign local system on the base circle; its derived cohomology has differential multiplication by -2, hence is zero over $\mathbb{Q}$. The projection from the Klein bottle is consequently a rational constant-coefficient equivalence although its circle fibres are not rationally aspherical. In a fibre square the tensor square of the orientation local system becomes untwisted, and the extra cohomology reappears. The commentary calculates the full complexes and the comparison maps, including the degree-zero term omitted from the source's abbreviated group-cohomology calculation.

The finite-group version replaces this computation by the averaging idempotent. It also shows why the proposed extension to truncated homotopy theories needs care: already the point mapping into BC_2 is an equivalence for the represented set-valued derivator, whereas locally constant C_2-sets retain the action. The corresponding assertions about complete equivalences cannot be inferred from connectedness.

One further correction is necessary in the proposed finite-poset model. The action on S^2 is antipodal and the action on S^1 is a reflection. Their product acts trivially on the top class of degree three. Hence

$$H^*((S^2 \times S^1)/C_2; \mathbb{Q}) \cong \mathbb{Q} \quad \text{in degrees } 0, 3,$$

and vanishes in other degrees. The base $\mathbb{RP}^2$ is rationally aspherical, but this total space is not. The example still disproves invariance under inclusion of a fibre; it does not supply the additional aspherical-total-space claim. An exact rational chain calculation for the thirteen-object base and fifty-two-object total poset independently confirms these Betti numbers. The orbit-category relations printed before this construction also require correction, which is explained separately in the commentary.

The final distinctions and their present interpretation.

The later definitions separate constant-coefficient equivalence, invariance of cohomology with every locally constant coefficient, and full faithfulness or equivalence on the entire category of locally constant objects. The last two are called 1- and 2-complete equivalence. A closed internal-Hom formalism identifies cohomological equivalence with 1-complete equivalence, under the stated compatibility assumptions. It does not identify ordinary constant-coefficient equivalence with either of them. The missing word "cohomological" in the sentence on manuscript page 122 is therefore a material slip.

The nerve-invariance assertions eventually formulated as the b' variants of **Der 3** have a precise later answer. Cisinski (2009, Theorem 1.3 and Corollary 3.8) proves invariance of locally constant coefficients under weak equivalences of indexing nerves, first in the model-category setting and then for derivators. In the space-valued case these coefficients can be read as coherent functors out of the fundamental infinity-groupoid. A weak equivalence changes

neither that groupoid nor its coefficient categories. This establishes the strong form of the lemma about HOT that Grothendieck had deferred; it does not collapse the distinctions for a fixed linear coefficient theory. The minimal-localizer theorem places the ordinary weak-equivalence class inside the broader framework that motivated these pages (Cisinski, 2004, Theorem 2.2.11).

Two developments in the commentary on these pages address unfinished questions directly. First, for a universally weak cohomological fibration, the formula for $f_* f^* F$ holds for every coefficient F on the base, with no local-constancy assumption. A factorization category for each arrow $z \to y$ reduces the left mate of the desired comparison to constant homology. The dual proof applies to $f_! f^* F$. As a consequence, a universal $\mathbb{D}$-equivalence induces fully faithful restriction on the whole coefficient category. The required closure of the diagram domain is stated explicitly. These arguments use the factorization method underlying homotopy exactness (Groth et al., 2014a, Theorem 3.8), and the full-faithfulness conclusion for universal weak equivalences of nerves also follows from the modern homotopical epimorphism criterion (Groth and Šťovíček, 2016, Theorem 6.10).

Second, the intermediate $\mathbb{D}$-fibrations are not closed under composition. Over $\mathbb{Q}$, take the opfibration over [1] with transition $e \to BC_2$ and then the covering determined by the regular C_2-set. Both factors satisfy the universal constant-coefficient condition. Their composite has transition from two discrete points to a contractible action groupoid, and fails already in degree-zero cohomology. The stronger cohomological and homological fibrations on manuscript page 121 control locally constant coefficients on the total category and are closed under composition. A separate example with transition $BC_2 \to e$ shows that this stronger relative theory has examples beyond HOT-fibrations. The proposed 2-complete variant also fails to compose for the represented set-valued derivator: a C_2-valued cocycle on the tetrahedral boundary face poset gives two functors between finite categories whose composite fails the required test. The obstructed restriction includes sets as trivial C_2-sets; its failure of essential surjectivity distinguishes this counterexample from the 1-complete case, which composes under the internal-Hom hypotheses.

The mathematical status is thus stated at the level of the actual assertions. The main cofinality and base-change theorems hold with their conventions and domain hypotheses made explicit; the false implications have concrete counterexamples; several stated questions admit the supplied proofs or the cited later theorems. A failure to locate a particular construction in the sources checked is recorded as such, never converted into a claim that it is an open problem or a new result. The label [ours] marks proofs and constructions given in the commentary; it carries no priority claim.

Chapter II

Cofinalité (à droite et à gauche) (préliminaire à la cofinalité cohomologique)

II-a · Chapter II, part 1 of 2 · manuscript pages 1–53

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre II
Cofinalité (à droite et à gauche)
(préliminaire à la cofinalité cohomologique)

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.
Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :
[**e-mail address removed**]

G. Maltsiniotis

1 Cofinalité cohomologique

p. 1
[1]

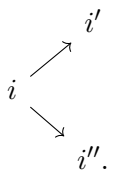
Soit

$$\alpha : C' \longrightarrow C$$

un foncteur de (petites) catégories, $\mathcal{M}$ une catégorie ayant les $\varprojlim$ pertinentes, F un préfaisceau (donc contravariant en i) sur C à valeurs dans $\mathcal{M}$. On a un homomorphisme canonique dans $\mathcal{M}$

$$\varprojlim_{C} F \longrightarrow \varprojlim_{C'} \underbrace{\alpha^*(F)}_{F\circ\alpha} \tag{1}$$

et on veut des conditions sur α pour que ce soit toujours un isomorphisme ([1]). On est ramené aussitôt au cas où $\mathcal{M} = (\mathrm{Ens})$. Disons qu'une catégorie I est *faiblement filtrante à gauche* si $I \neq \emptyset$ et si $\forall\, i', i'' \in \mathrm{Ob}\, I$, $\exists\, i \in \mathrm{Ob}\, I$ et des flèches



Cela signifie aussi que $\mathrm{Ob}\, I$ en tant qu'ensemble ordonné est filtrant à gauche.

Commentary · Exposition [ms. 1]

The presheaf represented by a change of index

Grothendieck begins again with ordinary limits because their elementary descent argument already contains the object which the cohomological theory must derive. We keep his variance throughout: a presheaf on C is a functor $C^{\mathrm{op}} \to \mathrm{Set}$, and c/α, also written $c\backslash C'$, has objects $(a, f : c \to \alpha a)$. The arrow in the first proposition is printed in the opposite direction; its intended direction is fixed both by this notation and by the proof on manuscript page 6. This is an edition slip, not a change of convention.

Put $K_\alpha(c) = \pi_0(c/\alpha)$. Precomposition with $c_0 \to c_1$ defines $K_\alpha(c_1) \to K_\alpha(c_0)$, so K_α is a presheaf. There is a natural bijection

$$\varprojlim_{C'^{\mathrm{op}}} \alpha^* F \;\cong\; \mathrm{Nat}(K_\alpha, F).$$

Indeed a compatible family (s_a) determines the map $[a, f] \mapsto F(f)(s_a)$. A morphism $(a, f) \to (b, g)$ in c/α gives $g = \alpha(h)f$, and compatibility gives $s_a = F(\alpha h)(s_b)$; hence the value is constant along every zigzag. Naturality in c follows by composition. Conversely, from $t : K_\alpha \to F$, put $s_a = t_{\alpha a}[a, 1_{\alpha a}]$. For $h : a \to b$, the arrow $(a, 1) \to (b, \alpha h)$ and naturality of t prove compatibility. These constructions are inverse.

Equivalently, $K_\alpha = \alpha_! 1_{C'}$. Thus the restriction map on limits is induced by the augmentation $K_\alpha \to 1_C$. Yoneda, applied in the presheaf category, proves the exact criterion:

$$\text{restriction preserves every presheaf limit} \quad\Longleftrightarrow\quad \pi_0(c/\alpha) = \{*\} \quad (c \in C).$$

The nonempty, connected comma categories are what matter. A common predecessor for each pair of objects is a stronger sufficient condition; it has no equalization requirement. For coefficients in a locally small category $\mathcal{M}$, apply $\mathrm{Hom}_{\mathcal{M}}(M, -)$ for every M. These functors preserve existing limits and jointly reflect isomorphisms, which justifies the reduction to sets without imposing extra completeness on $\mathcal{M}$.

In current covariant terminology, α is a final functor, while α^{op} is initial; the latter is the functor acting on the diagrams whose limits are written here. SGA 4 uses the cofinal terminology followed in the manuscript (Artin et al., 1972, Exposé I, §8.1). Proposition 3.2 below returns to precisely this criterion. We shall use the representing presheaf again there, rather than repeat the elementwise argument.

1. On dit alors que α est *cofinal,* cf. SGA 4, I 8.1, ou 3.3, p. 45.

Proposition. *Pour que* (1) *soit toujours un isomorphisme* (α *'anodin pour* $\varprojlim$ *'), il suffit que pour tout* $x_0 \in \mathrm{Ob}\, C$, *la catégorie* $x_0\backslash C'$ *des couples* (x', u), *avec* $x' \in \mathrm{Ob}\, C'$, $u : \alpha(x') \longrightarrow x$, *soit faiblement filtrante à gauche* ([2], [3]).

Ceci n'implique pas la conclusion plus forte

$$F \underbrace{\xrightarrow{\sim}}_{\text{adj.}} \alpha_* \alpha^* F$$

que le morphisme d'adjonction canonique soit un isomorphisme. Pour qu'il soit ainsi, il suffit que $\forall\, x_0 \in C$, le foncteur $\alpha/x_0 : C'/x_0 \longrightarrow C/x_0$ induit par α soit 'anodin pour les $\varprojlim$ ', ce qui signifie aussi que pour toute flèche $u : x_1 \longrightarrow x_0$ dans C, la catégorie $C'_{\vee u}$ formée des triples (x', ξ, η) avec $x' \in \mathrm{Ob}\, C'$ et $\xi : x_1 \longrightarrow \alpha(x')$, $\eta : \alpha(x') \longrightarrow x_0$ rendant commutatif le diagramme [2] p. 2

$$\begin{array}{ccc} x_1 & \xrightarrow{\quad u \quad} & x_0 \\ & {\scriptstyle\xi}\searrow \quad \nearrow{\scriptstyle\eta} & \\ & \alpha(x'), & \end{array}$$

est faiblement filtrante à gauche.

Plus généralement, considérons

$$\begin{array}{ccc} X' = C' & \xrightarrow{\quad \alpha \quad} & C = X \\ & {\scriptstyle f'}\searrow \quad \swarrow{\scriptstyle f} & \\ & S & \end{array}$$

et posons nous la question quand l'homomorphisme fonctoriel, par rapport à un préfaisceau F variable sur X,

$$f_*(F) \longrightarrow f'_*(\alpha^*(F))$$

est toujours un isomorphisme. (Le cas précédent est celui où $S = X$, $f = \mathrm{id}_X$.) On trouve qu'il suffit que $\forall\, s_0 \in \mathrm{Ob}\, S$, le foncteur

$$\alpha/s_0 : X'/s_0 \longrightarrow X/s_0$$

soit 'anodin pour les $\varprojlim$ ', et pour ceci, il suffit que pour `[tout]` objet (x_0, u_0) de X/s_0 (où

$$f(x_0) \xrightarrow{u_0} s_0)$$

la catégorie $(x_0, u_0)\backslash(X'/s_0)$ soit faiblement filtrante à gauche.

Cette catégorie s'interprète comme celle des triples (x', ξ, u) avec

$$\begin{cases} x' \in \mathrm{Ob}\, C' \\ u : g(x') = f(\alpha(x')) \longrightarrow s_0 \\ \xi : x_0 \longrightarrow \alpha(x') \end{cases}$$ [3]

`[plutôt` $f'(x')$ `au lieu de` $g(x')$`]` de façon à rendre commutatif le diagramme

$$\begin{array}{ccc} f(x_0) & \xrightarrow{\quad u_0 \quad} & s_0 \\ & {\scriptstyle f(\xi)}\searrow \quad \nearrow{\scriptstyle u} & \\ & f(\alpha(x')). & \end{array}$$

Je voudrais dégager des variantes de ces énoncés dans le cadre cohomologique, les foncteurs $\varprojlim$ étant remplacés par des $R\Gamma$ (de préfaisceaux abéliens p.ex. - mais on pense aussi aux 'cas non commutatifs', définis par des dérivateurs notamment.)

2. Cf. amélioration des remarques p. 5.
3. Cf. 3.2, p. 43-44, pour conditions nécessaires et suffisantes de cofinalité.

Si I est une catégorie *ordonnée finie*, la condition `[faiblement]` filtrante à gauche équivaut à l'existence d'un objet initial, ce qui implique que I est contractile. Dans les énoncés précedents, il se pourrait que la bonne philosophie soit de remplacer partout 'faiblement filtrante à gauche' par 'contractile'. Si p.ex. C est la catégorie finale, alors remplacer la condition que C' soit filtrante à gauche, par l'acyclicité. C'est en tous cas en accord avec l'esprit de l'axiome Der_5 (cas où J est la catégorie finale). En effet, dans le cas où $C = e$, la condition que α soit anodin pour les $\varprojlim$ équivaut visiblement à dire que le foncteur $\alpha^* : \mathcal{M} \longrightarrow \underline{\mathrm{Hom}}(C'^o, \mathcal{M})$, associant à tout $F \in \mathcal{M}$ le préfaisceau constant de valeur F sur C', donne un morphisme d'adjonction [4]

$$\varprojlim_e F = F \longrightarrow \alpha_*(\alpha^*(F)) = \varprojlim_{C'} \alpha^* F$$

qui est un isomorphisme. Et l'axiome Der_5 assure qu'il doit en être ainsi si C' est contractile.

Commentary · Exposition [ms. 3]

Why the relative problem has two endpoints

The passage from global sections to a direct image introduces a second endpoint. For $f : X \to S$ and a presheaf F, the pointwise right Kan extension formula is

$$(f_*F)(s) = \lim_{(f/s)^{\mathrm{op}}} F(x), \qquad (f/s) = \{(x, f(x) \to s)\}.$$

This is the ordinary comma-category formula for a Kan extension (Riehl, n.d., Theorem 6.2.1 and Corollary 6.2.9). If $f' = f\alpha$, the comparison at s is restriction along α/s. Its comma category at (x_0, u_0) consists exactly of factorizations

$$f(x_0) \xrightarrow{f(\xi)} f(\alpha a) \xrightarrow{u} s, \qquad uf(\xi) = u_0.$$

Connectedness and nonemptiness of these categories therefore suffice. The weak cofiltering condition written here also suffices, but the phrase "ce qui signifie aussi" on manuscript page 1 should not turn that sufficient condition into a necessary one.

When $S = X$ and $f = 1_X$, the test concerns factorizations of each arrow $x_0 \to x_1$ through αa. It is stronger than the test involving only x_0/α. This is the same distinction already isolated in the commentary on Chapter I, manuscript p. 49 ("The factorization category controls base change"); Proposition 4.1 will identify its ordinary categorical form. Keeping the two endpoints visible prevents an erroneous inference from invariance of global sections to full faithfulness of restriction.

La condition filtrante dans la proposition est d'ailleurs une condition *suffisante* d'anodinité, mais nullement nécessaire. Si p.ex. C est la catégorie finale, alors la condition d'anodinité pour $C' \longrightarrow C$ est simplement que C' soit connexe non vide, i.e. $\pi_0(C') = \{0\}$. (Condition qui visiblement néglige les π_i supérieurs.)

Soit $\boxed{\alpha : X' \longrightarrow X}$ un foncteur de petites catégories. La condition suggérée par la proposition, que les $x_0\backslash X'$ soient contractiles, signifie que α est une hot-équivalence co-stricte, i.e. que $\alpha^o : X'^o \longrightarrow X^o$ est une hot-équivalence 'localement sur X^o', donc aussi globalement. Si α lui-même était une hot-équivalence stricte, au lieu de α^o, alors on aurait par Der_5, [5] que pour $F \in \mathbf{D}(X)$, $F \xrightarrow{\sim} \alpha_*\alpha^*F$ ($\mathbf{D}$ dérivateur à gauche), ce qui est bien plus fort que $\Gamma_{\mathbf{D}}(X, F) \xrightarrow[\sim]{\alpha^*} \Gamma_{\mathbf{D}}(X', \alpha^*F)$. La condition duale implique (si $\mathbf{D}$ `[est un]` dérivateur à droite) $\alpha_!\alpha^*F \xrightarrow{\sim} F$. Mais est-il raisonnable de s'attendre qu'elle implique aussi la condition 'du côté cohomologie, au lieu de l'homologie',

$$\Gamma_{\mathbf{D}}(X, F) \xrightarrow{\sim} \Gamma_{\mathbf{D}}(X', \alpha^*(F)) \quad ?$$

Le premier cas, qui devrait emporter conviction, est celui du dérivateur DerAb, i.e. des catégories dérivées des faisceaux abéliens :

Question. *Soit $\alpha : X' \longrightarrow X$ une* hot-*équivalence co-stricte, F un préfaisceau abélien sur X. A-t-on alors $H^i(X, F) \xrightarrow[\sim]{\alpha^*} H^i(X', \alpha^*(F))$ pour tout $i \in \mathbf{Z}$?*

Si X admet un objet initial $\emptyset$, alors $\emptyset\backslash X' \simeq X'$, donc X' contractile, donc c'est OK pour F constant.

Commentary · Later developments [ms. 5]

Cohomological cofinality and the derived weight

Grothendieck's question about abelian coefficients has an affirmative answer. Its proof can follow the argument he has just discovered, provided that the presheaf K_α is replaced by its chain-level predecessor. We give the argument for a commutative ring k; $k = \mathbb{Z}$ answers the stated question. Write $D(C; k)$ for the derived category of k-module presheaves and use $\mathbf{R}\operatorname{Hom}_{C;k}$ for the *global* derived Hom, valued in $D(k)$.

Derived adjunction gives, naturally in F,

$$\mathbf{R}\Gamma(C', \alpha^* F) \simeq \mathbf{R}\operatorname{Hom}_{C;k}(\mathbf{L}\alpha_! k_{C'}, F).$$

For clarity, this can be constructed without assuming the disputed internal formula on manuscript page 10. Resolve $k_{C'}$ by the augmented simplicial presheaf of free modules whose degree n is

$$P_n = \bigoplus_{a_0 \to \cdots \to a_n} k \operatorname{Hom}_{C'}(-, a_0).$$

At a, its augmented chain complex is the nerve complex of a/C', contracted by its initial object 1_a. Each P_n is projective. Its normalized complex, in nonpositive cohomological degrees, is therefore a projective resolution. Since restriction is exact, $\alpha_!$ takes these projectives to projectives. Evaluating $\alpha_! P_\bullet$ at c gives the free chain complex of c/α. In particular

$$(\mathbf{L}\alpha_! k_{C'})(c) \simeq C_*(N(c/\alpha); k),$$

where a chain degree n is placed in cohomological degree $-n$. The usual formulation using the opposite comma category gives the same complex up to signed reversal.

If every $N(c/\alpha)$ is weakly contractible, its augmented complex is quasi-isomorphic to k. Consequently $\mathbf{L}\alpha_! k_{C'} \to k_C$ is an isomorphism. Applying derived Hom yields the desired comparison in every degree, and indeed for every complex F. This proof needs only k-acyclicity of these nerves; weak contractibility is the uniform condition which works for homotopy types as well. The two conditions must not be confused with mere connectedness. For example, $BG \to e$ is cofinal for ordinary presheaf limits whenever G is a group, but for $G = \mathbb{Z}$ and the constant coefficient $\mathbb{Z}$ one has $H^1(BG; \mathbb{Z}) = \mathbb{Z}$.

For spaces, the corresponding weight is $c \mapsto B(c/\alpha)$ in $\operatorname{Fun}(C^{\mathrm{op}}, \mathcal{S})$. The same mapping-object adjunction proves cofinality; the general higher categorical criterion is Lurie (2009, Theorem 4.1.3.1, together with Proposition 4.1.1.8), applied with the appropriate opposites. Its corollary 4.1.3.3 is Quillen's Theorem A. These results settle the question on page 5, without requiring the stronger unit or counit assertions in its preceding paragraph. Those assertions are conditional there on the manuscript's invocation of Der_5; interpreted as consequences of comma-category asphericity under present derivator axioms, they are false. The endpoint calculation in the next block makes the distinction explicit, continuing the caution about the changing axioms in the commentary on Chapter I, manuscript p. 43 ("What the localization axioms do and do not say").

Remarque. Je m'aperçois que dans la proposition 1, au lieu de la condition que les $x\backslash C'$ soient faiblement filtrantes à gauche, il suffirait qu'elles soient non vides et connexes, i.e.

0-connexes ([4]). Le fait que les $x\backslash C'$ non vide implique que $\varprojlim_C \longrightarrow \varprojlim_{C'}$ soit *injectif*. Pour p. 4
la sujectivité, si $\xi' \in \varprojlim_{C'}$ (cas ensembliste), on définit $\xi \in \varprojlim_C$ par les $\xi(x) \in F(x)$ à l'aide de [6]

$$x \xrightarrow{u} \alpha(x')$$

4. Cela signifie aussi que α est W_0-coasphérique. Pour un traitement ab ovo, cf. prop. 3.2, page 43.

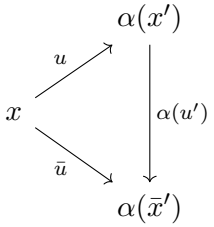
en prenant $u^*(\xi(x'))$ (où $\xi(x') \in (\alpha^*(F))(x') = F(\alpha(x'))$), le point essentiel est de montrer que les $u^*(\xi(x'))$ ne dépendent que de x, pas du choix de $(x', u) \in x\backslash C'$. Or pour (x', u) variable dans $x\backslash C'$, l'application

$$\begin{array}{rcl} \varphi : \mathrm{Ob}(x\backslash C') & \longrightarrow & F(x) \\ (x', u) & \longmapsto & u^*(\xi(x')) \end{array}$$

est telle que s'il existe une flèche

$$(x', u) \xrightarrow{u'} (\bar{x}', \bar{u}) \qquad \text{où} \qquad u' : x' \longrightarrow \bar{x}'$$

(donc le diagramme



est commutatif), alors $\varphi(x', u) = \varphi(\bar{x}', \bar{u})$. Donc φ est constant sur les composantes connexes de $x\backslash C'$, donc constant si $x\backslash C'$ est 0-connexe, d'où $\xi(x)$. On prouve alors aussitôt que pour x variable,

$$\xi \overset{\text{déf}}{=} (\xi(x))_{x \in \mathrm{Ob}\, C} \simeq \varprojlim F, \qquad \texttt{[plutôt} \in \varprojlim F, \texttt{]}$$

et que $\alpha^*(\xi) = \xi'$, cqfd.

On peut dire que pour $x_0 \in \mathrm{Ob}\, C_0$ fixé, si on regarde le préfaisceau *constant* sur $x_0\backslash C'$ de valeur $F(x_0)$, soit Φ_{x_0}, l'application

$$(x', u) \longmapsto \xi^*(u') : x_0\backslash C' \longrightarrow F(x_0)$$

peut être regardée comme une section de ce préfaisceau sur $x\backslash C'$, défini par $\xi' \in \varprojlim_{C'} \alpha^* F$, donc on a

$$\varprojlim_{C'} \alpha^* F \xrightarrow{c_{x_0}} \Gamma(x_0\backslash C', \Phi_{x_0}) \qquad (\xleftarrow{\sim} F(x_0) \text{ si } x_0\backslash C \text{ 0-connexe}).$$

Si on a $\boxed{x_0 \xrightarrow{v} x_1}$ dans C', on en déduit

$$x_1\backslash C' \xrightarrow{V} x_0\backslash C'$$

(les $x\backslash C'$ sont les fibres d'une *catégorie fibrée* $C\backslash C'$ sur C …) et on a $F(x_1) \longrightarrow F(x_0)$, d'où $\boxed{\Phi_{x_1} \longrightarrow V^*\Phi_{x_0}}$ pour les faisceaux constants correspondants sur $x_0\backslash C'$ resp. $x_1\backslash C'$ et on a commutativité dans

p. 5
[7]

$$\begin{array}{ccc} \varprojlim_{C'} \alpha^* F & \xrightarrow{c_{x_1}} & \Gamma(x_0\backslash C', F(x_0)) \simeq \mathrm{Hom}(\pi_0(x_0\backslash C'), F(x_0)) \\ {\scriptstyle c_{x_0}}\downarrow & & \downarrow{\scriptstyle V^*} \\ \mathrm{Hom}(\pi_0(x_1\backslash C'), F(x_1)) \simeq \Gamma(x_1\backslash C', F(x_1)) & \xrightarrow[(F(x_1)\to F(x_0))]{v^*} & \Gamma(x_1\backslash C', F(x_0)) \simeq \mathrm{Hom}(\pi_0(x_1\backslash C'), F(x_0)) \end{array}$$

$$\begin{array}{ccc} \pi_0(x_1\backslash C') & \xrightarrow{\pi_0(V)} & \pi_0(x_0\backslash C') \\ \vdots & & \vdots \\ F(x_1) & \xrightarrow{F(v)} & F(x_0) \end{array} \qquad \text{commutatif.}$$

On considère le préfaisceau d'ensembles

$$x \longmapsto \pi_0(x\backslash C') \qquad \text{sur } C,$$

appelons-le $\pi_0(C'/C)$, on aura un morphisme canonique

$$\varinjlim_{C'} \alpha^*(F) \xrightarrow{\sim} \mathrm{Hom}_{\substack{\text{préfaisceaux} \\ \text{d'ensembles sur } C}}(\pi_0(C'/C), F),$$

sans doute toujours [un] isomorphisme.

Si on travaille dans la catégorie dérivée des préfaisceaux des k-modules (k un anneau), la formule précédente suggère conjecturalement

$$\mathbf{R}\Gamma(C', \alpha^*F) \simeq \mathbf{R}\underbrace{\mathrm{Hom}}_{\text{Hom internes}}(\underbrace{\mathbf{L}\alpha_!(k_{C'})}_{\substack{\text{homologie} \\ \text{relative de} \\ C' \text{ sur } C}}, F).$$

L'augmentation canonique [8]

$$\mathbf{L}\alpha_!(k_{C'}) \longrightarrow k_C$$

redonne, en prenant les $\mathbf{R}$Hom dans F,

$$\mathbf{R}\Gamma(C, F) \longrightarrow \mathbf{R}\,\mathrm{Hom}(\mathbf{L}\alpha_!(k_{C'}), F) \xleftarrow{\sim} \mathbf{R}\Gamma(C', \alpha^*F)$$

l'homomorphisme canonique de 'contravariance' de la cohomologie.

Si par contre on travaille avec le dérivateur non additif HOT, donc F un 'type d'homotopie relatif' sur C, on doit avoir

$$\mathbf{R}\Gamma_{\mathrm{HOT}}(C', \alpha^*F) \xrightarrow{\sim} \mathbf{R}\Gamma(C\,,\ \underbrace{\mathrm{Hom}}_{\substack{\text{Hom interne} \\ \text{de la catégorie des} \\ \text{types d'homotopie} \\ \text{relatifs sur } C}}(\ \alpha_!\ (\ \underbrace{e_{C'}}_{\substack{\text{objet final} \\ \text{de HOT}(C'), \\ \text{type d'homotopie} \\ \text{relatif final} \\ \text{sur } C'}}\ ),\ F)).$$

Commentary · Examples [ms. 8]

Global adjunction does not identify an internal Hom

The proposed formula is useful because its global version is right. The difficulty begins only when its two sides are compared as objects over C. For presheaves of spaces, the correct equivalence is

$$\mathrm{Map}_{\mathcal{P}(C')}(1, \alpha^*F) \simeq \mathrm{Map}_{\mathcal{P}(C)}(\alpha_!1, F), \qquad \mathcal{P}(C) = \mathrm{Fun}(C^{\mathrm{op}}, \mathcal{S}).$$

Since $\mathcal{P}(C)$ is cartesian closed, the right side is also $\mathbf{R}\Gamma(C, \underline{\mathrm{Hom}}(\alpha_!1, F))$. This proves exactly the globally sectioned formula on page 8. The analogous abelian formula on page 7 takes values in $D(k)$ when "Hom internes" means the k-enriched global Hom. A sheaf-valued internal Hom would require an additional $\mathbf{R}\Gamma(C, -)$.

To see what global sections forget, take $C = [1] = (0 \to 1)$ and $i : e \to C$ with $i(*) = 1$. A presheaf is an arrow $F_1 \to F_0$. Direct calculation of all three adjoints gives

$$i^*F = F_1, \qquad i_!A = (A \xrightarrow{1} A), \qquad i_*A = (A \to *).$$

Here the displayed order is always the restriction map from value at 1 to value at 0. Every c/i is terminal, so i passes even the contractible-comma criterion; $\Gamma(C, F) = F_1$ and the comparison with $\Gamma(e, i^*F)$ is an identity. Nevertheless the unit has component $F_0 \to *$ at 0, and the counit $i_!i^*F \to F$ has component $F_1 \to F_0$ there. Neither is generally invertible.

Moreover, $i_!1 = 1_C$, hence

$$\underline{\mathrm{Hom}}_C(i_!1, F) = F, \qquad i_*i^*F = (F_1 \to *).$$

Taking $F = (\{*\} \to \{0, 1\})$ disproves the internal identification. Applying global sections to both sides gives $\{*\}$, as it should. With $j(*) = 0$ instead, every j/c is terminal, whereas $j_*j^*F = (F_0 \xrightarrow{1} F_0)$;

thus the opposite comma orientation also fails to imply the universal unit assertion. The same examples work in $D(k)$, replacing $*$ by the zero module and choosing an arrow which is not an isomorphism.

The shift argument on page 9 establishes the expected Hom groups once the derived adjunction is available. Recovering an equivalence of enriched Hom objects requires the enriched adjunction itself, or a constructed comparison and a conservativity argument. It is not supplied merely by a bijection on π_0 in the space-valued case. No such extra step is needed for the explicit bar-resolution proof above.

En fait, ces formules sont quasi-tautologiques, si on admet l'existence des foncteurs $\alpha_!$ p. 6
comme adjoints *à gauche* des α^*. Dans le cas abélien, disons, l'isomorphie des H^0 des deux [9]
membres est la formule d'adjonction

$$\mathrm{Hom}_{C'}(k_{C'}, \alpha^* F) \simeq \mathrm{Hom}_C(\alpha_!(k_{C'}), F),$$

d'où l'isomorphie pour les H^i des deux membres, par décalage. De même dans le cas de HOT, l'isomorphie canonique des π_0 des deux membres est la formule d'adjonction. Pour avoir l'isomorphisme plus fin dans $\mathbf{D}(e)$, la catégorie fondamentale du dérivateur envisagé, en termes de Hom internes, on doit admettre l'existence d'objets unité (les k_X dans un cas, les e_X `[plutôt` ε_X`]` dans l'autre), compatibles avec images inverses, de telle façon `[que]` $\mathrm{Hom\,int}(\varepsilon_X, \underbrace{F}_{\in \mathbf{D}_X}) \simeq F$. On aura

$$\Gamma(C', \alpha^* F) \overset{\text{déf}}{=} f'_*(\alpha^*(F)) \simeq f_*(\alpha_* \alpha^*(F)),$$

et la question est donc de donner une interprétation de $\alpha_* \alpha^* F$ comme

$$\alpha_* \alpha^* F \simeq \mathrm{Hom\,int}(\alpha_!(\varepsilon_{C'}), F) \qquad \text{dans } \mathbf{D}(C).$$

Pour ceci, on va écrire

$$\alpha^*(F) \simeq \mathrm{Hom\,int}(\varepsilon_{C'}, F),$$

et la formule doit résulter d'une 'formule d'adjonction interne'

$$? \qquad \boxed{\alpha_*(\mathrm{Hom\,int}_{C'}(G', \alpha^* F)) \simeq \mathrm{Hom\,int}_C(\alpha_! G', F).} \qquad [10]$$

(*Formule d'adjonction 'interne',* pour $\alpha_!$ et α^*, valable en principe pour *tout* $\alpha : C' \longrightarrow C$ dans Diag, tout G' dans $\mathbf{D}(C')$ et tout F dans $\mathbf{D}(C)$.)

Commentary · Exposition [ms. 9]

The projection formula is an extra property

Grothendieck correctly reduces the proposed internal adjunction to a projection formula. What fails is the expectation that every change of indexing category has this property. In a closed symmetric monoidal setting, write $L = \alpha_!$, $R = \alpha^*$ and $U = \alpha_*$. The canonical map supplied by the adjunction $L \dashv R$ has direction

$$L(RG \otimes G') \longrightarrow G \otimes LG'.$$

It is adjoint to $RG \otimes G' \to RG \otimes RLG'$, using the unit and the monoidal structure of R. The arrow printed on page 10 points in the inverse direction; such an inverse is available only after invertibility has been established.

For every H, the adjunctions give

$$\mathrm{Hom}(H, U\underline{\mathrm{Hom}}(G', RF)) \cong \mathrm{Hom}(L(RH \otimes G'), F),$$
$$\mathrm{Hom}(H, \underline{\mathrm{Hom}}(LG', F)) \cong \mathrm{Hom}(H \otimes LG', F).$$

Thus the canonical internal comparison $\underline{\mathrm{Hom}}(LG', F) \to U\underline{\mathrm{Hom}}(G', RF)$ is invertible for all F, G' exactly when the projection maps are invertible for all H, G'. This proves the reduction, and also locates the additional hypothesis precisely.

For cartesian products this property is called Frobenius reciprocity. An essential geometric morphism need not satisfy it. Moreover, Frobenius reciprocity itself is weaker than its validity in every slice, the latter being the condition appropriate to a locally connected geometric morphism. Garner and Streicher (2021, §1) explicitly distinguish these conditions; they also recall the equivalence between Frobenius reciprocity and preservation of exponentials by the right adjoint of an adjunction between cartesian closed categories. These later distinctions explain why existence of both Kan extensions does not settle the question. They should not be replaced by an unrestricted "projection formula for derivators".

Comment prouver cette formule ? En prenant l'opération produit $\otimes$ associé à l'opération Hom int, on est ramené (en prenant les foncteurs contravariants associés) à une formule

$$G \otimes \alpha_! G' \xrightarrow{\sim} \alpha_!(\alpha^* G \otimes G').$$

On trouve par adjonction un homomorphisme canonique, il faut voir s'il est toujours [un] isomorphisme.

Commentary · Development (commentators') [ms. 10]

The exact obstruction for a point inclusion

The walking-arrow example can be sharpened to measure exactly which restriction maps obstruct the projection formula. This makes the question on page 10 concrete even before a general theory of Frobenius adjunctions is introduced.

Point-inclusion criterion. [ours] Let d be an object of a small category C, and let $i : e \to C$ select d. For a set-valued presheaf G, the projection map

$$i_!(i^* G \times A) \longrightarrow G \times i_! A$$

is invertible for every set A if and only if $G(f) : G(d) \to G(c)$ is bijective for every arrow $f : c \to d$. Consequently the formula holds for all G, A if and only if every arrow with codomain d is an isomorphism.

Proof. The left adjoint to evaluation at d is $i_! A = A \times h_d$, where $h_d(c) = C(c, d)$. At c the projection map is

$$A \times G(d) \times C(c,d) \longrightarrow A \times G(c) \times C(c,d), \qquad (a, s, f) \longmapsto (a, G(f)s, f).$$

For $A = 1$ it is a disjoint union, indexed by f, of the maps $G(f)$; it is bijective precisely when each of them is. This also suffices for every A. If every f is invertible, every $G(f)$ is bijective. Conversely, suppose the formula holds for every presheaf. Taking $G = h_z$ gives a bijection $C(d, z) \to C(c, z)$ by precomposition with f, for every z. With $z = c$, obtain $g : d \to c$ such that $gf = 1_c$. With $z = d$, injectivity and $(fg)f = f = 1_d f$ imply $fg = 1_d$. Thus f is invertible. □

This includes the object 1 of [1], where the noninvertible incoming arrow is exactly the obstruction. It also explains why the analogous formula for a group action behaves differently: all indexing arrows are invertible. For a fixed G, local constancy along the arrows entering d is the precise repair; no conclusion about all coefficients follows from connectedness of a comma category.

We checked the pointwise Kan-extension treatment of Riehl (n.d., §6.2) and the Frobenius discussion of Garner and Streicher (2021, §1), and searched for the point-evaluation formulation. We could not find this elementary criterion stated there in this form. It is recorded as our worked continuation of the calculation, with no claim of priority.

p. 7
[11]

Commentary · Context [ms. 10]

Why the argument passes through toposes

The change of subject at the beginning of §2 is part of the cofinality problem. Grothendieck wants conditions on indexing categories which survive passage to categories of coefficients and which force the relevant homotopy types to disappear. A formulation in terms of the presheaf topos can detect all objects over the category, whereas the underlying preorder remembers only whether an arrow exists.

This is a small-scale instance of the program described in Grothendieck (1983, §69, especially the discussion of diagram categories on pp. 1–3 of the excerpt): the derived category of coefficients is accompanied by derived categories of diagrams of every shape. The concern there comes from operations on coherent diagrams which a single triangulated category does not retain. The present calculation asks what changing the shape does to sections and their derived functors. In parallel, Heller (1988, Chapter II, §1; Chapter III, §2) axiomatizes homotopy theories and develops the computation of homotopy Kan extensions. These references explain the setting, not a historical claim that one author's argument was taken from the other.

The additional tensor language on pages 7–10 needs separate structure. Heller's Chapter IV treats tensor and cotensor operations; the preceding projection counterexample shows why that structure cannot be inferred from the mere existence of left and right Kan extensions. We now return to a question for which the ordinary presheaf topos is already sufficient: which part of the filtering conditions is visible in the behavior of its connected objects?

2 Les catégories localement filtrantes sont les catégories localement totalement 0-connexes

Soit $\mathcal{T}$ un topos. On dit que $\mathcal{T}$ est *irréductible* si l'ensemble ordonné $\mathrm{Ouv}(\mathcal{T})$ des sous-objets de $e_{\mathcal{T}}$ a la propriété suivante : a) $\mathrm{Ouv}(\mathcal{T})$ n'est pas réduit à un élément, i.e. $\emptyset_{\mathcal{T}} \longrightarrow e_{\mathcal{T}}$ n'est pas un isomorphisme ($\mathcal{T}$ n'est pas identique à son sous-topos vide). b) Si U', U'' sont deux objets 'non vides', i.e. non initiaux, de $\mathrm{Ouv}(\mathcal{T})$, leur intersection est 'non vide'. Il revient au même pour b) de dire que si $\mathcal{T}'$, $\mathcal{T}''$ sont deux sous-topos fermés distinct de $\mathcal{T}$, alors leur supremum $\mathcal{T}' \cup \mathcal{T}''$ est distinct de $\mathcal{T}$, ou encore que tout ouvert de $\mathcal{T}$ est connexe. On dit que $\mathcal{T}$ est *localement irréductible* si a) $\mathcal{T}$ est localement connexe (i.e. chaque faisceau sur $\mathcal{T}$ est somme de faisceaux 0-connexes), et si b) pour tout faisceau 0-connexe U sur $\mathcal{T}$, le topos induit $\mathcal{T}/U$ est irréductible (i.e. l'ensemble ordonné $\mathrm{Ouv}(U)$ des sous-objets de U a la propriété b) ci-dessus (a) signifie simplement que $U \not\simeq \emptyset_{\mathcal{T}}$, i.e. que U n'est pas objet initial dans $\mathcal{T}^{\wedge}$; ou plus simplement) : tout sous-faisceau d'un faisceau connexe est connexe ([5]) ([6]).

Commentary · Exposition [ms. 11]

Irreducibility of an object and of its support

Grothendieck is distinguishing two ways to forget morphisms. The space $\mathrm{Esp}(X)$ remembers downward-closed sets of objects; the topos $\widehat{X} = \mathrm{Fun}(X^{\mathrm{op}}, \mathrm{Set})$ also remembers every category of elements. The first sees a common predecessor of two objects. The second must see common predecessors compatible with prescribed arrows.

Here an object is *connected* if it has no decomposition into two noninitial disjoint summands; "0-connected" adds noninitiality. Thus the assertion that every subobject of a connected object is connected allows the initial subobject. An object A is irreducible when it is noninitial and any two noninitial subobjects of A meet. Equivalently, every subobject is connected: a disconnected subobject gives two disjoint nonempty subobjects, while disjoint subobjects have a disconnected union. Applied to $A = 1$, this gives the definition of an irreducible topos. Applied to every nonempty connected A, after assuming local connectedness, it gives the local definition.

For $\widehat{X}$, subterminal objects are precisely sieves of objects: if x belongs and $y \to x$ exists, then y belongs. Any two nonempty such sieves meet exactly when each pair of objects has a common predecessor. This proves

$$\widehat{X} \text{ irreducible} \quad \Longleftrightarrow \quad X \neq \varnothing \text{ and } (\forall x, y)(\exists z)(z \to x, z \to y).$$

It is a condition on the associated preorder, not on equalities of arrows. A least class in that preorder need not give an initial object: uniqueness of arrows has been discarded. If the preorder quotient is

5. **NB** Irréductible n'implique pas localement irréductible, pas plus que connexe n'implique pas localement connexe etc.

6. $\mathcal{T}$ est localement irréductible si et seulement s'il a un ensemble des générateurs (U_α), avec les U_α irréductibles (donc 0-connexes, donc $\mathcal{T}$ est localement connexe).

artinian, a minimal class exists, and common lower bounds force it below every other class; this proves the assertion on page 13. Finite principal lower sets suffice as well. "Locally finite" here must have that meaning, as indicated by the finite slices X/x; finiteness of intervals alone would fail, for example for the ordered set $\mathbb{Z}$.

The list of equivalences on pages 12–13 is provisional. Its claim about components being weakly cofiltered is too weak, and its subsequent inclusion of PS 2 is too strong. The annotations and Proposition 2.1 correct these in opposite directions. The surviving characterization is PS 1 alone. We prove it below, retaining the successive formulations as part of the text's development.

Quand $\mathcal{T}$ est de la forme $\mathrm{Top}(T)$, où T est un espace topologique, alors $\mathcal{T}$ irréductible signifie que T est irréductible (car alors $\mathrm{Ouv}(\mathcal{T}) \simeq \mathrm{Ouv}(T)$). Cela implique déjà trivialement que T est localement connexe (tout ouvert est somme disjoint d'ouverts connexes), donc $\mathrm{Top}(\mathcal{T})$ est localement connexe. De plus, cela implique aussi que $\mathrm{Top}(T)$ est localement irréductible ([7]). (**NB** Les objets 0-connexes de $\mathrm{Top}(T)^\wedge$, i.e. les faisceaux d'ensembles 0-connexes sur T, [12] sont ceux dont la 'fibre générique' a exactement un point …) Aussi $\mathcal{T} = \mathrm{Top}(T)$ est localement irréductible si et seulement si T `[est]` localement connexe, et les composantes connexes de T sont irréductibles, i.e. si et seulement si $T = \coprod_i T_i$, les T_i irréductibles.

Commentary · Examples [ms. 12]

The generic fibre without a generic point

The topological case supplies the intuition for the definition, but it also calls for a precise interpretation of "generic fibre". An irreducible space need not possess a generic point. For a sheaf F on a nonempty irreducible space T, define instead

$$\gamma(F) = \operatorname*{colim}_{\varnothing \neq U \subseteq T} F(U),$$

where transition maps are restrictions as U shrinks. Nonempty opens form a filtering system under intersection. If a generic point exists, this is its stalk; otherwise it is the stalk at the corresponding point of the soberification. The following direct argument avoids requiring that point.

For every nonempty open U, a section of $F(U)$ has a well-defined generic germ. Because U is connected, the constant sheaf with value $\gamma(F)$ has sections $\gamma(F)$ on U. We obtain a map $F \to \underline{\gamma(F)}$ and a decomposition

$$F \cong \coprod_{g \in \gamma(F)} F_g$$

into its fibres. Every F_g is noninitial because g has a representative section. If A, B are two noninitial subobjects of F_g, choose sections $a \in A(U)$ and $b \in B(V)$ on nonempty opens. They have the same generic germ, so on some nonempty $W \subseteq U \cap V$ their restrictions agree. This supplies a section of $A \cap B$, proving that F_g is irreducible.

Consequently these fibres are exactly the connected components of F, and F is 0-connected if and only if $\gamma(F)$ is a singleton. This proves both the assertion about the generic fibre and the missing implication in footnote 7: the sheaf topos of an irreducible space is locally irreducible. More generally, if T is a disjoint union of open irreducible spaces, the conclusion follows component by component. Conversely, a locally irreducible sheaf topos is locally connected; its terminal object decomposes into connected subterminal objects, which are disjoint open subsets of T, and each is irreducible. This proves the topological characterization stated here.

For a general presheaf topos, no generic point of the underlying preorder can replace this argument. Parallel arrows produce different objects in slices, even when they have the same source and target. The next example shows exactly where the analogy with topological spaces ceases to suffice.

Si $\mathcal{T}$ est de la forme $\mathrm{Top}(X)$, X une catégorie, on sait que $\mathcal{T}$ est localement connexe (et même localement ∞-connexe). On voit d'autre part que $\mathcal{T}$ est irréductible si et seulement si $\mathrm{Esp}(X)$ (l'espace topologique associé, dont l'ensemble sous-jacent est $\mathrm{Ob}\, X$) est irréductible

7. expliciter la démonstration.

([8]); et on dit alors que X est irréductible ([9]). Comme les composantes connexes de X et celles de $\mathrm{Esp}(X)$ ou $\mathrm{Top}(X) = \mathcal{T}$ se correspondent, on voit donc que les conditions suivantes sont équivalentes : a) le topos $\mathrm{Top}(X)$ est localement irréductible, b) $\mathrm{Esp}(X)$ est localement irréductible, c) les composantes connexes de X sont irréductibles, d) les X/x [sont] irréductibles. On dit alors que X est localement irréductible ([10]). p. 8

Cela implique donc que pour tout X' étale sur X, avec X' 0-connexe, X' est irréductible.

Dire conditions équivalentes.

a) $\mathrm{Top}(X)$ est localement irréductible.

b) Les X/x [sont] irréductibles.

c) Tout objet 0-connexe de $X^\wedge$ est irréductible.

d) Les composantes connexes de X sont faiblement cofiltrantes (i.e. faiblement filtrantes à gauche).

e) Condition PS 1_g de SGA 4, I 2.

Si X est [une] catégorie, X irréductible signifie aussi que $X \neq \emptyset$ et $\forall\, x, y \in \mathrm{Ob}\, X$, [il] existe [un] s avec $s \nearrow x$, $s \searrow y$. (Cela signifie aussi que X est *faiblement filtrante* ([11]) à gauche, ou faiblement cofiltrante.) X localement irreductible signifie que pour x, y *appartenant à une même composante connexe,* [il] existe [un] s avec $s \nearrow x$, $s \searrow y$, i.e. $X/[x] \cap X/[y] \neq \emptyset$ (où $[x]$ est considéré comme élément de l'ensemble ordonné $\mathrm{Ob}\, X$). Il revient encore au même de dire que les catégories induites X/x sont irréductibles ([12]). [13]

Exemple. Une catégorie *ayant un objet initiale,* ou seulement un 'plus petit objet', est irréductible. Inversement, si l'ensemble *ordonné* $\mathrm{Ord}(X)$ associé à X (quotient de $\mathrm{Ob}(X)$ par la relation d'équivalence associé à son préordre) est fini, p.ex. X fini, ou seulement localement fini, p.ex. X localement fini (i.e. les catégories X/x finies), alors si X est irréductible, $\mathrm{Ob}\, X$ a un plus petit élément, i.e. X a un plus petit objet. C'est vrai plus généralement encore, si $\mathrm{Ord}(X)$ est artinien, i.e. toute suite décroissante dans $\mathrm{Ord}(X)$ est stationnaire.

Mais la condition intéressante est la condition topossique de *locale* irréductibilité, qui équivaut : I pseudo-filtrante à gauche, i.e. les composantes connexes de I filtrantes, i.e. les conditions PS 1, PS 2

PS 1 Tout diagramme $x \searrow$, $y \nearrow$ s se complète en $z \nearrow x$, $z \searrow y$, $x \searrow$, $y \nearrow$ s *commutatif.*

PS 2 Pour toute double flèche $x \rightrightarrows y$, [il] existe un égalisateur $z \longrightarrow x \rightrightarrows y$.

En résumé, on a la p. 9

Proposition 2.1. *Soit X un objet de* Cat. *Conditions équivalentes :* [14]

---

8. car les ouverts de $\mathrm{Top}(X)$, ceux de $\mathrm{Esp}(X)$ et ceux de X se correspondent.

9. Cela signifie aussi que pour x, y dans X, $\exists\, z$ avec $z \nearrow x$, $z \searrow y$, i.e. $\mathrm{Ord}(X)$ est filtrant à gauche.

10. Faux, on a (a $\iff$ d) $\implies$ (b $\iff$ c) X irréductible n'implique *pas* que les X/x le soient, exemple $X = (0 \overset{\alpha}{\underset{\beta}{\rightrightarrows}} 1)$ $(\alpha \neq \beta)$, on a $X/1 = 0' \searrow$, $0'' \nearrow 1$, qui n'est *pas* irréductible, ni localement irréductible puisqu'il est connexe. Pourtant il est étale sur X.

11. Il est mieux l'appeler *pseudo-filtrante,* et réservons faiblement filtrante à : localement faiblement filtrante + connexe.

12. faux, cf. annotation marginale plus haut.

a) *Le topos associé* $\mathrm{Top}(X)$ *est localement irréductible, i.e. (étant localement connexe) pour tout objet* F *de* $X^\wedge$ *et tout sous-objet* F' *de* F*, si* F *est connexe,* F' *l'est.*

a′) *Pour tout* X' *étale sur* X*, si* X' *est connexe, tout ouvert de* X' *est connexe.*

a″) *Tout ouvert d'une catégorie localisée* X/a *(*$a \in \mathrm{Ob}\, X$*) est connexe (i.e.* X/a *est une catégorie irréductible, i.e. le topos* $\mathrm{Top}(X/a)$ *l'est).*

b) *Pour tout objet* a *de* X*,* $\mathrm{Esp}(X/a)$ *est un espace topologique irréductible.*

c) *Les catégories* X/a *sont faiblement filtrantes à gauche (ou faiblement cofiltrantes), i.e. pour deux objets* ξ*,* η *d'une telle catégorie,* `[il]` *existe un objet* ζ *et deux flèches*

$$\begin{array}{ccc} & & \xi \\ & \nearrow & \\ \zeta & & \end{array} \qquad \begin{array}{ccc} & & \\ & \searrow & \\ & & \eta \end{array}$$

(i.e. l'ensemble ordonné Ord(X/a*) associé à* X/a *est filtrant à gauche)*

c′) *Pour tout diagramme en traits pleins dans* X*,* $\begin{array}{ccccc} & & x & & \\ & \cdots\nearrow & & \searrow & \\ z & & & & s \\ & \cdots\searrow & & \nearrow & \\ & & y & & \end{array}$ *, on peut le compléter en un carré* commutatif *comme indiqué. (Condition* PS 1 *des catégories localement filtrantes.)*

Ces conditions impliquent que toute composante connexe X_α de X est irréductible, ou ce qui revient au même, que $\mathrm{Esp}(X_\alpha)$ est irréductible,
mais ne sont pas impliquées par celle-ci. [15]

Contrexemple, la catégorie 0-connexe $0 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 1$, avec $\alpha \neq \beta$. On a $X/1 \simeq \begin{array}{ccc} 0 & & \\ & \searrow & \\ & & 1 \\ & \nearrow & \\ 0' & & \end{array}$, qui n'est *pas* irréductible, puisque $\Phi\backslash\{1\} \simeq \varepsilon$ est un ouvert disconnexe `[`$\Phi = X/1$`,` $\varepsilon = \bullet\ \bullet$`]`.

J'aimerais une condition 'topossique' sur X, équivalente à X 'pseudo-filtrante à gauche', i.e. les composantes connexes filtrantes, i.e. PS 1^{opp} (condition c′) ci-dessus) + PS 2^{opp} (condition de la double flèche), et qui de plus implique que tout objet 0-connexe de $X^\wedge$ (i.e. tout objet X' de Cat étale sur X et 0-connexe) soit W_∞-asphérique. Et je songe à la notion de 'totale asphéricité' de Pursuing Stacks.

Commentary · Exposition [ms. 15]

Completing Proposition 2.1

Proposition 2.1 isolates the part of filtering which can be expressed without equalizing parallel arrows. We give its proof because it explains both the corrected statement and the mechanism which will later be strengthened.

Recall that PS 1 says that every cospan $a \to x \leftarrow b$ admits a commuting square with some upper-left object c; no universal property is requested. Suppose PS 1 holds. In any category with this property, the relation "a and b have a common predecessor" is reflexive and symmetric. It is transitive: given $a \leftarrow r \to b$ and $b \leftarrow s \to d$, apply PS 1 to $r \to b \leftarrow s$ and compose the resulting arrows. Every arrow connects objects related in this sense. Hence any two objects in the same connected component have a common predecessor.

For a presheaf F, its category of elements $E = \int_X F \to X$ is a discrete fibration, the meaning of "étale" here. Every arrow into the image of an object has a unique lift into that object. Thus a PS 1 square in X lifts to a PS 1 square in E: the two lifts into the common target agree by uniqueness. Therefore E also satisfies PS 1. Components of E correspond to coproduct components of F. If F is 0-connected, E is nonempty and connected, so the previous paragraph gives common predecessors for every pair of its objects. Subobjects of F correspond to sieves in E; any two nonempty such sieves meet. Thus F is irreducible, and $\widehat{X}$ is locally irreducible.

Conversely, suppose every 0-connected presheaf is irreducible. The representable h_x is 0-connected, since its category of elements X/x has a terminal object. Given arrows $a \to x$ and $b \to x$,

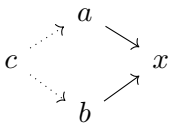
their generated subobjects of h_x are noninitial. Their intersection is noninitial, so it contains some element represented by arrows $c \to a$ and $c \to b$ with equal composite to x. This is PS 1. The descriptions using X/x, its associated space, and open subcategories are equivalent by the preceding sieve argument. This proves all the equivalences of Proposition 2.1.

The counterexample is the category with two objects and distinct arrows $\alpha, \beta : 0 \rightrightarrows 1$. Its objects have the common predecessor 0, so its underlying preorder is the irreducible two-point chain. But in $X/1$, the objects $(0, \alpha)$ and $(0, \beta)$ have no common predecessor: the only endomorphism of 0 is its identity. Their two singleton lower sets are disjoint nonempty subobjects of h_1. Thus $\widehat{X}$ is not locally irreducible. The mistaken converse on page 16 is precisely the assertion that the preorder remembers this equality of composites; the footnotes already retract it.

Définition 2.2. ([13]) Soit X une catégorie.

a) On dit que X est *faiblement filtrante* ([14]) si $X \neq \emptyset$, et si pour $a, b \in \operatorname{Ob} X$, $\exists\, c$ et des flèches $c \to a$, $c \to b$, i.e. $X/[a] \cap X/[b] \neq \emptyset$. (**NB** X est faiblement filtrante ([15]) à gauche si et seulement si $\operatorname{Ord}(X)$ l'est, i.e. est filtrante à gauche. **ReNB** X faiblement filtrante $\Longrightarrow$ X 0-connexe.)

b) On dit que X est *localement faiblement filtrante à gauche* ou plus simplement préfiltrante à gauche si les catégories X/x ($x \in \operatorname{Ob} X$) sont faiblement filtrantes à gauche, i.e. si X satisfait la condition : $\forall$ diagrammes $a \to x \leftarrow b$, $\exists\, c$ dans X et des flèches de c dans a, b rendant *commutatif* le carré



([16]). (**NB** Cette condition implique que $\operatorname{Ord}(X)$ est localement faiblement filtrant, mais n'équivaut pas à cette dernière. Exemple, $X = 0 \overset{\alpha}{\underset{\beta}{\rightrightarrows}} 1$, $X/1 = 0' \to 1 \leftarrow 0''$ n'est pas faiblement filtrante à gauche.)

p. 10

[16]

NB Si X localement faiblement filtrante à gauche, tout X' étale sur X itou.

NB ([17]) X localement faiblement filtrante à gauche $\Longleftrightarrow$ les composantes connexes sont faiblement filtrantes à gauche, ou encore si X est 0-connexe, X localement faiblement filtrante à gauche $\Longleftrightarrow$ X est faiblement filtrante ([18]) à gauche. On a en effet $\Longleftarrow$ tautologiquement ([19]), et $\Longrightarrow$ en montrant que si PS 1 `[est]` satisfait, alors $PS(a, b)$

13. Cf. tableau récapitulatif pages 97, 98 et surtout 99.
14. dire plutôt *pseudo-filtrante*, réserver 'faiblement filtrante' à 'localement faiblement filtrante' (ou *irréductible*) + connexe, condition PS $1'_{\mathrm{g}}$.
15. pseudo-filtrante.
16. Condition PS 1_{g}.
17. **Faux.**
18. pseudo-filtrante.
19. **faux,** ainsi $0 \overset{\alpha}{\underset{\beta}{\rightrightarrows}} 1$ est 0-connexe et faiblement filtrante à gauche, mais non pas localement faiblement filtrante à gauche car $0 \xrightarrow{\alpha} 1 \xleftarrow{\beta} 0$ ne se complète pas en un carré *commutatif.*

défini par $X/[a] \cap X/[b] \neq \emptyset$ est une relation d'équivalence, impliquée par la relation $C(a,b) : \{\mathrm{Hom}(a,b) \neq \emptyset$ *ou* $\mathrm{Hom}(b,a) \neq \emptyset\}$, donc la relation d'équivalence (homotopie) engendrée par $C(a,b)$ est plus fine que $PS(a,b)$.

c) On dit que X est *filtrante à gauche*, ou *cofiltrante,* si elle est faiblement filtrante ([20]) à gauche (donc *non vide*), et satisfait (en plus de la condition PS $1'_{\mathrm{g}}$ de a)) la condition PS 2_{g} suivante :

Pour toute double flèche $a \overset{u}{\underset{v}{\rightrightarrows}} b$, `[il]` existe `[une]` flèche $c \xrightarrow{w} a$ telle que $uw = vw$ ([21]).

Il revient au même de dire que X satisfait les trois conditions a′) X est 0-connexe (plus forte que $X \neq \emptyset$), b′) PS 1, i.e. X localement faiblement filtrante à gauche et c′) PS 2 comme avant.

d) On dit que X est *localement filtrante à gauche* si les composantes connexes sont filtrantes à gauche, i.e. si X satisfait les conditions PS 1 et PS 2 ('pseudo-cofiltrante' dans la terminologie de SGA 4). **NB** Si X est localement filtrante à gauche, il en est de même de tout X' étale sur X, car c'est évident pour PS 1 qui concerne les X/x, et pour PS 2 cela résulte du fait que $X' \longrightarrow X$ est fidèle. Donc si X est localement cofiltrante, les X/a sont filtrantes, mais la réciproque est fausse.

Commentary · Examples [ms. 16]

The equalization condition survives beyond every slice

The definitions now separate PS 1 from the additional condition PS 2. In the latter, "égalisateur" means an *equalizing arrow*: for $u, v : a \rightrightarrows b$ there is $w : c \to a$ with $uw = vw$. It does not mean that a categorical equalizer exists. This is the usual cofiltered condition, dual to SGA 4's PS 2 (Artin et al., 1972, Exposé I, §2.7).

To see why all slices can be cofiltered without the original category being locally cofiltered, take the one-object category BG for a nontrivial group G. It satisfies PS 1: a pair f, g can be completed by 1 and $g^{-1}f$. For its unique object $*$, the slice $BG/*$ has objects $g \in G$ and exactly one arrow from g to h, namely $h^{-1}g$. It is equivalent to the terminal category and is cofiltered. Yet distinct $u, v \in G$ cannot be equalized on the right, since $uw = vw$ and invertibility of w imply $u = v$. Hence BG fails PS 2.

This example also makes the topos distinction visible. Every connected G-set is transitive and therefore irreducible, so the presheaf topos is locally irreducible. Products of connected objects need not remain connected: for the regular right G-set, the diagonal right action on $G \times G$ has one orbit for each value of hg^{-1}. The product condition introduced next detects the missing equalization information. No defect in Proposition 2.1 is involved; its purpose was exactly to isolate PS 1.

On a les notions duales : *faiblement filtrante à droite* (ou simplement faiblement filtrante), et *filtrante à droite* ou simplement filtrante. (Notions duales de a), c) ci-dessus.) Pour la notion duale de b), savoir que les $x\backslash X$ soient faiblement filtrantes, on ne peut plus l'appeler 'localement faiblement filtrante', mais ce serait *co*localement faiblement filtrante. p. 11

Exemple. $X = \left(0 \xrightarrow{\gamma} 1 \overset{\alpha'}{\underset{\beta'}{\rightrightarrows}} 2 \overset{\alpha}{\underset{\beta}{\rightrightarrows}} 3\right)$ avec $(\alpha\alpha' = \beta\beta') \neq (\alpha\beta' = \beta\alpha')$.

Commentary · Development (commentators') [ms. 16]

The finite example and its homotopy type

The example between Definition 2.2 and Proposition 2.3 makes the loss of information in $\mathrm{Ord}(X)$ concrete. We read the displayed graph and its two relations as a presentation of X; in particular, no additional relation is imposed on composites with γ. Put

$$r = \alpha\alpha' = \beta\beta', \qquad s = \alpha\beta' = \beta\alpha', \qquad r \neq s.$$

20. pseudo-filtrante.

21. i.e. trois conditions a) $X \neq \emptyset$ b) PS 1′ habituelle c) PS 2 (double flèche).

Then $\mathrm{Ord}(X)$ is the chain $0 < 1 < 2 < 3$. Every pair of objects has both a common predecessor and a common successor. Nevertheless, 0 is not initial: the arrows $\alpha'\gamma, \beta'\gamma : 0 \rightrightarrows 2$ are distinct. Likewise, 3 is not terminal, since $r, s : 1 \rightrightarrows 3$ are distinct.

The same calculation locates the exact failure of PS 1. A commuting square for the cospan $1 \xrightarrow{\alpha'} 2 \xleftarrow{\beta'} 1$ would require two arrows from an object into 1. Their common source can only be 1 or 0, and both arrows would be respectively 1_1 or γ. Neither choice makes the square commute. Thus the left PS 1 fails, and the pair α', β' also has no equalizing arrow on the left. Dually, a commuting cocone over $3 \xleftarrow{\alpha} 2 \xrightarrow{\beta} 3$ can only use identity arrows out of 3, so the right PS 1 fails; the pair α, β cannot be equalized on the right either. The category is consequently neither locally cofiltered nor locally filtered, although its associated poset satisfies both conditions. This is the point of the paragraph immediately following the example.

The crossed composition relations also admit a useful homotopical interpretation which the manuscript does not develop.

The homotopy type of the finite example. [ours] *For the category presented above, $|NX| \simeq \mathbb{R}P^2$. In particular, its contractible reachability poset does not determine even the fundamental group of its nerve.*

Proof. Let C be the full subcategory on $1, 2, 3$. The functor $q : X \to C$ sends 0 to 1, sends γ to 1_1, and fixes the remaining generators. It respects the defining relations. For the inclusion $j : C \hookrightarrow X$, one has $qj = 1_C$ and a natural transformation $1_X \to jq$ whose component at 0 is γ and whose other components are identities. The resulting homotopy of nerves fixes NC, so $|NX| \simeq |NC|$.

Identify each set $C(i, j)$, for $1 \le i < j \le 3$, with $\mathbf{Z}/2$: label α', α, r by 0 and β', β, s by 1. The two displayed relations say exactly that composition adds these labels. Let P be the poset with objects (i, ε), where $i \in \{1, 2, 3\}$ and $\varepsilon \in \mathbf{Z}/2$, with $(i, \varepsilon) < (j, \delta)$ precisely when $i < j$. Its involution changes every ε to $\varepsilon + 1$. There is an invariant functor $P \to C$ sending (i, ε) to i and an arrow to the label $\delta - \varepsilon$. A chain of arrows in C lifts uniquely after choosing the lift of its first vertex, and the two choices are exchanged by the involution. This proves, degree by degree and compatibly with faces and degeneracies,

$$NP/(\mathbf{Z}/2) \simeq NC.$$

The order complex of P is the join of three two-point sets. Sending their vertices to $\{\pm e_1, \pm e_2, \pm e_3\}$ identifies it with the boundary of the octahedron. Radial projection identifies its realization with S^2, and the involution with the antipodal map. Realization commutes with the quotient, giving

$$|NC| \simeq |NP|/(\mathbf{Z}/2) \cong S^2/(x \sim -x) = \mathbb{R}P^2.$$

Since the sphere is simply connected, this double covering also gives $\pi_1(|NX|) \cong \mathbf{Z}/2$. □

The join and antipodal descriptions are classical (Hatcher, 2002, p. 9 and Example 1.43). Our addition is their explicit identification with this particular presentation. The bounded literature searches did not locate that identification; the result is marked as our calculation, without a claim of priority.

[17] Seule la notion a) (faiblement filtrante ([22]) à gauche) ne dépend que de $\mathrm{Ord}\, X$ - mais bien sûr si X est *pré*ordonnée, toutes trois ne dépendent que de $\mathrm{Ord}\, X$. Notons que dire que ça ne dépend que de $\mathrm{Ord}\, X$, ou que de $\mathrm{Esp}\, \mathrm{sep} X$, revient au même, puisque les deux se déterminent mutuellement. La condition a) signifie que $\mathrm{Esp}\, X$, ou aussi $\mathrm{Top}(X)$, est *irréductible.* On dira que X est *irréductible* (au lieu de pseudo-filtrante). Les trois autres notions ne peuvent s'exprimer en termes de $\mathrm{Esp}\, X$ ou de $\mathrm{Ord}\, X$ seul, mais elles s'expriment en termes du $\mathrm{Topos}\, X$. La condition b) (X localement faiblement filtrante) signifie que $\mathrm{Top}(X)$ est *localement irréductible,* et on dira aussi que X est *localement irréductible* (au lieu de localement faiblement filtrante). Ça signifie que pour tout X' étale sur X et 0-connexe, X' est faiblement filtrante à gauche, i.e. est irréductible.

Reste à exprimer la condition filtrante elle-même en termes de $\mathrm{Top}\, X$. Je rappelle à ce propos :

22. pseudo-filtrant.

Commentary · Exposition [ms. 17]

Products of connected objects

Grothendieck now asks for a condition on the topos which remembers the equalization of parallel arrows. Irreducibility alone cannot do this: it records whether two subobjects meet, whereas equalization concerns two maps with the same source. Proposition 2.3 first isolates the intermediate condition that the component functor preserve products. Throughout the following topos discussion, "connected object" means noninitial and connected, that is, 0-connected; the topos is locally connected. Thus every object is the coproduct of its connected components, and π_0 is left adjoint to the constant-object functor.

Write $\mathcal{E}$ for the category of objects of the topos and let S be the full generating subcategory of connected objects used in the text. The precise statement of 2.3(b) is

$$\pi_0(V \times U) \longrightarrow \pi_0(V) \quad \text{is bijective whenever } U \text{ is connected.}$$

The edition prints $\pi_0(U)$ on the right, although the displayed projection has target V. This is a slip of the edition, retained in the source. Condition (a) implies (c) by decomposing U and V into connected components: products distribute over coproducts in a topos, and each resulting product is connected. Conversely, (c) applied to connected objects gives (a); the same decomposition proves the equivalence with (b).

The reduction to generators deserves the proof which the footnote requests. For $F \in \mathcal{E}$, let S/F have objects $s \to F$, with $s \in S$. The canonical presentation

$$F \simeq \operatorname*{colim}_{(s \to F) \in S/F} s$$

holds in $\mathcal{E}$; it is the density presentation for the canonical site of generators. Applying π_0 gives $\pi_0(F) \simeq \pi_0(S/F)$, since every s is connected. If $s \times t$ is connected for every $s, t \in S$, separate preservation of colimits by products gives

$$\pi_0(F \times G) \simeq \operatorname*{colim}_{S/F \times S/G} \{*\} \simeq \pi_0(S/F) \times \pi_0(S/G).$$

This proves (d)⇒(c). Equivalently, one may use the clopen-partition argument in our block on Lemma 2.9 below; this avoids any choice of a presentation.

Applying (c) to the terminal object shows that the diagonal of $\pi_0(1)$ is a bijection onto its square. Consequently $\pi_0(1)$ has at most one element. The nondegenerate case has exactly one, which explains the separate nonemptiness clause in Definition 2.4. Nothing in this argument annihilates higher homotopy groups. The inference to W_∞-asphericity immediately following 2.3 is explicitly withdrawn in the manuscript's annotation; the stronger hereditary hypothesis introduced in 2.6 is what will support the cohomological argument.

Proposition 2.3. *Les conditions suivantes sur un topos $\mathcal{X}$ sont équivalentes (quand S (comme 'site') désigne une partie génératrice de* $\mathrm{Fais}(\mathcal{X}) = S^\wedge$, formée d'objets 0-connexes ([23])*.)*

a) Si U, V sont deux objets 0-connexes de $\mathrm{Ob}\,\mathcal{X}$*, $U \times V$ est 0-connexe.*

b) Si U est un objet 0-connexe de $\mathcal{X}^\wedge$, alors $U \longrightarrow e$ est 'W₀-asphérique', i.e. pour tout V, la projection $V \times U \longrightarrow V$ induit $\pi_0(U \times V) \xrightarrow{\sim} \pi_0(U)$.

c) Si U, V `[sont]` *dans $\mathcal{X}^\wedge$, alors*

$$\pi_0(U \times V) \xrightarrow{\sim} \pi_0(U) \times \pi_0(V).$$

d) Comme a), avec U, V dans S. (Mais on suppose les objets de S 0-connexes.)

([24]).

De plus, ces conditions impliquent que $\mathcal{X}$ est connexe, donc 0-connexe si $\mathcal{X} \neq \emptyset$. Cela implique [18] de plus que $\mathcal{X}$ est W_∞-asphérique, cf. Pursuing Stacks §35 (Corollary p. 52) ([25]).

23. donc $\mathcal{X}$ est localement 0-connexe.
24. Il faudrait expliciter la démonstation avec celle de 2.9 ci-dessous (p. 25).
25. **Faux,** loc. cit. ne dit rien de tel.

C'est essentiellement Proposition 1 de Pursuing Stacks §65, cf. loc. cit. p. 170-171 ([26]). Cf. aussi Pursuing Stacks, §35, pages 50-55.

p. 12

Définition 2.4. On dit que $\mathcal{X}$ est *totalement* 0*-connexe* s'il est localement connexe, non vide et satisfait les conditions équivalentes de 2.3. (NB c'est un cas particulier de la notion 'totalement W-asphérique' de loc. cit. On a ainsi la notion de totalement n-connexe.)

Notons que pour un objet X de Cat, dire que $\mathrm{Top}(X)$ est totalement 0-connexe revient à dire que $X \neq \emptyset$, et que pour $x, y \in \mathrm{Ob}\, X$, $x \times y$ dans $X^\wedge$ est 0-connexe, i.e. $X/x \times y$ est 0-connexe ([27]). C'est une condition très forte. On dira aussi que X est *totalement* 0*-connexe.* Notons ici

Commentary · Later developments [ms. 18]

The ordinary sifted condition and its variance

The product condition has acquired a useful name, provided that we keep track of the opposite category. For $\widehat{X} = \mathrm{Fun}(X^{\mathrm{op}}, \mathbf{Set})$, the left adjoint to the constant-presheaf functor is

$$\pi_0(F) = \operatorname*{colim}_{X^{\mathrm{op}}} F = \pi_0(X/F).$$

Thus $\widehat{X}$ is totally 0-connected precisely when X^{op} is *sifted*: its colimits in sets commute with finite products. In the ordinary categorical characterization, the indexing category is nonempty and its diagonal is final. For the conventions of these pages, this says that $X \to X \times X$ is co-cofinal. The comma category at (x, y) consists of common cones $z \to x$, $z \to y$, and is exactly $X/(h_x \times h_y)$.

This characterization also explains why binary products suffice. Repeated use of finality of the diagonal gives finality of every finite nonempty diagonal; nonemptiness handles the nullary product. The terminology and its relation to algebraic theories are developed by Adámek et al. (2010). The distinction from filteredness is structural: filtered colimits preserve finite limits, while sifted colimits are required to preserve finite products. Equalizers are precisely the operation absent from the latter requirement.

There is a corresponding distinction in higher category theory, but one must not transfer a statement by changing its vocabulary alone. In the higher version the relevant comma objects must be weakly contractible, rather than merely nonempty and connected. The characterization of filtered colimits by preservation of finite limits in spaces is Lurie (2009, Proposition 5.3.3.3); sifted colimits and geometric realizations are treated in his §5.5.8. Our present identification concerns *ordinary* set-valued colimits. The higher homotopy obstruction which survives it is exhibited later in the commentary on 2.19–2.20.

Proposition 2.5. *Soit $\mathcal{X}$ un topos. Si $\mathcal{X}$ est totalement* 0*-connexe, alors $\mathcal{X}$ est irréductible. (En fait la conclusion vaut, tautologiquement, en supposant seulement que le produit de deux objets* 0*-connexes de $\mathcal{X}^\wedge$ est non vide, et il suffit même de le supposer pour des sous-objets de l'objet final $e_{\mathcal{X}}$.) En particulier, si $\mathcal{X} = Top(X)$, où X est dans* Cat*, alors X est faiblement filtrante à gauche.*

Mais bien sûr, cela n'implique *pas* que $\mathcal{X}$ soit localement irréductible.

[19]

Par exemple la catégorie $\mathbf{\Delta}$, ou toute petite catégorie stable par produits binaires, est totalement 0-connexe, mais $\mathbf{\Delta}$ n'est pas localement irréductible (les catégories induites $\mathbf{\Delta}/\mathbf{\Delta}^n$, pour $n > 0$, ne sont visiblement *pas* irréductibles). (Mais rappelons que si $\mathcal{X} = \mathrm{Top}(T)$, T un espace topologique, alors T irréductible implique que $\mathcal{X}$ est aussi *localement* irréductible. Donc T totalement 0-connexe implique $\mathrm{Top}(T)$ globalement *et* localement irréductible, i.e. pour tout T' étale sur T, et 0-connexe, T' est irréductible.)

26. Mais je m'y borne aux topos de la forme $\mathrm{Top}(A)$, avec A dans Cat.

27. Cela signifie aussi que $\mathrm{diag}_X : X \longrightarrow X \times X$ est co-cofinale, i.e. $X^o \longrightarrow X^o \times X^o$ est cofinale, et implique qu'il en est de même de $X \longrightarrow X^E$ pour tout ensemble *fini* E.

Commentary · Examples [ms. 19]

The simplex category separates the two conditions

Grothendieck's example shows exactly where a global product condition loses the information needed locally. We spell it out because the word "visibly" hides two different computations. Let Δ have the nonempty finite ordinals $[n] = \{0, \dots, n\}$ and monotone maps. For any small category X with binary products, the product of two representables is representable; its category of elements has a terminal object. Such a nonempty X is therefore totally 0-connected. The separate assertion about Δ does not say that Δ has binary products: the ordered square $[1] \times [1]$ is not an ordinal.

For Δ, the product $h_{[m]} \times h_{[n]}$ is the simplicial set $\Delta^m \times \Delta^n$. Its vertices are the pairs (i, j). An edge joins (i, j) to $(i + 1, j)$ whenever the latter exists, and similarly joins (i, j) to $(i, j + 1)$. Hence its vertex graph is connected. Every simplex is joined in its category of elements to each of its vertices by a face map. The category of elements is consequently nonempty and connected, which verifies the generator test of 2.3 for all m, n.

Local irreducibility fails already over [1]. In $\Delta/[1]$, consider the two objects $[0] \xrightarrow{0} [1]$ and $[0] \xrightarrow{1} [1]$. A common predecessor would be a nonempty ordinal $[r]$ mapping to both copies of [0], with equal composites to [1]. Those composites are respectively constant at 0 and 1, so equality is impossible. The two principal sieves have empty intersection. In particular, $\Delta/[1]$ is not irreducible, and Δ is not cofiltered. Equivalently, the parallel vertex maps $[0] \rightrightarrows [1]$ admit no equalizing map into [0].

This example gives an explicit positive answer to the request in footnote 31: total 0-connectedness does not imply local irreducibility. Its opposite also supplies an ordinary sifted category which is not filtered. Neither conclusion requires an argument about derived limits.

Théorème 2.6. *Soit $\mathcal{X}$ un topos localement connexe, $S \subseteq \mathcal{X}^\wedge$ une petite partie génératrice formée d'objets* 0*-connexes. Conditions équivalentes (*sauf *c, c′, qui semblent* ([28]) *plus faibles, cf. démonstration).)*

a) Tout topos induit $\mathcal{X}/U$ sur U 0*-connexe dans $\mathcal{X}^\wedge$ est totalement* 0*-connexe.*

a′) Tout produit fibré $U \times_W V$ d'objets 0*-connexes dans $\mathcal{X}^\wedge$ est* 0*-connexe.*

b) $\mathcal{X}$ est localement irréductible, et ses composantes connexes sont totalement 0*-connexes.*

c) Comme a), avec U dans S, i.e. les $\mathcal{X}/U$ ($U \in \mathrm{Ob}\, S$) sont totalement 0*-connexes.*

c′) Comme a′), avec U, V, W dans S.

d) S est localement filtrante à gauche, i.e. S satisfait les deux conditions

PS* 1 *Tout diagramme* $\begin{array}{ccc} a & & \\ & \searrow & \\ & & x \\ & \nearrow & \\ b & & \end{array}$ *dans S s'insère dans* $\begin{array}{ccc} & a & \\ c & \swarrow\!\searrow & x \\ & b & \end{array}$ *commutatif.*

PS* 2 *Tout diagramme $a \rightrightarrows b$ s'insère dans un diagramme commutatif $c \longrightarrow a \rightrightarrows b$.*

([29]).

Ces conditions impliquent que les objets 0-connexes de $\mathcal{X}^\wedge$ (en particulier les éléments de S) sont ∞-connexes. p. 13 [20]

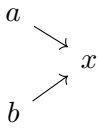
Corollaire 2.7 (de l'équivalence b $\iff$ d). *Soit X un objet de* Cat. *Pour que X soit filtrant à gauche il faut et il suffit que $Top(X)$ soit totalement* 0*-connexe et localement (et donc aussi globalement) irréductible (ou encore que X soit totalement* 0*-connexe et que pour tout $x \in \mathrm{Ob}\, X$, la catégorie localisée X/x soit irréductible (elle est même totalement* 0*-connexe),*

28. **sont.**

29. NB On a équivalence de (a), a′), b), d),) qui impliquent c), c′), lesquels sont équivalents. Mais il semble ([30]) que c), c′) soient strictement plus faible. Cf. cependant les variantes e), e′) dans 2.12 (p. 28, 29).

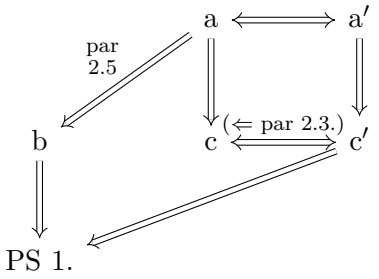
30. Il s'avère. C'est bien vrai, c) et c′) n'impliquent *pas* les autres conditions, cf. annotation marginale p. 28 (cas de $X = B_G$).

ou encore que pour deux objets a, b dans X, $a \times b \in X^\wedge$ soit 0*-connexe, et pour*



dans X, $a \times_x b$ (dans $X^\wedge$) soit 0*-connexe)* ([31]).

Démonstration de 2.6. On a le diagramme d'implications ± tautologiques



Montrons quand même les deux implications c′) $\Longrightarrow$ PS 1 et b $\Longrightarrow$ PS 1.

Commentary · Exposition [ms. 20]

What the hereditary condition adds

Theorem 2.6 makes the product condition stable under passage to a connected object as base. Its surviving equivalence is

$$(a) \iff (a') \iff (b) \iff (d), \qquad (c) \iff (c'),$$

with an implication from the first group to the second. Here the three objects U, V, W in (a') are all connected. Connectedness of W is essential: even in sets, two singleton objects mapping to different elements of a two-element set have empty pullback.

The equivalence of (a) and (a') is the same product test inside $\mathcal{E}/W$. Condition (a) implies local irreducibility by 2.5 and gives the assertion about components in (b). To see the decisive step (b)⇒(d), first use 2.8 for PS 1. Given $u, v : a \rightrightarrows b$ in S, both objects lie over the same connected component of 1. By (b), $b \times b$ is connected. The equalizer is the pullback

$$E = a \times_{b\times b} b, \qquad a \longrightarrow b \times b = (u, v), \quad b \longrightarrow b \times b = \mathrm{diag}\,.$$

Local irreducibility, applied over the connected object $b \times b$, makes E noninitial. A generator mapping to E supplies an equalizing arrow, which is PS 2. The text's "c) est vérifié" at manuscript p. 24 means (b); its reference to $\mathcal{X} \times \mathcal{X}$ should be read as an application of 2.8 in $\mathcal{X}$ itself.

There is a concise exactness interpretation after restricting to any connected component of the topos. Under (a'), decomposition into connected components gives, for arbitrary U, V, W in that component,

$$\pi_0(U \times_W V) \simeq \pi_0(U) \times_{\pi_0(W)} \pi_0(V).$$

Indeed, the summands of the pullback are precisely the products $U_i \times_{W_k} V_j$ for which both factors map to the same component W_k; each is connected by (a'). Since $\pi_0(1) = \{*\}$, π_0 preserves finite limits on this connected topos. We shall use this stronger fact in the cohomological argument. For a disconnected topos one applies it component by component; its global π_0 does not preserve the terminal object.

[21]

C'est contenu dans le

Corollaire 2.8. *Soient $\mathcal{X}$, S comme dans 2.6. Conditions équivalentes :*

31. Exemple de X totalement 0-connexe, mais non cofiltrant, i.e. non localement irréductible ? La catégorie $\mathbf{\Delta}$ des simplexes type est totalement 0-connexe, mais n'est pas cofiltrante car $\mathbf{\Delta}^0 = e \overset{\partial_0}{\underset{\partial_1}{\rightrightarrows}} \mathbf{\Delta}^1$ ne peut être égalisé ! Donc `[la catégorie opposée]` $\mathbf{\Delta}^o = I$ est telle que $I \longrightarrow I^E$ soit cofinal pour tout ensemble fini E, mais I non filtrante.

a) $\mathcal{X}$ est localement irréductible, i.e. les topos induits $\mathcal{X}/U$, où U est 0-connexe, sont irréductibles.

b) Tout produit fibré $U \times_W V$ d'objets 0-connexes U, V est 'non vide'.

b′) Tout produit fibré $U \times_W V$, avec U, V 'non vide', W 0-connexe, est non vide.

c) Les $\mathcal{X}/x$ avec $x \in S$ sont irréductibles.

d) Comme b), avec U, V, W dans S.

e) S est localement faiblement filtrante à gauche, i.e. satisfait PS 1.*

Ici les implications ± évidentes sont

p. 14

$$\begin{array}{ccccc} \mathrm{b}' & \Longrightarrow & \mathrm{b} & \Longrightarrow & \mathrm{a} \\ & & \Downarrow & & \Downarrow \\ & & \mathrm{d} & \Longleftrightarrow \mathrm{e} \Longleftrightarrow & \mathrm{c}. \end{array}$$

Prouvons les implications de la ligne du bas. Considérons d). Si $\begin{smallmatrix} & a & \searrow \\ & & & x \\ & b & \nearrow \end{smallmatrix}$ [est] dans S, dire que le produit fibré dans $\mathcal{X}$ est 'non vide' signifie, puisque S est un système de générateurs et en particulier *non vide,* qu'il existe c dans S et $c \longrightarrow a \times_x b$, ou ce qui revient au même, $\begin{smallmatrix} & a & \\ c & & \\ & b & \end{smallmatrix}$, rendant commutatif le diagramme $\begin{smallmatrix} & a & \searrow & \\ c & & & x \\ & b & \nearrow & \end{smallmatrix}$. Donc d) équivaut à PS 1, i.e. à e) ($\mathcal{X}$ localement faiblement filtrant à gauche). Considérons la condition c) : les $\mathcal{X}/a = (S/a)^\sim$ sont irréductibles. Cela signifie que pour deux sous-objets *non vides* $U, V \subseteq a$ dans $\mathcal{X}^\wedge$, on a $U \cap V \neq \emptyset$, ou encore que pour deux cribles

$\underline{U}$, $\underline{V}$ de S/x, si $\underline{U}$, $\underline{V}$ non vides (ce qui signifie que les sous-objets de x dans $\mathcal{X}^\wedge$ qu'ils définissent sont 'non vides', ou que les objets de S sont 'non vides' dans $\mathcal{X}^\wedge$), alors $\underline{U} \cap \underline{V}$ est non vide. On peut supposer que $\underline{U}$, $\underline{V}$ est engendré chacun par une seule flèche $a \xrightarrow{u} x$, $b \xrightarrow{v} x$. Le crible intersection est formé des objets $c \longrightarrow x$ de S/x tels que $c \longrightarrow x$ se factorise par un $\begin{smallmatrix} & a & \\ c & & \\ & b & \end{smallmatrix}$, donc dire que ce crible n'est pas vide, signifie que PS 1 est satisfait pour les deux flèches données u, v. Donc c) équivaut à la condition PS 1[, i.e. à] e. [22]

Ainsi on a établi

$$\mathrm{b}' \Longrightarrow \mathrm{b} \Longrightarrow \mathrm{a} \Longrightarrow (\mathrm{c} \Longleftrightarrow \mathrm{d} \Longleftrightarrow \mathrm{e}),$$

reste à prouver que c $\Longrightarrow$ b′. Soit donc $\begin{smallmatrix} U & & V \\ & \searrow \quad \swarrow & \\ & W & \end{smallmatrix}$ un diagramme d'objets 'non vides', avec W 0-connexe, il faut prouver que le produit fibré est non vide. Considérant $a \longrightarrow U$ et $b \longrightarrow V$ avec $a, b \in S$, on est ramené au cas où $U = a$, $V = b$ dans S. Prouvons que si $a \times_W b$ est 'vide', alors W est disconnexe. Pour ceci, considérons la sous-catégorie C de S/W formée des $x \longrightarrow W$ tels que $a \times_W x$ soit 'non vide'. Désignant par $w(a)$, $w(x)$ l'image de a, x dans W, cela signifie que $w(a) \cap w(x) \neq \emptyset$.

Notons que tout quotient dans $\mathcal{X}^\wedge$ d'un objet irréductible est irréductible, donc $w(a)$, $w(x)$ sont des objets irréductibles. Cela montre que pour x dans C et x' dans S/W, on a x' dans C, i.e. $w(a) \cap w(x') \neq \emptyset$ si et seulement si $w(x) \cap w(x') \neq \emptyset$. (Lemme direct : dans un objet W d'un topos $\mathcal{X}$, si on a [des] sous-objets W_1, W_2, W_3 irréductibles, si l'un des deux [plutôt trois] rencontre les deux autres, ceux-ci se rencontrent.) Cela montre que C est un crible, puisque $x' \longrightarrow x$ implique $w(x') \subseteq w(x)$, donc $w(x')$ rencontre $w(x)$, donc aussi $w(a)$. Mais alors le co-crible $C' = (S/W) \backslash C$ est également un crible, car si c est dans C', i.e. [si] $w(c)$ ne rencontre pas $w(x)$, a fortiori pas tout x' au-dessus de x, pour lequel on aura $w(x') \subseteq w(x)$. Ainsi, S/W est décomposé en somme de deux ouverts C, C', non vides puisque l'un contient a, l'autre b. Par suite, W est décomposé en somme de deux sous-objets non vides, donc W est non 0-connexe. [23]

p. 15

On a prouvé 2.8, donc aussi (revenant à 2.6) les implications vers PS 1 dans le diagramme page 20. Il reste à prouver

$$\begin{cases} \text{b} \Longrightarrow \text{PS 2},\ \text{c}' \Longrightarrow \text{PS 2} \\ \text{PS 1} + \text{PS 2 (i.e. d)} \Longrightarrow \text{a}' \end{cases}$$

([32]).

Commentary · Exposition [ms. 23]

How irreducible generators control an arbitrary slice

The proof of 2.8 explains why irreducibility, although weaker than filteredness, is genuinely local in the topos. The missing hypothesis in its printed condition (b) is that W be connected. The assertion used in the proof is (b'): noninitial U and V over a connected W have noninitial pullback.

Here is the passage from the generating objects to such a W. Suppose every $s \in S$ is irreducible, and consider the images $W_s \subseteq W$ of all maps $s \to W$. These images cover W and remain irreducible: inverse images of two nonzero subobjects under an epimorphism are nonzero, and their nonzero intersection has nonzero image. On this covering family, declare two images related when they meet. The relation is reflexive and symmetric. It is transitive because, if A and C both meet an irreducible B, the two nonzero subobjects $A \cap B$ and $B \cap C$ of B meet. Thus the equivalence classes give pairwise disjoint unions of subobjects covering W.

These unions are a coproduct decomposition in the topos. Since W is connected, there is only one class. Every pair of generating images therefore meets. Choose a generator mapping into each of the noninitial objects U and V. Their images in W meet, and epimorphisms remain epimorphisms after pullback; consequently $U \times_W V$ is noninitial. This proves the step (c)⇒(b') without assuming that every object of a site is representable as a presheaf on that site.

Finally, the condition that $s \times_x t$ be noninitial is equivalent to the existence of a generator r mapping to that pullback. Such a map is exactly a commutative square completing $s \to x \leftarrow t$, namely PS 1. This proves the equivalence with (d) and (e). The implication (b')⇒(a) follows by taking two nonzero subobjects of a connected object. The remaining implications are restrictions of this same test. Grothendieck's argument is therefore sound after restoring the condition on the base; the diagram of implications is not concealing an additional homotopical assumption.

[24] Nous nous plaçons donc dans l'hypothèse où PS 1 est satisfait (puisque b), c') et d) impliquent PS 1), donc où on est sous les conditions équivalentes de 2.8 - $\mathcal{X}$ est localement irréductible. Considérons alors la condition PS 2, et prouvons qu'elle est impliquée tant par b), que par c'), et qu'elle implique a'). Elle signifie que pour une double flèche $a \overset{u}{\underset{v}{\rightrightarrows}} b$ dans S, le conoyau `[plutôt noyau]` dans $S^\wedge$, ou, ce qui revient au même, dans $\mathcal{X}^\wedge = S^{\widetilde{}} \subseteq S^\wedge$, est 'non vide'. On peut l'écrire sous la forme d'un produit fibré

$$\begin{array}{ccc} a & \xrightarrow{(u,v)} & b \times b \\ \uparrow & & \uparrow{\scriptstyle \text{diag}} \\ \text{Ker}(u,v) & \longrightarrow & b. \end{array}$$

Ici par hypothèse a et b sont des objets 0-connexes de $\mathcal{X}^\wedge$ (ceux de S l'étant), c) est vérifié (($\mathcal{X}$ localement irréductible et) ses composantes connexes `[sont]` totalement 0-connexes), alors $b \times b$ est 0-connexe comme produit de deux objets 0-connexes sur une *même composante connexe,* laquelle est totalement 0-connexe. Comme $\mathcal{X} \times \mathcal{X}$ est localement irréductible, il s'ensuit que le produit fibré est non vide (2.8), donc b) ⟹ d). Prouvons d) ⟹ a), i.e. si

[25] S est localement cofiltrante, alors les topos induits $\mathcal{X}/W$, où W est 0-connexe, sont totalement 0-connexes. Ce topos induit est décrit par le site S/W, i.e. S/W est un système de générateurs. Or si S est localement filtrante, S/W (étale sur S) l'est aussi. Donc on peut supposer $W = e$, et à prouver que si S est localement cofiltrante et $\mathcal{X}$ 0-connexe, alors $\mathcal{X}$ est

32. NB finalement on ne prouve pas c' ⟹ PS 2.

totalement 0-connexe. Notons que $\mathcal{X}$ 0-connexe équivaut à S 0-connexe - il faudrait quand même l'expliciter :

Lemme 2.9. *Soit $\mathcal{X}$ un topos, $S \subseteq \mathcal{X}^\wedge$ une sous-catégorie pleine génératrice formée d'objets* 0-connexes. *Alors $\mathcal{X}$ est* 0*-connexe si et seulement si S l'est - plus généralement, l'ensemble ordonné des sous-objets à la fois ouverts et fermés de S dans* Cat *(ou encore des sous-objets ouverts = cribles de S qui sont aussi fermés, i.e. des cocribles) est isomorphe à* p. 16 *celui des sommandes directs de $\mathcal{X}$.*

Je l'ai utilisé tacitement dans la démonstration de 2.8, pour le topos induit $\mathcal{X}/W$. Je n'explicite pas la démonstration.

Commentary · Exposition [ms. 25]

Connectedness in a category of generators

Lemma 2.9 is what permits the categorical and topos-theoretic arguments to meet. Its content is stronger than the statement that a generating family covers the terminal object: a partition of the generators must descend to an actual coproduct decomposition of that object.

Let $S \subseteq \mathcal{E}$ be full, generating, and formed of connected objects. A decomposition $1 = A \amalg B$ assigns every $s \in S$ entirely to one summand, since the two pullbacks decompose s. Every arrow preserves this assignment. It thus gives a partition $S = S_A \amalg S_B$ into full subcategories with no arrows between them. Conversely, suppose such a partition is given. Let A and B be the unions in 1 of the images of the objects in the respective parts. They cover 1, because S generates. If $A \cap B$ were noninitial, some generator would map into it. Pulling back the epimorphic families defining A and B, and choosing a generator in a noninitial member, would produce an object $t \in S$ with maps to an object of S_A and one of S_B. This contradicts the partition. Hence $A \cap B = 0$, and $A \amalg B \simeq 1$.

The two constructions are inverse: a connected generator whose image lies in A cannot belong to the other part, and the images of all generators assigned to A jointly cover A. The same argument works with any set of parts and proves

$$\pi_0(S) \simeq \pi_0(1_{\mathcal{E}}).$$

It also applies to S/F, which is a full generating subcategory of $\mathcal{E}/F$. An object $s \to F$ is connected in the slice exactly when its domain s is connected, since coproducts in a slice are computed on domains. We obtain the repeatedly used identity

$$\pi_0(S/F) \simeq \pi_0(F).$$

This supplies the proof omitted at manuscript p. 25 and justifies the earlier use of the lemma in 2.8. It also explains why the category of elements, rather than only the set of objects of a site, controls connectedness.

Ainsi (revenant à la démonstration de d) $\Longrightarrow$ c) dans 2.6.) S étant 0-connexe et localement cofiltrante,

`[elle]` est en fait cofiltrante. Donc pour deux objets a, b de S, $\exists$ c avec $c \longrightarrow a$, $c \longrightarrow b$ [26] (condition PS 1′, plus forte que PS 1). Pour prouver que $\mathcal{X}$ est totalement 0-connexe, on est réduit à prouver par 2.3 que les $a \times b$ dans $\mathcal{X}^\wedge$ sont 0-connexes. Par 2.9, il revient au même de prouver qu'ils sont 0-connexes dans $S^\wedge$. Donc on est ramené à prouver le

Lemme 2.10. *Soit S une catégorie cofiltrante. Si a, b sont deux objets de S, $a \times b$ est un objet* 0*-connexe de $S^\wedge$.*

On voit de suite qu'il suffit de prouver (et en fait il revient au même, compte tenu que $a \times b$ est localement irréductible) que si on a deux flèches

$$\begin{array}{ccc} x & \longrightarrow & a \times b \\ & & \uparrow \\ & & y, \end{array}$$

alors $x \times_{a\times b} y \neq \emptyset$, i.e. $\exists\ z$ dans S rendant commutatif

$$\begin{array}{ccc} x & \longrightarrow & a \times b \\ \uparrow & & \uparrow \\ z & \longrightarrow & y. \end{array}$$

Or par PS 1′ il existe bien un z s'envoyant dans x et dans y, mais les deux flèches composées

$$z \rightrightarrows a \times b,$$

i.e. chacune des double flèches

$$(*) \qquad z \rightrightarrows a$$

$$(*') \qquad z \rightrightarrows b$$

peuvent n'être pas égales. Mais par PS 2,
on peut trouver

$$z' \longrightarrow z \qquad [27]$$

égalisant la double flèche $(*)$, donc on peut supposer que celle-ci est déjà triviale. Appliquant encore une fois PS 2, par un $z'' \longrightarrow z'$, on égalise de même la double flèche $(*')$, et on gagne. p. 17
On a donc prouvé, finalement,

$$\begin{array}{ccccccc} \mathrm{a} & \Longleftrightarrow & \mathrm{a}' & \Longrightarrow & \mathrm{b} & \Longleftrightarrow & \mathrm{d}\ (= \text{PS } 1 + \text{PS } 2) \\ \Downarrow & & & & & & \\ \mathrm{c} & \Longleftrightarrow & \mathrm{c}', & & & & \end{array}$$

mais pas l'implication inverse. En d'autres termes, si les topos induits $\mathcal{X}/a$, pour a dans S, sont totalement 0-connexes, je n'ai pas prouvé qu'il en est de même des $\mathcal{X}/W$, W un objet 0-connexe *quelconque* de $\mathcal{X}^\wedge$. Ainsi, on en est au point suivant

Commentary · Exposition [ms. 27]

Where both equalizations are used

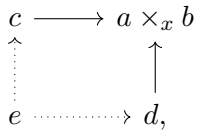
The completion of 2.6 is a useful instance of the finite-cone argument underlying filteredness. It is not enough to find a common predecessor of two objects: one must also arrange that the resulting maps agree in both factors. This is why the proof of Lemma 2.10 invokes PS 2 twice.

Suppose S is cofiltered. An object of $S/(h_a \times h_b)$ is a triple $(x, r : x \to a, s : x \to b)$. It exists by the common-predecessor axiom. Given two such triples with domains x and y, choose $z \to x$ and $z \to y$. Equalize the two resulting maps $z \rightrightarrows a$ by a map $z' \to z$, then equalize the two maps $z' \rightrightarrows b$ by $z'' \to z'$. The second operation preserves the first equality. The object with domain z'' is a common predecessor in $S/(h_a \times h_b)$. This proves its connectedness, as required in 2.10. In fact it is cofiltered: parallel arrows in it can be equalized after projection to S, and the resulting arrow still belongs to the category of elements.

Now let S merely be locally cofiltered, and let W be connected in the topos. The projection $S/W \to S$ is a discrete fibration. PS 1 lifts along a discrete fibration: once a common predecessor is chosen downstairs, lifting one leg determines the object upstairs, and equality of the two composites gives the other leg. PS 2 lifts for the same reason, using uniqueness of lifts. Thus S/W is locally cofiltered. Lemma 2.9 says it is connected, so it is cofiltered. Applying the preceding argument in this generating category and then the generator test of 2.3 shows that $\mathcal{E}/W$ is totally 0-connected. This is (d)⇒(a).

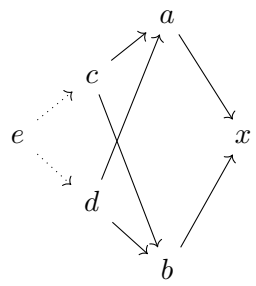
The proof establishes the announced equivalence of (a), (a'), (b), and (d). It does not establish (c')⇒PS 2, and the annotation correctly removes that assertion. Restricting the base of a pullback to a representable object has lost information: the object $b \times b$ required for the diagonal argument need not be among the chosen generators. The following group example makes this loss explicit.

Perplexité 2.11. *Soit X dans* Cat. *Si X est localement filtrante à gauche, alors les produits fibrés $a \times_x b$ d'objets dans X sont des objets 0-connexes de $X^\wedge$, et cette condition à son tour implique PS 1, i.e. que X est localement irréductible. Mais en présence de cette hypothèse PS 1, ou X localement irréductible, la condition sur les produits fibrés $a \times_x b$ équivaut à la suivante : pour tout diagramme* `[en]` *traits pleins* [28]



avec x, a, b, c, d dans X,

on peut compléter en un diagramme commutatif, *avec e dans X. En d'autre termes, tout diagramme commutatif à traits pleins ci-dessous se complète en y rajoutant un 'objet initial' e*



([33]).

Cette condition est donc impliquée par X localement cofiltrante, et elle est satisfaite aussi par ce qui précède plus généralement si et seulement si ([34]) les X/x sont cofiltrantes ([35]). Mais la réciproque est-elle vraie - cette condition de totale 0-connexité des X/x implique-t-elle que X soit totalement 0-connexe ? Mais je n'ai pas construit d'exemple où les X/x sont cofiltrantes sans que X soit localement cofiltrante (ou ce qui revient au même, de X 0-connexe avec les X/x cofiltrantes sans que X en soit cofiltrante)([36]) ; ni d'exemple où les X/x sont totalement 0-connexes, sans que les X/x soient cofiltrantes. Mais notons le p. 18

Commentary · Examples [ms. 28]

A groupoid invisible to its representable slices

The marginal answer to Perplexity 2.11 is decisive, and it also explains the weaker condition that remains. Let G be a nontrivial group and let BG be its one-object category. The objects of $BG/*$ are the elements $g \in G$, viewed as arrows to the unique object. For any g, h, there is exactly one arrow from g to h, namely the element k satisfying $hk = g$. Hence $BG/*$ is a nonempty indiscrete groupoid, equivalent to the terminal category, and is cofiltered.

The category BG itself is not cofiltered. If $g \neq h$ are parallel arrows, an equalization $gk = hk$ would imply $g = h$ because k is invertible. Its presheaf topos is the category of right G-sets. The representable is the regular G-set G; it is connected. Its square with diagonal action has

$$\pi_0(G \times G) \simeq G, \qquad [(a, b)] \longmapsto ba^{-1},$$

33. Il suffit d'exiger pour e la commutativité des deux carrés (e c b d) et (e d a c).

34. si et seulement si ?

35. Elle signifie que X satisfait PS 1, et la condition

PS′ 2 Tout diagramme commutatif

$$a \rightrightarrows b \longrightarrow x$$

s'insère dans un diagramme commutatif

$$y \longrightarrow a \rightrightarrows b \longrightarrow x.$$

36. Mais il y a un *contre-exemple* évident, si X est un groupoïde 0-connexe, les X/x sont $\approx e$, donc cofiltrantes, mais X n'est cofiltrant (i.e. `[mot(s) illisible(s)]`) que si X est 1-connexe. P.ex. si C est un groupoïde $\neq 1$, alors $X = B_G$ est tel que les X/x sont cofiltrantes, mais X non cofiltrant.

after choosing the corresponding convention for the right action. Thus the square is not connected. This verifies directly that the topos is locally irreducible but is not totally 0-connected, although the slice over the representable is equivalent to sets. The image of the regular object onto the terminal G-set is exactly the quotient missed by the generator-only test.

There is, however, no further example in which *every* X/x is totally 0-connected but some X/x is not cofiltered. Indeed, total 0-connectedness of all X/x makes all these categories irreducible, so X is locally irreducible. For $y \to x$, the slice $(X/x)/(y \to x)$ is isomorphic to X/y, and is therefore irreducible. Hence X/x is both totally 0-connected and locally irreducible. Corollary 2.7, whose proof has just been completed, makes it cofiltered.

In elementary terms, the surviving weaker condition is PS 1 together with

$$a \underset{v}{\overset{u}{\rightrightarrows}} b \xrightarrow{h} x, \quad hu = hv \quad \Longrightarrow \quad \exists\, t : c \to a \text{ with } ut = vt.$$

This is PS′ 2 in footnote 35. It equalizes parallel arrows only when they already become equal after a common outgoing arrow. In a groupoid that antecedent forces equality, so the condition is automatic. PS 2 has no such antecedent. The two questions in the paragraph are therefore settled differently: the first has the groupoid counterexample already supplied by Grothendieck; the second has a negative existence answer from his preceding theorem.

Corollaire 2.12 : *Les conditions équivalentes a), a′), b), d) (à l'exclusion de c), c′), plus faibles apparemment) du théorème 2.6*

équivalent encore à la suivante [29]

e) ∀ *x dans S, le topos induit* $\mathcal{X}/x$ *et le topos induit* $\mathcal{X}/x \times x$ *est irréductible* ([37]),

où encore à la suivante

e′) ∀ *x dans S,* $S^\wedge/x$ *et* $S^\wedge/x{\times}x$ *sont irréductibles, i.e.* S/x *et* $S/x{\times}x$ *sont irréductibles* ([38]).

C'est la forme 'topologique' la plus faible en apparence que je peux trouver pour la condition S localement cofiltrante, i.e. $\mathcal{X}$ (ou Top(S), au choix) localement totalement 0-connexe.

Commentary · Exposition [ms. 29]

Why the square of one generator detects equalizers

Corollary 2.12 is not merely another presentation of a long list of equivalent conditions. It isolates the diagonal as the part of the product test which forces equalization. We can read its criterion entirely in the topos, without treating the chosen site as a space.

Suppose that $s \times s$ is irreducible for every $s \in S$. The diagonal $s \hookrightarrow s \times s$ is a noninitial subobject. Every noninitial subobject of an irreducible object is irreducible, so each s is irreducible and 2.8 gives PS 1. For two maps $u, v : a \rightrightarrows b$ in S, the image of $(u, v) : a \to b \times b$ is noninitial. It meets the diagonal, because $b \times b$ is irreducible. Pullback stability of epimorphisms then says that

$$\mathrm{Eq}(u, v) = a \times_{b \times b} b$$

is noninitial. A generator mapping into this equalizer supplies PS 2. Conversely, PS 1 and PS 2 give local cofilteredness; the category of generators over $b \times b$ is cofiltered by the proof of 2.10, so this product is irreducible.

This also proves the comment that the separate hypothesis on $\mathcal{E}/s$ is redundant. The diagonal is a nonzero open subobject in the language of slice topoi, and irreducibility passes to such an open. Saying instead that s is irreducible and its diagonal is dense gives the same test: density means that the diagonal meets the image of every noninitial object mapping to $s \times s$. The pair (u, v) then gives the equalizer test just used. The relevant "open immersion" is the morphism of topoi associated with a monomorphism in $\mathcal{E}$; it is not an assertion that an arbitrary topological diagonal is open.

37. NB $\mathcal{X}/x$ inutile.

38. S/x inutile, car $x \xrightarrow{\text{diag}} x \times x$ est une immersion ouverte non vide, donc si $x \times x$ est irréductible, x l'est aussi. Mais il revient au même de dire que les S/x sont irréductibles (condition PS* 1, cf. 2.8, p. 21) *et* que les $x \xrightarrow{\text{diag}} x \times x$ sont dominants (ce que implique en effet que $x \times x$ est irréductible). Ceci redonne immédiatement la forme PS* 2 de la condition cofiltrante.

Definition 2.13 packages these equivalent conditions as local total 0-connectedness. The generator category is allowed to vary: once the condition is expressed for all connected slices of the topos, the proof of 2.6 applies to every full generating subcategory of connected objects.

Définition 2.13. On dit qu'un topos $\mathcal{X}$ est *localement totalement* 0-*connexe* s'il satisfait aux conditions équivalentes a, a′, b, d du théorème 2.6, i.e. si $\mathcal{X}$ est localement connexe et si de plus tout topos induit 0-connexe est totalement 0-connexe - le produit fibré de deux objets 0-connexes sur un troisième est 0-connexe. Il revient au même de dire que toute (ou que une) sous-catégorie pleine génératrice S de $\mathcal{X}^\wedge$ formée d'objets 0-connexes est localement cofiltrante (en supposant toujours $\mathcal{X}$ localement connexe, ou ce qui revient au même, l'existence d'un S formé d'objets 0-connexes).

On n'a pas terminé encore la démonstration de 2.6 - il reste à prouver que si $\mathcal{X}$ (supposé localement totalement 0-connexe) est 0-connexe, il est ∞-connexe - ou encore que les composantes connexes de $\mathcal{X}$, supposé localement totalement 0-connexe, sont ∞-connexes, i.e. W_∞-asphériques.

Corollaire 2.14. *Notons quand même que si $\mathcal{X}$ est de la forme Top(T), T un espace topologique, alors $\mathcal{X}$ est localement totalement* 0-*connexe si et seulement si il est localement irréductible, i.e. si et seulement si T est localement irréductible.* [30]

On est ramené à prouver que cette dernière condition implique la condition d) de 2.6, et p. 19
on peut prendre pour S l'ensemble ordonné des ouverts connexes non vides de S, cela devient alors évident.

Ceci dit, il faut se rappeler que la théorie cohomologique des espaces topologiques irréductibles n'est nullement triviale - comme nous le montre FAC de Serre ! D'autre part, c'est Serre qui m'avait fait observer que pour des coefficients *constants,* la cohomologie d'un espace irréductible est triviale.

Commentary · Examples [ms. 30]

Irreducible spaces still have nonconstant cohomology

The reference to Serre separates two coefficient questions which must remain distinct. On an irreducible nonempty topological space T, every nonempty open is connected. For an abelian group M, the constant sheaf $\underline{M}$ consequently has sections M on every nonempty open, with identity restriction maps. Restrictions to the empty open are also surjective. Thus $\underline{M}$ is flabby and

$$H^q(T, \underline{M}) = 0 \quad (q > 0), \qquad H^0(T, \underline{M}) = M.$$

The same argument on each irreducible component applies when the space is a disjoint union of irreducible open components, which is the local condition used in 2.14.

For a concrete contrast, take the Zariski space $T = \mathbf{P}^1_k$, which is irreducible, and the line bundle $\mathcal{O}(-2)$. On the standard affine charts with coordinate t and t^{-1}, its degree-one Čech quotient is

$$\frac{k[t, t^{-1}]}{k[t] + t^{-2}k[t^{-1}]} = k\,[t^{-1}].$$

The first summand removes all nonnegative powers and the second all powers at most -2; exactly the coefficient of t^{-1} remains. The affine charts and their intersection are acyclic for this coherent sheaf, so the quotient is $H^1(\mathbf{P}^1_k, \mathcal{O}(-2)) \simeq k$. The affine-cover comparison used here is Serre (1955, no. 47, Theorem 4); the displayed computation gives the particular example in full.

For 2.14 itself, choose the connected nonempty opens as generators. If the space is locally irreducible in Grothendieck's sense, each such open is irreducible. Two nonempty opens inside it meet, giving PS 1; the generator category is a poset, so parallel arrows are already equal and PS 2 imposes nothing further. The converse follows from local total 0-connectedness implying local irreducibility. This is why the two notions coincide for spaces while the group example separates them for general topoi.

Dans le cas actuel, prouvons déjà, si $\mathcal{X}$ est 0-connexe et localement totalement 0-connexe, que tout revêtement de l'objet final de $\mathcal{X}$ est trivial. Il suffit de voir que si U est 0-connexe,

$U \longrightarrow e$ est un isomorphisme. Mais considérons $U \hookrightarrow U \times U$, on sait que c'est un sommande direct, et comme $U \times U$ est connexe, c'est un isomorphisme. Donc $U \longrightarrow e$ [est un] monomorphisme, donc sommande direct, et comme $U \neq \emptyset$, on a $U \longrightarrow e$ isomorphisme.

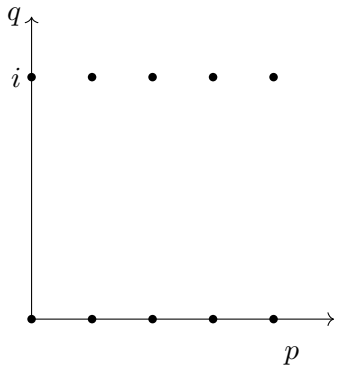
Reste à prouver que pour tout faisceau abélien constant F sur $\mathcal{X}$, on a $H^i(X, F) = 0$ [31] pour $i > 0$. On procède par récurrence sur i, c'est déjà fait pour $i = 1$. Supposons que ce soit prouvé pour les $j < i$ (i fixé) pour tout topos *totalement* 0-*connexe.* Soit $\xi \in H^i(\mathcal{X}, F)$, soit $U \longrightarrow e$ [un] épimorphisme tel que $\xi|U = 0$, et considérons la suite spectrale

$$E_2^{pq} = H^p(C(U^*, \mathcal{H}^q(F))) \Longrightarrow H^*(\mathcal{X}, F).$$

On a par hypothèse totalement 0-connexe

$$(*) \qquad \pi_0(U^q) \simeq \pi_0(U)^q,$$

et les composantes connexes de U^q sont totalement 0-asphériques,



donc par hypothèse de récurrence

$$E_1^{pq} = 0 \qquad \text{si } 0 < q < i,$$

donc on trouve une suite exacte

$$0 \longrightarrow H^i(C(U^*, F)) \longrightarrow H^i(\mathcal{X}, F) \longrightarrow H^0(\underbrace{H^i(U^*, F)}_{\substack{\text{complexe} \\ \text{co-semisimplicial} \\ \text{pour } i \\ \text{variable}}}) = \mathrm{Ker}(H^i(U, F) \rightrightarrows H^i(U^2, F)).$$

Commentary · Exposition [ms. 31]

The hereditary hypothesis in the cohomological proof

Grothendieck's induction proves the intended assertion after keeping its hypothesis fixed: the topos must be connected and *locally* totally 0-connected. The word "localement" disappears in the sentence introducing the induction at manuscript p. 31, but it is needed when the argument passes to components of powers of a covering object. Those components define connected slices of the original topos, and hence remain in the same class by 2.6. Mere total 0-connectedness does not provide this inheritance.

First consider a connected locally constant object L. Its diagonal in $L \times L$ is complemented, as one checks after a cover trivializing L. The product is connected, so this noninitial summand is the whole product. Hence $L \to 1$ is monic. Its image is also complemented, by the same local triviality; connectedness of the base and nonemptiness of L give $L \simeq 1$. Every covering object is therefore a coproduct of copies of 1. In the source paragraph, U must still denote a connected component of a *covering*, not an arbitrary connected sheaf.

For the higher cohomology, assume the vanishing in degrees $0 < q < n$ for every connected topos in this hereditary class. Given $\xi \in H^n(\mathcal{E}, \underline{M})$, choose an epimorphism $U \to 1$ killing ξ. Such a cover

exists because a positive-degree cohomology class represented in an injective resolution becomes a boundary locally. The augmented Čech nerve gives the descent spectral sequence

$$E_1^{pq} = H^q(\mathcal{E}/U^{p+1}, \underline{M}) \implies H^{p+q}(\mathcal{E}, \underline{M}).$$

Each U^{p+1} is a coproduct of connected objects whose slices remain in the class; thus the rows $0 < q < n$ vanish. The edge sequence gives

$$0 \longrightarrow H^n(C^\bullet) \longrightarrow H^n(\mathcal{E}, \underline{M}) \longrightarrow \ker\big(H^n(U, \underline{M}) \rightrightarrows H^n(U^2, \underline{M})\big).$$

The last image of ξ is zero by choice of U. If $I = \pi_0(U)$, product preservation identifies

$$C^p = \mathrm{Map}(I^{p+1}, M).$$

Choose $i_* \in I$, which exists since $U \to 1$ is epic. Inserting i_* as first coordinate defines $(sc)(i_0, \dots, i_{p-1}) = c(i_*, i_0, \dots, i_{p-1})$. For the alternating deletion differential, direct cancellation gives $ds + sd = 1$ in positive degrees, with the usual augmentation in degree zero. Thus $H^n(C^\bullet) = 0$ and $\xi = 0$. The "trivial" complex in the manuscript is this contractible augmented complex; its individual groups and differentials are not zero.

p. 20

Par hypothèse sur U, l'image de ξ dans le troisième terme est nulle, donc ξ appartient au premier terme. Mais pour $(*)$, le complexe $C(U^*, F)$ est le complexe *trivial* des puissances de $I = \pi_0(U)$, à coefficients dans M (si $F = M_{\mathcal{X}}$). Sa cohomologie en degrés > 0 est nulle, donc [32] $\xi = 0$, q.e.d.

Commentary · Later developments [ms. 32]

From connected components to the hypercover homotopy type

The stronger homotopical conclusion of 2.6 can also be proved directly, without deducing weak contractibility from an unspecified version of a cohomological Whitehead theorem. On a connected locally totally 0-connected topos, the component functor preserves finite limits, as shown above, and preserves epimorphisms: epimorphisms in a topos are regular, and the left adjoint π_0 preserves their coequalizer presentations. Let $U_\bullet \to 1$ be a hypercover by objects of this topos. Its matching objects are finite limits. Therefore

$$\pi_0\big(M_n(U_\bullet)\big) \simeq M_n\big(\pi_0(U_\bullet)\big),$$

and the hypercover epimorphisms become surjections of sets. The simplicial set $\pi_0(U_\bullet)$ consequently has a filler for every boundary $\partial\Delta^n \to \pi_0(U_\bullet)$, including the nonemptiness condition in degree zero. Its map to a point is a trivial Kan fibration, and it is weakly contractible.

In the hypercover construction of the homotopy type, whose classical setting is Artin and Mazur (1969, §§8–9), every stage is therefore weakly contractible. This proves the W_∞-asphericity meant here, as well as the corresponding assertion for each connected object by passage to its slice. For a presheaf topos it says concretely that every connected category of elements is weakly contractible.

There is also a short algebraic verification of the constant-coefficient calculation. Since π_0 preserves finite limits, it induces an exact functor on abelian group objects, left adjoint to the constant-abelian-sheaf functor. The latter therefore preserves injectives. Applying it to an injective resolution of M, and then taking global sections, recovers the same resolution of M, because the topos is connected. Thus $R\Gamma(\mathcal{E}, \underline{M}) \simeq M[0]$. These proofs make explicit why the hereditary condition repairs the false inference after 2.3. They impose no claim that the topos itself is equivalent to sets; the preceding projective-line example shows that arbitrary coefficients retain substantial information.

Proposition 2.15. ([39]) *Soit X un objet de* Cat. *Conditions équivalentes :*

a) X est filtrant, i.e. X est 0-connexe et X^o satisfait les conditions équivalentes du théorème 2.6 ($X^{o,\wedge}$ est localement totalement 0-connexe).

b) L'objet final de $X^\wedge$ est ind-représentable.

39. C'est déjà dans SGA 4, I 8.

Démonstration. On sait qu'on a

$$e_{X^\wedge} \simeq \varinjlim_{x\in X} x, \tag{2.15.1}$$

limite inductive prise dans $X^\wedge$. En effet, cela signifie aussi que pour tout objet a dans X, on a

$$\varinjlim_{x\in X} \underbrace{x(a)}_{\mathrm{Hom}_X(a,x)} \xrightarrow{\sim} e \qquad \text{(ensemble ponctuel)}$$

Mais considérons l'objet $X' = a\backslash X \longrightarrow X$ coétale sur X, qui est donc Cat-cofibré sur X, on sait que l'on a

$$\pi_0(X') \simeq \varinjlim_{x\in X} \pi_0(X'_x),$$

où $X'_x = \mathrm{Hom}(a, x)$, donc on trouve

$$\varinjlim_{x\in X} \mathrm{Hom}(a, x) = \pi_0(a\backslash X) = e$$

car $a\backslash X$ est 0-connexe.

Commentary · Exposition [ms. 32]

The terminal presheaf as an ind-object

The passage to ind-objects replaces the cone conditions by a representability statement. The general form, recorded in 2.18, is the familiar criterion from Artin et al. (1972, Exposé I, Theorem 8.3.3): a presheaf F on a small category X is a filtered colimit of representables if and only if its category of elements X/F is filtered. We give the argument, since it also supplies all the details of 2.15.

For every presheaf there is the density presentation

$$F \simeq \operatorname*{colim}_{(x,s)\in X/F} h_x.$$

Objectwise, the element represented by a pair $a \to x$, $s \in F(x)$ maps to its pullback in $F(a)$; it has the representative (a, t) with identity map, and the relations identify every other representative with this one. If X/F is filtered, this is the required ind-presentation.

Conversely, write $F = \mathrm{colim}_{i\in I} h_{x_i}$ with I filtered. The category of elements is nonempty, since the image of 1_{x_i} supplies an object. Two elements $s \in F(a)$ and $t \in F(b)$ have representatives at stages i, j; passage to a common stage k supplies arrows from (a, s) and (b, t) to $(x_k, \overline{1}_{x_k})$. Given parallel arrows $u, v : (a, s) \rightrightarrows (b, t)$, represent t by $b \to x_i$. The two composites $a \rightrightarrows x_i$ have the same image in $F(a)$. Equality in a filtered colimit of sets is witnessed at a common later stage. The resulting map $(b, t) \to (x_j, \overline{1}_{x_j})$ equalizes u, v. Thus X/F is filtered.

For $F = 1$, its category of elements is X itself, giving 2.15. The formula $1 = \mathrm{colim}_{x\in X} h_x$ is valid even when X is not filtered: at a its value is $\pi_0(a\backslash X) = \{*\}$ because $(a, 1_a)$ is initial. The condition lies in the possibility of a *filtered* presentation. The reindexing by a directed poset used in the manuscript is Deligne's result, Artin et al. (1972, Exposé I, Proposition 8.1.6); the proof above does not need this extra reindexing.

La formule (2.15.1) montre que si X est filtrant, $e_{X^\wedge}$ est limite inductive filtrante de foncteurs représentables. On sait d'ailleurs qu'on peut réindexer par une catégorie *ordonnée* filtrante (cf. SGA 4, I 8.1.6 (c'est dû à P. Deligne !)). [33]

On a prouvé a $\Longrightarrow$ b, prouvons b $\Longrightarrow$ a. Supposons donc

$$e_{X^\wedge} = \varinjlim_{i\in I} x_i,$$

où I est un (petit) ensemble ordonné filtrant, et $I \longrightarrow X$ un foncteur. Prouvons PS 1′ (forme droite), i.e. pour a, b dans X, `[il]` existe $\begin{matrix} a & \searrow & \\ & & x \\ b & \nearrow & \end{matrix}$. En effet, comme $\varinjlim_I \mathrm{Hom}(a, x_i) = e$, il

s'ensuit que $\mathrm{Hom}(a, x_i) \neq \emptyset$ pour i grand, de même $\mathrm{Hom}(b, x_i) \neq \emptyset$, donc il suffit de prendre $\begin{matrix} a & \searrow \\ & x_i \\ b & \nearrow \end{matrix}$. Prouvons PS 2_{d}, i.e. que pour `[une]` double flèche $a \overset{u}{\underset{v}{\rightrightarrows}} b$, `[il]` existe $b \longrightarrow x$ qui égalise.

Soit ξ_{i_0} dans $\mathrm{Hom}(b, x_{i_0})$, ses deux images dans $\mathrm{Hom}(a, x_{i_0})$ ne sont pas nécessairement égales, mais le deviennent dans la limite inductive, réduite à e, donc deviennent égales en composant avec $x_{i_0} \longrightarrow x_i$ pour i assez grand, donc le composé $b \longrightarrow x_{i_0} \longrightarrow x_i$ p. 21 [34]

$$a \rightrightarrows b \longrightarrow x_{i_0} \longrightarrow x_i$$

égalise la double flèche.

Corollaire 2.16. *Soit W un localiseur fondamental satisfaisant l'axiome des limites* Loc (8). *Si X dans* Cat *est une catégorie filtrante ou cofiltrante, alors X est W-asphérique.*

Plus généralement et agréablement :

Commentary · Later developments [ms. 34]
Filtered categories and the minimal localizer

Corollary 2.16 needs only the standard contractibility of a filtered category, together with the comparison between its weak equivalences and the chosen localizer. Here is the finite-data argument behind the first assertion. In a filtered category, any finite collection of objects has maps into one object. Given finitely many arrows between those objects, successive applications of the parallel-arrow axiom modify the common target so that all the required triangles commute. Thus every finite diagram of objects and arrows occurring in a finite simplicial subset of the nerve admits a compatible cone.

Every map from a finite sphere into the realization of the nerve can, after simplicial approximation, be represented in a finite such subset. The cone makes that map nullhomotopic. The nerve is nonempty and connected, so all its homotopy groups vanish. Reversing the category does not change the homotopy type of its nerve, and gives the cofiltered assertion as well.

The manuscript formulates the conclusion for a fundamental localizer satisfying its axiom Loc(8). A later theorem permits a sharper modern statement: the nerve weak equivalences W_∞ form the smallest fundamental localizer. This is Cisinski (2004, Theorem 2.2.11, p. 136). With the standard modern definition of fundamental localizer, every filtered or cofiltered category is consequently W-aspherical without an additional filtered-limit axiom. This is a later strengthening of the hypothesis in the manuscript, rather than a reason to alter its historical formulation. The finite-cone argument also underlies the higher categorical characterization cited earlier: filtered colimits of spaces commute with finite limits.

Corollaire 2.17. *Soit W comme dans 2.16. Alors toute catégorie filtrante est* totalement W-asphérique ([40]), *i.e. pour toute flèche $f : X \longrightarrow Y$ dans* Cat, *l'objet $f_!(e_X)$ de e_Y (défini aussi comme $y \longmapsto \pi_0(y\backslash X)$, et intervenant dans la factorisation canonique de f en ip, avec p étale et les cofibres* `[?]` *de p 0-connexes) est W-asphérique. En fait, $Y/f_!(e_{X^\wedge})$ est même une catégorie filtrante.* [35]

En effet, le foncteur $f_!$ commute aux limites inductives, donc $f_! e_{X^\wedge}$ est limite inductive filtrante des $f_!(x_i) = f(x_i)$. Donc $Y/f_!(e_{X^\wedge})$ est limite inductive filtrante des $Y/f(x_i)$. Donc c'est même une catégorie filtrante, a fortiori elle est W-asphérique.

Donc finalement, on trouve des exemples non triviaux de catégories totalement W-asphériques, i.e. des exemples donc des catégories sans objet final.

Commentary · Exposition [ms. 35]
Left Kan extension carries the ind-presentation

40. cf. XII, §V pour cette notion, que j'ai beaucoup peiné à dégager et à comprendre !

The reason behind 2.17 is that an ordinary left Kan extension sends a representable to a representable and preserves colimits. If $f : X \to Y$ and X is filtered, then

$$f_!1_X \simeq \operatorname*{colim}_{x\in X} f_!h_x \simeq \operatorname*{colim}_{x\in X} h_{f(x)}.$$

The result is an ind-representable presheaf on Y, so $Y/(f_!1_X)$ is filtered by the criterion just proved. This gives the precise argument for the last assertion of 2.17 and therefore its asphericity conclusion. The proof does not require f to be fully faithful or cofinal.

The canonical factorization mentioned in the statement has the following form:

$$X \xrightarrow{p} Y/(f_!1_X) \xrightarrow{i} Y.$$

Here i is the discrete fibration associated with the presheaf, and

$$(f_!1_X)(y) = \pi_0(y\backslash X).$$

The first map sends x to the component represented by $(x, 1_{f(x)})$. For an object (y, c) of the category of elements, the comma category $(y, c)\backslash X$ is exactly the component c of $y\backslash X$; it is nonempty and connected. Thus p is cofinal in the terminology of these pages. The edition assigns "étale" to p, but it is the second map i which is étale; the connected cofibres belong to p. The phrase "objet $\cdots$ de e_Y" similarly means an object of $\widehat{Y}$.

For the simplest example without a terminal object, take $X = Y = \mathbf{N}$ with its usual order and $f = 1_X$. Its terminal presheaf is the increasing colimit of the representables, and its category of elements is again $\mathbf{N}$. This is filtered and weakly contractible, although no natural number is terminal. It supplies exactly the kind of example for which Grothendieck introduces the ind-object formulation. Corollary 2.18 is the same argument with an arbitrary presheaf in place of the terminal one.

Corollaire 2.18. ([41]) *Soit X dans* Cat*, F dans $X^\wedge$. Pour que F soit ind-représentable il faut et il suffit que X/F soit filtrante.*

En effet, par (2.15), X/F est filtrante si et seulement si l'objet final F de X/F provient d'un ind-objet de X/F, mais cela signifie aussi que F est la limite inductive dans $X^\wedge$ d'un ind-objet de X, i.e. que F est ind-représentable.

Question. *Si X dans* Cat *est totalement W_∞-asphérique, i.e. les $f_!(e_{X^\wedge})$ sont W_∞-asphériques pour toute flèche $f : X \longrightarrow Y$ dans* Cat*, ou encore pour toute telle flèche, avec les $y\backslash Y$ 0-connexes ($y \in \operatorname{Ob} Y$), Y est W_∞-asphérique - cela implique-t-il que X soit filtrante ?* Ça ne m'étonnerait pas - et même en supposant seulement les $f_!(e_{X^\wedge})$ 1-connexes ([42]) (ils sont 0-connexes, dès que X l'est). Mais je ne veux pas m'attarder à présent sur cette question. [36]

Commentary · Exposition [ms. 36]

The quantifier in total asphericity

Grothendieck is asking which indexing categories remain homotopically harmless after an ordinary change of index. The adjective "total" concerns every possible target of a functor; it does not merely strengthen the contractibility of the nerve of the original category. It is useful to isolate the precise condition before reading the proposed answer. Write $BA = |NA|$, and write $T(A)$ for the assertion that BE is weakly contractible whenever $A \longrightarrow E$ is an ordinary cofinal functor. Here and below "cofinal" has the orientation fixed in Section 3: each category $e\backslash A$ must be nonempty and connected.

This formulation is equivalent to the one using $f_!1_A$ on the page. Indeed, for $f : A \longrightarrow Y$, put $P = f_!1_A$, so that

$$P(y) = \pi_0(y\backslash A).$$

Let Y/P be its category of elements. There is a canonical factorization

$$A \xrightarrow{q} Y/P \longrightarrow Y, \qquad a \longmapsto (fa, [a, 1_{fa}]).$$

41. déjà dans SGA 4, I 8.

42. Ils le sont toujours (cf. XII, p. 101), c'est 2-*connexe* qu'il faut lire ici !

For an object (y, c) of Y/P, the comma category $(y, c)\backslash A$ is precisely the connected component c of $y\backslash A$. Thus q is cofinal. Conversely, if f itself is cofinal, then $P = 1_Y$ and $Y/P = Y$. Consequently $T(A)$ says exactly that the category of elements of every ordinary presheaf $f_!1_A$ is weakly contractible. The occurrence of $y\backslash Y$ in the printed question is a slip: it is $y\backslash A$ that must be connected.

A filtered category satisfies T. Its ordinary cofinal images are filtered by Corollary 3.4 below, and their nerves are weakly contractible. The difficult direction is whether this test forces the equalization of parallel arrows. The counterexample inserted on pages 36bis–36ter withdraws the attempted negative answer; it does not prove the converse. The converse holds for monoids and for categories with finitely many objects and fails in general; both are proved in the commentary "Total asphericity is not filteredness" below.

There is a terminological precaution for the later literature. In Maltsiniotis (2020, Definition 1.7.2 and Proposition 1.7.1), a category is called totally aspherical when it is aspherical and its diagonal is aspherical, equivalently when products of representable presheaves are aspherical. This is a different quantifier from $T(A)$. In particular, the fact that the simplex category is a strict test category does not settle the question on this page. We retain $T(A)$ only inside this commentary, to keep the question itself unchanged.

La réponse à la question précédente est négative, comme on voit par la :

Proposition 2.19. ([43]) *Soit X dans* Cat *telle que $diag_X : X \longrightarrow X \times X$ soit cofinal* ([44])*, i.e. telle que X^o soit totalement* 0*-connexe. Alors X est totalement W_∞-asphérique (donc aussi totalement W-asphérique pour tout localiseur fondamental W tel que $W \supseteq W_\infty$).* p. 22

Soit en effet un foncteur cofinal $f : X \longrightarrow Y$, il faut prouver que Y est W_∞-asphérique. Or on sait que totalement 0-connexe implique W_∞-asphérique ([45]) (cf. proposition 2.3, p. 17-18), donc il suffit de voir que Y^o est totalement 0-connexe, i.e. $Y \xrightarrow{d_Y} Y \times Y$ cofinal. Or on a le diagramme commutatif

$$\begin{array}{ccc} X & \xrightarrow{d_X} & X \times X \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f\times f} \\ Y & \xrightarrow{d_Y} & Y \times Y, \end{array}$$

où par hypothèse f, donc aussi $f \times f$, et d_X sont cofinaux. Donc $\varphi = (f \times f) \circ p_X$ [plutôt $\varphi = (f \times f) \circ d_X$] est cofinal, et comme $\varphi = d_Y f$ et f est aussi cofinal, on conclut que d_Y l'est, q.e.d.

Corollaire. $\mathbf{\Delta}$ *étant totalement* 0*-connexe, $\mathbf{\Delta}^o$ est totalement W_∞-asphérique, pourtant elle n'est pas filtrante.*

Question. *X totalement W_∞-asphérique $\Longrightarrow$ X^o totalement* 0*-connexe ? Ou du moins implique PS 1′ ?*

[36bis]

Contrexemple 2.20 à la proposition fausse 2.19.

Soit X un objet de Cat, rajoutons-lui la catégorie finale e comme fermé, i.e. considérons un objet X' de Cat dans lequel X soit un sous-objet ouvert, e le fermé complémentaire :

$$X \overset{i}{\hookrightarrow} X' \overset{j}{\hookleftarrow} e.$$

On sait (cf. XII) qu'une telle opération est déterminée, à isomorphisme près, par la donnée d'un objet

$$H \in \mathrm{Ob}\, X^\wedge$$

en posant

$$H(x) = \mathrm{Hom}_{X'}(x, e),$$

43. **Faux.**
44. cf. définition 3.3, p. 45.
45. C'est **faux.**

la catégorie X/e s'identifiant (à isomorphisme près) à X/H. Elle est donc 0-connexe si et seulement si l'objet H de $X^\wedge$ est 0-connexe. Cela signifie aussi que le morphisme i est cocofinal, i.e. que $i^o : X^o \longrightarrow X'^o$ est cofinal. Supposons maintenant X W_∞-asphérique et totalement 0-connexe, i.e. l'actuel Loc (6), cf. XVI. La proposition 2.19 affirme que, sous ces conditions, X' (qui est lui aussi totalement 0-connexe) est même W-asphérique. Il s'ensuivrait donc que $i \in W_\infty$. Mais par XII, th. 4, cor. 4 (p. 149), il s'ensuivrait que X/e est W_∞-asphérique, i.e. p. 23
H W_∞-asphérique. Ainsi, si X est totalement 0-asphérique, tout objet 0-connexe de $X^\wedge$ serait [36ter]
∞-connexe. Mais c'est manifestement faux, p.ex. si on prend pour X une catégorie-test pour W_∞, p.ex. $X = \mathbf{\Delta}$.

Commentary · Examples [ms. 36]

The retracted proposition and a visible sphere

The useful part of Proposition 2.19 is its elementary calculation with cofinal functors: the ordinary cofinal image of a nonempty sifted category is sifted. The unsupported step is the passage from that ordinary connectedness condition to weak contractibility. The edition marks this step and the proposition as false. The appeal in Counterexample 2.20 to Chapter XII also needs its hypothesis restored: the reflection of a weak equivalence to the attaching category uses the strong form of W(7bis), explicitly required in Grothendieck (1990–1991d, Chapter XII, Section V.3.4, Corollary 4, note 104). That reflection is not valid for arbitrary weak homotopy equivalences. We can make the intended counterexample entirely explicit without using it.

The omitted reflection condition has a concrete obstruction. Choose a finite connected acyclic noncontractible simplicial complex K; a finite presentation complex with this homotopy type is given in Hatcher (2002, Example 2.38 and Theorem 2C.5). Its suspension is simply connected and acyclic, hence contractible by the homology Whitehead theorem. The face posets of the two cones on K are open subcategories covering the face poset of ΣK, with intersection the face poset of K. The two cones and their union are weakly contractible, but their intersection is not. Thus the weak equivalence from one cone into the union does not reflect to a weak equivalence from the intersection into the other cone. This contradicts the strong biconditional W(7bis), while leaving the direct homotopy-pushout theorem intact.

Let $H = \Delta[1]/\partial\Delta[1]$, the simplicial circle, and form the category E with full subcategory Δ, one additional object z, and

$$E([n], z) = H_n, \qquad E(z, [n]) = \varnothing, \qquad E(z, z) = \{1_z\}.$$

Composition into z is the simplicial action of H. Thus $\Delta/z = \Delta/H$, the category of simplices of H. Its classifying space has the homotopy type of $|H| = S^1$; this is the usual category-of-simplices realization, or the test-category equivalence for Δ (Maltsiniotis, 2020, Proposition 1.7.14). Since Δ has terminal object [0], its classifying space is contractible.

There is a direct description of BE. An n-simplex of NE that does not lie in $N\Delta$ consists of a chain in Δ followed by an arrow into z, with possible repetitions of z. The arrow into z and all its composites are exactly a simplex of $N(\Delta/H)$. These simplices glue to the mapping cone of the forgetful map $B(\Delta/H) \longrightarrow B\Delta$. Hence

$$BE \simeq \operatorname{hocolim}\big(B\Delta \longleftarrow B(\Delta/H) \longrightarrow *\big) \simeq \Sigma S^1 \simeq S^2.$$

The cone description can also be read degree by degree: adjoining the last vertex z supplies the cone direction, and deleting that vertex applies the forgetful map. It therefore identifies the attaching maps, not merely the vertices of the two spaces.

The inclusion $i : \Delta \hookrightarrow E$ is ordinary initial: Δ/e has a terminal object when $e \in \Delta$, while $\Delta/z = \Delta/H$ is nonempty and connected. Thus $i^{\mathrm{op}} : \Delta^{\mathrm{op}} \longrightarrow E^{\mathrm{op}}$ is cofinal. The source Δ^{op} is sifted: for $[m], [n]$, its category of common targets is opposite to $\Delta/(\Delta[m] \times \Delta[n])$, which is connected because the product of the two simplices is contractible. The target E^{op} is consequently sifted as well, whereas $B(E^{\mathrm{op}}) \simeq S^2$. This supplies both a counterexample to the implication used in Proposition 2.19 and a direct counterexample to its asserted corollary about Δ^{op}.

The distinction is the same one encountered earlier in the chapter between products of connected objects and their higher homotopy. Here its failure has a specific witness, a two-dimensional sphere. It is not a failure of the composition or cancellation properties of ordinary cofinal functors.

Cela me fait soupçonner que la conjecture

$$X \text{ totalement } W_\infty\text{-asphérique} \Longrightarrow X \text{ filtrante}$$

pourrait être valable.

Commentary · Development (commentators') [ms. 36]

A necessary directedness condition

Grothendieck asks, before returning to cofinality, whether his condition at least forces the existence of common successors. This weaker question admits an elementary answer. The argument below also settles his full question for preorders. We did not find this particular implication isolated in the consulted treatments of sifted categories, asphericity, or lax epimorphisms; the following is our continuation of the argument, with no assertion of priority.

Proposition (the common-successor obstruction). [ours] *If $T(A)$ holds, then A is nonempty and any two objects of A admit arrows to a common object. Consequently, for a preorder A, the condition $T(A)$ is equivalent to filteredness.*

Proof. The identity of A is cofinal, so $T(A)$ implies that BA is weakly contractible. In particular A is nonempty and connected. Suppose that objects a, b have no common successor. Let U be the full subcategory of objects receiving an arrow from a, and let V be the full subcategory of objects receiving an arrow from b. These subcategories are disjoint and closed under passage to successors.

Let P be the three-object category with precisely the nonidentity arrows

$$0 \xrightarrow{p} 1, \qquad 0 \xrightarrow{q} 2.$$

Define $f : A \longrightarrow P$ by sending U to 1, V to 2, and all other objects to 0. Closure under successors makes this a functor: an arrow cannot leave U or V, and there are no arrows between them. The category $0\backslash A$ computed for f is A itself. The other two comma categories are U and V. They are connected, since every object of U receives an arrow from a, and every object of V receives one from b. They need not have initial objects; the existence of those arrows is all that is used. Hence f is cofinal.

We next give P a cofinal image with nonzero second homology. Adjoin an object z and arrows

$$r, s : z \rightrightarrows 0, \qquad v : z \longrightarrow 1, \qquad w : z \longrightarrow 2,$$

subject to $pr = ps = v$ and $qr = qs = w$, and no other nonidentity arrows. Call the resulting category Q. The inclusion $j : P \hookrightarrow Q$ is cofinal. For an object of P, the relevant comma category has an initial object. For z, the category $z\backslash P$ has four objects r, s, v, w and four arrows

$$r \longrightarrow v, \quad r \longrightarrow w, \quad s \longrightarrow v, \quad s \longrightarrow w;$$

it is nonempty and connected.

To see exactly what obstructs contractibility, use normalized simplicial chains of NQ. There are four nondegenerate two-simplices, denoted $(r,p), (s,p), (r,q), (s,q)$, and no nondegenerate simplices in higher dimensions. Their boundaries are

$$\partial(r,p) = p - v + r, \quad \partial(s,p) = p - v + s,$$
$$\partial(r,q) = q - w + r, \quad \partial(s,q) = q - w + s.$$

Therefore

$$c = (r,p) - (s,p) - (r,q) + (s,q)$$

is a nonzero cycle. Since there are no three-simplices, it is not a boundary. More precisely, the coefficients of p, q, r, s, v, w in the boundary show that every two-cycle is an integral multiple of c, so $H_2(BQ; \mathbf{Z}) \cong \mathbf{Z}$. The composite jf is cofinal, contradicting $T(A)$. This proves the common-successor assertion.

In a preorder, parallel arrows are already equal. Nonemptiness and common successors therefore imply filteredness. The converse follows from preservation of filteredness by ordinary cofinal images, as in Corollary 3.4. This proves the last assertion. □

For general categories the argument leaves precisely the parallel-arrow equalization problem. The distinction is substantive: a one-object group category satisfies the common-successor condition for every group, while filteredness forces that group to be trivial. The next commentary settles it: the implication $T(A) \Rightarrow A$ filtered holds when some object receives an arrow from every object, and fails for the category of finite ordinals and increasing injections.

Commentary · Development (commentators') [ms. 36]

Total asphericity is not filteredness

The question whether $T(A)$ forces A to be filtered has a negative answer in general and a positive one for monoids and for categories with finitely many objects. Both rest on a description of $T(A)$ in which no target category appears. For a functor $H : A \to \mathrm{Set}$ write $\mathrm{El}(H)$ for its category of elements: the objects are the pairs (a, h) with $h \in H(a)$, and an arrow $(a, h) \to (a', h')$ is an arrow $v : a \to a'$ with $H(v)h = h'$. The statements below were not found in the sources consulted, except where a reference is given; no priority is claimed.

Theorem (a criterion on set-valued functors). [ours] *For a small category A the following are equivalent: (i) $T(A)$; (ii) A is nonempty and connected, and for every functor $H : A \to \mathrm{Set}$ every connected component of $\mathrm{El}(H)$ has the integral homology of a point; (iii) the same with "is weakly contractible". If they hold, every connected component E of every $\mathrm{El}(H)$ satisfies $T(E)$.*

Proof. Let $K : A \to \mathrm{Set}$ have nonempty connected $\mathrm{El}(K)$, and let A_K be A with one object x adjoined, $A_K(x, a) = K(a)$, no arrows into x but its identity, and composition given by the action of A on K. For the inclusion $A \to A_K$ the category $x\backslash A$ is $\mathrm{El}(K)$, and $a\backslash A$ has an initial object for $a \in A$; so the inclusion is cofinal. A chain of arrows of A_K not contained in A consists of x (possibly repeated) followed by a chain of $\mathrm{El}(K)$. Hence $N(A_K)$ is NA with the cone on $N\,\mathrm{El}(K)$ attached along the forgetful map, and

$$\widetilde{H}_{j+1}(BA_K) \longrightarrow \widetilde{H}_j(B\,\mathrm{El}(K)) \longrightarrow \widetilde{H}_j(BA)$$

is exact. Under (i) both BA and BA_K are weakly contractible, so $\mathrm{El}(K)$ has the homology of a point. A component of $\mathrm{El}(H)$ for an arbitrary H is $\mathrm{El}(K)$ for the subfunctor K of the elements lying in it. This proves (i)⇒(ii).

(ii)⇒(iii). Let E be a component of $\mathrm{El}(H)$ and $L : E \to \mathrm{Set}$ a functor. Then $\mathrm{El}(L) \cong \mathrm{El}(H')$ for the functor $H'(a) = \coprod_h L(a, h)$ on A, the sum over the $h \in H(a)$ with $(a, h) \in E$. So E again has property (ii). Let ΠE be the groupoid obtained by inverting all arrows of E, fix an object e_0, and identify $G = \mathrm{Aut}_{\Pi E}(e_0)$ with $\pi_1(BE, e_0)$. For a subgroup $\Gamma \subseteq G$ the functor $L_\Gamma(e) = \mathrm{Hom}_{\Pi E}(e_0, e)/\Gamma$ has bijective transition maps, and $\mathrm{El}(L_\Gamma) \to E$ is the connected covering with fundamental group Γ. By (ii) for E its first homology Γ^{ab} vanishes. Taking Γ cyclic gives $G = 1$. A simply connected CW complex with the homology of a point is weakly contractible.

(iii)⇒(i). Let $f : A \to B$ be cofinal. For $b \in B$ the category $b\backslash A$ is $\mathrm{El}(a \mapsto B(b, f(a)))$; it is nonempty and connected, hence weakly contractible by (iii), and Quillen's Theorem A makes Bf a weak equivalence. The constant functor with value a point shows that BA is weakly contractible. So is BB. The last assertion was shown in the second step. □

In particular the integral-homology form of $T(A)$ (every cofinal image has the homology of a point) is equivalent to $T(A)$. With coefficients in a nonzero commutative ring R the same proof, with the homology form of Theorem A, gives: every cofinal image of A has the R-homology of a point if and only if A is nonempty and connected and the components of all $\mathrm{El}(H)$ do. For $R = \mathbf{Q}$ this is weaker: a finite group satisfies it.

Proposition (exact colimits suffice). [ours] *If A is nonempty and connected and $\varinjlim_A$ is exact on functors $A \to \mathrm{Ab}$, then $T(A)$.*

Proof. For $H : A \to \mathrm{Set}$ and $F : \mathrm{El}(H) \to \mathrm{Ab}$ put $(\Sigma_H F)(a) = \bigoplus_{h \in H(a)} F(a, h)$. The functor Σ_H is exact and $\varinjlim_{\mathrm{El}(H)} F = \varinjlim_A \Sigma_H F$. So colimits over $\mathrm{El}(H)$ are exact, and so are colimits over a component E (extend by zero). The left derived functors of $\varinjlim_E$ on the constant functor $\mathbf{Z}$ are the groups $H_j(BE)$; they vanish for $j > 0$. Criterion (ii) applies. □

Theorem (a category with T that is not filtered). [ours] *Let Δ' be the category of the ordered sets $[n] = \{0, \dots, n\}$, $n \geq 0$, and strictly increasing maps. Then $T(\Delta')$ holds and Δ' is not filtered; it is not sifted either.*

Proof. The two maps $[0] \to [1]$ are not equalized by any injection, so Δ' is not filtered. A pair of maps $[0] \to [N] \leftarrow [0]$ is two points of $[N]$, and whether the first is smaller than, equal to or larger than the second is preserved by increasing injections; so the category of such pairs has three components and the diagonal of Δ' is not cofinal.

For exactness we show: *for every n there are M and an integral combination $t = \sum_g t_g\, g$ of maps $g : [n] \to [M]$ of Δ', with $\sum_g t_g = 1$, such that $t \circ h$ is the same combination for all maps $h : [m] \to [n]$ with the same m.* For $n = 1$, $M = 2$, one can take $t = (0, 1) - (0, 2) + (1, 2)$, naming a map by its values. In general, a map $g : [n] \to [M]$ is determined by its gaps $e_0 = g(0)$, $e_i = g(i) - g(i-1) - 1$, $e_{n+1} = M - g(n)$, that is, by a monomial z^e of degree $M - n$ in $z_0, \dots, z_{n+1}$;

a combination t becomes a homogeneous polynomial P_t, and $\sum t_g = P_t(1,\dots,1)$. Let D_i be the substitution $z_{i+1} \mapsto z_i$ followed by renumbering to variables $y_0,\dots,y_n$. Omitting the value $g(i)$ merges two gaps and the omitted point, so $P_{t\delta^i} = y_i D_i(P_t)$ for the map $\delta^i : [n-1] \to [n]$ that omits i. We construct, for all $n \ge 0$ and $k \ge 1$, a homogeneous polynomial $C_{n,k}$ with integer coefficients, $C_{n,k}(1,\dots,1) = 1$, such that $y_i^k D_i(C_{n,k})$ does not depend on i. Put $C_{0,k} = 1$. Given $Q = C_{n-1,2k}$ in the variables $y_0,\dots,y_n$, we want $D_i(C_{n,k}) = F_i := Q\prod_{l\ne i} y_l^k$ for all i; then $y_i^k D_i(C_{n,k}) = Q\prod_l y_l^k$. In the coordinates $x_i = z_i - z_{i+1}$ $(0 \le i \le n)$, $u = z_{n+1}$, the map D_i is reduction modulo x_i. The prescriptions agree modulo (x_i, x_j): for $j = i+1$ both give $D_i(Q)\prod_l w_l^k$, and for $j \ge i+2$ they give $w_{j-1}^{2k} D_{j-1}(Q)\prod_{l\ne i,j-1} w_l^k$ and $w_i^{2k} D_i(Q)\prod_{l\ne i,j-1} w_l^k$, which are equal by the property of Q. For a nonempty set S of indices let Φ_S be the common value of the F_i, $i \in S$, after setting $x_i = 0$ for all $i \in S$ (in homogeneous lifts written in the coordinates x, u). Then

$$C_{n,k} = \sum_{S\ne\varnothing} (-1)^{|S|+1}\Phi_S$$

reduces to F_j modulo x_j, because the terms S and $S\cup\{j\}$, $j \notin S$, cancel modulo x_j and the term $S = \{j\}$ remains. It is homogeneous, and its value at $(1,\dots,1)$, a point where all x_i vanish, is $F_0(1,\dots,1) = 1$. Now let t correspond to $C_{n,1}$. Then $s = t\delta^i$ is independent of i, and from $\delta^{b+1}\delta^a = \delta^a\delta^b$ $(a \le b)$ we get $s\delta^a = s\delta^b$; by induction on n, $s \circ h'$ depends only on the source of h', and every non-identity $h : [m] \to [n]$ factors through some δ^i. This proves the claim.

Let $F \subseteq F'$ be functors $\Delta' \to \mathrm{Ab}$. Any two objects of Δ' map to a third, so every element of $\varinjlim F$ is the class of one element $\xi \in F([n])$. Suppose its image in $\varinjlim F'$ is zero. The colimit is the quotient of $\bigoplus_m F'([m])$ by the elements $F'(v)\eta - \eta$, so $\xi = \sum_k (F'(v_k)\eta_k - \eta_k)$ in this direct sum, for finitely many $v_k : [s_k] \to [r_k]$ and $\eta_k \in F'([s_k])$. Choose N and maps $d_m : [m] \to [N]$ for the finitely many m that occur; applying $F'(d_m)$ on each summand gives $F'(d_n)\xi = \sum_k \big(F'(d_{r_k}v_k) - F'(d_{s_k})\big)\eta_k$ in $F'([N])$. Choose t for N and write $F'(t) = \sum_g t_g F'(g)$. Since $t \circ d_{r_k} v_k = t \circ d_{s_k}$, applying $F'(t)$ gives $F'(\tau)\xi = 0$ for $\tau = t \circ d_n$, a combination of maps $[n] \to [M]$ with coefficient sum 1. As $F \subseteq F'$, also $F(\tau)\xi = 0$. The class of $F(g)\xi$ in $\varinjlim F$ is the class of ξ for every g, so the class of $F(\tau)\xi$ is that of ξ; it is zero. So $\varinjlim_{\Delta'}$ preserves monomorphisms and is exact; Δ' is nonempty and connected, and the proposition applies. □

That colimits of abelian groups over this category are exact is attributed to Isbell and Mitchell in Adámek et al. (1981, A.5), together with the general criterion: $\varinjlim_A$ is exact on abelian groups if and only if each component of the category with the same objects and the integral combinations of arrows of coefficient sum 1 as morphisms is filtered. The withdrawn corollary on this page claimed T for Δ^{op}, which is false; the category Δ' is an example of what that corollary was after.

Theorem (monoids; finitely many objects). [ours] *Let A have an object q that receives an arrow from every object. Then $T(A)$ holds if and only if A is filtered. This applies to every monoid, and to every category with finitely many objects, where both $T(A)$ and filteredness provide such an object q.*

Proof. Assume $T(A)$. *Step 1: for parallel arrows $a, b : p \to q'$ there are an object r and an integral combination t of arrows $q' \to r$, with coefficient sum 1, such that $ta = tb$.* Let $P_c = \mathbf{Z}[A(c,-)]$, let $d : P_{q'} \to P_p$ be $c \mapsto ca - cb$, with kernel K and image J, and let H be the quotient of $A(p,-)$ by the relations $ca \sim cb$, so that $0 \to J \to P_p \to \mathbf{Z}[H] \to 0$ is exact. The functors P_c are projective with $\varinjlim P_c = \mathbf{Z}$, and $\varinjlim d = 0$. So $L_1 \varinjlim \mathbf{Z}[H]$ is the kernel of $\varinjlim J \to \varinjlim P_p$; this map is zero, because $\varinjlim P_{q'} \to \varinjlim J$ is surjective and $\varinjlim d = 0$. Hence $L_1 \varinjlim \mathbf{Z}[H] = \varinjlim J = \operatorname{coker}(\varinjlim K \to \mathbf{Z})$, the image of $\varinjlim K$ being generated by the coefficient sums of the elements of the groups $K(r)$. The left side is $H_1(B\,\mathrm{El}(H))$, and $\mathrm{El}(H)$ is connected (every object receives an arrow from the class of id_p); by the criterion it vanishes. So 1 is a sum of coefficient sums of finitely many elements of groups $K(r_i)$; pushing them to a common successor r of the r_i (the common-successor proposition of the preceding commentary) and adding gives t.

Step 2: the monoid $M = A(q,q)$ is filtered. For $x, m \in M$, Step 1 for the pair x, xm, followed by an arrow $r \to q$, gives $t \in \mathbf{Z}[M]$ with coefficient sum 1 and $tx = txm$. The element $\nu = tx$ of $\mathbf{Z}[Mx]$ is invariant under right multiplication ρ by m. Let S be its finite support. Each point of S has a ρ-preimage in S, distinct points have distinct preimages, so ρ permutes S and ν is constant on its cycles; the coefficient sum 1 is thus an integral combination of cycle lengths. Among all points of Mx that are ρ-periodic choose one, w, of least period π. If $\pi > 1$, let ℓ be a prime dividing π and apply the same to w in place of x: some cycle in Mw has length prime to ℓ, and its points w' satisfy $w'm^\pi = w'$ too, so their period is a proper divisor of π, a contradiction. Hence there is y

with $yxm = yx$. Given $m, m' \in M$, take y with $ym = y$ (case $x = 1$), then y' with $y'ym' = y'y$; then $y'ym = y'y = y'ym'$.

Step 3. Let $a, b : p \to q'$ be parallel, choose $g : q' \to q$, and by Step 1 an element $t = \sum n_i c_i \in \mathbf{Z}[M]$ of coefficient sum 1 with $tga = tgb$. In the filtered monoid M there is y with $yc_i = y$ for all i. Then $yga = \sum n_i yc_i ga = \sum n_i yc_i gb = ygb$. With common successors this shows that A is filtered. The converse is the fact that filtered categories satisfy T. □

So Grothendieck's expectation is right in every finite case and for all monoids, and the smallest counterexamples have infinitely many objects. For the monoid $\{1, e, f\}$ with $uv = v$ for $u, v \in \{e, f\}$, which is cofiltered with contractible classifying space, the theorem says that T fails; a cofinal image of the homotopy type of S^2 is obtained by adjoining a source x with two arrows $h, i : x \to *$ on which e and f act by the constant map with value i. Whether $T(A)$ implies exactness of $\varinjlim_A$ on abelian groups, the converse of the proposition, is not decided here.

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[37]

3 Retour sur la cofinalité

Il est instructif de relier les questions de cofinalité (pour une flèche $u : I' \longrightarrow I$ dans Cat) au dérivateur HOT_{W_0}, où $W_0 \subseteq \mathrm{Fl}\,\mathrm{Cat}$ est le plus grossier des localiseurs dans Cat satisfaisant l'axiome de connexité Loc 5, i.e.

$$W_0 = \{f \in \mathrm{Fl}\,\mathrm{Cat} \mid \pi_0(f) \text{ bijectif}\}. \tag{3.1}$$

On trouve alors, pour I dans Cat, une équivalence canonique, 2-fonctorielle en I,

$$\mathrm{HOT}_{W_0}(I) \xrightarrow{\sim} I^\wedge \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(I, (\mathrm{Ens})), \quad \texttt{[plutôt } \underline{\mathrm{Hom}}(I^o, \mathrm{Ens})\texttt{]} \tag{3.2}$$

en d'autres termes, on a une équivalence

$$\mathrm{HOT}_{W_0} \simeq \mathbf{D}_{(\mathrm{Ens})}. \tag{3.3}$$

Quand HOT_{W_0} est défini par localisation à partir de la catégorie $\mathrm{Cat}(I) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(I^o, \mathrm{Cat})$, la flèche (3.2) est celle déduite, par localisation, de la flèche

$$\begin{array}{rcl} \pi_0^I : \underline{\mathrm{Hom}}(I^o, \mathrm{Cat}) & \longrightarrow & \underline{\mathrm{Hom}}(I^o, (\mathrm{Ens})) \\ X & \longmapsto & \pi_0 \circ X, \end{array} \tag{3.4}$$

où

$$\pi_0 : \mathrm{Cat} \longrightarrow (\mathrm{Ens}) \tag{3.5}$$

est le foncteur 'composantes connexes'. Si on décrit $\mathrm{HOT}_{W_0}(I)$ comme localisée de Cat/I, (3.2) est déduit de la flèche

$$\left\{ \begin{array}{rcl} \mathrm{Cat}/I & \longrightarrow & I^\wedge = \underline{\mathrm{Hom}}(I^o, (\mathrm{Ens})) \\ X & \longmapsto & \underline{\pi_0}(X/I) = (i \longmapsto \pi_0(i\backslash X)). \end{array} \right. \tag{3.6}$$

Ainsi, les foncteurs familiers, pour une flèche [38]

$$u : I' \longrightarrow I$$

dans Cat

$$\left\{ \begin{array}{ccc} & \xrightarrow{u_!} & \\ I'^\wedge & \xleftarrow{u^*} & I^\wedge, \\ & \xrightarrow{u_*} & \end{array} \right. \tag{3.7}$$

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s'interprètent comme les foncteurs de même nom pour le dérivateur HOT_{W_0}, ou plus directement, pour le dérivateur (qu'on pourrait faussement considérer comme canularesque) $\mathbf{D}_{(\mathrm{Ens})}$.

Commentary · Exposition [ms. 37]

The represented derivator at dimension zero

Grothendieck now uses sets to separate the genuinely categorical part of cofinality from its higher homotopical refinements. The equivalence $\mathrm{HOT}_{W_0}(I) \simeq \mathrm{Fun}(I^{\mathrm{op}}, \mathrm{Set})$ has a short proof that explains why no hidden higher information survives. Let $d : \mathrm{Set} \longrightarrow \mathrm{Cat}$ assign the discrete category, and let π_0 be connected components. There is an adjunction $\pi_0 \dashv d$, with $\pi_0 d = 1$. For each category C, the unit $C \longrightarrow d\pi_0 C$ is a W_0-equivalence. Applied pointwise to an I^{op}-diagram, these units give a natural objectwise W_0-equivalence from that diagram to its discrete component diagram. Thus the two functors become inverse after localization. Since the construction commutes with restriction, it gives the stated equivalence of diagram theories.

The description by Cat/I requires the relative, rather than the global, component construction. To $X \longrightarrow I$ one associates the presheaf $i \longmapsto \pi_0(i\backslash X)$. Localizing the slice merely at functors that induce a bijection on the components of their total categories would give a different category. The relevant class is detected in all these comma categories, exactly as in the relative localization recalled in the commentary on Chapter I, manuscript p. 43 ("What the localization axioms do and do not say").

The large table then has an economical interpretation. The words "aspherical" and "coaspherical" become ordinary initiality and finality; the proper and smooth conditions say that specified strict lift categories are nonempty and connected. Right and left Kan extensions retain their ordinary meanings. Nothing in this reduction licenses replacing connected lift categories by contractible ones. Conversely, it makes the degree-zero assertions accessible by elementary calculations with connected components, without invoking a construction of general homotopy Kan extensions.

Cela donne un sens à la cohorte des notions cohomologiques associées à un dérivateur, ou plus particulièrement, à un localiseur fondamental W sur Cat (donnant naissance au dérivateur HOT_W). On les désignera par le nom de W_0-fourbi (au lieu de HOT_{W_0}-fourbi, ou

$\mathbf{D}_{(\mathrm{Ens})}$-fourbi), notamment pour une flèche $u : I' \longrightarrow I$ de Cat telle la notion

(3.8)

- W_0-équivalence $\Big(\iff$ $\pi_0(u)$ bijectif.$\Big)$
- W_0-asphérique $\Big(\iff$ les I'/i pour $i \in \mathrm{Ob}\, I$ sont 0-connexes.$\Big)$
- W_0-coasphérique $\Big(\iff$ les $i\backslash I'$ pour $i \in \mathrm{Ob}\, I$ sont 0-connexes.$\Big)$
- W_0-propre ([46]) $\Big(\iff$ les inclusions $I'_i \longrightarrow I'/i$ $(i \in \mathrm{Ob}\, I)$ sont W_0-coasphériques, i.e. les fibres des colocalisées $i'\backslash I' \longrightarrow i\backslash I$ $(i' = u(i))$ `[plutôt` $(i = u(i'))$] sont 0-connexes.$\Big)$

`[page 39]`

- W_0-lisse ([47]) $\Big(\iff$ les inclusions $I'_i \longrightarrow i\backslash I'$ sont W_0-asphériques, i.e. les fibres des flèches localisées $I'/i' \longrightarrow I/i$ $(i = u(i'))$ sont 0-connexes.$\Big)$
- W_0-parfait $\Big(\iff$ W_0-propre et W_0-lisse.$\Big)$
- W_0-équivalence universelle ([48]) $\Big(\iff$ π_0 bijectif et le reste par tout changement de base. Ce sont aussi les W_0-fibrations (ou W_0-fibrations faibles à fibres W_0-asphériques, i.e. 0-connexes.$\Big)$
- W_0-fibration $\Big(\iff$ $u^* : \mathrm{Cat}/I \longrightarrow \mathrm{Cat}/I'$ transforme W_0-équivalences en W_0-équivalences.$\Big)$
- W_0-fibration faible ([49]) $\Big(\iff$ le changement de base $u^* : \mathrm{Cat}/I \longrightarrow \mathrm{Cat}/I'$ transforme homotopismes $J_1 \longrightarrow J_2$ en W_0-équivalences - et il suffit de l'exiger pour l'inclusion d'un objet final (ou au choix initial) et pour l'inclusion d'un point dans $\mathbf{\Delta}^1$.$\Big)$

`[Nous avons une]` implication *stricte* (et comment) p. 26

$$W_0\text{-équivalence universelle} \Longrightarrow W_0\text{-fibration}$$

`[et une]` implication *stricte* (et comment !) ([50])

$$W_0\text{-fibration} \Longrightarrow W_0\text{-fibration faible.}$$

46. Exemple : les Cat-cofibrations.
47. Exemple : les Cat-fibrations.
48. Exemple : les flèches W_0-lisses ou W_0-propres, à fibres 0-connexes.
49. Exemple : les flèches W_0-lisses telles que $\pi_0(I'/I) = (i \longmapsto \pi_0(i\backslash I')) = u_!(e_{I'^\wedge})$ soit localement constant sur I, i.e. $\underline{\pi_0}(I'/I)$ soit un *revêtement* étale de I, ou si u^{opp} satisfait cette condition.
50. Mais si les fibres sont 0-connexes, les deux notions de W_0-fibrations sont équivalentes et signifient qu'on a une W_0-équivalence universelle …

Signalons, comme exemple d'utilisation de ces notions, que pour un diagramme cartésien dans Cat

$$\begin{array}{ccc} I' & \xleftarrow{p'} & J' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ I & \xleftarrow{p} & J, \end{array} \tag{3.9}$$

la flèche de changement de base pour u_*, v_* pour les préfaisceaux est [un] isomorphisme si u est W_0-propre, ou p W_0-lisse, et la flèche de changement de base pour $u_!$, $v_!$ est [un] isomorphisme si u est W_0-lisse ou p W_0 propre ([51]). Je signale à ce propos que ces énoncés, qui a priori se formulent pour les préfaisceaux à valeurs dans (Ens), restent valables pour [40] des préfaisceaux à valeurs dans une catégorie $\mathcal{M}$ quelconque (sous réserve d'existence des operations d'images directes cohomologiques ou homologiques envisagées). Cela se voit p.ex. en notant que

$$W_0\text{-équivalence} \implies \mathbf{D}_{\mathcal{M}}\text{-équivalence} \tag{3.10}$$

pour toute $\mathcal{U}$-catégorie $\mathcal{M}$, d'où résulte que toute ([52]) 'W_0-notion' implique la $\mathbf{D}_{\mathcal{M}}$-notion correspondante, et permet donc d'appliquer à $\mathbf{D}_{\mathcal{M}}$ les résultats le concernant. (Il y a bien sûr un argument direct, moins savant, consistant à se ramener au cas où $\mathcal{M}$ est petite, et à utiliser alors le plongement pleinement fidèle $\mathcal{M} \hookrightarrow \mathcal{M}^\wedge$ resp. $\mathcal{M} \hookrightarrow ((\mathcal{M}^o)^\wedge)^o$.)

Je rappelle pour mémoire une (troisième ?) caractérisation des flèches $u : I' \longrightarrow I$ W_0-propres (resp. W_0-lisses) : ce sont celles pour lesquelles le foncteur changement de base $u^* : \mathrm{Cat}/I \longrightarrow \mathrm{Cat}/I'$ transforme flèches W_0-coasphériques (resp. W_0-asphériques) en itou. Cela gagne en intérêt, du fait que la notion de foncteur W_0-coasphérique (resp. W_0-asphérique) s'interprète comme celle de foncteur cofinal (resp. co-cofinal), cf. plus bas.

La notion de W_0-fibration apparaît comme extrêmement forte - beaucoup plus forte à p. 27 certains égards que celle (disons) de W_∞-fibration ou de W_ω-fibration. Elle équivaut à ceci [41] (comme on voit aussitôt) : pour toute composante connexe I'_α de I', désignant par I_α la composante connexe correspondante de I, et pour tout J_α 0-connexe et au dessus de I_α, le produit fibré $I'_\alpha \times_{I_\alpha} J_\alpha$ est encore 0-connexe. (Ça ressemble aux histoires de totale 0-connexité d'un topos !) Mais cela signifie aussi que $I'_\alpha \longrightarrow I_\alpha$ est une W_0-équivalence universelle. Cela implique que ses fibres sont 0-connexes, et la réciproque est vraie si $u : I' \longrightarrow I$ est W_0-propre ou W_0-lisse. Ainsi, la condition essentielle pour une W_0-fibration est celle que les fibres des flèches induites $I'_\alpha \longrightarrow I_{u(\alpha)}$ [plutôt $I'_\alpha \longrightarrow I_\alpha$] soient 0-connexes. En fait, on a ceci

Commentary · Exposition [ms. 40]

The lift categories behind ordinary base change

The base-change assertion is an application of the lift-category criterion already developed in the commentary on Chapter I, manuscript pp. 68–70 ("The category of lifts behind properness" and "Completing the tests in the proper base-change theorem"). Its present specialization is useful because all the relevant tests are tests of connectedness. For $u : A \longrightarrow B$, properness means that, for every $a \in A$ and arrow $\beta : u(a) \longrightarrow b$, the category

$$L(a, \beta) = \{a \xrightarrow{h} a' \mid u(a') = b, u(h) = \beta\}$$

of strict lifts, with arrows over 1_b, is nonempty and connected. Smoothness is the dual condition on arrows into $u(a)$. An opfibration supplies initial objects in the displayed lift categories; a fibration supplies the dual terminal objects. Connectedness is the exact weakening that suffices for ordinary Kan extensions.

51. Sauf erreur, ca donne *deux caractérisations* des morphismes W_0-propres, ou des morphismes W_0-lisses.

52. Pas tout à fait exacte pour *toute* 'W_0-notion', il faut excepter les deux notions de W_0-fibration forte, et faible.

For a strict pullback square as in (3.9), the comparison of the comma categories computing a right Kan extension is final if u is proper or if the horizontal base-change functor is smooth. The ordinary cofinality theorem therefore gives the right base-change isomorphism. Reversing arrows gives the left formula. The same argument works in any coefficient category in which the requisite comma limits exist, so that these Kan extensions are computed pointwise. One can apply $\mathrm{Hom}(m, -)$ to the limits, or dually $\mathrm{Hom}(-, m)$ to the colimits, and reduce to the set-valued cofinality theorem. This avoids imposing unnecessary completeness assumptions on the coefficient category.

The passage from a W_0-equivalence to an equivalence measured by constant coefficients has a similarly elementary explanation. A limit of a constant diagram is a product indexed by the connected components of its shape, and a colimit is the corresponding coproduct, whenever those objects exist. A bijection on components preserves both. The printed footnote correctly excludes the two fibration notions from the blanket transfer of terminology: preserving a larger class of equivalences under every base change is a stronger requirement, even when the equivalences themselves have a simpler definition. The next proposition measures this effect exactly.

Proposition 3.1. *Soit $u : I' \longrightarrow I$ une flèche dans* Cat.

a) Pour que ce soit une W_0-équivalence universelle, il faut et il suffit que pour toute $\mathbf{\Delta}^1 \longrightarrow I$, i.e. toute flèche α de I, la catégorie $I' \times_I \mathbf{\Delta}^1 \overset{déf}{=} I'|\alpha$ soit 0*-connexe, ou encore que les fibres de I soient* 0*-connexes, et que pour toute flèche* non identique *α de I, $I'|\alpha$ soit* 0*-connexe.*

b) Pour que $u : I' \longrightarrow I$ soit une W_0-fibration, il faut et il suffit que pour toute composante connexe $I'_{\alpha'}$ de I', posant $\alpha = u(\alpha')$, la flèche induite $I'_{\alpha'} \longrightarrow I_\alpha$ soit une W_0-équivalence universelle (cf. critère a) pour cette notion). [42]

c) Pour que ce soit une W_0-fibration faible, il faut *que $\forall\, i \in I$, l'inclusion $I'_i \hookrightarrow i\backslash I'$ soit une W_0-équivalence, i.e. $\pi_0(I'_i) \xrightarrow{\sim} \pi_0(i\backslash I')$ soit bijectif,* et *que $i \longmapsto \pi_0(i\backslash I')$ soit un préfaisceau localement constant sur I (i.e. $\underline{\pi_0}(I'/I)$ soit un* revêtement *étale de I). Cette condition est aussi suffisante, si u est W_0-lisse. De plus (pour mémoire) u est une W_0-fibration faible si et seulement si u^{opp} l'est.*

Commentary · Exposition [ms. 42]

Why arrows detect universal connectedness

Proposition 3.1 says that universal preservation of connected components can be checked on vertices and single arrows. Its proof is a useful model for reading the much stronger componentwise condition in part (b). We give the details because the proposition is stated, rather than proved, here.

Suppose first that $u : A \longrightarrow B$ is a universal W_0-equivalence. Pullback along any $[1] \longrightarrow B$ gives a connected nonempty category, since $[1]$ is connected. Conversely, assume this arrow test. Applying it to an identity arrow at b shows that $A_b \times [1]$, and hence A_b, is connected and nonempty. Let $v : J \longrightarrow B$ be arbitrary. Every object of J has a nonempty fibre in $A \times_B J$, so the map on components is surjective. If two objects of the pullback project to objects joined by a zigzag in J, each edge of that zigzag has a connected pullback by hypothesis. Its endpoints can be joined through that pullback, and the connected vertex fibres join any two choices of endpoints. Concatenating these finite zigzags proves injectivity. Thus

$$\pi_0(A \times_B J) \xrightarrow{\sim} \pi_0(J)$$

for every J, which is the required universal equivalence. Notice that the test concerns strict pullbacks, not comma categories.

Now let A_λ be a connected component of A, mapping to the component B_μ. If pullback by u preserves every W_0-equivalence, apply it to $J \amalg \coprod_{\nu \neq \mu} B_\nu \longrightarrow B$, where $J \longrightarrow B_\mu$ is any connected nonempty category over B_μ. The resulting map on components forces $A_\lambda \times_{B_\mu} J$ to be connected and nonempty. Decomposing an arbitrary category over B_μ into its components then shows that $A_\lambda \longrightarrow B_\mu$ is a universal W_0-equivalence. Conversely, if all these component maps are universal equivalences, then for any $J \longrightarrow B$ there is a natural bijection

$$\pi_0(A \times_B J) \cong \pi_0(A) \times_{\pi_0(B)} \pi_0(J).$$

This formula immediately shows that pullback by u preserves every bijection on connected components. It proves part (b), including components of B not met by A.

The necessary conditions in part (c) come from the same tests with categorical homotopy equivalences. The inclusion of the initial object 1_b into $b\backslash B$ is such an equivalence; its pullback identifies $A_b \longrightarrow b\backslash A$ on components. Testing both endpoints of an arrow in B makes the resulting component presheaf locally constant. When u is smooth, ordinary left base change identifies the component presheaf after every base change with the restriction of this local system. Its colimit depends only on the fundamental groupoid of the indexing category, and categorical homotopy equivalences induce equivalences of those groupoids. This proves the stated sufficiency. The opposite-category assertion follows directly because categorical homotopies and π_0 are unchanged by reversal of arrows.

La plus grande différence entre la notion de W_0-fibration, et celle de W_0-fibration faible, me paraît que les morphismes W_0-parfaits (i.e. W_0-lisses et W_0-propres), et a fortiori les W-parfaits pour un localiseur fondamental $W \subseteq W_0$, sont forcément des W-fibrations faibles, mais (*même si* $W = W_\omega$, le plus fin de tous les localiseurs fondamentaux) ne sont pas des W-fibrations. L'exemple le plus frappant à cet égard est sans doute celui des revêtements étales, qui sont W_ω-parfaits (donc des W_ω-fibrations faibles et a fortiori des W_0-fibrations faibles), mais pas des W_0-fibrations ; car dans le cas où I', I sont 0-connexes, il faudrait que les fibres soient 0-connexes (cf. b) ci-dessus), i.e. que u soit un isomorphisme ! [43]

Commentary · Examples [ms. 43]

Monodromy explains the strictness

The covering example isolates what the componentwise test excludes. This is the same monodromy issue as in the commentary on Chapter I, manuscript p. 98 ("Sets already detect the missing monodromy"), now applied to the two precise definitions in (3.8). Let G be a nontrivial group. Its translation groupoid EG has object set G and exactly one arrow between each ordered pair of objects. The functor

$$EG \longrightarrow BG$$

sends $g \longrightarrow h$ to hg^{-1}. It is a discrete fibration and a discrete opfibration, so the strict lift categories are singletons and the map is proper and smooth. Both total and base categories are connected, but the strict fibre is the discrete set G.

The inclusion of the sole object $* \longrightarrow BG$ is a W_0-equivalence. Its pullback along $EG \longrightarrow BG$ is the map $G_{\mathrm{disc}} \longrightarrow EG$, whose map on components is $G \longrightarrow *$. This proves directly that the covering is not a W_0-fibration. It is a weak W_0-fibration by the smooth criterion of Proposition 3.1(c): the relative component presheaf is the locally constant G-set with regular action. Thus weak fibrations permit monodromy between different fibre components, whereas the strong W_0 condition treats each connected component of the total category separately.

There is a subscript caution in the preceding printed paragraph. The conclusion established by this example is "not necessarily a W_0-fibration." It does not assert that coverings fail to preserve weak homotopy equivalences under pullback. In fact their nerves are simplicial coverings, and they do preserve weak homotopy equivalences. Similarly a W_0-perfect map gives a weak W_0-fibration; a W-perfect map gives the corresponding weak W-fibration. These are distinct assertions when $W \subsetneq W_0$.

Une propriété sympathique des morphismes W_0-lisses (resp. W_0-propres), c'est que ce sont des applications *ouvertes* (resp. fermées) pour les espaces topologiques sous-jacents, ce qui signifie en effet que les fibres locales, i.e. des flèches localisées en haut $I'/i' \longrightarrow I/i$ (resp. colocales, i.e. les flèches colocalisées en bas $i'\backslash I' \longrightarrow i\backslash I$), sont surjectives, i.e. à fibres non-vides (ce qui est impliqué par la condition que celles-ci soient 0-connexes).

Dans tout ça je n'ai encore guère parlé des flèches W_0-asphériques resp. co-asphériques, qui pourtant sont ici une motivation principale pour introduire W_0 ! p. 28

Proposition 3.2. *Conditions équivalentes sur la flèche* $u : I' \longrightarrow I$ *dans* Cat :

a) *Pour toute $\mathcal{U}$-catégorie $\mathcal{M}$ et tout foncteur $F : I \longrightarrow \mathcal{M}$, la flèche canonique (sous réserve que les petites limites inductives soient représentables dans $\mathcal{M}$),*

$$\varinjlim_{I'} u^*(F) \longrightarrow \varinjlim_{I} F \tag{3.2.1}$$

est [un] *isomorphisme.* [44]

a′) *Pour toute $\mathcal{U}$-catégorie $\mathcal{M}$ où les petites limites projectives sont représentables et tout $F \in \mathcal{M}(I) = \underline{\mathrm{Hom}}(I^o, \mathcal{M})$, i.e. tout préfaisceau sur I à valeurs dans $\mathcal{M}$, la flèche canonique*

$$\varprojlim_{I} F \longrightarrow \varprojlim_{I'} u^*(F) \tag{3.2.2}$$

est [un] *isomorphisme.*

a″) ([53]) *Comme a′, avec $\mathcal{M} = (\mathrm{Ens})$, i.e. $F \in \mathrm{Ob}\, I^\wedge$.* ([54]).

b) *La flèche u est W_0-coasphérique, i.e. pour tout $i \in I$, $i\backslash I'$ est 0-connexe.*

L' equivalence de a, a′, a″ est ± tautologique en termes des définitions (cf. SGA 4, I, 8…). La forme a″ s'écrit, posant $\mathbf{D} = \mathrm{HOT}_{W_0}$, que $\forall\, F \in \mathbf{D}(I)$, on veut que

$$\underbrace{H^\bullet_{\mathbf{D}}(I, F)}_{\overset{\text{déf}}{=} p_{I,*}(F)} \longrightarrow \underbrace{H^\bullet_{\mathbf{D}}(I', u^*F)}_{\overset{\text{déf}}{=} p_{I',*}(F)} \qquad \texttt{[plutôt} \quad \underbrace{H^\bullet_{\mathbf{D}}(I', u^*F)}_{\overset{\text{déf}}{=} p_{I',*}(u^*F)} \texttt{]} \tag{3.2.3}$$

($p_I : I \longrightarrow e$ et $p_{I'} : I' \longrightarrow e$ les foncteurs canoniques) soit [un] isomorphisme. On sait que cela équivaut à u $\mathbf{D}$-coasphérique. La démonstration est un jeu amusant entre $u_!$, u^*, u_* à la fois. Il faut exprimer que le Hom d'un objet M de $\mathbf{D}(e)$ (= (Ens) en l'occurence) dans la flèche (3.2.3) est bijective. Ici, il

est plus naturel de prendre $M = e$ (ensemble ponctuel) et d'interpréter la flèche (3.2.3) comme p. 29 [45]

$$\mathrm{Hom}(e_I, F) \longrightarrow \mathrm{Hom}(e_{I'}, u^*F) \tag{3.2.4}$$

et par adjonction de $u_!$, u^* le deuxième membre s'interprète comme $\mathrm{Hom}(u_!(e_{I'}), F)$, et la flèche est déduite de

$$u_! e_{I'} = u_! u^*(e_I) \xrightarrow{\text{adj.}} e_I \tag{3.2.5}$$

en prenant le Hom dans F. Donc dire que (3.2.4) est [un] isomorphisme pour tout F signifie que (3.2.5) est [un] isomorphisme. Passant aux valeurs des deux membres en i, on trouve $\pi_0(i\backslash I') \xrightarrow{\sim} 1$, i.e. que $i\backslash I'$ est 0-connexe, q.e.d.

---

53. C'est une condition sur les morphismes de topos, savoir que si F [est] dans $\mathcal{X}^\wedge$, on a

$$H^0(\mathcal{X}, F) \xrightarrow{\sim} H^0(\mathcal{X}', u^*F).$$

Il y a [une] notion similaire pour les H^i ($i \leq n$), à dégager. Notion de morphismes W_0-coasphériques de topos ! C'est dans la nature d'une condition de 0-connexité relative, duale de la notion W_0-asphérique, qui signifie simplement que pour U 0-connexe dans $U^\wedge$ [plutôt $\mathcal{X}^\wedge$], $f^*(U)$ [plutôt $u^*(U)$] est 0-connexe. De même, on a sans doute des notions ('duales' des précédentes) de W_n-coasphéricité (bien comprise en principe si $\mathcal{X}$ est localement n-asphérique) (La W_n-coasphéricité, étant celle évoquée plus haut.)

54. **NB** On peut introduire

a‴) Comme a), mais avec $\mathcal{M} = (\mathrm{Ens})$.

a^{IV}) Comme a), avec le foncteur canonique $I' \longrightarrow I'^\wedge$.

Commentary · Exposition [ms. 45]

What the ordinary cofinality theorem tests

Grothendieck has now proved the criterion announced on the opening page, with the sufficient directedness condition replaced by the exact connectedness condition. The proof is the representability calculation already explained in "The presheaf represented by a change of index": the functor $F \longmapsto \Gamma(I', u^*F)$ is represented by $u_!1_{I'}$, and

$$(u_!1_{I'})(i) = \pi_0(i\backslash I').$$

Consequently the comparison on all set-valued presheaves is an isomorphism precisely when $u_!1_{I'} \longrightarrow 1_I$ is an isomorphism. Applying ordinary Hom functors reduces arbitrary existing limits and colimits to this case. This is the classical theorem of SGA 4, Exposé I, Section 8.1, in the variance used throughout these pages (Artin et al., 1972).

Two small examples distinguish its quantifiers. The inclusion $\{1\} \hookrightarrow [1]$ is cofinal, because each $i\backslash\{1\}$ is a point. For a presheaf $F(1) \longrightarrow F(0)$, its limit is therefore $F(1)$. But the restriction functor forgets $F(0)$ and is not fully faithful. On the other hand, for any nonempty connected category C, the projection $C \longrightarrow *$ is cofinal even when BC has nontrivial homotopy groups. Ordinary cofinality is therefore stronger than a bijection on components of arbitrary source and target, weaker than full faithfulness of restriction, and weaker than homotopy cofinality in a different direction. These comparisons refer to different properties of a functor, not to alternative conventions for the same word.

Footnote 53 distinguishes two degree-zero conditions on a morphism of topoi $f : \mathcal{E}' \longrightarrow \mathcal{E}$. Its coasphericity condition preserves global sections of every object. Its asphericity condition instead requires f^*U to be noninitial and connected whenever U is. For locally connected topoi, the latter is equivalent to preservation of component sets: decompose U into its noninitial connected components and use preservation of coproducts by f^*. For the morphism induced by $u : I' \longrightarrow I$, the restricted representable u^*h_i has category of elements I'/i. Thus this asphericity condition is ordinary initiality of u, by the cofinality theorem with arrows reversed; the section condition is finality. The two endpoints of [1] distinguish them: inclusion of 0 is initial, while inclusion of 1 is final.

The supplementary Yoneda test in footnote 54 should use the functor $I \longrightarrow I^\wedge$. Its colimit is 1_I, whereas its restriction along u has colimit $u_!1_{I'}$. The printed $I' \longrightarrow I'^\wedge$ has the wrong domain for the test as stated. The further suggestion in footnote 53 about higher cohomology should be read with coefficients specified: preservation of all set-valued sections, preservation of abelian cohomology through a fixed degree, and equivalence on all homotopy-valued diagrams detect different truncations. The earlier cohomological cofinality block supplies the derived comparison; the ordinary theorem here establishes its degree-zero model.

Définition 3.3.

a) Quand les conditions équivalentes de 3.2 sont satisfaites, donc quand les $i\backslash I'$ ($i \in \mathrm{Ob}\, I$) sont 0-connexes, on dit que $u : I' \longrightarrow I$ est un foncteur *cofinal.* On dit que u est *co-cofinal* si $u^o : I'^o \longrightarrow I^o$ est cofinal, i.e. si les localisées I'/i de I' sur I ($i \in \mathrm{Ob}\, I$) sont 0-connexes.

b) Si I' est une *sous-catégorie* de I, on dit que I' *est cofinale dans* I si le foncteur d'inclusion est cofinal, *co-cofinale* si ce foncteur est co-cofinal.

Corollaire 3.4. ([55])

a) Pour que u soit cofinal, il faut *que les catégories colocalisées $i\backslash I'$ soient non-vides, i.e. qu'on ait*

F 0) $\pi_0(f)$ `[plutôt` $\pi_0(u)$`]` *est bijectif.* [46]

F 1) Pour tout $i \in \mathrm{Ob}\, I$, `[il]` *existe* `[un]` $i' \in I'$ *et une flèche* $i \longrightarrow u(i')$ *dans* I.

([56]).

55. **NB** Pour un énoncé complémentaire de cofinalité, cf. XVIII prop. 4.9.6.3 (p. 224-225, 228-242). Il faudrait un énoncé qui coiffe les deux.

56. **NB** F 1 implique déjà `[la]` surjectivité de π_0, donc sa bijectivité si I' est 0-connexe - dans ce cas F 0 résulte de F 1.

b) Si I' est faiblement filtrante, pour que u soit cofinale, il suffit *qu'elle satisfasse F 1, ainsi que la condition :*

F 2) Pour toute double flèche $i \overset{\beta}{\underset{\gamma}{\rightrightarrows}} u(i')$ (avec $i \in \mathrm{Ob}\, I$, $i' \in \mathrm{Ob}\, I'$), `[il]` *existe une flèche $i' \xrightarrow{\alpha'} j'$ dans I', de telle façon que $u(\alpha')$ égalise (β, γ), i.e. $u(\alpha')\beta = u(\alpha')\gamma$.*

et ces conditions F 1, F 2 impliquent que les $i\backslash I'$ sont faiblement filtrantes, et I aussi ([57]).

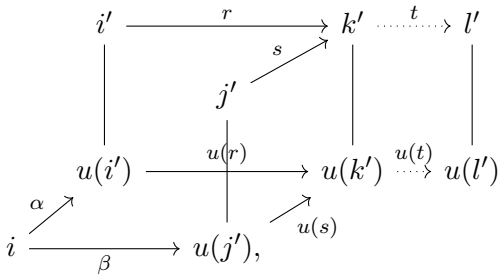
p. 30

c) Si I' est filtrante, alors u est cofinal si et seulement s'*il satisfait les deux conditions F1, F2, ou encore si et seulement si les $i\backslash I'$ sont* filtrantes *(et non seulement* 0*-connexes). Et cela implique que I' est également filtrante* ([58]).

d) Si I est préfiltrante (= localement faiblement filtrante, i.e. PS 1), et u pleinement fidèle, *pour que u soit cofinal, il* faut *et il* suffit *que u satisfasse la condition F 1 plus haut. Et alors I' est également préfiltrante, et les $i\backslash I'$ sont faiblement filtrantes. Si de plus I est filtrante, les $i\backslash I'$ et I' le sont également. (Comparer SGA 4, I 8.1.3.)*

Démonstration. a) F 1 est évident, car 0-connexe $\Longrightarrow$ non vide. F 0 car W_0-coasphérique [47] $\Longrightarrow$ W_0-équivalence, i.e. $\pi_0(u)$ bijectif.

Pour *prouver* b), il suffit de prouver que F 1, F 2 (si I' est faiblement filtrante) impliquent que les $i\backslash I'$ sont faiblement filtrantes (a fortiori 0-connexes) et de plus que I l'est également. On sait déjà qu'ils sont non vides par F 1, reste à voir que si (i', α), (j', β) sont dans $i\backslash I'$, alors ils sont 'majorés' par un troisième objet de $i\backslash I'$. Comme I' est faiblement filtrante, on trouve $\begin{matrix} i' & \overset{r}{\searrow} & \\ & & k' \\ j' & \underset{s}{\nearrow} & \end{matrix}$, d'où $u(r)$, $u(s)$ dans I (cf. diagramme)



appliquant F 2 à la double flèche $u(r)\alpha, u(s)\beta : i \rightrightarrows u(k')$, on trouve $k' \xrightarrow{t} l'$ (en pointillés sur `[le]` diagramme) telle que $(u(t)u(r))\alpha = (u(t)u(s))\beta$, soit γ, ce qui montre que l'objet $(l', \gamma : i \longrightarrow u(l'))$ majore (i', α) et (j', β).

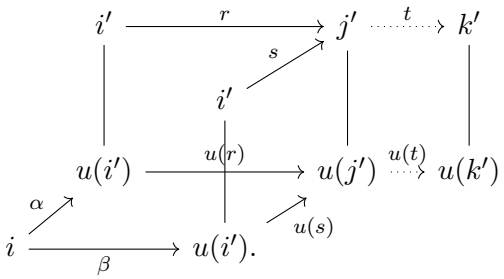
La démonstration que si I' est faiblement filtrante et u satisfait F 1, alors I est faiblement filtrante, est immédiate.

Prouvons c). Il reste à prouver que si I' est *filtrante* (non seulement faiblement filtrante), alors si u `[est]` cofinal, alors F 2 est vérifiée, et de plus les $i\backslash I'$ et I sont filtrantes (pas seulement faiblement filtrantes). Mais notons tout de suite que $i\backslash I' \longrightarrow I'$ est coétale, donc I' filtrante, donc localement filtrante implique que $i\backslash I'$ est aussi localement filtrante ; comme par hypothèse les $i\backslash I'$ sont 0-connexes, ils sont donc filtrantes. La condition F 2 sort facilement à présent :

p. 31

57. Si I' est localement faiblement filtrante (= préfiltrante), alors pour la cofinalité de u, il suffit qu'on ait F [48] 0, F 1, F 2, et alors les $i\backslash I'$ sont faiblement filtrantes, et I est préfiltrante tout comme I'.

58. Si I' est localement filtrante, u est cofinal si et seulement si elle satisfait F 0, F 1, F 2, i.e. si et seulement si les $i\backslash I'$ sont filtrantes, et alors I est localement filtrante.



Comme $i\backslash I'$ est filtrante, donc faiblement filtrante (PS 1), on peut trouver $i' \overset{r}{\underset{s}{\rightrightarrows}} j'$ telles que

$$u(r)\alpha = u(s)\beta,$$

et comme I' est filtrante, on peut trouver $t : j' \longrightarrow k'$ qui égalise la double flèche (r, s) :

$$tr = ts(\overset{\text{déf}}{=} \tau),$$

alors

$$\underbrace{u(t)u(r)}_{u(tr)}\,\alpha = \underbrace{u(t)u(s)}_{u(ts)}\,\beta \qquad \text{i.e.} \qquad u(\tau)\alpha = u(\tau)\beta,$$

i.e. on a vérifié F 2.

Reste à prouver que I est filtrante (si I' l'est et `[si]` $u : I' \longrightarrow I$ est cofinal). On sait déjà que I est faiblement filtrante par b), reste à vérifier la condition de la double flèche

$$i \overset{\alpha}{\underset{\beta}{\rightrightarrows}} j$$

(PS 2). Soit i' dans I' et $j \longrightarrow u(i')$, et considérons

$$i \overset{\gamma\alpha}{\underset{\gamma\beta}{\rightrightarrows}} u(i'),$$

par hypothèse F 2 $\exists\, r : i' \longrightarrow j'$ dans I' telle que $u(r)(\gamma\alpha) = u(r)(\gamma\beta)$, donc $u(r)\gamma$ égalise (α, β), q.e.d. [49]

La vérification de d) par AQT `[abréviation dont le sens échappe aux éditeurs]`.

Commentary · Development (commentators') [ms. 49]

The filtered criterion and eventual lifting

Corollary 3.4 converts connectedness of comma categories into the two concrete operations needed for filtered colimits: reaching the image of u, and equalizing parallel arrows after reaching it. The proof of parts (b) and (c) is present on the page. There is one harmless slip in the statement of (c): its last assertion concerns filteredness of I, not the already assumed filteredness of I'. Part (d) is left under the abbreviation AQT. Its proof uses full faithfulness in two different places.

One variance qualification is essential. Definition 2.2 called the completion of a cospan by a common predecessor PS 1 and explicitly called its right-hand dual "colocalement faiblement filtrante." Here part (d) and footnote 57 instead use "préfiltrante" or "localement faiblement filtrante" for the right-hand condition: a span must admit a commuting square with a common successor. The proof requires that interpretation. Under the earlier left-hand reading, part (d) would be false: the poset $I = (0 \longrightarrow 1,\ 0 \longrightarrow 2)$ satisfies left PS 1, and the full inclusion $\{1, 2\} \hookrightarrow I$ satisfies F 1, but its comma category under 0 is a discrete two-point category. Thus this is a necessary clarification of the printed terminology, not a variance-neutral use of PS 1.

Assume that I satisfies the right-hand Ore condition PS 1, that $u : I' \longrightarrow I$ is fully faithful, and that F 1 holds. Given objects $(a, \alpha : i \longrightarrow ua)$ and $(b, \beta : i \longrightarrow ub)$ of $i\backslash I'$, PS 1 supplies arrows $r : ua \longrightarrow j$ and $s : ub \longrightarrow j$ with $r\alpha = s\beta$. Choose $j \longrightarrow uc$ by F 1. Fullness lifts the two resulting

arrows $ua \longrightarrow uc$ and $ub \longrightarrow uc$ to arrows in I', and they give a common successor of the original two objects in $i\backslash I'$. F 1 also makes that comma category nonempty. It is therefore weakly filtered, hence connected, and u is cofinal.

To transfer PS 1 itself to I', start with a span $a \longrightarrow b$, $a \longrightarrow c$ in I'. Complete its image in I to a commuting square, reach an object in the image of u by F 1, and lift the two new arrows by fullness. Faithfulness ensures that the lifted square commutes. Thus I' satisfies PS 1. If I is filtered, a parallel pair $r, s : a \rightrightarrows b$ in I' can likewise be equalized in I, followed by an arrow to uc and lifted to an equalizer arrow $b \longrightarrow c$ in I'. Faithfulness again supplies the equality. Nonemptiness follows from F 1, so I' is filtered. The same reasoning, or the discrete-opfibration projection $i\backslash I' \longrightarrow I'$, makes each comma category filtered.

For locally filtered categories, the additional condition F 0 in footnotes 57–58 ensures that objects whose images lie in the same component already lie in the same source component. The arguments can then be carried out in that component. Without this component condition, F 1 and F 2 do not prevent several disconnected copies of one indexing category from being mapped onto it. This is why the global filtered statement and its local variant have different hypotheses.

Footnote 55 asks for a statement encompassing this corollary and Chapter XVIII. The exact later result is Lemma 4.9.6.3, part 1, manuscript p. 224, whose proof is in §4.9.6.7. Its additional condition is eventual lifting:

$$\text{F 3:}\quad \theta : u(a) \longrightarrow u(b) \implies \exists\, v : b \longrightarrow c,\ w : a \longrightarrow c, \qquad u(v)\theta = u(w).$$

For a faithful functor into a filtered category, the lemma identifies F 1 and F 3 with filteredness of the source together with cofinality (Grothendieck, 1990–1991n, Lemma 4.9.6.3(1), pp. 81–82). The following finite formulation combines the two arguments, retaining only the right Ore hypothesis where that suffices. It is our proof-level synthesis of the cited passages, without a claim of priority.

A finite comparison with eventual lifting. [ours] *If the target satisfies right PS 1, F 1 and F 3 make u cofinal. If u is faithful, its source also satisfies right PS 1 and is filtered whenever the target is filtered.*

Indeed, given (a, α) and (b, β) under i, right PS 1 followed by F 1 produces $\rho_a : u(a) \to u(c)$ and $\rho_b : u(b) \to u(c)$ with $\rho_a\alpha = \rho_b\beta$. Apply F 3 first to ρ_a, obtaining $v : c \to d$, $w : a \to d$, and then to $u(v)\rho_b$, obtaining $t : d \to e$, $z : b \to e$. The equality

$$u(tw)\alpha = u(t)u(v)\rho_a\alpha = u(t)u(v)\rho_b\beta = u(z)\beta$$

gives a common successor in the comma category. F 1 gives nonemptiness, hence cofinality. For a span in the source, this construction makes its two composites equal after applying u; faithfulness makes them equal before applying it. If the target is filtered, equalize the images of any parallel pair, reach another image object by F 1, and lift the resulting arrow eventually by F 3. Faithfulness then gives an equalizing arrow in the source. Cofinality supplies connectedness and nonemptiness, completing filteredness. Conversely, when the source is filtered and u cofinal, the already proved filteredness of $u(a)\backslash I'$ gives F 3 by taking a common successor of $(a, 1)$ and (b, θ).

Full faithfulness makes F 3 automatic, recovering part (d). The source-filtered case is the F 1–F 2 criterion in part (c). Thus the finite comparison requested in footnote 55 is explicit. Part 2 of the later lemma strengthens F 3 to simultaneous lifting for families of bounded cardinality and imposes the corresponding larger filtering hypotheses; that cardinal refinement is a separate assertion, not a consequence of the finite conditions alone.

p. 32

4 Foncteurs biasphériques

Soit encore

$$u : I' \longrightarrow I$$

une flèche de Cat, soit $\mathcal{M}$ une catégorie, d'où

$$u^* : \mathcal{M}(I) \longrightarrow \mathcal{M}(I'); \tag{4.1}$$

nous cherchons des conditions sur u, $\mathcal{M}$ pour que ce foncteur soit pleinement fidèle. Si le foncteur adjoint à gauche $u_!$ (resp. adjoint à droite u_*) existe, on sait qu'il revient au même

de dire que le morphisme d'adjonction

$$u_! u^* F \longrightarrow F \tag{4.2}$$

resp.

$$u_* u^* F \longleftarrow F \tag{4.2 bis}$$

soit [un] isomorphisme pour tout F dans $\mathcal{M}(I)$. Disons alors que u est $\mathcal{M}$-connexe, ou $\mathcal{M}$-biasphérique, cette terminologie sous-entendant que c'est là une propriété qui se réfère au dérivateur $\mathbf{D}_{\mathcal{M}} = (S \longmapsto \mathcal{M}(S))$ sur Cat défini par $\mathcal{M}$. Notons que

(4.3) Si u est $\mathcal{M}$-connexe, alors il est aussi $\mathcal{M}(S)$-connexe, pour tout S dans Cat, puisque $\mathcal{M}(S)(I) \simeq \mathcal{M}(I)(S) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(S^o, \mathcal{M}(I))$,

et [50]

(4.4) Si $\mathcal{M}' \hookrightarrow \mathcal{M}$ est pleinement fidèle, alors $\mathcal{M}$-connexe implique $\mathcal{M}'$-connexe,

enfin, inversement,

(4.5) pour que u soit $\mathcal{M}$-connexe, il faut et il suffit que pour toute *petite* sous-catégorie pleine $\mathcal{M}'$ de $\mathcal{M}$, u soit $\mathcal{M}'$-connexe.

Comme toute petite $\mathcal{M}'$ se plonge dans $\mathcal{M}'^{\wedge} = \mathrm{Ens}(\mathcal{M}')$, il résulte de (4.3), (4.4), (4.5) que

(4.6) Si u est Ens-connexe, il est $\mathcal{M}$-connexe pour toute catégorie $\mathcal{M}$,

car c'est le cas si $\mathcal{M}$ est petite par (4.3), donc pour tout $\mathcal{M}$ par (4.5).

Disons qu'un foncteur u dans Cat est $\mathcal{M}$-*cofinal* ou $\mathcal{M}$-coasphérique si l'homomorphisme fonctoriel

$$\varprojlim_{I} F \longrightarrow \varprojlim_{I'} u^*(F) \tag{4.7}$$

(flèche dans $\underline{\mathrm{Hom}}(\mathcal{M}(I), \mathcal{M})$ si les $\varprojlim$ de types I, I' existent dans $\mathcal{M}$, sinon flèche dans $\underline{\mathrm{Hom}}(\mathcal{M}(I), \mathcal{M}^{\wedge})$) est un isomorphisme. En se ramenant au cas de petites catégories $\mathcal{M}$, chacune plongée dans $\mathcal{M}^{\wedge}$, on voit par le critère (4.2 bis) que

(4.8) u est $\mathcal{M}$-connexe si pour tout $i \in \mathrm{Ob}\, I$,

le foncteur induit $I'/i \longrightarrow I/i$ est $\mathcal{M}$-cofinal. La réciproque n'est pas claire, car u est $\mathcal{M}$-connexe si et seulement si pour tout $i \in \mathrm{Ob}\, I$ et tout $\mathcal{M}$-préfaisceau F_i sur I/i *induit par un préfaisceau F sur I,* l'homomorphisme $\varprojlim_{I/i} F_i \longrightarrow \varprojlim_{I'/i} (u/i)^*(F/i)$ est [un] isomorphisme. p. 33 [51]

On ne peut donc affirmer d'emblée que cela implique itou pour *tout* F_i sur I/i, que si on sait que $(u/i)^* : \mathcal{M}(I) \longrightarrow \mathcal{M}(I/i)$ est essentiellement surjectif. Il en est ainsi pour tout i quand I est *ordonné,* car alors $I/i \longrightarrow I$ est pleinement fidèle, donc $\mathcal{M}(I/i) \longrightarrow \mathcal{M}(I)$ [est] pleinement fidèle, donc $\mathcal{M}(I) \longrightarrow \mathcal{M}(I/i)$ [est] un foncteur de localisation, en particulier essentiellement surjectif.

Mais quand $\mathcal{M} = (\mathrm{Ens})$, cette perplexité disparaît. On trouve la [52]

Commentary · Exposition [ms. 51]

The missing pointwise limit hypothesis

Grothendieck isolates a real difference between testing every coefficient on a slice and testing only coefficients restricted from the ambient category. There is also a second issue here: full faithfulness for coefficients in $\mathcal{M}$ does not imply full faithfulness after replacing $\mathcal{M}$ by its presheaf completion. The sufficient implication (4.8) is valid. The converse discussed on page 51, including its asserted ordered case, requires hypotheses ensuring that the relevant pointwise Kan-extension limits exist in $\mathcal{M}$.

Take $I = [1] = (0 \longrightarrow 1)$, $I' = \{1\}$, and let $\mathcal{M}$ be the discrete category on two objects m, n. Every $\mathcal{M}$-valued presheaf on I is constant, so restriction $\mathcal{M}(I) \longrightarrow \mathcal{M}(I')$ is an equivalence. Nevertheless, at $i = 0$ the induced functor on slices is $\varnothing \longrightarrow *$. In $\mathcal{M}^\wedge \simeq \mathrm{Set} \times \mathrm{Set}$, the Yoneda image of m is $(1, \varnothing)$, whereas the limit of the empty diagram is $(1, 1)$. Thus the comparison

$$\varprojlim_{I/0} y(m) = (1, \varnothing) \longrightarrow \varprojlim_{I'/0} y(m) = (1, 1)$$

is not invertible. This coefficient on $I/0$ even extends to I. Hence the intermediate assertion that $\mathcal{M}$-connectedness is equivalent to the slice comparison for restricted coefficients fails without pointwise limit hypotheses.

Existence of an abstract adjoint alone is insufficient: in this example restriction is an equivalence and has both adjoints. What fails is the formula that would compute its right adjoint by an empty limit inside $\mathcal{M}$. Completeness of $\mathcal{M}$, or existence of all the particular comma limits involved, supplies the needed repair.

The extension observation for posets is independently correct. Identify I/i with $A = \{x \mid x \leq i\}$. The map $r : I \longrightarrow A$ sending x to itself when $x \leq i$ and to i otherwise is order preserving: no arrow can enter A from outside, and an arrow leaving A is sent to $x \leq i$. With $j : A \hookrightarrow I$, one has $rj = 1_A$, so every coefficient on A extends by r^{op}. Under the pointwise limit hypotheses this proves the ordered-case converse. It does not remove those hypotheses. Proposition 4.1 subsequently quantifies over all coefficient categories and can use sets from the start; its intended theorem is unaffected by this coefficient-specific error.

Proposition 4.1. *Soit $u : I' \longrightarrow I$ une flèche dans* Cat. *Conditions équivalentes :*

a) Pour toute catégorie $\mathcal{M}$, $u^*_{\mathcal{M}} : \mathcal{M}(I) \longrightarrow \mathcal{M}(I')$ est pleinement fidèle (i.e. u est $\mathcal{M}$-connexe).

a′) Pour toute catégorie $\mathcal{M}$ où les limites inductives de type I, I' existent, la flèche d'adjonction pour F dans $\mathcal{M}(I)$

$$u_! u^* F \xrightarrow{\sim} F$$

est [un] *isomorphisme.*

a″) Dual de a′), avec [la] *flèche d'adjonction*

$$u_* u^* \xleftarrow{\sim} F.$$

b) b′) b″) Comme a) a′) a″), avec $\mathcal{M} =$ (Ens).

c) Comme b′), avec F représentable, i.e. de la forme $j \longmapsto \mathrm{Hom}_I(j, i)$.

d) Pour tout I dans I, $I'/i \longrightarrow I/i$ est cofinal.

d′) Pour tout i dans I, $i\backslash I' \longrightarrow i\backslash I$ est cofinal.

d″) Pour toute flèche $i_0 \xrightarrow{\alpha} i_1$ dans I, la catégorie $i_0\backslash\underset{\alpha}{I'}/i_1 \overset{\text{déf}}{=} I' \times_I \underbrace{i_0\backslash\underset{\alpha}{I}/i_1}_{\substack{\text{catégorie biïnduite} \\ (i_0,\alpha)\backslash(I/i_1) \simeq (i_0\backslash I)/(i_1,\alpha)}}$

est

0-*connexe.*

[53]

Démonstration. On sait déjà, par les préliminaires, l'équivalence de a, a′, a″, b, b′, b″, soit A cette condition - nous dirons que u est W_0-*connexe* ou W_0-*biasphérique,* si ces conditions équivalentes sont satisfaites. On a de plus l'autodualité

(4.8) u est W_0-connexe si et seulement si u^o l'est,

car on a évidemment

(4.9) u est $\mathcal{M}$-connexe si et seulement si u^o est $\mathcal{M}^o$-connexe.

Commentary · Later developments [ms. 53]

Biasphericity, lax epimorphisms, and factorization categories

The new question in Section 4 is whether a diagram and all its natural transformations can be recovered after restriction. The answer is governed by factorizations of arrows, just as ordinary cofinality was governed by objects of comma categories. This is the ordinary version of the factorization test encountered in the commentary on Chapter I, manuscript p. 49 ("The factorization category controls base change"). The proof of Proposition 4.1 continues beyond these pages; we give the complete argument for its corrected statement here.

For $u : A \longrightarrow B$ and $\gamma : b_0 \longrightarrow b_1$, let $\mathcal{F}_u(\gamma)$ have objects

$$(a, p, q), \qquad b_0 \xrightarrow{p} ua \xrightarrow{q} b_1, \quad qp = \gamma,$$

and arrows $h : a \longrightarrow a'$ satisfying $u(h)p = p'$ and $q'u(h) = q$. Then the following conditions are equivalent: restriction on presheaves is fully faithful for every coefficient category; it is fully faithful for sets; and every $\mathcal{F}_u(\gamma)$ is nonempty and connected.

To prove this, first work with sets, where both Kan extensions exist. Full faithfulness of u^* is equivalent to the counit $u_!u^*F \longrightarrow F$ being an isomorphism. Both sides preserve colimits in F, and every presheaf is a colimit of representables, so it suffices to take $F = h_{b_1} = B(-, b_1)$. At b_0 the pointwise colimit formula gives

$$(u_!u^*h_{b_1})(b_0) \cong \coprod_{\gamma \in B(b_0, b_1)} \pi_0 \mathcal{F}_u(\gamma).$$

Indeed, the generators are pairs $b_0 \longrightarrow ua \longrightarrow b_1$, and the colimit relations are exactly the zigzags induced by arrows of $\mathcal{F}_u(\gamma)$. Composition labels every such relation by the same γ. The counit sends a connected component to that composite. It is bijective if and only if each indicated category is nonempty and connected.

The set-valued criterion implies the version for arbitrary coefficients. It first implies full faithfulness for coefficients in every presheaf category, by applying the set-valued statement in each variable. A small category $\mathcal{M}$ embeds fully faithfully into its presheaf category by Yoneda; the full-faithfulness property descends to this full subcategory. For a possibly large locally small $\mathcal{M}$, any two diagrams on the small category B take values in a small full subcategory. Applying the preceding argument there descends every natural transformation and proves uniqueness. This establishes the reductions (4.3)–(4.6), with the usual universe enlargement when necessary.

For fixed b_1, the comma category of the induced functor $A/b_1 \longrightarrow B/b_1$ under (b_0, γ) is $\mathcal{F}_u(\gamma)$. Thus condition (d) is equivalent to the factorization criterion. Its dual says that

$$b_0\backslash A \longrightarrow b_0\backslash B \quad \text{is } \textit{co-cofinal}\text{, that is, initial.}$$

The edition instead prints "cofinal" in (d′). That cannot be retained as a mathematical equivalence. The inclusion $u : \{1\} \hookrightarrow [1]$ satisfies the printed condition: for $i = 0$ it induces the final-vertex inclusion into $[1]$, and for $i = 1$ it induces the identity of a point. Yet restriction on presheaves is not faithful: take $F(1) = \varnothing$, $F(0) = \{0, 1\}$; its identity and the automorphism transposing $0, 1$ become the same map after restriction. The corrected dual condition has comma categories opposite to $\mathcal{F}_u(\gamma)$, so it gives exactly the required connectedness. The two adjunction formulations and self-duality now follow, proving all the intended equivalences.

In later terminology these functors are *lax epimorphisms*, also called absolutely dense functors. Adámek et al. (2001) identify them by full faithfulness of set-valued restriction and by connected

categories of factorizations; the latter characterization is restated explicitly in Lucatelli Nunes and Sousa (2022, Theorem 4.1). For derivators, Groth and Šťovíček (2016, Theorem 6.10) replace connectedness by weak contractibility of the same factorization categories and obtain *homotopical epimorphisms*. Thus the arrowwise obstruction persists; the homotopy theory changes the required strength of its vanishing.

A quotient group displays the difference concretely. For a surjective homomorphism $\rho : H \twoheadrightarrow G$, restriction of G-sets to H-sets is fully faithful, because equivariance under H is equivalent to equivariance under its image G. For any $g \in G$, the category $\mathcal{F}_{B\rho}(g)$ is the transitive action groupoid with object set G, on which H acts through ρ; its stabilizer is $\ker \rho$. It is therefore equivalent to $B \ker \rho$. The functor $B\rho$ is an ordinary lax epimorphism, but it is a homotopical epimorphism only when $\ker \rho$ is trivial. With $\rho : \mathbf{Z} \twoheadrightarrow \mathbf{Z}/n\mathbf{Z}$, the obstruction has the homotopy type of a circle. The full-faithfulness statement about ordinary diagrams is correct; it simply does not recover that circle.

Overview · II-a

Cofinality, connected objects, and the recovery of diagrams.

Chapter II begins by asking when a change of indexing category leaves sections unchanged (Grothendieck, 1990–1991b). It ends this first part with the stronger question of when restriction recovers a whole diagram and every natural transformation between diagrams. Between these questions Grothendieck develops a topos-theoretic account of filtering, examines the homotopy information lost by ordinary connectedness, and specializes the language of derivators to sets. The movement is deliberate. Ordinary cofinality is not merely an elementary preliminary to the derived theory: its proof identifies the object which must be derived, and its limitations identify the higher information which a derivator must retain.

The conventions are essential. A coefficient on C is a contravariant functor, and the category denoted $c\backslash C'$ has objects $c \to \alpha(c')$. Thus a cofinal functor in these pages is final in the usual covariant terminology; its opposite is initial for the diagrams whose limits are being taken. Under this convention the first question has a representability answer. The presheaf

$$K_\alpha(c) = \pi_0(c\backslash C') = (\alpha_! 1_{C'})(c)$$

represents $F \mapsto \Gamma(C', \alpha^* F)$. The comparison with $\Gamma(C, F)$ is induced by $K_\alpha \to 1_C$. It is an isomorphism for all F exactly when every indicated comma category is nonempty and connected. This is the classical cofinality theorem of SGA 4, Exposé I, §8.1 (Artin et al., 1972). The initial weak cofiltering hypothesis is sufficient, and Grothendieck replaces it by the exact connectedness criterion as the argument develops.

The same proof suggests its own derived form. For a commutative ring k, replace the component presheaf by the derived weight $\mathbf{L}\alpha_! k_{C'}$. Its value at c is represented by chains on $N(c\backslash C')$. Derived adjunction then gives

$$\mathbf{R}\Gamma(C', \alpha^* F) \simeq \mathbf{R}\operatorname{Hom}_{C;k}(\mathbf{L}\alpha_! k_{C'}, F).$$

Consequently weak contractibility of all the comma nerves implies cohomological cofinality, answering the question on manuscript page 5 affirmatively. For these fixed linear coefficients, their k-acyclicity already suffices. For spaces the weight is the corresponding presheaf of classifying spaces, and the mapping-space adjunction gives the homotopical statement. Modern higher category theory places this argument within the general criterion for cofinality (Lurie, 2009, Theorem 4.1.3.1). The commentary constructs the linear comparison by a bar resolution, so the answer does not depend on the internal-Hom formula subsequently proposed in the manuscript.

That formula is a genuine point of failure. The global derived Hom belongs to the category of coefficients at a point; a sheaf-valued internal Hom belongs to the diagram category. Applying global sections can identify two such objects without identifying the objects themselves. For the terminal-vertex inclusion $i : \{1\} \hookrightarrow [1]$, a presheaf is an arrow $F_1 \to F_0$, and

$$i_! A = (A \xrightarrow{1} A), \qquad i_* A = (A \to *).$$

The comma categories for cofinality are points, and global sections are F_1. Nevertheless $i_* i^* F$ need not be F, while $\underline{\operatorname{Hom}}(i_! 1, F) = F$. This disproves the unrestricted internal identification in sets, hence already before derived complications arise. The correct internal comparison is equivalent to an additional projection-formula property, whose canonical arrow has the opposite direction from the one displayed in the manuscript. Our point-inclusion calculation identifies its obstruction exactly: for a fixed presheaf, every restriction along an arrow into the chosen object must be invertible. The formula holds for all coefficients precisely when all those incoming arrows are isomorphisms.

The turn to toposes is motivated by a different problem: how to recognize the equalization part of filtering without inspecting individual arrows. An irreducible presheaf topos records

common predecessors of objects, a condition visible in the underlying preorder. Local irreducibility records common predecessors compatible with a prescribed pair of arrows, namely PS 1. It still does not supply PS 2, the equalization of arbitrary parallel arrows. The parallel-arrow category $0 \rightrightarrows 1$ separates the first two conditions. A nontrivial group category BG separates local irreducibility from local cofilteredness: every representable slice is equivalent to a point, but cancellation prevents the equalization of distinct group elements. Several preliminary identifications in the manuscript are explicitly withdrawn by its annotations. The audit preserves this sequence of correction and distinguishes it from errors of transcription.

Total 0-connectedness adds preservation of products by the component functor. For $\widehat{X} = \mathrm{Fun}(X^{\mathrm{op}}, \mathrm{Set})$, it says that X^{op} is an ordinary sifted category. Sifted colimits preserve finite products; filtered colimits preserve finite limits, including equalizers (Adámek et al., 2010). The simplex category supplies the instructive distinction. The products of its representables have connected categories of elements, but the two vertices of [1] admit no common predecessor over [1]. Thus its presheaf topos is totally 0-connected without being locally irreducible. Nothing about this ordinary product test alone makes all the relevant nerves weakly contractible.

The surviving theorem in 2.6 makes the product condition hereditary under connected slices. Its equivalent formulations are (a), (a'), (b), and (d); the generator-slice conditions (c) and (c') are strictly weaker. The diagonal $b \to b \times b$ explains the distinction. Once $b \times b$ is connected and the topos is locally irreducible, its pullback against $(u, v) : a \to b \times b$ is noninitial and yields an equalizing arrow. Conversely, PS 1 and PS 2 lift to categories of elements, giving connected products in every connected slice. The commentary supplies the omitted generator-component lemma and the successive equalizations in this argument. It also restores connectedness of the base where the printed formulation of 2.8 omits it.

On a connected topos satisfying this hereditary condition, π_0 preserves finite limits. This makes the later cohomological argument precise. The induction must run over connected, locally totally 0-connected toposes, since it repeatedly passes to connected components of slice toposes. The degree-zero row of the covering spectral sequence is the augmented complex of functions $I^{p+1} \to M$, where I is the component set of a covering object. Inserting a chosen element of I contracts that complex. The stronger homotopical conclusion follows from hypercovers: applying the exact component functor to their matching-object diagrams gives boundary fillers in every degree, so the resulting simplicial sets are weakly contractible. This connects the argument to the classical hypercover construction of homotopy types (Artin and Mazur, 1969, §§8–9). It concerns constant coefficients and shape. It does not trivialize arbitrary sheaf cohomology, as the worked calculation $H^1(\mathbf{P}^1_k, \mathcal{O}(-2)) \cong k$ illustrates.

The ind-object reformulation in 2.15–2.18 returns to a classical theorem: a presheaf is a filtered colimit of representables exactly when its category of elements is filtered (Artin et al., 1972, Exposé I, Theorem 8.3.3). Taking the terminal presheaf recovers filteredness of the indexing category. Taking its ordinary left Kan extension along an arbitrary functor shows that the resulting category of elements is again filtered. The modern minimal-localizer theorem supplies the asphericity conclusion for every fundamental localizer, without the additional limit axiom used in the historical formulation (Cisinski, 2004, Theorem 2.2.11).

The converse question on page 36 has a stronger quantifier than later uses of the phrase "totally aspherical". In the commentary we denote by $T(A)$ the condition that every ordinary cofinal target of A have weakly contractible nerve. Filtered categories satisfy it. The attempted proof that ordinary siftedness suffices is retracted on pages 36bis–36ter; our explicit completion of the counterexample uses the simplicial circle and a one-point collage whose classifying space is S^2. We also prove a necessary common-successor condition for $T(A)$. Its obstruction is a four-object category with a cofinal three-object subcategory and a nonzero integral two-cycle. This settles $T(A) \Leftrightarrow A$ filtered for preorders. For general categories the remaining issue is equalization of parallel arrows. We did not locate a resolution under this exact universal

quantifier in the sources checked; this is a report of that bounded search, not a declaration of a recognized open problem. Results developed here are marked as the commentators' work and carry no claim of priority.

Section 3 shows how much of the earlier derivator terminology can be tested with sets. Pointwise localization of category-valued diagrams at component equivalences yields the represented derivator of sets. Properness and smoothness become connectedness of strict lift categories. A universal W_0-equivalence can be tested over vertices and arrows; a strong W_0-fibration is characterized component by component by those universal equivalences. Coverings explain why the weak condition is weaker: monodromy can connect different fibre components without making each fibre connected. The comparison with the commentary on Chapter I is made at the precise lifting and monodromy arguments, rather than repeating its whole discussion.

The final transition is from preservation of global sections to full faithfulness of restriction. Proposition 4.1 characterizes the latter by nonempty connected categories of factorizations

$$b_0 \longrightarrow u(a) \longrightarrow b_1$$

of every arrow $b_0 \to b_1$. These are the lax epimorphisms, or absolutely dense functors, of the later literature (Adámek et al., 2001). The homotopical version requires weakly contractible factorization categories (Groth and Šťovíček, 2016, Theorem 6.10). A quotient of groups gives a particularly concrete distinction: the factorization category has the homotopy type of the classifying space of the kernel. Two qualifications belong to the audit. The printed coslice condition in 4.1(d') must be initial, not final. Also the coefficient-specific discussion on page 51 invokes pointwise Kan-extension formulas without the necessary comma limits; a discrete coefficient category on two objects gives a counterexample even when the indexing category is ordered. The corrected theorem quantified over all coefficient categories is established by its set-valued case and Yoneda embeddings.

The historical setting is the program of coherent diagram categories described in Grothendieck (1983, §69), alongside the systematic homotopy Kan extensions of Heller (1988). The commentary keeps the successive formulations, counterexamples, and marginal revisions visible. It distinguishes a proof supplied here from a proof actually written in these pages, and distinguishes a settled theorem from an unsuccessful search for a later treatment.

Ceci montre l'équivalence de d, d′, qui se voit aussi par l'equivalence de chacune de ces deux conditions avec d″, par le critère de cofinalité du paragraphe précedent. Soit D cette condition sur u. On a par (4.8) D $\Longrightarrow$ A, donc p. 34

$$\mathrm{D} \Longrightarrow \mathrm{A} \Longrightarrow \mathrm{c},$$

il reste à prouver que c $\Longrightarrow$ D, plus précisément c $\Longrightarrow$ d″ et c′ $\Longrightarrow$ d″ [c′?]. Explicitons le morphisme d'adjonction (pour i dans I)

$$u_! \underbrace{u^* i} \longrightarrow i,$$

i.e. pour j dans I

$$\underbrace{(u_! u^* i)(j)}_{\pi_0(j\backslash(I'/u^*(i)))} \longrightarrow i(j) = \mathrm{Hom}(j, i).$$

Or $I'/u^*(i) \simeq I'/i$, donc on a la flèche [page 54]

$$(u_! u^*(i))(j) \simeq \pi_0(j\backslash(I'/i)) \longrightarrow \mathrm{Hom}_I(j, i). \tag{4.10}$$

Or on voit aussitôt que pour $\alpha : j \longrightarrow i$ dans $\mathrm{Hom}(j, i)$, la fibre de (4.10) en u s'identifie à $\pi_0(j\backslash\underset{\alpha}{I'}/i\,)$, de façon plus précise, $j\backslash\underset{\alpha}{I'}/i$ est la sous-catégorie ouverte et fermée de $J = j\backslash(I'/i)$ correspondant à la partie du $\pi_0(J)$ qui est au dessus de α dans (4.10). Cela achève la démonstration de 4.1.

Commentary · Exposition [ms. 54]

Reading the two-sided criterion

Grothendieck has now converted full faithfulness of restriction into a condition on individual arrows. This is the point at which the cofinality theorem becomes a tool for reconstructing an entire diagram. The commentary at manuscript p. 53 ("Biasphericity, lax epimorphisms, and factorization categories") already discusses these pages; we retain its notation and add two clarifications needed by the argument that follows.

For $u : A \to B$ and $\gamma : b_0 \to b_1$, write $\mathrm{Fact}_u(\gamma)$ for the category whose objects are triples (a, r, s) with $r : b_0 \to u(a)$, $s : u(a) \to b_1$, and $sr = \gamma$. A morphism to (a', r', s') is $h : a \to a'$ with $u(h)r = r'$ and $s'u(h) = s$. The map of connected components in (4.10) has precisely $\pi_0 \mathrm{Fact}_u(\gamma)$ as its fibre over γ. Its being a bijection therefore says that every arrow has a factorization through u, unique up to a finite zigzag of such factorizations. This is stronger than merely obtaining a connected comma category at each object.

The direction in 4.1(d′) must accordingly be read with care. For B/b_1, the relevant comma category is $\gamma\backslash(A/b_1)$, so its projection must be *cofinal* in the convention of Definition 3.3. For $b_0\backslash B$, the relevant category is $(b_0\backslash A)/\gamma$, so the dual projection must be *co-cofinal*. The edition prints "cofinal" in both places. Taken literally, the second condition is too weak: the inclusion $\{1\} \to [1]$ satisfies it, although restriction of presheaves to the terminal object is not fully faithful. The corrected orientation is also the one used in Proposition 4.4(b).

There is a separate existence issue in 4.1(a′), already isolated in the commentary at manuscript p. 51 ("The missing pointwise limit hypothesis"). The left Kan extension requires the colimits over the appropriate comma categories, not merely colimits of shapes I and I'. Working in a cocomplete coefficient category is sufficient. Once the adjoints exist, the full-faithfulness equivalences are formal, and the representable counit test is valid because both $u_! u^*$ and the identity preserve colimits. No comparable argument makes the representable *unit* test sufficient: its exact meaning is different.

NB Une autre façon de prouver que c $\Longrightarrow$ A c'est de noter que les deux membres dans le morphisme d'adjonction $u_! u^* F \longrightarrow F$ dans $I^\wedge$ sont des foncteurs $I^\wedge \longrightarrow I^\wedge$ qui commutent aux $\varinjlim$ quelconques. Donc pour vérifier que c'est [un] isomorphisme, il suffit de le vérifier pour $\overrightarrow{F}$ dans I, puisque I engendre $I^\wedge$ par $\varinjlim$. Par contre, un argument de ce type ne semble [page 55]

pas prouver que pour vérifier que l'on a un *isomorphisme* $F \longrightarrow u_* u^* F$, il suffise de le faire pour F représentable, i.e. de vérifier b″ pour F représentable.

Il resterait à voir si c′ $\Longrightarrow$ d″ [c′?].

Explicitons le morphisme d'adjonction

$$i \longrightarrow u_* u^* i$$

(pour $i \in \mathrm{Ob}\, I$) par ses valeurs sur j

$$(*) \qquad \underbrace{i(j)}_{\mathrm{Hom}(j,i)} \longrightarrow \Gamma(I'/j, u^* i).$$

Soit donc

$$u_j : I'/j \longrightarrow I,$$

on a

$$\Gamma(I'/j, \underbrace{u^*(i)|I'/j}_{u_j^*(i)}) \simeq \mathrm{Hom}(e_{I'/j}, u^*(i)) \qquad \text{p. 35}$$

$$\overset{\mathrm{adj}}{\simeq} \mathrm{Hom}(u_{j,!} e_{I'/j}, i),$$

et la flèche $(*)$, fonctorielle en i, est déduite d'une flèche

$$(**) \qquad u_{j,!}(e_{I'/j}) \longrightarrow j.$$

On voit tout de suite, comme ci-dessus, que $(**)$ est [un] isomorphisme si et seulement si pour tout $\alpha : k \longrightarrow j$, la catégorie $k \backslash \underset{\alpha}{I'/j}$ est 0-connexe, et si on l'exige pour tout j, k, α, c'est la condition d″. Mais il n'est pas clair que pour qu'on ait $(**)$ isomorphisme, il suffise que le Hom des deux membres à valeurs dans tout i dans I (et non tout F dans $I^\wedge$) soit [un] isomorphisme. La difficulté provient du fait que $u_{j,!}(e_{I'/j})$ n'est pas forcément représentable, et qu'un F quelconque dans $I^\wedge$ n'est pas forcément $\varinjlim$ de foncteurs représentables. Donc il est douteux que c′ [c′?] suffise à entraîner que u est $\overline{W_0}$-connexe. [page 56]

Commentary · Examples [ms. 56]

The unit on representables tests density

Grothendieck's hesitation on pages 54–56 identifies a genuine distinction. He asks whether knowing every representable presheaf can be recovered by right Kan extension already recovers all presheaves. The condition has a familiar modern interpretation: it says that u is dense. Density reconstructs the objects of the original category from their restricted Yoneda images; it does not reconstruct its entire presheaf category.

Indeed, let $y_B(b) = B(-, b)$. By adjunction, the component at c of the unit on $y_B(b)$ is the map

$$B(c, b) \longrightarrow \mathrm{Nat}\big(B(u-, c), B(u-, b)\big).$$

Thus the unit is invertible on all representables exactly when the restricted Yoneda functor $B \to \mathrm{Set}^{A^{\mathrm{op}}}$ is fully faithful. This is the definition of a dense functor; see Riehl (n.d., Definition 6.5.11). It has a different quantifier from the full faithfulness of u^* on *all* presheaves.

A finite example settles the question without any existence qualifications. Let B be the full subcategory of sets on the objects $1 = \{*\}$ and $2 = \{0, 1\}$, and let $u : e \to B$ select 1. For $b, c \in \{1, 2\}$, evaluation on elements gives

$$B(c, b) \xrightarrow{\sim} \mathrm{Set}\big(B(1, c), B(1, b)\big).$$

Both sides are simply the set of functions $c \to b$, and the displayed map is the identity under the indicated identifications. Hence every unit on a representable is an isomorphism. On the other hand, $\mathrm{Fact}_u(\mathrm{id}_2)$ is empty: any composite $2 \to 1 \to 2$ is constant, whereas id_2 is not. Proposition 4.1 therefore shows that u is not W_0-connected.

One can see the failure directly at the counit. The presheaf $u_! u^* y_B(2)$ evaluated at 2 parametrizes factorizations $2 \to 1 \to 2$; its image in $B(2, 2)$ consists of the two constant functions and misses the

identity and the transposition. The representable unit has nevertheless passed every test. The two adjunction maps carry different information.

This is an instance of the standard distinction between density and absolute density. Adámek et al. (2001) identify lax epimorphisms with absolutely dense functors and with the connected-factorization condition. The construction above applies that distinction to the question left here; we do not claim a new density theorem. It also explains the failed proof: representables generate a presheaf category by colimits, whereas the proposed right-adjoint argument would require an appropriate cogeneration statement, which is absent.

Corollaire 4.2. *Pour que u soit W_0-connexe, il* suffit *qu'il soit une W_0-équivalence universelle, i.e. (cf. 3.1) que les fibres soient 0-connexes, et que pour toute flèche non identique dans I, définissant $\mathbf{\Delta}^1 \longrightarrow I$, $I' \times_I \mathbf{\Delta}^1$ soit 0-connexe.*

4.3. Cette condition de W_0-équivalence universelle est cependant *strictement* plus forte que la W_0-connexité, comme on le voit sur l'exemple où u est une *équivalence de catégories* non surjective sur les objets (p.ex. l'inclusion d'un sommet dans un groupoïde 1-connexe ayant plus d'un objet). Plus généralement, tout foncteur de localisation est W_0-connexe, alors que ses fibres ne sont pas forcément connexes, et a fortiori ce n'est pas une W_0-équivalence universelle.

Cependant, si tout endomorphisme p d'un i dans I, tel que $p^2 = p$ (projecteur) est l'identité, et si $I' \longrightarrow I$ est 'transportable', p.ex. si dans I tout isomorphisme est une identité, alors u [page 57]
W_0-connexe implique que les fibres de u sont 0-connexes, comme on voit en exprimant d'' pour α flèche identique. Les conditions envisagées sur I sont satisfaites si I est ordonnée. Mais même si I' et I sont toutes deux ordonnées, u peut être W_0-connexe, sans être une W_0-équivalence universelle. Voici un exemple avec $I = \mathbf{\Delta}^2$:

$$I' = \left\{ \begin{array}{ccc} & & 1' \longrightarrow 2 \\ & & \downarrow \\ 0 & \longrightarrow & 1 \end{array} \right\} \qquad \downarrow$$

$$I = \left\{ 0 \longrightarrow 1 \longrightarrow 2 \right\}.$$

Les catégories bilocalisées de I sont les trois sommets $\{0\}$, $\{1\}$, $\{2\}$, et les trois segments p. 36
$[0,1]$, $[1,2]$ et $[0,2] = I$, les images inverses sont respectivement

$$\bullet\,, \quad \downarrow\,, \quad \bullet\,, \quad \begin{array}{c} \longrightarrow \\ \longrightarrow\downarrow \end{array}\,, \quad \begin{array}{c} \longrightarrow \\ \downarrow \end{array}\,, \quad I'\,,$$

elles sont toutes 0-connexes, i.e. W_0-asphériques, et même contractiles donc W_ω-asphériques (donc u est même W_ω-connexe, i.e. W_ω-biasphérique). Mais ce n'est pas une W_0-équivalence universelle, car l'image inverse de I' par $\mathbf{\Delta}^1 \xrightarrow{\partial} \mathbf{\Delta}^2$, $\partial(0) = 0$, $\partial(1) = 2$, est la catégorie disconnexe $\varepsilon = \{\bullet \quad \bullet\}$. [page 58]

J'avais espéré un moment que les foncteurs W_0-connexes $I' \xrightarrow{u} I$ seraient ceux qui se factorisent en $I' \xrightarrow{u_0} I'' \xrightarrow{v} I$, avec u_0 une équivalence, et v une W_0-équivalence universelle. Mais l'exemple précédent le met en défaut, avec u même W_ω-connexe. En effet, I'' serait déduit d'un ensemble préordonné, et v se factoriserait par l'ensemble ordonné correspondant, savoir I', donc serait de la forme

$$I'' \longrightarrow I' \xrightarrow{u} I,$$

où $I'' \longrightarrow I'$ est un passage au quotient, et l'image inverse de $\mathbf{\Delta}^1 \xrightarrow{\partial} I$ serait encore disconnexe (un π_0 de cardinal 2). Donc c'est vraiment sans espoir.

Commentary · Examples [ms. 58]

A contractible replacement which loses an edge

Grothendieck next tests whether full faithfulness might conceal a stronger, universally preserved equivalence. His four-object example separates these requirements even for finite posets. It deserves to be read as a complete computation, since the obstruction lies in a strict pullback which the homotopy type of the total category cannot detect.

Write A for the poset with precisely the nonidentity arrows

$$0 \longrightarrow 1, \qquad 1' \longrightarrow 1, \qquad 1' \longrightarrow 2,$$

and let $u : A \to [2]$ identify $1'$ with 1. For the three identity arrows of [2], the factorization categories are respectively the point $\{0\}$, the interval $1' \to 1$, and the point $\{2\}$. For $0 \to 1$ and $1 \to 2$, they are the full subposets $0 \to 1 \leftarrow 1'$ and $1 \leftarrow 1' \to 2$. The former has a terminal object and the latter an initial object. For $0 \to 2$, the factorization category is all of A: its nerve has no nondegenerate simplices above dimension one, and its realization is the tree with successive vertices $0, 1, 1', 2$. All six nerves are contractible. Thus this is a homotopical epimorphism, and restriction is fully faithful for every derivator; the later criterion is Groth and Ševčík (2016, Theorem 6.10).

Now pull back along the edge $[1] \to [2]$ selecting $0 \to 2$. The only objects left are 0 and 2, and A contains no arrow between them. The pullback is the discrete two-point category, whose map to [1] is not a W_0-equivalence. Full faithfulness has survived every factorization test and failed this base-change test.

The same computation rules out the proposed factorization through an equivalence followed by a universal equivalence. Suppose $A \xrightarrow{v} C \xrightarrow{w} [2]$ were such a factorization, with v an equivalence. Since A is a poset, C is a preorder, and its poset of isomorphism classes identifies with A. Any functor from C to [2] is constant on isomorphism classes. The inverse image of the edge $0 \to 2$ consequently has two nonempty components, corresponding to the objects 0 and 2 of A, and no morphism joining them. It cannot be connected. This argument uses the skeletal quotient only to count components; it does not assume that an arbitrary strict pullback preserves an equivalence.

Cependant

Corollaire 4.3. *Supposons que u soit W_0-lisse, ou W_0-propre, ou une W_0-fibration faible. Alors u est W_0-connexe si et seulement si c'est une W_0-équivalence universelle, i.e. si et seulement si a) ses fibres sont 0-connexes et b) $I' \times_I \mathbf{\Delta}^1$ [est] 0-connexe pour tout $\mathbf{\Delta}^1 \longrightarrow I$ (correspondant à une flèche non identique).*

Dans les trois cas on sait que u est une W_0-équivalence universelle si et seulement si ses fibres sont W_0-connexes. Or la W_0-lissité implique que $\forall i \in \mathrm{Ob}\, I$, l'inclusion [page 59]

$$I'_i \longrightarrow i\backslash I'$$

est une W_0-équivalence, d'autre part $i\backslash I' \longrightarrow i\backslash I$ étant W_0-asphérique, donc une W_0-équivalence par hypothèse, $i\backslash I'$ est 0-connexe, donc I'_i aussi. Le cas u W_0-propre est dual. Dans le cas [où] u [est] une W_0-fibration faible, on sait encore que $I'_i \longrightarrow i\backslash I'$ est une W_0-équivalence, et on conclut comme dans le cas lisse (ou, au choix, via $I'_i \longrightarrow I'/i$).

Mais l'idée me vient que les foncteurs W_0-connexes pourraient être exactement les foncteurs de localisation ? Il revient au même de dire qu'un foncteur $u : I' \longrightarrow I$ *conservatif* (i.e. $u(\alpha)$ isomorphisme $\Longrightarrow \alpha$ isomorphisme) qui est W_0-connexe, est une équivalence. Mais c'est faux, comme on voit en prenant $I = e$, I' un groupoïde 0-connexe et non 1-connexe : $I' \longrightarrow e$ est W_0-connexe (et même une W_0-équivalence universelle), mais ce n'est pas une équivalence (et pas un foncteur de localisation).

Commentary · Later developments [ms. 59]

The quantifier in homotopical epimorphisms

Grothendieck has separated three operations that are easy to conflate: localizing a category, embedding its diagram theory by restriction, and preserving an equivalence after every strict base change. The example $BG \to e$ at the end of this page separates the first two in dimension zero.

Constant G-sets form a full subcategory of G-sets, so restriction from sets is fully faithful; yet a localization of a groupoid at any collection of its arrows is equivalent to that groupoid, since all those arrows are already invertible. For $G \neq 1$, its projection to e is therefore not a localization.

The modern term *homotopical epimorphism* adds a quantifier absent from the fixed-$\mathbf{D}$ problem about to begin: restriction must be fully faithful for *every* derivator. Groth and Šťovíček (2016, Definition 6.1 and Theorem 6.10), in the arXiv v3 numbering, characterize this condition by weak contractibility of all the factorization categories. Their Remark 6.3(ii) expressly distinguishes full faithfulness for one specified derivator. The space-valued theory detects the universal condition; a particular theory of coefficients can forget more.

This development follows the path from Heller's homotopy theories and Cisinski's comparison results to a calculus of homotopy exact squares. For the present pages, the decisive feature is concrete: connected components in 4.1 are replaced by weakly contractible nerves. The application in tilting theory is equally concrete: a homotopical epimorphism identifies a category of coherent diagrams with a full subcategory of another one, described by unit or counit conditions. No claim that every such functor is an ordinary localization is required.

Je voudrais généraliser la notion de flèche W_0-biasphérique au cas d'un dérivateur $\mathbf{D}$ quelconque.

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[page 60]

Commentary · Exposition [ms. 60]

Recovering a global diagram from its slices

Grothendieck now wants the factorization criterion to work with an arbitrary theory of coefficients. The difficulty is no longer the adjunction: a right adjoint is fully faithful exactly when its counit is invertible, and a left adjoint exactly when its unit is invertible. The issue is whether the tests visible from global diagrams detect every diagram on a slice. Proposition 4.4 isolates a sufficient detection condition instead of assuming such a property implicitly.

Put $q = u/i : A/i \to B/i$, $\alpha : B/i \to B$, and let ξ lie in $\mathbf{D}(e)$. With the variance of the manuscript, the comparison of homotopy limits on B/i and A/i is induced, after applying $\mathrm{Hom}(\xi, -)$, by the counit

$$c_\xi : q_! q^*(\xi_{B/i}) \longrightarrow \xi_{B/i} \qquad \text{in } \mathbf{D}(B/i).$$

The subscripts in the edition's $(***)$ should be B/i on the object to which q^* is applied and on the target. The printed I'/i in both places makes that display ill-typed; the argument itself determines the correction.

If all c_ξ are invertible, then for every $G \in \mathbf{D}(B/i)$ the maps $\mathrm{Hom}(c_\xi, G)$ are bijections. By adjunction and Yoneda, the induced comparison of homotopy limits is invertible. Taking $G = \alpha^* F$ and using the pointwise right Kan extension formula proves that $F \to u_* u^* F$ is invertible at i. As this holds for every i, the derivator axiom detecting isomorphisms pointwise gives full faithfulness. This proves the sufficient direction without requiring every slice diagram to extend globally.

For the converse, full faithfulness supplies only

$$\mathrm{Hom}_{\mathbf{D}(B/i)}(c_\xi, \alpha^* F) \quad \text{bijective for every } F \in \mathbf{D}(B).$$

Adjunction turns this into bijectivity of $\mathrm{Hom}_{\mathbf{D}(B)}(\alpha_! c_\xi, F)$ for every F, so Yoneda says exactly that $\alpha_! c_\xi$ is invertible. Conservativity of $\alpha_!$ then proves that c_ξ is invertible. This is condition 3°; the proof needs conservativity only on this family of counits, although the stronger formulation is easier to use.

Condition 2° implies 3° by an elementary adjunction argument. If $L \dashv R$ and R is essentially surjective, then an arrow h for which Lh is invertible induces bijections on Hom into every Rd, hence into every object; therefore h is invertible. If B is a preorder, $B/i \to B$ is fully faithful. Its left Kan extension is fully faithful, so the unit $G \to \alpha^* \alpha_! G$ is invertible and α^* is essentially surjective. Thus 1° implies 2°, which implies 3°.

The hypothesis on coproducts in 4.7 will make 3° available in many coefficient theories, including pointed ones. It is a property of those coefficients, not a formal property of every derivator. The modern axioms used here—pointwise calculation of Kan extensions and pointwise detection of isomorphisms—are recorded in Groth (2013, Definition 1.5); their different roles will matter for the parenthetical assertion about faithfulness.

Proposition 4.4. *Soit* $\mathbf{D}$ *un dérivateur sur* Diag, *et* $u : I' \longrightarrow I$ *une flèche dans* Diag. *Supposons que* u_* *et* $u_!$ *existent, et se calculent de façon standard, pour tout flèche* u *dans* Diag. *(En fait, il suffit que ce soit le cas pour* u *et ses localisées* $u/i : I'/i \longrightarrow I/i$, *ou* u *et ses colocalisées* $i\backslash u : i\backslash I' \longrightarrow i\backslash I$.*) Considérons les conditions :*

a) $u^* : \mathbf{D}(I) \longrightarrow \mathbf{D}(I')$ `[est]` *pleinement fidèle.*

a′) $\forall$ F *dans* $\mathbf{D}(I)$, *la flèche*

$$u_! u^* F \longrightarrow F$$

est un isomorphisme.

a″) $\forall$ F *dans* $\mathbf{D}(I)$, *la flèche*

$$u_* u^* F \longleftarrow F$$

est un isomorphisme.

b) $\forall$ i *dans* I, *la flèche* $i\backslash I' \longrightarrow i\backslash I$ *est* $\mathbf{D}$*-asphérique.*

b′) $\forall$ i *dans* I, *la flèche* $I'/i \longrightarrow I/i$ *est* $\mathbf{D}$*-coasphérique.*

b″) $\forall$ $\alpha : i_0 \longrightarrow i_1$ *dans* I, *la catégorie* $i_0\backslash I'/i_1$ *est* $\mathbf{D}$*-asphérique.*

On a alors les implications

$$(b \Longleftrightarrow b' \Longleftrightarrow b'') \Longrightarrow (a \Longleftrightarrow a' \Longleftrightarrow a'').$$

On a $a \Longrightarrow b$ *(i.e. toutes les conditions licites envisagées sont équivalentes) dans chacun des cas suivants.* [page 61]

1° I *est préordonné.*

2° $\forall$ i *dans* I, $\alpha_i^* : \mathbf{D}(I) \longrightarrow \mathbf{D}(I/i)$ *est essentiellement surjectif.*

3° $\forall$ i *dans* I, *considérant* $\alpha_i : I/i \longrightarrow I$, *le foncteur* $\alpha_{i,!} : \mathbf{D}(I/i) \longrightarrow \mathbf{D}(I)$ *est conservatif.*

4° $\mathcal{D} = \mathrm{HOT}_{W_0} \approx \mathbf{D}_{\mathrm{Ens}}$.

DÉMONSTRATION. L'équivalence de a, a′, a″ est claire, soit A. De même l'equivalence de b, b′, b″, soit B.

Prouvons

$$B \Longrightarrow A$$

en supposant p.ex. l'existence de u_*. On admet tacitement, avec l'existence de $u_* : \mathbf{D}(I') \longrightarrow \mathbf{D}(I)$, le mode de calcul des fibres de $u_*(-)$. Donc A, via a″, équivaut à

$$H^\bullet_{\mathbf{D}}(I'/i, u^*F) \overset{iso}{\longleftarrow} H^\bullet_{\mathbf{D}}(I/i, F) \simeq F(i)$$

pour tout F dans $\mathbf{D}(I)$, tout i dans I. p. 38

Soit

$$\alpha_i : I/i \longrightarrow F, \qquad F_i = \alpha_i^*(F),$$

on doit donc exprimer que pour tout F_i provenant d'un F, on a

$$(*) \qquad H^\bullet_{\mathbf{D}}(I/i, F_i) \xrightarrow{\text{iso}} H^\bullet_{\mathbf{D}}(I'/i, (u/i)^*(F_i)).$$

Or sous réserve d'existence de $(u/i)_!$, dire que $(*)$ est `[un]` isomorphisme pour *tout* F_i de $\mathbf{D}(I/i)$, signifie que

$$u/i : I'/i \longrightarrow I/i$$

est $\mathbf{D}$-coasphérique, d'où b′ $\Longrightarrow$ a″. Il est clair que l'implication inverse est valable si α_i^* est essentiellement surjectif, c'est le cas 2°. Or 2° est satisfait si I est préordonné, d'où le cas 1°. [page 62]

Le cas 4° a déjà été traité (4.1). Il faut encore traiter le cas 3°, qui contient d'ailleurs le cas 2°, donc le cas 1° (et aussi le cas 4°, cf. plus bas).

Il faut exprimer l'isomorphie de la flèche $(*)$ pour F_i de la forme $\alpha_i^*(F)$. Soit ξ [un] objet de $\mathcal{A} = \mathcal{D}(e)$, il faut exprimer l'isomorphie de $\mathrm{Hom}(\xi, -)$ dans les deux membres de $(*)$, i.e. de

$$(**) \qquad \begin{array}{ccc} \mathrm{Hom}(I/i, F_i) & \longrightarrow & \mathrm{Hom}(\xi'_{I'/i}, (u/i)^*(F_i)) \\ & \overset{\text{adj.}}{\simeq} & \mathrm{Hom}((u/i)_!(\xi'_{I'/i}), F_i), \end{array}$$

[plutôt $\xi_{I'/i}$] flèche qui se déduit de

$$(***) \qquad (u/i)_!(u/i)^*(\xi_{I'/i}) \xrightarrow{\text{adj.}} \xi_{I'/i}$$

en prenant le Hom dans F_i. La condition u/i coasphérique (i.e. b′) signifie que $(***)$ est un isomorphisme pour tout ξ dans $\mathcal{A}$. *Suffit*-il pour ceci que son Hom dans un F_i soit bijectif, quand on restreint F_i dans $\mathbf{D}(I/i)$ à être dans l'image de $\mathbf{D}(I)$ par α_i^* ? Or cette flèche $(**)$, pour $F_i = \alpha_i^* F$, s'explicite par la formule d'adjonction relative au foncteur α_i, comme [page 63]

$$\mathrm{Hom}(\alpha_{i,!}(\xi_{I/i}), F) \longrightarrow \mathrm{Hom}(\alpha_{i,!}(\ldots), F).$$

Dire que c'est un isomorphisme pour tout F, signifie donc *exactement* que la transformée de la flèche $(***)$ par $\alpha_!$ est un isomorphisme. Pour en conclure que $(***)$ elle-même est [un] isomorphisme, la condition naturelle à imposer est donc que $\alpha_{i,!} : \mathbf{D}(I/i) \longrightarrow \mathbf{D}(I)$ soit conservatif. C'est le cas 3°. Cela achève la démonstration de 4.4.

4.5. Montrons que la condition 3° sur I n'est pas toujours satisfaite, dans le cas de $\mathbf{D} = (\mathrm{HOT}_{W_0})^{\mathrm{opp}} \approx (\mathbf{D}_{\mathrm{Ens}})^{\mathrm{opp}}$, bien que dans ce cas, il soit pourtant vrai que pour toute I' sur

I satisfaisant la condition a″ (autoduale), la condition b′ (elle aussi autoduale) est aussi satisfaite (par cas 4° ci-dessus). Je dis donc qu'il existe I et i dans I, tels que p. 39

$$\beta_{i,*} : (i\backslash I)^\wedge \longrightarrow I^\wedge$$

ne soit pas conservatif. (**NB** Par contre $\alpha_{i,!} : (I/i)^\wedge \longrightarrow I^\wedge$ est toujours conservatif, ce qui redonne le cas 4° comme cas particulier de 3°). Comme $\beta_i : i\backslash I \longrightarrow I$ est W_0-propre (et même W_ω-propre !), $\beta_{i,*}$ 'se calcule fibre par fibre', donc on a [page 64]

$$\beta_{i,*}(G)(j) \simeq \prod_{\alpha \in \mathrm{Hom}(i,j)} G(\alpha),$$

la fibre en question en j étant discrète. Prenons

$$I = 0 \underset{\beta}{\overset{\alpha}{\rightrightarrows}} 1, \qquad \text{avec } \alpha \neq \beta,$$

alors $0\backslash I$ est $\Psi = \begin{array}{ccc} & \overset{\alpha'}{\nearrow} & 1 \\ 0 & & \\ & \underset{\beta'}{\searrow} & 1' \end{array}$, et $\Psi^\wedge$ est la catégorie des diagrammes d'ensembles $G = \begin{array}{ccc} & \overset{\lambda}{\swarrow} & G_1 \\ G_0 & & \\ & \underset{\mu}{\nwarrow} & G_{1'} \end{array}$. Le foncteur β_* déduit de $\beta : i\backslash I \longrightarrow I$ transforme ce diagramme en

$$G_1 \times G_{1'} \underset{\lambda\,\mathrm{pr}_2}{\overset{\lambda\,\mathrm{pr}_1}{\rightrightarrows}} G_0.$$

[plutôt $\mu\,\mathrm{pr}_2$]. Si on se borne aux diagrammes G pour lesquels $G_{1'} = \emptyset$, β_* sur ceux-ci donne

$$\emptyset \rightrightarrows G_0,$$

donc la restriction de β_* à cette sous-catégorie (équivalente à $(\mathbf{\Delta}^1)^\wedge$) n'est pas fidèle, ni conservative, ne tenant compte que de la composante G_0, et non pas de G_1.

Commentary · Examples [ms. 64]

A product erases a component

Grothendieck's example explains why the conservative-slice hypothesis cannot simply be imposed on one chosen side of every derivator. A right Kan extension over a discrete fibre is a product, and the empty set can erase every other factor. This is an obstruction in ordinary presheaf theory, before any passage to homotopy categories.

Let B have two distinct arrows $\alpha, \beta : 0 \rightrightarrows 1$, and let $p : 0\backslash B \to B$ be the projection. Its domain has the shape $a \to b$, $a \to c$, with $p(a) = 0$ and $p(b) = p(c) = 1$. A presheaf G is a diagram $G_b \xrightarrow{\lambda} G_a \xleftarrow{\mu} G_c$, and

$$(p_*G)(0) = G_a, \qquad (p_*G)(1) = G_b \times G_c.$$

The two restriction maps are $\lambda\,\mathrm{pr}_1$ and $\mu\,\mathrm{pr}_2$. Take $G_a = 1$, $G_b = \varnothing$, $G_c = \varnothing$, and $H_a = 1$, $H_b = 1$, $H_c = \varnothing$. The evident map $G \to H$ is not an isomorphism, but $p_*G \to p_*H$ is the identity of the presheaf $\varnothing \rightrightarrows 1$. To see nonfaithfulness as well, take $G_b = 1$, $H_b = 2$ while keeping the a-values equal to 1 and the c-values empty; the two distinct maps on the b-component have identical direct images.

For the other variance, $B/i \to B$ is a discrete fibration and its left Kan extension is a disjoint union over each fibre. In sets, a map between two such unions that preserves the indexing is bijective exactly when each component is bijective. This proves the conservativity used in 4.4 for set coefficients. Dualizing the derivator exchanges these two mechanisms, which is why failure of the particular sufficient condition does not refute the equivalence of full faithfulness and biasphericity for sets or their opposite theory.

Definition 4.6. *On dit que $u : I' \longrightarrow I$ dans* Diag *est* **D**-biasphérique *si elle satisfait les conditions équivalentes b, b′, b″ de 4.4,* **D**-connexe *si elle satisfait la condition a, ou encore (sous réserve d'existence de $u_!$ resp. de u_*) les conditions équivalentes a′, a″.* [page 63bis]

Ainsi, **D**-biasphérique implique **D**-connexe, et la réciproque est vraie quand I satisfait (par rapport à **D**) la condition 3° de 4.4 (donc aussi, si elle satisfait l'une des conditions plus faibles 2°, 1°), ou la condition duale de celle-ci (vu que les notions **D**-biasphérique, **D**-connexe sont autoduales). À cet égard, notons le

Corollaire 4.7. *Supposons que $\mathcal{A} = \mathbf{D}(e)$ satisfasse la condition suivante : le foncteur*

$$\begin{cases} \mathcal{A} \times \mathcal{A} \longrightarrow \mathcal{A} \\ (\xi \quad , \quad \eta) \longrightarrow \xi \amalg \eta \end{cases}$$

est conservatif (resp. fidèle). Alors pour toute flèche étale $u : J \longrightarrow I$ dans Diag *(et en particulier pour $I/i \longrightarrow I$), le foncteur $u_! : \mathbf{D}(J) \longrightarrow \mathbf{D}(I)$ est conservatif (resp. fidèle)* ([1]). *Si cette condition, ou la condition duale, impliquant les produits $\xi \times \eta$ (ξ, η dans $\mathcal{A}$), est satisfaite, alors pour toute flèche $u : I' \longrightarrow I$ dans* Diag, *u est* **D***-connexe si et seulement si u est* **D***-biasphérique.* p. 40 [page 64bis]

Démonstration. L'hypothèse de conservativité, ou de fidélité se transporte aux sommes indexées par un ensemble quelconque (sous réserve d'existence), comme on voit en écrivant $\coprod_\lambda \xi_\lambda = \xi_{\lambda_0} \amalg \coprod_{\lambda \neq \lambda_0} \xi_\lambda$. L'assertion faite sur $u_!$ résulte alors du calcul de $u_!$ fibre par fibre (qui est valable vu que u est étale donc lisse), en tenant compte que les fibres sont discrètes.

1. **NB** Cette condition est satisfaite si $\forall \xi, \eta \in \mathcal{A}$ on a $\mathrm{Hom}(\xi, \eta) \neq \emptyset$, (car on se ramène à ceci : si u_1, u_2 sont deux flèches dans Ens entre ensembles *non vides,* et si $u_1 \times u_2$ est [un] isomorphisme, u_1, u_2 sont [des] isomorphismes). Cela s'applique en particulier si $\mathcal{A}$ admet un objet neutre, p.ex. si $\mathcal{A}$ est additive.

Commentary · Development (commentators') [ms. 64bis]

Faithfulness does not follow from the fibre calculation

Grothendieck uses disjoint fibres to promote a property of coproducts in $\mathbf{D}(e)$ to a property of $u_!$ on coherent diagrams. For conservativity, the argument is complete: an invertible $u_!h$ gives an invertible component h_j at every j, and the derivator axiom then makes h invertible. For faithfulness, pointwise evaluation does not provide the analogous conclusion. A small derived example shows that the parenthetical assertion is false.

An étale direct image killing a nonzero extension. [ours] Let k be a field and take $\mathbf{D}(K) = D(\mathrm{Vect}_k^{K^{\mathrm{op}}})$. There is a discrete fibration $p : J \to I$ for which $p_!$ is not faithful, although binary coproducts in $\mathbf{D}(e) = D(k)$ are faithful.

Proof. Take $I = (0 \to 1)$ and let J have objects a, b, c and the two nonidentity arrows $a \to b$, $a \to c$. Send a to 0 and both b, c to 1. The unique lifting condition for arrows with a fixed target verifies that p is a discrete fibration. A presheaf on J is written $(B \to A \leftarrow C)$. Its left Kan extension is

$$p_!(B \xrightarrow{f} A \xleftarrow{g} C) = (B \oplus C \xrightarrow{(f,g)} A).$$

This functor is exact, since it is computed by finite direct sums at each vertex. Its derived functor is consequently given by the same formula on complexes.

Consider the three representations

$$M = (k \to 0 \leftarrow 0), \qquad N = (0 \to k \xleftarrow{\mathrm{id}} k), \qquad E = (k \xrightarrow{\mathrm{id}} k \xleftarrow{\mathrm{id}} k).$$

There is a short exact sequence $0 \to N \to E \to M \to 0$: the inclusion is the identity at a, c and zero at b, and the quotient is the identity at b and zero elsewhere. It does not split. A section $M \to E$ would have to be the identity at b and zero at a; commutativity along $a \to b$ for the contravariant diagrams would then assert $\mathrm{id}_k = 0$. Therefore its class

$$\delta \in \mathrm{Ext}^1_{\mathrm{Vect}_k^{J^{\mathrm{op}}}}(M, N) = \mathrm{Hom}_{\mathbf{D}(J)}(M, N[1])$$

is nonzero. The equality is the usual identification of extension classes with degree-one derived morphisms; here all objects are representations of a finite acyclic quiver.

After applying $p_!$, the middle representation is $(k \oplus k \xrightarrow{(1,1)} k)$ and the quotient is $(k \to 0)$. The map at the source sending t to $(t, -t)$, together with the zero map at the target, gives a section of that quotient. Hence the image extension splits and $p_!\delta = 0$. Binary direct sum on $D(k)$ is faithful because each summand of a morphism is recovered by composing with the canonical inclusion and projection. This proves the assertion. □

The example concerns only "resp. fidèle". It leaves the conservative conclusion, and therefore the final equivalence in 4.7, intact. For represented derivators the evaluation functors are jointly faithful and the parenthetical argument does work; it also works under an explicit joint-faithfulness hypothesis on $\mathbf{D}(J) \to \mathbf{D}(e)^{\mathrm{Ob}\, J}$. We checked the distinction between coherent and incoherent diagram categories in Groth (2013, Introduction and Definition 1.5) and the treatment of homotopical epimorphisms in Groth and Šťovíček (2016, §6); we did not find this particular counterexample to 4.7 there. The calculation is included as a repair of the statement, without a claim of priority.

Je ne connais pas d'exemple d'un couple $(\mathbf{D}, u)$ d'un dérivateur $\mathbf{D}$ et d'une flèche u dans Diag, pour lesquels u soit $\mathbf{D}$-connexe sans être $\mathbf{D}$-asphérique. On a donné des conditions sur $\mathbf{D}$, qui assurent équivalence des deux notions. Si $\mathbf{D}$ est de la forme $\mathbf{D}_{\mathcal{M}}$, et si $\mathcal{M}$ satisfait la condition dite dans 4.7 ou sa duale, alors par 4.7 c'est OK. Donc un contrexemple avec un $\mathbf{D}_{\mathcal{M}}$ devrait en tenir compte. Je ne vais pas m'obstiner pour tirer au clair ce qui en est.

Commentary · Development (commentators') [ms. 64bis]

A connected restriction with an empty factorization category

Grothendieck asks whether the sufficient conditions just proved leave a genuine distinction between $\mathbf{D}$-connectedness and $\mathbf{D}$-biasphericity. The edition says "sans être $\mathbf{D}$-asphérique", but that literal question has an immediate negative answer: full faithfulness makes the unit invertible on constant objects, hence implies asphericity. The surrounding discussion and the next diagram concern *biasphericity*. A represented derivator supplies the requested separating example.

A two-valued coefficient theory. [ours] Let $L = \{0 < 1\}$ be the complete lattice, regarded as a category, and let $\mathbf{D}_L(K) = \mathrm{Fun}(K^{\mathrm{op}}, L)$. Let B be the free category on the directed cycle $x \xrightarrow{s} y \xrightarrow{t} x$, and let $u : e \to B$ select y. Then u is $\mathbf{D}_L$-connected and is not $\mathbf{D}_L$-biaspherical.

Proof. The complete lattice L has every small limit and colimit, given respectively by infima and suprema. Thus its represented prederivator is a derivator: restriction has both pointwise Kan extensions, disjoint unions of indexing categories give products of functor categories, and natural transformations are invertible exactly when their components are. It is also strong, since currying gives the required coherent-arrow identifications. These are the standard represented examples recalled in Groth (2013, Examples 1.2(1), 1.6 and Definitions 1.5, 1.8), with the manuscript's contravariant diagram convention.

An object F of $\mathbf{D}_L(B)$ gives inequalities $F(y) \le F(x)$ and $F(x) \le F(y)$. Antisymmetry forces equality. All arrows of B then have the unique endomorphism of that value as image. Every diagram is therefore constant, and a natural transformation between the two possible diagrams is precisely an inequality in L. Evaluation at y identifies $u^* : \mathbf{D}_L(B) \to L$ with an isomorphism of categories. In particular it is fully faithful.

For the identity of the omitted object, however,

$$\mathrm{Fact}_u(\mathrm{id}_x) = \varnothing.$$

Indeed, a factorization through y would be a path $x \to y$ followed by a path $y \to x$. Both have positive length, and in a free category their composite cannot be the length-zero identity. The empty category is not $\mathbf{D}_L$-aspherical: its homotopy colimit of the constant value 1 is the initial object 0, and the comparison $0 \to 1$ is not invertible. Equivalently, $L \to \mathbf{D}_L(\varnothing) = e$ is not fully faithful. Condition 4.4(b″) fails. □

Both potential escapes supplied by 4.7 genuinely fail here. Coproduct is maximum, so the map $(0,1) \to (1,1)$ becomes an identity after taking coproducts although its first component is not invertible. Product is minimum, so $(0,0) \to (1,0)$ gives the dual failure. The counterexample therefore does not contradict the conservative-coproduct theorem, nor the preorder-base case: the indexing category B is a free category with nontrivial endomorphisms, not a preorder.

The example isolates the information lost by the coefficient theory. It sees mutual reachability of objects but cannot distinguish the identity from a nonidentity cycle; the factorization condition still distinguishes them. This coefficient theory is also the classical derivator of truth values, or homotopy (-1)-types; see Shulman (2023, p. 33). We checked that discussion, the density and lax-epimorphism literature cited above, the fixed-derivator qualification in Groth and Šťovíček (2016, Remark 6.3(ii)), and searches for biasphericity in the derivator literature. We could not find this counterexample there. The claim here is the explicit proposition and proof, rather than an assertion that it has no earlier occurrence.

Corollaire 4.8. *Une flèche u de* Diag *qui est une* **D***-équivalence universelle est* **D***-biasphérique, et* a fortiori **D***-connexe. Inversement, si u est soit* **D***-lisse, soit* **D***-propre, soit une* **D***-fibration faible, et si u est* **D***-biasphérique, c'est une* **D***-équivalence universelle.* [page 65]

Commentary · Exposition [ms. 65]

What the seven classes retain

Grothendieck's diagram organizes the preceding tests by the amount of coefficient information they preserve. Its arrows remain useful, but the two observations below are needed to read the diagram after the preceding counterexample.

First, **D**-connectedness is stable under composition. If $u : A \to B$ and $v : B \to C$ are **D**-connected, then

$$(vu)^* \simeq u^* v^*$$

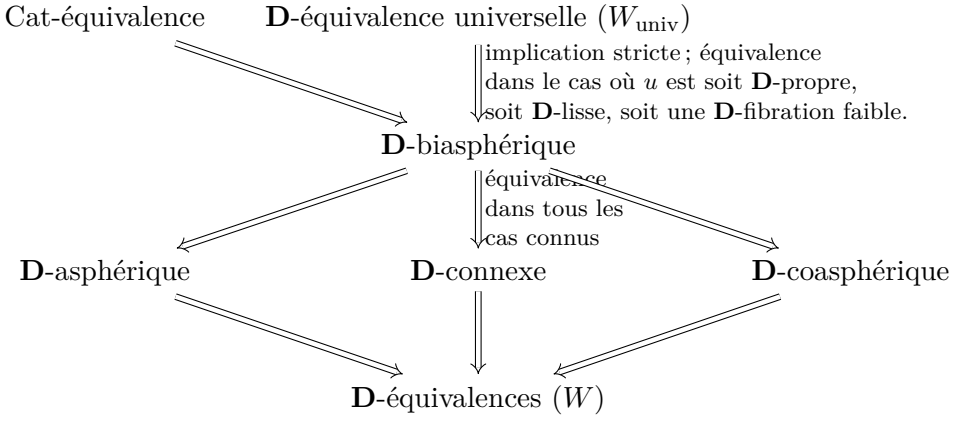
is a composite of fully faithful functors and is fully faithful. Thus the exception printed in the footnote cannot be an exception to closure under composition. This is a formal fact, independent of any Kan-extension existence hypothesis. The footnote's further assertion that all seven classes depend only on $W_{\mathbf{D}}$ requires an independent argument for the fixed-coefficient connectedness class; it does not follow from the stated factorization criterion, whose converse has just failed.

Second, connectedness always implies both asphericity and coasphericity. The adjunction unit and counit are invertible on every $F \in \mathbf{D}(B)$, hence on constant F in particular. The distinction

established above is between this global full faithfulness and the two-sided slice tests defining biasphericity. The label "équivalence dans tous les cas connus" records the author's experience at this stage, not an additional theorem.

For universal equivalences the factorization test still gives the implication in 4.8. The category of factorizations of a fixed arrow $\gamma : b_0 \to b_1$ through arbitrary objects of B has an initial object $(b_0, \mathrm{id}, \gamma)$ and a terminal object $(b_1, \gamma, \mathrm{id})$. Pulling u back to that category produces $\mathrm{Fact}_u(\gamma)$. A universal $W_{\mathbf{D}}$-equivalence therefore makes this category $\mathbf{D}$-aspherical. Under the smooth, proper, or weak-fibration hypotheses, the fibre-to-comma comparisons give the converse exactly as in 4.3. These hypotheses are what connect strict base changes to the homotopy calculation; the four-object example shows why they cannot simply be omitted.

La démonstration est celle du cas particulier 4.3. On a donc un diagramme d'implications :



([2]). Rappelons que pour $u : I' \longrightarrow I$ donné

a) u est $\mathbf{D}$-asphérique si et seulement si $\forall\, F$ *constant* dans $\mathbf{D}(I)$

$$(*) \qquad F \xrightarrow[\sim]{\text{adj.}} u_* u^* F \qquad \text{est } \texttt{[un]} \text{ isomorphisme.}$$

p. 41

b) u est $\mathbf{D}$-coasphérique si et seulement si $\forall\, F$ *constant* dans $\mathbf{D}(I)$

$$(**) \qquad F \xleftarrow[\sim]{\text{adj.}} u_! u^* F \qquad \text{est } \texttt{[un]} \text{ isomorphisme.}$$

c) u est $\mathbf{D}$-connexe si et seulement si pour *tout* F dans $\mathbf{D}(I)$ (pas forcément constant, ni localement constant), $(*)$ est `[un]` isomorphisme, ou ce qui revient au même, pour tout F $(**)$ est `[un]` isomorphisme.

Rappelons que la classe W_{asph} `[des]` flèches W-asphériques et celle, Liss_W, des flèches W-lisses, se déterminent mutuellement, ainsi [page 66]

$$(4.10) \qquad \begin{cases} (f : X \longrightarrow Y) \in W_{\mathrm{asph}} & \iff & \forall\, (Y' \xrightarrow{u} Y) \in \mathrm{Liss}_W, \\ & & X' = X \times_Y Y' \longrightarrow X' \text{ est } \in W. \\ (f : Y' \longrightarrow Y) \in \mathrm{Liss}_W & \iff & \text{Le foncteur } X \longmapsto X' = X \times_Y Y' \\ & & \text{de } \mathrm{Cat}/Y \text{ dans } \mathrm{Cat}/Y' \text{ transforme} \\ & & \text{flèches dans } W_{\mathrm{asph}} \text{ dans } \texttt{[plutôt en]} \\ & & \text{flèches dans } W_{\mathrm{asph}}. \end{cases}$$

2. **NB** Ces sept classes de morphismes dans Cat sont stables par composition sauf la notion de $\mathbf{D}$-connexité, toutes ces notions ne dépendent que de l'ensemble $W = W_{\mathbf{D}}$ des $\mathbf{D}$-équivalences.

Pour toute partie A de Fl(Cat) stable par isomorphismes, on peut associer une partie $\mathbf{L}(A)$ de Fl(Cat), par la relation

$$(u : Y' \longrightarrow Y) \in \mathbf{L}(A) \iff \begin{array}{l} u^* : \mathrm{Cat}/Y \longrightarrow \mathrm{Cat}/Y' \\ \text{transforme flèches} \\ \in A \text{ en flèches } \in A. \end{array} \tag{4.11}$$

On voit que $\mathbf{L}(A)$ contient les isomorphismes, est stable par composition, et est stable aussi par changement de base (comme c'est le cas de Liss_W). Inversement, à une partie L de Fl(Cat), stable par isomorphismes, on associe une partie $\mathbf{A}_W(L)$ par

$$(f : X \longrightarrow Y) \in \mathbf{A}_W(L) \iff \begin{array}{l} \forall\, (Y' \longrightarrow Y) \text{ dans } L, \\ X' = X \times_Y Y' \longrightarrow Y' \\ \text{est dans } W. \end{array} \tag{4.12}$$

Si L est stable par changement de base, alors $\mathbf{A}_W(L)$ est stable par composition. Il contient les isomorphismes (car W les contient), est contenu dans W si L contient les identités. Mais aucune raison qu'il soit stable par changement de base quelconque (et en fait, W_{asph} ne l'est pas). [page 67]

Commentary · Exposition [ms. 66]

What survives the cancelled argument

Grothendieck is seeking a way to recover the classes of smooth and proper maps from the weak equivalences alone. The two tests in (4.10) already suggest an order-theoretic correspondence, but they use different notions of preservation. This distinction explains the cancellation on the next page; it does not make every formula in the cancelled passage false.

To see exactly what is involved, write $S(A)$ for the stabilizer of A: a map $p : Y' \to Y$ belongs to $S(A)$ when its pullback functor on slices sends every arrow of A to another arrow of A. Write $N(L) = \mathbf{A}_W(L)$ for the arrows whose pullbacks along maps of L belong to W. Here W contains isomorphisms and is closed under composition. Classes are invariant under isomorphism of arrows, and the ambient category has pullbacks. The quantifier over arrows *in the slice* matters: one tests an arrow $X_1 \to X_2$ together with any structure map $X_2 \to Y$, not merely arrows with codomain exactly Y.

The class $S(A)$ contains isomorphisms and is closed under composition and pullback, by the identity and composition laws for pullback functors. If L is closed under pullback, then $N(L)$ is closed under composition: pull back a composite gf along $p \in L$, and factor its pullback as the pullback of f along a pullback of p, followed by the pullback of g along p. Both factors lie in W. If L also contains identities, $N(L) \subseteq W$; its isomorphisms belong automatically. These are the assertions preceding the cancellation, with their implicit hypotheses made explicit.

For $A \subseteq W$, there are also two useful inclusions:

$$A \subseteq N(S(A)), \qquad L \subseteq S(N(L)),$$

where the second requires L to be closed under pullback and composition. For its proof, let $p \in L$ and $f \in N(L)$ be over the target of p. The pullback p^*f lies in W. Every further test of p^*f along $q \in L$ is a test of f along the composite of q with the appropriate pullback of p. That composite is in L, so $p^*f \in N(L)$.

For classes L containing identities, these inclusions even prove the two displayed idempotence conclusions on page 67. Since N reverses inclusions,

$$NSNS(A) \subseteq NS(A), \qquad NSN(L) \subseteq N(L),$$

and the reverse inclusions follow from $A \subseteq NS(A)$, applied to $NS(A)$ and to $N(L)$; the identity test ensures $N(L) \subseteq W$. Thus NS is idempotent and $SNSN = SN$. However, S need not reverse inclusions. Consequently these calculations do not supply the order-reversing adjunction that the next section needs. In particular, the stronger justification printed there, $S(A)^\flat = S(A)$, does not follow from them and can fail. The repair in §6 is to replace $S(A)$ by a transporter into the fixed class W. That operation is genuinely order-reversing.

There is a further empty-class detail in the cancelled argument. Formula (4.14) does not explicitly require identities in L, whereas (4.12) does not explicitly restrict $N(L)$ to W. For $L = \varnothing$, one has $N(L) = \mathrm{Fl}(\mathrm{Cat})$, so the auxiliary assertion $N(L)^\natural = N(L)$ is false when W is proper: its left side is W_{univ}. Section 6 repairs the domain issue by inserting $f \in W$ into (6.6), and later by requiring isomorphisms in L.

The idempotence of SN itself nevertheless survives in Cat for a fundamental localizer and every class L closed under pullback and composition. If every member of L has empty domain, every test is an identity of the empty category, so $N(L)$ is all functors and $SN(L)$ is all functors; applying SN again gives the same class, since universal W-equivalences are stable under all pullbacks. Otherwise choose $p \in L$ with nonempty domain and pull it back to an object of its target lying in its image. This gives $C \to e$ in L with C nonempty. Further pullback gives $B \times C \to B$ in L for every B. Thus $f \in N(L)$ implies $f \times C \in W$. Choosing an object of C makes f a retract of $f \times C$ in the arrow category, so $f \in W$ by closure under retracts; for fundamental localizers this is Cisinski (2006b, Proposition 4.2.4, p. 183). Hence $N(L) \subseteq W$, and the earlier proof applies. This records the surviving conclusion without endorsing either incorrect auxiliary justification.

([3]). On a, pour un $A \subseteq W \subseteq \mathrm{Fl}(\mathrm{Cat})$ donné, stable par isomorphismes,

$$A \subseteq \mathbf{A}_W(\mathbf{L}(A)) \overset{\text{déf}}{=} A^\natural \subseteq W, \qquad \begin{array}{l}\text{partie stable par composition,}\\ \text{contient [les] isomorphismes,}\end{array} \tag{4.13}$$

et pour toute partie L de Fl(Cat) qui est stable par changement de base, et par composition p. 42

$$L \subseteq \mathbf{L}(\mathbf{A}_W(L)) \overset{\text{déf}}{=} L^\flat, \qquad \begin{array}{l}\text{partie de Fl Cat stable}\\ \text{par composition et par changement}\\ \text{de base, contenant les isomorphismes.}\end{array} \tag{4.14}$$

On a

$$\begin{cases} A^{\natural\natural} &= A^\natural, \quad \text{car } \mathbf{A}_W(L)^\natural = \mathbf{A}_W(L) \\ L^{\flat\flat} &= L^\flat, \quad \text{car } \mathbf{L}(A)^\flat = \mathbf{L}(A). \end{cases}$$

Exemples :

1°) $L = \text{Ét}$, $\mathbf{A}_W(L) = W_{\mathrm{asph}}$, $\mathbf{L}(\mathbf{A}_W(L)) = L^\flat = \mathrm{Liss}_W$.

2°) $L = \text{Coét}$, $\mathbf{A}_W(L) = W_{\mathrm{coas}}$, $\mathbf{L}(\mathbf{A}_W(L)) = L^\flat = \mathrm{Prop}_W$.

3°) $A = W$, $\mathbf{L}(A) = \mathrm{Fib}_W$, $A^\natural = W$.

4°) $L = \mathrm{Fl}\,\mathrm{Cat}$, $\mathbf{A}_W(L) = W_{\mathrm{univ}}$, $L^\flat = L$.

[page 68 vide]

3. Canulé [jusqu'à la fin du §4], cf. §6.

Commentary · Development (commentators') [ms. 67] p. 43 [page 69]

A cancelled justification tested in categories

The cancelled lines are best understood by testing the precise identity used in their justification. The following example keeps the ambient category equal to Cat and uses a fundamental localizer, so the difficulty is not an artefact of passing to an unrelated category.

A stabilizer need not be a closed right-hand class. [ours] Let W be the maximal fundamental localizer, consisting of all functors, and let $A = \mathrm{Equ}$ be the class of equivalences of categories. With S and N as above, $S(A)^\flat \neq S(A)$.

Indeed, every pullback functor takes functors to functors, so $N(L) = \mathrm{Fl}(\mathrm{Cat})$ for every L. Consequently

$$S(A)^\flat = S\big(N(S(A))\big) = S\big(\mathrm{Fl}(\mathrm{Cat})\big) = \mathrm{Fl}(\mathrm{Cat}).$$

To show that $S(A)$ is smaller, let J be the groupoid with two objects $0, 1$ and precisely one morphism between each ordered pair. Each point inclusion $e_i \to J$ is fully faithful and essentially surjective, hence an equivalence. Their strict pullback is empty, because its objects would have to map to both 0 and 1. Pulling $e_1 \to J$ back along $e_0 \to J$ therefore gives $\varnothing \to e_0$, which is not essentially surjective. Thus $e_0 \to J \notin S(A)$, proving the claim.

The same calculation shows why the anticipated polarity fails: although $A \subseteq B = \mathrm{Fl}(\mathrm{Cat})$, one has $S(A) \subsetneq S(B)$, whereas an order-reversing correspondence would require the opposite inclusion. This does not contradict the idempotence of either composite proved in the preceding block. For the present W those composites are particularly degenerate: NS is constant with value all functors, and SN is likewise constant.

This is an elementary diagnostic for the manuscript's cancellation. We checked the subsequent transporter construction in §6 and the preservation definition of smooth functors in Maltsiniotis (2005, Definition 4.6); we did not find this particular counterexample stated there. The mark above records our verification, not a claim of priority. The later block "The stabilizer of categorical equivalences" identifies the whole class $S(\mathrm{Equ})$.

5 Dualité formelle

Soient $\mathcal{P}$, $\mathcal{Q}$ deux ensembles ordonnés. J'essaye de dégager une notion de 'dualité' (formelle) reliant $\mathcal{P}$ et $\mathcal{Q}$. Ce sera une relation entre $\mathcal{P}$ et $\mathcal{Q}$

$$\mathcal{R} \subseteq \mathcal{P} \times \mathcal{Q} \tag{5.1}$$

ayant les propriétés suivantes

(D) Pour toute partie A de $\mathcal{P}$, l'ensemble

$$A^\cup \overset{\text{déf}}{=} \{q \in \mathcal{Q} \mid (p,q) \in \mathcal{R} \text{ pour tout } p \in A, \ \text{ i.e. } A \times \{q\} \subseteq \mathbf{R}\}$$

est une partie ouverte de $\mathcal{Q}$ ayant un plus grand élément $\beta(A) : A^\cup = \mathcal{Q}/\beta(A)$. Dualement, pour toute partie B de $\mathcal{Q}$, l'ensemble

$$B^\cap \overset{\text{déf}}{=} \{p \in \mathcal{P} \mid (p,q) \in \mathcal{R} \text{ pour tout } q \in B, \ \text{ i.e. } \{p\} \times B \subseteq \mathcal{R}\}$$

est une partie ouverte de $\mathcal{P}$ ayant un plus grand élément $\alpha(B)$, i.e. $B^\cap = \mathcal{P}/\alpha(B)$.

Le plus souvent, $\mathcal{P}$ et $\mathcal{Q}$ seront stables par sup quelconques (donc aussi par inf quelconques). Dans ce cas, (D) équivaut, du côté des $A^\cup$ (pour $A \in \mathfrak{P}(\mathcal{P})$) à ceci :

(D_0) Pour tout $p \in \mathcal{P}$, $\mathcal{R}(p) = \{q \in Q \mid (p,q) \in \mathcal{R}\}$ est une partie ouverte *filtrante* de $\mathcal{Q}$, ou ce qui revient au même, une partie non vide ($\Longleftrightarrow$ contenant le plus petit élément $\emptyset_\mathcal{Q}$ de $\mathcal{Q}$) stable par sup de deux éléments. De plus, $\mathcal{R}(p)$ est stable par sup filtrants, ou ce qui revient au même (étant lui-même filtrant), il contient un plus grand élément. Il revient au même de dire que $\mathcal{R}(p)$ est un *ouvert* de $\mathcal{Q}$ stable par sup quelconques (ce qui implique qu'il contient $\emptyset_\mathcal{Q}$). Dualement, pour tout $q \in \mathcal{Q}$, ${}^s\mathcal{R}(q) = \{p \in \mathcal{P} \mid (p,q) \in \mathcal{R}\}$ est une partie *ouverte* de $\mathcal{P}$, stable par sup quelconques, i.e. une partie ouverte ayant un plus grand élément. [page 70]

Commentary · Exposition [ms. 69]

The relation comes before the order

Grothendieck pauses to isolate the formal mechanism that a successful transporter construction must satisfy. The first part of the section concerns an arbitrary relation, even though the initial presentation starts with ordered sets. Separating those two levels makes the notation more transparent and prevents the later confusion between a closed subset and an element of an ordered set.

For a relation $R \subseteq P \times Q$, put

$$A^{\cup} = \{q \mid (p,q) \in R \text{ for every } p \in A\}, \qquad B^{\cap} = \{p \mid (p,q) \in R \text{ for every } q \in B\}.$$

The decisive equivalence is

$$B \subseteq A^{\cup} \quad \Longleftrightarrow \quad A \times B \subseteq R \quad \Longleftrightarrow \quad A \subseteq B^{\cap}.$$

Both operations reverse inclusions. Every $p \in A$ is related to every member of $A^{\cup}$, so $A \subseteq A^{\cup\cap}$. Applying the inclusion-reversing operation gives $A^{\cup\cap\cup} \subseteq A^{\cup}$; applying extensivity on the other side gives the reverse inclusion. Thus the triple operation equals the single operation, and the double operation is idempotent. This proves (5.5)–(5.7) without any order on P or Q.

The fixed subsets form complete lattices. Arbitrary intersections of fixed subsets are fixed: if $A = \bigcap_i A_i$, monotonicity gives $A^{\cup\cap} \subseteq A_i$ for every i, and extensivity gives the opposite inclusion $A \subseteq A^{\cup\cap}$. Their join is the closure of their union. The two polar operations exchange these closed-set lattices by inverse order-reversing bijections, proving Proposition 5.1 and Corollary 5.2. Empty families cause no exception: the largest closed subset of P is P, while the smallest is $Q^{\cap}$, which need not be empty.

The ordered version asks for more. An "open" subset in this passage is a lower set, and the additional requirement is that each polar lower set be principal. For complete lattices P, Q, this gives maps $\beta_0 : P \to Q$ and $\alpha_0 : Q \to P$ characterized by

$$q \leq \beta_0(p) \quad \Longleftrightarrow \quad p \leq \alpha_0(q).$$

Equivalently, $\alpha_0 : Q \to P^{\mathrm{op}}$ is left adjoint to $\beta_0 : P^{\mathrm{op}} \to Q$. It follows that β_0 sends joins in P to meets in Q, including the empty join. Conversely, if it has this property, set

$$\alpha_0(q) = \bigvee \{p \in P : q \leq \beta_0(p)\}.$$

The set under the join is a lower set and is closed under arbitrary joins, so its join belongs to it; this proves the displayed equivalence and constructs the adjoint. The ensuing fixed-point correspondence is the usual antitone Galois connection. A systematic later account is Ganter and Wille (1999, §0.4, Propositions 7–8 and Theorem 2); the arguments needed here are supplied above, independently of terminology.

En d'autres termes, quand on postule la condition (D) pour A resp. B réduit à un élément, alors elle reste vérifiée pour toute partie. Cela provient des formules

$$\text{(5.2)} \qquad \begin{cases} (\bigcup_i A_i)^{\cup} &= \bigcap_i A_i^{\cup} \\ (\bigcup_j B_j)^{\cap} &= \bigcap_j B_j^{\cap}, \end{cases}$$

et du fait que l'intersection de parties ouvertes de la forme $\mathcal{P}/\alpha_i$ dans $\mathcal{P}$ est de la forme $\mathcal{P}/\alpha$, où $\alpha = \inf_i \alpha_i$, et de même dans $\mathcal{Q}$. Ainsi on trouve

$$\text{(5.3)} \qquad \begin{cases} \beta(\bigcup_i A_i) &= \inf_i \beta(A_i) \\ \alpha(\bigcup_j B_j) &= \inf_j \alpha(B_j). \end{cases}$$

On posera, pour une partie A de $\mathcal{P}$ resp. B de $\mathcal{Q}$

$$\text{(5.4)} \qquad A^{\natural} = (A^{\cup})^{\cap}, \qquad B^{\flat} = (B^{\cap})^{\cup}$$ p. 44

de sorte qu'on aura

$$\text{(5.5)} \qquad A \subseteq A^{\natural}, \qquad B \subseteq B^{\flat}$$

On dira que A est $\mathcal{R}$-*clos* si $A = A^{\natural}$ (ou encore si A est de la forme $B'^{\cap}$ pour $B' \subseteq \mathcal{Q}$, cf. 5.6), que B est $\mathcal{R}$-*clos* si $B = B^{\flat}$. On aura pour tout $A \subseteq \mathcal{P}$, $B \subseteq \mathcal{Q}$ [71]

$$(5.6) \qquad \begin{cases} (A^{\cup})^{\flat} &= A^{\cup}, \quad \text{i.e. } A^{\cup} \text{ est } \mathcal{R}\text{-clos}, \\ (B^{\cap})^{\natural} &= B^{\cap}, \quad \text{i.e. } B^{\cap} \text{ est } \mathcal{R}\text{-clos}. \end{cases}$$

En particulier

$$(5.7) \qquad (A^{\natural})^{\natural} = A^{\natural}, \qquad (B^{\flat})^{\flat} = B^{\flat},$$

i.e. la $\mathcal{R}$-clôture $A^{\natural}$ de $A \subseteq \mathcal{P}$ est $\mathcal{R}$-clos, et de même la $\mathcal{R}$-cloture $B^{\flat}$ de $B \subseteq \mathcal{Q}$.

En fait, je m'aperçois que je suis en train de mélanger deux espèces de faits différents. Les relations (5.2), et (5.4) à (5.7), ne dépendent pas de la donné de relations d'ordre sur $\mathcal{P}$, $\mathcal{Q}$, ni d'axiomes reliant celles-ci à la relation donnée $\mathcal{R}$, mais seulement de cette relation, laquelle n'a pas à être soumise à aucune condition. Notons que

$$(5.8) \qquad \emptyset^{\cup} = \mathcal{Q}, \qquad \emptyset^{\cap} = \mathcal{P},$$

tandis que

$$(5.9) \qquad \emptyset^{\natural} = \mathcal{Q}^{\cap}, \qquad \emptyset^{\flat} = \mathcal{P}^{\cup}$$

sont respectivement la plus petite partie $\mathcal{R}$-close de $\mathcal{P}$ et de $\mathcal{Q}$.

L'application

$$(5.10) \qquad A \longmapsto A^{\natural}$$

est une application *croissante* (et *idempotente*) de $\mathfrak{P}(\mathcal{P})$ dans lui-même, ayant donc la propriété [72]

$$(\bigcup_i A_i)^{\natural} = (\bigcup_i A_i^{\natural})^{\natural}.$$

Pour tout $A \subseteq \mathcal{P}$, $A^{\natural}$ est la plus petite partie $\mathcal{R}$-close de $\mathcal{P}$ contenant A, et de même pour p. 45
tout $B \subseteq \mathcal{Q}$, $B^{\flat}$ est la plus petite partie $\mathcal{R}$-close de Q contentant B. On a de plus (c'était même à dire avant de dire que $A \longmapsto A^{\natural}$ est croissante)

$$(5.11) \qquad \begin{cases} A \subseteq A' &\Longrightarrow A'^{\cup} \subseteq A^{\cup} \\ B \subseteq B' &\Longrightarrow B'^{\cap} \subseteq B^{\cap}, \end{cases}$$

i.e. les applications

$$(5.12) \qquad \mathfrak{P}(\mathcal{P}) \underset{\alpha}{\overset{\beta}{\rightleftarrows}} \mathfrak{P}(\mathcal{Q}) \qquad \begin{matrix} A \longmapsto A^{\cup} \\ B \longmapsto B^{\cap} \end{matrix}$$

sont décroissantes (ce qui implique que les deux composés $\alpha\beta = (A \longmapsto A^{\natural})$ et $\beta\alpha = (B \longmapsto B^{\flat})$ sont croissantes). Leurs restrictions aux sous-ensembles $\mathfrak{P}_{\mathcal{R}}(\mathcal{P})$ et $\mathfrak{P}_{s\mathcal{R}}(\mathcal{Q})$ des parties $\mathcal{R}$-closes de $\mathcal{P}$ resp. de $\mathcal{Q}$, sont, par définition de celles-ci comme

$$\begin{aligned} \mathfrak{P}_{\mathcal{R}}(\mathcal{P}) &= \{A \subseteq \mathcal{P} \mid A^{\natural} \ (\overset{\text{déf}}{=} \alpha\beta(A)) = A\} \\ \mathfrak{P}_{s\mathcal{R}}(\mathcal{Q}) &= \{B \subseteq \mathcal{Q} \mid B^{\flat} \ (\overset{\text{déf}}{=} \beta\alpha(B)) = B\}, \end{aligned}$$

inverse l'une de l'autre. On trouve ainsi

Proposition 5.1. *Soient $\mathcal{P}$, $\mathcal{Q}$ deux ensembles, $\mathcal{R} \subseteq \mathcal{P} \times \mathcal{Q}$ une relation. Alors les applications α, β de (5.12) induisent des bijections inverses l' une de l'autre, entre l'ensemble* [73]

$Cl_{\mathcal{R}}(\mathcal{P})$ des parties $\mathcal{R}$-closes de $\mathcal{P}$, et l'ensemble $Cl_{^s\mathcal{R}}(\mathcal{Q})$ des parties ${}^s\mathcal{R}$-closes (nous dirons $\mathcal{R}$-closes par abus de langage) de $\mathcal{Q}$. Ces bijections sont décroissantes, donc des antiisomorphismes pour les relations d'ordre naturelles (d'inclusion).

Corollaire 5.2. *Soit $(A_i)_i$ une famille de parties $\mathcal{R}$-closes de $\mathcal{P}$. Alors $\bigcap A_i$ est une partie $\mathcal{R}$-close de $\mathcal{P}$, et*

$$(\bigcap_i A_i)^{\cup} = \underbrace{(\bigcup_i A_i^{\cup})^{\flat}}_{\substack{\text{sup des} \\ \text{parties } \mathcal{R}\text{-closes} \\ A_i^{\cup} \text{ de } \mathcal{Q}}} \quad . \tag{5.2.1}$$

Mais la relation $\mathcal{R}$ se 'prolonge' en une relation $\bar{\mathcal{R}}$ entre

$$\bar{\mathcal{P}} = \mathfrak{P}(\mathcal{P}), \qquad \bar{\mathcal{Q}} = \mathfrak{P}(\mathcal{Q}) \tag{5.13}$$

par la prescription

$$\underbrace{\bar{\mathcal{R}}}_{\subseteq \bar{\mathcal{P}} \times \bar{\mathcal{Q}}} = \{(A, B) \in \bar{\mathcal{P}} \times \bar{\mathcal{Q}} \mid B \subseteq A^{\cup}, \text{ i.e. } A \subseteq B^{\cap}, \text{ i.e. } A \times B \subseteq \mathcal{R}\}. \tag{5.14}$$

Les ensembles $\bar{\mathcal{P}}$, $\bar{\mathcal{Q}}$ sont stables par sup quelconques (réunion de parties de $\mathcal{P}$ resp. $\mathcal{Q}$), et la relation $\bar{\mathcal{R}}$ sur ces ensembles ordonnés satisfait à la condition (D_0) ci-dessus (p. 69), de façon évidente : pour A fixé, les B tels que $(A, B) \in \bar{\mathcal{R}}$ sont ceux qui sont contenus dans $A^{\cup}$, donc l'ensemble $\bar{\mathcal{R}}(A)$ n'est autre que $\bar{\mathcal{Q}}/A^{\cup}$, et dualement ${}^s\bar{\mathcal{R}}(B)$ n'est autre que $\bar{\mathcal{P}}/B^{\cap}$. Ainsi p. 46 [74]

$$\beta(A) = A^{\cup}, \qquad \alpha(B) = B^{\cap}. \tag{5.14}$$

Si j'ai songé à introduire l'axiome (D_0), pour un couple d'ensembles ordonnés quelconques, c'est à cause des cas où ceux-ci ne sont *pas* de la forme $\mathfrak{P}(\mathcal{P})$, $\mathfrak{P}(\mathcal{Q})$, et (alors même qu'ils le seraient) où la relation de base n'est pas déduite comme ci-dessus d'une relation quelconque entre ensembles $\mathcal{P}$, $\mathcal{Q}$.

Je reviens donc à l'axiomatique du début, avec $\mathcal{P}$, $\mathcal{Q}$ ordonnés, et $\mathcal{R}$ satisfaisant (D). Dans tous les cas auxquels je peux songer, $\mathcal{P}$ et $\mathcal{Q}$ sont stables par sup quelconques, donc (D) équivaut à (D_0). Notons que l'application

$$\mathcal{P} \longrightarrow \mathfrak{P}(\mathcal{P}), \qquad p \longmapsto \mathcal{P}/p$$

est injective, et de même pour $\mathcal{Q} \longrightarrow \mathfrak{P}(\mathcal{Q})$. Ainsi, la connaisance de $A \longmapsto A^{\cup}$ équivaut à celle de

$$\begin{array}{rcl} A & \longmapsto & \beta(A) \\ \beta : \mathfrak{P}(\mathcal{P}) & \longrightarrow & \mathcal{Q}, \end{array}$$

et de même celle de $B \longmapsto B^{\cap}$ équivaut à celle de [75]

$$\alpha : \mathfrak{P}(\mathcal{Q}) \longrightarrow \mathcal{P}.$$

En fait, il suffit de se donner

$$\begin{array}{rcccl} \beta_0 : & \mathcal{P} & \longrightarrow & \mathcal{Q} & \text{applications} \\ \alpha_0 : & \mathcal{Q} & \longrightarrow & \mathcal{P} & \textit{décroissantes} \end{array} \tag{5.15}$$

définies par

$$\left\{ \begin{array}{rcl} \beta_0(p) & = & \beta(\{p\}) \\ \alpha_0(q) & = & \alpha(\{q\}), \end{array} \right.$$

puisqu'on en déduit β, α par

$$\left\{\begin{array}{lcl}\beta(A) & = & \inf_{p\in A}\beta_0(p)\\ \alpha(B) & = & \inf_{q\in B}\alpha_0(q).\end{array}\right. \tag{5.16}$$

La relation $\mathcal{R}$ se déduit de la donnée de β_0, ou au choix de α_0, par

$$(p,q)\in\mathcal{R} \iff q\leq\beta_0(p) \iff p\leq\alpha_0(q). \tag{5.17}$$

Cela implique donc que α_0, β_0 se déterminent mutuellement, par les formules p. 47

$$\left\{\begin{array}{lcl}\alpha_0(q) & = & \sup\underbrace{\{p\in\mathcal{P}\mid q\leq\beta_0(p)\}}_{\{q\}^\cap}\\ \beta_0(p) & = & \sup\underbrace{\{q\in\mathcal{Q}\mid p\leq\alpha_0(q)\}}_{\{p\}^\cup}.\end{array}\right. \tag{5.18}$$

Reste à se demander quelles sont les applications $\alpha_0:\mathcal{P}\longrightarrow\mathcal{Q}$, ou $\beta_0:\mathcal{Q}\longrightarrow\mathcal{P}$, qui correspondent à une *'relation dualisante'* $\mathcal{R}$ sur $\mathcal{P}\times\mathcal{Q}$. Voici les conditions sur $\beta_0:\mathcal{P}\longrightarrow\mathcal{Q}$: [76]

ⓐ β_0 est *décroissante*,

ce qui implique que la relation

$$\underbrace{\mathcal{R}_{\beta_0}}_{\subseteq\mathcal{P}\times\mathcal{Q}}=\{(p,q)\in\mathcal{P}\times\mathcal{Q}\mid q\leq\beta_0(p)\}$$

satisfait à la condition (D_0) du côté gauche, et il faut encore la vérifier du côté droit, i.e. regarder, pour q fixé, l'ensemble

$$\{q\}^\cap=\{p\in\mathcal{P}\mid q\leq\beta_0(p)\}.$$

Comme β_0 est décroissant, on voit que $p\in\{q\}^\cap$, et $p'\leq p\Longrightarrow p'\in\{q\}^\cap$, i.e. $\{q\}^\cap$ est *ouvert.* On doit vérifier encore (supposant l'existence des sup quelconques dans $\mathcal{P}$) que cet ensemble est stable par sup quelconques - il en résultera qu'il est de la forme $\mathcal{P}/\alpha(q)$. Cela équivaut à la condition

ⓑ β_0 transforme sup quelconques dans $\mathcal{P}$ en inf dans $\mathcal{Q}$.

Si on regarde β_0 comme un foncteur contravariant $\mathcal{P}^o\longrightarrow\mathcal{Q}$, cela signifie qu'il commute aux limites projectives, ou encore qu'il a un adjoint à gauche $\mathcal{Q}\longrightarrow\mathcal{P}^o$, lequel commute aux limites inductives, et correspond à l'application décroissante $\alpha_0:\mathcal{Q}\longrightarrow\mathcal{P}$, transformant sup en inf. [77]

Ici il est commode de considérer $\natural$ et $\flat$ comme des opérations sur $\mathcal{P}$ resp. $\mathcal{Q}$, par

$$p^\natural=\alpha_0\beta_0(p),\qquad q^\flat=\beta_0\alpha_0(q). \tag{5.19}$$

La relation entre ces endomorphismes, et l'opération de même nom sur $\mathfrak{P}(\mathcal{P})$, $\mathfrak{P}(\mathcal{Q})$, est donnée par

$$\left\{\begin{array}{lcl}p^\natural & = & \text{Plus grand élément de } \{p\}^\natural\\ A^\natural & = & \sup_{p\in A}p^\natural,\end{array}\right. \tag{5.20}$$

et symétriquement du côté de $\mathcal{Q}$.

On dit que p est $\mathcal{R}$-*clos,* ou β_0-clos, ou α_0-clos, si $p=p^\natural$, et de même $q\in\mathcal{Q}$ est dit $\mathcal{R}$-*clos* etc. si $q=q^\flat$. Les applications p. 48

$$\left\{\begin{array}{lcl}p\longmapsto p^\natural : \mathcal{P}\longrightarrow\mathcal{P}\\ q\longmapsto q^\flat : \mathcal{Q}\longrightarrow\mathcal{Q}\end{array}\right. \tag{5.21}$$

[78]

sont *croissantes* et *idempotentes.* On peut caractériser $p^\natural$ comme le plus petit élément clos de $\mathcal{P}$ majorant p, et de même pour $q^\flat$. On trouve alors, par (5.19), que α_0 et β_0 induisent des bijections inverses l'une de l'autre, entre l'ensemble des éléments clos de $\mathcal{P}$, et l'ensemble des éléments clos de $\mathcal{Q}$

$$\mathcal{P}^\natural \underset{\bar{\alpha}_0}{\overset{\bar{\beta}_0}{\underset{\sim}{\rightleftarrows}}} \mathcal{Q}^\flat. \tag{5.22}$$

Mais en pratique, les applications α_0, β_0 ne sont pas obtenues directement, mais on dispose de la relation $\mathcal{R}$, satisfaisant la condition (D_0), dont on déduit α_0, β_0. Dans tous les cas auxquels je pense ici, ou bien $\mathcal{P}$, $\mathcal{Q}$ sont de la forme $\mathfrak{P}(P_0)$, $\mathfrak{P}(Q_0)$, et $\mathcal{R}$ se déduit d'une relation $R_0 \subseteq P_0 \times Q_0$ comme ci-dessus (de loin le plus fréquent, il me semble), ou bien $\mathcal{P}$ et $\mathcal{Q}$ apparaissent comme sous-ensembles convenables d'ensembles $\mathfrak{P}(P_0)$, $\mathfrak{P}(Q_0)$, munis de l'ordre induit, et la relation $\mathcal{R}$ sur $\mathcal{P} \times \mathcal{Q}$ s'obtient d'une façon plus complexe.

Commentary · Examples [ms. 77]

A join of closed elements need not be closed

The ordered version allows the closure of a subset to be represented by one element, and Grothendieck wants to pass freely between these descriptions. Formula (5.20) loses a closure in making this passage. The difficulty is mathematical, not a transcription error: the edition prints the second equality exactly as transcribed.

Let $c = \alpha_0\beta_0$ and let $A \subseteq P$. The precise identities are

$$\{p\}^{\cup\cap} = {\downarrow}c(p), \qquad A^{\cup\cap} = {\downarrow}c\left(\bigvee_{p\in A} p\right) = {\downarrow}c\left(\bigvee_{p\in A} c(p)\right).$$

Here $\downarrow x$ denotes the set of elements below x. To prove the middle equality, the polar $A^\cup$ has greatest element $\bigwedge_{p\in A}\beta_0(p) = \beta_0(\bigvee A)$, and its polar has greatest element $\alpha_0\beta_0(\bigvee A)$. For the last equality, $p \le c(p) \le c(\bigvee A)$ for every $p \in A$; apply monotonicity and idempotence to the two resulting inequalities. Thus the supremum can be taken in the *closed-element lattice*, where it means closure of the ambient supremum. It cannot in general be taken unchanged in P.

A small example displays both the omitted operation and its importance. Let $E = \mathbf{F}_2^2$, and take $P = \mathfrak{P}(E)$, $Q = \mathfrak{P}(E^*)$, with the relation that every vector in one subset is annihilated by every functional in the other. Double annihilation sends a subset to its linear span. To check this directly, a linear functional vanishing on a set vanishes on its span; if a vector lies outside that span, extend a basis of the span by that vector and prescribe a functional taking value 1 on it and 0 on the original basis.

Let e_1, e_2 be a basis and put $p_1 = \{0, e_1\}$, $p_2 = \{0, e_2\}$. Each p_i is already closed, but their supremum in P is

$$p_1 \cup p_2 = \{0, e_1, e_2\},$$

which omits $e_1 + e_2$ and is not closed. For the family $A = \{p_1, p_2\} \subseteq P$, the greatest element of $A^{\cup\cap}$ is all of E, whereas the supremum printed in (5.20) is only the three-element union. Moreover, $A^{\cup\cap}$ itself is a subset of P, namely all subsets of E, rather than an element of P. These two distinctions are exactly what the corrected formula records.

This correction does not disturb the subsequent statements that c is monotone, extensive and idempotent, or that the closed elements on the two sides correspond anti-isomorphically. It supplies the correct join operation on those elements and will be needed again when the elements of P are themselves classes of morphisms.

Exemples.

1) Relations d'*orthogonalité* entre parties de deux modules sur un anneau, muni d'un accouplement de dualité. Dans le cas des `[espaces]` vectoriels de dimension finie et d'une vraie dualité, les parties closes sont les sous-espaces vectoriels, dans le cas infini, ce sont les sous-espaces vectoriels fermés pour la 'topologie faible' définie par l'accouplement. [79]

2) Relations d'*orthogonalité* entre parties de deux groupes topologiques commutatifs localement compacts duales l'une de l'autre, au sens de Pontrjagin.

   Les parties closes sont les sous-groupes fermés.

3) Pour deux espaces vectoriels 'en dualité' sur un corps valué complet, tel **R** où **C**, la relation

$$|\langle x, x'\rangle| \leq 1.$$

   Dans le cas réel ou complexe, les parties closes sont les parties convexes cerclées [?] faiblement fermées.

4) Pour les espaces vectoriels réels, la relation de semi-polarité

$$\langle x, x'\rangle \geq -1.$$

   Les parties closes sont les parties convexes faiblement fermées contenant l'origine.

Commentary · Examples [ms. 79]

What the double polar closes

The examples identify the content of a formal closure: the relation supplies tests, and the double polar retains everything that those tests cannot distinguish. The algebraic and analytic identifications require their own separation results; the abstract Galois calculation alone does not prove them.

For a separating dual pair of vector spaces E, F over a field k, the annihilator of $A \subseteq E$ is $A^{\perp} = \{\varphi \in F : \langle a, \varphi\rangle = 0 \text{ for all } a \in A\}$. Its double annihilator is the closure of $\operatorname{span}(A)$ for the weak topology of the pairing, taking k discrete in the purely algebraic formulation. Here is a direct finite-coordinate proof. If x is not in the weak closure of a subspace V, there are $\varphi_1, \ldots, \varphi_n \in F$ for which the vector of values of x is outside the image of V in k^n. A linear functional on k^n vanishes on that image but not on this vector. The corresponding linear combination of the φ_i separates x from V. Conversely every annihilating functional is weakly continuous, so its kernel contains the weak closure of V. For real or complex weak topologies the same argument applies because finite-dimensional linear subspaces are closed. In finite-dimensional perfect duality all subspaces are closed, giving the familiar double-annihilator identity.

For a Pontryagin dual pair $G, \widehat{G}$, the annihilator of $A \subseteq G$ consists of the characters trivial on A. It is unchanged upon replacing A by the closed subgroup it generates. If x lies outside that closed subgroup H, the locally compact abelian quotient G/H has a continuous character nontrivial on xH. Composing it with the quotient map gives a character annihilating A but not x. Thus the double annihilator is $\overline{\langle A\rangle}$. The separation step is the substantive theorem; Morris (1977, Chapter 7, Proposition 38) gives this annihilator statement and its proof from character separation.

For the absolute polar over **R** or **C**, the answer is instead the weakly closed absolutely convex hull of $A \cup \{0\}$. One inclusion follows because each inequality $|\langle -, \varphi\rangle| \leq 1$ defines a weakly closed absolutely convex set containing 0. For the other, separate a point outside that hull by a weakly continuous linear functional; the absolute convexity allows multiplication by a scalar of modulus one, and rescaling gives a functional with absolute value at most 1 on the hull and greater than 1 at the point. The weak continuous dual is precisely the specified testing space. The Hahn–Banach separation theorem therefore supplies the desired test. In the real semi-polar convention $\langle x, \varphi\rangle \geq -1$, negate the separating functional; the double polar is the weakly closed convex hull of $A \cup \{0\}$, without the balancedness condition. These are the two forms of the bipolar theorem; compare Schaefer and Wolff (1999, Chapter IV, §1, Theorem 1.5).

The origin must not disappear from the statement. Under conventions allowing the empty set to count as convex and balanced, the list of absolute-polar closed sets must explicitly exclude it: $\varnothing^{\circ\circ} = \{0\}$ in a separating pair. In $\mathbf{R}^2$, for example, the absolute polar of $\{e_1, e_2\}$ is the square $\{|a| \leq 1, |b| \leq 1\}$, and its polar is the diamond $\{|x_1| + |x_2| \leq 1\}$. The inequality follows by maximizing $ax_1 + bx_2$ over the square; choosing the signs of a, b gives equality. This makes the distinction between linear span, convex hull and absolutely convex hull visible in the same two-dimensional setting.

5) Soit K un corps (pour simplifier), L une extension galoisienne de groupe G. Relation entre L et G

$$\gamma \cdot x = x, \qquad \text{i.e. } x \in L^{\gamma}.$$

Les parties closes de L sont les sous-extensions, les parties closes de G les sous-groupes de G.

Commentary · Examples [ms. 79]

The topology in the Galois example

The fixed-point relation offers another instructive instance of the same issue: the relation always gives a closure, but the description of its fixed subsets changes when the extension becomes infinite. For a Galois extension L/K, the polar of a subset $S \subseteq L$ is the group of automorphisms fixing every element of S. Its double polar is the intermediate field $K(S)$. For a subset $H \subseteq G = \mathrm{Gal}(L/K)$, the double polar is the closure of the subgroup generated by H in the Krull topology. Thus all subgroups are closed when the extension is finite; for an arbitrary Galois extension the manuscript's assertion must read *closed subgroups*. See Milne (2022, Proposition 7.12 and Theorem 7.13), where both directions of this correspondence are proved.

There is a concrete obstruction to omitting that qualification. Take $L = \overline{\mathbf{F}}_p$ and $K = \mathbf{F}_p$. The Galois group is $\widehat{\mathbf{Z}}$, with the Frobenius automorphism corresponding to 1. The cyclic subgroup generated by Frobenius is isomorphic to $\mathbf{Z}$; its image surjects onto every finite quotient $\mathbf{Z}/n$, so it is dense, but it is not all of the uncountable group $\widehat{\mathbf{Z}}$. An element fixed by Frobenius satisfies $x^p = x$, hence lies in $\mathbf{F}_p$. The double polar of this cyclic subgroup is consequently the whole Galois group. It is not a closed subset for the formal polarity, even though it is a subgroup.

This correction is already part of classical infinite Galois theory, rather than a later repair peculiar to derivators. The nearby allusion to inseparable analogues requires a different testing object, such as a suitable restricted Lie algebra in a height-one purely inseparable extension. No precise inseparable theorem is stated here, so the audit records an allusion, not an unrestricted extension of the Galois correspondence.

Il y a des analogues divers dans des cas inséparables (à coups de p-algèbres de Lie par exemple), et des énoncés de descente en tous genres.

Tous ces exemples sont de 1ère espèce, les ensembles ordonnés qui interviennent sont de la [page 80]
forme $\mathfrak{P}(P_0)$, $\mathfrak{P}(Q_0)$, et la relation entre eux est déduite d'une relation R_0 entre P_0 et Q_0. Voici maintenant des exemples plus complexes.

6) Soit $\mathcal{M}$ une catégorie ('grosse', le plus souvent …). On prend

$$\mathcal{P} = \mathcal{Q} = \mathfrak{P}(\mathrm{Fl}(\mathcal{M})).$$

La relation $\mathcal{R} \subseteq \mathcal{P} \times \mathcal{Q}$ est la relation de Quillen, que j'écris, pour $\Phi, \Psi \in \mathfrak{P}$

$$\Psi \subseteq \Phi_*,$$

ou encore de façon équivalente

$$\Phi \subseteq \Psi^*,$$

(Ψ a la LLP `[left lifting property]` pour Ψ, Φ la RLP `[right lifting property]` pour Ψ). Donc les applications α_0, β_0 sont données par

$$\beta_0(\Phi) = \Phi_*, \qquad \alpha_0(\Psi) = \Psi^*.$$

L'idée ici ne viendrait à personne d'introduire $\mathfrak{P}(\mathcal{P})$, $\mathfrak{P}(\mathcal{Q})$ - c'est déjà assez gros comme ça ! La notation pour $\Phi^\natural$ est ici $\tilde{\Phi}$, pour $\Psi^\flat$ c'est $\bar{\Psi}$. Les parties closes sont celles que j'appelle Q-closes à gauche et à droite.

La relation $\mathcal{R}$, que j'écris

$$\Phi \longleftrightarrow \Psi,$$ [page 81]

n'est pas déduite d'une relation entre flèches de $\mathcal{M}$, i.e. d'une partie de $\mathrm{Fl}(\mathcal{M}) \times \mathrm{Fl}(\mathcal{M})$, mais est de nature plus complexe. Pour la décrire, il faut introduire l'ensemble $\mathcal{D}$ des diagrammes carrés commutatifs (traits pleins) de la forme

(∗)
$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow{\scriptstyle i} & \overset{f}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \longrightarrow & Y \end{array}$$

(qu'on peut interpréter comme $\mathrm{Fl}(\underline{\mathrm{Fl}}(\mathcal{M}))$, ensemble des flèches dans $\underline{\mathrm{Fl}}(\mathcal{M})$). On a une partie $\mathcal{D}_0$ de $\mathcal{D}$, formée des diagrammes pour lesquels `[il]` existe `[un]` f avec deux triangles commutatifs (c'est donc l'image de $\underbrace{\mathrm{Fl}_3(\mathcal{M})}_{\mathrm{Hom}(\mathbf{\Delta}^3,\mathcal{M})}$ dans $\mathrm{Hom}(\mathbf{\Delta}^1 \times \mathbf{\Delta}^1, \mathcal{M})$, transposée de $\mathbf{\Delta}^1 \times \mathbf{\Delta}^1 \longrightarrow \mathbf{\Delta}^3$ donné par $(0,0) \longmapsto 0$, $(0,1) \longmapsto 1$, $(1,0) \longmapsto 2$, $(1,1) \longmapsto 3$), et une application p. 50

$$\mathcal{D}_0 \subseteq \mathcal{D} \xrightarrow{\varphi} \mathrm{Fl}(\mathcal{M}) \times \mathrm{Fl}(\mathcal{M}),$$

associant à (∗) le couple (i,j) `[plutôt` (i,p)`]` (source et but de (∗), considéré comme flèche dans $\underline{\mathrm{Fl}}(\mathcal{M})$ de i dans p). La relation $\mathcal{R}$ est alors déduite de φ, $\mathcal{D}_0$ par la condition [page 82]

$$\Phi \longleftrightarrow \Psi \qquad \Longleftrightarrow \qquad \varphi^{-1}(\Phi \times \Psi) \subseteq \mathcal{D}_0.$$

Mais à vrai dire, je m'aperçois que c'est quand-même déduit d'une relation entre éléments de $\mathrm{Fl}(\mathcal{M})$, et cette relation entre $i : A \longrightarrow B$ et $p : X \longrightarrow Y$ peut s'écrire ainsi :

(∗∗) pour toutes flèches $A \longrightarrow X$, $B \longrightarrow Y$ rendant commutatif le diagramme (∗), la flèche φ pointillée dans (∗) existe.

Cela me fait penser que si on a une relation $\mathcal{R}$ entre $\mathfrak{P}(P)$ et $\mathfrak{P}(Q)$, satisfaisant (D_0), donc décrite par une application décroissante

$$\mathfrak{P}(P) \longrightarrow \mathfrak{P}(Q)$$

transformant sup en inf, la donné d'une telle application équivaut à celle d'une application quelconque

$$P \longrightarrow \mathfrak{P}(Q),$$

ou ce qui revient au même, d'une relation $\mathcal{R}$ sur $P \times Q$. Donc cette structure doit être regardée comme 'de première espèce'. La situation change à partir du moment où pour décrire la relation $\mathcal{R}$ avec les propriétés voulues, on se trouve contraint de remplacer $\mathfrak{P}(P)$, $\mathfrak{P}(Q)$ par des sous-ensembles convenables - quand on ne regarde que certaines parties de P, Q. C'est ce qu'on va faire au § suivant. Le cas que je vais y traiter est le seul de son espèce que j'ai [page 83] rencontré - un cas de 'dualité formelle' qui s'introduit de façon naturelle, et qui ne soit 'de 1ère espèce'.

Commentary · Later developments [ms. 82]

Lifting gives a polarity before it gives factorization

Grothendieck's second thought about the lifting example is the correct one. A lifting problem involves a square, but the assertion that *every* square with a fixed left edge i and right edge p has a filler is a relation on the pair (i,p). Write $i \,\square\, p$ for that assertion. If Ψ is a class of left maps and Φ a class of right maps, then

$$\Psi \subseteq {}^{\square}\Phi \quad \Longleftrightarrow \quad \Phi \subseteq \Psi^{\square}.$$

The maps are order-reversing and their composites are closure operators by the preceding calculation. The text switches the order of the two factors in the square-based description: if φ records the pair

(i,p), the corresponding condition is $\varphi^{-1}(\Psi\times\Phi)\subseteq\mathcal{D}_0$. This is a notational slip in the printed source, not an obstruction to the construction.

This closure formalism isolates the lifting half of Quillen's framework. A mutually closed pair $(\mathcal{L},\mathcal{R})$ need not supply a factorization of every arrow. That additional existence condition is what makes it a weak factorization system. In the notation used by Riehl (2014, Definition 11.2.1), one requires both $\mathcal{L}={}^{\square}\mathcal{R}$, $\mathcal{R}=\mathcal{L}^{\square}$, and a factorization of every map as a left map followed by a right map. The small object argument supplies factorizations under separate generation and smallness hypotheses; its conclusion is not contained in the order-theoretic anti-isomorphism.

The distinction can be checked in a category with very few arrows. Let $\mathcal{M}$ be the ordered set with objects $0<1$ and its unique nonidentity arrow $a:0\to 1$. A square with both vertical edges a and identity horizontal edges has no filler, so a is not left orthogonal to itself. Identities lift against every map. It follows by checking the only remaining pairs that $(\mathrm{Fl}(\mathcal{M}),\mathrm{Is}(\mathcal{M}))$ and $(\mathrm{Is}(\mathcal{M}),\mathrm{Fl}(\mathcal{M}))$ are mutually closed; their factorizations happen to be the trivial ones. This example illustrates the elementary relation, without using any homotopy theory. In a general category one must still establish factorization independently.

Finally, every map from a powerset to a powerset that turns arbitrary unions into intersections is determined by its values on singletons:

$$\beta(A)=\bigcap_{p\in A}\beta(\{p\}).$$

Defining pRq by $q\in\beta(\{p\})$ recovers the relation. This is the exact reason the lifting example returns to Grothendieck's "first species". The transporter construction differs because its admissible right-hand classes must already be stable under pullback, while its left-hand classes will have to be stable under composition.

p. 51

6 Transporteurs, noyaux, stabiliseurs relatifs à W

Soit $\mathcal{M}$ une catégorie, munie d'une partie

$$W\subseteq\mathrm{Fl}(\mathcal{M}). \tag{6.1}$$

Je pense au cas d'un localiseur, mais n'introduirai d'hypothèse sur W qu'au besoin. Pour simplifier, je suppose $\mathcal{M}$ stable par produits fibrés.

Je pose

$$\text{(6.2)}\qquad \begin{cases} ({}^{4})\mathcal{P} &= \mathfrak{P}(W)=\{A\in\mathfrak{P}(\mathrm{Fl}(\mathcal{M}))\mid A\subseteq W\}\\ \mathcal{Q} &= \{L\in\mathfrak{P}(\mathrm{Fl}(\mathcal{M}))\mid \begin{array}{l}L \textit{ stable} \text{ par changement de}\\ \text{base (par qui je sous-entends aussi,}\\ \text{stable par isomorphie)}\\ \text{et par composition. }\}.\end{array}\end{cases}$$

J'introduis une relation

$$\mathcal{R}\subseteq\mathcal{P}\times\mathcal{Q}$$

de la façon suivante. Pour $A\in\mathcal{P}$, $L\in\mathcal{Q}$, on pose

$$\text{(6.3)}\qquad (A,L)\in\mathcal{R}\iff \text{Pour tout diagramme cartésien}$$
$$\begin{array}{ccc} X & \longleftarrow & X'\\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'}\\ Y & \xleftarrow[p]{} & Y'\end{array}$$
avec $f\in A$, $p\in L$, on a $f'\in W$
(i.e. A est stable par changement de base par $p\in L$).

[page 84]

Quand j'ordonne $\mathcal{P}, \mathcal{Q} \subseteq \mathfrak{P}(\mathrm{Fl}(\mathcal{M}))$ par la relation d'ordre induite par l'inclusion de parties de $\mathrm{Fl}(\mathcal{M})$, on trouve des sous-ensembles ordonnés de $\mathfrak{P}(\mathrm{Fl}(\mathcal{M}))$, stables par réunions quelconques, et aussi par intersections quelconques, avec la seul réserve que dans le cas de $\mathcal{P}$, la famille d'indices doit être $\neq \emptyset$ (puisqu'alors on trouve comme intersection $\mathrm{Fl}(\mathcal{M})$ tout entier, qui n'est pas en général contenu dans W).

Commentary · Exposition [ms. 83]

Why the admissible classes must be corrected

Grothendieck now replaces the moving target A in the stabilizer by the fixed target W. The resulting relation is suitable for a Galois connection, but the first choice of its two domains is too restrictive. Formula (6.2) requires the classes on the right to be closed under both pullback and composition. The transporter constructed on page 85 is closed under pullback, but it need not be closed under composition. The correction at (6.7) therefore removes composition from the definition of the right-hand domain and places it in the left-hand domain.

There is also a smaller order-theoretic point on page 84. Classes closed under composition are not in general closed under unions. In the category $0 < 1 < 2$, let L_1 consist of the identities and $0 \to 1$, and let L_2 consist of the identities and $1 \to 2$. Both classes are closed under pullback and composition, but their union omits $0 \to 2$. Thus the assertion about arbitrary unions is false with the initial definition of $\mathcal{Q}$. Intersections do preserve those properties; the appropriate join is obtained by closing a union under the required operations.

After (6.7), the framework is as follows. Assume W contains isomorphisms and is closed under composition. Let $\mathcal{P}$ consist of the composition-closed classes A with $\mathrm{Is}(\mathcal{M}) \subseteq A \subseteq W$. Let $\mathcal{Q}$ consist of the pullback-stable classes L containing all isomorphisms. Both are complete lattices. Meets are intersections, taking the empty meet to be W in $\mathcal{P}$ and all arrows in $\mathcal{Q}$. A join in $\mathcal{P}$ is the class of finite composites of arrows from the union; it remains in W. A nonempty join in $\mathcal{Q}$ is simply a union, while its empty join is the isomorphisms. These last empty-family conventions are necessary because every admissible class contains the isomorphisms.

For the size of these lattices, fix a universe large enough to contain the chosen collection of arrows, or work in a larger universe than that used for $\mathcal{M}$. The arguments concern classes invariant under isomorphism of arrows and universal quantification over pullback squares. This also explains the footnote on page 85: before invariance of W has been assumed, choosing one representative of each pullback is insufficient. Quantifying over every cartesian square avoids that choice. Once W contains isomorphisms and is composition-closed, invariance follows, and the familiar pullback-functor notation becomes harmless.

On va vérifier la condition (D_0) de la page 69, pour $\mathcal{R}$ et les relations d'ordre. Il y a deux vérifications à faire.

1°) `[où est '2^o' ?]` Donnons nous $A \in \mathcal{P}$, i.e. $A \subseteq W$, et considérons l'ensemble $\mathcal{R}(A)$ des $L \in \mathcal{Q}$ tels que l'on ait $(A, L) \in \mathcal{R}$. Il est clair que c'est un ouvert (et de même, que ${}^s\mathcal{R}(L)$ est un ouvert de $\mathcal{Q}$ pour tout L dans $\mathcal{Q}$) - i.e. $L \in \mathcal{R}(A)$, $L' \subseteq L$ implique $L' \in \mathcal{R}(A)$. Reste p. 52
à trouver un plus grand élément de $\mathcal{R}(A)$. Notons que si $L \in \mathcal{R}(A)$, alors pour tout $p \in L$, $p : Y' \longrightarrow Y$, le changement de base

$$p^* : \mathrm{Cat}/Y \longrightarrow \mathrm{Cat}/Y' : X \longmapsto X \times_Y Y'$$

doit transformer flèches $f \in A$ en flèches $g \in W$, comme il résulte du fait que L est stable par changement de base.

$$\begin{array}{ccc} X_1 & \xleftarrow{q_1} & X_1' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ X_2 & \xleftarrow{q_2} & X_2' \\ \downarrow & & \downarrow \\ Y & \xleftarrow{p} & Y'. \end{array}$$

[page 85]

Donc $L \subseteq \mathbf{L}_W(A)$, où on pose

$$\mathbf{L}_W(A) = \{(p : Y' \longrightarrow Y) \in \mathrm{Fl}(\mathcal{M}) \quad | \quad p^* : \mathrm{Cat}/Y \longrightarrow \mathrm{Cat}/Y' \text{ transforme flèches de } A \text{ en flèches } \textit{de } W.\}$$

([5]).

$$\begin{array}{ccc} X_1 & \longleftarrow & X_1' \\ {\scriptstyle f}\downarrow & \text{cart.} & \downarrow{\scriptstyle f'} \\ X_2 & \longleftarrow & X_2' \\ \downarrow & \text{cart.} & \downarrow \\ Y & \xleftarrow{p} & Y' \end{array}$$

$$f \in A \implies f' \in W.$$

Il est clair d'autre part que cette partie de $\mathrm{Fl}(\mathcal{M})$ est stable par changement de base, donc $\mathbf{L}_W(A) \in \mathcal{Q}$, et que $(A, \mathbf{L}_W(A)) \in \mathcal{R}$. Donc $\mathbf{L}_W(A)$ est un plus grand élément de $\mathcal{R}(A)$, donc on trouve

$$\mathcal{R} = \{(A, L) \in \mathcal{P} \times \mathcal{Q} \mid L \subseteq \mathbf{L}_W(A)\}. \tag{6.5}$$

p. 53

$$\begin{array}{ccc} X_1 & \xleftarrow{q_1} & X_1' \\ {\scriptstyle f}\downarrow & \text{cart.} & \downarrow{\scriptstyle f'} \\ X_2 & \xleftarrow{q_2} & X_2' \\ \downarrow & \text{cart.} & \downarrow \\ Y_1 & \xleftarrow{p_1} & Y_1' \\ \downarrow & \text{cart.} & \downarrow \\ Y & \xleftarrow{p} & Y'. \end{array}$$

Inversement, partons de $L \in \mathcal{Q}$, et montrons que l'ouvert ${}^s\mathcal{R}(L)$ de $\mathcal{P}$ a un plus grand élément. Pour ceci, posons

$$\mathbf{A}_W(L) = \{(f : X \longrightarrow Y) \in W \subseteq \mathrm{Fl}(\mathcal{M}) \quad | \quad \forall \quad \begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & \textit{cart.} & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y', \end{array} \text{ avec } p \in L, \text{ on a } f' \in W\}. \tag{6.6}$$

(NB Quand L contient les identités, comme c'est le cas notamment s'il est de la forme $\mathbf{L}_W(A)$ - alors il contient les isomorphismes, du moins si A les contient - alors $\mathbf{A}_W(L)$ est l'ensemble de $f \in \mathrm{Fl}(\mathcal{M})$ satisfaisant la condition énoncée après |, sans avoir à spécifier $f \in W$.) On a $\mathbf{A}_W(L) \in \mathcal{P}$, et on vérifie que $(\mathbf{A}_W(L), L) \in \mathcal{R}$ (en utilisant la stabilité de L par composition), enfin il est clair que $(A, L) \in \mathcal{R}$ implique $A \subseteq \mathbf{A}_W(L)$ (en utilisant $A \subseteq W$), donc $\mathbf{A}_W(L)$ est un plus grand élément de ${}^s\mathcal{R}(L)$.

Notons que si W *est stable par composition* (ce que nous allons supposer désormais), il en est de même de $\mathbf{A}_W(L)$ (ce qui utilise encore le fait que L soit stable par changement de [page 86]

4. cf. rectification p. 86.

5. NB le dire plutôt en termes de diagrammes cartésiens, vu qu'on n'a pas fait d'hypothèse de stabilité de W par isomorphismes.

base). De plus, si W *contient les isomorphismes* (ce que nous supposerons également), $\mathbf{A}_W(L)$ les contient aussi. Quant à $\mathbf{L}_W(A)$, il contient les isomorphismes *pourvu que* A soit stable par isomorphismes ; quand A est stable par composition et contient les isomorphismes, cette condition sera automatiquement satisfaite. Nous tenons à cette condition, c'est pourquoi du côté de A, dans la définition de $\mathcal{P}$, je vais introduire cette condition supplémentaire que A soit *stable par isomorphismes.* Donc je pose, en corrigeant (6.2),

$$(6.7)\quad \begin{cases} \mathcal{P} &= \{A \in \mathfrak{P}(\mathrm{Fl}(\mathcal{M})) \quad | \quad \mathrm{Is}(\mathcal{M}) \subseteq A \subseteq W, \quad A \text{ stable par composition}\} \\ \mathcal{Q} &= \{L \in \mathfrak{P}(\mathrm{Fl}(\mathcal{M})) \quad | \quad \mathrm{Is}(\mathcal{M}) \subseteq L, \quad L \text{ stable par changement de base}\}. \end{cases}$$

Commentary · Exposition [ms. 85]

The transporter and its adjoint

The corrected construction works because a larger testing class produces a smaller transporter. It is useful to give the complete argument, including the distinction between testing over one codomain and testing all arrows in a slice.

For $A \in \mathcal{P}$, define $T_W(A)$ to consist of the maps $p : Y' \to Y$ such that every arrow $f : X_1 \to X_2$ in A, equipped with a map $X_2 \to Y$, has $p^*f \in W$. Pullback pasting shows that $T_W(A)$ is stable under arbitrary base change: a test after pulling p back along $Z \to Y$ is already a test for p with its structure map factoring through Z. It contains the isomorphisms because $A \subseteq W$ and W is invariant under isomorphism. Thus $T_W(A) \in \mathcal{Q}$.

For $L \in \mathcal{Q}$, define $N_W(L)$ to consist of the maps $f \in W$ whose pullbacks along all maps of L into their codomain belong to W. The explicit requirement $f \in W$ is redundant here, since identities belong to L. Isomorphisms are in $N_W(L)$. For composable $f, g \in N_W(L)$ and $p \in L$ into the codomain of g, factor $p^*(gf)$ as the pullback of f along g^*p, followed by p^*g. Pullback stability gives $g^*p \in L$, so both factors lie in W. Hence $N_W(L)$ is composition-closed and belongs to $\mathcal{P}$.

The defining relation is now exactly

$$A \subseteq N_W(L) \quad \Longleftrightarrow \quad L \subseteq T_W(A).$$

For the forward direction, test $p \in L$ against $f \in A$ over Y. The base change $X_2 \times_Y Y' \to X_2$ is in L, so the assumption on f applies. Conversely, if p is a map of L into the actual codomain of f, this is a special case of the transporter test. This proves both implications and the maximality assertions on pages 85–86. Closure of L under composition is unnecessary for this adjunction; its appearance in the text's proof of $(N_W(L), L) \in \mathcal{R}$ is redundant.

Both operations reverse inclusions, so the usual polarity argument now applies without qualification. Their composites are monotone, extensive and idempotent; their fixed classes correspond anti-isomorphically. The extreme pairs in (6.9)–(6.10) follow directly: $T_W(\mathrm{Is})$ is all arrows, $N_W(\mathrm{Fl}) = W_{\mathrm{univ}}$, $N_W(\mathrm{Is}) = W$, and $T_W(W) = \mathrm{Fib}_W$. In particular, every closed left class contains all universal W-equivalences, and every closed right class contains Fib_W. This is the relation-theoretic content of §6, established with only the hypotheses actually imposed there.

On est bien sûr intéressé surtout aux éléments clos de $\mathcal{P}$, $\mathcal{Q}$, i.e. les parties A, L de $\mathrm{Fl}(\mathcal{M})$, contenant les isomorphismes, stables par composition (et L stable aussi par changement de base, $A \subseteq W$) telles que l'on ait p. 54

$$(6.8)\qquad A = \mathbf{A}_W(L), \qquad L = \mathbf{L}(A)$$

`[plutôt` $\mathbf{L}_W(A)$`]`.

Je garderai les notations $\natural$, $\flat$ du paragraphe précédent. Quels sont les deux couples extrêmes ? [page 87]

Partant de $A_0 = \mathrm{Is}(\mathcal{M})$ (plus petit des éléments de $\mathcal{P}$), on trouve $\mathbf{L}_W(A_0) = \mathrm{Fl}(\mathcal{M})$ (plus grand élément de $\mathcal{Q}$), et $A_0^\natural = \mathbf{A}_W(\mathrm{Fl}(\mathcal{M})) = W_{\mathrm{univ}}$, ensemble des flèches qui sont dans W et le restent par tout changement de base :

$$(6.9)\quad A_0^\natural = W_{\mathrm{univ}}, \qquad \beta_0(A_0) \overset{\text{déf}}{=} \mathbf{L}_W(A_0) = \mathrm{Fl}(\mathcal{M}), \qquad \text{où } A_0 = \mathrm{Is}(\mathcal{M}) \text{ plus petit élément de } \mathcal{P}.$$

Inversement, partons de $L_0 = \mathrm{Is}(\mathcal{M})$, le plus petit élément de $\mathcal{Q}$. On a $\mathbf{A}_W(L_0) = W$ (W étant stable par composition et contenant les isomorphismes), et $L_0^\flat = \mathbf{L}_W(W) = \mathrm{Fib}_W$, les W-fibrations :

$$L_0^\flat = \mathrm{Fib}_W, \qquad \alpha_0(L_0) = W, \qquad \text{où } L_0 = \mathrm{Is}(\mathcal{M}), \text{ plus petit élément de } \mathcal{Q}. \tag{6.10}$$

Donc les parties $\mathcal{R}$-closes à gauche de $\mathrm{Fl}(\mathcal{M})$ se promènent entre W_{univ} et W, les parties $\mathcal{R}$-closes à droite entre Fib_W et $\mathrm{Fl}(\mathcal{M})$.

Proposition 6.1. *Soient $A \in \mathcal{P}$, $L \in \mathcal{Q}$ tels que $A = \alpha_0(L)$, $L = \beta_0(A)$ (i.e. $A = \mathbf{A}_W(L)$, $L = \mathbf{L}_W(A)$). Pour que L soit stable par composition, il faut et il suffit que A soit stable par changement de base par les flèches $f \in L$. (Plus généralement, si $L \in \mathcal{Q}$ est un élément quelconque de $\mathcal{Q}$, et si on suppose de plus que L est stable par composition, alors $A = \mathbf{A}_W(L)$ est stable par changement de base par L.)* [page 88]

Commentary · Exposition [ms. 87]

When a transporter becomes a stabilizer

Proposition 6.1 identifies the extra condition that makes the formal correspondence compatible with iterated changes of base. Suppose (A, L) is a closed pair, so $A = N_W(L)$ and $L = T_W(A)$. If L is closed under composition, then A is stable under base change along L: for $f \in A$ and $p \in L$, its pullback belongs to W, and each further test along $q \in L$ is a test of f along $pq \in L$. Therefore $p^*f \in A$. When f is an arrow over the target of p, use the pullback of p along its codomain map; it belongs to L by pullback stability. This proves the more general assertion in parentheses, since it uses only $A = N_W(L)$.

Conversely, assume A is stable under base change along L. Given composable maps $q : Z \to Y'$ and $p : Y' \to Y$ in L, and an arrow $f \in A$ over Y, first pull back along p. The result is in A, and its pullback along q is again in $A \subseteq W$. Thus $(pq)^*f \in W$, whence $pq \in T_W(A) = L$. This proves the proposition completely.

For a closed pair one can sharpen its wording without adding a hypothesis:

$$L \text{ is composition-closed} \quad \Longleftrightarrow \quad L = \mathrm{Stab}(A).$$

Indeed, stability of A under L gives $L \subseteq \mathrm{Stab}(A)$, while the reverse inclusion always follows from $A \subseteq W$ and $L = T_W(A)$. Conversely every stabilizer is composition-closed. Thus the two names designate the same class exactly at the closed pairs satisfying Proposition 6.1. Without this extra condition the transporter may contain maps that preserve the W-tests but fail to preserve A itself.

The word "i.e." before (6.8) must accordingly be read with care. Closed pairs do not automatically have a composition-closed right member. Grothendieck is particularly interested in those that do; Proposition 6.1 states the criterion. The following finite example verifies that the distinction is real under the abstract hypotheses of this section.

Commentary · Development (commentators') [ms. 87]

A closed transporter which does not compose

The formal correction of $\mathcal{Q}$ in (6.7) becomes necessary in a four-object category. The example also separates the universal quantifier in a transporter from the stronger demand that it stabilize its source class.

A closed pair with a non-composition-closed right member. [ours] Let $\mathcal{M}$ be the chain $0 < 1 < 2 < 3$, with pullbacks given by minima. Write ij for its unique arrow $i \to j$ and I for its identities. Set

$$W = I \cup \{02, 03\}, \qquad A = I \cup \{03\}.$$

Then W and A contain the isomorphisms and are closed under composition. Moreover,

$$T_W(A) = \mathrm{Fl}(\mathcal{M}) \setminus \{13\}, \qquad N_W(T_W(A)) = A.$$

Thus $(A, T_W(A))$ is a closed pair, but 12 and 23 belong to its right member while their composite 13 does not.

To verify the transporter formula, the only nonidentity arrow of A to test is 03. It can be an arrow over y only when $y = 3$. Pullback along $x3$ gives $0x$, which lies in W precisely for $x = 0, 2, 3$. Therefore 13 is the unique excluded map; all identity tests succeed. This proves the first formula, including all slice tests.

For the kernel formula, the identities and 03 survive by the adjunction. The only remaining member of W is 02. Its pullback along 12, a member of the transporter, is 01, which is not in W. Hence 02 does not survive, and the second formula follows. Finally the right member is closed as well: applying the triple-operation identity for the established Galois connection gives $T_W N_W T_W(A) = T_W(A)$. This completes all the claims.

Here W is an abstract class of arrows with exactly the composition and isomorphism assumptions of §6; it is not asserted to be a fundamental localizer on Cat, or even to satisfy two-out-of-three. We checked the finite calculation exhaustively with the supplied script and proved it above. We did not find this specific example in the checked transporter and smooth-functor sources. It serves to explain the actual generality of Proposition 6.1, without making a claim about the special localizers used elsewhere in the manuscript.

Commentary · Context [ms. 87]

Intrinsic fibrations and sharp maps

Grothendieck wants the admissible changes of base to be determined by the weak equivalences themselves. This is why the extreme pair in (6.10) matters more than an isolated exercise with closure operations. Once W has been fixed, a morphism $p : E \to B$ belongs to Fib_W precisely when

$$p^* : \mathcal{M}/B \longrightarrow \mathcal{M}/E$$

preserves the arrows whose underlying morphisms belong to W. Both the test arrow and its map to B vary. Restricting the test to arrows with target B would omit the further base changes built into the definition.

The letter to Thomason of 2 April 1991 makes this intention explicit. In its first numbered discussion, Grothendieck proposes replacing a suitable proper fibration structure by the class determined by this preservation property; existence of enough such fibrations remains a separate factorization requirement (Grothendieck, 1991a, §1, printed p. 4). Thus the canonical class is available before one knows that it supports a useful model structure. The historical point is the choice of what should be intrinsic, not a rejection of the computational value of lifting and factorization axioms.

In a model category the same preservation property is the definition of a *sharp map* in Rezk (1998, §2). In a right proper model category every model fibration is sharp, and sharpness is equivalent to every strict pullback square along the map being homotopy cartesian (Rezk, 1998, Propositions 2.2 and 2.7). These statements explain both the usefulness and the limitations of the comparison: Grothendieck's Fib_W asks for a property of pullback with respect to W; a chosen Quillen fibration class supplies maps with that property under the additional hypothesis of right properness. An arbitrary transporter $\mathbf{L}_W(A)$ only promises to send the smaller class A into W. A second pullback can therefore leave the class of arrows on which that promise applies. The extra stability in Proposition 6.1 is exactly what permits the second step.

Terminologie : pour $A \in \mathcal{P}$, i.e. A une partie de $\mathrm{Fl}(\mathcal{M})$ contenue dans W contenant les isomorphismes et stable par composition, $\mathbf{L}_W(A)$ s'appelle le *transporteur de A dans W*. (Il y a une notion duale de *cotransporteur*.) C'est un élement de $\mathcal{Q}$, donc une partie de $\mathrm{Fl}(\mathcal{M})$ contenant les isomorphismes et stable par changement de base. D'autre part, pour L dans $\mathcal{Q}$, les $f \in \mathbf{A}_W(L)$ s'appellent flèches *W-L-asphériques,* ou les *W-L-équivalences,* ou les *W-équivalences L-universelles,* et $\mathbf{A}_W(L)$ s'appelle le *L-noyau de W*. On pourrait le noter plus simplement W_L (quand cette notation ne conflicte pas avec $W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$ pour $S \in \mathrm{Ob}\,\mathcal{M}$). C'est une partie de W qui (comme W lui-même) contient les isomorphismes et est stable par composition.

Enfin, il y a lieu aussi d'introduire le *stabiliseur* $\mathrm{Stab}(A)$, ou transporteur de A dans A, qui ne dépend pas de la donné de W, mais seulement de la partie A de $\mathrm{Fl}(\mathcal{M})$, contenant les isomorphismes et stable par composition. On pourrait aussi le noter Fib_A, et appeler

ses éléments des *A-fibrations* (A jouant à présent le rôle de W). On a $\mathrm{Stab}(A) \subseteq \mathbf{L}_W(A)$, et il semblerait qu'en pratique, ce sont ces stabiliseurs, plus que les transporteurs $\mathbf{L}_W(A)$, qui sont importants (quand les deux ne coïncident pas). C'est ainsi que dans Cat, si W est un localiseur fondamental, les morphismes W-lisses (resp. W-propres) Liss_W (Prop_W), qui sont les stabiliseurs des ensembles W_{as} (W_{coas}) des ensembles de morphismes W-asphériques (resp. W-coasphériques), jouant un rôle essentiel, alors que je ne sais pas caractériser de façon sensible les flèches des transporteurs $\mathbf{L}_W(W_{\mathrm{as}})$, $\mathbf{L}_W(W_{\mathrm{coas}})$, lesquels contiennent, en plus des morphismes W-lisses (resp. W-propres) les W-fibrations Fib_W. Je ne sais pas comment caractériser une classe de morphismes qui apparaît aussi hétéroclite! De même, il y a lieu d'introduire et de caractériser le stabiliseur de W_{bias} (ensemble des flèches W-biasphériques), comme les morphismes 'W-nets' Net_W, par une description similaire à celle des Liss_W, Prop_W, utilisant les bilocalisés $\backslash\underset{u}{X}/$, $\backslash\underset{v}{Y}/$, au lieu des localisés X/x, Y/y ou des colocalisés $x\backslash X$, $y\backslash Y$. Il y a lieu enfin de déterminer le stabiliseur de l'ensemble Equ des Cat-équivalences (ce qui ne dépend pas de la donné d'un W - mais on a $\mathrm{Equ} \subseteq \mathcal{P}(W)$ pour tout W fondamental).

p. 55
[page 89]

Commentary · Present state [ms. 89]

The scope of the remaining transporter questions

The questions at the end of the section ask for usable local tests for classes already defined by universal preservation properties. For smooth maps, that programme was developed explicitly: Maltsiniotis (2005, Definition 4.6, Theorem 4.13, and Example 4.17) defines preservation of asphericity under pullback, gives equivalent local conditions, and compares the result with smoothness for a fundamental localizer. Properness is obtained by opposite categories. This identifies a later theorem corresponding to the stabilizer side of Grothendieck's discussion.

The transporter $T_W(W_{\mathrm{asph}})$ asks only that pullbacks of aspherical maps be W-equivalences; its target condition is weaker than preservation of asphericity. The inclusions on page 89 follow directly: it contains $\mathrm{Stab}(W_{\mathrm{asph}})$ because $W_{\mathrm{asph}} \subseteq W$, and it contains $\mathrm{Stab}(W)$ because the latter preserves every arrow of W. The corresponding statements for coasphericity are dual. They do not identify a union of those classes, nor imply that the transporter is composition-closed.

We checked Maltsiniotis's asphericity paper and searched the literature under the manuscript's transporter, stabilizer and "net" terminology. We did not locate a theorem identifying the general biaspherical stabilizer with precisely the proposed bilocal lifting condition, or a full intrinsic description of both mixed transporters. This is a statement about the scope of our verification, not a claim that those problems remain open in every formulation. The separate question about the stabilizer of categorical equivalences has a definite answer, proved in the block "The stabilizer of categorical equivalences". The history of the intrinsic class Fib_W is discussed in "Intrinsic fibrations and sharp maps".

p. 56
[page 90]

Commentary · Later developments [ms. 89]

The stabilizer of categorical equivalences

The last question on this page has a concrete answer. The obstruction to preserving an equivalence under a strict pullback is an isomorphism of objects in the base which cannot be lifted. The relevant maps are the *isofibrations*, the fibrations of the usual model structure on categories; see Rezk (1996, §2 and Theorem 3.1). We give the argument in the notation of the manuscript, so that the answer also specifies the meaning of its stabilizer.

Proposition. Let Equ be the class of equivalences of small categories. A functor $p : E \to B$ belongs to $\mathrm{Stab}(\mathrm{Equ})$ if and only if, for every $e \in E$ and every isomorphism $\gamma : p(e) \to b$ in B, there are an object e' and an isomorphism $\widetilde{\gamma} : e \to e'$ with $p(e') = b$ and $p(\widetilde{\gamma}) = \gamma$.

Proof. Suppose first that this lifting property holds. Let $F : C \to D$ be an equivalence over B, with structure map $h : D \to B$. The pullback

$$F_E : C \times_B E \longrightarrow D \times_B E$$

is fully faithful: a morphism in either pullback is a pair of morphisms with equal images in B, and full faithfulness of F lifts the first member uniquely without changing that equality. To prove

essential surjectivity, take (d, e) in the target. Choose $c \in C$ and an isomorphism $\theta : F(c) \to d$. Apply the isomorphism-lifting property to $(h\theta)^{-1} : p(e) \to hF(c)$ and invert the resulting lift. This gives an isomorphism $\eta : e' \to e$ with $p(e') = hF(c)$ and $p(\eta) = h\theta$. The pair (θ, η) is an isomorphism $F_E(c, e') \to (d, e)$. Thus F_E is an equivalence, as required for the stabilizer.

Conversely, suppose $p \in \mathrm{Stab}(\mathrm{Equ})$. Let J be the groupoid with two objects $0, 1$ and exactly one morphism between each ordered pair. The data (e, γ) define a functor $J \to B$ taking 0 to $p(e)$, 1 to b, and $0 \to 1$ to γ. The inclusion $\{1\} \to J$ is an equivalence over B. Hence

$$E \times_B \{1\} \longrightarrow E \times_B J$$

is an equivalence. Essential surjectivity at $(e, 0)$ supplies $(e', 1)$ and an isomorphism between it and $(e, 0)$. Orienting that isomorphism from $(e, 0)$ to $(e', 1)$ gives the required $\widetilde{\gamma}$. This proves the converse. □

The proof also explains why the answer is sensitive to strict pullbacks. A point inclusion into J is itself an equivalence, but pulling it back along the inclusion of the other point gives $\emptyset \to e$. An isofibration supplies the missing representatives over isomorphic objects. No assertion about the fundamental localizer W is needed for this calculation. In particular, it answers the question about Equ without identifying the more delicate stabilizer of W-biaspherical maps with a familiar fibration class.

7 Morphismes cofiltrants et localement cofiltrants

Mais je ne veux pas m'attarder à présent sur ces développements, qui ne me paraissent pas urgents, et voudrais revenir aux questions liées à la notion de catégories filtrantes ou cofiltrantes.

Commentary · Exposition [ms. 90]

The conditions that pass through a comma category

Grothendieck returns to filteredness because it supplies concrete conditions under which the cofinality arguments can be repeated after restricting a diagram. The adjective "local" has a precise force here. It does not mean merely that the set of objects in each connected component is directed. The commutativity of the triangles must survive the construction of a common predecessor.

Write $P(C)$ for the following condition: every cospan $a \xrightarrow{r} c \xleftarrow{s} b$ admits arrows $u : d \to a$, $v : d \to b$ such that $ru = sv$. Write $E(C)$ for the equalization condition: every parallel pair $r, s : a \to b$ admits $t : d \to a$ with $rt = st$. Finally, write $I(C)$ for nonemptiness together with the existence of a common predecessor for every pair of objects. These are respectively PS_g1, PS_g2, and PS_g1' in the text. Then C is cofiltered precisely when $I(C)$ and $E(C)$ hold, and locally cofiltered precisely when $P(C)$ and $E(C)$ hold. The latter condition allows the empty category. Equivalently, every connected component is cofiltered; the elementary decomposition underlying this statement is recorded, with the opposite variance, in The Stacks Project Authors (2026e, Lemmas 4.19.6 and 4.19.8).

The weaker terms need an explicit convention. In the working proof, "weakly cofiltered" means $I(C)$, whereas "locally weakly cofiltered" means $P(C)$. Footnote 64 proposes reserving the former term for nonempty connectedness together with $P(C)$. These conditions differ: the category with two parallel arrows $0 \rightrightarrows 1$ satisfies I but fails P. The closing diagrams recognize the distinction and call I irreducibility. We therefore retain I and P in statements where the terminology changes. Proposition 7.3 and its composition corollary hold with either weak convention, provided it is used consistently.

Here is the argument behind the proposition, including its three independent assertions. Suppose first that P holds in Y and in every $y\backslash X$. Given a cospan in X, complete its image in Y to a commutative square with lower vertex y. The three objects of the cospan, equipped with the resulting maps from y, form a cospan in $y\backslash X$. Applying P there gives the required square in X. If instead I holds in Y and in the comma categories, two objects of X first acquire a common predecessor for their images in Y, and then acquire one in the corresponding comma category. Nonemptiness follows by choosing $y \in Y$ and an object of $y\backslash X$.

For equalization, start with $r, s : a \to b$ in X. Choose $v : y \to f(a)$ with $f(r)v = f(s)v$, using $E(Y)$. Then r, s are parallel arrows in $y\backslash X$. An equalizing arrow there equalizes them in X. These arguments prove the cofiltered and locally cofiltered cases as well. For the revised weak convention, $P(Y)$ and connectedness imply $I(Y)$: the relation "has a common predecessor" is transitive by P, and contains every arrow and its reverse. The same observation applies to the comma categories, so the preceding argument also proves connectedness of X.

The composition statement now has a transparent reason. A discrete opfibration $Y' \to Y$ identifies $y'\backslash(X \times_Y Y')$ with $y'_Y\backslash X$, by unique lifting of arrows out of y'. This proves the coétale base-change assertion of Remark 7.2. Apply it to $z\backslash Y \to Y$, and then apply Proposition 7.3 to $z\backslash X \to z\backslash Y$. The printed assertion that *étale* maps are locally "filtrants" has the variance reversed; Lemma 7.5 gives the correct locally cofiltered assertion below.

Definition 7.1. Soit $f : X \longrightarrow Y$ une flèche dans Cat. On dit que f est *cofiltrante* (resp. *localement cofiltrante*, resp. *faiblement cofiltrante* ([64]), resp. *localement faiblement cofiltrante* ou encore *précofiltrante*) si pour tout y dans Y, la catégorie colocalisé $y\backslash X$ est cofiltrante (resp. …). (Dualement, on dit que f est *filtrante*, resp. …, si les catégories X/y sont filtrantes, resp. …, i.e. si $f^o : X^o \longrightarrow Y^o$ est cofiltrante resp. ….)

Remarque 7.2. Notions stables par changement de base coétale. Contient les isomorphismes et même les Cat-équivalences. Les morphismes *étales* sont localement filtrants.

Proposition 7.3. *Soit $f : X \longrightarrow Y$ un morphisme cofiltrant (resp. localement cofiltrant, resp. faiblement cofiltrant, resp. précofiltrant), et supposons que Y soit cofiltrant (resp. localement cofiltrant, resp. faiblement cofiltrant, resp. précofiltrant). Alors X lui-même est cofiltrant (resp. …).*

Démonstration. Il s'agit, dans les quatre cas envisagés, de vérifier pour X la condition $\underbrace{\mathrm{PS}_g1' \,/\, \mathrm{PS}_g1}_{\text{deux derniers cas}} \,/\, \mathrm{PS}_g1' + \mathrm{PS}_g2 \,/\, \mathrm{PS}_g1 + \mathrm{PS}_g2$. Mais pour chacune des trois conditions PS_g1, PS_g1', PS_g2 séparément, on voit immédiatement que si elle est vérifiée pour Y et pour les $y\backslash X$, elle l'est pour X, d'où la proposition. [page 91]

Corollaire 7.3. *Les quatre notions envisagées dans (7.1) sont stables par composition.*

Démonstration immédiate, en utilisant la remarque (7.2).

$$\begin{array}{ccc} X & \longleftarrow & z\backslash X \\ \downarrow & & \downarrow \\ Y & \longleftarrow & z\backslash Y \\ \downarrow & & \downarrow \\ Z & \longleftarrow & z\backslash Z. \end{array}$$

Remarque 7.4. p. 57

1) Notons que la notion de morphisme cofiltrant ou faiblement cofiltrant n'est *pas* stable par changement de base quelconque, ni même par une immersion ouverte. Exemple : $e \longrightarrow \mathbf{\Delta}^1$, appliquant e en 1, est cofiltrant,

$$\begin{array}{ccc} & & e \\ & & \downarrow \\ 0 & \longrightarrow & 1 \end{array}$$

mais sa fibre en 0 est vide, donc n'est ni cofiltrante, ni faiblement cofiltrante. Je n'ai pas vu d'exemple montrant que la notion de flèche *localement* cofiltrante resp. précofiltrante

64. J'ai modifié la terminologie : maintenant faiblement cofiltrant = 0-connexe + localement faiblement cofiltrant, *stable par passage* à X' sur X, X' étale et ∞-connexe. Revoir la terminologie dans le cas faiblement cofiltrant, qui est une notion traîtresse, faute d'être stable par passage [à] un X' étale sur X.

n'est pas stable par changement de base, et ne voudrais pas m'y attarder, n'ayant pas besoin de connaître la réponse.

2) Pour que X soit cofiltrante (resp. …), il faut et il suffit que $X \longrightarrow e$ le soit. Ainsi la notion absolue est ramenée à une notion relative.

3) Si $f : X \longrightarrow Y$ est cofiltrante, elle est W-coasphérique (donc $\in W$) pour tout W satisfaisant l'axiome Loc (8) des limites inductives. En particulier, elle est W_0-asphérique, i.e. un foncteur cofiltrant (et même seulement faiblement cofiltrant) est *cofinal.* [page 92]

La raison principale pour m'intéresser aux flèches cofiltrantes est contenue dans la

Commentary · Development (commentators') [ms. 91]

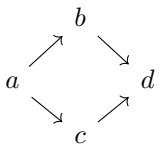
A finite answer to the base-change question

Grothendieck's example separates cofilteredness from its local form only by making a fibre empty. To answer the question he leaves beside it, the pullback must instead create a connected comma category whose two branches have lost their common predecessor.

A four-object counterexample. [ours]

The class of locally cofiltered functors, and the class of locally weakly cofiltered functors, are not stable under arbitrary base change, even among finite posets.

Proof. Let $f : \{1\} \hookrightarrow [1]$ select the final object of the walking arrow. Both $0\backslash\{1\}$ and $1\backslash\{1\}$ are singleton categories, so f is cofiltered. Let D be the diamond poset



and define $v : D \to [1]$ by $v(a) = 0$ and $v(b) = v(c) = v(d) = 1$. The pullback of f is the full inclusion $j : U \hookrightarrow D$, where $U = \{b, c, d\}$ and $b < d$, $c < d$. Since a is initial in D, there is an isomorphism $a\backslash U \cong U$. This category is connected, but the cospan $b \to d \leftarrow c$ has no common predecessor in U. Thus $P(a\backslash U)$ fails; in particular this comma category is not locally cofiltered. The same pullback disproves both proposed stability assertions. □

This example answers the question for strict categorical pullbacks, which are the pullbacks used in the source. It does not conflict with Remark 7.2: v is not a discrete opfibration, since the arrow $0 \to 1$ has several lifts with source a. Nor does it conflict with Proposition 7.4, which concerns an *étale change of the source* followed by a factorization through an *étale target.*

The same diamond makes the variance slip in Remark 7.2 visible. The lower open subposet $V = \{a, b, c\} \subset D$ defines a discrete fibration. Its comma category V/d is V, which is connected and not filtered: b and c have no common upper bound there. Thus an *étale* functor need not be locally filtered.

We checked the discussions of representable flatness in Tendas (2024, Remark 2.4) and Shulman (2011), and the explicit filteredness criteria in The Stacks Project Authors (2026e, Section 4.19). We did not locate this finite pullback construction there. The label records our supplied answer to the manuscript's question, without claiming priority for the example.

Commentary · Exposition [ms. 92]

The image that belongs in Proposition 7.4

The proposition is intended to compare sections or cosections after passing to an open part of a slice. The target must therefore be the part reached by that open, rather than the part reached by the whole slice. In (c) the printed definition of V omits the restriction to U. Its correct form is

$$V = \{z \in Y/f(x) \mid z\backslash U \neq \varnothing\}.$$

Equivalently, V is the lower open subcategory generated by $(f/x)(U)$. This is already forced by $U = \varnothing$: the desired target is then empty, whereas the target defined in print is all of $Y/f(x)$, because $(f/x)(x) = f(x)$ is its terminal object. More generally, the proof's assertion that $z\backslash U$ is nonempty

"par construction de V" uses exactly the corrected definition. This is a slip in the edition's text, not a mistranscription; the correction is consequently explained here and retained as a recorded slip in the source.

Once this target is specified, the two comparison formulas have their expected variance. With $g : U \to V$, a covariant diagram $G : V \to \mathcal{M}$ has

$$\operatorname*{colim}_U Gg \;\cong\; \operatorname*{colim}_V G,$$

whereas a presheaf $F : V^{\mathrm{op}} \to \mathcal{M}$ has

$$\lim_{V^{\mathrm{op}}} F \;\cong\; \lim_{U^{\mathrm{op}}} Fg^{\mathrm{op}}.$$

Both statements follow from the same nonempty-connected comma criterion for finality. The second does not assert invariance of limits of *covariant* diagrams under a final functor. The needed limits or colimits must exist in $\mathcal{M}$. The letter printed as q in (c) is legible in the edition; it denotes the map called g in the proof and in these formulas.

The homotopical qualification also has a precise scope. The cofiltered comma categories are W-aspherical under the stated axiom Loc (8), so g is W-coaspherical. Weak cofilteredness by itself gives connectedness, hence ordinary finality, but supplies no general assertion of contractibility. Footnote 65 correctly reserves the stronger conclusion for the cofiltered hypothesis. The phrase "W_0-asphérique" in Remark 7.4(3) must read "W_0-coasphérique" in the convention of this chapter.

Proposition 7.4. *Soit $f : X \longrightarrow Y$ dans* Cat *localement cofiltrant (resp. localement faiblement cofiltrant). Alors :*

a) Pour tout X' étale sur X, le composé $f' : X' \longrightarrow X \longrightarrow Y$ est localement cofiltrant (resp. …), et pour toute factorisation de f' en $X' \xrightarrow{g'} Y' \longrightarrow Y$ avec $Y' \longrightarrow Y$ étale, g' est localement cofiltrant (resp. …).

b) $\forall$ x dans X, le morphisme sur les localisés

$$f/x : X/x \longrightarrow Y/x \qquad (y = f(x))$$

est cofiltrant (resp. faiblement cofiltrant).

c) Si x, y comme dans b), et U est un ouvert de X/x, désignant par V l'ouvert de Y/y engendré par l'image de f/x (i.e. l'ouvert formé des z dans Y/y tels que $z\backslash(X/x)$ soit non vide), la flèche induite $q : U \longrightarrow V$ `[est]` *cofiltrante (resp. faiblement cofiltrante), a fortiori cofinale (et même W-coasphérique, pour tout W satisfaisant l'axiome des limites Loc (8))* ([65]).

Donc pout tout cofaisceau *G sur V, à valeurs dans une catégorie $\mathcal{M}$ avec $\varinjlim$, on a* `[un]` *isomorphisme* [page 93]

$$\varinjlim_U g^*(G) \xrightarrow{\sim} \varinjlim_V G,$$

et pour tout préfaisceau F sur V, à valeurs dans une catégorie $\mathcal{M}$ avec $\varprojlim$, on a `[un]` *isomorphisme* p. 58

$$\varprojlim_U g^*(F) \xleftarrow{\sim} \varprojlim_V F.$$

Démonstration. Je vais d'abord expliciter le

Lemme 7.5. *Soit $h : X' \longrightarrow X$ un morphisme étale dans* Cat. *Alors h est localement cofiltrant. De façon plus précise, pour tout x dans X, on a*

$$\pi_0(x\backslash X') \xrightarrow{\sim} \mathrm{Ob}\, X'_x,$$

65. Il suffit ici qu'on soit dans l'hypothèse respée (*f faiblement* localement cofiltrant) sauf pour le NB ici `[`le NB ici' est `(et même ...Loc (8))']`.

toute composante connexe de $x\backslash X'$ contient un élément et un seul de la catégorie discrète $X'_x \subseteq x\backslash X'$. Si x' est dans X'_x (i.e. x' objet de X' tel que $h(x') = x$), c'est un objet initial *de la composante connexe qui le contient. (Donc cette composante est cofiltrante.)*

Ceci prouve (et au delà) la première assertion de 7.4, compte tenu de 7.3 (stabilité par composition), ou parce qu'une catégorie étale (en l'occurrence $y\backslash X'$) au dessus d'une catégorie localement cofiltrante (resp. localement faiblement cofiltrante) (en l'occurrence $y\backslash X$) est itou. [page 94]

$$\begin{array}{ccc} y\backslash X' & \longrightarrow & X' \\ \downarrow{\scriptstyle \text{ét.}} & \text{cart.} & \downarrow{\scriptstyle \text{ét.}} \\ y\backslash X & \longrightarrow & X \\ \downarrow & \text{cart.} & \downarrow \\ y\backslash Y & \longrightarrow & Y. \end{array}$$

Pour la deuxième assertion dans a), on peut maintenant supposer $X' = X$. Il faut prouver que pour tout y' dans Y', $y'\backslash X \overset{\text{déf}}{=} X \times_{Y'} y'\backslash Y'$ est localement cofiltrante (resp. localement faiblement cofiltrante). Soit y dans Y l'image de y', et considérons le diagramme de trois carrés cartésiens ci-contre.

$$\begin{array}{ccccc} y'\backslash X & \longrightarrow & y\backslash X & \longrightarrow & X \\ \downarrow{\scriptstyle \text{ét.}} & \textit{cart.} & \downarrow{\scriptstyle \text{ét.}} & \textit{cart.} & \downarrow{\scriptstyle \text{ét.}} \\ y'\backslash Y' & \longrightarrow & y\backslash Y' & \longrightarrow & Y' \\ & & \downarrow & \textit{cart.} & \downarrow \\ & & y\backslash Y & \longrightarrow & Y \end{array}$$

Le Lemme 7.5 implique que $y'\backslash Y' \longrightarrow y\backslash Y'$ est l'inclusion dans $y\backslash Y'$ d'une composante connexe de $y\backslash Y'$, c'est donc une immersion ouverte. Donc $y'\backslash X \longrightarrow y\backslash X$ est une immersion ouverte. Comme $y\backslash X$ est localement cofiltrant (resp. localement faiblement cofiltrant) par hypothèse, il en est donc de même de $y\backslash X$. Cela prouve d). p. 59

Appliquant a) à la situation $X/x \longrightarrow Y/y$ dans b), on voit que les colocalisés $z\backslash(X/x)$ de X/x sur Y/y ($z \in \mathrm{Ob}\,(Y/y)$) sont localement cofiltrants (resp. faiblement). Mais comme X/x a un objet final au dessus de l'objet final de Y/y, il s'ensuit que $z\backslash(X/x)$ a un objet final, donc est 0-*connexe.* !

Donc ces catégories, étant localement cofiltrantes (resp. faiblement), sont en fait cofiltrantes (resp. faiblement cofiltrantes). [page 95]

Prouvons c). Notons que pour z dans V, $z\backslash U$ s'identifie à l'image inverse, par $z\backslash(X/x) \longrightarrow X/x$, de l'ouvert U de X/x, c'est donc un *ouvert* de $z\backslash(X/x)$, non vide par construction de V. Mais un ouvert non vide d'une catégorie cofiltrante (resp. faiblement cofiltrante) est encore cofiltrant (resp. faiblement cofiltrant). Cela signifie que $g : U \longrightarrow V$ est cofiltrant (resp. faiblement cofiltrant), a fortiori cofinal (et W-coasphérique, pour tout W satisfaisant Loc (8)). Cela achève la démonstration.

Commentary · Exposition [ms. 95]

Why the restriction argument works

The mechanism of Proposition 7.4 is the decomposition in Lemma 7.5, and it is useful to see why that decomposition is canonical. An *étale* functor $h : E \to C$ is a discrete fibration. Given $(e, \alpha : c \to h(e))$ in $c\backslash E$, lift α uniquely to an arrow $e_0 \to e$ with $h(e_0) = c$. This assigns an object e_0 of the discrete fibre to each object of the comma category. Every arrow of the comma

category preserves e_0, by uniqueness of the lift. For each fixed e_0, the object $(e_0, 1_c)$ is initial in the corresponding full subcategory. Thus

$$c\backslash E \;\cong \coprod_{e_0 \in h^{-1}(c)} (e_0\backslash E),$$

and each displayed summand has an initial object. This proves all assertions of the lemma, including the empty-fibre case.

For the first assertion of 7.4(a), $y\backslash X' \to y\backslash X$ is a discrete fibration. Conditions P and E pass to its total category: lift a commutative cone in the base to the prescribed object above its target; uniqueness makes the lifted arrows commute. For E, faithfulness of a discrete fibration makes the lifted equalization an actual equality. This proves preservation of local cofilteredness and its P-only variant.

For the second assertion, let $X \xrightarrow{g} Y' \xrightarrow{h} Y$ be the factorization, with h a discrete fibration. An object $y' \in Y'$ over y determines the summand of $y\backslash Y'$ whose initial object is $(y', 1_y)$. Its inverse image under $y\backslash X \to y\backslash Y'$ is exactly $y'\backslash X$. Hence $y'\backslash X$ is a union of connected components of $y\backslash X$, so it inherits P and E. This argument also explains the second diagram: the labels "ét." on its three vertical arrows are not justified, since $g : X \to Y'$ is arbitrary. The relevant map is the horizontal component inclusion and its pullback. The claimed result does not use those incorrect vertical labels.

Now set $y = f(x)$. The category X/x is *étale* over X, and Y/y is *étale* over Y. Part (a) therefore makes $f/x : X/x \to Y/y$ locally cofiltered. For $z = (z_0 \to y)$, its comma category $z\backslash(X/x)$ has a terminal object: take $(x, 1_x)$ with the specified arrow $z_0 \to y$. Every other object includes its unique morphism to this object as part of its structure. The comma category is consequently nonempty and connected. Combined with local cofilteredness this proves (b), and the same reasoning gives connectedness plus P in the weaker case.

For (c), use the corrected V above. The category $z\backslash U$ is a nonempty lower open full subcategory of $z\backslash(X/x)$. A nonempty lower open subcategory of a cofiltered category is cofiltered: for two objects, any common predecessor still lies in the open, and the same is true of the source of an equalizing arrow. For connectedness plus P, the common-predecessor argument proves connectedness and the square condition restricts. These observations prove (c), and also prove Remark 7.5(1) for an arbitrary cofiltered $f : X \to Y$ and an open $U \subset X$. In that remark the defining condition for V must likewise be $y\backslash U \neq \varnothing$, as its preceding words "engendré par $f(U)$" already require.

Remarque 7.5.

1) Le même argument que dans c) montre que si $f : X \longrightarrow Y$ est cofiltrant (resp. faiblement cofiltrant), alors pour tout ouvert U dans X, si on désigne par V l'ouvert de Y engendré par $f(U)$, i.e. l'ouvert formé des y dans Y tels que $y\backslash X \neq \emptyset$, alors la flèche induite $g : U \longrightarrow V$ est cofiltrante (resp. faiblement cofiltrante), et a fortiori g est cofinal (et plus généralement, si f est cofiltrant, g est W-coasphérique pour tout W satisfaisant la condition Loc (8)).

2) Un cas particulier intéressant de 7.4 est celui où on suppose que les $y\backslash X$ ($y \in \mathrm{Ob}\ Y$) sont sommes directes de catégories ayant chacune un *objet initial.* Alors on voit que cette condition est stable par les opérations de 7.4 a) [?], et par conséquent, les colocalisés $z\backslash(X/x)$ des morphismes localisés en haut $X/x \longrightarrow Y/y$ sont des catégories à objet initial. Ceci implique alors qu'il en est de même des colocalisés $z\backslash U$ ($z \in \mathrm{Ob}\ V$) dans $g : U \longrightarrow V$. Il s'ensuit que g est W-coasphérique pour *tout* localiseur fondamental W (sans avoir à se soucier de l'axiome des limites). [page 96]

   La condition envisagée sur $f : X \longrightarrow Y$ est satisfaite notamment dans les deux cas suivants (qui me paraissent de loin les plus importants) : a) f est étale et b) X est discrète.

Commentary · Exposition [ms. 96] p. 60 [page 97]

Initial objects remove the limit axiom

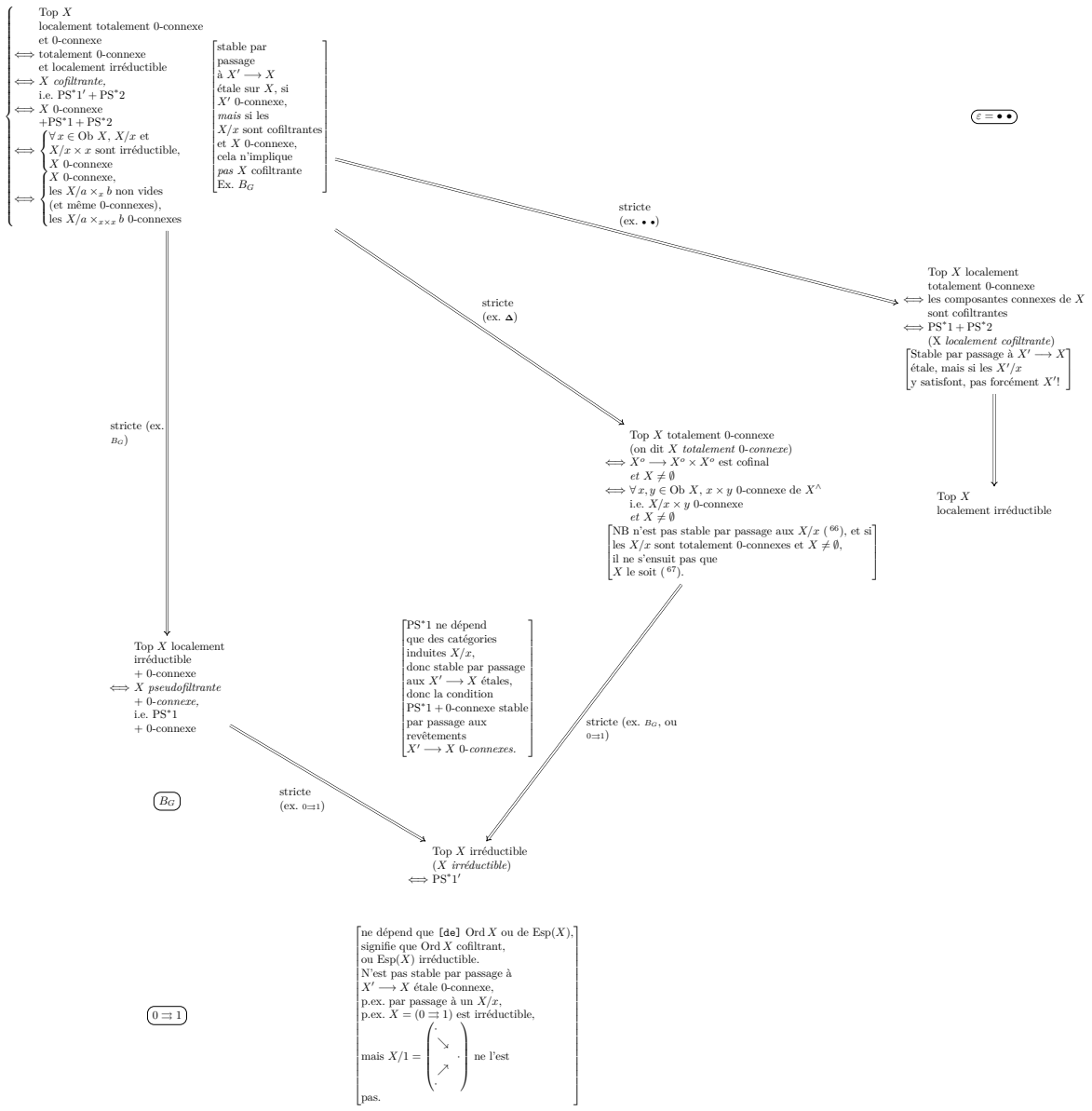
Grothendieck singles out components with initial objects because here the cofinality argument needs no passage to filtered limits or colimits. The bracketed question in Remark 7.5(2) can be settled directly. Suppose each $y\backslash X$ is a disjoint union of categories with initial objects. If $E \to C$ is a discrete fibration and every component of C has initial object c_0, each object of E receives a unique lift of the arrow from the relevant c_0. The lifts with sources over c_0 partition E into full subcategories, each having its source as an initial object. Consequently the initial-object condition is preserved by the first operation in 7.4(a). The second operation selects a union of components, by the preceding decomposition, and therefore preserves it as well.

For $f/x : X/x \to Y/f(x)$ the comma categories have terminal objects and hence just one component. Their initial-object decomposition therefore consists of a single summand, so they actually have initial objects. Passing to a nonempty lower open $z\backslash U$ preserves this initial object: if an open contains any object, it contains the source of every arrow into that object, in particular the initial object. This proves the strengthened assertion in full.

The examples in the text follow without an additional hypothesis. For a discrete fibration f, Lemma 7.5 supplies the initial objects. If X is discrete, every $y\backslash X$ is discrete, so all of its components are singleton categories. The strengthened conclusion can therefore apply to a functor which is far from final before one restricts to the open image.

For a fundamental localizer W, a category with initial object is W-aspherical. One can justify the initial-object version from the terminal-object axiom: the unique map to the terminal category and its section have composites related to the identity by the natural transformation from the constant initial-object functor. Fundamental localizers identify naturally transformed functors after passage to homotopy, as follows from the interval projection $C \times [1] \to C$ and the terminal-object axiom. Thus all the relevant comma categories are W-aspherical for every fundamental localizer. This is exactly where the appeal to Loc (8) disappears in the manuscript's argument; mere weak cofilteredness would not give this conclusion.

The later theory removes the same qualification from the full cofiltered case. A cofiltered category has weakly contractible nerve: any finite collection of simplices admits a compatible cone after finitely many common-predecessor and equalization operations, giving the required null homotopies. Cisinski (2004, Theorem 2.2.11, p. 136) proves that the weak equivalences detected by nerves form the minimal fundamental localizer. They therefore belong to every fundamental localizer. Consequently every cofiltered comma category is W-aspherical for every fundamental W, and a cofiltered functor is W-coaspherical without an additional Loc (8) assumption. The source's conditional assertion remains correct; the stronger present statement uses this later minimality theorem.



66. **Ex.** $\boxed{\Delta}$.
67. **Ex.** B_G.

Commentary · Examples [ms. 97] p. 61 [page 98]

Reading the six classes without the changing terminology

The diagrams collect the distinctions developed earlier in the chapter. Their content is more stable than their terminology: "pseudo-filtrante" and "pseudo-cofiltrante" sometimes omit the direction or change meaning. We use P, E, I as above and write T for the assertion that C^{op} is sifted. Explicitly, T means $C \neq \varnothing$ and the category of cones $c \to a$, $c \to b$ is connected for every a, b. This is Grothendieck's total 0-connectedness, as discussed already in the commentary at manuscript p. 18 ("The ordinary sifted condition and its variance"). The six classes are

C_{cof}	$I + E$ (equivalently $P + E +$ connected, nonempty)
C_{loc}	$P + E$
C_{weak}	$P +$ connected, nonempty
C_{pre}	P
C_{irr}	I
C_{tot}	T.

There are inclusions $C_{\mathrm{cof}} \subset C_{\mathrm{loc}}, C_{\mathrm{weak}}, C_{\mathrm{tot}}$, $C_{\mathrm{loc}} \subset C_{\mathrm{pre}}$, $C_{\mathrm{weak}} \subset C_{\mathrm{pre}}, C_{\mathrm{irr}}$, and $C_{\mathrm{tot}} \subset C_{\mathrm{irr}}$. The two middle classes C_{weak} and C_{tot} meet exactly in C_{cof}, by the criterion established earlier in the chapter. The following table makes the five separating examples completely explicit; G is nontrivial and A is the parallel-arrow category.

	C_{cof}	C_{loc}	C_{weak}	C_{pre}	C_{irr}	C_{tot}
B_G	0	0	1	1	1	0
$\mathbf{\Delta}$	0	0	0	0	1	1
$A = (0 \rightrightarrows 1)$	0	0	0	0	1	0
$\varepsilon = \{0, 1\}$	0	1	0	1	0	0
$B_G \amalg e$	0	0	0	1	0	0

For B_G, every cospan can be completed because its arrows are invertible: for $r, s \in G$, use $u = r^{-1}s$ and $v = 1$, obtaining $ru = sv$. Equalization fails for a nonidentity g and 1, since $gt = t$ would imply $g = 1$. The common-cone category of the unique object with itself has objects $(r, s) \in G^2$, with simultaneous right multiplication by $t \in G$ as morphisms. The quantity sr^{-1} labels its connected components, giving $|G|$ components. Thus T fails. This is the same obstruction as the diagonal action on $G \times G$; see The Stacks Project Authors (2026e, Lemma 4.19.4). Every slice $B_G/*$ nevertheless has exactly one arrow between any pair of its objects and is cofiltered. This verifies the diagram's warning that cofiltered slices do not imply a locally cofiltered category.

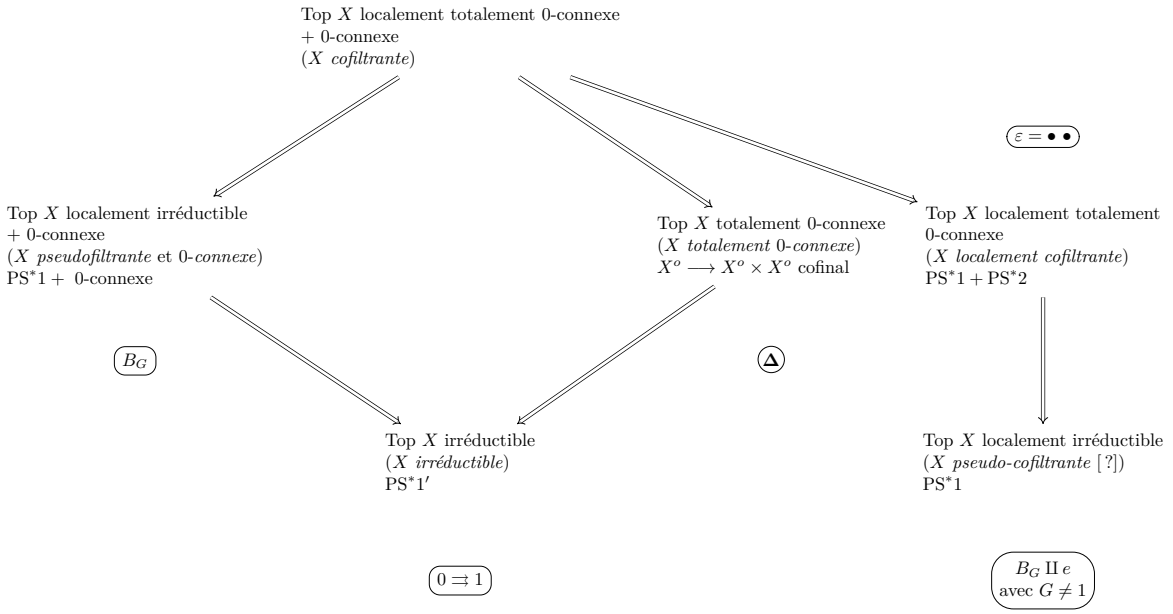
For A, the object 0 is a common predecessor of any pair of objects, so I holds. The cospan given by the two distinct arrows $r, s : 0 \to 1$ cannot be completed to a commutative square: the only arrow into 0 is its identity. Thus P fails. The cone category for the pair $(0, 1)$ has exactly two objects and no nonidentity arrows, so T fails too. The slice $A/1$ has the shape $r \to 1 \leftarrow s$ and is not irreducible. This simultaneously explains the failure of irreducibility to survive a connected *étale* change of source.

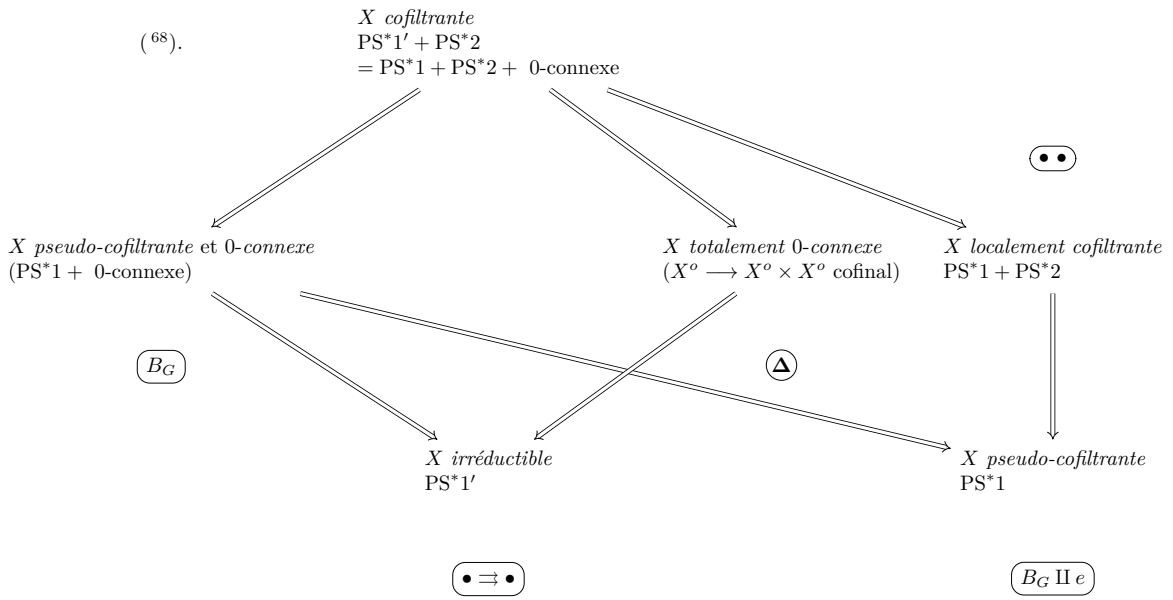
For $\mathbf{\Delta}$, the category of nonempty finite ordinals and monotone maps, every pair admits a common predecessor $[0]$. A cone from $[k]$ to $[m]$ and $[n]$ is a finite monotone string in the rectangular poset $[m] \times [n]$. It is connected, in its cone category, to each of its vertices by the corresponding map $[0] \to [k]$. Any two vertices can be joined by moving horizontally and vertically through the rectangle, each move being represented by a $[1]$-string. Thus the cone category is nonempty and connected, proving T. The two endpoint maps $[0] \rightrightarrows [1]$ cannot be equalized even by two different maps into their sources: every nonempty ordinal has a unique map to $[0]$, and the two resulting maps to $[1]$ remain distinct. Hence P fails. In particular $\Delta/[1]$ is not irreducible: the two endpoint objects have no common predecessor. Total connectedness therefore does not pass to slices.

For ε, the conditions P and E hold separately on each singleton component, while all four classes requiring connectedness fail. For $B_G \amalg e$, P still holds componentwise, E fails in the group component, and connectedness fails globally. These calculations also settle the abbreviated "Itou pour ε" on printed page 62: it can refer to exclusion from the four connected classes, but cannot refer to failure of local cofilteredness, since ε is locally cofiltered.

Finally, the product notation on page 97 takes place in the presheaf category. Thus $C/(a \times_x b)$ means the category of elements of $h_a \times_{h_x} h_b$, and $C/(x \times x)$ means that of $h_x \times h_x$; it is not a product of a slice with an object. The irreducibility of every C/x is P. Together with irreducibility

of every $C/(x \times x)$ it forces E: compare the object defined by $r, s : a \to x$ with $(x, 1_x, 1_x)$ and take a common predecessor. Conversely a cofiltered category provides cones and equalizations for these finite families, so the displayed categories of elements are cofiltered. This explains the strengthened representable tests in the first diagram, without repeating the longer topos arguments of the commentary on the first part of the chapter.

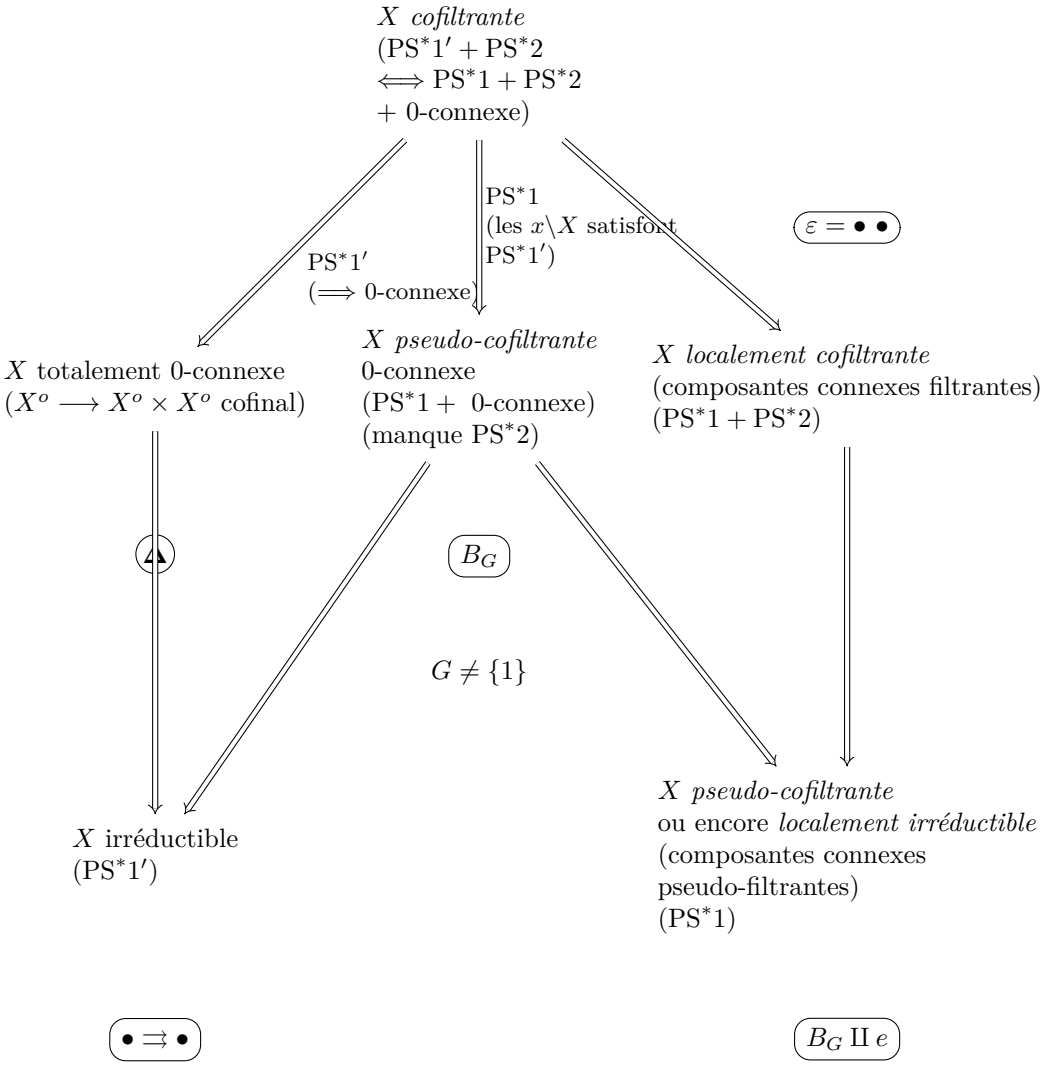




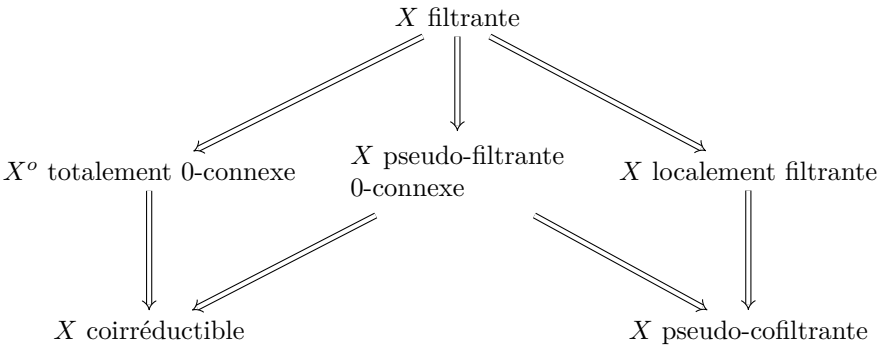
68. **NB** Les cinq catégories typiques B_G ($G \neq 1$), $\mathbf{\Delta}$, $0 \rightrightarrows 1$, $\varepsilon = \bullet\,\bullet$, $B_G \amalg e$ ($G \neq 1$) sont telles qu'elles appartiennent à la classe envisagée C d'objets de Cat, mais à aucune des cinq autres classes sauf celles qui sont reliées à C par $C \Longrightarrow C'$ sur le diagramme d'implications (ou par une implication composée).

$\bullet \rightrightarrows \bullet$	n'est ni pseudo-cofiltrante, ni totalement 0-connexe.	p. 62
B_G	n'est ni localement cofiltrante, ni totalement 0-connexe.	
$\mathbf{\Delta}$	n'est pas pseudo-filtrante.	
$B_G \amalg e$	n'est pas localement cofiltante, car B_G ne l'est pas, ni appartient à une des 4 classes d'objets connexes.	
Itou pour	$\varepsilon = \bullet\ \bullet$	

[page 99]



et dualement on a


X filtrante
X^o totalement 0-connexe
X pseudo-filtrante 0-connexe
X localement filtrante
X coirréductible
X pseudo-cofiltrante

donc on a une terminologie directe sauf pour 'X^o totalement 0-connexe' - je n'ose dire 'X totalement co-connexe'!! p. 63

Commentary · Later developments [ms. 99]

Representable flatness and the pointwise Kan extension

Grothendieck's cofiltered functors have a standard modern name: they are *representably flat*. For each $y \in Y$, the covariant set-valued functor $X \to \mathrm{Set}$, $x \mapsto Y(y, f(x))$, has category of elements $y\backslash X$; requiring this category to be cofiltered is precisely the definition. The adjective matters, since flatness for a functor into a topos has another, internal interpretation. Tendas (2024, Remark 2.4) explicitly distinguishes these notions, following the discussion of Shulman (2011).

One consequence places the closing section directly in the theory of presheaf topoi. For $F : X^{\mathrm{op}} \to \mathrm{Set}$, the left Kan extension along f^{op} is

$$(f_!F)(y) = \operatorname*{colim}_{(y\backslash X)^{\mathrm{op}}} F(x).$$

If f is cofiltered, the indexing category is filtered. Finite limits of presheaves are computed pointwise, and filtered colimits of sets commute with finite limits; hence $f_!$ preserves finite limits. Consequently the adjunction $f_! \dashv f^*$ is a geometric morphism in the direction determined by this inverse-image functor. This is an additional geometric adjunction alongside the usual $f^* \dashv f_*$.

If f is only locally cofiltered, the same formula is a coproduct, indexed by $\pi_0(y\backslash X)$, of filtered colimits. It preserves finite *connected* limits. To check the distinction, a compatible tuple in a connected finite diagram of such coproducts must lie throughout in one summand; within that summand the filtered-colimit argument applies. The terminal presheaf instead has image

$$(f_!1)(y) \cong \pi_0(y\backslash X),$$

which need not be a singleton. This separates the local condition from representable flatness exactly where the manuscript separates local cofilteredness from cofilteredness. The general connected-limit calculation appears in The Stacks Project Authors (2026e, Lemma 4.19.9).

The last diagrams thus organize three different demands on colimits: finite limits, finite connected limits, and finite products. The product condition is the sifted condition on X^{op}, whose later systematic treatment is Adámek et al. (2010, Section 1). The displayed examples show why none of these demands can be substituted for another without its accompanying connectedness or square condition. The missing nonemptiness qualification in the abbreviated diagonal criteria on pages 98–99 should always be read from page 97: the empty category has a final diagonal vacuously, but is not sifted.

Overview · II-b

Reconstruction, preservation, and the geometry of indexing categories.

These pages complete the second chapter by asking how much of a category is visible to its diagram theories. The starting point is the cofinality theorem: a restriction of diagrams preserves a specified limit or colimit when the appropriate comma categories are connected. The next question is more demanding. When does restriction embed the whole category of diagrams fully faithfully? Grothendieck then turns from individual functors to classes of functors. He tries to recover the classes controlling change of base from a chosen class of weak equivalences, and develops the order-theoretic machinery needed to state such a recovery. The closing discussion of cofiltered categories gives concrete tests and counterexamples for this programme. These are successive parts of one investigation: which local tests retain exactly the information that a theory of coefficients can detect?

For ordinary presheaves, the answer to the full-faithfulness question is a two-sided cofinality criterion. Given $u : A \to B$ and an arrow $\gamma : b_0 \to b_1$, form the category of factorizations $b_0 \to u(a) \to b_1$ whose composite is γ. Restriction on set-valued presheaves is fully faithful precisely when every such category is nonempty and connected. The counit evaluated on a representable presheaf has these connected-component sets as its fibres, and representables generate all presheaves by colimits. This is the decisive argument of Proposition 4.1. In later terminology the functors are lax epimorphisms, or absolutely dense functors; the equivalence of these formulations is established by Adámek et al. (2001). The commentary on the first part of the chapter already treats the general criterion. Here the commentary supplies the missing distinction between its two adjunction tests and checks the direction of the comma-category maps.

The right-adjoint unit behaves differently. Its being invertible on every representable says that the restricted Yoneda functor $B \to \mathrm{Set}^{A^{\mathrm{op}}}$ is fully faithful: this is density, in the usual sense (Riehl, n.d., Definition 6.5.11). It does not imply full faithfulness of restriction on all presheaves. The full category of sets on a singleton and a two-element set, with u selecting the singleton, already gives a counterexample. All maps of the original category are recovered from maps of sets of elements, but the identity of the two-element set cannot factor through the singleton. Thus the hesitation on manuscript pages 54–56 concerns a genuine obstruction. The commentary also computes the four-object example on pages 57–58: every factorization category is contractible, while one strict pullback is a disconnected two-point category. The example separates reconstruction of diagrams from universal preservation under strict base change.

For an arbitrary derivator $\mathbf{D}$, one must distinguish full faithfulness for that particular coefficient theory from full faithfulness for every derivator. Modern homotopical epimorphisms impose the latter quantifier. Groth and Šťovíček (2016, Definition 6.1 and Theorem 6.10, arXiv v3) characterize them by weak contractibility of the same factorization categories. Grothendieck's fixed-$\mathbf{D}$ problem is subtler. Global diagrams test only certain diagrams on a slice, and his proof of Proposition 4.4 carefully isolates the additional detection needed to recover all the slice tests. A conservative left Kan extension from the slice suffices; essential surjectivity of restriction to the slice is a stronger sufficient condition. The pointwise Kan-extension and detection axioms of a derivator perform different jobs in this argument (Groth, 2013, Definition 1.5).

The developments in the commentary make two limits of these sufficient conditions explicit. First, in the derivator represented by the complete lattice $\{0 < 1\}$, diagrams on the free category generated by a directed two-cycle are constant. Evaluation at one vertex is therefore fully faithful. Nevertheless, the identity of the other vertex has no factorization through it, and its factorization category is empty. This answers the intended question about $\mathbf{D}$-connectedness versus $\mathbf{D}$-biasphericity. Second, the parenthetical faithfulness assertion in Corollary 4.7 fails for coherent derived diagrams. A discrete fibration from a three-object fork to an interval

sends a nonsplit extension of quiver representations to a split extension. Direct sums in the coefficient derived category are faithful, but the Kan extension is not. The conservativity assertion survives: pointwise detection proves it. Each development is given with a complete proof and a statement of the sources checked, without a claim of priority.

The cancelled passage at the end of §4 must be read with comparable care. Let $S(A)$ be the maps whose pullback functors preserve a class A, and let $N(L)$ be the maps whose pullbacks along a testing class L lie in W. Under the requisite identity and stability hypotheses, the inclusions and the two idempotence calculations in the passage survive. The missing ingredient is order reversal of S: enlarging the preserved class changes both the inputs and the required outputs of a pullback test. It need not give a smaller stabilizer. With W the maximal localizer on categories and A the categorical equivalences, two point inclusions into the walking isomorphism directly refute the stronger identity used in the printed justification. This diagnosis preserves the valid calculations while explaining why they do not constitute a Galois connection.

Section 5 supplies that connection in its proper generality. Any relation between two sets defines two antitone polar operations. The composites are closure operators, and their closed subsets form anti-isomorphic complete lattices. When the sets are ordered, a further condition makes polar lower sets principal and hence represents their closures by elements. Formula (5.20) slips between these levels: a closed subset is not its greatest element, and a join of closed elements need not be closed in the ambient lattice. The corrected formula takes the closure of the join. Double annihilation in $\mathbf{F}_2^2$ gives a short counterexample to the printed equality. The classical examples also require their proper separation or topological hypotheses. In infinite Galois theory the fixed subgroups are the Krull-closed subgroups; the dense cyclic Frobenius subgroup in $\widehat{\mathbf{Z}}$ shows why the qualification cannot be dropped (Milne, 2022, Proposition 7.12 and Theorem 7.13). A lifting relation is another polarity, but a mutually closed pair of lifting classes still requires a separate factorization theorem to become a weak factorization system (Riehl, 2014, Definition 11.2.1).

Section 6 repairs the earlier construction by fixing its target class. The transporter $T_W(A)$ consists of maps whose pullback functors send arrows of A to arrows of W. Testing includes every arrow in the slice, with any structure map from its codomain to the base. It is not enough to test arrows whose codomain happens to equal the base. With A closed under composition and L stable under pullback, both containing isomorphisms, the equivalence

$$A \subseteq N_W(L) \quad \Longleftrightarrow \quad L \subseteq T_W(A)$$

is a genuine antitone adjunction. The correction at (6.7) removes composition closure from the right-hand domain, where it is not automatic. A four-object chain in the commentary exhibits a closed transporter that does not compose, under exactly the abstract assumptions of §6. The example is not claimed to use a fundamental localizer on Cat. Proposition 6.1 then identifies the additional condition: for a closed pair, the right class is composition-closed exactly when it equals the stabilizer of its left class.

This formalism has a direct historical continuation. Grothendieck's letter to Thomason singles out a fibration class defined intrinsically from weak equivalences, while treating factorization as another question (Grothendieck, 1991a, §1, pp. 3–4). Rezk's sharp maps use the same preservation condition. In a right proper model category they characterize the maps all of whose strict pullback squares are homotopy cartesian (Rezk, 1998, Propositions 2.2 and 2.7). Maltsiniotis develops local tests for the corresponding preservation of aspherical maps (Maltsiniotis, 2005, Definition 4.6 and Theorem 4.13). One question on page 89 has a particularly concrete answer: the stabilizer of categorical equivalences is exactly the isofibrations. The commentary gives both directions of the proof, using the walking isomorphism for necessity, and relates the result to the categorical model structure (Rezk, 1996, §2 and Theorem 3.1). For the general mixed transporters and the proposed bilocal description of the

biaspherical stabilizer, what was verified is kept apart from what was not. Failure to find a precise matching theorem is not presented as proof that the questions are open today.

The final section turns the abstract machinery into the geometry of predecessors. A cofiltered functor is tested by cofiltered comma categories; the local version allows several components and asks each to be cofiltered. Over a poset, these conditions become statements about lower open subsets. Proposition 7.4 is essentially sound, but its target lower open must be generated by the image of the chosen open subset: taking the whole principal lower set makes the printed condition false even for the empty subset. The commentary proves the corrected criterion and the initial-object assertion of Remark 7.5. It also exhibits a cofiltered map whose arbitrary base change is not even locally weakly cofiltered, while respecting the stated preservation under co-étale base change. For fundamental localizers, the later minimality theorem removes the extra Loc(8) qualification attached here to the asphericity of cofiltered categories (Cisinski, 2004, Theorem 2.2.11).

The final diagrams distinguish common predecessors, the filling of cospans, equalization of parallel arrows, and connectedness of the categories of cones over every finite family of objects. Their implications become transparent when these requirements are kept separate. Groups, the simplex category, a parallel-arrow category, and disjoint unions give the required separating examples. The modern interpretation of total connectedness is siftedness of the opposite category; cofilteredness adds the equalization needed for commuting with all finite limits, rather than only finite products (Adámek et al., 2010; The Stacks Project Authors, 2026e; Tendas, 2024). In this way the chapter ends where its larger project began: the shape of an indexing category governs which constructions a theory of diagrams can perform and which information restriction can retain. The derived-diagram motivation already appears in *Pursuing Stacks*, §69 (Grothendieck, 1983); here it is being refined into local, verifiable conditions on functors.

Chapter III

Hom externes dans les dérivateurs

III-V · Chapter III · manuscript pages 1–12

p. –

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre III

Hom externes dans les dérivateurs

p. –

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Relation des dérivateurs avec la catégorie Hot

p. 1
[page 1]

Soit $\mathbf{D}$ un dérivateur, $\mathcal{A}$ sa catégorie fondamentale,

$$\mathcal{A} = \mathbf{D}(e).$$

Je me propose de définir, si faire se peut, des foncteurs

$$(1)\qquad \begin{cases} \mathrm{Hot} \times \mathcal{A} \xrightarrow{\pi_{\mathbf{D}}} \mathcal{A} & (\xi, M) \longmapsto \xi \times M \text{ ou } \xi \otimes M \quad \text{(suivant les cas)} \\ \mathrm{Hot}^o \times \mathcal{A} \xrightarrow{\varpi_{\mathbf{D}}} \mathcal{A} & (\xi, M) \longmapsto \hom(\xi, M). \end{cases}$$

Ces foncteurs doivent s'échanger par dualité : si on prend les foncteurs π^o, ϖ^o sur les catégories opposées, on trouve

$$(1')\qquad \begin{cases} \mathrm{Hot} \times \mathcal{A}^o \xrightarrow{\varpi^o_{\mathbf{D}}} \mathcal{A}^o \\ \mathrm{Hot}^o \times \mathcal{A}^o \xrightarrow{\pi^o_{\mathbf{D}}} \mathcal{A}^o, \end{cases}$$

et on doit avoir

$$\begin{cases} \varpi^o_{\mathbf{D}} = \pi_{\mathbf{D}^o} \\ \pi^o_{\mathbf{D}} = \varpi_{\mathbf{D}^o}. \end{cases}$$

À vrai dire, considérant (1) comme des foncteurs

$$\begin{array}{ll} \mathrm{Hot} & \xrightarrow{\pi_{\mathbf{D}}} \ \underline{\mathrm{Hom}}(\mathcal{A}, \mathcal{A}) \\ \mathrm{Hot}^o & \xrightarrow{\varpi_{\mathbf{D}}} \ \underline{\mathrm{Hom}}(\mathcal{A}, \mathcal{A}), \end{array}$$

on s'attend à ce que ces foncteurs s'étendent en

$$(2)\qquad \begin{cases} \mathrm{Hot} \xrightarrow{\pi_{\mathbf{D}}} \underline{\mathrm{Hom}}(\mathbf{D}, \mathbf{D}) \\ \mathrm{Hot}^o \xrightarrow{\varpi_{\mathbf{D}}} \underline{\mathrm{Hom}}(\mathcal{A}, \mathcal{A}) \end{cases}$$

[plutôt deux fois $\underline{\mathrm{Hom}}(\mathbf{D}, \mathbf{D})$]. Cela signifie, essentiellement, que pour $\xi \in \mathrm{Ob}\,\mathrm{Hot}$ donné, les opérations $\xi \times_{\mathbf{D}} M$ et $\underline{\mathrm{Hom}}_{\mathbf{D}}(\xi, M)$ [plutôt $\hom_{\mathbf{D}}(\xi, M)$] sont définies non seulement dans $\mathcal{A} = \mathbf{D}(e)$, mais dans tous les $\mathbf{D}(I)$, et ceci de façon à commuter aux foncteurs f^* :

Commentary · Exposition [ms. 1]

Spaces acting on coefficients

Grothendieck is looking for the part of ordinary homotopy theory that must act on every theory of coefficients. A space should provide an instruction for assembling copies of an object, while the same space, with the variance reversed, should provide an instruction for taking their limit. This is the reason for placing the two operations together before attempting to construct an external Hom.

Write $BX = |NX|$ for the classifying space of a small category, and temporarily assume that all the categories used below belong to the permitted diagram category. If $p_X : X \to e$, put

$$T_X(M) = (p_X)_! p_X^* M, \qquad C_X(M) = (p_X)_* p_X^* M.$$

These expressions are already defined by the derivator. The problem is to show that they depend on X only through BX, and then to understand their compatibility as X and M vary. In the derivator of spaces they are $BX \times M$ and the derived function space $\mathrm{Map}(BX, M)$. In a derived category they are derived tensor and Hom constructions with the integral chains of BX. The common formalism therefore sees more than the set of components of X.

There are three different Hom constructions in play. The ordinary set $\mathrm{Hom}_{\mathbf{D}(e)}(M, N)$ records morphisms after localization. The cotensor $C_X(N)$ is another coefficient object, hence belongs to $\mathbf{D}(e)$. The proposed external Hom $\hom\mathrm{ext}(M, N)$ is a homotopy type. Its intended property is

$$\mathrm{Hom}_{\mathbf{D}(e)}(T_X M, N) \cong [BX, \hom\mathrm{ext}(M, N)].$$

In a model presentation this last object is the derived mapping space. It is not the internal Hom of a monoidal category of coefficients: no multiplication of two coefficient objects has been assumed. For example, the derived category of modules over a noncommutative ring still has an external mapping space, although a tensor product of two left modules over that ring has no such general meaning.

The extension to every $\mathbf{D}(I)$ is essential. With $p : X \times I \to I$, the same formulas define $T_{X,I} = p_! p^*$ and $C_{X,I} = p_* p^*$. Base change along products supplies their compatibility with restriction f^*. Thus the request in (2) is for morphisms of derivators, including their pseudonaturality isomorphisms, rather than unrelated endofunctors of the individual categories. The passage from $\mathbf{D}(e)$ to the shifted derivator $\mathbf{D}_I$ is the mechanism already used in the commentary on Chapter I, manuscript p. 4 (*From derived coefficients to diagram categories*); here its new role is to make the action itself coherent in the diagram variable.

For a represented derivator, the calculation reduces to an instructive limiting case:

$$T_X(M) = \coprod_{\pi_0 X} M, \qquad C_X(M) = \prod_{\pi_0 X} M.$$

Indeed a cone on a constant ordinary diagram is a family of maps constant on every connected component. The richer dependence on BX belongs to the homotopy theory, not to the notation $p_! p^*$ alone.

[page 2]

Si

$$f : I' \longrightarrow I,$$

alors pour $F \in \mathrm{Ob}\,\mathbf{D}(I)$, on doit avoir des isomorphismes canoniques p. 2

$$(3) \qquad \begin{cases} f^*(\xi \times_{\mathbf{D}} M) & \simeq & \xi \times_{\mathbf{D}} f^*(M) \\ f^* \mathrm{hom}_{\mathbf{D}}(\xi, M) & \simeq & \mathrm{hom}_{\mathbf{D}}(\xi, f^*(M)), \end{cases}$$

avec les compatibilités habituelles pour un composé fg, relatif à $I'' \xrightarrow{g} I' \xrightarrow{f} I$.

Mais je me place de préférence dans $\mathcal{A}$, ce qui devrait impliquer les constructions similaires dans tous les $\mathbf{D}(I)$, quitte à appliquer ce qui a été dit au dérivateur $\mathbf{D}_I$.

L'intuition algébrique fondamentale est celle-ci, en désignant par

$$X \longmapsto \mathrm{hot}(X) : (\mathrm{Cat}) \longrightarrow (\mathrm{Hot})$$

le foncteur de localisation de (Cat) dans (Hot), obtenu en inversant les flèches dans (Cat) qui sont des 'équivalences faibles'. Supposons X dans Diag, soit

$$p : X \longrightarrow e$$

le foncteur à valeurs dans la catégorie ponctuelle. On veut que

$$(4) \qquad \begin{cases} \mathrm{hot}(X) \times_{\mathbf{D}} M & \simeq & p_!(p^*(M)) & = & H^{\mathbf{D}}_{\bullet}(X, M) \\ \mathrm{hom}_{\mathbf{D}}(\mathrm{hot}(X), M) & \simeq & p_*(p^*(M)) & = & H^{\bullet}_{\mathbf{D}}(X, M) \end{cases}$$

(isomorphismes canoniques).

Ceci donnerait un principe de définition pour les opérations cherchées. Une difficulté vient [page 3] du fait qu'en général, Diag n'est pas égal à (Cat), et que les types d'homotopie provenant de Diag ne donnent pas (même à isomorphisme près) *tous* les types d'homotopie. Intuitivement, on ne trouve que des types d'homotopie 'assez petits'. Mais on peut localiser Diag par rapport aux équivalences faibles (il vaut mieux dire 'quasi-isomorphismes'), on trouve alors une catégorie $(\mathrm{Hot})_0$, et un foncteur canonique

$$\mathrm{Hot}_0 \longrightarrow \mathrm{Hot},$$

si on a de la chance, ce foncteur sera pleinement fidèle, et on essayera de décrire les foncteurs (1) ou (2) en y remplaçant Hot par la sous-catégorie pleine des 'petits' types d'homotopie. On

peut aussi oublier (provisoirement!) Hot, et travailler avec Hot_0. C'est ce qu'on va faire pour le moment.

Pour pouvoir définir les foncteurs

$$(5)\qquad \begin{cases} \mathrm{Hot}_0 \times \mathcal{A} \longrightarrow \mathcal{A} \quad (\xi, F) \longmapsto \xi \times_{\mathbf{D}} F \\ \mathrm{Hot}_0^o \times \mathcal{A} \longrightarrow \mathcal{A} \quad (\xi, F) \longmapsto \hom_{\mathbf{D}}(\xi, F), \end{cases}$$

je note que pour F [plutôt M] fixé, X variable dans Diag, les derniers membres dans (4) p. 3 [page 4]

$$(6)\qquad H^{\mathbf{D}}_{\bullet}(X, M), \qquad H^{\bullet}_{\mathbf{D}}(X, M)$$

sont bien, l'un covariant, l'autre contravariant en X. Donc il y a , à isomorphisme unique près, une seule paire de bifoncteurs (5) satisfaisant (bifonctoriellement) les isomorphismes (4), si et seulement si les expressions (6), considérées comme fonctorielles en X, transforment quasi-isomorphismes en isomorphismes. Cela équivaut donc à la validité du

Théorème (?) ([1]). *Soit $f : X' \longrightarrow X$ dans Diag. Si f est un quasi-isomorphisme, c'est une* **D***-équivalence.*

Quand X est la catégorie ponctuelle, ce n'est là autre que l'axiome Der 3 (p. 67). J'ignore si c'est une conséquence des axiomes Der 1 à Der 5 (d). Si ce n'est pas le cas, il faudrait l'ajouter aux axiomes, je crois. Je n'ai pas même prouvé qu'une équivalence de catégories dans Diag (au sens des catégories!) est une **D**-équivalence!

Commentary · Present state [ms. 4]

The descent theorem and its domain

The question marked *Théorème (?)* is precisely the obstruction to making the preceding formulas into operations of homotopy types. It has a positive answer for derivators on all small categories with the usual homotopy Kan extension axioms. The scope of the indexing category matters: that answer must not be silently substituted for a theorem about every restricted Diag occurring in the manuscript.

First, the claimed equivalence with descent is formal. Let W be the nerve weak equivalences in Diag. A bifunctor $(X, M) \mapsto T_X M$ factors through $\mathrm{Diag}[W^{-1}]$ exactly when every natural transformation $T_f : T_{X'} \to T_X$, $f \in W$, is invertible. The localization universal property applies also to natural transformations, giving the factorization with its specified comparison uniquely up to unique compatible isomorphism. The same argument applies contravariantly to C_X. The two conditions are equivalent because $T_X \dashv C_X$, and the comparison transformations are mates: a natural isomorphism between left adjoints has an invertible right mate, and conversely.

The easy case that Grothendieck says he has not yet checked needs no nerve theorem. If $f : X' \to X$ is an equivalence of categories, a quasi-inverse and its invertible unit and counit give, by the prederivator structure, an equivalence $f^* : \mathbf{D}(X) \to \mathbf{D}(X')$. It carries $p_X^* M$ to $p_{X'}^* M$. Hence it induces bijections

$$\mathrm{Hom}_{\mathbf{D}(X)}(p_X^* M, p_X^* N) \longrightarrow \mathrm{Hom}_{\mathbf{D}(X')}(p_{X'}^* M, p_{X'}^* N),$$

which, through the two adjunctions, imply exactly the required comparison isomorphisms. This argument is already available for a prederivator possessing the displayed adjoints.

The passage from categorical equivalences to all nerve weak equivalences is substantial. For a derivator on all small categories, its constant-coefficient equivalences form a fundamental localizer (Cisinski, 2008, Proposition 3.7); Cisinski (2004, Theorem 2.2.11) proves that nerve weak equivalences form the smallest fundamental localizer. Their combination is precisely Cisinski (2008, Corollary 3.10): every nerve weak equivalence is a derivator equivalence. The more structural answer is the same paper's Theorem 3.24 and Corollary 3.26: the derivator of spaces is freely generated under homotopy colimits by a point. Its universal action yields the tensor construction; duality yields the cotensor construction. These results settle the modern full-domain version of the question.

The universal property also identifies the operation, not merely its existence. In its cocontinuous form, evaluation at the point identifies cocontinuous morphisms from the space derivator to **D** with

1. Nouvel axiome ?

objects of $\mathbf{D}(e)$. Let F_M correspond to M. The classifying space BX is the homotopy colimit of the constant point diagram over X. Preservation of that homotopy colimit forces

$$F_M(BX) \cong (p_X)_! p_X^* M.$$

The equivalence of categories makes this construction functorial in maps $M \to N$, and the statement for shifted derivators supplies its diagram versions. This explains why the modern theorem answers the particular question on the page: the descended action is constrained to have Grothendieck's formula (4). It also explains why full-domain hypotheses enter the literature. The universal object is the entire space derivator; replacing its diagram domain by an arbitrary subcategory is an additional operation whose effect must be checked.

For a restricted Diag, two further assertions must be kept separate. The internal localization Hot_0 exists under the usual size convention, but its comparison with Hot need not be fully faithful merely by its definition. Even if it is fully faithful, it need not contain all homotopy types. One can work perfectly well with the action of Hot_0 once descent has been established there. What one cannot do is invoke Yoneda with test objects outside that category. This is exactly the limitation that returns on manuscript pages 7–10.

Notons la *formule d'adjonction*

$$\mathrm{Hom}(\xi \times_{\mathbf{D}} M, N) \xrightarrow{\sim} \mathrm{Hom}(M, \mathrm{hom}_D(\xi, N)). \tag{7}$$

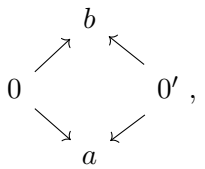
Si ξ provient de $X \in \mathrm{Diag}$, alors cela s'écrit

$$\mathrm{Hom}(p_! p^* M, N) \xrightarrow{\sim} \mathrm{Hom}(M, p_* p^* N), \tag{8}$$

or par adjonction les deux membres sont canoniquement isomorphes à [page 5]

$$\mathrm{Hom}_{\mathbf{D}(X)}(p^* M, p^* N).$$

Exemple. Prenant pour ξ le type d'homotopie du cercle standard (représenté p.ex. par l'ensemble ordonné



où on convient de prendre 0 comme origine, et d'orienter de 0 vers a), les deux foncteurs en question sont (sauf erreur) les foncteurs suspension Σ et 'cosuspension' ou foncteur 'espace des lacets' Ω. Sous réserve de vérifier que ΩF coïncide bien avec ce que j'ai décrit ailleurs, à propos des carrés substantiels cartésiens dans $\mathcal{A}$.

Commentary · Examples [ms. 5]

What the circle actually gives

Grothendieck wants the circle action to recover the suspension and loop constructions, thereby linking the abstract action to the elementary squares of the next chapter. The four-object poset drawn here does have the homotopy type of a circle: its nerve has four vertices and four nondegenerate edges and no higher nondegenerate simplices. The difficulty is therefore not a defective model of S^1. It is the distinction between an unpointed tensor and a reduced one, which he himself identifies on page 12.

In spaces,

$$T_{S^1}(M) = S^1 \times M, \qquad C_{S^1}(N) = \mathrm{Map}(S^1, N).$$

The second space is the free loop space. Choosing $s \in S^1$ gives the evaluation map $\mathrm{Map}(S^1, N) \to N$; its homotopy fibre over a chosen point n is $\Omega_n N$. A basepoint in the indexing circle alone does not turn every free loop into a loop based at n.

The algebraic calculation makes the excess term completely visible. With cohomological grading and a chosen orientation,

$$C_*(S^1;\mathbf{Z}) \simeq \mathbf{Z}[0] \oplus \mathbf{Z}[1].$$

Here $\mathbf{Z}[1]$ has its nonzero term in degree -1. Consequently, in $D(k\text{-Mod})$,

$$T_{S^1}(M) \simeq M \oplus M[1], \qquad C_{S^1}(M) \simeq M \oplus M[-1].$$

The inclusion of the basepoint and the augmentation split off M. Their reduced complements are $M[1]$ and $M[-1]$, the suspension and loop functors of this stable derivator. Taking $M = k \neq 0$, the extra degree-zero copy of k proves that the unqualified identification on the present page is false.

In a pointed homotopy theory the corresponding operation uses the smash tensor with a based circle; its cotensor is the based mapping object. The ordinary unpointed action is recovered by adjoining a disjoint basepoint to the space first. Formula (14) is therefore correct with Σ, Ω understood as the unpointed circle tensor and cotensor, but it is not then the familiar reduced suspension-loop identity. With the reduced meanings, the right side is the loop space of the pointed external Hom at its zero morphism. The explicit Kan extension construction of those reduced functors appears in Groth (2013, Definition 3.16 and Proposition 3.17); Chapter IV is already its essential diagrammatic pattern.

Dans Hot_0 on a une opération produit, en fait ça doit être le produit catégorique dans Hot_0. Mais sur les modèles il est donné par

$$\mathrm{hot}_0(X) \times \mathrm{hot}_0(Y) \;\simeq\; \mathrm{hot}_0(X \times Y) \tag{8}$$

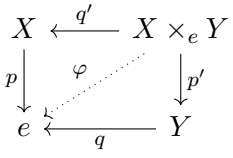
`[deux fois` (8)`]`. Ceci posé, on a les isomorphismes canoniques, pour $\xi, \eta \in \mathrm{Ob}\,\mathrm{Hot}_0$, p. 4

$$\begin{cases} \xi \times_{\mathbf{D}} (\eta \times_{\mathbf{D}} M) \;\simeq\; (\xi \times \eta) \times_{\mathbf{D}} M \\ \hom_{\mathbf{D}}(\xi, \hom_{\mathbf{D}}(\eta, M)) \;\simeq\; \hom_{\mathbf{D}}(\xi \times \eta, M). \end{cases} \tag{9}$$

Si ξ, η sont représentés par X, Y, la première formule p.ex. s'établit en calculant le [page 6] deuxième membre comme

$$\varphi_!(M_{X\times Y}) \simeq p_!(\; \underbrace{q'_!(M_{X\times Y})}_{\substack{\simeq\, q_!(M_Y)_X \\ \text{car } q \text{ est } \mathbf{D}\text{-lisse}}} \;) \simeq \xi \times_{\mathbf{D}} (\eta \times_{\mathbf{D}} M),$$

O.K.



J'ai aussi envie de prouver :

Commentary · Exposition [ms. 6]

Associativity and exactness from product squares

Grothendieck next checks that the action has the algebra expected of an action. The calculation is a Fubini argument for homotopy Kan extensions, and its force is that it works before any mapping-space enrichment has been supplied. We retain his convention that coefficients on I have underlying diagrams of shape I^o.

First, the product on Hot_0 mentioned just above can be justified on the stated models. Assume that Diag is closed under binary products. The nerve preserves those products, and weak equivalences of spaces are preserved by taking a product with a fixed space. Hence the product functor sends a pair of nerve weak equivalences to a nerve weak equivalence and descends to the localization. The diagonal functor descends too. Before localization the diagonal is left adjoint to the binary product; their unit and counit descend as natural transformations, and still satisfy the triangle identities.

The product functor on the localization is therefore right adjoint to its diagonal, which is exactly the universal property of a categorical product. This proves the sentence beginning *en fait ça doit être*, under the required closure assumption, without assuming that $\mathrm{Hot}_0 \to \mathrm{Hot}$ is fully faithful. Existence of this product should not be confused with closure under the right adjoints to products, which becomes relevant to (13).

For $X, Y \in \mathrm{Diag}$, compose the projections $X \times Y \to X \to e$. Pseudofunctoriality of left Kan extension identifies their composite with the extension to e. Base change for the product square identifies the first intermediate extension of a constant coefficient with the constant coefficient $T_Y M$ on X. Thus

$$T_X T_Y M \cong T_{X\times Y} M.$$

The terminal category supplies $T_e \cong \mathrm{id}$. The analogous right Kan calculation gives $C_X C_Y \cong C_{X\times Y}$, with the harmless interchange of X and Y supplied by the product symmetry. The associativity and unit comparisons are canonical mates. Their coherence follows by pasting product squares and using the composition constraints for Kan extension; it is not a second independent choice of associators.

The same argument proves more than preservation of finite colimits or finite limits. Fix X, let $f : I \to J$, and write $p : X \times I \to I$, $q : X \times J \to J$, $f' = 1_X \times f$. The product square is homotopy exact in both directions. Therefore

$$\begin{aligned} f_* C_{X,I} F &= f_* p_* p^* F \\ &\cong q_* f'_* p^* F \\ &\cong q_* q^* f_* F = C_{X,J} f_* F. \end{aligned}$$

The dual calculation gives $f_! T_{X,I} \cong T_{X,J} f_!$. Every Kan extension permitted by Diag is covered. This is the precise content of *exact à gauche* and *exact à droite* here.

For completeness, the decisive product comparison can be checked directly by the pointwise formula. At (x, j), the comma category computing the appropriate Kan extension is a product of the comma category for f with a slice of X having an extremal object. Homotopy Kan extension over the latter factor reduces to evaluation. The resulting isomorphism is the mate of the product square, so its pointwise verification also identifies the comparison that must be used. This is the special case of base change for Grothendieck fibrations and opfibrations formalized in Groth (2013, Proposition 1.24).

Finally,

$$\mathrm{Hom}(T_X M, N) \cong \mathrm{Hom}_{\mathbf{D}(X)}(p_X^* M, p_X^* N) \cong \mathrm{Hom}(M, C_X N)$$

proves the tensor-cotensor adjunction without representability in Hot. This separation is useful: (7), (9), and Proposition 1 are consequences of the Kan formalism once descent is available; the external Hom sought next is an additional representation problem.

Proposition 1. *Soit* $\xi \in \mathrm{Ob}\,\mathrm{Hot}_0$. *Alors* $F \longmapsto \xi \times_{\mathbf{D}} F$, *en tant que morphisme de dérivateurs* $\mathbf{D} \longrightarrow \mathbf{D}$, *est* exact à droite, *et* $F \longmapsto \hom_{\mathbf{D}}(\xi, F)$ *est exact à gauche.*

On peut supposer que ξ est défini par X dans Diag. Soit $f : I \longrightarrow J$, soit F dans $\mathbf{D}(I)$, il faut prouver

$$f_*(\hom_{\mathbf{D}}(\xi, F)) \;\simeq\; \hom_{\mathbf{D}}(\xi, f_*(F)).$$

Considérons donc

$$\begin{array}{ccc} X \times I & \xrightarrow{f'} & X \times J \\ {\scriptstyle p=(x,i)\longmapsto i}\downarrow & & \downarrow{\scriptstyle q=(x,j)\longmapsto j} \\ I & \xrightarrow{\quad f \quad} & J. \end{array}$$

On a

$$\hom_{\mathbf{D}}(\xi, F) = p_*(p^* F),$$

donc

$$f_*(\hom_{\mathbf{D}}(\xi, F)) = \underbrace{f_* p_*}_{(fp)_*}(p^*(F)) = (qf')_* p^*(F) \simeq q_*(f'_* p^*(F)).$$

Mais

$$f'_* p^*(F) \;\simeq\; q^* f_*(F),$$

formule de changement de base des f_*, pour le changement de base q, qui est lisse, car déduit de $X \longrightarrow e$, qui est lisse, par changement de base $J \longrightarrow e$. On trouve donc p. 5 [page 7]

$$f_*(\hom_{\mathbf{D}}(\xi, F)) \simeq q_* q^*(f_*(F)) \overset{\text{déf}}{=} \hom_{\mathbf{D}}(\xi, f_*(F)),$$

q.e.d. L'énoncé pour $F \longmapsto \xi \times_{\mathbf{D}} F$ est dual de celui qu'on vient de prouver.

Reprenons, pour $\xi \in (\mathrm{hot})_0$ [$= \mathrm{Hot}_0$], la formule d'adjonction

$$(*) \qquad \mathrm{Hom}(\xi \times_{\mathbf{D}} M, N) \simeq \mathrm{Hom}(M, \hom_{\mathbf{D}}(\xi, N)) \overset{\text{déf}}{=} \mathrm{Hom}_\xi(M, N).$$

Si ξ est défini par X dans Diag, les deux membres de $(*)$ s'identifient à

$$\mathrm{Hom}_{\mathbf{D}(X)}(p^* M, p^* N),$$

où $p : X \longrightarrow e$ est le foncteur structural.

Fixons M, N, et considérons l'expression $(*)$ comme contrafoncteur

$$(\mathrm{Hot}_0)^o \longrightarrow \mathrm{Ens}.$$

Il semblerait que dans tous les cas connus, ce foncteur soit 'représentable' par un objet, sinon de Hot_0 lui-même, du moins un objet dans Hot. Si on ne peut le définir comme objet H de Hot_0, le fait qu'on ait un isomorphisme canonique

$$(10) \qquad \mathrm{Hom}_\xi(M, N) \simeq \mathrm{Hom}_{\mathrm{Hot}}(\xi, H),$$

fonctoriel en $\xi \in \mathrm{Ob}\,\mathrm{Hot}_0$, ne [le] définit pas forcément à isomorphisme unique près. En fait, quand [page 8]

$$\mathbf{D} = \mathbf{D}_{\mathcal{M},\Sigma}$$

est défini par une catégorie de modèles $(\mathcal{M}, \Sigma)$, alors il y a une définition naturelle d'une opération hom ext (Hom 'externe') à valeurs dans Hot, savoir un foncteur

$$(11) \qquad \begin{cases} \mathcal{A}^o \times \mathcal{A} & \longrightarrow & \mathrm{Hot} \\ (M, N) & \longmapsto & \hom\mathrm{ext}(M, N), \end{cases}$$

où $\mathcal{A} = \mathcal{M}\Sigma^{-1}$ est la catégorie fondamentale de $\mathbf{D} = \mathbf{D}_{\mathcal{M},\Sigma}$. Il est vrai que je n'ai pas vérifié, si p. ex. $(\mathcal{M}, \Sigma)$ est sous-jacente à une catégorie de modèles quillénienne, de sorte que (par Thomason) $\mathbf{D}$ est bien un dérivateur, que cette notion de hom ext satisfait bien à une formule de la forme (10). Ici je vais le poser [?], au besoin, ces données supplémentaires : donnée de (11), et d'un isomorphisme fonctoriel

$$(12) \qquad \underbrace{\mathrm{Hom}_\xi(M, N)}_{\substack{= \mathrm{Hom}(\xi \times_{\mathbf{D}} M, N) \\ = \mathrm{Hom}(M, \hom_{\mathbf{D}}(\xi, N))}} \simeq \mathrm{Hom}_{\mathrm{Hot}}(\xi, \hom\mathrm{ext}(M, N)).$$

Commentary · Later developments [ms. 8]

External Hom as a derived mapping space

Grothendieck is now trying to recover the space of maps from the already available operations on coefficients. In a model category this space had been constructed independently: the hammock localization of Dwyer and Kan (1980b, Corollary 4.2, Proposition 4.4, and Corollary 4.7) supplies mapping complexes, identifies their components with localized morphisms, and compares them with the enrichment of a simplicial model category. This work predates the manuscript. Chapter V will approach the same issue through categories of zigzags.

Here is the verification of (12) in the simplicial case. Choose a cofibrant representative QM and a fibrant representative RN. A simplicial set K representing ξ is cofibrant in the usual model structure. The simplicial tensor-Hom adjunction gives

$$\mathrm{Map}_{\mathcal{M}}(K \otimes QM, RN) \cong \mathrm{Map}_{\mathrm{sSet}}(K, \mathrm{Map}_{\mathcal{M}}(QM, RN)).$$

Taking components identifies the left side with $\mathrm{Hom}_{\mathrm{Ho}(\mathcal{M})}(K \otimes^{\mathbf{L}} M, N)$. The right side is $[K, \mathrm{Map}_{\mathcal{M}}(QM, RN)]$, because the inner mapping complex is fibrant. Moreover $K \otimes^{\mathbf{L}} M$ is the homotopy colimit of the constant M-diagram indexed by the space K. For $K = BX$, the simplicial replacement of the constant X^o-diagram gives exactly that tensor; its simplices are copies of QM indexed by strings of arrows in X^o. Reversal identifies its realized indexing space with BX. This proves the formula with the particular tensor defined in (4).

The cotensor identity follows either from the dual model construction or from the tensor-cotensor adjunction already proved. Derived function complexes and framings give the corresponding statement for a general Quillen model category; the tensor, cotensor, and enrichment compatibilities are treated systematically in Riehl (2014, Sections 10.2–10.3). Thus the extra datum is supplied by the model presentation and is independent, up to homotopy equivalence, of the chosen resolutions.

For a presentable ∞-category $\mathcal{C}$, the same calculation has the particularly transparent form

$$\mathrm{Map}_{\mathcal{C}}\big(\operatorname*{colim}_{K} M, N\big) \simeq \mathrm{Map}_{\mathcal{S}}\big(K, \mathrm{Map}_{\mathcal{C}}(M, N)\big).$$

It is the universal property of the colimit indexed by the space K. Passing to homotopy categories gives (12); no new choice of external Hom is needed once $\mathcal{C}$ is given. The universal role of space-valued presheaves is made precise in Lurie (2009, Theorem 5.1.5.6).

For a completely abstract derivator, one should distinguish an action by spaces from a closed action possessing these Hom objects. Groth and Shulman (2017, Section 1, pp. 3–4, and Definition 4.2) explicitly distinguish the two structures. The cited results settle the model and presentable ∞-categorical cases; they do not justify erasing Grothendieck's representability hypothesis for every restricted diagram domain considered here.

Si maintenant η est un deuxième élément de Hot_0, on a envie d'écrire p. 6 [page 9]

$$(13) \qquad \begin{cases} \hom \mathrm{ext}(\eta \times_{\mathbf{D}} M, N) \simeq \hom \mathrm{ext}(M, \hom_{\mathbf{D}}(\eta, N)) \\ \qquad\qquad\qquad\quad \simeq \hom_{\mathrm{HOT}}(\eta, \hom \mathrm{ext}(M, N)), \end{cases}$$

où dans le deuxième membre, $\hom_{\mathrm{HOT}}$ est l'opération (1) dans le cas où $\mathbf{D} = \mathrm{HOT}$, et qui est défini aussi en termes du produit cartésien $\xi \times \eta$ dans Hot, par

$$\mathrm{Hom}_{\mathrm{Hot}}(\xi, \hom_{\mathrm{HOT}}(\eta, \zeta)) \simeq \mathrm{Hom}_{\mathrm{Hot}}(\xi \times \eta, \zeta),$$

en d'autres termes, pour η fixé, le foncteur

$$\zeta \longmapsto \hom_{\mathrm{HOT}}(\eta, \zeta)$$

est défini comme le foncteur adjoint à droite du foncteur

$$\xi \longmapsto \xi \times \eta.$$

Pour vérifier heuristiquement la formule (13) (vérification valable 'en forme' si les trois objets envisagés sont en fait définis comme objets de Hot_0 ...), on prend $\mathrm{Hom}_{\mathrm{Hot}}(\xi, -)$ à valeurs dans les trois membres, pour ξ donné dans Hot_0. Pour le premier, on trouve

$$\begin{aligned} \mathrm{Hom}_{\xi}(\eta \times_{\mathbf{D}} M, N) &= \mathrm{Hom}(\xi \times_{\mathbf{D}} (\eta \times_{\mathbf{D}} M), N) \\ &= \mathrm{Hom}((\xi \times \eta) \times_{\mathbf{D}} M, N). \end{aligned}$$

Pour le deuxième

$$\begin{aligned} \mathrm{Hom}_{\xi}(M, \hom_{\mathbf{D}}(\eta, N)) &= \mathrm{Hom}_{\mathcal{A}}(\xi \times_{\mathbf{D}} M, \hom_{\mathbf{D}}(\eta, N)) \\ &\simeq \mathrm{Hom}_{\mathcal{A}}(\eta \times_{\mathbf{D}} (\xi \times_{\mathbf{D}} M), N) \\ &\simeq \mathrm{Hom}_{\mathcal{A}}((\eta \times \xi) \times_{\mathbf{D}} M, N), \end{aligned}$$

[page 10]

et enfin pour le troisième

$$\begin{aligned}\mathrm{Hom}_{\mathrm{Hot}}(\xi \times \eta, \hom\mathrm{ext}(M,N)) &\simeq \mathrm{Hom}_{\xi\times\eta}(M,N)\\ &\simeq \mathrm{Hom}((\xi\times\eta)\times_{\mathbf{D}} M, N),\end{aligned}$$

donc toujours la même chose ! Pas de doute, ça doit marcher.

S'il faut se donner le hom ext (11) comme donnée supplémentaire, avec de plus un isomorphisme trifonctoriel (12), il faut aussi se donner les isomorphismes trifonctoriels (13), '*compatibles*' avec l'isomorphisme (12) au sens du calcul précédent.

Commentary · Exposition [ms. 10]

What the Yoneda calculation proves

Grothendieck's three calculations are intended to identify the external Hom of a tensor with the cotensor of an external Hom. The calculation is correct. His qualification about the test category is also correct, and is part of the argument rather than an unnecessary hesitation.

Suppose the external Hom represents (12), and suppose that all three candidate objects in (13) belong to Hot_0. For every $\xi \in \mathrm{Hot}_0$, tensor associativity and the tensor-cotensor adjunction give natural identifications of their represented functors with

$$\xi \longmapsto \mathrm{Hom}_{\mathcal{A}}((\xi\times\eta)\times_{\mathbf{D}} M, N).$$

Yoneda in Hot_0 then gives a unique isomorphism respecting these identifications. Naturality in η, M, N follows from the same uniqueness. The last candidate additionally requires the relevant function object to exist in Hot_0; closure under products alone does not imply that closure under function objects.

If the candidates lie only in Hot, the calculation instead identifies their *restricted* represented functors. Such an identification need not determine the objects. For a concrete example, take Hot_0 to be the full subcategory of discrete spaces. For any discrete set S and any nonempty connected space H, the set $[S,H]$ has one element: a map is a choice of a component for each element of S, and there is only one component. Consequently the point and S^1 define isomorphic functors on this test category, although they are not isomorphic homotopy types. This example isolates the exact logical failure that would result from applying unrestricted Yoneda here.

There are two legitimate completions. One may assume the restricted Yoneda functor is fully faithful on the particular objects under consideration, and then the displayed calculation suffices. Alternatively, a model or ∞-categorical construction gives the actual comparison of mapping spaces before passage to homotopy types, proving (13) directly. The manuscript follows the first course conditionally and otherwise proposes to retain the comparisons as part of the additional structure. No gap should be charged to a proof that explicitly states this restriction.

Par exemple, prenant $\eta =$ type d'homotopie du cercle S^1, les formules (13) se spécialisent en p. 7

$$\begin{aligned}\hom\mathrm{ext}(\Sigma M, N) &\simeq \hom\mathrm{ext}(M, \Omega N)\\ &\simeq \Omega\hom\mathrm{ext}(M,N)\\ &\overset{\text{déf}}{=} \hom_{\mathrm{HOT}}(S^1, \hom\mathrm{ext}(M,N)).\end{aligned} \tag{14}$$

Mais cette formule dans Hot n'est sans doute pas tout à fait la bonne, car le bon formalisme de Ω, Σ se fait dans les catégories *ponctuées,* et en particulier dans la catégorie homotopique ponctué $\mathrm{Hot}_\bullet$.

Je vais expliciter un exemple, en prenant le dérivateur

$$\mathrm{Dér}(A)$$

'dérivé' d'une catégorie abélienne A. Donc $\mathbf{D}(e) =$ catégorie dérivée de A. Pour éviter des difficultés accessoires ici, je vais supposer que A est la catégorie des k-modules, k un anneau [page 11]

quelconque.

À tout $\xi \in \operatorname{Ob}\mathrm{Hot}$ est associé un 'abélianisé'

$$\xi_{\mathrm{ab}} \overset{\text{déf}}{=} H_\bullet(\xi) \in \underbrace{D(\mathrm{Ab})}_{\text{ou } \mathbf{D}(\mathbf{Z}\text{-Mod})}, \qquad \text{catégorie dérivée des groupes abéliens,}$$

objet qui n'a de cohomologie qu'en degré ≤ 0. Ceci dit, si M est dans $D(k\text{-Mod})$, on aura

$$\begin{cases} \xi \times_{\mathbf{D}} M \;\simeq\; \xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} M \\ \hom_{\mathbf{D}}(\xi, M) \;\simeq\; \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\xi_{\mathrm{ab}}, M), \end{cases} \tag{15}$$

formule qui est en cheville avec la formule de Künneth pour l'homologie

$$(\xi \times \eta)_{\mathrm{ab}} \simeq \xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} \eta_{\mathrm{ab}}, \tag{16}$$

ce qui redonne un aspect familier aux formules (9), qui se réduisent à

$$\begin{cases} \xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} (\eta_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} M) \;\simeq\; (\xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} \eta_{\mathrm{ab}}) \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} M \\ \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\xi_{\mathrm{ab}}, \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\eta_{\mathrm{ab}}, M)) \;\simeq\; \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(\xi_{\mathrm{ab}} \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}} \eta_{\mathrm{ab}}, M). \end{cases} \qquad ([2])$$

D'autre part, existe-t-il bien un $\hom\operatorname{ext}(M, N)$, si M, N dans $D(k\text{-Mod})$? On pense à $\mathbf{R}\mathrm{Hom}_k(M, N)$, mais pour en faire un type d'homotopie, on doit 'tronquer en degrés > 0'. On pourrait donc prendre

$$\hom\operatorname{ext}(M, N) \;\simeq\; \tau_{\leq 0}(\mathbf{R}\mathrm{Hom}_k(M, N)).$$

Commentary · Examples [ms. 11]

Integral chains and the truncation of derived Hom

Grothendieck turns to complexes because they make the external Hom computable. We can verify every part of the proposed formula, provided that *abélianisation* means the derived free abelian-group construction and that the truncation is the canonical one.

For a simplicial set K, let

$$A_K = N\mathbf{Z}[K]$$

be its normalized free chain complex, written cohomologically in degrees at most zero. This is the object denoted $H_\bullet(\xi)$ in the manuscript. It is a chain complex up to quasi-isomorphism, not merely its graded homology groups. The free simplicial abelian-group functor is left adjoint to the forgetful functor. After deriving and applying the Dold–Kan equivalence, it gives

$$[K, U\,\mathrm{DK}(B)] \cong \mathrm{Hom}_{D(\mathbf{Z})}(A_K, B), \qquad B \in D(\mathbf{Z})^{\leq 0}.$$

Here DK means denormalization with the change from cohomological to homological grading, and U forgets the abelian-group structure. The underlying equivalence is The Stacks Project Authors (2026c, Theorem 14.24.3). It is this derived adjunction that answers the question in footnote 3; being only an abelian-group object in the ordinary homotopy category would not by itself specify the requisite strict or coherent abelian structure.

Put $R = \mathbf{R}\mathrm{Hom}_k(M, N)$, viewed as an object of $D(\mathbf{Z})$. For a noncommutative k, this target is still correct: there is generally no left k-module structure on this Hom complex. Since $A_K \in D(\mathbf{Z})^{\leq 0}$, the truncation adjunction gives

$$\begin{aligned} [K, U\,\mathrm{DK}(\tau_{\leq 0}R)] &\cong \mathrm{Hom}_{D(\mathbf{Z})}(A_K, \tau_{\leq 0}R) \\ &\cong \mathrm{Hom}_{D(\mathbf{Z})}(A_K, R) \\ &\cong \mathrm{Hom}_{D(k)}(A_K \otimes^{\mathbf{L}}_{\mathbf{Z}} M, N). \end{aligned}$$

2. Expliciter aussi la formule d'adjonction (7) (p. 4).

The final equality is the derived tensor-Hom adjunction; k acts on the factor M. This proves (10) and (12) for the claimed external Hom, with the otherwise suppressed Dold–Kan and forgetful functors restored.

The truncation is the smart truncation: its degree-zero term is $\ker(d : R^0 \to R^1)$, and its negative terms are those of R. Simply deleting the positive degrees would replace degree-zero cocycles by all of R^0, generally giving a wrong set of components. The homotopy groups of the resulting simplicial abelian group are

$$\pi_n \hom\operatorname{ext}(M,N) \cong H^{-n}R \cong \operatorname{Hom}_{D(k)}(M, N[-n]), \qquad n \geq 0.$$

For instance, with $k = \mathbf{Z}$, $M = \mathbf{Z}$, and $N = \mathbf{Z}[r]$, $r \geq 0$, one obtains $K(\mathbf{Z}, r)$. The adjunction then reads $[K, K(\mathbf{Z}, r)] \cong H^r(K; \mathbf{Z})$, including degree zero with its usual unreduced interpretation.

The other formula in (15) follows from the same adjunction:

$$T_K M = A_K \otimes^{\mathbf{L}}_{\mathbf{Z}} M, \qquad C_K M = \mathbf{R}\operatorname{Hom}_{\mathbf{Z}}(A_K, M).$$

The Eilenberg–Zilber comparison identifies $A_{K\times L}$ with $A_K \otimes^{\mathbf{L}}_{\mathbf{Z}} A_L$ (Eilenberg and Zilber, 1953, Section 1, theorem, pp. 200–201); normalized free chains are bounded above complexes of free abelian groups in this grading, so the ordinary tensor already computes the derived tensor. Formula (16) is this chain-level equivalence. A formula involving only the tensor product of homology groups would lose the Tor terms and would not suffice for Grothendieck's argument.

p. 8
[page 12]

La formule fondamentale (10) est-elle alors valable ? Le deuxième membre est en effet

$$\operatorname{Hom}_{\mathrm{Hot}}(\xi, H) \;\simeq\; \operatorname{Hom}_{\mathrm{Hotab}}(\xi_{\mathrm{ab}}, H) \qquad ({}^{3})$$

car H est un objet abélien de Hot (cf. le tapis sur l'abélianisation …), à partir de là c'est gagné :

$$\operatorname{Hom}_{\underbrace{\mathrm{Hotab}}_{\substack{\approx\, D(\mathbf{Z}\text{-Mod})^{\leq 0},\\ \text{sous-catégorie } \textit{pleine}\\ \text{de } D(\mathbf{Z}\text{-Mod})}}}(\xi_{\mathrm{ab}}, \tau_{\leq 0}\mathbf{R}\operatorname{Hom}_k(M,N)) \;\simeq\; \operatorname{Hom}_{D(\mathbf{Z}\text{-Mod})}(\xi_{\mathrm{ab}}, \mathbf{R}\operatorname{Hom}_k(M,N))$$

$$\simeq\; \operatorname{Hom}_{D(k\text{-Mod})}(\xi_{\mathrm{ab}} \times M, N),$$

O.K.

D'autre part, le bifoncteur

$$(M,N) \;\longmapsto\; \hom\operatorname{ext}(M,N) \;=\; \tau_{\leq 0}\mathbf{R}\operatorname{Hom}_k(M,N)$$

se factorise canoniquement par Hotab et a fortiori par $\mathrm{Hot}_{\bullet}$. Le 'bon' $\hom\operatorname{ext}(M,N)$ serait ici plutôt celui dans Hotab, voire même dans la catégorie $D^-(\mathbf{Z}\text{-Mod})$, déduite de celle là 'en rendant le foncteur translation inversible'. Il est évident d'ailleurs qu'ici le $\xi_{\mathrm{ab}} \times_{\mathbf{D}} M$, où $\xi = S^1$, n'est pas 'la bonne' suspension, qui doit être le décalage des degrés – ξ_{ab} ne devrait avoir de la cohomologie qu'en degré -1, et non aussi en degré 0.

Commentary · Present state [ms. 12]

Keeping all degrees of the mapping complex

Grothendieck's last suggestion seeks an external Hom that remembers the linear structure and the information restored by suspension. That is the correct direction, but the proposed bounded-above target requires a qualification when the coefficient category is the unbounded derived category.

The homotopy category of simplicial abelian groups is equivalent to $D(\mathbf{Z})^{\leq 0}$. Allowing finite shifts of its objects produces $D^-(\mathbf{Z})$: an object in the union has cohomology bounded above, and conversely any bounded-above complex can be shifted into nonpositive degrees. This observation

3. ? ? est-ce vrai ? ? ?

correctly describes the stated formal inversion of translation at the level of objects and morphisms. It does not imply that every derived Hom complex belongs to this union.

A two-term algebra already shows the difficulty. Let F be a field, let $k = F[\epsilon]/(\epsilon^2)$, and let $L = k/(\epsilon)$. A projective resolution of L is

$$\cdots \xrightarrow{\epsilon} k \xrightarrow{\epsilon} k \xrightarrow{\epsilon} k \longrightarrow L \longrightarrow 0.$$

It is exact because the kernel and image of multiplication by ϵ are both the ideal (ϵ). Applying $\mathrm{Hom}_k(-, L)$ gives one copy of F in every nonnegative cohomological degree and zero differentials. Thus

$$H^n \mathbf{R}\mathrm{Hom}_k(L, L) = \mathrm{Ext}^n_k(L, L) \cong F \qquad (n \geq 0).$$

The full derived Hom is not in $D^-(\mathbf{Z})$. The present external mapping space is nevertheless well defined: its nonpositive truncation is the discrete group F. The omitted extension groups reappear as $\pi_0 \mathrm{hom}\,\mathrm{ext}(L, L[n])$, for every $n \geq 0$. This example explains why a single mapping space and the whole stable mapping object are different levels of information.

For complexes, the full enhancement is by the unbounded complex $\mathbf{R}\mathrm{Hom}_k(M, N)$; in a general stable homotopy theory it is by a mapping spectrum. The classical differential graded formulation and its relation to triangulated categories are discussed by Keller (2006, Sections 2–3). The modern theory of monoidal and enriched derivators distinguishes the unpointed, pointed, and stable actions and the existence of their right adjoints (Groth et al., 2014b, Sections 3 and 8–9).

The correction here concerns the generality of a tentative proposal, not a theorem whose proof has collapsed. The formulas involving $\tau_{\leq 0}\mathbf{R}\mathrm{Hom}$ are valid. The earlier circle identification is repaired by reduced chains. An additional boundedness hypothesis, such as suitable finite projective dimension and boundedness on the inputs, is needed only if one insists that the entire linear external Hom take values in $D^-(\mathbf{Z})$.

Chapter IV

Diagrammes substantiels

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p.

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre IV

Diagrammes substantiels

p.

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.
Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :
[**e-mail address removed**]

G. Maltsiniotis

[Il n'y a pas de pages 1-72]

p. 1
[page 73]

Construction de diagrammes substantiels pour un dérivateur.

Soit $\mathbf{D}$ un prédérivateur, I dans Diag, considérons
$\varphi_I : \mathcal{D}(I) \longrightarrow \underline{\mathrm{Hom}}(I^o, \mathcal{A})$, où $\mathcal{A} = \mathbf{D}(e)$ est la catégorie fondamentale des coefficients.

Les objets de $\underline{\mathrm{Hom}}(I^o, \mathcal{A})$ s'interprètent comme des *diagrammes de type* I^o dans $\mathcal{A}$. Ce langage est surtout commode si I est ordonnée. Notons, pour $F \in \mathrm{Ob}\,\mathbf{D}(I)$, par $(F_i)_{i\in \mathrm{Ob}\, I}$ le I^o-diagramme associé dans $\mathcal{A}$. Par l'axiome Dér 2, le foncteur

$$\varphi : F \longmapsto (F_i)_{i\in \mathrm{Ob}\, I}$$

est conservatif. Mais il n'a aucune envie d'être fidèle, encore moins pleinement fidèle. Quand on connaît le diagramme $(F_i)_i$, on est loin de connaître F, pas même à isomorphisme non unique près, je présume. Il n'est nullement clair que si F, F' dans $\mathbf{D}(I)$ sont tels que les diagrammes associés soient isomorphes, ils soient eux-mêmes isomorphes.

Commentary · Context [ms. 73]

Why the skeleton is insufficient

Grothendieck's term *diagramme substantiel* names the information on which the whole project depends: a diagram before its coherent relations have been discarded. The point is not that an ordinary commutative diagram has been drawn inaccurately. It is that several different homotopies and compatibilities can produce the same drawing in the homotopy category.

In the model example,

$$\mathbf{D}(I) = \mathcal{M}^{I^o}[(W^{I^o})^{-1}], \qquad \varphi_I : \mathbf{D}(I) \longrightarrow (\mathrm{Ho}\mathcal{M})^{I^o}.$$

Localization is performed on the category of whole diagrams. The target instead localizes the values first and subsequently records arrows and their equations. These operations need not commute. Axiom Der 2 says that a morphism of coherent diagrams which is an isomorphism at every vertex is invertible. It says neither that every skeletal morphism lifts nor that two lifts agree. The distinction was illustrated in the commentary on Chapter I, manuscript p. 11 (*Conservative is not faithful*); there is no need to repeat that counterexample here.

For the walking arrow one can make a useful positive statement. A *strong* derivator satisfies fullness and essential surjectivity of

$$\mathbf{D}(I \times [1]) \longrightarrow \mathbf{D}(I)^{[1]^o}$$

for every I. This is the usual modern Der 5, not an identification of the numbered axioms appearing earlier in the manuscript (Groth, 2013, Definition 1.8). Fullness together with conservativity implies that two coherent arrows whose skeletal arrows are isomorphic are themselves isomorphic: lift the skeletal isomorphism by fullness, then use its invertibility at both endpoints. It does not imply a unique lift of that isomorphism. Thus Grothendieck's question about objects with isomorphic skeletons has a different answer for a single arrow in a strong derivator than for an unrestricted diagram shape.

The historical motivation is particularly close to the present construction. In *Pursuing Stacks*, Grothendieck (1983, Section 69, pp. 1–2 of the edited extract) recalls the need, arising around SGA 5, to supplement a derived category by derived categories of triangles and filtered complexes. Here he seeks a procedure that constructs the supplements and their maps from Kan extension itself. Heller's parallel treatment packages homotopy theories as systems of diagram categories (Heller, 1988, Introduction and Chapter I). Neither perspective asks one to manufacture a functorial cone on the bare homotopy category.

The two examples pursued below are deliberately chosen to retain a universal property throughout the construction. The loop square is determined by a zero extension followed by a right Kan extension. The unit and counit arrows are determined by a category that encodes the functor f. This is why canonical coherent versions of these particular arrows can be obtained even though a general skeletal arrow has no specified canonical lift.

Pour insister sur ce point, les objets de $\mathbf{D}(I)$ s'appelleront les '*diagrammes substantiels* de type I^o dans $\mathcal{A}$' (ou $\mathbf{D}$-substantiel, s'il y a risque de confusion). Pour étayer l'intuition et simplifier les explications, un objet F de $\mathbf{D}(I)$ ($\mathbf{D}$-coefficients sur I) sera représenté graphiquement par le diagramme 'faible' ou 'squelettique' correspondant – on le regarde comme une sorte de (pâle) 'squelette' du diagramme substantiel dont il provient. P. ex. si [page 74] $I = \mathbf{\Delta}_0 : 0 \longrightarrow 1$ `[plutôt` $\mathbf{\Delta}_1$`]`, on note un diagramme substantiel de type $(\mathbf{\Delta}_0)^o$ `[plutôt` $(\mathbf{\Delta}_1)^o$`]` par le même graphisme, que le diagramme ordinaire :

$$F_0 \longleftarrow F_1.$$

Si

$$I = \begin{array}{ccc} a & \longrightarrow & c \\ \uparrow & & \uparrow \\ 0 & \longrightarrow & b, \end{array}$$

les diagrammes substantiels de type I se notent

$$\begin{array}{ccc} F_a & \longleftarrow & F_c \\ \downarrow & & \downarrow \\ F_0 & \longleftarrow & F_b. \end{array}$$

Il faudra bien sûr préciser chaque fois si ce graphisme désigne un diagramme squelettique, ou substantiel. Souvent on part d'un squelette, et on essaye de 'l'étoffer' avec un peu de chair, pour en faire un diagramme substantiel – si possible, de façon canonique. C'est là un point de philosophie dérivatique essentiel pour la pratique. On procédera souvent de proche en proche, en 'substantifiant' des sous-diagrammes correspondants à une suite strictement croissante $I_0 \subset I_1 \subset \cdots \subset I_n = I$ de I, au mieux de la situation particulière. p. 2

En voici un exemple typique, en supposant $\mathbf{D}$ un dérivateur *ponctué* (les $\mathbf{D}(I)$ sont ponctuées, et les f^* transforment objets nuls en objets nuls). Il s'agit de la définition du foncteur 'cosuspension' ou 'espace de lacets' dans $\mathcal{A}$, [page 75]

$$\Omega : \mathcal{A} \longrightarrow \mathcal{A}.$$

Il est entendu que pour M dans $\mathcal{A}$, $\Omega(M)$ doit s'insérer dans un carré commutatif

(C)
$$\begin{array}{ccc} e & \longleftarrow & \Omega M \\ \downarrow & & \downarrow \\ M & \longleftarrow & e, \end{array}$$

où e désigne les éléments unités. Ce carré cependant devrait être substantiel, plus précisément, être un carré substantiel '$\mathbf{D}$-*cartésien*'. Étant entendu qu'un objet de $\mathbf{D}(I)$ (carré substantiel dans $\mathcal{A}$) est dit $\mathbf{D}$-cartésien, s'il est dans l'image essentielle de $\mathbf{D}(I_1)$ par i_{1*}, où

$$i_1 : I_1 \hookrightarrow I,$$

I_1 étant la sous-catégorie ouverte de I définie par

$$I_1 = \begin{array}{ccc} & & a \\ & & \uparrow \\ 0 & \longrightarrow & b. \end{array}$$

Commentary · Exposition [ms. 75]

Constructing the loop square without choosing homotopies

Grothendieck wants to obtain a functor Ω, not merely an object which happens to have the expected homotopy type. The two successive Kan extensions supply that functoriality. The direction of the arrows is governed by the presheaf convention: the indexing square has initial vertex 0 and final vertex c, while its coefficient diagram has arrows pointing toward 0.

Let $I_0 = \{0\}$, let $I_1 = \{0, a, b\}$ with $0 \to a, 0 \to b$, and let I be the completed square. Denote the inclusions by $i_0 : I_0 \to I_1$ and $i_1 : I_1 \to I$, as in the text. Since I_0 is a sieve in I_1, the *left* Kan extension in the presheaf convention is extension by the initial object outside I_0. Pointedness identifies this initial object with zero. Hence

$$(i_0)_! M|_0 = M, \qquad (i_0)_! M|_a = (i_0)_! M|_b = 0.$$

The pointwise assertion follows from empty comma categories at a, b, and the assertion at 0 follows from full faithfulness of the inclusion. Thus the cospan $0 \to M \leftarrow 0$ has been made coherent without choosing representatives or nullhomotopies.

Apply the right Kan extension $(i_1)_*$. Its value at c is the homotopy limit of that cospan. Set

$$\Omega M = c^*(i_1)_*(i_0)_! M.$$

The object $(i_1)_*(i_0)_! M$ carries, at once, the two maps from ΩM to zero and the specified coherent commutativity with the two zero maps to M. It is cartesian by construction. Every morphism $M \to N$ induces a morphism of these squares by the functoriality of the composite, so the construction really produces an endofunctor.

The stronger equivalence stated on page 77 follows with little extra work. Let $\mathcal{Q} \subset \mathbf{D}(I)$ be the full subcategory of cartesian squares with values zero at a, b. The functor

$$L = (i_1)_*(i_0)_! : \mathbf{D}(e) \longrightarrow \mathcal{Q}$$

is fully faithful, since both Kan extensions are. It is essentially surjective as well. If $Q \in \mathcal{Q}$, cartesianness gives $Q \cong (i_1)_* i_1^* Q$. The counit $(i_0)_! i_0^* i_1^* Q \to i_1^* Q$ is an isomorphism at 0, and at a, b both sides are zero. Der 2 therefore makes it an isomorphism. Substitution gives $Q \cong L(Q_0)$, and restriction to 0 is a quasi-inverse.

The phrase *unique à isomorphisme unique près* has its usual universal-property meaning: after specifying the identification of the input with M, a comparison is unique when required to induce that identification. A square may have automorphisms induced by automorphisms of M. Also, one should not add a requirement that the induced comparison fix an independently chosen identification of the fourth vertex. The theorem determines that vertex as part of the construction.

For complexes, the homotopy fibre of $0 \to M$ is $M[-1]$, so this recovers the reduced loop functor, with no unwanted summand M. This is the construction needed to repair the circle calculation of Chapter III.

[page 76]

La restriction du carré substantiel cherché (C) à la sous-catégorie I_1 est donc un diagramme substantiel de squelette

(Sq)
$$\begin{array}{ccc} & & e \\ & & \downarrow \\ M & \longleftarrow & e, \end{array}$$

dont il se déduit (à isomorphisme canonique près) en appliquant i_* à ce dernier, de façon p. 3
à le compléter en $\begin{array}{ccc} e & \leftarrow & \Omega M \\ \downarrow & & \vdots \\ M & \longleftarrow & e \end{array}$. Il reste donc à décrire une substantiation canonique du squelette (Sq), en termes de M. Cela nous conduit à regarder la sous-catégorie ouverte

$$I_0 = \{0\} : \quad \begin{array}{ccc} & \cdots\cdots\rightarrow & \\ \uparrow & & \uparrow \\ 0 & \cdots\cdots\rightarrow & \end{array}$$

de I_1 (donc aussi de I) réduite à l'objet initial, et l'inclusion

$$I_0 \overset{i_0}{\hookrightarrow} I_1,$$

et de substantifier le squelette (Sq) par $(i_0)_!(M)$.

On peut dire a posteriori que, si on connaissait le carré substantiel, il n'y a, à isomorphisme unique près, qu'une seule façon de l'obtenir comme un $i_{1*}i_{0!}(?)$, avec ? dans $\mathbf{D}(I_0) \simeq \mathcal{A}$, car $i_{0!}$ et i_{1*} étant pleinement fidèles, il en est de même de leur composé.

En fait, on voit que le foncteur [page 77]

$$i_{1*}i_{0!} : \ \mathcal{A} \simeq \mathbf{D}(I_0) \longrightarrow \mathbf{D}(I)$$

induit une équivalence de $\mathcal{A}$ avec la sous-catégorie pleine de $\mathbf{D}(I)$ formée des carrés **D**-cartésiens dont les fibres en a et b sont nulles. (Le foncteur quasi-inverse évident est i^*, où $i : U_0 \hookrightarrow I$ `[plutôt` $I_0 \hookrightarrow I$`]`.) Cela signifie donc que dans ce cas, la donnée du squelette de nature très particulière ($F_a \simeq F_b \simeq 0_{\mathcal{A}}$), plus la contrainte supplémentaire **D**-cartésienne, sur le carré cherché, détermine celui-ci à isomorphisme unique près en termes de son squelette : il y a une substantiation non seulement 'canonique' de celui-ci, mais même unique à isomorphisme unique près, à cause de la contrainte **D**-cartésienne.

Je voudrais commencer par regarder le cas plus simple encore où $I = \mathbf{\Delta}_1 = (0 \longrightarrow 1)$, donc les squelettes envisagés sont simplement les flèches

$$F_1 \longrightarrow F_0 \qquad \text{dans } \mathcal{A}.$$

La philosophie nous oblige, pour les flèches les plus fondamentales dans la théorie des dérivateurs $\mathbf{D}$ (flèches dans $\mathcal{A}$, ou par extension dans une $\mathbf{D}(I) \simeq \mathcal{A}_{\mathbf{D}_I}$, catégorie fondamentale du dérivateur induit $\mathbf{D}_I$) de se demander s'il n'y a pas

une façon 'canonique', 'naturelle', 'voulue par Dieu', de le 'remplumer' en un diagramme substantiel. Les flèches les plus évidentes qui se présentent sont relatives à un foncteur p. 4 [page 78]

$$f : I \longrightarrow J$$

dans Diag, et aux deux morphismes d'adjonction pour le couple (f^*, f_*), savoir

$$\begin{aligned} G &\longrightarrow f_* f^* G \quad \text{dans } D(J) \\ f^* f_* F &\longrightarrow F \qquad \text{dans } D(I). \quad (^2) \end{aligned}$$

Je vais essayer de remplumer ces deux flèches, et vais généraliser en envisageant la situation ci-contre `[ci-dessous]`

$$\begin{array}{ccc} I & & F \\ {\scriptstyle f}\downarrow & & \uparrow{\scriptstyle u} \\ J & & G, \end{array}$$

où

$$\mathrm{Hom}^f(G, F) \overset{\text{déf}}{=} \mathrm{Hom}(f^*G, F) \overset{\text{adj.}}{\simeq} \mathrm{Hom}(G, f_*F).$$

Donc le même u s'interprète par *deux* flèches squelettiques

$$\text{(1)} \qquad \begin{cases} f^*(G) \longrightarrow F & \text{dans } \mathbf{D}(I) \\ G \longrightarrow f_*(F) & \text{dans } \mathbf{D}(J). \end{cases}$$

Les remplumer, c'est les interpréter (si faire se peut) en termes de deux objets (si possible canoniques) dans $\mathbf{D}(I \times \mathbf{\Delta}_1)$ et $\mathbf{D}(J \times \mathbf{\Delta}_1)$ respectivement.

Je vais me guider sur le cas particulier d'un dérivateur trivial

$$\mathbf{D} = \mathbf{D}_{\mathcal{M}},$$

où $\mathcal{M}$ est une catégorie de modèles 'nue' ($\Sigma = \emptyset$).

Pour f fixé, on va construire une catégorie auxiliaire $K(f)$, de telle façon que pour toute catégorie $\mathcal{M}$, les systèmes (G, F, u) d'un $\mathcal{M}$-préfaisceau G sur J, d'un autre F sur I, et d'un f-homomorphisme $G \longrightarrow F$, s'identifient (fonctoriellement) aux $\mathcal{M}$-préfaisceaux sur $K(f)$. Ceci dit, on va essayer d'interpréter les flèches (1) squelettiques, mais peu importe ici – tous les $\mathbf{D}$-squelettes sont maintenant remplumés) en termes de la géométrie de $K(f)$. [page 79] p. 5

$$\begin{cases} \mathrm{Ob}\, K(f) = \mathrm{Ob}\, K(I) \amalg \mathrm{Ob}\, K(J) \qquad \texttt{[plutôt Ob } I \amalg \mathrm{Ob}\, J \texttt{]} \\ I,\ J \text{ s'identifiant à des sous-catégories pleines de } K(f), \\ \text{il n'y a pas des flèches d'un } j \in J \text{ dans un } i \in I. \\ \text{(Donc } I \text{ est ouverte dans } K(f),\ J \text{ son complémentaire fermé.)} \\ \text{Pour } i \in \mathrm{Ob}\, I,\ j \in \mathrm{Ob}\, J, \text{ on pose } \mathrm{Hom}_{K(f)}(i, j) = \mathrm{Hom}_J(f(i), j). \end{cases}$$

2. Bien sûr, problème dual pour $(f_!, f^*)$.

La composition des flèches dans $K(f)$ est évidente. On va définir une rétraction canonique

$$\bar{f} : K(f) \longrightarrow J$$

de $K(f)$ sur sa sous-catégorie fermée J. Donc $\bar{f}$ est connu sur $\operatorname{Ob} J$, et sur $\operatorname{Ob} I$ ce sera f :

$$\begin{cases} \bar{f}(j) = j & \text{si } j \in \operatorname{Ob} J \\ \bar{f}(i) = f(i) & \text{si } i \in \operatorname{Ob} I. \end{cases}$$

Il reste à définir, pour $x, y \in \operatorname{Ob} K(f)$, [page 80]

$$\operatorname{Hom}_{K(f)}(x, y) \longrightarrow \operatorname{Hom}_J(\bar{f}(x), \bar{f}(y)).$$

Si $x, y \in \operatorname{Ob} J$, c'est l'identité, si $x, y \in I$, c'est f. Si $x \in I$, $y \in J$, on doit définir une flèche

$$\operatorname{Hom}(f(x), y) \longrightarrow \operatorname{Hom}(f(x), y),$$

et on prend encore l'identité. On a

$$\begin{array}{ccccc} J & \xrightarrow{\ j\ } & K(f) & \xleftarrow{\ i\ } & I \\ & \searrow & \downarrow{\scriptstyle \bar{f}} & & \\ & & J & & \end{array}$$

$$\begin{cases} \bar{f} \circ j = \mathrm{id}_J \\ j \circ \bar{f} = \begin{cases} j & \text{sur } J \\ jf & \text{sur } I, \end{cases} \end{cases} \qquad \text{i.e.} \quad \boxed{\begin{array}{ll} j\bar{f}j = j & \text{sur } J \\ j\bar{f}i = jf & \text{sur } I \end{array}},$$

et on a $[K = K(f)]$

$$\begin{aligned} \alpha :\ & \mathrm{id}_K \longrightarrow j\bar{f} \qquad\qquad \text{i.e. homomorphisme fonctoriel} \\ & x \xrightarrow{\alpha(x)} j(\bar{f}(x)), \quad x \in \operatorname{Ob} K, \end{aligned}$$

défini ainsi :

p. 6

$$\begin{cases} \text{Si } x \in J,\ \alpha(x) = \mathrm{id}_x \ :\ x \xrightarrow{\sim} j(\bar{f}(x)) = j(x) = x \\ \text{Si } x \in I,\ \alpha(x) : \underbrace{x \longrightarrow f(x)}_{\text{dans } K(f)},\ \text{ou } \underbrace{f(x) \longrightarrow f(x)}_{\text{dans } J},\ \alpha(x) = \mathrm{id}_{f(x)}. \end{cases}$$

On doit vérifier la fonctorialité pour x variable dans K …

Commentary · Exposition [ms. 79]

The collage records a relative morphism

Grothendieck introduces $K(f)$ to turn a morphism between coefficients on two different bases into an ordinary diagram on one base. This is a categorical form of the mapping cylinder, or more precisely the collage of the profunctor $\operatorname{Hom}_J(f(-), -)$. The construction encodes the direction of the relative morphism in its mixed arrows.

For clarity, distinguish the copy of I from the copy of J, even when they have the same objects. Internal arrows are those of I and J; a mixed arrow $x \to y$, with $x \in I, y \in J$, is an arrow $f(x) \to y$ in J; there are no mixed arrows in the other direction. If $a : x' \to x$ lies in I, $b : y \to y'$ lies in J, and $\beta : f(x) \to y$ represents a mixed arrow, composition is

$$\beta \circ a = \beta \circ f(a), \qquad b \circ \beta = b\beta$$

in J. Associativity follows directly from associativity in I, J and functoriality of f. These formulas also prove that I is a sieve and J is its complementary cosieve.

Here is the universal property in full. Given presheaves $F : I^o \to \mathcal{M}$, $G : J^o \to \mathcal{M}$, and a natural transformation $u : f^*G \to F$, define $\Phi : K(f)^o \to \mathcal{M}$ to equal F and G on the two pieces. On the mixed arrow represented by $\beta : f(x) \to y$, set

$$\Phi(\beta) = \big(G(y) \xrightarrow{G(\beta)} G(f(x)) \xrightarrow{u_x} F(x)\big).$$

Naturality of u proves compatibility with composition on the I-side; functoriality of G proves compatibility on the J-side. Conversely, restriction gives F, G, and evaluation on the mixed arrow $x \to f(x)$ represented by $\mathrm{id}_{f(x)}$ gives u_x. The two constructions are inverse.

A morphism between triples is a pair $r : F \to F'$, $s : G \to G'$ with $r\,u = u'\,f^*s$. This equation is precisely naturality on the distinguished mixed arrows, and hence on every mixed arrow by the formula above. We have therefore obtained an isomorphism of categories, natural in $\mathcal{M}$:

$$\mathrm{Fun}(K(f)^o, \mathcal{M}) \cong \{(F, G, u : f^*G \to F)\}.$$

The isolated superscript o on the category of triples on manuscript page 82 is inconsistent with the preceding displayed universal property if the same variance is retained. One may either omit it in the modern formulation or dualize both sides. The printed reading is preserved in the transcription and recorded in the corrections table.

The retraction $\bar f : K(f) \to J$ and transformation $\alpha : \mathrm{id} \to j\bar f$ need no additional coherence data. In fact $\bar f \dashv j$: for $x \in K(f), y \in J$, the equality $\mathrm{Hom}_K(x, jy) = \mathrm{Hom}_J(\bar f x, y)$ follows from the defining cases. Its unit is precisely α, and its counit is the identity. This proves the naturality left for verification on page 80 and explains why the identities involving $\bar f$ fit together.

Soit Φ un $\mathcal{M}$-préfaisceau sur K. Il définit

$$\begin{aligned} F &= i^*(\Phi) \quad \in \mathrm{Ob}\,\mathbf{D}_{\mathcal{M}}(I) \\ G &= j^*(\Phi) \quad \in \mathrm{Ob}\,\mathbf{D}_{\mathcal{M}}(J). \end{aligned}$$

D'autre part $f^*(G)$ s'interprète comme $(j\bar f)^*(\Phi)|I$, car

$$\begin{aligned} f^*(G)(i) &= G(\underbrace{f(i)}_{=\bar f(i) \in J}) \\ &= j^*(\Phi)(\bar f(i)) \\ &= \Phi(j\bar f(i)) \\ &= (j\bar f)^*(\Phi)(i). \end{aligned}$$

Donc

$$f^*(G) \simeq (j\bar f)^*(\Phi)|I. \qquad \text{[page 81]}$$

Or la flèche

$$\alpha : \mathrm{id}_K \longrightarrow j\bar f$$

donne

$$\underbrace{\mathrm{id}_K(\Phi)}_{=\Phi} \xleftarrow{\Phi(\alpha)} (j\bar f)^*(\Phi), \qquad \text{fonctoriel en } \Phi,$$

d'où, en prenant les restrictions à I,

$$i^*(\Phi) \overset{\text{déf}}{=} F \xleftarrow{u} f^*G.$$

En fait, sans tellement se tortiller, on devrait pouvoir montrer que pour toute catégorie $\mathcal{M}$, la catégorie $\underline{\mathrm{Hom}}(K^o, \mathcal{M})$ des $\mathcal{M}$-préfaisceaux sur K est isomorphe à celle des triples (F, G, u), avec $F \in \mathrm{Ob}\,\underline{\mathrm{Hom}}(I^o, \mathcal{M})$, $G \in \underline{\mathrm{Hom}}(J^o, \mathcal{M})$, et $u : f^*(G) \longrightarrow F$. Ainsi, K apparaît comme solution d'un problème universel

$$\underline{\mathrm{Hom}}(K^o, \mathcal{M}) \longrightarrow \{F,\ G,\ u \text{ comme dessus}\}(\mathcal{M}).$$

Mais pour y préciser la flèche $\longrightarrow$ de façon conceptuelle simple, il est bon de se donner K avec les structures supplémentaires $I \underset{\bar{f}}{\overset{j}{\rightleftarrows}} K \xleftarrow{\ i\ } I$, $\bar{f}j = \mathrm{id}_J$, $\mathrm{id}_K \xrightarrow{\alpha} j\bar{f}$. Et ce système p. 7
$\{K,\ i,\ j,\ \bar{f},\ \alpha\}$ est (sauf erreur) défini à isomorphisme canonique près par la condition de [page 82]
représenter le foncteur $\mathcal{M} \longmapsto \{F,\ G,\ u\}^o(\mathcal{M})$ …

En particulier, le système défini par $G \in \mathrm{Ob}\,\mathbf{D}_{\mathcal{M}}(J)$

$$(f^*(G),\ G,\ \mathrm{id} : f^*(G) \longrightarrow f^*(G))$$

est défini sauf erreur par $\bar{f}^*(G)$ sur $K(f)$. On se propose [?] de trouver une interprétation similaire pour

$$(F,\ f_*(F),\ f^*f_*F \longrightarrow F)$$

($F \in \mathrm{Ob}\,\mathbf{D}(I)$). Le candidat naturel serait ici $\Phi = i_*(F)$. Admettons ce point (i.e. surtout la vérification du fait que $j^*(i_*(F)) \simeq f_*(F)$). Tout ceci montre que si $\mathbf{D}$ est à nouveau un dérivateur quelconque, on peut, par les formules précédentes, trouver un objet canonique Φ de $\mathbf{D}(K)$, en termes soit d'un $\mathbf{D}$-coefficient F sur I (en prenant $\Phi = i_*(F)$), soit d'un $\mathbf{D}$-coefficient G sur J (en prenant $\Phi = \bar{f}^*(G)$). D'autre part, on peut espérer définir canoniquement

$$\mathbf{D}(K) \longrightarrow \mathbf{D}(I \times \mathbf{\Delta}_1)$$
$$\mathbf{D}(K) \longrightarrow \mathbf{D}(J \times \mathbf{\Delta}_1)$$

dont les composés avec les foncteurs 'squelettes'

$$\mathbf{D}(I \times \mathbf{\Delta}_1) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{\Delta}_1^o, \mathbf{D}(I))$$
$$\mathbf{D}(J \times \mathbf{\Delta}_1) \longrightarrow \underline{\mathrm{Hom}}(\mathbf{\Delta}_1^o, \mathbf{D}(J))$$

redonnent (pour $\Phi \in \mathrm{Ob}\,\mathbf{D}(K)$) les flèches squelettiques [page 83]

$$\begin{cases} f^*(G) \xrightarrow{u} F \\ G \xrightarrow{v} f_*F \end{cases}$$

associées à (F, G, u). De façon précise, $f^*(G) \longrightarrow F$ se définit en termes de $\mathbf{D}$, et $\Phi \in \mathrm{Ob}\,\mathbf{D}(K)$, comme ci-dessus, et v s'en déduit par adjonction.

Commentary · Exposition [ms. 82]

The missing Kan extension comparison

Grothendieck's candidate i_*F is correct, and the comparison $j^*i_*F \cong f_*F$ follows directly from the geometry of the mixed arrows. This is the point at which the collage begins to do mathematical work: a right Kan extension along an arbitrary functor is recovered from one along an inclusion.

For $y \in J$, the comma category (i/jy) has objects $x \in I$ together with arrows $i(x) \to j(y)$ in $K(f)$. By definition, such an arrow is an arrow $f(x) \to y$ in J. Morphisms are the same commuting triangles in both descriptions. Hence there is a canonical isomorphism

$$(i/jy) \cong (f/y)$$

commuting with the projections to I, naturally in y. The pointwise right Kan extension axiom in the presheaf convention computes both $(i_*F)_{j(y)}$ and $(f_*F)_y$ as the homotopy limit of F over $(f/y)^o$. The comparison is therefore an isomorphism at every y, and Der 2 makes it an isomorphism in $\mathbf{D}(J)$.

There is a precise canonical map behind this pointwise calculation. The transformation $i \to jf$ induced by α yields

$$f^*j^*i_*F \longrightarrow i^*i_*F \cong F.$$

Its mate is $j^*i_*F \to f_*F$. The comma-category identification proves that this mate is the isomorphism just computed. Consequently the displayed arrow on I, after that identification, is exactly the counit

$f^*f_*F \to F$; agreement on objects is not being used as a substitute for agreement of the relevant natural transformations.

The other candidate is immediate but equally canonical. If $\Phi = \bar{f}^*G$, then $j^*\Phi = G$, $i^*\Phi = f^*G$, and the transformation on I is the identity, since $\bar{f}(\alpha_x) = \mathrm{id}_{\bar{f}(x)}$. These assertions hold for a derivator whenever the required indexing shapes are permitted. They use only restriction, right Kan extension, and the prederivator action on natural transformations.

In modern language the comma square expressing this argument is homotopy exact. The general calculus of such squares, with the comparison specified as a mate, is developed in Groth (2013, Proposition 1.26); the relevant earlier convention was fixed in the commentary on Chapter I, manuscript p. 31 (*The base-change map as a mate*). The argument above completes the local proof rather than appealing to a stronger base-change statement with unchecked hypotheses.

Commentary · Development (commentators') [ms. 83]

Completing both coherent adjunction arrows

Grothendieck leaves the chapter just after finding the characteristic map $K(f) \to [1]$. His construction can be completed explicitly. The point is to use that map together with the retraction to J, rather than to select a coherent representative of an arbitrary arrow in a homotopy category. We give the completion because it explains why both members of an adjoint pair can be treated by the same category.

Collage completion. [ours] Let $\mathbf{D}$ be a derivator in the presheaf convention, and let $f : I \to J$. Assume its diagram category contains $K = K(f)$, the products $I \times [1]$, $J \times [1]$, and the comma categories needed for the indicated right Kan extensions. There are functors

$$H : \mathbf{D}(K) \longrightarrow \mathbf{D}(I \times [1]), \qquad V : \mathbf{D}(K) \longrightarrow \mathbf{D}(J \times [1])$$

whose skeletal arrows are, respectively, $u : f^*G \to F$ and its adjoint $v : G \to f_*F$, for $F = i^*\Phi$, $G = j^*\Phi$, and u induced by α.

Proof. Define $h : I \times [1] \to K$ by

$$h(x,0) = i(x), \qquad h(x,1) = j(f(x)).$$

On an arrow $a : x \to x'$ at level 0 use $i(a)$, and at level 1 use $j(f(a))$. A mixed arrow $(x,0) \to (x',1)$, determined by a, is sent to the mixed arrow represented by $f(a)$. The definition of composition in the collage shows that this is a functor. Its vertical arrow at x is $\alpha_{i(x)}$. Therefore $H = h^*$ has restrictions F, f^*G, and its arrow in the presheaf direction is exactly $u : f^*G \to F$. This proves the first assertion without any Kan extension.

To construct the adjoint arrow, let $\chi : K \to [1]$ be 0 on I and 1 on J, and set

$$q = (\bar{f}, \chi) : K \longrightarrow J \times [1], \qquad V = q_*.$$

For $y \in J$, the category $(q/(y,0))$ contains only objects from I; its definition identifies it with (f/y). Thus

$$(q_*\Phi)|_{J\times\{0\}} \cong f_*F.$$

The category $(q/(y,1))$ has the terminal object given by $j(y)$ and the identity of $(y,1)$. Indeed every object of that comma category specifies precisely one arrow to $j(y)$, either an internal arrow of J or the mixed arrow represented by its J-component. In the presheaf convention the right Kan formula takes a homotopy limit over the opposite of this comma category, so the terminal object becomes an initial object and the limit is evaluation there. Consequently

$$(q_*\Phi)|_{J\times\{1\}} \cong G.$$

These comparisons are natural, and their pointwise invertibility implies invertibility as coefficient diagrams.

It remains to identify the arrow, since its two endpoints alone would not determine it. Put $E = q_*\Phi$, let $t : E_1 \to E_0$ be its skeletal arrow, and use the counit $\varepsilon : q^*E \to \Phi$. Its restriction to J gives an isomorphism $\lambda : E_1 \to G$. Its restriction to I gives $f^*E_0 \to F$; let $\theta : E_0 \to f_*F$ be its

mate. The preceding comma calculations say exactly that θ is an isomorphism. Naturality of ε with respect to the mixed arrows $x \to f(x)$ gives

$$(f^{*}E_{0} \longrightarrow F) \circ f^{*}t = u \circ f^{*}\lambda.$$

After substituting the definitions of θ and the counit of $f^{*} \dashv f_{*}$, this says that $\theta\, t\, \lambda^{-1} : G \to f_{*}F$ has adjoint u. The adjunction bijection identifies it with v. This proves the second assertion and the proposition. □

Taking $\Phi = \bar{f}^{*}G$, the first arrow is the identity on $f^{*}G$, so the second is the unit $G \to f_{*}f^{*}G$. Taking $\Phi = i_{*}F$, the first is the counit $f^{*}f_{*}F \to F$, while the second is the identity on $f_{*}F$. Hence both adjunction transformations have canonical coherent realizations, natural in their coefficients. Applying the construction after a shift by A gives the corresponding statement in every $\mathbf{D}(A)$; the compatibility with restriction in A follows from the product-square base change already used in Chapter III.

No strongness axiom was used. The input was the coherent Φ, and for the unit and counit that input was itself constructed by a specified Kan extension or restriction. The result does not supply a canonical section of the skeleton functor on all arrows.

The collage and the mate calculus are standard constructions. We checked the treatments of homotopy exact squares in Groth (2013, Section 1), enriched derivator adjunctions in Groth et al. (2014b, Sections 8–9), and pointwise Kan extension through modules in Riehl and Verity (2017, Section 5). We did not find in these sources the precise pair h^{*}, q_{*} written as a completion of this unfinished chapter. The marker above identifies our explicit completion and proof; it makes no claim of priority.

L'idée naturelle ici, serait de ramener le problème de l'étoffement des flèches d'adjonction sqelettiques, au cas des immersions ouvertes ou fermées. Notons que les sous-catégories ouvertes de K correspondent justement aux foncteurs $K \longrightarrow \mathbf{\Delta}_1$, en prenant les images inverses de l'ouvert générique. Il y a donc un unique

$$\varphi :\ K(f) \longrightarrow \mathbf{\Delta}_1$$

avec $I = \varphi^{-1}(\{0\})$. Donc `[fin du chapitre]`

Commentary · Examples [ms. 83]

A diagonal and its two projections

Grothendieck's geometric construction can be seen completely in the case where $I = \{a, b\}$ is discrete, $J = e$, and f is the unique functor. Here $K(f)$ is the cospan $a \to * \leftarrow b$. Its presheaf coefficients have the opposite shape: an object Φ is a coherent pair of arrows from G to F_a, F_b.

The right Kan extension $f_{*}F$ is the product $F_a \times F_b$ in $\mathbf{D}(e)$. Indeed Der 1 identifies $\mathbf{D}(I)$ with $\mathbf{D}(e) \times \mathbf{D}(e)$; the right adjoint to the diagonal restriction is the categorical product. The map $q : K(f) \to [1]$ sends a, b to 0 and $*$ to 1. Its right Kan extension sends the coherent pair to the single coherent arrow

$$G \longrightarrow F_a \times F_b$$

whose two components are the original arrows. This proves directly, in this example, that the second construction is the adjoint of the first.

For $\Phi = \bar{f}^{*}G$, both original arrows are identities. The output is the diagonal $G \to G \times G$, as a specified object of $\mathbf{D}([1])$. For $\Phi = i_{*}F$, the value at $*$ is $F_a \times F_b$, and restriction along h gives the two projection arrows. The same collage has therefore produced the unit and counit of the diagonal-product adjunction, including their coherent representatives.

The empty case checks the direction of every construction. If $I = \varnothing$, then $K(f) = J$, q is the level-1 inclusion, and $q_{*}\Phi$ has value Φ at level 1 and the terminal coefficient at level 0. Its presheaf arrow is $\Phi \to 1$, the unit of the adjunction between restriction to the empty category and its right Kan extension. Interchanging the two endpoints would give the wrong arrow even in this simplest case.

Chapter V

Catégories de chemins et localisation

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p.

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre V

Catégories de chemins et localisation

p.

Ce texte a été déchiffré et transcrit en LaTeX 2$_\varepsilon$ par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 La catégorie des chemins

p. 1
[page 1]

Soit Φ la catégorie suivante :

$$\begin{cases} \operatorname{Ob}\Phi &= \mathbf{Z} \\ \operatorname{Fl}\Phi &= \operatorname{Ob}\Phi \sqcup \mathbf{Z} \qquad (\operatorname{Ob}\Phi \leftrightarrow \text{identités}) \end{cases}$$

On note $\{\alpha\}$, ou simplement α, l'objet qui correspond à $\alpha \in \mathbf{Z}$. On note u_α la flèche qui correspond à l'entier $\alpha \in \mathbf{Z}$.

$$\begin{aligned} s(u_\alpha) &= \alpha \qquad \text{si } \alpha \text{ pair}, \quad \alpha+1 \quad \text{si } \alpha \text{ impair} \\ b(u_\alpha) &= \alpha+1 \qquad \text{si } \alpha \text{ pair}, \quad \alpha-1 \quad \text{si } \alpha \text{ impair} \end{aligned}$$

[plutôt $b(u_\alpha) = \alpha+1$ si α pair, α si α impair]

Il n'y pas à définir de compositions, car si deux flèches sont composables, elles sont l'identité. En fait, Φ est une catégorie *ordonnée*, $\{\alpha\} < \{\beta\}$ si et seulement si α pair, $\beta = \alpha+1$ [plutôt $\beta = \alpha+1$ ou $\beta = \alpha-1$].

On note, pour $n \in \mathbf{N}$,

$$\Phi_n \subseteq \Phi, \quad \operatorname{Ob}\Phi_n = \{\alpha \in \operatorname{Ob}\Phi \mid -n \le \alpha \le +n\},$$

la sous-catégorie pleine de Φ dont les objets sont les $\{\alpha\}$ avec $-n \le \alpha \le +n$. Ainsi

$$\Phi_1 = \{-1 \longleftarrow 0 \longrightarrow 1\},$$

la duale est

$$\Psi_1 = \Phi_1^o = \{-1 \longrightarrow 0 \longleftarrow 1\}.$$

On peut considérer Φ comme somme amalgamée d'une suite ordonnée infinie (de deux côtés) de copies de Φ_1, ou aussi d'une suite similaire de copies de Ψ_1.

$$\cdots \quad \overbrace{\longleftarrow -4 \longrightarrow -3 \longleftarrow -2 \longrightarrow -1 \longleftarrow 0 \longrightarrow 1 \longleftarrow 2 \longrightarrow 3 \longleftarrow 4 \longrightarrow}^{\Phi_1 \quad \Phi_1 \quad \Phi_1 \quad \Phi_1 \quad \Phi_1} \quad \cdots$$
$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}_{\Psi_1 \quad \Psi_1 \quad \Psi_1 \quad \Psi_1}$$

Commentary · Exposition [ms. 1]

A path remembers the arrows that will be inverted

Grothendieck is preparing a geometric description of localization. A morphism in a category of fractions is represented by a finite word of arrows, some traversed in their original direction and some traversed backwards. The alternation imposed here is a way of giving all such words a common shape. Inserting identity arrows makes any finite zigzag alternating; adding identities at its ends gives an interval symmetric about zero. The identities are bookkeeping, whereas the arrows traversed backwards carry the mathematical restriction imposed by the chosen class of denominators.

After the edition's correction of the target of u_α, the category Φ is the poset on $\mathbf{Z}$ whose only strict relations are

$$2i < 2i-1, \qquad 2i < 2i+1.$$

Every even vertex is a source and every odd vertex a sink. There is no composite of two nonidentity arrows. The sentence saying that two composable arrows "sont l'identité" must therefore mean that at least one of them is an identity. Composing an identity with an arbitrary arrow supplies the elementary exception to its literal wording.

The subcategory Φ_n has $2n$ edges, not n. Thus the manuscript's later references to the parity of a path concern the index n, or half its number of edges. This distinction explains the repair on manuscript page 14. A path of index 1 has shape $\cdot \leftarrow \cdot \rightarrow \cdot$; two such paths joined end to end have

the opposite orientation from Φ_2. The correct assertion at the end of the present page is that Φ_n, for even $n > 0$, is assembled from copies of $\Psi_1 = \Phi_1^o$. The printed Ψ_n has the wrong orientation. We keep that reading in the source and use the corrected statement in the mathematics.

The nerve of every finite Φ_n is an ordinary subdivided interval. Nevertheless the category itself, including the directions of its arrows, matters: a functor from it to X is an actual zigzag in X, and its natural transformations are actual commutative ladders. Passing immediately to the geometric interval would discard precisely the information needed to distinguish denominators from the other arrows.

Les Φ_n, pour n impair, sont de même somme de copies de Φ_1 mises bout à bout, et les Ψ_n, pour n pair > 0, somme de copies de Ψ_1.

Les Φ_n forment un système inductif avec foncteurs de transition pleinement fidèles, dont Φ est la limite inductive p. 2

$$\Phi = \varinjlim_n \Phi_n.$$

Mais pour la suite on va faire des Φ_n un système *projectif*, en définissant pour tout $n \leq m$ une *rétraction* $\varphi_{n,m}$ de Φ_m sur Φ_n, par

[page 2]

$$\left\{\begin{array}{l} \varphi_{n,m} : \Phi_m \longrightarrow \Phi_n \subset \Phi_m \\ \varphi_{n,m}(\{\alpha_i\}) = \left\{\begin{array}{lcll} \alpha_i & = & i & \in \operatorname{Ob}\Phi_n, \quad \text{i.e.} \quad -n \leq i \leq n \\ \alpha_i & = & n & \text{si} \quad n \leq i \leq m \\ \alpha_i & = & -n & \text{si} \quad -m \leq i \leq -n \end{array}\right. \end{array}\right.$$

[plutôt

$$\begin{array}{rcl} \varphi_{n,m} : \Phi_m & \longrightarrow & \Phi_n \subset \Phi_m \\ \varphi_{n,m}(\{i\}) & = & \left\{\begin{array}{lll} \{i\} & \texttt{si } i \in \operatorname{Ob}\Phi_n, & \texttt{i.e} \quad -n \leq i \leq n \\ \{n\} & & \texttt{si} \quad n \leq i \leq m \\ \{-n\} & & \texttt{si} \quad -m \leq i \leq -n \texttt{]} \end{array}\right. \end{array}$$

On note le système projectif d'ensembles ordonnés ainsi obtenu par

$$\boldsymbol{\Phi} = (\Phi_n)_{n \in \mathbf{N}} \in \operatorname{Pro}(\mathrm{Cat}).$$

Soit X une catégorie, alors pour $n \in \mathbf{N}$

$$\underline{\mathrm{Ch}}_n(X) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(\Phi_n, X)$$

est la catégorie des diagrammes de X de la forme

$$(*) \qquad A_{-n} \cdots \longleftarrow A_{-2} \xrightarrow{u_{-2}} A_{-1} \xleftarrow{u_{-1}} A_0 \xrightarrow{u_0} A_1 \xleftarrow{u_1} A_2 \xrightarrow{u_2} A_3 \longleftarrow \cdots A_n$$

avec $2n$ flèches bout à bout u_i $(-n \leq i \leq n)$ [plutôt $(-n \leq i \leq n-1)$], deux flèches consécutives allant toujours en sens opposé,

$$\text{soit} \ \cdot \longrightarrow \cdot \longleftarrow \cdot, \quad \text{soit} \ \cdot \longleftarrow \cdot \longrightarrow \cdot.$$

Le foncteur canonique déduit de $\varphi_{n,m} : \Phi_m \longrightarrow \Phi_n$

$$\varphi_X^{m,n} : \underline{\mathrm{Ch}}_n(X) \longrightarrow \underline{\mathrm{Ch}}_m(Y), \qquad n \leq m,$$

[plutôt $\underline{\mathrm{Ch}}_m(X)$] est le 'foncteur de dégénérescence', consistant à compléter un diagramme $(*)$, par $m-n$ termes A_{-n} avec des identités à gauche, et par $m-n$ termes $[A_n]$ avec des identités à droite, donc pour n pair on trouve p. 3

$$\underbrace{A_{-n} \cdots \xrightarrow{\mathrm{id}} A_{-n}}_{m-n \text{ termes}} \xleftarrow{\mathrm{id}} A_{-n} \longrightarrow A_{-(n-1)} \cdots\cdots A_{n-1} \longleftarrow A_n \xrightarrow{\mathrm{id}} \underbrace{A_n \xleftarrow{\mathrm{id}} A_n \longrightarrow \cdots A_n}_{m-n \text{ termes}}.$$

C'est un foncteur pleinement fidèle (ce qui correspond au fait que $\varphi_{n,m}$ est surjectif sur les objets et sur les flèches.

On pose

[page 3]

$$\underbrace{\underline{\mathrm{Ch}}_\infty(X)}_{\text{ou simplement } \underline{\mathrm{Ch}}(X)} = \underline{\mathrm{Hom}}(\boldsymbol{\Phi}, X) \overset{\text{déf}}{=} \varinjlim_n \underline{\mathrm{Ch}}_n(X)$$

([3]) ([4]). On a une inclusion pleinement fidèle

$$\underline{\mathrm{Ch}}_\infty(X) \hookrightarrow \underline{\mathrm{Hom}}(\Phi, X),$$

où la deuxième catégorie est celle des diagrammes infinis (dans les deux directions)

$$\cdots \longleftarrow A_{-2} \xrightarrow{u_{-2}} A_{-1} \xleftarrow{u_{-1}} A_0 \xrightarrow{u_0} A_1 \xleftarrow{u_1} A_2 \xrightarrow{u_2} \cdots$$

et où $\underline{\mathrm{Ch}}_\infty(X)$ s'identifie à la sous-catégorie pleine formées des objets pour lesquels les flèches u_i sont des identités, pour $|i|$ assez grand.

On appellera $\underline{\mathrm{Ch}}_\infty(X)$ la *catégorie des chemins* dans X, ses objets seront les chemins dans X. Ceux qui proviennent de $\underline{\mathrm{Ch}}_n(X)$ seront appelés *chemins standards de longueur* $2n$. (Le terme 'standard' s'applique ici à nos conventions de notations. Ainsi nous ne considérons que des chemins de longueur paire, et pour une longueur déterminée, nous choisissons l'une parmi les deux espèces raisonnables, par exemple pour les chemins de longueur 2 nous avons pris seulement

$$A_{-1} \longleftarrow A_0 \longrightarrow A_1$$

et non pas

$$A_{-1} \longrightarrow A_0 \longleftarrow A_1.)$$

Commentary · Exposition [ms. 3]

The pro-interval and the eventually constant diagram

Grothendieck wants paths of arbitrary finite length without making their length part of the final object. He obtains them by extending a finite diagram with identity arrows at both ends. The projective system of intervals and the inductive system of path categories describe the same operation from opposite sides.

Write $r_{n,m} : \Phi_m \to \Phi_n$ for the numerical clamping map

$$r_{n,m}(i) = \max(-n, \min(i, n)), \qquad n \le m.$$

It is a functor: an edge inside $[-n, n]$ is unchanged, and every other edge is sent to an identity. The equality $r_{n,m} r_{m,l} = r_{n,l}$ is literal, so precomposition gives a strict inductive system. Its transition functors are fully faithful, for a reason slightly more precise than mere surjectivity on objects and arrows. If $F, G : \Phi_n \to X$ and $\eta : F r_{n,m} \to G r_{n,m}$, naturality on each collapsed edge says that the two adjacent components of η agree. They therefore agree throughout either constant tail. Restriction to Φ_n determines η uniquely and yields an arbitrary natural transformation $F \to G$. This proves the required bijection on Hom sets.

Surjectivity alone would not prove this assertion for an arbitrary functor. The map from a discrete two-point category to the point is surjective on objects and arrows, while precomposition gives the diagonal $X \to X \times X$, which need not be full. What makes clamping work is the chain of edges sent to identities inside each fibre.

The filtered colimit is consequently realized as the full subcategory of $\mathrm{Fun}(\Phi, X)$ consisting of diagrams whose edges are identities outside a finite interval. A natural transformation between two such diagrams also becomes constant in both tails: choose one interval containing the nonidentity edges of both diagrams and apply naturality to the identity edges beyond it. Thus both objects and morphisms descend to a common finite stage. No assertion about commuting an arbitrary functor category with filtered colimits is being used here.

The notation $\mathrm{Fun}(\mathbf{\Phi}, X)$ is a convenient enriched version of the usual pro-object formula: it means $\mathrm{colim}_n \mathrm{Fun}(\Phi_n, X)$ as a category. Its set of objects is not a space of arbitrary infinite paths. The eventual identity condition is essential, and differs even from eventual invertibility.

Reflection $i \mapsto -i$ is the unique nonidentity automorphism of Φ_n when $n > 0$, since an automorphism of the underlying finite path is either the identity or reflection and both preserve the alternating orientation. For $n = 0$ the automorphism group is trivial. This harmless boundary case will also matter when the two endpoints coincide.

Je me rends compte que mes notations u_i ne sont pas commodes pour la dualité, je vais les changer, en u_i pour les flèches entre objets $\{i\}$ avec $i \geq 0$, et v_i la flèche symétrique de u_i par rapport à l'origine. (Le groupe des automorphismes des Φ_n est réduit au groupe à deux éléments, le seul automorphisme non trivial étant la symétrie $\{i\} \longrightarrow \{-i\}$. Donc nos chemins seront notés désormais

$$(*) \qquad A_{-2} \xrightarrow{v_1} A_{-1} \xleftarrow{v_0} A_0 \xrightarrow{u_0} A_1 \xleftarrow{u_1} A_2 \longrightarrow \cdots .$$

(Ce n'est pas non plus enthousiasmant …) p. 4

Soit

$$\varepsilon = \{-1, +1\}$$

l'ensemble ordonné *discret,* réduit aux deux éléments -1, $+1$, et définissons des foncteurs d'inclusion (pleinement fidèles)

$$\alpha_n : \varepsilon \longrightarrow \Phi_n \quad \alpha_n(-1) = -n \qquad \alpha_n(1) = n,$$

ces foncteurs sont compatible avec les $\varphi_{n,m}$, d'où

$$\alpha_\infty : \varepsilon \longrightarrow \mathbf{\Phi} \in \mathrm{ProCat},$$

d'où pour tout X un foncteur

$$\kappa : \underline{\mathrm{Hom}}(\mathbf{\Phi}, X) \longrightarrow \underline{\mathrm{Hom}}(\varepsilon, X) \simeq X \times X$$

i.e.

[page 5]

$$\underline{\mathrm{Ch}}(X) \longrightarrow X \times X \qquad c \longrightarrow (\sigma(x), \beta(x)).$$

[plutôt $(\sigma(c), \beta(c))$] Pour tout chemin c dans X i.e. un diagramme $(*)$ (satisfaisant à la condition de dégénérescence à l'infini), on a une source ou origine du chemin, et un but ou une *extrémité*

$$\begin{cases} \sigma(x) = A_i & \text{pour } i \leq -\mathrm{long}(x) \\ \beta(x) = A_i & \text{pour } i \geq \mathrm{long}(x) \end{cases}$$

[remplacer x par c; pour long(c) défini convenablement]. La fibre du foncteur précédent en x, y sera notée

$$\underline{\mathrm{Ch}}(X; x, y) \qquad \text{ou simplement } \underline{\mathrm{Ch}}(x, y),$$

catégorie des chemins de x à y.

Commentary · Examples [ms. 5]

Fixing the endpoints also fixes the vertical arrows

Grothendieck's source–target functor allows a path to become a candidate morphism from a specified object to another. Its fibre over (x, y) is a strict categorical fibre. An object is an eventually constant zigzag with tails x and y; a morphism is a natural transformation whose two tail components are 1_x and 1_y. This last condition is indispensable for the interpretation by a Hom set.

There are two different full-subcategory statements in these pages. The category of Σ-paths is full in the category of all paths, because it imposes a condition only on the objects. Likewise its fixed-endpoint fibre is full in the fixed-endpoint fibre of all paths. But a fixed-endpoint fibre is generally not full in the entire functor category. On pages 15–16 and 24 the expressions "sous-catégorie pleine"

3. L'ensemble des objets de cette catégorie se notera $\mathrm{Ch}_\infty(X)$ ou $\mathrm{Ch}(X)$, de même on définit les $\mathrm{Ch}_n(X)$.
4. dans cette formule: $\mathbf{\Phi}$ procatégorie

must be read with this correction: the endpoint components of a natural transformation are required to be identities.

For a concrete test take the one-object category BG of a nontrivial group G, and let Σ contain only the identity. The paths representing 1 and $g \neq 1$ may be written, at index 2,

$$(1,1,1,1), \qquad (g,1,1,1),$$

where the entries label the four alternating arrows. There is a natural transformation from the first diagram to the second with vertex components $(1,g,g,g,g)$. It satisfies all four naturality squares, but its final component is g, not the identity. Thus it is not a morphism in the strict fibre. If one used the full subcategory on objects with the prescribed endpoints, this transformation would identify 1 with g on connected components, although localization at identities leaves BG unchanged.

For an ordinary arrow category, the same convention makes the fibre over (x,y) discrete: a morphism from $f : x \to y$ to $g : x \to y$ with both vertical sides identities exists precisely when $f = g$, and is then unique. This is the precise assertion used on page 8.

The map $\varepsilon \to \Phi_n$ is fully faithful for $n > 0$. At $n = 0$ it sends both endpoint labels to the same object and is not fully faithful. It still defines the source–target functor, whose index-zero stage is the diagonal $X \to X \times X$. Consequently a path of index zero occurs in the fibre only when $x = y$, exactly as the identity construction requires.

p. 5

2 Calcul d'une catégorie de fractions $X\Sigma^{-1}$

Avant de continuer le formalisme des chemins (composition etc.), je vais tester l'utilité de la notion de catégorie des chemins, pour le 'calcul' des $X\Sigma^{-1}$, où

$$\Sigma \subset \mathrm{Fl}(X), \qquad \Sigma \text{ contient les identités}$$

est une partie autrement *quelconque* de $\mathrm{Fl}(X)$. On sait que $\mathrm{Ob}\, X\Sigma^{-1} = \mathrm{Ob}\, X$, et tout revient à définir les

$$\mathrm{Hom}_{X\Sigma^{-1}}(x,y), \qquad x,y \in \mathrm{Ob}\, X,$$

et pour $x,y,z \in \mathrm{Ob}\, X$ la composition [page 6]

$$\mathrm{Hom}_\Sigma(x,y) \times \mathrm{Hom}_\Sigma(y,z) \xrightarrow{(u,v)\mapsto vu} \mathrm{Hom}_\Sigma(x,z),$$

(où on note Hom_Σ au lieu de $\mathrm{Hom}_{X\Sigma^{-1}}$, pour alléger les notations).

Grâce à Σ, on définit une sous-catégorie pleine

$$\underline{\mathrm{Ch}}_\Sigma(X) \hookrightarrow \underline{\mathrm{Ch}}(X),$$

([3]) par la condition sur $c \in \mathrm{Ch}(X)$: Pour i impair, `[plutôt pair]` i.e. $i-1 \xleftarrow{u_i} i$ (u_i est dite 'rétrograde'), $c(u_i) \in \Sigma$, i.e. les 'tronçons' rétrogrades dans le chemin c doivent être dans Σ. On définit de même les sous-catégories pleines

$$\underline{\mathrm{Ch}}_\Sigma(X;x,y) = \{c \in \underline{\mathrm{Ch}}(X;x,y) \mid c \in \underline{\mathrm{Ch}}_\Sigma(X)\},$$

ou encore la fibre en (x,y) du foncteur source-but pour $\underline{\mathrm{Ch}}_\Sigma(X)$

$$\underline{\mathrm{Ch}}_\Sigma(X) \longrightarrow X \times X.$$

3. L'ensemble de ses objets est noté $\mathrm{Ch}(X)$. Ses éléments sont les Σ-chemins de x à y `[de x à y?]`.

Commentary · Exposition [ms. 6]

Why a commutative ladder gives equality after localization

Grothendieck places no calculus-of-fractions hypothesis on Σ: it is any set of arrows containing the identities. Its closure under composition is unnecessary for the path construction. The hypothesis concerns only the backwards horizontal arrows; the vertical arrows of a morphism of paths may be arbitrary arrows of X.

To see why this is sufficient, let $Q : X \to X[\Sigma^{-1}]$ be the localization and consider a square around a backwards edge,

$$\begin{array}{ccc} A_{i-1} & \xleftarrow{\ s\ } & A_i \\ {\scriptstyle a_{i-1}}\downarrow & & \downarrow{\scriptstyle a_i} \\ B_{i-1} & \xleftarrow{\ t\ } & B_i \end{array} \qquad s, t \in \Sigma.$$

Its commutativity gives $Q(a_{i-1})Q(s) = Q(t)Q(a_i)$, hence

$$Q(a_i)Q(s)^{-1} = Q(t)^{-1}Q(a_{i-1}).$$

The square therefore remains commutative after reversing its horizontal arrows. Apply this to every backwards edge and then compose the resulting forward squares. The endpoint components give a commutative square between the two composite arrows in the localization. In a strict endpoint fibre those components are identities, so the two composites are equal.

This proves the functor to the arrow category and the induced function on π_0 without assuming that any vertical arrow becomes invertible. It also explains the deliberate use of connected components: an equality may be witnessed by a finite zigzag of natural transformations rather than a single transformation. The later reparametrization argument is designed to produce just such zigzags.

The parity notation changes during the construction. It is safest to number the edges consecutively from the left whenever discussing a fixed even-index interval; its even-numbered edges are then the backwards ones. This numerical convention must not be confused with the original label u_α on the doubly infinite interval, or with the symmetric u_i, v_i notation introduced on page 4. The invariant definition is simply that an edge traversed right-to-left belongs to Σ.

On définit une application canonique

$$\mathrm{Ch}_\Sigma(X, x, y) \longrightarrow \mathrm{Hom}_\Sigma(x, y),$$

en associant 'a tout chemin de x dans y le Σ-homomorphisme de x dans y obtenu en *inversant* (dans $X\Sigma^{-1}$) les tronçons rétrogrades (ce qui est possible, puisqu'ils sont dans Σ), de sorte qu'on trouve dans $X\Sigma^{-1}$ un 'chemin direct' [page 7]

$$x = A_{-n} \longrightarrow \cdots \longrightarrow A_{-1} \longrightarrow A_0 \longrightarrow A_1 \longrightarrow A_2 \cdots A_n = A_{n+1} = \cdots = y.$$

et on compose tous ces morphismes.

En fait, on trouve aussitôt que l'on définit ainsi un *foncteur*

$$\underline{\mathrm{Ch}}_\Sigma(X) \longrightarrow \underline{\mathrm{Fl}}(X\Sigma^{-1}),$$

[Fl catégorie des flèches] du fait que si on a un diagramme

$$\begin{array}{ccc} A_{i-1} & \xleftarrow{u_i} & A_i \\ \downarrow{\scriptstyle f_{i-1}} & & \downarrow{\scriptstyle f_i} \\ A'_{i-1} & \xleftarrow{u'_i} & A'_i \end{array} \qquad (i \text{ pair}) \qquad \text{p. 6}$$

avec u_i, u'_i dans Σ, ce diagramme, en tant que diagramme dans $X\Sigma^{-1}$, est commutatif avec flèches horizontales des *isomorphismes,* et reste donc commutatif en inversant lesdites flèches. D'où un foncteur induit

$$\underline{\mathrm{Ch}}_\Sigma(X; x, y) \longrightarrow \underline{\mathrm{Fl}}(\underbrace{X\Sigma^{-1}}_{X'}; x, y). \qquad [\text{page 8}]$$

Mais pour toute catégorie X', et des objets x, y dans X', la catégorie $\underline{\mathrm{Fl}}(X', x, y)$ est une catégorie *discrète,* ayant comme ensemble d'objets $\mathrm{Hom}_{X'}(x, y)$. Ainsi le foncteur précédant définit en fait une application canonique

$$\boxed{\pi_0 \underline{\mathrm{Ch}}_\Sigma(X; x, y) \longrightarrow \mathrm{Hom}_\Sigma(x, y).}$$

Je dis que cette application est toujours bijective. Pour le voir, nous allons *constuire* 'from scratch' la catégorie $X\Sigma^{-1}$, en prenant comme objets ceux de X, et en *définissant*

$$\boxed{\mathrm{Hom}_\Sigma(x, y) = \pi_0 \underline{\mathrm{Ch}}_\Sigma(X; x, y).}$$

Il faut donc définir la composition

$$\mathrm{Hom}_\Sigma(x, y) \times \mathrm{Hom}_\Sigma(y, z) \longrightarrow \mathrm{Hom}_\Sigma(x, z).$$

Pour ceci, on va définir des foncteurs 'composition de chemins'

$$\begin{aligned} \underline{\mathrm{Ch}}_m(X; x, y) \times \underline{\mathrm{Ch}}_n(X; y, z) &\longrightarrow \underline{\mathrm{Ch}}_{m+n}(X; x, z) \\ (c, c') &\longmapsto c' \circ c, \end{aligned}$$

mais en nous bornant pour simplifier au cas où m, n ont *la même parité,* de telle façon qu'en 'mettant bout à bout' c et c' (dans cet ordre), on trouve bien un chemin de type $n + m$.

Commentary · Context [ms. 8]

Localization without a short-fraction hypothesis

Grothendieck now proposes to recover the ordinary localization directly from categories of paths. The basic localization by composable words and inverse symbols is classical; compare Gabriel and Zisman (1967, Chapter I, pp. 6–21) and the explicit word formulation in Dwyer and Kan (1980c, §3.1–3.3). His additional object of study is the category of presentations of a morphism, whose connected components recover those words modulo relations.

A short fraction such as $x \leftarrow x' \rightarrow y$ need not represent every localized morphism unless one imposes extra hypotheses. Arbitrary finite zigzags do. The use of all indices thus avoids an Ore condition, a model structure, and any saturation assumption on the original set Σ. Enlarging Σ to the subcategory it generates changes neither the ordinary localization nor the final set of components, but the unlocalized path categories can change. They should not be silently identified.

There are three tasks in the proposed proof. First, concatenation must descend from particular finite representatives to their classes. Second, the original arrows of X must determine a functor to those classes, and arrows in Σ must become invertible. Finally, every functor inverting Σ must factor uniquely by evaluating zigzags. The first task occupies most of the combinatorics. The second is where a generalized interval lemma is temporarily assumed. We supply its relative version below and also give direct witnesses for the particular identities needed here. The universal property then follows by an explicit argument at the end of the section.

Cette opération est 'transposée' du couple de foncteurs [page 9]

$$\begin{aligned} \Phi_n &\xrightarrow{\alpha} \Phi_{n+m} \quad i \longmapsto (i - m) \\ \Phi_m &\xrightarrow{\beta} \Phi_{n+m} \quad i \longmapsto (i + n), \end{aligned}$$

où on aura

$$\alpha(\{n\}) = \beta(\{-m\}) = n - m, \qquad \text{p. 7}$$

et on aura

$$\Phi_{n+m} \simeq \Phi_n \sqcup_e \Phi_m,$$

i.e. Φ_{n+m} est la 'somme amalgamée' dans (Cat), obtenue en 'recollant' Φ_n et Φ_m, par identification de l''élément but' de Φ_n avec l''élément source' de Φ_m. Le foncteur composition

de chemins induit [dans ce qui précède, à partir du début de la page 9, il faut inverser m et n]

$$\underline{\mathrm{Ch}}_{m,\Sigma}(x,y) \times \underline{\mathrm{Ch}}_{n,\Sigma}(y,z) \longrightarrow \underline{\mathrm{Ch}}_{m+n,\Sigma}(x,z),$$

d'où en passant aux π_0

$$\mathrm{Hom}_{\Sigma,m}(x,y) \times \mathrm{Hom}_{\Sigma,n}(y,z) \longrightarrow \mathrm{Hom}_{\Sigma,m+n}(x,z),$$

où on dénote, pour $a, b \in \mathrm{Ob}\, X$, par

$$\mathrm{Hom}_{\Sigma,n}(a,b) \overset{\text{déf}}{=} \pi_0(\underline{\mathrm{Ch}}_{m,\Sigma}(a,b)).$$

[plutôt $\underline{\mathrm{Ch}}_{n,\Sigma}$]
Pour en déduire une application sur les $\varinjlim$ [page 10]

$$\underbrace{\mathrm{Hom}_\Sigma(x,y)}_{=\varinjlim \mathrm{Hom}_{\Sigma,m}(x,y)} \times \underbrace{\mathrm{Hom}_\Sigma(y,z)}_{=\varinjlim \mathrm{Hom}_{\Sigma,n}(y,z)} \longrightarrow \underbrace{\mathrm{Hom}_\Sigma(x,z)}_{=\varinjlim \mathrm{Hom}_{\Sigma,n+m}(x,z)}$$

il faut prouver, pour $n \le n'$, $m \le m'$, la commutativité dans

$$\begin{array}{ccc} \mathrm{Hom}_{\Sigma,m}(x,y) \times \mathrm{Hom}_{\Sigma,n}(y,z) & \longrightarrow & \mathrm{Hom}_{\Sigma,m+n}(x,y) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\Sigma,m'}(x,y) \times \mathrm{Hom}_{\Sigma,n'}(y,z) & \longrightarrow & \mathrm{Hom}_{\Sigma,m'+n'}(x,y) \end{array}$$

[plutôt $\mathrm{Hom}_{\Sigma,m+n}(x,z)$, $\mathrm{Hom}_{\Sigma,m'+n'}(x,z)$] En d'autres termes, que pour $c : x \longrightarrow y$ et $c' : y \longrightarrow z$ deux Σ-chemins composables de longueur m et n respectivement, le composé $c'c : x \longrightarrow z$ ne change pas, 'à homotopie près' dans $\underline{\mathrm{Ch}}_\Sigma(x,z)$, quand on remplace c, c' par des chemins $\bar{c}$, $\bar{c}'$ de longueur m', n' déduits de c et c' par 'dégénérescence aux bouts'.

Plus précisement,

$$\bar{c}'\bar{c} \sim \overline{c'c},$$

le chemin composé est homotope à celui déduit de $c'c$ par 'dégénérescence aux bouts'.

Je viens de regarder pour $n = m = 1$, $n' = m' = 2$, et de constater qu'il n'y a pas de flèche naturelle entre $\bar{c}'\bar{c}$ et $\overline{c'c}$ - l'homotopie cherchée doit se définir à l'aide d'un 'chemin' dans $\underline{\mathrm{Ch}}_{\Sigma,m',n'}(X;x,z)$. Dans le cas général, on va en fait trouver une homotopie p. 8 [page 11]

$$h = h_{(n,m),(n',m')}$$

'rendant commutatif' le diagramme de foncteurs

$$\begin{array}{ccc} \mathcal{A} = \underline{\mathrm{Ch}}_{\Sigma,m}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma,n}(X;y,z) & \xrightarrow{*_{m,n}} & \underline{\mathrm{Ch}}_{\Sigma,m+n}(X;x,z) \\ \downarrow{\scriptstyle i_{m,m'} \times j_{n,n'}} & \overset{h}{\swarrow} & \downarrow{\scriptstyle k_{m+n,m'+n'}} \\ \underline{\mathrm{Ch}}_{\Sigma,m'}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma,n'}(X;y,z) & \xrightarrow{*_{m',n'}} & \underline{\mathrm{Ch}}_{\Sigma,m'+n'}(X;x,z) = \mathcal{B}, \end{array}$$

où les $i_{\alpha,\beta}$, $j_{\alpha,\beta}$, $k_{\alpha,\beta}$ sont les foncteurs de transition

$$\underline{\mathrm{Ch}}_{\Sigma,\alpha}(X;a,b) \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\beta}(X;a,b)$$

($\alpha \le \beta$) pour les (x,y), (y,z) et (x,z) (en guise de (a,b)) respectivement. L'homotopie h sera un chemin dans la catégorie $\underline{\mathrm{Hom}}(\mathcal{A},\mathcal{B})$, allant de

$$k_{m+n,m'+n'} \circ *_{m+n} \longrightarrow *_{m'+n'} \circ (i_{m,m'} \times j_{n,n'}).$$ [page 12]

L'existence d'une telle homotopie signifie ni plus ni moins que le diagramme carré ci-dessus est commutatif, en tant que diagramme dans la catégorie (Cat hot) des 'catégories à homotopie près,' dont les objets sont les (petites) catégories, et où on définit

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(\mathcal{A},\mathcal{B}) \overset{\text{déf}}{=} \underbrace{\pi_0(\underline{\mathrm{Hom}}(\mathcal{A},\mathcal{B}))}_{\substack{\text{classes d'homotopie}\\ \text{de foncteurs de } \mathcal{A}\\ \text{dans } \mathcal{B}}} .$$

C'est cette conclusion qui sera pour nous utile, et non la description explicite du chemin h. Cette existence admise (cf. ci-dessous la preuve), on voit donc que si on regarde, pour $a, b \in \mathrm{Ob}\, X$, le système des catégories $\underline{\mathrm{Ch}}_{\Sigma,r}(X; a, b)$ ($r \in \mathbf{N}$) comme un *ind-objet de* Cat hot, qu'on désigne par

$$\underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X; a, b) \in \mathrm{Ob}\,\mathrm{Ind}(\mathrm{Cat\,hot}),$$

on aura un morphisme canonique dans Ind(Cat hot)

$$\underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X; x, y) \times \underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X; y, z) \xrightarrow{*} \underline{\mathrm{Ch}}^{\mathrm{ind}}_{\Sigma}(X; x, z),$$

appelé *'ind-morphisme de composition de chemins.'* On aurait aimé le déduire d'un honnête *foncteur* 'composition de chemins'

$$\underline{\mathrm{Ch}}_{\Sigma}(X; x, y) \times \underline{\mathrm{Ch}}_{\Sigma}(X; y, z) \xrightarrow{?} \underline{\mathrm{Ch}}_{\Sigma}(X; x, z),$$
$$\text{qu'on noterait } (c, c') \longmapsto c' \circ c \text{ ou } c' \cdot c,$$

mais je doute qu'un tel foncteur existe,

Commentary · Present state [ms. 12]

What composition in an ind-category actually asserts

Grothendieck has reached a useful level of coherence before obtaining a single composition functor on the eventual path categories. For even indices m, n, concatenation gives an actual functor at a finite stage. Increasing either index changes its comparison with the larger-stage concatenation by a finite zigzag of natural transformations. This is exactly the relation imposed in Cat hot, where

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(A, B) = \pi_0\,\mathrm{Fun}(A, B).$$

Products descend to this category because one can multiply each natural transformation in a zigzag by an identity transformation.

For ind-objects the assertion is governed by the formula

$$\mathrm{Hom}_{\mathrm{Ind}(C)}((A_i), (B_j)) = \lim_i \operatorname*{colim}_j \mathrm{Hom}_C(A_i, B_j).$$

Take $A_{m,n} = \mathrm{Ch}_{\Sigma,m}(x, y) \times \mathrm{Ch}_{\Sigma,n}(y, z)$ and $B_r = \mathrm{Ch}_{\Sigma,r}(x, z)$, with all indices even. The stagewise concatenation maps land in B_{m+n}. Their compatibility as elements of the displayed inverse limit is precisely what the reparametrization lemma proves. The passage to components then gives an ordinary associative composition law.

A functor between the ordinary filtered unions would be additional data. Even if such a functor exists, its restriction to one source stage need not land in a bounded target stage: the source stage can have infinitely many objects. The parenthesis on page 13 explicitly adds the necessary boundedness condition. The equivalence asserted there is therefore conditional on this extra factorization, not a formal property of every functor between filtered colimits.

We do not infer a negative theorem from Grothendieck's doubt. The strict composition demanded on his particular category with symmetric end-padding is not constructed in these pages, and we did not locate a resolution of that exact requirement in the localization sources consulted. Other choices of path presentations do support strict composition. Their existence does not by itself provide the requested functor on this fixed category. The distinction is comparable to the one already emphasized in the commentary on Chapter I, manuscript p. 7 ("Coherent squares and what the axioms remember"): an equality after passage to a homotopy category does not specify its coherent witness.

p. 9
[page 13]

qui par passage aux ind-objets redonne le morphisme précédent, ou, ce qui revient au même, tel que pour tout $m, n \in \mathbf{N}$ (de même parité), il existe un $N \geq m+n$ assez grand tel que l'on ait un diagramme

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma,m}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma,n}(X;y,z) & \xrightarrow{*_{m,n}} & \underline{\mathrm{Ch}}_{\Sigma,m+n}(X;x,z) \\ \downarrow & \searrow & \downarrow \\ & & \underline{\mathrm{Ch}}_{\Sigma,N}(X;x,z) \\ & & \downarrow \\ \underline{\mathrm{Ch}}_{\Sigma}(X;x,y) \times \underline{\mathrm{Ch}}_{\Sigma}(X;y,z) & \xrightarrow{?} & \underline{\mathrm{Ch}}_{\Sigma}(X;x,z) \end{array}$$

et où le triangle supérieur est *commutatif dans* Cat hot. (Il faut supposer que pour N assez grand, le composé $\downarrow\!\!\!\rightarrow$ se factorise par $\underline{\mathrm{Ch}}_{\Sigma,N}(X;x,z)$ …) Pour voir que les foncteurs composition $*_{m,n}$ définissent un homomorphisme des ind-objets dans (Cat hot), il suffit de le prouver dans les cas particuliers $(m',n') = (m+1, n+1)$. On va faire la figure en supposant m, n *impairs.*

Je m'aperçois ici que je me suis canulé stupidement, la composition de deux chemins de longueur *impaire* m, n *n'est*

pas un chemin de longueur $m+n$, i.e. un diagramme de type Φ_n, mais un diagramme de type *dual* $\Phi_n^o = \Psi_n$. Par exemple, pour $m = n = 1$ [page 14]

$$\underbrace{A_0 \longleftarrow A_1 \longrightarrow A_2}_{c} = \underbrace{B_0 \longleftarrow B_1 \longrightarrow B_2}_{c'}$$

est de type Φ_2^o! Donc pour ne pas trop compliquer la vie, on va dans les développements précédents se borner à m, n *pairs,* donc des chemins de longueur au moins 4. C'est un peu ridicule! On est ramené à prouver la commutativité à homotopie près dans le cas où on laisse c' fixé et où on envoie $\underline{\mathrm{Ch}}_{\Sigma,m}(X;x,y)$ dans $\underline{\mathrm{Ch}}_{\Sigma,m+2}(X;x,y)$, ou inversement c fixe et $\underline{\mathrm{Ch}}_{\Sigma,n}(X;y,z)$ dans $\underline{\mathrm{Ch}}_{\Sigma,n+2}$ `[lire:`$\underline{\mathrm{Ch}}_{\Sigma,n+2}(X;y,z)$`]`. Prenons le premier cas, et notons un chemin c sans indices négatifs (ce n'est pas une notation bien heureuse!)

$$A_0 \longrightarrow A_1 \longleftarrow A_2 \cdots \longleftarrow A_{2m}$$

(NB à cause de la condition n pair, tous nos chemins commencent désormais par une flèche de A_0 dans A_1, jamais en sens inverse).

Je vois que l'explicitation de l'homotopie, sans offrir de difficulté, est très fastidieuse à expliquer. Je vais essayer de donner une démonstration conceptuelle, sans 'calcul' fastidieux.

Commentary · Exposition [ms. 14]

Even indices and reparametrization relative to the ends

Grothendieck's correction on this page is part of the proof. The earlier statement that equal parity suffices for concatenation is false for two odd indices. Translation by an odd integer reverses the source–sink parity in Φ, so the proposed embeddings of the two odd intervals into Φ_{m+n} are not functors with the advertised orientation. Restricting to even indices repairs the construction and loses no path, since every finite index is bounded by an even one.

For even m, n use the embeddings

$$\Phi_m \longrightarrow \Phi_{m+n}, \quad i \longmapsto i-n, \qquad \Phi_n \longrightarrow \Phi_{m+n}, \quad j \longmapsto j+m.$$

They identify the final vertex of the first interval with the initial vertex of the second. At the join both incident arrows have the same source–sink type, so no new composite of two nonidentity arrows appears. Consequently the resulting pushout in categories is Φ_{m+n}. Restricting a functor on that pushout is the same as giving the two composable diagrams with their common object literally

identified. Natural transformations with identity components at the exterior endpoints concatenate as well.

To compare padding before concatenation with padding afterwards, regard the two resulting diagrams as precompositions with maps from a larger interval to a smaller one. Both maps are nondecreasing in the ordinary numerical order of the vertices and fix the two endpoint labels. Their categorical orders are still the alternating orders. This second monotonicity condition, explicitly restored in the edition's footnote on page 17, prevents a backwards source edge from travelling forwards in the smaller interval.

The boundary equation is $h\alpha' = \alpha$, with α' the endpoint map into the larger interval and α the endpoint map into the smaller one. The corrected Hom expression on page 16 fixes the order of those intervals, but its description of the restriction as α' still has the wrong codomain; the displayed equation is the precise interpretation.

The intermediate maps in a homotopy must have that same property. If c is a Σ-path and h is a numerically nondecreasing functor between alternating intervals, each backwards edge is sent either to a backwards edge or to an identity. Hence ch remains a Σ-path. Every natural transformation between such h's induces a ladder with identity endpoint components. The forthcoming connectedness argument therefore proves compatibility inside the required path category, with the denominator condition intact.

The reduction to simultaneous increments $(m, n) \mapsto (m + 1, n + 1)$ before the correction is insufficient for arbitrary independent changes of the two indices. After restriction to even indices, separate increments by two in the first or second variable generate all transitions. Grothendieck's revised formulation on this page makes exactly this repair.

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[page 15]

Les deux foncteurs composés

$$f = k \circ *_{m,n} \ , \quad g = *_{m',n'} \circ (i \times j)$$

dans le carré page 11, dont il faut prouver qu'ils sont homotopes dans $\underline{\mathrm{Hom}}(\mathcal{A}, \mathcal{B})$, peuvent être vus comme obtenus de la façon suivante, en termes de deux foncteurs

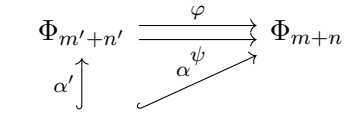
$$\varphi, \psi : \Phi_{m'+n'} \rightrightarrows \Phi_{m+n}, \tag{1}$$

d'où deux foncteurs 'transposés'

$$\underline{\mathrm{Hom}}(\Phi_{m,n}, X) \underset{G}{\overset{F}{\rightrightarrows}} \underline{\mathrm{Hom}}(\Phi_{m',n'}, X) \tag{2}$$

[plutôt Φ_{m+n}, $\Phi_{m'+n'}$] lesquels foncteurs 'induisent' f et g. De façon précise, désignant par a, b les objets source et but de Φ_{m+n}, par c l'objet qui correspond au but du chemin c de longueur m [trop de c's], à la source du chemin c' de longueur n qu'il s'agit de composer, enfin par a_0, b_0 les éléments source et but de $\Phi_{m'+n'}$, les foncteurs envisagés (1) appliquent a_0 dans a, b_0 dans b, donc définissent un diagramme commutatif

(3)



$\varepsilon = \{a_0, b_0\}$.
catégorie discrète

Ceci dit, la catégorie but $\mathcal{B}$ pour f, g, à savoir $\underline{\mathrm{Ch}}_{\Sigma,m'+n'}(X; x, z)$, s'interprète comme la sous-catégorie pleine de $\underline{\mathrm{Hom}}(\Phi_{m+n}, X)$ formée

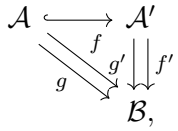
des foncteurs $c' : \Phi_{m'+n'} \longrightarrow X$ dont le composé avec α' est le foncteur [page 16]

$$\beta : \varepsilon \longrightarrow X \qquad \begin{matrix} a_0 \longrightarrow x \\ b_0 \longrightarrow z, \end{matrix} \tag{$*$}$$

et de même la catégorie source $\mathcal{A}$ s'interprète comme la sous-catégorie pleine de $\underline{\mathrm{Hom}}(\Phi_{m+n}, X)$ formée des $c : \Phi_{m+n} \longrightarrow X$ dont la restriction à $\{a, b, c\}$ est le foncteur

$$a \longrightarrow x,\ b \longrightarrow y,\ c \longrightarrow z.$$

[plutôt $b \longrightarrow z$, $c \longrightarrow y$] Elle est contenue dans la sous-catégorie pleine $\mathcal{A}'$ des foncteurs dont la restriction à $\{a, b\}$ soit $a \longrightarrow x,\ b \longrightarrow z$, i.e. dont le composé avec α soit le foncteur $\beta : \varepsilon \longrightarrow X$ déjà envisagé ci-dessus. Ainsi les foncteurs $f, g : \mathcal{A} \rightrightarrows \mathcal{B}$ se factorisent par $\mathcal{A}'$ p. 11



foncteurs déduits de φ, ψ ci-dessus, donnant lieu aux triangles commutatifs (3), et du foncteur $\beta : \varepsilon \longrightarrow X$, de la façon dite. Pour prouver que f et g sont homotopes, il suffit de le prouver pour f', g'. Et pour ceci, il suffit de montrer que φ et ψ sont homotopes en tant qu'objets de la catégorie $\underline{\mathrm{Hom}}(\Phi_{m+n}, \Phi_{m'+n'}; \alpha)$ formée des foncteurs dont la restriction à ε est α [plutôt de la catégorie $\underline{\mathrm{Hom}}(\Phi_{m'+n'}, \Phi_{m+n}; \alpha')$ formée des foncteurs dont la restriction à ε est α'].

D'ailleurs, les deux assertions (sur f', g', quels que soient X, x, y, z, [plutôt x, z] et sur φ, ψ, α [plutôt α']) sont en fait équivalentes comme on voit en faisant $X = \Phi_{m+n}$, $x = a$, $y = c$, $z = b$ [plutôt $x = a$, $z = b$]. Ceci sera contenu dans le lemme plus général. ([4]) [page 17]

Lemme. *Soient k et l deux entiers. Alors la sous-catégorie pleine de $\underline{\mathrm{Hom}}(\Phi_k, \Phi_l)$ formée des foncteurs $\Phi_k \longrightarrow \Phi_l$ appliquant source en source, but en but, est connexe.* (NB Elle est non vide si et seulement si $k \geq l$.)

On va décrire entièrement ces foncteurs (= applications croissantes) $\Phi_k \xrightarrow{\varphi} \Phi_l$, en identifiant $\mathrm{Fl}(\Phi_k)$ [plutôt l'ensemble des flèches non identiques de Φ_k] à l'intervalle d'entiers $[1, 2k]$, et en associant à φ la partie

$$E_\varphi \subset \mathrm{Fl}(\Phi_k) = [1, 2k] \qquad E_\varphi = \{u \in \mathrm{Fl}(\Phi_k) \mid \varphi(u) \text{ une identité}\}$$

formée des $u \in \mathrm{Fl}(\Phi_k)$ qui sont 'contractées' par φ en une identité. Si E est un ensemble d'entiers, on y introduit la notion de composantes connexes (par la relation d'équivalence dans E engendrée par la relation de succession $y = x + 1$). On voit aussitôt que les composantes connnexes de E_φ qui ne contiennent ni la première ni la dernière flèche de Φ_k sont paires, et que

$$\underbrace{\mathrm{card}\,\mathrm{Fl}(\Phi_k)}_{2k} = \mathrm{card}\,(E_\varphi) + \underbrace{\mathrm{card}\,\mathrm{Fl}(\Phi_l)}_{2l},$$

Commentary · Exposition [ms. 17]

A contraction is determined by the edges it erases

Grothendieck reduces a diagrammatic homotopy problem to finite combinatorics. Number the vertices of Φ_k from 0 to $2k$, preserving their numerical order; the vertex numbered t is the original vertex $t - k$. Number the edges from 1 to $2k$. For a numerically nondecreasing endpoint-preserving functor $f : \Phi_k \to \Phi_l$, adjacent values differ by zero or one: the target has no nonidentity arrow joining nonadjacent vertices. Every noncollapsed edge advances the numerical position once. Therefore

$$f_E(t) = -l + t - \big|E \cap [1, t]\big|, \qquad 0 \leq t \leq 2k,$$

where the values are written in the original target numbering and E is the set of collapsed edges. This formula both proves uniqueness and makes the prefix and suffix observations on page 20 transparent.

4. Dans les considérations qui suivent, on considère en fait tacitement que les foncteurs envisagés sont *croissants* sur les sommets pour l'ordre de leur numérotation. Donc il faudrait écrire $\underline{\mathrm{Hom}}_{cr}$ au lieu de $\underline{\mathrm{Hom}}$, dans les énoncés qui suivent, y compris dans le lemme de connexité ci-contre.

When $l > 0$, the admissibility conditions have a direct explanation. Exactly $2l$ edges survive, so $|E| = 2(k - l)$. Between successive surviving edges, an even number of erased edges is necessary and sufficient for their alternating orientations to agree. The same condition holds at an endpoint when the initial orientations agree. If k and l have opposite parity, the initial and terminal erased runs must both be odd. The parity of the terminal run follows from the total cardinality together with the parity of all interior runs; the edition rightly inserts both conditions (a) and (b) at that point.

Conversely, these conditions make each consecutive pair $f_E(t-1), f_E(t)$ either equal or the endpoints of the correctly oriented target edge. They thus define a functor. It is surjective on objects and nonidentity arrows, because it starts at $-l$, ends at l, and advances by steps of at most one. Its only identifications are the consecutive strings of vertices joined by edges of E, so it is the stated quotient. This establishes necessity and sufficiency, not merely the numerical count.

For $l = 0$ there is one constant functor and $E = [1, 2k]$. No initial target direction exists, so the parity condition must be omitted in that case. For $k = l = 0$ the same convention gives $E = \varnothing$. Endpoint-preserving functors exist exactly when $k \geq l$: necessity follows from counting surviving edges, while sufficiency follows from the clamping functor $\Phi_k \to \Phi_l$.

The footnote's restriction to numerical monotonicity is essential for the description by E. A general functor can retrace a target edge, and its noncollapsed edges need not number $2l$. Later we prove contractibility also for the full endpoint fibre, but that stronger statement does not make the intervening classification of contractions valid without its hypothesis.

For instance, number the source vertices of Φ_3 from 0 to 6, and retain the target numbering $-1, 0, 1$ for Φ_1. The row

$$(-1, 0, -1, 0, -1, 0, 1)$$

defines an endpoint-preserving functor. At the odd-numbered source vertices, which are sources in this numbering, its value is 0; every generating arrow is therefore sent to one of the two nonidentity arrows leaving 0 in Φ_1. No edge is erased. Nevertheless the difference in the indices is $3 - 1 = 2$, so the proposed cardinality would be four. This example isolates the precise role of the footnote: endpoint preservation is sufficient for the later contractibility theorem, but it is insufficient for this particular description by a set of contractions.

i.e. [page 18]

$$\operatorname{card}(E_\varphi) = 2(k - l).$$

Il est immédiat qu'un foncteur φ est surjectif sur les objets et sur les flèches, donc identifie p. 12
Φ_l à une catégorie quotient de Φ_k - et la relation d'équivalence dans Φ_k définie par φ est celle obtenue en identifiant tous les objets qui sont source ou but de flèches d'une même composante connexe I de E_φ

$$\cdots \longleftarrow \underbrace{\longrightarrow \longleftarrow \longrightarrow \longleftarrow}_{I} \longrightarrow \cdots$$

et les flèches dans cette composante à l'identité correspondante. Les conditions énoncées sur E (indépendamment d'un φ qui lui donnerait naissance assurent que la catégorie quotient est isomorphe (avec *préservation* de source et de but) soit à Φ_l, soit à la duale $\Psi_l = \Phi_l^o$, suivant la direction de la première flèche de $\bar{\Phi}_k$ (quotient de Φ_k). Cette 'direction initiale' est la même que celle pour Φ_k si la première flèche de Φ_k n'appartient pas à E, ou bien si elle y appartient et sa composante connexe dans E est de cardinal pair. Dans le cas inverse ($u \in E$ et sa composante connnexe dans E est de cardinal impair), la direction initiale de $\bar{\Phi}_k$ est opposée à celle de Φ_k.

On voit donc que E doit satisfaire encore à une condition supplémentaire, pour que le [page 19]
$\bar{\Phi}_k$ correspondant soit isomorphe à Φ_l, i.e. pour que E provienne d'un foncteur φ. Je résume toutes les conditions

a) $\operatorname{card} E = 2(k - l)$.

b) Les composantes connexes de E qui ne contiennent pas de flèche initiale ou finale sont de cardinal pair.

c) Si Φ_k et Φ_l ont même direction initiale (k et l de même parité), alors l'éventuelle composante de E contenant la flèche initiale doit être elle aussi de cardinal pair (ce

qui, à cause de a) [plutôt a) et b)], équivaut à la même condition pour l'éventuelle composante connexe contenant la flèche finale de Φ_k). Si Φ_k et Φ_l ont des directions initiales différentes, il faut que la flèche initiale de Φ_k appartienne à une composante connexe de cardinal *impair* de E (ce qui équivaut à itou pour la flèche finale).

D'autre part, le foncteur φ est uniquement déterminé par la connaissance de E, car il y a un *unique* isomorphisme $\bar{\Phi}_k \xrightarrow{\sim} \Phi_l$ compatible avec sources et buts, à cause du

Lemme (immédiat) : *Un automorphisme de Φ_l qui est l'identité sur l'objet source (ou sur l'objet but) est l'identité partout.*

Les foncteurs φ, ψ de tantôt correspondaient au choix, l'un de $E_\varphi = \{1, 2, k-1, k\}$ [plutôt $E_\varphi = \{1, 2, 2k-1, 2k\}$], l'autre de $E_\psi = \{1, 2, i, i+1\}$, 2 composantes connexes dans un cas, trois [plutôt deux] dans l'autre.) [page 20]

Dans le cas général, il y a lieu d'expliciter la relation d'ordre de l'ensemble $\underline{\mathrm{Hom}}^!(\Phi_k, \Phi_l)$ p. 13 (! signifie : compatible avec source et but), en termes des ensembles représentatifs E_φ, E_ψ. Je ne vais pas expliciter cette relation d'ordre, mais seulement une propriété de celle-ci qui nous sera suffisante. Soit φ_E le foncteur défini par une partie $E \subset [1, 2k]$. Tout d'abord :

Lemme 1 (immédiat) : *Soient $E, E' \subset [1, 2k]$ admissibles, et soit $j \in [1, 2k]$ tel que $E \cap [1, j] = E' \cap [1, j]$. Alors φ_E et $\varphi_{E'}$ coïncident sur le but de u_j (j une flèche de Φ_k) et sur tous les objets de Φ_k antérieurs à celle-ci (au sens de la numérotation des objets, et non de la relation d'ordre de la catégorie Φ_k bien sûr.) Dualement, si $E \cap [j, 2k] = E' \cap [j, 2k]$, alors φ_E et $\varphi_{E'}$ coïncident sur la source de y, et sur tous les objets de Φ_k postérieurs à celui-ci.*

Considérons maintenant une partie connexe I de E (pas forcément une composante connexe), $I = [r, s] \subset [1, 2k]$ (r le plus petit élément de I, s son plus grand élément) et supposons que

a) card$I$ est pair (i.e. $s - r$ est pair) [plutôt $s - r$ impair]

b) $r \neq 1$, et $r - 1 \notin E$ (i.e. $r - 1 \in [1, 2k] \backslash E$)

et soit

$$E' = (E \backslash \{s\}) \cup \{r-1\} = E \backslash I \cup \overbrace{[r-1, s-1]}^{I'}$$

déduit de I en 'translatant' sur la gauche d'un cran.

Il est immédiat que si E est admissible, E' aussi (et inversement). Ceci dit, on a le [page 21]

Lemme 2. *E et E' étant comme ci-dessus, il existe une homomorphisme de foncteurs entre φ_E et $\varphi_{E'}$* (je ne dis pas dans quel sens va la flèche!).

Corollaire. Énoncé dual, quand E' est déduit de E en translatant à droite d'un cran un intervalle connexe de longueur paire $I = [r, s]$ de E, $s \neq 2k$ et $s + 1 \notin E$.

En effet, dans ce cas E est déduit de E' par une opération du type explicité dans le lemme 2.

La connexité de la catégorie $\underline{\mathrm{Hom}}^!(\Phi_k, \Phi_l)$ est maintenant claire. Si k et l ont même parité, en poussant les composantes connexes de E vers la gauche, on écrit E sous la forme 'normale'

$$E_0 = [1, 2(k-l)].$$

Si k et l sont de parité inverse, on ne peut pousser toute la composante connexe de E qui contient l'élément dernier $2k$, car elle est de cardinal impair. Il faut laisser $2k$ dans E et pousser le reste, et on ramène E à la forme normale

$$E_0 = [1, 2(k-l)-1] \cup \{2k\}.$$

Commentary · Exposition [ms. 21]

Sliding an even block gives a natural transformation

Grothendieck's elementary move explains why different padding conventions become equivalent. The formula just obtained gives a short proof of the move, including the natural transformation he leaves implicit. Suppose $I = [r, s] \subset E$ has even cardinality, $r > 1$, and $r - 1 \notin E$, and put $E' = (E \setminus \{s\}) \cup \{r - 1\}$. Then

$$|E' \cap [1, t]| - |E \cap [1, t]| = \begin{cases} 1, & r - 1 \le t \le s - 1, \\ 0, & \text{otherwise.} \end{cases}$$

The map f_E is constant, say equal to a, on the vertices $r - 1, \dots, s$, because all intervening edges belong to E. Thus $f_{E'}$ differs from f_E only by replacing a with $a - 1$ at the vertices $r - 1, \dots, s - 1$. These two target vertices are joined by one arrow. Its orientation determines a natural transformation in one direction between the two functors; outside the displayed interval its components are identities. Naturality is automatic in a poset once the component inequalities hold.

The endpoints are unchanged, since $r - 1 > 0$ and $s - 1 < 2k$. Admissibility is preserved: removing an even initial block from a collapsed run preserves the parity of its remainder, and translating that even block preserves the parity of every run it meets. The initial and terminal parity conditions are therefore unchanged. This also proves the right-moving version by reversing the move.

Repeated left-moving operations terminate: the sum of the indices in E strictly decreases at each move. If the endpoint orientations agree, every collapsed run is even and all runs can be joined to the leftmost one, yielding $[1, 2(k - l)]$. If they disagree, the initial and terminal runs are odd. Keep the final edge $2k$ in place and move the remaining even part of the final run, together with the interior runs, to the left. The normal form is

$$[1, 2(k - l) - 1] \cup \{2k\}.$$

Each move is a morphism in one of the two directions, so every object is connected to the same normal form. This is a proof of connectedness, which is all the composition argument needs.

For example, with $k = 4, l = 2$, the set $E = \{1, 2, 7, 8\}$ successively becomes $\{1, 2, 6, 7\}$, $\{1, 2, 5, 6\}$, $\{1, 2, 4, 5\}$, and $\{1, 2, 3, 4\}$. The associated functors form a zigzag of four natural transformations relative to the endpoints. This gives a visible instance of the homotopy comparing padding at an exterior end with padding near a join. The proof does not assert that there is always one natural transformation between the original two functors.

cqfd!

Ce qui précède justifie notre raisonnement, qui donnait par composition des chemins des p. 14
applications de composition bien définies

$$\underbrace{\mathrm{Hom}_\Sigma(x,y)}_{=\pi_0\underline{\mathrm{Ch}}_\Sigma(x,y)} \times \underbrace{\mathrm{Hom}_\Sigma(y,z)}_{=\pi_0\underline{\mathrm{Ch}}_\Sigma(y,z)} \longrightarrow \underbrace{\mathrm{Hom}_\Sigma(x,z)}_{=\pi_0\underline{\mathrm{Ch}}_\Sigma(x,z)}.$$

Je dis que cette composition est associative. Ceci résulte du fait que si on a trois chemins [page 22]
(d'indices *pairs*!)

$$x \xrightarrow{c} y \xrightarrow{c'} z \xrightarrow{c''} t,$$

on a (sans avoir à effectuer des homotopies …) associativité stricte

$$c''(c'c) = (c''c')c.$$

Je dis qu'il y a des identités. On définit

$$1_x \in \mathrm{Hom}_\Sigma(x, x)$$

par le chemin d'indice 0 (qui est pair!), i.e. par le diagramme réduit à x, sans flèche (ou si on veut interpréter le diagramme comme un foncteur $e \longrightarrow X$, il y a la flèche identique sous-entendue). Nos raisonnements n'excluaient pas le cas $n = 0$ ou $m = 0$, et il est tautologique

que pour un chemin

$$x \xrightarrow{c} y$$

on a

$$c \cdot 1_x = 1_y \cdot c = c.$$

Donc on a bien défini une catégorie, que je vais appeler X_Σ. Ja vais définir un foncteur canonique

$$\theta : X \longrightarrow X_\Sigma$$

qui sera l'identité sur les objets, il faut le définir sur les flèches

$$\mathrm{Hom}_X(x, y) \longrightarrow \mathrm{Hom}_{X_\Sigma}(x, y) = \pi_0 \mathrm{Ch}_\Sigma(x, y)$$

[plutôt $\pi_0 \underline{\mathrm{Ch}}_\Sigma(x, y)$]. Malheureusement, une flèche

$$x \xrightarrow{u} y$$

ne peut s'interpréter directement comme un chemin à notre sens, il faut payer nos fuites! Mais on va à une telle flèche associer le chemin [page 23]

$$x \longrightarrow y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y,$$

élément de $\underline{\mathrm{Ch}}_{\Sigma,2}(x, y)$, ce qui définit l'application cherchée. Il faut vérifier la compatibilité avec les identités et sous la composition. À la flèche $1_x : x \longrightarrow x$ est associé l'élément de p. 15
$\Phi_2(x, x)$

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x,$$

il faut prouver que dans $\pi_0(\underline{\mathrm{Ch}}_{\Sigma,2}(x, x))$ il devient égal à l'élément déduit de $1_x \in \Phi_{\Sigma,0}(x, x)$, par le foncteur de transition. Mais cet élément n'est autre justement que

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x,$$

c'est le même (sans avoir même à passer au π_0)!

Commentary · Examples [ms. 22]

A denominator set need not be closed under composition

Grothendieck's use of arbitrarily long paths is already visible in the simplest one-object example. Let X have one object and endomorphism monoid $(\mathbf{N}, +)$, writing its generator multiplicatively as t. Take

$$\Sigma = \{1, t\}.$$

This contains the identity but is not closed under composition: $t^2 \notin \Sigma$. A backwards edge may carry t, but not t^2. For a path c, let $a(c)$ be the sum of the exponents on its forward edges and let $b(c)$ count the backwards edges labelled t. Its evaluation is determined by the integer

$$\nu(c) = a(c) - b(c).$$

A commutative ladder with identity endpoints preserves this integer. Indeed, write each vertical component as t^{h_i} with $h_0 = h_{2n} = 0$. For a forward edge of exponent a_i, naturality gives $h_i + a_i = a_i' + h_{i-1}$; for a backwards edge the analogous equality has the horizontal exponents reversed. Summing the signed equalities cancels all the h_i, leaving $\nu(c) = \nu(c')$.

Conversely, every integer is represented. A nonnegative integer r is represented by the ordinary arrow t^r with identity padding. For $-r < 0$, concatenate r backwards copies of t, separating them by identities, then pad to an even index. The cancellation ladders below show that a forward t followed by a backwards t represents the identity, in either order. Splitting a forward t^a into a successive copies of t is allowed by the functoriality calculation for ordinary arrows. These two operations reduce any path to the representative of its signed exponent. Thus

$$\pi_0 \, \mathrm{Ch}_\Sigma(X; *, *) \cong \mathbf{Z},$$

with concatenation corresponding to addition.

The resulting localization is the one-object category of the group $\mathbf{Z}$. Its universal property says exactly that a chosen endomorphism T becomes invertible: the image of the integer r must be T^r. Negative integers require arbitrarily many backwards edges when the denominator set is the original $\{1, t\}$. Replacing it by all powers of t permits a single backwards edge to represent them, but this changes the categories of presentations. The agreement after passage to connected components is the invariant conclusion.

Pour la compatibilité avec la composition

$$x \xrightarrow{u} y \xrightarrow{v} z,$$

on a d'une part

$$x \xrightarrow{vu} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z,$$

d'autre part

$$x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y \xrightarrow{v} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z.$$

On complète le premier chemin par dégénérescence aux deux bouts, pour le comparer au deuxième en tant qu'élément de $\underline{\mathrm{Ch}}_4(x, z)$,

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{vu} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z \xrightarrow{\mathrm{id}} z \xleftarrow{\mathrm{id}} z.$$

Commentary · Examples [ms. 23]

Two ordinary arrows and one comparison diagram

Grothendieck needs to show that sending an ordinary arrow to its padded path respects composition. This particular comparison is smaller than the general interval problem and can be completed explicitly. Let $x \xrightarrow{u} y \xrightarrow{v} z$, and regard it as a functor $T : [2] \to X$ with $T(0) = x$, $T(1) = y$, and $T(2) = z$.

Number the nine vertices of Φ_4 from 0 to 8. The concatenated path for u and v, and the padded path for vu, arise from the following functors to the ordered category [2]:

i	0	1	2	3	4	5	6	7	8
$f(i)$	0	1	1	1	1	2	2	2	2
$g(i)$	0	0	0	2	2	2	2	2	2
$h(i)$	0	0	0	1	1	2	2	2	2.

Here $h(i) = \min(f(i), g(i))$. Every row is a functor: along each forward edge its values are nondecreasing, while the values on the two ends of every backwards edge are equal. Pointwise inequalities give natural transformations

$$f \longleftarrow h \longrightarrow g.$$

Their endpoint components are identities, because all three rows take the values 0 and 2 at the two ends. Applying T produces the desired two-step ladder in $\mathrm{Ch}_{\Sigma,4}(X; x, z)$. No nonidentity arrow is placed on a backwards edge, so the proof works for any Σ containing the identities.

This calculation establishes $\theta(v)\theta(u) = \theta(vu)$ after passage to components. For identities the two diagrams already agree at the finite stage, as Grothendieck observes. Hence θ is a functor without assuming the general lemma. The display $\Delta_1 \sqcup \Delta_1 \simeq \Delta_2$ on the following manuscript page must mean amalgamation over the common object, $[1] \sqcup_{[0]} [1] \cong [2]$; the ordinary disjoint union has four objects.

The table also records why an unqualified homotopy in a functor category would be insufficient. An arbitrary map $\Phi_4 \to [2]$ may send a backwards edge to a nonidentity arrow. Its image under T would then require u, v, or vu to lie in Σ, which has not been assumed. Our three rows avoid that problem by making every backwards edge constant.

Il faut prouver que les deux sont homotopes. On a donc deux foncteurs

$$(*) \qquad f,g : \underbrace{\overbrace{\underline{\mathrm{Fl}}(X;x,y)}^{\text{catégorie discrète}} \times \overbrace{\underline{\mathrm{Fl}}(X;y,z)}^{\text{catégorie discrète}}}_{\mathcal{A}} \overset{f}{\underset{g}{\rightrightarrows}} \underbrace{\underline{\mathrm{Ch}}_{4,\Sigma}(X;x,z)}_{\mathcal{B}},$$

on veut prouver qu'ils sont homotopes. Pour ceci, on les regarde (comme tantôt) comme 'transposés' des deux foncteurs

$$\begin{array}{ccc} \mathbf{\Delta}_1 \sqcup \mathbf{\Delta}_1 \simeq \mathbf{\Delta}_2 & \underset{\psi}{\overset{\varphi}{\leftleftarrows}} & \Phi_4 \\ & \overset{\alpha}{\nwarrow} & \uparrow \alpha' \\ & & \varepsilon, \end{array}$$

vu que les catégories $\mathcal{A}$, $\mathcal{B}$ des deux membres de (*) s'interprètent comme des sous-catégories pleines de $\underline{\mathrm{Hom}}(\mathbf{\Delta}_2, X)$ et $\underline{\mathrm{Hom}}(\Phi_4, X)$ respectivement; à savoir les sous-catégories pleines formées des foncteurs dont le composé avec α resp. α' est donné par x et par z, et de plus (pour $\mathcal{A}$) dont la valeur en l'objet médian de $\mathbf{\Delta}_2$ soit y. On peut encore remplacer $\mathcal{A}$ par la sous-catégorie pleine plus grande $\mathcal{A}'$ de $\underline{\mathrm{Hom}}(\mathbf{\Delta}_2, X)$, obtenue en ignorant cette dernière condition, et il suffit de prouver que les deux foncteurs 'plus gros' p. 16

$$\mathcal{A}' \overset{f'}{\underset{g'}{\rightrightarrows}} \mathcal{B}'$$

sont homotopes. Et pour ceci, il suffit de voir encore que φ et ψ sont homotopes dans la [page 25] catégories des foncteurs $\Phi_4 \longrightarrow \mathbf{\Delta}_2$ qui sont compatibles avec la source, et le but. On voit donc que notre lemme d'homotopie de tantôt n'était pas assez général. Il faudrait définir les *catégories intervalles généralisés,* de longueur n quelconque (n = nombre de flèches bout à bout), une catégorie intervalle généralisé *standard* de longueur n a pour ensemble d'objets l'intervalle d'entiers

$$[0,n],$$

et pour tout couple d'entiers consécutifs $i-1$, i parmi les objets ($1 \leq i \leq n$), il y a exactement une flèche qui relie i, $i+1$ `[plutôt i − 1, i]`. On affecte l'intervalle $[i, i+1]$ du signe $+$ si $i \longrightarrow i+1$ (flèche en sens direct), du signe i `[plutôt −]` si $i \longleftarrow i+1$ (flèche en sens rétrograde) - donc les catégories standards sont définies, pour n fixé, par un système de signes $+$ ou $-$, dont on affecte les n 'tronçons' consécutifs $A_i := [i-1, i]$ ($1 \leq i \leq n$) de l'intervalle d'entiers $[0,n]$. Il y en a donc exactement 2^n.

On s'attend à ce que l'énoncé suivant soit vrai :

Commentary · Development (commentators') [ms. 25]

Contracting relative functor categories into an interval

Grothendieck's question asks whether the choices used to change a path's parametrization form a contractible category. We can prove a slightly more flexible statement that also accommodates the cancellation diagrams on page 27, where both endpoints have the same image. The proof uses the numerical order of an interval only to define minimum and maximum; the categorical order retains the prescribed directions of all its edges.

Relative interval contraction. [ours] Let J be the free category on an oriented path with vertices $0, \ldots, r$. Let $j : B \to A$ be a functor and $b : B \to J$ a fixed diagram. The strict fibre

$$\mathcal{F}_b = \{f : A \to J \mid fj = b\}$$

of restriction, with transformations restricting to 1_b, is either empty or categorically contractible: if $g \in \mathcal{F}_b$, its identity endofunctor is joined to the constant endofunctor at g by a finite zigzag of natural transformations. If A is itself an oriented interval and the fibre is restricted to functors

nondecreasing in the numerical vertex orders, the same assertion holds whenever that restricted fibre is nonempty.

Proof. The category J is a poset. Its order relation $a \leq_J a'$ means that the unique numerical segment between a and a' is oriented all the way from a to a'. For fixed c, each of the numerical clamping maps

$$a \longmapsto \min(a, c), \qquad a \longmapsto \max(a, c)$$

is a functor $J \to J$: the image of a directed segment is either a point or a directed subsegment with the same orientation. Consequently numerical minimum and maximum are functors $J \times J \to J$. Indeed they are order preserving in either variable separately, and a product-order comparison factors into one comparison in each variable.

Assume that $g \in \mathcal{F}_b$. For each vertex t of J define endofunctors of $\mathcal{F}_b$ by pointwise operations

$$F_t(f) = \min\bigl(f, \max(g, t)\bigr), \qquad G_t(f) = \min\bigl(g, \max(f, t)\bigr),$$

where t denotes the constant functor $A \to J$. These are functors on diagrams and on natural transformations because minimum and maximum are functors. They preserve the prescribed restriction: at an object of B, both f and g have the same value a, and

$$\min(a, \max(a, t)) = a.$$

Since J is a poset, the component of any resulting transformation at such an object is the identity. Thus the construction really stays inside the strict fibre.

Each generating arrow $t \to t'$ of J induces natural transformations $F_t \to F_{t'}$ and $G_t \to G_{t'}$. If an edge points in the opposite direction, both induced transformations point in that opposite direction. We therefore obtain finite zigzags as t runs numerically from 0 to r. At the ends,

$$F_r = 1_{\mathcal{F}_b}, \qquad F_0 = \min(-, g) = G_0, \qquad G_r = \mathrm{const}_g\,.$$

Traversing the F zigzag backwards and the G zigzag forwards contracts the identity to g. Every functor in the zigzag fixes g itself. On taking nerves and geometric realizations, natural transformations give homotopies, so $|N\mathcal{F}_b|$ is contractible as well. If $r = 0$, the nonempty fibre has one object and the conclusion is immediate.

For the numerical-monotonicity variant, the pointwise minimum or maximum of two nondecreasing numerical functions is nondecreasing, and constant functions are nondecreasing. Hence all F_t and G_t preserve the indicated full subcategory. Their restrictions give the same contraction. This completes the proof. □

The construction gives a uniform bound on the length of the contracting zigzag: there are r comparisons in the F family and r in the G family. This bound depends only on the target interval, not on the number of objects or arrows of A. In particular, no finite-source hypothesis was concealed in the proof. When $r = 1$, the contraction has the particularly transparent form

$$1_{\mathcal{F}_b} \longleftarrow \min(-, g) \longrightarrow \mathrm{const}_g$$

if the target edge points from 0 to 1; both arrows reverse if it points from 1 to 0. The fixed-boundary identity used above makes this a contraction inside the fibre even when j identifies several boundary labels. That includes the two coincident endpoint labels of an excursion.

Taking B to be the two endpoint labels proves both the proposed connectedness and the stronger contractibility of the generalized-interval functor category. Taking a fixed pair of equal endpoint values gives the version needed for an excursion followed by its return. For alternating intervals, the numerically monotone version strengthens the contraction-set argument on pages 17–21 and preserves the condition on backwards arrows when used for reparametrization.

The minimum–maximum preservation property is familiar in the theory of ordered digraphs; compare the definition of a Min–Max ordering in Hell and Rafiey (2012, p. 1577). We checked that account and the localization sources cited in this section, together with searches for relative functor categories of oriented intervals. We did not locate this fixed-restriction contraction in those sources. The result is presented as a direct continuation of the manuscript, with its full proof, and carries no claim of priority.

Lemme. *Soient I, I' deux intervalles généralisés, munis d'objets source et objet but*

(de sorte que I, I' soient canoniquement *isomorphes aux intervalles genéralisés standards correspondants). Soit* $\underline{\mathrm{Hom}}^!(I, I')$ *la catégorie des foncteurs de I dans I', compatibles avec source et but. Si cette catégorie est $\neq \emptyset$, elle est* 0*-connexe* (et, je présume, même contractile?). [page 26]

Je vais admettre provisoirement qu'il en est ainsi, du moins dans le cas particulier de Φ_4 (intervalle généralisé de longueur 8) et $\mathbf{\Delta}_2$. Cela donne donc un foncteur

$$\theta : X \longrightarrow X_\Sigma.$$

On a gagné, car la propriété universelle de ce foncteur est pratiquement évidente. Mais j'ai oublié de vérifier que θ transforme éléments de Σ en flèches inversibles. Soit donc

$$x \xrightarrow{u} y$$

avec $u \in \Sigma$, on va définir un chemin en sens inverse

$$y \xrightarrow{c} x$$

qui définira dans X_Σ une flèche inverse de u. L'idée naturelle est de prendre le 'chemin'

$$y \xleftarrow{u} x$$

de y dans x, mais ce n'est pas un chemin au sens de nos définitions, on va le compléter en

$$c : y \xrightarrow{\mathrm{id}} y \xleftarrow{u} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x. \qquad \text{p. 17}$$

Je dis que c fait l'affaire. On doit prouver deux relations d'homotopie, de $c \cdot \bar{u}$ et de $\bar{u} \cdot c$ avec les chemins identiques en x `[resp. en` y`]` de longueur 8 (indice 4, i.e. dans $\underline{\mathrm{Ch}}_{\Sigma,4}(X; x, x)$ resp. $\underline{\mathrm{Ch}}_{\Sigma,4}(X; y, y)$), où $\bar{u}$ est le chemin associé à u

$$x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y.$$

[page 27]

On a donc à regarder

$$x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{u} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x$$

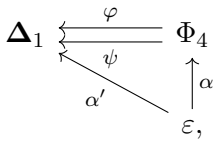
et voir qu'il est homotope au chemin trivial

$$x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} \cdots x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x$$

(des identités partout), et itou pour

$$y \xrightarrow{\mathrm{id}} y \xleftarrow{u} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xrightarrow{u} y \xleftarrow{\mathrm{id}} y \xrightarrow{\mathrm{id}} y \xleftarrow{\mathrm{id}} y$$

et le chemin trivial en y. Cette fois, on a encore deux foncteurs



compatibles avec objets sources et objets buts, et on doit prouver qu'ils sont homotopes dans $\underline{\mathrm{Hom}}^!(\Phi_4, \mathbf{\Delta}_1)$. Cela doit donc résulter du même lemme général. On a gagné - par une interprétation simple de $X\Sigma^{-1}$, et en particulier des $\mathrm{Hom}_{X\Sigma^{-1}}(x, y)$ et de leur composition, en termes de la géométrie des espaces de chemins :

$$\begin{cases} \mathrm{Hom}_{X\Sigma^{-1}} \simeq \pi_0(\underline{\mathrm{Ch}}_\Sigma(X; x, y)) \\ \text{Composition des homomorphismes dans } X\Sigma^{-1} \text{ correspond à} \\ \text{la composition des } \Sigma\text{-chemins dans } X \end{cases}$$

Commentary · Examples [ms. 27]

Cancellation uses the same endpoint twice

Grothendieck has written down the correct inverse path. The only remaining work is to make its two cancellation witnesses explicit. Let $u : x \to y$ belong to Σ, and write $T : [1] \to X$ for the corresponding functor. In the numerical ordering $0 < 1$, the first eight-edge excursion on the page is the image of the row

$$a = (0, 1, 1, 1, 1, 1, 0, 0, 0),$$

whereas the second is the image of

$$b = (1, 1, 0, 0, 0, 1, 1, 1, 1).$$

The source is Φ_4, numbered from 0 to 8. Both rows are functors: on a forward source edge the value can increase, and on a backwards edge it can decrease. Their endpoint values are respectively $(0, 0)$ and $(1, 1)$.

There are pointwise natural transformations

$$(0, 0, 0, 0, 0, 0, 0, 0, 0) \longrightarrow a, \qquad b \longrightarrow (1, 1, 1, 1, 1, 1, 1, 1, 1).$$

Applying T gives a transformation from the constant path at x to the first excursion, and one from the second excursion to the constant path at y. Their endpoint components are identities. Each horizontal backwards edge in these diagrams is either an identity or u, so it belongs to Σ. Thus both transformations are morphisms in the required Σ-path fibres, and the two cancellation identities follow immediately on π_0.

There is a small mismatch in the manuscript's invocation of the general lemma. That lemma is worded for maps taking the initial endpoint to the initial endpoint and the final endpoint to the final endpoint. These cancellation diagrams instead take both endpoints to 0, or both to 1. The relative-fibre theorem above includes them, but the two displayed transformations already prove exactly what is needed. This is a missing specification of the boundary condition, with an elementary repair, rather than a failure of localization.

Commentary · Exposition [ms. 27]

Completing the universal property

Grothendieck's path category now gives the promised localization. We record the complete conclusion because the construction uses less structure than many familiar fraction calculi.

Path presentation of ordinary localization. Let X be a small category and let Σ contain all its identity arrows. There is a canonical isomorphism of categories over X

$$X[\Sigma^{-1}] \cong X_\Sigma, \qquad \mathrm{Hom}_{X_\Sigma}(x, y) = \pi_0 \,\mathrm{Ch}_\Sigma(X; x, y),$$

where the path categories use strict endpoint fibres and composition is induced by concatenation at even indices.

Proof. The reparametrization argument proves independence of padding. More explicitly, the maps that compare padding before and after concatenation are numerically monotone endpoint-preserving functors between alternating intervals. The connectedness proved by even-block slides supplies a zigzag between them. After precomposition with a Σ-path, every backwards edge is still a backwards edge of that path or an identity. Thus the comparison is a zigzag inside the specified path category. Morphisms of the factors concatenate to morphisms of the result, so the product descends to components. Associativity follows from strict associativity of finite concatenation, after choosing common even indices; the zero-length paths give identities.

The preceding table proves that $\theta : X \to X_\Sigma$ preserves composition, and the cancellation ladders prove that it inverts Σ. Let $F : X \to Y$ invert Σ. Evaluate each finite path in Y by using F on forward arrows and inverses of $F(s)$ on backwards arrows. Identity padding leaves the result unchanged. Naturality squares, with the backwards ones inverted as above, show that endpoint-fixed ladders give equal arrows of Y. Evaluation therefore defines maps on the proposed Hom sets. It takes concatenation to composition and constant paths to identities, so it defines a functor $\overline{F} : X_\Sigma \to Y$ with $\overline{F}\theta = F$.

To prove uniqueness, one must know that a path class is the composite of its elementary arrows and inverse arrows. Pad the original path to an even index. Replace each forward arrow by its four-edge representative under θ, and each backwards arrow by the four-edge inverse representative. Concatenating these representatives gives the original list of nonidentity edges with additional identities inserted. It is the pullback of the original path by a numerically monotone contraction from a larger alternating interval. The usual end-padding is another such pullback. The two contractions belong to the same connected endpoint-preserving category, so the two paths have the same component. Hence every path class is the required composite, and any extension of F must evaluate it as $\overline{F}$ does.

Thus θ has the defining universal property of localization. Applying the two universal properties gives inverse functors between X_Σ and $X[\Sigma^{-1}]$ that agree with the identity on X. Their composites are identities by uniqueness. □

The proof explains the exact status of the manuscript: its principal theorem is correct, and its proof becomes complete after the endpoint-fibre convention, the even-index repair already made by the author, and the supplied interval comparisons. No Ore condition or closure of Σ under composition has entered.

Commentary · Later developments [ms. 27]

From components of path categories to mapping spaces

Grothendieck has recovered a set of morphisms by taking connected components of a category of presentations. The next question is whether one can retain higher homotopy information and compose it coherently. The relevant historical comparison already existed before these notes: the two papers of Dwyer and Kan appeared in 1980.

Their standard simplicial localization has the ordinary localization as its category of components (Dwyer and Kan, 1980c, §4.2). The hammock construction gives a more diagrammatic model: its simplices are commutative ladders of variable length, with all vertical arrows and all leftward horizontal arrows in the chosen subcategory of weak equivalences. Composition is concatenation followed by reduction (Dwyer and Kan, 1980a, Definition 2.1 and Proposition 3.1). These are close relatives of the present paths, but the vertical-arrow requirement is different: Grothendieck permits arbitrary vertical arrows. Therefore the formula for π_0 proved here does not itself identify the nerve of his path category with a derived mapping space.

There is also a preliminary distinction in the input data. The classical Dwyer–Kan construction starts with a subcategory W containing all objects, whereas the present Σ need contain only identities. To compare the two ordinary localizations one may take W to be the subcategory generated by Σ: every functor inverting Σ inverts its finite composites, and the converse is immediate. This proves equality of their universal properties. It does not identify the intervening categories of paths. The example with $\Sigma = \{1, t\}$ above shows concretely how adjoining composite denominators changes the allowed horizontal presentations before any homotopy comparison is proved.

The comparison becomes sharper under additional hypotheses. A homotopy calculus of fractions allows certain short zigzag categories to compute hammock mapping spaces; the exact reduction results are Dwyer and Kan (1980a, Proposition 6.2). Their hypotheses concern the homotopy types of diagram categories, not just the existence of short ordinary fractions. Consequently a reduction that is sufficient for Hom sets need not preserve all higher homotopy groups.

One elementary check shows the difference in what is being sought. If every arrow of a category X is inverted, the ordinary localization records only its groupoid completion. A higher localization retains the homotopy type of the classifying space: in the connected case the mapping spaces have the homotopy type of its loop space (Dwyer and Kan, 1980c, §5.5). Take the poset of nonempty proper faces of a tetrahedron: its nerve is the barycentric subdivision of the boundary sphere S^2. Its groupoid completion is equivalent to the terminal groupoid, since the sphere is connected and simply connected, while the higher endomorphism space has the nontrivial homotopy type ΩS^2. This example does not decide the homotopy type of Grothendieck's particular path category; it explains why agreement on its components is only the first comparison one would require.

The later development of simplicial and higher localizations gives a precise home for the ambition of retaining those spaces of presentations. It does not retroactively turn every category of zigzags into the same model. The admissible horizontal arrows, the vertical arrows, the allowed reparametrizations, and the endpoint conditions are part of the theorem to be proved.

p. 18

[page 28]

3 Passage aux catégories de diagrammes

Nous aurions dû noter la catégorie $X\ \mathcal{M}$, car par la suite elle jouera le rôle d'une 'catégorie de modèles'. On désignera par contre par X les 'catégories d'indices' sur lesquelles on veut construire les 'catégories de diagrammes' de type X

$$\mathcal{M}(X) = \underline{\mathrm{Hom}}(X, \mathcal{M}).$$

On se donne donc

$$\underbrace{\Sigma}_{\text{quasi-isomorphismes}} \subseteq \mathrm{Fl}(\mathcal{M}), \qquad \Sigma \supset \text{ensemble des flèches identiques},$$

et pour toute catégorie X on définit

$$\Sigma_X \subset \mathrm{Fl}\,\mathcal{M}(X)$$

ainsi: si $F, G \in \mathrm{Ob}\,\mathcal{M}(X)$ $(F, G : X \rightrightarrows \mathcal{M})$, une flèche $u : F \longrightarrow G$ (homomorphisme de foncteurs) est dans Σ_X si et seulement si $\forall x \in X$, $u(x) : F(x) \longrightarrow G(x)$ est dans Σ ('critère de quasi-isomorphie fibre par fibre', pour l'homomophisme entre les copréfaisceaux F, G envisagés).

Commentary · Exposition [ms. 28]

The parameter is a diagram shape

Grothendieck now asks the path construction to do more than compute a single localized Hom-set. He wants the same calculation after replacing the coefficient category by every category of diagrams. This is the point at which a category of paths begins to behave like an external Hom: its test objects are the indexing categories themselves. The change of letters is deliberate. In this section $\mathcal{M}$ supplies coefficients, whereas X is allowed to vary as a parameter.

There is no model-category structure in the hypotheses. We have a category $\mathcal{M}$ and a class Σ containing the identities; for the localization calculation even closure of Σ under composition is unnecessary. One may enlarge Σ to the subcategory it generates without changing the localization. We work in fixed universes, so that the localizations and the Hom-sets under discussion exist in the chosen larger universe. The designation "quasi-isomorphism" means membership in this specified class, rather than a statement about chain complexes.

Write $\mathcal{M}^X = \mathrm{Fun}(X, \mathcal{M})$ and let Σ_X consist of natural transformations whose components belong to Σ. Restriction along $f : X \to Y$ preserves these transformations and therefore induces

$$f^* : \mathcal{M}^Y[\Sigma_Y^{-1}] \longrightarrow \mathcal{M}^X[\Sigma_X^{-1}].$$

The two objects of special interest are the constant diagrams a_X, b_X. Thus the functor to be understood is

$$H_{a,b}(X) = \mathrm{Hom}_{\mathcal{M}^X[\Sigma_X^{-1}]}(a_X, b_X), \qquad H_{a,b} : \mathrm{Cat}^{\mathrm{op}} \to \mathrm{Ens}.$$

These are constant diagrams before localization. We are not replacing an arbitrary diagram by its values in $\mathcal{M}[\Sigma^{-1}]$; that would discard precisely the coherent information retained in Chapters I and II. See the commentary on Chapter I, manuscript p. 13 ("Localizing diagrams and deriving their Kan extensions") for that distinction.

For a fixed finite path shape Φ_n, the elementary exponential law

$$\mathrm{Fun}(\Phi_n, \mathrm{Fun}(X, \mathcal{M})) \cong \mathrm{Fun}(X, \mathrm{Fun}(\Phi_n, \mathcal{M}))$$

already contains the calculation. Its objects agree because both sides are functors $X \times \Phi_n \to \mathcal{M}$; its arrows agree because both sides are natural transformations of such functors. The restrictions on backward arrows are checked at each object of X, and the restrictions on the endpoints are also objectwise. The difficulty is therefore not the exponential law. It is the passage from one finite path shape to all path lengths, where one must retain a uniform bound.

Soient

$$a, b \in \mathrm{Ob}\,\mathcal{M}$$

et considérons les foncteurs constants correspondants sur X

$$a_X, b_X \in \mathrm{Ob}\,\mathcal{M}(X).$$

On se propose d'appliquer la théorie du §2 au 'calcul' de [page 29]

$$\mathrm{Hom}_{\Sigma_X}(a_X, b_X),$$

on a donc une bijection canonique

$$\boxed{\mathrm{Hom}_{\Sigma_X}(a_X, b_X) \simeq \pi_0 \underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X); a_x, b_x).}$$

On se propose d'expliciter la catégorie de chemins du second membre. On a, abstraction faite des données de Σ, a, b dans $\mathcal{M}$, l'expression pour la catégorie de *tous* les chemins dans $\mathcal{M}(X)$ p. 19

$$\begin{aligned}
\underline{\mathrm{Ch}}(\mathcal{M}(X)) &\simeq \underline{\mathrm{Hom}}\Big(\underbrace{\mathbf{\Phi}}_{\substack{\text{pro-catégorie}\\(\Phi_n)_{n\in\mathbf{N}}}}, \underbrace{\mathcal{M}(X)}_{\underline{\mathrm{Hom}}(X,M)}\Big) \\
&\overset{\text{déf}}{=} \varinjlim_n \underline{\mathrm{Hom}}(\Phi_n, \mathcal{M}(X)) \\
&\simeq \varinjlim_n \underline{\mathrm{Hom}}(\Phi_n \times X, \mathcal{M}) \\
&\simeq \varinjlim_n \underline{\mathrm{Hom}}(X, \underline{\mathrm{Hom}}(\Phi_n, \mathcal{M})) \\
&\overset{\text{déf}}{=} \varinjlim_n \underline{\mathrm{Hom}}(X, \underline{\mathrm{Ch}}_n(\mathcal{M})) \\
&\simeq \underline{\mathrm{Hom}}^!(X, \underline{\mathrm{Ch}}(\mathcal{M})),
\end{aligned}$$

où le signe ! signifie qu'on prend la sous-catégorie pleine des foncteurs

$$X \longrightarrow \underline{\mathrm{Ch}}(\mathcal{M}),$$

qui se factorisent par une des sous-catégories pleines $\underline{\mathrm{Ch}}_n(\mathcal{M})$. On peut dire aussi que si on [page 30] considère le système inductif des catégories $\underline{\mathrm{Ch}}_n(\mathcal{M})$ comme un ind-objet de Cat, qu'on va noter

$$\underline{\underline{\mathrm{Ch}}}(\mathcal{M}) \qquad\qquad \text{ind-objet de Cat,}$$

on a

$$\underline{\mathrm{Ch}}(\mathcal{M}(X)) \xrightarrow{\sim} \underline{\mathrm{Hom}}(X, \underline{\underline{\mathrm{Ch}}}(\mathcal{M})).$$

Si on en vient à Σ, Σ_X, on trouve aussitôt qu'un chemin dans $\mathcal{M}(X)$ est un Σ_X-chemin, si et seulement si le foncteur correspondant $X \longrightarrow \underline{\mathrm{Ch}}(\mathcal{M})$ se factorise par $\underline{\mathrm{Ch}}_\Sigma(\mathcal{M})$. Ainsi on trouve

$$\underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X)) \simeq \underline{\mathrm{Hom}}^!(X, \underline{\mathrm{Ch}}_\Sigma(\mathcal{M})),$$

où le signe ! a le même sens que plus haut. Si on introduit le ind-objet $\underline{\underline{\mathrm{Ch}}}_\Sigma(\mathcal{M})$ défini par les $\underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M})$, on trouve donc aussi

$$\underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X)) \xrightarrow{\sim} \underline{\mathrm{Hom}}(X, \underline{\underline{\mathrm{Ch}}}_\Sigma(\mathcal{M})).$$

Commentary · Exposition [ms. 30]

Why the exclamation mark is necessary

Grothendieck introduces the exclamation mark to remember that a path of diagrams has one length for the entire diagram. An object of $\mathrm{Fun}(X, \mathrm{Ch}_\Sigma(\mathcal{M}))$ may assign a finite path to every $x \in X$ without these lengths being bounded as x varies. Such a functor need not come from a path in $\mathcal{M}^X$. This is a genuine distinction even before taking connected components or weak equivalences.

Let K_n denote the fixed-endpoint path category at stage n, using the fully faithful identity-padding inclusions, and put $K_\infty = \bigcup_n K_n$. Then

$$\mathrm{Fun}^!(X, K_\infty) = \bigcup_n \mathrm{Fun}(X, K_n)$$

is the full subcategory on bounded functors. Fullness deserves a verification. If F and G factor through K_n, every component of a natural transformation $F \Rightarrow G$ lies in K_n, because $K_n \subseteq K_\infty$ is full. Its naturality equations are the same equations in either category. Thus no additional bound is needed for arrows between already bounded objects.

The failure for an infinite source can be seen without any homotopy theory. Take K_n to be the discrete category on $\{0, \dots, n\}$ and X the discrete category on $\mathbf{N}$. The functor $x \mapsto x$ from X to K_∞ lies in no $\mathrm{Fun}(X, K_n)$. It cannot even be isomorphic to a bounded functor, since the target is discrete. The ind-object (K_n) records a uniform finite stage which the ordinary union has forgotten.

Connected components do commute with the filtered union used here. A component of the union has a representative object at some stage. If two objects become connected, a finite zigzag of arrows witnesses this, and all its objects and arrows occur at one sufficiently large stage. This proves directly that

$$\varinjlim_n \pi_0 \,\mathrm{Fun}(X, K_n) \xrightarrow{\sim} \pi_0 \left(\varinjlim_n \mathrm{Fun}(X, K_n) \right).$$

Nothing in this argument permits one to replace the category inside the final parentheses by the unrestricted functor category. The exclamation mark and the double underline serve the same purpose: the first describes bounded functors, while the second retains the system of bounds as an ind-object.

[page 31]

Soit c un chemin dans $\mathcal{M}(X)$, et

$$X \longrightarrow \underline{\mathrm{Ch}}(\mathcal{M})$$

le foncteur qu'il définit, composons-le avec les foncteurs source-but

$$\underline{\mathrm{Ch}}(\mathcal{M}) \longrightarrow \mathcal{M} \times \mathcal{M},$$

on trouve un foncteur

$$X \longrightarrow \mathcal{M} \times \mathcal{M},$$

i.e. un objet (F_0, F_1) de $\mathcal{M}(X) \times \mathcal{M}(X)$. Je dis que F_0 et F_1 sont respectivement la source et le but du chemin c. De façon un peu plus précise, on a commutativité (stricte) dans le diagramme de foncteurs p. 20

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma_X}(\mathcal{M}(X)) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}^!(X, \underline{\mathrm{Ch}}_\Sigma(\mathcal{M})) \\ \downarrow{\scriptstyle \sigma_{\mathcal{M}(X)}, \beta_{\mathcal{M}(X)}} & & \downarrow{\scriptstyle \text{déduit de } \sigma_{\mathcal{M}}, \beta_{\mathcal{M}}} \\ \mathcal{M}(X) \times \mathcal{M}(X) & \xrightarrow{\sim} & \underline{\mathrm{Hom}}(X, \mathcal{M} \times \mathcal{M}), \end{array}$$

où les foncteurs σ sont les foncteurs source, les foncteurs β les foncteurs but.

Si on se donne arbitrairement les objets F_0, F_1 dans $\mathcal{M}(X)$, donc un foncteur

[page 32 vide]

$$X \longrightarrow \mathcal{M} \times \mathcal{M},$$ [page 33]

$\underline{\mathrm{Ch}}_{\Sigma_X}(F_0, F_1)$ s'identifie donc à la catégorie des foncteurs 'admissibles' (i.e. se factorisant par un $\underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M})$) $X \xrightarrow{C} \underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M})$ qui *relèvent* le foncteur $(\sigma_{\mathcal{M}}, \beta_{\mathcal{M}})$ `[plutôt` (F_0, F_1)`]` précédent, i.e. qui donnent lieu à un triangle commutatif (strict)

$$\begin{array}{ccc} & & \underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M}) \\ & \overset{C}{\nearrow} & \big\downarrow {\scriptstyle (\sigma_{\mathcal{M}}, \beta_{\mathcal{M}})} \\ X & \xrightarrow[(F_0,F_1)]{} & \mathcal{M} \times \mathcal{M}. \end{array}$$

Mais si les foncteurs F_0, F_1 sont constants, de valeur a, b comme tantôt, i.e. (F_0, F_1) constant de valeur (a, b), alors la catégorie des foncteurs C en question est tautologiquement isomorphe à celle des foncteurs de X dans la fibre de $(\sigma_{\mathcal{M}}, \beta_{\mathcal{M}})$ en (a, b), d'où l'isomorphisme de catégories fondamental

$$\boxed{\underline{\mathrm{Ch}}_{\Sigma_X}(X; a_X, b_X) \simeq \underline{\mathrm{Hom}}^{!}(X, \underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M}; a, b)).}$$

On va expliciter le contenu essentiel de cette formule en une

Commentary · Exposition [ms. 33]

The precise meaning of ind-representation

The scholium isolates what survives every arbitrary choice in the path calculation. If Cat hot denotes the category with small categories as objects and

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(X, Y) = \pi_0 \,\mathrm{Fun}(X, Y),$$

then the result says that $H_{a,b}$ is a filtered colimit of representable presheaves on Cat hot. The homotopy relation in this formula is generated by natural transformations, with arrows in either direction. Composition is well defined because whiskering preserves natural transformations.

For clarity, the fully faithful embedding of ind-objects in presheaves can be proved in the notation needed here. To an ind-object $K = (K_i)$ of a locally small category C associate

$$h_K(X) = \varinjlim_i \mathrm{Hom}_C(X, K_i).$$

If $L = (L_j)$ is another ind-object, Yoneda and the universal property of a colimit give

$$\mathrm{Nat}(h_K, h_L) \cong \varprojlim_i \mathrm{Nat}(h_{K_i}, h_L) \cong \varprojlim_i \varinjlim_j \mathrm{Hom}_C(K_i, L_j) = \mathrm{Hom}_{\mathrm{Ind}(C)}(K, L).$$

The limits and colimits occur in that order. No interchange theorem has been used. A natural isomorphism of the presheaves consequently gives a unique isomorphism of the corresponding ind-objects, compatible with the specified representations. This is the exact sense of the uniqueness statement on manuscript p. 34.

To obtain the scholium, apply the preceding path-localization theorem inside $\mathcal{M}^X$, then apply the exponential law with fixed endpoints, and finally commute π_0 with the filtered union. This yields

$$H_{a,b}(X) \cong \varinjlim_n \mathrm{Hom}_{\mathrm{Cat\,hot}}(X, K_n) = h_{(K_n)}(X).$$

The notation in the boxed formula just before the scholium has X where the ambient path category should be $\mathcal{M}(X)$; the edition prints this slip. The preceding endpoint square determines the intended meaning unambiguously.

The result has a useful limit. It gives uniqueness in Ind(Cat hot), not uniqueness of the ordinary union K_∞ in Cat hot. A colimit of a chosen sequence of actual categories contains information about the representatives and their compatibility which an ind-object in a homotopy category need not retain. The questions at the end of the chapter begin exactly at this boundary.

Scholie. Soient $\mathcal{M}$ une catégorie avec un ensemble Σ de flèches de localisation, $a, b \in \mathrm{Ob}\,\mathcal{M}$. Alors on peut trouver un objet $\underline{\underline{K}} = K(a, b)$, $K = (K_n)$, de Ind Cat (indexé par $\mathbf{N}$ mais

peu importe ce fait), et un isomorphisme canonique pour toute catégorie X (isomorphisme fonctoriel en X)

$$\begin{aligned}\mathrm{Hom}_{\Sigma_X}(a_X, b_X) &\simeq \pi_0(\underline{\mathrm{Hom}}(X, \underline{\underline{K}}))\\ &\overset{\text{déf}}{=} \pi_0(\varinjlim \underline{\mathrm{Hom}}(X, K_n))\\ &\simeq \varinjlim \pi_0(\underline{\mathrm{Hom}}(X, K_n)).\end{aligned}$$

Je vais épiloguer sur ce résultat, sans me soucier pour l'instant de la façon particulière dont la ind-catégorie K [plutôt $\underline{\underline{K}}$] a été obtenue en termes de $\mathcal{M}$, Σ, a, b. p. 21 [page 34]

Comme les $\pi_0\underline{\mathrm{Hom}}(X, K_n) = \mathrm{Hom}_{\mathrm{Cat\,hot}}(X, K_n)$ ne dépendent que de X et de K_n en tant qu'objets de Cat hot, je vais interpréter $\underline{\underline{K}}$ comme un ind-objet de Cat hot, et écrire la formule fondamentale sous la forme

$$\mathrm{Hom}_{\Sigma_X}(a_X, b_X) \simeq \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(X, \underline{\underline{K}}).$$

Ainsi, l'énoncé du scholie implique deux choses :

a) Le contrafoncteur $X \longmapsto \mathrm{Hom}_{\Sigma_X}(a_X, b_X)$ sur Cat se factorise canoniquement par Cat hot, via un foncteur canonique

$$\mathrm{Cat\,hot}^o \xrightarrow{H = H^{\mathcal{M},\Sigma}_{a,b}} \mathrm{Ens}\ ;$$

b) Ce dernier contrafoncteur sur Cat hot est *ind-représentable* par un ind-objet $\underline{\underline{K}}$ de Cat hot.

C'est cet ind-objet de Cat hot qui est défini à isomorphisme unique près par $\mathcal{M}$, Σ, a, b, indépendamment de tout artifice de calcul pour expliciter les $\mathrm{Hom}_{\Sigma_X}(a_X, b_X)$.

Notons aussi la conclusion suivante, nullement évidente à priori :

Notons d'abord (ceci étant implicite dans le scholie) que toute flèche dans Cat [page 35]

$$X \xrightarrow{f} Y$$

définit

$$\mathcal{M}(Y) \xrightarrow{\mathcal{M}(f)} \mathcal{M}(X) \qquad \text{appliquant } \Sigma_Y \text{ dans } \Sigma_X,$$

d'où

$$\mathcal{M}(Y)\Sigma_Y^{-1} \overset{\text{déf}}{=} \mathbf{D}_{\mathcal{M},\Sigma}(Y) \longrightarrow \mathcal{M}_X\Sigma_X^{-1} = \mathbf{D}_{\mathcal{M},\Sigma}(X),$$

[plutôt $\mathcal{M}(X)$] d'où, pour tout $a, b \in \mathrm{Ob}\,\mathcal{M}$, vu que $\mathcal{M}(f)$ transforme a_Y, b_Y en a_X, b_X :

$$\boxed{\mathrm{Hom}_{\Sigma_Y}(a_Y, b_Y) \longrightarrow \mathrm{Hom}_{\Sigma_X}(a_X, b_X).}$$

Ceci posé, on a :

Corollaire : *Si f est un homotopisme, alors quelque soient $a, b \in \mathcal{M}$, la flèche précédente est un isomorphisme.*

La question naturelle, c'est de savoir si la même conclusion est encore valable, en supposant seulement que f soit un '*quasi-isomorphisme*' (ou 'équivalence faible'), ici que f définisse un isomorphisme dans la catégorie de fractions Hot de Cat. (À ne pas confondre avec Cat hot, nettement plus fine). S'il en était ainsi, le contrafoncteur p. 22 [page 36]

$$X \longmapsto \mathrm{Hom}_{\Sigma_X}(a_X, b_X)$$

se factoriserait non seulement par Cat hot, mais même par la catégorie de fractions plus grossière Hot. Et dès lors se poserait la question si ce contrafoncteur hypothétique

$$\mathrm{Hot}^o \longrightarrow \mathrm{Ens}$$

est encore ind-représentable; et de préférence ([5]) ind-représentable par le ind-objet image de $\underline{\underline{K}}$ dans Ind Hot par

$$\mathrm{Ind}(\mathrm{Cat}\,\mathrm{hot}) \longrightarrow \mathrm{Ind}(\mathrm{Hot}),$$

déduit du foncteur de localisation Cat hot $\longrightarrow$ Hot. (Car cet ind-objet est un candidat tout trouvé.)

Commentary · Exposition [ms. 35]

Two homotopy invariance questions

Grothendieck has proved invariance under categorical homotopy equivalence, and now asks for invariance under weak equivalence of nerves. The first assertion is immediate from the ind-representation. A natural transformation $f \Rightarrow g$ induces, by precomposition, a natural transformation between the two functors on each $\mathrm{Fun}(Y, K_n)$. These functors therefore induce the same map on π_0. A finite zigzag gives the same conclusion, and a homotopy inverse to f gives inverse maps on $H_{a,b}$.

The second assertion has additional content. A weak equivalence Nf need not admit an inverse represented by a functor and a finite zigzag of natural transformations. Nor does the construction of $H_{a,b}$, for an arbitrary class Σ, supply a derived mapping space. Thus the corollary proves factorization through Cat hot; it does not prove factorization through Hot. This is the same difference that appeared in the commentary on Chapter I, manuscript p. 102 ("What the completion requires"), now applied to a particular family of coefficients.

Once that factorization is established, however, the subsequent question about ind-representability has a formal affirmative answer. It does not require the proposed right adjoint to Cat hot $\to$ Hot. More generally let $\varphi : C \to D$ be a localization, let $H = \varinjlim_i h_{K_i}$ on C, and suppose $H = \varphi^* H'$ for a presheaf H' on D. Restriction φ^* is fully faithful by the universal property of localization. Put $L = \varinjlim_i h_{\varphi K_i}$ on D. For every presheaf F on D, Yoneda gives natural bijections

$$\mathrm{Nat}(L, F) \cong \varprojlim_i F(\varphi K_i) \cong \mathrm{Nat}(H, \varphi^* F) \cong \mathrm{Nat}(H', F).$$

Yoneda, now in the presheaf category, proves $L \cong H'$. Equivalently, left Kan extension along φ^{op} sends h_K to $h_{\varphi K}$, preserves the filtered colimit, and recovers a descended presheaf; this is the usual Kan-extension formalism (Riehl, 2014, §§1.1–1.4). Thus, conditional only on weak-equivalence invariance, the image of (K_i) in Ind(Hot) represents the descended functor. What remains conditional on the later right adjoint is the characterization by a Thomason–Kan envelope, not this descent of ind-representability.

A useful sufficient hypothesis is that localization of the strict diagram categories presents the expected infinity-categories of diagrams. More explicitly, suppose a homotopy theory $\mathcal{C}$ satisfies, naturally in small X,

$$\mathcal{M}^X[\Sigma_X^{-1}] \simeq h\,\mathrm{Fun}(NX, \mathcal{C}).$$

If $Z = \mathrm{Map}_{\mathcal{C}}(a, b)$, the space of natural transformations between constant diagrams is

$$\mathrm{Map}_{\mathrm{Fun}(NX,\mathcal{C})}(a_X, b_X) \simeq \mathrm{Map}_{\mathcal{S}}(|NX|, Z).$$

Indeed, a coherent transformation chooses a point of Z at each object, a path for each arrow, and the compatible higher homotopies for composable strings; these are exactly the simplices of a map from NX into Z. Taking π_0 proves $H_{a,b}(X) \cong [|NX|, Z]$. Under this hypothesis the functor is represented on Hot, rather than merely ind-represented there.

Combinatorial simplicial model categories provide a familiar setting for this statement. The mapping space must be derived, using a cofibrant source and a fibrant target. The model-independent function-complex construction is due to Dwyer and Kan; their comparison theorem explains why the resulting homotopy type is independent of auxiliary model structures (Dwyer and Kan, 1980b, §4). The displayed hypothesis is stated explicitly here because it is not a consequence of an arbitrary choice of Σ.

5. ce sera automatique si on a ind-représentabilité comme foncteur sur Hot, cf. ci-dessous.

Commentary · Examples [ms. 36]

An external Hom that is an ordinary homotopy type

Grothendieck's candidate can be made completely concrete for simplicial sets with their usual weak equivalences. Take $a = \Delta^0$ and let $b = Z$ be a Kan complex. For a small category X, a projectively cofibrant replacement of the constant point diagram is supplied by the usual bar construction. At $x \in X$ its value is $N(X/x)$; this is contractible because X/x has the terminal object id_x. The diagram is assembled from the covariant representables by attaching simplices along their boundaries, so it is projectively cofibrant. The constant diagram Z_X is projectively fibrant.

A natural transformation from this resolution to Z_X is equivalent to a simplicial map from its colimit to Z. The colimit of the diagram $x \mapsto N(X/x)$ is NX: a simplex represented by a string ending in an arrow to x is identified with the same string at its last vertex. Consequently the derived simplicial mapping space between the two constant diagrams is Z^{NX}. Since the source simplicial set is cofibrant and Z is Kan,

$$\mathrm{Hom}_{\mathrm{Ho}(\mathbf{sSet}^X)}(*_X, Z_X) = \pi_0 Z^{NX} = [NX, Z].$$

This is the bar-construction calculation underlying the abstract assertion in the preceding block; see Riehl (2014, §§4.4–4.5 and 5.1) for the construction and its homotopical justification.

For $Z = K(A, m)$ with $m \geq 1$ and A abelian, it gives $H_{*,Z}(X) = H^m(NX; A)$, with constant coefficients. If X is the one-object category attached to the cyclic group C_r, the familiar periodic resolution has alternating differentials $g - 1$ and $1 + g + \cdots + g^{r-1}$. After applying $\mathrm{Hom}_{\mathbf{Z}[C_r]}(-, A)$ for the trivial action, these differentials become 0 and multiplication by r. Hence

$$H^{2q}(BC_r; A) = A/rA \quad (q \geq 1), \qquad H^{2q+1}(BC_r; A) = A[r] \quad (q \geq 0).$$

The category BC_r has finitely many objects and arrows, but its nerve has nondegenerate simplices in arbitrarily high dimensions. Thus "finite category" in the closing question must not be silently replaced by "finite CW complex". The same example also shows that successful homotopy-theoretic coefficients often produce an actually representable functor on Hot. This is distinct from essential constancy in Ind(Cat hot), the subject of the pessimistic parenthesis on manuscript p. 46. Passing from the first assertion to the second would require precisely the right adjoint discussed next.

Je vais faire quelques rappels catégoriques, généralisant la situation du foncteur canonique [page 37]
de localisation

$$\mathrm{Cat\,hot} \xrightarrow{\varphi} \mathrm{Hot},$$

dont on sait qu'il admet un adjoint à droite `[si un lecteur a une démonstration de ce fait, prière de contacter G. Maltsiniotis : [e-mail address removed]]`

$$\mathrm{Cat\,hot} \xleftarrow{\psi} \mathrm{Hot}, \qquad \mathrm{Hom}_{\mathrm{Hot}}(\varphi X, Y) \simeq \mathrm{Hom}_{\mathrm{Cat\,hot}}(X, \psi Y),$$

où ψ est le foncteur 'enveloppe de Kan' ou 'représentant de Kan'. On va donc considérer la situation plus générale

$$C \underset{\varphi}{\overset{\psi}{\leftrightarrows}} C' \qquad \mathrm{Hom}_{C'}(\varphi X, X') \simeq \mathrm{Hom}_C(X, \psi X')$$

i.e.

$$\varphi, \psi \text{ couple de foncteurs adjoints.}$$

On suppose de plus

$$\begin{cases} \varphi \text{ un foncteur localisation, i.e. } \psi \text{ pleinement fidèle,} \\ \text{i.e. } \varphi\psi \xrightarrow[\sim]{\mathrm{adj}} \mathrm{id}_{C'} \quad . \end{cases}$$

Soit $W = W_\varphi \subset \mathrm{Fl}\, C$, défini par $\varphi : W_\varphi = \{u \in \mathrm{Fl}\, C \mid \varphi(u) \text{ isomorphisme}\}$. Voici la proposition que j'avais oubliée, et que j'ai eu du mal à reconstituer, hélas!

Commentary · Development (commentators') [ms. 37]

The reflection does not exist

Grothendieck wants to convert homotopy invariance of a represented functor into a condition on its representing object. For a reflective localization this is an elementary and powerful argument, and Propositions 1 and 2 below are correct formal statements about any reflective localization. The edition questions the right adjoint to $\varphi : \mathrm{Cat\,hot} \to \mathrm{Hot}$ on which their application rests. It does not exist.

Theorem. [ours] *The functor $\varphi : \mathrm{Cat\,hot} \to \mathrm{Hot}$ has no right adjoint. More precisely, there is no small category K with bijections $\pi_0 \mathrm{Fun}(X, K) \cong [BX, K(\mathbf{Z}/2, 2)]$ natural in the small category X.*

Proof. A right adjoint ψ would give such a K, namely ψ of the homotopy type $Y = K(\mathbf{Z}/2, 2)$, since morphisms $X \to K$ in $\mathrm{Cat\,hot}$ are the connected components of $\mathrm{Fun}(X, K)$. Suppose K is given. Taking $X = e$ shows that $\pi_0 K = [\mathrm{pt}, Y]$ is one point.

Let Ω be an infinite set of cardinality greater than that of the set of arrows of K, and G the group of even permutations of Ω with finite support. It is the union of the alternating groups of the finite subsets of Ω, so it is simple, and its cardinality is that of Ω. A functor from the one-object category $\mathsf{B}G$ to K is an object k with a homomorphism $G \to \mathrm{Aut}(k)$. Its kernel is a normal subgroup; it is not trivial because $\mathrm{Aut}(k)$ has fewer elements than G; so the homomorphism is trivial. A natural transformation between two such constant functors is a single arrow of K. Hence $\mathrm{Fun}(\mathsf{B}G, K) \cong K$ and $\pi_0 \mathrm{Fun}(\mathsf{B}G, K)$ is one point.

On the other side $[BG, K(\mathbf{Z}/2, 2)] = H^2(G; \mathbf{Z}/2)$ classifies the central extensions of G by $\mathbf{Z}/2$, and one of them does not split. Let V be the real vector space with orthonormal basis $(e_\omega)_{\omega \in \Omega}$ and $\mathrm{Cl}(V)$ its Clifford algebra, with $v^2 = -|v|^2$. For $\omega \neq \omega'$ the unit vector $\tau = (e_\omega - e_{\omega'})/\sqrt{2}$ is invertible in $\mathrm{Cl}(V)$ and $v \mapsto -\tau v \tau^{-1}$ is the reflection of V that exchanges e_ω and $e_{\omega'}$. Let $\widehat{G}$ be the group of products of an even number of such vectors. Sending such a product to the corresponding product of transpositions is a surjective homomorphism $\widehat{G} \to G$; its kernel is contained in $\{\pm 1\}$, the kernel of the map from the Pin group to the orthogonal group, and $-1 = (\tau\tau')^2 \in \widehat{G}$ for two orthogonal such vectors τ, τ'. So $\widehat{G}$ is a central extension of G by $\mathbf{Z}/2$. The element $g = (\omega_1\, \omega_2)(\omega_3\, \omega_4)$ has order two, and its two lifts $\pm\tau\tau'$ have square -1; a splitting would lift g to an element of order two. So $H^2(G; \mathbf{Z}/2) \neq 0$, and the two sets cannot be in bijection. □

The argument shows more than the statement: a category K represents on $\mathrm{Cat\,hot}$ only what homomorphisms into its automorphism groups and its arrows can see, and a group that is large and simple is invisible to it. The functors $X \mapsto [BX, Y]$ that are representable on all of $\mathrm{Cat\,hot}$ are those with Y a 1-type, represented by a groupoid (the proof for a general Y, which replaces G by large simple groups with the homology of spheres, is not reproduced here); the fundamental groupoid $X \mapsto \Pi_1 X$ is a reflection of $\mathrm{Cat\,hot}$ onto groupoids modulo natural isomorphism, and it inverts the functors that are bijective on π_0 and on all π_1, a larger class than the weak equivalences. Propositions 1 and 2 apply to that reflection.

Thomason's model structure gives a fibrant replacement $K \to RK$ and an equivalence between its homotopy category and Hot (Thomason, 1980, Theorem 4.9 and Corollary 5.5). It does not identify $\pi_0 \mathrm{Fun}(X, RK)$ with $[NX, NK]$ for every X: mapping formulas in a model category need a cofibrant source, and the cofibrant objects of that structure are posets. Below, references to an envelope or to a reflective localization of $\mathrm{Cat\,hot}$ onto homotopy types are to be read as hypotheses that cannot be met; references to Thomason fibrant replacements mean the model-category construction and carry only its conclusions.

Proposition 1. *Soit $K \in Ob C$. Les conditions suivantes sur K sont équivalentes :*

(i) *Le foncteur* p. 23

$$X \longmapsto K(X) \overset{\text{déf}}{=} \mathrm{Hom}_C(X, K) \qquad C^o \longrightarrow \mathrm{Ens}$$

transforme flèches de W ('quasi-isomorphismes') en bijections.

(i′) *Même condition pour les flèches d'adjonction*

$$X \longrightarrow \psi\varphi X \qquad (\forall X \in \mathrm{Ob\ C})$$

(i″) *Même condition pour la flèche d'adjonction* ([6])

$$K \longrightarrow \psi\varphi K.$$

6. N.B. Il suffit même que l'élément id_K de $F_K(K)$ provienne de $F_K(\psi\varphi K)$ [où $F_K = K(-) = \mathrm{Hom}(-, K)$],

(ii) *K appartient à l'image essentielle de ψ*

(ii′) *$K \longrightarrow \psi\varphi K$ (adjonction) est un isomorphisme.*

(ii″) *L'homomorphisme fonctoriel en X*

$$\mathrm{Hom}_C(X,K) \longrightarrow \mathrm{Hom}_{C'}(\varphi X, \varphi K)$$

est un isomorphisme.

Démonstration. On a d'abord comme quasi-tautologies (ii′)$\Longrightarrow$ (ii)$\Longrightarrow$ (ii″)$\Longrightarrow$ (ii′), i.e. l'équivalence des trois dernières conditions. En effet, (ii′)$\Longrightarrow$ (ii) tautologique, (ii)$\Longrightarrow$ (ii″) car si $K = \psi(K')$ on a par adjonction [page 38]

$$\mathrm{Hom}_C(X,K) = \mathrm{Hom}_C(X,\psi(K')) \overset{\mathrm{adj}}{\simeq} \mathrm{Hom}_{C'}(\varphi X, K'),$$

or $\varphi(K) = \varphi\psi(K') \xrightarrow{\sim} K'$ (adjonction, et ψ pleinement fidèle).

$$\begin{array}{ccc} \mathrm{Hom}_C(X,K) & \xrightarrow{\quad\sim\quad} & \mathrm{Hom}_C(\varphi X, K') \\ & {\scriptstyle\varphi_{X,K}}\searrow \quad \nearrow{\scriptstyle\sim}\ \mathrm{adj} & \\ & \mathrm{Hom}_{C'}(\varphi X, \underbrace{\varphi K}_{=\varphi\psi(K')}) & \end{array} \qquad \text{qui se complète en}$$

ce qui prouve que $\varphi_{X,K}$ est un isomorphisme.

(ii″) $\Longrightarrow$ (ii′). Car la flèche

p. 24

$$\begin{array}{ccc} \mathrm{Hom}_C(X,K) & \xrightarrow{\alpha_X} & \mathrm{Hom}_{C'}(\varphi X, \varphi K) \\ & \searrow & \wr \Big| \mathrm{adj} \\ & & \mathrm{Hom}_C(X, \psi\varphi K) \end{array}$$

est déduite de la flèche d'adjonction

$$K \longrightarrow \psi\varphi K$$

en prenant $\mathrm{Hom}_C(X,-)$, donc α_X est un isomorphisme pour tout X si et seulement si cette flèche d'adjonction est un isomorphisme. Donc

$$\text{(ii)} \Longleftrightarrow \text{(ii′)} \Longleftrightarrow \text{(ii″)}.$$

D'autre part,

$$\text{(ii″)} \Longrightarrow \text{(i)} \Longrightarrow \text{(i′)} \Longrightarrow \text{(i″)}$$

sont tautologiques, en tenant compte pour (i) $\Longrightarrow$ (i) `[probablement (i) ⟹ (i′)]` que les flèches d'adjonction

$$X \xrightarrow{u} \psi\varphi X \qquad\qquad \text{dans } C$$

sont bien dans W, car on a un triangle commutatif

$$\begin{array}{ccc} \varphi(X) & \xrightarrow{\varphi(u)} & \varphi(\psi\varphi X) = (\varphi\psi)(\varphi X) \\ & \searrow^{\mathrm{id}_{\varphi(X)}} & \Big\downarrow \wr \text{ adj } \varphi\psi \longrightarrow 1, \text{ qui est un} \\ & & \text{isomorphisme car } \psi \text{ est pleinement fidèle} \\ & & \varphi(X), \end{array}$$

donc $\varphi(u)$ est bien un isomorphisme.

Reste à prouver (i″) $\Longrightarrow$ (ii′), i.e. on suppose [page 39]

que le foncteur $X \longrightarrow \mathrm{Hom}(X, K)$ transforme le morphisme d'adjonction particulier

$$K \xrightarrow{i} \psi\varphi K$$

en

$$\begin{array}{ccc} u & \longrightarrow & u \circ i \\ \mathrm{Hom}(\psi\varphi(K), K) & \longrightarrow & \mathrm{Hom}(K, K), \end{array}$$

qui est une bijection. Donc id_K est dans l'image, soit p un élément du premier membre qui lui donne naissance

p. 25

$$K \underset{p}{\overset{i}{\leftrightarrows}} \psi \underbrace{\varphi(K)}_{K'} \qquad \text{donc } p\circ i = \mathrm{id}_K$$

Considérons

$$ip = q : \psi(K') \longrightarrow \psi(K') \qquad (\text{NB } q^2 = q),$$

comme ψ est pleinement fidèle, il existe un unique

$$q_0 : K' \longrightarrow K' \qquad (K' \overset{\text{déf}}{=} \varphi(K))$$

tel que l'on ait

$$q = \psi(q_0).$$

La relation $pi = \mathrm{id}_K$ donne, en appliquant φ,

$$\varphi(p)\varphi(i) = \mathrm{id},$$

donc comme $\varphi(i)$ est un isomorphisme (c'est

$$\begin{array}{ccc} \varphi(K) & \xrightarrow{\sim} & \varphi(\psi\varphi)(K) = (\varphi\psi)(\varphi K) \\ & \overset{\sim}{\searrow}{}^{\mathrm{id}} & \wr \downarrow \mathrm{adj} \\ & & \varphi(K), \end{array}$$

rendant commutatif le triangle ci-contre), $\varphi(p)$ l'est. Donc $\varphi(q) = \varphi(i)\varphi(p)$ l'est, puisque $\varphi(i)$ et $\varphi(i)$ `[plutôt` $\varphi(p)$`]` le sont. Mais

$$\varphi(q) = \varphi\psi(q_0) \qquad \underbrace{\overset{\sim}{\longrightarrow}}_{\text{adj } \varphi\psi \overset{\sim}{\to} \mathrm{id}} \qquad q_0,$$

donc q_0 est un isomorphisme, donc aussi $q = \psi q_0$. Comme $q^2 = q$, cela implique $q = \mathrm{id}$. Donc i et p sont des isomorphismes inverses l'un de l'autre, ce qui établit (ii″) `[plutôt (ii′)]` et achève la démonstration.

Commentary · Exposition [ms. 39]

A retract of a local object is local

Grothendieck's proof of Proposition 1 identifies the precise amount of surjectivity needed: the identity of K must lift across its unit. This is stronger than a convenient trick. It says that the full subcategory of local objects is closed under retracts, and explains why checking every weak equivalence can be reduced to one particular map.

Let $\varphi : C \rightleftarrows C' : \psi$ be an adjunction with fully faithful right adjoint, and write $T = \psi\varphi$ and $\eta : 1 \to T$ for its idempotent localization monad. An object K is local exactly when η_K is invertible.

i.e.

(i‴) Il existe une rétraction de $\psi\varphi X$ sur X `[plutôt` $\psi\varphi K$ `sur` K`]` (NB Celle-ci sera même un isomorphisme).

Local objects invert W_φ under $\mathrm{Hom}_C(-,K)$ by adjunction. Conversely, if precomposition with η_K is surjective at the identity of K, choose $p : TK \to K$ with $p\eta_K = 1_K$.

Here is a short verification of the remaining step. Since $\varphi\eta_K$ is invertible by the triangle identity, φp is its inverse. Thus $\varphi(\eta_K p) = 1_{\varphi TK}$. The functor φ is faithful on the full subcategory of local objects: on that subcategory it is quasi-inverse to ψ. Both source and target of $\eta_K p$ are the local object TK, so $\eta_K p = 1_{TK}$. Hence η_K and p are inverse isomorphisms. This proves the implication from the weakest condition in the footnote and therefore all the equivalences in the proposition.

The representability corollary is then exact. If $H = H'\varphi$, every W_φ is sent to a bijection. When $H \cong \mathrm{Hom}_C(-,K)$, the proposition makes K local, and adjunction identifies H' with $\mathrm{Hom}_{C'}(-,\varphi K)$. In the other direction, a representative K' for H' pulls back to the representative $\psi K'$. Essential surjectivity of φ onto C' ensures that the resulting natural isomorphisms are determined by the original factorization. The repeated X in the footnote's first formulation is a printed slip; the object at this step is K, as the edition itself corrects.

Corollaire : *Considérons des foncteurs* $C^o \xrightarrow{H}$ Ens, *et* $C'^o \xrightarrow{H'}$ Ens, *tels que* $H \simeq H' \circ \varphi$. *Pour que* H *soit représentable, il faut et il suffit que* H' *le soit, et dans ce cas, l'objet* K *de* C *qui représente* H *satisfait aux conditions équivalentes de la proposition. Si* K, K' *les représentent, on a des isomorphismes canoniques* $K' \simeq \varphi K$, $K \simeq \psi K'$. [page 40]

Si H' est représentable par $K' \in \mathrm{Ob}\, C'$, on aura donc

$$H' \circ \varphi(X) \simeq \mathrm{Hom}_{C'}(\varphi X, K') \overset{\mathrm{adj}}{\simeq} \mathrm{Hom}_C(X, \psi(K'))$$

donc H est représentable par $K = \psi(K')$, satisfaisant la condition (ii) de la proposition. p. 26
Inversement, si H est représentable par un objet K, alors K satisfait à la condition (i), donc par (ii″) on a

$$\mathrm{Hom}_C(X,K) \xrightarrow{\sim} \mathrm{Hom}_{C'}(\varphi X, \varphi K),$$

donc le foncteur $H' : C^o \longrightarrow$ Ens qui factorise H s'explicite comme $X' \longmapsto \mathrm{Hom}_{C'}(X', \varphi K)$, donc est représentable par φK.

Passons aux ind-objets. Les foncteurs φ, ψ passent aux ind-objets, d'où des foncteurs

$$\mathrm{Ind}\, C \underset{\Phi=\mathrm{Ind}\,\varphi}{\overset{\Psi=\mathrm{Ind}\,\psi}{\rightrightarrows}} \mathrm{Ind}\, C'.$$

On aura encore la formule d'adjonction pour $X = (X_i)_{i\in I}$ dans $\mathrm{Ind}\, C$, $X' = (X'_{i'})_{i'\in I'}$ dans $\mathrm{Ind}\, C'$, car

$$\begin{aligned}\mathrm{Hom}(X, \Psi X') &= \varprojlim_i \varinjlim_{i'} \mathrm{Hom}(X_i, \psi(X'_{i'})) \\ &= \varprojlim_i \varinjlim_{i'} \mathrm{Hom}(\varphi X_i, X'_{i'}) \\ &= \mathrm{Hom}(\Phi X, X').\end{aligned}$$

On prouve de même que Ψ est pleinement fidèle, car si $X' = (X'_i)_{i\in I}$, $Y' = (Y'_j)_{j\in J}$ sont des [page 41]
objets de $\mathrm{Ind}\, C'$, on aura

$$\begin{array}{ccc} \mathrm{Hom}(X',Y') & \simeq \varprojlim_i \varinjlim_j & \mathrm{Hom}_{C'}(X'_i, Y'_j) \\ \downarrow{\scriptstyle \Psi_{X',Y'}} & & \downarrow{\scriptstyle \psi_{X'_i,Y'_j}} \\ \mathrm{Hom}(\Psi X', \Psi Y') & \simeq \varprojlim_i \varinjlim_j & \mathrm{Hom}_C(\psi(X'_i), \psi(Y'_j)), \end{array}$$

donc $\Psi_{X',Y'}$ est un isomorphisme si les $\psi_{X'_i,Y'_j}$ le sont.

Donc la proposition 1 peut s'appliquer à la situation Φ, Ψ entre ind-objets. Je vais l'expliciter sous la forme la plus commode :

Proposition 2. *Soit $K = (K_i)_{i\in I}$ un ind-objet de C, et soit H le foncteur*

$$\begin{cases} H : X & \longmapsto & \underbrace{\mathrm{Hom}_C(X,K)}_{K(X)} \overset{\text{déf}}{=} \varinjlim_i \mathrm{Hom}_C(X,K_i) \\ C^o & \longrightarrow & \mathrm{Ens} \end{cases}$$

qu'il ind-représente. Conditions équivalentes

(i) *Le foncteur H se factorise (à isomorphisme près) par C' via φ, i.e. transforme les $u \in W_\varphi$ en isomorphismes.* p. 27

(i′) *Même condition pour les flèches d'adjonction $X \longrightarrow \psi\varphi(X)$ (lesquelles sont dans W).*

(i″) *Même condition pour la flèche d'adjonction $K \longrightarrow \psi\varphi(K)$.*

(ii) *K appartient à l'image essentielle de* $\mathrm{Ind}\, C'$ *par Ψ.*

(ii′) *$K \longrightarrow \Psi\Phi(K)$ est un* isomorphisme *dans* $\mathrm{Ind}\, C$.)

(ii″) *L'homomorphisme fonctoriel*

$$K(X) = \mathrm{Hom}_C(X,K) \longrightarrow \mathrm{Hom}_{C'}(\varphi(X),\varphi(K))$$

`[plutôt` $\mathrm{Hom}_{\mathrm{Ind}(C)}(X,K) \longrightarrow \mathrm{Hom}_{\mathrm{Ind}(C')}(\Phi(X),\Phi(K))$`, en identifiant l'objet` X `de` C `à son image dans` $\mathrm{Ind}(C)$`]` *est un isomorphisme.*

Démonstration. L'équivalence de (ii), (ii′), (ii″) est la proposition 1 appliquée à Φ et Ψ, à cela près qu'il faut noter que (ii″) reste évidemment valable si on y remplace X par un ind-objet $(X_j)_{j\in J}$ quelconque. On a [page 42]

$$(\mathrm{ii}'') \Longrightarrow (\mathrm{i}) \Longrightarrow (\mathrm{i}')$$

tautologiquement, il reste à prouver que (i′) implique (i″) et (i″) (ii′). On ne peut pas appliquer directement la proposition 1, car l'hypothèse (i′) ici ne porte que sur les objets de $\mathrm{Ind}\, C$ qui proviennent de C, et il n'est pas évident à priori qu'elle passe aux objets quelconques. Mais cela l'est dans le cas d'un homomorphisme de ind-objets (X_i), (Y_i) sur le même ensemble d'indices, donné par un homomorphisme ordinaire (terme à terme) de systèmes inductifs sur I, $X_i \longrightarrow Y_i$: si les $X_i \longrightarrow Y_i$ sont dans W, l'hypothèse (i) implique que $H(Y) \longrightarrow H(X)$ est un isomorphisme, et de même pour (i′) pour $\underbrace{X}_{=(X_i)_i} \longrightarrow \underbrace{\Psi\Phi(X)}_{=(\psi\varphi(X_i))_i}$, où l'homomorphisme d'adjonction est donné terme à terme par les homomorphismes d'adjonction $X_i \longrightarrow \psi\varphi(X_i)$. Donc l'hypothèse (i′) ici implique l'hypothèse (i′) de la proposition 1 pour Φ et Ψ, donc aussi (ii′) (par la proposition 1), et on gagne.

Commentary · Exposition [ms. 42]

Why the argument passes to ind-objects

Grothendieck is careful at the one point where the extension to ind-objects is not a formal substitution. The hypothesis tests units on objects of C, whereas Proposition 1, applied to $\mathrm{Ind}(C)$, would test units on all ind-objects. The missing bridge is a limit, and it needs no exactness assumption.

Put $\Phi = \mathrm{Ind}(\varphi)$ and $\Psi = \mathrm{Ind}(\psi)$. For filtered systems $X = (X_i)$ and $Y' = (Y'_j)$,

$$\mathrm{Hom}(X, \Psi Y') = \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i, \psi Y'_j) \cong \varprojlim_i \varinjlim_j \mathrm{Hom}_{C'}(\varphi X_i, Y'_j).$$

This proves $\Phi \dashv \Psi$. Applying the same formula to two systems from C' proves that Ψ is fully faithful. The upper arrow in the edition's displayed pair of ind-functors points the wrong way; these types determine its direction.

For a fixed ind-object K, write $\widehat{H}(X) = \mathrm{Hom}_{\mathrm{Ind}(C)}(X, K)$. Its restriction to C is the functor H of the proposition. Suppose each unit $X_i \to \psi\varphi X_i$ induces a bijection under H. The unit of an arbitrary system X is represented term by term by these maps. Therefore

$$\widehat{H}(\Psi\Phi X) = \varprojlim_i H(\psi\varphi X_i) \xrightarrow{\sim} \varprojlim_i H(X_i) = \widehat{H}(X).$$

A limit of a natural diagram of bijections is a bijection: its inverse is obtained by taking the componentwise inverses, which remain compatible. There is no interchange of a filtered colimit with an inverse limit here. Proposition 1 now applies in $\mathrm{Ind}(C)$ and gives $K \cong \Psi\Phi K$.

Condition (i″) needs this extension $\widehat{H}$ to have a meaning, because K is not in general an object of C. The text uses $\psi\varphi K$ as shorthand for $\Psi\Phi K$; the corrected statement is a test at the unit of an ind-object. Likewise condition (ii″), when tested only on $X \in C$, extends to arbitrary ind-objects by the displayed limit formula. These are not extra hypotheses or repairs to the argument. They make explicit the conventions under which its proof is complete.

Revenons à la situation où $C = \mathrm{Cat}\,\mathrm{hot}$, $C' = \mathrm{Hot}$, φ le foncteur `[de]` localisation. Si $K \in \mathrm{Ob}\,\mathrm{Cat}\,\mathrm{hot}$, alors $\psi\varphi(K)$ s'appelle *l'enveloppe de Thomason-Kan* de K - si K est une catégorie, son enveloppe $\bar{K}$ et la flèche $K \longrightarrow \bar{K}$ ne sont définis que dans la catégorie $\mathrm{Cat}\,\mathrm{hot}$, i.e. $\bar{K}$ n'est défini qu'à 'homotopisme près'.
Mais on peut toujours *choisir* un objet $\bar{K}$ de Cat, et une flèche [page 43]

$$i : K \longrightarrow \bar{K} \qquad\qquad \textit{dans } \mathrm{Cat},$$

pas seulement dans $\mathrm{Cat}\,\mathrm{hot}$, de telle façon de plus que i soit une '*cofibration* de Thomason' p. 28
(et en particulier une immersion ouverte?). Il en résulte que si on a un système inductif

$$K = (K_n)_{n \in \mathbf{N}}$$

indexé par $\mathbf{N}$, et si on choisit pour chaque n une enveloppe de Thomason-Kan

$$K_n \overset{\alpha_n}{\hookrightarrow} \bar{K}_n$$

de cette façon, on peut trouver des foncteurs de transition

$$\bar{K}_n \xrightarrow{\bar{i}_n} \bar{K}_{n+1}$$

tels que les carrés

$$\begin{array}{ccc} K_n & \xhookrightarrow{\alpha_n} & \bar{K}_n \\ {\scriptstyle i_n}\downarrow & & \downarrow{\scriptstyle \bar{i}_n} \\ K_{n+1} & \xhookrightarrow{\alpha_{n+1}} & \bar{K}_{n+1} \end{array}$$

soient commutatifs, de telle sorte que l'on ait donc un homomorphisme de *systèmes inductifs de* Cat

$$K \longrightarrow \bar{K}$$

et a fortiori un homomorphisme de ind-objets de Cat. Ceci posé, les conditions (i), (ii′) de la proposition 2 montrent que le foncteur ind-représenté par K *dans* $C = \mathrm{Cat}\,\mathrm{hot}$ transforme quasi-isomorphismes

Commentary · Later developments [ms. 43]

What Thomason replacement supplies

Grothendieck needs replacements which can be chosen together with the transition maps, so that he still has a sequence of actual categories. This part of the argument has a direct model-category realization. Given a sequence $K_n \to K_{n+1}$, choose trivial cofibrations $\alpha_n : K_n \to R_n$ with R_n

Thomason fibrant. The square with top map α_n, bottom object R_{n+1} and right map $R_{n+1} \to *$ admits a lift. It gives $R_n \to R_{n+1}$ extending $\alpha_{n+1} i_n$. Since the indexing category is the freely generated chain, no relations beyond these adjacent squares need to be imposed.

The parenthetical question about open immersions also has a precise answer. Thomason cofibrations are sieves, hence full inclusions with no arrows from outside to inside. A technical correction matters here: Thomason originally asserted that retracts of Dwyer maps remain Dwyer maps. Cisinski (1999) gives a counterexample and proves that every cofibration is instead a pseudo-Dwyer map; the corrected assertion retains the sieve property and the proper model structure. This repairs that specific later proof, not a gap in Grothendieck's formal Propositions 1–2.

One can also replace any sequence of fibrant objects by a sequence of fibrant objects with cofibration transitions, up to a levelwise weak equivalence. Start with $T_0 = R_0$. If $g_n : T_n \to R_n$ has been constructed, factor the composite $T_n \to R_n \to R_{n+1}$ as

$$T_n \hookrightarrow T_{n+1} \xrightarrow[\sim]{g_{n+1}} R_{n+1},$$

where the first map is a cofibration and the second a trivial fibration. Then T_{n+1} is fibrant and the g_n form a strict map of sequences. The transition inclusions are fully faithful. This proves a useful model-theoretic version of the rearrangement proposed on manuscript p. 46. It gives a levelwise weak equivalence and therefore an isomorphism of ind-objects in Hot. Calling it an isomorphism in Ind(Cat hot) requires a further comparison; that conclusion is not supplied by the factorization axiom.

ordinaires (équivalences faibles) en bijections, *si et seulement si* l'homomorphisme précédent est un isomorphisme de ind-objets de Ind Cat hot. [page 44]

Ce sera donc cela la condition naturelle à imposer aux ind-catégories de chemins

$$\underline{\underline{\mathrm{Ch}}}_{\Sigma}(\mathcal{M}; a, b) \qquad\qquad (a, b \in \mathrm{ob}\,\mathcal{M})$$

provenant d'une 'catégorie de modèles' $\mathcal{M}$, Σ.

Bien sûr, disposant d'un système inductif explicite de catégories (pas seulement d'objets de Cat hot)

$$K_n = \underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M}; a, b),$$

il est tentant de passer à la limite inductive

$$K_\infty = \varinjlim_n \underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M}; a, b) \overset{\text{déf}}{=} \underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M}; a, b),$$

et ceci d'autant plus que les foncteurs de transition sont des monomorphismes et pleinement fidèles, de sorte que les K_n s'identifient à une suite de sous-catégories pleines de

K_∞, dont la réunion est K_∞. Mais d'autre part je ne sais donner aucune interprétation convaincante de K_∞, comme de celle du ind-objet de Cat hot defini par $\underline{\underline{K}} = (K_n)_{n\in\mathbf{N}}$, qui ind-représente le foncteur remarquable p. 29

$$H = H_{\mathcal{M};x,y} : \mathrm{Cat\,hot}^o \longrightarrow \mathrm{Ens}$$

factorisant le foncteur

$$X \longmapsto H_{\mathcal{M};x,y}(X) \overset{\text{déf}}{=} \mathrm{Hom}_{\mathcal{M}(X)\Sigma_X^{-1}}(x_X, y_X), \qquad \text{[page 45]}$$

par la formule donc

$$H(X) = \varinjlim_n \underbrace{\pi_0 \underline{\mathrm{Hom}}(X, K_n)}_{\mathrm{Hom}_{\mathrm{Cat\,hot}}(X, K_n)}.$$

Je ne sais donner d'interprétation de ce type ni de la catégorie K_∞ elle-même, ni de son image dans Cat hot, ni de son image dans Hot. Il est vrai que si je restreins le foncteur H à la sous-catégorie pleine de (Cat) formée des catégories *finies* (Ob X et Fl X finis), alors on aura

$$\underline{\mathrm{Hom}}^!(X, K_\infty) = \underline{\mathrm{Hom}}(X, K_\infty),$$

et l'isomorphisme précédant s'interprète comme

$$\mathrm{Hom}_{\mathcal{M}(X)\Sigma_X^{-1}}(a_X, b_Y) \overset{\text{déf}}{=} H(X) \simeq \pi_0 \underline{\mathrm{Hom}}(X, K_\infty) \overset{\text{déf}}{=} \mathrm{Hom}_{\mathrm{Cat\,hot}}(X, K_\infty).$$

[plutôt (a_X, b_X)] Cela prouve du moins que la connaissance de K_∞ est utile. Mais cette formule, pour X restreint aux catégories *finies,* ne définira pas K_∞ comme objet de Cat hot, à isomorphisme unique près dans cette catégorie - *sauf* dans le cas où K_∞ est elle-même homotope à une catégorie finie. (Je ne me rends pas bien compte si ça arrive très souvent … Mais on ne peut fonder une théorie générale des dérivateurs et des catégories de modèles 'dérivables' sur une hypothèse de ce genre.) [page 46]

Commentary · Development (commentators') [ms. 45]

Finitely many objects already give a uniform stage

Grothendieck uses finite categories to remove the exclamation mark. The proof is controlled by the number of objects, because fullness has already supplied every arrow between objects that have appeared. We record the resulting refinement to make this source of finiteness explicit.

Finite-object factorization. [ours] Let $K_0 \subseteq K_1 \subseteq \cdots$ be full subcategories, let K_∞ be their union, and let X have finitely many objects, with no restriction on its sets of arrows. Then the canonical functor

$$\varinjlim_n \mathrm{Fun}(X, K_n) \longrightarrow \mathrm{Fun}(X, K_\infty)$$

is an isomorphism of categories. In particular, the path-localization formula on manuscript p. 45 holds for such X.

Proof. Given $F : X \to K_\infty$, choose a stage containing each of its finitely many object-values, and take the largest of these stages. Every arrow of X has endpoints whose images lie in that stage, so fullness supplies its image arrow there. The composition and identity equations hold because inclusion is faithful. Thus F factors through this one stage. If F, G are two functors represented at possibly different stages, put them in a common stage. All components of every transformation $F \Rightarrow G$ then belong to that stage by fullness; its naturality equations are unchanged. Conversely, inclusions of stages introduce no identifications between different transformations. This proves both bijectivity on objects and bijectivity on every Hom-set. □

There is a version invariant under equivalence of the source. If X has only finitely many isomorphism classes of objects, choose a finite-object skeleton S and an equivalence $S \to X$. Precomposition gives equivalences of functor categories into every K_n and into K_∞. Passing to π_0 therefore proves

$$\varinjlim_n \pi_0 \,\mathrm{Fun}(X, K_n) \cong \pi_0 \,\mathrm{Fun}(X, K_\infty).$$

The strict equality of bounded and unrestricted functor categories need not hold for this larger class: an actual functor may send infinitely many isomorphic source objects to different target objects appearing at unbounded stages. The statement on components is the appropriate invariant one.

The qualification at the end of manuscript p. 45 also needs its domain of comparison. Ordinary Yoneda gives uniqueness among objects of Cat hot homotopy equivalent to finite categories. It does not give uniqueness among all objects merely because one candidate is finite. The same elementary fence which will make the later length obstruction visible already separates these assertions.

Failure of unrestricted uniqueness from finite tests. [ours] Let B be the poset $0 \to 1 \leftarrow 2 \to 3 \leftarrow \cdots$, and let B_n be its full subcategory on $\{0, \ldots, n\}$. The restrictions of $\mathrm{Hom}_{\mathrm{Cat\,hot}}(-, B)$ and $\mathrm{Hom}_{\mathrm{Cat\,hot}}(-, *)$ to finite categories are naturally isomorphic, although B and $*$ are not isomorphic in Cat hot.

Proof. The maps $r_j(i) = \min(i, j)$, for $0 \leq j \leq n$, are endofunctors of B_n: they preserve every fence arrow below j and collapse every arrow above it to an identity. For even j there is a transformation $r_j \Rightarrow r_{j+1}$; for odd j there is a transformation in the reverse direction. Thus r_0, the constant functor at 0, and $r_n = 1_{B_n}$ are joined by a zigzag. Postcomposition contracts $\mathrm{Fun}(X, B_n)$ for every X. When X is finite, the preceding factorization lemma places each functor $X \to B$ in some B_n. It is therefore homotopic to the constant functor at 0, and $\pi_0 \,\mathrm{Fun}(X, B)$ is a singleton. These singleton identifications are natural in X.

If 1_B were joined to the constant functor at 0 by L transformations, evaluation at $L+1$ would give a zigzag of L arrows connecting $L+1$ to 0. Each fence arrow changes the index by at most one, which is impossible. Hence B is not categorically contractible. □

The nerve of B realizes to an infinite ray and is contractible. This counterexample concerns Cat hot; the last-page obstruction in Hot will require a different example. It gives a precise reason to restrict the finite-object exception to comparisons within the finite subcategory, if that is the intended reading of the parenthesis.

We checked the general filtered-colimit and functor-category framework in Riehl (2014, Chapters 1 and 7) and the specific argument in the chapter. We did not find this exact refinement stated there. It is an elementary continuation of the printed proof, with no claim of priority. Its relevance is practical: a one-object category with infinitely many endomorphisms still satisfies the lemma, even though it is not a finite category in the author's stated sense.

Mais supposons que le foncteur H envisagé se factorise même par Hot, i.e. qu'il transforme quasi-isomorphismes en bijections. On a vu que cela équivaut à la condition qu'il puisse se représenter par un système inductif sur $\mathbf{N}$ formé de catégories de Thomason-Kan. Il doit être facile de plus, si on y tient, de s'arranger pour que les foncteurs de transition soient des monomorphismes et pleinement fidèles, pour avoir une $\varinjlim$ sympathique. Ce sera bien sûr une catégorie de Thomason-Kan. Cette catégorie est-elle déterminée par le foncteur $H : \mathrm{Cat\,hot}^o \longrightarrow \mathrm{Ens}$ à isomorphisme unique près dans Cat hot? Bien sûr, il n'est pas question qu'elle représente le foncteur H, sauf `[si]` celui-ci est bel et bien représentable, i.e. si le ind-objet $\underline{\underline{K}}$ de Cat hot est essentiellement constant (ce qui ne doit pratiquement jamais être le cas). Notons déjà que si u est un homomorphimse de ind-objets *dans* Cat, ou, ce qui revient p. 30 maintenant au même, un homomorphisme dans Cat [page 47]

$$K_\infty \xrightarrow{u_\infty} K'_\infty$$

tel que $\forall n$, $u_\infty(K_n) \subseteq K'_{n'}$ pour $n' = \tau(n)$ convenable, et si on suppose que l'homomorphisme de ind-objets induit un isomorphisme pour les foncteurs ind-représentables *sur* Cat hot (par exemple, si $\underline{\underline{K}}$ et $\underline{\underline{K}}'$ ind-représentent le même foncteur H, et l'homomorphisme de ind-objets est celui qui correspond à id_H), alors u_∞ est forcément un isomorphisme *dans* Hot, donc aussi (sauf erreur) dans Cat hot puisque K_∞, K'_∞ sont des catégories de Thomason-Kan. Mais ce qui n'est pas évident, c'est l'existence d'un homomorphisme u_∞ comme `[ci-]`dessus, et s'il existe, qu'il soit unique (ou du moins canonique?) comme flèche de Cat hot.

Commentary · Exposition [ms. 46]

Fibrancy passes to the union

Grothendieck expects the union of his replacement categories to remain a good representative of a homotopy type. For Thomason fibrancy this expectation is correct and has a finite combinatorial proof. A category K is Thomason fibrant when $\mathrm{Ex}^2 NK$ is a Kan complex. For a filtered system (K_i), the nerve commutes with its colimit: an n-simplex is a finite composable string, and both the string and the finitely many equations expressing a morphism of such strings occur at one stage. Moreover

$$(\mathrm{Ex}^2 S)_n = \mathrm{Hom}_{\mathbf{sSet}}(\mathrm{Sd}^2 \Delta^n, S)$$

commutes with filtered colimits, since $\mathrm{Sd}^2 \Delta^n$ is a finite simplicial set. Finally, a horn in a filtered colimit of Kan complexes lies in one stage, and has a filler at that stage. Consequently

$$\mathrm{Ex}^2 N(\varinjlim_i K_i) \cong \varinjlim_i \mathrm{Ex}^2 NK_i$$

is Kan whenever all the terms are Kan. This proof even dispenses with fully faithful transition maps.

Fibrancy does not answer the next uniqueness question. There are three different conclusions available from three different kinds of information. A presheaf determines its representing ind-object in Ind(Cat hot). A coherent map of strict diagrams which is a weak equivalence at every stage gives a weak equivalence on the appropriate homotopy colimits. A specified categorical homotopy

between maps of unions requires actual natural transformations, with a finite common length. These conclusions are not interchangeable.

In particular, an isomorphism in the ind-category of a homotopy category is only eventual equality of homotopy classes at each stage. It does not come with compatible choices of the homotopies. The claim on manuscript p. 47 that a corresponding union map must be a weak equivalence is false, even for full inclusions of finite fibrant stages with fibrant union; the example is in the commentary at ms. 49. The verified model-category replacement construction in the preceding block avoids this difficulty by giving actual coherent maps and levelwise weak equivalences. The raw categorical-homotopy strictification questions remain separately recorded in the audit.

Pour l'existence d'un u_∞, on est ramené à prouver que si on a un diagramme à flèches pleines

$$\begin{array}{ccc} K_n & \xrightarrow[\text{cofib.}]{i_n} & K_{n+1} \\ \downarrow{\scriptstyle u_n} & & \searrow \\ K'_{\tau(n)} & \xrightarrow[\text{transition}]{} & K'_N, \end{array}$$

et si une flèche u_{n+1} est donnée 'en pointillés' *dans* Cat hot, de façon à rendre le carré commutatif, alors on peut la réaliser par une flèche *dans* Cat qui rende le diagramme [page 48] commutatif pour de bon (dans Cat). On est ramené au même problème dans une situation

$$X \underset{i}{\overset{p}{\leftrightarrows}} Y,$$

où X est une catégorie de Thomason, i une cofibration de Thomason, p une rétraction de Y dans X *dans* Cat hot, on veut la réaliser par une vraie rétraction ([7]).

En admettant que ça soit le cas, donc que tout homomorphisme des ind-objets $\underline{\underline{K}}$, $\underline{\underline{K}}'$ *dans* Cat hot se représente par un homomorphisme entre les ind-objets *dans* Cat, donc aussi par un foncteur $K_\infty \longrightarrow K'_\infty$, reste la question si l'homomorphisme correspondant dans Cat hot est unique. Donc si on a deux foncteurs f, g qui induisent le même homomorphisme sur les foncteurs ind-représentés sur Cat hot, si f et g sont homotopes. Donc

Question. Deux foncteurs $f, g : K_\infty \longrightarrow K'_\infty$ tels que pour tout $n \in \mathbf{N}$, f_n, g_n qui s'en déduisent par restriction à K_n sont homotopes (et même homotopes en tant que foncteurs de K_n dans un K'_N convenable), sont-ils homotopes? ([8])

Je présume que non - qu'il peut arriver que la longueur minimum requise $l(n)$ pour un p. 31 chemin dans $\underline{\mathrm{Hom}}(K_n, K'_{\tau(n)})$ reliant f_n et g_n, tende vers l'infini si $n \longrightarrow \infty$. L'hypothèse que [page 49] les K'_n sont des catégories de Thomason y change-t-il quelque chose (deux objets homotopes dans $\underline{\mathrm{Hom}}(X, K')$, si K' est de Thomason, sont-ils reliés par une flèche?)

Commentary · Development (commentators') [ms. 49]

A complete instance of the length obstruction

Grothendieck identifies a concrete mechanism by which stagewise homotopies could fail to assemble: their lengths may become unbounded. The following example realizes that mechanism with full inclusions at every stage. It deliberately leaves the additional target-fibrancy condition aside, so that the obstruction itself can be seen without a model-category comparison.

Unbounded categorical homotopies. [ours] There are full filtrations $A_0 \subset A_1 \subset \cdots$ and $B_0 \subset B_1 \subset \cdots$, and functors $f, g : A_\infty \to B_\infty$, such that $f|_{A_n}$ and $g|_{A_n}$ are homotopic in $\mathrm{Fun}(A_n, B_n)$ for every n, but f and g are not homotopic in $\mathrm{Fun}(A_\infty, B_\infty)$.

Proof. Take A_n to be discrete on $\{0, \ldots, n\}$. Take B_n to be the full subcategory on the same vertices of the infinite alternating fence

$$0 \longrightarrow 1 \longleftarrow 2 \longrightarrow 3 \longleftarrow 4 \longrightarrow \cdots .$$

7. [un ? dans la marge]
8. K, K' catégories de Thomason, les K_n, K'_n aussi.

There are no nonidentity composites, so this is a poset. Define $f(i) = 0$ and $g(i) = i$. For $0 \le j \le n$, put $h_j(i) = \min(i, j)$ on A_n. Since the source is discrete, these assignments are functors. If j is even there is a transformation $h_j \Rightarrow h_{j+1}$, and if j is odd there is a transformation $h_{j+1} \Rightarrow h_j$: each component is either the identity or the indicated fence arrow. Hence

$$h_0 = f|_{A_n}, \quad h_1, \quad \dots, \quad h_n = g|_{A_n}$$

is a categorical homotopy of length n, contained in B_n.

Suppose a zigzag of L natural transformations connected f and g globally. Evaluate it at the object $L + 1$ of A_∞. Each nonidentity arrow in B_∞ changes the vertex by exactly one, so a zigzag of L arrows cannot connect 0 to $L + 1$. This contradiction proves the assertion; it also proves that the displayed stagewise length n is minimal. □

Both filtrations consist of full subcategories, and the source stages are discrete, hence Thomason fibrant. We have not shown that the target stages or their union are Thomason fibrant; this construction therefore does not answer the stronger question with that additional hypothesis. The next commentary does. It proves exactly the length mechanism suggested in the first sentence of manuscript p. 49. The argument was developed directly from that sentence. We checked the surrounding categorical-homotopy and model-category references cited in this section and did not find this exact fence formulation; it is presented as an elementary illustration, not a priority claim.

Commentary · Development (commentators') [ms. 49]

The same ind-morphism, two different maps of the unions

With Thomason fibrancy imposed on every stage and on the unions, the questions of ms. 47–49 still have negative answers. The example is a strict form of the symmetric-group shift used at ms. 52.

Proposition. [ours] *There are a sequence $T_0 \subset T_1 \subset \cdots$ of full inclusions of finite categories, each T_n Thomason fibrant and isomorphic in* Cat hot *to a group, with Thomason-fibrant union T, and a functor $s : T \to T$ with $s(T_n) \subseteq T_{n+1}$, such that:*

(a) *for every n, the functor $s|_{T_n}$ and the inclusion $T_n \to T_{n+1}$ are joined by a zigzag of three natural transformations, so s and* id_T *induce the same endomorphism of the ind-object (T_n) of* Cat hot*;*

(b) *Bs is not a weak equivalence; in particular s and* id_T *are not homotopic;*

(c) *there is a weak equivalence $r : T \to \mathsf{B}G$ onto a group, between Thomason-fibrant categories, that is not an isomorphism in* Cat hot*.*

Proof. Let G_j be the group of permutations of $\{0, \dots, j\}$, with $G_j \subset G_{j+1}$ by fixing $j + 1$, and $G = \bigcup G_j$. Let T_n have the objects $0, \dots, n$, with $\mathrm{Hom}(i, j) = G_j$ if $i \le j$ and empty otherwise, and composition by multiplication in the group of the target. The inclusions $T_n \subset T_{n+1}$ are full. The functor $r_n : T_n \to \mathsf{B}G_n$ that forgets the objects and the functor $j_n : \mathsf{B}G_n \to T_n$ onto the object n satisfy $r_n j_n = \mathrm{id}$, and the arrows $i \to n$ labelled by the unit form a natural transformation $\mathrm{id} \Rightarrow j_n r_n$. So $T_n \cong \mathsf{B}G_n$ in Cat hot.

Fibrancy. A category is Thomason fibrant when it has the right lifting property for the functors $c\,\mathrm{Sd}^2\Lambda \to c\,\mathrm{Sd}^2\Delta$ associated with the horn inclusions, and these are sieves (Thomason, 1980, Proposition 4.2). Let $F : A \to T_n$ be given on such a sieve $A \subset B$. The functor $r_n F$ extends to B, because the nerve of a groupoid is a Kan complex. Extend F by sending every object of B outside A to n and every arrow that meets the complement of A to the arrow with the label given by the extension of $r_n F$. No arrow leads from the complement into A, and $\mathrm{Hom}(i, n) = G_n$ for all i, so this is defined, and it is a functor because composition in T_n multiplies labels. So T_n is fibrant, and the union T is fibrant by the preceding commentary on ms. 46.

The shift. For $g \in G_j$ let $\sigma(g) \in G_{j+1}$ fix 0 and act on $\{1, \dots, j + 1\}$ as g does on $\{0, \dots, j\}$; let $s : T \to T$ send i to $i + 1$ and the arrow labelled g to the arrow labelled $\sigma(g)$. On T_n both s and the inclusion into T_{n+1} map, by the arrows labelled by the unit, to the functors with the same labels and constant value $n + 1$ on objects, and these two are isomorphic by the cycle $c = (0\,1 \cdots n + 1)$, since $cgc^{-1} = \sigma(g)$ for $g \in G_n$. This is (a).

The functors r_n are compatible and give $r : T \to \mathsf{B}G$; the nerve and homotopy groups commute with the union, so Br is a weak equivalence and BT is a $K(G, 1)$. Under this identification s induces $\sigma : G \to G$ on π_1, which is not surjective (its image fixes 0). This is (b). For (c): if r were

invertible in Cat hot, there would be a functor $j : \mathsf{B}G \to T$ with rj joined to the identity by natural transformations, that is, with $rj : G \to G$ an inner automorphism. But j sends the object to some i and G into $\mathrm{Hom}(i,i) = G_i$, so the image of rj lies in $G_i \neq G$. □

Consequences for these pages. The assertion of ms. 47, that a morphism of such systems inducing an isomorphism of ind-objects induces a weak equivalence of the unions, is false: s induces the identity of the ind-object. The question of ms. 48, whether two maps of unions that are homotopic on every stage inside a later stage are homotopic, has the answer no, with homotopies of length three at every stage; what fails is their compatibility, not the growth of their length as in the fence example above. The union of fibrant stages is fibrant, but the union T of stages each isomorphic in Cat hot to a groupoid is not isomorphic to one, by (c). And a weak equivalence between Thomason-fibrant categories need not be an isomorphism in Cat hot (ms. 47, "sauf erreur"): fibrancy is not the envelope property.

Commentary · Examples [ms. 49]

Fibrancy does not reduce every homotopy to one arrow

The parenthetical question asks a sharper thing than a uniform bound: for a Thomason fibrant target, must two homotopic functors already be joined by a single natural transformation? A three-object category gives a negative answer, and the fibrancy can be checked directly against the generating horns.

Every category C with a terminal object t is Thomason fibrant. Indeed, Thomason (1980, Propositions 2.5 and 4.2) identify the lifting test with inclusions

$$c\,\mathrm{Sd}^2\,\Lambda^k[n] \hookrightarrow c\,\mathrm{Sd}^2\,\Delta[n],$$

which are sieves. More generally, if $A \subseteq D$ is a sieve and $F : A \to C$ is a functor, extend F by assigning t to every object outside A. There are no arrows from an outside object to an object of A. Send arrows between outside objects to 1_t, and each arrow from inside to outside to the unique map into t. Composition is respected by uniqueness of maps to the terminal object. This constructs the required lift against every displayed horn, so $C \to *$ is a Thomason fibration.

Now take C to be $0 \to 1 \leftarrow 2$, whose terminal object is 1, and take $X = *$. The functors selecting 0 and 2 lie in the same component of $\mathrm{Fun}(*, C) = C$, connected by the two-arrow zigzag through 1. There is no arrow $0 \to 2$ or $2 \to 0$, so there is no single transformation in either direction between them. The target is fibrant by the preceding proof.

This answers the exact one-arrow question. It does not say that minimal lengths can grow without bound for a fixed fibrant target, and it does not establish a counterexample to the preceding question with fibrant full filtrations. A claim of bounded shortening, or of compatibility of stagewise homotopies, has different quantifiers and needs a separate argument.

Bien sûr, on peut prétendre que les catégories $\underline{\mathrm{Ch}}_{\Sigma,n}(\mathcal{M}; a, b)$ ont été définies 'canoniquement', et leur limite inductive $\underline{\mathrm{Ch}}_{\Sigma}(\mathcal{M}; a, b)$ aussi. Mais la définition était assez arbitraire et abracadabrante - je suis convaincu que dix mathématiciens, se proposant de développer cette même idée des catégories de chemins, dans le même but que moi, arriveraient à dix définitions différentes! Pourtant, il n'est pas exclu que l'on finisse par trouver un système transitif d'isomorphismes dans Cat hot, entre les catégories de chemins 'infinis' (i.e. de longueur non précisée) $K_\infty = K_\infty(\mathcal{M}; a, b)$ que nous aurons construites, dans le but principal (disons) [page 50] de représenter le foncteur H *sur les catégories finies.* Et que les isomorphismes obtenus seront, bien entendu, des homomorphismes dans Ind(Cat hot). J'ai l'impression de ne pas avoir encore axiomatisé à fond la notion même de chemins, qui après tout a un sens géométrique assez évident et important par lui-même. Donc par ce qu'il faut entendre par '*une* théorie des chemins dans une catégorie'. Une telle théorie $\mathcal{T}$ associerait donc à $\mathcal{M}$, Σ, a, b un système inductif de catégories $\underline{\mathrm{Ch}}^{\mathcal{T}}_{n,\Sigma}(\mathcal{M}; a, b)$. Et les contraintes géométriques naturelles à imposer à de telles théories, en vue aussi de leur utilisation pour expliciter

$$\mathrm{Hom}_{\mathcal{M}(X)\Sigma_X^{-1}}(a_X, b_X) \xrightarrow{\sim} \pi_0 \underline{\mathrm{Hom}}(X, \underline{\underline{\mathrm{Ch}}}^{\mathcal{T}}(\mathcal{M}; a, b))$$

donnerait un système transitif d'isomorphismes non seulement entre les ind-objets de Cat hot, mais peut-être aussi entre les $\underline{\mathrm{Ch}}^{\mathcal{T}}_{\infty,\Sigma}(\mathcal{M}; a, b)$, pour des théories differentes. Si on dispose d'un tel énoncé, la catégorie $\underline{\mathrm{Ch}}_{\infty,\Sigma}(\mathcal{M}; a, b)$, ou plus exactement ses images dans Cat hot et Hot, prendraient un sens convaincant. [page 51]

Commentary · Later developments [ms. 51]

What a canonical mapping space requires

Grothendieck wants a theory of paths whose result is independent of its combinatorial presentation. The preceding scholium already supplies such independence at the level it actually computes: different constructions representing the same $H_{a,b}$ give uniquely isomorphic ind-objects in Cat hot. To obtain a canonical homotopy type of paths, one needs information beyond the connected-component calculation.

Dwyer and Kan's simplicial localization provides the standard stronger construction. For a relative category $(\mathcal{M}, W)$ it gives simplicial mapping objects with the localized Hom-sets as their connected components (Dwyer and Kan, 1980a, §2 and Proposition 3.1). Their hammock construction uses diagrams of zigzags and coherent higher comparisons; in a model category its mapping objects recover the usual derived function complexes (Dwyer and Kan, 1980b, Proposition 4.4 and Corollary 4.7). Thus composition and its higher compatibility are retained while allowing different presentations of the same homotopy theory.

There is a precise caution in comparing it with these pages. Grothendieck's K_∞ is the ordinary union of full categories of fixed-endpoint paths, and the morphisms of those categories are the transformations specified in sections 1–2. The theorem proved here identifies π_0 of bounded diagram categories with an ordinary localized Hom-set. That theorem alone is not a weak-equivalence comparison between NK_∞ and a Dwyer–Kan function complex. Replacing vertical morphisms, changing allowed zigzags, or using a homotopy colimit may change the higher homotopy type while leaving this π_0 calculation intact.

Accordingly, the modern theory answers the demand for an invariant mapping-space object, but does not automatically verify each proposed comparison between the particular unbounded path categories in the manuscript. In the favorable model case discussed above, the desired functor $H_{a,b}$ is represented on Hot by the derived mapping space. A direct identification of a chosen K_∞ with that object is a separate theorem to be proved under stated assumptions. This distinction explains both the force of the scholium and the legitimacy of the questions which follow it.

Il y a d'autre part un autre point de vue, qui peut-être pourrait 'canonifier' les catégories $\underline{\mathrm{Ch}}_\infty$ en tant qu'objet de Cat hot et surtout de Hot (on passerait à Cat hot en appliquant le foncteur ψ, enveloppe de Thomason-Kan …). En effet, on est intéressé à construire, en termes de $\mathcal{M}$, Σ, un 'dérivateur' (ou un prédérivateur) $\mathbf{D}_{\mathcal{M},\Sigma}$ surtout sur le 'domaine' des catégories finies. L'étendre à un domaine plus large est peut-être important dans un 'deuxième souffle', mais peut être considéré comme du luxe dans le premier mouvement de fondation d'une théorie des dérivateurs. Ceci dit, on peut se poser des questions du type suivant.

Soit

$$\mathrm{Hot}_0 \subset \mathrm{Hot}$$

la sous-catégorie pleine de Hot, définie par les catégories finies ([9]) - appelons les types d'homotopie 'de type fini' (ou de *présentation* finie?). On a un foncteur évident p. 32 [page 52]

$$\begin{aligned} \mathrm{Hot} &\longrightarrow \mathrm{Hot}_0^\wedge = \underline{\mathrm{Hom}}(\mathrm{Hot}_0^o, (\mathrm{Ens}))\,, \\ X &\longmapsto \tilde{X} = (\xi \longmapsto \mathrm{Hom}_{\mathrm{Hot}}(\xi, X)). \end{aligned}$$

Ce foncteur est *conservatif* `[cela semble faux]`, puisque un homomorphisme $X \longrightarrow X'$ dans Hot est un isomorphisme s'il induit des isomorphismes pour les foncteurs $\tilde{X} \longrightarrow \tilde{X}'$, et même seulement pour leurs valeurs pour les sphères S^n $(n \geq 0)$. (Sauf erreur. Y a-t-il des canulars, parce que le critère d'équivalence faible par les π_i ne s'applique pas directement,

9. *NB* Hot_0 est équivalente à une petite catégorie.

en l'absence de points bases. Ça ne doit pas être très sérieux … [cela semble en fait sérieux]). Mais est-il peut-être même *pleinement fidèle* ([10])? De sorte que Hot se réaliserait comme sous-catégorie pleine de $\mathrm{Hot}_0^{\wedge}$, contenant les foncteurs représentables?

Commentary · Examples [ms. 52]

Finite unbased tests miss a symmetric-group shift

Grothendieck's final question concerns a specific restricted Yoneda functor, and here a concrete counterexample settles both the conservativity assertion and the stronger full-faithfulness question. The mechanism is that every finite test sees a conjugating element, but no one element works globally. The finite-support symmetric-group example is the classical case recorded in Matumoto et al. (1984, Proposition 4.1); the attribution and its relation to the later general result are confirmed by Arlin and Christensen (2023, Remark 2.3). We give the elementary verification in the exact form needed for these pages.

Let G be the group of permutations of $\mathbf{N} = \{0, 1, \dots\}$ with finite support. Define a homomorphism $s : G \to G$ by

$$s(\sigma)(0) = 0, \qquad s(\sigma)(n+1) = \sigma(n) + 1.$$

It is injective but is not surjective, since every image permutation fixes 0, whereas the transposition $(0\ 1)$ belongs to G. Therefore $Bs : BG \to BG$ is not a weak homotopy equivalence: on fundamental groups it is s.

Now let P be any finitely generated group and $\rho : P \to G$ a homomorphism. Choose a finite generating set. The union of the supports of the images of those generators is finite, so every element of $\rho(P)$ fixes all integers greater than some m. Let $c = (0\ 1\ \cdots\ m+1)$. Conjugation by c carries the action on $\{0, \dots, m\}$ to the shifted action on $\{1, \dots, m+1\}$ and fixes 0. Directly, for every $p \in P$,

$$c\rho(p)c^{-1} = s(\rho(p)).$$

Hence $s\rho$ and ρ are conjugate homomorphisms.

For a connected CW complex T, homotopy classes of maps to $BG = K(G, 1)$ correspond to homomorphisms $\pi_1 T \to G$ modulo conjugation. One way to verify this description is to choose a vertex, extend a chosen group homomorphism across the 2-skeleton, and then use the vanishing higher homotopy groups of BG for existence and uniqueness of further extensions. Forgetting the chosen basepoint changes the homomorphism by conjugation. If T is the realization of the nerve of a finite category, each component has finitely presented fundamental group: its finite 2-skeleton gives generators and relations. Applying the preceding conjugacy calculation on each component therefore shows

$$(Bs)_* : [T, BG] \longrightarrow [T, BG] \quad \text{is the identity}$$

for every $T \in \mathrm{Hot}_0$. The same holds for every sphere. This proves that the restricted Yoneda functor is not conservative.

It is also not faithful. The maps 1_{BG} and Bs are distinct in Hot, since the latter is not an equivalence and the former is. Nevertheless they induce the same natural endomorphism of the presheaf $T \mapsto [T, BG]$ on Hot_0: both are the identity at each test object. Thus the proposed full faithfulness fails even at injectivity on morphisms. The two editorial warnings at the end of the chapter are justified; the missing basepoint information cannot be recovered merely by adding more finite unbased tests.

Commentary · Present state [ms. 52]

The modern scope of the final obstruction

The counterexample above addresses exactly the finite-category tests in the manuscript. A much stronger theorem is now available: Arlin and Christensen (2023, Theorem 2.1) prove that no set of objects of the unpointed homotopy category jointly reflects isomorphisms. Their proof enlarges the symmetric-group construction to a sufficiently large regular cardinal. Consequently the obstruction is not removed by choosing a different small family of unbased tests. The same paper distinguishes this theorem from an earlier argument of Heller whose asserted boundedness property had a gap.

10. [un ? dans la marge]

> The loss occurs at the passage to sets of homotopy classes. Retaining more structure changes the answer: Arlin and Christensen (2023, Theorem 4.3) show that spheres jointly detect equivalences in the homotopy 2-category of spaces, whose Hom-categories are fundamental groupoids of mapping spaces. Already the automorphisms in the mapping groupoid out of a point retain the fundamental group which a set of unbased components forgets. At the infinity-categorical level, mapping spaces provide the full Yoneda data. These results give a precise modern reason for the insistence, throughout these chapters, on keeping coherent diagrams and their transformations together.

Overview · III-V

These chapters study how a homotopy theory can retain operations that ordinary homotopy classes conceal. They approach the problem from three directions. Chapter III asks for an action of homotopy types on coefficients and for an external Hom that represents the resulting families of morphisms. Chapter IV constructs coherent diagrams by extending coefficients along carefully chosen inclusions. Chapter V returns to the elementary presentation of a localization by zigzags, then asks how much of its homotopy theory those zigzags remember. The order matters. The external Hom is a desired object; coherent diagrams provide a way to construct operations on it; the categories of paths are a candidate model. What is established at each stage must be distinguished from what the next stage would require.

These chapters have three independent manuscript numberings. Chapter III contains manuscript pages 1–12, Chapter IV contains 73–83 (the edition explicitly says that pages 1–72 are absent), and Chapter V contains 1–52, with page 32 marked blank. We therefore cite manuscript locations with a chapter numeral. In the edition, the mathematical text occupies forty-seven pages, besides titles and editorial notices. The inserted comments preserve the edition's corrections in square brackets. The corrections to the transcription restore what that edition actually prints; mathematical repairs are stated in the commentary instead of being silently substituted for the historical text.

Chapter III begins with a concrete requirement. If $p : X \to e$ and M is a coefficient, a tensor by the homotopy type of X should be $p_!p^*M$, and a cotensor should be p_*p^*M. Their adjunction follows directly from the two adjunctions involving p^*:

$$\mathrm{Hom}(p_!p^*M, N) \cong \mathrm{Hom}_{\mathbf{D}(X)}(p^*M, p^*N) \cong \mathrm{Hom}(M, p_*p^*N).$$

The issue is descent in X: a weak homotopy equivalence of indexing categories must induce an isomorphism on constant-coefficient homology and cohomology. There is also a size issue. A derivator defined on a restricted class of shapes cannot be presumed to see every homotopy type. These are separate questions, and neither is settled by the formal adjunction alone.

The modern action of spaces on derivators gives a precise setting for the first construction, while model presentations provide derived mapping spaces. The distinction between an action and a closed enrichment remains essential: the former is not, by itself, a representability theorem for the external Hom. We relate the discussion to the universal property of the derivator of spaces and to the later account of homotopy-theoretic module structures (Cisinski, 2008; Groth and Shulman, 2017). For familiar derived categories one can see all the operations explicitly. With cohomological grading,

$$K \otimes M = C_*(K; \mathbf{Z}) \otimes^{\mathbf{L}}_{\mathbf{Z}} M, \qquad M^K = \mathbf{R}\mathrm{Hom}_{\mathbf{Z}}(C_*(K; \mathbf{Z}), M).$$

The mapping space from M to N is obtained from the good truncation $\tau_{\leq 0}\mathbf{R}\mathrm{Hom}_k(M, N)$ by the Dold–Kan construction and forgetting the abelian structure. Thus its nth homotopy group at zero is $\mathrm{Hom}_{D(k)}(M, N[-n])$. A positive Ext group appears as a higher homotopy group after the corresponding shift of the target; it is not a positive homotopy group of this unshifted mapping space merely because it is an Ext group.

The circle calculation explains why pointedness enters the manuscript. Unreduced chains on S^1 give $\mathbf{Z} \oplus \mathbf{Z}[1]$, hence $S^1 \otimes M \simeq M \oplus M[1]$ in the derived category. The suspension is the reduced summand. Grothendieck's first identification of the tensor with suspension is therefore corrected by his own closing observation; it should not be reported as an unresolved defect of his final construction. His suggestion of a bounded-above enhancement, however, requires a boundedness hypothesis on derived Hom. An elementary example over the dual numbers shows why arbitrary coefficients do not supply one.

Chapter IV explains why these operations must be constructed on coherent diagrams. The underlying-diagram functor is conservative, but it discards information about coherent

compositions and homotopies. The commentary on Chapter I already examines this failure in detail. Here the new point is constructive: Kan extension along a suitable sequence of inclusions supplies the coherence that a diagram drawn in the fundamental category does not specify. The loop object is obtained by extending a single coefficient by zero to a punctured square and then taking its right Kan extension to the whole square. Full faithfulness of these extensions explains both the existence and the correctly qualified uniqueness of the resulting cartesian square. This is the pattern developed systematically for pointed and stable derivators in Groth (2013).

The same method makes the adjunction maps visible. For $f : I \to J$, Grothendieck forms the collage $K(f)$ with mixed morphisms $\mathrm{Hom}_{K(f)}(i, j) = \mathrm{Hom}_J(f(i), j)$. A presheaf on $K(f)$ is exactly a pair of presheaves together with a map $f^*G \to F$. The elementary description identifies the required units and counits before the derived case is attempted. We complete the construction by giving the indexing functors and checking the pointwise Kan-extension comparison. The benefit is a canonical coherent realization of these particular adjunction maps. It does not assert a canonical lift of every arrow drawn in a homotopy category.

Chapter V begins with alternating finite paths, extended by identities at their two ends. A backward segment is required to belong to Σ; the connected components of the resulting category with fixed endpoints then give the morphisms of $\mathcal{M}[\Sigma^{-1}]$. Here "fixed" includes the morphisms: transformations in the fibre have identity components at the two endpoints. Replacing this fibre by the full subcategory on paths with those endpoint objects would change the construction. We give the quotient and composition argument with this convention throughout.

The manuscript's parity correction is also substantive. Concatenating two of the initially chosen odd-indexed standard paths reverses the standard orientation. Passing to the cofinal subsystem of even indices repairs the problem. Grothendieck's analysis of admissible contractions establishes the necessary compatibility with padding. His more general interval lemma is left as an assumption, with contractibility proposed as a strengthening. The development supplied here proves that strengthening: for a finite oriented path as target, every nonempty category of functors with a prescribed restriction is categorically contractible. Numerical minimum and maximum give explicit endofunctors and a zigzag between the identity and a constant functor. The fixed-restriction version also covers the cancellation diagrams whose two endpoints have the same image. We give the full argument and state the extent of the literature search; the label *ours* records responsibility for this proof, not a claim of priority.

Passing to diagrams exposes a uniformity condition that a bare union of path categories forgets. A path in $\mathcal{M}^X$ has one finite length bound valid at every object of X. Consequently the natural representing object is an ind-category (K_n), with

$$\mathrm{Hom}_{\mathcal{M}^X[\Sigma_X^{-1}]}(a_X, b_X) \cong \operatorname*{colim}_n \pi_0 \,\mathrm{Fun}(X, K_n).$$

This is an unconditional statement about ordinary localizations. It is not yet an identification of NK_∞ with a derived mapping space. The later simplicial-localization theory supplies canonical mapping spaces and comparison theorems under their stated hypotheses; the commentary keeps those theorems separate from what this particular path model proves.

The formal propositions on reflective localizations and their ind-categories can be preceded by a simpler affirmative result. If $H_{a,b}$ inverts nerve weak equivalences, the image of its representing ind-object already ind-represents the descended functor on Hot. Indeed, left Kan extension along the opposite of a localization carries representable presheaves to representable presheaves and preserves filtered colimits; restriction along the localization is fully faithful. This conclusion needs no right adjoint between the original categories.

The stronger formal propositions on reflective localizations and their ind-categories are sound when the fully faithful right adjoint is supplied. Their application to $\mathrm{Cat}\,\mathrm{hot} \to \mathrm{Hot}$ needs that additional premise. Here $\mathrm{Cat}\,\mathrm{hot}$ means categories and functors modulo finite

zigzags of natural transformations, whereas Hot inverts weak equivalences of nerves. The editorial request for a proof of the right adjoint is consequently material. Thomason's model structure does not alone establish the asserted adjunction with this particular source category. We leave that premise unresolved in this audit and explain the valid replacement-based conclusions without promoting an unsuccessful search to a claim that the problem is open.

The last page admits a definite answer. Testing an unpointed homotopy type by every finite category does not give a conservative or a faithful restricted Yoneda functor. Let G be the group of permutations of $\mathbf{N}$ with finite support, and let $s : G \to G$ shift a permutation one place to the right while fixing zero. The induced Bs is not an equivalence. Every map from a finite category's nerve into BG, however, uses a finitely generated fundamental group and hence only finitely many permutation supports. On those supports the shift is conjugation by one finite permutation. Thus Bs induces the identity on all the proposed tests. The same example separates Bs from the identity despite their identical restricted Yoneda transformations. The modern general theorem says that no set of unpointed spaces jointly reflects isomorphisms in Hot (Arlin and Christensen, 2023, Theorem 2.1 and Remark 2.3). Retaining mapping groupoids changes this conclusion: the sphere tests do reflect equivalences in the homotopy 2-category, as their Theorem 4.3 proves. The obstruction on the closing page is precisely information discarded by passing too early to sets of unpointed homotopy classes.

Chapter VI

HOT

VI-a · Chapter VI, part 1 of 3 · manuscript pages 1–70

p. [sans numéro]

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre VI

HOT

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

p. [sans numéro]

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[e-mail address removed]

G. Maltsiniotis

Le dérivateur HOT

14.11.90

(Les premières feuilles, d'il y a deux semaines, ont été larguées, dépassées par VII)

Je définis Hot à partir de la catégorie de modèles

$$\mathcal{M} = \mathrm{Cat},\ W \subseteq \mathrm{Fl}(\mathrm{Cat}),$$

W étant en principe les Hot-équivalences. Mais je vais dans la mesure du possible axiomatiser plus, en supposant donné un ensemble W plus général, appelé 'localisateur fondamental' (basic localiser), p. ex. $W = W_{\mathbf{D}}$, où $\mathbf{D}$ est un dérivateur déjà donné (mais en supposant que son domaine soit (Cat) tout entier). Nous allons noter au fur et à mesure de quelles propriétés de W nous aurons besoin. Aux fins de référence ultérieure, je vais rappeler ici celles que j'ai dégagées dans Pursuing Stacks.

L1 Saturation de W

a) *Les identités* `[sont]` *dans* W.

b) *Si deux parmi* gf, f, g `[sont]` *dans* W, *le troisième aussi,* $X \xrightarrow{f} Y \xrightarrow{g} Z$.

c) *Si* gf *et* fg *sont dans* W, *alors* f *et* g *aussi,* $X \underset{g}{\overset{f}{\rightleftarrows}} Y$.

L2 $\mathbf{\Delta}^1$ *est universellement* W*-asphérique, i.e.* $\mathbf{\Delta}^1 \times X \longrightarrow X$ *est dans* W *pour tout* X *dans* Cat. Cela implique

L2 bis $\mathbf{\Delta}^1$ *est* W*-asphérique.* (C'était L3′ dans Pursuing Stacks.)

L3 *Si* X *a un objet final,* X *est* W*-asphérique.* ([1])

Je note ici `[des]` variantes renforcées.

L3 bis *Si* X *est* Hot-*asphérique, il est* W*-asphérique.*

L3 ter *Si* X *est* Hot-*asphérique, il est universellement* W*-asphérique, i.e.* $X \times Y \xrightarrow{\mathrm{pr}_2} Y$ *est dans* W *pour tout* Y *dans* Cat. ([2])

1. Il y a un L3′ dual de L3.
2. Suffit pour entraîner L2, L3.

L3 quater *Toute* Hot-*équivalence est dans* W. p. 2
L3 quater $\Longrightarrow$ L3 ter $\Longrightarrow$ L3 bis $\Longrightarrow$ L3.
L3 ter $\Longrightarrow$ L2 $\Longrightarrow$ L2 bis.

[page 2]

Commentary · Context [ms. 1]

The axioms are being found

Grothendieck wants to construct the homotopy theory of diagrams from categories themselves. The expression "catégorie de modèles" here initially means a category supplied with distinguished weak equivalences; no classes of Quillen cofibrations and fibrations have yet been specified. We write W∞ for functors inducing weak homotopy equivalences of nerves, and Q : Cat $\to$Cat[W -1] for ordinary localization. Categories and localizations are understood in fixed, sufficiently large universes. This avoids mistaking the construction of Q for a smallness theorem about its hom sets.

The changing list of axioms is part of the argument. In particular, the L4 of manuscript page 2 is the absolute assertion that an aspherical functor is a weak equivalence. The relative assertion for a map X $\to$Y over an arbitrary third category S appears only as L4 bis on page 44. Later proofs require this stronger formulation. The modern definition uses the relative statement from the outset; compare Cisinski (2004, Definition 1.1.3 and Remark 1.1.4). Similarly, the L5 on page 2 concerns descent of fibrewise equivalences, whereas the L5 on page 43 concerns recovery of fibrewise equivalences from a total equivalence. They are different assertions, with different strength. The interval [1] explains why these axioms can remember homotopy. A natural transformation F $\Rightarrow$G is a functor A $\times$ [1] $\to$B with prescribed endpoint restrictions. L2 makes both endpoints inverse to the projection after localization. Thus a natural transformation becomes an equality between the induced maps of localized categories. This observation will repeatedly turn an elementary adjunction into a weak equivalence without requiring a model structure.

The optional L3 quater also has a precise later history. Under the modern relative axioms, the inclusion W$\infty\subseteq$W is a theorem, established by Cisinski (2004, Theorem 2.2.11). That theorem does not license replacing the initial absolute L4 by its relative form without explanation. In these pages the additional hypothesis is explicitly adopted at manuscript page 7, and we retain the distinction when tracing the proofs.

L4 *Si* $f : X \longrightarrow Y$ *est* W*-asphérique (i.e. les* $X/y \longrightarrow Y/y$ *dans* W*, i.e. (si on a L3) si les* X/y *sont* W*-asphériques), alors* f *aussi.* (Axiome de localisation.) ([3]) `[`aussi' = dans` $W$`]`

L5 (Axiome de fibration). *Si*

$$\begin{array}{ccc} X & \xrightarrow{\ f\ } & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

avec X *et* Y *fibrées sur* S *et* f *cartésien, si pour tout* $s \in S$ $f_s : X_s \longrightarrow Y_s$ *est dans* W*,* f *aussi.*

Dans Pursuing Stacks j'ai supposé toujours $\boxed{\text{L1, L3, L4}}$ (impliquant L2) - c'étaient ces conditions qui définissaient la notion de 'localisateur' (fondamental) dans Cat. ([4]) D'autre part, $\boxed{\text{L1, L2 bis, L5}}$ impliquent L1,2,3,4,5.
On va supposer ici aussi qu'on ait au moins L1, L3, L4 (i.e. W un localisateur). En appliquant la construction de I, p. 72-**75,** on voit que $f : X \longrightarrow Y$ est dans W si et seulement si $f^o : X^o \longrightarrow Y^o$ l'est. Donc L1, L3, L4 impliquent L3′, L4′ (axiomes duaux de L3, L4).
Il y avait des axiomes complémentaires

La $f \in W \Longrightarrow \pi_0(f)$ *bijectif.*

3. il y a un L4′ dual de L4.

4. Si W `[est]` un localisateur, alors les catégories contractiles sont W-asphériques, les homotopismes sont dans W, les applications homotopes sont égales dans $(\mathrm{Cat})W^{-1} \overset{\text{déf}}{=} \mathrm{Hot}_W$.

Lb *Le foncteur canonique de localisation*

$$\mathrm{Cat} \longrightarrow (\mathrm{Cat})W^{-1} \overset{\text{déf}}{=} \mathrm{Hot}_W$$

commute aux sommes finies.

Lc *Itou pour produits finis.*

Il faudrait quand même dégager des conditions concrètes sur W pour la validité de Lb, Lc. p. 3
Les conditions L3 quater et La s'écrivent respectivement

$$W_\infty \subseteq W, \quad W \subseteq W_0.$$

[page 3]

Commentary · Exposition [ms. 2]

What the first fibration axiom already gives

Grothendieck's short implication from L1, L2 bis and the original L5 can be expanded into a useful collection of elementary arguments. It also clarifies why that first L5 becomes dispensable after relative descent has been imposed. First apply L5 to the cartesian functor [1] × A →A between fibrations over A. Its fibre map is [1] →{∗}, so L2 follows. For the projection p : A × [1] →A and its sections e0, e1, two-out-of-three gives e0, e1 ∈W. If H : A × [1] →B is a natural transformation between F and G, it follows that F ∈W if and only if G ∈W: both are equivalent to H ∈W. Moreover, Qe0 = Qp−1 = Qe1, whence QF = QG. An adjoint pair F ⊣G has GF and FG connected to the corresponding identities by its unit and counit. Both composites belong to W, and L1(c) gives F, G ∈W. Applying this to the terminal-object adjunction proves L3, and the initial-object case follows as well.

There is a particularly concrete way to obtain invariance under opposites without assuming it. Let T(A) have objects the arrows a →b of A, with a morphism from a →b to a′ →b′ given by a′ →a and b →b′ making the evident rectangle commute. The two projections

T(A)op −→A,

T(A)op −→Aop are fibrations. Their fibres are respectively (a\A)op and A/b, each with a terminal object. L5, applied to each projection as a cartesian functor to the identity fibration, makes both projections elements of W. Their naturality in A and two-out-of-three now imply

T(f)op ∈W

f op ∈W. f ∈W ⇐⇒ ⇐⇒ Here the fibration assertion can be checked directly: a change of the first endpoint is made by precomposition before taking opposites, and a change of the second endpoint by postcomposition. These are the specified cartesian lifts after reversal.

Now let f : X →Y be a map over S such that every f/s : X/s →Y/s belongs to W. The construction Ψ on manuscript pages 17–20 produces split opfibrations with these maps as their fibre maps, and Ψ(f) preserves the chosen cocartesian lifts. After taking opposites, L5 applies, so Ψ(f) ∈W. The natural maps X →Ψ(X) and Y →Ψ(Y) are homotopy equivalences by the adjunction exhibited below; hence two-out-of-three gives f ∈W. This proves even L4 bis, and therefore L4. Each ingredient is independent of the later, much stronger L5. The argument verifies the implication recorded here while anticipating only constructions that Grothendieck gives explicitly in the next few pages.

Leur conjonction est l'axiome des plus raisonnables

$$W_\infty \subseteq W \subseteq W_0.$$

Bien sûr, l'axiome L5 a maintenant l'air un peu étriqué, i.e. faiblard. On va voir très vite de quoi on aura besoin!
On veut donc définir un dérivateur

$$\mathrm{HOT}_W$$

à partir de la catégorie de modèles

$$(\mathrm{Cat}, W \subseteq \mathrm{Fl}(\mathrm{Cat})).$$

Donc pour définir

$$\mathrm{HOT}_W(S) \qquad\qquad (S \in \mathrm{Ob}\,\mathrm{Cat}),$$

la récette générale est de prendre la catégorie

$$\mathcal{M}(S) = \mathrm{Cat}(S) = \underline{\mathrm{Hom}}(S^o, (\mathrm{Cat})),$$

d'y introduire l'ensemble de flèches

$$\begin{cases} W_S \subseteq \mathrm{Fl}(\mathcal{M}(S)) \\ u \in W_S \iff u_s \in W \quad \forall\, s \in \mathrm{Ob}\, S. \end{cases}$$

On doit poser

$$\mathrm{HOT}_W(S) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(S^o, \mathrm{Cat}) W_S^{-1}.$$

Mais la catégorie $\underline{\mathrm{Hom}}(S^o, \mathrm{Cat})$ s'interprète, à isomorphisme canonique près, comme celle

$$\underline{\mathrm{Fibsc}}\, S$$

des catégories X Cat-fibrées sur S, munies d'un 'scindage', i.e. d'un choix de foncteurs changement de base $u^* : X_t \longrightarrow X_s$ (pour $t \overset{u}{\to} s$ dans S) [plutôt $u^* : X_s \longrightarrow X_t$], avec de plus *transitivité stricte*

$$(vu)^* = u^* v^* \qquad \text{si } \; t \overset{u}{\to} s \overset{v}{\to} r \;\text{ dans } S$$

[page 4]

Commentary · Exposition [ms. 3]

Products and sums survive the localization

Grothendieck asks for concrete conditions ensuring Lb and Lc because a homotopy theory should still allow objects to be assembled and compared. A useful answer is available as soon as relative L4 bis is admitted. We give the argument, including the passage from closure of weak equivalences to the universal property of products and coproducts.

Suppose W satisfies L1, L2 and L4 bis. If f : A →B lies in W, then f × idC lies in W for every C. View this map over C. Its comma map at c is f × idC/c. Since C/c has a terminal object, the two projections A × (C/c) →A and B × (C/c) →B are homotopy equivalences: the terminal object defines sections and a contraction natural transformation. They are in W by the preceding block. Two-out-of-three makes the comma map a weak equivalence, and L4 bis gives the assertion. The same axiom, applied over a discrete indexing category I, shows that ‘ i fi ∈W whenever every fi ∈W. The latter closure is also Cisinski (2004, Proposition 1.1.5).

For completeness, localization of a product of relative categories is the product of their localizations. One can see this directly from the generators: (w, id) and (id, w′) become invertible, their commutation squares still commute after inversion, and a pair of finite zigzags can be interleaved to give a zigzag in the product. Consequently both the diagonal and the binary product functor descend. The adjunction Δ⊣× descends with its original unit and counit, since those are natural transformations between relative functors and the triangle identities remain identities. Its descended right adjoint is therefore a binary product in Cat[W −1]. The same argument with ‘ ⊣Δgives binary coproducts. The adjunctions defining the initial and terminal objects descend without any closure hypothesis beyond identities belonging to W.

Thus finite products and coproducts are preserved by Q under these concrete assumptions. This is an ordinary localization argument, not a claim that arbitrary homotopy limits or colimits agree with ordinary ones. In particular, it does not supply the derived Kan extensions needed for the derivator. Those require the diagram categories whose construction begins here.

et non seulement transitivité à isomorphisme canonique près. Ainsi on peut écrire aussi p. 4

$$(*) \qquad \mathrm{HOT}_W(S) \xrightarrow{\sim} (\underline{\mathrm{Fibsc}}\, S) W_S^{-1},$$

où cette fois W_S est interprété comme

$$W_S \subseteq \mathrm{Fl}(\underline{\mathrm{Fibsc}}\, S),$$

savoir

$$W_S = \{u \in \mathrm{Fl}(\underline{\mathrm{Fibsc}}\, S) \mid u_s \in W \text{ pour tout } s \in \mathrm{Ob}\, S\}$$

(foncteurs strictement cartésiens, i.e. commutant aux scindages, qui induisent des W-équivalences entre les catégories fibres).

La flèche dans $(*)$ est bel et bien un isomorphisme de catégories, pas seulement une équivalence. Mais on va définir d'autres catégories, plus commodes à certains égards, qui seront *équivalentes* (on l'espère), mais sûrement pas isomorphes, à $\mathrm{HOT}_W(S)$. (On s'occupera plus tard de la variance de $\mathrm{HOT}_W(S)$ en S.)

Les catégories $\underline{\mathrm{Hom}}(S^o, \mathrm{Cat})$, $\underline{\mathrm{Fibsc}}\, S$, ainsi que les variantes que nous allons introduire maintenant, sont en fait des 2-catégories. Mais pour décrire $\mathrm{HOT}_W(S)$, seule les structures de 1-catégories sont utilisées. On a l'impression que c'est bien grossier, qu'on perd quelque [page 5]
chose ce faisant. Mais ce qui est perdu va se retrouver, et surabondamment, dans la structure des $\underline{\mathrm{Hom}}$ internes dans les $\mathrm{HOT}(S)$, si on se 'rappelle' que ces $\underline{\mathrm{Hom}}$ internes, donnant lieu à des 'types d'homotopie relatifs' $\underline{\mathrm{Hom}}_{\mathrm{HOT}_W}(S; \xi, \eta)$ (si ξ, η en sont), peuvent s'interpréter en termes de ∞-champs en groupoïdes sur S, avec toute la richesse de structure qui leur appartient. Mais ceci est une autre histoire…

Commentary · Exposition [ms. 4]

A diagram written as its category of elements

Grothendieck passes to fibrations in order to replace an externally indexed diagram by one category carrying a projection to its index. This is the same organizing construction used in descent; the relevant background is Grothendieck (1971, Exposé VI, §§8–9). It is useful to write out the variance because both directions of the construction will soon be needed.

For F : Sop →Cat, its Grothendieck construction has objects (s, x) with x ∈F(s). An arrow (s, x) →(t, y) consists of u : s →t and a : x →F(u)y in F(s). Composition is

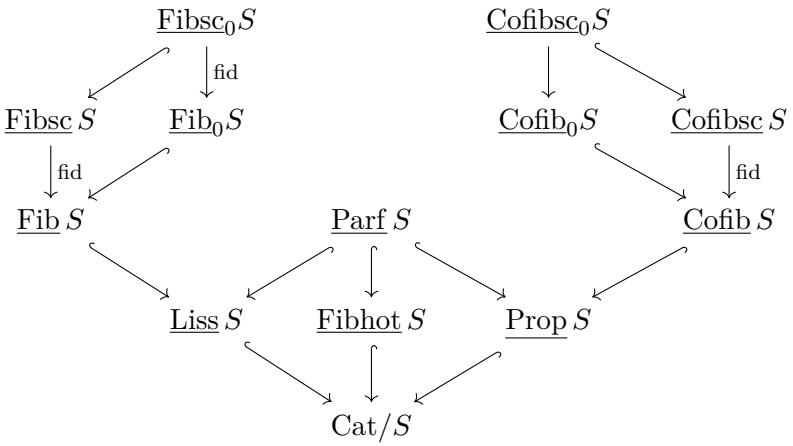
(v, b) ∘(u, a) = (vu, F(u)(b) ∘a).

The chosen cartesian lift of u at (t, y) is (u, id) : (s, F(u)y) →(t, y). The strict functoriality of F makes these choices compose strictly. Conversely, a normalized splitting of a fibration gives strict pullback functors u∗, with (vu)∗= u∗v∗and identity pullbacks equal to identities. These equations, rather than the existence of unspecified isomorphisms, are what is meant by a splitting.

A natural transformation F →G induces a functor of the total categories that preserves the specified lifts. Conversely, such a functor restricts to the components of a strict natural transformation. Therefore the weak equivalences in the diagram category correspond exactly to fibrewise weak equivalences among splitting-preserving functors. With fibrations encoded by their fibre families, this is the identification Grothendieck calls an isomorphism. If arbitrary underlying object sets for total categories are allowed, the usual construction and reconstruction are canonically isomorphic over S, and the comparison is expressed as an equivalence. No homotopical argument depends on this bookkeeping convention.

The subsequent forgetful functor to Fib S changes the morphisms: there one allows all functors over S. Forgetting a splitting is therefore not merely forgetting redundant labels. The natural comparison maps used later must be checked in the category whose morphisms they actually preserve. This is the source of the obstruction on manuscript pages 32–39, and of the strictification work beginning on page 52.

En fait, je vois un diagramme de 13 catégories de 'categories sur S', à examiner à présent



où Fib, Cofib désignent les catégories Cat-fibrées, Cat-cofibrées sur S, Liss et Prop les catégories lisses resp. propres sur S, Parf celles qui sont à la fois lisses et propres (fibrations parfaites), enfin Fibhot les catégories Hot-fibrées sur S. Toutes les flèches sont des foncteurs strictement pleinement fidèles, à l'exception des deux (plutôt quatre) flèches verticales supérieures, qui sont fidèles sans plus. Je viens d'en rajouter encore 4, de catégories : Fibsc$_0$, Fib$_0$, Cofibsc$_0$, Cofib$_0$ sont les sous-catégories pleines des catégories correspondantes sans indice, formées des catégories fibrées resp. cofibrées X sur S, pour lesquelles les foncteurs changement de base resp. cochangement de base p. 5 [page 6]

$$u^* : X_t \longrightarrow X_s, \qquad \text{resp.} \qquad u_* : X_s \longrightarrow X_t,$$

pour $s \xrightarrow{u} t$ dans S, sont des Hot-équivalences. En termes de foncteurs

$$S^o \xrightarrow{F} \mathrm{Cat} \qquad \text{resp.} \qquad S \xrightarrow{F'} \mathrm{Cat},$$

cela signifie que ce foncteur transforme flèches quelconques dans S en éléments de W dans Cat, i.e. qu'on a un diagramme commutatif

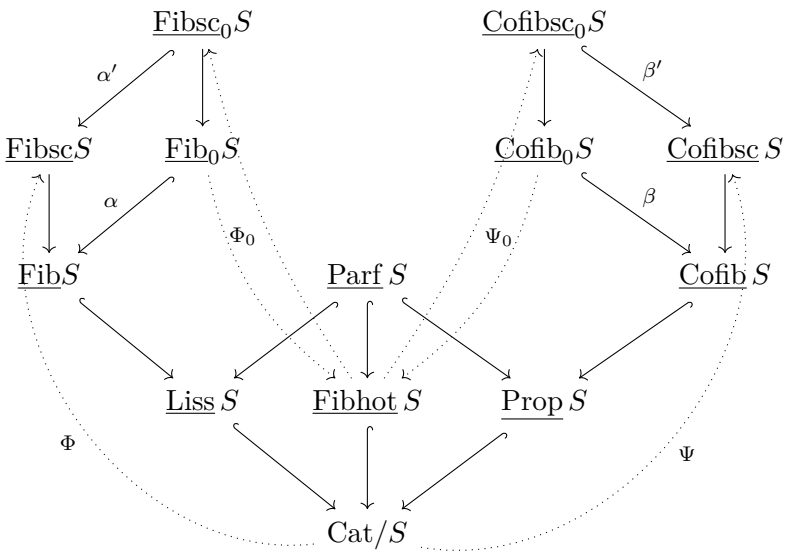
$$\begin{array}{ccc} S^o & \xrightarrow{\qquad F \qquad} & \mathrm{Cat} \\ {\scriptstyle \mathrm{can}}\downarrow & & \downarrow{\scriptstyle \mathrm{can}} \\ \Pi(S^o) \simeq (\Pi S)^o = S^o(\mathrm{Fl}\, S^o)^{-1} & \xrightarrow[\bar F]{} & \mathrm{Hot}_W = (\mathrm{Cat})W^{-1} \end{array}$$

(et de même pour F'), i.e. qu'on a un 'système local' $\bar F$ sur S, à valeurs dans Hot$_W$, défini à l'aide de F. Je rappelle qu'on trouve également un tel système local dans le cas d'une catégorie Hot-fibrée sur S (et même un système local à valeurs dans Hot lui-même, au lieu de Hot$_W$ - on utilise ici tacitement que $W_\infty \subseteq W$, i.e. L3 quater ([5])). Et il en est de même a fortiori pour les objets de Parf S. Pour Liss S resp. Prop S, on trouve simplement un foncteur [page 7]

$$S^o \xrightarrow{\bar F} \mathrm{Hot} \qquad \text{resp.} \qquad S \xrightarrow{\bar F'} \mathrm{Hot},$$

d'où, en composant avec Hot $\longrightarrow$ Hot$_W$, des foncteurs à valeurs dans Hot$_W$.

Dans chacune de ces 14 [13?] catégories, on introduit un ensemble de flèches, qui serviront à localiser, et on le désignera toujours par la même notation W_S. Dans toutes ces catégories sauf seulement Cat/S, W_S désigne la W-équivalence 'fibre par fibre'. Dans Cat/S d'autre part, ce sera la W-équivalence sans plus. Toutes les flèches du diagramme sont compatibles avec les ensembles localisants de flèches. p. 6 [page 8]



Commentary · Examples [ms. 7]

A local system in the homotopy category misses twisting

Grothendieck's passage from F to F$^-$: Π(Sop) →HotW records monodromy, but he also signals on page 5 that richer coherence remains. A familiar pair of bundles makes the difference visible. We use ordinary spaces to display the phenomenon; replacing spaces by their simplex categories gives categorical models with the same homotopy information.

Consider the trivial circle bundle S2 × S1 →S2 and the Hopf bundle (Hatcher, 2002, Examples 4.45 and 4.51)

S3 = {(z0, z1) ∈C2 : |z0|2 + |z1|2 = 1} −→CP1 ∼= S2, (z0, z1) 7−→[z0 : z1]. On the chart z0 = 0, with coordinate w = z1/z0, a trivialization is

λ

λ ∈S1. (w, λ) 7−→ 1 + |w|2 (1, w), p The other chart is analogous. Thus both maps are locally trivial bundles with fibre S1, and hence fibrations. Since the base is simply connected, their parallel transport determines the same functor from the fundamental groupoid to the homotopy category: a constant local system with value S1. Nevertheless they are not fibre homotopy equivalent. Such an equivalence would induce a weak equivalence of total spaces by the homotopy exact sequences, whereas

H1(S2 × S1; Z) = Z.

H1(S3; Z) = 0, The missing information is the coherence of transport around two-dimensional homotopies, not an additional choice of monodromy around loops.

In contemporary language the appropriate datum is a functor from the entire homotopy type of the base, considered as an ∞-groupoid, to spaces. Passing to the fundamental groupoid of the base and the ordinary homotopy category of spaces discards part of that datum. Straightening and unstraightening provide the relevant modern comparison; Lurie (2009, Theorems 2.2.1.2 and 3.2.0.1) treat respectively space-valued and category-valued versions. Here this explains the direction of Grothendieck's remark about ∞-groupoid stacks, without asserting that a single ordinary category HOTW (S) already determines every higher structure he envisages. The commentary on the earlier chapters, on what a diagram theory remembers, remains applicable.

There is also a small logical distinction in the definition on page 6. If the subscript 0 means that all transition functors lie in W∞, their images lie in W once W∞⊆W. The converse need not hold for a larger W. A W-locally constant diagram and an ordinarily homotopically locally constant diagram must therefore be distinguished whenever W is still variable.

`[Dans ce diagramme, il y a aussi` Liss$_0$ `et` Prop$_0$ `mais sans connexion aux autres objets.]`

5. On va le supposer vrai dans toute la suite.

Je suis quasiment sûr du

Theorème. ([6]) *Tous les foncteurs précédents induisent des équivalences entre les catégories localisées.*

Comme il y a 16 foncteurs, ça a l'air d'un exercise de patience. Le principe général, c'est qu'on arrive à définir des foncteurs en sens inverse, compatibles avec les W_S, de telle façon que les composés soient reliées aux foncteurs identiques par des flèches qui sont dans W_S, de sorte que par passage aux localisés ces foncteurs induisent des (équivalences) quasi-inverses des foncteurs déduits de ceux du diagramme. Tous ces foncteurs en sens opposé résultent essentiellement par composition, de trois foncteurs bien familiers. Deux de ceux-ci sont indiqués en pointillés sur le diagramme, comme Φ, Ψ.

Le troisième est un foncteur canonique [page 9]

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S,$$

qu'on va expliciter également.

Commentary · Examples [ms. 8]

The walking arrow detects the failed equivalence

Grothendieck's first comparison theorem would identify all the categories in the large diagram after the specified localizations. The later marginal note withdraws it. The obstruction already occurs over the smallest base with a noninvertible arrow and explains why the distinction between fine and coarse equivalence occupies the rest of the chapter.

Let S = [1] = (0 →1) and take W = W∞. Consider the contravariant diagrams

F(1) = ∅,

F(0) = {∗}, G(0) = G(1) = {∗}. There is a unique natural transformation F →G. Its Grothendieck construction is the functor {0} ,→[1] over S. Both source and target are split fibrations, and both total categories have contractible nerve, so this map is a coarse weak equivalence. Its fibre at 1 is ∅→{∗}, which is not a weak equivalence.

This difference persists after localization; it is not merely a difference between two presenting classes of arrows. Evaluation at 1 descends from the category of strict diagrams localized at objectwise weak equivalences to Hot. The displayed map has image ∅→{∗} there and therefore cannot become an isomorphism. Its coarse image is invertible by definition. Thus the forgetful comparison from fibrewise localization to coarse localization is not an equivalence. The same argument applies whenever W ⊆W0, since then empty and nonempty categories cannot become isomorphic after applying $\pi 0$.

The locally constant subcategory is also genuinely smaller. If a diagram H(1) →H(0) has a weak equivalence as its transition and were isomorphic to F after objectwise localization, evaluation would make H(1) empty and H(0) contractible in Hot, contradicting the invertibility of its transition. Reversing all arrows gives the corresponding example for split opfibrations. This does not invalidate the formal localization lemma on the next pages: the missing hypothesis is that its comparison transformations lie in the particular weak-equivalence classes being used.

Le foncteur Φ (resp. Ψ) appliqué à $\underline{\mathrm{Fib}}_0 S$ (resp. $\underline{\mathrm{Cofib}}_0 S$), induit un foncteur p. 7

$$(*) \qquad \underline{\mathrm{Fib}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}_0 S \qquad \text{resp.} \qquad \underline{\mathrm{Cofib}}_0 S \longrightarrow \underline{\mathrm{Cofibsc}}_0 S$$

(ce qu'il faudra vérifier). Cela fournit toutes les flèches 'inverses' dont on a besoin, à l'exception seulement de α, β, α', β'. Mais si on a prouvé, par la méthode indiquée, que toutes les flèches du diagramme induisent des équivalences sur les catégories localisées, pour prouver qu'il en est encore de même pour α, β, α', β', il suffit de le prouver pour les deux foncteurs

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fib}}\, S, \qquad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofib}}\, S,$$

6. C'est totalement canulé, cf. commentaires page 32 ff., solution des perplexités pages 41 ff. Voir surtout le Théorème-Scholie récapitulatif p. 70-81.

donc il reste à définir les foncteurs en sens inverse

$$\underline{\mathrm{Fib}}\, S \longrightarrow \underline{\mathrm{Fibsc}}_0 S, \qquad \underline{\mathrm{Cofib}}\, S \longrightarrow \underline{\mathrm{Cofibsc}}_0 S,$$

et comme les sources sont des sous-catégories pleines de Cat/S, il suffira de définir

$$\Phi_0 : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}}_0 S, \qquad \Psi_0 : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Cofibsc}}_0 S.$$

Pour ceci, on posera

$$(**) \qquad \Phi_0 = \Phi r C, \qquad \Psi_0 = \Psi r C,$$

où $C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S$, Φ et Ψ sont les 3 foncteurs 'rétrogrades' inversés plus haut, et $r : \underline{\mathrm{Parf}}(S) \hookrightarrow \mathrm{Cat}/S$ est l'inclusion. Il faut donc vérifier que Φ_0, Ψ_0 définis par $(**)$, qui a priori ont comme catégories but $\underline{\mathrm{Fibsc}}\, S$ et $\underline{\mathrm{Cofibsc}}\, S$, prennent leur valeur dans les sous-catégories avec indice 0. [page 10]

Je vais donner un énoncé d'équivalence unique, qui couvrira (presque) tous les 16 cas à la fois :

Corollaire. *Considérons le diagramme*

$$\begin{array}{ccccc} & & \underline{\mathrm{Parf}}\, S & & \\ & & \downarrow{\scriptstyle r} & & \\ \underline{\mathrm{Fibsc}}_0 S & \overset{p}{\searrow} & & \overset{q}{\swarrow} & \underline{\mathrm{Cofibsc}}_0 S \\ & & \mathrm{Cat}/S & & \end{array}$$

et soit $\mathcal{F}$ une sous-catégorie pleine de Cat/S, telle que l'un des foncteurs p, q, r se factorise par $\mathcal{F}$. Soit $W_{\mathcal{F}}$ induit sur $\mathcal{F}$ par W_S dans Cat/S. Alors $\mathcal{F} \hookrightarrow \mathrm{Cat}/S$ induit une équivalence sur les catégories localisées.

DÉMONSTRATION. Je dis qu'il suffit de prouver, pour les trois foncteurs p, q, r, et pour les foncteurs en sens inverse [page 11] p. 8

$$\begin{array}{ccccc} & & \underline{\mathrm{Parf}}\, S & & \\ & & \uparrow{\scriptstyle C} & & \\ \underline{\mathrm{Fibsc}}_0 S & \overset{\Phi_0}{\nwarrow} & & \overset{\Psi_0}{\nearrow} & \underline{\mathrm{Cofibsc}}_0 S \\ & & \mathrm{Cat}/S & & \end{array}$$

que les six composés

$$p\Phi_0, \Phi_0 p, q\Psi_0, \Psi_0 q, rC, Cr$$

sont reliés aux foncteurs identiques (dans la catégorie pertinente) par une flèche qui est dans le localisateur correspondant. Pour chacun de ces trois couples de foncteurs (p, Φ_0), (q, Ψ_0), (r, C), cela prouvera déjà qu'en passant aux localisés, ils induisent des équivalences quasi-inverses l'une de l'autre. Mais on a mieux à l'aide du lemme formel suivant.

Lemme. *Soient (C, W), (C', W') deux catégories avec localisateurs $W \subseteq \mathrm{Fl}\, C$, $W' \subseteq \mathrm{Fl}\, C'$. Soit*

$$p : C \longrightarrow C'$$

un foncteur compatible avec les localisateurs, et supposons qu'il existe un foncteur $\varphi : C' \longrightarrow C$ également compatible, et tel que l'on ait des flèches (à direction non précisée)

$$\varphi p \xrightarrow{u} id_C, \qquad p\varphi \xrightarrow{v} id_{C'},$$

qui soient dans W resp. W' 'argument par argument'. [page 12]

Alors p et φ induisent des équivalences quasi-inverses l'une de l'autre en passant aux catégories localisées (elles n'ont pas de merite). De plus, pour toute sous-catégorie pleine C'_0 de C', telle que p se factorise (en $p_0 : C \longrightarrow C'_0$) par C'_0, si on désigne par W'_0 le localisateur induit dans C'_0 par C', alors $p_0 : C \longrightarrow C'_0$ et $\varphi_0 = \varphi|C'_0 : C' \longrightarrow C$ [plutôt $C'_0 \longrightarrow C$] induisent également des équivalences quasi-inverses l'une de l'autre sur les localisés.

En effet, $\varphi_0 p_0 = (\varphi i)p_0 = \varphi(ip_0) = \varphi p \xrightarrow{u} \mathrm{id}_C$ (ici $i : C'_0 \hookrightarrow C'$ désigne l'inclusion), et $F \overset{\text{déf}}{=} p_0\varphi_0 = p_0\varphi i$ satisfait à

$$iF = \underbrace{ip_0}_{p}\varphi i = (p\varphi)i.$$

Pour définir $F \xrightarrow{v_0} \mathrm{id}_{C'_0}$, qui soit dans W'_0 argument par argument, il suffit de définir [page 13]

$$\underbrace{iF}_{=(p\varphi)i} \xrightarrow{\bar{v}_0} i,$$

qui soit dans W' argument par argument, puisque W'_0 induit par W' et que $F \longmapsto iF$

$$\underline{\mathrm{Hom}}(C'_0, C'_0) \longrightarrow \underline{\mathrm{Hom}}(C'_0, C')$$

Commentary · Exposition [ms. 12]

The localization lemma keeps its full force

Grothendieck separates a sound formal mechanism from the geometric assertions to which he hopes to apply it. The lemma is useful precisely because its hypotheses record both kinds of information: preservation of weak equivalences by the functors, and membership of the comparison maps in the specified weak-equivalence classes.

Let P : (C, W) →(D, V) and R : (D, V) →(C, W) be relative functors. Suppose natural transformations, in either direction, connect RP with idC and PR with idD, with all components respectively in W and V . Their images under localization are natural isomorphisms. The induced functors P^- and R^- are therefore quasi-inverse equivalences. One needs neither a calculus of fractions nor triangle identities for the original transformations. Quasi-inverse equivalences, unlike a prescribed adjunction, require only the two natural isomorphisms.

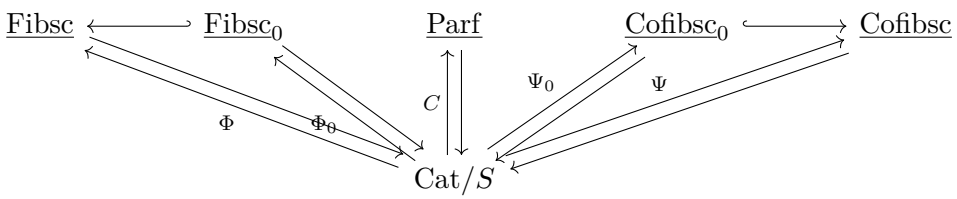
Now suppose P factors through a full subcategory i : D0 ,→D, written P = iP0, and put V0 = V ∩Mor(D0) and R0 = Ri. The first comparison restricts to R0P0 = RP. The second restricts to a transformation between iP0R0 and i. Because both of its endpoints lie in D0, fullness gives a unique transformation between P0R0 and idD0; faithfulness verifies its naturality. Every component belongs to V0. Consequently P0 and R0 also give inverse equivalences after localization. Applying the same argument to i shows that D0[V −1] →D[V −1] is an equivalence.

This explains the apparently strong corollary about an arbitrary full subcategory containing the image of a replacement functor. The weak-equivalence class on that full subcategory must be the induced one. For example, a full subcategory of Cat/S containing all perfect replacements may calculate the coarse localization. Nothing in the lemma permits changing its induced coarse class to the fibrewise class without proving their agreement. Nor does fullness apply to the passage from splitting-preserving maps to arbitrary maps over S: that functor is generally faithful but not full. These are the two points at which the first comparison diagram asks for additional mathematics.

est pleinement fidèle (i l'étant). Or il suffit de prendre p. 9

$$\bar{v}_0 = v * i : (p\varphi) \circ i \text{ —} \mathrm{id}_{C'} \circ i.$$

Pour résumer, il faut montrer que le composé des couples de foncteurs quasi-inverses marche dans les cinq cas suivants



D'ailleurs par dualité, on est ramené aux trois cas de $\underline{\text{Fibsc}}$, $\underline{\text{Fibsc}}_0$, $\underline{\text{Parf}}$. (Peut-être le cas $\underline{\text{Fibsc}}_0$, avec Φ_0 défini en termes de Φ et de C, se ramènera-t-il aux deux cas qui l'entourent, de $\underline{\text{Fibsc}}$ et $\underline{\text{Parf}}$.)

Je vois que je dois rectifier le tir, tout d'abord parce que j'ai oublié que les foncteurs Φ, Ψ [page 14] que j'avais en vue ne sont *pas* compatibles avec les localiseurs `[= localisateurs]` - une flèche $f : X \longrightarrow Y$ de Cat/S telle que la flèche correspondante dans Cat soit dans W, ne va pas induire une flèche dans $\underline{\text{Fibsc}}\, S$ ou $\underline{\text{Cofibsc}}\, S$ qui soit dans W sur chaque fibre. Ce n'est pas grave, car on peut remplacer les foncteurs par $\Phi' = \alpha'\Phi_0$, $\Psi' = \beta'\Psi_0$, qui sauf erreur doivent marcher. Mais en vue du Lemme précédent, il y a une perplexité, car j'ai l'air d'avoir prouvé que si $\mathcal{F}$ est une sous-catégorie pleine de Cat/S dont je suppose seulement qu'elle contient l'image essentielle d'une des trois catégories $\underline{\text{Fib}}_0 S$, $\underline{\text{Cofib}}_0 S$, ou $\underline{\text{Parf}}\, S$, alors l'inclusion dans Cat/S induit une équivalence des catégories localisées, en prenant dans $\mathcal{F}$ comme localisateur celui induit par Cat/S, i.e. $f : X \longrightarrow Y$ (triangle sur S) est dans $W_{\mathcal{F}}$ si et seulement si en tant que flèche de Cat, f est dans W. Or ceci coïncide bien (via un théorème non trivial sur lequel je devrai revenir dans le cadre d'un localisateur fondamental général W) avec l'ensemble de flèches de localisation 'fibre par fibre', dans le cas de $\mathcal{F} = \underline{\text{Parf}}\, S$, $\underline{\text{Fibhot}}\, S$, $\underline{\text{Fib}}_0 S$, $\underline{\text{Cofib}}_0 S$.

Par contre, ce n'est nullement le cas en prenant $\mathcal{F} = \underline{\text{Fib}}\, S, \underline{\text{Liss}}\, S$ ou $\underline{\text{Cofib}}\, S$. Ayant choisi [page 15] dans ces catégories comme localisateurs la notion de W-équivalences 'fibre par fibre', j'ai donc l'air de dire que je trouve pour ces quatre catégories les *mêmes* localisées (à équivalence près) que si je prenais le localisateur beaucoup plus grossier hérité par Cat/S. Cela semble difficile à croire, et il faudra que je fasse très attention en démontrant les énoncés-clef.

Pour voir que les flèches en traits pleins du grand diagramme p. 6 sont toutes compatibles avec les localis(at)eurs `[sic]`, donc aussi les flèches composées au moyen de celles-ci, j'ai à utiliser déjà ceci (pour le cas des flèches $\underline{\text{Liss}} \longrightarrow \text{Cat}/S$, $\underline{\text{Prop}} \longrightarrow \text{Cat}/S$):

① Si X, Y lisses ou propres sur S, et $f : X \longrightarrow Y$ un S-foncteur, si $f_s : X_s \longrightarrow Y_s$ est p. 10 dans W pour tout $s \in S$, alors $f \in W$. ([7]).

Cela renforce considérablement la condition L5 sur W, où on supposait que f est un foncteur *cartésien* entre catégories *fibrées* sur S. (À dégager, pour les différentes propriétés dont on a besoin pour W, à quelles propriétés standard on peut les ramener. Tout sera OK si $W = W_\infty$.)

I) **Définition de Φ, Ψ.** [page 16]

Commentary · Exposition [ms. 15]

The precise local-to-global argument

Grothendieck needs fibrewise equivalence to imply total equivalence without requiring a map of fibrations to preserve cartesian arrows. Smoothness or properness provides the comparison between strict fibres and comma categories that makes this possible. The treatment of these notions in the commentary on Chapter I explains the lifting categories behind those comparisons; here their role is descent for a map of total categories.

For a proper functor p : X →S, the canonical functor Xs →X/s is a W∞-equivalence; for a smooth functor, the corresponding functor is Xs →s\X. These assertions, in the appropriate aspherical form, are the criterion developed in Maltsiniotis (2005, Theorem 1.2 and its dual). Suppose X, Y are proper over S and f : X →Y is any functor over S. Naturality gives

Xs

Ys

X/s

Y/s. Under W∞⊆W, the vertical maps belong to W. The two-out-of-three property implies that fs ∈W if and only if f/s ∈W. If this holds for every s, relative L4 bis gives f ∈W. For smooth functors the proof uses the undercategories and the dual relative axiom. No cartesian or cocartesian condition on f occurs.

7. OK si on suppose L3 quater et L4.

The argument also explains the eventual universal statement on pages 41–43. Proper and smooth functors are stable under change of base. For S′ →S, the strict fibre of the pulled-back map at s′ is fs′ = fp(s′). Applying the same proof over S′ shows that a fibrewise equivalence between two proper objects, or between two smooth objects, remains a total equivalence after every base change.

The parenthetical assertion here that L3 quater and L4 suffice suppresses one hypothesis: the version of Theorem A actually used is relative L4 bis. Absolute L4 directly covers the case Y = S, since S/s has a terminal object. For general Y it is the stronger formulation that is needed in the displayed proof. Grothendieck explicitly identifies this distinction on manuscript page 44. We follow that correction rather than assigning an unproved consequence to the initial axiom.

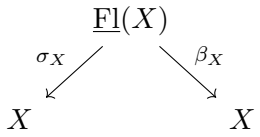
On va calquer la construction canonique de $\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$ (ou $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$) factorisant $X \xrightarrow{\mathrm{diag}} X \times X$, et la factorisation canonique correspondante d'un $f : X' \longrightarrow X$ en $X' \xrightarrow{i} \bar{X}' \xrightarrow{\bar{f}} Y$ [plutôt $\bar{X}' \xrightarrow{\bar{f}} X$]. On remplacera simplement $\underline{\mathrm{Ch}}(X)$ par la sous-catégorie $\underline{\mathrm{Ch}}_+(X)$, laquelle n'est autre que la catégorie des flèches de X

$$\underline{\mathrm{Fl}}(X) = \underline{\mathrm{Hom}}(\mathbf{\Delta}_1, X),$$

muni des foncteurs

$$X \xrightarrow{i_X} \underline{\mathrm{Fl}}(X) \xrightarrow{p_X=(\sigma_X,\beta_X)} X \times X,$$

où i_X est la flèche $x \longmapsto \mathrm{id}_x$, et σ_X, β_X sont les foncteurs source et but. La fibre de p_X en $(a, b) \in \mathrm{Ob}(X \times X)$ est la catégorie discrète définie par l'ensemble $\mathrm{Hom}(a, b)$, ça dépend grandement de (a, b) et pas seulement de sa composante connexe dans $X \times X$ - donc p_X n'a rien d'une Hot-fibration. Il n'est en géneral ni propre ni lisse ni rien, par contre les deux flèches composées avec pr_1 et pr_2



sont sympathiques, on a en effet

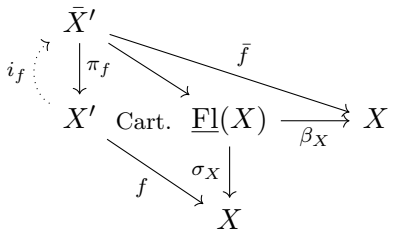
Lemme. *On a* [page 17]

$$\sigma_X^{-1}(\{x\}) \simeq x\backslash X, \qquad \beta_X^{-1}(\{x\}) \simeq X/x.$$

Le foncteur σ_X est Cat*-fibrant à fibres contractiles (donc asphériques, donc W-asphériques), le foncteur β_X cofibrant à fibres contractiles (donc asphériques, donc W-asphériques). Ce sont donc aussi des* Hot*-équivalences, a fortiori des W-équivalences* ([8]).

Corollaire 1. *Le foncteur $i_X : X \longrightarrow \underline{\mathrm{Fl}}(X)$ est une* Hot*-équivalence* (puisque son composé avec la Hot-équivalence σ_X est l'identité).

Voici la construction générale de la factorisation de $f : X' \longrightarrow X$, correspondant au choix précédent d'une 'catégorie de chemins' $\mathcal{C}(X) = \underline{\mathrm{Fl}}(X)$ p. 11



8. En fait, ce sont même des homotopismes, cf. remarque marginale plus bas (plus générale - on retrouve le cas de σ_X en faisant $X' = X$.)

où i_f est déduit de la section i [plutôt i_X] de $\underline{\mathrm{Fl}}(X)$ sur X ([9]). Ainsi, $\bar{X}'$ est la catégorie des couples

$$\{(x', u) \mid x' \in \mathrm{Ob}\, X', \quad \underbrace{u}_{\in \mathrm{Fl}\,(X)} : f(x') \longrightarrow y\},$$

le foncteur $\bar{f}$ est défini par

$$\bar{f}(x', u) = \mathrm{but}(u) = y$$

et i_f par

$$i_f(x') = (x', \mathrm{id}_{f(x')} : f(x') \xrightarrow{\mathrm{id}} f(x')),$$

donc on a bien

$$f(x') = \bar{f} i_f(x').$$

Corollaire 2. *Le foncteur* [page 18]

$$\bar{f} : \bar{X}' \longrightarrow X$$

a comme fibres les catégories X'/x, il est cofibrant, les foncteurs de changement de base sont les foncteurs $X'/x \longrightarrow X'/y$ associés aux flèches $x \longrightarrow y$ dans X.

Commentary · Exposition [ms. 17]

The arrow category has two different good endpoints

Grothendieck introduces a path construction which remembers the direction of an arrow. Its endpoint maps are excellent replacement tools separately, although the pair of endpoints is not a homotopy-fibration model of the diagonal. The elementary adjunctions make all of these assertions transparent. Write Ar(A) = Fun([1], A), and let i(a) = ida, σ(a →b) = a, β(a →b) = b. A commutative square from i(x) to a →b is determined by its arrow x →a. A square from a →b to i(y) is determined by b →y. Therefore

$\beta \dashv$i $\dashv\sigma$.

In particular σi = βi = id, and the squares associated with an arrow a →b give natural transformations iσ ⇒id and id ⇒iβ. Both endpoint projections and i are categorical homotopy equivalences, hence belong to every W satisfying L1 and L2.

The fibration properties can be checked without choosing anything. Pulling a →b back along a′ →a means precomposing to obtain a′ →b; the resulting square is cartesian for σ. Pushing it forward along b →b′ means postcomposing to obtain a →b′; that square is cocartesian for β. The fibres are a\A and A/b, with initial and terminal identities respectively. This example already belongs to the original descent formalism; see Grothendieck (1971, Exposé VI, §11(a)).

For the combined map (σ, β) : Ar(A) →A × A, however, the fibre at (a, b) is the discrete set HomA(a, b). Take A = [1]. Its fibre at (1, 0) is empty, while that at (0, 0) is a point, although the base is connected. Thus it is not a Hot-fibration. It is neither smooth nor proper: at (1, 0) its undercategory has the object arising from (1, 0) →(1, 1), and its overcategory has the object arising from (0, 0) →(1, 0), whereas its strict fibre is empty. In fact each of these comma categories is a point. The necessary fibre-to-comma weak-equivalence condition fails in both directions. This distinction is why the directed arrow construction gives Φ and Ψ, while the two-sided replacement C requires the longer path category Ch∞(S).

En fait, on voit que cette catégorie est non seulement cofibrée, mais canoniquement *scindée.* ([10]).

Appliquant ceci à des $f : X \longrightarrow S$, on trouve une construction

$$\mathrm{Cat}/S \xrightarrow{\Psi} \underline{\mathrm{Cofibsc}}\, S,$$

9. **NB** $\pi_f i_f = \mathrm{id}$, et $i_f \pi_f \longrightarrow \mathrm{id}$, donc i_f, π_f sont des homotopismes inverses l'un de l'autre dans Cat hot. Cela précise [le] cor. 3, page 18.

10. **Corollaire 3.** *La flèche $i_f : X' \longrightarrow \bar{X}'$ est un X-morphisme (si X' sur X par f, $\bar{X}'$ par $\bar{f}$), i.e. $f = \bar{f} \circ i_f$, et c'est une* Hot-*équivalence* (car $\pi_f : \bar{X}' \longrightarrow X'$ est Cat-fibrant, à fibres contractiles, donc une Hot-équivalence).

c'est bel et bien fonctoriel en X variant dans Cat/S. Si

$$\psi : X \longrightarrow Y$$

dans Cat/S, alors ψ induit des p. 12

$$\psi/s : X/s \longrightarrow Y/s \ ,$$

pour $s \in S$, et ceci de façon compatible avec les foncteurs 'cochangement de base' associés aux $u : s \longrightarrow t$ dans S [page 19]

$$\begin{array}{ccc} X/s & \xrightarrow{u_*^X} & X/t \\ {\scriptstyle \psi/s}\downarrow & & \downarrow{\scriptstyle \psi/t} \\ Y/s & \xrightarrow{u_*^Y} & Y/t \end{array} \quad .$$

Cela définit donc un morphisme de catégories scindées sur S

$$\Psi(X) \longrightarrow \Psi(Y),$$

c'est lui!

On a un morphisme fonctoriel en X

$$X \xrightarrow{i_X} \Psi(X)$$

dans Cat/X, et c'est une Hot-équivalence, donc $\in W$. Sur les fibres, le morphisme i_X induit

$$X_s \xrightarrow{i_{X,s}} X/s,$$

l'inclusion canonique.

Corollaire 1. *Si* $X \xrightarrow{f} Y$ *dans* Cat/S *est dans* W*, alors* $\Psi(X) \xrightarrow{\Psi(f)} \Psi(Y)$ *aussi.* Mais on se gardera de croire que c'est dans W_S au sens de Cofib S ou Cofibsc S. [page 20]

Commentary · Exposition [ms. 19]

A canonical opfibration and its two projections

Grothendieck's replacement Ψ packages all the overcategories X/s into one category. Writing down its arrows shows both why it is strictly functorial and why a projection used to prove homotopy invariance need not be a morphism over the original base.

For p : X →S, an object of Ψ(X) is (x, u : p(x) →s). A morphism to (y, u′ : p(y) →t) is a pair vu = u′p(a).

(a : x →y, v : s →t), The structural projection q sends this object to s and this morphism to v. For v : s →t, the morphism (idx, v) : (x, u) →(x, vu) is the specified cocartesian lift. Composition of these lifts is literal composition, so the opfibration is split and its fibre at s is X/s.

There is also the forgetful projection π : Ψ(X) →X, which sends (x, u) to x, and the functor i : X →Ψ(X) sending x to (x, idp(x)). A map from i(x) to (y, u′) is uniquely determined by a : x →y, because its base arrow must be u′p(a). Hence i ⊣π, with πi = idX and counit

−−−−→(x, u).

iπ(x, u) = (x, idp(x)) This proves that i and π are inverse up to categorical homotopy. The identity qi = p makes i a functor over S, but pπ(x, u) = p(x) generally differs from q(x, u) = s. Thus π generally is not a functor over S. The printed statement that i is a morphism in Cat/X is nonetheless true when Ψ(X) carries π: i is a section. Both descriptions of i are legitimate, and neither licenses using π as an inverse in Cat/S.

For a functor f : X →Y over S, define Ψ(f)(x, u) = (f(x), u) and Ψ(f)(a, v) = (f(a), v). This preserves the specified cocartesian lifts and is strictly functorial in f. The maps i form a natural transformation in Cat/S. Their being in W shows, by the resulting commutative square, that Ψ preserves and reflects coarse weak equivalences. On fibres, however, its effect is f/s. This last observation specifies the stronger class for which Ψ becomes a replacement preserving fibrewise weak equivalences.

Car sur les fibres de $\Psi(X)$, $\Psi(Y)$ on trouve les flèches

$$f/s : X/s \longrightarrow Y/s$$

induits par f, qui n'ont aucune raison d'être dans W. Qu'elles le soient signifie que $f : X \longrightarrow Y$ est dans W 'localement au dessus de S' ([11]), ce qui (pour les W qui nous intéressent) est beaucoup plus fort que d'être dans W. (Si $Y = S$, cela signifie en fait que $p : X \longrightarrow S$ est W-asphérique, beaucoup plus fort que $p \in W$.)

Donc pour prouver que

$$\underline{\mathrm{Cofib}}\, S \longrightarrow \mathrm{Cat}/S \qquad \text{ou} \qquad \underline{\mathrm{Cofibsc}}\, S \longrightarrow \mathrm{Cat}/S$$

induisent des équivalences pour les catégories localisées, le foncteur Ψ directement ne sera pas utilisable. Cependant : p. 13

Corollaire 2. *Supposons X, Y dans* $\underline{\mathrm{Prop}}\, S$ *ou dans* $\underline{\mathrm{Fibhot}}\, S$*, et considérons*

$$\Psi(f) : \Psi(X) \longrightarrow \Psi(Y),$$

pour que ce soit une W-équivalence dans $\underline{\mathrm{Cofibsc}}\, S$*, i.e. une W-équivalence fibre par fibre, il faut et il suffit que f soit une W-équivalence fibre par fibre* (et cela doit impliquer ([12]) que f est une W-équivalence). [page 21]

Commentary · Exposition [ms. 20]

Which comma construction repairs which variance

Grothendieck now isolates the extra condition that makes Ψ a relative replacement for fibrewise homotopy theory. Its fibre map is exactly f/s, so the condition is that every overcategory map lie in W. When the target is S itself this says that f is W-aspherical, since S/s has a terminal object. For a general target it is a relative condition, later called fine equivalence on the left.

The interval calculation after manuscript page 8 already shows why total equivalence cannot suffice. The complete comparison of both endpoint inclusions, given alongside manuscript page 32, also separates the two comma tests. On a smooth object, the strict fibre compares to the undercategory s\X; on a proper object, to the overcategory X/s. Consequently the relevant fibrewise replacement for a fibration is Φ, and for an opfibration it is Ψ. This is the geometric reason the later fine-right equivalence belongs on the fibration side of the large diagram, while fine-left equivalence belongs on the opfibration side. The direction names refer to the comma constructions, not to the position of the objects in that diagram.

En effet, dans le diagramme commutatif

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ \downarrow & & \downarrow \\ X/s & \xrightarrow{f/s} & Y/s \end{array}$$

les flèches verticales sont dans W, car X, Y `[sont]` propres sur S (ou X, Y dans $\underline{\mathrm{Fibhot}}\, S$), donc f_s est dans W si et seulement si f/s l'est.

`[Le foncteur]` Φ se définit de façon duale à Ψ, ce qui revient à échanger les rôles de σ_X, β_X dans le diagramme page 17.

([13])

11. cf. (1) page 15.
12. cf. (1) page 15.
13. **Corollaire 3.** *Si $X \in Ob\, \underline{\mathrm{Fibhot}}\, S$, alors $\Psi(X)$ est dans* $\underline{\mathrm{Fibsc}}_0 S$.

Cela signifie en effet que les $X/s \longrightarrow X/t$ sont dans W, or l'hypothèse sur X implique que c'est dans W_∞. OK.

II) Définition de $C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S$

Cette fois, c'est la factorisation des flèches $p : X \longrightarrow S$ dans Cat, en

$$X \xrightarrow{j_{X/S}} \bar{X} \xrightarrow{\bar{p}} S,$$

où i_X `[plutôt` $j_{X/S}$`]` est une Hot-équivalence (donc dans W), et $\bar{p}$ parfait. Cela se construit formellement comme tantôt, en utilisant $\underline{\mathrm{Ch}}_\infty(S)$ au lieu de $\underline{\mathrm{Fl}}(S)$, et la factorisation

$$S \xrightarrow{j_S} \underline{\mathrm{Ch}}_\infty(S) \xrightarrow{p_S=(\sigma_S,\beta_S)} S \times S$$

de $\mathrm{diag}_S : S \longrightarrow S \times S$. La situation est plus jolie, car p_S est déjà 'parfait', ce qui implique formellement que le $\bar{p}$ qui s'en déduit, pour factoriser $p : X \longrightarrow S$, est lui-même parfait.

Ici encore, la construction est fonctorielle, on trouve le foncteur

p. 14

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}\, S$$ [page 22]

qu'il faut composer au foncteur en sens inverse

$$r : \mathrm{Cat}/S \longleftarrow \underline{\mathrm{Parf}}\, S.$$

Tout d'abord, vérifions que C, tout comme r, est compatible avec les localiseurs. Soit $f : X \longrightarrow Y$ dans Cat/S, supposons f est dans W, prouvons $C(f)$ est dans W_S. On a commutativité dans

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle j_X}\downarrow & & \downarrow{\scriptstyle j_Y} \\ CX & \xrightarrow{Cf} & CY \end{array}$$

et j_X, j_Y sont dans W, donc f est dans W si et seulement si Cf l'est. Mais dans le cas $W = W_\infty$ on sait que cela équivaut à ce que Cf soit dans W_∞ fibre par fibre. Admettant ① p. 15 pour W, il reste à s'assurer d'une réciproque dans $\underline{\mathrm{Parf}}$, i.e.

Commentary · Later developments [ms. 21]

From comma fibres to homotopy fibres

Grothendieck introduces the longer path category because he now wants a replacement whose transport is homotopically invertible in both directions. The guiding theorem is Quillen's Theorem B: if every transition X/s →X/t induced by an arrow s →t is a weak homotopy equivalence, then N(X/s) computes the homotopy fibre of NX →NS over s. A precise restatement appears in Cisinski (2004, Theorem 2.3.2). This is the point at which locally constant comma data become actual fibre data for the homotopy type of a map.

For an object already Hot-fibred over S, the comparison Xs →X/s is a weak equivalence, and the transitions between these comma categories are weak equivalences. Consequently Ψ(X) is a split opfibration with homotopically invertible transition functors. The footnote's printed target Fibsc0S is therefore a slip: the target is Cofibsc0S. Its dual statement concerns Φ and split fibrations. The slip does not affect the argument in the adjacent square.

The functor C uses results about Ch∞(S) established outside these pages. The input here is a natural factorization X −−→C(X) →S with jX ∈W∞and C(X) →S perfect. Given this input, all subsequent uses of its functoriality are formal. For instance, the naturality square of j implies that f ∈W if and only if C(f) ∈W, provided W∞⊆W. What is not formal is replacing the last total condition by a condition on fibres for an arbitrary larger W. That is the question on pages 22–24, and the stronger L5 introduced later addresses exactly it. The correct comparison therefore depends on which localizer is being studied, not only on the quality of the replacement object.

② Soit $\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$ dans $\underline{\mathrm{Parf}}\, S$ (ou seulement dans $\underline{\mathrm{Fibhot}}\, S$). Si f est dans W, alors les $f_s : X_s \longrightarrow Y_s$ aussi. [page 23]

Essayons de le prouver, en calquant la démonstration dans le cas $C = C_\infty$ (XI, th. 2). On peut supposer S connexe ([14]), en utilisant La (i.e. $W \subseteq W_0$) et en admettant le

Lemme. *Soit $f : X \longrightarrow Y$ dans* Cat. *Alors $f \in W$ si et seulement si $\pi_0(f)$ est bijectif, et pour toute composante connexe X_i de X, si Y_i est la composante connexe correspondante de Y, $f_i : X_i \longrightarrow Y_i$ induit par f est dans W.*

Utilisons alors l'existence de

$$S' \longrightarrow S$$

Hot-fibration avec S' Hot-asphérique ([15]). On peut supposer même que $S' \longrightarrow S$ est parfait. Supposons qu'on sache qu'un changement de base par de telles fibrations parfaites transforme W-équivalences en itou. (Ça doit être vrai pour les W_i, mais on a vu que c'était faux pour l'équivalence k-homologique, cf. I p. 87 ff.) Alors $X' \xrightarrow{f'} Y'$ sur S' est une W-équivalence. Or pour $s' \in S'$, comme $\{s'\} \longrightarrow S'$ est une Hot-équivalence (S' Hot-asphérique), et X', Y' étant Hot-fibrés sur S, il s'ensuit que $X'_{s'} \longrightarrow X'$, $Y'_{s'} \longrightarrow Y'$ sont des Hot-équivalences, donc aussi $X'_{s'} \xrightarrow{f'_{s'}} Y'_{s'}$, comme on voit sur le carré commutatif p. 15

$$\begin{array}{ccc} X'_{s'} & \xrightarrow{f'_{s'}} & Y'_{s'} \\ \wr\downarrow & & \wr\downarrow \\ X' & \xrightarrow{f'} & Y'. \end{array}$$

Comme $S' \longrightarrow S$ est surjectif, cela montre que les $f_s : X_s \longrightarrow Y_s$ sont dans W, qed. [page 24]

Commentary · Exposition [ms. 23]

The contractible-base argument without a component reduction

Grothendieck's strategy is to inspect a chosen fibre after making the base contractible. Its force is easiest to see when the required base-change assumption is stated before the reduction. It then becomes clear that the componentwise lemma is unnecessary for this particular argument, as his marginal annotation already observes.

Assume W∞⊆W, two-out-of-three, and the following additional property: pullback along a perfect functor preserves W-equivalences. Let f : X →Y be a functor over S, with X, Y Hot-fibred over S, and suppose its total functor lies in W. Fix s ∈S. Use the perfect replacement of the point {s} →S to obtain

−→jS′ −→rS, {s} j ∈W∞, where r is perfect and S′ is weakly contractible. The distinguished object s′ = j(s) lies over s. This construction is allowed even when S is disconnected; its image lies in the relevant component.

Put X′ = X ×S S′ and Y ′ = Y ×S S′. The extra pullback hypothesis gives f ′ : X′ →Y ′ in W. Base change preserves the Hot-fibration property. Since S′ is weakly contractible, the inclusions of the fibres at s′ into X′ and Y ′ are weak homotopy equivalences. Equivalently, their homotopy-fibre squares are pulled back along the weak equivalence {s′} →S′. In the commutative square

Xs

Ys

X′

Y ′, three sides lie in W, so fs ∈W. This proves the fibre-detection assertion under precisely the stated additional hypothesis. It does not prove fs ∈W∞from f ′ ∈W.

The proof uses neither La nor a decomposition of f into connected components. Its genuinely substantial input is stability of W under these perfect base changes, later called L6. The speculation here that truncated homotopy equivalences should satisfy that stability is withdrawn in the later annotations; the obstruction is discussed at manuscript pages 43–45. The proof itself is valid as a conditional argument, and its dependence on L6 is the mathematical information to retain.

14. ne sert à rien

15. **NB** Il suffit d'avoir une Hot-fibration $S' \longrightarrow S_0 \hookrightarrow S$, où S_0 est la composante connexe de S contenant l'élément $s \in \mathrm{Ob}\, S$ envisagé.

Ainsi, (2) sera conséquence de ②. Soit $p : Y' \longrightarrow Y$ une Hot-fibration (ou même un foncteur parfait, si ça peut aider …). Soit $f : X \longrightarrow Y$ une S-flèche (dans Cat/S), alors $f' : X' \longrightarrow Y'$ par changement de base. Si $f \in W$, $f' \in W'$

$$\begin{array}{ccc} X' & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & \text{cart.} & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{p} & Y \end{array} \qquad f \in W \text{ et } p \text{ parfait} \Longrightarrow f' \in W$$

En tous cas, il nous faut absolument ② pour pouvoir conclure que C est compatible avec les localiseurs.

Considérons $rC : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$, on a une flèche

$$\begin{cases} \mathrm{id}_{\mathrm{Cat}/S} \xrightarrow{j} rC \\ j_X : X \longrightarrow CX \quad \text{la flèche canonique } i_{X/S} \texttt{ [plutôt } j_{X/S}\texttt{]} \text{ (p. 21),} \end{cases}$$

et j_X est toujours dans W.

Considérons $Cr : \underline{\mathrm{Parf}}\, S \longrightarrow \underline{\mathrm{Parf}}\, S$, i.e. appliquons la construction C à un X parfait sur S. Alors on a encore

$$\begin{cases} \mathrm{id}_{\underline{\mathrm{Parf}}\, S} \xrightarrow{j'} Cr \\ j'_X = j_{r(X)} : X = r(X) \xrightarrow{j_X} CX. \end{cases}$$

Ce morphisme est dans W, donc aussi une W-équivalence fibre par fibre, puisque source et but sont parfaits, donc des Hot-fibrés sur S.

Cela prouve déjà que les foncteurs r et C induisent des équivalences quasi-inverses l'une de l'autre sur les catégories localisées. Par le lemme p. 11, cela montre que pour toute sous-catégorie pleine $\mathcal{F}$ de Cat/S contenant $\underline{\mathrm{Parf}}\, S$, l'inclusion de $\mathcal{F}$ dans Cat/S induit une équivalence sur les localisés, en prenant comme localiseur dans $\mathcal{F}$ le localiseur 'grossier' induit par Cat/S. Cela s'applique notamment aux sous-catégories pleines [page 25] p. 16

$$\underline{\mathrm{Liss}}\, S, \quad \underline{\mathrm{Fibhot}}\, S, \quad \underline{\mathrm{Prop}}\, S,$$

mais en prenant dans $\underline{\mathrm{Liss}}\, S$ et $\underline{\mathrm{Prop}}\, S$ non pas le localiseur fin 'fibre par fibre', comme je l'avais annoncé à tort, mais le localiseur grossier induit par Cat/S. (Pour $\underline{\mathrm{Fibhot}}\, S$, il n'y a pas de risque d'ambiguïté). Il faudrait que je me convainque que si je prends la W-équivalence fine (dans le cas $W = W_\infty$, disons), je ne trouve *pas* les 'bons' localisés, i.e. le foncteur des localisés, vers le localisé de Cat/S, n'est pas une équivalence. Ceci en fait suppose que dans $\underline{\mathrm{Fib}}\, S$ et $\underline{\mathrm{Cofib}}\, S$ également, c'est l'équivalence *grossière* qu'il convient de prendre. (Dans $\underline{\mathrm{Fib}}_0 S$ et $\underline{\mathrm{Cofib}}_0 S$, ou $\underline{\mathrm{Fibsc}}_0 S$ et $\underline{\mathrm{Cofibsc}}_0 S$, les deux équivalences sont identiques, puisque ces quatre catégories sont contenues dans $\underline{\mathrm{Fibhot}}\, S$.) La suite nous montrera s'il en est bien ainsi. Par contre, dans $\underline{\mathrm{Fibsc}}\, S$ et $\underline{\mathrm{Cofibsc}}\, S$, il n'y a pas de doute, suivant les principes généraux de construction d'un dérivateur à partir d'un $(\mathcal{M}, W)$, qu'il faut prendre l'équivalence fine, fibre par fibre. À ce sujet, il faut faire attention que les foncteurs verticaux du diagramme p. 8 [page 26]

$$\underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S \longrightarrow \underline{\mathrm{Cofib}}\, S,$$

quoique essentiellement surjectifs et fidèles, ne sont pas pleinement fidèles. Ils le seraient, et mêmes des équivalences de catégories, si dans $\underline{\mathrm{Fib}}\, S$ resp. $\underline{\mathrm{Cofib}}\, S$ je me bornais aux S-foncteurs *cartésiens* resp. *cocartésiens*. Donc on peut interpréter, à équivalence près, $\underline{\mathrm{Fibsc}}\, S$ et $\underline{\mathrm{Cofibsc}}\, S$ comme $\underline{\mathrm{Fib}}\, S$ et $\underline{\mathrm{Cofib}}\, S$, où on aurait restreint les morphismes aux seuls S-foncteurs *cartésiens* resp. *cocartésiens*. Il semblerait donc qu'en restreignant ainsi les flèches, mais en restreignant de façon plus draconienne encore le localiseur, on trouve des catégories localisées équivalentes.

Commentary · Exposition [ms. 24]

What the path replacement actually proves

Grothendieck wants a category over S whose strict fibres already have the homotopy types of the homotopy fibres. The replacement jX : X −→C(X) achieves this geometrically: jX is a weak homotopy equivalence, is natural over S, and C(X) −→S is perfect. The difficulty is to decide which weak equivalences this replacement preserves. Two assertions must be separated, because one is formal and the other detects a substantive property of the coefficient theory.

Write W tot for maps over S whose underlying functor lies in W. Assuming W∞⊆W and two-outof-three, naturality gives

f ∈W tot

⇐⇒C(f) ∈W tot C(f)jX = jY f, S . Thus no base-change axiom is needed to make C a functor between the coarse relative categories. If F ⊆ Cat/S is full and contains every perfect object, C factors through F. The two natural transformations j on Cat/S and its restriction to F become invertible after coarse localization. Consequently the inclusion of F induces an equivalence. This proves the assertion of manuscript page 25 under the stated elementary hypotheses, independently of the conjectural pullback assertion on the present page.

The stronger assertion is that C(f) is a W-equivalence on every fibre whenever f ∈W tot S . This does require the implication later named L5 on manuscript page 43. The preceding argument with a contractible covering base explains a sufficient condition precisely. For a chosen s ∈S, take a perfect S′ −→S with contractible total category and an object over s; it suffices to cover the component containing s. If pullback along this map preserves W, the resulting total map X′ −→Y ′ belongs to W. For homotopy fibrations the inclusions of the chosen fibres into X′ and Y ′ are weak homotopy equivalences. Two-out-of-three then gives the desired fibre map in W. The final comparison gives a Wequivalence, not necessarily a weak homotopy equivalence: the sentence immediately before manuscript page 24 compresses these two conclusions.

For W = W∞this is the familiar invariance of a homotopy fibre under equivalence of total spaces over a fixed base. For a coarser W it is an additional demand. The covering calculation at manuscript page 43 below will show its failure even when both maps to the base are discrete fibrations and discrete opfibrations. In particular the phrase “il nous faut absolument” concerns compatibility with the fibrewise localizer, not the already established compatibility with the coarse one.

C’est assez étrange - on va voir si c’est bien vrai! [page 27]

Commentary · Context [ms. 26]

Strict splittings and Cartesian maps

The comparison with diagram categories depends on what the morphisms preserve. The definition on manuscript page 4 uses functors preserving the chosen Cartesian lifts strictly. A Cartesian functor between split fibrations need only carry a chosen lift to some Cartesian lift; the latter is isomorphic to the chosen lift at the image object. These conditions give strict and pseudonatural transformations, respectively. They are different conditions on actual functors, before localization.

Here is a concrete failure of fullness. Put S = [1] and let G = C2 = {1, c}. The projection S ×BG −→S has its constant splitting, with chosen lift (u, 1) of u : 0 −→1. Define an endofunctor over S which fixes the two fibre groupoids and sends a cross-fibre arrow (u, g) to (u, cg). This is a functor: composition with a vertical arrow is compatible with the formula because G is abelian. Every arrow over u is Cartesian, since its group component is invertible. Our endofunctor is therefore Cartesian, but it does not preserve the splitting, since (u, 1) is sent to (u, c). Restricting the target category to Cartesian functors does not make the forgetful functor from strictly split morphisms full.

There is a separate qualification concerning objects. The functor BC4 −→BC2 induced by reduction modulo 2 is a Grothendieck fibration: each base arrow has a lift and every arrow is Cartesian. A splitting would be a homomorphic section C2 −→C4. Such a section cannot exist, because each lift of the nonidentity element has order 4. Hence this fibration is not isomorphic over BC2 to an object with a splitting. This is the group-extension example already explained in Grothendieck (1971, Exposé VI, §9). An equivalence over the base to a split replacement is the appropriate statement; essential surjectivity in the ordinary category Cat/S would ask for an isomorphism of categories over the base.

These distinctions do not damage the intended homotopical comparison. Equivalences of fibre categories belong to W∞, and strict replacements can be connected by such equivalences. What

needs proof is precisely that replacing coherent choices by strict ones induces the correct localized comparison. Grothendieck encounters that issue directly in the failure of the proposed unit on manuscript page 30 and returns to it on pages 55–60. The underlying classical language comes from the descent formalism of SGA 1; the new task here is to control the weak equivalences during this passage. We retain the printed assertion and read it with this necessary qualification, rather than changing the category of morphisms silently.

III) Construction de Φ_0, Ψ_0 : $\mathrm{Cat}/S \to \underline{\mathrm{Fibsc}}_0 S$, $\mathrm{Cat}/S \to \underline{\mathrm{Cofibsc}}_0 S$.

On doit avoir

$$\Phi_0(X) = \Phi rC(X) \qquad\qquad \Psi_0(X) = \Psi rC(X),$$

où

$$r : \underline{\mathrm{Parf}}\, S \longrightarrow \mathrm{Cat}/S$$

est l'inclusion canonique. En tant que foncteurs à valeurs dans Fibsc resp. Cofibsc, cela définit déjà Φ_0, Ψ_0 par

$$\Phi_0 = \Phi rC \qquad\qquad \Psi_0 = \Psi rC,$$

mais il faut voir que Φ_0, Ψ_0 prennent leurs valeurs dans les sous-catégories pleines à indice 0. p. 17
Cela équivaut donc au Lemme suivant, appliqué à $Y = rC(X)$:

Lemme. *Soit Y parfait sur S (ou seulement un* Hot*-fibré sur S). Alors $\Phi(Y)$ est dans $\underline{\mathrm{Fibsc}}_0 S$, $\Psi(Y)$ dans $\underline{\mathrm{Cofibsc}}_0 S$.*

Il faut prouver (pour $\Psi(Y)$) que si $s \longrightarrow t$ est une flèche dans S, alors

$$\Psi(Y)_s \longrightarrow \Psi(Y)_t$$

est une W-équivalence. Or cette flèche s'identifie à

$$Y/s \longrightarrow Y/t$$

et on sait bien que c'est une Hot-équivalence, par définition des Hot-fibrés, donc c'est une [page 28]
W-équivalence.

On a donc les foncteurs

$$\underline{\mathrm{Fibsc}}_0 S \underset{\Phi_0}{\overset{p}{\rightleftarrows}} \mathrm{Cat}/S,$$

il faut voir qu'ils satisfont aux conditions générales du lemme p. 11 (et ce sera pareil pour q, Ψ_0). On sait déjà que p est compatible avec les localiseurs. Prouvons le pour Ψ_0. Soit

$$f : X \longrightarrow Y$$

dans Cat/S une W-équivalence, alors $rC(f)$ l'est aussi. Il suffit de prouver ceci :

Soit

$$q : X' \longrightarrow Y'$$

un morphisme dans $\underline{\mathrm{Fibhot}}\, S$ (par exemple un morphisme de $\underline{\mathrm{Parf}}\, S$), qui soit W-équivalence, donc `[qui induit]` des W-équivalences sur les fibres. Alors $\Psi(q) : \Psi(X') \longrightarrow \Psi(Y')$ est une W-équivalence fibre par fibre, i.e.

$$X'/s \longrightarrow Y'/s \qquad \text{[page 29]}$$

une W-équivalence pour tout $s \in S$ (a été vu dans le corollaire 2, page 20).

Considérons

$$p\Phi_0 : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S,$$

ça transforme donc X en $\Phi C(X)$, regardé comme objet de Cat/S sans plus. On a un morphisme fonctoriel

$$X \xrightarrow{j_X} C(X) \qquad\qquad j : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C,$$

d'où un morphisme fonctoriel p. 18

$$\Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X) \qquad\qquad \Phi * j : \Phi \longrightarrow \Phi_0$$

et également un morphisme fonctoriel

$$X \xrightarrow{i_X} \Phi(X),$$

d'où en composant

$$X \xrightarrow{i_X} \Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X), \qquad X \xrightarrow{\alpha_X} \Phi C(X)$$

i.e.

$$\alpha : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow p\Phi_0.$$

Pour X fixé dans Cat/S, on sait que j_X est dans W, donc aussi $\Phi(j_X)$ (quoique non fibre par fibre), cf. cor. 1 page 20. On sait aussi que i_X est dans W, donc aussi le composé α_X.

Considérons

$$\Phi_0 p : \underline{\mathrm{Fibsc}}_0\, S \longrightarrow \underline{\mathrm{Fibsc}}_0\, S \quad .$$

On part de X dans $\underline{\mathrm{Fibsc}}_0\, S$, on oublie le scindage et le fait même qu'il soit Cat-fibrant (on a appliqué p), on lui applique Φ_0, i.e. on regarde ΦCX comme objet de $\underline{\mathrm{Fibsc}}\, S$ (et même de $\underline{\mathrm{Fibsc}}_0\, S$). On veut définir encore [page 30]

$$X \xrightarrow{\beta_X} \Phi_0 p(X) = \Phi C(X) \qquad \textbf{dans } \underline{\mathrm{Fibsc}}_0\textbf{,}$$

fonctoriel en X. Or on a comme tantôt

$$X \xrightarrow{j_X} C(X),$$

d'où en appliquant Φ

$$\Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X),$$

mais regardé cette fois comme flèche dans $\underline{\mathrm{Fibsc}}_0\, S$ (N.B. on a bien $\Phi(X)$ dans $\underline{\mathrm{Fibsc}}_0\, S$, car X, provenant de $\underline{\mathrm{Fibsc}}_0\, S$, est dans $\underline{\mathrm{Fibhot}}\, S$, et on applique la remarque 'Cor. 3' page 21.) D'autre part, on a une flèche dans Cat/S

$$X \xrightarrow{i_X} \Phi(X),$$

est-elle une flèche dans $\underline{\mathrm{Fibsc}}_0(S)$, i.e. est-elle cartésienne? On voit tout de suite que non, que l'on a seulement, pour une flèche $u : s \longrightarrow t$ dans S, un homomorphisme de commutation (pas un isomorphisme) dans

p. 19

$$\begin{array}{ccc} X_s & \xrightarrow{u_*^X} & X_t \\ \downarrow{\scriptstyle i_s} & \overset{\alpha}{\nearrow} & \downarrow{\scriptstyle i_t} \\ X/s & \xrightarrow[=u_*^{\Phi(X)}]{(\mathrm{id}_X)/u} & X/t \end{array}$$

[page 31]

Donc il me faut remballer! On peut considérer i_X au mieux comme une flèche de $\underline{\mathrm{Fib}}_0\, S$, et en prenant les composés on trouve une flèche dans $\underline{\mathrm{Fib}}_0\, S$:

$$X \xrightarrow{i_X} \Phi(X) \xrightarrow{\Phi(j_X)} \Phi C(X),$$
$$\beta_X (= \alpha_X)$$

i.e. (regardant maintenant Φ_0 comme foncteur à valeurs dans $\underline{\mathrm{Fib}}_0\, S$) un homomorphisme de foncteurs

$$\beta : \mathrm{id}_{\underline{\mathrm{Fib}}_0} S \longrightarrow \Phi_0 p_0,$$

où $p_0 : \underline{\mathrm{Fib}}_0 S \hookrightarrow \mathrm{Cat}/S$ est l'inclusion canonique. Ce β est dans W argument par argument, et de même pour α page 29, interprété cette fois comme

$$\mathrm{id}_{\mathrm{Cat}/S} \longrightarrow p_0 \Phi_0.$$

Ce qu'on a prouvé finalement est ceci:

Commentary · Exposition [ms. 30]

Why the obvious unit fails to be Cartesian

Grothendieck has found a natural weak equivalence of total categories and now asks it to be a morphism of coherent diagrams. The request is stronger. For the fibration construction Φ(X) the fibre at s is s\X, whereas the obvious map sends x ∈Xs to (x, 1s). A Cartesian lift u∗y −→y in X, lying over u : s −→t, is sent to an arrow whose source is (u∗y, 1s). The chosen Cartesian lift in Φ(X) ending at (y, 1t) instead starts at (y, u). These two objects are connected by the lift in X, but in general they are not isomorphic in s\X.

The smallest example is X = S = [1] with the identity projection. Every arrow of X is Cartesian over S. In 0\[1] the two objects 10 : 0 −→0 and u : 0 −→1 are joined by a noninvertible arrow. Thus the image of the Cartesian arrow u under iX is not Cartesian in Φ(X). Indeed, in the Grothendieck construction of a strict diagram, an arrow over u is Cartesian exactly when its vertical comparison component is invertible. Here that comparison component is the displayed noninvertible arrow. The obstruction remains even though both comma categories are contractible.

The square printed across manuscript pages 30–31 uses X/s, X/t and covariant symbols u∗. Those are the symbols of the dual construction Ψ, while the surrounding discussion concerns Φ. The intended argument is correct with s\X, t\X and contravariant transport, or after dualizing the entire discussion. This variance slip should not be turned into a purported failure of the comma construction itself.

Passing to the full category Fib0S permits the map iX, because its morphisms are arbitrary functors over S. The natural weak equivalence X −→ΦC(X) then proves the coarse comparison in the "théorème-remballage". The theorem does not yet compare strict coherent diagrams by fibrewise weak equivalences. This is the exact boundary between what the argument has proved and what the next attempt must establish. The later Cartesian comparison is directed towards X, rather than away from it; its construction supplies the missing information about coherence.

Théorème-remballage. *Soit $\mathcal{F}$ une sous-catégorie pleine de* Cat/S *qui contient une des trois sous-catégories pleines* $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Parf}}\, S$, $\underline{\mathrm{Cofib}}_0 S$, *et munissant $\mathcal{F}$ du localiseur W 'grossier' (induit par celui de* Cat/S*). Alors l'inclusion*

$$\mathcal{F} \longrightarrow \mathrm{Cat}/S$$

induit une équivalence sur les localiseurs.

Par contre, les catégories localisées des quatre catégories de fibrés scindés [page 32]

$$\underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S, \quad \underline{\mathrm{Fibsc}}_0 S, \quad \underline{\mathrm{Cofibsc}}_0 S$$

échappent à nos arguments. *Et pour cause!* Il est totalement ouvert, sur S connexe disons (p. ex. $S = \mathbf{\Delta}^1$), que pour un type d'homotopie relatif ξ sur S, les types induits ξ_s ($s \in S$)

forment un système local sur S, à valeurs dans Hot_W, comme ce serait le cas si le théorème annoncé au début était vrai. Au contraire, *toute* flèche de Hot_W doit provenir d'un objet de $\mathrm{Hot}_W(\mathbf{\Delta}^1)$. Les localisés des 9 autres catégories du diagramme p. 8, et de toutes celles dont il est question dans le théorème qu'on vient de prouver, doivent correspondre 'visiblement' aux sous-catégories pleines de $\mathrm{HOT}_W(S)$ formées des HOT-coefficients localement constants sur S.

Aussi, je me rends compte que c'était absurde de m'attendre à ce que $\underline{\mathrm{Fibsc}}\, S$ et $\underline{\mathrm{Cofibsc}}\, S$ donnent des catégories localisées canoniquement équivalentes. Car les éléments ξ de la première des deux catégories donnent naissance chacun à un foncteur contravariant sur S p. 20 [page 33]

$$\bar{\xi} : S^o \longrightarrow \mathrm{Hot}_W,$$

ceux η de la seconde (qui n'est autre que $(\underline{\mathrm{Fibsc}}(S^o))^o$) donnent naissance chacun à un foncteur *covariant*

$$\bar{\eta} : S \longrightarrow \mathrm{Hot}_W,$$

ce qui n'est pas pareil! Ce n'est 'pareil' que si ce sont des coefficients locaux, ce qui signifie exactement que ξ est dans $\underline{\mathrm{Fibsc}}_0 S$ resp. η dans $\underline{\mathrm{Cofibsc}}_0 S$. On s'attend donc bien $\textcircled{1^o}$ que les flèches

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fib}}_0 S, \quad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofib}}_0 S$$

induisent des équivalences sur les localisés (chose que j'ai été incapable de prouver tantôt) ([16]). Par contre, il est pire que déraisonnable de s'attendre à ce qu'il en soit de même pour les foncteurs

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofibsc}}\, S,$$

ça devrait donner par localisation les foncteurs pleinement fidèles d'inclusion, des catégories d'objets localement constants dans $\mathrm{HOT}_W(S)$ resp. dans $(\mathrm{HOT}_W(S^o))^o$. Et de même pour les foncteurs induits sur les localisés par [page 34]

$$(*) \qquad \underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S \longrightarrow \underline{\mathrm{Cofib}}\, S.$$

Comme les localisés des catégories but de ces deux foncteurs ont tout l'air de s'identifier aux sous-catégories précédentes d'objets localement constants, il semblerait donc que : $\textcircled{2^o}$ Les foncteurs précédents définissent des rétractions de la catégorie $\mathrm{HOT}_W(S)$ resp. $\mathrm{HOT}_W(S^o)^o$ sur la sous-catégorie des objets localement constants sur S, les foncteurs localisés des foncteurs d'inclusion

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}_0 S \longrightarrow \underline{\mathrm{Cofibsc}}\, S$$

étant pleinement fidèles et ayant comme images essentielles les sous-catégories en question.

Commentary · Examples [ms. 32]

A relative type which is not locally constant

Grothendieck's reversal becomes transparent on the walking arrow. Let S = [1] = {0 −→1}, let X be the one-object category over 0, and let Y = S with its identity projection. The inclusion

f : {0} −→[1]

is a map of split fibrations over S. The corresponding contravariant diagrams are

X1 = ∅−→X0 = ∗,

Y1 = ∗−→Y0 = ∗. Both total categories are contractible, so f ∈W∞. But the fibre at 1 is ∅−→∗, which is not a weak homotopy equivalence. Fibre evaluation is a functor on the localization by objectwise weak equivalences, and sends isomorphisms there to isomorphisms of homotopy types. Therefore the image of f in that localization is not invertible. Since its image in the coarse localization is invertible, the comparison between these two localizations is not an equivalence.

All four tests of manuscript page 41 can be computed here. For the overcategories,

16. Les quasi-inverses étant bel et bien donnés par Φ_0, Ψ_0 - car [?] par quoi d'autre donc ???

X/0 = ∗,
Y/0 = ∗, X/1 = ∗, Y/1 = [1], so f is a local, or left-fine, weak equivalence. For the undercategories,
1\X = ∅,
0\X = ∗, 0\Y = [1], 1\Y = ∗. Thus it fails the colocal, or right-fine, test. Pulling back along the point 1 −→S gives the same failure, so it is not a universal weak equivalence. The example lies inside the category of split fibrations, hence also inside the category of smooth functors. Its opposite supplies the corresponding example for opfibrations and proper functors. This will also resolve the contradictory sentence on manuscript page 47.

An object equivalent to a locally constant diagram must have isomorphic fibre homotopy types along every base arrow: invertibility is invariant under isomorphism in the homotopy category of diagrams. The first diagram above does not have this property. Accordingly, the full subcategory of locally constant objects is proper even when the classifying space of the indexing category is contractible. Contractibility of the base makes local systems constant up to equivalence; it does not force every coherent diagram to be locally constant.

This gives a definite negative answer to the question on manuscript page 25 for the fibrewise localization of smooth or fibred objects. It also explains why the initial thirteen-category comparison could not have been correct with a single choice of weak equivalences. A derived category of diagrams must retain the noninvertible transition ∅−→∗, whereas the total homotopy type in this example forgets it.

Cette rétraction m'apparait comme une chose tout à fait mystérieuse. C'est lié au fait que pour le dérivateur très spécial $\mathbf{D} = \mathrm{HOT}$, quand on a un $\xi \in \mathbf{D}(S)$, on en déduit (à HOT/S-équivalence près) un type de diagramme X au dessus de S, qu'on peut regarder alors [page 35]
(via la construction C) comme définissant un type d'homotopie relatif *localement constant* sur S.

p. 21

Donc pour élucider la situation, i.e. les relations entre les 13 catégories localisées provenant du diagramme de la page 8, il reste à regarder si les assertions hypothétiques que je viens de faire sont bien valables, savoir

(1°) Le foncteur déduit de

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fib}}_0 S$$

par passage aux localisés est une équivalence de catégories (et on peut dans cette question remplacer $\underline{\mathrm{Fib}}_0 S$ par n'importe quelle autre sous-catégorie pleine de Cat/S la contenant - cela revient au même), et

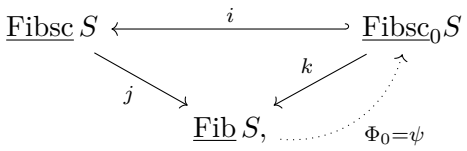
(2°) Le foncteur déduit de

$$\underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibsc}}\, S$$

par localisation est pleinement fidèle. (Que son image soit exactement formée des objets localement constants du premier localisé, interprété comme $\mathrm{HOT}_W(S)$, est tautologique.)

Il n'y a plus lieu de se soucier du côté cofibrés, puisque tout ce qu'on pourra en dire revient [page 36]
à ce que je viens d'expliciter, pour X^o au lieu de X.

Dans la direction de (1°), nos arguments précédents à coups de Φ_0, p nous donnent déjà un renseignement utile. Mais il sera plus commode à présent de laisser tomber $\underline{\mathrm{Fib}}_0 S$ ou plutôt, de le remplacer par $\underline{\mathrm{Fib}}\, S$ (ce qui est licite, on l'a dit dans (1°)), de sorte qu'on a maintenant un diagramme commutatif (traits pleins)



où i est une inclusion (strictement) pleinement fidèle, j (donc aussi k) fidèle. Enfin, on a un foncteur canonique ψ $(= \Phi_0)$ (en pointillés). Les localiseurs sur la ligne du haut sont les

localiseur fins (pour $\underline{\mathrm{Fibsc}}_0 S$ c'est pareil que la W-équivalence grossière), sur $\underline{\mathrm{Fib}}\, S$ par contre on prend la W-équivalence grossière. (On sait que le localisé est le même, à équivalence près, que si on se restreint à la sous-catégorie pleine $\underline{\mathrm{Fib}}_0 S$, ce qui est assez remarquable.

Tous les quatre foncteurs i, j, k, Φ_0 sont compatibles aux localiseurs, donc passent aux localisés [page 37] p. 22

$$\begin{array}{ccc} \mathrm{HOT}_W(S) & \xleftarrow{\quad\bar{\imath}\quad} & \mathrm{HOT}^?_W(S) \\ & {\scriptstyle\bar{j}}\searrow \quad \swarrow{\scriptstyle\bar{k}} \quad \uparrow{\scriptstyle\bar{\psi}} & \\ & \mathrm{HOT}^{\mathrm{lc}}_W(S), & \end{array}$$

où l'exposant ? dans la notation signifie que je ne sais interpréter cette catégorie, sauf si je prouve que $\bar{k}$ est une équivalence, et l'exposant lc dans la catégorie du bas dois suggerer l'intuition qu'il s'agit en tous cas d'une sorte de 'types d'homotopie relatifs' (dans un sens modifié) *localement constants* sur S. Le problème est de comprendre ce diagramme. La conjecture naturelle est

1°) $\bar{k}$ est une équivalence. (Ce qui implique que $\bar{\psi}$ est l'équivalence quasi-inverse, que $\bar{k}$, $\bar{\psi}$ permettent essentiellement d'identifier $\mathrm{HOT}^?_W$ et $\mathrm{HOT}^{\mathrm{lc}}_W$, de sorte que $\bar{\imath}$ et $\bar{\jmath}$ deviennent des foncteurs entre HOT_W et $\mathrm{HOT}^{\mathrm{lc}}_W$.)

2°) $\bar{\imath}$ est pleinement fidèle (NB l'image essentielle est bien comprise de toutes façons.)

En direction de 1°), je sais pour l'instant que l'on a [page 38]

$$\bar{k}\bar{\psi} \simeq \mathrm{id}_{\mathrm{HOT}^{\mathrm{lc}}_W(S)},$$

donc $\bar{k}$ est essentiellement surjectif (sur les objets), et induit des *surjections* sur les $\mathrm{Hom}(-,-)$, tandis que $\bar{\psi}$ est injectif sur les classes d'isomorphismes d'objets, et sur les $\mathrm{Hom}(-,-)$ (i.e. *fidèle*).

Si les relations entre $\mathrm{HOT}^?_W$ et $\mathrm{HOT}^{\mathrm{lc}}_W$ s'avèrent imbitables (p. ex. $\bar{\psi}$ pas même pleinement fidèle, i.e. $\bar{k}$ pas même fidèle), alors la catégorie $\mathrm{HOT}^?_W$ mérite d'être larguée dans la corbeille, et on ne gardera que

$$\begin{array}{c} \mathrm{HOT}_W(S) \\ {\scriptstyle\bar{\jmath}}\downarrow \quad \uparrow{\scriptstyle\bar{\psi}'} \\ \mathrm{HOT}^{\mathrm{lc}}_W(S), \end{array}$$

où on définit $\bar{\psi}'$ comme

$$\bar{\psi}' = \bar{\imath}\bar{\psi} = \overline{i\psi}.$$

Et la question (plus modeste) devient à présent :

3°) $\bar{\psi}'$ est-il pleinement fidèle? (L'image *essentielle* est la même que celle de $\underline{\mathrm{Fib}}_0$ ($\approx \underline{\mathrm{Fib}}$), elle est formée sauf erreur des objets localement constants - à vérifier au besoin …)

Il est difficile d'imaginer que $\bar{k}$, ou ce qui revient au même $\bar{\psi}$, puisse être une équivalence, sans qu'il existe au moins une 'W_S-homotopie' p. 23 [page 39]

$$\mathrm{id}_{\underline{\mathrm{Fibsc}}_0 S} \;\leftrightarrow\leftrightarrow^h\leftrightarrow\leftrightarrow\; \psi k$$

dans la catégorie des endofoncteurs de $\underline{\mathrm{Fibsc}}_0 S$. À vrai dire, je me rends compte maintenant que la description de $\bar{\psi}$ à partir de $\psi : X \longmapsto \Phi_0 X = \Phi C X$ est inutilement compliquée, car on avait fait le détour par $C(X)$ uniquement pour être sûr de tomber dans $\underline{\mathrm{Fibsc}}_0 S$, pas seulement dans $\underline{\mathrm{Fibsc}}\, S$. Mais si on laisse tomber $\underline{\mathrm{Fibsc}}_0 S$, on s'en fout, et le foncteur $\Phi : X \longmapsto \Phi(X)$ est beaucoup plus naturel. Et si on ne laisse pas tomber, il n'y a pas d'inconvénient de remplacer

la source $\underline{\mathrm{Fib}}\, S$ par $\underline{\mathrm{Fib}}_0 S$ (qui a même localisé, à équivalence près), et sur $\underline{\mathrm{Fib}}_0 S$ à nouveau il n'y a pas lieu de s'encombrer de CX, pour le plaisir de lui appliquer Φ, car $\Phi(X)$ lui-même est déjà dans $\underline{\mathrm{Fibsc}}_0 S$.

Commentary · Examples [ms. 38]

What a one-sided inverse does and does not give

Grothendieck has a useful piece of the comparison: a natural isomorphism $\mathrm{k}^- \psi^- \simeq 1$. It proves essential surjectivity of k^- and faithfulness of ψ^-, and it shows that ψ^- is injective on isomorphism classes of objects. The statement about surjectivity on all morphism sets requires an additional hypothesis.

To locate the issue, let A be the discrete category on two objects a0, a1, and let B = ∗be terminal. There is a unique functor k : A −→B; let ψ pick a0. Then kψ = 1B strictly. Yet

HomA(a0, a1) = ∅−→HomB(∗, ∗) = {1}

is not surjective. Thus a functor with a section need not be full. In this example ψ is even fully faithful, so the parenthetical suggestion that failure of faithfulness of k is equivalent to failure of full faithfulness of ψ cannot be justified solely from the retract relation. For that converse take instead A = B $\coprod$BM, with B = ∗and a nontrivial monoid M, let k be the unique functor to ∗and let ψ select the first summand. Then ψ is fully faithful while k is not faithful.

The precise positive statement is easy and sufficient for the progress claimed so far. Choose −→1. For b, b′ ∈B and h : b −→b′, the morphism ψ(h) maps under k, followed by conjugation ε : kψ with ε, to h. Hence the map

HomA(ψb, ψb′) −→HomB(b, b′)

is split surjective. If ψ(h) = ψ(h′), applying k and conjugating gives h = h′, proving faithfulness. If ψb ≃ψb′, the same argument gives b ≃b′. None of these statements identifies an arbitrary a ∈A with ψka.

A natural zigzag of weak equivalences between 1 and ψk, sought on the following pages, would supply exactly that missing identification after localization. The full comparison theorem can therefore remain correct although the present formal inference is too strong. We record an error in this intermediate inference, with its corrected scope, rather than a gap in the later comparison once that zigzag has been constructed.

Donc, la question est celle-ci : Si X varie [page 40]

dans $\underline{\mathrm{Fibsc}}_0 S$, y-a-t'il une homotopie dans $\underline{\mathrm{Fibsc}}_0 S$, 'naturelle', i.e. fonctorielle en X,

$$X \xrightarrow{\quad c \quad} \Phi(X) = \text{ Catégorie fibrée scindée de fibres les } s\backslash X$$

Mais un chemin qui soit formé de flèches

$$X = X_0 \longrightarrow X_1 \longrightarrow X_2 \cdots \longrightarrow X_n = \Phi(X)$$

qui soient toutes cartésiennes (et de plus, bien sûr, dans W!). Cela signifie donc qu'on aurait des catégories fibrées scindées sur S, X_1, …, X_{n-1}, dépendant *fonctoriellement* de X, qui relient

$$X = X_0 \qquad \text{à} \qquad \Phi(X) = X_n$$

i.e.

$$X_s \qquad \text{à} \qquad s\backslash X \qquad ???$$

Je ne vois rien à l'horizon.

p. 24
[page 41]

Commentary · Present state [ms. 40]

The locally constant comparison is now a theorem

Grothendieck is asking for morphisms that retain coherent transport while forgetting arbitrary choices of Cartesian lifts. His unsuccessful unit is not evidence that the locally constant subcategory fails to embed fully. It identifies a defect of that particular representative of the comparison. The

arguments on manuscript pages 55–60 repair the comparison, and pages 65–69 establish the relevant reflection.

The higher coherence retained by local constancy was illustrated at manuscript page 7. The later theory treats the local constancy question for a general derivator. Cisinski (2009, Proposition 3.10 and Theorem 3.13) proves that locally constant objects form a full subderivator and that its inclusion has both a left and a right adjoint. These are the published numbering; the earlier arXiv version numbers the corresponding statements 3.9 and 3.12. The theorem establishes the positive status of the full embedding and the existence of the two adjoints, without identifying all the coarse localizations of this chapter for an arbitrary W.

That last restriction matters. If W is homological or a finite homotopy truncation, a coarse Wequivalence of total spaces need not be a W-equivalence on homotopy fibres. The local constancy construction in a derivator still makes sense, but the additional identification with the coarse localization of Cat/S uses the stronger hypothesis isolated below. For W∞the categorical path model and the modern description by coherent space-valued diagrams fit together; the commentary on manuscript pages 65–69 gives the precise reflector. This question is therefore settled in its intended homotopy-theoretic setting, while its unqualified extension to every localizer is false.

15.11.90

Je crois qu j'ai enfin compris, cette nuit, la situation exacte. Il y a lieu, dans Cat/S et dans les autres catégories du diagramme p. 8 (s'envoyant toutes dans Cat/S) d'envisager quatre sortes de 'W-équivalences'.

$$\begin{array}{ccc} f : X & \longrightarrow & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

a) La *W-équivalence 'grossière'* : dans Cat/S est une W-équivalence grossière si en tant que flèche de Cat, c'est une W-équivalence. ([17]).

b) La *W-équivalence 'par fibres'* (sur S) : $\forall\ s \in S$, $f_s : X_s \longrightarrow Y_s$ est une W-équivalence. ([18]).

c) La *W-équivalence 'locale sur S'* : $\forall\ s \in S$, $f/s : X/s \longrightarrow Y/s$ est une W-équivalence. Signifie que c'est une W-équivalence, et le reste par tout changement de base $S' \longrightarrow S$ qui est un isomorphisme local. ([19]).

c′) La *W-équivalence 'colocale sur S'* : $\forall\ s \in S$, $s\backslash f : s\backslash X \longrightarrow s\backslash Y$ est une W-équivalence. Signifie que c'est une W-équivalence, et le reste par tout changement de base $S' \longrightarrow S$ qui est un co-isomorphisme local. ([20]).

Je vais résumer dans un énoncé récapitulatif les relations entre ces quatre notions de W-équivalence, en pensant au cas $W = W_\infty$. Une fois la chose faite, je me préoccuperai de dégager les conditions minimales sur un W général qui assurent que c'est vrai pour W.

$$\begin{array}{ccc} f : X & \longrightarrow & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

Proposition 1. *Soit* *dans* Cat/S.

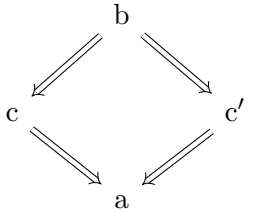
1) *La W-équivalence locale sur S, et la W-équivalence colocale sur S, impliquent la W-équivalence grossière :*

$$\begin{array}{ccc} \mathrm{c} & & \mathrm{c}' \\ & \searrow \quad \swarrow & \\ & \mathrm{a} & \end{array}$$

p. 25

17. stable par changement de base à la fois lisse et propre, plus généralement si $S' \longrightarrow S$ est une Hot-fibration.
18. stable par tout chagement de base $S' \longrightarrow S$
19. devrait être stable par changement de base *lisse* $S' \longrightarrow S$
20. devrait être stable par changement de base *propre* sur S

2) *Si X, Y sont propres sur S, ou X, Y lisses sur S, ou X, Y des* Hot-*fibrés sur S, alors la W-équivalence par fibres implique que c'est une W-équivalence* universelle *sur S (W-équivalence, et le reste par* tout *changement de base $S' \longrightarrow S$), donc dans ces trois cas, on a*



([21]).

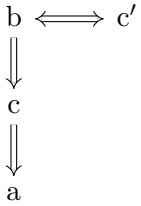
Mais on peut préciser chacun de ces trois cas :

(2 fib) *Si X, Y sont* Hot-*fibrés sur S, alors la W-équivalence grossière de f implique que c'est une W-équivalence par fibres, donc les quatre notions de W-équivalence sont équivalentes pour f.*

$$a \Longleftrightarrow c \Longleftrightarrow c' \Longleftrightarrow b$$

([22])

(2 lisse) *Si X, Y sont lisse sur S, alors la W-équivalence par fibres sur S équivaut à la W-équivalence colocale sur S et implique la W-équivalence* universelle *sur S, donc dans ce cas* [page 42]



Commentary · Exposition [ms. 41]

Four tests on a map over the base

Grothendieck now separates the total homotopy type from the two possible diagram directions. For p : X −→S there are three kinds of fibre to consider. The strict fibre Xs remembers only objects and arrows mapping to s and 1s. The comma category X/s also allows arrows p(x) −→s in the base; it varies covariantly with s. The comma category s\X allows arrows s −→p(x) and varies contravariantly. Thus for f : X −→Y over S the tests are

total

f ∈W,

fibrewise fs ∈W (s ∈S),

local, or left-fine f/s ∈W (s ∈S),

colocal, or right-fine s\f ∈W (s ∈S). The word "local" here refers to these particular comma tests, not to a chosen Grothendieck topology. The names "left" and "right" are the manuscript's conventions. They must not be interchanged merely because the fibrations are drawn on the left side of the thirteen-category diagram.

For the ordinary weak homotopy equivalences, the role of the comma tests is to detect the maps of coherent diagrams produced by Ψ and Φ. A map can preserve the total homotopy type while changing one of those diagrams, as the walking-arrow example at manuscript page 32 proves. Conversely, strict fibres alone do not detect a total equivalence without some condition on the maps to S. For example, the functor from the discrete two-object category {0, 1} to [1], over [1], is an isomorphism on both strict fibres but changes the total space from

21. Il suffit X, Y W-propres sur S, ou W-lisses sur S, ou W-fibrés sur S.

22. a $\Longrightarrow$ b seulement moyennant L5, voir ci-dessous p. 43.

two components to a contractible space. It also fails both comma tests: over 1, and under 0, respectively, it compares a discrete two-point category to a contractible one.

The assertion concerning arbitrary base change of a fibrewise equivalence is formal: the fibre of a pullback over s′ ∈S′ is the original fibre over its image in S. The assertion concerning base change of a coarse equivalence along every homotopy fibration is not formal for a general W; it is exactly the additional stability later called L6 bis. Footnote 17 is valid for W∞, the setting explicitly adopted for the proposition, and must not be read as a theorem about every localizer.

The local-isomorphism formulation has an elementary explanation in the usual categorical convention of discrete fibrations. If T −→S is a discrete fibration, then

(X ×S T)/t ∼= X/p(t).

An object on the right determines the object of T above its source by the unique lifting property. Thus local equivalence is preserved by this pullback, and relative axiom L4 bis gives total equivalence afterward. Conversely the representable discrete fibrations S/s −→S recover the comma tests themselves. Dualizing gives the colocal statement. This explains both the strength and the orientation of the local tests without equating them to arbitrary base-change invariance.

p. 26

([23]).

(2 prop) *Si X, Y sont propres sur S, alors la W-équivalence par fibres sur S équivaut à la W-équivalence locale sur S :*

$$\begin{array}{ccc} \mathrm{b} & \Longleftrightarrow & \mathrm{c} \\ \Downarrow & & \\ \mathrm{c}' & & \\ \Downarrow & & \\ \mathrm{a} & & \end{array}$$

([24]).

Démonstration 1) n'est autre que L4 ([25]) et son dual L4′ pour un localiseur fondamental W, et on a vu que si on suppose L1, L3, alors L4 équivaut à L4′, et dans la notion de localiseur fondamental on supposera toujours que c'est vérifié. Également, nous supposons que L3 est vrai même sous la forme plus forte L3 quater (p. 1) : $W_\infty \subseteq W$, i.e. toute Hot-équivalence est une W-équivalence.

(2 lisse) Si X, Y `[sont]` lisses sur S, alors les $X_s \longrightarrow s\backslash X$, $Y_s \longrightarrow s\backslash Y$ sont des W_∞-équivalences, donc des W-équivalences (il suffirait même que X, Y soient W-lisses/S), donc le diagramme commutatif

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ {\scriptstyle i_s^X}\downarrow & & \downarrow{\scriptstyle i_s^Y} \\ s\backslash X & \xrightarrow{s\backslash f} & s\backslash Y \end{array}$$

nous montre que f_s est une W-équivalence si et seulement si $s\backslash f$ l'est, d'où b $\Longleftrightarrow$ c′. Pour voir que la W-équivalence par fibres est même une W-équivalence *universelle* sur S, on est ramené à montrer que c'est une W-équivalence grossière (puisque la lissité `[est]` stable par changement de base sur S), et on gagne puisque c′ $\Longrightarrow$ a par 1).

Le cas (2 propre) est dual.

(2 fib) Si X, Y sont des Hot-fibrés sur S, alors les inclusions p. 27

[page 43]

$$X_s \longrightarrow s\backslash X, \qquad X_s \longrightarrow X/s,$$

23. il suffit même X, Y W-lisses sur S.
24. Il suffit même X, Y W-propres sur S.
25. Non, il y a confusion, L4 est plus faible, cf. p. 44 l'axiome L4 bis.

et de même pour Y, sont des W_∞-équivalences, donc des W-équivalences (il suffirait même que X, Y soient des W-fibrations sur S ([26]), i.e. par définition que les foncteurs précédents soient des W-équivalences, et que la même condition soit satisfaite après tout changement de base sur S), l'argument dans le cas lisse resp. propre montre déjà que

$$\begin{array}{ccccc} \mathrm{c} & \Longleftarrow & \mathrm{b} & \Longrightarrow & \mathrm{c}' \\ & & \Downarrow & & \\ & & \mathrm{a}, & & \end{array}$$

et que la W-équivalence par fibres (ou la W-équivalence locale sur S, resp. colocale sur S) implique déjà la W-équivalence *universelle* sur S. ([27])

Commentary · Exposition [ms. 42]

When strict fibres compute a comma fibre

Grothendieck's proposition is useful because it states exactly when a strict fibre is a valid homotopical substitute for one of the comma categories. The discussion of smoothness and properness in the commentary on Chapter I, manuscript pages 52–59, supplies the geometric input; here we use that input to compare weak equivalences of objects over the same base.

If X −→S is smooth, the canonical inclusion Xs −→s\X is a weak homotopy equivalence. When both X and Y are smooth, the commutative square

Xs

Ys

s\X

s\Y has vertical arrows in W if W∞⊆W. Two-out-of-three proves the equivalence of the fibrewise and right-fine tests. For proper objects the same argument uses Xs −→X/s and proves equivalence with the left-fine test. The proof requires no comparison of total spaces at this stage. The same argument works with W-smooth or W-proper objects when the vertical maps are directly known to be in W.

To obtain universality, make any change of base T −→S. Smoothness and properness are preserved, and the fibrewise maps remain in W. The preceding comparison gives a colocal or local W-equivalence over T, respectively. Relative axiom L4 bis, or its dual, now makes the total map over T a W-equivalence. The base change was arbitrary, so the map is universally a W-equivalence. This fills the quantifier in the proof; it is not a new appeal to L6. Here base change acts on a map already known to be a fibrewise equivalence between suitably regular objects.

For homotopy fibrations both comma comparisons are weak homotopy equivalences, universally on the base. Hence the fibrewise, local and colocal conditions coincide and imply universal total equivalence. The reverse implication from a total equivalence is the extra statement later named L5. It holds for W∞ by invariance of homotopy fibres, and the next examples show why it cannot be appended for every W.

The modern extension to asphericity structures is given by Maltsiniotis (2005, Corollary 4.16 and Theorem 5.4): Grothendieck fibrations are smooth, and smoothness is characterized by the appropriate base-change property. This is the direct continuation of the present comparison argument. In highercategorical straightening one likewise distinguishes equivalence in the diagram direction from equivalence of underlying total spaces; the latter agrees with the former for locally constant fibrations, not for every right or left fibration.

La question a $\Longrightarrow$ b est par contre nettement plus délicate, et on a vu que ce n'est nullement le cas pour tout localisateur fondamental - sans doute pas pour celui associé au dérivateur Der($\mathbf{Z}$-Mod), en tous cas pas à celui associé à Der(k-Mod), quand 2 est inversible dans k. (Il suffirait même que $\exists\, n \in \mathbf{N}$, $n \geq 2$, tel que n inversible dans k - le contrexemple typique est $X = S^1 \xrightarrow{n} S^1 = Y = S$, le revêtement étale d'ordre n du cercle.)

L'implication a $\Longrightarrow$ b est donc une condition très forte sur W, je vais la noter L5 (l'ancienne L5, cf. page 2, ne méritant pas de survivre, car c'est une côte mal taillée).

26. comparer déf. 3, p. 117, dans (I).

27. valable en supposant seulement X, Y des W-fibrés sur S.

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

L5 *Soit* *dans* Cat*, avec X, Y des* Hot-*fibrés sur S. Si f est une W-équivalence ('grossière' par rapport à S), alors c'est une W-équivalence par fibres* ([28]).

On a vu qu'à un ε près ('lemme' p. 23), L5 est une conséquence de

L6 *Soit $f : X \longrightarrow Y$ une W-équivalence, $Y' \longrightarrow Y$ un morphisme parfait (i.e. lisse et propre), alors $f' : X' = X \times_Y Y' \longrightarrow Y'$ est une W-équivalence.*

Notons la variante bien naturelle [page 44]

Commentary · Examples [ms. 43]

The circle covering as a categorical counterexample

Grothendieck's covering of the circle gives a complete obstruction to L5 for homology coefficients in which its degree is invertible. It is useful to realize it inside Cat, so that no passage from topological coverings to categorical fibres is left implicit.

Fix n ≥2. Let S = BZ, the one-object category with endomorphism group Z. Let X be the action groupoid for translation of Z on Z/n. Its objects are residues a, and a morphism a −→b is an integer m with b = a + m modulo n; composition is addition. The projection p : X −→S sends that morphism to m. Unique lifting of each integer with specified source or target makes p both a discrete fibration and a discrete opfibration. Its strict fibre is the discrete set Z/n.

The groupoid X is connected. The automorphism group of the object 0 is nZ, so the inclusion B(nZ) −→X is an equivalence. Under this identification, p is the homomorphism nZ ,→Z. Thus both classifying spaces are circles, and the induced map on H1(−; k) is multiplication by n after choosing the usual generators. It is the identity on H0 and both sides have zero homology in higher degrees. If k is a nonzero commutative ring in which n is invertible, p is a k-homology equivalence. Regard it now as the map from X −→S to S −→1S. On strict fibres it induces

H0(−; k) : kn −→k.

Z/n −→∗, The latter map adds coordinates and has nonzero kernel. Hence it is not a fibrewise homology equivalence, although both objects over S are perfect.

The same example disproves L6 itself, not only the fibre-detection formulation. Let E be the translation groupoid of Z acting on itself, and let E −→BZ be the evident discrete fibration and opfibration. There is exactly one arrow between each ordered pair of objects of E, so E is contractible. The objects of X ×S E are pairs (a, z); simultaneous translation preserves a −z modulo n. Each of the resulting n components has exactly one arrow between any two of its objects and is contractible. Therefore

X ×S E −→E

induces kn −→k on H0 and fails to be a homology equivalence. The base-change map is perfect, as required by L6. The argument explains why replacing "homotopy fibration" by "perfect" cannot repair the homological version.

Commentary · Development (commentators') [ms. 43]

An integral-coefficient version of the obstruction

Grothendieck also suspects failure for integral homology, where a nontrivial finite covering of the circle no longer gives a homology equivalence. The following explicit construction settles that case. Its ingredients are standard: a semidirect product and the mapping-torus homology sequence. We checked the discussion of homological local constancy in Cisinski (2009, Scholium 1.26) and the mapping-torus calculation in Hatcher (2002, Example 2.48); we did not find this particular categorical application to the sentence here in those sources. The marking records the application and proof supplied by us, not a priority claim for the group or its homology.

Proposition (integral homology does not detect the fibre). [ours]There is a perfect functor p : X −→S which induces an isomorphism on integral homology in every degree, while its strict fibre has nonzero first integral homology. Consequently the localizer of integral homology equivalences fails L5.

28. Dans XII (p. 12) on voit que si W satisfait à L1, L3 quater, L4 bis (cf. page suivante), L5, alors $\boxed{W = W_\infty}$. Donc c'est parfaitement illusoire, si on suppose L5 de continuer à traîner un W.

Proof. Put A = Z[1/2], let h : A −→A be multiplication by 2, and put G = A ⋊h Z. Take X = BG, S = BZ and let p be induced by the quotient G ↠Z. Since this homomorphism is surjective, every base arrow admits lifts with either prescribed source or prescribed target; all arrows in these groupoids are invertible and hence Cartesian and cocartesian. Thus p is both a Grothendieck fibration and an opfibration, and is perfect. Its strict fibre is BA.

We compute the fibre's homology first. The group A is the filtered union of the cyclic subgroups 2−rZ. The nerve, its free simplicial chain complex and homology all commute with this filtered union: each finite tuple of group elements occurs at a finite stage, and filtered colimits of abelian groups are exact. Since BZ has the homotopy type of a circle, this gives

(i ≥2).

H0(BA; Z) = Z, H1(BA; Z) = A, Hi(BA; Z) = 0 The map induced by h acts by 2 on H1 and by the identity on H0.

The group automorphism h induces a homeomorphism of the realization of the nerve of A. Its mapping torus Th is a bundle over the circle. Its fundamental group is A ⋊h Z after choosing the orientation of the circle, and its higher homotopy groups vanish by the homotopy sequence, since the fibre is K(A, 1). Hence Th is a classifying space for G, compatibly with the projection to BZ. The mapping-torus homology sequence has the terms

· · · −→Hi(BA)

−−−→Hi(BA) −→Hi(Th) −→Hi−1(BA) −→ · · · . In degree 1, 1 −h∗= −1 on A is invertible; in degree 0 it is zero on Z. Exactness therefore gives Hi(Th) = 0 for i ≥2 and an isomorphism H1(Th) −→Z. Naturality with respect to the map of fibres BA −→∗identifies this last isomorphism with the homology map induced by the projection to the circle. Both spaces are connected, so the projection is also an isomorphism on H0. This proves that p is an integral homology equivalence. Its map of fibres to the fibre of 1S is BA −→∗, which fails on H1 = A = 0. □

The example distinguishes absolute homology of the total category from homology after changing the base. The monodromy multiplies the fibre's homology by 2, and 1 −2 is invertible; its contribution disappears from the total homology without disappearing from the fibre. Thus the obstruction is present even with integral coefficients, independently of inverting the degree of a finite covering.

L6 bis *Comme L6, mais en supposant seulement $Y' \longrightarrow Y$ une* Hot-*fibration.*

Il y a des chances que pratiquement, quand L6 (ou seulement L5) est verifié, L6 bis l'est aussi. p. 28

On va introduire un axiome

L7 a) *Soit $(f_i : X_i \longrightarrow Y_i)_{i\in I}$ une famille de flèches dans* Cat*, soit $f : \coprod X_i \longrightarrow \coprod Y_i$ la flèche correspondante sur les catégories somme. Pour que $f \in W$, il faut et il suffit que $f_i \in W$ pour tout $i \in I$.*

b) *Soit $f : X \longrightarrow Y$ dans W. Si Y est* 0*-connexe, X aussi.*

Cet axiome équivaut à la conjonction de La ($f \in W \Longrightarrow \pi_0 f$ bijectif) et de la validité du 'Lemme' page 23, que je remets ici pour mémoire :

Lemme. *L'axiome L7 précédent sur le localiseur fondamental W équivaut à la condition suivante : $f : X \longrightarrow Y$ est dans W si et seulement si $\pi_0(f)$ est bijectif, et f induit des W-équivalences $f_i : X_i \longrightarrow Y_i$ sur les composantes connexes.*

Il me faut revenir encore sur le 1) de la proposition, il n'est pas vrai que 'c'est' la condition L4 sur W (que j'admets, ai-je dit), c'est L4 dans le cas particulier où $Y = S$. Réflexion faite, je le vois comme une variante renforcée de L4, qui ne semble pas devoir découler de L4 :

$$\begin{array}{ccc} X & \xrightarrow{\ f\ } & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

L4 bis *Soit* *dans* Cat*, supposons que f soit une W-équivalence locale sur S (i.e. $f/s : X/s \longrightarrow Y/s$ une W-équivalence $\forall\ s \in S$), alors f est une W-équivalence* ([29]).

Par la suite, par un 'localiseur fondamental', nous entendrons un ensemble [page 45]

$$W \subseteq \mathrm{Fl}\,\mathrm{Cat}$$

29. équivaut à l'axiome dual, en termes des $s\backslash f$

satisfaisant (tout au moins) les axioms

L1, L3 quater, L4 bis, L7,

i.e.

Commentary · Exposition [ms. 44]

Componentwise locality and relative Theorem A

Grothendieck separates two axiomatic matters here. The first concerns disjoint components; the second concerns descent over a base that need not be the target of the map being tested. Keeping them separate clarifies which part of the preceding argument each axiom repairs.

The component criterion is exactly equivalent to L7. To prove the nontrivial direction, let f : X −→Y belong to W and decompose Y = ‘ j Yj into its nonempty connected components. Put Xj = f −1(Yj), a union of components of X. Then f = ‘ j fj, so L7(a) implies fj ∈W for every j. L7(b) implies that every Xj is nonempty and connected. Hence $\pi 0$(f) is bijective, and the fj are precisely the component maps. Conversely, if $\pi 0$(f) is bijective and every component map belongs to W, L7(a) gives f ∈W. The empty target creates no exception: a functor into the empty category has empty source.

To recover L7 from the criterion, consider a family (fi). The component map of ‘ i fi is just a component map of one fi; moreover its map on components is bijective exactly when each $\pi 0$(fi) is bijective. The criterion therefore gives both directions of L7(a). If the target of a W-equivalence is connected, its component bijection makes the source connected, giving L7(b). This is a complete proof of the lemma, including the reflection direction in the coproduct assertion.

For the second matter, the original L4 applies to a map X −→Y when the overcategories of its fibres over objects of Y are aspherical. In Proposition 1, however, one compares X/s and Y/s for an independent base S. These overcategories need not be aspherical; only the map between them is assumed to be a W-equivalence. Thus the printed sentence identifying the proposition directly with the original L4 does not establish it. L4 bis is exactly the required relative formulation. The dual condition uses s\X −→s\Y .

The accepted modern definition of a fundamental localizer includes the relative formulation; see Cisinski (2004, §1.1). Its minimal member is W∞by that paper's Theorem 2.2.11. This gives a precise later framework for the corrected argument. We make no claim here that the absolute and relative formulations are logically independent under every possible package of the earlier axioms; what the text has exhibited is the need for the relative assertion in this proof. Opposite-category invariance then transfers the relative assertion to its dual, using the category-of-simplices comparison discussed earlier in the chapter.

a) *Saturation* de W (conditions a) b) c) p. 1)

b) $W_\infty \subseteq W$ [$\subseteq W_0$] ([30]) ([31]).

c) Si $f : X \longrightarrow Y$ sur S est localement sur S une W-équivalence, alors $f \in W$ (L4 bis)

d) Pour que $f : X \longrightarrow Y$ soit dans W, il faut et il suffit que $\pi_0(f)$ soit bijectif, et que les $f_i : X_i \longrightarrow Y_i$ induits sur les composantes connexes soient dans W. p. 29

Ces axiomes sont vérifiés par tous les W qui je connais, et qu' on aurait envie de regarder comme localiseurs. Par contre, L5, L6, L6 bis (p. 43, 44) sont des axiomes de nature beaucoup plus spéciale. J'ai envie de dire qu'ils distinguent les localiseurs de nature proprement *'homotopique'* de ceux qui sont de nature *'homologique'* (ou 'cohomologique'). Il faudrait que je m'assure qu'ils sont satisfaits pour les W_i (i-équivalence d'homotopie) ([32]).

Le fait que L4 bis, i.e. c) ci-dessus, implique l'axiome dual, résulte du fait que L4 bis implique ceci

30. Je ne suis plus sûr s'il est vraiment utile d'imposer $W_\infty \subseteq W$. Prendre au lieu de ça l'axiome de l'objet final, soit b′). Si on veut a) b′) c) + L5, alors on aura $W \subseteq W_\infty$. **NB** a) b) c) + L5 implique $W = W_\infty$, cf. annotation page 43.

31. cf. d)

32. C'est *faux*, cf. annotation ci-dessus.

Proposition 2. *Soit $f : X \longrightarrow Y$ dans* Cat*, alors $f \in W$ si et seulement si $f^o : X^o \longrightarrow Y^o$ est dans W.*

Se voit par l'argument de I, page 75 (cor. 2) - n'utilise que L4, pas même L1, L3.

[page 46]

Commentary · Present state [ms. 45]

Why the new fibration axiom selects full homotopy theory

Grothendieck's second L5 is much stronger than the first axiom with that label. The first passed from equivalences of fibres to an equivalence of totals. The new one attempts the reverse implication for homotopy fibrations. The annotations correctly recognize that this is incompatible with the proposed range of examples, but the displayed uniqueness claim needs the standing connectedness condition W ⊆W0. Without it the maximal class of all functors satisfies L1, L3 quater, L4 bis and L5, so those four conditions alone do not force W = W∞.

Here is the mechanism of the uniqueness assertion, retaining W∞⊆W ⊆W0, two-out-of-three, the new L5, and the path replacement already available in the chapter. Suppose first that T −→∗ belongs to W. Its classifying space is connected by W ⊆W0. For a chosen object t form the perfect replacement P = CT (∗) −→T of the inclusion of that object. Its total category is weakly contractible. Two-out-of-three implies P −→T ∈W. Applying L5 to this map between the homotopy fibrations P −→T and T −→1T shows that its fibre F −→∗belongs to W. The fibre models the based loop space of BT. Repeating the argument with F and successive fibres gives categorical models for all iterated loop spaces, each W-aspherical and hence connected. Therefore

πn(BT, t) = π0(ΩnBT) = 0

(n ≥1). Together with connectedness this says that T is weakly contractible.

Now let f : X −→Y belong to W. Replace it by the perfect map CY (X) −→Y . Its total map is still in W, and L5 makes every strict fibre W-aspherical. What we have just proved makes those fibres weakly contractible. Since this is a homotopy fibration, its homotopy fibres are weakly contractible, so its total map is a weak homotopy equivalence; the replacement X −→CY (X) is also one. Thus f ∈W∞. This proves W = W∞. The argument uses the previously established homotopy-fibre property of the path replacement, not a new assertion that arbitrary strict fibres compute homotopy fibres. The later abstract characterization in Cisinski (2004, Theorem 2.3.6) likewise includes preservation of connected components.

One can see the failure for every finite truncation directly. Let W≤n mean maps inducing isomorphisms on πi for 0 ≤i ≤n, with all basepoints understood. Choose a nonzero abelian group A and a categorical model S of K(A, n + 1). The perfect replacement P = CS(∗) −→S has contractible total space, so belongs to W≤n, while its fibre models K(A, n) and is not W≤n-aspherical. For n = 0 that fibre is the nontrivial discrete group A. This disproves L5. Pulling P −→S back along itself yields a total space weakly equivalent to ΩBS, because the base P is contractible and the map is a homotopy fibration. The projection P ×S P −→P therefore fails to belong to W≤n, disproving L6 even for this perfect base change. The finite truncations are homotopical theories, but they do not satisfy the unrestricted fibre-detection requirement of this L5.

Je vais essayer de résumer ce que je crois avoir compris :

1) La W-équivalence 'par fibres' n'a d'intérêt (pour construire des catégories localisées) que dans le cas où on sait qu'elle implique déjà, soit la W-équivalence locale, soit la W-équivalence colocale (soit même la W-équivalence universelle) sur S, donc quand on suppose X, Y pour le moins tous deux soit W-lisses, soit W-propres, soit W-fibrés sur S.
2) On pourrait appeler la W-équivalence locale sur S la *'W-équivalence fine à gauche'* sur S, et la W-équivalence colocale sur S la *'W-équivalence fine à droite'* sur S. Elles coïncident dans $\underline{\mathrm{Fibhot}}\, S$, a fortiori dans $\underline{\mathrm{Parf}}\, S$ et dans les quatre catégories (cf. diagramme p. 8) $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Fibsc}}_0 S$, $\underline{\mathrm{Cofib}}_0 S$, $\underline{\mathrm{Cofibsc}}_0 S$ ([33]). Par contre, elles ne coïncident pas sur les catégories *bordantes* du diagramme, notamment sur $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Prop}}\, S$, $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Cofib}}\, S$, $\underline{\mathrm{Fibsc}}\, S$, $\underline{\mathrm{Cofibsc}}\, S$. À cet égard, c'est la *W-équivalence fine* **droite** (en termes des $s\backslash X$) qui est intéressante sur le côté (hélas) gauche du diagramme, i.e. pour $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Fibsc}}\, S$, alors que c'est la *$W$-équivalence fine* **gauche** (en termes des X/s) qui est pertinente du côté droit, pour $\underline{\mathrm{Prop}}\, S$,

33. On pourrait les appeler les six catégories *internes* du diagramme.

Cofib S, Cofibsc S. On n'aura jamais à regarder simultanément les deux équivalences fines sur une même catégorie du diagramme, à la seule exception de Cat/S tout entier.

[page 47]

3) La W-équivalence grossière, elle, est 'bonne partout' - mais ça dépend pour qui. Pour les 6 catégories qui sont au dessus de Fibhot S, i.e.

$$\text{Fibhot}\, S, \quad \text{Parf}\, S, \quad \text{Fib}_0 S, \quad \text{Fibsc}_0 S, \quad \text{Cofib}_0 S, \quad \text{Cofibsc}_0 S,$$

elle coïncide avec toutes les autres équivalences (moyennnant l'axiome L5), pour les trois catégories bordantes supérieures gauches p. 30

$$\text{Liss}\, S, \quad \text{Fib}\, S, \quad \text{Fibsc}\, S$$

du diagramme, elle coïncide avec la W-équivalence fine droite, et dualement sur

$$\text{Prop}\, S, \quad \text{Cofib}\, S, \quad \text{Cofibsc}\, S$$

elle coïncide avec la W-équivalence fine gauche. Quant à Cat/S tout entier, c'est la seule des catégories du diagramme où il y ait *trois* notions intéressantes différentes de W-équivalence, à savoir : la grossière, et les deux fines (à gauche, à droite). Sur les autres six catégories bordantes du diagramme, il y en a en principe deux : la grossière, et une spécifiée des deux fines (défendu de regarder l'autre - à ses risques et périls …) : la droite du côté gauche, la gauche du côté droit. (NB Je devrais refaire le diagramme, décidément, en mettant les fibrations à droite, pas à gauche.) Quant aux six catégories internes au diagramme, dans elles il n'y a plus qu'une seule notion de W-équivalence (puisque sur elles les quatre notions coïncident).

(**NB** il n'y a pas de page 48)

[page 48 (vide)]

[page 49]

Commentary · Exposition [ms. 47]

Reading the corrected classification

Grothendieck is now trying to separate the homotopy theory of a diagram from the homotopy type of its total category. This separation is the reason for retaining three localizations. Write WS for equivalences of total categories, W d S for the condition that every s\X →s\Y belongs to W, and W g S for the analogous condition on X/s →Y/s. The superscripts record the two comma directions, not the placement of a category in the drawing. One sentence in the present paragraph must be corrected mathematically. The assertion that WS coincides with W d S on the smooth or fibred side, and dually with W g S on the proper or cofibred side, is false. The edition prints it without a negation. On the fibred side it is fibrewise equivalence that coincides with W d S; total-category equivalence remains a weaker condition. The walking-arrow example in the commentary at manuscript p. 32 proves the distinction: {0} ,→[1], viewed over [1], is an equivalence of total homotopy types, but its fibres over 1 are respectively empty and nonempty. The same paragraph subsequently says that there are two notions on each border, and p. 49 uses the correct distinction.

On the six internal categories the equality with WS uses the new L5 of p. 43. This is different from the earlier axiom carrying that name. Under L5, a map between homotopy fibrations which is an equivalence on total categories is an equivalence on fibres. The ordinary replacement arguments compare all the coarse localizations without this additional axiom, and comma resolutions independently compare the fine theories. L5 is needed when we identify the coarse class on internal objects with their fine class, and thereby realize the coarse theory as a fully faithful locally constant part of the fine theory. This distinction prevents the corrected theorem from inheriting the false equality in its provisional summary.

Je résume maintenant ce que je crois avoir compris sur les catégories localisées correspondantes. Il y en a, à équivalence près, *trois* différentes seulement en tout et pour tout, quand on prend les localiseurs 'admissibles' dans les catégories, comme je viens de le dire (c'étant ça en somme l'intérêt objectif des prescriptions!) De façon précise :

1°) La *W-équivalence grossière* donne 'la même' catégorie localisée pour les 13 catégories du diagramme, i.e. toutes les flèches de transition du diagramme induisent des équivalences pour cette W-équivalence grossière. Le localisé sera noté $\boxed{\mathrm{HOT}^{\mathrm{lc}}_W(S)}$, où l'exposant lc signifie: (W-types d'homotopie relatif sur S) *'localement constants.'* (Il ne me reste qu'un petit doute pour $\underline{\mathrm{Fibsc}}\, S$ et $\underline{\mathrm{Cofibsc(S)}}$ ([34]), si l'équivalence grossière y est bel et bien raisonnable. On verra en faisant la démonstration.)

2°) La *W-équivalence fine à droite* donne le même résultat pour toutes les catégories du diagramme où elle est considérée comme admise, *et où elle ne coïncide pas avec la W-équivalence grossière* déjà considérée, donc sur les quatre catégories bordantes gauches

$$\mathrm{Cat}/S, \quad \underline{\mathrm{Liss}}\, S, \quad \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Fibsc}}\, S$$

([35]). Le localisé est noté $\boxed{\mathrm{HOT}_W(S)}$.

[page 50]
p. 31

$2'^o$) Dualement, la *W-équivalence fine gauche* donne le même résultat sur toutes les catégories du diagramme où elle est admise, et où elle ne coïncide pas avec la grossière, i.e. sur les 4 catégories

$$\mathrm{Cat}/S, \quad \underline{\mathrm{Prop}}\, S, \quad \underline{\mathrm{Cofib}}\, S, \quad \underline{\mathrm{Cofibsc}}\, S.$$

Il y a lieu de noter la catégorie des fractions commune par le sigle

$$\mathrm{HOT}_W(S^o)^o \qquad (\text{ou aussi } \mathrm{HOT}^o_W(S)),$$

puisqu'elle est isomorphe à celle notée par ce sigle.

Si on veut des notations sans ambiguïté, désignant des catégories bien définies, et non pas seulement 'à équivalence près', je choisirai

$$\mathrm{HOT}_W(S) \overset{\text{déf}}{=} \underline{\mathrm{Fibsc}}(S)(W^{\mathrm{d}}_S)^{-1}$$

d'où

$$\mathrm{HOT}_W(S^o) \underbrace{\simeq}_{\text{isomorphisme de catégories}} \underline{\mathrm{Cofibsc}}(S)(W^{\mathrm{g}}_S)^{-1},$$

où W^{d}_S, W^{g}_S désignent les ensembles de flèches correspondants à la W-équivalence fine droite resp. gauche. Pour $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, je préfère une définition aussi symétrique que possible en S, S^o, et de plus une catégorie aussi petite que possible, ce qui me conduit à choisir

$$\mathrm{HOT}^{\mathrm{lc}}_W(S) \overset{\text{déf}}{=} (\underline{\mathrm{Parf}}\, S)W_S^{-1},$$

où on a noté par W_S les W-équivalences dans $\underline{\mathrm{Parf}}\, S$. (Inutile cette fois de préciser lesquelles - toutes les quatre `[notions]` coïncident dans $\underline{\mathrm{Parf}}\, S$!)

[page 51]

Commentary · Exposition [ms. 50]

The covariant theory and the extra opposite

Grothendieck wants a notation which makes the symmetry between fibrations and opfibrations immediate. There is a genuine symmetry, but the edition alternates between two incompatible formulas. With the definition fixed on manuscript p. 3, the precise identities are

[Cofibsc(S)][(W g

HOTW (S) = [Sop, Cat][W −1 S)−1] ≃[S, Cat][W −1 obj] = HOTW (Sop). obj], Here Wobj means objectwise membership in W. There is no opposite on the value category in the last expression. The displayed isomorphism in the middle of p. 50 has this correct form; the preceding prose and the

34. Ce doute n'est pas fondé, tout marche parfaitement!

35. **NB** Sur les trois dernières cette équivalence est la W-équivalence par fibres.

editorial correction at p. 51 insert an additional opposite. We retain those readings as part of the source, while using the comma-category models to determine the variance of every assertion.

Indeed, taking opposite total categories sends a fibration over Sop to an opfibration over S. It sends a functor X →Y to a functor Xop →Y op in the same direction. Thus it defines a covariant isomorphism of the categories of structured objects; it does not reverse their morphisms. The same observation justifies the correctly stated symmetry of the perfect-fibration localization at p. 51.

The extra opposite cannot be repaired by an unspecified equivalence. Take S = e and W = W∞. The category of homotopy types is not equivalent to its opposite. Its initial object ∅is strict: a morphism T →∅exists only when T ≃∅. Such an equivalence would force the initial object of the opposite category, namely the one-point homotopy type, to be strict there. This would say that every map ∗→T is an isomorphism, contradicted by the inclusion of one point into a two-point set. The external opposite is therefore a substantive variance slip, even though both fine localizations themselves are correctly constructed.

The notation HOTo W (S) may still be used as a label for the covariant theory, provided it is defined by the left-fine localization. It should not be interpreted here as taking the opposite of HOTW (Sop). This convention also gives the stated direction to the functor j in the diagram which follows.

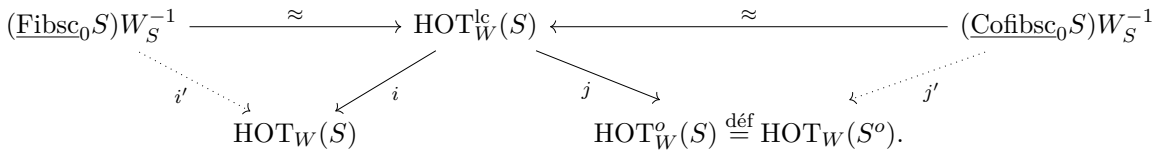
Notons l'isomorphisme

$$\underline{\mathrm{Parf}}(S^o) \simeq \underline{\mathrm{Parf}}\, S$$

(provenant du fait que $f : X \longrightarrow Y$ dans Cat est une fibration parfaite si et seulement si f^o l'est). Cet isomorphisme $X \longmapsto X^o$ est compatible avec les localiseurs (car $f \in W \iff f^o \in W$), d'où

$$\mathrm{HOT}^{\mathrm{lc}}_W(S^o) \simeq \mathrm{HOT}^{\mathrm{lc}}_W(S)$$

(isomorphisme de catégories). On aura deux foncteurs canoniques entre les trois catégories de W-types d'homotopie relatifs sur S



[plutôt $\overset{\text{déf}}{=} \mathrm{HOT}_W(S^o)^o$] Ces flèches i, j sont suggerées par les flèches en pointillés i', j', provenant des inclusions p. 32

$$\underline{\mathrm{Fibsc}}_0 S \hookrightarrow \underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}_0 S \hookrightarrow \underline{\mathrm{Cofibsc}}\, S,$$

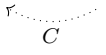
qui définiraient i, j à l'aide du choix de foncteurs quasi-inverses pour les flèches horizontales. Mais i, j ne seraient définis qu'à isomorphisme (unique) près. Il sera plus joli de choisir les quasi-inverses de façon canonique, à l'aide des flèches

$$\underline{\mathrm{Fibsc}}_0 S \xleftarrow{\Phi_0} \underline{\mathrm{Parf}}\, S \xrightarrow{\Psi_0} \underline{\mathrm{Cofibsc}}_0 S$$

(où $\Phi_0 = \Phi|\underline{\mathrm{Parf}}\, S$, $\Psi_0 = \Psi|\underline{\mathrm{Parf}}\, S$) du diagramme p. 8. La question principale qui reste non élucidée, c'est si ces foncteurs i, j sont bien *pleinement fidèles,* comme je m'y attends certes. [page 52]

Je veux maintenant prouver les affirmations précédentes sur les localisés.

① Le localisé pour la W-équivalence grossière. Il suffira, à un ε près, d'établir les conditions du lemme p. 11, dans les trois cas suivants. $\underline{\mathrm{Parf}}\, S \xrightarrow{r} \mathrm{Cat}/S$ (ça donnera le cas de



$\underline{\mathrm{Fibhot}}\, S$, $\underline{\mathrm{Liss}}\, S$ et $\underline{\mathrm{Prop}}\, S$ sur Cat/S) et $\underline{\mathrm{Fibsc}}_0 S \xrightarrow{p} \mathrm{Cat}/S$, $\underline{\mathrm{Cofibsc}}_0 S \xrightarrow{q} \mathrm{Cat}/S$,

qui donneront respectivement les cas de

$$\underline{\mathrm{Fib}}_0 S,\ \underline{\mathrm{Fib}}\, S,\ \underline{\mathrm{Liss}}\, S \qquad \text{et} \qquad \underline{\mathrm{Cofib}}_0 S,\ \underline{\mathrm{Cofib}}\, S,\ \underline{\mathrm{Prop}}\, S \qquad \text{sur } S.$$

Mais l'ennui pour p et q, c'est que les foncteurs en sens inverse que j'avais construit (et noté encore Φ_0, Ψ_0 (mais c'étaient essentiellement $\Phi \circ r \circ C$ et $\Psi \circ r \circ c$, à ne pas confondre avec les foncteurs Φ_0, Ψ_0 en bas de la page précédente, et marqués aussi dans le diagramme p. 8), notons les plutôt Φ', Ψ', marchent bien du côté Cat/S, i.e. $p\Phi'$, $q\Psi'$ reçoivent le foncteur $\mathrm{id}_{\mathrm{Cat}/S}$ par une W-flèche, mais pas du côté des catégories but - on n'arrive pas à trouver une flèche naturelle entre $\Phi' p$ resp. $\Psi' q$, et les foncteurs identiques de $\underline{\mathrm{Fibsc}}_0 S$ resp. $\underline{\mathrm{Cofibsc}}_0 S$. Je ne vois pas, en effet, qu'il y en ait - la difficulté, par cette voie, parait irrémédiable. Par contre, [page 53]
si on regarde Φ', Ψ' comme foncteurs à valeurs dans $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Cofib}}_0 S$ respectivement (donc dans les catégories où on n'exige pas que les flèches soient catésiennes resp. cocartésiennes), on trouve encore des W-morphismes

$$\mathrm{id} \longrightarrow \Phi' p, \qquad \mathrm{id} \longrightarrow \psi' q,$$

ce qui liquide au moins déjà le cas de toutes les catégories des diagrammes p. 8 qui sont des sous-catégories pleines de Cat/S ([36]). Il ne reste à traiter que le cas des quatre catégories de p. 33
structures 'scindées'

$$\underline{\mathrm{Fibsc}}\, S, \quad \underline{\mathrm{Fibsc}}_0 S \qquad \text{et} \qquad \underline{\mathrm{Cofibsc}}\, S, \quad \underline{\mathrm{Cofibsc}}_0 S.$$

Bien sûr il suffit de le faire d'un côté, disons du côté des fibrés, par dualité ($S \longmapsto S^o$) cela établira le résultat symétrique. On va donc regarder d'un peu plus près le carré de foncteurs

$$\begin{array}{ccc} \underline{\mathrm{Fibsc}}\, S & \xleftarrow{\alpha'} & \underline{\mathrm{Fibsc}}_0 S \\ {\scriptstyle\varphi}\big\downarrow{\scriptstyle p} & & {\scriptstyle p_0}\big\downarrow{\scriptstyle\varphi_0} \\ \underline{\mathrm{Fib}}\, S & \xleftarrow{\alpha} & \underline{\mathrm{Fib}}_0 S \end{array}$$

([37], [38]). Le foncteur

$$\Phi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}}\, S$$

induit des foncteurs en sens inverse de p, p_0 sur les sous-catégories pleines $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Fib}}_0 S$ de Cat/S, foncteurs marqués en pointillés comme φ, φ_0.

[page 54]

Commentary · Context [ms. 53]

Why the free split fibration is the right comparison

The difficulty with the first attempt is a difficulty about morphisms, not about the existence of enough good objects. A path completion puts every object into a particularly well-behaved class, but its unit need not preserve a chosen splitting. The comma construction Φ has a stronger universal property: it freely supplies the selected Cartesian lifts. That property gives the reverse comparison map with exactly the morphism convention required by Fibsc(S).

This construction was already part of the theory of fibred categories before the present manuscript. Gray's treatment constructs the free split fibration, and the modern formulation identifies split fibrations and splitting-preserving functors with strict algebras for the comma-category 2-monad on Cat/S (Gray, 1966; Emmenegger et al., 2024). In the latter reference, Theorem 2.6 separates strict algebras from pseudo-algebras; Theorem 5.4 treats the two adjoints used to split an already fibred category. These are distinct operations. The present Φ is the free construction on an arbitrary category over S.

36. Mais c'est idiot de prendre des foncteurs si compliqués, parce qu'on veut tout ramener à Cat/S. Il est de plus simple de comparer $\underline{\mathrm{Fibhot}}\, S$ directement à $\underline{\mathrm{Fibsc}}_0 S$ et à $\underline{\mathrm{Cofibsc}}_0 S$. Φ_0 et Ψ_0 (qui sont en fait définis sur $\underline{\mathrm{Fibhot}}$, pas seulement sur $\underline{\mathrm{Parf}}$), et on trouve comme ci-dessous $\mathrm{id} \xrightarrow{W} \pi_0\Phi_0$ et $\Phi_0\pi_0 \xrightarrow{W} \mathrm{id}$ (itou pour Ψ_0 et $\underline{\mathrm{Cofibsc}}_0 S$), où $\pi_0 : \underline{\mathrm{Fibsc}}_0 S \longrightarrow \underline{\mathrm{Fibhot}}\, S$.

37. NB Il s'agit de prouver seulement que p, p_0 induisent des équivalences pour les localisés (pour la W-équivalence grossière). On sait déjà qu'il en est ainsi pour α, et ça en résultera pour α'.

38. inutile de regarder ce carré, cf. annotations marginales ci-dessus (haut de la page)

Its importance for localization is elementary. Let U : Fibsc(S) →Cat/S forget the splitting. There are transformations

η : id −→UΦ,

γ : ΦU −→id. The first lives in Cat/S; the second lives in the category with splitting-preserving morphisms. Both become isomorphisms after the relevant localization. Consequently U and Φ induce inverse equivalences. One does not need a comparison transformation in a category with an incorrectly enlarged or restricted class of arrows.

For coarse equivalences, this conclusion requires only that W contain categorical homotopy equivalences and satisfy two-out-of-three. The ordinary functor forgetting the initial base object gives UΦ(X) →X, with a section ηX and a natural contraction. Although the forgetful map is usually not over S, the contraction proves that ηX is an equivalence of total homotopy types. The counit is then a coarse equivalence because its composite with the unit on UY is the identity. The more informative fibrewise verification comes on pp. 58–60.

Il faut seulement vérifier que si X/S est dans $\underline{\mathrm{Fib}}_0 S$, alors $\varphi(X) = \Phi(X)$ est dans la sous-catégorie pleine $\underline{\mathrm{Fibsc}}_0 S$ de $\underline{\mathrm{Fibsc}}\, S$. Mais on a vu que c'est le cas, pour peu seulement que X/S soit dans $\underline{\mathrm{Fibhot}}\, S$, car les morphismes de changement de base dans $\Phi(X)$, associés aux flèches $u : s \longrightarrow t$ dans S

$$u^*_{\Phi_X} : t\backslash X \longrightarrow s\backslash X$$

sont des W_∞-équivalences si X est Hot-fibré sur S. D'autre part, pour X quelconque dans Cat/S, on a une flèche, fonctorielle en X,

$$X \longrightarrow \Phi(X),$$

i.e. $\mathrm{id}_{\mathrm{Cat}/S} \longrightarrow \pi\Phi$ où $\pi : \underline{\mathrm{Fibsc}}\, S \longrightarrow \mathrm{Cat}/S$ est le foncteur canonique, qui est une W_∞-équivalence (même un homotopisme), donc une W-équivalence. A fortiori les foncteurs $p\varphi$, $p_0\varphi_0$ donnent lieu ($\underline{\mathrm{Fib}}\, S$ et $\underline{\mathrm{Fib}}_0 S$ étant des sous-catégories pleines de Cat/S) à des flèches

$$\mathrm{id} \longrightarrow p\varphi, \qquad \mathrm{id} \longrightarrow p_0\varphi_0.$$

Il reste à trouver une flèche (et W-équivalence grossière)

p. 34

$$\varphi p \longrightarrow \mathrm{id},$$

laquelle induira automatiquement

$$\varphi_0 p_0 \longrightarrow \mathrm{id}$$

[page 55]

puisque $\underline{\mathrm{Fibsc}}_0 S$ est [une] sous-catégorie pleine de $\underline{\mathrm{Fibsc}}\, S$. En fait, on va définir $\varphi p \longrightarrow \mathrm{id}$, donc

$$\boxed{\gamma_X : \Phi(X) \longrightarrow X}$$

quand X est fibrée scindée sur X - une flèche *cartésienne,* donc (si on veut) compatible avec les scindages de $\Phi(X)$, et de X. En fait, on aura une telle flèche, fonctorielle en X, pour X variable dans $\underline{\mathrm{Fib}}\, S$ (donc sans se borner à des S-morphismes cartésiens entre catégories fibrées sur S), mais la flèche $\longrightarrow$ étant bien sûr *cartésienne.* (Ça a un sens, sans choisir de scindage sur X, et sans qu'il en existe forcément.) La définition de cette flèche γ fait appel de façon essentielle à l'hypothèse que X soit Cat-fibrant sur S. Bien sûr, composé à la flèche d'inclusion évidente

$$X \xrightarrow{i_X} \Phi(X)$$

on va avoir

$$\gamma_X i_X = \mathrm{id}_X$$

([39]). On définit γ fibre par fibre,

$$\gamma_{X,s} : \Phi(X)_s \overset{\text{déf}}{=} s\backslash X \longrightarrow X_s,$$

39. NB Comme on sait que $i_X \in W$, il en résultera bien $\gamma_X \in W$ comme on le veut.

donc par une *rétraction* de $s\backslash X$ sur la sous-catégorie pleine X_s. Un objet de $s\backslash X$ est un couple (x, u), $x \in \mathrm{Ob}\, X$, $u : s \longrightarrow p_X(x)$ ($p_X : X \longrightarrow S$ le morphisme structural de X). On pose

$$\gamma(x, u) = u_X^*(x) \in \mathrm{Ob}\, X_s,$$

ce qui suppose donc qu'on a choisi des foncteurs image inverse $u_X^* : X_t \longrightarrow X_s$, pour la catégorie Cat-fibrée X/S. (Mais à isomorphisme unique près, γ ne dépend pas de ces choix.) Je me dispense de la vérification fastidieuse que cette loi est bien 'fonctorielle en s', i.e. définit bien une flèche cartésienne [page 56]

$$\gamma_X : \Phi(X) \longrightarrow X,$$

et de vérifier que celle-ci est fonctorielle en X.

Peut-être est-il plus élégant de montrer que $\pi : \underline{\mathrm{Fibsc}}\, S \longrightarrow \mathrm{Cat}/S$ et $\Phi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}}\, S$ sont adjoints l'un de l'autre : p. 35

$$\boxed{\mathrm{Homcart}_S(\Phi(\underbrace{X}_{\in \mathrm{Cat}/S}), \underbrace{Y}_{\in \underline{\mathrm{Fibsc}}}) \simeq \mathrm{Hom}_S(X, \pi Y),}$$

d'où résulteront des flèches d'adjonction

$$\Phi\pi \longrightarrow \mathrm{id}_{\underline{\mathrm{Fibsc}}}, \qquad \pi\Phi \longleftarrow \mathrm{id}_{\underline{\mathrm{Fib}}},$$

[plutôt $\pi\Phi \longleftarrow \mathrm{id}_{\mathrm{Cat}/S}$] et on aura gagné. Mais pour prouver cette formule d'adjonction, j'ai l'impression qu'il faut utiliser $\varphi p \longrightarrow \mathrm{id}$ … Il y a des vérifications fastidieuses …

Finalement, je vois maintenant qu'on a cinq cas à traiter, et trois si on joue sur les symétries : [page 57]

Commentary · Exposition [ms. 56]

The free split adjunction, with its arrows specified

Grothendieck's proposed adjunction is the efficient way to finish the comparison, once "Cartesian" in the Hom-set on the left is read as preserving the selected Cartesian lifts. We give its construction because the same calculation resolves the naturality issue in the preceding paragraph.

For p : X →S, an object of ΦX is (s, x, u) with u : s →p(x). A morphism

(s, x, u) −−→(t, y, v)

is a pair (a, f) with a : s →t, f : x →y, and p(f)u = va. The chosen lift along a with target (t, y, v) is

−−−−−→(t, y, v).

(s, y, va) Identity and composition laws for these chosen lifts hold strictly. The unit sends x to (p(x), x, id). Let q : Y →S be split, and let h : X →UY be any functor over S. Write cu,z : u∗z →z for the chosen lift. Define

bh(s, x, u) = u∗h(x).

For (a, f) as above, define bh(a, f) as the unique arrow over a such that

cv,h(y) bh(a, f) = h(f)cu,h(x).

The Cartesian property of cv,h(y) gives existence and uniqueness. These same uniqueness statements prove the identity and composition laws: both possible images of a composite have the required base arrow and the same composite with the terminal chosen lift. The strict splitting laws show that a selected lift (a, id) is taken to the selected lift in Y . Thus bh is splitting-preserving.

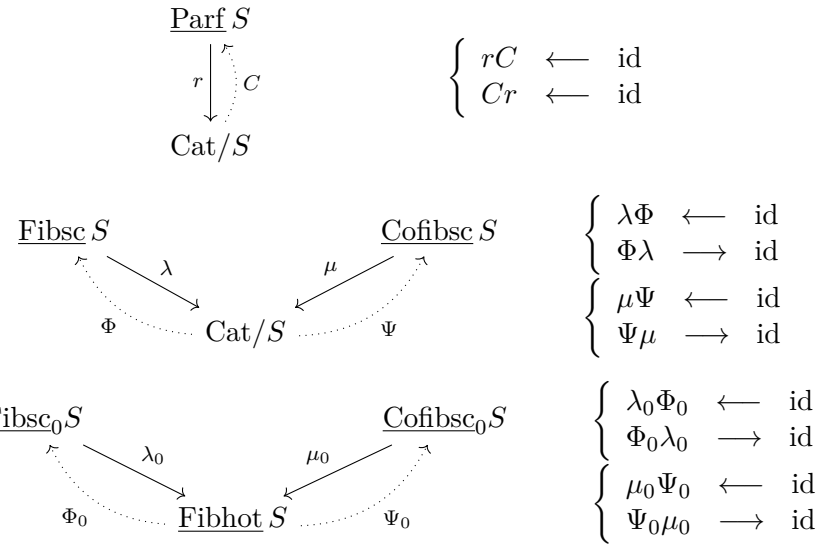
Conversely, every splitting-preserving H : ΦX →Y is forced by h = HηX. The chosen arrow (s, x, u) →(p(x), x, id) forces its object value to be u∗h(x), and Cartesian uniqueness forces its value on every morphism. The two constructions are inverse and natural, proving the actual adjunction

Homsplit/S(ΦX, Y) ∼= HomS(X, UY).

Its counit is γY (s, y, u) = u∗y, and γY ηUY = idUY . Its naturality is strict for splitting-preserving functors, which are precisely the morphisms needed on this side of the comparison. This is the free-algebra adjunction underlying the 2-monad description cited above.

For a merely cloven fibration the same formula still gives a Cartesian functor ΦX →X, using the coherence isomorphisms of the cleavage. Different choices yield canonically isomorphic functors

compatible with their Cartesian comparison arrows. But an arbitrary functor over S need not preserve Cartesian arrows, so the formula is not strictly natural on the full category Fib(S). Nor can "Cartesian" alone imply preservation of the selected splitting. Those stronger assertions in the source are unnecessary; the strict adjunction just proved gives the comparison in Fibsc(S), and the next block explains what survives after localization for all functors over S.



([40])

NB Les quatre derniers homomorphismes de foncteurs, relatifs au troisième diagramme (avec les indices 0) sont simplement induits par ceux relatifs au diagramme qui le précède. Donc les vérifications-clef sont relatives aux deux premiers diagrammes, et par symétrie, on n'a finalement qu'à regarder les *deux* couples de foncteurs (r, C), (λ, Φ), et pour chacun de ces couples, exhiber les deux homomorphismes fonctoriels, des deux composés avec les foncteurs identiques.

p. 36
[page 58]

Commentary · Development (commentators') [ms. 57]

Naturality recovered after localization

Grothendieck wants the comparison to accommodate all functors over the base, since the categories Fib(S) and Liss(S) are full subcategories of Cat/S. The raw pullback construction does not have that naturality. The localization nevertheless supplies exactly the required repair. We spell out the following elementary elaboration of his argument; the "ours" label concerns this formulation and proof, not priority. We checked the free-fibration and strictification formulations in Emmenegger et al. (2024, Sections 2 and 5). They distinguish strict and pseudo preservation of cleavages but do not state the following repair for arbitrary functors over the base. We make no broader claim about its absence from the literature.

Localized naturality of pullback retractions. [ours]Suppose W∞⊆W. Choose normalized cleavages on fibrations X →S. Then the functors γX : ΦX →X determine a natural isomorphism between the functors induced by UΦ and the identity on the right-fine localization of the full category Fib(S). The same conclusion holds for coarse localization.

Proof. First, any natural transformation over S between functors F, G : A →B makes their images equal in (Cat/S)[(W d S)−1]. It is represented by a functor H : A × [1] →B over S. The projection r : A × [1] →A is right-fine: for every s its comma functor is the projection

(s\A) × [1] −→s\A,

a categorical homotopy equivalence. Its two sections become the same inverse of r, hence F = Hj0 and G = Hj1 become equal. The identical proof applies to coarse localization. If A is fibred, A × [1] is fibred, so this argument also takes place within the indicated full subcategory.

40. **NB** Tous les foncteurs marqués sont compatibles avec la W-équivalence grossière, et des dix homomorphismes fonctoriels sont des W-équivalences grossières.

Now let f : X →Y be any functor over S, without a Cartesian hypothesis. For (s, x, u) define a vertical arrow

θf,(s,x,u) : f(u∗

Xx) −→u∗ Y f(x) by the equation

cY

u,f(x)θf,(s,x,u) = f(cX u,x). Existence and uniqueness follow from the Cartesian property of the lift in Y . This defines a natural transformation fγX ⇒γY Φ(f) over S. To check naturality, compose the two candidate arrows with the chosen lift to the terminal object of a morphism (a, g) in ΦX. Both composites are f(g)f(cX u,x), u,x) = f(gcX and both have the same base arrow; Cartesian uniqueness identifies them. The first paragraph of the proof therefore gives naturality after localization.

The comparison maps are right-fine equivalences. On a fibre, γX,s is right adjoint to Xs ,→s\X: an arrow y →x over u : s →p(x) factors uniquely through u∗ Xx. Every adjunction gives a homotopy equivalence of nerves. Since source and target of γX are fibrations, right-fine equivalence is equivalent to fibrewise equivalence. Thus the localized natural transformation is an isomorphism. Its coarse version follows either from the same argument or from γXηX = id and coarse invertibility of ηX. □

The defect before localization is real. Take S = [1], X = S, Y = S × [1], and f(s) = (s, s). At the object (0, 1, 0 →1) of ΦX, the two functors in the comparison square have values (0, 0) and (0, 1). These objects are not isomorphic. The comparison θ is the noninvertible arrow between them. Thus the source's strict naturality assertion cannot simply be replaced by a claim of naturality up to isomorphism; the localization, or the restriction to splitting-preserving morphisms, performs an essential part of the repair.

② Les localisés pour la *W-équivalence fine droite.* Il n'y a plus lieu, on l'a dit, de regarder les catégories internes du diagramme p. 8, où ladite équivalence coïncide avec la W-équivalence grossière, pour laquelle on connait déjà les localisés. Il reste à regarder les catégories

$$\mathrm{Cat}/S, \quad \underline{\mathrm{Liss}}\, S, \quad \underline{\mathrm{Fib}}\, S, \quad \underline{\mathrm{Fibsc}}\, S.$$

Le lemme de la page 11 nous dit qu'il suffit de montrer que dans la situation

$$\underline{\mathrm{Fibsc}}\, S \xrightarrow{\lambda} \mathrm{Cat}/S,$$

il y a un foncteur en sens inverse, avec les propriétés specifiées. Il en résultera que toute sous-catégorie pleine $\mathcal{F}$ de Cat/S qui contient $\underline{\mathrm{Fib}}\, S$, quand on la munit du localiseur de la W-équivalence fine droite, donne un localisé équivalent, par le foncteur induit par l'inclusion, au localisé correspondant de Cat/S, i.e. essentiellement à $\mathrm{HOT}_W(S)$.

Comme foncteur en sens inverse, il s'impose de prendre encore Φ. Il y a à vérifier trois choses :

a) Φ [est] compatible avec les localiseurs, i.e. si $\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$ sur S est une W-équivalence fine droite, i.e. les $s\backslash X \longrightarrow s\backslash Y$ sont des W-équivalences, alors $\Phi(f)$ aussi. Mais ladite équivalence coïncide dans $\underline{\mathrm{Fibsc}}\, S$ (comme aussi dans $\underline{\mathrm{Fib}}\, S$ et $\underline{\mathrm{Liss}}\, S$) avec la W-équivalence par fibres, et d'autre part $\Phi(f)_s$ n'est autre que la flèche précédente $s\backslash X \longrightarrow s\backslash Y$. [page 59] Donc il est évident que

$$f \in \underbrace{W_S^{\mathrm{d}}}_{W\text{-équivalence droite}} \iff \Phi(f) \in \underbrace{W_S^{\mathrm{f}}}_{W\text{-équivalence par fibres}}.$$

b) Que $X \xrightarrow{i_X} \lambda\Phi(X)$ est dans W_S^{d}, pour tout X dans Cat/S. Cela signifie donc que i_X induit

$$s\backslash X \longrightarrow s\backslash\Phi(X)$$

qui est une W-équivalence. Ce serait O.K. s'il était clair que $X \longmapsto \Phi(X)$ commute à tout changement de base propre sur S (donc en particulier à $s\backslash S \longrightarrow S$), puisqu'on sait que

$X \longrightarrow \Phi(X)$ est une W-équivalence. C'est sans doute trop demander, mais du moins que ça commute au changement de base qui est un co-isomorphisme local (telles les $s\backslash S \longrightarrow S$).

Pour un tel changement de base $S' \xrightarrow{p} S$, vérifier que la fibre de $\Phi(X')$ en s' est isomorphe à celle de $\Phi(X)$ en $s = p(s')$, je m'en contenterai. Or cette fibre est $s'\backslash X'$, déduit de X' sur S' par le changement de base $s'\backslash S' \longrightarrow S'$, donc déduit de X par $s'\backslash S' \longrightarrow S$. Or $s'\backslash S' \longrightarrow s\backslash S$ est un isomorphisme par hypothèse sur p (co-isomorphisme local), donc le changement de base $s'\backslash S' \longrightarrow S$ s'identifie à $s\backslash S \longrightarrow S$, et l'image inverse est bien isomorphe à $s\backslash X$, OK. p. 37 [page 60]

Commentary · Exposition [ms. 59]

The comma resolution is right-fine

Grothendieck has reduced the fine comparison to checking that the unit and counit are fine equivalences. This is the decisive point at which the comma categories preserve the relative diagram, rather than merely its total homotopy type. A direct contraction proves the assertion for every X →S and makes the proposed base-change verification unnecessary.

Fix s ∈S. An object of s\ΦX consists of

−→vt

−→up(x). s Define R : s\ΦX →s\X by composition, R(v, u, x) = (x, uv). A morphism represented by (a, f) satisfies av = v′ and p(f)u = u′a; it is sent to f, which satisfies p(f)uv = u′v′. Thus R is a functor. The functor J = s\ηX sends (x, w) to the factorization

−→p(x)

−→p(x). s We have RJ = id. The arrow (u, idx) gives a natural transformation

ids\ΦX −→JR.

It is over s, since uv is the composite arrow of the target factorization. These identities prove that J and R are inverse up to categorical homotopy. Hence ηX belongs to W d S whenever W contains W∞.

For a split fibration Y , the fibre of the counit is

(y, u) 7−→u∗y.

γY,s : s\Y −→Ys, If K : Ys ,→s\Y is the inclusion, Cartesian factorization gives

Homs\Y (Kz, (y, u)) ∼= HomYs(z, u∗y).

Thus K ⊣γY,s, with identity unit for a normalized splitting. In particular both functors are W∞equivalences. The counit is therefore fibrewise, and hence right-fine, an equivalence. Finally, Φ(f)s is literally s\f; this verifies compatibility of Φ with the two localizers without an additional argument. All parts of the comparison now fit together. The unit and counit induce inverse natural isomorphisms between the localized composites. We obtain

S)−1] ≃Fibsc(S)[W −1

(Cat/S)[(W d obj]. If F is any full subcategory of Cat/S containing the fibrations, the resolution UΦ lands in F and its unit is a morphism of F on restriction. The same argument proves the asserted equivalence with F[(W d S)−1]. Reversing the comma directions gives the corresponding statement for opfibrations and W g S.

c) Que de même

$$\Phi\lambda \longrightarrow \mathrm{id}$$

est non seulement dans W_S (grossier), mais que c'est une W-équivalence par fibres. Donc on part de X fibrée sur S. On regarde

$$\Phi(X) \longrightarrow X,$$

pour voir que c'est une W-équivalence par fibres, il suffit de voir qu'il en est ainsi de $X \longrightarrow \Phi(X)$,

$$X_s \longrightarrow s\backslash X$$

est une W-équivalence pour tout $s \in S$. Mais ceci est déjà vrai dès que X est lisse sur S.

Comme le cas de la W-équivalence fine gauche se ramène au cas actuel par $S \longmapsto S^o$, on a donc prouvé tout ce qu'était attendu, à la seule exception de la pleine fidèlité des foncteurs i, j de la page 51. [page 61]

Mais dans le résumé prospectif de la situation pour les localiseurs, p. 49-51, j'ai oublié d'autres foncteurs que i, j, de nature si possible encore plus triviale. Ce sont ceux qu' on peut décrire à partir du foncteur identique, sur une des catégories du diagramme où aient cours deux, voire trois, notions de W-équivalence différentes. Ce sont, je le rappelle

Commentary · Exposition [ms. 60]

The hypotheses of the fine comparison

The comparison just established has survived the earlier changes of axioms. Its proof uses the inclusion W∞⊆W, the elementary saturation properties, and the smooth-fibre criterion already checked on p. 42. In particular it does not use the new L5. On split fibrations, right-fine equivalences are exactly objectwise W-equivalences of the associated strict diagrams; the analogous statement holds on split opfibrations. This is the part of the construction which remains useful for homological and truncated equivalences.

The pure coarse comparison with the internal categories also needs no new L5. The path replacement C produces homotopy fibrations, its unit X →C(X) is a total homotopy equivalence, and naturality together with two-out-of-three makes C preserve coarse equivalences. L5 enters in the additional identification of coarse equivalences between these replacements with fibrewise, hence fine, equivalences. This is what makes C induce the functor back into the fine category used for the reflection. The later annotation correctly says that this requirement is highly restrictive. The commentary on pp. 43–45 explains why it must not be silently transferred to arbitrary homological localizers.

For W = W∞, the modern literature gives a precise model for the coarse part: small categories, nerve weak equivalences, and smooth proper functors form a category of fibrant objects; perfect objects over a base suffice to compute the corresponding localization. See Cisinski (2009, Proposition 4.4 and Corollary 4.5). This places Grothendieck's path replacement inside a developed homotopical framework. The explicit comma proof above is still useful, since it shows exactly which comparison remains valid when the coefficient equivalences are enlarged.

1°) Les quatre catégories

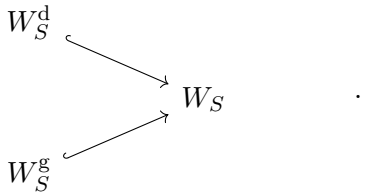
$$(*) \qquad \underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S \longrightarrow \underline{\mathrm{Liss}}\, S \longrightarrow \mathrm{Cat}/S, \qquad W_S^{\mathrm{d}} \subseteq W$$

avec la W-équivalence fine droite d'une part, grossière de l'autre, et dualement

$$(*') \qquad \underline{\mathrm{Cofibsc}}\, S \longrightarrow \underline{\mathrm{Cofib}}\, S \longrightarrow \underline{\mathrm{Prop}}\, S \longrightarrow \mathrm{Cat}/S, \qquad W_S^{\mathrm{g}} \subseteq W$$

pour la W-équivalence fine gauche d'une part, la grossière de l'autre, et de plus

2°) Cat/S, qui était inclue dans les deux suites précédentes, et qui est la seule catégorie du diagramme donnant lieu à trois notions de W-équivalence différentes



Dans chacun des quatre cas (C, munie de deux localiseurs $\Sigma \subseteq \Sigma'$) de la suite ($*$), le foncteur p. 38

$$C\Sigma^{-1} \longrightarrow C\Sigma'^{-1}$$

est essentiellement 'le même', à cause des théorèmes de comparaison. Il suffit de regarder ce foncteur dans un seul das, p. ex. un des deux extrêmes $\underline{\mathrm{Fibsc}}\, S$ ou Cat/S. Même remarque pour la suite ($*'$). Donc finalement il reste deux foncteurs remarquables à examiner, qui n'ont [page 62] pas l'air du tout de provenir des foncteurs du diagramme p. 8 ou de quelque composé (ça ne

donne que des équivalences canoniques ou bien les foncteurs i, j de page 51, quand on va des catégories internes au diagramme vers les catégories bordantes). Le plus joli est sans doute de les regarder à partir de Cat/S, puisque c'est là seulement qu'on les a tous les deux à la fois

$$\begin{array}{ccc} (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} \approx \mathrm{HOT}_W(S) & & \\ & \overset{\rho}{\searrow} & \\ & & (\mathrm{Cat}/S)W_S^{-1} \approx \mathrm{HOT}^{\mathrm{lc}}_W(S). \\ & \underset{\sigma}{\nearrow} & \\ (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1} \approx \mathrm{HOT}^{o}_W(S) & & \end{array}$$

Par contre, les foncteurs (qu'on a envie d'appeler 'foncteurs d'inclusion') i, j de la page 51, en sens inverse de ρ, σ, sont ceux qui proviennent, par localisation W_S^{d} ou W_S^{g} selon le cas, d'un des foncteurs du diagramme page 8 (ou d'un composé de tels foncteurs), allant d'une des six catégories internes (où toutes les W-équivalences envisagées coïncident) vers l'une des sept catégories bordantes, sauf quand cette dernière est Cat/S, est déterminé déjà avec quelle W-équivalence, W_S^{d} ou W_S^{g}, on va travailler `[structure de cette phrase?]`. La détermi-nation peut-être la plus jolie est par [page 63]

$$(*) \qquad \underline{\mathrm{Parf}}\, S \longrightarrow \mathrm{Cat}/S$$

qui donne le foncteur i quand on travaille avec W_S^{d}, le foncteur j quand on travaille avec W_S^{g}. On voit alors tout de suite que si on compose avec le foncteur pertinent, soit ρ soit σ, défini par l'inclusion $W_S^{\mathrm{d}} \subseteq W_S$ resp. $W_S^{\mathrm{g}} \subseteq W_S$ (comme parties de $\mathrm{Fl}(\mathrm{Cat}/S)$), on trouve le foncteur

$$(**) \qquad (\underline{\mathrm{Parf}}\, S)W_S^{-1} \xrightarrow{\approx} (\mathrm{Cat}/S)W_S^{-1}$$

qui est l'*équivalence canonique* entre ces deux catégories. Ainsi le foncteur correspondant à i, j page 51, exprimé en termes de Cat/S comme foncteur en sens inverse de ρ, σ (p. 62), est-il le composé de l'équivalence quasi-inverse de $(**)$, laquelle s'explicite par le foncteur C, et du foncteur sur le localisé W_S^{d} ou W_S^{g} déduit de $(*)$, i.e.

p. 39

$$\begin{array}{ccccc} & & & & \mathrm{HOT}_W(S) \approx (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} \\ & & \overset{i}{\nearrow} & & \uparrow \text{car } W_S = W_S^{\mathrm{d}} \text{ sur } \underline{\mathrm{Parf}} \\ \mathrm{HOT}^{\mathrm{lc}}_W(S) \approx (\mathrm{Cat}/S)W_S^{-1} & & \xrightarrow{\quad C \quad} & & (\underline{\mathrm{Parf}}\, S)W_S^{-1} \\ & & \underset{j}{\searrow} & & \downarrow \text{car } W_S = W_S^{\mathrm{g}} \text{ sur } \underline{\mathrm{Parf}} \\ & & & & \mathrm{HOT}^{o}_W(S) \approx (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}. \end{array}$$

On voit par là qu'on a des isomorphismes canoniques [page 64]

$$\rho i \simeq \mathrm{id}, \qquad \sigma j \simeq \mathrm{id}$$

dans la catégorie des endofoncteurs de $(\mathrm{Cat}/S)W_S^{-1}$ ($\approx \mathrm{HOT}^{\mathrm{lc}}_W(S)$), isomorphisme qui s'explicite ici par le fait que ρi, σj sont déduits du même foncteur C par passage aux localisés W_S^{-1} (donc en fait $\rho i = \sigma j$!), et qu'on a une flèche fonctorielle en X

$$X \xrightarrow{i_X} C(X),$$

qui est dans W pour tout X. Ainsi, ρ et σ sont à isomorphisme près des *rétractions* des catégories $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^{o}_W(S)$ sur la catégorie $\mathrm{HOT}^{\mathrm{lc}}_W(S)$. De plus, ces rétractions sont ce qu'on appelle des foncteurs de localisation (comme on voit sur le diagramme de la page 62).

(On dit que $C \xrightarrow{F} C'$ est un *foncteur de localisation* si, désignant par $\Sigma \subseteq \mathrm{Fl}\, C$ l'ensemble des flèches de C transformées en des isomorphismes par F, le foncteur

$$C\Sigma^{-1} \xrightarrow{\bar{F}} C'$$

qui factorise F est une équivalence de catégories.)

On sait d'ailleurs que si un foncteur f a un adjoint g (à gauche ou à droite), alors f est un foncteur [de] localisation si et seulement si l'adjoint g est pleinement fidèle. Pour prouver que i, j sont pleinement fidèles, il serait donc tentant de prouver qu'ils sont adjoints des foncteurs ρ, σ. [page 65]

Commentary · Exposition [ms. 64]

What the section of localization still has to prove

Grothendieck has now produced the same locally constant category from both comma directions. The identity of Cat/S induces

S)−1] −→(Cat/S)[W −1

ρ : (Cat/S)[(W d S], and path replacement induces a functor in the reverse direction. The equation ρi ≃id says that the coarse category is a retract. It does not yet say that i is fully faithful; this is the distinction already encountered at manuscript p. 38.

The required additional datum is now visible. There is a natural arrow X →C(X) in the fine category, and it becomes invertible in the coarse category. Its direction suggests a unit for ρ ⊣i, not for the opposite adjunction. The inverse coarse arrow C(X) →X then supplies a counit. Grothendieck's change of orientation on p. 67 is consequently forced by where the unit is invertible.

A localization with a fully faithful right adjoint is a reflection onto a full subcategory. In this case its image consists of the relative homotopy types whose transport along every base arrow is invertible. The adjective "locally constant" therefore describes an intrinsic condition on a diagram, not constancy of a chosen presentation of its total category. The path construction supplies a representative satisfying that condition and a universal map to it. The argument on the next pages establishes that universality through the two triangle identities, so that no fullness conclusion is extracted merely from a retraction.

Pour nous y reconnaître, je vais poser

$$\begin{cases} \mathcal{A} = \mathrm{Cat}/S, \quad \Sigma_0 = W_S^{\mathrm{d}}, \quad \Sigma = W_S & (\subseteq \mathrm{Fl}(\mathcal{A})) \\ C : \mathcal{A} \longrightarrow \mathcal{A} \quad \text{le foncteur familier, qui a la vertu} \\ \qquad\qquad C(\Sigma) \subseteq \Sigma_0 \quad . \end{cases}$$

On a donc la situation p. 40

$$(*) \qquad (\mathcal{A}, \Sigma_0) \underset{C}{\overset{\rho_0 = \mathrm{id}_{\mathcal{A}}}{\rightleftarrows}} (\mathcal{A}, \Sigma),$$

induisant par passage aux localisés

$$\mathcal{A}\Sigma_0^{-1} \underset{\bar{C}}{\overset{\rho = \bar{\rho}_0}{\rightleftarrows}} \mathcal{A}\Sigma^{-1},$$

on veut vérifier que ρ et C [plutôt $\bar{C}$] sont adjoints l'un de l'autre, C [plutôt $\bar{C}$] *sûrement à droite,* ce qui signifie que l'on doit avoir des homomorphismes fonctoriels

$$\begin{cases} \mathrm{id}_{\mathcal{A}\Sigma^{-1}} & \rightleftarrows & \rho\bar{C} & (= \overline{\rho_0 C}) \\ \mathrm{id}_{\mathcal{A}\Sigma_0^{-1}} & \rightleftarrows & \bar{C}\rho & (= \overline{C\rho_0}) \end{cases}$$

([41]) [les flèches en bas vont en direction corrigée, mais les flèches en haut ne sont pas effacées] satisfaisant deux relations de compatibilité sur lesquelles on reviendra. On peut espérer trouver ces homomorphismes fonctoriels à partir de

$$\begin{cases} \mathrm{id}_{\mathcal{A}} & \xrightarrow{i} & \rho_0 C & = & C \\ \mathrm{id}_{\mathcal{A}} & \xleftarrow{p} & C\rho_0 & = & C \end{cases}$$

et dans notre cas on dispose bel et bien de [page 66]

$$i : \mathrm{id}_{\mathcal{A}} \longrightarrow C \qquad \text{i.e. de } \; i_X : X \longrightarrow C(X)$$

fonctoriel en $X \in \mathrm{Ob}\,\mathcal{A}$ (et de plus i_X est dans Σ). Il n'y a pas cependant de flèche naturelle en sens inverse p_X

$$C(X) \xrightarrow{?} X.$$

J'avais songé à la projection $\pi_X : C(X) \longrightarrow X$, déduite de la construction $C(X) \simeq X \times_S (\underline{\mathrm{Ch}}_\infty(S), \sigma)$, mais ce n'est malheureusement pas un S-morphisme. Comme on cherche une flèche dans $\mathcal{A}\Sigma_0^{-1}$, on pourrait se contenter d'avoir une flèche fonctorielle en sens inverse, du moment qu'elle soit dans Σ_0, donc puisse s'inverser quand on travaille dans $\mathcal{A}\Sigma_0^{-1}$. Le candidat evident est $i_X : X \longrightarrow C(X)$ dont on vient de parler. L'ennui, c'est que i_X est seulement dans Σ, pas dans Σ_0 ([42]). Pourtant, quand X est lui-même de la forme $C(Y)$ (plus généralement, s'il est un Hot-fibré sur S), alors le fait que $X \longrightarrow C(X)$ soit dans Σ implique déjà qu'il soit dans Σ_0. Notons donc ceci

$$\begin{cases} i : \mathrm{id}_{\mathcal{A}} \longrightarrow C \text{ tel que} & 1^o) & i_X : X \longrightarrow C(X) \textit{ dans } \Sigma \\ & & \text{pour tout } X \in \mathrm{Ob}\,\mathcal{A} \\ & 2^o) & i_X \text{ dans } \Sigma_0 \text{ si } X \text{ de la} \\ & & \text{forme } C(Y) \end{cases}$$

L'idée me vient, après des essais merdouillants [sic], que $\bar{C}$ pourrait être l'adjoint à droite, et non à gauche [c'est la version corrigée (droite et gauche échangées) cf. p. 65], i.e. que les flèches cherchées pourraient aller en sens inverse p. 41 [page 67]

$$\begin{cases} \mathrm{id}_{\mathcal{A}\Sigma^{-1}} & \xleftarrow{p} & \rho\bar{C} & (= \overline{\rho_0 C}) \\ \mathrm{id}_{\mathcal{A}\Sigma_0^{-1}} & \xrightarrow{i} & \bar{C}\rho & (= \overline{C\rho_0}) \end{cases}$$

(j'inverse du coup les notations pour p et i), donc on est amené si possible à chercher des flèches

$$\begin{array}{lcll} \mathrm{id}_{\mathcal{A}} & \xleftarrow{p} & \rho_0 C & = C \\ \mathrm{id}_{\mathcal{A}} & \xrightarrow{i} & C\rho_0 & = C, \end{array}$$

donc des flèches fonctorielles en X

$$X \xrightarrow{i_X} C(X) \xrightarrow{p_X} X.$$

On tient bel et bien une i_X fonctorielle - toujours la même! Pour p_X, on n'en a pas plus qu'avant, mais se rappelant qu'il suffit de le trouver dans $\mathcal{A}\Sigma^{-1}$ (et non plus $\mathcal{A}\Sigma_0^{-1}$!), il suffit

41. mais si C est à *droite,* les flèches ici doivent aller en sens inverse et c'est bien à quoi je dois me résoudre p. 67.

42. S'il n'en était pas ainsi, le type d'homotopie relatif sur S défini par X/S serait localement constant (puisqu'il en est ainsi pour celui défini par $C(X)$), ce qui n'est pas toujours le cas.

d'inverser la flèche i_X, qui est en effet un *isomorphisme* dans $\mathcal{A}\Sigma^{-1}$ (mais non dans $\mathcal{A}\Sigma_0^{-1}$). On posera donc

$$p_X = i_X^{-1} : C(X) \longrightarrow X \qquad \text{dans } \mathcal{A}\Sigma^{-1}.$$

Il faut prouver deux compatibilités seulement, pour les foncteurs $\bar{C}$ et ρ entre $\mathcal{A}\Sigma_0^{-1}$ et $\mathcal{A}\Sigma^{-1}$ (et j'écrirai C au lieu de $\bar{C}$) : les deux composés

$$\begin{cases} \rho \xrightarrow{\rho * i} \rho C \rho \xrightarrow{p * \rho} \rho \overset{\text{id}}{\cdots\cdots} \rho & \text{(relation pour foncteurs à valeurs dans } \mathcal{A}\Sigma^{-1}) \\ C \xrightarrow{i * C} C\rho C \xrightarrow{C * p} C \overset{\text{id}}{\cdots\cdots} C & \text{(relation pour foncteurs à valeurs dans } \mathcal{A}\Sigma_0^{-1}) \end{cases}$$

doivent être l'identité id_ρ resp. id_C.

La première relation se vérifie tautologiquement, elle revient à dire que dans $\mathcal{A}\Sigma^{-1}$, on a [page 68] un composé identité des

$$X \xrightarrow{i_X} C(X) \xrightarrow{i_X^{-1}} X \qquad !$$

La seconde signifie que pour tout X, on a un composé identité dans $\mathcal{A}\Sigma_0^{-1}$

$$C(X) \xrightarrow{i_{C(X)}} C(C(X)) \xrightarrow{C(p_X)} C(X).$$

Or $p_X : C(X) \longrightarrow X$ avait été défini dans $A\Sigma^{-1}$ comme inverse i_X^{-1} de i_X, d'autre part le foncteur C sur $\mathcal{A}$ définit par hypothèse $\mathcal{A}\Sigma^{-1} \longrightarrow \mathcal{A}\Sigma_0^{-1}$, plus précisément $C(\Sigma) \subseteq \Sigma_0$ ainsi p. 42 $C(i_X)$ est inversible dans $\mathcal{A}\Sigma_0^{-1}$, et $C(p_X) = C(i_X)^{-1}$. Donc la compatibilité revient à la relation

$$C(i_X)^{-1} i_{C(X)} = \mathrm{id} \qquad \textit{dans } \mathcal{A}\Sigma_0^{-1},$$

i.e.

$$\boxed{i_{C(X)} = C(i_X)} : C(X) \longrightarrow C^2(X).$$

En d'autre termes, il faut prouver le :

Commentary · Exposition [ms. 67]

A precise reflection lemma

The remaining argument is formal once the two classes of arrows and the two uses of the unit are distinguished. We state the lemma in the form needed here; it also shows that the iterated-unit equality can be established without further geometry. This is the ordinary categorical form of the recognition principle for reflective localizations; compare Lurie (2009, Proposition 5.2.7.4) for its higher categorical formulation.

Lemma. Let Σ0 ⊆Σ be classes of arrows of a category A, and suppose the localizations below exist. Let C : A →A and η : id →C satisfy

C(Σ) ⊆Σ0,

ηX ∈Σ, ηCX ∈Σ0 (X ∈A). Then the identity and C induce functors

P : A0 = A[Σ−1

0] −→A1 = A[Σ−1], R : A1 −→A0, with P ⊣R and R fully faithful. The unit is induced by η, and the counit is the inverse of its image in A1.

Since C(Σ0) ⊆Σ0, the endofunctor C and its natural transformation descend to an Proof. endofunctor T = RP and a transformation e : idA0 →T. Naturality extends across the formal inverses because both functors invert the required arrows. At every object x, both

a = eT x : Tx →T 2x,

b = T(ex) : Tx →T 2x are isomorphisms: the first by the third hypothesis, the second by the first two. We claim that a = b.

Naturality at ex gives bex = aex. Apply T and cancel the isomorphism b on the right to obtain T(b) = T(a). Naturality at a gives

T(a)a = eT 2xa,

so T(a) = eT 2x. Naturality at b gives
T(b)a = eT 2xb.

Substitute the preceding equalities and cancel the isomorphism T(a). This proves a = b. Notice that we have never cancelled ex, which need not be invertible in A0.

Let ϵ : PR →idA1 be the inverse of the transformation induced by η in A1. Its first triangle identity is ϵP xP(ex) = idP x, true by construction. For the second, every object of A1 is represented by an object X of A. On its image under R, the composite R(ϵX)eRX is b−1a, which is the identity by the calculation above. Hence P ⊣R. The counit is an isomorphism, so R is fully faithful. □

The full image is exactly the class of x for which ex is invertible. Objects in the image satisfy this by the third hypothesis; conversely, invertibility identifies x with RPx. Applied to Σ0 = W d S and Σ = WS, this proves the desired full faithfulness of i. The left-comma version proves the corresponding assertion for j, with the covariant convention explained at p. 50. Grothendieck's concatenation argument is a particularly transparent geometric verification of the same identity, and remains useful for understanding the replacement itself.

Lemme. *Soit X dans* Cat/S*, considérons les deux flèches*

$$i_{C(X)},\ C(i_X) : C(X) \rightrightarrows C(C(X)).$$

Ces flèches sont égales, sinon au sens absolu, du moins dans $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$.

Ici $f_X : X \longrightarrow S$ est la projection standard et

$$C(X) \;=\; \{(x,c) \;\mid\; x \in \mathrm{Ob}\, X,\ c : f_X(x) \underbrace{\longrightarrow}_{\text{chemin dans } S} t\}$$

$$C^2(X) \;=\; \{(x,c,c') \;\mid\; x \in \mathrm{Ob}\, X,\ \underbrace{f_X(x) \xrightarrow{c} t \xrightarrow{c'} t'}_{c,\ c' \text{ chemins dans } S}\},$$

les projections dans S sont données par

$$\begin{aligned} (x,c) &\longmapsto t = \text{but } c \\ (x,c,c') &\longmapsto t' = \text{but } c' \end{aligned}$$

On trouve [page 69]

$$i_{C(X)}(x,c) = (x,c,1_t) : \quad \begin{array}{c} x \\ \downarrow \\ f_X(x) \xrightarrow{c} t \underset{\text{chemin vide}}{\xrightarrow{1_t}} t \end{array}$$

$$C(i_X)(x,c) = (x,1_{f_X(x)},c) : \quad \begin{array}{c} x \\ \downarrow \\ f_X(x) \underset{\text{chemin vide}}{\xrightarrow{1_{f_X(x)}}} f_X(x) \xrightarrow{c} t \end{array}$$

Mais on a le choix entre deux théories de chemins, pour construire $C(X)$: soit avec les $\underline{\mathrm{Ch}}_\infty(S)$, soit avec les $\underline{\mathrm{Ch}}(S)$. 'Ça donne les mêmes résultats', mais suivant les situations l'un est plus commode que l'autre. Je vais travailler avec les $\underline{\mathrm{Ch}}(S)$, auquel cas les 1_S sont des unités à gauche et à droite, et on a une loi de composition pour les chemins. Cette loi nous permet de définir

$$\begin{aligned} C^2(X) &\xrightarrow{\gamma} C(X) \\ (x,c,c') &\longmapsto (x, c' \circ c), \end{aligned}$$

et les composés de $i_{C(X)}$ et de $C_{i(X)}$ avec cette flèche sont $\mathrm{id}_{C(X)}$ dans les deux cas p. 43

$$C(X) \rightrightarrows C^2(X) \xrightarrow{\gamma} C(X). \qquad \text{(id)}$$

[page 70]

Donc pour montrer que $i_{C(X)} = C(i_X)$ dans $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$, il suffit de prouver que γ est dans W_S^{d}. Mais on sait que le foncteur C transforme W_S en W_S^{d}, donc $C(i_X)$ est dans W_S^{d}, et comme $\gamma \circ C(i_X) = \mathrm{id}$, γ est dans W_S^{d}, OK!

J'ai envie de grouper ce qui a été prouvé dans un énoncé récapitulatif.

Commentary · Exposition [ms. 69]

Concatenation and the two iterated units

Grothendieck's path model makes the reflection tangible. An object of C(X) is an object x together with a path from p(x) to its new base point. Iterating C adds a second path. The two units place the empty path in different positions, so there is no reason for the resulting functors to be literally equal before localization.

Using the finite-path version with strict empty-path units, concatenation gives a functor over S (x, c, c′) 7−→(x, c′c).

μX : C2(X) −→C(X), On morphisms it concatenates the two compatible path diagrams; compatibility of their endpoints makes this well-defined, and composition is respected because it is respected in each path segment. The two empty-path laws give the strict equations

μXηC(X) = idC(X),

μXC(ηX) = idC(X). The relevant homotopical input is C(WS) ⊆W d S. Since ηX ∈WS, we have C(ηX) ∈W d S. Two-out-ofthree applied to the second equation yields μX ∈W d S. After fine localization both ηC(X) and C(ηX) are therefore the inverse of the same arrow μX, which proves their equality.

This proof does not cancel the unit X →C(X) in the fine category. That would erase precisely the distinction the chapter has worked to establish. Instead it cancels a map between already completed objects. Nor does the argument require the associativity of concatenation: the two unit identities and two-out-of-three suffice for the triangle equation. Associativity equips C with the familiar monadic structure, but the adjunction needs only the part displayed here.

The formal lemma in the preceding block shows that the equality also follows from the stated hypotheses on pp. 65–66. Thus there is no outstanding coherence gap in this part of the proof. The choice of finite paths supplies an explicit common inverse and exhibits why adjoining paths becomes idempotent after the appropriate localization.

Commentary · Later developments [ms. 69]

Path completion and the modern category of local systems

The theorem reaches its most familiar modern form when W = W∞. Let S denote the infinitycategory of spaces, and let

ℓ: N(S)op −→Π∞N(S) be groupoid completion; we identify the completion of N(S)op with that of N(S) by reversing paths. Then the fine theory models homotopy coherent contravariant space-valued diagrams, and the locally constant theory models diagrams which carry every base arrow to an equivalence:

HOTW∞(S) ≃Ho Fun(N(S)op, S),

HOTlc

W∞(S) ≃Ho Fun(Π∞N(S), S). The first comparison can be justified through an actual model equivalence. Thomason's adjunction

c Sd2 : sSet ⇄Cat : Ex2 N

is a Quillen equivalence for nerve weak equivalences (Thomason, 1980, Theorem 4.9 and the introduction). Apply both functors objectwise to Sop-diagrams with their projective model structures. The right adjoint preserves fibrations and trivial fibrations objectwise. If F is projectively cofibrant and G projectively fibrant, the criterion for an adjoint pair of maps to be weak equivalences is checked at each object using the original Quillen equivalence; projective cofibrancy implies objectwise cofibrancy. Thus the diagram adjunction is again a Quillen equivalence. The simplicial-set diagram model then presents the coherent functor category by Lurie (2009, Proposition 4.2.4.4). This explains why strict category-valued diagrams, after objectwise localization, retain coherent space-valued diagrams.

Here Ho is taken after forming the category of coherent diagrams. It is not the category of ordinary functors into Ho(S). The higher transport information discussed earlier in the chapter is retained by this order of operations.

Restriction $\ell*$is fully faithful by the universal property of groupoid completion. Its left adjoint is left Kan extension $\ell!$, and the reflector on diagrams is $\ell*\ell!$. This identifies the universal property of Grothendieck's C, rather than asserting that the particular path-category construction is literally a chosen infinity-categorical Kan extension. The standard straightening comparison and the adjunction for Kan extension are given in Lurie (2009, Theorem 2.2.1.2 and Proposition 4.3.3.7); the reflection for a left proper combinatorial model category of coefficients is Cisinski (2009, Corollary 2.5).

A local system of spaces on the infinity-groupoid Π∞N(S) can equivalently be viewed as a space over |N(S)|, by taking its total space and homotopy fibres. Consequently the coarse theory has the interpretation

HOTlc

W∞(S) ≃Ho(S/|N(S)|). For a direct model-categorical comparison see Cisinski (2009, Example 1.7, equation (1.7.1)). This is a homotopy category of a slice, and not generally the ordinary slice of a homotopy category. The distinction already matters for a base with nontrivial fundamental group, as the following example shows.

These interpretations explain what became of the full-faithfulness question: for spaces it is settled by the reflector, and more general coefficient theories have their own locally constant subcategories and reflection results. They do not justify identifying an arbitrary W-coarse localization with that subcategory. Such an identification requires the additional exactness properties discussed on pp. 43–45. The existence of a locally constant reflection and the adequacy of a particular coarse test are separate assertions.

Commentary · Exposition [ms. 70]

Three theories within the comparison diagram

Grothendieck now wants the many presentations to express one intelligible distinction. The proof that supplies it has already been explained in the commentary at manuscript pp. 64, 67, and 69: the path replacement defines a section of the coarse localization, and the two iterated units agree after fine localization. Here the new work is to organize the consequences and identify the image of that section.

For a map $v : X \to Y$ over S, the three tests are

$$\begin{array}{l|l} W_S & v \in W \text{ on total categories,} \\ W_S^{\mathrm{d}} & s\backslash X \longrightarrow s\backslash Y \in W \text{ for every } s, \\ W_S^{\mathrm{g}} & X/s \longrightarrow Y/s \in W \text{ for every } s. \end{array}$$

The relative Theorem A supplies the inclusions of both fine classes in W_S. Thus the identity on Cat/S gives ρ and σ; these are quotient functors between homotopy categories, before any assertion about adjoints. In the other direction one must first replace X: the functor C sends a coarse equivalence to both kinds of fine equivalence. This is why the inclusion of locally constant objects is represented by a nontrivial construction on total categories.

The replacement by a split fibration does something different. Its fibre at s is $s\backslash X$, so it preserves the right-fine information already present in X. Dually, the split opfibration has fibre X/s. Neither construction forces its transition maps to become equivalences. The path replacement does force this, and consequently belongs to the upper part of the diagram. Confusing these two purposes would make the distinction between $\mathrm{HOT}_W(S)$ and $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ disappear before it has been proved.

Part I is therefore a statement about coarse localization of all the presentations. Part II identifies the appropriate fine localization along each side. Part III says when a strict fibre is a valid test and when the two fine tests agree. The hypothesis L5 is used at precisely the passage from a coarse equivalence between homotopy fibrations to a fibrewise equivalence. It should not be silently dropped when the diagram is reused. The subsequent remarks at manuscript pp. 81–82 explicitly reopen this dependency.

Théorème-Scholie. On considère le diagramme de catégories de la page 8 ([43]).

Ⓘ Quand on munit toutes les 13 catégories du diagramme du localiseur W_S (W-équivalence grossière, i.e. W-équivalence dans Cat, sans plus), *tous* les foncteurs du diagramme induisent

43. On suppose sur W les conditions a) b) c) de p. 45, de plus L5 (p. 43). Cf. commentaires p. 81.

des équivalences sur les catégories localisées ([44]). Désignons par

$$\mathrm{HOT}^{\mathrm{lc}}_{W}(S)$$

'la' catégorie localisée commune.

(II) Munissons les quatre catégories bordantes gauches

$$(*) \qquad \mathrm{Cat}/S \longleftarrow \underline{\mathrm{Liss}}\, S \longleftarrow \underline{\mathrm{Fib}}\, S \longleftarrow \underline{\mathrm{Fibsc}}\, S$$

du localiseur W^{d}_S de la W-équivalence fine *droite* sur S ($\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \quad \swarrow & \\ & S & \end{array}$ est dans W^{d}_S si et seulement si les $s\backslash X \longrightarrow s\backslash Y$ le sont `[plutôt, sss ils sont dans` W`]`, pour tout $s \in \mathrm{Ob}\, S$). Les foncteurs du diagramme, i.e. ceux de $(*)$, entre ces catégories, donc aussi leurs composés, induisent des équivalences entre les localisées ([45]). Désignons par

$$\mathrm{HOT}_{W}(S)$$

'la' catégorie localisée commune.

Enoncé dual pour

$$(*') \qquad \mathrm{Cat}/S \longleftarrow \underline{\mathrm{Prop}}\, S \longleftarrow \underline{\mathrm{Cofib}}\, S \longleftarrow \underline{\mathrm{Cofibsc}}\, S,$$

Commentary · Examples [ms. 70]

Two computations of the reflection

The reflection records how the base arrows must become invertible. For the interval S = [1], a contravariant diagram is an arrow B −−→uA. Since |N[1]| is contractible, locally constant diagrams are equivalent to constant diagrams. The reflector is

−−→uA) 7−−→(A

−−→A), (B with unit given by u at 1 and the identity at 0. To verify its universal property at the coherent level, a map from B →A to the constant diagram on K consists of maps A →K, B →K, and a specified homotopy relating the second to the composite through A. The space of choices of the second map together with such a homotopy is contractible over the first. Thus the mapping space is naturally equivalent to Map(A, K), exactly as required. This argument also includes empty values.

For example, the diagram ∅→∗becomes the constant point diagram. Its total Grothendieck construction is the category supported at 0, while that of the reflected diagram is [1]. Their total nerves are equivalent, although the unit fails to be an objectwise equivalence at 1. This is the earlier counterexample viewed as the unit of an adjunction. The inclusion of constant diagrams also has a right adjoint, sending B →A to B: the dual mapping-space computation gives Map(K, B). The left and right adjoints are distinct constructions even for this smallest base.

For a second computation take the one-object groupoid S = BZ. Every arrow of the base is already invertible, so every coherent diagram on S is locally constant and the reflection is an equivalence. Nevertheless the coarse slice contains more morphisms than the ordinary slice of Ho(S). Consider the object z : ∗→BZ given by a base point. In the infinity-categorical slice,

EndHo(S/BZ)(z) ∼= Z.

Map/BZ(z, z) ≃ΩBZ, Indeed, the underlying map ∗→∗is unique, but being a map over BZ includes a path from the chosen base point to itself. Homotopy classes of such paths are the integers. In the ordinary category Ho(S)/BZ, the same object has only its identity endomorphism, because the over-condition is imposed merely as an equality of homotopy classes.

44. De plus, si $\mathcal{F}$ est une sous-catégorie pleine de Cat/S qui contient une des deux sous-catégories $\underline{\mathrm{Fib}}_0 S$, $\underline{\mathrm{Cofib}}_0 S$ (ce qui inclut le cas des 8 catégories des diagrammes, autres que Cat/S elle-même), l'inclusion $(\mathcal{F}, W_S|\mathcal{F})$ dans $(\mathrm{Cat}/S, W_S)$ induit une équivalence pour les localisés.

45. Plus généralement, ça marche pour toute sous-catégorie pleine de Cat/S qui contient $\underline{\mathrm{Fib}}\, S$ (et dualement, avec W^{g}_S au lieu de W^{d}_S, si $\mathcal{F}$ contient $\underline{\mathrm{Cofib}}\, S$).

The associated local system has fibre Z with its regular translation action. A translation-equivariant map Z →Z is determined by the image n of 0 and is k 7→k + n; composition adds the integers. This recovers the endomorphism calculation directly from local systems. Path completion of the point over BZ supplies the categorical analogue of its universal-cover replacement, recovering exactly the transport that the ordinary slice would forget.

Overview · VI-a

The problem of the chapter. Grothendieck starts from a category of coefficients, here the category of small categories itself, and asks how its homotopy theory should vary with a diagram shape. The value at a small category S is to be

HOTW (S) = Fun(Sop, Cat)[W −1

obj], where a natural transformation belongs to Wobj when all its components belong to W. The essential operation is to localize the category of diagrams before forgetting their coherence. A functor from Sop to the already localized category of coefficients generally retains less information. This is the same motivation that led to the diagram formulation in Grothendieck (1983, §69), and that the letter to Thomason subsequently places at the centre of the derivator programme (Grothendieck, 1991). The discussions of coherence and evaluation in the commentary on Chapter I, manuscript pages 7, 17–18 and 27, remain the background; the present chapter supplies categorical constructions rather than repeating those objections.

The first seventy manuscript pages, dated 14–15 November 1990, record the discovery of the correct comparison theorem. Thirteen categories of objects over S initially appear to offer interchangeable models. They include split fibrations, arbitrary fibrations, smooth and proper functors, homotopy fibrations, perfect functors, and all of Cat/S. The objects alone do not decide the homotopy theory. The permitted morphisms and the chosen weak equivalences matter just as much. The first theorem overlooks that distinction, is expressly withdrawn, and is replaced by a comparison between three theories. The manuscript's revisions are part of the proof's development, not noise to remove from it. The axioms and the four tests. The axiom list also changes. The original L4 is an absolute Theorem A: an aspherical functor is a weak equivalence. L4 bis, introduced on manuscript page 44, is the relative assertion about a map X →Y over a third category S, tested on every map X/s →Y/s. It is the latter formulation that supports the relative descent arguments. The initial L5 passes from fibrewise weak equivalences of Cartesian maps between fibrations to weak equivalences of total categories. The new L5 on page 43 goes in the opposite direction for maps between homotopy fibrations. Their common label must not obscure their different content. The commentary proves the initial claim that L1, L2 bis and the original L5 already imply the required elementary axioms, including relative descent, by a twisted-arrow argument for opposite invariance.

For a map f : X →Y over S, the decisive tests are

coarse

f ∈W,

fibrewise fs : Xs →Ys belongs to W for every s,

left-fine

f/s : X/s →Y/s belongs to W for every s,

right-fine

s\f : s\X →s\Y belongs to W for every s. Over a smooth object, the strict fibre compares to the undercategory; over a proper object, it compares to the overcategory. These comparisons and relative descent explain when fibrewise equivalence implies a universal equivalence of totals. The modern asphericity formulation retains precisely this mechanism (Maltsiniotis, 2005, Theorem 1.2, Corollary 4.16 and Theorem 5.4). Coarse equivalence alone does not give either fine condition on the external borders of the diagram. Over [1], the inclusion {0} →[1] is a coarse equivalence of split fibrations, but its fibre at 1 is the map from the empty category to a point. Evaluation at that fibre proves that the two localizations themselves differ. The two comma resolutions. The functor Φ replaces X →S by the split fibration whose fibre at s is s\X; Ψ gives the split opfibration with fibre X/s. Their units are categorical homotopy equivalences on total categories. More precisely, the unit of Φ is right-fine, and the unit of Ψ is left-fine. Composition of the two arrows in a comma object gives an explicit contraction proving this

stronger fact. Consequently the right-fine localization of Cat/S computes the contravariant diagram theory, while the left-fine localization computes its covariant counterpart.

The comparison with chosen splittings needs a genuine repair of the argument, but not a change in the desired theorem. A Cartesian functor need not preserve the selected Cartesian lifts strictly, and an arbitrary functor over the base need not preserve Cartesian arrows at all. The actual free split adjunction is

Homsplit/S(ΦX, Y) ∼= HomS(X, UY).

Its counit supplies the comparison in the strict category. For arbitrary functors between fibrations the pullback retraction has a canonical, possibly noninvertible natural comparison. Natural transformations over S induce equality after fine localization, since the cylinder projection A × [1] →A is fine. This yields the natural isomorphism needed after localization. The exposition constructs the adjunction and the localized comparison in full; the historical and modern strictification references are separated from this particular elaboration (Gray, 1966; Emmenegger et al., 2024).

The covariant theory is HOTW (Sop). The additional outer opposite that appears in several passages of the edition is incorrect with the initial definition above: taking opposite total categories preserves the direction of functors between them. The corrected variance is used throughout the commentary, while both the original statements and the editorial brackets are preserved in the source. The strength of path completion. The two-sided path replacement gives a natural map X →C(X) which is a weak homotopy equivalence, with C(X) →S perfect. This immediately proves comparisons for the coarse localizations, without the new L5. To identify coarse and fibrewise equivalences on the internal categories, and then identify their common theory with a full subcategory of the fine theory, is stronger. That step is what the new L5 supplies.

Its restriction on W is severe. A degree-n covering of the circle is an equivalence for homology with coefficients in a ring where n is invertible, but its fibre has n components. The commentary realizes the covering as an action groupoid, so that perfectness and the strict fibre can be checked

directly. For integral homology it gives a different explicit example:

Z[1/2] ⋊2 Z

−→BZ. B The fibre has first homology Z[1/2]. The mapping-torus sequence shows that the total projection is nevertheless an integral homology equivalence, because 1 −2 acts invertibly on that fibre homology. This calculation settles the tentative integral-coefficient assertion on manuscript page 43. The development label marks the supplied application and proof, with a bounded account of the literature checked; it claims no priority for the group or the homology calculation.

Finite homotopy truncations fail as well: a path fibration over K(A, n + 1) has contractible total space and fibre K(A, n). Retaining the standing condition W∞⊆W ⊆W0, the new L5 actually forces W = W∞. Iterating path fibres shows that every W-aspherical category has connected iterated loop spaces, hence is weakly contractible; applying this to the fibres of a replacement of an arbitrary W-map proves the conclusion. The shortened annotation omitting the component condition cannot stand literally, since the maximal class of all functors is a counterexample. Cisinski's later characterization explicitly includes the corresponding condition on components (Cisinski, 2004, Theorem 2.3.6). The reflection and its modern meaning. The final argument proves full faithfulness through an adjunction. Put

Σ0 = W d

A = Cat/S, S ⊆Σ = WS. Path completion carries Σ into Σ0, its unit ηX belongs to Σ, and ηCX belongs to Σ0. These conditions give

P : A[Σ−1

0] ⇄A[Σ−1] : R, P ⊣R, with R fully faithful. The unit is induced by η, and the counit is its inverse in the coarse category. The commentary verifies both triangle identities. In

particular, the equality C(ηX) = ηCX after fine localization follows formally from naturality and the stated invertibilities. Grothendieck's concatenation of paths gives an additional geometric proof by providing a common inverse. There is no need to cancel ηX in the fine category, where it need not be invertible.

For W = W∞, the resulting interpretation is

HOTW∞(S) $\simeq$Ho Fun(N(S)op, S),

HOTlc

W∞(S) $\simeq$Ho Fun($\Pi\infty$N(S), S) $\simeq$Ho(S/|N(S)|). Here S is the infinity-category of spaces, and the homotopy category is taken after forming coherent diagrams or the slice. The reflector is restriction after left Kan extension along groupoid completion. The commentary explains the comparison through Thomason's model and straightening (Thomason, 1980; Lurie, 2009). It also computes the interval reflector and the endomorphisms of the point over BZ, which distinguish the homotopy category of the slice from the ordinary slice of the homotopy category.

The locally constant embedding and its two adjoints are now theorems in the derivator setting (Cisinski, 2009, Proposition 3.10 and Theorem 3.13). Their existence does not identify an arbitrary coarse W-localization with that subcategory. The surviving qualification is about hypotheses, not an unresolved version of the full-faithfulness problem for spaces. We also do not label the

relationship between the weakest absolute and relative axiom packages as a current open problem: this audit establishes the implications used here and records where a weaker package is insufficient for the printed proof. The construction of the path categories themselves is an input from outside these pages. The first part of the chapter ends in the opening clauses of the theorem–scholium on manuscript page 70; the continuation on pages 71–81 is commented in the second part.

et les localiseurs W_S^{g} dans ces catégories ($f \in W_S^{\mathrm{g}}$ si et seulement si les $f/s : X/s \longrightarrow Y/s$ sont dans W). On trouve une 'même catégorie localisée', notée p. 44

$$\mathrm{HOT}_W^o(S),$$

et on a tautologiquement

$$\mathrm{HOT}_W^o(S) \approx \mathrm{HOT}_W(S^o). \qquad \text{[page 71]}$$

[plutôt $(\mathrm{HOT}_W(S^o))^o$ cf. p. 50]

(III) Sur les six catégories internes du diagramme, les trois notions de W-équivalence, en termes de W_S, W_S^{d}, W_S^{g} coïncident, et coïncident avec la W-équivalence par fibres W_S^{f}. Sur les trois catégories de $(*)$ autres que Cat/S, les localiseurs W_S^{f}, W_S^{d} coïncident; sur les trois catégories de $(*')$ autres que Cat/S, les localiseurs W_S^{f}, W_S^{g} coïncident. Mais en général ([46]), sur ces six catégories, ces localiseurs W_S^{f} ne sont pas égaux à W_S, plus précisément donnent des localisées non équivalentes, elles sont donc munies chacune de *deux* localiseurs essentiellement distincts, W_S^{f} et W_S. La catégorie Cat/S est la seule catégorie du diagramme sur laquelle les localiseurs W_S, W_S^{d} et W_S^{g} sont tous les trois essentiellment distincts. C'est la seule qui permette d'obtenir simultanément les trois catégories localisées, comme

$$\begin{array}{ccccc} & & \mathrm{HOT}_W^{\mathrm{lc}}(S) \simeq (\mathrm{Cat}/S)W_S^{-1} & & \\ & \overset{\rho}{\nearrow} & & \overset{\sigma}{\nwarrow} & \\ \mathrm{HOT}_W(S) \simeq (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} & & & & \mathrm{HOT}_W^o(S) \simeq (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1} \end{array}$$

Commentary · Exposition [ms. 71]

Opposite categories and opposite derivators

The dual statement is meant to recover the covariant theory from the contravariant one. At the level of the ordinary localized categories used here, its variance can be checked directly. Taking opposites gives a functor

$$O_S : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S^{\mathrm{op}}, \qquad (X \to S) \longmapsto (X^{\mathrm{op}} \to S^{\mathrm{op}}),$$

which sends $v : X \to Y$ to $v^{\mathrm{op}} : X^{\mathrm{op}} \to Y^{\mathrm{op}}$, with the same direction. Moreover

$$s\backslash(X^{\mathrm{op}}) \cong (X/s)^{\mathrm{op}}.$$

For the opposite-invariant localizer under consideration, O_S interchanges the left-fine and right-fine classes. Its square is the identity. Consequently the localization definitions of this Scholie give

$$(\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1} \simeq (\mathrm{Cat}/S^{\mathrm{op}})(W_{S^{\mathrm{op}}}^{\mathrm{d}})^{-1},$$

with no opposite on the resulting category.

The editorial insertion on this page asks for an additional outer opposite. That insertion belongs to the different convention of an *opposite derivator*, whose value is $\mathbb{D}(S^{\mathrm{op}})^{\mathrm{op}}$; it is not compatible with identifying both sides here with the displayed ordinary localizations. We preserve the edition's insertion, and distinguish the conventions rather than altering its words. See also manuscript p. 50, where the same issue first appears. The two diagram functors at manuscript p. 73 confirm the present interpretation: their values lie in Hot_W, with the index category reversed, rather than in $\mathrm{Hot}_W^{\mathrm{op}}$.

A small check prevents a significant error later. At $S = e$ and for ordinary weak homotopy equivalences, both fine localizations are Ho(Spaces). This category is not equivalent to its opposite: its initial object is strict, since a map to the empty space can only have empty source, whereas its terminal object is not costrict: a map $* \to S^0$ need not be invertible. These properties are preserved by equivalences and exchanged by passage to the opposite. In particular dualizing the present comparison does not reverse the order of the two adjoints: both $\rho \dashv i$ and $\sigma \dashv j$ have the order printed in Part IV. The later hesitation at manuscript p. 84 comes from precisely this variance distinction.

46. sans doute dans tous les cas où S n'est pas équivalente à une catégorie dicrète.

Commentary · Development (commentators') [ms. 71]

The groupoid test for collapse of the comparison

Grothendieck asks in footnote 46 whether the distinction between coarse and fine localization persists for every base not equivalent to a discrete category. A group with one object already shows that the proposed dividing line is too narrow. The relevant issue is the presence of noninvertible arrows, not the presence of nontrivial automorphisms.

Proposition (the canonical comparison). [ours] For $W = W_\infty$, the ordinary weak homotopy equivalences of categories, the canonical localization

$$\rho : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S)$$

is an equivalence if and only if S is a groupoid. The same assertion holds for σ.

We use the established interpretation of the right-fine theory as coherent space-valued presheaves and its locally constant subcategory. Equivalently, one can use Part V and the split-fibration presentation; the proof below needs only discrete representables for the difficult implication. If S is a groupoid, every functor defined on S^{op} sends arrows to equivalences. Hence every object is locally constant, and the fully faithful right adjoint i is essentially surjective. Its left adjoint ρ is therefore an equivalence.

Conversely, suppose ρ is an equivalence. For each object z, consider the discrete presheaf $h_z(s) = \mathrm{Hom}_S(s, z)$ and its category of elements over S. It defines an object of $\mathrm{HOT}_W(S)$. Since this object must be locally constant, precomposition by every $u : a \to b$ induces a weak equivalence of discrete spaces

$$\mathrm{Hom}_S(b, z) \xrightarrow{-\circ u} \mathrm{Hom}_S(a, z).$$

A weak equivalence between discrete spaces is a bijection. With $z = a$, surjectivity gives $v : b \to a$ with $vu = 1_a$. With $z = b$, injectivity applied to uv and 1_b gives $uv = 1_b$, because both become u after precomposition by u. Thus every arrow is invertible. Reversing the argument proves the covariant assertion.

For example, take $S = BG$ with $G \neq 1$. Its unique object has automorphism group G, so BG is not equivalent to a discrete category; nevertheless both canonical comparisons are equivalences. This disproves the printed conjecture even for the ordinary localizer. The proposition concerns the specified comparison functor: detecting a nonconstant representable proves that this functor fails to be an equivalence, without asserting an unrelated classification of possible abstract equivalences between the two categories.

This is our explicit application of Yoneda to the question in the footnote, not a claim of priority. We checked the locally constant comparison in Cisinski (2009, §§1–2) and the presheaf interpretation in Lurie (2009, Theorem 2.2.1.2); neither is being claimed as a new result. We did not find the footnote's proposed criterion stated in the passages checked.

Ici, ρ et σ sont des foncteurs de localisation (comme il est évident sur la définition).

Tout foncteur ou foncteur composé du diagramme page 8, allant d'une catégorie interne au diagramme vers une catégorie bordante, induit un foncteur pour les localisées pour W_S^{d} (quand la catégorie but fait partie des quatre gauches, rappelées dans (∗), p. 70), soit pour W_S^{g} (quand [page 72] ces catégories font partie des quatre du côté droit, rappelées dans (∗′)), lesquels localiseurs coïncident aussi avec W_S^{f} sauf dans le seul cas de la catégorie Cat/S. Tous les foncteurs ainsi obtenus se réduisent, à équivalence près, à deux, suivant que la catégorie localisée but est $\mathrm{HOT}_W(S)$ (cas de la localisation W_S^{d}), ou qu'elle est $\mathrm{HOT}^o_W(S)$ (cas de la localisation W_S^{g}). On trouve ainsi des foncteurs (en sens inverse de ρ et de σ) p. 45

$$\begin{array}{ccc} & \mathrm{HOT}^{\mathrm{lc}}_W(S) & \\ {\scriptstyle i}\swarrow & & \searrow{\scriptstyle j} \\ \mathrm{HOT}_W(S) & & \mathrm{HOT}^o_W(S). \end{array}$$

Les foncteurs ρ, σ s'interchangent par dualité $S \longmapsto S^o$, et de même pour i, j.

(IV) Les foncteurs ρ, i sont adjoints l'un de l'autre dans cet ordre, i.e. on a pour deux

objets $\xi \in \mathrm{ObHOT}_W(S)$, $\eta \in \mathrm{ObHOT}^{\mathrm{lc}}_W(S)$

$$\mathrm{Hom}(\rho(\xi), \eta) \simeq \mathrm{Hom}(\xi, i(\eta))$$

(isomorphisme de bifoncteurs). Et dualement les foncteurs σ, j sont adjoint l'un de l'autre dans cet ordre : pour $\xi \in \mathrm{ObHOT}^{o}_W(S)$, $\eta \in \mathrm{ObHOT}^{\mathrm{lc}}_W(S)$

$$\mathrm{Hom}(\xi, j(\eta)) \simeq \mathrm{Hom}(\sigma(\xi), \eta).$$

Il s'ensuit notamment que *les foncteurs i, j sont pleinement fidèles.* [page 73]

Ⓥ On a des foncteurs évidents

$$\begin{array}{lccc} (*) & \mathrm{HOT}_W(S) & \longrightarrow & \underline{\mathrm{Hom}}(S^o, \mathrm{Hot}_W) \\ (*') & \mathrm{HOT}^{o}_W(S) & \longrightarrow & \underline{\mathrm{Hom}}(S, \mathrm{Hot}_W). \end{array}$$

Le premier s'interprète sur les quatre catégories $(*)$ (bordantes gauches) de page 70, qui servent indifférement à décrire $\mathrm{HOT}_W(S)$, par

$$s \longmapsto \mathrm{hot}_W(X_s)$$

dans le cas des trois catégories distinctes de Cat/S, et par

$$s \longmapsto \mathrm{hot}_W(s\backslash X)$$

dans le cas de Cat/S. La dépendence fonctorielle par rapport à s est évidente dans le cas de Cat/S (donnée par $s \longmapsto \mathrm{hot}_W(s\backslash X)$), et dans le cas de $\underline{\mathrm{Fibsc}}\, S$, $\underline{\mathrm{Fib}}\, S$, où on dispose, pour $u : s \longrightarrow t$ dans S, du changement de base

$$u^* : X_t \longrightarrow X_s.$$

Dans le cas de $\underline{\mathrm{Liss}}\, S$, la dépendence fonctorielle se déduit de celle de $\mathrm{hot}_W(s\backslash X)$, grâce au fait que l'inclusion

$$X_s \longrightarrow s\backslash X$$

est alors une W-équivalence.

On a une interprétation duale de l'application $(*)'$, `[mot illisible]` p. 46

$$s \longmapsto \mathrm{hot}_W(X_s) \qquad \text{ou} \qquad s \longmapsto \mathrm{hot}_W(s\backslash X), \quad \text{[page 74]}$$

avec la loi fonctorielle en s déduite du foncteur $t\backslash X \longrightarrow s\backslash X$ `[plutôt` $X/s \longrightarrow X/t$`]` (pour $u : s \longrightarrow t$) dans le cas du Cat/S, du foncteur cochangement de base dans les cas de $\underline{\mathrm{Cofib}}\, S$ et $\underline{\mathrm{Cofibsc}}\, S$, et de la variante pour Cat/S et de la W-équivalence

$$X_s \longrightarrow X/s$$

dans le cas de $\underline{\mathrm{Prop}}\, S$.

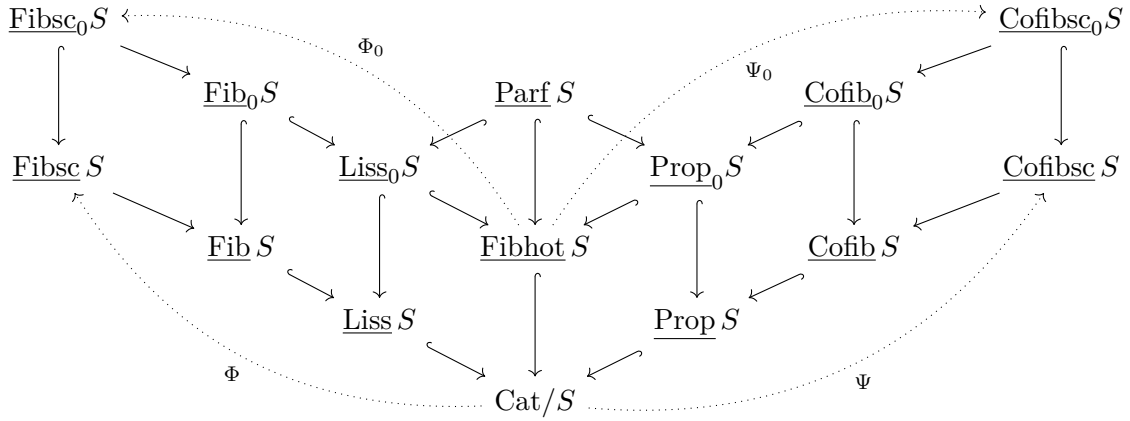
On trouve alors ceci : pour qu'un objet ξ de $\mathrm{HOT}_W(S)$ (resp. $\mathrm{HOT}^{o}_W(S)$) soit dans l'image essentielle de $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, il faut et il suffit que le foncteur associé $S^o \longrightarrow \mathrm{Hot}_W$ (resp. $S \longrightarrow \mathrm{Hot}_W$) soit un système local ([47]), i.e. transforme toute flèche de S en un isomorphisme, i.e. se factorise par $(\Pi S)^o \longrightarrow \mathrm{Hot}_W$ resp. $\Pi S \longrightarrow \mathrm{Hot}_W$ ([48]). Donc pour un X dans Cat/S, cela signifie soit que les foncteurs

$$t\backslash X \longrightarrow s\backslash X$$

associés à des $u : s \longrightarrow t$ dans S soient tous des W-équivalences (cas de $\mathrm{HOT}_W(S)$), soit que les foncteurs

$$X/s \longrightarrow X/t \qquad \text{[page 75]}$$

le soient (cas de $\mathrm{HOT}^o_W(S)$). (NB. ces conditions ne sont nullement équivalentes. Mais si X [est] lisse [sur] S, la première implique la seconde, et l'inverse est vrai si X est propre sur S, et dans ce cas, la condition équivaut à ce que X soit dans $\underline{\mathrm{Fibhot}}$. Enfin, si X est déjà dans $\underline{\mathrm{Fibhot}}\,S$, alors les deux conditions sont automatiquement satisfaites ([49]).



([50]). De plus, si deux éléments $\xi \in \mathrm{ObHOT}_W(S)$, $\xi' \in \mathrm{ObHOT}^o_W(S)$ proviennent d'un même p. 47
$\eta \in \mathrm{ObHOT}^{\mathrm{lc}}_W(S)$, alors les deux foncteurs correspondants

$$(\Pi S)^o \xrightarrow{\varphi} \mathrm{Hot}_W \xleftarrow{\varphi'} \Pi S$$

sont *associés,* i.e. chacun est isomorphe à 'l'inverse' de l'autre.

$$\begin{cases} \varphi^o(s) &= \varphi(s) \\ \varphi^o(u) &= \varphi(u)^{-1}. \end{cases}$$

Commentary · Exposition [ms. 74]

Completing the test for local constancy

The fibre diagram is being used here to recognize objects in an already constructed full subcategory. This is weaker than claiming that an arbitrary diagram in Hot_W reconstructs the coherent object, an issue treated in the commentary at manuscript p. 7. The recognition statement nevertheless needs the verification requested in footnote 47.

Represent ξ by $X \to S$, and replace it by the split fibration $\Phi X \to S$. Its strict fibre at s is $s\backslash X$, and the transition along $u : s \to t$ is precomposition

$$u^* : t\backslash X \longrightarrow s\backslash X.$$

The replacement map is right-fine, so X and ΦX represent the same object of $\mathrm{HOT}_W(S)$. If all the u^* belong to W, this split fibration belongs to $\underline{\mathrm{Fib}}_0 S$: it is a categorical fibration with weakly constant fibres. By the comparison already proved, it belongs to $\underline{\mathrm{Fibhot}}S$. Its class is therefore in the image of i. This proves sufficiency without selecting transport maps in an arbitrary nonsplit representative.

For necessity, choose a homotopy-fibration representative $Y \to S$ of an object in the image. Its comma-fibre transition maps are weak equivalences by the defining local-constancy property. The diagram functor sends an isomorphism in $\mathrm{HOT}_W(S)$ to an isomorphism of diagrams in Hot_W. Whether every structural arrow of such a diagram is invertible is invariant under an isomorphism of diagrams. Hence the property holds for every representative of the same class. The covariant argument uses ΨX and the transitions $X/s \to X/t$.

47. il faudra encore écrire la vérification
48. où ΠS est le groupoïde fondamental de S
49. On aurait dû dans le diagramme introduire aussi $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Prop}}_0 S$!
50. Le diagramme de *quinze* catégories remarquables.

To pass from "invertible in Hot_W" to "belongs to W", one uses strong saturation. It holds for W_∞ and, in the standard theory, for every basic localizer; see Cisinski (2006c, Proposition 4.2.4). This makes explicit a logical input that the sentence "cela signifie" suppresses. Without saturation, the invariant statement uses the saturation of W, rather than arbitrary membership in the originally chosen class.

Finally, the factorization through ΠS in the text concerns only the underlying homotopy-category-valued diagram. Its existence follows from the ordinary universal property of groupoid completion. It does not replace the coherent local system by a representation of the fundamental groupoid. For space-valued coefficients, higher homotopies of the base can still act on the fibres. This leaves the argument above intact: invertibility of all arrows detects local constancy, while the object being detected still retains its coherence.

(VI) NB Dans la nouvelle version du diagramme, où j'ai rajouté des catégories $\underline{\mathrm{Liss}}_0 S$ et $\underline{\mathrm{Prop}}_0 S$ pour rendre le tracé plus transparent, les catégories appelés 'bordantes' deviennent le 'rez-de-chausseée', et les catégories 'internes', le 'premier étage' ou 'étage'. Tous les 7 carrés du diagramme sont catésiens. Toutes les flèches sont des inclusions pleinement fidèles, sauf les quatre flèches horizontales extérieures des catégories de Cat-fibrés ou de Cat-cofibrés *scindés*, avec morphismes exclusivement cartésiens ou cocartésiens et même compatibles avec les scindages, vers les catégories correspondantes de structures non scindées.

Commentary · Exposition [ms. 75]

What the fifteen categories remember

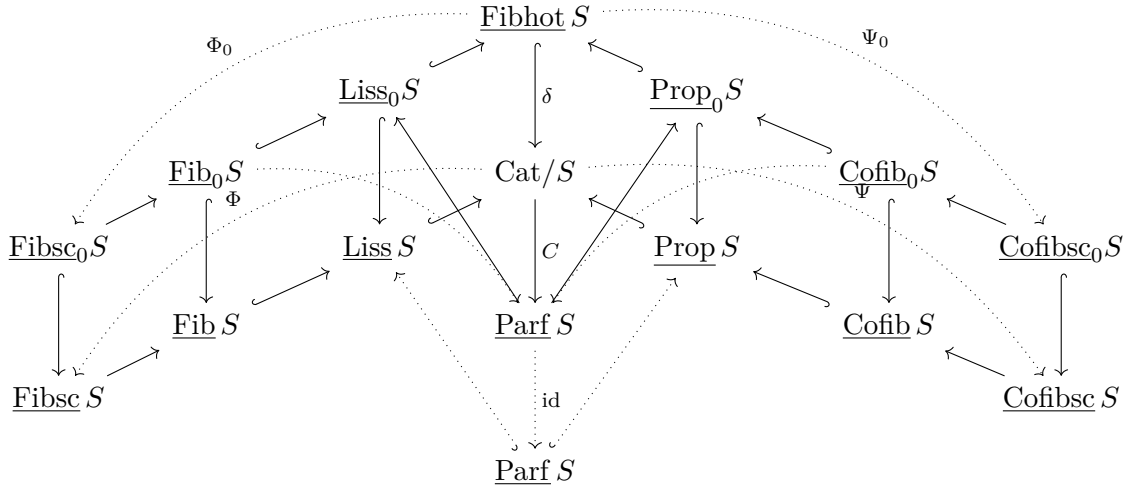
The expanded diagram separates two choices which are often bundled together in contemporary notation: a property of the projection $X \to S$, and a restriction on the morphisms allowed between such projections. This separation is responsible for both its size and its usefulness.

Except for the categories of split fibrations and split opfibrations, the displayed categories are full subcategories of Cat/S. A map between two fibred objects is therefore an arbitrary functor over S; it need not preserve Cartesian arrows. For the split categories, Grothendieck imposes preservation of the splitting, and hence preservation of the chosen Cartesian or cocartesian arrows. Their forgetful functors are not full. The commentary at manuscript p. 30 explains why the naive unit for a splitting construction need not lie in this narrower morphism category.

The seven Cartesian squares assert that the relevant conditions on objects and morphisms are simultaneous conditions. In particular, the square with $\underline{\mathrm{Parf}} S$ expresses the conjunction of the left and right properties, while the outer squares impose the locally constant condition on a split or nonsplit fibration. These assertions should be read in the ordinary category-of-categories sense in which the given subcategories and forgetful functors have been specified. They are not additional assertions that the corresponding nerves form homotopy pullback squares.

The explicit comma replacements remove the possible difficulty caused by the narrow morphism categories. Applied to a functor over S, the rule $s \mapsto s\backslash X$ gives a strictly natural transformation of strict Cat-valued diagrams, and therefore a splitting-preserving map of their categories of elements. This supplies a functor even when the original map did not preserve Cartesian arrows. The equivalence after localization follows from the natural comparison zigzags established earlier, not from pretending that all the forgetful functors were already fully faithful. A later systematic treatment of such comma and category-of-elements comparisons is Maltsiniotis (2005, Theorem 3.8).

[page 76]



Commentary · Examples [ms. 75]

Maps over a group retain the action

The common locally constant category still remembers the map to the base. A completely discrete example makes this visible without any higher coherence. Let $G = C_2 = \{1, \tau\}$, let $S = BG$, and let $E = G$ with its regular right action. Its category of elements has two objects, with one arrow between any ordered pair; the arrow from e to e' lies over the unique g with $e' = eg$. Its total category is equivalent to e, while its projection to BG records the free transitive action.

The associated fibre diagram has one value, the discrete two-point space E, with the nontrivial element of G acting by transposition. An endomorphism over the base is an equivariant map $E \to E$. Such a map is determined by the image of 1, and either choice gives a map: $e \mapsto ae$ for $a \in G$. Thus there are exactly two endomorphisms. Since these diagrams have discrete values, a coherent natural transformation has no nontrivial higher homotopies; their derived mapping space is the same discrete set of two equivariant maps. Consequently the endomorphism set is also two-element in each of the equivalent coarse and fine theories over BG.

Forgetting the base would instead replace the total category by a point, whose endomorphism set in Ho(Spaces) is a singleton. The coarse theory over BG therefore cannot be obtained by simply computing maps between the total homotopy types. It localizes maps *over* BG, and the permitted zigzags continue to remember that structure.

The same computation can be made with spaces. The object is the universal covering $EG \to BG$, with contractible total space and discrete fibre G. Its derived space of self-maps over BG is $\Omega BG \simeq G$. The ordinary space of self-maps of its contractible total space is contractible. This is the relative distinction behind the Hom formula at manuscript p. 81. It also explains why a groupoid base is not a degenerate case of the theory, even though the fine-to-coarse comparison is an equivalence for such a base.

J'ai tracé en pointillés verts `[couleur dans l'original seulement]` les cinq foncteurs Φ, Ψ, Φ_0, Ψ_0, C qui par passage aux localisés donnent des quasi-inverses pour toutes les équivalences de catégories énoncées dans Ⓘ Ⓘ du présent Scholie. De façon précise, le foncteur

$$\Phi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibsc}}\, S$$

associe à toute X dans Cat/S la catégorie scindée sur S

$$s \longmapsto s\backslash X$$

sur S. Ce foncteur, et ses restrictions à $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Fib}}\, S$ et, plus généralement, à toute sous- p. 48
catégorie pleine $\mathcal{F}$ de Cat/S contenant $\underline{\mathrm{Fib}}\, S$, est compatible avec les localiseurs W_S d'une

part (sur $\mathcal{F}$ et $\underline{\mathrm{Fibsc}}\, S$), W_S^{d} d'autre part, et passe donc aux localisés correspondants, pour fournir le quasi-inverse cherché de

$$(\underline{\mathrm{Fibsc}}\, S)\Sigma^{-1} \longrightarrow \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ ou } W_S^{\mathrm{d}})$$

On peut aussi, pour toute sous-catégorie pleine $\mathcal{F}'$ intermédiaire

$$\underline{\mathrm{Fib}}\, S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S,$$

regarder Φ comme un foncteur $\mathcal{F} \longrightarrow \mathcal{F}'$ (plus précisément, prendre le composé

$$\mathcal{F} \xrightarrow{\Phi|\mathcal{F}} \underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S \hookrightarrow \mathcal{F}'\ ),$$

soit $\Phi_{\mathcal{F}',\mathcal{F}}$, et de ce qui vient d'être dit résulte que les deux foncteurs

$$\mathcal{F}' \underset{\Phi_{\mathcal{F}',\mathcal{F}}}{\overset{}{\hookrightarrow}} \mathcal{F}$$

établissent des équivalences quasi-inverses l'une de l'autre [page 77]

$$\mathcal{F}'\Sigma^{-1} \underset{\Phi_{\mathcal{F}',\mathcal{F}}}{\overset{\approx}{\longrightarrow}} \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ ou } W_S^{\mathrm{d}})$$

Énoncé dual pour le foncteur

$$\Psi : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Cofibsc}}\, S,$$

associant à X dans Cat/S la catégorie cofibrée scindée sur S

$$s \longmapsto X/s.$$

Il donne des couples de foncteurs quasi-inverses

$$(\underline{\mathrm{Cofibsc}}\, S)\Sigma^{-1} \underset{\Psi_{\mathcal{F}}}{\overset{}{\longrightarrow}} \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ ou } W_S^{\mathrm{g}})$$

pour tout sous-catégorie pleine $\mathcal{F}$ de Cat/S telle que

$$\underline{\mathrm{Cofib}}\, S \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S,$$

et des couples de foncteurs quasi-inverses p. 49

$$\mathcal{F}'\Sigma^{-1} \underset{\Psi_{\mathcal{F}',\mathcal{F}}}{\overset{\approx}{\longrightarrow}} \mathcal{F}\Sigma^{-1} \qquad (\Sigma = W_S \text{ ou } W_S^{\mathrm{g}})$$

pour deux sous-catégories pleines $\mathcal{F}'$, $\mathcal{F}$

$$\underline{\mathrm{Cofib}}\, S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S.$$

Les foncteurs Φ, Ψ envoient la sous-catégorie pleine $\underline{\mathrm{Fibhot}}\, S$ de 'l'étage', dans les sous-catégories $\underline{\mathrm{Fibsc}}_0 S$, $\underline{\mathrm{Cofibsc}}_0 S$, respectivement, donc induisent à l'étage des foncteurs Φ_0, Ψ_0

$$\underline{\mathrm{Fibsc}}_0 S \xleftarrow{\Phi_0} \underline{\mathrm{Fibhot}}\, S \xrightarrow{\Psi_0} \underline{\mathrm{Cofibsc}}_0 S, \qquad \text{[page 78]}$$

jouant à l'étage exactement le même rôle que Φ, Ψ au rez-de-chaussée. La seule différence, c'est que à l'étage on a $W_S = W_S^{\mathrm{d}} = W_S^{\mathrm{g}}$ $(= W_S^{\mathrm{f}})$, donc il n'y a qu'une suite de catégories localisées à examiner. On trouve ainsi

$$(\underline{\mathrm{Fibsc}}_0 S)W_S^{-1} \xrightarrow{\approx} \mathcal{F}W_S^{-1} \quad \text{(} \approx \text{ via } \Phi_{0,\mathcal{F}}\text{)}$$

$$\mathcal{F}'W_S^{-1} \xrightarrow{\approx} \mathcal{F}W_S^{-1} \quad \text{(} \approx \text{ via } \Phi_{0,\mathcal{F}',\mathcal{F}}\text{)}$$

pour $\mathcal{F}$ (ou $\mathcal{F}$, $\mathcal{F}'$) sous-catégorie(s) pleine(s) de $\underline{\mathrm{Fibhot}}\, S$ telle(s) que

$$\underline{\mathrm{Fib}}_0 S \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S \qquad \text{resp.} \qquad \underline{\mathrm{Fib}}_0 S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S,$$

et de même du côté droit pour les sous-catégories pleines $\mathcal{F}$ ou $\mathcal{F}$, $\mathcal{F}'$ de $\underline{\mathrm{Fibhot}}\, S$ satisfaisant

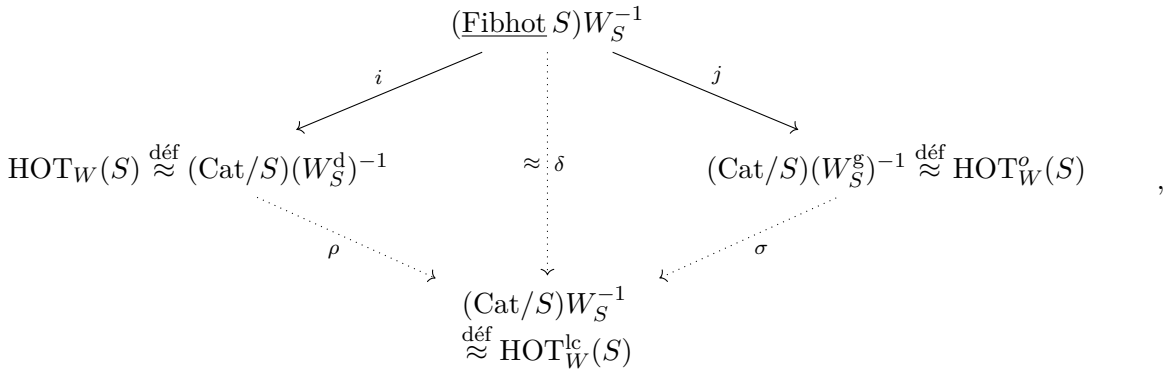
$$\underline{\mathrm{Cofib}}_0 S \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S \qquad \text{resp.} \qquad \underline{\mathrm{Cofib}}_0 S \subseteq \mathcal{F}' \subseteq \mathcal{F} \subseteq \underline{\mathrm{Fibhot}}\, S,$$

NB. La catégorie $\underline{\mathrm{Parf}}\, S$ contient à la fois $\underline{\mathrm{Fib}}_0 S$ et $\underline{\mathrm{Cofib}}_0 S$, et fait donc partie des catégories admissibles tout comme $\mathcal{F}$, ou [?] $\mathcal{F}'$.

Ainsi, les quatre foncteurs Φ, Ψ, Φ_0, Ψ_0 donnent tous les quasi-inverses qu'on peut souhaiter au rez-de-chaussée d'une part (Φ et Ψ), à l'étage de l'autre (Φ_0, Ψ_0). Ils ramènent donc toutes les localisées pertinentes aux quatre suivantes

$$(\mathrm{Cat}/S)W_S^{-1}, \quad (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}, \quad (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1} \qquad \text{(au rez-de-chaussée)}$$
$$(\underline{\mathrm{Fibhot}}\, S)W_S^{-1} \qquad \text{(à l'étage)}.$$

(VII) Il reste à comparer la localisée à l'étage avec les localisées au rez-de-chaussée. Les flèches verticales descendantes de l'étage au rez-de-chaussée, sont compatibles avec W_S^{d} du côté gauche du diagramme, avec W_S^{g} du côté droit, donc elles induisent essentiellement *deux* foncteurs différents i, j sur $(\underline{\mathrm{Fibhot}}\, S)W_S^{-1}$, dans $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$ et $(\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}$: p. 50 [page 79]



,

de façon à faire commuter le carré complété par les pointillés, représentant les foncteurs de localisation ρ, σ. On a bien sûr

$$(*) \qquad \rho i = \sigma j : (\underline{\mathrm{Fibhot}}\, S)W_S^{-1} \longrightarrow (\mathrm{Cat}/S)W_S^{-1}$$
induit par l'inclusion
$\delta : \underline{\mathrm{Fibhot}}\, S \hookrightarrow \mathrm{Cat}/S$

Nous avons énoncé dans (I) que cette flèche est une équivalence, de sorte que ρ et σ peuvent être interprétées, à isomorphisme de foncteurs près, comme des rétractions de $\mathrm{HOT}_W(S)$ et de

$\mathrm{HOT}^o_W(S)$ sur une sous-catégorie commune, s'interprétant soit comme $(\underline{\mathrm{Fibhot}}\, S)W_S^{-1}$, soit comme $(\mathrm{Cat}/S)W_S^{-1}$. Il reste à décrire le foncteur quasi-inverse de $(*)$, ce qui donnera en même temps les quasi-inverses de tous les foncteurs verticaux (de l'étage vers le rez-de-chaussée) dans le diagramme p. 76, plus exactement pour leurs localisés, pour W_S.

Commentary · Examples [ms. 78]

A finite comma diagram with an infinite reflection

The comma replacement and the path replacement can have very different fibres. To see the distinction explicitly, let S have two objects a, b and exactly two nonidentity arrows $u, v : a \rightrightarrows b$. Its nerve realizes to a circle. Take $X = e$ over b.

The right comma diagram $F(s) = s\backslash X$ is discrete, with

$$F(b) = \{*\}, \qquad F(a) = \{u, v\}, \qquad F(u)(*) = u, \quad F(v)(*) = v.$$

Its category of elements has three objects, two over a and one over b, with one arrow from each object over a to the object over b. The latter is terminal, so the total category is contractible. This split fibration is right-fine equivalent to X, but it is not locally constant: neither map from a singleton to the two-point set is an equivalence. Dually, X/s is empty at a and a singleton at b, since there is no arrow $b \to a$.

We can exhibit a locally constant replacement of F without writing the entire category of paths. Put

$$L(a) = L(b) = \mathbb{Z}, \qquad L(u)(n) = n, \qquad L(v)(n) = n + 1,$$

viewing L as a presheaf of discrete spaces. Define $F \to L$ by sending $*$ to 0, u to 0, and v to 1. The two naturality squares commute. All transition maps of L are bijections. Its category of elements has objects a_n, b_n and arrows

$$a_n \longrightarrow b_n, \qquad a_{n+1} \longrightarrow b_n \qquad (n \in \mathbb{Z}).$$

There are no nonidentity composable pairs. Its nerve is the infinite line subdivided into edges, hence is contractible. The functor from the three-object category of elements of F to this infinite category is therefore a coarse weak equivalence. Both its source and target are nonempty and contractible, which verifies the assertion directly rather than by a fibrewise test.

Since L is locally constant, the established reflection identifies its class with $i\rho(F)$. The fibre of this reflection is $\mathbb{Z}$, whereas the original comma fibres had at most two points. Transport around the generating loop of the circle is translation by one, up to the choice of orientation. Geometrically, the total locally constant object is the universal cover of the circle. The reflection must add all iterates of transport; it cannot be computed by merely declaring the two original transition maps invertible within their finite sets. This example gives a concrete role to the path construction in a place where the comparison diagram alone can conceal it.

[page 80]

C'est ce qui est obtenu à l'aide du foncteur 'montant' (du rez-de-chaussée vers l'étage) C, qu'on peut regarder comme un foncteur $\mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fibhot}}\, S$, mais qui en fait prend ses valeurs dans $\underline{\mathrm{Parf}}\, S$. (C'est là un détail sans grande importance, me semble-t-il.)

$$\mathrm{Cat}/S \xrightarrow{C} \underline{\mathrm{Parf}}\, S \hookrightarrow \underline{\mathrm{Fibhot}}\, S.$$

Ce foncteur est compatible avec les localiseurs W_S des deux côtés, et établit des équivalences (quasi-inverses des équivalences

$$(\underline{\mathrm{Parf}}\, S)W_S^{-1} \xrightarrow{\sim} (\mathrm{Cat}/S)W_S^{-1} \qquad \text{et} \qquad (\underline{\mathrm{Fibhot}}\, S)W_S^{-1} \xrightarrow{\sim} (\mathrm{Cat}/S)W_S^{-1}),$$
$$(\mathrm{Cat}/S)W_S^{-1} \xrightarrow{\sim} (\underline{\mathrm{Parf}}\, S)W_S^{-1} \left(\xrightarrow{\sim} (\underline{\mathrm{Fibhot}}\, S)W_S^{-1}\right).$$

Donc, quand les trois catégories HOT pour S sont interprétées en termes de localisées de Cat/S, c'est le foncteur C (plus exactement le composé $\mathrm{Cat}/S \xrightarrow{C} \underline{\mathrm{Parf}}\, S \hookrightarrow \mathrm{Cat}/S$) qui p. 51

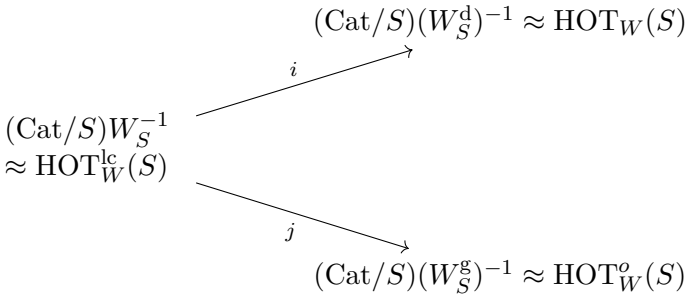
fournit à la fois le foncteur d'inclusion (!) i, et le foncteur d'inclusion j dans $\mathrm{HOT}_W(S)$ et $\mathrm{HOT}^o_W(S)$ respectivement. Le foncteur en question

$$C' : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$$

satisfait bien

$$C'(W_S) \subseteq W_S^{\mathrm{d}} \cap W_S^{\mathrm{g}}$$

et définit les deux foncteurs



Cela nous permet d'expliciter les formules d'adjonction de (IV), quand on y interprète ξ, η comme provenant de X, Y dans Cat/S : [page 81]

$$\boxed{\mathrm{Hom}_{W_S^{\mathrm{d}}}(X, CY) \simeq \mathrm{Hom}_{W_S^{\mathrm{g}}}(X, CY) \simeq \mathrm{Hom}_{W_S}(X, Y),}$$

où $\mathrm{Hom}_{W_S^{\mathrm{d}}}$, $\mathrm{Hom}_{W_S^{\mathrm{g}}}$ et Hom_{W_S} désignent respectivement les Hom dans les catégories localisées pour W_S^{d}, W_S^{g} et W_S, i.e. dans $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^o_W(S)$ et $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ respectivement ([51]).

Commentaire. Toutes les constructions et les énoncés de (I) à (VI), à l'exception du fait que les foncteurs descendants induisent des équivalences pour les localisés *grossiers*, et des formules d'adjonction de (IV), sont de nature quasiment soritale, et n'exigent d'autres conditions sur le localiseur fondamental $W \subseteq \mathrm{Fl}\,\mathrm{Cat}$ que L1, L3 quater, L4 bis, récapitulées comme a) b) c) de la page 45. (On n'a même pas à utiliser d).) Il est vrai qu'en l'absence de la condition L5 (page 43), on ignore si à l'étage les deux localiseurs pertinents, W_S et W_S^{f} (égal à W_S^{d} du côté gauche, à W_S^{g} du côté droit) sont égaux. Donc à priori on trouve à l'étage *deux* catégories localisées essentiellement distinctes, et non pas une seule, savoir $(\underline{\mathrm{Fibhot}}\,S)\Sigma^{-1}$ avec $\Sigma = W_S$ ou $\Sigma = W_S^{\mathrm{f}}$ $(= W_S^{\mathrm{d}} = W_S^{\mathrm{g}})$.

Par contre, le fait que les foncteurs de descente induisent des équivalences entre les localisés grossiers est plus profond, il exige (me semble-t-il) qu'on connaisse un foncteur 'montant' C, disons de Cat/S vers $\underline{\mathrm{Fibhot}}\,S$, compatible avec W_S, et muni de p. 52

$$X \xrightarrow{i_X} C(X),$$

qui soit dans W_S pour X quelconque. Quand on sait que Cat est une catégorie de Quillen, même sans avoir les propriétés des foncteurs $\underline{\mathrm{Ch}}_\infty$ ou $\underline{\mathrm{Ch}}$ ([52]), on doit sans doute pouvoir se débrouiller. Mais la formule d'adjonction me semble rester inattendue et profonde, même en [page 82]

51. Cette formule suggère que si $X \in \mathrm{Ob}\,\mathrm{Cat}/S$, $Y \in \mathrm{Ob}\,\underline{\mathrm{Fibhot}}\,S$, alors

$$\mathrm{Hom}_{W_S^{\mathrm{d}}}(X, Y) \simeq \mathrm{Hom}_{W_S^{\mathrm{g}}}(X, Y) \simeq \mathrm{Hom}_{W_S}(X, Y),$$

mais ça résulte de la formule d'adjoinction et de $Y \longrightarrow C(Y)$ dans $W_S^{\mathrm{d}} \cap W_S^{\mathrm{g}}$ pour Y dans $\underline{\mathrm{Fibhot}}\,S$.

52. ni même être sûr de l'existence d'une factorisation *fonctorielle* de $X \xrightarrow{f} S$ en $X \xrightarrow{i_X} C(X) \xrightarrow{\bar{f}} S$ avec $i_X \in W$ et $\bar{f}$ parfait.

admettant les résultats de Quillen et Thomason ([53]). Elle parait liée ici à une propriété assez particulière du foncteur C, savoir le lemme p. 68

$$i_{C(X)} \underset{\uparrow}{=} C(i_X) : C(X) \longrightarrow C(C(X)),$$

$$\text{dans } \underbrace{(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}}_{\mathrm{HOT}_W(S)} \text{ et dans } \underbrace{(\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}}_{\mathrm{HOT}^o_W(S)}$$

ce qui n'a pas l'air purement formel. Il serait intéressant d'examiner dans quelle mesure cette formule garde un sens dans une 'catégorie de modèles' $(\mathcal{M}, W)$ plus ou moins quelconque, ou provenant d'un triple de Quillen $(W, \mathrm{Cof}, \mathrm{Fib})$ dans $\mathcal{M}$. Il est clair que la factorisation $X \xrightarrow{f} S$ en $X \xrightarrow{i_X} C(X) \xrightarrow{\bar{f}} S$ ($i_X \in W$, $\bar{f} \in \mathrm{Parf}$), jouant le rôle des 'fibrations' dans le contexte Quillen, a un sens intrinsèque indépendant du choix d'une construction particulière, chose qu'on peut préciser de façon parfaite. C'est la description des localiseurs W_S^{d}, W_S^{g} dans $\mathcal{M}/S$ (pour $S \in \mathrm{Ob}\,\mathcal{M}$) qui n'est pas comprise de façon générale, en dehors du cas $\mathcal{M} = \mathrm{Cat}$. En d'autre termes, je ne comprends pas comment construire des catégories $\mathrm{HOT}_{\mathcal{M},W}(S)$, $\mathrm{HOT}^o_{\mathcal{M},W}(S)$, même si $(\mathcal{M}, W)$ provient d'un triple de Quillen, je vois seulement la description d'une catégorie $\mathrm{HOT}^{\mathrm{lc}}_{\mathcal{M},W}(S) = (\mathcal{M}/S)(W_S^{\text{grossière}})^{-1}$, et un foncteur de localisation $\mathcal{M}/S \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_{\mathcal{M},W}(S)$. Il faut arriver à décrire dans $\mathcal{M}/S$ des W-équivalences (*'relatives à S'*) plus fines que la grossière donnée par W sur la source X sans plus. Dans Cat j'y arrive en utilisant soit la famille des isomorphismes locaux (pour décrire W_S^{d}), soit celle des isomorphismes colocaux (pour décrire W_S^{g}) - en tous les cas, il y a une notion de 'localisation' (soit sur S, soit sur S^o …), avec toute la richesse des intuitions liées à une telle notion. Peut-être, sans le savoir ai-je [page 83]
toujours traité Cat non comme une catégorie 'nue', mais comme un site, ou plutôt (puisqu'il y p. 53
a l'involution $X \longmapsto X^o$ qui ne peut être éludée), comme un 'bi-site' - on y a *deux* manières de localiser, soit par les X/x (topologie de Zariski!) soit par les $x\backslash X$ (topologie co-Zariskienne). Il y a aussi la topologie lisse (toute proche sans doute de l'intuition de la cohomologie étale), et la topologie propre - et on s'attend que la lisse donne essentiellement les mêmes résultats que la Zariskienne (vue que la topologie étale lui est égale - il n'y a pas lieu d'en faire un plat …); et la topologie propre les mêmes résultats (p. ex. la même cohomologie?) que la topologie co-Zariskienne.

Commentary · Exposition [ms. 81]

The target controls the comparison of morphisms

The formula on this page is a universal property of the replacement of the *target*. It is stronger than saying that X and $C(X)$ have the same total homotopy type, and it explains why homotopy fibrations are the right objects in which to receive maps.

Write $Q_{\mathrm{d}}, Q_{\mathrm{g}}, Q$ for the three localization functors out of Cat/S. The adjunction $\rho \dashv i$ identifies

$$\mathrm{Hom}_{\mathrm{HOT}_W(S)}(Q_{\mathrm{d}}X, iQY) \cong \mathrm{Hom}_{\mathrm{HOT}^{\mathrm{lc}}_W(S)}(QX, QY).$$

The chosen model for iQY is $Q_{\mathrm{d}}CY$, and the analogous model for jQY is $Q_{\mathrm{g}}CY$. This gives both isomorphisms printed in the box, with their naturality in X and Y. One must retain CY on the fine sides unless Y is already local.

For $Y \in \underline{\mathrm{Fibhot}}S$, the map $i_Y : Y \to CY$ is both right-fine and left-fine. Postcomposition by this map identifies the corresponding Hom sets with targets Y and CY. Combining these identifications with the box proves footnote 51. In particular, the fine and coarse localizations have the same maps *into a local object*, although they can have different maps between arbitrary objects. This is the familiar defining property of a reflective full subcategory, here recovered from explicit relative-category models.

53. Voir cependant le théorème scholie 2, p. 108-113. Ce n'est finalement pas si profond qu'il me semblait, mais quasiment sorital!

There is also a useful way to recognize the arrows inverted by ρ. A map $f : A \to B$ in $\mathrm{HOT}_W(S)$ is inverted if and only if, for every local object L, precomposition induces a bijection

$$\mathrm{Hom}(B, L) \longrightarrow \mathrm{Hom}(A, L).$$

One implication follows from the adjunction. Conversely these bijections say that precomposition by ρf is a bijection against every object of the coarse category; Yoneda implies that ρf is invertible. The fibre diagram alone does not supply these Hom sets. The regular-action example above shows why even objects with contractible total spaces can have nontrivial relative endomorphisms.

Commentary · Exposition [ms. 82]

Coherence already follows from the two invertibilities

Grothendieck is trying to decide which part of the reflection depends on the special geometry of paths. Concatenation gives an attractive proof of the equality of the two iterated units, but under the hypotheses already written at manuscript p. 66 that equality is also formal. The distinction matters for the abstraction that follows.

Let T be an endofunctor of an ordinary category and $\eta : 1 \to T$ a natural transformation. Assume that both $T\eta_X$ and η_{TX} are invertible for every X. Put $a = T\eta_X$ and $b = \eta_{TX}$. Naturality at η_X gives $a\eta_X = b\eta_X$. Applying T and cancelling the invertible map a gives $Ta = Tb$. Next apply naturality to the two maps $a, b : TX \to T^2X$:

$$\eta_{T^2X} a = (Ta)b, \qquad \eta_{T^2X} b = (Tb)b.$$

The right sides agree, and η_{T^2X} is invertible. Hence $a = b$. Notice that the proof never cancels η_X, which need not be invertible.

Now work in $\mathcal{A}\Sigma_0^{-1}$. The functor C induces an endofunctor T because $\Sigma_0 \subseteq \Sigma$ and $C(\Sigma) \subseteq \Sigma_0$. Its coaugmentation is the image of i. The map Ti_X is invertible because $i_X \in \Sigma$, and i_{TX} is invertible by the second condition at manuscript p. 66. The lemma therefore gives $i_{C(X)} = C(i_X)$ in the fine localization. The first triangle identity is already tautological, and this proves the second. The same argument applies to the left-fine localization.

This is the ordinary-category case of the standard characterization of a localization by an idempotent coaugmented endofunctor; compare Lurie (2009, Proposition 5.2.7.4). The displayed proof is included to make the precise cancellation visible. No new geometric property of C is required at this step once both invertibilities are known. Establishing those invertibilities, and identifying the essential image of C, remain mathematical work. The source's later Theorem-Scholie 2, at manuscript pp. 108–113, puts exactly these inputs into a general replacement theorem.

Il semble bien que dans toutes les catégories de modèles utilisées jusqu'à présent pour décrire Hot comme une localisée, on dispose de notions de localisation : (Top), $\mathbf{\Delta}^\wedge$ (ou $A^\wedge$, avec A catégorie-test quelconque), (Cat); et dans Cat on dispose même de *deux* telles notions. Je pense par contre aux ∞-groupoïdes, c'est peut-être la seule catégorie de modèles à tel point proche déjà de Hot elle-même, qu'on n'y discerne pas de notion de localisation, et que l'idée même de vouloir s'y localiser (si ce n'est en 'localisant' Hot lui-même) paraît quasiment saugrenue. [page 84]

Commentary · Context [ms. 83]

The extra structure hidden in a categorical base

The word "bi-site" is Grothendieck's way of isolating information that a bare presentation of a homotopy theory does not remember. The categories S/s and $s\backslash S$ retain the orientation of arrows in S. Localizing the total category alone can destroy that orientation. Consequently a construction which is invariant under weak homotopy equivalence of the base cannot, without additional input, recover both of the directed diagram theories.

The labels also need a small correction in the surrounding prose. A local isomorphism is a discrete fibration, modelled by a slice projection $S/s \to S$; these test W_S^{g}. A co-local isomorphism is a discrete opfibration, modelled by $s\backslash S \to S$; these test W_S^{d}. The sentence at manuscript p. 82

interchanges the two superscripts. The conventions at pp. 83, 94, and 135 agree with the comma definitions above. The edition's wording is retained, and the correction register records the swap. This is a notational slip, not a change in the comparison theorem.

There is a useful historical continuity here. In Grothendieck (1983, §69, pp. 1–3 of the edited extract), Grothendieck describes the passage from one derived category to the family of derived categories of diagrams, relates the motivation to the SGA 5 trace formalism, and then formulates the two Kan-extension adjoints. The present passage asks for a converse kind of explanation: which geometric structure on a category of models permits its objects themselves to serve as indexing bases? For Cat, the answer uses comma categories and the two orientations. It is not supplied merely by knowing that its localization presents spaces.

This also explains the distinction from a slice of a model category. For $S \in \mathcal{M}$, the ordinary slice model structure, when available, localizes maps by their total weak equivalences. That is the coarse theory in these notes. A category of diagrams indexed by an ordinary category S, with objectwise weak equivalences, has a different role: it retains the directed indexing shape. In Cat the same symbol S can play both roles, which is precisely what makes their comparison possible and initially deceptive.

No Grothendieck topologies or covering axioms for the four adjectives are defined in this paragraph. The expectations about equal cohomology should therefore be recorded as a programme, rather than as a theorem about a specified pair of sites. In particular, the remark identifying "étale" and "Zariski" belongs to these categorical local-isomorphism intuitions; it is not an assertion that the Čech or sheaf theories of the *scheme-theoretic* ètale and Zariski topologies coincide. The precise comparison statements available in these pages are the comma and fine-localization equivalences already established.

Cette rapide digression ne convainc à présent que la compatibilité dite pour la construction de $C(X)$ et $i_X : X \longrightarrow C(X)$, ne peut en aucun cas être purement formelle, une pure histoire de catégories de Quillen disons. Ça donnera au mieux la compatibilité dans $(\mathcal{M}/S)W_S^{-1}$, ce qui est ridicule! Mais si on prend comme localiseur Σ dans $\mathcal{M}/S$ les isomorphismes sans plus, la relation de compatibilité va devenir fausse à tous les coups! Comme ensemble de flèches Σ un peu plus raisonnable, qui soit encore très petit, on peut songer p. ex. à la W-équivalence *universelle* sur S (nettement plus fine encore que W_S^{d}, W_S^{g} dans le cas particulier de Cat). Aurait-on encore compatibilité pour cet ensemble de flèches? Je suis aussi un peu perplexe par le fait (pas bien compris) que les formules d'adjonction pour les topologies Zariskienne et co-Zariskienne soient en sens opposé. (Ou me serai-je trompé en dualisant à la va-vite?) Mais à vrai dire, cette compatibilité à elle seule, même quand on utilise une structure de site sur $\mathcal{M}$, ne suffit pas à écrire une formule d'adjonction. On a deux catégories localisées,

$$\mathrm{HOT}^{\mathrm{lc}}_{\mathcal{M}} = (\mathcal{M}/S)W_S^{-1}, \qquad \mathrm{HOT}_W(S) = (\mathcal{M}/S)(W_S^{\mathrm{f}})^{-1},$$

où W_S^{f} désigne une W-équivalence 'fine', correspondant à la notion de localisation, plus exactement aux changements de base 'admissibles' qu'on a choisis, et on a un foncteur de localisation [page 85]

$$\mathrm{Hot}_W(S) \longrightarrow \mathrm{Hot}^{\mathrm{lc}}_W(S),$$

mais pas de foncteur en sens inverse. Mais si, mais si, en travaillant avec la catégorie similaire Fibhot S, ça doit garder un sens tout au moins quand on a une bonne notion de 'fibration'. Je vois un petit travail de dégrossissage conceptuel à faire, avec peut-être une formule d'adjonction à la clef, mais je ne vais pas m'y lancer à présent.

Commentary · Development (commentators') [ms. 84]

A universal version of the coherence calculation

For the categorical replacement used here, the question has an affirmative answer under the standing hypotheses of the Theorem-Scholie. Put $U_S = \{u \mid u \times_S T \in W_T \text{ for every } T \to S\}$. Part III of the Scholie identifies W_S with the fibrewise class W_S^{f} on the full subcategory of Hot-fibrant objects. These objects are stable under arbitrary strict base change, as the source states

explicitly at manuscript p. 119. If $u : P \to Q$ is a coarse weak equivalence between such objects and $f : T \to S$ is arbitrary, the fibre of its pullback at $t \in T$ is precisely

$$(u \times_S T)_t = u_{f(t)}.$$

It therefore belongs to W. The pulled-back source and target remain Hot-fibrant, so Part III over T gives $u \times_S T \in W_T^{\mathrm{f}} = W_T$. This proves that U_S and W_S agree between Hot-fibrant objects; the converse uses the identity base change.

The values of C are perfect, hence Hot-fibrant. Naturality of $\eta : 1 \to C$ and two-out-of-three imply $C(W_S) \subseteq W_S$, and the preceding comparison strengthens this to $C(W_S) \subseteq U_S$. Likewise η_{CX} is a coarse weak equivalence between Hot-fibrant objects and therefore lies in U_S. Both iterated units are now invertible in $(\mathrm{Cat}/S)U_S^{-1}$, so the cancellation lemma at manuscript p. 82 proves

$$C\eta_X = \eta_{CX} \quad \text{in } (\mathrm{Cat}/S)U_S^{-1}.$$

The unit and counit construction below then gives the corresponding reflection. Thus the source's particular question is answered using its fibrewise comparison and base-change stability; no additional identification of strict pullbacks with homotopy pullbacks is needed for that application.

There is also a useful sufficient criterion for other replacements, formulated directly in homotopy-theoretic terms. Here its additional hypothesis is explicit.

Proposition (universal comparison for good replacements). [ours] Fix S and $W = W_\infty$. Let $\mathcal{H} \subseteq \mathrm{Cat}/S$ be a full subcategory such that, for every $P \to S$ in $\mathcal{H}$ and every $T \to S$, the canonical square obtained by applying nerves to $P \times_S T$ is a homotopy pullback square. Suppose $C : \mathrm{Cat}/S \to \mathcal{H}$ has a natural map $\eta : 1 \to C$ whose components belong to W_S. Let U_S consist of those maps f over S for which $f \times_S T$ is a weak equivalence for every $T \to S$. Then

$$C(W_S) \subseteq U_S, \qquad C\eta_X = \eta_{CX} \quad \text{in } (\mathrm{Cat}/S)U_S^{-1}.$$

Moreover the canonical functor

$$(\mathrm{Cat}/S)U_S^{-1} \longrightarrow (\mathrm{Cat}/S)W_S^{-1}$$

has a fully faithful right adjoint induced by C.

To prove the assertion, first consider a coarse weak equivalence $f : P \to Q$ between objects of $\mathcal{H}$. For any $T \to S$, its base change is represented, up to weak equivalence, by the map between homotopy pullbacks

$$NP \times^h_{NS} NT \longrightarrow NQ \times^h_{NS} NT.$$

This is an equivalence because homotopy pullback preserves equivalences of diagrams. Thus $f \in U_S$. The converse holds by taking $T = S$. Consequently U_S and W_S agree on maps between objects of $\mathcal{H}$.

Naturality and two-out-of-three imply that C preserves W_S. Its values lie in $\mathcal{H}$, so the preceding paragraph gives $C(W_S) \subseteq U_S$. Also η_{CX} is a coarse equivalence between two objects of $\mathcal{H}$, hence belongs to U_S. In $(\mathrm{Cat}/S)U_S^{-1}$ the two iterated units are therefore invertible. The cancellation lemma at manuscript p. 82 proves their equality. The unit of the proposed adjunction is the image of η, and its counit in the coarse localization is the inverse of that image. The two triangle identities follow from this equality and from $\eta^{-1}\eta = 1$. Since the counit is invertible, the right adjoint is fully faithful. This proves all assertions.

This second criterion applies only after its homotopy-pullback hypothesis has been established, for example by a relative Theorem B argument after each base change. It does not infer that hypothesis solely from the word "fibration", and it does not assert that every category equipped only with weak equivalences has such a replacement. The later Theorem-Scholie 2, especially manuscript pp. 111–112, supplies the abstract comparison for any finer class agreeing on fibrant objects; the opening argument verifies that agreement for the source's universal class. Our contribution here is the explicit verification for universal equivalences by the two stated methods. We checked the localization criterion in Lurie (2009, Proposition 5.2.7.4) and the source's later abstract theorem; no priority is claimed for either ingredient.

p. 54

16.11.90

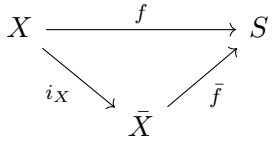
Je vais quand même essayer de faire un travail 'de dégrossissage', car le diagramme des 15 catégories a l'air un peut touffu, finalement! Je me donne

$$(\mathcal{M}, W) \qquad W \subseteq \mathrm{Fl}\,\mathcal{M},$$

un $S \in \mathrm{Ob}\,\mathcal{M}$, d'où $\mathcal{M}/S$, avec $W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$ induit par W. On va poser

$$\mathrm{Ho}^{\mathrm{lc}}(S) = (\mathcal{M}/S)W_S^{-1},$$

ça jouera le rôle des types d'homotopie relativement constants. Je suppose d'autre part que l'on a distingué une classe $F \subseteq \mathrm{Fl}(\mathcal{M})$ de flèches, jouant le rôle de 'fibrations' (ce seront les morphismes Hot-fibrants dans Cat, donnant lieu à la sous-catégorie $\underline{\mathrm{Fib}}\,\mathrm{hot}\,S$ de Cat/S). Je prévois qu'il faudra supposer, outre que $\mathcal{M}$ est stable par $\varprojlim$ finie, que F est stable par changement de base quelconque, et stable par composition, qu'il contient les isomorphismes. Et surtout qu'on a un théorème de factorisation [page 86]



$$i_X \in W,\ \bar{f} \in F$$

pout toute flèche f dans $\mathcal{M}$. Chez Quillen on exige même que i_X ait la propriété de relèvement à droite relativement à F, mais ça ne devrait pas être essentiel, puisque je n'ai pas eu à utiliser ce fait dans Cat. Quand je travaille dans la catégorie $\mathcal{M}/S$, où f est sous-entendu pour un objet X de ladite, je noterai (par abus de langage) $C(X)$ pour $\bar{X}$, bien qu'il n'est pas dit qu'on ait un choix canonique de $C(X)$, encore moins que $C(X)$ dépende fonctoriellement de X.

Commentary · Present state [ms. 85]

The locally constant inclusion has two adjoints

The programme has a sharper established form than the reflection proved here. For a derivator $\mathbb{D}$ and a small category A, Cisinski (2009, Proposition 3.10 and Theorem 3.13) proves that the full inclusion of locally constant coherent diagrams has both a left and a right adjoint, and that, for fixed A, these subcategories form a derivator in an additional diagram variable. This is not the assertion that the varying-base assignment $S \mapsto \mathrm{HOT}^{\mathrm{lc}}_W(S)$ is a derivator. These are the theorem numbers in the published 2009 article; its earlier preprint uses different numbering. The model-category construction is given there by the local structures of §2, with the fibrant objects characterized in Proposition 2.3.

For spaces, the reason is transparent in the language of homotopy-coherent functors. Let $q : NS \to \Pi_\infty S$ be groupoid completion. Restriction

$$q^* : \mathrm{Fun}((\Pi_\infty S)^{\mathrm{op}}, \mathrm{Spaces}) \longrightarrow \mathrm{Fun}((NS)^{\mathrm{op}}, \mathrm{Spaces})$$

is fully faithful and identifies its image with the functors carrying all arrows to equivalences. Its left and right Kan extensions give the two adjoints, since spaces admit small homotopy colimits and limits. Passing to homotopy categories gives the corresponding statement here.

The second adjoint of this one inclusion should not be identified with the functor j in the Scholie. The latter embeds the coarse category into a *different*, covariant fine theory. This keeps the later general theorem distinct from Grothendieck's two parallel reflections. The vague site comparison of the preceding paragraph requires a precise choice of sites before it acquires a similarly definite present status.

Désignons par

$$(*) \qquad \underline{\mathrm{Fib}}\,S \subseteq \mathcal{M}/S$$

la sous-catégorie de $\mathcal{M}/S$ formée des X dont le morphisme structural $f_X : X \longrightarrow S$ est dans F. Désignons (par abus de notation) par W_S le localiseur induit sur $\underline{\mathrm{Fib}}\, S$ par W_S, alors l'inclusion $(*)$ définit

$$(\underline{\mathrm{Fib}}\, S)W_S^{-1} \xrightarrow{\delta} (\mathcal{M}/S)W_S^{-1} \overset{\text{déf}}{=} \mathrm{Ho}_W^{\mathrm{lc}}(S).$$

Une première question, c'est de voir que c'est une équivalence. On voit que c'est essentiellement surjectif grâce à l'existence des $C(X)$, i_X. C'est peu! On va essayer de définir une flèche en sens inverse [page 87]

$$\bar{C} : (\mathcal{M}/S)W_S^{-1} \longrightarrow (\underline{\mathrm{Fib}}\, S)W_S^{-1},$$

i.e. un *foncteur* p. 55

$$\gamma : \mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\, S)W_S^{-1}$$

qui transforme $W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$ en isomorphismes. On choisira, pour X dans $\mathcal{M}/S$, un $C(X)$ dans $\underline{\mathrm{Fib}}\, S$ et une

$$i_X : X \longrightarrow C(X)\ , \qquad i_X \in W_S,$$

et on posera

$$\gamma(X) = C(X).$$

La difficulté, c'est de montrer que ça ne dépend pas des choix faits, à isomorphisme canonique près, et que c'est fonctoriel en X - et le deuxième implique le premier.

Pour voir que ça marche, visiblement il faut supposer quelque chose de plus. Je vois deux hypothèses qui, l'une et l'autre, pourraient faire marcher : l'une, c'est qu'on se donne déjà C comme foncteur et i_X fonctoriel en X (c'est ce qui était le cas dans Cat tantôt). L'autre, c'est que les i_X ont, par rapport à F, la propriété de relèvement à là Quillen. [page 88]

Commentary · Exposition [ms. 87]

Objectwise replacement and functorial replacement

The restart of 16 November isolates a familiar but consequential distinction. A weak equivalence $X \to C(X)$ with fibrant target gives a representative of each localized object; it does not by itself define a functor on localized morphisms. Grothendieck correctly stops at essential surjectivity before adding a hypothesis that controls maps.

In the functorial case the proof uses much less than a full model structure. Put $\mathcal{A} = \mathcal{M}/S$, let $\mathcal{H} = \underline{\mathrm{Fib}}S$ be full, and let $D : \mathcal{H} \hookrightarrow \mathcal{A}$ be its inclusion. Suppose $C : \mathcal{A} \to \mathcal{H}$ is a functor and $\eta : 1_{\mathcal{A}} \to DC$ has components in W_S. For $u : X \to Y$, naturality gives

$$DC(u)\eta_X = \eta_Y u.$$

If $u \in W_S$, two-out-of-three implies $C(u) \in W_S$. Hence both functors descend to the localizations. The transformation η becomes an isomorphism $1 \cong \overline{D}\,\overline{C}$. Restricting it to $\mathcal{H}$ gives $1 \cong \overline{C}\,\overline{D}$, since fullness ensures that the same components are maps in $\mathcal{H}$. Thus $\overline{C}$ and $\overline{D}$ are quasi-inverse equivalences.

For any full $\mathcal{F}$ with $\mathcal{H} \subseteq \mathcal{F} \subseteq \mathcal{A}$, the same proof applies to $C|_{\mathcal{F}}$ and the two inclusions. This proves the intermediate-category assertion at manuscript p. 89. No lifting axiom, finite-limit hypothesis, or base-change stability is used in this functorial branch. Those properties become relevant in the other branch, where lifts have to replace a functorial choice.

The word "fibrant" is relative to the chosen base: the objects of $\mathcal{H}$ are maps $X \to S$ in the distinguished class F. It does not assert that X is fibrant as an object of $\mathcal{M}$, unless S is terminal or an additional argument relates the two conditions. This distinction will also matter when path objects are formed inside the slice in the following pages.

Je commence par l'hypothèse que C est déjà un foncteur. Le diagramme

$$\begin{array}{ccc} X & \xrightarrow{i_X} & CX \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle C(u)} \\ X' & \xrightarrow{i_{X'}} & CX', \end{array}$$

où $i_X,\ i_{X'} \in W$, montre que C transforme W_S en W_S, donc en des isomorphismes de $\mathrm{Ho}^{\mathrm{lc}}_W(S)$. Donc on tient le foncteur cherché γ. On a en fait des foncteurs

$$\underline{\mathrm{Fib}}\, S \underset{C}{\overset{\overset{\text{incl.}}{D}}{\hookleftarrow}} \mathcal{M}/S$$

compatibles avec les localiseurs W_S, induisant δ et γ, et on a

$$\mathrm{id}_{\mathcal{M}/S} \longrightarrow DC, \qquad \text{dans } W_S \text{ argument par argument,}$$

car on a

$$X \xrightarrow{i_X} DC(X) = C(X)$$

avec $i_X \in W_S$. De même, on a

$$\mathrm{id}_{\underline{\mathrm{Fib}}\, S} \longrightarrow CD \qquad \text{dans } W_S \text{ argument par argument,}$$

avec le même p. 56

$$X \xrightarrow{i_X} CX \qquad (\text{cette fois } X \text{ dans } \underline{\mathrm{Fib}}\, S).$$

Donc on a sans larmes les deux foncteurs δ, γ comme quasi-inverses l'un de l'autre. Et le [page 89] lemme page 11 montre même que pour *toute* sous-catégorie pleine $\mathcal{F}$ de $\mathcal{M}/S$, telle que

$$\underline{\mathrm{Fib}}\, S \subseteq \mathcal{F} \subseteq \mathcal{M}/S,$$

les inclusions induisent des équivalences sur les localisés W_S

$$(\underline{\mathrm{Fib}}\, S)W_S^{-1} \xrightarrow{\sim} \mathcal{F}W_S^{-1} \xrightarrow{\sim} \underbrace{(\mathcal{M}/S)W_S^{-1}}_{\overset{\text{déf}}{=}\mathrm{Ho}^{\mathrm{lc}}_W(S)}.$$

Je vais traiter maintenant le cas de type Quillen, pour me convaincre que l'existence du foncteur C, et le choix particulier qu'on peut en faire, est irrelevant. (Ce dernier point est d'ailleurs évident a posteriori, quand on sait que δ est une équivalence, car le quasi-inverse γ sera alors unique à isomorphisme unique près, et le choix d'un foncteur C, et d'un homomorphisme fonctoriel $i : \mathrm{id} \longrightarrow C$, n'est qu'une façon parmi d'autres d'expliciter le quasi-inverse.)

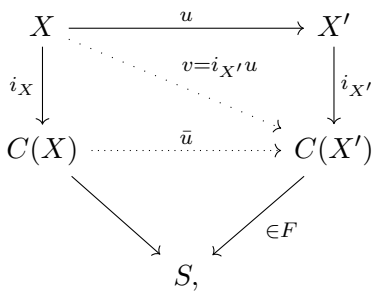
On se place donc dans l'hypothèse quillénienne sur les i_X. On choisit $C(X)$, i_X pour [page 90] chaque X dans $\mathcal{M}/S$, et on pose comme tantôt

$$\gamma(X) = C(X) \qquad \text{comme objet de } (\mathcal{M}/S)W_S^{-1}.$$

Montrons que c'est fonctoriel en X. Soit

$$u : X \longrightarrow X'$$

dans $\mathcal{M}/S$, considérons le diagramme anodin (commutatif)



complétons le par la flèche en gros pointillés, $v = i_{X'}u$ (rendant le triangle supérieur commutatif). ([54]). L'hypothèse sur i_X implique que la flèche $\bar{u}$ en pointillés fins existe, de façon à avoir encore deux triangles commutatifs. Mais ça signifie exactement qu'il y a un *S-morphisme* $\bar{u}$, i.e. une flèche dans $\mathcal{M}/S$, rendant commutatif le carré p. 57

$$\begin{array}{ccc} X & \xrightarrow{u} & X' \\ {\scriptstyle i_X}\downarrow & & \downarrow{\scriptstyle i_{X'}} \\ C(X) & \xrightarrow{\bar{u}} & C(X'), \end{array}$$

et il s'impose de prendre [page 91]

$$\gamma(u) = \text{ image de } \bar{u} \text{ dans } \mathrm{Fl}((\underline{\mathrm{Fib}}\, S)W_S^{-1}).$$

Il faut prouver que cette flèche ne dépend pas du choix de $\bar{u}$. On doit dégager un Lemme, où on désigne par Cof_0 l'ensemble des flèches de $\mathcal{M}$ qu'ont la propriété de relèvement par rapport aux éléments de $\mathrm{Fib} = F$. (Cof_0 joue le rôle des cofibrations triviales.)

Commentary · Exposition [ms. 89]

What makes a comparison canonical

The desired independence of replacement is an independence with specified comparison maps. This qualification explains the otherwise too compressed phrase "unique à isomorphisme unique près" without introducing a new gap in the argument.

Suppose $\delta : \mathcal{H} \to \mathcal{A}$ is an equivalence, and γ, γ' are quasi-inverses equipped with isomorphisms $\epsilon : \delta\gamma \to 1$ and $\epsilon' : \delta\gamma' \to 1$. Full faithfulness of δ gives a unique natural transformation $\theta : \gamma \to \gamma'$ satisfying

$$\delta\theta = (\epsilon')^{-1}\epsilon.$$

Its components are invertible because their images are, so this is the unique comparison compatible with the specified identifications. This is the precise meaning needed for replacement functors. If the identifications are forgotten, a quasi-inverse can have nontrivial natural automorphisms; for the identity equivalence of the one-object category BC_2, its natural automorphisms are the two central elements of C_2.

At manuscript p. 90 the chosen map i_X has the *left* lifting property with respect to fibrations in current terminology: it is the left vertical side of the square, while $C(X') \to S$ is the right vertical side. The printed phrase "relèvement à droite" on p. 86 is resolved by this diagram. The lift supplies a possible $\bar{u} : C(X) \to C(X')$. Its uniqueness has not been asserted. What the next lemma proves is that different lifts give the same localized arrow. Once that is proved, identity and composition follow by choosing the identity lift and the composite lift. This is how a nonfunctorial choice of fibrant objects becomes a genuinely functorial construction after localization.

Lemme. *Soit un diagramme en traits pleins dans* $\mathcal{M}/S$

$$\begin{array}{ccc} X & \xrightarrow{u} & X' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i'} \\ C & \overset{v}{\cdots\!\rightarrow} & C' \end{array}$$

54. Lemme : soit

$$\begin{array}{ccc} X & \xrightarrow{u} & X' \\ {\scriptstyle i}\downarrow & & \downarrow \\ C & \cdots\!\rightarrow & C' \end{array}$$

dans $\mathcal{M}/S$ avec $i \in \mathrm{Cof}_0$, $(C' \longrightarrow S) \in F = \mathrm{Fib}$, alors la flèche en pointillés existe dans $\mathcal{M}/S$, de façon à rendre le carré commutatif. `[cf. Lemme p. 91]`

avec $i \in Cof_0$, et $C' \in Ob\,\underline{\mathrm{Fib}}\,S$ i.e. $(C' \longrightarrow S) \in Fib$. Alors v existe, de façon à rendre le carré commutatif. De plus, si $i \in W$ et $C \in \mathrm{Ob}\,\underline{\mathrm{Fib}}\,S$, la classe de v comme flèche de $(\underline{\mathrm{Fib}}\,S)W_S^{-1}$ en est indépendante du choix fait.

Il reste à prouver cette indépendance, ce qui résultera aussitôt du corollaire : [page 92]

Commentary · Context [ms. 91]

Choosing replacements without choosing a functor

Grothendieck is separating the existence of good representatives from the possibility of doing mathematics with them. A factorization of every object over S gives a representative in $\underline{\mathrm{Fib}}\,S$, but it does not yet identify the morphisms of the two localizations. That is the work of the next lemma. The question is narrower than the construction of a model category: we require enough lifting to extend a map into a fibrant object, and enough paths to make two extensions agree after localization.

The lifting class is

$$\mathrm{Cof}_0 = {}^{\perp}\mathrm{Fib}.$$

Thus its maps have the *left* lifting property against fibrations. Neither the notation nor the definition says that every member of this class lies in W; this is why the factorization hypothesis subsequently asks for $i_X \in \mathrm{Cof}_0 \cap W$. The manuscript's occasional phrase "relèvement à droite" describes the wrong side of the lifting square, as the editors explicitly correct on p. 108. For the extension problem here, the right vertical arrow is the structural map $C' \to S$. The square with top arrow $X \to C'$ and bottom arrow $C \to S$ has a lift because $i : X \to C$ belongs to ${}^{\perp}\mathrm{Fib}$. No choice of a cleavage or of a functorial factorization is involved.

What must become canonical is the *class* of the extension, not an actual extension. Once its class is independent of the lift, a composite of two chosen lifts is itself an allowable lift of the composite. Independence therefore supplies compatibility with composition. This is an instructive instance of a construction becoming functorial only after passing to a homotopy category.

In modern language the other route, hypothesis A, is a right deformation: an endofunctor C with a natural weak equivalence $\mathrm{id} \to C$ and values in a full subcategory of good objects. The equivalence of localizations follows from this data, without a model structure; compare the dual of Riehl (2014, Definition 2.2.1 and Remark 2.2.2). Hypothesis B replaces this strictly natural data by lifting and a proof of independence. Its advantage is that a model category supplies it even when we do not wish to choose a functorial factorization. Its cost is the relative path-object argument that Grothendieck now gives.

Corollaire. *Soit un diagramme dans $\mathcal{M}/S$*

$$X \xrightarrow{\ i\ } C \underset{v_2}{\overset{v_1}{\rightrightarrows}} C',$$

avec C et C' dans $\underline{\mathrm{Fib}}\,S$ (i.e. $C \longrightarrow S$ et $C' \longrightarrow S$ dans Fib), et $i \in Cof_0$ et $i \in W$. Si $v_1 i = v_2 i$, alors v_1 et v_2 définissent la même flèche de $(\underline{\mathrm{Fib}}\,S)W_S^{-1}$.

Démonstration. Soit [p. 58]

$$v = (v_1, v_2) : C \longrightarrow C' \times C',$$

[plutôt $C' \times_S C'$] qui rend commutatif par hypothèse le diagramme

$$\begin{array}{ccc} X & \xrightarrow{\ w\ } & C' \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle \mathrm{diag}_{C'}} \\ C & \xrightarrow{\ v\ } & C' \times_S C', \end{array}$$

où $w = v_1 i = v_2 i$. Factorisons $\mathrm{diag}_{C'}$ en

$$C' \xrightarrow{\ i'\ } \bar{C}' \xrightarrow{\ \delta'\ } C' \times_S C', \qquad \text{avec } \delta' \in \mathrm{Fib},$$

et $i' \in W$. On trouve un carré commutatif

$$\begin{array}{ccccc} X & \xrightarrow{w'=i'w} & \bar{C}' & & \\ {\scriptstyle i}\downarrow & \overset{\bar{v}}{\nearrow} & \downarrow{\scriptstyle \delta'} & & \\ C & \xrightarrow{v} & C' \times_S C' & \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} & C' \end{array}$$

avec $i \in \mathrm{Cof}_0$, $\delta' \in \mathrm{Fib}$, d'où l'existence de la flèche en pointillés, rendant les deux triangles commutatifs. On veut prouver $\mathrm{pr}_1 \circ v = \mathrm{pr}_2 \circ v$ dans $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$, comme $v = \delta'\bar{v}$ il suffit de prouver que l'on a $\mathrm{pr}_1\delta' = \mathrm{pr}_2\delta'$ dans $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$.

Il faut seulement vérifier d'abord (pour justifier l'argument) que cette assertion a un [page 93] sens, i.e. que $\bar{C}'$ est lui aussi dans $\underline{\mathrm{Fib}}\, S$. Mais les hypothèses de stabilité de $F = \mathrm{Fib}$ par changement de base et composition assurent que 1°) $C' \times_S C'$ est dans $\underline{\mathrm{Fib}}\, S$ (qui est stable par produit), et par suite, 2°) que $\bar{C}'$ l'est aussi, puisque $\delta' \in \mathrm{Fib}$. Comme C' est aussi dans $\underline{\mathrm{Fib}}\, S$ et $C' \xrightarrow{i'} \bar{C}'$ est dans W_S, pour vérifier $\mathrm{pr}_1\delta' = \mathrm{pr}_2\delta'$ dans le localisé de $\underline{\mathrm{Fib}}\, S$, il suffit de voir que $(\mathrm{pr}_1\delta')i' = (\mathrm{pr}_2\delta')i'$, or comme $\delta'i' = \mathrm{diag}_{C'}$, on a l'égalité même dans $\underline{\mathrm{Fib}}\, S$ (sans localiser).

On a donc défini $\gamma(X)$, $\gamma(u)$ pour objets et flèches de $\mathcal{M}/S$. La transitivité pour $\gamma(u)$ est claire, et aussi $\gamma(\mathrm{id}) = \mathrm{id}$. Donc on tient le foncteur

$$\gamma : \mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\, S)W_S^{-1}.$$

Soit maintenant $\mathcal{F}$ une sous-catégorie pleine de $\mathcal{M}/S$, avec

$$\underline{\mathrm{Fib}}\, S \subseteq \mathcal{F} \subseteq \mathcal{M}/S,$$

considérons γ comme foncteur à valeurs dans $\mathcal{F}W_S^{-1}$. (J'ai oublié de vérifier que γ transforme W_S dans W_S, mais c'est tautologique.) Ainsi on a des foncteurs

p. 59

$$\mathcal{F}W_S^{-1} \underset{\gamma}{\overset{\delta}{\rightleftarrows}} (\mathcal{M}/S)W_S^{-1},$$

il faut voir qu'ils sont quasi-inverse l'une de l'autre. [page 94]

Commentary · Exposition [ms. 93]

The path object makes the choices agree

Grothendieck's proof works inside the subcategory whose localization we want to understand. This point prevents a circular argument: knowing that two lifts become equal in $(\mathcal{M}/S)[W_S^{-1}]$ would not yet show that they become equal in $(\underline{\mathrm{Fib}}\, S)[W_S^{-1}]$, since full faithfulness of the inclusion is precisely what remains to be proved.

Write $\mathcal{A} = \mathcal{M}/S$ and $\mathcal{F} = \underline{\mathrm{Fib}}\, S$. Suppose that $v_1, v_2 : C \to C'$ agree after precomposition with $i : X \to C$, where i has the left lifting property against fibrations and C, C' lie in $\mathcal{F}$. Factor the relative diagonal as

$$C' \xrightarrow{j} P \xrightarrow{(d_0,d_1)} C' \times_S C', \qquad j \in W, \quad (d_0, d_1) \in \mathrm{Fib}.$$

The object $C' \times_S C'$ is fibrant over S: either projection is a pullback of $C' \to S$, and its composite with the other structural map is a fibration. The same is true of P. The equalities $d_0 j = d_1 j = \mathrm{id}_{C'}$ therefore take place in $\mathcal{F}$. Since j is inverted in $\mathcal{F}[W^{-1}]$, we obtain $d_0 = d_1 = j^{-1}$ there.

The square with left side i, right side (d_0, d_1), bottom side (v_1, v_2), and top side $jv_1i = jv_2i$ has a lift $h : C \to P$. Consequently $v_1 = d_0h = d_1h = v_2$ in $\mathcal{F}[W^{-1}]$. This is the whole independence argument. In fact, the additional condition $i \in W$ is not needed for this particular conclusion. It is needed when the chosen i_X are used to compare an arbitrary object with its replacement.

For completeness, let $u : X \to Y$ and choose a lift $\bar{u} : CX \to CY$ satisfying $\bar{u}i_X = i_Y u$. Define $\gamma(u) = [\bar{u}]$. The identity of CX is a permissible lift for id_X, and $\bar{v}\bar{u}$ is a permissible lift for vu.

Independence proves the functor laws. If $u \in W$, the equation $\bar{u} i_X = i_Y u$ and two-out-of-three imply $\bar{u} \in W$. Thus γ descends through the localization at W_S.

The same squares finish the naturality checks left on p. 94. In the coarse localization they give a natural isomorphism $\mathrm{id} \xrightarrow{\sim} D\gamma$, where D denotes inclusion. On any full intermediate subcategory $\mathcal{F} \subseteq \mathcal{G} \subseteq \mathcal{A}$, both X and CX belong to $\mathcal{G}$, so the same maps give the second natural isomorphism $\mathrm{id} \xrightarrow{\sim} \gamma D$. These two isomorphisms establish the comparison theorem without identifying arbitrary fractions by hand. The proof also records exactly where closure of Fib under pullback and composition is used: in constructing the path object *inside* $\mathcal{F}$.

Du côté de $\mathcal{M}/S$: on doit comparer X et $\delta\gamma(X) = C(X)$, c'est fait par

$$i_X : X \longrightarrow C(X)$$

qui est dans W_S, d'où l'isomorphisme dans le localisé

$$X \simeq \delta\gamma(X) \qquad \text{dans } (\mathcal{M}/S)W_S^{-1}.$$

Il faut vérifier que c'est fonctoriel en X.

Du côté de $\mathcal{F}$, comparer X et $\gamma\delta(X)$, i.e. X et $C(X)$. Même tabac, on choisit $i_X : X \simeq C(X) = \gamma\delta(X)$. À vérifier la fonctorialité.

Il s'agit maintenant d'introduire un localisé plus fin de $\mathcal{M}/S$. Pour ceci, on suppose donné une autre classe de morphismes, je l'appelle $\mathcal{L}$ (comme 'localisation')

$$\mathcal{L} \subseteq \mathrm{Fl}(\mathcal{M}),$$

p. ex. dans le cas de $\mathcal{M} = \mathrm{Cat}$, $\mathcal{L}$ sera formée des isomorphismes locaux ([55]) (s'il s'agit de définir W_S^{g}), ou des isomorphismes colocaux ([56]) (s'il s'agit de définir W_S^{d}). On supposera que $\mathcal{L}$ contient les isomorphismes, et $\mathcal{L}$ stable par composition ([57]), mais non par changement de base. On définit alors

$$W_S^{\mathcal{L}} \subseteq \mathrm{Fl}(\mathcal{M}/S)$$

par

$$f \in W_S^{\mathcal{L}} \overset{\text{déf}}{\Longleftrightarrow} (f \times_S S') \in W, \quad \forall (S' \longrightarrow S) \in \mathcal{L}.$$

[page 95]

Comme $\mathcal{L}$ contient id_S, on aura

$$W_S^{\mathcal{L}} \subseteq W_S,$$

d'où un foncteur canonique

$$\underbrace{(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}}_{\overset{\text{déf}}{=}\mathrm{Ho}_W^{\mathcal{L}}(S)} \xrightarrow{\rho} \underbrace{(\mathcal{M}/S)(W_S)^{-1}}_{\overset{\text{déf}}{=}\mathrm{Ho}_W^{\mathrm{lc}}\mathcal{L}(S)}$$

qui est un foncteur de localisation. Je voudrais définir un foncteur en sens inverse. Pour ceci, p. 60
il est naturel de supposer (en analogie avec la situation de Cat)

Hypothèse 1. *Sur* $\underline{\mathrm{Fib}}\, S$, $W_S = W_S^{\mathcal{L}}$, *i.e. si* $f : X \longrightarrow Y$ *est dans* W *et* X, Y *dans* $\underline{\mathrm{Fib}}\, S$, *alors pour tout changement de base* $S' \longrightarrow S$ *qui est dans* $\mathcal{L}$, $f' : X_{S'} \longrightarrow Y_{S'}$ *est encore dans* W.

Dans le contexte de Quillen, cette hypothèse a une nette tendance à être vérifiée quelque soit $\mathcal{L}$, i.e. pout *tout* changement de base $S' \longrightarrow S$ sans restriction. Il suffit en fait que la structure de Quillen soit 'propre', plus précisément (et moins restrictivement), qu'un `[morphisme]` $\in W$ reste tel après changement de base par un morphisme $\in$ Fib. Dégageons ceci comme un Lemme.

[page 96]

Lemme. *L'hypothèse est vérifiée si on suppose que $F = Fib$ et W sont reliés par la condition supplémentaire (en plus de la factorisation i_X) : pour tout carré cartésien*

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & S', \end{array}$$

si f est dans Fib et p dans W, alors $q \in W$.

Commentary · Exposition [ms. 95]

Testing equivalences after change of base

The class $W_S^{\mathcal{L}}$ is designed to remember information that a total-space equivalence can lose. It asks that an equivalence survive every observation of the base allowed by $\mathcal{L}$. At this stage $\mathcal{L}$ is a collection of tests, not a topology and not a system of weak equivalences. Its role is entirely expressed by pullback.

For a fixed base S, the definition can be written

$$W_S^{\mathcal{L}} = \bigcap_{(g:T\to S)\in\mathcal{L}} (g^*)^{-1}(W_T).$$

Each inverse image contains the isomorphisms and satisfies two-out-of-three; so does their intersection. The identity test gives $W_S^{\mathcal{L}} \subseteq W_S$. These statements use neither closure of $\mathcal{L}$ under composition nor closure under base change. The doubt in footnote 57 therefore has a positive answer for this part of the construction.

Composition becomes useful when we move the base. If $g : T \to S$ lies in $\mathcal{L}$ and $\mathcal{L}$ is composition-closed, then

$$g^*(W_S^{\mathcal{L}}) \subseteq W_T^{\mathcal{L}}.$$

Indeed, for a further test $h : U \to T$ in $\mathcal{L}$, the pullback h^*g^*u is canonically isomorphic to $(gh)^*u$, and gh is an allowed test over S. This proves contravariance along the chosen class. Contravariance along an arbitrary g would need the stronger condition that gh be allowed whenever h is allowed; it does not follow from the definition alone.

The localization functor ρ forgets the extra tests. The question about a right adjoint concerns a special class of objects on which those tests no longer distinguish additional weak equivalences. Hypothesis 1 says exactly that: two fibrations over S which are equivalent on total spaces remain equivalent after every allowed base change. It permits the inclusion of fibrant objects to descend to the finer localization. It does *not* say that every map $X \to CX$ is a fine equivalence; that would make the reflection an equivalence and erase the very distinction being introduced.

For the categorical localizers earlier in the chapter, local and colocal tests retain opposite directions of a diagram. Under the standing fundamental-localizer hypotheses, the two suggestions in footnotes 55–56 follow from the base-change theorem discussed at p. 122 below. Every proper base change preserves W_S^{d}, and hence preserves the total weak equivalence of such a map. Conversely, the coslice projections $s\backslash S \to S$ are proper: they are discrete opfibrations. Testing against all proper maps therefore includes the defining coslice tests and gives exactly W_S^{d}. The dual argument identifies testing against smooth maps with W_S^{g}. When properness or smoothness is taken relative to a specified W, the corresponding relative base-change criterion is the needed hypothesis. These conclusions do not extend to an arbitrary test class in an arbitrary relative category. The commentary on Chapter I, manuscript pp. 52–58, and on this chapter, pp. 41–44, explains the geometric criteria; the abstract argument here needs only equality of the two localizers on $\underline{\mathrm{Fib}}\, S$.

Démonstration. Soit

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$$

55. Ou aussi, des morphismes *lisses*. Ça doit donner le même résultat.
56. Ou aussi des morphismes propres.
57. Il n'en est peut-être pas besoin.

avec p, $q \in \mathrm{Fib}$, $f \in W$, montrons que pour *tout* changement de base $g : S' \longrightarrow S$, $f_{S'} : X_{S'} \longrightarrow Y_{S'}$ est encore dans W. On factorise g en

$$S' \xrightarrow{i} \bar{S}' \xrightarrow{\bar{g}} S, \qquad\qquad i \in W,\ g \in \mathrm{Fib},$$

[plutôt $\bar{g} \in$ Fib] et on est ramené à prouver que l'assertion est vraie dans les deux cas particuliers où $g \in \mathrm{Fib}$, ou $g \in W$. Le cas $g \in \mathrm{Fib}$ résulte aussitôt de l'hypothèse du Lemme (appliquée au carré cartésien

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow{f} & Y, \end{array}$$

où $p \in \mathrm{Fib}$.) p. 61

Dans le cas où $S' \longrightarrow S$ est dans W, il s'ensuit par l'hypothèse du Lemme appliquée à

$$\begin{array}{ccc} Y & \xleftarrow{p} & Y' \\ \downarrow{\scriptstyle \in \mathrm{Fib}} & & \downarrow \\ S & \longleftarrow & S' \end{array}$$

que $p : Y' \longrightarrow Y$ est $\in W$, et de même pour $q : X' \longrightarrow X$, et $f \in W$ implique encore $f' \in W$. [page 97]

Revenons à l'hypothèse p. 95. On aura donc un foncteur

$$(\underline{\mathrm{Fib}}\, S, W_S = W_S^{\mathcal{L}}) \longrightarrow (\mathcal{M}/S, W_S^{\mathcal{L}}),$$

d'où pour les localisés un foncteur horizontal

$$\begin{array}{ccc} (\underline{\mathrm{Fib}}\, S)W_S^{-1} & \xrightarrow{\quad i_0 \quad} & (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1} \overset{\text{déf}}{=} \mathrm{Ho}_W^{\mathcal{L}}(S) \\ \downarrow{\scriptstyle \approx} & \nearrow & \\ \mathrm{Ho}_W^{\mathrm{lc}}(S) \overset{\text{déf}}{=} (\mathcal{M}/S)W_S^{-1} & & \end{array}$$

qui factorise le foncteur vertical induit par l'inclusion $(\underline{\mathrm{Fib}}\, S, W_S) \longrightarrow (\mathcal{M}/S, W_S)$ - foncteur dont on a vu (dans les deux cas envisagés) que c'est une équivalence. En inversant ce dernier et le composant avec i_0, on trouve donc un foncteur

$$\underbrace{\mathrm{Ho}_W^{\mathrm{lc}}(S)}_{\overset{\text{déf}}{=}(\mathcal{M}/S)W_S^{-1}} \xrightarrow{\ i\ } \underbrace{\mathrm{Ho}_W^{\mathcal{L}}(S)}_{\overset{\text{déf}}{=}(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}} ,$$

tel que

$$\rho i \simeq \mathrm{id},$$

donc une *rétraction* de $\mathrm{Ho}_W^{\mathcal{L}}(S)$ sur $\mathrm{Ho}_W^{\mathrm{lc}}(S)$. On s'attend que i soit pleinement fidèle, et plus précisément encore, que ρ et i soient adjoints, i étant l'adjoint à droite de ρ : [page 98]

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \underset{i}{\overset{\rho}{\leftrightarrows}} \mathrm{Ho}_W^{\mathcal{L}}(S),$$

i.e. on cherche p. 62

$$i\rho \quad \underbrace{\xleftarrow{\ i\ }}_{\textit{pas}\ \text{iso}} \quad \text{id} \quad \text{dans } \mathrm{Ho}_W^{\mathcal{L}} \quad (\text{localisation par } W_S^{\mathcal{L}})$$

$$\rho i \quad \underbrace{\xrightarrow{\ p\ }}_{\text{iso}} \quad \text{id} \quad \text{dans } \mathrm{Ho}_W^{\mathrm{lc}} \quad (\text{localisation par } W_S),$$

`[trop de i's]` donc pour X dans $\mathcal{M}/S$, on cherche

$$\begin{aligned} i_X : \quad X \quad &\longrightarrow \quad C(X) \quad \text{tout au moins dans } (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1} \\ p_X : \quad X \quad &\longleftarrow \quad C(X) \quad \text{tout au moins dans } (\mathcal{M}/S)(W_S)^{-1}. \end{aligned}$$

On choisira pour i_X la flèche 'canonique' de X dans $C(X)$, définie dans $\mathcal{M}/S$ lui-même, sans avoir à localiser. Mais dans le 'cas Quillen' (quand C n'est pas donné comme foncteur $\mathcal{M}/S \longrightarrow \underline{\mathrm{Fib}}\, S$), il faut voir que cet i_X est fonctoriel en X, quand on le regarde dans le localisé $(\mathcal{M}/S)W_S^{-1}$. Mais c'est contenu dans la construction de C comme *foncteur*

$$\mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\, S)W_S^{-1}.$$

On notera que i_X est dans W_S sans plus, pas dans $W_S^{\mathcal{L}}$, donc ne définit pas en général un isomorphisme de $\mathrm{Ho}_W^{\mathcal{L}}$.

Commentary · Exposition [ms. 97]

Why the properness argument works

Grothendieck wants an easily checked condition which guarantees that a weak equivalence between fibrations remains an equivalence after a change of base. The lemma gives a sufficient condition in his sparse axiomatic setting. Its two cases are worth keeping distinct, because they apply the same properness axiom to different squares.

Let $u : X \to Y$ be a weak equivalence over S, with $X \to S$ and $Y \to S$ fibrations, and let $g : T \to S$. Factor g as

$$T \xrightarrow{a} \bar{T} \xrightarrow{b} S, \qquad a \in W, \quad b \in \mathrm{Fib}.$$

First pull back along b. The map $Y \times_S \bar{T} \to Y$ is a fibration, being a pullback of b. Right properness, applied to u along this fibration, says that $u_{\bar{T}}$ lies in W. The structural maps of its source and target over $\bar{T}$ are still fibrations.

Now pull back along a. Right properness applied to those two structural fibrations says that both projections

$$X \times_S T \longrightarrow X \times_S \bar{T}, \qquad Y \times_S T \longrightarrow Y \times_S \bar{T}$$

lie in W. In their commutative square the lower horizontal arrow is $u_{\bar{T}} \in W$. Two-out-of-three proves $u_T \in W$. This completes the argument, with no assumption that g itself be a fibration or a weak equivalence.

The sufficient condition is stronger than necessary in a full Quillen model category. For every map $g : T \to S$, the pullback functor between the slice model categories preserves fibrations and trivial fibrations. It is right adjoint to postcomposition with g, and is therefore a right Quillen functor. The dual of Ken Brown's lemma shows that it preserves weak equivalences between objects fibrant in the slice. Thus Hypothesis 1 holds for all base changes in any model category, without global right properness. The base S need not itself be fibrant: fibrancy of $X \to S$ is fibrancy as an object of the slice model category. The corresponding statement for a category of fibrant objects is proved as Lemma C.1.11 of Riehl and Verity (2022), using Lemma C.1.10. This does not invalidate Grothendieck's proof: his preceding axioms are weaker than the full model-category axioms, so a separate sufficient condition is useful.

Right properness concerns pulling back an arbitrary weak equivalence *along a fibration*. The conclusion here concerns pulling back a weak equivalence *between two fibrations over the same base* along an arbitrary map. The factorization argument explains the passage from one arrangement to the other.

On définit p_X à l'aide de i_X, comme l'inverse de i_X *dans* $(\mathcal{M}/S)W_S^{-1}$, qui existe car i_X est dans W_S, donc `[est un]` isomorphisme dans $(\mathcal{M}/S)W_S^{-1} = \mathrm{Ho}_W^{\mathrm{lc}}$, et p est donc `[un]` [page 99] isomorphisme. On retrouve donc $\rho i \simeq \mathrm{id}$, donc que ρ est une *rétraction* de $\mathrm{Ho}_W^{\mathcal{L}}$ sur $\mathrm{Ho}_W^{\mathrm{lc}}$.

Pour prouver l'adjonction, en procédant comme dans le cas de Cat, on est ramené à prouver que pour tout X,

$$\begin{cases} C(i_X) = i_{C(X)} : C(X) \rightrightarrows C(C(X)) \\ \textit{dans } \mathrm{Ho}_W^{\mathcal{L}} \text{ (localisation par } W_S^{\mathcal{L}}). \end{cases}$$

J'y étais arrivé dans Cat, en utilisant un morphisme

$$C(C(X)) \xrightarrow{c} C(X)$$

(essentiellement, un morphisme de composition des chemins) tel que $c \in W_S^{\mathcal{L}}$, et que $c \circ C(i_X) = c \circ i_{C(X)}$. Ça avait l'air très délicat, car c'était à grand peine que j'étais arrivé à construire un foncteur $\underline{\mathrm{Ch}}$ ('chemins') raisonnable, où on puisse bel et bien composer les chemins! Je renonce pour le moment à axiomatiser les propriétés du foncteur $\underline{\mathrm{Ch}}$ dont j'ai besoin pour faire marcher la machine (je devrais y revenir tantôt). Je vais commencer plutôt par le 'cas Quillen', avec des $i_X \in \mathrm{Cof}_0$.

On reprend la description du 'foncteur' C (à valeurs dans $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$, et alors sans guillemets), on trouve que $C(i_X)$, où p. 63 [page 100]

$$X \xrightarrow{u=i_X} X' = C(X)$$

est défini simplement (dans $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$) par 'la' flèche $C(u)$ rendant commutatif le diagramme (dans $\underline{\mathrm{Fib}}\, S$)

$$\begin{array}{ccc} X & \xrightarrow{u\ (=i_X)} & X'\ (=C(X)) \\ {\scriptstyle i_X}\downarrow & & \downarrow{\scriptstyle i_{X'}\ (=i_{C(X)})} \\ C(X) & \xrightarrow[C(u)\ (=C(i_X))]{} & C(X')\ (=C(C(X))), \end{array}$$

de sorte qu'on a dans $\mathcal{M}/S$ le diagramme

$$X \xrightarrow{\overbrace{i_X}^{\in \mathrm{Cof}_0 \cap W}} \overbrace{C(X)}^{\in \mathrm{Ob}\,\underline{\mathrm{Fib}}\, S} \underset{C(i_X)}{\overset{i_{C(X)}}{\rightrightarrows}} \overbrace{C(C(X))}^{\in \mathrm{Ob}\,\underline{\mathrm{Fib}}\, S},$$

commutatif, i.e. $i_{C(X)} i_X = C(i_X) i_X$. Le corollaire page 92 nous dit que les flèches $i_{C(X)}$ et $C(i_X)$ sont égales dans $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$. Mais par hypothèse (p. 95), dans $\underline{\mathrm{Fib}}\, S$ on a $W_S = W_S^{\mathcal{L}}$, donc les flèches sont égales dans $(\underline{\mathrm{Fib}}\, S)(W_S^{\mathcal{L}})^{-1}$, et a fortiori dans $(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}$, et on gagne.

Voyons ce qu'on peut faire dans le cas où on dispose d'un *foncteur* C. On suppose bien sûr que ce foncteur dans $\mathcal{M}/S$ n'est pas lié au S particulier dans $\mathcal{M}$ qu'on a choisi comme base, que sa construction provient à la Cartan-Serre d'un foncteur [page 101]

$$X \longmapsto \mathrm{Ch}(X) : \mathcal{M} \longrightarrow \mathcal{M},$$

('objet des chemins') et de morphismes fonctoriels

$$X \xrightarrow{\alpha_X} \mathrm{Ch}(X) \xrightarrow{p_X} X \times X, \qquad \boxed{p_X \alpha_X = \mathrm{diag}_X}$$

([58]). Je présume qu'il faudra supposer surtout ceci

Commentary · Exposition [ms. 100]

Naturality in the finer localization

The proposed unit of the reflection has to exist where fewer arrows have been inverted. This is the issue hidden by the notation i_X: in the coarse localization i_X is invertible, whereas in the finer localization it usually is not. The construction by lifts supplies the required stronger naturality, although the sentence on p. 98 mentions only the coarse category.

Put $\mathcal{A} = \mathcal{M}/S$, $\mathcal{F} = \underline{\mathrm{Fib}}\, S$, and write W_0 for either $W_S^{\mathcal{L}}$ or the more general fine localizer of Theorem-Scholie 2. We assume $W_0 \subseteq W_S$ and equality on arrows of $\mathcal{F}$. For a map $u : X \to Y$, a chosen lift satisfies the strict identity

$$\bar{u} i_X = i_Y u$$

58. **NB** On pose $\mathrm{pr}_1 \circ p_X = \sigma_X$, $\mathrm{pr}_2 \circ p_X = \beta_X$ (flèches 'source' et 'but' pour l'objet X).

in $\mathcal{A}$. Two choices of $\bar{u}$ agree in $\mathcal{F}[W_S^{-1}]$, hence also after mapping into $\mathcal{A}[W_0^{-1}]$, because W_S and W_0 coincide on $\mathcal{F}$. The displayed identity therefore defines a natural transformation in the finer localization. Naturality for formal inverses of W_0-arrows follows by multiplying a naturality square by those inverses; this is precisely the universal property of localization for natural transformations.

For $u = i_X$, both $\overline{i_X}$ and i_{CX} extend the map $i_{CX}i_X$. The path-object lemma identifies them in $\mathcal{F}[W_S^{-1}]$. This is stronger than the equality needed in $\mathcal{A}[W_0^{-1}]$. Let $\eta_X = [i_X]$ be the fine unit, and let the coarse counit be $[i_X]^{-1}$. One triangle identity is cancellation of $[i_X]$ with its inverse. The other is cancellation after the equality

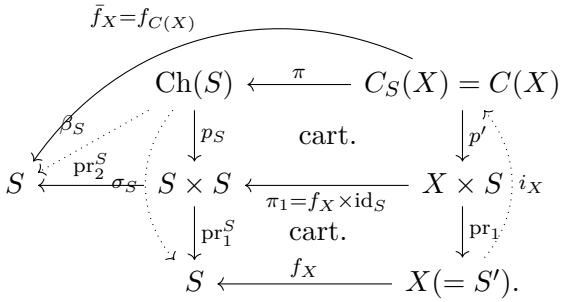
$$[C(i_X)] = [i_{CX}]$$

just proved. Consequently $\rho \dashv i$, and the counit is invertible, so i is fully faithful.

No strict functor $C : \mathcal{A} \to \mathcal{F}$ has been manufactured in this argument. The expression $C(i_X)$ means a class represented by an allowable lift. The remark on p. 100 that the entire displayed square lies in $\underline{\mathrm{Fib}}\, S$ is a small source slip: its upper-left object X need not be fibrant. The square lies in $\mathcal{M}/S$; the two arrows whose classes are compared have fibrant sources and targets, which is all the proof uses.

Hypothèse 2. $\alpha_X \in W$, $p_X \in \mathrm{Fib}$.
Pour un X sur S, on construira $C_S(X)$ ou $C(X)$ ainsi

p. 64



Il est l'image inverse de $(\mathrm{Ch}(S), \mathrm{pr}_1^S p_S = \sigma_S)$ par $X \xrightarrow{f_X} S$, mais on a explicité la factorisation de σ_S sur le diagramme pour réinterpréter la flèche structurale $\bar{f}_X = f_{C(X)}$, définie comme

$$f_{C(X)} = \beta_S \pi (= \mathrm{pr}_2^S \underbrace{p_S \pi}_{= \pi_1 p'})$$

par la formule

$$f_{C(X)} = \underbrace{(\mathrm{pr}_2^S \pi_1)}_{\text{projection } \mathrm{pr}_2 \text{ de } X \times S \text{ dans } S} p'.$$

Or [page 102]

$p' \in \mathrm{Fib}$, car déduit de $p_S \in \mathrm{Fib}$ par changement de base π_1,

$(\mathrm{pr}_2 : X \times S \longrightarrow S) \in \mathrm{Fib}$ pourvu que $X \longrightarrow \underbrace{e}_{\text{objet final de } \mathcal{M}}$ soit dans Fib (même raison),

donc si $X \in \underline{\mathrm{Fib}}\, e$, on a $f_{C(X)} \in \mathrm{Fib}$. Comme c'est une propriété essentielle qu'on veut avoir pour *tout* X dans $\mathcal{M}/S$, on va faire l'hypothèse supplémentaire

Hypothèse 3. $\forall\, X \in Ob\, \mathcal{M}$, $X \longrightarrow e$ *est dans Fib.*

Quant à

$$i_X : X \longrightarrow C(X),$$

il est déduit de la section α_S de $(\mathrm{Ch}(S), \sigma_S)$ sur S par changement de base $X \longrightarrow S$. Notons maintenant que

$$\mathrm{Ch}(S) \underset{\beta_S}{\overset{\sigma_S}{\rightrightarrows}} S \qquad \text{sont dans } W,$$

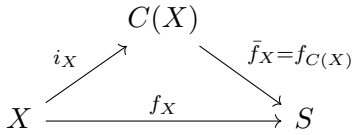
car leurs composés avec $S \xrightarrow{\alpha_S} \mathrm{Ch}(S)$ sont l'identité, et $\alpha_S \in W$. D'autre part, $\sigma_S = \mathrm{pr}_1^S \circ p$ est dans Fib, comme composé de deux éléments dans Fib (NB $S \longrightarrow e$ dans Fib par Hypothèse 3). Ainsi σ_S, β_S sont dans $\mathrm{Fib} \cap W$. On aura besoin de

Hypothèse 4. *L'ensemble de flèches Fib $\cap$ W est stable par tout changement de base.*

Cela implique donc que p. 65 [page 103]

$$C(X) \xrightarrow{\mathrm{pr}_1 \pi'} X$$

`[plutôt` $\mathrm{pr}_1 p'$`]` est $\in W$ (étant déduit de $\sigma_S : \mathrm{Ch}(S) \longrightarrow S$ par changement de base), donc toute section $i_X : X \longrightarrow C(X)$ de $C(X)$ sur X est dans i_X `[plutôt dans` W`]`. On a donc bien une factorisation, fonctorielle en X



avec

$$i_X \in W, \qquad f_{C(X)} \in \mathrm{Fib}.$$

C'est tout ce qu'on avait utilisé pour établir

$$(\underline{\mathrm{Fib}}\ S)W_S^{-1} \xrightarrow{\sim} (\mathcal{M}/S)W_S^{-1}.$$

Mais il s'agit maintenant, sous l'hypothèse où on introduit un $\mathcal{L}$, de prouver que pour tout X dans $\mathcal{M}/S$, les deux flèches

$$C(X) \underset{i_{C(X)}}{\overset{C(i_X)}{\rightrightarrows}} C(C(X))$$

sont égales dans $(\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}$. Comme par hypothèse on a $W_S^{\mathcal{L}} = W_S$ dans $\underline{\mathrm{Fib}}\ S$ (Hypothèse 1), et que $C(X),\ C(C(X)) \in \mathrm{Ob}\,\underline{\mathrm{Fib}}\ S$, il suffit de prouver que les deux flèches sont égales *dans* $(\underline{\mathrm{Fib}}\ S)W_S^{-1}$. Cet énoncé est indépendant de la donnée de $\mathcal{L}$, et on a envie de le prouver. [page 104]

Commentary · Exposition [ms. 103]

The mapping-path factorization and its hypotheses

Grothendieck now wants a replacement with visible geometry. A point of $C_S(X)$ consists of a point of X together with a path from its image to a new point of S. The new structural map remembers the endpoint. This changes the map to S while preserving the total homotopy type, and explains why the two projections of the path object play different roles.

Assume finite limits, pullback and composition stability of Fib, and natural maps

$$S \xrightarrow{\alpha_S} P(S) \xrightarrow{(\sigma_S, \beta_S)} S \times S$$

with composite the diagonal, $\alpha_S \in W$, and $(\sigma_S, \beta_S) \in \mathrm{Fib}$. Define

$$C_S(X) = X \times_{S,\sigma_S} P(S), \qquad q : C_S(X) \to X, \qquad \bar{f} = \beta_S \,\mathrm{pr}_{P(S)}\,.$$

The unit sends x to $(x, \alpha_S f(x))$, so $qi_X = \mathrm{id}_X$ and $\bar{f} i_X = f$. These formulas describe genuine morphisms: the generalized-element description is justified by the universal property of the pullback and the Yoneda embedding, not by an assumption that the category consists of sets.

To see that $\bar{f}$ is a fibration, rewrite the pullback as

$$C_S(X) = (X \times S) \times_{S \times S} P(S),$$

where $X \times S \to S \times S$ is $f \times \mathrm{id}$. Its projection $p' : C_S(X) \to X \times S$ is a fibration. The second projection $X \times S \to S$ is a pullback of $X \to e$, so it is a fibration if X is fibrant. Hypothesis 3 makes this true for every X. Thus $\bar{f} = \mathrm{pr}_2\, p'$ is a fibration.

To see that i_X is a weak equivalence, first note that $\sigma_S\alpha_S = \mathrm{id}$ makes σ_S a weak equivalence. It is a fibration since S is fibrant and (σ_S, β_S) is a fibration. Hypothesis 4 then makes its pullback q a trivial fibration. Since $qi_X = \mathrm{id}$, two-out-of-three gives $i_X \in W$. The proof uses pullback stability of *trivial fibrations*; it does not require arbitrary weak equivalences to survive arbitrary pullbacks.

This is the mapping-path mechanism behind Brown's factorization lemma. Its persistence in higher category theory is especially explicit in Riehl and Verity (2022, Lemma 1.2.19), where a map in an ∞-cosmos factors as a section of a trivial fibration followed by an isofibration. The categorical hypotheses and the meaning of equivalence differ, but the pullback calculation is the same.

The concise condition A′ on p. 112 must be read as the advertised *strengthening of A*. In particular, it retains the conclusion $i_X \in W$ already required in A. If its path data are instead used as an independent recipe for proving A, the pullback-stability step just exhibited must be supplied, for example by Hypothesis 4. This distinguishes an abbreviated restatement from a missing hypothesis in the theorem as actually formulated.

Peut-être y a-t-il moyen d'expliciter un langage géométrique, pour s'y reconnaître. Identifions les objets X de $\mathcal{M}$ aux foncteurs $\mathcal{M}^o \longrightarrow$ (Ens) qu'ils définissent,

$$X \longmapsto (T \longmapsto X(T) = \mathrm{Hom}_{\mathcal{M}}(T, X))\,.$$

Ainsi, pout tout T

$$\mathrm{Ch}(X)(T)$$

est un ensemble, appelé *l'ensemble des chemins dans* $X(T)$. Il s'envoie (par p_X) fonctoriellement (en T) dans $(X \times X)(T) = X(T) \times X(T)$, i.e. chaque chemin c dans $X(T)$ a une origine $\sigma(c)$, et un but $\beta(c)$. De plus, par α_X, $X(T)$ s'envoie (fonctoriellement en T) dans l'ensemble des chemins de $X(T)$: pour chaque élément x dans $X(T)$, il y a donc un chemin (noté 1_x ou α_x) bien défini de source et but x, qu'on appelle le chemin trivial, ou vide. Ceci dit,

pour X dans $\mathcal{M}/S$, $C(X)$ représente le contra-foncteur suivant dans $\mathcal{M}$: pour tout T dans $\mathcal{M}$, p. 66 [page 105]

$$C(X)(T) = \left\{(x, c) \;\middle|\; x \in X(T),\ c : f_X(x) \xrightarrow{c} s \ \text{ un chemin dans } S(T),\ \text{de source } f_X(x)\right\},$$

où $f_X : X \longrightarrow S$ est le morphisme structural.

Les 'calculs' faits, il y a deux (?) jours montrent qu'on a de même une interprétation

$$C(C(X))(T) = \left\{(x, c, c') \;\middle|\; x \in X(T),\ f_X(x) \xrightarrow{c} s \xrightarrow{c'} t,\ c \text{ un chemin de source } f_X(x),\ c' \text{ un chemin de source égale à } \beta(c)\right\}$$

et les applications $i_{C(X)}$ et $C(i_X)$ s'identifient respectivement à

$$\begin{array}{lcll} (x, c) & \longmapsto & (x, 1_{f(x)}, c) & f(x) \xrightarrow{1_x} f(x) \xrightarrow{c} s \\ (x, c) & \longmapsto & (x, c, 1_{\beta(c)}) & f(x) \longrightarrow s \xrightarrow{1_s} s \end{array}$$

[plutôt $f(x) \xrightarrow{1_{f(x)}} f(x) \xrightarrow{c} s$]. Pour les comparer, on a une envie irrésistible d'écrire

$$c \circ 1_x = 1_s \circ c = c,$$

mais a-t-on le droit? Il faudrait pouvoir composer les chemins, et c'est très, très difficile!

Mais on peut toujours axiomatiser. On suppose donc que pour X dans $\mathcal{M}$, $\mathrm{Ch}(X)$ n'est pas seulement une structure de (soi-disants) 'chemins' dans X amorphe, seulement la donnée de la source et du but des chemins et des chemins triviaux (ces structures sur $\mathrm{Ch}(X)$ [page 106]

dépendant fonctoriellement de X), mais qu'on a une composition des chemins quand ceux-ci sont 'composables', i.e. une flèche d'un produit fibré

$$(c, c') \longmapsto c' \circ c : (\mathrm{Ch}(X), \beta_X) \times_X (\mathrm{Ch}(X), \sigma_X) \xrightarrow{\pi_X} \mathrm{Ch}(X),$$

i.e. on peut composer des chemins dans $X(T)$

$$x \xrightarrow{c} y \xrightarrow{c'} z$$

si et seulement si $\mathrm{but}\, c = \mathrm{source}\, c'$, et de plus on exige bien sûr que $\mathrm{source}(c' \circ c) = \mathrm{source}\, c$, $\mathrm{but}(c' \circ c) = \mathrm{but}\, c'$, ce qui s'exprime par un diagramme commutatif

$$\begin{array}{ccc} (\mathrm{Ch}(X), \beta_X) \times_X (\mathrm{Ch}(X), \sigma_X) & \longrightarrow & \mathrm{Ch}(X) \\ & \searrow \quad \swarrow_{p_X} & \\ & X \times X. & \end{array}$$

Commentary · Examples [ms. 105]

A groupoid in which both inserted paths are visible

Grothendieck's element notation is most useful when it allows us to see the two maps $CX \rightrightarrows C^2X$ before localizing. A groupoid gives a completely algebraic example, including a small correction to the order of the formulas on p. 105.

Let G be a group, let $S = BG$ be its one-object groupoid, and let $X = e \to BG$ select the object. Work in groupoids with equivalences as weak equivalences and isofibrations as fibrations. Take $P(S)$ to be the arrow groupoid: its objects are $g \in G$, and a morphism $g \to g'$ is a pair (a, b) with

$$bg = g'a.$$

Source and target send this pair to a and b. The constant-path functor sends the unique object of BG to $1 \in G$ and an arrow a to (a, a). It is fully faithful, and every g is isomorphic to 1, so it is an equivalence. The endpoint functor $P(S) \to S \times S$ is an isofibration: prescribed endpoint isomorphisms (a, b) lift by choosing $g' = bga^{-1}$.

The pullback defining $C_S(e)$ fixes the source component of a morphism to be the identity. Its objects are still $g \in G$, but between g and g' there is exactly one morphism, whose image in BG is

$$b = g'g^{-1}.$$

Thus $C_S(e)$ is a contractible groupoid. Its projection to BG is an isofibration: an arrow h with source the image of g lifts to the unique arrow $g \to hg$. The inclusion $e \to C_S(e)$ chooses 1 and is an equivalence. The strict fibre over the object of BG, on the other hand, is the discrete set G, since $g'g^{-1} = 1$ precisely when $g' = g$. The replacement has preserved the total type while encoding the nontrivial fibre needed over BG.

Objects of $C_S^2(e)$ are pairs (g, h), representing successive paths. The two maps are

$$i_{C_S(e)}(g) = (g, 1), \qquad C_S(i_e)(g) = (1, g).$$

The first inserts a constant path at the *endpoint*; the second first replaces the original object e and therefore inserts a constant path at the *source*. These are the reverse of the labels assigned by "respectivement" in the edition. The formulas themselves are present there; their attribution is interchanged. We retain that source slip and use the correctly labelled formulas in the commentary.

Composition gives $\gamma(g, h) = hg$. It is also a functor on morphisms. The unique morphism $(g, h) \to (g', h')$ has image

$$d = h'g'g^{-1}h^{-1}$$

in BG, which is exactly the image of the unique morphism $hg \to h'g'$ in $C_S(e)$. Therefore $\gamma i_C = \gamma C(i) = \mathrm{id}$ already before localization. Since the two insertion maps are weak equivalences between fibrant objects, they become the same inverse of γ after localization. This example exhibits

the geometric comparison which the abstract argument needs; it does not identify categorical fibrations with model-category fibrations in general.

The same example makes a noninvertible fine unit concrete. Let W_0 on groupoids over BG consist of equivalences on each strict fibre. It is a two-out-of-three class, obtained by applying the fibre functors. It is contained in the total equivalences for maps between isofibrations, and there it agrees with the total equivalences. To obtain a class contained in the total equivalences on all objects, take its intersection with W; the agreement on isofibrations remains true. If G is nontrivial, the unit $e \to C_S(e)$ is a total equivalence but is not an equivalence on the strict fibre: there it is the inclusion of the singleton $\{1\}$ into the discrete groupoid G. The fibre functor, followed by localization at equivalences of groupoids, detects that this unit cannot become invertible in the fine localization. Thus the fine and coarse localizations differ even though their equivalences between fibrant objects coincide. This is precisely the situation of the reflection theorem.

p. 67

La pensée naturelle, c'est qu'avec cette composition et ces identités, les $X(T)$, $\mathrm{Ch}(T)$ deviennent les objets et les flèches d'une *catégorie* - i.e. que $(X, \mathrm{Ch}(X), \alpha_X, p_X, \pi_X)$ définissent une structure de 'catégorie dans $\mathcal{M}$'. Mais pour ce qu'on veut faire maintenant, on n'a sans doute pas besoin de tout, i.e. de l'axiome pertinent d'associativité de la composition et celui [page 107] sur les unités, mais seulement que les 1_x soient des unités bilatères pour la composition.

En supossant donné ce supplément de structure sur le foncteur Ch, et revenant à la considération du foncteurs associé

$$C : \mathcal{M}/S \longrightarrow \underline{\mathrm{Fib}}\ S,$$

on trouve un morphisme canonique

$$C(C(X)) \xrightarrow{\gamma_X} C(X)$$

par

$$(x, c, c') \longmapsto (x, c' \circ c),$$

et l'axiome des unités bilatères nous assure aussitôt que

$$(*) \qquad \gamma_X \circ i_{C(X)} = \gamma_X \circ C(i_X) = \mathrm{id}_{C(X)}.$$

Comme $i_{C(X)} \in W$, cela prouve déjà que $\gamma_X \in W$, donc $\gamma_X \in W_S$. Donc γ_X devient un isomorphisme dans $(\underline{\mathrm{Fib}}\ S)W_S^{-1}$, et la relation $(*)$ implique bien $i_{C(X)} = C(i_X)$ dans le localisé, q.e.d.

Je vais à nouveau résumer ce que j'ai dégagé.

Théorème-Scholie 2. Soit $\mathcal{M}$ une catégorie, munie de deux ensembles de flèches [page 108]

$$W,\ \mathrm{Fib} \subseteq \mathrm{Fl}(\mathcal{M}).$$

On fait les hypothèses suivantes.

Commentary · Exposition [ms. 107]

What composition contributes to the reflection

Grothendieck asks for a composition of paths because it contracts the two-step replacement back to a one-step replacement. For this purpose there is no triple of paths to compare. Associativity therefore has no role in the proof: the only identities used are $c \circ 1 = c$ and $1 \circ c = c$. This explains the deliberately weaker formulation on pp. 106–107.

With the corrected order of the two unit insertions, the formula

$$\gamma_X(x, c, c') = (x, c' \circ c)$$

gives a common left inverse of i_{CX} and $C(i_X)$. Two-out-of-three makes γ_X a weak equivalence, because $\gamma_X i_{CX} = \mathrm{id}$ and $i_{CX} \in W$. Both its source and target are fibrant over S, so Hypothesis 1

also makes it a fine weak equivalence. Its inverse in the fine localization is therefore both i_{CX} and $C(i_X)$. The path calculation of manuscript p. 69 has now been isolated from its construction in Cat.

An internal category would require associative composition in addition to the endpoint and unit identities, and its set of arrows on a test object T would be $P(X)(T)$. The edition's $\mathrm{Ch}(T)$ in the paragraph spanning pp. 106–107 is a type error: $\mathrm{Ch}(T)$ is an object of $\mathcal{M}$, whereas the discussion concerns a set of arrows between elements of $X(T)$. The intended expression is $\mathrm{Ch}(X)(T)$.

There is also a shorter route once the comparison of localizations has been proved. That comparison can transfer the required equality from the coarse localization back to the localized fibrant subcategory. We give the argument at p. 111. It shows that the path composition is a useful geometric witness, while the formal reflection theorem itself needs only the functorial replacement of A. This strengthens the stated theorem without changing the validity of the path proof.

1° W contient les isomorphismes, si f, g sont composables, si deux parmi f, g, gf sont dans W, le troisième aussi.

2° $\mathcal{M}$ est stable par $\varprojlim$ finies, Fib contient les isomorphismes, est stable par composition et par changement de base ([59]).

Pour S dans $\mathcal{M}$, $\boxed{\underline{\mathrm{Fib}}\ S}$ désigne la sous-catégorie pleine de $\mathcal{M}/S$ formée des $X \longrightarrow S$ qui sont dans Fib. On distingue deux versions différentes d'un axiome de factorisation d'une $f : X \longrightarrow S$ quelconque dans $\mathcal{M}$ en $X \xrightarrow{i} \bar{X} \xrightarrow{\bar{f}} S$, avec $i \in W$, $\bar{f} \in \mathrm{Fib}$.

A) On se donne, pour tout S dans $\mathcal{M}$, un foncteur p. 68

$$\mathcal{M}/S \longrightarrow \mathcal{M}/S \qquad\qquad X \longmapsto C(X) \ \ (= \bar{X})$$

et un morphisme de foncteurs en X

$$X \xrightarrow{i_X} C(X)$$

avec

$$i_X \in W, \qquad\qquad C(X) \in \mathrm{Ob}\,\underline{\mathrm{Fib}}\ S.$$

B) Désignant par

$$\mathrm{Cof}_0 \subseteq \mathrm{Fl}\,\mathcal{M}$$

l'ensemble des flèches de $\mathcal{M}$ qui possèdent la propriété de relèvement à droite `[plutôt à gauche]` par rapport à Fib, on suppose 'à la Quillen', que toute flèche $f : X \longrightarrow S$ se factorise en

$$X \xrightarrow{i_X} C(X) \xrightarrow{\bar{f}} S,$$

avec

$$i \in \mathrm{Cofib}_0 \cap W, \qquad\qquad \bar{f} \in \mathrm{Fib}.$$

`[où` $\mathrm{Cof}_0 = \mathrm{Cofib}_0$`]`.

Donnons nous S dans $\mathcal{M}$, considérons

$$W_S \subseteq \mathrm{Fl}(\mathcal{M}/S)$$

[page 109]

$$f : X \longrightarrow Y \qquad \searrow \quad \swarrow \qquad S$$

l'ensemble de flèches induit par W (est dans W_S si et seulement si il l'est en oubliant S), et désignons par W_S (par abus de notation) l'ensemble de flèches induit par W_S sur $\underline{\mathrm{Fib}}\ S$, ou plus généralement sur toute sous-catégorie pleine $\mathcal{F}$ de $\mathcal{M}/S$, avec

$$\underline{\mathrm{Fib}}\ S \subseteq \mathcal{F} \subseteq \mathcal{M}/S.$$

59. Ces hypothèses de stabilité sur Fib ne serviront que dans II, et encore seulement dans le cas A), repris in toto sous la forme A′ plus bas (p. 112).

Commentary · Exposition [ms. 108]

Reading the two versions of the abstract theorem

Theorem-Scholie 2 organizes the preceding work around two independent questions. The first is whether fibrant representatives compute the coarse localization. The second is whether those representatives form a reflective subcategory of a finer localization. Keeping the two questions separate makes the scope of each hypothesis transparent.

For I under A, we need a full subcategory $\mathcal{F} \subseteq \mathcal{A}$, a functor $C : \mathcal{A} \to \mathcal{F}$, and a natural transformation $\mathrm{id}_{\mathcal{A}} \to C$ whose components are in W. Two-out-of-three shows that C preserves W. Restricting the transformation to $\mathcal{F}$ gives the second comparison with the identity, and hence an equivalence of localizations. Finite limits, fibrations, and lifting properties are irrelevant to this formal argument. Their role is to provide the replacement in particular examples.

Under B, the functor only appears after localization. The left lifting property supplies its arrows, and the path-object lemma supplies independence. In the proof as written, finite limits and stability of Fib under pullback and composition are used at pp. 92–93 to keep the relative path object fibrant over S. Footnote 59 says that these stability hypotheses occur only in II, and only in A′. That description of the proof is incorrect. It does not affect the theorem, because the hypotheses are included in its statement.

The condition in II is equally precise:

$$W_0 \subseteq W, \qquad W_0 \cap \mathrm{Fl}(\mathcal{F}) = W \cap \mathrm{Fl}(\mathcal{F}).$$

It is an equality on arrows between good objects, not on all arrows with one good endpoint. In particular, the replacement map $X \to CX$ need not belong to W_0. The resulting unit in the fine localization is generally noninvertible, while its image in the coarse localization is invertible. That asymmetry is exactly what permits a nontrivial reflection.

The term “localement constant” in the abstract theorem is motivated by the application to Cat/S. For an arbitrary relative category $\mathcal{M}$, the localization $(\mathcal{M}/S)[W_S^{-1}]$ has been defined without a notion of transport along arrows of S. Its identification with local systems is an additional interpretation supplied by a model of homotopy types. The formal theorem establishes the reflection; it does not by itself define monodromy or a theory of locally constant diagrams.

Ⓘ Le foncteur

$$(*) \qquad (\underline{\mathrm{Fib}}\ S)W_S^{-1} \xrightarrow{\sim} (\mathcal{M}/S)W_S^{-1}$$

induit par l’inclusion est une équivalence de catégories, plus généralement pour toute $\mathcal{F}$ comme dessus

$$(\underline{\mathrm{Fib}}\ S)W_S^{-1} \longrightarrow \mathcal{F}W_S^{-1} \longrightarrow (\mathcal{M}/S)W_S^{-1}$$

sont des équivalences de catégories. Un foncteur quasi-inverse est obtenu (dans les deux cas A) et B)) à l’aide du foncteur

$$\gamma : \mathcal{M}/S \longrightarrow (\underline{\mathrm{Fib}}\ S)W_S^{-1}$$

qui est donné sur les objets par p. 69

$$\gamma(X) = C(X),$$

et qui sur les flèches se définit de façon évidente dans le cas A) (fonctorialité de $C(X)$ en X), et dans le cas de B) se construit ad hoc, comme la seule loi fonctorielle pour laquelle les homomorphismes

$$i_X : X \longrightarrow C(X)$$

(interprétés comme flèches dans $(\mathcal{M}/S)W_S^{-1}$) soient eux-mêmes fonctoriels en X. [page 110]

On désigne par

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \qquad \text{(lc comme ‘localement constant’)}$$

la catégorie des fractions $(\mathcal{M}/S)W_S^{-1}$, équivalente à $(\underline{\mathrm{Fib}}\, S)W_S^{-1}$.

(II) Donnons nous un localiseur W_S^0 dans $\mathcal{M}/S$

$$W_S^0 \subseteq W_S \subseteq \mathrm{Fl}(\mathcal{M}/S),$$

avec l'hypothèse formelle similaire à celle de W (cf. 1°). On suppose que

$3^o)$ $W_S^0 \cap \mathrm{Fl}(\underline{\mathrm{Fib}}\, S) = W_S|\underline{\mathrm{Fib}}\, S$.

(Cf. pages 94-96 pour une façon générale d'obtenir des tels localiseurs. Il faut bien voir que pour définir des 'types d'homotopie' relatifs sur S le localiseur W_S est beaucoup trop grossier, moralement il ne donne qu'une catégorie des types d'homotopie relatifs localement constants, et c'est justement ça qui est montré par l'equivalence de (I), où $\underline{\mathrm{Fib}}\, S$ est visualisé comme formé des 'fibrations localement triviales' sur S.) On a alors un deuxième localisé

$$\mathrm{Ho}_{W_S^0}(S) \overset{\text{déf}}{=} (\mathcal{M}/S)(W_S^0)^{-1},$$

et deux foncteurs

$$(*) \qquad \mathrm{Ho}_{W_S^0}(S) \underset{i}{\overset{\rho}{\rightleftarrows}} \mathrm{Ho}_W^{\mathrm{lc}}(S),$$

où ρ est le foncteur de localisation (strict) évident, dû à l'inclusion $W_S^0 \subseteq W_S$, et où i est essentiellement le foncteur déduit de l'inclusion [page 111]

$$(\underline{\mathrm{Fib}}\, S, \underbrace{W_S|\underline{\mathrm{Fib}}\, S}_{=W_S^0|\underline{\mathrm{Fib}}\, S}) \longrightarrow (\mathcal{M}/S, W_S^0)$$

par localisation *au moyen de* W_S^0, donc le foncteur

$$(**) \qquad (\underline{\mathrm{Fib}}\, S)(W_S^0)^{-1} \longrightarrow (\mathcal{M}/S)(W_S^0)^{-1},$$

compte tenu de l'équivalence $(*)$ de (I), i.e. i est le composé du quasi-inverse de $(*)$, avec le p. 70
foncteur $(**)$ précédent. Explicitement, il est défini par

$$i(X) = \text{classe de C(X) dans } (\mathcal{M}/S)(W_S^0)^{-1},$$

en tenant compte du fait que $X \longmapsto C(X)$ transforme W_S en W_S^o, grâce à la condition $3^o)$ sur W_S^0.

Ceci posé, *sous l'hypothèse* B ('Quillen'), on a : dans le couple de foncteurs (ρ, i) de $(*)$, page 110, *i est adjoint à droite de ρ,* i.e. on a pour $\xi \in \mathrm{Ob}\,\mathrm{Ho}_{W_S^0}(S)$, $\eta \in \mathrm{Ob}\,\mathrm{Ho}_W^{\mathrm{lc}}(S)$

$$\mathrm{Hom}_{W_S^0}(\xi, i(\eta)) \simeq \mathrm{Hom}_{W_S}(\rho(\xi), \eta),$$

ce qui s'explicite, pour X, Y dans $\mathcal{M}/S$, par

$$\boxed{\mathrm{Hom}_{W_S^0}(X, C(Y)) \simeq \mathrm{Hom}_{W_S}(X, Y)}$$

([60]). Il s'ensuit que *le foncteur i est pleinement fidèle.*

La même formule d'adjonction, et la pleine fidélité de i, sont valables dans le cas A, quand [page 112]
on renforce l'hypothèse A (relative à un S particulier) en une hypothèse plus restrictive et plus compréhensive, que voici.

60. **NB** Le premier membre est indépendant du choix de W_S^0! (Pourvu seulement que l'hypothèse $W_S^0 \cap \mathrm{Fl}\,\underline{\mathrm{Fib}}\, S = W_S \cap \mathrm{Fl}\,\underline{\mathrm{Fib}}\, S$, en plus de $W_S^0 \subseteq W_S$, soit satisfaite.)

Commentary · Later developments [ms. 111]

Functorial replacement already gives the reflection

Grothendieck's construction has supplied more than the final formulation uses. Hypothesis A alone suffices for II; the additional path composition in A′ is unnecessary for the adjunction. The missing observation is that I has already proved full faithfulness in the coarse localization. We can use that result to establish coherence without cancelling a noninvertible unit in the fine localization.

Here is the precise argument. Let $\mathcal{F} \subseteq \mathcal{A}$ be full, let $W_0 \subseteq W$ contain the isomorphisms and satisfy two-out-of-three, and assume that they agree on $\mathcal{F}$. Suppose given a functor $C : \mathcal{A} \to \mathcal{F}$ and a natural transformation $\eta : \mathrm{id}_{\mathcal{A}} \to C$ with $\eta_X \in W$. Put

$$\mathcal{D} = \mathcal{A}[W_0^{-1}], \qquad \mathcal{B} = \mathcal{A}[W^{-1}], \qquad \mathcal{K} = \mathcal{F}[W^{-1}].$$

By I, the inclusion $J : \mathcal{K} \to \mathcal{B}$ is an equivalence. The inclusion into the fine localization gives $k : \mathcal{K} \to \mathcal{D}$. The functor C induces $q : \mathcal{D} \to \mathcal{K}$, since it carries $W_0 \subseteq W$ into W between objects of $\mathcal{F}$. The transformation η descends to $\mathrm{id}_{\mathcal{D}} \to kq$.

We first compare the iterated units in $\mathcal{K}$. Naturality in $\mathcal{A}$ gives

$$C(\eta_X)\eta_X = \eta_{CX}\eta_X.$$

In $\mathcal{B}$, η_X is invertible, so $C(\eta_X) = \eta_{CX}$. Both arrows are arrows of $\mathcal{F}$. Since J is faithful, their classes are equal in $\mathcal{K}$ itself. There is no appeal here to the desired full faithfulness of k; the faithful functor is the coarse comparison J, whose equivalence was established before W_0 entered the argument.

For $Y \in \mathcal{F}$, the arrow $\eta_Y : Y \to CY$ is a weak equivalence in $\mathcal{F}$. Define a natural counit $\varepsilon : qk \to \mathrm{id}_{\mathcal{K}}$ by $\varepsilon_Y = [\eta_Y]^{-1}$. Naturality follows by inverting the natural isomorphism induced by $\eta|_{\mathcal{F}}$. The triangle identities are now explicit:

$$k(\varepsilon_Y)\eta_{kY} = \mathrm{id}_{kY}, \qquad \varepsilon_{qX}\, q(\eta_X) = [\eta_{CX}]^{-1}[C(\eta_X)] = \mathrm{id}_{qX}.$$

Thus $q \dashv k$. Its counit is an isomorphism, so k is fully faithful. Composing on the left with the equivalence J identifies q with the canonical localization $\rho : \mathcal{D} \to \mathcal{B}$ and gives

$$\rho \dashv kJ^{-1}.$$

The functor kJ^{-1} is Grothendieck's i. In particular,

$$\mathrm{Hom}_{\mathcal{D}}(X, CY) \simeq \mathrm{Hom}_{\mathcal{B}}(X, Y)$$

under A without A′, and for every choice of W_0 with the stated agreement property.

The bijection can also be written without invoking the adjunction as a black box. For $Y \in \mathcal{K}$, it sends $f : X \to kY$ to $\varepsilon_Y q(f) : qX \to Y$. Its inverse sends $h : qX \to Y$ to $k(h)\eta_X$. Substituting one formula into the other, naturality of η or ε moves the comparison map past f or h, and the corresponding triangle identity cancels the remaining pair. These formulas establish both naturality variables and show that the comparison with the coarse localization is the canonical one. In particular, no new choice of a map from X into a fibrant target has been made.

This strengthening belongs to the general theory of derived adjunctions, rather than being a new phenomenon. Apply the deformable-adjunction theorem, Riehl (2014, Theorem 2.2.11), to the identity adjunction between $(\mathcal{A}, W_0)$ and $(\mathcal{A}, W)$: its left functor is already homotopical, and its right functor admits the right deformation C, on whose image the two classes agree. The proof above is included to make the application independent of a stronger axiomatization of homotopical categories, and to show precisely how the reflection lemma of manuscript p. 67 receives its coherence condition here.

A′) Le foncteur $C : \mathcal{M}/S \longrightarrow \mathcal{M}/S$ et le morphisme fonctoriel $i_X : X \longrightarrow C(X)$ sont construits 'à la Cartan-Serre' à partir d'un foncteur défini sur $\mathcal{M}$ entier

$$\mathcal{M} \xrightarrow{\mathrm{Ch}} \mathcal{M},$$

du type 'chemins', i.e. muni en plus des données suivantes

a)
$$X \xrightarrow{\alpha_X} \mathrm{Ch}(X) \xrightarrow{p_X=(\sigma_X,\beta_X)} X \times X$$
(structure de source et but des chemins, et de chemins triviaux), et

b) Une composition des chemins, sans exiger ici l'associativité de la composition mais seulement que les chemins triviaux soient unités bilatères pour la composition. De plus, on suppose que pout tout X
$$\begin{cases} \alpha_X \in W,\ p_X \in \mathrm{Fib} \\ (X \longrightarrow e) \in \mathrm{Fib}. \end{cases}$$

Corollaire. Soient X dans $\mathcal{M}/S$, Y dans $\underline{\mathrm{Fib}}\, S$, alors p. 71 [page 113]
$$\mathrm{Hom}_{W_S^0}(X,Y) \xrightarrow{\sim} \mathrm{Hom}_{W_S}(X,Y).$$

Démonstration. Par la formule d'adjonction (p. 111), le deuxième membre est $\mathrm{Hom}_{W_S^0}(X, CY)$. Mais comme Y et $C(Y)$ sont dans $\underline{\mathrm{Fib}}\, S$, où W_S et W_S^0 coïncident, $Y \xrightarrow{i_Y} C(Y)$ est non seulement dans W_S, mais dans W_S^0, donc $\mathrm{Hom}_{W_S^0}(X,Y) \xrightarrow{\sim} \mathrm{Hom}_{W_S^0}(X, CY)$, d'où le résultat.

Revenons à Cat, Cat$/S$. La digression précédente sur $(\mathcal{M}, W, \mathrm{Fib})$ a permis de mieux situer ce qui concerne le foncteur C, et les relations entre les catégories localisées de
$$\mathrm{Cat}/S \ \ (\text{pour } W_S,\ W_S^{\mathrm{d}},\ W_S^{\mathrm{g}}) \qquad \text{et} \qquad \underline{\mathrm{Parf}}\, S \ \ (\text{pour } W_S^{\mathrm{f}} = W_S = W_S^{\mathrm{d}} = W_S^{\mathrm{g}})$$
ainsi que pour les catégories intermédiaires
$$\underline{\mathrm{Parf}}\, S \subseteq \mathcal{F} \subseteq \mathrm{Cat}/S$$
pleines de Cat$/S$ (sur le diagramme p. 76, il y en a cinq autres, savoir $\underline{\mathrm{Fibhot}}\, S$, $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Prop}}\, S$, $\underline{\mathrm{Prop}}_0\, S$). Pour ce qui concerne les relations entre les localisés de ces sept catégories, qui se trouvent au centre du diagramme, les énoncés du théorème-scholie 1 (p. 70-81) ne donnent rien de plus que le contenu du théorème-scholie 2 (p. 108-113). Pour les établir, on a donc le choix entre les deux hypothèses A) ou B) dudit théorème. Soit utiliser un foncteur C_S explicite, à l'aide d'un foncteur $X \longmapsto \underline{\mathrm{Ch}}(X)$ ('chemins dans X') sur Cat, donc les structures [page 114] de source, but et composition pour les chemins, et les chemins triviaux (agissant comme unités bilatères pour la composition). La construction d'un tel foncteur C n'a par l'air trivial et semble ne pas être connue - j'ai peiné pour y arriver! (Avec, en plus, l'associativité de la composition …). Ou bien utiliser, à la Quillen, l'existence d'une classe de morphismes dits Q-fibrants, avec des bonnes propriétés de factorisation. Ce n'est pas non plus évident dans Cat, et c'est établi par Thomason, par une voie un peu compliquée. Peut-être serai-je amené à revenir dessus - mais seulement en cas de besoin!

Commentary · Examples [ms. 113]

Different fine localizers can have the same fibrant targets

Grothendieck's corollary says more than full faithfulness on pairs of fibrant objects. The source X is arbitrary: it is only the *target* Y that must be fibrant. The canonical map of Hom sets is bijective because $Y \to CY$ is already inverted in the finer localization, and the reflection computes maps into CY. This is why the Hom set in footnote 60 is independent of the chosen fine localizer.

A finite-set example isolates the statement from homotopical geometry. Let $\mathcal{A}$ be a skeleton of finite sets, let W consist of all maps, let $\mathcal{F}$ consist only of a singleton 1, and put $C(X) = 1$. The unique maps $X \to 1$ form a natural transformation in W. The coarse localization is equivalent to the terminal category: after all arrows are inverted, $X \to 1$ is a natural isomorphism to the constant functor. The full subcategory $\mathcal{F}$ also localizes to the terminal category.

First take W_0 to be the bijections. Its localization is just the category of finite sets. The reflection to the terminal category has right adjoint selecting 1, since

$$\mathrm{Hom}_{\mathcal{A}}(X, 1) = \{*\}$$

for every finite X, including the empty set. It plainly does not follow that the fine localization is terminal: for example $\mathrm{Hom}(1, \{0, 1\})$ has two elements.

For a second fine class, let W_1 consist of the maps which preserve whether a finite set is empty. Equivalently, W_1 contains the identity of the empty set and every map between nonempty finite sets. It is the inverse image of the isomorphisms under the functor

$$E : \mathcal{A} \to [1], \qquad E(\varnothing) = 0, \quad E(X) = 1 \quad (X \neq \varnothing),$$

so it satisfies two-out-of-three. The localization at W_1 is equivalent to [1]. Indeed, the functor $j : [1] \to \mathcal{A}$ selecting $\varnothing$ and 1 satisfies $Ej = \mathrm{id}$, and the natural maps $X \to jE(X)$, identity on the empty set and the unique map to 1 otherwise, all lie in W_1. These exhibit quasi-inverse functors after localization.

Thus the two fine categories are FinSet and [1], which are very different. Nevertheless, both give a singleton set of maps from any object into the distinguished fibrant target. Both W_0 and W_1 agree with W on $\mathcal{F}$, so the corollary predicts exactly this agreement. The example also makes the direction of the result unmistakable: maps *out of* the distinguished object are not fixed. In FinSet there are two maps $1 \to \{0, 1\}$; in [1] their images coincide.

Par contre, les assertions concernant les catégories $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Fibsc}}\, S$, ou $\underline{\mathrm{Cofib}}\, S$, $\underline{\mathrm{Cofibsc}}\, S$, et leurs variantes à indices 0 (donc les huit catégories-ailes du diagramme), et leurs relations à Cat/S d'une part, à $\underline{\mathrm{Fibhot}}\, S$ de l'autre (pour les catégories à indice 0), ne peuvent être contenues dans un tapis général sur des catégories de modèles $(\mathcal{M}, W)$, me semble-t-il, vu la nature très spéciale des quatre catégories désignées ci-dessus, de leur rôle très particulier dans le formalisme catégorique. Or tout au moins les catégories 'scindées' $\underline{\mathrm{Fibsc}}\, S$, $\underline{\mathrm{Cofibsc}}\, S$, et par suite aussi leurs variantes à indice 0, jouent un rôle essentiel pour la théorie des $\mathrm{HOT}(S)$, vu que c'est elles qui interviennent dualement quand on applique la recette générale [page 115] de construction d'un dérivateur à partir d'une catégorie de modèles $(\mathcal{M}, W)$. Donc je ne vois pas de simplification substantielle à apporter au théorème-scholie 1. Tout au plus pourrait-on p. 72 élaguer le diagramme de catégories envisagées, en laissant tomber $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Prop}}\, S$, $\underline{\mathrm{Prop}}_0 S$, qui n'y jouent pas un rôle vraiment utile. Pour ce qui concerne les résultats énoncés, ce qui les concerne est couvert par ce qui est dit pour les sous-catégories pleines $\mathcal{F}$ de Cat/S qui contiennent (disons) $\underline{\mathrm{Parf}}\, S$ (ou au choix celles qui contiennent $\underline{\mathrm{Fib}}\, S$ ou $\underline{\mathrm{Cofib}}\, S$).

Commentary · Later developments [ms. 114]

The model-category route in Cat

Grothendieck invokes Thomason to explain that the abstract replacement argument is not restricted to topological or simplicial objects. Thomason's model structure has as weak equivalences the functors whose nerves are weak homotopy equivalences, and as fibrations the functors detected as Kan fibrations by $\mathrm{Ex}^2 N$; its existence is Theorem 4.9 of Thomason (1980). Factorization into a trivial cofibration followed by a fibration supplies hypothesis B. The theorem concerns the ordinary homotopy theory of nerves; it should not be cited as constructing a model structure for an unspecified class W.

There is a historical correction relevant to the preceding properness discussion. Thomason's original proof asserted that Dwyer maps are closed under retracts and hence that every cofibration is a Dwyer map. Cisinski (1999, pp. 228–229) gives a counterexample. His Lemma 4 and Proposition 2 on p. 231 replace Dwyer maps by pseudo-Dwyer maps and recover properness. The model-category existence theorem is unaffected. Thus the later correction repairs an auxiliary assertion in the cited literature, not the validity of Grothendieck's abstract comparison theorem.

The two uses of "fibration" in this chapter must remain distinct. A Grothendieck fibration over S carries cartesian lifts of arrows and presents a contravariant diagram of categories. A Thomason fibration is defined by a lifting condition after applying $\mathrm{Ex}^2 N$. A categorical comparison between

> their localized presentations is a theorem, not a consequence of the shared terminology. This is the reason the eight outer categories in the large comparison diagram do not disappear merely because one has a model structure on Cat: they encode strict or split diagrammatic data which the abstract factorization axioms have not named.
>
> The return to those categories is therefore mathematically necessary. The abstract detour has made the localizations and reflection intelligible; the split-fibration presentation will now make variation in S intelligible. These are complementary uses of the same models, and neither replaces the other.

La description des catégories du type $\mathrm{HOT}(S)$, pour S fixé, me paraît à présent parfaitement bien comprise. Il est temps de faire varier S, et de comprendre le caractère contravariant en S. On se donne donc

$$S' \xrightarrow{f} S,$$

et on se propose de décrire les foncteurs image inverse pour les trois types de catégories $\mathrm{HOT}(S)$. Mais nous allons nous borner à présent aux $\mathrm{HOT}_W(S)$ et $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, puisque tout ce qui concerne $\mathrm{HOT}^o_W(S) = \mathrm{HOT}_W(S^o)$ `[plutôt` $\mathrm{HOT}_W(S^o)^o$`, cf. p. 50]` s'en déduira par dualité $S \longmapsto S^o$. Donc il s'agit de décrire [page 116]

$$(*) \qquad f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$$

et le foncteur induit

$$f^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

Quand on décrit les catégories $\mathrm{HOT}(S)$, conformément à la recette générale, à l'aide de foncteurs

$$F : S^o \longrightarrow \mathrm{Cat}$$

(préfaisceaux en catégories sur S), ou, ce qui revient au même, de catégories scindées, la description de la flèche $(*)$ est évidente : elle revient à la loi

$$\left|\begin{array}{ccc} F & \longmapsto & F \circ f^o \\ \underline{\mathrm{Hom}}(S^o, \mathrm{Cat}) & \longrightarrow & \underline{\mathrm{Hom}}(S'^o, \mathrm{Cat}), \end{array}\right.$$

en passant aux localisés pour la W-équivalence argument par argument. Dans le langage des catégories fibrées, il est bien connu que cette opération triviale sur les préfaisceaux en catégories correspond à l'opération *image inverse* de catégories sur S - laquelle opération passe aux scindages. Cela nous donne donc

$$f^*(\mathrm{hot}_{W,S}(X)) = \mathrm{hot}_{W,S'}(\underbrace{X \times_S S'}_{X'}),$$

pour toute X fibrée scindée sur S. Or on sait que le scindage de X ne joue aucun rôle, finalement, dans la description de la classe

$$\mathrm{hot}_{W,S}(X) \in \mathrm{HOT}_W(S)$$

associée à X. On peut oublier le scindage, et même (pour étudier les $\mathrm{Hom}_{\mathrm{Hot}_W(S)}(X,Y)$) travailler avec des S-foncteurs *quelconque* entre catégories fibrées sur S - pas la peine même de les supposer cartésiens, voir même compatibles avec des scindages. Cela me paraît une chose très importante pour le formalisme des $\mathrm{HOT}_W(S)$, et très spéciale au cas de la catégorie de modèles Cat. C'est un peu tacite dans le théorème-scholie - mais mes doutes initiaux concernant la catégorie Fibsc elle-même montrent bien que je sentais qu'il y avait là quelque chose de pas évident, ni même (presque) `[?]` de vraisemblable. Et pourtant l'existence du foncteur Φ montre qu'il en est bien ainsi. p. 73 [page 117]

Commentary · Exposition [ms. 116]

Pullback is precomposition in the split presentation

Grothendieck returns to split fibrations because a change of base is completely visible in that presentation. For a strict functor $F : S^{\mathrm{op}} \to \mathrm{Cat}$, write $\int_S F$ for its Grothendieck construction. An object is a pair (s, x) with $x \in F(s)$; an arrow $(s, x) \to (t, y)$ consists of $a : s \to t$ and a map $x \to F(a)y$ in $F(s)$. The projection to S is the associated split fibration.

If $f : S' \to S$, there is a canonical isomorphism over S'

$$S' \times_S \int_S F \;\cong\; \int_{S'} (F f^{\mathrm{op}}).$$

On objects it sends (s', x) with $x \in F(f(s'))$ to the same pair. On an arrow $a : s' \to t'$ it uses the same map $x \to F(f(a))y$. Identities and composition agree because F and f are functors. The chosen cartesian lifts are carried to the chosen cartesian lifts, so the isomorphism also respects the splittings.

An objectwise weak equivalence of functors remains objectwise a weak equivalence after precomposition: its component at s' is the original component at $f(s')$. This proves that the operation descends to the relevant localization and gives the formula for f^* on split representatives. If $g : S'' \to S'$ is another functor, precomposition satisfies

$$(F f^{\mathrm{op}}) g^{\mathrm{op}} = F(fg)^{\mathrm{op}}$$

strictly. After changing to equivalent presentations by arbitrary fibrations or by categories over S, these equalities are expressed by the canonical comparison isomorphisms for iterated pullbacks. The coherence comes from the same strict composition law, rather than from a new choice of replacement for each composite.

This paragraph only establishes the formula on the specified good representatives. Extending it to other objects or deciding when an arbitrary pullback computes the derived inverse image requires the base-change results on the following pages. The categorical calculation here explains why Grothendieck keeps the split models in the theory even after showing that their localized category is independent of the splitting.

On peut donc interpréter f^* en termes de $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Fib}}\, S'$, par le foncteur changement de base

$$f^* : \underline{\mathrm{Fib}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S', \qquad X \longmapsto X \times_S S'/S',$$

lequel foncteur préserve évidemment la W-équivalence par fibres W_S^{f}, $W_{S'}^{\mathrm{f}}$, qu'est celle pertinente ici

$$f^*(W_S^{\mathrm{f}}) \subseteq W_{S'}^{\mathrm{f}},$$

de sorte que f^* induit bien

$$(\underline{\mathrm{Fib}}\, S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}\, S')(W_S^{\mathrm{f}})^{-1}$$

[plutôt $(\underline{\mathrm{Fib}}\, S')(W_{S'}^{\mathrm{f}})^{-1}$], i.e. le foncteur $(*)$ [page 118]

$$f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S').$$

D'autre part, $\mathrm{HOT}_W(S)$ peut être décrit aussi comme localisé de Cat/S tout entier, par rapport au localiseur W_S^{d}, lequel sur $\underline{\mathrm{Fib}}\, S$, et même sur $\underline{\mathrm{Liss}}\, S$, coïncide avec W_S^{f}. Mais on se bute ici à la difficulté que le foncteur image inverse

$$f^* : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S', \qquad X \longmapsto X \times_S S'/S',$$

n'est *pas* compatible en général (si S n'est isomorphe à la catégorie ponctuelle, pour S' quelconque) avec les localiseurs W_S^{d}, $W_{S'}^{\mathrm{d}}$. On le voit p.ex. en prenant $S' = \{s\}$, et en se rappelant que W_S^{d} n'implique *pas*, sur Cat/S tout entier, la W-équivalence par fibres.

Commentary · Exposition [ms. 117]

Restriction must be calculated on a suitable model

Grothendieck now makes the base category vary. The comparison theorem has done more than identify several presentations of one category: it provides presentations on which an operation can be performed before localization. For inverse image the useful presentation consists of fibrations, or more generally smooth categories over the base. The calculation then reduces to the familiar operation on fibres.

Write $q_S : \mathrm{Cat}/S \to \mathrm{HOT}_W(S)$ for localization at W_S^{d}. If $X \to S$ is smooth, the strict fibre X_s computes the same homotopy type as the comma category $s\backslash X$. For $f : S' \to S$ and $s' \in S'$, the strict fibre of $X \times_S S'$ at s' is exactly $X_{f(s')}$. Consequently a map between smooth objects which is a W-equivalence on comma categories remains so after this pullback. The comparison with split fibrations identifies the resulting localized operation with restriction of category-valued presheaves. This proves Proposition 1, including its naturality in maps between smooth objects.

The qualification on the objects matters even when the morphism being pulled back already belongs to W_S^{d}. Its membership says something about all comma fibres; it says nothing by itself about its strict fibres. Pullback along the inclusion of an object extracts a strict fibre. Thus arbitrary pullback cannot act directly on the entire relative category $(\mathrm{Cat}/S, W_S^{\mathrm{d}})$. The commentary at manuscript p. 42 ("When strict fibres compute a comma fibre") isolates precisely the comparison required here.

There is also a useful distinction between the arrows allowed in this presentation and the arrows needed to describe transport. For calculating the category $\mathrm{HOT}_W(S)$, all functors over S between fibrations are allowed; Cartesian functors form a smaller presentation which gives the same localization. A chosen cleavage is therefore not part of a relative homotopy type. When a natural transformation between changes of base is considered on ms. 128–130, a cleavage will nevertheless provide a convenient formula for the induced map. Independence of that auxiliary choice is a separate assertion, not an additional restriction on the morphisms of $\mathrm{HOT}_W(S)$.

Ces ennuis n'ont pas lieu sûr $\underline{\mathrm{Liss}}\, S$, où W_S^{d} et W_S^{f} coïncident. On peut donc décrire f^* sur les HOT aussi à l'aide de

$$f^* : \underline{\mathrm{Liss}}\, S \longrightarrow \underline{\mathrm{Liss}}\, S', \qquad X \longmapsto X \times_S S'/S',$$

en passant aux localisés

$$f^* : (\underline{\mathrm{Liss}}\, S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Liss}}\, S')(W_{S'}^{\mathrm{f}})^{-1}.$$

En d'autres termes :

Proposition 1. *Si X dans Cat/S est lisse sur S, alors on a un isomorphisme canonique*

$$f^*(\mathrm{hot}_{W,S}(X)) \simeq \mathrm{hot}_{W,S'}(X'), \qquad \text{où } X' = X \times_S S'/S'.$$

Une remarque similaire s'applique à $\underline{\mathrm{Fibhot}}\, S$, où $W_S^{\mathrm{d}} = W_S^{\mathrm{f}}$ également. Le foncteur changement de base induit p. 74 [page 119]

$$\underline{\mathrm{Fibhot}}\, S \longrightarrow \underline{\mathrm{Fibhot}}\, S', \qquad X \longmapsto X \times_S S'/S',$$

compatible avec les localiseurs relatifs pertinents (tous ceux envisagés jusqu'à présent coïncident sur ces catégories, avec les hypothèses faites sur W), d'où

$$f^* : (\underline{\mathrm{Fibhot}}\, S)W_S^{-1} \longrightarrow (\underline{\mathrm{Fibhot}}\, S')W_{S'}^{-1}.$$

Les localisés cette fois ne sont pas les HOT_W, mais les $\mathrm{HOT}_W^{\mathrm{lc}}$, donc on trouve une description sympathique de

$$f^* : \mathrm{HOT}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}}(S'),$$

plus avantageuse encore qu'en termes de $\underline{\mathrm{Liss}}_0 S$, $\underline{\mathrm{Liss}}_0 S'$, car s'appliquant à des catégories un peu plus vastes (NB $\underline{\mathrm{Liss}}_0 \subseteq \underline{\mathrm{Fibhot}}$). Notons donc le [page 120]

Corollaire 1. *La formule de la proposition 1 (p. 118) reste valable si X est supposé* Hot-*fibrant sur S (au lieu de lisse sur S).* (On trouve alors des types d'homotopie relatifs *localement constants* sur S, S'.)

On a donc une description de f^* qui s'applique à toutes les catégories du côté gauche du diagramme [page] 76, à la seule exception de Cat/S elle-même. Si X est dans Cat/S, sans être ni lisse ni Hot-fibrée sur S, comment trouver l'image inverse dans $\mathrm{HOT}_W(S')$? On a le choix entre Φ et C pour améliorer la situation! On sait que X définit la même classe dans $\mathrm{HOT}_W(S)$ que $\Phi(X)$, $C(X)$, qui sont l'une lisse (et même fibrée scindée), l'autre Hot-fibrée (et même parfaite) sur S. Donc

Commentary · Present state [ms. 120]

The two replacements do different work

Grothendieck wants one formula for the inverse image of an arbitrary category over S. The comma replacement Φ_S supplies it. The path replacement C_S supplies a formula for a different object, namely the reflection into locally constant relative types. At this point the text briefly identifies those two tasks. Both the sentence preceding Corollary 2 and its second displayed isomorphism are present in the edition; the discrepancy is not introduced by the transcription.

The distinction is already established in the source. The natural map

$$X \longrightarrow \Phi_S(X)$$

belongs to W_S^{d}, whereas $X \longrightarrow C_S(X)$ is generally only in W_S. Accordingly the valid general formula is

$$f^* q_S(X) \simeq q_{S'}\big(\Phi_S(X) \times_S S'\big).$$

If $i_S : \mathrm{HOT}_W^{\mathrm{lc}}(S) \to \mathrm{HOT}_W(S)$ denotes the inclusion, the formula involving C_S is

$$f^* i_S \rho_S q_S(X) \simeq q_{S'}\big(C_S(X) \times_S S'\big).$$

Here restriction preserves locally constant objects, so that $f^* i_S$ identifies with $i_{S'} f_{\mathrm{lc}}^*$. These equations become the same only when $q_S(X)$ is already locally constant.

A small calculation makes the error decisive without repeating the examples given in the commentary at manuscript pp. 8 and 32. Take $W = W_\infty$, $S = (0 \to 1)$ and $X = \{0\} \to S$. The comma fibres of X are a point at 0 and the empty category at 1. Its relative type is therefore the presheaf of spaces $(\varnothing \to *)$. The local-constant reflection is the constant point: maps from this presheaf to a locally constant arrow are determined by the value at 0, and evaluation at 0 identifies locally constant arrows with spaces. Pullback to 1 consequently gives $\varnothing$ from X and $*$ from $C_S(X)$. Corollary 2 cannot identify them.

This is a localized mistake in the calculation, not a defect in the comparison theorem. The later discussion on ms. 123–127 correctly distinguishes $\rho_S q_S(X)$ from $q_S(X)$, and the construction of the adjunction on ms. 139–145 consistently uses Φ_S. Keeping the printed error visible is useful: it shows exactly why the two replacement functors were needed.

Corollaire 2. *Si X dans* Cat/S *est quelconque, alors*

$$f^*(\mathrm{hot}_{W,S}(X)) \simeq \mathrm{hot}_{W,S'}(\Phi(X) \times_S S') \simeq \mathrm{hot}_{W,S'}(C(X) \times_S S').$$

On peut s'attendre, d'autre part, que pour certains changements de base particuliers, la formule de la proposition 1 soit valable sans restriction sur X dans Cat/S. Il doit revenir ± au même que le foncteur image inverse

$$X \longmapsto X \times_S S'$$

soit compatible avec les localiseurs W_S^{d}, $W_{S'}^{\mathrm{d}}$. La condition de W-équivalence 'colocale sur S' doit être vue comme une condition de nature 'homologique', pour le dérivateur HOT. p. 75
Elle s'associe, pour un dérivateur quelconque $\mathbf{D}$, à la condition (pour $f : X \longrightarrow Y$ dans [page 121]

Cat/S) que les $s\backslash X \longrightarrow s\backslash Y$ sont des **D**-équivalences, ce qui s'écrit ici élégamment en termes d'homologie relative, par la condition que $\forall\ a \in \mathcal{A} = \mathbf{D}(e)$, la flèche induit par f

$$\underbrace{H_\bullet^{\mathbf{D}}(X/S, a_X)}_{\overset{\text{déf}}{=}(p_X)_!(a_X)} \longrightarrow \underbrace{H_\bullet^{\mathbf{D}}(Y/S, a_Y)}_{\overset{\text{déf}}{=}(p_Y)_!(a_Y)}$$

soit [un] isomorphisme. (Utiliser le 'calcul' des faisceaux d'homologie relative $(p_X)_!(a_X)$, $(p_Y)_!(a_Y)$, par la caractérisation des fibres dans Der 4 bis). Or, les changements de base **D**-*propres* $S' \longrightarrow S$ commutent aux $p_!$, de sorte qu'on trouve le

Lemme. *La notion d'équivalence* **D**-*colocale sur* S *dans* Cat/S *est stable par tout changement de base* $S' \longrightarrow S$ **D**-*propre. (Quel que soit le dérivateur* **D**.*)*

On se rappelera que les $s\backslash S \longrightarrow S$ sont justement des changements des base propres! On peut dire que l'équivalence **D**-colocale peut se décrire en exigeant que $f : X \longrightarrow Y$ soit transformé en W-équivalences par certains types de changement de base $S' \longrightarrow S$, en prenant au choix l'un des trois types

a) Les changements de base $s\backslash S \longrightarrow S$ (ce qui redonne la définition que nous avions prise).

b) Les changements de base qui sont des co-isomorphismes locaux.

c) Les changements de base propres.

d) Les changements de base **D**-propres.

On ne peut appliquer ces résultats qu'au dérivateur HOT, faute de l'avoir encore construit, et verifié, les axiomes pertinents. On peut poser simplement la [page 122]

Commentary · Exposition [ms. 121]

A homological test for colocal equivalence

Grothendieck introduces relative homology here because it packages all the comma-category tests into a single morphism of coherent diagrams. Let $p_X : X \to S$, $p_Y : Y \to S$, and let $u : X \to Y$ be a functor over S. For $a \in \mathbf{D}(e)$, write a_X for the constant coefficient. The map of interest is

$$(p_X)_! a_X \longrightarrow (p_Y)_! a_Y.$$

Its component at s is the map on homotopy colimits of the constant coefficient over $s\backslash X$ and $s\backslash Y$. This is the pointwise Kan-extension formula in the contravariant convention used throughout these pages. If isomorphisms are detected by evaluation, requiring the displayed map to be invertible for every a is exactly the condition that all these comma-category maps be **D**-equivalences.

Now suppose $g : T \to S$ is **D**-proper in the sense that the relevant base-change comparisons for homological direct image are invertible. Pulling back the displayed map gives

$$(p_{X_T})_! a_{X_T} \longrightarrow (p_{Y_T})_! a_{Y_T}, \qquad X_T = X \times_S T, \quad Y_T = Y \times_S T.$$

The first map was invertible, and restriction preserves isomorphisms; hence the new map is invertible. Evaluating at every object of T gives the claimed stability of colocal equivalence. This proves the lemma without assuming that strict fibres compute comma fibres: the base-change theorem supplies that comparison at the level of relative homology.

The four classes of test maps listed in the text are genuinely nested. A representable coslice projection is a discrete opfibration, hence a co-local isomorphism; those maps are proper, and absolute properness implies **D**-properness. Testing against the largest class therefore implies testing against the first. Conversely the lemma shows that the first condition survives the largest class, and a colocal equivalence is in particular a global **D**-equivalence. The source's phrase "three types" precedes four entries and has no mathematical consequence.

The historical terminology keeps the geometrical motivation visible. Properness and smoothness were designed to distinguish the two positions in a base-change square, following the analogous cohomological theorems in algebraic geometry. For the categorical definitions, including their dual formulations by taking mates, see Maltsiniotis (2005, Definition 1.1). Here the lesson is particularly concrete: the general derivator theorem predicts the correct class of changes of base, but cannot yet be used to establish the axioms of the very derivator under construction.

Conjecture (sûre!) : *Soit $p : S' \longrightarrow S$ une flèche propre (sous-entendu :* HOT*-propre) dans* Cat*. Alors le foncteur changement de base*

$$\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S', \qquad X \longmapsto X \times_S S'/S'$$

est compatible avec les localiseurs W_S^{d}, $W_{S'}^{\mathrm{d}}$. (Pour un foncteur $S' \longrightarrow S$ lisse, le changement de base serait compatible avec W_S^{g}, $W_{S'}^{\mathrm{g}}$.) ([61]).

Mais pout tout changement de base $S' \longrightarrow S$ (propre ou non) qui a la propriété précédente, donc qui induit un foncteur

$$\underbrace{(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}}_{\approx \mathrm{HOT}_W(S)} \longrightarrow \underbrace{(\mathrm{Cat}/S')(W_{S'}^{\mathrm{d}})^{-1}}_{\approx \mathrm{HOT}_W(S')} ,$$

il est immédiat, à l'aide du théorème de comparaison (p.ex. avec $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Liss}}\, S'$) sur S et sur S', que ce foncteur n'est autre (à isomorphisme canonique près) que le foncteur f^* image inverse des types d'homotopie relatifs. Donc énoncons le p. 76

Corollaire 3. *La formule de la proposition 1 reste valable sans hypothèse aucune sur X dans* Cat$/S$*, pourvu que la flèche de changement de base $f : S' \longrightarrow S$ soit telle que le foncteur $X \longmapsto X \times_S S'$ de* Cat$/S$ *dans* Cat$/S'$ *transforme W-équivalences colocales sur S en W-équivalences colocales sur S'.* (Ce qui devrait être le cas si (et seulement si?) f est W-propre, donc aussi si f est ('absolument') propre ([62]).) [page 123]

Commentary · Later developments [ms. 122]

The proper base-change conjecture has an elementary proof

The conjecture on ms. 122 is not an open question. Grothendieck himself proves it later in Chapter VI, ms. 168–169, printed p. 103 of the complete edition; the footnote points to that proof. The argument avoids the circular appeal to a derivator whose axioms are still being established (Grothendieck, 1990b, ms. 168–169).

For clarity, isolate its mechanism. If $v : T \to U$ is a universal W-equivalence and $a : A \to B$ is a map over U, there is a pullback square

$$\begin{array}{ccc} A \times_U T & \xrightarrow{a_T} & B \times_U T \\ \downarrow & & \downarrow \\ A & \xrightarrow{a} & B. \end{array}$$

Both vertical maps belong to W. The two-out-of-three property consequently gives $a_T \in W$ if and only if $a \in W$.

Apply this with $g : S' \to S$ proper and an object $s' \in S'$, putting $s = g(s')$. The properness criterion says that

$$s'\backslash S' \longrightarrow s\backslash S$$

is a universal homotopy equivalence; in the relative variant it is a universal W-equivalence. A map in W_S^{d} is a W-equivalence after pullback to $s\backslash S$. The preceding square shows that it remains so after further pullback to $s'\backslash S'$. This is exactly the definition of its pullback belonging to $W_{S'}^{\mathrm{d}}$. Passing to opposite categories gives the smooth, left-fine version.

Corollary 3 therefore gives the expected strict formula for proper changes of base. Its proof does not depend on choosing a preferred representative: the functor defined on all of Cat$/S$ agrees, on the smooth presentation, with the previously constructed restriction functor. The equivalence of the two presentations then identifies their descents. The later proof establishes the sufficient condition. The converse requires the additional test calculation given below, and its formulation depends on the chosen localizer.

61. NB Cette 'conjecture' est prouvé sans lemme à la page 168-169, sans avoir besoin de rien qui ne soit déjà connu ici. Cf. théorème 4, cas b) p. 162.

62. cf. I, pages 55-59.

On n'a travaillé jusqu'ici qu'avec les localiseurs W_S^{d}, $W_{S'}^{\mathrm{d}}$, à l'exclusion de W_S, $W_{S'}$. (Sauf sur les catégories contenues dans $\underline{\mathrm{Fibhot}}\, S$, où les deux coïncident.) Quand on ne fait pas d'hypothèse draconienne sur $p : S' \longrightarrow S$, du type Hot-fibration, on n'a aucune chance pour que $X \longmapsto X \times_S S'$ transforme W-équivalences grossières dans itou, donc aucune chance d'attraper l'image inverse du type d'homotopie relatif 'grossier' de X, en prenant celui de $X' = X \times_S S'$. Cela est lié au fait que pour un $S' \longrightarrow S$ général, les foncteurs de localisation

$$\mathrm{HOT}_W(S) \xrightarrow{\rho_S} \mathrm{HOT}_W^{\mathrm{lc}}(S), \qquad \mathrm{HOT}_W(S') \xrightarrow{\rho_{S'}} \mathrm{HOT}_W^{\mathrm{lc}}(S')$$

n'ont aucune envie de commuter au foncteur image inverse f^* sur les HOT_W. On peut hasarder la

Proposition 2. *Conditions équivalentes sur le foncteur $S' \longrightarrow S$:*

a) Le diagramme de foncteurs

$$\begin{array}{ccc} \mathrm{HOT}_W(S) & \xrightarrow{\rho_S} & \mathrm{HOT}_W^{\mathrm{lc}}(S) \\ \downarrow{\scriptstyle f^*} & & \downarrow{\scriptstyle f^*_{\mathrm{lc}}} \\ \mathrm{HOT}_W(S') & \xrightarrow{\rho_{S'}} & \mathrm{HOT}_W^{\mathrm{lc}}(S') \end{array}$$

commute (à isomorphisme de foncteurs composés près).

b) Le foncteur changment de base [page 124]

$$\underline{\mathrm{Liss}}\, S \longrightarrow \underline{\mathrm{Liss}}\, S', \qquad X \longmapsto X \times_S S'/S'$$

transforme W_S en $W_{S'}$. p. 77

b′) Énoncé analogue pour $\underline{\mathrm{Fib}}\, S \longrightarrow \underline{\mathrm{Fib}}\, S'$.

b″) Énoncé analogue pour $\underline{\mathrm{Fibsc}}\, S \longrightarrow \underline{\mathrm{Fibsc}}\, S'$.

([63], [64]).

Démonstration. Si $\mathcal{F}$ est une des trois catégories $\underline{\mathrm{Liss}}\, S$, $\underline{\mathrm{Fib}}\, S$ ou $\underline{\mathrm{Fibsc}}\, S$, le foncteur ρ_S peut se décrire comme le foncteur de localisation canonique

$$\mathcal{F}(W_S^{\mathrm{d}})^{-1} \longrightarrow \mathcal{F}W_S^{-1}$$

(ça marcherait pour n'importe quelle sous-catégorie pleine $\mathcal{F}$ de Cat/S qui contient $\underline{\mathrm{Fib}}\, S$). Or si b), b′) ou b″) est satisfait, on trouve un diagramme commutatif de foncteurs induits par le changement de base

$$\begin{array}{ccc} \mathcal{F}(W_S^{\mathrm{d}})^{-1} & \xrightarrow{\rho_S} & \mathcal{F}(W_S)^{-1} \\ {\scriptstyle f^*_{W_S^{\mathrm{d}}}}\downarrow & & \downarrow{\scriptstyle f^*_{W_S}} \\ \mathcal{F}'(W_{S'}^{\mathrm{d}})^{-1} & \xrightarrow{\rho_{S'}} & \mathcal{F}'(W_{S'})^{-1}, \end{array}$$

sous réserve, il est vrai, dans le cas $\mathcal{F} = \mathrm{Cat}/S$, que le foncteur image inverse soit aussi compatible avec W_S^{d}, $W_{S'}^{\mathrm{d}}$ (ce qui est automatique si $\mathcal{F} = \underline{\mathrm{Fib}}\, S$ ou $\underline{\mathrm{Fibsc}}\, S$). On sait que $f^*_{W_S^{\mathrm{d}}}$ décrit le foncteur f^* sur les types d'homotopie relatifs, et pour prouver a) comme conséquence de b) (renforcé), b′ ou b″, il reste à s'assurer que le foncteur $f^*_{W_S}$ décrit de même le foncteur induit par le précédent sur les types d'homotopie relatifs localement constants. Or on a les inclusions canoniques $\mathcal{F}_0 \subseteq \mathcal{F}$, $\mathcal{F}'_0 \subseteq \mathcal{F}'$, et sur $\mathcal{F}_0$, $\mathcal{F}'_0$ `[on a]` $W_S = W_S^{\mathrm{d}} = W_S^{\mathrm{f}}$, et les [page 125]

63. Pas prouvée, seulement que b, b′, b″ impliquent a). Pour l'implication inverse, il faudrait savoir que toute flèche de $\mathcal{F}'$ qui devient `[un]` isomorphisme dans $\mathcal{F}'W_{S'}^{-1}$, est dans $W_{S'}$ (saturation forte de $W_{S'}$ dans $\mathcal{F}'$), chose que je ne sais pas. Mais je crois que ce doit être vrai.

64. **NB** Si on suppose p propre, on doit pouvoir prendre aussi $\mathcal{F} = \mathrm{Cat}/S$.

catégories sont OK pour décrire $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ et $\mathrm{HOT}^{\mathrm{lc}}_W(S')$, et le foncteur f^* induit entre eux par images inverses ([65]) (**NB** Le foncteur changement de base $X \longmapsto X \times_S S'$ est compatible avec les localiseurs sur ces catégories, car c'est trivial sous la forme W^{f}_S.) On a alors, par images inverses $X \longmapsto X \times_S S'$, un diagramme commutatif

$$\begin{array}{ccc} \mathcal{F}_0(W_S)^{-1} & \xrightarrow{\approx} & \mathcal{F}(W_S)^{-1} \\ {\scriptstyle (f^*_{W_S})_0}\downarrow & & \downarrow{\scriptstyle f^*_{W_S}} \\ \mathcal{F}'_0(W_{S'})^{-1} & \xrightarrow{\approx} & \mathcal{F}'(W_{S'})^{-1}, \end{array}$$

où les flèches horizontales sont des *équivalences* de catégories, par le théorème-scholie 1 Ⓘ . p. 78
Cela prouve que $f^*_{W_S}$ lui aussi est bon pour calculer les images inverses de types d'homotopie relatifs.

On a prouvé que b, b′, b″ impliquent chacun a), prouvons que a) les implique. Or a) dit exactement que les images inverses f^* des éléments de $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ sont, à isomorphisme près, [page 126]
obtenus par le foncteur image inverse sur $\mathcal{F}$ tout entier, en interprétant $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ comme $\mathcal{F}W_S^{-1}$. Mais cela implique bien que $\mathcal{F} \longrightarrow \mathcal{F}'$ doit être compatible avec les localiseurs. Mais je charrie - on trouve seulement que si $u : X \longrightarrow Y$ dans $\mathcal{F}$ est dans W_S, alors la flèche $u' : X' \longrightarrow Y'$ donne un isomorphisme dans $\mathcal{F}'(W_{S'})^{-1}$. Je ne sais pas, d'autre part, si cela implique que u' est dans $W_{S'}$.

Corollaire 1. *Supposons que $p : S' \longrightarrow S$ soit une* Hot*-fibration. Alors le foncteur image inverse $X \longmapsto X \times_S S'$ de* Cat$/S$ *dans Cat$/S'$ applique W_S dans $W_{S'}$, donc le foncteur $p^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$ commute aux foncteurs de localisation ρ_S, $\rho_{S'}$ dans ces catégories.* (**NB** Il faut faire l'hypothèse adéquate sur W, savoir L6 bis page 44. C'est OK pour W_∞.)

Le fait que $X \longmapsto X \times_S S'$ soit compatible avec W_S, $W_{S'}$ est une tautologie, compte tenu de l'hypothèse L6 bis. Il ne semble pas qu'il soit aussi compatible à W^{d}_S, $W^{\mathrm{d}}_{S'}$ (je devrais faire un contrexemple, avec bien sûr S' non propre sur S). Donc on est sous les conditions de b), qui implique a).

[page 127]

Commentary · Exposition [ms. 126]

Saturation supplies the missing converse

Grothendieck wants to recognize when restriction commutes with the reflection into locally constant objects. His forward implication is sound: preservation of coarse weak equivalences on any of the indicated smooth or fibred presentations lets the square of localizations descend. For the converse he notices precisely what has not yet been justified. An arrow becoming invertible after localization need not, for an arbitrary chosen class of arrows, have belonged to that class.

There is a short repair once W is strongly saturated in Cat. Let $\mathcal{F}$ be any category of categories over a fixed base, with any chosen class of allowed morphisms, and let $W_{\mathcal{F}}$ consist of those allowed morphisms whose underlying functors belong to W. The forgetful functor descends to

$$\mathcal{F}[W_{\mathcal{F}}^{-1}] \longrightarrow \mathrm{Cat}[W^{-1}].$$

If an actual arrow u of $\mathcal{F}$ becomes invertible on the left, its underlying functor becomes invertible on the right. Strong saturation gives $u \in W$. The converse is part of localization. Thus the restriction of a strongly saturated coarse class remains strongly saturated, even when $\mathcal{F}$ is not full.

Assume now that the square in condition a) commutes. For $u \in W_S$ in a smooth or fibred presentation, its image under ρ_S is an isomorphism. Commutativity says that f^*u becomes an isomorphism in the coarse localization over S'. The preceding argument then gives $f^*u \in W_{S'}$. This proves each of b), b′), and b″). For W_∞, the needed saturation follows immediately by applying the nerve and the homotopy category of simplicial sets. In the later theory it holds for every fundamental localizer; see Cisinski (2006c, Proposition 4.2.4 in the author PDF). The gap belongs to the proof available at this point, not to the resulting equivalence.

65. NB $(\mathrm{Cat}/S)_0 = \underline{\mathrm{Fibhot}}\, S$, pour les autres cas, c'est clair.

Commutation itself is a real restriction. In the interval calculation above, let $f : \{1\} \to (0 \to 1)$ and let $F = (\varnothing \to *)$. Then

$$\rho_{\{1\}} f^* F = \varnothing, \qquad f^*_{\mathrm{lc}} \rho_S F = *.$$

This supplies a concrete obstruction without confusing it with the separate failure of arbitrary strict pullback to preserve W_S^{d}. Grothendieck's Hot-fibration hypothesis is sufficient because it ensures preservation of the coarse class by pullback. Strong saturation closes his converse in Proposition 2; it does not assert that every functor satisfies its equivalent conditions.

Commentary · Development (commentators') [ms. 126]

Properness and a three-object counterexample

The question on ms. 123 and the proposed counterexample on ms. 126 can both be settled concretely. The useful characterization of W-properness is the dual of Maltsiniotis (2005, Theorem 1.2(b)). For $f : A \to S$, $a \in A$, and $h : f(a) \to b$ in S, let $L_f(a, h)$ have objects arrows $k : a \to a'$ with $f(a') = b$ and $f(k) = h$. A morphism is an arrow $\ell : a' \to a''$ with $f(\ell) = \mathrm{id}_b$ and $\ell k = k'$. Then f is W-proper exactly when every $L_f(a, h)$ is W-aspherical. These are the outgoing comma categories for the inclusion $A_b \to A/b$.

Proposition (detecting properness by inverse image). [ours] For $W = W_\infty$, strict pullback along a fixed $f : A \to S$ preserves right-fine weak equivalences on all categories over S if and only if f is proper. For an arbitrary fundamental localizer W, W-properness is equivalent to this preservation property holding after every change of base of f.

Proof. Properness implies preservation by the preceding block, and properness is stable under base change. To prove the first converse, test the assumed preservation on the unit

$$\{b\} \longrightarrow \Phi_S(\{b\}) = S/b.$$

This is right-fine: its comma category at s maps the discrete set $\mathrm{Hom}_S(s, b)$ into the category of factorizations of those arrows, and each factorization category has an initial and a terminal object.

After pullback along f, take the comma category at a. In the target, objects are triples

$$(a', k : a \to a', r : f(a') \to b).$$

The composite $rf(k)$ defines a decomposition indexed by $h \in \mathrm{Hom}_S(f(a), b)$. The component indexed by h has initial object (a, id_a, h). The corresponding source component is exactly $L_f(a, h)$. A weak equivalence of these disjoint unions to the contractible target components forces every $L_f(a, h)$ to be weakly contractible. The cited criterion gives properness.

For the general converse, suppose preservation holds after every base change. Given $h : f(a) \to b$, pull back f along the functor $[1] \to S$ representing h. Apply the preservation hypothesis to $\{1\} \to [1]$ over $[1]$, which is the same unit test. At the object $(a, 0)$ of $A \times_S [1]$, its source comma category is $L_f(a, h)$ and its target comma category has an initial object. Since the map belongs to W, two-out-of-three implies $L_f(a, h) \to e \in W$. The criterion again gives W-properness. □

Example (why arbitrary localizers require base change). [ours] The distinction between the two formulations cannot be omitted for arbitrary fundamental localizers. Let W_{ne} consist of functors whose source and target are either both empty or both nonempty. This is a fundamental localizer: weak saturation and the terminal-object axiom are immediate, and the relative theorem-A axiom follows because an object of a nonempty target gives a nonempty target comma category, hence a nonempty source comma category. Take a nontrivial group G and $f : e \to BG$. Over the one-object category BG, every comma category of $X \to BG$ is nonempty exactly when X is nonempty. Right-fine W_{ne}-equivalence therefore detects only nonemptiness. Strict pullback along f does not change the object set of X, so it preserves that class. Yet $L_f(*, g)$ is empty for every nonidentity $g \in G$, and f is not W_{ne}-proper. This example concerns the broader localizer question, not a weakening of the stronger hypotheses used elsewhere in this chapter.

Example (a Hot-fibration which is not proper). [ours] Let E be the three-object poset with nonidentity arrows $u \to v$ and $u \to w$, and put

$$p : E \longrightarrow (0 \to 1), \qquad p(u) = p(v) = 0, \quad p(w) = 1.$$

This is a Grothendieck fibration. The lift $u \to w$ of $0 \to 1$ is Cartesian: for an object over 0, its arrows to u and its arrows to w are both singletons when the object is u, and both empty when it is

v. The two fibres are $(u \to v)$ and $\{w\}$; their transition functor chooses u and is a weak equivalence. Thus p lies in $\underline{\mathrm{Fib}}_0$, hence in $\underline{\mathrm{Fibhot}}$ under the standing comparison hypotheses.

There is also a direct verification of its particularly strong homotopical behaviour. The section choosing u over 0 and w over 1 has a natural transformation $sp \Rightarrow \mathrm{id}_E$ over the base, whose only nonidentity component is $u \to v$. This deformation retraction survives every strict base change. Every projection $E \times_S T \to T$ is therefore a nerve weak equivalence. Two-out-of-three in a square over S shows that pullback along p preserves all coarse weak equivalences.

Nevertheless p is not proper, since $L_p(v, 0 \to 1)$ is empty. More explicitly, the right-fine weak equivalence $\{1\} \to [1]$ pulls back to $\{w\} \to E$. Its comma functor at v is $\varnothing \to e$, so it is not right-fine. This is precisely the counterexample requested on ms. 126. We checked the properness criterion just cited; the elementary tests and examples here are explicit completions of the manuscript's questions, without a priority claim.

Je ne me sens pas capable d'exhiber un foncteur autre qu'une Hot-fibration (ou tout au moins une W-fibration), pour lequel le foncteur image inverse sur HOT_W soit compatible aux foncteurs de localisation ρ_S, $\rho_{S'}$. Bien sûr, si X est quelconque dans Cat/S, rien n'empêche de regarder son localisé pour W_S, i.e. $\rho_S(\mathrm{hot}_{W,S}(X))$, et de demander comment trouver son image inverse. Si $p : S' \longrightarrow S$ est une Hot-fibration, alors il suffit de prendre $\mathrm{hot}_{W,S'}(X')$, $X' = S \times_S S'$. Dans tout autre cas (j'entends, sans hypothèse sur X/S, plus précisément sans supposer que X est lui-même Hot-fibré sur S), cette formule est canulée sans espoir. On doit, pour calculer $f^*\rho_S(\mathrm{hot}_{W,S}(X))$, 'calculer' $\rho_S(\mathrm{hot}_{W,S}(X))$ en 'montant à l'étage', comme $\mathrm{hot}_{W,S}(C(X))$. On trouve alors l'image inverse $\mathrm{hot}_{W,S'}(C(X) \times_S S')$ (par `[le]` corollaire 1, p. 119).

Maintenant que la loi contravariante de $\mathrm{HOT}_W(S)$ en $S \in \mathrm{Ob}\,\mathrm{Cat}$ semble bien comprise, il s'agit de voir qu'on a bien un dérivateur HOT. Je ne m'attends pas à ce que les $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ définissent eux aussi un dérivateur, mais il faudrait m'en assurer expressément. (C'est pour l'existence de $f_!$, f_* que je doit chercher.) [page 128]

À dire vrai, il faudrait, pour bien faire, regarder aussi la structure de 2-*foncteur* de

$$S \longmapsto \mathrm{HOT}_W(S) \qquad \mathrm{Cat}^o \longrightarrow \mathrm{Cat},$$

donc pour S, S' données dans Cat, définir p. 79

$$\underline{\mathrm{Hom}}(S', S)^o \longrightarrow \underline{\mathrm{Hom}}(\mathrm{HOT}_W(S), \mathrm{HOT}_W(S')).$$

Cela revient surtout, pour

$$f, g : S' \rightrightarrows S$$

deux flèches de S' dans S, et un homomorphisme de foncteurs

$$u : f \longrightarrow g,$$

à décrire un homomorphisme de foncteurs

$$u^* : g^* \longrightarrow f^*$$

entre $\mathrm{HOT}_W(S)$ et $\mathrm{HOT}_W(S')$. C'est d'ailleurs chose faite, dans le cadre général des $(\mathcal{M}, W)$, quand on décrit $\mathrm{HOT}_W(S)$ et $\mathrm{HOT}_W(S')$ par localisation de $\mathrm{Hom}(S^o, \mathrm{Cat})$ et $\mathrm{Hom}(S'^o, \mathrm{Cat})$. Ces catégories s'interprètent respectivement comme $\underline{\mathrm{Fibsc}}\, S$ et $\underline{\mathrm{Fibsc}}\, S'$, donc comme des catégories de catégories sur S resp. S', on voudrait donc interpréter u^* en termes de telles catégories, et tant qu'à faire, pour des catégories X sur S aussi générales que possible. Nous nous attendrons alors à devoir faire des hypothèses, soit sur X, soit sur f et sur g, qui assurent que les images de $\mathrm{hot}_{W,S}(X)$ par f^* et g^* ne sont autres que $\mathrm{hot}_{W,S}(f^*X)$, $\mathrm{hot}_{W,S}(g^*X)$ respectivement, où on pose [page 129]

$$f^*(X) = X \times_S (S', f), \qquad g^*(X) = X \times_S (S', g).$$

Donc on devra sans doute se placer dans l'une des hypothèses suivantes:

a) X est lisse sur S.

b) X est Hot-fibrée sur S.

c) f et g sont propres (et sous réserve de la validité de la conjecture page 122).

On s'attend alors à ce que u^* soit associé à un homomorphisme

$$u^* : g^*(X) \longrightarrow f^*(X) \qquad \textit{dans } \mathrm{Cat}/S'$$

défini en termes de u. Y a-t-il moyen de le définir bel et bien, sans supposer X au moins fibrée sur S? Prenons $S' = e$, donc f, g correspondant à des objets $s, t \in \mathrm{Ob}\, S$, et u à une fléche

$$u : s \longrightarrow t,$$

les images inverses de X/S par f, g sont les fibres X_s, X_t, et on cherche

$$u^* : X_t \longrightarrow X_s.$$

On n'a aucune chance d'en trouver canoniquement, si on ne suppose X fibrée sur S! Donc ce ne sont pas a), b) ni c) qui peuvent nous servir, mais uniquement l'hypothèse $X \in \mathrm{Ob}\, \underline{\mathrm{Fib}}\, S$. Donc on ne peut espérer trouver une description de u^* entre $\mathrm{HOT}_W(S)$ et $\mathrm{HOT}_W(S')$ en termes d'opérations sur les catégories sur S, S', comme suggéré à l'instant, qu'en se bornant à des 'modèles' X qui sont dans $\underline{\mathrm{Fib}}\, S$, tout au moins. Et je ne doute pas en effet que ça marche sur $\underline{\mathrm{Fib}}\, S$ pas moins bien que directement sur $\underline{\mathrm{Fibsc}}\, S$, en termes de $\underline{\mathrm{Hom}}(S^o, \mathrm{Cat})$.

p. 80 [page 130]

Commentary · Development (commentators') [ms. 128]

Local constancy and the missing comma formula

Grothendieck asks whether the locally constant part, as the base varies, is itself a derivator. His expectation that it is not is correct for ordinary homotopy types, but the suggested obstruction to the existence of adjoints is not the obstruction. It is the pointwise formula for those adjoints that fails. We give the small calculation because it separates this question from the comparison theorem.

Proposition (the interval obstruction). [ours] Let $\mathcal{L}(S)$ be the full subcategory of the space derivator on S consisting of locally constant objects, with the induced restriction functors. Then $S \mapsto \mathcal{L}(S)$ does not satisfy the pointwise left Kan-extension axiom.

Proof. We use the source's presheaf convention. Let $S = (0 \to 1)$, let $a : \{0\} \to S$, and let $b : \{1\} \to S$. An object of $\mathcal{L}(S)$ is a coherent arrow $F_1 \to F_0$ which is an equivalence. Evaluation at 0 is an equivalence

$$a^* : \mathcal{L}(S) \longrightarrow \mathcal{L}(e) = \mathrm{Hot}.$$

Indeed the constant-arrow functor is a quasi-inverse: the diagram map from $F_1 \to F_0$ to $F_0 \xrightarrow{\mathrm{id}} F_0$ has components $F_1 \to F_0$ and id_{F_0}, both weak equivalences. This construction is natural in F, before taking the homotopy category. Thus a left adjoint $a_!^{\mathcal{L}}$ exists and is the constant-arrow functor, up to the unique isomorphism between adjoints.

Take the one-point space $*$. We obtain

$$b^* a_!^{\mathcal{L}}(*) \simeq *.$$

On the other hand, the comma category $1\backslash\{0\}$ of arrows $1 \to a(0)$ is empty. The pointwise formula for a left Kan extension in the presheaf convention would therefore identify this value with the homotopy colimit of the empty diagram, namely $\varnothing$. The categories $\mathcal{L}(e) = \mathrm{Hot}$ and $\mathcal{L}(\varnothing) = e$ have the usual initial-object adjunction, so that this last value is forced. Since $\varnothing$ is not isomorphic to $*$, the required base-change map cannot be invertible. □

This proof also identifies what the unrestricted direct image does: in the full space derivator $a_!(*)$ is the representable presheaf $S(-, 0)$, with values $*$ at 0 and $\varnothing$ at 1. It is not locally constant. The locally constant left adjoint first takes this extension and then reflects it. The reflection changes the value at 1, and the comma formula is lost.

Corollary (commutation beyond Hot-fibrations). [ours] The inclusion $a : \{0\} \to (0 \to 1)$ commutes with local-constant reflection for spaces, but is not a Hot-fibration.

Proof. The reflection of any arrow $F_1 \to F_0$ is the constant arrow on F_0. To see its universal property, map to a constant arrow on K; the space of maps is $\mathrm{Map}(\mathrm{hocolim}_{S^{\mathrm{op}}} F, K)$, and 0 is terminal in S^{op}, so this homotopy colimit is F_0. Evaluation a^* at 0 therefore commutes with reflection. Nevertheless the weak equivalence $\{1\} \to S$ pulls back along a to $\varnothing \to \{0\}$, which is not a weak equivalence. Hence a is not a Hot-fibration. This answers the question on ms. 127. □

The later general result is stronger than is needed for the example. Cisinski (2009, Corollary 3.11) proves that restriction on locally constant coefficients has both adjoints for every derivator. His Theorem 3.13 concerns a different construction: one fixes a category A and varies an additional diagram category I, retaining arbitrary coherent I-diagrams of locally constant A-coefficients. That construction is a derivator. It does not say that the varying-base assignment $S \mapsto \mathcal{L}(S)$ is one.

We checked those statements, the pointwise formula in Groth (2013, Proposition 1.26), and the localization description in Cisinski's paper. We did not find this exact interval calculation stated there. The marked result records our explicit completion of the question in these pages; it makes no claim of priority for the obstruction.

Pour les axiomes essentiels sur les dérivateurs, je renvoie à I p. 67-68 et p. 111-112. On a vu que Der 1, (Der 2) ([66]) sont satisfaits pour tout prédérivateur défini par un couple $(\mathcal{M}, W \subseteq \mathrm{Fl}\,\mathcal{M})$, donc c'est OK ici pour HOT_W. Il s'ensuit que c'est OK aussi pour $\mathrm{HOT}^{\mathrm{lc}}_W$, vu la caractérisation des $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ en tant que sous-catégories de $\mathrm{HOT}_W(S)$. Pour Der 1, ça marche-t-il même pour des sommes quelconque de catégories, i.e. a-t-on

$$\mathrm{HOT}_W(\coprod_i S_i) \xrightarrow{\sim} \prod_i \mathrm{HOT}_W(S_i)\,?$$

On a en tous cas

$$\underline{\mathrm{Fib}}(\overbrace{\coprod_i S_i}^{S}) \xrightarrow{\sim} \prod_i \underline{\mathrm{Fib}}(S_i),$$

et le localiseur W^{f}_S dans $\underline{\mathrm{Fib}}(S)$ s'identifie bel et bien au produit des localiseurs $W^{\mathrm{f}}_{S_i}$ dans les $\underline{\mathrm{Fib}}\,S_i$. Mais dans le cas d'une famille peut-être infinie de catégories $\mathcal{F}_i$, munies de localiseurs W_i, si on pose [page 131]

$$\mathcal{F} = \prod \mathcal{F}_i, \qquad W = \prod W_i \subseteq \mathrm{Fl}(\mathcal{F}),$$

le foncteur canonique

$$\mathcal{F}W^{-1} \longrightarrow \prod_i (\mathcal{F}_i W_i^{-1})$$

est-il toujours une équivalence? Même dans `[`Même dans' barré]` le cas I fini il faut (il me semble) un argument assez soigneux, soit par la propriété universelle par récurrence sur $\mathrm{card}\, I$ (on est ramené au cas de deux foncteurs), soit à coups de catégories de chemins, en notant que

$$(*) \qquad \underline{\mathrm{Ch}}^W_\infty(\mathcal{F}; (s_i), (t_i)) \xrightarrow{\sim} \prod_i \underline{\mathrm{Ch}}^{W_i}_\infty(\mathcal{F}_i; s_i, t_i),$$

et en passant aux π_0, qui donne

$$\mathrm{Hom}_W((s_i), (t_i)) \xrightarrow{\sim} \prod_i \mathrm{Hom}_{W_i}(s_i, t_i).$$

Mais la formule $(*)$, visiblement, ne marche que pour I fini. Dans le cas général, on trouve la sous-catégorie $\prod^!_i$ de $\prod_i$ formée des systèmes $(c_i)_{i\in I}$ de chemins qui sont de longueur bornée.

66. Canulé pour Der 2, je ne sais si c'est vrai même en supposant W *fortement* saturé, i.e. tel que $\mathrm{hot}_W(f)$ isomorphisme $\Longrightarrow f \in W$.

Commentary · Exposition [ms. 130]

A natural transformation is realized by transport

Grothendieck has explained restriction on objects; a prederivator also remembers how restriction varies with a natural transformation of indexing functors. For $u : f \Rightarrow g : S' \to S$, the presheaf convention requires a transformation $u^* : g^* \Rightarrow f^*$. A pullback square alone supplies no arrow in this direction. A fibration supplies it through Cartesian lifting.

Let $p : X \to S$ be a Grothendieck fibration. For every $a \in S'$ and $x \in X_{g(a)}$, choose a Cartesian arrow

$$\lambda_{a,x} : u_a^* x \longrightarrow x$$

over $u_a : f(a) \to g(a)$. Send (a, x) in g^*X to $(a, u_a^* x)$ in f^*X. For an arrow $(h, v) : (a, x) \to (b, y)$ in g^*X, the equality

$$g(h)u_a = u_b f(h)$$

and the Cartesian property of $\lambda_{b,y}$ give a unique arrow $w : u_a^* x \to u_b^* y$ over $f(h)$ satisfying

$$\lambda_{b,y} w = v \lambda_{a,x}.$$

Uniqueness proves preservation of identities and composition. This constructs the requested functor over S'.

Different choices of Cartesian lifts are related by unique vertical isomorphisms. The resulting transport functors are therefore naturally isomorphic. The same uniqueness gives the comparison between transport along a composite transformation and successive transport, together with its coherence. A splitting can make selected comparison maps identities, but the localized operation is independent of the splitting.

For an arbitrary functor $H : X \to Y$ over S, transport need not commute strictly with H: H need not preserve Cartesian arrows. Factoring $H(\lambda_{a,x})$ through a chosen Cartesian lift in Y gives a vertical natural transformation between the two resulting functors. A vertical natural transformation is a homotopy over S': it is represented by a functor from the cylinder $X \times [1]$, whose projection to X is a weak equivalence on fibres. Thus the two maps agree after localization. This supplies the naturality needed when all functors over the base, rather than just Cartesian functors, are used in the fibred presentation.

The source's negative observation is accordingly about finding a direct functor on insufficiently structured representatives. It does not obstruct the existence of the 2-functor. That 2-functor is already defined by precomposition on coherent presheaves; the lifting calculation explains how to recognize its 2-cells geometrically.

Commentary · Development (commentators') [ms. 130]

Strong saturation does not imply pointwise detection

Footnote 66 withdraws the assertion that every relative category defines a prederivator satisfying Der 2, even when its weak equivalences are strongly saturated. There is a distinction between two statements. If an actual natural transformation has invertible evaluations after localization, strong saturation makes each component a weak equivalence; that transformation is therefore inverted in the diagram category. An arbitrary morphism of the localized diagram category, however, need not have an actual natural transformation as representative. The following construction shows that this distinction can defeat pointwise detection. It also connects this footnote to the unbounded-length obstruction discussed on ms. 131.

Proposition (a strongly saturated counterexample to Der 2). [ours] There exist a small category M, a strongly saturated class W, and a countable discrete category I such that

$$M^I[(W^I)^{-1}] \longrightarrow \prod_{i \in I} M[W^{-1}]$$

is not conservative. In fact, the source has a morphism whose evaluations are all isomorphisms but has no morphism in the reverse direction. The category M can be chosen to be a disjoint union of finite categories.

Proof: the finite categories. Let R be the walking retraction, with objects $0, 1$ and arrows

$$i : 0 \longrightarrow 1, \qquad r : 1 \longrightarrow 0, \qquad ri = \mathrm{id}_0, \qquad e = ir \neq \mathrm{id}_1.$$

Its five arrows are $\mathrm{id}_0, \mathrm{id}_1, i, r, e$; in particular, its only isomorphisms are its identities. For example, this is the subcategory of sets generated by the inclusion of a singleton into a two-element set and its retraction.

Fix a positive odd integer n. Define a finite poset Q_n with elements

$$x_0, \dots, x_{n-1}, \quad y_0, \dots, y_{n-1}, \quad z.$$

For writing its order relations only, put $x_n = y_n = z$. Generate the order by the two fences

$$x_0 \longleftarrow x_1 \longrightarrow x_2 \longleftarrow \cdots \longleftarrow x_n, \qquad y_0 \longleftarrow y_1 \longrightarrow y_2 \longleftarrow \cdots \longleftarrow y_n,$$

together with $x_j \le y_j$ and $z \le y_j$ for $0 \le j < n$. An arrow in these displays means an order relation from its source to its target. The generated relation is antisymmetric: no y_j with $j < n$ reaches an x_k or z, and the nonidentity arrows among the x's have odd source and even target. Among the x's the only nonidentity comparisons are the fence edges. In particular, $z = x_n$ reaches just x_{n-1} among the other x's. The full subposet on the y's has initial object $y_n = z$.

Let C_n be the full subcategory of the finite product category $R \times Q_n$ on the objects

$$X_j = (0, x_j), \quad Y_j = (1, y_j) \quad (0 \le j < n), \qquad X_n = (0, z), \quad Y_n = (1, z).$$

Thus X_n and Y_n are distinct objects, despite having the same second coordinate. Write $F_n : C_n \to R$ for projection and set

$$W_n = F_n^{-1}(\mathrm{Iso}\, R).$$

This class is strongly saturated. Indeed, F_n inverts W_n and hence factors through $C_n[W_n^{-1}]$. If an original arrow becomes invertible in that localization, its image under F_n is an isomorphism, which is exactly the condition defining W_n. This argument also confirms that the section, the retraction and the idempotent e do not inadvertently become weak equivalences.

A short morphism with a long inverse. There are weak arrows $c_j : Y_n \to Y_j$ given by $(\mathrm{id}_1, z \le y_j)$, with $c_n = \mathrm{id}$, and arrows $i_j : X_j \to Y_j$ given by $(i, x_j \le y_j)$. The product category also supplies $r_n : Y_n \to X_n$, with

$$r_n i_n = \mathrm{id}_{X_n}.$$

For every fence edge $a : X_s \to X_t$, let $b : Y_s \to Y_t$ be the corresponding weak arrow. The following equations hold strictly:

$$i_t a = b i_s, \qquad b c_s = c_t.$$

In the localization put $t_j = c_j^{-1} i_j : X_j \to Y_n$. These equations give $t_t a = t_s$. Let $p_n : X_0 \to X_n$ be the morphism represented by the entire weak fence, traversed from index 0 to index n. Since every fence edge is weak, p_n is invertible. Transporting the preceding equation along its edges gives

$$i_n p_n = t_0 = c_0^{-1} i_0.$$

Consequently the three-arrow word

$$f_n = r_n c_0^{-1} i_0 : X_0 \longrightarrow X_n$$

equals p_n and is an isomorphism.

Nevertheless every zigzag representing a morphism $X_n \to X_0$ has at least n arrows. To prove this, assign the heights

$$h(X_j) = j, \qquad h(Y_j) = n \qquad (0 \le j \le n).$$

Every original forward arrow lowers height by at most one. For arrows between X objects this follows from the fence description. Arrows from an X to a Y cannot lower it. Arrows between Y objects preserve it. Finally, a Y object can map to an X object only from Y_n, and then the target is X_n or X_{n-1}; these are exactly the comparisons in Q_n with a y source and an x target. Such an arrow also lowers height by at most one.

A weak arrow has the same R coordinate at its two ends. Between Y objects it preserves height, and between X objects its two indices differ by at most one. Its formal inverse therefore lowers height by at most one as well. Every letter in a localization zigzag, whether a forward arrow or an inverted weak arrow, lowers height by at most one. A word from height n to height 0 consequently has length at least n. This argument concerns every possible representative and allows all the extra endomorphisms supplied by fullness in $R \times Q_n$.

The countable diagram. Take

$$M = \coprod_{n \text{ positive odd}} C_n, \qquad W = \coprod_{n \text{ positive odd}} W_n,$$

and let I be the discrete category of positive odd integers. The projections F_n together give $F : M \to R$, and $W = F^{-1}(\operatorname{Iso} R)$ is strongly saturated by the same argument. Define four I-diagrams by choosing, in the nth component, X_0, Y_0, Y_n, X_n, respectively. The families i_0, c_0, r_n give a single diagram-category zigzag

$$\mathcal{X} \longrightarrow \mathcal{U} \xleftarrow{W^I} \mathcal{V} \longrightarrow \mathcal{Z}.$$

Its localized morphism $f : \mathcal{X} \to \mathcal{Z}$ evaluates to f_n and is therefore pointwise invertible.

Suppose that any morphism $\mathcal{Z} \to \mathcal{X}$ existed in $M^I[(W^I)^{-1}]$. Choose a finite zigzag representing it, of length L. At the coordinate indexed by any odd $n > L$, evaluation gives a zigzag $X_n \to X_0$ with at most L letters. All its objects remain in C_n, since the disjoint union has no arrows between its components. The height bound gives $n \leq L$, a contradiction. There is thus no reverse morphism at all, and f is not invertible. □

The use of infinitely many coordinates is essential to this proof; it makes no assertion about the same question restricted to finite nondiscrete shapes. It refutes the unrestricted statement in footnote 66, including its strongly saturated version. It also strengthens the failure of the infinite-product comparison: that comparison can fail to reflect isomorphisms, not merely fail to be full.

For comparison, a precise sufficient hypothesis is available without a model structure. Suppose W is strongly saturated and every morphism in $M^I[(W^I)^{-1}]$ has a representative

$$X \xleftarrow{w} X' \xrightarrow{a} Y' \xleftarrow{v} Y, \qquad w, v \in W^I.$$

If its evaluations are isomorphisms, then every a_i becomes invertible in $M[W^{-1}]$. Saturation gives $a_i \in W$, so the displayed word is invertible. A left or right calculus of fractions is a special case, allowing one of the outside arrows to be an identity. The same argument proves the Der 2 part of Lagkas Nikolos (2018, Proposition 4.2.8, pp. 85–86); Remark 4.2.19, p. 95, explicitly discusses the role of two- and three-arrow descriptions in localization of prederivators.

The counterexample above was constructed for this commentary. The literature checks located the stated fraction criterion and the usual model-category results, but did not locate this particular finite-category construction. Its proof is included in full; the marking records our contribution to the present audit and does not assert historical priority for the general failure of localization to commute with infinite products.

Donc la question de la validité de Der 1 sans hypothèse de finitude sur l'ensemble d'indices I reste en suspens pour l'instant. (Je soupçonne que c'est vrai, mais il faudrait, pour le prouver, tenir une description des flèches dans $\mathrm{HOT}_W(S)$ par des chemins de longueur bornée, disons de longueur deux ou trois. C'est une question intéressante, mais que je réserve pour plus tard, si le besoin m'y contraint . . .) p. 81 [page 132]

Commentary · Development (commentators') [ms. 131]

Unbounded zigzags obstruct infinite products

Grothendieck identifies the right difficulty in commuting localization with an infinite product: a morphism in a localization is represented by a finite word, and the lengths needed in different coordinates may be unbounded. The following construction turns that observation into a counterexample, even when the localizing classes are strongly saturated.

Proposition (a product of finite fences). [ours] For $n \geq 1$, let P_n be the finite poset with objects $0, 1, \ldots, 2n$ and nonidentity relations

$$0 < 1 > 2 < 3 > \cdots < 2n-1 > 2n.$$

Let W_n consist of every arrow. Then each $P_n[W_n^{-1}]$ is equivalent to the terminal category, but

$$\left(\prod_{n\geq 1} P_n\right)\left[\left(\prod_{n\geq 1} W_n\right)^{-1}\right] \longrightarrow \prod_{n\geq 1} P_n[W_n^{-1}]$$

is not full. More precisely, two objects $x = (x_n)$ and $y = (y_n)$ in the category on the left can be connected by a morphism if and only if $\sup_n |x_n - y_n| < \infty$.

Proof. The only nonidentity arrows in P_n join adjacent integers; there are no composable pairs of nonidentity arrows. Inverting these arrows gives the free groupoid on the undirected path with vertices $0, \dots, 2n$. Between any two vertices there is exactly one reduced path. Hence this groupoid is equivalent to the terminal groupoid.

Put $P = \prod_n P_n$. Since every arrow of P is inverted, an arrow in its localization is represented by a finite zigzag in P. One step in such a zigzag changes the nth coordinate by at most one, for every n. A zigzag with k steps therefore implies $|x_n - y_n| \leq k$ for all n. This proves necessity of the bound.

Conversely, suppose $|x_n - y_n| \leq K$ for all n. In each fence take the unique shortest undirected path from x_n to y_n. Its arrow directions alternate. If its first arrow has the wrong orientation for a common alternating pattern beginning with a forward arrow, prepend an identity. After the path finishes, append identities. Every coordinate now has a zigzag of the same length $K + 1$ and the same pattern of directions. Taking coordinatewise products gives a zigzag in P. This proves sufficiency.

Take $x_n = 0$ and $y_n = 2n$. There is no arrow between them in $P[W^{-1}]$, while their images have exactly one arrow between them in the product of the localized categories. Thus the comparison is not full. The localizing classes are all-arrow classes, so they are strongly saturated. □

The original question for HOT_{W_∞} nevertheless has a positive answer. Diagram categories of simplicial sets carry model structures with pointwise weak equivalences. For a disjoint union of shapes, the diagram model category is the product of the diagram model categories of the components. Its cofibrant-fibrant replacements are chosen coordinatewise. Between cofibrant-fibrant objects, every homotopy-category morphism has an actual representative, and homotopy can be tested coordinatewise using coordinatewise cylinder objects. Therefore the homotopy category of this product is the product of the homotopy categories, with no bound on the number of factors. The common model-category description provides the uniform presentations absent from the fence example; compare Groth (2013, Proposition 1.30).

We checked the finite-word construction of localization in Gabriel and Zisman (1967, Chapter I, §1) and the model-category derivator result just cited. We did not find this exact fence formulation in those sources. It is an elementary elaboration of the obstruction explicitly singled out in the manuscript, not a claim that the general phenomenon is new. In particular, it does not support replacing Grothendieck's tentative general argument with an assertion that all relative categories satisfy the coproduct axiom.

Plus importante est la question de la validité de Der 3 et de ses variantes (I, p. 111-112). Le minimum qu'on demande d'un dérivateur $\mathbf{D}$, c'est que si $S' \longrightarrow S$ dans Diag est une Hot-équivalence, ce soit une $\mathbf{D}$-équivalence, i.e. (dans le cas actuel) que le foncteur

$$f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$$

soit pleinement fidèle sur la sous-catégorie des objets *constants* dans $\mathrm{HOT}_W(S)$. Cette condition devient plus tangible quand on dispose déjà soit des foncteurs cohomologie $H^\bullet_{\mathbf{D}}(S, -)$, $H^\bullet_{\mathbf{D}}(S', -)$, soit de l'homologie $H^{\mathbf{D}}_\bullet(S, -)$, $H^{\mathbf{D}}_\bullet(S', -)$. En termes de cette dernière, elle signifie que

$$p_! p^*(a) \xrightarrow{\sim} p'_! p'^*(a) \qquad \texttt{[est un]}\ \text{isomorphisme},$$

pour tout a dans $\mathcal{A} = \mathbf{D}(e)$. (Où $p : S \longrightarrow e$, $p' : S' \longrightarrow e$ sont les morphismes structuraux sur la catégorie ponctuelle.) On a vu la formule heuristique générale [page 133]

$$p_! \underbrace{p^*(a)}_{a_S} = \mathrm{hot}(S) \times a,$$

où $\operatorname{hot} S$ est la classe de $S \in \operatorname{Ob}\operatorname{Cat}$ dans $\mathrm{Hot} = (\mathrm{Cat})W_\infty^{-1}$, et où le produit $\times$ est une opération externe de Hot sur $\mathcal{A} = \mathbf{D}(e)$

$$\mathrm{Hot} \times \mathcal{A} \longrightarrow \mathcal{A} \ .$$

Dans le cas actuel, $\mathcal{A} = \mathrm{Hot}$, et l'opération

$$\mathrm{Hot} \times \mathrm{Hot} \longrightarrow \mathrm{Hot}$$

ne doit être autre que le foncteur produit cartésien, lequel est déduit du foncteur similaire dans Cat. Quand on dispose de ce formalisme, le résultat cherché pour un $f : S' \longrightarrow S$ se réduit simplement à dire que si f est dans W, alors il en est de même de $f \times \mathrm{id}_X$ pour tout X dans Cat (en prenant X tel que $\mathrm{hot}(X) \simeq a$). Or là c'est un fait bien connu, résultant des axiomes mis sur W.

J'ai donc l'impression que je n'ai aucune chance de prouver Der 3a′ (p. 111), ni même Der 3a) (cas où S = e), sans avoir une maîtrise du formalisme des $p_!$. Et j'ai bon espoir que cetter maîtrise va s'obtenir à bon compte, alors que la question des p_* semble beaucoup plus délicate. p. 82 [page 134]

Commentary · Exposition [ms. 133]

Constant coefficients explain the turn to homology

Grothendieck turns from inverse image to homological direct image because the latter makes homotopy invariance of constant coefficients visible. For the space derivator, the homotopy colimit of a constant space K over S^{op} is

$$\mathrm{hocolim}_{S^{\mathrm{op}}} K \simeq |N(S^{\mathrm{op}})| \times K \simeq |NS| \times K.$$

One way to calculate it is the simplicial replacement: in degree n it is a coproduct of copies of K, one for every string of n composable arrows of S^{op}. Realization therefore gives the displayed product. The reversal of a string identifies the realizations of the nerves of a category and its opposite.

For $f : S' \to S$, the canonical homological comparison runs from the expression for S' to the expression for S and is represented by

$$|Nf| \times \mathrm{id}_K : |NS'| \times K \longrightarrow |NS| \times K.$$

When $f \in W_\infty$, this is a weak equivalence. The isomorphism written in the source may consequently be read in either direction after localization; before inversion the functorial map has the direction just displayed. For a general coefficient derivator the corresponding operation is tensoring with a homotopy type, rather than an internal Cartesian product of coefficients.

Adjunction now relates the calculation to the desired fully faithful restriction on constant objects. For constant coefficients a, b,

$$\mathrm{Hom}_{\mathbf{D}(S)}(p^*a, p^*b) \simeq \mathrm{Hom}_{\mathbf{D}(e)}(p_!p^*a, b).$$

If the homological comparison is an isomorphism for every a, it induces a bijection on these Hom sets for every b. Conversely, bijectivity for every b identifies the comparison by Yoneda. This is why the direct-image formalism gives exactly the statement Grothendieck wants, rather than merely an invariant of isomorphism classes.

The axiom numbers in these pages belong to the evolving manuscript. The condition called Der 3 here should not be confused with the usual modern axiom bearing that number, which asks for both Kan adjoints. Likewise the footnote retracting the unrestricted claim about Der 2 records a genuine limitation of the argument for arbitrary relative categories. The development at manuscript p. 130 gives a counterexample even when W is strongly saturated, using a countably discrete diagram shape. It does not address the restriction of that question to finite shapes. The model-category theorem establishes the pointwise detection axiom for the space derivator; it does not vindicate the cancelled general assertion.

Je vais donc me tourner d'abord vers l'axiome Der 4′ d'existence de

$$f_! : \mathrm{HOT}_W(S') \longrightarrow \mathrm{HOT}_W(S),$$

comme adjoint à gauche de f^*. Je vais essayer de construire ad-hoc un foncteur $f_!$, puis les deux flèches d'adjonction

$$\begin{array}{ccc} \mathrm{id} & \longrightarrow & f^* f_! \\ \mathrm{id} & \longleftarrow & f_! f^*, \end{array}$$

et enfin, de vérifier les deux compatibilités pertinentes.

L'idée géométrique 'évidente', c'est pour $X' \in \mathrm{Cat}/S'$, représentant $\xi' \in \mathrm{Ob}\,\mathrm{HOT}_W(S')$, de définir $f_!(\xi')$ par

$$f_!(\mathrm{hot}_{S',W}(X')) = \mathrm{hot}_{S,W}(X'),$$

où dans le second membre, X' est regardé comme objet de Cat/S, grâce à $f : S' \longrightarrow S$ et au foncteur

$$f_! : \mathrm{Cat}/S' \longrightarrow \mathrm{Cat}/S$$

qu'il induit. Le seul problème, c'est de voir si ce foncteur est compatible avec les localiseurs [page 135]
$W^{\mathrm{d}}_{S'}$, W^{d}_S, et sinon, s'il l'est tout au moins sur la sous-catégorie $\underline{\mathrm{Fib}}\,S'$ ou $\underline{\mathrm{Fibsc}}\,S'$, qui suffit pour calculer $\mathrm{HOT}_W(S')$ par localisation. Il faut donc voir si un $X'_1 \longrightarrow X'_2$ dans Cat/S', qui est dans $W^{\mathrm{d}}_{S'}$, i.e. qui est dans W et le reste par tout changement de base de la forme $s'\backslash S' \longrightarrow S'$, plus généralement pour tout co-isomorphisme local, est aussi dans W^{d}_S, i.e. si les changements de base de la forme $s\backslash S \longrightarrow S$ le transforment en W-équivalences (grossières). Or les co-isomorphismes locaux sont stables par *tout changement de base* (cf. I, p. 55 dans le cas dual des isomorphismes locaux), donc on gagne!

$$\begin{array}{ccc} \overline{S'} = s\backslash S' & \longrightarrow & S' \\ \downarrow & & \downarrow \\ \overline{S} = s\backslash S & \longrightarrow & S \end{array} \quad .$$

Je vais quand même le vérifier encore. Si

$$\begin{array}{ccc} \overline{S'} & \longrightarrow & S' \\ \downarrow & & \downarrow \\ \overline{S} & \longrightarrow & S \end{array}$$

est un carré cartésien dans Cat, on a dit dans loc. cit., essentiellement, que pour $\overline{s'}$ dans $\overline{S'}$, p. 83
ayant des images s', $\overline{s}$, s dans S', $\overline{S}$, S respectivement, le carré des colocalisés

$$\begin{array}{ccc} \overline{s'}\backslash\overline{S'} & \longrightarrow & s'\backslash S' \\ \downarrow & & \downarrow \\ \overline{s}\backslash\overline{S} & \longrightarrow & s\backslash S \end{array}$$

est encore cartésien. Je viens encore le de revérifier à l'instant, pour les objets tout au [page 136]
moins. Dire que $\overline{S} \longrightarrow S$ est un isomorphisme local, signifie que les $\overline{s}\backslash\overline{S} \longrightarrow s\backslash S$ sont `[des]` isomorphismes. Mais le carré cartésien précédent montre qu'alors les $\overline{s'}\backslash\overline{S'} \longrightarrow s'\backslash S'$ sont aussi `[des]` isomorphismes, on gagne!

Théorème 1. *Pour toute flèche $f : S' \longrightarrow S$ dans* Cat*, le foncteur induit*

$$f_! : \mathrm{Cat}/S' \longrightarrow \mathrm{Cat}/S$$

est compatible avec les localiseurs $(W^{\mathrm{d}}_{S'}, W^{\mathrm{d}}_{S})$*, donc dualement avec les localiseurs* $(W^{\mathrm{g}}_{S'}, W^{\mathrm{g}}_{S})$*, et trivialement avec les localiseurs* $(W_{S'}, W_S)$*. Il induit donc un diagramme commutatif de foncteurs sur les localisés*

$$\begin{array}{ccccc} \mathrm{HOT}_W(S') & \xrightarrow{\rho_{S'}} & \mathrm{HOT}^{\mathrm{lc}}_W(S') & \xleftarrow{\sigma_{S'}} & \mathrm{HOT}^{o}_W(S') \\ \downarrow{\scriptstyle f_!} & & \downarrow{\scriptstyle f^{\mathrm{lc}}_!} & & {\scriptstyle f^o_!}\downarrow \\ \mathrm{HOT}_W(S) & \xrightarrow{\rho_S} & \mathrm{HOT}^{\mathrm{lc}}_W(S) & \xleftarrow{\sigma_S} & \mathrm{HOT}^{o}_W(S). \end{array}$$

Définition 1. *On a défini ainsi les foncteurs verticaux* $f_!$*,* $f^{\mathrm{lc}}_!$*,* $f^o_! = (f^o)_!$*, appelés foncteurs image directe homologique, ou foncteurs d'intégration pour* f *(et les types d'homotopie relatifs).*

On a obtenu en même temps, gratuitement, la compatibilité de ces foncteurs aux foncteurs [page 137]
[de] localisation $\rho_{S'}$, $\sigma_{S'}$ pour $\mathrm{HOT}(S')$, et ρ_S, σ_S pour $\mathrm{HOT}(S)$. Par contre, on ne s'attendra pas qu'ils commutent avec les foncteurs pleinement fidèles d'inclusion des $\mathrm{HOT}^{\mathrm{lc}}_W$ dans HOT_W. En effet, si on ne suppose pas f dans Fibhot ([67]), il est tout à fait immoral de s'attendre à ce que $f_!$ dans $\mathrm{HOT}_W(S')$ transforme types localement constants en itou. La situation est donc exactement inverse de celle pour le foncteur f^*, qui était compatible avec les foncteurs d'inclusion i_S, $i_{S'}$ et j_S, $j_{S'}$ - c'était à tel point trivial qu'il n'a pas été question même de le dire, voire en faire une proposition! La situation est

Commentary · Exposition [ms. 136]

Changing the structure map gives direct image

Grothendieck's proposed direct image has a particularly economical description: keep the total category and compose its map to the base. The substantial point is that this operation respects right-fine equivalence even though inverse image did not respect it on arbitrary representatives. The proof uses a class of test changes of base which is itself stable under pullback.

Let $u : X \to Y$ be in $W^{\mathrm{d}}_{S'}$ and let $f : S' \to S$. For an object $s \in S$, form

$$E = (s\backslash S) \times_S S' \longrightarrow S'.$$

This projection is a discrete opfibration: an object is an arrow $s \to f(s')$, and an arrow out of s' has a unique lift obtained by composing that arrow in S. Right-fine equivalences remain global W-equivalences after pullback along such maps. Applying this characterization to u shows that

$$s\backslash(X/S) \longrightarrow s\backslash(Y/S)$$

is in W. Since this holds for every s, the same underlying functor, regarded over S, belongs to W^{d}_S.

The characterization can also be checked directly from the comma resolution. For a discrete opfibration $q : E \to S'$ and $e \in E$, unique lifting gives

$$e\backslash(X \times_{S'} E) \cong q(e)\backslash X.$$

Thus the map between the comma resolutions over E is a weak equivalence on fibres. The localizer's fibrewise-to-total property makes it a global weak equivalence, and the comparison from each comma resolution to its total category is a weak equivalence. Two-out-of-three gives the desired global conclusion. This uses only the comparison machinery already available.

The square of coslices considered in the source is Cartesian on morphisms as well as objects: an arrow is a pair of commuting triangles whose images in the bottom-right coslice agree. The universal property of the original pullback gives exactly one arrow in the top-left coslice with those data. Therefore the author's object check extends immediately to a proof of the asserted categorical pullback. The term "isomorphisme local" on ms. 136 must be read in its co-local sense, since it is outgoing coslices that occur throughout this argument.

We obtain

$$f_! q_{S'}(X) = q_S(X \to S' \xrightarrow{f} S).$$

Composition of structure maps gives the composition law for these functors. Their compatibility with ρ and σ is equally formal: the routes around the displayed square apply the same total-category functor and then invert a larger class. Neither observation says that direct image preserves locally constant objects. The interval example above shows that it need not.

67. Ce cas doit être regardé avec soin ultérieurement.

même encore meilleure pour le foncteur $f_!$ entre les HOT, puisque la description de ce foncteur peut se faire directement sur Cat/S' tout entier, sans avoir à se borner à des sous-catégories plus ou moins astucieuses, où le foncteur induit serait compatible avec les localiseurs ($\underline{\mathrm{Liss}}\, S$ et $\underline{\mathrm{Fibhot}}\, S$ dans le cas de f^*). p. 84 [page 138]

Commentary · Examples [ms. 137]

Integration and the local-constant reflection

Grothendieck distinguishes two compatibility statements here because integration changes the support of a relative type. The inclusion $a : \{0\} \to S = (0 \to 1)$ already exhibits the phenomenon. The one-point object over $\{0\}$ is locally constant. Its direct image in the full relative category has comma fibres

$$(0\backslash\{0\}) \simeq e, \qquad (1\backslash\{0\}) = \varnothing.$$

It is therefore not locally constant on S. Its reflection is the constant point, as calculated in the earlier block on ms. 128. Thus $a_! i_{\{0\}}$ and $i_S a_!^{\mathrm{lc}}$ cannot be identified in general.

The commutation with reflection goes in the direction stated in the theorem:

$$\rho_S f_! \simeq f_!^{\mathrm{lc}} \rho_{S'}.$$

There is a second explanation once the adjunctions have been established. The inclusions i_S commute with restriction, and ρ_S is left adjoint to i_S. Both functors in this display are consequently left adjoint to the same composite restriction followed by inclusion. The uniqueness of adjoints identifies them. In contrast, preserving the essential image of $i_{S'}$ under $f_!$ is a further property of f.

For ordinary weak equivalences, the Hot-fibration hypothesis in footnote 67 supplies precisely that property. If $s \to t$ is an arrow of S, the induced functor $t\backslash S' \to s\backslash S'$ is a weak equivalence. For a locally constant coefficient F, the pointwise value $(f_! F)(s)$ is the homotopy colimit of the pulled-back coefficient over $(s\backslash S')^{\mathrm{op}}$. A weak equivalence of indexing categories preserves this homotopy colimit with locally constant coefficients. Thus every transition of $f_! F$ is an equivalence: $f_!$ carries locally constant objects to locally constant objects, and $f_! i_{S'} \simeq i_S f_!^{\mathrm{lc}}$. This is the space-valued case of Cisinski (2009, Proposition 1.24, applied to the opposite indexing convention); the proof there makes the same pointwise homotopy-colimit comparison. It gives the affirmative answer sought in the footnote, while retaining the hypothesis that the preceding interval example shows to be substantive.

For the terminal map $p : S \to e$, composition of structure maps simply forgets the base. If a category $X \to S$ represents a coherent family of homotopy types, then $p_! q_S(X)$ is the homotopy type of the total category X. For a category of elements this is the homotopy-colimit formula expressed geometrically. It explains the word "integration": the coherent fibres, with their transport, are assembled into their total homotopy type. One should retain the word "coherent" here, since an ordinary diagram in the homotopy category alone does not determine that total type.

On a de façon immédiate la transitivité de $f_!$, $f_!^{\mathrm{lc}}$, $f_!^o$, i.e.

$$(gf)_! = g_! f_!, \qquad (gf)_!^{\mathrm{lc}} = g_!^{\mathrm{lc}} f_!^{\mathrm{lc}}, \qquad (gf)_!^o = g_!^o f_!^o.$$

Mais on ne va pas faire bouger S, S', f pour le moment, car il faut maintenant, avant toute autre chose, essayer d'établir l'adjonction de $f_!$ et de f^* (dans cet ordre).

Pour le moment, occupons-nous de HOT_W seulement - le cas de HOT_W^o s'en déduira en appliquant ce qu'on saura à f^o. Et on s'occupera de $\mathrm{HOT}_W^{\mathrm{lc}}$ plus tard.

1) *L'homomorphisme*

$$\alpha : \mathrm{id}_{\mathrm{HOT}_W(S')} \longrightarrow f^* f_!.$$

Si X' est dans Cat/S', $f_!(\mathrm{hot}_{S',W}(X'))$ est $\mathrm{hot}_{S,W}(X')$, mais on ne peut trouver son image inverse f^* sur S' en prenant l'image inverse catégorique. Car même si on fait des hypothèses draconiennes sur X' en tant qu'objet de Cat/S' (p.ex. X' fibrée sur S'), celles-ci seront réduites à néant quand on aura composé avec f, sur lequel nous ne voulons ici faire aucune hypothèse. Il faut donc d'abord appliquer le foncteur Φ_S à X' sur S, puis prendre son image inverse sur S' : [page 139]

$$f^* f_!(\mathrm{hot}_{S',W}(X')) = \mathrm{hot}_{S',W}(\Phi_S(X'/S) \times_S S').$$

Il faut donc trouver une flèche

$$\alpha_{X'} : X' \longrightarrow \Phi_S(X'/S) \times_S S'$$

dans le localisé $(\mathrm{Cat}/S')(W^{\mathrm{d}}_{S'})^{-1}$, voire même une flèche dans Cat/S' lui-même. Mais par les propriétés d'adjonction des foncteurs $f_!$ et f^* au *niveau catégorique,* une telle flèche revient à une flèche

$$f_!(X') \longrightarrow \Phi_S(X'/S)$$

dans Cat/S, i.e. à une flèche qui rend commutatif le carré

$$\begin{array}{ccc} X' & \longrightarrow & \Phi_S(X'/S') \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S. \end{array}$$

Or on a une telle flèche canonique, savoir $i_{X'/S}$. Elle est fonctorielle en X' sur S (sans avoir même à se soucier si X' sur S provient d'un X' sur S', i.e. si la flèche structurale $X' \longrightarrow S$ se factorise par $S' \longrightarrow S$). p. 85

2) *L'homomorphisme* [page 140]

$$\beta : \mathrm{id}_{\mathrm{HOT}_W(S)} \longleftarrow f_! f^*.$$

Commentary · Exposition [ms. 139]

The unit comes from the comma replacement

The unit must compare an object over S' with the result of forgetting down to S and then deriving inverse image. Forgetting the intermediate base can destroy the useful smoothness or fibration hypothesis. This is why Φ_S appears between those two operations, even if the original category was already fibred over S'.

It is helpful to describe the construction on objects. For $p : X \to S$, the category $\Phi_S(X)$ has objects $(s, x, v : s \to p(x))$ and is projected to s. A morphism from (s, x, v) to (t, y, w) is a pair $(h : s \to t, k : x \to y)$ satisfying $p(k)v = wh$. The unit i_X sends x to $(p(x), x, \mathrm{id}_{p(x)})$. Thus for $p' : X' \to S'$ the unit of the desired adjunction is represented by

$$\alpha_{X'} : X' \longrightarrow \Phi_S(X'/S) \times_S S', \qquad x \longmapsto \big((fp'(x), x, \mathrm{id}_{fp'(x)}), p'(x)\big).$$

The same formula on arrows uses $fp'(k)$ in the first coordinate and $p'(k)$ in the second. It is a functor over S' and is natural in every functor $X' \to Y'$ over S'.

Equivalently, it is uniquely characterized by its two projections:

$$\mathrm{pr}_1 \alpha_{X'} = i_{X'/S}, \qquad \mathrm{pr}_2 \alpha_{X'} = p'.$$

This characterization is the ordinary slice adjunction between composition with f and pullback along f, applied before localization. The target of $\alpha_{X'}$ represents $f^* f_! q_{S'}(X')$ because $\Phi_S(X'/S)$ is a split fibration over S. Hence the natural map descends to $\mathrm{id} \to f^* f_!$.

Naturality before localization is enough here, but it is important to use the correct base in the replacement. The top-right object in the source's square on ms. 139 is printed as $\Phi_S(X'/S')$. Its structure map and the preceding formula show that $\Phi_S(X'/S)$ is intended. This is a source slip, retained in the transcription and registered separately. No replacement over S' can repair the loss of smoothness caused by composing its structure map with an arbitrary f.

Commentary · Exposition [ms. 139]

The replacement must be taken over the new base

Grothendieck now has a particularly economical candidate for direct image: a category over S' is regarded as a category over S. The difficulty is to show that this operation has the expected universal property after localization. Its prospective right adjoint is not arbitrary strict pullback. It

is pullback after a replacement which remembers arrows in the base. The next seven manuscript pages make that distinction precise enough to construct an adjunction, including its two triangle identities.

Write $Q_S = \Phi_S$ and $i_X : X \longrightarrow Q_S X$. For a category $p : X \longrightarrow S$, an object of $Q_S X$ is a triple $(s, x, a : s \longrightarrow p(x))$. A morphism from (s, x, a) to (t, y, b) consists of $v : s \longrightarrow t$ and $u : x \longrightarrow y$ satisfying $p(u)a = bv$. The projection to s is the split fibration associated with the contravariant functor $s \longmapsto s\backslash X$. The map i_X sends x to $(p(x), x, \mathrm{id}_{p(x)})$. This recalls the comma replacement discussed at manuscript p. 20 ("Which comma construction repairs which variance"), but here its role is more demanding: it supplies the unit of a derived adjunction.

For $f : S' \longrightarrow S$, denote by L the functor induced by composition with f, and by R derived inverse image. If $[X]$ denotes $\mathrm{hot}_{S,W}(X)$, their formulas are

$$L[X'] = [X'/S], \qquad R[X] = [Q_S X \times_S S'].$$

The replacement in the second formula is over S. Even when $X' \longrightarrow S'$ is already a fibration, its composite with an arbitrary f need not be one. Replacing over S' would therefore solve the wrong problem. This is the reason for the otherwise rather insistent occurrences of X'/S in the argument. The occurrence of $Q_S(X'/S')$ in the square on manuscript page 139 is a slip already present in the edition; its intended meaning is $Q_S(X'/S)$.

The unit can be written before localization. Its component is the unique S'-functor

$$a_{X'} : X' \longrightarrow Q_S(X'/S) \times_S S'$$

whose first projection is $i_{X'/S}$ and whose second projection is the structure map of X'. Explicitly, it sends x' to the triple with base object $f(p'(x'))$ and identity arrow, together with $p'(x') \in S'$. The construction is natural in X'. Passing through the localization therefore gives $\alpha : \mathrm{id} \longrightarrow RL$.

The counit is slightly different. If $X \longrightarrow S$ is smooth, the projection $X \times_S S' \longrightarrow X$ represents $\beta_{[X]} : LR[X] \longrightarrow [X]$. For an arbitrary representative, the formula is instead

$$\beta_{[X]} = [i_X]^{-1}[\mathrm{pr}_1 : Q_S X \times_S S' \longrightarrow Q_S X].$$

Both arrows in this fraction lie over S, and the arrow inverted is one already known to belong to W_S^{d}. The forgetful functor $Q_S X \longrightarrow X$ on underlying categories does not generally lie over S: the two structure maps at (s, x, a) have values s and $p(x)$. The fraction is thus doing necessary work.

The formulas also explain why naturality survives the change of representatives. For an arrow $u : X \longrightarrow Y$ over S, the equalities $Q_S(u)i_X = i_Y u$ and $Q_S(u)\mathrm{pr}_1 = \mathrm{pr}_1(Q_S(u) \times_S S')$ give the naturality square for β after the i's have been inverted. The construction uses the localization universal property, rather than a choice of inverses in the original category. This is precisely the amount of structure needed for the adjunction proof that follows.

On peut décrire $\mathrm{HOT}_W(S)$ par $(\underline{\mathrm{Fib}}\, S)(W_S^{\mathrm{d}})^{-1}$ ou $(\underline{\mathrm{Liss}}\, S)(W_S^{\mathrm{d}})^{-1}$ par exemple, de sorte que f^* a une description simple par

$$f^*(\mathrm{hot}_{S,W}(X)) = \mathrm{hot}_{S',W}(X \times_S S')$$

(pour X lisse sur S). On a ensuite

$$f_! f^*(\mathrm{hot}_{S,W}(X)) = \mathrm{hot}_{S,W}(X \times_S S'),$$

mais attention, $X \times_S S'$ n'est pas a priori dans $\underline{\mathrm{Liss}}\, S$, seulement dans Cat/S sans plus. Mais cela n'empêche qu'il donne ce qu'il faut, par localisation $(W_S^{\mathrm{d}})^{-1}$ dans Cat/S. Pour définir β_X, il suffit donc de trouver une flèche

$$\beta_X : X \longleftarrow X \times_S S'$$

dans Cat/S, et on voit bien pr_1^X ([68]). On a donc

$$\begin{cases} \alpha : \mathrm{id} & \longrightarrow & f^* f_! \\ \beta : \mathrm{id} & \longleftarrow & f_! f^* \end{cases}$$

68. quand $\xi \in \mathrm{HOT}_W(S)$ est défini par X dans Cat/S non nécessairement lisse, on doit trouver $\Phi_S(X) \times_S S' \longrightarrow X$ dans $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$, on prend $(i_X)^{-1}\mathrm{pr}_1^{\Phi_S(X)}$ (cf. page 143).

3) *La compatibilité*

$$f_! \xrightarrow{f_!*\alpha} f_!f^*f_! \xrightarrow{\beta*f_!} f_!.$$
$$\text{id}$$

On doit donc prendre X' sur S', et expliciter les flèches à partir de l'objet

$$\xi' = \mathrm{hot}_{S',W}(X').$$

On a p. 86

$$f_!(\xi') = \mathrm{hot}_{S,W}(X'/S), \qquad \text{[page 141]}$$

et pour calculer $f^*f_!(\xi')$, il faut comme tantôt prendre $\Phi_S(X')$ et on trouve

$$f^*f_!(\xi') = \mathrm{hot}_{S',W}(\Phi_S(X') \times_S S').$$

Enfin,

$$f_!f^*f_!(\xi') = \mathrm{hot}_{S,W}(\Phi_S(X') \times_S S').$$

Il faut expliciter les flèches

$$\underbrace{f_!(\xi')} \xrightarrow{f_!(\alpha_{\xi'})} \underbrace{f_!f^*f_!(\xi')} \xrightarrow{\beta_{f_!(\xi')}} \underbrace{f_!(\xi')}.$$

(∗)
$$X'/S \xrightarrow[\text{déduit de } i_{X'}]{\alpha_{X'}} \Phi_S(X') \times_S S' \xrightarrow{\beta_{X'}} X'/S$$

sur S

$\alpha_{X'}$ est déduit de $i_{X'} : X' \longrightarrow \Phi_S(X')$ comme l'unique S'-morphisme rendant commutatif le diagramme

(∗∗)
$$\begin{array}{ccccc} X' & \xrightarrow{\alpha_{X'}} & \Phi_S(X') \times_S S' & \xrightarrow{\mathrm{pr}_1} & \Phi_S(X') \\ \downarrow & \overset{i_{X'}}{\nearrow} & & & \downarrow \\ S' & & \xrightarrow{\qquad\qquad} & & S. \end{array}$$

Pour décrire $\beta_{f_!(\xi')}$, où $f_!(\xi') = \mathrm{hot}_{S,W}(X')$, il faut donner une description générale de β_ξ qui soit valable pour *tout* représentant X de ξ, pas seulement dans le cas X lisse sur S. Pour calculer $f^*(\xi)$, il faut donc prendre le $\mathrm{hot}_{S',W}$ de $\Phi(X) \times_S S'$, puis pour $f_!f^*(\xi)$, il faut prendre le $\mathrm{hot}_{S,W}$ du même $\Phi_S(X) \times_S S'$. Il faut définir, si on peut, une flèche [page 142]

$$\Phi_S(X) \times_S S' \longrightarrow X \qquad \text{dans } \mathrm{Cat}/S,$$

ou du moins dans $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$. Mais on a

$$i_X : X \longrightarrow \Phi_S(X)$$

qui est dans W_S^{d}, et qui par produit cartésien définit

$$i_X \times_S \mathrm{id}_{S'} : X \times_S S' \longrightarrow \Phi_S(X) \times_S S'$$

(en d'autres termes on a pris l'image inverse de i_X par $S' \longrightarrow S$). Cette flèche est-elle encore dans W_S^{d}? ([69]). Si oui, on a le diagramme

69. Non, elle ne l'est pas, car si $S' = \{s\}$, cela signifierait que $X'_s \longrightarrow s\backslash X$ est dans W, ce qui est faux en général. p. 87

$$\begin{array}{ccc} \Phi_S(X) \times_S S' & \xrightarrow{\beta_X ?} & X \\ {\scriptstyle i_X \times_S \mathrm{id}_{S'}}\uparrow & \nearrow_{\mathrm{pr}_1^X} & \\ X \times_S S' & & \end{array} \quad = \pi_X$$

et on définit la flèche cherchée β_X *dans* $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$ par la commutativité de ce triangle, donc comme

$$\beta_X = \mathrm{pr}_1^X \circ \underbrace{\pi_X^{-1}}_{\text{inverse dans } (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}} .$$

Mais ça ne marche pas - cf. annotation! On va faire alors l'inverse [page 143]

$$\begin{array}{ccc} \Phi_S(X) \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} & \Phi_S(X) \\ & \searrow^{\beta_X ?} & \uparrow{\scriptstyle i_X} \quad \text{dans } W_S^{\mathrm{d}} \\ & & X \end{array}$$

Commentary · Examples [ms. 142]

A pullback can destroy the replacement equivalence

Grothendieck's first attempt to construct the counit asks whether the map i_X can be inverted after an arbitrary pullback. The walking arrow shows exactly where that attempt breaks. We work with ordinary weak homotopy equivalences, $W = W_\infty$, so that an empty category cannot be equivalent to a point.

Let $S = [1] = (0 \longrightarrow 1)$ and let $X = \{1\} \longrightarrow S$. An object of $\Phi_S X$ is an arrow of S with target 1. There are two such objects, $0 \longrightarrow 1$ and id_1, and one nonidentity morphism between them. The projection identifies $\Phi_S X$ with S over itself. Consequently

$$i_X : \{1\} \longrightarrow S$$

is right-fine: for $s = 0$ and $s = 1$, the functor $s\backslash X \longrightarrow s\backslash S$ is a weak equivalence between contractible categories.

Now pull back along $j : \{0\} \longrightarrow S$. The resulting map is

$$X \times_S \{0\} = \varnothing \longrightarrow \Phi_S X \times_S \{0\} = \{0\}.$$

It is not in W, nor is it a right-fine equivalence over the point. Regarded again as a map over S, it still fails the right-fine test at 0. Thus the proposed inverse of $\pi_X = i_X \times_S \mathrm{id}_{S'}$ need not exist in either localization relevant to the attempted argument.

The calculation also distinguishes the two inverse images without any abstract machinery. Strict pullback of X to 0 is empty, whereas derived pullback of $[X]$ to 0 is a point. The missing object records the arrow $0 \longrightarrow 1$. A comma fibre retains that arrow; a strict fibre does not.

There is no defect in the replacement itself. The map i_X becomes an isomorphism over S, exactly as intended. The invalid move was to apply a functor which had not been shown to preserve that class of equivalences on all of Cat/S. On manuscript page 143 Grothendieck repairs the argument himself: he first projects to $\Phi_S X$, and then inverts i_X in the localization over S. No pullback of an arbitrary denominator is required.

([70]), on définit β_X comme $(i_X)^{-1}\mathrm{pr}_1^{\Phi_S(X)}$, où $(i_X)^{-1}$ est défini dans $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$.

Pour prouver la commutativité de $(*)$ (p. 141), comme $i_{X'} : X' \longrightarrow \Phi_S(X')$ est dans W_S^{d}, il suffit de prouver, au lieu de $\beta_{X'}\alpha_{X'} = \mathrm{id}_{X'}$, la relation déduite en composant avec $i_{X'} : X' \longrightarrow \Phi_S(X')$

$$\underbrace{i_{X'}\beta_{X'}}_{\mathrm{pr}_1^{\Phi_S(X')}} \alpha_{X'} = i_{X'},$$

70. Description de β_ξ pour $\xi = \mathrm{hot}_{S,W}(X)$, avec X quelconque dans Cat/S, à expliciter dans définition en forme.

i.e.

$$\mathrm{pr}_1^{\Phi_S(X')} \circ \alpha_{X'} = i_{X'},$$

ce qui n'est autre que la relation de définition de $\alpha_{X'}$ (cf. (∗∗) p. 141), on gagne!

4) *Compatibilité*

$$f^* \xrightarrow{\alpha * f^*} f^* f_! f^* \xrightarrow{f^* * \beta} f^*.$$
$$\text{id}$$

Partons donc d'un $\xi \in \mathrm{ObHOT}_W(S)$, qu'on peut supposer défini par $X \in \mathrm{Ob}\underline{\mathrm{Fib}}\, S$. Alors p. 88 [page 144]

$$\begin{aligned} f^*(\xi) &= \mathrm{hot}_{S',W}(X \times_S S'), \\ f_! f^*(\xi) &= \mathrm{hot}_{S,W}(X \times_S S'), \\ f^* f_! f^*(\xi) &= \mathrm{hot}_{S',W}(\Phi_S(X \times_S S') \times_S S'). \end{aligned}$$

Il faut expliciter maintenant

$$\underbrace{f^*(\xi)}_{\mathrm{hot}_{S',W}(X')} \xrightarrow{\alpha_{f^*(\xi)}} \underbrace{f^* f_! f^*(\xi)}_{\mathrm{hot}_{S',W}(\Phi_S(X') \times_S S')} \xrightarrow{f^*(\beta_\xi)} \underbrace{f^*(\xi)}_{\mathrm{hot}_{S',W}(X')},$$

et vérifier que le composé est l'identité. On a posé

$$X' = X \times_S S',$$

et le diagramme est donc aussi

$$X' \xrightarrow{\alpha_{X'}} \Phi_S(X') \times_S S' \xrightarrow{f^*(\beta_\xi)} X', \qquad \mathrm{id}_{X'} : X' \to X'$$

dans $(\mathrm{Cat}/S')(W_{S'}^{\mathrm{d}})^{-1}$. Pour vérifier la commutativité, on peut composer avec

$$X \times_S S' = X' \xrightarrow{i_X \times_S \mathrm{id}_{S'}} \Phi_S(X') \times_S S' \qquad (\text{où } i_X : X \longrightarrow \Phi_S(X))$$

qui est dans $W_{S'}^{\mathrm{d}}$ ([71]), or on vérifie, sur la construction générale de $\beta_\xi = \beta_X$ (page précédente), que l'on a commutativité dans

$$\begin{array}{ccc} \Phi_S(X') \times_S S' & \xrightarrow{f^*(\beta_\xi)} & X' \\ & \searrow^{\Phi_S(\mathrm{pr}_1^X) \times_S S'} & \downarrow{\scriptstyle i_X \times_S S'} \\ & & \Phi_S(X) \times_S S'. \end{array}$$

Commentary · Exposition [ms. 143]

The counit is a fraction over the original base

Grothendieck's first formula for the counit is correct when the representative $X \to S$ is smooth: strict pullback represents derived inverse image, and projection

$$X \times_S S' \longrightarrow X$$

represents $f_! f^* q_S(X) \to q_S(X)$. To check the first triangle identity, however, one needs the counit on $f_! q_{S'}(X')$, whose natural representative need not be smooth over S. The extension to arbitrary representatives is the small but essential issue being worked out on ms. 141–143.

71. **NB** Comme X est lisse/S, et $\Phi_S(X)$ aussi, $X \longrightarrow \Phi_S(X)$ reste une W-équivalence colocale par tout changement de base.

For arbitrary X, put $R(X) = \Phi_S(X) \times_S S'$. The replacement map $i_X : X \to \Phi_S(X)$ is invertible in $\mathrm{HOT}_W(S)$, so the valid formula is

$$\beta_X = [i_X]^{-1}[\mathrm{pr}_1] : q_S(R(X)) \longrightarrow q_S(X).$$

Both arrows in this fraction are considered over S. Their naturality follows from the naturality of i and of projection. If X is smooth, the map $i_X \times_S S'$ is a right-fine equivalence over S', and the natural square with the two projections identifies this general formula with the simple projection formula above.

The abandoned attempt instead inverts $i_X \times_S S'$ without a hypothesis making that pullback a weak equivalence. A different interval calculation shows the problem directly. Take $S = (0 \to 1)$, $X = \{1\} \to S$, and $S' = \{0\}$. The strict fibre of X over 0 is empty, while the fibre of $\Phi_S(X)$ is $0\backslash X \simeq e$. The pullback of i_X is therefore $\varnothing \to e$. It is not a weak equivalence, either over S' or after regarding it over S; its comma fibre at 0 already fails. There is no inverse with which to form the proposed fraction.

The distinction is between an inverse in the localization over S, which is available, and the pullback of that inverse on arbitrary representatives, which is not represented by pulling back its denominator. The corrected roof leaves the denominator over S and uses projection from a replacement on which inverse image can be computed. This is the same discipline that governs total derived functors: apply the operation to a suitable replacement, then use the comparison isomorphism in the category where it is known to exist.

On est ramené à la commutativité de [page 145]

(∗)
$$\begin{array}{ccc} X \times_S S' = X' & \xrightarrow{\alpha_{X'}} & \Phi_S(X') \times_S S' \\ & \overset{?}{\searrow}_{i_X \times_S S'} & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X) \times_S S'} \\ & & \Phi_S(X) \times_S S' \end{array}$$

Commentary · Exposition [ms. 145]

The last square is naturality

The argument has reached the last square before the second triangle identity. No new homotopical input is required. The remaining equality is the naturality of the comma-replacement map, and making that equality explicit finishes the adjunction on these pages.

For the first triangle, start with $X' \to S'$, regarded over S after composition. The defining relation for the unit is

$$\mathrm{pr}_1 \alpha_{X'} = i_{X'}.$$

The general formula for the counit gives $\beta_{X'} = i_{X'}^{-1} \mathrm{pr}_1$ in the localization over S. Consequently $\beta_{X'} \alpha_{X'} = \mathrm{id}_{X'}$. This is the identity $\beta f_! \circ f_! \alpha = \mathrm{id}_{f_!}$, and its proof applies without any smoothness hypothesis on X' over S.

For the second triangle we may represent ξ by a fibration $X \to S$. Put $X' = X \times_S S'$ and $r = \mathrm{pr}_1 : X' \to X$. Since X is smooth, the counit at ξ is represented by r. To calculate its derived inverse image on the possibly nonsmooth source $X' \to S$, replace that source by $\Phi_S(X')$. Functoriality of Φ_S then gives the relation

$$(i_X \times_S S')\, f^*(\beta_\xi) = \Phi_S(r) \times_S S'$$

in the localization over S'. Here $i_X \times_S S'$ is invertible because both X and $\Phi_S(X)$ are smooth over S.

It remains to prove

$$(\Phi_S(r) \times_S S') \alpha_{X'} = i_X \times_S S'.$$

These are maps $X' \to \Phi_S(X) \times_S S'$. Their second projections both equal the structure map $X' \to S'$. Their first projections are respectively

$$\Phi_S(r) i_{X'} \qquad \text{and} \qquad i_X r,$$

which are equal by the naturality of $i : \mathrm{id} \to \Phi_S$. The universal property of the pullback proves the displayed equality already in Cat/S'. Composing the preceding relation with $\alpha_{X'}$ and cancelling $i_X \times_S S'$ now gives

$$f^*(\beta_\xi)\alpha_{f^*\xi} = \mathrm{id}_{f^*\xi}.$$

The two triangle identities establish $f_! \dashv f^*$.

There is a minor but consequential printed index in the intervening display on ms. 144. The map labelled $i_X \times_S \mathrm{id}_{S'}$ has target $\Phi_S(X) \times_S S'$, as in the next triangle, rather than $\Phi_S(X') \times_S S'$. The latter is the target of $\alpha_{X'}$ and the source of $f^*(\beta_\xi)$; identifying them would obscure the naturality square just used. The edition prints the prime, so it is preserved as a source slip. The proof given here completes this last square and makes no claim of originality.

Overview · VI-b

Relative homotopy types, reflection, and integration.

These pages occupy a hinge in the construction of HOT. Grothendieck has accumulated several ways of representing a homotopy type over a small category S: a presheaf of categories, a split or unsplit fibration, a smooth category over S, or simply an arbitrary functor $X \to S$. He now wants to know exactly which localizations make these presentations interchangeable. Once that question is settled, the base may vary and the categories so obtained can begin to carry the operations of a derivator. The movement from a comparison at fixed S to inverse and direct images explains both the long abstract digression and the care devoted to the two triangular identities of an adjunction.

The indispensable distinction is between forgetting weak equivalences in the total category and testing them relative to the base. For a map $u : X \to Y$ over S, write

$$\begin{aligned} u \in W_S &\iff u \in W, \\ u \in W_S^{\mathrm{d}} &\iff (s\backslash u) \in W \quad \text{for every } s \in S, \\ u \in W_S^{\mathrm{g}} &\iff (u/s) \in W \quad \text{for every } s \in S. \end{aligned}$$

Here $s\backslash X = (s \downarrow X)$ and $X/s = (X \downarrow s)$ are comma categories, not strict fibres. The two fine classes retain respectively contravariant and covariant variation along arrows of S. On suitably smooth or proper representatives, the corresponding comma categories may be replaced by strict fibres. On homotopically locally constant representatives, the coarse and both fine tests coincide under the additional fibration hypothesis used in this part of the manuscript. The names of the fifteen categories in the large diagram serve this comparison; they do not describe fifteen different homotopy theories.

Two replacements organize the whole argument. The comma replacement $\Phi_S X$ is a split fibration with fibre $s\backslash X$; its natural map from X is a right-fine equivalence. It therefore changes the representative while retaining the relative homotopy type. The path replacement $C_S X$ has locally constant fibres and is connected to X by a coarse equivalence. It retains the total homotopy type but generally changes the relative one. The distinction is expressed by a reflective localization

$$\rho_S : \mathrm{HOT}_W(S) \rightleftarrows \mathrm{HOT}_W^{\mathrm{lc}}(S) : i_S, \qquad \rho_S \dashv i_S, \qquad \rho_S i_S \simeq \mathrm{id}.$$

The unit $X \to C_S X$ need not be invertible in $\mathrm{HOT}_W(S)$. This is why the second equivalence in Corollary 2 on manuscript page 120, which offers C_S as an alternative to Φ_S for computing the inverse image of an arbitrary relative type, must be corrected. Its C_S expression computes the inverse image after locally constant reflection. The commentary on the first part of Chapter VI already explains the basic distinction; the present commentary applies it precisely where the manuscript temporarily loses it.

The opening overlap with VI-a completes the adjunction by comparing $i_{C(X)}$ with $C(i_X)$. Concatenation of paths gives a common retraction, and hence equality after localization. There is also a formal reason: for a coaugmented endofunctor, invertibility of both iterated units forces their compatibility and identifies the essential image as a reflective subcategory. This familiar recognition principle is stated in higher categorical form by Lurie (2009, Proposition 5.2.7.4). Thus the path construction supplies particularly useful representatives, but its composition law is stronger data than the adjunction itself needs.

The abstract discussion beginning on manuscript page 85 isolates precisely this point. In a category $\mathcal{M}$ with weak equivalences W and a class of fibrations, Grothendieck compares the localization of $\mathcal{M}/S$ with that of its full subcategory of fibrant objects. A functorial replacement gives the equivalence directly. A replacement obtained only by lifting properties requires a separate argument: two choices of lift are compared through a factorization of a

relative diagonal, and become equal after localization. The manuscript proves this comparison, rather than merely assuming that choices disappear. Theorem–Scholie 2 then introduces a finer class $W_S^0 \subseteq W_S$ agreeing with W_S between fibrant objects and constructs the associated reflection. Our exposition writes out the naturality checks and the independence of choices, and explains why the functorial-replacement case already yields the adjunction without the additional path-composition hypothesis A'. This is an application of established deformation theory, not a new localization theorem; compare Riehl (2014, Theorem 2.2.11).

There are several distinct uses of the word fibration here. A Grothendieck fibration describes transport of objects in a category. A homotopical fibration controls comma categories and strict fibres. A fibration in a Quillen model structure is part of a lifting and factorization system. They interact, but the manuscript's abstract argument does not identify them. In particular, invoking Thomason's model structure gives a method for ordinary nerve weak equivalences on categories; it does not by itself establish every assertion for every proposed fundamental localizer. Likewise, weak saturation, meaning the formal closure axioms initially imposed on W, must be distinguished from strong saturation, which detects membership in W by invertibility in $\mathrm{Cat}[W^{-1}]$. The latter is supplied for fundamental localizers by Cisinski (2006c, Proposition 4.2.4 in the cited author version). It closes the explicit saturation reservation in the converse of Proposition 2 on manuscript pages 124–126.

Once the base varies, raw pullback of an arbitrary $X \to S$ is insufficient. To calculate f^* one first takes a representative, such as $\Phi_S X$, on which strict pullback respects the relevant weak equivalences. Smoothness of $X \to S$ supplies one sufficient condition; properness of the change of base supplies another. The "sure conjecture" on manuscript page 122 is not left open in the work: Grothendieck proves it on manuscript pages 168–169 of the same chapter. The commentary separates this later proof from what has actually been established in these pages. For ordinary weak homotopy equivalences, testing the comma replacement of a point also proves the converse: preservation of right-fine equivalences by strict pullback characterizes properness. A three-object span gives the nonproper homotopy fibration requested on manuscript page 126; it preserves coarse equivalences but fails to preserve a particular fine equivalence. These computations make the difference between the two base-change conditions explicit.

A natural transformation $f \Rightarrow g$ also requires attention: it induces $g^* \Rightarrow f^*$ in the presheaf convention, but a literal functor between arbitrary strict fibres need not exist. Split fibrations make that action explicit; localization then removes the dependence on the chosen splitting.

The questions about the locally constant theory lead to a useful distinction in the modern answer. Restriction between categories of locally constant diagrams has both adjoints; this holds for arbitrary derivator coefficients by Cisinski (2009, Corollary 3.11). Nevertheless, the varying-base assignment $S \mapsto \mathrm{HOT}^{\mathrm{lc}}(S)$ does not form a derivator for ordinary homotopy types. The pointwise Kan extension formula already fails for a point included in the walking arrow. Fixing a base S and varying an additional diagram category is a different operation: that construction does give the derivator of locally constant coefficients, as established in the same paper's Theorem 3.13. The distinction is visible without any higher machinery and prevents an attractive but incorrect explanation of Grothendieck's hesitation on manuscript page 128.

Another reservation concerns infinite disjoint unions of indexing categories. The corresponding product statement for arbitrary localizations is false. The bounded-length obstruction that Grothendieck notices can be realized by a countable family of finite fence categories: each factor becomes the terminal groupoid when all arrows are inverted, while the product has objects that no finite zigzag can connect. A refinement using finite categories inside a product with the walking retraction also answers footnote 66: strong saturation alone does not ensure that evaluation detects isomorphisms in localized diagram categories. A uniform three-arrow morphism is invertible at every coordinate, but its coordinate inverses require unbounded word lengths. The diagram shape in this counterexample is countably discrete; no

finite-shape conclusion is asserted. The space derivator nevertheless satisfies both pointwise detection and the unrestricted disjoint-union axiom. Model structures and controlled fraction presentations supply the additional uniformity that the abstract questions lack.

The final pages construct integration. Postcomposition with $f : S' \to S$ sends $X' \to S'$ to the same total category over S, and preserves the fine weak equivalences. This yields $f_!$ before any choice of fibrant replacement. The unit uses the comma replacement over the new base; the counit for arbitrary X is the localized composite

$$\Phi_S X \times_S S' \longrightarrow \Phi_S X \xrightarrow{i_X^{-1}} X.$$

The order matters. Pulling back i_X and then trying to invert it can fail, because strict pullback need not preserve a fine equivalence between arbitrary representatives. Grothendieck notices and repairs that attempt on manuscript pages 142–143. At manuscript page 145 the argument has reached the last square in the second triangle; the commentary completes its verification from naturality. In the case of spaces this integration is homotopy left Kan extension, and for the structural map it recovers homotopy colimit. The present derivator formalism provides its axiomatic setting; see Groth (2013).

The modern picture may be expressed as coherent diagrams of spaces on S^{op}, with locally constant objects corresponding to diagrams on the homotopy type of S. It is the entire homotopy type, not just the ordinary fundamental groupoid together with a functor into the homotopy category, that retains all the coherence. In particular, factoring the underlying diagram through ΠS detects local constancy but does not reconstruct its coherent object. The comma replacements and their categories of elements retain the additional data needed to form the homotopy colimit and to calculate its maps.

dans $(\mathrm{Cat}/S')(W^{\mathrm{d}}_{S'})^{-1}$. Essayons de prouver que c'est commutatif dans Cat/S', pas seulement dans le localisé par $W^{\mathrm{d}}_{S'}$. On peut le regarder comme diagramme dans Cat/S et prouver la commutativité à ce titre. Cette relation de commutativité $vu = w$ se décompose alors en deux, suivant les deux facteurs $\Phi_S(X)$ et S' de l'objet but $\Phi_S(X) \times_S S'$. Suivant le facteur S', la relation signifie simplement que $\alpha_{X'}$ est un S'-morphisme, c'est OK. Pour le facteur $\Phi_S(X)$, on prolonge le diagramme $(*)$ à droite par le carré p. 89

$$\begin{array}{ccc} X' & \xrightarrow{\alpha_{X'}} & \Phi_S(X') \times_S S' \xrightarrow{\mathrm{pr}_1^{\Phi_S(X')}} \Phi_S(X') \\ & \searrow & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1 X)\times_S S'} \qquad \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X)} \\ & & \Phi_S(X) \times_S S' \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} \Phi_S(X), \end{array}$$

et compte tenu que le composé de la première ligne horizontale est $i_{X'}$, par construction de $\alpha_{X'}$, on est ramené à la commutativité de [page 146]

$$(*) \qquad \begin{array}{ccc} X \times_S S' = X' & \xrightarrow{i_{X'}} & \Phi_S(X') \\ {\scriptstyle i_X \times_S S'}\downarrow & & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X)} \\ \Phi_S(X) \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} & \Phi_S(X). \end{array}$$

Or le morphisme $i_X : X \longrightarrow \Phi_S(X)$ dans Cat/S donne lieu au carré commutatif (comme toute flèche $i : X \longrightarrow Y$)

$$\begin{array}{ccc} X \times_S S' & \xrightarrow{\mathrm{pr}_1^X} & X \\ {\scriptstyle i_X \times_S S'}\downarrow & & \downarrow{\scriptstyle i_X} \\ \Phi_S(X) \times_S S' & \xrightarrow{\mathrm{pr}_1^{\Phi_S(X)}} & \Phi_S(X), \end{array}$$

de sorte que le composé inférieur de $(*)$ n'est autre que celui de $(**)$, donc égal au composé supérieur de ce dernier, et la commutativité de $(*)$ revient à celle de

$$\begin{array}{ccc} X \times_S S' = X' & \xrightarrow{i_{X'}} & \Phi_S(X') \\ {\scriptstyle \mathrm{pr}_1^X}\downarrow & & \downarrow{\scriptstyle \Phi_S(\mathrm{pr}_1^X)} \\ X & \xrightarrow{i_X} & \Phi_S(X), \end{array}$$

qui est évidente par fonctorialité de $Z \xrightarrow{i_Z} \Phi_S(Z)$, appliquée à la flèche $\mathrm{pr}_1^X : X' \longrightarrow X$ de Cat/S.

On a ainsi prouvé que les foncteurs

p. 90

$$(*) \qquad \mathrm{HOT}_W(S') \underset{f^*}{\overset{f_!}{\rightleftarrows}} \mathrm{HOT}_W(S) \qquad \text{[page 147]}$$

sont adjoints l'un de l'autre, avec des morphismes d'adjonction

$$\begin{cases} \mathrm{id} & \xrightarrow{\alpha} & f^* f_! \\ \mathrm{id} & \xleftarrow{\beta} & f_! f^* \end{cases}$$

précisés plus haut. Je présume que la même démonstration, en travaillant avec les localiseurs grossiers W_S, $W_{S'}$, au lieu de W^{d}_S, $W^{\mathrm{d}}_{S'}$, donnerait que les foncteurs correspondants

$$(**) \qquad \mathrm{HOT}^{\mathrm{lc}}_W(S') \underset{f^*_{\mathrm{lc}}}{\overset{f^{\mathrm{lc}}_!}{\rightleftarrows}} \mathrm{HOT}^{\mathrm{lc}}_W(S)$$

sont adjoints l'un de l'autre, avec essentiellement 'les mêmes' morphismes d'adjonction α, β. Je n'ai pas envie de reprendre les raisonnements précédents et préfère le *déduire* de l'adjonction dans la situation $(*)$. Cela me donnera l'occasion de mieux comprendre les traits formels essentiels de la situation. Je poserai pour simplifier [page 148]

$$\begin{array}{rclcrcl} \mathrm{HOT}_W(S) & = & \mathcal{A}, & & \mathrm{HOT}_W(S') & = & \mathcal{A}', \\ \mathrm{HOT}^{\mathrm{lc}}_W(S) & = & \mathcal{A}_0, & & \mathrm{HOT}^{\mathrm{lc}}_W(S') & = & \mathcal{A}'_0, \end{array}$$

et j'utilise les couples de foncteurs adjoints

$$\mathcal{A} \underset{i}{\overset{\rho}{\rightleftarrows}} \mathcal{A}_0, \qquad \mathcal{A}' \underset{i'}{\overset{\rho'}{\rightleftarrows}} \mathcal{A}'_0,$$

où ρ, ρ' sont des foncteurs de localisation, i.e. i, i' des foncteurs pleinement fidèles. Je dispose d'un couple de foncteurs adjoints

$$\mathcal{A}' \underset{f^*}{\overset{f_!}{\rightleftarrows}} \mathcal{A},$$

et je suppose de plus

Commentary · Exposition [ms. 146]

The two triangles reduce to ordinary naturality

The adjunction proof is complete, although the second triangle occupies several diagrams. Its purpose is to turn the chosen representative formulas into a genuine universal property. We give both checks together, keeping the distinction between identities in a category of categories and identities after localization.

For X' over S', put $Z = \Phi_S(X'/S)$ and let $a : X' \longrightarrow Z \times_S S'$ be the unit representative. The first triangle is immediate from the defining equality $\mathrm{pr}_1 a = i_{X'/S}$:

$$\beta_{L[X']} L(\alpha_{[X']}) = [i_{X'/S}]^{-1}[\mathrm{pr}_1][a] = \mathrm{id}_{L[X']}.$$

This calculation uses a denominator over S, where its invertibility was established. In particular, it does not revive the failed pullback argument of page 142.

For the second triangle we may represent an arbitrary object of $\mathrm{HOT}_W(S)$ by a smooth $X \longrightarrow S$. Put $X' = X \times_S S'$ and $p = \mathrm{pr}_1 : X' \longrightarrow X$. The map

$$r_X = i_X \times_S \mathrm{id}_{S'} : X' \longrightarrow \Phi_S X \times_S S'$$

is now right-fine over S'. Here the smoothness of both X and $\Phi_S X$ is essential: right-fine equivalences between these good representatives are preserved by arbitrary base change. This is a property of the representatives, not a property of arbitrary arrows in Cat/S.

Under these identifications, the description of the derived counit says

$$[r_X] \circ R(\beta_{[X]}) = [\Phi_S(p) \times_S \mathrm{id}_{S'}].$$

The unit $a_{X'} : X' \longrightarrow \Phi_S X' \times_S S'$ therefore satisfies

$$[r_X] \circ R(\beta_{[X]})\alpha_{R[X]} = [\Phi_S(p) \times_S \mathrm{id}_{S'}][a_{X'}].$$

The right side is $[r_X]$. To see this before localization, compare its two projections. Projection to S' is the structure map of X'. Projection to $\Phi_S X$ gives $\Phi_S(p) i_{X'}$, which equals $i_X p$ by naturality of i. These are exactly the projections of r_X. Since $[r_X]$ is invertible, cancellation proves

$$R(\beta_{[X]})\alpha_{R[X]} = \mathrm{id}_{R[X]}.$$

The target of r_X is $\Phi_S(X) \times_S S'$. In the displayed line immediately preceding the first triangle on manuscript page 144 the edition prints $\Phi_S(X') \times_S S'$ instead. The adjacent diagram and the expression $i_X \times_S \mathrm{id}_{S'}$ determine the intended target unambiguously. This is a typographical slip, not a gap in the naturality argument.

We can now use $L \dashv R$ without further representative choices. Its Hom bijection sends $h : L\xi' \longrightarrow \eta$ to $R(h)\alpha_{\xi'}$, and its inverse sends $k : \xi' \longrightarrow R\eta$ to $\beta_\eta L(k)$. The two equations just proved say exactly that these transformations are inverse. They also make the construction independent of an accidental presentation of a morphism by a fraction. The result is stronger than the assertion that direct and inverse image have compatible formulas on objects.

Commentary · Context [ms. 147]

Adjunction is the organizing operation

Grothendieck's insistence on constructing both adjunction arrows belongs to the original motivation of derivators. In *Pursuing Stacks*, §69, printed pp. 1–2, he describes the diagram categories as a way to recover structure lost by a single derived category, and identifies the two adjoints to restriction as the operations replacing direct and inverse limits (Grothendieck, 1983). The present calculation supplies one of those adjoints in the categorical model of homotopy types.

Its immediate use is already visible. Once the unit and counit exist, a natural transformation between diagrams of indexing categories can be transported across adjunctions. This produces the comparison maps for Kan extensions, rather than merely plausible isomorphisms between their values. The commentary on Chapter I, manuscript p. 31 ("The base-change map as a mate") explained that formal operation; manuscript pages 138–146 have now constructed the particular adjunction to which it will be applied.

There is a further gain. An adjunction may pass to a reflective subcategory even when its two functors are not constructed there in the same manner. Direct image descends through a localization, while inverse image restricts to local objects. The next argument separates these two mechanisms. That separation will become decisive when the chapter later distinguishes $\mathrm{HOT}^{\mathrm{lc}}_{W}$ from $\mathrm{HOT}^{\mathrm{lc!}}_{W}$.

a) $f_!$ est compatible avec les localiseurs de $\mathcal{A}'$, $\mathcal{A}$ définis par ρ', ρ respectivement, donc définit un foncteur (bien déterminé à isomorphisme unique près)

$$f_!^0 : \mathcal{A}'_0 \longrightarrow \mathcal{A}_0,$$

donnant donc lieu à un diagramme commutatif (à isomorphisme fonctoriel près)

p. 91

(I)
$$\begin{array}{ccc} \mathcal{A}' & \xrightarrow{\rho'} & \mathcal{A}'_0 \\ {\scriptstyle f_!}\downarrow & & \downarrow{\scriptstyle f_!^0} \\ \mathcal{A} & \longrightarrow & \mathcal{A}_0. \end{array}$$

[plutôt $\mathcal{A} \xrightarrow{\rho} \mathcal{A}_0$]

b) 'Dualement', $f^* : \mathcal{A} \longrightarrow \mathcal{A}'$ est compatible avec les sous-catégories pleines images essentielles des foncteurs pleinement fidèles i, i', et induit alors un foncteur, défini à isomorphisme unique près : [page 149]

$$f_0^* : \mathcal{A}_0 \longrightarrow \mathcal{A}'_0,$$

donnant lieu à un carré de foncteurs, commutatif à isomorphisme fonctoriel canonique près

(II)
$$\begin{array}{ccc} \mathcal{A}_0 & \xrightarrow{i} & \mathcal{A} \\ {\scriptstyle f_0^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathcal{A}'_0 & \xrightarrow{i'} & \mathcal{A}'. \end{array}$$

Sous ces conditions, le couple de foncteurs

$$\mathcal{A}_0 \underset{f_0^*}{\overset{f_!^0}{\rightleftarrows}} \mathcal{A}'_0, \qquad [\text{plutôt } \mathcal{A}'_0 \underset{f_0^*}{\overset{f_!^0}{\rightleftarrows}} \mathcal{A}_0,]$$

est un couple de foncteurs adjoints. (NB quand je parle de couples de foncteurs adjoints, celui écrit au-dessus de l'autre est l'adjoint à gauche.)

Démonstration. Soient

$$\eta_0 \in \mathrm{Ob}\mathcal{A}_0, \qquad \xi'_0 \in \mathrm{Ob}\mathcal{A}'_0,$$

il faut définir une bijection bifonctorielle

$$\mathrm{Hom}_{\mathcal{A}_0}(f_!^0(\xi_0'), \eta_0) \simeq \mathrm{Hom}_{\mathcal{A}_0'}(\xi_0', f_0^*(\eta_0))$$

Pour cela, on va écrire ξ_0' sous la forme

$$\xi_0' = \rho'(\xi')\,, \qquad \xi' \in \mathrm{Ob}\mathcal{A}',$$

il suffira d'établir la bijection bifonctorielle en remplaçant ξ_0' par $\rho'(\xi')$, i.e.

$$\mathrm{Hom}_{\mathcal{A}_0}(f_!^0(\rho'(\xi')), \eta_0) \simeq \mathrm{Hom}_{\mathcal{A}_0'}(\rho'(\xi'), f_0^*(\eta_0))$$

(En effet, la formule d'adjonction s'interprète comme un isomorphisme de bifoncteurs, ou, ce qui revient au même, de foncteurs $\mathcal{A}_0' \underset{\psi}{\overset{\varphi}{\rightrightarrows}} \mathcal{A}_0^\wedge$, or p. 92 [page 150]

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(\mathcal{A}_0', \mathcal{A}_0^\wedge) & \longrightarrow & \underline{\mathrm{Hom}}(\mathcal{A}', \mathcal{A}_0^\wedge) \\ \varphi & \longmapsto & \varphi \circ \rho' \end{array}$$

est pleinement fidèle, puisque ρ' est un foncteur de localisation, donc il revient au même de se donner un isomorphisme $\varphi \simeq \psi$, ou $\varphi \circ \rho' \simeq \psi \circ \rho'$.) On doit donc transformer l'expression

$$\begin{array}{ccccc} & \mathrm{Hom}_{\mathcal{A}_0}(\underbrace{f_!^0(\rho'(\xi'))}_{\rho f_!(\xi')}, \eta_0) & \underbrace{\simeq}_{\text{commutativité de (I)}} & \mathrm{Hom}_{\mathcal{A}_0}(\rho f_!(\xi'), \eta_0) & \\ \underbrace{\simeq}_{\text{adjonction } \rho,\, i} & \mathrm{Hom}_{\mathcal{A}}(f_!(\xi'), i(\eta_0)) & \underbrace{\simeq}_{\text{adjonction } f_!,\, f^*} & \mathrm{Hom}_{\mathcal{A}'}(\xi', \underbrace{f^* i(\eta_0)}_{i'(f_0^*(\eta_0))}) & \\ \underbrace{\simeq}_{\text{commutativité de (II)}} & \mathrm{Hom}_{\mathcal{A}'}(\xi', i'(f_0^*(\eta_0))) & \underbrace{\simeq}_{\text{adjonction } \rho',\, i'} & \mathrm{Hom}_{\mathcal{A}_0'}(\underbrace{\rho'(\xi')}_{\xi_0'}, f_0^*(\eta_0)), & \end{array}$$

ce qui est la bijection annoncée.

Je résume l'essentiel des résultats obtenus sur $f_!$, $f_!^{\mathrm{lc}}$:

Théorème 2. *Soit $f : S' \longrightarrow S$ une flèche dans* Cat*, considérons les foncteurs*

$$\mathrm{HOT}_W(S') \underset{f^*}{\overset{f_!}{\rightleftarrows}} \mathrm{HOT}_W(S)$$

$$\mathrm{HOT}_W^{\mathrm{lc}}(S') \underset{f_{\mathrm{lc}}^*}{\overset{f_!^{\mathrm{lc}}}{\rightleftarrows}} \mathrm{HOT}_W^{\mathrm{lc}}(S),$$

où les foncteurs $f_!$, $f_!^{\mathrm{lc}}$ sont décrits dans le théorème 1 *(p. 136). Les morphismes d'adjonction* [page 151]

$$\begin{cases} \mathrm{id} & \longrightarrow & f^* f_! \\ \mathrm{id} & \longleftarrow & f_! f^* \end{cases}$$

sont décrits p. 138 à 140.

Commentary · Exposition [ms. 150]

Passing an adjunction through two reflections

Grothendieck wants the locally constant adjunction without repeating the categorical calculations. The abstract argument is correct, and it is useful to state its hypotheses independently of any identification of local homotopy theories. Suppose

$$\rho : \mathcal{A} \rightleftarrows \mathcal{A}_0 : i, \qquad \rho' : \mathcal{A}' \rightleftarrows \mathcal{A}_0' : i'$$

are adjunctions with i, i' fully faithful, and $L : \mathcal{A}' \rightleftarrows \mathcal{A} : R$ is an adjunction. Let Σ and Σ' be the classes inverted by ρ and ρ'. Assume that $L(\Sigma') \subseteq \Sigma$ and that R carries the essential image of i into that of i'. We then have functors $\bar{L}$ and $\bar{R}$, with natural isomorphisms

$$\bar{L}\rho' \simeq \rho L, \qquad i'\bar{R} \simeq Ri.$$

The desired adjunction follows from the natural chain

$$\begin{aligned} \mathrm{Hom}_{\mathcal{A}_0}(\bar{L}\rho' x', y) &\simeq \mathrm{Hom}_{\mathcal{A}_0}(\rho L x', y) \\ &\simeq \mathrm{Hom}_{\mathcal{A}}(Lx', iy) \\ &\simeq \mathrm{Hom}_{\mathcal{A}'}(x', Riy) \\ &\simeq \mathrm{Hom}_{\mathcal{A}'}(x', i'\bar{R}y) \\ &\simeq \mathrm{Hom}_{\mathcal{A}'_0}(\rho' x', \bar{R}y). \end{aligned}$$

Every object of $\mathcal{A}'_0$ is isomorphic to one of the form $\rho' x'$. The resulting bijection also descends naturally in that object. Precomposition with a localization is fully faithful on functor categories: a naturality square for an inverted arrow is obtained from the original square by inversion, and then naturality holds for every composite of ordinary arrows and formal inverses. This supplies the step which cannot be replaced by essential surjectivity alone. If one wishes to avoid conventions about the notation $\mathcal{A}_0^{\wedge}$ in the text, the relevant object is simply the bifunctor on $(\mathcal{A}'_0)^{\mathrm{op}} \times \mathcal{A}_0$ appearing in this display.

The two compatibility assumptions actually describe the same condition. This standard observation clarifies why Grothendieck calls them dual. An object $z \in \mathcal{A}$ belongs to the reflective subcategory if and only if

$$\mathrm{Hom}(b, z) \longrightarrow \mathrm{Hom}(a, z)$$

is bijective for every $a \longrightarrow b$ in Σ. One direction follows from adjunction. For the converse, apply the condition to the unit $z \longrightarrow i\rho z$. It gives a retraction $i\rho z \longrightarrow z$. Since $i\rho z$ is already local, the same orthogonality shows that the other composite is the identity. Thus the unit is invertible.

If L preserves localization equivalences and y is local, the adjunction identifies

$$\mathrm{Hom}(b', Ry) \longrightarrow \mathrm{Hom}(a', Ry)$$

with $\mathrm{Hom}(Lb', y) \longrightarrow \mathrm{Hom}(La', y)$. This is a bijection for every $a' \longrightarrow b' \in \Sigma'$, so Ry is local. Conversely, if R preserves local objects, the same argument shows that $\rho L(a' \longrightarrow b')$ induces bijections into every object of $\mathcal{A}_0$. Yoneda then makes it invertible. In particular, one need not verify two unrelated conditions in applications. The basic facts about reflections and fully faithful adjoints are recorded in Riehl (n.d., Lemma 4.5.11 and Definition 4.5.12); the displayed Hom calculations give the particular descent argument needed here.

This formal lemma should be carried unchanged into the later revision of the chapter. What changes there is its application. A coarse localization of categories over S does not automatically identify with the full subcategory of locally constant objects. The latter comparison requires the additional hypotheses examined on manuscript pages 171–204. Theorem 2 therefore gives the fine adjunction without that difficulty, and gives the locally constant adjunction whenever the asserted reflections and their compatibility have been established. A correct formal argument does not supply a missing identification of its input categories.

On voudrait expliciter la formule sympathique, obtenue en passant

$$\boxed{f_! f^*(\xi) = \mathrm{hot}_{S,W}(S'/S) \times_S \xi}\ ,$$

pour ceci il faut expliciter le deuxième membre. Notons la

Proposition 4. *Il existe un unique foncteur*

$$(\xi, \eta) \longmapsto \xi \times_S \eta : \mathrm{HOT}_W(S) \times \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S)$$

avec la propriété que pour X, Y variables dans $\underline{\mathrm{Liss}}\, S$, on ait un isomorphisme bifonctoriel en X, Y

p. 93

$$(*) \qquad \mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y) \simeq \mathrm{hot}_{S,W}(X \times_S Y)$$

([72]).

C'est formel, en utilisant seulement que sur $\underline{\mathrm{Liss}}\, S$, le bifoncteur

$$(X, Y) \longrightarrow X \times_S Y$$

est compatible avec les localiseurs $W_S^{\mathrm{d}} \times W_S^{\mathrm{d}}$ et W_S^{d}. Pour le voir, on est ramené à voir que si $X \xrightarrow{u} X'$ dans $\underline{\mathrm{Liss}}\, S$ est dans W_S^{d}, alors il en est de même de $X \times_S Y \xrightarrow{u \times_S \mathrm{id}_Y} X' \times_S Y$, pour tout Y dans $\underline{\mathrm{Liss}}\, S$. En fait, on n'utilise que la lissité de X, X', pas celle de Y, et c'est immédiat, car si $X \longrightarrow X'$ est dans W_S^{d}, $X \times_S Y \longrightarrow X' \times_S Y$ est dans W_Y^{d}, donc aussi dans W_S^{d} (cf. théorème 1 appliqué à $Y \longrightarrow S$). Cette remarque suggère en même temps : [page 152]

Commentary · Exposition [ms. 151]

Deriving the product in both variables

The product appears because Grothendieck wants to understand the endofunctor $f_! f^*$. Before identifying it with multiplication by a relative homotopy type, he has to decide which fibre products survive localization. Proposition 4 makes the safe construction: take both representatives smooth over the base.

For smooth $X, Y \longrightarrow S$, their fibre product is again smooth. If $u : X \longrightarrow X_1$ is a right-fine equivalence between smooth objects, pulling it back along an arbitrary $Y \longrightarrow S$ gives a right-fine equivalence over Y. Forgetting from Y to S preserves that class. Hence

$$X \times_S Y \longrightarrow X_1 \times_S Y$$

is right-fine over S. The same assertion holds in the other variable when the objects varied are smooth. Localizing separately in the two variables therefore produces a bifunctor on $\mathrm{HOT}_W(S)$. The uniqueness asserted in the proposition is uniqueness with its comparison isomorphisms, rather than literal equality of all choices of localized categories.

For arbitrary representatives its value is

$$[X] \times_S [Y] \simeq [\Phi_S X \times_S \Phi_S Y].$$

If X is smooth, $\Phi_S X$ may be replaced by X in this expression. That fact does not yet allow $\Phi_S Y$ to be replaced by Y. The two replacements concern different maps. The first is an equivalence between smooth objects, to which the base-change statement applies. The second would require arbitrary pullback along $X \longrightarrow S$ to preserve all right-fine equivalences. Smoothness of X alone does not say this.

Associativity, symmetry and the unit are transported with their coherence maps. Their diagrams commute already for strict fibre products of categories; localization preserves those equalities. The unit is the class of $S \longrightarrow S$. Equivalently, the diagonal–product adjunction and the terminal-object adjunction descend, so the multiplication is the categorical product in the localized category. In the usual homotopy theory its diagram interpretation is simply the pointwise product of space-valued presheaves.

The boxed formula preceding the proposition must consequently be read provisionally. A formula

$$f_! f^*(\xi) \simeq [S'/S] \times_S \xi$$

for arbitrary f would identify a construction deriving only the varying input with one deriving both factors. Grothendieck notices the discrepancy almost immediately. The following counterexample is therefore part of the proof's development, and the corrected corollary is its actual conclusion. Treating the crossed-out assertion as a theorem would obscure the main mathematical distinction these pages establish.

Corollaire 1. (Faux) *La formule bifonctorielle* $(*)$ *reste valable pour* X *dans* $\underline{\mathrm{Liss}}\, S$, Y *quelconque dans* Cat/S, *et aussi dans le cas inverse.* ([73])

72. **NB** La multiplication $\times_S$ dans $\mathrm{HOT}_W(S)$ est associative, commutative, unitaire *etc., etc.*, puisque c'est le cas dans $\underline{\mathrm{Liss}}\, S$.

73. Cet énoncé est sûrement *faux!* Contrexemple p. 153, cf. corollaire 1 rectifié page 154-156.

Dans le premier cas, considérons

$$Y \xrightarrow{i_Y} \Phi(Y)$$

dans W_S^{d}, donc un isomorphisme dans $(\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}$, on aura donc

$$\begin{array}{ccc} \mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y) & \xrightarrow{\sim} & \mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(\Phi(Y)) \\ & \underbrace{\xrightarrow{\sim}}_{\text{car } X \text{ et } \Phi(Y) \text{ lisses sur } S} & \mathrm{hot}_{S,W}(X \times_S \Phi(Y)), \end{array}$$

or je dis que

$$X \times_S Y \xrightarrow{X \times_S i_Y} X \times_S \Phi(Y)$$

est encore dans W_S^{d}. Il suffit de voir que c'est dans W_X^{d} (cf. théorème 1), et pour ceci, que le changement de base lisse $X \longrightarrow S$ induit $\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/X$ compatible avec les localiseurs W_S^{d}, W_X^{d}. Mais c'est *faux* (prendre $X = \{s\}$).

Montrons que les corollaire est faux. Je vais prendre $Y = \{s\}$, catégorie ponctuelle en s, alors $\Phi(Y)$ est la catégorie fibrée à fibres discrètes définie par le foncteur représentable F [page 153]
défini par s, donc c'est S/F et, en l'occurence S/s:

$$\Phi(\{s\}) \simeq S/s \qquad \text{comme catégorie sur } S.$$

Si p.ex. s est élément final e de S, on trouve

$$\Phi(\underset{S}{e}) \simeq S, \qquad \text{p. 94}$$

donc

$$\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(e) = \mathrm{hot}_{S,W}(X),$$

d'autre part

$$\mathrm{hot}_{S,W}(X \times_S e) = \mathrm{hot}_{S,W}(X_e).$$

Si les deux étaient isomorphes dans $\mathrm{HOT}_W(S)$ par la flèche canonique

$$X_e \longrightarrow X,$$

en appliquant le foncteur $p_!$ (où $p : S \longrightarrow e$), il s'ensuiverait que $X_e \longrightarrow X$ est dans W. Si par [154]
exemple $S = \mathbf{\Delta}^1$, cela impliquerait que pour tout X lisse sur S, $X_1 \longrightarrow X$ est dans W, alors que normalement la lissité implique que $X_0 \longrightarrow X$ est dans W. Donc ça reviendrait à dire que $\underline{\mathrm{Liss}}\ \mathbf{\Delta}^1 = \underline{\mathrm{Liss}}_0 \mathbf{\Delta}^1$, d'où aussitôt $\underline{\mathrm{Liss}}\ S = \underline{\mathrm{Liss}}_0 S$, pour tout S dans Cat, ça se saurait! Le contrexemple le plus trivial est de prendre $X = \{0\} \hookrightarrow S = \mathbf{\Delta}^1$, l'inclusion de l'ouvert 'générique', immersion ouverte, donc lisse, la fibre X_1 est vide!

Corollaire 1 rectifié. *Si X est dans Cat/S, pas nécessairement lisse sur S, alors le foncteur*

$$\mathrm{mult}_{X/S} : Y \longmapsto Y \times_S X : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$$

induit un foncteur

$$\underline{\mathrm{Liss}}\ S \longrightarrow \mathrm{Cat}/S$$

compatible avec les localiseurs W_S^{d}, W_S^{d}, donc induit un foncteur

$$(\underline{\mathrm{Liss}}\ S)(W_S^{\mathrm{d}})^{-1} \longrightarrow (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1},$$

ou encore (compte tenu de l'équivalence induite par l'inclusion

$$(\underline{\mathrm{Liss}}\, S)(W_S^{\mathrm{d}})^{-1} \xrightarrow{\sim} (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} \quad),$$

un foncteur, encore noté $\mathrm{multhot}_{X/S}$,

$$\mathrm{multhot}_{X/S} : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S). \qquad [155]$$

Commentary · Examples [ms. 154]

The false corollary on every fibre

Grothendieck chooses the smallest base on which the two product constructions can differ. We expand his example because it also identifies the endofunctor which replaces the false formula. Let $S = [1] = (0 \longrightarrow 1)$, let $X = \{0\} \longrightarrow S$, and let $Y = \{1\} \longrightarrow S$.

The inclusion of 0 is a discrete fibration: an arrow into 0 can only be its identity. Thus X is smooth. Its comma fibres are

$$0\backslash X = e, \qquad 1\backslash X = \varnothing.$$

For Y both comma fibres are points, and $\Phi_S Y \cong S$. The relative homotopy type $[Y]$ is therefore the terminal object of $\mathrm{HOT}(S)$. It follows that

$$[X] \times_S [Y] \simeq [X].$$

On the other hand, $X \times_S Y = \varnothing$, so

$$[X \times_S Y] = [\varnothing].$$

Evaluation at 0 distinguishes the two objects: it gives a point on the first side and the empty homotopy type on the second. One smooth factor has not made the strict fibre product into the derived product.

There is a useful computation for an arbitrary coefficient. Write a space-valued presheaf on $[1]$ as a map $b : B \longrightarrow A$, with values $F(1) = B$ and $F(0) = A$. It may be represented by a smooth category over S, or calculated in the usual space-valued diagram model. For the inclusion $j_1 : \{1\} \longrightarrow S$, restriction gives B, and left Kan extension returns the constant diagram on B. Thus

$$j_{1!}j_1^* F = (B \xrightarrow{\mathrm{id}} B), \qquad [Y] \times_S F = F = (B \xrightarrow{b} A).$$

The counit has components id_B at 1 and b at 0. The candidate product formula is an equivalence precisely when b is a weak equivalence. For $F = [X]$, this is the map $\varnothing \longrightarrow e$.

For comparison, the inclusion $j_0 : \{0\} \longrightarrow S$ gives

$$j_{0!}j_0^* F = (\varnothing \longrightarrow A) \simeq [X] \times_S F.$$

The difference between the endpoints has a categorical explanation. The object 0 defines a sieve, so left Kan extension of presheaves from it is extension by the empty object. The object 1 instead has an incoming arrow from outside its image, and that arrow transports B into the value at 0.

This also proves a sharper warning contained in the corrected corollary. The map $Y \longrightarrow S$ is a right-fine equivalence, so $[Y] \simeq [S]$. Nevertheless the two fixed-factor endofunctors are different: multiplication by the category S/S is the identity, whereas $\mathrm{multhot}_{Y/S}$ sends $B \longrightarrow A$ to the constant diagram on B. Thus $\mathrm{multhot}_{Y/S}$ depends on the chosen category Y over S, not only on its class in $\mathrm{HOT}(S)$.

Ce foncteur satisfait donc à la relation

$$\mathrm{multhot}_{X/S}(\mathrm{hot}_{S,W}(Y)) \underbrace{\simeq}_{\text{isomorphisme canonique}} \mathrm{hot}_{S,W}(X \times_S Y)$$

pour Y *lisse sur* S, et pour Y quelconque dans Cat/S,

$$\mathrm{multhot}_{X/S}(\mathrm{hot}_{S,W}(Y)) \underbrace{\simeq}_{\text{isomorphisme canonique}} \mathrm{hot}_{S,W}(X \times_S \Phi_S(Y)).$$

p. 95

Quand X est lui-même lisse sur S, ce foncteur n'est autre que la multiplication par $\mathrm{hot}_{S,W}(X)$:

$$\mathrm{multhot}_{X/S} = \mathrm{mult}_{\mathrm{hot}_{S,W}(X)} \qquad \text{si } X \textit{ lisse sur } S.$$

Mais sa valeur pour $\mathrm{hot}_{S,W}(Y)$, quand Y n'est *pas* lisse sur S, ne doit pas être confondue avec $\mathrm{hot}_{S,W}(X \times_S Y)$.

Ainsi, si X et Y sont dans Cat/S, considérons

$$X \times_S Y \qquad \text{et} \qquad \boxed{\mathrm{hot}_{S,W}(X \times_S Y).}$$

a) Si ni X ni Y ne sont supposés lisses sur S, cet élément 'n'a aucune interprétation naturelle en termes d'opérations sur les catégories $\mathrm{HOT}_W(S)$' ([74]).

[156]

Commentary · Development (commentators') [ms. 155]

A precise boundary for the product formula on finite posets

The walking-arrow example suggests asking which inclusions permit the formula for every coefficient. In a finite poset the answer can be read directly from the order. The following elementary development extends Grothendieck's example to a complete criterion in that setting. We use ordinary homotopy types throughout, so $W = W_\infty$.

Proposition (full subposets and the universal product formula). [ours] Let S be a finite poset, regarded as a category, and let $u : U \hookrightarrow S$ be the inclusion of a full subposet. In space-valued presheaves put $P_U = u_!1$, where 1 is the terminal presheaf on U. The canonical map

$$\theta_F : u_!u^*F \longrightarrow P_U \times F$$

is an equivalence for every $F : S^{\mathrm{op}} \longrightarrow \mathsf{Spaces}$ if and only if U is a lower ideal: whenever $s \leq t$ and $t \in U$, one has $s \in U$.

Here $u_!$ means homotopy left Kan extension along u^{op}. The two components defining θ_F are $u_!u^*F \longrightarrow u_!1$, induced by the terminal map, and the counit $u_!u^*F \longrightarrow F$. This specifies the comparison whose invertibility is claimed.

Proof. For $s \in S$, write

$$U_{\geq s} = \{t \in U \mid s \leq t\}.$$

The pointwise formula gives

$$(u_!G)(s) \simeq \operatorname*{hocolim}_{(U_{\geq s})^{\mathrm{op}}} G(t), \qquad P_U(s) \simeq B(U_{\geq s}),$$

where B denotes the nerve's homotopy type. Replacing a category by its opposite does not change this homotopy type.

Suppose first that U is a lower ideal. If $s \notin U$, the category $U_{\geq s}$ is empty, so both $(u_!u^*F)(s)$ and $(P_U \times F)(s)$ are empty. If $s \in U$, the poset $U_{\geq s}$ has initial object s, and its opposite has terminal object s. The homotopy colimit is therefore $F(s)$; the same argument with constant value e gives $P_U(s) \simeq e$. Under these identifications $\theta_F(s)$ is the identity of $F(s)$. This proves the sufficiency, including the comparison map, at every object of S.

For necessity, suppose that $s \notin U$ but $s \leq t_0$ for some $t_0 \in U$. Take the representable discrete presheaf

$$F = h_s, \qquad h_s(t) = \mathrm{Hom}_S(t, s).$$

For every $t \in U_{\geq s}$, one has $s \leq t$ and $t \neq s$. Antisymmetry then excludes $t \leq s$, so $h_s(t) = \varnothing$. The diagram computing $(u_!u^*h_s)(s)$ is constantly empty, and its homotopy colimit is empty. By contrast,

$$(P_U \times h_s)(s) \simeq B(U_{\geq s}) \times e$$

is nonempty, since t_0 is an object of $U_{\geq s}$. Hence $\theta_{h_s}(s)$ is not an equivalence. This contradicts the asserted universal formula and proves necessity. □

The test uses only a representable presheaf. Consequently, in this finite-poset setting the universal formula can already be tested on the collection of representables h_s. For a point inclusion $\{t\} \hookrightarrow S$, it holds for every coefficient exactly when t is minimal. It holds for every point inclusion exactly when S is discrete.

There is also a distinction from absolute weak equivalence. In the chain $0 < 1 < 2$, take $U = \{1, 2\}$. Both U and S have contractible nerves, and $P_U \simeq 1$: each nonempty upper intersection $U_{\geq s}$ has a least element. Nevertheless u^*h_0 is empty, so $u_!u^*h_0$ is empty while $P_U \times h_0 \simeq h_0$ is not. Equivalence of the total homotopy types of the indexing categories does not remove the variation of an arbitrary coefficient diagram.

The positive sieve calculation is standard Kan-extension calculus; compare Groth (2013, Proposition 1.23, with the indexing categories replaced by their opposites). We checked the pointwise and sieve formulations in the cited derivator literature and searched for the

74. Sauf dans le cas où l'un des deux soit dans Fibhot S, cf. annotation page suivante.

> corresponding finite-poset converse under projection formula and Frobenius reciprocity. We did not locate this exact elementary criterion there. The proof is included as a continuation of the example, without a claim of priority or of a new general projection-formula theorem.

b) Si X et Y sont lisses sur S, cet élément s'interprète comme

$$\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y)$$

([75]).

c) Si l'un des deux X, Y est lisse sur S, sans rien supposer sur l'autre, on a une interprétation en termes de mult_X (si Y lisse), ou mult_Y (si X lisse), notamment

$$\begin{aligned} \mathrm{hot}_{S,W}(X \times_S Y) &\simeq \mathrm{multhot}_X(\mathrm{hot}_{S,W}(Y)) \quad \text{si } Y \text{ lisse sur } S \\ &\simeq \mathrm{multhot}_Y(\mathrm{hot}_{S,W}(X)) \quad \text{si } X \text{ lisse sur } S. \end{aligned}$$

Corollaire 2. *Soit $f : S' \longrightarrow S$ une flèche dans* Cat*, d'où $f_!$, f^* entre* $\mathrm{HOT}_W(S')$ *et* $\mathrm{HOT}_W(S)$*. On a la relation*

$$f_! f^*(\xi) \simeq \mathrm{multhot}_{S'}(\xi),$$

donc si S' est lisse sur S,

$$f_! f^*(\xi) \simeq \mathrm{hot}_{S,W}(S') \times_S \xi.$$

Nous sommes maintenant en mesure de vérifier les axiomes Der 4′b, Der 5′ relatifs aux images directes (I, p. 67, 68), et de plus Der 3b (I, p. 111). L'axiome Der 4′ n'est autre que l'existence des adjoints à gauche $f_!$ des f^*. L'axiome Der 5′ permet le 'calcul' des fibres d'un objet $f_!(\xi)$. Il dit qu'une certaine flèche du changement de base, relatif au diagramme commutatif dans Cat

$$\begin{array}{ccc} S' & \xleftarrow{q} & s\backslash S' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & e \simeq \{s\} \end{array} \qquad (s \in \mathrm{Ob}\, S),$$

p. 96 [157]

savoir

$$p^*(f_!(\xi')) \longrightarrow f'_!(q^*(\xi'))$$

est un isomorphisme (pour $\xi' \in \mathbf{D}(S')$). Soit donc

$$\xi' \in \mathrm{Ob}\,\mathrm{HOT}_W(S'), \qquad \xi' = \mathrm{hot}_{S',W}(X')$$

avec X' dans Cat/S'. Comme

$$f_!(\xi') = \mathrm{hot}_{S,W}(X'),$$

et que X' n'a aucune raison d'être lisse sur S (si on ne suppose pas S' lui-même lisse sur S, auquel cas il pourrait être utile de prendre X' lisse sur S', donc aussi sur S), pour calculer $p^*(f_!(\xi'))$ il n'est pas question de prendre la fibre X'_s, mais il faut prendre $s\backslash X' = \Phi_S(X')_s$:

$$p^*(f_!(\xi')) \simeq \mathrm{hot}_W(s\backslash X').$$

$$\begin{array}{ccc} X' & \longleftarrow & s\backslash X' \\ \downarrow & & \downarrow \\ S' & \xleftarrow{q} & s\backslash S' \\ \downarrow & & \downarrow \\ S & \longleftarrow & s\backslash S \end{array}$$

75. Mais aussi si l'*un* des deux éléments est dans $\underline{\mathrm{Fibhot}}\, S$ (et moyennant l'axiome L6 bis sur W).

Commentary · Exposition [ms. 156]

Fixed-factor multiplication is a direct-image composite

The corrected corollary does more than impose an extra hypothesis on the false formula. It isolates an operation which remains available for every category $x : X \longrightarrow S$. Its varying argument is derived, while the category X is held fixed:

$$M_X([Y]) = [X \times_S \Phi_S Y].$$

The base-change assertion used in Proposition 4 makes this independent of the chosen smooth representative of $[Y]$. The example just calculated shows why it is not generally independent of the representative of $[X]$.

This operation has a direct adjunction-theoretic description:

$$M_X \simeq x_! x^*.$$

Indeed, for smooth Y the inverse image $x^*[Y]$ is represented over X by $Y \times_S X$. Direct image along x regards the same total category as lying over S. Thus both sides give $[Y \times_S X]$, naturally in Y. Since smooth objects present the localization, the identification holds on all of $\mathrm{HOT}_W(S)$. Corollary 2 is the special case $X = S'$.

There is a canonical comparison with the derived product. The map $i_X : X \longrightarrow \Phi_S X$ induces, for smooth Y,

$$[X \times_S Y] \longrightarrow [\Phi_S X \times_S Y] \simeq [X] \times_S [Y].$$

The right side derives both inputs; the left side has only derived the varying input. If X is smooth, the comparison is invertible, because it is the pullback of a right-fine equivalence between smooth representatives. Without that hypothesis the comparison can fail, as it does for $X = \{1\}$ over the walking arrow. The later discussion of homotopical fibrations and properness will supply further sufficient hypotheses. Nothing in the present corollary establishes those extensions in advance.

The identity $M_X \simeq x_! x^*$ also supplies structure which is easy to miss if one sees only the fibre-product formula. It is the underlying endofunctor of a comonad. Its counit is the projection map already used in the adjunction proof, and its comultiplication is

$$x_! x^* \xrightarrow{x_! \alpha x^*} x_! x^* x_! x^*.$$

The comonad laws follow from the triangle identities. When X is smooth, this is multiplication by $[X]$ with the diagonal coalgebra structure coming from the cartesian product. This interpretation explains the order of the argument: the adjunction is established first, and the formula for its composite is then refined according to the geometry of the fixed factor.

Thus the three cases listed on manuscript pages 155–156 have different information content. With two smooth factors, one obtains the product of two objects of the homotopy category. With only the varying factor smooth, one obtains the value of M_X. For two arbitrary strict representatives, their strict fibre product need not be determined by either localized class. These are statements about the operations available from particular representatives, rather than about different possible definitions of the same product.

Considérons le diagramme de changement de base par $s\backslash S \longrightarrow S$ ci-contre `[i.e. ci-dessus]`, à carrés cartésiens. On voit d'abord que $q^*(\xi')$ n'est autre que la classe dans $\mathrm{HOT}_W(s\backslash S')$ de l'image inverse de X' par le changement de base $s\backslash S' \longrightarrow S'$. Pour l'assurer, il suffirait de prendre X' lisse sur S' (ce qui est licite) - alors la description de l'image inverse [158]
est OK pour tout changement de base. Mais on a vu que dans le cas d'un co-isomorphisme local comme $s\backslash S' \longrightarrow S'$, il n'y a pas même besoin de cet artifice - la formule est OK sans restriction de lissité. Ainsi

$$q^*(\xi') = \mathrm{hot}_{s\backslash S', W}(s\backslash X'),$$

d'où p. 97

$$f'_!(q^*(\xi')) \simeq \mathrm{hot}_{S,W}(s\backslash X') = \mathrm{hot}_W(s\backslash X').$$

Donc on a bien trouvé

$$p^* f_! \simeq f'_! q^*.$$

Commentary · Exposition [ms. 157]

The comma square and its actual comparison map

Grothendieck now verifies the pointwise formula for direct image. This is where the construction becomes a derivator calculation: evaluation at an object of the base is expressed by a homotopy colimit over a category of arrows. The comma replacement makes the formula almost tautological, but the comparison map and the variance deserve to be stated.

Fix $s \in S$ and write $C = s\backslash S'$. Its objects are pairs $(s', a : s \longrightarrow f(s'))$, and a morphism from (s', a) to (t', b) is $v : s' \longrightarrow t'$ such that $f(v)a = b$. Let $q : C \longrightarrow S'$ forget a, let $r : C \longrightarrow e$, and let $p : e \longrightarrow S$ select s. The square drawn in the text is not strictly commutative. It carries the canonical natural transformation

$$\tau : pr \Longrightarrow fq, \qquad \tau_{(s',a)} = a.$$

The two cartesian squares in the larger diagram involving $s\backslash S$ are, by contrast, strict pullback squares. These two kinds of square should not be identified.

With the presheaf convention $\mathrm{HOT}(S) = \mathrm{Ho}(\mathsf{Spaces}^{S^{\mathrm{op}}})$, τ gives $q^*f^* \longrightarrow r^*p^*$. The standard left-adjoint mate is

$$r_!q^* \xrightarrow{r_!q^*\alpha} r_!q^*f^*f_! \longrightarrow r_!r^*p^*f_! \xrightarrow{\beta} p^*f_!.$$

It points in the reverse direction to the arrow displayed on manuscript page 157. Once exactness is proved, that displayed arrow can be read as its inverse. The direction is fixed by the units, counits and structural 2-cell; it is not a further hypothesis on W. For the general mate construction, see Riehl (n.d., Proposition 4.3.8).

Let $X' \longrightarrow S'$ represent ξ'. The right side $p^*f_!\xi'$ is represented by $s\backslash X'$. On the left, q is a discrete opfibration: an arrow $v : s' \longrightarrow t'$ has the unique lift taking (s', a) to $(t', f(v)a)$. The pullback rule for these co-isomorphisms permits us to compute $q^*\xi'$ by strict pullback even for arbitrary X'. That pullback has objects $(a : s \longrightarrow f(s'), x')$ with $p'(x') = s'$, and is canonically isomorphic to $s\backslash X'$. Applying $r_!$ then forgets its map to C, so

$$r_!q^*\xi' \simeq \mathrm{hot}_W(s\backslash X') \simeq p^*f_!\xi'.$$

These identifications also calculate the mate. An object represented by (a, x') is carried by the unit into the comma replacement with its identity arrow; the structural transformation then supplies a; the counit forgets the redundant intermediate object. The resulting object is precisely (a, x') in $s\backslash X'$, and the same description sends a morphism to itself. Thus the mate is the identity under these categorical identifications. This proves the canonical comparison is invertible, rather than merely exhibiting an unspecified isomorphism between its source and target.

In modern diagram notation the formula reads

$$(f_!F)(s) \simeq \operatorname*{hocolim}_{(s\backslash f)^{\mathrm{op}}} F(s').$$

The opposite is forced by the presheaf convention. Modern presentations using covariant diagrams write the corresponding over-category formula instead. The pointwise axiom and its comma-square formulation appear in Groth (2013, Definition 1.5 and Proposition 1.26); the calculation here supplies them concretely for the categorical model.

On peut maintenant prouver aussi Der 3b. On a vu qu'une fois qu'on dispose des adjoints à gauche $f_!$ des f^*, cet axiome s'énonce sous *forme homologique* ainsi : Si

$$f : S' \longrightarrow S$$

est dans W, il faut prouver que pour tout

$$\xi \in \mathrm{HOT}_W(e) \overset{\text{déf}}{=} \mathrm{Hot}_W,$$

la flèche correspondante

$$(*) \qquad p'_!p'^*(\xi) \longrightarrow p_!p^*(\xi)$$

est [un] isomorphisme dans Hot_W, où $p : S \longrightarrow e$, $p' : S' \longrightarrow e$ sont les flèches structurales. Or comme tout objet de Cat est lisse sur e, le corollaire 2 précédent permet d'identifier cette flèche $(*)$ à

$$\eta' \times_S \xi \longrightarrow \eta \times_S \xi$$

déduite de [159]

$$\mathrm{hot}_W(f) : \underbrace{\eta'}_{\overset{\text{déf}}{=} \mathrm{hot}_W(S')} \longrightarrow \underbrace{\eta}_{\overset{\text{déf}}{=} \mathrm{hot}_W(S)} .$$

Comme celle-ci est [un] isomorphisme par hypothèse, on gagne.

Commentary · Exposition [ms. 158]

Constant coefficients detect the total homotopy type

Having obtained direct images, Grothendieck can express an equivalence of indexing categories by its effect on homology with constant coefficients. This assertion concerns the total homotopy type of the base. It should be kept separate from equivalence of all categories of diagrams over the base.

Let $p : S \longrightarrow e$ and represent $\xi \in \mathrm{Hot}_W$ by a category T. Since every category is smooth over a point, the preceding product computation gives

$$p_! p^*(\xi) \simeq \mathrm{hot}_W(S) \times \xi.$$

For $f : S' \longrightarrow S$, with $p' = pf$, the induced comparison is therefore

$$\mathrm{hot}_W(S') \times \xi \longrightarrow \mathrm{hot}_W(S) \times \xi.$$

The subscript S on the multiplication signs in the edition's display on page 158 is a slip: both objects belong to $\mathrm{Hot}_W = \mathrm{HOT}_W(e)$, so this is the product over e.

If $f \in W$, its image in Hot_W is invertible; applying the product functor proves the desired comparison is invertible for every ξ. Conversely, if every such comparison is invertible, take $\xi = e$, the product unit. Then $\mathrm{hot}_W(f)$ itself is invertible. Thus the precise condition detected is membership in the strong saturation $\overline{W}$, the class of arrows inverted by $\mathrm{Cat} \longrightarrow \mathrm{Hot}_W$. When W is already strongly saturated this is simply $f \in W$. The later restatement on manuscript page 204 makes this distinction explicit.

The walking arrow again illustrates the scope. The inclusion $\{1\} \longrightarrow [1]$ is a weak equivalence of nerves. It therefore gives the same homology of constant coefficients as $[1] \longrightarrow e$. Restriction of arbitrary presheaves along that inclusion, however, remembers only B from a diagram $B \longrightarrow A$. It cannot be an equivalence of all diagram categories: the diagrams $\varnothing \longrightarrow e$ and $\varnothing \longrightarrow \varnothing$ have the same restriction but are distinguished at 0.

The stronger statement about *locally constant* diagrams is different. On [1], local constancy requires $B \longrightarrow A$ to be an equivalence, so evaluation at either endpoint loses no information up to coherent equivalence. This small example explains why the next page strengthens the conclusion only on $\mathrm{HOT}^{\mathrm{lc}}_W$. It also explains why the axiom controlling the relation between coarse equivalence and fibrewise equivalence will enter that proof. The elementary product argument for constant coefficients does not itself establish that stronger assertion.

Donc on a prouvé

Théorème 3. *Le prédérivateur* HOT *satisfait aux axiomes* Der 1, Der 2 ([76]) *(pour mémoire), et* Der 3b ([77]), Der $4'$, Der $5'$.

On a envie de prouver aussi les variantes renforcées de Der 3, la forme la plus forte étant celle-ci (Der $3b'_2$) : si $f \in W$, alors f^* induit une *équivalence* de catégories

$$f_0^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

76. *pas prouvé*, le 'résultat' auquel je pensais est canulé, cf. complément proposition 5, page 164.

77. Et même la variante la plus forte Der $3b'_2$ de Der 3, cf. démonstration page suivante. **NB** Pour Der $3b'_2$, on a utilisé l'axiome L6 sur W - et c'est essentiel. On voit que la validité de Der $3b'_2$ implique L6, et même la forme plus forte L6 bis, et même si on suppose seulement Der $3b'_1$, i.e. f_0^* pleinement fidèle.

La pleine fidélité du foncteur f_0^* doit résulter de la variante naturelle Der $3b''_{coh}$ de Der 3b, une fois qu'on aura développé le formalisme des Hom internes dans les $\mathrm{HOT}_W(S)$. Mais pour donner un sens à Der $3b''_{coh}$, on a besoin des foncteurs *cohomologie,* i.e. de p_* et p'_* (au lieu de $p_!$, $p'_!$) pour HOT, donc c'est encore prématuré pour l'instant. La construction des f_* et celle des Hom internes se fera d'ailleurs sûrement ± conjointement.

Mais je m'aperçois à l'instant qu'on trouve quasiment 'à l'œil' le fait que f_0^* soit bien une *équivalence* de catégories, et que $f_!^0$ soit l'équivalence quasi-inverse, en utilisant l'adjonction des foncteurs $f_!^0$, f_0^*, et se rappelant que pour prouver que ce sont des équivalences quasi-inverses l'une de l'autre, il revient au même de montrer que les morphismes d'adjonction p. 98 [160]

$$\begin{array}{ccc} \mathrm{id} & \xrightarrow{\alpha_0} & f_0^* f_!^0 \\ \mathrm{id} & \xleftarrow{\beta_0} & f_!^0 f_0^* \end{array}$$

sont [des] isomorphismes. Je vais essayer de décrire ces flèches d'adjonction dans le cas général (sans supposer $f \in W$), puis de montrer qu'elles sont dans W si f l'est.

Commentary · Present state [ms. 159]

What Theorem 3 establishes at this point

Grothendieck has proved the left-adjoint and pointwise formulas, and has checked homological invariance for constant coefficients. These are substantial conclusions. The annotations to Theorem 3 also record two separate qualifications which must remain part of its statement.

First, the claim of Der 2 is withdrawn in footnote 76. A family of evaluations is conservative if a morphism in the localized diagram category is invertible whenever all its evaluations are invertible. Defining weak equivalences objectwise does not, by itself, prove this statement for arbitrary morphisms of a localization. One still has to control how such morphisms are represented, or compare the localization with a model in which objectwise detection can be proved. The supplement on manuscript page 164 belongs to that unfinished part of the argument. It would be inaccurate either to count the parenthetical "pour mémoire" as a proof or to infer that conservativity is false in ordinary homotopy theory.

For the standard model the question is settled. Cisinski (2003, Theorem 6.11, including its proof of Der 2) constructs homotopy right Kan extensions for model categories with small limits and proves the corresponding derivator axioms; his Proposition 6.13 constructs the space-valued external Hom. For combinatorial model categories, Groth (2013, Proposition 1.30) gives a strong derivator. These results justify the usual space-valued diagram interpretation used in the examples above.

For every basic localizer, Der 2 now follows from Cisinski (2006a, Propositions 3.1.30 and 4.2.7): conservativity for presheaf localizers transfers through a diagramwise equivalence using Δ (4.1.29). Thus the revised relative $\mathrm{W}(1,2,3)$ suffice, without accessibility or $\mathrm{W}(4)$. This later repair preserves the historical withdrawal; it makes no assertion for arbitrary weak basic localizers.

Second, the strengthened assertion that a W-equivalence of bases induces an equivalence of locally constant categories uses additional control of fibres. The annotation names the earlier axiom L6, and the following pages examine exactly where it is needed. For ordinary weak equivalences, homotopy invariance of locally constant model-category diagrams is proved by Cisinski (2009, Theorem 1.3).

The final paragraph about constructing f_* together with internal Hom is therefore a programme at this place in the text. The adjunction proved above supplies $f_!$, but it supplies no right adjoint to f^* by formal symmetry: taking opposites of indexing categories is not the same as changing a left adjoint into a right adjoint in the same coefficient theory. The later model-category constructions give both operations in their stated settings. The localizer-by-localizer analysis in the remainder of this chapter remains necessary to determine what follows from Grothendieck's chosen axioms alone.

[1) Soit] $\xi = \mathrm{hot}^{\mathrm{lc}}_{S,W}(X)$ dans $\mathrm{HOT}_W(S)$. On peut supposer $X \in \mathrm{Ob}\,\underline{\mathrm{Parf}}\,S$. Comme ['Comme' barré] X lisse sur S, on a

$$f_0^*(\xi) = \mathrm{hot}^{\mathrm{lc}}_{S',W}(X \times_S S'), \qquad f_!^0 f_0^*(\xi) = \mathrm{hot}^{\mathrm{lc}}_{S,W}(X \times_S S').$$

Donc

$$\beta_{0,\xi} : f_!^0 f_0^*(\xi) \longrightarrow \xi$$

est une flèche

$$X \times_S S' \longrightarrow X \qquad \text{dans } (\mathrm{Cat}/S)(W_S)^{-1}$$

et je présume que c'est pr_1 ([78]). Est-ce dans W si $f : S' \longrightarrow S$ l'est? Oui (car X est parfait
sur S), sous réserve que W satisfait l'axiome L6 (cf. page 43). [161]

2) Soit

$$\xi' = \mathrm{hot}^{\mathrm{lc}}_{S',W}(X') \qquad \text{dans } \mathrm{HOT}_W(S'),$$

on a

$$f^0_!(\xi') = \mathrm{hot}^{\mathrm{lc}}_{S,W}(X'),$$

mais pour expliciter f_0^* de cet élément, comme X' n'a aucune raison d'être un Hot-fibré sur S (auquel cas il suffirait de prendre $f^*(X') \overset{\text{déf}}{=} X \times_S S'$), il faut d'abord prendre $C_S(X')$, puis appliquer f^* :

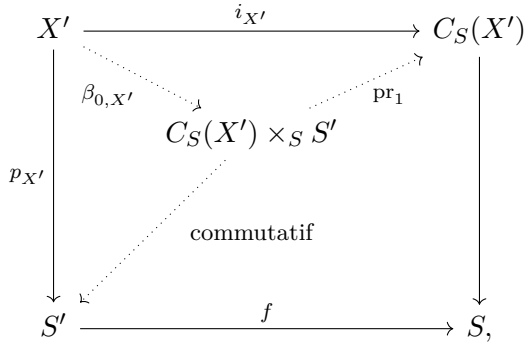
$$f_0^* f^0_!(X') = \mathrm{hot}^{\mathrm{lc}}_{S',W}(C_S(X') \times_S S').$$

Il faut donc définir

$$X' \xrightarrow{\beta_{0,X'}} C_S(X') \times_S S' \qquad \text{dans } (\mathrm{Cat}/S')W^{-1}_{S'}.$$

Or on trouve une flèche canonique dans Cat/S', déduite du S-morphisme $i_{X'} : X' \longrightarrow C_S(X')$, i.e. du carré commutatif en traits pleins

p. 99



comme l'unique flèche pointillée $\beta_{0,X'}$ qui soit un S'-morphisme (triangle gauche commutatif) et qui rende commutatif le triangle supérieur.

Si $f \in W$, alors pr_1, déduit de f par changement de base $C_S(X') \longrightarrow S$, qui est parfait, est dans W par L6, on sait qu'il en est de même de $i_{X'}$, donc aussi de $\beta_{0,X'}$, on gagne.

Pour terminer les propriétés générales des foncteurs $f_!$ pour HOT_W, je voudrais donner les énoncés raisonnables de changement de base. Ce serait un [page 162]

Commentary · Exposition [ms. 161]

When a change of base becomes an equivalence

Grothendieck now wants to know how much of the base a locally constant coefficient can remember. The adjunction is already available; the task is to determine when its two comparison maps become invertible. This is a stronger assertion than the agreement of cohomology with constant coefficients discussed just before it. A locally constant family has transport, and changing the base must preserve the whole family if restriction is to be an equivalence.

Write $f : S' \to S$, and retain the hypotheses under which the locally constant adjunction was constructed. For a perfect representative $X \to S$, the counit is the projection

$$\varepsilon_X : X \times_S S' \longrightarrow X.$$

78. **NB** On voit que l'axiome L6 est ± *nécessaire* pour que le résultat sur f_0^*, $f^0_!$ soit valable.

The notation β_0 used for this map agrees with the preceding adjunction. To prove that ε_X is a W-equivalence from the hypothesis $f \in W$, one must know that this particular change of base preserves W. This is the role of L6. The fact that a product with a fixed category preserves W is insufficient: $X \to S$ need not be a product projection.

The unit has a different presentation because an arbitrary $X' \to S'$ need not remain a good representative after composition with f. Put $P = C_S(X')$. The structural map $P \to S$ is perfect, and the natural map $j : X' \to P$ is a total W-equivalence. The map

$$a : X' \longrightarrow P \times_S S', \qquad \mathrm{pr}_1\, a = j,$$

is the unit on the chosen representative. If $f \in W$, L6 gives $\mathrm{pr}_1 \in W$, and two-out-of-three gives $a \in W$. Thus both the unit and the counit are inverted. The triangle identities then make $f^0_!$ and f^*_0 mutually quasi-inverse equivalences. The label $\beta_{0,X'}$ in the edition is a slip: this second comparison is $\alpha_{0,X'}$. The same pages twice place $\mathrm{hot}^{\mathrm{lc}}$ in an ambient category without the superscript lc; the locally constant category is intended.

The footnote about necessity also has a precise reading. Suppose that the adjunction on locally constant objects is an equivalence for every $f \in W$. Its counit then shows that $X \times_S S' \to X$ is inverted by the relevant localization for every perfect $X \to S$. Applying the total-space functor shows that it is inverted in Hot_W. If W is strongly saturated, it belongs to W. Consequently the asserted invariance forces the perfect-pullback stability used in the proof. Without strong saturation the same argument gives membership in the strong saturation of W, which explains why the manuscript cautiously writes "± nécessaire".

This argument will remain useful after the axioms are reorganized on manuscript p. 171. It identifies exactly the property needed at this point, without treating an earlier choice of notation for that property as permanent. It also distinguishes two operations which the succeeding pages repeatedly separate: replacing the total category by a better object over S, and testing whether that replacement is still an equivalence on each fibre.

Théorème 4. *Soit*

$$\begin{array}{ccc} X & \xleftarrow{\ q\ } & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{\ p\ } & S' \end{array}$$

un diagramme cartésien dans Cat*, considérons la flèche de changement de base (dans* $\underline{\mathrm{Hom}}(\mathrm{HOT}_W(X), \mathrm{HOT}_W(S'))$*) :*

$$p^* f_! \longleftarrow f'_! q^*.$$

C'est un isomorphisme dans chacun des deux cas suivants.

a) f est lisse.

b) p est propre.

Démonstration. *Cas a).* On teste avec ξ dans $\mathrm{HOT}_W(X)$, on peut supposer ξ de la forme

$$\xi = \mathrm{hot}_{X,W}(Z) \qquad \text{avec } Z \in \mathrm{Ob}\,\underline{\mathrm{Liss}}\, X.$$

Notons que Z est alors aussi lisse sur S, puisque f est lisse.

$$f_!(\xi) = \mathrm{hot}_{S,W}(Z), \qquad p^* f_!(\xi) = \mathrm{hot}_{S',W}(\underbrace{Z \times_S S'}_{Z',\ \text{cf. diagramme}})$$

$$q^*(\xi) = \mathrm{hot}_{X',W}(Z'), \qquad f'_! q^*(\xi) = \mathrm{hot}_{S',W}(Z'),$$

OK!

$$\begin{array}{ccc} Z & \longleftarrow & Z' \\ \downarrow & & \downarrow \\ X & \xleftarrow[q]{} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow[p]{} & S' \end{array}$$

Cas b). Il faut admettre que le foncteur changement de base par p : [page 163]

$$Z \longmapsto Z \times_S S' : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$$

est compatible avec les localiseurs W_S^{d}, $W_{S'}^{\mathrm{d}}$ (cf. plus bas) - c'est ça, formellemet, la propriété pertinente pour p ([79]). Il en résultera que

$$p^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$$

se calcule, pour un élément dans $\mathrm{HOT}_W(S)$ donné par un Z *quelconque* dans Cat/S (pas nécessairement lisse sur S), par

$$p^*(\mathrm{hot}_{S,W}(Z)) = \mathrm{hot}_{S',W}(Z \times_S S').$$

Partons alors, comme dans a), d'un élément ξ de $\mathrm{HOT}_W(X)$ donné par un Z de Cat/X (inutile cette fois de supposer Z lisse), le calcul immédiat précédent établit encore la formule de commutation.

Mais il ne semble pas que la propriété de compatibilité de $X \longmapsto X \times_S S'$, avec l'équivalence colocale sur S et sur S', résulte de façon ± évidente de la condition que $p : S' \longrightarrow S$ soit propre ([80]). J'ai envie de le déduire de fourbis généraux sur le formalisme des dérivateurs, à savoir du théorème de changement de base des $f_!^{\mathbf{D}}$ pour un morphisme $\mathbf{D}$-propre.

Sous forme duale d'un théorème de changement de base pour $f_*^{\mathbf{D}}$, par changement de [page 164] base $\mathbf{D}$-lisse, c'est fait dans I, p. 56 à 58, et on doit pouvoir l'appliquer ici. Notons que dans la démonstration du théorème loc. cit., on n'utilise que L4 [plutôt Der 4], i.e. l'existence des $f_*^{\mathbf{D}}$, à l'exclusion des $f_!^{\mathbf{D}}$ (L4′ [plutôt Der 4′]), plus L5 [plutôt Der 5] bien sûr, donc la forme duale ne fait appel à l'existence des $f_*^{\mathbf{D}}$, seulement des $f_!^{\mathbf{D}}$. On a donc une bonne chance de pouvoir l'appliquer à HOT_W (où je n'ai pas encore établi l'existence des f_*^{HOT}).

Commentary · Exposition [ms. 163]

The canonical map in the base-change theorem

The purpose of Theorem 4 is to turn the concrete model for $f_!$ into a calculational rule. A pullback square of categories gives two ways of transporting a coefficient from X to S'. An isomorphism between their values would not by itself suffice: one needs the particular comparison determined by the adjunction, since subsequent compositions of squares use its compatibility with pasting.

For the square in the text, the comparison is the composite

$$f'_! q^* \xrightarrow{f'_! q^* \eta_f} f'_! q^* f^* f_! \cong f'_! f'^* p^* f_! \xrightarrow{\varepsilon_{f'} p^* f_!} p^* f_!.$$

Here $q^* f^* \cong f'^* p^*$ is induced by the equality $fq = pf'$. This formula fixes the orientation. In the smooth case choose $Z \to X$ smooth. Then $Z \to S$ is smooth, and all inverse images in this expression can be computed literally. The unit introduces the relevant pullback and the counit projects from it; their composite is the canonical identification between the two copies of $Z \times_S S'$. Thus the calculation on p. 162 proves that the canonical map is invertible, and not merely that its source and target happen to be isomorphic.

79. Ça doit être *équivalent* à p propre. Comparer avec le corollaire 1, page 166.

80. Mais si, cf. démonstration page 168. Les développements des quatre pages qui suivent sont inutiles par cette démonstration.

The proper case uses a different good feature. If literal pullback along p carries W^{d}_S into $W^{\mathrm{d}}_{S'}$, then for every $Z \to S$ the comparison

$$Z \times_S S' \longrightarrow \Phi_S(Z) \times_S S'$$

is right-fine. Hence literal pullback computes p^* even for a representative which is not smooth over S. To finish, one may still choose $Z \to X$ smooth in order to compute q^*; this costs nothing, because every coefficient has such a representative. Both sides of base change are again represented by $Z \times_S S'$, and the displayed mate gives its identity after the canonical comparisons. The direct argument on pp. 168–169 proves precisely the preservation statement needed here.

A small example shows why smoothness concerns transport rather than constancy. Let $S = (0 \to 1)$ and take the presheaf of sets F with

$$F(0) = \{a, b\}, \qquad F(1) = \{c\}, \qquad c \longmapsto a.$$

Its category of elements X has the arrow $a \to c$ and the isolated object b, with the evident map $f : X \to S$. This is a discrete Grothendieck fibration, hence smooth. For the terminal coefficient on X, the object $f_!1_X$ is represented by $X \to S$. Pullback to 0 is the two-point category, whereas pullback to 1 is the one-point category. Both values agree with base change along the corresponding point inclusion. The transition between the fibres is not a weak equivalence, and no local constancy has been used.

The terminology is inherited from the base-change theorems of algebraic geometry, but the hypotheses here are conditions on categories of lifts. The precise categorical characterization is recorded by Maltsiniotis (2005, Theorem 1.2); its proper version follows by opposites. In the modern derivator formalism, Groth (2013, Propositions 1.24 and 1.26) treats the corresponding exact squares for Grothendieck fibrations, opfibrations and comma categories. These references clarify the later language without supplying the missing Der 2 argument in the manuscript's first attempted proof.

Notons d'abord p. 101

Proposition 5. *Soit $f : S' \longrightarrow S$ une flèche dans* Cat. *Alors f est une* HOT_W*-équivalence* si et seulement si $\mathrm{hot}_W(f)$ *est* `[un]` *isomorphisme. Si W est fortement saturé, cela signifie donc que $f \in W$ (et en tous cas $f \in W \Longrightarrow f$* `[est]` *une* HOT_W*-équivalence)* ([81]).

C'est contenu dans la démonstration de Der 3b tantôt. Je rappelle que f est une HOT-équivalence, par définition, si et seulement si pour $a, b \in \mathrm{Hot}_W (= \mathrm{HOT}_W(e))$,

$$(*) \qquad \mathrm{Hom}_{\mathrm{HOT}_W(S)}(a_S, b_S) \longrightarrow \mathrm{Hom}_{\mathrm{HOT}_W(S')}(a_{S'}, b_{S'})$$

est `[un]` isomorphisme, où a_S etc. désigne $p_S^*(a)$ etc. ($p_S : S \longrightarrow e$). Les formules d'adjonction de $p_{S!}$, p_S^* et de $p_{S'!}$, $p_{S'}^*$ permettent d'interpréter la flèche $(*)$ comme

$$\mathrm{Hom}_{\mathrm{HOT}_W(S)}(p_{S!}(a_S), b) \longrightarrow \mathrm{Hom}_{\mathrm{HOT}_W(S')}(p_{S'!}(a_{S'}), b),$$

déduit par 'transposition' $\mathrm{Hom}(-, b)$ de la flèche

[page 165]

$$p_{S'!}(a_{S'}) \longrightarrow p_{S!}(a_S)$$

induite par f - donc il faut prouver que celle-ci est un isomorphisme (c'est la forme que nous avions utilisé tantôt). Or cette flèche s'identifie à

$$\eta' \times a \longrightarrow \eta \times a$$

induite par

$$\mathrm{hot}_W(f) : \underbrace{\eta'}_{=\mathrm{hot}_W(S')} \longrightarrow \underbrace{\eta}_{=\mathrm{hot}_W(S)},$$

et dire que c'est `[un]` isomorphisme quel que soit a dans Hot_W revient à dire que $\eta' \longrightarrow \eta$ l'est (puisque la multiplication dans Hot_W a une élément unité …).

81. devrait remonter vers le théorème 3, p. 159.

Maintenant je me rappelle que pour avoir l'axiome Der 2 des dérivateurs, je devrais également supposer W *fortement* saturé (i.e. égal au localiseur défini par le foncteur canonique $\mathrm{Cat} \longrightarrow (\mathrm{Cat})W^{-1} \overset{\text{déf}}{=} \mathrm{Hot}_W$). (C'est ce que je supposais d'emblée, en partant d'un $(\mathcal{M}, W)$ général pour définir un prédérivateur, pour avoir l'axiome Der 2.)

En appliquant le dual du théorème de I, p. 56, on trouve donc le théorème de changement de base pour $f_!$, pour changement de base $f : S' \longrightarrow S$ *W-propre.*

Mais la démonstration utilise sûrement Der 2 (critère d'isomorphisme par fibres, dans $\mathrm{HOT}_W(S)$), que je n'ai pas prouvé encore. Elle est incomplète pour le moment. Mais je peux noter dès maintenant le [page 166] ‖

Commentary · Exposition [ms. 165]

What constant coefficients detect

Grothendieck interrupts the base-change argument to identify the absolute equivalences detected by HOT_W. This is a useful move: it prevents the phrase "HOT_W-equivalence" from concealing a new, unexplained class of maps. The proof uses the adjunction just established and the already available product formula for a constant coefficient.

Let $p_S : S \to e$ and put $\eta = \mathrm{hot}_W(S)$. For $a \in \mathrm{Hot}_W$, write $a_S = p_S^* a$. The representing-category description gives a natural isomorphism

$$p_{S!}(a_S) \cong \eta \times a.$$

Indeed, if $a = \mathrm{hot}_W(T)$, then a_S is represented by $S \times T \to S$ and its left direct image is represented by $S \times T$. This also identifies the morphism induced by $f : S' \to S$: it is $\mathrm{hot}_W(f) \times \mathrm{id}_a$.

Adjunction therefore turns the defining map for a HOT_W-equivalence into

$$\mathrm{Hom}_{\mathrm{Hot}_W}(\eta \times a, b) \longrightarrow \mathrm{Hom}_{\mathrm{Hot}_W}(\eta' \times a, b), \qquad \eta' = \mathrm{hot}_W(S').$$

The base-dependent subscripts in the edition's displayed Hom formula are slips; both Hom sets here are in Hot_W. If $\mathrm{hot}_W(f)$ is an isomorphism, every displayed map is a bijection. Conversely, assume these maps are bijections for all a, b, and take a to be the terminal object 1. The product identifications $\eta \times 1 \cong \eta$ and $\eta' \times 1 \cong \eta'$ reduce the assertion to bijectivity of

$$\mathrm{Hom}_{\mathrm{Hot}_W}(\eta, b) \longrightarrow \mathrm{Hom}_{\mathrm{Hot}_W}(\eta', b) \quad \text{for every } b.$$

Yoneda gives invertibility of $\mathrm{hot}_W(f)$. More explicitly, taking $b = \eta'$ supplies a left inverse of $\mathrm{hot}_W(f)$, and injectivity for $b = \eta$ forces it to be a right inverse. This proves Proposition 5.

Strong saturation enters only when the result is translated back from the localized category to the original class W. The proposition without saturation says that the detected maps form $\overline{W}$, the class inverted by $\mathrm{Cat} \to \mathrm{Hot}_W$. If $W = \overline{W}$, this is exactly W. The proof does not require the detection of all isomorphisms in $\mathrm{HOT}_W(S)$ by point evaluations. In particular, it must not be counted as a proof of Der 2. That latter statement concerns arbitrary arrows of a localized category, including zigzags, whereas Proposition 5 identifies the image of a given functor of indexing categories.

The source itself recognizes the distinction on p. 166. The intended appeal to the abstract proper-base-change theorem still needs Der 2, so the proof by that route is incomplete. Nothing is wrong with Proposition 5, and nothing forces Theorem 4 to wait for a global repair of the derivator axioms: the direct proof on pp. 168–169 works before that repair. Keeping this order of dependence is the reason to retain both the interrupted argument and the author's later replacement of it.

Corollaire 1 (du théorème 4, p. 162). *Si la flèche $S' \xrightarrow{p} S$ dans* Cat *donne lieu (pour tout $f : X \longrightarrow S$ dans* Cat/S*) à la formule de changement de base pour* p. 102

$$f_! : \mathrm{HOT}_W(X) \longrightarrow \mathrm{HOT}_W(S),$$

alors le foncteur $p^* : X \longmapsto X \times_S S'$ de Cat/S *dans* Cat/S' *est compatible avec les localiseurs W_S^{d}, $W_{S'}^{\mathrm{d}}$.* (L'implication inverse a été vue déjà, cf. pages 161-163.)

Cela résultera du

Lemme. *Soit $u : X' \longrightarrow X$ une flèche dans* Cat/S*. Pour que $u \in W_S^{\mathrm{d}}$, il faut et il suffit (si W* `[est]` *fortement saturé) que pour tout $\xi \in \mathrm{Hot}_W$, la flèche correspondante*

$$u_* : p'_!(\xi_{X'}) \longrightarrow p_!(\xi_X)$$

soit `[un]` *isomorphisme dans* $\mathrm{HOT}_W(S)$*.*

C'est suffisant, sous réserve de saturation forte de W, car si u_* est `[un]` isomorphisme, passant aux fibres, on trouve par Der $5'$

$$H_\bullet^{\mathrm{HOT}_W}(s\backslash X', \xi) \longrightarrow H_\bullet^{\mathrm{HOT}_W}(s\backslash X, \xi),$$

si c'est `[un]` isomorphisme, alors $s\backslash X' \longrightarrow s\backslash X$ est par définition une HOT-équivalence, ce qui implique que c'est dans W (proposition 5).

C'est nécessaire, car si $\xi = \mathrm{hot}_W(T)$, donc $\xi_S = \mathrm{hot}_{S,W}(T_S)$ (où $T_S = T \times S/S$), et on a donc [page 167]

$$p'_!(\xi_{X'}) = \mathrm{hot}_{S,W}(X' \times T), \qquad p_!(\xi_X) = \mathrm{hot}_{S,W}(X \times T),$$

et il faut prouver que si $u : X' \longrightarrow X$ est dans W_S^{d}, alors $\underbrace{X' \times T}_{X' \times_S T_S} \longrightarrow \underbrace{X \times T}_{X \times_S T_S}$ aussi, ce qui signifie simplement que

$$(*) \qquad s\backslash(X' \times T) \longrightarrow s\backslash(X \times T)$$

est dans W pour tout $s \in S$. Mais le changement de base $s\backslash S \longrightarrow S$ commute aux produits fibrés, donc transforme $X \times T = X \times_S T_S$ en $(s\backslash X) \times_{s\backslash S} T_{(s\backslash S)} = s\backslash X \times T$, et de même pour X'. Donc la flèche s'identifie à

$$(s\backslash X') \times T \longrightarrow (s\backslash X) \times T,$$

déduite de

$$s\backslash X' \longrightarrow s\backslash X$$

par le foncteur $? \times T$. Comme ce foncteur transforme W en W, on gagne.

Commentary · Exposition [ms. 167]

A single coefficient already tests right-fine maps

The lemma on pp. 166–167 is meant to extract a statement about actual functors from a base-change isomorphism. Its family of constant coefficients is natural from the cohomological viewpoint. Once the pointwise formula for $f_!$ is known, however, the terminal coefficient already detects the maps under consideration. This observation shortens the proof and makes its independence from Der 2 explicit.

Assume W is strongly saturated, and let $u : X' \to X$ be a morphism over S. Then

$$u \in W_S^{\mathrm{d}} \quad \Longleftrightarrow \quad \mathrm{hot}_{S,W}(u) \text{ is an isomorphism.}$$

The forward implication is the definition of the localization. For the converse, apply the evaluation functor s^* to the asserted isomorphism. The pointwise formula identifies its value with

$$\mathrm{hot}_W(s\backslash X') \longrightarrow \mathrm{hot}_W(s\backslash X).$$

Strong saturation makes $s\backslash u$ a member of W for every s, which is exactly the definition of W_S^{d}. This argument applies evaluation to an isomorphism already known to exist; it does not infer a global isomorphism from a family of pointwise isomorphisms. Consequently it uses no unproved conservativity axiom.

Now let $p : X \to S$ and $p' : X' \to S$. The image of the terminal coefficient under left direct image is $p_! 1_X = \mathrm{hot}_{S,W}(X)$, and the map induced by u is $\mathrm{hot}_{S,W}(u)$. Thus the preceding equivalence is

precisely the manuscript's test with $\xi = 1$. For a general constant coefficient represented by T, the induced map is represented by $u \times \mathrm{id}_T$. Since

$$s\backslash(X \times T) \cong (s\backslash X) \times T,$$

product stability of W proves the implication from $u \in W_S^{\mathrm{d}}$ to the test for every ξ. The larger family of tests is therefore equivalent to the single one, rather than an additional hypothesis.

This gives Corollary 1 with its quantifiers visible. Suppose that base change along $p : S' \to S$ is valid for left direct image along every $X \to S$. If $u : X_1 \to X_2$ belongs to W_S^{d}, its map of pushed-forward terminal coefficients is invertible. Pulling that map back along p preserves invertibility. Naturality of base change identifies the resulting map with the left direct image of the terminal-coefficient map induced by

$$u' : X_1 \times_S S' \longrightarrow X_2 \times_S S'.$$

The single-coefficient test over S' implies $u' \in W_{S'}^{\mathrm{d}}$. Strong saturation is used at this last conversion to membership in W; without it the argument establishes the corresponding statement for the class detected in the localization.

This is a useful instance of the distinction between a criterion for a representative and a criterion for an arbitrary localized morphism. The former is enough for the present base-change theorem. The latter is the broader problem of Der 2, which remains separate.

Finalement, je m'aperçois que j'ai eu un blocage stupide, pour la question que le changement de base propre soit compatible à la W-équivalence colocale. Soit p. 103 [page 168]

$$\begin{array}{ccccc} X & \xrightarrow{\ f\ } & Y & & \\ & \searrow & \swarrow & & \\ & & S & \xleftarrow{\ g\ } & S', \end{array}$$

avec $f \in W_S^{\mathrm{d}}$ et g propre, il faut voir que $f' : X' \longrightarrow Y'$ déduit de f par changement de base $S' \longrightarrow S$ est encore dans $W_{S'}^{\mathrm{d}}$. Cela signifie que tout changement de base de la forme $s'\backslash S' \longrightarrow S'$ le transforme en $\in W$. Mais soit $s = g(s')$, on a le carré commutatif

$$\begin{array}{ccc} S & \longleftarrow & S' \\ \uparrow & & \uparrow \\ \underbrace{s\backslash S}_{=Z} & \longleftarrow & \underbrace{s'\backslash S'}_{=Z'}, \end{array}$$

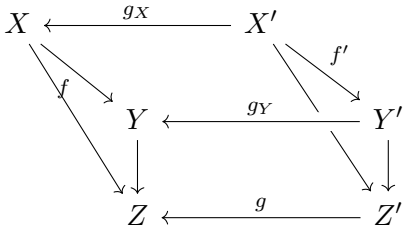
et il suffit de voir que pour toute flèche $\varphi : \mathcal{X} \longrightarrow \mathcal{Y}$ dans Cat/Z qui est dans W, son image inverse par changement de base $Z' \longrightarrow Z$ est itou. (En appliquant ceci au cas $\mathcal{X} = X \times_S (s\backslash S)$, $\mathcal{Y} = Y \times_S (s\backslash S)$, $\varphi = s\backslash f$.) Or ceci résulte du fait que, par hypothèse de propreté, $Z' \longrightarrow Z$ est propre et à fibres Hot-asphériques, donc c'est une Hot-équivalence universelle. Il suffirait même que g soit W-propre, pour impliquer que $Z' \longrightarrow Z$ est une W-équivalence universelle, i.e. qui reste telle après tout changement de base. Or on a le [page 169]

Lemme. Si $g : Z' \longrightarrow Z$ est une W-équivalence universelle, alors le foncteur

$$g^* : X \longmapsto X \times_Z Z' : \mathrm{Cat}/Z \longrightarrow \mathrm{Cat}/Z'$$

est compatible avec les localiseurs W_Z, $W_{Z'}$, i.e. si $f : X \longrightarrow Y$ dans Cat/Z est dans W, de

même $f' : X' \longrightarrow Y'$ dans Cat/Z'.



Cela est évident sur le carré supérieur du diagramme ci-dessus, où g_X, g_Y sont dans W par l'hypthèse sur g, donc f est dans W si et seulement si f' l'est, cqfd.

[page 170 manque]

[page 170 manque]
p. 104
[page 171]

Commentary · Exposition [ms. 169]

Proper base change without a pointwise conservativity axiom

The short proof on pp. 168–169 succeeds because it turns a relative question into an absolute one over a suitable comma category. Properness supplies a comparison of bases which remains a W-equivalence after every pullback. Two-out-of-three can then transport the desired equivalence across that comparison.

We first isolate the elementary part. Let $v : T' \to T$ be a universal W-equivalence, meaning that for every $A \to T$ the projection $A \times_T T' \to A$ belongs to W. If $h : A \to B$ is a map over T, the commutative square

$$\begin{array}{ccc} A \times_T T' & \xrightarrow{h'} & B \times_T T' \\ {\scriptstyle v_A}\downarrow & & \downarrow{\scriptstyle v_B} \\ A & \xrightarrow[h]{} & B \end{array}$$

has both vertical arrows in W. If $h \in W$, then $hv_A \in W$ and $v_B h' = hv_A$ forces $h' \in W$. The reverse implication is the same calculation in the other direction. Literal pullback along a universal equivalence therefore preserves and reflects W. No localization, no adjunction and no pointwise detection is involved in this lemma.

Take now $g : S' \to S$ proper, or more generally W-proper, and fix $s' \in S'$ with $s = g(s')$. Properness gives a universal W-equivalence

$$v : s'\backslash S' \longrightarrow s\backslash S.$$

Its fibre over an arrow $s \to t$ is the category of its lifts starting at s'. The defining asphericity conditions for properness, together with their stability after base change, give the stated universality. This is the geometric content of the proof; it is not a consequence of the existence of initial objects in the two comma categories alone. Two contractible categories can be connected by a functor which fails to be a universal equivalence.

For $u : X \to Y$ over S, form $u_s : s\backslash X \to s\backslash Y$, a map over $s\backslash S$. If $u \in W_S^{\mathrm{d}}$, then $u_s \in W$. Pullback along v identifies u_s with

$$s'\backslash(X \times_S S') \longrightarrow s'\backslash(Y \times_S S').$$

The elementary lemma makes this map a member of W. Since s' was arbitrary, literal pullback along g preserves right-fine equivalences. This completes the proper case of Theorem 4 by the model calculation explained above.

For a discrete opfibration, the role of the lift categories is especially transparent. An arrow $g(s') \to t$ has a unique lift with prescribed source s'. Hence the induced functor on undercategories is an isomorphism; the preceding comparison needs no homotopy argument at all. For an ordinary Grothendieck opfibration, a chosen cocartesian lift is initial in the corresponding category of lifts, so its nerve is contractible. The general proper case replaces this initial object by the weaker asphericity condition required by W.

The manuscript calls the earlier obstruction a "blocage". Mathematically, the change is a change of test objects: the universal comparison over $s\backslash S$ carries exactly the information needed for right-fine equivalence. The abandoned four-page route is not a counterexample to proper base change, nor does it expose a defect in the construction of $f_!$.

Nouvelle version de la théorie de HOT_W.

Je me suis aperçu que l'hypothèse $W \supseteq W_\infty$ (L3 quater, p. 1), que j'avais fini par mettre sur W, est artificielle, et que l'hypothèse L5 (p. 43), voire L6, L6 bis (p. 43-44), est inutile pour une grande partie de la théorie de HOT_W. J'ai été ainsi amené à revoir la théorie (i.e. ce que j'en ai fait jusqu'à présent) sous les hypothèses suivantes sur W (cf. XII) :

W(1) (*Saturation* de W) : comme avant (L1 page 1)

W(2) (localisation) : c'est l'ancien L4 bis ([82]).

W(3) (objet final) : Si X a un objet final, X est W-asphérique (ancien L3).

Ces trois axiomes sont suffisants pout tout ce que j'avais fait dans Pursuing Stacks, et dans une grande partie de ce qui a été fait ici sur HOT. Parfois cependant il faut l'ancien L 5, que je réénonce ici (comparer XII, théorème `[numéro du théorème manque]`) :

Commentary · Exposition [ms. 171]

Axioms for the localization and axioms for its fibres

Grothendieck's new beginning separates the construction of a homotopy theory from the stronger rules which make its total spaces determine its fibres. Much of the preceding work survives under three localizer axioms. The extra fibration axiom enters at identifiable comparison maps, and the next ten pages are an effort to locate those maps rather than to replace the entire construction.

The localization axiom W(2) is a relative Theorem A statement. For a triangle $X \to Y \to S$, it says that membership of $X/s \to Y/s$ in W for every s implies membership of $X \to Y$ in W. Together with the terminal-object axiom and the saturation rules, it gives the familiar local-to-global mechanism. The weakened W(2 bis) keeps only the case in which all the relevant comma categories are W-aspherical. The distinction matters when the minimum class W_ω is introduced: it is defined using this weaker axiom, not by simply repeating the complete list W(1,2,3).

There is a second distinction in the word "fibration". The ordinary category $\underline{\mathrm{Fib}}S$ consists of Grothendieck fibrations over S. The category $\underline{\mathrm{Fib}}_W S$ in the revised diagram consists of weak W-fibrations and is defined by a homotopical condition. One equivalent test is this: for every categorically contractible $T \to S$ and every $t \in T$, the inclusion

$$X_t \longrightarrow X \times_S T$$

belongs to W. This is the formulation of Chapter XII, Theorem 3, manuscript pp. 51–52, to which Grothendieck refers here (Grothendieck, 1990–1991f). The word "contractible" in this test cannot be replaced silently by "W-aspherical". Doing that asks for a stronger condition, whose relation to the weak condition is precisely part of the fibration-axiom problem.

For an ordinary Grothendieck fibration, the weak condition has a concrete interpretation: each transport functor $u^* : X_{s'} \to X_s$ must belong to W. Thus an ordinary fibration can be smooth without being weakly W-fibred, as in the two-fibre example above. Conversely, a weak W-fibration need not possess chosen cartesian lifts. The comma replacement Φ is what supplies an ordinary split fibration representing its homotopical data.

W(4b) compares two tests for a morphism between such weakly fibred objects. The global test asks whether $X \to Y$ lies in W as a functor of total categories. The fibre test asks whether each $X_s \to Y_s$ lies in W. The implication from the fibre test to the global test already follows from W(2), after comparing fibres with comma fibres. W(4b) asserts the converse. It is therefore a cancellation rule for total spaces over a fixed base, not merely a restatement of two-out-of-three.

The first version of W(4b′) restricts that rule to W_ω-perfect objects. This does prove equality of the two tests inside that smaller category. It does not yet compare an arbitrary weakly fibred X with its perfect replacement $C(X)$, because X need not belong to the smaller category. The footnote retracting the claimed sufficiency points to exactly this missing scope in the available argument. The retraction does not establish that the implication is false. With the later minimality and filtered-colimit results explained below, the alternating replacement on pp. 192–195 supplies the missing fine unit; the original restricted condition does then imply W(4b). The commentary at p. 195 gives the transfer argument. Later a different condition receives the same label W(4b′); the two formulations must be kept separate throughout the audit.

82. i.e. le L5 de la page 2. L'axiome est repris (sous une forme voisine et équivalente) au bas de la page 43.

W(4b) Soient X, Y W-fibrés sur S, $f : X \longrightarrow Y$ un S-morphisme. Si $f \in W$, alors $\forall s \in \mathrm{Ob}\, S$, $f_s : X_s \longrightarrow Y_s$ est dans W.

Pour la notion de W-fibré (*faible*), j'ai été long à trouver enfin *la* bonne notion, bonne pour un W ne satisfaisant que W(1, 2, 3) (et même seulement W(1, 2 bis, 3), où W(2 bis) est l'ancien L4). Je renvoie pour cela à (XII, théorème 3, p. 51), où la notion est bien cernée à coups de conditions équivalentes. En fait, pour les applications principales à HOT_W, vues jusqu'à présent, il suffit ([83]) d'une variante affaiblie de W(4b) :

W(4b′) Comme W(4b), mais en exigeant de plus que X, Y soient dans $\underline{\mathrm{Parf}}_\omega\, S$.

Pour la notion de morphismes W-propres, W-lisses, W-parfaits dans Cat, je renvoie à XII, partie III (p. 22-52), et les notations $\underline{\mathrm{Liss}}_W S$, $\underline{\mathrm{Prop}}_W S$, $\underline{\mathrm{Parf}}_W S$ (sous-catégories de Cat/S), et $\mathrm{Parf}_W \subseteq \mathrm{Fl(Cat)}$. Dans l'énoncé de W(4b′) on a posé [page 172]

$$\underline{\mathrm{Parf}}_\omega S = \underline{\mathrm{Parf}}_{W_\omega} S, \qquad \text{où } W_\omega \subseteq \mathrm{Fl(Cat)} \text{ est la plus petite partie de Fl(Cat) satisfaisant W}(1, 2\text{bis}, 3).$$

Je note de même

$$W'_\omega \subseteq \mathrm{Fl(Cat)}, \qquad \text{plus petite partie de Fl(Cat) satifaisant W}(1, 2, 3, 4\text{b}).$$

L'hypothèse $W \supseteq W_\infty$ que nous avions fini par imposer sur W, et qui est apparu artificielle au cours de la réflexion du XII, a avantage à être remplacée par p. 105

$$W \supseteq W'_\omega \qquad (\text{ou par } W \supseteq \overline{W'_\omega},\ \text{où } \overline{W'_\omega} \text{ est le saturé fort de } W'_\omega)$$

([84]). Mais nous ne l'imposons pas comme condition impérative, au même titre que W(1, 2, 3), pas plus que W(4b). Le rôle que j'ai fait jouer à W_∞ dans les semaines passées, semble revenir plus naturellement à W'_ω ou $\overline{W'_\omega}$. Il en est ainsi notamment aussi, il me semble, dans l'énoncé des conditions du type Der 3 sur un dérivateur (I, p. 111-112), qu'il faudrait revoir dans cette lumière. Mais tout dépend, si (comme j'en ai l'impression) on peut faire vraiment de l'algèbre homotopique substantielle, avec notamment les suites exactes longues de fibration et de suspension, pour HOT_W, avec des W qui sont *strictement* contenus dans W_∞. Avant de même être assuré, il est trop tôt encore pour me prononcer sur la 'bonne' formulation de Der 3,

Commentary · Later developments [ms. 172]

The proposed theory below ordinary weak equivalence

Grothendieck is trying to remove an externally imposed reference to classical homotopy. His question about a localizer strictly smaller than W_∞ concerns the strength of the axioms themselves: perhaps their formal consequences already contain more classical homotopy theory than he has yet extracted.

For the modern basic-localizer axioms, this question is settled. Cisinski (2004, Definition 1.1.3 and Theorem 2.2.11) proves that every basic localizer contains the nerve weak equivalences W_∞. Thus no strictly smaller basic localizer satisfies that complete package. The weaker list defining W_ω requires a further theorem. Cisinski (2009, Theorem 1.10 and Remark 1.11) proves the corresponding minimality among weak basic localizers, using the absolute Theorem A condition. It follows that $W_\omega = W_\infty$ as well. To check the saturation hypothesis, if $ri = \mathrm{id}$ and $ir \in W$, the manuscript's condition on the two composites ri, ir gives $i, r \in W$; this is the retract clause in the weak-basic-localizer formulation. W(2 bis) and W(3) supply its other two clauses. The two axiom lists remain different even though their minimum classes now have the same description.

83. C'est finalement une erreur, semble-t-il à présent. Je crois que l'axiome W(4b′) ne sert finalement à strictement rien (si ce n'est en fait à entretenir des idées fausses.)

84. $\overline{W'_\omega}$ est le plus petit sous-ensemble *fortement saturé* de Fl(Cat), satisfaisant W(4b), et il satisfait même la condition W′(4a), variante la plus forte des axiomes du type W(4).

This result changes the status of the proposed example, but it leaves the reason for the revision intact. Removing an explicit hypothesis $W_\infty \subseteq W$ can still clarify which finite arguments use only the localizer axioms and which use a separate fibre-detection condition. Indeed, a class larger than W_∞ can satisfy the basic axioms while failing W(4b). The connected-components localizer used below gives an elementary instance. The ordinary theory is minimal in the order of classes of equivalences; further localization is entirely possible, and it can forget information that a fibrewise test retains.

The same minimum is obtained after adjoining W(4b): $W'_\omega = W_\infty$. Indeed, Quillen's Theorem B identifies the fibres of a weak W_∞-fibration with its homotopy fibres (Cisinski, 2004, Theorem 2.3.2). A total weak equivalence over a fixed base therefore induces weak equivalences on the fibres. Thus W_∞ satisfies W(4b) and is an admissible candidate for W'_ω; basic minimality gives the reverse inclusion. Its strong saturation is the same class. The homotopy-pullback description also supplies the strongest base-change property intended in footnote 84.

This identifies the minimum within the standing W(1,2,3) framework. The inclusion $W'_\omega \subseteq W$ does not, by itself, imply that W inherits every closure property appearing in the defining list: the notion of weak W-fibration changes when W changes. The fibre-cancellation axiom is not simply an upward-closed property of subsets of Fl(Cat). This explains why Grothendieck treats the suggested containment and the assertion W(4b) as different conditions in the paragraph itself.

By Cisinski (2006a, Corollary 4.2.22), W(6 bis) is automatic for basic localizers; the theorem assumes relative W(2).

[page 173]

qui est l'axiome mettant en relation le dérivateur **D** avec W_∞, ou avec quelque autre localiseur fondamental.

Je vais faire à présent un résumé de la théorie de HOT_W, revue dans l'optique que je viens de dire : on suppose pour W *les conditions* $\mathrm{W}(1,2,3)$ *sans plus,* sauf mention explicite.

① **Localiseurs principaux** sur Cat/S et sur les principales catégories de S-catégories (cf. diagramme p. 76).

Ils seront tous induits par les trois localiseurs familiers

$$W_S^{\mathrm{d}} \qquad W_S^{\mathrm{g}}$$
$$\searrow \quad \swarrow$$
$$W_S$$

sur Cat/S. Il y a lieu aussi de regarder W_S^{f}, mais jamais sur Cat/S tout entier, mais sur d'autres catégories du diagramme, et toujours dans des cas où W_S^{f} coïncide soit avec W_S^{d}, soit avec W_S^{g}. Mais le point de vue W_S^{f} est souvent plus directement commode, et a l'avantage de plus d'être symétrique.

ⓐ Dans $\underline{\mathrm{Liss}}_W S$, on a

p. 106

$$\begin{array}{l} W_S^{\mathrm{f}} \quad = W_S^{\mathrm{d}} = W_S^{\mathrm{univ}} \\ \Downarrow \\ W_S^{\mathrm{g}} \\ \Downarrow \\ W_S \end{array}$$

Dualement,

ⓑ dans $\underline{\mathrm{Prop}}_W S$, on a

$$\begin{array}{ccc} W_S^{\mathrm{f}} & = W_S^{\mathrm{g}} = W_S^{\mathrm{univ}} & \\ \Downarrow & & \\ W_S^{\mathrm{d}} & & \\ \Downarrow & & \\ W_S, & & \end{array}$$

enfin, [page 174]

ⓒ dans $\underline{\mathrm{Fib}}_W S$, on a

$$\begin{array}{ccccccc} W_S^{\mathrm{f}} & \Longleftrightarrow & W_S^{\mathrm{d}} & \Longleftrightarrow & W_S^{\mathrm{g}} & \Longleftrightarrow & W_S^{\mathrm{univ}} \\ \Downarrow & & & & & & \\ W_S. & & & & & & \end{array}$$

Cette implication $W_S^{\mathrm{f}} \Longrightarrow W_S$ est stricte si on ne fait pas sur W l'axiome W(4b), dont l'utilité ici est exactement d'assurer

> $W_S^{\mathrm{f}} \Longleftrightarrow W_S$ dans $\underline{\mathrm{Fib}}_W S$ *si* W(4b) `[est]` satisfait
>
> et alors les cinq localiseurs envisagés ci-dessus
>
> dans $\underline{\mathrm{Fib}}_W S$ sont tous égaux.

Les relations entre [?] localiseurs dans $\underline{\mathrm{Liss}}_W S$, $\underline{\mathrm{Prop}}_W S$, $\underline{\mathrm{Fib}}_W S$ se transportent automatiquement aux sous-catégories de celles-ci, plus généralement à toutes celles qui s'envoient par un foncteur d'inclusion dans une de celles-ci. Dans le diagramme p. 76, toutes les catégories, à la seule exception de Cat/S tout entier, sont dans ce cas. p. 107

Commentary · Exposition [ms. 174]

Five tests and the role of good objects over the base

The diagram of localizers is the bookkeeping on which the revised theory depends. Grothendieck is distinguishing the information carried by a strict fibre, a comma fibre, and a total category. To read its equalities, every test must refer to the same map $u : X \to Y$ over the same base S.

The five conditions are

$$\begin{aligned} u \in W_S &\iff u \in W \text{ on total categories,} \\ u \in W_S^{\mathrm{f}} &\iff u_s : X_s \to Y_s \in W \quad (s \in S), \\ u \in W_S^{\mathrm{d}} &\iff s\backslash X \to s\backslash Y \in W \quad (s \in S), \\ u \in W_S^{\mathrm{g}} &\iff X/s \to Y/s \in W \quad (s \in S), \\ u \in W_S^{\mathrm{univ}} &\iff X \times_S T \to Y \times_S T \in W \quad (T \to S). \end{aligned}$$

In the last line universality is relative to S. It is not the assertion that $u : X \to Y$ remains a W-equivalence after every map into its total target Y. The difference is already visible when $S = e$.

If X and Y are W-smooth over S, the inclusions of strict fibres into under-comma fibres give a commutative square

$$\begin{array}{ccc} X_s & \xrightarrow{u_s} & Y_s \\ \downarrow & & \downarrow \\ s\backslash X & \xrightarrow[s\backslash u]{} & s\backslash Y \end{array}$$

whose vertical arrows lie in W. Two-out-of-three identifies the fibre and right-fine tests. Smoothness survives change of base. Thus the fibre test, applied after every $T \to S$, gives a right-fine map over

T and hence a total W-equivalence by W(2). This proves $W_S^{\mathrm{f}} \subseteq W_S^{\mathrm{univ}}$ on smooth objects. The reverse inclusion follows by taking $T = \{s\}$. Finally, taking $T = S/s$ gives $W_S^{\mathrm{univ}} \subseteq W_S^{\mathrm{g}}$, and the relative localizer axiom gives $W_S^{\mathrm{g}} \subseteq W_S$. These are exactly the first chain of implications in the text. Opposites give the proper chain.

For weak W-fibrations, the defining universal comparisons apply to both the initial and the terminal comma categories. Hence both vertical comparisons, for $s\backslash X$ and for X/s, are in W, and remain so after changing the base. The same proof identifies all four fine or universal tests. W(4b) adds the reverse implication from the total test. Its statement is equivalent to equality of all five classes on $\underline{\mathrm{Fib}}_W S$ for every S.

A two-object example explains the restriction to good objects. Let $S = (0 \to 1)$, let X be the discrete category on $0, 1$ mapped identically on objects to S, and let $Y = S$ with its identity structural map. The functor $u : X \to Y$ is the identity on objects. Every strict fibre map is the identity of a point, so $u \in W_S^{\mathrm{f}}$ for ordinary weak equivalences. But X has two connected components and Y has one, so $u \notin W_S$. At 0, the under-comma category $0\backslash X$ consists of two isolated objects, whereas $0\backslash Y$ is an interval. The right-fine test detects exactly the missing arrow. This is why the text avoids using the strict-fibre localizer on all of Cat/S.

The phrase that the inclusion is strict "si on ne fait pas" W(4b) should be read as a statement about what the basic axioms do not force. For a fixed W and a fixed S, absence of an additional assumption is not a proof that the classes differ. The exact universal assertion is the equivalence with W(4b); the example below supplies a localizer and a base where the difference is real.

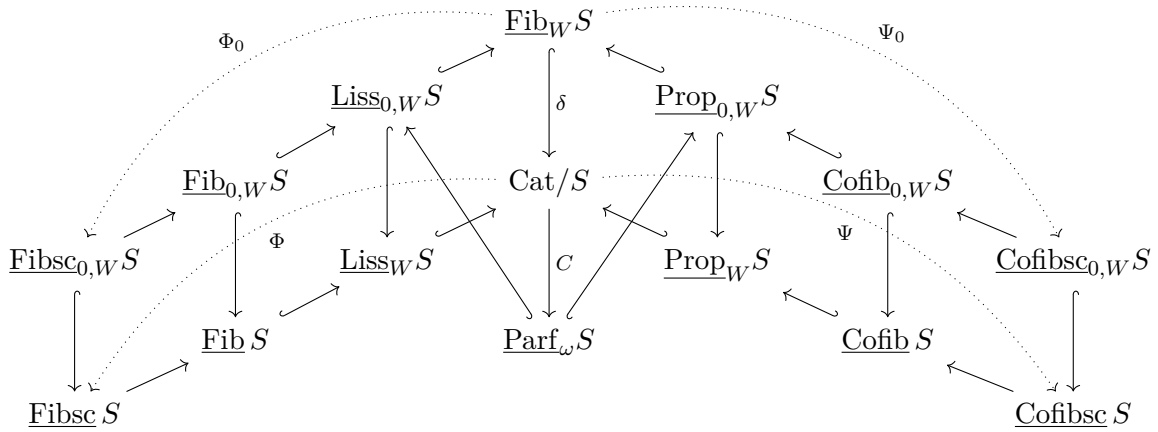
Pour les résultats principaux développés jusqu'à présent sur HOT_W, quand l'axiome W(4b) est utilisé, la forme faible W(4b′) suffit. On a

> $W_S^{\mathrm{f}} \Longleftrightarrow W_S$ dans $\underline{\mathrm{Parf}}_\omega S$ *si* W(4b′) `[est]` satisfait
>
> et alors les cinq localiseurs W_S^{f}, W_S^{d}, W_S^{g}, W_S^{univ}, W_S
>
> sont égaux dans $\underline{\mathrm{Parf}}_\omega S$.

② **Principales catégories de catégories sur S.**

Ce sont, à des modifications d'éclairage près, celles du diagramme p. 76. Les catégories $\underline{\mathrm{Fibsc}}\, S$, $\underline{\mathrm{Fib}}\, S$, $\underline{\mathrm{Cofibsc}}\, S$, $\underline{\mathrm{Cofib}}\, S$ et leurs variantes à indice 0, sont inchangées - leur rôle est d'être aussi petites

que possible pour remplir leur fonction (de description de $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^o_W(S)$, $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, ou $\mathrm{HOT}^{\mathrm{lc!}}_W(S)$ ci-dessous). La catégorie $\underline{\mathrm{Parf}}\, S$ du diagramme remplissait une fonction du même type, c'est pourquoi nous la remplaçons à présent (en surenchérissant) par $\underline{\mathrm{Parf}}_\omega S$. Les autres catégories du diagramme sont, en plus de Cat/S, $\underline{\mathrm{Liss}}\, S$ et $\underline{\mathrm{Prop}}\, S$ et leurs variantes à indice 0, ainsi que $\underline{\mathrm{Fibhot}}\, S$. (Toutes ces catégories, ainsi que $\underline{\mathrm{Parf}}\, S$, étaient donc construites par référence à W_∞, non pas à W.) Leur rôle est d'être aussi grandes que possible pour donner lieu à certaines propiétés et relations. Or il se trouve qu'ici la référence à W_∞ était artificielle, et il y a lieu de mettre plutôt les catégories $\underline{\mathrm{Liss}}_W S$, $\underline{\mathrm{Prop}}_W S$, $\underline{\mathrm{Fib}}_W S$, et en plus les variantes $\underline{\mathrm{Liss}}_{0,W} S$, $\underline{\mathrm{Prop}}_{0,W} S$. Ainsi le diagramme devient [page 175]



([85]) (tous les carrés [sont] cartésiens, les foncteurs notés $\hookrightarrow$ sont pleinement fidèles),

où j'ai encore marqué en vert [dans l'original seulement] les cinq *'foncteurs d'inversion'* p. 108 [page 176]

$$\Phi,\ \Psi,\ \Phi_0,\ \Psi_0,\ C,$$

qui fournissent des quasi-inverses des foncteurs sur les catégories localisées déduits par les *foncteurs d'inclusion.*

③ **Les quatre types de catégories** HOT_W**.**

Sur chacune des catégories de l'etage, il y a *deux* localiseurs à regarder,

$$\begin{array}{c} W_S^{\mathrm{f}} \quad = W_S^{\mathrm{g}} = W_S^{\mathrm{d}} = W_S^{\mathrm{univ}} \\ \Big\Downarrow \\ W_S, \end{array}$$

donnant lieu à deux localisés distincts, notés

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) \qquad \text{et} \qquad \mathrm{HOT}_W^{\mathrm{lc!}}(S),$$

il se trouve que (grâce à Φ_0 et Ψ_0) ces localisés 'sont les mêmes' pour les 8 catégories de l'étage, à l'exception de $\underline{\mathrm{Parf}}_\omega S$; plus exactement, les foncteurs d'inclusion (à l'exception peut-être de ceux ayant pour source $\underline{\mathrm{Parf}}_\omega S$) induisent des équivalences sur les deux types de localisés, les équivalences quasi-inverses étant obtenues par Φ_0 (aile gauche) ou Ψ_0 (aile droite). Quand on localise *pour* W_S (à l'exclusion de W_S^{f}), on trouve aussi des quasi-inverses par $C|\underline{\mathrm{Liss}}_{0,W}S$ resp. $C|\underline{\mathrm{Prop}}_{0,W}S$, pour les deux inclusions

$$\underline{\mathrm{Liss}}_{0,W}S \longleftarrow \underline{\mathrm{Parf}}_\omega S \longhookrightarrow \underline{\mathrm{Prop}}_{0,W}S$$

([86]). Par raison de symétrie, le plus joli semble de définir

$$\left\{ \begin{array}{lcl} \mathrm{HOT}_W^{\mathrm{lc}}(S) & = & (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1} \\ \mathrm{HOT}_W^{\mathrm{lc!}}(S) & = & (\underline{\mathrm{Fib}}_W S)(W_S)^{-1}, \end{array} \right.$$

et on a un foncteur de localisation

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) \xrightarrow{\lambda} \mathrm{HOT}_W^{\mathrm{lc!}}(S)$$

induit par l'inclusion

$$W_S^{\mathrm{f}} \subseteq W_S.$$

Commentary · Exposition [ms. 176]

Why four localizations appear

Grothendieck's large diagram organizes choices of representatives. The smaller categories make constructions explicit; the larger ones make stability properties easier to state. The object represented should not depend on a convenient replacement, but that independence must be proved for the particular class of arrows one inverts.

There are two first choices. Localizing all of Cat/S by the under-comma test gives $\mathrm{HOT}_W(S)$; using the over-comma test gives $\mathrm{HOT}_W^o(S)$. The replacements Φ and Ψ make these two categories

85. NB la nouvelle notation serait plutôt $\underline{\mathrm{Fibf}}_W S$, car il s'agit de W-fibrations qu'à présent j'appelle 'faibles' (et définies seulement dans Cat, à l'opposé des W-fibrations, définissables dans toute catégorie de modèles).

86. cf. plus bas, ce qu'on peut dire pour les localisés pour W_S^{f}.

accessible through smooth and proper models respectively. Their symmetry comes from taking opposite categories, not from an identification of the two comma tests on arbitrary objects over S.

The weakly fibred objects make a second pair of choices possible. On $\underline{\mathrm{Fib}}_W S$, all the fine tests agree, so one may form

$$\mathcal{L}_S = \underline{\mathrm{Fib}}_W S[(W_S^{\mathrm{f}})^{-1}].$$

This retains the homotopy type of each fibre together with its transport. One may instead invert every map of weakly fibred objects whose total functor belongs to W:

$$\mathcal{G}_S = \underline{\mathrm{Fib}}_W S[W_S^{-1}].$$

The notation in the text is $\mathcal{L}_S = \mathrm{HOT}_W^{\mathrm{lc}}(S)$ and $\mathcal{G}_S = \mathrm{HOT}_W^{\mathrm{lc!}}(S)$. The exclamation mark records that this second construction may identify more arrows. It does not add an extra adjunction or an additional kind of pointwise constancy.

Since $W_S^{\mathrm{f}} \subseteq W_S$ on weakly fibred objects, the identity functor induces $\lambda : \mathcal{L}_S \to \mathcal{G}_S$. This is a localization by its universal property. Namely, a functor out of $\mathcal{L}_S$ factors through λ precisely when it inverts the images of all total W-equivalences between weakly fibred objects. No full faithfulness of λ follows from this description.

The replacement C explains why the coarse category also has the presentation

$$\mathcal{G}_S \simeq (\mathrm{Cat}/S)[W_S^{-1}].$$

It sends every object into $\underline{\mathrm{Parf}}_\omega S \subseteq \underline{\mathrm{Fib}}_W S$, and its natural unit is a total W-equivalence. Therefore both composites of the inclusion and replacement become naturally isomorphic to the identity after coarse localization. The same proof applies to a full intermediate category containing the image of C. For fine localization, however, the proof requires a fibrewise unit; the coarse unit cannot be used in its place.

For Φ_0 and Ψ_0 the necessary finer comparison is available on $\underline{\mathrm{Fib}}_W S$. This is why the two wings of the upper diagram still describe $\mathcal{L}_S$, whereas the perfect vertex is temporarily excluded from that assertion. The pattern is not an arbitrary list of exceptions. Each exception records a precise natural transformation whose known class is too coarse for the desired localization.

The diamond on p. 178 consequently says $\rho i = \lambda = \sigma j$, up to the canonical identifications of presentations. Calling i and j inclusions expresses the intended role of locally constant objects inside the two directional theories. Their full faithfulness still requires an argument. The reflection formulas established after W(4b) is imposed provide such an argument; the shape of the diamond alone does not.

p. 109
[page 177]

Pour toute sous-catégorie pleine

$$\mathcal{F} \subseteq \underline{\mathrm{Fib}}_W S$$

qui contient l'une des trois sous-catégories

$$\underline{\mathrm{Fib}}_{0,W} S,\ \underline{\mathrm{Parf}}_\omega S,\ \underline{\mathrm{Prop}}_{0,W} S,$$

l'inclusion de $\mathcal{F}$ dans $\underline{\mathrm{Fib}}_W S$ définit des équivalences

$$\mathcal{F}(W_S)^{-1} \approx \mathrm{HOT}_W^{\mathrm{lc!}}(S),$$

et si $\mathcal{F}$ contient l'une des deux sous-catégories $\underline{\mathrm{Fib}}_{0,W} S$, $\underline{\mathrm{Prop}}_{0,W} S$ (à l'exclusion de $\underline{\mathrm{Parf}}_\omega S$), on trouve aussi

$$\mathcal{F}(W_S^{\mathrm{f}})^{-1} \approx \mathrm{HOT}_W^{\mathrm{lc}}(S).$$

Le rez-de-chaussée contribue deux autres variantes de HOT, en localisant soit par W_S^{d} (aile gauche), soit par W_S^{g} (aile droite). Les foncteurs d'inclusion définissent des équivalences pour les localisées soit pour W_S^{d} (aile gauche), soit pour W_S^{g} (aile droite). Le plus joli (symétrie) semble de définir ces deux localisés par Cat/S

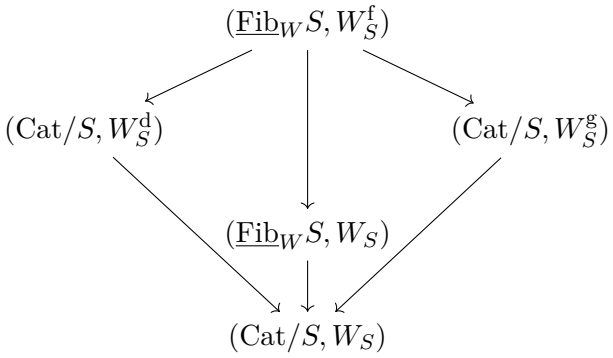
$$\mathrm{HOT}_W(S) = (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1}, \qquad \mathrm{HOT}_W^{o}(S) = (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}.$$

Les quasi-inverses pour les équivalences de catégories localisées, induites par les foncteurs d'inclusion, sont donnés soit par Φ (aile gauche), soit par Ψ (aile droite).

Mais Φ et Ψ fournissent des quasi-inverses également quand on localise pour W_S, la W-équivalence grossière au lieu de la fine - pareil qu'à l'étage. On trouve donc au rez-de-chaussée un (unique) troisième localisé, qu'on peut décrire plus joliment par

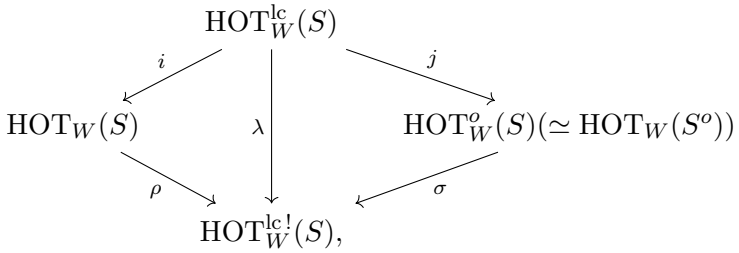
$$(\mathrm{Cat}/S)(W_S)^{-1},$$

mais il est équivalent à $(\underline{\mathrm{Fib}}_W S)W_S^{-1}$ par le foncteur induit par l'inclusion. (Un quasi-inverse étant fourni par C.) C'est donc $\mathrm{HOT}_W^{\mathrm{lc}!}(S)$. On trouve ainsi, par le diagramme commutatif de catégories avec localiseurs [page 178]



(où la flèche verticale courte induit une équivalence sur les localisés), le diagramme commutatif entre les catégories HOT_W liées à S : p. 110

$(*)$



où ρ, σ, λ sont des foncteurs de localisation, et où i, j sont ressentis comme des foncteurs d'inclusion. Ce diagramme $(*)$ semble épuiser les foncteurs naturels qui relient entre elles les quatre catégories envisagées.

Les propriétés des foncteurs Φ, Ψ, C qui ont été utilisées, pour fournir tous les quasi-inverses voulus de foncteurs (entre catégories localisées) induits par les foncteurs d'inclusion du diagramme, sont les suivantes. [page 179]

Commentary · Development (commentators') [ms. 178]

An explicit failure of faithfulness for the coarse comparison

Grothendieck's four-category diagram raises a question about arrows, not only about which objects become isomorphic. The coarse localization can erase distinct automorphisms of a locally constant family. We make that loss explicit for the connected-components localizer. The covering obstruction discussed at manuscript p. 43 already distinguished a total equivalence from a fibre equivalence; the calculation here follows the comparison functor λ itself and determines the localization of a whole class of examples.

Let W_0 be the class of functors inducing bijections on connected components. It satisfies the basic localizer axioms. Saturation and the terminal-object condition follow directly from the functor $\pi_0 : \mathrm{Cat} \to \mathrm{Set}$. For the relative axiom, given $X \to S$, there is a natural bijection

$$\pi_0 X \cong \operatorname*{colim}_{s \in S} \pi_0(X/s).$$

To see this without a homotopy-colimit theorem, send the class of an object $(x, p(x) \to s)$ to the component of x. Every component occurs by using $(x, \mathrm{id}_{p(x)})$. The relations induced by morphisms in a slice, and by arrows of S, preserve that component. Conversely, every object $(x, p(x) \to s)$ is identified in the colimit with $(x, \mathrm{id}_{p(x)})$, and every arrow $x \to x'$ gives the corresponding identification through the slice over $p(x')$. These observations give inverse maps. A pointwise bijection of the slice-component diagrams therefore induces a bijection on total components, proving W(2).

Proposition (orbit collapse).[ours] Let G be a nontrivial group and $S = BG$. For $W = W_0$, the functor

$$\lambda : \mathrm{HOT}^{\mathrm{lc}}_{W_0}(BG) \longrightarrow \mathrm{HOT}^{\mathrm{lc!}}_{W_0}(BG)$$

is not faithful. On the full category of discrete fibrations over BG associated with right G-sets, its fine and coarse source presentations are respectively the category of right G-sets and the category of sets; the induced functor is the orbit functor.

Proof. For a right G-set E, let $\int E \to BG$ be its category of elements. Its objects are elements of E, and a morphism over $g \in G$ with target e' has source $e'g$. Unique lifting, and the invertibility of g, make the projection both a discrete fibration and a discrete opfibration. The categories of prescribed lifts are singletons, so these objects are W_ω-perfect and in particular weakly W_0-fibred. Functors over BG between them correspond exactly to equivariant maps of G-sets.

The strict fibre of $\int E$ is the discrete set E, while its set of total connected components is E/G. It follows that a map induced by $h : E \to F$ is fine precisely when h is a bijection, and coarse precisely when $E/G \to F/G$ is a bijection. A bijective equivariant map is already an isomorphism. Hence fine localization of this category changes nothing.

For the coarse localization, write $O(E) = E/G$, and give every set A the trivial G-action $T(A)$. The natural equivariant map

$$q_E : E \longrightarrow T(O(E))$$

induces the identity on orbit sets. Thus every q_E is inverted by coarse localization, while $OT = \mathrm{id}_{\mathrm{Set}}$. The induced functors O and T are consequently quasi-inverse equivalences after that localization: TO is naturally isomorphic to the identity through the inverted q_E. Under this equivalence the fine-to-coarse functor is exactly O.

It remains to verify that the loss of faithfulness persists in the full categories of the proposition, rather than only in the restricted example category. Take $E = G$ with its regular right action. For each $h \in G$, left multiplication $L_h(g) = hg$ is equivariant and defines an endomorphism of $\int G$ over BG. Evaluation on the strict fibre followed by π_0 is a functor

$$\underline{\mathrm{Fib}}_{W_0}(BG) \longrightarrow \mathrm{Set}$$

which inverts every fine equivalence. It therefore factors through $\mathrm{HOT}^{\mathrm{lc}}_{W_0}(BG)$. Its values on the L_h are distinct set maps, so these endomorphisms remain distinct in the fine localization. On the other hand, $q_G : G \to \{*\}$ is coarse, because both orbit sets are singletons. The equalities $q_G L_h = q_G$ and invertibility of q_G in the coarse localization imply that every $\lambda(L_h)$ is the identity. Thus λ is not faithful. □

This also gives a completely explicit failure of W(4b), even on the perfect objects used in its first weakened variant: $\int G \to BG$ is a total W_0-equivalence, whereas its fibre map $G \to \{*\}$ is not. The proposition does not assert that the full coarse category is always a category of sets. The equivalence with sets was proved for the displayed category of discrete fibrations; nonfaithfulness in the full category was established separately by fibre evaluation.

We checked the localizer discussion in (Cisinski, 2004) and the locally constant formalism in (Cisinski, 2009). We did not find this exact calculation of the manuscript's λ there. The argument is an elementary development of familiar action-groupoid and localization constructions, with no claim of priority.

ⓐ On a un homomorphisme fonctoriel dans Cat/S

$$X \xrightarrow{i_X} \Phi(X), \qquad \underline{i_X \in W_S^{\mathrm{d}}}.$$

Quand X est dans $\underline{\mathrm{Fib}}_W S$, $\Phi(X)$ est dans $\underline{\mathrm{Fibsc}}_0 S$ `[plutôt` $\underline{\mathrm{Fibsc}}_{0,W} S$`, cf. diag. p. 175]`, donc $i_X \in W_S^{\mathrm{f}}$ ($= W_S^{\mathrm{d}}$ sur $\underline{\mathrm{Fib}}_W S$). Ainsi Φ induit le foncteur Φ_0 du diagramme.

Dualement, on a

$$X \xrightarrow{i_X} \Psi(X), \qquad \underline{i_X \in W_S^{\mathrm{g}}},$$

et quand X est dans $\underline{\mathrm{Fib}}_W S$, $\Psi(X)$ est dans $\underline{\mathrm{Cofibsc}}_0 S$ [plutôt $\underline{\mathrm{Cofibsc}}_{0,W} S$], d'où le foncteur Ψ_0.

ⓑ On a un homomorphisme fonctoriel

$$\left|\begin{array}{l} X \xrightarrow{i_X} C(X), \qquad \underline{i_X \in W_S} \\ \text{et } C(X) \in \mathrm{Ob}\,\underline{\mathrm{Parf}}_\omega S. \end{array}\right.$$

||| Mais on fera attention qu'en l'absence de l'axiome W(4b) sur W, on ne peut *pas* affirmer $i_X \in W_S^{\mathrm{f}}$, même si X est dans $\underline{\mathrm{Fib}}_W S$ (voire, dans $\underline{\mathrm{Parf}}_\omega S$), seul cas (de toutes façons) où on est en droit de s'y attendre. Je note à ce propos le diagramme *cartésien,* pour $f : X \longrightarrow Y$ dans Cat/S p. 111

$$\begin{array}{ccc} C(Y)_s & \xleftarrow{C(f)_s} & C(X)_s \\ {\scriptstyle \pi^s_Y}\downarrow & & \downarrow{\scriptstyle \pi^s_X} \\ Y & \xleftarrow{\quad f \quad} & X, \end{array} \qquad \text{flèches verticales dans } \underline{\mathrm{Parf}}_\omega \subseteq \underline{\mathrm{Fib}}_W$$

qui montre par exemple que l'on ne peut pas s'attendre que $f \in W_S \implies C(f) \in W_S^{\mathrm{f}}$ sans [page 180] disposer de l'axiome W(4a′) [qu'est-ce que c'est?] (ou d'un axiome de force comparable, tel W(4b)).

Ce sont ces faits négatifs (en l'absence de W(4b)) qui sont cause que je ne peux rien dire de particulier sur les foncteurs induits par *localisation fine* $(W_S^{\mathrm{f}})^{-1}$ des deux foncteurs d'inclusion

$$\underline{\mathrm{Liss}}_0 S \longleftarrow \underline{\mathrm{Parf}}_\omega S \longrightarrow \underline{\mathrm{Prop}}_0 S,$$

ou, ce qui revient au même, de celui induit par le composé commun de ceux-là avec les inclusions dans $\underline{\mathrm{Fib}}_W S$

$$\begin{array}{ccccc} & & \underline{\mathrm{Parf}}_\omega S & & \\ & \swarrow & \downarrow & \searrow & \\ \underline{\mathrm{Liss}}_0 S & & & & \underline{\mathrm{Prop}}_0 S \\ & \searrow & \downarrow & \swarrow & \\ & & \underline{\mathrm{Fib}}_W S, & & \end{array}$$

en d'autres termes, je ne peux rien dire sur les propriétés de fidélité du foncteur

$$(\underline{\mathrm{Parf}}_\omega S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1},$$

et encore moins que ce soit une équivalence de catégories. Donc, en l'absence de W(4b), pour les sous-catégories pleines $\mathcal{F}$ de $\underline{\mathrm{Fib}}_W S$ qui contiennent $\underline{\mathrm{Parf}}_\omega S$, sans contenir ni $\underline{\mathrm{Liss}}_0 S$ ni $\underline{\mathrm{Prop}}_0 S$, il se pourrait a priori que les localisées

$$\mathcal{F}(W_S^{\mathrm{f}})^{-1}$$

soient toutes différentes. Je pense notamment à $\underline{\mathrm{Parf}}_W S$, aussi intéressante en elle-même, après tout, que $\underline{\mathrm{Parf}}_\omega S$. Donc en plus des quatre localisées du diagramme (∗), p. 178, il y a toute une série infinie 'tératologique' de localisées qui refusent de se laisser 'boucler' [?] par [page 181] Φ_0 ou Ψ_0, et pour lesquelles même C n'est d'aucun secours, faute de l'axime W(4b).

Commentary · Exposition [ms. 180]

Which replacement actually gives a quasi-inverse

Grothendieck now tests the tools used to draw the preceding diagram. The decisive information is carried by the natural maps into the replacements. A replacement whose value belongs to a desirable subcategory gives an equivalence of localizations only when its comparison map lies in the localizer used on both sides.

For Φ the comparison $X \to \Phi(X)$ is right-fine on all objects. If X is weakly W-fibred, its value is a split locally constant fibration, and the equivalence of fine tests on weakly fibred objects makes the comparison fibrewise. The same reasoning for Ψ uses the left-fine comparison and the proper side. These statements explain why the upper wings remain equivalent after fine localization. They are applications of the localization comparison lemma already discussed in the commentary at manuscript p. 12, rather than a new assertion that localization preserves arbitrary full embeddings. At the split vertices one also uses the free split-fibration adjunction discussed in the commentary at manuscript pp. 53–59; the original unit need not preserve the chosen splitting.

The map $X \to C(X)$ comes with a weaker recorded guarantee: it belongs to W_S. Its target is perfect, but perfection of the target alone does not make the comparison fine. The Cartesian square displayed on p. 179 isolates the issue even more sharply. For a total W-equivalence $u : X \to Y$, the map $C(u)_s$ is obtained by pulling u back along the perfect map $C(Y)_s \to Y$. To conclude that $C(u)_s \in W$ one needs a suitable preservation theorem for that pullback. Two-out-of-three for the total units only proves $C(u) \in W_S$; it supplies no missing fibre comparison.

There are two different questions here. One asks whether C turns coarse equivalences into fine ones. The other asks whether the inclusion of perfect objects induces an equivalence after fine localization. Failure of the first proposed proof does not establish a negative answer to the second question. Another replacement, or an additional limit argument, could still produce the required fine comparisons. This is exactly what the alternating construction on pp. 192–197 is designed to investigate.

Accordingly, the "tératologique" family of localizations is a possibility recorded during the investigation, not a theorem that infinitely many pairwise inequivalent categories have been constructed. The source supplies no family of counterexamples on these pages. Modernly, the alternating replacement settles this particular question: the results explained at p. 172 make its values W_ω-perfect and its units fine on every weakly W-fibred input. Every full intermediate category containing $\underline{\mathrm{Parf}}_\omega S$ therefore has the same fine localization as $\underline{\mathrm{Fib}}_W S$, by the common replacement and its units. The proof is given at p. 195. This does not identify fine and coarse localization: the connected-components counterexample still applies to λ.

④ Cas où l'axiome W(4b) est satisfait.

Alors les perplexités que je viens de dire disparaissent. On a aussitôt, en vertu de W(4b) : p. 112

$$\Big|\Big| \qquad \text{Si } X \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_W S, \text{ alors } i_X : X \longrightarrow C(X) \text{ est dans } W_S^{\mathrm{f}} \quad (\textit{en supposant l'axiome } \mathrm{W}(4\mathrm{b}))$$

et C fournit des quasi-inverses pour les foncteurs 'd'inclusion' entre localisés fins (pour W_S^{f}) envisagés à la page précédente. Il en fournit `['Il en fournit' barré]` un également pour le foncteur λ envisagé plus haut ((∗) page 178)

$$\underbrace{(\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1}}_{\overset{\text{déf}}{=}\mathrm{HOT}^{\mathrm{lc}}_W(S)} \longrightarrow \underbrace{(\mathrm{Cat}/S)(W_S)^{-1}}_{\simeq \mathrm{HOT}^{\mathrm{lc}\,!}_W(S)},$$

lequel est donc une équivalence. Une autre façon, plus évidente, de dire la même chose, c'est que sur chacune des catégories de l'étage les deux localiseurs W_S^{f} et W_S sont désormais identiques (puisqu'elles sont toutes contenues dans $\underline{\mathrm{Fib}}_W S$, où elles `[plutôt ils]` le sont justement par l'axiome W(4b)), donc le foncteur λ entre ces deux localisés (définition originale de λ, dans le cas $\mathcal{F} = \underline{\mathrm{Fib}}_W S$, cf. page 176) est même le *foncteur identique,* les deux catégories localisées étant *égales :*

$$\mathrm{HOT}^{\mathrm{lc}}_W(S) = \mathrm{HOT}^{\mathrm{lc}\,!}_W(S) \qquad !$$

Ainsi le diagramme (∗) p. 178 prend la forme

(∗)
$$\begin{array}{ccccc} & & \mathrm{HOT}^{\mathrm{lc}}_W(S) & & \\ & \overset{\rho}{\nearrow}\!\!\underset{i}{\swarrow} & & \overset{\sigma}{\nwarrow}\!\!\underset{j}{\searrow} & \\ \mathrm{HOT}_W(S) & & & & \mathrm{HOT}^{o}_W(S). \end{array}$$

Commentary · Exposition [ms. 181]

The fibre axiom closes the comparison

The force of W(4b) is now visible in one line. For a weakly W-fibred object X, both X and $C(X)$ are in $\underline{\mathrm{Fib}}_W S$, and the natural map $i_X : X \to C(X)$ is a total W-equivalence. W(4b) makes it a fibrewise W-equivalence. This is the missing comparison in the preceding block.

It is useful to check that this gives a functor on the desired localizations, not only good objects. If $u : X \to Y$ is fine, naturality gives

$$C(u)i_X = i_Y u.$$

On each fibre, i_X and i_Y are W-equivalences. Two-out-of-three therefore makes $C(u)$ fine. The functor C thus descends to fine localizations. On the full category of weakly fibred objects and on every indicated intermediate full subcategory containing its perfect image, the two composites with inclusion are naturally isomorphic to the identity through the same units. This proves the claimed equivalences.

For λ an even shorter argument is available. The two presentations are localizations of the very same category $\underline{\mathrm{Fib}}_W S$, and W(4b) identifies their two classes of denominators. Choosing the same localization construction on each side makes λ literally the identity. Without that choice, the invariant conclusion is its canonical equivalence. Grothendieck's exclamation mark after the displayed equality expresses this formal simplification, not an additional theorem about particular models of localization.

The reduction from four categories to three consequently has an exact hypothesis. The connected-components example above shows that it cannot be asserted for every localizer satisfying only the basic axioms. The next step is to identify the remaining maps as adjoints and to prove that the locally constant category sits fully faithfully inside the two directional theories. Those assertions require the mapping-property calculation on pp. 182–183; they do not follow from drawing a triangle in place of the diamond.

Quand on interprète non seulement les deux catégories latérales $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^o_W(S)$, mais aussi celle du milieu, comme des localisées de Cat/S, pour les localiseurs W^{d}_S, W^{g}_S et W_S respectivement, les foncteurs ρ, σ sont ceux déduits des deux inclusions [page 182]

$$\begin{array}{ccc} W^{\mathrm{d}}_S & & W^{\mathrm{g}}_S \\ & \searrow \quad \swarrow & \\ & W_S & \end{array} \qquad \text{comme parties de } \mathrm{Fl}(\mathrm{Cat}/S),$$

tandis que les foncteurs i, j sont déduits, par passage aux localisés, d'un seul et même foncteur

$$\mathrm{Cat}/S \xrightarrow{C} \mathrm{Cat}/S,$$

ce foncteur ayant la vertu d'appliquer W_S dans $W^{\mathrm{d}}_S \cap W^{\mathrm{g}}_S$,

$$f \in W_S \overset{\text{moyennant W(4b)}}{\Longrightarrow} C(f) \in W^{\mathrm{d}}_S \cap W^{\mathrm{g}}_S \qquad (\text{et même } \in W^{\mathrm{univ}}_S).$$

(Et pour cause, puisque C se factorise par $\underline{\mathrm{Fib}}_W S$ (et même par $\underline{\mathrm{Parf}}_\omega S$), où W_S est identique à W^{d}_S, W^{g}_S, W^{univ}_S. Que $f \in W_S$ implique $C(f) \in W$ est indépendant de l'axiome W(4b), et se voit sur le diagramme p. 113

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ {\scriptstyle i_X}\downarrow & & \downarrow{\scriptstyle i_Y} \\ C(X) & \xrightarrow{C(f)} & C(Y), \end{array}$$

où les flèches i_X, i_Y sont dans W_S.)

Ceci dit, il est vrai que les couples

$$(\rho, i),\ (\sigma, j)$$

sont des couples de foncteurs adjoints, donc (ρ et σ étant des foncteurs de localisation) que i et j sont *pleinement fidèles.* La formule d'adjonction peut se lire comme

$$(*) \qquad \boxed{\mathrm{Hom}_{W_S^{\mathrm{g}}}(X, CY) \simeq \mathrm{Hom}_{W_S^{\mathrm{d}}}(X, CY) \simeq \mathrm{Hom}_{W_S}(X, Y)}$$

pour X, Y dans Cat/S sans plus. Un Hom_Σ indique le Hom dans $(\mathrm{Cat}/S)\Sigma^{-1}$. [page 183]

Corollaire. *Si X dans Cat/S, Y dans $\underline{\mathrm{Fib}}_W S$, alors*

$$\begin{array}{ccc} \mathrm{Hom}_{W_S^{\mathrm{g}}}(X, Y) & & \mathrm{Hom}_{W_S^{\mathrm{d}}}(X, Y) \\ & \searrow^{\sim} \quad \swarrow_{\sim} & \\ & \mathrm{Hom}_{W_S}(X, Y) & \end{array}$$

sont bijectifs.

En effet, Y dans $\underline{\mathrm{Fib}}_W(S)$ implique que $Y \xrightarrow{i_Y} C(Y)$ est dans $W_S^{\mathrm{univ}} \subseteq W_S^{\mathrm{d}} \cap W_S^{\mathrm{g}}$, donc

$$\mathrm{Hom}_{W_S^{\mathrm{g}}}(X, Y) \xrightarrow{\sim} \mathrm{Hom}_{W_S^{\mathrm{g}}}(X, CY) \underbrace{\xrightarrow{\sim}}_{\text{adjonction}} \mathrm{Hom}_{W_S}(X, Y),$$

et itou pour $\mathrm{Hom}_{W_S^{\mathrm{d}}}(X, Y)$.

⑤ Perplexités sur les variantes affaiblies de l'axiome W(4b).

Que reste-t-il de toutes ces belles choses (plus que trois catégories HOT, au lieu de $4 + \infty$!) quand on ne suppose plus que W(4b′) au lieu de W(4b), i.e. l'égalité de W_S et W_S^{f} dans $\underline{\mathrm{Parf}}_\omega S$? Finalement, il me semble (contrairement à ce que j'avais cru voir) qu'il ne reste presque rien ([87]). On ne sait alors toujours pas, pour X dans $\underline{\mathrm{Fib}}_W S$, si p. 114

$$i_X : X \longrightarrow C(X)$$

n'est dans W_S^{f}, et non seulement dans W_S; on ne le sait que pour $X \in \underline{\mathrm{Parf}}_\omega S$, mais dans ce cas on n'en a pas besoin, car l'utilité du foncteur $C(X)$ et de l'inclusion $i_X : X \longrightarrow C(X)$, c'est 'd'améliorer' X et de le rendre parfait, voire même W-parfait. Mais s'il l'est déjà, c'est plus la peine. Il y aura, comme avant, 4 localisées essentiellement distinctes HOT_W, HOT_W^o, $\mathrm{HOT}_W^{\mathrm{lc}}$, $\mathrm{HOT}_W^{\mathrm{lc}!}$, plus une infinité d'autres $\mathcal{F}(W_S^{\mathrm{f}})^{-1}$ pour $\mathcal{F}$ entre $\underline{\mathrm{Parf}}_\omega S$ et $\underline{\mathrm{Fib}}_W S$ (cf. page 180), avec le seul bénéfice qu'on peut maintenant exclure le cas $\mathcal{F} = \underline{\mathrm{Parf}}_\omega S$, car dans ce cas $\mathcal{F}(W_S^{\mathrm{f}})^{-1} = \mathcal{F}(W_S)^{-1} = \mathrm{HOT}_W^{\mathrm{lc}!}(S)$ fait partie des quatre canoniques. La chose décisive, c'est qu'on n'a toujours aucun moyen de prouver que [page 184]

$$\lambda : \mathrm{HOT}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}!}(S)$$

est une équivalence, et je doute que ce soit le cas. Il faudrait regarder avec $W = W_{\mathbf{D}}$ la $\mathbf{D}$-équivalence, où $\mathbf{D}$ est un dérivateur $\mathrm{Der}(k\text{-Mod})$, k un anneau donné (p.ex. $k = \mathbf{Z}$, ou k un corps $\mathbf{Z}/p\mathbf{Z}$ ou $\mathbf{Q}$ …). Je ne vais pas le faire maintenant cependant. (Il est vrai que l'axiome W(4b′) n'est pas non plus satisfait dans ce cas …)

87. **NB** Si W satisfait l'axiome des limites, alors W(4b′) implique W(4b), cf. théorème récapitulatif dans XVI 4.4.13 (p. 58-62), notamment 2°). La condition W(4b) est la condition B de cet énoncé, et W(4b′) la condition D. (Elle est impliquée par W(4a) = C, ce que je soupçonnais nullement en écrivant XII …)

Commentary · Exposition [ms. 183]

Local objects absorb the extra denominators

Grothendieck has reached the point where the three remaining homotopy categories can be compared by their maps, rather than by successive descriptions of their objects. This is the purpose of the boxed formula. Its force is that the source X is arbitrary: local constancy is imposed on the target, where it supplies the extension property needed to invert the additional arrows.

Write $\mathcal{H}_d(S) = (\mathrm{Cat}/S)[(W_S^{\mathrm{d}})^{-1}]$ and $\mathcal{H}_c(S) = (\mathrm{Cat}/S)[W_S^{-1}]$. Under W(4b), the functor C represents a fully faithful right adjoint i to the localization $\rho : \mathcal{H}_d(S) \to \mathcal{H}_c(S)$. For X, Y in Cat/S, the adjunction gives

$$\mathrm{Hom}_{\mathcal{H}_d(S)}(X, CY) \simeq \mathrm{Hom}_{\mathcal{H}_c(S)}(X, Y).$$

This formula is natural in both variables. In particular, it identifies the map induced by localization, after the canonical identification of CY with Y in $\mathcal{H}_c(S)$; it is not merely an abstract bijection between two sets of the same size. The dual localization gives the other bijection in the text.

To see what changes when Y is already W-fibred, use the comparison $i_Y : Y \to CY$. It belongs to W_S^{f}, and on the category of W-fibred objects this is also W_S^{d} and W_S^{g}. Thus postcomposition by i_Y is a bijection on Hom sets in either lateral category. Composing with the displayed adjunction proves the corollary exactly as stated. Notice that no replacement of the source is required. An arbitrary object can map into a local object, and this map already incorporates all the relations that the coarse localization would introduce.

There is a useful intrinsic reformulation. Let $v : A \to B$ be a coarse weak equivalence, and regard it as a map in $\mathcal{H}_d(S)$. For every W-fibred Y, precomposition induces a bijection

$$v^* : \mathrm{Hom}_{\mathcal{H}_d(S)}(B, Y) \longrightarrow \mathrm{Hom}_{\mathcal{H}_d(S)}(A, Y).$$

Indeed the two sides are the corresponding Hom sets in $\mathcal{H}_c(S)$, where v is invertible. This is the categorical meaning of being local for the additional denominators. It is this property, rather than the mere existence of a map $Y \to CY$, that makes a reflection useful. The distinction was prepared in the commentary at manuscript pp. 64–69; here it becomes an operational formula for computing morphisms.

The hypothesis W(4b) enters at a precise place. Naturality of $X \to CX$ and two-out-of-three show, without W(4b), that C preserves total W-equivalences. To conclude that $C(v)$ is also right-fine, one must pass from a total equivalence between fibred objects to equivalences on their fibres. That is exactly what W(4b) supplies. The following pages investigate whether a smaller family of test objects can supply the same passage.

Si dans l'énoncé de l'axiome, on remplaçait $\underline{\mathrm{Parf}}_\omega S$ par $\underline{\mathrm{Parf}}_W S$, rien d'essentiel ne serait changé - tout au plus je remplacerais $\underline{\mathrm{Parf}}_\omega S$ par $\underline{\mathrm{Parf}}_W S$ dans le brillant diagramme p. 175. Finalement, je suis à présent convaincu qu'il faut tout ou rien : ou bien *quatre* catégories $\mathrm{HOT}(S)$ pour W, en se contentant de W(1, 2, 3), ou bien rien que trois, i.e. pas de $\mathrm{HOT}_W^{\mathrm{lc}!}(S)$ - et pour ça, il ne semble pas qu'on puisse se contenter de moins de W(4b).

Il suffirait, il est vrai, de supposer que $W_S = W_S^{\mathrm{f}}$ sur l'une quelconque des sept catégories [page 185] de l'étage autres que $\underline{\mathrm{Parf}}_\omega S$. Cela impliquerait alors que dans les six autres, on aura $\overline{W_S} = \overline{W_S^{\mathrm{f}}}$ (mêmes saturés forts dans $\mathrm{Fl}(\mathcal{F})$). Un des deux les plus faibles des axiomes en question serait :

Commentary · Present state [ms. 184]

The later annotation changes the question

Grothendieck is asking how much fibre detection is needed to identify the two locally constant theories. The note added to this passage belongs to a later stage of the investigation. In Chapter XVI, Theorem 4.4.13, especially part 2°, he proves an equivalence of the relevant fibration conditions when the localizers concerned are stable under sequential colimits. Its conditions B, C, and D are the later counterparts of the fibre-detection condition, the base-change condition, and the restriction to perfect objects discussed here. The proof uses the alternating replacement of manuscript pages 192–195, together with a factorization argument (Grothendieck, 1991b, §4.4, especially Corollaries 4.4.10–4.4.12 and Theorem 4.4.13).

Thus the apparent weakness of a perfect-object test must be read together with the limit hypothesis. The test cannot be used on an arbitrary weak fibration until a suitable replacement

has been constructed and its unit has been proved fine. These are precisely the tasks undertaken later in the chapter. The annotation supplies a conditional resolution in its historical setting. The modern automatic-colimit result discussed after manuscript p. 195 supplies the limit hypothesis for the standing basic localizers, while the restricted fibre-detection test remains necessary. The references to "XVI 4.9" in notes 95–96 should be located through the statements: in the current electronic edition the relevant replacement results occur in §4.4.

Si $f : X \longrightarrow Y$ est un foncteur *cartésien* de catégories Cat-fibrées et W-fibrées ([88]) sur une autre S, si $f \in W$, alors $f_s \in W$ pour tout $s \in S$,

(et l'autre est la version duale, pour les foncteurs cocartésiens de catégories cofibrées). Sauf erreur, cet axiome doit déjà impliquer que $\overline{W}$ satisfait la version W(4a) la plus forte de toutes (a fortiori W(4b)) des axiomes W(4). Cet axiome me paraît donc à présent beaucoup plus raisonnable que W(4b′), et ceci d'autant plus qu'il se réfère directemment à la description de $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ en termes de foncteurs (ou de 'pseudofoncteurs', si on ne veut supposer les catégories scindées et les foncteurs compatibles aux scindages) $S^o \longrightarrow \mathrm{Cat}$. D'ailleurs, bien qu'il n'en ait pas l'air, cet axiome *est* autodual, puisque la version duale, pour S fixée, est la version directe pour S^o! Cela rend cet axiome encore plus appétissant. Donc pour formuler W(4a′), j'ai envie à présent de laisser tomber $\underline{\mathrm{Parf}}_\omega$, et de faire pareil. Si on garde deux versions W(4a′) (l'ancienne) et W(4a″) (la nouvelle), l'une et l'autre ont la vertu d'impliquer W(4b). Et W(4b′) (nouvelle version) doit impliquer W(4a) (version ultraforte), sinon pour W, du moins pour $\overline{W}$. p. 115

Mes idées étaient tenacement [?] embrouillées jusqu'à aujourd'hui même, car j'avais l'idée que, grâce au mirifique foncteur C, la catégorie $\underline{\mathrm{Parf}}_\omega S$, comme les sept autres catégories de l'étage, pourrait servir à définir $\mathrm{HOT}^{\mathrm{lc}}_W(S)$. Or il n'en est rien, semble-t-il, à moins de supposeer l'axiome W(4b), ou la variante W(4b′) (nouvelle version). L'ancienne version de W(4b′) s'avère parfaitement inutilisable. [page 186]

Commentary · Exposition [ms. 185]

A smaller test family must still contain replacements

Grothendieck is trying to economize the fibre-detection axiom without losing the reflection. The difficulty is structural. Testing total equivalences only between perfect objects says nothing about the map which first makes a given fibred object perfect, unless that map is already known to be a fibrewise equivalence. This is why the first version of W(4b′) does not immediately do the work expected of it.

The issue can be stated without the changing notation. Let $\mathcal{F}$ be a category of objects over S, and let $E \subseteq V$ be two classes of arrows, thought of as fibrewise and total equivalences. Suppose $\mathcal{P} \subseteq \mathcal{F}$ is a full subcategory on which E and V agree. This equality alone does not identify $\mathcal{F}[E^{-1}]$ with $\mathcal{F}[V^{-1}]$. What is additionally needed is a functor $R : \mathcal{F} \to \mathcal{P}$ with a natural map $\eta : \mathrm{id} \to R$ whose components belong to E. Once such a functor exists, the conclusion follows, and in fact the argument proves more. For $u : X \to Y$ in V, naturality and two-out-of-three imply $R(u) \in V$. Since RX, RY lie in $\mathcal{P}$, it follows that $R(u) \in E$. In the square

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ {\scriptstyle \eta_X}\downarrow & & \downarrow{\scriptstyle \eta_Y} \\ RX & \xrightarrow[R(u)]{} & RY \end{array}$$

the other three arrows therefore lie in E, so u does also. Thus $E = V$ throughout $\mathcal{F}$, provided these are the actual two-out-of-three classes under consideration; if only invertibility in the localization was assumed, the conclusion is equality of their strong saturations.

In the present application, $\mathcal{F} = \underline{\mathrm{Fib}}_W S$, $E = W^{\mathrm{f}}_S$, and $V = W_S$. The old perfect-object test does not initially provide the required fine unit. By contrast, a test on split fibrations with W-equivalent

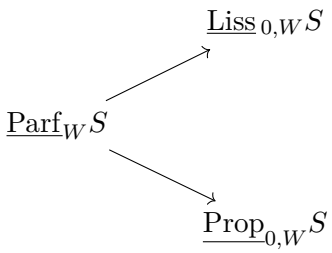
88. 'W-fibrées' signifiant ici que pour $u \in \mathrm{Fl}\, S$, les foncteurs u^* de changement de base $X_{s'} \longrightarrow X_s$, $Y_{s'} \longrightarrow Y_s$ sont dans W.

transport does: the functor Φ takes a W-fibred object into that class, and its unit is fibrewise in W. The infinite alternating construction below provides another such replacement, now into $\underline{\mathrm{Parf}}_W S$, assuming the sequential-colimit axiom.

This explains both the frustration and the eventual simplification. There are two different statements to prove: enough good objects exist, and coarse equivalences between these objects are fine. The first is a replacement statement, the second a detection statement. Neither follows merely from writing down a localization. The proposition on manuscript p. 196 combines them; the omitted adjunction check on p. 197 can then be completed by transferring the already established reflector, as we shall do there.

À ce propos, je n'ai toujours pas répondu clairement pourtant, pas un 'oui' ou un 'non', à la question :

Question : *les inclusions*



induisent-elles, même en l'absence de W(4b) (ou de ses variantes affaiblies) des équivalences pour les localisées fines (par W_S^{f})? Même question pour $\underline{\mathrm{Parf}}_\omega S$ au lieu de $\underline{\mathrm{Parf}}_W S$ ([89]). Même question pour

$$\underline{\mathrm{Bifib}}\, S \subseteq \underline{\mathrm{Parf}}_\omega S,$$

la catégorie des catégories sur S qui sont à la fois Cat-*fibrées et* Cat-*cofibrées.*

Je ne me rends pas compte, même en admettant (cette fois) l'axiome W(4a), si $\underline{\mathrm{Bifib}}\, S$ est assez grosse pour pouvoir décrire tous le types d'homotopie relatifs localement constants sur S, i.e. si

$$(\underline{\mathrm{Bifib}}\, S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1}$$

est au moins essentiellement surjectif, sinon une équivalence de catégories. Il manque un foncteur

$$\mathrm{Cat}/S \longrightarrow \underline{\mathrm{Bifib}}\, S \qquad \text{ou du moins} \qquad \underbrace{\underline{\mathrm{Fib}}_W S}_{\text{ou } \underline{\mathrm{Fib}}_\omega S\,?} \longrightarrow \underline{\mathrm{Bifib}}\, S,$$

qui puisse jouer le rôle de Φ et de Ψ *simultanément.* Je n'ai pas même vérifié que Φ ne transforme *pas* objets de $\underline{\mathrm{Prop}}_W S$ ou de $\underline{\mathrm{Prop}}_{0,W}(S)$ en objets W-propres (donc W-parfaits, étant de toutes façons W-lisse et même W_ω-lisse) sur S ([90]). Si c'était pourtant vrai sur $\underline{\mathrm{Prop}}_W S$, ça montrerait que l'image essentielle de $\underline{\mathrm{Prop}}_W S$ dans $\mathrm{HOT}_W(S)$ $(= (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1})$ est contenu dans $\mathrm{HOT}_W^{\mathrm{lc}}(S)$. Prenant $S = \mathbf{\Delta}^1$, cela signifierait que si dans X on a une sous-catégorie fermée X_1 telle que $X_1 \overset{i}{\hookrightarrow} X$ soit W_ω-asphérique, alors $1\backslash X \longrightarrow 0\backslash X$ est dans W - or $0\backslash X \simeq X$, $1\backslash X \simeq X_1$, la flèche est $X_1 \longrightarrow X$, qui est bien dans W étant W-asphérique! C'est étrange et quasiment immoral ([91]). Mais à force de gribouillis je viens de me convaincre que même si X est cofibrée sur S (donc aussi W-propre), $\Phi(X)$ n'est pas W-propre (ce qui impliquerait W-parfait), ni même un W-fibré, ce qui signifierait que

p. 116 [page 187] [page 188]

89. OK en présence de l'axiome W(6bis) sur les $\varinjlim$, cf. p. 192 ff.
90. Voir réponse à cette intéressante question page 192.
91. On observe plus bas que c'est ainsi chaque fois que S a un objet final.

$$s'\backslash X \longrightarrow s\backslash X \qquad \text{associé à } s \longrightarrow s' \text{ dans } S$$

serait toujours dans W. Posant $Z' = s'\backslash S$, $Z = s\backslash S$, cela signifierait que si on a un *co-isomorphisme local*

$$i : \underbrace{Z'}_{\simeq z_0\backslash Z} \longrightarrow Z,$$

où Z', Z ont des éléments initiaux, et si X est cofibrée sur Z, X' son image inverse sur Z', alors $X' \longrightarrow X$ est dans W. En fait, si Z a aussi un élément *final* e (comme c'était le cas tantôt pour $\mathbf{\Delta}^1$), alors $Z' \simeq z_0\backslash Z$ en a aussi en e', qui s'envoie sur e, et on aura le diagramme commutatif

$$\begin{array}{ccc} X' & \xrightarrow{\ j\ } & X \\ {\scriptstyle W}\uparrow & & \uparrow{\scriptstyle W} \\ X'_{e'} & \xrightarrow{\ \sim\ } & X_e, \end{array}$$

d'où $j \in W$. Donc pour faire un contrexemple, il faut prendre Z sans élément final. [page 189]

Commentary · Present state [ms. 186]

What the bifibration question asks

Grothendieck now asks for a replacement which has both kinds of categorical lifting at once. The question is stronger than the existence of a W-perfect replacement. A Grothendieck bifibration has actual Cartesian and coCartesian lifts. Smoothness and properness ask for suitable categories of lifts to be W-aspherical. The latter condition can hold without a distinguished universal lift, so an equivalence using $\underline{\mathrm{Parf}}_W S$ does not by itself answer the question about $\underline{\mathrm{Bifib}}S$.

One case admits a complete answer with the constructions already at hand. Suppose the base S is a groupoid. Every Grothendieck fibration $p : E \to S$ is then also an opfibration. To prove this, let $u : s \to t$ and $x \in E_s$. Choose a Cartesian lift of u^{-1} with codomain x. Its domain lies over t, and the lift is an isomorphism because its base arrow is invertible: Cartesian lifting of the inverse gives a two-sided inverse by the uniqueness in the Cartesian universal property. The inverse of this lift is therefore a coCartesian arrow over u with domain x. This supplies the required coCartesian lifts for all u and x.

Consequently $\Phi(X) \to S$ is a bifibration for every $X \to S$. If X is W-fibred, its unit $X \to \Phi(X)$ is in W_S^{f}. The functor Φ preserves fine equivalences between such objects: in each fibre it gives a comma functor, and the two fibre-inclusion equivalences identify this with the original fibre map. The same unit, restricted to bifibrations, stays inside that full subcategory. Passing to the fine localizations therefore gives mutually inverse equivalences

$$(\underline{\mathrm{Bifib}}S)[(W_S^{\mathrm{f}})^{-1}] \simeq (\underline{\mathrm{Fib}}_W S)[(W_S^{\mathrm{f}})^{-1}] \qquad (S \text{ a groupoid}).$$

No colimit axiom is used. This includes discrete bases, where the assertion is simply the independent comparison on every fibre.

For an arbitrary base, the proof breaks at exactly the attempted inversion of u. A fibration classified by $F : S^{\mathrm{op}} \to \mathrm{Cat}$ is a bifibration when its transport functors have the appropriate left adjoints, with the coherent structure coming from the liftings. Knowing only that these transport functors lie in W does not produce such adjoints. The sequential construction below proves the fine-localization comparison for $\underline{\mathrm{Parf}}_W S$; it does not equip its output with Cartesian and coCartesian lifts, and therefore does not establish the general $\underline{\mathrm{Bifib}}S$ comparison.

A relevant categorical construction is now available: Clarke et al. (2025, Theorem 1.17) constructs the free cloven bifibration on any functor $p : X \to S$, with a natural unit $X \to \mathrm{Bif}(p)$. Its universal property uses functors preserving the chosen lifts. Propositions 1.23–1.24 make the unit fully faithful and give it a left inverse on total categories. Those conclusions do not make the unit a fibrewise W-equivalence on weakly W-fibred inputs, which is the missing assertion for the present application. The antecedent path construction in Lamarche (2013, §4, pp. 22–23) explicitly distinguishes its own weak equivalences from ordinary nerve weak equivalences. Thus these free constructions cannot be cited as a verified homotopical replacement for this question, even at $W = W_\infty$.

We did not find the precise arbitrary-base, arbitrary-localizer bifibration comparison in the literature checked. This is a limit of the present audit, not an assertion that the problem is open.

> The modern theory of locally constant coefficients gives a homotopy invariant category of such coefficients; for example, Cisinski (2009, Theorem 1.3 and Corollary 1.28) proves invariance under nerve weak equivalences and its detection by locally constant coefficients. That theorem does not assert the strict bifibration presentation asked for here.

Mais prenons Z `[un]` ensemble ordonné, avec un objet initial 0, mais sans objet final, et prenons pour z_0 un élément *maximal* de Z, de sorte que $Z' = z_0\backslash Z$ soit isomorphe à $\{z_0\}$. Il faut donner dans Cat un exemple où l'inclusion de X_{z_0} dans X n'est pas dans W. On va prendre

p. 117

$$Z = \quad 0 \nearrow a \quad 0 \searrow b = z_0,$$

donc la donnée de X cofibrée coscindée sur Z équivaut à un diagramme dans Cat

$$X_0 \xrightarrow{\alpha} X_a, \qquad X_0 \xrightarrow{\beta} X_b.$$

Je pense que sous des conditions assez fortes (p.ex. si α, β sont des immersions ouvertes, ou des immersions fermées - à vérifier!), la catégorie X est isomorphe dans Hot à la somme amalgamée dans Cat

$$X_a \coprod_{X_0} X_b,$$

et il n'y `[a]` aucune raison (sauf si on suppose $\alpha \in W$, et encore …) que $X_b \longrightarrow X_a \coprod_{X_0} X_b$ soit dans W. Prenons $X_0 = X_b = e$, $\beta = \mathrm{id}_e$, donc α est donné par la donnée d'un élément t de $\mathrm{Ob}(X_a = T)$. La somme amalgamée est T, qui n'est pas isomorphe à e dans Hot_W, si T n'a pas été prise W-asphérique. Mais je vais essayer de montrer directement que l'inclusion

[page 190]

$$T = X_a \longrightarrow X$$

est $\in W$. Pour ceci je regarde la sous-catégorie ouverte Z'' de Z, complémentaire de la sous-catégorie pleine fermée Z', et je note que $X'' = X|Z''$ étant propre sur Z'' et a étant objet final de Z'', on a

$$X_a \longrightarrow X'' \qquad \text{dans } W,$$

il reste à voir qu'il en est de même de $X'' \longrightarrow X$. Or X'' est déduite de X en ajoutant un unique élément ξ (celui de X_b), lequel ne s'envoie dans aucun élément de X'', et qui ne reçoit aucun élément de X_a - par contre pour l'unique élément ξ_0 de X_0 il y a une unique flèche de celui-ci dans ξ. On a donc une rétraction de X sur X'', envoyant ξ sur ξ_0 et la flèche précédente sur id_{ξ_0}. Est-ce un homotopisme? Il suffit de `[montrer]` qu'en tant qu'endomorphisme φ de X, c'est $\sim \mathrm{id}_X$. Or on a bien $\mathrm{id}_X \longleftarrow \varphi$, dont la valeur sur X'' est l'identité, et la valeur sur ξ est l'unique $\varphi(\xi) = \xi_0 \longrightarrow \xi$. C'est un peu bref, mais ça doit être vrai. (De toutes façons, X est la réunion de deux ouverts X'', et de l'ouvert $X|\{0, b\} = X'''$, avec $X''' \simeq \mathbf{\Delta}^1$, et $X'' \cap X''' \simeq e$, donc au moins pour la cohomologie à coefficients localement constants de X je suis à l'aise par Mayer-Vietoris, et sais bien que c'est celle du morceau X'', donc celle de X_0 `[ou $X_a$, indice illisible]` nullement celle de e!)

On a donc vu au moins ceci : Si X est cofibrée donc W-propre sur S, il ne s'ensuit nullement que $\Phi(X)$ soit encore W-propre sur S ([92]), ni que ce soit un W-fibré. Mais que se passe-t-il si

p. 118
[page 191]

92. voir résultat plus précis page suivante!

$X \in \underline{\mathrm{Prop}}_{0,W} S$, par exemple s'il provient de $\underline{\mathrm{Cofibsc}}_0 S$? Alors X est dans $\underline{\mathrm{Fib}}_W S$, donc on sait déjà que $\Phi(X)$ est dans $\underline{\mathrm{Fibsc}}_{0,W} S$, donc ce n'est pas par là qu'on fera un contrexemple. Mais considérons encore X cofibrée sur

$$S = \quad \begin{array}{ccc} & & a \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & b, \end{array}$$

donnée par le diagramme

$$\begin{array}{ccc} & & X_a \\ & \overset{u}{\nearrow} & \\ X_0 & & \\ & \underset{v}{\searrow} & \\ & & X_b \end{array}$$

dans Cat, en supposant cette fois $u, v \in W$ ($X \in \underline{\mathrm{Cofibsc}}_{0,W} S$). Alors $\Phi(X)$ est la catégorie fibrée scindée sur S définie par le diagramme contravariant sur S

$$\begin{array}{ccc} & & X_a \\ & \overset{i}{\swarrow} & \\ X_0 & & \\ & \underset{j}{\nwarrow} & \\ & & X_b, \end{array}$$

où i, j sont les inclusions (qui sont des immersions fermées). Dire que $\mathcal{X} = \Phi(X)$ est propre sur S signifie aussi que ses restrictions au dessus des sous-catégories ouvertes $\{0, a\}$ et $\{0, b\}$ de S le sont (car toute flèche $\mathbf{\Delta}^1 \longrightarrow S$ se factorise par l'une ou l'autre). Cela signifie donc aussi que l'inclusion des fibres fermées [page 192]

$$X_a \longrightarrow X, \qquad X_b \longrightarrow X$$

sont coasphériques. Mais $X_a \longrightarrow X$ ne peut être coasphérique que si $X_b = \emptyset$, car pour $x \in X_b \subseteq X$, $x \backslash X_a$ (ainsi que X_a / x) est vide, donc *pas* W-asphérique (si on ne suppose $W = \mathrm{Fl}(\mathrm{Cat})$, cas que nous excluons). Donc $\Phi(X)$ n'est propre sur S *que* si X_a et $X_b = \emptyset$ (résultat valable pour *toute* X sur S, qu'elle soit cofibrée sur S ou W-propre, ou non). Si on veut $u, v \in W$, cela implique donc que $X_0 = \emptyset$, donc on a prouvé :

Commentary · Examples [ms. 190]

A branching base defeats one-sided completion

Grothendieck needs a counterexample which separates transport in one direction from transport in the other. A base with a terminal object hides this distinction: an opfibration over such a base has the homotopy type of its terminal fibre. The branching poset $S = (0 \to a,\ 0 \to b)$ removes that shortcut.

Take a pointed category (T, t), and form the split opfibration classified by

$$e \xrightarrow{t} T, \qquad e \xrightarrow{\mathrm{id}} e.$$

Its total category X has the objects of T, together with objects ξ_0 over 0 and ξ_b over b. The additional arrows are a single arrow $\xi_0 \to \xi_b$ and, for every $z \in T$, arrows $\xi_0 \to z$ indexed by

$\mathrm{Hom}_T(t,z)$. Composition with arrows of T is the evident one. There are no arrows from T to ξ_0 or ξ_b, and none from ξ_b to the other objects.

Let X'' be the full subcategory obtained by deleting ξ_b. There is a retraction $r : X \to X''$ which sends ξ_b to ξ_0 and the arrow $\xi_0 \to \xi_b$ to an identity. If $j : X'' \hookrightarrow X$ denotes the inclusion, then $rj = \mathrm{id}$ and there is a natural transformation $jr \Rightarrow \mathrm{id}_X$, equal to the identity on X'' and to $\xi_0 \to \xi_b$ at ξ_b. Naturality needs checking only on the new arrow, where both composites are that arrow. Thus j is a categorical homotopy equivalence and belongs to every localizer in use here.

There is a second retraction $r_a : X'' \to T$, sending ξ_0 to t and fixing T. In fact r_a is left adjoint to the inclusion $j_a : T \hookrightarrow X''$: the only nontrivial adjunction formula is

$$\mathrm{Hom}_T(r_a\xi_0, z) = \mathrm{Hom}_T(t,z) = \mathrm{Hom}_{X''}(\xi_0, j_a z).$$

Hence $T \to X'' \to X$ belongs to W. The fibre over b is nevertheless e. If $T \to e$ is not in W, the map $X_b \to X$ is not in W, by two-out-of-three. Since $\Phi(X)_0 \simeq X$ and $\Phi(X)_b \simeq e$, transport along $0 \to b$ in the completed fibration is not a W-equivalence. This proves that $\Phi(X)$ need not even be W-fibred, although $X \to S$ is an opfibration and hence proper.

For the ordinary weak-homotopy localizer, a completely finite choice is $T = \{t, t'\}$ discrete. Then X is the disjoint union of the three-object poset $\xi_b \leftarrow \xi_0 \to t$ and the isolated object t'. Its nerve has two contractible components, whereas X_b has one. No homology calculation or pushout theorem is required. For a general W, the argument requires a pointed T which is not W-aspherical; the nondegeneracy assumption $W \neq \mathrm{Fl}(\mathrm{Cat})$ alone should not be substituted for that requirement in this particular example.

The parenthetical pushout question has a positive answer under the suggested symmetric hypotheses. If $A \hookrightarrow B$ and $A \hookrightarrow C$ are both full sieves, their pushout P contains no arrows from $B \setminus A$ to $C \setminus A$, or conversely. Every composable chain lies in one side, so $NP = NB \cup_{NA} NC$. The nerve inclusions are monomorphisms, and this union is the homotopy pushout of the two inclusions. Equivalently one can check the canonical functor from the total category of $B \leftarrow A \to C$ by comma categories: for $p \in B \setminus A$, retract the copy of A/p lying on the C-side onto the middle copy. The resulting category has terminal object p on the B-side. The case $p \in C \setminus A$ is symmetric, and for $p \in A$ the comma category contracts to the three copies of p joined by a span. Theorem A therefore proves the comparison is in W. For two cosieves, apply the dual argument. The special example above is simpler still, since the explicit retractions determine the homotopy type for every allowed W.

p. 119

||| Soit $X \in \mathrm{Cat}/S$, $S = \begin{array}{ccc} & & a \\ 0 & \nearrow & \\ & \searrow & \\ & & b \end{array}$. Si $\Phi(X)$ est propre sur S, alors $X = \emptyset$ (si on suppose $W \neq \mathrm{Fl}(\mathrm{Cat})$).

Mais chemin faisant est apparue une idée bizarre. Considérons le diagramme infini, pour X dans Cat/S :

$$X \longrightarrow \Phi(X) \longrightarrow \Psi\Phi(X) \longrightarrow \Phi\Psi\Phi(X) \longrightarrow \cdots$$

Si X est quelconque, ces flèches sont alternativement dans W_S^{d} et W_S^{g}, donc leurs composées sont dans W_S sans plus. Mais si X est dans $\underline{\mathrm{Fib}}_W S$, comme cette catégorie est *stable* par Φ, Ψ, et que sur $\underline{\mathrm{Fib}}_W S$ $W_S^{\mathrm{d}} = W_S^{\mathrm{g}} = W_S^{\mathrm{f}} = W_S^{\mathrm{univ}}$, on voit que dans ce cas les flèches sont toutes dans W_S^{f}. On a envie maintenant de poser

$$\Phi_\infty(X) = \varinjlim \text{ de tout ça}$$

et on définirait symétriquement [page 193]

$$\Psi_\infty(X) = \varinjlim(X \longrightarrow \Psi(X) \longrightarrow \Phi\Psi(X) \longrightarrow \cdots).$$

Commentary · Development (commentators') [ms. 192]

The obstruction persists on any base with two maximal branches

Grothendieck strengthens the preceding counterexample: over the three-object branching poset, a single application of Φ cannot be proper unless the original category is empty. The obstruction lies in the absence of arrows between the two maximal branches. We can isolate that reason and extend the statement without introducing additional homotopical machinery.

Branching obstruction. [ours] Let S be a poset with a least element 0 and two distinct maximal elements a, b. Let W be a localizer satisfying the standing axioms and suppose that $\varnothing \to e$ does not belong to W. For every $X \to S$, if $\Phi_S(X) \to S$ is W-proper, then $X = \varnothing$.

Proof. The fibre diagram of $\Phi_S(X)$ is $s \mapsto s\backslash X$. Since 0 is least, $0\backslash X$ identifies with X. Since a and b are maximal, $a\backslash X = X_a$ and $b\backslash X = X_b$. The two transport functors to the fibre at 0 are therefore the inclusions $X_a \hookrightarrow X$ and $X_b \hookrightarrow X$.

Properness forces each of these inclusions to be coaspherical. Here is the precise test being used. Pull the fibration back along the arrow $0 \to a$. For an object x of its fibre over 0, the category of arrows from x to the fibre over a is the comma category $x\backslash X_a$. Properness requires it to be W-aspherical. This is the elementary lifting-category criterion for properness, applied to this one arrow; the same statement holds for b.

If $x \in X$ existed, the category $x\backslash X_a$ would be W-aspherical and hence nonempty. Choose one of its objects, an arrow $x \to x_a$ with $x_a \in X_a$. Properness along $0 \to b$, applied now to the object x_a of X, would require $x_a\backslash X_b$ to be W-aspherical. But this category is empty: an arrow $x_a \to x_b$ would project to an arrow $a \to b$, impossible for distinct maximal elements of a poset. This contradicts the hypothesis on the empty category. Thus X has no objects. Conversely, the empty category over S is proper, since all the lifting tests with a specified source object are vacuous. □

For the standing localizers, the nondegeneracy condition in the text implies the displayed hypothesis on $\varnothing \to e$. Indeed, if $\varnothing \to e$ were in W, product stability would put $\varnothing \to A$ in W for every category A. Two-out-of-three would then put every functor in W.

This proof also fills the final step in the boxed assertion on manuscript p. 192. Showing that $X_a = X_b = \varnothing$ is not yet, by itself, a proof that $X_0 = \varnothing$ for an arbitrary $X \to S$. One must use coasphericity once more: $\varnothing \to X$ cannot be coaspherical when X has an object. The argument above incorporates this step directly. The preceding contravariant diagram prints X_0 at its centre; its mathematical value is $X = 0\backslash X$, as both the following inclusions and the proof require. That printed slip is retained in the source and recorded in the correction ledger.

We checked the lifting criteria in Maltsiniotis (2005, Proposition 4.3, Theorem 4.13, and Example 4.17); we did not find this particular branching formulation there. The proposition is an elementary extension of the present manuscript argument, with no claim of priority.

Pour que ce soit utilisable, il faut poser l'axiome W(6 bis) (comparer XII, pages 14, 15). La limite inductive d'une suite de morphismes $f_i : X_i \longrightarrow Y_i$, pour un morphisme de systèmes inductifs dans Cat indexés par $\mathbf{N}$, est dans W si les f_i le sont.

Cet axiome est voisin de W(6) de loc. cit., mais n'est impliqué par lui que moyennant W(4b). Il est plus fort dans le cas de systèmes inductifs indexés par $\mathbf{N}$ (donc aussi, plus généralement, de systèmes inductifs sur des ensembles ordonnés filtrants ayant un ensemble cofinal dénombrable), mais ne dit rien sur les systèmes inductifs sur des ensembles ordonnés d'indices plus généraux.

Cet axiome implique aussi qu'une limite inductive filtrante de morphismes f_i est W-lisse, resp. W-propre, si les f_i le sont. On trouve alors que $\Phi_\infty(X)$ est à la fois W-lisse (comme limite inductive des $\Phi(X)$, $\Phi\Psi\Phi(X)$, …, $\Phi(\Psi\Phi)^n(X)$, lesquelles sont W-lisses sur S) et W-propre (comme limite inductive des $\Psi\Phi(X)$, …, $(\Psi\Phi)^n(X)$ …). D'autre part, que $X \longrightarrow \Phi_\infty(X)$ est dans W_S^{f} - car W_S^{f} est stable par limites inductives filtrantes dénombrables dans Cat/S. On a donc trouvé un foncteur [page 194]

$$\Phi_\infty : \underline{\mathrm{Fib}}_W S \longrightarrow \underline{\mathrm{Parf}}_W S \subseteq \underline{\mathrm{Fib}}_W S$$

et un morphisme fonctoriel

$$i_X : X \longrightarrow \Phi_\infty(X), \qquad \boxed{i_X \in W_S^{\mathrm{f}}} \qquad \text{si } X \in \underline{\mathrm{Fib}}_W S.$$

Ce foncteur permet alors, en l'absence de W(4b), de faire ce que le foncteur C refusait absolument. Il est vrai que C a l'avantage d'être défini sur Cat/S tout entier, et d'y être raisonnable et même fort utile, quand on a W(4b). Mais Φ_∞ est, lui aussi, défini sur Cat/S tout entier, ainsi que i_X p. 120

$$\Phi_\infty : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Parf}}_W S$$
$$i : \mathrm{id} \longrightarrow \Phi_\infty \qquad \text{i.e.} \begin{cases} i_X : X \longrightarrow \Phi_\infty(X) \\ i_X \in W_S, \end{cases}$$

avec le seul handicap (inévitable) que si X n'est pas dans $\underline{\mathrm{Fib}}_W S$, on ne peut dire mieux que $i_X \in W_S$, à l'exclusion de $i_X \in W_S^{\mathrm{f}}$. Mais pour $C(X)$ c'était forcément pareil. Donc on a l'impression que Φ_∞, au choix, Ψ_∞, bonnet blanc, blanc bonnet, joue en présence de W(6 bis) le même rôle que C en présence de W(4b). Tout au plus (en présence des deux axiomes à la fois) le foncteur C a-t-il l'avantage esthétique de prendre ses valeurs dans $\underline{\mathrm{Parf}}_\omega S$, au lieu de $\underline{\mathrm{Parf}}_W S$ seulement ([93]). Mais il s'impose de se rendre compte à présent que Φ_∞ a un [page 195] avantage exactement du même ordre sur C, en introduisant

$$W''_\omega = \text{plus petite partie } \Sigma \text{ de } \mathrm{Fl}(\mathrm{Cat}) \text{ satisfaisant } \mathrm{W}(1, 2 \text{ bis}, 3, 6 \text{ bis})$$

et noter que

$$\Phi_\infty(X) \in \mathrm{Ob}\, \underline{\mathrm{Parf}}_{W''_\omega} S, \qquad i_x \in W''_\omega$$

- mais là je m'aperçois qu'on a

$$W_\omega \subseteq W''_\omega,$$

donc C est quand même un peu meilleur à cet égard …

Commentary · Exposition [ms. 194]

Why alternating completion works at the limit

The failure of one-sided completion suggests Grothendieck's next move. Instead of asking Φ to preserve properness, he restores properness with Ψ, then restores smoothness with Φ, and continues. No finite stage has to possess both properties. At the limit, the odd and even stages are cofinal and can establish the two properties separately.

Put $X_0 = X$, $X_1 = \Phi X$, $X_2 = \Psi\Phi X$, and continue alternately, with the canonical unit at each step. Let $RX = \varinjlim_n X_n$. W(6 bis) means that an objectwise W-equivalence between two sequential diagrams induces a W-equivalence on their ordinary colimits in Cat. Its quantifier is sequential. The unqualified words "limite inductive filtrante" in the next paragraph must be read with this restriction; an arbitrary filtered indexing category is not covered by the axiom as written.

The passage from this axiom to smoothness and properness of $RX \to S$ can be made explicit by the lifting-category criterion. For a functor $p : E \to S$, a base arrow $u : s \to t$, and $y \in E_t$, let $L_p(u, y)$ have objects arrows $v : x \to y$ with $p(v) = u$ and morphisms commuting triangles whose arrow between sources lies over id_s. Smoothness is equivalent to $L_p(u, y)$ being W-aspherical for every u, y. Properness is the dual condition on arrows with a specified source. These are the familiar fibre categories of $E/y \to S/t$ and their duals.

Suppose first that $E = \varinjlim_n E_n$ over the fixed base S, with every $E_n \to S$ smooth. An object y of E is represented by $y_n \in E_n$ at some stage. After restricting to that tail of the sequence, there is a canonical isomorphism

$$L_p(u, y) \simeq \varinjlim_{m \geq n} L_{p_m}(u, y_m).$$

To verify it, represent an arrow to y at a sufficiently late stage; its image in S is already the required fixed arrow u. A morphism of the lifting category is represented by one further arrow and one equality of composites, and that equality holds at a later stage. The same argument detects equality of two such morphisms. Thus the displayed functor is bijective on objects and arrows after taking

93. *En fait* $\Phi_\infty = C$, si C est défini via des chemins infinis d'un seul côté simplement. Si on définit C via les chemins *finis* de tous types c, alors il prend en effet ses valeurs dans $\underline{\mathrm{Parf}}_\omega S$.

the colimit. Every stage maps to e by a member of W, so W(6 bis) makes the colimit W-aspherical. This proves sequential stability of smoothness. Reversing arrows proves stability of properness.

Now the odd subsequence $X_1, X_3, \dots$ consists of smooth categories over S, and the even subsequence $X_2, X_4, \dots$ consists of proper ones. Both have colimit RX, because each is cofinal in the original sequence. Hence RX is W-perfect. Every transition $X_n \to X_{n+1}$ belongs to W_S, so every composite $X \to X_n$ does. Apply W(6 bis) to the natural transformation from the constant sequence at X to (X_n): its colimit is $X \to RX$, proving that this map lies in W_S for arbitrary X.

When X is W-fibred, all stages remain W-fibred and the transitions are fibrewise W-equivalences. Fibres over a fixed s commute with these colimits: their objects and arrows are defined by fixed equations in S, and any equality in a filtered colimit is witnessed at a later stage. Applying W(6 bis) in each fibre yields

$$X_s \longrightarrow (RX)_s \quad \text{in } W \qquad \text{for every } s \in S.$$

Thus the same construction supplies both kinds of unit advertised in the text: a total equivalence for arbitrary objects and a fine equivalence for fibred objects. It is the latter statement that was missing for the earlier proposed perfect-object replacement. Nothing in this argument asserts that $RX \to S$ is a Grothendieck bifibration: a sequential colimit of lifting categories known to be aspherical does not supply actual universal lifts.

Quand on dispose de Φ_∞ avec les propriétés qu'on vient d'expliciter, on trouve que l'inclusion

$$\underline{\mathrm{Parf}}_W S \longrightarrow \underline{\mathrm{Fib}}_W(S)$$

induit une équivalence non seulement pour les localisés pour W_S, mais aussi pour les localisés fins

$$(\underline{\mathrm{Parf}}_W S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W(S))(W_S^{\mathrm{f}})^{-1} \overset{\text{déf}}{=} \mathrm{HOT}^{\mathrm{lc}}_W(S)$$

([94]). Mais on ne peut pour autant réduire le nombre des localisés utiles à trois, en l'absence de W(4b). Tout au plus, ayant un foncteur Φ_∞ qui est plus simple que C, en l'absence de W(4b), du fait qu'on sait de toutes façons que

$$i_X : X \longrightarrow \Phi_\infty(X)$$

est dans W_S^{f} si X est dans $\underline{\mathrm{Fib}}_W S$, peut on espérer s'en tirer avec la variante affaiblie de W(4b) obtenue en se bornant dans l'énoncé de W(4b) (p. 171) au cas où X, Y sont dans $\underline{\mathrm{Parf}}_{W''_\omega} S$. Cette fois, ça suffit en effet à impliquer que le foncteur p. 121 [page 196]

$$\lambda : (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\underline{\mathrm{Fib}}_W S)(W_S)^{-1}$$

est une équivalence de catégories, puisque on peut le 'calculer' aussi bien en utilisant $\underline{\mathrm{Parf}}_{W''_\omega} S$ au lieu de $\underline{\mathrm{Fib}}_W S$, or l'hypothèse W(4b($\mathrm{W}''_\omega$)) implique que sur $\underline{\mathrm{Parf}}_{W''_\omega} S$ les deux localiseurs W_S^{f}, W_S sont *égaux.*

Commentary · Exposition [ms. 195]

Fine replacement gives an equivalence of localizations

Grothendieck can now return to the question on manuscript p. 186 with an actual construction in hand. The essential point is stronger than the existence of a perfect object with the right total homotopy type: for a fibred input, the comparison is an equivalence in every fibre. This is what allows the replacement to control morphisms as well as objects.

Today W(6 bis) is automatic: every basic localizer is closed under filtered colimits (Cisinski, 2006a, Corollary 4.2.22). This applies to the standing relative W(1, 2, 3), without accessibility; we do not extend it here to arbitrary weak basic localizers.

94. On peut y remplacer même $\underline{\mathrm{Parf}}_W S$ par $\underline{\mathrm{Parf}}_{W''_\omega} S$, *ou* par toute autre sous-catégorie pleine de $\underline{\mathrm{Fib}}_W S$ contenant $\underline{\mathrm{Parf}}_{W''_\omega} S$.

Let $R = \Phi_\infty$, let $\mathcal{F} = \underline{\mathrm{Fib}}_W S$, and let $\mathcal{P} = \underline{\mathrm{Parf}}_W S$. Under W(6 bis), the unit $\eta_X : X \to RX$ is fine for every $X \in \mathcal{F}$. The functor R preserves fine equivalences on $\mathcal{F}$. Indeed, if $u : X \to Y$ is fine, then in each fibre the equality

$$(Ru)_s(\eta_X)_s = (\eta_Y)_s u_s$$

and two-out-of-three show that $(Ru)_s$ lies in W. Thus both the inclusion $J : \mathcal{P} \hookrightarrow \mathcal{F}$ and R induce functors after fine localization. The units give natural isomorphisms

$$\mathrm{id} \xrightarrow{\sim} \overline{J}\,\overline{R} \quad \text{on } \mathcal{F}[(W_S^{\mathrm{f}})^{-1}], \qquad \mathrm{id} \xrightarrow{\sim} \overline{R}\,\overline{J} \quad \text{on } \mathcal{P}[(W_S^{\mathrm{f}})^{-1}].$$

For the second assertion, both endpoints of the unit belong to $\mathcal{P}$, and the subcategory is full. These two natural isomorphisms prove that the induced functors are quasi-inverse equivalences. In particular, full faithfulness is part of the result; there is no separate assumption about roofs or a calculus of fractions.

The same proof works with any full intermediate subcategory containing the image of R. The more precise claim in footnote 94 follows because the construction already lands in $\underline{\mathrm{Parf}}_{W''_\omega} S$. One must keep the two minimal classes distinct at this stage of the manuscript. The inclusion $W_\omega \subseteq W''_\omega$ does not, on its own, put the output in $\underline{\mathrm{Parf}}_\omega S$. It puts it in the class named in the footnote.

The minimality comparison above does, however, allow us to settle the stronger annotation in footnote 89. It identifies W_ω with the nerve weak-homotopy equivalences W_∞. This class is stable under sequential colimits. Indeed, the nerve commutes with filtered colimits because each finite ordinal category $[n]$ is finitely presented. Weak equivalences of simplicial sets are stable under such colimits: apply Ex^∞, which commutes with them since every subdivided simplex is finite; a filtered colimit of Kan complexes is Kan, and maps and homotopies from finite simplicial spheres are witnessed at finite stages, giving the required isomorphisms on all homotopy groups and on components. Thus W_∞ satisfies W(6 bis). Minimality gives

$$W_\omega \subseteq W''_\omega \subseteq W_\infty = W_\omega.$$

Consequently the replacement can indeed be taken in $\underline{\mathrm{Parf}}_\omega S$, with the fine unit just proved. This conclusion uses the modern identification of the minimal class; it was not a consequence of the single inclusion written on manuscript p. 195.

The original restricted W(4b′) now suffices as well. Given a total W-equivalence $u : X \to Y$ between weakly fibred objects, naturality of the unit makes Ru a total W-equivalence between $\underline{\mathrm{Parf}}_\omega$-objects. The restricted hypothesis makes Ru fine, and the two fine units then make u fine by two-out-of-three in each fibre. This proves full W(4b), and therefore the fine-to-coarse comparison. It explains precisely how the historical retraction acquires a positive answer once the later colimit and minimality theorems are available.

This replacement equivalence still does not identify the fine and coarse localizations. It allows either of them to be computed on a smaller category; it does not say that the two classes of denominators agree there. That remaining assertion is exactly the restricted W(4b) hypothesis in the following proposition. Grothendieck's distinction between replacing objects and comparing equivalences is therefore preserved even after the successful infinite construction.

Il faut une conclusion pratique utilisable de toutes ces perplexités et observations. Essayons

Proposition. *Supposons que l'on soit sous l'une des conditions suivantes.*

1°) *L'axiome* W(4b) *est satisfait dans le cas de catégories X, Y qui sont dans* $\underline{\mathrm{Fibsc}}_{0,W} S$ ([95]) *(cette condition sur W est en fait autoduale).*

2°) *L'axiome* W(4b) *est satisfait dans le cas de catégories X, Y qui sont dans* $\underline{\mathrm{Parf}}_{W''_\omega} S$*, où W''_ω est le plus petit ensemble $\Sigma \subseteq Fl(\mathrm{Cat})$ satisfaisant les conditions* W(1, 2 bis, 3, 6 bis). *(On peut aussi se contenter de la condition, plus forte mais plus compréhensible, que* W(4b) *est OK si X, Y sont dans* $\underline{\mathrm{Parf}}_W S$ ([96])*.) De plus, on suppose que W satisfait* W(6 bis).

95. mais c'est *équivalent* à W(4b)! Cf. XVI 4.9. loc. cit.

96. Mais cela implique aussi W(4b), cf. loc. cit.

Sous ces conditions, le foncteur correspondant [?] λ

$$\mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc!}}_W(S)$$

est une équivalence de catégories (cf. diagramme p. 178), donc (ρ, i), (σ, j) *peuvent être regardés comme des couples de foncteurs entre deux catégories (cf.* $(*)$ *p. 181).* Ce sont des couples de foncteurs adjoints, *donc* i, j *sont pleinement fidèles, et on a la formule d'adjonction* $(*)$*, page 182.*

Pour prouver l'adjonction de (ρ, i) (ou dualement de (σ, j)), on exprime encore les deux catégories en présence $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ comme $(\mathrm{Cat}/S)(W^{\mathrm{d}}_S)^{-1}$, $(\mathrm{Cat}/S)(W_S)^{-1}$, le foncteur ρ étant la localisation canonique allant du premier vers le second, et le foncteur i étant donné par un foncteur d'inversion [page 197]

$$\gamma : (\mathrm{Cat}/S, W_S) \longrightarrow (\mathrm{Cat}/S, \overline{W^{\mathrm{d}}_S})$$

qui est C dans le cas 1°, Φ_∞ dans le cas 2°. Les morphismes d'adjonction

$$\rho i \longrightarrow \mathrm{id}, \qquad i\rho \longleftarrow \mathrm{id}$$

doivent se décrire alors comme dans le cas où on suppose tout bonnement W(4b), et les compatibilités se voient de même : on est ramené à prouver que deux flèches p. 122

$$\gamma(X) \underset{\gamma(i_X)}{\overset{i_{\gamma(X)}}{\rightrightarrows}} \gamma(\gamma(X))$$

sont égales dans $(\mathrm{Cat}/S)(W^{\mathrm{d}}_S)^{-1}$. La vérification dans le cas 1°), où $\gamma = C$, ne doit pas être changée. Dans le cas 2°) où $\gamma = \Phi_\infty$, je ne l'ai pas fait. Je ne le ferai que sous la pression des besoins.

⑥ Loi contravariante de $\mathrm{HOT}_W(S)$, $\mathrm{HOT}^{\mathrm{lc}}_W(S)$.

Il y a pour

$$f : S' \longrightarrow S$$

un homomorphisme du diagramme

$$\begin{array}{ccc} \mathrm{HOT}^{\mathrm{lc}}_W(S') & & \\ \swarrow \quad \searrow & & \\ \mathrm{HOT}_W(S') & & \mathrm{HOT}^{o}_W(S') \end{array} \qquad \texttt{[Plutôt} \qquad \begin{array}{ccc} \mathrm{HOT}^{\mathrm{lc}}_W(S) & & \\ \swarrow \quad \searrow & & \\ \mathrm{HOT}_W(S) & & \mathrm{HOT}^{o}_W(S) \texttt{]} \end{array}$$

dans le diagramme similaire pour S `[plutôt` S'`]`, par des foncteurs f^* qui sont définis respectivement sur [page 198]

$$\begin{array}{rcccc} \underline{\mathrm{Liss}}_W S & \text{pour} & \mathrm{HOT}_W(S) & \longrightarrow & \mathrm{HOT}_W(S') \\ \underline{\mathrm{Prop}}_W S & \text{pour} & \mathrm{HOT}^{o}_W(S) & \longrightarrow & \mathrm{HOT}^{o}_W(S') \\ \underline{\mathrm{Fib}}_W S & \text{pour} & \mathrm{HOT}^{\mathrm{lc}}_W(S) & \longrightarrow & \mathrm{HOT}^{\mathrm{lc}}_W(S'), \end{array}$$

par l'intermédiaire des foncteurs

$$f^* : X \longmapsto X \times_S S' : \begin{cases} \underline{\mathrm{Liss}}_W S & \longrightarrow & \underline{\mathrm{Liss}}_W S' \\ \underline{\mathrm{Prop}}_W S & \longrightarrow & \underline{\mathrm{Prop}}_W S' \\ \underline{\mathrm{Fib}}_W S & \longrightarrow & \underline{\mathrm{Fib}}_W S', \end{cases}$$

lequel foncteur est compatible avec les localiseurs pretinents, qui sont W^{f}_S dans tous les cas. Par contre, en l'absence d'hypothèses draconiennes sur W ou sur S, on ne peut dire que

f^* soit compatible avec W_S, $W_{S'}$ sur une quelconque des catégories du diagramme p. 175, permettant de définir $\mathrm{HOT}^{\mathrm{lc!}}_W(S)$, de façon à nous donner un foncteur

$$\mathrm{HOT}^{\mathrm{lc!}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc!}}_W(S').$$

En l'absence d'axiomes sur W, le seul cas qui me soit connu est celui où f est *une W-équivalence universelle,* i.e. une W-fibration à fibres W-asphériques, auquel cas il est évident que pour *toute* flèche $u : X \longrightarrow Y$ dans Cat/S, $u \in W_S$ si et seulement si $(u' : X' \longrightarrow Y') \in W_{S'}$. On trouve alors un homomorphisme du diagramme complet $(*)$ p. 176 `[plutôt p. 178]` pour S, dans le diagramme correspondant pour S'.

Commentary · Exposition [ms. 197]

Completing the omitted compatibility of the two units

Grothendieck has proved enough to compare the localizations, but deliberately leaves the triangle check for Φ_∞ unfinished. There is a short way to complete it which uses the fine replacement already constructed. It also explains the later annotation that the proposed weakened hypotheses recover the full fibre-detection axiom.

First consider any S-map $u : X \to Y$ between W-fibred categories, with $u \in W$. Under condition 1°, put $R = \Phi$; under condition 2°, put $R = \Phi_\infty$. In either case η_X and η_Y are fine equivalences. Naturality and two-out-of-three show $Ru \in W$. In the first case RX, RY are split fibrations with W-equivalent transports; in the second they lie in $\underline{\mathrm{Parf}}_{W''_\omega} S$. The restricted fibre-detection hypothesis therefore gives $Ru \in W^{\mathrm{f}}_S$. Applying two-out-of-three in each fibre to

$$(Ru)_s(\eta_X)_s = (\eta_Y)_s u_s$$

gives $u_s \in W$. Thus each displayed hypothesis implies W(4b) itself. In the first case, it is enough to impose the test only on Cartesian maps, since $\Phi(u)$ is Cartesian. This is the replacement argument formalized later in Grothendieck (1991b, Lemma 4.4.9 and Corollary 4.4.10); the calculation here uses only the constructions already present in Chapter VI.

Under condition 1°, the established reflector C already proves the proposition. We now work under condition 2°, retaining W(6 bis), and compare $R = \Phi_\infty$ with C. Write $\alpha_X : X \to CX$ for the unit of C. Work in $\mathcal{H}_d(S)$, where the established reflection gives $C \simeq i\rho$. Let $R = \Phi_\infty$ and retain $\eta_X : X \to RX$ for its unit. The object RX is W-fibred, so α_{RX} is an isomorphism in $\mathcal{H}_d(S)$. Also $\eta_X \in W_S$, hence $C(\eta_X)$ is right-fine under W(4b). We therefore have a natural isomorphism

$$\theta_X = (\alpha_{RX})^{-1} C(\eta_X) : CX \xrightarrow{\sim} RX.$$

Naturality of α gives the exact identity

$$C(\eta_X)\alpha_X = \alpha_{RX}\eta_X, \qquad \text{hence} \qquad \theta_X \alpha_X = \eta_X.$$

Thus R represents the same reflection, with its given unit. This is stronger than an unspecified isomorphism between the two endofunctors: it identifies the units, which are what determine the adjunction.

It remains to extract the equality Grothendieck asks for. Since R^2X is a local object, the universal property of the reflection says that precomposition by η_X is a bijection

$$\mathrm{Hom}_{\mathcal{H}_d(S)}(RX, R^2X) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{H}_d(S)}(X, R^2X).$$

The two maps η_{RX} and $R(\eta_X)$ have the same image under this bijection, because naturality of η gives

$$\eta_{RX}\eta_X = R(\eta_X)\eta_X.$$

Therefore $\eta_{RX} = R(\eta_X)$ in $\mathcal{H}_d(S)$, as required. Both maps are isomorphisms: the first because RX is fibred, the second because R sends coarse equivalences to fine ones after the fibre-detection argument. Inverting this common map supplies the multiplication of the idempotent reflector, and the triangle identities follow.

The missing verification is consequently a genuine omission in the written proof, but no additional axiom is needed to repair it. The repair is not the generally invalid cancellation of η_X from an equality of composites; η_X need not be invertible in the lateral category. Cancellation is justified here by its proven universal property against the local target R^2X.

Si on suppose W(4b), alors $W_S = W_S^{\mathrm{f}}$ sur $\underline{\mathrm{Fib}}_W S$, donc sur cette catégorie-là le foncteur f^* est compatible (sans mérite!) avec les localiseurs W_S, $W_{S'}$, mais ce n'est pas nouveau vraiment par rapport à ce qui a été dit plus haut, pour $\underline{\mathrm{Fib}}_W S$, $\underline{\mathrm{Fib}}_W S'$, et pour les $\mathrm{HOT}_W^{\mathrm{lc}}$. p. 123

Mais si on suppose W(4a), et $f \in \underline{\mathrm{Fib}}_W$, alors f^* sur Cat/S tout entier est compatible [page 199] avec les localiseurs W_S, $W_{S'}$, et on trouve un calcul de f^* sur $\mathrm{HOT}_W^{\mathrm{lc!}}(S) \approx \mathrm{HOT}_W^{\mathrm{lc}}(S)$ par

$$f^*(\mathrm{hot}_{S,W}^{\mathrm{lc}}(X)) = \mathrm{hot}_{S',W}^{\mathrm{lc}}(X \times_S S').$$

Donc c'est là un cas, exceptionellement, où les foncteurs f^* pour les HOT_W sont compatibles non seulement aux foncteurs d'inclusion i, j (de la page 176 [plutôt p. 178]), mais aussi aux foncteurs de localisation ρ, σ.

Oubli : Si $p : S' \longrightarrow S$ est W-propre, alors $p^* : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$ est compatible avec W_S^{d}, $W_{S'}^{\mathrm{d}}$, et $p^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$ peut se 'calculer' par la formule

$$p^*(\mathrm{hot}_{S,W}(X)) = \mathrm{hot}_{S',W}(X \times_S S')$$

pour *tout* X dans Cat/S.

⑦ Produit dans $\mathrm{HOT}_W(S)$.

Le foncteur

$$(X, Y) \longmapsto X \times_S Y : \underline{\mathrm{Liss}}_W S \times \underline{\mathrm{Liss}}_W S \longrightarrow \underline{\mathrm{Liss}}_W S$$

est compatible avec les localiseurs $W_S^{\mathrm{f}} \times W_S^{\mathrm{f}}$ et W_S^{f}, donc passe aux localisés en

$$\begin{cases} \mathrm{HOT}_W(S) \times \mathrm{HOT}_W(S) & \longrightarrow & \mathrm{HOT}_W(S) \\ (\xi, \eta) & \longmapsto & \xi \times_S \eta \end{cases}$$

([97]). Ce produit est associatif, unitaire, commute avec toutes les compatibilités habituelles, comme il résulte du fait que c'est comme ça dans $\underline{\mathrm{Liss}}_S$. L'élément unité est la classe de $S \xrightarrow{\mathrm{id}} S$. La sous-catégorie $\mathrm{HOT}_W^{\mathrm{lc}}(S)$ est stable par le produit, qui induit [page 200]

$$\mathrm{HOT}_W^{\mathrm{lc}}(S) \times \mathrm{HOT}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}}(S),$$

lui-même induit par le foncteur produit $(X, Y) \longrightarrow X \times_S Y$ dans $\underline{\mathrm{Fib}}_W S$.

Commentary · Exposition [ms. 199]

Restriction depends on the chosen equivalences

Grothendieck's three inverse-image functors are built from the same categorical pullback, but they are defined on different useful representatives. This is why the functorial diagram is more informative than an undifferentiated assertion that pullback preserves weak equivalences.

For a smooth representative $X \to S$, its pullback along $f : S' \to S$ is again smooth, and fibrewise equivalences remain fibrewise equivalences. Since fine and right-fine equivalences agree on smooth objects, this constructs f^* on HOT_W. The dual argument uses proper representatives for $\mathrm{HOT}_W^{\mathrm{o}}$. On $\underline{\mathrm{Fib}}_W S$, the same pullback preserves the class of objects and the fine equivalences, and therefore constructs f^* on $\mathrm{HOT}_W^{\mathrm{lc}}$. In each case the composition and identity constraints are those of categorical pullback, passed through localization. Their coherence is inherited from the coherent pullback identifications before localization.

For the coarse localization, this argument is unavailable until total equivalences are controlled. If f is universally in W, every projection $X \times_S S' \to X$ belongs to W. Given $u : X \to Y$, the natural square with these projections proves

$$u \in W \quad \Longleftrightarrow \quad u \times_S S' \in W$$

97. **NB** Si X, Y sont *quelconques* dans Cat/S, on a $\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y) = \mathrm{hot}_{S,W}(\Phi_S(X) \times_S \Phi_S(Y))$. Si X ou Y est lisse sur S, on peut y remplacer $\Phi(X)$ resp. $\Phi(Y)$ par X resp. Y.

by two-out-of-three. This proves both preservation and reflection of total equivalences; no good-representative restriction is needed. Under W(4a), the same preservation statement holds when f is W-fibred, by the defining universal preservation property of that axiom. It then computes the coarse restriction directly on every $X \to S$.

There is a second, differently placed improvement. If $p : S' \to S$ is proper, pullback preserves right-fine equivalences on all of Cat/S. The displayed formula for p^* on an arbitrary representative is therefore legitimate. The condition is on the map changing the base, whereas the first construction required smoothness of the object being pulled back. Either condition can make the ordinary pullback compute the desired derived inverse image, but the two arguments test different arrows.

These distinctions also explain the compatibility claims. Restriction on fine representatives commutes with the inclusions of locally constant objects into the lateral theories. Commutation with the coarse localization requires preservation of the coarse denominators, and therefore the extra hypotheses just discussed. A diagram of functors cannot be promoted from one assertion to the other merely because the object formulas look identical.

Si X est quelconque dans Cat/S, le foncteur

$$\mathrm{mult}_{X/S} : Y \longmapsto X \times_S Y : \underline{\mathrm{Liss}}_W S \longrightarrow \mathrm{Cat}/S$$

est compatible avec les localiseurs W_S^{d}, car ce foncteur peut s'écrire comme le composé du foncteur correspondent *à valeurs dans* $\underline{\mathrm{Liss}}_W X$, lequel est compatible avec les localiseurs W_S^{d}, W_S^{d}, et le foncteur 'oubli de X' de $\underline{\mathrm{Liss}}_W X$ (même de Cat/X) dans Cat/S, lequel est compatible avec W_X^{d} et W_S^{d}. On trouve ainsi un foncteur p. 124

$$\begin{cases} \mathrm{mult}_{X/S} : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S) \\ \mathrm{mult}_{X/S}(\mathrm{hot}_{S,W}(Y)) \overset{\text{déf}}{=} \mathrm{hot}_{S,W}(X \times_S Y) \qquad \text{si } Y \textit{ lisse} \text{ sur } S \end{cases}$$

([98]), puisque $\mathrm{HOT}_W(S)$ peut s'obtenir comme localisé tant de $\underline{\mathrm{Liss}}_W S$ que de Cat/S, pour W_S^{d}. Quand X est lui-même W-lisse sur S, ce foncteur n'est autre que

$$\begin{cases} \mathrm{mult}_\xi : \eta \longmapsto \xi \times_S \eta, \\ \text{où} \quad \xi = \mathrm{hot}_{S,W}(X). \end{cases}$$

Mais il n'en est pas de même si X n'est pas lisse, i.e. il faut distinguer dans ce cas, même pour Y W-lisse sur S,

$$\mathrm{hot}_{S,W}(X \times_S Y) \qquad \text{et} \qquad \mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y).$$

Le deuxième terme peut s'expliciter comme $\mathrm{hot}_{S,W}(\Phi(X) \times_S Y)$ qui reçoit $\mathrm{hot}_{S,W}(X \times_S Y)$, i.e. on a un homomorphisme canonique [page 201]

$$\underbrace{\mathrm{hot}_{S,W}(X \times_S Y)}_{=\mathrm{mult}_{X/S}(Y)} \longrightarrow \underbrace{\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y)}_{=\mathrm{hot}_{S,W}(\Phi(X) \times_S Y)},$$

induit par

$$X \times_S Y \longrightarrow \Phi(X) \times_S Y,$$

mais cette dernière flèche n'a aucune raison d'être dans W_S^{d}, et n'induit donc pas nécessairement un isomorphisme dans $\mathrm{HOT}_W(S)$. (Contrexemple page 154.)

98. Cette formule est également valable si Y est dans $\underline{\mathrm{Fib}}_W S$, cf. (∗), p. 202.

Commentary · Exposition [ms. 200]

Multiplication by a category over the base

Grothendieck introduces $\mathrm{mult}_{X/S}$ because an arbitrary category over the base still acts on relative homotopy types, even when its strict fibre product does not compute the internal product. The action remembers the map $X \to S$ before replacement.

Let $p : X \to S$. For a smooth representative $Y \to S$, pullback computes $p^*[Y]$ as $[X \times_S Y]$ over X. Postcomposition then gives

$$\mathrm{mult}_{X/S} = p_! p^*.$$

This identity already explains why the functor is well defined for arbitrary X: the right-fine equivalences are respected in the two stages where Grothendieck factors the construction. The repeated W_S^{d} at the start of printed p. 124 is a source slip; the pullback stage has source class W_S^{d} and target class W_X^{d}.

By contrast, the internal product is obtained by replacing both factors by smooth categories:

$$[X] \times_S [Y] = [\Phi X \times_S \Phi Y].$$

The map $X \to \Phi X$ induces a comparison from the action to multiplication by $[X]$. For smooth Y, it is represented by $X \times_S Y \to \Phi X \times_S Y$. Its failure to be right-fine in general is the counterexample already analyzed on manuscript pp. 153–156; there is no need to repeat it here. What changes in the present discussion is the proposed hypothesis that X is W-fibred.

The argument given on pp. 201–202 correctly proves the action formula when X is fibred. Indeed the unit $X \to \Phi X$ is fine; after arbitrary base change to Y it is fine between W-fibred objects over Y, hence right-fine there. Forgetting the intermediate base preserves right-fine equivalences. This proves that the action of Y/S on $[X]$ can be computed by the strict fibre product even for arbitrary Y.

There remains an apparent asymmetry: identifying that value with $[X] \times_S [Y]$ also requires replacing Y. Grothendieck proves this when X is proper and suspects that the extra condition is necessary. The next development supplies the missing comparison for a W-fibred X by using the transports in its split-fibration replacement. The argument will identify the canonical comparison map itself, so it also gives the natural compatibility needed for products and inverse images.

Il faut regarder à part le cas *où X est dans* $\underline{\mathrm{Fib}}_W S$, j'ai envie de dire que dans ce cas on a

$$(*) \qquad \mathrm{hot}_{S,W}(X \times_S Y) \xrightarrow{\sim} \underbrace{\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y)}_{= \mathrm{hot}_{S,W}(\Phi(X) \times_S \Phi(Y))}$$

([99]) pour *tout* Y dans Cat/S (pas nécessairement W-lisse), a fortiori que dans ce cas

$$(**) \qquad \begin{cases} \mathrm{mult}_{X/S} = \mathrm{mult}_\xi & \text{dans } \mathrm{HOT}_W(S) \\ \text{où } \xi = \mathrm{hot}_{S/W}(X/S) \text{ (si } X \text{ dans } \underline{\mathrm{Fib}}_W S). \end{cases}$$

Dans le cas où Y est lisse sur S (ce qui suffit pour établir $(**)$), il suffit de se rappeler que $X \longrightarrow \Phi(X)$ est dans W_S^{f} (résulte du fait que X est W-fibré sur S ([100]), donc $X_s \longrightarrow s\backslash X$ est dans W, $\forall\ s \in S$), donc $\underbrace{X \times_S Y}_{X_Y} \longrightarrow \underbrace{\Phi(X) \times_S Y}_{\Phi(X)_Y}$ est aussi dans W_Y^{f}, donc (comme X_Y, $\Phi(X)_Y$ sont W-fibrés sur Y) dans W_Y^{d}, donc aussi dans W_S^{d}. En fait, même sans supposer Y lisse, ceci montre que l'on a p. 125

$$(*) \qquad \mathrm{hot}_{S,W}(X \times_S Y) \xrightarrow{\sim} \mathrm{mult}_{Y/S}(\mathrm{hot}_{S,W}(X)) \qquad \text{si } X \text{ est dans } \underline{\mathrm{Fib}}_W S. \qquad \text{[page 202]}$$

Pour avoir $(*)$ pour Y quelconque, on peut par ce qui précède supposer X lisse (quitte à le remplacer par $\Phi(X)$), donc X dans $\underline{\mathrm{Liss}}_{0,W} S$. Il faut voir que

$$X \times_S Y \longrightarrow X \times_S \Phi(Y)$$

99. Cette formule n'est établie qu'en supposant ou bien Y W-lisse (en plus de X W-fibré sur S), ou bien X W-propre, p.ex. X W-parfait.

100. il suffit même W-prélisse sur S

est encore dans W_S^{d}. Or si X est W-*propre* sur S, le changement de base $X \longrightarrow S$ est compatible avec W_S^{d}, W_X^{d} (sur Cat/S tout entier), et on gagne. (Ça doit être faux sans hypothèse de W-propreté sur X, *ou* de W-lissité sur Y …)

Quand on a

$$f : S' \longrightarrow S,$$

alors

$$f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$$

est compatible avec les structures de produit.

⑧ Les foncteurs $f_!$: théorème d'adjonction, propriétés du dérivateur HOT_W.

Soit

$$f : S' \longrightarrow S$$

dans Cat, alors le foncteur induit

$$f_! : \mathrm{Cat}/S' \longrightarrow \mathrm{Cat}/S$$

est compatible avec les localiseurs $(W_{S'}^{\mathrm{d}}, W_S^{\mathrm{d}})$, ou $(W_{S'}^{\mathrm{g}}, W_S^{\mathrm{g}})$, ou $(W_{S'}, W_S)$. On trouve donc un foncteur induit pour les localisées, et `[un]` diagramme commutatif [page 203]

$$\begin{array}{ccccc}
\mathrm{HOT}_W(S') & & & & \mathrm{HOT}^o_W(S') \\
\downarrow{\scriptstyle f_!} & \searrow^{\rho'} & & \swarrow_{\sigma'} & \downarrow{\scriptstyle f^o_!} \\
\mathrm{HOT}_W(S) & & \mathrm{HOT}^{\mathrm{lc}\,!}_W(S') & & \mathrm{HOT}^o_W(S). \\
& \searrow_{\rho} & \downarrow{\scriptstyle f^{\mathrm{lc}\,!}_!} & \swarrow^{\sigma} & \\
& & \mathrm{HOT}^{\mathrm{lc}\,!}_W(S) & &
\end{array}$$

Commentary · Development (commentators') [ms. 202]

A fibred factor suffices for the product formula

Grothendieck's desired formula on manuscript p. 201 is stronger than the proof immediately following it. The remaining difficulty can be resolved by calculating over a comma category: the arrow from its fixed starting object trivializes the transport of a locally constant fibration. This gives the unrestricted second factor that the manuscript leaves undecided.

Product comparison for a fibred factor. [ours] Assume $\mathrm{W}(1,2,3)$. If $X \to S$ is W-fibred in the weak, universally stable sense used in this chapter, then for every $Y \to S$ the canonical map

$$[X \times_S Y] \longrightarrow [X] \times_S [Y] \quad \text{in } \mathrm{HOT}_W(S)$$

is an isomorphism. No properness assumption on X or smoothness assumption on Y is required.

Proof. We first treat a split Grothendieck fibration $X = \int_S F$ with $F : S^{\mathrm{op}} \to \mathrm{Cat}$ and $F(u) \in W$ for every arrow u of S. Fix $s \in S$ and put $B = s\backslash Y$. An object of B is (y, a), with $a : s \to q(y)$, where $q : Y \to S$ is the structure map. The projection

$$E = s\backslash(X \times_S Y) \longrightarrow B$$

is the split fibration classified by $(y, a) \mapsto F(qy)$. A morphism $(y, a) \to (y', a')$ is an arrow $h : y \to y'$ with $qh\, a = a'$, and its transport is $F(qh)$.

The maps $F(a) : F(qy) \to F(s)$ form a natural transformation to the constant diagram at $F(s)$, because

$$F(a)F(qh) = F(qh\, a) = F(a').$$

They therefore define a Cartesian functor

$$T : E \longrightarrow F(s) \times B, \qquad (x, y, a) \longmapsto (F(a)x, (y, a)).$$

Each fibre map of T belongs to W. Hence $T \in W$: a fibrewise W-equivalence between Grothendieck fibrations is right-fine, as seen by the fibre-to-comma adjunctions, and the relative localization axiom sends right-fine equivalences to total ones. This implication uses the fibrewise-to-total direction; it does not invoke W(4b).

We check that this proves the canonical comparison, not just an equivalence between its endpoints. Since X is smooth, $[X] \times_S [Y]$ is represented by $X \times_S \Phi Y$. Let

$$k_s : s\backslash(X \times_S Y) \longrightarrow s\backslash(X \times_S \Phi Y)$$

be induced by the unit of Y. The target is a comma category of a split fibration, so pullback to its fibre at s defines a functor

$$r_s : s\backslash(X \times_S \Phi Y) \longrightarrow F(s) \times (s\backslash Y).$$

The inclusion of that fibre is left adjoint to r_s: an arrow from a fibre object to an object over t, lying over $c : s \to t$, is uniquely an arrow in the fibre to its Cartesian pullback. Thus $r_s \in W$. On an object (x, y, a) of the source of k_s, the unit of Y introduces the identity arrow at $q(y)$, and r_s pulls back along a. Consequently $r_s k_s = T$, on objects and on morphisms. Since $r_s, T \in W$, two-out-of-three gives $k_s \in W$. This holds for every s, proving

$$X \times_S Y \longrightarrow X \times_S \Phi Y \quad \text{belongs to } W_S^{\mathrm{d}}.$$

For a general W-fibred X, its replacement ΦX is a split fibration with W-equivalent transports. The unit $X \to \Phi X$ is fine. After pullback to any Y, it is fine between W-fibred objects over Y, because this class of objects is stable under arbitrary base change. Fine and right-fine equivalences agree on that class, and forgetting Y preserves right-fine equivalences. Hence

$$X \times_S Y \longrightarrow \Phi X \times_S Y \quad \text{belongs to } W_S^{\mathrm{d}}.$$

Apply the split case to ΦX. Composing the two maps gives a right-fine equivalence from $X \times_S Y$ to $\Phi X \times_S \Phi Y$, whose image is exactly the canonical product comparison. This proves the assertion.□

The universal stability used in the last paragraph is part of the weak-fibration notion, not a hidden strengthening of it; see Grothendieck (1990–1991f, Chapter XII, Part III, Theorem 3, manuscript pp. 51–52). The construction also shows naturality in both X and Y, since each map was induced by the functorial comma replacement or by its Cartesian transport.

In particular, the proposed general formula $(*)$ on manuscript p. 201 is valid under its stated fibred hypothesis. The suspicion at the end of p. 202 that properness of X or smoothness of Y is necessary is not borne out. We checked the comma-category and base-change treatments in Maltsiniotis (2005, §§4–5) and the locally constant framework in Cisinski (2009); we did not locate this exact formulation there. The proof is included as a continuation of the manuscript's argument, without a claim that the result is new.

On n'a pas de foncteur $f_!^{\mathrm{lc}}$ au niveau des $\mathrm{HOT}_W^{\mathrm{lc}}$, sauf si on fait une hypothèse, telle W(4b), qui assure que $\mathrm{HOT}_W^{\mathrm{lc}!} \approx \mathrm{HOT}_W^{\mathrm{lc}}$. p. 126

Le théorème principal, que le foncteur $f_!$ est adjoint à gauche de f^*, est valable *sous la seule hypothèse* W(1, 2, 3) ([101]). Donc dualement pour $f_!^o$, f_o^*. Mais comme on n'a pas de foncteur $f_{\mathrm{lc}\,!}^*$, la question d'adjonction ne se pose pas pour f général, et sans hypothèse W(4b) sur W. Elle ne se pose, apparemment, que pour f un W-fibré trivial (cf. fin de ⑥, p. 198), et dans ce cas l'adjonction ne doit pas faire de difficultés. Mais tant que je n'ai pas de sentiment bien net au sujet de la signification des $\mathrm{HOT}_W^{\mathrm{lc}!}$, ce n'est pas là un résultat bien bandant [sic].

L'adjonction de $f_!$, f^* pour HOT_W permet déjà de montrer que le prédérivateur HOT_W satisfait les conditions Der 1, Der 4′, Der 5′ des dérivateurs. De plus, on a ce qu'on peut [page 204]

101. i.e. avec ces nouvelles notations Loc(1, 2, 3 bis) (cf. XVI).

appeler la propriété Der 3b pour HOT_W en tant que W-dérivateur, i.e.

> *Si $f \in W$, alors f est une* HOT_W*-équivalence* ([102]) (moyennant les seuls axiomes $\mathrm{W}(1,2,3)$ sur W). Plus précisément, f est une HOT_W-équivalence si et seulement si $\mathrm{hot}_W(f)$ est `[un]` isomorphisme, i.e. $f \in \overline{W}$, le saturé fort de W.

Si W *satisfait* W(4b), alors on a également un couple de foncteurs adjoints

$$\mathrm{HOT}^{\mathrm{lc}}_W(S') \underset{f^*_{\mathrm{lc}}}{\overset{f^{\mathrm{lc}}_!}{\rightleftarrows}} \mathrm{HOT}^{\mathrm{lc}}_W(S)$$

pout toute flèche $S' \longrightarrow S$ dans Cat. On voit aussi dans ce cas que

> si $f : S' \longrightarrow S$ est dans W, alors f^*_{lc} et $f^{\mathrm{lc}}_!$ entre $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ et $\mathrm{HOT}^{\mathrm{lc}}_W(S')$ sont des équivalences quasi-inverses l'une de l'autre (moyennant W(4b)),

ce qui est l'axiome Der $3b_2$ 'version W-dérivateurs'. On ne peut s'attendre à mieux, du côté Der 3!

Oubli : Soit $f : S' \longrightarrow S$, on a alors $\boxed{f_! f^*} : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S)$ est égal à $\mathrm{mult}_{S'/S}$ (cf. p. 200). Il est donc égal à mult_ξ, où $\xi = \mathrm{hot}_{S,W}(S')$, si on suppose ou bien S' W-lisse sur S, ou bien S' W-fibré sur S (cf. (∗∗), p. 201).

⑨ Théorème de changement de base pour $f_!$ dans HOT_W.

Si on a p. 127

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{g} & S', \end{array}$$

avec f W-lisse *ou* g W-propre `[la suite manque]`

Commentary · Later developments [ms. 204]

The final adjunction and the surviving base-change fragment

Grothendieck closes the revision by separating the constructions that need only the three standing axioms from the stronger locally constant comparison. Postcomposition $f_!$ respects the three relative equivalence classes, and the earlier unit and counit prove $f_! \dashv f^*$ for HOT_W without W(4b). The locally constant adjunction requires the additional comparison that identifies the fine theory with the coarse localization.

For $f \in W$ under W(4b), the locally constant unit and counit are isomorphisms; hence $f^{\mathrm{lc}}_!$ and f^*_{lc} are inverse equivalences. This assertion concerns locally constant coefficients. It does not say that restriction on all coherent diagrams is an equivalence whenever the indexing categories have the same homotopy type. For instance, the terminal category and the walking arrow have equivalent nerves, whereas the latter still indexes a genuine coherent morphism. The distinction is already built into the separate lateral and locally constant categories.

The constant-coefficient claim has an elementary calculation. If $p : S \to e$ and $T \in \mathrm{Cat}$, the constant coefficient represented by T is represented over S by $S \times T$. Its homotopy colimit is therefore represented by $S \times T$ as an absolute category. A functor $f : S' \to S$ acts by $f \times \mathrm{id}_T$. Product stability gives the implication $f \in W \Rightarrow f \times T \in W$. Conversely, testing $T = e$ detects whether $[f]$ is invertible in $\mathrm{Cat}[W^{-1}]$. This explains the appearance of the strong saturation $\overline{W}$ in the final statement, rather than an assertion that every chosen class W was strongly saturated from the outset.

The last paragraph of the source displays a square and the hypotheses "f smooth or g proper", then stops. Its missing continuation is a source lacuna, not an unproved conclusion which we may

102. On le prouve en interprétant la notion de HOT_W-équivalence par la HOT_W*-homologie.* Cela revient finalement à ceci : Si $f : S' \longrightarrow S$ est dans W, alors $\forall T$, $S' \times T \longrightarrow S \times T$ aussi!

attribute to Grothendieck. If the square is Cartesian, the expected statement can be read against the base-change result proved earlier in the chapter: the canonical mate

$$f'_! q^* \longrightarrow g^* f_!$$

is invertible for such a square. The surviving fragment does not itself repeat the Cartesian hypothesis; it is necessary in this reconstruction. In the smooth-f case, represent the coefficient by $Z \to X$ smooth over X. Its composite to S is smooth, so both sides are represented by the same category $Z \times_S S'$ over S'. In the proper-g case, ordinary pullback along g computes inverse image for every representative, and its pullback q is proper as well; the same equality of representatives proves the formula. The map is the standard mate built from the unit and counit, so the identification agrees with the canonical base-change comparison rather than merely producing some isomorphism between the functors.

This is the categorical form of the base-change mechanism isolated in Maltsiniotis (2005, Proposition 5.3, Theorem 5.4, and Remark 5.5), with the opposite-category convention adjusted to the present $s\backslash X$ model. The surviving text is left unchanged at its stopping point; the preceding calculation belongs to the commentary.

Overview · VI-c

Chapter VI, manuscript pages 138–204: operations, fibres, and the choice of equivalences.

These pages bring the construction of HOT_W to a point where it can support a calculus of operations (Grothendieck, 1990b). The principal achievement is an adjunction. For a functor $f : S' \to S$, composition of structural maps already gives a direct-image functor on categories over the base. After localization, this becomes the left adjoint $f_!$ of the inverse-image functor f^*. The work lies in making the inverse image homotopically meaningful, identifying the unit and counit, and proving that their two composites are identities. The ensuing discussion of products, base change, and locally constant objects repeatedly returns to the choice of a concrete representative and to the changes of representative that survive the intended operation.

The notation needs a fixed orientation. We follow the manuscript's presheaf convention. A split category fibred over S represents a functor $S^{\mathrm{op}} \to \mathrm{Cat}$. For $X \to S$, the comma replacement $\Phi_S(X)$ has fibre $s\backslash X$ at s; its unit $i_X : X \to \Phi_S(X)$ is a right-fine equivalence. Consequently

$$f^*\mathrm{hot}_{S,W}(X) \simeq \mathrm{hot}_{S',W}\big(\Phi_S(X) \times_S S'\big), \qquad f_!\mathrm{hot}_{S',W}(X') = \mathrm{hot}_{S,W}(X').$$

The second formula remembers the change of structural map. It does not assert that the representative remains fibred or smooth over the new base. This is the reason the first formula cannot generally be replaced by strict pullback of X.

The adjunction proof on pages 138–146 is unusually revealing. The unit is the S'-map induced by $i_{X'}$ and the structural map $X' \to S'$. For the counit one first tries to invert the pullback of i_X; the attempt fails because that pullback need not belong to the relevant class of equivalences. The correct formula is

$$\beta_X = (i_X)^{-1}\,\mathrm{pr}_1^{\Phi_S(X)}.$$

Its inverse is taken in the localization over S, where i_X is known to be invertible. One triangle identity is then the definition of the unit, and the other is naturality of i. Our commentary writes out this reduction, with the domains and codomains kept visible. The later passage to locally constant objects is a formal adjunction argument between reflective subcategories, once the reflection and the appropriate commutation squares are available.

Products and the operation attached to a representative.

The next issue is more subtle than the construction of a multiplication law might suggest. Two smooth representatives have a strict product over S that computes the product in $\mathrm{HOT}_W(S)$. With arbitrary representatives one must use

$$\mathrm{hot}_{S,W}(X) \times_S \mathrm{hot}_{S,W}(Y) \simeq \mathrm{hot}_{S,W}\big(\Phi_S(X) \times_S \Phi_S(Y)\big).$$

The first corollary on page 152 claims that one smooth factor suffices to omit both replacements. Grothendieck retracts it immediately. Over the walking arrow $0 \to 1$, the smooth inclusion of 0 and the terminal-point inclusion of 1 have empty strict intersection, whereas replacing the latter by its comma fibration changes the calculation. This small example determines the correct scope of every later product formula.

What remains for an arbitrary $X \to S$ is the endofunctor $\mathrm{multhot}_{X/S}$, obtained by multiplying smooth representatives by this particular X. It gives the unconditional identity

$$f_!f^* \simeq \mathrm{multhot}_{S'/S}\,.$$

An identification with multiplication by the localized object $\mathrm{hot}_{S,W}(S')$ requires a further argument. The distinction concerns functorial dependence on the first representative, rather

than a failure of associativity or of the usual coherence of products. We return to it at the end of the chapter, where a proof using the Grothendieck construction extends the calculation for weak W-fibrations. The proof and its precise assumptions are given in the development block, independently of an appeal to a general projection formula.

The formula for a left Kan extension at an object is obtained in the same concrete way. Its value at s is represented by $s\backslash X'$, rather than by the strict fibre X'_s. The comma square carries its canonical natural transformation; it is not in general a strictly commutative square. With that transformation and the presheaf variance made explicit, the base-change mate has the standard orientation, and the computation proves it invertible. This supplies the pointwise left-Kan-extension axiom. The smooth/proper base-change theorem then follows by choosing representatives on which strict pullback is legitimate. The initial detour through a general derivator theorem is incomplete because it invokes pointwise detection of isomorphisms. The direct argument on pages 168–169 removes that dependence.

Why four localizations appear.

The revision beginning on page 171 is structural. Grothendieck separates the basic localizer axioms from the additional principle that a total equivalence between weak fibrations is an equivalence on every fibre. For a map $u : X \to Y$ over S, the classes used in the discussion are

$$\begin{aligned} W_S &: u \in W \text{ on the total categories,} \\ W_S^{\mathrm{f}} &: u_s \in W \text{ for every strict fibre,} \\ W_S^{\mathrm{d}} &: s\backslash u \in W \text{ for every } s, \\ W_S^{\mathrm{g}} &: u/s \in W \text{ for every } s. \end{aligned}$$

On weak fibrations, the fine conditions agree; their implication to total equivalence is available under the basic axioms. The converse is W(4b). Without it, the fine localization of weak fibrations and their coarse localization cannot be identified merely because both admit useful replacements. This yields HOT_W, its opposite-variance counterpart, $\mathrm{HOT}_W^{\mathrm{lc}}$, and $\mathrm{HOT}_W^{\mathrm{lc!}}$, with a localization λ between the last two.

The path functor C exposes the difficulty. Its unit is a total equivalence and its values are perfect over the base. These properties give coarse equivalences of presentations. They do not, by themselves, prove that the unit is fine on a weak fibration. W(4b) supplies exactly that step, identifies the two locally constant theories, and allows the reflection formulas to be read as genuine adjunctions with fully faithful right adjoints. The audit keeps this hypothesis attached to each use of that conclusion. A calculation with action groupoids makes the distinction visible on morphisms: coarse localization can identify maps that remain distinct in the fine theory.

This is also the place to separate historical uncertainty from present knowledge. The minimality theorem for the usual homotopy localizer, in the modern basic-localizer framework, is due to Cisinski (2004). It answers the proposed search for a smaller basic localizer; it does not identify every generalized coefficient theory with ordinary homotopy theory. The manuscript's varying axioms and its changing use of "fibration" must still be translated before a later theorem is applied.

Alternating replacement and the questions it answers.

One application of Φ cannot produce all the desired geometry. The fork-shaped base with two incomparable terminal branches gives a particularly strong obstruction: for a nondegenerate localizer, the comma replacement of a nonempty category over that base cannot be proper. The commentary verifies the categorical retraction in the preceding example, repairs the central object of the printed diagram, and gives the missing last step in the emptiness argument. A separately marked extension treats a class of partially ordered bases by the same mechanism. These arguments concern properness of a particular replacement, not the nonexistence of any satisfactory presentation.

Grothendieck then alternates the two comma constructions and takes the colimit of the resulting sequence. The odd and even subsequences respectively provide smooth and proper presentations. The proof uses W(6 bis): closure of W under colimits of sequences of arrows. Its historical formulation does not assert arbitrary filtered closure. The limiting object is perfect, and its unit is fine when the original object is a weak fibration. This gives a fine equivalence between the weak-fibration and perfect-object presentations without using W(4b). It does not identify fine and coarse equivalences.

The later theorem of Cisinski (2006a, Corollary 4.2.22) makes W(6 bis) automatic for basic localizers. With the identification $W_\omega = W'_\omega = W''_\omega = W_\infty$ established in the commentary, the replacement therefore answers the perfect-presentation question affirmatively under the standing axioms. The same fine units transfer the original restricted W(4b′) test to all weak fibrations. These conclusions concern the relative localization axiom W(2); they do not silently extend a colimit theorem to every weaker axiom system considered in the manuscript.

The practical proposition on pages 196–197 combines this construction with restricted fibre detection. Its last adjunction verification is expressly postponed. We supply it by first transporting fibre detection along the functorial replacement and then comparing the resulting unit with the established C-reflection. The later Chapter XVI confirms the role of replacement in relating the fibration axioms; the brief status block locates its theorem and keeps its historical hypotheses visible. No priority claim is attached to the elementary arguments supplied here.

The reach of the construction.

For the ordinary homotopy theory of spaces, the contemporary description of locally constant diagrams uses homotopy Kan extensions and localization of diagram categories; Cisinski (2009) gives a precise treatment. The model-category construction of derivators supplies a standard framework (Cisinski, 2003; Groth, 2013). More generally, Der 2 holds for every basic localizer (Cisinski, 2006a, Propositions 3.1.30 and 4.2.7). The commentary after manuscript p. 159 explains the comparison which permits that conclusion. The unrestricted strict-bifibration presentation over an arbitrary base remains outside what we have established: a free bifibration alone does not provide a fine equivalence from the given weak fibration. Over groupoid bases we give the complete comparison directly.

The surviving final recap deliberately lists Der 1, Der 4′, Der 5′, and the appropriate W-relative form of Der 3b. Der 2 has been withdrawn from the earlier list pending a proof; its later resolution does not supply a missing argument to the historical text. The audit also preserves the missing manuscript page 170 and the incomplete final base-change paragraph. What can be proved from the surviving material is supplied in commentary; the missing text is not reconstructed.

Chapter VII

Catégories de chemins (2)

VII-a · Chapter VII, part 1 of 3 · manuscript pages 1–57

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre VII

Catégories de chemins (2)

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

Comme le formalisme ‘simplifié’ des chemins que j’ai voulu dégager à la va-vite s’est révélé d’une lourdeur insupportable, de nature à le rendre quasiment inutilisable, je vais essayer de dégager un formalisme souple, collant à l’intuition, au prix seulement d’un effort conceptuel supplémentaire. p. 1 [page 1]

1 Intervalles et leur variantes

On appelle *catégorie intervalle standard* une catégorie définie par un ensemble ordonné, dont l’ensemble des élements est un intervalle d’entiers $[0, n]$ ($n \in \mathbf{N}$, on n’exclut pas le cas $n = 0$, i.e. de la catégorie ponctuelle $\{0\}$) :

$$\mathrm{Ob}\, I = [0, n] \subseteq \mathbf{N},$$

et satisfaisant l’hypothèse suivante :

a) pour toute paire de deux sommets consécutifs $i-1$, i de I ($1 \leq i \leq n$), il y a une flèche qui les relie, donc soit $i \longrightarrow i+1$, soit $i \longleftarrow i+1$ [plutôt $i-1 \longrightarrow i$, $i-1 \longleftarrow i$] (cette flèche est notée $\underline{u}_i$,

$$(i-1) \xrightarrow{\underline{u}_i} i \qquad \text{ou} \qquad (i-1) \xleftarrow{\underline{u}_i} i \ \ ).$$

b) La catégorie I est engendrée par l’ensemble de ses flèches $\underline{u}_i$ (appelées *flèches de transition*).

La flèche de transition est dite *directe* si elle va de $i-1$ à i, *rétrograde* dans le sens inverse. Si on affecte l’intervalle $[i-1, i]$ du signe + ou − suivant que $\underline{u}_i$ est directe ou rétrograde, on voit donc que les intervalles standard de longueur donnée n correspondent biunivoquement aux systèmes de signes + ou − sur les n intervalles de transition $[i-1, i]$ ($1 \leq i \leq n$), il y en a donc 2^n. Le sommet 0 est l’*origine* ou la *source* de l’intervalle, le sommet n l’*extrémité* ou le *but* de l’intervalle. [page 2]

Commentary · Exposition [ms. 1]

A combinatorial interval carries two orders

Grothendieck begins again because a path should be easy to concatenate and easy to change without losing its endpoints. The extra conceptual effort announced here consists in making its shape an object of study. Before forming a category of all paths, he isolates the finite categories which can serve as their domains. The commentary on Chapter VI, manuscript p. 69 (“Concatenation and the two iterated units”) discusses the preceding path construction; the present chapter reorganizes its combinatorics.

A standard interval is the free category on an oriented path graph. Write its transition signs as

$$\tau = (\varepsilon_1, \dots, \varepsilon_n), \qquad \varepsilon_i \in \{+, -\}.$$

There is one object at each integer from 0 to n. A positive sign supplies the generating arrow $i-1 \to i$ and a negative sign supplies $i \to i-1$. The free category is thin: between two fixed vertices there is at most one directed path, since the underlying undirected graph has no cycle. Thus the description as a category associated with a partially ordered set introduces no additional relations. Conversely, the two conditions in the text reconstruct precisely this free category.

For later use we record the complete hom-set calculation. If $i < j$, then

$$\begin{aligned} I_\tau(i, j) \neq \varnothing \quad &\Longleftrightarrow \quad \varepsilon_{i+1} = \cdots = \varepsilon_j = +, \\ I_\tau(j, i) \neq \varnothing \quad &\Longleftrightarrow \quad \varepsilon_{i+1} = \cdots = \varepsilon_j = -. \end{aligned}$$

Every nonempty hom-set contains exactly one arrow. These formulas prove both the classification by sign words and the count 2^n. They also explain why a functor out of an interval will be specified freely by its values on the transition arrows.

The numerical order and the categorical partial order have different jobs. The first tells us in which order the vertices are visited; the second tells us which composites exist. For the shape $0 \to 1 \leftarrow 2$, the designated source of the interval is 0 and its designated target is 2, although neither statement says that 0 is initial or 2 terminal. A path of this shape travels along the first arrow and against the second. Confusing these two orders would already make the later distinction between an opposite path and an opposite category disappear.

Length counts transitions. The interval of length zero has one vertex and no transition, and the interval of length n has $n+1$ vertices. We keep this convention throughout; it will identify the isolated equality $n = \operatorname{card} I$ on ms. 13 as a printed slip rather than a competing convention.

Un intervalle standard est *orienté,* du fait qu'on à la relation d'ordre *total* naturelle sur l'ensemble de ses sommets, savoir celle des entiers dans $[0, n]$. Quand on parlera de la relation d'ordre sur I, qu'un sommet i en *'précède'* ou *'suit'* un autre j ($i \leq j$ ou $i \geq j$ respectivement), c'est de cette relation d'ordre qu'on voudra parler. Celle associée à la structure de catégorie ne sera jamais regardée comme une relation d'ordre. Il y a également une relation d'ordre total naturelle sur l'ensemble des flèches de transition $\underline{u}_i$ ($1 \leq i \leq n$), donnée par la relation d'ordre des entiers i, $1 \leq i \leq n$, qui les indexent. On dira qu'une flèche i `[plutôt` $\underline{u}_i$`]` de transition *précède* ou *suit* une autre `[`$\underline{u}_j$`]`, ou qu'elle est *successeur immédiat* ou *prédécesseur immédiat* d'une autre (si $i = j+1$ ou $i = j-1$).

Si on a un ensemble d'entiers p. 2

$$E \subseteq \mathbf{Z},$$

on considérera dans E la relation d'équivalence engendrée par la relation entre deux éléments (i, j) de E : j est successeur immédiat de i *dans* $\mathbf{Z}$ (i.e. $j = i+1$). Deux éléments distinct i, j de E sont congrus pour cette relation si et seulement si l'intervalle d'entiers qu'ils déterminent est $\subseteq E$, donc

$$\text{si } i < j,\ i \text{ et } j \text{ sont } \textit{'connectés'} \text{ dans } E \text{ si et seulement si } [i, j] \subseteq E.$$

Quand on parlera des 'composantes connexes' de E, il s'agira toujours des classes suivant cette relation d'équivalence. [page 3]

Nous identifierons l'ensemble des flèches de transition $\operatorname{Fl}\operatorname{tr}(I)$ de I à l'intervalle d'entiers $[1, n]$, donc à une partie de $\mathbf{Z}$. Si on a une partie $E \subseteq \operatorname{Fl}\operatorname{tr}(I)$, on y a donc la notion de flèches de transition 'connectées' dans E, et la notion de connexité pour E. Si $i < j$, $\underline{u}_i$ et $\underline{u}_j$ sont connectées si et seulement si toutes les flèches de transition intermédiaires entre $\underline{u}_i$ et $\underline{u}_j$ sont dans E. Et E est connexe si et seulement s'il est soit vide, soit formée d'une suite de flèches de transition dont chacune est successeur immédiat de celle qui la précède,

$$E = \{\underline{u}_i, \underline{u}_{i+1} \ldots, \underline{u}_{i+l-1}\},$$

où l est le cardinal de E. Une telle E est appelée une *chaîne* de flèches de transition, et l est sa longueur. Une chaîne est dite *composable* si le composé $\underline{u}_{i+l-1} \circ \cdots \circ \underline{u}_{i+1} \circ \underline{u}_i$ `[ou` $\underline{u}_i \circ \underline{u}_{i+1} \circ \ldots \circ \underline{u}_{i+l-1}$`]` a un sens, i.e. si elles sont toutes de même sens, soit directe, soit rétrograde.

On désigne par $\operatorname{Fl}\operatorname{tr}^+(I)$, $\operatorname{Fl}\operatorname{tr}^-(I)$ l'ensemble des flèches de transition directes, ou rétrogrades, respectivement. Leurs composantes connexes ne sont autres que les chaînes composables maximales. Ainsi, les chaînes composables maximales forment une partition de l'ensemble des flèches de transition, en intervalles d'entiers mutuellement disjoints. Les chaînes forment elles-mêmes un ensemble totalement ordonné

$$C_1,\ C_2,\ \ldots\ C_r \qquad (1 \leq r \leq n).$$

L'entier r s'appelle la *'longueur utile'* de l'intervalle I, c'est donc le cardinal de l'ensemble de ses chaînes composables maximales. [page 4]

Deux chaînes composables maximales [consécutives] sont toujours de signe opposé, si l'une est directe l'autre est rétrograde, et inversement. La première chaîne composable maximale C_1 (disons simplement *'chaîne structurale'*) a le même signe que $\underline{u}_1$; [on] dit que la *direction initiale* de I est directe, resp. rétrograde, s'il en est ainsi de $\underline{u}_1$, ou ce qui revient au même, de C_1. De même on définit la *direction finale* de I comme celle de $\underline{u}_n$, ou encore celle de C_r. Direction initiale et finale sont égales si la longueur utile r est impaire, inégales si elle est paire. (**NB** Ce n'est que pour $n = 0$, i.e. quand $I = \{0\}$ est la catégorie ponctuelle, que direction initiale et direction finale ne sont pas définies.) p. 3

Les chaînes composables correspondent aux sous-catégories pleines de I qui sont isomorphes à une catégorie $\mathbf{\Delta}_i$ (où i est la longueur de la chaîne). Notons que $\mathbf{\Delta}_i$ n'a pas d'autre automorphisme que l'identité, donc les chaînes maximales de longueur donnée $i \geq 1$ correspondent exactement aux foncteurs

$$\mathbf{\Delta}_i \hookrightarrow I$$

qui sont injectifs sur les sommets. Un tel foncteur est pleinement fidèle, et un monomorphisme. Ce foncteur définit une application strictement croissante ou strictement décroissante sur les sommets, suivant que la chaîne est directe ou rétrograde. [page 5]

Commentary · Exposition [ms. 4]

Runs, simplices, and the meaning of maximality

Grothendieck next separates the places where a path can be composed from the places where its direction turns. A maximal constant-sign run of the word τ is his structural chain. If the run lengths are $l_1, \ldots, l_r$, then

$$n = l_1 + \cdots + l_r,$$

and the sign of the ath run is $(-1)^{a-1}\varepsilon_1$, when signs are written as ± 1. This proves the parity rule for initial and final directions. The useful length r records the number of changes of compositional regime, rather than the number of arrows which happen to have been written down.

The paragraph identifying chains with simplex subcategories needs a small but substantive separation. A chain was defined on ms. 3 to be a contiguous set of transitions. An object-injective functor $\Delta_k \to I$ may skip vertices. For example,

$$\Delta_1 \longrightarrow \Delta_2, \qquad 0 \longmapsto 0, \quad 1 \longmapsto 2,$$

is fully faithful, but its image is not the domain of one transition of Δ_2; nor does it describe its maximal structural chain, whose length is two. The words "chaînes maximales" and the unrestricted injectivity condition both occur in the edition. They therefore remain in the transcription, with their correction supplied here.

The exact statement is useful in its own right. For $k > 0$, an injective functor from Δ_k picks $k+1$ vertices in a single structural interval, in their categorical order. Indeed its first and last image vertices are connected by an arrow, so every vertex between them in the numerical order belongs to the same constant-sign run. All intermediate image vertices lie between these two in that run's categorical order. Conversely, any such selection determines an injective functor. It is fully faithful because the target is thin and its order on those vertices is exactly the simplex order. Its object function is strictly increasing in the external orientation for a positive run and strictly decreasing for a negative run.

To recover a chain in the earlier sense, impose numerical contiguity of the image. To recover a maximal structural chain, require the image to be the whole structural interval. These distinctions do not disturb the intended decomposition into runs. They identify which extra condition the printed correspondence omitted, while preserving its correct full-faithfulness assertion.

Notons que tout simplexe type $\mathbf{\Delta}_i$, $i \geq 0$, peut être vu comme un intervalle, de longueur i, de longueur utile 1 (si $i \geq 1$, 0 si $i = 0$). Son origine est 0, son extrémité est i. Mais appelons une catégorie quelconque une *catégorie intervalle* si elle est isomorphe à un intervalle standard. Notons ici

Proposition 1. *Soit I un intervalle standard de longueur $n \geq 1$.*

a) Soit $i \in I$. Pour que $i \in \{0, n\}$, i.e. pour que i soit un des deux 'bouts' *(ou* 'extrémités'*) (source ou but) de I, il faut et il suffit que l'une des deux catégories I/i ou $i\backslash I$ soit la catégorie ponctuelle et que l'autre soit isomorphe à un $\mathbf{\Delta}_j$ ([1]). (Caractérisation intrinsèque des bouts d'un intervalle.) Pour que i soit la source, il faut et il suffit que $i\backslash I$* `[phrase incomplète]`

b) Tout foncteur $I \longrightarrow I$ qui fixe les deux extrémités de I est l'identité.

Corollaire. *Le groupe des automorphismes de I a au plus deux éléments. S'il en a deux, l'automorphisme non trivial échange les deux extrémités. (On dit alors que l'intervalle est* symétrique.*)*

Commentary · Exposition [ms. 5]

The endpoint determines an orientation, not an endofunctor

The purpose of Proposition 1 is to make orientation intrinsic enough to transport it through isomorphisms. Its first assertion is correct. At an endpoint, one of the two slices is a point and the other is the linearly ordered category running along the unique adjacent structural chain. An interior vertex at which the signs agree has both a nonidentity incoming and a nonidentity outgoing arrow, so neither slice is a point. At a turning vertex, one slice is a point, but the other has two nonempty branches meeting at their common source or common target. In the source case it has two incomparable maximal objects, and in the target case two incomparable minimal objects. It cannot be a simplex category. This gives the stated characterization of the two endpoints.

The second assertion is false with the word "foncteur" which the edition prints. On Δ_2 the monotone map

$$0 \longmapsto 0, \qquad 1 \longmapsto 0, \qquad 2 \longmapsto 2$$

defines a nonidentity endofunctor fixing both endpoints. The arrow $1 \to 2$ is sent to the composite $0 \to 2$; there is no requirement that a functor preserve a generating transition as a generating transition. This is exactly the flexibility which the later category of types is intended to accommodate.

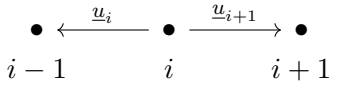
What the subsequent argument proves is that an *automorphism* fixing the endpoints is the identity. Here is a direct proof which also supplies the uniqueness statements used on ms. 7–9. A nonidentity arrow of an interval is indecomposable into two nonidentity arrows if and only if it is a transition: the displayed hom-set formula proves this, by factoring every longer directed segment at an intermediate vertex. Isomorphisms preserve indecomposability. The graph of indecomposable arrows, after forgetting directions, is the ordinary path with vertices $0, \ldots, n$. A graph automorphism fixing 0 fixes its unique neighbour, then that neighbour's still unvisited neighbour, and so on. It fixes every vertex and hence every arrow. Every graph automorphism is consequently the identity or vertex reversal. The categorical automorphism group is a subgroup of this two-element group.

This is also the intrinsic reconstruction of the external total order: choose one endpoint, call it s_0, and successively take the unique neighbour not already visited. Grothendieck's slice argument implements the same reconstruction in categorical terms. In its inductive paragraph the slices must be taken in the remaining full tail I' at s_i; the printed $i\backslash I$ and I/i have dropped that qualification. Neither reconstruction is preserved by arbitrary functors, which may collapse or lengthen transitions.

The unfinished sentence about characterizing the *source* of the interval should not be completed by guessing. The underlying category determines the pair of endpoints, but a symmetric interval has an automorphism interchanging them. Selecting one of them is precisely the additional orientation. After the definition of a reduced interval on ms. 12 we prove a sharp version of the stronger rigidity assertion: all endpoint-fixing endofunctors are identities exactly for reduced intervals.

[page 6]

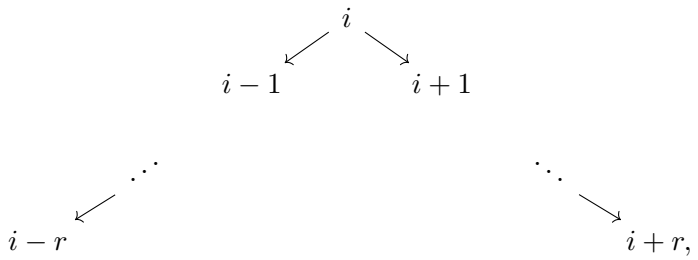
NB Si i n'est pas un bout, et si néanmoins l'une des deux catégories $i\backslash I$, I/i est ponctuelle, disons I/i, alors au voisinage immédiat de i, I a la forme



1. où j est la longueur de l'unique chaîne structurale qui 'touche' i (i.e. telle que i soit dans la sous-catégorie pleine définie par ladite chaîne).

($\underline{u}_i$ rétrograde, $\underline{u}_{i+1}$ directe - si c'était $i\backslash I$ la ponctuelle, alors $\underline{u}_i$ serait directe, $\underline{u}_{i+1}$ rétrograde - en tous les cas ces deux flèches sont de sens opposé, i.e. non composables). On trouve donc que $i\backslash I$ est une catégorie de la forme

p. 4



où r est la longueur de la chaîne structurale contenant $\underline{u}_i$, s celle de la chaîne structurale contenant $\underline{u}_{i+1}$. Cette catégorie est une somme amalgamée $\mathbf{\Delta}_r \amalg \mathbf{\Delta}_s$ (recollées suivant leur origine), mais n'est pas isomorphe à une $\mathbf{\Delta}_j$, p. ex. faute d'avoir un objet final.

Cette observation prouve a). Pour prouver b), il suffit de prouver que l'ordre total sur I est déterminé de façon intrinsèque, par le choix de *l'origine* s_0 de I (parmi ces deux extrémités). Supposons que l'on ait déjà déterminé intrinsèquement les sommets s_0, s_1, …, s_i ($i \geq 0$). Comment déterminer s_{i+1}? (Quand $i < n$.) Soit I' la sous-catégorie pleine de I formée des [page 7]
$j \in I$ qui ne sont pas parmi les s_j `[plutôt` s_k`]` déjà construits, sauf le dernier s_i (donc $s_i \in I'$, $\operatorname{card} I' \geq 2$). Alors I' est isomorphe à un intervalle standard, avec s_i correspondant à l'origine. Une des catégories $i\backslash I$, I/i est ponctuelle, l'autre isomorphe à une catégorie $\mathbf{\Delta}_j$, et de plus, i est objet initial ou final de $\mathbf{\Delta}_j$. Si c'est l'objet initial 0, on prend pour s_{i+1} le successeur immédiat de i dans $\mathbf{\Delta}_j$. Si c'est l'objet final j, on prend pour s_{i+1} le prédécesseur immédiat $j - 1$.

Si on était parti du but n de I, au lieu de partir de la source, on aurait trouvé la relation d'ordre total opposée. Affectant l'élément but du numéro d'ordre 0, on trouve une renumération de I, i devenant $n - i$, et les flèches de transition qui étaient directes pour la relation d'ordre initiale, deviennent rétrogrades dans la relation d'ordre opposée, et inversement. On voit ainsi :

Proposition 2. *Si I est un intervalle standard, il existe un unique couple (I', θ), où I' est un intervalle standard, et θ un isomorphisme $I \xrightarrow{\sim} I'$ qui transforme le but de I dans la source de I' (et la source de I en le but de I').*

Cet intervalle I' est appelé l'intervalle *opposé* de I. Il faut le voir, intrinsèquement, comme la 'même' categorie I, mais avec une 'orientation' opposée.

Une façon plus claire de voir ceci est la suivante. Soit J une catégorie intervalle, i.e. [page 8]
isomorphe à un intervalle standard. Alors il y a exactement (si $I \not\simeq e$) *deux* couples (I, θ) d'un intervalle standard I et d'un isomorphisme $\theta : I \xrightarrow{\sim} J$. Ces deux isomorphismes induisent sur J deux ordres totaux, opposés l'un de l'autre. Chaque couple est déterminé de façon unique par ledit ordre total, qu'on appelera une *orientation* de l'intervalle J. Ainsi J a deux orientations, et chacune détermine un *isomorphisme orienté* (i.e. respectant les orientations) p. 5
de J avec un intervalle standard ([2]). Ces deux intervalles sont les mêmes si et seulement s'il existe un automorphisme de J (nécessairement unique) qui échange ses deux orientations, ou ce qui revient au même, qui échange entre elles les deux extrémités. (Ainsi les catégories opposées l'une de l'autre



2. **NB** On parlera de l'intervalle *orienté opposé* I^{opp} d'un intervalle *orienté* I, qu'il faut se garder de confondre avec la catégorie opposée I^o. Ici I^{opp} a même catégorie sous-jacente que I, mais avec une orientation opposée.

sont chacunes des intervalles symétriques.) Choisir une orientation revient aussi ‘a choisir l'origine parmi les deux extrémités.

Notons que la *catégorie opposée* à un intervalle est un intervalle, et la catégorie opposée à un intervalle standard est l'intervalle standard qui s'en déduit en changeant tous les signes sur les intervalles de transition $[i-1, i]$ $(1 \leq i \leq n)$. On se gardera de confondre la propriété $I \simeq I^o$, avec la propriété de symétrie.

Ainsi les deux catégories (Ψ_1 et Φ_1) ci-dessus sont symétriques, mais opposées l'une de l'autre sans être isomorphes. D'autre part, les $\mathbf{\Delta}_i$ ne sont pas symétriques, mais chacune isomorphe à la catégorie opposée. [page 9]

Si I est isomorphe à la catégorie opposée I^o, et si θ est un tel isomorphisme (nécessairement unique), alors il est immédiat que θ renverse l'orientation (sinon ce serait l'identité sur les sommets, et les flèches de transition seraient à la fois directes et indirectes), donc $\theta(0) = n$, $\theta(1) = n-1$, et on veut que $\underline{u}_1$ et $\underline{u}_n$ sont de même signe. Mais si I est symétrique, $\underline{u}_1$ et $\underline{u}_n$ sont de signe opposé. Donc I ne peut être à la fois symétrique et isomorphe à l'intervalle opposé `[plutôt à la catégorie opposée; exception :` $I = \{0\}$`]`.

Pour toute catégorie $\mathbf{\Delta}_i$, celle-ci étant un intervalle, il y a (si $i > 0$) *deux orientations.* En tant qu'intervalles orientés, il y a $\mathbf{\Delta}_i^+$ (avec l'orientation ordinaire, donné par l'ordre ordinaire sur $\mathrm{Ob}\,\mathbf{\Delta}_i = [0, i]$), et $\mathbf{\Delta}_i^-$ avec l'orientation opposée.

p. 6
[page 10]

Commentary · Examples [ms. 9]

Reversal and categorical duality in a sign word

Grothendieck's two uses of “opposite” describe operations which can be written down independently. This makes the examples of symmetry and self-duality entirely explicit. For a sign word $\tau = (\varepsilon_1, \ldots, \varepsilon_n)$, put

$$R\tau = (-\varepsilon_n, \ldots, -\varepsilon_1),$$
$$D\tau = (-\varepsilon_1, \ldots, -\varepsilon_n).$$

The first is the standard model of the same category with its external orientation reversed; the standardizing isomorphism sends i to $n-i$. The second is the categorical opposite with the original numerical orientation. The two involutions commute, and $RD\tau$ is the word read backwards without changing its signs.

For $n > 0$, the interval is symmetric precisely when

$$\varepsilon_i = -\varepsilon_{n+1-i} \qquad (1 \leq i \leq n).$$

Necessity follows because a nonidentity automorphism must be reversal, by the preceding proof; sufficiency follows by checking each transition. This condition forces n to be even. Likewise $I \cong I^o$ precisely when

$$\varepsilon_i = \varepsilon_{n+1-i} \qquad (1 \leq i \leq n).$$

The isomorphism must reverse the numerical order: an order-preserving bijection of the vertex line would be the identity and would have to reverse a nonidentity transition without changing its endpoints. These two sign conditions cannot hold together for a nonempty word.

For instance, $(+,-)$ and $(-,+)$ satisfy the first condition. They give the cospan and span, each symmetric, but the two categories are not isomorphic: one has a terminal object and two minimal objects, the other an initial object and two maximal objects. The constant word $(+,+,+)$ satisfies the second condition and gives the self-dual category Δ_3 with no nonidentity automorphism. A less uniform self-dual example is $(+,-,+)$; its middle sink and source are exchanged under the anti-isomorphism. For a reduced alternating word, symmetry holds at positive even length and self-duality at odd length.

The point is separate: it has a unique orientation and only the identity automorphism, and both operations leave it unchanged. Keeping this case explicit avoids construing its coincident endpoints as two different choices.

2 Sous-catégories pleines d'un intervalle

Soit I une catégorie intervalle orientée, de longueur $n \geq 1$. Nous identifierons $\mathrm{Ob}\, I$ à l'ensemble des entiers $[0, n]$. Pour une chaîne $\underline{u}_i$, $\underline{u}_{i+1}$, …, $\underline{u}_{i+l-1}$ de flèches de transition dans I, on appelle *domaine* de la chaîne, l'ensemble des sources et de buts de ces éléments. C'est donc un intervalle d'entiers $[i-1, i+l-1]$, de longueur l égale à celle de la chaîne, dont l'origine est déterminé par la première flèche $\underline{u}_i$ de la chaîne (c'est sa source, si $\underline{u}_i$ est directe, et son but si $\underline{u}_i$ est rétrograde), et dont l'extrémité est déterminée par la dernière $\underline{u}_{i+l-1}$. Les intervalles d'entiers (vus comme des sommets de I) correspondant aux chaînes structurales sont appelés les *intervalles structuraux* ou *intervalles critiques* de I. Contrairement aux chaînes structurales, ces intervalles ne forment pas une partition de I, puisque deux intervalles structuraux peuvent avoir une extrémité commune. Si les chaînes structurales sont

$$C_1,\ C_2,\ \ldots\ ,\ C_r \qquad (r = \text{ longueur utile de } I)$$

de longueurs

$$l_1,\ l_2,\ \ldots\ ,\ l_r,$$

alors les intervalles structuraux sont

$$\underbrace{[0, l_1]}_{I_1},\ \underbrace{[l_1, l_1+l_2]}_{I_2},\ \ldots\ ,\ \underbrace{[l_1+\cdots+l_{r-1}, l_1+\cdots+l_{r-1}+l_r]}_{I_r},$$

une suite d'intervalles d'entiers dont deux consécutifs ont une extrémité commune, et tels que si $i < j$, chaque élément de I soit $\leq$ chaque élément de I_j. On appelle *sommet critique* de I les extrémités des intervalles structuraux.

Proposition 3. *Soit $s \in I$, soit $I' = I/s$ (sous-catégorie pleine de I des s' tels qu'il existe $s' \longrightarrow s$), $I'' = I/s$* `[plutôt` *$I'' = s\backslash I$*`]`*, $I'_0 = (I/s)\backslash\{s\}$, $I''_0 = (s\backslash I)\backslash\{s\}$ ([3]). Pour que s soit critique, il faut et il suffit que $I\backslash\{s\}$ soit catégorie somme de I'_0 et I''_0.*

Commentary · Exposition [ms. 10] [page 11]

Critical vertices and the correct deletion test

Grothendieck wants to retain exactly the vertices which a path cannot bypass by composing arrows. These are the endpoints of the structural intervals. If their positions are

$$0 = c_0 < c_1 < \cdots < c_r = n,$$

then $[c_{a-1}, c_a]$ is one constant-sign run, and each interior c_a is a change of sign. In the categorical partial order, the critical vertices are exactly the minimal and maximal vertices. This observation explains the two slices in Proposition 3, but it also exposes an error in its global formulation.

Let

$$A = (I/s) \setminus \{s\}, \qquad B = (s\backslash I) \setminus \{s\}.$$

These are respectively the strict predecessors and strict successors of s in the categorical order. They do not in general exhaust the other vertices. In $0 \to 1 \leftarrow 2$, take $s = 0$. It is critical; A is empty and B consists of 1, whereas $I \setminus \{0\}$ still contains 2. Thus the printed equality with all of $I \setminus \{s\}$ cannot be right. The edition prints the same reading.

The footnote gives the valid local test: there is no arrow between A and B. If s is critical, it is minimal or maximal and one of these categories is empty. If s is not critical, its two adjacent transitions have the same sign. One adjacent vertex is a strict predecessor and the other a strict successor; their composite through s is an arrow between A and B. Equivalently, the full subcategory on the vertices comparable with s, after removing s, is the coproduct $A \amalg B$ exactly when s is critical.

For the application to subintervals on the next page, the appropriate *global* test uses the numerical order. Suppose $0 < s < n$, and let L and H be the full subcategories on $[0, s-1]$ and $[s+1, n]$. Then

$$I \setminus \{s\} = L \amalg H \quad \Longleftrightarrow \quad s \text{ is critical.}$$

3. i.e. qu'on ait : il n'y a pas de flèches entre un élément de I'_0 et un élément de I''_0.

If s is a turning vertex, an arrow from one side to the other would traverse two transitions of opposite signs, which is impossible. If it is not a turning vertex, the two neighbouring vertices are connected by their composite through s, an arrow still present in the full subcategory obtained by deleting s. This corrected statement proves the separation property used in Proposition 5. At a numerical endpoint, deleting it leaves a connected interval; its criticality is already accounted for by its being an endpoint.

Proposition 4. *Soient i, j dans I, avec $i < j$. Conditions équivalentes :*

a) Il existe une flèche entre i et j (soit $i \longrightarrow j$, soit $j \longrightarrow i$).

b) i et j appartiennent à un même intervalle structural (nécessairement unique).

c) La sous-catégorie pleine de I définie par l'ensemble d'objets $[i, j]$ est isomorphe à une $\mathbf{\Delta}_\alpha$ (on aura $\alpha = j - i$).

On dira alors que i et j sont *directement connectés* dans I. (Et on utilisera la même terminologie dans une catégorie quelconque : i et j sont directement connectés si et seulement si $\mathrm{Hom}_I(i, j)$ *ou* $\mathrm{Hom}_I(j, i)$ est non vide.) p. 7

Proposition 5. *Soit $J \subseteq I$ une partie de I, considérons la sous-catégorie pleine de I qu'elle définit. Alors :*

a) Les composantes connexes J_α de J sont des intervalles et l'ordre induit sur un J_α par celui de I est une orientation (dite orientation induite*) de J_α.*

b) Supposons $\operatorname{card} J > 1$. *Soit i le plus petit élément de J, j le plus grand élément, donc $i < j$. Pour que J lui-même soit un intervalle, il faut et il suffit que tout sommet critique dans $[i, j]$ soit élément de J.*

([4]).

NB Si $s \in [i, j]$ est un sommet critique de I qui n'est pas dans J, alors par la proposition 3 il disconnecterait J, i.e. J serait somme directe de deux sous-catégories J' et J'' non vides ($i \in J'$, $j \in J''$), donc J ne serait pas un intervalle.

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Commentary · Examples [ms. 11]

Connected full subcategories and the vertices they must retain

Grothendieck is preparing a reduction which composes consecutive arrows without identifying a change of direction. The distinction between a full subcategory and an arbitrary subcategory matters here. A full subcategory retains every composite whose endpoints remain, even when an intermediate vertex has been removed. Thus numerical gaps do not necessarily disconnect it.

Let $J = \{j_0 < \cdots < j_m\}$ be a nonempty subset of the vertices of I, with its full induced category. Two successive chosen vertices j_a, j_{a+1} are joined by an arrow if and only if the signs between them are constant. By Proposition 4 this is equivalent to there being no critical vertex strictly between them. Whenever this condition fails, there can be no arrow from any chosen vertex at or before j_a to any chosen vertex at or after j_{a+1}, in either direction: such an arrow would cross the same change of sign. The connected components are therefore the maximal consecutive blocks of the list $(j_0, \ldots, j_m)$ in which successive entries are directly connected.

Within one such block, make a generating transition between each successive pair, in the direction of its arrow in I. The free category on this oriented line is exactly the induced category. To verify fullness, an arrow between nonconsecutive retained vertices exists in I precisely when the signs on the whole intervening segment agree. That is also precisely the condition that the successive arrows of the block compose. This proves Proposition 5(a), including the claim about the induced orientation. It proves (b) as well: the whole chosen list is connected exactly when every ambient critical vertex between its extreme vertices was retained.

For a concrete calculation take

$$I : \quad 0 \longrightarrow 1 \longrightarrow 2 \longleftarrow 3 \longleftarrow 4 \longrightarrow 5.$$

4. On appelle *sous-intervalle* de I une sous-categorie pleine connexe, i.e. qui est elle-même un intervalle. Notons d'ailleurs que *toute* sous-catégorie I' de I qui est un intervalle, est nécessairement pleine, cf. plus bas `[paragraphe]` des foncteurs entre intervalles.

Its critical vertices are $0, 2, 4, 5$. The full category on $\{0, 2, 4, 5\}$ is the reduced interval

$$0 \longrightarrow 2 \longleftarrow 4 \longrightarrow 5.$$

Deleting 1 or 3 has merely replaced a pair of arrows by its composite. The full category on $\{0, 1, 4, 5\}$ has two components, $\{0, 1\}$ and $\{4, 5\}$, because the omitted turning vertex 2 separates them. The least subinterval containing 1 and 5 is $\{1, 2, 4, 5\}$: the endpoints must remain, the two intervening turns must remain, and the preceding calculation shows that nothing else is required. This is the corollary on ms. 12.

The last sentence of footnote 4 asserts more than fullness permits: an interval subcategory need not be full. Inside Δ_2, keep all three objects and the arrows $0 \to 1$ and $0 \to 2$, but omit $1 \to 2$. This is a subcategory, because its two nonidentity arrows are not composable. Abstractly it is the span interval with vertex order $1, 0, 2$. It is not full in Δ_2. Thus fullness must remain part of the definition of a subinterval; it cannot be inferred solely from the abstract shape of the subcategory.

Finally, for a full subinterval with at least two vertices, an interior retained vertex is a turn in the induced shape exactly when it was an ambient critical vertex. This proves Proposition 6. The reduced interval is consequently the smallest full connected part of I which contains its endpoints, and its length is the useful length of I.

Corollaire. *L'ensemble des sommets critiques de I est aussi l'ensemble des sommets du plus petit sous-intervalle de I contenant les extrémités. Plus généralement, si $i < j$ sont deux sommets de I, le plus petit sous-intervalle de I contenant i, j est formé de i, j et des sommets critiques s tels que $i < s < j$.*

Définition. Un intervalle est dit *réduit* (ou *irrédondant*) si tous ses sommets sont critiques, ou ce qui revient au même, si ses chaînes structurales sont de longueur 1.

Un intervalle orienté *réduit* est connu, à isomorphisme près, quand on connaît

a) sa longueur n,

b) sa direction initiale + (directe) ou − (rétrograde).

NB Cette direction est la même que la direction finale si n est impair, elle est opposée si n est pair.

Ainsi, suivant les deux cas de direction initiale, on trouve

$$0 \longrightarrow 1 \longleftarrow 2 \longrightarrow \cdots \qquad \cdots\ n \qquad (\text{Avec} \quad \begin{array}{ll} n-1 \longrightarrow n & \text{si } n \text{ impair,} \\ n-1 \longleftarrow n & \text{si } n \text{ pair.)} \end{array}$$

$$0 \longleftarrow 1 \longrightarrow 2 \longleftarrow \cdots \qquad \cdots\ n \qquad (\text{Avec} \quad \begin{array}{ll} n-1 \longleftarrow n & \text{si } n \text{ impair,} \\ n-1 \longrightarrow n & \text{si } n \text{ pair.)} \end{array}$$

Proposition 6. *Un sous-intervalle J, ayant i et j $(i < j)$ comme plus petit et plus grand sommet, est réduit si et seulement si tout $s \in J$ distinct de i et de j est critique.* p. 8

Corollaire. *La sous-catégorie pleine de I formée des sommets critiques est un sous-intervalle. On l'appelle l'intervalle réduit associé à I, noté $I_{\text{réd}}$.*

p. 9

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Commentary · Development (commentators') [ms. 12]

The critical-vertex inclusion has two adjoints

The reduced interval retains every turn, but Grothendieck has not yet described how to return to it from an arbitrary vertex. There are two particularly useful choices, one following the categorical source of a run and one following its categorical target. They relate reduction to the endpoint-preserving transformations which a path calculus needs.

Two canonical collapses. [ours] Let $j : R = I_{\text{red}} \hookrightarrow I$ be the full inclusion of the critical vertices of an oriented interval. There are retractions $r_s, r_t : I \to R$ with

$$r_t \dashv j \dashv r_s, \qquad r_s j = r_t j = \mathrm{id}_R.$$

Both retractions fix the endpoints. At a noncritical vertex x, $r_s(x)$ is the categorical source of its structural interval and $r_t(x)$ is that interval's categorical target. At a critical vertex both are the identity. For the point we set $R = I$ and take all three functors to be the identity.

Proof. A noncritical vertex lies strictly inside a unique structural interval. On a positive run $[a, b]$, the source-collapse sends every interior vertex to a, leaves b at b, and is a monotone map of its categorical chain. On a negative run the source is b and the same argument applies with the categorical order reversed. The target-collapse is obtained by exchanging source and target. The run-wise maps agree at shared critical vertices. Every arrow belongs to one run, so these formulas give functors on the whole interval.

To verify the adjunctions rather than merely exhibit homotopies, let q be a critical vertex. If x is itself critical, fullness of j gives the required hom-set bijections. If x is interior to a run, the only critical vertices comparable with x are its source a and target b, with $a \to x \to b$ in the categorical order. Indeed an arrow extending beyond either endpoint of that run would have to pass a change of sign. Consequently

$$I(jq, x) \neq \varnothing \iff q = a \iff R(q, r_s x) \neq \varnothing,$$
$$I(x, jq) \neq \varnothing \iff q = b \iff R(r_t x, q) \neq \varnothing.$$

For the middle equivalences we use that a source endpoint is a minimal vertex of I and a target endpoint a maximal vertex; hence the same holds in R. All these hom-sets are empty or singletons, so the bijections are unique and natural. They establish $j \dashv r_s$ and $r_t \dashv j$. The two retraction identities follow directly from the definition. The counit and unit are the unique arrows

$$j r_s(x) \longrightarrow x, \qquad x \longrightarrow j r_t(x),$$

which are identities whenever x is critical. This also verifies the triangle identities directly. □

These natural transformations are categorical deformations relative to all critical vertices, in particular relative to both endpoints. Applying the nerve, a natural transformation $F \Rightarrow G$ supplies a simplicial homotopy through the functor $I \times \Delta_1 \to I$ which it defines. Thus Nj has homotopy inverses Nr_s and Nr_t. Nothing in this argument asserts that the interval categories themselves are equivalent: a skeletal interval with an interior noncritical vertex has strictly more objects than its reduction.

The effect on diagrams gives a more informative comparison. For every category X, precomposition yields

$$r_s^* \dashv j^* \dashv r_t^*, \qquad j^* r_s^* = j^* r_t^* = \mathrm{id}_{\mathrm{Fun}(R,X)}.$$

This reversal of adjunctions under precomposition is the standard calculation of Riehl (n.d., Proposition 4.3.7); it can also be read directly from the two displayed natural transformations by applying each diagram $I \to X$. The source extension inserts identity arrows before the final arrow of each directed run, and the target extension inserts them after its first arrow. Restriction to R composes each run.

For $I = \Delta_2$, $R = \{0, 2\}$ and $X = \Delta_1$, consider the two diagrams with vertex values

$$F = (0, 0, 1), \qquad G = (0, 1, 1).$$

They have the same restriction $(0, 1)$ to R. There is no natural transformation $G \to F$, since its middle component would have to be $1 \to 0$ in Δ_1, whereas their restrictions have an identity map. Hence j^* is not full and is not an equivalence. If X is a groupoid, every component of both comparison transformations is invertible, and j^* is an equivalence, with either section as an inverse. Reduction therefore loses factorization data in general, but loses no such data when all arrows of the target are already invertible.

The general precomposition-adjunction statement is known and cited above. The critical-vertex reduction itself also appears in Arakawa and Cnossen (2026, Lemma 4.16). For the additional explicit adjoints and the sharp rigidity criterion below, we checked that construction, the local interval arguments here, and the adjunction calculus in Riehl (n.d., §4.3); targeted searches for critical-vertex reductions and their adjoints did not locate these particular formulations. We supply the extra deductions as elementary developments of the present definitions, without a priority claim.

Commentary · Development (commentators') [ms. 12]

The exact scope of endpoint rigidity

The false generality of Proposition 1(b) suggests asking precisely which part of an interval allows an endpoint-preserving functor to move a vertex. The preceding collapses give one implication, while a finite displacement argument gives the other.

Endpoint rigidity criterion. [ours] For a finite oriented interval I, the following are equivalent: I is reduced; every endofunctor of I fixing its two designated endpoints is the identity. We include the point as a reduced interval.

Proof. Suppose I is reduced, of length $n > 0$, and identify its vertices with $[0, n]$. A nonidentity arrow of I is then a single transition: two consecutive transitions have opposite signs and cannot compose. If $f : I \to I$ is any functor, the image of a transition is therefore either an identity or a transition. Its object function satisfies

$$-1 \le f(i) - f(i-1) \le 1 \qquad (1 \le i \le n).$$

If $f(0) = 0$ and $f(n) = n$, summation gives

$$n = f(n) - f(0) = \sum_{i=1}^{n} \big(f(i) - f(i-1)\big) \le n.$$

Equality forces every summand to be 1. Thus $f(i) = i$ for all i, and thinness forces f to be the identity on arrows as well. We have made no monotonicity assumption on f; it is a consequence of the equality. The point case is immediate.

Conversely, if I is not reduced, a structural interval has an interior vertex. The endofunctor jr_s of the preceding result fixes every critical vertex and hence both endpoints, but sends that interior vertex to the source of its run. It is not the identity. □

The obstruction is therefore exactly a nontrivial factorization inside a constant-sign run. Automorphisms cannot change such a factorization because they preserve the graph of indecomposable arrows. General functors can replace it by an identity followed by a longer composite. This is why the automorphism proof on ms. 6–7 is sound while its printed extension to all endofunctors fails.

3 Chemins. Catégorie de chemins d'un type donné τ.

Soient I un intervalle orienté, et X une catégorie. On appelle *chemin de type I dans X* tout foncteur $I \longrightarrow X$, et *catégorie de chemins de type I dans X* la catégorie

$$\underline{\mathrm{Ch}}_I(X) = \underline{\mathrm{Hom}}(I, X).$$

Comme I est *canoniquement* isomorphe à une catégorie intervalle standard I_0, on peut identifier les chemins de type I, au de type I_0. Le type τ d'un chemin le longueur n ($= \operatorname{card} I$) dans X peut donc être décrit par une suite de n signes + ou −

$$\tau_1,\ \tau_2,\ \ldots\ ,\ \tau_n \qquad (\tau_i \in \{-1, +1\} \text{ pour } 1 \le i \le n).$$

À cause du fait que I est engendré pas ses flèches de transition, un foncteur $c : I \longrightarrow X$ est déterminé quand on connaît

a) Les $c(i)$ ($i \in I$), i.e. une suite de $n+1$ objets de X :

$$a_0,\ a_1,\ \ldots\ ,\ a_n\ ([5]);$$

b) Les $c(\underline{u}_i)$ (où les $\underline{u}_i$ sont les flèches de transition dans I), i.e. pour tout $i \in [1, n]$ une flèche u_i dans X entre a_{i-1} et a_i, allant de a_{i-1} dans a_i si $\underline{u}_i$ est directe ($\tau_i = +1$), et de a_i dans a_{i-1} si $\underline{u}_i$ est rétrograde ($\tau_i = -1$),

5. a_0 est dit la *source* du chemin, a_n son but.

et à toute telle donnée a), b) correspond un foncteur. Ainsi on pourra identifier les chemins de type I dans X aux diagrammes du type précisé dans X. Et les morphismes entre chemins correspondent de même aux morphismes entre diagrammes a et a', une flèche $a \xrightarrow{\alpha} a'$ étant donc une suite de $n+1$ flèches [page 14]

$$\alpha_i : a_i \longrightarrow a'_i, \qquad i \in [0, n],$$

satisfaisant la condition de commutativité pour les n carrés

$$\begin{array}{ccc} a_{i-1} & \xrightarrow{u_i} & a_i \\ {\scriptstyle \alpha_{i-1}}\downarrow & & \downarrow{\scriptstyle \alpha_i} \\ a'_{i-1} & \xrightarrow{u'_i} & a'_i \end{array} \qquad \text{(cas } \underline{u}_i \text{ directe)}$$

ou

$$\begin{array}{ccc} a_{i-1} & \xleftarrow{u_i} & a_i \\ {\scriptstyle \alpha_{i-1}}\downarrow & & \downarrow{\scriptstyle \alpha_i} \\ a'_{i-1} & \xleftarrow{u'_i} & a'_i. \end{array} \qquad \text{(cas } \underline{u}_i \text{ rétrograde)}$$

p. 10

Si on a une flèche entre deux sommets quelconque i, j dans I, avec $i < j$ et j non successeur immédiat de i, alors i, j appartiennent à un même intervalle critique, et dans I on aura

$$i \xrightarrow{\underline{u}_{i+1}} i+1 \xrightarrow{\underline{u}_{i+2}} \cdots\ j-1 \xrightarrow{\underline{u}_j} j \qquad \text{une chaîne de longueur } j-i \geq 2$$

ou

$$i \longleftarrow i+1 \longleftarrow \quad \cdots\ j-1 \longleftarrow j,$$

et la flèche correspondante entre a_i et a_j s'obtient en composant les flèches u_i correspondantes.

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Commentary · Exposition [ms. 14]

A fixed path type is a freely generated diagram

Grothendieck now uses the work on intervals to make path data and path morphisms completely explicit. Fixing the type does not impose any compatibility among the arrows of the path beyond their sources and targets. This is the benefit of having chosen a free path category as the domain.

For example, let the type be $(+,+,-)$. A diagram consists of

$$a_0 \xrightarrow{u_1} a_1 \xrightarrow{u_2} a_2 \xleftarrow{u_3} a_3.$$

The only nonidentity arrow of the domain which is not a transition is $0 \to 2$, and its image must be u_2u_1. Given a second diagram $a'_0 \to a'_1 \to a'_2 \leftarrow a'_3$, a map between them consists of four arrows $\alpha_i : a_i \to a'_i$ satisfying

$$\alpha_1 u_1 = u'_1 \alpha_0, \qquad \alpha_2 u_2 = u'_2 \alpha_1, \qquad \alpha_2 u_3 = u'_3 \alpha_3.$$

Naturality for the longer arrow follows by calculation:

$$\alpha_2(u_2u_1) = (u'_2\alpha_1)u_1 = u'_2(u'_1\alpha_0) = (u'_2u'_1)\alpha_0.$$

Every longer constant-sign run is handled by the same induction, with the index order reversed when its arrows are retrograde. Thus the displayed transition squares are both necessary and sufficient for a natural transformation. Identities and composition are taken vertex by vertex; composing commutative squares verifies that the result is again a morphism of paths.

This also gives a useful exact model of the fixed-type category. Let $\mathrm{Arr}(X) = \mathrm{Fun}(\Delta_1, X)$, with ordinary domain and codomain functors d, c. For a positive transition its ordered endpoint pair is (d, c); for a negative one it is (c, d). Then $\mathrm{Fun}(I_\tau, X)$ is the iterated strict fibre product of n copies of $\mathrm{Arr}(X)$, matching the second ordered endpoint of one to the first ordered endpoint of the next. On objects this identifies the shared vertices. On arrows it identifies the corresponding vertical

components of the naturality squares. The two descriptions are therefore isomorphic as categories, not merely bijective on objects. At length zero the category is X itself.

The construction retains all factorizations and all commuting ladders between them. It has not inverted the retrograde arrows in X and has not identified paths related by a ladder. Those operations would require additional choices or a localization. Here a retrograde transition simply says that the arrow of X points against the chosen order of traversal. Keeping this distinction explicit will matter when the category of types starts to vary.

Exemples.

1. $I = \mathbf{\Delta}_0 = e$, $\underline{\mathrm{Ch}}_I(X) \simeq X$. Les chemins de longueur 0 s'identifient aux objets a de X. L'unique élément id_e de $\mathrm{Fl}(I)$ donne id_a dans X.
2. $I = \mathbf{\Delta}_1^+$, $\underline{\mathrm{Ch}}_I(X) \simeq \underline{\mathrm{Fl}}(X)$. Les chemins de type $\mathbf{\Delta}_1^+$ sont ceux de longueur 1 et *directs,* i.e. donnés par une flèche
$$a_0 \longrightarrow a_1$$
dont la source est aussi la source a_0 du chemin. Par contre, les chemins de type $\mathbf{\Delta}_1^-$ sont donnés par des diagrammes
$$a_0 \longleftarrow a_1,$$
i.e. par des flèches dont le but est considéré comme source du chemin. Bien sûr, les catégories $\underline{\mathrm{Ch}}_{\mathbf{\Delta}_1^+}(X)$ et $\underline{\mathrm{Ch}}_{\mathbf{\Delta}_1^-}(X)$ sont canoniquement isomorphes, mais par un isomorphisme qui *échange source et but du chemin.* Nous nous garderons donc bien de confondre les deux chemins précédents.
3. $\mathbf{\Delta}_2^+$, $\mathbf{\Delta}_2^-$, correspondant aux diagrammes [page 16]
$$a_0 \longrightarrow a_1 \longrightarrow a_2$$
resp.
$$a_0 \longleftarrow a_1 \longleftarrow a_2,$$
on peut faire une remarque similaire que pour $\mathbf{\Delta}_1^+$, $\mathbf{\Delta}_1^-$.
4. Φ_1, Ψ_1, ce sont respectivement les intervalles p. 11
$$\Phi_1 :\ 0 \xleftarrow{u_0} 1 \xrightarrow{u_1} 2 \qquad \text{ou} \qquad \begin{array}{ccc} & & 0 \\ & \nearrow & \\ 1 & & \\ & \searrow & \\ & & 2 \end{array},$$
et
$$\Psi_1 :\ 0 \xrightarrow{u_0} 1 \xleftarrow{u_1} 2 \qquad \text{ou} \qquad \begin{array}{ccc} & & 0 \\ & \swarrow & \\ 1 & & \\ & \nwarrow & \\ & & 2 \end{array}.$$
Ils correspondent respectivement aux diagrammes
$$a_0 \xleftarrow{u_0} a_1 \xrightarrow{u_1} a_2$$
et
$$a_0 \longrightarrow a_1 \longleftarrow a_2.$$
Les catégories Φ_1, Ψ_1 sont opposées l'une de l'autre, et ne sont pas isomorphes. Aussi n'y a-t-il en général aucun isomorphisme entre $\underline{\mathrm{Ch}}_{\Phi_1}(X)$ et $\underline{\mathrm{Ch}}_{\Psi_1}(X)$.

Commentary · Context [ms. 16]

Zigzags before taking a localization

Grothendieck's paths belong to the same categorical language as fractions, while retaining information which a category of fractions has already identified. In Gabriel and Zisman (1967, Chapter I, §1.1), localization adjoins inverses to a specified collection of arrows and imposes the relations needed for functoriality and inversion. A span $x \leftarrow z \rightarrow y$ can consequently represent a morphism from x to y when its left arrow is among those to be inverted. The fixed-type path category written here makes no such selection: the objects and every component of a natural transformation are ordinary categorical data.

The two-dimensional comparison is equally concrete. The reduced hammocks of Dwyer and Kan (1980a, Definition 2.1) consist of commutative ladders of zigzags, but their vertical arrows and their left-pointing arrows are required to lie in a specified subcategory of weak equivalences; identity columns are removed and neighbouring columns of the same direction are composed. Their Proposition 3.1 identifies connected components of the resulting mapping spaces with ordinary localized morphisms. Grothendieck's full functor category of one fixed interval is therefore an ingredient of a comparable calculus, not by itself a hammock mapping space. The later commentary on the varying category of types explains the precise relation; here the historical comparison tells us why it is useful to retain ladders rather than only strings of arrows.

Soit I un intervalle orienté, et considérons l'*intervalle opposé* I^{opp}, ayant même catégorie sous-jacente mais l'orientation opposé. Comme la définition de la catégorie des chemins du type I ne fait pas intervenir l'orientation de I, on a donc un isomorphisme canonique

$$\underline{\mathrm{Ch}}_I(X) \xrightarrow{\sim} \underline{\mathrm{Ch}}_{I^{\mathrm{opp}}}(X).$$

[page 17]

Commentary · Examples [ms. 16]

A span and a cospan give different diagram categories

Grothendieck emphasizes that the direction of a turn changes the fixed-type category. The difference can be seen without any localization or homological algebra. We give a finite example which also distinguishes equivalence of categories, not only strict isomorphism.

Let X be the three-object poset

$$a \longleftarrow b \longrightarrow c.$$

A span diagram in X is a triple (x_0, x_1, x_2) with $x_1 \leq_X x_0$ and $x_1 \leq_X x_2$. If its middle value is b, each endpoint can independently be a, b, or c, giving nine diagrams. If the middle value is a, both endpoints must be a; if it is c, both must be c. Thus

$$\left|\mathrm{Ob}\,\mathrm{Fun}(\Phi_1, X)\right| = 3^2 + 1^2 + 1^2 = 11.$$

A cospan diagram has $x_0 \leq_X x_1$ and $x_2 \leq_X x_1$. For middle value b there is one possibility. For middle value a, both endpoints independently lie in $\{b, a\}$, giving four; for middle value c, both independently lie in $\{b, c\}$, again giving four. Hence

$$\left|\mathrm{Ob}\,\mathrm{Fun}(\Psi_1, X)\right| = 1^2 + 2^2 + 2^2 = 9.$$

Morphisms in both functor categories are completely determined: there is one from (x_0, x_1, x_2) to (y_0, y_1, y_2) exactly when $x_i \leq_X y_i$ for each i. The naturality squares then commute automatically because X is thin. These categories are posets and have no distinct isomorphic objects. An equivalence between posets is an isomorphism, since essential surjectivity is actual surjectivity and full faithfulness reflects their order. Their different object counts therefore rule out an equivalence.

Orientation reversal is a separate operation. For the span, interchanging x_0 and x_2 gives another span and an automorphism of its diagram category. It leaves the middle vertex and both arrow directions of the underlying span shape intact, while exchanging the two endpoint evaluations. Passing from the span to the cospan reverses the categorical arrows at the turn. The preceding count detects exactly that change.

Commentary · Later developments [ms. 16]

When a change of orientation becomes a stable equivalence

The span–cospan distinction remains real for the ordinary functor categories which Grothendieck has just defined. A later theorem explains why analogous diagram categories can nevertheless become equivalent after one passes to a stable homotopy theory. For a finite oriented tree Q, Groth and Šťovíček (2016, Theorem 9.19 and Corollary 9.23, arXiv v3) construct reflection equivalences and deduce strong stable equivalence between any two orientations of the same underlying tree. In particular, for a stable derivator $\mathbb{D}$,

$$\mathbb{D}^{\Phi_1} \simeq \mathbb{D}^{\Psi_1}.$$

These are equivalences of shifted derivators, natural with respect to exact morphisms. They are constructed using stable operations such as fibres, cofibres, and biproducts; they are not induced by an isomorphism of the span and cospan categories. The finite example above therefore presents no obstruction. It concerns arbitrary ordinary targets, whereas the reflection theorem uses stability to transform the values and arrows of the diagram. This gives a precise later setting in which changing the signs of an interval preserves a diagram theory even though it does not preserve its indexing category.

Néanmoins on se gardera de confondre un chemin de type I avec le chemin opposé de type I^{opp}, la source de l'un étant le but de l'autre et inversement. C'est dire qu'un chemin doit être vu comme *plus* qu'un simple foncteur

$$I \longrightarrow X,$$

il faut de plus se donner l'*orientation de I* comme faisant partie de la structure de chemin. Donc la définition formelle raisonnable, qui colle à l'intuition du chemin comme allant d'une *source* bien définie vers un *but* également bien défini, c'est qu'un chemin (défini en termes d'une catégorie intervalle I) est la donné d'un couple (ω, c), où ω est une orientation de I, et c un foncteur $c : I \longrightarrow X$. Quand on regarde la catégorie des chemins d'un 'type' déterminé, ce *type* est (I, ω), et non I. C'est quand ω est fixé une bonne fois que cette catégorie s'identifie impunément à $\underline{\mathrm{Hom}}(I, X)$ sans plus.

Il faut voir cette catégorie de chemins comme munie de la structure essentielle

$$\begin{array}{c} \underline{\mathrm{Ch}}_{(I,\omega)}(X) \\ \big\downarrow{\scriptstyle (\sigma,\beta)} \\ X \times X \end{array} \qquad \begin{array}{l} c \longmapsto (\sigma(c), \beta(c)) \\ \text{ou } \sigma(c) \text{ est la } \textit{source}, \\ \qquad \beta(c) \text{ le } \textit{but} \text{ de } c. \end{array}$$

p. 12
[page 18]

Commentary · Exposition [ms. 17]

The endpoints are part of the fixed-type object

Grothendieck's insistence on orientation is most transparent over the category of endpoint pairs. The category attached to an oriented type (I, ω) is not merely $\mathrm{Fun}(I, X)$; it comes with the functor

$$(\sigma, \beta) : \mathrm{Fun}(I, X) \longrightarrow X \times X.$$

On a natural transformation $\alpha : c \Rightarrow d$, its value is (α_0, α_n). Thus a morphism of paths may move the endpoints, and the displayed functor records exactly how. Its strict fibre over (x, y) consists of paths with endpoints x, y and natural transformations which are the identity at both endpoints. This fibre is a concrete category of fixed-endpoint ladders.

Let ω' be the reverse orientation on the same underlying interval category. The identity on that underlying category identifies the two ordinary functor categories; under this identification the endpoint pair changes from $(c(0), c(n))$ to $(c(n), c(0))$. Consequently the comparison is a functor over the transposition $s(x, y) = (y, x)$. If $x \neq y$, it carries the fibre over (x, y) to the fibre over (y, x), not to itself. This explains why equality of the underlying functor categories does not erase the distinction between a path and its reverse.

The endpoint map $\alpha : \{0,1\}_{\mathrm{disc}} \to I$ packages the same structure as restriction. For positive length its two images are distinct. For length zero it is the fold onto the sole vertex, so the word "inclusion" on the next page needs that qualification; restriction is then the diagonal $X \to X \times X$. All the formulas still hold. The subsequent discussion of symmetric types compares this identity-on-underlying-diagram reversal with the additional comparison induced by a reflection of the shape. Their difference belongs to the structure over $X \times X$, which is why the base category has been kept visible.

Ce foncteur s'identifie au foncteur de composition

$$c \longmapsto c \circ \alpha$$

où α est le foncteur d'inclusion

$$\begin{cases} \varepsilon \xrightarrow{\alpha} I \\ \varepsilon = \{0,1\}, & \text{catégorie discrète à deux éléments,} \\ \alpha \text{ donné par} & \alpha(0) = \text{ source de } I \\ & \alpha(1) = \text{ but de } I. \end{cases}$$

Si ω' est l'orientation opposée de I, on a un diagramme commutatif

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{(I,\omega)}(X) & \xrightarrow[\sim]{\varphi} & \underline{\mathrm{Ch}}_{(I,\omega')}(X) \\ \downarrow{\scriptstyle (\sigma,\beta)} & & \downarrow{\scriptstyle (\sigma',\beta')} \\ X \times X & \xrightarrow{\ s\ } & X \times X, \end{array}$$

où s est le foncteur de symétrie

$$s(x,y) = (y,x)$$

([6]). Donc l'isomorphisme φ ne respecte pas les flèches structurales 'source-but' (σ, β).

Il y a cependant un embarras quand il existe un isomorphisme d'intervalles *orientés*

$$\theta : (I,\omega) \xrightarrow{\sim} (I,\omega'),$$

i.e. quand I est symétrique. Ce n'est pas le cas pour les $\mathbf{\Delta}_i$ $(i \geq 1)$ (ces catégories n'ont pas d'autres automorphismes que l'identité), mais c'est le cas pour les catégories Φ_1, Ψ_1 de l'exemple 4°, plus généralement pour les catégories réduites de longueur *paire.* Comme [page 19]
chaque fois qu'on a un isomorphisme θ d'intervalles *orientés,* on trouve un isomorphisme entre les catégories de chemins de type (I,ω) et $(I,\omega)^{\mathrm{opp}} = (I,\omega')$, *compatible* cette fois avec les foncteurs (σ,β) :

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{(I,\omega)}(X) & \xrightarrow[\sim]{\Psi_\theta} & \underline{\mathrm{Ch}}_{(I,\omega')}(X) \\ & {\scriptstyle (\sigma,\beta)} \searrow \quad \swarrow {\scriptstyle (\sigma',\beta')} & \\ & X \times X. & \end{array}$$

Il est bon de regarder un *type* de chemins comme une donné non catégorique, mais p. 13
combinatoire, une simple suite τ de n signes τ_i $(1 \leq i \leq n)$ + ou − (n = longueur des chemins [page 20]
envisagés). Cette donné combinatoire définit un intervalle orienté standard, I_τ. Quand on a un intervalle orienté quelconque (I,ω), il définit un type τ, et on l'identifie canoniquement au type I_τ. Il vaut mieux écrire $\underline{\mathrm{Ch}}_\tau(X)$, plutôt que $\underline{\mathrm{Ch}}_{(I,\omega)}(X)$, pour bien indiquer que pour

6. Le foncteur φ est celui qui associe à tout chemin c, de type τ, le *chemin opposé* c^{opp} ou c^{-1}, de type τ^{opp}. Si $\tau = \tau^{\mathrm{opp}}$, c'est donc un chemin *de même type,* et φ est un *automorphisme* de $\underline{\mathrm{Ch}}_\tau(X)$. Cet automorphisme (fonctoriel en X) est déduit d'un automorphisme θ de I, cf. ci-dessous.

X donné, c'est là une catégorie totalement 'épinglée', sans automorphismes liés à la nature particulière de τ (à des automorphismes de τ!).

On peut préciser ce point ainsi. Soit τ' un autre type d'intervalles, et proposons-nous de déterminer des 'lois' qui permettent à tout chemin de type τ dans une catégorie arbitraire X, d'associer un chemin de type τ' de cette même catégorie, et ceci de façon 'fonctorielle', i.e. compatible avec les applications horizontales

$$\begin{array}{ccc} \mathrm{Ch}_\tau(X) & \xrightarrow{\mathrm{Ch}_\tau(f)} & \mathrm{Ch}_\tau(X') \\ {\scriptstyle L_X}\downarrow & & \downarrow{\scriptstyle L_{X'}} \\ \mathrm{Ch}_{\tau'}(X) & \xrightarrow{\mathrm{Ch}_{\tau'}(f)} & \mathrm{Ch}_{\tau'}(X') \end{array}$$

associées à un foncteur $f : X \longrightarrow X'$.

[page 21]

Commentary · Exposition [ms. 20]

Operations on paths are maps of their shapes

Grothendieck wants to do the combinatorics once, in a category small enough that every operation can be seen. A path in an arbitrary category carries many labels, but an operation valid for every category cannot depend on accidental properties of those labels. Its entire content is already present in the universal path. The passage from an oriented interval to its sign word makes this universal object unambiguous. It does not assert that a particular category of paths has no automorphisms; it removes the ambiguity coming from presenting the same unoriented interval in two different ways.

Write $P_I(X) = \mathrm{Hom}_{\mathrm{Cat}}(I, X)$ for the *set* of paths of shape I, keeping $X^I = \underline{\mathrm{Hom}}(I, X)$ for their category. The distinction between these two expressions matters in the displayed Yoneda argument. If

$$L_X : P_I(X) \longrightarrow P_J(X)$$

is natural in all functors $X \to Y$, put $\varphi = L_I(\mathrm{id}_I) : J \to I$. Applying naturality to the functor $c : I \to X$ gives

$$L_X(c) = c \circ L_I(\mathrm{id}_I) = c\varphi.$$

Thus every such operation is precomposition with exactly one φ, and every φ defines such an operation. This is the covariant Yoneda lemma, with the representing objects appearing in the reverse order; see Riehl (n.d., Theorem 2.2.4 and Corollary 2.2.8). No hypothesis about interval categories is needed for this argument.

There is then a canonical operation on morphisms of paths. If $v : c \Rightarrow d$, set $(L_X v)_j = v_{\varphi(j)}$. For an arrow $a : j \to k$ of J, its naturality square is the naturality square of v at $\varphi(a)$. Identities and composites are preserved component by component. Consequently the operation lifts to the functor

$$\varphi^* : X^I \longrightarrow X^J.$$

This proves the extension asserted on manuscript page 23, without assuming a separate coherence theorem. Natural transformations between changes of shape have an equally concrete meaning: $\alpha : \varphi \Rightarrow \psi$ gives $c\alpha : c\varphi \Rightarrow c\psi$, naturally in c and X.

The special advantage of intervals now enters. They are posets, so a functor into an interval is determined by its object function, and a natural transformation between two such functors exists precisely when the corresponding pointwise arrows exist. If it exists, it is unique. This uniqueness makes every required diagram of *shape transformations* commute. It does not make the categories X^I thin when X is an arbitrary category: those categories still remember all morphisms of X.

For fixed endpoints the same universal-path argument is even more economical. Suppose I and J have specified sources and targets. The operation $c \mapsto c\varphi$ preserves both endpoints for every X and c exactly when $\varphi(0_J) = 0_I$ and $\varphi(1_J) = 1_I$. Necessity follows by taking $X = I$ and $c = \mathrm{id}_I$; sufficiency is direct evaluation. This is the reason the first category of all changes of shape will soon have to be restricted.

Par exemple, prenant le type $\mathbf{\Delta}_2^+$ (i.e. $+1, +1$), correspondant aux chemins

$$a_0 \xrightarrow{u_1} a_1 \xrightarrow{u_2} a_2,$$

on peut associer à un tel chemin le chemin de type $[\mathbf{\Delta}_1^+]$

$$a_0 \xrightarrow{u_2u_1} a_2,$$

ou aussi le chemin de type $\mathbf{\Delta}_2^+$

$$a_0 \xrightarrow{u_2u_1} a_2 \xrightarrow{\mathrm{id}} a_2,$$

etc. Comme le foncteur

$$\begin{cases} X & \longmapsto & \mathrm{Ch}_\tau(X) & (= \mathrm{Ob}\,\underline{\mathrm{Ch}}_\tau(X)) \\ \mathrm{Cat} & \longrightarrow & (\mathrm{Ens}) \end{cases}$$

est *représenté* par l'objet I_τ de X [plutôt de Cat], le sorite des foncteurs représentables nous dit que de telles lois de formation de chemins de type τ à partir des chemins de type τ' [plutôt τ' à partir des chemins de type τ] correspondent biunivoquement aux flèches dans Cat

$$I_{\tau'} \xrightarrow{\varphi} I_\tau\ ,$$

en prenant simplement la 'loi' correspondante à φ p. 14

$$c \longmapsto c \circ \varphi \qquad \text{pour } c : I_\tau \longrightarrow X.$$

De façon plus précise, l'application naturelle

$$\underbrace{\mathrm{Hom}_{\mathrm{Cat}}(I_{\tau'}, I_\tau)}_{=\mathrm{Ob}\,\underline{\mathrm{Hom}}(I_{\tau'}, I_\tau)} \longrightarrow \mathrm{Hom}(\mathrm{Ch}_\tau, \mathrm{Ch}_{\tau'}),$$

où le deuxième Hom est considéré comme un Hom d'objets dans $\mathrm{Cat}^\vee = \underline{\mathrm{Hom}}(\mathrm{Cat}, \mathrm{Ens})$. Ainsi, dans les deux exemples plus haut, où $\tau = \mathbf{\Delta}_2^+$ et $\tau' = \mathbf{\Delta}_1^+$ ou $\mathbf{\Delta}_2^+$, les foncteurs en sens inverse sont [page 22]

$$\begin{array}{lll} \mathbf{\Delta}_2^+ & \{ & 0 \longrightarrow 1 \longrightarrow 2 \\ & & \uparrow \qquad\quad \uparrow \\ \mathbf{\Delta}_1^+ & \{ & 0 \longrightarrow\quad 1 \end{array}$$

et

$$\begin{array}{lll} \mathbf{\Delta}_2^+ & \{ & 0 \longrightarrow 1 \longrightarrow 2 \\ & & \uparrow \quad \nearrow \quad \uparrow \\ \mathbf{\Delta}_2^+ & \{ & 0 \longrightarrow 1 \longrightarrow 2\,. \end{array}$$

C'est là un principe général pour ramener toutes les 'opérations' $\pm$ canoniques sur des chemins dans des catégories arbitraires, et des énoncés les concernant, à des questions où X disparaît et où on travaille dans les catégories intervalles elles-mêmes. Un des avantages techniques, c'est que ce sont des catégories fort petites, donnant lieu de plus à une intuition topologique sûre (qui risque de se volatiliser par moments en travaillant dans des catégories de chemins), et aussi que dans de telles catégories, donc aussi dans les catégories de foncteurs entre elles, il y a toujours au plus *une seule flèche* allant d'un objet donné vers un autre. En d'autres termes, dans de telles catégories tous les diagrammes sont commutatifs. [page 23]

Bien sûr, on voit aussi que si on a une loi de formation L de chemins de type τ' à partir de chemins de type τ (avec la condition de fonctorialité par rapport à la catégorie X où on prend les chemins), cette loi est nécessairement *fonctorielle par rapport aux chemins :* si on a un homomorphisme

$$c \xrightarrow{v} c' \qquad \text{dans } \underline{\mathrm{Ch}}_\tau(X), \qquad \text{p. 15}$$

alors la loi L 'se prolonge' à de tels v pour donner

$$L(c) \xrightarrow{L(v)} L(c'),$$

avec les relations habituelles

$$L(\mathrm{id}_C) = \mathrm{id}_{L(c)}, \qquad L(v'v) = L(v')L(v).$$

Cela résulte de la description de L comme une opération L_φ en termes d'un foncteur bien déterminé

$$I_{\tau'} \xrightarrow{\varphi} I_\tau,$$

lequel à son tour donne pour toute X un *foncteur*

$$\begin{cases} \underline{\mathrm{Ch}}_\tau(X) & \longrightarrow & \underline{\mathrm{Ch}}_{\tau'}(X) \\ c & \longmapsto & c \circ \varphi \end{cases}$$

plus précis que simplement un *application* $\mathrm{Ch}_\tau(X) \longrightarrow \mathrm{Ch}_{\tau'}(X)$ entre les ensembles d'objets correspondants.

Commentary · Examples [ms. 23]

Composition and insertion of an identity

The two examples on manuscript pages 21–22 illustrate how a familiar operation becomes a finite calculation before any coefficient category is chosen. Take $I = [2] = (0 \to 1 \to 2)$ and $J = [1]$. The unique endpoint-preserving functor $\varphi : J \to I$ sends 0 to 0 and 1 to 2. A path $c = (a_0 \xrightarrow{u} a_1 \xrightarrow{v} a_2)$ therefore becomes the arrow $vu : a_0 \to a_2$.

A morphism from c to $d = (b_0 \xrightarrow{u'} b_1 \xrightarrow{v'} b_2)$ is a triple (h_0, h_1, h_2) satisfying $h_1 u = u' h_0$ and $h_2 v = v' h_1$. The induced morphism between the composite arrows is the square with sides h_0, h_2. Its commutativity follows from the explicit computation

$$h_2 vu = v' h_1 u = v' u' h_0.$$

The middle component has done its work and can be forgotten. The reversed variance is essential: an operation which reduces the length of a path is represented here by a functor from the shorter interval to the longer one.

For the second operation let $J = I = [2]$ and take the nondecreasing object function $\psi = (0, 2, 2)$. Then $\psi^* c = (a_0 \xrightarrow{vu} a_2 \xrightarrow{1} a_2)$, and $\psi^* h = (h_0, h_2, h_2)$. The first naturality square is the square just computed; the second is $h_2\, 1 = 1\, h_2$. Since $\psi\psi = \psi$, the resulting operation on paths is strictly idempotent. This equality is checked on the three objects of $[2]$, rather than on all categories and all pairs of composable arrows.

Endpoint preservation alone has not made such maps rigid. On $[2]$ the three nondecreasing endpoint-preserving endofunctors have object functions $(0, 0, 2)$, $(0, 1, 2)$, and $(0, 2, 2)$. They respectively place the composite in the second edge, leave the path unchanged, and place the composite in the first edge. This also supplies a concrete reminder of the correction to Proposition 1(b): fixing the endpoints does not force an arbitrary endofunctor of a nonreduced interval to be the identity. The rigidity there belongs to isomorphisms, while the present theory deliberately needs the noninvertible maps.

On a le choix maintenant entre

a) Une étude plus ou moins systématique des foncteurs entre intervalles. Je préfère attendre pour cela d'avoir besoin de ceci et de cela, et de ne faire que le strict minimum pour les constructions que j'ai en vue. [page 24]

b) Une description de la compositions des chemins.

c) Un formalisme des catégories de chemins *de type non précisé.*

Comme mon objectif est surtout de décrire de telles catégories $\underline{\mathrm{Ch}}(X)$ (sans indice τ!), avec des bonnes propriétés, je commencerai par c), regarderai ensuite b) comme un test crucial pour l'utilisabilité des notions obtenues, et enfin m'occuperai de a) quand je saurai de quoi j'aurai en besoin. p. 16

4 Catégorie de chemins $\underline{\mathrm{Ch}}(X)$

Je veux 'inclure' toutes les catégories $\underline{\mathrm{Ch}}_\tau(X)$ de chemins de tous les types qu'on veut, dans une même catégorie $\underline{\mathrm{Ch}}(X)$. Rappelons

$$\underline{\mathrm{Ch}}_\tau(X) = \underline{\mathrm{Hom}}(I_\tau, X),$$

où I_τ est l'intervalle (orienté) standard de type τ.

L'ensemble de tous les types d'intervalles peut être identifié au monoïde libre Ty à deux [page 25] générateurs Δ^+ et Δ^-, dont les éléments sont les suites finies quelconques (eventuellement vides) d'objets alternativement égales à Δ^+ et à Δ^-. Si le premier élément est Δ^+, on a un type de chemin à direction initiale directe, rétrograde dans le cas contraire. La décomposition de I_τ, ou de $\mathrm{Fl}\,(I_\tau)$, en intervalles critiques correspond à l'écriture réduite d'un mot comme

$$(\Delta^+)^{l_1}(\Delta^-)^{l_2}\ \cdots\ (\Delta^?)^{l_r} \qquad (\Delta^? = \Delta^+ \text{ si } r \text{ impair}, \Delta^? = \Delta^- \text{ si } r \text{ pair}),$$

ou comme

$$(\Delta^-)^{l_1}(\Delta^+)^{l_2}\ \cdots\ (\Delta^?)^{l_r} \qquad (\Delta^? = \Delta^- \text{ si } r \text{ impair}, \Delta^? = \Delta^+ \text{ si } r \text{ pair}).$$

Les l_i sont les longueurs des intervalles critiques successifs (ou des chaines structurales successives), leur nombre est la longueur réduite de l'intervalle.

On va considérer Ty comme l'ensemble des objets d'une catégorie $\underline{\mathrm{Ty}}$:

$$\begin{cases} \mathrm{Ob}\,\underline{\mathrm{Ty}} = \mathrm{Ty} = \text{ monoïde libre à deux générateurs } \Delta^+,\ \Delta^- \text{ précédents.} \\ \text{Si } \tau, \tau' \in \mathrm{Ty},\ \mathrm{Hom}(\tau, \tau') = \mathrm{Hom}_{\mathrm{Cat}}(I_\tau, I_{\tau'}). \end{cases}$$

([7]). On notera qu'on a non seulement un ensemble $\mathrm{Hom}_{\mathrm{Cat}}(I_\tau, I_{\tau'})$, mais même une catégorie $\underline{\mathrm{Hom}}(I_\tau, I_{\tau'})$, ce qui revient ici à dire que $\mathrm{Hom}(I_\tau, I_{\tau'})$ (i.e. $\mathrm{Hom}(\tau, \tau')$) peut être vu comme un ensemble *ordonné*. Mais pour l'instant nous allons ignorer cette relation d'ordre.

La composition des morphismes est la composition des foncteurs entre catégories I_τ. Ainsi [page 26] on a un foncteur *pleinement fidèle* ([8]) canonique

$$\tau \longmapsto I_\tau : \underline{\mathrm{Ty}} \xrightarrow{\rho} (\mathrm{Cat}) \qquad ([9]),$$

par quoi la catégorie I_τ `[plutôt` $\underline{\mathrm{Ty}}$`]` pourrait s'identifier à une sous-catégorie pleine de (Cat) (celle des I_τ). Mais je préfère ne pas identifier $\tau \in \mathrm{Ty}$ (donnée combinatoire) avec I_τ (donnée catégorique). p. 17

Commentary · Exposition [ms. 25]

Three successive categories of types

Grothendieck is testing which changes of parameter should count as morphisms of paths. The changes of definition in these pages are purposeful, but a notation retained through all three stages can obscure which fullness assertion remains true. We distinguish the stages here, without altering the source's notation. All three categories have the finite sign words as objects, including the empty word e.

Let T_{all} have every functor $I_\tau \to I_\upsilon$ as a morphism; let T_∂ retain just the functors preserving the two chosen endpoints; and let $T_\uparrow$ retain those which also preserve the external, linear order of vertices. Thus

$$T_\uparrow \subseteq T_\partial \subseteq T_{\mathrm{all}}.$$

7. Définition un peu canulée, il faut se borner aux homomorphismes de I_τ dans $I_{\tau'}$ compatibles avec source et but - cf. p. 29.

8. faux, cf. annotation marginale page précédente - on ne prend pas *tous* les foncteurs dans Cat entre I_τ et $I_{\tau'}$.

9. foncteur *réalisation* d'un type (combinatoire) d'intervalles.

The category eventually denoted by $\underline{\mathrm{Ty}}$ in these pages is $T_{\uparrow}$. On page 25 it is initially T_{all}, while pages 29–31 pass through T_{∂}. For T_{all}, realization in Cat is fully faithful by definition. For T_{∂}, realization is fully faithful into $\varepsilon\backslash\mathrm{Cat}$, where ε is the discrete category on two objects; forgetting ε loses fullness. For $T_{\uparrow}$ even the embedding into $\varepsilon\backslash\mathrm{Cat}$ is generally not full. The footnotes retracting fullness on page 26 and the assertion of fullness on page 31 are therefore compatible when read at their respective stages.

Here is a small witness to the last distinction. From the alternating interval I_{+-+} to I_+ there is a functor with object function

$$(0,1,2,3) \longmapsto (0,1,0,1).$$

Every forward transition goes to $0 \to 1$ and the backward transition, whose categorical direction is from its right vertex to its left vertex, also goes to $0 \to 1$. Both endpoints are preserved, but the vertex function is not nondecreasing. Thus it belongs to T_{∂} and not to $T_{\uparrow}$. The monotonicity condition excludes this backtracking.

The sign words themselves form a free monoid on $+$ and $-$. Arbitrary adjacent repetitions are allowed; the alternating feature is the alternation of signs between *maximal constant runs*. A nonempty word has a unique expression $\varepsilon_1^{l_1} \cdots \varepsilon_r^{l_r}$ with $l_i > 0$ and $\varepsilon_{i+1} = -\varepsilon_i$. Its length is $\sum l_i$, its useful length is r, and its critical intervals are these runs with their endpoint vertices restored. The empty word has length and useful length zero and no initial sign. This last exception should remain in force whenever the text speaks of the first generator.

In the undercategory description, $\varepsilon \to I_e$ identifies the two objects of ε. It is a structure map, not an embedding; the hooked arrows drawn on page 29 should not be read as asserting injectivity in this degenerate case. Keeping e is indispensable: it will be the unit for concatenation.

Soit X une (petite?) catégorie. On a un contrafoncteur

$$\begin{cases} S & \longmapsto & \underline{\mathrm{Hom}}(S, X) \\ (\mathrm{Cat})^o & \longrightarrow & (\mathrm{Cat}) \end{cases}$$

défini par X, d'où en composant avec ρ un contrafoncteur

$$\underline{\mathrm{Ty}}^o \xrightarrow{\mathrm{CH}_X} \mathrm{Cat}.$$

C'est le contrafoncteur

$$\begin{cases} \tau \longmapsto \underline{\mathrm{Hom}}(I_\tau, X) \overset{\text{déf}}{=} \underline{\mathrm{Ch}}_\tau(X), \\ \text{i.e. } \mathrm{CH}_X(\tau) = \underline{\mathrm{Ch}}_\tau(X). \end{cases}$$

Mais un contrafoncteur sur une petite catégorie $\underline{\mathrm{Ty}}$ donne lieu à une *catégorie fibrée* (et même scindée) sur celle-ci.

Je vais désigner cette catégorie fibrée par $\underline{\mathrm{Ch}}(X)$. (Provisoirement - si elle s'avère trop grosse, je la restreindrai …) Ainsi on a [page 27]

$$\begin{array}{c} \underline{\mathrm{Ch}}(X) \\ \big\downarrow{\scriptstyle \pi_X} \\ \underline{\mathrm{Ty}}\ . \end{array}$$

La fibre de $\underline{\mathrm{Ch}}(X)$ en $\tau \in \mathrm{Ob}\,\underline{\mathrm{Ty}}$ est la catégorie $\underline{\mathrm{Ch}}_\tau(X)$ des chemins de type τ dans X. Cela nous dit donc quelles sont les flèches dans $\underline{\mathrm{Ch}}(X)$ au dessus des flèches identiques dans $\underline{\mathrm{Ty}}$. Mais que sont les flèches plus générales? On se donne une flèche dans $\underline{\mathrm{Ty}}$

$$\varphi : \tau' \longrightarrow \tau, \qquad \text{i.e. } I_{\tau'} \xrightarrow{\varphi} I_\tau,$$

quelles sont les flèches de $\underline{\mathrm{Ch}}(X)$ au dessus de φ? Par le sorite général des catégories fibrées, si

$$\begin{aligned} c \in \underline{\mathrm{Ch}}(X)_\tau &= \underline{\mathrm{Ch}}_\tau(X) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(I_\tau, X) \\ c' \in \underline{\mathrm{Ch}}(X)_{\tau'} &= \underline{\mathrm{Ch}}_{\tau'}(X) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(I_{\tau'}, X) \end{aligned}$$

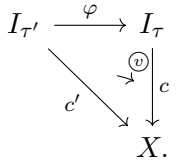
on a

$$\boxed{\mathrm{Hom}_{\varphi}(c', c) \simeq \mathrm{Hom}(c', \underbrace{\varphi^*(c)}_{=c\circ\varphi}).}$$

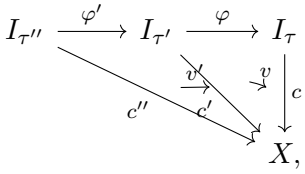
Donc : les homomorphismes du chemin c' de type τ' dans X, dans le chemin c de type τ, 'au dessus' du morphisme φ de τ' dans τ, i.e. du foncteur p. 18 [page 28]

$$\varphi : I_{\tau'} \longrightarrow I_{\tau},$$

est un homomorphisme v de c' dans $c \circ \varphi$, i.e.



La composition des flèches est évidente :



on a

$$v : \quad c' \quad \longrightarrow \quad c \circ \varphi, \quad \text{d'où } c' \circ \varphi' \xrightarrow{v*\varphi'} \underbrace{(c \circ \varphi) \circ \varphi'}_{=c\circ(\varphi\varphi')}$$

$$v' : \quad c'' \quad \longrightarrow \quad c' \circ \varphi'$$

et

$$v'' = (v * \varphi') \circ v', \qquad \text{donc} \qquad c'' \xrightarrow{v'} c' \circ \varphi' \xrightarrow{v*\varphi'} c \circ (\varphi\varphi'). \quad (c'')$$

Mais a-t-on des foncteurs source et but

$$\underline{\mathrm{Ch}}(X) \underset{\beta}{\overset{\sigma}{\rightrightarrows}} X \qquad \begin{array}{rcl} c & \longmapsto & \mathrm{source}(c) \\ c & \longmapsto & \mathrm{but}(c) \end{array} \qquad ?$$

Il faudrait donc qu'un homomorphisme v comme dessus

$$v : c' \longrightarrow c \circ \varphi$$

définisse

$$\begin{array}{rrcl} \sigma(c) : & \mathrm{source}(c') & \longrightarrow & \mathrm{source}(c) \\ \beta(c) : & \mathrm{but}(c') & \longrightarrow & \mathrm{but}(c). \end{array}$$

Commentary · Context [ms. 27]

The split fibration of all paths

Grothendieck now uses the machinery of descent to assemble paths of different shapes. The construction is the one in Grothendieck (1971a, Exposé VI, §§8–9): a strict contravariant functor to categories supplies a category with a chosen, strictly composable family of Cartesian lifts. The commentary on Chapter VI, manuscript pages 26 and 53–56, discussed splittings and their universal constructions. Here the splitting needs no choices, because it is given by literal precomposition.

For any of the three shape categories T, put $F_X(\tau) = X^{I_\tau}$. An object of its Grothendieck construction is (τ, c), and a morphism

$$(\tau', c') \longrightarrow (\tau, c)$$

is a pair (φ, v) with $\varphi : \tau' \to \tau$ and $v : c' \Rightarrow c\varphi$. If $(\varphi', v') : (\tau'', c'') \to (\tau', c')$, their composite is

$$(\varphi, v)(\varphi', v') = \big(\varphi\varphi', (v\varphi') \circ v'\big).$$

At $i \in I_{\tau''}$ the second component is $v_{\varphi'(i)} v'_i$. Threefold composition is associative because both parenthesizations have component $v_{\varphi'\varphi''(i)} v'_{\varphi''(i)} v''_i$ and the same shape map. The identity is $(1_\tau, 1_c)$. These checks explain the word "*évidente*" in the source without suppressing the variance.

The chosen lift of $\varphi : \tau' \to \tau$ at (τ, c) is

$$(\varphi, 1_{c\varphi}) : (\tau', c\varphi) \longrightarrow (\tau, c).$$

To verify its Cartesian property, let a map from (ω, d) to (τ, c) have shape $\varphi\theta$ and transformation $w : d \Rightarrow c\varphi\theta$. There is exactly one factor over θ through the displayed lift, namely (θ, w). The chosen lifts compose on the nose, since $(c\varphi)\theta = c(\varphi\theta)$. More generally (φ, v) is Cartesian exactly when v is invertible: it factors as the vertical arrow $(1, v)$ followed by this chosen lift, and a vertical Cartesian arrow is an isomorphism.

For $T = T_\partial$ or $T_\uparrow$ the endpoint functors send (φ, v) to $v_{0_{\tau'}}$ and $v_{1_{\tau'}}$. Their targets are $c(0_\tau)$ and $c(1_\tau)$ because φ preserves endpoints. The formula for composition immediately gives functoriality. Likewise a functor $H : X \to Y$ acts by $(\tau, c) \mapsto (\tau, Hc)$ and $(\varphi, v) \mapsto (\varphi, Hv)$, preserving the chosen Cartesian lifts. Restricting the permitted shape maps is literally base change of this split fibration, which proves the fiber-product assertion on page 30.

Size causes no mathematical obstruction here. Fix universes $\mathcal{U} \in \mathcal{V}$. The set of finite words and their finite interval functors is $\mathcal{U}$-small. If X is a $\mathcal{V}$-small category, each X^{I_τ} and the total construction are $\mathcal{V}$-small. One may therefore take $X = \mathrm{Cat}_{\mathcal{U}}$ after working in a sufficiently larger universe. This supplies the convention which the source deliberately leaves implicit.

p. 19
[page 29]

Mais comme

$$\begin{cases} \sigma(c) = c(0_\tau),\ \beta(c) = c(1_\tau) \\ \text{où } 0_\tau,\ 1_\tau \text{ sont les objets source et but de l'intervalle orienté standard } I_\tau\ , \end{cases}$$

et de même pour τ', v n'induira bien

$$\underbrace{\sigma(c')}_{=c'(0_{\tau'})} \longrightarrow c(\varphi(0_{\tau'})) \stackrel{?}{=} \underbrace{\sigma(c)}_{c(0_\tau)},$$

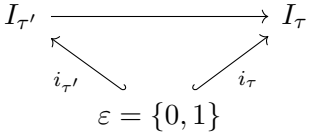
et itou pour $\beta(c')$ et $\beta(c)$, que si on impose

$$\boxed{\begin{cases} \varphi(0_{\tau'}) &= 0_\tau \\ \varphi(1_{\tau'}) &= 1_\tau, \end{cases}}$$

i.e. que le foncteur envisagé φ soit *compatible avec la source et le but.* C'est ce visiblement qu'il fallait imposer dans la définition de

$$\mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau', \tau) = \underline{\mathrm{Hom}}^{\textcircled{m}t}(I_{\tau'}, I_\tau) \quad \text{-}$$

il faut prendre les foncteurs φ respectant les structures d'intervalle, i.e. rendant commutatif le diagramme



définissant l'orientation de $I_{\tau'}$ et de I_τ ([10]). Cette rectification faite, j'ai bon espoir que la catégorie $\underline{\mathrm{Ch}}(X)$ qu'on vient de construire est 'la bonne'. On pourrait désigner par $\underline{\mathrm{CH}}(X)$ la catégorie plus grande, sur la catégorie $\underline{\mathrm{Typ}}$ ayant mêmes objets que $\underline{\mathrm{Ty}}$, mais plus de morphismes : à savoir [page 30]

$$\begin{aligned} \mathrm{Hom}_{\underline{\mathrm{Typ}}}(\tau,\tau') &= \mathrm{Hom}_{\mathrm{Cat}}(I_\tau, I_{\tau'}) \\ \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau,\tau') &= \mathrm{Hom}^{\mathrm{int}}_{\mathrm{Cat}}(I_\tau, I_{\tau'}) \\ &\quad \text{(homomorphismes respectant la structure d'intervalle).} \end{aligned}$$

On aura

$$\underline{\mathrm{Ch}}(X) \simeq \underline{\mathrm{CH}}(X) \times_{\underline{\mathrm{Typ}}} \underline{\mathrm{Ty}},$$

i.e. $\underline{\mathrm{Ch}}(X)$ est la *restriction* de la catégorie fibrée $\underline{\mathrm{CH}}(X)$ sur $\underline{\mathrm{Typ}}$, à la sous-catégorie (non-pleine) $\underline{\mathrm{Ty}}$ de $\underline{\mathrm{Typ}}$. p. 20

Commentary · Examples [ms. 28]

Changing shape need not have a direct image

The projection to the category of types is a fibration because paths can always be restricted. Extending a path to a new shape asks a different question, and generally requires colimits in its coefficient category. This is the usual distinction between a fibration and a bifibration; compare Grothendieck (1971a, Exposé VI, §10 and §11(f)).

Consider the interval $J = I_{-+} = (0 \leftarrow 1 \rightarrow 2)$ and its unique collapse $q : J \to e$. This collapse belongs even to $T_\uparrow$. Its inverse-image functor is the diagonal

$$q^* : X \longrightarrow X^J,$$

sending an object to its constant span. If an opCartesian lift over q existed at every span c, its target $q_!c$ would represent the functor

$$x \longmapsto \mathrm{Nat}(c, \mathrm{const}_x).$$

Indeed every map from c over q to a constant path would factor uniquely through the opCartesian lift. This is precisely the universal property of the pushout of the span $c(0) \leftarrow c(1) \rightarrow c(2)$.

Take $X = J$ itself and $c = 1_J$. There is no cocone on this diagram: neither 0 nor 2 maps to the other, and neither maps to 1. Thus $q_!c$ does not exist, and the path projection is not an opfibration in this example. This does not reveal a gap in Grothendieck's assertion, which is only that the construction is split fibrational. It identifies precisely what would be added by a stronger assertion.

For $X = \mathrm{Set}$ the same calculation has a complete answer. Let $A \leftarrow C \rightarrow B$ be a span. Then $q_!c$ is the quotient of $A \amalg B$ by the equivalence relation generated by identifying the two images of each $z \in C$. A map from this quotient to S is exactly a pair $A \to S$, $B \to S$ which agrees on C, so it is exactly a natural transformation $c \Rightarrow \mathrm{const}_S$. For a general change of shape φ and a cocomplete X, left Kan extension supplies the corresponding left adjoint to φ^*. Such extra adjoints come from X; the split fibration itself is available without them.

Commentary · Development (commentators') [ms. 29]

How much endpoint preservation is forced

The correction on page 29 gives exactly what is needed for endpoint maps obtained by taking components of a natural transformation. There is a slightly larger possibility if one allows those components to be followed by canonical arrows inside the target interval. Describing it clarifies why endpoint preservation is a structural choice, and why concatenation will justify the stricter choice.

Endpoint evaluation criterion. [ours] Let T be a subcategory of T_{all} with all sign words as objects. Its path construction admits functors $\sigma_X, \beta_X : \int F_X \to X$, natural in X, restricting in each

10. En fait, il faut sans doute restreindre encore plus les morphismes en exigeant qu'il soient *croissants* sur les objets - cf. page 31.

fiber to evaluation at the specified source and target, if and only if every $\varphi : \tau' \to \tau$ in T admits arrows

$$a_\varphi : \varphi(0_{\tau'}) \longrightarrow 0_\tau, \qquad b_\varphi : \varphi(1_{\tau'}) \longrightarrow 1_\tau \quad \text{in } I_\tau.$$

When they exist, these endpoint functors are unique. The permitted functors form a category T_{ev} containing T_∂, and it is the largest such subcategory of T_{all}.

Proof. For necessity take $X = I_\tau$ and the universal path $c = 1_{I_\tau}$. The Cartesian arrow

$$(\varphi, 1_\varphi) : (\tau', \varphi) \longrightarrow (\tau, 1_{I_\tau})$$

must be sent by σ_X to an arrow $\varphi(0_{\tau'}) \to 0_\tau$, and by β_X to an arrow $\varphi(1_{\tau'}) \to 1_\tau$. Each is unique when it exists, because an interval is thin. For composable maps $\tau'' \xrightarrow{\psi} \tau' \xrightarrow{\varphi} \tau$, the arrows

$$a_\varphi \circ \varphi(a_\psi), \qquad b_\varphi \circ \varphi(b_\psi)$$

have precisely the endpoints required for $\varphi\psi$. Identities satisfy the requirement as well. Hence the permitted functors form a category.

To construct the endpoint maps, define on $(\varphi, v) : c' \to c$

$$\sigma_X(\varphi, v) = c(a_\varphi) \circ v_{0_{\tau'}}, \qquad \beta_X(\varphi, v) = c(b_\varphi) \circ v_{1_{\tau'}}.$$

For a second morphism $(\psi, w) : c'' \to c'$, naturality of v at a_ψ gives

$$v_{0_{\tau'}} c'(a_\psi) = c\varphi(a_\psi) v_{\psi(0_{\tau''})}.$$

After composing with $c(a_\varphi)$ and $w_{0_{\tau''}}$, this says that the product of the two source arrows is

$$c(a_{\varphi\psi}) v_{\psi(0_{\tau''})} w_{0_{\tau''}},$$

which is the source arrow assigned to the composite. The same argument with b_ψ proves the target identity. Identity morphisms and naturality in X follow directly from the formulas.

Finally, every (φ, v) factors as a vertical morphism $(1, v)$ followed by a chosen Cartesian arrow. The endpoint functors are prescribed on vertical arrows. Naturality in X applies to $c : I_\tau \to X$ and forces their values on the Cartesian arrow to be $c(a_\varphi)$ and $c(b_\varphi)$. Thus the formulas are forced, proving uniqueness and maximality. □

For example, the functor $e \to I_+$ selecting 0 belongs to T_{ev}: its source comparison is an identity and its target comparison is $0 \to 1$. It does not preserve both endpoints. The functor selecting 1 does not belong to T_{ev}, since there is no arrow $1 \to 0$. This already prevents endpoint evaluation on T_{all}.

The enlargement does not solve the concatenation problem. Concatenate two copies of the admissible map $e \to I_+$ selecting 0. The common vertex of $e * e = e$ would have to map both to vertex 0 of the first target interval and to vertex 1, the starting vertex of the second interval in I_{++}. The two prescribed maps therefore do not glue. Strict endpoint preservation makes precisely these images agree. The theorem is an elementary specialization of the general description of functors out of a split fibration in Grothendieck (1971a, Exposé VI, Proposition 12.1). We checked that description and the Yoneda treatment cited above; we did not find this particular maximal class of interval maps stated there. The marker records this explicit deduction and proof, without a claim of priority.

NB Les catégories fibrées $\underline{\mathrm{CH}}(X)$, $\underline{\mathrm{Ch}}(X)$ sur $\underline{\mathrm{Typ}}$, $\underline{\mathrm{Ty}}$ dépendent fonctoriellement de X, variant dans (Cat), comme il se doit. Comme de juste, on s'y intéresse surtout quand X *n'est pas* dans (Cat), i.e. quand X est une 'grosse' catégorie (telle que (Cat) elle-même). Je me dispense d'expliciter ici les questions pertinentes d'Univers.

Deux tests essentiels, pour s'assurer si la notion est bonne

a) La théorie des $\mathcal{M}\Sigma^{-1}$.

b) La composition des chemins doit bien marcher.

On a le diagramme de foncteurs canoniques

$$\begin{array}{ccc} \underline{\mathrm{Ty}} & \hookrightarrow & \varepsilon\backslash\mathrm{Cat} \\ \downarrow & & \downarrow \\ \underline{\mathrm{Typ}} & \hookrightarrow & \mathrm{Cat}, \end{array}$$

où les flèches horizontales sont pleinement fidèles.

Commentary · Present state [ms. 30]

The fraction test and hammock localization

Grothendieck's two tests concern different levels of the construction. Concatenation should be an operation on actual paths and their morphisms; localization should arise after paths are subjected to specified identifications or homotopical comparisons. The category $\underline{\mathrm{Ch}}(X)$ constructed here has not yet been asserted to satisfy the universal property of a localization.

For a category $\mathcal{M}$ and a specified class Σ of arrows, ordinary localization is already constructed by Gabriel and Zisman (1967, Chapter I, §1.1 and Lemma 1.2). Its arrows are composable words of forward arrows and formal inverses of arrows in Σ, modulo the category relations and inverse relations. A functor out of the quotient is consequently equivalent to a functor from $\mathcal{M}$ sending Σ to isomorphisms. This existence statement does not require the Ore hypotheses needed for a calculus with uniformly short fractions.

The homotopical comparison already available before these notes is quite specific. In Dwyer and Kan (1980a, Definition 2.1 and Propositions 2.2 and 3.1), a hammock is a finite grid of zigzags, with backward horizontal arrows and vertical arrows required to be weak equivalences. Identity columns are removed and adjacent columns pointing in the same direction are composed. Their mapping simplicial sets are equivalent to the standard simplicial localization, and their sets of components recover ordinary localization:

$$\pi_0 L^H(\mathcal{M}, \Sigma)(x, y) \cong \mathrm{Hom}_{\mathcal{M}[\Sigma^{-1}]}(x, y).$$

Thus there is a proven construction in which changing the length and subdivision of zigzags is accompanied by the higher comparison data needed for mapping spaces.

This is a comparison of mathematical mechanisms, not an identification of the constructions. The present $\underline{\mathrm{Ch}}(X)$ allows arbitrary vertical natural transformations, has not specified which backward arrows may be inverted, and uses a single ordinary category of types. The hammock conditions, the fixed endpoints, and the higher simplices must all be supplied before a localization theorem can be invoked. The source's demand that the fraction theory "work" is a design requirement at this point, not a research conjecture left open in these pages. Its exact implementation in Grothendieck's subsequent text should be checked on its own definitions.

p. 21

5 Propriétés des morphismes d'intervalles

Je me suis plus ou moins convaincu que pour se prêter à des calculs et à des raisonnements simples, il faut encore plus restreindre les morphismes dans la définition de la catégorie $\underline{\mathrm{Ty}}$. Je vais exiger que les morphismes $\tau \longrightarrow \tau'$ soient des foncteurs

$$w : I_\tau \longrightarrow I_{\tau'}$$

qui sont compatibles avec la structure d'*intervalles orientés* de I_τ, $I_{\tau'}$, i.e. qu'ils transforment source en source, but en but, mais de plus qu'ils soient *croissants* sur les objets (pour la relation d'ordre de progression des chemins). C'est une restriction qui géométriquement me paraît assez naturelle. (Elle est similaire à la restriction analogue, dans la définition pré-catégorique des morphismes entre $\boldsymbol{\Delta}_i$ …) C'est cette hypothèse qui permet de faire une théorie de structure assez simple des morphismes d'intervalles. Mais pour l'instant je vais distinguer 'morphismes' et 'morphismes croissants'.

Commentary · Later developments [ms. 31]

The same type category in later zigzag constructions

Grothendieck's final restriction on changes of shape has a precise counterpart in the later literature on localization. The correspondence can be checked directly, before making any assertion about homotopy theory. Given a word τ, mark the backward transitions of I_τ and their composites, together with the identities. This makes I_τ a relative category. An increasing endpoint-preserving functor automatically respects this marking: a nonconstant image of a backward transition is a composite of backward transitions. Conversely, forgetting the marking of an increasing endpoint-preserving relative functor gives exactly a morphism of $\underline{\mathrm{Ty}}$. The objects, morphisms, and concatenation are therefore unchanged by this decoration.

Under $+ \leftrightarrow A$ and $- \leftrightarrow W^{-1}$, this identifies $\underline{\mathrm{Ty}}$ with the category Z of Mazel-Gee (2018, Definition 3.5, Remark 3.6, and footnotes 8–10, pp. 341–342). Their weak-equivalence labels restrict which diagrams in a coefficient category are allowed; they do not impose an additional restriction on these monotone shape maps. The empty word remains the monoidal unit. This is an identification of the finite indexing categories, rather than an inference from a resemblance between pictures.

The image factorization implicit in the manuscript also explains a useful structure. If $f : I_\tau \to I_\upsilon$ is increasing, the full subcategory on its image is an interval. Factor f through its standard oriented model:

$$I_\tau \xrightarrow{q} I_\omega \xhookrightarrow{j} I_\upsilon.$$

The first map is surjective on vertices and the second injective. Their uniqueness follows from the unique increasing enumeration of the image. Nonidentity surjections lower the length, and nonidentity injections raise it. Thus length is a Reedy degree. The reduced interval gives the smallest image of an endpoint-preserving injection: Proposition 5 says that every such image contains all critical vertices. If the interval is not reduced, its reduced inclusion is consequently initial among the proper injections into it; for a reduced interval there are none. These statements follow from the full-subcategory argument already given, without a new hypothesis on coefficients.

This Reedy structure and this exact use of critical vertices occur in Arakawa and Cnossen (2026, Example 4.12 and Lemma 4.16), where the category is denoted Zig_∂. Their work is an August 2026 preprint. We cite these known applications, rather than present the Reedy interpretation as an original result. What remains to be added to the present manuscript is the relative and homotopical diagram theory over this category of shapes.

Soit [page 32]

$$f : I_\tau \longrightarrow I_{\tau'}$$

un morphisme d'intervalles orientés. Considérons les sous-catégories pleines, isomorphes à des Δ_i^+ ou Δ_i^- comme intervalles orientés, définis par les intervalles critiques de I_τ. Ils forment une suite Σ_τ de sous-catégories de I_τ

$$\Sigma_\tau = (\mathbf{\Delta}_{l_1}^{\varepsilon_1}(\tau), \mathbf{\Delta}_{l_2}^{\varepsilon_2}(\tau), \ldots, \mathbf{\Delta}_{l_r}^{\varepsilon_r}(\tau))$$

dont I_τ est la somme amalgamée (on reviendra sur ce point plus `[bas]`), r étant la longueur utile de τ. On les appelle les *sous-intervalles critiques* de I_τ. Ainsi f se décompose en une suite de morphismes

$$f_i : \mathbf{\Delta}_{l_i}^{\varepsilon_i} \longrightarrow I_{\tau'}, \qquad i \in [1, r],$$

tels que

a) pour $i \in [1, r-1]$, le but de f_i soit égal à la source de f_{i+1}.
De plus,

b) f_1 transforme source en source, et f_r transforme but en but.

Inversement, quand on se donne des f_i satisfaisant ces deux conditions, ils définissent un morphisme d'intervalles f unique. (Et si la condition b) n'est pas exigée, on trouve une description des morphismes de catégories $I_\tau \longrightarrow I_{\tau'}$.)

On va d'abord regarder un foncteur quelconque [page 33]

$$f : \mathbf{\Delta}_i^\varepsilon \longrightarrow I_{\tau'} \qquad (\varepsilon \in \{\pm 1\}) \quad .$$

Commentary · Exposition [ms. 32]

A functor cannot cross a turn in one run

Grothendieck begins the structure theorem by cutting an interval where its direction changes. Each piece is a chain, and the image of a chain remains comparable even when its length changes. This elementary observation is what allows a map of long zigzags to be described by a finite amount of ordered data.

For a nonempty word τ let

$$0 = s_0 < s_1 < \cdots < s_r = n$$

be its critical vertices, and let $K_i = [s_{i-1}, s_i]$ be its maximal constant-sign runs, regarded as full subcategories. The category I_τ is obtained by gluing the K_i at their common endpoint vertices. To see the universal property directly, choose functors $f_i : K_i \to Y$ agreeing at shared endpoints. They determine the image of every generating transition of I_τ. Each nonidentity arrow of I_τ is the unique composite of the transitions along a single run, so its image is forced to be the corresponding composite in Y. The only composition equations are associativity along those runs, and these hold in Y. Thus the f_i extend uniquely to a functor. For $Y = I_\upsilon$, imposing the values of $f_1(s_0)$ and $f_r(s_r)$ additionally imposes endpoint preservation, exactly as on page 32.

Now let K be a critical run of the source and suppose $f|K$ is nonconstant. The image vertices are pairwise comparable in I_υ: any two source vertices are joined by an arrow in one direction, whose image joins their images. Let $a < b$ be the smallest and largest distinct image vertices in the external order. Their comparability means that all target transitions between a and b have one sign. They therefore lie in a unique maximal run J', and so do all the image vertices between them. This proves both the existence and uniqueness in Proposition 7.

If the source run has sign ε and J' has sign η, the object function of $f|K$ is nondecreasing when $\varepsilon = \eta$ and nonincreasing when $\varepsilon = -\eta$. A nonconstant function which is nondecreasing in the external order cannot be of both sorts. Hence an increasing functor sends every nonconstant source run into a target run of the same sign. This proves the sign assertion as well; no endpoint hypothesis has been used.

The preceding paragraph of the source overstates the constant case. A vertex strictly inside a long target run belongs to one critical interval; an interior *critical* vertex belongs to two; an endpoint of a positive-length interval belongs to one. For instance, the constant map with value 1 in $I_{++} = [2]$ factors through its sole critical interval, although 1 is not an endpoint. This is a printed slip, retained in the text, and it does not affect Proposition 7 because that proposition assumes nonconstancy. There is also a harmless zero-length exception to the footnote's comparable-subset lemma: I_e has one vertex and no critical runs. A nonconstant map cannot have this target, so the proposition itself remains valid without an added exception.

p. 22

Supposons pour fixer les idées que $\varepsilon = +1$, i.e. que le premier membre soit un intervalle critique direct … On trouve que f se factorise par un sous-intervalle critique de $I_{\tau'}$, soit $\mathbf{\Delta}_j^{\varepsilon'}$. Il est donc défini par une application croissante de $\mathbf{\Delta}_i^{\varepsilon}$ dans $\mathbf{\Delta}_j^{\varepsilon'}$ (si ε, ε' `[sont]` de même signe), une application décroissante dans le sens inverse ([11]). Mais si on exige que f soit croissant, alors il faut que *ou bien* f soit constant (auquel cas, si sa valeur n'est pas une extrémité de $I_{\tau'}$, il y a exactement *deux* intervalles critiques de $I_{\tau'}$ par lesquels se factorise f) *ou bien* f non constant et $\varepsilon = \varepsilon'$ (et en tous cas, si f non constant, alors l'intervalle critique par lequel se factorise f est unique). Pour résumer :

Proposition 7. *Soient I_τ, $I_{\tau'}$ deux intervalles orientés,*

$$f : I_\tau \longrightarrow I_{\tau'}$$

un foncteur. Alors pour tout intervalle critique $J \subseteq I_\tau$ tel que $f|J$ soit non constant, il existe un unique intervalle critique $J' \subseteq I_{\tau'}$ tel que $f|J$ se factorise par J', i.e. tel que $f(J) \subseteq J'$. Si f est croissant, alors J et J' sont de même signe, i.e. tous deux directs, ou tous deux rétrogrades. [page 34]

11. **Lemme.** *Si $I_{\tau'}$ est un intervalle, toute partie J de $I_{\tau'}$ telle que pour deux éléments de J, il existe toujours une flèche de l'un dans l'autre, est contenue dans un intervalle critique (unique si* $\operatorname{card} J \geq 2$*).*

Ainsi, si on regarde la suite $s_0, s_1, \ldots, s_n$ des sommets critiques de τ, alors

$$t_0 = f(s_0),\ t_1 = f(s_1),\ \ldots\ ,\ t_n = f(s_n)$$

forment une suite de sommets de $I_{\tau'}$, telle que deux consécutifs soient reliés par une flèche de $I_{\tau'}$. Donc la sous-catégorie pleine de $I_{\tau'}$ définie par ces t_i est connexe, c'est donc un sous-intervalle. Si f est croissant, ce sous-intervalle est $[t_0, t_n]$. Si f est un morphisme d'intervalles, il contient tous les sommets critiques de $I_{\tau'}$. Dans tous les cas, prenant t_i, t_j $(i < j)$ et appliquant ce qui précède à $f|J$, où J est le sous-intervalle de I d'origine s_i, d'extrémité s_j, on voit que tout sommet critique de $I_{\tau'}$ compris entre t_i et t_j est parmi les t_k $(i \leq k \leq j)$. Pour résumer :

Corollaire. *Soient $s_i \leq s_j$ deux sommets critiques de $I_{\tau'}$. Alors tout sommet critique de $I_{\tau'}$ compris entre $f(s_i) = t_i$ et $f(s_j) = t_j$, est l'image d'un sommet critique s_k compris entre s_i et s_j (i.e. $i \leq k \leq j$). En particulier, si f est un morphisme d'intervalles,*

$$f(I_{\tau\,\text{réd}}) \supseteq I_{\tau'\,\text{réd}}$$

(où $I_{\tau\,\text{réd}}$ désigne l'ensemble des sommets critiques de I_τ ou aussi le sous-intervalle réduit qu'il définit).

Commentary · Exposition [ms. 34]

Critical vertices are crossed, not necessarily preserved

The next step records which turns in the target must be encountered by a map. This is weaker than preservation of all critical vertices, and the distinction is needed when the target is not reduced. It also prevents an ordinary interval of integers from being confused with a connected full subcategory of an interval category.

Let $s_0, \ldots, s_r$ be the critical vertices of the source and put $t_i = f(s_i)$. Consecutive t_i are comparable, since they are images of the endpoints of a source run. Thus the full subcategory on the *set* $\{t_0, \ldots, t_r\}$ is connected. Suppose t_i and t_j lie on opposite sides of an interior critical vertex z of the target. If no t_k, $i \leq k \leq j$, were equal to z, some consecutive pair in this finite sequence would lie strictly on opposite sides of z. Such a pair cannot be comparable: the two transitions adjoining an interior critical vertex have opposite signs. This contradiction proves that $z = t_k$ for some k in the indicated range. The argument permits backtracking and repetitions; it does not require monotonicity.

If f preserves the two endpoints, take $i = 0$ and $j = r$. Every target critical vertex is consequently the image of a source critical vertex. This is the content of the corollary, whose first occurrence of $I_{\tau'}$ should be read as I_τ when it describes the s_i. The stronger inclusion in the other direction is false. In the displayed example $I_{+-+} \to I_{++}$, the object function is

$$(0, 1, 2, 3) \longmapsto (0, 1, 1, 2).$$

All four source vertices are critical. Their image contains the two target endpoints and also its noncritical vertex 1; the middle source transition collapses. The example therefore gives exactly the one-sided containment asserted in the text.

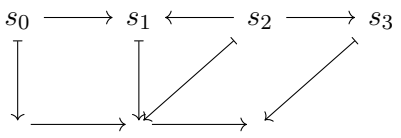
There is a second printed overstatement just before the corollary. Even for increasing f, the full subcategory on the t_i need not have *every* integer vertex between t_0 and t_r. Take the identity of I_{++}. Its source critical vertices are $0, 2$, so their image is $\{0, 2\}$, a full subcategory isomorphic to $[1]$, whereas the ambient interval $[0, 2]$ has three vertices. The correct statement is that the image subcategory has least and greatest vertices t_0, t_r, its order-convex hull is $[t_0, t_r]$, and it contains every ambient critical vertex in that hull. This is all that the argument uses.

One can now see why the image of $\mathrm{Cr}(\tau, f) \to \mathrm{Cr}(\upsilon)$ is an interval in the ordered set of target runs. Join successive numbers t_i by real line segments, obtaining a continuous walk in $[0, m]$. Each nonconstant segment is contained in the unique target run attached to K_i. If the walk reaches two distinct target runs, it must traverse every intervening run, and each such traversal occurs on at least one nonconstant segment. Conversely, a target run not encountered by the walk cannot be attached to any K_i. The same argument shows that every target run meeting the segment between $f(0)$ and $f(n)$ in a positive-length segment is encountered. In particular, endpoint preservation makes the run map surjective. This completes the proofs of Corollary 2(a)–(c), including the parts for maps which backtrack.

On fera attention que même si f est croissant, on *n'a pas* en général $f(I_{\tau\,\text{réd}}) \subseteq I_{\tau'\,\text{réd}}$, i.e. [page 35]
f ne transforme pas forcément sommets critiques en sommets critiques, Exemple minimal : p. 23

Φ_3^+ :

$\mathbf{\Delta}_2^+$:



Φ_3^+ est réduite, les deux sommets critiques transformés en le sommet non critique 1.

Désignons par $\mathrm{Cr}(\tau)$ l'ensembles des intervalles critiques de I_τ, en correspondance biunivoque (respectant l'ordre total) avec l'ensemble des flèches de transition de $I_{\tau\,\text{réd}}$,

$$\mathrm{Cr}(\tau) \simeq \mathrm{Fl\,tr}\,(I_{\tau\,\text{réd}}).$$

Soit, pour un foncteur

$$f : I_\tau \longrightarrow I_{\tau'},$$

$\mathrm{Cr}(\tau, f)$ l'ensemble des intervalles critiques de I_τ sur lesquels f ne soit pas constant. On a alors une application

$$(*) \qquad \mathrm{Cr}(\tau, f) \longrightarrow \mathrm{Cr}(\tau')$$

induite par f via la proposition 7, associant à tout $J \in \mathrm{Cr}(\tau, f)$ l'unique $J' \in \mathrm{Cr}(\tau')$ tel que $f(J) \subseteq J'$. On a de plus

Corollaire 2.

a) L'image de $(*)$ *est un segment de l'ensemble totalement ordonné* $\mathrm{Cr}(\tau')$*, i.e. si deux intervalles critiques* J'*,* J'' *dans* $I'_{\tau'}$ *sont dans l'image, il en est de même de tous ceux qui sont compris entre eux.* [page 36]

b) Cette image contient aussi l'intervalle critique de $I_{\tau'}$ *(il y en a au plus un) qui contient* $f(0_\tau)$ *(resp.* $f(1_\tau)$*) et dont l'intersection avec le segment* $[f(0_\tau), f(1_\tau)]$ *est de cardinal* ≥ 2*, ainsi que tous les intervalles critiques contenus dans ledit segment.*

c) Si f *est un morphisme d'intervalles orientés, alors l'application induite*

$$\mathrm{Cr}(f) : \mathrm{Cr}(\tau, f) \longrightarrow \mathrm{Cr}(\tau')$$

est surjective. De façon plus précise, soit $J' \subseteq \mathrm{Cr}(\tau')$*, et soit* $J = f^{-1}(J')$*. Alors* $\mathrm{Cr}(f)^{-1}(\{J'\})$ *est formé des intervalles critiques dans* $\mathrm{Cr}(\tau, f)$ contenus dans J*. On a de plus,* si f est croissant,

$$\mathrm{long}(J') = \sum_{K \in \mathrm{Cr}(f)^{-1}(\{J'\})} \pi_{\mathrm{n}}(K, f),$$

où $\pi_{\mathrm{n}}(K, f)$ *est la 'partie' (numérique) de* $f|K$*, i.e. la longueur de l'intervalle* $[f(0_K), f(1_K)]$*. (***NB** *Si* K *est un intervalle critique de* I *tel que* $K \notin Cr(\tau, f)$*, i.e.* $f|K$ *est constant, on posera* $\pi_{\mathrm{n}}(K, f) = 0$*. Donc la formule précédente se lit aussi*

$$\mathrm{long}(J') = \sum_{\underbrace{K \in \mathrm{Cr}(I_\tau),\ K \subseteq J}_{\text{(i.e. } K \in \mathrm{Cr}(f^{-1}(J')))}} \pi_{\mathrm{n}}(K, f).\ )$$

Corollaire 3. *Si* f *est un morphisme* croissant *d'intervalles orientés, alors pour tout* $J' \in \mathrm{Cr}(\tau')$*, on a*

$$(0 \leq)\ \ \mathrm{card}\,\mathrm{Cr}(f)^{-1}(\{J\}) \longrightarrow \mathrm{long}(J'), \qquad \text{p. 24}$$

l'égalité n'étant obtenue que si les parties de f *sur les intervalles critiques de* I_τ *qui sont au dessus de* J' *sont toutes égales à* 0 *ou* 1 *(donc si et seulement si sur les* $J \in \mathrm{Cr}(f)^{-1}(\{J'\})$*, elles sont égales à* 1*).* [page 37]

Commentary · Exposition [ms. 36]

The length budget over a target run

Monotonicity turns the preceding coverage statement into an additive formula. A map may pause, but it cannot spend the same portion of the target twice. This is the numerical mechanism behind Corollaries 3–5.

Let $f : I_\tau \to I_\upsilon$ be increasing and preserve endpoints. For a source critical run $K_i = [s_{i-1}, s_i]$, write

$$d_i = f(s_i) - f(s_{i-1}) \geq 0.$$

When $d_i > 0$, Proposition 7 associates to K_i the unique target run containing its image; this is $\mathrm{Cr}(f)(K_i)$. Fix a target run $J' = [a, b]$. The nonconstant segments $[f(s_{i-1}), f(s_i)]$ attached to J' cover $[a, b]$, meet only at consecutive endpoints, and are encountered in their natural order. Coverage follows because $f(0) = 0$, $f(n) = m$ and every intervening critical vertex is attained; disjointness of interiors follows from the monotonicity of the sequence $f(s_i)$. Their lengths therefore add to $b - a$:

$$b - a = \sum_{\substack{K_i \in \mathrm{Cr}(\tau, f) \\ \mathrm{Cr}(f)(K_i) = J'}} d_i.$$

This is the formula for $\pi_{\mathrm{n}}(K_i, f)$ in the source. It is valid even if f skips noncritical target vertices, because length measures the whole segment spanned by the image, not the number of image vertices.

Every summand displayed is a positive integer. Consequently, if $k_{J'} = |\mathrm{Cr}(f)^{-1}(J')|$, then

$$1 \leq k_{J'} \leq b - a,$$

and equality on the right holds precisely when every positive d_i in the sum is 1. The printed arrow in Corollary 3 is a typesetting slip for $\leq$, and its unprimed J in the singleton should be J'. The subsequent discussion of equality gives the intended reading independently of our proof.

For a first calculation take $I_{+-+} \to I_{++}$ with vertex function $(0, 1, 1, 2)$. Its three source runs have lengths of image $1, 0, 1$; the single target run has length 2. Thus $k_{J'} = 2$ and the upper bound is attained. For the identity $I_{++} \to I_{++}$, the sole source run has image length 2, so $k_{J'} = 1 < 2$. Both maps are increasing and preserve endpoints. What is counted is the number of nonconstant *runs*, not the number of nonidentity generating arrows.

The alternative summation on page 36 may harmlessly include every global source run contained in $f^{-1}(J')$: the added terms have zero length. Its parenthetical identification of these runs with all critical runs of the preimage is, however, too literal. A preimage can contain a truncated piece of a neighboring global run on which f is constant. For example, take

$$f : I_{++--} \longrightarrow I_{+-}, \qquad (0, 1, 2, 3, 4) \longmapsto (0, 1, 1, 1, 2).$$

For $J' = [0, 1]$, its preimage has vertices $0, 1, 2, 3$ and critical runs $[0, 2]$ and $[2, 3]$. Only $[0, 2]$ is a full global source run contained in the preimage; $[2, 3]$ is a proper part of the global run $[2, 4]$. The extra piece is constant with value 1, hence contributes zero. The equality of sums survives, but equality of the indexing sets does not. One should either keep the global-run indexing or explicitly allow such zero-contribution boundary pieces.

Corollaire 4. *Supposons de plus* I' $[= I_{\tau'}]$ *réduit. Alors*

$$Cr(f) : Cr(\tau, f) \longrightarrow Cr(\tau') \ (\simeq \mathrm{Fl}\,\mathrm{tr}(\tau'))$$

est bijective.

Donc dans ce cas, τ' réduit, on obtient une application en sens inverse

$$\begin{array}{ccc} \mathrm{Tr}(\tau') \simeq \mathrm{Cr}(\tau') & \xrightarrow{\quad \mathrm{Cr}^*(f) \quad} & \mathrm{Cr}(\tau) \\ & {\scriptstyle \mathrm{Cr}(f)^{-1}} \searrow \quad \nearrow {\scriptstyle \text{inclusion}} & \\ & \mathrm{Cr}(\tau, f), & \end{array}$$

application qui transforme l'intervalle critique J' $(= \{i-1, i\})$ en l'*unique* intervalle critique J de I_τ tel que

$$f(J) = J'.$$

Mais $J \simeq \mathbf{\Delta}_i^\varepsilon$, et on a donc un morphisme *surjectif* [page 38]

$$f_0 : \mathbf{\Delta}_i^\varepsilon \longrightarrow J' \simeq \mathbf{\Delta}_1^\varepsilon.$$

On voit tout de suite qu'il existe alors une *unique* flèche de transition u de $\mathbf{\Delta}_i^\varepsilon$, telle que $f(u) \neq \mathrm{id}$. Donc on trouve le corollaire suivant, plus précis que le corollaire 4 :

Corollaire 5. *Soit $f : I_\tau \longrightarrow I_{\tau'}$ un morphisme croissant d'intervalles orientés, avec $I_{\tau'}$ réduit, et soit $\mathrm{Tr}(\tau, f)$ l'ensemble des flèches de transition $\underline{u}_i$ de I_τ telle que $f(\underline{u}_i) \neq \mathrm{id}$, i.e. telles que $\pi(u_i, f) = 1$. Alors f induit une application*

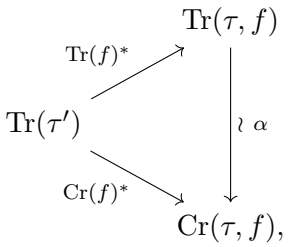
$$\mathrm{Tr}(\tau, f) \longrightarrow \mathrm{Tr}(\tau')$$

qui est bijective, et (bien sûr) croissante, donc un isomorphisme d'ensembles ordonnés.

Ainsi l'application $\mathrm{Tr}(\tau') \longrightarrow \mathrm{Cr}(\tau, f)$ de tantôt se précise par une application bijective

$$\mathrm{Tr}(f)^* : \mathrm{Tr}(\tau') \longrightarrow \mathrm{Tr}(\tau, f),$$

reliée à $\mathrm{Cr}(f)^*$ plus haut par le triangle commutatif



l'application α associant à toute $u_i \in \mathrm{Tr}(\tau, f)$ l'unique intervalle critique qui 'contienne' u_i p. 25
(i.e. tel que u_i soit une flèche dudit intervalle, ou encore, fasse partie de la 'chaîne critique' [page 39]
associée …).

Commentary · Examples [ms. 38]

A reduced target remembers the surviving transitions

Grothendieck next specializes the length formula to a target with no redundant vertices. Every target run then has length one. There is exactly one nonconstant source critical run over it, and that source run can cross its two target vertices only once. This converts a numerical constraint into the combinatorial invariant used in Proposition 8.

More explicitly, on a source run $K = [s, t]$ mapping onto the two vertices $a, a+1$, the nondecreasing vertex function has the form

$$f(s) = \cdots = f(i-1) = a, \qquad f(i) = \cdots = f(t) = a+1$$

for exactly one i with $s < i \leq t$. Hence exactly one transition in K has nonidentity image. The run and its target have the same sign, so this transition maps to the target's unique transition in its categorical direction, including when that direction is backward. Performing this construction for all target runs gives the bijection of Corollary 5. It is increasing because the source runs and their crossings occur in the same external order as the target runs. The triangle with α commutes because α forgets which transition was selected and remembers its containing source run.

Take $\tau = ++--++$ and $\upsilon = +-+$. Choose transitions $2, 4, 6$ in the source. The associated vertex map is

$$(0, 1, 2, 3, 4, 5, 6) \longmapsto (0, 0, 1, 1, 2, 2, 3).$$

The first source run maps across the first forward target edge at transition 2; the second maps across the backward target edge at transition 4; the third crosses the final forward edge at transition 6. The other three transitions become identities. If a path of type υ in X is

$$x_0 \xrightarrow{a} x_1 \xleftarrow{b} x_2 \xrightarrow{c} x_3,$$

restriction along this map is the path

$$x_0 \xrightarrow{1} x_0 \xrightarrow{a} x_1 \xleftarrow{1} x_1 \xleftarrow{b} x_2 \xrightarrow{1} x_2 \xrightarrow{c} x_3.$$

In each of the three source runs either of its two transitions could instead be the surviving one. These choices give $2^3 = 8$ different maps to this reduced target, all explicitly determined by their three crossing positions. The general theorem on the next pages will allow arbitrary sign words and express this determination as a signed-subsequence bijection.

When the target is e, there are no target runs and no nondegenerate transitions. The unique endpoint-preserving map $I_\tau \to I_e$ collapses every transition. Both sides of the bijection in Corollary 5 are empty, so the assertion includes this case. When the source is e and the target has positive length, no endpoint-preserving map exists. These two degenerate cases explain why the later subset description needs the empty word as well as the alternating nonempty words.

On a mieux encore :

Proposition 8. *Soient $\tau, \tau' \in \mathrm{Ty}$, τ'* réduit. *Pour tout morphisme croissant d'intervalles orientés*

$$f : I_\tau \longrightarrow I_{\tau'}$$

(on va dire simplement que f est une flèche dans $\underline{\mathrm{Ty}}$, et écrire $f \in \mathrm{Hom}_{\underline{\mathrm{Ty}}}(I_\tau, I'_\tau)$), associons lui la partie $\mathrm{Tr}(\tau, f)$ des flèches de transition de I_τ non dégénérées pour f. Ces flèches forment un ensemble totalement ordonné affecté de signes + ou − (caractère direct ou rétrograde des flèches), et f induit une bijection d'ensembles totalement ordonnés

$$\underbrace{E_f}_{\subseteq \mathrm{Tr}(\tau)} = \mathrm{Tr}(\tau, f) \simeq \mathrm{Tr}(\tau'),$$

et même un isomorphisme d'ensembles totalement ordonnés 'affectés de signes'. Ainsi, à toute f on associe une partie E_f de $\mathrm{Tr}(\tau)$, et telle que son 'type' soit égal à τ'. Désignons par

$$\mathfrak{P}_{\tau'}(\mathrm{Tr}(\tau)) \subseteq \mathfrak{P}(\mathrm{Tr}(\tau))$$

l'ensemble des parties E de $\mathrm{Tr}(\tau)$ dont le type soit τ'.

Alors l'application [page 40]

$$f \longmapsto \mathrm{Tr}(\tau, f) : \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \tau') \longrightarrow \mathfrak{P}_{\tau'}(\mathrm{Tr}(\tau))$$

est bijective *: se donner un morphisme $f : \tau \longrightarrow \tau'$ dans $\underline{\mathrm{Ty}}$,* revient au même *que de se donner une suite strictement croissante de flèches de transition*

$$\underline{u}_{i_1},\ \underline{u}_{i_2},\ \ldots\ ,\underline{u}_{i_{n'}} \qquad (n' = \log \tau')$$

dans I_τ ($i_1 < i_2 < \cdots < i_{n'}$), telle que la suite des signatures correspondante

$$(\varepsilon_{i_1}, \varepsilon_{i_2}, \ldots, \varepsilon_{i_{n'}}) = \tau'$$

soit égale à τ'.

Commentary · Exposition [ms. 40]

Choosing the transitions that survive

The reduced target allows Grothendieck to replace a functor by a selection of edges. This is the first form of the normal form that will make concatenation manageable: each edge of the target has one identifiable ancestor in the source. The hypothesis that the target is reduced is doing exactly the work needed here.

Write the source signature as $\tau = (\varepsilon_1, \dots, \varepsilon_n)$ and the reduced target as $\tau' = (\varepsilon'_1, \dots, \varepsilon'_m)$. An increasing endpoint-preserving functor has a nondecreasing object map

$$f : [0, n] \longrightarrow [0, m], \qquad f(0) = 0, \quad f(n) = m.$$

The image of the ith transition is either an identity or a nonidentity arrow joining $f(i-1)$ to $f(i)$. In a reduced interval every nonidentity arrow is a transition: two successive transitions have opposite signs and cannot compose. Consequently

$$f(i) - f(i-1) \in \{0, 1\}.$$

Exactly m differences are 1. If their indices are $i_1 < \cdots < i_m$, preservation of arrow direction gives $\varepsilon_{i_j} = \varepsilon'_j$. Conversely such a signed subsequence defines

$$f(k) = \#\{j \mid i_j \le k\}.$$

On a selected transition this map uses the corresponding target transition; on every other transition it uses an identity. These choices extend uniquely to a functor because the source is the free category on its oriented linear graph. They preserve the two endpoints and recover the selected subsequence. This proves Proposition 8, including its empty-target case: the empty subsequence gives the unique functor to the point.

For example, from the word $++-+$ to the reduced word $+-+$ there are exactly two morphisms. The negative transition must be the third source edge, the last positive transition must be the fourth, and the first positive transition may be either the first or the second. Their object maps are respectively

$$(0, 1, 1, 2, 3), \qquad (0, 0, 1, 2, 3).$$

The two paths retain different occurrences of the first positive step. Thus the type records positions, not merely the numbers of positive and negative transitions.

For a single-edge target the same argument gives the stated representability of positive and negative transitions. A positive edge at position i determines the map to Δ_+ which is 0 before that edge and 1 after it; a negative edge determines the analogous map to Δ_-. Naturality remains to be explained because an edge can be stretched over several target edges. The next construction supplies exactly that explanation. When the target is not reduced, the selection alone is insufficient: the unique morphism $\Delta_+ \to \Delta_+^2$ stretches one edge across two, so the size of each surviving image must also be retained.

Corollaire 1. *Pour tout τ, on a des bijections canoniques*

$$\begin{cases} \mathrm{Tr}^+(\tau) & \simeq & \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \Delta^+) \\ \mathrm{Tr}^-(\tau) & \simeq & \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \Delta^-), \end{cases}$$

fonctorielles en τ *(variant dans* $\underline{\mathrm{Ty}}$*).*

Pour donner un sens au corollaire, il faut définir p. 26

$$\tau \longmapsto \mathrm{Tr}^+(\tau), \qquad \tau \longmapsto \mathrm{Tr}^-(\tau)$$

comme *foncteurs* contravariant en τ, donc pour un morphisme

$$f : I_{\tau_2} \longrightarrow I_{\tau_1}$$

dans $\underline{\mathrm{Ty}}$, définir des applications [page 41]

$$\begin{array}{rcl} \mathrm{Tr}^+(f)^* : \mathrm{Tr}^+(\tau_1) & \longrightarrow & \mathrm{Tr}^+(\tau_2) \\ \mathrm{Tr}^-(f)^* : \mathrm{Tr}^-(\tau_1) & \longrightarrow & \mathrm{Tr}^-(\tau_2), \end{array}$$

en d'autres termes, une application

$$\mathrm{Tr}(f)^* : \mathrm{Tr}(\tau_1) \quad \longrightarrow \quad \mathrm{Tr}(\tau_2)$$

'compatible avec les signatures'. Dans le corollaire 5 plus haut, on a défini une telle application seulement pour τ' réduit $[\tau_1 = \tau',\ \tau_2 = \tau]$ ([12]). Pour une flèche u de I_τ, appelons *partie ensembliste* de f sur u, où de u pour f, la partie ensembliste de la flèche $f(u)$ dans $I_{\tau'}$, i.e. le segment $[f(i-1), f(i)] \subseteq I_{\tau'}$ (ici $u = \underline{u}_i$). Soit

$$\underline{u}_{i_1},\ \ldots\ ,\underline{u}_{i_N} \qquad \text{ou } (v_1, v_2, \ldots, v_n)$$

la suite croissante des éléments de $\mathrm{Tr}(\tau, f)$, on voit alors par récurrence sur i que pour $i \in [1, N-1]$, on a

$$\begin{array}{rcl} f(\mathrm{but}(v_i)) & = & f(\mathrm{source}(v_{i+1}))\ \ (= t_i), \\ f(\mathrm{source}(v_1)) & = & 0_{\tau'}, \\ f(\mathrm{but}(v_N)) & = & 1_{\tau'}. \end{array}$$

Considérons la suite

$$\{\overbrace{0_{\tau'}}^{=t_0}, t_1, t_2, \ldots, t_{N-1}, \overbrace{1_{\tau'}}^{=t_N}\},$$

elle est définie par le sous-ensemble correspondant de $I_{\tau'}$, qu'on appelera l'ensemble des *sommets critiques de $I_{\tau'}$ pour f*. C'est une suite contenant $0_{\tau'}$, $1_{\tau'}$, et telle que deux éléments consécutifs sont reliés dans $I_{\tau'}$ par une flèche, ce qui équivaut à dire qu'elle contient les sommets critiques (intrinsèques) de $I_{\tau'}$. D'autre part, pour tout intervalle de transition $[i-1, i]$ de $I_{\tau'}$ ($i \in [1, n']$), il existe un et un seul intervalle de transition $[t_{j-1}, t_j]$ pour cette suite ($j \in [1, N]$), telle qu'on ait $[i-1, i] = [t_{j-1}, t_j]$. C'est dire qu'il existe une et une seule $v_j \in \mathrm{Tr}(\tau, f)$ telle que la partie de f sur v_j (savoir justement $[j-1, j]$) contienne (celle de) $\underline{u}_i$. Pour résumer : p. 27 [page 42]

Corollaire 2. *Soit $f : \tau \longrightarrow \tau'$ dans* Ty. *Alors pour toute $u \in \mathrm{Tr}(\tau') = \mathrm{Tr}(I_{\tau'})$, il existe une et une seule $v \in \mathrm{Tr}(\tau)$ telle que la partie de f sur v contienne u. De plus, les signatures de u et de v sont égales.*

C'est ainsi qu'on obtient donc une application

$$\mathrm{Tr}(f)^* : \mathrm{Tr}(\tau') \quad \longrightarrow \quad \mathrm{Tr}(\tau) \qquad (\text{se factorisant par } \mathrm{Tr}(\tau, f))$$

induisant

$$\begin{array}{rcl} \mathrm{Tr}^+(f)^* : \mathrm{Tr}^+(\tau') & \longrightarrow & \mathrm{Tr}^+(\tau) \\ \mathrm{Tr}^-(f)^* : \mathrm{Tr}^-(\tau') & \longrightarrow & \mathrm{Tr}^-(\tau). \end{array}$$

La transitivité pour un composé

$$\tau \xrightarrow{f} \tau' \xrightarrow{g} \tau''$$

ou

$$I_\tau \xrightarrow{f} I_{\tau'} \xrightarrow{g} I_{\tau''}$$

est immédiate. Si u, u', u'' sont des flèches de transition dans I_τ, $I_{\tau'}$, $I_{\tau''}$ respectivement, telles que

$$\begin{array}{rcll} \pi(g, u') & \supseteq & \pi(u''), & \text{i.e. } u' = \mathrm{Tr}(g)^*(u'') \\ \pi(f, u) & \supseteq & \pi(u'), & \text{i.e. } u = \mathrm{Tr}(f)^*(u'), \end{array}$$

alors on a

$$\pi(gf, u) \ \supseteq\ \pi(u''), \qquad \text{i.e. } u = \mathrm{Tr}(gf)^*(u'').$$

12. Ce qui suit ne fait pas intervenir les signatures - c'est un sorite sur les applications croissantes entre ensembles finis totalement ordonnés. [page 43]

Si les flèches en question ont comme partie intrinsèque $[i-1,i]$, $[i'-1,i']$, $[i''-1,i'']$ respectivement, l'hypothèse est

$$\begin{array}{rcl} g(i'-1) & \leq & i''-1, \quad g(i') \geq i'' \\ f(i-1) & \leq & i'-1, \quad f(i) \geq i' \end{array}$$

et comme g est croissant, on a déduit

$$\begin{array}{rcll} gf(i-1) & \leq & g(i'-1) & (\leq i''-1) \\ gf(i) & \geq & g(i') & (\geq i''), \end{array}$$

c.q.f.d.

Commentary · Examples [ms. 42]

A stretched edge and the containment convention

Grothendieck is assigning every target transition to the unique source transition whose image crosses it. One small example fixes both the meaning of this assignment and two printed slips immediately before Corollary 2.

Let $f : \Delta_+ \to \Delta_+^2$ have object map $f(0) = 0$, $f(1) = 2$. There is one surviving source transition, so $N = 1$, $t_0 = 0$ and $t_1 = 2$. The two target transitions have underlying vertex segments $[0,1]$ and $[1,2]$. Both are contained in $[t_0, t_1] = [0,2]$, and neither equals it. Thus the printed relation

$$[i-1,i] = [t_{j-1}, t_j]$$

must be read mathematically as $[i-1,i] \subseteq [t_{j-1}, t_j]$. The parenthesis identifying the part of f on v_j as $[j-1,j]$ has the same problem: that part is $[t_{j-1}, t_j]$. Both readings occur in the edition; they are retained in the source and repaired here. Corollary 2 itself uses containment and is correct.

The intervals $[t_{j-1}, t_j]$ may meet at a vertex, but their sets of transitions are disjoint. Since $t_0 = 0$, $t_N = m$ and the t_j are strictly increasing, these transition sets partition $\{1, \ldots, m\}$. A transition cannot lie in two of them: its two endpoints would then have to belong to their intersection, which contains at most one vertex. This proves uniqueness without assuming that the target is reduced. Functoriality forces every edge in the block assigned to v_j to have the sign of v_j.

The words "source" and "but" in the preceding display concerning the v_j must here be read as the first and last vertices in the traversal order. On a retrograde transition they are the reverse of its categorical domain and codomain. Writing $f(i_j - 1)$ and $f(i_j)$ avoids this ambiguity. If the target is the point there are no target transitions and the pullback map of transition sets is the unique map from the empty set; no first or last surviving transition should be invoked.

p. 28

Ainsi, le corollaire 1 plus haut obtient un sens précis et il est démontré du même coup, vu que les correspondances du corollaire 1 sont données par

$$\underbrace{f}_{\in \mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \Delta^\varepsilon)} \longmapsto \mathrm{Tr}(f)^*(\underline{u}),$$

où $\underline{u}$ est l'unique flèche de transition de Δ^ε.

Commentary · Exposition [ms. 43]

Edges reverse the direction of a map

The contravariant transition map contains more information than its notation initially suggests. It determines the whole functor, and gives a particularly economical modern description of the final category of types. This observation explains why the later calculations require only ordered lists and nonnegative integers.

Let $\mathcal{O}_\pm$ have as objects finite linearly ordered sets equipped with a sign at each element, including the empty set, and as arrows the nondecreasing maps which preserve signs. We use the skeletal

version with underlying sets $[1,n]$. For $f:\tau\to\tau'$ in $\underline{\mathrm{Ty}}$, where τ and τ' have lengths n and m, define

$$q_f:[1,m]\longrightarrow[1,n],\qquad q_f(j)=\min\{i\mid j\le f(i)\}.$$

The endpoint conditions ensure that this minimum exists when $m>0$; its defining inequalities are

$$f(q_f(j)-1)<j\le f(q_f(j)).$$

This is Grothendieck's $\mathrm{Tr}(f)^*$. It is nondecreasing, and it preserves signs because the image of a source transition is a monochromatic string of target transitions of that same sign.

Conversely, suppose $q:[1,m]\to[1,n]$ is nondecreasing and sign-preserving. Put

$$F_q(i)=\#q^{-1}([1,i]),\qquad 0\le i\le n.$$

The fibre $q^{-1}(i)$ is a consecutive block, possibly empty, whose target signs are all ε_i. Hence the consecutive vertices $F_q(i-1)$ and $F_q(i)$ are joined in the target by an arrow of the required direction, or coincide when the fibre is empty. Sending the ith source transition to this arrow defines a unique functor. Its object map is nondecreasing, begins at 0, and ends at m. The inequalities above show $q_{F_q}=q$, while counting the target edges crossed by the first i source edges gives $F_{q_f}(i)=f(i)$. These constructions are inverse, including the empty cases. There is no map from the point to a nonpoint interval in $\underline{\mathrm{Ty}}$, just as there is no function from a nonempty edge set to the empty one.

If $g:\tau'\to\tau''$, a target edge of τ'' has a unique ancestor in τ' and then a unique ancestor in τ. Equivalently, applying the displayed inequalities twice gives

$$q_{gf}=q_fq_g.$$

Identities induce identities. We therefore have an isomorphism of skeletal categories

$$\underline{\mathrm{Ty}}\ \cong\ \mathcal{O}_{\pm}^{\mathrm{op}}.$$

The reversal of variance is essential: a functor stretches a source edge into a target block, whereas q_f sends each member of that block back to its source edge. The signs can be forgotten for the construction on ordered sets, as Grothendieck observes, but must be restored to say when the resulting vertex map is a functor between the oriented interval categories.

This identification also establishes the naturality promised in Corollary 1. If an edge of sign ε represents $h:\tau'\to\Delta_\varepsilon$, the edge representing hf is q_f of the edge representing h. It is precisely ordinary precomposition in the representable presheaf $\mathrm{Hom}_{\underline{\mathrm{Ty}}}(-,\Delta_\varepsilon)$. The underlying single-sign calculation is the familiar ordinal description of the simplicial category; see Aguiar and Mahajan (2010, Appendix D.1, pp. 713–715). The signed formulation follows by retaining the labels throughout the inverse constructions above.

On va utiliser ces sorites pour donner, pour tout $\tau\in\underline{\mathrm{Ty}}$, une description combinatoire simple des catégories $\underline{\mathrm{Ty}}/\tau$ et $\tau\backslash\underline{\mathrm{Ty}}$, ou du moins de leurs ensembles d'objets.

1° Description de $\mathrm{Ob}\,(\tau\backslash\underline{\mathrm{Ty}})$**.**

À $f\in\mathrm{Ob}(\tau\backslash\underline{\mathrm{Ty}})$, i.e.

$$f:\tau\longrightarrow\tau'\qquad\text{i.e. } f:I_\tau\longrightarrow I_{\tau'}$$

dans $\underline{\mathrm{Ty}}$, est associé $E=E_f=\mathrm{Tr}(\tau,f)\subseteq\mathrm{Tr}(\tau)$, l'ensemble des flèches de transition critiques (i.e. non dégénérées) pour f. À chaque $u\in E$, associons l'entier $\mathrm{long}(\pi(f,u))$, la *partie* de f sur u. Alors (τ',f) est déterminé de façon unique par le couple [page 44]

$$(E,\pi),\qquad E\in\mathfrak{P}(\mathrm{Tr}(\tau)),\qquad \underbrace{\pi}_{\text{pondération}}:E\longrightarrow\underbrace{\mathbf{N}^*}_{\substack{=\text{ensemble des}\\ \text{entiers}\ge 1}}.$$

Ou si on préfère, par l'application $\pi=\pi_f$

$$\mathrm{Tr}(\tau)\xrightarrow{\pi}\mathbf{N},\qquad u\longmapsto\pi(u)=\pi(u,f)=\mathrm{long}(\pi_{\tau'}(f(u)))$$

(qui redonne E comme

$$E=\pi^{-1}(\mathbf{N}^*)\qquad).$$

Cette application n'est soumise à aucune restriction,

$$\mathrm{Ob}\,(\tau\backslash\underline{\mathrm{Ty}}) \xrightarrow[\sim]{\text{bij.}} \mathrm{Hom}_{\mathrm{Ens}}(\mathrm{Tr}(\tau), \mathbf{N}).$$

On récupère τ' à partir de π ainsi. On prend $E = \pi^{-1}(\mathbf{N}^*)$ (flèches de transition non dégénérées pour π), E est totalement ordonné par l'ordre induit par celui de $\mathrm{Tr}(\tau)$, il est de plus signaturé [sic] par la signaturation [sic] de $\mathrm{Tr}(\tau)$, soit

$$(\varepsilon_1, n_1),\ \ldots\ ,\ (\varepsilon_r, n_r)$$

la suite des signatures et pondérations, de la suite des éléments de E ($\varepsilon_i \in \{\pm 1\}$, $n_i \in \mathbf{N}^*$ pour $i \in [1, r]$). Alors,

$$\tau' = \Delta_{\varepsilon_1}^{n_1} \Delta_{\varepsilon_2}^{n_2} \cdots \Delta_{\varepsilon_r}^{n_r}, \qquad r = \mathrm{card}\, E,$$

dans le monoïde libre Ty, où on a désigné par Δ_+, Δ_- (au lieu de Δ^+, Δ^-) les deux générateurs. L'application

$$I_\tau \longrightarrow I_{\tau'}$$

est décrite de façon unique en disant qu'elle est *constante* sur chacune des 'composantes connexes' (au sens de la numérotation des sommets de I_τ) de l'ensemble $E' = E'(f) \subseteq I_\tau = [0, n]$ des sommets qui sont sources ou buts de flèches de transition dégénérées pour f, i.e. dans $\mathrm{Tr}(\tau)\backslash E$. Et sur le α-ème élément de E ($\alpha \in [1, r = \mathrm{card}\, E]$), d'indice i_α dans $\mathrm{Tr}(\tau)$, f induit le foncteur

p. 29 [page 45]

$$\mathbf{\Delta}_{\varepsilon_\alpha} \simeq \underbrace{J_{i_\alpha}}_{\substack{\text{intervalle de transition d'indice } i_\alpha \\ \text{dans } I_\tau}} \longrightarrow I_{\tau'}$$

donné par *l'unique* Ty-*flèche* dans I_τ de J_{i_α} dans le α-ème facteur, de type $\Delta_{\varepsilon_\alpha}^{n_\alpha}$, de τ', donné par $(*)$

$$\Delta_{\varepsilon_\alpha}^{n_\alpha} \simeq \text{type de } \mathbf{\Delta}_{n_\alpha}^{\varepsilon_\alpha} \quad \ldots$$

Commentary · Examples [ms. 44]

Weight vectors and composition by block sums

The outgoing classification now has coordinates which are useful for actual calculations. For a fixed source $\tau = (\varepsilon_1, \ldots, \varepsilon_n)$, put

$$a_i = f(i) - f(i-1) = \#q_f^{-1}(i), \qquad A_i = \sum_{h=1}^{i} a_h, \quad A_0 = 0.$$

Every a_i is a nonnegative integer. The target is the word

$$\tau' = \varepsilon_1^{a_1} \varepsilon_2^{a_2} \cdots \varepsilon_n^{a_n}, \qquad f(i) = A_i,$$

where exponent zero means omission. These equations prove the claimed unrestricted parametrization of $\mathrm{Ob}(\tau\backslash\underline{\mathrm{Ty}})$: they construct an outgoing morphism from every weight vector and recover the vector from every outgoing morphism. When adjacent source signs agree, their target blocks may merge into one critical interval, but their individual lengths remain part of the morphism.

Take the source $+ - + + -$ and the weights $(2, 0, 1, 3, 0)$. Its target is $+ + + + + +$, and its vertex map is

$$(0, 2, 2, 3, 6, 6).$$

The pullback on target transitions is $(1, 1, 3, 4, 4, 4)$. Both negative source edges have been collapsed. The two consecutive positive source edges at positions 3 and 4 survive separately, even though their target images lie in the same positive run.

To see composition, give the six target edges the further weights

$$(b_1, \ldots, b_6) = (1, 0, 2, 1, 0, 3).$$

The second map has vertex values $(0,1,1,3,4,4,7)$ and target $+^7$. The composite has weights $(1,0,2,4,0)$, since its ith weight is

$$c_i = \sum_{j=A_{i-1}+1}^{A_i} b_j.$$

Thus its vertex map is $(0,1,1,3,7,7)$, which is also obtained by direct substitution into the two displayed object maps. An empty block contributes zero; a block of length greater than one contributes a sum. This is the numerical content of composition in Section 6.

The formula gives a useful direct verification of associativity. Given a third weight vector, first replacing every b_j by the sum over its descendants and then summing over the a_i-block is the same finite sum as first merging the b_j-blocks and then summing over the merged block. The order and boundaries of the summands are unchanged. No cancellation, choice of representative, or equivalence relation enters this calculation.

2° Détermination de $\mathrm{Ob}\,(\underline{\mathrm{Ty}}/\tau')$.

Si $(\tau, f) \in \mathrm{Ob}\,(\underline{\mathrm{Ty}}/\tau')$, donc

$$f : I_\tau \longrightarrow I_{\tau'},$$

on lui associe tout d'abord la partie

$$\mathrm{Cr}(\tau', f) = E'_f = E' \subseteq I_{\tau'}$$

formée des sommets (dits 'critiques pour f', cf. page 41) de $I_{\tau'}$ qui sont source ou but d'une flèche $f(u)$, où $u \in \mathrm{Tr}(\tau, f)$, i.e. $f(u)$ n'est pas dégénérée ([13]). E' contient $f(I_{\tau\,\mathrm{réd}})$ (i.e. les images par f des sommets critiques de I_τ), mais peut être plus grand. (P.ex. si $\tau = \tau'$, $f = \mathrm{id}$, [page 46]
$E' = I_{\tau'}$, qui est donc strictement plus grand que $I_{\tau'\,\mathrm{réd}}$ si τ' n'est pas réduit.) Pour tout $i \in E'$, considérons $f^{-1}(\{i\}) \subseteq I_\tau$. Comme f `[est]` croissant, c'est un intervalle de l'ensemble totalement ordonné I_τ, donc c'est un sous-intervalle de la catégorie intervalle I_τ. Comme tel, il a une signature $\tau_i \in \mathrm{Ty}$. On a donc une application

$$\eta_f \text{ ou } \eta : E' \longrightarrow \mathrm{Ty}.$$

Je dis que $(\tau, f) \in \mathrm{Ob}\,(\underline{\mathrm{Ty}}/\tau')$ est déterminé de façon unique par ce couple

$$(E', \eta) \begin{cases} E' \in \mathfrak{P}(I_{\tau'}) = \mathfrak{P}([0, n']) \\ \eta : E' \longrightarrow \mathrm{Ty}\ . \end{cases}$$

De plus, celui-ci est soumis à la seule restriction que

$$E' \supseteq \mathrm{Cr}(\tau') = \text{ensemble des sommets critiques de } I_{\tau'}$$

(où ce qui revient au même, que E' contient $0_{\tau'}$, $1_{\tau'}$ et que deux éléments consécutifs de E' p. 30
sont reliés par une flèche de $I_{\tau'}$) :

$$\begin{cases} \mathrm{Ob}\,(\mathrm{Ty}/\tau') \simeq \text{ensemble des couples } (E', \eta), \text{ avec} \\ \qquad\qquad E' \subseteq [0, \underbrace{n'}_{=\log \tau'}], \quad E' \supseteq \mathrm{Cr}(\tau'), \\ \qquad\qquad \eta : E' \longrightarrow \mathrm{Ty}\ . \end{cases}$$

Soit

$$0 = t_0 < t_1 < \cdots < t_r = n'$$

13. dégager un lemme f croissante inutile.

la suite des éléments de E', soit pour $\boxed{i \in [1, N]}$

$$\left\| \begin{array}{l} l_i = t_i - t_{i-1} \\ \varepsilon_i \quad \text{signature de } [t_{i-1}, t_i] \text{ dans } I_{\tau'} \ , \end{array} \right.$$

et pour $\boxed{i \in [0, N]}$, soit

$$\| \quad \tau_i = \eta(t_i).$$

[page 47]

Alors le type τ est donné par

$$(*) \qquad \tau = \tau_0 \Delta_{\varepsilon_1} \tau_1 \Delta_{\varepsilon_2} \cdots \Delta_{\varepsilon_N} \tau_N,$$

[de] longueur $N + \sum_{i=0}^{N} \underbrace{\operatorname{long}(\tau_i)}_{\lambda_i \geq 0}$.

Commentary · Exposition [ms. 45]

Fibres are components of the collapsed-edge graph

Grothendieck wants a geometric reconstruction of the same weight vector. The intended reconstruction is correct once the graph of collapsed transitions is kept, but the printed description in terms of a set of vertices loses information that is needed to find its components.

Consider $\tau = +++$ with weights $(0, 1, 0)$. The corresponding map to $+$ is

$$f(0) = f(1) = 0, \qquad f(2) = f(3) = 1.$$

The vertices incident to a collapsed transition form the whole set $\{0, 1, 2, 3\}$. As a subset of the integers this set is connected. Consequently the assertion that f is constant on its integer-connected components would make f constant everywhere, contrary to the displayed functor. This is a printed mathematical slip, not a defective proof of the weight classification.

Instead, take the graph with all source vertices and only those edges $[i-1, i]$ for which $a_i = 0$. Its components are consecutive strings of zero-weight edges, together with any isolated vertices. In the example they are $\{0, 1\}$ and $\{2, 3\}$. Since

$$f(j) - f(i) = \sum_{h=i+1}^{j} a_h \qquad (i \leq j),$$

nonnegativity shows that two vertices have the same image if and only if every intervening edge has weight zero. The graph components are therefore exactly the nonempty vertex fibres of f. On each surviving transition the endpoint values are determined by the cumulative sum of the earlier weights. This proves both existence and uniqueness of the proposed geometric reconstruction after the correction.

The same observation produces a canonical factorization. First collapse the zero-weight edges, leaving the signed subword consisting of the active source edges; then stretch its successive edges by their positive weights. The first vertex map is surjective, the second strictly increasing, and both preserve the endpoints. Uniqueness follows either from the fibre components or from the unique image factorization of q_f in the category of signed ordered sets. This factorization will let us distinguish a combinatorial contraction from a contraction which has an inverse on one side.

3° Description de $\underline{\mathrm{Fl}(\mathrm{Ty})}$.

Si

$$f : \tau \longrightarrow \tau' \qquad \texttt{[est]} \text{ dans } \underline{\mathrm{Ty}} \ ,$$

on lui associe comme tantôt un entier $\underline{N \geq 0}$ ($N = \operatorname{card} \mathrm{Cr}(\tau', f) - 1$, $N = 0$ si et seulement si $\tau' = 1$, type de l'intervalle ponctuel), des applications

$$\left\{ \begin{array}{lcll} [1, N] & \longrightarrow & [+1, -1] \times \mathbf{N}^*, & i \longmapsto (\varepsilon_i, l_i) \\ [0, N] & \longrightarrow & \mathrm{Ty} & i \longmapsto \tau_i \end{array} \right.$$

([14]). La flèche f dans $\underline{\mathrm{Ty}}$ est determinée par ces données. Le type τ est donné par la formule $(*)$ ci-dessus, le type τ' par

$$(*') \qquad \tau' = \Delta_{\varepsilon_1}^{l_1}\Delta_{\varepsilon_2}^{l_2}\ \cdots\ \Delta_{\varepsilon_N}^{l_N}, \qquad \text{longueur } n' = \textstyle\sum l_i,$$

et l'application croissante de $I_\tau = [0, N + \sum \mathrm{long}(\tau_i)]$ dans $I_{\tau'} = [0, \sum l_i]$ est donnée ainsi : p. 31

constante sur $[0, 0+\lambda_0]$	$\longmapsto$	0
constante sur $[\lambda_0+1, \lambda_0+1+\lambda_1]$	$\longmapsto$	l_1
constante sur $[\lambda_0+\lambda_1+2, \lambda_0+\lambda_1+2+\lambda_2]$	$\longmapsto$	l_2
$\dots$		
constante sur $[\lambda_0+\lambda_1+\cdots+\lambda_i+(i+1), \lambda_0+\cdots+\lambda_i+\lambda_{i+1}+(i+1)]$	$\longmapsto$	l_i.

p. 32

Commentary · Exposition [ms. 47] [page 48]

Reading the morphism from its target

The incoming classification organizes a morphism by the places where its nonconstant pieces meet. It is the same normal form viewed from the target: choose the visited vertices, and at each visited vertex record the entire interval that has been collapsed there. This description is useful because the intervening noncollapsed transitions are then forced.

Let $f : I_\tau \to I_{\tau'}$ be increasing and endpoint-preserving, with target length m. The uniform definition is

$$E' = \mathrm{im}(f) \subseteq [0, m].$$

When $m > 0$, it agrees with the printed definition by endpoints of nondegenerate images. Indeed, take a vertex value t assumed by f. If not every source vertex has value t, connectedness of the underlying source path supplies an edge from the nonempty set $f^{-1}(t)$ to its complement. That edge is nondegenerate and one of its image endpoints is t. Conversely every such endpoint is a value of f. This argument does not require monotonicity, and supplies the elementary lemma requested in footnote 13, provided the functor is nonconstant.

For $m = 0$ all transition images are identities, so the printed endpoint definition would instead give $E' = \varnothing$. The later claim that E' contains the endpoint, and the definition $N = \#E' - 1$, then fail. Taking $E' = \mathrm{im}(f) = \{0\}$ repairs the nullary case and agrees with the explicit description of $N = 0$ on manuscript page 47. This is an omitted boundary convention; it does not change the nonconstant construction.

Write the distinct elements of E' as

$$0 = t_0 < t_1 < \cdots < t_N = m.$$

Every two consecutive values are joined by a target arrow: the source transition between their two adjacent nonempty fibres has exactly these image endpoints. Hence the segment from t_{j-1} to t_j is monochromatic, with sign ε_j. Equivalently, E' contains the intrinsic critical vertices of the target, namely its endpoints and every place where the signs change. The notation $\mathrm{Cr}(\tau')$ has changed meaning at this point in the manuscript: here it denotes these vertices, whereas earlier it denoted critical intervals. Keeping these two sets distinct removes a potential confusion in the formula.

The fibre over t_j is a nonempty consecutive interval of vertices; let its signature be τ_j and its length λ_j. Reading the source from left to right gives exactly

$$\tau = \tau_0\Delta_{\varepsilon_1}\tau_1\Delta_{\varepsilon_2}\cdots\Delta_{\varepsilon_N}\tau_N.$$

Conversely, choose any subset E' containing the critical vertices and any words $\tau_0, \dots, \tau_N$. Form this source word. Collapse the τ_j-block to t_j, and map the transition between consecutive blocks to the target arrow joining t_{j-1} and t_j, in the direction prescribed by ε_j. This defines an increasing endpoint-preserving functor; its image is exactly E', and its fibres have the prescribed words. Each

14. **NB** Si $N = 0$, $[1, N] = \emptyset$, et il n'y a pas de ε_i, l_i; l'application $\underbrace{[0, N]}_{=\{0\}} \longrightarrow$ Ty se réduit à la donnée de $\tau_0 \in$ Ty.

arrow is uniquely determined by its endpoints in an interval category, so there is no additional gluing datum. The two constructions are inverse. For a point target they reduce to the arbitrary word τ_0 and its unique map to the point.

For use in computations, set

$$l_j = t_j - t_{j-1}, \qquad B_j = j + \sum_{h<j} \lambda_h.$$

The correct full formula is

$$f(k) = t_j = \sum_{h=1}^{j} l_h \quad \text{for } B_j \le k \le B_j + \lambda_j, \qquad 0 \le j \le N.$$

The source's explicit table gives l_2 on the third fibre, where the value must be $l_1 + l_2$; its final row similarly needs the corresponding cumulative sum. The formula above repairs all rows at once. For example, take target $++-$, visited vertices $E' = \{0, 2, 3\}$, and fibre words $\tau_0 = -$, $\tau_1 = +$, τ_2 empty. Then the source is $-++-$ and the object map is $(0, 0, 2, 2, 3)$. The first positive transition crosses the two target positive edges, the next is collapsed, and the final negative transition crosses the final target edge. The two jumps have sizes 2 and 1, while the final vertex value is their sum 3.

Commentary · Development (commentators') [ms. 47]

Which contractions admit a section

The normal form makes it possible to ask precisely when a deleted collection of transitions can be restored functorially. This is useful for the later path formalism: a surjection of underlying vertex sets need not be a split epimorphism in the category of signed types. The signs can obstruct every possible section.

Section criterion for a contraction. [ours] Let $\tau = (\varepsilon_1, \dots, \varepsilon_n)$ and let $E = \{e_1 < \cdots < e_k\}$ be a subset of its transition positions. Let

$$p_E : \tau \longrightarrow \tau|_E$$

be the contraction with weights 1 at the positions in E and 0 elsewhere. If $k = 0$, it has a section exactly when $n = 0$. If $k > 0$, it has a section exactly under the following conditions. Every sign before e_1 equals ε_{e_1}, and every sign after e_k equals ε_{e_k}. In each open gap (e_j, e_{j+1}), the sequence of signs is a possibly empty string of ε_{e_j} followed by a possibly empty string of $\varepsilon_{e_{j+1}}$. When these conditions hold, the number of sections is

$$\prod_{\substack{1 \le j < k \\ \varepsilon_{e_j} = \varepsilon_{e_{j+1}}}} (e_{j+1} - e_j).$$

An empty product is 1.

Proof. Under the signed-edge anti-isomorphism, p_E corresponds to the inclusion $E \hookrightarrow [1, n]$. A section of p_E therefore corresponds exactly to a nondecreasing sign-preserving retraction

$$r : [1, n] \longrightarrow E, \qquad r(e_j) = e_j.$$

If E is empty such a map exists exactly when its domain is empty. Assume henceforth $k > 0$. Monotonicity forces every index before e_1 to map to e_1, and every index after e_k to map to e_k. This proves the two exterior conditions, and these parts of r are unique.

For $e_j < i < e_{j+1}$, monotonicity forces $e_j \le r(i) \le e_{j+1}$. Since there is no element of E strictly between them, $r(i)$ is one of these two endpoints. Moreover the indices mapped left must all precede the indices mapped right. There is therefore a single cut in the gap. Sign preservation is equivalent to the signs before the cut being ε_{e_j} and those after it being $\varepsilon_{e_{j+1}}$. This proves necessity of the gap condition. If the endpoint signs differ, an admissible cut is unique. If they agree, every sign in the gap must have that common value and all $d + 1$ cuts are admissible, where $d = e_{j+1} - e_j - 1$.

Conversely, choose an admissible cut independently in every gap, use the forced exterior values, and fix every e_j. The resulting map is nondecreasing across each gap and its boundary, hence globally

nondecreasing; it preserves signs by construction and retracts onto E. These choices describe every retraction exactly once. Multiplying their independent numbers gives the displayed product. The signed-edge correspondence turns each retraction back into a unique section of p_E, completing the proof.

For $\tau = +-$ and $E = \{1\}$ the trailing negative sign violates the exterior condition. Thus the surjective vertex map $+- \to +$ has no section. For $\tau = +++$ and $E = \{1, 3\}$ the gap has one omitted positive edge, and there are exactly two sections; their vertex maps $[0, 2] \to [0, 3]$ are $(0, 1, 3)$ and $(0, 2, 3)$. This distinguishes the collapse of an edge from a simplicial degeneracy with an available section.

We checked the ordinal description and comonoid interpretation in Aguiar and Mahajan (2010, Appendix D), the zig-zag and Reedy structures of Arakawa and Cnossen (2026, Definition 1.1 and Example 4.12), and searched for signed or coloured ordinal retractions and sections of Grothendieck's interval maps. We could not find this exact interval-form criterion and count in those sources. The result is an elementary consequence of the normal form, recorded here as a proved continuation of the calculation, without a claim of priority.

6 Composition dans $\underline{\mathrm{Ty}}$

On a une loi de composition (partout définie) dans $\mathrm{Ob}\,(\underline{\mathrm{Ty}}) = \mathrm{Ty}$, c'est celle du monoïde libre à deux générateurs Δ_+, Δ_-. Cette loi de composition s'interprète, au niveau catégorique (et non plus seulement combinatoire) comme une somme amalgamée

$$I_{\tau\tau'} = I_\tau \amalg_{(e, \underline{1}_\tau, \underline{0}_{\tau'})} I_{\tau'},$$

où e est la catégorie unité, et où $\underline{1}_\tau$, $\underline{0}_{\tau'}$ représentent maintenant les morphismes $e \longrightarrow I_\tau$, $e \longrightarrow I_{\tau'}$, ayant comme image l'élément but et l'élément source respectivement de l'intervalle orienté envisagé. On a donc un carré cocartésien dans (Cat)

$$\begin{array}{ccc} & e & \\ {\scriptstyle \underline{1}_\tau}\swarrow & & \searrow{\scriptstyle \underline{0}_{\tau'}} \\ I_\tau & \text{cocart.} & I_{\tau'} \\ {\scriptstyle i_{\tau,\tau'}}\searrow & & \swarrow{\scriptstyle j_{\tau,\tau'}} \\ & I_{\tau\tau'}, & \end{array}$$

où toutes les flèches sont des foncteurs mono et pleinement fidèles. Que ce diagramme (évidemment commutatif) soit bien cocartésien dans (Cat) revient, par définition, à dire que pour toute X dans Cat, le diagramme transposé

$$\begin{array}{ccccc} & & X & & \\ & {\scriptstyle \text{(but)}\ \beta_\tau}\nearrow & & \nwarrow{\scriptstyle \sigma_{\tau'}\ \text{(source)}} & \\ \underline{\mathrm{Ch}}_\tau(X) & & & & \underline{\mathrm{Ch}}_{\tau'}(X) \\ & \nwarrow & & \nearrow & \\ & & \underline{\mathrm{Ch}}_{\tau\tau'}(X), & & \end{array}$$

`[est cartésien]`

ce qui revient à la tautologie suivante qu'un chemin C dans X de type $\tau\tau'$ 'n'est autre chose' qu'un couple (c, c') d'un chemin c de type τ et d'un chemin c' de type τ' 'mis bout à bout' (dans cet ordre), i.e. tels que [page 49]

$$\mathrm{but}(c) = \mathrm{source}(c')$$

(et on a alors

$$\begin{aligned} \mathrm{source}(C) &= \mathrm{source}(c) \\ \mathrm{but}(C) &= \mathrm{but}(c') \end{aligned}$$

). p. 33

Il se pose ici le sempiternel problème si on va noter ce chemin $c{\cdot}c'$, ce qui semble à première vue s'imposer, ou plutôt $c' \circ c$, regardant le chemin c de $x = \text{source}(c)$ vers $y = \text{but}(c)$ comme une sorte de morphisme de x dans y (cf. plus bas le formalisme des catégories de fractions), c' comme un morphisme de $y = \text{source}(c')$ vers $z = \text{but}(c')$, et en interprétant le composé C comme le composé de ces deux morphismes c, c' (dans cet ordre du point de vue des opérations à effectuer), qu'on note traditionellement $c' \circ c$ ou $c' \cdot c$, et non $c \circ c'$ ou $c \cdot c'$, hélas! C'est cette notation que nous allons suivre plus bas.

Mais pour l'instant on ne va pas se soucier des chemins dans des catégories quelconques X, mais revenir au formalisme des τ et des I_τ eux-mêmes. L'interprétation catégorique de $\tau\tau'$ comme une somme amalgamée nous assure que cette opération de composition est fonctorielle : [page 50]

$$\tau_1 \xrightarrow{f} \tau_2, \quad \tau_1' \xrightarrow{f'} \tau_2' \qquad \text{donne } \tau_1\tau_1' \xrightarrow{f*f'} \tau_2\tau_2'$$

(on n'écrira pas $f \cdot f'$ pour la flèche notée $f * f'$, à cause de la confusion possible avec la composition des flèches dans $\underline{\text{Ty}}$). On a transitivité de façon évidente : si de plus $\tau_2 \xrightarrow{g} \tau_3$, $\tau_2' \xrightarrow{g'} \tau_3'$, `[on a]`

$$(gf) * (g'f') = (g * g') \circ (f * f').$$

Donc on a un bifoncteur de composition

$$\underline{\text{Ty}} \times \underline{\text{Ty}} \longrightarrow \underline{\text{Ty}} \qquad (c, c') \longmapsto c \cdot c' \text{ ou } c * c'.$$

(En fait, on a aussi un bifoncteur

$$\underline{\text{Typ}} \times \underline{\text{Typ}} \longrightarrow \underline{\text{Typ}},$$

mais on préfère oublier maintenant la catégorie $\underline{\text{Typ}}$, qui donne un formalisme nettement moins simple.)

Commentary · Context [ms. 49]

Concatenation is a pushout with chosen endpoints

Grothendieck's reason for introducing types now becomes visible: the concatenation of words should encode the actual gluing of paths, including their transformations. An interval is a free category on an oriented linear graph. Gluing the last vertex of the first graph to the first vertex of the second produces the graph of the concatenated word, and the free-category construction preserves this pushout. Therefore

$$I_{\tau\tau'} = I_\tau \amalg_e I_{\tau'}$$

is a pushout in Cat with the specified endpoint maps. Any directed path between two vertices of one summand stays in that summand: entering the other would require crossing the gluing vertex twice in an underlying linear graph. Thus the inclusions are fully faithful as well as injective on objects.

The corresponding assertion about path categories is also literal. Restriction gives an isomorphism

$$\text{Fun}(I_{\tau\tau'}, X) \cong \text{Fun}(I_\tau, X) \times_X \text{Fun}(I_{\tau'}, X),$$

where the fibre product uses evaluation at the two endpoints being identified. On objects, two functors agree at the common vertex exactly when their two paths concatenate. On morphisms, two natural transformations glue exactly when their components at that vertex are equal. Naturality along each generating edge then gives naturality on every composite. This verifies the categorical pullback, rather than only the corresponding assertion about sets of paths.

The convention on the order of composition now has a concrete meaning. The word $\tau\tau'$ is read first through τ and then through τ', whereas the corresponding categorical composition is written $c' \circ c$. This harmless reversal of written order should be fixed before discussing fractions. It is not an opposite-category operation and does not reverse any transition.

The endpoint data are indispensable. The parenthetical assertion on manuscript page 50 that concatenation also gives a bifunctor on $\underline{\text{Typ}}$, as defined on page 30 using *all* functors, is false. The one-arrow categories I_+ and I_- are isomorphic in $\underline{\text{Typ}}$. A bifunctor whose object operation is word

concatenation would carry this isomorphism, together with id_{I_+}, to an isomorphism between I_{++} and I_{-+}. The former is a three-object chain; the latter has one initial object and two incomparable maximal objects. They are not isomorphic. Thus the claimed bifunctor cannot exist on that larger category. It does exist on endpoint-preserving functors, even without monotonicity, because their restrictions agree at the glued vertex. The final category $\underline{\mathrm{Ty}}$ satisfies this requirement and remains closed under concatenation. This correction isolates precisely the structure that the path formalism needs.

Cette composition est *strictement* associative et *strictement unitaire* (l'unité étant bien sûr l'élément unité 1 du monoïde libre Ty, correspondant à la catégorie ponctuelle $\{0\}$), et pas seulement 'à isomorphisme près'. Et pour cause, puisque dans $\underline{\mathrm{Ty}}$ les isomorphismes sont des identités. *Donc* $\underline{\mathrm{Ty}}$ *est un monoïde dans la catégorie* (Cat) des petites catégories!

Cela implique que non seulement $\mathrm{Ob}\,\underline{\mathrm{Ty}} = \mathrm{Ty}$, mais aussi $\mathrm{Fl}(\underline{\mathrm{Ty}})$, et plus généralement tous les ensembles de la forme $\mathrm{Hom}_{\mathrm{Cat}}(X, \underline{\mathrm{Ty}})$ sont des monoïdes, et ceci fonctoriellement en X, par exemple les composantes $\mathrm{Hom}_{\mathrm{Cat}}(\mathbf{\Delta}_i, \underline{\mathrm{Ty}})$ de l'ensemble semi-simplicial 'nerf' de $\underline{\mathrm{Ty}}$, [page 49′ (il y a un décalage dans la numérotation)]

$$\mathrm{Nerf}_i(\underline{\mathrm{Ty}}) = \mathrm{Hom}_{\mathrm{Cat}}(\mathbf{\Delta}_i, \underline{\mathrm{Ty}}),$$

sont des monoïdes, et les applications semi-simpliciales entre elles sont des homomorphismes de monoïdes : le nerf de $\underline{\mathrm{Ty}}$ est un *monoïde semi-simplicial.*

On va décrire ces monoïdes $\mathrm{Nerf}_i(\underline{\mathrm{Ty}})$, et constater que ce sont tous des monoïdes *libres,* tout comme $\mathrm{Nerf}_0(\underline{\mathrm{Ty}}) = \mathrm{Ob}\,\underline{\mathrm{Ty}} = \mathrm{Ty}$. Les applications semi-simpliciales entre ces monoïdes sont donc connus quand on les connaît sur les *générateurs* de ces monoïdes libres (lesquels générateurs sont aussi les éléments indécomposables, donc définis intrinsèquement en termes de la structure multiplicative). p. 34

Pour $\mathrm{Nerf}_0(\underline{\mathrm{Ty}}) = \mathrm{Ty}$, c'est une chose vue, les générateurs sont

$$\Delta_+, \Delta_-,$$

correspondant respectivement aux intervalles orientés

$$\begin{aligned} \mathbf{\Delta}_+ &: \quad 0 \quad \longrightarrow \quad 1 \\ \mathbf{\Delta}_- &: \quad 0 \quad \longleftarrow \quad 1. \end{aligned}$$

Pour $\mathrm{Nerf}_1(\underline{\mathrm{Ty}}) = \mathrm{Fl}(\underline{\mathrm{Ty}})$, i.e. pour les flèches, le résultat $3^o)$ du paragraphe précédent (détermination de $\mathrm{Fl}(\underline{\mathrm{Ty}})$) s'interprète aussi en disant que $\mathrm{Fl}(\underline{\mathrm{Ty}})$ est le monoïde libre engendré par un système de générateurs infini [page 50′]

$$\mathfrak{f}_{\varepsilon,n}\ , \qquad (\varepsilon, n) \in \{+1, -1\} \times \mathbf{N},$$

indexé par l'ensemble produit $\{+1, -1\} \times \mathbf{N}$. Ici $\mathfrak{f}_{\varepsilon,n}$ désigne l'unique flèche dans $\underline{\mathrm{Ty}}$

$$\mathfrak{f}_{\varepsilon,l} : \Delta_\varepsilon \longrightarrow \Delta_\varepsilon^l$$

(où bien sûr l'exposant l désigne une exponentation pour la loi de composition dans $\underline{\mathrm{Ty}}$

$$\Delta_\varepsilon^l = \underbrace{\Delta_\varepsilon \circ \Delta_\varepsilon \circ \ \cdots \ \circ \Delta_\varepsilon}_{l \text{ copies}},$$

et non la puissance cartésienne dans Cat, où dans $\underline{\mathrm{Ty}}$ (où elle n'existe que si $l \in \{0, 1\}$)). Dans la description de $\mathrm{Fl}(\underline{\mathrm{Ty}})$ dans le paragraphe précédent, on a écrit le 'terme général' du monoïde libre en groupant ensemble les groupes de foncteurs successifs de la forme $f_{+,0}$, $f_{-,0}$, i.e. corréspondant aux deux flèches de dégénérescence indécomposables (pour $*$)

$$\begin{cases} f_{+,0} : \quad \Delta_+ \quad \longrightarrow \quad e \\ f_{-,0} : \quad \Delta_- \quad \longrightarrow \quad e \end{cases}$$

([15]). Le sous-monoïde libre de Fl(Ty) engendré par ces deux éléments est le monoïde des flèches de dégénérescence (pure) quelconques dans Ty : p. 35 [page 51]

$$f_\tau : I_\tau \longrightarrow e \qquad (\tau \in \mathrm{Ty}),$$

et on a

$$f_\tau * f_{\tau'} = f_{\tau\tau'},$$

i.e. l'application

$$\tau \longmapsto f_\tau : \quad \underbrace{\mathrm{Ob}\,(\underline{\mathrm{Ty}})}_{=\mathrm{Ty}=\mathrm{Nerf}_0(\underline{\mathrm{Ty}})} \longrightarrow \underbrace{\mathrm{Fl}(\underline{\mathrm{Ty}})}_{\mathrm{Nerf}_1(\underline{\mathrm{Ty}})}$$

est un homomorphisme de monoïdes, transformant Δ_ε ($\varepsilon \in \{+1,-1\}$) dans $\mathfrak{f}_{\varepsilon,0}$. Ce n'est pas l'application de dégénérenscence de Nerf_0 dans Nerf_1, i.e. l'application $\tau \longmapsto \mathrm{id}_\tau$. Celle-ci est donnée par

$$\begin{array}{ccc} \mathrm{Ob}\,(\underline{\mathrm{Ty}}) & \longrightarrow & \mathrm{Fl}(\underline{\mathrm{Ty}}) \\ \tau & \longmapsto & \mathrm{id}_\tau \end{array} \quad : \quad \left| \begin{array}{lcll} & \Delta_\varepsilon & \longmapsto & \mathfrak{f}_{\varepsilon,1} \qquad (\varepsilon \in \{+1,-1\}) \\ \text{i.e.} & \Delta_+ & \longmapsto & \mathfrak{f}_{+,1} \\ & \Delta_- & \longmapsto & \mathfrak{f}_{-,1}\ . \end{array} \right.$$

Quant aux homomorphismes source et but

$$\mathrm{Fl}(\underline{\mathrm{Ty}}) \underset{\beta\ (\mathrm{but})}{\overset{\sigma\ (\mathrm{source})}{\rightrightarrows}} \mathrm{Ob}\,(\underline{\mathrm{Ty}}),$$

ils sont évidents sur les générateurs

$$\left\{ \begin{array}{rcl} \sigma(\mathfrak{f}_{\varepsilon,l}) & = & \Delta_\varepsilon \\ \beta(\mathfrak{f}_{\varepsilon,l}) & = & \Delta_\varepsilon^l. \end{array} \right.$$

Commentary · Later developments [ms. 50']

The type category is generated by two comonoids

The strict monoidal structure places Grothendieck's calculation beside a classical universal property of the simplicial category. In the single-sign case, endpoint-preserving maps of finite intervals are opposite to monotone maps of finite ordered edge sets. The latter category, with ordinal sum, contains the universal monoid; its opposite contains the universal counital coassociative comonoid. An explicit account, with the weak-composition coordinates used here, is Aguiar and Mahajan (2010, Appendix D.1–D.2, especially Proposition D.2, pp. 713–716), which traces the universal property to Mac Lane's Proposition VII.5.1. The two signs simply keep two independent copies of this structure in the same ordered word.

In Ty put

$$\delta_\varepsilon = \mathfrak{f}_{\varepsilon,2} : \Delta_\varepsilon \longrightarrow \Delta_\varepsilon^2, \qquad e_\varepsilon = \mathfrak{f}_{\varepsilon,0} : \Delta_\varepsilon \longrightarrow 1, \qquad \varepsilon \in \{+,-\}.$$

Both ways of applying δ_ε twice give the unique endpoint-preserving map from Δ_ε to Δ_ε^3. Applying e_ε to either output of δ_ε gives the identity, since its surviving weight is 1. Thus each Δ_ε is a counital coassociative comonoid. There is no specified relation between the positive and negative comonoids.

More precisely, a strict monoidal functor from Ty to a strict monoidal category $(\mathcal{C}, \otimes, \mathbf{1})$ is uniquely specified by a pair of counital coassociative comonoids C_+ and C_- in $\mathcal{C}$. Here is the construction, which also proves the assertion. For either comonoid, let $\delta^{(a)} : C \to C^{\otimes a}$ denote the iterated comultiplication, with $\delta^{(0)}$ its counit and $\delta^{(1)} = \mathrm{id}$. Coassociativity and the counit laws give, for every list $b_1, \ldots, b_a$ of nonnegative integers,

$$\big(\delta^{(b_1)} \otimes \cdots \otimes \delta^{(b_a)}\big)\delta^{(a)} = \delta^{(b_1+\cdots+b_a)}.$$

15. **NB** Le 'mot' vide de Fl(Ty) correspond à la flèche id_1, où 1 est l'élément unité de Ty, réalisant l''intervalle' ponctuel.

For completeness, the binary identity $(\delta^{(p)} \otimes \delta^{(q)})\delta^{(2)} = \delta^{(p+q)}$ follows by induction on positive p, q using coassociativity; the cases $p = 0$ or $q = 0$ are the counit identities, including $p = q = 0$. Applying this binary identity successively to the first $a - 1$ factors and the last factor proves the displayed formula by induction on a. The case $a = 0$ says that the identity of the unit composed with the counit is the counit.

Send the word $\varepsilon_1 \cdots \varepsilon_n$ to $C_{\varepsilon_1} \otimes \cdots \otimes C_{\varepsilon_n}$ and send its outgoing morphism of weights $(a_1, \ldots, a_n)$ to

$$\delta_{\varepsilon_1}^{(a_1)} \otimes \cdots \otimes \delta_{\varepsilon_n}^{(a_n)}.$$

The block-sum formula for composition and the preceding identity prove that this assignment is a functor. The all-one vector gives identities; the empty word gives the unit; concatenation of vectors gives the tensor product of maps. It is therefore strict monoidal. Conversely every outgoing morphism is the tensor product of its one-edge factors, and each such factor is an iterated comultiplication or counit. A strict monoidal functor must have the displayed values, establishing uniqueness.

This is a reinterpretation of the known ordinal/comonoid universal property, not a new homotopy theorem. It explains the two apparently different generators in the manuscript: $\mathfrak{f}_{\varepsilon,0}$ erases a transition, whereas $\mathfrak{f}_{\varepsilon,2}$ subdivides one. Their iterates produce every allowed reparametrization. No braiding is present: the order of the source transitions is part of the structure and cannot be permuted.

Je vais décrire maintenant $\mathrm{Nerf}_2(\underline{\mathrm{Ty}})$, i.e. les systèmes

$$\tau \xrightarrow{f} \tau' \xrightarrow{g} \tau''.$$

Se donner $f \in \mathrm{Fl}(\underline{\mathrm{Ty}})$ revient à se donner une suite ordonnée de générateurs $f_{\varepsilon,l}$, donc une suite [page 52]

$$\mathfrak{f}_{\varepsilon_1,l_1},\ \mathfrak{f}_{\varepsilon_2,l_2},\ \ \cdots\ \ ,\mathfrak{f}_{\varepsilon_n,l_n}$$
$$((\varepsilon_i, l_i))_{1\leq i\leq n} \in [\{\pm 1\} \times \mathbf{N}]^n.$$

On a alors

$$x = \Delta_{\varepsilon_1}\Delta_{\varepsilon_2}\cdots\Delta_{\varepsilon_n} \qquad \log x = n$$
$$y = \Delta_{\varepsilon_1}^{l_1}\Delta_{\varepsilon_2}^{l_2}\cdots\Delta_{\varepsilon_n}^{l_n} \qquad \log y = \textstyle\sum_{i\in[1,n]} l_i.$$

Se donner $q : y \longrightarrow z$, i.e. un objet de $y\backslash\underline{\mathrm{Ty}}$, revient à se donner une application de l'ensemble des $m = \sum l_i$ flèches de transition de y dans $\mathbf{N}$. En fait, on a p. 36

$$\mathrm{Tr}(y) = \mathrm{Tr}(\Delta_{\varepsilon_1}^{l_1}) \amalg \mathrm{Tr}(\Delta_{\varepsilon_2}^{l_2}) \amalg \cdots \amalg \mathrm{Tr}(\Delta_{\varepsilon_n}^{l_n})$$

('ensemble somme', i.e. réunion disjointe), donc

$$\mathrm{Hom}_{\mathrm{Ens}}(\mathrm{Tr}(y)) \simeq \prod_{i\in[1,n]} \mathrm{Hom}_{\mathrm{Ens}}(\underbrace{\mathrm{Tr}(\Delta_{\varepsilon_i}^{l_i})}_{[1,l_i]}, \mathbf{N}),$$

donc la donnée de g équivaut à la donnée d'un système de n applications

$$\begin{cases} \pi_i : [1, l_i] \longrightarrow \mathbf{N} & \text{(pondération de } \Delta_{\varepsilon_i}^{l_i}) \\ \text{pour } i \in [1, n]. \end{cases}$$

Mais pour $i \in [1, n]$, considérons l'élément de Nerf_2

$$\Delta_{\varepsilon_1} \xrightarrow{\mathfrak{f}_{\varepsilon_i,l_i}} \Delta_{\varepsilon_i}^{l_i} \xrightarrow{g_{\varepsilon,\pi_i}} \underbrace{\Delta_{\varepsilon_i}^{\pi_i(1)}\Delta_{\varepsilon_i}^{\pi_i(2)}\cdots\Delta_{\varepsilon_i}^{\pi_i(l_i)}}_{\Delta_{\varepsilon_i}^{\sum_j \pi_i(j) \overset{\text{déf}}{=} \log(\pi_i)}}, \tag{1}$$

où g_{ε_i,π_i} est la flèche définie en termes de π_i au paragraphe précédent. Mais ce n'est autre aussi que [page 53]

$$g_{\varepsilon_i,\pi_i} = \mathfrak{f}_{\varepsilon_i,\pi_i(1)} * \mathfrak{f}_{\varepsilon_i,\pi_i(2)} * \cdots * \mathfrak{f}_{\varepsilon_i,\pi_i(l_i)}. \tag{2}$$

L'élément de Nerf_2 envisagé ne dépend que de

$$(\varepsilon_i, l_i, \pi_i), \qquad \text{où } (\varepsilon_i, l_i) \in \{\pm 1\} \times \mathbf{N},\ \pi_i : [1, l_i] \longrightarrow \mathbf{N}.$$

(**NB** Si $l_i = 0$, $[1, l_i] = [1, 0] = \emptyset$ par convention, et π_i est l'unique application $\emptyset \longrightarrow \mathbf{N}$, donc la donnée $(\varepsilon_i, l_i, \pi_i)$ revient alors à (ε_i, l_i).)

Si

$$(\varepsilon, l, \pi) \ \text{ comme dessus}, \qquad (\varepsilon, l) \in \{\pm 1\} \times \mathbf{N},\ \pi : [1, l] \longrightarrow \mathbf{N},$$

désignons par

$$\mathfrak{f}_{\varepsilon,l,\pi}$$

l'élément de Nerf_2 donné par

$$\Delta_\varepsilon \xrightarrow{\mathfrak{f}_{\varepsilon,l}} \Delta_\varepsilon^l \xrightarrow{\mathfrak{f}_{\varepsilon,\pi(1)} \ * \cdots * \ \mathfrak{f}_{\varepsilon,\pi(l)}} \Delta_\varepsilon^{\pi(1)} \Delta_\varepsilon^{\pi(2)} \cdots \Delta_\varepsilon^{\pi(l)}.$$

On peut formuler la construction précédente en disant que tout élément de Nerf_2 peut se décomposer de façon unique en un produit

$$(f, g) = \mathfrak{f}_{\varepsilon_1,l_1,\pi_1} * \mathfrak{f}_{\varepsilon_2,l_2,\pi_2} * \cdots * \mathfrak{f}_{\varepsilon_n,l_n\pi_n},$$

ce qui prouve que $\mathrm{Nerf}_2(\underline{\mathrm{Ty}})$ est le monoïde libre sur les générateurs précédents $\mathfrak{f}_{\varepsilon,l,\pi}$. (**NB** à la suite vide correspond

$$1 \xrightarrow{\mathrm{id}_1} 1 \xrightarrow{\mathrm{id}_1} 1 \qquad .)$$

Si l'objet de Nerf_2 est

$$x \xrightarrow{f} y \xrightarrow{g} z,$$

p. 37
[page 54]

la construction précédente nous dit, en termes de la donnée combinatoire, i.e. de la suite de données

$$(\varepsilon_1, l_1, \pi_1),\ \ldots\ ,\ (\varepsilon_n, l_n, \pi_n),$$

quels sont x, y, z : chacun d'eux est le produit dans $\mathrm{Nerf}_0 = \mathrm{Ty}$ des n facteurs x_i, y_i ou z_i respectivement, pour $i \in [1, n]$, donnés dans la définition des $\mathfrak{f}_{\varepsilon_i,l_i,\pi_i}$, i.e. de

$$\Delta_{\varepsilon_i},\ \Delta_{\varepsilon_i}^{l_i},\ \text{ et } \Delta_{\varepsilon_i}^{\pi_i(1)} \Delta_{\varepsilon_i}^{\pi_i(2)} \cdots \Delta_{\varepsilon_i}^{\pi_i(l_i)} = \Delta_{\varepsilon_i}^{\mathrm{long}(\pi_i)}$$

(où $\mathrm{long}(\pi_i) = \sum_{j \in [1,l_i]} \pi_i(j)$). Donc

$$\begin{cases} x &= \prod_{i\in[1,n]}^{\mathrm{Ty}} \Delta_{\varepsilon_i} \\ y &= \prod_{i\in[1,n]}^{\mathrm{Ty}} \Delta_{\varepsilon_i}^{l_i} \\ z &= \prod_{i\in[1,n]}^{\mathrm{Ty}} \Delta_{\varepsilon_i}^{\mathrm{long}(\pi_i)} \end{cases}$$

([16]), et on trouve de même f, g comme des produits (au sens $*$) de n facteurs f_i resp. g_i ($i \in [1, n]$) explicités dans les formules (1), (2) (p. 52, 53) :

$$\begin{aligned} f_i &= \mathfrak{f}_{\varepsilon_i,l_i} : \ \Delta_{\varepsilon_i} \longrightarrow \Delta_{\varepsilon_i}^{l_i} \\ g_i &= g_{\varepsilon_i,l_i} : \ \Delta_{\varepsilon_i}^{l_i} \longrightarrow \Delta_{\varepsilon_i}^{\mathrm{long}(\pi_i)}, \end{aligned}$$

g_i lui-même étant représenté comme un $*$-produit de l_i facteurs indécomposables, de sorte que f et g apparaissent sous forme de $*$-produits de n, resp. de $\sum_{i\in[1,n]} l_i$ facteurs de la forme $\mathfrak{f}_{\varepsilon,\nu}$. Pour terminer de tenir en mains l'objet semi-simplicial tronqué $(\mathrm{Nerf}_i(\underline{\mathrm{Ty}}))_{0\le i\le 2}$, il reste à

[page 55]

16. attention, les produits doivent être pris en respectant l'ordre des facteurs donné par la notation indicielle en i.

décrire les deux applications de dégénérescence $\mathrm{Nerf}_1 \longrightarrow \mathrm{Nerf}_2$ (cas où f ou g est une flèche identique) et surtout l'application de *composition* des flèches

$$\mathrm{Nerf}_2(\underline{\mathrm{Ty}}) \xrightarrow{\gamma} \mathrm{Nerf}_1(\underline{\mathrm{Ty}}) = \mathrm{Fl}(\underline{\mathrm{Ty}}).$$

Il suffit de donner γ sur les générateurs, et on a tautologiquement, par définition des $\mathfrak{f}_{\varepsilon,l,\pi}$

$$\gamma(\mathfrak{f}_{\varepsilon,l,\pi}) = \mathfrak{f}_{\varepsilon,\mathrm{long}(\pi)}.$$

(**NB** Il n'existe au plus qu'un seul morphisme de Δ_ε (de Δ_+ ou de Δ_-) dans un intervalle I_τ. Il existe si et seulement si τ est de la forme Δ^l_ε, $l \in \mathbf{N}$.)

Commentary · Exposition [ms. 53]

Free monoids in degrees one and two

Grothendieck uses the nerve to keep composition together with concatenation. A string of composable reparametrizations can still be cut along the original source edges. The resulting factors are independent, and this is the reason the monoid in each nerve degree is free.

In degree one, a morphism with source word $\varepsilon_1 \cdots \varepsilon_n$ and weights $(a_1, \ldots, a_n)$ has the unique decomposition

$$f = \mathfrak{f}_{\varepsilon_1,a_1} * \cdots * \mathfrak{f}_{\varepsilon_n,a_n}.$$

The factors are indexed by source edges, even when their weights vanish. A generator $\mathfrak{f}_{\varepsilon,0}$ is therefore not the unit of this free monoid: its source has one edge. The monoid unit is the empty factorization, namely the identity of the point. Nor is $\mathfrak{f}_{\varepsilon,1}$ the monoid unit; it is the identity of a one-edge object and is indecomposable under concatenation. This separates the categorical identity operation from the unit of the tensor product.

Now consider $\tau \xrightarrow{f} \tau' \xrightarrow{g} \tau''$. The ith original edge produces an ordered block of a_i edges after f. Let $\pi_i : [1, a_i] \to \mathbb{N}$ record the weights that g assigns to that block. All descendants have the original sign ε_i. Consequently the pair has the unique factorization

$$(f, g) = \prod_{i=1}^{n,*} \mathfrak{f}_{\varepsilon_i,a_i,\pi_i}.$$

Existence comes from cutting every intermediate word into its descendant blocks; uniqueness comes from the fixed order of the original edges and the unique weight vector of each restriction. The nullary case $a_i = 0$ has one empty function π_i, so it contributes a legitimate generator whose two targets are the point. For $n = 0$ the whole pair is the identity string at the point and is the monoid unit. This proves the claimed freeness in degree two without discarding extinct blocks.

Writing $N_r = N_r(\underline{\mathrm{Ty}})$, the three faces of a degree-two generator are

$$\begin{aligned} d_2(\mathfrak{f}_{\varepsilon,a,\pi}) &= \mathfrak{f}_{\varepsilon,a}, \\ d_1(\mathfrak{f}_{\varepsilon,a,\pi}) &= \mathfrak{f}_{\varepsilon,\sum_{j=1}^{a} \pi(j)}, \\ d_0(\mathfrak{f}_{\varepsilon,a,\pi}) &= \mathfrak{f}_{\varepsilon,\pi(1)} * \cdots * \mathfrak{f}_{\varepsilon,\pi(a)}. \end{aligned}$$

The last line is the empty product when $a = 0$. The two degeneracies of a degree-one generator insert the identity before or after its map:

$$s_0(\mathfrak{f}_{\varepsilon,a}) = \mathfrak{f}_{\varepsilon,1,(a)}, \qquad s_1(\mathfrak{f}_{\varepsilon,a}) = \mathfrak{f}_{\varepsilon,a,1_a}.$$

Here (a) is the one-entry weight vector and 1_a the all-one vector of length a. These formulas recover precisely the composition and degeneracies described on manuscript pages 54–56, while displaying the other two faces as well.

The word "semi-simplicial" in this edition includes degeneracies; in current terminology the nerve is a simplicial monoid. The faces and degeneracies are monoid homomorphisms because concatenation is a bifunctor and is strictly associative and unital. This statement concerns the whole degreewise free monoid. Its distinguished basis need not itself be a simplicial set: d_0 can send one generator to a product of several generators, or to the unit.

p. 38

Le cas $g = \mathrm{id}$ correspond au cas où

$$\pi_i(j) = 1 \qquad \forall j \in [1, l_i],$$

les $\mathfrak{f}_{\varepsilon_i, l_i, \pi_i}$ sont alors détérminés en fonction des $\mathfrak{f}_{\varepsilon_i, l_i}$, et l'application de dégénérescence est

$$\mathfrak{f}_{\varepsilon,l} \longmapsto \mathfrak{f}_{\varepsilon,l,1_l} \qquad (\text{où } 1_l \text{ est la fonction constante de valeur 1 sur } [1,l]).$$

Le $f = \mathrm{id}$ est celui où les l_i sont égaux à 1. Alors chaque $\pi_i : [1,1] \longrightarrow \mathbf{N}$ peut être vu comme un élément de $\mathbf{N}$, la donnée de $(\varepsilon_i, l_i, \pi_i)$ revient à celle des $(\varepsilon_i, n_i) \in \{\pm 1\} \times \mathbf{N}$, i.e. la donnée d'une suite de n générateurs parmi les $\mathfrak{f}_{\varepsilon,l}$. L'application de dégénérescence $\mathrm{Nerf}_1 \longrightarrow \mathrm{Nerf}_2$ est ici donnée sur les générateurs par [page 56]

$$\mathfrak{f}_{\varepsilon,l} \longmapsto \mathfrak{f}_{\varepsilon,1,l}.$$

Heureusement, on n'a plus à prouver l'associativité de la composition.

NB La simplicité de ces descriptions renforce mon sentiment que la catégorie $\underline{\mathrm{Ty}}$ définie à présent, avec les *morphismes d'intervalles orientés croissants* $I_\tau \longrightarrow I_{\tau'}$ comme flèches de $\underline{\mathrm{Ty}}$, est bien 'la bonne'. On va déjà voir si elle donne une description sympa des catégories de fractions $X\Sigma^{-1}$, sans tortillements comme précédemment.

Mais on a promis de déterminer $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$, pour tout $r \in \mathbf{N}$. Le générateur général de Nerf_r est de la forme (pour $\varepsilon \in \{\pm 1\}$ fixé)

$$\underbrace{\Delta_\varepsilon}_{=\Delta_\varepsilon^{l_0} \text{avec } l_0=1} \xrightarrow{f_1} \Delta_\varepsilon^{l_1} \xrightarrow{f_2} \Delta_\varepsilon^{l_2} \longrightarrow \cdots \xrightarrow{f_r} \Delta_\varepsilon^{l_r},$$

où les $f_i : \Delta_\varepsilon^{l_{i-1}} \longrightarrow \Delta_\varepsilon^{l_i}$, $i \in [1,r]$, sont donnés par des applications

$$\pi_{i-1} : [1, l_{i-1}] \longrightarrow \mathbf{N} \qquad \text{telles que} \quad \underbrace{\mathrm{long}(\pi_{i-1})}_{\overset{\text{déf}}{=} \Sigma_{j \in [1,l_i]} \pi_i(j)} = l_i.$$

(NB $\pi_0 : [1,1] = \{1\} \longrightarrow \mathbf{N}$ est identifié à $\mathrm{long}(\pi_0) = \pi_0(1) = l_1$.)

Pour ne pas faire dépendre la définition du domaine des π_i ($i \in [0, r-1]$) des entiers l_i, on peut aussi considérer les π_i comme des applications [page 57]

$$\mathbf{N}^* \xrightarrow{\pi} \mathbf{N} \qquad (\mathbf{N}^* = \{i \in \mathbf{N} \mid i \geq 1\})$$

à supports finis (i.e. tels que $\pi(i) = 0$ pour i grand), et exiger de façon précise

$$(**) \qquad \begin{aligned} &\mathrm{supp}\,\pi_0 \subseteq \{1\} \\ &\mathrm{supp}\,\pi_1 \subseteq [1, \mathrm{long}(\pi_0)] \\ &\qquad \cdots \\ &\boxed{\mathrm{supp}\,\pi_i \subseteq [1, \mathrm{long}(\pi_{i-1})]}, \qquad \text{pour } \boxed{i \in [0, r-1]} \quad . \end{aligned}$$

Commentary · Exposition [ms. 57]

All nerve degrees as ordered forests

The final assertion of manuscript page 57 is that the same decomposition works for arbitrarily long strings of reparametrizations. It is useful to complete the argument now, since it explains both the support conditions at the bottom of the page and the one face map that behaves differently from the others.

Fix $r \geq 0$. Call an r-step string *rooted* when its initial type has one transition. Every intermediate type is then a power of the same sign, and the string has the form

$$\Delta_\varepsilon^{l_0} \longrightarrow \Delta_\varepsilon^{l_1} \longrightarrow \cdots \longrightarrow \Delta_\varepsilon^{l_r}, \qquad l_0 = 1.$$

For $0 \le a < r$ its ath weight vector is

$$\pi_a : [1, l_a] \longrightarrow \mathbb{N}, \qquad l_{a+1} = \sum_{j=1}^{l_a} \pi_a(j).$$

One can regard this as an ordered rooted tree with r levels of edges: a transition at level a has $\pi_a(j)$ ordered descendants at level $a+1$. A vertex may have no descendants. Once a whole level is empty, all subsequent levels are empty because there is no morphism from the point to a nonpoint type.

For a general string, let its initial word have transitions $1, \dots, n$. At the first stage the descendants of each original transition form a consecutive monochromatic block. At the next stage the descendants of all transitions in one such block again form a consecutive block: concatenate their individual image blocks, allowing zero lengths. Induction gives such a block at every level. Restricting all maps to the descendants of a fixed original edge produces a rooted string, and concatenating the n rooted strings reconstructs the original string. Each boundary is determined by cumulative weights, so this factorization is unique. A rooted string cannot be a product of two nonunit strings, since the lengths of their initial words would be positive integers summing to one. Conversely a string whose initial word has at least two edges splits after its first edge. Thus rooted strings are exactly the indecomposables, and

$$N_r(\underline{\mathrm{Ty}}) \cong \text{the free monoid on rooted } r\text{-step strings.}$$

For $r = 0$ there are exactly two generators, the two signs. For an empty initial word there is only the all-identity string at the point, which is the monoid unit. These boundary cases establish the assertion for every r.

The simplicial operators can now be read directly on these forests. An inner face removes one level and replaces the two successive weight vectors by the sums over consecutive descendant blocks. The last face forgets the last level. A degeneracy repeats a level, inserting weight 1 at each of its transitions. These operations preserve a single root. The first face removes the original root level: a rooted string with l_1 first-generation transitions becomes the concatenation of their l_1 rooted descendant strings. This is the higher-dimensional version of the formula for d_0 above. Since these are exactly restriction and composition of actual functors $[r] \to \underline{\mathrm{Ty}}$, they satisfy all simplicial identities; the unique forest decomposition identifies their formulas with those categorical operators.

For $r > 0$, to use fixed-domain functions as on manuscript page 57, extend each π_a by zero to $\mathbb{N}^*$. The exact conditions are

$$\mathrm{supp}(\pi_0) \subseteq \{1\}, \qquad \mathrm{supp}(\pi_a) \subseteq [1, \mathrm{long}(\pi_{a-1})] \quad (1 \le a < r).$$

The source's final boxed range starts at $a = 0$, although π_{-1} is undefined; it must begin at 1. The preceding display defining $\mathrm{long}(\pi_{i-1})$ likewise needs the indices $j \in [1, l_{i-1}]$ and $\pi_{i-1}(j)$. These printed indexing slips do not affect the argument once the successive domains are kept in view.

One should not infer from this result that the simplicial monoid is freely generated by a simplicial set of indecomposables: the first-face calculation rules that out for this particular basis. Nor does the algebraic freeness make its classifying space complicated. The point is terminal in $\underline{\mathrm{Ty}}$, since every weight vector admits the unique further collapse to zero. Appending that terminal object gives the usual contraction of the nerve. The free presentations describe how all reparametrizations fit together, while this terminal object describes their ordinary homotopy type.

Commentary · Present state [ms. 57]

What the shape calculus does for localization

The free monoids just constructed organize subdivision and concatenation. To obtain mapping spaces, a second ingredient is needed: comparisons of labelled paths must carry the specified weak-equivalence information. For a relative category (C, W) and fixed x, y, let $Z_\tau(C, W)_{x,y}$ be the category of paths of type τ from x to y, with backward arrows in W, and natural transformations whose components lie in W and whose endpoint components are identities. Precomposition makes these categories contravariant in $\underline{\mathrm{Ty}}$. Their nerves, or their groupoid completions, retain comparisons that the set of path components alone would discard.

The analogous construction for relative infinity-categories is developed by Mazel-Gee (2018, Definitions 3.9 and 3.17). Under a homotopical three-arrow calculus, his Theorem 4.4 identifies the geometric realizations of its hammock mapping objects with the groupoid completions of three-arrow zigzag categories. This is an additional hypothesis, not a consequence of the finite normal form in the manuscript. Independently, Mazel-Gee (2019, Theorem 3.8) proves that completing the relative Rezk nerve presents localization. That universal property supplies the criterion with which a proposed path model must be compared.

For the general mapping-space comparison, Arakawa and Cnossen (2026, Theorem 1.9 and Corollary 1.11) prove in their preprint that

$$\mathrm{Map}_{C[W^{-1}]}(x,y) \simeq \operatorname*{colim}_{\tau \in \underline{\mathrm{Ty}}^{\mathrm{op}}} \left| Z_\tau(C,W)_{x,y} \right|,$$

where the colimit is taken in spaces and $|-|$ is groupoid completion. The displayed formula uses the ordinary relative categories just defined; the passage from their strict endpoint fibres to the invariant formula is supplied by Convention 4.2, Definition 4.3, Theorem 4.7, and Proposition 4.8(b) of the same paper. The general infinity-categorical statement uses the corresponding infinity-categories of relative diagrams and their endpoint fibres. Their Corollary 5.7 supplies the localization identification for Mazel-Gee's hammock construction. These references distinguish the established Rezk-nerve theorem from the more recent explicit comparison.

The comparison makes the scope of the present pages clear. They provide the shapes, their functorial changes, and strict concatenation. They do not yet choose W or impose the conditions on vertical arrows used above. No new open conjecture is required to finish their finite combinatorics. The subsequent localization construction still has to be checked with its own morphisms and universal property; the contractibility of an interval, by itself, does not perform that check.

Overview · VII-a

Chapter VII, manuscript pages 1–57: the shapes of paths.

The opening of Chapter VII is a restart. Grothendieck has found the earlier path formalism too cumbersome to serve as a foundation for localization, and he now separates a path from the finite combinatorial object on which it is parametrized. The distinction gives these pages their unity. First one understands intervals independently of any coefficient category; next one regards every operation on paths as a change of interval; finally one assembles all types and makes their concatenation strictly functorial. The later theory of fractions is a test of the construction, not yet a theorem established in these pages. The commentary on localization and path completion in Chapter VI, especially at manuscript pages 64–70, supplies the background; the present chapter changes the finite apparatus on which such arguments are to rest.

An interval of length n is the free category on an oriented path with vertices $0, \dots, n$. Its signature is a word $\tau = (\varepsilon_1, \dots, \varepsilon_n)$ in two signs. There are two orders to remember. The progression order is the linear order of the vertices; the categorical order records the existence of arrows. A path proceeds from vertex 0 to vertex n even when some of its arrows point backwards. Reversing this progression order changes the signature to $(-\varepsilon_n, \dots, -\varepsilon_1)$, whereas taking the opposite category changes it to $(-\varepsilon_1, \dots, -\varepsilon_n)$. Their distinction is already needed to understand why a span is symmetric as an interval but is not isomorphic to its opposite cospan.

The maximal constant runs of the signature are the structural chains. Their endpoints are the critical vertices; retaining just these vertices gives the reduced interval. These notions organize all full subcategories of an interval. A nonempty full subcategory has interval components, and a subset with extreme vertices $i < j$ is connected precisely when it retains every critical vertex between them. The proof is elementary and decisive: two consecutive retained vertices can be joined by an arrow exactly when no change of sign has been omitted between them. Thus reduction remembers the turns while suppressing subdivisions inside a composable run. It does not mean that arbitrary diagrams on the interval may forget the intermediate objects.

Several literal statements near the beginning need repair. Proposition 1(b) asserts that every endpoint-fixing endofunctor is the identity. On [2], the function $(0, 1, 2) \mapsto (0, 0, 2)$ is a counterexample. The proof on the following pages establishes rigidity of endpoint-fixing *isomorphisms*, which is sufficient for the classification of orientations. In fact, the stronger endofunctor assertion holds exactly for reduced intervals. Proposition 3's deletion criterion also needs the vertices incomparable with the deleted vertex to be separated from its incoming and outgoing links. Finally, an interval-shaped subcategory need not be full. All these readings occur in the edition and remain in the source text. The accompanying blocks give counterexamples and prove the repaired assertions, so that subsequent correct arguments do not inherit an unnecessary uncertainty.

Reduction has a useful additional interpretation. For the full inclusion $j : I_{\mathrm{red}} \hookrightarrow I$, send an interior vertex of each run to that run's categorical target or categorical source, and fix all critical vertices. The resulting maps satisfy

$$p_{\mathrm{target}} \dashv j \dashv p_{\mathrm{source}}.$$

The commentary proves these adjunctions by comparing the relevant hom-sets. They give categorical deformation retractions and explain exactly why reduction is harmless for groupoid-valued paths. For a general category X, restriction $X^I \to X^{I_{\mathrm{red}}}$ need not be an equivalence. This is the first place where the geometry of the shape and the information in a labelled diagram must be kept distinct.

Once the shape is fixed, a path of type τ in X is a functor $I_\tau \to X$, and its morphisms are natural transformations. The category is X^{I_τ}; its set of objects is the represented functor

$\mathrm{Hom}_{\mathrm{Cat}}(I_\tau, X)$. Yoneda then says that every operation on paths natural in the coefficient category is precomposition by a unique functor between their shapes, in the reverse direction. The proof evaluates the operation on the identity path in I_τ. This explains both composition of successive arrows and insertion of identities, and supplies their action on morphisms without an additional coherence choice. The manuscript's distinction between the set and the category of paths must therefore survive the typography.

The category of types changes twice while Grothendieck tests it. Initially it permits all functors between intervals; then it retains functors preserving the two endpoints; finally it also requires their vertex functions to be nondecreasing in the progression order. Only the first version is fully faithful in Cat. The second is fully faithful in the category of categories under the discrete two-point category. The final version, denoted $\underline{\mathrm{Ty}}$, is generally not full there either. These are successive definitions, not contradictory assertions about a fixed category. The final restriction preserves directed progress through a path and excludes changes of parameter which backtrack.

For any chosen version T, the strict contravariant functor $\tau \mapsto X^{I_\tau}$ gives a split Grothendieck fibration over T. A morphism from (τ', c') to (τ, c) is a pair

$$\varphi : \tau' \to \tau, \qquad \upsilon : c' \Rightarrow c\varphi.$$

Composition is whiskering followed by composition of transformations. This is the machinery of Grothendieck (1971a, Exposé VI, §§8–10), specialized to restriction along finite shapes. Endpoint-preserving shape maps make evaluation at the two ends into functors on the total category. The commentary also determines the maximal enlargement for which natural endpoint evaluations can still exist, allowing comparison arrows at the ends. That enlargement illustrates why the subsequent requirement of strict gluing is stronger. No colimits in X are needed for the fibration; extending paths in the opposite direction would require Kan extensions, and already a collapsed span asks for a pushout.

The main calculation of the second half has a particularly short modern form. Give the transition set $\mathrm{Tr}(\tau) = \{1, \ldots, n\}$ its order and its signs. If $f : I_\tau \to I_\upsilon$ is a morphism of the final type category, every target transition belongs to the image interval of exactly one source transition. This defines a monotone, sign-preserving map

$$f^* : \mathrm{Tr}(\upsilon) \longrightarrow \mathrm{Tr}(\tau).$$

Conversely, given such a map h, recover the vertex function by

$$f(i) = \#\{j : h(j) \leq i\}.$$

These assignments are inverse and reverse composition. Thus the type category is opposite to the category of finite signed linear orders with monotone sign-preserving maps. Its many normal forms are descriptions of this one correspondence.

For a reduced target, f^* is injective, so a map is specified by a signed subsequence of the source word. In general, a map out of a fixed source is specified by nonnegative integer weights: an edge of sign ε and weight a becomes a run of a edges of sign ε. Weight zero collapses that edge. A map into a fixed target is instead described by its image vertices and by the signatures of the fibers over them. The latter description must use cumulative sums for the target coordinates, and collapsed fibers are components of the graph of collapsed edges, not components of the union of their endpoint vertices with every ambient adjacency restored. The point target is treated separately. These corrections preserve the two classification results.

Concatenation glues the final vertex of one interval to the initial vertex of the next. On standard words and endpoint-preserving maps it is strictly associative and unital, and makes $\underline{\mathrm{Ty}}$ a strict monoidal category. It cannot extend with this object operation to the category allowing *all* interval functors: there $+$ and $-$ are isomorphic, whereas $++$ and $-+$ are not. The

restriction on morphisms is therefore essential to the source's own construction. Through the signed-edge description, Ty is the free strict monoidal category on two counital coassociative comonoids. The one-generator background is the opposite of the augmented simplex category; see Aguiar and Mahajan (2010, Appendix D, Proposition D.2). The two-generator version follows by ordered concatenation, without a symmetry interchanging the letters.

The nerve calculation is now transparent. A chain of r composable type maps splits uniquely according to the edges of its initial word. Each factor begins at a single signed edge; its later stages are successive ordered expansions of that edge, allowing complete disappearance. Consequently every $N_r(\mathrm{Ty})$ is a free monoid. The face deleting the initial object can send a generator to several generators, or to the empty word, so this assertion does not identify the nerve with the free simplicial monoid on its degreewise generators. The commentary gives every face and degeneracy by the corresponding operation on ordered forests and resolves the final support-indexing slip.

The later comparison can be made at the level of the indexing category itself. Marking the negative transitions identifies Ty with the zigzag type category in Mazel-Gee (2018, Definition 3.5). The closing status block distinguishes this finite identification from the additional construction of localization mapping spaces, including the recent comparison in Arakawa and Cnossen (2026, Theorem 1.9 and Corollary 1.11). The development blocks prove explicit consequences left unworked here: the reduction adjunctions and sharp rigidity criterion, the endpoint evaluation criterion, and a criterion with a counting formula for sections of contractions. Their markers record our proofs and limited literature searches, not a claim of priority. The unresolved task at manuscript page 57 is the promised construction of fractions with its precise morphisms; the finite combinatorics considered in these pages has been settled in the commentary.

D'autre part, posant $l_i = \mathrm{long}(\pi_{i-1})$ (pour $i \in [1, r]$), p. 39

$$f_i : \Delta_\varepsilon^{l_{i-1}} \longrightarrow \Delta_\varepsilon^{l_i \ (=\mathrm{long}(\pi_{i-1}))}$$

est défini comme

$$f_i = \mathfrak{f}_{\varepsilon,\pi_{i-1}(1)} * \mathfrak{f}_{\varepsilon,\pi_{i-1}(2)} * \cdots * \mathfrak{f}_{\varepsilon,\pi_{i-1}(l_{i-1})},$$

de sorte que l'élément $(*)$ de $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$ est déterminé de façon unique par la donnée des $\varepsilon \in \{\pm 1\}$ et de

$$\pi_\bullet = (\pi_i)_{i\in[0,r-1]} \in (\mathrm{Hom}_{\mathrm{Ens}}(\mathbf{N}^*, \mathbf{N}))^r$$

satisfaisant les conditions $(**)$. Soit $F_{\varepsilon,\pi_\bullet}$ cet élément de $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$, et soit N_r l'ensemble des $\pi_\bullet$ qu'on vient d'expliciter. Alors

$$(F_{\varepsilon,\pi_\bullet})_{(\varepsilon,\pi_\bullet)\in\{\pm1\}\times N_r}$$

est un système de générateurs libre du monoïde $\mathrm{Nerf}_r(\underline{\mathrm{Ty}})$.

Si $l_i = 0$ (i.e. $\boxed{\pi_{i-1} = 0}$), on doit avoir $\mathrm{supp}(\pi_{i+1}) = \emptyset$, i.e. $\pi_{i+1} = 0$, d'où $l_{i+1} = 0$, et de même pour les l_j et π_j suivants. Géométriquement l'hypothèse s'exprime par la condition que le but de f_i est l'élément 1 de Ty (correspondant à l'intervalle ponctuel), et comme $\mathrm{Hom}(1, \tau) = \emptyset$ pour $\tau \neq 1$, cela signifie que les flèches qui suivent f_i doivent être toutes égales à la flèche $\mathrm{id}_1 : 1 \longrightarrow 1$. [page 58]

Commentary · Exposition [ms. 57]

Degreewise free is a precise assertion

Grothendieck has now made the combinatorics of a changing path shape completely explicit. The commentary at manuscript pages 53–57 already explains the free generators in every degree and counts refinement stages. One additional distinction helps us use that calculation correctly: freeness in each degree does not mean that the intrinsic generators themselves form a simplicial set.

To see the distinction, take the generator of degree two represented by

$$\Delta_+ \longrightarrow \Delta^2_+ \xrightarrow{\mathrm{id}} \Delta^2_+.$$

Its first face, which forgets the initial object and first arrow, is $\mathrm{id}_{\Delta^2_+} = f_{+,1} * f_{+,1}$. This is a product of two nonunit elements in the degree-one monoid. It is therefore not a generator. In the notation of the text, the general formula is

$$d_0(f_{\varepsilon,l,\pi}) = f_{\varepsilon,\pi(1)} * \cdots * f_{\varepsilon,\pi(l)},$$

where the empty product for $l = 0$ is the unit. An isomorphism between free monoids preserves their intrinsic indecomposables. Consequently $N_\bullet\underline{\mathrm{Ty}}$ cannot be the degreewise free monoid on a simplicial set: such a construction would send each generator under every face to a generator. The monoid structure maps are determined by generators, as Grothendieck says; their images can be arbitrary words.

There is also a useful boundary case in his finite-support description. The condition on π_0 is separate, namely $\mathrm{supp}\,\pi_0 \subseteq \{1\}$. The recursive condition must then run over $1 \leq i \leq r-1$, since π_{-1} has not been defined. If the length at stage i is zero, the next weighting is $\pi_i = 0$, and every later stage remains the empty interval. The edition instead writes π_{i+1} at this step. Both index slips are retained in the source and registered in the corrections table. They change neither the free decomposition nor its geometric meaning: a path shape which has disappeared cannot reappear under an endpoint-preserving interval map.

Remarque. Dans la description combinatoire de $\underline{\mathrm{Ty}}$ et de $\mathrm{Nerf}_\bullet(\underline{\mathrm{Ty}})$, l'ensemble des 'signatures' $\{\pm 1\}$ joue un rôle assez bidon. Il peut être remplacé par n'importe quel ensemble Σ (dit 'ensemble des signatures'), et on construit alors une catégorie monoïdale (ou un

monoïde catégorique) ayant comme ensemble d'objets le monoïde libre engendré par Σ, interprétés comme des intervalles d'entiers $[0, n]$, munis de 'signatures' dans Σ pour chaque intervalle de transition, ou ce qui revient au même, des 'graphes' orientés Σ-pondérés de type combinatoire un intervalle $\mathbf{\Delta}_i$, ce qui revient à la donnée d'une suite (ordonnée) de n éléments $\varepsilon_1,\ \varepsilon_2,\ \dots\ ,\ \varepsilon_n \in \Sigma$, i.e. d'un 'mot' $\varepsilon_1\varepsilon_2 \dots \varepsilon_n$, qu'on préfère écrire $\Delta_{\varepsilon_1}\Delta_{\varepsilon_2}\cdots\Delta_{\varepsilon_n}$. On peut définir de façon évidente les homomorphismes *croissants* de tels graphes pondérés orientés (sous-entendu : avec l'élément source et l'élément but dans la structure, à respecter par les morphismes), sans avoir besoin d'une interprétation catégorique. La fonctorialité de l'opération de composition des intervalles pondérés, pour cette notion de morphisme, est évidente. Le choix $\Sigma = \{\pm 1\}$ est à ma connaissance le seul utile, à cause de son lien avec la notion de chemins et le calcul de π_0 d'un objet de X `[plutôt d'un objet` X `de` Cat`]`, etc. [page 59]

Commentary · Exposition [ms. 59]

The localization test

Grothendieck is about to test his category of types against a construction whose universal property is already settled: the category obtained by formally inverting specified arrows. This is a demanding first test because literal concatenation gives too much information. It remembers the length of a zigzag, every inserted identity, and every subdivision of a composite. A localization must identify presentations which differ in these ways, while retaining their common source and target.

Let $P(X)$ denote the category whose arrows are literal finite zigzags in X. Length is additive under composition. Thus an invertible arrow of $P(X)$ must have length zero. Moreover the assignment $u \mapsto c(u)$ from arrows of X to one-edge paths does not preserve composition: $c(vu)$ has one edge and $c(v)c(u)$ has two. It does not preserve identities either, since $c(\mathrm{id}_a)$ has length one. These observations concern the literal path construction uniformly in X. For particular categories there can of course be useful functors in one direction: if X is a groupoid, every zigzag has an effect in X, and evaluation is a functor $P(X) \to X$.

The category $\underline{\mathrm{Ch}}(a, b)$ supplies the missing identifications as morphisms. A type map replaces a string of composable arrows by its composite or inserts identity arrows through pullback. A natural transformation compares two paths of the same type by a ladder of commuting squares. Since the endpoints are fixed, the components at a and b must be identities. Passing to π_0 identifies objects joined by a finite zigzag of these morphisms. A single directed natural transformation is enough for this purpose; it need not be invertible.

The desired category therefore has

$$\mathrm{Hom}(a, b) = \pi_0\underline{\mathrm{Ch}}(a, b).$$

Its composition comes from concatenation before taking components. This is legitimate because $\pi_0(A \times B) = \pi_0(A) \times \pi_0(B)$ for categories: a zigzag in a product projects to the factors, while zigzags in the factors can be traversed one coordinate at a time. Associativity and units are then inherited, rather than imposed as further equivalence relations.

The relative version allows only a chosen class Σ of arrows to be traversed backwards. Its aim is the formula

$$\pi_0\underline{\mathrm{Ch}}_\Sigma(a, b) \cong \mathrm{Hom}_{X[\Sigma^{-1}]}(a, b).$$

At this stage the assertion concerns ordinary hom-sets. The homotopy type of the whole path category is a separate, stronger question, and the later pages of the chapter turn precisely to the additional choices and coherence needed for it.

p. 40

7 Explicitation des catégories de fractions $X\Sigma^{-1}$ en termes des $\underline{\mathrm{Ch}}(X; a, b)$

Si X est une catégorie, on désigne par $\underline{\mathrm{Ch}}(X)$ la catégorie des chemins dans X (cf. §4), et pour $a, b \in \mathrm{Ob}\, X$, par

$$\underline{\mathrm{Ch}}(X; a, b) \qquad \text{ou simplement } \underline{\mathrm{Ch}}(a, b)$$

celle des chemins de a vers b, i.e. de source a, but b. L'intuition géométrique (qu'on explicitera), c'est qu'un tel chemin c définit un chemin 'direct' de a vers b (i.e. un chemin de type τ' 'direct', i.e. toutes les flèches de transition sont directes, i.e. τ' de la forme Δ_+^n), de même longueur que le chemin initial c, obtenu en inversant toutes les flèches rétrogrades dans le chemin c. Cette opération est effectivement possible dans X lui-même si et seulement si lesdites flèches sont inversibles. On obtient alors une suite composable de flèches, dont le composé total (qu'on peut appeler 'l'effet utile' du chemin) est une flèche de a vers b. Dans le cas général, les chemins de a vers b peuvent être vus comme une sorte de 'flèches généralisées' allant de a vers b, dont la composition serait la composition évidente, et les identités les 'chemins vides' relatifs aux $a \in \mathrm{Ob}\, X$. On trouve d'ailleurs bien ainsi une catégorie, ayant mêmes objets que X, mais où les [page 60]

$$\mathrm{Hom}_{\mathrm{Ch}}(a,b) = \mathrm{Ch}(a,b) = \mathrm{Ob}\,\underline{\mathrm{Ch}}(a,b)$$

sont prohibitivement gros. Cela s'exprime notamment par le fait que les seules flèches inversibles dans X_{Ch} sont les identités ! On n'a pas de foncteur naturel entre X_{Ch} et X qui soit l'identité sur les objets. Car à un chemin $a \xrightarrow{c} b$ on ne peut associer de X-morphismes $a \longrightarrow b$ (sauf si c `[est]` de type Δ_+), et à un X-morphisme $a \xrightarrow{u} b$ on peut associer un chemin $c(u)$ de type Δ_+, mais si $a \xrightarrow{u} b \xrightarrow{v} c$, $c(vu)$ est encore de type Δ_+ alors que $c(v)c(u)$ est de type Δ_+^2, ils ne sont pas égaux. Par contre, il y a une flèche naturelle entre ces deux éléments de $\underline{\mathrm{Ch}}(a,c)$, flèche au dessus de l'unique homomorphisme

$$\Delta_+ \xrightarrow{\mathfrak{f}_{+,2}} \Delta_+^2.$$

On a alors

$$\mathfrak{f}_{+,2}^*(\underbrace{a \xrightarrow{u} b \xrightarrow{v} c}_{c(v)c(u)}) = (\underbrace{a \xrightarrow{vu} c}_{c(vu)}), \qquad \text{[page 61]}$$

d'où une flèche canonique

$$c(vu) \longrightarrow c(v)c(u) \qquad \text{dans } \underline{\mathrm{Ch}}(a,c).$$

Cela montre que $c(vu)$ et $c(v)c(u)$ sont tout au moins dans une même composante connexe de $\underline{\mathrm{Ch}}(a,c)$, ou (comme on dit encore) qu'ils sont 'homotopes'. Cela suggère ici de modifier la définition de $\mathrm{Hom}_{\mathrm{Ch}}(a,b)$ en

$$\pi(a,b) = \pi_0(\underline{\mathrm{Ch}}(a,b)).$$

Le foncteur de composition des chemins p. 41

$$\begin{array}{ccccc} \underline{\mathrm{Ch}}(a,b) & \times & \underline{\mathrm{Ch}}(b,c) & \longrightarrow & \underline{\mathrm{Ch}}(a,c) \\ (c & , & c') & \longmapsto & c' \circ c \end{array}$$

`[il y a trop de` c`]` ([17]) induit une application sur les π_0, ce qui donne une loi de composition

$$\pi(a,b) \times \pi(b,c) \longrightarrow \pi(a,c)$$

sur les flèches. Il est immédiat qu'elle est associative (la composition des chemins l'étant), et unitaire. Elle admet comme flèche identique en x soit le 'lacet vide en x', lequel est (directement) homotope dans $\underline{\mathrm{Ch}}(x,x)$ au chemin 'dégénéré' $x \xrightarrow{\mathrm{id}} x$ de type Δ_+, ou aussi au chemin dégénéré de n'importe quel type τ, disons [page 62]

$$x \xleftarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xrightarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x \xleftarrow{\mathrm{id}} x,$$

17. Ce foncteur est déduit des foncteurs

$$\underline{\mathrm{Ch}}_\tau(a,b) \times \underline{\mathrm{Ch}}_{\tau'}(b,c) \longrightarrow \underline{\mathrm{Ch}}_{\tau\tau'}(a,c)$$

évidents, cf. page 77.

un tel chemin n'étant autre que $p_\tau^*(x)$, où

$$p_\tau : \tau \longrightarrow 1$$

est le morphisme de dégénérescence (l'unique morphisme dans $\underline{\mathrm{Ty}}$ vers 1, qui est objet final), d'où

$$p_\tau^* : \underline{\mathrm{Ch}}_1(x,x) = \{x\} \longrightarrow \underline{\mathrm{Ch}}_\tau(x,x) \subseteq \underline{\mathrm{Ch}}(x,x).$$

La catégorie $\Pi(X)$ ainsi obtenu doit s'appeler le *groupoïde fondamental* de X. En effet,

Commentary · Examples [ms. 61]

A span and its fraction

The construction becomes concrete as soon as reversing one arrow creates a morphism which the original category did not contain. Consider the category X generated by the span

$$a \xleftarrow{s} x \xrightarrow{t} b$$

with no nonidentity arrows besides s and t, and put $\Sigma = \{\mathrm{id}_a, \mathrm{id}_x, \mathrm{id}_b, s\}$. The zigzag from a to b is a Σ-path although $\mathrm{Hom}_X(a,b)$ is empty. Its effect after localization is the fraction ts^{-1}.

We can compute the entire localization. It has the same three objects and the following hom-sets, with rows denoting sources:

	a	x	b
a	$\{\mathrm{id}_a\}$	$\{s^{-1}\}$	$\{ts^{-1}\}$
x	$\{s\}$	$\{\mathrm{id}_x\}$	$\{t\}$
b	$\varnothing$	$\varnothing$	$\{\mathrm{id}_b\}$.

Composition is determined by $s^{-1}s = \mathrm{id}_x$, $ss^{-1} = \mathrm{id}_a$, and $(ts^{-1})s = t$. For any functor $F : X \to Y$ with $F(s)$ invertible, there is exactly one extension to this category: it sends s^{-1} to $F(s)^{-1}$ and ts^{-1} to $F(t)F(s)^{-1}$. This verifies the universal property directly.

The same calculation can be read inside the path categories. A path starting at b can only stay at b, because t is not permitted as a backward arrow. A nonconstant path ending at b reaches it by a final occurrence of t; before that it moves between a and x using s in either direction. Adjacent traversals of s in opposite directions cancel by the homotopies on manuscript page 63. After removing identity edges, only one of the entries in the displayed table remains. This proves directly that the connected components of the relative path categories give exactly those hom-sets.

The two objects a and x are isomorphic in the result. Choosing a as their representative identifies a skeleton with the walking arrow $a \to b$. Thus localization has changed the hom-set from a to b, while it has introduced no path from b to a. This is the elementary reason that a relative localization is generally a category rather than a groupoid.

a) toutes les flèches de $\Pi(X)$ sont inversibles, i.e. $\Pi(X)$ est bien un groupoïde, et

b) le foncteur canonique

$$X \longrightarrow \Pi(X)$$

qui est l'identité sur les objets, et qui est défini sur les flèches par $u \longmapsto c(u)$ comme ci-dessus, est *universel* (dans Cat) pour les foncteurs (i.e. les Cat-morphismes) de X vers des *groupoïdes.* (Plus généralement encore, il est universel pour les foncteurs $X \xrightarrow{F} Y$ tels que F transforme toute flèche de X en `[une]` flèche inversible.)

Pour prouver que toute flèche dans $\Pi(X)$ est inversible, comme tout chemin non vide est composé de chemins 'élémentaires' de type Δ_+ ou Δ_-, il suffit de prouver que ces chemins sont inversibles. En fait, ils se correspondent biunivoquement, p. 42 [page 63]

$$\underline{\mathrm{Ch}}_{\Delta_+}(a,b) \simeq \underline{\mathrm{Ch}}_{\Delta_-}(b,a),$$

en associant au chemin de a vers b

$$c(u) : a \xrightarrow{u} b \qquad \text{de type } \Delta_+$$

le chemin

$$c(u)^o : b \overset{u}{\leftarrow} a \qquad \text{de type } \Delta_-$$

de b vers a. Je dis que

$$\begin{array}{rcl} c(u)^o c(u) & \sim & 1_a \\ c(u)c(u)^o & \sim & 1_b \end{array}$$

$$\left.\begin{array}{ccccc} a & \xrightarrow{u} & b & \xleftarrow{u} & a \\ \uparrow{\scriptstyle \mathrm{id}} & & \uparrow{\scriptstyle u} & & \uparrow{\scriptstyle \mathrm{id}} \\ a & \xrightarrow{\mathrm{id}} & a & \xleftarrow{\mathrm{id}} & a \end{array}\right\} \text{ on a un homomorphisme dans } \underline{\mathrm{Ch}}_{\Delta_+\Delta_-}(a,a) \text{ de } 1_a \text{ dans } c(u)^o c(u).$$

$$\left.\begin{array}{ccccc} b & \xleftarrow{u} & a & \xrightarrow{u} & b \\ \uparrow{\scriptstyle u} & & \uparrow{\scriptstyle \mathrm{id}} & & \uparrow{\scriptstyle u} \\ a & \xleftarrow{\mathrm{id}} & a & \xrightarrow{\mathrm{id}} & a \end{array}\right\} \text{ itou dans } \underline{\mathrm{Ch}}_{\Delta_-\Delta_+}(b,b), \text{ de } 1_b \longrightarrow c(u)c(u)^0$$

`[plutôt` $1_b \longleftarrow c(u)c(u)^0$`, viz.`

$$\begin{array}{ccccc} b & \xleftarrow{u} & a & \xrightarrow{u} & b \\ \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle \mathrm{id}} \\ b & \xleftarrow{\mathrm{id}} & b & \xrightarrow{\mathrm{id}} & b \end{array}$$ `].`

Notons que, plus généralement, pour tout chemin p. 43

$$c : a \longrightarrow b, \qquad \text{de type } \tau, \qquad \text{[page 64]}$$

on définit un *chemin opposé*

$$c^o : b \longrightarrow a \qquad \text{de type } \tau^o,$$

où

$$\tau \longmapsto \tau^o : \mathrm{Ty} \longrightarrow \mathrm{Ty}$$

est l'antiïnvolution du monoide libre Ty qui échange les deux générateurs Δ_-, Δ_+, et qui est donc donné sur l'élément général par

$$(\Delta_{\varepsilon_1}\Delta_{\varepsilon_2}\cdots\Delta_{\varepsilon_n})^o = \Delta_{-\varepsilon_n}\Delta_{-\varepsilon_{n-1}}\cdots\Delta_{-\varepsilon_1}.$$

Par récurrence sur la longueur de c ([18]), en utilisant le résultat précédent, que $c(u)$, $c(u)^o$ sont inverses l'un de l'autre dans $\Pi(X)$, on trouve

$$\left.\begin{array}{rcl} c^o c & = & \mathrm{id}_a \\ c c^o & = & \mathrm{id}_b \end{array}\right\} \text{dans } \Pi(X).$$

18. inutile, résulte directement de la formule (∗) `[de la]` page suivante, en décomposant c en composés de chemins 'élémentaires' de type Δ_+ ou Δ_-.

Commentary · Exposition [ms. 63]

Two cancellation homotopies with fixed endpoints

Grothendieck wants reversal of a single edge to give an inverse after taking components. The proof requires two different directions of natural transformation, because the two loops have different shapes. Keeping those directions visible also explains the editors' correction on this page.

For $u : a \to b$, the loop at a is the diagram $L_a = (a \xrightarrow{u} b \xleftarrow{u} a)$. Let K_a be the constant $+-$ diagram at a. The components

$$(\mathrm{id}_a, u, \mathrm{id}_a) : K_a \longrightarrow L_a$$

form a natural transformation. Both endpoint components are identities, so this is a morphism in the fixed-endpoint fibre $\underline{\mathrm{Ch}}_{+-}(a,a)$. The collapse of the type $+-$ to the empty type gives a cartesian arrow from K_a to the empty path at a. We have therefore joined L_a to the empty path by a zigzag in the total fixed-endpoint category.

For the other loop, $L_b = (b \xleftarrow{u} a \xrightarrow{u} b)$, the appropriate constant diagram is K_b at b. This time the transformation goes in the other direction:

$$(\mathrm{id}_b, u, \mathrm{id}_b) : L_b \longrightarrow K_b.$$

It is a morphism in $\underline{\mathrm{Ch}}_{-+}(b,b)$, and K_b again maps to the empty path through the type collapse. The uncorrected diagram in the edition has a constant diagram at a below L_b. Its endpoint components are u, so it is not a morphism in this fixed-endpoint fibre. The bracketed correction supplies exactly the downward transformation just described.

Two conventions are doing work here. First, a constant diagram of nonempty type is an object of its fibre, whereas the empty path belongs to the fibre over the unit type. When the source writes 1_a or 1_b in a fibre, it is temporarily using the constant diagram as a representative of the unit component. Second, connectedness forgets the direction of the arrows used to connect objects. Nothing requires both cancellation homotopies to point the same way.

The relative argument is identical when $u \in \Sigma$: every nonidentity backward edge in these diagrams is then an allowed denominator, and all remaining backward edges are identities. This proves that $\kappa(u)$ is invertible in $\Pi_\Sigma(X)$. There is no condition on a vertical component of a path transformation beyond naturality and the identity endpoint condition.

Exemple. Si

$$\begin{aligned} \tau &= \Delta_+\Delta_-\Delta_+^2\Delta_- \\ \tau^o &= \Delta_+\Delta_-^2\Delta_+\Delta_-, \end{aligned}$$

et si on a un chemin

$$c : a_0 \xrightarrow{u_1} a_1 \xleftarrow{u_2} a_2 \xrightarrow{u_3} a_3 \xrightarrow{u_4} a_4 \xleftarrow{u_5} a_5$$

de type τ, on en déduit le chemin

$$c^o : a_5 \xrightarrow{u_5} a_4 \xleftarrow{u_4} a_3 \xleftarrow{u_3} a_2 \xrightarrow{u_2} a_1 \xleftarrow{u_1} a_0$$

de type τ^o. L'opération $c \longmapsto c^o$ satisfait en effet [pour]

$$\begin{array}{ccccc} a & \xrightarrow{\gamma} & b & \xrightarrow{\gamma'} & c \\ c & \xrightarrow{\gamma'^o} & b & \xrightarrow{\gamma^o} & a \end{array}$$ [page 65]

la relation

$$(*) \qquad (\gamma'\gamma)^o = \gamma^o\gamma'^o,$$

Commentary · Examples [ms. 64]

A circle in a category with two arrows

Grothendieck's reversal operation removes backtracking without destroying genuine loops. A small example distinguishes the two phenomena. Let X have two objects a, b and two distinct nonidentity arrows $f, g : a \to b$, and invert every arrow. The zigzag

$$a \xrightarrow{f} b \xleftarrow{g} a$$

represents the loop $h = g^{-1}f$ at a.

All morphisms in the resulting groupoid have unique expressions of the following forms, with $n \in \mathbf{Z}$:

	a	b
a	h^n	gh^n
b	$h^n g^{-1}$	$gh^n g^{-1}$.

For existence, use g as a chosen path from a to b and replace f by gh. After cancelling adjacent $g^{-1}g$ and gg^{-1}, a word has exactly the indicated form. For uniqueness, consider the functor from X to the one-object group $\mathbf{Z}$ which sends f to 1 and g to 0. Since its target is a groupoid, it extends to the groupoid completion, and it sends h^n to n. Thus different integers cannot give equal arrows. Composition in the table adds the integer exponents, with the displayed g and g^{-1} providing the changes of endpoint.

It follows that $\mathrm{Aut}_{\Pi(X)}(a) \cong \mathbf{Z}$ and $\pi_0 \underline{\mathrm{Ch}}(a,a) \cong \mathbf{Z}$. The nerve supplies a geometric check: the only nondegenerate positive-dimensional simplices of NX are the two edges f, g. There are no composable pairs of nonidentity arrows. Its realization is therefore the graph consisting of two vertices joined by two edges, which is a circle.

If instead Σ consists only of identities, every backward edge is an identity and the effect of a path is already defined in X. The relative localization is then X itself. In particular the two arrows f, g remain distinct and there is no nonidentity loop at a. The difference lies entirely in which arrows may be traversed backwards, not in a choice of representative for the same unrestricted path space.

ce qui est en accord avec l'intuition que l'opération $\gamma \longmapsto \gamma^o$ joue le rôle d'un passage à l'inverse ('à homotopie près'), et on a la formule p. 44

$$(\gamma'\gamma)^{-1} = \gamma^{-1}\gamma'^{-1}$$

pour deux flèches inversibles γ, γ' composables (dans l'ordre γ', γ).

Pour la propriété universelle de

$$X \longrightarrow \Pi(X)$$

(qui permettra d'écrire

$$\Pi(X) \simeq X\Sigma^{-1}, \qquad \text{avec } \Sigma = \mathrm{Fl}(X) \ \ ),$$

autant l'établir tout de suite dans le cadre général où on se donne une partie

$$\Sigma \subseteq \mathrm{Fl}(X),$$

et on définit, pour $\tau \in \underline{\mathrm{Ty}}$,

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \subseteq \underline{\mathrm{Ch}}_{\tau}(X)$$

comme la sous-catégorie pleine de $\underline{\mathrm{Ch}}_{\tau}(X)$ formée des chemins dans X dont *toutes les flèches rétrogrades*

sont dans X `[plutôt dans Σ]`. On notera que pour qu'un chemin c et son opposé c^o soient tous les deux des Σ-*chemins,* il faut et il suffit que *toutes* les flèches du chemin c soient dans Σ (on dira alors que c est un Σ-*chemin stricte*), puisque les flèches rétrogrades de c^o sont les flèches directes de c. [page 66]

Pour que les sous-catégories pleines $\underline{\mathrm{Ch}}_{\tau,\Sigma}(X)$ soient stables par les foncteurs images inverses

$$f^* : \underline{\mathrm{Ch}}_{\tau'}(X) \longrightarrow \underline{\mathrm{Ch}}_{\tau}(X)$$

associés aux flèches

$$f : \tau \longrightarrow \tau'$$

dans $\underline{\mathrm{Ty}}$, ou simplement pour qu'y ait stabilité pour les deux foncteurs f^* associés à un $f : \Delta_- \longrightarrow 1$ et $\Delta_- \longrightarrow \Delta^2_-$, il faut et il suffit que Σ soit stable par composition et contienne toutes les flèches identiques

$$\begin{cases} x \in \mathrm{Ob}\, X \implies \mathrm{id}_x \in \Sigma \\ u, v \in \Sigma, \quad u,\, v \text{ composables} \Longrightarrow uv \in \Sigma. \end{cases}$$

En effet, les flèches dans un chemin $f^*(c') \in \underline{\mathrm{Ch}}_\tau(X)$ sont soit des flèches identiques (qui correspondent aux flèches de transition de I_τ dégénérées pour f), soit des flèches parmi celles de τ, ou des composés d'une suite de flèches consécutives composables de τ' (qui correspondent aux flèches de transition dans $I_{\tau'}$ incluses dans la partie d'une flèche de transition de I_τ), avec conservation des signatures.

p. 45

Commentary · Exposition [ms. 66] [page 67]

Why denominators must form a subcategory

Grothendieck imposes exactly the closure needed to let every interval map act on relative paths. A backward source edge may cover several consecutive backward target edges, or it may collapse to a vertex. Pullback must therefore allow composition of denominators and insertion of identity denominators. These are the two tests singled out in the text.

The map $\Delta_- \to 1$ pulls the empty path at x back to the one-edge path $x \xleftarrow{\mathrm{id}_x} x$. Stability forces $\mathrm{id}_x \in \Sigma$ for every x. Next, for composable denominators $v : z \to y$ and $u : y \to x$, the backward path

$$x \xleftarrow{u} y \xleftarrow{v} z$$

pulls back along $\Delta_- \to \Delta^2_-$ to $x \xleftarrow{uv} z$. Thus stability forces $uv \in \Sigma$. Conversely, assume these two closure properties. The image of a source edge under an endpoint-preserving increasing type map is either a vertex or a consecutive monochromatic block. A backward source edge consequently becomes either an identity or a composite of backward target arrows. Both belong to Σ. This proves stability under every type map, including maps with both collapses and subdivisions.

The condition says that Σ is the arrow set of a wide subcategory of X. It does not require an Ore condition, saturation, or a two-out-of-three property. If we begin with an arbitrary collection S of arrows to invert, we may first take its closure under identities and finite composition. A functor inverts this closure exactly when it inverts S, so the intended universal localization is unchanged. Saturation is a later operation: it adds every arrow whose image in the localization happens to be invertible.

The adjective "full" in $\underline{\mathrm{Ch}}_{\Sigma,\tau}(X)$ deserves equal attention. The restriction is on backward *horizontal* arrows of the path. Given two such paths, every natural transformation between them remains available, with identity components at prescribed endpoints. Its internal vertical arrows need not lie in Σ. This fullness is consistent with the ordinary localization theorem, as the calculation at manuscript page 68 will show. It must be remembered when making a higher-homotopical comparison with constructions which restrict vertical arrows as well.

Finally, reversal of an individual path remains allowed precisely when its forward arrows also belong to Σ. Closure of the entire path system under reversal therefore forces $\Sigma = \mathrm{Fl}(X)$. This statement concerns literal permitted paths; it does not say that a localization can be a groupoid only in that case. Arrows outside Σ may already be invertible in X, or become invertible after localization.

On suppose ces conditions satisfaites, de sorte que les $\underline{\mathrm{Ch}}_{\tau,\Sigma}(X)$ forment les fibres d'une sous-catégorie fibrée pleine de $\underline{\mathrm{Ch}}(X)$ sur $\underline{\mathrm{Ty}}$, qu'on dénotera par

$$\underline{\mathrm{Ch}}_\Sigma(X) \hookrightarrow \underline{\mathrm{Ch}}(X).$$

Il est évident que cette sous-catégorie est stable par l'opération de composition des chemins (mais non pas, en général, on l'a dit, par passage de c à c^o (il faudrait pour cela que Σ contienne *toutes* les flèches de X, puisque pour toute $u : a \longrightarrow b$, qui est donc dans $\underline{\mathrm{Ch}}_{\Delta_+,\Sigma}(X)$, u^o donné par $b \xleftarrow{u} a$ dans $\underline{\mathrm{Ch}}_{\Delta_-}(X)$ devrait être un Σ-chemin, d'où $u \in \Sigma$)).

On va définir une catégorie canonique

$$\Pi_\Sigma(X),$$

ayant même objets que X, et un foncteur canonique

$$X \longrightarrow \Pi_\Sigma(X)$$

qui est l'identité sur les objets, et qui transforme les $u \in \Sigma$ en flèches inversibles. Pour cela on calque la description du groupoïde fondamental (mais $\Pi_\Sigma(X)$ ne sera pas en général un groupoïde), en posant

$$\mathrm{Hom}_\Sigma(x,y) = \pi_0(\underline{\mathrm{Ch}}_\Sigma(x,y)),$$

et en définissant la composition des 'Σ-flèches' en termes de la composition des chemins. L'argument qui prouvait, dans le cas de $\Pi(X)$, que *toute* flèche de X, vue comme chemin c de type Δ_+, était inversible dans $\Pi(X)$, ayant comme inverse c^o de type Δ_-, est encore valable, pourvu que c^o soit encore bien un Σ-chemin, ce qui exige que $u \in \Sigma$. [page 68]

On va l'interpréter en termes d'un foncteur canonique

$$(*) \qquad \kappa : X \longrightarrow \Pi_\Sigma(X) \qquad ([19])$$

qui se décrit encore comme dans le cas du groupoïde fondamental, en associant à toute flèche u de X la classe d'homotopie du chemin $c(u)$ qu'elle définit. On voit donc que ce foncteur canonique transforme les $u \in \Sigma$ en flèches inversibles.

Quand les flèches de Σ sont déjà inversibles dans X, je dis que le foncteur $(*)$ est un isomorphisme, i.e. que pour tout $a, b \in \mathrm{Ob}\, X$,

$$\alpha : \mathrm{Hom}_X(a,b) \longrightarrow \pi_0 \underline{\mathrm{Ch}}_\Sigma(a,b), \qquad u \longmapsto \underbrace{\mathrm{ch}_\Sigma(u)}_{=\mathrm{ch}(u)}$$

est bijectif. Pour le voir, on va définir

$$\beta : \pi_0(\underline{\mathrm{Ch}}_\Sigma(a,b)) \longrightarrow \mathrm{Hom}_X(a,b),$$

qui sera inverse de β. On définit

p. 46

Commentary · Exposition [ms. 68]

The effect of a zigzag descends to components [page 69]

Grothendieck now uses evaluation in a category where the denominators are already invertible. Before checking that evaluation is inverse to α, we should verify that it is defined on a connected component of the path category. This is a short verification, but it explains why the internal vertical arrows of a path transformation need not be invertible.

The useful statement is slightly more general. Let $F : X \to Y$ send every arrow of Σ to an isomorphism. For a Σ-path $c = (x_0, \ldots, x_n)$, let $e_i : F(x_{i-1}) \to F(x_i)$ be $F(u_i)$ if its ith edge is forward and $F(u_i)^{-1}$ if that edge is backward. Define

$$E_F(c) = e_n \cdots e_1, \qquad E_F(\text{empty path at } a) = \mathrm{id}_{F(a)}.$$

This formula is defined solely by the inversion hypothesis.

First compare paths c, c' of the same type by a natural transformation with components $\eta_i : F(x_i) \to F(x'_i)$. For a forward edge the commuting square reads $\eta_i e_i = e'_i \eta_{i-1}$. For a backward edge the original square is

$$\eta_{i-1} F(u_i) = F(u'_i)\eta_i.$$

Multiplication by the inverses of $F(u_i)$ and $F(u'_i)$ gives exactly the same formula

$$\eta_i e_i = e'_i \eta_{i-1}.$$

Successively composing these equations yields

$$\eta_n E_F(c) = E_F(c')\eta_0.$$

At fixed endpoints, η_0 and η_n are identities. Hence the two effects agree. No inverse of any η_i has been taken.

19. Sur les flèches, κ est défini, pour $u : a \longrightarrow b$, par $\kappa(u)$ = classe d'homotopie de $\mathrm{ch}(u)$ dans $\underline{\mathrm{Ch}}_\Sigma(a,b)$, où $\mathrm{ch}(u)$ est le chemin de type Δ_+ de a vers b défini par u.

We must also compare different types. If $q : \tau \to \tau'$ and c' is a path of type τ', then each edge of q^*c' is a composite over the corresponding consecutive block of edges of c', or an identity for an empty block. A forward block has the same effect as its composite. A backward block does too, because inversion reverses the order of composition. The blocks partition the target edges in their traversal order, so

$$E_F(q^*c') = E_F(c').$$

A general morphism in the total path category factors into a morphism within a fibre followed by a cartesian morphism of this form. We have therefore proved invariance under every morphism and hence under every connecting zigzag. Evaluation defines a map on $\pi_0\underline{\mathrm{Ch}}_\Sigma(a,b)$.

Taking $F = \mathrm{id}_X$ gives the missing descent check for β in the text. The identity $\beta\alpha = \mathrm{id}$ is immediate. For the other identity, a backward traversal of an invertible $u \in \Sigma$ represents the inverse of $\kappa(u)$; this is also $\kappa(u^{-1})$, since κ is a functor. Replacing each backward edge in this way gives the all-forward path $c^\natural$, and repeated composition identifies its class with the elementary path of its composite. These equalities take place in $\Pi_\Sigma(X)$. The omitted subscripts in equations (1) and (2) on manuscript page 69 should not lead us to infer equality in a relative localization merely from equality in its groupoid completion.

$\beta(c)$, pour une Σ-flèche de type τ, le longueur n, en inversant toutes les flèches rétrogrades (ce qui est possibles, puisqu'elles sont dans Σ), de façon à obtenir un chemin $c^\natural$ de type Δ_+^n, et en prenant le composé (dans X) des flèches de ce chemin $c^\natural$. Il est évident alors que $\beta\alpha = \mathrm{id}$, il faut prouver que l'on a aussi $\alpha\beta = \mathrm{id}$, i.e. que dans $\Pi_\Sigma(X)$, les flèches $u = \int c^\natural$ et c sont égales, i.e. u et c sont homotopes dans $\underline{\mathrm{Ch}}_\Sigma(a,b)$. Considérons le chemin $c^\natural$, de type Δ_+^n :

$$a = a_0 \xrightarrow{u_1} a_1 \xrightarrow{u_2} a_2 \longrightarrow \cdots \longrightarrow a_{n-1} \xrightarrow{u_n} a_n = b,$$

et soit u_i ($i \in [1,n]$) une flèche de ce chemin, correspondant à une flèche de transition $\underline{u_i}$ *rétrograde* de τ, donc par construction

$$u_i = v_i^{-1} \qquad \text{dans } X, \text{ avec } v_i \in \Sigma$$

la flèche d'indice i dans le chemin initial c de type τ. Si $\underline{u_i}$ par contre est directe, on aura

$$u_i = v_i.$$

(NB On désigne par v_i la i-flèche dans le chemin c.) Or on a

$$(*) \qquad c = c_n * c_{n-1} * \cdots * c_1 \qquad \text{dans } \Pi_\Sigma(X)$$

et le facteur c_i est soit la classe d'homotopie de $\mathrm{ch}(v_i) = \mathrm{ch}(u_i)$ (cas où la flèche de transition $\underline{u_i}$ de τ est directe), soit celle du chemin *opposé* au chemin $\mathrm{ch}(v_i) : a_i \xrightarrow{v_i} a_{i-1}$ de type Δ_+, donc le chemin $\mathrm{ch}(v_i)^o : a_{i-1} \xleftarrow{v_i} a_i$ de type Δ_-. Je dis que

$$(1) \qquad \boxed{\mathrm{ch}(v_i)^o = \mathrm{ch}(u_i) \qquad \textit{dans } \Pi(X),}$$

ce qui prouve que dans $\Pi_\Sigma(X)$, c est égal au composé des $\mathrm{ch}(u_i)$ ($i \in [1,n]$), donc à $c^\natural$. Il faut prouver de plus que

$$(2) \qquad c^\natural = \mathrm{ch}(u) \qquad (u = u_n u_{n-1} \cdots u_1) \qquad \text{dans } \Pi(X).$$

Commentary · Context [ms. 69] [page 70]

Relations before higher homotopies

Grothendieck's path construction is testing the same universal problem as the classical category of fractions, while retaining the witnesses for its elementary relations. In Gabriel and Zisman (1967, Chapter I, §1.1 and Lemma 1.2), localization is constructed from paths with formal reverse arrows by

imposing composition, identity, and inverse relations. Lemma 1.2 also gives the universal property on functor categories and natural transformations. Here composition and identity relations come from type changes; the inverse relations come from the two fixed-endpoint ladders. The evaluation calculation above shows that arbitrary path ladders respect these relations. The universal property on the following pages will prove that taking components introduces no additional identifications.

This is also the appropriate scale on which to read the connection with the derivator programme. In Grothendieck (1983, §69, pp. 1–4 of the separately edited text), the diagramwise construction

$$I \longmapsto \mathrm{Fun}(I, M)[W_I^{-1}]$$

retains a whole family of localized diagram categories, where W_I consists of pointwise weak equivalences. The present calculation makes morphisms in such ordinary localizations accessible by diagrams of arrows and commuting squares. It does not by itself supply the homotopy Kan extensions or the higher comparisons required for a derivator.

The chronological comparison with Dwyer and Kan is particularly instructive, since their simplicial localization papers appeared in 1980, before these manuscript pages. Their hammock construction restricts every vertical arrow to the chosen subcategory of weak equivalences, as well as every backward horizontal arrow; its component category is the ordinary localization, and it is weakly equivalent to their standard simplicial localization (Dwyer and Kan, 1980a, Definition 2.1 and Propositions 2.2, 3.1). Grothendieck's full relative path fibres allow arbitrary internal vertical arrows. Agreement of component categories is therefore a common starting point, not an automatic comparison theorem for the entire nerves. The stronger assertions later in the chapter must be justified with the actual path category and its actual indexing maps.

Donc il y a deux vérifications à faire (où je suppose à nouveau Σ quelconque) :

1°) Soit $u \in \Sigma$, $u : a \longrightarrow b$, supposons Σ inversible dans X. Alors

$$\mathrm{ch}(u)^o = \mathrm{ch}(u^{-1}) \qquad \text{dans } \Pi_\Sigma(X),$$

i.e. les deux chemins de b vers

$$\begin{array}{lll} b & \xleftarrow{u} & a \quad \text{de type } \Delta_- \\ b & \xrightarrow{u^{-1}} & a \quad \text{de type } \Delta_+ \end{array}$$

sont homotopes dans $\underline{\mathrm{Ch}}_\Sigma(b, a)$. Or on sait déjà, comme $\kappa : X \longrightarrow \Pi_\Sigma(X)$ est un foncteur, que p. 47

$$\mathrm{hot}\,\mathrm{ch}(u^{-1}) = (\mathrm{hot}\,\mathrm{ch}(u))^{-1} \qquad \text{dans } \Pi_\Sigma(X)$$

où $\mathrm{hot}(\bullet)$ désigne la classe d'homotopie d'une chaine (ici classe dans $\underline{\mathrm{Ch}}_\Sigma(b, a)$ et dans $\underline{\mathrm{Ch}}_\Sigma(a, b)$ respectivement). Donc l'énoncé à prouver c'est que $\mathrm{hot}(\mathrm{ch}(u)^o)$ est égal à l'*inverse* de $\mathrm{hot}(\mathrm{ch}(u))$ dans $\Pi_\Sigma(X)$. C'est donc ce qu'on a prouvé pour *tout* $u \in \mathrm{Fl}(X)$, dans le cas de $\Pi(X)$. L'argument doit marcher, du moment qu'il a un sens, i.e. que $\mathrm{ch}(u)^o$ est bel et bien un Σ-chemin, i.e. $u \in \Sigma$.

2°) La formule (2) signifie que pour tout chemin c de type Δ_+^n (donc tautologiquement un Σ-chemin), allant de a vers b, sa classe dans $\mathrm{Hom}_\Sigma(a, b)$ n'est autre que celle de $\mathrm{ch}(u)$, où u est le composé $u_n u_{n-1} \cdots u_1$. Mais on a [page 71]

$$c = c_1 * c_2 * \cdots * c_n,$$

d'où dans $\Pi_\Sigma(X)$

$$\mathrm{hot}(c) = \mathrm{hot}(c_n)\mathrm{hot}(c_{n-1}) \cdots \mathrm{hot}(c_1),$$

or $\mathrm{hot}(c_i) = \mathrm{hot}\,\mathrm{ch}(u_i) = \kappa(u_i)$, donc

$$\begin{aligned} \mathrm{hot}(c) &= \kappa(u_n)\kappa(u_{n-1}) \cdots \kappa(u_1) \\ &= \kappa(\underbrace{u_n u_{n-1} \cdots u_1}_{u}), \end{aligned}$$

puisque κ est un foncteur, donc

$$\mathrm{hot}(c) = \mathrm{hot}\,\mathrm{ch}(u),$$

c.q.f.d.

Notons maintenant que si (X', Σ') est un deuxième couple d'une catégorie X' et d'un ensemble *'admissible'* de flèches de X', et si

$$f : X \longrightarrow X'$$

est un foncteur tel que

$$f(\Sigma) \subseteq \Sigma',$$

alors il existe un *unique* foncteur

$$\bar{f} = \Pi_\Sigma(f) : \Pi_\Sigma(X) \longrightarrow \Pi_{\Sigma'}(X')$$

qui 'prolonge' f, i.e. tel que le carré p. 48

$$\begin{array}{ccc} X & \xrightarrow{\kappa_X} & \Pi_\Sigma(X) \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \bar{f}=\Pi_\Sigma(f)} \\ X' & \xrightarrow{\kappa_{X'}} & \Pi_{\Sigma'}(X') \end{array} \qquad \text{[page 72]}$$

soit commutatif. L'unicité résulte du fait que sur les objets $\bar{f}$ doit être donné par f, et que sur les flèche la commutativité du carré implique que $\bar{f}$ est connu sur les flèches de la forme $\kappa_X(u)$, ainsi donc que sur les inverses de telles flèches, quand ces inverses sont inversibles, donc en particulier sur les $\kappa_X(u)^{-1}$ pour $u \in \Sigma$. Or toute flèche de $\Pi_\Sigma(X)$, provenant d'un chemin, est composé d'une suite de flèches de cette forme. Pour l'existence de $\bar{f}$, on note que f induit des foncteurs

$$\underline{\mathrm{Ch}}_\tau(X; a, b) \longrightarrow \underline{\mathrm{Ch}}_\tau(X'; f(a), f(b))$$

qui commutent aux opérations φ^* sur les chemins associées aux foncteurs

$$\varphi : \tau \longrightarrow \tau',$$

d'où un homomorphisme de catégories fibrées sur $\underline{\mathrm{Ty}}$

$$f_{a,b} : \underline{\mathrm{Ch}}(X; a, b) \longrightarrow \underline{\mathrm{Ch}}(X'; f(a), f(b)).$$

D'où une application sur les π_0, i.e.

$$\bar{f}_{a,b} : \mathrm{Hom}_\Sigma(a, b) \longrightarrow \mathrm{Hom}_\Sigma(f(a), f(b)), \qquad \text{[page 73]}$$

donnée par

$$\bar{f}_{a,b}(\mathrm{hot}(c)) = \mathrm{hot}(f_{a,b}(c)).$$

On a évidemment, pour deux chemins $\gamma : a \longrightarrow b$ et $\gamma' : b \longrightarrow c$,

$$f_{a,c}(\gamma\gamma') = f_{b,c}(\gamma')f_{a,b}(\gamma),$$

ce qui prouve que $\bar{f}$ est compatible avec la composition. Il est trivial qu'il est compatible avec les identités (car $f_{a,a}$ transforme le chemin vide en a en le chemin vide en $f(a)$), donc on a un foncteur $\bar{f}$, et il rend bel et bien le carré commutatif.

On peut maintenant prouver la propriété universelle de $\Pi_\Sigma(X)$ et de p. 49

$$X \xrightarrow{\kappa} \Pi_\Sigma(X), \qquad \text{[page 74]}$$

savoir que tout foncteur $f : X \longrightarrow X'$ tel que $u \in \Sigma \Longrightarrow f(u)$ inversible, se factorise *de façon unique* par $\Pi_\Sigma(X)$. L'unicité se voit comme tantôt ([20]). Pour l'existence, on note que f applique Σ dans l'ensemble Σ'

des flèches inversibles de X' (lequel ensemble satisfait bien aux deux conditions d'admissibilité). Donc par ce qui précède on a un diagramme commutatif

$$\begin{array}{ccc} X & \xrightarrow{\kappa} & \Pi_\Sigma(X) \\ {\scriptstyle f}\downarrow & \overset{\tilde{f}}{\searrow} & \downarrow{\scriptstyle \bar{f}=\Pi_\Sigma(f)} \\ X' & \xrightarrow[\sim]{\kappa'} & \Pi_{\Sigma'}(X'), \end{array}$$

où $\bar{f}$ est uniquement déterminé. Comme κ' est un isomorphisme (comme on a vu tantôt, puisque Σ' `[est]` formé des flèches inversibles), l'existence et l'unicité de $\bar{f}$ équivaut à la conclusion annoncée,

c.q.f.d.

Encore un sorite sur les classes d'homotopie d'éléments d'une catégorie X. De façon générale, on voit que X et $\Pi_\Sigma(X) = X\Sigma^{-1}$ ont mêmes composantes connexes, car si a et b dans X sont reliés par une flèche de $X\Sigma^{-1}$, i.e. par un Σ-chemin, alors ils sont dans une même composante connexe de X. Mais l'inverse est également vrai, comme on voit p. ex. en prenant $\Sigma = \mathrm{Fl}(X)$, et en disant que dans le *groupoïde* $\Pi(X)$, où deux sommets sont dans une même composante `[connexe]` si et seulement si $\exists\ x \longrightarrow y$ (relation qui est non seulement transitive et reflexive, mais aussi symétrique dans un groupoïde). Un chemin [page 75]

$$a \xrightarrow{c} b$$

sera appelé aussi une *homotopie* de a vers b. On dit que a et b sont *homotopes* s'il existe une homotopie c de a vers b, auquel cas il existe aussi une homotopie 'inverse' c^o de b vers a. Cela signifie aussi que a et b sont dans la même composante connexe de X, i.e. définissent le même élément de $\pi_0(X)$.

Commentary · Exposition [ms. 74]

The universal property includes natural transformations

Grothendieck has now brought the path construction back to the familiar category of fractions. The work with arbitrary lengths and their changes of parameter pays for a universal property requiring only that the denominators contain identities and be closed under composition. No condition allowing fractions to be shortened to spans has entered the argument. This is the general localization of Gabriel and Zisman (1967, Chapter I, §1.1 and Lemma 1.2); a calculus of left or right fractions is additional structure that can simplify its representatives.

Write $Q : X \longrightarrow L = \Pi_\Sigma(X)$. A functor $F : X \longrightarrow Y$ carrying Σ into isomorphisms must send a zigzag

$$a = x_0 \xrightarrow{u_1} x_1 \xleftarrow{s_2} x_2 \xrightarrow{u_3} \cdots \xleftarrow{s_n} x_n = b$$

to the corresponding composite of the $F(u_i)$ and $F(s_j)^{-1}$, in their order of traversal. The preceding discussion of the evaluation map proves that this prescription is constant on components of the path category. It therefore defines $\overline{F} : L \longrightarrow Y$. Every arrow of L is represented by a zigzag, so these prescribed values also prove uniqueness. The factorization through $\Pi_{\mathrm{Iso}(Y)}(Y) \simeq Y$ used in the text packages precisely this argument.

There is a useful categorical refinement, already part of the classical universal property. Precomposition induces an isomorphism

$$Q^* : \mathrm{Fun}(L, Y) \xrightarrow{\sim} \mathrm{Fun}_\Sigma(X, Y),$$

where the right side is the full subcategory of $\mathrm{Fun}(X, Y)$ on the functors inverting Σ. The statement concerns all natural transformations, not merely natural isomorphisms. To see the extra assertion

20. C'est inutile, cf. ci-dessous.

directly, let $\eta : F \Rightarrow G$ be a natural transformation between such functors. Since Q is the identity on objects, the only possible extension has components $\overline{\eta}_x = \eta_x$. Naturality for $s : x \to y$ gives

$$\eta_y F(s) = G(s)\eta_x, \qquad \eta_x F(s)^{-1} = G(s)^{-1}\eta_y.$$

The second equality supplies naturality for the formal inverse of s. Naturality then passes to every composite representing an arrow of L. These components define the required transformation, and uniqueness follows again from the identity on objects. This also explains why relative functors induce functors on localizations without further choices: both their action on arrows and their action on transformations are determined before any presentation of a fraction is chosen.

For the present chapter, this refinement matters because a path whose vertices are themselves functors is a diagram of natural transformations. The later passage to diagram categories will repeatedly use the same distinction between determining objects and controlling the morphisms between them. The localization established here already supplies both at the ordinary categorical level.

L'homotopie sera dite *simple* si c est de type Δ_+ ou Δ_-, i.e. si elle est donnée par une flèche $u : a \longrightarrow b$ ou $v : b \longrightarrow a$. Si on cherche une homotopie entre a et b, on essaiera de la trouver la plus 'courte' possible. Si on a une homotopie c de longueur l de a vers b, alors en remplaçant τ par $\tau_{\text{réd}}$, donc I_τ par $I_{\tau\,\text{réd}}$, le sous-intervalle réduit correspondant, on trouve une homotopie de a à b de type $\tau_{\text{réd}}$, dont la longueur est égale à la *'longueur utile'* du type τ de c (ce qui justifie la terminologie). Notons que si [page 76]

$$\tau = \Delta_\varepsilon^{l_1}\Delta_{-\varepsilon}^{l_2}\Delta_\varepsilon^{l_3}\cdots\Delta_{(-1)^r\varepsilon}^{l_r},$$

alors

$$\tau_{\text{réd}} = \Delta_\varepsilon\Delta_{-\varepsilon}\Delta_\varepsilon\cdots\Delta_{(-1)^r\varepsilon},$$

`[plutôt deux fois` $(-1)^{r-1}$`]` où r est la longueur utile de τ (la longueur pas utile étant $n = \sum_{i\in[1,r]} l_i$). p. 50

Quand on part d'une homotopie entre *foncteurs*

$$X \longrightarrow Y,$$

il est entendu qu'il s'agit d'homotopies dans la catégorie $\underline{\mathrm{Hom}}(X, Y)$. (Ou, suivant le contexte, dans une sous-catégorie précisée de $\underline{\mathrm{Hom}}(X, Y)$.)

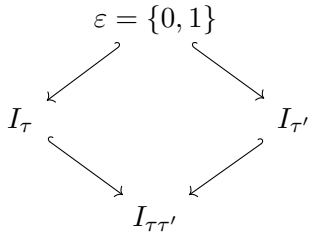
À la page 61, j'ai invoqué l'existence d'un *foncteur* de *'composition des chemins'* [page 77]

$$\underline{\mathrm{Ch}}_\Sigma(X; a, b) \times \underline{\mathrm{Ch}}_\Sigma(X; b, c) \longrightarrow \underline{\mathrm{Ch}}_\Sigma(X; a, c),$$

qui sur la fibre du premier membre au dessus de $(\tau, \tau') \in \mathrm{Ob}(\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}})$, serait donné par

$$(*) \qquad \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X; b, c) \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(X; a, c),$$

la composition des chemins de types précisés τ, τ'. Qu'on ait là un *foncteur,* pas seulement une application sur les ensembles d'objets, est assez évident, et peut s'exprimer par exemple à l'aide du diagramme d'amalga`[ma]`tion



en le transformant par $\underline{\mathrm{Hom}}(\cdot, X)$ (au lieu de le transformer par $\mathrm{Hom}_{\mathrm{Cat}}(\cdot, X)$ seulement), du moins dans le cas particulier où $\Sigma = \mathrm{Fl}(X)$ (qui était celui de la page 61). Mais le passage aux flèches est assez évident - pour deux composés $c_1' \circ c_1$ et $c_2' \circ c_2$, de

$$\begin{array}{ccccc} a & \xrightarrow{c_1} & b & \xrightarrow{c_1'} & c \\ \| & & \| & & \| \\ a & \xrightarrow[c_2]{} & b & \xrightarrow[c_2']{} & c \end{array} \qquad \begin{array}{c} \\ w \\ \\ w' \\ \\ \end{array}$$

[on] voit que les homomorphismes w'' du premier dans le second, dans $\underline{\mathrm{Ch}}_{\Sigma}(X; a, c)$, qui sont l'identité sur le sommet d'indice $n+1$ des deux chemins (où $n = \log \tau$), correspondant à 1_τ et à $0_{\tau'}$, 'sont' exactement les couples formés d'un homomorphisme $w : c_1 \longrightarrow c_2$ et p. 51
$w' : c_1' \longrightarrow c_2'$ dans $\underline{\mathrm{Ch}}_{\Sigma,\tau}(a, b)$ et $\underline{\mathrm{Ch}}_{\Sigma,\tau'}(b, c)$ respectivement, et si w'', w, w' se correspondent ainsi, on posera [page 78]

$$w'' = w * w'.$$

Il y a une transitivité évidente à vérifier pour voir que ça définit un *foncteur* $(*)$. On peut exprimer la situation à l'aide du diagramme

$$\begin{array}{ccccc} & & \underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}} & \xrightarrow{\pi \text{ (composition)}} & \underline{\mathrm{Ty}} \\ & \swarrow^{p_1} & & \searrow_{p_2} & \\ \underline{\mathrm{Ty}} & & & & \underline{\mathrm{Ty}}, \end{array}$$

où p_1, p_2 sont les deux projections. On considère sur $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ les trois catégories fibrés

$$p_1^*(\underline{\mathrm{CH}}_{\Sigma,\tau}(a, b)), \quad p_2^*(\underline{\mathrm{CH}}_{\Sigma,\tau'}(b, c)), \quad \pi^*(\underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a, c))$$

[plutôt $\pi^*(\underline{\mathrm{CH}}_{\Sigma,\tau\tau'}(a, c))$], dont les fibres en (τ, τ') sont respectivement

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(a, b), \quad \underline{\mathrm{Ch}}_{\Sigma,\tau'}(b, c), \quad \underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a, c).$$

(On écrit $\underline{\mathrm{CH}}$ au lieu de $\underline{\mathrm{Ch}}$, pour indiquer que l'on considère les catégories envisagées comme des catégories (fibrées) *au dessus de* ($\underline{\mathrm{Ty}}$) ([21]).) Au dessus de $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$, on a un homomorphisme (cartésien) de catégories fibrées

$$(**) \qquad p_1^*(\underline{\mathrm{CH}}_{\tau}(a, b)) \times_{\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}} p_2^*(\underline{\mathrm{CH}}_{\tau}(b, c)) \longrightarrow \pi^*(\underline{\mathrm{CH}}_{\tau}(a, c))$$

[plutôt $p_1^*(\underline{\mathrm{CH}}_{\tau}(a, b)) \times_{\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}} p_2^*(\underline{\mathrm{CH}}_{\tau'}(b, c)) \longrightarrow \pi^*(\underline{\mathrm{CH}}_{\tau\tau'}(a, c))$], car c'est OK sur les fibres ($(*)$ page 77), et si

$$(\varphi, \varphi') : (\tau_1, \tau_1') \longrightarrow (\tau, \tau') \qquad (\varphi : \tau_1 \longrightarrow \tau,\ \varphi' : \tau_1' \longrightarrow \tau') \qquad \text{[page 79]}$$

est une flèche dans $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$, on en déduit un diagramme commutatif

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma,\tau}(a, b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(b, c) & \xrightarrow{\text{comp.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau}(a, c) \\ \downarrow{\scriptstyle \varphi_{a,b}^* \times (\varphi'_{b,c})^*} & & \downarrow{\scriptstyle (\varphi * \varphi')^*_{a,c}} \\ \underline{\mathrm{Ch}}_{\Sigma,\tau_1}(a, b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau_1'}(b, c) & \xrightarrow{\text{comp.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau_1\tau_1'}(a, c) \end{array}$$

[plutôt $\underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a, c)$], i.e. la composition des chemins d'extrémités fixés est compatible avec les images inverses de chemins par flèches entre les types de chemins.

p. 52

21. Mais il s'agit pour p_1^*, p_2^* et π^* de produits cartésiens ordinaires dans Cat, il n'y a pas à avoir peur et à écrire des $\underline{\mathrm{CH}}$ …

Commentary · Examples [ms. 76]

Connectedness does not impose an orientation

Grothendieck next isolates the least information carried by his paths: whether two objects belong to one component. For any admissible Σ, the map $Q : X \to X[\Sigma^{-1}]$ induces a bijection on connected components. Indeed, every arrow of X supplies an edge between its two objects in the localization, so a zigzag in X remains a zigzag there. Conversely, an arrow of the localization is represented by a Σ-zigzag in X; forgetting which of its arrows were designated denominators gives an ordinary zigzag in X. Applying this observation to the finitely many arrows of a zigzag in the localization proves the converse on components.

The orientation must be kept separate from this argument. Let $X = [1]$, with its single nonidentity arrow $0 \to 1$, and let Σ contain only identities. Then $X[\Sigma^{-1}] = X$ and $0, 1$ belong to the same component, although there is no arrow $1 \to 0$ in the localization. Thus the phrase about the converse on manuscript page 75 is read as a statement about connected components. It would be false if read as the assertion that connectedness supplies a directed Σ-path in every prescribed direction. When $\Sigma = \mathrm{Fl}(X)$, reversal is available, and the groupoid formulation does permit that stronger assertion.

Reduction of a type has an equally concrete meaning. A string

$$a_0 \longrightarrow a_1 \longrightarrow a_2 \longleftarrow a_3 \longleftarrow a_4 \longrightarrow a_5$$

can be replaced by

$$a_0 \longrightarrow a_2 \longleftarrow a_4 \longrightarrow a_5$$

by composing the arrows in each maximal run. The signs of the three remaining edges are $+, -, +$. For r runs beginning with ε, their signs are $(-1)^{j-1}\varepsilon$, $1 \le j \le r$; this is the reason for the edition's correction of the last exponent on pages 76–77. If the original backward arrows belong to Σ, closure under composition ensures that the reduced backward arrows still do. The same construction in $\mathrm{Fun}(A, B)$ composes successive natural transformations of the same direction. Here a homotopy between functors means a finite zigzag of natural transformations; its endpoints need not be naturally isomorphic.

Mais l'homomorphisme cartésien $(**)$ de catégories fibrées sur $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ n'est pas tout à fait la même chose qu'un anodin foncteur

$$(*) \qquad \underline{\mathrm{Ch}_\Sigma}(a,b) \times \underline{\mathrm{Ch}_\Sigma}(b,c) \longrightarrow \underline{\mathrm{Ch}_\Sigma}(a,c).$$

La formule $(*)$ définit en tout cas une application sur les ensembles d'objets, puisque les ensembles d'objets de chacune des trois catégories envisagées se décompose en somme d'ensembles $\underline{\mathrm{Ch}_{\Sigma,\tau}}(\bullet,\bullet)$, où τ parcourt $\mathrm{Ty} = \mathrm{Ob}\,\underline{\mathrm{Ty}}$. Mais cela se prolonge-t-il de façon naturelle aux morphismes?

Mais voilà que je me rappelle que ce fameux produit cartésien $p_1^*(\text{fourbi}) \times p_2^*(\text{machin})$ n'est autre que le produit cartésien tout bête [page 80]

$$\underline{\mathrm{Ch}_\Sigma}(a,b) \times \underline{\mathrm{Ch}_\Sigma}(b,c),$$

envoyé sur $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ par le produit cartésien des deux foncteurs de projection

$$\begin{aligned} p_{a,b} : \underline{\mathrm{Ch}_\Sigma}(a,b) &\longrightarrow \underline{\mathrm{Ty}} \\ p_{b,c} : \underline{\mathrm{Ch}_\Sigma}(b,c) &\longrightarrow \underline{\mathrm{Ty}}, \end{aligned}$$

d'où

$$\begin{array}{c} \underline{\mathrm{Ch}_\Sigma}(a,b) \times \underline{\mathrm{Ch}_\Sigma}(b,c) \\ \big\downarrow{\scriptstyle p_{a,b} \times p_{b,c}} \\ \underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}} \end{array}$$

qui est bel et bien fibrant (comme produit cartésien de deux foncteurs fibrants) et a les fibres [page 81]

qu'il faut! On trouve donc bien, par le sorite général sur les foncteurs fibrants (ici sur $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$) un foncteur de catégories fibrées sur $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$

$$\underline{\mathrm{Ch}_\Sigma}(a,b) \times \underline{\mathrm{Ch}_\Sigma}(b,c) \longrightarrow \underbrace{\pi^*(\underline{\mathrm{CH}_\Sigma}(a,c))}_{=(\underline{\mathrm{Ty}}\times\underline{\mathrm{Ty}})\times_{\underline{\mathrm{Ty}}}\underline{\mathrm{Ch}_\Sigma}(a,c)} \quad (22) \qquad .$$

Comme on a d'autre part un foncteur du dernier membre (produit fibré) vers le facteur $\underline{\mathrm{Ch}_\Sigma}(a,c)$, on gagne. (On l'aurait retrouvé par le raisonnement direct que je m'apprêtais de faire, et que j'ai biffé - mais je suis plus heureux comme ça!)

Pour bien faire, il faudrait prouver l'*associativité* des foncteurs de composition, pas seulement au niveau des objets, par des arguments comme ceci, mais ça revient simplement à une relation du type p. 53

$$(w'' * w') * w = w'' * (w' * w)$$

pour un système de trois flèches dans

$$\underline{\mathrm{Ch}_\Sigma}(a,b),\ \underline{\mathrm{Ch}_\Sigma}(b,c) \text{ et } \underline{\mathrm{Ch}_\Sigma}(c,d)$$

respectivement. Ça ne peut être que trivialement vrai, mais fastidieux à expliciter.

[page 82]

Commentary · Exposition [ms. 81]

The total composition and its associativity

Grothendieck returns to composition because the localization argument uses a functor on the total path categories, including morphisms that change the type. The commentary at manuscript pages 48–50 describes the fixed-type operation. The additional verification here is that this operation respects the arrows of the Grothendieck construction. Writing these arrows explicitly also settles the associativity left to the reader on pages 81–82.

For fixed endpoints, put $P_{a,b}(\tau) = \underline{\mathrm{Ch}_{\Sigma,\tau}}(X; a, b)$. An arrow in its total category,

$$(\tau, c) \longrightarrow (\sigma, d),$$

is a pair (φ, η) with $\varphi : \tau \to \sigma$ and an endpoint-fixed natural transformation $\eta : c \Rightarrow d\varphi$. If a second arrow is $(\psi, \theta) : (\sigma, d) \to (\rho, e)$, its composite with the first is

$$(\psi\varphi, (\theta\varphi)\eta).$$

This is the concrete form of the fibred-category formalism used in the text.

To concatenate two such arrows, concatenate the type maps and glue their transformations at the common endpoint:

$$((\varphi, \eta), (\varphi', \eta')) \longmapsto (\varphi * \varphi', \eta * \eta').$$

Both components at the joining object are the identity. Hence the two definitions there agree, and the naturality square for every edge comes from exactly one of the two given naturality squares. The resulting objects remain Σ-paths because concatenation does not change an edge or its direction. A vertical component of a transformation is unrestricted: $P_{a,b}(\tau)$ is a full subcategory on its allowed objects, with the endpoint condition imposed on transformations.

Compatibility with composition is checked at vertices. On the first interval the two possible composites have component $(\theta\varphi)_i\eta_i$; on the second they have the corresponding primed component; at the joining vertex both have the identity. They are therefore equal. The base maps agree by functoriality of concatenation in $\underline{\mathrm{Ty}}$. Thus the formula defines the desired functor on total categories. Equivalently,

$$p_1^*\underline{\mathrm{Ch}_\Sigma}(a,b) \times_{\underline{\mathrm{Ty}}\times\underline{\mathrm{Ty}}} p_2^*\underline{\mathrm{Ch}_\Sigma}(b,c) \cong \underline{\mathrm{Ch}_\Sigma}(a,b) \times \underline{\mathrm{Ch}_\Sigma}(b,c),$$

and the cartesian functor to the pullback along concatenation can be followed by the projection to $\underline{\mathrm{Ch}_\Sigma}(a,c)$. This is the recognition made on page 80.

For associativity, number the vertices of three paths of lengths n, m, l by successive blocks in $[0, n+m+l]$. The first block keeps its indices, the second is shifted by n, and the third by $n+m$.

22. produit fibré par $\pi : \underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}} \longrightarrow \underline{\mathrm{Ty}}$.

Either parenthesization produces those same indices, objects, and arrows. The same statement holds for the components of a glued transformation and for the underlying type map. Consequently both parenthesizations agree as functors, and the length-zero path is a strict unit. With these standard representatives, no later strictification is required.

One diagrammatic slip deserves explicit repair. The amalgamation on page 77 is

$$I_\tau \amalg_{\{1_\tau = 0_{\tau'}\}} I_{\tau'},$$

a pushout over a singleton category. The displayed $\varepsilon = \{0, 1\}$ would glue two pairs of endpoints. Already for two forward one-edge intervals, that different pushout has two parallel arrows between two objects, rather than the two successive arrows of $I_{\Delta^2_+}$. The joining vertex has index n in the numbering $0, \dots, n+m$; the printed $n+1$ uses the wrong index. Neither slip affects the correct componentwise construction.

Les sueurs froides sont terminées!

En résumé, le formalisme des $\underline{\mathrm{Ch}}_\Sigma(X)$ et des $\underline{\mathrm{Ch}}_\Sigma(X; a, b)$ développé jusqu'à présent donne satisfaction complète sur les points suivants :

a) Bon formalisme de composition des chemins (qui était dégueulasse avant).

b) Traitement élégant de la construction des catégories de fractions $\mathcal{M}\Sigma^{-1}$. Mais je m'attends que ce sera moins joli qu'avant, quand il s'agira de remplacer un $(\mathcal{M}, \Sigma)$ par des $(\underbrace{\mathcal{M}(X)}_{=\underline{\mathrm{Hom}}(X,\mathcal{M})}, \Sigma_X)$ (X une catégorie d'indices) `[plutôt` $\underline{\mathrm{Hom}}(X^o, \mathcal{M})$`]`. C'est pour y pallier qu'il me faudra développer des énoncés de comparaison, entre la catégorie $\underline{\mathrm{Ch}}_\Sigma$ construite ici, et d'autres dont le type dans Hot se prête aux calculs de façon plus directe, comme dans l'état premier de la question. De plus, on sera sûrement obligé d'en venir à des Ind-catégories, pour pouvoir suivre la variance de $\mathcal{M}(X)\Sigma_X^{-1}$ pour X variable.

c) Bon formalisme des 'homotopies' et des compositions des homotopies (ce qui est une autre façon de parler des chemins et de leur composition).

p. 54

Commentary · Context [ms. 82] [page 83]

Why the category of diagrams returns

The reservation at the end of this section brings the calculation back to the purpose of a derivator. Localizing $\mathcal{M}$ answers the question at one point. To vary coefficients over an indexing category A, one must also localize

$$\mathcal{M}(A) = \mathrm{Fun}(A^{\mathrm{op}}, \mathcal{M}), \qquad \Sigma_A = \{\eta \mid \eta_a \in \Sigma \text{ for every } a \in A\}.$$

The opposite in the edition's correction records the presheaf convention. It is a variance convention, not an alteration of the localization theorem: the latter applies to either functor category once its convention is fixed.

There is a small but decisive issue of uniformity. An object of a path category for $\mathcal{M}(A)$ is a finite zigzag of natural transformations with one common word of signs over all $a \in A$. An arbitrary family of zigzags in $\mathcal{M}$ may use a different word, or require an unbounded length, as a varies. Thus a description that works perfectly for individual objects still has work to do before it describes localization naturally in the parameter A. The ind-objects considered later organize precisely the freedom to enlarge a finite path type while retaining its functorial meaning.

This interpretation agrees with Grothendieck's subsequent account in his letter to Thomason of 2 April 1991. In §2, printed page 9, he explicitly defines the associated prederivator by $A \mapsto \mathcal{M}(A)[\Sigma_A^{-1}]$, with presheaf diagrams and objectwise denominators; on printed page 8 he emphasizes the transformations between restriction functors as part of the structure (Grothendieck, 1991a). The present concern about types is a concrete instance of that requirement. It asks for controlled variation of the entire diagram theory, beyond the universal property of a single localization.

8 Le lemme d'homotopie pour les morphismes d'intervalles

C'est le suivant :

Lemme 1. *Soient $\tau, \tau' \in \mathrm{Ty}$, et $f, g : \tau \rightrightarrows \tau'$ deux morphismes dans $\underline{\mathrm{Ty}}$. Alors f, g sont homotopes dans la catégorie $\underline{\mathrm{Hom}}\,\underline{\mathrm{int}}(I_\tau, I_{\tau'})$ des morphismes croissants d'intervalles orientés de I_τ dans $I_{\tau'}$.*

Cette catégorie pourrait licitement se désigner par $\underline{\mathrm{Hom}}_{\underline{\mathrm{Ty}}}(\tau, \tau')$, ou $\underline{\mathrm{Hom}}(\tau, \tau')$. C'est une sous-catégorie de la catégorie $\underline{\mathrm{Hom}}(I_\tau, I_{\tau'})$ de *tous* les foncteurs de I_τ dans $I_{\tau'}$, et une sous-catégorie pleine de la catégorie $\underline{\mathrm{Hom}}^\varepsilon(I_\tau, I_{\tau'})$ des foncteurs f qui sont compatibles avec la donnée des sources et des buts, i.e. qui rendent commutatif le diagramme

$$\begin{array}{ccc} & \varepsilon = \{0,1\} & \\ {\scriptstyle (\sigma_\tau, \beta_\tau)} \swarrow & & \searrow {\scriptstyle (\sigma_{\tau'}, \beta_{\tau'})} \\ I_\tau & \xrightarrow{\quad f \quad} & I_{\tau'}. \end{array}$$

Si f, g sont dans cette catégorie (pas forcément morphismes croissants d'ailleurs), une flèche de f dans g est donc la donnée, pour tout $i \in I_\tau$, d'une flèche $f(i) \xrightarrow{w_i} g(i)$ dans $I_{\tau'}$, avec les deux conditions :

a) Si $i \in \{0_\tau, 1_\tau\}$, $w_i = \mathrm{id}$, et

b) le carré commutatif habituel pour une flèche $i \longrightarrow j$ dans I_τ. (Il suffit d'avoir cette compatibilité pour les flèches de transition $\underline{u_i} : i - 1 \longrightarrow i$ dans I_τ $(i \in [1, \log \tau])$.)

Une autre façon de décrire cette catégorie $\underline{\mathrm{Hom}}^\varepsilon$, c'est de dire que c'est la catégorie

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$$\underline{\mathrm{Ch}}_\tau(I_{\tau'}; 0_{\tau'}, 1_{\tau'}) \qquad (\simeq \underline{\mathrm{Hom}}^\varepsilon(I_\tau, I_{\tau'})),$$

mais $\underline{\mathrm{Hom}}_{\underline{\mathrm{Ty}}}(\tau, \tau')$ est la sous-catégorie pleine de cette catégorie formée des *chemins croissants* dans $I_{\tau'}$. Le lemme d'homotopie signifie aussi que cette catégorie $\underline{\mathrm{Hom}}_{\underline{\mathrm{Ty}}}(\tau, \tau')$ est toujours *connexe.* Elle n'est d'ailleurs pas forcément 0-connexe, car elle peut être vide. Notons ici, pour plus tard, la

Proposition 1 : *Soient $\tau, \tau' \in \mathrm{Ty}$, soient r, r' leurs longueurs utiles. Pour que $\mathrm{Hom}_{\underline{\mathrm{Ty}}}(\tau, \tau')$ soit non vide, il faut et il suffit que l'une ou l'autre des deux conditions suivantes soit satisfaite :*

a) $r \geq r' + 1$

b) $r = r'$, *et (si $r \neq 0$) τ et τ' ont même direction initiale (de sorte que $\tau_{\text{réd}} = \tau'_{\text{réd}}$).*

p. 55

(Démonstration ± immédiate.)

Notons que si τ, τ' n'ont pas même direction initiale, alors tout morphisme $\tau \longrightarrow \tau'$ est forcément dégénéré sur le premier intervalle critique de I_τ. On aura donc $\tau = \Delta_\varepsilon^{l_0} \cdot \tau_0$ (où l_0 est la longueur du premier intervalle critique, et où τ_0 a `[la]` direction initiale opposée de celle de τ), et un isomorphisme

$$\underline{\mathrm{Hom}}(\tau, \tau') \simeq \underline{\mathrm{Hom}}(\tau_0, \tau'),$$

où cette fois τ_0, τ' ont même direction initiale. On supposera donc désormais que *la direction initiale de τ, τ' est la même.* La condition $\underline{\mathrm{Hom}}(\tau, \tau') \neq \emptyset$ équivaut maintenant à $\underline{r \geq r'}$, ce que nous pouvons supposer. Je vais alors décrire un élément canonique F_0 de $\underline{\mathrm{Hom}}(\tau, \tau')$, ainsi. Soient

[page 85]

$$I_1, I_2, \ldots, I_r$$

les intervalles critiques de I_τ, et

$$I'_1, I'_2, \ldots, I'_{r'} \qquad (r' \leq r)$$

ceux de $I_{\tau'}$. Le foncteur F_0 proviendra de morphismes d'intervalles orientés (comme

$$\left\{\begin{array}{lcl} I_1 & \xrightarrow{f_1} & I'_1 \\ I_2 & \xrightarrow{f_2} & I'_2 \\ & \cdots & \\ I_{r'} & \xrightarrow{f_{r'}} & I'_{r'} \end{array}\right. \quad)$$

et sera donc forcément trivial sur les intervalles critiques I_α pour $r' < \alpha \leq r$ (il y en a si et seulement si $r > r'$). Comme I_τ, $I_{\tau'}$ ont même direction initiale, il s'ensuit que pour $\alpha \in [1, r']$, I_α et I'_α ont même signature (à savoir $\varepsilon_\alpha = (-1)^{\alpha+1}\varepsilon_1$, où ε_1 est la signature commune de I_1 et I'_1). Donc

$$I_\alpha = \Delta^{l_\alpha}_{\varepsilon_\alpha}, \qquad I'_\alpha = \Delta^{l'_\alpha}_{\varepsilon_\alpha},$$

et on définit f_α comme

$$f_\alpha = \mathfrak{f}_{\varepsilon_\alpha, l'_\alpha} * \mathrm{triv}_\alpha,$$

où

$$\mathfrak{f}_{\varepsilon_\alpha, l'_\alpha} : \Delta_{\varepsilon_\alpha} \longrightarrow \Delta^{l'_\alpha}_{\varepsilon_\alpha}$$

et où

$$\mathrm{triv}_\alpha : \Delta^{l_{\alpha-1}}_{\varepsilon_\alpha} \longrightarrow 1.$$

En d'autres termes, pour $\varepsilon = +1$ (p. ex.) on doit définir un morphisme des intervalles type $\underbrace{\mathbf{\Delta}^+_{l_\alpha}}_{=[0,l_\alpha]} \longrightarrow \underbrace{\mathbf{\Delta}^+_{l'_\alpha}}_{=[0,l'_\alpha]}$, et on pose (avec les numérotations internes à $\mathbf{\Delta}_{l_\alpha}$, $\mathbf{\Delta}_{l'_\alpha}$)

$$f_\alpha(0) = 0, \qquad f_\alpha(i) = l'_\alpha \ \text{ pour } i \in [1, l_\alpha].$$

Commentary · Exposition [ms. 84]

The two orders in the interval lemma

Grothendieck now wants the choice of a type map to disappear after homotopy. The statement is about a particularly constrained mapping category, and its constraints explain both the nonemptiness criterion and the proof. Number the source and target vertices by $[0, n]$ and $[0, m]$, and write their edge signs as ε_i and δ_j. An object is a nondecreasing function $F : [0, n] \to [0, m]$ satisfying $F(0) = 0$, $F(n) = m$, and

$$F(i-1) < F(i) \quad \Longrightarrow \quad \delta_{F(i-1)+1} = \cdots = \delta_{F(i)} = \varepsilon_i.$$

The condition says that one source edge can traverse a run of target edges only when their directions agree. It cannot cross a change of direction in the target. An arrow $F \to G$ is instead defined by the categorical order of the target: for every i, there must be an arrow $F(i) \to G(i)$ in $I_{\tau'}$. Such arrows, when present, are unique, and their naturality is automatic because the target is a poset. In particular the ordering of maps is generally different from pointwise numerical comparison.

Let r, r' count the maximal constant-sign runs. Each target run must be traversed using at least one source edge of the same sign. Distinct target runs require distinct source runs in the same order, because their signs alternate and a source run cannot pass a target turning vertex. This proves $r \geq r'$ when the initial signs agree. If the initial signs disagree, the whole first source run is forced to be constant at the initial target vertex, and the necessary inequality is $r - 1 \geq r'$. Conversely, under either inequality, map the first edge of each of the earliest matching source runs across the corresponding whole target run, collapse its remaining edges, and collapse every unused source run at the final target vertex. This constructs an admissible map and proves Proposition 1. If the target has no edges, there is exactly one map from every source; if the source has no edges and the target has an edge, there is none.

These observations also clarify the word "connected" in the text. Its assertion is that any two objects are connected by a finite zigzag of transformations; it allows the mapping category to be empty. For a nonempty mapping category, the lemma asserts that its nerve is connected.

Grothendieck then asks whether a stronger contraction exists. The following example separates that question from the proposed terminal-object argument, and the development after it gives a uniform contraction.

L'idée donc, c'est d'être dans les dispositions d'un voyageur qui, tenu à un plan de voyage par étapes décrit par τ, voudrait arriver aussi vite que possible à $1_{\tau'}$, en partant de $0_{\tau'}$. Il va dès la première étape $\Delta_\varepsilon \subseteq I_1$, se porter jusqu'au bout de I_1', regrettant ne pas pouvoir aller plus loin (à cause du plan de voyage). Il doit rester inactif jusqu'à ce qu'il soit arrivé au moment marqué par l'extrémité finale de I_1 (si celle-ci est au delà de celle de sa première flèche de transition, i.e. si $l_1 > 1$), puis il fait son deuxième grand pas jusqu'à l'extrémité terminale de I_2' etc. Il peut espérer, à chaque moment de son voyage, être en avance (ou du moins pas en retard) sur les autres voyageurs, qui auraient suivi de tactiques de progression différentes. Si cet espoir était justifié, cela signifierait que F_0 serait un plus grand élément de $\underline{\mathrm{Hom}}(\tau, \tau')$ (**NB** toutes ces catégories $\underline{\mathrm{Hom}}$ entre intervalles sont 'ordonnées', i.e. il y a au plus une flèche entre deux objets), i.e. un objet final. Cela impliquerait que $\underline{\mathrm{Hom}}(\tau, \tau')$ serait même contractile - chose peut-être vraie, mais sûrement pas de façon si simpliste je crains. Je me suis d'ailleurs canulé : F_0 est sans doute bien un 'plus grand' élément, mais pour l'ordre défini par la numérotation des sommets de $I_{\tau'}$ (donc $F \leq G$ si et seulement si $F(i) \leq G(i)$ $\forall i \in I_\tau$), et nullement pour l'ordre défini par la structure de catégorie de $\underline{\mathrm{Hom}}(\tau, \tau')$.

p. 56

[page 86]

L'idée pour prouver la connexité de $\underline{\mathrm{Hom}}(\tau, \tau')$, c'est de ramener tout $F \in \mathrm{Hom}(\tau, \tau')$, par une suite d'homotopies simples, à la forme normale. Rappelons que F est donné par une suite croissante de flèches de transition

[page 87]

$$\underline{u}_{i_1},\ \underline{u}_{i_2},\ \dots\ ,\ \underline{u}_{i_s}, \qquad i_1 < i_2 < \cdots < i_s,$$

et une suite d'entiers ≥ 1

$$\lambda_1,\ \lambda_2,\ \dots\ ,\ \lambda_s$$

avec la condition que

$$(*) \qquad \Delta_{\varepsilon_{i_1}}^{\lambda_1} \Delta_{\varepsilon_{i_2}}^{\lambda_2} \cdots \Delta_{\varepsilon_{i_s}}^{\lambda_s} = \tau'.$$

Les $\underline{u}_{i_\alpha}$ sont les flèches de transition de I_τ *critiques* pour F, i.e. non dégénérées pour F, et λ_α est la *partie* de $\underline{u}_{i_\alpha}$ pour F. Le foncteur normalisé ('pressé') F_0 est distingué par les conditions suivantes a) et b) `[et c)]`. Tout d'abord :

Commentary · Examples [ms. 86]

A quickest map that is neither initial nor terminal

The normal form reaches each target turning vertex as early as the source permits. It maximizes numerical progress, but the categorical direction reverses whenever the target changes direction. A four-object mapping category makes the distinction visible without any large diagram.

Take

$$\tau = \Delta_+^2 \Delta_-^2, \qquad \tau' = \Delta_+ \Delta_-.$$

The target is $0 \to 1 \leftarrow 2$. Every increasing endpoint-preserving functor has vertex list

$$F_{a,b} = (0, a, 1, b, 2), \qquad a \in \{0, 1\}, \quad b \in \{1, 2\}.$$

Indeed the middle source vertex must map to the target turning vertex 1: it cannot lie below 1, since the remaining negative edges could not reach 2, and it cannot lie above 1, since the initial positive edges could not reach it. The choices of a, b are then unrestricted in the displayed ranges. The exact arrow relation is

$$F_{a,b} \longrightarrow F_{a',b'} \quad \Longleftrightarrow \quad a \leq a' \ \text{ and } \ b \geq b',$$

where the inequalities on the right are numerical. The reversal of the second inequality is forced by $2 \to 1$ in the target.

The quickest functor is $F_0 = F_{1,2}$. It is numerically greater than or equal to every other functor, but the terminal object of this mapping category is $F_{1,1}$ and its initial object is $F_{0,2}$. In particular $F_{1,1}$ has no arrow to F_0, and F_0 has no arrow to $F_{0,2}$. The objection on manuscript page 86 is therefore exact, even in this small case.

Nevertheless a contraction to F_0 is immediate once we normalize one target run at a time. Define $R(F_{a,b}) = F_{1,b}$. There is a natural transformation $\mathrm{id} \Rightarrow R$, and another $\mathrm{const}_{F_0} \Rightarrow R$. Hence

$$\mathrm{id} \implies R \impliedby \mathrm{const}_{F_0}.$$

This is the pattern of the general argument: each step advances across only one constant-sign target run, so all its components have one common categorical direction. The entire numerical maximum need not define a single transformation.

Commentary · Development (commentators') [ms. 86]

A uniform contraction of the interval mapping category

Grothendieck's proposed normal form can be made into a contraction by advancing across one target run at a time. This answers the question on page 86 for the increasing endpoint-preserving mapping categories used in this section. We did not find this particular argument in the literature checked for the present commentary: the source's §8 proves connectedness, while the institutional abstract of Fernandes et al. (2019) concerns order-preserving self-transformations of a fence, without these simultaneous endpoint and linear-order constraints. We checked that abstract and metadata; its manuscript could not be retrieved. The result below is an elementary supplement, without a claim of priority.

Proposition (run-by-run contraction).[ours] *Let $P(\tau, \tau')$ be the category of nondecreasing endpoint-preserving functors $I_\tau \to I_{\tau'}$, with endpoint-fixed natural transformations. If $P(\tau, \tau')$ is nonempty, its identity functor is connected to the constant functor at the quickest map by a finite zigzag of natural transformations. In particular its nerve is contractible. One can use at most r' transformations, where r' is the number of target runs.*

Proof. We induct on r'. When the target has no edges, every functor is the same constant functor and $P(\tau, \tau')$ is the terminal category. Assume that the target is nontrivial. The source is then nontrivial too. If its initial sign differs from the target's, every map is constant on the entire first source run, as proved above. Every transformation is the identity at those vertices. Restriction past that run is consequently an isomorphism of mapping categories. We may use this isomorphism to assume that the first source and target runs have the same sign ε.

Let the first source run have vertices $0, \dots, a$, and the first target run have vertices $0, \dots, b$, where $a, b \geq 1$. The numerical clamping map

$$h : [0, m] \longrightarrow [0, m], \qquad h(j) = \max\{j, b\},$$

is a functor of the target interval category. On its first run it collapses every arrow to an identity at b; on every later edge it is the identity. These checks on generating edges prove functoriality, including at the turning vertex b.

For each $F \in P(\tau, \tau')$, define

$$(RF)(0) = 0, \qquad (RF)(i) = h(F(i)) \quad (1 \leq i \leq n).$$

The first source run cannot pass the first target turning vertex. Thus $F(i) \leq b$ for $0 \leq i \leq a$, and $(RF)(i) = b$ for $1 \leq i \leq a$. The new first edge is an allowed traversal from 0 to b of sign ε; all later edges are obtained by applying the functor h to their old images. This proves that RF is a functor. Its vertex map remains nondecreasing, it fixes both endpoints, and hence it is again an object of $P(\tau, \tau')$.

For a transformation $F \Rightarrow G$, apply h to its components at vertices $i \geq 1$ and use the identity at vertex 0. The resulting components define a transformation $RF \Rightarrow RG$. Indeed both functors are already valid interval functors, their target is thin, and each component exists by functoriality of h. Identities and composites are preserved componentwise. We have therefore constructed an endofunctor R of $P(\tau, \tau')$.

The comparison with the identity has one uniform direction. If $\varepsilon = +$, the unique arrows $j \to h(j)$ give $\mathrm{id}_P \Rightarrow R$; if $\varepsilon = -$, the arrows $h(j) \to j$ give $R \Rightarrow \mathrm{id}_P$. At vertex 0 the component is

the identity, and at the final vertex it is also the identity. Naturality in the source interval and in F follows from thinness of the target and of the mapping category. Only values in the first target run have changed, which is why this comparison exists despite the failure of a global numerical-maximum argument.

Let $Q \subseteq P$ be the full subcategory of maps with values $0, b, \dots, b$ on the first source run. The endofunctor R takes values in Q and restricts to the identity there. Delete the first source run, retain its last vertex as the new initial vertex, and do the same for the target. Restriction identifies Q with the endpoint-preserving mapping category of these two remaining intervals. To verify this identification on morphisms, observe that every component over the deleted prefix is forced to be an identity; restriction and the prescribed extension are inverse functors. The remaining target has $r' - 1$ runs. Since Q contains RF for every F, it is nonempty, so induction applies.

Compose the inductive zigzag on Q with $R : P \to Q$ and the inclusion $Q \hookrightarrow P$. Adjoining the single comparison between id_P and R gives a zigzag to the constant functor at the map that crosses the first target run on the first available source edge and then follows the inductively quickest map. This is exactly Grothendieck's normal form, after the forced initial deletion when necessary. The number of comparisons increases by one, hence is at most r'. A natural transformation induces a simplicial homotopy after taking nerves, so this categorical contraction contracts the nerve as asserted. □

The proposition supplies more than a path joining two chosen maps: its choices of intermediate functors are simultaneous and natural in the object of the mapping category. It concerns the increasing subcategory. No conclusion about all endpoint-preserving functors, without numerical monotonicity, is being used here.

a) La suite des signes $\varepsilon_{i_1}, \varepsilon_{i_2}, \dots, \varepsilon_{i_s}$ est 'alternée', i.e. $\varepsilon_{i_\alpha} \neq \varepsilon_{i_{\alpha+1}}$ ($\alpha \in [1, s-1]$). (Cela traduit le fait que le voyageur pressé ne fait pas en plusieurs pas ce qu'il pourrait faire en un seul.) Comme le premier de ces signes est forcément $\varepsilon'_1 = \varepsilon_1$, cela signifie que la suite des signes est la même que celle des ε_1, ε_2, ..., $\varepsilon_{r'}$ (à la longueur près). Mais mieux, que l'expression $(*)$ de τ' est son unique expression réduite, donc que $s = r'$ (longueur utile de τ'), et que les λ_i sont les longueurs l'_i des intervalles critiques de $I_{\tau'}$.
Soit donc $\alpha \in [1, s = r']$, et considérons la flèche de transition F-critique $\underline{u}_{i_\alpha}$, soit j_α l'indice marquant l'intervalle critique de I_τ contenant la partie de $\underline{u}_{i_\alpha}$ dans I_τ, p. 57 [page 88]

$$\underline{u}_{i_\alpha} \in \mathrm{Fl}(I_{j_\alpha}),$$

alors F induit des morphismes (croissants) d'intervalles orientés

$$F_\alpha = F|I_{j_\alpha} : I_{j_\alpha} \longrightarrow I'_\alpha$$

et est trivial sur les autres intervalles critiques (ceux pas de la forme I_{j_α}). La donnée des F_α détermine donc F sur les I_{j_α}, donc sur *tous* les I_j, donc partout (par l'opération $*$ de composition par recollement). De plus la condition a) assure que chaque F_α est de nature assez particulière : une seule flèche de transition critique $\underline{u}_{i_\alpha}$. De plus, pour qu'on ait $F = F_0$, il faudra :

b) $\forall \alpha \in [1, r']$, $\underline{u}_{i_\alpha}$ est la *première* flèche de transition de l'intervalle critique I_{j_α} qui la contient. Et enfin :

c) La suite des intervalles critiques de I_τ

$$I_{j_1},\ I_{j_2},\ \dots\ ,\ I_{j_{r'}} \qquad \text{(de signes alternés } \varepsilon_1,\ -\varepsilon_1,\ \varepsilon_1,\ \dots)$$

associés aux flèches F-critiques, n'est autre que celle

$$I_1,\ I_2,\ \dots\ ,\ I_{r'}$$

des r' premiers intervalles critiques de I_τ, i.e.

$$(j_1, j_2, \dots, j_{r'}) = (1, 2, \dots, r'). \qquad \text{[page 89]}$$

Ceci vu, l'idée naturelle pour 'déformer' F en F_0, c'est de trouver une suite d'homotopies qui transforment F de façon à lui faire acquérir successivement les caractéristiques a), b), c) du voyageur pressé. Au besoin, pour aller plus vite au but, on utilisera l'hypothèse de récurrence sur $n = \log \tau$, que le théorème de connexité est déjà prouvé quand I est de longueur $\leq n-1$. (NB Ce théorème est trivial pour $n \leq 1$, car alors les catégories $\underline{\mathrm{Hom}}(\tau, \tau')$ sont toutes ou bien vides, ou bien ponctuelles.)

1) On veut assurer a). Pour ceci, on utilise l'hypothèse de récurrence, pour remplacer un groupe de `[mot illisible]` flèches critiques consécutives de même signe, par une seule flèche critique. Cette méthode ne foire que si les flèches critiques sont toutes de même signe ε et incluent $\underline{u}_1$ et $\underline{u}_n$. Mais l'hypothèse de récurrence appliquée à $I = [1, n-1] \subseteq I_\tau$ nous ramène au cas où il n'y a que deux flèches critiques $\underline{u}_1$ et $\underline{u}_n$. Si le premier intervalle critique n'est pas réduit à Δ_ε, i.e. soit de longueur 2, alors l'hypothèse de récurrence appliquée à $I = [1, n] \subseteq I_\tau$ nous ramène au cas où les deux flèches critiques sont $\underline{u}_1$ et $\underline{u}_2$, et on est alors ramené au cas où $\boxed{I = \Delta^2_\varepsilon,\ \tau' = \Delta^2_\varepsilon.}$ Sinon, au cas où $\boxed{\tau = \Delta_\varepsilon \Delta^{l_2}_{-\varepsilon} \Delta_\varepsilon,\ \tau' = \Delta^2_\varepsilon,}$ et où les deux flèches critiques sont encore la première et la dernière, et $\tau' = \Delta^2_\varepsilon$. Ce sont là les deux cas irréductibles, qu'il faudra traiter à part.

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Ces cas admis, on a donc moyen pour définir F de façon à satisfaire a). Ensuite on veut :

2) Assurer que la suite des intervalles critiques I_{j_1}, I_{j_2}, …, $I_{j_{r'}}$ qui contient les flèches critiques $\underline{u}_{i_1}$, …, $\underline{u}_{i_{r'}}$, soit la suite I_1, …, $I_{r'}$. C'est encore immédiat par l'hypothèse de récurrence, le seul cas irréductible étant celui où $r' = 1$ (une seule flèche critique), et où il s'agit de faire descendre cette flèche critique jusqu'à $\underline{u}_1$. Si I_1 est de longueur > 1, alors on fait descendre en $\underline{u}_2$ par hypothèse de récurrence, et le cas irréductible est celui où $\boxed{\tau = \Delta^2_\varepsilon,\ \tau' = \Delta_\varepsilon,}$ une flèche critique $\underline{u}_2$, la faire descendre en $\underline{u}_1$. Si par contre I_1 `[est]` de longueur 1, le cas irréductible est $\boxed{\tau = \Delta_\varepsilon \Delta^{l_2}_{-\varepsilon} \Delta_\varepsilon,\ \tau' = \Delta_\varepsilon,}$ une flèche critique $\underline{u}_{l_2+2}$, la faire descendre en $\underline{u}_1$.

Sous réserve de traiter les deux nouveaux cas irréductibles, on peut déformer F en un foncteur qui satisfait à a) et à c). Reste à normaliser chacun des $F_\alpha : I_\alpha \longrightarrow I'_\alpha$ ($\alpha \in [1, r']$). Donc on est ramené à $\Delta^n_\varepsilon \longrightarrow \Delta_\varepsilon$: une seule flèche critique $\underline{u}_\alpha$, $\alpha \in [1, n]$, à faire descendre en $\underline{u}_1$. Si $n \geq 2$, par hypothèse de récurrence elle descend en $\underline{u}_2$, cas supposé traité en 2°). Donc on gagne, sous réserve de traiter les 4 cas irréductibles.

A) Cas $\tau = \Delta^2_\varepsilon$, $\tau' = \Delta^2_\varepsilon$ ou Δ_ε. [page 92]

La catégorie $\underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \Delta_\varepsilon)$ a deux éléments, les morphismes $\Delta^2_\varepsilon \longrightarrow \Delta_\varepsilon$ correspondent aux flèches de transition de Δ^2_ε ($= \mathbf{\Delta}^+_2$ ou $\mathbf{\Delta}^-_2$). Dans le cas $\varepsilon = +1$, ce sont les deux applications de $\{0, 1, 2\}$ dans $\{0, 1\}$, qui appliquent 0 en 0, 2 en 1, et 1 soit en 1 (F_0), soit en 1 `[plutôt en 0]` (F_1).

$$0 \longrightarrow 1 \longrightarrow 2$$

$$0 \longrightarrow 1$$

On a évidemment une flèche $F_1 \longrightarrow F_0$ (le voyageur pressé arrive plus vite …). Similaire pour $\varepsilon = -1$,

$$\underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \Delta_\varepsilon) \simeq \mathbf{\Delta}^1, \qquad \text{catégorie 0-connexe.}$$

La catégorie $\underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \Delta^2_\varepsilon)$ a trois éléments, s'identifiant à l'ensemble des parties non vide de l'ensemble $\mathrm{Tr}(\tau)$ à deux éléments des flèches de transition $\underline{u}_1$, $\underline{u}_2$ de τ : il y a $F_{\underline{u}_1} = F_0$, $F_{\underline{u}_2}$, $F_{\underline{u}_1, \underline{u}_2}$. Si $\varepsilon = +1$, il s'agit des applications croissantes (sans plus) de $\{0, 1, 2\}$ dans $\{0, 1, 2\}$, appliquant 0 en 0, 2 en 2, et 1 en ce qu'elle voudra.

Donc $1 \longmapsto 2$ correspond à F_0 p. 59

$1 \longmapsto 1$ correspond à F_1

$1 \longmapsto 0$ correspond à F_2

et on a bien sur $F_2 \longrightarrow F_1 \longrightarrow F_0$,

$$\underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \Delta^2_\varepsilon) \simeq \mathbf{\Delta}^2, \qquad \text{catégorie 0-connexe.}$$

Dans ces deux cas, deux éléments de $\mathrm{Hom}(\tau, \tau')$ sont reliés par une homotopie *simple.*
B) Cas $\tau = \Delta_\varepsilon \Delta^l_{-\varepsilon} \Delta_\varepsilon$, $\tau' = \Delta^2_\varepsilon$ ou Δ_ε.
Notons que l'on a, pour $\tau, \tau' \in \underline{\mathrm{Ty}}$ quelconque, un isomorphisme de catégories [page 93]

$$\underline{\mathrm{Hom}}(\tau, \tau') \longrightarrow \underline{\mathrm{Hom}}(\tau'^o, \tau^o),$$

exprimant simplement que

$$\tau \longmapsto \tau^o$$

est un `[2]`-foncteur de $\underline{\mathrm{Ty}}$ dans $\underline{\mathrm{Ty}}$, et même un automorphisme de cette `[2]`-catégorie. Donc dire que $\underline{\mathrm{Hom}}(\tau, \tau')$ est connexe revient à dire que $\underline{\mathrm{Hom}}(\tau^o, \tau'^o)$ l'est. Donc on peut supposer ci-dessus que $\varepsilon = +1$ (et pareil dans A!).

$$\tau : \quad \longrightarrow \underbrace{\longleftarrow \longleftarrow \cdots \longleftarrow}_{l} \longrightarrow$$

$$\tau' : \quad \longrightarrow \longrightarrow \quad \text{ou} \quad \longrightarrow .$$

Mais tous les morphismes $\tau \longrightarrow \tau'$ sont triviaux sur l'intervalle critique I_2 (rétrograde), ce qui implique que

$$\underline{\mathrm{Hom}}(\tau, \tau') \xleftarrow{\sim} \underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \tau'),$$

via le foncteur de contraction canonique

$$I_\tau \longrightarrow I_{\Delta^2_\varepsilon} = \mathbf{\Delta}^2.$$

Or dans A on a vu que $\underline{\mathrm{Hom}}(\Delta^2_\varepsilon, \tau')$ est connexe (pour τ' de la forme Δ_ε ou Δ^2_ε).
On a gagné!

Commentary · Exposition [ms. 93] p. 60

The four small cases and the completed proof [page 94]

Grothendieck's induction reduces the normalization to two monotone cases and two cases with one intervening run of the opposite sign. The local calculations are sound; the reversal formula printed at the start of case B needs its arguments restored to their original order. Reversal of the numbered interval gives

$$P(\tau,\tau') \cong P(\tau^o,\tau'^o), \qquad F \longmapsto r_{\tau'} F r_\tau^{-1},$$

where $r_\tau : I_\tau \to I_{\tau^o}$ reverses the vertex numbering. It is a covariant isomorphism of the underlying categories, since the reversed word also changes every sign. The displayed $P(\tau'^o,\tau^o)$ in the edition would reverse the direction of a functor and is not this operation. The next sentence in the source already has the correct order.

For $\tau = \Delta^2_+$ and target Δ^k_+ with $k = 1$ or 2, a map is determined by $F(1) = j$, $0 \le j \le k$. There is a unique transformation from j to j' exactly when $j \le j'$. Thus the two mapping categories are the chains [1] and [2]. With negative signs the order reverses, producing isomorphic chains. In particular any two objects are joined by one simple homotopy. These are the calculations on pages 92–93, including the edition's correction of the repeated value 1 in the first example.

For $\tau = \Delta_+\Delta^l_-\Delta_+$ and a positive target, every edge of the negative middle run must collapse: a nondecreasing map cannot carry a negative source edge to a nonconstant positive target arrow. It therefore factors uniquely through the contraction $I_\tau \to I_{\Delta^2_+}$. The factorization also holds for transformations, because all components over the collapsed middle run coincide. This proves the displayed isomorphism of mapping categories and reduces the remaining cases to the preceding chains. Two small corrections clarify the longer induction on page 90. A first source run that is not a singleton has length at least two; the printed "length two" omits that qualification. More substantially, to concentrate all activity except the final edge, the inductive subinterval must be $[0, n-1]$, not the printed $[1, n-1]$. The latter fixes both interior boundary values, and cannot eliminate a nonzero traversal between them: the identity on a positive three-edge interval already demonstrates the problem. On the prefix $[0, n-1]$, induction instead leaves its first edge active and its remaining edges constant, while the final edge is unchanged. The following use of $[1, n]$ then moves the tail activity to the second edge when the first run has length at least two. With these corrections the stated reductions have their intended meaning. The run-by-run contraction above gives a full alternative proof of their connectedness conclusion.

Commentary · Later developments [ms. 93]

The path model and simplicial localization

The interval lemma makes changes of parameter homotopically harmless in a precise local sense. It is natural to compare the resulting path constructions with Dwyer–Kan localization, but the categories being compared must first be specified. In the present text, $\underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b)$ is full on the diagrams whose backward edges lie in Σ; its interior vertical transformations are arbitrary arrows of X. In the hammock construction, the vertical arrows are required to lie in the chosen weak-equivalence subcategory as well. These prescriptions agree when $\Sigma = \mathrm{Fl}(X)$, and can differ for a proper subcategory of denominators.

Dwyer and Kan (1980a, Definition 2.1, Proposition 2.2 and Proposition 3.1) construct the hammock mapping simplicial sets, compare them with simplicial localization, and identify their components with ordinary localization. Their definition also permits strict composition by concatenation followed by reduction. This gives a useful historical comparison: their work appeared in 1980, before these notes of 1990. It supplies an established construction retaining homotopical information about fractions, but its comparison theorem does not identify the nerve of the particular full path category in this manuscript without an additional argument.

Here the elementary result is already valuable on its own. It says that a category of allowed parameter maps carries a contraction whenever it has an object. The next section transports natural transformations between such maps through precomposition on path diagrams. That step explains why Grothendieck needed the homotopies to remain inside the increasing endpoint-preserving category, rather than merely to exist between maps of the underlying realizations.

9 Théorème d'homotopie pour les catégories de chemins

Soient X une catégorie, $\Sigma \subseteq \mathrm{Fl}(X)$ `[un]` ensemble admissible de flèches dans X, $a, b \in \mathrm{Ob}\, X$. Considérons pour $\tau \in \mathrm{Ty}$ variable, les catégories

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(X), \qquad \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b),$$

qui forment les fibres des catégories Cat-fibrées

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma}(X) & \xleftarrow{\text{incl.}} & \underline{\mathrm{Ch}}_{\Sigma}(X; a, b) \\ & \searrow \quad \swarrow & \\ & \underline{\mathrm{Ty}} & \end{array}$$

sur la catégorie $\underline{\mathrm{Ty}}$. Ainsi, pour $\tau, \tau' \in \mathrm{Ty}$, et pour toute

$$\varphi : \tau \longrightarrow \tau',$$

on a un carré commutatif d'homomorphismes

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) & \xleftarrow{\text{incl.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X; a, b) \\ {\scriptstyle \varphi_X^*}\downarrow & & \downarrow{\scriptstyle \varphi_{a,b}^*} \\ \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) & \xleftarrow{\text{incl.}} & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b), \end{array}$$

avec transitivité pour φ variable (composition $\psi \circ \varphi$ de flèches dans $\underline{\mathrm{Ty}}$).

En fait, pour φ variable dans $\underline{\mathrm{Hom}}(\tau, \tau')$, les foncteurs images inverses φ_X^*, $\varphi_{a,b}^*$ dépendent *fonctoriellement* de φ; i.e. on a des *foncteurs*

$$(*) \qquad \underline{\mathrm{Hom}}(\tau, \tau') \longrightarrow \begin{cases} \underline{\mathrm{Hom}}(\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X), \underline{\mathrm{Ch}}_{\Sigma,\tau}(X)) \\ \underline{\mathrm{Hom}}(\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X; a, b), \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b)) \end{cases}$$

p. 61

Commentary · Exposition [ms. 94]

[page 95]

Why a homotopy of types acts on every path

Grothendieck now uses the interval calculation to remove a choice from the theory: the operation induced on paths should depend on the source and target types, rather than on the particular subdivision or collapse chosen between them. The relevant assertion concerns natural transformations, not merely the induced maps of connected components. This stronger form is what will allow a diagram of categories to descend to a diagram in Cat hot.

Write $P_\tau = \underline{\mathrm{Ch}}_{\Sigma,\tau}(X)$ and let $\mathcal{T}(\tau,\tau')$ be the category of endpoint-preserving type maps used in the preceding section. If $\eta : \varphi \Rightarrow \psi$ is a morphism of this category and $F : I_{\tau'} \to X$ is a permitted path, the transformation

$$F\eta : F\varphi \Longrightarrow F\psi$$

is a morphism in P_τ. Indeed, both its objects are permitted paths because φ and ψ carry backward arrows to backward arrows or identities. The fibre P_τ is the *full* subcategory of the functor category on these objects: the components of $F\eta$ need not belong to Σ. Naturality in F is precisely the interchange law for natural transformations. Thus whiskering gives a functor

$$\mathcal{T}(\tau,\tau') \longrightarrow \underline{\mathrm{Hom}}(P_{\tau'}, P_\tau), \qquad \varphi \longmapsto \varphi^*.$$

This is the elementary two-dimensional mechanism behind the theorem.

The endpoint condition does a second piece of work. For a transformation of type maps its two endpoint components are identities, so $(F\eta)_0 = \mathrm{id}_{F(0)}$ and $(F\eta)_1 = \mathrm{id}_{F(1)}$. The displayed functor therefore lands in the category of functors and vertical transformations over $X \times X$. Restriction to the strict fibre at (a,b) consequently preserves the entire homotopy. An arbitrary homotopy of functors over an unspecified base would not have this property. The words "au dessus de X" on manuscript page 97 should here be understood as "over $X \times X$", as the following diagram makes explicit.

Finally, connectedness of $\mathcal{T}(\tau,\tau')$, proved by the interval homotopy lemma, supplies a finite zigzag from φ to ψ. Applying the whiskering functor supplies the required finite zigzag of natural transformations. Taking nerves turns each transformation into a simplicial homotopy. No assertion that the individual pullback functors are equivalences is made: the theorem compares parallel operations, and this is exactly what the next construction needs.

Cela provient de l'inclusion

$$\underline{\mathrm{Hom}}(\tau,\tau') \hookrightarrow \underline{\mathrm{Hom}}^{\varepsilon}(I_\tau, I_{\tau'})$$

et de la description des catégories $\underline{\mathrm{Ch}}_\tau$, $\underline{\mathrm{Ch}}_{\tau'}$ comme catégories de foncteurs, telle

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \simeq \underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma}(I_\tau, X),$$

où Σ_τ désigne l'ensemble des flèches dans I_τ qui sont soit inversibles, soit *rétrogrades.* (Cet ensemble de flèches est admissible, car stable par composition.) On désigne par $\underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma}$ la sous-catégorie pleine de $\underline{\mathrm{Hom}}$ formée des foncteurs qui appliquent Σ_τ dans Σ. Or de façon générale, si

$$(I,\Sigma_I), \quad (I',\Sigma_{I'}), \quad (X,\Sigma)$$

sont trois catégories, munies d'ensembles admissibles de flèches, considérant les foncteurs $\varphi : I \longrightarrow I'$ avec $\varphi(\Sigma_I) \subseteq \Sigma_{I'}$, alors le foncteur familier

$$\underline{\mathrm{Hom}}(I,I') \longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}(I',X), \underline{\mathrm{Hom}}(I,X))$$

(transposé du foncteur de composition

$$\begin{array}{ccccc} \underline{\mathrm{Hom}}(I,I') & \times & \underline{\mathrm{Hom}}(I',X) & \longrightarrow & \underline{\mathrm{Hom}}(I,X) \\ (\varphi & , & F') & \longmapsto & F'\circ\varphi \quad), \end{array}$$

se généralise en un foncteur

$$(*) \qquad \underline{\mathrm{Hom}}_{\Sigma_I,\Sigma_{I'}}(I,I') \longrightarrow \underline{\mathrm{Hom}}(\underline{\mathrm{Hom}}_{\Sigma_{I'},\Sigma}(I',X), \underline{\mathrm{Hom}}_{\Sigma_I,\Sigma}(I,X)),$$

transposé du foncteur de composition

$$\underline{\mathrm{Hom}}_{\Sigma_I,\Sigma_{I'}}(I,I') \times \underline{\mathrm{Hom}}_{\Sigma_{I'},\Sigma}(I',X) \longrightarrow \underline{\mathrm{Hom}}_{\Sigma_I,\Sigma}(I,X)$$

induit par $(*)$.

Comme toute

$$\varphi : \tau \longrightarrow \tau' \qquad \text{[page 96]}$$

induit

$$\varphi : I_\tau \longrightarrow I_{\tau'},$$

transformant flèches rétrogrades en flèches identiques ou rétrogrades, i.e.

$$\varphi(\Sigma_\tau) \subseteq \Sigma_{\tau'},$$

on trouve le premier des deux foncteurs $(*)$ annoncés page 94, qui est associé par transposition au foncteur γ p. 62

$$(*) \qquad \begin{array}{ccc} \underline{\mathrm{Hom}}(\tau,\tau') \times \underbrace{\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X)}_{\simeq \underline{\mathrm{Hom}}_{\Sigma_{\tau'},\Sigma}(I_{\tau'},X)} & \xrightarrow{\quad\gamma\quad} & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \\ {\scriptstyle \mathrm{can.}}\downarrow & & \| \\ \underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma_{\tau'}}(I_\tau,I_{\tau'}) \times \underline{\mathrm{Hom}}_{\Sigma_{\tau'},\Sigma}(I_{\tau'},X) & \xrightarrow{\quad\gamma_0\quad} & \underline{\mathrm{Hom}}_{\Sigma_\tau,\Sigma}(I_\tau,X) \end{array}$$

déduit du foncteur composition γ_0 dans ce diagramme : $\gamma = \gamma_0 \circ \mathrm{can}$. D'autre part, ce foncteur de composition γ induit visiblement un foncteur sur les sous-catégories de chemins à source et but fixés

$$\underline{\mathrm{Hom}}(\tau,\tau') \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X;a,b) \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\tau}(X;a,b),$$

d'où la deuxième des deux flèches annoncées dans $(*)$ p. 94.

Cela montre donc que si

$$\varphi_1, \varphi_2 : \tau \rightrightarrows \tau'$$

sont deux morphismes de τ dans τ' homotopes dans $\underline{\mathrm{Hom}}(\tau,\tau')$, alors les foncteurs induits entre catégories de Σ-chemins ou entre catégories de Σ-chemins à extrémités précisés :

$$\begin{array}{rrcl} \varphi_{1\,X}^*, \varphi_{2\,X}^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) & \rightrightarrows & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X), \\ (\varphi_1)_{a,b}^*, (\varphi_2)_{a,b}^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X;a,b) & \rightrightarrows & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X;a,b) \end{array}$$

sont également homotopes. [page 97]

Or le lemme d'homotopie nous assure que l'hypothèse $\varphi_1 \sim \varphi_2$ est *toujours* satisfaite. Donc :

Théorème d'homotopie sur les opérations universelles sur les chemins. *Soient X, Σ, a, b comme dessus, et $\tau, \tau' \in \mathrm{Ty}$ telles que $\mathrm{Hom}(\tau,\tau') \neq \emptyset$. Alors les foncteurs φ_X^*, $\varphi_{X;a,b}^*$ induits par un $\varphi \in \mathrm{Hom}(\tau,\tau')$*

$$\begin{cases} \varphi_X^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) & \longrightarrow & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) \\ \varphi_{X;a,b}^* : & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X;a,b) & \longrightarrow & \underline{\mathrm{Ch}}_{\Sigma,\tau}(X;a,b) \end{cases}$$

ne dépendent pas de φ, à homotopie près.

On peut préciser cet énoncé, de telle façon que l'énoncé pour les $\varphi_{X;a,b}^*$ apparaisse, comme il devrait, comme corollaire pour celui sur φ_X^*. Pour ceci, on va regarder les catégories de chemins dans X comme catégories *au dessus de X,* par les foncteurs 'source et but' :

$$\begin{array}{ccc} \underline{\mathrm{Ch}}_{\Sigma,\tau}(X) & & \underline{\mathrm{Ch}}_{\Sigma,\tau'}(X) \\ & {\scriptstyle (\sigma_\tau,\beta_\tau)} \searrow \quad \swarrow {\scriptstyle (\sigma_{\tau'},\beta_{\tau'})} & \\ & X \times X & \end{array}$$

Alors le foncteur $(*)$ (p. 94) se factorise par la sous-catégorie $\underline{\mathrm{Hom}}_{\mathrm{Cat}/X\times X}(\cdot,\cdot)$ de $\underline{\mathrm{Hom}}_{\mathrm{Cat}}(\cdot,\cdot)$ formée des foncteurs qui commutent aux foncteurs de projection dans $X \times X$, et les homomorphismes entre tels foncteurs qui s'effectuent par des flèches 'verticales' de la catégorie but (i.e. par des flèches au dessus de flèches identiques de la catégorie base $X \times X$) : p. 63 [page 98]

$$\underline{\mathrm{Hom}}(\tau,\tau') \longrightarrow \underline{\mathrm{Hom}}_{\mathrm{Cat}/X\times X}(\underline{\mathrm{Ch}}_{\Sigma,\tau'}(X), \underline{\mathrm{Ch}}_{\Sigma,\tau}(X)).$$

Donc dans l'énoncé du théorème d'homotopie, on peut préciser que l'homotopie se fait dans cette sous-catégorie, i.e. s'effectue par une suite d'homotopies simples qui sont des morphismes de foncteurs entre catégories au dessus de $X \times X$. Cela implique alors l'énoncé d'homotopie sur les $\varphi^*_{X;a,b}$.

p. 64

10 Question de comparaison pour les catégories Cat-fibrées satisfaisant au 'théorème d'homotopie' (base filtrante) [page 99]

Nous allons généraliser maintenant au cas d'une catégorie (Cat)-fibrée sur une autre

$$\begin{array}{c} C \\ {\scriptstyle\pi}\big\downarrow \\ T, \end{array}$$

en pensant surtout au cas $T = \underline{\mathrm{Ty}}$, $C = \underline{\mathrm{Ch}}_{\Sigma}(X; a, b)$. L'hypothèse essentielle est la suivante :

(Hypothèse d'homotopie) Si $t, t' \in \mathrm{Ob}\, T$ sont tels que $\mathrm{Hom}(t,t') \neq \emptyset$, alors les foncteurs $\varphi^* : C_{t'} \longrightarrow C_t$ associés aux $\varphi \in \mathrm{Hom}(t,t')$ sont homotopes.

Ils définissent donc une seule et même flèche dans Cat hot. Je désigne par T_{ord} ou T_{o} l'ensemble $\mathrm{Ob}\,(T)$, avec comme relation d'ordre celle déduite de la structure de catégorie :

$$t \leq t' \qquad \Longleftrightarrow \qquad \mathrm{Hom}_T(t,t') \neq \emptyset,$$

de sorte qu'on a un foncteur

$$T \longrightarrow T_{\mathrm{o}}$$

qui est *universel* pour les foncteurs de T dans des catégories 'ordonnées' (i.e. où toute flèche est monomorphique [non, cf. • ⇉ •; plutôt, où toujours $\#\mathrm{Hom}(a,b) \leq 1$ (admettant des isomorphismes non identiques)]) ou pour les foncteurs F sur T tels que F soit constant sur tout $\mathrm{Hom}(t,t')$. L'hypothèse d'homotopie peut s'exprimer en disant que le foncteur composé

$$T^o \longrightarrow \mathrm{Cat} \longrightarrow \mathrm{Cat\,hot}$$

$$t \longmapsto C_t$$

se factorise (évidemment de façon unique) en un foncteur

$$H_C : T_{\mathrm{o}}^o \longrightarrow \mathrm{Cat\,hot},$$

[page 100]

i.e. qu'on a un diagramme commutatif

p. 65

$$\begin{array}{ccc} T^o & \xrightarrow{t \longmapsto C_t} & \mathrm{Cat} \\ \big\downarrow & & \big\downarrow \\ T_{\mathrm{o}}^o & \xrightarrow{H_C} & \mathrm{Cat\,hot}. \end{array}$$

Commentary · Exposition [ms. 99]

The preorder remembers existence, not multiplicity

Grothendieck's purpose is to retain the common value of all operations between two fibres while discarding their multiplicity. Let $q : T \to P$ be the identity on objects, with

$$P(t,t') = \begin{cases} \{*\}, & T(t,t') \neq \varnothing, \\ \varnothing, & T(t,t') = \varnothing. \end{cases}$$

The category P is a preorder; distinct objects may be mutually comparable. Its universal property is among thin categories, whose hom-sets have at most one element. Having every arrow monic is a different condition: the category with two parallel arrows already separates the two notions, as the editors' correction observes.

Let $F : T^{\mathrm{op}} \to \mathrm{Cat}$ describe the fibration. In this discussion

$$\mathrm{Cat\,hot}(A,B) = \pi_0 \underline{\mathrm{Hom}}(A,B)$$

means functors modulo finite zigzags of natural transformations. It is not the localization Hot of categories at weak equivalences of nerves. Natural transformations become equalities in Cat hot, and hence also in Hot, but not every weak equivalence of nerves has a functorial homotopy inverse. The distinction matters throughout the comparison problem.

The homotopy hypothesis says exactly that the composite of F with $\mathrm{Cat} \to \mathrm{Cat\,hot}$ factors through P^{op}. If P is nonempty and any two objects have a common lower bound, P^{op} is filtered and the resulting diagram defines an ind-object H_C. Here nonemptiness is part of the usual convention for "filtrant". The displayed pairwise condition in the manuscript must be read with that convention; otherwise the empty base would satisfy it vacuously and would have no cofinal sequence of the proposed form.

One must keep this modest hypothesis distinct from T being cofiltered. A cofiltered category must also admit a common precomposition equalizing any pair of parallel arrows. Nothing of that kind is assumed here. The usual definition and its role in finite-limit arguments are recorded in The Stacks Project Authors (2026i, Categories, Definition 4.19.1 and Lemma 4.19.2). For $\underline{\mathrm{Ty}}$, concatenation supplies common lower bounds because either factor can be collapsed to the empty type. It does not identify the different ways to perform a collapse. The homotopy theorem, rather than strict equalization, is what allows those different ways to be forgotten in H_C.

Nous allons supposer de plus :

(Hypothèse de filtration) L'ensemble ordonné T_{o} associé à T est *filtrant* décroissant (filtrant à gauche), i.e. si $t, t' \in \mathrm{Ob}\, T$, il existe $t'' \in \mathrm{Ob}\, T$ et des flèches $t'' \longrightarrow t$, $t'' \longrightarrow t'$.

Cette hypothèse est bien satisfaite dans $\underline{\mathrm{Ty}}$, car pour $\tau, \tau' \in \mathrm{Ob}\, \underline{\mathrm{Ty}} = \mathrm{Ty}$, il suffit de prendre $\tau\tau'$, qui s'envoie dans τ par $\mathrm{id}_\tau * p_{\tau'}$, dans τ' par $p_\tau * \mathrm{id}_{\tau'}$, où $p_\tau : \tau \longrightarrow 1$ est le morphisme de τ dans l'objet final.

Ainsi, H_C définit un ind-objet de Cat hot, que nous allons désigner par $\underline{\underline{C}}$. Si on suppose de plus :

(Hypothèse de dénombrabilité, toujours satisfaite si $\mathrm{card}\, T_{\mathrm{o}} \leq \aleph_0$) T_{o} admet un sous-ensemble *dénombrable* cofinal à gauche.

alors cet ind-objet de Cat hot peut être défini, de beaucoup de façons, en termes d'un système inductif dans Cat (pas seulement dans Cat hot) [page 101]

$$(C_n)_{n \in \mathbf{N}},$$

d'où on récupère l'ind-objet $\underline{\underline{C}}$ dans Cat hot en passant au système inductif correspondant dans Cat hot, puis à l'ind-objet associé.

Pour construire le système inductif C_n, il suffit de prendre une chaîne infinie

$$t_0 \longleftarrow t_1 \longleftarrow t_2 \longleftarrow \cdots \longleftarrow t_n \longleftarrow t_{n+1} \longleftarrow \cdots$$

de morphismes dans T, telle que l'ensemble des t_i soit cofinal à gauche dans T_o. On va supposer de plus C *strictement fibrée* sur T (i.e. provenant d'un véritable foncteur $T^o \longrightarrow$ Cat, i.e. on peut choisir les foncteurs images inverses

$$\varphi^* : C_{t'} \longrightarrow C_t$$

de façon transitive stricte, pas seulement transitive à isomorphisme près) - cas auquel on peut toujours se ramener d'ailleurs par l'astuce de Giraud. Cette hypothèse est d'ailleurs vérifiée dans le cas $\underline{\mathrm{Ch}}_{\Sigma}(X; a, b)$ sur $\underline{\mathrm{Ty}}$. Ainsi, les

$$(C_{t_i})_{i\in\mathbf{N}}$$

forment un système inductif de catégories indexées par $\mathbf{N}$, le morphisme de transition

$$C_n \longrightarrow C_{n'} \qquad \text{i.e. } C_{t_n} \longrightarrow C_{t_{n'}}$$

étant le foncteur image inverse $\varphi^*_{n,n'}$, où p. 66

$$\varphi_{n,n'} : t_{n'} \longrightarrow t_n \qquad \text{[page 102]}$$

est le composé des $t_{n'} \longrightarrow t_{n'-1} \longrightarrow \cdots \longrightarrow t_n$. (Si on n'avait pas transitivité stricte, on définirait $\varphi^*_{n,n'}$ comme le composé des foncteurs image inverse par des flèches de transition successives $t_{n'} \longrightarrow t_{n'-1} \longrightarrow \cdots \longrightarrow t_{n+1} \longrightarrow t_n$, ce ne serait $\varphi^*_{n,n'}$ qu'à isomorphisme canonique près.)

Commentary · Context [ms. 101]

Making a sequence strict

Grothendieck wants an actual category as a sequential colimit, so he temporarily makes the reindexing functors strictly composable. This is a familiar issue from the theory of fibred categories: a cleavage supplies canonical comparison isomorphisms, whereas a splitting makes those comparisons identities. The distinction is already developed in Grothendieck (1971b, Exposé VI, §§8–9). The replacement invoked here is an equivalence of fibred categories; it is not a claim that a chosen cleavage on the original total category can always be made split without changing that category.

There is a concrete construction behind the reference to Giraud. For $t \in T$, let $\widetilde{F}(t)$ be the category of cartesian functors $T/t \to C$ over T, with vertical natural transformations. Precomposition along $T/s \to T/t$, induced by $s \to t$, is strictly associative. Evaluation at the terminal object id_t gives an equivalence $\widetilde{F}(t) \simeq C_t$: a cartesian functor is reconstructed from its value there by cartesian lifts, and any morphism between those values extends uniquely to a vertical transformation. Thus $\widetilde{F}$ is a strict replacement of the original pseudofunctor. The right splitting and its equivalence property are treated explicitly in Emmenegger et al. (2024, Theorem 5.4 and Remark 5.5); this gives a precise modern reference for the "astuce de Giraud".

Countability has a different role. Enumerate a cofinal family $s_0, s_1, \ldots$ in P. Choose t_0 below s_0, then choose t_{n+1} below both t_n and s_{n+1}, together with an arrow $t_{n+1} \to t_n$. The resulting chain is cofinal in the preorder. Pullback produces

$$C_{t_0} \longrightarrow C_{t_1} \longrightarrow C_{t_2} \longrightarrow \cdots .$$

Even without replacing the entire fibration, this particular sequence can be made strict simply by defining its long transition maps as composites of its adjacent ones. There are no competing paths in the freely generated chain to compare. The arrow $C_{t_n} \to C_{t_{n-1}}$ printed on manuscript page 102 runs in the wrong direction; the intended transition is $C_{t_{n-1}} \to C_{t_n}$.

Soit

$$C_\infty = \varinjlim_n C_n = \varinjlim_n C_{t_n}$$

la catégorie $\varinjlim$. (Elle sera surtout sympathique si on a pu choisir le système $(t_n)_{n\in\mathbf{N}}$ de telle façon que les foncteurs de transition

$$C_{t_n} \longrightarrow C_{t_{n-1}}$$

soient pleinement fidèles, et peut-être même [des] monomorphismes, de telle façon que les C_n s'identifient à des sous-catégories pleines de leur limite inductive C_∞.) Je voudrais comprendre les relations précises entre les trois genres d'objets suivants :

ⓐ C elle-même (comme objet dans Cat, pas de Cat/T), et surtout les objets de Cat hot et de Hot qui lui correspondent.

ⓑ C_∞, et les objets de Cat hot et de Hot qui lui correspondent. Dans quelle mesure ces deniers sont-ils indépendant du choix du système projectif cofinal à gauche $(t_n)_{n\in\mathbf{N}}$ dans T? Quel est le rôle de C_∞? Est-elle canoniquement isomorphe à C dans Cat hot, ou dans Hot? (Auquel cas la question de son 'rôle' me semblerait résolue!) [page 103]

ⓒ Le ind-objet H_C dans Cat hot, et l'ind-objet de Hot qu'il définit.

Je note que l'ind-objet H_C, et aussi les catégories C_∞ construites comme dessus, ne changent pas quand on remplace T par une sous-catégorie T' cofinale à gauche, ou aussi par une catégorie de la forme T/t_0 ($t_0 \in \mathrm{Ob}\, T$), qui donne lieu à l'ensemble ordonnée associé

$$(T/t)_\mathrm{o} = T_\mathrm{o}/t = \{t \in T_\mathrm{o} \mid t \leq t_0\}$$

[plutôt $(T/t_0)_\mathrm{o} = T_\mathrm{o}/t_0 = \{t \in T_\mathrm{o} \mid t \leq t_0\}$]. Si la relation entre les objets dans a), et ceux dans b) et dans c), est très étroite, on en concluerait que le 'type' de C dans Cat hot, ou dans Hot suivant le cas, ne change pas en remplaçant C par $C \times_T T'$ ou par $C/t_0 \simeq C \times_T T/t_0$. Bien mieux, elle ne changerait pas en remplaçant T par $(\mathbf{N})^o$, la catégorie (ordonnée) opposée à $\mathbf{N}$ (ayant donc un unique objet final), et la catégorie fibrée C sur T par la catégorie $C_{\underline{t}}$ induite par C sur $(\mathbf{N})^o$, via le foncteur p. 67 [page 104]

$$\underline{t} : (\mathbf{N})^o \longrightarrow T$$

défini par le système projectif $\underline{t} = (t_n)_{n\in\mathbf{N}}$ dans T. Cela signifierait donc que pour comprendre le type homotopique (au sens strict, i.e. dans Cat hot, ou large, i.e. dans Hot) de C, on peut le 'nettoyer' de façon draconienne, la base T étant vue comme une sorte de gangue un peu vaseuse, ou du moins pléthorique.

Commentary · Development (commentators') [ms. 103]

A slice does not have the same preorder

Grothendieck proposes restricting the construction to types below a chosen type. This is sound for a lower ideal of the preorder, but a slice category retains the actual arrow to the chosen object. The equality of preorders printed here therefore needs a correction that affects the proposed argument.

A slice obstruction. [ours] There is a finite category T whose associated preorder is cofiltered, while the associated preorder of T/t_0 is not cofiltered.

Proof. Take two objects s, t_0, their identities, and exactly two further arrows $f, g : s \rightrightarrows t_0$. The preorder P has $s < t_0$, so s is a lower bound for every pair. The slice T/t_0 has three objects

$$(s, f), \qquad (s, g), \qquad (t_0, \mathrm{id}_{t_0}),$$

with one arrow from each of the first two objects to the last. An arrow between the first two would have to be id_s and would force $f = g$. There is no fourth object that could map to both. Its preorder is consequently not cofiltered, although the lower ideal $P/t_0 = P$ is. This proves the assertion. □

The example also prevents interpreting the displayed equality as an equivalence after removing duplicate isomorphic objects: the two incomparable objects remain distinct in the skeletal

preorder. It does not dispute the connectedness of the slice, which has a terminal object, nor the separate construction of H_C over P^{op}.

There are two precise repairs. If only the ind-object is wanted, use the full subcategory $T_{\leq t_0} \subseteq T$ on objects admitting some arrow to t_0. Its preorder is exactly the lower ideal of P. Its inclusion is cofinal on associated preorders: given t, choose a common lower bound of t and t_0. It is cofiltered as a preorder for the same reason. Restriction therefore leaves H_C unchanged. The actual slice T/t_0 can instead be used when T is cofiltered in the categorical sense. Given $u : a \to t_0$ and $v : b \to t_0$, first choose $r \to a, b$, then precompose once more to equalize the resulting two arrows $r \rightrightarrows t_0$. This supplies a common predecessor in the slice. Parallel arrows in the slice can likewise be equalized in T, and the resulting equality still lies over t_0. Thus the slice is then cofiltered.

We checked the standard filtering results in The Stacks Project Authors (2026i, Categories, §4.19) and the homotopy-colimit comparison discussed below; we did not find this particular correction to manuscript page 103 stated there. The example and the two repairs are recorded as a local development of the argument, without a claim of priority.

ⓓ De toutes façons, si T' est une autre catégorie filtrante à gauche, et

$$\alpha : T' \longrightarrow T$$

un foncteur cofinal à gauche, de sorte que C sur T, et $C' = C \times_T T'$ sur T', définissent essentiellement le 'même' objet de Ind Cat hot, et que la catégorie C'_∞ pour C' sur T' fait partie des catégories C_∞ pour C sur T - on peut se demander si le foncteur canonique

[page 105]

$$C' \longrightarrow C$$

est également toujours un isomorphisme dans Cat hot, ou sinon, du moins dans Hot.

Un intérêt particulier s'attache à $\pi_0(C)$, et de plus à sa variance pour C sur T variable de la façon suivante. Si S est dans Cat, on regarde les catégories

$$\underline{\mathrm{Hom}}_{\mathrm{Cat}}(S, C_t) \qquad (t \in T),$$

on trouve un foncteur

$$t \longmapsto \underline{\mathrm{Hom}}_{\mathrm{Cat}}(S, C_t), \qquad T^o \longrightarrow (\mathrm{Cat}),$$

d'où une catégorie fibrée sur T, que je vais désigner par

$$\underline{\mathcal{H}om}_T(S_T, C)$$

(Sauf erreur, elle s'interprète comme la catégorie fibrée sur T correspondante au champ sur $\mathrm{Top}(T) = T^\wedge$ du morphisme de T-champs de S_T dans C, où $S_T = S \times T$ (considérée comme catégorie fibrée sur T à fibre essentiellement constante S) dans C. Ici j'identifie catégorie fibrée sur T, et champs sur le topos $T^\wedge$.) Il est immédiat que la propriété d'homotopie (p. 99) est stable par passage de C à $\underline{\mathcal{H}om}_T(S_T, C)$. On s'intéresse à la variation de

$$\pi_0(\underline{\mathcal{H}om}_T(S_T, C))$$

comme contrafoncteur en S,

p. 68

$$H : S \longmapsto \pi_0 \underline{\mathcal{H}om}_T(S_T, C) \qquad (\mathrm{Cat})^o \longrightarrow (\mathrm{Ens}),$$

[page 106]

et à une éventuelle expression de ce contrafoncteur H en termes de l'ind-objet H_C de Cat hot, par la formule hypothétique

$$H(S) \simeq \mathrm{Hom}_{\mathrm{Ind}(\mathrm{Cat\,hot})}(\gamma(S), H_C),$$

où

$$\gamma : \mathrm{Cat} \longrightarrow \mathrm{Cat\,hot}$$

est le foncteur canonique. Si on avait un tel isomorphisme canonique fonctoriel en S, cela donnerait une description (à isomorphisme unique près) de l'ind-objet H_C en termes du foncteur précédent H sur Cat.

Il me vient un doute ici, en vue de l'exemple $T = \underline{\mathrm{Ty}}$, $C = \underline{\mathrm{Ch}}_\Sigma(X; a, b)$. Les trois types d'objets a) b) c) ont le même π_0, et il est sûr que 'c'est le bon'. D'autre part, pour une catégorie paramètre variable S, il est sûr aussi que le bon foncteur en S est

$$S \longmapsto \pi_0(\underline{\mathrm{Hom}}_T(S_T, C)) \simeq \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(S, H_C),$$

isomorphe aussi, *si S est finie,* à

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(S, C_\infty),$$

ce qui montre que, du moins pour des arguments $S \in (\mathrm{Cat})$ qui sont des catégories *finies*, les objets C_∞ et H_C sont 'bons', il nous permettent de décrire un foncteur important qui domine la situation envisagée. Moyennant une hypothèse très forte, mais satisfaite dans les cas importants, on peut dans ces deux formules remplacer Cat hot par Hot. [page 107]

Commentary · Exposition [ms. 106]

The functor represented by the ind-object

Grothendieck's parameter category S restores information that the connected components of a single fibre cannot carry. The proposed formula for $H(S)$ is valid under the hypotheses already imposed, and its proof explains both the ind-object and the restriction to finite S in the next formula.

For any strict diagram $A : T^{\mathrm{op}} \to \mathrm{Cat}$, the connected components of its fibred total category are

$$\pi_0\left(\int_T A\right) \cong \operatorname*{colim}_{T^{\mathrm{op}}} \pi_0 A(t).$$

Indeed, the total category is generated by vertical arrows and cartesian transport arrows. Vertical arrows impose exactly the component relation inside each $A(t)$; a cartesian arrow imposes exactly the relation identifying a class with its restriction along an arrow of T. Those are the generators and relations defining the colimit of sets, with no further relation. Apply this observation to $A(t) = \underline{\mathrm{Hom}}(S, C_t)$. A zigzag between two reindexing functors induces a zigzag after postcomposition with functors from S, so the diagram of component sets factors through P^{op}. We obtain, naturally in S,

$$\begin{aligned} H(S) &= \pi_0 \underline{\mathcal{H}om}_T(S_T, C) \\ &\cong \operatorname*{colim}_{P^{\mathrm{op}}} \mathrm{Cat\,hot}(S, C_t) = \mathrm{Hom}_{\mathrm{Ind(Cat\,hot)}}(S, H_C). \end{aligned}$$

The last equality is the defining formula for maps from a constant object into an ind-object. The usual fully faithful embedding of ind-objects into set-valued presheaves then gives the asserted uniqueness of H_C. Setting $S = e$ proves the equality of the three component sets in the text.

To compare with C_∞, suppose S has finitely many objects and arrows. A functor $S \to \mathrm{colim}_n C_n$ involves finitely many objects, arrows, and composition equations. Represent its objects and arrows at a common stage, then move to a later stage where all the finitely many required equations hold. It therefore descends to some C_n. A natural transformation between two such functors also consists of finitely many arrows and naturality equations and descends in the same way. Finally, if two descended functors lie in the same component, a witnessing homotopy is a *finite* zigzag of such transformations; the whole zigzag descends to one later stage. This proves the bijection

$$\operatorname*{colim}_n \mathrm{Cat\,hot}(S, C_n) \xrightarrow{\sim} \mathrm{Cat\,hot}(S, C_\infty).$$

It is an instance of the finite-presentation argument underlying filtered colimits, not a fibrancy assertion.

The finiteness cannot simply be omitted. Let C_n be the discrete category on $\{0, \dots, n\}$ and let S be the discrete category on $\mathbf{N}$. Then C_∞ is discrete on $\mathbf{N}$. The left side consists of functions $\mathbf{N} \to \mathbf{N}$ with bounded image; the right side consists of all functions. The identity function lies only in the latter. The ind-object retains the information that a whole S-family must occur at one stage, while the ordinary colimit allows the stage to vary with the individual element of S.

Par contre, sauf pour le cas où S est réduit à un point et où $\mathrm{Hom}_{\mathrm{Cat\,hot}}$ ou $\mathrm{Hom}_{\mathrm{Hot}}$ de S dans C n'est autre que $\pi_0(C)$ ($\simeq \pi_0(C_\infty) \simeq \pi_0(H_C)$), donc effectivement 'bon', je ne sais pas si

$$\underbrace{\mathrm{Hom}_{\mathrm{Cat\,hot}}(S, C)}_{=\pi_0 \underline{\mathrm{Hom}}(S,C)} \qquad \text{ou} \qquad \mathrm{Hom}_{\mathrm{Hot}}(S, C)$$

est lié de façon raisonnable aux ensembles précédents (déterminés par H_C, resp. par C_∞), p. ex. s'ils lui sont isomorphes de façon naturelle, et s'ils ont une signification géométrico-topologique raisonnable. Je commence à suspecter que *non* - donc qu'on aurait ici un cas où une relation de type très général concernant un π_0, ici donc

$$\pi_0(C) \simeq \pi_0(C_\infty) \simeq \pi_0(H_C),$$

ne se prolonge pas en un isomorphisme sur les π_i supérieurs, et en une propriété de Hot-équivalence.

Je ne trouve d'ailleurs aucune flèche directe entre C et l'un des objets C_∞, H_C; tout ce que je vois dans ce sens, c'est p. 69 [page 108]

$$\begin{array}{ccc} C_{\mathbf{N}^o} & \xrightarrow{\substack{\text{foncteur de} \\ \text{localisation}}} & C_\infty \\ \downarrow & & \\ C, & & \end{array}$$

où la flèche verticale est la flèche de projection de $C_{\mathbf{N}^o} = C \times_T \mathbf{N}^o$ (avec $\mathbf{N}^o \xrightarrow{t} T$). Ceci n'établit une relation topologique étroite entre C (en tant qu'objet de Cat hot ou de Hot simplement) et C_∞, que si les deux flèches précédentes sont tout au moins des Hot-équivalences. Et le cas-type (problématique) où la question de la relation entre C et C_∞ se pose, est donc celui où $T = \mathbf{N}^o$, i.e. celui d'un système inductif quelconque de (petites) catégories, indexé par $\mathbf{N}$, et interprété comme défini par une catégorie fibrée $C = C_{\mathbf{N}^o}$ sur $\mathbf{N}^o$. Dans ce cas particulier, on a du moins une flèche canonique

$$C \longrightarrow C_\infty$$

sur laquelle on peut se poser la question si c'est une Hot-équivalence. Sinon, cela prouvera que dans le cas général, le Hot-type de C (et a fortiori sa classe d'isomorphie dans Cat hot) n'a pas de sens géométrique intéressant. C'est à dire que C est intéressante uniquement *en tant que catégorie fibrée sur* T, permettant notamment de construire l'ind-objet H_T de Cat hot ou de Hot, mais pas en tant que 'catégorie nue'. [page 109]

Commentary · Examples [ms. 107]

A finite category need not be a finite homotopy probe

Grothendieck explicitly reserves the passage from Cat hot to Hot for an additional hypothesis. The reservation is necessary. The finite-presentation proof above counts objects, arrows, and equations in a category. Compactness in the homotopy category of spaces concerns the whole nerve, and a finite category can have nondegenerate simplices in arbitrarily high degrees.

Let $S = \mathbf{B}(\mathbf{Z}/2)$, which has one object and two arrows, and put $K = NS$, $K_n = \mathrm{sk}_n K$. Let $D_n = \Delta/K_n$ be the category of simplices of K_n. Every simplex of K belongs to one skeleton, so

$$D_\infty = \operatorname*{colim}_n D_n = \Delta/K.$$

The simplex-category resolution gives $|ND_n| \simeq |K_n|$ and $|ND_\infty| \simeq |K|$, naturally for the inclusions. The precise last-vertex statement for an arbitrary simplicial set is Kerodon (2026, Corollary 11.9.11.5). Thus the filtered colimit has the homotopy type of $B(\mathbf{Z}/2)$, while each stage has an n-dimensional realization.

The identity of $B(\mathbf{Z}/2)$ cannot factor in Hot through any $|K_n|$. Such a factorization would make its identity map on cohomology factor through a group vanishing in degrees greater than n. But

$$H^*(B(\mathbf{Z}/2); \mathbf{F}_2) \cong \mathbf{F}_2[u], \qquad |u| = 1.$$

For completeness, the normalized group cochains with trivial $\mathbf{F}_2$ coefficients have one basis element in each degree: a normalized cochain is determined by its value on the string consisting entirely of the nonidentity group element. Its differential is zero, since the two end terms cancel and every interior product gives the identity, on which a normalized cochain vanishes. The degree-one generator and its cup powers take value one on these strings. This proves the displayed calculation and the obstruction.

Consequently the canonical map

$$\operatorname*{colim}_n \mathrm{Hom}_{\mathrm{Hot}}(S, D_n) \longrightarrow \mathrm{Hom}_{\mathrm{Hot}}(S, D_\infty)$$

is not surjective, although S is finite as a category. There is no contradiction with the preceding Cat hot formula. Finite CW homotopy types are compact probes for filtered homotopy colimits, but the nerve of this finite group is not one. This is the point at which the two meanings of "finite" must be separated when reading the notation $\mathrm{Hot}_0^\wedge$ later on these pages.

Commentary · Later developments [ms. 108]

The sequential total category is a homotopy colimit

Grothendieck has reduced one part of the comparison to a single sequence. That part has a uniform answer: the canonical functor from its fibred total category to its ordinary filtered colimit is a weak equivalence of nerves. No full-faithfulness assumption on the transition functors is needed.

Write $E = \int_{\mathbf{N}^{\mathrm{op}}} C_n$ and $D = \operatorname{colim}_n C_n$. With the variance of this manuscript, an arrow $(m, y) \to (n, x)$ over $m \to n$, where $m \geq n$, is an arrow $y \to F_{n,m}x$ in C_m. The canonical functor $q : E \to D$ sends it to the corresponding arrow in D. Cartesian transport arrows become identities. The localization description in the text is understood in the usual categorical sense: after inverting those arrows, E becomes *equivalent* to D. Literal equality of its object set with that of D is neither needed nor generally true.

The homotopy-colimit theorem of Thomason (1979, Theorem 1.2), in its dual form for a fibration, identifies $|NE|$ with $\operatorname{hocolim}_n |NC_n|$. The convention for a contravariant diagram and the comparison map are displayed explicitly in Cegarra (2020, §3, equation (9)). Two elementary filtered-colimit facts complete the argument. First,

$$ND \cong \operatorname*{colim}_n NC_n :$$

an r-simplex is a finite string of composable arrows, and finitely many representatives and equalities can all be realized at one stage. Second, for a filtered diagram of simplicial sets K_n, the augmentation

$$\operatorname{hocolim}_n K_n \longrightarrow \operatorname*{colim}_n K_n$$

is a weak equivalence. One way to see the second fact is to use the simplicial replacement. In each fixed simplicial degree, the fibre over a simplex of the colimit is the category of its representatives at the different stages. Any two representatives eventually agree, and finitely many such agreements hold simultaneously at a later stage. This category is nonempty and filtered, hence its nerve is contractible: every finite diagram admits a cone, which contracts each finite simplicial sphere. Thus

the augmentation is a weak equivalence in each row of the bisimplicial replacement, and diagonal realization preserves these rowwise weak equivalences. The composite is the map induced by q.

This identifies the genuine issue in the span on manuscript page 108. The horizontal arrow is always a weak equivalence. Whether the vertical arrow is one depends on how the chosen sequence sits inside T, including the homotopy carried by its parallel arrows. Passing to the preorder has deliberately forgotten that information.

Commentary · Exposition [ms. 108]

The total category and the telescope

Grothendieck wants a path category that carries composition and at the same time has the intended homotopy type. The two objects in this opening comparison solve different parts of that problem. The total category remembers how the types of paths vary; the filtered colimit forgets the stages at which paths were represented. We write $BC = |NC|$, and call a functor a weak equivalence when its nerve is a weak homotopy equivalence. This is the meaning of an isomorphism in Hot. It is stronger than a bijection on connected components and different from an isomorphism in Cat hot, where morphisms are connected components of ordinary functor categories.

Here is a proof of the assertion announced on the following page. Let $F_0 \to F_1 \to \cdots$ be a strict sequence of small categories, let $E \to \mathbf{N}^o$ be its fibred Grothendieck construction, and let $L = \operatorname{colim}_n F_n$. The canonical functor $q : E \to L$ is a weak equivalence. For clarity, a morphism $(n, x) \to (m, y)$ in E, with $n \geq m$, is a morphism $x \to F_{m,n}y$ in F_n.

To prove the assertion, fix $z \in L$ and consider the comma category q/z. Any finite collection of its objects, arrows, and commuting triangles can be represented at one sufficiently large stage r: filtered colimits of categories are computed from finite diagrams and their eventual relations. We may also represent z at stage r. Transporting the finitely many objects to that stage gives a diagram in F_r/z_r. Its projection to q/z has a natural transformation to the original diagram, supplied by the cartesian arrows in E. It has a natural transformation to the constant diagram $(r, z_r, 1)$, supplied by the arrows to z_r. Thus every finite simplicial diagram in $N(q/z)$ contracts after passing to a stage. The same argument with a chosen vertex proves connectedness and vanishing of all homotopy groups; the comma category is nonempty because z has a representative. Quillen's Theorem A now gives the assertion (Quillen, 1973, §1, Theorem A). This argument does not require the transition functors to be faithful or injective on objects.

The second leg of the displayed roof is a separate matter. Take a nontrivial group G, let $T = BG$ be its one-object category, and take $C = T$ over itself. The preorder obtained by forgetting the distinction between parallel arrows is a point. A sequence constant on the identity arrow is therefore cofinal in that preorder, whereas the comparison is $* \to BG$. Its fundamental group is $1 \to G$, so it is not a weak equivalence. This proves exactly the limitation noticed on manuscript page 109: cofinality in the preorder of objects does not account for the automorphisms of the indexing category. It does not predict failure for the particular category of interval types, which has additional structure.

Mais je viens de me convaincre que $C_{\mathbf{N}^o} \longrightarrow C_\infty$ est bien une Hot-équivalence. Donc la flèche $C_{\mathbf{N}^o} \longrightarrow C$ définit une flèche *dans* Hot

$$C_\infty \longrightarrow C$$

(en composant avec l'inverse dans Hot de la précédente). Est-ce un isomorphisme? Donc pour cette question, on est ramené à la question pour $C_{\mathbf{N}^o} \longrightarrow C$. Mais si on prend $C = T$, cela pose déjà la même question pour $\mathbf{N}^o \longrightarrow T$, donc la question si T est asphérique, sous la seule hypothèse que T_o soit filtrante à gauche et dénombrable à l'infini (à gauche). Mais bien sûr c'est faux (p. ex. si T n'a qu'un seul objet, il n'est pas pour autant asphérique !).

Cela me convainc donc qu'à défaut d'hypothèse supplémentaire draconienne sur T, la catégorie C 'nue' n'offre pas d'intérêt particulier, à part son π_0 (qui est donné aussi par C_∞, et par H_C). Il n'est pas dit, certes, que Ty ne satisfasse pas à ces 'conditions draconiennes'. Mais comme C 'nue' ne s'insère pas dans un formalisme sympathique (p.ex. on n'a pas d'interprétation géométrique de $\operatorname{Hom}_{\text{Cat hot}}(S, C)$, ou de $\operatorname{Hom}_{\text{Hot}}(S, C)$), je crois que par la suite je peux en faire abstraction. Exit a) !

p. 70

[page 110]

Commentary · Examples [ms. 109]

What a one-object base can retain

Grothendieck's one-object example can be made completely explicit, and it shows exactly which information is absent from the ind-object. Let G be a nontrivial discrete group and let $T = \mathbf{B}G$ be the category with one object and endomorphism group G. Take $C = T$ with its identity projection. This is a split fibration with terminal fibre and identity reindexing functors. Its homotopy hypothesis holds strictly; its associated preorder is the one-point category, which is cofiltered and has a countable cofinal family.

Choose the constant chain at that object, with identity transition arrows. Every C_n and C_∞ is the terminal category, and H_C is the constant terminal ind-object. The total category remains $\mathbf{B}G$. Consequently the comparison in Hot is

$$* \longrightarrow BG,$$

where $BG = |N\mathbf{B}G|$. Its source has trivial fundamental group and its target has fundamental group G. It is a bijection on components and is not a weak equivalence. More generally the constant fibration with fibre D has total category $T \times D$, so it retains the product $BG \times |ND|$ while its sequential colimit is D.

There is also a direct answer to the mapping-set question of manuscript page 107. Set $G = \mathbf{Z}/2$ and $S = \mathbf{B}G$. Since the fibres of $C \to T$ are terminal, $H(S)$ is a singleton. The identity and the trivial functor $S \to C$ are not homotopic in Cat hot: natural transformations between functors of one-object groupoids express conjugacy of the corresponding homomorphisms, and these two homomorphisms are not conjugate. They remain distinct in Hot, because their maps on π_1 differ. Thus neither of the proposed mapping sets into the bare total category recovers $H(S)$ in this generality.

The example concerns the abstract hypotheses of §10. It does not decide the special path-category comparison over $\underline{\mathrm{Ty}}$. That case has additional geometric structure, used later in the chapter. Nor does it make C topologically meaningless: C remembers the base monodromy, which is precisely the datum suppressed by the thin reflection. The warning is against identifying two different invariants.

5.11.90

J'ai quand même passé la journée entière hier à essayer de comprendre la relation de C et de son type dans Hot, à celui de C_∞. En effet, comme on a finalement une flèche dans Hot

$$C_\infty \longrightarrow C,$$

déduite du diagramme dans (Hot)

$$\begin{array}{ccc} C_{\mathbf{N}^o} & \xrightarrow{\sim} & C_\infty \\ \downarrow & & \\ C, & & \end{array}$$

de prouver que $C_\infty \longrightarrow C$ est un isomorphisme dans Hot semble la façon la plus simple, et peut-être unique, de prouver que l'image de C_∞ dans Hot ne dépend pas, à isomorphisme [page 111] unique près, des choix faits pour construire C_∞. Sans compter que dans le cas motivant, $C = \underline{\mathrm{Ch}}_\Sigma(X)$ ou $\underline{\mathrm{Ch}}_\Sigma(X; a, b)$, $T = \underline{\mathrm{Ty}}$, c'est la catégorie des chemins C qui donne lieu à une loi de composition des chemins impeccable, fonctorielle, associative et tout, alors qu'il n'en est rien pour C_∞. On ne trouve pas une loi de composition

$$C_\infty \times C_\infty \longrightarrow C_\infty$$

p. 71

dans Cat, ni même dans Cat hot ou dans Hot - ou plutôt, si on arrive pourtant à construire une flèche $C_\infty \times C_\infty \longrightarrow C_\infty$ dans Cat, c'est encore laborieusement à coups de choix ± arbitraires, [page 112]

cran par cran, et sans savoir si ce qu'on obtient a une propriété d'unicité, ne serait ce que dans Hot; on ne trouve une interprétation géométrique tangible de cette composition que dans $\mathrm{Hot}_0^\wedge$ (où on a une véritable loi de groupoïde), ou dans H_T en tant qu'objet de $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$ ou de $\mathrm{Ind}\,\mathrm{Hot}$. (Donc la limite inductive dans $\mathrm{Hot}_0^\wedge$ n'est autre que l'élément de $\mathrm{Hot}_0^\wedge$ défini par C_∞.)

Mes efforts pour prouver que

$$C_\infty \longrightarrow C$$

était un isomorphisme dans Hot, ou ne serait-ce qu'un 1-isomorphisme (i.e. un isomorphisme non seulement pour les π_0, ce qui est toujours vrai, mais aussi pour les π_1 de toutes les composantes), ont échoués. J'ai trouvé qu'il suffisait que le foncteur

$$T \longrightarrow T_{\mathrm{o}}$$

soit ou bien lisse et coasphérique, ou bien propre et asphérique - donc tout est OK si $T = T_{\mathrm{o}}$, i.e. si T est ordonnée. Mais dans le cas $T = \underline{\mathrm{Ty}}$, le foncteur $T \longrightarrow T_{\mathrm{o}}$ n'est ni propre, ni lisse (pas même 0-propre ou 0-lisse).

Finalement j'en suis arrivé à la conviction que dans le cas qui nous occupe ($T = \underline{\mathrm{Ty}}$, $C = \underline{\mathrm{Ch}}_\Sigma(X; a, b)$), la flèche $C_\infty \longrightarrow C$ n'est en général *pas* un isomorphisme dans Hot. En effet, dans le cas $a = b$, si cela `[était vrai]`, on aurait une description du Hot-groupe $\Omega(X, a) = \underline{\mathrm{Ch}}(X, a)$ ([23]) en termes d'une catégorie C munie d'une loi de composition *associative* (et unitaire) *dans* Cat, redonnant la loi de groupe de Ω en passant de (Cat) à Hot. Prenant le nerf de C, on trouverait un monoïde semi-simplicial qui, dans Hot, donnerait la loi de groupe de $\Omega(X, a)$. Or je me rappelle ([24]) qu'on m'a expliqué qu'en général, la loi de groupe de $\Omega(X, a)$ ne pourrait pas être obtenue par une loi *associative* sur un ensemble semi-simplicial. Ainsi, l'insuccès de mes efforts semble tenir, non à ma maladresse, mais à la nature des choses. [page 113]

Commentary · Exposition [ms. 112]

The cofinality condition that controls total categories

Grothendieck is seeking a hypothesis that allows the base to be replaced without changing the total homotopy type. The appropriate general test is a condition on comma categories, stronger than cofinality of the associated preorders. It gives a precise sufficient criterion without requiring every reindexing functor between fibres to be a weak equivalence.

Let $\alpha : J \to T$ be a functor. If every comma category α/t has contractible nerve, then for every Grothendieck fibration $C \to T$ the projection

$$C \times_T J \longrightarrow C$$

is a weak equivalence of nerves. To prove this, fix $x \in C_t$ and consider the comma category of the displayed projection over x. Its objects are triples consisting of $j \in J$, $y \in C_{\alpha j}$, and an arrow $y \to x$ over some $u : \alpha j \to t$. Forgetting y gives a functor to α/t. Cartesian lifting supplies a right adjoint: send (j, u) to the cartesian arrow $u^* x \to x$. The adjunction follows from the unique factorization of an arrow through a cartesian lift. Hence these two comma categories have equivalent nerves. The dual of Quillen's Theorem A now applies; see Lurie (2009, Corollary 4.1.3.3), applied to the opposite functor, for a precise modern formulation.

The condition is also necessary if the conclusion is required for *every* fibration over T. Use the discrete fibration $T/t \to T$. Its total category has a terminal object, and its pullback is α/t. Universal invariance of total homotopy types therefore forces α/t to be contractible. This elementary converse explains why a criterion expressed only in terms of which pairs of objects admit arrows cannot suffice.

For a cofiltered preorder P and a decreasing cofinal sequence t_n, the category α/t is the nonempty tail of indices for which $t_n \leq t$. It has a terminal object, the least such index. Thus the criterion

23. On fait ici $\Sigma = \mathrm{Fl}(X)$.

24. C'est un erreur - ça doit être pour d'*autres* objets groupe de Hot que les groupes de lacets. J'ai vu dans Quillen que $\Omega(X, x)$ doit se décrire par un groupe semi-simplicial (quand X est un ensemble semi-simplicial, x un sommet). Donc le problème si $C_\infty \simeq C$ dans Hot reste entier.

proves the positive assertion in the text when $T = P$. For $T = \mathbf{B}G$ and the constant identity sequence, by contrast, α/t is a disjoint union of copies of $\mathbf{N}^{\mathrm{op}}$ indexed by G, so it is not contractible when $G \neq 1$. These are the same two cases that the preceding examples separate. This comma criterion characterizes invariance for every fibration. The properness and smoothness conditions in the source instead exploit its hypothesis on parallel pullbacks, as the following proof shows; they need not imply the universal comma criterion.

Commentary · Later developments [ms. 112]

Why the smooth and proper criteria suffice

The two sufficient conditions asserted here do follow from the earlier homotopy hypothesis. They are conditions on the base together with this particular class of coefficients; they need not make the chosen sequence cofinal for every fibration. The distinction matters because the hypothesis on parallel pullbacks supplies exactly the local constancy needed in the proof.

Write $q : T \to P = T_{\mathrm{o}}$, let $\alpha : \mathbf{N}^{\mathrm{op}} \to T$ be the chosen sequence, and replace the fibration by a strict diagram $F : T^{\mathrm{op}} \to \mathrm{Cat}$ as above. Put $K = NF$, regarded as a diagram of spaces, and $k = q^{\mathrm{op}}$. The fibre $E_p = q^{-1}(p)$ has the single object p and endomorphism monoid $\mathrm{End}_T(p)$. Every endomorphism $u : p \to p$ is parallel to id_p. Hence $F(u)$ is homotopic to the identity, so its nerve is a weak equivalence. Thus $K|_{E_p^{\mathrm{op}}}$ is a locally constant diagram: every one of its structure maps is a weak equivalence.

Use smoothness and properness for the localizer of weak equivalences of nerves. The fibre criterion of Maltsiniotis (2005, Theorem 1.2(b), printed p. 4) says that smoothness makes

$$j_p : E_p \longrightarrow p\backslash q$$

aspherical, meaning that every j_p/z has contractible nerve. By the smooth–proper duality on pp. 4–5, properness makes

$$j_p : E_p \longrightarrow q/p$$

coaspherical, meaning that every $z\backslash j_p$ has contractible nerve. In the first case coasphericity of q says that $p\backslash q$ is contractible; in the second case asphericity of q says that q/p is contractible. Quillen's Theorem A and its dual therefore show that E_p is contractible in either case.

We shall use the following precise local-constancy fact. If I has contractible nerve and $L : I \to \mathrm{sSet}$ sends every arrow to a weak equivalence, then for every $i \in I$ the canonical maps

$$L(i) \longrightarrow \mathrm{hocolim}_I L, \qquad \mathrm{holim}_I L \longrightarrow L(i)$$

are weak equivalences. The first is Cisinski (2009, Corollary 1.18, printed p. 599); the second is its dual, applied to the opposite index category and opposite model category. The pointwise homotopy Kan-extension formula is Proposition 1.14 of the same paper, with the right-extension formula obtained by the same duality. These are statements about coherent diagrams, so no choice of mutually compatible homotopies between parallel pullbacks is required.

Smooth and coaspherical. Let $G = \mathbf{L}k_!K$ be the left homotopy Kan extension. Its value at p is

$$G(p) \simeq \mathrm{hocolim}_{(p\backslash q)^{\mathrm{op}}} K \simeq \mathrm{hocolim}_{E_p^{\mathrm{op}}} K.$$

Here the second equivalence follows because j_p^{op} is homotopy final: its under-comma categories are the opposites of the contractible categories j_p/z. The local-constancy fact shows that the adjunction unit $K \to k^*G$ is an objectwise weak equivalence. Let $v = (q\alpha)^{\mathrm{op}} : \mathbf{N} \to P^{\mathrm{op}}$. Cofinality of the chosen sequence in the preorder makes v homotopy final: for $p \in P$, the category p/v is the nonempty tail of indices for which $t_n \leq p$, and it has a least element. Consequently

$$\mathrm{hocolim}_{\mathbf{N}} K\alpha^{\mathrm{op}} \xrightarrow{\sim} \mathrm{hocolim}_{\mathbf{N}} v^*G \xrightarrow{\sim} \mathrm{hocolim}_{P^{\mathrm{op}}} G \simeq \mathrm{hocolim}_{T^{\mathrm{op}}} K.$$

The last equivalence is composition of left homotopy Kan extensions. By the unit and its adjunction identity, this composite is the canonical comparison of homotopy colimits.

Proper and aspherical. Instead take $G = \mathbf{R}k_*K$. The pointwise formula and the proper fibre criterion give

$$G(p) \simeq \mathrm{holim}_{(q/p)^{\mathrm{op}}} K \simeq \mathrm{holim}_{E_p^{\mathrm{op}}} K \xrightarrow{\sim} K(p).$$

Now j_p^{op} is homotopy initial, which justifies restricting the homotopy limit, and the final arrow is the local-constancy equivalence. These maps are the components of the counit $k^*G \to K$, so this counit is an objectwise weak equivalence. Asphericity of q also makes k homotopy final, since $p/k \cong (q/p)^{\mathrm{op}}$. Both k and $v = k\alpha^{\mathrm{op}}$ therefore give homotopy-colimit equivalences to $\mathrm{hocolim}_{P^{\mathrm{op}}} G$. It follows by two-out-of-three that

$$\mathrm{hocolim}_{\mathbf{N}}\, \alpha^{\mathrm{op}*} k^* G \longrightarrow \mathrm{hocolim}_{T^{\mathrm{op}}}\, k^* G$$

is a weak equivalence. The counit transfers this conclusion to K.

Finally, the natural fibred form of Thomason's homotopy-colimit theorem identifies the comparison just proved with $C \times_T \mathbf{N}^{\mathrm{op}} \to C$. The sequential comparison already proved identifies its source with C_∞ in Hot. Thus each of the source's two conditions indeed makes its canonical $C_\infty \to C$ an isomorphism in Hot.

This proof also identifies the exact limit of the statement. It used local constancy on E_p^{op}, supplied by the homotopy hypothesis on F. For example, let T be the one-object category of the monoid $\{1, 0\}$ with absorbing zero. The natural transformation from the identity functor to the constant functor whose component is 0 contracts its nerve. The map $q : T \to *$ is both smooth and proper, by the fibre criterion, and both aspherical and coaspherical. But for the constant identity sequence α, the category α/t is a disjoint union of two copies of $\mathbf{N}^{\mathrm{op}}$. Thus these conditions do not imply the universal comma criterion for arbitrary fibrations; the coefficient hypothesis cannot be dropped from the argument.

Commentary · Examples [ms. 112]

Two disconnected lifting categories

The assertion about the projection to the preorder has a small, concrete explanation. Passing to the preorder remembers that a change of type exists and forgets how many such changes there are. For smoothness or properness, those forgotten choices must be connected by commuting triangles of lifts. Natural transformations between interval functors do not by themselves provide such triangles.

Write $q : \underline{\mathrm{Ty}} \to T_o$ for the projection, with T_o retaining the same objects and having one arrow whenever the corresponding Hom set in $\underline{\mathrm{Ty}}$ is nonempty. A necessary condition for 0-smoothness is the following connected lifting property. Given $a \in \underline{\mathrm{Ty}}$, $b \in T_o$, and an arrow $b \to q(a)$, consider the category of arrows $x \to a$ with $q(x) = b$ lying over that arrow; morphisms are commuting triangles over id_b. This category must be nonempty and connected. The dual condition, with fixed source, is necessary for 0-properness. These are the W_0 specializations of the smoothness criterion and its dual, where W_0 consists of functors inducing bijections on connected components (Maltsiniotis, 2005, Theorem 1.2(b), p. 4).

For smoothness take $b = + - +$ and $a = +$. The interval of type b has four vertices $0, 1, 2, 3$, and the interval of type a has two. An increasing endpoint-preserving functor $b \to a$ is forced to collapse the middle, negative edge. There are exactly two such functors, with vertex lists

$$(0, 0, 0, 1) \qquad \text{and} \qquad (0, 1, 1, 1).$$

On the other hand, the only endpoint-preserving increasing endofunctor of $+ - +$ is its identity. To see this, its first internal value must be at most 1, since a nonconstant image of the first positive edge cannot cross the following negative edge. Its second internal value must be at least 2, by the same argument at the last positive edge. The middle negative edge can then be preserved only when these two values are 1 and 2. As q is the identity on objects, the strict fibre over b has this one object and only its identity arrow. The lifting category therefore has precisely two objects and only their identities. It is disconnected, proving that q is not 0-smooth.

For properness take the fixed source $a = ++$ and target $b = +$. There are exactly two endpoint-preserving increasing functors $a \to b$, with vertex lists

$$(0, 0, 1) \qquad \text{and} \qquad (0, 1, 1).$$

The only endpoint-preserving increasing endofunctor of $+$ is the identity. The fixed-source lifting category is consequently another discrete category with two objects. Thus q is not 0-proper.

Both examples are compatible with the earlier contractibility theorem for nonempty categories of interval functors. In that theorem the morphisms are endpoint-fixed natural transformations.

Here they are factorizations through endofunctors of a fixed type. The distinction explains why connectedness of every mapping category cannot supply the lifting property required on this page.

Commentary · Examples [ms. 112]

What the preorder forgets about a lift

Grothendieck's proposed sufficient criterion fails for the category of types for a concrete reason: its preorder identifies distinct changes of parameter which cannot be connected inside a lifting category. The definition on manuscript page 99 retains the same objects in $\underline{\mathrm{Ty}}_o$ and replaces every nonempty set of morphisms by a singleton. It does not identify different objects whose images happen to be isomorphic in that preorder.

Take

$$S = (0 \longrightarrow 1 \longleftarrow 2 \longrightarrow 3), \qquad T = (0 \longrightarrow 1).$$

An endpoint-preserving monotone type map $S \to T$ changes its vertex value from 0 to 1 at exactly one of the two forward edges. There are precisely two such maps, with vertex lists

$$(0,1,1,1), \qquad (0,0,0,1).$$

Both S and T have only the identity endomorphism in $\underline{\mathrm{Ty}}$. For S, a monotone endomorphism can advance by at most one vertex on each edge, since its alternating target has no nonidentity composite of two consecutive edges. Its three increments must sum to three because the endpoints are fixed, so each increment is one. The assertion for T is immediate.

Let $p : \underline{\mathrm{Ty}} \to \underline{\mathrm{Ty}}_o$ and consider the unique base arrow $pS \to pT$. An incoming lift with target T has source exactly S, and its morphisms are endomorphisms of S commuting with the lifted arrow. Its lifting category is therefore the discrete category on the two displayed maps. The outgoing lifting category with source S is likewise discrete on those maps, now because its morphisms are endomorphisms of T. Neither category is connected. The incoming and outgoing lift criteria consequently show that p is neither 0-smooth nor 0-proper, and hence neither smooth nor proper. This verifies the source's assertion; failure of a sufficient criterion still does not imply failure of the geometric comparison itself.

Je vais résumer les faits positifs auxquels je suis arrivé dans le diagramme suivant, qui a lieu dans la situation générale $C \longrightarrow T$ envisagée au début du présent paragraphe :

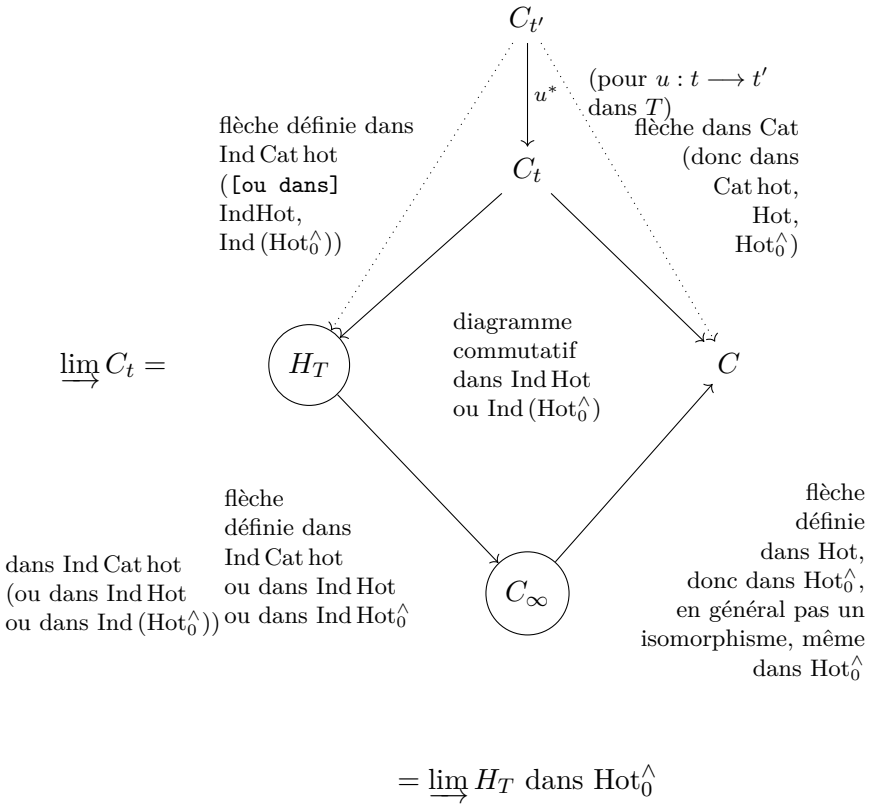
Commentary · Context [ms. 113]

The obstruction withdrawn in the footnote

Grothendieck is trying to explain why an associative composition is easy on all finite paths but elusive on a chosen infinite refinement. The proposed explanation on this page is withdrawn in footnote 24, and the withdrawal is mathematically decisive. A based loop space does admit a strictly associative model. One elementary topological model consists of Moore loops: a loop is a pair (ℓ, γ), where $\ell \geq 0$, $\gamma : [0, \ell] \to X$, and both endpoints are the basepoint. Concatenation adds lengths and joins the maps without rescaling their domains. Its associativity is an equality, and the loop of length zero is a strict unit. Reversal is an inverse up to homotopy, which is exactly the group law required in Hot; a strict monoid model does not require strict inverses.

The simplicial statement is stronger. For a reduced simplicial set K, Kan's loop-group construction gives a simplicial group GK and a weak equivalence $K \to \overline{W}GK$. Consequently $|GK|$ models the based loop space of $|K|$. We checked the precise reducedness hypothesis and the unit equivalence in Stevenson (2012, Theorem 21), which gives a modern proof of Kan's theorem. The word "semi-simplicial" in this part of the manuscript includes the simplicial formalism with degeneracies; it must not be read as a claim about an arbitrary semisimplicial object without them.

These constructions settle the alleged obstruction, but do not by themselves identify either of Grothendieck's particular categories with the model. That identification is the work still to be done. For $\Sigma = \mathrm{Fl}(X)$, section 13 of the present fragment eventually compares the all-types category with $\underline{\mathrm{Ch}}_\infty$ and restores the strict concatenation model. The footnote should therefore be read as a correction to an inference, not as an incidental qualification of a theorem. The unsuccessful search described in the surrounding prose is part of the discovery of the later proof. [page 114] p. 72



Commentary · Present state [ms. 113]

Strict loop multiplication and the later comparison

Grothendieck wants to retain the strictly associative composition already constructed on the total category of paths. His hesitation here concerns whether that useful algebraic structure can live on a correct model of the loop space. Footnote 24 retracts the proposed obstruction, and the retraction is mathematically decisive. A multiplication that is associative only in the homotopy category need not come equipped with the coherent data needed for a strict model. A loop space, however, already has those data; its origin as a space of loops supplies much more than an isolated group object in the homotopy category.

There is a precise classical replacement. For a reduced simplicial set K, Kan constructs a simplicial group GK and a principal GK-bundle over K with weakly contractible total space (Kan, 1958, Definition 6.3 and Theorem 7.1). After passage to a reduced Kan model, this identifies GK with the based loop homotopy type. Its multiplication and inverses are strict in every simplicial degree. Thus the existence of a strictly associative simplicial model cannot disprove the comparison proposed here. This assertion concerns the existence of a model, rather than a claim that a particular category of paths is that model; the latter still requires a comparison argument.

The same manuscript subsequently supplies that argument for unrestricted paths. In Chapter VII, §13, manuscript pages 157–170, Grothendieck returns to the case $\Sigma = \mathrm{Fl}(X)$ and obtains the comparison through the two projections from the sequential Grothendieck construction. The endpoint functors and their fibres are essential; a weak equivalence of total categories alone would not justify a statement about their strict fibres. The elementary lemma that constant paths induce a homotopy equivalence $X \to \mathrm{Fun}(I_\tau, X)$ is left without a written proof at manuscript page 170. The doubts recorded on the present page therefore belong to the progress of the argument, not to its final conclusion (Grothendieck, 1990–1991o, Ch. VII, §13).

That remaining lemma has a direct proof. Delete the last vertex of a nontrivial oriented interval, retract it to its neighbour, and let j denote the inclusion of the remaining interval. The retraction r is a functor, rj is the identity, and either $jr \Rightarrow \mathrm{id}$ or $\mathrm{id} \Rightarrow jr$ is a natural transformation, according to the direction of the deleted edge. Iterating gives a finite zigzag between the identity of I_τ and its constant functor at the initial vertex. Precomposition gives the corresponding zigzag between the identity of $\mathrm{Fun}(I_\tau, X)$ and the composite of evaluation at that vertex followed by the constant-path

functor. The other composite is strictly the identity of X. Thus constant paths give a categorical homotopy equivalence, and hence a weak equivalence of nerves.

For the audit this distinction has two consequences. The inference from an alleged impossibility of strict loop multiplication is false as written, and is explicitly withdrawn by the author. The unrestricted path comparison must be recorded with its later positive treatment, whereas the unrestricted *general-fibration* comparison of §10 still has the counterexamples discussed above. These statements have different hypotheses. Neither footnote 24 nor the later path argument removes the base-category obstruction for an arbitrary $C \to T$.

`[Les annotations dans ce diagramme concernent les flèches en traits pleins.]`

Les objets importants sont H_T et C_∞, alors que C (sur T) et les C_t prennent un rôle technique intermédiaire, pour construire H_T et C_∞. On peut considérer C_∞ comme une catégorie (objet de (Cat)), mais dont la construction dépend de nombreux arbitraires, à l'inverse de H_T dans Ind Cat hot. C'est l'objet de $\mathrm{Hot}_0^\wedge$ déduit de C_∞ qui a un sens canonique, et peut être décrit comme la limite inductive de H_T comme objet de $\mathrm{Ind}\,\mathrm{Hot}_0^\wedge$, i.e. comme représentant le même foncteur sur Hot_0 que H_T.

Dans le cas géométrique $T = \underline{\mathrm{Ty}}$, et C de la forme $\underline{\mathrm{Ch}}_\Sigma(X; a, b)$, donnant lieu à $C_\infty(a,b)$, on *n'a pas* de définition naturelle d'un bifoncteur de composition des chemins [page 115]

$$C_\infty(a,b) \times C_\infty(b,c) \longrightarrow C_\infty(a,c).$$

On peut définir, certes, des accouplements

$$\underline{\mathrm{Ch}}_{\Sigma,\tau}(a,b) \times \underline{\mathrm{Ch}}_{\Sigma,\tau'}(b,c) \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\tau\tau'}(a,c),$$

ou, avec les notations relatives à la donnée de

$$\underline{t} : \mathbf{N}^o \longrightarrow \underline{\mathrm{Ty}},$$
$$\underline{\mathrm{Ch}}_n(a,b) \times \underline{\mathrm{Ch}}_m(b,c) \longrightarrow \underline{\mathrm{Ch}}_{n+m}(a,c),$$

mais ces accouplements *ne sont pas* compatibles avec les morphismes de transition des trois p. 73
systèmes inductifs de catégories $(\underline{\mathrm{Ch}}_n)_{n\in\mathbf{N}}$, ils ne le sont *qu'à homotopie près.* Cela suffit cependant pour avoir des accouplements parfaits

$$H_T(a,b) \times H_T(b,c) \longrightarrow H_T(a,c),$$

avec associativité, et (si $\Sigma = \mathrm{Fl}(X)$ tout entier) existence de *vrais* inverses, i.e. de flèches (dans Ind Cat hot) de 'passage au chemin opposé'

[page 116]

$$(*) \qquad H_T(a,b) \longrightarrow H_T(b,a), \qquad c \longmapsto c^o,$$

donnant lieu au diagramme commutatif exprimant que c'est bien un inverse pour la composition. Une autre façon de le dire, c'est que pour tout objet Y de Cat hot (ou de Ind Cat hot), les ensembles

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, H_T(a,b)) \overset{\text{déf}}{=} H_T^Y(a,b)$$

forment, pour $a, b \in X$ variables, les flèches d'un *groupoïde* ([25]) $\mathrm{Ch}_\Sigma^Y(X)$ ayant pour ensemble de sommets $\mathrm{Ob}\,X$, et $H_T^Y(a,b)$ étant formé des flèches de a vers b. L'application

$$H_T^Y(a,b) \longrightarrow H_T^Y(b,a), \qquad c \longmapsto c^o,$$

25. Si $\Sigma = \mathrm{Fl}(X)$. Si Σ est plus petit, au lieu d'un groupoïde, on trouve une *catégorie* sans plus.

déduite du foncteur $(*)$ plus haut n'est autre que l'application de passage à l'inverse. Cette catégorie n'est d'ailleurs autre que celle déduit de

$$X(Y)\Sigma_Y^{-1}, \qquad \text{où} \begin{cases} X(Y) = \underline{\mathrm{Hom}}_{\mathrm{Cat}}(Y, X) \\ \Sigma_Y \subseteq \mathrm{Fl}(X(Y)), \\ \text{défini en termes de } \Sigma \text{ 'argument par argument'}, \end{cases}$$

à l'aide du foncteur

$$X \longrightarrow X(Y), \qquad a \longmapsto a_Y,$$

associant à tout $a \in X$ le foncteur *constant* sur Y de valeur a, en prenant

$$H_T(a, b) \simeq \mathrm{Hom}_{X(Y)\Sigma_Y^{-1}}(a_Y, b_Y) \ . \qquad \text{[page 117]}$$

Commentary · Development (commentators') [ms. 115]

The stage of a concatenation

Grothendieck's ind-object accommodates concatenation because a sufficiently refined type can dominate any two given types and their concatenation. The notation $n + m$ is convenient for a multiplicative standard sequence, but does not follow from an arbitrary cofinal sequence alone. The following small calculation locates the missing convention.

A slow cofinal sequence. [ours] There is a cofinal decreasing sequence of types t_n for which no type map $t_{n+m} \to t_n t_m$ exists for some n, m.

Proof. Put $w = \Delta_+\Delta_-$ and

$$t_n = w^{k(n)}, \qquad k(n) = \lfloor \log_2(n + 1) \rfloor.$$

Use identity maps when $k(n + 1) = k(n)$ and collapse the final copies of w otherwise. This gives $t_{n+1} \to t_n$. It is cofinal: every finite signed word is obtained from a sufficiently long alternating word by deleting unwanted edges and expanding the selected edges into its maximal constant-sign runs. For $n = m = 3$, however, $t_6 = w^2$, whereas $t_3 t_3 = w^4$. A type map from a word to a reduced alternating target must use at least one source edge for every target edge, since the image of a single source edge is a constant-sign run. Four source edges cannot cover the eight edges of w^4. Thus there is no such map. □

The obstruction is visible on path categories themselves. Take $X = I_{w^4}$, with endpoints a, b, and let c be its middle vertex. The inclusions of its two four-edge halves give objects of $\underline{\mathrm{Ch}}_{t_3}(X; a, c)$ and $\underline{\mathrm{Ch}}_{t_3}(X; c, b)$. The category $\underline{\mathrm{Ch}}_{t_6}(X; a, b)$ is empty: every nonidentity arrow of this alternating interval joins adjacent vertices, so four transitions cannot cover the graph distance eight between its endpoints. Hence a pairing with target stage 6 cannot exist in this example, even with $\Sigma = \mathrm{Fl}(X)$.

For a general chain the correct target is a chosen $k = k(n, m)$ admitting $t_k \to t_n t_m$. Cofinality guarantees such a k, and passage to a larger stage or a different chosen map makes no difference in $\mathrm{Cat}_{\mathrm{hot}}$, by the homotopy theorem. Equivalently, one may choose the standard sequence $t_n = w^n$, where $t_n t_m = t_{n+m}$. We checked the type-map description in the preceding part of this chapter and the usual filtered-colimit formalism; we did not find this particular indexing correction recorded in the literature. Its purpose is to make the intended ind-construction exact, not to alter that construction.

Commentary · Exposition [ms. 116]

Parameters turn the ind-object back into localization

Grothendieck ends this part by explaining why the ind-object is geometrically useful: its values on parameter categories recover ordinary localization of diagram categories. The assertion has a direct proof, and it supplies associativity and inverses at the same time.

For an actual small category Y, write $X(Y) = \underline{\mathrm{Hom}}(Y, X)$ and let Σ_Y consist of natural transformations whose components belong to Σ. It contains identities and is closed under composition. A functor

$$Y \longrightarrow \underline{\mathrm{Ch}}_{\Sigma,\tau}(X; a, b)$$

is the same thing as a functor $Y \times I_\tau \to X$ whose endpoints are the constant functors a_Y, b_Y, with identity endpoint components on morphisms in Y, and whose backward arrows belong pointwise to Σ. By exchanging the two variables, it is exactly a Σ_Y-path of type τ in $X(Y)$ from a_Y to b_Y. This identification respects natural transformations and every type-reindexing functor. The two total categories over $\underline{\mathrm{Ty}}$ are therefore isomorphic.

Taking connected components and applying the localization theorem of §7 gives

$$\begin{aligned} H_T^Y(a,b) &:= \mathrm{Hom}_{\mathrm{Ind}(\mathrm{Cat\,hot})}(Y, H_T(a,b)) \\ &\cong \mathrm{Hom}_{X(Y)[\Sigma_Y^{-1}]}(a_Y, b_Y). \end{aligned}$$

This corrects two notational slips in the printed formulas: the ambient category on manuscript page 116 is Ind(Cat hot), and the left side on page 117 needs the superscript Y. No finiteness of Y is required for this formula. The uniform type of a Y-family is retained by the ind-object; replacing it by C_∞ is the separate step for which finite presentation was needed.

Concatenation on either side is the same operation on functors $Y \times I_\tau \to X$, so this bijection respects composition and units. It follows that the sets $H_T^Y(a,b)$ are the hom-sets of the full subcategory of $X(Y)[\Sigma_Y^{-1}]$ on the constant diagrams. They form a groupoid when Σ is all arrows: the reversed path represents the inverse, and the backtracking contractions of §7 are natural in Y. Yoneda in the ind-category turns these pointwise identities into the associativity, unit, and inverse identities of the ind-valued path system. Finite products of ind-objects are represented by the corresponding product diagrams, so the pairings have exactly the domain stated in the manuscript.

If Y itself is an ind-object, the abstract hom-set on the first line still makes sense. For a presentation $Y = (Y_i)$ it is $\lim_i H_T^{Y_i}(a,b)$. The expression $\underline{\mathrm{Hom}}(Y,X)$ on the second line is only the stated functor category when Y is an actual small category; it should not be silently applied to a formal ind-object. The natural extension is the displayed inverse limit on source indices.

Commentary · Exposition [ms. 117]

What the ind-object represents

The ind-object retains a uniform bound on the complexity of a family of paths. This is why it can encode localization before a single category has acquired a satisfactory geometric interpretation. With $[Y,D]_{\mathrm{cat}} = \pi_0 \underline{\mathrm{Hom}}(Y,D)$, the formula is

$$H_T^Y(a,b) = \operatorname*{colim}_n [Y, C_n(a,b)]_{\mathrm{cat}} \cong \mathrm{Hom}_{X(Y)[\Sigma_Y^{-1}]}(a_Y, b_Y).$$

Here Σ_Y consists of natural transformations whose components belong to Σ, and the right side uses the constant diagrams. A morphism on the right is represented by a finite zigzag of natural transformations. Its finite length is a bound shared by every object of Y; it therefore gives a map into one stage on the left. Refinement and the elementary relations of localization give the identifications in the colimit. This is the mechanism already discussed in the commentary on Chapter V, manuscript pages 30–33, concerning the parameter category and ind-representation. The formula printed on page 117 omits the superscript Y from H_T^Y; without it an ind-object is being equated to a set.

For finite Y, the natural map

$$\operatorname*{colim}_n [Y, C_n]_{\mathrm{cat}} \longrightarrow [Y, \operatorname*{colim}_n C_n]_{\mathrm{cat}}$$

is bijective. Indeed a functor uses finitely many objects, arrows, and relations, all of which lift to one stage. A finite zigzag of natural transformations uses finitely many more data, so equality also occurs at one stage. Finite presentation, rather than any topological fibrancy, is doing the work. For infinite Y, a functor to the colimit can have an unbounded range of stages, and this argument stops.

Composition of localized zigzags is associative because juxtaposition of finite strings is associative. When all arrows of X are inverted, reversal supplies the inverse string. Hence the family of sets $H_T^Y(a,b)$ is a groupoid with object set $\mathrm{Ob}(X)$. For a general Σ it is only a category. Passing from this set-valued representation to mapping *spaces* is an additional step: the notation does not identify the ordinary nerve of every functor category with a derived mapping object.

Dans le même cas géométrique, je me suis convaincu d'ailleurs que C_∞ était, en tant

qu'objet de Hot (pas seulement de $\mathrm{Hot}_0^{\wedge}$) *unique à* Hot-*isomorphisme près,* mais (chose étrange) peut-être pas à Hot-isomorphisme *unique* près. De façon générale, revenant aux

Commentary · Later developments [ms. 117]

From path components to mapping spaces

The last formula brings the discussion back to its original purpose: families of paths should describe morphisms after localization, and should continue to do so when an indexing category is introduced. At the level of sets of components, the preceding arguments give a complete description. To retain higher homotopies, a modern treatment keeps the diagram of zigzag categories together with the coherences in its changes of shape.

One precise formulation is the hammock formula. For a relative infinity-category $(\mathcal{C}, \mathcal{W})$, let $\mathrm{Zig}_Z(\mathcal{C}, \mathcal{W})_{x,y}$ denote the infinity-category of zigzags of shape Z with fixed endpoints and pointwise weak-equivalence transformations. Write $|-|$ for groupoid completion, which for an ordinary category is represented by the realization of its nerve. Then, with colimit taken in spaces,

$$\mathrm{Map}_{\mathcal{C}[\mathcal{W}^{-1}]}(x, y) \simeq \operatorname*{colim}_{Z \in \mathrm{Zig}_\partial^{\mathrm{op}}} \left|\mathrm{Zig}_Z(\mathcal{C}, \mathcal{W})_{x,y}\right|.$$

Here Zig_∂ has endpoint-preserving changes of shape. Arakawa and Cnossen prove this formulation in their 2026 preprint, Definition 1.7, Theorem 1.9 and Corollary 1.11; Remark 1.10 identifies composition with concatenation (Arakawa and Cnossen, 2026). Their §4 compares the construction with Dwyer–Kan's original hammock localization.

The notation must not obscure a difference in definitions. Grothendieck's $\underline{\mathrm{Ch}}_{\Sigma,\tau}$ on these pages is a *full* subcategory of the functor category: membership constrains the backward arrows of an object, while morphisms are arbitrary natural transformations with the specified endpoint condition. The hammock construction just quoted also constrains the vertical components of those transformations. A comparison of the two models for general Σ is an additional theorem, not a consequence of recognizing the same words of signs. When all arrows are marked this difference disappears, and the later comparison within Chapter VII is directly relevant. This qualification is particularly useful when passing from X to $X(Y)$: it prevents an equality of localized morphism *sets* from silently being promoted to an identification of mapping *spaces*.

Commentary · Development (commentators') [ms. 117]

Independence of the sequential homotopy type

Grothendieck's final distinction between an isomorphism and a uniquely determined isomorphism can be made precise without solving the comparison with the total category C. The two questions separate: a bad base can obstruct $C_\infty \to C$ while all the sequential models still have the same weak homotopy type. The following argument gives the latter conclusion under the abstract hypotheses of §10. Its ingredients are standard telescope arguments; we include their application here because we did not locate this exact reachability-preorder statement in the sources checked. For the usual sequential homotopy-colimit construction see Dugger (n.d., §14.9, Proposition 14.10 and Corollary 14.11).

Proposition (independence of choices). [ours] Let T be a nonempty small category and let $F : T^{\mathrm{op}} \to \mathrm{Cat}$ be a strict functor. Suppose that $F(u)$ and $F(v)$ are joined by a finite zigzag of natural transformations whenever u, v are parallel arrows of T. Let

$$t_0 \longleftarrow t_1 \longleftarrow \cdots, \qquad s_0 \longleftarrow s_1 \longleftarrow \cdots$$

be two chains whose objects are cofinal to the left in the reachability preorder of T. Then

$$\varinjlim_n F(t_n) \quad \text{and} \quad \varinjlim_m F(s_m)$$

have isomorphic images in Hot. The assertion includes arbitrary choices of the arrows in either chain. It does not assert a preferred isomorphism.

Proof. We first record precisely which properties of sequential homotopy types will be used. For a sequence of small categories A_n, the nerve commutes with its filtered colimit. Indeed, a simplex is

a finite string of objects and arrows subject to finitely many incidence equations; these data, and equalities between two presentations of them, occur at some common stage. Hence

$$N\big(\varinjlim_n A_n\big) = \varinjlim_n NA_n.$$

For a sequence of simplicial sets K_n, the canonical map

$$\mathrm{Tel}(|K_n|) \longrightarrow \Big|\varinjlim_n K_n\Big|$$

is a weak equivalence. Here the telescope glues the upper end of $|K_n| \times [0,1]$ to the next space by its transition map. To check this assertion on homotopy groups, replace simplicial sets by $\mathrm{Ex}^\infty K_n$. This replacement commutes with filtered colimits: $\mathrm{Sd}\,\Delta^r$ is finite for each r, so Ex commutes with them, and the sequential iteration does also. A filtered colimit of Kan complexes is Kan, because every horn is finite. Maps and homotopies defining a homotopy group use finite simplicial data, so homotopy groups, with a basepoint represented at some stage, commute with this filtered colimit. On the telescope side a map or a homotopy from a compact sphere or cylinder has bounded height. Each bounded initial telescope retracts to its final space. Its homotopy groups therefore give the same direct limits. The canonical map induces these identifications, including the statement about components, and is a weak equivalence.

We also need invariance when each transition map is replaced by a homotopic map. Suppose $f_n, g_n : A_n \to A_{n+1}$ are continuous maps with homotopies H_n from g_n to f_n. A map from $\mathrm{Tel}(f)$ to $\mathrm{Tel}(g)$ is the identity on every level A_n. On the cylinder for f_n, it first traverses the cylinder for g_n, then follows H_n in level A_{n+1}. More explicitly it sends (x, r), for $0 \le r \le \frac{1}{2}$, to $(x, 2r)$ in the g_n cylinder, and for $\frac{1}{2} \le r \le 1$ to $H_n(x, 2r-1)$ in level A_{n+1}. The endpoint at $r = 1$ is $f_n(x)$, so these maps glue. Reversing the homotopies gives a map back. Their composites are homotopic to the identities, fixing the levels, by cancelling a path immediately followed by its reverse and reparametrizing the cylinders. These cancellations are defined separately on each cylinder and agree at its ends; they consequently glue on the entire telescope. No compatibility between different H_n is required. Finally, passing to a cofinal subsequence preserves the weak homotopy type of a telescope, by the same bounded-height calculation of homotopy groups.

We now compare the chains. Choose indices, increasing without bound, and arrows in T to obtain an interleaving

$$t_{n_0} \longleftarrow s_{m_0} \longleftarrow t_{n_1} \longleftarrow s_{m_1} \longleftarrow \cdots .$$

At each step this is possible by cofinality of the opposite chain; after finding an index, moving farther down that chain allows the indices to increase. Apply F and take nerves and realizations. The even subsequence has objects $|NF(t_{n_i})|$. Its bonding map is obtained from the composite of two of the newly chosen arrows. That composite and the original transition $t_{n_{i+1}} \to t_{n_i}$ are parallel arrows of T. By hypothesis their pullbacks are connected by natural transformations; taking nerves turns these into a homotopy of continuous maps. The preceding telescope construction therefore identifies the homotopy type of this even subsequence with that of the original t-sequence. The odd subsequence has the corresponding property for the s-sequence. Both are cofinal subsequences of the same interleaved telescope. Combining these comparisons with the telescope-to-colimit weak equivalences proves the proposition. □

Overview · VII-b

Paths, localization, and the information in an indexing category

These pages occupy the middle of Chapter VII. They begin with the end of the combinatorial analysis of path types, establish the ordinary category of fractions by means of path categories, and then examine whether a category containing all shapes of paths has the same homotopy type as a sequentially enlarged model. Their central concern is a compatibility problem. Concatenation is completely satisfactory before taking limits: it is functorial, associative and unital. The sequential models make other calculations accessible, but their transition maps need not respect that composition strictly. Grothendieck is trying to preserve both advantages in one construction.

The category $\underline{\mathrm{Ty}}$ consists of finite oriented intervals, equipped with their spatial order and their two endpoints. Its arrows preserve that order, the categorical direction of each traversed edge, and the endpoints. Concatenation makes it a strict monoidal category. The opening calculation describes its nerve degree by degree as a free monoid: a composable string of type maps splits uniquely over the edges of its initial interval. The data attached to one initial edge are successive finite weightings recording how many edges replace it at each stage. The commentary on the first part of the chapter explains this calculation through manuscript page 57. The commentary here adds the distinction needed for its later use: a face can send an indecomposable generator to a product. Degreewise freeness is therefore not freeness on a simplicial set.

For an ordinary category X, a path of type τ is a functor $I_\tau \to X$. Between paths of the same type one keeps natural transformations, and between different types one also keeps pullback along interval maps. These categories assemble into a category fibred over $\underline{\mathrm{Ty}}$. Fixing endpoints a, b means taking the strict fibre of the endpoint functor: the components of a natural transformation at both ends must be identities. This requirement is responsible for several of the correct diagrams in the text. Dropping it would change the hom-sets obtained after passing to components.

If Σ is a wide subcategory of X, we allow backward edges only when they lie in Σ, and write $\underline{\mathrm{Ch}}_\Sigma(X; a, b)$ for the resulting total fixed-endpoint path category. Closure of Σ under identities and composition is exactly what makes this construction stable under every type map. A source edge can collapse, or cover a consecutive block of target edges; the two possibilities require precisely these two closure conditions. No calculus-of-fractions hypothesis is needed at this stage. The fibres are full subcategories of functor categories, so their internal vertical transformations remain unrestricted.

The first decisive result is

$$\mathrm{Hom}_{X[\Sigma^{-1}]}(a, b) \cong \pi_0 \underline{\mathrm{Ch}}_\Sigma(X; a, b).$$

Concatenation induces the composition on the right. A one-edge path followed by its reversal is connected to a constant path through a ladder with identity endpoint components, and the constant path is connected to the empty path by a type collapse. Thus every arrow of Σ becomes invertible. For a functor $F : X \to Y$ which inverts Σ, evaluation of a zigzag in Y is unchanged under either a ladder or a type change. In a ladder, the equations for its successive squares telescope to

$$\eta_n E_F(c) = E_F(c')\eta_0;$$

the endpoint identities give equality of effects. The argument uses inverses of the backward horizontal edges but never inverses of the internal vertical components. It supplies the omitted well-definedness check for the map β on manuscript page 68. The resulting evaluation has the required unique factorization property. This is the classical localization problem of Gabriel and Zisman (1967, Chapter I, §1), with the witnesses for its relations retained as categorical data.

Two examples make the theorem tangible. Inverting one arm of a span creates the fraction across the span and produces a category with a walking-arrow skeleton. Inverting both arrows of the category $a \rightrightarrows b$ produces a connected groupoid with automorphism group $\mathbf{Z}$; the nerve of the original category is a circle. The complete calculations distinguish cancellation of backtracking from the survival of a genuine loop. They also show why a relative localization need not be a groupoid.

The next task is to establish that concatenation really acts on all morphisms of the total path categories. A ladder can be glued to a second ladder because their common endpoint components are identities. A change of type can be glued to a second change of type because the common endpoint has a specified image. Writing a total morphism as a type map together with a vertical transformation makes the functoriality and strict associativity transparent. The commentary gives this calculation explicitly. It also repairs an edition diagram which glues over the two-element set $\{0, 1\}$ where the construction requires a single identified endpoint.

Sections 8 and 9 then explain why the particular interval map inducing a pullback ceases to matter after categorical homotopy. The relevant homotopy relation joins functors by finite zigzags of natural transformations. Grothendieck proves that every nonempty category of endpoint-preserving interval maps is connected, and asks whether it is contractible. We give a complete categorical contraction. After a forced initial collapse when the first signs disagree, the first target run can be filled by the first source edge. Clamping the remaining vertex images to the end of that run defines an idempotent functor and a natural transformation in the direction determined by the run's sign. The retract is a smaller interval-mapping category. Induction contracts it to a point. This is marked as a development of the present commentary, without a claim of priority.

Precomposition transports these natural transformations to the path categories. Because the endpoint components remain identities, the homotopy takes place over $X \times X$ and restricts to every fixed-endpoint fibre. In modern notation, the diagram of fibres becomes constant on parallel arrows after passage to Cat hot, the category whose morphisms are categorical-homotopy classes of functors. This is the bridge to the abstract discussion of a fibration $C \to T$ in §10.

Here three objects must be distinguished. There is the total category C, the ind-object determined by its fibres in Cat hot, and an ordinary sequential colimit C_∞ formed after choosing a chain cofinal in the reachability preorder T_{o}. The source assumes that this preorder is cofiltered. It does not assume that parallel arrows of T can be equalized. Forgetting that difference would conceal the principal obstruction: for the constant point-valued diagram over a one-object group BG, the ind-object and every sequential model are points, while the total category is BG.

There is a further failure on manuscript page 103. The reachability preorder of a slice T/t_0 need not be the lower ideal of t_0 in T_{o}. The category with two parallel arrows already disproves that assertion. The lower ideal may still be used by restricting to the full subcategory of objects which admit an arrow to t_0, but this is only cofinality on reachability preorders. Comparison of total categories requires the actual comma categories. The commentary formulates and proves the corresponding sufficient-and-necessary condition for base change to preserve the homotopy type of every Grothendieck fibration, using Quillen's Theorem A (Quillen, 1973, §1). The source's separate properness and smoothness criteria are also proved under its hypothesis on parallel pullbacks. Contractibility of the strict fibres of $T \to T_{\mathrm{o}}$, together with local constancy of the coefficients there, allows homotopy Kan extension to descend the diagram to the preorder. This argument explains why those sufficient criteria can hold without the universal comma condition (Maltsiniotis, 2005; Cisinski, 2009).

The sequential colimit itself is less problematic. The nerve preserves filtered colimits, and for a sequence its homotopy colimit maps by a weak equivalence to its ordinary colimit. We prove that two cofinal chains give isomorphic objects of Hot under the source's homotopy

hypothesis: interleave cofinal subsequences, compare their bonding maps by homotopies, and compare the resulting telescopes. This answers the existence part of the independence question. The proof neither chooses a preferred isomorphism nor promotes it to a categorical homotopy equivalence in Cat hot.

Finite probes explain part of the remaining caution. A finite category is finitely presented for ordinary categorical filtered colimits, which gives the asserted formula for Cat hot-morphisms into C_∞. Its nerve need not have the homotopy type of a finite CW complex. The one-object category of the group of order two is a concrete counterexample to unrestricted compactness after passage to Hot: its mod-two cohomology persists in every degree, whereas each finite-dimensional skeleton of its nerve has bounded cohomological dimension. Replacing Cat hot by Hot therefore needs a separate argument. The mapping-set formulas at the end of manuscript page 117 are audited with their parameter Y and their ambient ind-category restored in the commentary.

The proposed obstruction from strict loop multiplication on manuscript page 113 is already withdrawn in footnote 24. Kan's loop-group theorem supplies strict simplicial group models (Kan, 1958, Theorem 7.1). For unrestricted paths, the specific comparison is taken up positively later in this same chapter, §13; it is not a current open problem. The constant-path lemma left unwritten at manuscript page 170 is supplied here by successively retracting the end vertices of an oriented interval. Modern hammock localization gives a separate, precise mapping-space theory. Its requirement that vertical arrows also be weak equivalences must be checked when comparing it with Grothendieck's full path fibres (Dwyer and Kan, 1980a; Arakawa and Cnossen, 2026).

conditions du début du paragraphe, C_∞ doit s'écrire $C_\infty(\underline{t})$, elle dépend du choix du foncteur cofinal p. 74

$$\underline{t} : \mathbf{N}^o \longrightarrow T.$$

Quand $\underline{t}$ varie dans la catégorie

$$\underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T),$$

les $C_\infty(\underline{t})$ varient de façon contravariante en $\underline{t}$ (donc définissent une catégorie fibrée sur $\underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)$). De plus, le foncteur

$$C_\infty(\underline{t'}) \longrightarrow C_\infty(\underline{t})$$

associé à un morphisme de systèmes projectifs

$$\underline{t} \longrightarrow \underline{t'}$$

est un *isomorphisme* dans Hot. Donc le foncteur [page 118]

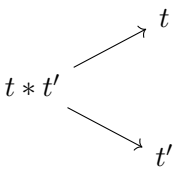
$$\underline{t} \longmapsto C_\infty(\underline{t}), \qquad \underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T) \longrightarrow (\mathrm{Hot})$$

se factorise par le groupoïde fondamental

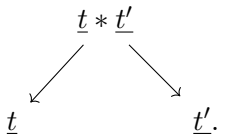
$$\Pi(\underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)) \longrightarrow (\mathrm{Hot}).$$

Quand ce groupoïde, i.e. quand la catégorie $\underline{\mathrm{Hom\,cof}}$ elle-même est connexe, on peut donc conclure que tous les $C_\infty(\underline{t})$ sont isomorphes entre eux dans Hot. Pour conclure même à un système *transitif* d'isomorphismes entre les $C_\infty(\underline{t})$, i.e. que $C_\infty(\underline{t})$ est 'unique à isomorphisme *unique* près', il faudrait savoir de plus que le groupoïde fondamental est équivalent à la catégorie ponctuelle (groupe fondamental trivial), ou, comme on dit, que $\underline{\mathrm{Hom\,cof}}$ soit *simplement connexe,* ou 1-connexe.

La connexité de $\underline{\mathrm{Hom\,cof}}$ est assurée quand la catégorie T a (comme c'est le cas pour $\underline{\mathrm{Ty}}$) une loi de composition $(t, t') \longmapsto t * t'$, avec des morphismes fonctoriels en t, t'



(cette dernière condition étant automatique si la loi de composition admet l'objet final de T, supposé existant, comme unité bilatère). Dans ce cas, $\underline{t} = (t_n)_n$ et $\underline{t'} = (t'_n)_n$ sont reliés entre p. 75
eux par $\underline{t} * \underline{t'} = (t_n * t'_n)_n$, puisqu'on a un morphisme de systèmes projectifs [page 119]



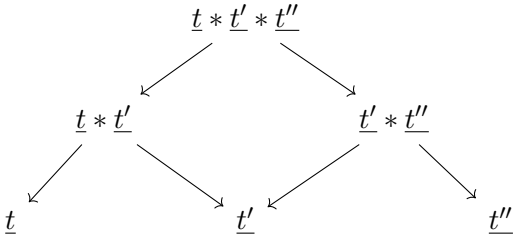
Je ne sais d'ailleurs pas si la flèche dans Hot

$$C_\infty(\underline{t}) \longrightarrow C_\infty(\underline{t'})$$

associée à $\underline{t'} \longrightarrow \underline{t}$ dans $\underline{\mathrm{Hom\,cof}}$ ne dépend que de $\underline{t}$, $\underline{t'}$, et pas du choix de $\underline{t'} \longrightarrow \underline{t}$. Ce qui est clair, c'est que son image dans $\mathrm{Hot}_0^\wedge$ est unique, c'est l'identité quand on identifie $C_\infty(\underline{t})$ et $C_\infty(\underline{t'})$ au seul et même élément de $\mathrm{Hot}_0^\wedge$ défini par $\varinjlim H_T$. [page 120]

Et pourtant, je m'aperçois à l'instant que, sous l'hypothèse de l'existence de la loi de composition et de son unité bilatère, et admettant de plus que la loi en question est associative à isomorphisme près, *on a transitivité* dans les isomorphismes (canoniques!) entre les $C_\infty(\underline{t})$ qu'on vient d'indiquer !

(Sans avoir pour autant à calculer le groupe fondamental de $\underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)$!) Si on a trois éléments $\underline{t}$, $\underline{t'}$, $\underline{t''}$ de $\underline{\mathrm{Hom\,cof}}$, il suffit de regarder le diagramme déduit du diagramme commutatif suivant



en lui appliquant le contrafoncteur $\underline{s} \longrightarrow C_\infty(\underline{s})$ de $\underline{\mathrm{Hom\,cof}}$ dans Hot.

Donc dans le cas des $C(a,b) = \underline{\mathrm{Ch}_\Sigma}(X; a, b)$, je n'ai plus aucun scrupule à considérer la catégorie $C_\infty(a,b)$ comme un objet *canonique* de Hot, relié à l'ind-objet canonique $H_T(a,b)$ dans Hot par un homomorphisme canonique

$$(*) \qquad C_\infty(a,b) \longleftarrow H_T(a,b),$$

qui dans $\mathrm{Hot}_0^\wedge$ donne un isomorphisme

$$(**) \qquad C_\infty(a,b) \simeq \varinjlim H_T(a,b),$$

i.e. la flèche $(*)$ induit, pour $\xi \in \mathrm{Ob\,Hot}$, une application ensembliste

$$\mathrm{Hom}_{\mathrm{Hot}}(\xi, C_\infty(a,b)) \longleftarrow \underbrace{\mathrm{Hom}_{\mathrm{Ind\,Hot}}(\xi, H_T(a,b))}_{\varinjlim_n \mathrm{Hom}_{\mathrm{Hot}}(\xi, C_n(a,b))},$$

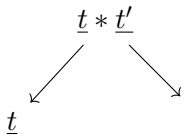
qui est *bijective* si ξ est dans Hot.

Bien entendu, les flèches

p. 76
[page 121]

$$(*) \qquad C_\infty(\underline{t'}) \longrightarrow C_\infty(\underline{t})$$

dans Hot, associés aux flèches $\underline{t} \longrightarrow \underline{t'}$ dans $\underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)$, sont en fait des flèches dans (Cat) elle-même - et même des foncteurs pleinement fidèles, dans le cas 'géométrique' et pour les deux flèches canoniques



Elles sont isomorphiques *dans* Hot, parce qu'elles définissent un homomorphisme sur les ind-objets associés de Cat hot, donc aussi pour les ind-objets associés dans Hot et dans $\mathrm{Hot}_0^\wedge$, d'où résulte que

$$\mathrm{Hom}_{\mathrm{Hot}}(Y, C_\infty(\underline{t'})) \longrightarrow \mathrm{Hom}_{\mathrm{Hot}}(Y, C_\infty(\underline{t}))$$

associé à $(*)$ est une bijection si Y *est finie* ([26]), cette flèche s'identifie alors à

$$\varinjlim_n \mathrm{Hom}_{\mathrm{Hot}}(Y, C_{t'_n}) \longrightarrow \varinjlim_n \mathrm{Hom}_{\mathrm{Hot}}(Y, C_{t_n}).$$

26. Ce qui suffit à assurer que $(*)$ est [un] isomorphisme dans Hot : Hot $\longrightarrow \mathrm{Hot}_0^\wedge$ est conservatif!

(Car $Z \longmapsto \mathrm{Hom}_{\mathrm{Hot}}(Y, Z)$ commute aux limites inductives filtrantes, si Y est une catégorie *finie.* Cela provient du fait qu'une $\varinjlim$ filtrante de catégories de Thomason-Kan est itou …) Mais j'ignore par contre si $(*)$ est $\overrightarrow{\texttt{[un]}}$ isomorphisme dans Cat hot. Seulement que pour une catégorie *finie* Y, elle induit

$$\mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, C_\infty(\underline{t}')) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, C_\infty(\underline{t})),$$

les deux membres s'identifiant alors à [page 122]

$$H_T(Y)_{\mathrm{Cat\,hot}} = \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(Y, H_T).$$

Pour résumer ces lumières et perplexités sur les $C_\infty(\underline{t})$:

Commentary · Development (commentators') [ms. 120]

Contracting the category of choices

Grothendieck seeks a coherent comparison between different cofinal sequences, so that changing a path parameter does not change the resulting operation in Hot. His three-sequence diagram supplies this comparison, but the natural projections already imply a stronger statement about the space of choices.

Contraction of the refinement category. [ours] Let T be a category with a bifunctor $*$ and natural maps $t * t' \to t$ and $t * t' \to t'$. Let $\mathcal{H}$ be the nonempty category of sequences $\mathbf{N}^o \to T$ cofinal in the preorder of objects. Then $N\mathcal{H}$ is contractible. Consequently every functor $\mathcal{H}^o \to \mathcal{A}$ taking all arrows to isomorphisms has a unique system of comparison isomorphisms induced by arrows and their inverses; these comparisons depend only on their two endpoints and compose transitively.

Proof. Fix $s \in \mathcal{H}$. Define $Q(t) = t * s$ pointwise, on both objects and morphisms of sequences. For every $a \in T$ there is an index n and a map $t_n \to a$. Composing $t_n * s_n \to t_n \to a$ shows that $Q(t)$ is still cofinal. The two natural projections give natural transformations

$$\mathrm{id}_{\mathcal{H}} \longleftarrow Q \longrightarrow \mathrm{const}_s.$$

The nerve of a natural transformation gives a homotopy of the associated maps on realizations: explicitly, a chain of arrows and a cut in that chain use the first functor before the cut, the transformation at the cut, and the second functor afterwards. Thus the identity of $B\mathcal{H}$ is homotopic to the constant map at s, proving contractibility.

A functor taking all arrows to isomorphisms factors through the localization of $\mathcal{H}$ at all its arrows, its fundamental groupoid. Contractibility makes this groupoid equivalent to a point: between any two of its objects there is exactly one morphism. Applying the functor to that morphism proves the uniqueness and transitivity assertions, including independence from the particular arrow of sequences used. The opposite variance changes the direction of the comparisons but not the argument. □

Associativity of $*$ and a terminal unit are unnecessary for this proof once the two natural projections are given. In the interval application, concatenation has these projections by collapsing the other interval, and a cofinal sequence exists. The proposition isolates the formal part of the argument. It does *not* prove that the actual functors $C_\infty(t') \to C_\infty(t)$ are weak equivalences; that premise requires the geometric comparison or a valid relative argument. In particular the conservativity assertion in footnote 26 cannot supply it in the stated generality. We checked the general nerve and cofinality mechanisms in Quillen's §1 and the smooth-functor formulation of Maltsiniotis (2005, Theorem 1.2); we did not locate this explicit application to Grothendieck's category of sequences in those sources. The result is an elementary continuation of his construction, without a claim of priority.

Commentary · Exposition [ms. 121]

Two different finiteness arguments

The comparison on this page must keep finite categorical data separate from a finite homotopy type. The preceding block on ind-representation proves a filtered-colimit formula in Cat hot whenever the source category is finitely presented. In Hot, the corresponding safe hypothesis is that the classifying space of the source has the homotopy type of a finite CW complex, or more generally

is compact for the filtered homotopy colimits in question. A finite poset has a finite nerve and meets this hypothesis. But Grothendieck explicitly defines a finite category by finiteness of both its objects and its arrows in Chapter V, manuscript page 45, and uses these categories to define Hot_0 on pages 51–52 (Grothendieck, 1990a). With that convention, the assertion is false.

For a concrete distinction, let Y be the one-object category of the group $\mathbf{Z}/2$. It is finite, but $BY \simeq \mathbf{RP}^\infty$. Consider the increasing skeleta $\mathbf{RP}^n$. The identity of $\mathbf{RP}^\infty$ cannot factor, up to homotopy, through one skeleton: such a factorization would make the identity on $H^k(\mathbf{RP}^\infty; \mathbf{F}_2)$ factor through zero for every $k > n$, whereas these cohomology groups are nonzero. Thus

$$\operatorname*{colim}_n [BY, \mathbf{RP}^n] \longrightarrow [BY, \mathbf{RP}^\infty]$$

is not surjective. The same example can be expressed entirely in categories by using compatible face-poset models for the finite skeleta, or the categories of simplices of their simplicial models. It is the size of the nerve in all dimensions, not merely the number of arrows, that matters here.

For a finite simplicial source K, the valid filtered-colimit argument is direct. Replace the targets by a functorial Kan replacement built using finite horn data. Such a replacement commutes with filtered colimits. Maps from K and simplicial homotopies between them involve finitely many simplices, so they appear at one stage. Finite CW sources follow by a finite simplicial model up to homotopy. The statement about filtered colimits of Thomason-fibrant categories is also valid, since the relevant horn domains are finite after applying the fixed subdivisions. It does not establish compactness for $B(\mathbf{Z}/2)$.

Accordingly, the literal last sentence of page 120 omits even the intended subscript zero; restoring that subscript would still leave the compactness error. A valid replacement restricts to homotopy-finite tests and thereby changes the stated test category. This repair does not alter the elementary finite-presentation calculation in Cat hot.

Commentary · Present state [ms. 121]

The unpointed detection claim

Grothendieck uses finite tests here to recognize a weak equivalence. In the unpointed homotopy category, the assertion that $\mathrm{Hot} \to \widehat{\mathrm{Hot}_0}$ is conservative is false. The editors already flag the claim in Chapter V, manuscript page 52, and the commentary there gives the symmetric-group shift example; we do not repeat it. The general theorem is stronger: no set of spaces jointly detects all isomorphisms in the unpointed homotopy category (Arlin and Christensen, 2023, Theorem 2.1). Their Theorem 4.3 shows why retaining one more level of data helps: spheres do detect equivalences in the homotopy 2-category, whose morphism categories retain homotopy classes of homotopies.

This defect in the inference does not disprove the particular path comparisons. They can be proved with based or relative information, or, in the all-arrows situation of section 13, by comparing the explicit total path categories over $X \times X$. An audit must separate that geometric theorem from the invalid general detection principle used on this page. For based spaces, all basepoints and their homotopy groups give the usual Whitehead test. For free homotopy classes, the fundamental-group information lost to conjugacy must first be restored.

Commentary · Development (commentators') [ms. 121]

A comparison that fails under the general hypotheses

The preceding distinction can be tested inside the exact fibred setting of section 10, rather than only in the abstract homotopy category. Grothendieck assumes there that parallel reindexing functors are homotopic, that the preorder of the base is cofiltered and has a countable coinitial subset, and that a strict cleavage has been chosen. These assumptions produce the same ind-object, but they do not force every map between the resulting colimits to be a weak equivalence.

A fibred symmetric-group example. [ours] There is a strict fibred category $C \to T$ satisfying these assumptions, with T also admitting an associative product with terminal unit and natural projections, for which a morphism of cofinal sequences induces an isomorphism in Ind Cat hot and a noninvertible map in Hot.

Proof. Let FI be the skeleton of finite sets and injections, with objects $[n] = \{0, \dots, n-1\}$, and set $T = \mathrm{FI}^o$. Write $\mathbf{B}G$ for the one-object category of a group G. Define a strict functor

$$F : T^o = \mathrm{FI} \longrightarrow \mathrm{Cat}, \qquad F([n]) = \mathbf{B}\mathfrak{S}_n.$$

For an injection $j : [n] \hookrightarrow [m]$, send a permutation σ to the permutation $j\sigma j^{-1}$ on $j([n])$, extended by the identity on its complement. Extension along a composite of injections agrees exactly with successive extension, so F is a functor. Its Grothendieck construction is strictly fibred over T.

If $j, k : [n] \hookrightarrow [m]$ are parallel injections, choose $h \in \mathfrak{S}_m$ with $hj = k$, by extending the resulting bijection of their images to the complements. Then $F(k)(\sigma) = hF(j)(\sigma)h^{-1}$ for every σ. Conjugation by h is a natural isomorphism between the associated one-object functors. Thus the homotopy hypothesis is satisfied. The preorder T_o is $\mathbf{N}^o$. Disjoint union provides the required associative operation and the empty set is its terminal unit in T; the two inclusions in FI become its natural projections.

Take $t'_n = [n]$ and $t_n = [n+1]$, with the opposite of the usual initial-segment inclusions as transitions. Both sequences are cofinal. The injections $\delta_n : [n] \hookrightarrow [n+1]$, $i \mapsto i+1$, commute with those inclusions and therefore define a morphism $t \to t'$ in $\mathcal{H}$. The two filtered colimits of the fibres identify with $\mathbf{B}G$, where G is the group of finite-support permutations of $\mathbf{N}$. The induced functor is the homomorphism

$$s : G \longrightarrow G, \qquad (s\sigma)(0) = 0, \quad (s\sigma)(i+1) = \sigma(i) + 1.$$

It is not surjective, since the transposition (0 1) is absent from its image. Consequently Bs does not induce an isomorphism on fundamental groups and is not a weak equivalence.

At each finite stage, however, δ_n and the standard inclusion are parallel injections, so their induced functors are naturally isomorphic. In $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$ the comparison is therefore the ordinary cofinal-shift isomorphism. This proves all assertions. □

The earlier Chapter V counterexample supplies the shift mechanism; the additional content here is its realization by a strict fibred category satisfying the hypotheses actually used on pages 117–123. We checked those hypotheses against the editors' complete Chapter VII, manuscript pages 99–101, and the detection theorem of Arlin and Christensen (2023); we did not find this indexed realization in those sources. No priority is claimed. The example does not satisfy the stronger full-faithfulness features of the geometric padding maps. It leaves the later all-arrows path theorem intact, while disproving the general inference that an ind-isomorphism of homotopy classes must remain an isomorphism after these ordinary colimits are formed.

a) À tout $\underline{t} = (t_n)_{n \in \mathbf{N}^o} \in \underline{\mathrm{Hom\,cof}}(\mathbf{N}^o, T)$ est associé un système inductif, indexé par $\mathbf{N}$, de catégories $C_n(\underline{t})$ $(= C_{t_n})$, et leur limite inductive

$$C_\infty(\underline{t}).$$

b) Ce système inductif, et aussi sa limite est [un] foncteur contravariant en $\underline{t}$. Si $\underline{t}, \underline{t'} \in \mathrm{Ob}\,\underline{\mathrm{Hom\,cof}}$, alors on peut comparer $C_\infty(\underline{t})$, $C_\infty(\underline{t'})$ et les systèmes inductifs correspondants, grâce à p. 77

$$\begin{array}{ccc} & \underline{t} * \underline{t'} & \\ \swarrow & & \searrow \\ \underline{t} & & \underline{t'}. \end{array}$$

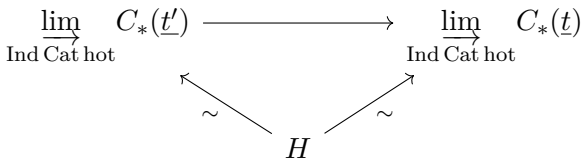
c) L'objet

$$(C_n(\underline{t}))_{n \in \mathbf{N}} \in \mathrm{Ob}\,\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$$

défini par le système inductif $(C_n(\underline{t}))_{n \in \mathbf{N}}$ de Cat est (à isomorphisme unique près) indépendant du choix de $\underline{t}$, soit H_C ou H. Les homomorphismes

$$(*) \qquad C_\infty(\underline{t'}) \xrightarrow{u^*} C_\infty(\underline{t}) \qquad \text{associé à } u : \underline{t} \longrightarrow \underline{t'}$$

sont compatibles avec cette donnée, i.e. on a commutativité dans



d) Il s'ensuit que les homomorphismes $(*)$ sont des *isomorphismes* dans Hot, donc tous les $C_\infty(\underline{t})$ sont isomorphes *dans* Hot. L'astuce des $t * t'$ (dans le cas où on a une loi $t * t'$ associative et admettant l'objet final comme objet unité) donne même des isomorphismes *canoniques* [page 123]

$$C_\infty(\underline{t}') \longrightarrow C_\infty(\underline{t}) \qquad \text{dans Hot},$$

et c'est là un système transitif d'isomorphismes. Donc on a dans Hot un objet canonique

$$C_\infty \in \mathrm{Ob}\,\mathrm{Hot},$$

dont la relation à $H \in \mathrm{Ob}\,\mathrm{Ind}\,\mathrm{Hot}$ est déduite de c), c'est une flèche canonique

$$H \longrightarrow C_\infty \qquad \text{dans Ind Hot } ([27]).$$

Cette flèche induit un isomorphisme pour les objets correspondants de $\mathrm{Hot}_0^\wedge$; mais en regardant H non comme ind-objet de $\mathrm{Hot}_0^\wedge$, mais prenant le foncteur H_0 induit sur Hot_0 par le foncteur ind-représentable H sur Hot. Ce foncteur induit est 'représentable' par l'objet C_∞ de la catégorie plus grande

$$H_0 \simeq C_\infty \qquad \text{dans } \mathrm{Hot}_0^\wedge.$$

(Si le foncteur canonique $\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^\wedge$ est pleinement fidèle, alors cette formule décrit C_∞ comme objet de Hot, à isomorphisme unique près, en termes de H_0, donc aussi en termes de $H \in \mathrm{Ob}\,\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$.)

e) J'ignore par contre si les flèches [p. 78] [page 124]

$$C_\infty(\underline{t}') \longrightarrow C_\infty(\underline{t})$$

sont des isomorphismes dans Cat hot, bien qu'elles soient définies en termes d'homomorphismes de systèmes inductifs de catégories, qui sont des isomorphismes dans Ind Cat hot. Il en est ainsi, même pour les flèches $C_\infty(\underline{t}), C_\infty(\underline{t}') \rightrightarrows C_\infty(\underline{t} * \underline{t}')$ utilisées pour comparer $C_\infty(\underline{t})$ et $C_\infty(\underline{t}')$ canoniquement. Ainsi, je ne vois pas comment définir des flèches en sens inverse dans Cat, qui seraient candidats à être des isomorphismes inverses dans Cat hot.

Commentary · Examples [ms. 123]

A colimit need not be the same ind-object

Grothendieck's warning in footnote 27 concerns the test objects allowed, not a defect of the colimit. It is already visible for sets. Put $D_n = \{0, \dots, n\}$, viewed as discrete categories, and $D = \mathbf{N}$. For every finite discrete K one has

$$\operatorname*{colim}_n \mathrm{Hom}(K, D_n) = \mathrm{Hom}(K, D),$$

27. **NB** H n'est pratiquement jamais essentiellement constant, donc la flèche ci-contre `[n'est]` pas un isomorphisme dans Ind Hot.

> because the image of a finite set is bounded. The map from the ind-object $(D_n)_n$ to the constant ind-object D is nevertheless not an isomorphism. Applying maps from the constant object $\mathbf{N}$, its source consists of bounded-image functions $\mathbf{N} \to \mathbf{N}$, while the target also contains the identity. These computations are unchanged in Hot: maps between discrete spaces are functions, and no homotopy identifies different functions.
>
> Thus restriction to the finite-test category can identify two presheaves that differ on larger tests. The ind-object records how a family is bounded; the colimit records its pointwise values after the bound has been forgotten. This is the precise content behind "pratiquement jamais essentiellement constant." It is not a universal prohibition: a sequence that is eventually constant is a constant ind-object, and more generally one can have an ind-isomorphism to a constant object without literal eventual constancy.

f) Dans le cas géométrique ($T = \underline{\mathrm{Ty}}$, situation décrite en termes de X, Σ, et $a, b \in \mathrm{Ob}\, X$, d'où des $C_\infty(\underline{t}; a, b)$, $C_\infty(\overline{a, b})$, $H(a, b)$ etc.), on *n'a pas* d'opération de composition de chemins au niveau des *catégories* $C_\infty(\underline{t}; a, b)$ et $C_\infty(\underline{t}; b, c)$, mais seulement au niveau des objets $H(a, b)$, $H(b, c)$ de Ind Cat hot ou de Ind Hot. Suis-je seulement sûr de savoir définir *dans* Hot

$$C_\infty(a, b) \times C_\infty(b, c) \longrightarrow C_\infty(a, c) \quad ?$$

Sûrement, il n'y a pas de problèmes pour les images dans $\mathrm{Hot}_0^\wedge$, puisque là ce sont les mêmes que celles de $H(a, b)$, $H(b, c)$, $H(a, c)$ respectivement, et on a des accouplements parfaits, associatifs, avec unité, et avec inverses là ou il doit y en avoir. Mais pas pour les $C_\infty(a, b)$ eux-mêmes? [page 125]

C'est un peu vexant qu'avec tout ce travail il semblerait qu'on n'ait pas même été foutu de définir l'analogue dans Cat du classique espace des lacets $\Omega(X, x)$ comme un groupe *dans* Hot. (Que fait Thomason?) Mais voilà une idée pensable [?]. Pour définir *quand-même*

$$C_\infty(a, b) \times C_\infty(b, c) \longrightarrow C_\infty(a, c)$$

dans Hot, on va décrire les deux foncteurs $C_\infty(a, b)$ et $C_\infty(b, c)$ par des systèmes projectifs différents $\underline{t}$, $\underline{t'}$. On peut l'expliciter en partant d'un τ_0 de longueur utile ≥ 2, et en prenant dans les deux cas

$$t_n = t'_n = \tau_0^n,$$

mais des morphismes de transition différents

$$(t_{n+p} \longrightarrow t_n) = \pi_p * \mathrm{id}_{\tau_0^n}$$
$$(t'_{n+p} \longrightarrow t'_n) = \mathrm{id}_{\tau_0^n} * \pi_p,$$

où $\pi_p : \tau_0^p \longrightarrow 1$ est la dégénérescence totale. On définira $C_\infty(a, c)$ en termes de $\underline{t''} = \underline{t} * \underline{t'}$, donc

$$\begin{cases} t''_n = \tau_0^{2n} \\ (t''_{n+p} \longrightarrow t''_n) = \pi_p * \mathrm{id}_{\tau_0^{2n}} * \pi_p. \end{cases}$$

Cette fois, les foncteurs de composition p. 79

$$C_n(\underline{t}; a, b) \times C_n(\underline{t'}; b, c) \longrightarrow C_n(\underline{t''}; a, c),$$

sont compatibles avec les foncteurs transition pour n variable, et on trouve [page 126]

$$C_\infty(\underline{t}; a, b) \times C_\infty(\underline{t'}; b, c) \longrightarrow C_\infty(\underline{t''}; a, c),$$

donc une flèche dans Hot

$$\mathrm{comp}_{\tau_0} : C_\infty \times C_\infty \longrightarrow C_\infty.$$

Il faut voir qu'elle ne dépend pas du choix de τ_0, mais si on a τ_0 et τ_0', on les compare par $\tau_0 \times \tau_0' \begin{array}{l} \longrightarrow \tau_0 \\ \longrightarrow \tau_0' \end{array}$, il suffit donc de voir que s'il existe une flèche $\tau_0 \longrightarrow \tau_0'$, alors $\mathrm{comp}_{\tau_0} = \mathrm{comp}_{\tau_0'}$. Mais une telle flèche induit des flèches $\underline{t}_{\tau_0} \longrightarrow \underline{t}_{\tau_0'}$, $\underline{t'}_{\tau_0} \longrightarrow \underline{t'}_{\tau_0'}$, $\underline{t''}_{\tau_0} \longrightarrow \underline{t''}_{\tau_0'}$, d'où un diagramme dans Cat

$$\begin{array}{ccc} C_\infty(\underline{t}_{\tau_0'}; a, b) \times C_\infty(\underline{t'}_{\tau_0'}; b, c) & \xrightarrow{\mathrm{comp}_{\tau_0'}} & C_\infty(\underline{t''}_{\tau_0'}; a, c) \\ \downarrow & & \downarrow \\ C_\infty(\underline{t}_{\tau_0}; a, b) \times C_\infty(\underline{t'}_{\tau_0}; b, c) & \xrightarrow{\mathrm{comp}_{\tau_0}} & C_\infty(\underline{t''}_{\tau_0}; a, c), \end{array}$$

dont on vérifie tout de suite qu'il est commutatif. Or ce diagramme peut se réécrire dans Hot comme

$$\begin{array}{ccc} C_\infty(a, b) \times C_\infty(b, c) & \xrightarrow{\mathrm{comp}_{\tau_0'}} & C_\infty(a, c) \\ \downarrow \mathrm{id} & & \downarrow \mathrm{id} \\ C_\infty(a, b) \times C_\infty(b, c) & \xrightarrow{\mathrm{comp}_{\tau_0}} & C_\infty(a, c), \end{array}$$

d'où $\mathrm{comp}_{\tau_0} = \mathrm{comp}_{\tau_0'}$.

[page 127]

Commentary · Exposition [ms. 126]

Padding at the two outer ends

Grothendieck's construction now solves the binary-composition problem by choosing the locations where new constant segments are inserted. A common stage in each factor is harmless because the diagonal $\mathbf{N} \to \mathbf{N} \times \mathbf{N}$ is cofinal. The location of the padding is what makes the squares commute.

Write a path in the first system as c_n and one in the second as d_n. Under $n \mapsto n + p$, the first becomes a string of p constant blocks followed by c_n, while the second becomes d_n followed by p constant blocks. Concatenation gives

$$(1_a^{\cdot p} * c_n) * (d_n * 1_c^{\cdot p}) = 1_a^{\cdot p} * (c_n * d_n) * 1_c^{\cdot p},$$

which is exactly the transition prescribed for the target system. The equality holds for the vertical transformations as well as for objects, because concatenation is a bifunctor. It therefore induces a functor on the filtered colimits. Products of categories commute with these filtered colimits, again because objects, arrows, and their defining equations are finite data.

A morphism $\tau_0 \to \tau_0'$ gives a commutative square before passage to homotopy; once the refinement maps have been established to be weak equivalences and coherently identified, the square gives equality of the two binary operations in Hot. To compare unrelated types, one can use $\tau_0 * \tau_0'$ and its two collapse maps. The printed $\tau_0 \times \tau_0'$ on this page cannot literally mean their categorical product as interval types: the product of two nontrivial intervals generally has a square rather than a linear underlying shape. Concatenation supplies the common refinement required by the argument.

The remaining associativity issue is real at the level of this proof. A third path has two joining positions, and the two binary constructions use different padding conventions at those positions. One must compare the resulting maps with coherent reparametrizations, or compare with a single strictly associative all-types model. Equality after restriction to finite tests cannot be substituted for that comparison. In the all-arrows case, the comparison established in section 13 is precisely the route back to the strict concatenation which motivated the discussion on page 111.

Je recule devant la vraie difficulté - celle de prouver l'associativité de cette loi de composition. Elle devient associative dans $\mathrm{Hot}_0^\wedge$ - mais pour que deux flèches dans Hot soient égales (ici donc

$$C_\infty(a, b) \times C_\infty(b, c) \times C_\infty(c, d) \rightrightarrows C_\infty(a, d) \quad),$$

suffit-il qu'il en soit ainsi de leurs images dans $\mathrm{Hot}_0^\wedge$ - en d'autres termes, le foncteur

$$\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^\wedge,$$

est-il (en plus d'être conservatif) au moins fidèle (sinon pleinement fidèle)?

Il y a aussi que cette loi de composition est compatible avec celle des $H(a,b)$ dans Ind Hot, p. 80
pour les flèches canoniques

$$H(a,b) \longrightarrow C_\infty(a,b) \qquad \text{dans Ind Hot},$$

si on saurait que ce sont des épimorphismes, et le restent par produits cartésiens, on aurait gagné.

Bien sûr, on saura que c'est associatif, une fois qu'on saura que la composititon définie ici est compatible avec la composition des chemins dans le cadre semi-simplicial. Mais comment fait-on dans ce cadre, pour composer les chemins de longueur non bornée? N'y a-t-il pas là exactement la même difficulté? À moins de s'en tirer en se ramenant tout de suite, pour la [page 128]
théorie des opérations Ω, aux complexes de Kan, et aux chemins de type Δ_+. Mais pour les composer *fonctoriellement,* il me semble y avoir toujours le même problème - on trouvera des chemins de type $\Delta_+\Delta_+ = \mathbf{\Delta}_2^+$, et non pas Δ_+, mais on peut alors composer les homotopies directes (revient à s'induire sur $\Delta_+ \simeq \{0,2\} \subseteq \mathbf{\Delta}_2^+$). Or j'ai fait le formalisme des catégories de chemins sans me soucier de conditions du type Kan. Je présume que dans le cadre semi-simplicial, les $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1, X)$ ne donnent *pas* le bon objet des chemins, pour X pas de Kan. (Car sinon, il semblerait que pour $X = X_0$ à composantes *finies,* le calcul des π_i devrait être fini, ce qui n'est pourtant pas le cas, m'a-t-on toujours assuré …)

Commentary · Examples [ms. 127]

A map invisible at every finite stage

Grothendieck asks whether finite tests determine a morphism and whether the ind-object maps epimorphically onto its colimit. A small telescope gives a negative answer to both questions. We give the calculation because it exhibits the obstruction without relying on an unspecified phantom-map example.

Let T be the mapping telescope of the degree-two maps

$$S^1 \xrightarrow{2} S^1 \xrightarrow{2} S^1 \xrightarrow{2} \cdots .$$

Its finite subtelescopes T_n retract onto circles. Its fundamental group is $\mathbf{Z}[1/2]$, and its higher homotopy groups vanish: a map or nullhomotopy from a compact sphere lies in a finite subtelescope. Thus T is a $K(\mathbf{Z}[1/2], 1)$. After contracting the ray of vertices, a cellular model has one vertex, one loop e_n for each $n \geq 0$, and one two-cell with boundary $e_n - 2e_{n+1}$. Cellular cohomology consequently gives

$$H^2(T;\mathbf{Z}) = \operatorname{coker}\left(\prod_{n\geq 0}\mathbf{Z} \longrightarrow \prod_{n\geq 0}\mathbf{Z}, \quad (b_n) \longmapsto (b_n - 2b_{n+1})\right).$$

This is the standard telescope computation of Hatcher (2002, Examples 3F.7 and 3F.9).

To see nonvanishing directly, take $a_{2j} = 1$ and $a_{2j+1} = 0$. If $a_n = b_n - 2b_{n+1}$, iteration would give

$$b_0 = \sum_{j=0}^{N} 2^j a_j + 2^{N+1} b_{N+1} \qquad (N \geq 0).$$

In the ring of 2-adic integers, this forces $b_0 = 1 + 4 + 16 + \cdots = -1/3$, impossible for an integer b_0. Hence a determines a nonzero cohomology class. By the representation of integral degree-two cohomology, it determines an essential map

$$f : T \longrightarrow K(\mathbf{Z}, 2)$$

(Hatcher, 2002, Theorem 4.57). Each restriction to T_n is nullhomotopic, since $H^2(T_n;\mathbf{Z}) = 0$. Every map from a finite CW complex into T has image in a finite subtelescope up to homotopy, so all finite homotopy tests identify f with the constant map.

Even if the test category includes nerves of all categories with finitely many objects and arrows, this example remains invisible. Each connected component of such a nerve has finitely generated

fundamental group. Any homomorphism from that group to $\mathbf{Z}[1/2]$ has cyclic image, because finitely many denominators have a common power of two. The associated map to $T = K(\mathbf{Z}[1/2], 1)$ therefore factors up to homotopy through $K(\mathbf{Z}, 1) = S^1$, or through a point. Its composite with f is nullhomotopic because $[S^1, K(\mathbf{Z}, 2)] = 0$.

Finally let $H = (T_n)_n$ in $\mathrm{Ind}\,\mathrm{Hot}$. The two maps $f, 0 : T \rightrightarrows K(\mathbf{Z}, 2)$ become equal after precomposition with $H \to T$, since

$$\mathrm{Hom}_{\mathrm{Ind}\,\mathrm{Hot}}(H, K(\mathbf{Z}, 2)) = \lim_n [T_n, K(\mathbf{Z}, 2)] = \{0\}.$$

They are distinct before precomposition. Hence $H \to T$ is not an epimorphism. This example refutes the proposed general inference; it does not contradict associativity of the particular path composition, which requires its own geometric proof.

Je présume pourtant, quand on prend $\Sigma = \mathrm{Fl}(X)$, que les objets $\underline{\mathrm{Ch}}_\infty(X; a, b)$ et $\underline{\mathrm{Ch}}_\infty(\bar{X}; a, b)$ sont isomorphes dans Hot, quand on immerge X dans une enveloppe de Thomason-Kan $\bar{X}$.

Commentary · Exposition [ms. 128] p. 81 [page 129]

Edges and the fibres of the path object

Grothendieck wants the categorical construction to work before fibrant replacement. The distinction needed here concerns the endpoint fibres. For a Kan complex K, the map $K^{\Delta^1} \to K \times K$ is a Kan fibration, and its fibre over two vertices models their space of paths. For an arbitrary simplicial set, the same strict fibre need not compute a homotopy fibre. This does not mean that the total space K^{Δ^1} has the wrong homotopy type: evaluation at an endpoint has the constant-path section and the simplicial interval contraction gives a homotopy from their composite to the identity.

A finite example makes the endpoint problem explicit. Let X be the category with objects $0, 1, 2$ and nonidentity arrows $0 \to 1 \leftarrow 2$. Its classifying space is contractible, so the topological space of paths from 0 to 2 is nonempty and contractible. But there is no morphism $0 \to 2$. The fibre of the functor category $\underline{\mathrm{Hom}}([1], X) \to X \times X$ over $(0, 2)$ is therefore empty. The two-edge zigzag already supplies the missing path, and the refinement formalism is designed to supply all the further homotopies between such representatives.

There is a related distinction in the discussion of composition. Two composable edges of a simplicial set specify the spine $\Delta^1 \cup_{\Delta^0} \Delta^1$, not an entire Δ^2. For a Kan complex a filler produces a third edge, with higher fillers controlling the dependence on choices. In an ordinary category the interval [2] already includes the composite arrow; this explains why the categorical notation for two direct segments can conceal the simplicial filling step. A simplicial loop group, a Moore-path model, or a coherent localization retains the data needed to compare repeated compositions. None requires an algorithm computing all homotopy groups of a finite simplicial set.

The last conjecture on this page is correct for the all-arrows path object: a weak equivalence of categories induces weak equivalences on the corresponding derived path spaces, with their endpoints carried along. The justification for identifying Grothendieck's specific categories with those spaces belongs to the geometric argument in sections 12–13, not to a calculation with one-edge paths.

11 Rapide retour sur les $(\mathcal{M}, \Sigma)$, dans l'optique des dérivateurs

Commentary · Exposition [ms. 129]

What an external Hom must represent

Grothendieck now asks what his paths would have to accomplish in order to serve a theory of derived diagrams. A path from a to b is no longer only a point of a topological mapping space. It must also describe a morphism between the constant diagrams a_Y and b_Y, after objectwise localization of Y-diagrams. This is the reason for testing the path construction against every parameter category Y.

Write

$$\mathbb{D}(Y) = \mathrm{Fun}(Y, \mathcal{M})[\Sigma_Y^{-1}], \qquad F_{a,b}(Y) = \mathrm{Hom}_{\mathbb{D}(Y)}(a_Y, b_Y).$$

The conventions on variance can be reversed together without changing this discussion. The first displayed condition says precisely that $F_{a,b}$ factors through the localization of categories at nerve weak equivalences. It is a necessary invariance property for the proposed representation by a space. It is not by itself a definition of a derivator: existence and pointwise computation of homotopy Kan extensions are additional requirements. For combinatorial model categories these requirements hold by Groth (2013, Proposition 1.30). For their locally constant diagrams, the invariance asserted here is a theorem of Cisinski (2009, Theorem 1.3); constant diagrams are particular locally constant diagrams, and the subcategory in that theorem is full.

The contemporary form of the intended answer is especially transparent when a relative category presents an ∞-category $\mathcal{C}$ and its localized strict diagram categories present $\mathrm{Fun}(NY, \mathcal{C})$. With $K = \mathrm{Map}_{\mathcal{C}}(a, b)$ one then has

$$F_{a,b}(Y) \cong \pi_0 \, \mathrm{Map}_{\mathcal{S}}(|NY|, K).$$

Indeed, a coherent transformation between the two constant diagrams is a coherent Y-indexed family of points of K; this is the limit of the constant K-diagram, hence the mapping space from the groupoid completion $|NY|$ to K. Taking components gives the formula. The diagram-presentation hypothesis is substantive: one cannot replace it by the existence of an arbitrary admissible set Σ.

The comparison between strict model-category diagrams and coherent diagrams is established, under the relevant model-category hypotheses, in Lurie (2009, Proposition 4.2.4.4).

Here the source means finite sets of objects and arrows by a finite category (Grothendieck, 1990a, ms. 45); such a category need not have a finite nerve. Consequently the earlier compactness argument does not establish the comparison for every finite category. The conditional assertion under the additional locality of H is not disproved merely by that observation.

This also clarifies the finite-stage issue. The ind-object $H(a,b)$ records a bound on the length of a zigzag that works simultaneously over the entire parameter. Passing to $\underline{\mathrm{Ch}}_\infty$ can forget this uniformity. It does not follow that modern derived mapping spaces must themselves be ind-objects: in the good diagram presentations just described, one space K represents the functor for every small Y. What fails for a general pair $(\mathcal{M},\Sigma)$ may be the identification of its localized strict diagrams with coherent diagrams, rather than representability of the latter. The example below separates these two issues with a six-object category.

Quand on s'intéresse au formalisme des $\underline{\mathrm{Ch}}_\Sigma(X;a,b)$ en vue d'un *dérivateur* $\mathbf{D}_{X,\Sigma}$ qu'on voudrait définir en termes de X, Σ, il semblerait que le prédérivateur qu'on obtient bel et bien ne mérite le nom de dérivateur que lorsqu'on suppose que pour toute flèche

$$Y \longrightarrow Y'$$

dans Cat qui est une Hot-équivalence, les flèches correspondantes

$$\mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(Y', H(a,b)) \longrightarrow \mathrm{Hom}_{\mathrm{Ind\,Cat\,hot}}(Y, H(a,b)),$$

ou encore

$$\varinjlim_n \mathrm{Hom}_{\mathrm{Cat\,hot}}(Y', \underline{\mathrm{Ch}}_{n,\Sigma}(a,b)) \longrightarrow \varinjlim_n \mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, \underline{\mathrm{Ch}}_{n,\Sigma}(a,b))$$

sont des bijections. J'ai regardé dans V ce que ça signifie pour l'objet $H(a,b) = (\underline{\mathrm{Ch}}_{n,\Sigma}(a,b))_n$ de Ind Cat hot. C'est cette condition seulement qui assure, du point de vue des dérivateurs, une interprétation convaincante de $\underline{\mathrm{Ch}}_{\infty,\Sigma}(a,b)$,

$$\underline{\mathrm{Ch}}_{\infty,\Sigma}(a,b) \;=\; \varinjlim_n \underline{\mathrm{Ch}}_{n,\Sigma}(a,b)$$

en tant qu'objet de Hot, ou plus exactement de $\mathrm{Hot}_0^\wedge$, car on a alors pour Y une catégorie *finie*

$$\underbrace{\mathrm{Hom}_{\mathrm{Cat\,hot}}(Y, H(a,b))}_{\simeq \mathrm{Hom}_{X(Y)\Sigma_Y^{-1}}(a_X, b_X)} \;\simeq\; \mathrm{Hom}_{\mathrm{Hot}}(Y, \underline{\mathrm{Ch}}_{\infty,\Sigma}(a,b)),$$

car je veux quand-même que le type dans Hot soit le bon, qu'il correspond bien à l'homotopie des espaces de chemins. Donc on voudrait qu'un morphisme

[page 130]

$$X \xrightarrow{f} X'$$

dans (Cat), qui est une Hot-équivalence, définisse

$$\underline{\mathrm{Ch}}_\infty(X;a,b) \longrightarrow \underline{\mathrm{Ch}}_\infty(X'; f(a), f(b)),$$

qui soient eux aussi des Hot-équivalences. D'autre part, c'est évidemment compatible avec la brillante composition des chemins, que j'ai réussi à définir entre ces objets de Hot. Donc (sous réserve d'avoir vérifié ce que je viens de dire) la vérification de l'associativité se ramenerait au cas où X est de Thomason-Kan. Mais dans ce cas, il y a des chances que, tout comme dans le cas semi-simplicial, les $\underline{\mathrm{Ch}}_\infty(X;a,b)$ peuvent se définir (comme objets de (Hot)) comme étant simplement les chemins de type Δ_+ de a à b, qu'on compose alors de la façon triviale. Mais non, ces catégories sont discrètes!

Je renonce pour l'instant à poursuivre plus la question d'associativité. Demander à Thomason pour les propriétés du foncteur $\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^\wedge$.

Cela signifie, à très peu de choses près, qu'on a une interprétation de $\underline{\mathrm{Ch}}_{\infty,\Sigma}(a,b)$, en tant qu'objet de Hot, comme un 'Hom externe' $\mathrm{Hom}\,\mathrm{ext}(a,b)$ à valeurs dans Hot, défini sur la catégorie $\mathcal{M} = X$ à l'aide du seul ensemble admissible de flèches $\Sigma \subseteq \mathrm{Fl}(\mathcal{M})$, opération satisfaisant la condition p. 82 [page 131]

$$(*) \qquad \boxed{\mathrm{Hom}_{\mathrm{Hot}}(Y, \mathrm{Hom}\,\mathrm{ext}(a,b)) \;\simeq\; \mathrm{Hom}_{\mathcal{M}(Y)\Sigma_Y^{-1}}(a_X, b_X)}$$

(qui peut s'interpréter aussi, soit comme

$$\mathrm{Hom}_{\mathcal{M}}(p_!(a_Y), b)$$

si $p_! : \underbrace{\mathcal{M}(Y)\Sigma_Y^{-1}}_{\mathbf{D}(Y)} \longrightarrow \mathcal{M}$ existe, soit comme

$$\mathrm{Hom}_{\mathcal{M}}(a, p_*(b_Y))$$

si $p_* : \underbrace{\mathcal{M}(Y)\Sigma_Y^{-1}}_{\mathbf{D}(Y)} \longrightarrow \underbrace{\mathcal{M}}_{=\mathbf{D}(*)}$ existe); avec le seul petit ennui que cette formule n'est valable que pour Y *finie.* Si on veut exprimer le second membre pour Y variable quelconque dans Cat, il ne suffit pas d'un objet de Hot ($\mathrm{Hom}\,\mathrm{ext}(a,b)$) dans le premier membre, mais il faut l'ind-objet $H(a,b) = (\underline{\mathrm{Ch}}_n(a,b))_{n\in\mathbf{N}}$ dans Hot. Si donc on tient à travailler dans un 'domaine' Diag du dérivateur $\mathbf{D} = \mathbf{D}_{\mathcal{M},\Sigma}$ aussi grand que possible, en se bornant à des catégories finies, et qu'on veut qu'une formule $(*)$ reste valable pour toute Y dans le domaine, il faut 'en toute rigueur' définir le $\mathrm{Hom}\,\mathrm{ext}(a,b)$ non comme objet de Hot, mais de Ind Hot. Il se décrirait donc alors comme [page 132]

$$\mathrm{Hom}\,\mathrm{ext}(a,b) = (\underline{\mathrm{Ch}}_{n,\Sigma}(a,b))_n \qquad \text{(ind-objet dans Hot)}\ ,$$

et l'objet $\underline{\mathrm{Ch}}_{\infty,\Sigma}$ dans Hot en est une sorte de limite inductive.

Commentary · Context [ms. 131]

The original category and its localized value

Grothendieck wants the same external Hom to explain both homology and cohomology with constant coefficients. This is an adjunction calculation once the ambient categories have been fixed. Put $p : Y \to e$. If the relevant adjoints exist, then

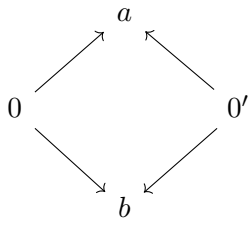
$$\begin{aligned}\mathrm{Hom}_{\mathbb{D}(Y)}(p^*a, p^*b) &\cong \mathrm{Hom}_{\mathbb{D}(e)}(p_!p^*a, b)\\ &\cong \mathrm{Hom}_{\mathbb{D}(e)}(a, p_*p^*b).\end{aligned}$$

Here $\mathbb{D}(e) = \mathcal{M}[\Sigma^{-1}]$, not the original $\mathcal{M}$ unless the notation has been changed. Likewise the a_X, b_X of the displayed formula mean a_Y, b_Y. These small notational adjustments matter because an adjoint to restriction of localized diagrams is a derived operation; an adjoint before localization need not compute it.

The historical motivation is unusually explicit. In the retrospective in *Pursuing Stacks*, Grothendieck (1983, §69, pp. 1–2) relates the inadequacy of an isolated derived category to the trace questions arising from SGA 5, and proposes retaining derived categories of all diagram shapes. The present pages carry that programme into a concrete question: can the dependence on the parameter shape itself be represented homotopically? The adjunction formulas explain why a positive answer belongs simultaneously to the homological and the cohomological sides of the theory. The subsequent calculations concern the geometric adequacy of the proposed representative, rather than another formal manipulation of these adjunctions.

Je me suis demandé si la condition préalable, savoir que les foncteurs $H(a,b)$ sur Cat transforment Hot-équivalences $Y' \longrightarrow Y$ en bijections, était bel et bien une (forte) restriction. Je me suis convaincu que c'est bien le cas, déjà dans le cas particulier où $\Sigma = \mathrm{Fl}(X)$. Il me semble que lorsque les composantes connexes de $\Omega(X,a)$ ne sont pas contractiles, ladite

condition n'est pratiquement jamais satisfaite. J'ai pensé p. ex. au cas où X a le type d'homotopie (dans Hot) d'une 2-sphère, de sorte que son $\Omega(X,a) = \underline{\mathrm{Ch}}_\infty(X,a,a)$ doit avoir un $\pi_1 \simeq \mathbf{Z}$. Prenons Y ayant le type dans Hot d'un cercle, disons

Y : 

,

donc les foncteurs p. 83
[page 133]

$$f : Y \longrightarrow \underline{\mathrm{Ch}}_\infty(X; \bullet, \bullet) \overset{\text{déf}}{=} \Omega(X, \bullet)$$

correspondent à *certains* 'lacets' de $\Omega(X, \bullet)$ (mais lacets à origine non précisée), ils ont un type d'homotopie au sens topologique ('nombre de tours' autour de la sphère X) donné par un entier $n \in \mathbf{Z}$ (le 'degré' du lacet λ). L'intuition évidente, c'est que l'ensemble des degrés δ qu'on peut obtenir ainsi, avec une catégorie *finie* donnée Y qui représente le cercle $S^1 \in (\mathrm{Hot})$, est fini - $|\delta|$ majoré sûrement par le nombre de ces flèches (donc au moins par 4 dans le cas envisagé, chaque flèche de Y faisant faire *au plus* un tour autour du pseudo-cercle $\Omega(S^2, \bullet)$) ([28]). D'autre part, quand on représente S^1 par des catégories Y_n arbitrairement grandes (disons déduites des intervalles $\Phi_n = (\Delta_+\Delta_-)^n$ en identifiant la source et le but), on doit trouver des degrés arbitrairement élevés. Or toute Y_n s'envoie dans $Y(= Y_2)$ par une flèche

$$Y_n = Y' \longrightarrow Y = Y_2$$

qui est de degré 1, donc qui est une Hot-équivalence. Mais l'application correspondante [page 134]

$$\pi_0\underline{\mathrm{Hom}}(Y, \Omega(X, \bullet)) \longrightarrow \pi_0\underline{\mathrm{Hom}}(Y', \Omega(X, \bullet))$$

n'est pas bijective. Ce genre d'argument me semble devoir marcher chaque fois que l'ensemble des groupes d'homotopie $\pi_j(X, x)$, pour $j \geq 2$ (donc l'ensemble des groupes d'homotopie π_i de $\Omega(X, x)$, pour $i \geq 1$) est *infini,* en prenant pour Y, Y' différents représentants Y_n de la sphère S^i, disons les Y_n déduites de Φ_n ($n \geq 2$) en prenant $\Phi_n^i = \Phi_n \times \Phi_n \times \cdots \times \Phi_n$ (qui est une sorte de i-cube combinatoire), et en identifiant son 'bord' à un seul point. Je présume que le nombre d'élements dans $\pi_i(\Omega(X, x))$ qu'on peut obtenir à partir de flèches $Y_n \longrightarrow \Omega(X, x)$ est limité par la taille de Y_n, mais on peut sûrement les obtenir chacun, en prenant n assez grand. Si donc $\xi \in \pi_i(\Omega)$ échappe à Y_{n_0} ($n_0 = 0$, disons), on prend un n' assez grand pour que ξ n'échappe pas à $Y_{n'}$, et on choisit

$$Y_{n'} \longrightarrow Y_n \ ,$$

[$n = n_0$?]

qui soit une Hot-équivalence (p. ex. à l'aide d'une rétraction de $\Phi_{n'}$ sur Φ_n, compatible avec sources et buts). [page 135]

Commentary · Development (commentators') [ms. 134]

A fixed circle already sees every degree

Grothendieck's intuition that localization of strict diagrams can depend on the chosen category presenting a homotopy type is correct. The proposed explanation by a bound on the degree of a four-arrow circle is not. A natural transformation between long paths contains an arbitrarily wide commuting ladder. Its contribution to a two-dimensional homotopy class is not bounded by the number of arrows in the parameter category. We give a finite example that both disproves the suggested bound and proves the desired failure of invariance.

A suspension calculation. [ours] Let Y be the four-object poset with minimal objects r_0, r_1, maximal objects q_0, q_1, and all four relations $r_i < q_j$. Let X be obtained by adjoining two incomparable objects s, n, each greater than every object of Y. Take all arrows of X as weak equivalences. Then $|NY|$ is a circle and $|NX|$ is its suspension, hence a 2-sphere. For the constant diagram s_Y the group

$$G_Y = \mathrm{Aut}_{\Pi\,\mathrm{Fun}(Y,X)}(s_Y)$$

contains an element of infinite order. Moreover, for one fixed four-arrow Y, functors $Y \to \underline{\mathrm{Ch}}_\infty(X; s, s)$ realize every integer in $\pi_1\Omega|NX|$.

Proof. Denote the inclusion by $j : Y \hookrightarrow X$, and choose a vertex $y_0 \in Y$. Since all objects of Y lie below both suspension vertices, there is a loop of natural transformations

(B1) $$s_Y \longleftarrow j \longrightarrow n_Y \longleftarrow (y_0)_Y \longrightarrow s_Y.$$

28. probablement on ne trouve tout au plus que les degrés δ entre -2 et $+2$.

This is a loop in the nerve of the functor category, based at s_Y. Under evaluation it gives a continuous family of loops in $|NX|$: at y the loop travels from s to n along the meridian through y, and returns along the fixed meridian through y_0.

To check its class, let γ be an oriented fundamental 1-cycle of $|NY|$. The two varying portions of the family sweep out the two cones on γ. The fixed return meridian contributes no 2-chain. Thus the image of the fundamental class of $|NY| \times S^1$ is, with a choice of orientation,

$$n * \gamma - s * \gamma,$$

which is the fundamental cycle of the suspension $|NX|$. In particular its degree is 1. Equivalently, collapse the fixed meridian in the suspension: the adjoint of the family becomes the usual suspension map of the identity of $|NY|$. The collapse of that contractible arc induces an isomorphism on H_2, so this gives the same calculation. Consequently (B1) is nontrivial and its k-fold power has degree k. A null-homotopy in the functor category would give a null-homotopy after evaluation and degree zero. All these powers are therefore distinct in G_Y.

The same construction takes place in one finite path stage. For $\rho = \Delta_-\Delta_+\Delta_-\Delta_+$ define

$$F(y) = \big(s \longleftarrow y \longrightarrow n \longleftarrow y_0 \longrightarrow s\big).$$

For an arrow $y \to y'$ use that arrow in the varying position and identities in the other four positions. All squares commute because X is a poset. This defines a functor $F : Y \to \underline{\mathrm{Ch}}_\rho(X; s, s)$. Concatenating k copies pointwise gives a functor into $\underline{\mathrm{Ch}}_{\rho^k}(X; s, s)$ of degree k. Insert identity arrows at the two ends to fit this alternating word into a sufficiently long symmetric stage t_N; this changes only parametrization, not the degree. Reversal gives negative k, and the constant family gives zero. This proves the assertion for $\underline{\mathrm{Ch}}_\infty$, without assuming that its concatenation has already been made into a globally defined bifunctor. □

The example also supplies a genuine change-of-parameter failure. Write $B\mathbb{Z}$ for the one-object category with automorphism group $\mathbb{Z}$. There is a functor $u : Y \to B\mathbb{Z}$ sending one of the four edges to $1 \in \mathbb{Z}$ and the other three to 0. Choose the sign so that the induced map on the circle has degree 1. Its nerve is a weak equivalence: both classifying spaces are $K(\mathbb{Z}, 1)$ and the induced map on fundamental groups is an isomorphism. Every functor $B\mathbb{Z} \to X$ is constant, since a poset has only identity automorphisms; every natural transformation between such functors is exactly one arrow of X. Therefore

$$\mathrm{Fun}(B\mathbb{Z}, X) \cong X, \qquad \mathrm{Aut}_{\prod \mathrm{Fun}(B\mathbb{Z},X)}(s) = 1.$$

Restriction along u maps this trivial group to G_Y, which contains all the powers of (B1). It is not surjective. By the localization interpretation of $H(s, s)$ used on ms. 129, this is precisely a failure of the required invariance of $H(s, s)$.

We checked the relevant diagram-localization and fibrancy results in Cisinski (2009), Thomason (1980), and Raptis (2010). We did not find this finite suspension calculation presented there. It is included as a direct completion of Grothendieck's proposed example, without a claim of priority. The failure proved here concerns localized *strict* diagrams in this relative category. It is consistent with the homotopy invariance of coherent diagrams explained above.

Ceci attire fortement l'attention sur le fait que si Y, C sont deux catégories, la catégorie $\underline{\mathrm{Hom}}(Y, C)$ *n'est pas* un représentant de $\underline{\mathrm{Hom}}_{\mathrm{Hot}}(\mathrm{hot}(Y), \mathrm{hot}(C))$, on a seulement une flèche canonique

$$(*) \qquad \mathrm{hot}(\underline{\mathrm{Hom}}(Y, C)) \longrightarrow \underline{\mathrm{Hom}}_{\mathrm{Hot}}(\mathrm{hot}(Y), \mathrm{hot}(C)),$$

i.e.

$$\underbrace{\mathrm{hot}(\underline{\mathrm{Hom}}(Y, C)) \times \mathrm{hot}(Y)}_{\mathrm{hot}(\underline{\mathrm{Hom}}(Y,C)\times Y)} \longrightarrow \mathrm{hot}(C),$$

p. 84

laquelle flèche est simplement déduite de

$$\underline{\mathrm{Hom}}(Y, C) \times Y \longrightarrow C.$$

Mais la flèche $(*)$ n'a aucune envie d'être `[un]` isomorphisme. C'est parce qu'elle ne l'est déjà pas dans le cas particulier $Y = \mathbf{\Delta}^1$, essentiellement, qui nous oblige, pour faire une théorie

des chemins dans les catégories qui soit topologiquement raisonnable, de ne pas nous borner à $\mathbf{\Delta}^1$ comme type de chemins, mais à prendre des intervalles composés $\tau \in \mathrm{Ty}$.

Commentary · Examples [ms. 135]

The absolute arrow category and its endpoint fibres

Grothendieck is trying to isolate the defect that forced him to lengthen paths. The defect is real, but the specific absolute comparison on this page does not exhibit it. For every category C, put $A = \mathrm{Fun}([1], C)$, let $c : C \to A$ send an object to its identity arrow, and let $e_0 : A \to C$ evaluate at the source. Then $e_0 c = \mathrm{id}_C$. For an arrow $f : x \to y$, the square with vertical arrows id_x and f defines a natural transformation

$$ce_0 \Longrightarrow \mathrm{id}_A.$$

After taking nerves, this is a homotopy from Bce_0 to the identity. Hence Be_0 is a homotopy equivalence. Evaluation at 0 also gives a homotopy equivalence $\mathrm{Map}(|N[1]|, BC) \to BC$, since the interval contracts to 0. The canonical comparison

$$B\,\mathrm{Fun}([1], C) \longrightarrow \mathrm{Map}(|N[1]|, BC)$$

commutes with these evaluations and is consequently a weak equivalence. No fibrancy assumption on C is involved.

What cannot be inferred is the corresponding assertion on *strict* endpoint fibres. The fibre over (a, b) of $A \to C \times C$ is the discrete category $\mathrm{Hom}_C(a, b)$: a square with identity arrows at its two ends can only relate an arrow to itself. The homotopy fibre of the topological path fibration is instead the full path space from a to b. The fork example discussed at ms. 128 already shows that these can differ even on components. Thus the motivation for alternating paths survives, while the sentence claiming failure of the absolute $Y = [1]$ comparison must be corrected.

There is also a direct example of absolute failure for another shape. Let P be any poset with $BP \simeq S^1$, and take $Y = B\mathbb{Z}$. As above, $\mathrm{Fun}(B\mathbb{Z}, P) \cong P$. Its classifying space is connected, whereas $\mathrm{Map}(S^1, S^1)$ has one component for every integer degree. The comparison misses every nonconstant degree. This pinpoints the missing derived dependence on the source, without confusing the total arrow category with its fibres.

[page 136]

Par contre, $(*)$ est [un] isomorphisme quand C est une catégorie de Thomason-Kan. Il suffit de voir que pour tout Z dans (Cat), le Hom de $\mathrm{hot}(Z)$ dans les deux membres est isomorphe par la flèche déduite de $(*)$. Or si C est de Thomason-Kan, de même $\underline{\mathrm{Hom}}(Y, C)$ (immédiat sur la définition de Thomason-Kan en termes de prolongements de $A_0 \xhookrightarrow{i} A$ avec $A_0 \to C$ et $A \dashrightarrow C$ ...([29])), donc

$$\mathrm{Hom}_{\mathrm{Hot}}(\mathrm{hot}(Z), \mathrm{hot}(\underline{\mathrm{Hom}}(Y, C))) \xleftarrow{\sim} \pi_0(\underbrace{\underline{\mathrm{Hom}}_{\mathrm{Cat}}(Z, \underline{\mathrm{Hom}}(Y, C))}_{\underline{\mathrm{Hom}}_{\mathrm{Cat}}(Z \times Y, C)}) \simeq \pi_0 \underline{\mathrm{Hom}}(Z \times Y, C),$$

et

$$\mathrm{Hom}_{\mathrm{Hot}}(\mathrm{hot}(Z), \underline{\mathrm{Hom}}_{\mathrm{Hot}}(\mathrm{hot}(Y), \mathrm{hot}(C))) \simeq \mathrm{Hom}_{\mathrm{Hot}}(\underbrace{\mathrm{hot}(Z) \times \mathrm{hot}(Y)}_{\mathrm{hot}(Z \times Y)}, \mathrm{hot}(C))$$

$$\underbrace{\xleftarrow{\sim}}_{\text{car } C \text{ est de Thomason-Kan}} \qquad \pi_0(\underline{\mathrm{Hom}}_{\mathrm{Cat}}(Z \times Y, C)),$$

O.K.

On voit donc que les catégories $\underline{\mathrm{Ch}}_{\infty,\Sigma}(X; a, b)$ n'ont aucune tendance à être des catégories de Thomason-Kan. Pourtant, son type dans Hot doit être 'le bon', sans avoir d'abord à remplacer X par une enveloppe de Thomason-Kan $\bar{X}$.

29. un peu vif! Il faut savoir que les applications cofibrantes i de Thomason le restent après multiplication cartésienne par Y ...

Commentary · Development (commentators') [ms. 136]

A fibrant target does not derive every source

Grothendieck's next attempt is to impose a Kan condition on the target. His footnote identifies one missing assertion about products of cofibrations, but the displayed proof also replaces arbitrary maps in the localized category by ordinary functors from an arbitrary source. Thomason fibrancy alone does not permit that replacement. The following example makes the issue visible already on components.

A fibrant-poset obstruction. [ours] There are a Thomason-fibrant category P and a finite category Y for which

$$B\,\mathrm{Fun}(Y,P) \longrightarrow \mathrm{Map}(BY,BP)$$

is not a weak equivalence. Here "finite" means finitely many objects and arrows, not a finite nerve.

Proof. Let $G = \mathbb{Z}/2$ and $Y = BG$. By Raptis (2010, Theorem 2.6), Thomason's model structure restricts to posets, with the same weak equivalences and fibrations. Start with a poset representing BG, for example the twofold subdivision construction $c\,\mathrm{Sd}^2\,N(BG)$, and take its fibrant replacement within posets. We obtain a poset P that is fibrant as a category and satisfies $BP \simeq BG$. Every functor $BG \to P$ sends the single object to some p and every element of G to id_p. Natural transformations are exactly the relations $p \le p'$. Hence

$$\mathrm{Fun}(BG,P) \cong P.$$

The source of the comparison is therefore connected. Its image consists of constant maps $BG \to BP$. But the target has at least two components: a constant map induces the zero homomorphism on π_1, whereas a homotopy equivalence $BG \to BP$ induces an isomorphism. These maps cannot be homotopic, even without specified basepoints, since conjugation cannot turn a trivial homomorphism into an isomorphism. The comparison fails on π_0. □

This also supplies an immediate test of the first isomorphism asserted in the proof on ms. 136: take its $Z = e$. The claimed bijection would identify the connected set $\pi_0\,\mathrm{Fun}(Y,P)$ with $[BY,BP]$, which has the two distinct classes just exhibited. One cannot repair that step merely by proving that the functor category is fibrant. In Thomason's model structure the source has to be handled as well; in particular every cofibrant category is a poset (Thomason, 1980, Proposition 5.7), so this BG is not cofibrant.

The literal product claim in footnote 29 would also be too strong for arbitrary cofibrations: multiplying the cofibration $\varnothing \to e$ by Y would make every category cofibrant. The next development also rules out preservation of all trivial cofibrations under product with arbitrary Y: the lifting adjunction would otherwise force every such functor category to be fibrant.

A safe model for the desired derived mapping space is the simplicial function complex

$$\mathrm{Map}_{\mathrm{sSet}}(NY, RNP),$$

where RNP is a Kan replacement. Every simplicial set is cofibrant, and this construction derives the target in the cartesian model category of simplicial sets. The ordinary nerve identity $N\,\mathrm{Fun}(Y,P) = (NP)^{NY}$ explains precisely what the uncorrected formula omits. We checked the model structure and the poset replacement in the cited papers; we did not find this particular $B(\mathbb{Z}/2)$ application there. The result is recorded as an elementary application, not a priority claim.

p. 85
[page 137]

Commentary · Development (commentators') [ms. 136]

Fibrancy is not preserved by strict diagram categories

The failure of the preceding mapping-space formula does not logically decide the separate assertion that $\mathrm{Fun}(Y,C)$ is fibrant whenever C is. That assertion also fails, but it needs a different example. An equivariant construction makes the obstruction precise: ordinary horn filling need not respect an action on the diagram.

A finite parameter with a nonfibrant functor category. [ours] There are a countable Thomason-fibrant category E and a finite category Y such that $\mathrm{Fun}(Y,E)$ is not Thomason-fibrant.

Proof. Let D be the four-object crown used above, so $BD \simeq S^1$. It is not Thomason-fibrant. Indeed, $\mathrm{Ex}^2\,ND$ has finitely many 1-simplices: they are the functors from the finite poset $c\,\mathrm{Sd}^2\,\Delta^1$ to D. In a Kan complex every element of a based fundamental group is represented by one loop

edge, since horn fillers compose and invert edge paths. If $\mathrm{Ex}^2\, ND$ were Kan, its fundamental group would therefore be finite, contrary to its being isomorphic to $\pi_1(BD) = \mathbb{Z}$.

Put $G = \mathbb{Z}/2$ and give D the trivial action. We first construct a G-poset Q whose underlying category is fibrant and whose fixed subposet is exactly D. Start with $Q_0 = D$. At step r, for every generating Thomason horn inclusion

$$A = c\,\mathrm{Sd}^2\,\Lambda^n_i \;\longrightarrow\; B = c\,\mathrm{Sd}^2\,\Delta^n$$

and every ordinary functor $F : A \to Q_r$, attach the free G-cell $G \times B$ along $G \times A$, using the equivariant attaching map $(g, a) \mapsto gF(a)$. Here G in a product denotes the discrete set of its two elements. Take all these pushouts simultaneously and put $Q = \varinjlim_r Q_r$.

These horn inclusions are Dwyer inclusions of finite posets. The pushout description in May et al. (2017, Lemmas 4.7–4.8) shows that the old poset stays full and that the new objects are exactly the disjoint sets $G \times (B \setminus A)$. All new objects have free G-action. Thus $Q^G = D$, with its original order. Every horn map into Q factors through some finite stage, because its domain is finite, and it obtains a filler at the next stage. Consequently Q is Thomason-fibrant. There are only countably many finite horn shapes and, at each countable stage, countably many maps from them. Hence Q is countable.

Define the transport category $E = Q \rtimes G$ as follows. Its objects are those of Q, and an arrow $x \to y$ labelled $g \in G$ exists exactly when $gx \le y$; composition multiplies labels. There is a projection $E \to BG$. We claim that E is Thomason-fibrant. Given a horn diagram $F : A \to E$, write g_{ab} for the label of an arrow $a \to b$. Since BA is contractible, its fundamental groupoid is equivalent to a point. The projected functor $A \to BG$ is therefore naturally isomorphic to a constant functor: choose labels h_a with

$$h_b g_{ab} = h_a.$$

Replacing $F(a)$ by $h_a F(a)$ turns every arrow label into 1 and gives a functor $A \to Q$. Extend it to $B \to Q$ by fibrancy of Q. Extend the assignments h_a arbitrarily to the remaining objects of B, and undo the replacement: at b use $h_b^{-1} F'(b)$, and label $b \to c$ by $h_c^{-1} h_b$. The inequalities defining F' make this a functor $B \to E$, and it restricts exactly to the original horn diagram. This proves the claim.

Finally set $Y = BG$. A functor $BG \to E$ induces a homomorphism $\phi : G \to G$, either trivial or the identity. The trivial case is the identity action at an arbitrary object of E. In the identity case, an endomorphism labelled by the nontrivial g requires $gx \le x$; applying g gives the reverse inequality, so $gx = x$. Its objects are therefore precisely $D = Q^G$. A natural transformation labelled by h must satisfy $h\phi(g) = \psi(g)h$. Since G is abelian, this forces $\phi = \psi$. For the identity case the remaining condition is just $x \le y$, and every $h \in G$ is allowed. We obtain an actual decomposition

$$\mathrm{Fun}(BG, E) \cong E \amalg (D \times BG).$$

The category D is a retract of this coproduct: project the second summand onto D, send the first constantly to a chosen object of D, and use identity labels for the section. A retract of a fibrant category is fibrant. Since D is not fibrant, neither is $\mathrm{Fun}(BG, E)$. □

Thus both claims on ms. 136 fail: arbitrary strict diagram categories need not preserve Thomason fibrancy, and even when a particular functor category is fibrant, it need not represent the derived mapping space. The equivariant cell and fixed-point ingredients above are standard and are cited at their use. We did not locate this application to strict diagram categories in the primary sources checked, including the published equivariant-poset paper; the argument is supplied here without a claim of priority.

12 Retour sur les ‘catégories de chemins’ dans Cat, comme catégorie de modèles pour Hot

Je vois exactement deux applications intéressantes du formalisme des chemins dans les catégories. L’une, qui m’a surtout forcé la main, est pour pouvoir construire des Hom ext, Hom *externes* à valeurs dans Hot, pour tout prédérivateur **D** défini par un couple $(\mathcal{M}, \Sigma)$ (et

je ne connais pas de prédérivateur qui ne soit défini ainsi).

L'autre application est pour avoir dans (Cat) une *construction fonctorielle,* en termes d'un objet X quelconque ou d'une flèche $X \longrightarrow Y$, de l'analogue du formalisme de Cartan-Serre sur les espaces de chemins. Dans ce cadre, il est entendu qu'on prend toujours $\Sigma_X = \mathrm{Fl}(X)$, ensemble de *toutes* les flèches de X. Pour construire fonctoriellement en X dans (Cat)

$$\begin{array}{c} \underline{\mathrm{Ch}}_\infty(X) \\ \downarrow \\ X \times X, \end{array}$$

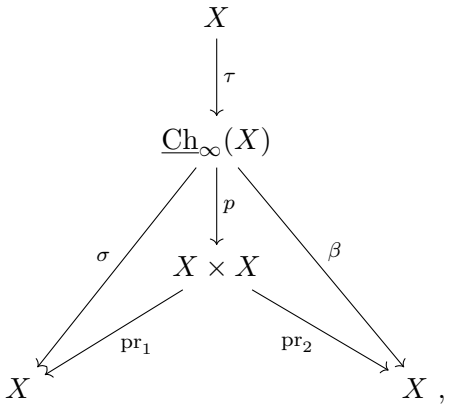
il faut avoir fixé un système projectif

$$\underline{t} = (t_n)_{n \in \mathbf{N}}$$

dans $\underline{\mathrm{Ty}}$ qui soit cofinal dans $\underline{\mathrm{Ty}}$, ce qui signifie simplement que parmi les t_n il y en a de [page 138] longueur utile arbitrairement grande (et comme

$$n \longmapsto \text{long.ut.}(t_n)$$

est croissante, cela implique que long.ut.(t_n) tend vers $+\infty$ pour $n \longrightarrow +\infty$). Les faits principaux à vérifier ensuite (si faire se peut) sont les propriétés essentielles des flèches dans le diagramme p. 86

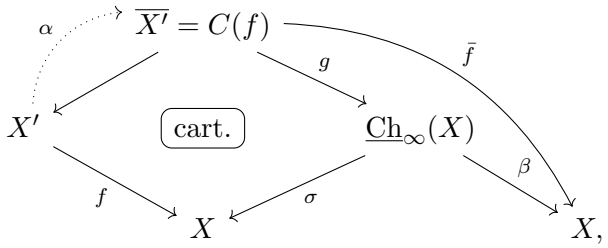


$$\left|\begin{array}{l} \sigma\tau = \beta\tau = \mathrm{id}_X \\ p\tau = \mathrm{diag}_X \end{array}\right.$$

et plus généralement, pour une flèche

$$X' \xrightarrow{f} X$$

dans Cat, des flèches dans le diagramme correspondant



$$\bar{f}\alpha = (\beta g)\alpha = f,$$

et plus particulièrement de βg. La flèche en pointillés α est la section de $\overline{X'}$ sur X' déduite de [page 139]

la section τ de $\underline{\mathrm{Ch}}_\infty(X)$ sur X qui réalise la décomposition de f en

$$X' \xrightarrow{\alpha} \overline{X'} \xrightarrow{\bar f} X,$$

où $\bar f$ devrait être une flèche *'fibrante'* et une Hot-équivalence, α une flèche *'cofibrante'* au sens de Thomason. (Mais s'il en est ainsi, est à voir!) Voici pour le moment les propriétés essentielles que je voudrais vérifier.

Commentary · Exposition [ms. 139]

The two maps in a mapping-path factorization

Grothendieck turns from representing an external Hom to replacing a map by its homotopy fibres. This change of purpose makes the endpoints decisive. Put $E = \underline{\mathrm{Ch}}_\infty(X)$, with source σ, target β, and constant-path section i. Then

$$C(f) = X' \times_{X,\sigma} E, \quad r : C(f) \longrightarrow X', \quad \bar f = \beta\,\mathrm{pr}_E, \quad \alpha(x') = (x', i(fx')).$$

The strict identities are $r\alpha = \mathrm{id}$ and $\bar f\alpha = f$. Geometrically, r forgets a path whose initial point is prescribed; its fibre should be contractible. The other map remembers the final point; its fibre should be the homotopy fibre of f. Thus the weak equivalence is α, whereas $\bar f$ is a weak equivalence exactly when f is. The sentence on ms. 139 that attaches the unconditional weak-equivalence assertion to $\bar f$ is contradicted by item 3° immediately below it.

Once r is known to be a weak equivalence, the rest of this assertion uses only two-out-of-three. Since $r\alpha = \mathrm{id}$, the section α is a weak equivalence. Since $\bar f\alpha = f$, the maps f and $\bar f$ are weak equivalences simultaneously. Neither inference says that α is a Thomason cofibration, or that $\bar f$ is a Thomason fibration. Smoothness and properness concern the homotopy exactness of base change, while Thomason's chosen fibrations are tested by $\mathrm{Ex}^2 N$. Their definitions must not be interchanged.

The cofibration expectation actually fails for the unaltered constant-path inclusion. Take $f = \mathrm{id}_X$ with $X = [1]$. A nonconstant alternating path from 0 to 1, padded by identity arrows, admits a natural transformation to the constant path at 1: at a vertex labelled 0 use $0 \to 1$, and at a vertex labelled 1 use its identity. Hence the image of $\alpha : X \to E$ is not a sieve. Every Thomason cofibration is a pseudo-Dwyer map and therefore a sieve (Raptis, 2010, §2, following Proposition 2.4). Thus α need not be a Thomason cofibration. This does not damage the intended homotopy-fibre replacement; it limits the proposed identification with a particular model-category factorization.

The expectation that $\bar f$ is always a Thomason fibration also fails directly. Take $X = e$. Since $\underline{\mathrm{Ch}}_\infty(e) = e$ strictly, the construction gives $C(f) = X'$ and $\bar f = f : X' \to e$. Choosing $X' = D$, the nonfibrant crown of the preceding development, gives a counterexample. The homotopical lifting properties proved later do not impose Thomason fibrancy on an arbitrary category mapping to e.

The properties in items 1°–3° are requirements at this stage of the manuscript. Their proofs begin only after the concrete symmetric model is fixed on ms. 145. We will use them in that order, rather than reading the list as an already established theorem for an unspecified projective system.

1°) p est lisse et propre, et est une HOT-fibration.

2°) Donc il en est de même de $\sigma = \mathrm{pr}_1 p$ et $\beta = \mathrm{pr}_2 p$. De plus, ce sont des HOT-fibrations *triviales,* i.e. à fibres asphériques.

3°) Plus généralement (pour ce qui concerne β), $\overline{X'} \xrightarrow{\bar f} X$ devrait encore être propre et lisse et une HOT-fibration, mais bien sûr en général pas triviale. (Elle devrait être triviale si et seulement si f est une Hot-équivalence.) ([30]). p. 87

Un mot sur la 'dualité'. Les énoncés sur σ (resp. β) se traduisent en énoncés sur β (resp. σ) par passage de $\underline{t}$ à $\underline{t}^o = (t^o_n)$, où $\tau \longmapsto \tau^o$ est le foncteur dans $\underline{\mathrm{Ty}}$ 'passage à l'intervalle opposé'. On a en effet

un isomorphisme fonctoriel en X (passage au chemin opposé)

$$\boxed{\begin{array}{ccc} c & \longmapsto & c^o \\ \underline{\mathrm{Ch}}^t_\infty(X) & \longrightarrow & \underline{\mathrm{Ch}}^{t^o}_\infty(X) \end{array}} \qquad ([31]) \qquad [\text{page } 140]$$

30. résulte formellement de 1° et 2°, cf. page 170.

qui 'échange source et but'. Bien sûr, on a intérêt à prendre $\underline{t}$ tel qu'on ait $\underline{t} = \underline{t}^o$ (c'est ce que j'avais fait, par un sain instinct, dès le début dans V), de sorte qu'on a un foncteur involutif canonique

$$\begin{array}{ccc} c & \longmapsto & c^o \\ \underline{\mathrm{Ch}}_\infty(X) & \longrightarrow & \underline{\mathrm{Ch}}_\infty(X) \end{array}$$

(je laisse tomber à nouveau l'exposant $\underline{t}$), qui 'échange source et but'. Donc on trouve, à isomorphisme canonique près, la 'même' factorisation de $f : X' \longrightarrow X$ en $\bar{f} \circ \alpha$, si on fait comme dans page 38 pour définir $\overline{X'}$ et α, ou si dans ce diagramme on intervertit les rôles de σ et de β.

Commentary · Exposition [ms. 140]

Two different reversals

Grothendieck needs two symmetries because two different operations are being compared: exchanging the endpoints of a path, and replacing the ambient category by its opposite. For a signed word $\tau = (\varepsilon_1, \dots, \varepsilon_r)$, with $\varepsilon_i \in \{+, -\}$, write

$$\tau^o = (-\varepsilon_r, \dots, -\varepsilon_1), \qquad \tau^- = (-\varepsilon_1, \dots, -\varepsilon_r).$$

The first operation reverses the ordered interval. It is an anti-automorphism of concatenation and interchanges source and target without changing X. The second reverses every categorical arrow while preserving the ordering of the vertices. It is an automorphism of concatenation and is the operation accompanying $X \mapsto X^{\mathrm{op}}$.

At a fixed stage the corresponding identities are obtained by reindexing diagrams and natural transformations:

$$\underline{\mathrm{Ch}}_\tau(X) \cong \underline{\mathrm{Ch}}_{\tau^o}(X), \qquad \underline{\mathrm{Ch}}_\tau(X^{\mathrm{op}}) \cong \underline{\mathrm{Ch}}_{\tau^-}(X)^{\mathrm{op}}.$$

The first identity is understood over the transposition of $X \times X$; the second is over $(X \times X)^{\mathrm{op}}$. Passing through the specified transition maps gives the identities for projective systems. For $\tau_0 = \Delta_+\Delta_-$ one has $\tau_0^o = \tau_0$ but $\tau_0^- = \Delta_-\Delta_+$. This explains why path reversal becomes a strict involution in the symmetric model, while categorical duality initially changes its parity. The translation introduced on ms. 147 supplies the remaining comparison. The commentary at manuscript p. 9 already distinguishes these operations; here their role is to transfer the endpoint and base-change assertions.

Enfin :

4°) Si

$$c : a \longrightarrow b$$

est un chemin quelconque dans X, on voudrait en conclure un isommorphisme dans Hot (de 'transport par c')

$$\overline{X'}_a \xrightarrow[\sim]{\bar{c}} \overline{X'}_b,$$

avec bien sûr transitivité pour la composition,
ce qui donnera aussi

$$\overline{c^{\mathrm{opp}}} = \bar{c}^{-1}$$ [page 141]

(en admettant que les chemins triviaux $a \xrightarrow{\mathrm{id}} a$ donnent l'identité $\overline{X'}_a \xrightarrow[\sim]{\mathrm{id}} \overline{X'}_a$!) ([32, 33]). Bien sûr, $\bar{c}$ ne doit dépendre que de la *classe d'homotopie* de $\bar{c}$ (dans $\underline{\mathrm{Ch}}_\tau(X; a, b)$, ou même dans $C = \underline{\mathrm{Ch}}(X; a, b)$, catégorie de tous les chemins dans X, de a vers b, de type non précisé). En d'autres termes, on doit avoir un foncteur p. 88

$$\begin{array}{ccc} \Pi(X) & \longrightarrow & \mathrm{Hot} \\ a & \longmapsto & \mathrm{hot}(\overline{X'}_a) \end{array}$$

31. cf. page 143 pour le passage de $\underline{t}$ à $\underline{t}^-$.
32. Surtout pas se borner au cas où c serait un des t_n!
33. décrire quand-même l'effet sur les π_0!

du groupoïde fondamental de X dans Hot. À vrai dire, ceci n'a rien à voir avec la situation particulière étudiée ici - ça doit marcher chaque fois qu'on a une HOT-fibration, ici $\overline{X'} \longrightarrow X$, mais aussi $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$ ([34]). Et pour une $\mathbf{D}$-fibration $X' \longrightarrow X$ ($\mathbf{D}$ un dérivateur quelconque), et $a \in \mathbf{D}(e)$, on doit avoir de même deux foncteurs

$$\Pi(X) \rightrightarrows \mathbf{D}(e)$$

correspondant à

$$\begin{cases} x & \longmapsto & H^\bullet_{\mathbf{D}}(X'_x, \bullet) \\ x & \longmapsto & H^{\mathbf{D}}_\bullet(X'_x, \bullet). \end{cases}$$

Le cas qui nous occupe ici serait celui où $\mathbf{D} = \mathrm{HOT}$, où on prend $H^{\mathbf{D}}_\bullet$, et $a =$ objet final de $\mathbf{D}(e) =$ Hot.) [page 142]

5°) Si 1°, 2°, 3° sont vérifiés, on est moralement sûr que la construction des $\underline{\mathrm{Ch}}_\infty(X)$ est bel et bien l'équivalent exact du formalisme de Cartan-Serre. Pour en avoir le cœur totalement net, i.e. raccrocher explicitement cette théorie aux constructions classiques, il faudrait prouver un *énoncé de comparaison* avec la théorie semi-simpliciale, qui dirait essentiellement ceci : Si on désigne, pour X dans Cat, par $\tilde{X}$ l'objet associé dans $\mathbf{\Delta}^\wedge$ (le 'nerf' de X), alors la flèche

$$\underline{\mathrm{Ch}}_\infty(X)^\sim \longrightarrow \underbrace{(X \times X)^\sim}_{\simeq \tilde{X} \times \tilde{X}}$$

dans $\mathbf{\Delta}^\wedge/(X \times X)^\sim$, est W-équivalente sur $(X \times X)^\sim$ à la flèche

$$\underline{\mathrm{Ch}}(\tilde{X}) \longrightarrow \tilde{X} \times \tilde{X}$$

de la théorie semi-simpliciale des chemins (à supposer qu'il y ait une telle flèche classique définie pour *tout* objet X de $\mathbf{\Delta}^\wedge$). Ou qu'on ait tout au moins des isomorphismes canoniques dans Hot

$$\mathrm{hot}_{\mathrm{Cat}}(\underline{\mathrm{Ch}}_\infty(X; a, b)) \;\simeq\; \mathrm{hot}_{\mathbf{\Delta}^\wedge}(\underline{\mathrm{Ch}}(\tilde{X}; \tilde{a}, \tilde{b})).$$

Commentary · Later developments [ms. 142]

Transport and the comparison with ordinary path spaces

Grothendieck wants the replacement to remember how its fibres vary, and his marginal request to compute π_0 is the right concrete test. For the ordinary mapping-path construction, a point in the homotopy fibre at a is a pair (x', γ) with $\gamma : f(x') \rightsquigarrow a$. Transport along $c : a \rightsquigarrow b$ sends it to

$$(x', \gamma * c).$$

Homotopic paths give homotopic maps of fibres; concatenation and reversal give composition and inverse in the homotopy category. This produces the functor from the fundamental groupoid requested in item 4°. It is a truncation of the coherent family of fibres. A functor into Hot alone need not retain all that family's higher twisting, a distinction already illustrated in the commentary on Chapter VI, manuscript pp. 7 and 74.

For a connected pointed map $f : (X', x'_0) \to (X, a)$, this transport has an explicit component formula. Set $G = \pi_1(X, a)$ and $H = \operatorname{im} \pi_1(f)$. Then

$$\pi_0 \operatorname{hofib}_a(f) \cong H \backslash G.$$

To see the formula, join x'_0 to x' by a path δ. The pair (x', γ) determines the loop $f\delta * \gamma$. A different δ multiplies its class on the left by an element of H. Conversely, a homotopy of pairs gives precisely this relation, by tracing the path of their first coordinates; any such relation is realized by that construction. A loop c at a therefore acts by right multiplication $Hg \mapsto Hg[c]$. For the inclusion $2\mathbb{Z} \subset \mathbb{Z}$, viewed as a map of one-object groupoids, there are two components and a generator of the

34. Ici aussi, il faudrait décrire l'effet du transport $\bar{c}$ sur les π_0 des fibres.

base interchanges them. This answers the request about π_0 without choosing representatives in a categorical lifting category.

There is a functorial simplicial path object for every simplicial set K, not only for a Kan complex. Choose a functorial Kan replacement $K \to RK$ and put

$$P(K) = (K \times K) \times_{RK \times RK} (RK)^{\Delta^1}.$$

Its endpoint map is a Kan fibration. The constant-path map $K \to P(K)$ is a weak equivalence: right properness first identifies $P(K) \to (RK)^{\Delta^1}$ as a weak equivalence, and evaluation identifies the latter with RK. Apply this with $K = NX$. Once $i : X \to \underline{\mathrm{Ch}}_\infty(X)$ has been proved a weak equivalence, the span

$$N\underline{\mathrm{Ch}}_\infty(X) \longleftarrow NX \longrightarrow P(NX)$$

is already a zigzag of weak equivalences over $NX \times NX$. This gives the first form of comparison in item 5°. Identifying the *strict* endpoint fibres with ordinary path spaces still requires the homotopy-fibre property of the categorical endpoint map; a weak equivalence of total spaces over a base does not by itself identify strict fibres.

For a general coefficient $A \in \mathbb{D}(e)$, the analogous homology transport is covariant; cohomology transport uses pullback along the inverse fibre equivalence. These constructions require the homotopy base-change isomorphisms appropriate to the stated $\mathbb{D}$-fibration.

The later theory of locally constant diagrams makes the distinction between a coherent family and its monodromy systematic (Cisinski, 2009, Theorem 1.3 and the discussion preceding it). The language of straightening provides a higher-categorical model of the family itself (Lurie, 2009, Theorem 2.2.1.2). These results explain the final form of the construction; the elementary fibre comparison remains the work that this chapter undertakes.

Dans 1° et 3°, on veut prouver que certaines flèches (p et $\bar{f}$) sont à la fois *propre* et *lisse* - ce qui est vraiment idéal ! Je dis qu'il suffit à chaque fois de prouver l'un des deux, que l'autre s'en déduit (comme 'cas particulier') par dualité. Pour ceci, notons p. 89 [page 143]

$$\tau \longmapsto \tau^-$$

l'automorphisme (*pas anti*-automorphisme) de $\underline{\mathrm{Ty}}$ qui consiste à changer de signe toutes les flèches de transition - c'est donc, au niveau des objets, l'unique automorphisme du monoïde libre Ty, qui échange Δ_+ et Δ_-. Au système projectif $\underline{t}$ dans $\underline{\mathrm{Ty}}$ correspond le système projectif $\underline{t}^-$. Ceci dit, on a un isomorphisme de catégories

$$\boxed{\underline{\mathrm{Ch}}^{\underline{t}}_\infty(X^o) \simeq (\underline{\mathrm{Ch}}^{\underline{t}^-}_\infty(X))^o} \tag{35}$$

et le diagramme commutatif

$$\begin{array}{ccc} \underline{\mathrm{Ch}}^{\underline{t}}_\infty(X^o) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}^{\underline{t}^-}_\infty(X)^o \\ \downarrow{\scriptstyle p^{\underline{t}}_{X^o}} & & \downarrow{\scriptstyle (p^{\underline{t}^-}_X)^o} \\ X^o \times X^o & = & (X \times X)^o. \end{array}$$

Ainsi, dire que $p^{\underline{t}}_X$ `[plutôt` $p^{\underline{t}^-}_X$`]` est lisse, i.e. que $(p^{\underline{t}}_X)^o$ `[plutôt` $(p^{\underline{t}^-}_X)^o$`]` est propre, équivaut à dire que $p^{\underline{t}}_{X^o}$ est propre.

Il nous reste donc, à ce sujet, que deux faits à vérifier : la lissité de $p : \underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$, et celle de $\bar{f} : \overline{X'} \longrightarrow X$. [page 144]

a) Lissité de $p : \underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$. On se donne une flèche

$$w : (a, b) \longrightarrow (a', b'), \qquad \text{i.e.} \quad \begin{cases} u : & a & \longrightarrow & a' \\ v : & b & \longrightarrow & b', \end{cases}$$

35. cf. page 140 pour le passage de $\underline{t}$ à $\underline{t}^o$.

sur la base, un objet c' au dessus du but, i.e.

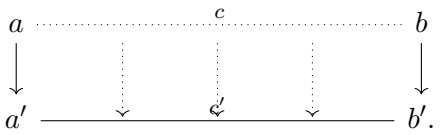
$$c' \in \underline{\mathrm{Ch}}_\infty(X; a', b'),$$

et on regarde la catégorie des flèches dans $\underline{\mathrm{Ch}}_\infty(X)$, de but c', qui relèvent w,

$$\begin{array}{ccc} \underset{\text{variable}}{c} & \xrightarrow{\varphi} & \underset{\text{fixe}}{c'} \\ \Big| & & \Big| \\ (a,b) & \xrightarrow{w} & (a',b'), \end{array}$$

i.e. la catégorie des couples (c, φ), où p. 90

$$\begin{cases} c \in \underline{\mathrm{Ch}}_\infty(X; a, b) \\ \varphi : c \longrightarrow c' \quad (\in \underline{\mathrm{Ch}}_\infty(X; a', b')) \\ \text{une flèche induisant} \quad \begin{array}{ll} u & \text{sur les sources} \\ v & \text{sur les buts,} \end{array} \end{cases}$$



[page 145]

9.11.90

Commentary · Exposition [ms. 144]

Why a category of lifts is the right test

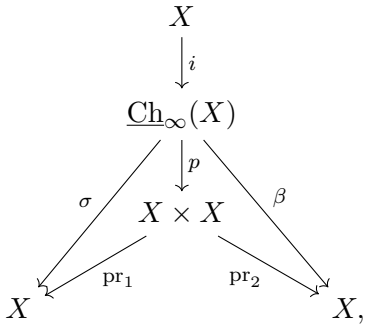
Grothendieck now replaces the geometric word "smooth" by a calculation that can be carried out on paths. Fix an arrow $w : b \to b'$ in a base B and an object e' above b'. For $p : E \to B$, the relevant category has objects (e, φ) with $p(e) = b$ and $\varphi : e \to e'$ satisfying $p(\varphi) = w$. A morphism from (e, φ) to (e_1, φ_1) is a vertical map $h : e \to e_1$ with $\varphi_1 h = \varphi$. Thus one must control both existence of lifts and the homotopy type of the space of choices.

This is exactly a comma category for the inclusion $E_b \to b\backslash E$. Requiring it to be contractible for every (e', w) says that this inclusion is aspherical. The equivalence with smoothness is the criterion of Maltsiniotis (2005, Theorem 1.2). In the present case $B = X \times X$ and $w = (u, v)$. The annotation placing φ inside $\underline{\mathrm{Ch}}_\infty(X; a', b')$ cannot be literal: its source has endpoints (a, b), so it is a morphism of the total path category over (u, v).

The next construction is designed to make these comma categories manageable. At a finite interval whose two endpoint arrows point into the interior, a lift is obtained by changing the two endpoints and composing with u and v. This is a Cartesian lift, hence a terminal object in the corresponding category of lifts. To deal with a path already lying in some stage, enlarge its support to a stage with the required endpoint parity. The filtered union then accommodates every finite diagram in the lifting category. This is why the two-index system introduced next is useful: it permits the same infinite path category to be approached by finite stages suited separately to incoming and outgoing lifts.

The proof of properness uses the opposite variance, as the duality formula on ms. 143 indicates. It does not follow from the mere existence of one lift, nor from a statement about the homotopy type of the total path category. The full calculation is carried out on ms. 150–153.

Pour la question précédente, relative aux flèches dans le diagramme



il faut préciser absolument la déscription de $\underline{\mathrm{Ch}}_\infty(X)$ *dans* Cat, donc il faut préciser le système projectif $\underline{t} = (t_n)_{n\in\mathbf{N}}$. Nous allons prendre en effet

$$t_n = (\underbrace{\Delta_+\Delta_-}_{\tau_0})^{2n},$$

avec les morphismes de transition décrits par ailleurs

$$\underbrace{t_{n+p}}_{=\tau_0^{2n+2p}} \longrightarrow \underbrace{t_n}_{\tau_0^n},$$

étant donnés par $\delta_p * \mathrm{id}_{\tau_o^{2n}} * \delta_p$, où $\delta_p : \tau_0^p \longrightarrow 1$ est la dégénérescence de τ_0^p. Ainsi on peut considérer $\underline{\mathrm{Ch}}_\infty(X)$ comme étant la catégorie des chemins biïnfinis dans X p. 91

$$\overbrace{\overbrace{\overbrace{\cdots \longrightarrow a_{-5} \longleftarrow a_{-4} \longrightarrow a_{-3} \xleftarrow{u_{-2}} a_{-2} \xrightarrow{u_{-1}} a_{-1} \xleftarrow{u_0} a_0 \xrightarrow{u_1} a_1 \xleftarrow{u_2} a_2 \xrightarrow{u_3} a_3 \longleftarrow a_4 \longrightarrow a_5 \longleftarrow \cdots,}^{\underline{\mathrm{Ch}}_0}}^{\underline{\mathrm{Ch}}_1}}^{\underline{\mathrm{Ch}}_2}$$

avec les u_i des identités pour $|i|$ assez grand.

La sous-catégorie (pleine)

$$\underline{\mathrm{Ch}}^W_{t_n}(X) \;=\; \underline{\mathrm{Ch}}^W_{(\Delta_+\Delta_-)^{2n}}$$

s'identifie à celle pour laquelle on a

> u_i une identité pour $i \notin [-2i+1, 2i]$, i.e. les seuls u_i 'significatifs' sont les $4i$ flèches u_{-2i+1}, u_{-2i+2}, …, u_{2i}.

Mais il sera plus commode de regarder $\underline{\mathrm{Ch}}_\infty(X)$ comme la limite inductive d'une famille à deux indices de sous-catégories pleines [page 146]

$$\underline{\mathrm{Ch}}_{n,m}(X), \qquad n, m \in \mathbf{Z},$$

en définissant cette catégorie comme *vide* si $n > m$, et pour $n \le m$ comme formée des objets

$$\begin{cases} (a_i, u_i)_{i\in\mathbf{Z}} \in \underline{\mathrm{Ch}}^W_{n,m}(X) \iff u_i \text{ une identité pour } i \notin [n, m-1] \\ \overset{\mathrm{id}}{\cdots} a_{n-1} \xrightarrow{\mathrm{id}} \underbrace{a_n \xrightarrow{u_n} a_{n+1} \cdots \longrightarrow a_{m-1} \xrightarrow{u_{m-1}} a_m}_{\text{'partie utile' du chemin}} \xrightarrow{\mathrm{id}} a_{m+1} \overset{\mathrm{id}}{\cdots}. \end{cases}$$

Donc pour $n = m$, on trouve la sous-catégorie pleine isomorphe à X, formée des chemins infinis 'totalement dégénérés', i.e. tels que *toutes* les u_i soient des identités. (Cette sous-catégorie de $\underline{\mathrm{Ch}}_\infty$ ne dépend donc pas du choix de $n = m$.)

$$\xrightarrow{\mathrm{id}} \underbrace{a_{-1}}_{=a} \xleftarrow{\mathrm{id}} \underbrace{a_0}_{=a} \xrightarrow{\mathrm{id}} \underbrace{a_1}_{=a} \xleftarrow{\mathrm{id}}$$

Tous les sommets d'un tel chemin sont égaux à un même objet de X, et les flèches de transition entre eux sont identiques.

La structure de $\underline{\mathrm{Ch}}^W_{n,m}(X) \simeq \underline{\mathrm{Ch}}^W_{\tau_{n,m}}(X)$ ne dépend que de la longueur $m-n$ de $\tau_{n,m}$ et de la *parité* de n. Si n est pair, la direction initiale de $\tau_{n,m}$ est positive, et

$$\tau_{n,m} = \underbrace{\Delta_+\Delta_-\cdots}_{m-n \text{ facteurs en alternance}} = \begin{cases} (\Delta_+\Delta_-)^{\frac{m-n}{2}} & \text{si } m \text{ pair} \\ (\Delta_+\Delta_-)^{\frac{m-n}{2}-1}\Delta_+ & \text{si } m \text{ impair} \end{cases}$$

[plutôt $\frac{m-n-1}{2}$ au lieu de $\frac{m-n}{2}-1$], tandis que pour n impair c'est juste l'inverse

p. 92
[page 147]

$$\tau_{n,m} = \underbrace{\Delta_-\Delta_+\cdots}_{m-n \text{ facteurs alternés}} = \begin{cases} (\Delta_-\Delta_+)^{\frac{m-n}{2}} & \text{si } m \text{ impair} \\ (\Delta_-\Delta_+)^{\frac{m-n}{2}-1}\Delta_- & \text{si } m \text{ pair} \end{cases}$$

[plutôt $\frac{m-n-1}{2}$ au lieu de $\frac{m-n}{2}-1$]. La direction initiale est conforme à la parité de n, i.e. égale à $(-1)^n$, tandis que la direction terminale est opposée à la parité de m, i.e. égale à $(-1)^m$ [plutôt $(-1)^{m+1}$].

Commentary · Exposition [ms. 147]

Finite support and endpoint parity

Grothendieck's two-sided sequence makes the eventual constancy of a path concrete. Its advantage is that support may be enlarged independently at the two ends. To avoid the shift in edge numbering between ms. 145 and ms. 146, write e_i for the edge joining a_i and a_{i+1}, directed forwards for even i and backwards for odd i. Thus e_i is the u_{i+1} of the large diagram on ms. 145. A path supported on $[n,m]$ has e_i equal to an identity outside $n \le i < m$.

For $n < m$, the source-end sign is $(-1)^n$ and the target-end sign is $(-1)^{m-1} = (-1)^{m+1}$. The four cases are

n mod 2	m mod 2	initial sign	terminal sign
0	0	+	−
0	1	+	+
1	0	−	−
1	1	−	+

For an odd length $m-n$, the number of full two-letter blocks is $(m-n-1)/2$; this is the correction already supplied by the editors in both displayed formulas. When $n = m$ there are no endpoint arrows and the subcategory is the constant-path copy of X; the sign table is not intended for that degenerate case.

The inclusion for $n' \le n \le m \le m'$ extends a diagram by identity arrows. It is fully faithful, since a natural transformation is forced to be constant on each identity tail. Every eventually constant path and every natural transformation between two such paths belongs to a sufficiently large common support interval. Hence the filtered colimit of these full subcategories is the asserted category of two-sided paths, on morphisms as well as objects. The even-even intervals form a cofinal family of support intervals, and so do the odd-odd intervals. The first family has both endpoint arrows pointing into the interval and is adapted to Cartesian lifts; the second has both pointing out and is adapted to coCartesian lifts. Cofinality here is the ordinary directedness of support intervals, not an assertion that a fixed inclusion of categories is a weak equivalence.

Je note que l'on a $\tau_0 = (\tau_0)^o$, d'où

$$\underline{t} = \underline{t}^o, \qquad \text{mais } \underline{t} \neq \underline{t}^-,$$

et pourtant on a des isomorphismes fonctoriels en X

$$\underline{\mathrm{Ch}}_{\underline{t}}(X) \simeq \underline{\mathrm{Ch}}_{\underline{t}^-}(X)$$

respectant source et but, à cause d'un isomorphisme

$$\underline{t} \simeq \underline{t}^- \qquad \text{dans } \mathrm{Pro}(\underline{\mathrm{Ty}}).$$

On peut dire p. ex. que le *foncteur translation* T^p d'amplitude p (pour $p \in \mathbf{Z}$) est défini sur $\underline{\mathrm{Ch}}^W_{\underline{t}}(X)$, par

$$\left\{ \begin{array}{l} T^p(((a_i),(u_i))_{i\in\mathbf{Z}}) = ((b_j),(v_j))_{j\in\mathbf{Z}} \\ \text{avec} \left\{ \begin{array}{lcl} b_j & = & a_{j-p} \\ v_j & = & u_{j-p} \end{array} \right. \end{array} \right.$$

et que

$$T^p : \left\{ \begin{array}{lcl} \underline{\mathrm{Ch}}_t(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_{-t}(X) \\ \underline{\mathrm{Ch}}_{-t}(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_t(X) \end{array} \right\} \text{ si } p \text{ impair}$$

$$T^p : \left\{ \begin{array}{lcl} \underline{\mathrm{Ch}}_t(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_t(X) \\ \underline{\mathrm{Ch}}_{-t}(X) & \xrightarrow{\sim} & \underline{\mathrm{Ch}}_{-t}(X) \end{array} \right\} \text{ si } p \text{ pair.}$$

En d'autres termes, on peut formellement regarder la catégorie somme [page 148]

$$\underline{\mathrm{Ch}}_{\pm}(X) = \underline{\mathrm{Ch}}_{+}(X) \amalg \underline{\mathrm{Ch}}_{-}(X)$$

et l'opération translation T comme un automorphisme qui échange les deux morceaux, et T^p n'est autre que son itéré p-ème, qui échange les morceaux $+$ et $-$ si p impair, qui les stabilise si p pair.

Il y a aussi l'automorphisme involutif de passage au chemin opposé, qui échange source et but, et que je définis (conformément à l'isomorphisme canonique identique p. 93

$$\underline{t} \xrightarrow{\sim} \underline{t}^o \quad)$$

comme la symétrie S

$$\left\{ \begin{array}{l} S((a_i),(u_i)) = ((b_j),(u_j)) \ \text{ avec} \\ \quad b_j \ = \ a_{-j} \\ \quad u_j \ = \ u_{-j+1}. \end{array} \right.$$

Si je le regarde comme opérant sur $\underline{\mathrm{Ch}}^W_{\pm}(X)$, il est lié au foncteur T par

$$\left\{ \begin{array}{rcl} S^2 & = & \mathrm{id} \\ STS^{-1} & = & T^{-1}, \end{array} \right.$$

de sorte que c'est le groupe semidirect de $\mathbf{Z}/2\mathbf{Z}$ par $\mathbf{Z}$ (groupe affine de dimension 1 de l'anneau $\mathbf{Z}$) qui opère sur $\underline{\mathrm{Ch}}_{\pm}$.

Commentary · Examples [ms. 148]

The dihedral action on a two-sided path

Grothendieck introduces translations to compare the two parity choices without changing either endpoint. In the edge convention of the preceding block, translation and reversal are

$$(T^p a)_j = a_{j-p}, \quad (T^p e)_j = e_{j-p}, \qquad (Sa)_j = a_{-j}, \quad (Se)_j = e_{-j-1}.$$

These formulas apply equally to the components of a natural transformation, so they define functors, not merely permutations of the objects. Since paths are eventually constant, translation leaves both eventual values unchanged; reversal exchanges them. Odd translation reverses parity, whereas even translation preserves it. Reversal preserves each parity convention because reversing the direction of an edge also reverses the parity of its index.

The group relations are identities of functors. On vertices,

$$(S^2 a)_j = a_j, \qquad (STSa)_j = a_{j+1} = (T^{-1}a)_j,$$

and substitution in the edge formula gives the same equalities on edges and natural transformations. The acting group is therefore

$$D_\infty = \mathbb{Z} \rtimes \mathbb{Z}/2, \qquad (\epsilon, k) : j \longmapsto \epsilon j + k, \quad \epsilon \in \{1, -1\}.$$

The integer subgroup is the translation subgroup and reflection acts on it by inversion. This specifies the semidirect-product convention independently of the order of the groups in the French phrase.

For example, a path with one nonidentity edge $e_0 : x \to y$ is carried by T^2 to the same edge at position 2, with two additional identity positions on one side and two fewer on the other. Both have source x and target y, but they are different objects of the strict path category. The formulas above prove an isomorphism of categories and the dihedral relations. They do not by themselves prove that T^2 is the identity after passage to Hot. This is the specific coherence question taken up on the next page. It should be resolved by a homotopy or a comparison of the relevant diagrams, not by treating coordinate translation as literal equality.

[page 149]

On est embarassé de trouver un automorphisme T^2 de la situation $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$, sans qu'il soit évident que dans Hot, ce foncteur induise l'identité! Je ne vois aucun moyen de relier un chemin c et T^2c par une flèche dans $\underline{\mathrm{Ch}}_\infty$, ni même pas un chemin 'canonique'. Mais je me dis que cet automorphisme est décrit par un automorphisme, non pas du système projectif

$$(\tau_{n,m})_{n,m \in \mathbf{Z}^o \times \mathbf{Z},\ n \le m}$$

(où $\mathbf{Z}^o$ désigne l'ensemble ordonné *opposé* de $\mathbf{Z}$), mais du *pro-objet* de $\underline{\mathrm{Ty}}$ qu'il définit (qui est bien sûr 'le même' pour tous les systèmes projectifs cofinaux à gauche dans $\underline{\mathrm{Ty}}$). Il en résulte déjà que T^2 induit l'identité sur l'ind-objet de Cat hot, $\underline{\underline{\mathrm{Ch}}}_\infty(X)$, et a fortiori sur l'ind-objet correspondant de Hot, ou de $\mathrm{Hot}_0^\wedge$, ou enfin pour l'objet de $\mathrm{Hot}_0^\wedge$ lui-même. De même pour les $\underline{\mathrm{Ch}}_\infty(X; a, b)$. Il s'ensuit, revenant à $\underline{\mathrm{Ch}}_\infty(X)$ lui-même et aux $\underline{\mathrm{Ch}}_\infty(X; a, b)$, que l'automorphisme induit par T^2 est l'identité dans $\mathrm{Hot}_0^\wedge$. J'espère que Hot $\longrightarrow \mathrm{Hot}_0^\wedge$ est tout au moins fidèle, de sorte que ça prouvera que T^2 induit l'identité dans Hot également. [page 150]

Je reviens aux questions 1° à 4° d'il y a quelques jours.

Pour 1°), il faut prouver que

$$p : \underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$$

est lisse et propre et une Hot-fibration. Par dualité, p propre résultera de p lisse. Mais $\underline{\mathrm{Ch}}_\infty(X)$ est $\varinjlim$ des $\underline{\mathrm{Ch}}_{n,m}(X)$, avec $n \longrightarrow -\infty$, $m \longrightarrow +\infty$, et où on peut se borner à n et m *pairs.* Or on voit que pour n, m pairs, i.e. $\tau_{n,m}$ à flèche initiale directe, à flèche finale

rétrograde, $\underline{\mathrm{Ch}}_{n,m}(X) \simeq \underline{\mathrm{Ch}}_{\tau_{n,m}}(X)$ est Cat-*fibrée* sur $X \times X$, donc lisse (i.e. HOT-lisse), p. 94
et cela passe aux $\varinjlim$ ([36]).

Ainsi $\underline{\mathrm{Ch}}_\infty(X) \xrightarrow{p} X \times X$ est à la fois propre et lisse ([37]). Dire que c'est une HOT-fibration, revient p. ex. à dire que pour

$$(a, b) \xrightarrow{(u,v)} (a', b')$$

dans $X \times X$, la flèche correspondante

$$\underline{\mathrm{Ch}}_\infty(X)/(a, b) \longrightarrow \underline{\mathrm{Ch}}_\infty(X)/(a', b')$$

est une Hot-équivalence. Or, comme p est propre, les inclusions des fibres en (a, b) et en (a', b') dans l'un et l'autre membre sont des Hot-équivalences : [page 151]

36. **NB** $\underline{\mathrm{Ch}}_\infty(X)$ n'est ni fibrée ni cofibrée sur $X \times X$ - car les foncteurs d'inclusion entre les $\underline{\mathrm{Ch}}_{n,m}$ ne sont pas cartésiens resp. cocartésiens.

37. 12.11. Mais ça implique déjà que c'est une fibration !

$$(*)\qquad \begin{array}{ccc} \underline{\mathrm{Ch}}_\infty(X)/(a,b) & \xrightarrow{\rho} & \underline{\mathrm{Ch}}_\infty(X)/(a',b') \\ \wr\uparrow & & \wr\uparrow \\ \underline{\mathrm{Ch}}_\infty(X;a,b) & \xrightarrow{\varphi} & \underline{\mathrm{Ch}}_\infty(X;a',b') \end{array}$$

Ainsi on trouve une flèche en pointillés dans Hot, dont il faut prouver que c'est une Hot-équivalence. Mais en considérant

$$u : a \longrightarrow a' \qquad v : b \longrightarrow b'$$

comme des *chemins* dans X, et

$$u^o : a' \longleftarrow a$$

comme un chemin rétrograde de a' à a, la théorie de variance générale (dans IX) nous donne bel et bien un isomorphisme dans (Hot) :

$$\underline{\mathrm{Ch}}_\infty(X;a,b) \longrightarrow \underline{\mathrm{Ch}}_\infty(X;a',b'),$$

on a envie de dire que c'est φ, i.e. qu'en l'insérant à la place de φ, le carré ci-dessus devient commutatif (ce qui prouverait ce qu'on veut). Je pense que si on travaille avec les ind-objets $\underline{\underline{\mathrm{Ch}}}_\infty(X)/(a,b)$, $\underline{\underline{\mathrm{Ch}}}_\infty(X;a,b)$ etc., pour lesquels on a des accouplements qui s'explicitent agréablement aux crans finis, et donnent naissance aux φ, cela devrait être facile (quoiqu'un peu fastidieux). Cela ne prouverait pas pour autant la commutativité de $(*)$ dans Hot, mais seulement dans $\mathrm{Hot}_0^\wedge$, ce qui n'est peut-être pas suffisant. Mais cela suffit pour prouver que la flèche ρ dans $(*)$ devient `[un]` isomorphisme dans $\mathrm{Hot}_0^\wedge$, donc elle est `[un]` isomorphisme. p. 95 [page 152]

Commentary · Exposition [ms. 150]

Why both parities give the endpoint fibration

Grothendieck wants the categorical path object to carry the same geometric information as an ordinary path fibration, before any fibrant replacement of X. The two ends of a finite zigzag already explain how this can happen. For a positive-length interval, when both endpoints are sources in the interval category, changing their values contravariantly only composes the two outer arrows; this gives a Grothendieck fibration over $X \times X$. When both endpoints are sinks, the analogous operation is covariant and gives an opfibration. The two parity choices, with positive length imposed, are cofinal among the finite supports of an eventually constant path. The zero-length case is the diagonal functor and must not be used in the fibration claim.

The passage to the filtered union deserves a proof, since the inclusion functors need not preserve Cartesian arrows. For a functor $q : E \to B$, an object $e \in E$, and an arrow $h : b \to q(e)$, put

$$L_q(e,h) = \{(d,\gamma) : q(d) = b,\ \gamma : d \to e,\ q(\gamma) = h\},$$

with morphisms over 1_b commuting with γ. Smoothness means that every such category is weakly contractible; properness has the dual outgoing-lift test. These are the fibre-to-comma criteria of Maltsiniotis (2005, Theorem 1.2 and its dual). Suppose $E = \varinjlim_i E_i$ over a filtered category, with all functors over B. Once e is represented at one stage, its lift category is the filtered colimit of the corresponding lift categories at later stages. Every object, arrow, and equation defining a finite simplex occurs at a stage, and every equality holds at some later stage. Nerves preserve these filtered colimits. A filtered colimit of weakly contractible simplicial sets is weakly contractible: after Ex^∞, which also commutes with filtered colimits, the assertions about based homotopy groups involve only finitely many simplices. Consequently smoothness and properness pass separately to this filtered colimit. Preserving Cartesian arrows is unnecessary.

It follows that $\underline{\mathrm{Ch}}_\infty(X) \to X \times X$ is smooth and proper. We can also justify directly the later marginal observation that these two conditions imply the required homotopy-fibre property. For $h : b \to b'$ in B, let M_h be the category of arrows $d \to d'$ in E over h, with its two projections to E_b and $E_{b'}$. Properness and smoothness show that these projections are weak equivalences. For example, the comma category under an object $d_0 \in E_b$ retracts, by precomposition, onto its outgoing-lift

category; the dual argument uses postcomposition. Quillen's Theorem A applies. The arrows defining M_h then give a natural transformation between its two maps into $b\backslash E$, the second obtained by precomposition from $b'\backslash E$. Since smoothness identifies each strict fibre with its under-comma category, precomposition

$$b'\backslash E \longrightarrow b\backslash E$$

is a weak equivalence. Quillen's Theorem B identifies the strict fibres with homotopy fibres (Quillen, 1973, §1, Theorem B). This proof also survives every base change. It establishes what the manuscript needs without the finite-test conservativity invoked on the following pages.

Commentary · Examples [ms. 151]

A homotopy fibration without Cartesian lifts

The warning in footnote 36 can be seen in the walking-arrow category $X = [1]$. It explains why a contractible category of choices is the appropriate conclusion, even when no best choice exists.

Consider the constant eventual path $c = 1$ and the base arrow $(0 \to 1, 1_1) : (0, 1) \to (1, 1)$. An incoming lift is precisely an eventual alternating path d from 0 to 1: its natural transformation to the constant path 1 is unique. If the endpoint map were a Grothendieck fibration, this lifting category would have a terminal object. Its arrows are pointwise inequalities between the 0–1 diagrams. For any candidate d, choose a direct edge far enough to the left that d is still constantly 0 there. Let e change from 0 to 1 across that edge and remain 1 afterwards. This is an allowed alternating path, because all its other edges are identities. At a vertex immediately after the chosen edge, e is 1 and d is 0, so there is no natural transformation $e \to d$. No terminal lift exists.

Dually, take the constant path 0 and the outgoing base arrow $(1_0, 0 \to 1)$. A cocartesian lift would give an initial object among paths from 0 to 1. For any candidate d, choose a single-step path whose direct transition is farther right than the point where d has become constantly 1. There is no arrow from d to this later-step path. Thus no initial lift exists either. The incoming and outgoing lift categories are nevertheless weakly contractible by the filtered-stage proof. This is the exact distinction between the two homotopical lifting properties proved here and an ordinary fibration or opfibration.

2°) Il faut montrer que les HOT-fibrations

$$\underline{\mathrm{Ch}}_\infty(X) \underset{\beta}{\overset{\sigma}{\rightrightarrows}} X$$

sont triviales, ou ce qui revient au même, qu'elles sont à fibres asphériques. Il suffit de le voir pour σ, donc que

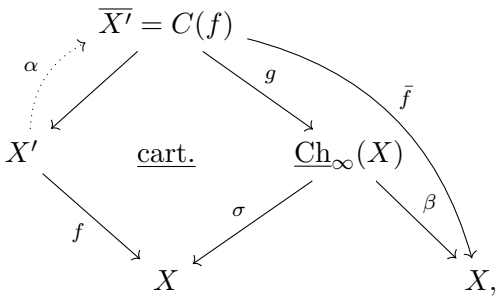
$$\underline{\mathrm{Ch}}_\infty(X; a, ?)$$

est asphérique. Mais cette catégorie est la limite inductive des $\underline{\mathrm{Ch}}_n(X; a, ?)$, i.e. des

$$\underline{\mathrm{Hom}}^{\text{source fixée}}(\tau^n, X).$$

Mais comme l'intervalle I_{τ^n} se rétracte par déformation sur son origine, il est immédiat que cette catégorie $\underline{\mathrm{Hom}}$ est même contractile.

3°) Considérons



[page 153]

je voudrais prouver que $\bar{f}$ est une **D**-fibration propre et lisse ([38]).

Commentary · Exposition [ms. 152]

The contraction with one endpoint fixed

The next task is to make a path starting at a disappear while keeping its starting point fixed. This is stronger than knowing merely that the interval has contractible nerve, because an arbitrary weak equivalence of interval categories need not induce a weak equivalence of strict functor categories. Here the interval has an explicit categorical deformation, and that is enough.

Write its vertices as $0, \dots, r$, in their geometric order, with either orientation on each edge. For $0 \le k \le r$, define $r_k(j) = \min(j, k)$. Each $r_k : I \to I$ is a functor. Successive r_k, r_{k+1} are connected by a natural transformation in the direction of the edge between k and $k+1$: on the tail $j \ge k+1$ use that edge, and elsewhere use identities. Naturality follows on the generating edges, hence on all arrows. Every r_k fixes 0, while r_0 is constant and $r_r = 1_I$. Precomposition therefore joins the identity of $\mathrm{Fun}(I, X)_{a,?}$ to the constant functor at the constant path a_I, through a finite zigzag of natural transformations which is relative to the source. Its nerve is contractible.

Filtered colimits of these one-endpoint categories are weakly contractible by the finite simplicial argument just given. Thus both σ and β have weakly contractible strict fibres. Since they are smooth and proper, their pullbacks along arbitrary functors are weak equivalences. In the mapping-path construction, this universal statement is precisely what allows the category X' to replace X at the source endpoint. No supposition that X itself is a Thomason fibrant category enters this argument.

$\overline{X'}$ est la catégorie des couples

$$(x', c) \ \in \ \mathrm{Ob}\, X' \times \mathrm{Ch}_\infty(X)$$

tels que

$$\sigma(c) = f(x')$$

et

$$\bar{f}(x', c) = \beta(c), \qquad \text{but de } c.$$

La fibre de $\overline{X'}$, $\bar{f}$ en $x_0 \in X$ est donc la catégorie des couples $(x', c) \in \mathrm{Ob}\, X' \times \mathrm{Ch}_\infty(X)$, tels que

$$\sigma(c) = f(x'), \qquad \beta(c) = x_0.$$

On peut regarder encore $\overline{X'}$ comme limite inductive des sous-catégories $\overline{X'}_{n,m}$, correspondante aux couples (x', c) avec $c \in \mathrm{Ch}_{n,m}(X)$. On trouve que pour m pair (direction terminale de $\tau_{n,m}$ rétrograde) les $\overline{X'}_{n,m}$ sont Cat-fibrées sur X, pour m impair (direction terminale de $\tau_{n,m}$ directe) elles sont Cat-cofibrées. Donc on conclut encore comme tantôt, $\overline{X'}$ étant (au choix) limite inductive filtrante de sous-catégories lisses sur X, ou propres sur X. p. 96

Il faut prouver enfin que $\overline{X'} \xrightarrow{\bar{f}} X$ est une **D**-fibration ([39]), ce qui revient p. ex. à dire que pour une flèche [page 154]

$$a \xrightarrow{v} b$$

dans X, la flèche induite

$$\overline{X'}/a \xrightarrow{\bar{v}} \overline{X'}/b$$

est une Hot-équivalence. On sait déjà comme tantôt que l'on a un diagramme à flèches verticales des Hot-équivalences

$$\begin{array}{ccc} \overline{X'}/a & \longrightarrow & \overline{X'}/b \\ \wr\uparrow & & \wr\uparrow \\ \overline{X'}_a & \dashrightarrow & \overline{X'}_b, \end{array}$$

38. Ça résulte formellement de ce qui a été déjà vu, cf. pages 170-171.
39. Cela résulte déjà du fait que c'est propre et lisse!

et tout revient à montrer qu'une certaine flèche entre $\overline{X'}_a$ et $\overline{X'}_b$ dans Hot est [un] isomorphisme. J'ai envie de définir directement une telle flèche dans Hot, voire dans Cat. Or pour les m impairs, les $\overline{X'}_{n,m}$ sont Cat-cofibrées sur X. Pour $\overline{X'}_{-m,m}$ (m impair) on trouve donc

$$(\overline{X'}_{-m,m})_a \xrightarrow{v_*} (\overline{X'}_{-m,m})_b,$$

l'ennui est qu'elles ne sont pas compatibles avec les morphismes de transition pour m variable. [page 155]
Elles le sont cependant *à homotopie près* sûrement, donc on trouvera au moins une flèche d'ind-objets dans Cat hot

$$\overline{\overline{X'}}_a \xrightarrow{v_*} \overline{\overline{X'}}_b,$$

ce qui par passage à $\mathrm{Hot}_0^\wedge$ donne quand-même une flèche

$$\overline{X'}_a \xrightarrow{v_*} \overline{X'}_b.$$

En travaillant avec les m impairs, on trouve de même des foncteurs en sens inverses

$$(\overline{X'}_{-m,m})_a \xleftarrow{v^*} (\overline{X'}_{-m,m})_b,$$

qui ne se recollent pas pour m variable, mais qui se recollent à homotopie près, pour donner p. 97
une flèche dans Ind Cat hot

$$\overline{\overline{X'}}_a \xleftarrow{v^*} \overline{\overline{X'}}_b,$$

d'où aussi

$$\overline{X'}_a \xleftarrow{v^*} \overline{X'}_b \qquad \text{dans } \mathrm{Hot}_0^\wedge.$$

D'autre part, on doit pouvoir montrer soritalement, cran par cran, que v_*v^* et v^*v_* dans [page 156]
Ind Cat hot sont égales à l'identité, donc v_* et v^* sont des isomorphismes inverses l'un de l'autre. On doit vérifier d'autre part soritalement que

$$\begin{array}{ccc} \overline{\overline{X'}}/a & \xrightarrow{\bar v} & \overline{\overline{X'}}/b \\ \uparrow & & \uparrow \\ \overline{\overline{X'}}_a & \xrightarrow{v_*} & \overline{\overline{X'}}_b, \end{array}$$

dans Ind Cat hot est commutatif. Passant au diagramme correspondant dans $\mathrm{Hot}_0^\wedge$, en prenant les limites inductives, on trouve un carré commutatif dans $\mathrm{Hot}_0^\wedge$

$$\begin{array}{ccc} \overline{X'}/a & \xrightarrow{\bar v} & \overline{X'}/b \\ \uparrow\wr & & \uparrow\wr \\ \overline{X'}_a & \xrightarrow[\sim]{v_*} & \overline{X'}_b, \end{array}$$

d'où résulte que $\bar v$ est [un] isomorphisme dans $\mathrm{Hot}_0^\wedge$, donc aussi dans Hot.

Commentary · Later developments [ms. 153]

Evrard's categorical cocylinder

Grothendieck is seeking an explicit factorization of an arbitrary functor by adjoining paths. The same purpose already appears in Evrard (1975, §I.1), whose variable-length zigzags carry both natural transformations and maps which insert identity segments. The indexing conventions differ from those used here; an identification of the two categories would therefore require a comparison functor, not just the observation that their objects look alike.

A precise modern theorem is Shoikhet (2016, Theorem 2.8): the categorical cocylinder gives a functorial factorization by a weak equivalence followed by a homotopy-fibred functor, and there is a

dual construction. Its relation with the manuscript is the role of the second map: after applying the nerve, ordinary fibres compute homotopy fibres. Shoikhet's Proposition 1.14 formulates a stronger, base-change-stable version through simplicial sharp maps, under the prefibration hypotheses stated there. We use Quillen's comma-category theorem directly when those strict prefibration hypotheses have not been established. These distinctions matter for the total category $\underline{\mathrm{Ch}}(X)$, whose type-changing arrows will obstruct properness below, even though its fixed-endpoint categories still have the correct homotopy types.

Pour vérifier que la HOT-fibration $\bar{f}$ est triviale si et seulement si f est une HOT-équivalence, il faudrait avoir développé dans Cat le formalisme de la suite exacte des lacets - on est tout près visiblement !

p. 98

Commentary · Exposition [ms. 156] [page 157]

When the replacement is trivial

The desired test for triviality is already formal once the mapping-path construction has been established. Put $r : C(f) \to X'$ for the projection which forgets the path. It is the pullback of the universally weakly equivalent source map σ, so r is a weak equivalence. Its section i_f is consequently a weak equivalence as well. The equality $\bar{f} i_f = f$ gives

$$\bar{f} \text{ is a weak equivalence} \quad \Longleftrightarrow \quad f \text{ is a weak equivalence}$$

by two-out-of-three. If triviality is expressed instead by weakly contractible fibres, the homotopy-fibre theorem gives the same criterion. One can see both directions by replacing the realized map by a Serre fibration: a weak equivalence has contractible homotopy fibres, and a fibration with such fibres induces isomorphisms on every based homotopy group and a bijection on components.

Thus an independently developed long exact sequence for categorical loop objects is not needed for this particular conclusion. Nor should the preceding finite-test inference bear its proof. The claimed conservativity of the unbased restricted Yoneda functor is false in the generality used in the manuscript; the commentary on the finite-test obstruction in Chapter V, manuscript p. 52, explains this. The two-out-of-three argument, together with the actual endpoint-fibration theorem, proves exactly the assertion required here.

13.11.90

13 Où on prouve enfin que $\underline{\mathrm{Ch}}(X; -, -)$ est aussi bon (voire meilleur) que $\underline{\mathrm{Ch}}_\infty(X; -, -)$.

Je crois finalement que les catégories $\underline{\mathrm{Ch}}(X)$, $\underline{\mathrm{Ch}}(X; a, b)$ (sans indice ∞!) sont 'bonnes', que les flèches canoniques dans Hot

$$\begin{cases} \underline{\mathrm{Ch}}_\infty(X) & \xrightarrow{\sim} \quad \underline{\mathrm{Ch}}(X) \\ \underline{\mathrm{Ch}}_\infty(X; a, b) & \xrightarrow{\sim} \quad \underline{\mathrm{Ch}}(X; a, b) \end{cases}$$

sont des isomorphismes. (La première est même [un] isomorphisme dans $\mathrm{HOT}(X \times X)$, ce qui implique la deuxième relation, en passant aux fibres …) Je crois que je suis à présent (après XI) en mesure de le prouver. Cela résultera des deux résultats-clef suivants :

1) Le foncteur canonique
$$X \longrightarrow \underline{\mathrm{Ch}}(X)$$
(associant à tout $x \in \mathrm{Ob}\, X$ le chemin vide de source et but X) est une Hot-équivalence, et :

2) Le foncteur source-but
$$\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$$
est *lisse* et *propre* (donc une 'Hot-fibration parfaite').

Cela doit donner ce qu'on cherche. Comme la $X \times X$-flèche $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow \underline{\mathrm{Ch}}(X)$ (et les flèches déduites $\underline{\mathrm{Ch}}_\infty(X;a,b) \longrightarrow \underline{\mathrm{Ch}}(X;a,b)$) a été définie par l'intermédiaire de la catégorie

$$\mathcal{C}_{\mathbf{N}^o} = \mathbf{N}^o \times_{\underline{\mathrm{Ty}}} \mathcal{C}$$

(où $\mathcal{C} = \underline{\mathrm{Ch}}(X)$ `[est]` considérée comme catégorie fibrée sur $\underline{\mathrm{Ty}}$, et $\mathbf{N}^o \longrightarrow \underline{\mathrm{Ty}}$ est le système projectif des $\underbrace{(\Delta_+\Delta_-)^{2n}}_{t_n}\dots$)

à l'aide des deux flèches canoniques [page 158]

$$\begin{array}{ccc} \mathcal{C}_{\mathbf{N}^o} & \longrightarrow & \underline{\mathrm{Ch}}_\infty(X) \\ \downarrow & & \\ \underline{\mathrm{Ch}}(X), & & \end{array}$$

Commentary · Development (commentators') [ms. 157]

A properness claim which the empty path disproves

The return to paths of arbitrary type is motivated by their strictly associative concatenation. Their geometric usefulness is real, but the proposed proof gives them a stronger lifting property than they possess. We isolate the error before using the comparison diagram.

The empty-path obstruction. [ours] For $X = [1] = (0 \xrightarrow{w} 1)$, neither

$$\underline{\mathrm{Ch}}(X) \longrightarrow X \times X \qquad \text{nor} \qquad \mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$$

is proper, with the morphisms defined in this chapter.

Proof. Let z be the empty path at 0, lying above $(0,0)$, and consider the base arrow $(1_0, w) : (0,0) \to (0,1)$. Properness requires the category of its outgoing lifts with source z to be weakly contractible. In $\underline{\mathrm{Ch}}(X)$, a morphism from a path of type e to a path of type τ lies over an endpoint-preserving type map $e \to \tau$. The single vertex of e would have to map to both endpoints of τ. Hence $\tau = e$. But an empty path has equal endpoints and cannot lie above $(0,1)$. The lift category is empty. For $\mathcal{C}_{\mathbf{N}^o}$ the same argument is even more immediate: an arrow from stage 0 can only end at stage 0, because its image in $\mathbf{N}^o$ has source 0. Again its target must have equal endpoints. This proves both failures. □

The opposite of a total Grothendieck construction reverses the arrows in its indexing category as well as the arrows inside its fibres. Reversing a path or replacing X by X^o does not perform that operation on $\underline{\mathrm{Ty}}$ or on $\mathbf{N}^o$. This is the missing distinction in the deductions of properness on pp. 161 and 169. The same lifting test is the dual of Maltsiniotis (2005, Theorem 1.2).

This counterexample does not contradict the comparison of fixed-endpoint homotopy types which opens the section. That conclusion has a repair: smoothness of $\underline{\mathrm{Ch}}(X)$, together with explicit cancellation of a path and its reverse, gives the homotopy-fibre theorem by Quillen B. The intermediary $\mathcal{C}_{\mathbf{N}^o}$ can then be compared with $\underline{\mathrm{Ch}}_\infty(X)$ using its filtered-colimit construction. We give the full argument below. We did not find this empty-path counterexample or this precise repair of the present pages in the primary sources checked, namely Evrard, Shoikhet, and Maltsiniotis; the mathematical ingredients of the repair are classical.

on sera obligé de prouver aussi les résultats qui correspondent à 1) 2) pour $\mathcal{C}_{\mathbf{N}^o}$, i.e. p. 99

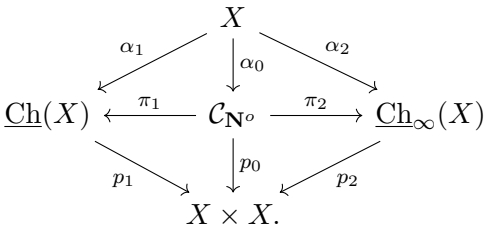
1′)

$$\begin{array}{ccc} X & \longrightarrow & \mathcal{C}_{\mathbf{N}^o} \\ \wr\downarrow & & \\ \mathcal{C}_{t_0} = \mathcal{C}_0 & & \end{array}$$

(l'inclusion canonique de $\mathcal{C}_0$, fibre de $\mathcal{C}_{\mathbf{N}^o}$ (fibrée sur $\mathbf{N}^o$) en $0 \in \mathbf{N}^o$) est une Hot-équivalence, et :

2′) $\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$ est propre et lisse.

Moyennant ces quatre résultats techniques, que je vais prouver tantôt (et 1′ et 2′ ne font aucune surprise pour moi), on gagne. Considérons en effet le diagramme



Par 1) et 1′) (concernant α_1 et α_0) et le résultat similaire déjà connu pour α_2, ces trois foncteurs sont des Hot-équivalences. Donc il en est de même de π_1, π_2. Comme p_1, p_0, p_2 sont des Hot-fibrations (étant propres et lisses), il s'ensuit que π_1 et π_2 induisent des Hot-équivalences sur les fibres [page 159]

$$\underline{\mathrm{Ch}}(X;a,b) \quad \xleftarrow{\pi_1(a,b)} \quad \mathcal{C}_{\mathbf{N}^o}(a,b) \quad \xrightarrow{\pi_2(a,b)} \quad \underline{\mathrm{Ch}}_\infty(X;a,b).$$

Je l'avais vérifié (sans mal) déjà pour π_2 et les $\pi_2(a,b)$, ce qui avait permis, en inversant π_2 dans Hot, de définir *dans* Hot

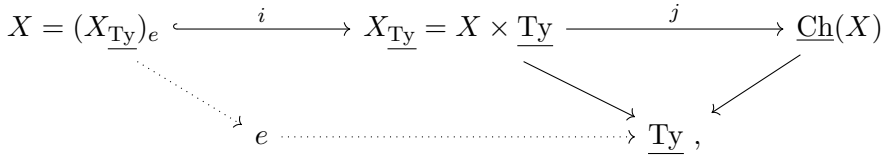
$$\begin{cases} \underline{\mathrm{Ch}}_\infty(X) & \longrightarrow & \underline{\mathrm{Ch}}(X) \\ \underline{\mathrm{Ch}}_\infty(X;a,b) & \longrightarrow & \underline{\mathrm{Ch}}(X;a,b), \end{cases}$$

et dire que ce sont des isomorphismes signifie justement que π_1 et les $\pi_1(a,b)$ sont des isomorphismes.

Je ne crois pas que $\mathcal{C}_{\mathbf{N}^o}$ ait en lui-même d'autre interêt, que de permettre de comparer les $X \times X$-catégories $\underline{\mathrm{Ch}}_\infty(X)$ et $\underline{\mathrm{Ch}}(X)$, à l'aide du diagramme précédent. Par contre, $\underline{\mathrm{Ch}}_\infty(X)$ et $\underline{\mathrm{Ch}}(X)$ m'aparaissent comme les deux ingrédients essentiels du formalisme des chemins dans Cat.

Je vais prouver 1, 2, 1′, 2′.

1) Considérons le diagramme p. 100 [page 160]



où $X_{\underline{\mathrm{Ty}}}$ et $\underline{\mathrm{Ch}}(X)$ sont considérées comme catégories Cat-fibrées sur $\underline{\mathrm{Ty}}$, et le foncteur j est déduit des foncteurs canoniques

$$X \longrightarrow \underline{\mathrm{Ch}}(X)_\tau \overset{\text{déf}}{=} \underline{\mathrm{Ch}}_\tau(X),$$

associant à $x \in \mathrm{Ob}\, X$ le chemin $p_\tau^*(x)$, où $p_\tau : \tau \longrightarrow e$ est la projection de τ vers l'élément unité (et élément final) $e = 1_{\underline{\mathrm{Ty}}}$ de $\underline{\mathrm{Ty}}$, et $\underline{\mathrm{Ch}}_e(X)$ est identifiée à X. Il faut vérifier le

Lemme. *Pour tout $\tau \in \underline{\mathrm{Ty}}$, le foncteur précédent est une* Hot-*équivalence (et même un homotopisme).*

Mais on sait que pour deux catégories $\mathcal{X}$, $\mathcal{C}$ lisses sur une autre, T, tout T-morphisme de $\mathcal{X}$ dans $\mathcal{C}$ qui induit des Hot-équivalences sur les fibres, est une Hot-équivalence. Ainsi j est est une Hot-équivalence. D'autre part, $\underline{\mathrm{Ty}}$, ayant un objet final, est asphérique, donc $X \times \underline{\mathrm{Ty}} \xrightarrow{\mathrm{pr}_1} X$ est une Hot-équivalence, d'où résulte aussitôt que i est une Hot-équivalence. Donc de même ji, or $ji = \alpha_1$, q.e.d. [page 161]

Commentary · Exposition [ms. 160]

Completing the constant-path lemma

The first comparison is a statement about total homotopy types, and here Grothendieck's argument works. Its useful feature is that it does not ask the chosen sequence of interval types to be homotopy cofinal in $\underline{\mathrm{Ty}}$.

For every finite oriented interval I, evaluation at the initial vertex and the constant-diagram functor give

$$X \xrightarrow{c} \mathrm{Fun}(I, X) \xrightarrow{\mathrm{ev}_0} X, \qquad \mathrm{ev}_0\, c = 1_X.$$

The truncation functors $r_k(j) = \min(j, k)$ constructed above join $c\,\mathrm{ev}_0$ to the identity by a finite zigzag of natural transformations. This proves the lemma promised here and left unwritten on p. 170: c is a categorical homotopy equivalence, not just a weak equivalence of nerves. The argument applies to every sign pattern and to the zero-length interval.

These constant-diagram functors commute strictly with precomposition by type maps. They therefore assemble to the displayed Cartesian functor

$$j : X \times \underline{\mathrm{Ty}} \longrightarrow \underline{\mathrm{Ch}}(X)$$

between Grothendieck fibrations over $\underline{\mathrm{Ty}}$, and each of its fibre functors is a weak equivalence. The standard fibrewise comparison theorem for Grothendieck constructions makes j a weak equivalence. This is also the objectwise invariance behind Thomason's homotopy-colimit theorem; the fibred convention is recalled explicitly in Shoikhet (2016, §1.5). Since $\underline{\mathrm{Ty}}$ has terminal object e, the inclusion $X \to X \times \underline{\mathrm{Ty}}$ at e is a weak equivalence. Their composite is the empty-path section $X \to \underline{\mathrm{Ch}}(X)$.

Exactly the same proof over $\mathbf{N}^o$ proves $X \to \mathcal{C}_{\mathbf{N}^o}$ is a weak equivalence. For $\underline{\mathrm{Ch}}_\infty(X)$ the constant-diagram functors assemble to a levelwise weak equivalence of filtered diagrams; invariance of weak equivalences under filtered colimits gives the corresponding assertion. Consequently the two comparison arrows out of $\mathcal{C}_{\mathbf{N}^o}$ are weak equivalences on total categories, by two-out-of-three. What remains is genuinely relative: a weak equivalence over $X \times X$ need not be a weak equivalence on its strict fibres until those fibres have been identified as homotopy fibres.

2) Prouvons p. ex. la lissité de $\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$, la propriété s'en déduira en appliquant ce résultat à X^{opp}. Soit donc

$$(a', b') \;\longrightarrow\; (a, b), \qquad \text{i.e. } a' \xrightarrow{u} a, \quad b' \xrightarrow{v} b,$$

une flèche dans $X \times X$, et c un objet de $\underline{\mathrm{Ch}}(X)$ au dessus de (a, b) :

$$\begin{array}{ccc} a' & \overset{c'}{\underset{\rightarrow}{\cdots\cdots\cdots}} & b' \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle v} \\ a & \xrightarrow[\rightarrow]{c} & b. \end{array}$$

On regarde la catégorie $\mathcal{C}(c; u, v)$ des couples (c', w) qui relèvent $(c, (u, v))$, où donc c' est un chemin (de type non précisé) de a' vers b', et

$$w : c' \longrightarrow c$$

une flèche dans $\underline{\mathrm{Ch}}(X)$ qui sur les objets source induise u, et sur les objets buts induise v (ainsi c' est un chemin de a' vers b'). Il faut prouver que cette catégorie $\mathcal{C}(c; u, v)$ est asphérique. Pour cela, on regarde le chemin composé p. 101 [page 162]

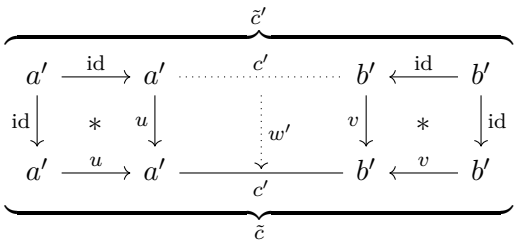
$$\tilde{c} = \mathrm{ch}(u)\ c\ \mathrm{ch}(v)^o,$$

où $\mathrm{ch}(u)$ est le chemin (direct) de a' vers a défini par $u : a' \longrightarrow a$, et $\mathrm{ch}(v)^o$ désigne donc le chemin (rétrograde) de b à b' défini par $v : b' \longrightarrow b$. Si c est de type τ, $\tilde{c}$ est donc de type $\Delta_+\tau\Delta_-$. On va comparer $\mathcal{C}(c; u, v)$ et la catégorie $\underline{\mathrm{Ch}}(X)/\tilde{c}$, à l'aide de deux foncteurs

$$\mathcal{C}(c; u, v) \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \underline{\mathrm{Ch}}(X; a', b')/\tilde{c}\ ,$$

dont on prouvera qu'ils définissent des flèches inverses l'une de l'autre dans Hot. Comme $\underline{\mathrm{Ch}}(X)/\tilde{c}$ est contractile (ayant un objet final), donc asphérique, il en est de même de $\mathcal{C}(c;u,v)$, et on gagnera.

On définit φ par $\boxed{\varphi(c',w')=(\tilde{c}',\tilde{w}'),}$ où (c' étant du type τ') $\tilde{c}'$ est le chemin de type $\Delta_+\tau'\Delta_-$ déduit de c' comme image inverse par la dégénérescence $\Delta_+\tau'\Delta_- \longrightarrow \tau'$, i.e. on prolonge c' d'un 'cran Δ_+' $a' \xrightarrow{\mathrm{id}} a'$ à gauche et d'un 'cran Δ_-' $b' \xleftarrow{\mathrm{id}} b'$ à droite, cf. figure.



On peut dire que [page 163]

$$\tilde{c}' = a'_{\Delta_+} \circ c' \circ b'_{\Delta_-},$$

où de façon générale, si $x \in X$ et $\tau \in \mathrm{Ty}$, x_τ désigne le 'chemin constant' de valeur x, type τ. On a d'autre part des morphismes évidents

$$a'_{\Delta_+} \longrightarrow \mathrm{ch}(u), \qquad b'_{\Delta_-} \longrightarrow \mathrm{ch}(v)^{\mathrm{opp}},$$

comme montré [?] par les flèches verticales $(\mathrm{id}_{a'}, u)$ et $(v, \mathrm{id}_{b'})$ dans les deux carrés marqués $*$ du diagramme, flèches qui sont l'une dans $\underline{\mathrm{Ch}}_{\Delta_+}(X)$, l'autre dans $\underline{\mathrm{Ch}}_{\Delta_-}(X)$. Ceci dit, on pose

$$\tilde{w}' : \tilde{c}' \longrightarrow \tilde{c}, \qquad \tilde{w} = \lambda(u) * w' * \mu(v),$$

où

$$\begin{cases} \lambda(u) : & a'_{\Delta_+} & \longrightarrow & \mathrm{ch}(u) \\ \mu(v) : & b'_{\Delta_-} & \longrightarrow & \mathrm{ch}(v)^{\mathrm{opp}} \end{cases}$$

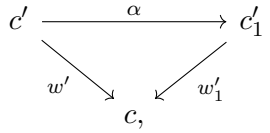
(cf. figure). Le couple p. 102

$$(\tilde{c}', \tilde{w}') \in \mathrm{Ob}\,\underline{\mathrm{Ch}}(X)/\tilde{c}$$

ainsi défini dépend fonctoriellement de (c', w') variable dans $\mathcal{C}(c;u,v)$: Si (c'_1, w'_1) est un deuxième objet de cette catégorie, une flèche du premier dans le second est une flèche

$$c' \xrightarrow{\alpha} c'_1$$

dans $\underline{\mathrm{Ch}}(X; a', b')$, telle qu'on ait commutativité dans

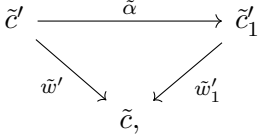


$w' = w'_1\alpha.$

Une telle flèche α définit

$$\tilde{\alpha} = a'_{\Delta_+} * \alpha * b'_{\Delta_-} : \tilde{c}' \longrightarrow \tilde{c}'_1$$

dans $\underline{\mathrm{Ch}}(X; a', b')$, et on vérifie aussitôt que l'on a commutativité dans [page 164]



i.e. $\tilde{w}' = \tilde{w}'_1\tilde{\alpha},$

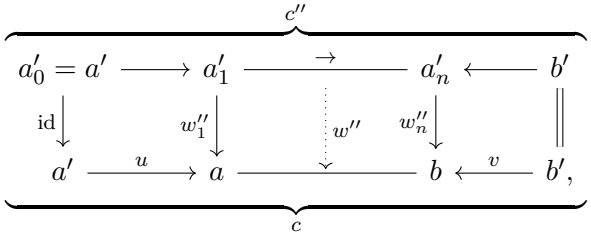
i.e. que $\tilde{\alpha}$ est bien une flèche dans $\underline{\mathrm{Ch}}(X; a', b')/\tilde{c}$. De même, que $\alpha \longmapsto \tilde{\alpha}$ définit une dépendance fonctorielle, i.e. $(\beta\alpha)\tilde{} = \tilde{\beta}\tilde{\alpha}$, est évident sur la définition de $\tilde{\alpha}$. Donc, on a bien défini

$$\mathcal{C}(c; u, v) \xrightarrow{\varphi} \underline{\mathrm{Ch}}(X; a', b')/\tilde{c},$$

définissons maintenant

$$\mathcal{C}(c; u, v) \xleftarrow{\psi} \underline{\mathrm{Ch}}(X; a', b')/\tilde{c}.$$

Soit (c'', w'') un objet de $\underline{\mathrm{Ch}}(X; a', b')/\tilde{c}$,



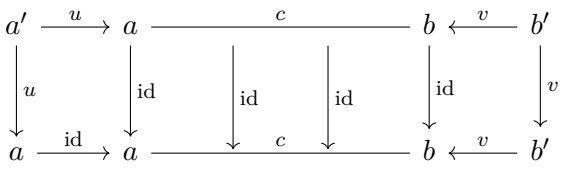
donc p. 103

$$w'' : c'' \longrightarrow \tilde{c}, \qquad \text{flèche dans } \underline{\mathrm{Ch}}(X; a', b'),$$

notons qu'on a une flèche de rétraction évidente

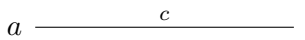
$$\tilde{c} \xrightarrow{\rho_c} c \qquad \text{associée à } \Delta_+\tau\Delta_- \longrightarrow \tau \text{ (rétraction)},$$

induisant u et v sur la source et le but :



$\Big\} \tilde{c} \qquad \in \underline{\mathrm{Ch}}_{\Delta_+\tau\Delta_-}(a', b')$

$\Big\} \delta^*_\tau(c) \qquad \in \underline{\mathrm{Ch}}_{\Delta_+\tau\Delta_-}(a, b)$



$\Big\} c \qquad \in \underline{\mathrm{Ch}}_\tau(a, b).$

Donc on définit un objet [page 165]

$$(c', w') = \psi(c'', w'') \ \in \mathrm{Ob}\,\mathcal{C}(c; u, v)$$

par

$$c' = c'', \qquad w' = \rho_c w'',$$

et c'est fonctoriel en (c'', w'').

Commentary · Exposition [ms. 164]

The lifting category is contracted relative to its endpoints

The long diagram calculation has a simple purpose: to replace the category of all incoming lifts of (u, v) by a slice with a terminal object. It is a proof of smoothness, and no properness assertion is needed to make it work.

Fix $u : a' \to a$, $v : b' \to b$, and a path $c : a \rightsquigarrow b$. Write $A = \underline{\mathrm{Ch}}(X; a', b')$ and $\widetilde{c} = \mathrm{ch}(u)c\mathrm{ch}(v)^o$. Its slice $A/\widetilde{c}$ has terminal object $1_{\widetilde{c}}$. The functor φ pads an incoming lift $w' : c' \to c$ with a constant edge at each end. Its new map to $\widetilde{c}$ has the components $(1_{a'}, u)$ on the left and $(v, 1_{b'})$ on the right. A morphism $\alpha : c' \to c'_1$ is padded by identities, so the equality $w'_1\alpha = w'$ becomes $\widetilde{w}'_1\widetilde{\alpha} = \widetilde{w}'$. Both functoriality and the slice condition follow from the interchange law for concatenation of natural transformations.

To recover a lift from a slice object, there is an arrow $\rho_c : \widetilde{c} \to c$ over the collapse $\Delta_+\tau\Delta_- \to \tau$, with endpoint components u, v. Define $\psi(c'', w'') = (c'', \rho_c w'')$. The identity which makes the argument relative is

$$\rho_c \widetilde{w}' = w' \rho_{c'}.$$

It holds on the central interval by the definition of $\widetilde{w}'$, and on the two added edges by the two commutative endpoint squares. The maps $\rho_{c'}$ are natural because collapsing constant collars commutes with every endpoint-preserving type map. They give a natural transformation $\psi\varphi \Rightarrow 1$ on the lifting category.

On the other side, both $\varphi\psi$ and the identity map naturally to the constant endofunctor with value $1_{\widetilde{c}}$, because the slice has a terminal object. Thus their nerves are homotopic. The two functors are homotopy inverses, and the lifting category is contractible. This proves smoothness of $\underline{\mathrm{Ch}}(X) \to X \times X$ completely. The wrong primes in the first padding diagram and the wrong final vertex in the diagram of ρ_c are printed errors; the corrected endpoint descriptions above specify the maps without ambiguity.

Le composé $\varphi\psi$ étant un endomorphisme de la catégorie contractile $\underline{\mathrm{Ch}}(X; a', b')/\tilde{c}$, est homotope à l'identité. Il faut vérifier qu'il en est de même de $\psi\varphi$, endomorphisme de $\mathcal{C}(c; u, v)$. Ce foncteur associe à (c', w'), $(\tilde{c}', \overline{w'})$, où $\tilde{c}'$ est défini comme plus haut, $\tilde{c}' = a'_{\Delta_+} \circ c' \circ b'_{\Delta_-}$, et où $\overline{w'}$ est le composé

$$\underbrace{\tilde{c}'}_{=\delta'^{*}_{\tau'}(c')} \quad \underbrace{\xrightarrow{\rho_{c'}}}_{\substack{\text{flèche}\\ \text{canonique}}} \quad c' \xrightarrow{w'} c.$$

On a donc

$$(\psi\varphi)(c', w') = (\tilde{c}', \overline{w'}) \xrightarrow{\rho_{c'}} (c', w'),$$

c'est bien un morphisme dans $\mathcal{C}(c; u, v)$, i.e. on a commutativité dans

$$\begin{array}{ccc} \tilde{c}' & \xrightarrow{\quad\rho_{c'}\quad} & c' \\ & {\scriptstyle \overline{w'}}\searrow \quad \swarrow{\scriptstyle w'} & \\ & c & \end{array} \qquad \text{(flèche de } \underline{\mathrm{Ch}}(X; a', b')\text{)}.$$

Ainsi, le foncteur $\psi\varphi$ est bien homotope au foncteur identique. On a gagné ! p. 104

Passons à 1′, 2′) pour $\mathcal{C}_{\mathbf{N}^o}$. L'assertion 1′) se démontre exactement de la même manière [page 166] que 1) pour $\underline{\mathrm{Ch}}(X)$, en regardant $\mathcal{C}_{\mathbf{N}^o}$ comme fibrée sur $\mathbf{N}^o$ (qui a un objet final 0), et en utilisant le diagramme

$$\begin{array}{ccccc} X = (X_{\mathbf{N}^o})_e & \xrightarrow{\quad i' \quad} & X_{\mathbf{N}^o} = X \times \mathbf{N}^o & \xrightarrow{\qquad j' \qquad} & \mathcal{C}_{\mathbf{N}^o} \\ & & & \searrow \quad \swarrow & \\ & & & \mathbf{N}^o\ , & \end{array}$$

où j' est (cartésien et) une Hot-équivalence sur chaque fibre d'après le lemme, donc une Hot-équivalence globalement, et i' est une Hot-équivalence car $\mathbf{N}^o$ est asphérique (ayant un objet final). Donc $j'i'$ est asphérique, q.e.d.

Il reste à prouver que

$$\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$$

est propre et lisse. Ici encore, c'est exactement le même argument que tantôt, je présume. Il y [page 167] a cependant une complication technique. Nous avions pris, pour définir

$$\underline{t} : \mathbf{N}^o \longrightarrow \underline{\mathrm{Ty}}, \qquad \underline{t} = (t_n)_n,$$

le choix

$$t_n = (\Delta_+\Delta_-)^{2n},$$

ou si on préfère

$$t_n = (\Delta_+\Delta_-)^n,$$

en tous cas les t_n sont à direction initiale $+$, terminale $-$, et ce sont des types réduits. Si on prend $\Delta_+ t_n \Delta_-$, on trouve un type non réduit, qui n'est pas parmi les t_n. Il ne servirait à rien de changer la définition de t_n en $t_n = (\Delta_-\Delta_+)^{\text{quelque chose}}$, car cette fois $\Delta_+ t_n \Delta_-$ sera bien réduit, mais n'aura pas les bons signes initial et terminal. Donc pour définir $\tilde{c}$ en termes de c, comme objet de $\mathcal{C}_{\mathbf{N}^o}$, il faudra, si c est de type $\tau = t_n$, prendre $\tilde{c}$ de type $\Delta_+\Delta_-\tau\Delta_+\Delta_- = t_{n+1}$ (en prenant $t_{n+1} = (\Delta_+\Delta_-)^{2n}$), et de même pour définir $\tilde{c}'$ en termes de c'. Les définitions pertinentes du passage de (c', w') à $(\tilde{c}', \tilde{w}')$ sont décrites dans la figure :

Commentary · Development (commentators') [ms. 166]

Recovering the path-space theorem without properness

The smoothness argument supplies enough control once we also know that transport along an arrow of $X \times X$ is invertible up to homotopy. Full path categories are particularly convenient for this second step, because their concatenation is a genuine functor.

Transport and homotopy fibres for all path types. [ours] Let $E = \underline{\mathrm{Ch}}(X)$ with endpoint functor $p : E \to B = X \times X$. For every (a, b), the natural square with upper left corner $\underline{\mathrm{Ch}}(X; a, b)$ and lower left corner $\{(a, b)\}$ is homotopy Cartesian. In particular,

$$B\underline{\mathrm{Ch}}(X; a, b) \simeq \mathrm{hofib}_{(a,b)}(BX \xrightarrow{\Delta} BX \times BX),$$

where $B(-)$ denotes the classifying space, so the right side is the ordinary path space from a to b up to weak equivalence.

Proof. For $h = (u, v) : (a', b') \to (a, b)$ define

$$S_h : \underline{\mathrm{Ch}}(X; a, b) \longrightarrow \underline{\mathrm{Ch}}(X; a', b'), \qquad c \longmapsto \mathrm{ch}(u)c\mathrm{ch}(v)^o.$$

The opposite paths define a candidate inverse. Its composites insert the two backtracking paths associated to u and v. For an arrow $u : x \to y$, cancellation is provided by the explicit diagrams

$$\mathrm{empty}_x \longleftarrow x_{\Delta_+\Delta_-} \longrightarrow (x \xrightarrow{u} y \xleftarrow{u} x),$$

$$(y \xleftarrow{u} x \xrightarrow{u} y) \longrightarrow y_{\Delta_-\Delta_+} \longrightarrow \mathrm{empty}_y.$$

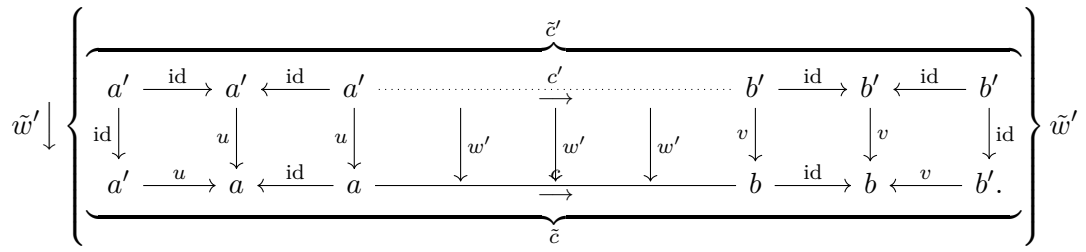
The first rightward arrow has components $(1_x, u, 1_x)$; the second display's first arrow has components $(1_y, u, 1_y)$. The remaining arrows collapse constant intervals. They are morphisms with fixed endpoints. Whiskering with an arbitrary path makes these diagrams natural in that path. Cancelling each of the two inserted backtracks consequently gives homotopies from both composites to the identity. Hence S_h is a homotopy equivalence.

Smoothness identifies the inclusion $i_z : E_z \to z\backslash E$ as a weak equivalence for every $z \in B$. The arrows $\rho_c : S_h(c) \to c$ over h give a natural transformation

$$i_{(a',b')} S_h \implies h^* i_{(a,b)}$$

in $(a', b')\backslash E$. The three known weak equivalences in this square force $h^* : (a, b)\backslash E \to (a', b')\backslash E$ to be a weak equivalence. Quillen's Theorem B now identifies E_z with the homotopy fibre of p. Finally the constant-path section $X \to E$ is a weak equivalence over $X \times X$, so it identifies that homotopy fibre with the homotopy fibre of the diagonal. The latter is equivalent to the space of paths from a to b: in the standard homotopy-fibre model, join the two paths from the chosen intermediate point, reversing the first. □

The use of Quillen B is classical (Quillen, 1973, §1); the explicit application above repairs the proof in these pages. We have checked the categorical cocylinder treatments of Evrard (1975) and Shoikhet (2016), but did not locate this precise correction of Grothendieck's total path category there. No claim of priority is attached to the argument.



Ainsi

$$\begin{cases} \tilde{c} &= \xi_u \circ c \circ \eta_v & \text{où} & \left\{ \begin{array}{lccccc} \xi_u : & a' & \xrightarrow{u} & a & \xleftarrow{\mathrm{id}} & a \\ \eta_v : & b & \xrightarrow{\mathrm{id}} & b & \xleftarrow{v} & b' \end{array} \right\} \text{ type } \Delta_+\Delta_- \\ \tilde{c}' &= a'_{\Delta_+\Delta_-} \circ c' \circ b'_{\Delta_+\Delta_-} & & & & \\ \tilde{w}' &= \lambda(u) * w * \mu(v) & \text{avec} & \left\{ \begin{array}{lccc} \lambda(u) : & a'_{\Delta_+\Delta_-} & \longrightarrow & \xi_u \\ \lambda(v) : & b'_{\Delta_+\Delta_-} & \longrightarrow & \eta_v \end{array} \right\} & \text{comme dans } \texttt{[la]} \text{ figure.} \end{cases}$$

On trouve ainsi, mutatis mutandis,

$$\mathcal{C}_{\mathbf{N}^o}(c; u, v) \xrightarrow{\varphi} \mathcal{C}_{\mathbf{N}^o}(a', b')/\tilde{c}, \qquad \varphi(c', w') = (\tilde{c}', \tilde{w}'),$$

et on définit le foncteur

$$\mathcal{C}_{\mathbf{N}^o}(c; u, v) \xleftarrow{\psi} \mathcal{C}_{\mathbf{N}^o}(a', b')/\tilde{c}$$

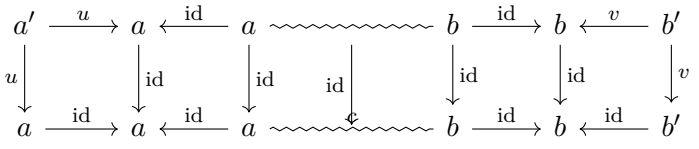
ainsi : à (c'', w'') dans $\mathcal{C}_{\mathbf{N}^o}(a', b')/\tilde{c}$, donc un chemin c'' de type $t_n = (\Delta_+\Delta_-)^{2n}$ de a' à b' et un homomorphisme $w'' : c'' \longrightarrow \tilde{c}$ à extrémités fixes, on associe (c', w') dans $\mathcal{C}_{\mathbf{N}^o}(c; u, v)$, avec

$$c' = c'', \qquad w' = \rho_c w'',$$

où

$$\rho_c : \tilde{c} \longrightarrow c$$

est défini de façon similaire que tantôt, par



$\tilde{c}$, type t_{n+1}.

$\varphi^*_{n,n+1}(c)$.

[page 169]

On doit seulement vérifier que l'endofoncteur $\psi\varphi$ de $\mathcal{C}_{\mathbf{N}^o}(c; u, v)$ est homotope à l'identité. Mais ce foncteur transforme (c', w') en $(\tilde{c}' = a'_{\Delta_+\Delta_-} c' b'_{\Delta_+\Delta_-}, \overline{w'})$, où $\overline{w'}$ est encore le composé

$$\tilde{c}' \xrightarrow{\text{can.}} c' \xrightarrow{w'} c.$$

p. 106

On a donc

$$\psi\varphi \longrightarrow \mathrm{id},$$

$\psi\varphi$ est homotope au foncteur identique.

Cela prouve la lissité de $\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$. Pour la propreté, il faut prouver la lissité de le cas de X^o, et du système projectif $\underline{t}^{\mathrm{opp}} \simeq (t_n^{\mathrm{opp}})_n$. Mais on a $\underline{t}^{\mathrm{opp}} = \underline{t}$, et il suffit d'appliquer ce qu'on a déjà prouvé au cas de X^o, pour voir que $\mathcal{C}_{\mathbf{N}^o} \longrightarrow X \times X$ est propre.

Il ne reste plus que le lemme à prouver. Ça doit être essentiellement trivial, et je renonce à rédiger une démonstration.

[page 170]

Commentary · Development (commentators') [ms. 169]

The sequential intermediary and the repaired comparison

Grothendieck's intermediary remembers a finite stage as part of an object. That memory explains both its value for comparison and the limitation of the padding argument just attempted. Put $D = \Delta_+\Delta_-$, $t_n = D^{2n}$, and denote the specified symmetric-collapse transition by $\delta_{m,n} : t_m \to t_n$. An arrow of $\mathcal{C}_{\mathbf{N}^o}$ over $m \to n$ has exactly this underlying type map. Padding both ends of an arbitrary such arrow produces

$$1_D * \delta_{m,n} * 1_D : t_{m+1} \longrightarrow t_{n+1},$$

which is generally different from $\delta_{m+1,n+1}$. The former retains the outermost blocks and collapses blocks immediately inside them; the latter collapses the outermost blocks. Thus the formula on p. 168 need not even define the promised object of the slice of $\mathcal{C}_{\mathbf{N}^o}$. This is a restriction of the chosen indexing arrows, independent of the corrected exponents and primes.

Smoothness of the sequential intermediary. [ours] Nevertheless $\mathcal{C}_{\mathbf{N}^o} \to X \times X$ is smooth.

Proof. Fix a target path c of stage n and an incoming base arrow $h : (a', b') \to (a, b)$. Its lift category is fibred over the tail $\{m \geq n\}^o$. The fibre at m is the incoming-lift category of h into $\delta^*_{m,n}c$ for the fixed-type endpoint functor

$$\mathrm{Fun}(t_m, X) \longrightarrow X \times X.$$

For $m > 0$ this functor is a Grothendieck fibration: both endpoints of t_m are sources, and a Cartesian lift is obtained by changing their objects and composing the first and last arrows with the specified endpoint maps. The corresponding lift category has that Cartesian lift as a terminal object. The only exception is $m = 0$, possible when $n = 0$. Its fibre is empty unless $a' = b'$ and the two components of h coincide; when these equalities hold it is a singleton. If it is empty, omit stage 0; otherwise retain it. In either case every fibre over the remaining tail is contractible, and that tail has a terminal object. The fibrewise comparison theorem for Grothendieck fibrations, applied to the projection to this tail, shows that the total lift category is weakly contractible. This is the smoothness criterion. □

This argument proves only smoothness. Properness is excluded by the empty-path example above. Likewise the word "asphérique" at the end of the proof of (1′) on p. 166 is stronger than what was proved: for a positive-stage object c, the comma category of $X \to \mathcal{C}_{\mathbf{N}^o}$ over c is empty, since there is no base arrow $0 \to n$ for $n > 0$. The section is a weak equivalence, as required.

Comparison on every fixed-endpoint fibre. [ours] The canonical span

$$\underline{\mathrm{Ch}}(X; a, b) \longleftarrow \mathcal{C}_{\mathbf{N}^o}(a, b) \longrightarrow \underline{\mathrm{Ch}}_\infty(X; a, b)$$

consists of weak equivalences.

Proof. Write $E_1 = \underline{\mathrm{Ch}}(X)$, $E_0 = \mathcal{C}_{\mathbf{N}^o}$, $E_2 = \underline{\mathrm{Ch}}_\infty(X)$, all over $B = X \times X$, and $\pi_i : E_0 \to E_i$. The constant-path lemma proves both π_i are weak equivalences on total categories. The filtered Grothendieck-to-colimit comparison, established in the commentary at manuscript pages 108–109, proves $\pi_2 : E_{0,z} \to E_{2,z}$ is a weak equivalence on each strict fibre: apply that comparison to the sequence of fixed-endpoint categories itself.

By the endpoint theorem for E_2, its strict fibre is its homotopy fibre over z. The natural diagram

$$\begin{array}{ccc} E_{0,z} & \xrightarrow{\quad\pi_2\quad} & E_{2,z} \\ \downarrow & & \downarrow\sim \\ \mathrm{hofib}_z(BE_0 \to BB) & \xrightarrow{\sim} & \mathrm{hofib}_z(BE_2 \to BB) \end{array}$$

is understood after applying B to the top row. Its lower arrow is a weak equivalence because the maps on total spaces are weak equivalences over the same base. The left vertical map is therefore a weak equivalence. The preceding block proves the same homotopy-fibre description for E_1. Repeating the square with π_1 shows that $E_{0,z} \to E_{1,z}$ is a weak equivalence. □

The conclusion of the section is thus recovered, although its claimed properness and its particular sequential padding construction cannot be retained. The repair uses based homotopy fibres, never a conservativity assertion about unbased finite tests. The literature check is the one stated in the preceding development block; these are proofs supplied here for the specified models.

Commentary · Development (commentators') [ms. 169]

Even translation acts trivially after localization

The comparison now also settles the question about T^2 left on pp. 149–150. Its resolution uses a visible change of finite support, rather than equality after restriction to finite test objects.

Translation invariance. [ours] The even translation of an eventually constant alternating path induces the identity in Hot, both on $\underline{\mathrm{Ch}}_\infty(X)$ and on each fixed-endpoint category.

Proof. Keep $D = \Delta_+\Delta_-$ and $t_n = D^{2n}$. Define R on the sequential Grothendieck category by sending a path of stage n to stage $n+1$, adding two constant D-blocks at its right end. The asymmetric collapse $q_n : t_{n+1} \to t_n$ defining this operation commutes with the symmetric transitions:

$$q_n \delta_{m+1,n+1} = \delta_{m,n} q_m \qquad (m \geq n).$$

Both sides collapse the same number of left and right blocks. Hence R is a functor. On the eventual bi-infinite realization, with the centred indexing of these pages this operation is $U = T^{-2}$, and $\pi_2 R = U\pi_2$.

The maps q_n are permitted in $\underline{\mathrm{Ty}}$. They give a natural transformation $\pi_1 R \Rightarrow \pi_1$ into the full path category, whose component is the Cartesian arrow from the padded path to the original path. By the comparison just proved, $k = \pi_1\pi_2^{-1}$ is an isomorphism in Hot. Therefore

$$kU\pi_2 = \pi_1 R = \pi_1 = k\pi_2,$$

and cancellation gives $U = 1$. The same construction fixes the endpoints, so the argument applies to every fibre. Taking inverses gives $T^2 = 1$ as well. □

We did not find this finite-support proof in the checked literature. It is a consequence of the repaired comparison, with no additional general theorem about the restricted Yoneda functor.

Commentary · Development (commentators') [ms. 169]

Why the comparison preserves composition

Grothendieck's original reason for retaining all finite path types was their strictly associative concatenation. The comparison of endpoint fibres should therefore identify that operation with ordinary path concatenation, not merely identify the underlying homotopy types. The constant-path sections determine this compatibility once composable pairs are treated as carefully as individual paths.

Compatibility with path concatenation. [ours] Put $E = \underline{\mathrm{Ch}}(X)$ and $K = BX$. The equivalences

$$u_{a,b} : B\underline{\mathrm{Ch}}(X; a, b) \longrightarrow P_{a,b}K \quad \text{in Hot},$$

obtained from the constant-path diagonal factorization, can be chosen canonically in the localized slice category and satisfy

$$u_{a,c}\, B(*) = *_{\mathrm{top}}(u_{a,b} \times u_{b,c}) \quad \text{in Hot}.$$

Here $P_{a,b}K$ is the ordinary space of paths from a to b, and the products have the indicated endpoint fibres as their domain. In particular the equivalence on loops respects the usual multiplication in the homotopy category.

Proof. Let $D = E \times_X E$, using the target of the first path and the source of the second. Its projection to $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$ is the fibred Grothendieck construction with fibre $\mathrm{Fun}(\tau * \upsilon, X)$ at (τ, υ): matching the two endpoint values is exactly the condition for gluing the diagrams on the two intervals. The constant-diagram functor from X to this fibre is a categorical homotopy equivalence, by the truncation homotopies proved at manuscript page 160. These functors commute with the type maps. The same fibrewise comparison used there, and the terminal object (e, e) of $\underline{\mathrm{Ty}} \times \underline{\mathrm{Ty}}$, show that the constant-pair section $j : X \to D$ is a weak equivalence.

We also need the relative assertion for D. Its three-endpoint functor $r : D \to X^3$ factors as

$$D \longrightarrow E \times X \longrightarrow X^3, \qquad (c, d) \longmapsto (c, \beta d) \longmapsto (\sigma c, \beta c, \beta d).$$

The first functor is the base change of $E \to X^2$ along $(c, z) \mapsto (\beta c, z)$; the second is its product with 1_X. Thus r is smooth, by stability under base change and composition (Maltsiniotis, 2005, Proposition 4.7). For a base arrow $(u, v, w) : (a', b', c') \to (a, b, c)$ the functor on strict fibres is

$$S_{(u,v)} \times S_{(v,w)}.$$

Both factors are homotopy equivalences by the explicit backtrack cancellations at page 166. The two collapse arrows ρ agree on their common middle endpoint, with component v, so together give a natural arrow over (u, v, w). Smoothness identifies each strict fibre with its under-comma category. The natural arrow then shows that every transition between those comma categories is a weak equivalence, exactly as in the proof for E. Quillen's Theorem B applies. Consequently

$$B\big(\underline{\mathrm{Ch}}(X; a, b) \times \underline{\mathrm{Ch}}(X; b, c)\big) \longrightarrow \mathrm{hofib}_{(a,b,c)}(BD \to K^3)$$

is a weak equivalence.

Now use ordinary spaces, with compactly generated mapping spaces. Put $P = K^{[0,1]}$ and $Q = P \times_K P$, and let $s : K \to P$ and $t : K \to Q$ be their constant-path sections. Both are weak equivalences: P and Q are the spaces of maps from the intervals $[0, 1]$ and $[0, 2]$, respectively, and evaluation at the initial point is a deformation retraction onto constants. Their endpoint maps, to K^2 and K^3, are fibrations. Write $i : K \to BE$ for the constant-path section and still write $j : K \to BD$ for its two-path version. In the homotopy categories of spaces over these endpoint bases, define

$$u = s\, i^{-1} : BE \longrightarrow P, \qquad v = t\, j^{-1} : BD \longrightarrow Q.$$

These are the unique comparison isomorphisms whose composites with the constant sections are s and t.

All maps in the remaining argument can be regarded over K^3. For concatenation the target is the pullback of $P \to K^2$ along $(x, y, z) \mapsto (x, z)$; for either projection it is the corresponding pullback along $(x, y, z) \mapsto (x, y)$ or (y, z). These pullbacks are just products with the unused copy of K, so they preserve weak equivalences and the displayed localized comparisons. Both composites comparing categorical concatenation with topological concatenation send j to the same constant path. Since j is invertible in this slice homotopy category, cancellation gives

$$u\, B(*) = *_{\mathrm{top}}\, v.$$

The identical argument with either path projection shows that v commutes with both projections under u.

Take the derived fibre over (a, b, c). The relative assertion for D, the endpoint theorem for E, and the ordinary path fibrations identify these derived fibres with the stated strict path spaces and their products. The two projection identities imply that the induced map of the product fibres is $u_{a,b} \times u_{b,c}$, since a map to a product in Hot is determined by its two projections. The concatenation identity is therefore the required formula. Constants give the corresponding unit identity. □

This argument supplies the compatibility omitted by a bare comparison of homotopy types. It uses Quillen's relative fibre theorem and localization at weak equivalences, not a conservativity or faithfulness assertion about finite tests. We checked the path constructions in Evrard (1975, Section I.1) and the diagonal and cocylinder comparisons in Shoikhet (2016, Sections 1.5 and 2.1, Theorem 2.8); we did not find this particular comparison through the category of composable pairs there. The ingredients are classical, and no priority is claimed.

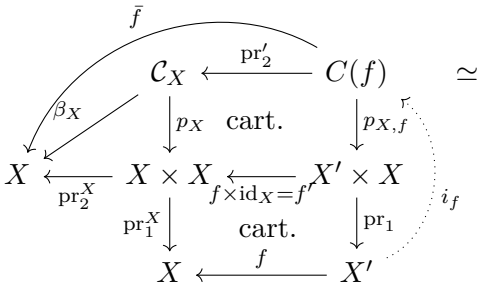
Je vais maintenant montrer 'formellement' que la factorisation 'canonique' d'une flèche de Cat $f : X' \longrightarrow X$ en

$$X' \xrightarrow{i_f} \overline{X'} \xrightarrow{\bar{f}} X, \qquad i_f \text{ Hot-équivalence},$$

avec (on l'espérait) $\bar{f}$ propre et lisse, à partir d'une factorisation similaire pour la flèche diagonale $X \xrightarrow{\mathrm{diag}_X} X \times X$ en

$$\begin{array}{ccc} & X & \\ & \big\downarrow{\scriptstyle i_X} & \text{Hot-équivalence} \\ \mathcal{C}_X & \text{'chemins'} & \\ & \big\downarrow{\scriptstyle p_X} & \\ & X \times X & \end{array}$$

est chose formelle. (Cf. page 139 pour les notations et [la] problématique.) On regarde le diagramme



([40]). Comme p_X est propre et lisse, i.e. 'parfait', son image inverse sur $X' \times X$, savoir $p_{X,f}$, l'est aussi; d'autre part $\mathrm{pr}_2^X \circ f'$ est la projection pr_2 de $X' \times X$ sur X, donc c'est aussi un morphisme parfait (puisque $X' \longrightarrow e$ est parfait, donc il en est de même du composé p. 107 [page 171]

$$(\mathrm{pr}_2^X \underbrace{f'}) \circ \underbrace{p_{X,f}} = \underbrace{\mathrm{pr}_2^X(p_X} \mathrm{pr}_2') = \beta_X \mathrm{pr}_2' \overset{\text{déf}}{=} \bar f \quad).$$

Commentary · Exposition [ms. 170]

The formal diagonal factorization, with its hypothesis retained

The closing argument extracts the reusable part of the path construction. Suppose a category C_X is given with a factorization

$$X \xrightarrow{i_X} C_X \xrightarrow{p_X} X \times X, \qquad p_X i_X = \Delta_X,$$

where i_X is a weak equivalence and p_X is smooth and proper. Put $\sigma_X = \mathrm{pr}_1 p_X$ and $\beta_X = \mathrm{pr}_2 p_X$. Both are smooth and proper, because these classes are closed under composition and base change; a projection is a base change of a map to the terminal category. The identity $\sigma_X i_X = 1_X$ implies that σ_X is a weak equivalence. Its strict fibres are homotopy fibres by the endpoint argument, hence weakly contractible. The lift-category criterion and Quillen A then show that every base change of σ_X is a weak equivalence.

For $f : X' \to X$, form the ordinary pullback $C(f) = X' \times_{X,\sigma_X} C_X$. Its projection $r : C(f) \to X'$ is a weak equivalence. The section i_f induced by i_X satisfies $r i_f = 1_{X'}$, so i_f is a weak equivalence. The map to the other endpoint factors as

$$C(f) \longrightarrow X' \times X \longrightarrow X.$$

The first arrow is the base change of p_X, and the second is a projection. Their composite $\bar f$ is therefore smooth and proper. The equation $\bar f i_f = f$ proves the requested factorization. This is exactly the formal argument of the manuscript, with the universal weak-equivalence step justified.

It applies to $C_X = \underline{\mathrm{Ch}}_\infty(X)$, whose two lifting properties were proved using the two cofinal parity systems. It cannot be applied with the asserted hypothesis to $C_X = \underline{\mathrm{Ch}}(X)$ or $\mathcal{C}_{\mathbf{N}^\circ}$: those endpoint maps fail properness. Their weak equivalence over $X \times X$ does not transfer a strict comma-category lifting condition. The formal result and the comparison theorem therefore coexist without identifying their hypotheses.

Je rappelle que la section i_f de $C(f)$ sur X' est définie à partir de la section i_X de $(\mathcal{C}_X, \sigma_X)$ sur X. C'est une Hot-équivalence, car $\sigma_X = \mathrm{pr}_1^X p_X$ est parfait à fibres asphériques, donc universellement une Hot-équivalence, donc $\mathrm{pr}_1 p_{X,f} : C(f) \longrightarrow X'$ est aussi une Hot-équivalence, et toute section en est donc une aussi.

40. NB $C(f)$ est défini comme l'image inverse sur X' du fibré C_X sur X (par $\sigma_X = \mathrm{pr}_1^X \circ p_X$) par $f : X' \longrightarrow X$.

Commentary · Examples [ms. 171]

A mapping-path fibre for a homomorphism of groups

The factorization records more than the kernel of a functor on components. For a homomorphism $\phi : H \to G$, write $\mathbf{B}H$ and $\mathbf{B}G$ for their one-object groupoids and consider the functor $\mathbf{B}\phi$ between them. The homotopy fibre has a particularly small categorical model, the action groupoid F_ϕ with objects $g \in G$ and arrows

$$h : g \longrightarrow g' \quad \Longleftrightarrow \quad g'\phi(h) = g.$$

If $h : g \to g'$ and $k : g' \to g''$, their composite is kh, since $g''\phi(kh) = g$. All arrows are invertible. This is the over-comma category $(\mathbf{B}\phi)/{*}$, where $*$ is the unique object of $\mathbf{B}G$. Every arrow in the base groupoid acts on it by an isomorphism, so Quillen B identifies BF_ϕ with the homotopy fibre.

Two objects lie in the same component precisely when they are in the same coset $g \operatorname{im}\phi$. An automorphism of any g is an h with $\phi(h) = 1$, hence the isotropy group is $\ker\phi$. Choosing one representative in each coset gives an equivalence of groupoids

$$F_\phi \simeq \coprod_{G/\operatorname{im}\phi} \mathbf{B}(\ker\phi).$$

Thus π_0 detects the failure of surjectivity and the fundamental group of each component detects the failure of injectivity. For multiplication by a nonzero integer m on $\mathbf{Z}$, the fibre has $|m|$ contractible components. For the map $\mathbf{Z} \to 1$, it is a circle up to homotopy. The corresponding fibre of Grothendieck's mapping-path factorization has exactly these homotopy types. In particular it is weakly contractible precisely when ϕ is an isomorphism, which illustrates the triviality criterion at p. 156 without computing any individual categorical path.

Overview · VII-c

Paths, comparison, and the information carried by a model.

These pages bring the long construction of categorical paths to its decisive test. A path should do more than give a word in arrows and their inverses. Its category should have the homotopy type of a space of paths, should vary with both endpoints, and should turn an arbitrary functor into a map whose strict fibres compute homotopy fibres. Grothendieck wants these properties before making any fibrant replacement of the category in which the paths lie. The problem is therefore one of comparison between concrete models, with their actual arrows and their actual endpoint maps (Grothendieck, 1990–1991o, ms. 108–171).

There are three models in play. The category $\underline{\mathrm{Ch}}(X)$ retains a finite interval type as part of a path, and permits changes of type. Concatenation is strictly associative there. A chosen sequence of longer alternating intervals gives categories C_n and a filtered colimit C_∞; identity arrows added at the ends become identifications in this colimit. Between them stands the Grothendieck construction $C_{\mathbf{N}^o}$, which retains the stage and the transition morphisms. Passing the diagram (C_n) through homotopy first gives yet another object, the ind-object H_T. None of these operations should be silently substituted for another. The opening pages, from manuscript page 108, are largely an investigation of what is lost by doing so.

For a filtered diagram the comparison from its Grothendieck construction to its ordinary colimit is a weak equivalence. Nerves commute with filtered colimits, and filtered homotopy colimits of simplicial sets map by weak equivalences to ordinary colimits. This explains one side of the comparison span on manuscript page 108. The other side depends on the indexing category and on the diagram. Grothendieck correctly observes that cofinality in the preorder of objects does not make the original indexing category contractible: a one-object group already distinguishes the two statements. The appropriate general machinery is the homotopy-colimit theorem for the Grothendieck construction and the comma-category criteria for homotopy cofinality (Thomason, 1979; Quillen, 1973).

The proposed independence of the chosen sequence is more delicate than the text acknowledges. An isomorphism of ind-objects in the ordinary homotopy category does not control the compatibility of the homotopies used at successive stages. Our finite-injection example makes this precise within the general hypotheses of manuscript pages 99–101. To a finite set we associate the one-object category of its permutation group. Injections act by extending permutations by the identity. Parallel injections induce naturally isomorphic functors, yet a morphism between two cofinal sequences induces, on the colimit, the non-surjective shift of the finitary permutation group. Thus the general comparison assertion on pages 117–121 is false. This does not invalidate the geometric path comparison proved later by a different argument.

The same distinction clarifies the questions about $\mathrm{Hot} \longrightarrow \mathrm{Hot}_0^\wedge$. In this notation the subscript 0 refers to a chosen category of finite tests; it is not a truncation to homotopy 0-types. Restricting unbased representable functors to finite tests is neither generally faithful nor generally conservative. The locally inner permutation-group example in the commentary on Chapter V, manuscript page 52, already treats conservativity. Here a telescope of degree-two maps of circles supplies an explicit failure of faithfulness: an essential map to $K(\mathbf{Z}, 2)$ is invisible on every finite stage. Its obstruction is calculated in the commentary, rather than attributed vaguely to a failure of coherence. The cellular and representability tools are classical (Hatcher, 2002, §3F and Theorem 4.57). In addition, a category with finitely many arrows need not have a homotopy-finite nerve. Compactness of a test in the homotopy category requires its own hypothesis. These qualifications matter whenever a displayed bijection is used to infer an isomorphism or an equality of maps.

External Hom and the role of the parameter category.

Section 11 returns to the reason that paths entered the theory of derivators. For a relative

category $(\mathcal{M}, \Sigma)$ one can always form the prederivator

$$A \longmapsto \mathcal{M}^A[\Sigma_A^{-1}],$$

where Σ_A is defined objectwise. Grothendieck asks when the morphisms between constant diagrams are represented by a homotopy type, with the expected dependence on the homotopy type of A. This is a substantive condition on a relative category. It does not follow merely from the existence of an ordinary localization, and the existence of a prederivator is not the existence of all the Kan extensions required of a derivator.

The discussion of a categorical two-sphere on pages 132–135 identifies a real problem but gives an incorrect explanation. A fixed finite circle-shaped parameter can already support arbitrarily high degrees when the length of the target path is allowed to increase: concatenate the same family of loops repeatedly. We work this out for the six-object suspension of the four-object circle poset. The same example gives a genuine failure of homotopy invariance by comparing that circle poset with the one-object category $B\mathbf{Z}$. Every functor from a group to a poset is constant; the resulting localization has lost the family that the circle-shaped parameter sees. The example separates the valid concern about parameter categories from the proposed bound on degrees.

There is a second distinction at page 135. The total arrow category $\mathrm{Fun}([1], C)$ has the homotopy type of C, for every C. Evaluation and the constant-diagram functor give the needed adjunction and natural transformation. What fails for an arbitrary category is the claim that its strict fixed-endpoint arrow sets are derived path spaces. Likewise, Thomason fibrancy of the target alone does not turn every ordinary functor category into a derived internal Hom. The commentary supplies a fibrant-poset counterexample and distinguishes a derived mapping-space construction from this incorrect assertion. The separate claim that ordinary functor categories preserve Thomason fibrancy is also false. An explicit equivariant horn-cell construction gives a fibrant category E for which $\mathrm{Fun}(B(\mathbf{Z}/2), E)$ has a nonfibrant retract; the proof uses the fixed-point pushout formulas of May et al. (2017, Lemmas 4.7–4.8). Simplicial localization provides a systematic external Hom for a relative category, while additional model-theoretic hypotheses justify comparisons with categories of ordinary diagrams (Dwyer and Kan, 1980c,a; Groth, 2013; Cisinski, 2009; Lurie, 2009).

What the path comparison establishes.

Sections 12 and 13 concern $\Sigma = \mathrm{Mor}(X)$. Alternating paths of unbounded length are the essential construction. By choosing the parity of their end arrows one obtains finite-stage Grothendieck fibrations or opfibrations over the endpoint category. The categories of lifts pass to filtered unions. This proves smoothness and properness of $\underline{\mathrm{Ch}}_\infty(X) \to X \times X$, although the union need not itself be a Grothendieck fibration or opfibration. The lift-category argument, not the existence of compatible Cartesian arrows at all stages, is what survives the passage to the union. The terminology comes from base-change phenomena: modern asphericity theory makes the categorical tests precise (Maltsiniotis, 2005).

For the category of all finite paths the source proves smoothness by a useful explicit construction. Given endpoint arrows and a path c, insert those arrows at the two ends to form $\widetilde{c}$. The category of lifts is compared with the slice over $\widetilde{c}$, which has a terminal object. Padding, retraction, and their naturality give the contraction. This is a sound argument once the notation in the diagrams is read correctly. Its claimed dual, however, does not prove properness. An empty path admits no outgoing change to a nonempty interval type. Over $X = [1]$, changing only one endpoint therefore gives an empty category of outgoing lifts. The same obstruction applies to $C_{\mathbf{N}^o}$. The assertion that both these endpoint projections are proper must remain false in the audit.

The desired comparison nevertheless survives. For $\underline{\mathrm{Ch}}(X)$, concatenating an endpoint arrow, or its reversed path, defines transport between fibres. The two composites are connected to the identity by the explicit backtracking-cancellation zigzags. Combining these equivalences

with the proved smoothness gives the hypotheses of the dual of Quillen's Theorem B. Hence strict endpoint fibres compute homotopy fibres. The known total and fibrewise equivalences $C_{\mathbf{N}^o} \to \underline{\mathrm{Ch}}_\infty(X)$ transfer the same property to the middle model. Finally the total equivalence $C_{\mathbf{N}^o} \to \underline{\mathrm{Ch}}(X)$ becomes a fibrewise equivalence by comparison of homotopy fibres. This repairs the concluding argument without assigning a false properness property to either total category.

It follows that the two geometric path models have equivalent endpoint fibres. Strict concatenation in the category of finite paths then furnishes the expected composition in Hot, including the loop-space multiplication. The diagonal factorization and the mapping-path construction on pages 170–171 work with the perfect model $\underline{\mathrm{Ch}}_\infty(X)$; their formal proof is expanded in the commentary. Evrard's earlier categorical cocylinder and Shoikhet's refinement give closely related factorization theorems, with precisely stated homotopy-fibration conclusions. They should not be cited as asserting that Grothendieck's particular maps are Thomason fibrations or that the inclusion of constant paths is a Thomason cofibration (Evrard, 1975; Shoikhet, 2016).

The larger programme is continuous with *Pursuing Stacks*, §69: homotopy theory is to retain the diagram categories needed to form and compare fibres, cofibres, and their operations. The letter to Thomason of 2 April 1991 explicitly returns to unbounded categorical paths and to constructions that avoid a preliminary Kan replacement (Grothendieck, 1983, 1991a). Here the useful boundary is precise. The geometric comparison and mapping-path factorization can be proved; the general finite-test and ordinary-internal-Hom assertions require correction. The developments marked [ours] give the explicit counterexamples and repairs used in the commentary. They record arguments given here, with the literature checks stated in their blocks, and make no claim of priority.

Chapter VIII

1-types d'homotopie relatifs : leur intégration ...

p.

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre VIII

1-types d'homotopie relatifs : leur intégration ...

p.

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Intégration des 1-types d'homotopie relatifs sur une catégorie.

p. 1
[page 1]

Les types d'homotopie relatifs sur une (petite) catégorie S sont définis par les *champs* (en catégories) sur $S^\wedge$, qui correspondent essentiellement aux contrafoncteurs $S^o \longrightarrow (\mathrm{Cat})$, ou encore aux catégories X sur S qui sont Cat-fibrées sur S. (Dans $(\mathrm{Cat}/S)_{\mathrm{fib}}$, on inverse les flèches qui sont des Hot-équivalences sur les fibres.) D'autre part, les 1-types d'homotopie relatifs sur S doivent se décrire par des *champs en groupoïdes* sur S. Si on a un type d'homotopie relatif, défini par

$$X \longrightarrow S, \qquad \text{catégorie Cat-fibrée sur } S,$$

ou par

$$\begin{array}{ccc} S^o & \longrightarrow & \mathrm{Cat} \\ s & \longmapsto & X_s\ , \end{array}$$

le 1-type d'homotopie associé est défini par

$$s \longmapsto \Pi(X_s), \qquad S^o \longrightarrow (\mathrm{Grp}), \qquad \text{catégorie des groupoïdes,}$$

où $\Pi(X_s)$ est le groupoïde fondamental de X_s. On note la catégorie fibrée correspondante sur S par $\boxed{\Pi(X/S)}$. Il est vrai que si

$$X \longrightarrow X'$$

est un homomorphisme de catégories fibrées sur S qui induise une Hot-équivalence sur les fibres, elle induit une équivalence fibrée

$$\Pi(X/S) \longrightarrow \Pi(X'/S),$$

mais non un isomorphisme. Il y aurait donc du travail à faire pour définir un foncteur

$$\underbrace{\mathrm{Hot}(S)}_{\substack{\text{catégorie des}\\ \text{types d'homotopie}\\ \text{relatifs sur } S}} \xrightarrow{\xi \longmapsto \Pi(\xi/S)} \underbrace{\mathrm{Grp}(S)}_{\substack{\text{catégorie des}\\ \text{groupoïdes}\\ \text{fibrés sur } S\\ \text{(ou catégories}\\ \text{des champs en}\\ \text{groupoïdes}\\ \text{sur } S^\wedge)}} .$$

Mais ce n'est pas pour l'instant mon propos. Plutôt, de donner une formule de transitivité en homologie non commutative, analogue à $R(gf)_* \simeq Rg_*Rf_*$ en cohomologie.

Commentary · Exposition [ms. 1]

Why the target is a 2-category

Grothendieck wants integration to retain the automorphisms of a homotopy type, and not merely its connected components. A groupoid is exactly the amount of structure needed for a homotopy 1-type. Its objects represent points, its arrows represent homotopy classes of paths, and the nerve has no homotopy groups above degree one. For a category C, the groupoid ΠC is its localization at *all* arrows. In particular, it is generally very different from the maximal subgroupoid of C.

The relative version must remember transport as well. We write $\mathsf{Gpd}(S)$ for the 2-category of categories fibred in groupoids over S, functors over S, and natural transformations over S. Every arrow of a category fibred in groupoids is cartesian; consequently every functor over the base between such categories preserves cartesian arrows, and every 2-arrow is invertible. A choice of cleavage gives a pseudofunctor $S^{\mathrm{op}} \to \mathsf{Gpd}$, including the isomorphisms comparing successive inverse images. This is the language of Grothendieck (1971b, Exposé VI, §§6–8). No topology imposing additional descent

has been specified here: “champs” is being used through the equivalence with fibred categories on the presheaf topos.

Taking fundamental groupoids fibre by fibre is compatible with these comparison isomorphisms. It therefore gives a relative truncation, rather than a mere collection of unrelated groupoids. A weak equivalence of fibres becomes an equivalence of groupoids, which explains Grothendieck’s hesitation about a functor to an ordinary category of actual groupoids: localization requires its inverted arrows to become *isomorphisms*. One can either pass to the homotopy category of $\mathsf{Gpd}(S)$, whose morphisms are isomorphism classes of cartesian functors, or retain the entire 2-category. In modern language the underlying operation is the pointwise truncation

$$\mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S}) \longrightarrow \mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S}_{\leq 1}).$$

Here $\mathcal{S}$ denotes spaces and $\mathcal{S}_{\leq 1}$ their full subcategory of 1-types. The composition law for integration will be an equivalence with coherent comparison isomorphisms, precisely the form in which it can be useful again under a further change of base.

The terminology continues the programme of Grothendieck (1983, §69, especially printed pp. 3–5): integration is the left adjoint to change of the indexing category. The present calculation isolates a particularly concrete part of that programme, where ordinary categories and their groupoid completions suffice to construct the adjoint.

p. 2
[page 2]

Si

$$f : S \longrightarrow T$$

est un foncteur, on va définir un foncteur

$$f_! : \underline{\mathrm{Grp}}(S) \longrightarrow \underline{\mathrm{Grp}}(T)$$

(image directe de 1-types d'homotopie relatifs). Mais notons d'abord que si

$$p : X \longrightarrow S$$

est une S-catégorie *quelconque* (pas nécessairement Cat-fibrée), on lui associe néanmoins un type d'homotopie relatif, via la catégorie fibrée $\tilde{X}^{(S)}$, ou $\tilde{X}$, ayant comme fibres les $s\backslash X$ (en notant que $s' \longrightarrow s$ dans S définit bien $s\backslash X \longrightarrow s'\backslash X$) ([1]). On a un S-foncteur canonique

$$X \xrightarrow{i} \tilde{X}, \qquad x \longmapsto (x, p(x), \underbrace{p(x)}_{=\,`s'} \xrightarrow{\mathrm{id}} p(x)).$$

C'est une S-Hot-équivalence si X était déjà fibrée. En fait, sur chaque fibre c'est un homotopisme car on définit alors $\tilde{X}_{s_0} \xrightarrow{j} X_{s_0}$ de façon évidente, de façon que

$$X_{s_0} \underset{j}{\overset{i}{\rightleftarrows}} \tilde{X}_{s_0}, \qquad ji \simeq \mathrm{id}, \qquad ij \underbrace{\xrightarrow{\text{homotopie simple}}}_{\substack{\text{NB cette flèche}\\ \text{n'est pas [un]}\\ \text{isomorphisme}\\ \text{en général.}}} \mathrm{id}.$$

Revenons alors à $f : S \longrightarrow T$, et considérons un groupoïde fibré sur S,

$$G \longrightarrow S,$$

ce n'est pas une catégorie fibrée sur T, mais on peut lui associer

$$t \longmapsto t\backslash G, \qquad T^o \longrightarrow (\mathrm{Cat}),$$

d'où une catégorie fibrée sur T, dont on prend le Π relatif. Mais il est plus simple de prendre directement

$$t \longmapsto \Pi(t\backslash G),$$

qui définit une catégorie fibrée en groupoïdes sur T.

Commentary · Exposition [ms. 2]

The comma category computes integration

The comma construction makes the variance visible. For $p : X \to S$, an object of $s\backslash_S X$ is a pair $(x, \alpha : s \to p(x))$; an arrow to (x', α') is an arrow $v : x \to x'$ with $p(v)\alpha = \alpha'$. Precomposing α gives the contravariant dependence on s. Thus the displayed formula is meaningful even when p has no cartesian lifts.

When p is fibred, those lifts explain the comparison with the actual fibre. Choose a normalized cleavage and send (x, α) to $\alpha^* x \in X_s$. If v is a comma arrow, its composite with the chosen cartesian arrow $\alpha^* x \to x$ factors uniquely through $\alpha'^* x' \to x'$. This defines a functor $j_s : s\backslash_S X \to X_s$. Its composite with the inclusion i_s is the identity, and the cartesian arrows give $i_s j_s \Rightarrow \mathrm{id}$. Applying the nerve yields a homotopy equivalence. The last natural transformation need not be an isomorphism, exactly as the manuscript warns; after applying Π, it becomes one. This also proves the asserted equivalence $H \simeq \Pi(H/S)$ when H is fibred in groupoids.

1. Ayant $\tilde{X}$, on définit $\Pi(X/S) \overset{\text{déf}}{=} \Pi(\tilde{X}/S)$.

For $f : S \to T$ and $G \to S$, the same construction gives

$$(f_!G)_t = \Pi(t\backslash_T G).$$

The subscript on the comma is essential: its objects use arrows $t \to f(s)$ in T. Equivalently, if $F : S^{\mathrm{op}} \to \mathsf{Gpd}$ describes G, then

$$(f_!F)(t) \simeq \tau_{\leq 1} \operatorname*{colim}_{(s,\alpha)\in(t\backslash f)^{\mathrm{op}}} F(s),$$

where the colimit on the right is the homotopy colimit in spaces. The total category is the concrete way to form this colimit before truncation. In particular, the $s\backslash G$ printed in the unit on manuscript page 4 must be read as $f(s)\backslash_T G$; this is a notational slip in the edition, not an error introduced by the transcription.

For orientation, let $S = [0 \to 1]$, and let F be the diagram $F(1) \to F(0)$. Integration over S gives $F(0)$ up to equivalence, whereas its groupoid of sections is $F(1)$. Indeed, the construction above sends an object over 1 to its inverse image over 0, and its cartesian arrow becomes invertible in ΠG. Sections are instead determined by their value over 1. The two endpoints already separate homology from cohomology.

Ainsi on trouve bien un foncteur (en fait, même un 2-foncteur de 2-catégories) p. 3 [page 3]

$$f_! : \underline{\mathrm{Grp}}(S) \longrightarrow \underline{\mathrm{Grp}}(T).$$

Si

$$u : G \longrightarrow G'$$

est un homomorphisme de groupoïdes fibrés sur S, qui induise des Cat-équivalences sur les fibres, alors

$$f_!(u) : f_!(G) \longrightarrow f_!(G')$$

est aussi une Cat-équivalence sur les fibres. (Cela est formel, à partir du fait que $f_!$ est un 2-foncteur, compte tenu que si on a une équivalence sur les fibres, alors il y a un quasi-inverse – c'est une histoire générale de catégories fibrées …)

Soit maintenant

$$p : X \longrightarrow S,$$

et prenons le cas $G = \Pi(X/S)$. Je dis qu'on a alors une T-équivalence

$$(*) \qquad \boxed{f_!(\Pi(X/S)) \simeq \Pi(X/T),}$$

en d'autres termes, pour connaître $\Pi(X/T)$ (à T-équivalence près), il suffit de connaître $\Pi(X/S)$, et de lui appliquer le foncteur $f_!$ d'intégration des 1-types d'homotopie relatifs, pour $f : S \longrightarrow T$.

Notons que l'on a trivialement

$$\Pi(X/S) = p_!(e_X),$$

où e_X désigne le X-groupoïde fibré trivial (toutes les fibres ponctuelles). Donc la formule précédente s'écrit comme une formule de transitivité

$$f_!(p_!(\Gamma)) \simeq (fp)_!(\Gamma),$$

où Γ est le groupoïde fibré trivial sur X. Cela suggère de généraliser la formule $(*)$ en [page 4]

$$\boxed{(fp)_!(\Gamma) \simeq f_!(p_!(\Gamma))}$$

pour *tout* groupoïde fibré sur X, dans une situation

$$X \xrightarrow{p} S \xrightarrow{f} T.$$

Ce sera formel, si on établit une propriété de 2-adjonction pour le foncteur $f_!$:

Théorème. *Soit $f : S \longrightarrow T$ une flèche dans* (Cat)*, G un champ en groupoïdes sur S, H un champ en groupoïdes sur T. On a alors une équivalence de catégories canonique* p. 4

$$\boxed{\underline{\mathrm{Hom}}_{\underline{\mathrm{Grp}}(S)}(G, f^*H) \simeq \underline{\mathrm{Hom}}_{\underline{\mathrm{Grp}}(T)}(f_!G, H).}$$

Les flèches d'adjonction

$$\begin{cases} f_!f^*H \longrightarrow H & H \in \mathrm{Ob}\,\underline{\mathrm{Grp}}(T) \\ f^*f_!G \longleftarrow G & G \in \mathrm{Ob}\,\underline{\mathrm{Grp}}(S) \end{cases}$$

devraient se définir sans problème. Pour la deuxième, il faut définir pour $s \in S$

$$G_s \longrightarrow (f^*(f_!(G)))_s = (f_!(G))_{f(s)} \overset{\text{déf}}{=} \Pi(s\backslash G),$$

or on a

$$G_s \longrightarrow s\backslash G,$$

d'où

$$\underbrace{\Pi(G_s)}_{\substack{= G_s \\ \text{car } G_s \\ \text{est un} \\ \text{groupoïde}}} \longrightarrow \Pi(s\backslash G),$$

O.K. C'est fonctoriel en s …

Pour la flèche $f_! f^* H \longrightarrow H$ sur T, il faut trouver pour tout $t \in \mathrm{Ob}\, T$, et fonctoriellement en t, [page 5]

$$\underbrace{(f_! f^* H)_t}_{\overset{\text{déf}}{=} \Pi(t\backslash f^*(H))} \longrightarrow \underbrace{H_t}_{= \Pi(H_t)},$$

il suffirait donc de définir

$$t\backslash f^*(H) \longrightarrow H_t,$$

mais on a le diagramme cartésien

$$\begin{array}{ccc} f^*H = H \times_S T & \longrightarrow & H \\ \downarrow & & \downarrow \\ S & \xrightarrow{\quad f \quad} & T \ni t \end{array}$$

`[plutôt` $H \times_T S$`]` induisant

$$f^* H \longrightarrow H,$$

d'où p. 5

$$t\backslash f^* H \longrightarrow t\backslash H \qquad \text{(au lieu de s'envoyer dans } H_t),$$

mais d'autre part, comme H est fibrée sur T, donc lisse sur T, l'inclusion

$$H_t \longrightarrow t\backslash H$$

est une Hot-équivalence. Ou disons plutôt : $f^* H \longrightarrow H$ est un morphisme de catégories sur T, induisant

$$\underbrace{\Pi(f^* H/T)}_{\overset{\text{déf}}{=} f_!(f^* H)} \longrightarrow \Pi(H/T),$$

or le T-foncteur canonique [page 6]

$$H \longrightarrow \Pi(H/T) \qquad (= \text{catégorie sur } T \text{ définie par } t \longmapsto \Pi(t\backslash H))$$

est une équivalence, parce que H est un fibré en groupoïdes sur T. Donc on peut trouver un T-quasi-inverse

$$\Pi(H/T) \longrightarrow H$$

et en composant on trouve

$$f_!(f^* H) \longrightarrow \Pi(H/T) \longrightarrow H$$

la flèche cherchée.

Ayant les 2-flèches d'adjonction, je m'arrête là dans les vérifications !

Commentary · Exposition [ms. 6]

Completing the adjunction and its transitivity

The proposed unit and counit are the right ones. To finish their verification without making incompatible choices of quasi-inverses, it is useful to prove a slightly more general universal property. For any category $X \to T$, put $R_T X = \Pi(X/T)$. If $H \to T$ is fibred in groupoids, restriction along $X \to R_T X$ induces an equivalence

$$\mathrm{Hom}_T(R_T X, H) \simeq \mathrm{Fun}_T(X, H). \tag{VIII.1}$$

Both sides are groupoids of functors and transformations over T.

Here is a direct construction of the inverse. Given $A : X \to H$ over T, choose cartesian lifts in H and send $(x, \alpha : t \to p(x))$ to $\alpha^* A(x)$. For a comma arrow v, the cartesian factorization of $A(v)$ gives an arrow between these two objects in H_t. All such arrows are invertible, so the functor extends uniquely across the localization $t\backslash_T X \to \Pi(t\backslash_T X)$. The comparison isomorphisms of the cleavage supply its dependence on t; their cocycle equation follows from the uniqueness of cartesian factorization. A transformation $A \Rightarrow B$ extends by applying α^* to its component at x. Thus this construction is a functor on the Hom-groupoids, not just a bijection between classes of objects.

Restricting the extension to $(x, \mathrm{id}_{p(x)})$ recovers A. Conversely, an over-T functor $B : R_T X \to H$ sends the cartesian arrow from (x, α) to $(x, \mathrm{id}_{p(x)})$ to a cartesian arrow in H. Cartesian factorization therefore gives a natural isomorphism between $B(x, \alpha)$ and $\alpha^* B(x, \mathrm{id})$. This identifies B with the extension of its restriction, naturally also in transformations. It proves (VIII.1), including all the coherence needed for an adjunction.

Apply (VIII.1) to the total category of G viewed over T. An over-T functor $G \to H$ is exactly an over-S functor $G \to f^* H$. We obtain the claimed equivalence of Hom-groupoids. The unit sends $g \in G_s$ to the class of $(g, \mathrm{id}_{f(s)})$. The counit sends the class of $((s, h), \alpha : t \to f(s))$ to $\alpha^* h$. The first triangle evaluates the latter formula at an identity arrow. The second uses the canonical cartesian arrow just described, so is the natural isomorphism between extension after restriction and the original functor. Their compatibility is the uniqueness assertion in the same cartesian factorization. Choosing different cleavages changes these formulas by a unique compatible comparison; it does not change the biadjunction.

Transitivity now follows at the same level of precision. For $X \xrightarrow{p} S \xrightarrow{f} T$,

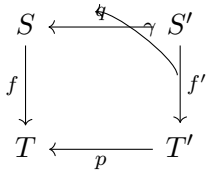
$$\begin{aligned} \mathrm{Hom}_T(f_! p_! \Gamma, H) &\simeq \mathrm{Hom}_S(p_! \Gamma, f^* H) \\ &\simeq \mathrm{Hom}_X(\Gamma, p^* f^* H) \\ &\simeq \mathrm{Hom}_T((fp)_! \Gamma, H). \end{aligned}$$

The bicategorical Yoneda argument identifies the two representing objects. For three composable maps, both parenthesizations represent the same composite restriction; the representing equivalences consequently satisfy the associativity constraint. Taking $\Gamma = e_X$ gives $p_! e_X = \Pi(X/S)$ and the first boxed transitivity formula. What the source leaves unchecked is therefore a coherence verification, not a missing hypothesis in the theorem.

p. 6

2 Questions de changement de base

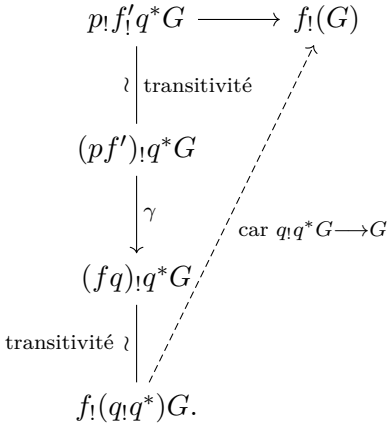
(∗)



$$p^* f_!(G) \longleftarrow f'_! q^*(G)$$

La flèche est définie, pourvu qu'on ait commutativité ou une flèche de commutativité $\gamma : pf' \longrightarrow fq$ (pas nécessairement [un] isomorphisme). Par adjonction, il revient au même

de donner



On se demande quand la flèche est [un] isomorphisme.

Le théorème suivant doit être vrai, et résulte de mes fourbis généraux. [page 7]

Théorème. *Si le carré* $(*)$ *est* cartésien *dans* (Cat)*,* γ *un isomorphisme, et si on suppose que* f *soit* 1*-lisse, ou* p 1*-propre, alors le* T'*-foncteur de changement de base est une équivalence de* T'*-catégories.*

Les 'fourbis' généraux seraient en termes d'un dérivateur **D** ad hoc, celui des 1-types d'homotopie relatifs. J'y reviendrai par la suite.

Commentary · Later developments [ms. 7]

Base change for truncated homotopy types

Grothendieck's next step asks whether integration can be performed after restricting the base. The natural transformation $\gamma : pf' \Rightarrow fq$ fixes the direction of the comparison: contravariance gives $q^* f^* \Rightarrow f'^* p^*$, whose mate is $f'_! q^* \Rightarrow p^* f_!$. At t', its effect before groupoid completion is the evident functor sending a lift $\delta : t' \to f'(s')$ to the composite $p(t') \xrightarrow{p\delta} pf'(s') \xrightarrow{\gamma_{s'}} fq(s')$.

The theorem is valid for the derivator of 1-types. More generally, set

$$\mathbb{D}_n(I) = h\,\mathrm{Fun}(NI^{\mathrm{op}}, \mathcal{S}_{\leq n}), \qquad n \geq -1.$$

Its left Kan extensions are homotopy left Kan extensions followed by n-truncation; its right Kan extensions are homotopy limits. The truncation is an accessible reflector, and the truncated category remains presentable (Lurie, 2009, Proposition 5.5.6.18 and Remark 5.5.6.21), so these Kan extensions exist for every small indexing category. An n-aspherical category is one whose n-truncation is a point, with the case $n = -1$ meaning nonemptiness. Smoothness and properness must be taken relative to this theory. They are conditions on the relevant comma categories and their changes of base, not merely on the homotopy types of the strict fibres.

A precise later formulation is Maltsiniotis (2005, Theorem 1.2, Corollary 4.16, Example 4.17, and §5). Theorem 1.2 identifies smoothness with the asphericity of the fibre-to-comma inclusions; §5 treats the corresponding base-change morphism. With his covariant convention replaced by the contravariant convention here, its two alternatives are exactly that the integrating map f is smooth or that the restricting map p is proper. Applying the statement to the localizer detected by fundamental groupoids gives equivalences of groupoids at every t', hence an equivalence of fibred groupoids. No information above the first homotopy group is needed for this particular comparison.

This distinction between a correct integration theorem and an insufficient local-constancy criterion will matter on manuscript page 9. Truncating the *values* of a diagram does not truncate all the coherence by which the values are assembled over its base.

On se pose la question quand le 2-foncteur associé à $f : S \longrightarrow T$

$$f^* : \underline{\underline{\mathrm{Grp}}}(T) \longrightarrow \underline{\underline{\mathrm{Grp}}}(S)$$

est 0-fidèle, 1-fidèle, 2-fidèle ([2]). Si G, H sont des fibrés en groupoïdes sur T, on considère p. 7

$$\underline{\underline{\mathrm{Hom}}}_{\mathrm{Grp}(T)}(G,H) \longrightarrow \underline{\underline{\mathrm{Hom}}}_{\mathrm{Grp}(S)}(f^*G, f^*H),$$

et suivant que ce foncteur est toujours fidèle, ou pleinement fidèle, ou une équivalence de catégories, on dit que f^* est 0-, 1-, ou 2-fidèle. Dans le cas 2-fidèle, cela implique que si K est un fibré en groupoïdes sur S, s'il est équivalent à un f^*G, alors G est déterminé à T-équivalence près, elle même déterminée à isomorphisme unique près.

Commentary · Examples [ms. 7]

Sections alone do not test full faithfulness

The three faithfulness conditions distinguish the lifting of transformations, of functors, and of their isomorphisms. The footnote proposes testing them using section groupoids alone. As an assertion for an arbitrary f, that reduction is false.

Let $T = [0 \to 1]$, $S = \{1\}$, and let f be the inclusion. For every contravariant groupoid diagram $H = (H_1 \to H_0)$, evaluation at 1 is an equivalence $\Gamma(T,H) \simeq H_1$: a section consists of h_1, h_0, and an isomorphism $h_0 \simeq u^*h_1$, and the choices of the latter pair form a contractible groupoid. Thus

$$\Gamma(T,H) \longrightarrow \Gamma(S, f^*H)$$

is an equivalence for every H.

Nevertheless, take $G_1 = H_1 = \varnothing$, $G_0 = *$, and $H_0 = B(\mathbf{Z}/2)$. The only structure maps from the empty groupoid exist uniquely. A functor $G \to H$ is now just a functor $* \to B(\mathbf{Z}/2)$, and its natural automorphism group is $\mathbf{Z}/2$. Consequently

$$\mathrm{Hom}_T(G,H) \simeq B(\mathbf{Z}/2) \longrightarrow \mathrm{Hom}_S(f^*G, f^*H) \simeq *$$

is not faithful. The proposed section test therefore fails even for 0-faithfulness, despite satisfying the strongest of its own three tests.

There is a useful exact replacement. If $\pi_G : G \to T$ denotes the *total category* of the first coefficient, then

$$\mathrm{Hom}_T(G,H) \simeq \Gamma(G, \pi_G^*H).$$

Indeed, a section of $G \times_T H \to G$ is precisely a functor $G \to H$ over T, and the same identification holds for transformations. After restriction, the base becomes $S \times_T G$. Testing sections on these changed bases, with the pulled-back coefficients, is equivalent to testing the required Hom-groupoids. The missing base change cannot in general be suppressed by invoking an internal Hom: arbitrary restriction of presheaves does not preserve that internal Hom.

Commentary · Development (commentators') [ms. 7]

Factorizations detect all three levels

Grothendieck's faithfulness question can be answered without imposing smoothness or properness. The appropriate test keeps both endpoints of an arrow of T, as did the factorization categories discussed in the commentary on Chapter II, manuscript page 53. In the untruncated setting, Groth and Ŝtovíček (2016, arXiv version, Definition 6.9 and Theorem 6.10) give the corresponding criterion for homotopical epimorphisms. We spell out its elementary three-level refinement for groupoids. We did not find this combined formulation in the sources checked; the argument uses standard density and adjunction, and carries no claim of priority.

For an arrow $\beta : t \to t'$ of T, let $\mathcal{F}_f(\beta)$ have objects

$$(s, a : t \to f(s), b : f(s) \to t'), \qquad ba = \beta,$$

and arrows $v : s \to s'$ satisfying $f(v)a = a'$ and $b'f(v) = b$.

2. La i-fidélité de f^* ($i \in \{0,1,2\}$) équivaut à celle des foncteurs (obtenus en prenant $G = e_T$)

$$\underline{\Gamma}(H/T) \longrightarrow \underline{\Gamma}(f^*H/S).$$

Proposition (the three tests). [ours] For $i \in \{0,1,2\}$, restriction $f^* : \mathsf{Gpd}(T) \to \mathsf{Gpd}(S)$ is i-faithful if and only if every $\mathcal{F}_f(\beta)$ is $(i-1)$-connected. Explicitly, these conditions are nonemptiness for $i=0$, nonempty connectedness for $i=1$, and $\Pi\mathcal{F}_f(\beta) \simeq *$ for $i=2$.

Proof. The counit makes the factorization categories appear. Write $h_{t'} = \mathrm{Hom}_T(-,t')$, viewed as a discrete groupoid-valued presheaf. The comma formula gives, at t,

$$(f_! f^* h_{t'})_t \simeq \coprod_{\beta : t \to t'} \Pi\mathcal{F}_f(\beta) \longrightarrow \mathrm{Hom}_T(t,t'). \tag{VIII.3}$$

The counit maps every summand to its indexing arrow. The biadjunction identifies restriction on $\mathrm{Hom}_T(G,H)$ with precomposition by $\varepsilon_G : f_! f^* G \to G$.

To prove necessity at a fixed t, use a coefficient $H = t_* K$, where t_* is the right biadjoint to evaluation at t and K is an arbitrary groupoid. The adjunction reduces the test for $G = h_{t'}$ to precomposition along (VIII.3), with values in K. If a summand is empty, take $K = B(\mathbf{Z}/2)$: an automorphism at that index is invisible after restriction, so faithfulness fails. If a summand has two connected components, the same K allows different automorphisms on those components, which cannot descend from the single index; full faithfulness fails. If a summand is connected but has nontrivial fundamental group, take K to be that summand's groupoid. Its identity functor, extended by constant functors on the other summands, is not isomorphic to a functor factoring through the discrete target. Essential surjectivity fails. These observations give precisely the three necessary conditions.

For sufficiency, first keep $G = h_{t'}$ and allow arbitrary H. Under the nonemptiness condition, a modification between two maps $h_{t'} \to H$ is determined by its pullback along $\varepsilon_{h_{t'}}$: at every t and every β, there is an object over β on which to evaluate it. Under connectedness, a modification between the pulled-back maps has equal components on all objects over a fixed β, since the two maps send every arrow of that groupoid to an identity. It therefore defines a unique component at β. These components are natural in t: reindex an object lying over β, and use naturality of the given modification. Thus the modification descends uniquely, proving full faithfulness. In the third case, (VIII.3) is itself an equivalence at every t, so precomposition is an equivalence of Hom-groupoids.

To pass from representables to an arbitrary G, one needs their coherent density, not just generation of connected components. Index the representables by the total category of G, sending (t', g) to $h_{t'}$. Their homotopy colimit in groupoid-valued presheaves, evaluated at t, is $\Pi(t\backslash_T G)$, hence is equivalent to G_t by the cartesian-lift argument already proved. Thus G is this homotopy colimit. Restriction preserves it, because such colimits are computed pointwise. Taking maps into H, respectively into f^*H, changes these colimits into homotopy limits of the corresponding Hom-groupoids. Faithful functors remain faithful under these limits, because a modification is determined by its components. Fully faithful functors remain fully faithful, because the components lift uniquely and their compatibility follows by faithfulness. Equivalences remain equivalences by the universal property of a homotopy limit. Equivalently, the homotopy fibres of faithful and fully faithful functors of groupoids are, respectively, 0-truncated and (-1)-truncated spaces; these conditions are preserved by limits. The representable tests therefore imply the required assertion for every G, H. This completes the proof.

At level zero the condition has a particularly concrete form: every t is a retract of some $f(s)$. Indeed, apply nonemptiness to id_t; conversely such a retraction supplies a factorization of every arrow with target t. For $f : Z \to e$, the sole factorization category is Z. The three tests become, respectively, that Z is nonempty, connected, or 1-connected. Even the strongest condition concerns maps between constant coefficients; it leaves room for the nonconstant diagram over S^2 constructed below.

[page 8]

Soit $i \in \mathbf{N}$. On dit que f est i-asphérique si $\forall t \in T$, $Z = S/t$ est i-asphérique, i.e.

$$(*) \qquad \pi_0(Z),\ \ldots,\ \pi_i(Z) \qquad \text{triviaux.}$$

On définit aussi (-1)-asphérique : les S/t sont non-vides. De même pour la notion de i-coasphérique.

Commentary · Exposition [ms. 8]

The valid descent assertions in Theorem 3

Theorem 3 contains a useful distinction: full faithfulness of restriction describes how a coefficient already on T is recovered, whereas its essential image describes which coefficients on S descend. The hypotheses in parts 1 and 2 do give these conclusions. The assertion in part 3 needs an additional degree of connectivity.

We give the argument for the valid assertions. Put $n = i - 1$. Under hypothesis a, the inclusions $S_t \to S/t$ are n-coaspherical; under hypothesis b, the inclusions $S_t \to t\backslash S$ are n-aspherical. This is the fibre-to-comma criterion for properness or smoothness, respectively. Combining it with the stated asphericity or coasphericity shows that every strict fibre S_t is n-connected. The same criterion says that the categories of lifts of an arrow of T, with its source fixed in the proper case or its target fixed in the smooth case, are n-connected.

For $i = 0$, every fibre is nonempty. A natural transformation between two functors $G \to H$ over T is determined by its components over each object of T; choosing one object of S_t detects each component. Restriction is therefore faithful on every Hom-groupoid.

For $i = 1$, every fibre is connected. Suppose a transformation between the restrictions of two functors $G \to H$ has been given. At $g \in G_t$, its component calculated using $s \in S_t$ is independent of s, because naturality along the vertical arrows of S_t, and then along zigzags of such arrows, identifies the components. These components are natural within G_t. To verify naturality across an arrow $a : t \to t'$, choose a lift of a with one endpoint fixed; its existence follows from the lift-category criterion. Naturality upstairs is the required naturality downstairs. The descended transformation is unique because every fibre is nonempty. This proves full faithfulness on Hom-groupoids.

For $i = 2$, base change computes the relevant adjunction on each strict fibre. Under hypothesis b, if H is on T, the counit has the fibrewise form

$$(f_! f^* H)_t \simeq \Pi(S_t) \times H_t \longrightarrow H_t.$$

Under hypothesis a, use the right biadjoint f_* instead: its unit is

$$H_t \longrightarrow (f_* f^* H)_t \simeq \mathrm{Fun}(\Pi(S_t), H_t).$$

The right biadjoint is the groupoid of cartesian sections over the comma category S/t; the proper base-change formula replaces that comma category by S_t. Since $\Pi(S_t) \simeq *$, the displayed arrows are equivalences. The adjunction equivalences on Hom-groupoids now prove that f^* is 2-faithful. Notice that this argument uses constant coefficients *on each fibre*; it does not assert that every locally constant coefficient there is constant.

The essential-image statement follows by applying the other adjunction arrow. Under hypothesis b, its restriction to a fibre is the unit

$$G|_{S_t} \longrightarrow \mathrm{const}_{S_t}\left(\Pi \int_{S_t} G\right).$$

If $G|_{S_t}$ is equivalent to a constant groupoid K, integration gives $\Pi(S_t) \times K \simeq K$, so this unit is an equivalence. Thus $G \to f^* f_! G$ is fibrewise an equivalence, and G descends. Under hypothesis a, use $f^* f_* G \to G$ and the same argument with sections: $\mathrm{Fun}(\Pi(S_t), K) \simeq K$. Necessity is immediate, because a coefficient pulled back from T restricts to a constant coefficient on each strict fibre. This proves part 2 with precisely its printed fibrewise-*constant* condition. Its replacement by equivalence of transport functors requires a separate argument and is where the next difficulty lies.

En fait, pour $i \in [-1, +\infty[\subseteq \mathbf{Z}$, on définit un dérivateur $\mathbf{D}_i$ des i-types d'homotopie relatifs (cf. plus bas). D'où les notions d'objet $\mathbf{D}_i$-asphérique (= i-asphérique), de $\mathbf{D}_i$-équivalence, de foncteur $\mathbf{D}_i$-asphérique et $\mathbf{D}_i$-coasphérique, de foncteur $\mathbf{D}_i$-lisse et $\mathbf{D}_i$-propre, de $\mathbf{D}_i$-fibration et de $\mathbf{D}_i$-fibration *triviale* (i.e. à fibres $\mathbf{D}_i$-asphériques). On doit avoir le

Théorème 3. *Soit $f : S \longrightarrow T$ dans* (Cat)*, $i \in \{0, 1, 2\}$. Supposons l'une des conditions suivantes satisfaite :*

a) f est $(i-1)$-propre, et $(i-1)$-asphérique (s'exprime par la condition que certaines catégories, telles les S/t, soient $(i-1)$-asphériques).

b) f est $(i-1)$-lisse, et $(i-1)$-coasphérique.

Alors :

1°) Le foncteur f^* image inverse

$$f^* : \underline{\mathrm{Grp}}(T) \longrightarrow \underline{\mathrm{Grp}}(S)$$

est i-fidèle.

*2°) Dans le cas $i = 2$ (f pleinement fidèle), on a de plus ceci : si $G \in \mathrm{Ob}\,\underline{\mathrm{Grp}}(S)$ est un fibré en groupoïdes sur S, alors G est équivalent à un fibré en groupoïdes de la forme $f^*(H)$ si et seulement si pour tout $t \in T$, la restriction G_t de G à S_t est équivalente à un fibré en groupoïdes constant sur S_t. (**NB** La condition a) ou b) implique déjà que S_t est 1-connexe, i.e. que son groupoïde fondamental est équivalent à e.)* p. 8 [page 9]

3°) Soit Z un objet 1-connexe dans (Cat)*, G un fibré en groupoïdes sur Z. Pour que G soit Z-équivalent à un groupoïde constant sur Z, il faut et il suffit que pour toute flèche u dans Z, le morphisme de groupoïdes u^* induit sur les fibres de G soit une Z-équivalence* ([3]).

Corollaire. *La condition nécessaire et suffisante de 2°) s'écrit aussi ainsi : pour toute flèche u de S 'verticale' sur T, i.e. telle que $f(u)$ soit une flèche identique, le foncteur u^* induit par u sur les fibres de G soit une équivalence de groupoïdes.*

Commentary · Examples [ms. 9]

A finite obstruction in degree two

Grothendieck wants a practical descent test that can be checked on the arrows of the fibres. The obstruction to this last simplification already appears over a finite poset, with every fibre the same two-element automorphism group. It makes the distinction from the preceding valid argument quite explicit. The general coherence issue was isolated in the commentary on Chapter VI, manuscript pages 7 and 74; here it produces a counterexample to a stated theorem.

Let Z be the poset of nonempty proper subsets of $\{0,1,2,3\}$, ordered by inclusion. It has fourteen objects. Its nerve is the barycentric subdivision of the boundary of a tetrahedron, so $|NZ| \cong S^2$ and $\Pi Z \simeq *$. Write $A = \mathbf{F}_2$. On the strictly increasing triples in Z, define

$$c(\sigma,\tau,\upsilon) = \begin{cases} 1, & (\sigma,\tau,\upsilon) = (\{0\},\{0,1\},\{0,1,2\}), \\ 0, & \text{otherwise.} \end{cases}$$

Extend c by zero to triples with a repeated object. There is no strictly increasing chain of four objects, so this normalized 2-cochain is a cocycle.

It is useful to verify directly that it is not a coboundary. Let z be the sum, with coefficients in $\mathbf{F}_2$, of the twenty-four flags vertex $\subset$ edge $\subset$ face of the tetrahedral boundary. Every edge of this order complex occurs in exactly two such flags: a vertex–edge pair extends to two faces, a vertex–face pair has two intervening edges, and an edge–face pair has two possible vertices. Hence $\partial z = 0$, whereas $\langle c, z\rangle = 1$. If $c = \delta b$, this pairing would equal $\langle b, \partial z\rangle = 0$, a contradiction.

Now define a category E_c over Z. There is one object over each σ, and for $\sigma \subseteq \tau$ the arrows over that inclusion are labelled by A; there are no other arrows. The identity has label zero, and composition is

$$(\tau \subseteq \upsilon, b) \circ (\sigma \subseteq \tau, a) = \big(\sigma \subseteq \upsilon, a + b + c(\sigma,\tau,\upsilon)\big). \tag{VIII.2}$$

The cocycle equation is exactly associativity; normalization gives the unit equations. Each fibre is BA. Every arrow is cartesian: composition with a fixed arrow carries the set of lifts of any preceding inclusion bijectively onto the set of lifts of its composite, since in (VIII.2) it adds a fixed element of A. Thus $E_c \to Z$ is a category fibred in groupoids. Choosing the lift labelled zero for each inclusion makes every inverse-image functor $BA \to BA$ the identity. Only the comparison isomorphisms for successive inverse images are nontrivial, and they are encoded by c. Passing to an equivalent split

3. On dit alors que G est *essentiellement localement constant* sur Z.

fibration gives the same counterexample in terms of a strict groupoid-valued functor; the obstruction does not arise from a choice of presentation.

Suppose that E_c were equivalent over Z to a constant fibred groupoid. Its fibre would have to be equivalent to BA, so the constant fibred groupoid would admit a section. An equivalence over Z and its quasi-inverse transport that section to a section of E_c. Such a section chooses a label $b(\sigma,\tau)$ on each inclusion; its functoriality says

$$b(\sigma,\upsilon) = b(\sigma,\tau) + b(\tau,\upsilon) + c(\sigma,\tau,\upsilon).$$

This is precisely $\delta b = c$, which has just been excluded. Therefore E_c is not constant, although Z is 1-connected and every transport functor is an equivalence. This disproves part 3.

It also disproves the corollary under its own hypotheses. Take $f : Z \to e$. A functor to e is both a Grothendieck fibration and an opfibration, hence smooth and proper for the localizer of 1-types. Its sole fibre is the 1-connected category Z, so hypotheses a and b for $i = 2$ both hold. Every arrow is vertical and satisfies the proposed transport test, but E_c is not in the image of f^*. Part 2 itself remains valid: its actual condition demands that $E_c|_Z$ be constant, which it is not.

Commentary · Present state [ms. 9]

The corrected local-constancy criterion

The missing datum is classical 2-monodromy. A locally constant diagram with fibre a groupoid K determines a classifying map

$$|NZ| \longrightarrow B\,\mathrm{Aut}(K),$$

where $\mathrm{Aut}(K)$ is the groupoid of self-equivalences of K and natural isomorphisms. The target is a 2-type: its first homotopy group records equivalence classes of self-equivalences, and its second is $\mathrm{Aut}(\mathrm{id}_K)$. A simply connected base removes the first kind of monodromy but can still support the second. In the finite example it is the nonzero class in $H^2(S^2;\mathbf{F}_2)$.

A precise classification is given by Polesello and Waschkies (2005, arXiv version, Theorem 2.2.5 and §2.3); their Corollary 2.2.4 gives constancy over a 2-connected base. Applied to the CW complex $|NZ|$, it proves the corrected statement: if Z is 2-connected, a groupoid diagram is equivalent to a constant one if and only if all its transport functors are equivalences. Alternatively, over a 1-connected base one must require its remaining 2-monodromy to vanish. This is a known obstruction, not a new conjecture or a priority claim. Strengthening the corollary by requiring 2-connected strict fibres, while retaining the hypotheses of part 2, makes the vertical-arrow criterion sufficient by applying this corrected statement separately to every fibre.

Chapter IX

Retour sur les catégories de fractions $\mathcal{M}W^{-1}$, comparaison avec Quillen

VIII-X · Chapter IX · manuscript pages 1–25

p. [non numérotée]

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre IX
Retour sur les catégories de fractions $\mathcal{M}W^{-1}$, comparaison avec Quillen

p. [non numérotée]

Ce texte a été déchiffré et transcrit en LaTeX 2ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Fonctorialité de $\mathrm{Ch}_\infty^\Sigma(a,b)$ en a, b dans $\mathcal{M}\Sigma^{-1}$

p. 1
[page 1]

On se donne une catégorie $\mathcal{M}$ avec [un] ensemble admissible de flèches $\Sigma \subseteq \mathrm{Fl}(\mathcal{M})$, d'où (en prenant p. ex. $\underline{t} = (t_n)$, avec $t_n = (\underbrace{\Delta_+\Delta_-}_{\tau_0})^{2n}$, et $\underbrace{t_{n+1}}_{=\tau_0^{2n+2}} \longrightarrow \underbrace{t_n}_{=\tau_0^{2n}}$ défini comme $p * \mathrm{id}_{\tau_0^{2n}} * p$, où $p : \tau_0 \longrightarrow 1$ la dégénérescence.) On a alors une théorie des $\underline{\mathrm{Ch}}_{\Sigma,\infty}(\mathcal{M},a,b)$, que je note simplement $\mathrm{Ch}_\infty(a,b)$ en tant qu'objet de Hot. Des accouplements

$$\mathrm{Ch}_\infty(a,b) \times \mathrm{Ch}_\infty(b,c) \longrightarrow \mathrm{Ch}_\infty(a,c) \qquad \text{dans Hot}$$

avec (nous allons l'admettre) *associativité* dans Hot. Cela définit donc des accouplements sur les $\pi_0(\mathrm{Ch}_\infty(a,b)) \overset{\text{déf}}{=} \pi_0(a,b)$, qui n'est autre que la composition des flèches dans $\mathcal{M}\Sigma^{-1}$. D'autre part, on a pour $a,b \in \mathrm{Ob}\,\mathcal{M}$

$$\mathrm{Hom}_{\mathcal{M}}(a,b) \longrightarrow \pi_0(\mathrm{Ch}_\infty(a,b)) = \pi_0(a,b),$$

compatible avec les compositions. Mais on peut l'ignorer pour l'instant, et regarder $\mathrm{Ob}\,\mathcal{M}$, ou simplement $\mathcal{M}_0$, avec la seule structure

$$\begin{cases} \mathcal{M}_0 \times \mathcal{M}_0 & \longrightarrow & \mathrm{Hot} & (a,b) & \longmapsto & \mathrm{Ch}_\infty(a,b) \\ \mathcal{M}_0 \times \mathcal{M}_0 \times \mathcal{M}_0 & \longrightarrow & \mathrm{Fl}\,\mathrm{Hot} & (a,b,c) & \longmapsto & (\mathrm{Ch}_\infty(a,b) \times \mathrm{Ch}_\infty(b,c) \underbrace{\longrightarrow}_{\substack{\text{flèche} \\ \text{dans Hot}}} \mathrm{Ch}_\infty(a,c)) \end{cases}$$

avec condition [d']associativité, et aussi une condition d'unités :

$$\mathrm{Ch}_\infty(a,a) \text{ est muni d'un } 1_a \in \pi_0\mathrm{Ch}_\infty(a,a) = \pi_0(a,a),$$

jouant le rôle d'une 'identité bilatère', dans le sens suivant. Notons que les accouplements

$$\begin{array}{ccccc} \mathrm{Ch}_\infty(a,b) & \times & \mathrm{Ch}_\infty(b,c) & \longrightarrow & \mathrm{Ch}_\infty(a,c) \\ P & \times & Q & \longrightarrow & R \end{array}$$

définissent [page 2]

$$\begin{array}{ccc} P & \longrightarrow & \underline{\mathrm{Hom}}_{\mathrm{Hot}}(Q,R) \\ Q & \longrightarrow & \underline{\mathrm{Hom}}_{\mathrm{Hot}}(P,R), \end{array}$$

d'où en passant aux π_0

$$\begin{array}{ccc} \pi_0(P) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(Q,R) \\ \pi_0(Q) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(P,R), \end{array}$$

en d'autres termes, p. 2

$$\begin{array}{ccc} \pi_0(a,b) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(b,c),\mathrm{Ch}_\infty(a,c)) \\ \pi_0(b,c) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(a,b),\mathrm{Ch}_\infty(a,c)). \end{array}$$

Commentary · Exposition [ms. 1]

Composition before functoriality

Grothendieck wants a morphism in the localized category to act on an entire space of paths, not merely on its set of components. This is why he temporarily forgets how the path categories were constructed. Once composition and its units are available, change of either endpoint is a formal operation. The hypothesis being made is substantial: associativity is assumed for the particular spaces $\mathrm{Ch}_\infty(a,b)$. The abstract argument that follows does not establish that hypothesis.

Here is the argument with all its variance visible. Write $C(a,b)$ for the proposed Hom object in a monoidal category $(\mathcal{H},\otimes,I)$, and suppose that

$$m_{abc} : C(a,b)\otimes C(b,c) \longrightarrow C(a,c), \qquad j_a : I \longrightarrow C(a,a)$$

satisfy associativity and the two unit equations, with the associator and unitors of $\mathcal{H}$ understood. This is enrichment, with the factors ordered in the direction in which a path is traversed; reversing the tensor convention gives the usual notation of Kelly (1982, §§1.1–1.3). Symmetry is not required. In particular, the Cartesian signs in the general displays on manuscript page 7 should be read as the specified monoidal product $*$.

The ordinary category C_0 has

$$C_0(a,b) = \mathrm{Hom}_{\mathcal{H}}(I, C(a,b)).$$

For $u : I \to C(a,b)$ and $v : I \to C(b,c)$ its composition is

$$I \xrightarrow{\sim} I\otimes I \xrightarrow{u\otimes v} C(a,b)\otimes C(b,c) \xrightarrow{m_{abc}} C(a,c).$$

The enriched associativity and unit equations give the ordinary ones after precomposition with I. Thus the j_a become the identities of C_0. Their uniqueness follows already in the endomorphism monoid: if j and j' are both two-sided units, then $j = j\circ j' = j'$. Existence as units on the Hom objects is a separate assumption; an element that is a unit only after applying $\mathrm{Hom}(I,-)$ need not satisfy the stronger enriched equations.

To change endpoints, let $u : a' \to a$ and $v : b \to b'$ belong to C_0. Insert the corresponding maps out of I on the two sides of $C(a,b)$ and compose twice. This gives

$$C(u,v) : C(a,b) \longrightarrow C(a',b').$$

For composable $u' : a'' \to a'$ and $v' : b' \to b''$, the two composites $C(u',v')C(u,v)$ and $C(uu',v'v)$ are the two parenthesizations of the same fivefold composition, with u', u on the left and v, v' on the right. Associativity identifies them. The unit equations identify $C(1_a,1_b)$ with the identity. Inserting only the left or only the right element also shows that the two actions commute. We have therefore obtained an ordinary bifunctor

$$C_0^{\mathrm{op}} \times C_0 \longrightarrow \mathcal{H}.$$

Applying $\mathrm{Hom}_{\mathcal{H}}(I,-)$ gives the ordinary Hom bifunctor, including its action on morphisms, rather than only objectwise bijections.

For $\mathcal{H} = \mathrm{Hot}$, I is a point and $\mathrm{Hom}(I,-)$ is π_0. One can obtain the two endpoint actions by currying in the Cartesian closed homotopy category, as Grothendieck does on page 2, but the insertion argument shows that closedness is unnecessary. This economy matters when he passes to ind-objects: it is the monoidal structure and its coherence that the construction uses.

On en déduit en particulier pour $a, b \in \mathcal{M}_0$

$$(*) \qquad \begin{cases} 1) & \pi_0(a,a) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(a,b), \mathrm{Ch}_\infty(a,b)) \\ 2) & \pi_0(a,a) & \longrightarrow & \mathrm{Hom}_{\mathrm{Hot}}(\mathrm{Ch}_\infty(b,a), \mathrm{Ch}_\infty(b,a)), \end{cases}$$

et un objet α_a de $\pi_0(a,a)$ est unité bilatère si ses deux images sont les homomorphismes identiques dans $\mathrm{Ch}_\infty(a,b)$ resp. $\mathrm{Ch}_\infty(b,a)$. Passant aux π_0 de ceux-ci, cela implique que α_a est unité bilatère pour la composition des $\pi_0(x,y)$, relativement à b. Si on veut que ce soit le cas pour *tout* b, cela signifie (prenant $b = a$) que α_a est un objet unité bilatère pour la structure de composition associative (donc de monoïde) dans $\pi_0(a,a)$, donc que $\pi_0(a,a)$ est un monoïde *unitaire,* ayant α_a pour objet unité. L'élément α_a est donc déterminé de façon unique. Et on veut que α_a existe pour tout $a \in \mathcal{M}_0$, donc $\mathcal{M}_0$, muni des $\pi_0(a,b)$ et de leur composition, devient une *catégorie,* soit $\mathcal{M}_\Pi$; et les flèches identiques de celle-ci doivent jouer le rôle d'unité bilatère également pour les 'Hom externes' $\mathrm{Ch}_\infty(a,b)$, à valeurs dans Hot. [page 3]

On peut dire que pour $a = a_0$ fixé, les $\mathrm{Ch}_\infty(a_0, b)$ (b variable) forment les valeurs d'un *foncteur* covariant

$$\mathcal{M}_\Pi \longrightarrow \mathrm{Hot} \qquad b \longmapsto \mathrm{Ch}_\infty(a_0, b),$$

en associant à

$$b \xrightarrow{u} b', \qquad u \in \pi_0(b, b'),$$

la flèche

$$\mathrm{Ch}_\infty(a, b) \longrightarrow \mathrm{Ch}_\infty(b')$$

[plutôt $\mathrm{Ch}_\infty(a_0, b) \longrightarrow \mathrm{Ch}_\infty(a_0, b')$] correspondante $*_1$ [?]. Que ce soit bien là une loi fonctorielle doit résulter formellement de l'associativité des accouplements sur les Ch_∞. De même, on a une loi fonctorielle, pour b fixé,

$$\mathcal{M}_\Pi^o \longrightarrow \mathrm{Hot}, \qquad a \longmapsto \mathrm{Ch}_\infty(a, b),$$

grâce à l'associativité. En fait, l'associativité impliquera aussi qu'on a un *bifoncteur*

$$\begin{array}{ccc} \mathcal{M}_\Pi^o \times \mathcal{M}_\Pi & \longrightarrow & \mathrm{Hot} \\ (a\,,\,b) & \longmapsto & \mathrm{Ch}_\infty(a, b), \end{array}$$

car si

$$a' \xrightarrow{u} a, \qquad b \xrightarrow{v} b',$$

p. 3
[page 4]

on a le diagramme

$$\begin{array}{ccc} \mathrm{Ch}_\infty(a, b) & \xrightarrow[v]{\text{comp. avec}} & \mathrm{Ch}_\infty(a, b') \\ \Big\downarrow{\scriptstyle \text{comp. avec } u} & & \Big\downarrow{\scriptstyle \text{comp. avec } u} \\ \mathrm{Ch}_\infty(a', b) & \xrightarrow[v]{\text{comp. avec}} & \mathrm{Ch}_\infty(a', b'), \end{array}$$

qui est commutatif grâce à l'associativité pour

$$\mathrm{Ch}_\infty(a', a),\ \mathrm{Ch}_\infty(a, b),\ \mathrm{Ch}_\infty(b, b').$$

Corollaire de cette dépendance bifonctorielle : *Si* $u : a' \longrightarrow a$, $v : b \longrightarrow b'$ *sont dans* $\mathcal{M}_\Pi$, *alors la flèche correspondante*

$$\mathrm{Ch}_\infty(a, b) \longrightarrow \mathrm{Ch}_\infty(a', b') \qquad \textit{dans } \mathrm{Hot}$$

est un isomorphisme.

Commentary · Examples [ms. 4]

An invertibility hypothesis already restored on page 8

The intended application is that weakly equivalent endpoints have the same Hom object. For this application u and v are invertible in the localization. The corollary as printed has omitted this condition; manuscript page 8 states the correct version explicitly.

A discrete example detects the omission without any homotopy theory. Enrich the ordered category $0 < 1$ in sets, regarded as discrete spaces, by its usual Hom sets. Change the first variable by the noninvertible map $u : 0 \to 1$ and fix the second variable at 0. The induced map is

$$C(1, 0) = \varnothing \longrightarrow C(0, 0) = \{1_0\}.$$

This is the endpoint action of $u : 0 \to 1$ with the second variable fixed at 0, and is not an isomorphism. If u and v are invertible, functoriality supplies the inverse $C(u^{-1}, v^{-1})$, proving the repaired corollary.

The following display also omits the opposite on the first factor: its domain is $\mathcal{M}_\Pi^{\mathrm{op}} \times \mathcal{M}_\Pi$. Both omissions occur in the edition, so they are retained in the source and entered separately in the audit. They do not invalidate the preceding construction of the bifunctor.

Les mêmes arguments devraient marcher si au lieu des Hom ext à valeurs dans Hot, on les prenait dans Ind Hot, ou dans Cat hot, ou dans Ind Cat hot – je pense bien sûr aux $\mathrm{CH}_\infty(a,b)$ (dans la théorie de la localisation) dans Ind Cat hot, qui peuvent être regardés comme provenant d'un bifoncteur (où $\mathcal{M}_\Pi = \mathcal{M}\Sigma^{-1}$)

$$\mathcal{M}_\Pi \times \mathcal{M}_\Pi \longrightarrow \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}.$$

[page 5]

Voici l'axiomatisation définitive (?). On remplace Hot par une catégorie

$\mathcal{H}$, catégorie avec produit $*$ associatif, commutatif (?) ([2]), unitaire, et toutes les compatibilités habituelles au besoin,

l'objet unité pour $*$ sera noté 1, et on notera

$$\begin{cases} \pi_0(\xi) & \overset{\text{déf}}{=} & \mathrm{Hom}_{\mathcal{H}}(1,\xi) \\ \pi_0 : \mathcal{H} & \longrightarrow & \mathrm{Ens}. \end{cases}$$

Si on a

$$P * Q \xrightarrow{u} R,$$

p. 4

on en déduit

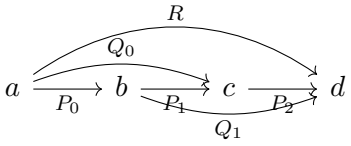
$$\begin{cases} \pi_0(P) & \xrightarrow{c^{\mathrm{l}}_u} & \mathrm{Hom}_{\mathcal{H}}(Q,R) \\ \pi_0(Q) & \xrightarrow{c^{\mathrm{r}}_u} & \mathrm{Hom}_{\mathcal{H}}(P,R) \end{cases}$$

ainsi :

$$\begin{array}{rrcl} \text{Si } \alpha \in \pi_0(P),\ \text{i.e. } 1 \xrightarrow{\alpha} P,\ \text{alors} & \alpha * \mathrm{id}_Q : Q \longrightarrow P * Q & & \\ \text{et} & c^{\mathrm{l}}_u(\alpha) & = & u \circ (\alpha * \mathrm{id}_Q) \\ \text{et dualement} & c^{\mathrm{r}}_u(\beta) & = & u \circ (\mathrm{id}_P * \beta). \end{array}$$

Si on a trois objets P_0, P_1, P_2 dans $\mathcal{H}$, des accouplements

$$\begin{cases} P_0 * P_1 & \xrightarrow{u_1} & Q_0 \\ P_1 * P_2 & \xrightarrow{u_2} & Q_1 \\ P_0 * Q_1 & \xrightarrow{u_3} & R \\ Q_0 * P_2 & \xrightarrow{u_4} & R \end{cases}$$



avec une relation d'associativité, disant que

[page 6]

$$\begin{array}{ccc} P_0 * P_1 * P_2 & \longrightarrow & Q_0 * P_2 \\ \downarrow & & \downarrow \\ P_0 * Q_1 & \longrightarrow & R \end{array}$$

est commutatif, alors on en déduit deux relations de transitivité et une de commutation, pour les opérations associées aux éléments de $\pi_0(P_0)$, $\pi_0(P_1)$, $\pi_0(P_2)$, qu'on n'explicitera pas, sauf l'une :

$$\begin{array}{rrcl} \text{Soit } \beta \in P_1,\ \gamma \in P_2, & & & \\ \text{d'où par } \beta : & c_{u_1}(\beta) : \ P_0 & \longrightarrow & Q_0, \\ \text{et par } \gamma : & c_{u_4}(\gamma) : \ Q_0 & \longrightarrow & R, \\ \text{et par } \beta *_{u_2} \gamma \in \pi_0(Q_1) : & c_{u_3}(\beta \ [*_{u_2}] \gamma) : \ P_0 & \longrightarrow & R, \end{array}$$

p. 5

[plutôt $\beta \in \pi_0(P_1)$, $\gamma \in \pi_0(P_2)$], et bien on a

$$\boxed{c(\beta *_{u_2} \gamma) \;=\; c(\gamma)c(\beta).}$$

Il faut encore définir $\beta *_{u_2} \gamma$, via l'accouplement $P_1 * P_2 \xrightarrow{u_2} Q_1$, on a en effet

$$\begin{array}{ccc} \pi_0(P_1) \times \pi_0(P_2) & \longrightarrow & \pi_0(P_1 * P_2) \\ (\beta\,,\,\gamma) & \longmapsto & \beta * \gamma, \end{array}$$

et on compose avec

$$\pi_0(u_2) : P_1 * P_2 \longrightarrow Q_1$$

[plutôt $\pi_0(u_2) : \pi_0(P_1 * P_2) \longrightarrow \pi_0(Q_1)$].

On se donne un ensemble $\mathcal{M}_0$, et [page 7]

$$\left\{ \begin{array}{rcll} \mathcal{M}_0 \times \mathcal{M}_0 & \xrightarrow{\text{Ch}} & \mathcal{H} \quad \text{noté} & (a,b) \longmapsto \text{Ch}(a,b) \text{ ou } \text{Hom ext}_{\mathcal{H}}(a,b) \\ \mathcal{M}_0 \times \mathcal{M}_0 \times \mathcal{M}_0 & \xrightarrow{\text{comp.}} & \text{Fl}\,\mathcal{H} & (a,b,c) \longmapsto (\text{Ch}(a,b) \times \text{Ch}(b,c) \longrightarrow \text{Ch}(a,c)) \end{array} \right.$$

avec deux axiomes

a) Associativité pour les compositions, pour $(a,b,c,d) \in \mathcal{M}_0^4$

$$\begin{array}{ccc} \text{Ch}(a,b) \times \text{Ch}(b,c) \times \text{Ch}(c,d) & \longrightarrow & \text{Ch}(a,c) \times \text{Ch}(c,d) \\ \downarrow & & \downarrow \\ \text{Ch}(a,b) \times \text{Ch}(b,d) & \longrightarrow & \text{Ch}(a,d). \end{array}$$

b) Existence d'objets unité bilatères

$$1_a \in \pi_0(\text{Ch}(a,a)) \qquad \text{pour tout } a \in \mathcal{M}_0$$

Alors les $\pi_0(\text{Ch}(a,b))$ et leur composition définissent sur $\mathcal{M}_0$ comme ensemble d'objets une structure de *catégorie* $\mathcal{M}_{0\,\Pi}$, ayant les 1_a comme identités, et on trouve que

$$(a,b) \longmapsto \text{Ch}(a,b)$$

peut s'interpréter à partir d'un *bifoncteur* canonique

$$\mathcal{M}_{0\,\Pi}^o \times \mathcal{M}_{0\,\Pi} \longrightarrow \mathcal{H},$$

dont le composé avec

$$\mathcal{H} \xrightarrow{\pi_0} \text{Ens}$$

n'est autre que le bifoncteur $(a,b) \longrightarrow \text{Hom}_{\mathcal{M}_{0\,\Pi}}(a,b)$,

$$\text{Hom}_{\mathcal{M}_{0\,\Pi}}(a,b) \;\simeq\; \pi_0(\text{Ch}(a,b)). \qquad \begin{array}{r}\text{p. 6}\\ \text{[page 8]}\end{array}$$

Cette relation est la *définition* de $\underline{\text{Hom}}_{\mathcal{M}_{0\,\Pi}}$, mais ici la relation est entendue comme une bijection *fonctorielle* en a, b.

On trouve notamment que si

$$u : a' \longrightarrow a, \qquad v : b \longrightarrow b'$$

sont des flèches *inversibles* dans $\mathcal{M}_{0\,\Pi}$, alors

$$\text{Ch}(a,b) \longrightarrow \text{Ch}(a',b')$$

2. On ne doit pas avoir besoin de la commutativité.

correspondant dans $\mathcal{H}$ est [un] *isomorphisme.*

On appliquera tout ceci surtout aux deux situations déduites d'une situation $(\mathcal{M}, \Sigma)$, avec

$$\begin{array}{llllll} 1^o) & \mathcal{H} & = & \mathrm{Hot}, & \mathrm{Ch}(a,b) & = & \mathrm{Ch}^{\Sigma}_{\infty}(\mathcal{M}; a, b) \\ 2^o) & \mathcal{H} & = & \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}, & \mathrm{Ch}(a,b) & = & \mathrm{CH}^{\Sigma}_{\infty}(\mathcal{M}; a, b)\ ([2]). \end{array}$$

Dans les deux cas, on a donc un foncteur sur $(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1})$, à valeurs soit dans Hot, soit dans $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$. (**NB** Les deux structures de catégorie sur $\mathcal{M}_0$ déduites des ces deux foncteurs sont canoniquement isomorphes.)

$$\begin{array}{ccc} & \nearrow\ \mathrm{Hot} & (a,b) \longmapsto \mathrm{Ch}^{\Sigma}_{\infty}(a,b) \\ (\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1}) & & \\ & \searrow\ \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot} & (a,b) \longmapsto \mathrm{CH}^{\Sigma}_{\infty}(a,b) \end{array}$$

Comme les foncteurs canoniques [page 9]

$$\begin{array}{ccccccc} \mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot} & \xrightarrow{\mathrm{Ind}\,\varphi} & \mathrm{Ind}\,\mathrm{Hot} & \xrightarrow{\mathrm{Ind}\,\psi} & \mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0 & \xrightarrow{\varinjlim} & \mathrm{Hot}^{\wedge}_0, \\ \uparrow & & \uparrow & & \uparrow & \nearrow_{\mathrm{id}} & \\ \mathrm{Cat}\,\mathrm{hot} & \xrightarrow{\varphi} & \mathrm{Hot} & \xrightarrow{\psi} & \mathrm{Hot}^{\wedge}_0 & & \end{array}$$

commutant aux produits finis, toute théorie de 'Hom externes' à valeurs dans une de ces catégories, en définit une à valeurs dans chacune de celles dans laquelle elle s'envoie. D'ailleurs, ces foncteurs ne sont pas tous pleinement fidèles (c'est en tous cas faux pour φ et $\mathrm{Ind}\,\varphi$, et j'ignore ce qui en est de ψ et $\mathrm{Ind}\,\psi$ – les flèches verticales par contre sont pleinement fidèles ([3]).) Mais ils induisent des bijections pour les π_0. Donc la catégorie $\mathcal{M}_{0\,\Pi}$ déduite d'une théorie de Hom externes à valeurs dans une de ces catégoires, et celle déduite dans une autre où elle s'envoie, est la même à isomorphisme canonique près. Pour ce qui est de cette définition de $\mathcal{M}_{0\,\Pi}$, rien n'est changé donc quand on regarde les CH_{∞} comme des objets de $\mathrm{Ind}\,\mathrm{Hot}$, ou seulement de $\mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0$, ou même seulement de $\mathrm{Hot}^{\wedge}_0$, ou quand on regarde les Ch_{∞} comme des objets dans $\mathrm{Hot}^{\wedge}_0$. Mais bien sûr, la donnée des accouplements de composition perd chaque fois en précision (sauf peut-être en passant de Hot à $\mathrm{Hot}^{\wedge}_0$, si ψ est pleinement fidèle …). Quand on prend les images des $\mathrm{CH}^{\Sigma}_{\infty}(a,b)$ et $\mathrm{Ch}^{\Sigma}_{\infty}(a,b)$ dans $\mathrm{Hot}^{\wedge}_0$ (la dernière catégorie du diagramme), on trouve des objets canoniquement isomorphes (fonctoriellement en a, b dans $\mathcal{M}_{0\,\Pi} \simeq \mathcal{M}\Sigma^{-1}$). Quand on les regarde comme étant respectivement dans Ind Hot et dans $\mathrm{Hot} \hookrightarrow \mathrm{Ind}\,\mathrm{Hot}$, on trouve un homomorphisme fonctoriel

p. 7

[page 10]

$$(*) \qquad \mathrm{CH}^{\Sigma}_{\infty}(a,b) \longrightarrow \mathrm{Ch}^{\Sigma}_{\infty}(a,b) \qquad \text{dans } \mathrm{Ind}\,\mathrm{Hot} \text{ ou dans } \mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0 \text{ au choix.}$$

La $\varinjlim$ de $\mathrm{CH}^{\Sigma}_{\infty}(a,b)$ dans Hot n'existe pas en général (sauf si l'ind-objet est essentiellement constant), mais elle existe dans $\mathrm{Hot}^{\wedge}_0$, et l'isomorphisme canonique

$$(**) \qquad \varinjlim_{\mathrm{Hot}^{\wedge}_0} \mathrm{CH}^{\Sigma}_{\infty}(a,b) \xrightarrow{\sim} \mathrm{Ch}^{\Sigma}_{\infty}(a,b) \qquad \text{dans } \mathrm{Hot}^{\wedge}_0$$

est la flèche déduite de $(*)$ dans $\mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0$.

2. $*$ dans Hot et $\mathrm{Ind}\,\mathrm{Cat}\,\mathrm{hot}$ est le produit cartésien sans plus.

3. et également, $\mathrm{Ind}\,\mathrm{Hot}^{\wedge}_0 \xrightarrow{\varinjlim} \mathrm{Hot}^{\wedge}_0$ n'est *pas* pleinement fidèle.

Commentary · Present state [ms. 9]

What the finite tests forget

Grothendieck is trying to recover a space of morphisms from its values on a manageable collection of diagram shapes. The attraction is clear: if those values determined both spaces and maps between them, a difficult coherence argument could be replaced by calculations on finite diagrams. The notation needs its earlier definition. In Chapter V, manuscript pages 51–52, Hot_0 is the full subcategory of homotopy types represented by categories with finitely many objects and arrows (Grothendieck, 1990d, Chapter V, printed pp. 31–32). It is not the subcategory of finite CW complexes. For example, the one-object category of a finite nontrivial group is finite, whereas its nerve need not have finite dimension.

Choose a small skeleton $\mathcal{A}$ of Hot_0. The functor in question is

$$\psi(X)(K) = [K, X], \qquad K \in \mathcal{A}.$$

It preserves products, and evaluation at the one-point category recovers $\pi_0 X$. These useful facts survive the negative answer to the full-faithfulness question. The commentary on Chapter V, manuscript p. 52, discusses the obstruction ("Finite unbased tests miss a symmetric-group shift" and "The modern scope of the final obstruction"). The general theorem is that no set of unpointed homotopy types jointly detects isomorphisms in Hot (Arlin and Christensen, 2023, Theorem 2.1). Applied to $\mathcal{A}$, this says that ψ is not even conservative, and hence is not fully faithful.

There is also a separate answer to the faithfulness question, which is what the hoped-for associativity argument actually needs. The presheaf category on $\mathcal{A}$ has a faithful set-valued functor

$$F \longmapsto \coprod_{K \in \mathrm{Ob}\,\mathcal{A}} F(K) :$$

a natural transformation is determined by its components, which this disjoint union retains. If ψ were faithful, the composite would make Hot concrete. Freyd's nonconcreteness theorem rules this out; his argument already applies to a full subcategory of simply connected, finite-dimensional CW complexes, so forgetting basepoints does not evade the obstruction (Freyd, 1970, pp. 1 and 5 of the 2004 reprint). Thus ψ is not faithful either. These are statements with the usual fixed-universe interpretation of small spaces and small sets; restricting the allowed spaces to a particular small category changes the question.

Consequently, agreement after passage to $\mathrm{Hot}_0^{\wedge}$ cannot in general prove equality of the two composites in Hot. It follows as well, by restriction to constant ind-objects, that $\mathrm{Ind}\,\psi$ is not fully faithful. None of this invalidates the underlying category of morphisms obtained by taking π_0; it identifies exactly the additional coherence that manuscript page 13 still has to supply.

Commentary · Examples [ms. 10]

Three meanings of a limiting space

Grothendieck keeps an ind-object because it records the stage at which a family of paths can be represented. Passing to its union, or to a colimit of its homotopy types, asks a different question. The phrase "n'existe pas en général" should be read as the absence of a general existence theorem here. Essential constancy is a sufficient condition, not a necessary condition for an ordinary colimit in Hot.

For a complete elementary example, let $E_n = \{0, \dots, n\}$, with the discrete topology and the evident inclusions, and let $E = \mathbf{N}$. For every space Y there are natural bijections

$$[E, Y] = (\pi_0 Y)^{\mathbf{N}} \simeq \varprojlim_n (\pi_0 Y)^{E_n} = \varprojlim_n [E_n, Y].$$

Hence E is the ordinary colimit in Hot. Nevertheless, the ind-object (E_n) is not essentially constant. If it were isomorphic to a constant object Z, its functor of maps to constant objects would imply $Z \simeq E$. An inverse morphism from the constant E to (E_n) factors through some E_n, by the defining formula for maps in an ind-category. Its composite with the cocone would make the identity of $\mathbf{N}$ factor through a finite set on π_0, which is impossible.

The same example explains the footnote about the colimit functor on presheaves. Let F_n be the constant presheaf with value E_n on $\mathcal{A}$, and F the constant presheaf with value $\mathbf{N}$. Then

$\operatorname{colim} F_n = F$. The identity of F has no lift to a morphism from the constant ind-object F to (F_n): such a lift would factor through one finite F_n, again contradicting evaluation at any object of $\mathcal{A}$. Thus

$$\mathrm{Ind}(\widehat{\mathcal{A}}) \longrightarrow \widehat{\mathcal{A}}, \qquad (F_n) \longmapsto \operatorname{colim} F_n,$$

is not full.

A further qualification is needed for the displayed comparison with Ch_∞. A filtered homotopy colimit $K = \operatorname{hocolim} K_n$ is tested by a *compact* homotopy type A through

$$\operatorname{colim}_n [A, K_n] \simeq [A, K].$$

Finite-category homotopy types are not all compact. Take $A = BC_2 = \mathbf{RP}^\infty$ and $K_n = \mathbf{RP}^n$. Their union is A, but the identity of A cannot factor, even up to homotopy, through any K_n: it would kill the nonzero mod-2 cohomology in degrees above n (Hatcher, 2002, Theorem 3.19). Thus the finite-category definition of Hot_0 alone does not justify the comparison in manuscript page 10. For the particular path system, one must either prove the required factorization on these test categories by its special categorical structure, or impose an appropriate finite-test/compactness hypothesis. We do not replace that missing model-specific argument by a general assertion about filtered homotopy colimits.

Dans les deux cas CH et Ch (qu'on va désigner par le symbole commun $C_\infty(a,b)$), la [page 11]
fonctorialité de cet objet $C_\infty(a,b)$ (même au sens le plus faible, comme objet de $\mathrm{Hot}_0^\wedge$) par rapport à a et b (même au sens le plus faible, comme foncteur $\mathcal{M}^o \times \mathcal{M} \longrightarrow \mathrm{Hot}_0^\wedge$ et pas même $(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1}) \longrightarrow \mathrm{Hot}_0^\wedge$) n'est pas évidente du tout sur la définition directe en termes du système inductif explicite dans Cat

$$C_\infty(a,b) \;=\; (\mathrm{Ch}_{\tau_0^{2n}}(a,b))_{n\in\mathbf{N}},$$

car une flèche $a' \longrightarrow a$ ou $b \longrightarrow b'$ dans $\mathcal{M}$, et même qu'elle serait dans Σ, ne définit aucunement une flèche dans Cat

$$\begin{array}{rcl} & C_\infty(a,b) & \longrightarrow \quad C_\infty(a',b) \\ \text{ou} \;\; & C_\infty(a,b) & \longrightarrow \quad C_\infty(a,b'). \end{array}$$

Cette difficulté rappelle celle qu'avait à surmonter Quillen, dans Homotopical Algebra, pour affronter la différence entre 'left homotopies' et 'right homotopies'. Je me rappelle comment on peut néanmoins définir de telles flèches, comme résultant d'une flèche (beaucoup plus riche) de 'composition' de chemins entre a' et a, et a et b, ou entre a et b, et b et b', soit en p. 8
général donc l'accouplement [page 12]

$$C_\infty(a,b) \times C_\infty(b,c) \longrightarrow C_\infty(a,c),$$

dont la définition directe n'est pas moins problématique, et ne peut en tous cas *pas* se donner comme une flèche dans Cat lui-même, ou dans Ind Cat. Interprétant p. ex. $\mathrm{Ch}_\infty(a,b)$ en termes de chemins *infinis* (des deux cotés) de a à b, en un sens évident (mais 'essentiellement finis' des deux cotés), on peut définir aussi

$$\mathrm{Ch}^-_\infty(a,b), \qquad \mathrm{Ch}^+_\infty(a,b)$$

comme les sous-catégories pleines formées respectivement de chemins qui sont totalement dégénérés en degrés ≥ 0 (i.e. $a_0 \longleftarrow a_1 \longrightarrow a_2 \longleftarrow a_3 \cdots$ sont tous des identités), ce qui revient aux chemins infinis seulement à gauche, i.e. s'arrêtant à a_0, et symétriquement pour $\mathrm{Ch}^+_\infty(a,b)$. Même tabac pour les CH^+, CH^-.

Ceci dit, on a ceci (revenant à la notion C_∞ ambivalente) :

a) les inclusions pleinement fidèles [page 13]

$$C^-_\infty(a,b) \hookrightarrow C_\infty(a,b) \hookleftarrow C^+_\infty(a,b)$$

sont des isomorphismes dans Ind Cat hot resp. dans Hot, et

b) On a bel et bien une composition des chemins

$$C_\infty^-(a,b) \times C_\infty^+(b,c) \longrightarrow C_\infty(a,c).$$

L'*associativité* dans le cas du CH n'offre d'ailleurs pas de difficulté, à cause de la possibilité de se ramener aux

$$C_m(a,b) \times C_n(b,c) \times C_p(c,d) \rightrightarrows C_N(a,d)$$

pour N grand, et prouver l'egalité comme flèches dans Cat hot – ce qui résultera du lemme d'homotopie sur les intervalles dans Cat. Dans le cas du Ch_∞ par contre je n'ai pas trouvé de démonstration. (Faute seulement d'avoir essayé sérieusement de m'y mettre? Je n'ai pas trop envie, vu qu'il se pourrait que Hot $\longrightarrow \mathrm{Hot}_0^\wedge$ soit pleinement fidèle, auquel cas le travail devient inutile …)

Commentary · Later developments [ms. 13]

The path model and the localization it is meant to represent

Grothendieck is trying to make concatenation independent of an arbitrary placement of the middle endpoint. Moving one path to the left and the other to the right produces an actual functor; the homotopy equivalences from these one-sided path categories then give composition in Hot. For the ind-object CH_∞, every finite collection of paths and reparametrizations can be handled at one larger stage. His interval homotopies therefore give the asserted associativity in Ind Cat hot. For the realized Ch_∞, the text explicitly leaves the associativity proof unfinished. An eventual equality at every finite stage is not, by itself, a specified compatible homotopy on the whole realization.

The commentary on Chapter V, manuscript p. 12 ("What composition in an ind-category actually asserts") explains the relevant ind-Hom formula. Here it gives the exact associativity argument. Put $\mathcal{A} = \mathrm{Cat\,hot}$ and write $C_m(a,b)$ for the finite stages. Morphisms from the triple product into $\mathrm{CH}_\infty(a,d)$ are computed by

$$\varprojlim_{m,n,p} \varinjlim_{N} \mathrm{Hom}_{\mathcal{A}}\big(C_m(a,b) \times C_n(b,c) \times C_p(c,d), C_N(a,d)\big).$$

For each fixed (m,n,p), concatenate the three finite words before returning to the chosen standard shape. Parenthesization does not change this concatenated word; it changes only the finite reparametrization and the inserted identity padding. At a common larger stage the interval-homotopy lemma identifies those reparametrizations; its relative endpoint condition is the one explained in "Even indices and reparametrization relative to the ends" and "Sliding an even block gives a natural transformation" in the commentary on Chapter V, manuscript pp. 14 and 21. Hence the two representatives have the same class in the displayed filtered colimit for every (m,n,p), which proves equality in the inverse limit. No choice of these finite homotopies compatible as (m,n,p) varies is required for this equality of ind-morphisms. Concatenating an identity path similarly adds only identity padding, so the same argument proves the strong unit equations for CH_∞. This is the precise advantage of the ind-object in the present argument.

The general target of this construction was already available in a different model. The hammock localization of Dwyer and Kan (1980a, §2; Propositions 2.2 and 3.1) is a simplicial category whose composition concatenates and reduces hammocks; its category of components is $\mathcal{M}[W^{-1}]$. Thus it supplies coherent mapping spaces, with strictly associative composition at that model level, for a relative category. These statements concern the hammock construction. They do not identify its simplicial sets with Grothendieck's fixed-type path categories, whose transformations and stabilization conventions must be compared separately.

This distinction continues the commentary on Chapter V, manuscript p. 27 ("From components of path categories to mapping spaces"). Equality of π_0, agreement on finite tests, and a weak equivalence of the mapping spaces are three different assertions. We have not located in the sources checked a comparison theorem stated for exactly the bilateral stabilization and unrestricted natural transformations used here. The model-specific comparison, including compatibility with concatenation, is consequently left identified as an unfinished verification in these pages, not advertised as a new open problem in homotopy theory. The abstract bifunctor construction above is complete once its stated associativity and unit hypotheses hold.

2 Comparaison avec Quillen

Il se pose un certain nombre de questions sur ce foncteur $(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1}) \longrightarrow \mathrm{Hot}$, et sa relation aux constructions classiques, notamment à la théorie de Quillen. Je vais maintenant noter

$$\mathrm{Ch}^{\Sigma}_{\infty}(a,b) \;=\; \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}}(a,b) \quad \text{ou simplement}$$
$$\mathrm{Hom}\,\mathrm{ext}(a,b).$$

1°). Soient $f, g : a \rightrightarrows b$ dans $\mathcal{M}$, alors (si $\mathcal{M}$ est une 'model category') Quilllen définit un ensemble

$$\pi_1(a,b;f,g) \qquad \text{ou} \qquad \pi_1(f,g)$$

des classes d'homotopie de f à g, de telle façon que $\mathrm{Hom}_{\mathcal{M}}(a,b)$ devient l'ensemble des objets d'un groupoïde, dont les flèches sont les éléments de $\pi_1(f,g)$, pour une notion de composition naturelle. Je suspecte que l'on a

$$\pi_1(f,g) \;=\; \mathrm{Hom}_{\underbrace{\Pi(\underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b))}_{\substack{\text{groupoïde}\\ \text{fodamental}\\ \text{de } \underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b)}}}(f,g),$$

quand f et g sont considérés comme des 'chemins infinis' (en insérant chaque $f : a \longrightarrow b$ dans

$$\cdots \underbrace{a_{-2}}_{=a} \xleftarrow{\mathrm{id}} \underbrace{a_{-1}}_{=a} \xrightarrow{f} \underbrace{a_0}_{=b} \xleftarrow{\mathrm{id}} \underbrace{a_1}_{=b} \longrightarrow \cdots \;)$$

Commentary · Exposition [ms. 14]

Homotopies as arrows in a mapping groupoid

Grothendieck's question asks for the first layer of the mapping space, where homotopies themselves become morphisms. Its modern formulation begins with a derived mapping space $\mathrm{Map}^h(a,b)$, and takes its fundamental groupoid. The objects corresponding to actual maps $a \to b$ form a full subgroupoid. They need not meet every component until the usual cofibrancy and fibrancy hypotheses, or replacements, have been imposed.

For a cofibrant object a and a fibrant object b in a simplicial model category, $\mathrm{Map}(a,b)$ is a Kan complex. A simplicial path from f to g represents a homotopy; two such paths give the same arrow of its fundamental groupoid precisely when they are homotopic relative to their endpoints. Concatenation in the groupoid is defined by filling an inner two-dimensional horn. Different fillers give the same class because the Kan conditions provide the requisite relative homotopy. Constant paths give identities and reversed paths give inverses. Consequently

$$\mathrm{Hom}_{\Pi\,\mathrm{Map}(a,b)}(f,g)$$

is the exact object Grothendieck wants, not the set of equivalence classes of maps $a \to b$.

Quillen's original construction has precisely the cofibrant-source and fibrant-target qualifications: see Chapter I, §2, Proposition 1.2.1 in the checked retypesetting; Chapter II, §2 identifies this groupoid with the fundamental groupoid of the simplicial mapping complex (Quillen, 1967). For a general model category, Dwyer and Kan (1980b, Propositions 4.4–4.5 and 4.8) compare resolutions with the hammock mapping spaces and with the simplicial enrichment when it exists. These results justify the same description using Map^h after replacements. The identification with $\Pi\mathrm{Ch}^W_{\infty}(a,b)$ still requires the particular path-model comparison just discussed. The source's conjectural equality should therefore be stated as a natural equivalence of groupoids after that comparison, not as a literal equality of two chosen constructions.

[page 15]

Donc la définition des $\pi_1(f,g)$ (si c'est bien ça) se prolonge de façon naturelle aux chemins infinis (essentiellement finis) arbitraires.

2°) Supposons que $\mathcal{M}$ admette un objet à la fois initial et final $0_{\mathcal{M}}$ ($\mathcal{M}$ 'ponctuée'). Je dis qu'alors toutes les catégories $\underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b)$ sont ponctuées de façon naturelle : à savoir, par le chemin infini de a à b défini (comme plus haut) à partir de $f = 0_{a,b} : a \longrightarrow b$, l'application nulle de a à b. On a envie alors de regarder les $\underline{\mathrm{Ch}}^{\Sigma}_{\infty}(a,b)$ comme définissant des éléments dans le $\mathrm{Hot}_\bullet$ ('Hot ponctué'). C'est plus fin que de dire qu'on se donne une classe neutre

$$e \longrightarrow \mathrm{Ch}(a,b) \qquad \text{dans Hot},$$

car le foncteur de $\mathrm{Hot}_\bullet$ dans Hot (ou dans $e\backslash\mathrm{Hot}$) n'est pas fidèle. Le problème, c'est de définir les flèches

$$\mathrm{Ch}(a,b) \times \mathrm{Ch}(b,c) \longrightarrow \mathrm{Ch}(a,c)$$

dans $\mathrm{Hot}_\bullet$, et pas seulement dans Hot, et avec associativité (et les unités) dans $\mathrm{Hot}_\bullet$ également. Si on applique la définition avec $\mathrm{Ch}^-(a,b)$ et $\mathrm{Ch}^+(b,c)$, le composé des éléments nuls $a \xrightarrow{0} b$ de $\mathrm{Ch}^-(a,b)$ et $b \xleftarrow{\mathrm{id}} b \xrightarrow{0} c$ de $\mathrm{Ch}^+(b,c)$ est l'élément de $\mathrm{Ch}_\infty(a,c)$ que voici p. 10 [page 16]

$$\begin{array}{ccccccccccccccc}
 & & a_{-2} & \longleftarrow & a_{-1} & \longrightarrow & a_0 & \longleftarrow & a_1 & \longrightarrow & a_2 & & \cdots & & \\
 & & & & & & \| & & & & & & & & \\
\mathrm{id} & \cdots & a & \xleftarrow{\mathrm{id}} & a & \xrightarrow[0]{} & b & \xleftarrow{\mathrm{id}} & b & \xrightarrow[0]{} & c & \xleftarrow{\mathrm{id}} & \cdots & \mathrm{id} &
\end{array}$$

ce n'est *pas* l'élément nul de $\mathrm{Ch}_\infty(a,c)$! Mais on a des flèches canoniques

$$e^-_{a,b} \circ e^+_{b,c} \rightleftarrows e_{a,c} \qquad \text{dans } \mathrm{Ch}_\infty(a,c)$$

qui sont l'identité sur tous les sommets sauf celui d'indice 0. Il y a un sorite à développer, comme quoi une paire

$$(f : K_\bullet \longrightarrow L_\bullet, c)$$

d'un foncteur entre catégories ponctuées, et d'un chemin $c : f(e_{K_\bullet}) \longrightarrow e_{L_\bullet}$, définit dans Hot une flèche canonique

$$\mathrm{hot}_\bullet(K_\bullet) \longrightarrow \mathrm{hot}_\bullet(L_\bullet)$$

avec transitivité si on a

$$(f' : L_\bullet \longrightarrow M_\bullet, c' : f'(e_{L_\bullet}) \longrightarrow e_{M_\bullet}),$$

Commentary · Exposition [ms. 16]

The path needed to preserve a basepoint

Grothendieck wants the zero map to remain a specified point throughout composition, because its loop group will define suspension and loops. A point chosen in the unpointed homotopy category does not contain the required information. For example, based maps $S^1 \to BG$ are classified by elements of a group G, whereas free homotopy identifies conjugate elements. For $G = S_3$, the two distinct transpositions (12) and (13) give different based classes and the same free class. This proves the asserted failure of faithfulness of the forgetful functor.

The proposed extra datum is exactly the right one at the level of spaces. For pointed CW spaces (K,k) and (L,l), a map $f : K \to L$ together with a path c from $f(k)$ to l defines a morphism in the pointed homotopy category. To see this without suppressing the path, attach an interval to K at k and point the resulting space K^+ at its free endpoint. Define $F : K^+ \to L$ by f on K and by c on the interval. The collapse $r : K^+ \to K$ is a pointed homotopy equivalence. The required morphism is $[F][r]^{-1}$. The same construction applies to nerves of pointed categories after geometric realization, because a vertex inclusion is a cofibration.

For a second pair (g,d), the composite is represented by $(gf, (gc) * d)$. Indeed, the space obtained by attaching two successive intervals to K maps to the target by f, then the first path followed

under g, then d. Collapsing the subdivision of the two intervals identifies this representative with the single attached interval carrying $(gc) * d$. Repeating this argument gives associativity: the increasing piecewise-linear reparametrizations of a concatenated interval are homotopic relative to the two endpoints. Homotopic choices of c relative endpoints give the same pointed morphism. Arbitrary paths with the same endpoints need not do so; their difference can act through the fundamental group.

The specific claim immediately before this construction is stronger and is false as written, but it has a simple canonical repair. Denote the displayed concatenation by P and the standard zero path from a to c by Q. Then P has b at vertices $0, 1$, while Q has c at both. A transformation cannot be the identity at every vertex except 0: the backward identity edge between 1 and 0 forces its components at those two vertices to agree. Instead define $\alpha : P \to Q$ by

$$\alpha_0 = \alpha_1 = 0_{b,c}, \qquad \alpha_i = \mathrm{id} \quad (i \neq 0, 1).$$

The square at $-1 \to 0$ commutes because a composite with a zero map is zero; the square at $1 \to 0$ compares two identity edges; and the square at $1 \to 2$ says $\mathrm{id}_c \, 0_{b,c} = 0_{b,c}$. Every other square is an identity square. Since transformations of paths are unrestricted, this is an admissible canonical comparison for every W containing identities. It supplies the needed basepoint path without a reparametrization or an additional choice. A reverse arrow with zero components would force $\mathrm{id}_c = 0_{c,c}$ at the last square and is not available in general. This repairs the single basepoint comparison; associativity and the other coherence conditions for the particular pointed composition still require verification.

[page 17]

d'où

$$f'' = f' \circ f : K_\bullet \longrightarrow M_\bullet, \qquad \text{et} \quad c'' : f''(e_{K_\bullet}) \longrightarrow e_{M_\bullet}),$$

définit ainsi

$$f'(f(e_{K_\bullet})) \xrightarrow{f'(c)} f'(e_{L_\bullet}) \xrightarrow{c'} e_{M_\bullet}$$
$$\underbrace{\qquad\qquad\qquad\qquad}_{c'' \text{ par définition}}$$

? Cela nous donne donc bel et bien des accouplements dans $\mathrm{Hot}_\bullet$ (et aussi dans $\mathrm{Cat}\,\mathrm{hot}_\bullet$?). Il est difficile de croire qu'ils pourraient ne pas être associatifs! D'ailleurs $\mathrm{Hot}_\bullet$, avec le produit cartésien ordinaire ([4]), satisfait à l'axiomatique de tantôt pour $\mathcal{H}$, et $\mathrm{Hot}_\bullet \longrightarrow \mathrm{Hot}$ est compatible avec ce produit ([5]), et avec π_0.

Commentary · Exposition [ms. 17]

Why the monoidal unit changes in the pointed theory

The two footnotes identify a genuine change of monoidal structure. In pointed spaces the Cartesian unit is the one-point space, and

$$\mathrm{Hom}_{\mathrm{Hot}_\bullet}(*, X) = \{*\}.$$

It cannot recover the components of X, so the Cartesian structure does not instantiate the earlier convention $\pi_0(X) = \mathrm{Hom}(I, X)$. With the smash product, the unit is S^0, and

$$\mathrm{Hom}_{\mathrm{Hot}_\bullet}(S^0, X) = \pi_0(X).$$

Thus the corrected enrichment uses smash composition

$$C(a, b) \wedge C(b, c) \longrightarrow C(a, c).$$

The identity is the pointed map $S^0 \to C(a, a)$ carrying the nonbasepoint to 1_a and the basepoint to zero. Composition must kill either zero input, not merely the pair of zero inputs. That is the additional requirement hidden by a single basepoint comparison for a Cartesian product. The standard pointed simplicial model of derived mapping spaces has this absorbing-zero property, hence

4. Non, il faut prendre le produit $\xi_\bullet \divideontimes \eta_\bullet$ ($\xi_\bullet \times \eta_\bullet$, avec $\xi_\bullet \vee \eta_\bullet$ contracté en e).
5. Non, mais on s'en tire quand même …

its composition factors through the smash product. This is the familiar monoidal example in Kelly (1982, §1.1).

Forgetting basepoints is not strong monoidal for smash and Cartesian products. Nevertheless there is a natural quotient map

$$U(X) \times U(Y) \longrightarrow U(X \wedge Y)$$

and a unit map $* \to U(S^0)$ selecting its nonbasepoint. These make U lax monoidal: both associativity composites collapse exactly those tuples with a zero coordinate, and both unit composites leave the other coordinate unchanged. Composing these maps with smash composition recovers associative unpointed composition. The footnote "on s'en tire quand même" can therefore be made precise without asserting preservation of products. The symbol in the edition is the crossed product ⁂, explained there as collapsing the wedge, and is faithfully transcribed; its meaning here is the smash product.

Si tout ceci marche, on aurait donc une théorie des $\operatorname{Hom}\operatorname{ext}_\bullet$ à valeurs dans $\operatorname{Hot}_\bullet$; et sans doute aussi pour les CH_∞ comme objets de $\operatorname{Ind}\operatorname{Hot}_\bullet$, et non seulement de $\operatorname{Ind}\operatorname{Hot}$. Mais de p. 11
toutes façons on ne peut en général laisser tomber Cat hot, qui reste importante – donc il faudrait avoir des objets $\mathrm{CH}_\infty(a,b)$ dans $\operatorname{Ind}\operatorname{Cat}\operatorname{hot}_\bullet$, pour avoir une côte bien taillée.

Les $\pi_0(C(a,b))$ seront à présent des ensembles ponctués, et ce sera par l'image de l'application nulle $a \xrightarrow{0} b$, i.e. le composé

$$a \longrightarrow 0 \longrightarrow b$$

[page 18]

qu'il faut composer avec l'elément déduit du chemin tronqué

$$a \longrightarrow \underbrace{0}_{=a_0} \longleftarrow b \qquad \dots$$

Il faudrait montrer que 0 est aussi un objet neutre de $\mathcal{M}\Sigma^{-1}$, i.e. que

$$\pi_0(a,0) = \pi_0(0,a) = \{0\} \qquad \text{pour tout } a \in \operatorname{Ob}\mathcal{M}$$

et interpréter l'élément nul dans $\pi_0(a,b)$ comme l'application nulle dans $\mathcal{M}\Sigma^{-1}$. De façon générale, il faut des conditions qui assurent (sans que $\mathcal{M}$ soit forcément ponctuée) que

$$\mathcal{M} \xrightarrow{\gamma} \mathcal{M}\Sigma^{-1}$$

commute aux sommes finies, aux produits finis (donc $\gamma(\varphi_{\mathcal{M}}) = \varphi_{\mathcal{M}\Sigma^{-1}}$, resp. $\gamma(e_{\mathcal{M}}) = e_{\mathcal{M}\Sigma^{-1}}$) ([6]).

Commentary · Exposition [ms. 18]

Zero objects and finite sums after localization

Grothendieck needs the distinguished component to be an actual zero morphism of the localized category. At the level of ordinary localization this requires no additional compatibility condition on W. If z is initial in $\mathcal{M}$, write $i_x : z \to x$ for its unique maps. For every ordinary arrow $u : x \to y$ we have $ui_x = i_y$. If u is inverted, the same equality gives $u^{-1}i_y = i_x$ in the localization. An induction along any finite zigzag starting at z therefore identifies it with i_x. Thus z remains initial. The reversed argument preserves terminal objects. In particular a zero object remains zero, and $a \to 0 \to b$ is the zero morphism in $\mathcal{M}[W^{-1}]$. This is also the nullary case of descent of adjunctions through localization (Krause, 2021, Lemma 1.1.6).

The displayed path $a \to 0 \leftarrow b$ needs a different caution: its backward edge belongs to W only when $b \to 0$ does. The two forward zero maps $a \to 0 \to b$ always give the required ordinary zero morphism; inserting backward identity edges puts this composite into the permitted zigzag shapes.

6. à faire.

The preservation of the zero object therefore does not depend on the inadmissible shortened path or on the unresolved pointed composition.

For binary products there is a useful sufficient condition. Suppose $\mathcal{M}$ has finite products and W contains identities and is closed under their finite products. The product functor and the diagonal then descend to the localizations, as do the unit and counit of $\Delta \dashv \prod$. Naturality for inverse generators follows by multiplying the original naturality squares by the inverses of their W-edges; the triangle identities remain equalities after localization. Finally

$$(\mathcal{M}^n)[(W^n)^{-1}] \simeq (\mathcal{M}[W^{-1}])^n \qquad (n < \infty) :$$

a tuple of finite zigzags is represented by one tuple-shaped zigzag after padding the finitely many words by identities, and the relations are coordinatewise. Consequently the descended diagonal still has the descended product as right adjoint. This proves that localization preserves finite products. Reversing arrows gives the corresponding statement for finite coproducts. One can weaken closure by requiring only that the relevant products or coproducts of W-arrows become invertible.

For an additive category, closure of W under finite direct sums is already enough for the ordinary localized category to be additive; a five-lemma hypothesis is unnecessary for this conclusion (Krause, 2021, Lemma 2.2.1). Indeed, products and coproducts survive and the original canonical comparison from their sum to their product remains an isomorphism, giving biproducts. Addition is therefore defined by the usual diagonal–biproduct–codiagonal formula. The morphism -1_x remains an inverse for the addition on x, and its naturality extends over inverted arrows, so every Hom monoid is a group. A central k-action similarly extends. These conclusions concern ordinary Hom groups; constructing the proposed complexes of higher Hom information is the further problem on the next page.

Supposons enfin que $\mathcal{M}$ soit *additive,* et même k-linéaire (k un anneau fixé), i.e. on a de plus un homomorphisme d'anneaux

$$k \longrightarrow \mathrm{End}(\mathrm{id}_{\mathcal{M}}).$$

On doit supposer quelque compatibilité de Σ à la structure additive, p. ex. que la somme directe de deux flèches dans Σ est dans Σ, et sans doute quelque chose de plus fort, du genre 'lemme des cinq' pour Σ. (Mais on ne suppose pas forcément $\mathcal{M}$ abélienne, donc on aurait dû mal à formuler un lemme des cinq?)

Mais peut-être ici faut-il un axiome plus subtil, p.ex. (si $\underbrace{\Sigma}_{\text{suspension}}$ et Ω existent ([7])) que [page 19] l'ensemble des morphismes d'adjonction $\Sigma\Omega \longrightarrow \mathrm{id}$, $\mathrm{id} \longrightarrow \Omega\Sigma$ soit `[formé d']`isomorphismes, ce qui implique que Ω resp. Σ est pleinement fidèle. En tous cas, l'attente naturelle ici, c'est que $\mathrm{Ch}^{\Sigma}_{\infty}(a,b)$ soit non seulement dans $\mathrm{Hot}_\bullet$, mais même dans la catégorie $D^{\leq 0}(k\text{-Mod})$, sous-catégorie pleine de la catégorie dérivée $D^{-}(k\text{-Mod})$ formée des K tels que $H^i(K) = 0$ si $i > 0$. On peut aussi la noter $D_{\geq 0}(k\text{-Mod})$, l'interprétant en termes de complexes de 'chaînes' p. 12 plutôt que de cochaînes. Ici les accouplements doivent correspondre à l'opération $* = \overset{\mathbf{L}}{\otimes}_k$ dans $D_{\geq 0}(k\text{-Mod})$, admettant k comme objet unité, et

$$\pi_0(K_\bullet) \;\simeq\; H_0(K_\bullet).$$

Donc on doit avoir

$$\mathrm{Hom\,ext}(a,b) \overset{\mathbf{L}}{\otimes}_k \mathrm{Hom\,ext}(b,c) \longrightarrow \mathrm{Hom\,ext}(a,c),$$

où cette fois Hom ext désigne l'objet de $D_{\geq 0}(k\text{-Mod})$, au lieu d'un objet de Hot seulement, ou de $\mathrm{Hot}_\bullet$. On utilise les foncteurs

$$D_{\geq 0}(k\text{-Mod}) \longrightarrow \mathrm{Hot}_\bullet \longrightarrow \mathrm{Hot}$$

pour faire le lien avec la théorie ponctuée dans $\mathrm{Hot}_\bullet$, ou `[la]` théorie générale dans Hot. Si

7. cf. plus bas ce que ça signifie. [page 20]

on désigne par

$$\gamma : \ D_{\geq 0}(k\text{-Mod}) \ \longrightarrow \ \mathrm{Hot}_\bullet$$

le premier de ces foncteurs, il ne commute pas aux produits $*$ ($\overset{\mathbf{L}}{\otimes}_k$ et $\times$ respectivement), mais on a, pour $P_\bullet$, $Q_\bullet$ variable dans $D_{\geq 0}(k\text{-Mod})$, une flèche fonctorielle canonique

$$\gamma(P_\bullet \overset{\mathbf{L}}{\otimes}_k Q_\bullet) \xleftarrow{can.} \gamma(P_\bullet) \times \gamma(Q_\bullet)$$

'k-bilinéaire', ce qui montre que de donner des accouplements

$$P_\bullet * Q_\bullet \xrightarrow{c_k} R_\bullet,$$

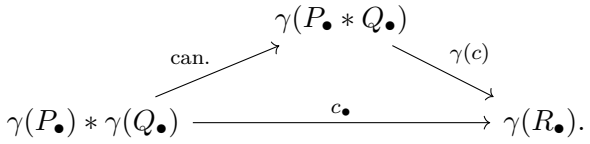
d'où

$$(*) \qquad \gamma(P_\bullet * Q_\bullet) \xrightarrow{\gamma(c_k)} \gamma(R_\bullet),$$

est l'un et l'autre plus fin que de se donner seulement

$$(*') \qquad \gamma(P_\bullet) * \gamma(Q_\bullet) \xrightarrow{c_\bullet} \gamma(R_\bullet),$$

vu qu'une donnée $(*)$ impliquera une donne $(*')$ par



Ceci vu, on s'attend bien à avoir une théorie *forte* des accouplements p. 13 [page 21]

$$\mathrm{Hom}\,\mathrm{ext}_k(a,b) \overset{\mathbf{L}}{\otimes}_k \mathrm{Hom}\,\mathrm{ext}_k(b,c) \ \longrightarrow \ \mathrm{Hom}\,\mathrm{ext}_k(a,c),$$

d'où on déduira les accouplements plus faibles dans $\mathrm{Hot}_\bullet$ (au lieu de $D_{\geq 0}(k\text{-Mod})$)

$$\mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(a,b) \times \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(b,c) \ \longrightarrow \ \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(a,c).$$

Commentary · Later developments [ms. 19]

A sufficient differential graded setting

Grothendieck now asks for more than additive Hom groups: he wants composition on their derived complexes before taking H_0. A precise sufficient setting is a differential graded model, for example complexes of modules with quasi-isomorphisms. Choose one cofibrant replacement $Qa \to a$ for each object, in the projective model structure; all complexes are fibrant. The complexes $\mathrm{Hom}^\bullet(Qa, Qb)$ then compose literally by differential graded composition, since the middle representative is the same in the two factors. This is the enhancement described in Keller (2006, §§2.2 and 3.2). The claim here is about this specified setting. The text does not state a set of hypotheses on an arbitrary additive relative category that would construct such an enhancement, and the existence of an adjoint equivalence $\Sigma \dashv \Omega$ alone does not give that construction.

For a commutative ring k, put

$$C(a,b) = \tau_{\leq 0}\,\mathrm{Hom}^\bullet_k(Qa, Qb) \simeq \tau_{\leq 0}\mathbf{R}\,\mathrm{Hom}_k(a,b).$$

The good truncation retains the negative degrees and the cycles in degree zero. It belongs to $D^{\leq 0}(k)$, or equivalently to $D_{\geq 0}(k)$ after reversing degrees. Differential graded composition gives

$$C(a,b) \overset{\mathbf{L}}{\otimes}_k C(b,c) \longrightarrow C(a,c),$$

using the Koszul interchange to put the factors into the usual composition order. Composing two cycles of degree zero gives a cycle, and the sum of two nonpositive degrees is nonpositive, so composition restricts to the good truncations. Replacing the two input complexes by projective resolutions provides the indicated derived tensor map. Its associativity is inherited from differential graded composition and the coherent tensor comparison; its unit is $k \to C(a,a)$ carrying 1 to the identity. Since k is projective over itself,

$$\mathrm{Hom}_{D(k)}(k, C(a,b)) = H^0 C(a,b) = \mathrm{Hom}_{D(k)}(a,b).$$

The unqualified "ring" in the source needs attention at exactly this tensor product. Two left modules over a noncommutative ring do not have the displayed tensor product with values in left modules. One may assume k commutative, as above, or use the appropriate bimodule category. Moreover, the image of $k \to \mathrm{End}(\mathrm{id}_{\mathcal{M}})$ is central: any two natural endomorphisms commute by applying the naturality of one to a component of the other. Thus the scalar action already factors through a commutative image.

The connective complex produces a pointed space by Dold–Kan and forgetting the simplicial module structure. The convention is

$$\pi_n \mathrm{Map}^h(a,b) = H^{-n}\mathbf{R}\,\mathrm{Hom}_k(a,b) = \mathrm{Hom}_{D(k)}(a[n], b), \qquad n \geq 0,$$

with the first equality at the zero basepoint for $n > 0$. The higher positive Ext groups of two degree-zero modules are not positive homotopy groups of this mapping space. This is the grading distinction explained in the commentary on Chapter III, manuscript p. 12 ("Keeping all degrees of the mapping complex").

For a concrete calculation, take $a = k[0]$ and $b = k[1]$, with cohomological convention $K[1]^i = K^{i+1}$. The source is projective, so the ordinary Hom complex already computes derived Hom and is $k[1]$, concentrated in degree -1. Dold–Kan gives $K(k,1)$; there is one component, its fundamental group is the additive group of k, and every higher homotopy group vanishes. On the other hand $\mathrm{Hom}_{D(k)}(k[1], k[1]) = k$, exactly the group predicted by suspension representability. This example makes visible the information that would disappear if one kept only the ordinary zero map $k \to k[1]$.

Commentary · Exposition [ms. 20]

The bilinear comparison is lax monoidal

Grothendieck correctly asks for a comparison map in the direction needed to transport composition. It need not be an isomorphism. Let P and Q be nonnegative chain complexes and choose nonnegative projective resolutions $P' \to P$ and $Q' \to Q$. Their Dold–Kan simplicial modules $A = \Gamma P'$ and $B = \Gamma Q'$ have a degreewise map of pointed sets

$$U(A) \times U(B) \longrightarrow U(A \otimes_k B), \qquad (a,b) \longmapsto a \otimes b.$$

The shuffle equivalence identifies the normalized complex of $A \otimes_k B$ with $P' \otimes_k Q'$, which computes $P \otimes_k^{\mathbf{L}} Q$. This yields the source's comparison in $\mathrm{Hot}_\bullet$. It factors through the smash product because either zero input is sent to zero. For coherence, use the unital associative Alexander–Whitney comparison $\Gamma P' \otimes_k \Gamma Q' \to \Gamma(P' \otimes_k Q')$; its normalized map is inverse to shuffle up to chain homotopy. The derived construction is independent of the chosen projective replacements. Dold–Kan itself is Theorem 14.24.3 of The Stacks Project Authors (2026c); the tensor comparisons, their identities, and the induced monoidal structure on Γ are checked in Schwede and Shipley (2003, §2.3, especially equation (2.8)).

Already with $P = Q = k[0]$, the map on components is

$$k \times k \longrightarrow k, \qquad (x,y) \longmapsto xy,$$

which is not injective for any nonzero ring, since $(0,0)$ and $(0,1)$ have the same image. It nonetheless carries a tensor pairing to a pointed or unpointed pairing by composition. Thus "more information" means that the bilinear factorization is specified, not that the forgetful functor is fully faithful or that its monoidal comparison is invertible. The displayed Cartesian product is legitimate for this weaker pairing; to apply the enriched unit convention in pointed spaces one uses its smash factorization, as explained above.

3°) Revenant au cas $\mathcal{M}$ ponctuée sans plus, Quillen définit, pour a, b dans $\mathcal{M}$,

$$\pi_1(a,b) \;=\; \pi_1(a,b\,;0_a,0_b)$$

comme bifoncteur sur $(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1})$ (avec les notations de 1°)). Si ce que j'avance sans preuve dans 1°) est vrai, il doit s'ensuivre qu'on a un isomorphisme canonique

$$\pi_1(a,b) \;\simeq\; \pi_1(\mathrm{Ch}_\infty^\Sigma(a,b)),$$

où $\mathrm{Ch}_\infty^\Sigma(a,b)$ est vu comme objet de $\mathrm{Hot}_\bullet$, donc son π_1 est défini. On a en tous cas, sans aucune référence à des cofibrations et des fibrations, un bifoncteur à valeurs dans Groupes, composé dans

$$(\mathcal{M}\Sigma^{-1})^o \times (\mathcal{M}\Sigma^{-1}) \xrightarrow{(a,b)\longmapsto \mathrm{Ch}_\infty^\Sigma(a,b)} \mathrm{Hot}_\bullet \xrightarrow{\pi_1} (\text{Groupes}).$$

Si pour a fixé, b variable,

$$b \;\longmapsto\; \pi_1(a,b)$$

est représentable, on note l'objet qui le représente par $\Sigma(a)$ (*suspension* – ne pas confondre ce Σ avec $\Sigma \subseteq \mathrm{Fl}(\mathcal{M})$, qu'il vaut mieux maintenant noter W, comme 'weak equivalence'). Donc

[page 22]

$$\mathrm{Hom}(\Sigma a,b) \;\simeq\; \pi_1(a,b),$$

et Σa dépend fonctoriellement de a, sous réserve d'existence. S'il existe toujours, on a

$$\Sigma : \; \mathcal{M}W^{-1} \;\longrightarrow\; \mathcal{M}W^{-1}.$$

On définit de même, sous réserve d'existence,

$$\Omega : \; \mathcal{M}W^{-1} \;\longrightarrow\; \mathcal{M}W^{-1}$$

par

$$\mathrm{Hom}(a,\Omega b) \;\simeq\; \pi_1(a,b),$$

et si tous les deux existent, on a la formule d'adjonction

$$\mathrm{Hom}(\Sigma a,b) \;\simeq\; \mathrm{Hom}(a,\Omega b) \qquad (\simeq \pi_1 \mathrm{Hom}\,\mathrm{ext}_{\mathrm{Hot}_\bullet}(a,b)).$$

Commentary · Exposition [ms. 22]

Representability and its higher counterpart

The representability argument is complete once the pointed mapping bifunctor has been constructed. Let $\mathcal{C} = \mathcal{M}[W^{-1}]$ and write $F(a,b) = \pi_1 \operatorname{Map}^h(a,b)$ at zero. If $F(a,-)$ is represented by Σa, then a morphism $a \to a'$ induces $F(a',-) \to F(a,-)$; Yoneda gives $\Sigma a \to \Sigma a'$. Uniqueness in Yoneda proves the functor laws. The same argument in the other variable constructs Ω, and the two representing bijections give $\Sigma \dashv \Omega$. The appearance of $0_a, 0_b$ in the source's page-21 expression must be read as two copies of $0_{a,b}$, since both endpoints of a path lie in the same mapping space.

Representability of π_1 alone supplies only a statement on components of loop spaces. To obtain the displayed equivalences of whole mapping spaces one requires homotopy limits and colimits in the enhanced category. In a pointed category with derived mapping spaces and a homotopy pullback defining $\Omega b = 0 \times^h_b 0$, the defining universal property gives

$$\operatorname{Map}^h(a,\Omega b) \simeq * \times^h_{\operatorname{Map}^h(a,b)} * = \Omega \operatorname{Map}^h(a,b).$$

Dually, the homotopy pushout $\Sigma a = 0 \amalg^h_a 0$ gives the other equivalence. Both respect loop multiplication because concatenation is induced by the same pointed path construction. This is the additional structure behind the source's test (c), rather than a consequence of an arbitrary bijection on π_0.

Quillen proves the group-valued bifunctor and its two representabilities for a pointed model category in Chapter I, §2; the checked retypesetting labels the result Theorem 1.2 (Quillen, 1967). The same construction is built into pointed derivators, where the coherent square, rather than a square in the underlying homotopy category, defines suspension and loops (Groth, 2013, Definition 3.16 and Proposition 3.17). The unit being invertible makes Σ fully faithful; the counit being invertible makes Ω fully faithful. If both are invertible, they form an equivalence. Requiring both is the stability condition, not a prerequisite for the existence of the two functors in a pointed model theory.

The following paragraph also uses existence of the appropriate finite products and coproducts. With those in place, a natural multiplication on $\mathrm{Hom}(-, \Omega b)$ is, by Yoneda, a map $\Omega b \times \Omega b \to \Omega b$; the group equations follow by the same argument. Dually Σa is a cogroup. In the enhanced setting, a derived mapping-space functor carries homotopy products in its second variable, and homotopy coproducts in its first, to products of spaces. These are the products and coproducts that justify the source's compatibility with group objects. Existence of ordinary products in the localized category, without a comparison to the enhanced products, would not by itself prove the mapping-space statement. On a pointed derivator Ω is right adjoint to Σ as a morphism of derivators, so it preserves homotopy right Kan extensions, and Σ preserves homotopy left Kan extensions; this supplies the left and right exactness anticipated in footnote 8.

L'existence de p. 14

$$\Omega : \mathcal{M}W^{-1} \longrightarrow \mathcal{M}W^{-1}, \quad \text{ou de}$$
$$\Sigma : \mathcal{M}W^{-1} \longrightarrow \mathcal{M}W^{-1},$$

voire même des deux à la fois, apparaît comme une hypothèse ⓐ extrêmement forte sur $(\mathcal{M}, W)$. Cette condition *plus* celle ⓑ que les $\underline{\mathrm{Ch}}_\infty(a, b)$ soient homotopes à des catégories de Thomason-Kan, ou (ce qui revient au même, à peu de choses près?) que $Y \longmapsto \mathrm{Hom}_{\mathcal{M}(Y)W_Y^{-1}}(a_Y, b_Y)$ transforme Hot-équivalences de Cat en bijections (ne serait-ce que pour Y fixé …), semble le test crucial pour savoir si '$(\mathcal{M}, \Sigma)$ `[plutôt` $(\mathcal{M}, W)$`]` se prête à une théorie homotopique 'dans $\mathcal{M}$' '. Je doute que

ces conditions puissent être satisfaites pour $\mathcal{M}$ finie, sauf dans le cas trivial où $\mathcal{M} = \emptyset$, où $\mathcal{M}\Sigma^{-1}$ `[plutôt` $\mathcal{M}W^{-1}$`]` est équivalente à la catégorie ponctuelle. [page 23]

Commentary · Development (commentators') [ms. 23]

A finite test of the proposed axioms

Grothendieck suspects that his tests already force an infinite supply of objects and operations. This would be natural for a model of a nontrivial stable homotopy theory, but the tests actually written permit a finite pointed category. The following calculation isolates the missing requirement. We checked the cited localization and model-category sources and the commentary on the earlier chapters; we did not find this particular example discussed as a response to this passage. Its role is diagnostic, without a claim of priority.

A finite relative category satisfying tests (a) and (b). [ours] Let $\mathcal{E}$ have objects $0, x$, with 0 a zero object, $\mathrm{End}(x) = \{1_x, z\}$, and z the composite $x \to 0 \to x$, so $z^2 = z \neq 1_x$. All other Hom sets are singletons. This is the full subcategory of pointed sets on a singleton and a two-element pointed set. Take W to consist of identities. Then $\mathcal{E}[W^{-1}] = \mathcal{E}$, which is neither empty nor equivalent to a point.

We first compute the actual path category. With all backward edges identities, a bilateral eventually constant path is an eventually constant forward chain from a to b, with duplicated vertices at the backward steps. Composition of its finite nonidentity part defines a functor

$$q : \mathrm{Ch}^W_\infty(a, b) \longrightarrow \mathrm{Hom}_{\mathcal{E}}(a, b)_{\mathrm{disc}}.$$

Naturality with fixed endpoints makes the composite invariant under morphisms of paths. Choose the section s that inserts a morphism at the forward edge $-1 \to 0$, with identities elsewhere. Define a third functor R by replacing the part strictly before 0 by the constant value a, retaining the part from 0 onward, and using the composite $a \to a_0$ at the joining edge. The eventual identity tails make that composite well-defined. There is a natural transformation $R \to \mathrm{id}$: its early components

are the composites $a \to a_i$, and its later components are identities. There is also $R \to sq$: its early components are identities and its later components are the composites $a_i \to b$. All the squares commute by composition of forward chains; duplicated backward vertices have identical components. Thus

$$\mathrm{id} \longleftarrow R \longrightarrow sq, \qquad qs = \mathrm{id}.$$

This proves a categorical homotopy equivalence to the discrete Hom category. In particular every component is contractible, every π_1 is zero, and the path categories are homotopy equivalent to discrete categories, which are Thomason–Kan fibrant. This verifies the first formulation of test (b).

The second formulation is equally direct. Since W_Y contains only identities, no localization changes $\mathcal{E}(Y)$. A natural transformation between the constant diagrams a_Y, b_Y assigns one arrow $a \to b$ to each connected component of Y, because naturality along every arrow forces equality. Hence

$$\mathrm{Hom}_{\mathcal{E}(Y)}(a_Y, b_Y) \cong \mathrm{Hom}_{\mathcal{E}}(a,b)^{\pi_0 Y}.$$

Every Hot-equivalence induces a bijection here. Finally $\Sigma a = 0 = \Omega a$ represents the identically trivial group-valued bifunctor $\pi_1(a,b)$ in both variables. Test (a) holds, as do the mapping-space equivalences of test (c): all their terms are points. The nontrivial endomorphisms of x survive in π_0, so this is not the terminal exception.

The finite-limit hypothesis introduced on the next page would rule this example out for an elementary reason. If a category has finitely many arrows, say N, and all finite powers, then two distinct maps $a \to b$ would give at least 2^n maps $a \to b^n$ for every n, contradicting $2^n > N$. Thus every Hom set has at most one element. If the category is pointed, the zero maps make every Hom set nonempty, so it is equivalent to the terminal category. This is the finite form of the usual small-completeness argument; see Leinster (2014, Exercise 6.3.23(d)). The distinction is therefore exact: the stated tests do not imply the finiteness obstruction, while a finite pointed category with finite products is already trivial before localization.

Un troisième test ⓒ qui me paraît important (mais peut-être ça résulte du reste), c'est les formules dans $\mathrm{Hot}_\bullet$

$$(*) \qquad \begin{cases} \mathrm{Hom\,ext}_\bullet(\Sigma a, b) \simeq \mathrm{Hom\,ext}_\bullet(a, \Omega b) \\ \qquad\qquad\qquad\quad \simeq \Omega \mathrm{Hom\,ext}_\bullet(a,b), \end{cases}$$

où le $\mathrm{Hom\,ext}_\bullet$ désigne le Hom externe à valeurs dans $\mathrm{Hot}_\bullet$. Il faut le voir séparément (sous condition d'existence soit de Ω, soit de Σ) comme deux formules 'duales' l'une de l'autre,

$$(**) \qquad \begin{cases} \mathrm{Hom\,ext}_\bullet(a, \Omega b) \simeq \Omega \mathrm{Hom\,ext}_\bullet(a,b) \\ \mathrm{Hom\,ext}_\bullet(\Sigma a, b) \simeq \Omega \mathrm{Hom\,ext}_\bullet(a,b), \end{cases}$$

isomorphismes de *groupes* dans $\mathrm{Hot}_\bullet$. (**NB** Ωb est un objet groupe de $\mathcal{M}W^{-1}$, car il représente un contrafoncteur à valeurs dans Gr, et de même Σa est un cogroupe de $\mathcal{M}W^{-1}$. Le foncteur $y \longmapsto \mathrm{Hom\,ext}(a,y)$ doit commuter aux produits finis, et donc transformer objets groupes en itou. Et de même $x \longmapsto \mathrm{Hom\,ext}(x,b)$ doit transformer sommes finies en produits finis, donc cogroupes en groupes ([8]).)

Je me suis convaincu que les isomorphismes $(**)$ n'ont de chance d'exister que si on a la condition préalable ⓑ.

8. Le foncteur Ω doit provenir d'un endomorphisme du dérivateur $\mathbf{D} = \mathbf{D}_{\mathcal{M},W}$, et celui-ci doit être exact à gauche, et dualement pour Σ.

3 Essai de construction du foncteur Ω

p. 15
[page 24]

Il n'y a sans doute guère de chance de prouver quelque chose sur le foncteur

$$\Omega : \mathcal{M}W^{-1} \longrightarrow \mathcal{M}W^{-1},$$

sans tenir en mains une *construction* plus ou moins explicite de celui-ci. Je vais essayer, en m'inspirant de Quillen, de dégager des conditions générales sur $(\mathcal{M}, W)$ (plus générales que celles de Quillen) qui assurent l'existence de Ω, et permettent de le construire. L'idée géométrique remonte à Serre-Cartan. Il s'agit, pour un objet S de $\mathcal{M}$, de construire un objet $\mathrm{Ch}(S)$ *dans* $\mathcal{M}$, jouant le rôle d'un 'espace de chemins', et qui doit s'inclure dans un diagramme de factorisation de $\mathrm{diag}_S : S \longrightarrow S \times S$ (on suppose que dans $\mathcal{M}$ les $\varprojlim$ finies, donc les produits finis, existent)

$$\begin{array}{ccc} & & S \\ & & \downarrow{\scriptstyle i_S} \\ \mathrm{diag}_S & \searrow & \mathrm{Ch}(S) \\ & & \downarrow{\scriptstyle p_S} \\ & & S \times S \longleftarrow e. \end{array}$$

La propriété essentielle, c'est que p soit une *'fibration'* (dans un sens convenable, à dégager).

Commentary · Exposition [ms. 24]

Which factorization computes loops

Grothendieck returns to an actual path object because representability must eventually be proved by a construction. In a pointed model category, choose a fibrant replacement of S and a factorization

$$S \xrightarrow{\sim} PS \xrightarrow{(d_0,d_1)} S \times S$$

of its diagonal, with the first map a weak equivalence and the second a fibration. The fibre over $(0,0)$ computes ΩS. Fibrancy is part of the useful hypothesis: it makes the endpoint maps fibrations to a fibrant object, so their pullbacks compute the intended homotopy fibres. The first map may be chosen to be a trivial cofibration, but its weak-equivalence property is essential. At this point the source is still seeking its axioms, so this is a completion of the proposed construction, not a correction of a theorem claimed to have been proved.

Merely asking for the second map to be a fibration, even adding a cofibration for the first, would be insufficient. In pointed simplicial sets take a nontrivial Kan complex $S = K(A,1)$, let $PS = S \times S$, and let the second map be the identity. The diagonal is a monomorphism, hence a cofibration, and S is fibrant. Nevertheless the indicated fibre is a point, whereas the true loop space has components A. What fails is precisely the omitted weak equivalence $S \to PS$.

The representation can be read directly from mapping spaces. In a pointed simplicial model category, let T be cofibrant. Applying $\mathrm{Map}(T,-)$ to the displayed factorization gives a path object for the Kan complex $K = \mathrm{Map}(T,S)$: its endpoint map is a fibration by the simplicial model axiom and its first map is a weak equivalence between fibrant objects. Mapping out of T preserves the pullback that defines the fibre. Therefore

$$\mathrm{Map}(T, \Omega S) \simeq \Omega K, \qquad [T, \Omega S] \cong \pi_1 \,\mathrm{Map}(T,S).$$

The latter bijection is natural in both variables after replacements. It is independent of the chosen path object because all such objects compute the same homotopy pullback $0 \times^h_S 0$. For arbitrary model categories the corresponding statement is Quillen's path-object argument, using homotopy classes of left and right homotopies; the checked version gives it as equation (5) in the proof of Theorem 1.2 (Quillen, 1967, Chapter I, §2).

To replace the mapping space by $\mathrm{Ch}^W_\infty(T,S)$ in the final display of the manuscript requires its comparison with the derived mapping space. A path object in $\mathcal{M}$ and an interval of zigzags in Cat play complementary roles; they are not interchangeable definitions. The audit leaves the specific comparison visibly conditional while giving the completed argument in the model setting.

Commentary · Context [ms. 24]

Path objects before the full model structure

Grothendieck is looking for precisely the axioms needed to construct loops. This route had already been isolated by Brown (1973, §1, pp. 420–421), whose categories of fibrant objects retain weak equivalences, fibrations, and path objects without specifying cofibrations. Brown requires pullbacks of fibrations and of trivial fibrations to exist and remain of the same kind; every object is fibrant. His path-object axiom factors the diagonal by a weak equivalence followed by a fibration. These are sufficient structural data for the factorization lemma proved there, which replaces an arbitrary map by a weak equivalence followed by a fibration.

The comparison clarifies the two tentative choices on the next manuscript page. Fibrancy of S supplies part of the working setting, while the essential property of $S \to \mathrm{Ch}(S)$ is its being a weak equivalence. A cofibration condition on that arrow is not needed for Brown's axioms. We use this established precedent to specify a sufficient framework, without suggesting that the manuscript cites Brown or that these axioms alone identify its particular bilateral path categories with derived mapping spaces.

[page 25]

À titre accessoire il peut être utile (peut-être) que i soit une 'cofibration' ([9]). On définit $\Omega(S)$ comme la fibre de $\mathrm{Ch}(S)$ sur $S \times S$ (par rapport à l'unique morphisme $e = e_{\mathcal{M}} \longrightarrow S \times S$, où $e_{\mathcal{M}}$ est l'objet nul de $\mathcal{M}$).

$$\begin{array}{ccc} \mathrm{Ch}(S) & \longleftarrow & \Omega S \\ \downarrow & & \downarrow \\ S \times S & \longleftarrow & e. \end{array}$$

9. et peut-être que $S \longrightarrow e$ soit également une fibration ?

C'est un objet de $\mathcal{M}$, qu'on regardera surtout comme un objet de $\mathcal{M}W^{-1}$. Il faut s'assurer qu'on a une bijection fonctorielle en T dans $\mathcal{M}W^{-1}$ p. 16

$$\underbrace{\pi_1 \underline{\mathrm{Ch}}^W_\infty(T,S)}_{\overset{\text{déf}}{=}\pi_1(T,S)} \simeq \underbrace{\mathrm{Hom}_{\mathcal{M}W^{-1}}(T,\Omega S)}_{\overset{\text{déf}}{=}\pi_0\underline{\mathrm{Ch}}^W_\infty(T,\Omega S)}.$$

Pour cela, il s'impose de comparer

$$\underline{\mathrm{Ch}}^W_\infty(T,\Omega S) \qquad \text{et} \qquad \underbrace{\Omega_{\mathrm{Hot}_\bullet}(\underline{\mathrm{Ch}}^W_\infty(T,S))}_{\overset{\text{'par définition' pour } \Omega \text{ de Hot}_\bullet}{=} \underline{\mathrm{Ch}}_\infty(\mathrm{Ch}^W_\infty(T,S);0,0)},$$

puisqu'il s'agit de construire une bijection entre leurs π_0.

Commentary · Examples [ms. 25]

The loop object of a complex, with its contraction

For complexes the proposed construction can be written down without any choices, and its fibre displays the sign in the loop shift. Use cochain complexes of k-modules with the model structure whose weak equivalences are quasi-isomorphisms and whose fibrations are degreewise surjections. Every complex is fibrant. For a complex B, define

$$(PB)^n = B^n \oplus B^n \oplus B^{n-1}, \qquad d(x,y,h) = (dx, dy, y - x - dh).$$

The third component of d^2 is $dy - dx - d(y - x - dh) = 0$, so this is a complex. The maps

$$i(b) = (b,b,0), \qquad p(x,y,h) = (x,y), \qquad r(x,y,h) = x$$

are chain maps, pi is the diagonal, and p is a degreewise surjection. To verify the required weak equivalence, define the degree -1 homotopy $H(x,y,h) = (0,h,0)$. Direct calculation gives

$$(dH + Hd)(x,y,h) = (0, y - x, h) = (\mathrm{id} - ir)(x,y,h), \qquad ri = \mathrm{id}.$$

Thus i is a chain homotopy equivalence. Its fibre over zero under p is the complex with terms B^{n-1} and differential $-d$, namely $B[-1]$. We have obtained $\Omega B = B[-1]$ and, dually, $\Sigma B = B[1]$.

For a cofibrant replacement $QT \to T$, the corresponding Hom complex satisfies

$$H^0 \mathrm{Hom}^\bullet(QT, B[-1]) = H^{-1} \mathrm{Hom}^\bullet(QT, B) = \pi_1 \mathrm{Map}^h(T,B),$$

which verifies the source's required bijection in this example. More than the bijection is visible: applying the good nonpositive truncation and Dold–Kan to these Hom complexes identifies the whole mapping space into $B[-1]$ with the loop of the mapping space into B. All maps used above commute with chain maps $B \to B'$, so the construction is functorial before localization. This is exactly the kind of explicit factorization that the closing paragraph asks us to find.

Chapter X

Comparaison de Cat avec $\mathbf{\Delta}^{\wedge}$

p.

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre X
Comparaison de Cat avec $\mathbf{\Delta}^{\wedge}$

p.

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Rappel des notions de D-asphéricité, et lien avec le 'localisateur fondamental $W_{\mathbf{D}}$'

p. 1 [page 1]

Pour un dérivateur **D**, il s'agit des notions suivantes (cf. I) :

Commentary · Exposition [ms. 1]

The homotopy shapes seen by a coefficient theory

Grothendieck now returns to the part of a derivator that can be expressed without choosing a particular category of coefficients. The relevant equivalences concern *constant* coefficients. Write $p_A : A \to e$ and $K_A = p_A^* K$. Then $u : A \to B$ belongs to $W_{\mathbf{D}}$ precisely when, for every $K, L \in \mathbf{D}(e)$, restriction gives a bijection

$$\mathrm{Hom}_{\mathbf{D}(B)}(K_B, L_B) \longrightarrow \mathrm{Hom}_{\mathbf{D}(A)}(K_A, L_A).$$

This recalls the distinction explained in the commentary on Chapter I, manuscript p. 61 ("Equivalence measured by constant coefficients"). It is not the assertion that $u^* : \mathbf{D}(B) \to \mathbf{D}(A)$ is an equivalence on arbitrary coherent diagrams. The distinction is what makes $W_{\mathbf{D}}$ a theory of homotopy *shapes*, while **D** still records the coefficients carried by those shapes.

The two comma categories retain different orientations. An object of A/b is an arrow $u(a) \to b$; an object of $b\backslash A$ is an arrow $b \to u(a)$. Accordingly, the fibre of $A/a \to B/u(a)$ over $\beta : b \to u(a)$ consists of factorizations of β by arrows $a' \to a$ lying over it. Its asphericity is the lifting criterion for smoothness, with the opposite criterion for properness. These are the categorical forms of the two base-change conditions, not assertions that the strict fibres A_b alone control base change. The later formulation is Maltsiniotis (2005, Theorem 1.2 and its dual).

(i) *Objet* **D**-*asphérique* de Cat (i.e. $Z \longrightarrow e$ est une **D**-équivalence). NB Je préfère dire '**D**-asphérique' au lieu de '**D**-contractile', qui prête parfois à confusion.

(ii) **D**-équivalence pour une flèche $f : X \longrightarrow Y$ dans Cat.

(iii) *Foncteur* $f : X \longrightarrow Y$ **D**-*asphérique* (i.e. les X/y sont **D**-asphériques) et **D**-*coasphérique* (les $y\backslash X$ sont **D**-asphériques). C'est ce que j'avais appelé '**D**-équivalence stricte' (ou 'costricte'), ou foncteur homologiquement **D**-cofinal (ou cohomologiquement **D**-cofinal), mais la terminologie 'asphérique' me semble la plus simple et la meilleure. (Pêchée dans PS, Pursuing Stacks.)

(iv) *Foncteur* $f : X \longrightarrow Y$ **D**-*lisse,* **D**-*propre* (i.e. tel que les fibres des foncteurs induits $X/x \longrightarrow Y/f(x)$, respectivement $x\backslash X \longrightarrow f(x)\backslash Y$, soient **D**-asphériques).

(v) **D**-*fibration* $f : X \longrightarrow Y$ (notion autoduale), i.e. telle que pour $Y'' \xrightarrow{u} Y'$ au dessus de Y, si Y'', Y' [sont] Hot-asphériques, alors $u_X : X'' \longrightarrow X'$ dans

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \longleftarrow & Y'' \end{array}$$

est une **D**-équivalence ([3]). La **D**-*fibration* est dite *triviale* (ou *asphérique*) si et seulement si ses fibres sont asphériques, ou encore si et seulement si f est asphérique, ou si et seulement si f est coaspérique. Cela signifie aussi que f est une **D**-équivalence *universelle,* i.e. que pour tout $Y' \longrightarrow Y$, $X' = X \times_Y Y' \longrightarrow Y'$ est une **D**-équivalence.

(vi) On a aussi des variantes affaiblies de la notion de **D**-fibrations, savoir les '**D**-*fibrations à gauche',* ou les f '*cohomologiquement* **D**-*fibrants',* et de même les '**D**-*fibrations à droite',* i.e. les f '*homologiquement* **D**-*fibrants'* ([4]). La notion 'gauche' ou cohomologique pour [page 2]

p. 2

3. On dira aussi que f est **D**-*fibrant.* (**NB** Il n'y a pas de notion '**D**-cofibrant' à ma connaissance.)

4. Nouvelle terminologie : fibrations cohomologiques (resp. homologiques) *faibles,* cf. I, pages 111-126 pour une révision corrigée des diverses notions de '**D**-fibrations'.

$f : X \longrightarrow Y$ signifie que les foncteurs $X_y \hookrightarrow X/y$ ($y \in \mathrm{Ob}\, Y$) et $X/y \longrightarrow X/y'$ (pour $(u : y \longrightarrow y') \in \mathrm{Fl}\, Y$) sont des **D**-équivalences. Et dualement avec les $X_y \hookrightarrow y\backslash X$, $y'\backslash X \longrightarrow y\backslash X$ pour la notion 'droite' ou 'homologique'. Les **D**-fibrations ne sont autres que les **D**-fibrations gauches (cohomologiques) universelles ou les **D**-fibrations droites (homologiques) universelles.

Commentary · Exposition [ms. 2]

Why triviality is a universal condition

The point of the fibration condition is to make the fibre calculations survive a change of base. The qualifier "universal" in the next sentence therefore carries real mathematical content; the earlier distinctions are reviewed in the commentary on Chapter I, manuscript p. 104 ("The universal criterion, with the quantifiers restored").

Here is a useful direct verification of the asserted triviality criterion. Suppose $f : X \to Y$ is a **D**-fibration and every X_y is $W_{\mathbf{D}}$-aspherical. After any change of base $B \to Y$, the same hypotheses hold. For $b \in B$, apply the definition to $e \to B/b$, selecting its final object. Both bases are ordinarily contractible, so

$$(X \times_Y B)_b \longrightarrow (X \times_Y B)/b$$

lies in $W_{\mathbf{D}}$. Hence every category on the right is $W_{\mathbf{D}}$-aspherical. The asphericity theorem gives $X \times_Y B \to B \in W_{\mathbf{D}}$. Conversely, a universal $W_{\mathbf{D}}$-equivalence has aspherical fibres by base change to points. It is a **D**-fibration: in its defining square the two projections to Y'' and Y' lie in $W_{\mathbf{D}}$, and $Y'' \to Y'$ does also because both bases are ordinarily contractible; two-out-of-three gives the required arrow between pullbacks. Applying the same comparison with Y/y and $y\backslash Y$ proves the two equivalent comma-category criteria. The ordinary contractibility of the test bases, specified in the manuscript, has not been replaced by $W_{\mathbf{D}}$-asphericity.

Finalement, toute cette kyrielle de notions se déduit de la seule notion de **D**-équivalence (ii). Si je désigne par

$$W_{\mathbf{D}} \subseteq \mathrm{Fl}(\mathrm{Cat})$$

l'ensemble de ces **D**-équivalences, on trouve ce que dans Pursuing Stacks j'ai appelé un 'localisateur' (localiseur) dans Cat ([5]). Dans Pursuing Stacks c'était une telle partie

$$W \subseteq \mathrm{Fl}(\mathrm{Cat}),$$

et non la donnée d'un dérivateur, qui était pris comme la donnée de départ fondamental [page 3]
pour faire de l'algèbre homotopique en tous genres. J'avais d'ailleurs développé une kyrielle de notions du même genre (i) à (vi), en partant de cette donnée W. Il se pourrait que les énoncés pour les notions que j'ai développé dans I, en termes d'un dérivateur, soient valables plus généralement en termes d'un localisateur W, dans la mesure du moins où ils gardent un sens. (Donc pour l'instant je mets à part les énoncés proprement cohomologiques ou homologiques, qui font directement appel à **D**.) Encore qu'il se pourrait que moyennant des conditions à expliciter sur W, qui impliquent que (Cat, W) définissent bel et bien un dérivateur $\mathbf{D}_{\mathrm{Cat},W} = \mathbf{D}$, on arrive encore à interpréter W comme un $W_{\mathbf{D}}$, de sorte que les énoncés cohomologiques et homologiques gardent un sens.

Commentary · Later developments [ms. 3]

Realizing a localizer by a derivator

Grothendieck asks for a coefficient theory realizing the proposed weak equivalences. A precise answer is available under an accessibility hypothesis. Fix a universe for small diagram shapes and a larger universe for coefficient categories, and call a fundamental localizer accessible when it is generated by a small set of functors. Cisinski (2006a, Theorem 5.2.15) constructs a cofibrantly

5. Mais attention, je suppose ici que le domaine Diag de **D** est Cat tout entier!

generated, left proper model structure on Cat with weak equivalences W. The model-category construction of derivators then gives

$$\mathbf{D}_W(A) = \mathrm{Cat}^{A^o}\big[(W^{A^o})^{-1}\big]$$

with both homotopy Kan extensions (Cisinski, 2003, Theorem 6.11 and its dual). The comparison with simplicial presheaves and the recovery of W from constant-coefficient tests are supplied by Cisinski (2006a, Lemma 4.2.13, Theorem 4.2.15, and Corollary 4.2.18). Thus $W_{\mathbf{D}_W} = W$. Accessibility is the hypothesis of this realization statement; it is not implicit in the definition of an arbitrary derivator.

Realization does not determine a pre-existing coefficient theory. For a complete and cocomplete category $\mathcal{C}$, the represented derivator satisfies

$$\mathrm{Hom}_{\mathcal{C}^{A^o}}(K_A, L_A) = \mathrm{Map}\big(\pi_0 A, \mathrm{Hom}_{\mathcal{C}}(K, L)\big).$$

Indeed, naturality says exactly that the component of a natural transformation is constant along every zigzag in A. For $\mathcal{C} = \mathrm{Set}$ or Ab, a Hom set with at least two elements detects whether $\pi_0(u)$ is bijective: a missing component prevents injectivity of precomposition, and two identified components prevent surjectivity. Both derivators consequently give the same localizer W_0 of component bijections. Their categories of coefficients are not equivalent, since Ab has a zero object and Set does not. The equality $W_{\mathbf{D}_W} = W$ answers the question on these pages without identifying the original $\mathbf{D}$ with $\mathbf{D}_W$.

J'ai envie d'abord de regarder dans Pursuing Stacks les axiomes que j'ai dégagé pour les localisateurs, et vérifier s'ils sont tous vérifiés pour $W_{\mathbf{D}}$. Ces conditions L 1 à L 5 sont [page 4] énoncées dans Pursuing Stacks, §64 (page 164).

L 1 Saturation de W_D. O.K.

Cela résulte formellement du fait qu'on a un foncteur

$$\mathrm{Cat} \xrightarrow{\varphi} C \qquad (= \mathrm{Cat}^o)$$

tel que pour $u \in \mathrm{Fl}(\mathrm{Cat})$, $u \in W_{\mathbf{D}} \iff \varphi(u)$ isomorphisme. (Prendre pour $\varphi(X)$ la sous-catégorie de $\mathbf{D}(X)$ formée des $\mathbf{D}$-coefficients *constants* sur X).

Commentary · Exposition [ms. 4]

Making the saturation argument literal

Grothendieck's argument is sound once the category of constants is represented with its coefficient labels retained. Define $C_{\mathbf{D}}(A)$ to have objects $K \in \mathbf{D}(e)$ and morphisms

$$C_{\mathbf{D}}(A)(K, L) = \mathrm{Hom}_{\mathbf{D}(A)}(K_A, L_A).$$

Restriction is the identity on these object labels. Thus $u \in W_{\mathbf{D}}$ if and only if $C_{\mathbf{D}}(u)$ is an *isomorphism* of categories, whereas the essentially full subcategory of constant objects gives the equivalent formulation using equivalences of categories. This avoids choosing representatives for coincident or isomorphic constant diagrams.

The construction proves strong saturation. If the image of u becomes invertible in $\mathrm{Cat}[W_{\mathbf{D}}^{-1}]$, the factorization of the contravariant functor $C_{\mathbf{D}}$ through that localization makes $C_{\mathbf{D}}(u)$ invertible; hence $u \in W_{\mathbf{D}}$. In *Pursuing Stacks*, §64, "saturation" is the weaker list of identities, two-out-of-three, and the two-sided-composite condition. The present argument proves more than that list requires (Grothendieck, 1983, §64, manuscript pp. 163–165).

L 2 "$\mathbf{\Delta}^1$ est W-asphérique sur e", i.e. pour tout X dans Cat, $X \times \mathbf{\Delta}^1 \longrightarrow X$ est dans W. Cela signifie aussi que $\mathbf{\Delta}^1 \longrightarrow e$ est une $\mathbf{D}$-fibration triviale, ce qui est bien le cas.

p. 3

L 3 Si $X \in \mathrm{Ob}\,\mathrm{Cat}$ a un objet final, alors $X \longrightarrow e$ est dans W.

En fait, nous exigeons pour les dérivateurs que ceci soit vrai en supposant seulement X asphérique (au sens absolu), et on en conclut même que $X \longrightarrow e$ est W-asphérique, i.e. $X \times Y \xrightarrow{\mathrm{pr}_2} Y$ est dans W pour tout Y dans Cat ([6]).

L 4 Si $f : X \longrightarrow Y$ est tel que $X/y \longrightarrow Y/y$ sont dans W, alors f est dans W.

C'est OK.

[page 5]

L 5 Si X, Y sont Cat-fibrées sur S, et $X \longrightarrow Y$ est un S-morphisme *cartésien,* induisant sur les fibres des $X_s \longrightarrow Y_s$ qui sont dans W, alors f est dans W.

Pour $W_{\mathbf{D}}$, c'est la proposition 6, page 71 de I.

Ces axiomes L 1 à L 5 (équivalent à L 1, L 3′ ($\mathbf{\Delta}^1 \longrightarrow e$ est dans W) et L 5) sont dans la nature d'axiomes de stabilité, assurant que W est assez grand. Il y a une autre condition qui limite W à ne pas être trop grand :

Commentary · Exposition [ms. 5]

Completing the reduction to three axioms

The shortened axiom list expresses a useful economy: integration of families, together with one interval, already forces the basic homotopy calculus. The reduction is sketched in *Pursuing Stacks*, §64. We spell it out, using the twisted-arrow construction of Chapter I rather than its mapping-cone presentation; compare the commentary on Chapter I, manuscript p. 73 ("The twisted-arrow category and its two projections").

Assume L 1, L 3′, and L 5. Applying L 5 to the constant fibrations $A \times [1] \to A$ gives L 2. Their two sections belong to W by two-out-of-three. A natural transformation $F \Rightarrow G$, viewed as a functor $A \times [1] \to B$, therefore shows that $F \in W$ if and only if $G \in W$. A category with a final or initial object retracts to that object, with its other composite naturally homotopic to the identity. Saturation gives L 3 and its opposite.

To pass between the two comma orientations, let $T(A)$ have objects arrows $a \to b$ and morphisms from $h : a \to b$ to $h' : a' \to b'$ pairs

$$r : a \to a', \qquad s : b' \to b, \qquad sh'r = h.$$

Both projections $T(A) \to A$ and $T(A) \to A^o$ are Grothendieck fibrations. Their fibres are $(a\backslash A)^o$ and A/b, respectively, and have final objects. L 5 consequently puts both projections in W. The construction is natural in A, so its two commutative squares and two-out-of-three give

$$u \in W \quad \Longleftrightarrow \quad u^o \in W.$$

This also establishes the final assertion L d).

Next let $u : A \to B$ have W-aspherical categories $b\backslash A$. The Grothendieck construction

$$E = \int_{b \in B} (b\backslash A) \longrightarrow B$$

is a fibration with these fibres, hence lies in W by L 5. Its objects are triples $(b, a, \alpha : b \to u(a))$. The forgetful functor $r : E \to A$ has the section $i(a) = (u(a), a, 1)$, satisfies $ri = 1$, and has a natural transformation $1_E \Rightarrow ir$ with component $(\alpha, 1_a)$. Thus r and i belong to W, and u does also. Opposite invariance proves the analogous assertion for A/b. Under the hypothesis of L 4, the final object of B/b makes every A/b aspherical by two-out-of-three, so L 4 follows. Conversely, L 1–L 5 include L 1 and L 5 and imply L 3′ by L 3. This proves the claimed equivalence of axiom systems. For the modern relative form, suppose $A \to B$ lies over C and every $A/c \to B/c$ is in W. Opposite invariance gives objectwise equivalences $(A/c)^o \to (B/c)^o$. These diagrams define fibrations over C^o, so L 5 applies to their Grothendieck constructions; their forgetful functors are deformation retractions to A^o and B^o. Thus u^o, and hence u, belongs to W. This yields the relative Theorem A axiom of Cisinski (2004, Definition 1.1.3). No model structure is used.

L a) Si $f \in W$, alors $\pi_0(f)$ est bijectif.

6. En fait L 2 $\Longrightarrow$ L 3.

Ceci n'est pas satisfait pour tous les $W_{\mathbf{D}}$ – p. ex. si $\mathbf{D}$ est le dérivateur *nul,* alors $W_{\mathbf{D}} = \mathrm{Fl}\,\mathrm{Cat}$! Mais pour les dérivateurs utiles en pratique, cela semble toujours satisfait.

Commentary · Examples [ms. 5]

A nonzero coefficient theory that forgets components

The additional condition L a) restricts the amount of homotopy theory the coefficients may forget. Even a nonzero derivator need not satisfy it. Take the complete lattice $\mathbf{2} = \{0 < 1\}$ and its represented derivator

$$\mathbf{D_2}(A) = \mathrm{Fun}(A^o, \mathbf{2}).$$

For nonempty A, the Hom sets between the two constant diagrams are the Hom sets of $\mathbf{2}$: all are singletons except $\mathrm{Hom}(1_A, 0_A) = \varnothing$. For $A = \varnothing$ there is one diagram and one morphism. Therefore

$$u : A \to B \in W_{\mathbf{D_2}} \quad \Longleftrightarrow \quad (A = \varnothing \iff B = \varnothing).$$

In particular the fold $e \amalg e \to e$ belongs to this localizer, although its map on components is not injective. This is the usual coarse localizer W_{-1} (Cisinski, 2006a, Examples 3.3.5–3.3.6). The terminal coefficient category gives the still larger localizer of all functors, exactly as Grothendieck observes. By contrast, coefficients with a Hom set of at least two elements detect components in the represented examples calculated above. His remark about useful examples is a comment on the intended coefficients, not an additional derivator axiom.

L b), c) Le foncteur

$$\mathrm{Cat} \longrightarrow \mathrm{Cat}\, W^{-1}$$

commute aux sommes finies et aux produits finis ([7]).

À examiner ce que ça signifie en termes concrets sur W – je n'ai pas vérifié que c'est O.K. pour les $W_{\mathbf{D}}$.

Commentary · Present state [ms. 5]

Finite products and sums already descend

The questions L b) and L c) have a formal answer under the axioms already retained. If $u : A \to B$ is in W, the map of constant fibrations over any C gives

$$u \times 1_C : A \times C \longrightarrow B \times C \quad \in W$$

by L 5. Applying L 5 over a discrete indexing category likewise shows that coproducts of arrows in W belong to W. These closure properties are also Cisinski (2004, Propositions 1.1.5 and 1.1.7).

Closure alone should be connected to the universal properties of the claimed products and sums. Put $H = \mathrm{Cat}[W^{-1}]$. The localization of $\mathrm{Cat} \times \mathrm{Cat}$ at the componentwise weak equivalences is $H \times H$: a pair of zigzags can be interleaved by holding the other coordinate fixed, and the commutation squares between the two coordinates remain valid after inversion. The diagonal and the product functor are homotopical, so their adjunction, including its unit and counit, descends to H. Thus the image of $A \times B$ represents the product of the images of A and B. Descending the adjunction between coproduct and diagonal proves the corresponding coproduct assertion. The nullary adjunctions give the initial and terminal objects as well. Consequently the canonical localization preserves finite sums and finite products. A Quillen structure is unnecessary here; the explicit closure argument supplies the homotopy invariance needed for the *ordinary* products and sums appearing in the question.

L d) Si $f : X \longrightarrow Y$ dans W, alors $f^o : X^o \longrightarrow Y^o$ aussi.
C'est le corollaire 2, page 75 de I.

7. Ça marche si (Cat, W) provient d'un triple de Quillen (avec des Q-fibrations, Q-cofibrations, Q-équivalences faibles).

Overview · VIII-X

Integration, spaces of morphisms, and the choice of weak equivalences.

These three chapters investigate how much of homotopy theory can be reconstructed from operations on ordinary categories, provided that one keeps the right forms of coherence. Chapter VIII isolates integration for relative 1-types. Chapter IX returns to the path categories introduced earlier, first to extract an abstract theory of composition and then to compare it with Quillen's constructions. The opening of Chapter X changes the point of departure: instead of beginning with a derivator and its coefficients, Grothendieck asks whether a class of weak equivalences of categories can itself generate the theory. The movement is from a concrete adjunction, through the structure of its spaces of morphisms, to the axioms that make such constructions possible.

These pages comprise all of Chapters VIII and IX and the opening of Chapter X, twenty-seven pages of the edition. Manuscript references must always include the chapter: they run through VIII.1–9, IX.1–25, and X.1–5. Our audit follows those separate sequences. Original statements, including the edition's retained slips and editorial interventions, remain distinguishable from the commentary. The two doubtful readings are settled against the edition; neither licenses a mathematical change to Grothendieck's text.

Relative groupoids and a genuine biadjunction.

The decisive construction of Chapter VIII is simple enough to write down on an object of the base. For $f : S \to T$ and a category G fibred in groupoids over S, set

$$(f_!G)_t = \Pi(t\backslash_T G).$$

The comma category records a point of G together with an arrow from t to its image in T; groupoid completion turns this record into a homotopy 1-type. Its use is not limited to an already fibred category. For any $X \to T$, the same construction gives a relative groupoid $R_T X$, and extension along $X \to R_T X$ is universal for functors into categories fibred in groupoids. Cartesian lifts construct the extension, while their uniqueness supplies the coherence.

This universal property completes the adjunction verification left unfinished in the manuscript. It proves an equivalence of Hom-groupoids, not only a bijection on isomorphism classes. Both triangle comparisons arise from the same extension-and-restriction construction. Transitivity of integration is then the uniqueness of the left biadjoint to composite restriction. The point of the 2-categorical formulation is thus practical: the automorphisms that would have been discarded by taking components are precisely what make the comparison usable in a further composition. The base-change theorem has the corresponding later expression in the theory of asphericity structures and smooth functors (Maltsiniotis, 2005, §5).

The final descent statements require a more careful separation. The proper or smooth hypotheses of Theorem 3 support its assertions about faithfulness and about the essential image described by coefficients that are *constant on each strict fibre*. They do not justify replacing this condition by the assertion that every vertical transport is an equivalence. A locally constant groupoid family can carry degree-two coherence even when its base is simply connected. We give a finite example over the fourteen-element face poset of the boundary of a tetrahedron. Every fibre is $B(\mathbf{Z}/2)$ and every transport is the identity, but a nonzero normalized 2-cocycle prevents the total category from admitting a section. Consequently it cannot be equivalent to a constant family. This disproves part 3 and the subsequent corollary as printed, while leaving the preceding fibrewise-constant criterion intact.

The repair is expressed by higher monodromy: equivalence classes of locally constant groupoid families with a fixed fibre are controlled by maps to its automorphism 2-type. A 1-connected base removes ordinary monodromy but can retain the degree-two class; 2-connectivity removes both (Polesello and Waschkies, 2005, Corollary 2.2.4 and Theorem 2.2.5, arXiv version). A separate, shorter counterexample shows that section groupoids alone do not

test all the proposed faithfulness conditions. The correct section tests must also be made after passing to the total category of the first coefficient. Both corrections concern the quantifiers and the coherence in descent, rather than the integration construction itself.

The general faithfulness question also admits an exact answer without smoothness or properness. For each arrow of T, form its category of factorizations through objects in the image of f. Restriction is faithful on transformations, fully faithful on Hom-groupoids, or an equivalence on Hom-groupoids precisely when these factorization categories are respectively nonempty, connected, or 1-connected. We prove this three-level criterion by the counit on representables and coherent density, alongside the known untruncated homotopical-epimorphism criterion (Groth and Šťovíček, 2016, Theorem 6.10, arXiv version).

What composition gives, and what it presupposes.

Chapter IX begins with a strong idea: composition of paths should explain endpoint functoriality. In a monoidal category $\mathcal{H}$, Hom objects $C(a, b)$ with associative composition and coherent units determine an ordinary category with morphisms $\mathrm{Hom}_{\mathcal{H}}(I, C(a, b))$. Inserting these elements into composition produces the bifunctor $C_0^{\mathrm{op}} \times C_0 \to \mathcal{H}$. The proof is formal once the hypotheses are in place. Neither symmetry nor an internal Hom is needed. This is the familiar passage from an enriched category to its underlying ordinary category (Kelly, 1982, §§1.1–1.3).

There are two limits to what this argument proves in the passage under review. First, the endpoint maps give isomorphisms only when those endpoint maps are invertible. The early corollary omits that hypothesis, and a discrete two-object example disproves the wording; the manuscript itself restores it on page 8. Second, the abstract argument assumes associative composition on the particular objects $\mathrm{Ch}_\infty(a, b)$. It does not construct that composition. Grothendieck sketches the finite-stage argument for the ind-object CH_∞ and explicitly leaves the corresponding realized-space verification unfinished.

The possible shortcut through finite tests does not solve this difficulty. The earlier definition of Hot_0 uses homotopy types of finite categories, not just finite CW complexes. The resulting restricted Yoneda functor is not conservative, by the general theorem on small families of unpointed tests, and it is not faithful, by nonconcreteness of the homotopy category (Arlin and Christensen, 2023, Theorem 2.1) and Freyd (1970, pp. 1, 5 of the reprint). Thus equality after passing to those presheaves cannot replace a compatible homotopy on the original mapping spaces. The commentary also distinguishes an ind-object, an ordinary colimit in Hot, and a homotopy colimit. An explicit increasing sequence of finite discrete sets has an ordinary colimit without being essentially constant as an ind-object. A finite-category test such as BC_2, on the other hand, is not compact with respect to all filtered homotopy colimits. These examples delimit the arguments that the comparison maps actually require.

Modern localization supplies the intended kind of mapping-space theory. The hammock localization has coherent mapping spaces and a component category equal to the ordinary localization (Dwyer and Kan, 1980a, §2). This identifies the modern object to which Grothendieck's construction should be compared; it does not by itself prove a comparison for his exact bilateral stabilization and transformation conventions. We keep that model-specific verification explicit. It is not described as a new open problem in the general theory of localization.

Pointed and linear refinements.

The later part of Chapter IX asks that the zero map become a specified basepoint in every mapping space. A basepoint in the unpointed homotopy category is insufficient: based homotopies remember data lost under free homotopy. The commentary constructs the pointed morphism associated to a map and a chosen path between basepoints, including its composition law. It also explains the significance of the manuscript's two corrective footnotes. The Cartesian unit in pointed spaces has only one map to every object; the unit that recovers all components is S^0 for the smash product. Composition must therefore annihilate either

zero input. Forgetting basepoints carries this structure to unpointed composition through a lax monoidal comparison.

Some questions at the level of ordinary localization have particularly direct answers. Initial and terminal objects survive localization without extra hypotheses, by naturality of their unique arrows along both ordinary and inverted generators. Suitable closure under finite products or sums lets the corresponding adjunctions descend. For an additive category, stability of the denominator class under finite direct sums already yields an additive ordinary localization (Krause, 2021, Lemma 2.2.1). The stronger request for complexes of morphisms requires an enhanced setting. We treat complexes of modules explicitly, with connective truncations of derived Hom, derived tensor composition, and the Dold–Kan comparison. This makes the grading visible: positive homotopy groups of the mapping space are negative cohomology groups of the Hom complex.

The suspension and loop objects are then representability statements. Representing the fundamental-group bifunctor yields an adjunction; the equivalences of whole mapping spaces arise from homotopy pushouts and pullbacks. An explicit path complex verifies the loop construction with its differential, its contraction, and its fibre. We also develop a finite counterexample to the proposed force of tests (a) and (b): the full category of pointed sets on one and two elements, with only identities inverted, satisfies both tests and even the displayed mapping-space identities. Its loop and suspension functors are zero, while its ordinary category is nontrivial. The missing strength is therefore not finiteness of the category alone. This calculation is marked as the commentators' development, without a priority claim.

Localizers and the information retained by coefficients.

The opening of Chapter X extracts a class $W_{\mathbf{D}}$ by testing maps of indexing categories against constant coefficients. It must be distinguished from the condition that restriction be an equivalence on all coherent diagrams. The localizer axioms express the stability of those constant-coefficient tests under interval homotopy, comma constructions, and integration of fibrations. We complete the reduction from the five displayed axioms to the shorter list, including the role of the twisted-arrow category in proving invariance under opposites. The finite-product and finite-sum questions are then settled by explicit closure arguments and descent of the product and coproduct adjunctions. No model structure is needed for these formal consequences.

Realization is a further theorem. For an accessible fundamental localizer, the later model structures and derivator constructions give a coefficient theory that realizes its weak equivalences (Cisinski, 2006a, Theorems 4.2.15 and 5.2.15); (Cisinski, 2003, Theorem 6.11 and its dual). This does not reconstruct an arbitrary original derivator from $W_{\mathbf{D}}$. The represented derivators of sets and of abelian groups have the same component-detecting localizer, although their coefficient categories are inequivalent. The represented truth-value lattice gives an equally concrete nonzero example whose localizer detects only emptiness, disproving any attempt to impose component detection on all nonzero derivators. These examples show precisely what Chapter X has isolated: the homotopy shapes visible to a theory, while its further algebra of coefficients remains additional structure.

Chapter XI

Hot-fibrations, foncteurs propres et foncteurs lisses etc. (dans Cat)

XI-XIIa · Chapter XI · manuscript pages 127–136

p. –

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XI

Hot-fibrations, foncteurs propres et foncteurs lisses etc. (dans Cat)

p. –

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.
Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :
[e-mail address removed]

G. Maltsiniotis

[Il n'y a pas de pages 1-126]

p. 1
[page 127]

1 Hot-fibrations dans Cat

Théorème 1. *Soit $f : X \longrightarrow Y$ dans* Cat. *Conditions équivalentes :*

Commentary · Exposition [ms. 127]

The universal meaning of a Hot-fibration

Grothendieck wants the ordinary pullback of categories to calculate a homotopy pullback whenever one of its sides belongs to a recognisable class. This is what makes a fibre a homotopically meaningful object, rather than an accident of a presentation. Write $BC = |NC|$, where N is the nerve, and let W_∞ consist of the functors inducing weak homotopy equivalences of classifying spaces. Condition a) says that

$$f^* : \mathrm{Cat}/Y \longrightarrow \mathrm{Cat}/X, \qquad (T \longrightarrow Y) \longmapsto (X \times_Y T \longrightarrow X),$$

preserves W_∞. The quantifier ranges over maps between *arbitrary* objects of Cat/Y, and not only over weak equivalences with target Y. It therefore includes every further change of base.

In the terminology of Rezk (1998, §2, Proposition 2.7), these are precisely the sharp maps in the Thomason model category. The identification uses the definition of sharpness and the right properness of that model category; it does not identify this class with the smaller class of its model fibrations. Every strict pullback square along such a map is homotopy cartesian. This interpretation also explains why composition and arbitrary base change will be available without choosing liftings.

There is a small but decisive quantifier in a') and a"). The point $e \longrightarrow Y'$ is arbitrary; it need not select the final, respectively initial, object whose existence is assumed. Over the interval $0 \longrightarrow 1$, both endpoint inclusions must therefore become weak equivalences after pullback. Asking only for the distinguished final endpoint, for every category having a final object, leads instead to the later notion of preproperness. Asking only over intervals is weaker again. These distinctions become the subject of Chapter XII, and they should not be erased by replacing every occurrence of "fibration" with a single modern definition.

a) Pour tout $Y'' \xrightarrow{p} Y'$ sur Y, d'où $X'' = X \times_Y Y'' \xrightarrow{q} X' = X \times_Y Y'$, si p est une Hot*-équivalence, q l'est aussi.*

a′) Comme a), avec $Y'' = e$, Y' ayant un objet final.

a″) Comme a), avec $Y'' = e$, Y' ayant un objet initial ([1]).

b′) Pour tout $y \in \mathrm{Ob}\, Y$, $X_y \longrightarrow X/y$ est une Hot*-équivalence, pour tout $y \longrightarrow y'$ dans Y, $X/y \longrightarrow X/y'$ est une* Hot*-équivalence; enfin ces conditions sont encore vérifiées pour $f' : X' \longrightarrow Y'$, pour tout changement de base $Y' \longrightarrow Y$ dans* Cat. *(En fait, il suffit de le demander pour la première des deux conditions, concernant $X_y \longrightarrow X/y$.)*

b″) Dual de b′), avec $X_y \longrightarrow y\backslash X$ et $y'\backslash X \longrightarrow y\backslash X$.

c) Pour tout faisceau d'ensembles localement constant F sur X, $f_*(F)$ est localement constant et se calcule 'fibre par fibre', de même pour $R^1 f_*(F)$ si F est un faisceau en groupes, et pour les $R^i f_*(F)$ si F est un faisceau abélien. De plus, on veut que ces conditions restent satisfaites après tout changement de base. ([2])

Démonstration. On a les implications ± tautologiques

$$\mathrm{a} \implies \mathrm{a}' \implies \mathrm{b}' \implies \mathrm{c},$$

1. garder a″) et b″), conditions duales de a′), b′) [a″) et b″) étaient barrés].

2. *c′) (version homologique). 1°) en termes d'images directes de groupoïdes localement constants, et 2°) en termes des $R^i f_!(L)$* [plutôt $L_i f_!(L)$], *pour L faisceau abélien localement constant.*

et il reste à prouver que c $\implies$ a. Comme a est autoduale, étant équivalente à a′, b′, elle est aussi équivalente aux conditions duales de celles-ci, a″ et b″ (cf. cor. 2, page 75). On est ramené à prouver que si c) est satisfait, et si on a

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow[p]{} & Y' \end{array}$$

cartésien dans Cat, si p est une Hot-équivalence, q aussi. On utilise le critère d'Artin-Mazur. p. 2

Commentary · Exposition [ms. 127]

Comma categories connect the geometric and coefficient tests

The comma categories in b') serve two purposes at once: they express a pointwise right Kan extension, and they supply canonical transition maps. We use $X/y = (f \downarrow y)$, with objects $(x, f(x) \longrightarrow y)$. Its strict fibre X_y is the full subcategory on identity arrows. A morphism $u : y \longrightarrow z$ gives a genuine functor $u_* : X/y \longrightarrow X/z$, so the coherence needed for transport is already present before localization.

Here is the short argument hidden in a') $\Rightarrow$ b'). Apply a') over Y/y, which has final object $(y, 1_y)$. The inclusion of the strict fibre over that object is $X_y \longrightarrow X/y$. Next work over Y/z and select its object (y, u). The resulting weak equivalence $X_y \longrightarrow X/z$ is the composite of $X_y \longrightarrow X/y$ and u_*. Two-out-of-three proves that $u_* \in W_\infty$. The same argument works after every base change.

The parenthesis in b') retains the transition condition on the original base; it says that only the fibre-to-comma condition needs to be imposed universally. Indeed, after $v : Y' \longrightarrow Y$, the comparison

$$X'/y' \longrightarrow X/v(y')$$

is a weak equivalence: both its source and target receive a weak equivalence from the same strict fibre $X_{v(y')}$. Comparing this square for an arrow of Y' with the corresponding arrow of Y then proves the required transition condition on Y'.

For a locally constant coefficient F, the value of the derived direct image at y is computed on X/y. Thus the preceding weak equivalences identify it with the cohomology of X_y, and identify its transition maps with isomorphisms. In degrees zero and one this means sections and torsors; for abelian coefficients it means every derived degree. The use of local coefficients is essential: untwisted integral homology alone cannot recognise the fundamental group or its action on higher information. The homological form of the categorical Whitehead criterion is stated explicitly in Thomason (1980, Proposition 2.4); the cohomological formulation used here is the criterion to which Grothendieck gives the name Artin–Mazur.

1°) Si F est un faisceau d'ensembles localement constant sur X, il faut prouver que $H^0(X, F) \xrightarrow{\sim} H^0(X', q^*F)$.
Mais [page 128]

$$H^0(X, F) \simeq H^0(Y, \underbrace{f_*(F)}_{\substack{\text{localement constant} \\ \text{par hypothèse c)}}}) \quad \underbrace{\xrightarrow{\qquad\sim\qquad}}_{\substack{\text{car } p \\ \text{Hot-équivalence}}} \quad H^0(Y', p^*f_*(F)),$$

mais

$$p^*f_*(F) \xrightarrow{\sim} f'_*(q^*(F)),$$

car $f_*(F)$ et $f'_*(q^*F)$ se calculent 'fibre par fibre' par hypothèse. Donc

$$H^0(X, F) \simeq H^0(Y', f'_*(q^*(F))) \simeq H^0(X', q^*(F)),$$

q.e.d.

2°) Si F est un faiscau en groupes localement constant sur X, il faut prouver

$$H^1(X,F) \xrightarrow{\sim} H^1(X',q^*F).$$

Le fait que c'est injectif résulte déjà de 1°), qui implique que le foncteur

$$\begin{array}{ccc} F\text{-torseurs sur } X & \longrightarrow & q^*(F)\text{-torseurs sur } X' \\ P & \longmapsto & f^*P \end{array}$$

est pleinement fidèle, par l'argument bien connu utilisant les faisceaux $\mathcal{H}om$. Il faut voir que ce foncteur est essentiellement surjectif.
On a un homomorphisme de suites exactes d'ensembles ponctués [page 129]

$$\begin{array}{ccccccccc} H^1(Y,f_*F) & \xrightarrow{u} & H^1(X,F) & \xrightarrow{v} & \overbrace{H^0(Y,R^1f_*(F))}^{\ni\eta} & \xrightarrow{w} & H^2(Y,f_*F) \\ \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\delta} \\ H^1(Y',f'_*F) & \xrightarrow{u'} & \underbrace{H^1(X',F')}_{\ni\xi'} & \xrightarrow{v'} & \underbrace{H^0(Y',R^1f'_*(F'))}_{\ni\eta'} & \xrightarrow{w'} & H^2(Y',f'_*F') \end{array}$$

([3]), où les faisceaux f_*F, $R^1f_*(F)$ sont localement constants par hypothèse et se calculent fibre par fibre, les faisceaux $f'_*(F')$ et $R^1f'_*(F')$ aussi, qui implique que ces derniers sont canoniquement isomorphes (par les flèches de changement de base) aux images inverses de leurs homologues $f_*(F)$, $R^1f_*(F)$. Comme p est une Hot-équivalence, cela implique que les flèches α, γ sont `[des]` isomorphismes. Donc si $\xi' \in H^1(X',F')$, son image η' dans $H^0(Y',R^1f'_*(F'))$ provient d'un $\eta \in H^0(Y,R^1f_*(F))$ bien déterminé. Il faut voir que cet η provient d'un $\xi \in H^1(X,F)$. Ceci (si je me rappelle bien) est possible si et seulement si une certaine obstruction $w(\eta)$ dans $H^2(Y,f_*(F))$ est dans le sous-ensemble neutre du H^2. **NB** Le champ sur $Y^\wedge$ des F-torseurs sur les $X^\wedge/U$ qui remontent η est une $\mathcal{F} = f_*F$-gerbe $\mathcal{G}_\eta$, et il faut voir qu'elle est neutre. Mais il y a mieux à faire : p. 3

Commentary · Exposition [ms. 129]

Torsor descent and the band of the obstruction

The difficult part of the coefficient argument is essential surjectivity for torsors. Full faithfulness is already secure: if P, Q are F-torsors, the sheaf $\mathcal{I} = \underline{\mathrm{Isom}}_F(P,Q)$ is locally constant, and

$$\mathrm{Hom}_F(P,Q) = H^0(X,\mathcal{I}).$$

An equivariant map between two torsors is invertible, and the degree-zero calculation identifies these Hom sets after pullback.

The displayed obstruction sequence should, however, be read as a provisional sketch. For a general nonabelian F, the gerbe of lifts of a section η of R^1f_*F need not have the fixed band of f_*F. Its band is obtained from automorphisms of local representatives of η, hence involves inner forms of F. For a direct check, take $f : BC_2 \to e$, constant $F = S_3$, and the torsor represented by the inclusion of a transposition subgroup $C_2 \hookrightarrow S_3$. Its automorphism group is $C_{S_3}(C_2) = C_2$, whereas $f_*F = S_3$. The component of this torsor therefore cannot have the fixed band S_3. Nor does nonabelian H^2 behave as a group with a distinguished zero: neutrality is a condition on a gerbe. The notation G in the next paragraph is not defined. These are reasons to replace that portion of the argument, not reasons to reject the theorem. The classical framework is the theory of bands and gerbes in Giraud (1971, Chapter IV); we give a direct repair that avoids choosing an obstruction class.

For each y, let $\mathcal{T}(y)$ be the groupoid of torsors under the restriction of F to X/y. Restriction along $X/y \longrightarrow X/z$ makes $\mathcal{T}$ a groupoid-valued contravariant diagram on Y. We claim that every

3. où on a posé $F' = q^*(F)$.

transition functor is an equivalence. Essential surjectivity follows from the local constancy of $R^1 f_* F$: its value at y is precisely the set of isomorphism classes of $\mathcal{T}(y)$. To prove full faithfulness for a transition $u : y \longrightarrow z$, take $P, Q \in \mathcal{T}(z)$. Their isomorphism sheaf is a locally constant set-valued coefficient on X/z. Apply the universally quantified degree-zero hypothesis to the base change

$$f_z : X/z \longrightarrow Y/z.$$

The values at $(z, 1_z)$ and (y, u) are respectively $\mathrm{Hom}(P, Q)$ and $\mathrm{Hom}(u^* P, u^* Q)$. Local constancy makes the restriction map bijective. This is the place where coefficients on a base-changed total category, rather than only coefficients originally chosen on X, are needed.

The same hypotheses identify $\mathcal{T}(y)$ with the groupoid of $F|_{X_y}$-torsors. On isomorphism classes this is the fibre formula for $R^1 (f_y)_* F$, applied over Y/y. On Hom sets it is the fibre formula for $(f_y)_* \mathcal{I}$, for the isomorphism sheaf just used. Thus the identification is an equivalence of groupoids, not merely a bijection on H^1.

Now let $p : Y' \longrightarrow Y$ be a weak equivalence, and form the cartesian square of the source. The natural comparison from $p^* \mathcal{T}$ to the torsor diagram $\mathcal{T}'$ on Y' is an objectwise equivalence: both values at y' restrict equivalently to the torsors on the same strict fibre $X_{p(y')}$. Finally,

$$\mathrm{Tors}_F(X) \simeq \operatorname*{holim}_{y \in Y^{\mathrm{op}}} \mathcal{T}(y).$$

This is ordinary descent for torsors, expressed on the representable objects Y/y. Explicitly, a compatible family of torsors on these slices determines its torsor over $x \in X$ by evaluation at $(x, 1_{f(x)})$; the comparison maps along $f(a)$, for $a : x \longrightarrow x'$, give its restriction maps. The descent cocycle is exactly their composition law. Pulling a torsor on X to all slices gives the inverse construction.

Homotopy limits of locally constant groupoid diagrams are invariant under a weak equivalence of indexing categories. One can express this as invariance of sections under equivalence of fundamental 2-groupoids, or obtain it by applying the homotopy-invariance theorem for locally constant model-category diagrams to groupoids and their derived right adjoint of constant diagrams; see Cisinski (2009, Theorem 1.3). Consequently

$$\mathrm{Tors}_F(X) \longrightarrow \mathrm{Tors}_{q^* F}(X')$$

is an equivalence. This proves the missing essential surjectivity and the required bijection on H^1. It also supplies the local-constancy argument for the lifting gerbe that the source compresses into a parenthesis.

On note que cette gerbe $\mathcal{G}$ est localement constante (puisqu'elle correspond à un lien $f_*(G)$ localement constant) et que $\mathcal{G}'$ est son image inverse. Mais comme p est une Hot-équivalence, il s'ensuit que le foncteur [page 130]

$$\underline{\Gamma}(Y, \mathcal{G}) \longrightarrow \underline{\Gamma}(Y', \mathcal{G}')$$

est une équivalence de catégories, donc ξ' provient (à isomorphisme près) d'un unique ξ de $\underline{\Gamma}(Y, \mathcal{G})$, q.e.d.

3°) Soit F faisceau abélien localement constant sur X, il faut voir [que]

$$(*) \qquad H^i(X, F) \xrightarrow{\sim} H^i(X', F') \qquad (F' = q^* F),$$

pour tout $i \in \mathbf{N}$. Mais les deux termes sont l'aboutissement des suites spectrales de Leray, de termes initiaux

$$E_2^{pq} \simeq H^p(Y, R^q f_*(F))$$
$$E_2'^{pq} \simeq H^p(Y', R^q f'_*(F')),$$

et $(*)$ est associé à un homomorphisme de suites spectrales, qui pour les termes initiaux est

$$(**) \qquad H^p(Y, R^q f_*(F)) \xrightarrow{\sim} H^p(Y', R^q f'_*(F')),$$

associé à

$$(***) \qquad p^*(R^q f_*(F)) \longrightarrow R^q f'_*(F').$$

Mais par hypothèse les $R^q f_*(F)$, $R^q f'_*(F')$ sont localement constants et se calculent fibre par fibre,
d'où il résulte que les homomorphismes de changement de base $(***)$ sont [des] isomorphismes, [page 131]
et comme il s'agit de faisceaux localement constants et que p est une Hot-équivalence, il s'ensuit que $(**)$ est [un] isomorphisme, donc aussi $(*)$ sur les aboutissements, q.e.d.

Commentary · Exposition [ms. 131]

The Leray comparison has finite diagonals

Grothendieck can now return to abelian coefficients, where the formal argument is both shorter and complete. For each q, the canonical comparison

$$p^* R^q f_* F \longrightarrow R^q f'_* q^* F$$

is an isomorphism because its values are the cohomology groups of identical strict fibres. Both sides are locally constant. Since p is a weak equivalence, it therefore induces an isomorphism on every $E_2^{p,q}$ term of the Leray spectral sequences.

There is no convergence obstruction here. These are first-quadrant cohomological spectral sequences, and in total degree n only the $n+1$ pairs $p+q=n$ occur. Comparison gives isomorphic associated graded groups of finite filtrations of $H^n(X,F)$ and $H^n(X', q^*F)$; induction through those filtrations gives the asserted isomorphism. No boundedness of the cohomological dimension of X or Y is required. The missing prime on F in the printed E_2-comparison is an edition slip, retained in the transcription and recorded separately. Together with the torsor argument, this completes the coefficient criterion.

Définition 1. Sous les conditions équivalentes du théorème 1, on dit que f est une Hot-*fibration.*

Proposition 1. *Cette notion est stable par changement de base (trivial), par composition (formel sur a)). Pour que f soit une* Hot*-fibration, il faut et il suffit que f* [plutôt f^o] *le soit.* p. 4

Commentary · Later developments [ms. 131]

From strict fibres to homotopy fibres

The stability assertions now have a geometric interpretation. For composable maps $X \longrightarrow Y \longrightarrow Z$, pulling a weak equivalence back along the composite is the succession of the two pullbacks. Opposite categories preserve strict pullbacks, and reversing a nerve simplex identifies the classifying spaces of C and C^{op}. Thus both composition and self-duality follow directly from a), without another coefficient argument.

The strict fibre also calculates the homotopy fibre. By b'), the categories X/y vary by weak equivalences along all arrows of Y, and $X_y \longrightarrow X/y$ is a weak equivalence. The dual form of Quillen's Theorem B therefore identifies

$$BX_y \simeq \mathrm{hofib}_y(BX \longrightarrow BY).$$

The exact citation is Quillen (1973, §1, Theorem B and its corollary); its hypotheses concern the comma-category transition functors. Merely knowing the homotopy types of the strict fibres, without the comparison to comma categories, would not suffice.

This perspective explains the force of the theorem while leaving room for Grothendieck's chosen constructions. Cisinski (2009, Proposition 4.4) later proves that weak equivalences of nerves together with the maps that are both smooth and proper form a category of fibrant objects, with functorial factorization. These smooth-and-proper maps are a particularly useful subclass of the present Hot-fibrations. The statement does not say that all Hot-fibrations are smooth and proper. Their advantages are the explicit categories of lifts and the simultaneous control of left and right homotopy Kan extensions.

Proposition 2. *Soit $f : X \longrightarrow S$ une* Hot*-fibration. Alors $s \longmapsto \mathrm{hot}(X_s)$ est un système local sur S, à valeurs dans* Hot*. Donc les X_s pour s dans une même composante connexe*

de S sont Hot*-isomorphes entre eux (mais pas canoniquement, si cette composante n'est pas* 1*-connexe).*

Commentary · Examples [ms. 131]

A group extension and its monodromy

The local system of fibre homotopy types already contains nontrivial information in a finite group example. Regard a group as its one-object category. If

$$1 \longrightarrow K \longrightarrow G \xrightarrow{\varphi} Q \longrightarrow 1$$

is exact, the functor from the one-object groupoid of G to that of Q is an isofibration. Its nerve is a Kan fibration. Pullbacks of weak equivalences along Kan fibrations are weak equivalences, and the nerve preserves pullbacks, so this functor is a Hot-fibration. Its strict fibre is the one-object groupoid K.

The comma category over the unique object of Q has objects $q \in Q$. A morphism from q to q' is an element $g \in G$ satisfying $q = q'\varphi(g)$, with a harmless reversal if the other convention for composition is chosen. Surjectivity of φ makes this a connected groupoid; the automorphism group of any object is K. Thus the inclusion of the strict fibre is an equivalence of categories. If φ is not surjective, its connected components are the corresponding cosets of $\varphi(G)$, and that inclusion is not a weak equivalence. This computes precisely why an arbitrary homomorphism of groups need not be a Hot-fibration.

For the nontrivial monodromy, take $G = \mathfrak{S}_3$, $K = \mathfrak{A}_3 = C_3$, and $Q = C_2$, with φ the sign. A lift of the nonidentity element of C_2 is a transposition. Conjugation by it sends a three-cycle to its inverse. A different lift changes this automorphism by an inner automorphism of K; inner automorphisms induce naturally isomorphic functors on the one-object groupoid and hence the same map in Hot. The action on the fibre is consequently well defined there. It is nontrivial, since inversion acts nontrivially on $\pi_1(BC_3) = C_3$.

With constant coefficient $\mathbb{F}_3$ on $B\mathfrak{S}_3$, the degree-one fibre cohomology is

$$H^1(C_3; \mathbb{F}_3) = \mathrm{Hom}(C_3, (\mathbb{F}_3, +)) \cong \mathbb{F}_3,$$

and the base generator acts by -1. Its invariant subgroup is zero. The low-degree Leray sequence gives $H^1(\mathfrak{S}_3; \mathbb{F}_3) = 0$: the other term $H^1(C_2; \mathbb{F}_3)$ is zero, and positive-degree cohomology of C_2 with these coefficients vanishes by the averaging operator $\frac{1}{2}(1+\tau)$. Independently, the same answer follows from $\mathfrak{S}_3^{\mathrm{ab}} = C_2$. The fibre cohomology is locally constant on the base, but it is not a constant coefficient with trivial monodromy. This is the distinction that the printed qualifier about noncanonical identifications is meant to preserve; a nonsimply connected base can also carry trivial monodromy, as a product shows.

Théorème 2. *Soit* $\begin{array}{ccc} X' & \xrightarrow{f} & X \\ & {\scriptstyle p'}\searrow \quad \swarrow{\scriptstyle p} & \\ & S & \end{array}$ *un diagramme commutatif dans* Cat*, avec p, p' des* Hot*-fibrations, et soit $S_0 \subseteq S$ une partie de S telle que $S_0 \longrightarrow \pi_0(S)$ soit surjectif. Pour que f soit une* Hot*-équivalence, il faut et il suffit que les $f_s : X_s \longrightarrow Y_s$ $(s \in S_0)$ le soient.*

a) La suffisance se prouve comme tantôt, avec le critère de Artin-Mazur.

Notons que l'hypothèse sur S_0 assure que si F, F' sont deux faisceaux d'ensembles localement constants sur S, et $u : F \longrightarrow F'$ un homomorphisme tel que $u_s : F_s \xrightarrow{\sim} F'_s$ pour $s \in S$, alors u est [un] isomorphisme. [page 132]

1°) Soit F un faisceau d'ensembles localement constant sur X, F' son image inverse sur X', il faut prouver

$$H^0(X, F) \xrightarrow{\sim} H^0(X', F').$$

Or

$$H^0(X, F) \simeq H^0(Y, \underbrace{f_*(F)}_{\substack{\text{localement} \\ \text{constant sur } S}}) \longrightarrow H^0(Y, f'_*(F')) \simeq H^0(X', F'),$$

il faut prouver que

$$(*) \qquad p_*(F) \xrightarrow{\sim} p'_*(F')$$

est [un] isomorphisme. Or par hypothèse sur p, p', les deux faisceaux sont localement constants et se calculent fibre par fibre, d'où résulte que $(*)$ est un isomorphisme pour les fibres en $s \in S_0$, donc un isomorphisme partout. OK.

2°) Il en résulte déjà, si F est un faisceau en *groupes,* que le foncteur

$$(F\text{-torseurs sur } X) \longrightarrow (F\text{-torseurs sur } X')$$

est pleinement fidèle, donc $H^1(X,F) \longrightarrow H^1(X',F')$ [est] injectif; il reste à prouver que ce foncteur est essentiellement surjectif. À cause de la pleine fidélité, c'est *local sur* S (**NB** On voit directement que l'hypothèse fait sur les $s \in S_0$ implique la même hypothèse pour tout $s \in S$, en utilisant le fait que p, p' sont des fibrations) ([4]). Donc on est ramené au cas où $S = S/s_0$ a un objet final s_0.

Commentary · Exposition [ms. 132]

Testing one fibre in each component

Theorem 2 is the fibrewise Whitehead theorem for these categorical fibrations. Its fibre map is $f_s : X'_s \longrightarrow X_s$; the displayed $X_s \longrightarrow Y_s$ is an edition slip. Transport is natural with respect to f, so if f_s is a weak equivalence at one point of a component then it is a weak equivalence at every point of that component. The choice of S_0 therefore loses no information.

For sufficiency, the source compares direct images of local coefficients. Its degree-zero comparison should be written

$$H^0(X,F) = H^0(S,p_*F) \longrightarrow H^0(S,p'_*F') = H^0(X',F').$$

Both direct images are locally constant, and their values on S_0 are isomorphic by the fibre hypothesis. A morphism between locally constant diagrams is an isomorphism everywhere if it is so at one object of each component: propagate the property along an unoriented chain of arrows, whose transition maps are invertible. This proves degree zero and, by the isomorphism-sheaf argument, full faithfulness on torsors.

The local reduction for essential surjectivity is legitimate once it is applied to these groupoids of torsors, rather than only their sets of isomorphism classes. On each representable S/s, both total categories still satisfy the hypotheses. The torsor diagrams constructed above are locally constant there, and S/s has a final object. Their groupoids of global sections are thus equivalent to their values at that object. Fibrewise equivalence supplies local preimages of any torsor on X'. Full faithfulness uniquely lifts the overlap isomorphisms between these preimages; their cocycle follows from its validity after pullback and faithfulness. Effective descent for torsors gives the global preimage. This justifies both the phrase "local on S" and the degree-one edge identifications on manuscript page 133. The abelian comparison follows from Leray as before.

For necessity there is a shorter modern proof, complementary to the printed path-category argument. The preceding block identifies both strict fibres with homotopy fibres. An equivalence $BX' \longrightarrow BX$ over BS induces equivalences on all homotopy pullbacks along $\{s\} \longrightarrow BS$, so every Bf_s is a weak equivalence. Conversely, maps of spaces over a fixed base are detected on their homotopy fibres: on each base component the long exact homotopy sequences give the higher homotopy isomorphisms, and the action of $\pi_1(BS,s)$ on the components of the fibre identifies the components of the total space with its orbit set. A fibrewise equivalence preserves that action and those orbit sets. This also treats disconnected or empty fibres, which a statement only about based groups would overlook.

On a alors

$$H^1(X,F) \simeq H^0(S,R^1p_*(F))$$
$$H^1(X',F') \simeq H^0(S,R^1p'_*(F')),$$

p. 5 [page 133]

4. Les hypothèses faites sont bel et bien locales sur S (du moins si $S_0 = S$).

et on voit encore que

$$R^1p_*(F) \longrightarrow R^1p'_*(F')$$

est [un] isomorphisme (car il se calcule fibre par fibre, et on utilise l'hypothèse sur les f_s), donc induit un isomorphisme

$$H^1(X,F) \longrightarrow H^1(X',F').$$

3°) Si F est même abélien, on doit prouver

$$H^i(X,F) \xrightarrow{\sim} H^i(X',F')$$

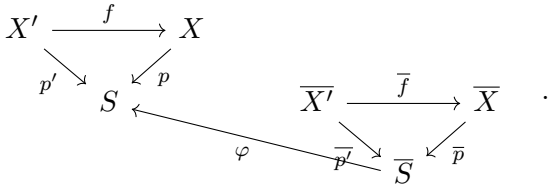
pour tout i, et cela résulte encore de la suite spectrale de Leray pour (p,F) et pour (p',F').

b) Inversement, supposons que f soit une Hot-équivalence, et prouvons que les $f_s : X'_s \longrightarrow X_s$ sont des Hot-équivalences. Je vais utiliser le

Lemme. *Soit $S \in \mathrm{Ob}\,\mathrm{Cat}$, S 0-connexe. Alors il existe un $\overline{S} \in \mathrm{Ob}\,\mathrm{Cat}$, $\overline{S}$ Hot-asphérique, et $\overline{S} \longrightarrow S$ une Hot-fibration.*

Voyons comment ceci implique le théorème 2 (partie réciproque). On peut supposer S 0-connexe, et on fait le changement de base par $\overline{S} \longrightarrow S$

comme dans le lemme : [page 134]



Alors $\overline{p'}$ et $\overline{p}$ sont des Hot-fibrations, et comme φ est une Hot-fibration, le changement de base par φ transforme la Hot-équivalence f en une Hot-équivalence $\overline{f}$. Considérons alors $\overline{s} \in \mathrm{Ob}\,\overline{S}$, et le diagramme

$$\begin{array}{ccc} \overline{X'} & \xrightarrow{\overline{f}} & \overline{X} \\ {\scriptstyle i'}\uparrow & & \uparrow{\scriptstyle i} \\ \overline{X'}_{\overline{s}} & \xrightarrow{\overline{f}_{\overline{s}}} & \overline{X}_{\overline{s}} \end{array} \quad .$$

Comme $\{\overline{s}\} \longrightarrow \overline{S}$ est une Hot-équivalence, la flèche i qui s'en déduit par le changement de base $\overline{p} : \overline{X} \longrightarrow \overline{S}$ l'est aussi, puisque $\overline{p}$ est une Hot-fibration, et de même pour i'. Ainsi dans le carré commutatif précédent, $\overline{f}$, i et i' sont des Hot-équivalences, donc aussi $\overline{f}_{\overline{s}}$. Mais si $s = \varphi(\overline{s})$, $\overline{f}_{\overline{s}}$ s'identifie à $f_s : X'_s \longrightarrow X_s$. D'ailleurs, il est immédiat que φ, étant une Hot-fibration p. 6

à source non-vide, et à but connexe, est surjective. (**NB** toutes les fibres d'une même composante connexe de S sont Hot-équivalentes, donc si l'une est non-vide, toutes le sont.) Cela achève la démonstration, modulo le lemme. [page 135]

Commentary · Exposition [ms. 135]

A contractible replacement without a derivator

The lemma is not asserting the existence of a contractible covering space: a general connected homotopy type has no such covering. It asks for a fibration whose fibre retains the entire loop homotopy type. The source's based path category is designed to do exactly this. The dependency it leaves open at the end of the chapter concerns the proof that this specific categorical path construction is both proper and smooth.

For the existential lemma alone, Thomason's theory already gives an independent proof. Choose $s_0 \in S$ and factor its inclusion in the Thomason model category as

$$e \xrightarrow{i} E \xrightarrow{p} S, \qquad i \text{ a trivial cofibration}, \quad p \text{ a model fibration}.$$

Then BE is weakly contractible. Since $\mathrm{Ex}^2 N$ preserves pullbacks and sends p and all its base changes to Kan fibrations, right properness of simplicial sets proves that p is a Hot-fibration in the sense of a). The existence of the factorization is Thomason (1980, Theorem 4.9); its weak equivalences and fibrations are Definitions 2.1–2.2, and properness is Corollary 5.5.

The map p is surjective on objects when S is nonempty and connected. The object $i(*)$ supplies one nonempty fibre. Pull back over an arrow of S; both endpoint fibres have the homotopy type of that pullback, so one is nonempty if and only if the other is. A zigzag in S reaches every object of its connected component. This is the elementary justification of the surjectivity used on manuscript page 135. It gives the printed proof of the converse in Theorem 2 without invoking any derivator: pull back along $E \longrightarrow S$, use that the inclusions of its fibres are weak equivalences, and apply two-out-of-three.

This replacement does not diminish the purpose of Chapter XII. The abstract model factorization proves existence, whereas Grothendieck needs constructions with explicit categories of lifts, duality, and control of both direct images. The later smooth-and-proper factorization quoted above addresses that stronger requirement.

Il reste à prouver celui-ci. Comme $S = \emptyset$ [plutôt $S \neq \emptyset$], je peux choisir une 'origine'

$$s_0 \in S,$$

et regarder la catégorie

$$\underline{\mathrm{Ch}}_\infty(S; s_0, -)$$

des chemins infinis dans S, d'origine s_0, fibre de $\underline{\mathrm{Ch}}_\infty(S)$ par le foncteur-source

$$\sigma : \underline{\mathrm{Ch}}_\infty(S) \longrightarrow S.$$

On sait que les fibres de σ sont Hot-asphériques (VII), et la catégorie construite ici n'est autre que (avec les notations de VII, p. 152) que la catégorie $\overline{S'}$ qui entre dans la factorisation canonique de

$$f : \{s_0\} = S' \longrightarrow S$$

en

$$\{s_0\} = S' \underbrace{\longrightarrow}_{\text{Hot-équivalence}} \overline{S'} \xrightarrow{\overline{f}} S,$$

où ici $\overline{f}$ s'interprète simplement comme le foncteur but (i.e. le foncteur induit sur $\underline{\mathrm{Ch}}_\infty(S; s_0, -)$ par le foncteur but [page 136]

$$\beta : \underline{\mathrm{Ch}}_\infty(S) \longrightarrow S \ ).$$

On a prouvé dans loc. cit. que $\overline{f}$ est une Hot-fibration ('propre' et 'lisse'), mais dans un contexte qui semblait faire appel tacitement à la théorie du dérivateur HOT. Donc pour mettre les résultats 'topologiques' de loc. cit. sur une base solide, il faut développer les notions pertinentes de foncteur propre, foncteur lisse, sans référence à une théorie HOT. C'est ce que j'avais d'ailleurs fait déjà en 1983, du temps de Pursuing Stacks, mais sans rédaction en forme à ce sujet. Il me faut donc à présent revenir sur ces notions, de façon un peu systématique. D'ailleurs, j'aurais dû mal à construire un seul foncteur Hot-fibrant (sauf les exemples triviaux, tels les isomorphismes, ou les produits $Y \times Z \xrightarrow{\mathrm{pr}_1} Y$), si je ne disposais d'une théorie des foncteurs lisses et des foncteurs propres ([5]).

5. J'exagère un peu quand même – après tout il doit être assez évident que les foncteurs de Thomason-Kan sont des Hot-fibrations (mais il y en a encore un peu trop peu !).

Commentary · Context [ms. 136]

Why the next chapter starts again

Grothendieck's return to proper and smooth functors is an effort to remove a dependency, not a rejection of the homotopy theory just obtained. A path construction had been used to produce fibrations while some of its asserted properties still appealed to the derivator that those fibrations were meant to construct. Categories of lifts offer a direct combinatorial foundation for those properties.

The historical background is visible in *Pursuing Stacks*, §69: the coefficient theory is to remember categories of coherent diagrams together with both adjoints of restriction, rather than merely a single homotopy category (Grothendieck, 1983). The letter to Thomason of 2 April 1991 makes a related distinction explicit: the weak equivalences carry the intrinsic homotopy theory, while classes of fibrations and cofibrations make its operations calculable (Grothendieck, 1991a, §1). We refer to the commentary on Chapter I, manuscript p. 126 ("The later letter and the meaning of a fibration"), for the broader historical discussion.

The next pages isolate two different issues. One is to describe the maps along which categorical pullback behaves well. The other is to decide how much of the usual homotopy theory is forced by the proposed axioms on W. The revisions of terminology in Chapter XII are necessary because an axiom suitable for the full homotopy type can be too strong for a localization that deliberately forgets higher homotopy groups.

Chapter XII

Caractérisation de W_∞. Foncteurs W-propres, W-lisses etc. Sommes amalgamées et carrés W-cocartésiens dans Cat

XI-XIIa · Chapter XII, sections 1–4 · manuscript pages 1–58

p. –

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XII

Caractérisation de W_∞. Foncteurs W-propres, W-lisses etc. Sommes amalgamées et carrés W-cocartésiens dans Cat**.**

p. –

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

19.11.90 p. 1
[page 1]

I Caractérisation de W_∞.

Je voudrais caractériser W_∞ en tant que partie de Fl(Cat), par ses propriétés comme localiseur, et sans faire appel à la connaissance de notions utilisant déjà W_∞, comme la propreté etc. Je voudrais tout au moins des propriétés sur un localiseur W, familières pour W_∞, qui soient si fortes qu'elles impliquent $W \subseteq W_\infty$. Bien sûr, dans les démonstrations je me permets d'utiliser tout ce que je sais sur W_∞.

Bien sûr, il faudra au moins une forme de saturation, peut-être très faible, je présume le suivant suffira ([6]).

W(1) (Axiome de saturation.) a) Les identités sont dans W, b) si f, g `[sont]` composables et deux parmi f, g, gf sont dans W, le troisième aussi, c) si $fg \in W$, $gf \in W$, alors $f, g \in W$.

Une propriété particulièrement cruciale, qui déjà distingue W_∞ de l'ensemble des homotopismes (notion trop stricte à beaucoup d'égards), est l'axiome que j'ai appelé L4 bis :

Commentary · Examples [ms. 1]

A contractible nerve without a categorical contraction

Grothendieck's insistence that homotopisms are too restrictive has a concrete explanation. Let P be the ordered set with objects $0, 1, 2, \ldots$ and nonidentity relations

$$0 < 1 > 2 < 3 > 4 < 5 > \cdots .$$

There are no composable pairs of nonidentity arrows. Its nerve therefore realizes to the half-line subdivided at the integers; in particular $P \to e$ belongs to W_∞.

Nevertheless $P \to e$ is not a homotopism in the sense used on the next manuscript page. Suppose that the identity of P could be joined to the constant functor at c by a zigzag of m natural transformations. Evaluating the zigzag at an integer n would give a path of at most m edges from n to c in the underlying undirected graph of P. Every arrow changes the integer by at most one. Thus $|n - c| \leq m$ for every n, which is impossible. A homotopy inverse to $P \to e$ would select precisely such a constant functor, so no homotopy inverse exists.

This example separates two finite conditions that are easily conflated. Each individual vertex can be connected to 0 by a finite path. A natural-transformation contraction would require one finite zigzag working simultaneously at every vertex. The localization axioms allow the classifying space to forget that uniform bound.

W(2) (Axiome de localisation.) Soit $f : X \longrightarrow Y$ une flèche dans Cat/S, telle que $\forall\ s \in \mathrm{Ob}\, S$, $f_s : X/s \longrightarrow Y/s$ soit dans W (on dira que f est dans W 'localement sur S'). Alors $f \in W$.

Commentary · Exposition [ms. 1]

The two directions of the characterization

Grothendieck wants the equivalences of ordinary homotopy theory to emerge from operations on categories. There are two logically different assertions to establish. The first says that the proposed axioms cannot identify too many maps: $W \subseteq W_\infty$. The second says that a sufficiently coherent theory cannot identify too few: $W_\infty \subseteq W$. This section attacks the first assertion by repeatedly taking fibres. The later minimal-localizer theorem supplies the second. Keeping these directions separate makes the successive changes of axioms intelligible.

Write $BA = |NA|$ for the classifying space of a small category, and let e be the terminal category. Here W_∞ consists of the functors u for which Bu is a weak homotopy equivalence. The expression W-aspherical means only $A \to e \in W$. It does not mean that A has vanishing higher homotopy

6. (non, j'ai dû le renforcer en 2 [?]).

groups until a comparison with W_∞ has been established. In particular, it is not the usual topological convention under which a connected $K(G,1)$ is called aspherical.

The local-to-global axiom W(2) tests a map $u : X \to Y$ over S on the comma categories $X/s \to Y/s$. These are not the strict fibres $X_s \to Y_s$. An object of X/s is a pair $(x, p(x) \to s)$, so the test retains all ways of approaching s. W(2 bis) is the absolute theorem-A form: contractibility, in the sense of W, of every X/y implies $u \in W$. The distinction matters: the source explicitly uses the relative form for Proposition 3, whereas the loop argument and the final nondegeneracy repair need only the absolute form.

The commentary on Chapter VI, manuscript pp. 44–45, explained why a fibre axiom begins to select ordinary homotopy theory. The present pages give the mechanism: any nonzero homotopy group of a W-aspherical category can be lowered, by looping, to a forbidden nontrivial set of components.

`[Les axiomes W(1) etc. sont appelés aussi ① etc. dans l'original.]`

Dans le cas $Y = S$, cela donne

Corollaire : *Si $f : X \longrightarrow Y$ est W-asphérique, i.e. $y\backslash X \longrightarrow y\backslash Y$* `[plutôt` $X/y \longrightarrow Y/y$`]` *dans W pour tout $y \in \mathrm{Ob}\, Y$, alors $f \in W$* ([7]).

Proposition 1. *Si W satisfait* W(1) *et* W(2)*, les propriétés suivantes sur W sont équivalentes :*

a) Pour tout X dans Cat*, $X \times \mathbf{\Delta}^1 \xrightarrow{\mathrm{pr}_1^X} X$ est dans W (i.e. $\mathbf{\Delta}^1 \longrightarrow e$ est 'universellement dans W ').*

b) Si X dans Cat *a un objet final, alors X est W-asphérique, i.e. $X \longrightarrow e$ est dans W.*

c) Tout homotopisme est dans W.

7. On peut considérer la validité du corollaire comme une variante affaiblie W(2 bis) de W(2) ci-dessus. Dans la proposition 1, on n'utilise que W(2 bis), forme faible de W(2).

Démonstration. b) $\Longrightarrow$ a) ([8]), car pour voir que $\mathrm{pr}_1 \in W$, en appliquant le corollaire ci-dessus, il faut prouver que $\forall\, x \in X$ p. 2

$$(*) \qquad (X \times \mathbf{\Delta}^1)/x \longrightarrow X/x$$

est dans W. Or

$$(X \times \mathbf{\Delta}^1)/x \simeq X/x \times \mathbf{\Delta}^1,$$

$$\begin{array}{ccccc} \mathbf{\Delta}^1 & \longleftarrow & X \times \mathbf{\Delta}^1 & \longleftarrow & X/x \times \mathbf{\Delta}^1 \\ \downarrow & & \downarrow & & \downarrow \\ e & \longleftarrow & X & \longleftarrow & X/x \end{array}$$

[la flèche verticale à droite est ajoutée] et les deux catégories X/x, $X/x \times \mathbf{\Delta}^1$ sont W-asphériques car elles ont un objet final, donc $(*)$ est dans W par saturation faible W(1).

a) $\Longrightarrow$ c). Je rappelle que

$$f : X \longrightarrow Y$$

est un homotopisme s'il existe [page 2]

$$g : Y \longrightarrow X$$

telle que

$$(*) \qquad gf \sim \mathrm{id}_X, \quad fg \sim \mathrm{id}_Y,$$

où la relation $(\sim)$ désigne l'homotopie dans $\underline{\mathrm{Hom}}(X, X)$ et $\underline{\mathrm{Hom}}(Y, Y)$ respectivement (définie par chemins allant d'un élément à l'autre). Prouvons le

Lemme. *Supposons que W satisfait* W(1) *et le* a) *de la proposition, i.e. $X \times \mathbf{\Delta}^1 \longrightarrow X$ dans W pour tout X dans* Cat. *Alors si X, Y* [sont] *dans* Cat *et f, g dans $\underline{\mathrm{Hom}}(X, Y)$ sont homotopes, on a*

$$\mathrm{hot}_W(f) = \mathrm{hot}_W(g).$$

De plus, $f \in W$ si et seulement si $g \in W$.

Commentary · Exposition [ms. 2]

Saturation at the point where it is used

The proof is carefully arranged so that equality after localization is not mistaken for membership in W. For a natural transformation $f \Rightarrow g$, let $F : X \times [1] \to Y$ be its corresponding functor and let $p : X \times [1] \to X$ be the projection. If $p \in W$, its sections i_0, i_1 lie in W by two-out-of-three. Since $pi_0 = pi_1 = \mathrm{id}_X$, their images in $\mathrm{Cat}[W^{-1}]$ are both p^{-1}. Thus $f = Fi_0$ and $g = Fi_1$ become equal there. Independently, two-out-of-three gives

$$f \in W \quad \Longleftrightarrow \quad F \in W \quad \Longleftrightarrow \quad g \in W.$$

Iteration handles a finite zigzag of transformations.

Now suppose f has a categorical homotopy inverse g. The preceding argument puts both gf and fg in W. The separate clause W(1c) then puts f and g in W. This last step would not follow merely from the existence of their inverse images in the localization unless strong saturation had already been proved.

Three usages of saturation should therefore remain distinct. Two-out-of-three is W(1b); the two-cycle condition is W(1c); strong saturation says that every map inverted by the localization already belongs to W. The modern weak-saturation axiom used for fundamental localizers asks, in addition to identities and two-out-of-three, only that $ri = \mathrm{id}$ and $ir \in W$ imply $r \in W$ (Cisinski, 2004, Definition 1.1.1). W(1c) implies this condition. Consequently the later results apply to the source's axioms without silently inserting strong saturation into this elementary proof.

8. n'utilise W(1) que sous forme faible, sans la condition de saturation c). NB Seul b) $\Longrightarrow$ c) utilise W(2 bis), les implications a) $\Longrightarrow$ c) $\Longrightarrow$ b) sont valables sous la seule hypothèse W(1).

Il suffit de le prouver quand il existe un homomorphisme $f \longrightarrow g$. Cela peut s'interpréter par l'existence d'une flèche

$$X \times \mathbf{\Delta}^1 \xrightarrow{F} Y$$

dont les restrictions à $X \times \{0\}$ et $X \times \{1\}$ respectivement soient f, g. Pour prouver que $\mathrm{hot}_W(f) = \mathrm{hot}_W(g)$, il suffit de prouver que

$$i_0, i_1 : X \longrightarrow X \times \mathbf{\Delta}^1 \qquad x \longmapsto (x, 0),\ x \longmapsto (x, 1)$$

sont égales dans Hot_W, puisque p. 3

$$f = F \circ i_0, \quad g = F \circ i_1.$$

Or cela résulte du fait que ce sont des sections de $X \times \mathbf{\Delta}^1$ sur X, et que dans Hot_W la projection $X \times \mathbf{\Delta}^1 \xrightarrow{p} X$ est [un] isomorphisme (donc $i_0 = i_1 = p^{-1}$ dans Hot_W). De plus, le diagramme

$$\begin{array}{ccc} X & \xrightarrow{i_0} & X \times \mathbf{\Delta}^1 \\ & \searrow^{f} & \downarrow{\scriptstyle F} \\ & & Y, \end{array}$$

où $i_0 \in W$, montre que $f \in W$ si et seulement si $F \in W$, et de même $g \in W$ si et seulement si $F \in W$, donc $f \in W \iff g \in W$.

Revenons à $(*)$, on voit que dans Hot_W $g \circ f = \mathrm{id}_X$, $f \circ g = \mathrm{id}_Y$, donc f et g sont des Hot_W-isomorphismes inverses l'un de l'autre. D'autre part, $gf \in W$, $fg \in W$, puisque $\mathrm{id}_X, \mathrm{id}_Y \in W$. Donc on gagne, si on a formulé W(1) de façon assez forte. On a prouvé donc a) $\Longrightarrow$ c).

c) $\Longrightarrow$ b) est trivial, puisque si X a un objet final, $X \longrightarrow e$ est un homotopisme.

On doit bien sûr par la suite supposer que W satisfait aux conditions équivalentes de la proposition. On prendra p. ex. la forme [page 3]

W(3) (Axiome de l'élément final.) Si X a un objet final, alors X est W-asphérique.

Commentary · Exposition [ms. 3]

Terminal objects, initial objects, and duality

The terminal-object axiom gives a usable supply of trivial categories. If t is terminal in A, the natural transformation $\mathrm{id}_A \Rightarrow \mathrm{const}_t$ makes $A \to e$ a homotopism; an initial object gives the transformation in the reverse direction. Proposition 1 therefore derives initial-object asphericity from the same axioms. No second axiom is being assumed.

For a functor $u : A \to B$, the category B/b has the terminal object (b, id_b). Hence two-out-of-three identifies the two formulations

$$A/b \longrightarrow B/b \in W \quad \Longleftrightarrow \quad A/b \longrightarrow e \in W.$$

This is why the corrected slice orientation in the source's first corollary is essential: it determines whether the subsequent notion is asphericity or coasphericity.

Proposition 2 next turns this one-sided language into a self-dual theory. The bridge is the twisted-arrow category $\Phi(A)$, with comparison functors to A and A^{op}. Their comma categories have explicit contractions, so absolute theorem A puts both comparisons in W. Naturality then compares u, $\Phi(u)$, and u^{op} by two-out-of-three. The construction and its variance were treated in the commentary on Chapter I, manuscript pp. 73–75, and are not repeated here. What is new in the present argument is its economy of hypotheses: the comparisons use W(2 bis), not the full relative W(2). Applying the same construction to u^{op} gives the reverse implication, so $u \in W$ if and only if $u^{\mathrm{op}} \in W$.

Corollaire 1. *Pour que $f : X \longrightarrow Y$ soit W-asphérique, il faut et il suffit que les X/y soient W-asphériques* ($y \in \mathrm{Ob}\, Y$).

Car pour que $X/y \longrightarrow Y/y$ soit dans W, comme Y/y est W-asphérique (objet final), il faut et il suffit que X/y le soit aussi.

Corollaire 2. *Si X dans* Cat *a un objet initial, il est W-asphérique.* (En effet, $X \longrightarrow e$ est un homotopisme, et on applique prop. 1.) ([9])

Je mettrai pour mémoire

Corollaire 3. *Tout homotopisme dans* Cat *est dans W. Si $f, g : X \rightrightarrows Y$ dans* Cat *sont homotopes, alors elles sont égales dans* Hot_W*, et f est dans W si et seulement si g l'est.*

Proposition 2. Soit $f : X \longrightarrow Y$ dans Cat. Si $f \in W$, $f^\circ \in W^\circ$ `[plutôt` $f^\circ \in W$`]`.

On utilise la démonstration de I, p. 75, à l'aide de la catégorie nommée $\Phi(X)$ ($\mathrm{Ob}\, \Phi(X) = \mathrm{Fl}(X)$, mais avec une notion de flèche dans $\Phi(X)$ un peu particulière). On trouve que $f \in W \iff \Phi(f) \in W$, et de même $f^\circ \in W \iff \Phi(f) \in W$. p. 4

NB. Dans la démonstration de la proposition 2, comme dans celle de la proposition 1 (pour l'implication b) $\Longrightarrow$ a)), on n'a utilisé la propriété de localisation W(2) que sous la forme faible du corollaire, i.e. sous la forme f W-asphérique $\Longrightarrow f \in W$.

Corollaire (de la prop. 2) : *Soit $W \subseteq \mathrm{Fl}(\mathrm{Cat})$ satisfaisant l'axiome* W(1) (saturation). *Alors la conjonction des axiomes* W(2), W(3) *équivaut à celle des axiomes duaux* W(2′), W(3′) ([5]).

Il suffit de prouver que 1,2,3 $\Longrightarrow$ 2′,3′.

Pour 2′ `[plutôt` 3′`]`, c'est le corollaire 2 ci-dessus. Pour 3′ `[plutôt` 2′`]`, cela résulte aussitôt de la proposition 2. (NB Cet énoncé resterait valable, si on aurait pris W(2) sous la forme faible seulement, cf. le NB page précédente.) [page 4]

Je dois encore mettre un axiome de la forme L5, L6 (cf. VI, p. 43-44). Mais l'énoncé de ces axiomes utilise la notion de Hot-fibrés, ou de foncteurs parfaits, i.e. fait appel à la connaissance de W_∞, ce que j'aimerais éviter (si je peux). Je dois donc introduire la notion de W-fibration, sur le modèle de ce que j'ai fait dans le contexte des dérivateurs, dans I (cf. p. 111-126). On introduit la notion de **D**-fibration à la page 117 (prop. 5), par `[des]` conditions équivalentes, dont certaines ont un sens en termes de la donnée de W sans plus. Soit donc

$$f : X \longrightarrow Y$$

dans Cat et considérons les conditions

Commentary · Exposition [ms. 4]

What the four fibration tests quantify over

Grothendieck is looking for a notion of family whose strict pullbacks compute homotopy-invariant changes of base. His strongest condition says that

$$f^* : \mathrm{Cat}/Y \longrightarrow \mathrm{Cat}/X$$

preserves the underlying W-maps. The intermediate condition tests this only between W-aspherical bases. The left-hand test asks that, for every $v : T \to Y$ whose domain T has a terminal object and every object $t \in T$, the inclusion of the strict fibre

$$X_{v(t)} \longrightarrow X \times_Y T$$

belong to W. The chosen t need not be the terminal object. The right-hand test uses bases with an initial object. This universal quantification over $T \to Y$ distinguishes the definition from an isolated comparison $X_y \to X/y$.

9. C'est donc l'énoncé dual de l'axiome W(3) de l'objet final.

5. Itou si on remplace dans cet énoncé W(2) par W(2 bis).

The arrows in the source's diagram follow at once because a terminal or initial category is W-aspherical. Their converses express additional mathematics. For W_∞, Chapter XI supplies them. For a general W, the step from a one-sided test to invariance under every W-map is precisely the proposed axiom W(4). One should not import it from the formalism of a model category: an arbitrary categorical functor is not a fibration of a model structure merely because it satisfies one of these tests.

The eventual terminology is settled only at XII, manuscript pp. 51–52. The early adjective *ultraforte* names preservation of all W-maps, and it is useful to retain that descriptive meaning while reading these pages. The later W-fibration is defined using categorical homotopisms in the base. In a truncated theory the distinction survives: the path family over an $(n+1)$-sphere passes the ordinary homotopy tests but does not preserve W_n-equivalences. The complete computation is given below at manuscript pp. 16–17.

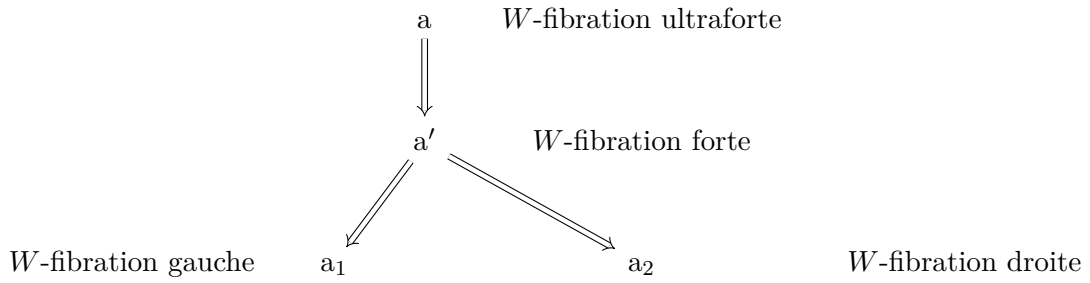
a) Si $Y'' \longrightarrow Y'$ dans Cat/Y est dans W, alors $X'' \longrightarrow X'$ aussi (le changement de base par $X \longrightarrow Y$ transforme W dans W).

a′) Comme a), mais avec Y'', Y' W-asphériques.

a_1) Comme a), dans le cas particulier où Y' a un objet final et Y'' est la catégorie ponctuelle.

a_2) Dual de a_1), avec objet initial de Y'.

On a donc



et dans le cas $W = W_\infty$ toutes ces conditions sont équivalentes. J'ai envie [de] prendre la plus forte de ces conditions, mais je dois me donner des moyens pour trouver des W-fibrations en un sens aussi fort. Par exemple j'aurais envie de savoir que si X est dans Cat, alors $\underline{\mathrm{Ch}}_\infty(X) \longrightarrow X \times X$ ou $\underline{\mathrm{Ch}}(X) \longrightarrow X \times X$ est une W-fibration. Dans les deux cas, et (disons) pour W_∞, je le prouve en prouvant que p est à la fois lisse, et propre. Il faudrait donc revoir les notions de lissité et de propreté dans le contexte de la donnée d'un W. p. 5 [page 5]

Si je me rappelle bien, la propriété a′) [plutôt a_1)] était quelque chose d'assez aisément vérifiable, il suffisait de supposer que les

$$X_y \longrightarrow X/y \qquad \text{et} \qquad X/y \longrightarrow X/y'$$

(pour $y \in \mathrm{Ob}\, Y$ resp. $u : y \longrightarrow y'$ dans $\mathrm{Fl}\, Y$) sont dans W, et que ça reste vrai par tout changement de base - ce qui revient à la condition a_1) sur f. On pourrait donc formuler une condition sur W ainsi :

W(4) (Axiome des W-fibrations ([6]).) Soit $f : X \longrightarrow Y$ dans Cat satisfaisant la condition de W-fibration a_1) ci-dessus (p. 4), alors elle satisfait même à a), i.e. le changement de base par f transforme flèches dans W en flèches dans W ([7]).

Grâce à la proposition 2, il s'ensuit que la condition duale a_2) sur f implique la condition de W-fibration forte, donc l'axiome W(4), en apparence asymétrique, signifie aussi que les quatre conditions de fibration sur f sont équivalentes, il est donc en fait symétrique.

6. Voir variante affaiblie suffisante, dans le NB p. 10. Voir encore [?] de variantes pages 13, 14 ci-dessous.
7. i.e. une W-fibration à gauche est une fibration ultraforte.

NB Je ne vois pas, sans axiomes supplémentaires sur W, comment passer de a_1 à a' seulement. La difficulté, c'est de montrer, moyennant a_1 seulement, que si Y est W-asphérique, l'inclusion $X_y \longrightarrow X$ (où $y \in \mathrm{Ob}\, Y$) est dans W. Dans le contexte des dérivateurs **D**, cela se voyait par voie **D**-cohomologique.

Cet axiome très fort suffit à impliquer que si X, Y sont W-fibrés sur S, et si $f : X \longrightarrow Y$ est un S-morphisme qui est dans W, alors $f_s : X_s \longrightarrow Y_s$ est dans W ($\forall\ s \in S$). Montrons déjà au moins l'inverse.

Proposition 3. *Soient X, Y dans* Cat/S, *$f : X \longrightarrow Y$ une S-flèche. Supposons (X, Y W-fibrés) que $\forall\ s \in S$, $f_s : X_s \longrightarrow Y_s$ soit dans W, alors $f \in W$* ([8]). [page 6]

Il s'impose d'utiliser l'axiome de localisation, donc de regarder les f/s induits par f p. 6

$$\begin{array}{ccc} X/s & \xrightarrow[f/s]{} & Y/s \\ {\scriptstyle i_s^X}\uparrow & & \uparrow{\scriptstyle i_s^Y} \\ X_s & \xrightarrow{f_s} & Y_s \end{array} .$$

Commentary · Exposition [ms. 6]

From strict fibres to the total category

Proposition 3 is a descent statement, and the comma categories supply the descent data. Suppose $p : X \to S$ and $q : Y \to S$ satisfy the left fibration test and $u : X \to Y$ lies over S. Pulling p back to S/s, with terminal object (s, id_s), shows that the canonical inclusion $X_s \to X/s$ is in W; similarly for Y. The square in the text therefore gives

$$u_s \in W \quad \Longleftrightarrow \quad u/s \in W.$$

If this holds at every s, the relative axiom W(2) gives $u \in W$. The dual proof applies when both maps satisfy the right test. The maps must satisfy a common variance; a mixed pair is not covered by this argument.

There is also a direct reason why one representative per connected component suffices. For an arrow $a : s \to t$, the left fibration condition makes $X/s \to X/t$ and $Y/s \to Y/t$ weak equivalences. Comparing the two horizontal maps in the resulting square shows that $u_s \in W$ exactly when $u_t \in W$. Propagate this statement along a finite zigzag of arrows joining the chosen representative to any other object.

None of these steps uses W(4). That distinction is the point of the proposition: fibrewise equivalences imply a total equivalence by local-to-global descent, while a total equivalence need not be detectable on fibres without a stronger hypothesis on the theory. The path-space example below will show the failure sharply.

La condition a_1 suffit à impliquer que i_s^X et i_s^Y sont dans W, donc f/s est dans W si et seulement si f_s l'est. On n'a pas eu à utiliser l'axiome W(4), et il suffisait de supposer que X, Y soient fibrés 'W-cohomologiques' (ou W-fibrés à gauche) sur S, ou dualement, des W-fibrés à droite.

De plus :

Corollaire ([9]). *Soit $S_0 \subseteq \mathrm{Ob}\, S$ une partie de S telle que $S_0 \longrightarrow \pi_0(S)$ surjectif. Alors dans la proposition 3 il suffit de supposer $f_s \in W$ pour $s \in S_0$.*

Il suffit de voir que si $u : s \longrightarrow s'$ est une flèche dans S, alors $f_s \in W \iff f_{s'} \in W$. On utilise le fait que $f_s \in W \iff f/s \in W$, et de même pour $f_{s'}$. Donc à voir que

8. *Sans* axiome W(4). C'est la première fois qu'on a besoin de W(2) sous sa forme forte, à l'exclusion de W(2 bis).

9. Sans axiome W(4).

$f/s \in W \iff f/s' \in W$. Or on a

$$\begin{array}{ccc} X/s & \xrightarrow{f/s} & Y/s \\ {\scriptstyle X/u}\downarrow & & \downarrow{\scriptstyle Y/u} \\ X/s' & \xrightarrow{f/s'} & Y/s' \end{array} ,$$

où X/u et Y/u sont dans W, d'où le résultat.

Mais revenons à la réciproque : si $f \in W$, alors a-t-on $f_s \in W_S$? [plutôt $f_s \in W$?] Il y a un argument dans ce sens dans VI, p. 23, que je vais reprendre. L'idée est celle-ci. On considère la composante connexe Y_0 de Y [plutôt S_0 de S] contenant s, et on construit un W-fibré (au *sens fort* a))

$$S' \longrightarrow S_0 \subseteq S \qquad \text{[page 7]}$$

avec S' *W-asphérique.* Comme $X \longrightarrow Y$ sur S est dans W, il en sera donc de même de $X' \longrightarrow Y'$ sur S'. Comme les fibres de X', Y' en s' s'identifient à celles de X, Y en l'image de s' dans S, on est ramené au cas où S est W-asphérique. Mais alors on a

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle i_s^X}\uparrow & & \uparrow{\scriptstyle i_s^Y} \\ X_s & \xrightarrow{f_s} & Y_s \end{array} ,$$

où i_s^X, i_s^Y sont dans W, car déduites de $\{s\} \longrightarrow S$ (qui est dans W, S étant W-asphérique) par les changements de base W-fibrants $X \longrightarrow S$, $Y \longrightarrow S$. Donc $f \in W \implies f_s \in W$. p. 7

NB On n'a utilisé la propriété de fibration pour X, Y qu'au sens a') ([10]).

Il faut donc prouver, si on peut, le

Lemme (?) ([11]). *Soit S un objet* 0*-connexe de* Cat. *Alors il existe un objet W-asphérique S' de* Cat*, et une W-fibration (au sens* fort *de a), page 4) $S' \longrightarrow S$.*

Commentary · Exposition [ms. 7]

A path cover makes the converse local

The converse fibre test needs a cover on which the base becomes aspherical without changing the homotopy type of the fibres. Fix s_0 in a connected component of S, and use the endpoint map

$$\beta : \underline{\mathrm{Ch}}(S; s_0, -) \longrightarrow S.$$

Its source is intended to model based paths with free endpoint. Every object in the chosen component is the endpoint of a finite zigzag starting at s_0, so this cover reaches each object of that component.

Assume the two properties under construction: $E = \underline{\mathrm{Ch}}(S; s_0, -)$ is W-aspherical and β is ultraforte. If $u : X \to Y$ over S is in W, then the pullback $u_E : X_E \to Y_E$ is in W. For any $e_0 \in E$, the point map $e \to E$ is in W, by two-out-of-three with $E \to e$. The strong-aspherical-base condition for X_E and Y_E now places both fibre inclusions in W. Their comparison square gives $u_{\beta(e_0)} \in W$.

This formulation avoids an unnoticed component restriction: one pulls back u along the actual composite $E \to S$. The inclusion $S_0 \to S$ of the selected component is a pullback of the point inclusion $e \to \pi_0 S$. A functor to a discrete category is perfect, so this inclusion is perfect as well; W(4) makes it preserve W under base change. The composite of this inclusion with the perfect path cover has the required property. No prior assertion that restriction of an arbitrary W-map to one component remains in W is needed here.

The dependency that remains is real but explicitly acknowledged: asphericity of E must be proved before the converse fibre theorem is used. Deriving it from the desired conclusion about fibres of $\underline{\mathrm{Ch}}(S) \to S$ would be circular. The next paragraph changes the base from S to the category of path types precisely to avoid that circle.

10. Et même, dans a') on pourrait exiger que $Y'' = e$, Y' W_ω-asphérique, où W_ω [est] défini page 14, $W_\omega \subseteq W \cap W_\infty \ldots$

11. Est prouvé moyennant l'axiome W(4) (+ des petites vérifications ultérieures).

L'idée naturelle, c'est de prendre, pour $s_0 \in \mathrm{Ob}\, S$ fixé,

$$S' = \underline{\mathrm{Ch}}(S; s_0, -) \qquad \text{ou} \qquad S' = \underline{\mathrm{Ch}}_\infty(S; s_0, -),$$

et la flèche

$$\beta : c \longmapsto \text{but } c : S' \longrightarrow S.$$

On sait que S' est W_∞-aphérique et que β est W_∞-fibrante (et même parfaite). Il faut voir que S' est aussi W-asphérique, et β W-fibrante. Il faut donc revoir la démonstration dans le cas de W_∞, et prouver qu'elle se transforme à W, moyennant les axiomes qu'il faut sur W. [page 8] J'ai envie de regarder d'abord le choix $S' = \underline{\mathrm{Ch}}(S; s_0, -)$, sans passage à une limite inductive qui m'obligerait à mettre un axiome de limites sur W. Les propriétés homotopiques délicates et essentielles de $\underline{\mathrm{Ch}}(S)$ sont établies dans VII, p. 157-169, savoir que dans le diagramme

$$S \xrightarrow{i_S} \underline{\mathrm{Ch}}(S) \xrightarrow{p_S} S \times S,$$

$i_S \in W_\infty$ et p_S est une W_∞-fibration (en fait, *propre* et *lisse* pour W_∞) ([12]). Le fait que $i_S \in W$ `[plutôt` $i_S \in W_\infty$`]` implique que les foncteurs source et but

$$\underline{\mathrm{Ch}}(S) \underset{\beta}{\overset{\sigma}{\rightrightarrows}} S \qquad \sigma = \mathrm{pr}_1 \circ p_S,\ \beta = \mathrm{pr}_2 \circ p_S$$

sont dans W_∞ (puisque $\sigma i_S = \beta i_S = \mathrm{id}_S$). Comme ce sont des Hot-fibrations, il en résulte que les fibres $\underline{\mathrm{Ch}}(S; s, -)$ resp. $\underline{\mathrm{Ch}}(S; -, s)$ sont Hot-asphériques. Mais même en admettant que σ, β sont dans W et sont des W-fibrations, je ne peux en conclure tout de suite que les fibres sont W-asphériques - il y aurait cercle vicieux! Il me faut donc une démonstration directe.

Considérons le foncteur

$$\begin{array}{c} \underline{\mathrm{Ch}}(S; s, -) \\ \Big\downarrow{\scriptstyle p} \\ \underline{Ty} \end{array},$$

c'est un foncteur Cat-fibrant, donc W-lisse (je devrai développer la notion de W-lissité ([13]) p. 8 avec soin, j'admettrai pour le moment les propriétés formelles familières, et les justifierai par la suite). À cause de la W-lissité, pour prouver $p \in W$, il suffit de prouver que les fibres de [page 9] p sont W-asphériques ([14]). Or les fibres sont les $Z_\tau = \underline{\mathrm{Ch}}_\tau(S; s, -)$, qui sont des catégories *contractiles* (i.e. $Z_\tau \longrightarrow e$ est un homotopisme), donc W-asphériques par W(3) (cf. cor. 3, page 3).

Commentary · Exposition [ms. 9]

Why finite path types avoid a limiting axiom

The projection to path types separates combinatorics from homotopy. For a fixed finite oriented interval τ, the category $\underline{\mathrm{Ch}}_\tau(S; s, -)$ consists of its diagrams in S whose initial endpoint is s. Replace the last vertex by its predecessor and the last edge by the identity. This defines a functor on the diagram category; the last arrow of each diagram, with identities at the other vertices, defines a natural transformation to or from the identity functor, according to the orientation of the edge. Its image is the subcategory of diagrams with final edge an identity, canonically identified with the diagram category for the shortened interval. Repeat there and include back into the original category. Induction on the number of edges gives a finite zigzag from the identity to the constant-path functor

12. Dans l'énoncé des 'propriétés essentielles' de $\underline{\mathrm{Ch}}(S)$, on peut travailler avec W_ω, la plus petite partie de Fl(Cat) satisfaisant les axiomes W(1), W(2 bis), W(3) (cf. page 14), donc $W_\omega \subseteq W \cap W_\infty$.

13. $f : X \longrightarrow Y$ est W-lisse si les $X_y \longrightarrow y\backslash X$ sont W-asphériques.

14. On utilise ici W(2 bis), pas $W(2)$.

at s. This supplies the fixed-type contraction which Chapter VII states and later leaves unwritten (Grothendieck, 1990–1991o, manuscript pp. 152, 160, 170).

The total category over $\underline{Ty}$ is a Grothendieck fibration. Therefore it is W-smooth: every required category of lifts with prescribed target has a terminal cartesian lift. Aspherical strict fibres then give a W-equivalence to $\underline{Ty}$, by the dual of absolute theorem A. The unit path type is terminal in $\underline{Ty}$, so W(3) makes this base W-aspherical as well (Grothendieck, 1990–1991o, manuscript pp. 160–161). A claim that the fibres alone make the total category aspherical would omit this base hypothesis. Thus the path cover is built from categorical contractions and theorem A, before invoking any stability of filtered colimits.

The same economy explains the lifting calculation for $\underline{\mathrm{Ch}}(S) \to S \times S$. For endpoint arrows $u : a' \to a$, $v : b' \to b$ and a path $c : a \rightsquigarrow b$, form the concatenation $\widetilde{c} = \mathrm{ch}(u)c\,\mathrm{ch}(v)^{\mathrm{op}}$. The category of lifts is compared with $\underline{\mathrm{Ch}}(S; a', b')/\widetilde{c}$, which has a terminal object. Chapter VII constructs both comparison functors and links their composites to the identities by natural transformations (Grothendieck, 1990–1991o, manuscript pp. 161–166). These are actual categorical homotopies, so W(1)–W(3) put the comparisons in W. Applying the construction to the opposite category supplies the other variance. This establishes properness and smoothness independently of W(4). Only afterwards does W(4) upgrade the family to preservation of arbitrary W-maps.

The lifting-category comparison is a proof by reference to Chapter VII; the audit checked the cited construction and retains that dependency. The mapping-path factorization then follows from base-change and composition of perfect maps, with its first map a section of a universally aspherical source projection (Grothendieck, 1990–1991o, manuscript pp. 170–171). The distinction between finite zigzags and the infinite completion $\underline{\mathrm{Ch}}_\infty$, already developed in the commentary on Chapter VII, manuscript pp. 157–170, explains why the present proof need not assume W(6).

D'autre part, si on admet que $\underline{\mathrm{Ch}}(S) \xrightarrow{p_S} S\times S$ est W-lisse (et W-propre), donc $\underline{\mathrm{Ch}}(S) \xrightarrow{\sigma_S} S$ aussi, étant à fibres W-asphériques comme on vient de voir, il s'ensuit que $\sigma_S \in W$, d'où $S \xrightarrow{i_S} \underline{\mathrm{Ch}}(S)$ aussi dans W (puisque $\sigma_S \circ i_S = \mathrm{id}_S$). Donc il reste à prouver seulement que

$$(*) \qquad \underline{\mathrm{Ch}}(S) \xrightarrow{p_S} S \times S$$

est W-propre et W-lisse. Il en résultera que

$$\underline{\mathrm{Ch}}(S; s_0, -) \xrightarrow{\beta} S$$

est lui aussi W-propre et W-lisse, car déduit de p_S par le changement de base

$$S \longrightarrow S \times S \qquad s \longmapsto (s_0, s).$$

Il faut ensuite voir que W-lisse et W-propre implique la propriété de W-fibration *sous sa forme la plus forte* a) (alors que je prévois qu'a priori on ne l'aura que sous les formes faibles a_1, a_2 - pas même sous la forme a'). C'est ici donc que l'axiome W(4) des W-fibrations sera essentiel ([15])!

Voici l'argument pour la lissité de $(*)$, dans loc. cit. p. 161 ff. Il faut prouver que pour $c \in \mathrm{Ob}\,\underline{\mathrm{Ch}}(S)$, et une flèche $u \in \mathrm{Fl}(S \times S)$ `[plutôt` $(u, v) \in$ `Fl(`$S \times S$`)]`, une certaine catégorie de 'relèvements de u de source c ' `[plutôt 'relèvements de` (u, v) `de but` c `']`, $\mathcal{C} = \mathcal{C}(c, u, v)$, est W-asphérique ([16]). Pour ceci, on la compare à une catégorie $\mathcal{C}'$ à objet final, donc contractile et a fortiori W-asphérique, à l'aide de deux foncteurs

$$\mathcal{C} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \mathcal{C}',$$

et des homotopies $\varphi\psi \sim \mathrm{id}_{\mathcal{C}'}$ (sans mérite, $\mathcal{C}'$ étant contractile) et $\psi\varphi \sim \mathrm{Id}_{\mathcal{C}}$ (on trouve même une flèche $\psi\varphi \longrightarrow \mathrm{id}_{\mathcal{C}}$). Ainsi φ est un homotopisme, donc une W-équivalence (cor. 3, p. 3), donc $\mathcal{C}$ est W-asphérique, $\mathcal{C}'$ l'étant. La propreté de p_S équivaut à la lissité de $p_{(S^\circ)}$, elle est donc prouvée du même coup. [page 10]

15. sous la forme affaiblie W(4a″) (p. 13).
16. En fait, on prouve que $\mathcal{C}(c, u, v)$ est *contractile*, donc elle est W_ω-asphérique. p. 9

Ainsi, le lemme de la page 7 est prouvé, moyennant l'axiome W(4) qui assure qu'une flèche qui est à la fois W-lisse et W-propre, satisfait à la condition très forte a) des W-fibrations.

NB En fait, il suffirait, au lieu de l'axiome W(4) sous sa forme forte, de supposer qu'une $f : X \longrightarrow Y$ qui est à la fois 'homotopiquement lisse' et 'homotopiquement propre', i.e. les fibres des $X/x \longrightarrow Y/y$ et des $x\backslash X \longrightarrow y\backslash Y$ sont contractiles (donc W-asphériques pour tout W satisfaisant W(1,2,3)), soit W-fibration forte. Énonçons quand même ce que nous avons prouvé.

Proposition 4. *Si W satisfait* W(1,2,3) ([17]) (nous le supposons toujours par la suite) *plus* W(4) (sous une forme un peu affaiblie suffisante, cf. remarque précédente), *on a ceci: Soit $f : X \longrightarrow Y$ une flèche dans* Cat$/S$*, avec X, Y des W-fibrés sur S. Si $f \in W$, alors $f_s : X_s \longrightarrow Y_s$ est dans W pour tout $s \in S$.*

Je me demande si j'en sais assez pour pouvoir prouver

$$W \subseteq W_\infty.$$

Il faut quand même éliminer des cas canularesques, comme $W = \mathrm{Fl}(\mathrm{Cat})$, donc mettre l'axiome

W(5) (Axiome de connexité ([18]).) Si $f : X \longrightarrow Y$ est dans W, $\pi_0(f) : \pi_0(X) \longrightarrow \pi_0(Y)$ `[est]` bijectif.

Je factorise alors f en ([19]) [page 11]

$$X \xrightarrow{i} \bar{X} \xrightarrow{\bar{f}} Y,$$

avec $\bar{f}$ une W_∞-fibration et une W-fibration ([20]), $i \in W \cap W_\infty$ - cela marche finallement, par la construction de VII, p. 170-171 (de Cartan-Serre), à partir des propiétés de $Y \longrightarrow \underline{\mathrm{Ch}}(Y) \longrightarrow Y \times Y$ sans plus. Comme $\bar{f}i = f \in W$ et $i \in W$, on en déduit $\bar{f} \in W$. Comme $i \in W_\infty$, on en conclut $f \in W_\infty \iff \bar{f} \in W_\infty$. Donc pour prouver que f est dans W_∞, on est ramené au cas où f est une W-fibration et une W_∞-fibration. Mais alors la proposition 4 nous montre que les fibres de f sont W-asphériques ([21]), et il suffit de voir qu'elles sont W_∞-asphériques (proposition 3 ([22]) appliquée à W_∞). Ainsi on est ramené au cas où $Y = e$, i.e. à prouver que si X dans Cat est W-asphérique, il est W_∞-asphérique.

Soit $x_0 \in \mathrm{Ob}\, X$, considérons comme tantôt

$$E = \underline{\mathrm{Ch}}(X, x_0, -) \xrightarrow{p} X,$$

on sait que E est $W \cap W_\infty$-asphérique, et que c'est une fibration pour W et pour W_∞. Comme p. 10
X est W-asphérique (comme E), $p \in W$, donc à nouveau par la proposition 4 ([23]), les fibres de p sont W-asphériques, en particulier (par W(5)) 0-connexes. On a donc prouvé le

Lemme. ([24]) *Soit (X, x_0) dans* Cat$_\bullet$. *Si X est W-asphérique, alors $\Omega(X, x_0) \overset{déf}{=}$* [page 12]
$\underline{\mathrm{Ch}}(X; x_0, x_0)$ *l'est aussi* (moyennant les axiomes W(1,2,3,4) sur W). *Donc (par itération) les $\Omega^n(X, x_0)$ sont W-asphériques* ($\forall\, n \in \mathbf{N}$).

Corollaire. *Si de plus W satisfait l'axiome* W(5) *de connexité, alors X est W_∞-asphérique.*

En effet, 'on sait' que cela signifie que tous les $\Omega^n(X, x_0)$ sont 0-connexes. Or l'axiome de connexité W(5) assure que W-asphérique implique 0-connexe.

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17. Pour W(2), il le suffit sous la forme W(2 bis).
18. Voir à ce sujet le corollaire page 22.
19. À nouveau, dans ces arguments on a intérêt à travailler avec W_ω.
20. en fait, $\bar{f}$ est à la fois W_∞-parfait (i.e. W_∞-lisse et propre) et W-parfait (en fait, W_ω-parfait).
21. on n'a utilisé ici que cette propriété sur les fibres, comme consequence de W(4). (On utilise W(4) sous sa forme la plus faible W(4b″), cf. pages 13, 14).
22. on n'utilise la proposition 3 que dans le cas $Y = S$, pour lequel W(2 bis) suffit.
23. en fait, la condition W(4b″) sur W.
24. Utilise W(1, 2 bis, 3, 4b″) sur W.

Commentary · Exposition [ms. 12]

The loop comparison that closes Theorem 1

Grothendieck has isolated the decisive argument: a W-aspherical category remains W-aspherical after forming its based loop category. To obtain the theorem, these loops must detect the ordinary homotopy groups. This is the comparison left open in the running text, and it has a short proof once the ordinary fibration properties already used are available.

Let $E = \underline{\mathrm{Ch}}(X; x, -)$ and $F = \underline{\mathrm{Ch}}(X; x, x)$. The endpoint map $E \to X$ is a W_∞-fibration. Its strict fibre over x maps by a weak equivalence to the corresponding comma category, and the comma categories vary by weak equivalences along arrows of X. Quillen's theorem B therefore identifies BF with the homotopy fibre of $BE \to BX$ (Quillen, 1973, Theorem B and its corollary, pp. 97–98). Since BE is weakly contractible, this gives a pointed equivalence

$$B\underline{\mathrm{Ch}}(X; x, x) \simeq \Omega_x BX.$$

At the next step choose the constant loop as basepoint. Repeating the same argument gives

$$B\Omega^r(X, x) \simeq \Omega^r_x BX, \qquad \pi_0\Omega^r(X, x) \cong \pi_r(BX, x) \quad (r \geq 1),$$

with the right side regarded as its underlying pointed set where only connectivity is used.

Now W(4b″) applies to the perfect path map $E \to X$: both total space and base are W-aspherical, so their map is in W, and its fibre is W-aspherical. Induction gives this for every $\Omega^r(X, x)$. W(5) makes each of them nonempty and connected, hence all positive homotopy groups of BX vanish; W(5) applied to $X \to e$ gives connectivity in degree zero. Thus $X \to e \in W_\infty$.

Finally, for $u : A \to B \in W$, the canonical perfect replacement $A \to \bar{A} \to B$ has first map in both W and W_∞. The second map has W-aspherical fibres by W(4b″); the preceding argument makes those fibres weakly contractible. Its ordinary homotopy-fibre sequence makes it a weak equivalence, and two-out-of-three gives $u \in W_\infty$. The conclusion is proved; the source presents it with the loop comparison still deferred.

Mais le 'on sait' est un tout petit peu léger ([25]). Pour le justifier, il faudrait, ou bien que je fasse le lien avec la théorie 1/2-simpliciale classique, et prouver que les Ω^n que je construis sont 'les bons', i.e. que leur nerf redonne bien (à isomorphisme près dans Hot) les Ω^n classiques (??). En particulier, que le $\pi_0(\Omega^n(X, x_0))$ est bien le $\pi_n(X, x_0)$ 'de tout le monde'. Ou bien que je prouve directement que la 0-connexité des $\Omega^n(X, x_0)$ équivaut à la W_∞-asphéricité de X, définie cohomologiquement à la Artin-Mazur. C'est donc un travail qu'il me faudra faire absolument. Mais ce n'est plus qu'un simple travail de routine, et je peux énoncer dès maintenant le

Théorème 1. ([26], [27], [28]) *Soit* $W \subseteq \mathrm{Fl(Cat)}$ *satisfaisant les conditions* W(1,2,3,4,5) (axiome de saturation, de localisation, de l'objet final, de fibration, de connexité). *Alors*

$$W \subseteq W_\infty.$$

Corollaire. *On peut caractériser* W_∞ *comme la plus grande partie de* Fl(Cat) *satisfaisant aux conditions* W(1) *à* W(5) *précédentes.* [page 13]

Bien sûr, on aimerait un axiome 6 plus intéressant, qui joint aux précédents assure que $W = W_\infty$ ([29]). Ou est-ce que ces propriétés assurent déjà que $W = W_\infty$? Pour le voir, il faudrait peut-être que j'en sache plus sur la structure des objets W_∞-asphériques? Si c'était faux, je songe à un axiome de 'descente homotopique', qui impliquerait qu'un objet W_∞-asphérique est aussi W-asphérique.

25. C'est prouvé fort simplement dans le corollaire du 'theorème 1', page 190.

26. Il suffit même de W(4) sous une forme affaiblie, avec des foncteurs 'fibrants' dans un sens 'absolu'. Cf. le NB p. 10 - et surtout, cf. pages 13, 14.

27. NB Au lieu de W(5), il suffit de supposer $W \neq \mathrm{Fl(Cat)}$, cf. th. 2, p. 20.

28. Les hypothèses utiles sur W sont finalement W(1), W(2 bis), W(3), W(4b″), et $W \neq \mathrm{Fl(Cat)}$.

29. Cf. axiome W(6) page suivante.

Commentary · Present state [ms. 13]

The conjectural minimum is now known

The question on this page has a positive answer without the proposed additional axiom W(6). Cisinski proved that W_∞ is the smallest fundamental localizer (Cisinski, 2004, Theorem 2.2.11). The refinement to the absolute theorem-A axioms identifies it also as the smallest *weak* fundamental localizer (Cisinski, 2006a, Theorem 6.1.18). The source's W(1c) implies the weak-saturation condition in that theorem. Hence W(1), W(2 bis), and W(3) already give $W_\infty \subseteq W$.

Combining this inclusion with Theorem 1 gives $W = W_\infty$ under W(4b″) and W(5). The nondegeneracy argument at manuscript pp. 20–22 replaces W(5) by $W \neq \mathrm{Fl}(\mathrm{Cat})$. Accordingly, the conjecture at manuscript p. 15 is no longer open, and adding closure under retracts is unnecessary.

There is a local notational change to observe. On manuscript p. 13, W_ω is introduced as the intersection of classes satisfying W(1)–W(5). On the next page it denotes the minimum satisfying only the basic localizer axioms, preferably W(2 bis). These are different definitions during the exploration. The later theorem makes both classes equal to W_∞, but that equality must not be used to conceal their different roles in the original argument.

The proof strategy here also reappears in Cisinski's characterization by a theorem-B property: a fundamental localizer preserving components and making every locally constant equivalence aspherical must be W_∞ (Cisinski, 2004, Theorem 2.3.6). Its loop-space argument is the modern counterpart of the preceding manuscript proof.

p. 11

Je note dès maintenant que si des W_i ($i \in I$) satisfont aux axiomes W(1) à W(5), il en est de même de $\bigcap_i W_i$ ([30]). Donc il y a un plus petit ensemble de flèches W_ω satisfaisant à ces conditions. S'il n'est égal à W_∞, i.e. si les axiomes W(1) à W(5) ne caractérisent déjà W_∞, ce serait un localiseur vraiment très intéressant.

Je me rends compte que je n'ai utilisé qu'une forme assez faible de l'axiome W(4) - essentiellement, la validité de la proposition 4. Je vais résumer différentes variantes de l'axiome de fibration.

W(4)

a$_1$) Si $f : X \longrightarrow Y$ dans Cat est une W-fibration gauche, c'est une W-fibration ultraforte.

a$_2$) Si $f : X \longrightarrow Y$ est une W-fibration droite, c'est une W-fibration ultraforte.

a′) Si $f : X \longrightarrow Y$ est W-parfait, i.e. à la fois W-propre et W-lisse ([31]), alors f est une W-fibration ultraforte.

a″) Si $f : X \longrightarrow Y$ est à la fois homotopiquement propre et homotopiquement lisse (cf. NB page 10), f est une W-fibration ultraforte. (Il vaut mieux dire : Si f est W_ω-parfait, c'est une W-fibration ultraforte.)

b) Soit $f : X \longrightarrow Y$ une flèche dans Cat/S, avec X et Y des W-fibrés sur S. Si $f \in W$, alors $\forall\, s \in S$, $f_s : X_s \longrightarrow Y_s$ est `[dans]` W.

b′) Soit $f : X \longrightarrow Y$ une flèche de Cat/S qui soit W-parfaite et W_∞-parfaite, alors $f \in W$ implique que les fibres de f sont W-asphériques. (Il vaut mieux dire : Si f est W_ω-parfait, il est une W-fibration ultraforte.) `[Plutôt : Soit` $f : X \longrightarrow Y$ `une flèche dans` Cat/S `telle que` $X \longrightarrow S$ `et` $Y \longrightarrow S$ `soient` W_ω`-parfaites. Si` $f \in W$`, alors` $\forall\, s \in S$`,` $f_s : X_s \longrightarrow Y_s$ `est dans` W`.]`

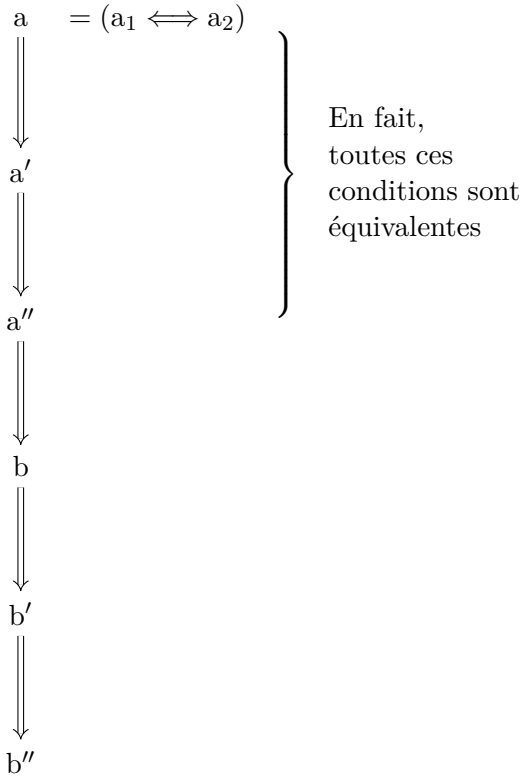
[page 14]

b″) Soit $W_\omega \subseteq W \cap W_\infty$ la plus petite partie de Fl(Cat) satisfaisant les conditions W(1,2,3) ([32]) (axiomes de saturation, de localisation, de l'objet final). Soit $f : X \longrightarrow Y$ une flèche dans Cat qui soit W_ω-parfaite. Si $f \in W$, alors les fibres de f sont W_ω-asphériques `[plutôt` W`-asphériques]`.

Entre ces propriétés de W, on a les implications tautologiques (sauf a″) $\Longrightarrow$ b), qui est essentiellement la proposition 4)

p. 12

30. C'est vrai pour tout ensemble d'axiomes dans la nature d'axiomes de stabilité.
31. ce qui implique que c'est une W-fibration gauche *et* droite.
32. Il vaut mieux prendre W(2 bis) au lieu de W(2), pour trouver W_ω la plus petite possible!



([33]).

Eh bien, dans le lemme page 12, il suffisait de supposer que W satisfait à la version la plus faible b″) de l'axiome des fibrations (sept versions!). En effet, les fibrations $\bar{X} \longrightarrow Y$ et $E \longrightarrow X$ utilisées dans la démonstration du lemme sont W_ω-parfaites, et il suffisait de savoir que si elles sont dans W, alors les fibres sont W-asphériques.

Commentary · Exposition [ms. 14]

What the weakest fibre axiom actually needs

The list of seven variants is an attempt to reduce the amount of fibration theory needed for the characterization. Its least demanding member, W(4b″), asks only this: a W-equivalence which is W_ω-perfect has W-aspherical strict fibres. The conclusion is relative to W, as the editors correct; it does not assert that the fibres are already W_ω-aspherical.

For the proof of Theorem 1 this is enough because the maps to which the axiom is applied have a special origin. The replacement $\bar{A} \to B$ and the based-path projection $\underline{\mathrm{Ch}}(X; x, -) \to X$ are W_ω-perfect before any fibration axiom for W is assumed. Two-out-of-three puts these maps in W, and W(4b″) then supplies exactly the asphericity required. Arbitrary W-fibrations need not be considered.

The distinction between preservation and detection also explains the middle variants. W(4a″) requires base change along an absolute perfect map to preserve W. W(4b′) concerns a map *between* two absolute perfect families over the same base. W(4b″) concerns one perfect map and compares it to the identity family on its target. The editorial replacement of the entire statement of b′ restores this intended progression.

The brace printed on this page spans the three a-conditions; it should not be read as a proof of every converse in the seven-term chain. Section IV returns to those converses and to the saturation needed for the localization arguments. In the modern setting, minimality and the nondegeneracy criterion below identify the two possibilities as W_∞ and the class of all maps, so these different rigidity axioms ultimately select the same possibilities. That later conclusion is stronger than the argument supplied by the diagram itself.

33. Je reviens sur les diverses variantes de W(4) plus bas, p. 53-56.

Mais je ne pense pas que même la version la plus forte W(4a) de l'axiome des fibrations, jointe aux trois axiomes précédents et à W(5) (connexité), implique $W = W_\infty$. Il faudrait tout au moins y joindre l'axiome

W(6) (Axiome des limites filtrantes.) Soit $X = \varinjlim_i X_i$ une limite inductive filtrante dans Cat. Si les X_i sont W-asphériques, X aussi.

Ceci devrait être équivalent à la propriété, en apparence plus forte, en termes de flèches de W :

Proposition 5. *Supposons qu'on ait* W(6) (en plus de W(1) à W(4) ([34])), *et soit* [page 15]

$$(f_i)_{i\in I} : (X_i)_{i\in I} \longrightarrow (Y_i)_{i\in I}$$

un homomorphisme de systèmes inductifs dans Cat*, indexé par un ensemble ordonné filtrant,* p. 13 *soient* $X = \varinjlim X_i$, $Y = \varinjlim Y_i$, $f = \varinjlim f_i$, $f : X \longrightarrow Y$. *Si* f_i *est dans* W *pour tout* i*, alors* $f \in W$.

On factorise chaque f_i en

$$X_i \xrightarrow{\alpha_i} \bar{X}_i \xrightarrow{\bar{f}_i} Y_i, \qquad \alpha_i \in W_\omega,\ \bar{f}_i \in \operatorname{Parf} W_\omega$$

([35]) suivant le système breveté. La construction étant fonctorielle, les $\bar{X}_i$ forment eux aussi un système inductif. De plus, comme le foncteur $S \longrightarrow \underline{\mathrm{Ch}}(S)$ qui donne naissance à la factorisation canonique des morphismes, commute aux $\varinjlim$ filtrantes (immédiat), il s'ensuit que, si on pose

$$\bar{X} = \varinjlim_i \bar{X}_i$$

et qu'on regarde les flèches

$$X \xrightarrow{\alpha} \bar{X} \xrightarrow{\bar{f}} Y, \qquad \alpha = \varinjlim \alpha_i,\ \bar{f} = \varinjlim \bar{f}_i,$$

avec $\bar{f}\alpha = \bar{f}$, cette factorisation de f est bien sa factorisation canonique. Donc $\alpha \in W$, $\bar{f} \in \operatorname{Parf} W_\omega$. Ceci vu, en vertu de prop. 3, $f \in W$, i.e. $\bar{f} \in W$, est vrai si les fibres de $\bar{f}$ sont W-asphériques. Mais les fibres de $\bar{f}$ s'interprètent comme limites inductives filtrantes de fibres des $\bar{f}_i$, lesquelles sont W-asphériques par la condition W(4b″) des fibrations. Donc l'axiome W(6) donne ce qu'il faut.

Commentary · Exposition [ms. 15]

Filtered fibres require a moving basepoint

Proposition 5 turns an axiom about aspherical objects into an axiom about maps by making the fibres of those maps explicit. Two finite-presentability assertions are implicit in the sketch, and both can be checked directly.

First, a path of a fixed finite type is a functor from a finite category, and a morphism between two such paths is specified by finitely many arrows and equations. If $S = \varinjlim_i S_i$ is a filtered colimit, any path or morphism of paths in S is represented at one stage, and an equality between representatives holds at some later common stage. Thus $\underline{\mathrm{Ch}}(-)$, including its source and target maps, commutes with filtered colimits. Since filtered colimits of sets commute with finite limits, the finite pullbacks defining the mapping-path replacement commute with them as well.

Second, a fibre over $y \in Y = \varinjlim_i Y_i$ is not the colimit of fibres over an unspecified fixed y. Choose a representative $y_i \in Y_i$, and let y_j be its image for $j \geq i$. There is a canonical isomorphism

$$(\bar{X})_y \ \cong\ \varinjlim_{j\geq i} (\bar{X}_j)_{y_j}.$$

34. W(4) sous la forme la plus faible W(4b″) .

35. W_ω introduit dans la condition W(4b″), page 14.

For objects, a representative x_j mapping to y has image equal to y_k at a sufficiently late stage k, so it becomes an object of the indicated fibre. For arrows, the condition that the image be the identity of y is likewise witnessed at a later stage. The same finite-stage argument proves injectivity and compatibility with composition. Passing to $j \geq i$ is cofinal and remains filtered.

Each map $\bar{f}_j$ belongs to W, since f_j and its replacement inclusion do. W(4b″) makes its fibres W-aspherical, W(6) makes their displayed colimit W-aspherical, and theorem A for a proper or smooth map gives $\bar{f} \in W$. Finally $\alpha \in W_\omega \subseteq W$, so $f = \bar{f}\alpha \in W$. The printed equation $\bar{f}\alpha = \bar{f}$ is a slip retained in the source.

For fundamental localizers in the modern sense, filtered-colimit stability is already a theorem, with no accessibility assumption (Cisinski, 2006a, Corollary 4.2.22). The calculation here remains useful because it shows exactly how a fibre argument reduces that stability to the object form.

Conjecture (hardie? sotte?). *Si* $W \subseteq \mathrm{Fl}(\mathrm{Cat})$ *satisfait les axiomes* W(1) *à* W(6) *(avec* W(4) *sous sa forme la plus faible* W(4b″) ([36])*), alors* $W = W_\infty$.

Mais il est bien possible qu'il faille renforcer un peu W(1) (saturation), en exigeant que tout rétracte d'un $f \in W$ soit dans W. [page 16]

Commentary · Present state [ms. 16]

The proposed retract axiom is unnecessary

The possible need for an extra retract axiom is resolved by the later theory. Every fundamental localizer with the relative theorem-A axiom W(2) is strongly saturated and consequently closed under retracts (Cisinski, 2006a, Proposition 4.2.4). In the weaker setting using W(2 bis), the specific equality conjectured here already follows from the weak minimal-localizer theorem and the fibre argument, so no independent retract hypothesis is required for that conclusion either. This does not assert strong saturation for every weak fundamental localizer before the fibre axiom is imposed.

This does not retroactively make every appeal to an isomorphism in Hot_W valid under two-out-of-three alone. The elementary proofs above deliberately establish membership in W directly. The later strong-saturation theorem explains when such arguments may be shortened; it should be cited at the point where that stronger conclusion is used.

Remarque. Considérons, pour $n \in \mathbf{N}$, l'ensemble $W_n \subseteq \mathrm{Fl}(\mathrm{Cat})$ des 'n-équivalences d'homotopie' dans Cat :

$$f \in W_n \iff \pi_0(f) \text{ bijectif, et } \pi_i(f,x) : \pi_i(X,x) \longrightarrow \pi_i(Y,f(x)) \text{ bijectif pour } 0 < i \leq n.$$

Je croyais que ça satisfaisait à tout ce qu'on peut demander à un localiseur, sans avoir pris la peine de l'écrire avec soin. Ma ça ne peut satisfaire à W(1) jusqu'à W(4b″), puisqu'on n'a pas $W_n \subseteq W_\infty$!

Commentary · Exposition [ms. 16]

Truncated equivalences really are localizers

The class W_n is not a defective homotopy theory. It is the ordinary theory observed through the first n homotopy groups. More precisely,

$$u \in W_n \quad \Longleftrightarrow \quad \tau_{\leq n} Bu \text{ is an equivalence,}$$

where $\tau_{\leq n}$ is Postnikov truncation. This functor is left adjoint to the inclusion of n-types (Lurie, 2009, Proposition 5.5.6.18). Its universal property gives a direct modern proof of the assertions the source recalls without repeating.

Identities, two-out-of-three, and the two-cycle condition hold because they hold for equivalences of n-types. Strong saturation follows too: the functor $\tau_{\leq n}B$, regarded as taking values in the homotopy category of n-types, factors through $\mathrm{Cat}[W_n^{-1}]$. Hence any map inverted there induces an equivalence of n-types and therefore was in W_n already. Categories with terminal objects have contractible nerves, giving W(3).

36. et peut-être W(2) sous la forme plus faible W(2 bis).

For the relative axiom W(2), let $X \to S$ be any functor. The simplicial bar construction on the covariant diagram $s \mapsto N(X/s)$ has bisimplices consisting of

$$x_0 \to \cdots \to x_q, \qquad p(x_q) \to s_0 \to \cdots \to s_r.$$

For a fixed q-simplex of NX, its r-direction is the nerve of $p(x_q)\backslash S$, which has an initial object. Realizing this direction therefore gives a weak equivalence of the diagonal bar construction with NX. In other words,

$$BX \simeq \operatorname*{hocolim}_{s \in S} B(X/s).$$

This is a homotopy colimit of spaces, retaining the arrows of S, rather than a colimit of their homotopy categories. If $X/s \to Y/s$ is in W_n for every s, apply $\tau_{\leq n}$ to the displayed colimits. As a left adjoint, truncation preserves them as colimits in the category of n-types. The resulting objectwise equivalence induces an equivalence of colimits, proving $X \to Y \in W_n$.

Finally, the nerve commutes with filtered colimits because each finite chain of arrows and each relation is witnessed at a finite stage. A Kan replacement such as Ex^∞ also commutes with filtered colimits: each simplex of Ex is represented by a map from a finite subdivision, and its sequential iteration commutes with colimits. Homotopy groups of Kan complexes are computed by finite simplicial spheres and finite homotopies. They consequently commute with filtered colimits, with basepoints represented at sufficiently late stages. This proves W(6) for W_n. What fails is the fibre axiom, as the next computation shows.

W(1) est trivial sur la définition (et on a sans larmes même la *saturation forte* de W_i, i.e. si f dans Cat est `[un]` isomorphisme dans $\mathrm{Hot}_{W_i} = (\mathrm{Cat})W_i^{-1}$, alors $f \in W_i$). Je crois me rappeler avoir vérifié W(2), par la même méthode que pour W_∞, en utilisant un critère cohomologique à la Artin-Mazur, pour exprimer $f \in W_i$, et en utilisant les suites spectrales de Leray pour $X \longrightarrow S$ et $Y \longrightarrow S$. La condition W(3) est trivialement satisfaite puisque $W_n \supseteq W_\infty$ et qu'elle l'est pour W_∞. Donc ça canule là où je m'y

attendais le moins : l'axiome des fibrations. Prenons la forme faible : Soit $f : X \longrightarrow Y$ p. 14
une W_ω-fibration et $f \in W_i$, les fibres X_s sont elles W_i-asphériques ([37])? On peut supposer X et Y 0-connexes, soit $x \in X_s$, la suite exacte d'homotopie donne

$$\pi_{i+1}(X) \xrightarrow[\text{si } i \leq n-1]{\text{iso}} \pi_{i+1}(Y) \longrightarrow \pi_i(X_s) \longrightarrow \pi_i(X) \xrightarrow[\text{si } i \leq n]{\text{iso}} \pi_i(Y),$$

donc

$$\pi_i(X_s) = 1 \qquad \text{si } 0 \leq i \leq n-1,$$

mais pour $i = n$ on trouve [page 17]

$$\pi_n(X_s) = \mathrm{Coker}\left(\pi_{n+1}(X) \longrightarrow \pi_{n+1}(Y)\right),$$

il n'y a aucune raison que ce soit nul, comme ça devrait si $X_s \longrightarrow e$ était dans W_n.

Commentary · Examples [ms. 17]

The missing group occurs one degree higher

Grothendieck's exact sequence identifies the precise loss of information. For a connected homotopy fibre F of an n-equivalence $E \to B$, with $n \geq 1$, exactness gives

$$\pi_i F = 0 \quad (0 \leq i < n), \qquad \pi_n F \cong \operatorname{coker}\bigl(\pi_{n+1}E \to \pi_{n+1}B\bigr).$$

At $n = 1$, the cokernel is still meaningful: the connecting homomorphism from the abelian group $\pi_2 B$ surjects onto $\pi_1 F$, so the latter group is abelian in this situation. The long exact sequence and its low-dimensional interpretation are recalled in Hatcher (2002, Theorem 4.41).

37. $\omega > \infty$, $W_\omega \subset W_\infty \subset W_n$

For an example entirely within Cat, take S to be the poset of nonempty proper faces of an $(n+2)$-simplex. Then BS is the barycentric subdivision of its boundary, hence has the homotopy type of S^{n+1}. Choose a vertex s, and use

$$E = \underline{\mathrm{Ch}}(S; s, -) \xrightarrow{\beta} S.$$

The total space BE is contractible, and β is W_ω-perfect by the construction recalled on manuscript pp. 8–11. Since the homotopy groups of the sphere vanish through degree n, $\beta \in W_n$. Its fibre $F = \underline{\mathrm{Ch}}(S; s, s)$, however, satisfies

$$BF \simeq \Omega S^{n+1}, \qquad \pi_n BF \cong \mathbb{Z}.$$

Thus $F \to e \notin W_n$, contradicting W(4b″). The same square shows that β does not preserve all W_n-equivalences under base change: $e \to S$ is in W_n, whereas its pullback $F \to E$ is not.

For $n = 0$, the sphere example remains valid, with $\pi_0 \Omega S^1 \cong \mathbb{Z}$, but the displayed *group cokernel* in the source requires correction. In general $\pi_0 F$ is the pointed coset set of the image of $\pi_1 E$ in $\pi_1 B$, not its quotient by the normal closure. Take $G = S_3$, $H = \langle (12) \rangle$, and the action groupoid $E = (G/H) /\!\!/ G$. The projection $E \to BG$ is both a Grothendieck fibration and an opfibration; its strict fibre is the discrete three-element set G/H. Both $E \simeq BH$ and BG are connected, so the projection is in W_0. Yet the group cokernel of $H \to G$ is trivial, since the conjugates of (12) generate S_3. The fibre has three components. This is a genuine degree-zero qualification, not a defect in the argument for positive n.

Il est vrai que parfois on renforce justement la condition pour W_n, disons qu'on définit $W'_n \subseteq W_n$ par

$$f \in W'_n \iff \begin{cases} \text{a)} & \pi_0(f) \text{ bijectif,} \\ \text{b)} & \pi_i(f,x) : \pi_i(X,x) \longrightarrow \pi_i(Y, f(x)) \text{ bijectif pour } 0 < i \leq n \text{ et } x \in X, \\ \text{c)} & \pi_{n+1}(f,x) : \pi_{n+1}(X,x) \longrightarrow \pi_{n+1}(Y, f(x)) \textit{ surjectif,} \end{cases}$$

c'est à dire aussi

$$f \in W'_n \iff \begin{cases} \text{les 'fibres homotopiques' } Z \text{ de } X \longrightarrow Y \text{ (i.e. les fibres de } \bar{X} \xrightarrow{\bar{f}} Y) \\ \text{sont } n\text{-connexes, i.e. 0-connexe et } \pi_i(Z_y, z) = 0 \text{ pour } 1 \leq i \leq n. \end{cases}$$

Alors X dans Cat sera W'_n-asphérique si et seulement s'il est W_n-asphérique, i.e. n-connexe. Donc par la forme 2 de la définition de W'_n, l'axiome W(4b″) est satisfait. Il faut croire que ça pèche ailleurs! Peut-être déjà la saturation.

Commentary · Examples [ms. 17]

Connectivity repairs fibres but loses cancellation

The strengthened class W'_n measures a connectedness condition on a map: its homotopy fibres are n-connected. The long exact sequence makes this equivalent to bijectivity on $\pi_0, \ldots, \pi_n$ and surjectivity on π_{n+1}. For the terminal map $X \to e$, the last surjectivity condition is automatic, so W_n and W'_n have exactly the same aspherical objects.

The failure of two-out-of-three can be seen with the same finite category S modeling S^{n+1}. Let $a : e \to S$ select a vertex and let $b : S \to e$ be the terminal functor. Then $ba = \mathrm{id}_e \in W'_n$. Also $b \in W'_n$: it is an isomorphism on homotopy groups through degree n, and the map $\mathbb{Z} = \pi_{n+1} S \to 0$ is surjective. But $a \notin W'_n$, because $0 \to \mathbb{Z}$ is not surjective. This works also for $n = 0$, using the circle.

Thus the problematic cancellation is exactly

$$b,\ ba \in W'_n \quad \not\Rightarrow \quad a \in W'_n.$$

The source's informal argument about surjective homomorphisms is correct; the example supplies actual functors and verifies every homotopy group involved. A class of connected maps can be stable under many geometric operations without being the class inverted by a localization. The modern theory treats connectedness as one side of a factorization system; it is not required to satisfy the cancellation law for equivalences.

Si gf et g sont dans W'_i, il n'est pas forcément de même de f,

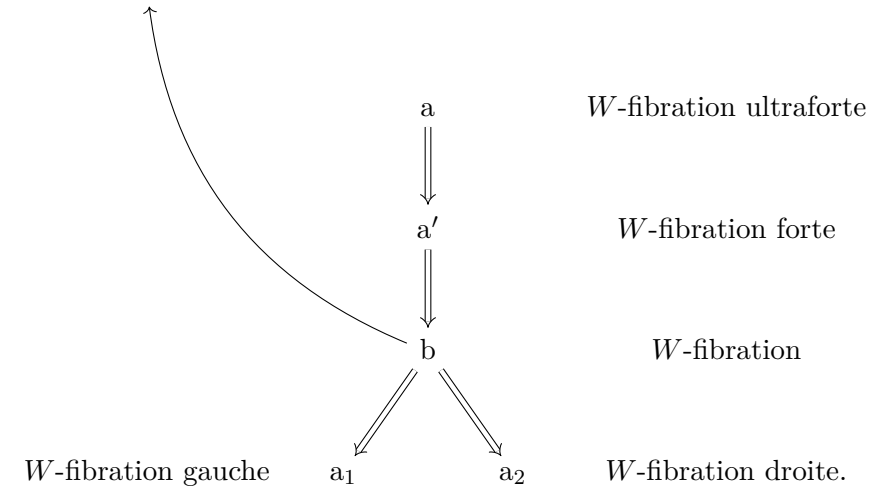
car si $\bar{g}\bar{f}$ (et donc $\bar{g}$) est surjectif, il n'en est par forcément de même pour $\bar{f}$! Donc W'_n ne mérite pas le nom de 'localiseur'.

Par contre, W_n me paraît excellent, si on ne lui demande pas l'impossible - de satisfaire à un 'axiome de fibration' qui semble bien (à peu de choses près) caractériser le seul W_∞ parmi les 'bons' localiseurs. En tous cas l'axiome de connexité W(5) est satisfait, et l'axiome des limites filtrantes W(6) aussi. Par ailleurs W(1) (et même la saturation forte), W(2), W(3) sont satisfaits par tout $W_{\mathbf{D}}$ associé à un dérivateur - ce qui confirme à nouveau leur importance. Les axiomes W(4) (l'axiome 'draconien' des fibrations, ± spécial à W_∞), W(5) (connéxité) et W(6) (limites) m'apparaissent comme de nature plus spéciale, je ne songerais à imposer aucun d'eux pour l'usage terminologique de 'localiseur fondamental'. p. 15 [page 18]

II Perplexités sur les axiomes et la terminologie. p. 16

Dans I et la plupart du temps, en travaillant avec les dérivateurs, j'ai supposé $\boxed{W_{\mathbf{D}} \supseteq W_\infty}$, hypothèse qui m'a parue essentielle (à tort ([38]) ou à raison). On voit donc que si $W_{\mathbf{D}}$ satisfait à l'axiome de fibrations W(4), même sous sa forme la plus faible W(4b″), il est *égal* à W_∞. Cela montre bien que cet axiome doit être vu comme pratiquement une *caractérisation* de W_∞ (parmi les localiseurs fondamentaux qui contiennent W_∞ – et ces localisateurs me paraissent les plus importants, et peut-être les seuls vraiment utiles).

La théorie des **D**-fibrations dans I, p. 111-126, a été faite dans l'esprit qu'on vient de dire, en supposant $W_{\mathbf{D}} \supseteq W_\infty$. La notion de **D**-fibrations à laquelle je m'étais finalement arreté était par suite différente des notions envisagées ici dans un esprit différent (page 4). Elle revient à tester la condition a) (trop forte presque toujours, sauf pour $W = W_\infty$), caractérisant les W-fibrations ultrafortes, par des $Y'' \longrightarrow Y'$, où Y'', Y' sont tous deux W_∞-asphériques (ce qui est plus exigeant que W-asphérique). C'est seulement quand Y'' et Y' sont W_∞-asphériques (au lieu de W-asphériques), qu'on demande que $X'' \longrightarrow X'$ soit dans W. On trouve donc une [page 19] notion moins exigeante que celle de W-fibration forte. Si on dénote ici cette condition par b), on trouve (sous réserve que $W_\infty \subseteq W$), avec les notations de la page 4 ([39])



a_1), a_2) étaient aussi appelées **D**-fibrations cohomologique resp. **D**-fibrations homologiques ([40]). Quand **D** satisfait Der 3a′ (I, p. 111) sur les **D**-coefficients localement constants sur les X W_∞-asphériques, alors les notions b), a_1), a_2) sont équivalentes. (Je présume que c'est le cas pour tous les **D** et tous les W vraiment intéressants.) Par contre, j'ai l'impression que la notion de W-fibration forte, voire ultraforte, n'est guère utilisable que dans le cas $W = W_\infty$, où elle coïncide avec les précédentes. C'est dire qu'on peut probablement oublier la terminologie, et se rappeler par contre que dans le cas $W = W_\infty$, les W-fibrations ont des propriétés d'une p. 17 force extrême, s'exprimant par les deux propriétés dégagées dans W(4a) et W(4b).

Commentary · Exposition [ms. 19]

Changing the test bases changes the theory

The revised hierarchy separates the homotopy type of the indexing category from the coefficients used to observe it. A W_∞-aspherical category is an absolutely contractible test shape. A W-aspherical category may instead be invisible only after truncation or after a chosen cohomological localization. Demanding preservation over the latter shapes is stronger when $W_\infty \subseteq W$.

38. à *tort* il s'avère maintenant.

39. Cf. la bonne notion de W-fibration dans un contexte où on ne suppose pas $W_\infty \subseteq W$, dans th. 3 et déf. 3, p. 51, 52.

40. Non, c'est une notion encore plus forte, à coups de coefficients localement constants, cf. I, p. 121.

For W_n, the sphere category used above is W_n-aspherical but not W_∞-aspherical. It is therefore an admissible base in the early strong test a', and the path family fails that test; it is not an admissible base in the new condition b. This is the concrete reason that the revised notion can accommodate truncated homotopy theories. The source's footnote 40 also corrects a different confusion: a condition tested using every locally constant coefficient is stronger than a condition on the underlying localizer alone. The two distinctions should not be merged.

The opening inference of section II needs the nondegeneracy hypothesis that Theorem 2 will make explicit. From $W_{\mathbf{D}} \supseteq W_\infty$ and W(4), one cannot conclude $W_{\mathbf{D}} = W_\infty$ without excluding $W_{\mathbf{D}} = \mathrm{Fl}(\mathrm{Cat})$. The terminal derivator, whose value on every indexing category is the terminal category, gives $W_{\mathbf{D}} = \mathrm{Fl}(\mathrm{Cat})$; all the fibration implications then hold trivially. With $W_{\mathbf{D}} \neq \mathrm{Fl}(\mathrm{Cat})$, the corrected inference follows from Theorems 1 and 2. This is the degenerate exception already recognized at manuscript p. 10, rather than a new obstruction.

[page 20]

Dans la théorie de HOT_W, influencé par la réflexion générale sur les dérivateurs et l'axiome Der 3b, j'ai travaillé sous l'hypothèse $W \supseteq W_\infty$, mais en utilisant un axiome de fibration L5 = W(4b) qui, il s'avère, implique finalement $W = W_\infty$. J'ai l'impression que je n'ai jusqu'à présent (propriétés de $f_!$ pour HOT) pas utilisé $W \supseteq W_\infty$, donc ça peut être utile, comme 'ça ne coûte pas plus cher' (et oblige d'être plus attentif à ce qu'on utilise), de laisser tomber $W \supseteq W_\infty$ (impliquant $W = W_\infty$), et de mettre par contre sur W des axiomes aussi forts (notamment de fibration) que les propriétés de W_∞, 'sans se priver en rien'. Je supposerai donc W(1), W(2), W(3), W(4a) (la forme la plus forte de l'axiome de fibration, cf. pages 13, 14) - c'est tout ce qui a été utile jusqu'à présent. Si on y joint W(5) (axiome de connexité) ça implique $W \subseteq W_\infty$. Mais pour l'instant je ne crois pas en avoir un besoin.

Commentary · Development (commentators') [ms. 20]

A fibre criterion that avoids component restriction

Grothendieck wants to deduce preservation of components from the nontriviality of the theory. The displayed proof begins by restricting a W-map to each component of its target; footnote 42 correctly observes that this invokes a stronger fibre-detection statement than W(4b″). The theorem can be proved with the weaker axiom. The following formulation isolates the repair. We did not find this exact formulation in the sources checked for this audit; it is an elementary reorganization of the fibre argument, with no claim of priority.

Proposition (two-step nondegeneracy criterion). [ours] Let V be a class of functors containing the categorical homotopisms and satisfying two-out-of-three and absolute theorem A: if every $A/b \to e$ lies in V, then $A \to B$ lies in V. Suppose also that for every $u : A \to B$ in V and every $b \in B$, its ordinary homotopy fibre has the weak homotopy type of a category F_b with $F_b \to e \in V$. Then either every map of V induces a bijection on components, or $V = \mathrm{Fl}(\mathrm{Cat})$.

Proof. We first record two consequences of the stated axioms. If $\emptyset \to e \in V$, then every $\emptyset \to C$ belongs to V by absolute theorem A, since all its comma categories are empty. Two-out-of-three applied to $\emptyset \to C \to e$ gives $C \to e \in V$. Applying it once more to $C \to D \to e$ gives every functor $C \to D \in V$.

If $D \to e \in V$, then $D \times D \to e \in V$. Indeed, for $d \in D$, the comma category of the second projection is $(D \times D)/d \cong D \times (D/d)$. The category D/d has a terminal object, so the projection $D \times (D/d) \to D$ is a categorical homotopism: its section at that terminal object has composite naturally joined to the identity. Hence $(D \times D)/d \to e \in V$. Absolute theorem A gives $D \times D \to D \in V$, and composition gives the assertion. When D is empty the assertion is immediate.

Now take $u \in V$ for which $\pi_0 u$ is not bijective. If a component of B is missed, choose b in it. The homotopy fibre of Bu at b is empty: a point in that fibre would supply a path from an image point to b. Consequently its categorical model F_b is empty as well, because a category with an object has nonempty realization. The preceding paragraph gives $V = \mathrm{Fl}(\mathrm{Cat})$.

Suppose instead that two distinct components of A map into the same component of B. Choose one object in each and paths from their images to some b. These give two points of the homotopy fibre which cannot lie in the same component, since projection to BA would connect the chosen objects. Its categorical model $D = F_b$ is therefore disconnected and V-aspherical. The product argument makes $D \times D$ V-aspherical, so its diagonal $d : D \to D \times D$ belongs to V. Choose objects d_0, d_1 in different components of D. The homotopy fibre of Bd at (d_0, d_1) is empty: it would

otherwise provide a point connected to both d_0 and d_1. Apply the fibre hypothesis to this diagonal. It again gives $\emptyset \to e \in V$, and hence $V = \mathrm{Fl(Cat)}$. This proves the proposition.

For the source's class W, the hypothesis about homotopy fibres follows from its canonical factorization $A \to \bar{A} \to B$. The first arrow belongs to $W_\omega \subseteq W \cap W_\infty$; the second is W_ω-perfect and lies in W, so W(4b″) makes its strict fibres W-aspherical. Ordinary theorem B identifies those fibres with the homotopy fibres of the original map. The proposition therefore proves Theorem 2 using W(1), W(2 bis), W(3), and W(4b″), including the weakened hypotheses asserted in footnote 28 of Theorem 1.

Mais méditant sur la question, je crois avoir trouvé le

Théorème 2 (!) ([41]) : *Soit* $W \subseteq \mathrm{Fl(Cat)}$ *satisfaisant* W(1,2,3,4b″)*. Si* W *ne satisfait pas à* W(5)*, alors* $W = \mathrm{Fl(Cat)}$.

Démonstration. Soit $f : X \longrightarrow Y$ dans Cat, tel que $\pi_0(f)$ ne soit pas bijectif. Soit I la catégorie discrète ayant pour ensemble d'objets l'ensemble $\pi_0(Y)$, on a donc un diagramme commutatif

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & I, & \end{array}$$

d'autre part un foncteur d'une catégorie dans une catégorie discrète est toujours W_ω-parfait, donc p, q sont des W-fibrations. Comme $f \in W$, il s'ensuit $f_i \in W$ pour tout $i \in I$ ([42]).

Si $\pi_0(f)$ n'était pas surjective, prenant i qui n'est pas dans l'image, on aurait $X_i = \emptyset \longrightarrow Y_i \neq \emptyset$, qui serait dans W. Soit donc [page 21]

$$f : X = \emptyset \longrightarrow Y$$

dans W. Alors $\forall$ Z dans Cat, $f \times \mathrm{id}_Z : \underbrace{X \times Z}_{=\emptyset} \longrightarrow Y \times Z$ est dans W (car Z est W-fibrée sur e), donc $\emptyset \longrightarrow Y \times Z$ est dans W. Donc aussi la projection $Y \times Z \longrightarrow Y$. Comme [p. 18]

$$\begin{array}{ccc} Y \times Z & \longrightarrow & Y. \\ & \nwarrow \quad \nearrow & \\ & \emptyset & \end{array}$$

cette projection est une W-fibration, il s'ensuit par W(4b″) que ses fibres sont W-asphériques. Comme $Y \neq \emptyset$, il y a une telle fibre, elle est $\simeq Z$. Donc Z est W-asphérique. Donc on trouve que *toute* catégorie Z dans Cat est W-asphérique. Donc toute flèche $Z \longrightarrow Z'$ dans Cat est dans W, q.e.d.

Supposons maintenant $\pi_0(f)$ surjective, mais non injective. Il y aurait une fibre de $\pi_0(X) \longrightarrow \pi_0(Y) = I$ qui serait de cardinal > 1. En prenant le f_i correspondant, on trouve $X_i \longrightarrow Y_i$ avec Y_i 0-connexe et $\mathrm{card}\,\pi_0(X_i) \geq 2$. Donc on est ramené au cas

$$f : X \longrightarrow Y \qquad \begin{cases} f \in W \\ Y \text{ 0-connexe} \\ \mathrm{card}\,\pi_0(X) \geq 2. \end{cases}$$

Considérons la factorisation canonique

$$X \xrightarrow{i_f} \bar{X} \xrightarrow{\bar{f}} Y,$$

avec $i_f \in W$, donc $\bar{f} \in W$. Donc par W(4b″) les fibres de $\bar{f}$ sont W-asphériques. Mais on voit tout de suite, en décomposant $\bar{X}$ suivant les composantes connexes X_i de X, que ces fibres ont un π_0 de cardinalité $\geq \mathrm{card}\,\pi_0(X) \geq 2$. Donc on est ramené au cas où $Y = e$, i.e.

$$X \text{ } W\text{-asphérique}, \qquad \mathrm{card}\,\pi_0(X) \geq 2.$$

41. Cf. pour un résultat plus fort XV 9.12, p. 133 bis. **NB** Le résultat est plus général, mais pas vraiment plus fort, car il ne donne pas $W = \mathrm{Fl(Cat)}$.

42. On utilise W(4) sous forme plus forte que W(4b″)!

Considérons $X \times X$, qui est aussi W-asphérique, et l'application diagonale [page 22]

$$d : X \longrightarrow X \times X = Y,$$

qui est donc dans W. Mais $\pi_0(d)$ s'identifie à l'application diagonale $\pi_0(X) \longrightarrow \pi_0(X) \times \pi_0(X)$, qui n'est pas surjective puisque $\operatorname{card} \pi_0(X) \geq 2$. Donc pour la première partie, $W = \mathrm{Fl}(\mathrm{Cat})$.

Donc sous forme plus positive :

Corollaire. *Soit* $W \subseteq \mathrm{Fl}(\mathrm{Cat})$ *qui satisfait à* W(1,2,3,4b″)*, et telle que* $W \neq \mathrm{Fl}(\mathrm{Cat})$. *Alors* W *satisfait l'axiome de connexité* W(5)*, et par suite (th. 1, page 12)* $W \subseteq W_\infty$.

p. 19

III Classes remarquables de foncteurs liés à W.

Finalement, j'appelerai *'localiseur fondamental'* une partie W de Fl(Cat) satisfaisant les trois conditions suivantes :

(i) Saturation W(1) (cf. rappel p. 1).

(ii) Axiome de l'élément final W(3) (page 3).

(iii) La variante affaiblie W(2 bis) de l'axiome de localisation W(2) donnée dans le corollaire page 1 (anciennement c'était L4′ dans Pursuing Stacks).

Quand j'aurai besoin de plus sur W, je le dirai explicitement. Ces trois axiomes ont suffi pour pratiquement tout ce que j'ai fait avec W dans Pursuing Stacks, la seule exception étant un usage (si je me rappelle) unique, quelque part, d'une variante affaiblie de W(2) (pour les S-foncteurs *cartésiens* entre S-catégories *fibrées* - si sur toute fibre il est dans W, il est lui-même dans W).

Commentary · Exposition [ms. 22]

The localizer retained for the geometric argument

Grothendieck now separates the part of his theory that depends on a notion of weak equivalence from the much stronger requirements imposed on fibrations. This is the point of retaining only W(1), W(3), and W(2 bis). The distinction is consequential: a map can have a well-behaved theory of lifting along arrows without preserving every weak equivalence under pullback. Throughout this section, a category is *W-aspherical* when its map to the terminal category belongs to W. For $W = W_\infty$, this means that its nerve has weakly contractible realization. The word does not mean that only its higher homotopy groups vanish; connectedness and trivial fundamental group are included.

The notation for comma categories fixes the direction of every assertion that follows. For $f : X \to Y$, an object of X/y is a pair $(x, \alpha : f(x) \to y)$, whereas an object of $y\backslash X$ is $(x, \alpha : y \to f(x))$. The strict fibre X_y contains the objects mapping *equal* to y and only those arrows mapping to 1_y. Even when several objects of Y are isomorphic to y, they do not enter this strict fibre. These three categories solve different problems. The comma categories allow a comparison with y; the strict fibre requires equality over y.

An aspherical functor has all X/y aspherical. W(2 bis) then says that it belongs to W. This is the absolute form of Quillen's Theorem A. W(2), by contrast, tests a map $X \to Y$ over an arbitrary common base S through all $X/s \to Y/s$. The formally stronger axiom will be named again at the corollary on manuscript page 30; it is not imported into the preceding elementary proofs. In later terminology, the distinction is between the absolute and relative Theorem A axioms: Maltsiniotis (2012, §1.1) calls the version with the absolute axiom a *weak* fundamental localizer and reserves "fundamental localizer" for the relative version. Cisinski (2006a, Theorem 6.1.18) proves that the minimal weak fundamental localizer is W_∞. That minimality theorem does not assert that every weak fundamental localizer satisfies the relative axiom. The manuscript's terminology is still settling here.

The next propositions convert these axioms into geometric tests. Their economy comes from arranging a comparison so that a category with a terminal or initial object appears, and then using two-out-of-three. No model structure on Cat, and no derived direct-image functor, is needed for that argument.

On suppose donc par la suite donné un localiseur fondamental

$$W \subseteq \mathrm{Fl}(\mathrm{Cat}).$$

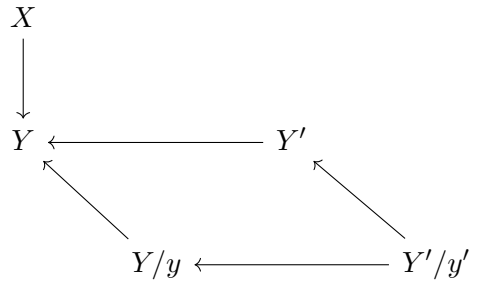
Proposition 1. *Soit $f : X \longrightarrow Y$ dans* Cat. *Conditions équivalentes :*

a) Pour tout isomorphisme local $Y' \longrightarrow Y$, $f' : X' \longrightarrow Y'$ est $\in W$.

b) Même condition, dans le cas particulier $Y' = Y/y \longrightarrow Y$, donc $\forall\ y \in Y$, $X/y \simeq X \times_Y (Y/y)$ est W-asphérique.

Définition 1. Sous ces conditions équivalentes a), b) de la proposition 1, on dit que f est *W-asphérique.* On dit [que] f est *W-coasphérique* si f° est W-asphérique, i.e. si les $y\backslash X$ sont W-asphériques (pour $y \in \mathrm{Ob}\, Y$). [page 23]

Démonstration. Il suffit de prouver b) $\Longrightarrow$ a). Par définition d'un isomorphisme local, on voit que la condition b) est stable par changement de base $Y' \longrightarrow Y$, avec Y' un isomorphisme local (puisque les $Y'/y' \xrightarrow{\sim} Y/y$, donc $X'/y' \xrightarrow{\sim} X/y$). Il suffit donc de voir que si f satisfait à b), alors $f \in W$. Mais c'est l'axiome W(2 bis).



Remarque. Si f a un adjoint à droite g, alors pour tout $y \in Y$, $X/y \simeq X/g(y)$, donc X/y a un objet final et est donc W-asphérique. Donc si f a un adjoint à droite (à gauche), alors f est W-asphérique (W-coasphérique). p. 20

Mais même si f n'a pas d'adjoint à droite, il y a quand même un adjoint généralisé

$$g : Y \longrightarrow X^\wedge$$

et on a

$$X/y \simeq X/g(y) \qquad \text{(pour l'inclusion pleinement fidèle } X \hookrightarrow X^\wedge),$$

donc $\mathrm{Top}(X/y)$ s'identifie au topos induit $\mathrm{Top}(X)/g(y)$. La condition f W-asphérique signifie donc que les faisceaux $g(y)$ dans $\mathrm{Top}(X)$ sont W-asphériques. Pour de nombreuses autres formulations équivalentes, je renvoie à Pursuing Stacks. [page 24]

Commentary · Exposition [ms. 23]

The generalized right adjoint is a presheaf of arrows

The passage through presheaves explains why the comma categories are the correct substitutes for an adjoint. Define

$$g(y)(x) = \mathrm{Hom}_Y(f(x), y), \qquad g : Y \longrightarrow [X^{\mathrm{op}}, \mathrm{Set}].$$

Precomposition makes $g(y)$ a presheaf on X, and postcomposition makes g covariant in y. Its category of elements has objects (x, α) and arrows $h : x \to x'$ satisfying $\alpha = \alpha' f(h)$. It is therefore exactly X/y. If $f \dashv r$, the adjunction identifies $g(y)$ with the representable presheaf $X(-, r(y))$; the category of elements then becomes $X/r(y)$ and has terminal object $1_{r(y)}$. The asphericity of a functor admitting a right adjoint is thus the representable case of the construction in the text.

The asserted identification of topoi can also be made explicit. For any presheaf P on X, a presheaf $A \to P$ determines on its category of elements the presheaf

$$(x, p) \longmapsto \{a \in A(x) : a \mapsto p\}.$$

Conversely, from a presheaf B on $\int_X P$, recover $A(x)$ as the disjoint union of $B(x,p)$ over all $p \in P(x)$. Restriction along $h : x \to x'$ uses the arrow $(x, P(h)p') \to (x', p')$. These operations are inverse up to their canonical isomorphisms and yield

$$[(\textstyle\int_X P)^{\mathrm{op}}, \mathrm{Set}] \simeq [X^{\mathrm{op}}, \mathrm{Set}]/P.$$

Taking $P = g(y)$ proves the statement about $\mathrm{Top}(X/y)$. Its use here is interpretive: the failure of an actual right adjoint is measured by the homotopy type of a presheaf of possible arrows, rather than by the absence of a distinguished representing object.

The commentary on Chapter I, manuscript pages 64–66, discusses local constancy and the Yoneda argument. Here the additional point is that an adjoint is sufficient but far from necessary: it supplies a terminal element of every comma category, while W-asphericity requires only their homotopical triviality.

Proposition 2. *Soit* $X \xrightarrow{f} Y \xrightarrow{g} Z$ *dans* Cat. *Si* f *et* g *sont* W*-asphériques (*W*-coasphériques), de même* gf*. Si* gf *et* f *sont* W*-asphériques (*W*-coasphériques),* g *aussi.*

Rappel. Un morphisme $p : S' \longrightarrow S$ dans Cat est un *isomorphisme local* si $\forall\ s' \in \mathrm{Ob}\, S'$, le morphisme induit

$$p_{s'} : S'/s' \xrightarrow{\sim} S/s \qquad (s = p(s'))$$

est un isomorphisme : p induit un isomorphisme sur les 'localisées' de S' et S en s' et $s = p(s')$ respectivement. Il est immédiat que cette notion est stable par composition, que si gf et g sont des isomorphismes locaux, alors f l'est aussi, que tout isomorphisme est un isomorphisme local. De plus, les isomorphismes locaux sont *stables par changement de base,* ce qui résulte du fait (trivial et important) que si on a un carré cartésien dans Cat

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & S', \end{array}$$

et si $x' \in \mathrm{Ob}\, X'$ a pour image x, s', s dans X, S', S, alors le diagramme

$$\begin{array}{ccc} X/x & \xleftarrow{q/x'} & X'/x' \\ {\scriptstyle f/x}\downarrow & & \downarrow{\scriptstyle f'/x'} \\ S/s & \xleftarrow{p/s'} & S'/s' \end{array}$$

est également cartésien. (Donc si p/s' est `[un]` isomorphisme, q/x' aussi.)

Commentary · Exposition [ms. 24]

Local isomorphisms and the scope of cancellation

The "local isomorphisms" in this argument are discrete fibrations. Indeed, an isomorphism $S'/s' \to S/p(s')$ says that every arrow ending at $p(s')$ has exactly one lift ending at s', and that its factorization triangles lift uniquely. Conversely, that unique-lifting property gives the displayed isomorphism of slice categories. The categories of elements of presheaves supply the basic examples. In particular, $Y/y \to Y$ is such a map.

This description makes Proposition 1 transparent. If every X/y is W-aspherical and $q : Y' \to Y$ is a local isomorphism, then, for $y' \in Y'$ with image y, strict pullback gives

$$(X \times_Y Y')/y' \cong X/y.$$

Theorem A applied over Y' puts $X \times_Y Y' \to Y'$ in W. Conversely, the particular base changes $Y/y \to Y$ recover the same condition because Y/y has a terminal object.

The cancellation in Proposition 2 uses precisely this class of allowed base changes. Given $X \xrightarrow{f} Y \xrightarrow{g} Z$ and a local isomorphism $Z' \to Z$, form the three pullbacks X', Y', Z'. Its pullback $Y' \to Y$ is again a local isomorphism. Thus asphericity of f gives $f' \in W$. If g is also aspherical, $g' \in W$ and hence $g'f' \in W$. If instead gf is aspherical, then $g'f' \in W$ and two-out-of-three gives $g' \in W$. Proposition 1 now gives the two assertions. The opposite-category argument proves the coaspherical forms.

There is no cancellation statement with g and gf implying asphericity of f. For example, let f select the terminal object of [1], and let $g : [1] \to e$. Both g and gf are aspherical, whereas the comma category of f over 0 is empty. For any W with $\varnothing \to e \notin W$, f is not aspherical. This elementary directional distinction is exactly what the local-isomorphism proof preserves.

p. 21
[page 25]

Revenons à la proposition 2. La stabilité de la notion de morphisme asphérique par composition résulte aussitôt de la forme a) de la condition. De même pour la condition que si gf et f sont W-asphériques, g aussi - il faut montrer que si $Z' \longrightarrow Z$ est `[un]` isomorphisme local, alors g' est dans W, or l'hypothèse implique que $g'f'$ et f' le sont.

$$\begin{array}{ccc} X & \xleftarrow{r} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{q} & Y' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g'} \\ Z & \xleftarrow{p} & Z' \end{array}$$

Corollaire 1. *L'image inverse d'un $f : X \longrightarrow Y$ qui est W-asphérique (resp. W-coasphérique) par un isomorphisme local (resp. par un coïsomorphisme local) est W-asphérique (W-coasphérique).*

$$\begin{array}{ccccc} X & \longleftarrow & X' & \longleftarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \underset{\text{iso local}}{\longleftarrow} & Y' & \underset{\text{iso local}}{\longleftarrow} & Y'' \end{array}$$

Résulte formellement de la définition par a) et du fait qu'un composé de deux isomorphismes locaux (resp. coïsomorphismes locaux) est itou.

NB Bien sûr, la notion de foncteur W-asphérique n'est pas stable par changement de base quelconque. Par exemple, l'inclusion $X = \{y_0\} \longrightarrow Y$ est W-asphérique si et seulement si les catégories discrètes $\mathrm{Hom}(y_0, y)$ sont W-asphériques, i.e. (sous réserve que l'on ait W(5), axiome de connexité) si et seulement si les $\mathrm{Hom}(y_0, y)$ sont de cardinal 1, i.e. y_0 *objet initial* de Y. Mais si $Y \neq \{y_0\}$, prenant y dans Y, $y \neq y_0$, et posant $Y' = \{y\} \longrightarrow Y$, on a $X' = X \times_Y Y' = \emptyset$ et $\emptyset \longrightarrow e$ n'est pas asphérique!

[page 26]

Par contre, on s'attend que les morphismes W-asphériques soient stables par changements de base W-*lisses* (qui généralisent les isomorphismes locaux), et les morphismes W-coasphériques par changements de base W-propres. Cf. plus bas.

Commentary · Exposition [ms. 26]

Properness asks for contractible categories of lifts

The anticipated base-change theorem needs a condition on the map along which one pulls back. Grothendieck finds it by inspecting every possible lift of an arrow. For $f : X \to Y$, an object $x \in X$, and $u : f(x) \to y$, write $L_f(x, u)$ for the category whose objects are arrows $a : x \to z$ with $f(a) = u$

and whose morphisms are arrows $h : z \to z'$ over 1_y satisfying $ha = a'$. Its morphisms matter as much as its objects: merely asking for at least one lift would lose the homotopy information.

Let $j_y : X_y \to X/y$ send z to $(z, 1_y)$. The object (x, u) of X/y gives an equality of categories

$$(x, u)\backslash j_y \cong L_f(x, u).$$

The same category is the strict fibre over u of $x\backslash X \to f(x)\backslash Y$. These identifications prove the equivalence of conditions (a), (b), and (d) in Proposition 3. The edition's intermediate list "a) b) c)" on manuscript page 27 should read "a) b) d)"; the remaining argument establishes (c).

To see that argument without suppressing an endpoint, pull f back along the arrow $u : [1] \to Y$. Write $E = X \times_Y [1]$. For an object in E_1, the comma category into $E_1 \hookrightarrow E$ has an initial object. For an object $(x, 0)$ in E_0, that comma category is $L_f(x, u)$. Consequently $E_1 \hookrightarrow E$ is coaspherical exactly when all these lift categories are aspherical. This is why a test over the walking arrow is sufficient for *properness*: the test asks for coasphericity of the inclusion, not merely that its total homotopy type be an equivalence.

The dual incoming-lift category consists of arrows $a : z \to x$ over a fixed $u : y \to f(x)$. Its morphisms are vertical factorizations. Requiring these categories to be W-aspherical defines smoothness. The terminology records two directions of the same lifting geometry. The commentary on Chapter I, manuscript pages 52 and 68, explains the coefficient-theoretic origin; here Grothendieck obtains the criterion directly from W.

Corollaire 2. *Soit $f : X \longrightarrow Y$ dans* Cat *une* équivalence *de catégories. Alors f est W-asphérique et W-coasphérique.*

En effet, f a un adjoint à gauche et un adjoint à droite, et on utilise la remarque page 23.

Proposition 3. *Soit $f : X \longrightarrow Y$ dans* Cat. *Conditions équivalentes :* p. 22

a) Pour tout $y \in \mathrm{Ob}\, Y$, le foncteur d'inclusion

$$(*) \qquad X_y \longrightarrow X/y$$

*est W-co*asphérique.*

b) Pour tout $x_0 \in X$, le morphisme induit pour les colocalisées

$$x_0\backslash X \longrightarrow y_0\backslash Y \qquad (\text{où } y_0 = f(x_0))$$

est à fibres W-asphériques.

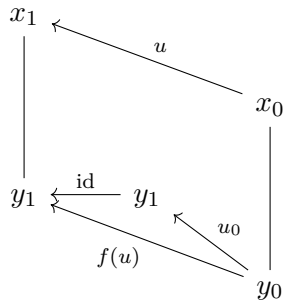
c) Pour tout morphisme $\mathbf{\Delta}^1 = Y' \longrightarrow Y$, considérons $X' = X \times_Y Y'$ sur $\mathbf{\Delta}^1$, l'inclusion $X'_1 \longrightarrow X'$ de la fibre fermée de X' (fibre sur le point fermé de Y') est W-coasphérique.

d) Pour toute flèche $u_0 : y_0 \longrightarrow y_1$ dans Y, et tout x_0 dans X_{y_0}, la catégorie $X(x_0; u_0)$ formée des flèches $u' : x_0 \longrightarrow x$ dans X de source x_0, et qui relèvent u `[plutôt` u_0`]`, *est W-asphérique.*

Démonstration. Pour expliciter la condition a), il faut expliciter la catégorie $\xi_0\backslash(X_y)$, [page 27] pour un objet quelconque $\xi_0 = (x_0, u_0)$ de X/y, $x_0 \in \mathrm{Ob}\, X$,

$$u_0 : y_0 \longrightarrow y_1 = y \quad \text{dans } Y, \text{ avec } y_0 = f(x_0).$$

Cette catégorie $\xi_0\backslash X_{y_1}$ s'explicite alors comme la catégorie des x_1 dans X_{y_1}, *munis* d'une flèche de ξ_0 dans l'image $\xi_1 = (x_1, \mathrm{id}_{f(x_1)} = \mathrm{id}_{y_1})$ de x_1 dans X/y_1 dans ξ_0 `[‘dans` ξ_0`’ à effacer]`, i.e. d'une flèche u de x_0 dans x_1, telle que $f(u)$ donne lieu au triangle commutatif dans Y, i.e. $f(u) = u_0$.



Donc c'est la catégorie notée $X(x_0; u_0)$ dans d). On voit tout de suite qu'elle est aussi isomorphe à la fibre de $x_0\backslash X \longrightarrow y_0\backslash Y$ en l'élément u_0 de $y_0\backslash Y$, dont il est question dans b). D'où l'équivalence de a) b) c). D'autre part, la donnée de u_0 dans Y équivaut à celle d'un foncteur $\mathbf{\Delta}^1 \longrightarrow Y$, comme dans c), et pour exprimer alors que $\underbrace{X'_1}_{\simeq X_{y_1}} \longrightarrow X'$ est coasphérique, il suffit d'exprimer que les $x'\backslash X'_1$ ($x' \in \mathrm{Ob}\, X'$) sont W-asphériques. Or c'est trivial si $x' \in \mathrm{Ob}\, X'_1$ (objet initial), donc il suffit de le vérifier pour x' dans $X'\backslash X'_1 = X'_0 \simeq X_{y_0}$. Alors x' s'identifie à un élément x_0 de X_{y_0}, et on vérifie tout de suite que $x_0\backslash X'_1$ est isomorphe à la catégorie $X(x_0; u_0)$ de d). Cela achève la démonstration.

Définition 2. Sous les conditions de la proposition 3, on dit que f est *W-propre.* On dit que f est *W-lisse* si f° est W-propre, i.e. si les morphismes d'inclusion $(*)$ `[plutôt` $X_y \longrightarrow y\backslash X$`]` sont W-asphériques etc. (conditions duales de celles de la proposition 3). p. 23 [page 28]

Remarque. La condition a) de la proposition 3 est satisfaite en particulier si l'inclusion $(*)$ a un adjoint à gauche (cf. remarque page 23). Cela signifie que les catégories $X(x_0; u)$ ont un objet initial. Cela signifie aussi que la catégorie X sur Y est Cat-cofibrée sur Y au sens faible, i.e. qu'il existe, pour $u_0 : y_0 \longrightarrow y_1$ flèche dans Y, et x_0 dans X_{y_0}, un objet $u_{0*}(x_0)$ dans X_{y_1} qui représente le foncteur covariant $X_{y_1} \longrightarrow$ (Ens), $x_1 \longmapsto \mathrm{Hom}_u(x_0, x_1)$. Ainsi, si X est Cat-cofibrée au sens faible sur Y, et a fortiori si elle est Cat-cofibrée, alors $f : X \longrightarrow Y$ est W-propre. Dualement, si f est faiblement Cat-fibrée, f est `[`W`-]`lisse.

Commentary · Examples [ms. 28]

Representable lifts and their homotopical relaxation

Grothendieck first places ordinary fibred categories inside the new theory. Fix $x \in X$ and $u : f(x) \to y$. An initial object $a : x \to u_*x$ of $L_f(x, u)$ gives, for every $z \in X_y$, a bijection

$$\mathrm{Hom}_{X_y}(u_*x, z) \xrightarrow{\sim} \{b : x \to z : f(b) = u\}, \qquad h \longmapsto ha.$$

Conversely, such a representing arrow is initial in $L_f(x, u)$. Choosing it for all (x, u) is the weak cofibred condition used here. It only asserts this factorization property for arrows over the fixed arrow u; the stronger condition on composition is part of the usual opfibration formalism. Either condition gives properness, since initial categories are W-aspherical. The dual argument gives smoothness for prefibrations.

A useful calculation shows how much weaker asphericity can be than representability. Let C be the poset with three objects b, d, e and relations $b < d$, $b < e$. Form its left cone $X = C^{\triangleleft}$ by adjoining an initial object a, and let $p : X \to [1]$ send a to 0 and C to 1. The outgoing-lift category from a over $0 \to 1$ is C itself. It is contractible because b is initial. This example is already a weak opfibration. To remove representability while retaining contractibility, replace C by the four-object fence

$$b < d, \qquad e < d, \qquad e < c.$$

Its nerve is the path $b - d - e - c$, hence contractible, but it has no initial object. In the corresponding cone, $L_p(a, 0 \to 1) \cong C$ is therefore aspherical without being representable by a universal lift. For every identity in the base, the lift category is a coslice in the appropriate fibre and has an initial object. There are no other base arrows to test. Thus this p is W_∞-proper but is not weakly opfibred.

The dual cone gives a smooth map without the prefibration property. Maltsiniotis (2005, Proposition 4.3 and Example 4.17) places these criteria in the broader language of asphericity

structures; for a structure coming from a fundamental localizer, the resulting weak smoothness criterion recovers W-smoothness. For an arbitrary asphericity structure those notions need not agree.

Proposition 4. ([43]) *La notion de morphisme W-propre est stable par composition, par changement de base. Les coisomorphismes locaux sont propres (trivial sur b)), a fortiori les isomorphismes sont propres, les immersions fermées sont propres.*

Énoncé dual pour les morphismes lisses : stabilité par changement de base, par composition, les isomorphismes locaux sont lisses, notamment les immersions ouvertes.

[page 29]

La stabilité par changement de base est immédiate sur la forme c). La stabilité par composition `[est]` moins triviale :

$$\begin{array}{lccccc} x & X & \longleftarrow & x\backslash X & \longleftarrow & \bar{X} \\ & \downarrow{\scriptstyle f} & & \downarrow & & \downarrow \\ y & Y & \longleftarrow & y\backslash Y & \longleftarrow & \bar{Y} \\ & \downarrow{\scriptstyle g} & & \downarrow & & \downarrow \\ z & Z & \longleftarrow & z\backslash Z & \longleftarrow & \{u\} \end{array}$$

Dans le diagramme à carrés cartésiens ci-contre `[plutôt dont les deux carrés de droite sont cartésiens]` (déduit de $x \in \mathrm{Ob}\, X$, d'où $y = f(x)$, $z = g(y) = gf(x)$, et de $u \in \mathrm{Ob}\, z\backslash Z$), il faut prouver que $\bar{X}$ est W-asphérique. On sait que $\bar{Y}$ l'est (g propre) et que $\bar{X} \longrightarrow \bar{Y}$ est à fibres W-asphériques. Il suffit donc d'utiliser la proposition 5 ci-dessous, et la remarque immédiate

Lemme. *Si f est propre, il en est de même de $x\backslash f : x\backslash X \longrightarrow y\backslash Y$, pour tout x dans X, $y = f(x)$.*

En effet, si u' est dans $x\backslash X$ et v' dans $y\backslash Y$, et si x', y' sont leurs images dans X, Y respectivement, alors

p. 24

$$u'\backslash(x\backslash X) \simeq x'\backslash X, \qquad v'\backslash(y\backslash Y) \simeq y'\backslash Y,$$

et la flèche entre les deux s'identifie à la flèche de localisation

$$x'\backslash X \longrightarrow y'\backslash Y.$$

Commentary · Exposition [ms. 29]

The composition proof has no circular dependence

The proof of composition proceeds by factoring a category of lifts through another category of lifts. It uses Proposition 5, but Proposition 5 itself needs only the definition of properness and the already established base-change stability. Thus its appearance after Proposition 4 does not create a circular proof.

First prove the part of Proposition 5 that will be used. If $q : A \to B$ is proper and every strict fibre A_b is aspherical, then $A_b \to A/b$ is coaspherical and hence belongs to W. Two-out-of-three makes A/b aspherical, and W(2 bis) gives $q \in W$. The same reasoning applies after every base change, because properness and the asphericity of the fibres persist. The smooth version follows by taking opposites.

Now let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be proper. Fix $x \in X$ and $u : gf(x) \to z$. There is a functor

$$L_{gf}(x, u) \longrightarrow L_g(f(x), u), \qquad a \longmapsto f(a).$$

43. Dire aussi : Si $X \xrightarrow{f} Y \xrightarrow{g} Z$, si gf et f W-lisses, alors g est lisse en les points de $\mathrm{Im}\, f$, idem pour propre.

It is the base change of $x\backslash f : x\backslash X \to f(x)\backslash Y$ along the fibre of $f(x)\backslash g$ over u. The coslice functor $x\backslash f$ is proper: taking the coslice at an arrow $x \to x'$ reduces its incoming data to $x'\backslash X \to f(x')\backslash Y$, precisely the original lifting test. Its displayed base change is therefore proper. The fibre over a lift $v : f(x) \to y'$ is $L_f(x, v)$, which is aspherical. Proposition 5 makes the displayed functor a W-equivalence. Since its target is aspherical by properness of g, its source is aspherical too. This proves properness of gf.

The same diagram proves the observation in footnote 43. If f and gf are proper, its source is aspherical, and the displayed proper functor still has aspherical fibres. Its target is therefore aspherical. This proves the lifting condition for g at every object $f(x)$ in the strict image of f; it does not assert anything about objects outside that image. Duality proves the smooth statement. These arguments require W(2 bis), not the full relative axiom W(2).

Proposition 5. *Soit $f : X \longrightarrow Y$ dans* Cat*, supposons f W-propre ou W-lisse, et à fibres W-asphériques, alors $f \in W$ (et même f est* universellement *dans W, i.e. il est dans W et le reste par tout changement de base).*

Comme les hypothèses sont stables par changement de base, il suffit de prouver $f \in W$. [page 30] Supposons p. ex. f propre, prouvons que f est *W-asphérique*, i.e. que les X/y ($y \in \mathrm{Ob}\ Y$) sont W-asphériques. Mais $X_y \longrightarrow X/y$, étant W-coasphérique par hypothèse de propreté, est dans W (par W(2 bis)), et comme X_y est W-asphérique par hypothèse, X/y aussi. Mais par W(2 bis), f W-asphérique $\Longrightarrow f \in W$. Le cas f lisse est dual.

Remarque. Ainsi, si f est propre, s'il est à fibres W-asphériques il est 'universellement W-asphérique', i.e. est et reste W-asphérique par tout changement de base. Inversement, si f est universellement dans W (pas forcément propre), alors en particulier ses fibres sont W-asphériques. Mais ça n'implique bien sûr pas que f soit propre, ou lisse ([44]).

Commentary · Examples [ms. 30]

A universal equivalence with neither lifting property

The qualification after Proposition 5 has a small model which makes the role of the lifting condition visible. Let P be the poset with objects u, v, w, z and exactly the strict relations

$$u < v, \qquad u < z, \qquad w < z.$$

Define $p : P \to [1]$ by $p(u) = p(v) = 0$ and $p(w) = p(z) = 1$. Both fibres are intervals, and the nerve of P is the path $v - u - z - w$. More is true than the contractibility of these three nerves: p has a contraction over its base.

The section $s : [1] \to P$ selects u and z. Let $R : P \to P$ send v to u and fix u, w, z. Each displayed relation is preserved, so R is a functor over $[1]$. The pointwise inequalities give natural transformations over $[1]$

$$\mathrm{id}_P \longleftarrow R \longrightarrow sp.$$

For the first transformation use $u \to v$ at v; for the second use $w \to z$ at w; all remaining nonidentity components are absent. Since $ps = \mathrm{id}_{[1]}$, these transformations exhibit p as a homotopy equivalence of categories over $[1]$. After any base change $T \to [1]$, the section and both transformations pull back, so $P \times_{[1]} T \to T$ is still a homotopy equivalence. It therefore belongs to every W considered in this section.

Nevertheless p is not W_∞-proper: the arrow $0 \to 1$ has no outgoing lift from v, so its lift category is empty. Nor is it W_∞-smooth: the same arrow has no incoming lift ending at w. Thus universal membership in W does not imply either geometric lifting property. The same failure holds for any W for which the empty category is not aspherical.

This also verifies the terminology in footnote 44. A universal W-equivalence is universally aspherical because, after any base change and any further local-isomorphism test, one is still taking a base change of the original map. The four-object example is universally aspherical and universally coaspherical without being proper or smooth. The latter properties inspect the lift category at each particular object, which a relative contraction may move.

44. Mais 'universellement dans W' implique déjà 'universellement W-asphérique', puisque la notion de W-asphéricité est définie par `[la]` proposition 1 a).

Corollaire. *Si W satisfait* W(2) (pas seulement W(2 bis))*, alors pour un S-morphisme $X \longrightarrow Y$ de catégories qui sont toutes deux W-propres sur S, ou toutes deux W-lisses, si $f_s : X_s \longrightarrow Y_s$ est dans W pour tout $s \in S$, alors $f \in W$.*

Supposons p. ex. f propre, considérons, pour $s \in \mathrm{Ob}\ S$ [page 31]

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ {\scriptstyle i_s^X}\downarrow & & \downarrow{\scriptstyle i_s^Y} \\ X/s & \xrightarrow{f/s} & Y/s, \end{array}$$

comme X, Y [sont] propres, les flèches verticales sont dans W (il suffirait de beaucoup moins que la propreté …), donc $f_s \in W \Longrightarrow f/s \in W$. Ainsi, '$f$ est dans W localement sur S', ce qui implique $f \in W$ par W(2).

Commentary · Exposition [ms. 31]

Fibrewise comparison uses the structural maps

The corollary compares two families over S; it does not assume that the comparison $f : X \to Y$ is itself proper. The phrase "Supposons p. ex. f propre" at the start of the proof is an edition slip. The case being considered is that both structural maps $X \to S$ and $Y \to S$ are proper. This is confirmed by the next sentence and by the two vertical arrows in the displayed square.

In fact, the proof needs only

$$X_s \longrightarrow X/s \in W, \qquad Y_s \longrightarrow Y/s \in W \quad \text{for every } s \in S.$$

If the fibre map f_s belongs to W, the commutative square and two-out-of-three imply $f/s \in W$. The remaining step from all these slice comparisons to $f \in W$ is exactly the relative axiom W(2). Replacing all slices by coslices gives the smooth case, using the opposite form of the same axiom.

This distinction will persist when properness is weakened to preproperness. The geometric hypothesis supplies comparisons between fibres and comma categories; the relative localization axiom assembles the comma comparisons over the whole base. There is no appeal to the converse assertion that a weak equivalence between total categories must be a weak equivalence on fibres. That substantially stronger assertion belongs to the fibration discussion later in this chapter.

p. 25

Proposition 6. *Soit*

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow[p]{} & S', \end{array}$$

cartésien dans Cat. *Si f* [est] *W-asphérique, p W-lisse, alors f' est W-asphérique.*

Il faut prouver [que] $X' \times_{S'} S'/s'$ est W-asphérique ($s' \in \mathrm{Ob}\ S'$). Soit $s = p(s')$, on a

$$\begin{array}{ccccc} S & \xleftarrow{p} & S' & & \\ & \nwarrow & & \nwarrow & \\ & & S/s & \xleftarrow[p_0]{} & S'/s', \end{array}$$

avec p_0 W-lisse (lemme p. 28 [plutôt 29]) et à fibres W-asphériques, donc une W-équivalence universelle (prop. 5). Comme l'image inverse de X sur S/s est W-asphérique (hypothèse pour f), il [s'ensuit]

que dans le carré cartésien ci-contre [ci-dessous] [page 32]

$$(*)\qquad \begin{array}{ccc} X/s & \xleftarrow{W} & X'/s' \\ \downarrow & & \downarrow \\ S/s & \xleftarrow{W} & S'/s' \end{array}$$

les flèches horizontales sont dans W, et la première flèche verticale l'est par hypothèse sur f, donc la seconde aussi, donc X'/s' `[est]` W-asphérique.

Commentary · Exposition [ms. 32]

Why smooth base change preserves asphericity

Grothendieck's anticipated stability theorem now follows from the same categories of lifts, with the two variables interchanged. For a smooth $p : S' \to S$ and $s' \in S'$, put $s = p(s')$ and consider

$$p_{s'} : S'/s' \longrightarrow S/s.$$

The dual of the coslice lemma makes $p_{s'}$ smooth. Its strict fibre over an arrow $v : t \to s$ is the category of incoming lifts of v ending at s'. Smoothness of p says that this category is aspherical. Proposition 5 therefore makes $p_{s'}$ a universal W-equivalence.

For an arbitrary $X \to S$, the square

$$\begin{array}{ccc} (X \times_S S')/s' & \longrightarrow & X/s \\ \downarrow & & \downarrow \\ S'/s' & \xrightarrow[p_{s'}]{} & S/s \end{array}$$

is a strict pullback. An object of either description of its upper-left corner is a pair (x, t') together with an arrow $t' \to s'$ and the equality $f(x) = p(t')$; the arrow conditions agree as well. Its top map belongs to W, independently of any assumption on X. If $X \to S$ is aspherical, X/s is aspherical, so the top map makes the upper-left category aspherical. This proves Proposition 6.

For the corollary, apply this same universal equivalence once to X and once to Y. If $f : X \to Y$ over S satisfies $f/s \in W$, the commutative comparison square with horizontal arrows in W gives $f'/s' \in W$. Thus smooth pullback preserves W_S^{g}, the class detected on all over-categories. Opposite categories give the preservation of W_S^{d} by proper pullback. Neither argument needs the extra inference $W_S^{\mathrm{g}} \subseteq W$; accordingly the proof works with W(2 bis) alone.

Corollaire (généralise `[1a]` prop. 6, obtenue en faisant $Y = S$). *Soit $p : S' \longrightarrow S$* `[un]` *morphisme W-lisse dans* Cat. *Alors le foncteur changement de base*

$$\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$$

est compatible avec les localiseurs W_S^{g}, $W_{S'}^{\mathrm{g}}$, i.e. transforme $f : X \longrightarrow Y$ qui est dans W localement *sur S, dans $f' : X' \longrightarrow Y'$ dans W localement sur S'.*

Cette fois, on regarde le diagramme similaire à $(*)$

$$\begin{array}{ccccc} X/s & \xleftarrow{W} & X'/s' & & \\ & \searrow^{f/s} & \downarrow & \searrow^{f'/s'} & \\ & & Y/s & \xleftarrow{W} & Y'/s' \\ \downarrow & \swarrow & \downarrow & \swarrow & \\ S/s & \xleftarrow{W} & S'/s', & & \end{array}$$

où les flèches horizontales sont dans W, donc $f/s \in W$ (hypothèse) implique $f'/s' \in W$.

Dualement, si p est W-propre, le foncteur image inverse est compatible avec les localiseurs W_S^{d}, $W_{S'}^{\mathrm{d}}$. p. 26 [page 33]

Commentary · Later developments [ms. 33]

The passage from lifting categories to exact squares

The geometry just proved is the ingredient that later formulations package as exactness of squares. The term "proper" is justified by a universal base-change property; "smooth" describes the corresponding property when it is the map of bases that varies. The connection with the geometric proper and smooth base-change theorems, already discussed in the commentary on I.72, is made explicit in Maltsiniotis (2005, Introduction and Theorem 1.2).

The precise categorical statement appears in Maltsiniotis (2012, Propositions 2.12–2.13): a cartesian square with proper right side or smooth bottom side is W-exact, and the universal square condition characterizes properness. Its derivator interpretation is given by Theorem 4.32 of that paper, where cohomological base change is characterized through the appropriate weak exactness condition. These assertions explain why the lifting categories in the present text are sufficient; they are the categories appearing in pointwise base-change comparisons.

There is a convention to keep visible when comparing modern derivator texts. Grothendieck's coefficients here are contravariant diagrams. If a derivator is written with covariant diagrams instead, opposite categories exchange fibrations and opfibrations and interchange the two directions of the statement. In that convention Groth (2013, Propositions 1.24 and 1.26) proves base change for cartesian squares with a Grothendieck fibration on the right or an opfibration at the bottom, and identifies this property with the pointwise Kan-extension axiom. The apparent reversal has this variance explanation.

These developments do not identify every strict pullback with a homotopy pullback. An exact square controls a specified comparison of Kan extensions. Here the point is that properness or smoothness makes the relevant comma-category comparison cofinal in the required direction. The construction can therefore be transported to other coefficient theories without choosing path objects in each one separately.

Si $f : X \longrightarrow S$ est W-propre, on va définir un foncteur canonique

$$\begin{array}{rccc} \mathrm{hot}_*^{X/S,W} : S & \longrightarrow & \mathrm{Hot}_W \overset{\text{déf}}{=} (\mathrm{Cat})W^{-1} \\ \text{donné sur les objets par} \quad s & \longmapsto & \mathrm{hot}_W(X_s). \end{array}$$

Il faut définir, pour une flèche u dans S, $u : s \longrightarrow s'$, une flèche

$$\mathrm{hot}_W(X_s) \longrightarrow \mathrm{hot}_W(X_{s'})$$

dans Hot_W - flèche que nous pouvons noter $\mathrm{hot}_*^{X/S,W}(u)$. On la définit par le diagramme

$$\begin{array}{ccc} X_s & \cdots\cdots\cdots\cdots\cdots\cdots\rightarrow & X_{s'} \\ \Big\downarrow{\scriptstyle i_s^X} & & \Big\downarrow{\scriptstyle i_{s'}^X} \\ X/s & \xrightarrow{X/u} & X/s', \end{array}$$

où les flèches verticales sont W-coasphériques, donc dans W. Ce sont donc des isomorphismes dans Hot_W, d'où l'unique flèche en pointillés dans Hot_W, rendant le carré commutatif. Sur cette définition il est immédiat que

$$u \longmapsto \mathrm{hot}_W(u)$$

est compatible avec composition, et unités.

Ce foncteur est donc défini en supposant seulement que les [page 34]

$$X_y \longrightarrow X/y$$

sont dans W, ce qui est nettement moins fort que propre. Si on suppose que cette hypothèse est satisfaite 'universellement' (i.e. reste satisfaite après tout changement de base $S' \longrightarrow S$)

([45]), alors on a donc, pour tout S' dans Cat/S, un foncteur

$$\mathrm{hot}_*^{X'/S',W} : S' \longrightarrow \mathrm{Hot}_W,$$

et on vérifie tout de suite que ce n'est autre que le composé dans

p. 27

$$(*) \qquad \begin{array}{ccc} S' & \xrightarrow{\quad p \quad} & S \\ & {\scriptstyle \mathrm{hot}_W^{X'/S'}} \searrow \quad \swarrow {\scriptstyle \mathrm{hot}_W^{X/S}} & \\ & \mathrm{Hot}_W & \end{array}$$

[plutôt $\mathrm{hot}_*^{X'/S',W}$ resp. $\mathrm{hot}_*^{X/S,W}$]. On a des observations duales si X est lisse sur S, plus généralement si les inclusions

$$X_s \longrightarrow s\backslash X$$

sont dans W. On trouve alors un contrafoncteur

$$\begin{cases} S^\circ & \longrightarrow & \mathrm{Hot}_W \\ s & \longmapsto & \mathrm{hot}_W(X_s) \\ u & \longmapsto & \mathrm{hot}^*_{X/S,W}(u) \text{ ou } \mathrm{hot}^*_{X/S}(u) \end{cases}$$

et une compatibilité similaire à $(*)$, si l'hypothèse est satisfaite *universellement* sur S, p. ex. f W-lisse. [page 35]

Commentary · Exposition [ms. 35]

Transport obtained by conjugating the comma functor

To construct transport, Grothendieck uses a functor that already exists before localization. Let $Q : \mathrm{Cat} \to \mathrm{Hot}_W$ be the localization. The assignment

$$s \longmapsto X/s, \qquad (u : s \to t) \longmapsto (X/u) : (x, \alpha) \longmapsto (x, u\alpha)$$

is a strict functor $S \to \mathrm{Cat}$. If $i_s : X_s \to X/s$ belongs to W for every s, define

$$\lambda(u) = Q(i_t)^{-1} Q(X/u) Q(i_s).$$

The existence of the inverse is the only use of the hypothesis. For $s \xrightarrow{u} t \xrightarrow{v} r$, multiplication cancels $Q(i_t)$ against its inverse and gives

$$\lambda(v)\lambda(u) = Q(i_r)^{-1} Q(X/v) Q(X/u) Q(i_s) = \lambda(vu).$$

The identity case follows in the same way. Thus the word "functor" is justified by the comma construction, not by a choice of lifts in the strict fibres.

For base change $q : S' \to S$, the canonical maps $X'/s' \to X/q(s')$ form a natural transformation of these comma diagrams, and the square with $X'_{s'} \cong X_{q(s')}$ and the two inclusions commutes. If both inclusions belong to W, two-out-of-three puts the comma comparison in W and the displayed formula gives precisely the compatibility asserted in the text. Requiring the inclusion hypothesis after every base change is the definition of W-preproperness in footnote 45. Opposite categories give W-presmoothness, called W-prélissité in the source, and a contravariant transport functor.

Properness is a practical sufficient condition because it says more: each i_s is coaspherical. Preproperness keeps the universal weak-equivalence conclusion and discards its local lifting explanation. Neither should be confused with the condition on the two endpoint inclusions over individual arrows considered on manuscript page 36. As the development below shows, those individual comparisons need not obey the displayed composition law.

Proposition 7. *Supposons $f : X \longrightarrow S$ à la fois W-propre et W-lisse, plus généralement que les morphismes*

$$X_s \longrightarrow X/s, \qquad X_s \longrightarrow s\backslash X \qquad (s \in S)$$

45. On dira que f est *W-préproper.*

soient dans W, et que cette hypothèse reste verifiée après tout changement de base. Alors les foncteurs correspondants

$$\lambda = \mathrm{hot}_*^{X/S,W} : \quad S \longrightarrow \mathrm{Hot}_W$$
$$\lambda^\circ = \mathrm{hot}^*_{X/S,W} : \quad S^\circ \longrightarrow \mathrm{Hot}_W$$

transforment toute flèche u de S en [un] *isomorphisme de* Hot_W*, et on a*

$$\lambda^\circ(u) = \lambda(u)^{-1} \qquad \textit{dans} \quad \mathrm{Hot}_W,$$

i.e. $\lambda(u)$ et $\lambda^c irc(u)$ sont inverses l'un de l'autre.

Démonstration. La donnée de $u : s_0 \longrightarrow s_1$ équivaut à celle de

$$S' = \mathbf{\Delta}^1 \longrightarrow S,$$

d'où

$$X' = X \times_S S' \qquad \text{sur } S' = \mathbf{\Delta}^1,$$

avec les deux fibres

$$X'_0 \simeq X_{s_0}, \qquad X'_1 \simeq X_{s_1}.$$

L'hypothèse assurant l'existence de λ s'exprime par la condition que

$$X'_1 \longrightarrow X'/1 \simeq X'$$

est dans W, de sorte que le diagramme

$$X'_0 \xrightarrow{i_0} X' \xleftarrow[W]{i_1} X'_1$$

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[page 36]

définit une flèche $\lambda(u) : X'_0 \longrightarrow X'_1$ dans Hot_W, savoir $\lambda(u) = i_1^{-1} i_0$. L'hypothèse qui assure l'existence de λ° s'exprime par la condition symétrique $i_0 \in W$, qui permet de définir la flèche $\lambda^\circ(u) : X'_1 \longrightarrow X'_0$ par $\lambda^\circ(u) = i_0^{-1} i_1$. Quand les deux conditions sont satisfaites simultanément, on voit donc que $\lambda(u)$, $\lambda^\circ(u)$ sont inverses l'un de l'autre.

Commentary · Examples [ms. 36]

Two directions of transport give monodromy

Proposition 7 says that the two canonical transports become inverse when both comma comparison hypotheses hold universally. Pulling back over $u : [1] \to S$ gives $E = X \times_S [1]$ and inclusions $i_0 : X_{s_0} \to E$, $i_1 : X_{s_1} \to E$. Preproperness makes i_1 a W-equivalence, and presmoothness makes i_0 one. The already constructed functors therefore satisfy

$$\lambda(u) = Q(i_1)^{-1} Q(i_0), \qquad \lambda^{\mathrm{op}}(u) = Q(i_0)^{-1} Q(i_1).$$

These formulas prove the claimed inverse relation without an additional saturation assumption. The covariant functor factors through the ordinary groupoid completion $S[\mathrm{Mor}(S)^{-1}]$. It records transport around loops, or monodromy, at the level of the homotopy category.

For a concrete example, let $C_2 = \{1, \sigma\}$ act on the discrete set $F = \{0, 1\}$ by interchange. Regard BC_2 as the one-object groupoid. The action groupoid E has objects $0, 1$ and an arrow labelled g from x to gx. The projection $E \to BC_2$ is both a fibration and an opfibration: every base arrow has a unique incoming and a unique outgoing lift once the appropriate endpoint is specified. Its strict fibre is F. The transport along σ interchanges the two points, in both directions, since $\sigma^{-1} = \sigma$.

The action is free and transitive, so there is exactly one arrow between each ordered pair of objects of E. Either object is consequently both initial and terminal, and BE is contractible. The base has fundamental group C_2, while the fibre consists of two points. This is the usual universal two-sheeted covering expressed entirely by categories. It illustrates why the fibre functor and its monodromy carry information that the homotopy type of the total category alone does not retain. It also illustrates the scope of Proposition 7: it gives a functor into Hot_W, not a claim that ordinary homotopy-category monodromy classifies all homotopy-coherent families.

Remarques. Pour définir λ 'universellement', il n'était pas indispensable de considérer les X/s et les inclusions $X_s \longrightarrow X/s$, au lieu de cela il suffisait de regarder les changements de base $S' = \mathbf{\Delta}^1 \xrightarrow{u} S$, et de faire l'hypothèse que pour tout tel changement de base, l'inclusion $X'_1 \xrightarrow{i_1} X' = X_{(\mathbf{\Delta}^1, u)}$ est dans W. Cette hypothèse est automatiquement stable par changement de base arbitraire. (La W-propreté de $f : X \longrightarrow S$ signifie que $X'_1 \longrightarrow X'$ est non seulement dans W, mais même W-coasphérique.) Dualement, pour la définition 'universelle' de λ°, il suffirait que pour tout changement de base $\mathbf{\Delta}^1 \longrightarrow S$ comme dessus, l'inclusion $X'_0 \xrightarrow{i_0} X'$ est dans W.

Commentary · Development (commentators') [ms. 36]

An eight-object obstruction to edge transport

Grothendieck proposes to replace the universal comma comparisons by endpoint comparisons over the walking arrow. Those tests do define a morphism for each arrow of the base, but they do not by themselves make these morphisms functorial. The distinction is already visible on connected components, and also answers the question about pseudofibrations on manuscript page 40.

Proposition (edge tests do not force coherence). [ours] There exists a functor $p : P \to [2]$ such that, for every $u : [1] \to [2]$, both endpoint inclusions into $P \times_{[2]} [1]$ belong to W_∞, but the resulting edge transports fail the composition law. Moreover, p does not satisfy the fibration test over contractible bases.

Proof. Indices will be taken modulo 2. Let P have eight objects

$$a_0, a_1, b_0, b_1, d_0, d_1, c_0, c_1,$$

and exactly the following nonidentity order relations:

$$a_i < d_i, \qquad b_i < d_i, \qquad b_i < c_i, \qquad a_i < c_{1-i} \quad (i = 0, 1).$$

Every relation starts at an a_i or a b_i and ends at a d_i or a c_i. Thus no two strict relations are composable, and the list already defines a transitive partial order. Define $p(a_i) = 0$, $p(b_i) = p(d_i) = 1$, and $p(c_i) = 2$. All listed relations preserve this order, so p is a functor.

The fibre P_0 is the discrete pair $\{a_0, a_1\}$, the fibre P_1 is the disjoint union of the two intervals $b_i < d_i$, and P_2 is the discrete pair $\{c_0, c_1\}$. For the nonidentity base arrow $0 \to 1$, the pullback is the disjoint union of the two posets $a_i < d_i > b_i$. Each component has terminal object d_i, and each endpoint fibre meets it in a nonempty contractible subcategory. Both endpoint inclusions are therefore weak homotopy equivalences. For $1 \to 2$, the two components are $d_i > b_i < c_i$; each has initial object b_i, and the same conclusion follows. For $0 \to 2$, the pullback consists of the two intervals $a_i < c_{1-i}$, so again both endpoint inclusions are weak equivalences. A constant functor $[1] \to [2]$ has pullback $P_j \times [1]$; its endpoint inclusions are categorical homotopy equivalences. These six functors exhaust all $[1] \to [2]$, proving every required edge test.

Label the two connected components of each fibre by i. The span through the $0, 1$ pullback identifies component i with component i; so does the span through the $1, 2$ pullback. The $0, 2$ pullback instead identifies i with $1 - i$. Thus the maps induced on π_0 satisfy

$$\pi_0\lambda(0 \to 1) = 1, \qquad \pi_0\lambda(1 \to 2) = 1, \qquad \pi_0\lambda(0 \to 2) = (0\ 1).$$

Since π_0 sends W_∞ to bijections, it factors through Hot, and these unequal functions prove that the proposed transports cannot form a functor into Hot.

For completeness, the nerve of P has eight vertices, eight nondegenerate edges, and no higher nondegenerate simplices. Those edges form exactly the cycle

$$a_0 - d_0 - b_0 - c_0 - a_1 - d_1 - b_1 - c_1 - a_0.$$

Its realization is therefore a circle. Each fibre has two connected components, whereas P is connected. Consequently none of the inclusions $P_j \to P$ is in W_∞, although the base $[2]$ has both an initial and a terminal object. This disproves the stronger fibration condition over contractible bases and completes the proof.

The counterexample also satisfies the edge tests for W_0, the localizer detected by connected components, and has the same failure of composition there. The issue is therefore not a higher-dimensional obstruction inside the fibres. The missing information is how transports for composable

arrows agree across a triangle in the base. The universal comma hypotheses used in Proposition 7 supply precisely the strict diagram from which that agreement was obtained. We checked the lift-category and smoothness treatments in Maltsiniotis (2005, §4) and Maltsiniotis (2012, §§1–2); we did not find this finite example there. This is a development of the manuscript's question, with no claim of priority.

La notion la plus faible raisonnable de W-fibré, me semble être celle où on suppose les [page 37]
deux choses à la fois. Les hypothèses qu'on a faites jusqu'à présent, à l'exclusion de la forme forte de W-propreté ou de W-lissité, reviennent toujours à supposer que le *changement de base par* $f : X \longrightarrow S$ transforme certaines flèches dans Cat/S en `[élément de]` W : Pour définir λ, ces flèches sont

$$\text{soit } \{s\} \longrightarrow S/s, \qquad \text{soit } \{1\} \longrightarrow \mathbf{\Delta}^1(\longrightarrow S),$$

pour définir λ°, ce sont

$$\text{soit } \{s\} \longrightarrow s\backslash S, \qquad \text{soit } \{0\} \longrightarrow \mathbf{\Delta}^1(\longrightarrow S),$$

le 'soit' suivant la définition qu'on a adoptée. Dans ces exemples, il s'agit toujours de l'inclusion de l'objet final d'une catégorie S', dans ladite, ou inversement de l'objet initial. Cela montre d'ailleurs (si on exige l'universalité), que la première hypothèse (assurant λ) est *exactement* cela (cas de l'inclusion d'un objet final s' - puisque pour un tel objet, on a $S'/s' \simeq S'$), tandis que la deuxième hypothèse exige moins, seulement pour le cas de $\mathbf{\Delta}^1$. Et dualement pour λ' `[plutôt` λ°`]`. Mais il convient de noter que pratiquement, la façon la plus commode, et de loin,

pour vérifier de telles propriétés, c'est soit par la W-propreté de f, soit par la W-lissité - ce qui ramène à vérifier que certaines catégories bien précisées sont W-asphériques, sans avoir [page 38]
à se soucier de changements de base et de vérifier que les flèches construites en termes de changements de base arbitraires sont dans W. Et l'emploi de la W-propreté et de la W-lissité est d'autant plus sympathique, que ces notions sont stables par composition,

alors que toutes les notions dérivées, ainsi que les notions de fibrations à la clef, ont une fâcheuse tendance à ne *pas* être stable par composition. p. 29

On pourrait appeler pseudo-W-propre resp. quasi-W-propre (un peu mieux que pseudo) ce qu'on vient de dégager - j'entends des ricanements! Et si on veut les deux à la fois - λ et λ° associés comme on a dit? Je vois le même principe de description avec des $S'' \longrightarrow S'$ sur S, d'exiger que $X'' \longrightarrow X'$ soit transformé en des `[éléments de]` W. Avec les choix suivants qui ont l'air sympathique

a$_1$) Y' a un objet final, $Y'' = e$. La définition revient donc à la condition que les [page 39]

$$X_s \longrightarrow X/s, \qquad X/s' \longrightarrow X/s$$

(pour $s \in \mathrm{Ob}\, S$ où $u : s' \longrightarrow s$ dans S) soient dans W, et que ça reste comme ça après tout changement de base $S' \longrightarrow S$.

a$_2$) Y' a un objet initial, $Y'' = e$, c'est la condition duale à coups de

$$X_s \longrightarrow s\backslash X, \qquad s\backslash X \longrightarrow s'\backslash X.$$

a$_0$) $Y' = \mathbf{\Delta}^1$, $Y'' = e$ ([46]).

b) Y' est contractile, Y'' aussi (ou au choix, Y' contractile, $Y'' = e$)

donc la condition revient à ceci : pour toute Y' sur Y *contractile* (notion indépendante de W, c'est ça son principal avantage) les inclusions des fibres de $X' = X \times_Y Y'$ sur Y' dans X' sont `[dans]` W.

46. NB a$_1$), a$_2$) sont notés pareil p. 4, mais a$_0$) a été oublié.

Commentary · Exposition [ms. 39]

The test objects explain the hierarchy

The proposed definitions differ by the category that is allowed to serve as a test base. For condition a_1, a base T with terminal object t is tested at *every* object $r \in T$, not only at t. Testing the terminal object gives preproperness. Once it is available, the unique arrow $r \to t$ identifies the comparison $X_r \to X_T$ with transport to the terminal fibre. Requiring that comparison to belong to W adds the demand that all such transports be weak equivalences. The dual reading gives a_2.

Condition a_0 restricts the base to [1] and allows either endpoint. The preceding eight-object example shows that it cannot be substituted for a_1 or a_2, even for ordinary weak homotopy equivalences. In particular, existence of individually invertible edge roofs is not a substitute for a functor of fibres. This is the reason for keeping the preproper or presmooth hypothesis in the equivalences proved later in the section.

For condition b, the equivalence of the two tests stated in the text has a short proof. Suppose all fibre inclusions are in W over every categorically contractible test base. If $v : T'' \to T'$ is a map between two such bases, choose an object t''; contractibility ensures that an object exists. In the pullback square, the fibre at t'' identifies with the fibre at $v(t'')$. The two inclusions from that common fibre into the total categories both belong to W, so their commutative triangle puts the total pullback map in W by two-out-of-three. The converse is the special case $T'' = e$.

Condition b' allows W-aspherical bases instead. Condition c allows every map in W as a change of base. Its stronger quantifier is substantial: it asks for preservation of a whole class of weak equivalences, whereas the earlier conditions test selected contractible shapes. The terminology at this point is provisional, but these quantifiers are exact and should be retained when comparing later definitions.

Je sens que même pour des W générales, b) reste du domaine du vérifiable, tandis qu'on mollit si on passe à l'extrême en généralité dans ce sens :

b′) Comme b), mais en supposant seulement Y' W-asphérique ([47]).

Enfin (la condition a) de la page 4)

c) Si $Y'' \longrightarrow Y'$ est `[dans]` W sans plus, $X'' \longrightarrow X'$ aussi.

La seule de ces notions qui ait la moindre envie d'être stable par composition, c'est la plus forte c). Par contre, la notion de flèche W-parfaite (cf. déf. 3) est stable spontanément par composition : p. 30 [page 40]

Définition 3. On dit que $f : X \longrightarrow S$ est *W-parfaite* si elle est à la fois W-propre et W-lisse.

Commentary · Exposition [ms. 40]

The interval test and the full fibration condition

Grothendieck is trying to retain the mobility of fibres without asking that every weak equivalence survive every pullback. The latter requirement is exceptionally strong when W forgets some homotopy groups. The useful intermediate condition asks instead that a categorically contractible base make the family constant up to W. Here "contractible" means that its identity functor is joined to a constant functor by a finite zigzag of natural transformations. This is the notion of homotopy already used in section I; it must not silently be replaced by weak contractibility of the nerve.

The weakest test in the diagram examines only $[1] \to S$. It gives, for each arrow $u : s \to t$, a span

$$X_s \longrightarrow X \times_S [1] \longleftarrow X_t$$

whose arrows belong to W. Inverting them produces an isomorphism between the two fibres in Hot_W. What it does not supply is the compatibility of this isomorphism with composition of arrows of S. The development "An eight-object obstruction to edge transport" above gives a finite poset over [2] for which all these endpoint tests are weak equivalences, whereas the transports along $0 \to 1 \to 2$ and $0 \to 2$ disagree already on connected components. Its total category has the homotopy type of

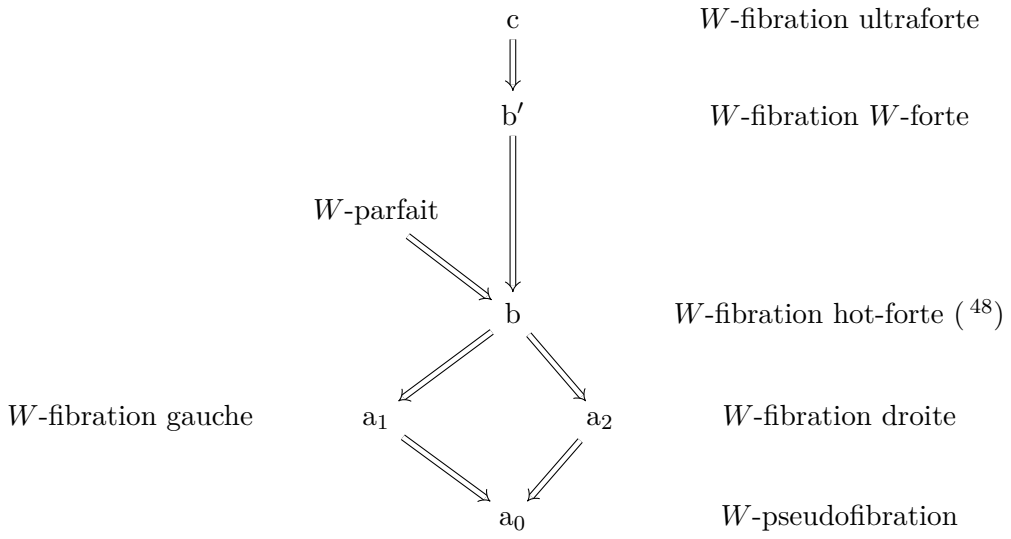
47. C'est essentiellement le a′) de p. 4

a circle and each fibre has two components. It therefore answers negatively the question about a_0 raised in the next paragraph.

Proposition 8 adds precisely the missing global control. Universal preproperness identifies a fibre with the appropriate comma category; the comma categories carry genuinely functorial transport, because postcomposition in S is an actual functor. This explains why preproperness together with the interval test succeeds where the interval test alone fails. Perfectness is a convenient sufficient condition, since it supplies both universal comma comparisons. It is stronger than the fibration condition being isolated here.

Dans le cas $W = W_\infty$, toutes les variantes de la page précédente, à l'exclusion seulement de a_0), sont équivalentes. Ce serait intéressant de dégager des conditions minimales sur W pour assurer qu'il en soit ainsi. (Et il se pose la question si dans le cas de W_∞, a_0) suffit pour impliquer le reste - je ne m'y attends guère …)

Dans le cas général, on a le diagramme d'implications (et encore je me suis limité)



[‘W-parfait $\Longrightarrow$ b’ encerclé].

Proposition 8. *Soit $f : X \longrightarrow S$ dans* Cat. *Les conditions suivantes sont équivalentes.* [page 41]

a) f est une W-fibration gauche, i.e. la condition a_1 *de la page 39 est vérifiée.*

b) f est W-prépropre, et $\forall\ s_0 \longrightarrow s_1$ dans S, $X/s_0 \longrightarrow X/s_1$ est dans W.

c) f est W-prépropre, et une W-pseudofibration, i.e. la condition a_0 de la page 39 est vérifiée.

([49]).

NB Ces conditions sont satisfaites si f est en même temps W-prélisse, cf. prop. 7. p. 31

Commentary · Exposition [ms. 41]

Completing the terminal-object test

Grothendieck wants a condition that can be checked on the original base, while remaining valid after every change of base. We spell out the argument because the compressed reference to Y/s and Y/s' hides the universal quantifier.

Write $E_u = X \times_S [1]$ for $u : s_0 \to s_1$. If the terminal-object test holds, apply it first to $T/t \to T \to S$, for an arbitrary $T \to S$ and $t \in T$. Its terminal fibre is X'_t and its total category is

48. Il semblerait que ce soit ça, finalement, la ‘bonne notion’ de W-fibration, *équivalente* à a_1 et a_2, lesquelles seraient équivalentes entre elles. Cf. prop. 9, p. 48.

49. **Corollaire.** *Quand f est* Cat*-cofibrant, alors ces conditions équivalent aussi à ceci : $\forall\ u : s_0 \longrightarrow s_1$ dans S, $u_* : X_{s_0} \longrightarrow X_{s_1}$ est dans W.*

X'/t, where $X' = X \times_S T$. Thus $X'_t \to X'/t$ belongs to W: this is universal preproperness. Next apply the same test to S/s_1. Both the object $(s_1, 1)$ and the object (s_0, u) give fibre inclusions into X/s_1 in W. Since the latter factors as

$$X_{s_0} \longrightarrow X/s_0 \longrightarrow X/s_1,$$

two-out-of-three gives $X/s_0 \to X/s_1 \in W$. This supplies the missing detail in $a) \Rightarrow b)$.

Assume now universal preproperness. The canonical functor $E_u \to X/s_1$ restricts on the terminal fibre to $X_{s_1} \to X/s_1$. Both this inclusion and $X_{s_1} \to E_u$ belong to W, so $E_u \to X/s_1$ belongs to W. Its restriction to the initial fibre is the composite displayed above. Consequently

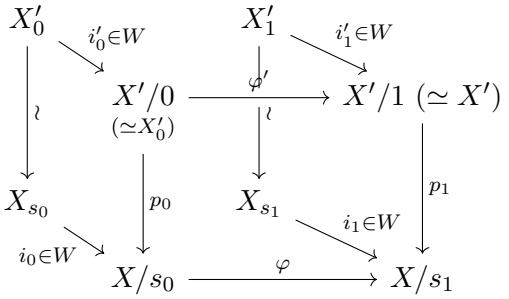
$$X_{s_0} \to E_u \in W \quad \Longleftrightarrow \quad X/s_0 \to X/s_1 \in W.$$

The other endpoint inclusion is already in W. This proves $b) \Longleftrightarrow c)$, including the asserted stability under arbitrary base change, since preproperness and the interval condition both have that stability by definition.

Finally, after any base change to a category T with terminal object t, the inclusion of the fibre over $s \in T$ factors through $X'/s \to X'/t = X'$. Both factors belong to W, proving $a)$. The dual argument uses initial objects and the categories $s\backslash X$. For an ordinary opfibration the inclusions $X_s \to X/s$ have adjoints and are homotopy equivalences; its postcomposition maps identify, up to these comparisons, with the pushforward functors u_*. This proves the corollary in footnote 49 with its stated variance.

DÉMONSTRATION. On a a) $\Longrightarrow$ b) trivialement (faire $Y'' = Y/s$, $Y'' = Y/s'$ `[lien avec l'assertion? Cf. p. 4.]`).

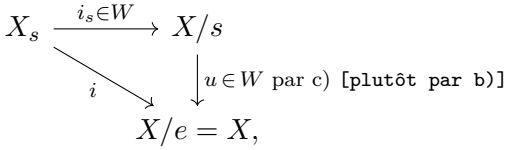
b) $\Longleftrightarrow$ c) par le raisonnement de la proposition 7, car en termes de $X' = X_u = X \times_S (\mathbf{\Delta}^1, u)$, la condition b) pour $u : s_0 \longrightarrow s_1$ donné s'interprète dans le diagramme commutatif



`[il y a quelques` W `redondants dans le diagramme original]` où i_0, i_1, i'_0, i'_1 sont dans W à cause de l'hypothèse de W-prépropreté (stable par changement de base).

Donc p_0, p_1 sont `[dans]` W, donc $\varphi \in W$ si et seulement si $\varphi' \in W$, or $\varphi' \in W$ est la condition a_0.

Prouvons enfin que b), c) impliquent a). Comme c) (donc aussi b)) est stable par changement [page 42] de base, il suffit de montrer ceci : Si S a un objet final e, alors pour tout $s \in \mathrm{Ob}\, S$, l'inclusion $X_s \xrightarrow{i} X$ est dans W. Or on a le diagramme commutatif



qui montre bien $i = ui_s \in W$.

Dualement, si f est W-prélisse, alors la condition a_0 (f une W-pseudofibration) est équivalente à a_2 (f une W-fibration droite), et signifie aussi que les

$$s_1\backslash X \longrightarrow s_0\backslash X$$

sont dans W. Mais la notion de 'W-fibration' sur laquelle on tombe (si on ne se contente du minable a_0!) est différente dans les deux cas duaux l'un de l'autre. On aimerait dans les deux cas trouver des W-fibrations 'hot-fortes' (condition b), tout au moins en mettant le paquet (s'il le faut) - i.e. pour f W-parfait. Ça revient donc à ceci.

Question : *Soient* $X \xrightarrow{f} S$, *avec* f W*-parfait,* S *contractile,* $s \in \mathrm{Ob}\, S$. *L'inclusion* $X_s \longrightarrow X$, *est-elle dans* W*?*

p. 32
[page 43]

On va supposer seulement que f est W-lisse et une W-fibration gauche [plutôt que f est simplement une W-fibration à droite, i.e que f est W-prélisse et une W-pseudofibration] (cf. prop. 8). On va se ramener au cas où X est même fibrant sur S, en introduisant $\Phi_S(X)$ [catégorie fibrée définie par le foncteur $S^\circ \longrightarrow \mathrm{Cat}$, $s \longmapsto s\backslash X$] et

$$X \xrightarrow{i_X} \Phi_S(X).$$

Commentary · Exposition [ms. 43]

The split replacement and its actual comparison maps

Grothendieck now replaces a family by a split fibration so that transport can be written before localization. The replacement is legitimate for a more concrete reason than an appeal to a previously constructed derivator.

An object of $\Phi_S(X)$ is a triple (s, x, a) with $a : s \to p(x)$. A morphism to (t, y, b) is a pair $(u : s \to t, v : x \to y)$ satisfying $p(v)a = bu$. The structural projection sends this object to s. Its fibre at s is $s\backslash X$, and an arrow $u : s \to t$ acts by precomposition. This is a strict functor $S^{\mathrm{op}} \to \mathrm{Cat}$, hence a split Grothendieck fibration.

The comparison i_X sends x to $(p(x), x, 1)$. The forgetful functor $r : \Phi_S(X) \to X$ satisfies $ri_X = 1_X$, and the arrows $(a, 1_x)$ define a natural transformation $1_{\Phi_S(X)} \Rightarrow i_X r$. In fact r is left adjoint to i_X. Thus i_X is a categorical homotopy equivalence and belongs to every W under consideration. On the fibre at s it is exactly

$$X_s \longrightarrow s\backslash X.$$

For a universally prelisse X/S, these fibre comparisons belong to W after every base change. The comparison of total categories and the comparison of individual fibres are therefore two separate, verified assertions.

If X/S is also a pseudofibration, the dual of Proposition 8 says that every precomposition $t\backslash X \to s\backslash X$ is in W. Consequently $\Phi_S(X)$ is a split fibration with transport in W. In a square comparing a fibre inclusion for X with the corresponding inclusion for $\Phi_S(X)$, both horizontal arrows are in W; two-out-of-three transfers the desired conclusion. No saturation argument is needed for this step. The familiar free-fibration construction and its variance were discussed in the commentary on Chapter VI, manuscript pp. 53–60; the additional point here is the universal fibre comparison that permits its use over an arbitrary new base.

L'ennui, c'est que j'ai fait la théorie de HOT_W jusqu'à maintenant avec l'hypothèse (qui s'avère idiote) que $W \supseteq W_\infty$ - je serai donc obligé de revoir rapidement sous quelles hypothèses plus raisonnables sur W les résultats familiers sont valables. Par exemple j'aimerais que $i_X \in W$ ([50]), de sorte qu'on ait

$$\begin{array}{ccc} X_s & \xrightarrow[\text{car } X \text{ } W\text{-lisse}]{i_s \in W} & \Phi_S(X)_s = s\backslash X \\ {\scriptstyle j_s^X}\downarrow & & \downarrow{\scriptstyle j_s^{\Phi_S(X)}} \\ X & \xrightarrow{i_X \in W} & \Phi_S(X), \end{array}$$

on voit donc que pour prouver que l'inclusion j_s^X est [dans] W, il suffit de prouver la même chose pour $\Phi_S(X)$, qui est fibrant (pas seulement W-lisse). De plus, l'hypothèse pseudo-

50. en fait, $i_X \in (W_\omega)_S^{\mathrm{d}} \subseteq W_S^{\mathrm{d}} \subseteq W_S$.

fibration sur X passe sûrement à $\Phi_S(X)$ (je réserve les deux vérifications à plus tard ([51])). On est donc ramené au cas de

$$X \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_0 S, \qquad S \text{ contractile},$$

à prouver que l'inclusion

$$X_s \longrightarrow X$$

est dans W. Je vais prouver ceci : [page 44]

Lemme. *Soient S dans* Cat, $X \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_0 S$, *et considérons, pour un S' dans* Cat,

$$\begin{array}{rcl} f & \longmapsto & f^*(X/S) = X \times_S (S', f) \\ \underline{\mathrm{Hom}}(S', S)^\circ & \longrightarrow & \underline{\mathrm{Fib}}_0 S'. \end{array}$$

(Ce foncteur est défini grâce au fait que X est fibrant sur S, et peut s'interpréter donc, à S-équivalence près, en termes d'un foncteur $S^\circ \longrightarrow$ Cat … ([52])). Les flèches de $\underline{\mathrm{Hom}}(S', S)$ sont transformées en flèches $\in W^{\mathrm{d}}_{S'}$.

En effet, en termes de foncteurs p. 33

$$S'^\circ \longrightarrow \mathrm{Cat},$$

si X est définie par

$$S^\circ \xrightarrow{\mathscr{X}} \mathrm{Cat},$$

alors $f^*(X)$ est définie par $\mathscr{X} \circ f$, et la flèche

$$\mathscr{X} \circ f \longrightarrow \mathscr{X} \circ g$$

définie par $u : f \longrightarrow g$ n'est autre que $\mathscr{X} * u$. Sa valeur en $s' \in \mathrm{Ob}\, S'$ est obtenue en prenant

$$u(s') : f(s') \longrightarrow g(s'),$$

d'où par $\mathscr{X}$

$$\mathscr{X}(u(s')) : \underbrace{\mathscr{X}(g(s'))}_{(\mathscr{X}\circ g)(s')} \longrightarrow \underbrace{\mathscr{X}(f(s'))}_{(\mathscr{X}\circ f)(s')}$$

([53]). Mais l'hypothèse $X \in \underline{\mathrm{Fib}}_0 S$ signifie que les $\mathscr{X}(v)$, $v \in \mathrm{Fl}(S)$, sont tous dans W. Donc $\mathscr{X}$ est dans W fibre par fibre, donc dans $W^{\mathrm{d}}_{S'}$.

[page 45]

Commentary · Exposition [ms. 45]

Transport along a natural transformation

The purpose of the lemma is to turn a homotopy of indexing functors into an actual zigzag of families. For this purpose it is useful to work with the split fibration just constructed. Let $F : S^{\mathrm{op}} \to \mathrm{Cat}$ be its classifying functor, with every $F(u)$ in W, and let $\alpha : f \Rightarrow g$ be a natural transformation of functors $T \to S$.

Contravariance gives the strict natural transformation

$$F(g(-)) \longrightarrow F(f(-)), \qquad t \longmapsto F(\alpha_t).$$

Its Grothendieck construction is a Cartesian functor $g^*X \to f^*X$ over T. Each component belongs to W. The relative localization axiom, applied in the dual comma convention, implies that this total functor belongs to W as well. Explicitly, the inclusion of the fibre of a Grothendieck fibration into its under-comma category has an adjoint, so the componentwise assertion gives an equivalence on those comma categories; the dual of W(2) then gives the total equivalence.

51. O.K.
52. **NB** $\Phi_S(X)$ était même déjà scindée, définie par le foncteur $s \longmapsto s\backslash X$.
53. à dégager un énoncé à part.

The first displayed arrow in the source proof points in the opposite direction from the correctly written component immediately below it. It is an edition slip: the transformation is $F \circ g^{\mathrm{op}} \Rightarrow F \circ f^{\mathrm{op}}$. Composition is nevertheless completely controlled: for $\beta : g \Rightarrow h$, the identity

$$F(\beta_t \alpha_t) = F(\alpha_t) F(\beta_t)$$

identifies the transport for $\beta\alpha$ with the composite of the two transports, in the contravariant order. Units are identities. After localization every such transport is invertible, and one may use its inverse when writing the covariant version of the functor in the next paragraph.

The factorization through the groupoid completion of $\underline{\mathrm{Hom}}(T, S)$ follows from its universal property. Connected functors therefore induce isomorphic pullbacks, but a connected component does not choose a particular path, and different paths may induce different isomorphisms. This is the precise reason for the parenthetical qualification "non canoniquement".

Corollaire 1. *Le foncteur du lemme, composé avec*

$$\underline{\mathrm{Fib}}_0 S' \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S') \overset{\text{déf}}{=} (\underline{\mathrm{Fib}}_0 S')(W^{\mathrm{d}}_{S'})^{-1}$$

se factorise par le groupoïde fondamental de $\underline{\mathrm{Hom}}$ *:*

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(S', S)^{\circ} & \longrightarrow & \underline{\mathrm{Fib}}_0 S' \\ \downarrow & & \downarrow \\ \prod(\underline{\mathrm{Hom}}(S', S))^{\circ} & \longrightarrow & \mathrm{HOT}^{\mathrm{lc}}_W(S'). \end{array}$$

Si f, g sont dans une même composante connexe de $\underline{\mathrm{Hom}}(S', S)$, alors f^*X et g^*X sont isomorphes (non canoniquement, en général) dans $\mathrm{HOT}^{\mathrm{lc}}_W(S')$.

Commentary · Examples [ms. 45]

A loop can act on the fibre even when the functor is fixed

Grothendieck's qualification about noncanonical isomorphisms concerns genuine monodromy. It can be seen without higher homotopy groups. Let $S = BC_2$, the one-object category with automorphism group $C_2 = \{1, \tau\}$, and let $F : S^{\mathrm{op}} \to \mathrm{Cat}$ take its value in the discrete category $D = \{0, 1\}$, with $F(\tau)$ exchanging the two objects. All transports are isomorphisms.

There is only one functor $e \to BC_2$, but it has two natural automorphisms. Pullback along this functor is the fibre D. The identity natural transformation induces 1_D, whereas the transformation τ induces the transposition of D. These are different even after localization at W_∞, because they induce different maps on $\pi_0(D)$. Thus "the two indexing functors are homotopic" cannot single out an isomorphism between their pullbacks.

The total category $\int F$ is equally explicit. Its objects are $0, 1$, and a morphism $d \to d'$ is an element $g \in C_2$ with $d = F(g)(d')$. There is exactly one such element for each ordered pair. The total category is therefore equivalent to e, whereas the fibre has two components. Its map to BC_2 is the categorical model of the universal two-sheeted covering. The inclusion of the fibre is not a weak equivalence; the base is not contractible. This example also shows why the word "contractile" in the next argument is a mathematical hypothesis, not a convenient choice of coordinates.

Il est plus astucieux de ne pas fixer X et de considérer

$$(*) \qquad \underline{\mathrm{Hom}}(S', S) \longrightarrow \underline{\mathrm{Hom}}(\mathrm{HOT}^{\mathrm{lc}}_W(S), \mathrm{HOT}^{\mathrm{lc}}_W(S'))$$

correspondant à

$$\begin{array}{ccccc} \underline{\mathrm{Hom}}(S', S) & \times & \mathrm{HOT}^{\mathrm{lc}}_W(S) & \longrightarrow & \mathrm{HOT}^{\mathrm{lc}}_W(S') \\ (f & , & \xi) & \longmapsto & f^*(\xi). \end{array}$$

p. 34
[page 46]

Le lemme dira que les flèches de $S' \longrightarrow S$ sont transformées par $(*)$ en isomorphismes, d'où

$$\prod \underline{\mathrm{Hom}}(S', S) \longrightarrow \underline{\mathrm{Hom}}(\mathrm{HOT}^{\mathrm{lc}}_W(S), \mathrm{HOT}^{\mathrm{lc}}_W(S')),$$

donc si $f, g : S' \rightrightarrows S$ sont deux foncteurs homotopes, alors

$$f^* \simeq g^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \rightrightarrows \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

Il s'ensuit aussitôt ceci

Corollaire 2. *Si f est un homotopisme, $f^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S')$ est une équivalence de catégories.*

En effet, prenant $g : S \longrightarrow S'$ telle que

$$fg \sim \mathrm{id}_S, \qquad gf \sim \mathrm{id}_{S'},$$

on trouve que

$$g^* f^* \simeq (\mathrm{id}_S)^* = \mathrm{id}_{\mathrm{HOT}^{\mathrm{lc}}_W(S)}$$
$$f^* g^* \simeq (\mathrm{id}_{S'})^* = \mathrm{id}_{\mathrm{HOT}^{\mathrm{lc}}_W(S')},$$

donc on trouve que (quitte à ajuster ces isomorphismes fonctoriels) f^* et g^* sont des équivalences quasi-inverses l'une de l'autre.

J'ai l'impression ([54]) que tout ça ne doit pas utiliser plus que les axiomes standard sur W comme localiseur fondamental, tout au plus W(2) au lieu de la forme faible W(2 bis).

Revenons à X lisse sur S contractile. Le corollaire 2 implique que

$$\mathrm{Hot}_W \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S)$$
$$\xi \longmapsto p_S^*(\xi),$$

où $p_S : S \longrightarrow e$, est une équivalence. Donc dans la catégorie $\mathrm{HOT}^{\mathrm{lc}}_W(S)$, X est isomorphe à un Cat-fibré de la forme $Z \times S \xrightarrow{\mathrm{pr}_2} S$, avec Z dans Cat. Or pour $X_1 = Z \times S \longrightarrow S$, [page 47]
il est immédiat que $X_{1\,s} \longrightarrow X_1$ est dans W, car un produit cartésien de flèches [dans] W (savoir id_Z et $s \longrightarrow S$) est dans W. (En fait, l'inclusion est même un homotopisme!) Notons maintenant que si X_0, X_1 sont deux objets dans $\underline{\mathrm{Liss}}_0 S$, tels qu'il existe une S-flèche $X_0 \longrightarrow X_1$ qui soit dans W^{d}_S, alors $X_{0\,s} \longrightarrow X_0$ est dans W si et seulement si $X_{1\,s} \longrightarrow X_1$ l'est. En effet, on a

$$\begin{array}{ccc} X_{0\,s} & \xrightarrow{\in W} & X_{1\,s} \\ {\scriptstyle i_s^{X_0}}\downarrow & & \downarrow{\scriptstyle i_s^{X_1}} \\ X_0 & \xrightarrow{\in W} & X_1 \end{array}$$

ici on utilise sans doute
W(2), non seulement W(2 bis),

qui prouve le résultat. p. 35

Commentary · Exposition [ms. 47]

The finite zigzag that makes contraction effective

Grothendieck correctly notices, immediately below, that an isomorphism in a localization does not by itself provide a zigzag consisting entirely of the specified weak equivalences. The contraction supplies such a zigzag directly.

54. O.K.

Suppose $F : S^{\mathrm{op}} \to \mathrm{Cat}$ sends arrows into W, and choose a finite zigzag

$$1_S = \varphi_0 \iff \varphi_1 \iff \cdots \iff \varphi_m = c_t$$

of natural transformations, with the displayed double arrows indicating a choice of one direction at each step. Set $X_j = \int (F \circ \varphi_j^{\mathrm{op}})$ over S. Transport along the chosen transformation gives, in the appropriate direction, a functor $X_j \to X_{j+1}$ over S which is in W on every fibre. By the argument of the preceding block it is also in W on total categories.

For every $s \in S$ the resulting square

$$\begin{array}{ccc} (X_j)_s & \longrightarrow & (X_{j+1})_s \\ \downarrow & & \downarrow \\ X_j & \longrightarrow & X_{j+1} \end{array}$$

has horizontal arrows in W; if the chosen natural transformation points backwards, reverse the horizontal arrows. Two-out-of-three says that one vertical arrow belongs to W exactly when the other does. At the last stage the family is $F(t) \times S \to S$. The map $e \to S$ selecting s is a categorical homotopy equivalence: a contraction to t, evaluated at s, supplies a zigzag between s and t, which joins the constant functors c_s and c_t. Multiplying this homotopy by $F(t)$ shows that

$$F(t) \cong F(t) \times \{s\} \longrightarrow F(t) \times S$$

is a categorical homotopy equivalence, hence is in W. Moving back through the finite zigzag proves $X_s \to X \in W$.

The replacement comparison at p. 43 transfers this conclusion to every prelisse pseudofibration; the dual argument treats preproper pseudofibrations. This proves the question of p. 42 and the contractible-base part of Proposition 9 under W(1,2,3). It uses the explicit contraction, componentwise transport, the relative localization axiom, and two-out-of-three. Strong saturation is not needed for this particular conclusion.

Il est vrai que le fait que $X = X_0$ et $X_1 = Z \times S$ soient isomorphes dans $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ n'implique pas, par lui-même, l'existence d'une chaîne d'isomorphismes provenant de flèches [page 48] dans W^{d}_S. Mais en fait, si dans l'énoncé du corollaire 2 ci-dessus on compare X dans $\underline{\mathrm{Fib}}_0 S$ à $g^*f^*(X)$, grâce à une homotopie $\mathrm{id}_S \sim fg$, on trouve bel et bien une suite de foncteurs $\varphi_i : S \longrightarrow S$, avec $\varphi_0 = \mathrm{id}_S$, $\varphi_N = fg$, et φ_i et φ_{i+1} reliés par une flèche dans $\underline{\mathrm{Hom}}(S, S)$, donc on a une suite d'objets $X_i = \varphi_i^*(X)$ de $\underline{\mathrm{Fib}}_0 S$ (avec $X_0 = X$, $X_N = g^*f^*X$), et X_i et X_{i+1} reliés par une flèche *dans* $\underline{\mathrm{Fib}}_0 S$ (pas seulement dans $\mathrm{HOT}^{\mathrm{lc}}_W(S)$), laquelle flèche est dans $W^{\mathrm{d}}_S = W^{\mathrm{f}}_S$. Donc on gagne, sous réserve de vérifications de routine diverses ([55]). Donc modulo celles-ci, on aura

Proposition 9 ([56], [57]). *Supposons que X dans* Cat/S *soit une W-fibration à gauche (cf. prop. 8). Alors c'est même une W-fibration (au sens de b), page 40, où c'est appelé 'W-fibration* hot*-forte'), en d'autres termes, si $S'' \longrightarrow S'$ est sur S avec S'', S' contractiles, alors $X'' \longrightarrow X'$ est dans W ([58]). On a sans doute mieux : si $S'' \longrightarrow S'$ est un homotopisme, alors $X'' \longrightarrow X'$ est dans W.*

Commentary · Exposition [ms. 48]

What the localization argument proves

Grothendieck next asks for invariance under an arbitrary categorical homotopy equivalence of bases. Here the argument changes: he identifies the map of total categories with a map of homotopy colimits and applies an adjunction. Let $v : T \to S$ be a homotopy equivalence and let ξ be a locally

55. O.K.
56. C'est ok en supposant que W est *fortement* saturé, i.e. $\mathrm{hot}_W(f)$ isomorphisme $\Longrightarrow f \in W$.
57. Donc W-fibration à gauche $\iff$ W-fibration à droite $\iff$ W-fibration.
58. on trouve seulement que $\mathrm{hot}_W(X'') \longrightarrow \mathrm{hot}_W(X')$ `[est un]` isomorphisme.

constant coefficient represented by the family. The contraction argument for the two composites of a homotopy inverse shows that restriction

$$v^* : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(T)$$

is an equivalence. Once the comparison with strict pullback and the adjunction $p_! \dashv p^*$ have been established, the displayed Hom calculation on p. 50 proves that the actual map of total categories becomes invertible in Hot_W.

This is an assertion about the specified map, rather than merely an abstract isomorphism between its source and target. The last inference to membership in W requires strong saturation. The footnote makes that distinction correctly. Later theory proves that every fundamental localizer in the modern, stronger sense is strongly saturated; see the status discussion at p. 16 and (Cisinski, 2006a, Proposition 4.2.4). Here the stronger sense includes the relative axiom W(2), not merely W(2 bis) in the terminology fixed at the beginning of section III. Theorem 3 explicitly assumes W(1,2,3), so that later theorem removes its additional saturation hypothesis. We are not asserting this consequence for an arbitrary weak fundamental localizer satisfying only W(2 bis).

The finite-zigzag argument in the preceding block already proves the contractible-base tests directly in W. Thus it is consistent for Theorem 3 to state those equivalences under W(1,2,3), while presenting its more general assertion *d*) first in the localized category.

[page 49]

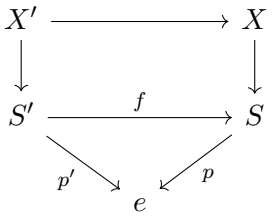
Prouvons (si nous pouvons) ce complément sympathique. Je précise que j'ai eu à faire l'hypothèse de W-prélissité (où de W-prépropreté) pour être sûr que l'image inverse $X \times_S S'$, par $f' : S' \longrightarrow S$, coïncide bien avec f^* sur les $\mathrm{HOT}^{\mathrm{lc}}_W$ - autrement je n'ai aucune chance. 'On sait' (tout est à revoir!) que c'est O.K. dans les trois cas X W-lisse, X W-propre, X une W-fibration (mais en quel sens???). La vérification à faire (quand on travaille avec les W^{d}_S) est celle-ci :

(∗) $X \xrightarrow{i_X} \Phi_S(X) = Y$ est dans W^{d}_S (sans mérite), et le reste par tout changement de base.

Mais pour des $X, Y \in \mathrm{Ob}\,\mathrm{Cat}/S$ qui sont W-prélisse, la W^{d}_S-équivalence coïncide avec W^{f}_S, laquelle est préservée par changement de base, ainsi que l'hypothèse sur X, Y. Donc on est tranquille.

On est ramené à prouver que si X est W-prélisse sur S et une W-pseudofibration, et si $S' \longrightarrow S$ est un homotopisme, alors $X' \longrightarrow X$ est dans W. Par l'argument de tantôt utilisant $X \longrightarrow \Phi_S(X)$, on se ramène au cas où X est fibrée scindée.

p. 36
[page 50]



La flèche $X' = f^*(X) \longrightarrow X$ s'interprète dans Hot_W comme

(∗) $$p'_!(f^*(\xi)) \longrightarrow p_!(\xi)\ , \qquad \text{où } \xi = \mathrm{hot}_{S,W}(X).$$

Je dis que cette flèche est [un] isomorphisme, i.e. que pour tout $\eta \in \mathrm{Hot}_W$, la flèche transposée

(∗∗) $$\mathrm{Hom}_W(p_!(\xi), \eta) \longrightarrow \mathrm{Hom}_W(p'_!(f^*(\xi)), \eta)$$

est [une] bijection. Admettant la formule d'adjonction, la flèche (∗∗) s'interprète comme

$$\mathrm{Hom}_W(\xi, \underbrace{p^*(\eta)}_{\eta_S}) \longrightarrow \mathrm{Hom}_W(f^*(\xi), \underbrace{p'^*(\eta)}_{f^*(\eta_S)}),$$

i.e.

$$\mathrm{Hom}_W(\xi, \eta_S) \longrightarrow \mathrm{Hom}_W(f^*(\xi), f^*(\eta_S)),$$

que ce soit [un] isomorphisme résulte donc de f^* pleinement fidèle sur les $\mathrm{HOT}^{\mathrm{lc}}_W$. Or on a vu que f^* est une équivalence (cor. 2, page 46).

Je vais reprendre les propositions 9 et 8, un peu mal foutues. [page 51]

Théorème 3. *On suppose que W satisfait à* W(1,2,3). *Soit $f : X \longrightarrow S$. Considérons les conditions suivantes.*

a) $\forall$ $S'' \longrightarrow S'$ *sur S, avec S'', S'* contractiles, *$X'' \longrightarrow X'$ est dans W.*

a′) $\forall$ *S' sur S, avec S' contractile, et $s' \in S'$, l'inclusion*

$$X_{s'} \hookrightarrow X' = X \times_S S'$$

est dans W.

a_1) Comme a′), mais en supposant que S' a un objet final (au lieu de S' contractile).

a_2) Comme a′), mais en supposant que S' a un objet initial.

*b_1) f est W-*prépropre *(i.e. a′) est satisfait quand s' est un objet final de S', sans autre condition sur S' dans* Cat$/S$*), et de plus pour toute flèche $s_0 \longrightarrow s_1$ dans S, la flèche induite $X/s_0 \longrightarrow X/s_1$ est dans W (i.e. a′* [plutôt a]*) est satisfait pour $Y/s_0 \longrightarrow Y/s_1$* [plutôt $S/s_0 \longrightarrow S/s_1$]*).* p. 37

*b_2) f est W-*prélisse *(i.e. a′) est satisfait quand s' est un objet initial de S', sans autre condition sur S'), et de plus les flèches $s_0\backslash X \longrightarrow s_1\backslash X$* [plutôt $s_1\backslash X \longrightarrow s_0\backslash X$] *(pour $s_0 \longrightarrow s_1$ dans S) sont dans W, (i.e. a′* [plutôt a]*) satisfait pour $s_1\backslash Y \longrightarrow s_0\backslash Y$* [plutôt $s_1\backslash S \longrightarrow s_0\backslash S$]*).*

c_1) f est W-prépropre et une W-pseudofibration, i.e. la condition a′) est satisfaite dans les deux cas suivants: 1°) s' est un objet final de S', 2°) $S' = \mathbf{\Delta}^1$.

c_2) f est W-prélisse et une W-pseudofibration, i.e. a′) dans les deux cas: 1°) s' est objet initial de S', 2°) $S' = \mathbf{\Delta}^1$.

Toutes ces 8 conditions sont équivalentes. Elles impliquent la suivante :

d) Soit $S'' \xrightarrow{u} S'$ un homotopisme dans Cat$/S$*, et considérons $u_X : X'' \longrightarrow X'$. Alors* $\mathrm{hot}_W(u_X)$ *est* [un] *isomorphisme. (Donc si W est fortement saturé, $u_X \in W$. Donc sous cette condition sur W, d) équivaut aux conditions précédentes.)*

Définition 4. On dit que $f : X \longrightarrow Y$ dans Cat est une *W-fibration*, si elle satisfait les conditions équivalentes a) à c_2) de la proposition 9. [page 52]

Commentary · Exposition [ms. 52]

Reading the eight equivalent conditions

Grothendieck has now isolated a usable fibration notion. Its eight formulations organize two independent requirements: strict fibres must compute comma fibres universally, and the latter must vary by weak equivalences. The reference to "proposition 9" in Definition 4 means the immediately preceding Theorem 3; Proposition 9 did not contain eight labelled conditions.

Here is the logical structure, including the point at which the contractible-base argument is used. Condition *a*) implies *a′*) by taking $S'' = e$. Conversely, for a map $u : T \to U$ between contractible categories over S, choose $t \in T$. The triangle of inclusions of the same strict fibre into X_T and X_U commutes. Both inclusions are in W by *a′*), so $X_T \to X_U \in W$. Contractible categories are nonempty; there is no missing empty-base case.

The implications *a′*) $\Rightarrow$ a_1), a_2) follow because categories with terminal or initial objects admit categorical contractions. Proposition 8 gives $a_1) \iff b_1) \iff c_1)$ and its dual gives $a_2) \iff b_2) \iff c_2)$. Starting with either c_1) or c_2), use the corresponding split replacement and the finite transport zigzag to obtain the fibre inclusion over every contractible base. This proves the reverse implication to *a′*) and therefore all eight equivalences.

For an ordinary Grothendieck fibration, the comma comparisons are automatic and the remaining test is simply that every pullback functor u^* lies in W; the opfibred statement uses u_*. For a family that is both preproper and prelisse, the two interval endpoint inclusions already lie in W, so the pseudofibration test is automatic. These observations prove the three corollaries. They do not identify a W-fibration with a strict Grothendieck fibration, nor with a perfect functor: the first supplies strict Cartesian lifts, and the second supplies stronger asphericity of lift categories.

Corollaire 1. *Si f est W-propre ou W-lisse, pour que f soit une W-fibration, il faut et il suffit que f soit une W-pseudofibration, i.e. que pour $S' = \mathbf{\Delta}^1 \longrightarrow S$, les inclusions des deux fibres*

$$X'_0 \xrightarrow{i_0} X' \xleftarrow{i_1} X'_1$$

de $X' = X \times_S S'$ soient dans W.

Corollaire 2. *Supposons f* Cat*-fibrant. Alors f est une W-fibration si et seulement si pout toute flèche $u : s \longrightarrow s'$ dans S, $u^* : X_{s'} \longrightarrow X_s$ est dans W. Énoncé dual si f est* Cat*-cofibrant.*

Corollaire 3. *Supposons f W-prépropre et W-prélisse (par exemple f W-parfait). Alors f est une W-fibration.*

Proposition 10. *Soit $f : X \longrightarrow Y$ une flèche dans $\underline{\mathrm{Fib}}_W S$ ([59]) (sous-catégorie pleine de* Cat*$/S$ formée des W-fibrés sur S). Si $\forall\ s \in S$, $f_s : X_s \longrightarrow Y_s$ est dans W, alors $f \in W$.*

Commentary · Exposition [ms. 52]

Fibrewise equivalence gives equivalence of total categories

Proposition 10 is a comparison theorem in one direction. It says that equivalent fibres give equivalent totals when the structural maps make those fibres compute the relevant comma categories. It does not yet assert that an equivalence of totals can be detected fibre by fibre; that converse is the separate axiom W(4b).

This argument uses the full relative axiom W(2), as assumed in Theorem 3; the weaker standing definition of section III alone is not the stated justification. Suppose first that X and Y are prelisse over S, and that $h : X \to Y$ is in W on every fibre. For each s the commutative square

$$\begin{array}{ccc} X_s & \xrightarrow{h_s} & Y_s \\ \downarrow & & \downarrow \\ s\backslash X & \longrightarrow & s\backslash Y \end{array}$$

has its upper and vertical arrows in W. Hence its lower arrow is in W. The dual form of W(2) gives $h \in W$. If the structural maps are preproper, use X/s and Y/s instead. This proves exactly the strengthening in footnote 59, without any pseudofibration hypothesis.

The source's replacement proof packages the same argument differently. The maps i_X, i_Y are homotopy equivalences on total categories and are fibrewise equivalences under prelissity. Their split replacements are smooth over S, so the corollary to Proposition 5 applies to the induced map. Referring to that corollary, rather than requiring the map between the replacements itself to be smooth, is the correct reading of the final sentence.

Considérons

$$\begin{array}{ccccc} X & & \xrightarrow{\quad f \quad} & & Y \\ & \searrow^{i_X} & & & \searrow^{i_Y} \\ & & \Phi_S(X) & \xrightarrow[\Phi_S(f)]{} & \Phi_S(Y), \end{array}$$

p. 38

où $i_X, i_Y \in W$, donc $\Phi_S(f) \in W \Rightarrow f \in W$. Or le fait que X `[est]` W-fibrée sur S implique que $i_X \in W_S^{\mathrm{f}}$ (il suffit même que X soit prépropre `[plutôt prélisse]` sur S), itou pour Y, i_Y. Donc l'hypothèse $f \in W_S^{\mathrm{f}}$ implique $\Phi_S(f) \in W_S^{\mathrm{f}}$, donc $\Phi_S(f) \in W$ par `[la]` proposition 5 (page 29).

p. 39

59. X et Y tous deux prépropres ou tous deux prélisses suffit.

[page 53]

IV Sur l'axiome W(4) et ses variantes.

([60]). J'ai passé des jours à essayer de prouver que toutes ces variantes sont équivalentes. Je vais donner une liste de celles, parmi ces variantes, qui avec le recul me paraissent encore avoir une utilité. (Avec des notations différentes de celles des pages 13-14.)

Commentary · Exposition [ms. 53]

The two directions of a fibration axiom

Grothendieck separates the two properties that ordinary homotopy theory makes look inseparable. The first concerns a change of base: a weak equivalence $T \to S$, pulled back along a suitable family $X \to S$, should remain a weak equivalence. The second concerns a map between families over one fixed base: if its total map is a weak equivalence, each fibre map should be one. These are W(4a) and W(4b), respectively. W(4c) is the special case of the second in which the second family is $S \to S$ itself.

The primes restrict the objects on which these properties are tested. In W(4a') and W(4c') the map is perfect for the minimal localizer; in W(4b') the structural families are split and the comparison preserves the splittings; in W(4b") the structural families are W-perfect. These restrictions are not interchangeable merely by renaming a fibration. In particular, the argument of section I used a stronger base-change condition than the eventual definition on p. 52. The editors' objection to the arrow $a' \Rightarrow b$ records a genuine dependence on that change of definition.

There is also a change of lettering. The weak perfect-fibre axiom called W(4b") in section I is now W(4c'). Keeping its content fixed is safer than carrying its old label into a later argument. The subsequent proof establishes one implication through locally constant coefficients; it does not prove every arrow that the surrounding discussion anticipates.

W(4a) Toute W-fibration (cf. th. 3, p. 51, déf. 3, p. 52) est ultraforte, ce qui revient à ceci: pour

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array}$$

cartésien dans Cat, si $f \in \underline{\mathrm{Fib}}_W$, $p \in W$, alors $q \in W$.

W(4a′) Comme a), en supposant $f \in \mathrm{Parf}_\omega$, i.e. f W_ω-parfait (donc une W_ω-fibration), où $W_\omega \subseteq W$ est la plus petits partie Σ de Fl(Cat) satisfaisant W(1, 2 bis, 3).

W(4b) Soit un diagramme commutatif $\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$ dans Cat, avec $p, q \in \mathrm{Fib}_W$, $f \in W$. Alors $\forall s \in S$, $f_s : X_s \longrightarrow Y_s$ `[est dans W]`.

W(4b′) Comme b), en supposant `[de plus]` X et Y fibrés scindés sur S, f compatible avec `[les]` scindages (donc foncteur cartésien).

W(4b″) Comme b), en supposant p, q *W-parfaits.*

W(4c) Soit $f : X \longrightarrow Y$, avec $f \in \mathrm{Fib}_W \cap W$, alors les fibres de f sont W-asphériques.

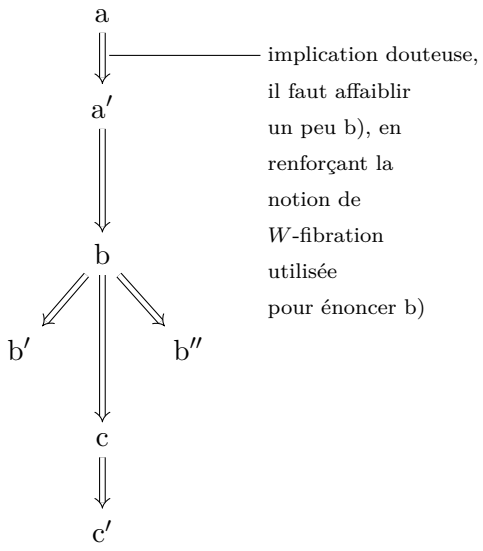
W(4c′) Comme c), en supposant $f \in \mathrm{Parf}_\omega \cap W$. (C'est l'ancien W(4b″), dépouillé du W_∞-parasite.)

J'ai cerclé les notions `[encadrées ici]` qui ont été vraiment utiles. Pratiquement, on [page 54]
peut parier que c'est la notion la plus forte, W(4a), qui est la seule importante. Voici les implications tautologiques (à l'exception de W(4a′) $\Longrightarrow$ W(4b), qui est démontré dans la démonstration de la proposition 4 (pages 6-10) ([61])).

p. 40

60. Cf. aussi partie VI (HOT_W), p. 186-196

61. Cette démonstration utilise une notion plus forte de W-fibration que celle du théorème 3, p. 51.



J'ai surtout sué pour montrer que c′) ⟹ c), ou c) ⟹ b), sans y arriver. L'effort m'a permis de me débarrasser de diverses idées fausses, et de les remplacer par des justes, cf. plus bas. Il y a quand même un résultat positif qui surnage :

Proposition 1. *La condition b) (et sans doute même b′) et b″)* ([62])*, mais je ne vais pas le prouver ici …) implique sans doute a) pour $\bar{W}$, le saturé fort de W. En tous cas, si W est fortement saturé, les conditions a), a′), b) (et sûrement aussi b′) et b″)) sont équivalentes.*

Commentary · Present state [ms. 54]

The later resolution of the weakest fibre axiom

Grothendieck's reservations here belong to the search for a proof from a deliberately small list of axioms. The minimal weak-localizer theorem gives $W_\omega = W_\infty$ even with W(2 bis), by Cisinski (2006a, Theorem 6.1.18). The proof in section I, completed by the nondegeneracy argument at p. 20, then shows that W(4c') forces

$$W = W_\infty \quad \text{or} \quad W = \mathrm{Fl}(\mathrm{Cat}).$$

The second possibility must be retained: all the fibration axioms hold vacuously when every functor is a weak equivalence. This argument uses the perfect replacement and iterated homotopy-fibre construction, together with the minimality theorem; it does not require filtered-colimit stability or strong saturation of an arbitrary weak localizer. Those properties follow after the displayed classification. For the nontrivial alternative, ordinary homotopy theory gives the full pullback and fibrewise comparison statements. A related published characterization under the full relative-localization hypothesis is Cisinski (2006a, Theorem 6.4.22 and Remark 6.4.23), formulated using locally constant functors.

The split comparison test b') also detects the weakest perfect-fibre test. If $E \to B$ is W_∞-perfect and belongs to W, compare the split replacements $\Phi_B(E) \to \Phi_B(B)$. Its total map is in W, since both replacement units are. Their transports are weak equivalences, and the comparison is strictly Cartesian. Thus b') makes each $b\backslash E \to b\backslash B$ a W-equivalence. The perfect fibre comparison $E_b \to b\backslash E$ and the initial object of $b\backslash B$ then show that E_b is W-aspherical. The implication from the perfect pullback test a') uses the mapping-path comparison already audited in section I, rather than the disputed reuse of an older fibration definition.

Accordingly the weak perfect test, the split test, and the full tests have the same eventual alternatives. This retrospective conclusion does not turn the unfinished implications in the manuscript into proofs written there. In the claim register their original qualifications remain conjectural or sketched, with the later resolution recorded separately.

62. et en supposant dans le cas de b″, que W satisfait la condition des limites inductives W(6 bis), cf. prop. 5 du 'corollaire' de la, page 14.

Démonstration de b) $\Longrightarrow$ a), quand W est fortement saturé. En fait, moyennant b) (et sans même supposer W fortement saturé), nous allons prouver que si on a

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{p} & Y' \end{array}$$

cartésien dans Cat, avec f une W-fibration, et $p \in W$, alors $\underline{q \in \bar{W}}$, i.e. $\mathrm{hot}_W(q)$ [est un] isomorphisme. Pour ceci, nous interprétons la flèche q dans Hot_W de la façon suivante. Soit [page 55]

$$\xi = \mathrm{hot}_{Y,W}(X/Y), \qquad \xi' = p^*(\xi),$$

qui dit que q s'identifie à la flèche canonique

$$(*) \qquad H_\bullet^{\mathrm{HOT}_W}(Y', \xi') \longrightarrow H_\bullet^{\mathrm{HOT}_W}(Y, \xi) \qquad \text{dans } \mathrm{Hot}_W = \mathrm{HOT}_W(e), \qquad \text{p. 41}$$

où on note de plus que $\xi \in \mathrm{HOT}_W(Y)$ est un coefficient *localement constant* sur Y.

Établissons d'abord cette interprétation de q. Notons que comme $X \in \underline{\mathrm{Fib}}_W Y$, l'image inverse $\xi' = p^*(\xi)$ n'est autre que $\mathrm{hot}_{Y',W}(X' = X \times_Y Y')$. D'autre part, si $g : Y \longrightarrow e$, $g' : Y' \longrightarrow e$ sont les flèches structurales, les deux membres de $(*)$ s'interprètent resp. comme $g'_!(\xi')$ et $g_!(\xi)$ (les $g'_!$ et $g_!$ au sens du prédérivateur HOT_W). Or le calcul de $g'_!$, $g_!$ nous donne que les deux membres s'identifient resp. à

$$\mathrm{hot}_W(X'), \qquad \mathrm{hot}_W(X),$$

et nous admettons que la flèche

$$\mathrm{hot}_W(X') \longrightarrow \mathrm{hot}_W(X)$$

n'est autre que $\mathrm{hot}_W(q)$. D'ailleurs, comme X est W-fibrée sur S, par définition pratiquement, ξ est dans $\mathrm{HOT}^{\mathrm{lc}}_W(S)$. Donc il nous suffira de prouver le

Corollaire. *Supposons que W satisfait à W(4b), et soit $p : S' \longrightarrow S$ dans W, $\xi \in \mathrm{HOT}^{\mathrm{lc}}_W(S)$, $\xi' = p^*(\xi)$. Alors*

$$H_\bullet(S', \xi') \xrightarrow{\sim} H_\bullet(S, \xi)$$

est un isomorphisme dans Hot_W. [page 56]

Commentary · Exposition [ms. 56]

Why fully faithful restriction controls integration

Grothendieck's proof is driven by the fact that a homotopy colimit is characterized by its maps into constant coefficients. We can state the argument in a form which makes every required comparison visible. Let $p : T \to S$, and suppose that restriction is fully faithful on locally constant coefficients. Let $q : S \to e$ and $r : T \to e$ be the structural maps, and suppose the adjunctions $q_! \dashv q^*$ and $r_! \dashv r^*$ are available. For a locally constant ξ, there is a canonical comparison

$$c : r_! p^* \xi \longrightarrow q_! \xi.$$

For every coefficient A at the point, adjunction identifies precomposition by c with

$$\mathrm{Hom}(\xi, q^* A) \longrightarrow \mathrm{Hom}(p^* \xi, p^* q^* A).$$

Both ξ and q^*A are locally constant, so full faithfulness makes this a bijection. Yoneda implies that c is an isomorphism. Essential surjectivity of restriction is stronger than this argument requires.

The application to a literal pullback $X_T \to X$ needs two further identifications: strict pullback must represent $p^*\xi$, and the chosen integration functors must send these representatives to their total categories, with c represented by the actual projection. Prelissity supplies the first identification; the construction of $q_!$ supplies the second. The source acknowledges these comparisons instead of proving them anew. The same formal Hom argument appears in Cisinski (2009, Proposition 1.10), and its Theorem 1.3 supplies ordinary weak-homotopy invariance for model-category-valued locally constant diagrams. These later results clarify the argument without making W(4b) an unmentioned consequence of W(1,2,3).

Démonstration. Il suffit de prouver que pour tout $\eta \in \mathrm{Ob}\,\mathrm{Hot}_W$ on a une bijection pour les $\mathrm{Hom}(-,\eta)$. Par la formule d'adjonction sur S, S', cette application entre les Hom dans Hot_W s'identifie [à]

$$\mathrm{Hom}_{\mathrm{HOT}_W(S)}(\xi,\eta_S) \longrightarrow \mathrm{Hom}_{\mathrm{HOT}_W(S')}(\xi',\eta_{S'}),$$

où $\eta_S = p^*(\eta)$, $\eta_{S'} = p'^*(\eta)$ [plutôt $\eta_S = g^*(\eta)$, $\eta_{S'} = g'^*(\eta)$, où $g : S \longrightarrow e$, $g' : S' \longrightarrow e$ sont les flèches structurales], donc $\eta_{S'} = p^*(\eta_S)$, et la flèche $(*)$ est la flèche sur les Hom induite par le foncteur

$$p^*_{\mathrm{lc}} : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S').$$

Or moyennant l'axiome W(4b) (que nous n'avons pas utilisé jusqu'à présent), ledit foncteur est une *équivalence de catégories* si $p \in W$ (cf. VI, p. 204).

q.e.d.

De peur de les oublier, je vais maintenant faire une liste des petites questions que j'ai élucidées chemin faisant en gribouillant.

Commentary · Examples [ms. 56]

Why preservation of Cartesian arrows matters

Grothendieck's first observation separates fibrewise smoothness from smoothness of the total functor. A small example makes the lost information explicit. Let C be the one-object category with endomorphism monoid $(\mathbf{N},+)$, and put $X = Y = C \times [1]$, both projected to $[1]$. Define an endofunctor $h : X \to Y$ over $[1]$ to be the identity on both fibres and to send a morphism crossing from 0 to 1, labelled $n \in \mathbf{N}$, to the morphism labelled $n+1$.

This is a functor. A composite crosses from 0 to 1 at most once, so adding 1 commutes with precomposition and postcomposition by fibre morphisms. Each fibre functor is the identity and hence is smooth and proper. Both structural projections are products, hence split fibrations. But h does not preserve Cartesian arrows: the Cartesian crossing labelled 0 is sent to the non-Cartesian crossing labelled 1.

To test smoothness of h, take the target object over 1 and the arrow in Y from the object over 0 labelled 0. A lift with this prescribed image would be a crossing morphism labelled n in X with $n+1 = 0$. There is none. Its lifting category is empty, so h is not W_∞-smooth. This is a fully explicit instance of the failure claimed in item 1. If the functor preserves the splitting, an arbitrary arrow over $u : s \to t$ factors through the chosen Cartesian lift, and its lift category is the corresponding lift category of the fibre functor at s. That identification proves the positive assertion in item 1.

1°) Si X, Y sont W-lisses sur S, et $f : X \longrightarrow Y$ un S-morphisme tel que $f_s : X_s \longrightarrow Y_s$ soit W-lisse pour tout $s \in \mathrm{Ob}\,S$, il ne s'ensuit *pas* que f soit W-lisse. Mais c'est pourtant vrai quand X et Y sont fibrées scindées sur S (ce qui implique déjà leur W-lissité sur S), et f compatible avec les scindages et a fortiori cartésien. Énoncé dual pour les catégories cofibrées scindées et les foncteurs strictement cartésiens entre elles qui sont W-propres fibre par fibre : ils sont W-propres. p. 42 [page 57]

2°) J'étais intéressé, dans un cas de catégories fibrées scindées X, Y sur S et d'un foncteur strictement cartésien $f : X \longrightarrow Y$, supposé W-parfait fibre par fibre, de conclure que f soit aussi W-parfait. (C'était un cas où de plus on sait que X, Y sont aussi des W-fibrés sur S, i.e. les fonteurs changement de base $X_{s'} \longrightarrow X_s$, $Y_{s'} \longrightarrow Y_s$ sont dans W. Mais ça n'a pas beaucoup servi …) La question qui restait, en 1°), était celle de la *W-propreté* de f. Mais c'est sans espoir (même pour $Y = S$), en vertu de la brillante

Proposition 2. *Supposons que* $S = 0 \begin{array}{c} \nearrow a \\ \searrow b \end{array}$ *, et soit Z une catégorie sur S. Alors $\Phi(Z)$ est propre sur S si et seulement si $Z = \emptyset$.*

Commentary · Exposition [ms. 57]

The fork obstructs outgoing lifts

The fork is being used for a second purpose here: a free split fibration supplies incoming Cartesian lifts, while properness requires the appropriate outgoing lift categories to be aspherical. The commentary on Chapter VI, manuscript pp. 190–192, gives the full fork calculation and the support obstruction; we refer to it rather than repeat the calculation.

For the present application, one may see the obstruction at an object $(0, z, a)$ of $\Phi_S(Z)$ whose label in S lies on one branch. The arrow from 0 to the other branch has no compatible outgoing lift from that object: it would require a morphism in the fork from the first terminal branch to the other. If Z has an object over either branch, this construction applies. If every object of the nonempty category Z lies over 0, then an outgoing lift towards either terminal branch is again impossible, because no object of Z can label its target. These two cases exhaust the possibilities. In each case the relevant lift category is empty.

Thus the assertion is correct for ordinary properness, and for W-properness whenever W is nontrivial: the empty category is then not W-aspherical. If $W = \mathrm{Fl}(\mathrm{Cat})$, every lift category, including the empty one, is W-aspherical and every functor is W-proper. That degenerate case must be excluded from the relative reading of "if and only if $Z = \varnothing$". Fibrewise perfectness of the structural map to S carries no information about this outgoing obstruction, since every map from a category to the point is perfect.

Prenons alors dans 2°) ci-dessus $X = \Phi(Z)$, $Y = S$, $f : \Phi(Y) \longrightarrow S$ [plutôt $f : \Phi(Z) \longrightarrow S$] le morphisme structural (donc les f_s sont W-parfaits), alors f n'est propre que si $X = \emptyset$. Pourtant, si on prend Z W-fibrée sur S, $\Phi(Z)$ l'est aussi, et on a un contre-exemple à 2°).

Mais on s'est quand même convaincu du résultat suivant :

Proposition 3. *Soient X, Y catégories fibrées scindées sur S, $f : X \longrightarrow Y$ S-foncteur strictement cartésien, tel que pour tout $s \in S$, $f_s : X_s \longrightarrow Y_s$ soit* W-fibrant. *Alors,* si W satisfait W(4a), *f est W-fibrant.* [page 58]

Commentary · Development (commentators') [ms. 58]

A missing hypothesis in the last proposition

Grothendieck is trying to lift a fibrewise fibration statement through the Grothendieck construction. The preceding paragraph mentions an additional assumption that both structural families are W-fibrations over S. That assumption is absent from Proposition 3 as printed. The literal statement is false, even for ordinary weak equivalences.

A strictly Cartesian counterexample. [ours] Take $S = [1]$. Let $F, G : S^{\mathrm{op}} \to \mathrm{Cat}$ be diagrams of discrete categories

$$F(0) = F(1) = e, \qquad G(0) = e, \quad G(1) = \{a, b\},$$

with the unique transition maps. Let η_0 be the identity and let $\eta_1 : e \to \{a, b\}$ select a. These components form a strict natural transformation. Its Grothendieck construction is the inclusion

$$h : \{o \to a\} \longrightarrow \{o \to a,\ o \to b\}.$$

Both source and target are split fibrations over [1], and h preserves the chosen Cartesian lifts: the only nonidentity chosen lift in the source is sent to $o \to a$. Each component of η is perfect. Indeed every functor between discrete categories is both a Grothendieck fibration and an opfibration; the only arrows to lift in its target are identities. It is consequently a W_∞-fibration.

Pull back h along the arrow $o \to b$ in the target. The resulting category has one object, lying over $0 \in [1]$. Its fibres are e at 0 and $\varnothing$ at 1. The second endpoint inclusion is $\varnothing \to e$, which is not a weak equivalence. Thus h is not even a W_∞-pseudofibration. This proves the counterexample. Notice that the target structural family is not a W_∞-fibration: its transport is $\{a, b\} \to e$. This explains exactly how the extra assumption mentioned in the preceding discussion would exclude the example.

Here is a sufficient repair for the fibrewise-perfect situation that motivated the proposition. It states explicitly the two comparison properties used in the proof rather than hiding them in an appeal to ordinary intuition.

A sufficient fibrewise-smooth repair. [ours] Assume W(4a) and W(4b). Let $F, G : S^{\mathrm{op}} \to \mathrm{Cat}$ send every arrow into W, and let $\eta : F \Rightarrow G$ be a strict natural transformation. Suppose every η_s is both W-smooth and a W-fibration; fibrewise perfectness is sufficient. Then $h = \int \eta$ is W-smooth and a W-fibration.

First, an arrow in $\int G$ from (s, y) to the image of (t, x) is given by $u : s \to t$ and $\beta : y \to \eta_s F(u)x$. Its lift category for h is exactly the lift category of β for η_s. This proves smoothness, because the latter category is W-aspherical.

Next fix $u : s \to t$. Form

$$P = G(t) \times_{G(s)} F(s).$$

The projection $P \to F(s)$ belongs to W by W(4a), since it is the pullback of $G(u) \in W$ along the W-fibration η_s. The naturality map $F(t) \to P$ composes with this projection to $F(u) \in W$, so it also belongs to W. Both $F(t) \to G(t)$ and $P \to G(t)$ are W-fibrations. W(4b) therefore shows that for every $z \in G(t)$ the induced relative-fibre transport

$$F(t)_z \longrightarrow F(s)_{G(u)z}$$

belongs to W.

To finish, factor an arbitrary arrow of $\int G$ into an arrow within a fibre followed by a chosen Cartesian arrow. For an arrow within $G(s)$, transport on the under-comma categories of h is in W because η_s is a W-fibration. The comparison with its total under-comma category is a homotopy equivalence: an object above t with base arrow $u : s \to t$ is sent to its Cartesian pullback $F(u)x$ above s, giving a right adjoint to the inclusion of the fibre under-comma category. For a chosen Cartesian arrow, the displayed relative-fibre transport and smoothness identify the corresponding under-comma map as an element of W. Composition now gives the assertion for every arrow of $\int G$. The dual form of Proposition 8, together with smoothness, proves that h is a W-fibration.

This repair covers the preceding discussion when its structural families and fibrewise-perfect hypotheses are retained. It does not claim that adding only the omitted structural-fibration hypothesis proves the entire printed formulation. We checked the later treatments of smooth functors and locally constant families cited in this commentary and did not locate this exact small counterexample or this explicit sufficient repair there. These are deductions given here, without a claim of priority.

(J'espérais avoir que f est W-fibrant 'à l'œil', pour prouver que W(4c) implique W(4b). Mais j'étais obligé de supposer W(4a) …)

Commentary · Later developments [ms. 58]

The coherence condition in a later gluing theorem

The last failed shortcut has a useful later analogue. Rezk (1998, Theorem 7.1) proves that the homotopy colimit of a diagram of sharp maps is sharp when every naturality square is homotopy Cartesian. This square condition is the coherence missing from a bare list of fibrewise fibration assertions. In the counterexample just given, the square of discrete fibres

$$\begin{array}{ccc} e & \longrightarrow & e \\ \downarrow & & \downarrow \\ \{a, b\} & \longrightarrow & e \end{array}$$

is not homotopy Cartesian: the homotopy pullback of the other three corners has two points.

This is a comparison of mathematical mechanisms, not an identification of all the models. Thomason's comparison between the nerve of a Grothendieck construction and a homotopy colimit is a weak equivalence; such a comparison alone does not transport the property that every *strict* pullback is homotopy Cartesian. To deduce a statement about the literal functor of Proposition 3, the comparison must also control the pullbacks that occur in its fibration test. That is why the source's dependence on W(4a), and the omitted control of structural transport, cannot be removed by a slogan about homotopy colimits.

Overview · XI-XIIa

Fibrations, localizers, and the coherence of fibres

The pages assembled here turn a useful geometric intuition into a problem of foundations. A family of categories should behave like a fibration when its strict fibres calculate homotopy fibres and vary coherently over the base. Grothendieck first characterizes that behaviour for ordinary weak homotopy equivalences. He then asks which axioms on an abstract class W of functors permit the same constructions, and discovers that several apparently natural requirements already force the whole ordinary homotopy theory. The resulting revisions are part of the argument: the word "fibration" changes its tests as the desired scope of the theory becomes clearer.

These pages comprise Chapter XI, manuscript pp. 127–136, and Chapter XII, manuscript pp. 1–58, through the end of section IV. These are separate manuscript sequences. The edition explicitly says that Chapter XI has no pp. 1–126; their absence is not a gap in the text. We write $BC = |NC|$ for the classifying space of a category and W_∞ for the functors whose nerves are weak homotopy equivalences.

The geometric theorem and its coefficient proof. Chapter XI begins with a property of the actual pullback functor:

$$f^* : \mathrm{Cat}/Y \longrightarrow \mathrm{Cat}/X \quad \text{preserves } W_\infty.$$

This is a universal condition on changes of base, not a condition merely on the homotopy types of the individual fibres. Theorem 1 identifies it with universal comma-category tests and with base-change and local-constancy statements for direct images of locally constant coefficients. In the terminology later used for model categories, these are sharp maps for the Thomason weak equivalences (Rezk, 1998, §2). Quillen's Theorem B explains the geometry of the comma tests: when their transition functors are weak equivalences, the strict fibre, compared with the appropriate comma category, represents the homotopy fibre (Quillen, 1973, §1).

Most of the coefficient argument is complete as written. The degree-zero comparison and the first-quadrant Leray comparison require no new finiteness hypothesis. The nonabelian passage needs more care. A component of a torsor groupoid has the automorphism group of a representative torsor; this group can be an inner form of the coefficient group, and its direct image is not generally the fixed group printed in the proposed obstruction sequence. We therefore replace that sequence by an argument with the groupoids of torsors on X/y. The universal degree-zero hypothesis gives full faithfulness of their transition functors; the degree-one hypothesis gives essential surjectivity. The same two hypotheses identify their values with the torsor groupoids on strict fibres. Descent reconstructs global torsors from this locally constant diagram, and homotopy invariance of such diagrams proves the required comparison (Cisinski, 2009, Theorem 1.3).

Theorem 2 then detects an equivalence between two such families on one fibre in each connected component of the base. The source's local reduction for torsors becomes precise through the same descent argument. Its converse uses a contractible replacement of the base. We give an independent construction by factoring a point in Thomason's model category (Thomason, 1980, Theorem 4.9), and also check the specific path-category dependency that Chapter XII recalls from Chapter VII. This separates the existence of a suitable replacement from the stronger task of constructing it explicitly with both properness and smoothness.

What the axioms force. Chapter XII starts with identities and cancellation, a theorem-A principle, and the asphericity of categories with a terminal object. Two forms of theorem A occur. W(2) is relative: equivalences on comma categories over every object of a third category imply equivalence of the total functor. W(2 bis) is absolute: aspherical comma

categories imply that a functor is an equivalence. The manuscript deliberately uses the latter in several arguments. Modern "fundamental localizer" normally includes the former. We keep the distinction throughout the audit.

The minimum conjectured in these pages is now known. Cisinski identifies W_∞ as the smallest fundamental localizer (Cisinski, 2004, Theorem 2.2.11), and as the smallest weak fundamental localizer with the absolute theorem-A axiom (Cisinski, 2006a, Theorem 6.1.18). Strong saturation and stability under filtered colimits are also established for full fundamental localizers; those results are not assertions about every class satisfying only the weaker axiom. An inference from "invertible in the localization" to "belongs to W" is justified at the point where its applicable hypothesis is available.

The fibration axioms add a rigidity that the basic localizer axioms do not possess. Their weakest perfect-fibre form, combined with nondegeneracy, makes homotopy fibres of W-maps W-aspherical. Iterating the loop construction then detects all homotopy groups. We repair the source's nondegeneracy argument without its unjustified restriction to a component: a missed component produces an empty homotopy fibre, while two identified components produce a disconnected aspherical fibre, whose diagonal in its square again has an empty homotopy fibre. Asphericity of the empty category collapses the theory. Thus the alternatives are W_∞ and the class of every functor. The second alternative cannot be discarded from the opening assertion about derivators; the terminal derivator realizes it.

The truncation examples explain why a more flexible fibration notion is needed. The class W_n retains homotopy groups only through degree n, and is a genuine localizer. But the based-path family of an $(n+1)$-sphere has an invisible total space and an invisible base while its fibre retains a nonzero n-th homotopy group, or a nontrivial component set when $n = 0$. Hence the strongest fibre axioms fail. In degree zero, the exact sequence must be read in pointed sets and with the appropriate group action: a cokernel of groups does not describe the component set in general.

Lift categories supply the missing structure. Section III develops proper and smooth functors through categories of lifts with prescribed source or target. Those categories turn universal comma comparisons into workable tests, prove stability under base change and composition, and explain why Grothendieck (op)fibrations furnish examples. They also support the passage from fibrewise equivalence to equivalence of total categories. The later formulations by Maltsiniotis describe the corresponding exact squares, while the derivator account explains the associated base-change transformations (Maltsiniotis, 2005, 2012; Groth, 2013).

The proposed weakening to endpoint equivalences over every arrow loses a necessary compatibility. We give a poset with eight objects over [2]. Its fibres have two contractible components, every interval pullback has both endpoint inclusions in W_∞, and transport along $0 \to 1$ and $1 \to 2$ preserves the component labels. Transport along $0 \to 2$ exchanges them. Its total nerve is an eight-cycle. Consequently the edge transports do not compose, and the pseudofibration question raised on manuscript p. 40 has a negative answer. The complete proof is independent of the accompanying finite computation.

The successful replacement uses universal preproperness or prelissity in addition to the interval test. Comma categories then provide actual functorial transport. A split replacement converts this into a diagram of categories whose arrows lie in W. For a categorically contractible base, a finite zigzag of natural transformations gives an explicit finite zigzag of fibrewise and total W-equivalences. This proves the contractible-base portion of Theorem 3 directly, without extracting a zigzag from an abstract isomorphism in a localization. Here "contractible" has the manuscript's categorical meaning; a weakly contractible nerve need not admit such a finite contraction.

The last proposition and the limits of the repair. Section IV returns to the relations among the fibration axioms, then proposes a fibrewise gluing statement. Its final Proposition 3

is false literally as printed. Over [1], take the strictly Cartesian inclusion of the split fibration $\{o \to a\}$ into $\{o \to a, o \to b\}$. Every fibre map is a functor between discrete categories and is perfect. Pulling the total functor back along $o \to b$ gives endpoint fibres e and $\varnothing$; it is not even a pseudofibration. The target structural family is not a W_∞-fibration, precisely the extra condition mentioned in the preceding discussion and omitted from the proposition.

We prove a sufficient repair for that preceding, fibrewise-perfect situation: the structural transports lie in W, the fibre maps are smooth W-fibrations, and W(4a) and W(4b) control the relevant pullback and fibre comparison. We do not claim that adding only the omitted structural condition repairs the full printed statement. Rezk's later gluing theorem has an analogous additional requirement that naturality squares be homotopy Cartesian (Rezk, 1998, Theorem 7.1); passing between models does not by itself remove the need to control strict pullbacks.

The marked developments give these deductions with complete proofs and bounded accounts of the literature examined, without a claim of priority. The registers distinguish what the source proves, sketches, assumes, or conjectures from the status supplied by the commentary. They preserve edition slips in the French text and apply only corrections of the transcription. Questions not decided under their exact weaker hypotheses are identified as limitations of this audit, rather than declared open in the literature.

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V Sommes amalgamées et carrés W-cocartésiens dans Cat

1 Généralités sur $U \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} T$, et les sommes amalgamées.

1.1 Données de recollement entre un ouvert et un fermé.

Soit X dans Cat, U un ouvert dans X, T le complémentaire fermé. (**NB** Les ouverts correspondent bien aux ouverts de $\mathrm{Ob}\, X$, topologisé par sa relation de préordre naturelle $x \leq y \iff \mathrm{Hom}_X(x,y) \neq \emptyset$, de même pour les fermés. Les topologies associées aux relations de préordre sont celles où toute intersection d'ouverts est encore ouverte, i.e. toute réunion de fermés est encore fermée. Donc les ouverts d'une telle topologie sont les fermés d'une topologie duale, correspondant à la relation de préordre opposée. La relation de préordre associée à la topologie est

$$x \leq y \qquad \iff \qquad x \in \overline{\{y\}},$$

la topologie associée au préordre sur I est celle pour laquelle les ouverts sont les réunions d'ensembles de la forme I/x (lesquels forment une base d'ouverts), ou encore les *'cribles'* ($x \in U$, $y \leq x \Longrightarrow y \in U$), les fermés sont les réunions d'ensembles $x\backslash I$, c'est-à-dire les cocribles. Pour toute partie A de I, il y a un plus petit ouvert contenant A, qui est aussi son plus petit voisinage, c'est la réunion des I/x pour $x \in A$. Et dualement il y a un plus grand fermé contenu dans A, dont le complémentaire est le plus petit ouvert contenant $B = I\backslash A$, donc $\bigcup_{x\in B} I/x$. L'adhérence de A est la réunion des adhérences $x\backslash I$ de ses points.)

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Commentary · Exposition [ms. 59]

An open–closed cut remembers a profunctor

Grothendieck begins with a cut of a category which permits arrows to cross in only one direction. This is the reason that the subsequent pushout calculation is manageable. An open subcategory $U \subset X$ is a full sieve: if an arrow has its target in U, its source belongs to U. The complementary full subcategory T is therefore a cosieve, and there are no arrows from T to U. Once the compositions internal to U and T are known, the remaining information is precisely the action of these compositions on the sets of arrows from U to T.

There is a reversal in the displayed specialization convention on this page. With $x \leq y$ meaning that $\mathrm{Hom}_X(x,y)$ is nonempty and with the opens taken to be downsets, one has

$$\overline{\{x\}} = \{y \mid x \leq y\}, \qquad x \leq y \iff y \in \overline{\{x\}}.$$

The edition really prints $x \in \overline{\{y\}}$. That formula is incompatible with its following description of opens as sieves; it is a source slip, not a transcription error. For the walking arrow $0 \to 1$, the opens are $\varnothing, \{0\}, \{0,1\}$ and the closure of $\{0\}$ is the whole space. All the subsequent formulas for closure and boundary use the downset convention and are correct with it. In particular, the smallest open containing A is its downward closure, while its ordinary closure is its upward closure. No antisymmetry is required: categorically isomorphic objects, and more generally objects connected by arrows in both directions, cannot be distinguished by these opens.

For completeness, the reconstruction in Proposition 1 can be written without a choice of compositions. Given

$$H : U^{\mathrm{op}} \times T \longrightarrow \mathrm{Ens},$$

form a category $\mathrm{Coll}(H)$ whose objects are the tagged disjoint union of those of U and T, whose hom sets within each part are the original ones, whose hom sets from T to U are empty, and whose hom sets from $u \in U$ to $t \in T$ are $H(u,t)$. Define composition with $a : u' \to u$ by $h \mapsto H(a,1)(h)$ and composition with $b : t \to t'$ by $h \mapsto H(1,b)(h)$. The two operations commute by bifunctoriality. Every composable string contains at most one arrow crossing from U to T, so these identities, together with associativity in the two pieces, prove associativity in the collage. The identities follow from the identity laws for H.

A functor preserving the two parts consists of functors f_U, f_T and a natural transformation $H \to H'(f_U^{\mathrm{op}} \times f_T)$. The naturality squares are exactly the equations expressing preservation of composition involving a crossing arrow. This proves full faithfulness of the correspondence of Proposition 1; the collage construction proves essential surjectivity. With tagged objects there is a specified inverse up to canonical isomorphism, which explains the editors' preference for an equivalence over an unqualified isomorphism of categories. Today H is called a profunctor and $\mathrm{Coll}(H)$ its collage. The point of the terminology is that a category can be assembled from its internal categories and a bimodule of arrows between them.

La situation (X, U, T) se reconstitue, à isomorphisme unique près, par la connaissance de

1°) Les deux catégories U, T.

2°) Le bifoncteur $\begin{cases} H : U^\circ \times T \longrightarrow \mathrm{Ens} \\ H(u,t) = \mathrm{Hom}_X(u,t). \end{cases}$

De plus, ces données 1°) 2°) en sont soumises à aucune restriction. On a, plus exactement

Proposition 1. *Un isomorphisme* `[plutôt une équivalence de catégories]` *entre la catégorie $\mathcal{T}$ dont les objets sont les triples (X,U,T), avec $X \in \mathrm{Ob}\,\mathrm{Cat}$, $U \in \mathrm{Ouv}(X)$, $T \in \mathrm{Ferm}(X)$, U et T complémentaires l'un de l'autre, les morphismes de (X,U,T) dans (X',U',T') étant les foncteurs $f : X \longrightarrow X'$ qui induisent $U \longrightarrow U'$, $T \longrightarrow T'$ (d'où $U = f^{-1}(U')$, $T = f^{-1}(T')$), avec la catégorie $\mathcal{T}'$ des triples (U,T,H), de deux catégories U, T et d'un bifoncteur* p. 44

$$H : U^\circ \times T \longrightarrow \mathrm{Ens}$$

(ou encore un préfaiscau $U^\circ \longrightarrow T^\vee$ sur U, donc un objet de $\underline{\mathrm{Hom}}(U^\circ, T^\vee)$)), un morphisme de (U,T,H) dans (U',T',H') étant un triple

$$(f_U, f_T, u) \qquad \begin{cases} \left.\begin{array}{rccc} f_U : & U & \longrightarrow & U' \\ f_T : & T & \longrightarrow & T' \end{array}\right\} \text{des foncteurs} \\ u : H \longrightarrow H' \circ (f_U^\circ \times f_T), \\ \text{i.e. } H(u,t) \longrightarrow H'(f_U(u), f_T(t)) \\ \text{un homomorphisme fonctoriel} \\ \text{(flèche dans } (U \times T^\circ)^\wedge \simeq \underline{\mathrm{Hom}}(U^\circ, T^\vee))), \end{cases}$$

composition des flèches évidente.

Si $U \hookrightarrow X \hookleftarrow T$ est un triple, et

$$f_{U'} : U' \longrightarrow U, \ f_{T'} : T' \longrightarrow T$$

des flèches dans Cat, il y a une façon privilégiée de 'recoller' U' et T', de façon à ce que l'objet (X',U',T') s'envoie dans (X,U,T) en induisant

$f_{U'}$, $f_{T'}$: [page 61]

$$\begin{array}{ccccc} U' & \xhookrightarrow{i'} & X' & \xhookleftarrow{j'} & T' \\ \downarrow{\scriptstyle f_{U'}} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_{T'}} \\ U & \xhookrightarrow{i} & X & \xhookleftarrow{j} & T, \end{array}$$

elle s'obtient à l'aide des bifoncteurs

$$\begin{cases} H'(u',t') \overset{\text{déf}}{=} H(f_{U'}(u'), f_{T'}(t')) \\ \text{i.e. } H' = H \circ (f_{U'}^\circ \times f_{T'}). \end{cases}$$

Ce triple, i.e. cet objet de $\mathcal{T}$, représente le foncteur suivant sur $\mathcal{T}$:

$$(X'',U'',T'') \longmapsto \mathrm{Hom}_{\mathrm{Cat}}(U'',U) \times \mathrm{Hom}_{\mathrm{Cat}}(T'',T),$$

i.e. on a pour tout objet (X'', U'', T'') de $\mathcal{T}$ une bijection canonique, fonctorielle en cet objet

$$(*) \qquad \mathrm{Hom}_{\mathcal{T}}((X'', U'', T''), (X', U', T')) \simeq \mathrm{Hom}_{\mathrm{Cat}}(U'', U) \times \mathrm{Hom}_{\mathrm{Cat}}(T'', T).$$

Nul doute qu'on a même p. 45

$$(**) \qquad \underline{\mathrm{Hom}}_{\mathcal{T}}((X'', U'', T''), (X', U', T')) \simeq \underline{\mathrm{Hom}}_{\mathrm{Cat}}(U'', U) \times \underline{\mathrm{Hom}}_{\mathrm{Cat}}(T'', T),$$

tout aussi fonctoriel, dont $(*)$ se déduit en passant aux ensembles d'objets. On dit que le triple (X', U', T') est le triple *induit* par le triple (X, U, T), via les foncteurs $f_{U'}$, $f_{T'}$.

NB Si $f_{U'}$, $f_{T'}$ sont pleinement fidèles, alors il en est de même de $f : X' \longrightarrow X$. Nous utiliserons cette opération de recollement privilégié surtout quand $U' = U$, $f_{U'} = \mathrm{id}_U$, ou inversement $T' = T$, $f_{T'} = \mathrm{id}_T$.

1.2 Relation avec le recollement des topos.

La situation d'un triple

$$(\mathcal{X}, \mathcal{U}, \mathcal{T})$$

d'un topos $\mathcal{X}$, d'un ouvert $\mathcal{U}$ de $\mathcal{X}$, et d'un

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Commentary · Exposition [ms. 61]

The induced collage has a relative universal property

The induced triple is meant to express change of the two categories on which the gluing data live. Its universal property has to remember a map to the original triple. Without that map the displayed formulas $(*)$ and $(**)$ are false; both formulas are reproduced faithfully from the edition.

Write $\mathcal{T}$ for the category of triples and $P : \mathcal{T} \to \mathrm{Cat} \times \mathrm{Cat}$ for $(X, U, T) \mapsto (U, T)$. Let $b = (f_{U'}, f_{T'}) : (U', T') \to (U, T)$ and let X' be the collage of b^*H. For any triple $Y = (Y, V, S)$ the correct bijection is

$$\mathrm{Hom}_{\mathcal{T}}(Y, X') \simeq \mathrm{Hom}_{\mathcal{T}}(Y, X) \underset{\mathrm{Hom}(V,U)\times\mathrm{Hom}(S,T)}{\times} \big(\mathrm{Hom}(V, U') \times \mathrm{Hom}(S, T')\big).$$

On the right one chooses a morphism $g : Y \to X$ and functors $a : V \to U'$, $c : S \to T'$ such that $g|_V = f_{U'}a$ and $g|_S = f_{T'}c$. The map on mixed hom sets of g then has type

$$H_Y \longrightarrow H(f_{U'}a-, f_{T'}c-) = (b^*H)(a-, c-).$$

It defines a unique lift $Y \to X'$, and restriction of a lift recovers exactly these data. Thus $X' \to X$ is a cartesian arrow of P, not an object representing arbitrary pairs of functors to U and T.

The same argument works on natural transformations. A transformation between two lifts is a pair of transformations on V, S satisfying the naturality equations for every mixed arrow, and these are precisely the equations in the corresponding strict pullback of functor categories. This is the correctly typed version of $(**)$. One may express the statement in the language of a split fibration of gluing data, using the literal restriction operation on bifunctors; the corresponding assertion for unlabelled collages is invariant under the canonical isomorphisms.

The missing datum cannot be supplied by merely inserting primes on the right of the printed formulas. Take $U = T = U' = T' = e$, both change-of-base functors the identity, and $H = \{a, b\}$. Then $X' = X$ is the category with two parallel arrows from the open object to the closed object. There are two part-preserving functors from the walking arrow to X', whereas there is only one pair of functors from its two singleton parts to U', T'. The choice of crossing arrow is precisely the information lost by the printed right-hand side. The following observation about full faithfulness is unaffected: on the two diagonal sorts of hom sets it follows from $f_{U'}, f_{T'}$, and on mixed hom sets it follows from the defining equality $H' = b^*H$.

sous-topos fermé complémentaire, est toute similaire (**NB** Si $\mathcal{X} = \mathrm{Top}(X)$, avec X dans Cat, alors $\mathrm{Ouv}(\mathcal{X}) \simeq \mathrm{Ouv}(X)$, puisque les ouverts correspondent aux sous-faisceaux du

faisceau final sur $\mathcal{X}$, i.e. du préfaisceau final sur X. Cette bijection est compatible avec le passage aux complémentaires.) La théorie des topos dit que les couples $(\mathcal{X},\mathcal{U},\mathcal{T})$ `[plutôt triples]` se reconstituent, à équivalence définie à isomorphisme unique près, à l'aide d'un triple

$$\begin{cases}\mathcal{U},\ \mathcal{T} \text{ deux topos}\\ \varphi : \mathrm{Fais}\,(\mathcal{U}) \longrightarrow \mathcal{T} \text{ un foncteur exact à gauche ('accessible')},\end{cases}$$

ce foncteur φ étant obtenu, quand on part de

$$\mathcal{U} \xrightarrow{i} \mathcal{X} \xleftarrow{j} \mathcal{T},$$

comme

$$\varphi = j^* i_*.$$

De plus, le triple $(\mathcal{U},\mathcal{T},\varphi)$ peut être choisi arbitrairement, avec la seule restriction (pratiquement anodine, car toujours satisfaite pour tous les foncteurs naturels qu'on rencontre) que φ soit (en plus d'être exact à gauche) 'accessible'.

Dans le cas actuel $\mathcal{X} = \mathrm{Top}(X)$, $\mathcal{U} = \mathrm{Top}(U)$, $\mathcal{T} = \mathrm{Top}(T)$, le foncteur $\varphi = j^* i_*$ est non seulement exact à gauche, mais même il commute aux $\varinjlim$ quelconques, puisqu'il en est ainsi (toujours) de i_*, et aussi (parce que le j topossique est induit par le $j : T \longrightarrow X$ catégorique) de j^*. En fait, on trouve ceci : p. 46 [page 63]

Commentary · Context [ms. 62]

The presheaf case is already present in SGA 4

Grothendieck's comparison with toposes recovers a particularly exact antecedent. In Artin et al. (1972, Exposé IV, Theorem 9.5.4), an open subtopos and its closed complement reconstruct the ambient topos through the accessible left exact gluing functor. Exercise 9.5.10 of the same exposé then treats presheaf toposes: its parts (a)–(c) identify opens with sieves, introduce the cross-hom bifunctor, and characterize the presheaf case by preservation of products by the gluing functor. Thus Proposition 2 is a return to a construction already formulated there, now serving a calculation of categorical pushouts and their homotopy properties. The letters i, j are interchanged between the SGA convention and the present pages.

The category of sheaves in the glued topos is the category of triples

$$(F, G, \theta), \qquad F \in \mathcal{U}, \quad G \in \mathcal{T}, \quad \theta : G \longrightarrow \varphi(F).$$

The comparison arrow records how a section on the closed side approaches the open side. For $X = \mathrm{Coll}(H)$, a presheaf E on X restricts to $F = E|_U$ and $G = E|_T$. Its values on the mixed arrows are maps

$$G(t) \times H(u,t) \longrightarrow F(u)$$

natural in u, t. Transposing in the first variable produces $G(t) \to \mathrm{Hom}_{U^\wedge}(H(-,t), F)$, exactly the required gluing arrow. Conversely these maps reconstruct the restriction maps of E on all arrows; their naturality supplies the mixed composition laws. This directly explains why the ordinary categorical collage and the topos gluing construction agree. The comparison is not an analogy between two shapes of diagrams: both constructions encode the same mixed restriction maps.

Proposition 2 : *Considérons une situation de recollement de deux topos $\mathcal{U}$, $\mathcal{T}$ par $\varphi : \mathrm{Fais}\,\mathcal{U} \longrightarrow \mathrm{Fais}\,\mathcal{T}$, foncteur exact à gauche. Supposons que $\mathcal{U} = \mathrm{Top}(U)$, $\mathcal{T} = \mathrm{Top}(T)$, avec U et T dans* Cat. *Pour que le topos recollé $\mathcal{X}$ soit équivalent à un topos* $\mathrm{Top}(X)$, *avec X dans* Cat, *il faut et il suffit que le foncteur de recollement φ commute aux $\varinjlim$ quelconques, ou ce qui revient au même, qu'il ait un adjoint à gauche :*

$$\psi : \underbrace{\mathrm{Fais}\,\mathcal{T}}_{=T^\wedge} \longrightarrow \underbrace{\mathrm{Fais}\,\mathcal{U}}_{=U^\wedge}.$$

On s'attend dès lors que la donnée de recollement φ, ou ce qui revient au même le foncteur ψ ayant un adjoint à droite, i.e. commutant aux $\varinjlim$ quelconques, revient à celle d'une donnée de recollement H

$$H : U^\circ \times T \longrightarrow \mathrm{Ens}$$

pour U et T. Mais on sait bien que les foncteurs $\psi : T^\wedge \longrightarrow \mathcal{C}$ dans une catégorie $\mathcal{C}$ avec $\varinjlim$ quelconques, qui commutent auxdites $\varinjlim$, forment une catégorie équivalente (par le foncteur $\psi \longmapsto \psi|T(\subseteq T^\wedge)$) à celle des foncteurs

$$\psi_0 : T \longrightarrow \mathcal{C} \quad (= \underbrace{U^\wedge}_{\underline{\mathrm{Hom}}(U^\circ,\mathrm{Ens})} \text{ en l'occurrence}),$$

et la donnée d'un tel foncteur ψ_0 revient bien à celle d'un H. Donc [page 64]

Corollaire. *La catégorie des données de recollement topossique $\varphi : U^\wedge \longrightarrow T^\wedge$ entre* $\mathrm{Top}(U)$, $\mathrm{Top}(T)$*, telles que le topos recollé soit un 'topos* paradisiaque*', i.e. équivalent à un topos de la forme $X^\wedge$ (avec X dans* Cat*), est équivalente à celle des données de recollement $H : U^\circ \times T \longrightarrow$ Ens entre U et T. La donnée topossique φ associée à H est le foncteur adjoint à droite de $T^\wedge \longrightarrow U^\wedge$ déduit de $T \xrightarrow{H} U^\wedge$ par prolongement commutant aux $\varinjlim$, ou encore elle est donnée par*

$$\varphi(\underbrace{F}_{\in U^\wedge})(t) \simeq \mathrm{Hom}_{U^\wedge}(H(t), F),$$

où dans le second membre H est interprété comme un foncteur $T \longrightarrow U^\wedge$. Le topos recollé par φ est canoniquement équivalent à $\mathrm{Top}(X)$*, où X est l'objet de* Cat *obtenu par recollement à l'aide de H.*

Du point de vue de la théorie des faisceaux, cela donne l'interprétation de H, comme exprimant le foncteur

$$\varphi = j^* i_* : U^\wedge \longrightarrow T^\wedge$$

par la formule précédente.

p. 47

Commentary · Exposition [ms. 64] [page 65]

The adjunction which characterizes the presheaf gluing

The extra preservation of limits isolates those gluing functors that come from actual sets of arrows. Let $H_t = H(-, t) \in U^\wedge$. Its colimit-preserving extension from the Yoneda image of T is

$$\psi(G) = \int^{t \in T} G(t) \times H_t, \qquad \varphi(F)(t) = \mathrm{Hom}_{U^\wedge}(H_t, F).$$

Here the coend is an ordinary colimit in $U^\wedge$. A map $\psi(G) \to F$ is a family of maps $G(t) \times H_t \to F$ compatible with the arrows of T. By the ordinary set-valued adjunction between product and hom, this is exactly a natural transformation $G \to \varphi(F)$. This proves $\psi \dashv \varphi$ and the displayed formula of the corollary. It also proves that φ preserves all small limits.

Conversely, suppose that an accessible gluing functor $\varphi : U^\wedge \to T^\wedge$ preserves all small limits. The adjoint functor theorem for locally presentable categories supplies a left adjoint ψ. Set $H_t = \psi(y_T t)$. Every presheaf is the colimit of its representables indexed by its category of elements; since ψ is a left adjoint, this determines ψ from H. The preceding adjunction then identifies its right adjoint with φ. Applying the explicit construction of the preceding block identifies the glued topos with $\mathrm{Coll}(H)^\wedge$. This proves the sufficient direction of Proposition 2 with the size assumption made explicit. The accessibility assumption belongs to the ambient gluing problem; it should not be silently discarded in an abstract claim that every limit-preserving functor has a left adjoint.

For the converse direction of the proposition, let the glued topos be equivalent to $C^\wedge$. Its specified open subtopos corresponds under this equivalence to a subterminal presheaf. Such a presheaf takes the value either $\varnothing$ or a singleton at each object, and its support is a sieve $C_U \subset C$. The complementary full cosieve C_T presents the closed complement. Restriction to C_T preserves all limits, as does the right adjoint to restriction to C_U. Their composite is the transported gluing functor, so φ preserves all limits. The argument uses the specified open and closed pieces but does

not require the given presentations U, T to coincide literally with C_U, C_T. Equivalences of presheaf categories preserve the limiting operations involved. This is precisely the content of the presheaf case of the SGA exercise just cited.

There is a variance convention in the word "category" in the corollary. For fixed U, T, a transformation $H \to K$ induces $\psi_H \to \psi_K$, but it induces a transformation $\varphi_K \to \varphi_H$ on the right adjoints. Thus the asserted equivalence of categories of gluing data uses reversed transformations on the right-adjoint side; with ordinary pointwise transformations it is an antiequivalence. The object classification and every displayed objectwise formula remain unchanged.

A useful test is $U = T = e$ and $H = S$ a set. The collage has two objects and one crossing arrow for each element of S. The functors are $\psi(G) = G \times S$ and $\varphi(F) = F^S$. A presheaf on the collage is a pair of sets F, G with an S-indexed family of maps $G \to F$. If $S = \varnothing$, there is no compatibility datum and the presheaf topos is the product of two copies of Ens. If S is a singleton, it is the category of arrows of sets. These limiting cases also verify the direction $G \to \varphi(F)$ of the gluing arrow.

1.3 Flèches dans Cat, de source ou but un objet recollé (X, U, T).

Quand (X, U, T) est donné par une donnée de recollement

$$U,\ T,\ H : U^\circ \times T \longrightarrow \mathrm{Ens},$$

les flèches

$$f : X' \longrightarrow X$$

dans Cat se décrivent immédiatement ainsi : la donnée de f définit d'abord

$$U' = f^{-1}(U), \qquad T' = f^{-1}(T),$$

couple d'un ouvert de X' et du fermé complémentaire, de sorte que cette donnée équivaut à celle de

1°) un couple (U', T') dans X'

2°) un morphisme (dans $\mathcal{T}$) de (X', U', T') dans (X, U, T), ce qui équivaut par 1.1 à un triple

$$\begin{cases} f_{U'} : & U' & \longrightarrow & U \\ f_{T'} : & T' & \longrightarrow & T \\ u : & H_{U',T'} & \longrightarrow & H \circ (f_{U'}^\circ \times f_{T'}), \end{cases}$$

où on a posé

$$\begin{cases} H_{U',T'} : U'^\circ \times T' \longrightarrow \mathrm{Ens} \\ H_{U',T'}(u', t') = \mathrm{Hom}_{X'}(u', t'), \end{cases}$$

i.e. $H_{U',T'}$ est la donnée de recollement qui conrrespond au triple (X', U', T') dans $\mathcal{T}$.

Nous serons plus intéressés par la situation inverse, i.e. par la question de déterminer les foncteurs [page 66]

$$\underbrace{X}_{=X_{U,T,H}} \longrightarrow X'$$

de X, obtenu par (U, T, H), dans un objet quelconque de Cat. On trouve tautologiquement

Proposition 3. $\mathrm{Hom}_{\mathrm{Cat}}(X, X')$ *est en correspondance bijective canonique, fonctorielle en* X' *dans* Cat*, avec l'ensemble des triples*

p. 48

[(∗)]
$$\begin{cases} \left.\begin{array}{l} f_U : U \longrightarrow X' \\ f_T : T \longrightarrow X' \end{array}\right\} \text{flèches dans Cat} \\ u : H \longrightarrow H_{X'}(f_U^\circ \times f_T),\ \text{flèche dans } \underline{\mathrm{Hom}}(U^\circ \times T, \mathrm{Ens}), \end{cases}$$

où pour tout X' dans Cat*, on a posé*

$$H_{X'} = ((x,y) \longmapsto \mathrm{Hom}_{X'}(x,y)) : X'^{\circ} \times X' \longrightarrow \mathrm{Ens}.$$

Sûrement on aura même un isomorphisme de catégories

$$\underline{\mathrm{Hom}}(X, X') \simeq \text{catégorie des triples } (f_U, f_T, u) \text{ comme dans } (*).$$

1.4 Construction de sommes amalgamées à partir d'une immersion ouverte où fermée.

Pour une construction d'une sorte de somme amalgamée, en termes d'un triple (X, U, T) + …, qui soit symétrique en U, T, j'ai envie de regarder d'abord la situation 'mixte', d'un diagramme en traits pleins [page 67]

$$(*) \qquad \begin{array}{ccccc} U & \xhookrightarrow{i} & X & \xhookleftarrow{j} & T \\ {\scriptstyle f_U}\downarrow & & \vdots\, {\scriptstyle f} & & \downarrow{\scriptstyle f_T} \\ U' & \xhookrightarrow{i'} & X' & \xhookleftarrow{j'} & T', \end{array}$$

qu'on aimerait compléter 'canoniquement', si faire se peut, en un morphisme de triples (i.e. en une flèche dans $\mathcal{T}$). L'idée naturelle, c'est que le X' qui s'y insère serait la limite inductive, dans Cat, du diagramme initial. Si $T' = T$, $f_T = \mathrm{id}_T$, cela signifie donc que X' s'identifie à la somme amalgamée $X \amalg_U U'$, et inversement si $U' = U$, $f_U = \mathrm{id}_U$.

Commentary · Exposition [ms. 66]

Functors out of a collage include the mixed naturality equation

The next construction will be recognized by mapping out of it. Proposition 3 is therefore the correct place to retain natural transformations, even though its first formulation only mentions sets of functors. A functor $\mathrm{Coll}(H) \to Y$ consists of $a : U \to Y$, $b : T \to Y$ and maps

$$\eta_{u,t} : H(u,t) \longrightarrow \mathrm{Hom}_Y(a(u), b(t))$$

compatible with precomposition in U and postcomposition in T. The argument constructing the collage proves both existence and uniqueness of the functor from these data.

A morphism from (a, b, η) to $(\bar{a}, \bar{b}, \bar{\eta})$ is a pair of natural transformations $r : a \Rightarrow \bar{a}$, $s : b \Rightarrow \bar{b}$ satisfying, for every $h \in H(u,t)$,

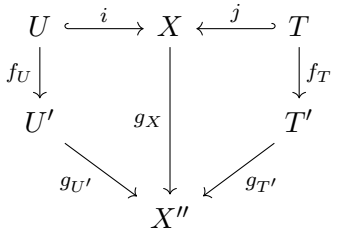
$$s_t \, \eta_{u,t}(h) = \bar{\eta}_{u,t}(h) \, r_u.$$

The components r_u and s_t define a natural transformation on the collage because its naturality is already known on arrows within either part, while this equation is its naturality on a crossing arrow. Conversely restriction recovers the pair and the equation. Composition and identities are componentwise. This gives the claimed isomorphism of functor categories, rather than merely an equivalence on their objects.

For a functor into a collage the procedure is different in presentation but just as exact. Pulling back the open part specifies a sieve in the source, and its complement specifies the cosieve; these are forced by the functor. The residual data are then a morphism of profunctors. This is why Proposition 3 does not require the images of a and b to be disjoint: the target is arbitrary, and a functor out of the collage may identify objects from the two parts. The universal property of the forthcoming ordinary pushout must allow precisely these functors.

Désignons par X' la somme amalgamée `[plutôt la limite inductive]`, et explicitons, pour X'' quelconque dans Cat, l'ensemble des flèches g de X' dans X'', comme foncteur en X''. C'est l'ensemble des triples

$$(*) \qquad (g_X, g_{U'}, g_{T'}) \qquad \begin{cases} g_X : X \longrightarrow X'' \\ g_{U'} : U' \longrightarrow X'' \\ g_{T'} : T' \longrightarrow X'' \end{cases} \quad \text{avec} \quad \begin{array}{l} g_X \circ i = g_{U'} f_U \\ g_X \circ j = g_{T'} f_T. \end{array} \qquad \text{p. 49}$$



Notons que pour un tel triple, g_X est partiellement connu déjà en termes de $g_{U'}$, $g_{T'}$, par ses restrictions $g_X \circ i$ et $g_X \circ j$ à U et T respectivement, par les formules

$$g_U = g_{U'} f_U, \qquad g_T = g_{T'} f_T.$$

Donc la donnée d'un triple $(*)$ équivaut à celle d'un triple [page 68]

$$(g_{U'}, g_{T'}, g_X) \qquad \begin{cases} g_X \text{ une flèche } X \longrightarrow X'' \text{ correspondant à} \\ g_U = g_{U'} \circ f_U \text{ et } g_T = g_{T'} \circ f_T \text{ déjà donnés (via } g_{U'},\ g_{T'}), \end{cases}$$

donc en vertu de 1.3. cela correspond à

$$(*) \qquad (g_{U'}, g_{T'}, u) \qquad \begin{cases} g_{U'} : U' \longrightarrow X'' \\ g_{T'} : T' \longrightarrow X'' \\ u : H \longrightarrow \underbrace{H_{X''}\left((g_{U'} \circ f_U)^\circ \times (g_{T'} \circ f_T)\right)}_{(u,t) \longmapsto \mathrm{Hom}_{X''}(g_{U'} f_U(u), g_{T'} f_T(t))}. \end{cases}$$

On va interpréter la donnée de u, en termes d'un foncteur

$$(**) \qquad F^* = (H' \longmapsto H' \circ F) : \underline{\mathrm{Hom}}(U'^\circ \times T', \mathrm{Ens}) \longrightarrow \underline{\mathrm{Hom}}(U^\circ \times T, \mathrm{Ens})$$

où

$$F : U^\circ \times T \longrightarrow U'^\circ \times T' \qquad \text{est donné par } F = f_U^\circ \times f_T.$$

On a un objet $H'' = H''_{g_{U'}, g_{T'}}$ de la catégorie source de $(**)$ déterminé en termes du couple $(g_{U'}, g_{T'})$, par

$$H''(u', t') = \mathrm{Hom}_{X''}(g_{U'}(u'), g_{T'}(t')),$$

et le bifoncteur but de la flèche u dans $(*)$ s'interprète comme

$$F^*(H'') = H'' \circ F : (u,t) \longmapsto \underbrace{H''(f_U(u), f_T(t))}_{= \mathrm{Hom}(g_{U'} f_U(u), g_{T'} f_T(t))}.$$

Ainsi, u s'interprète comme une flèche

$$u : H \longrightarrow F^*(H'').$$

Or le foncteur F^* a un adjoint à gauche $F_!$, i.e. on a

$$(***) \qquad \mathrm{Hom}(H, F^*(H'')) \simeq \mathrm{Hom}(F_!(H), H'').$$

avec

$$F_! : \underline{\mathrm{Hom}}(U^\circ \times T, \mathrm{Ens}) \longrightarrow \underline{\mathrm{Hom}}(U'^\circ \times T', \mathrm{Ens}).$$

Posons

$$H' = F_!(H) : U'^\circ \times T' \longrightarrow \mathrm{Ens},$$

c'est un bifoncteur bien déterminée par la seule donnée du diagramme initial $(*)$ p. 67, ou ce qui revient au même, en termes de [p. 50] [page 69]

$$(U, T, H, f_U : U \longrightarrow U', f_T : T \longrightarrow T').$$

Commentary · Exposition [ms. 68]

Transporting the crossing arrows by a Kan extension

The two categories U, T can now be changed independently. Their new internal arrows are already present in U', T'. What must be supplied are all composites crossing from the first new category to the second, subject to the relations forced by old compositions. This is what the left Kan extension supplies.

Put $F = f_U^{\mathrm{op}} \times f_T$. The ordinary pointwise formula can be written in a form that displays every generator:

$$H'(u', t') = \int^{u \in U,\, t \in T} \mathrm{Hom}_{U'}(u', f_U u) \times H(u, t) \times \mathrm{Hom}_{T'}(f_T t, t').$$

The notation means the quotient of the disjoint union of these products by the relations for composition in U and T. An element is represented by a triple

$$u' \xrightarrow{\alpha} f_U u, \qquad h \in H(u, t), \qquad f_T t \xrightarrow{\beta} t'.$$

For $a : u_0 \to u_1$ in U and $b : t_0 \to t_1$ in T, the defining relations are

$$[f_U(a)\alpha, h, \beta] = [\alpha, H(a, 1)h, \beta], \qquad [\alpha, H(1, b)h, \beta] = [\alpha, h, \beta f_T(b)],$$

whenever the displayed terms are typed. This is the coend formula for $\mathrm{Lan}_F H$; the equivalent comma-category formula is the pointwise Kan-extension theorem (Riehl, n.d., Theorem 6.2.1).

To prove the pushout assertion directly, form $C = \mathrm{Coll}(H')$ and send a crossing arrow h of X to $[1, h, 1]$. The relations make this a functor $X \to C$, and the inclusions of U', T' agree with it on U, T. Now take compatible functors $X \to Y$, $U' \to Y$, $T' \to Y$. On the generator above there is only one possible extension: it must take the value $\beta\, h\, \alpha$, after applying the respective functors. The two relations hold because the original functors preserve composition. Thus there is a unique functor $C \to Y$ inducing all three given functors. This proves the strict colimit universal property, and also identifies its canonical map from X.

Since C was constructed as a collage, U' is a full sieve and T' its full complementary cosieve. Their internal hom sets have not been enlarged by the quotient: no string can leave the open part and subsequently return to it. This supplies the structural content of Theorem 1 beyond existence of a colimit in Cat. It also clarifies the distinction from the earlier induced triple: restriction of H gave a cartesian lift, whereas this left Kan extension gives the universal forward transport. The labels "pullback" and "pushout" refer to these different universal properties, not to a formal substitution of primes in the same formula.

Et on voit que la donnée d'une flèche $X' \longrightarrow X''$, X'' objet variable de Cat, équivaut à celle de

$$g_{U'} : U' \longrightarrow X'',\ g_{T'} : T' \longrightarrow X'',\ u : H' \longrightarrow H'' = H_{X''}(g_{U'}^{\circ} \times g_{T'}),$$

c'est à dire, en vertu de 1.3., à celle d'une flèche

$$X_0' \longrightarrow X'',$$

où X_0' désigne l'objet de Cat déduit de U', T' par recollement au moyen de H'. Cela montre donc que la limite inductive cherchée n'est autre que X_0'. On a ainsi trouvé le

Théorème 1. *La limite inductive X' dans* Cat *du diagramme $(*)$ page 67, défini en termes d'un triple (X, U, T) dans $\mathcal{T}$ et de flèches $f_U : U \longrightarrow U'$, $f_T : T \longrightarrow T'$ dans* Cat*, existe (sans mérite) et les flèches canoniques* [page 70]

$$U' \xhookrightarrow{i'} X' \xhookleftarrow{j'} T'$$

sont des immersions ouvertes et fermées complémentaires l'une de l'autre. La donnée de recollement qui définit X' via U', T' est donnée par

$$H' = (\underbrace{f_U^{\circ} \times f_T}_{F})_!(H) : U'^{\circ} \times T' \longrightarrow \mathrm{Ens},$$

où H est la donnée de recollement qui définit X, i.e. $H : U^\circ \times T \longrightarrow$ Ens est le bifoncteur $(u,t) \longmapsto \mathrm{Hom}_X(u,t)$.

Ainsi on a, pour u' dans U' et t' dans T'

$$\mathrm{Hom}_{X'}(u',t') = F_!(H)(u',t'),$$

et il reste à expliciter $F_!(H)$. Ici $F = f_U^\circ \times f_T$ est un foncteur covariant sur $U^\circ \times T$, on peut l'interpréter comme un préfaisceau sur $U \times T^\circ$, et écrire dès lors

$$(*) \qquad \underbrace{F_!(H)(u',t')}_{\mathrm{Hom}_{X'}(u',t')} = \varinjlim_{(u',t')\backslash U\times T^\circ} \underbrace{\mathrm{Hom}_X(u,t)}_{H(u,t)} .$$

La catégorie $(u',t')\backslash U \times T^\circ$ sur laquelle on prend la $\varinjlim$ s'explicite comme un produit p. 51

$$(u',t')\backslash U \times T^\circ \simeq (u'\backslash U) \times (T/t')^\circ.$$

Les applications

$$\mathrm{Hom}_X(u,t) \longrightarrow \mathrm{Hom}_{X'}(u',t') \qquad (= F_!(H)(u',t'))$$

déduites de cette formule ne sont autres que les suivantes : u, t sont données comme objets [page 71]
de $u'\backslash U$, T/t', i.e. avec des flèches

$$u' \xrightarrow{\alpha} f_U(u), \qquad f_T(t) \xrightarrow{\beta} t',$$

d'où

$$\mathrm{Hom}_X(u,t) \xrightarrow{\text{via } f=f_X:X\to X'} \mathrm{Hom}_{X'}(\underbrace{f(u)}_{f_U(u)}, \underbrace{f(t)}_{f_T(t)}) \xrightarrow{\text{via } \alpha,\ \beta} \mathrm{Hom}_{X'}(u',t').$$

Notons pour mémoire que pour tout objet Z de Cat, et tout préfaisceau G sur Z, on a la formule

$$\varinjlim_Z G \simeq \pi_0(Z/G),$$

où Z/G s'interprète en termes de l'immersion canonique $Z \longrightarrow Z^\wedge$ ($G \in \mathrm{Ob}\, Z^\wedge$). C'est donc une catégorie fibrée sur Z à fibres discrètes, et on a en fait un isomorphisme entre la sous-catégorie pleine de Cat/Z formée de telles catégories fibrées à fibres discrètes, avec $Z^\wedge$. Dans ce dictionnaire parfait entre préfaisceaux d'ensembles et catégories fibrées à fibres discrètes, le foncteur $\varinjlim$ sur $Z^\wedge$ correspond simplement au foncteur π_0 sur la catégorie des objets au dessus de Z dans Cat, lui-même induit par le foncteur (indépendant de Z)

$$\mathrm{Cat} \xrightarrow{\pi_0} \mathrm{Ens}.$$

On peut donc réécrire la formule fondamentale (∗) p. 70, en termes du foncteur H [page 72]
définissant le recollement X de U et de T, par

$$(*) \qquad \boxed{\underbrace{\mathrm{Hom}_{X'}(u',t')}_{H'(u',t')} \simeq \pi_0(\underbrace{(u',t')\backslash U \times T^\circ/H}_{((u'\backslash U)\times(T/t')^\circ)/H}).}$$

Par la suite, nous n'aurons sans doute qu'à utiliser le cas où on a soit $T' = T$, $f_T = \mathrm{id}_T$, soit l'inverse. Nous travaillerons surtout dans le premier cas, en déduisant ce dont nous aurons besoin pour le cas dual, par principe de dualité (automorphisme $X \longmapsto X^\circ$ de la catégorie p. 52
Cat). Dans ce cas, dans la formule fondamentale (∗) l'élément t' de T' s'écrit comme un élément t de T, et la formule se retranscrit comme

$$(**) \qquad \boxed{\mathrm{Hom}_{X'}(u',t) \simeq \pi_0(\ \underbrace{u'\backslash U/t}_{\text{catégorie des triples } (u,\alpha,\beta'),\ \begin{cases} u \in \mathrm{Ob}\, U \\ \alpha : u \longrightarrow t \\ \beta' : u' \longrightarrow f_U(u) \end{cases}}\ ).}$$

C'est la formule que j'avais trouvée directement dans ce cas, sans me soucier encore de faire varier T en même temps que U. Je voudrais faire le lien entre les deux formules $(*)$ et $(**)$, par voie topologique, donc établir une bijection canonique, fonctorielle etc.

$$\pi_0(u'\backslash U/t) \xrightarrow{\sim} \pi_0(((u'\backslash U) \times (T/t)^\circ)/H).$$

On a sûrement un foncteur [page 73]

$$u'\backslash U/t \longrightarrow ((u'\backslash U) \times (T/t)^\circ)/H$$

qui associe à un triple

$$(u, \alpha, \beta') : u \in \mathrm{Ob}\, U,\ \alpha : u \longrightarrow t,\ \beta' : u' \longrightarrow f_U(u)$$

le triple $(\bar{u}, \bar{t}, \xi)$ $(\xi \in H(\bar{u}, \bar{t}))$ dans le second membre, avec $\bar{u}$ dans $u'\backslash U$, $\bar{t}$ dans T/t, défini ainsi :

$$\bar{u} = (u, \beta'),\ \bar{t} = (t, \mathrm{id}_t), \underbrace{\xi}_{=\alpha} \in H(\bar{u}, \bar{t}) = \mathrm{Hom}(u, t).$$

En somme

$$u'\backslash U/t = \text{fibre de } ((u'\backslash U) \times (T/t)^\circ)/H \text{ en l'objet initial } \bar{t} \text{ de } (T/t)^\circ$$

par la projection naturelle, flèche composée

$$\begin{array}{c} ((u'\backslash U) \times (T/t)^\circ)/H \\ \big\downarrow \\ (u'\backslash U) \times (T/t)^\circ \\ \big\downarrow \pi \\ (T/t)^\circ. \end{array}$$

Commentary · Exposition [ms. 72]

Why connected components give the ordinary hom set

Grothendieck wants a topological description of a purely ordinary quotient. For a presheaf $G : Z^{\mathrm{op}} \to \mathrm{Ens}$, the category of elements Z/G has objects (z, g) and arrows

$$(z, g) \longrightarrow (z', g') \quad \text{given by} \quad a : z \longrightarrow z', \qquad G(a)(g') = g.$$

The colimit of G is the disjoint union of its values modulo the identifications $G(a)(g') \sim g'$. These are exactly the identifications imposed on objects when one passes to connected components of Z/G. Hence

$$\operatorname*{colim}_{Z^{\mathrm{op}}} G \simeq \pi_0(Z/G).$$

The source writes $\varinjlim_Z G$ for this presheaf colimit. The opposite belongs in the modern indexing category; it is not a correction to the numerical value of the formula.

The same convention explains the indexing category in the preceding page. If H is regarded as a covariant functor on $U^{\mathrm{op}} \times T$, its ordinary left Kan extension is indexed by

$$F/(u', t') \simeq (u'\backslash f_U)^{\mathrm{op}} \times (f_T/t').$$

Equivalently, regard H as a presheaf on $U \times T^{\mathrm{op}}$ and take its presheaf colimit over the opposite category

$$(u'\backslash f_U) \times (f_T/t')^{\mathrm{op}}.$$

This is the convention used in the edition. Its sentence "Ici $F = \cdots$ ... on peut l'interpréter comme un préfaisceau" contains a harmless but real letter slip: it is H, not the indexing functor F, that is the set-valued presheaf.

When $T' = T$, the resulting objects can be compressed to factorizations

$$(u, \alpha : u' \to p(u), \gamma : u \to t).$$

An arrow from (u, α, γ) to (v, α', γ') is $a : u \to v$ with $p(a)\alpha = \alpha'$ and $\gamma' a = \gamma$. Its connected component is the crossing arrow of the pushout represented by $\gamma\alpha$. Thus the factorization category keeps track of all presentations of a new arrow and all relations between those presentations. Taking π_0 intentionally forgets its higher homotopy. The later asphericity conditions ask precisely when this forgetting is harmless for the homotopy problem, and do not follow from the existence of the ordinary colimit alone.

Il faut voir que l'inclusion de la fibre donne une bijection sur les π_0, ou ce qui revient au même, que l'inclusion de la fibre $u'\backslash U$ *de* π en l'objet initial (t, id_t) de $(T/t)^\circ$ donne une bijection sur les $\varinjlim$ du foncteur H. Donc l'inclusion a intérêt à être asphérique, ou du moins co-cofinalle.

Mais l'inclusion d'un objet initial est toujours asphérique, et le reste par produit avec un objet quelconque (i.e. $u'\backslash U$) de Cat (et même par tout changement de base lisse …), d'où la conclusion. p. 53 [page 74]

Dans la suite, nous nous intéressons surtout à la situation de sommes amalgamées (où on change de notation un peu, à présent)

$$(*) \qquad \begin{array}{ccccc} U & \xhookrightarrow{i} & X & \xleftarrow{j} & T \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots \\ U' & \xhookrightarrow{i'} & X' & \leftarrow & T' \simeq T. \end{array}$$

Le fermé T' complémentaire de U' dans X' est alors canoniquement isomorphe au fermé T complémentaire de U dans X, l'isomorphisme étant induit par q. Souvent nous identifierons T' et T, ce qui est licite en regardant X' comme déduit du recollement de U' et de t `[plutôt T]`, par le bifoncteur $U'^{\circ} \times T \longrightarrow$ Ens explicité dans la formule $(**)$ p. 72. Nous nous intéressons surtout à dégager des conditions qui assurent les deux implications suivantes

a) $i \in W \Longrightarrow i' \in W$ (et même $\Longleftrightarrow$).

b) $p \in W \Longrightarrow q \in W$ (et même $\Longleftrightarrow$).

Proposition 4. *Soit $t \in \mathrm{Ob}\, T$. Pour que $t \in \mathrm{Ob}\, \bar{U}$, il faut et il suffit que $q(t) \in \mathrm{Ob}\, \overline{U'}$. En d'autres termes, le 'bord' $\dot{T} = T \cap \bar{U}$ de T dans X est 'le même' que son bord dans X'.* [page 75]

Commentary · Exposition [ms. 74]

The one-sided formula has an explicit deformation retraction

The comparison of the two factorization formulas admits a direct argument which also verifies Grothendieck's cofinality claim with its variance intact. Write E for the category of elements occurring in the general formula when $T' = T$. Its objects are

$$(u, \alpha : u' \to p(u), v, \beta : v \to t, h : u \to v).$$

An arrow to $(u_1, \alpha_1, v_1, \beta_1, h_1)$ consists of

$$a : u \to u_1, \quad b : v_1 \to v, \qquad p(a)\alpha = \alpha_1, \quad \beta b = \beta_1, \quad h = bh_1 a.$$

Let $D = u'\backslash U/t$ be the category in the boxed one-sided formula. Define

$$R : E \longrightarrow D, \qquad R(u, \alpha, v, \beta, h) = (u, \alpha, \beta h).$$

On arrows R takes (a, b) to a; the equality $\beta_1 h_1 a = \beta b h_1 a = \beta h$ proves that this is an arrow of D. The inclusion $J : D \to E$ takes (u, α, γ) to $(u, \alpha, t, 1_t, \gamma)$, and $RJ = 1_D$.

For each object of E, the pair $(1_u, \beta)$ gives an arrow

$$JR(u, \alpha, v, \beta, h) \longrightarrow (u, \alpha, v, \beta, h).$$

The equations above verify naturality, so $JR \Rightarrow 1_E$. Consequently J and R induce inverse homotopy equivalences on nerves, and in particular inverse bijections on π_0. This proves more than the displayed comparison requires. The construction is natural in u' and t, because precomposition of α and postcomposition of β commute with βh.

The reference to the initial object $(t, 1_t)$ of $(T/t)^{\mathrm{op}}$ is therefore sound. An initial object is cofinal for the colimit of a *presheaf*; in covariant colimit notation one has passed to the opposite and it becomes terminal. The proof does not assert that an initial object computes arbitrary covariant colimits. This is the same variance discipline needed in the commentary on Chapter I, manuscript p. 47 ("Cofinality retains an orientation"), but here the explicit retraction identifies the whole category of presentations rather than only invoking the abstract cofinality criterion.

Démonstration. Dire que $t \in \mathrm{Ob}\,\bar{U}$ signifie aussi

$$U/t \neq \emptyset$$

et implique, par continuité, que $q(t) \in \overline{q(U)} \subseteq \overline{U'}$. Inversement, si $q(t) \in \overline{U'}$, i.e. s'il existe $u' \in U'$ tel que

$$\mathrm{Hom}(u', q(t))(= \pi_0(u'\backslash U/t)) \neq \emptyset,$$

il en résulte que $u'\backslash U/t \neq \emptyset$, a fortiori U/t (qui reçoit $u'\backslash U/t) \neq \emptyset$, q.e.d.

Proposition 5. *Soit*

$$\begin{array}{ccc} Y & \xrightarrow{i} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ Y' & \xrightarrow{i'} & X' \end{array}$$

un carré cocartésien dans Cat. *Si i est une immersion ouverte (resp. fermée), alors i' aussi, et le carré est aussi cartésien. Idem pour p et q. Si i et p sont des immersions ouvertes ([1])* p. 54 *(resp. …), alors la situation est isomorphe à celle d'un objet X' de* Cat, *de deux ouverts (resp. fermés) X, Y' de X ayant pour réunion X', pour intersection Y. Inversement, dans une telle situationm le carré correspondant est cocartésien, i.e. X* `[plutôt $X'$]` *s'identifie à la somme amalgamée*

$$X' \simeq X \amalg_Y Y'.$$

Donc aussi on a la proposition 15, page 158, et ses corollaires.

p. 55

2 Sommes amalgamées et W-équivalences.

2.1 Axiome du carré cocartésien et axiome du cube (ou 'Lemme des cinq').

Nous supposons à présent donné un localiseur fondamental

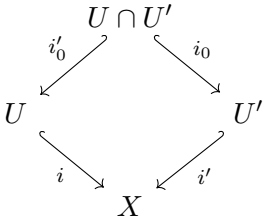
$$W \subseteq \mathrm{Fl}(\mathrm{Cat}),$$

donc W satisfait à W(1, 2 bis, 3). Nous ne supposons pas pour le moment pour les autres axiomes standard W(4) (axiome de fibration), W(5) (axiome de connexité), W(6) (axiome des limites inductives). Mais nous allons introduire à présent deux axiomes de stabilité supplémentaire, de nature différente, concernant les sommes amalgamées.

W(7 bis) (Axiome du carré cocartésien.) Soient X un objet de Cat, U, U' deux ouverts de X tels que $U \cup U' = X$, considérons le diagramme cocartésien correspondant (cf. prop. 5 en marge de la page `[ci-dessus]`) :

1. **NB** Nous utilisons *'immersion ouverte'*, *'immersion fermée'* au *sens strict* : c'est un *monomorphisme* pleinement fidèle dont l'image est une sous-catégorie *pleine,* soit ouverte, soit fermée. [page 76]

(∗)



Commentary · Examples [ms. 75]

Orbits of crossing arrows and the information discarded

The pushout formula can be seen in a small calculation before it is used homotopically. Let U be the discrete category on u_0, u_1, let $T = e$ with object t, and let $H(u_i, t)$ be a singleton. Thus X is the span $u_0 \to t \leftarrow u_1$. Collapse U to $U' = e$ by p. The category $u'\backslash U/t$ has exactly two objects and no nonidentity arrows, so the pushout $X' = X \amalg_U e$ has two objects and *two* distinct parallel arrows $u' \rightrightarrows t$. Identifying the two objects of U does not identify arrows which have no relation between them. Its nerve realizes to a circle, whereas the nerve of X is contractible. Here p is not a weak homotopy equivalence, so this is an illustration of the formula, not a counterexample to any asserted stability under a weak equivalence.

A group action makes the quotient and its lost homotopy particularly explicit. Put $U = BG$, let $T = e$, and take a right G-set S as the presheaf $H(-, t)$. On pushing BG to e, the mixed hom set becomes

$$\mathrm{Hom}_{X'}(u', t) = S/G.$$

Indeed the category of elements has objects $s \in S$ and an arrow labelled g from s to s' when $s = s'g$. All arrows are invertible, and two objects lie in the same connected component exactly when they lie in the same orbit. If $S = K\backslash G$ is a transitive right action, the automorphism group of the object K is K. Inclusion of that object is fully faithful and essentially surjective, hence the category of elements is equivalent to BK.

Thus a single crossing arrow in the quotient may stand for a factorization category with homotopy type BK. Choosing K nontrivial shows that a singleton ordinary colimit says nothing about the contractibility of its category of presentations. If $S = G$ with the free right action, the stabilizer is trivial and the category of elements is contractible. If S is the singleton with trivial action, it is BG. Both have the same orbit set. The distinction is exactly the one that motivates the total asphericity conditions later in this section.

Alors

$$i_0 \in W \qquad \Longleftrightarrow \qquad i \in W$$

(et symétriquement

$$i_0' \in W \qquad \Longleftrightarrow \qquad i' \in W).$$

NB Par dualité, cet axiome est équivalent à l'axiome similaire, avec des *fermés* T, T' de X de réunion X.

Corollaire. *Supposons* W(7 bis). *Considérons les conditions suivantes sur le carré précédent.*

a) $U \cap U'$ W-asphérique.

b) U, U' W-asphériques.

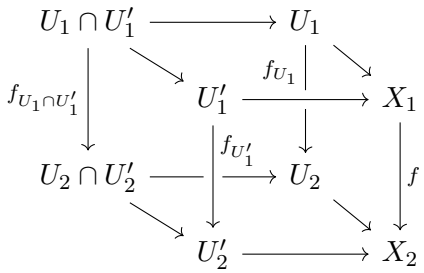
c) X W-asphérique.

Alors (a et b) implique c, et (b et c) implique a.

NB Moralement, il faudrait aussi que (c et a) implique b, i.e. si $U \cap U'$ et X sont W-asphériques, il en est de même de U et de U'. Mais il ne semble pas que l'axiome du carré cocartésien permette de le prouver. Par contre, l'axiome qui suit impliquera le corollaire précédent sous sa forme complète.

p. 56

W(7) ('Axiome du cube', ou 'Lemme des cinq'.) Soit un diagramme commutatif cubique



représentant un morphisme (flèches verticales) d'un

[page 77]

carré cocartésien $\begin{array}{ccc} U_1 \cap U_1' & \longrightarrow & U_1 \\ \downarrow & & \downarrow \\ U_1' & \longrightarrow & X_1 \end{array}$ du type envisagé dans W(7 bis) (X_1 réunion de deux ouverts U_1, U_1') dans un autre (à indices 2). Considérons les conditions suivantes.

Commentary · Exposition [ms. 76]

Ordinary covers yield homotopy pushouts, with one direction of excision

Grothendieck's new axioms try to turn a cover by two categorical opens into an excision principle. The underlying geometric fact is especially clean. Suppose $X = U \cup V$ with U, V full sieves and $A = U \cap V$. A simplex of NX is a composable chain $x_0 \to \cdots \to x_n$. Its last vertex lies in U or in V; since these are sieves, all earlier vertices then lie in the same one. Hence

$$NX = NU \cup_{NA} NV$$

is an actual pushout of simplicial sets. The two maps out of NA are monomorphisms. On geometric realization they are inclusions of CW subcomplexes, so this pushout also computes the homotopy pushout. For cosieves the same proof uses the first vertex of the chain.

This proves the direct implication of W(7 bis) for W_∞: if $A \to V$ is a weak equivalence then $U \to X$ is a weak equivalence. It also proves the direct cube assertion (a and b) $\Rightarrow c$: an objectwise weak equivalence of the two spans induces a weak equivalence of their homotopy pushouts. This invariance is the property of a derived colimit. It may be checked using the double mapping-cylinder model, or by mapping into a fibrant space, where the universal mapping space is the homotopy pullback of the three mapping spaces of the span.

The reverse implication requested in W(7 bis), and the two additional implications requested in W(7), are strictly stronger assertions. The formal corollaries drawn here from the proposed axioms are valid as conditional statements, as is Proposition 6's implication W(7)⇒W(7 bis). However, Theorem 2 on manuscript page 78 is false for the full printed axiom. A homotopy pushout can destroy the fundamental group of its intersection. The next block gives a finite categorical counterexample, with the ordinary and homotopy pushouts agreeing throughout. This is not a failure of the nerve to preserve the relevant pushout, and is not repaired by using a derived pushout instead.

a) $f_{U_1 \cap U_1'} \in W$.
b) f_{U_1} et $f_{U_1'}$ sont `[dans]` W.

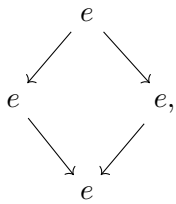
c) $f \in W$.

Alors deux parmi ces conditions impliquent la troisième. En d'autres termes, quand l'une de ces conditions est satisfaite, les deux autres sont équivalentes entre elles.

NB Cet axiome est équivalent à l'axiome dual, en termes d'objets X_1, X_2 représentés chacun comme réunion de deux fermés.

Corollaire. *Supposons* W(7) *satisfait. Alors le corollaire à* W(7 bis) *(axiome qu'on n'a pas à utiliser ici), sous sa forme complétée (cf. le NB qui le suit) est valable : deux parmi les conditions a), b), c) impliquent la troisième.*

Il suffit de regarder l'unique morphisme du diagramme (∗), p. 76, dans le diagramme trivial

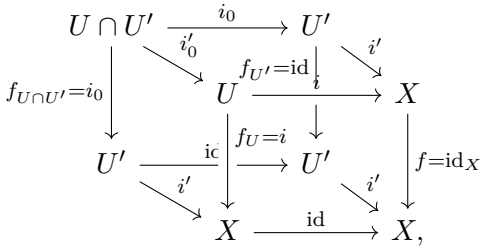


correspondant au cas où X est la catégorie finale e, et U, U' égaux à X. Alors le corollaire n'est autre que W(7) appliqué à ce cas particulier.

Cet argument fait sentir que W(7) est sans doute plus fort que W(7 bis). Incité par ce sentiment, je m'aperçois avec plaisir qu'il est fondé :

Proposition 6. *L'axiome* W(7) *(lemme des cinq) implique* W(7 bis) *(axiome du carré cocartésien).* p. 57 [page 78]

On regarde le diagramme cubique suivant



du carré cocartésien donné dans le carré cocartésien déduit de X et de ses deux ouverts X, U', d'intersection U'. Ici les deux flèches verticales f et $f_{U'}$ sont des isomorphismes, donc [dans] W. Avec les notations de W(7), on a donc c) ($f \in W$), donc a) et b) sont équivalentes, i.e. $i_0 \in W$ et $i \in W$ sont équivalentes.

q.e.d.

Théorème 2. *Le localiseur fondamental W_∞ satisfait à l'axiome* W(7).

C'est sûrement une conséquence du critère cohomologique d'Artin-Mazur, compte tenu de la suite exacte de Mayer-Vietoris pour l'un et l'autre des deux carrés cocartésiens, et les variantes non commutatives de Mayer-Vietoris. Je n'ai pas écrit la vérification en forme, et devrais sans doute réparer cette omission.

Commentary · Development (commentators') [ms. 78]

A finite counterexample to the asserted reverse excision

Grothendieck hopes that a five-lemma argument will recover the homotopy type of an intersection from the two pieces and their union. The obstruction can be isolated using the familiar acyclic noncontractible complex of Hatcher (2002, Example 2.38). We give the categorical consequence in full; the input space and its properties are classical. We checked that example, the finite simplicial replacement theorem, the subsequent distinction between the direct and reverse axioms in Chapter XVI, and the model-independent homotopy-pushout argument. We did not find this exact finite-poset formulation attached to Theorem 2 in the sources checked; no priority is claimed.

Proposition (finite failure of W(7 bis) and W(7)). [ours] *There is a finite poset X, covered by two full sieves U, V, such that NU, NV, NX are contractible but $N(U \cap V)$ is not weakly contractible. Consequently W_∞ does not satisfy either printed axiom.*

Proof. Start with a connected CW complex K_0 with one vertex, two oriented edges a, b, and two two-cells attached along

$$a^5 b^{-3}, \qquad b^3 (ab)^{-2}.$$

Its cellular differential from dimension two to dimension one is

$$d_2 = \begin{pmatrix} 5 & -2 \\ -3 & 1 \end{pmatrix}.$$

Its determinant is -1, so d_2 is an isomorphism. The differential d_1 is zero and there are no higher cells. It follows that $\widetilde{H}_*(K_0; \mathbb{Z}) = 0$.

Here is an explicit verification that its fundamental group is nontrivial. With permutations composed from right to left, send

$$a \longmapsto (1\ 4\ 3\ 2\ 5), \qquad b \longmapsto (1\ 2\ 3).$$

The product ab is $(1\ 5)(3\ 4)$, so the images satisfy $a^5 = b^3 = (ab)^2 = 1$ and define a homomorphism from the presented group to A_5. The image of a is nonidentity, proving nontriviality. We need neither a computation of the full group nor asphericity of the presentation complex.

By the finite case of Hatcher (2002, Theorem 2C.5), choose a finite simplicial complex K homotopy equivalent to K_0. Let A be the poset of its nonempty simplices ordered by inclusion. Its nerve is the barycentric subdivision of K, so $|NA|$ is connected and acyclic, and has nontrivial fundamental group. Adjoin two incomparable elements u, v above every element of A, and call the resulting finite poset X. Put

$$U = A \cup \{u\}, \qquad V = A \cup \{v\}.$$

Both are downsets of X, hence full sieves, and $U \cap V = A$. They have terminal objects u and v, respectively. Their nerves are the two cones on NA. A chain in X cannot contain both u and v, and therefore $|NX|$ is the unreduced suspension of $|NA|$.

The suspension of a connected space is simply connected, by van Kampen applied to its two thickened cones. Mayer–Vietoris gives

$$\widetilde{H}_n(|NX|; \mathbb{Z}) \cong \widetilde{H}_{n-1}(|NA|; \mathbb{Z}) = 0.$$

Thus $|NX|$ is a simply connected acyclic CW complex. It is contractible by the homological Whitehead theorem (Hatcher, 2002, Corollary 4.33). The inclusion $U \to X$ is consequently a weak homotopy equivalence. The inclusion $A \to V$ is not, since its source has nontrivial fundamental group and its target is contractible. This contradicts the reverse implication of W(7 bis). The map from the cover square of X to the constant square on e satisfies conditions b, c of W(7) but not a. It therefore contradicts W(7) as well. □

The proposed cohomological proof fails at a specific point. Local systems on the intersection need not extend to either piece or to the union. In the example, the nontrivial representation of $\pi_1(NA)$ in A_5 cannot extend across either cone. Mayer–Vietoris for coefficients restricted from the ambient contractible space only sees constant coefficients on the intersection, for which its acyclicity suppresses the obstruction. The nonabelian comparison required by a weak-equivalence criterion cannot be recovered from that restricted coefficient class. This is the same general difficulty as in the commentary on Chapter I, manuscript p. 81 (“The coefficient lost in the attempted proof”), now realized in a finite categorical cover.

[page 79]

Remarque. La version duale (au sens de Quillen) de l'axiome W(7 bis) du carré cocartésien, i.e. l'axiome correspondant pour les carrés cartésiens, n'est pas valable pour W_∞. L'énoncé pertinent serait le suivant : soit

$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{p} & S' \end{array}$$

un carré cartésien, avec f une *W-fibration.* Alors $f \in W \iff f' \in W$. Si W satisfait à W(4c) (ce qui est le cas pour W_∞), alors $f \in W$ implique bien que ses fibres sont W-asphériques, donc

que c'est une W-équivalence universelle, i.e. $f' \in W\ \forall p : S' \longrightarrow S$. Mais pour l'implication inverse $f' \in W \Longrightarrow f \in W$, il faut supposer visiblement que

$$\pi_0(p) : \pi_0(S') \longrightarrow \pi_0(S)$$

est *surjectif*. (P. ex. si $S' = \emptyset$, on a $X' = \emptyset$, donc $f' \in W$, sans que ça implique $f \in W$!) p. 58
Le fait que dans le cas cocartésien de W(7 bis), $i_0 \in W \Longrightarrow i \in W$ doit être considéré comme 'allant de soi' - autrement le carré ne mérite pas, du point de vue W-homotopique, le nom de 'cocartésien'. Par contre, l'implication inverse $i \in W \Longrightarrow i_0 \in W$ m'apparaît comme un trait très particulier du dérivatuer HOT_W (quand il le possède), et notamment de $\mathrm{HOT} = \mathrm{HOT}_{W_\infty}$. Le début de la remarque signifie simplement que le dérivateur dual HOT° ne possède [page 80]
pas cette même propriété. Je dois renvoyer à ce sujet aux deux axiomes pour les dérivateurs correspondants à W(7 bis), W(7), ce que je ferai sans doute sous peu, de façon à donner un sens plus précis aux présents commentaires.

Commentary · Present state [ms. 79]

What survives of the cube principle

The distinction between direct and reverse excision is already visible in Grothendieck's later reorganization. Chapter XVI separates the cube conditions into Loc(6a), Loc(6b), and Loc(6c); its Commentary 3.4 distinguishes the direct theory of homotopy cofibres from the extra reversibility property (Grothendieck, 1991c, §3, manuscript pp. 16–27). For the pages at hand the reliable assertion for ordinary homotopy types is the direct one: weak equivalences of all three entries of a span give a weak equivalence of its homotopy pushout. A reverse five lemma cannot be added without hypotheses that exclude the preceding example. Theorem 2 has no repair with its stated hypotheses; its direct part is a theorem.

The pullback remark makes a different, correct observation. Base change need not meet every connected component of the base. Take $S = \{0, 1\}$ discrete, and the discrete fibration $X \to S$ with one point over 0 and two points over 1. Pulling back along $\{0\} \to S$ gives an isomorphism, while $X \to S$ is not even bijective on connected components. This is a nonempty version of the empty-base example printed in the source.

Under the intended fibration hypothesis, contractibility of homotopy fibres is the correct test. If base change meets every component and the pullback is a weak equivalence, its homotopy fibres are contractible. Fibre transport along paths then shows that every homotopy fibre of the original fibration is contractible, so the original map is a weak equivalence. This argument explains the appearance of surjectivity on π_0; it must be applied where the source's W-fibration axioms really identify categorical fibres with homotopy fibres. Merely calling an arbitrary functor a fibration does not supply that identification. The corresponding discussion of categorical fibres was developed in the commentary on Chapter VI, manuscript p. 45 ("Why the new fibration axiom selects full homotopy theory").

The final contrast with the dual derivator should therefore be read in two parts. Its warning about detection after pullback is justified. Its attribution of unrestricted reverse excision to HOT depends on the false Theorem 2. Both qualifications are needed before applying the proposed axioms to the less special pushouts that follow.

Commentary · Development (commentators') [ms. 79]

What the symmetric cube rule can detect

The failed implication in Theorem 2 has a structural source. A homotopy pushout can erase information in its intersection. It is useful to determine whether passing to a bounded homotopy theory removes that obstruction, and to compare the answer with the homological argument suggested in the manuscript. The following finite family gives a uniform negative answer to the first question. We did not locate this precise family of tests in the source, its Appendix, or the later papers checked; the observation is elementary and is offered without a priority claim.

Finite truncation obstruction. [ours] For an integer $n \geq 0$, let $W_{\leq n}$ consist of functors whose nerves induce a bijection on components and isomorphisms on all homotopy groups in degrees

$1, \dots, n$, at every basepoint. There is a finite poset covered by two sieves for which the reverse implication of W(7 bis) fails for $W_{\le n}$.

Proof. Let A_n be the poset of nonempty proper subsets of a set with $n+2$ elements, ordered by inclusion. A nondegenerate simplex of NA_n is a strictly increasing chain of such subsets. Thus NA_n is the barycentric subdivision of the boundary of the $(n+1)$-simplex, and $|NA_n|$ is homeomorphic to S^n. Adjoin two distinct elements $+$ and $-$, each greater than every element of A_n, with no order relation between them. Call the resulting poset D_n, and put $U_+ = A_n \cup \{+\}$ and $U_- = A_n \cup \{-\}$. These are sieves, their intersection is A_n, and each has a greatest element. Their nerves are cones on NA_n. Since a chain of D_n can contain at most one of $+$ and $-$, their union is exactly ND_n, whose realization is the suspension of $|NA_n|$, hence S^{n+1}.

It follows that $U_+ \hookrightarrow D_n$ belongs to $W_{\le n}$: both spaces are connected, and all their homotopy groups in degrees at most n vanish. On the other hand, $A_n \hookrightarrow U_-$ does not belong to $W_{\le n}$. For $n = 0$ it identifies two components. For $n \ge 1$ it kills the nonzero group $\pi_n(S^n) \simeq \mathbb{Z}$. This is precisely the forbidden reverse implication in the cover square. □

The example for $n = 1$ has eight objects in the union: six objects forming the barycentric subdivision of a triangle boundary and the two cone points. These tests do not replace the acyclic example at p. 78, since the latter is needed for untruncated weak equivalences. Together they show that the issue is not repaired by merely stopping the homotopy groups at some finite degree.

For comparison, consider the class of functors inducing isomorphisms on nerve homology with coefficients in a fixed ring R. The full symmetric cube rule *does* hold for that class. Here is the complete chain-level reason. If $X = U \cup U'$ is a union of two sieves and $A = U \cap U'$, every chain of composable arrows in X belongs to one of the two subcategories: choose a subcategory containing its last vertex and use the sieve condition. The same statement holds for nondegenerate chains. Consequently the normalized chain complexes have the natural short exact sequence

$$0 \longrightarrow C_*(NA; R) \xrightarrow{(\iota, -\iota')} C_*(NU; R) \oplus C_*(NU'; R) \xrightarrow{\jmath + \jmath'} C_*(NX; R) \longrightarrow 0.$$

Injectivity, exactness in the middle, and surjectivity can each be checked on the basis of chains: those in the intersection occur twice, all others once. For a morphism between two such covers, take the mapping cones of the three vertical chain maps. The result is again a short exact sequence of complexes. Its long exact homology sequence implies that if any two of these cones are acyclic, then so is the third. The middle cone is the direct sum of the cones for the maps on U and U', so its acyclicity is equivalent to the acyclicity of both summands. This proves all three implications requested in W(7). For cosieve covers, the first vertex supplies the same proof.

The manuscript's proposed Mayer–Vietoris justification therefore has an exact domain of validity: it works for fixed-coefficient homology equivalences. The passage from those equivalences to weak homotopy equivalences is the step that cannot be supplied. The acyclic intersection in the preceding counterexample makes that distinction visible without any ambiguity about the existence or computation of the pushout.

2.2 Application aux carrés cocartésiens déduits d'une immersion ouverte ou fermée.

Les axiomes W(7), W(7 bis) (du cube, et du carré cocartésien) concernent des carrés cocartésiens de nature très particulière. Nous allons voir à présent comment en déduire des énoncés du même type pour des carrés cocartésiens beaucoup plus généraux, y compris pour des carrés qui n'ont rien de cocartésien (dans Cat) - et que nous appelerons 'homotopiquement cocartésiens' (relativement à W), ou simplement W-cocartésiens. (Il y aura également des carrés W-cartésiens un jour …) Dans le cas où $W = W_\infty$, je les appelerai aussi Hot-cocartésiens.

Je commencerai par le cas particulier d'un carré cocartésien associé à une immersion ouverte (diagramme (∗), p. 74) :

$$(*) \qquad \begin{array}{ccccc} U & \xrightarrow{\ i\ } & X & \xleftarrow{\ j\ } & T \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots \\ U' & \xrightarrow{\ i'\ } & X' & \xleftarrow{\ j'\ } & T' \simeq T. \end{array}$$

[page 81]

Le grand traquenard ici qu'il faut arriver à exorciser, c'est qu'il peut arriver que $i \in W$, sans qu'on ait $i' \in W$, et aussi $p \in W$ sans qu'on ait $q \in W$. Je m'en étais convaincu en 1983 en écrivant Pursuing Stacks (durant les semaines de 'scratchwork'), et n'ai pas envie de rechercher un contre-exemple ici, s'il ne se présente de lui-même (ce qu'il fera sans doute).

Je vais donner un cas important où ces ennuis disparaissent :

Commentary · Exposition [ms. 81]

Separating the boundary from the ambient category

Grothendieck wants to isolate the part of a pushout which can change its homotopy type. The complement T is carried isomorphically to T', whereas the arrows entering it from U are changed by a Kan extension. The useful intermediate category is therefore the closure $\bar{U}$, which consists of U and the objects of T reached by at least one arrow from U. Write $A = \bar{U} \cap T$. We then have a square

$$\begin{array}{ccc} A & \longrightarrow & T \\ \downarrow & & \downarrow \\ \bar{U} & \longrightarrow & X \end{array}$$

of cosieve inclusions. It is both Cartesian and cocartesian. Every arrow of X lies in one of these two full subcategories: if its source lies in $\bar{U}$, its target does too; otherwise its source lies in T, and its target lies in T. The same assertion for composable strings shows that the nerve is the union of the two corresponding simplicial subsets. Its realization is consequently a homotopy pushout, since inclusions of simplicial subsets realize as cofibrations.

The formula for the pushed-out gluing presheaf shows that an object $t \in T$ receives an arrow from U' after pushout exactly when it received an arrow from U before pushout. Indeed, a nonempty Kan-extension value is represented by an original arrow and a morphism in U'; conversely, an original arrow supplies such a value at the image of its source. Thus $A \to A'$ is an isomorphism. All the movement has been confined to $\bar{U} \to \overline{U'}$.

This explains the architecture of Proposition 6. If $U \to \bar{U}$ and $U' \to \overline{U'}$ are weak equivalences, two-out-of-three replaces the question about p by the question about $\bar{p}$. The resulting map of closed-cover squares is the identity on A and T. The ordinary homotopy gluing lemma therefore proves, for W_∞, the useful implication

$$p \in W_\infty \quad \Longrightarrow \quad q \in W_\infty.$$

This argument requires only the usual forward gluing statement.

The biconditionals in the printed proposition have stronger hypotheses. They explicitly invoke W(7 bis) and W(7), with their reverse implications as formulated on pp. 76–77. The counterexample discussed at p. 78 prevents us from treating those axioms as properties of W_∞. Proposition 6 remains a valid formal deduction under its stated axioms; its proof is not a proof that ordinary weak homotopy equivalences can be detected backwards from every such pushout. This distinction will matter when the one-point test is proposed on p. 85.

Proposition 6 : *Considérons les adhérences $\bar{U}$ et $\overline{U'}$ de U et U' dans X, X' respectivement, et supposons que les inclusions*

$$U \xrightarrow{i_0} \bar{U}, \qquad U' \xrightarrow{i'_0} \overline{U'}$$

soient dans W ([64]). Alors

p. 59

$$\begin{cases} \text{ⓐ } i \in W & \Longleftrightarrow \quad i' \in W \quad \textit{si on a } \text{W(7 bis)} \\ \text{ⓑ } p \in W & \Longleftrightarrow \quad q \in W \quad \textit{si on a } \text{W(7)}. \end{cases}$$

Démonstration. Considérons d'abord le cas ⓐ de i, i'. Soit

$$k : \bar{U} \hookrightarrow X$$

64. cf. th. 3 p. 107 pour des conditions générales assurant que cette hypothèse est satisfaite.

l'inclusion. Le triangle $U \xrightarrow{i_0} \bar{U} \xrightarrow{k} X$ $\quad\Big(\text{où } i = k \circ i_0\Big)$, où $i_0 \in W$, montre que

$$i \in W \iff k \in W.$$

D'autre part, X est la réunion des fermés $\bar{U}$, T, donc on a un carré cocartésien dans Cat

$$(*) \qquad \begin{array}{ccc} \bar{U} \cap T = \dot{T} & \xrightarrow{k_0} & T \\ {\scriptstyle j_0}\downarrow & & \downarrow{\scriptstyle j} \\ \bar{U} & \xrightarrow{k} & X. \end{array}$$

L'axiome W(7 bis) (sous sa forme duale (i.e. appliqué au carré formé par les catégories opposées …)) implique que [page 82]

$$k \in W \iff k_0 \in W,$$

d'où

$$i \in W \iff (k_0 : \dot{T} \longrightarrow T) \in W.$$

De même on trouve, en appliqunt ceci à (X', U') au lieu de (X, U)

$$i' \in W \iff (k'_0 : \dot{T}' \longrightarrow T') \in W.$$

Or on a $T \xrightarrow{\sim} T'$ (isomorphisme), et par la proposition 4 cet isomorphisme transforme $\dot{T}$ en $\dot{T}'$. Donc

$$k_0 \in W \iff k'_0 \in W \qquad \text{(trivialement)},$$

d'où a).

Prouvons b). Le carré commutatif induit par $(*)$, p. 80, p. 60

$$\begin{array}{ccc} U & \xrightarrow{i_0} & \bar{U} \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \bar{p}} \\ U' & \xrightarrow{i'_0} & \overline{U'}, \end{array}$$

où $i_0, i'_0 \in W$ par hypothèse, montre que

$$p \in W \iff \bar{p} \in W,$$

donc tout revient à prouver que

$$\bar{p} \in W \overset{?}{\iff} q \in W.$$

Mais q induit un homomorphisme du carré $(*)$ déduit de $(X, \bar{U}, T)$ (avec $X = \bar{U} \cup T$), dans le carré similaire pour X'

$$\begin{array}{ccc} \overline{U'} \cap T' = \dot{T}' & \xrightarrow{k'_0} & T \\ {\scriptstyle j'_0}\downarrow & & \downarrow{\scriptstyle j'} \\ \overline{U'} & \xrightarrow{k'} & X', \end{array}$$

les flèches verticales étant notées $q_{\dot{T}}$, q_T, $\bar{p}$, q. On sait que $q_{\dot{T}}$, q_T sont des isomorphismes, [page 83] donc dans W, donc le 'lemme des cinq' W(7) nous assure que $\bar{p} \in W \iff q \in W$, q.e.d.

Commentaire. Cette proposition nous amène à chercher des conditions sur un ouvert U d'un X, qui assurent que l'inclusion $i_0 : U \longrightarrow \bar{U}$ soit `[dans]` W, et *que cette propriété soit conservée par tout cochangement de base* $U \longrightarrow U'$. De plus, nous recherchons des conditions qui soient de nature locale sur $\bar{U}$, plus exactement sur $\bar{U} \cap T = \dot{U} = \dot{T}$. Nous pouvons motiver

ce souci (a posteriori) en disant que nous voudrions que la propriété $i_0 \in W$ soit conservée, non seulement par cochangement de base quelconque $U \longrightarrow U'$, mais aussi, de plus, par passage de X à toute sous-catégorie pleine X_1 contenant U, donnant donc naissance à une situation

$$U \overset{i_Y}{\hookrightarrow} X_1, \qquad U \xrightarrow{(i_Y)_0} \bar{U}^{X_1} = \bar{U} \cap X_1$$

[$Y = X_1$?], similaire à la situation de départ. Plus généralement encore, nous voudrions qu'elle soit conservée en passant du triple (X, U, T) à un triple "induit" (X_1, U, T_1), par une flèche quelconque $g : T_1 \longrightarrow T$ de but T dans Cat, donnant naissance au couple de flèches

$$U \overset{\mathrm{id}}{\to} U,\ g : T_1 \longrightarrow T$$

(cf. pages 60, 61). Le cas précédent est celui où T_1 est une sous-catégorie pleine de T, g la flèche d'inclusion, Notons à ce sujet la proposition soritale p. 61 [page 84]

Proposition 7. *Conditions équivalentes sur l'ouvert U d'un objet X de* Cat (moyennant W(7 bis)) :

a) L'inclusion $i_0 : U \longrightarrow \bar{U}$ est W-asphérique, i.e. pour tout $t \in \dot{U} = \bar{U} \backslash U$, la catégorie U/t est W-asphérique.

b) L'inclusion $i_0 : U \longrightarrow \bar{U}$ est dans W, et cette condition reste satisfaite, quand on remplace (X, U) par (X_1, U), où (X_1, U) est déduit de (X, U) au moyen d'une flèche $T_1 \longrightarrow T (= X \backslash U)$ dans Cat*, comme rappelé ci-dessus.*

b′) Comme b), dans le cas où T_1 est la catégorie ponctuelle e, et où de plus son image est dans $\dot{U}$.

Démonstration. Il est évident que si dans b), b′) on remplace la condition "$i_0 \in W$" par la condition plus forte 'i_0 est W-asphérique', les quatre [plutôt trois] conditions a) à b′) sont équivalentes. Il reste à prouver que la condition que les U/t soient W-asphériques ($t \in \bar{U}$ - mais c'est trivial pour $t \in U$, donc il suffit de l'exprimer pour $t \in \bar{U} \backslash U = \dot{U}$) est déjà impliquée par la plus faible b′ et on devine qu'on a ceci :

Corollaire. (Moyennant W(7 bis).) *Soit X dans* Cat*, U un ouvert de X tel que $X \backslash U = \overline{\{t\}}$ soit la catégorie finale. Alors l'inclusion $U \overset{i}{\to} X$ est dans W si et seulement si elle est W-asphérique, i.e. si et seulement si U/t est W-asphérique (où t est l'unique élément de T).* [page 85]

Nous démontrons ultérieurement ce corollaire ([65]). La proposition 7 ne nous servira pas d'ici là - elle n'aura qu'un rôle purement esthétique à jouer, pour justifier notre propos délibéré de chercher des conditions sur $i : U \longrightarrow X$ (pour assurer que $U \longrightarrow \bar{U}$ est dans W et le reste, quand on remplace (X, U) par d'autres couples (X_1, U_1) qui s'en déduisent par certaines opérations …) qui soient de *nature locale* sur $\bar{U}$ (en fait, en les points de $\dot{U}$), et qui en tout état de cause entrainent la *W-asphéricité* de $U \longrightarrow \bar{U}$. [page 86]

Commentary · Exposition [ms. 85]

The one-point test and the direction of Theorem A

The purpose of Proposition 7 is to turn a condition on the whole inclusion into a condition at each new point. For a sieve $i : U \hookrightarrow X$, the comma category U/u has final object (u, id_u) whenever $u \in U$. All possible failures of asphericity are therefore concentrated at the complementary objects. Replacing T by the point selecting t preserves precisely the presheaf $u \mapsto \mathrm{Hom}_X(u, t)$, and hence precisely the category U/t. This proves the equivalence of the three versions in which every occurrence of "belongs to W" is replaced by "is W-aspherical." The remaining implication is exactly the deferred corollary, not a consequence of the definition.

For ordinary weak homotopy equivalences, the corollary's reverse implication is false. To see its geometry without altering the source, take a connected acyclic category A whose nerve is not weakly contractible, as in the counterexample at p. 78. Form U by adjoining a final object u_+ to A.

65. cf. cor. 4 de la proposition 15, page 149.

Form X by adjoining instead two distinct final cone points u_+ and t, with no morphisms between them. Then $U \hookrightarrow X$ is a sieve and its complement is the one-object category $\{t\}$. Every object of A has its unique arrow to t, while u_+ has none, so

$$U/t \simeq A.$$

On the other hand, BU is contractible and $BX \simeq \Sigma BA$. The suspension of a connected acyclic CW complex is simply connected and acyclic, hence weakly contractible by the Hurewicz and Whitehead theorems. Thus $U \to X$ is a weak homotopy equivalence although it is not aspherical.

The missing distinction is between the homotopy type of an attachment and the homotopy type of its attaching category. A cone attachment can erase a nontrivial perfect fundamental group without leaving any homology behind. Theorem A supplies the implication from contractible comma categories to weak equivalence; it does not supply the converse. In modern language this is also the distinction between a weak equivalence of classifying spaces and a homotopy initial functor: the comma categories here are $U/t = (i \downarrow t)$. Compare the dual of Lurie (2009, Theorem 4.1.3.1 and Corollary 4.1.3.3), applied to i^{op}. The source has retained W(7 bis) in the corollary, and the audit retains it too. Its subsequent use for W_∞ needs an independent argument in each application.

J'avais espéré que cette condition d'asphéricité (suggérée par la notion 'd'immersion de Dwyer') serait stable par cochangement de base quelconque $U \longrightarrow U'$, ce qui aurait assuré en même temps qu'après une telle opération, l'immersion ouverte $i' : U' \longrightarrow X'$ obtenue satisfait encore à la condition que $i'_0 : U' \longrightarrow \overline{U'}$ est dans W. Nous verrons que cette attente n'est pas fondée - la condition de W-asphéricité n'est *pas* stable par cochangement de base, et elle ne suffit *pas* même à assurer qu'après un cochangement de base, $i'_0 : U' \longrightarrow \overline{U'}$ soit dans W. Il faudra une notion plus forte, qui est suggérée par le cas où $T = \{t\}$ est la catégorie finale et contenue dans $\bar{U}$ (cas auquel nous avons essayé de nous ramener dans la proposition 7). Dans ce cas, on l'a dit (corollaire précédent), la condition que $i : U \longrightarrow \bar{U} = X$ soit dans W, équivaut à la W-asphéricité de i, i.e. à la W-asphéricité de U/t, et de même pour $i' : U' \longrightarrow \overline{U'} = X'$: elle est dans W si et seulement si U'/t' est W-asphérique, où t' est l'unique élément de $T'(\simeq T) = X' \backslash U'$. Donc il faut des conditions sur U/t qui assurent que non seulement U/t, mais aussi toutes les catégories similaires U'/t `[plutôt` U'/t'`]`, soient W-asphériques. Dans le cas général où T est quelconque, c'est cette condition-là que nous allons imposer aux U/t ($t \in \mathrm{Ob}\, T$), et il s'avérera que cette fois-ci, elles sont stables `[plutôt elle est stable]` par tout cochangement de base $U \xrightarrow{p} U'$, i.e. que les U'/t' ($t' \in \mathrm{Ob}\, T'$) satisferont alors à la même condition. C'est cette condition que nous allons à présent reformuler.

p. 62

[page 87]

Commentary · Exposition [ms. 87]

A boundary presheaf and all its direct images

Grothendieck now strengthens the boundary condition in the way dictated by the actual pushout formula. The extra quantifier is the essential feature. Given $H : U^{\mathrm{op}} \to \mathbf{Set}$, let $\int_U H = U/H$ denote its category of elements. Its objects are pairs (u, h) with $h \in H(u)$, and a morphism $(u, h) \to (v, k)$ is an arrow $a : u \to v$ satisfying $H(a)(k) = h$. If X is obtained by adjoining a complementary point t with $\mathrm{Hom}_X(u, t) = H(u)$, then $U/t = \int_U H$ literally records all ways to enter that point.

For $p : U \to U'$, the new boundary is $p_! H$, where $p_!$ is the ordinary left Kan extension along p^{op}. It is useful to calculate it through the category of elements, rather than to lose the variance in an informal colimit. With $\alpha : \int_U H \to U$ the projection,

$$(p_! H)(u') \cong \pi_0\big(u' \downarrow p\alpha\big).$$

An object on the right is $(u, h, \beta : u' \to p(u))$. Its connected component is the generator of the Kan extension represented by that triple. A map $a : u \to v$ with $H(a)(k) = h$ identifies the generator (u, h, β) with $(v, k, p(a)\beta)$; these are exactly the relations in the pointwise colimit. This gives both the formula and its functoriality in u'.

The ordinary asphericity of H only tests $p = \mathrm{id}_U$. Total asphericity requires the entire category of elements of every $p_! H$ to be aspherical. The operation takes connected components before the resulting presheaf is assembled over U'. It can therefore destroy homotopy information inside each

comma category while creating nontrivial topology in the assembled category. The finite example at p. 100 will make this effect visible in degree two.

The definition has a deliberate stability property. If H is totally W-aspherical and $p : U \to U'$ is any functor, then $p_!H$ is totally W-aspherical: for every $q : U' \to U''$ there is a canonical isomorphism $q_!p_!H \cong (qp)_!H$. Thus the strengthened condition is preserved by the very operation against which it was designed to protect the boundary. This is the formal reason the later definition of a W-Dwyer immersion uses these presheaves.

The editors' warning about terminology is necessary. The earlier test-category usage of "totally aspherical" concerns products or diagonal functors. Here the quantifier runs over all outgoing functors and their set-valued left Kan extensions. The commentary on Chapter II, manuscript p. 36 ("The quantifier in total asphericity") already treats this quantifier; the present pages apply it to a concrete gluing problem. Likewise the filtered examples should be read together with the commentary on Chapter II, manuscript pp. 32–35, rather than as a new assertion that contractibility alone controls all direct images.

2.3 Objets totalement W-asphériques de Cat, et morphismes totalement W-asphériques.

Reprenons la situation

$$U \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} T, \qquad \text{avec } T = \{t\}, \text{ catégorie finale, } t \in \bar{U}, \text{ i.e. } X = \bar{U},$$

et celle d'un cochangement de base $U \longrightarrow U'$, donnant naissance à un carré cocartésien dans Cat

$$(*) \qquad \begin{array}{ccccc} U & \overset{i}{\hookrightarrow} & X & \overset{j}{\dashleftarrow} & T = \{t\} \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots\,{\scriptstyle \text{iso}} \\ U' & \overset{i'}{\hookrightarrow} & X' & \overset{j'}{\dashleftarrow} & T' = \{t'\}. \end{array}$$

La donnée de recollement H entre U et T, $H : \underbrace{U^\circ \times T}_{\simeq U^\circ} \longrightarrow \mathrm{Ens}$, peut s'interpréter comme un préfaisceau sur U, $H \in \mathrm{Ob}\, U^\wedge : U^\circ \longrightarrow \mathrm{Ens}$, et on a [page 88]

$$U/t \simeq U/H,$$

où U/H est interprété en termes du plongement canonique

$$U \hookrightarrow U^\wedge.$$

De même, le recollement entre U' et T' sera donné par un préfaisceau

$$H' \in \mathrm{Ob}\, U'^\wedge : U'^\circ \longrightarrow \mathrm{Ens},$$

et on aura p. 63

$$U'/t' \simeq U/H'.$$

La relation entre H' et H a été explicitée au §1, comme

$$H' = p_!H, \qquad \text{où } p_! : U^\wedge \longrightarrow U'^\wedge$$

est l'adjoint à gauche de $p^* : U'^\wedge \longrightarrow U^\wedge$, donné par

$$p^*F' = F' \circ p.$$

On a

$$p_!(H)(u') \simeq \pi_0(u'\backslash U/H). \qquad [\text{page 89}]$$

Définition 1. ⓐ Soit U un objet de Cat, $H \in \mathrm{Ob}\, U^\wedge$ un préfaisceau sur U. On dit que H est *totalement W-asphérique* si pour toute flèche $p : U \longrightarrow U'$ de Cat, $p_!(H)$ dans $U'^\wedge$ est un objet W-asphérique, i.e. $U'/p_!H$ est W-asphérique. ⓑ On dit qu'un objet V de Cat est *totalement W-asphérique,* si le foncteur final e_V sur V ($v \longmapsto \{\emptyset\}$) est totalement W-asphérique, i.e. si pour toute flèche $f : V \longrightarrow V'$ dans Cat, le préfaisceau

$$\pi_0(f) \overset{\text{déf}}{=} f_!(e_V) \overset{\text{déf}}{=} (v' \longmapsto \pi_0(v'\backslash V))$$

sur V' est W-asphérique, i.e. si la catégorie

$$\underline{\pi}_0(f) \quad \text{ou} \quad \underline{\pi}_0(V/V') \overset{\text{déf}}{=} V'/\pi_0(f)$$

est W-asphérique. [Cette notion est différente de la notion de totale asphéricité de Pursuing Stacks.]

Notons que la définition des foncteurs $p_!$ (ou $f_!$) comme adjoints, et leur 'calcul' en termes de π_0 de certaines catégories, ne dépend pas du localiseur W. Donnons quelques précisions à ce sujet. On a un morphisme d'adjonction dans $V^\wedge$

$$f^* f_!(F) \longleftarrow F,$$

d'où

$$(*) \qquad V/F \longrightarrow V/f^* f_!(F) \simeq V \times_{V'} (V'/f_!(F)),$$

le dernier isomorphisme provenant du fait que l'on a commutativité, à isomorphisme canonique près, dans le diagramme de foncteurs

$$\begin{array}{ccc} V'^\wedge & \xrightarrow{f^*=(F' \longmapsto F' \circ f)} & V^\wedge \\ {\scriptstyle F' \longmapsto V'/F'}\downarrow & & \downarrow{\scriptstyle F \longmapsto F/V} \\ \mathrm{Cat}/V' & \xrightarrow{f^*_{\mathrm{Cat}}=(W' \longmapsto W' \times_{V'} V)} & \mathrm{Cat}/V, \end{array}$$

en d'autres termes, l'opérateur f^* sur les préfaisceaux s'interprète, en termes géométriques, comme l'opération $W' \longmapsto W' \times_{V'} V$ de changement de base par $V \longrightarrow V'$, sur les catégories au dessus de V' qui sont fibrées à fibres discrètes (propriété qui se conserve par changement de base). p. 64 [page 90]

Ceci dit, la donnée d'une flèche $(*)$ équivaut, par la propriété universelle du changement de base, à celle d'une flèche (canonique en l'occurrence)

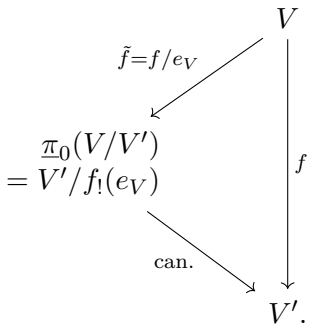
$$f/F : V/F \longrightarrow V'/f_!(F)$$

rendant commutatif le carré

$$(*) \qquad \begin{array}{ccc} V/F & \xrightarrow[\text{ou foncteur source}]{\text{oubli de } F} & V \\ {\scriptstyle f/F}\downarrow & & \downarrow{\scriptstyle f} \\ V'/(F' = f_!(F)) & \xrightarrow{\text{oubli de } F'} & V'. \end{array}$$

Dans le cas qui nous intéresse surtout ici, $F = e_V$, $V/F \longrightarrow V$ est un isomorphisme, et on obtient une factorisation canonique de foncteurs [page 91]

(∗∗)



Explicitons le V'-morphisme $\tilde{f}$ sur les fibres,

$$(***) \qquad \tilde{f}_{v'} : V_{v'} \longrightarrow \underline{\pi}_0(V/V')_{v'} \overset{\text{déf}}{=} \pi_0(v'\backslash V),$$

où dans le dernier membre, on a identifié l'ensemble $\pi_0(v'\backslash V)$ à la catégorie discrète qu'il définit. Pour expliciter cette flèche $(***)$, notons qu'on a une inclusion canonique

$$V_{v'} \longrightarrow v'\backslash V$$

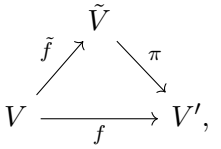
et pour toute catégorie W, un morphisme canonique (d'adjonction) $W \longrightarrow \pi_0(W)$, d'où un composé

$$V_{v'} \longrightarrow v'\backslash V \longrightarrow \pi_0(v'\backslash V),$$

qui n'est autre que $\tilde{f}_{v'}$ dans $(***)$.

Il doit être facile de montrer que la factorisation $(**)$ de f est caractérisée, à isomorphisme unique près, en tant que factorisation dans Cat/V'

p. 65
[page 92]



par les propriétés suivantes :

a) π est fibrée à fibres discrètes, i.e. un isomorphisme local (i.e. $\tilde{V}$ est isomorphe dans Cat/V' à une catégorie V/F', avec F' dans $V'^{\wedge}$).

b) $\tilde{f}_!(e_V) = e_{\tilde{V}}$, i.e. pout tout $\tilde{v} \in \mathrm{Ob}\,\tilde{V}$, la catégorie $\tilde{v}\backslash V$ est 0-connexe : les colocalisés de V au dessus de $\tilde{V}$ sont 0-connexes.

Notons ici, en passant, que pour *tout* morphisme $\pi : W' \longrightarrow W$ dans Cat qui soit fibré à fibres discrètes, i.e. tel que W' soit isomorphe dans Cat/W à une catégorie de la forme W/F ($F \in \mathrm{Ob}\, W^{\wedge}$), on a

$$(*) \qquad \pi_!(e_{W'}) = F, \qquad \text{i.e. } W' \underbrace{\simeq}_{\text{dans Cat}/W} W/\pi_!(e_{W'}).$$

Il faut vérifier que le foncteur

$$w \longmapsto \pi_0(\underbrace{w\backslash(W/F)}_{\overset{\text{déf}}{=} w\backslash W/F})$$

est canoniquement isomorphe à F. Mais l'inclusion

$$\underbrace{W'_w}_{=F(w)\ \text{(catégorie discrète)}} \longrightarrow w\backslash W'$$

définit bien comme tantôt [page 93]

$$F(w) = W'_w \longrightarrow \pi_0(w\backslash W'),$$

il faut prouver que c'est bijectif. Or, W' étant W_∞-lisse sur W, on sait que l'inclusion $W'_w \hookrightarrow w\backslash W'$ est dans W_∞, a fortiori induit un isomorphisme pour les π_0.

Ceci posé, si on a une factorisation de $f : V \longrightarrow V'$ en $f = \pi\tilde{f}$ ayant les propriétés a) b) de la page précédente, on voit déjà que l'on a

$$f_!(e_V) \simeq \pi_!(\underbrace{\tilde{f}_!(e_V)}_{\simeq e_{\tilde{V}} \text{ par b)}}) \simeq \pi_!(e_{\tilde{V}}),$$

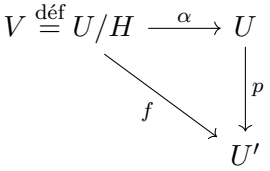
ce qui déjà détermine $\tilde{V}$ à isomorphisme canonique près (grâce à a) comme

$$\tilde{V} \simeq V'/f_!(e_V) \overset{\text{déf}}{=} \underline{\pi}_0(V'/V),$$

et il ne doit pas être bien dur de se convaincre que $\tilde{f}$ n'est autre que la flèche canonique de la page 91. p. 66

Proposition 8. *Pour qu'un objet H de $U^\wedge$ soit totalement W-asphérique, il suffit que l'objet U/H de* Cat *soit totalement W-asphérique.*

Supposons en effet cette dernière condition satisfaite et prouvons que H est totalement W-asphérique, i.e. que pour tout $p : U \longrightarrow U'$, $U'/p_!(H)$ soit W-asphérique. [page 94]



Soit f le foncteur composé du diagramme ci-contre `[ci-dessus]`, on a $H \simeq \alpha_!(e_V)$ (par $(*)$ p. 92), d'où

$$p_!(H) \simeq p_!\alpha_!(e_V) \simeq (p\alpha)_!(e_V) \simeq f_!(e_V),$$

qui est W-asphérique par hypothèse sur $V = U/H$.

Commentary · Exposition [ms. 93]

The connected-component factorization, with its proof

Grothendieck's relative $\underline{\pi}_0$ turns a map of categories into a genuine family of sets over its target. This is the right construction for understanding what ordinary Kan extension forgets. Let $f : A \to B$, put

$$F(b) = \pi_0(b \downarrow f), \qquad E = B/F,$$

and let $\pi : E \to B$ be the projection. Precomposition with a morphism $b' \to b$ gives the restriction map $F(b) \to F(b')$, so F is a presheaf. The factorization map

$$j : A \longrightarrow E, \qquad j(a) = \big(f(a), [(a, \mathrm{id}_{f(a)})]\big)$$

is functorial: a morphism $a \to a'$ identifies the evident two objects in $f(a) \downarrow f$. We have $f = \pi j$.

To prove the missing connectedness assertion, fix $e = (b, c) \in E$. An object of $e \downarrow j$ is an $a \in A$ together with $\beta : b \to f(a)$ such that the component of (a, β) is c. Its morphisms are exactly the morphisms of $b \downarrow f$ between those objects. Thus

$$e \downarrow j \;\cong\; (b \downarrow f)_c,$$

the component indexed by c. It is nonempty and connected by definition. In the usual ordinary-category terminology, j is a *final functor*. No higher contractibility is being asserted.

We also need to recover a discrete fibration from this procedure. Suppose $\pi : E \to B$ is the category of elements of a presheaf F. For $(e, \beta : b \to \pi e)$ in $b \downarrow \pi$, the unique lift of β into e has source $(b, F(\beta)(e))$. This assignment gives a functor to the discrete fibre E_b. Its composite with the

fibre inclusion is naturally connected to the identity by the lifted arrows, and its restriction to the fibre is the identity. Consequently

$$\pi_0(b \downarrow \pi) \cong F(b), \qquad \pi_! 1_E \cong F.$$

The same argument proves a homotopy equivalence between the comma category and the discrete fibre, which explains the appeal to smoothness on p. 93 without requiring that theory here.

Finally, suppose $f = \rho k$ is another factorization with $\rho : D \to B$ discrete and $k : A \to D$ final. By the Kan-extension formula $k_! 1_A = 1_D$, and hence

$$f_! 1_A \cong \rho_! k_! 1_A \cong \rho_! 1_D.$$

The preceding recovery calculation identifies D over B with $B/F = E$. This identification also respects the first factor. Indeed, for any presheaf G on B, a functor $A \to B/G$ over B is the same as a natural transformation $1_A \to f^*G$, and adjunction identifies these with natural transformations $f_! 1_A \to G$. For $G = F$, the canonical first factor corresponds to id_F; the identification obtained from $\rho_! 1_D \cong F$ carries k to that same map. This proves uniqueness with its compatibility, rather than only determining the middle category as an abstract object.

The result is the discrete-fibration form of the comprehensive factorization of Street and Walters (1973). A complete direct proof in the opposite variance is in Kelly (1982, §4.7, pp. 73–75, especially (4.77)–(4.79)); Kelly also identifies an invalid Kan-extension inference in the original proof and replaces it by the component calculation used above. The theorem predates these manuscript pages. Its role here is to distinguish the final first factor from a homotopy final map, whose comma categories would have to be weakly contractible.

Commentary · Later developments [ms. 93]

From comprehensive factorization to homotopy families

The geometric interpretation of $\underline{\pi}_0$ has continued to be useful independently of derivators. Berger and Kaufmann (2017, §1.3, Lemma 1.6, and Theorem 1.7) place the two ordinary comprehensive factorizations in a general theory of comprehension schemes: presheaves supply discrete fibrations, and the unit of comprehension supplies their universal approximation. Their Proposition 1.10 and Remark 1.11 explain the associated base-change and Frobenius properties. These are structural statements about the factorization, not claims that its first factor is a weak homotopy equivalence.

For the comparison with higher categories, the decisive change is to keep the homotopy type of each comma category instead of only its set of components. The homotopy-finality criterion requires $B(b \downarrow f)$ to be contractible for every b, as in Lurie (2009, Theorem 4.1.3.1). Here j only makes these spaces connected. Thus comprehensive factorization is a useful discrete approximation, but it cannot replace a construction of a homotopy Kan extension. The examples below measure precisely what is lost when this approximation is used before forming a total category.

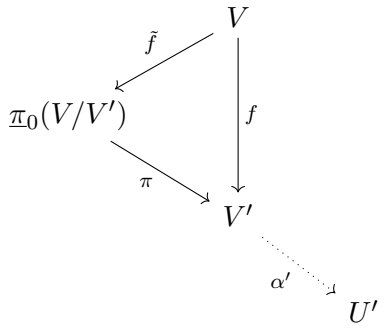
La réciproque de `[la]` prop. 8 est-elle valable, i.e. si H est totalement W-asphérique, en est-il de même de $V = U/H$? La difficulté provient du fait que cela signifie que pour *toute* flèche $f : V \longrightarrow V'$ dans Cat de source V, on ait $f_!(e_V)$ W-asphérique, alors que l'hypothèse équivaut à le postuler seulement pour les flèches qui se factorisent par U. Notons que si on ne fait pas cette hypothèse sur f, mais qu'on suppose au lieu de cela que f s'insère dans un carré commutatif [page 95]

$$\begin{array}{ccc} V & \xrightarrow{\alpha} & U \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle p} \\ V' & \xrightarrow{\alpha'} & U', \end{array}$$

avec α' lui aussi un isomorphisme local, alors on peut conclure que $\underline{\pi}_0(f) = \underline{\pi}_0(V/V')$ est bien asphérique. En effet, on a un isomorphisme canonique dans Cat/U',

$$\underline{\pi}_0(\alpha' f) \simeq \underline{\pi}_0(f), \qquad \text{i.e. } \underline{\pi}_0(V/V') \simeq \underline{\pi}_0(V/U'), \qquad \text{p. 67}$$

du fait p. ex. que la factorisation canonique de f



fournit une factorisation du même type pour $\alpha' f$, en

$$\alpha' f = (\alpha' \pi) \circ \tilde{f}.$$

Ceci dit, on peut appliquer l'hypothèse à p et on trouve bien la W-asphéricité de $\underline{\pi}_0(f) \simeq \underline{\pi}_0(\alpha' f) = \underline{\pi}_0(p\alpha)$.

L'idée naturelle, pour une réciproque à `[1a]` prop. 8, serait de prendre pour U' la somme amalgamée de U et V' sur V, ce qui marchera pourvu qu'on sache que α' est bel et bien un isomorphisme local tout comme α. C'est bien le cas p. ex. si $V \longrightarrow U$ est une immersion ouverte, i.e. H un sous-préfaisceau du préfaisceau final. Mais je crains que ce ne soit pas vrai en général - ça a l'air un peu trop beau pour être vrai! C'est en effet déjà faux dans le cas [page 96]

$$\begin{array}{ccc} V = E_G & \xrightarrow{\alpha \text{ can}} & B_G = U \\ \Big\downarrow{\scriptstyle f} & & \\ V' = B_{G'}, & & \end{array}$$

dans le cas p. ex. où $G = \{\pm 1\}$, groupe à deux éléments, et où l'élément α de G', image par f de l'une des deux flèches non identiques de E_G est $\neq 1$ `[plutôt de carré` $\neq 1$`]`. Si H' est le sous-groupe invariant engendré par α `[plutôt par` α^2`]` dans G', la somme amalgamée s'identifie à $B_{G'/H'}$, or $B_{G'} \longrightarrow B_{G'/H'}$ n'est *pas* un isomorphisme local (car $(B_{G'})/t_0 \longrightarrow (B_{G'/H'})/t'_0$ n'est pas bijectif sur les objets, s'identifiant à $G' \xrightarrow{q} G'/H' \ \ldots$).

Mais d'autre part la condition de totale W-asphéricité sur un objet H de $U^\wedge$ paraît déjà à tel point exigeante, que je me vois mal construire un exemple où elle est satisfaite, sans que U/H elle-même ne soit totalement W-asphérique. [page 97]

Commentary · Exposition [ms. 96]

The converse for subterminal presheaves and its obstruction

Proposition 8 compares two quantifiers. Write $V = U/H$ and $\alpha : V \to U$. Total asphericity of V tests every outgoing functor $f : V \to C$. Total asphericity of H tests the special family $p\alpha$, with $p : U \to B$. The proposition follows because $\alpha_! 1_V = H$, but reversing it requires a way of placing an arbitrary f in that special family without changing its comprehensive middle category.

The square criterion on p. 95 does this whenever one can complete

$$\begin{array}{ccc} V & \xrightarrow{\alpha} & U \\ {\scriptstyle f}\big\downarrow & & \big\downarrow{\scriptstyle p} \\ C & \xrightarrow[\beta]{} & B \end{array}$$

with β a discrete fibration. To justify the assertion completely, factor f as a final functor $V \to E$ followed by a discrete fibration $E \to C$. Since discrete fibrations compose, the same E is the

comprehensive middle category of $\beta f = p\alpha$. Its category is therefore W-aspherical when H is totally W-aspherical.

If H is subterminal, α is a sieve inclusion. Form the ordinary categorical pushout $B = U \amalg_V C$. Theorem 1 preserves the sieve inclusion on the other side, so $\beta : C \to B$ is again a sieve, in particular a discrete fibration. The preceding argument applies to every $f : V \to C$. This supplies the full proof of the source's subterminal case: for such H, the total asphericity of H and of U/H are equivalent.

The groupoid example identifies why the same proof cannot be extended merely by replacing "sieve" with "discrete fibration." Let $G = C_2$ and let E_G be the indiscrete groupoid on two objects. Its projection to BG sends both nonidentity arrows to the unique nontrivial element g. A functor $E_G \to BG'$ sends one of these arrows to an element $a \in G'$ and the inverse arrow to a^{-1}. In a cocone on this span, the two objects must become equal, and compatibility with the projection to BG forces

$$a = a^{-1}, \quad \text{equivalently } a^2 = 1.$$

There are no additional relations. The pushout is therefore

$$B\big(G'/\langle\!\langle a^2 \rangle\!\rangle\big).$$

For example, with $G' = \mathbb{Z}$ and a its generator, the induced map is $B\mathbb{Z} \to BC_2$. A discrete fibration between one-object groupoids must lift every arrow uniquely into the sole object, so its map of morphism sets must be bijective. This quotient map is not. The bracketed corrections in the edition, replacing a by a^2, are thus mathematically necessary.

This counterexample defeats the proposed pushout method; it does not by itself refute the general converse of Proposition 8. That converse is false: a presheaf with three elements on a monoid with two elements is given in the commentary at ms. 97. The subterminal case is proved, and the obstruction to the broader argument is explicit.

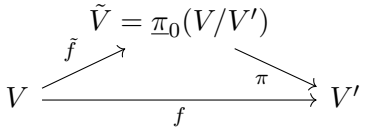
L'existence d'objets totalement W-asphériques dans Cat n'est pas tellement évidente a priori. Qu'il y en ait un bon paquet est assuré par la

Proposition 9. p. 68

a) Tout objet V de Cat *ayant un objet final est totalement W-asphérique - donc tout préfaisceau d'ensembles H* représentable *sur une catégorie U (la représentabilité signifiant justement que U/H a un objet final) est totalement W-asphérique.*

b) ([66]) (Supposons W(7 bis) *satisfait.) Soient V dans* Cat, *V', V'' deux ouverts de réunion V. Si V', V'', $V' \cap V''$ sont totalement W-asphériques, il en est de même de V.*

Démonstration de a). Soit V avec objet final et $f : V \longrightarrow V'$, d'où la factorisation



de f. On démontre sans mal ([67]) (histoire de cofinalité) ceci :

Corollaire. *L'image par $\tilde{f}$ d'un objet final de V est un objet final de $\tilde{V} = \underline{\pi}_0(V/V')$.*

Comme un objet de Cat avec objet final est W-asphérique par W(3), cela établit a).

Prouvons b). J'ai envie de dire que si on a $f : V \longrightarrow W$ dans Cat, alors $\underline{\pi}_0(V/W)$ est réunion de deux ouverts isomorphes à $\underline{\pi}_0(V'/W)$, $\underline{\pi}_0(V''/W)$, dont l'intersection est isomorphe à $\underline{\pi}_0(V' \cap V''/W)$. Cela signifierait ni plus ni moins que ceci : $f_!$ transforme les monomorphismes [page 98]

66. !! Sans doute *faux* tel quel, cf. rectification page suivante.

67. Voici une démonstration simple : On a pour toute V dans Cat $e_V = \varinjlim_{v \text{ dans } V} v$ (limite inductive du foncteur d'inclusion de V dans $V^\wedge$), car $\forall\, x$ dans V

$$\varinjlim_{v \text{ dans } V} \mathrm{Hom}(x, v) \simeq \pi_0(x\backslash V) = e.$$

Comme $f_!$ commute aux $\varinjlim$, on a donc $f_!(e_V) = \varinjlim_{v \text{ dans } V} f(v)$. Si v_0 est un objet final de V, cela montre que $f_!(e_V) = f(v_0)$, q.e.d.

$V' \longrightarrow V$, $V'' \longrightarrow V$ (interprétés comme flèches dans $V^\wedge$, de but e_V) en monomorphismes, et le produit fibré $V' \cap V'' = V' \times_V V''$ en le produit fibré correspondant dans $W^\wedge$. Mais le foncteur $f_!$ commute aux $\varinjlim$ quelconques (étant un adjoint à gauche), et je ne vois aucune raison a priori pour qu'il commute (disons) aux produits fibrés (ce qui impliquerait aussi qu'il transforme monomorphismes en itou). Si je me rappelle bien, c'est ultrafaux en général. Donc je doute fort à présent que l'énoncé b) soit vrai, sans hypothèse supplémentaire. Pour assurer la monomorphie, on songe à imposer l'existence de *rétractions* de V sur V', V'' et de V' ou V'' sur $V' \cap V''$.

Commentary · Development (commentators') [ms. 97]

Total asphericity tested on final images

Grothendieck's search becomes more transparent if we express the quantifier through the first half of comprehensive factorization. The following elementary consequence is our formulation of what the construction already permits; it is not a claim of priority. We checked the factorization accounts of Street–Walters, Kelly, and Berger–Kaufmann, which establish the factorization but do not state this characterization of the manuscript's total-asphericity class.

Final-image criterion. [ours] For any localizer W for which asphericity is invariant under isomorphism, a small category A is totally W-aspherical if and only if every ordinary final functor $A \to C$ has W-aspherical target. Moreover, the class of totally W-aspherical categories is closed under ordinary final functors.

Proof. If $u : A \to C$ is final, then $u_! 1_A = 1_C$. Hence the category of elements used in the definition of total asphericity is $C/1_C \cong C$, proving the necessity. Conversely, factor an arbitrary $f : A \to B$ comprehensively as

$$A \xrightarrow{j} E = B/(f_! 1_A) \xrightarrow{\pi} B.$$

The first factor is final, so the assumed condition says that E is W-aspherical. This is exactly the required test for f.

For the closure assertion, suppose A is totally W-aspherical and $u : A \to C$ is final. For every $g : C \to D$,

$$g_! 1_C \cong g_! u_! 1_A \cong (gu)_! 1_A.$$

Its category of elements is W-aspherical by the hypothesis on A. Thus C is totally W-aspherical.□

There is an equivalent extremal description which is sometimes useful for constructing examples. Among subclasses of the W-aspherical categories which are closed under taking targets of ordinary final functors, the totally W-aspherical categories form the largest one. They themselves have that closure property, and they are aspherical by taking $f = \mathrm{id}$. Conversely, if A belongs to any such subclass, every comprehensive middle category of an outgoing map from A belongs to it, hence is aspherical. The criterion just proved applies.

The result also separates two kinds of positive examples. If A has final object a_0, then for every $f : A \to B$,

$$f_! 1_A \cong B(-, f(a_0)), \qquad B/(f_! 1_A) \cong B/f(a_0),$$

whose final object is $(f(a_0), \mathrm{id})$. This is Proposition 9(a) in an explicit form. Filtered categories provide the further examples already explained in the commentary on Chapter II, manuscript pp. 32–35, through filtered colimits of representables. For instance $(\mathbb{N}, \leq)$ has no final object, so the conjecture in footnote 73 cannot survive even in ordinary homotopy theory. The new criterion explains what both examples must control: all final images, not just their own classifying spaces.

Commentary · Development (commentators') [ms. 97]

A total presheaf whose category of elements is not total

Grothendieck expects that Proposition 8 has no converse outside the subterminal case, and proposes to look for a totally W-aspherical presheaf H whose category of elements is not totally W-aspherical. Such a presheaf exists on the smallest category that is not a preorder and not a group. We found no source for the example; it is given without a claim of priority.

Proposition (the converse of Proposition 8 fails). [ours] *Let U be the category with one object $*$ and $\mathrm{End}(*) = \{1, e\}$, $e^2 = e$. Let H be the presheaf on U with $H(*) = \{i, h_1, h_2\}$, on which e acts*

by the constant map with value i. Then H is totally W-aspherical for every fundamental localizer W. The category U/H is not totally W-aspherical for $W = W_\infty$, for the homology equivalences W_R with coefficients in a nonzero commutative ring R, and for the n-equivalences with $n \geq 2$.

Proof. Let $R = U(-, *)$ be the representable presheaf. Its elements are 1 and e, and e acts on them by $x \mapsto xe$, the constant map with value e. The endomorphism $\varepsilon : x \mapsto ex$ of R is idempotent with image the subpresheaf $Q = \{e\}$. The presheaf H is the pushout $R \amalg_Q R$ of two copies of R along Q: the element e is i, and the two copies of 1 are h_1, h_2.

Let $p : U \to C$ be any functor to a small category, and put $c = p(*)$, $d = p(e)$, so that $d^2 = d$. The functor $p_!$ preserves colimits, sends R to $C(-, c)$, and sends ε to the endomorphism $\alpha \mapsto d\alpha$. Every functor preserves the splitting of an idempotent. Hence $p_!Q$ is the subpresheaf

$$Q_d(v) = \{\alpha : v \to c \mid d\alpha = \alpha\} \subseteq C(v, c),$$

with its inclusion, and

$$p_!H \cong C(-, c) \amalg_{Q_d} C(-, c).$$

This pushout is formed objectwise along injections. The category of elements $E' = C/p_!H$ is therefore the union of two full sieves S_1, S_2, each isomorphic to C/c, with $S_1 \cap S_2 = C/Q_d$. A chain of arrows of E' lies in the sieve that contains its last object, so $NE' = NS_1 \cup NS_2$ and $NS_1 \cap NS_2 = N(C/Q_d)$. Each S_j has the terminal object (c, id_c). The category C/Q_d has the object (c, d), and the arrows $\alpha : (v, \alpha) \to (c, d)$ are the components of a natural transformation from the identity functor to the constant functor at (c, d): for an arrow $g : (v, \alpha) \to (w, \beta)$ one has $\beta g = \alpha$. Thus BS_1, BS_2 and their intersection are contractible, and so is BE'. Since p was arbitrary, H is totally W_∞-aspherical. Every fundamental localizer contains W_∞ (Cisinski, 2004, Théorème 2.2.11), which gives the first assertion.

The category $E = U/H$ has the three objects i, h_1, h_2 and, besides identities, the arrows

$$e_i : i \to i, \qquad u_1 : i \to h_1, \qquad u_2 : i \to h_2, \qquad e_i^2 = e_i, \quad u_j e_i = u_j.$$

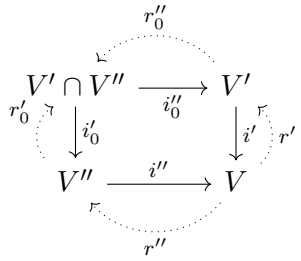
Let $q : E \to \Psi$ be the functor to the outgoing fork $\Psi = (a \leftarrow z \to b)$ of the commentary at ms. 100 which sends i, h_1, h_2 to z, a, b and e_i to the identity. It is bijective on objects and full. For an object y of Ψ with preimage $\tilde{y}$, every object $(x,\ y \to q(x))$ of $y\backslash E$ receives an arrow from $(\tilde{y}, \mathrm{id}_y)$, obtained by lifting $y \to q(x)$. So q is final. The inclusion $f : \Psi \to C$ into the four-object category of the same commentary is final and $BC \simeq S^2$. The composite $fq : E \to C$ is a final functor whose target is not aspherical for any of the localizers named. By the final-image criterion above, U/H is not totally aspherical for them. □

The objects h_1, h_2 of U/H have no common successor; this is the property that fails, and it cannot be seen by functors that factor through U. For the localizers W_0 and W_1 (bijections on π_0; bijections on π_0 and isomorphisms on all π_1) the converse of Proposition 8 does hold. A final functor induces a bijection on connected components, and a final image of a 1-connected category is 1-connected (the argument with coverings at ms. 101). If H is totally W_0-aspherical, the test $p = \mathrm{id}$ shows that U/H is nonempty and connected, and then so is every final image of U/H; for W_1 the same reasoning applies with 1-connectedness.

Je viens de vérifier la chose, et donne le **Corollaire 1** (rectification de `[1a]` prop. 9, partie b) ([68]) : *L'énoncé est valable si on suppose qu'il existe une* rétraction r'_0 *de* V'' *sur* $V' \cap V''$, p. 69

68. NB Il y a un énoncé légèrement plus général, en partant d'une catégorie U et d'un diagramme
$$\begin{array}{ccc} V' \cap V'' & \longrightarrow & V' \\ \downarrow & & \downarrow \\ V'' & \longrightarrow & V \end{array}$$
dans $U^\wedge$: si V', V'', $V' \cap V''$ sont totalement W-asphériques, il en est de même de V.

et de même une rétraction r''_0 de V' sur $V' \cap V''$. ([69])



(**NB** Nous verrons plus tard qu'il suffit de l'existence d'*une* de ces rétractions pour assurer la conclusion.)

Démonstration. Comme V est somme amalgamée de V' et V'', il existe une rétraction r' de V sur V' (donnée par $\mathrm{id}_{V'}$ sur V', par $i''_0 r'_0$ sur V'') qui induit r'_0 sur V'', et de même il existe une rétraction r'' de V sur V'', induisant r''_0 sur V'. L'existence de ces rétractions [page 99] assure déjà que pour tout foncteur $f : V \longrightarrow W$, le diagramme transformé de

$$\begin{array}{ccc} V' \cap V'' & \longrightarrow & V' \\ \downarrow & & \downarrow \\ V'' & \longrightarrow & V \end{array}$$

(intérprété comme diagramme dans $V^\wedge$) est formé de monomorphismes ([70])

$$\begin{array}{ccc} f_!(V' \cap V'') & \hookrightarrow & f_!(V') \\ \downarrow & & \downarrow \\ f_!(V'') & \hookrightarrow & f_!(V), \end{array}$$

et de plus que $f_!(V') \cap f_!(V'')$ (qui contient $f_!(V' \cap V'')$) lui est en fait *égal.* Il en résulte que l'on a un diagramme à la fois cartésien et cocartésien d'immersions ouvertes dans Cat p. 70

$$\begin{array}{ccc} W/f_!(V' \cap V'') & \hookrightarrow & W/f_!(V') \\ \downarrow & & \downarrow \\ W/f_!(V'') & \hookrightarrow & W/\Big(f_!(V) \overset{\text{déf}}{=} f_!(e_V)\Big), \end{array}$$

lequel s'identifie aussi à

$$\begin{array}{ccc} \underline{\pi}_0(V' \cap V''/W) & \hookrightarrow & \underline{\pi}_0(V'/W) \\ \downarrow & & \downarrow \\ \underline{\pi}_0(V''/W) & \hookrightarrow & \underline{\pi}_0(V/W). \end{array}$$

Commentary · Exposition [ms. 98]

Why a categorical retraction cannot be sent through this Kan extension

Grothendieck tries to protect a union from collapsing under $f_!$ by splitting its inclusions. Splitting is indeed preserved by every functor, but the retraction must be a morphism in the category on

69. Corollaire rectifié faux également - les rétractions ne sont pas des flèches de $V^\wedge$, et ne passent dans $p_!$.
70. **faux,** déjà dans le cas où $V = \Phi$, V', V'' ses deux ouverts canoniques isomorphes à $\mathbf{\Delta}^1$, et $V' \cap V'' = e$.

which that functor acts. Here $f_!$ acts on $V^\wedge$, whereas the retractions just constructed are functors between full subcategories of V.

For a sieve $S \subseteq V$, the associated subterminal presheaf $1_S \subseteq 1_V$ takes the value one on objects of S and the empty set elsewhere. If S omits an object of V, there cannot be a map $1_V \to 1_S$: at that object it would have to be a function from a singleton to the empty set. Thus a proper sieve, even if it admits a categorical retraction, does not define a split subobject of the terminal presheaf. A functor $r : V \to S$ moves the base object; a morphism in $V^\wedge$ must stay over it. This is exactly the defect identified by the source's corrective footnote.

What would suffice is a genuinely different hypothesis. Given a union $H = H_1 \cup H_2$ in a presheaf category, suppose that for every p the images of its three inclusion maps remain monomorphisms and the square remains Cartesian. Since $p_!$ preserves the pushout, the resulting categories of elements then form a union of two sieves with the expected intersection. The ordinary forward homotopy gluing lemma proves total W_∞-asphericity of H when the other three presheaves are totally W_∞-aspherical. This is the precise exactness condition the proposed proof needs. Categorical retractions provide none of it.

L'hypothèse sur V', V'', $V' \cap V''$ implique que les trois sommets du carré autres que [page 100]
$\underline{\pi}_0(V/W)$ sont W-asphériques, donc par W(7 bis) il en est de même de ce dernier, q.e.d.

Corollaire 2 ([71]). *Moyennant* W(7 bis)*, la catégorie* $\Psi = \begin{array}{ccc} & & a \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & b \end{array}$ *(ou* $0 \longleftarrow 1 \longrightarrow 2$*)*

est totalement W-asphérique ([72]).

On applique le corollaire 1 à Ψ et à ses deux sous-catégories ouvertes, Ψ/a, Ψ/b (isomorphes à $\mathbf{\Delta}^1$), et à leur intersection $\{0\} \simeq \mathbf{\Delta}^0 = e$. Comme ces trois catégories ont un objet final (contrairement à Ψ elle-même, qui n'a qu'un objet initial …), elles sont totalement W-asphériques. Le corollaire s'applique, l'existence des rétractions r'_0, r''_0 étant ici évidente.

C'est surtout la partie b) (rectifiée) de la page 9 qui assure l'existence de catégories totalement W-asphériques autres que celles ayant un objet final ([73]). Il est vrai qu'en sens inverse il faudrait quand même s'assurer qu'un objet X W-asphérique de Cat n'est pas *toujours* totalement W-asphérique! Le contre-exemple n'est pas tellement évident finalement - j'ai [page 101]
passé des heures avant d'en trouver un et d'y voir un peu clair! Il suffit en somme d'exhiber un exemple d'une flèche dans Cat

$$f : X \longrightarrow Y$$

telle que

p. 71

a) $\underline{\pi_0}(f) \overset{\text{déf}}{=} \underline{\pi}_0(X/Y) \xrightarrow{\sim} Y$, i.e. les $\pi_0(y\backslash X)$ `[plutôt les` $y\backslash X$`]` sont 0-connexes (pour $y \in \operatorname{Ob} Y$),

Commentary · Examples [ms. 100]

The finite fork counterexample, computed completely

Grothendieck eventually finds a very small witness to the failure of the preceding gluing claims. The construction is in his Appendix to Chapter XII, manuscript pp. 10–12, where the final corrective footnote identifies the homotopy type as S^2 (Grothendieck, 1990–1991p). We work it out here because it explains at once the failure of total asphericity and the failed use of retractions.

Let C have four objects x, z, a, b and the nonidentity morphisms generated by

$$\begin{array}{ccccc} x & \underset{\beta}{\overset{\alpha}{\rightrightarrows}} & z & \xrightarrow{\ r\ } & a \\ & & \downarrow{\scriptstyle s} & & \\ & & b & & \end{array} \qquad r\alpha = r\beta, \quad s\alpha = s\beta.$$

71. Plus généralement, si I est une catégorie *ordonnée finie,* ayant un objet initial, alors
72. **Faux,** Ψ n'est *pas* totalement W-asphérique.
73. Je conjecture à présent que ce sont les seules! ([74])
74. Faux, il y a les catégories filtrantes (si W satisfait l'axiome des limites).

Besides the four displayed arrows there are just the two composites $x \to a$ and $x \to b$. Let A be its full subcategory on z, a, b, the outgoing fork Ψ, and let $f : A \hookrightarrow C$ be the inclusion. The category A has initial object z, so BA is contractible.

We first compute $f_!1_A$. At a and b the outgoing comma category is a point. At z it has initial object (z, id). At x it has four objects

$$(z, \alpha),\ (z, \beta),\ (a, r\alpha),\ (b, s\alpha),$$

and one arrow from each of the first two to each of the last two. Its nerve is a four-edge circle. In particular it is nonempty and connected. Thus every $c \downarrow f$ is connected, and

$$f_!1_A = 1_C, \qquad \underline{\pi}_0(A/C) = C.$$

The map f is final in the ordinary sense, although its comma category at x is not contractible.

The homotopy type of C can be read without a spectral sequence. Let K be the full subcategory on x, z, with its two parallel arrows. Its nerve realizes as a circle. The full subcategories C_a on x, z, a and C_b on x, z, b each adjoin a final object to K. Their nerves are therefore cones on NK. They cover NC, with intersection NK, because there are no arrows between a and b. Consequently

$$BC = BC_a \cup_{BK} BC_b \cong \Sigma BK \simeq S^2.$$

Here the union description is literal at the level of nerves. Hence A is not totally W_∞-aspherical: its direct image along f has category of elements C, with $H_2(BC; \mathbb{Z}) \cong \mathbb{Z}$.

Now return to the proposed retraction repair. Inside A, put $A_a = \{z \to a\}$ and $A_b = \{z \to b\}$. Both have final objects, their intersection is $\{z\}$, and both admit retractions to that intersection. Proposition 9(a) therefore gives total asphericity of all three pieces. Nevertheless their union is A, which we have just shown is not total. The false step can be inspected at a single value. Write $y_c = C(-, c)$. Since left Kan extension carries a representable to the corresponding representable,

$$f_!(1_{\{z\}}) = y_z, \quad f_!(1_{A_a}) = y_a, \quad f_!(1_{A_b}) = y_b.$$

The equality $1_{A_a} = A(-, a)$ follows because a receives exactly one arrow from z and none from b, and similarly for b. Evaluating the first inclusion after Kan extension at x yields

$$\{\alpha, \beta\} = C(x, z) \longrightarrow C(x, a) = \{r\alpha = r\beta\}.$$

It is not injective. The other leg has the same defect. Their pushout is the singleton $1_C(x)$, exactly as preservation of colimits predicts. The putative intersection square has failed before any homotopy argument is applied.

The same finite construction also explains the two kinds of pushout failure announced on p. 81. Adjoin a final object to A, giving $A^\triangleright$. Pushing out $A \subset A^\triangleright$ along f adjoins a final object to C, because $f_!1_A = 1_C$. The upper inclusion is an aspherical weak equivalence; the lower inclusion $C \to C^\triangleright$ is not, since $BC \simeq S^2$.

To make the other leg a weak equivalence as well, replace the target of f by $C^\triangleright$, with final point c_+. The composite $p : A \to C^\triangleright$ is a weak equivalence between contractible categories. Its direct image of 1_A is the subterminal presheaf supported on C: it is one on every object of C and empty at c_+. The resulting pushout P is C with two distinct final cone points, c_+ and the newly attached point. Hence

$$BP \simeq \Sigma BC \simeq S^3.$$

Both $A \to A^\triangleright$ and p are weak equivalences, while neither map from $C^\triangleright$ or $A^\triangleright$ to P is. This proof uses the explicit nerve, so it does not depend on the invalid reverse implication of the one-point corollary on p. 85.

Commentary · Development (commentators') [ms. 100]

Testing the retracted assertions under their stated axiom

The fork example proves that ordinary weak contractibility is insufficient for total asphericity. There is a logical point still to check: several of the source's retracted conclusions explicitly assume W(7 bis), whereas W_∞ does not satisfy that axiom. To refute those conditional conclusions

themselves, rather than only their proposed proof, we need an example in a weak equivalence class for which the axiom holds. Fixed-coefficient homology provides such a test. The homological tools are standard; the application below is our explicit check of these conditional assertions, with no claim of priority.

A homological counterexample with W(7). [ours] Let R be a nonzero commutative ring, and let W_R consist of functors inducing isomorphisms on all homology groups of their nerves with coefficients in R. Then W_R is a fundamental localizer satisfying the full W(7). Nevertheless a union of two totally W_R-aspherical sieves with totally W_R-aspherical intersection need not be totally W_R-aspherical, even when both pieces retract onto their intersection.

Verification of the test class. Identities, two-out-of-three, and closure under retracts follow by applying homology. A category with a terminal object has contractible nerve and hence maps to the terminal category by an arrow in W_R. We spell out the relative Theorem A axiom, since that is the remaining condition needed here. Suppose $A \xrightarrow{u} B \to C$ is a triangle and every induced map $A/c \to B/c$ is an R-homology equivalence. For the composite $f : A \to C$, form the bisimplicial set with (p, q)-simplices

$$\big(a_0 \to \cdots \to a_p, \quad f(a_p) \to c_0 \to \cdots \to c_q\big).$$

Deleting the last a-vertex composes the crossing arrow with the image of that last arrow under f; deleting the first c-vertex composes on its other side. These rules, with the ordinary remaining faces and identity degeneracies, make the two simplicial directions commute. Take free R-modules, normalize in each direction, and use the usual sign in the total differential.

For each fixed a-chain, the second direction is the nerve of $f(a_p)\backslash C$, a category with initial object $\mathrm{id}_{f(a_p)}$. Its positive homology vanishes and its degree-zero homology is R. The spectral sequence that first takes this homology therefore identifies the homology of the total complex with $H_*(NA; R)$. In the other direction, for each fixed chain $c_0 \to \cdots \to c_q$, the first simplicial direction is $N(A/c_0)$. Consequently the map induced by u gives an isomorphism on that first homology page by the hypothesis on the comma categories. The comparison of the first-quadrant spectral sequences gives an isomorphism of total homology, and thus $H_*(NA; R) \cong H_*(NB; R)$. Convergence is sufficient here without any finiteness assumption: there are only finitely many bidegrees in any given total degree. This proves the relative localizer axiom. Finally the normalized-chain argument after ms. 79 proves W(7) for precisely this class W_R.

The counterexample. Use the finite functor $f : \Psi \to C$ computed in the preceding block, with $\Psi = (z \to a, z \to b)$. It is ordinary final, so $f_! e_\Psi = e_C$. But $BC \simeq S^2$, and

$$H_2(BC; R) = R \neq 0.$$

Hence Ψ is not totally W_R-aspherical. Its two full sieves $\{z \to a\}$ and $\{z \to b\}$, and their intersection $\{z\}$, each have a terminal object. Proposition 9(a), whose proof does not use W(7), makes all three totally W_R-aspherical. The constant functors from the two arrows to $\{z\}$ are retractions of the inclusions of the intersection. Thus both the union assertion and its proposed retraction repair fail, and the conclusion asserting totality of the outgoing fork fails as well. All this occurs in a fundamental localizer satisfying W(7), so the conditional qualifications in those statements do not save them. □

The same calculation keeps two different defects apart. The proof's use of left Kan extension to preserve a monomorphism is invalid before any weak equivalence class is chosen. The example above additionally shows that the resulting total-asphericity conclusion can be false after all its stated homotopical axioms have been imposed.

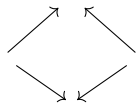
et telle que de plus

b) X est W-asphérique, et Y non.

La difficulté est la suivante, quand on cherche des contre-exemples 'idiots' : c'est que l'hypothèse a) implique déjà, quand X est W_∞-asphérique (ou seulement 1-connexe) que Y est 1-connexe : le foncteur

$$Y' \longmapsto X' = X \times_Y Y'$$

de la catégorie des revêtements étales Y' de Y vers celle des revêtements étales X' de X est pleinement fidèle (vérification immédiate), d'où le résultat annoncé. Donc un contre-exemple [page 102]

doit faire appel à l'existence d'un $\pi_i(Y,y)$ non nul, avec $i \geq 2$. Cela implique qu'il n'y a pas de contre-exemple avec des catégories de la forme $Y = Z/F$ (déduites d'un foncteur $q : X \longrightarrow Z$ en prenant $F = q_!(e_X) = \pi_0(X/Z)$), où Z est une catégorie telle que $\mathbf{\Delta}^1$ (?), Φ, Ψ, ou  (ayant le type d'homotopie du cercle) etc., car les composantes connexes de telles catégories Y sont des $K(\pi,1)$ - les π_i supérieures ($i \geq 2$) sont nuls.

Commentary · Exposition [ms. 101]

Why the first obstruction occurs above the fundamental group

Grothendieck's assertion about coverings explains why a counterexample with a circle as target cannot work. Let $f : A \to B$ be final, so every $b \downarrow f$ is nonempty and connected. Let $L, M : B \to \mathbf{Set}$ be local systems, meaning that every arrow is sent to a bijection. Given a natural transformation $\theta : Lf \to Mf$, choose any object $(a, \beta : b \to f(a))$ of $b \downarrow f$ and define

$$\bar{\theta}_b = M(\beta)^{-1}\theta_a L(\beta).$$

If $h : a \to a'$ is a morphism in that comma category, then $f(h)\beta = \beta'$ and the naturality of θ identifies the expressions obtained from (a,β) and (a',β'). Connectedness makes the expression independent of all choices. For $\gamma : b' \to b$, the object $(a, \beta\gamma)$ computes $\bar{\theta}_{b'}$, proving naturality in b. Its restriction to A is θ, using (a, id), and the same formula proves uniqueness. Thus restriction along f is fully faithful on local systems.

Finality also preserves the set of connected components: it preserves the colimit of the constant singleton diagram, which is π_0. Suppose now that BA is simply connected. Then BB is connected, and every pulled-back local system on A is constant. Fix a fibre of a local system L on B. Each element of that fibre determines a section of the pulled-back system; full faithfulness, applied to maps from the singleton local system, makes it descend uniquely to a global section of L. Every monodromy action is therefore trivial. Applying this to the regular action of $\pi_1(BB)$ shows that the group itself is trivial. This supplies the omitted verification on p. 101.

The four-object example above is consistent with this restriction: $BC \simeq S^2$ is simply connected but not contractible. Its critical comma category is a circle, and taking π_0 discards exactly that loop before rebuilding the total category. By contrast, if the target category is a discrete fibration over one of the height-one categories drawn on p. 102, it has no composable nonidentity arrows. Its nerve is a graph. Each connected component is a $K(\pi,1)$, so simple connectivity would force it to be contractible. The source's exclusion of these examples is thus correct for the displayed bases; it is not a statement about arbitrary categories merely sharing their homotopy type.

In the modern cofinality theorem, connectedness of the comma categories is strengthened to their weak contractibility (Lurie, 2009, Theorem 4.1.3.1). That stronger condition controls all homotopy colimits and implies weak equivalence of classifying spaces. The ordinary finality used here controls ordinary colimits and already constrains coverings, but it leaves the higher homotopy obstruction exposed.

Cette difficulté vue, le contre-exemple devient évident. Je pars d'un objet 1-connexe *quelconque* Y de X `[plutôt de` Cat`]`, et je prends $y_0 \in \mathrm{Ob}\, Y$,

$$X = \underline{\mathrm{Ch}}(Y; y_0, -) \xrightarrow{f} Y.$$

On sait que X est W_∞-asphérique, et que f est W_∞-parfait, donc les inclusions

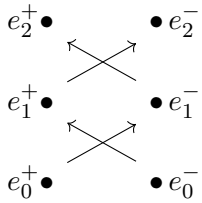
$$X_y \longrightarrow y\backslash X$$

sont des W_∞-équivalences. L'hypothèse a) que les $y\backslash X$ soient 0-connexes équivaut donc à la même hypothèse sur les fibres X_y, ou encore sur une seule fibre (f étant parfait, donc une W_∞-fibration, et Y 0-connexe), soit [page 103]

$$X_{y_0} \simeq \Omega(Y, y_0).$$

Sa 0-connexité équivaut à l'hypothèse de 1-connexité de Y (on n'a utilisé encore que celle de 0-connexité). Donc pour un contre-exemple, il suffit de prendre Y 1-connexe et non ∞-connexe. Le cas le plus simple est celui où Y a le type d'homotopie de la sphère S^2, réalisé p. ex. par la catégorie ordonnée ci-contre `[ci-dessous]` à 6 sommets.

p. 72



`[de plus,` $e_0^+ \leq e_1^+ \leq e_2^+$`,` $e_0^- \leq e_1^- \leq e_2^-$`]` ([75]). Je résume le résultat obtenu.

Proposition 10. *Soit Y une petite catégorie 1-connexe ($y_0 \in \mathrm{Ob}\, Y$). Alors $\underline{\mathrm{Ch}}(Y; y_0, -)$ est W_∞-asphérique, mais n'est pas totalement W_∞-asphérique si Y elle-même n'est ∞-connexe, i.e. W_∞-asphérique.* ([76]).

On peut se demander si Y W_∞-asphérique implique que $\underline{\mathrm{Ch}}(Y; y_0, -)$ est totalement W_∞-asphérique. J'en doute fort!

Revenons maintenant à la situation du début de la section, diagramme $(*)$, p. 87, [page 104]

$$(*) \qquad \begin{array}{ccccc} U & \xhookrightarrow{i} & X & \xleftarrow{j} & T = \{t\} \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots\ \text{iso} \\ U' & \xhookrightarrow{i'} & X' & \xleftarrow{j'} & T' = \{t'\}. \end{array}$$

Nous pouvons donner maintenant un exemple où i est dans W (et même W-asphérique) sans que i' le soit :

$$(**) \qquad i \in W, \qquad i' \notin W.$$

On peut même supposer que t soit objet final de X (ce qui signifie que l'objet H de $U^\wedge$ est isomorphe à e_U, l'objet final), et t' objet final de X' (i.e. $p_!(e_U) = e_{U'}$, i.e. les $u'\backslash U$ sont 0-asphériques). Dans ce cas, comme X, X' sont W-asphériques, les relations $(**)$ équivalent à

$$(***) \qquad U \text{ est } W\text{-asphérique, et } U' \text{ non,}$$

ce qui est justement la situation que nous venons de construire (p. 101-103).

Commentary · Exposition [ms. 103]

The information lost by taking components

Grothendieck has found the obstruction that the preceding attempts were missing: an ordinary direct image of a set-valued presheaf remembers connected components of comma categories, whereas a weak equivalence of nerves must preserve all homotopy groups. The path construction makes the difference particularly visible. Put $P = \underline{\mathrm{Ch}}(Y; y_0, -)$ and let $f : P \to Y$ be its endpoint functor. The earlier path theory supplies the two facts used here: BP is contractible and the homotopy fibre of Bf over y is a based path space from y_0 to y. We write $BA = |NA|$ for the classifying space of a category. The comparison of strict fibres with comma categories is the additional consequence of the perfectness hypothesis; contractibility of BP alone would not supply that comparison.

75. C'est en somme cet exemple (à homotopie topologique près) que je réalise (sans d'abord m'en apercevoir dans l'Appendice p. 10-12, mais avec des catégories finies).

76. Mais un exemple plus percutant est $\Psi = \begin{array}{ccc} & & b \\ a & \nearrow & \\ & \searrow & \\ & & c \end{array}$, qui est W_∞-asphérique, mais pas totalement!

For connected Y, the components of each such path space form a torsor for $\pi_1(BY, y_0)$. Thus simple connectivity gives

$$(f_!e_P)(y) = \pi_0(y\backslash P) = \{*\}.$$

Consequently $f_!e_P = e_Y$, and the category of elements of this direct image is Y itself. If BY is not contractible, this single functor f disproves total W_∞-asphericity of P. This argument uses precisely the universal quantifier in Definition 1: the homotopy type of P cannot determine how all its ordinary direct images behave.

The six-element ordered set displayed in the text is a useful finite model of the required target. Its elements occur in three levels $\{e_k^+, e_k^-\}$, $k = 0, 1, 2$, and every element of a lower level precedes every element of a higher level. Its order complex is therefore

$$S^0 * S^0 * S^0 \cong S^2.$$

There are six vertices, twelve edges, and eight triangular faces. Each simplex chooses at most one vertex in each level, so these are exactly the faces of the boundary of an octahedron. The two chains supplied in the editors' brackets are essential: the diagonal arrows alone would not specify this ordered set.

The finite fork construction in the commentary at ms. 100 supplies the same failure without a path category: a contractible Ψ admits a functor $\Psi \to C$ with connected undercategories and $BC \simeq S^2$. We shall use it below. The question about $\underline{\mathrm{Ch}}(Y; y_0, -)$ when BY is contractible is a different question. Proposition 10 gives no implication in that case, and a replacement of the path category by some equivalent contractible category would not answer it. The next commentary answers it, negatively, with this same fork.

Commentary · Development (commentators') [ms. 103]

The path category over a contractible base need not be total

The question that closes this page, whether $\underline{\mathrm{Ch}}(Y; y_0, -)$ is totally W_∞-aspherical when Y is W_∞-aspherical, has a negative answer, and the witness is the category that Grothendieck names in the footnote to Proposition 10.

Proposition. [ours] *Let $Y = \Psi$ be the outgoing fork $a \leftarrow z \rightarrow b$ and $y_0 = z$. Then BY is contractible, and $P = \underline{\mathrm{Ch}}(Y; z, -)$ is W_∞-aspherical and not totally W_∞-aspherical. It is not totally W_R-aspherical for any nonzero commutative ring R either.*

Proof. The object z is initial, so BY is contractible; in particular Y is 1-connected. Proposition 10 applies: P is W_∞-aspherical, and, as recalled in the preceding commentary, the endpoint functor $f : P \to Y$ satisfies $f_!e_P = e_Y$, that is, every $y\backslash P$ is nonempty and connected. So f is final. The inclusion $g : \Psi \to C$ of the commentary at ms. 100 is final, and $BC \simeq S^2$. Final functors compose, because $(gf)_!e_P = g_!f_!e_P = g_!e_Y = e_C$. The category of elements of $(gf)_!e_P$ is C, which is neither W_∞-aspherical nor W_R-aspherical. □

The proof uses only two facts: the endpoint functor of the path category over a 1-connected base is final, and total asphericity passes to final images. Hence $\underline{\mathrm{Ch}}(Y; y_0, -)$ is totally W_∞-aspherical only if Y is. Since total asphericity of Y is far stronger than asphericity (for a category with finitely many objects it means that Y is filtered; see the commentary on Chapter II, ms. 36), the path construction does not produce totally aspherical categories from aspherical ones.

On aimerait aussi trouver un exemple où dans (∗) on ait $p \in W$, $q \notin W$.

Je prendrai encore $(X, U, \{t\})$ tel que t soit objet final de X, mais si on veut $q \notin W$, on ne peut supposer t' élément final de X'. Au lieu de cela, je vais supposer p. 73 [page 105]

a) U, U' (et aussi X, ayant objet final t) W-asphériques, donc $i \in W$ (et même i W-asphérique), $p \in W$, mais

b) X' non W-asphérique, donc $i', q \notin W$.

Pour ceci, notons que

$$U'/t' \simeq U'/p_!(e_U) \overset{\text{déf}}{=} \underline{\pi}_0(U/U').$$

Le corollaire de la proposition 7 (cf. p. 85), non encore démontré, nous dit que $i' \in W$ (i.e. ici, X' W-asphérique) si et seulement si i' est W-asphérique, i.e. U'/t' est W-asphérique. Nous allons l'admettre ici. Tout revient donc à trouver une flèche

$$p : U \longrightarrow U'$$

dans Cat avec les propriétés suivantes.

a) U et U' sont W-asphériques,

b) $\underline{\pi}_0(U/U')$ n'est pas W-asphérique.

Nous allons reprendre l'exemple précédent

$$f : X \longrightarrow Y$$

(p. 101-103), en prenant

$$U = X, \qquad \underline{\pi}_0(U/U') = Y,$$

donc tout revient, à partir de l'exemple donné $X \xrightarrow{f} Y$, à construire une flèche [page 106]

$$(*) \qquad \pi : Y \longrightarrow U'$$

dans Cat qui satisfasse les conditions

a) π est un isomorphisme local,

b) U' est W-asphérique.

On prendra alors

$$p = \pi f.$$

Mais par la façon dont X a été construit à partir d'un Y 1-connexe et non ∞-connexe, par ailleurs quelconque, on est ramené finalement à trouver $(*)$ satisfaisant a), b), et de plus

c) Y est 1-connexe, mais non ∞-connexe. p. 74

(NB On suppose maintenant $W = W_\infty$.) Il revient au même de trouver

U' objet *asphérique* de Cat, $F \in \mathrm{Ob}\, U'^{\wedge}$ tels que U'/F soit 1-connexe et non ∞-connexe.

On songe par exemple à prendre simplement une sous-catégorie ouverte U'_0 de U', qui soit 1-connexe et non ∞-connexe. Partons d'une telle catégorie U'_0 - par exemple une réalisation homotopique dans Cat, de la sphère S^2. On l'immerge comme ouvert dans un U' asphérique, en lui ajoutant un objet final. On a gagné.

Faisons un énoncé en forme. [page 107]

Proposition 11. *On peut trouver une situation* $(*)$ (p. 124 `[plutôt 104]`) *avec*

a) t objet final de X.

b) U, U' (et X) W_∞-asphériques, donc $i, p \in W_\infty$ (et même i W_∞-asphérique).

c) X' non W_∞-asphérique, donc $i', q \notin W_\infty$.

Il suffit pour cela de partir d'un objet 1-connexe Y de Cat, de lui adjoindre un objet final u'_0, ce qui donne U', de prendre $U = \underline{\mathrm{Ch}}(Y; y_0, -)$, $p : U \longrightarrow U'$ le composé $U \xrightarrow{\text{but}} Y \xhookrightarrow[\text{can.}]{\text{incl.}} U'$, et X déduit de U en lui ajoutant un objet final t. On a alors $U'/t' \simeq Y$, d'où

$$X' \ W_\infty\text{-asphérique (i.e. } i' \in W_\infty) \Longleftrightarrow Y \ W_\infty\text{-asphérique.}$$

On a le contre-exemple cherché en prenant Y (1-connexe mais) non ∞-connexe.

Commentary · Examples [ms. 107]

Two cones and the two failures of pushout invariance

Grothendieck now converts the failure of total asphericity into failures of ordinary categorical pushouts. The two constructions deserve to be separated because they test different edges of the square. We can make both finite by using the functor $f : \Psi \to C$ constructed at ms. 100. Here $\Psi = (z \to a, z \to b)$ has an initial object, $BC \simeq S^2$, and every category $c\backslash\Psi$ is connected and nonempty. Hence

$$f_! e_\Psi = e_C.$$

The crucial point is that this is an equality of ordinary presheaves; f is not a weak equivalence.

For the first square take $U = \Psi$, $U' = C$, and let $X = \Psi^\triangleright$ be obtained by adjoining a new terminal object t. The gluing weight for $U \subset X$ is e_Ψ. The pushout formula proved in §1 says that $X' = X \amalg_\Psi C$ has gluing weight $f_! e_\Psi = e_C$. Thus $X' = C^\triangleright$, with its new terminal object t'. Both X and X' are contractible. The inclusion $i : \Psi \to \Psi^\triangleright$ is a weak equivalence and is even W_∞-aspherical: its slices over old objects have terminal objects and its slice over t is Ψ. But $i' : C \to C^\triangleright$ is not a weak equivalence, because its source has the homotopy type of S^2. This proves the assertion at ms. 104 without any converse to the gluing lemma.

For the second square first adjoin a terminal object s to C, and write $U' = C^\triangleright$. Let $j : C \hookrightarrow U'$ be the resulting sieve and put $p = jf$. Now U, U' and $X = U^\triangleright$ are all contractible, so both i and p are weak equivalences. Since j is a sieve, extension by $j_!$ sends e_C to the subterminal presheaf which is a singleton on C and empty at s. Indeed, $c\backslash C$ has an initial object for $c \in C$, whereas $s\backslash C$ is empty. Therefore

$$p_! e_U = j_! e_C, \qquad (p_! e_U)(c) = \{*\}, \quad (p_! e_U)(s) = \varnothing.$$

The pushout $X' = X \amalg_U U'$ is consequently obtained from C by adjoining two distinct objects s, t' above every object of C, with no arrow between s and t'. It is the union of two terminal cones on C.

This description computes the homotopy type directly. Every composable string in X' lies in one of the two cones, because a string cannot contain both of their incomparable apexes. Hence

$$NX' = N(C^\triangleright_s) \amalg_{NC} N(C^\triangleright_{t'}).$$

The two inclusions of NC are simplicial monomorphisms, so this ordinary union is a homotopy pushout. Its realization is the unreduced suspension of BC, and therefore

$$BX' \simeq \Sigma BC \simeq S^3.$$

Neither $i' : U' \to X'$ nor $q : X \to X'$ is a weak equivalence. This proves Proposition 11 at ms. 107 and shows exactly how an arbitrary sieve can destroy pushout invariance even when the other three categories are contractible.

The text invokes the converse in Proposition 7 to justify the second construction. That converse depends on the stronger gluing axiom discussed at ms. 78–85 and cannot be used as an unconditional theorem for W_∞. The suspension computation repairs this use of it. More generally, for the simply connected Y of the manuscript, ΣBY is contractible exactly when BY is contractible: suspension shifts reduced integral homology, and a simply connected acyclic CW complex is contractible by the Hurewicz and Whitehead theorems. The hypothesis of simple connectivity is doing real work in this last equivalence.

2.4 Application aux immersions ouvertes dwyériennes.

Donnons quand même l'enoncé *positif* auquel nous en avions.

Théorème 3. *Soient (X, U, T) un triple $\in \mathcal{T}$ (U ouvert de X dans* Cat*, $T = X\backslash U$), et pour toute flèche $p : U \longrightarrow U'$ flèche de* Cat*, le carré cocartésien dans* Cat

$$\begin{array}{ccccc} U & \xhookrightarrow{i} & X & \dashleftarrow & T \\ {\scriptstyle p}\downarrow & & {\scriptstyle q}\downarrow & & \vdots\,\text{iso} \\ U' & \xhookrightarrow{i'} & X' & \dashleftarrow & T'. \end{array}$$

p. 75
[page 108]

Supposons que pour tout $t \in \dot{T} = \dot{U}$, le préfaisceau d'ensembles

$$i^*(t) = (u \longmapsto \mathrm{Hom}(u,t)) \in U^\wedge$$

soit totalement W-asphérique (déf. 1, p. 88) ([77]) - p. ex. que la catégorie U/t soit totalement W-asphérique (prop. 8, page 93). (NB le cas où les U/t ont un objet final, cf. prop. 9 a), i.e. celui où l'inclusion $U \longrightarrow \bar{U}$ admet un adjoint à droite, est celui envisagé par Dwyer, celui des immersions ouvertes de Dwyer ([78]).) Alors pour toute $p : U \longrightarrow U'$, l'immersion ouverte

$$i'_0 : U' \longrightarrow \overline{U'}$$

est W-asphérique (et même, tout comme i_0, strictement W-asphérique), et a fortiori elle est [dans] *W. Par suite (prop. 6* [p. 81]*) on a ceci.*

ⓐ $\quad i \in W \iff i' \in W \quad$ *moyennant* W(7 bis).
ⓑ $\quad p \in W \iff q \in W \quad$ *moyennant* W(7). ([79])

Commentary · Exposition [ms. 108]

A condition on weights that survives every direct image

After the counterexamples, Grothendieck seeks a hypothesis expressed in the actual gluing data. For a sieve $i : U \hookrightarrow X$, write $\overline{U}$ for the cosieve generated by U, and put $\dot{T} = \overline{U} \setminus U$. An object t belongs to $\dot{T}$ precisely when at least one arrow $u \to t$ exists with $u \in U$. Its gluing weight is

$$H_t = \mathrm{Hom}_X(i(-), t) \in \widehat{U}.$$

Only these objects need testing. At $u \in U$ the weight is representable and therefore totally aspherical; outside $\overline{U}$ it is empty and no assertion of asphericity is intended.

The hypothesis of Theorem 3 is that every H_t, $t \in \dot{T}$, remains aspherical after every ordinary left Kan extension. A pushout along $p : U \to U'$ replaces this weight by $p_!H_t$. Its total asphericity follows without a gluing axiom: for a second functor $r : U' \to V$,

$$r_!(p_!H_t) \cong (rp)_!H_t,$$

and the right-hand side is aspherical by the original hypothesis. This calculation explains both the terminology "strictly aspherical" for $U \to \overline{U}$ and its stability under arbitrary pushout.

The support is preserved as well. If H_t is nonempty, a representative $h \in H_t(u)$ produces a class at $p(u)$ in $p_!H_t$; a colimit of sets cannot delete that class. If H_t is empty, its direct image is empty. Thus the boundary objects of the two pushouts identify with the same $\dot{T}$, and the preceding test covers exactly the new boundary. It follows that

$$i'_0 : U' \longrightarrow \overline{U'}$$

is strictly W-aspherical and hence W-aspherical. The claim at the start of the proof on ms. 112 that i' itself is aspherical must read i'_0. A W-dwyérienne inclusion need not be a weak equivalence: the empty sieve into a nonempty category already demonstrates the distinction.

The two equivalences printed as (a) and (b) have a different logical status from this weight calculation. They use respectively W(7 bis) and W(7), exactly as stated. Those additional axioms must remain visible; they are not automatic properties of weak homotopy equivalences of nerves. The modern Dwyer pushout theorem gives the useful forward implications in its own hypotheses, as discussed in the later-development block. The intrinsic statement here, valid before imposing either gluing axiom, is preservation of the condition on all the weights.

77. **NB** On dit sous ces conditions que $i_0 : U \longrightarrow \bar{U}$ est *strictement W-asphérique* (cf. déf. 2 page suivante).

78. Je ne connais pas d'autre cas où th. 3 s'applique - faute de connaître de catégorie totalement W-asphérique qui n'ait un objet final, ni d'objet F d'une catégorie $U^\wedge$ qui soit totalement W-asphérique sans être représentable, i.e. U/F a un objet final. À présent, je sais que les catégories *filtrantes* sont totalement W-asphériques si W satisfait l'axiome des limites inductives.

79. Démonstration page 112.

Commentary · Examples [ms. 108]

A boundary with no last object

The footnote about filtered categories points beyond the original adjoint criterion. The commentary on Chapter II, manuscript pp. 35–36 ("Left Kan extension carries the ind-presentation" and "The quantifier in total asphericity"), explains why filtered presentations make this universal condition tractable. Here their role can be seen in one concrete inclusion.

Let U be the ordered set $\mathbb{N}$ with its usual order, and let $X = \mathbb{N} \cup \{\infty\}$ with $n < \infty$ for every n. The inclusion $i : U \hookrightarrow X$ is a sieve and $\overline{U} = X$. For an old object n, the slice U/n has terminal object n. For the new object ∞, the slice is U itself. For any $p : \mathbb{N} \to B$, one has

$$F := p_! e_{\mathbb{N}} \cong \varinjlim_{n} \mathrm{Hom}_B(-, p(n)).$$

Write $p_{nm} : p(n) \to p(m)$ for the transition map when $n \leq m$, and let $E = B/F$. This category is nonempty, since it contains $(p(0), [\mathrm{id}_{p(0)}])$. Given finitely many objects (b_j, x_j) of E, choose representatives $g_j : b_j \to p(n_j)$ of their elements. For $m \geq n_j$ for every j, the arrows $p_{n_j m} g_j$ give morphisms from all these objects to $(p(m), [\mathrm{id}_{p(m)}])$.

For parallel morphisms $h, k : (b, x) \rightrightarrows (c, y)$ in E, choose a representative $g : c \to p(n)$ of y. The defining condition on these morphisms gives $[gh] = x = [gk]$ in $F(b)$. Equality in this filtered colimit means that, for some $m \geq n$,

$$p_{nm} gh = p_{nm} gk.$$

Thus $p_{nm} g : (c, y) \to (p(m), [\mathrm{id}_{p(m)}])$ equalizes h and k. These are exactly the nonemptiness, finite-cocone, and parallel-arrow tests for a filtered category. A filtered category has weakly contractible nerve: a cocone on the finite data of a simplicial sphere map contracts that map. Thus every such E is W_∞-aspherical.

It follows that i is totally W_∞-dwyérienne. It is also a couniversal weak equivalence, without using either additional gluing axiom. After any pushout its slice over the new object is the filtered category of elements just computed, while its slices over the old objects have terminal objects. Quillen's Theorem A therefore applies to every pushout inclusion. It is not a Dwyer immersion in the original adjoint sense. A right adjoint $r : X \to U$ to i would require

$$n \leq r(\infty) \quad \text{for every } n \in \mathbb{N},$$

which is impossible. This distinguishes the two classes even though both $\mathbb{N}$ and $\mathbb{N}^{\triangleright}$ are contractible.

The negative answer also holds within the full gluing framework assumed in the source's question. Let $W_{\mathbb{Z}}$ denote integral-homology equivalences of nerves. Every pushout just considered is a weak homotopy equivalence and therefore an integral-homology equivalence, so i is couniversally a $W_{\mathbb{Z}}$-equivalence. Every tested category of elements is already weakly contractible, so i is also totally $W_{\mathbb{Z}}$-dwyérienne. The development at ms. 100, *Testing the retracted assertions under their stated axiom*, verifies that $W_{\mathbb{Z}}$ is a fundamental localizer satisfying full W(7). The failure of the right-adjoint condition is independent of the choice of weak equivalences. Thus the proposed recovery of an original Dwyer immersion fails even with W(7) imposed. The distinct converse from an arbitrary homotopical cofibration to Grothendieck's condition on every boundary weight is not established by this example.

Corollaire 1. *Supposons* W(7). *Si $i : U \hookrightarrow X$ est une immersion ouverte telle que $i_0 : U \hookrightarrow \bar{U}$ soit totalement* `[il suffit strictement]` *W-asphérique, alors $i : U \longrightarrow X$ est une W-cofibration ('ultraforte'), par quoi j'entends ceci : pour toute flèche $U' \xrightarrow{f} U''$ dans $U\backslash\mathrm{Cat}$,*

telle que $f \in W$, la flèche $g : X' \longrightarrow X''$ qui s'en déduit par cochangement de base à partir de l'objet X de $U\backslash\mathrm{Cat}$, est aussi dans W. [page 109]

Démonstration du corollaire : on applique le théorème 3 à l'immersion ouverte $U' \xhookrightarrow{i'} X'$ déduite de $U \xhookrightarrow{i} X$ par le cochangement de base $U \xrightarrow{f} U'$, en notant de plus que l'hypothèse du théorème 3 sur i est préservée en passant de i à i'.

Il faut quand même introduire une terminologie adéquate :

Définition 2 :

a) Une flèche $f : X \longrightarrow Y$ est dite *strictement W-asphérique* si pour toute $y \in \mathrm{Ob}\, Y$, $f^*(y) = (x \longmapsto \mathrm{Hom}_Y(f(x), y))$ est un objet totalement W-asphérique de $X^\wedge$ (cf. déf. 1, page 88). Elle est dite *totalement W-asphérique* si pour tout $y \in \mathrm{Ob}\, Y$, X/y ($\simeq X/f^*(y)$) est un objet totalement W-asphérique de Cat.

b) Une immersion ouverte $i : U \longrightarrow X$ est dite *W-dwyérienne* (resp. *totalement* `[W-]` *dwyérienne*) si l'immersion induite $i_0 : U \longrightarrow \bar{U}$ est strictement (resp. totalement) W-asphérique. Une immersion fermée $j : T \longrightarrow X$ est dite *W-dwyérienne* (resp. *totalement W-dwyérienne*) si et seulement si $j^\circ : T^\circ \longrightarrow X^\circ$ est une immersion ouverte W-dwyérienne (resp totalement W-dwyérienne). p. 76

c) Une flèche $i : U \longrightarrow X$ de Cat (pas nécessairement une immersion) est appelée une *W-cofibration,* si pour toute flèche $f : U' \longrightarrow U''$ dans $U\backslash$Cat, la flèche $g : X' \longrightarrow X''$ qui s'en déduit par le cochangement de base par X (objet de $U\backslash$Cat) est dans W quand f l'est. [page 110]

NB Cette notion de W-cofibration est stable par composition et par cochangement de base, et les isomorphismes de Cat sont des W-cofibrations. Si f est une W-cofibration, f° l'est et inversement.

(Le corollaire s'énonce en disant qu'une immersion ouverte ou fermée qui est W-dwyérienne est une W-cofibration.)

d) Une flèche $i : U \longrightarrow X$ dans Cat est appelée une *W-cofibration triviale*, ou une *W-équivalence couniverselle,* si pour toute U' dans $U\backslash$Cat, i.e. toute flèche $p : U \longrightarrow U'$ dans Cat, la flèche déduite de i,

$$i' : U' \longrightarrow X' = U' \amalg_U X' \quad \texttt{[plutôt } U' \amalg_U X\texttt{]} \quad ,$$

est dans W. (NB Il est immédiat que cela implique bien que i est une W-cofibration.)

(Cette notion de W-cofibration triviale est elle aussi stable par composition et par cochangement de base, et les isomorphismes dans Cat en sont.)

Commentary · Exposition [ms. 110] [page 111]

What the cofibration condition quantifies over

Grothendieck's definition concerns the entire undercategory, and this is why he calls the condition "ultraforte". For $i : U \to X$, let

$$i_! : U\backslash\mathrm{Cat} \longrightarrow X\backslash\mathrm{Cat}, \qquad (U \to A) \longmapsto (X \to X \amalg_U A).$$

The condition is that this functor preserve the weak equivalences of underlying categories. In particular it tests a weak equivalence $A \to B$ after an arbitrary preliminary change of base $U \to A$. Merely requiring preservation of weak equivalences whose source is U would be weaker. The modern terminology and its relation to model-category cofibrations are treated in the later-development block.

The elementary closure assertions follow from the universal properties of pushouts. If $U \to X \to Z$ are two such maps, pushout along the composite is the composite of their pushout functors, up to the canonical isomorphism, so it preserves weak equivalences. If $U \to V$ is arbitrary and $i' : V \to X \amalg_U V$ is the pushout of i, then for $A \to B$ under V its two pushouts along i' are canonically the pushouts along i of the same map considered under U. This proves stability under change of base. Isomorphisms give equivalences of the undercategories. Passage to opposite categories proves the duality assertion provided that W is invariant under opposites, as in the standing localizer framework.

The adjective "trivial" is defined by a separate universal property: every pushout $A \to X \amalg_U A$ of i belongs to W. This implies the preceding preservation condition by two-out-of-three. Indeed the natural square

$$\begin{array}{ccc} A & \longrightarrow & X \amalg_U A \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle g} \\ B & \longrightarrow & X \amalg_U B \end{array}$$

has both horizontal maps in W; when $f \in W$, commutativity and two-out-of-three give $g \in W$. Stability of these universal weak equivalences under composition and pushout follows by factoring each new pushout into two pushouts of the original maps.

Without an additional hypothesis one should not replace this definition by "a W-cofibration which belongs to W". Corollary 2 establishes that implication for the W-dwyérienne sieves considered there, using Theorem 3 and the stated gluing assumptions. This is the reason for retaining the full definition even though its name resembles the terminology of a model category.

Commentary · Later developments [ms. 110]

The later theory of homotopy cofibrations

Grothendieck wants a property of an attachment that is visible from weak equivalences alone. This is exactly the property now called an *h-cofibration*: the functor of cobase change on undercategories preserves weak equivalences. The terminology and a general model-categorical treatment appear in Batanin and Berger (2017, Definition 1.1), who explicitly acknowledge the antecedent in Grothendieck's manuscripts. In a left proper model category their Proposition 1.6 and Corollary 1.9 identify this condition with the requirement that every pushout along the map compute a homotopy pushout. Their Lemma 1.10 also proves that an h-cofibration which is a weak equivalence is couniversal. These are precise later replacements for the useful direction of the present gluing programme; none requires the reverse implication of W(7 bis). The same undercategory definition is retained as "W-cofibration" by Manetti and Meazzini (2019, Definition 2.1 and Lemma 2.4).

The definition here must be kept distinct from the cofibrations of a chosen Quillen model structure. For example, for every small category C, the map $\varnothing \to C$ is a W_∞-cofibration: its cobase-change functor is $A \mapsto A \amalg C$, and the nerve of a disjoint union is a disjoint union of nerves. A weak equivalence on the first summand and the identity on the second induce a weak equivalence on the union, component by component. Thus every category is cofibrant for Grothendieck's *relative* notion. This statement does not assert that every category is cofibrant in Thomason's model structure. The distinction explains why a satisfactory description of these attachments can be broader than the particular generating cofibrations used to construct a model category.

Corollaire 2. (Moyennant W(7).) *Soit $i : U \longrightarrow X$ une immersion ouverte W-dwyérienne. Pour que i soit une W-cofibration triviale, il faut et il suffit que $i \in W$, ou ce qui revient au même* (cf. page 82), *que $\dot{T} \hookrightarrow T$ soit dans W.*

C'est évidemment nécessaire, et la suffisance résulte du théorème 3 a).

Questions ouvertes.

a) J'ignore si une immersion ouverte i qui est une W-cofibration, voire une W-cofibration triviale, est forcément W-dwyérienne ([80]). Mais par le corollaire de la proposition 7 (p. 85), je sais que c'est le cas si on suppose T isomorphe à la catégorie finale e. Il s'ensuit toute au moins ceci : Pour que non seulement $i : U \longrightarrow X$, mais aussi toutes les immersions ouvertes $i_1 : U \longrightarrow X_1$ qui s'en déduisent par le procédé de prop. 7 b) au moyen de flèches quelconques $T_1 \longrightarrow T$ de but T dans Cat, soient des W-cofibrations, il faut et il suffit que i soit dwyérienne.

b) Y-a-t-il une bonne notion d' immersion dwyérienne, applicable à toutes les immersions p. 77 [page 112]

Commentary · Development (commentators') [ms. 111]

The open questions of this page

Question a) asks whether a sieve that is a W-cofibration, or even a trivial W-cofibration, must be W-dwyérienne, and asserts this when the complement T is the final category. Three small examples separate the notions. In each of them $U = \{u\}$ is a single object, so a functor $p : U \to A$ is the choice of an object $c = p(u)$ of A.

Proposition (a trivial cofibration that is not dwyérienne). [ours] *Let X have objects u, t, s and nonidentity arrows $\alpha, \beta : u \to t$, $\gamma : t \to s$, $\delta : u \to s$, with $\gamma\alpha = \gamma\beta = \delta$. The*

80. et peut-être même une immersion de Dwyer?

sieve $U = \{u\} \hookrightarrow X$ is a trivial W-cofibration for every fundamental localizer W. It is not W-dwyérienne when W is W_∞, a homology localizer W_R with $R \neq 0$, or the class W_n of n-equivalences for some $n \geq 0$.

Proof. The presheaf $v \mapsto \mathrm{Hom}_X(v,t)$ on U has two elements, so U/t is a discrete category with two objects; the test $p = \mathrm{id}$ shows that this presheaf is not totally W-aspherical for the localizers named. Let $p : U \to A$ be arbitrary, $c = p(u)$, and $P = X \amalg_U A$. Its objects are those of A and t, s; for $v \in A$,

$$P(v,t) = A(v,c) \times \{\alpha, \beta\}, \qquad P(v,s) = A(v,c), \qquad P(t,s) = \{\gamma\},$$

composition with γ forgets the second coordinate, and there are no arrows from t or s to A. Let $j : A \to P$ be the inclusion and $r : P \to A$ the functor that is the identity on A, sends t, s to c, projects the two crossing sets to $A(v,c)$ and sends γ to id_c. Let $F : P \to P$ be the identity on A, send t and s to s, each crossing arrow to its composite into s, and γ to id_s. There are natural transformations

$$\mathrm{id}_P \Longrightarrow F \Longleftarrow jr,$$

both the identity on A; the first has components γ at t and id_s at s, the second has at t and at s the arrow $c \to s$ corresponding to id_c. Naturality at a crossing arrow (g, α) or (g, β) is the relation $\gamma\alpha = \gamma\beta = \delta$. Since $rj = \mathrm{id}_A$, the map Bj is a homotopy equivalence. So every cobase change of $U \to X$ is in $W_\infty \subseteq W$. □

This answers the first sentence of question a) negatively; the footnote's stronger hope fails with it, since a Dwyer immersion is dwyérienne.

Proposition (the case $T \simeq e$). [ours] *Let X_0 have objects u, t and exactly two nonidentity arrows $\alpha, \beta : u \to t$, and let $U = \{u\}$, $T = \{t\}$. For $W = W_\infty$ and for $W = W_R$, the sieve $U \hookrightarrow X_0$ and every sieve $U \hookrightarrow X_1$ deduced from it by a change of complement $T_1 \to T$ (Proposition 7 b)) is a W-cofibration, and none of them with $T_1 \neq \varnothing$ is W-dwyérienne.*

Proof. The category X_1 has the objects u and those of T_1, the arrows of T_1, and two arrows $\alpha_x, \beta_x : u \to x$ for every $x \in T_1$, compatible with the arrows of T_1. Each crossing presheaf has two elements, so no U/x is connected. For $p : U \to A$ with $c = p(u)$, the pushout $P_A = X_1 \amalg_U A$ has $P_A(v,x) = A(v,c) \times \{\alpha, \beta\}$ for $v \in A$, $x \in T_1$. A chain of arrows of P_A either lies in A, or lies in T_1, or consists of a chain in A, a crossing arrow (g, λ) with $\lambda \in \{\alpha, \beta\}$, and a chain in T_1; the chain in A together with g is a chain of A/c. Write $J = N(A/c) * NT_1$ for the join, and $J_\alpha \cup J_\beta$ for two copies of J glued along NT_1. Then NP_A is the pushout of simplicial sets

$$NA \longleftarrow N(A/c) \amalg N(A/c) \hookrightarrow J_\alpha \cup J_\beta,$$

in which the left map forgets the arrow to c and the right map is a monomorphism; it is therefore a homotopy pushout. A functor $h : A \to A'$ under U induces a map of these diagrams. The categories A/c and $A'/h(c)$ have terminal objects, so their nerves and the joins J are contractible, and the maps on $N(A/c) \amalg N(A/c)$ and on $J_\alpha \cup J_\beta$ are weak equivalences whatever h is. If $h \in W_\infty$, the map $NP_A \to NP_{A'}$ is then a weak equivalence. For W_R one uses the Mayer–Vietoris sequences of the same pushouts. □

So the assertion "je sais que c'est le cas si on suppose T isomorphe à la catégorie finale e" is false for W-cofibrations, and with it the equivalence stated at the end of question a), also under the axiom W(7), which the localizers W_R satisfy. The corollary of Proposition 7 concerns membership in W of one inclusion, not the cofibration property. What it does give is the statement for *trivial* cofibrations and homology:

Proposition. [ours] *Let $U \hookrightarrow X$ be a sieve whose complement is one object t with only its identity, and suppose that $H = \mathrm{Hom}_X(-,t)|_U$ is not empty. For a nonzero commutative ring R, the inclusion is a trivial W_R-cofibration if and only if H is totally W_R-aspherical, that is, if and only if it is W_R-dwyérienne.*

Proof. After a cobase change $p : U \to A$ the complement is still $\{t\}$ and the crossing presheaf is $p_!H$. A chain of arrows of $X \amalg_U A$ that does not lie in A is a chain of $A/p_!H$ followed by t, so $N(X \amalg_U A)$ is NA with the cone on $N(A/p_!H)$ attached along the forgetful map, and $B(X \amalg_U A)$ is the mapping cone of $B(A/p_!H) \to BA$. The inclusion of BA is an R-homology isomorphism exactly when the reduced R-homology of $A/p_!H$ vanishes, that is, when $A/p_!H$ has the R-homology of a point. Quantifying over p gives the statement. □

For W_∞ the same computation shows that a one-point sieve is a trivial W_∞-cofibration if and only if every $A/p_!H$ is nonempty and integrally acyclic and the cone attachment does not change fundamental groups. Whether this forces H to be totally W_∞-aspherical is not decided here: a counterexample would be a presheaf H all of whose direct images have acyclic categories of elements, one of them with a nontrivial (necessarily perfect) fundamental group.

et incluant le cas des immersions ouvertes et des immersions fermées?

Démonstration du théorème 3 (p. 107). Il suffit de prouver que si i est W-dwyérienne (i.e. i_0 strictement W-asphérique). alors i' l'est également - car il en résultera que i' est W-asphérique et a fortiori $\in W$. Il faut donc vérifier que pour tout $t' \in \dot{U}' = \dot{T}$, l'élément $i^*(t') = (u' \longmapsto \mathrm{Hom}_{X'}(u', t'))$ de $U'^\wedge$ est totalement W-asphérique. Or t' provient d'un unique élément t de $\dot{T}$, et on sait qu'on a

$$i'^*(t) = p_!(i^*(t)) \quad \texttt{[plutôt } i'^*(t') = p_!(i^*(t))\texttt{]}, \qquad \text{où } \; p_! : U^\wedge \longrightarrow U'^\wedge.$$

Par hypothèse, $F = i^*(t)$ est un élément totalement W-asphérique de $U^\wedge$, ce qui signifie que pour toute flèche $p : U \longrightarrow U'$ dans Cat de souce U, $p_!(F)$ est un élément W-asphérique de U'. En fait, on a le lemme immédiat (par transitivité $(p'p)_! \simeq p'_! p_!$).

Lemme. *Soit U dans* Cat*, et F dans $U^\wedge$. Si F est totalement W-asphérique (resp. si U/F est totalement W-asphérique), alors pour toute flèche $p : U \longrightarrow U'$ dans* Cat *de source U, $F' = p_!(F)$ est un objet totalement W-asphérique de $U'^\wedge$ (resp.*

U'/F' est un objet totalement W-asphérique de Cat*.)* [page 113]

Corollaire 3 (du théorème 3). *Si $i : U \longrightarrow X$ est une immersion ouverte ou fermée W-dwyérienne (resp. totalement W-dwyérienne), alors pour toute flèche $p : U \longrightarrow U'$ dans* Cat*, $i' : U' \longrightarrow X' = U' \amalg_U X'$ [plutôt $U' \amalg_U X$] l'est aussi.*

(NB Le cas non respé est essentiellement le théorème 3. Il ne dépend d'aucun axiome supplémentaire sur W, mais n'a d'intérêt qu'en vue des équivalences ⓐ et ⓑ (p. 108) qui en résultent, et qui sont valables moyennant W(7 bis) resp. W(7) seulement.)

Commentary · Exposition [ms. 113]

The stronger clause uses comprehensive factorization

The lemma proves more than stability of a condition on a presheaf. Its parenthesized assertion says that the category of elements after direct image is itself totally aspherical, so that one may test it by arbitrary functors leaving that category. This does not follow merely by repeating the formula for a second direct image on the original base. The canonical comprehensive factorization supplies the missing step.

Write $V = U/F$, let $a : V \to U$ be the projection, and set $b = pa : V \to U'$. The identity $a_! e_V = F$ gives

$$b_! e_V = p_! F = F'.$$

Let $E = U'/F'$. The canonical functor $v : V \to E$ appearing in the comprehensive factorization of b is final in the ordinary sense: for every $e \in E$, the category $e\backslash V$ is connected and nonempty. Equivalently,

$$v_! e_V = e_E.$$

This is the factorization established at ms. 91–93; it involves connectedness, not contractibility.

Now suppose that V is totally W-aspherical, and let $r : E \to Z$ be any functor. Transitivity of ordinary left Kan extension gives

$$r_! e_E \cong r_! v_! e_V \cong (rv)_! e_V.$$

The last presheaf is W-aspherical by totality of V, so E is totally W-aspherical. This proves the parenthesized clause with its full quantifier. The unparenthesized clause is simpler: for $s : U' \to Z$, one has $s_! F' \cong (sp)_! F$, to which the definition applies directly.

The same argument yields a useful formal principle. If $v : V \to E$ is an ordinary final functor and V is totally W-aspherical, then E is totally W-aspherical. The converse is not being asserted.

In particular, an arbitrary final functor need not be a weak homotopy equivalence; the finite example at ms. 100 makes that failure explicit. Here totality of the source is precisely what permits the final functor to be used in every subsequent direct-image calculation.

Corollary 3 follows by applying the two clauses to the gluing weights of each boundary object, with the support calculation already explained after Theorem 3. This proof establishes stability under pushout. It supplies no composition theorem for arbitrary strictly or totally W-aspherical functors: in a composite one pulls back a weight of the second functor, whereas the lemma controls left Kan extension. The distinction identifies the mathematical content of the doubt expressed on ms. 113–114. Composition is treated in the commentary at ms. 114: both classes of functors are stable under it, by a different argument.

Commentaire de motivation : les immersions ouvertes ou fermées W-dwyériennes, ou totalement W-dwyériennes, sont stables par cochangement de base (d'après le corollaire précédent), mais n'ont pas du tout l'air de l'être par composition. À cet égard, les 'immersions de Dwyer' (où on suppose l'existence de l'adjoint à droite de $i_0 : U \longrightarrow \bar{U}$, dans le cas d'une immersion ouverte) se comportent mieux, étant stables par composition. (**NB** Il ne semble pas non plus clair que les flèches strictement W-asphériques, ou les totalement W-asphériques, soient stables par composition - sorry!) Ma motivation pour m'échiner sur la notion W-dwyérienne est double. [page 114]

a) J'aimerais avoir des conditions à la fois compréhensibles, et aussi générales que possible, sur une immersion (ouverte p. ex), pour que ce soit une W-cofibration, propriété visiblement importante (toute comme la notion duale de W-fibration).

p. 78

b) J'ai envie, pour des applications ultérieures, de disposer d'une classe d'immersions ouvertes ayant les propriétés suivantes :

   1°) Ce sont des W-cofibrations, et

   2°) Si $U \overset{i}{\hookrightarrow} X$ est dans la classe, et si $\varepsilon \overset{j}{\to} I$ est l'inclusion 'source-but' de la catégorie discrète $\{0,1\}$ dans l'intervalle

$$\Phi_n = 0 \longrightarrow 1 \longleftarrow 2 \longrightarrow \dots 2n-1 \longleftarrow 2n$$

   ($j(0) = 0$, $j(1) = 2n$), $n \geq 2$, l'inclusion α de $(U \times I) \cup (X \times \varepsilon)$ dans $X \times I$ est dans la classe.

Or cette dernière propriété de stabilité n'est *pas* vérifiée par les immersions de Dwyer. (Prendre $i : \{0\} \longrightarrow \mathbf{\Delta}^1 = \{0,1\}$, l'inclusion α n'est pas immersion de Dwyer.)

[page 115]

Commentary · Examples [ms. 114]

The interval test fails because a slice has two maxima

The interval condition asks for a class of admissible inclusions that can also attach both ends of a homotopy. Its categorical form is a pushout product. The original Dwyer condition is too rigid for this purpose, and the example in the text can be checked at a single object.

Take $X = [1] = (0 < 1)$, $U = \{0\}$, and

$$I = \Phi_n = (0 \to 1 \leftarrow 2 \to \cdots \leftarrow 2n), \qquad n \geq 2.$$

Let $A = (\{0\} \times I) \cup ([1] \times \{0, 2n\})$, considered as a full subposet of $[1] \times I$. The inclusion $A \hookrightarrow [1] \times I$ is a sieve: both end vertices of I are sources, and the entire lower row is already in A. Its generated cosieve is the whole product, since every upper-row object $(1, k)$ receives an arrow from $(0, k)$.

If this were an original Dwyer immersion, the inclusion into that cosieve would admit a right adjoint. For an inclusion of ordered sets, this means that for every object x of the larger poset the lower set A/x has a greatest element. Test $x = (1, 1)$. Its lower set inside A contains exactly

$$(0,0), \quad (0,1), \quad (0,2), \quad (1,0).$$

The strict relations are

$$(0,0) < (0,1), \qquad (0,2) < (0,1), \qquad (0,0) < (1,0).$$

Thus $(0,1)$ and $(1,0)$ are distinct incomparable maximal elements, and neither is above the other. There is no greatest element. This proves the failure of the adjoint condition for every $n \geq 2$.

The calculation also explains why ordinary contractibility of this slice is insufficient to recover the missing adjoint: its nerve is an interval with three edges, hence contractible. An adjoint selects a universal object, a much stronger requirement. Whether another class provides the desired interval stability is a structural question about the interaction of pushout and homotopy, rather than a reason to weaken *terminal object* to *contractible* in this particular criterion. The subsequent use of integration replaces a difficult ordinary pushout by a category designed to retain the homotopy of the entire diagram.

Commentary · Exposition [ms. 114]

A terminal fibre controls all the ordinary direct images

The last subsection returns to a situation in which the apparent loss of higher information can be controlled. For a W-proper functor over a category with a terminal object, the terminal-fibre inclusion is W-coaspherical. Under W(5) it is ordinary final, which is the property needed for total asphericity. Making this explicit shortens the proof and explains the editors' repeated suggestion to replace W-properness by W_0-properness.

Let $f : X \to I$ be W-proper, let i_0 be terminal in I, and write $k : X_{i_0} \hookrightarrow X$. The properness criterion says that k is W-coaspherical: every undercategory $x\backslash X_{i_0}$ is W-aspherical. If W(5) holds, a W-aspherical category is nonempty and connected. Hence k is an ordinary final functor, and the pointwise formula gives

$$(k_! e_{X_{i_0}})(x) = \pi_0(x\backslash X_{i_0}) = \{*\}.$$

For every $p : X \to Y$ it follows that

$$(pk)_! e_{X_{i_0}} \cong p_! k_! e_{X_{i_0}} \cong p_! e_X. \tag{C1}$$

If X_{i_0} is totally W-aspherical, the left side of (C1) is an aspherical presheaf on Y for every p, proving Proposition 11 of ms. 115. This proposition has the same number as the counterexample on ms. 107; the manuscript page is needed to distinguish them. The proof uses no W(7) or W(7 bis).

The general component formula in Proposition 12 explains how (C1) fits into a family. W-properness together with W(5) identifies $\pi_0(X_i)$ with $\pi_0(X/i)$, naturally in i, and therefore supplies the transition maps even when f is not an opfibration. For an arbitrary functor f, there is a canonical bijection

$$\varinjlim_{i \in I} \pi_0(X/i) \;\cong\; \pi_0(X). \tag{C2}$$

One can see it without choosing a cleavage. A representative $(x, \beta : f(x) \to i)$ in X/i has the same image in the colimit as $(x, \mathrm{id}_{f(x)})$, by the transition along β. A morphism $x \to x'$ identifies the latter representatives through the induced object of $X/f(x')$. Thus the colimit imposes exactly the zigzag relation defining components of X. Conversely each morphism in a comma category, and every transition between comma categories, preserves that zigzag component. These two observations construct mutually inverse maps in (C2).

Replacing $\pi_0(X/i)$ by $\pi_0(X_i)$ proves Proposition 12. The dual proof printed in the manuscript is equally valid, but after passing to presheaves its colimit is indexed by I^{op}; the printed I at ms. 117 suppresses this reversal. The precise proper/smooth criteria, their duality and their stability under composition and base change are given in Maltsiniotis (2005, Theorem 1.2 and Proposition 4.7). The result here concerns ordinary components and ordinary Kan extensions, so the connectivity hypothesis cannot be omitted just because the discussion takes place in a homotopical setting.

Commentary · Later developments [ms. 114]

Dwyer attachments and higher categorical pushouts

The adjoint condition belongs to the Dwyer maps of Thomason (1980, Definition 4.1 and Proposition 4.3). One subsequent correction matters when comparing them with model-category cofibrations. Classical Dwyer maps are not closed under retracts, and some Thomason cofibrations are not Dwyer maps. Cisinski (1999, Definition 1, Lemma 4 and Proposition 2) repairs the relevant argument with pseudo-Dwyer maps: these are retract-stable and include every Thomason cofibration. Thus the manuscript's search for a broader attachment condition addresses a distinction that cannot be removed by simply identifying the original Dwyer maps with all cofibrations.

The desired stability properties concern whether an ordinary attachment retains the information needed after passage to a homotopy theory. For classical Dwyer maps, a stronger result is now available. Hackney et al. (2024, Theorem 1.5) prove that, for a pushout $D = B \amalg_A C$ along a Dwyer inclusion $A \hookrightarrow B$, the comparison

$$NB \amalg_{NA} NC \longrightarrow ND$$

is a weak *categorical* equivalence in the Joyal model structure. Their general Theorem 1.6, in its expanded statement on p. 2179, treats discretely flat cofibrations. These are defined by requiring the corresponding functor of discrete simplicial categories to preserve Bergner weak equivalences under cobase change, together with injectivity on objects. Proposition 3.3 supplies this property for Dwyer maps; Example 3.5 includes co-Dwyer maps and pseudo-Dwyer maps. This gives one modern class that accommodates both orientations requested on p. 111.

This higher categorical assertion concerns the comparison between the ordinary categorical pushout and a pushout in the category of $(\infty, 1)$-categories. The weak homotopy type of a category discards all noninvertible-arrow information, whereas its image among $(\infty, 1)$-categories retains that information. Consequently the two ambient weak equivalence classes must be specified before comparing definitions of flatness. The theorem just cited does not identify discretely flat inclusions with the manuscript's W-Dwyer immersions, nor does it by itself establish the particular interval-product stability requested here. Those further identifications would require additional arguments. The concrete calculations in the preceding blocks keep the original question at the level at which it is posed.

Commentary · Development (commentators') [ms. 114]

Composition of aspherical functors and of dwyérienne immersions

Grothendieck doubts that strictly or totally W-aspherical functors, and the two classes of dwyérienne immersions, are stable under composition. For functors the doubt is unfounded; for immersions it is justified for some localizers and not for others. The statements below were not found in the sources consulted; no priority is claimed.

Theorem (functors compose). [ours] *Let W be a fundamental localizer.*

(a) *If $f : X \to Y$ is strictly W-aspherical and H is a totally W-aspherical presheaf on Y, then f^*H is a totally W-aspherical presheaf on X. Consequently strictly W-aspherical functors are stable under composition.*

(b) *Totally W-aspherical functors are stable under composition.*

(c) *If B is a totally W-aspherical category and $D : B \to \mathrm{Cat}$ a functor with totally W-aspherical values, the cofibred category $\int_B D$ is totally W-aspherical.*

Proof. (a) Let $p : X \to X'$ be arbitrary. For $y \in Y$ put $Q_y = p_! \operatorname{Hom}_Y(f(-), y)$, a presheaf on X', covariant in y. Each Q_y is totally W-aspherical, by the hypothesis on f and the lemma of ms. 112. Let Z be the category with full subcategories X' and Y, with $\operatorname{Hom}_Z(x', y) = Q_y(x')$ and no arrows from Y to X'; let $j : Y \hookrightarrow Z$ be the inclusion and $F = j_! H$. Then $F|_Y = H$ and

$$F|_{X'} = \int^{y \in Y} Q_y \times H(y) \cong p_! \Big(\int^{y} \operatorname{Hom}_Y(f(-), y) \times H(y) \Big) = p_! f^* H =: K.$$

Since H is total, Z/F is W-aspherical. The inclusion $l : X'/K \hookrightarrow Z/F$ is a sieve. Its comma category over an object (x', k) of X'/K has a terminal object. Its comma category over (y, h), $h \in H(y)$, has as objects the pairs (x', q) with $q \in Q_y(x')$ (the element of $K(x')$ is then the class of (q, h)) and as

arrows the arrows of X' that respect q; it is X'/Q_y, which is W-aspherical. By the relative form of Theorem A that defines a fundamental localizer, $l \in W$, and so X'/K is W-aspherical. As p was arbitrary, f^*H is total. If $g : Y \to Z_0$ is also strictly aspherical, apply this to $H = \mathrm{Hom}_{Z_0}(g(-), z)$, whose restriction along f is $\mathrm{Hom}_{Z_0}(gf(-), z)$.

(b) Let $f : X \to Y$ and $g : Y \to Z_0$ be totally aspherical and $z \in Z_0$. The functor $f_z : X/z \to Y/z$, $(x, \gamma) \mapsto (f(x), \gamma)$, has comma category over (y, β) canonically isomorphic to X/y, because the arrow to z is determined as $\beta g(h)$ by $h : f(x) \to y$. So f_z is totally aspherical, hence strictly aspherical by Proposition 8. The final presheaf of Y/z is total because g is totally aspherical; by (a) its restriction, the final presheaf of X/z, is total. That is, X/z is a totally W-aspherical category.

(c) Put $E = \int_B D$, with objects (b, x), $x \in D(b)$, and projection $\pi : E \to B$. For $b \in B$ the functor $D(b) \to E/b$, $x \mapsto (b, x, \mathrm{id}_b)$, has the left adjoint $(b', x, \beta) \mapsto D(\beta)(x)$. A right adjoint is final. So E/b is a final image of $D(b)$ and is total; π is a totally aspherical functor, as is $B \to e$, and (b) applies. □

Proposition (immersions: two counterexamples). [ours] *For $W = W_0$ and for $W = W_1$ neither the W-dwyérienne nor the totally W-dwyérienne sieves are stable under composition.*

Proof. Let V be the discrete category on two objects for W_0, and the category with two objects and two parallel arrows for W_1; BV is S^0, respectively S^1. Let $B = V^{\triangleleft}$ be V with an initial object o adjoined. Let X consist of B and of a second copy V_0 of V, with

$$X(v_0, b) = B(v, b), \qquad X(b, v_0) = \varnothing \qquad (v \in V,\ b \in B),$$

and composition induced by B. The sieve $V_0 \hookrightarrow X$ generates the cosieve $V_0 \cup V$, on which the inclusion of V_0 has the right adjoint $v \mapsto v_0$; it is a Dwyer immersion. The functor $X \to B$ forgetting the copy is left adjoint to the inclusion $B \subset X$, so $BX \simeq BB$ is contractible. A final functor is bijective on π_0 and surjective on π_1; hence X is totally W_0- and totally W_1-aspherical. Let $Y = X^{\triangleright}$ be X with a terminal object t adjoined. The sieve $X \hookrightarrow Y$ is totally W-dwyérienne, its only boundary category being $X/t = X$. For the composite sieve $V_0 \hookrightarrow Y$ the boundary category at t is $V_0/t \simeq V$ and the boundary presheaf is the final presheaf of V_0; neither is W-aspherical. □

The targets Y have six objects and 15, respectively 21, arrows.

Theorem (total immersions for W_∞). [ours] *For $W = W_\infty$ the totally W-dwyérienne sieves are stable under composition.*

Proof. We use the criterion of the commentary on Chapter II, ms. 36: a category A is totally W_∞-aspherical if and only if it is nonempty and connected and every connected component of the category of elements of every functor $A \to \mathrm{Set}$ is weakly contractible; and then those components are themselves totally W_∞-aspherical and any two objects of A have a common successor. A nonempty cosieve S of such an A is the category of elements of the functor with value a point on S and empty elsewhere; it is connected, because two of its objects have a common successor in A, which lies in S. So S is totally W_∞-aspherical.

Let $U \subset X \subset Y$ be totally W_∞-dwyérienne sieves and $y \in Y$ with $U/y \neq \varnothing$. Then X/y is total (it has a terminal object if $y \in X$). The objects (x, β) of X/y with $U/x \neq \varnothing$ form a nonempty cosieve S_y, total by the preceding paragraph, and the functor $(x, \beta) \mapsto U/x$ on S_y has total values. By part (c) of the theorem its cofibred category G is total. The functor $G \to U/y$ that composes $u \to x \to y$ has the left adjoint $(u, \gamma) \mapsto ((u, \gamma), (u, \mathrm{id}_u))$, so it is final, and U/y is total. □

The same proof works for the homology localizers W_R, with the homological form of the criterion (R-acyclic in place of weakly contractible). For the strict class, the W_R-dwyérienne sieves are also stable under composition: the restriction of a totally W_R-aspherical presheaf to a cosieve on which it is nonempty is totally W_R-aspherical, after which part (a) applies; the proof of the restriction statement is not reproduced here. For W_∞-dwyérienne sieves the question is undecided. It comes down to the question left open in the commentary at ms. 111: does a presheaf all of whose direct images have integrally acyclic categories of elements have only weakly contractible ones?

Commentary · Development (commentators') [ms. 114]

The endpoint product is not a cofibration

The class asked for in b) does not exist for W_∞: the inclusion α built from the simplest sieve is not a W_∞-cofibration at all. This is stronger than the observation that α is not a Dwyer immersion.

Theorem. [ours] *Let $n \geq 2$, $I = \Phi_n$, $\varepsilon = \{0, 2n\} \subset I$, and let $\alpha : A \hookrightarrow P$ be the inclusion of $A = (\{0\} \times I) \cup ([1] \times \varepsilon)$ in $P = [1] \times I$, deduced from $\{0\} \hookrightarrow [1]$. Let C be the four-object category*

of the commentary at ms. 100, of the homotopy type of S^2.

(a) *If W is a fundamental localizer for which a category of the homotopy type of S^3 is not W-aspherical (for instance W_∞, W_R with $R \neq 0$, or the m-equivalences with $m \geq 3$), then α is not a W-cofibration.*

(b) *If C is not W-aspherical (which includes the 2-equivalences), then α, which belongs to W, is not a trivial W-cofibration.*

Proof. Write $l_k = (0, k)$ and $u_k = (1, k)$. The objects of A are $l_0, \dots, l_{2n}, u_0, u_{2n}$; its nonidentity arrows are those of the lower copy of I and $l_0 \to u_0$, $l_{2n} \to u_{2n}$, and no two of them are composable. Both A and P have contractible classifying spaces, so $\alpha \in W_\infty \subseteq W$.

Recall that C has objects x, z, a, b and arrows $\alpha', \beta' : x \to z$, $r : z \to a$, $s : z \to b$ with $r\alpha' = r\beta'$, $s\alpha' = s\beta'$. Let D be either C or the category $C^{\triangleright}$ obtained by adjoining a terminal object ω. Define $p : A \to D$ by

$$p(l_0) = z, \qquad p(u_0) = b, \qquad p(l_k) = a \ (1 \leq k \leq 2n), \qquad p(u_{2n}) = a,$$

sending $l_0 \to l_1$ to r, $l_0 \to u_0$ to s and all other arrows to id_a. Let $Q = P \amalg_A D$. Since A is a sieve in P, the category Q consists of D, of the objects $u_1, \dots, u_{2n-1}$ with the arrows of the upper copy of I between them, and of the crossing sets $Q(v, u_k) = (p_! H_k)(v)$, where $H_k = \mathrm{Hom}_P(-, u_k)|_A$ (Theorem 1).

The presheaf H_1 is the final presheaf of the sieve $A_1 = \{l_0, l_1, l_2, u_0\}$ of A and is empty elsewhere. So $(p_! H_1)(v) = \pi_0(v \backslash A_1)$, the comma category being formed with the restriction $A_1 \to D$ of p. For $v = z$ it is A_1 itself, a tree; for $v = a$ it is $l_2 \to l_1$; for $v = b$ it is the point u_0; for $v = x$ it has the five objects $(l_0, \alpha'), (l_0, \beta'), (l_1, r\alpha'), (l_2, r\alpha'), (u_0, s\alpha')$ and is connected; for $v = \omega$ it is empty. Hence the object u_1 receives exactly one arrow from each object of C and none from ω. For $2 \leq k \leq 2n-1$ the category A/u_k is nonempty and connected and p is constant on it with value a, so $p_! H_k = D(-, a)$; the arrows between $u_2, \dots, u_{2n-1}$ induce identities of $D(-, a)$, and $u_2 \to u_1$ induces the unique map $D(-, a) \to p_! H_1$.

Let $R \subset Q$ be the full subcategory on D and u_1. The functor $\rho : Q \to R$ that is the identity on R, sends u_k to a for $k \geq 2$, is the identity on the crossing sets $D(v, a)$, sends the arrows between these u_k to id_a and $u_2 \to u_1$ to the arrow $a \to u_1$, satisfies $\rho|_R = \mathrm{id}_R$; and the arrows $a \to u_k$ corresponding to id_a are the components of a natural transformation from ρ, followed by the inclusion of R, to id_Q. So $BQ \simeq BR$.

(a) Take $D = C^{\triangleright}$. Then p is a functor between categories with contractible classifying spaces, so $p \in W_\infty \subseteq W$; it is a morphism of $A \backslash \mathrm{Cat}$ from id_A to p, and its image under cobase change along α is $P \to Q$. The category R is C with the two terminal objects ω and u_1 adjoined, with no arrow between them: its nerve is the union of two cones on NC, so $BR \simeq \Sigma BC \simeq S^3$. Thus P is W-aspherical and Q is not, and $P \to Q$ is not in W.

(b) Take $D = C$. Then R is C with one terminal object adjoined and $BQ \simeq BR$ is contractible. The cobase change $C \to Q$ of α along p goes from a category that is not W-aspherical to one that is, so it is not in W. □

For $n = 2$ the category Q of (a) has 8 objects and 30 arrows. Part (b) also excludes the cofibration property whenever a W-cofibration that belongs to W is a trivial W-cofibration. For W_∞ this is Batanin and Berger (2017, Lemma 1.10), applied to Thomason's model structure, which is left proper; there it adds nothing to (a).

Corollary (no class of the kind requested). [ours] *Let W be as in (a) and $n \geq 2$. If a class $\mathcal{K}$ of sieves consists of W-cofibrations and contains, with $i : U \hookrightarrow X$, the inclusion of $(U \times I) \cup (X \times \varepsilon)$ in $X \times I$, then every member of $\mathcal{K}$ is an isomorphism. The same holds for W as in (b) and classes of trivial W-cofibrations.*

Proof. W-cofibrations, and trivial W-cofibrations, are stable under retracts in the category of arrows, because a retract of a functor in W is in W. Let $i : U \hookrightarrow X$ be in $\mathcal{K}$. If there is an arrow $u \to x$ with $u \in U$, $x \notin U$, then $\{0\} \hookrightarrow [1]$ is a retract of i (by the functor $X \to [1]$ with fibre U over 0, and the functor $[1] \to X$ choosing that arrow), the endpoint product α of $\{0\} \hookrightarrow [1]$ is a retract of that of i, and the theorem is contradicted. If there is no such arrow, then $X = U \amalg V$; if $V \neq \varnothing$, the endpoint product of i is the inclusion of $(U \times I) \amalg (V \times \varepsilon)$ in $(U \amalg V) \times I$, which is in $\mathcal{K}$ and has the crossing arrow $(v, 0) \to (v, 1)$, so the first case applies to it. □

At the other extreme, for W_0 every functor is a W_0-cofibration, because $\pi_0 : \mathrm{Cat} \to \mathrm{Set}$ preserves pushouts, and all sieves form a class with the two properties. The same is true for W_1 and for every fundamental localizer containing W_1: the pushout along a functor injective on objects has the same

fundamental groupoid as the cofibred category of the span; this proof is not reproduced here. By Cisinski (2006a, Proposition 9.2.17) a fundamental localizer contains W_1 exactly when a category of the homotopy type of S^2 is aspherical for it. So for the localizers of homotopy theory proper the endpoint product has to be replaced by a homotopy-invariant construction, which is what the integration of the following pages provides.

2.5 Complément : critère d'asphéricité totale d'une catégorie W-propre sur une autre.

Commentary · Examples [ms. 115]

Why the interval test exceeds Dwyer maps

The preceding paragraph asks for enough cofibrations to build homotopies with prescribed endpoints. Its product with the zigzag interval is therefore a structural test, rather than an incidental counterexample. We can make the obstruction completely visible in a finite ordered set.

Write $P = [1] \times \Phi_n$ and

$$A = (\{0\} \times \Phi_n) \cup ([1] \times \{0, 2n\}).$$

Here $[1] = \{0 < 1\}$, and the even vertices of Φ_n map to their adjacent odd vertices. The inclusion $A \subset P$ is a sieve: a predecessor of an object in its bottom row is again in that row, and a predecessor of either of its two top endpoints is already in A. Every object $(1, k)$ receives a map from $(0, k)$, so the cosieve generated by A is all of P.

For a full inclusion of ordered sets, the existence of a right adjoint is particularly concrete. If r is right adjoint to $A \hookrightarrow P$, then $r(p)$ must be the greatest element of $\{a \in A : a \leq p\}$. At $p = (1, 1)$, this set has the two incomparable maximal elements $(1, 0)$ and $(0, 1)$; for $n \geq 2$ the other top endpoint cannot lie below p. There is consequently no greatest element, hence no right adjoint. This is exactly the obstruction in the definition of a Dwyer map, since the relevant cosieve is already P. The original inclusion $\{0\} \hookrightarrow [1]$ is Dwyer, with right adjoint the constant functor to 0. Thus the endpoint product operation genuinely leaves that class.

The definition and generated-cosieve criterion are those of Thomason (1980, Definition 4.1 and Lemma 5.2). There is a second, distinct limitation in the history of this class: retracts of Dwyer maps need not be Dwyer. Cisinski (1999, pp. 228–231, especially Proposition 2) gives the counterexample and repairs the properness argument for Thomason's model structure using pseudo-Dwyer maps. The interval calculation above concerns a different closure operation. Both facts explain why Grothendieck seeks a larger homotopically defined class rather than treating the adjoint criterion as the final definition of a cofibration.

Soit

$$f : X \longrightarrow I$$

une catégorie propre sur une catégorie d'indices I. Si I a un objet final i_0, alors X/i_0 s'identifie à X, et par suite l'inclusion

$$X_{i_0} \longrightarrow X$$

de la fibre de X en i_0 est W-asphérique `[plutôt W-coasphérique]`, et a fortiori est dans W. Donc X est asphérique si et seulement si X_{i_0} l'est. Nous voulons étendre ce résultat en un critère de *totale* W-asphéricité, de façon précise prouver la

Proposition 11 : *Soit $f : X \longrightarrow I$ W-propre, et soit i_0 un objet final de I. Si X_{i_0} est totalement W-asphérique* (déf. 1, page 88), *il en est de même de X.* (Sous réserve que W satisfasse l'axiome de connexité W(5).)

La démonstration sera donnée plus bas, après quelques préliminaires ayant un intérêt indépendant. Tout d'abord, rappelons que si X est W-propre sur I, alors il donne naissance à un foncteur

$$\begin{array}{ccc} I & \longrightarrow & \mathrm{Hot}_W \\ i & \longmapsto & \mathrm{hot}_W(X_i)(\simeq \mathrm{hot}_W(X/i)); \end{array}$$

en le composant avec le foncteur

$$\pi_0 : \mathrm{Hot}_W \longrightarrow \mathrm{Ens}$$

(sous réserve que l'axiome de connexité W(5) (p. 10) soit satisfait) on trouve un foncteur p. 79 [page 116]

$$i \longmapsto \underbrace{\pi_0(X_i)}_{\simeq \pi_0(X/i)} : I \longrightarrow \mathrm{Ens}.$$

Ceci posé, on a le résultat sorital

Commentary · Exposition [ms. 115]

Components of a proper family

Grothendieck wants to transfer a condition on every ordinary left Kan extension from the last fibre to the whole category. A weak equivalence of categories alone would not accomplish this: the functor $p_!$ here is the underived left Kan extension between categories of set-valued presheaves. The useful consequence of properness is that it controls the components of all the relevant comma categories.

We use $\widehat{Y} = \mathrm{Fun}(Y^{\mathrm{op}}, \mathrm{Set})$ and write 1_X for the terminal presheaf. For $p : X \to Y$ one has

$$(p_! 1_X)(y) = \pi_0(y\backslash X),$$

where an object of $y\backslash X$ is $(x, a : y \to p(x))$. In this chapter, "totally W-aspherical" means that $Y/(p_! 1_X)$ is W-aspherical for every Y and every $p : X \to Y$. This is the definition on manuscript pp. 88–89; the editors explicitly distinguish it from the earlier use of total asphericity in *Pursuing Stacks* (Grothendieck, 1990–1991d, Definition 1). It is this presheaf condition that Proposition 11 transfers.

The calculation of components has a direct proof that also fixes its variance. For any $f : X \to I$, without a properness hypothesis, the categories X/i form a covariant diagram by postcomposition. There is a canonical bijection

$$\operatorname*{colim}_{i \in I} \pi_0(X/i) \longrightarrow \pi_0(X). \tag{A1}$$

Indeed, every object x occurs as $(x, \mathrm{id}_{f(x)})$ in $X/f(x)$, so the map is surjective. The class of an arbitrary $(x, a : f(x) \to i)$ is identified in the colimit with the class of (x, id), by the transition functor associated to a. If $v : x \to x'$ is a morphism of X, the two resulting classes are equal: move the first object along $f(v)$ into $X/f(x')$, where v becomes a morphism to (x', id). These are precisely the relations defining $\pi_0(X)$. This proves injectivity as well.

If f is W_0-proper, the fibre inclusion $X_i \to X/i$ induces a bijection on components. Transporting the covariant structure through these bijections supplies the functor $i \mapsto \pi_0(X_i)$ and turns (A1) into Proposition 12. There is no choice of transport functors on the fibres in this argument. Under W(5), the inclusion $W \subseteq W_0$ makes W-properness sufficient. The two footnotes recommending W_0-properness identify the exact hypothesis needed for this calculation.

For the relative formula, apply the same argument to $y\backslash X \to I$. Its fibre over i is $y\backslash X_i$. The projection $y\backslash Y \to Y$ is a discrete opfibration, hence proper; properness is preserved by base change and composition. Consequently

$$(p_! 1_X)(y) \cong \operatorname*{colim}_{i \in I} (p_{i!} 1_{X_i})(y). \tag{A2}$$

Precomposition by a map $y' \to y$ commutes with every functor used to construct this bijection. Thus (A2) is an isomorphism of presheaves, not merely a collection of unrelated bijections. This supplies both the naturality and the universal property asserted in Corollary 1.

Proposition 12 : *Soit $f : X \longrightarrow I$ W-propre, et supposons que W satisfasse l'axiome de connexité* W(5) (sinon, la W-propreté ne nous sera d'aucune utilité …) ([81]). *On a alors une*

81. Il vaut mieux dire : Soit f W_0-propre, où W_0 est le localiseur de la 0-connexité …

bijection canonique

$$(*) \qquad \pi_0(X) \xleftarrow{\sim} \varinjlim_I \pi_0(X_i),$$

où la flèche est induite par les applications canoniques $\pi_0(X_i) \longrightarrow \pi_0(X)$.

Démonstration. Passant à $f^\circ : X^\circ \longrightarrow I^\circ$, on est ramené à prouver le même résultat pour X *W-lisse* sur I, avec la seule différence que le foncteur

$$i \longmapsto \pi_0(X_i)$$

sur I sera contravariant, i.e. un préfaisceau, ce qui offre l'avantage de l'intuition attachée au formalisme des préfaisceaux. Soit

$$g : I \longrightarrow e, \qquad \text{d'où } gf = p : X \longrightarrow e,$$

il est immédiat, par le calcul standard

$$p_!(F) = \pi_0(X/F) \qquad (F \in \mathrm{Ob}\, X^\wedge)$$

(où on a identifié $e^\wedge$ à (Ens)), qu'on a

$$\pi_0(X) = p_!(e_X) = g_!(f_!(e_X)).$$

Or le préfaisceau

$$i \longmapsto \pi_0(X_i) \qquad (\simeq \pi_0(i\backslash X), \text{ car } X \text{ } W\text{-lisse sur } I)$$

n'est autre que $f_!(e_X)$. D'autre part, pour un préfaisceau quelconque G sur I, on a

[page 117]

$$g_!(G) \simeq \pi_0(I/G) \simeq \varinjlim_I G(i),$$

et appliquant ceci à $f_!(e_X)$, on trouve la formule $(*)$ de la proposition 12.

Revenons à la situation avec f *W-propre,* et considérons un foncteur

$$p : X \longrightarrow Y,$$

nous sommes intéressés à 'calculer'

$$p_!(e_X) = (y \longmapsto \pi_0(y\backslash X)) \in \mathrm{Ob}\, Y^\wedge.$$

Considérons pour ceci le diagramme suivant

p. 80

$$\begin{array}{ccccc} y\backslash X_i & \longrightarrow & y\backslash X & \longrightarrow & y\backslash Y \\ \downarrow & & \downarrow & & \downarrow \\ X_i & \longrightarrow & X & \xrightarrow{p} & Y \\ \downarrow & & \downarrow f & & \\ \underset{\simeq e}{\{i\}} & \longrightarrow & I & & \end{array},$$

où les trois carrés sont cartésiens. Comme $y\backslash Y \longrightarrow Y$ est W-propre, il en est de même de $y\backslash X \longrightarrow X$ par changement de base, donc du composé $y\backslash X \longrightarrow X \longrightarrow I$ par transitivité. Le diagramme montre d'autre part que la fibre de $y\backslash X$ en $i \in \mathrm{Ob}\, I$ s'identifie à $y\backslash X_i$.

La proposition 12 nous donne donc la formule

[page 118]

$$\pi_0(y\backslash X) = \varinjlim_I \pi_0(y\backslash X_i),$$

laquelle formule ne peut s'empêcher d'être fonctorielle en y, tout comme celle de prop. 12 l'est dans $\underline{\mathrm{Prop}}_W I$. Or pour y variable dans Y, le préfaisceau sur Y

$$y \longmapsto \pi_0(y\backslash X_i)$$

n'est autre que $p_{i!}(e_{X_i})$, où

$$p_i : X_i \longrightarrow Y$$

est le composé $X_i \longrightarrow X \xrightarrow{p} Y$ (induit par p). On a ainsi prouvé

Corollaire 1 (de prop. 12). *Soit $f : X \longrightarrow I$ W-propre ([82]), et $p : X \longrightarrow Y$ un foncteur quelconque. Soit, pour tout $i \in \mathrm{Ob}\, I$, $p_i : X_i \longrightarrow Y$ le foncteur induit sur la fibre X_i de X en i. On a alors une loi* fonctorielle *naturelle*

$$i \longmapsto p_{i!}(e_{X_i}) : I \longrightarrow Y^\wedge,$$

et on a la formule

$$(*) \qquad p_!(e_X) \xleftarrow{\sim} \varinjlim_I p_{i!}(e_{X_i}),$$

où la flèche est induite par les morphismes canoniques

$$p_{i!}(e_{X_i}) \longrightarrow p_!(e_X)$$

induites par les inclusions $X_i \longrightarrow X$. p. 81

La loi fonctorielle de $p_{i!}(e_{X_i})$ en i a été explicitée en cours de démonstration. [page 119]

Notons que lorsque X sur I est cofibrée et scindée, i.e. provient d'un foncteur

$$i \longmapsto X_i, \quad I \longrightarrow \mathrm{Cat},$$

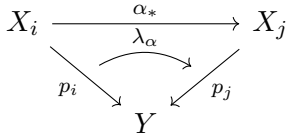
alors la loi fonctorielle de $p_{i!}(e_{X_i})$ en i s'explicite de façon particulièrement agréable : à une flèche

$$\alpha : i \longrightarrow j, \qquad \text{d'où } \alpha_* : X_i \longrightarrow X_j,$$

est associé

$$p_!(\alpha) : p_{i!}(e_{X_i}) \longrightarrow p_{j!}(e_{X_j}),$$

en notant qu'on a un diagramme



avec une flèche de commutativité

$$\lambda_\alpha : p_i \longrightarrow p_j \alpha_*$$

(pas `[un]` isomorphisme en général), d'où

$$p_{i!}(e_{X_i}) \xrightarrow{(\lambda_\alpha)_!} p_{j!}\underbrace{\alpha_{*!}(e_{X_i})}_{\longrightarrow e_{X_j}} \longrightarrow p_{j!}(e_{X_j}),$$

plus de doute c'est lui! ([83]).

82. dire W_0-propre …
83. Cette flèche `[la flèche` $(\lambda_\alpha)_!$`]` devrait être explicitée avec soin!

Commentary · Exposition [ms. 119]

The comparison map requested in footnote 83

The point of the split-opfibration calculation is to see the transition map before passing to components. This also answers the request in footnote 83 without relying on a convention for mates of adjunctions.

Let $\alpha : i \to j$, let $\alpha_* : X_i \to X_j$ be the chosen transport, and let $\lambda_x : p_i(x) \to p_j(\alpha_* x)$ be the image under p of the chosen cocartesian arrow over α. For every $y \in Y$ there is a functor

$$y\backslash X_i \longrightarrow y\backslash X_j, \qquad (x, a) \longmapsto (\alpha_* x, \lambda_x a). \tag{A3}$$

A morphism $v : (x, a) \to (x', a')$ is sent to $\alpha_* v$. This is a morphism of the target comma category because

$$p_j(\alpha_* v)\lambda_x a = \lambda_{x'} p_i(v) a = \lambda_{x'} a'.$$

Passing to components gives the displayed map $p_!(\alpha)$ in the manuscript. It is natural in y because both routes for a map $b : y' \to y$ replace a by ab. For two composable base arrows, the split identities and the cocartesian composition law identify the two composites of (A3). Hence these maps form the asserted covariant diagram of presheaves.

To recover the factorization written by Grothendieck, first keep the intermediate object $\alpha_{*!} 1_{X_i}$ and apply the natural transformation of left Kan extensions induced by $\lambda : p_i \Rightarrow p_j \alpha_*$. Then use the counit-induced map $\alpha_{*!} 1_{X_i} \to 1_{X_j}$. The first arrow is the mate of $(p_j \alpha_*)^* \Rightarrow p_i^*$, whose component on a presheaf F is $F(\lambda_x)$. Formula (A3) proves that this mate has the stated direction. An invertible λ is unnecessary.

If I has terminal object i_0, the colimit in (A2) is its value at i_0. We therefore obtain, naturally for every $p : X \to Y$,

$$p_! 1_X \cong p_{i_0 !} 1_{X_{i_0}}.$$

If X_{i_0} is totally W-aspherical, the category of elements of the right-hand presheaf is W-aspherical, which proves Proposition 11. The printed $p_{i_0 !}(X_{i_0})$ on the next page is a source slip: the argument must be $1_{X_{i_0}}$. Taking $X = Z \times I$ proves Corollary 3, since the product projection is a split opfibration with final fibre Z. The argument in fact requires only W_0-properness of f together with the stated total W-asphericity of its final fibre.

Corollaire 2 : *Sous les conditions du corollaire 1, supposons que I ait un objet final i_0.* [page 120]
Alors

$$(p_{i_0})_!(e_{X_{i_0}}) \longrightarrow p_!(e_X)$$

est un isomorphisme dans $Y^\wedge$.

Commentary · Exposition [ms. 119]

The transition map on representatives

The footnote asks that the arrow induced by λ_α be written out. This matters because the triangle is only laxly commutative: replacing its natural transformation by an equality would erase the data that moves a fibre over i to a fibre over j.

Suppose $X \to I$ is the split opfibration associated to a strict functor $D : I \to \mathrm{Cat}$, and write $X_i = D(i)$. An arrow $\alpha : i \to j$ gives $\alpha_* : X_i \to X_j$ and chosen opcartesian arrows $x \to \alpha_* x$ in X. Applying $p : X \to Y$ gives a natural transformation

$$\lambda_\alpha : p_i \longrightarrow p_j \alpha_*.$$

For $y \in Y$, an object of $y\backslash X_i$ is a pair $(x, \beta : y \to p_i(x))$. Define

$$L_{\alpha,y}(x, \beta) = \big(\alpha_* x, \lambda_{\alpha,x}\beta\big) \quad \text{in } y\backslash X_j. \tag{C3}$$

If $h : x \to x'$ is a morphism in the first comma category, so that $p_i(h)\beta = \beta'$, its image is $\alpha_* h$. Naturality of λ_α gives

$$p_j(\alpha_* h)\lambda_{\alpha,x}\beta = \lambda_{\alpha,x'} p_i(h)\beta = \lambda_{\alpha,x'}\beta',$$

which verifies the comma-category condition. Hence (C3) is a functor. Taking connected components yields the value at y of the required presheaf map

$$(p_i)_! e_{X_i} \longrightarrow (p_j)_! e_{X_j}.$$

Precomposition with a map $y' \to y$ commutes strictly with (C3), proving naturality in y.

This also identifies the two arrows in the manuscript's displayed composite. First keep the object x in X_i and replace β by $\lambda_{\alpha,x}\beta$. That is the map induced by λ_α from $(p_i)_! e_{X_i}$ to $(p_j\alpha_*)_! e_{X_i}$. Next apply α_* to x; under transitivity of Kan extension this is the counit map

$$(p_j)_!(\alpha_*)_! e_{X_i} \longrightarrow (p_j)_! e_{X_j}.$$

Their composite is exactly (C3), so its direction is fixed without an appeal to an unspecified mate convention.

Functoriality in i follows from the split identities. For $\alpha : i \to j$ and $\gamma : j \to k$ one has

$$(\gamma\alpha)_* = \gamma_*\alpha_*, \qquad \lambda_{\gamma\alpha,x} = \lambda_{\gamma,\alpha_* x}\lambda_{\alpha,x}.$$

Substitution in (C3) gives $L_{\gamma,y}L_{\alpha,y} = L_{\gamma\alpha,y}$; identity arrows act identically. Thus the transition maps constitute an actual functor $I \to \widehat{Y}$, not merely objectwise maps in a localization. With a non-split opfibration the comparison isomorphisms of a cleavage make the same formulas agree after taking components.

For a general W-proper functor, still assuming W(5), the manuscript obtains these transitions through the comma-category construction. The projection $y\backslash Y \to Y$ is a discrete opfibration, hence proper; base change and composition make $y\backslash X \to I$ proper. Its fibre over i is $y\backslash X_i$. Proposition 12 applied there gives

$$\pi_0(y\backslash X) \cong \varinjlim_{i\in I} \pi_0(y\backslash X_i).$$

The construction is natural in y and so proves the presheaf isomorphism of Corollary 1. If I has a terminal object, the colimit is its value at that object, recovering (C1). In the product case $X = Z \times I$, the terminal fibre is Z, which proves Corollary 3 as well.

D'où aussitôt la proposition 11 : si X_{i_0} est totalement W-asphérique, alors X l'est, car pour tout $p : X \longrightarrow Y$, $p_!(e_X) \simeq p_{i_0\,!}(X_{i_0})$, et ce dernier élément de $Y^\wedge$ est W-asphérique par hypothèse sur X_{i_0}.

Notons le cas particulier :

Corollaire 3. *Soit Z un objet totalement W-asphérique dans* Cat. *Alors son produit $Z \times I$ par toute I dans* Cat *ayant un objet final est aussi totalement W-asphérique.* p. 82

Commentary · Development (commentators') [ms. 120]

Total asphericity depends on more than homotopy type

The transfer result just proved has a stronger content than invariance of the classifying space. The finite construction developed below, in "A strict cocone can repair the wrong pushout", makes this distinction explicit.

Corollary (failure of homotopy invariance). [ours] Total W_∞-asphericity in the sense of this chapter is not invariant under weak homotopy equivalences of categories.

Proof. The cited construction supplies finite face posets E, S and a functor $f : E \to S$ such that BE is contractible, $BS \simeq S^2$, and every E/s is nonempty and connected. Their construction and all these properties are proved there. Set $p = f^{\mathrm{op}} : E^{\mathrm{op}} \to S^{\mathrm{op}}$. For each $s \in S$,

$$(p_! 1_{E^{\mathrm{op}}})(s) = \pi_0(s\backslash E^{\mathrm{op}}) = \pi_0((E/s)^{\mathrm{op}}) = \{*\}.$$

Thus $p_! 1_{E^{\mathrm{op}}}$ is the terminal presheaf on S^{op}, whose category of elements is S^{op}. Its classifying space is a two-sphere, so E^{op} is not totally W_∞-aspherical. Nevertheless $E^{\mathrm{op}} \to e$ is a weak homotopy equivalence. The category e is totally W_∞-aspherical: along any functor selecting $y \in Y$, its terminal presheaf is sent by left Kan extension to the representable $Y(-, y)$, and the category of elements Y/y has a terminal object. □

This corollary is a consequence of the explicit finite calculation below, not an assertion that the underlying topological phenomenon is new. It explains why Proposition 11 uses properness to identify ordinary Kan extensions rather than inferring total asphericity merely from $X_{i_0} \to X \in W$.

Commentary · Exposition [ms. 120]

A span is remembered by its transition arrows

The next construction turns an arbitrary span into a category in which the gluing instructions remain visible as morphisms. The ordinary pushout identifies their images. Integration keeps those images separate and inserts comparison arrows between them. This small change is what retains the homotopy information of the gluing.

Write $\Phi = (1 \to 0 \leftarrow 2)$ and let the span $X_1 \xleftarrow{\alpha_1} X_0 \xrightarrow{\alpha_2} X_2$ be the functor $\Phi^{\mathrm{op}} \to \mathrm{Cat}$. Its fibred Grothendieck construction $\mathcal{X}$ has the disjoint union of the three sets of objects. Within a fibre the morphisms are unchanged. The only morphisms between distinct fibres go from a side fibre to the middle fibre, and are

$$\mathrm{Hom}_{\mathcal{X}}(a, x) = \mathrm{Hom}_{X_k}(a, \alpha_k x), \qquad a \in X_k,\ x \in X_0,\ k = 1, 2. \tag{A4}$$

Composition on the source side is ordinary composition in X_k; composition with $v : x \to x'$ on the target side uses $\alpha_k(v)$. These rules satisfy associativity because α_k is a functor. They also show directly that $\mathcal{X} \to \Phi$ is a split fibration.

The identity of $\alpha_k x$ determines a morphism

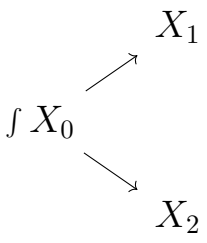
$$i_k \alpha_k(x) \longrightarrow i_0(x).$$

Naturality follows from the composition rule in (A4). These two transformations are the missing sides of the apparently noncommutative square on manuscript p. 123. Their sources and targets are usually different objects, so the square is not strictly commutative. Natural transformations give homotopies on nerves; for every fundamental localizer the resulting functors become equal after localization. The assertion that the square commutes in Hot_W is therefore correct, but does not replace the two transformations that constructed it.

This is the variance of the *smooth* integral used in this section. An opfibred construction of the covariant span places the transition arrows in the opposite direction. Grothendieck compares those two constructions later in §3.3. Reversing only the base while leaving the fibres untouched is not an innocuous notational change.

For $W = W_\infty$, realization of the nerve of $\mathcal{X}$ is a homotopy pushout of $BX_1 \leftarrow BX_0 \to BX_2$, where $BC = |NC|$. The closed-piece calculation below gives a direct proof in this particular shape. The general Grothendieck-construction theorem is due to Thomason (1979, Theorem 1.2); a primary modern proof and an explicit comparison map are given in Cegarra (2020, §3, equation (9) and Theorem 3). The manuscript expresses the corresponding fact intrinsically as a left Kan extension in the category of coherent diagrams. Its footnote 84 correctly retains the stronger localization hypothesis needed by the preceding construction of that adjoint.

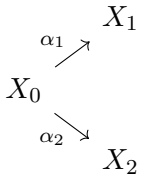
3 Sommes amalgamées et intégration



3.1 Réduction canonique au cas d'un diagramme d'immersions fermées

Considérons un diagramme

(∗)



dans Cat. Nous pouvons le regarder comme un contrafoncteur

$$\Phi^\circ \longrightarrow \mathrm{Cat},$$

où

$$\Phi = \begin{pmatrix} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{pmatrix},$$

donc comme définissant une catégorie fibrée (scindée) $\mathscr{X}$ sur Φ. Il est parfois commode de représenter cette catégorie $\mathscr{X}$ par le 'signe d'integration'

$$\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ \int X_0 & & \\ & \searrow & \\ & & X_2, \end{array}$$

et la regarder comme une sorte de 'somme amalgamée homotopique' de X_1 et X_2 sous X_0, ou comme une 'limite inductive homotopique' du diagramme de départ $(*)$.

Il faut entendre par là que (pour $W \subseteq \mathrm{Fl}(\mathrm{Cat})$ donné) l'image $\mathrm{hot}_W(\mathscr{X})$ de $\mathscr{X}$ dans Hot_W [page 122]
a un 'sens homotopique' remarquable en termes de $(*)$, de façon précise, le diagramme $(*)$ définit un objet ξ de $\mathrm{HOT}_W(\Phi)$ (qui est une catégorie de fractions de la catégorie de tous les diagrammes du type $(*)$ …), et si

$$\pi : \Phi \longrightarrow e$$

désigne la projection absolue de Φ, on a

$$\pi_!(\xi) = \mathrm{hot}_W(\mathscr{X}),$$

Commentary · Context [ms. 121]

Integration as a calculus of gluing

Grothendieck's integral is designed to make a family and its transition maps into one geometric object. Grothendieck's discussion of integration in Chapter VI, manuscript pages 81 and 146, gives the general background. Here the additional issue is quite specific: the total category must retain enough information to decide whether a given cocone is a homotopy pushout.

In *Pursuing Stacks*, §69, Grothendieck (1983) connects diagram categories to the trace problems of SGA 5, then describes integration by a total cofibred category. He also singles out presheaf categories as a setting where actual amalgamated sums are available. These two concerns meet in the present pages. The integral supplies the homotopical comparison; the last theorem shows that a pushout of presheaves along a monomorphism gives an actual categorical pushout with the required homotopy type. The intervening analysis of open immersions is the bridge between the two constructions.

The letter to Thomason of 2 April 1991 gives a closely related explanation of why one diagram category cannot suffice: a derived category "ne vient jamais seule" (Grothendieck, 1991a, §2). In §1 of the same letter, Grothendieck places weak equivalences and localization before the auxiliary classes used in model structures. This explains the order of the present argument. He constructs a comparison map first and asks whether it belongs to W; cofibration criteria are then ways of proving that assertion.

The source is also a record of the distinctions being worked out. A natural transformation provides a homotopy without becoming an equality of functors; a weak equivalence of total categories need not preserve every ordinary Kan extension; a gluing theorem need not reflect equivalences backwards. The proofs below keep the transition arrows, the presheaf adjunctions, and the excision hypotheses visible at precisely the steps where each is used.

Commentary · Exposition [ms. 122]

Integration retains the homotopy of the span

The shift to integration changes what is being glued. An ordinary pushout identifies the common category immediately. The integrated category instead retains a copy of that category and arrows connecting it to both sides. These arrows provide the homotopies needed for a derived colimit.

For the span $X_1 \xleftarrow{\alpha_1} X_0 \xrightarrow{\alpha_2} X_2$, the manuscript uses the indexing category $\Phi = (1 \to 0 \leftarrow 2)$ and a contravariant functor on it. The associated fibred category $\mathscr{X}$ has objects (k, x) with $x \in X_k$. It contains each X_k fully, and its additional morphisms are

$$\operatorname{Hom}_{\mathscr{X}}((k,x),(0,z)) = \operatorname{Hom}_{X_k}(x, \alpha_k z), \qquad k = 1, 2.$$

There are no arrows between the 1- and 2-fibres, nor from the 0-fibre to either of them. Composition with an arrow in X_k is precomposition; composition with an arrow $z \to z'$ in X_0 is postcomposition by its image under α_k. These rules give the split Grothendieck construction with precisely the variance used by Grothendieck. This continues the discussion of variance in Chapter VI, manuscript p. 81, rather than replacing it by a covariant convention without explanation.

For this span one can prove the homotopy-pushout assertion directly. Let M_k be the full subcategory of $\mathscr{X}$ on its 0- and k-fibres. It is a categorical mapping cylinder for α_k. The functor

$$r_k : M_k \longrightarrow X_k$$

is the identity on X_k and sends $z \in X_0$ to $\alpha_k z$. If $s_k : X_k \hookrightarrow M_k$ is the inclusion, then $r_k s_k = \mathrm{id}$, and the canonical arrow $\alpha_k z \to z$ in M_k gives a natural transformation $s_k r_k \to \mathrm{id}_{M_k}$, equal to the identity on X_k. Thus Br_k is a homotopy equivalence, and its restriction to BX_0 is $B\alpha_k$.

Every composable string in $\mathscr{X}$ lies in M_1 or M_2, and their intersection is exactly X_0. Therefore

$$N\mathscr{X} = NM_1 \amalg_{NX_0} NM_2.$$

Both maps out of NX_0 are monomorphisms, so this is a homotopy pushout of simplicial sets. The two homotopy equivalences Br_k, together with the identity on BX_0, now identify its homotopy type with

$$B\mathscr{X} \simeq \operatorname{hocolim}\big(BX_1 \leftarrow BX_0 \to BX_2\big).$$

This is the span instance of Thomason (1979, Theorem 1.2), whose statement is also recorded by Chachólski and Scherer (2001, Corollary 26.9); the direct proof records the orientation and the connecting morphisms needed in these pages.

For example, take X_0 to be two discrete objects and $X_1 = X_2 = e$. The ordinary pushout is e. The integrated category has four objects and four nonidentity arrows: each of the two side objects maps to each of the two middle objects. Its nerve is a square with no two-dimensional nondegenerate simplices, hence has realization S^1. The integration remembers the circle formed by the two independent identifications; the ordinary pushout has forgotten it.

The final formula $\pi_!(\xi) = \mathrm{hot}_W(\mathscr{X})$ expresses this construction in the derivator already under discussion. It is a canonical isomorphism in the localized category, rather than a literal equality of chosen representatives. For W_∞ the preceding argument gives its concrete meaning. For another W the formula presupposes the standing hypotheses under which the diagram localization and its homotopy Kan extension are available; an arbitrary class of arrows would not by itself justify that notation. The advertised reduction to a diagram of closed immersions begins here; it is not completed on manuscript page 122.

Overview · XIIb-a

Attachments, ordinary direct images, and homotopy.

These pages, Chapter XII, section V, manuscript pp. 59–122, begin with a concrete problem: how to attach one category to another along a full subcategory, and how to recognize when the attachment respects weak equivalences. The first part answers the categorical question completely. The second discovers how much more is needed for the homotopical question. Along the way Grothendieck introduces total asphericity, isolates the Dwyer condition, and reaches the Grothendieck construction as a replacement for an ordinary pushout. The text is exploratory in a mathematically consequential sense. Some tentative arguments are explicitly retracted; one proposed theorem survives in the text even though its asserted reverse implication is false. The commentary keeps these different situations separate.

An "open" subcategory $U \subseteq X$ is a sieve: an arrow whose target lies in U has its source there as well. Its complementary full subcategory T is a cosieve. There are no arrows from T to U, so the information not already contained in these two categories is exactly the bifunctor

$$H : U^{\mathrm{op}} \times T \longrightarrow \mathrm{Set}, \qquad H(u,t) = \mathrm{Hom}_X(u,t).$$

Its two functorial actions encode composition on either side of a crossing arrow. Conversely any such bifunctor constructs a category on the disjoint union of the object sets. This is the collage of a profunctor. Maps and natural transformations out of the collage have equally explicit descriptions. The formulas on manuscript p. 61 must be read relatively: changing U and T by pullback of H gives a cartesian lifting over the category of pairs of categories, not the unqualified representability asserted in the printed displays. The corrected universal property is proved in the commentary.

This description also explains the relation with open–closed gluing of topoi. A presheaf on the collage is a pair of presheaves on U and T, together with the restriction data across H. The gluing functor is

$$\varphi(F)(t) = \mathrm{Hom}_{\widehat{U}}(H(-,t), F).$$

It preserves limits and is right adjoint to the cocontinuous extension of $t \mapsto H(-,t)$. The converse recovers the collage from an accessible gluing functor preserving small limits. This is already the construction of SGA 4, Exposé IV, Exercise 9.5.10; the comparison is exact, including the characterization of the resulting topos as a presheaf topos (Artin et al., 1972). Natural transformations between right adjoints reverse the direction of transformations between their left adjoints, a convention that must be retained when stating the category-level equivalence. The historical connection helps explain why the categorical attachment problem first appears here in the language of an open and its closed complement.

The advantage of the collage is that a pushout can be computed by an ordinary Kan extension. Given $p : U \to U'$ and $r : T \to T'$, the new crossing bifunctor is

$$H' = \mathrm{Lan}_{p^{\mathrm{op}} \times r}\, H.$$

Its pointwise coend description records a crossing arrow together with paths in the new categories to its two ends, modulo the relations imposed by composition. When only U changes, this becomes

$$\mathrm{Hom}_{X'}(u',t) \cong \pi_0(u' \backslash U / t), \qquad X' = X \amalg_U U'.$$

The category on the right remembers factorizations through U; the ordinary pushout retains only their connected components. This formula proves that U' is again a sieve, that the complement remains T, and that the support of the gluing weight is unchanged. It is also the source of the later difficulty: taking components does not preserve the higher homotopy of a factorization category.

The gluing theorem that survives, and the reverse implication that fails.

If a category is the union of two sieves, every simplex of its nerve lies in one of them. Its nerve is therefore the simplicial union along the intersection, and the square is a homotopy pushout. Consequently weak equivalences on the intersection and on both pieces induce a weak equivalence on the union. This is the valid gluing direction. It does not imply the symmetric rule W(7), which asks that any two among equivalence on the intersection, equivalence on both pieces, and equivalence on the union imply the third.

Theorem 2 on manuscript p. 78 asserts that this stronger rule holds for W_∞. A complete counterexample uses a connected acyclic noncontractible finite complex. Take a finite poset A with that homotopy type and adjoin two incomparable terminal cone points. The two one-cone subcategories are sieves, each contractible, and their union has the homotopy type of the suspension of BA. The suspension is simply connected and acyclic, hence contractible, whereas BA remains noncontractible. Thus the map from a cone to the union is a weak equivalence, but the opposite map from the intersection to the other cone is not. The counterexample refutes both the reverse part of W(7 bis) and the full W(7); it does not merely exhibit a missing proof.

The commentary gives the classical two-cell input explicitly, verifies its homology and a nontrivial fundamental-group quotient, and passes to a finite simplicial model and its face poset (Hatcher, 2002, Example 2.38 and Theorem 2C.5). A further development supplies finite tests against every bounded homotopy-group version of the axiom. It also explains why the proposed Mayer–Vietoris argument succeeds for fixed-coefficient homology equivalences: the three mapping-cone complexes form a short exact sequence. The failure occurs in passing from that homological statement to the asserted unstable homotopical converse.

This finding changes how the subsequent pages must be read. Assertions deduced under W(7) or W(7 bis) remain conditional assertions. Their application to W_∞ cannot use Theorem 2. In particular, the proposed one-point converse to asphericity on pp. 84–85 is not an unconditional theorem about weak homotopy equivalences. The audit records the actual hypotheses and, where possible, supplies a direct proof of a useful consequence that avoids the false reverse principle.

Total asphericity measures ordinary images.

For a presheaf F on U, total W-asphericity means that the category of elements of $p_!F$ is W-aspherical for every functor $p : U \to U'$. These are ordinary set-valued Kan extensions. For a category V, the definition applies to its terminal presheaf. It is substantially stronger than $V \to e \in W$, and it is not interchangeable with other uses of "totally aspherical" in the literature on test categories.

The relevant ordinary factorization sends $f : V \to V'$ to

$$V \longrightarrow V'/f_!e_V \longrightarrow V'.$$

The second map is a discrete fibration; the first is final in the ordinary categorical sense. Its undercategories are nonempty and connected, rather than necessarily contractible. This is the comprehensive factorization. The commentary proves its universal property and relates it to the treatment of Street–Walters in Kelly's account (Kelly, 1982, §4.7). It yields a useful formulation developed here: a category is totally W-aspherical exactly when every target of an ordinary final functor out of it is W-aspherical. In particular the class of totally aspherical categories is closed under ordinary final images. This reformulation identifies the quantified condition without confusing ordinary finality with homotopy finality.

Grothendieck's attempted stability under unions fails because a left Kan extension need not preserve the monomorphisms describing a cover. His attempted use of retractions does not repair that defect: a categorical retraction does not supply a splitting of the associated inclusion of presheaves. A finite example makes both issues explicit. A final functor from the contractible outgoing three-object fork lands in a four-object category whose classifying

space is S^2. Its relevant comma category is a square, connected but not contractible. Two distinct representatives of a Kan-extension value become equal after adjoining one branch. The construction was located in Grothendieck's Appendix XII and is credited there. It also produces the pushout failures on pp. 104–107; adjoining a further cone gives a classifying space S^3, so their validity can be checked independently of the disputed one-point converse.

The converse from totality of a presheaf to totality of its category of elements is established here for subterminal presheaves, as the source proposes. A general answer was not located in the primary literature checked. The same limitation applies to certain composition and classification questions posed later in these pages. An unsuccessful search is not turned into a claim of present-day openness.

Good attachments and the passage to integration.

A W-Dwyer sieve asks that every nonempty crossing weight be totally W-aspherical. Ordinary Kan-extension transitivity proves that this condition is preserved by pushout. The stronger version, expressed through totality of the categories of elements themselves, uses the final first map of the comprehensive factorization. Classical Dwyer immersions are the representable case: the inclusion of U into the cosieve it generates admits a right adjoint. Filtered examples show why total asphericity is broader than the existence of a terminal object. The interval-product calculation on p. 114, however, shows directly why the classical adjoint condition is too rigid for every intended homotopy attachment.

The manuscript's W-cofibrations have since been studied as h-cofibrations, defined by preservation of weak equivalences under cobase change on undercategories. In left proper model categories they characterize attachments whose pushouts compute homotopy pushouts; the theory also recovers the equivalence between trivial h-cofibrations and couniversal weak equivalences (Batanin and Berger, 2017, Proposition 1.6, Corollary 1.9 and Lemma 1.10). This is a precise later form of the useful gluing principle. For classical Dwyer maps, the nerve comparison is now known to be a weak categorical equivalence, with a generalization to discretely flat cofibrations (Hackney et al., 2024, Theorems 1.5–1.6). These later notions are compared with the definitions here without identifying their different ambient weak equivalences.

The final properness criterion is independent of the failed cube axiom. Under the necessary connectivity hypothesis, the inclusion of a terminal fibre is ordinary final. Its terminal presheaf therefore has direct image equal to the terminal presheaf of the total category. Composing Kan extensions proves that total asphericity of that fibre implies total asphericity of the total category. The commentary gives the general component-colimit formula and writes out the natural transformation requested in the footnote on p. 119, including its action on representatives and its compatibility with composition.

The text then begins a new construction. Integrating a span retains its middle category and the arrows connecting it to both sides. For two points mapping to a point on each side, the ordinary pushout is a point, whereas the integrated category has classifying space a circle. Two categorical mapping cylinders give a direct proof that the integrated span models the homotopy pushout. The final displayed identity $\pi_! \xi \cong \mathrm{hot}_W(\mathscr{X})$ states the intended derivator interpretation. The announced reduction to closed immersions continues after that identity, beyond manuscript page 122.

d'après le calcul connu du foncteur $\pi_!$, adjoint à gauche du foncteur π^* p. 83

$$\mathrm{HOT}_W(e) \overset{\text{déf}}{=} \mathrm{Hot}_W \longrightarrow \mathrm{HOT}_W(\Phi)$$

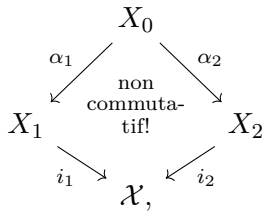
([84]). Or pour tout objet I dans Cat, de projection absolue $p_I : I \longrightarrow e$, le foncteur

$$p_{I!} : \mathrm{HOT}_W(I) \longrightarrow \mathrm{HOT}_W(e) = \mathrm{Hot}_W$$

doit être regardé comme l'équivalent 'homotopique' du foncteur limite inductive $\varinjlim_I$ des diagrammes de type I° dans la catégorie Hot_W - lesquelles limites inductives *n'existent pas* dans Hot_W (sauf pour des catégories très particulières - les seules que je connaisse sont les catégories discrètes (qu'il faut peut-être même supposer finies en plus)). [page 123]

Notons qu'on a un diagramme de foncteurs canoniques

(∗)



lequel diagramme n'a rien de commutatif dans Cat. *Il est cependant commutatif dans* Hot_W (quel que soit le localiseur fondamental W), et même dans Cat hot, car si on désigne par

$$i_0 : X_0 \longrightarrow \mathcal{X}$$

l'inclusion canonique, on a des homomorphismes canoniques

$$i_1\alpha_1 \longrightarrow i_0 \longleftarrow i_2\alpha_2,$$

ce qui montre que $i_1\alpha_1$ et $i_2\alpha_2$ sont homotopes, donc définissent la même flèche dans Hot_W.

Considérons dans Φ les deux fermés isomorphes à $\mathbf{\Delta}^1$ [page 124]

$$\begin{aligned} F_1 = 1\backslash\Phi \quad &= \{0,1\} \\ F_2 = 2\backslash\Phi \quad &= \{0,2\}, \end{aligned}$$

on a

$$F_1 \cap F_2 = 0\backslash\Phi = \{0\}, \qquad F_1 \cup F_2 = \Phi.$$

Commentary · Examples [ms. 123]

A circle hidden by an ordinary pushout

The distinction between a homotopy colimit and a colimit in the homotopy category is already visible in a span of finite discrete categories. Take $X_0 = \{a, b\}$ and $X_1 = X_2 = e$, with both structure maps the unique functor. Call the two side objects l, r. Formula (A4) gives precisely four nonidentity arrows:

$$l \to a, \quad l \to b, \quad r \to a, \quad r \to b.$$

No two nonidentity arrows are composable. The nondegenerate nerve is consequently a graph with four vertices and four edges, namely a four-cycle. Its realization is a circle. One can compute this without any homotopy-colimit theorem: choose three of the edges as a spanning tree and collapse

84. Cela suppose qu'on a W(2), non seulement W(2 bis).

that tree; the remaining edge is one loop. Thus $\pi_1(B\mathcal{X}) \cong \mathbb{Z}$, $H_1(B\mathcal{X};\mathbb{Z}) \cong \mathbb{Z}$, and all reduced homology in other degrees vanishes.

The ordinary pushout $e \amalg_{\{a,b\}} e$ in Cat is e. Indeed, a strictly compatible pair of functors from its two copies of e to Y chooses a single object of Y; both points of X_0 impose the same equality. The canonical comparison $\mathcal{X} \to e$ therefore collapses the circle and is not in W_∞. This example will fail Definition 3 exactly where it should.

The ordinary pushout of the same diagram also exists in the homotopy category of spaces and is a point. For a space T, a homotopy class $e \to T$ is a component of T. Compatibility of two such classes along the discrete two-point space forces them to be the same component, so the set of compatible pairs is $\pi_0(T)$, represented by e. A homotopy pushout instead remembers two paths implementing the two compatibilities; their difference produces the circle. Thus even existence of the ordinary colimit does not give the desired answer.

The parenthetical uncertainty on manuscript p. 123 about infinite discrete indexing categories should not be carried into the modern statement: the homotopy category of spaces has all small coproducts, represented by disjoint unions of CW models. What fails is a general ordinary-colimit calculus that recovers coherent homotopy colimits. The four-object calculation isolates that failure without making an unnecessary nonexistence assertion.

p. 84

Soient

$$\begin{aligned} 1\backslash\mathcal{X} = \mathcal{X} \times_\Phi 1\backslash\Phi &= \mathcal{X} \times_\Phi F_1 \\ 2\backslash\mathcal{X} = \mathcal{X} \times_\Phi 2\backslash\Phi &= \mathcal{X} \times_\Phi F_2 \end{aligned}$$

les images inverses de F_1, F_2 dans $\mathcal{X}$, on a donc

$$1\backslash\mathcal{X} \cap 2\backslash\mathcal{X} = X_0, \qquad 1\backslash\mathcal{X} \cup 2\backslash\mathcal{X} = \mathcal{X},$$

donc le diagramme

(∗∗)

$$\begin{array}{ccccc} & & X_0 & & \\ & {\scriptstyle\alpha'_1}\swarrow & & \searrow{\scriptstyle\alpha'_2} & \\ 1\backslash\mathcal{X} & & & & 2\backslash\mathcal{X} \\ & {\scriptstyle i'_1}\searrow & & \swarrow{\scriptstyle i'_2} & \\ & & \mathcal{X}, & & \end{array}$$

déduit du diagramme similaire

$$\begin{array}{ccccc} & & \{0\} & & \\ & \swarrow & & \searrow & \\ F_1 & & & & F_2 \\ & \searrow & & \swarrow & \\ & & \Phi, & & \end{array}$$

est (comme ce dernier) un carré d'immersions fermées à la fois cartésien et cocartésien. D'autre part, on a un homomorphisme canonique du diagramme (∗) dans (∗∗), qui est l'identité sur les sommets X_0 et $\mathcal{X}$, et qui sur X_1, X_2 est donné par les inclusions canoniques [page 125]

$$\underbrace{X_1}_{\mathcal{X}_1 \text{ fibre de } \mathcal{X} \text{ en } 1} \longrightarrow 1\backslash\mathcal{X}, \qquad \underbrace{X_2}_{\mathcal{X}_2 \text{ fibre de } \mathcal{X} \text{ en } 2} \longrightarrow 2\backslash\mathcal{X}.$$

Ces inclusions, on le sait, sont W-asphériques (elles ont même un adjoint à droite), a fortiori elles sont dans W. Donc le diagramme non commutatif (∗) est isomorphe, en tant que diagramme dans Hot_W, au diagramme commutatif cartésien et cocartésien (∗∗) formé d'immersions fermées. Comme $\mathcal{X}$ est bel et bien somme amalgamée de ses deux fermées $1\backslash\mathcal{X}$, $2\backslash\mathcal{X}$ sous leur intersection X_0, cela étaye l'intuition qui voit $\mathcal{X}$ comme une 'somme amalgamée homotopique'. Donnons tout de suite l'application suivante :

Commentary · Exposition [ms. 125]

Closed pieces give a genuine gluing diagram

Grothendieck now replaces the informal square by a strict cover, so that an excision axiom can actually be applied. The construction is a categorical version of replacing the two legs of a span by mapping cylinders.

Let C_k be the full subcategory of $\mathcal{X}$ on $X_k \amalg X_0$. This is $k\backslash\mathcal{X}$ in the manuscript. It is a cosieve: an arrow starting in its side fibre can only remain there or enter X_0, and arrows starting in X_0 remain in X_0. We have

$$C_1 \cap C_2 = X_0, \qquad C_1 \cup C_2 = \mathcal{X}, \qquad \mathcal{X} = C_1 \amalg_{X_0} C_2. \tag{A5}$$

The last equality holds in Cat, not just on objects. A string of composable arrows cannot pass from one side fibre to the other; it can enter the common fibre but cannot leave it. Giving a functor out of $\mathcal{X}$ is therefore exactly giving functors out of the two pieces that agree on X_0. The intersection statement also makes the square cartesian.

The inclusion $j_k : X_k \to C_k$ has right adjoint r_k, defined by $r_k(a) = a$ on X_k and $r_k(x) = \alpha_k x$ on X_0. On the mixed morphisms it uses (A4). Indeed, for every $a \in X_k$ and $z \in C_k$,

$$\mathrm{Hom}_{C_k}(j_k a, z) = \mathrm{Hom}_{X_k}(a, r_k z).$$

The unit is the identity and the counit is the canonical transition arrow on the middle fibre. Thus Bj_k and Br_k are mutually inverse homotopy equivalences, and both functors belong to every fundamental localizer.

For spaces, (A5) proves the promised homotopy-pushout calculation. Every simplex of $N\mathcal{X}$ lies in NC_1 or NC_2, and their intersection is NX_0. Hence it is the pushout of these nerves. The two inclusions of NX_0 are simplicial monomorphisms, whose realizations are CW-subcomplex inclusions, so this ordinary union computes the homotopy pushout of $BC_1 \leftarrow BX_0 \to BC_2$. The retractions r_k give a strictly commuting map of this span to $BX_1 \leftarrow BX_0 \to BX_2$, because $r_k|_{X_0} = \alpha_k$; all three components are homotopy equivalences. Homotopy invariance of the double mapping cylinder proves the result.

The phrase "homomorphism of diagrams" on p. 125 must retain this distinction: the inclusions from the original side fibres form a comparison with specified natural transformations, and become a comparison of commutative squares in Hot_W. The retractions just written give a strict map of spans in the other direction. No strict commutativity of the original square is being asserted or needed. For a general W, the same replacement reduces Proposition 13 to precisely the closed-cover form of the stated axiom W(7 bis).

Proposition 13. *Soit un diagramme* $X_1 \xleftarrow{\alpha_1} X_0 \xrightarrow{\alpha_2} X_2$ *dans* Cat, *d'où le diagramme* (∗), *p. 123,*

p. 85

$$\begin{array}{ccccc} & & X_0 & & \\ & \swarrow^{\alpha_1} & & \searrow^{\alpha_2} & \\ X_1 & & & & X_2 \\ & \searrow_{i_1} & & \swarrow_{i_2} & \\ & & \mathcal{X}, & & \end{array}$$

commutatif dans Hot_W *(mais non dans* Cat, *sauf si* $X_0 = \emptyset$*). Supposons que* W *satisfait* W(7 bis) (axiome du carré cocartésien, cf. p. 75). *Alors on a* [page 126]

$$\begin{cases} \alpha_1 \in W \iff i_2 \in W \\ \alpha_2 \in W \iff i_1 \in W. \end{cases} (^{85})$$

Démonstration. Utilisant l'homomorphisme précédent de (∗) dans (∗∗), on voit que chacune

85. **NB** Pour l'implication $\Longrightarrow$ on n'a besoin que de la forme directe de W(7 bis), i.e. (avec les notations de

des flèches α_1, α_2, i_1, i_2 est dans W, si et seulement si la flèche correspondante α'_1, α'_2, i'_1 ou i'_2 l'est. Donc les équivalences ci-dessus équivalent à

$$\alpha'_1 \in W \iff i'_2 \in W$$
$$\alpha'_2 \in W \iff i'_1 \in W,$$

lesquelles sont vraies en vertu de W(7 bis) (formulation duale) appliqué à $\mathcal{X}$ et aux deux fermés $1\backslash\mathcal{X}$, $2\backslash\mathcal{X}$ de $\mathcal{X}$.

Commentary · Development (commentators') [ms. 126]

Why converse excision is an additional axiom

Proposition 13 is correctly proved under its stated hypothesis. The strength of that hypothesis nevertheless matters: the displayed equivalences contain converse excision, whereas the usual invariance of a homotopy pushout under weak equivalences gives only the forward implication. A suspension makes the difference precise.

Proposition (the acyclic test for converse excision). [ours]Let W contain the weak homotopy equivalences of small categories and satisfy two-out-of-three. Suppose that, whenever $C = C_1 \cup C_2$ is a union of two full cosieves with intersection A, the implication

(A6) $$C_2 \longrightarrow C \text{ belongs to } W \quad\Longrightarrow\quad A \longrightarrow C_1 \text{ belongs to } W$$

holds. Then every category A whose classifying space is connected and integrally acyclic is W-aspherical.

Proof. Adjoin to A two distinct initial vertices l, r, with one arrow from each vertex to every object of A, and no arrows between l and r. Call the resulting category C and its two one-vertex enlargements C_1, C_2. The full subcategories C_1, C_2 are cosieves, and their intersection is A. Their classifying spaces are cones on BA; the classifying space of C is their union along BA, namely the unreduced suspension ΣBA.

Since BA is connected, van Kampen applied to the two slightly enlarged cones makes ΣBA simply connected. The reduced Mayer–Vietoris sequence gives $\widetilde{H}_n(\Sigma BA;\mathbb{Z}) \cong \widetilde{H}_{n-1}(BA;\mathbb{Z}) = 0$ for every n. A simply connected acyclic CW complex is contractible: if it had a first nonzero homotopy group, the Hurewicz theorem would identify it with a nonzero homology group; with all homotopy groups zero, Whitehead's theorem applies. Equivalently, use Hatcher (2002, Corollary 4.33). Thus $C_2 \to C$ is a weak homotopy equivalence, and lies in W. By (A6), $A \to C_1$ lies in W. Since $C_1 \to e$ also lies in W, two-out-of-three gives $A \to e \in W$. □

For completeness, an explicit finite CW complex shows why this conclusion is too strong for W_∞. Let K have one vertex, three oriented edges x, y, z, and three two-cells with relators

$$x^2(xyz)^{-1}, \qquad y^3(xyz)^{-1}, \qquad z^5(xyz)^{-1}.$$

Its cellular boundary $C_2(K;\mathbb{Z}) \to C_1(K;\mathbb{Z})$ has, in these bases, the matrix

$$M = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 4 \end{pmatrix}, \qquad \det M = -1.$$

It is an isomorphism. The other reduced cellular groups are zero, so K is acyclic. Its fundamental group is nontrivial: with permutation multiplication read from right to left, send

$$x \longmapsto (1\,2)(3\,4), \quad y \longmapsto (1\,3\,5), \quad z \longmapsto (1\,2\,5\,3\,4).$$

These permutations have orders $2, 3, 5$ respectively and product $xyz = 1$, so they satisfy the three relators and define a homomorphism with nontrivial image in A_5. A finite simplicial complex homotopy equivalent to K exists by Hatcher (2002, Theorem 2C.5); let A be its poset of nonempty faces. The nerve of A is its barycentric subdivision. Thus BA is connected and acyclic but has nontrivial fundamental group.

Apply the construction in the proof to this finite poset. Then C_1, C_2, and C are weakly contractible, while A is not. Hence $C_2 \to C$ belongs to W_∞ but $A \to C_1$ does not. This disproves

XVI) de Loc(6′a).

full W(7 bis) for W_∞. Mapping this closed-cover square to the trivial square of one-point categories also disproves the implication $b\,\&\,c \Rightarrow a$ in full W(7). The same example disproves the corresponding converses in Proposition 13 and the corollaries if their extra hypotheses are deleted. It does not contradict the propositions as actually stated.

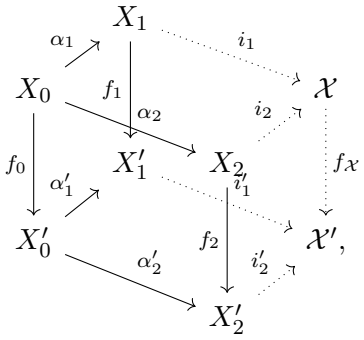
The topology behind the example is classical; compare the acyclic presentation complex in Hatcher (2002, Example 2.38). We checked that example, the manuscript's definitions of W(7 bis) and W(7), and the standard nerve and suspension arguments. We did not find this exact formulation as a test of these manuscript axioms in the references inspected. The proposition records the deduction needed here, without a claim of priority. In particular the assertion earlier in the chapter that W_∞ satisfies full W(7) cannot be justified by a Mayer–Vietoris argument alone: local systems on the contractible union restrict to constant systems on A, and do not test its nontrivial fundamental group.

There is a precise positive form of the five-lemma intuition. For a closed cover, every nondegenerate nerve simplex belongs to one of the two pieces, so normalized integral chains form a natural short exact sequence

$$0 \longrightarrow C_*(NA) \longrightarrow C_*(NC_1) \oplus C_*(NC_2) \longrightarrow C_*(NC) \longrightarrow 0. \tag{A7}$$

The first map has signs $(+,-)$ and the second adds the two inclusions. Exactness follows separately on each simplex, including degree zero. Given a map of two covers, if any two of the maps on the three terms of (A7) are quasi-isomorphisms, the third is a quasi-isomorphism by the associated long exact homology sequences. For the middle term this is equivalent to being a homology isomorphism on each piece. Thus all three implications of the cube axiom hold for integral homology equivalences of closed covers. The passage from this homological statement to weak homotopy equivalences is exactly the passage that the counterexample prevents.

Corollaire : *Considérons* deux *diagrammes de type* Φ°*, et un homomorphisme* $f = (f_0, f_1, f_2)$ *de l'un dans l'autre (cf. diagramme en traits pleins ci-contre* `[ci-dessous]`*):*



d'où un homomorphisme [page 127]

$$f_{\mathcal{X}} = \int f : \quad \mathcal{X} = \int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array} \qquad \longrightarrow \qquad \mathcal{X}' = \int \begin{array}{ccc} & & X_1' \\ X_0' & \nearrow & \\ & \searrow & \\ & & X_2', \end{array}$$

s'insérant dans un diagramme cubique (cf. diagramme précédent complété par des pointillés). p. 86
Considérons les conditions suivantes.

a) $f_0 \in W$.

b) f_1 *et* f_2 *sont* `[dans]` W.

c) $f_{\mathcal{X}} = \int (f_0, f_1, f_2) \in W$.

Supposons l'axiome W(7) ('lemme des 5', cf. page 76) *satisfait. Alors deux parmi les conditions a), b), c) impliquent la troisième*

$$\begin{cases} a \,\&\, b & \Longrightarrow & c \\ b \,\&\, c & \Longrightarrow & a \\ c \,\&\, a & \Longrightarrow & b. \end{cases}$$

En effet, on a un homomorphisme du diagramme cubique ci-dessus dans le diagramme similaire, où X_1, X_2, X'_1, X'_2 sont remplacés respectivement pas les fermés $1\backslash\mathcal{X}$, $2\backslash\mathcal{X}$ de $\mathcal{X}$, et $1\backslash\mathcal{X}'$, $2\backslash\mathcal{X}'$ de $\mathcal{X}'$, les deux carrés supérieur et inférieur étant des inclusions d'ouverts `[plutôt de fermés]` de $\mathcal{X}$ resp. de $\mathcal{X}'$. Ainsi, les conditions a), b), c) sont équivalentes respectivement aux conditions similaires pour le nouveau diagramme cubique (a) et c) n'ayant d'ailleurs pas [page 128] changé, et b) donnant

b′) $f'_1 : \mathcal{X}/1 \longrightarrow \mathcal{X}'/1$ et $f'_2 : \mathcal{X}/2 \longrightarrow \mathcal{X}'/2$ sont dans W.

`[plutôt` $f'_1 : 1\backslash\mathcal{X} \longrightarrow 1\backslash\mathcal{X}'$ `et` $f'_2 : 2\backslash\mathcal{X} \longrightarrow 2\backslash\mathcal{X}'$`]`). Alors la conclusion du corollaire est équivalente à W(7) (formulation duale) dans le cas du nouveau diagramme cubique, q.e.d.

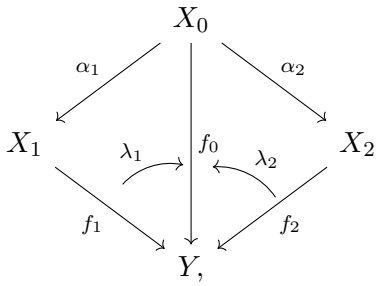
3.2 Comparaison de $\int X_0 \begin{matrix} \nearrow X_1 \\ \searrow X_2 \end{matrix}$ avec une somme amalgamée ordinaire.

Carrés W-homotopiquement cocartésiens.

La catégorie intégrée $\mathcal{X} = \int X_0 \begin{matrix} \nearrow X_1 \\ \searrow X_2 \end{matrix}$ est solution du problème universel suivant,

variante 2-catégorique de celui qui définit la limite inductive des diagrammes. Il s'agit, pour Y objet variable dans Cat, de considérer les systèmes de trois foncteurs f_0, f_1, f_2 et des deux homomorphismes fonctoriels λ_1, λ_2 : p. 87

(∗)



$$\begin{aligned} \lambda_1 : \quad f_1\alpha_1 &\longrightarrow f_0 \\ \lambda_2 : \quad f_2\alpha_2 &\longrightarrow f_0. \end{aligned}$$

Se donner une flèche de Cat [page 129]

$$\mathcal{X} \longrightarrow Y$$

'revient au même' que de se donner un système $(f_0, f_1, f_2, \lambda_1, \lambda_2)$ comme dessus, et ceci 'fonctoriellement en Y'. C'est là le cas particulier de la détermination générale des foncteurs de la catégorie fibrée 'intégrée' d'un contrafoncteur

$$I^\circ \longrightarrow \mathrm{Cat}, \qquad \begin{aligned} i &\longmapsto X_i \\ u &\longmapsto u^* \end{aligned}$$

vers une catégorie Y, en termes d'un système de foncteurs

$$X_i \xrightarrow{f_i} Y,$$

et d'homomorphismes fonctoriels

$$f_i \circ u^* \xrightarrow{\lambda_u} f_j$$

pour $u : i \longrightarrow j$, d'où $u^* : X_j \longrightarrow X_i$, avec des conditions de transitivité pour des flèches composées, dont on n'a pas à tenir compte dans le cas actuel, vu qu'il n'y a pas de flèches composées à regarder dans $I = \Phi$ (sauf des composées avec une flèche identique, qu'on peut se dispenser de regarder ici …).

Commentary · Exposition [ms. 129]

The universal property keeps its two-cells

The universal property on these pages explains why integration is a useful replacement for an ordinary colimit. It represents functors equipped with comparison transformations. Those transformations are part of the input and are allowed to be noninvertible.

We can prove the assertion directly from (A4). Suppose given $f_i : X_i \to Y$ and $\lambda_k : f_k\alpha_k \Rightarrow f_0$ for $k = 1, 2$. Define $F : \mathcal{X} \to Y$ to be f_i on each fibre. A mixed morphism from $a \in X_k$ to $x \in X_0$, represented by $h : a \to \alpha_k x$, must be sent to

$$f_k(a) \xrightarrow{f_k(h)} f_k(\alpha_k x) \xrightarrow{\lambda_{k,x}} f_0(x). \tag{A8}$$

This rule respects precomposition in X_k. It respects postcomposition by $v : x \to x'$ because naturality says

$$f_0(v)\lambda_{k,x} = \lambda_{k,x'} f_k(\alpha_k v).$$

There are no composites of two mixed arrows, so these checks prove functoriality. Conversely, restrict F to the fibres and apply it to the mixed arrows represented by identities. This recovers the f_i and the λ_k, and every mixed arrow factors as an internal arrow followed by one of these identities. The constructions are inverse.

The assertion also holds at the level of categories of functors. A natural transformation $F \Rightarrow G$ corresponds to transformations $\theta_i : f_i \Rightarrow g_i$ such that, for $k = 1, 2$,

$$\theta_0\lambda_k = \mu_k(\theta_k\alpha_k),$$

where μ_k are the comparison transformations for G. Necessity is naturality on the canonical mixed arrows; sufficiency follows from their factorization just used. Thus the universal property is a natural isomorphism of categories with the category of these lax cocones. The word "lax" here is fixed by the displayed direction $f_k\alpha_k \to f_0$, so it carries no hidden variance convention.

For a general strict functor $I^{\mathrm{op}} \to \mathrm{Cat}$, the additional condition is explicit. If $u : i \to j$ and $v : j \to k$, with maps $u^* : X_j \to X_i$ and $v^* : X_k \to X_j$, then

$$\lambda_{vu} = \lambda_v \circ (\lambda_u v^*), \qquad \lambda_{\mathrm{id}_i} = \mathrm{id}_{f_i}.$$

For the two-legged shape these conditions have no nonidentity instances. This is why the construction can be given here without a separate coherence argument.

When the original square is strictly commutative, set $f_0 = f_k\alpha_k$ and take both λ_k to be identities. Formula (A8) then defines φ_Q on every morphism and shows immediately that $\varphi_Q i_i = f_i$. Every chosen cartesian transition arrow is sent to an identity. This supplies the requested construction in footnote 87; the comma-category calculations announced there are carried out in the later lemmas on pp. 152–154. Definition 3 asks whether this particular comparison belongs to W. A bare commutative square in Hot_W would not determine the comparison, because it has forgotten the transformations from which it was made.

Si en particulier on a un diagramme *commutatif* dans Cat [page 130]

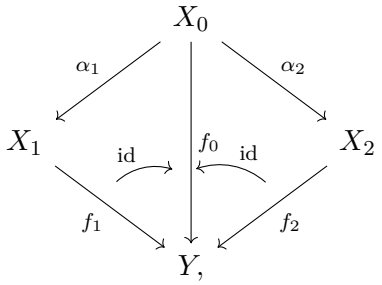
p. 88

$$Q \quad = \quad \begin{array}{ccccc} & & X_0 & & \\ & \overset{\alpha_1}{\swarrow} & & \overset{\alpha_2}{\searrow} & \\ X_1 & & & & X_2 \\ & \underset{f_1}{\searrow} & & \underset{f_2}{\swarrow} & \\ & & Y, & & \end{array}$$

on peut le compléter par

$$f_0 \overset{\text{déf}}{=} f_1\alpha_1 = f_2\alpha_2$$

en un diagramme

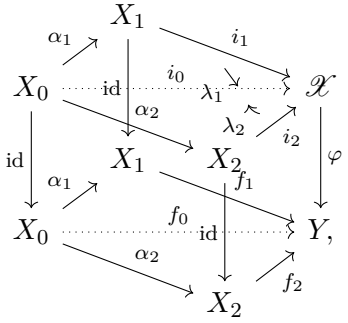


ce qui montre qu'on a une donnée du type $(*)$ ci-dessus, définissant donc une flèche

$$\varphi_Q : \quad \mathscr{X} = \int \begin{array}{c} X_1 \\ \nearrow \\ X_0 \\ \searrow \\ X_2 \end{array} \qquad \longrightarrow \qquad Y$$

qui donne lieu à un diagramme cubique

[page 131]



commutatif sauf le carré supérieur, i.e. on a p. 89

$$\varphi i_1 = f_1, \qquad \varphi i_2 = f_2, \qquad \text{et de plus} \quad \varphi i_0 = f_0,$$

φ transformant de plus les homomorphismes fonctoriels

$$\lambda_1 : i_1\alpha_1 \longrightarrow i_0, \qquad \lambda_2 : i_2\alpha_2 \longrightarrow i_0$$

en les homomorphismes fonctoriels identiques pertinents.

Définition 3. On dit que le diagramme commutatif

$(*)$

$$\begin{array}{ccccc} & & X_0 & & \\ & \overset{\alpha_1}{\swarrow} & & \overset{\alpha_2}{\searrow} & \\ X_1 & & & & X_2 \\ & \underset{f_1}{\searrow} & & \underset{f_2}{\swarrow} & \\ & & Y & & \end{array}$$

est W-*homotopiquement cocartésien* ([86]) si l'homomorphisme canonique

$$\mathscr{X} = \int \begin{array}{ccc} & \nearrow & X_1 \\ X_0 & & \\ & \searrow & X_2 \end{array} \xrightarrow{\varphi} Y$$

qu'il définit est dans W. ([87])

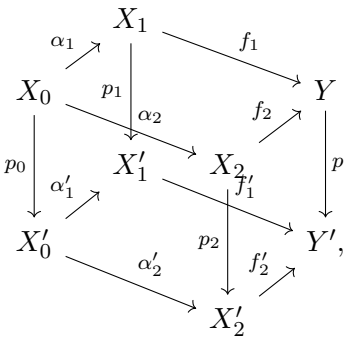
On a alors le : [page 132]

Corollaire 1. *Supposons* W(7 bis). *Si le carré* $(*)$ *est* W*-homotopiquement cocartésien, alors*

$$\begin{cases} \alpha_1 \in W & \Longleftrightarrow \quad f_2 \in W \\ \alpha_2 \in W & \Longleftrightarrow \quad f_1 \in W. \end{cases}$$

Cela résulte aussitôt du diagramme cubique de la page précédente, où toutes les flèches verticales sont dans W, et de l'énoncé similaire concernant la face supérieure du cube, donné dans la proposition 13, p. 125.

Corollaire 2. *Supposons* W(7). *Considérons un diagramme cubique* p. 90



où les deux faces inférieure et supérieure du cube sont des carrés W*-homotopiquement cocartésiens. Considérons les conditions* [page 133]

a) $p_0 \in W$.

b) p_1 *et* $p_2 \in W$.

c) $p \in W$.

Alors deux de ces conditions impliquent la troisième.

Résulte aussitôt de la définition 3, et du résultat similaire (cor. page 126) pour le diagramme cubique similaire, avec Y, Y' remplacées par $\mathscr{X} = \int \begin{array}{ccc} & \nearrow & X_1 \\ X_0 & & \\ & \searrow & X_2 \end{array}$ et $\mathscr{X}' = \int \begin{array}{ccc} & \nearrow & X'_1 \\ X'_0 & & \\ & \searrow & X'_2 \end{array}$.

86. ou simplement W-cocartésien, ou qu'il définit Y comme somme amalgamée W-homotopique de etc.

87. Introduire φ dans une proposition en forme, et donner ses propriétés essentielles : flèches strictement cartésiennes sont transformées en flèches *identiques* + lemmes 1 et 2 des p. 152, 154.

3.3 Intégration propre versus intégration lisse. Énoncé d'équivalence entre les deux points de vue. Commentaires sur le cas des limites W-homotopiques quelconques.

La définition 3, et les corollaires précédents, n'ont d'intérêt que dans la mesure où on dispose de critères maniables, permettant d'affirmer qu'un carré commutatif est bel et bien homotopiquement cocartésien. C'est ce qu'on va essayer de faire tantôt. Il y a aussi une perplexité, concernant le rôle privilégié joué, dans la théorie développée ici, par les immersions fermées, à l'exclusion des immersions ouvertes. Nous avons en somme ramené, du point de vue homotopique, les diagrammes quelconques $\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$ dans Cat, au cas d'un diagramme d'immersions fermées, où la somme amalgamée prend des allures familières. Mais pourquoi pas aussi bien le ramener au cas d'immersions ouvertes? On considérerait alors le diagramme de départ comme définissant une catégorie *cofibrée* sur

p. 91 [page 134]

$$\Phi^\circ = \Psi = \begin{array}{ccc} & & 1 \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & 2, \end{array}$$

soit $\tilde{\mathscr{X}}$. On a dans Ψ les deux ouverts

$$\underbrace{\Psi/1}_{=\{0,1\}} = U_1, \qquad \underbrace{\Psi/2}_{=\{0,2\}} = U_2, \qquad U_1 \cap U_2 = \{0\},$$

d'où dans $\tilde{\mathscr{X}}$ deux ouverts images inverses

$$\tilde{\mathscr{X}}/1, \qquad \tilde{\mathscr{X}}/2, \qquad \text{avec } \tilde{\mathscr{X}}/1 \cap \tilde{\mathscr{X}}/2 = X_0,$$

d'où un diagramme commutatif d'inclusions ouvertes

(∗)
$$\begin{array}{ccccc} & & X_0 & & \\ & \swarrow^{\alpha'_1} & & \searrow^{\alpha'_2} & \\ \tilde{\mathscr{X}}/1 & & & & \tilde{\mathscr{X}}/2 \\ & \searrow_{i'_1} & & \swarrow_{i'_2} & \\ & & \tilde{\mathscr{X}}, & & \end{array}$$

Commentary · Exposition [ms. 133]

The other orientation of integration

Grothendieck wants the definition of a homotopy pushout to belong to the span itself, without retaining the auxiliary choice of open or closed inclusions used to construct it. The question is quite concrete. The smooth integral introduced above has transition arrows pointing from the two outer fibres towards the central fibre. The proper integral has transition arrows in the other direction. Neither construction is obtained by merely changing the names of the objects.

Write $A = X_0$, $B = X_1$, $C = X_2$, and let $a : A \to B$ and $b : A \to C$ be the given functors. The proper integral P has the objects of A, B, and C, retained as three disjoint sets. In addition to the arrows within those categories, its mixed hom-sets are

$$P(x, y) = B(a(x), y), \qquad P(x, z) = C(b(x), z) \quad (x \in A,\ y \in B,\ z \in C).$$

There are no mixed arrows in the opposite directions, or between B and C. Composition is composition in B and C, with a and b applied when necessary. Thus a functor $P \to Y$ consists of functors f_A, f_B, f_C and transformations

$$f_A \longrightarrow f_B a, \qquad f_A \longrightarrow f_C b.$$

This is the reverse of the transformation data described in "A span is remembered by its transition arrows". For a strictly commuting cocone these transformations are identities, so both constructions have a canonical comparison with Y.

The full subcategory P_1 on A and B is a sieve in P, as is the corresponding P_2 on A and C. Their intersection is A, and their union is P. The inclusion $j : B \to P_1$ has a left adjoint r: it is the identity on B and sends $x \in A$ to $a(x)$. The identity

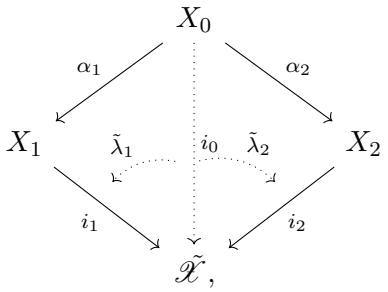
$$P_1(x, jy) = B(r(x), y)$$

proves the adjunction, including its mixed case. Consequently j is a weak equivalence for every fundamental localizer. The same argument applies to $C \to P_2$. This explains the parallel with the closed-cover replacement of the smooth integral: the two replacements have the same three vertex homotopy types, but their transition arrows have opposite orientations.

The equivalences asserted on manuscript page 135 inherit the hypotheses of their earlier counterparts. In particular, the reverse implication from an outer inclusion being in W to the corresponding original leg being in W uses the strong form of W(7 bis). The cube statement uses W(7). These are the additional excision assumptions analysed in "Why converse excision is an additional axiom". The comparison between the two integrals, which is the object of Proposition 14, does not require either converse-excision principle.

[page 135]

lequel reçoit le diagramme non commutatif



p. 92

les flèches

$$X_1 \longrightarrow \tilde{\mathscr{X}}/1, \qquad X_2 \longrightarrow \tilde{\mathscr{X}}/2$$

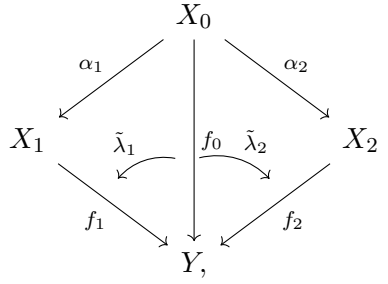
étant dans W (et même W-coasphériques - elles ont même un adjoint à gauche).

Donc on trouve, comme précédemment

$$\begin{cases} \alpha_1 \in W & \Longleftrightarrow \quad i_2 \in W \\ \alpha_2 \in W & \Longleftrightarrow \quad i_1 \in W, \end{cases}$$

et de même on peut formuler un 'lemme des cinq' comme le corollaire page 125 `[plutôt 126]`, mais avec $\tilde{\mathscr{X}}$, $\tilde{\mathscr{X}}'$ au lieu de $\mathscr{X}$, $\mathscr{X}'$.

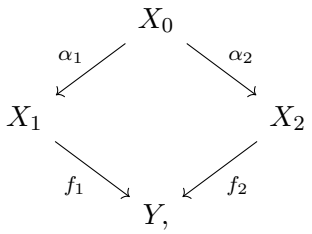
De plus, $\tilde{\mathscr{X}}$ est la solution d'un problème universel dans Cat, en termes de données



[page 136]

la seule différence avec celui formulé à la page 128 étant la direction opposée des flèches λ_1, λ_2. Pour tout carré commutatif

$(*)$



on trouve encore une flèche canonique

$$\tilde{\varphi} : \tilde{\mathscr{X}} \longrightarrow Y.$$

Il serait tout aussi raisonnable de dire que le carré $(*)$ est W-homotopiquement cocartésien, lorsque $\tilde{\varphi}$ est `[dans]` W, que d'adopter la définition 3, page 131 - on trouverait exactement les mêmes corollaires. Les deux définitions seraient elles équivalentes??

Je note que $\tilde{\mathscr{X}}$ étant cofibrée sur $\Psi = \Phi^\circ$, $\tilde{\mathscr{X}}^\circ$ est fibrée sur $\Psi^\circ = \Phi$. De façon précise, on aura p. 93

$$\tilde{\mathscr{X}}^\circ = \int \begin{array}{ccc} & \overset{\alpha_1^\circ}{\nearrow} & (X_1)^\circ \\ (X_0)^\circ & & \\ & \underset{\alpha_2^\circ}{\searrow} & (X_2)^\circ, \end{array}$$

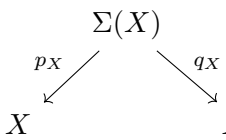
et il s'agit donc de comparer les deux intégrales [page 137]

$$\mathscr{X} = \int \begin{array}{ccc} & \overset{\alpha_1}{\nearrow} & X_1 \\ X_0 & & \\ & \underset{\alpha_2}{\searrow} & X_2 \end{array}, \qquad \tilde{\mathscr{X}}^\circ = \int \begin{array}{ccc} & \overset{\alpha_1^\circ}{\nearrow} & (X_1)^\circ \\ (X_0)^\circ & & \\ & \underset{\alpha_2^\circ}{\searrow} & (X_2)^\circ \end{array},$$

notamment comme objets de $\mathrm{HOT}_W(\Phi)$. Pour cela, je reprends la construction du foncteur

$$X \longmapsto \Phi(X) : \mathrm{Cat} \longrightarrow \mathrm{Cat}$$

de I, p. 72-75, mais je préfère ici noter ce foncteur $\Sigma(X)$ (car Φ désigne déjà une petite catégorie $\begin{array}{ccc} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{array}$). On utilisera les homomorphismes fonctoriels en X :



qui sont W-asphériques, donc $\in W$. De plus, $\Sigma(X^\circ) = \Sigma(X)$, $p_{X^\circ} = q_X$, $q_{X^\circ} = p_X$. Ainsi, si

$$X_* \quad = \quad \begin{array}{ccc} & & X_1 \\ & \overset{\alpha_1}{\nearrow} & \\ X_0 & & \\ & \underset{\alpha_2}{\searrow} & \\ & & X_2 \end{array} \qquad \in \quad \mathrm{Hom}(\Phi^\circ, \mathrm{Cat})$$

$$X_*^\circ \quad = \quad \begin{array}{ccc} & & (X_1)^\circ \\ & \overset{\alpha_1^\circ}{\nearrow} & \\ (X_0)^\circ & & \\ & \underset{\alpha_2^\circ}{\searrow} & \\ & & (X_2)^\circ \end{array} \qquad \in \quad \mathrm{Hom}(\Phi^\circ, \mathrm{Cat}),$$

on trouve [page 138]

$$\Sigma(X_*) = \begin{pmatrix} & & \Sigma(X_1) \\ & \overset{\Sigma(\alpha_1)}{\nearrow} & \\ \Sigma(X_0) & & \\ & \underset{\Sigma(\alpha_2)}{\searrow} & \\ & & \Sigma(X_2) \end{pmatrix}$$

et les flèches p. 94

$$\begin{array}{ccc} & \Sigma(X_*) & \\ \overset{p_{X_*}}{\swarrow} & & \overset{q_{X_*}}{\searrow} \\ X_* & & X_*^\circ, \end{array}$$

dans $\underline{\mathrm{Hom}}(\Phi^\circ, \mathrm{Cat})$. Ces flèches sont dans W terme à terme, donc les homomorphismes dans Cat

$$\begin{array}{ccc} & \int \Sigma(X_*) & \\ \overset{\int p_{X_*}}{\swarrow} & & \overset{\int q_{X_*}}{\searrow} \\ \mathscr{X} = \int X_* & & \int X_*^\circ = \tilde{\mathscr{X}}^\circ \end{array}$$

sont dans W.

Commentary · Exposition [ms. 137]

A common resolution of a category and its opposite

Grothendieck's comparison uses a resolution that reverses the category without reversing the direction of the functors between categories. This distinction is responsible for the utility of Σ. Taking opposites is a covariant operation on Cat: a functor $u : A \to B$ gives $u^\circ : A^\circ \to B^\circ$ in the same direction. Applied objectwise, it therefore preserves the shape of the span.

Here is a concrete convention for $\Sigma(C)$. Its objects are arrows $h : x \to y$ of C. An arrow from $h : x \to y$ to $h' : x' \to y'$ is a pair

$$r : x \to x', \qquad s : y' \to y, \qquad h = sh'r.$$

The projections $p_C(h) = x$ and $q_C(h) = y$ take values in C and C°, respectively. This is the opposite of one common convention for the twisted-arrow category. Reversing every displayed arrow identifies $\Sigma(C^\circ)$ with $\Sigma(C)$ and interchanges p and q. The equality written in the manuscript may be read as this specified canonical identification.

For completeness, the asphericity of the projections can be seen from their fibres. The projection p_C is a split fibration: over $r : x \to x'$ the inverse-image functor sends $h' : x' \to y'$ to $h'r : x \to y'$. Its fibre over x is $(x\backslash C)^\circ$, which has the final object id_x. To test asphericity, pull back along $C/c \to C$. The resulting category $\Sigma(C)/c$ is again split fibred, now over C/c, with those same aspherical fibres.

Since C/c has a final object, the fibre theorem gives that $\Sigma(C)/c$ is W-aspherical. Thus p_C is W-aspherical. Applying the identical argument to C° proves the assertion for q_C.

The importance of the resolution is its strict functoriality. For a diagram $F : I \to \mathrm{Cat}$, it supplies a diagram of natural transformations

$$F \xleftarrow{p_F} \Sigma F \xrightarrow{q_F} F^\circ$$

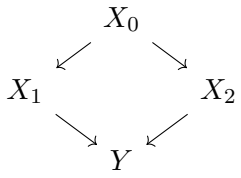
whose components lie in W. Integrating the two transformations again gives weak equivalences. At the level of total categories this compares the smooth integral of F with the smooth integral of F°, the latter being canonically the opposite of the proper integral of F. Applying the same resolution once more to that total category removes the final opposite.

For a strict cocone Q into Y, all these maps commute with the comparison maps into Y, $\Sigma(Y)$, or Y°. The calculation on manuscript page 140 is therefore a calculation in an arrow category, not merely a list of equivalences between their sources. In each commuting square, both horizontal maps belong to W; two-out-of-three gives that one vertical comparison belongs to W precisely when the other does. This proves the equivalence of the four conditions in Proposition 14(b), and the same zigzags prove the stronger assertions about coherent diagrams in parts (c) and (d).

This proof is also useful for its scope. It compares the images after inverting the chosen weak equivalences; it does not assert that C and C° are equivalent categories. The fork $0 \to 1$, $0 \to 2$ already separates those statements, since it has an initial object and no final object, while its opposite has a final object and no initial object.

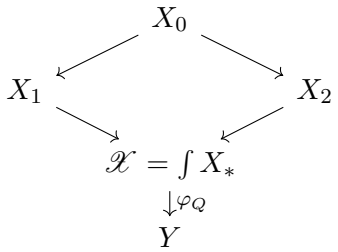
Finalement, l'innocent carré commutatif

(∗) Q =



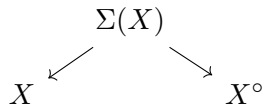
(et le diagramme correspondant

(∗∗) $\widehat{Q}$ =



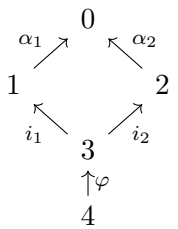
)

donne naissance, en le transformant par $X \longmapsto X^\circ$ et par $X \longmapsto \Sigma(X)$, et en tenant compte des homomorphismes fonctoriels en X [page 139]



, à trois diagrammes du type (∗∗), i.e. modelés sur la catégorie opposée de

Γ =



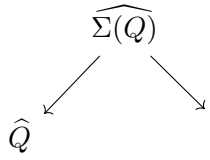
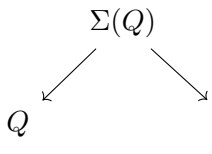
(i.e. $\mathbf{\Delta}^1 \times \mathbf{\Delta}^1$ ([88]), auquelle on a rajouté un objet initial)

([89]), savoir

88. faux, car le carré n'est pas commutatif

$$\widehat{Q}, \qquad \widehat{\Sigma(Q)}, \qquad \widehat{Q^\circ},$$

et deux homomorphismes de diagrammes ([dans] $\underline{\mathrm{Hom}}(\Gamma^\circ, \mathrm{Cat})$)

$(***)$  (déduit de ),

lesquels sont dans W terme à terme (i.e. sont dans $W_\Gamma \subseteq \mathrm{Fl}(\underline{\mathrm{Hom}}(\Gamma^\circ, \mathrm{Cat}))$). Explicitement, en les sommets $i = 0, 1, 2$, resp. 4 de Γ, $(***)$ n'est autre que

$$D_i = \left(\begin{array}{ccc} & \Sigma(X_i) & \\ {\scriptstyle p_{X_i}} \swarrow & & \searrow {\scriptstyle q_{X_i}} \\ X_i & & X_i^\circ \end{array} \right) \qquad \text{resp.} \qquad \begin{array}{ccc} & \Sigma(Y) & \\ {\scriptstyle p_Y} \swarrow & & \searrow {\scriptstyle q_Y} \\ Y & & Y^\circ, \end{array}$$

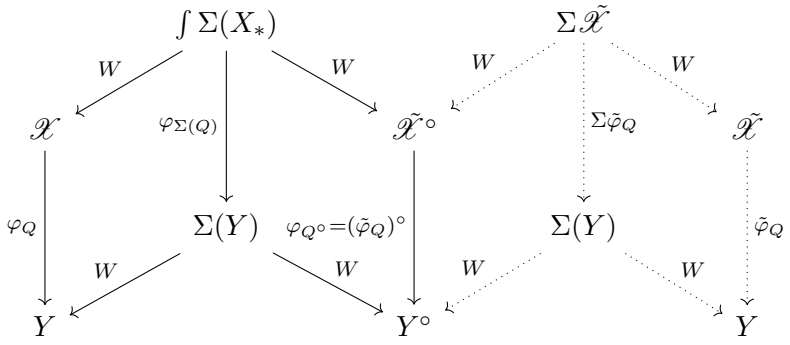
en le sommet 3 c'est

$$\begin{array}{ccc} & \int \Sigma(X_*) & \\ {\scriptstyle \int p_{X_*}} \swarrow & & \searrow {\scriptstyle \int q_{X_*}} \\ \mathscr{X} = \int X_* & & \int X_*^\circ = \tilde{\mathscr{X}}^\circ, \end{array}$$

obtenu par intégration des flèches

$$\begin{array}{ccc} & \Sigma(X_*) & \\ {\scriptstyle p_{X_*}} \swarrow & & \searrow {\scriptstyle q_{X_*}} \\ X_* & & X_*^\circ. \end{array}$$

Regardant la restriction des deux morphismes de diagrammes $(***)$ aux parties des trois diagrammes de type Γ° au dessus de la queue $\{3, 4\}$ de Γ on trouve le diagramme en traits pleins que voici [page 140]



qu j'ai complété par un deuxième volet en pointillés. Dans ce diagramme de quatre carrés accolés, les côtés horizontaux des carrés représentent des flèches de Cat qui sont dans W. On trouve ainsi : p. 96

$$\varphi_Q \in W \iff \varphi_{\Sigma Q} \in W \iff (\varphi_{Q^\circ} = \tilde{\varphi}_Q^\circ) \in W \iff \Sigma\tilde{\varphi}_Q \in W \iff \tilde{\varphi}_Q \in W.$$

On trouve aussi que les cinq flèches verticales sont isomorphes, en tant que flèches de Hot_W. Pour résumer :

89. On a $\alpha_1 i_1 \neq \alpha_2 i_2$, mais la relation $\alpha_1 i_1 \varphi = \alpha_2 i_2 \varphi$.

Commentary · Exposition [ms. 139]

The indexing category must retain both paths

Grothendieck keeps the entire comparison diagram because the two routes from X_0 to the integral become equal only after applying the map to Y. The correction in footnotes 88 and 89 is essential to that purpose. Before reaching Y, the routes even land in different fibres of the integral. They cannot be made equal by calling the drawing a commutative square.

Let $J = \Gamma^\circ$. A precise presentation has vertices $0, 1, 2, 3, 4$, arrows

$$0 \xrightarrow{a_1} 1 \xrightarrow{j_1} 3 \xrightarrow{v} 4, \qquad 0 \xrightarrow{a_2} 2 \xrightarrow{j_2} 3,$$

and the single relation $vj_1a_1 = vj_2a_2$. The two arrows $j_1a_1, j_2a_2 : 0 \to 3$ remain distinct. Thus 4 is terminal, but the full subcategory on $0, 1, 2, 3$ is the free diamond, not the ordered square $[1] \times [1]$. This is exactly the relation stated by footnote 89, with the orientation reversed.

To include the dotted central inclusion mentioned in footnote 91, add a generator $r : 0 \to 3$ and the relation $vr = vj_1a_1$. The natural transformations connecting j_1a_1 and j_2a_2 with r are additional two-dimensional data; they are not relations in this ordinary indexing category. Every resolution used in Proposition 14 also preserves r, so adding it causes no difficulty for the objectwise weak-equivalence argument. It simply records more of the construction.

Proposition 14. *Considérons un diagramme (carré commutatif)*

$$Q \quad = \quad \begin{array}{ccc} & X_0 & \\ \swarrow & & \searrow \\ X_1 & & X_2 \\ \searrow & & \swarrow \\ & Y & \end{array}$$

dans Cat*, et le diagramme 'opposé' déduit par* $X \longmapsto X^\circ$

$$Q^\circ \quad = \quad \begin{array}{ccc} & X_0^\circ & \\ \swarrow & & \searrow \\ X_1^\circ & & X_2^\circ \\ \searrow & & \swarrow \\ & Y^\circ, & \end{array}$$

et les flèches correspondantes [page 141]

$$\begin{cases} \varphi_Q : \quad \mathscr{X}_Q \quad = \quad \left(\int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array} \right) \quad \longrightarrow \quad Y \\ \varphi_{Q^\circ} : \quad \mathscr{X}_{Q^\circ} \quad = \quad \left(\int \begin{array}{ccc} & & X_1^\circ \\ X_0^\circ & \nearrow & \\ & \searrow & \\ & & X_2^\circ \end{array} \right) \quad \longrightarrow \quad Y^\circ. \end{cases}$$

Enfin, considérons aussi les flèches $\tilde{\varphi}_Q$, $\tilde{\varphi}_{Q^\circ}$ *canoniques, relative à l'intégrale propre* $\int_{\mathrm{pr}}$ *(au* p. 97

lieu de l'intégrale lisse $\int = \int_{\mathrm{lis}}$)

$$\begin{cases} \tilde{\varphi}_Q : \ \tilde{\mathscr{X}}_Q \ = \ \left(\int_{\mathrm{pr}} X_0 \begin{array}{c} \nearrow X_1 \\ \searrow X_2 \end{array} \right) \longrightarrow Y \\ \tilde{\varphi}_{Q^\circ} : \ \tilde{\mathscr{X}}_{Q^\circ} \ = \ \left(\int_{\mathrm{pr}} X_0^\circ \begin{array}{c} \nearrow X_1^\circ \\ \searrow X_2^\circ \end{array} \right) \longrightarrow Y^\circ. \end{cases}$$

On a alors ce qui suit :

ⓐ $\varphi_{Q^\circ} \simeq (\tilde{\varphi}_Q)^\circ$, $\varphi_Q \simeq (\tilde{\varphi}_{Q^\circ})^\circ$ *(d'où résulte déjà que* $\begin{cases} \varphi_Q \in W & \Longleftrightarrow & \tilde{\varphi}_{Q^\circ} \in W \\ \varphi_{Q^\circ} \in W & \Longleftrightarrow & \tilde{\varphi}_Q \in W \end{cases}$ *([90]))*.

ⓑ *Les relations*

$$\varphi_Q \in W, \qquad \varphi_{Q^\circ} \in W, \qquad \tilde{\varphi}_Q \in W, \qquad \tilde{\varphi}_{Q^\circ} \in W$$

sont équivalentes, en d'autres termes : pour que le carré soit W-homotopiquement cocartésien (version intégrale lisse), il faut et il suffit que le carré opposé Q° le soit, et ces conditions équivalent aussi aux conditions correspondantes en termes des intégrales propres (au lieu des intégrales lisses).

ⓒ *Les quatre flèches φ_Q, φ_{Q°, $\tilde{\varphi}_Q$, $\tilde{\varphi}_{Q^\circ}$ de* Cat *en tant que flèches dans* Hot_W*, sont isomorphes dans* $\underline{\mathrm{Fl}}(\mathrm{Hot}_W)$*. Plus précisément, en tant que flèches de* Cat*, elles définissent des objets isomorphes de* $\mathrm{HOT}_W(\mathbf{\Delta}^1)$*.* [page 142]

ⓓ *Plus précisément encore, les quatre diagrammes dans* Cat *de type* Γ°*, où*

$$\Gamma = \begin{array}{ccccc} & & 1 & & \\ & \overset{\alpha_1}{\swarrow} & & \overset{i_1}{\nwarrow} & \\ 0 & & & & 3 \overset{\varphi}{\longleftarrow} 4, \\ & \underset{\alpha_2}{\nwarrow} & & \underset{i_2}{\swarrow} & \\ & & 2 & & \end{array}$$

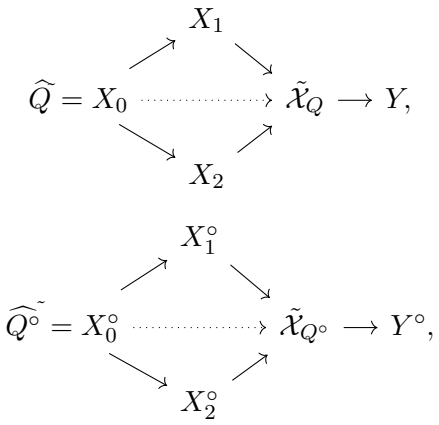
$\alpha_1 i_1 \varphi = \alpha_2 i_2 \varphi$, dans lesquels s'insèrent les quatre flèches qu'on vient d'envisager, savoir

$$\widehat{Q} = \begin{array}{ccccc} & \overset{\alpha_1}{\nearrow} & X_1 & \overset{i_1}{\searrow} & \\ X_0 & \cdots\cdots & & \cdots\cdots\rightarrow & \mathscr{X}_Q \overset{\varphi_Q}{\longrightarrow} Y, \\ & \underset{\alpha_2}{\searrow} & X_2 & \underset{i_2}{\nearrow} & \end{array}$$

p. 98

$$\widehat{Q^\circ} = \begin{array}{ccccc} & \overset{\alpha_1^\circ}{\nearrow} & X_1^\circ & \overset{i_1'}{\searrow} & \\ X_0^\circ & \cdots\cdots & & \cdots\cdots\rightarrow & \mathscr{X}_{Q^\circ} \longrightarrow Y^\circ, \\ & \underset{\alpha_2^\circ}{\searrow} & X_2^\circ & \underset{i_2'}{\nearrow} & \end{array}$$

90. et que les flèches φ_Q et $\tilde{\varphi}_{Q^\circ}$ d'une part, φ_{Q° et $\tilde{\varphi}_Q$ d'autre part, sont isomorphes comme flèches dans Hot_W.



([91]) *sont isomorphes dans* $\mathrm{Hot}_W(\Gamma)$. *De façon plus précise encore, on a les isomorphismes dans* $\underline{\mathrm{Hom}}(\Gamma^\circ, \mathrm{Cat})$

$$(\widetilde{Q})^\circ \simeq \widetilde{Q^\circ},$$

d'où en l'appliquant à Q° *au lieu de* Q,

$$(\widetilde{\widehat{Q^\circ}})^\circ \simeq \widehat{Q},$$

ce qui implique (via les isomorphismes canoniques valables dans $\mathrm{HOT}_W(S)$ *pour tout objet* S *dans* Cat *et tout* $X_* \in \mathrm{Ob}\,\underline{\mathrm{Hom}}(S, \mathrm{Cat})$, $X_* \simeq X_*^\circ$ ([92])*, qu'on a les isomorphismes canoniques*

$$\left.\begin{array}{c} \widehat{Q} \simeq \widetilde{\widehat{Q^\circ}} \\ \widehat{Q^\circ} \simeq \widehat{\widetilde{Q}} \end{array}\right\} \qquad \textit{dans } \mathrm{HOT}_W(\Gamma),$$

mais aussi directement

$$\widehat{\widetilde{Q}} \simeq \widehat{\widetilde{Q^\circ}}, \qquad \widetilde{\widehat{Q}} \simeq \widetilde{\widehat{Q^\circ}} \qquad \textit{dans } \mathrm{HOT}_W(\Gamma)$$

(et il reste à vérifier qu'on a ainsi un système transitif *d'isomorphismes canoniques dans* $\mathrm{HOT}_W(\Gamma)$*, entre les quatre objets envisagés, ce qui doit être essentiellement sorital …).* [page 143]

Commentary · Development (commentators') [ms. 143]

The four comparisons form a coherent system

Grothendieck leaves a small but real task after Proposition 14: the canonical comparisons should compose transitively. Without this check, choosing a route through the four constructions could leave an unspecified automorphism. The following argument supplies the check directly in the localized category of diagrams. We did not find this particular four-comparison verification stated separately in the literature consulted; it is a formal completion of the construction on these pages, with no claim of priority.

Coherence of the four integration comparisons. [ours] Let Q be a strictly commuting cocone on a small category-valued diagram, and use the same expanded indexing category J for the integral, the cocone, and their structural inclusions. In the localization

$$\mathcal{L}_J = \mathrm{Fun}(J, \mathrm{Cat})[W_J^{-1}],$$

the comparisons furnished by Σ between the smooth and proper integrals of Q and of Q° form a transitive system of isomorphisms. The assertion includes the arrows from the integral to the cocone vertex and the optional central inclusions.

Proof. Write O for objectwise passage to opposite categories. It preserves W_J, so it acts on $\mathcal{L}_J$. The resolution of the preceding block gives a natural isomorphism

$$\eta_D : D \xrightarrow{\sim} OD, \qquad \eta_D = q_D p_D^{-1}.$$

91. **NB** On devrait encore compléter ces diagrammes (non commutatifs) par les flèches en pointillés …
92. via le foncteur Σ utilisé plus haut …

Because $\Sigma(OD)$ is identified with $\Sigma(D)$ with its projections exchanged,

$$\eta_{OD} = \eta_D^{-1}.$$

Naturality holds first for actual maps of diagrams and then for their inverses in the localization; hence it holds for every arrow in $\mathcal{L}_J$. Applying it to η_D gives

$$O(\eta_D)\eta_D = \eta_{OD}\eta_D, \qquad O(\eta_D) = \eta_D^{-1}.$$

Thus the opposite operation and its canonical comparison already satisfy the required involution identity.

Now put $A = \widehat{Q}_{\mathrm{lis}}$ and $B = \widehat{Q^\circ}_{\mathrm{lis}}$. Integrating the objectwise resolution of the original diagram, while using $\Sigma(Y)$ at the cocone vertex, gives

$$\alpha_Q : A \xrightarrow{\sim} B.$$

The same resolution for Q° interchanges its two projections, including at the integrated vertex. Consequently $\alpha_{Q^\circ} = \alpha_Q^{-1}$. By the strict descriptions of the two Grothendieck constructions, the other two objects are canonically

$$C = \widehat{Q}_{\mathrm{pr}} = OB, \qquad D = \widehat{Q^\circ}_{\mathrm{pr}} = OA.$$

Proper integration is the composite of objectwise opposite, smooth integration, and objectwise opposite again. Since $O(\eta_Q) = \eta_Q^{-1}$ also holds in the localized category of input cocones, applying this composite to the resolution shows that its direct proper comparison from C to D is $O(\alpha_Q^{-1})$. Naturality of η for α_Q says exactly

$$\eta_B\alpha_Q = O(\alpha_Q)\eta_A,$$

or, equivalently,

$$O(\alpha_Q^{-1})\eta_B\alpha_Q = \eta_A.$$

This is the commutativity of the square formed by the four constructions. Together with the involution identities, it shows that the composite around the square, in either direction, is the identity. Every route between two vertices can be reduced by cancelling inverse adjacent edges and using this square identity. The resulting comparison is independent of the route, and its composites are transitive. All equalities take place in $\mathcal{L}_J$, so they also hold after restricting to the comparison arrow or to any other subdiagram. □

The argument is unchanged for an arbitrary index category. For $F : I \to \mathrm{Cat}$, one may define J by freely adjoining a vertex v and arrows $r_i : i \to v$, then a terminal vertex y, so that the composites $i \to v \to y$ satisfy the relations of the given strict cocone. The arrows $r_j u$ and r_i are not identified for $u : i \to j$; their images in the integral are linked by transition transformations. With this presentation, the arbitrary-shape claim on manuscript pages 143–144 follows from the same proof. The phrase "without commutation relations" describes the first, free adjunction of v; only the subsequent vertex y is terminal. Keeping these two operations distinct prevents the indexing error already corrected on page 139.

Remarques.

1) Ce qu'on vient de dire dans le cas des diagrammes dans Cat de type $\Phi^\circ = \Psi$, a l'air de garder un sens pour les diagrammes de type quelconque

$$X_* : S^\circ \longrightarrow \mathrm{Cat}, \qquad s \longmapsto X_s$$

(S un objet de Cat). On peut l'intégrer de façon lisse, $\int X_* = \int_{\mathrm{lis}} X_*$, obtenant une catégorie fibrée (scindée) sur S, ou de façon propre, $\int_{\mathrm{pr}} X_*$, obtenant une catégorie cofibrée scindée sur S°. On peut également considérer X_*°, composé de X_* avec l'automorphisme $Y \longmapsto Y^\circ$ dans Cat, et l'intégrer de façon soit propre, soit lisse. Dans tous ces cas, on trouve un diagramme de type $\widehat{S}^\circ$, où $\widehat{S}$ est déduit de S en ajoutant un élément qui p. 99

s'envoie par une flèche 'canonique' (donnée) dans chacun des autres (*sans* relations de commutation!) Un *système inductif* de morphismes dans Cat [page 144]

$$f \qquad X_s \xrightarrow{f_s} Y$$

donne alors un morphisme

$$\int_{\text{lis}} X_* \xrightarrow{\varphi_Q} Y,$$

où Q désigne le diagramme, de type $\widehat{S}^\circ$, formé des X_s, de Y, des $u^* : X_t \longrightarrow X_s$ (pour $u : s \longrightarrow t$ dans S) et des f_s. De même on aura

$$\int_{\text{prop}} X_* \xrightarrow{\tilde{\varphi}_Q} Y,$$

et si

$$\Gamma = (\widehat{S} + \text{objet initial}),$$

on trouve des diagrammes $\widehat{Q}$, $\widetilde{Q}$ de type Γ° dans Cat. Sauf erreur, les relations entre $\widehat{Q}$, $\widetilde{Q}$, $\widehat{Q^\circ}$, $\widetilde{\widehat{Q}^\circ}$ sont les mêmes que celles explicitées dans la proposition 14 ci-dessus. Je ne vais pas m'arrêter là-dessus avec le soin requis, sauf en cas de besoin. On dira qu'un diagramme Q définit Y comme $\varinjlim_I W$-homotopique [où $I = S^\circ$] si $\varphi_Q \in W$. Il revient au même de dire que $\varphi_{Q^\circ} \in W$, ou que $\tilde{\varphi}_Q \in W$ ou que $\tilde{\varphi}_{Q^\circ} \in W$ … [page 145]

Commentary · Later developments [ms. 144]

The comparison map in later formalisms

The definition by a specified map from an integral has a direct later interpretation. For ordinary homotopy types, put

$$\mathbb{D}(J) = \mathrm{Fun}(J, \mathrm{Cat})[W_{\infty,J}^{-1}] = \mathrm{HOT}_{W_\infty}(J^{\mathrm{op}}).$$

The opposite in the last expression corrects for the manuscript's presheaf convention. A square is an object of $\mathbb{D}(\square)$, where $\square = [1] \times [1]$. Let $i : \ulcorner \hookrightarrow \square$ omit the terminal vertex. In the derivator formulation a square Q is cocartesian when the counit

$$i_! i^* Q \longrightarrow Q$$

is an isomorphism (Groth, 2013, Definition 3.9). At the three vertices of $\ulcorner$ its components are already isomorphisms. At the last vertex, the component is the homotopy-colimit comparison represented here by φ_Q. This identifies Definition 3 with the later coherent-square definition. It also explains why a square merely drawn in $\mathbb{D}(e)$ is insufficient: it need not specify the coherent object to which $i_!$ applies.

The composition and cancellation theorem for adjacent cocartesian squares is Groth (2013, Proposition 3.13). Its hypotheses concern squares inside one coherent rectangle. They do not assert that an arbitrary map between homotopy pushouts reflects an equivalence on the intersection from equivalences on the pieces and the union. The latter is the stronger cube assertion tested by the acyclic example above.

There is a second distinction when the fibres are retained as categories. A Cartesian fibration classified by a diagram of ∞-categories models its colimit only after its Cartesian transition edges have been inverted; see Lurie (2009, Corollary 3.3.4.3 and Warning 3.3.4.4). Passing further to the groupoid completion inverts every edge. The corresponding space-valued statement is Proposition 3.3.4.5 of the same reference. This is compatible with the classifying-space interpretation of the manuscript's integral, while preserving the stronger categorical information before localization.

The constant terminal-category diagram over [1] makes the distinction concrete. Its ordinary total category is [1]. The colimit of the diagram in categories is e, and its colimit in Cat_∞ is also e. The unique nonidentity arrow of [1] is a transition arrow; inverting it gives a

> category equivalent to e. Before that inversion, [1] and e are not equivalent categories: the two objects of [1] are not isomorphic. Their classifying spaces are nevertheless both contractible. Thus the three assertions—a total category, a categorical colimit, and a colimit of homotopy types—agree only after the indicated operations. The present text is computing the last of them and retaining a strict representative for its map to the proposed target.

2) Le cas 'universel', pour $\begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array}$ donné, d'un carré commutatif $Q = \begin{array}{ccccc} & & X_1 & & \\ X_0 & \nearrow & & \searrow & \\ & & & & Y \\ & \searrow & & \nearrow & \\ & & X_2 & & \end{array}$

dans Cat, donnant naissance à

$$\varphi_Q : \left(\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ \int X_0 & & \\ & \searrow & \\ & & X_2 \end{array} \right) \longrightarrow Y,$$

est celui de $Y = X_1 \amalg_{X_0} X_2$, donnant naissace à

p. 100

$$Q_0 = Q(X_*) = \begin{array}{ccccc} & & X_1 & & \\ X_0 & \nearrow & & \searrow & \\ & & & & X_1 \amalg_{X_0} X_2. \\ & \searrow & & \nearrow & \\ & & X_2 & & \end{array}$$

Le cas général s'y ramène, car on aura un diagramme commutatif

$$\begin{array}{ccccc} \left(\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ \int X_0 & & \\ & \searrow & \\ & & X_2 \end{array} \right) & & \xrightarrow{\varphi_{Q_0} = \varphi_{Q(X_*)}} & & X_1 \amalg_{X_0} X_2 \\ & \searrow^{\varphi_Q} & & & \downarrow{\scriptstyle \alpha_Q} \\ & & & & Y. \end{array}$$

Mais il doit y avoir des cas intéressants où φ_{Q_0} n'est *pas* dans W, i.e. le carré Cat-cocartésien $\begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array}$ $\begin{array}{ccc} & & \\ & \searrow & \\ X_1 \amalg_{X_0} X_2 & & \\ & \nearrow & \\ & & \end{array}$ n'est pas W-homotopiquement cocartésien, et où pourtant on peut trouver un carré commutatif Q complétant X_*, de telle façon que ce carré soit W-homotopiquement commutatif `[plutôt cocartésien]`, i.e. que $\varphi_Q = \alpha_Q \varphi_{Q_0}$ soit dans W. C'est d'autant plus plausible que dans Cat, les sommes amalgamées ont tendance à

[page 146]

ne pas avoir des bonnes propriétés homotopiques, même si l'une des flèches α_1, α_2 est [un] monomorphisme (voire une immersion ouverte), contrairement à ce qui se passe dans $\mathbf{\Delta}^\wedge$, et plus généralement dans les catégories de la forme $Z^\wedge$ (Z dans Cat).

Commentary · Development (commentators') [ms. 146]

A strict cocone can repair the wrong pushout

Grothendieck asks for more than a strict pushout that fails to be a homotopy pushout. He asks whether its failure can disappear after mapping that pushout to a different target, while retaining the original span. The following finite example answers his question affirmatively for W_∞, even when one leg is an open immersion. The sources checked on preservation of pushouts, including the Dwyer-map criterion and its refinement in Hackney et al. (2024, Theorems 1.5–1.6, arXiv v3), do not supply this particular construction. We give a complete argument rather than a priority claim.

A finite repair example. [ours] There exist finite posets $E \subset B$, with E a sieve in B, such that BE and BB are contractible, while the categorical pushout $P = B \amalg_E e$ has classifying space S^3. Consequently the canonical comparison

$$c : \int_{\mathrm{lis}} (B \leftarrow E \to e) \longrightarrow P$$

is not in W_∞, but its composite with $P \to e$ is in W_∞. Here $BD = |ND|$ denotes the classifying space of a category D.

Proof. We first construct a map from a contractible face poset to a spherical one whose comma categories are connected. Let K be the finite simplicial complex with six vertices a, b, c, A, B, C and maximal faces

$$abc, \quad abB, \quad aBA, \quad bcC, \quad bCB, \quad caA, \quad cAC.$$

It is the disk obtained by placing the triangle abc inside the triangle ABC and triangulating the surrounding annulus with the remaining six triangles. In particular it is contractible. This description can also be checked without a drawing: successive free-face collapses remove

$$(AB, aAB),\ (AC, cAC),\ (aA, acA),\ (A, cA),\ (BC, bBC),$$
$$(aB, abB),\ (B, bB),\ (bC, bcC),\ (C, cC),$$
$$(ab, abc),\ (a, ac),\ (b, bc),$$

leaving the vertex c. In each pair the first simplex is a free codimension-one face of the second at the moment it is removed.

Let L be the boundary of the tetrahedron on vertices a, b, c, n. The vertex map fixing a, b, c and sending A, B, C to n is simplicial from K to L: the seven triangles have images abc, abn, an, bcn, bn, can, cn. Let E and S be the posets of nonempty faces of K and L, and let $f : E \to S$ send a face to its image. Thus BE is a subdivided disk and BS a subdivided two-sphere.

For every face s of L, the comma category E/s is the face poset of the subcomplex consisting of faces whose image is contained in s. If $n \notin s$, this subcomplex is the simplex on s, so it is connected and nonempty. If $n \in s$, it contains the entire outer boundary $AB \cup BC \cup CA$. Every inner vertex allowed by s attaches to that boundary by a radial edge: a by aA, b by bB, and c by cC. Every remaining simplex contains vertices of this same connected subcomplex. It follows that

$$\pi_0(E/s) = \{*\} \quad \text{for every } s \in S.$$

Only connectedness is claimed: for $s = \{n\}$ the comma category is the face poset of the outer boundary circle.

Adjoin an initial object o to S and call the resulting poset T. Define a poset B on the disjoint union of E and T. Keep their internal orders and add

$$e \le s \quad \Longleftrightarrow \quad f(e) \le s \qquad (e \in E,\ s \in S).$$

There are no relations from T to E, and none from E to o. Transitivity follows from monotonicity of f. The subposet E is a sieve. The functor $r : B \to T$ which is the identity on T and sends e to $f(e)$ is left adjoint to the inclusion $j : T \to B$, because

$$B(b, jt) = T(r(b), t)$$

for every $b \in B$, $t \in T$, including the empty hom-sets when $t = o$ and $b \in E$. Hence Bj is a homotopy equivalence. The initial object o makes BT, and therefore BB, contractible.

It remains to compute the categorical pushout, since a topological quotient would give the wrong answer. Collapsing the sieve E to one object $*$ introduces no arrows from T to $*$ and does not change arrows within T. Any arrow from $*$ to $s \in S$ is represented by a relation $e \leq s$ in B. The relations after collapsing E identify two representatives exactly along zigzags of comparable elements in E/s. Thus

$$\mathrm{Hom}_P(*, s) = \pi_0(E/s) = \{*\}, \qquad \mathrm{Hom}_P(*, o) = \emptyset.$$

For clarity, this description also verifies the universal property: a functor out of B that is constant on E assigns the same arrow to any two representatives in one such connected component, and therefore factors uniquely through the displayed category. Conversely every functor from that category supplies exactly such compatible data on B and e. Composition is induced by the maps $E/s \to E/s'$ for $s \leq s'$.

Consequently P is the poset with two incomparable minimal objects $o, *$, both lying below every object of S. Its order complex is exactly the simplicial join

$$NP = N\{o, *\} * NS, \qquad BP \cong S^0 * |\operatorname{sd} L| \cong S^3.$$

The three categories in the original span $B \leftarrow E \to e$ are contractible. Their smooth integral is a split fibration over the contractible indexing fork, with contractible fibres, and hence is contractible. Its canonical map c cannot be a weak equivalence to P, whose third homology is $\mathbb{Z}$. On the other hand the unique functor from the integral to e is a weak equivalence, and it is precisely the composite $P \to e$ after c. This proves the assertion. □

The construction has 25 objects in E, 14 in S, 40 in B, and 16 in P. The accompanying finite computation checks the face lists, the elementary collapses, all fourteen connected comma categories, and the homology of NP; the join calculation above identifies its entire homotopy type. The mechanism is visible in the proof: the ordinary categorical pushout keeps only $\pi_0(E/s)$ in each mixed hom-set. It replaces a family containing a circular comma category by a family of singletons, and the resulting total category acquires the additional sphere. Sending that sphere to a point supplies the strict cocone envisaged on manuscript pages 145–146.

3.4 Critères pour les carrés homotopiquement cocartésiens.

Les critères que j'ai dégagé sont tous contenus dans l'énoncé suivant :

Théorème 4 ([93], [94]). *Considérons un diagramme carré commutatif dans* Cat

(∗)
$$Q \quad = \quad \begin{array}{ccc} X_0 & \xrightarrow{\alpha_1} & X_1 \\ {\scriptstyle \alpha_2}\downarrow & \overset{f_0}{\searrow} & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow[f_2]{} & Y, \end{array}$$

et considérons le morphisme canonique

p. 101

$$\varphi_Q : \quad \mathcal{X} = \Big(\int \quad \begin{array}{ccc} & & X_1 \\ & \overset{\alpha_1}{\nearrow} & \\ X_0 & & \\ & \underset{\alpha_2}{\searrow} & \\ & & X_2 \end{array} \Big) \quad \longrightarrow \quad Y,$$

l'intégrale étant l'intégrale lisse. *(Il y a un énoncé dual concernant l'intégrale propre de*

[page 147]

93. **NB** Dans ce théorème, et les corollaires 1, 2, 3, on n'a besoin que de la *forme faible* de W(7 bis), savoir $i_0 \in W \Longrightarrow i \in W$ (cf. p. 79) (Loc(6′a) avec les notations de XVI).

94. **NB** Au total, le théorème (partie directe) est valable sans hypothèse W(7 bis) ou W(7 bis faible) sur W, et il en est de même de ses corollaires 1 à 3 (à l'exclusion des corollaires 4, 5, qui eux utilisent W(7 bis) dans toute sa force).

$$\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & .) \\ & \searrow & \\ & & X_2 \end{array}$$

a) Pour que φ_Q soit W-lisse, il faut et il suffit que f_1, f_2 et $f_0 = f_1\alpha_1 = f_2\alpha_2$ le soient. (Moyennant W(7 bis) ([95])).

b) La fibre de φ_Q en un élément y de Y est isomorphe à

$$(**) \qquad \int \begin{array}{ccc} & & X_{1\,y} \\ & \nearrow & \\ X_{0\,y} & & \\ & \searrow & \\ & & X_{2\,y} \end{array} \qquad (\simeq \mathcal{X}_y).$$

Corollaire 1. ([97]) *Supposons f_1, f_2 et $f_1\alpha_1 = f_2\alpha_2$ W-lisses, et les catégories intégrées $(**)$ W-asphériques pour tout $y \in \mathrm{Ob}\, Y$. (NB Il s'agit encore de l'intégrale lisse, mais peu importe au fond, car on a vu (cf. partie supérieure du diagramme p. 140, qui implique que $\mathcal{X}$* `[est]` *W-asphérique si et seulement si $\tilde{X}$ l'est) que l'une est W-asphérique si et seulement si l'autre l'est.) Alors $\varphi_Q \in W$, i.e. le carré $(*)$ est W-homotopiquement cartésien* `[plutôt cocartésien]`*. Par dualité, même conclusion si on suppose f_1, f_2, $f_1\alpha_1 = f_2\alpha_2$* propres *au lieu de lisses. (Car Q W-homotopiquement trivial* `[plutôt cocartésien]`*, si et seulement si Q° l'est, par* `[la]` *prop. 14.)*

En effet, un morphisme dans Cat qui est W-lisse et à fibres W-asphériques est dans W (et c'est même une W-équivalence universelle …). [page 148]

Commentary · Exposition [ms. 147]

Smoothness and the two different uses of a fibre

Grothendieck now turns the comparison into a criterion that can be checked on a given square. There are two separate operations in Theorem 4. The fibre formula is an identity of categories: imposing that the comparison map has the value y imposes the same condition in each of its three pieces. Smoothness is a local lifting condition, tested on slices at objects of the integral. The latter condition is what permits a strict fibre to determine the homotopy-theoretic conclusion.

It is useful to read the annotations as a refinement of the hypotheses, rather than as several competing versions of the theorem. If f_0, f_1, f_2 are W-smooth, then φ_Q is W-smooth for every fundamental localizer. The reverse implication for f_0 is asserted under the strong form of W(7 bis). No extra excision axiom is required for the fibre identity

$$(\mathcal{X}_Q)_y \cong \int_{\mathrm{lis}} (X_{1y} \leftarrow X_{0y} \rightarrow X_{2y}).$$

The calculations on manuscript pages 152–157 establish these assertions; their local content is explained below at the two lemmas. In particular, the annotations do not justify removing the strong hypothesis from the converse.

The last step of Corollary 1 deserves to be made explicit. If $u : E \to Y$ is W-smooth, the fibre inclusion $E_y \to y\backslash E$ is W-aspherical. If every E_y is W-aspherical, two-out-of-three gives that every $y\backslash E$ is W-aspherical. The dual of the fundamental Theorem A then yields $u \in W$. This proof is stable under arbitrary base change: smoothness is stable under base change and the new strict fibres

95. Pour la partie directe (f_0, f_1, f_2 lisses $\Longrightarrow$ φ_Q lisse), on n'a besoin que de W(7 bis) ([96]), forme faible. Ce n'est que pour la réciproque (qui ne m'a encore jamais servie) qu'on a besoin de W(7 bis), dans toute sa force.

96. On n'a pas même besoin de W(7 bis) faible, cf. annotation page 156. On utilise une hypothèse restrictive sur W dans la réciproque seulement (φ_Q lisse $\Longrightarrow$ f_0, f_1, f_2 lisses).

97. Valable sans hypothèse sur W (autre que Loc(1,2,3 bis)).

are the old ones. Hence u is in fact a universal W-equivalence. The same argument with properness uses E/y instead. These are the fibre criteria formalized in Maltsiniotis (2005, Theorem 1.2).

Consequently the forward criterion in Corollary 1 has precisely the strength needed later: smoothness of all three structure maps, together with asphericity of the integrated fibres, proves the square homotopy cocartesian, and even proves the comparison a universal weak equivalence. The qualifications in the first two footnotes to Theorem 4 correctly anticipate this result.

Corollaire 2. *On suppose que W satisfait* W(7 bis) (axiome du carré cocartésien) ([98]). *Supposons le carré Q cocartésien dans* Cat ([99]), *et que α_1 soit une immersion ouverte (resp.* p. 102 *fermée) et f_1 lisse (resp. propre). Alors Q est W-homotopiquement cocartésien.* ([100])

Il suffit d'établir le résultat non respé, l'autre en résultera par dualité (puisque Q est W-cocartésien si et seulement si Q° l'est). Par hypothèse f_1 est W-lisse, d'autre part α_1 et f_2 sont des immersions ouvertes, donc W-lisses, donc $f_1\alpha_1$ est W-lisse aussi. (D'ailleurs α_2 aussi, car le carré est aussi cartésien, donc α_2 déduit de f_1 (qui est W-lisse par hypothèse) par changement de base $X_2 \longrightarrow Y$.) Donc φ_Q est [W-]lisse, et il suffit de voir que ses fibres sont W-asphériques. Le calcul $(**)$ donne, pour y dans $Y\backslash X_2$,

$$\mathcal{X}_y \simeq \int \begin{array}{ccc} & & e \\ \emptyset & \nearrow & \\ & \searrow & \\ & & \emptyset \end{array} \simeq e, \qquad \text{O.K.}$$

Et pour y dans X_2

$$\mathcal{X}_y \simeq \int \begin{array}{ccc} & & X_{1\,y} \\ X_{0\,y} & \overset{\sim}{\nearrow} & \\ & \searrow & \\ & & e \end{array} .$$

Comme $X_{0\,y} \longrightarrow X_{1\,y}$ est un isomorphisme, donc est dans W, il s'ensuit que $e \longrightarrow \mathcal{X}_y$ l'est [page 149] aussi (prop. 13, p. 125), moyennant W(7 bis), donc $\mathcal{X}_y$ est W-asphérique, O.K. ([101]).

Corollaire 3. (Moyennant W(7 bis) ([102]).) *Soit X dans* Cat, *U_1, U_2 deux ouverts de X tels que $X = U_1 \cup U_2$, $U_0 = U_1 \cap U_2$ l'ouvert intersection. Alors le carré d'immersions ouvertes*

$$\begin{array}{ccc} U_0 & \hookrightarrow & U_1 \\ \downarrow & & \downarrow \\ U_2 & \hookrightarrow & X \end{array}$$

est W-cocartésien. (Énoncé dual pour deux fermés T_1, T_2 tels que $T_1 \cup T_2 = X$, avec $T_0 = T_1 \cap T_2$.)

On applique le corollaire 2. On sait que le carré est cocartésien dans Cat ([103]), de plus $U_1 \hookrightarrow X$ est bien W-lisse, étant une immersion ouverte.

p. 103

98. inutile.

99. NB On n'utilise pas vraiment que Q soit cocartésien, plutôt que Q est *cartésien,* et f_2 (donc aussi α_1) une immersion ouverte (resp. fermée), et que pour $y \in Y\backslash X_2$, $X_{1\,y}$ soit W-asphérique. (Dans le cas où Q est cocartésien, $X_{1\,y}$ est $\simeq e$.)

100. On a besoin ici de W(7 bis), forme directe, i.e. de Loc(6′a) (notation de XVI).

101. On n'a besoin que de la forme directe de W(7 bis), i.e. (notation de XVI) de Loc(6′a).

102. inutile.

103. On n'a pas même besoin de le savoir, cf. annotation marginale du corollaire 2.

Corollaire 4. (Moyennant W(7 bis) ([104]).) *Soit $U \overset{i}{\hookrightarrow} X$ une immersion ouverte telle que $T = X\backslash U$ soit isomorphe à la catégorie ponctuelle e. Pour que $i \in W$, il faut et il suffit que i soit W-asphérique, i.e. que U/t (où t est l'unique objet de T) soit W-asphérique.*

Il suffit bien sûr de prouver que

$$i \in W \Longrightarrow U/t \text{ } W\text{-asphérique.}$$

Considérons le carré cartésien [page 150]

$$\begin{array}{ccc} U/t & \xhookrightarrow{i/t} & X/t \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U & \xhookrightarrow{i} & X, \end{array}$$

nous allons prouver qu'il est W-cocartésien. Moyennant W(7 bis), il s'ensuivra que $i \in W$ implique $i/t \in W$, et comme X/t est W-asphérique (ayant un objet final), cela signifie que U/t est W-asphérique, ce qu'on veut prouver.

Commentary · Examples [ms. 149]

Why the cover criterion needs no converse excision

Grothendieck's cover criterion is meant to provide a large supply of actual homotopy pushouts. The annotations correctly remove W(7 bis) from Corollaries 2 and 3, although the displayed proof still invokes it. The missing simplification is elementary: one of the maps in the only nontrivial fibre is an isomorphism, and that special case admits a direct contraction.

For a category A, consider

$$E = \int_{\mathrm{lis}} (A \xleftarrow{\mathrm{id}} A \longrightarrow e).$$

Keep the central copy A_0 and the outer point $*$, and call their full subcategory K. There is a unique arrow $* \to a$ for every $a \in A_0$, so $*$ is initial in K. Define $R : E \to K$ to be the identity on K and to send the outer copy $a \in A_1$ to its central copy in A_0. On mixed arrows it uses the corresponding arrow of A. The canonical transition arrows $a_1 \to a_0$ give a natural transformation $\mathrm{id}_E \to jR$, where $j : K \to E$ is the inclusion, and $Rj = \mathrm{id}_K$. Thus j and R are homotopy inverse after taking nerves. Since K has an initial object, E is W-aspherical for every fundamental localizer. This proof also covers $A = \emptyset$.

In Corollary 2, a point outside X_2 has integrated fibre e. At a point of X_2, cartesianness identifies $X_{0y} \to X_{1y}$ with an isomorphism, and the fibre is of the form just contracted. The three maps to Y are smooth: f_1 by hypothesis, f_2 because it is a sieve inclusion, and $f_0 = f_1\alpha_1$ by composition. The forward part of Theorem 4 therefore proves the conclusion without W(7 bis). This also establishes the stronger version recorded in footnote 99: strict cocartesianness may be replaced by cartesianness, the sieve condition on f_2, smoothness of f_1, and W-asphericity of X_{1y} outside X_2. In that version the exterior fibres need not be singletons.

For a cover $X = U_1 \cup U_2$ by sieves, there is a second, particularly transparent argument in ordinary homotopy theory. Every simplex $x_0 \to \cdots \to x_n$ in NX lies in a member of the cover containing x_n, because a sieve contains every predecessor of its objects. Therefore

$$NX = NU_1 \cup_{N(U_1 \cap U_2)} NU_2$$

is an equality of simplicial sets. The two inclusions are monomorphisms, so its realization is a pushout along subcomplex inclusions and computes the homotopy pushout. For cosieves one uses the first vertex x_0 instead. This calculation explains why categorical open and closed covers are especially well adapted to excision.

As a worked example, let X be the poset of nonempty proper faces of a tetrahedron. Let U_1 consist of the faces of one facet, and let U_2 consist of the faces of the other three facets. Both are downward closed, and their union is X. Their intersection is the face poset of the boundary of the

104. Ici on a besoin de W(7 bis) forme forte, i.e. $i_0 \in W \iff i \in W$ (cf. annotation th. 4).

first facet. The order complexes are barycentric subdivisions: BU_1 is one triangle, BU_2 is the disk made of the other three triangles, and $B(U_1 \cap U_2)$ is their common boundary circle. The integral therefore has homotopy type

$$D^2 \overset{h}{\underset{S^1}{\cup}} D^2 \simeq S^2,$$

and the canonical comparison identifies this sphere with BX, the subdivided tetrahedral boundary. The comparison is determined by the actual inclusions of face posets; it is not merely an abstract equivalence between two spaces known to be spheres.

Il suffit, en vertu du corollaire 2, de prouver que le carré est cocartésien dans Cat, puisque i/t est une immersion ouverte (i l'étant), et $X/t \longrightarrow X$ est un isomorphisme local, donc W-lisse. On voit que $(X/t)\backslash(U/t) \simeq e$ (catégorie ponctuelle), d'ailleurs l'unique élément de $(X/t)\backslash(U/t)$ est l'élément final $t_0 : t \xrightarrow{\mathrm{id}_t} t$ de X/t. Ainsi $H_{t_0} \simeq e_{U/t}$ (préfaisceau final), et $p_!(H_{t_0}) \simeq H'_t \overset{\text{déf}}{=} (u \longmapsto \mathrm{Hom}_X(u,t))$, ce qui caractérise la somme amalgamée dans Cat …

Le corollaire 4 n'est autre que le corollaire non démontré de la proposition 7 (p. 85).

Voici une variante du corollaire 4.

Corollaire 5. (Moyennant W(7 bis) ([105]).) *Soit $i : U \hookrightarrow X$ une immersion ouverte, soit $t_0 \in \mathrm{Ob}\, T$ un point* fermé *de $T = X\backslash U$, tel que $\{t_0\}$ soit isomorphe à la catégorie ponctuelle,* [page 151] *i.e.* $\mathrm{Hom}_X(t_0,t_0) = \{\mathrm{id}\}$. *Supposons que pour tout $t \in T\backslash\{t_0\}$, U/t soit W-asphérique. Pour que l'on ait $i \in W$, il faut et il suffit que i soit asphérique, i.e. que U/t_0 soit également W-asphérique.*

Il faut prouver que $i \in W$ implique que U/t_0 est W-asphérique. Considérons l'inclusion

$$i_0 : U \longrightarrow U' \overset{\text{déf}}{=} X\backslash\{t_0\},$$

l'hypothèse sur les U/t, pour $t \in T\backslash\{t_0\}$, signifie que i_0 est W-asphérique. Donc i_0 est dans W, et comme le composé $i : U \xrightarrow{i_0} U' \xrightarrow{i'} X$ est dans W, cela implique que $i' : U' \longrightarrow X$ p. 104 est dans W. Nous sommes maintenant dans la situation du corollaire 4, qui implique que i' est W-asphérique. Donc $i = i'i_0$ est W-asphérique, comme composé de deux morphismes W-asphériques, q.e.d.

Commentary · Exposition [ms. 151]

A single missing point and the role of the strong axiom

Grothendieck uses the preceding criterion to finish a result deferred earlier in the chapter. When only one object is missing, the slice over that object records every possible attachment to the open subcategory. The square on manuscript page 150 is consequently both a strict pushout and a homotopy pushout. The remaining implication is a reflection statement, and this is where the strong form of W(7 bis) enters.

Here is the strict pushout assertion in detail. Suppose $U \subset X$ is a sieve, its complement consists of one object t, and $\mathrm{End}_X(t) = \{\mathrm{id}_t\}$. Let $F : U \to Z$ and $G : X/t \to Z$ agree on U/t after projection to U. Define $H : X \to Z$ on U to be F, and set $H(t) = G(\mathrm{id}_t)$. For an arrow $a : u \to t$, define $H(a)$ to be the image under G of the canonical arrow from (u,a) to (t,id_t) in X/t. The source of this image is $F(u)$ by the agreement assumption. If $b : u' \to u$ lies in U, the factorization of $(u',ab) \to (t,\mathrm{id}_t)$ through (u,a) gives

$$H(ab) = H(a)F(b).$$

There are no arrows from t into U, since U is a sieve, and the only endomorphism at t is its identity. Thus these rules define a functor, and they determine it uniquely. This proves the pushout universal property of the slice square, without any appeal to the presheaf notation H_{t_0} used in the manuscript.

The projection $X/t \to X$ is a discrete fibration, hence smooth. Corollary 2 proves that this square is homotopy cocartesian without an extra excision hypothesis. If the lower inclusion $U \to X$

105. forme forte.

is in W, however, deducing that the upper inclusion $U/t \to X/t$ is in W is exactly a converse-excision step. Under strong W(7 bis), Proposition 13 supplies it. The target X/t has a final object, so U/t is then W-aspherical. Conversely, if U/t is W-aspherical, the inclusion $U \to X$ is W-aspherical: its slices at objects of U have final objects, and its only remaining slice is U/t. Theorem A gives the weak equivalence.

The reduction in Corollary 5 is now precise. Because t_0 is closed in $X \setminus U$ and U is a sieve, t_0 has no arrow to any other object of X. Hence $U' = X \setminus \{t_0\}$ is again a sieve. The hypotheses make $U \to U'$ aspherical. If $U \to X$ is in W, two-out-of-three makes $U' \to X$ a weak equivalence; Corollary 4 makes it aspherical. Composition then makes $U \to X$ aspherical, which supplies the remaining slice condition at t_0. The argument is complete under its stated strong hypothesis. It should not be read as a general converse to Theorem A for ordinary weak homotopy equivalences.

Démonstration de th. 4.

ⓐ Critère de lissité de $\varphi_Q : \mathcal{X} = \int \begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \longrightarrow Y. \\ & \searrow & \\ & & X_2 \end{array}$

Comme $\{1\}$, $\{2\}$ sont des ouverts de $\Phi = \begin{array}{ccc} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{array}$, $i_1 : X_1 \longrightarrow \mathcal{X}$ et $i_2 : X_2 \longrightarrow \mathcal{X}$ sont des immersions ouvertes. Comme $\varphi_Q \circ i_1 = f_1$, $\varphi_Q \circ i_2 = f_2$, on voit déjà que [page 152]

f_1 et f_2 lisses $\iff$ φ_Q lisse sur l'ouvert réunion de $i_1(X_1)$ et $i_2(X_2)$, i.e. sur $\mathcal{X} \setminus i_0(X_0)$.

Il reste à exprimer la lissité de φ_Q en les points x_0 de $i_0(X_0)$. Soient x_1, x_2, y les images de x_0 dans X_1, X_2, Y respectivement. Il faut exprimer que les fibres de

$$(*) \qquad (\varphi_Q)/x_0 : \mathcal{X}/x_0 \longrightarrow Y/y$$

sont asphériques. Nous allons pour cela expliciter $\mathcal{X}/x_0$ et la flèche $(*)$ en termes du carré commutatif

$$(**) \qquad \begin{array}{ccc} X_0/x_0 & \xrightarrow{\alpha_1/x_0} & X_1/x_1 \\ {\scriptstyle \alpha_2/x_0}\downarrow & & \downarrow{\scriptstyle f_1/x_1} \\ X_2/x_2 & \xrightarrow{f_2/x_2} & Y/y, \end{array}$$

induit par Q. Je dis

Lemme 1. *Soit un carré commutatif*

$$Q \quad = \quad \begin{array}{ccc} X_0 & \xrightarrow{\alpha_1} & X_1 \\ {\scriptstyle \alpha_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow{f_2} & Y \end{array}$$

dans Cat, $x_0 \in \mathrm{Ob}\, X_0$, *$x_1$, x_2, y ses images dans X_1, X_2, Y respectivement, Q/x_0 le carré* $(**)$ *ci-dessus, induit par Q sur les localisées X_0/x_0 etc., $\mathcal{X} = \int \begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$. On a alors un* p. 105 [page 153]

isomorphisme canonique

$$\mathcal{X}/x_0 \quad \simeq \quad \int \begin{array}{ccc} & & X_1/x_1 \\ X_0/x_0 & \nearrow & \\ & \searrow & \\ & & X_2/x_2 \end{array} \quad ,$$

et la flèche $(\varphi_Q)/x_0$, localisée de $\varphi_Q : \mathcal{X} \longrightarrow Y$ en x_0, s'identifie à la flèche $\varphi_{(Q/x_0)}$:

$$\varphi_{(Q/x_0)} : \quad \int \begin{array}{ccc} & & X_1/x_1 \\ X_0/x_0 & \nearrow & \\ & \searrow & \\ & & X_2/x_2 \end{array} \quad \longrightarrow \quad Y/y.$$

Démonstration du Lemme. On sait que

$$\mathcal{X}/x_0 \longrightarrow \mathcal{X}$$

est Cat-fibrée (scindée) (à fibres discrètes), et comme

$$\mathcal{X} \longrightarrow \Phi$$

est aussi Cat-fibrée (scindée), il s'ensuit qu'il en est de même de la flèche composée

$$\mathcal{X}/x_0 \longrightarrow \Phi.$$

Donc pour 'calculer' $\mathcal{X}/x_0$, en tant que catégorie sur Φ, il faut expliciter ses trois fibres, et les deux foncteurs de changement de base qui les relient. On trouve aussitôt que ces fibres sont canoniquement isomorphes à

$$X_0/x_0, \qquad X_1/x_1, \qquad X_2/x_2,$$

et on admettra (!) que les foncteurs changement de base

$$X_0/x_0 \longrightarrow X_1/x_1, \qquad X_0/x_0 \longrightarrow X_2/x_2$$

(définis à isomorphisme unique près) ne sont autres que les localisés α_1/x_0 et α_2/x_0 de $\alpha_1 : X_0 \longrightarrow X_1$ et $\alpha_2 : X_0 \longrightarrow X_2$. Cela montre donc qu'on a un isomorphisme [page 154]

$$\mathcal{X}/x_0 \quad \simeq \quad \int \begin{array}{ccc} & & X_1/x_1 \\ X_0/x_0 & \nearrow & \\ & \searrow & \\ & & X_2/x_2 \end{array} \quad ,$$

et l'on admettra aussi que la flèche $(\varphi_Q)/x_0 : \mathcal{X}/x_0 \longrightarrow Y/y$ s'identifie à $\varphi_{(Q/x_0)}$.

Pour exprimer la W-lissité de φ_Q en $x_0 \in \mathrm{Ob}\, X_0 \subseteq \mathrm{Ob}\, \mathcal{X}$, on est donc ramené à exprimer la W-asphéricité des fibres de

$$\varphi_{(Q/x_0)} : \quad \int \begin{array}{ccc} & & X_1/x_1 \\ X_0/x_0 & \nearrow & \\ & \searrow & \\ & & X_2/x_2 \end{array} \quad \longrightarrow \quad Y/y. \qquad \text{p. 106}$$

Pour expliciter ces fibres, on a besoin d'un autre lemme auxiliaire

Commentary · Exposition [ms. 154]

The slice of the integral, with its arrows

Grothendieck is reducing smoothness to a calculation at one object. The reduction succeeds because the smooth integral remembers the arrows into the central stratum, rather than merely the three categories as disjoint pieces. We make this calculation explicit; it supplies both identifications which the proof asks us to admit.

Write E for the smooth integral of $X_0 \xrightarrow{\alpha_i} X_i$, $i = 1, 2$. Its objects are the tagged objects of the three categories. Arrows within a stratum are the original arrows, and the additional hom-sets are

$$\mathrm{Hom}_E((i,b),(0,a)) = \mathrm{Hom}_{X_i}(b, \alpha_i a), \qquad i = 1, 2.$$

There are no arrows in the reverse direction or between the two outer strata. Composition is composition in X_i, using α_i when a central arrow is involved. Thus the cartesian arrow over $i \to 0$ has the direction

$$(i, \alpha_i a) \longrightarrow (0, a).$$

The arrows printed in the opposite direction on manuscript page 155 are an edition slip, not a different convention for this integral. The functor $\varphi_Q : E \to Y$ sends an additional arrow represented by $h : b \to \alpha_i a$ to $f_i(h)$, whose target is $f_i\alpha_i(a) = f_0(a)$. In particular it sends the cartesian arrow just displayed to an identity. This is the property which the source notices it had left implicit.

Fix $x_0 \in X_0$ and put $x_i = \alpha_i x_0$. An object of E/x_0 in the central stratum is precisely an arrow $g : a \to x_0$ in X_0. An object in stratum i is an arrow $h : b \to x_i$ in X_i. A cross-stratum morphism from h to g is an arrow $k : b \to \alpha_i a$ satisfying

$$(\alpha_i g)k = h.$$

This is exactly a morphism from h to the image of g in X_i/x_i. Within each stratum we have the ordinary slice morphisms. These descriptions identify objects, hom-sets, identities, and compositions, giving the asserted canonical isomorphism

$$E/x_0 \cong \int_{\mathrm{li}} (X_0/x_0 \longrightarrow X_1/x_1,\ X_0/x_0 \longrightarrow X_2/x_2).$$

Under it the functor to Y/y, $y = f_0 x_0$, sends g to $f_0 g$ and h to $f_i h$; on a cross-stratum arrow it is $f_i k$. This proves the compatibility with φ_{Q/x_0} as well as the isomorphism of the source categories.

The same description proves Lemma 2 without a further coherence argument. The strict fibre over y consists of objects mapping to y and arrows mapping to 1_y. A cross-stratum arrow belongs to this fibre exactly when its representing arrow in X_i does. Consequently

$$E_y \cong \int_{\mathrm{li}} (X_{0y} \longrightarrow X_{1y},\ X_{0y} \longrightarrow X_{2y}).$$

Strict fibres need not be full subcategories. For instance, the strict fibre of the identity functor of a one-object group category over its object is the terminal category, whereas the full subcategory on that object is the whole group category. Keeping the identity condition on arrows is essential in the present calculation.

For an arrow $u : y' \to y$, set

$$A_i(u) = (X_i/x_i)_u.$$

Its objects are arrows $a \to x_i$ whose image is exactly u; its morphisms are commuting triangles whose remaining edge maps to $1_{y'}$. The preceding two identifications now give, with no choices,

$$(E/x_0)_u \cong \int_{\mathrm{li}} (A_0(u) \longrightarrow A_1(u),\ A_0(u) \longrightarrow A_2(u)).$$

This is the precise local input to the smoothness criterion. For the later systematic treatment, the locality of smoothness under passage to slices is Proposition 4.10 of Maltsiniotis (2005); the argument here retains the additional explicit description needed for this particular integral.

Lemme 2 ([106]). *Soit un carré commutatif dans* Cat

$$Q \quad = \quad \begin{pmatrix} X_0 & \xrightarrow{\alpha_1} & X_1 \\ {\scriptstyle\alpha_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow{f_2} & Y \end{pmatrix},$$

d'où

$$\varphi_Q : \quad \left(\int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array} \right) \quad \longrightarrow \quad Y,$$

et soit $y \in \operatorname{Ob} Y$. Alors la fibre de φ_Q en y [page 155]

$$\mathcal{X}_y = \varphi_Q^{-1}(\underbrace{\{y\}}_{\text{catégorie ponctuelle}})$$

est donnée par (comme catégorie au dessus de Φ)

$$\mathcal{X}_y \quad \simeq \quad \int \begin{array}{ccc} & & X_{1\,y} \\ X_{0\,y} & \nearrow & \\ & \searrow & \\ & & X_{2\,y} \end{array} .$$

Démonstration. Il est immédiat qu'on a

$$\mathcal{X}_y \cap \underbrace{\mathcal{X}_s}_{X_s} \simeq (X_s)_y \qquad \text{pour } s \in \operatorname{Ob}\Phi = \{0,1,2\},$$

et (le hasard fait bien les choses) les foncteurs changement de base

$$X_0 \longrightarrow X_1, \qquad X_0 \longrightarrow X_2$$ p. 107

envoient la sous-catégorie (non pleine) $X_{0\,y}$ de X_0 dans les sous-catégories (non pleines) $X_{1\,y}$ et $X_{2\,y}$ de X_1 et X_2 respectivement.

$$\begin{array}{ccccc} X_{s\,y} & \longrightarrow & \mathcal{X}_y & \longrightarrow & \{y\} \\ \downarrow & \text{cart.} & \downarrow & \text{cart.} & \downarrow \\ X_s & \longrightarrow & \mathcal{X} & \longrightarrow & Y \\ \downarrow & \text{cart.} & \downarrow & & \\ \{s\} & \longrightarrow & \Phi & & \end{array}$$

On a donc une sous-catégorie fibrée de $\mathcal{X}$, sous réserve de vérifier que les flèches strictement cartésiennes

$$x_0 \longrightarrow \alpha_1(x_0), \qquad x_0 \longrightarrow \alpha_2(x_0)$$

dans $\mathcal{X}$, pour x_0 dans $X_{0\,y}$, sont dans $\mathcal{X}_y$, i.e. sont au-dessus de la flèche identique de y. Or c'est là la propriété essentielle du foncteur $\varphi_Q : \mathcal{X} \longrightarrow Y$, qu'on a oublié d'expliciter dans 3.2.

106. NB C'est le b) du th. 4. [page 156]

Ça a vraiment l'air de marcher (comme sur des roulettes!)

Appliquons le Lemme 2 au diagramme carré Q/x_0 et à $\varphi_{(Q/x_0)}$ (page 154), on voit donc que les fibres à examiner en les $u \in \mathrm{Ob}\, Y/y$ $(u : y' \longrightarrow y)$ peuvent s'écrire comme les intégrales

$$\mathcal{X}(x_0,u) \overset{\text{déf}}{=} \int \begin{array}{ccc} & & (X_1/x_1)_u \\ & \nearrow & \\ (X_0/x_0)_u & & \\ & \searrow & \\ & & (X_2/x_2)_u \end{array} .$$

Utilisons maintenant le fait que f_1, f_2 sont W-lisses (exprimant que φ_Q est W-lisse dans $\mathcal{X}\backslash X_0$), donc que $(X_1/x_1)_u$ et $(X_2/x_2)_u$ sont W-asphériques. Dans le carré W-cocartésien

$$\begin{array}{ccc} & (X_1/x_1)_u & \\ {\scriptstyle \alpha_1(x_0,u)} \nearrow & & \searrow {\scriptstyle i_1(x_0,u)} \\ (X_0/x_0)_u & & (\mathcal{X}/x_0)_u \simeq \mathcal{X}(x_0,u) \\ {\scriptstyle \alpha_2(x_0,u)} \searrow & & \nearrow {\scriptstyle i_2(x_0,u)} \\ & (X_2/x_2)_u & \end{array}$$

([107]), il faut exprimer que $(\mathcal{X}/x_0)_u$ est W-asphérique, sachant que $(X_1/x_1)_u$ et $(X_2/x_2)_u$ le sont. Donc cela équivaut à p. 108

$$i_1(x_0,u) \in W \qquad (\text{ou } i_2(x_0,u) \in W),$$

donc aussi (moyennant W(7 bis)) à [page 157]

$$\alpha_1(x_0,u) \in W, \qquad \text{i.e. } (X_0/x_0)_u \text{ } W\text{-asphérique.}$$

Dire que c'est vrai pour tout u dans Y/y, signifie que $f_0 : X_0 \longrightarrow Y$ est lisse en u `[plutôt en` x_0`]`.

Donc on a prouvé a), sous la forme plus précise :

Commentary · Exposition [ms. 157]

Repairing the noncommuting fibre square

The aim of the calculation on page 156 is to reconstruct smoothness at the central object from smoothness on the three pieces. The displayed diamond cannot itself be used as a commutative square: its two composites land in different outer strata of the integral. They are related to the inclusion of the central stratum by cartesian natural transformations. Footnote 107 correctly withdraws that use of the square.

Here is the direct argument, with its hypotheses visible. Suppose f_i is W-smooth at x_i for $i = 0, 1, 2$. For every $u : y' \to y$, all three categories $A_i(u)$ from the preceding block are W-aspherical. Their smooth integral L_u is a split fibration over $\Phi = (1 \to 0 \leftarrow 2)$, hence is W-smooth. For each $s \in \Phi$, smoothness identifies its strict fibre, up to a W-equivalence, with the comma category $s\backslash L_u$. These comma categories are therefore W-aspherical. The dual form of the relative Theorem A gives $L_u \to \Phi$ in W. Since 0 is final in Φ, L_u is W-aspherical. The slice calculation identifies L_u with $(E/x_0)_u$, proving smoothness of φ_Q at x_0. No pushout reflection axiom has entered.

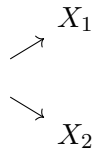
107. attention, ce n'est pas un carré commutatif. Dire plutôt que $(\mathcal{X}/x_0)_u \simeq \int \begin{array}{cc} \nearrow & \\ & \searrow \end{array}$ est lisse sur Φ à fibres W-asphériques (si f_0, f_1, f_2 sont lisses), donc W-asphérique puisque Φ l'est.

For the converse, suppose the two outer maps are smooth at x_1, x_2, and retain the *strong* form of W(7 bis) stipulated in the source. Then $A_1(u)$ and $A_2(u)$ are W-aspherical. Proposition 13, applied to their integral and its closed mapping-cylinder replacement, gives

$$A_1(u) \longrightarrow L_u \text{ in } W \quad \Longleftrightarrow \quad A_0(u) \longrightarrow A_2(u) \text{ in } W.$$

The left side is equivalent to asphericity of L_u, and the right side to asphericity of $A_0(u)$, by two-out-of-three. This proves the asserted equivalence. It applies Proposition 13 to the correctly constructed integral; it does not promote the original noncommuting diamond to a strict square. The reverse implication is a reflection statement, stronger than the preservation statement used in the direct proof. The counterexample to strong reflection discussed earlier in the commentary on this chapter explains why it cannot be transferred without its hypothesis to ordinary weak homotopy equivalences.

At an outer object x_i, every arrow with target x_i already belongs to X_i, since that stratum is open. Hence $E/x_i = X_i/x_i$, and local smoothness is exactly that of f_i. At a central object the hypotheses on both outer objects remain necessary for the converse equivalence just established. The final sentence after Corollary 6 drops them, as the editorial annotation observes; the following finite example measures precisely what is lost.

Corollaire 6. *Soit* $x \in \mathrm{Ob}\,\mathcal{X}$ *(où* $\mathcal{X} = \int$  *.) On veut exprimer la* W*-lissité de*

$$\varphi_Q : \mathcal{X} \longrightarrow Y$$

en x*.*

a) Si $x \in \mathrm{Ob}\,X_i$ *(*$i \in \{1,2\}$*),* φ_Q *est* W*-lisse en* x *si et seulement si* $f_i : X_i \longrightarrow Y$ *l'est.*

b) Supposons W(7 bis) ([108])*. Soit* $x_0 \in \mathrm{Ob}\,X_0$*, et supposons* f_1*,* f_2 *lisses en* x_1*,* x_2*. Alors* φ_Q *est* W*-lisse en* x_0 *si et seulement si* $f_0 : X_0 \longrightarrow Y$ *l'est.*

Donc (moyennant W(7 bis)) *dans les trois cas* $x \in \mathrm{Ob}\,X_i$ *(*$i \in \{0,1,2\}$*) on trouve que* φ_Q *est* W*-lisse en* x *si et seulement si* $f_i : X_i \longrightarrow Y$ *l'est* `[pas vraiment pour` $i = 0$`]`.

ⓑ Quant à b) du théorème 4, c'est le lemme 2 ci-dessus. Donc tout est prouvé!

[page 158]

Commentary · Development (commentators') [ms. 157]

A finite witness for the missing outer hypothesis

Grothendieck has isolated a useful local criterion, but the final formulation would make the middle map alone control the whole neighbourhood. We give a small witness showing why the two outer hypotheses must stay in the statement. We did not find this particular finite witness in the smoothness discussion of Maltsiniotis (2005) or the integration discussion of Cisinski (2006a, §3.3); it is an elementary application of their criteria, not a claim of priority.

Proposition (failure of the unconditional central criterion). [ours] Let W be a fundamental localizer satisfying W(5), so that every W-equivalence induces a bijection on connected components. There is a commutative square of finite posets for which f_0 is smooth and f_2 is smooth, but the smooth integral map φ_Q is not W-smooth at an object of the central stratum.

Proof. Let $Y = X_0 = X_2 = [1] = (0 \to 1)$, with $f_0 = f_2 = \alpha_2 = 1$. Let X_1 be the poset with objects a, b, c and only nonidentity relations $a < c$ and $b < c$. Define

$$f_1(a) = f_1(b) = 0, \qquad f_1(c) = 1, \qquad \alpha_1(0) = a, \quad \alpha_1(1) = c.$$

Both nonidentity arrows of X_1 map to $u : 0 \to 1$. These formulas define functors and give $f_1\alpha_1 = 1_Y$, so the required square commutes. Identity functors are smooth; this proves the assertions about f_0 and f_2.

108. Inutile pour la partie directe, en supposant `[phrase incomplète]`.

Take $x_0 = 1$, $x_1 = c$, $x_2 = 1$ and the arrow $u : 0 \to 1$ in Y. The three strict lift categories are

$$A_0(u) = e, \qquad A_1(u) = \{a \to c,\ b \to c\}_{\mathrm{disc}}, \qquad A_2(u) = e.$$

The first map chooses $a \to c$, and the second is the unique map between terminal categories. Their smooth integral has four objects: a central object z, outer objects a_*, b_* in stratum 1, and d_* in stratum 2. Its only nonidentity arrows are

$$a_* \longrightarrow z \longleftarrow d_*.$$

The object b_* is isolated. Thus its connected-component set has two elements. By W(5) it is not W-aspherical. The slice identification gives this category as $(E/x_0)_u$, which violates the defining local smoothness test. □

The failure occurs in a missing lift relation at the first outer stratum. It involves no question about strictification and no exotic homotopy type. In Corollary 6(b), the assumption that f_1 is smooth at x_1 excludes exactly this additional component.

3.5 Applications aux sommes amalgamées dans une catégorie $X^\wedge$ (avec $X \in \mathrm{Ob}\,\mathrm{Cat}$).

D'abord une caractérisation des sommes amalgamées impliquant une immersion ouverte :

Proposition 15. *Soit* p. 109

$$Q : \qquad \begin{array}{ccc} U & \xrightarrow{\ i\ } & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U' & \xrightarrow{\ i'\ } & X' \end{array}$$

un carré commutatif dans Cat. *Pour que ce carré soit cocartésien dans* Cat, *avec i (donc i') une immersion ouverte, il faut et il suffit que les conditions suivantes soient satisfaites :*

a) Q est cartésien.

b) i' est une immersion ouverte (donc i aussi).

c) Si $T' = X'\backslash U'$, et $T = X \times_{X'} T' = X\backslash U$, $T \longrightarrow T'$ est un isomorphisme.

d) Pour tout $t \in T$, si t' est son image dans T', le foncteur induit par p

$$p/t : \underbrace{U/t}_{=V_t} \longrightarrow \underbrace{U'/t'}_{=V'_{t'}}$$

satisfait à $(p/t)_!(e_{U/t}) \simeq e_{U'/t'}$, ou ce qui revient au même, les 'fibres colocales' de p/t, $v'\backslash V_t$, avec v' dans $V'_{t'}$, sont 0-connexes.

Démonstration immédiate - ça devrait plutôt remonter au §1. [page 159]

Commentary · Exposition [ms. 158]

Gluing the morphisms across an open immersion

Grothendieck now changes scale: the homotopical assertion will rest on an ordinary categorical pushout. An open immersion $U \hookrightarrow X$ is a full sieve. Its complementary full subcategory T is a cosieve; there are no arrows from T to U. Thus X is specified by U, T, and the bifunctor of crossing arrows

$$H : U^{\mathrm{op}} \times T \longrightarrow \mathrm{Set}, \qquad H(u,t) = \mathrm{Hom}_X(u,t).$$

Precomposition and postcomposition encode every composition involving a crossing arrow. This description explains the four conditions in Proposition 15: the first three identify the two pieces, while the fourth identifies the arrows joining them.

Assume first that Q is cartesian, that $U' \hookrightarrow X'$ is open, and that the induced map $T \to T'$ is an isomorphism. Identify T with T' by this specified map. For $t \in T$, let H_t and H'_t be the presheaves $\mathrm{Hom}_X(-,t)$ and $\mathrm{Hom}_{X'}(-,t)$ on U and U'. The square supplies a natural comparison

$$p_! H_t \longrightarrow H'_t.$$

The universal category obtained by pushing $U \hookrightarrow X$ along p has pieces U' and T and crossing-arrow presheaf $p_! H_t$. Indeed, a functor from this category to Z is a pair of functors on U' and T, together with compatible images of crossing arrows; by the defining adjunction of $p_!$, these data are exactly a functor $X \to Z$ and a functor $U' \to Z$ agreeing on U. Composition with arrows of T is preserved because H_t varies functorially with t. This proves the universal property, not merely an objectwise description. Consequently Q is a pushout precisely when all the displayed comparisons are isomorphisms.

We can test these comparisons by explicit factorization categories. An element of $(p_! H_t)(u')$ is represented by a pair

$$b : u' \longrightarrow p(u), \qquad a : u \longrightarrow t,$$

modulo the relation generated by morphisms $h : u \to v$ with $a = a_v h$ and $p(h)b = b_v$. Its image is the composite $q(a)b : u' \to t$ in X'. Fix such a composite $a' : u' \to t$. The category of its representatives and these morphisms is exactly

$$(u', a')\backslash(U/t),$$

where $U/t \to U'/t$ is induced by p. Thus the fibre of the comparison over a' is the set of connected components of this category. The comparison is bijective for every u' if and only if every such category is nonempty and connected. This proves condition (d), including the equivalence with $(p/t)_! e = e$.

Conversely, the explicit universal category just constructed has U' as a full sieve, has complement exactly T, and its inverse image of U' is exactly U. Hence a pushout along $U \hookrightarrow X$ necessarily satisfies (a)–(c); its crossing-arrow comparison is an isomorphism, so it also satisfies (d). This completes both directions of Proposition 15. The argument is entirely about ordinary categories and ordinary left Kan extension.

There is a small example showing why the last condition cannot be omitted. Take $U = U' = e$, X the discrete category on u, t, and $X' = (u \to t)$, with q the identity on objects and $p = 1$. The square is cartesian; both open subcategories are $\{u\}$, and the complements are the same terminal category. But U/t is empty whereas $U'/t = e$. The missing crossing arrow makes the factorization category empty, so (d) fails. The pushout along the identity p is X, not X'. The condition about connected factorization categories is exactly the information which the object sets and the two strata do not contain.

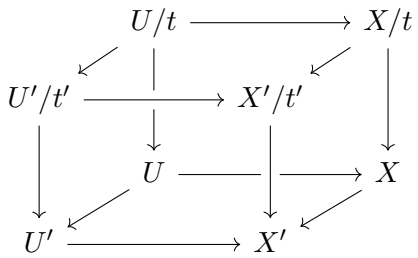
Corollaire 1. *Soit Q un carré* cartésien (cf. page 158), *avec i' une* immersion ouverte *(donc i aussi), et q W-*lisse *(donc p aussi). On suppose que W satisfait l'axiome* W(5) *de connexité* ([109]). *Soit $T' = X'\backslash U'$, $T = X\backslash U$, supposons que $T' \longrightarrow T$ soit un isomorphisme. Alors Q est cocartésien.*

Il suffit de prouver la condition d) de prop. 15. Considérons le carré

$$(*) \qquad \begin{array}{ccc} U/t & \xrightarrow{i/t} & X/t \\ {\scriptstyle p/t}\downarrow & & \downarrow{\scriptstyle q/t} \\ U'/t' & \xrightarrow{i'/t'} & X'/t', \end{array}$$

je dis qu'il est *cartésien.* Cela résulte de la seule hypothèse que Q soit cartésien, sans hypothèse de lissité sur q ni que i' soit une immersion ouverte, et se voit p. ex. en regardant le diagramme cubique : p. 110

109. Donc il suffit q W_0-lisse, où $W_0 = \{f \in \mathrm{FlCat} \mid \pi_0(f) \text{ bijectif}\}$.



([110]). Dans ce diagramme, la face antérieure et postérieure du cube est cartésienne, par définition de U/t et de U'/t' respectivement, et la base l'est par hypothèse. Il en résulte que [page 160] la face supérieure du cube est également un carré cartésien.

Ceci vu, commme q est lisse, il en est de même de q/t dans le carré cartésien $(*)$, donc aussi de p/t par changement de base. D'autre part, la lissité de q implique que q/t est à fibres W-asphériques, donc par changement de base il en est de même de p/t (qui est donc une W-équivalence universelle, i.e. une W-fibration triviale). De plus, à cause de la lissité de p/t, pour v' dans U'/t', l'inclusion

$$(U/t)_{v'} \longrightarrow v'\backslash(U/t)$$

est dans W, donc (par W(5)) définit une bijection sur les π_0. Comme $(U/t)_{v'}$ est W-asphérique par ce qui précède, il s'ensuit (par W(5)) que son π_0 est de cardinal 1, donc il en est de même de $v'\backslash(U/t)$, i.e. celui-ci est 0-connexe. Or c'est là la condition d) de la proposition 15, qu'il fallait établir.

[page 161]

Corollaire 2. *Soit Q un carré cartésien* (p. 158), *avec i' (donc aussi i) une immersion ouverte, et q (donc aussi p) un* isomorphisme local. *Soit $T = X\backslash U$, $T' = X'\backslash U'$ (donc $T \simeq X \times_{X'} T'$), et supposons $T' \longrightarrow T$ isomorphisme. Alors Q est cocartésien.*

C'est un cas particulier du corollaire 1, mais sans référence explicite à un localiseur W. On peut pendre $W = W_\infty$, ou $W = W_0$, pour se ramener au corollaire 1. Mais il est plus raisonnable de dire que dans le carré cartésien $(*)$, l'hypothèse que q soit un isomorphisme local implique que q/t est un isomorphisme, donc p/t l'est aussi. Il en résulte bien que les $v'\backslash(U/t)$ sont 0-connexes, cette catégorie étant isomorphe à $v'\backslash(U'/t')$, donc a un objet initial.

Commentary · Exposition [ms. 161]

From smoothness to the connected factorization test

The two corollaries of Proposition 15 replace its explicit factorization condition by familiar geometric hypotheses. Their force is that a condition on the ambient map q controls every crossing arrow simultaneously.

For a cartesian square Q and an object $t \in X$, the square of slices displayed on page 159 is cartesian. An object of its pullback is an arrow $x \to t$ in X together with a chosen lift of its image-source to U'; the cartesian property of Q identifies that choice with an object of U lying above x. The same description of morphisms gives the isomorphism with U/t. No smoothness is needed for this step.

If q is W-smooth, then $q/t : X/t \to X'/t'$ is W-smooth and has W-aspherical strict fibres. The first assertion is locality of smoothness on slices; the second is its defining lift-category test at t. Both statements are inherited by the pullback p/t. For $v' \in U'/t'$, smoothness gives a W-equivalence

$$(U/t)_{v'} \longrightarrow v'\backslash(U/t).$$

Its source is W-aspherical, so its target is too. W(5) converts this into the required statement that the target has exactly one connected component. Proposition 15 then proves that Q is cocartesian. Only the connectedness part of smoothness is used: as footnote 109 says, W_0-smoothness is sufficient.

For a local isomorphism q, the entire slice map q/t is an isomorphism, and so is p/t. In that case every factorization category $v'\backslash(U/t)$ has an initial object. This gives the second corollary without any localizer. The later formulation that local isomorphisms are smooth is Maltsiniotis

110. NB Ici t est un élément quelconque de $\mathrm{Ob}\, X$, t' son image dans X'.

(2005, Proposition 4.9), but the direct slice argument also explains the stronger ordinary conclusion needed here.

The arrows on complements in the statements of both corollaries are printed $T' \to T$. The map induced by the square is $T \to T'$, as in Proposition 15(c) and in the application on page 166. The condition is that this induced map be an isomorphism. Its inverse then exists, but no inverse map is part of the initial data. We retain the printed direction in the text and record the correction in the register.

Soit maintenant X un objet de Cat, et considérons un diagramme cocartésien dans $X^\wedge$:

$$(*) \qquad \begin{array}{ccc} G & \xrightarrow{\ i\ } & F \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ G' & \xrightarrow{\ i'\ } & F', \end{array}$$

où i est un monomorphisme. On sait que le caractère cocartésien du carré se vérifie 'argument par argument' par le caractère cocartésien de p. 111 [page 162]

$$(*) \qquad \begin{array}{ccc} G(x) & \xrightarrow{\ i(x)\ } & F(x) \\ {\scriptstyle p(x)}\downarrow & & \downarrow{\scriptstyle q(x)} \\ G'(x) & \xrightarrow{\ i'(x)\ } & F'(x), \end{array}$$

dans Ens, $\forall\, x$ dans X. Or un tel carré cocartésien dans Ens, avec $i(x)$ [un] monomorphisme, est aussi cartésien. Cela implique que le carré Q dans $X^\wedge$ est également cartésien.

Considérons, d'autre part, le foncteur canonique

$$\varphi_X : F \longmapsto X/F : X^\wedge \longrightarrow \mathrm{Cat}.$$

Il vaut mieux, d'ailleurs, le regarder comme un foncteur

$$\varphi_X : X^\wedge \longrightarrow \mathrm{Cat}/X,$$

et je rappelle, pour mémoire, que ce foncteur est pleinement fidèle, et que son image essentielle est formée des catégories X' sur X telles que

$$X' \longrightarrow X$$

soit un isomorphisme local, ou ce qui revient au même, que X' soit *fibrée* sur X à fibres discrètes : [page 163]

$$\varphi_X : X^\wedge \xrightarrow{\ \sim\ } \underbrace{\underline{\text{Fib disc}}\, X}_{\substack{\text{catégories fibrées} \\ \text{à fibres discrètes}}} .$$

Notons que toute flèche $X' \longrightarrow X''$ dans $\underline{\text{Fib disc}}\, X$ est un isomorphisme local (tautologique sur $X'/x' \longrightarrow X''/x''$), ce qui montre que si on a un diagramme

$$\begin{array}{ccc} X'/x' & \longrightarrow & X''/x'' \\ & {\scriptstyle \text{iso.}}\searrow \quad \swarrow{\scriptstyle \text{iso.}} & \\ & X/x & \end{array}$$

$$\begin{array}{ccc} X_1 & & X_2 \\ & \searrow \quad \swarrow & \\ & X_0 & \end{array}$$

dans $\underline{\text{Fib disc}}\, X$, le produit fibré $X_3 = X_1 \times_{X_0} X_2$ dans Cat est tel que $X_3 \longrightarrow X_2$ est un isomorphisme local, donc aussi $X_3 \longrightarrow X$. C'est-à-dire $\underline{\text{Fib disc}}(X)$ est stable par produits p. 112 [page 164]

fibrés. Ceci implique que les produits fibrés existent dans $\underline{\text{Fib disc}}\, X$, et que ce sont les produits fibrés dans Cat, ou dans Cat/X.

De même pour les produits finis dans $\underline{\text{Fib disc}}\, X$, ce sont les produits finis dans Cat/X. Bref

Lemme 1. *La sous-catégorie pleine* $\underline{\text{Fib disc}}\, X$ *de* Cat/X *est stable par* $\varprojlim$ *finies, donc les* $\varprojlim$ *existent dans* $\underline{\text{Fib disc}}\, X$ *et le foncteur d'inclusion*

$$\underline{\text{Fib disc}}\, X \longrightarrow \text{Cat}/X$$

y commute, et en particulier transforme carrés cartésiens de $\underline{\text{Fib disc}}\, X$ *en carrés cartésiens de* Cat/X*, ou ce qui revient au même, de* Cat*. De plus, en vertu de l'équivalence*

$$\varphi_X : X^\wedge \xrightarrow{\sim} \underline{\text{Fib disc}}\, X \hookrightarrow \text{Cat}/X,$$

le foncteur φ_X *de* $X^\wedge$ *dans* Cat/X *est exacte à gauche, en particulier il transforme carrés cartésiens de* $X^\wedge$ *en carrés cartésiens de* Cat/X*, ou encore de* Cat*.*

Notons aussi qu'étant exact à gauche, le foncteur φ_X de $X^\wedge$ dans Cat/X transforme monomorphismes en monomorphismes. D'autre part : [page 165]

Commentary · Later developments [ms. 164]

The exactness of the category of elements

Grothendieck's finite-limit lemma is the first part of a stronger known statement. The relative category-of-elements functor

$$j_X : \widehat{X} \longrightarrow \text{Cat}/X, \qquad F \longmapsto (X/F \to X),$$

preserves all small limits and colimits. This is Proposition 3.2.12 of Cisinski (2006a), in the published version, page 141. The absolute functor $i_X : F \mapsto X/F$ preserves colimits and pullbacks, and has the right adjoint stated in Corollary 3.2.13, page 142. We spell out the mechanism because it will also explain the pushout in Theorem 5.

An object of X/F is (x, a) with $a \in F(x)$. An arrow $(x, a) \to (y, b)$ is an arrow $u : x \to y$ such that $F(u)b = a$. Thus an n-simplex of its nerve is uniquely determined by a chain in X and the element at its last vertex:

$$N(X/F)_n \cong \coprod_{x_0 \to x_1 \to \cdots \to x_n} F(x_n).$$

All earlier elements are recovered by restriction. The final face uses the last restriction map of F; the other face and degeneracy maps change the underlying chain. The displayed bijections therefore define the nerve functor, not just its degreewise cardinalities.

Colimits of presheaves are evaluated pointwise, and coproducts commute with colimits. It follows that $N(X/F)$ preserves every small colimit in F. A cocone of nerves whose colimit is again a nerve is also a colimit cocone in categories: apply the fully faithful nerve to maps into an arbitrary target category. This proves the colimit assertion for i_X, and the cocones retain their maps to X, proving it for j_X as well. For limits over X, work over the fixed simplicial set NX. The fibre in degree n above a specified chain is $F(x_n)$, so limits are again calculated pointwise. Since the nerve creates limits, this proves the limit assertion for j_X.

For completeness, the absolute right adjoint is

$$R_X(C)(x) = \text{Fun}(X/x, C),$$

where the right side is the set of functors. For $u : x \to y$, restriction is precomposition with $X/x \to X/y$. A functor $H : X/F \to C$ assigns to $a \in F(x)$ the composite

$$X/x \longrightarrow X/F \xrightarrow{H} C, \qquad (z \xrightarrow{v} x) \longmapsto (z, F(v)a).$$

These assignments form a map $F \to R_X(C)$. Conversely, given such a map, evaluate its functor at 1_x to obtain the image of (x, a). An arrow $u : (x, a) \to (y, b)$ is sent to the arrow in the assigned functor on X/y from u to 1_y. Naturality gives the correct source and ensures preservation of identities and compositions. The two constructions are inverse. This proves the adjunction directly.

The relative target matters for products: $X/(F \times G)$ is the fibre product $(X/F) \times_X (X/G)$. It is generally not the ordinary product of the two total categories. The absolute functor sends the terminal presheaf to X, not to the terminal category. The manuscript correctly formulates finite exactness in Cat/X.

Lemme 2. *Les monomorphismes qui sont des isomorphismes locaux sont exactement les immersions ouvertes. Donc dans* Fib disc X*, les monomorphismes sont les immersions ouvertes, et le foncteur $\varphi_X : F \longmapsto X/F$ transforme monomorphismes en immersions ouvertes.*

Revenons alors au diagramme cocartésien $(*)$ (p. 161), et au diagramme correspondant dans Cat (et même dans Fib disc X)

$$\begin{array}{ccccc} X/G & \xrightarrow{X/i} & X/F & \dashleftarrow & T \\ {\scriptstyle X/p}\downarrow & & \downarrow{\scriptstyle X/q} & & \vdots \\ X/G' & \xrightarrow{X/i'} & X/F' & \dashleftarrow & T'. \end{array}$$

Comme $(*)$ est cartésien, le carré précédent est cartésien, par ce qui précède. Je veux prouver qu'il est également cocartésien (dans Cat/X, ou dans Cat, cela revient au même), et pour ceci [page 166] j'applique le corollaire 2 de la proposition 15 (p. 161), compte tenu que X/q est un isomorphisme local, et X/i' (étant un isomorphisme local `[et un]` monomorphisme) une immersion ouverte. Il suffit de prouver que, T' et T étant les sous-catégories fermées complémentaires des ouverts X/G' de X/F' et de X/G de X/F, que $T \longrightarrow T'$ est `[un]` isomorphisme.

`[Maintenant,` F `et` G`, et` F' `et` G'`, mais pas` p `et` q`, échangent leur rôles.]`

La vérification est immédiate : T est la catégorie des flèches $x \longrightarrow G$ dans $X^\wedge$ (de source x dans X) qui ne se factorisent pas par G `[plutôt par` F`]`, T' celle des flèches $x \longrightarrow G'$ qui ne se factorisent pas par F', le foncteur $T \longrightarrow T'$ étant le foncteur composition de $x \longrightarrow G$ avec $G \xrightarrow{q} G'$ p. 113

$$\begin{array}{ccc} F & \hookrightarrow & G \\ \downarrow & & \downarrow \\ F' & \hookrightarrow & G' \end{array}$$

(NB Le composé ne se factorise pas par F', car autrement, à cause du caractère cartésien de Q, $x \longrightarrow G$ se factoriserait par F.) Mais pour x donné, la donnée de $x \longrightarrow G$ ne se factorisant pas par F est celle d'un élément de $G(x) \backslash F(x)$, et à cause du caractère cocartésien de Q, [page 167]

$$(G(x) \backslash F(x)) \longrightarrow (G'(x) \backslash F'(x))$$

est bijectif, i.e. 'il revient au même' de se donner $x \longrightarrow G$ ne se factorisant pas par F, ou $x \longrightarrow G'$ ne se factorisant pas par F'.

Commentary · Exposition [ms. 167]

Why the complement calculation is enough

Grothendieck wants to transport a pushout of presheaves to a pushout of categories, where the homotopical criterion is already available. We use throughout the notation of the final theorem, $F \hookrightarrow G$ and $F' \hookrightarrow G'$. The source changes the letters immediately before this calculation; the editorial note marks that change, and no mathematical reversal is intended.

At each $x \in X$, the pushout of sets has the concrete description

$$G'(x) \cong F'(x) \amalg \big(G(x) \setminus F(x)\big).$$

The map q_x sends an element of $F(x)$ through p_x and leaves the complementary elements distinct. Hence $F'(x) \to G'(x)$ is injective and its inverse image in $G(x)$ is exactly $F(x)$. This proves

simultaneously that i' is a monomorphism and that the square of presheaves is cartesian. It also gives a bijection between the two objectwise complements.

A monomorphism $F \hookrightarrow G$ induces a full sieve $X/F \hookrightarrow X/G$: if (y, b) belongs to X/F and $u : (x, a) \to (y, b)$ is an arrow of X/G, then $a = G(u)b$ belongs to $F(x)$. Conversely, a monomorphism which is a local isomorphism is a full sieve. To see this, lift uniquely every arrow into an object in its image using the isomorphism on slices; injectivity on objects and arrows then identifies the lift with the given arrow. This is Lemma 2 with its categorical content made explicit.

It remains to check that the bijection of complementary objects respects all arrows in both directions. Let (x, a) and (y, b) be outside X/F. An arrow between their images in X/G' has an underlying arrow $u : x \to y$ satisfying

$$G'(u)q_y(b) = q_x(a).$$

By naturality its left side is $q_x G(u)b$. The element $G(u)b$ cannot belong to $F(x)$, since then its image would belong to $F'(x)$, whereas $q_x(a)$ does not. Both $G(u)b$ and a are therefore in the complement where q_x is injective. They are equal. This gives precisely one arrow in the original complement, with the same underlying u. Identities and compositions are preserved because both complement functors lie over X. The functor $T \to T'$ is thus an isomorphism of categories, not merely a bijection on objects. Proposition 15, Corollary 2, now proves the categorical pushout.

The stronger exactness result in the previous block gives an alternative: i_X preserves this pushout even without the monomorphism hypothesis. The monomorphism is needed for the homotopical conclusion, and for the simultaneous pullback assertion. Indeed, the nerve formula sends the square to a simplicial pushout along a monomorphism. Such a pushout computes the homotopy pushout of the nerves. Together with the Grothendieck-construction comparison, this proves that its canonical smooth-integral map is in W_∞, and hence in every fundamental localizer, by Cisinski (2006a, Corollary 4.2.19). Thus the W(7 bis) proviso in the homotopy-pushout clause of Theorem 5 can be removed. It must still be retained for the *reflection* implications in the following corollary.

Ainsi on a prouvé le

Théorème 5. *Soit X dans* Cat*, et*

$$Q : \quad \begin{array}{ccc} F & \xrightarrow{\ i\ } & G \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ F' & \xrightarrow{\ i'\ } & G' \end{array}$$

un carré cocartésien dans $X^\wedge$, avec i [un] *monomorphisme. Alors i' est* [un] *monomorphisme, le carré Q est aussi cartésien, et le carré correspondant dans* Cat$/X$ *ou dans* Cat

$$\begin{array}{ccc} X/F & \xrightarrow{X/i} & X/G \\ {\scriptstyle X/p}\downarrow & & \downarrow{\scriptstyle X/q} \\ X/F' & \xrightarrow{X/i'} & X/G' \end{array}$$

est à la fois cartésien et cocartésien. De plus, si W est un localiseur fondamental dans Cat*, satisfaisant* W(7 bis) (en plus de W(1,2,3))*, alors le carré X/Q dans* Cat *est W-cocartésien.* [page 168]

Ce dernier énoncé résulte de ce qui précédait et du corollaire 2 du théorème 4, compte tenu que X/q est un isomorphisme local, donc est W-lisse. p. 114

Commentary · Examples [ms. 168]

An attachment over the walking arrow

Theorem 5 can be seen without any large category of simplices. Let $X = [1] = (0 \to 1)$. A presheaf on X is a map of sets $F(1) \to F(0)$. Take $G(1) = \{r, s\}$, $G(0) = \{a, b, c\}$, with restrictions

$r \mapsto a$, $s \mapsto b$. Let F be the subpresheaf with $F(1) = \{r\}$ and $F(0) = \{a, b\}$, and let $F' = e$ be the terminal presheaf. The unique map $F \to e$ gives the pushout

$$G'(1) = \{*, s\}, \qquad G'(0) = \{*, c\}, \qquad * \mapsto *, \quad s \mapsto *.$$

The category X/G has five objects and two nonidentity arrows,

$$(0, a) \longrightarrow (1, r), \qquad (0, b) \longrightarrow (1, s),$$

with $(0, c)$ isolated. Its open subcategory X/F has three objects: the first arrow and the isolated object $(0, b)$. The category X/e is [1]. In the pushout, $(1, r)$ is identified with $(1, *)$, and both $(0, a)$ and $(0, b)$ with $(0, *)$. The resulting category has exactly the two arrows

$$(1, *) \longleftarrow (0, *) \longrightarrow (1, s)$$

and the isolated object $(0, c)$. This is X/G'. There are no further nonidentity compositions to add.

The two complementary closed subcategories are both the discrete category on $(1, s)$ and $(0, c)$. The new attachment changes the crossing arrow into $(1, s)$ but leaves the complement itself unchanged, exactly as Proposition 15 predicts. Their nerves give a homotopy pushout. On connected components the four corners have respectively $2, 3, 1, 2$ components, and the last set is the pushout of the first three component sets. This also checks concretely that the word "cartesian" for the presheaf square does not mean that either vertical map is an equivalence: the left vertical map merges two components, and the right one merges the corresponding two components after attachment.

Corollaire : ([111]) *On a* (moyennant W(7 bis)) *les équivalences*

$$\begin{aligned} i \in W_X \quad &\Longleftrightarrow \quad i' \in W_X \\ p \in W_X \quad &\Longleftrightarrow \quad q \in W_X, \end{aligned}$$

où

$$W_X \subseteq Fl(X^\wedge)$$

est défini par (cf. Pursuing Stacks …)

$$u \in W_X \qquad \Longleftrightarrow \qquad X/u : X/F \longrightarrow X/G \quad \textit{est dans } W.$$

Ainsi, dans une catégorie $X^\wedge$, les sommes amalgamées impliquant un monomorphisme se comportent de façon parfaite du point de vue homotopique, plus exactement pour les localiseurs W_X, pourvu seulement que W satisfasse à W(7 bis). Notamment, les monomorphismes dans $X^\wedge$ sont des 'W_X-cofibrations', en un sens similaire de celui de la définition 2 c) (pages 99, 100 [plutôt 109, 110]). [page 169]

[page 170 n'existe pas.]

Commentary · Present state [ms. 169]

From attachment stability to model structures

The final assertion concerns a class of cofibrations in the sense developed in this chapter: pushing a W_X-equivalence out along a monomorphism again gives a W_X-equivalence. This preservation property survives for ordinary homotopy types. The two reverse implications displayed in the corollary are different assertions and depend on the strong W(7 bis) hypothesis. The earlier failure of pushout reflection applies here as well after representing spaces by simplicial presheaves.

It would go beyond the text to infer that, for every small X, the class W_X is the weak-equivalence class of a model structure whose cofibrations are all monomorphisms. Already $X = e$ shows the missing condition. For $W = W_\infty$, a presheaf is a set and X/F its discrete category, so W_X consists of bijections. If a model structure on sets had these weak equivalences and all monomorphisms as cofibrations, every trivial fibration would be a bijection. Factoring a noninjective map as a

111. **NB** Si on veut seulement $i \in W_X \Longrightarrow i' \in W_X$, $p \in W_X \Longrightarrow q \in W_X$, il suffit que l'on ait W(7 bis) *faible*, i.e. $i_0 \in W \Longrightarrow i \in W$ dans l'énoncé de W(7 bis), p. 76 (Loc(6′a) avec les notations de XVI).

cofibration followed by a trivial fibration would make it injective, a contradiction. Attachment stability alone does not supply the factorization axiom.

The later theory identifies the additional hypotheses. For an accessible fundamental localizer and a W-local test category, the induced presheaf equivalences admit a cofibrantly generated model structure with monomorphisms as cofibrations; this is Cisinski (2006a, Corollary 4.2.18). In the ordinary case, the local test condition is characterized by the requirement that maps with the right lifting property against all monomorphisms be weak equivalences; see Cisinski (2006a, Theorem 4.1.19). The general existence theorem for accessible presheaf localizers, Theorem 1.4.3 of the same work, supplies the factorizations. The present calculation establishes the attachment mechanism on which that construction depends, while the test-category condition ensures that it models the intended homotopy theory.

Overview · XIIb-b

Chapter XII, section 5, manuscript pages 115–169.

These pages develop a concrete calculus for deciding when an ordinary gluing diagram of categories represents a homotopy gluing. Its central object is a comparison map. Given a strictly commuting square with initial corner X_0, adjacent corners X_1, X_2, and terminal corner Y, Grothendieck constructs a category $\mathcal{X}_Q$ from the three-category span and a canonical functor

$$\varphi_Q : \mathcal{X}_Q \longrightarrow Y.$$

The square is W-homotopically cocartesian when this particular functor belongs to the chosen weak-equivalence class W. The definition remembers the cocone, not only its four vertex homotopy types. The work that follows explains why the construction is independent of its two orientations, how smoothness can test the comparison on fibres, and why monomorphisms of presheaves provide a reliable supply of such squares.

The opening argument concerns a different but related exactness problem. Total W-asphericity here is the requirement that, for every $p : X \to Y$, the category of elements of the ordinary left Kan extension $p_!1_X$ be W-aspherical. It is the definition of this chapter, explicitly distinguished by the editors from the earlier use of the same phrase in *Pursuing Stacks*. W_0-properness, or W-properness under W(5), allows Grothendieck to compute

$$p_!1_X \cong \operatorname*{colim}_{i\in I} p_{i!}1_{X_i}$$

for a family $X \to I$. A terminal object of I reduces this colimit to the last fibre. The commentary supplies the generators-and-relations proof on components and the precise functor on comma categories requested in footnote 83. It also proves that total W_∞-asphericity is not invariant under weak homotopy equivalence. The properness hypothesis controls ordinary Kan extensions; equivalence of classifying spaces alone does not do that work.

The smooth integral of $X_1 \leftarrow X_0 \to X_2$ keeps the three sets of objects disjoint. Its mixed arrows run from the outer fibres to the middle fibre, with

$$\operatorname{Hom}_{\mathcal{X}_Q}(a, x) = \operatorname{Hom}_{X_i}(a, \alpha_i x) \qquad (a \in X_i,\ x \in X_0).$$

Its universal property therefore includes natural transformations $f_i\alpha_i \Rightarrow f_0$. They need not be invertible. Strict commutativity of the given square supplies identity transformations and hence the canonical φ_Q. Dividing the integral into two full closed pieces gives a genuine categorical pushout and pullback. Each piece retracts onto its outer fibre by an adjunction. The nerve of this closed cover proves directly that the total category represents the homotopy pushout of classifying spaces. The general theorem is Thomason's; the bibliography distinguishes verification of the original metadata from verification of an independent primary proof (Thomason, 1979; Cegarra, 2020).

Even the smallest examples explain why this construction is needed. For the span $e \leftarrow \{a, b\} \to e$, the integral is a four-object category whose nerve is a circle. Its ordinary categorical pushout is a point. The ordinary pushout of the same diagram in the homotopy category is also a point: it retains equality of components but loses the two paths implementing compatibility. The integral retains those paths as transition arrows. This is the concrete content, on these pages, of the insistence on forming diagram categories before localization.

The excision hypotheses require particular care. Proposition 13 and its cube corollary are valid conditional assertions under the full W(7 bis) and W(7) printed in the manuscript. Those axioms include reflection statements stronger than the usual preservation of weak equivalences by homotopy pushout. They fail for W_∞. The commentary proves this with a connected acyclic finite complex having nontrivial fundamental group. Its suspension is contractible, so the union of its two cones and both cones are contractible although their

intersection is not. A map of this closed-cover square to the point square violates the converse cube implication. The usual forward gluing theorem survives. A normalized-chain calculation also explains exactly why the complete three-way implication works for integral homology equivalences. The earlier chapter's claim about full W(7) for W_∞ is therefore not a valid justification of the converses in the present portion.

The proper integral reverses the transition arrows. Grothendieck compares the two constructions by the functorial resolution

$$C \longleftarrow \Sigma(C) \longrightarrow C^{\mathrm{op}}.$$

Both projections are aspherical, and the same comparison can be integrated. Proposition 14 proves agreement of the four versions obtained from smooth or proper integration and from the span or its opposite. The commentary writes out the chosen twisted-arrow convention and completes the deferred transitivity verification. The natural comparison with the opposite operation is involutive; its naturality makes the square of four integration comparisons commute. The indexing category must keep its two paths distinct until they reach Y. Replacing that free diamond by a commutative square would erase precisely the information the proof is using.

The question on manuscript pages 145–146 asks whether a bad ordinary pushout can become a good homotopy cocone after mapping to another target. We answer it with finite posets. A collapsible disk with seven triangles maps to the tetrahedral two-sphere so that all associated comma categories are nonempty and connected. Its face poset E embeds as a sieve in a contractible mapping-cylinder poset B. The categorical pushout

$$P = B \amalg_E e$$

has two incomparable minimal vertices below the face poset of that sphere. Its nerve is their join with the subdivided sphere, hence has classifying space S^3. The span $B \leftarrow E \to e$ has contractible integral. Consequently the comparison to P is not a weak equivalence, whereas its composite with $P \to e$ is. The proof includes the categorical pushout universal property; a separate exact finite calculation checks the collapses, comma connectivity, and rational homology. This is a worked development, with no claim of priority. The same disk-to-sphere map supplies the promised counterexample to homotopy invariance of total asphericity.

Theorem 4 turns from construction to recognition. Its forward assertion says that smoothness of the three maps f_i implies smoothness of φ_Q. The reverse assertion for the middle map retains strong W(7 bis). Two slice calculations identify the strict fibres relevant to local smoothness with integrals of lift categories. Footnote 107 correctly observes that the apparent square in the original proof is not strictly commutative. The repair uses the split fibration of the integral over the fork and its aspherical fibres. This proves the forward direction without the strong excision axiom. An explicit contraction removes the same unnecessary hypothesis from the open-cover corollaries. The final unconditional sentence after Corollary 6, however, omits essential outer smoothness assumptions; a four-object lift category with an isolated component gives a finite counterexample.

The last part returns to ordinary category theory. A full sieve $U \subset X$ and its complement T are glued by the presheaves $\mathrm{Hom}_X(-, t)$ on U. Proposition 15 says that a square along $U \subset X$ is a pushout exactly when it identifies the complementary categories and transports these crossing-arrow presheaves by left Kan extension. The connected factorization categories in its statement are the fibres of that comparison of presheaves. This gives a full proof of the criterion and explains why an agreement on objects and strata would be insufficient.

For a presheaf F on X, the formula

$$N(X/F)_n = \coprod_{x_0 \to \cdots \to x_n} F(x_n)$$

then makes the final gluing theorem transparent. It proves preservation of colimits by the category-of-elements construction; the relative construction into Cat/X also preserves all small limits, the known strengthening recorded by Cisinski (2006a, Proposition 3.2.12 and Corollary 3.2.13). A pushout of presheaves along a monomorphism becomes a simplicial pushout along a monomorphism. Thus the positive homotopy-pushout clause of Theorem 5 holds for every fundamental localizer. Its subsequent reflection implications still require the strong hypothesis. Model structures require further conditions, supplied later by accessibility and local test categories; attachment stability alone is insufficient.

The later derivator definition identifies φ_Q with the terminal component of the coherent left-Kan-extension counit (Groth, 2013, Definition 3.9). The comparison with ∞-categories requires another localization: a total category models the categorical colimit after its transport arrows are inverted, while classifying spaces invert all arrows (Lurie, 2009, Corollary 3.3.4.3). These formulations clarify the target of the present calculations without replacing their elementary categorical proofs. The equivalence of the two integration conventions and the strict-cocone question are resolved, as are the specified coherence and local-smoothness gaps. The opening request for a sufficiently large class of cofibrations closed under the stated interval operation is not settled here in its full generality. We record that limitation without declaring it a present-day open problem.

ms. page 171

ed. 115

VI Critère de W_∞-asphéricité pour les $\Omega^n(X,x)$.

1 Cogitations sur W_n.

On va définir, pour $n \in \mathbf{N}$,

$$W_n \subseteq \mathrm{Fl}(\mathrm{Cat})$$

par les conditions suivantes (où $f : X \longrightarrow Y$ est une flèche dans Cat).

$f \in W_0 \iff$ a) $\pi_0(f)$ bijectif.

$\iff$ b) Pour tout préfaisceau d'ensembles F *constant* sur Y, $H^0(Y,F) \longrightarrow H^0(X, f^*F)$ est bijectif.

$\iff$ c) Le foncteur $F \longmapsto f^*F$ des préfaisceaux d'ensembles *constants* sur Y vers les faisceaux d'ensembles constants sur X est pleinement fidèle.

`[Les termes de faisceau ou de préfaisceau sont utilisés ici indifféremment.]`

NB Si F_0 est un ensemble, $F_{0\,Y}$ le faisceau d'ensembles constant défini par F sur Y, on a

$$H^0(Y, F_{0\,Y}) \simeq \mathrm{Hom}_{\mathrm{Ens}}(\pi_0(Y), F_0) \qquad \text{(fonctoriel en } Y)$$
$$\mathrm{Hom}_{Y^\wedge}(F_{0\,Y}, G_{0\,Y}) \simeq \mathrm{Hom}_{\mathrm{Ens}}(\pi_0(Y), \mathrm{Hom}_{\mathrm{Ens}}(F_0, G_0)) \qquad \text{(fonctoriel en } Y),$$

ce qui implique que a) $\Longrightarrow$ c) $\overset{\text{tautologiquement}}{\underset{\substack{\text{en prenant}\\ \mathrm{Hom}_{Y^\wedge}(e_Y, F)}}{\Longrightarrow}}$ b) $\Longrightarrow$ a).

$f \in W_1 \iff$ a) $\pi_0(f)$ bijectif, et pour tout $x \in \mathrm{Ob}\, X$, $\pi_1(f,x)$: $\pi_1(X,x) \longrightarrow \pi_1(Y, f(x))$ est bijectif. (NB Il suffit de le vérifier pour *un* x dans chaque composante connexe de X.)

$\iff$ a′) $\Pi(X) \xrightarrow{\Pi f} \Pi(Y)$ est une équivalence de catégories.

$\iff$ b) $H^0(Y,F) \longrightarrow H^0(X, f^*F)$ bijectif pour tout faisceau d'ensembles *localement constant* sur Y, et $H^1(Y,F) \longrightarrow H^1(X, f^*F)$ bijectif si de plus F est un faisceau en groupes.

$\iff$ b′) Comme b), mais pour le H^1 on se borne à F *constant*, et on exige $H^1(f)$ *surjectif* seulement.

$\iff$ c) Le foncteur $F \longmapsto f^*F$ de la catégorie des faisceaux d'ensembles localement constants sur Y vers celle des faisceaux localement constants sur X est une équivalence de catégories.

Commentary · Exposition [ms. 171]

What the first two tests measure

Grothendieck is trying to recognize finite homotopy information without leaving the category of small categories. The two initial cases explain why locally constant coefficients, rather than arbitrary presheaves, are the appropriate witnesses. Write $BX = |NX|$ for the classifying space of X. A locally constant set-valued presheaf is a functor on the fundamental groupoid $\Pi(BX)$, with the variance reversed if one retains the presheaf convention. In this block all changes of variance are absorbed by inversion in the groupoid. The word "constant" on p. 171 means globally constant, whereas "locally constant" permits monodromy. For a disconnected category these are substantially different conditions.

For a set A, a section of the constant presheaf A_X chooses one element of A on each connected component. Thus $H^0(X, A_X) = \mathrm{Map}(\pi_0 X, A)$. A natural transformation $A_X \to B_X$ similarly chooses one map $A \to B$ per component. These formulas prove all three tests for W_0. In the converse direction, if precomposition with $\pi_0 f$ is bijective for every target set, Yoneda gives that $\pi_0 f$ is a bijection. The empty category is included: the map from a two-element set to its set of sections on the empty category cannot be bijective.

There is a useful elementary explanation of the apparently weak condition b′) for W_1. After using constant sets to identify components, work with connected X, Y and put $\alpha : H = \pi_1(X, x) \to G = \pi_1(Y, f(x))$. Sections of the local system corresponding to a G-set A are A^G, and sections after pullback are A^H. Their equality for every A forces α to be surjective. Indeed, let $K = \alpha(H)$ and take the transitive G-set G/K. The coset K is fixed by H; if every H-fixed element is G-fixed, then $K = G$. Constant nonabelian H^1 has the familiar description

$$H^1(X, L_X) = \mathrm{Hom}(H, L)/L,$$

where L acts by conjugation. Surjectivity for $L = H$ says that the class of the identity of H is represented by $\rho\alpha$ for some $\rho : G \to H$. Conjugating ρ makes $\rho\alpha = 1_H$, so α is injective as well. If there are several components, choose the identity representation on the chosen component and trivial representations on the others. This proves b′)⇒a), including the componentwise assertion.

An equivalence of fundamental groupoids conversely identifies every category of local systems, its global sections, and its torsors. This proves a)⇒b), c). To see directly that c) also forces the groupoid equivalence, full faithfulness recovers global sections as maps from the terminal object and hence forces α to be surjective by the same coset argument; essential surjectivity makes the regular H-set the restriction of a G-set, so $\ker\alpha$ acts trivially on a faithful H-set and is trivial. The torsor interpretation is the first step of the comparison between cohomology and homotopy types recalled in Lurie (2009, Preface, pp. vii–viii). Nothing here requires a choice of a universal covering category: the fundamental groupoid already contains precisely the needed information.

NB Le groupe $\pi_1(X, x)$ est défini via la théorie du groupoïde fondamental. On sait qu'il est en bijection canonique avec $\pi_0(\Omega(X, x))$, où $\Omega(X, x) = \underline{\mathrm{Ch}}(X; x, x)$ (ce qui permet d'avoir une loi de composition monoïdale sur $\Omega(X, x)$, donnant la loi de groupe sur le π_0), ou au choix $\Omega(X, x) = \underline{\mathrm{Ch}}_\infty(X; x, x)$ (plus commode dans certaines situations, mais ne donnant pas directement la loi de composition dans $\pi_0(\Omega(X, x))$).

Je vais poser, pour $n \in \mathbf{N}$,

$$\pi_n(X, x) = \pi_0(\Omega^n(X, x)),$$

Ω^n désignant l'itéré n-ième du foncteur $(X, x) \longmapsto \Omega(X, x)$ de la catégorie $\mathrm{Cat}_\bullet$ des objets ponctués de Cat dans elle-même. Je préfère ici travailler avec la théorie $\underline{\mathrm{Ch}}$ au lieu de $\underline{\mathrm{Ch}}_\infty$. Tôt ou tard il faudra donner un énoncé de comparaison …

Théorème 1 ([112]). *Soit $f : X \longrightarrow Y$ dans* Cat*, et soit $n \in \mathbf{N}$, $n \geq 2$, S une partie de* $\mathrm{Ob}\, X$ *telle que le composé $S \longrightarrow \mathrm{Ob}\, X \longrightarrow \pi_0(X)$ soit* `[un]` *épimorphisme. Les conditions suivantes sont équivalentes.*

112. Sans doute *faux*. On doit avoir a) $\Longrightarrow$ b), mais b) $\Longrightarrow$ a) `[est]` sans doute faux déjà pour $n = 2$ (sauf le cas X, Y 1-connexes). Cf. énoncé rectifié `[p. 178]`.

a) $\pi_i(f,x)$ *isomorphe pour tout* $x \in \mathrm{Ob}\, X$ *et tout* $i \in [0,n]$.

a′) *Comme a), mais avec* $x \in S$.

b) *On a*

$$\begin{array}{lll} H^0(Y,F) & \xrightarrow{\sim} & H^0(X,f^*F) \quad \textit{pour } F \textit{ faisceau d'ensembles} \\ & & \textit{localement constant sur } Y. \\ H^1(Y,F) & \xrightarrow{\sim} & H^1(X,f^*F) \quad \textit{si de plus } F \textit{ faisceau en groupes.} \\ H^i(Y,F) & \xrightarrow{\sim} & H^i(X,f^*F) \quad \textit{si de plus } F \textit{ faisceau abélien, } i \in [2,n]. \end{array}$$

ms. page 173

Commentary · Exposition [ms. 172]

Loops, finite types, and the scope of the proposed theorem

The iterated path construction is being asked to do two jobs. First it supplies the groups $\pi_i(X,x)$ intrinsically in Cat; subsequently it must show that these groups are the usual homotopy invariants of BX. The later identification of the minimal fundamental localizer with ordinary weak homotopy equivalences removes the distinction between W_ω and W_∞ in the modern interpretation; see Cisinski (2004, Theorem 2.2.11). It should not be used to erase Grothendieck's effort to obtain the result from the categorical path construction itself.

Theorem 1 contains three separate assertions. The equivalence a)⇔a′) is the basepoint transport statement established below. The implication a)⇒b) is correct, and the argument on pp. 180–185 supplies its proof after the twisted-coefficient evaluation map is made explicit. The converse b)⇒a) is false in general. The first failure, however, is in degree 3, not in degree 2 as footnote 112 suggests. The degree-two case survives even with nontrivial fundamental group. We prove this distinction below, because it materially changes the interpretation of the ensuing "rectification".

It is also necessary to distinguish an equivalence of n-types from a map with n-connected homotopy fibres. For the former, $\pi_0 f$ is bijective and $\pi_i f$ is an isomorphism for $1 \le i \le n$. For the latter one needs, in addition, surjectivity on π_{n+1}. Grothendieck identifies this additional condition correctly on p. 179, but calls the fibre condition the "good localizer" on p. 187. That last sentence cannot stand as a definition of a class of weak equivalences: the fibre condition fails the two-out-of-three axiom. The distinction does disappear for a map to a point, which explains why the argument of Theorem 2 is unaffected.

NB L'équivalence de a) et a′) devrait être triviale, mais pour cela il faut avoir défini les $\pi_n(X,x)$, pour x variable dans $\mathrm{Ob}\, X$, comme système local sur X, chose que je n'ai pas faite encore. Je vais essayer de le faire dans un courant d'air. Tout d'abord, les $\Omega(X,x)$ sont les fibres d'une W_ω-fibration parfaite ([113]), s'insérant dans le carré cartésien

$$\begin{array}{ccc} \underline{\mathrm{Ch}}(X) & \longleftarrow & \Omega(X,-) \\ {\scriptstyle p_X=(\sigma_X,\beta_X)}\downarrow & \nearrow_{i_X} & \downarrow \\ X \times X & \xleftarrow{\mathrm{diag}_X} & X. \end{array}$$

113. **NB** W_ω est le plus petit localiseur de Cat (satisfaisant W(1, 2 bis, 3)).

ed. 117 On sait alors que l'on a un foncteur

$$(*) \qquad \begin{aligned} \Pi X &\longrightarrow \mathrm{Hot}_{W_\omega} \\ x &\longmapsto \mathrm{hot}_{W_\omega}(\Omega(X,x)). \end{aligned}$$

Or on a, pour $i \geq 1$,

$$\pi_i(X,x) = \pi_{i-1}(\underbrace{\Omega(X,x)}_{\text{objet ponctué}}).$$

Cela donne le système local cherché des π_i, si seulement on avait un foncteur un peu plus précis que $(*)$, savoir

$$\Pi X \longrightarrow \mathrm{Hot}_{\bullet W_\omega} \qquad \text{(catégorie } W_\omega\text{-homotopique } \textit{ponctuée})$$

([114]). Or la flèche $i_X : X \longrightarrow \underline{\mathrm{Ch}}X$ relevant p_X donne une section canonique de $\Omega(X,-)$ sur X - celle justement qui associe à tout $x \in X$ le point marqué 1_x de $\Omega(X;x,x)$ ('lacet vide en x'). La question est donc celle-ci : ms. page 174

> On a une W-fibration $\Omega \longrightarrow X$, avec une section donné σ. Cela définit-il un foncteur canonique
>
> $$X \longrightarrow \mathrm{Hot}_{\bullet W},$$
>
> transformant toute flèche en `[un]` isomorphisme?

L'idée naturelle est de prendre

$$x \longmapsto \mathrm{hot}_{\bullet W}(\Omega/x)$$

en ponctuant Ω/x par le point marqué $\sigma(x)$ de $\Omega_x \hookrightarrow \Omega/x$. L'ennui, c'est que les morphismes de transition $\Omega/x \longrightarrow \Omega/x'$ (pour $x \xrightarrow{u} x'$ dans X) ne sont pas compatibles avec les points marqués. Mais on a une flèche (savoir $\sigma(u)$) allant de l'image $\sigma(x)/x'$ du point marqué de Ω/x vers le point marqué $\sigma(x')/x'$ de Ω/x'. Ceci doit suffire pour définir la flèche ms. page 175

$$(\Omega/x, \sigma(x)/x) \longrightarrow (\Omega/x', \sigma(x')/x')$$

comme flèche dans $\mathrm{Hot}_{\bullet W}$, avec transitivité pour une composée $x \xrightarrow{u} x' \xrightarrow{v} x''$ dans X. Je vais
ed. 118 pour l'instant me borner à une indication, quitte à faire des vérifications circonstanciées par ms. page 176 la suite. Au besoin, j'utiliserai dès maintenant que pour $i \in \mathbf{N}$, $i \geq 1$,

$$\pi_i(f,x) : \pi_i(X,x) \longrightarrow \pi_i(Y, f(x))$$

peuvent s'interpréter comme provenant d'un homomorphisme de systèmes locaux

$$\underset{\sim i}{\pi}(X,-) \longrightarrow f^*(\underset{\sim i}{\pi}(Y,-))$$

sur X. La condition a) du théorème 1 s'interprète en disant que cet homomorphisme est un *isomorphisme* pour $i \in [1,n]$, et la condition a') se borne alors à rappeler qu'il suffit de tester qu'il en est ainsi pour *un* point de chaque composante connexe de X.

---

114. Mais il faut aussi vérifier que les $\pi_n(X,x)$ $(n \geq 1)$ sont des foncteurs sur $\mathrm{Hot}_{\bullet W_\infty}$, i.e. transforment W_∞-équivalences en itou. Cela résulte de W(4a) pour W_∞, qui implique que $(X,x) \mapsto \Omega(X,x)$ se factorise par $\mathrm{Hot}_{\bullet W_\infty}$ (en tant que foncteur à valeurs dans $\mathrm{Hot}_{\bullet W_\infty}$).

Commentary · Development (commentators') [ms. 175]

Making pointed transport explicit

Grothendieck needs to transport a marked object along an arrow even when the underlying transition functor does not preserve that object strictly. The discrepancy is useful data: the arrow supplied by the section tells us exactly which change of basepoint to make. The following construction fills in the indicated verification using only arrow categories and localization. We checked the natural-homotopy and adjunction lemmas in Cisinski (2004, §1.1, especially Proposition 1.1.9 and Corollary 1.1.10); we did not locate a discussion of this particular pointed comma-category formula in the consulted sources. The result is an elementary completion of the manuscript's construction, with no assertion of priority.

Pointed transport lemma. [ours] Suppose W contains every functor admitting a left or a right adjoint and satisfies two-out-of-three. In the localization of pointed categories at functors whose underlying functors lie in W, an arrow $a : d \to e$ of a category D defines a canonical isomorphism

$$t_a : (D, d) \longrightarrow (D, e).$$

It is natural in functors $D \to D'$, satisfies $t_{1_d} = 1$, and satisfies $t_{ba} = t_b t_a$ for composable arrows a, b.

Proof. Regard a as the marked object of $D^{[1]}$. The two evaluation functors give a pointed span

$$(D, d) \xleftarrow{e_0} (D^{[1]}, a) \xrightarrow{e_1} (D, e).$$

The constant-diagram functor is left adjoint to e_0 and right adjoint to e_1, so both evaluations belong to W. Define $t_a = [e_1][e_0]^{-1}$. For the identity arrow the constant-diagram functor is a pointed common section, proving the unit assertion. For a composable pair, mark $D^{[2]}$ by the diagram $d \xrightarrow{a} e \xrightarrow{b} h$. Evaluation at every vertex belongs to W. For the endpoints this again follows from adjunctions. For the middle vertex use the natural transformation $e_0 \Rightarrow e_1$ and the following elementary consequence of the assumptions: naturally related functors belong to W simultaneously. Indeed, represent a natural transformation by $H : A \times [1] \to B$. The projection $A \times [1] \to A$ has an adjoint and lies in W; each endpoint section lies in W by two-out-of-three. Thus $Hi_0 \in W$ if and only if $H \in W$, if and only if $Hi_1 \in W$. Restriction to the three edges commutes with vertex evaluation and is therefore in W by two-out-of-three. On this common pointed source,

$$[e_2][e_0]^{-1} = \big([e_2][e_1]^{-1}\big)\big([e_1][e_0]^{-1}\big).$$

This is the composition assertion. Applying a functor to arrow diagrams commutes strictly with all evaluations and restrictions, proving naturality. □

Now let $q : E \to X$ be the fibration under discussion, with section σ. For $u : x \to x'$ write $u_* : E/x \to E/x'$ for postcomposition. Its marked source maps to $u_*\sigma(x)$, and the section gives an arrow

$$a_u : u_*\sigma(x) \longrightarrow \sigma(x')$$

in E/x'. Define the pointed transition to be $t_{a_u}[u_*]$. For $v : x' \to x''$, the equality $a_{vu} = a_v\, v_*(a_u)$, naturality of t, and the composition formula prove the functor law. When the comma transitions belong to W, these maps are isomorphisms and the functor factors through ΠX. The pointed inclusions $E_x \to E/x$ identify it with the family of strict fibres whenever those inclusions belong to W. Applied to the loop fibration, this gives the local systems of higher homotopy groups and proves the equivalence of the one-basepoint and every-basepoint formulations. It gives ordinary transport in the pointed homotopy category; a coherent family of pointed spaces contains further information, which is not claimed to follow from this ordinary functor alone.

L'énoncé du théorème 1 garde un sens pour $n = 1$, et dans ce cas a été 'rappelé' précédemment (et est ± tautologique). Cela va nous permettre de procéder par récurrence sur n, en supposant que le théorème est prouvé pour $n - 1$. Cela signifie que l'on peut supposer que les conditions a) *et* b) sont satisfaites par f pour $i \in [1, n-1]$, et se borner à prouver que, sous ces conditions, la condition a) (ou a′), pour $i = n$, équivaut à la condition b) pour $i = n$.

Donc il faut prouver simplement ceci : ms. page 177

Lemme ([115]). *Supposons que* $\pi_i(f, x)$ `[est un]` *isomorphisme pour* $i \in [0, n-1]$ *et b)*

115. Sans doute faux.

satisfait pour $i \in [0, n-1]$, où $n \in \mathbf{N}$, $n \geq 2$, est fixé. Alors les conditions suivantes sont équivalentes :

a) $\pi_n(f,x) : \pi_n(X,x) \longrightarrow \pi_n(Y, f(x))$ `[est un]` *isomorphisme pour tout $x \in \operatorname{Ob} X$.*

b) $H^n(Y,F) \xrightarrow{\sim} H^n(X, f^*F)$ `[est un]` *isomorphisme pour tout faisceau abélien localement constant sur Y.*

Notons que f induit une flèche dans $\mathrm{Cat}_\bullet$,

$$\Omega(f,x) : \Omega(X,x) \longrightarrow \Omega(Y, f(x)),$$

d'où

$$\begin{array}{ccc} \pi_i(\Omega(X,x)) & \longrightarrow & \pi_i(\Omega(Y,f(x))) \\ \wr| & & \wr| \\ \pi_{i+1}(X,x) & \longrightarrow & \pi_{i+1}(Y,f(x)), \end{array}$$

s'identifiant à $\pi_{i+1}(f,x)$:

$$\pi_i(\Omega(f,x)) \simeq \pi_{i+1}(f,x).$$

ed. 119 L'hypothèse a) du lemme implique que $\pi_i(\Omega(X,x)) \xrightarrow{\sim} \pi_i(\Omega(Y,y))$ pour $i \in [0, n-1]$. Il s'agit ici des π_i basés en les points unités. On devra sans doute prouver que, du fait qu'il s'agit d'un morphisme de monoïdes dans Cat (satisfaisant certaines conditions?) ça reste vrai pour les π_i en *tout* point ξ de $\Omega(X,n)$. Par l'hypothèse de récurrence, on saura donc que $\Omega(X,x) \longrightarrow \Omega(Y,y)$ induit un isomorphisme pour la cohomologie à coefficients localement ms. page 178
constants, jusqu'en dimension $n-1$ inclus.

Finalement, je me suis ± convaincu que l'énoncé du théorème 1 est faux tel quel. Voici un énoncé que j'éspère vrai, et que je vais essayer de prouver dans un esprit 'utilitaire', en utilisant tout ce que je sais en homotopie et cohomologie (et pas seulement ce que j'ai développé 'from scratch' dans le modélisateur Cat).

Théorème 1 rectifié. *Considérons, pour un entier $n \geq 2$ donné, les conditions suivantes sur une flèche $f : X \longrightarrow Y$ dans* Cat *(ou une application continue d'espaces topologiques, ou un morphisme dans $\mathbf{\Delta}^\wedge$ etc. etc.) :*

a_n) Si Z est une 'fibre homotopique' de f (NB il y en a essentiellement une par composante connexe de Y), alors Z est n-connexe, i.e. (p. ex) $\pi_i(Z) = 0$ pour $i \in [0,n]$. ms. page 179

Commentary · Examples [ms. 178]

The first genuine failure and the role of loop multiplication

The withdrawn induction fails because ordinary cohomology in degree n need not see all of π_n when lower homotopy is already present. A concrete example makes the failure exact. Take

$$f : S^2 = \mathbf{CP}^1 \longrightarrow \mathbf{CP}^\infty = K(\mathbf{Z}, 2).$$

Both spaces are simply connected. Their integral cellular chain complexes have $\mathbf{Z}$ in degree 0 and degree 2, and no groups in degrees 1 and 3; $\mathbf{CP}^\infty$ has additional even-dimensional cells starting in degree 4. Thus for every abelian group M, the map induces isomorphisms on $H^i(-;M)$ for $0 \leq i \leq 3$: the nonzero groups in this range are $H^0 = M$ and $H^2 = M$. Nonabelian H^1 is trivial on both spaces and every local system is constant. Yet

$$\pi_3(S^2) = \mathbf{Z} \longrightarrow \pi_3(K(\mathbf{Z},2)) = 0$$

is not an isomorphism. The source group is detected by the Hopf fibration, and the target calculation follows from $S^1 \to S^\infty \to \mathbf{CP}^\infty$; both computations appear in Hatcher (2002, Examples 4.50–4.51). This disproves Theorem 1 and the lemma on p. 177 for $n = 3$, even with simply connected source and target. Applying the category-of-simplices construction to singular simplicial models gives an actual functor of small categories with the same failure; the nerve/category-of-simplices equivalence is recorded in Cisinski (2004, Lemma 2.1.15).

> The proposed passage from the unit component of a loop space to all components is nevertheless sound. If M is a topological monoid with $\pi_0 M$ a group, multiplication by any $m \in M$ is a homotopy equivalence. Choose m' representing its inverse component. Paths from mm' and $m'm$ to the unit show that multiplication by m' is a homotopy inverse, using associativity. A homomorphism of such monoids commutes with these translations. Hence its effect on homotopy groups in every component is determined by its effect at the unit, together with the bijection on components. Realization of a monoidal category has this property when its monoid of components is a group, as it is for Grothendieck's loop category. The monoidal argument repairs the specific verification requested at the top of p. 178; it does not repair the false cohomological converse used by the abandoned induction.

b_n) $\pi_0(f)$ *est bijectif, et* $\forall\ x \in X$, $\pi_i(f,x) : \pi_i(X,x) \longrightarrow \pi_i(Y, f(x))$ *est* `[un]` *isomorphisme pour* $i \in [1,n]$. *(***NB** *Il suffit de le dire pour* un *point* x *de chaque composante connexe de* X*.)*

c_n) $H^i(Y,F) \longrightarrow H^i(X, f^*F)$ `[est un]` *isomorphisme pour* F *faisceau localement constant sur* Y *(avec* F *abélien si* $i \geq 2$, F *faisceau en groupes si* $i = 1$*).*

On a alors

$$a_n \implies b_n \implies ({}^{116})\ c_n \implies ({}^{117})\ a_{n-1}$$

(et je présume que chacune de ces implications est stricte).

Démonstration. $\mathrm{a}_n \Longrightarrow \mathrm{b}_n$ est conséquence immédiate de la suite exacte d'homotopie pour une Hot-fibration $\bar{X} \xrightarrow{\bar{f}} Y$ qui factorise $X \xrightarrow{f} Y$ en $f = \bar{f} \circ i$, $i \in W_\infty$. En fait, a_n *équivaut* à b_n + la surjectivité des $\pi_{n+1}(X,x) \longrightarrow \pi_{n+1}(Y, f(x))$, laquelle bien sûr ne résulte *pas* de b_n, i.e. de la bijectivité des $\pi_i(f,x)$ pour $i \leq n$ (prendre des $K(\pi, n+1)$). Donc l'implication est bien stricte.

ed. 120 $\mathrm{b}_n \Longrightarrow \mathrm{c}_n$. On peut supposer que f est une Hot-fibration (et même Hot-parfait si ça nous chante …). On sait déjà que pour ms. page 180

$$(*) \qquad H^i(Y,F) \longrightarrow H^i(X, f^*F)$$

tout est O.K. si $i \in \{0,1\}$. Donc on peut supposer F abélien. Si ça nous avance, on peut supposer, par récurrence, (que l'implication $\mathrm{b}_{n-1} \Longrightarrow \mathrm{c}_{n-1}$ est connue, donc) que $(*)$ est `[un]` isomorphisme pour $i < n$. Reste à le voir pour $i = n$. On est ramené bien sûr, comme $\pi_0(f)$ est bijectif, au cas où X, Y sont 0-connexes, donc il y a essentiellement une seule fibre Z. La suite exacte d'homotopie nous apprend que

$$\pi_i(Z,z) = 0 \qquad \text{pour } i \leq n-1 \ (Z \text{ est } (n-1)\text{-connexe})$$

et de plus

$$\pi_n(Z,z) \longrightarrow \pi_n(X,z) \qquad \text{est nul}$$

(car $\pi_n(X,z) \longrightarrow \pi_n(Y,y)$ `[est un]` isomorphisme donc `[un]` monomorphisme), donc

$$\pi_{n+1}(Y,y) \longrightarrow \pi_n(Z,z) \qquad \texttt{[est un]}\ \text{épimorphisme.}$$

Comme Z est $(n-1)$-connexe (donc aussi 1-connexe), on a

$$(*) \qquad H^i(Z,G) = 0\ , \qquad \text{si } 0 < i < n\ ,$$

pour G localement constant (donc constant) sur Z.

---

116. si X, Y 1-connexes, c'est vraiment prouvé, sinon, il y a des vérifications à faire (qui ont l'air soritales …).

117. **NB** Même si X, Y 1-connexes, il n'est pas clair que $\mathrm{c}_n \Longrightarrow \mathrm{b}_n$, car il faut montrer que $\pi_n(Z) \longrightarrow \pi_n(X)$ est nul, et on trouve seulement que le composé avec $\pi_n(X) \longrightarrow H_n(X)$ est nul. Je présume qu'il y a des contre-exemples.

Commentary · Exposition [ms. 179]

Three conditions and the extra degree

The corrected theorem separates fibre connectivity, homotopy isomorphisms, and cohomological isomorphisms. This separation is the main conceptual gain of the first section. In condition c_n the printed text has omitted the range of i; the intended range is $0 \le i \le n$, as both proof and notation require. We use singular local coefficients on classifying spaces, or on CW replacements when discussing spaces. A statement about arbitrary topological spaces with their ordinary sheaf cohomology would require additional hypotheses and is not supplied by the argument here.

For a homotopy fibre Z over $y = f(x)$, the exact homotopy sequence gives

$$\pi_{n+1}(X,x) \longrightarrow \pi_{n+1}(Y,y) \longrightarrow \pi_n(Z,z) \longrightarrow \pi_n(X,x) \longrightarrow \pi_n(Y,y).$$

Together with the preceding terms, this proves

$$a_n \iff b_n \text{ and } \pi_{n+1}(X,x) \twoheadrightarrow \pi_{n+1}(Y,y)$$

for one basepoint in every component, with nonempty connected fibres understood. In contemporary terminology a_n says that f is $(n+1)$-connected when a k-connected map is defined to have $(k-1)$-connected fibres. Authors also use a convention shifted by one; specifying the fibres avoids that ambiguity.

The map $* \to K(A, n+1)$, for $A \neq 0$, satisfies b_n but not a_n, since its homotopy fibre is $K(A,n)$. At the other end, $S^n \to *$ has $(n-1)$-connected fibre, but fails c_n: with $M = \mathbf{Z}$, its degree-n cohomology map is $0 \to \mathbf{Z}$. These examples prove strictness of the first and last arrows in the chain. The middle arrow is strict for $n = 3$ by the preceding example, but it is an equivalence for $n = 2$. Thus the conjecture that *each* implication is strict for every $n \ge 2$ is false. This does not affect the implication chain itself.

The lower boundary in the fibre statement matters. A bijection on components and isomorphisms on π_i through degree n yield only $(n-1)$-connected fibres; the remaining group $\pi_n Z$ can be the quotient of $\pi_{n+1}Y$ by the image of $\pi_{n+1}X$. It is precisely this group that enters the next low-degree exact sequence. The proof succeeds by showing that its restriction map from the cohomology of X is zero, rather than by claiming that the group itself vanishes.

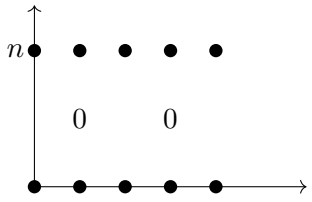
ms. page 181

Écrivons la suite spectrale de Leray pour $f : X \longrightarrow Y$ et $f^*F = F_X$:

$$H^*(X, f^*F) \Longleftarrow E_2^{pq} = H^p(Y, H^q(Z, f^*F)),$$

où '$H^q(Z)$' doit être vu comme un système local sur Y. Donc par $(*)$ on a

$E_2^{pq} = 0$ si $0 < q < n$,



,

donc on trouve la suite exacte

$$0 \longrightarrow H^n(Y,F) \longrightarrow H^n(X,F_X) \xrightarrow{\alpha} H^0(Y, \underbrace{H^n(Z, \overbrace{F_Z}^{M_Z})}_{= \mathrm{Hom}(\pi_n(Z,z),M) \text{ par [1e] th. de Hurewicz}}) \longrightarrow H^{n+1}(Y,F).$$

ed. 121 Ici je fixe un point x de X, Z désigne la fibre de f en $y = f(x)$, et M celle de F en y, $M = F_y$. Le groupe $\pi_1(Y)$ y opère, d'ailleurs $\pi \overset{\text{déf}}{=} \pi_1(X,x) \simeq \pi_1(Y,y)$ aussi. Le $H^0(Y, H^n(Z, F_Z))$ peut s'interpréter, sauf erreur, comme

$$\mathrm{Hom}_\pi(\pi_n(Z,x), M),$$

en tenant compte de l'opération de $\pi = \pi_1(X, x)$ sur $\pi_n(Z, x)$ également. (À vérifier.) Il faut interpréter la flèche

$$H^n(X, F_X) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z, x), M),$$

via l'application composée [où $z = x$, vu comme point de Z]

$$(*) \qquad \pi_n(Z, z) \longrightarrow \pi_n(X, x) \longrightarrow H_n(X, \mathbf{Z}).$$

Si F est *constant* sur Y (p. ex. si Y [est] 1-connexe), donc de valeur M, on a les flèches canoniques

$$(*') \qquad H^n(X, F_X) = H^n(X, M) \longrightarrow \mathrm{Hom}(H_n(X, \mathbf{Z}), M) \longrightarrow \mathrm{Hom}(\pi_n(Z, z), M)(^{118})$$

[lire : Hom_π],

et sûrement (en dehors de toute hypothèse de type b_n ou autre) l'application canonique directe 'de localisation sur Y', suivi de …, ms. page 182

$$(**) \qquad H^n(X, F_X) \longrightarrow H^0(Y, H^n(Z, F_Z)) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z), M)$$

n'est autre que la composée des applications canoniques ci-dessus. Dans le cas actuel où on suppose b_n, la deuxième flèche dans $(**)$ est [un] isomorphisme, donc la composée 'n'est autre' que la flèche de localisation elle-même. D'autre part $\pi_n(Z, z) \longrightarrow \pi_n(X, x)$ dans $(*)$ est nul, donc aussi $\pi_n(Z, z) \longrightarrow H_n(X, \mathbf{Z})$, donc aussi la flèche composée $(*')$ qui s'en déduit. Revenons alors à la suite exacte de Leray (page précédente), on trouve que $H^n(Y, F) \xrightarrow{\sim} H^n(X, F_X)$.

Commentary · Exposition [ms. 181]

The exact sequence behind the cohomological implication

The use of a path replacement now lets Grothendieck import the classical fibration argument into his categorical model. Assume b_n, reduce to a connected component, and replace f by a fibration with fibre Z. Then Z is $(n-1)$-connected, and the homotopy sequence also says that $\pi_n Z \to \pi_n X$ is zero. For an abelian local system F on Y with fibre M at y, the restriction $F|_Z$ is constant. The cohomological Serre–Leray spectral sequence has

$$E_2^{p,q} = H^p(Y; \mathcal{H}^q(Z; M)) \quad \Longrightarrow \quad H^{p+q}(X; f^*F).$$

Here $\mathcal{H}^q(Z; M)$ includes both the transport of the fibre and that of M; replacing it by an untwisted constant group would lose the content of the argument. The historical spectral sequence is Serre (1951, Chapter II, §8, Proposition 6); for the indexing and filtration in the untwisted presentation see Hatcher (2004, Theorem 1.14).

Since $H^q(Z; M) = 0$ for $0 < q < n$, the only contributions in total degrees at most n are the bottom row and $E_2^{0,n}$. No differential reaches the bottom row before d_{n+1} from $(0, n)$ to $(n+1, 0)$. Reading the filtration therefore gives the natural exact sequence

$$\begin{aligned} 0 \longrightarrow H^n(Y; F) \longrightarrow H^n(X; f^*F) \xrightarrow{\alpha} \mathrm{Hom}_{\mathbf{Z}[\pi]}(\pi_n Z, M) \\ \longrightarrow H^{n+1}(Y; F) \longrightarrow H^{n+1}(X; f^*F), \end{aligned}$$

where $\pi = \pi_1 Y = \pi_1 X$. Only exactness through the penultimate group is needed. The identification of $E_2^{0,n}$ uses the natural Hurewicz isomorphism $\pi_n Z \cong H_n Z$ and the universal coefficient theorem, followed by taking monodromy invariants. The fibre is simply connected, so no ambiguity from inner automorphisms survives in its degree-n group. The map α is restriction to a fibre, viewed as an invariant homomorphism.

There is an efficient chain-level way to finish the twisted case. If T is connected with fundamental group π, a local system G with fibre M is represented by a left $R = \mathbf{Z}[\pi]$-module. A cocycle representing $c \in H^n(T; G)$ is an R-linear map

$$z : C_n(\widetilde{T}) \longrightarrow M, \qquad z\partial = 0.$$

118. grâce à la composition avec $(*)$.

Evaluation on cycles gives a homomorphism $H_n(\widetilde{T}) \to M$; changing z by a coboundary has zero effect on cycles. Composing with the equivariant Hurewicz homomorphism produces

$$\varepsilon_G : H^n(T; G) \longrightarrow \mathrm{Hom}_R(\pi_n(T, t), M), \qquad n \geq 2.$$

This is the desired spherical evaluation. Its additivity follows from the homomorphism property of Hurewicz, or from the pinch map of the sphere; its equivariance follows directly from R-linearity. Naturality follows by lifting a pointed map to universal covers and applying the chain map. The local-coefficient chain model is described in Hatcher (2002, §3H, pp. 327–334); the verification above supplies the particular application the manuscript leaves open.

For $j : Z \to X$, choose the lift $Z \to \tilde{X}$ corresponding to the marked point. Because Z is simply connected and $(n-1)$-connected, naturality of evaluation and Hurewicz identify α with

$$H^n(X; f^*F) \xrightarrow{\varepsilon_{f^*F}} \mathrm{Hom}_R(\pi_n X, M) \xrightarrow{j^*} \mathrm{Hom}_R(\pi_n Z, M).$$

The last map is zero because $j_* : \pi_n Z \to \pi_n X$ is zero. Hence $\alpha = 0$, and exactness proves that $H^n(Y; F) \to H^n(X; f^*F)$ is an isomorphism. In lower degrees it is already an isomorphism by the zero rows of the spectral sequence. Degrees 0 and nonabelian 1 are controlled by the fundamental groupoid. This completes $b_n \Rightarrow c_n$ for all the coefficients actually stated, without an assumption of simple connectivity.

Comment récupérer le cas où F n'est *pas* constant? Dans le cas $n = 2$, cela marchait, en passant aux revêtements universels $\tilde{X}$, $\tilde{Y}$ de X, Y (par rapport aux points x, y choisis), et en écrivant les suites spectrales

$$\begin{array}{ccccc} H^*(Y, F) & \Longleftarrow & E_2^{pq} & = & H^p(\pi, H^q(\tilde{Y}, F_{\tilde{Y}})) \\ \downarrow & & \downarrow & & \\ H^*(X, F_X) & \Longleftarrow & E_2'^{pq} & = & H^p(\pi, H^q(\tilde{X}, F_{\tilde{X}})). \end{array}$$

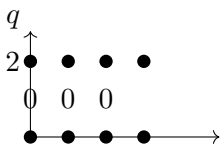
L'hypothèse b_n est conservée en passant de $f : X \longrightarrow Y$ à $\tilde{f} = \tilde{f}_x : \tilde{X} \longrightarrow \tilde{Y}$, donc on sait
ed. 122 que

ms. page 183

$$(*) \qquad E_2^{pq} \xrightarrow{\sim} E_2'^{pq} \qquad \texttt{[est un]} \text{ isomorphisme si } q \leq n.$$

Dans le cas $n = 2$, utilisant que

$$E_2^{p1} = E_2'^{p1} = 0 \qquad \text{car } H^1(\tilde{X}) = H^1(\tilde{Y}) = 0,$$



,

on trouve deux suites exactes

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^2(\pi, H^0(\tilde{Y})) & \longrightarrow & H^2(Y) & \longrightarrow & H^0(\pi, H^2(\tilde{Y})) & \longrightarrow & H^3(\pi, H^0(\tilde{Y})) \\ & & \downarrow \text{iso} & & \downarrow & & \downarrow \text{iso} & & \downarrow \text{iso} \\ 0 & \longrightarrow & H^2(\pi, H^0(\tilde{X})) & \longrightarrow & H^2(X) & \longrightarrow & H^0(\pi, H^2(\tilde{X})) & \longrightarrow & H^3(\pi, H^0(\tilde{X})), \end{array}$$

et un homomorphisme de l'une dans l'autre, qui permet de conclure par le lemme des 5 que $H^2(Y, F) \longrightarrow H^2(X, F_X)$ est `[un]` isomorphisme. Mais pour n quelconque, la relation $(*)$ sur les termes E_2 (pour $q \leq n$) ne me semble pas impliquer la relation similaire pour les E_n supérieures, et de là pour les termes E_∞, ce qui permettrait de conclure.

Je reviens donc à la suite exacte p. 181, avec F faisceau *tordu.* J'ai envie de trouver une application canonique

$$H^n(X, G) \longrightarrow \overbrace{\mathrm{Hom}_\pi(\pi_n(X, x), G_x)}^{\text{Homs } \textit{additifs} \text{ bien sûr, i.e. de } \mathbf{Z}[\pi]\text{-modules}},$$

où G est un faisceau abélien localement constant quelconque sur X, et $\pi = \pi_1(X,x)$. Si j'interprète $\pi_n(X,x)$ par des classes d'homotopie d'applications ponctuées ms. page 184

$$(S^n, x_0) \xrightarrow{\varphi} (X,x)$$

j'ai bien, pour une telle application φ,

$$\varphi^* : H^n(X,G) \longrightarrow H^n(S^n, \overbrace{\varphi^*G}^{\simeq (G_x)_{S^n}}) \simeq G_x$$

(tenant compte que S^n est 1-connexe, et que sa n-cohomologie à valeurs dans un faisceau constant M est M.) Sûrement l'application

$$H^n(X,G) \longrightarrow G_x$$

ed. 123 ainsi définie ne dépend que de la classe d'homotopie ponctuée de φ, d'où

$$\pi_n(X,x) \times H^n(X,G) \longrightarrow G_x,$$

et il faut de plus prouver a) que c'est compatible avec les opérations de $\pi = \pi_1(X,x)$, et b) que c'est additif en $\pi_n(X,x)$, pour avoir l'application cherchée

$$H^n(X,G) \longrightarrow \mathrm{Hom}_\pi(\pi_n(X,x), G_x).$$

Si j'admets ceci, il n'y a qu'un pas pour admettre aussi que la flèche qui nous intéresse vraiment (dans la suite exacte p. 181) ms. page 185

$$H^n(X,F_X) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z,z), \underbrace{M}_{=F_y})$$

n'est autre que la composée de la flèche précédente (page précédente) pour $G = F_X$, avec

$$\mathrm{Hom}_\pi(\pi_n(X,x), M) \longrightarrow \mathrm{Hom}_\pi(\pi_n(Z,x), M)$$

déduite de

$$\pi_n(Z,x) \longrightarrow \pi_n(X,x).$$

Or cette flèche est nulle, par l'hypothèse b_n. Donc la flèche α dans la suite exacte page 181 est nulle, donc $H^n(Y,F) \longrightarrow H^n(X,F_X)$ est `[un]` isomorphisme, q.e.d. (mais il y a pas mal de travail de vérification à faire …).

Commentary · Development (commentators') [ms. 183]

Completing the universal-cover comparison

Grothendieck briefly abandons the universal-cover method because he does not see how its comparison of E_2 rows controls later pages. The method can be completed directly at the level of complexes. It is useful to record the argument, since the needed range is finite and one should not invoke an unrestricted comparison theorem without checking its boundary. We use the ordinary truncation of complexes and the local-coefficient complex in Hatcher (2002, §3H). We did not find this particular completion of p. 183 in the consulted literature; its ingredients are standard homological algebra.

Finite-range descent lemma. [ours] Let π be a group and let $u : C \to D$ be a map of nonnegative cochain complexes of $\mathbf{Z}[\pi]$-modules which induces isomorphisms on cohomology in degrees $0 \le q \le n$. Then

$$H^k(R\Gamma(\pi, C)) \longrightarrow H^k(R\Gamma(\pi, D))$$

is an isomorphism for $0 \le k \le n$.

Proof. The smart truncation $\tau_{\le n}C$ agrees with C below n, has $\ker(d : C^n \to C^{n+1})$ in degree n, and is zero above n. The natural inclusion fits into the truncation triangle

$$\tau_{\le n}C \longrightarrow C \longrightarrow \tau_{\ge n+1}C.$$

Derived invariants carry a complex whose cohomology vanishes below $n+1$ to one with the same lower bound. Indeed, choose an injective resolution starting in degree $n+1$, or inspect the first-quadrant spectral sequence with terms $H^p(\pi, H^q(-))$: all its potentially nonzero terms have $p \ge 0$ and $q \ge n+1$. Applying $R\Gamma$ to the triangle and taking its long exact cohomology sequence shows that

$$H^k(R\Gamma(\pi, \tau_{\le n}C)) \xrightarrow{\sim} H^k(R\Gamma(\pi, C)), \qquad k \le n,$$

because the adjacent terms of degrees $k-1$ and k for $\tau_{\ge n+1}C$ both vanish. The same holds for D. Finally $\tau_{\le n}u$ is a quasi-isomorphism, and derived invariants preserve quasi-isomorphisms. The two truncation comparisons prove the assertion. □

Apply the lemma to the equivariant cochain complexes of the universal covers with constant fibre M and its given π-action. Derived invariants compute the cohomology with the corresponding local coefficients on the quotient: one may use a free resolution of $\mathbf{Z}$ over $\mathbf{Z}[\pi]$ and the usual total double complex. Indeed, each group of covering chains is free over $\mathbf{Z}[\pi]$, so each cochain term is a product of coinduced modules and is acyclic for the invariants functor. Its ordinary invariant complex is the local-coefficient cochain complex. The simply connected case of $b_n \Rightarrow c_n$ gives the required cohomology isomorphisms of the cover complexes through degree n, and the lemma gives the desired twisted comparison on X and Y. Thus the alternative proposed on p. 182 does work. The spherical-evaluation proof remains closer to the geometric reason the edge map vanishes.

$\mathrm{c}_n \Longrightarrow \mathrm{a}_{n-1}$. On peut procéder par récurrence sur n. (NB L'enoncé a un sens même pour $n = 1$, où il est prouvé, et même $\mathrm{c}_1 \Longrightarrow \mathrm{b}_1$, et $\mathrm{b}_1 \Longrightarrow \mathrm{a}_0$, plus généralement $\mathrm{b}_n \Longrightarrow \mathrm{a}_{n-1}$, immédiat par `[la]` suite exacte d'homotopie.) Donc on sait déjà, en plus de l'hypothèse c_n, qu'on a a_{n-2}, i.e. Z est $(n-2)$-connexe. Il faut prouver qu'il est même $(n-1)$-connexe, i.e. $\pi_{n-1}(Z) = 0$. Le cas $n = 2$ demande peut-être une attention particulière, car on ne sait pas même encore que Z est 1-connexe. Mais on sait que c_2 implique c_1, qui équivaut `[?]` à b_1, donc on a ms. page 186

$$\pi_1(X, x) \longrightarrow \pi_1(Y, y) \qquad \text{isomorphisme,}$$

donc

$$\pi_1(Z, x) \longrightarrow \pi_1(X, x) \qquad \texttt{[est]}\ \textit{nul,}$$

donc

$$\pi_2(Y, y) \longrightarrow \pi_1(Z, x) \qquad \texttt{[est un]}\ \text{épimorphisme,}$$

ce qui implique tout au moins que $\pi_1(Z, x)$ est *abélien.* Il faut le voir comme un π-module, et prouver sa nullité comme tel (moyennant c_2).

Commentary · Exposition [ms. 185]

Recovering the first missing homotopy group

For the converse direction Grothendieck changes the use of the spectral sequence: instead of annihilating the edge map, he chooses coefficients in which the first possible fibre group can test itself. This is why all local systems must be admitted. A single constant coefficient group need not detect a nontrivial module with monodromy.

Assume c_n. The degree-0 and degree-1 conditions give $\pi_0 X = \pi_0 Y$ and identify $\pi_1 X$ with $\pi_1 Y = \pi$. Induction gives that the homotopy fibre Z is $(n-2)$-connected. For $n \ge 3$ its first possible nonzero group $A = \pi_{n-1}Z$ is abelian and Hurewicz identifies it with $H_{n-1}Z$. For $n = 2$ one must first justify the same assertion differently: exactness gives a surjection $\pi_2 Y \twoheadrightarrow \pi_1 Z$, since $\pi_1 X \to \pi_1 Y$ is an isomorphism. Thus $\pi_1 Z$ is abelian, and its abelianization map to $H_1 Z$ is already an isomorphism. In both cases transport makes A a π-module; the possible ambiguity in degree one is inner conjugation in an abelian group and therefore vanishes.

For the local system determined by a π-module M, the first nonzero fibre row gives

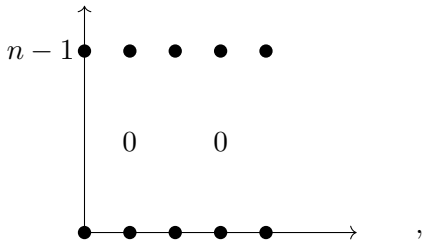
$$0 \longrightarrow H^{n-1}(Y;M) \longrightarrow H^{n-1}(X;f^*M) \longrightarrow \mathrm{Hom}_{\mathbf{Z}[\pi]}(A,M) \longrightarrow H^n(Y;M) \longrightarrow H^n(X;f^*M).$$

For $n=2$, the degree-one groups in this display use abelian coefficients, a case included in c_n. The first comparison is surjective and the last is injective. Exactness therefore forces $\mathrm{Hom}_{\mathbf{Z}[\pi]}(A,M)=0$. Every π-module occurs as a coefficient system on Y, so take $M=A$. The identity of A must vanish, and hence $A=0$. This proves $c_n \Rightarrow a_{n-1}$, with the endpoint case checked rather than hidden under an appeal to Hurewicz.

This proof uses less than full c_n at its final step: after the lower-degree induction, surjectivity in degree $n-1$ and injectivity in degree n suffice. The full hypothesis is a convenient uniform formulation, not an optimal list of separate tests. The source has found the correct mechanism; the qualifications about coefficients and the degree-one fibre group complete its sketch.

ed. 124 Passons au cas général. On a pour F localement constant abélien sur Y

$$H^*(X,F_X) \Longleftarrow \underbrace{E_2^{pq} = H^p(Y,H^q(Z,F_Z))}_{\substack{=0 \text{ si } 0<q<n-1 \\ \text{car } Z \text{ est } (n-2)\text{-connexe}}},$$



D'où

$$0 \longrightarrow H^{n-1}(Y,F) \longrightarrow H^{n-1}(X,F_X) \longrightarrow \underbrace{H^0(Y,H^{n-1}(Z,F_Z))}_{\substack{\mathrm{Hom}_\pi(\pi_{n-1}(Z,x),F_x) \\ \text{par Hurewicz}}} \longrightarrow H^n(Y,F) \longrightarrow H^n(X,F_X).$$

L'hypothèse c_n implique qu'on a ms. page 187

$$\mathrm{Hom}_\pi(\pi_{n-1}(Z,x), \underbrace{M}_{=F_x}) = 0$$

et ceci *quel que soit le π-module M,* puisque $\pi = \pi_1(X,x) \xrightarrow{\sim} \pi_1(Y,y)$. Cela implique (en prenant p. ex. $M = \pi_{n-1}(Z,x)$!) que $\pi_{n-1}(Z,x)=0$. (Mais en faisant attention, dans le cas $n=2$, qu'on sait déjà que $\pi_1(Z,x)$ est abélien).

Donc le théorème rectifie page 178 a bien l'air vrai. Il est un peu triste, puisque dans chacune des implications

$$a_n \implies b_n \implies c_n \implies a_{n-1}$$

on perd un chouia, et en fin de compte on sort avec la magnifique implication

$$a_n \implies a_{n-1}.$$

De toutes façons, il est ± clair que le "bon" localiseur W_n, celui qui doit définir les "n-types d'homotopie" comme objets de Hot_{W_n}, est défini par la condition a_n.

ms. page 188

2 Critère de W_n-asphéricité.

Je vais essayer de faire mieux quand Y est réduit à un point. Cette fois, je
ed. 125 veux être strict, et donner (si je peux) des démonstrations complètes dans le cadre où je me place à présent, celui du modéliseur Cat, et en n'utilisant que ce que je sais y démontrer directement à présent, sans me référer à la théorie semi-simpliciale ou topologique.

Commentary · Development (commentators') [ms. 187]

Degree two and the localizer correction

Two brief remarks in the manuscript deserve a more precise disposition. The suggestion of a degree-two counterexample is stronger than the examples support, and the choice of the fibre condition as the localizer confuses a useful generating class with its saturation. Both can be settled without changing the successful implication arguments.

The degree-two converse. [ours] For maps of classifying spaces, the condition c_2 implies b_2. Consequently b_2 and c_2 are equivalent without a simple-connectivity hypothesis.

Proof. The degree-one argument already identifies fundamental groupoids, so work on one component and write $R = \mathbf{Z}[\pi]$. Let

$$I = \mathrm{Hom}_{\mathbf{Z}}(R, \mathbf{Q}/\mathbf{Z}), \qquad (r\varphi)(s) = \varphi(sr).$$

This is a left R-module and hence a permitted local system on Y. Evaluation at $1 \in R$ gives a natural adjunction isomorphism

$$\mathrm{Hom}_R(C_*(\widetilde{Y}), I) \cong \mathrm{Hom}_{\mathbf{Z}}(C_*(\widetilde{Y}), \mathbf{Q}/\mathbf{Z}),$$

and likewise for X. Its inverse sends a homomorphism λ to the map $c \mapsto (r \mapsto \lambda(rc))$. Since $\mathbf{Q}/\mathbf{Z}$ is injective as an abelian group, evaluation on homology gives

$$H^2(Y; I) \cong \mathrm{Hom}_{\mathbf{Z}}(H_2\widetilde{Y}, \mathbf{Q}/\mathbf{Z}).$$

The isomorphism is natural in the lifted map $\widetilde{f}$. Hypothesis c_2 therefore says that the dual of $H_2(\widetilde{f})$ is an isomorphism. Dualization into $\mathbf{Q}/\mathbf{Z}$ is exact and detects nonzero abelian groups: a nonzero element generates a cyclic subgroup with a nonzero homomorphism to $\mathbf{Q}/\mathbf{Z}$, and injectivity extends this homomorphism to the whole group. Applying this observation to kernel and cokernel shows that $H_2(\widetilde{f})$ itself is an isomorphism. The universal covers are simply connected, so the natural Hurewicz isomorphisms $\pi_2\widetilde{X} \cong H_2\widetilde{X}$ and $\pi_2\widetilde{Y} \cong H_2\widetilde{Y}$ prove that $\pi_2 f$ is an isomorphism. □

The algebraic ingredients are the standard injective-cogenerator argument, checked in The Stacks Project Authors (2026f, Tags 01D8 and 08WE), together with Hatcher (2002, Theorem 4.32). We did not find a source addressing footnote 112 specifically; this is our application of those familiar tools to its question. It is not a new form of the Hurewicz theorem. The same argument shows that the degree-j part of c_n detects the map on H_j of universal covers. For $j = 2$ these homology groups are exactly π_2; for $j = 3$ the Hopf class in the earlier example explains why the conclusion changes.

Now consider the last sentence of p. 187. If $A \neq 0$, the maps

$$* \xrightarrow{s} K(A, n+1) \xrightarrow{r} *$$

satisfy $rs = 1$. The fibre of r is n-connected and the fibre of rs is contractible, but the fibre of s is $K(A, n)$, which is not n-connected. Thus the class defined by a_n fails two-out-of-three. The class defined by b_n, equivalently the maps made into equivalences by nth Postnikov truncation, is the correct weak-equivalence class for n-types. The modern truncation adjunction is given in Lurie (2009, Proposition 5.5.6.18).

There is a charitable precise reading of Grothendieck's intuition.

Saturation of the fibre class. [ours] In spaces, a_n *generates* the same localization as b_n.

Proof. Every Postnikov map $X \to P_nX$ has n-connected fibres: it is an isomorphism on π_i through n, and the target has zero π_{n+1}. Given a b_n-map $f : X \to Y$, its natural square with $P_nf : P_nX \to P_nY$ has vertical maps in a_n, and its lower map is a weak homotopy equivalence, hence also lies in a_n. Two-out-of-three puts f in the saturation of a_n. Conversely, b_n is saturated and contains a_n. This proves the equality. □

The proof is a direct application of the standard truncation adjunction just cited; the formulation addresses the specific localizer sentence, without a priority claim. It leaves the literal definition on p. 187 marked as false. For $X \to *$, the two conditions agree already before saturation, so the next theorem needs no change of indexing.

Théorème 2 (?). *Soit X dans* Cat, $n \in \mathbf{N}$ *un entier* ≥ 1. *Alors les conditions suivantes sont équivalentes :*

a_n) X est 0-connexe, et si x est dans X, $\pi_i(X,x) = 0$ pour $i \in [1,n]$ (condition qui ne dépend pas du choix de x), i.e. les $\Omega^i(X,x)$ pour $i \in [1,n]$ sont 0-connexes.

b_n) Pour tout groupe G, $G \longrightarrow H^0(X, G_X)$ est un isomorphisme, et $H^1(X, G_X) = 0$. De plus, si G est abélien, on a aussi $H^i(X, G_X) = 0$, pour $i \in [1,n]$.

Démonstration. Pour $n = 1$ cette équivalence est ± tautologique et a été rappelée (sous une forme légèrement différente) au début (p. 71). On procède par récurrence. On peut donc supposer que $n \geq 2$, que X satisfait a_{n-1}, b_{n-1}, et prouver qu'alors a_n, b_n sont équivalentes, i.e. $\pi_n(X,x) = 0$ équivaut à $H^n(X,M) = 0$ pour tout groupe abélien M.

On utilise la W_∞-fibration (parfaite) ms. page 189

$$\begin{array}{ccc} E = \underline{\mathrm{Ch}}(X; x, -) & & E \ W_\infty\text{-asphérique} \\ \big\downarrow & & \\ X & & \end{array}$$

de fibre Z en x égale à $\Omega(X,x)$.

Commentary · Context [ms. 188]

Why the path fibration is the right final test

Grothendieck now asks the categorical path construction to reproduce an argument whose classical power came from the contractible total space of based paths. Serre's thesis developed precisely this exchange between the homology of a space, that of its loop space, and their homotopy groups; see Serre (1951, Chapter IV, §5, Proposition 7). The point of repeating the argument in Cat is not a new calculation of the groups. It is to show that the categorical paths already carry enough geometry to recognize the ordinary weak homotopy type, before appealing to a comparison with topological paths.

For $E = \underline{\mathrm{Ch}}(X; x, -) \to X$, the total category is W_∞-aspherical and the fibre is $Z = \Omega(X,x)$. Under the induction hypothesis, both base and fibre have long vanishing ranges. The spectral sequence then has only its bottom row and one possible first fibre row. The vanishing of the total cohomology forces those two rows to communicate by transgression. This is the reason the proof over a point is sharper than the preceding relative theorem: there is no independent cohomology of a nontrivial target that can absorb the first obstruction.

The condition in degree zero should be read literally for every group. Taking a nontrivial group G, the diagonal map $G \to G^{\pi_0 X}$ is bijective only when $\pi_0 X$ has one element. It excludes the empty category as well. Once X is connected, the condition $H^1(X, G_X) = *$ for every group G forces $\pi_1 X = 1$: choose $G = \pi_1 X$ and the conjugacy class of its identity representation. This is why abelian coefficients alone would not prove the initial case; a perfect nontrivial fundamental group could otherwise survive.

Supposons a_n) pour X, donc a_{n-1}) pour $\Omega(X,x)$. Par hypothèse de récurrence, on a donc

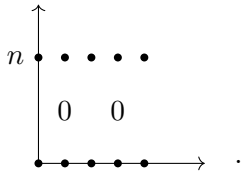
$$H^i(Z, M) = 0 \qquad \text{pour } i \in [1, n-1],$$

M groupe abélien. Donc dans la suite spectrale de Leray,

$$H^*(E, M) \Longleftarrow E_2^{pq} = H^p(X, H^q(Z, M))$$

on a

$$E_2^{pq} = 0 \qquad \text{si } q \in [1, n-1],$$



ed. 126 On a donc la suite exacte

$$0 \longrightarrow H^n(X, M) \longrightarrow H^n(E, M)$$

et comme E est W_∞-asphérique, on a $H^n(E, M) = 0$, donc $H^n(X, M) = 0$.

Inversement, supposons b_n), prouvons a_n). On sait déjà qu'on a a_{n-1}), donc a_{n-2} pour $\Omega(X, x)$, ce qui implique

$$H^i(Z, M) = 0 \qquad \text{pour } i \in [1, n-2],$$

donc

$$E_2^{pq} = 0 \qquad \text{si } q \in [1, n-2].$$

On a donc la suite exacte

ms. page 190

$$0 \longrightarrow \underbrace{H^{n-1}(X, M)}_{=0} \longrightarrow \underbrace{H^{n-1}(E, M)}_{=0} \longrightarrow H^0(X, H^{n-1}(Z, M)) \longrightarrow \underbrace{H^n(X, M)}_{=0} \longrightarrow H^n(E, M),$$

donc

$$H^0(X, \quad \underbrace{H^{n-1}(Z, M)}_{\substack{\text{système local} \\ \textit{constant} \text{ sur } X, \\ \text{car } X \text{ est 1-connexe}}} \quad) = 0,$$

donc

$$H^{n-1}(Z, M) = 0,$$

ce qui, par hypothèse de récurrence, implique $\pi_{n-1}(Z, z) = 0$, donc $\pi_n(X, x) = 0$, q.e.d.

Corollaire. *Pour que X soit W_∞-asphérique, i.e. pour qu'on ait $G \xrightarrow{\sim} H^0(X, G)$, $H^1(X, G) = 0$* `[pour]` *tout groupe* `[G]`*, et $H^i(X, G) = 0$ si $i \geq 2$ pour G abélien, il faut et il suffit que* 1°) *X soit 0-connexe, et (x étant un point de X)* 2°) *les $\Omega^i(X, x)$* `[soient]` *0-connexes pour $i \geq 1$.*

ms. page 191

Commentary · Exposition [ms. 190]

The loop criterion, including its degree-two endpoint

Theorem 2 is proved by the displayed induction, with one small verification at its lower endpoint. Suppose first that X is n-connected. Then $Z = \Omega X$ is $(n-1)$-connected, so the induction hypothesis annihilates $H^q(Z; M)$ for $1 \leq q < n$. In the path-fibration spectral sequence this gives an injection

$$H^n(X; M) \hookrightarrow H^n(E; M) = 0.$$

The lower-degree groups have already vanished, and the degree-zero and torsor conditions follow from connectedness and simple connectivity. This proves the forward implication exactly in the form stated.

Conversely, assume the cohomological conditions through degree n. The induction hypothesis makes X $(n-1)$-connected and Z $(n-2)$-connected. For $n \geq 2$, X is simply connected, so the fibre-cohomology local systems on X are constant. The low-degree exact sequence for $E \to X$ gives

$$H^0(X; \mathcal{H}^{n-1}(Z; M)) = 0, \qquad \text{hence} \qquad H^{n-1}(Z; M) = 0$$

for every abelian M. If $n \geq 3$, Z is simply connected. Its nonabelian degree-one conditions and all its lower abelian cohomology conditions are already satisfied; the induction hypothesis applied to Z therefore gives $\pi_{n-1} Z = 0$.

For $n = 2$, one cannot immediately apply that same sentence: Z is only known to be connected, and the displayed computation gives vanishing of $H^1(Z; M)$ only for abelian groups M. The loop multiplication supplies the missing information. The two compositions of loops in the connected loop

space, concatenation of based loops and the multiplication induced from the loop-space multiplication, have the same unit and satisfy the interchange law. The Eckmann–Hilton argument makes $\pi_1 Z$ abelian. Thus

$$H^1(Z; M) = \mathrm{Hom}(\pi_1 Z, M),$$

and choosing $M = \pi_1 Z$ forces $\pi_1 Z = 0$. Equivalently, this is the familiar fact that $\pi_1 \Omega X = \pi_2 X$ is abelian. It closes the induction without requiring an additional theorem about nonabelian cohomology of Z.

The corollary follows by allowing every finite n. Connectedness of X together with connectedness of every iterated loop category says that all homotopy groups of BX vanish. Since a realization of a nerve has CW type, Whitehead's theorem makes BX weakly equivalent, indeed homotopy equivalent, to a point; see Hatcher (2002, Theorem 4.5). Here "aspherical" means weakly contractible, not merely the modern usage $\pi_i = 0$ for $i > 1$. For example $B\mathbf{Z} \simeq S^1$ has no higher homotopy groups, but its first loop category has components indexed by $\mathbf{Z}$, so it fails this criterion immediately.

The later minimal-localizer theorem identifies this test with Grothendieck's original axiomatic weak equivalences. Its local characterization also makes the loop method visible in the final proof: Cisinski (2004, Theorem 2.3.6, especially 2.3.6.3–2.3.6.4) passes from a weakly aspherical object to its loops and then uses vanishing of homotopy groups. Thus the categorical construction has reached its intended target. The distinction between different choices of categorical paths has become a question of canonical comparison, which is exactly the subject of the following pages.

3 Sur la philosophie des fibres homotopiques.

Le critère d'asphéricité (pour W_∞) que je viens d'obtenir, grâce aux foncteur $\underline{\mathrm{Ch}}(X)$ (ou au choix, $\underline{\mathrm{Ch}}_\infty(X)$) et ses dérivés $\Omega(X, x)$ et les itérés, enlève tout doute (s'il pouvait en rester encore) que les 'espaces de chemins' que j'ai construit (assez péniblement!) sont bel et bien 'les bons'. Je voudrais maintenant me rendre compte de façon la plus précise possible de leur 'canonicité', dans quelle mesure ces constructions sont indépendant d'un choix (p.ex. celui entre la théorie $\underline{\mathrm{Ch}}$, et $\underline{\mathrm{Ch}}_\infty$).

Tout revient, finalement, à des questions de factorisation d'une flèche quelconque

$$f : X \longrightarrow Y$$

ed. 127 de Cat en

$$X \xrightarrow{i_f} \bar{X} \xrightarrow{\bar{f}} Y,$$

avec

$$i_f \in W, \qquad \bar{f} \in W\text{-fib}$$

(le fait que $\bar{f}$ puisse même être choisie W_ω-parfait, ne me paraissant pas essentiel ici).

C'est bien de cela qu'il s'agit pour construire
un objet 'chemins dans X'

$$\underline{\mathrm{Ch}}(X),$$

ms. page 192

en factorisant l'application diagonale

$$X \xrightarrow{\mathrm{diag}_X} X \times X,$$

et aussi quand on construit $\Omega(X, x)$ comme fibre en x d'une W-fibration

$$E(X, x) \longrightarrow (X, x)$$

qui factorise l'inclusion

$$\{x\} \longrightarrow X$$

(donc $E(X, x)$ étant W-asphérique).

Commentary · Context [ms. 191]

From a model for paths to an intrinsic operation

The question of canonicity belongs to Grothendieck's original reason for introducing derivators. In *Pursuing Stacks*, §69, he recalls the inadequacy of a single derived category for some of the diagrammatic constructions arising from the SGA 5 trace formalism. He proposes retaining derived categories for all diagram shapes, with their variance in the indexing category. The two three-object shapes for spans and cospans already encode homotopy fibres and cofibres (Grothendieck, 1983, §69, printed pp. 1–3). The move on these pages, from a particular category of paths to a property of localized slices, is a local instance of that larger programme.

The letter to Thomason of 2 April 1991 makes the distinction especially explicit. Its first numbered discussion treats W as the essential homotopical datum and fibrations or cofibrations as additional means of proving existence and calculating operations; it then describes the intrinsic class of pullback-preserving W-fibrations (Grothendieck, 1991a, point 1, pp. 3–4). This is a retrospective clarification of the position being explored here, not evidence that every provisional axiom of the present pages has already been checked. In particular, the retract qualification discussed below remains necessary under the literal two-out-of-three assumptions. The calculation of paths serves the intrinsic operation by providing enough useful representatives on which the operation becomes ordinary pullback.

On peut être plus ou moins exigeant, et regarder, soit pour un 'but' Y (ou plutôt un S) fixé dans Cat, une factorisation fonctorielle

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fib}}_W S \hookrightarrow \mathrm{Cat}/S$$

du morphisme structural $X \longrightarrow S$ d'un objet, en

$$X \xrightarrow{i_X} C(X) \longrightarrow S,$$

i.e. on se donne le foncteur C, *plus*
le morphisme fonctoriel ms. page 193

$$i_C : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C$$

avec

$$(i_X : X \longrightarrow C(X)) \in W \qquad \forall\ X \text{ dans } \mathrm{Cat}/S,$$

soit être encore plus exigeant et faire varier en même temps la source et le but de $f : X \longrightarrow Y(= S)$, et demander un foncteur

$$\begin{array}{rcl} C : \underline{\mathrm{Fl}}(\mathrm{Cat}) & \longrightarrow & \mathrm{Cat} \\ f & \longmapsto & C(f), \end{array}$$

ed. 128 avec des morphismes fonctoriels en f

$$\underbrace{s(f)}_{=X} \xrightarrow{i_f} \underbrace{C(f)}_{=\bar{X}} \xrightarrow{p_f} \underbrace{b(f)}_{=Y}$$

(i, p des flèches dans $\underline{\mathrm{Hom}}(\underline{\mathrm{Fl}}(\mathrm{Cat}), \mathrm{Cat})$), avec

$$i_f \in W, \qquad p_f \in W\text{-fib}$$

pour toute flèche f dans Cat.

Commentary · Exposition [ms. 192]

A replacement must remember its comparison map

Grothendieck now asks what the lengthy construction of paths has bought him beyond a convenient calculation. The preceding connectivity criterion identifies its homotopical effect; it does not yet identify the construction independently of its presentation. The change of viewpoint is to regard a path category as a factorization of a structural map. A factorization of the diagonal gives paths, and a factorization of a point inclusion gives based paths. Once the second factor has the required base-change property, its ordinary fibres can serve as homotopy fibres. The same question can therefore be asked without mentioning intervals or zigzags.

For a fixed base put $\mathcal{A} = \mathrm{Cat}/S$, and let $\mathcal{F} \subseteq \mathcal{A}$ be the full subcategory of the permitted fibrations over S. The data are a functor $C : \mathcal{A} \to \mathcal{F}$ and a natural transformation $i : \mathrm{id}_{\mathcal{A}} \to jC$, where j is inclusion and every i_X belongs to W_S. The transformation is part of the data. An arbitrary choice of an equivalent object for every X would neither provide the arrows $C(f)$ nor specify which comparison is to be called canonical. Naturality supplies both controls at once. Indeed,

$$C(f)i_X = i_Y f$$

and two-out-of-three show that $f \in W_S$ if and only if $C(f) \in W_S$. Thus the homotopicality of C is a consequence of its coaugmentation, not an extra axiom.

There are two levels of functoriality in the text. A replacement on $\mathcal{A}$ controls variation of the object over one base. A replacement on the arrow category also controls maps between bases, although it need not commute with ordinary pullbacks. The latter distinction matters: a functorial factorization of arrows does not, merely by being functorial, identify $C(X \times_S T)$ with $C(X) \times_S T$. Derived base change will be obtained from a localization argument. It is not asserted to be an identity between these raw constructions.

This is the same elementary geometry that underlies Brown's mapping-path factorization. In a category of fibrant objects, a factorization of the diagonal of Y yields a factorization of $f : X \to Y$ by pulling that path object back along f. Brown requires no functorial choice of path objects for this argument (Brown, 1973, Part I, §1, pp. 420–422). Grothendieck deliberately asks for a more explicit choice here so that comparison maps can be written in the original category. The two approaches answer different parts of the same problem: existence of the derived construction and a convenient formula for computing it.

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Dans les deux points de vue, dans quel sens a-t-on *unicité* d'une telle construction?

Prenons d'abord le premier point de vue, factorisation des flèches à but fixé S, donc `[c'est une]` question dans Cat/S. J'ai oublié une propriété importante, mais elle résulte de ce qui a été dit : le foncteur

$$X \longmapsto C(X) : \qquad \begin{array}{ccc} \mathrm{Cat}/S & \longrightarrow & \mathrm{Cat}/S \\ & \searrow \quad \nearrow & \\ & \underline{\mathrm{Fib}}_W S & \end{array}$$

transforme `[les]` *flèches* `[de]` W *en itou*, comme on voit sur le diagramme commutatif

$$\begin{array}{ccc} X & \xrightarrow{i_X} & C(X) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle C(f)} \\ Y & \xrightarrow{i_Y} & C(Y), \end{array}$$

où i_X, i_Y sont dans W, donc

$$f \in W \qquad \Longleftrightarrow \qquad C(f) \in W.$$

Soit

$$(C', i')$$

ms. page 195

un deuxième foncteur, $\mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$,

muni d'un morphisme

$$i' : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C'$$

qui est dans W argument par argument, et tel que C' se factorise par $\underline{\mathrm{Fib}}_W S$. On peut comparer C et C' moyennant $C \circ C'$, car

$$\begin{array}{ccccc} & & C \circ C' & & \\ & {\scriptstyle C*i'} \nearrow & & \nwarrow {\scriptstyle i*C'} & \\ C = C \circ \mathrm{id} & & & & C' = \mathrm{id} \circ C', \end{array}$$

ed. 129 i.e. on a, pour tout X dans Cat/S,

$$\begin{array}{ccccc} & & C(C'(X)) & & \\ & {\scriptstyle C(i'_X)} \nearrow & & \nwarrow {\scriptstyle i_{C'(X)}} & \\ C(X) & & & & C'(X), \end{array}$$

le deux flèches étant dans W (celle de gauche parce que $C(W) \subseteq W$). Ce sont des flèches dans $\underline{\mathrm{Fib}}_W S$, et si on suppose que W satisfait W(4b), elles induisent des W-équivalences en chaque fibre. Cela montre donc déjà

qu'à isomorphisme (peut-être non-canonique) près, le foncteur composé défini par C, ms. page 196

$$X \longmapsto C(X) \longmapsto (s \longmapsto \mathrm{hot}_W(C(X)_s))$$

$$\begin{array}{ccccc} \mathrm{Cat}/S & \longrightarrow & \underline{\mathrm{Fib}}_W S & \longrightarrow & \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W), \\ & \searrow & & \nearrow & \\ & & {\scriptstyle \varphi_C} & & \end{array}$$

ne dépend pas du choix particulier de C, i.e. deux tels foncteurs 'fibres homotopiques' (correspondant à deux choix C, C') sont toujours isomorphes. Mais au lieu d'utiliser $C \circ C'$ pour définir un tel isomorphisme, on aurai pu utiliser $C' \circ C$, ou quelque autre procédé. On aimerait comprendre s'il y a unicité véritable, une caractérisation intrinsèque de ces isomorphismes $\varphi_C \xrightarrow{\sim} \varphi_{C'}$, indépendamment de telle construction ou de telle autre.

Je présume que $C \circ C'$ est lui aussi muni de

$$\mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C \circ C',$$

et même a priori de deux façons possible

ms. page 197

$$\begin{array}{ccccc} & & C(C'(X)) & & \\ & {\scriptstyle C(i'_X)} \nearrow & & \nwarrow {\scriptstyle i_{C'(X)}} & \\ C(X) & & & & C'(X) \\ & {\scriptstyle i_X} \nwarrow & & \nearrow {\scriptstyle i'_X} & \\ & & X, & & \end{array}$$

soit par $C(i'_X) \circ i_X$, soit par $i_{C(X)} \circ i'_X$, heureusement elles sont égales, car pour toute flèche $X \xrightarrow{i'} X'$ dans Cat/S, le carré

Commentary · Exposition [ms. 196]

Two replacements and one normalized comparison

The composite replacement is a useful device because it compares two constructions without asking either to map directly to the other. Write the two coaugmentations as $i : \mathrm{id} \to C$ and $i' : \mathrm{id} \to C'$. For every X the diagram gives

$$C(X) \xrightarrow{C(i'_X)} CC'(X) \xleftarrow{i_{C'(X)}} C'(X).$$

Both arrows are weak equivalences: the first because C preserves W_S, and the second because it is a component of i. Their passage to fibrewise equivalences uses the additional fibre axiom invoked in the text. This last step must be retained. A weak equivalence of total objects over a base need not induce weak equivalences of their ordinary fibres when the objects have not been suitably replaced.

There is already a precise answer to the question about the order CC' versus $C'C$. Let $q : \mathcal{A} \to \mathcal{A}[W_S^{-1}]$ be localization. The required comparison between the two localized endofunctors is

$$\theta_{C,C'}(X) = q(i'_X)q(i_X)^{-1}.$$

It is the unique comparison compatible with the specified maps from $q(X)$. Naturality of i, applied to i'_X, gives

$$C(i'_X)i_X = i_{C'(X)}i'_X.$$

After localization this proves

$$q(i_{C'(X)})^{-1}q(C(i'_X)) = q(i'_X)q(i_X)^{-1}.$$

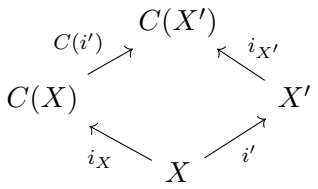
The common-refinement construction therefore gives exactly the normalized comparison. Reversing the order of the replacements gives the same answer. For three replacements the cocycle identity is literal:

$$\theta_{C',C''}\theta_{C,C'} = \theta_{C,C''}.$$

The apparent ambiguity disappears as soon as compatibility with the coaugmentations is made explicit.

When this calculation is transported to the category of fibres, one uses the full faithfulness of the localization comparison proved below. In that category the formula is not an instruction to apply an ordinary fibre functor to i_X: its source X need not be fibrant. Instead, it characterizes the comparison after passing through the equivalence of localized categories. This is the categorical content of the word "canonical" here. It does not say that the underlying functors admit no other automorphisms after their specified comparison maps have been forgotten.

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est commutatif (par fonctorialité de $i_X : X \longrightarrow C(X)$), et on applique ceci au cas de $X' = C'(X)$, $i' = i'_X$.

Cette remarque nous encourage à faire un saut dans l'abstraction, et d'introduire la catégorie ms. page 198

$$(*) \qquad \mathcal{C} = \mathcal{C}_S = \begin{cases} \text{catégorie des couples } (C, i), \text{ avec} \\ C : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S \text{ se factorisant par } \underline{\mathrm{Fib}}_W S, \\ i : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C \text{ tel que } i_X \in W \ \forall\ X \text{ dans } \mathrm{Cat}/S, \end{cases}$$

une flèche

$$(C, i) \longrightarrow (C', i')$$

étant une flèche

$$C \xrightarrow{u} C' \qquad (\text{dans } \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{Cat}/S))$$

rendant commutatif le carré `[lire : le diagramme]`

$$\begin{array}{ccc} & \mathrm{id}_{\mathrm{Cat}/S} & \\ {\scriptstyle i}\swarrow & & \searrow{\scriptstyle i'} \\ C & \xrightarrow{\quad u \quad} & C'. \end{array}$$

En d'autres termes, $\mathcal{C}$ est défini comme une sous-catégiorie pleine de

$$\mathrm{id}_{\mathrm{Cat}/S} \backslash \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{Cat}/S),$$

à savoir celle dont les objets (C, i) satisfont les deux conditions $(*)$ plus haut (impliquant l'une et l'autre le localiseur W).

Ces conditions impliquent ceci ms. page 199

a) $\forall\ X$ dans Cat/S, $(u_X : C(X) \longrightarrow C'(X)) \in W$, puisque

$$\begin{array}{ccc} & X & \\ {\scriptstyle i_X \in W}\swarrow & & \searrow{\scriptstyle i'_X \in W} \\ C(X) & \xrightarrow{\quad u_X \quad} & C'(X), \end{array}$$

ed. 131 et

b) si W satisfait W(4b), de sorte que $u_X \in W_S^{\mathrm{f}}$, alors ce définit un *isomorphisme* entre $\varphi_C(X)$ et $\varphi_{C'}(X)$

$$\begin{array}{ccc} s & \longmapsto & \mathrm{hot}_W(C(X)_s) \\ & & \downarrow u_{X,s} \\ s & \longmapsto & \mathrm{hot}_W(C'(X)_s). \end{array}$$

Mieux encore, dans ce cas, le foncteur composé

$$\mathcal{C} \times \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fib}}_W S \longrightarrow \mathrm{HOT}_W^{\mathrm{lc}}(S)$$

donne naissance à

$$\mathcal{C} \longrightarrow \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}_W^{\mathrm{lc}}(S))$$

qui transforme toute flèche de $\mathcal{C}$ en `[un]` *isomorphisme,* donc se factorise par le groupoïde fondamental de $\mathcal{C}$,

$$\Pi\mathcal{C} \longrightarrow \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}_W^{\mathrm{lc}}(S)).$$

Quant au foncteur envisagé tantôt ms. page 200

$$\begin{array}{ccc} \mathcal{C} & \longrightarrow & \underbrace{\underline{\mathrm{Hom}}\,(\mathrm{Cat}/S, \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W))}_{\simeq \underline{\mathrm{Hom}}((\mathrm{Cat}/S)\times S^\circ, \mathrm{Hot}_W)} \\ C & \longmapsto & ((X,s) \longmapsto \mathrm{hot}_W(C(X)_s)\,), \end{array}$$

il se déduit du précédent en composant avec

$$\begin{array}{ccc} \mathrm{HOT}_W(S) & \longrightarrow & \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W), \\ & \searrow \quad \nearrow & \\ & \underline{\mathrm{Hom}}(\Pi S^\circ, \mathrm{Hot}_W) & \end{array}$$

plus exactement, avec le foncteur qui s'en déduit

$$\underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}_W(S)) \longrightarrow \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \underline{\mathrm{Hom}}(S^\circ, \mathrm{Hot}_W)).$$

On voit poindre ici un principe plausible de canonicité d'isomorphismes, en ce qui concerne les foncteurs

$$C_W : \mathrm{Cat}/S \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S)$$

définis par les $C \in \mathrm{Ob}\,\mathcal{C}$, par la relation

$$\Pi\mathcal{C} \approx \text{ catégorie ponctuelle, i.e. } \mathcal{C} \text{ est 1-connexe.}$$

De plus, les arguments utilisés jusqu'ici sont de nature à tel point formelle, que les aspects
ed. 132 spécifiques du Modéliseur Cat semblent n'y jouer pratiquement aucun rôle, ou un rôle très ms. page 201
effacé.

Commentary · Exposition [ms. 200]

The category of choices is contractible

Grothendieck's category $\mathcal{C}$ packages the compatibility just used. Its arrows are natural transformations of replacement functors that commute with their coaugmentations. Every such arrow is pointwise in W_S, by two-out-of-three. Hence any construction which sends these weak equivalences to isomorphisms factors through $\Pi\mathcal{C}$. The remaining question is whether this groupoid has exactly one arrow between any two objects. The natural common refinements give more: provided a replacement exists, the nerve of $\mathcal{C}$ is contractible.

Here is a direct proof. Fix $(R, r) \in \mathcal{C}$ and define

$$T(C, i) = \big(CR, (Cr)i\big).$$

The displayed unit is also $(iR)r$, by naturality of i. The composite CR takes values in $\mathcal{F}$, and its coaugmentation is a weak equivalence. Thus T is an endofunctor of $\mathcal{C}$. On an arrow $u : C \to D$ put $T(u) = uR$. There are natural transformations

$$\mathrm{id}_{\mathcal{C}} \xrightarrow{\alpha} T \xleftarrow{\beta} \mathrm{const}_{(R,r)}, \qquad \alpha_C = Cr, \quad \beta_C = iR.$$

Both are arrows under the identity functor because $(Cr)i = (iR)r$. To check their naturality, use, respectively, naturality of u at r_X and the equation $ui = j$ for a map $(C, i) \to (D, j)$. These are equalities before localization.

A natural transformation $F \to G$ gives a functor $\mathcal{D} \times [1] \to \mathcal{E}$, hence a homotopy between the maps of nerves induced by F and G; this is the elementary nerve calculation used in Quillen (1973, §1, Proposition 2). The two transformations above consequently homotope the identity of $|N\mathcal{C}|$ to a constant map. With the usual change of universe if necessary, $|N\mathcal{C}|$ is contractible. In particular, $\Pi\mathcal{C}$ is equivalent to the terminal groupoid. Any two zigzags of comparisons between the same replacements induce the same isomorphism in a target in which all arrows of $\mathcal{C}$ become invertible.

This argument also locates the imprecision in the word *filtrante* on the next page. A common target for two objects is only one of the conditions for a filtered category; parallel arrows must also become equal after postcomposition. The common-refinement square does not establish that second condition. Nor do the natural maps into a binary product, even with all the displayed coherence, establish it in general. For example, the category of nonempty finite ordered sets and order-preserving injections has ordered concatenation and natural inclusions from both factors. It is not filtered: the two injections from a singleton into a two-element ordered set remain distinct after every injection. Nevertheless its nerve is contractible by the same argument. This example refutes the inference being used, not a separately proved assertion that this particular $\mathcal{C}$ fails to be filtered. Contractibility supplies precisely the canonicity needed here, so no strict equalization claim is required.

Le carré de la page 197 montre surtout que la catégorie $\mathcal{C}$ est filtrante - (C, i), (C, i') s'envoient l'un et l'autre dans (C'', i''), avec $C'' = C \circ C'$, et $i'' = (C * i') \circ i = (i * C') \circ i'$. Il

y a même une loi de composition dans $\mathcal{C}$, nullement commutative ni unitaire c'est sûr, mais sûrement associative, et (si cette composition est notée $*$) avec des morphismes fonctoriels

$$\xi \xrightarrow{\alpha_{\xi,\xi'}} \xi * \xi' \xleftarrow{\beta_{\xi,\xi'}} \xi'$$

qui assurent le caractère filtrant croissant de $\mathcal{C}$. Ces flèches semblent remplacer en quelque sorte l'unité bilatère absente, et on s'attend à ce qu'elles aient les mêmes excellentes propriétés que si on avait une telle unité, concernant les composés doubles $(\xi * \xi') * \xi'' \simeq \xi * (\xi' * \xi'')$,

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$$(\xi * \xi') * \xi'' \xrightarrow{\sim} \xi * (\xi' * \xi'').$$

Il y a deux façons d'envoyer un produit binaire

$$\xi * \xi', \qquad \xi' * \xi'', \qquad \xi * \xi''$$

dans le produit ternaire, et bien elles doivent dans chacun des trois cas, être égales. Et de même il y a, ceci admis, encore deux façons d'envoyer ξ dans le produit ternaire, en factorisant soit par $\xi * \xi'$, soit par $\xi * \xi''$, et de même pour les deux autres foncteurs ξ', ξ''; et là encore, ces deux façons doivent être égales, dans chacun de ces trois cas - donc $3 + 3 = 6$ égalités en tout à écrire et à vérifier - on a vu pire!

La philosophie calculatoire devrait être celle-ci : Si on a un produit n-uple

$$\xi_1 * \xi_2 * \cdots * \xi_n$$

(où je me dispense de mettre des parenthèses ...), pour tout produit m-uple partiel ($m \in [1, n-1]$)

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$$\xi_{i_1} * \xi_{i_2} * \cdots * \xi_{i_m} \qquad (i_1 < i_2 < \cdots < i_m),$$

en utilisant à tour de bras les isomorphismes d'associativité et les $\alpha_{\xi,\xi'}$, $\beta_{\xi,\xi'}$, il n'y a pourtant qu'une seule façon de l'envoyer dans le produit n-uple. J'ai bien l'impression que ça doit être vrai. (Est-ce que ça implique déjà que $\mathcal{C}$ est 1-connexe?)

En fait, il y a bien une unité bilatère, mais elle n'est pas dans $\mathcal{C}$, mais dans la catégorie plus grande

$$\mathrm{id}_{\mathrm{Cat}/S} \backslash \underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{Cat}/S),$$

qui elle, comme toute catégorie

$$\mathrm{id}_{\mathcal{X}} \backslash \underline{\mathrm{Hom}}(\mathcal{X}, \mathcal{X}),$$

ed. 133 est une catégorie monoïdale, i.e. avec produit associatif *et* unitaire. Les flèches $\alpha_{\xi,\xi'}$, $\beta_{\xi,\xi'}$ sont les flèches qui résultent de la présence de cette unité (vu que $\mathcal{C}$ est une sous-catégorie pleine), et alors il doit devenir clair qu'on a les compatibilités dites.

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Mais même une catégorie monoïdale, est elle 1-connexe? Par exemple B_G, pour G un groupe disons, est bien une catégorie monoïdale et même 'groupale', une G-catégorie (tout ce qu'il y a de strict), et pourtant elle n'est *pas* 1-connexe!

Commentary · Exposition [ms. 203]

Insertion of units accounts for all six equalities

The associativity discussion is an effort to make the comparison procedure usable without remembering a particular sequence of refinements. It can be completed inside the ordinary monoidal category of endofunctors. An object of the coslice $\mathrm{id}_{\mathcal{A}} \backslash \mathrm{Fun}(\mathcal{A}, \mathcal{A})$ is a pointed endofunctor (C, i). Set

$$(C, i) * (D, j) = (CD, (Cj)i) = (CD, (iD)j).$$

The equality is naturality of i. Horizontal composition of natural transformations makes this a bifunctor. Its associativity is strict when functor composition is taken strictly, and its unit is $(\mathrm{id}, \mathrm{id})$.

That unit is also initial in the coslice: the unique arrow to (C, i) is i itself. Although the unit need not lie in the replacement subcategory, its arrows supply α and β there.

For an ordered list $(C_1, i_1), \dots, (C_n, i_n)$ and a subset $J \subseteq \{1, \dots, n\}$, insert an identity endofunctor in every position not in J. The map from this partial composite to $C_1 \cdots C_n$ is the horizontal composite of the identity transformations in the retained positions and the transformations $i_k : \mathrm{id} \to C_k$ in the omitted positions. This gives a specified map for every inclusion of ordered subsets. Its compatibility with a second insertion follows from the interchange law for horizontal and vertical composition of natural transformations. In components that law is just the naturality square which permits two adjacent insertions to change order. Any two orders of insertion differ by adjacent interchanges, so they give the same map. This proves the three binary-to-ternary and the three unary-to-ternary equalities requested in the manuscript, and proves the general ordered-subword assertion at the same time.

The concern raised using B_G has a valid lesson but needs a restriction on G. The one-object category with automorphism group G is a strict monoidal category of this form only when G is abelian. Indeed, the tensor product and composition share the same unit and obey interchange, so

$$a \otimes b = (a \circ 1) \otimes (1 \circ b) = (a \otimes 1) \circ (1 \otimes b) = a \circ b;$$

interchanging the two factorizations gives $a \circ b = b \circ a$. Conversely, an abelian group defines such a strict monoidal category by using its group operation for tensor product. Taking $G = \mathbf{Z}$ therefore gives a valid example with nontrivial fundamental group. What contracts the replacement category is the natural pair of maps into $C * R$, not monoidality alone. The source's broad assertion about an arbitrary group is left as printed and is registered separately from the repaired example.

Bien sûr, on peut court-circuiter la question de la 1-connexité de $\mathcal{C}$, en notant qu'après tout on a déjà a priori un objet privilégié dans

$$\underline{\mathrm{Hom}}(\mathrm{Cat}/S, \mathrm{HOT}^{\mathrm{lc}}_W(S)),$$

i.e. un foncteur canonique

$$\mathrm{Cat}/S \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S),$$

savoir le foncteur composé

$$(*) \qquad \mathrm{Cat}/S \underbrace{\longrightarrow}_{\substack{\text{localisation} \\ \text{par } W_S}} (\mathrm{Cat}/S)W_S^{-1} \longrightarrow \underbrace{\mathrm{HOT}^{\mathrm{lc}}_W(S)}_{\overset{\text{déf}}{=}(\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1}} \quad ,$$

où la deuxième flèche est définie comme le quasi-inverse de

$$(**) \qquad (\underline{\mathrm{Fib}}_W S)(W_S^{\mathrm{f}})^{-1} \longrightarrow (\mathrm{Cat}/S)W_S^{-1}$$

déduit de l'inclusion ms. page 205

$$(\underline{\mathrm{Fib}}_W S, W_S^{\mathrm{f}}) \hookrightarrow (\mathrm{Cat}/S, W_S).$$

Le fait que $(*)$ soit une équivalence, se prouve justement par le fait de l'existence d'un foncteur $C : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S$ et d'un $i : \mathrm{id}_{\mathrm{Cat}/S} \longrightarrow C$, ayant les propriétés qu'on sait, i.e. du fait que $C \neq \emptyset$. De plus, la donnée d'un objet C de $\mathcal{C}$ donne un moyen commode d'expliciter le quasi-inverse dudit foncteur. Mais ce quasi-inverse a une existence canonique et indépendante du choix d'un C, i.e. d'un mode de calcul. Le fait que le foncteur composé φ dans $(*)$ puisse se 'relever' (à isomorphisme près) en un foncteur

$$C : \mathrm{Cat}/S \longrightarrow \underline{\mathrm{Fib}}_W S$$

entre $\varphi(X) = \mathrm{hot}_{W,S}(X)$ et $\mathrm{hot}_{W,S}(C(X))$ et que l'isomorphisme puisse s'expliciter, lui aussi, par une flèche X-fonctorielle

$$i_X : X \longrightarrow C(X), \qquad i_X \in W, \qquad \text{ms. page 206}$$

ed. 134 dans Cat/S lui-même (et non seulement dans le localisé (Cat/S)W_S^{-1}), est une vertu particulière du [mot illisible], qui ne doit pas pour autant faire oublier le caractère intrinsèque des flèches (∗), indépendamment de tout mode de calcul.

Commentary · Exposition [ms. 205]

Why the inclusion becomes an equivalence

Grothendieck's second answer to the canonicity problem identifies the category being computed. We give the localization argument explicitly, since its hypotheses explain both its strength and its limits. Let $(\mathcal{A}, W)$ be a category with weak equivalences satisfying two-out-of-three, let $j : \mathcal{F} \hookrightarrow \mathcal{A}$ be full, and suppose $C : \mathcal{A} \to \mathcal{F}$ is equipped with $i : \mathrm{id}_{\mathcal{A}} \to jC$ whose components lie in W. Give $\mathcal{F}$ the induced weak equivalences. Then

$$\overline{j} : \mathcal{F}[W^{-1}] \longrightarrow \mathcal{A}[W^{-1}]$$

is an equivalence. Indeed, naturality and two-out-of-three show that C preserves weak equivalences, so it induces $\overline{C}$. The transformation i induces an isomorphism $\mathrm{id} \simeq \overline{j}\,\overline{C}$. Its restriction to $\mathcal{F}$ has all its components in $\mathcal{F}$, because that subcategory is full, and gives $\mathrm{id} \simeq \overline{C}\,\overline{j}$. These two natural isomorphisms prove the assertion. This is the same replacement mechanism used earlier in the commentary, notably on Chapter VI, manuscript p. 112; here it proves an equivalence with the entire localized slice.

For the particular functor (∗∗) the weak equivalences on $\underline{\mathrm{Fib}}_W S$ have been written W_S^{f} rather than as the induced W_S. One therefore also uses their agreement on this subcategory: the fibre axiom supplies the implication from total to fibrewise equivalence, while the converse is the relevant local-to-global property. Nonemptiness of $\mathcal{C}$ alone, without this compatibility of weak-equivalence classes, is not the whole statement. Once the compatibility is available, the preceding proof applies exactly as written.

The printed sentence on manuscript p. 205 calls (∗) an equivalence. The equivalence proved by the argument is (∗∗); (∗) starts in the unlocalized category and includes a localization functor. This is an edition-level slip, not a defect in the replacement argument. For example, even over the terminal base, localization identifies distinct natural-homotopic maps of categories, so the original localization functor cannot generally be fully faithful.

There is also a precise uniqueness statement for the inverse. If $J : \mathcal{B} \to \mathcal{H}$ is an equivalence, and $P, P' : \mathcal{H} \to \mathcal{B}$ are supplied with counits $\epsilon : JP \xrightarrow{\sim} \mathrm{id}$ and $\epsilon' : JP' \xrightarrow{\sim} \mathrm{id}$, full faithfulness of J gives a unique natural isomorphism $\vartheta : P \to P'$ satisfying

$$\epsilon'\, J\vartheta = \epsilon.$$

Its image under J is $(\epsilon')^{-1}\epsilon$, which proves existence, uniqueness, naturality, and the cocycle law. Thus the intrinsic object is the inverse equipped with its comparison to the given equivalence. A formula using paths is one realization of that object. In the more general setting of derivable categories, Cisinski (2010, Proposition 1.8) obtains the analogous equivalence between a category and its fibrant subcategory even when the chosen replacements are organized by a cleavage rather than by a strict functor into the original category.

‖ La clef de la situation, et des questions de canonicité, me semble être dans le fait que (∗∗) est une équivalence de catégories - et il faut avouer que, tout familier et quasiment banal qu'il soit finalement devenu pour moi, c'est là un fait tout à fait remarquable et nullement évident a priori. (Mais sans doute peut-on l'avoir aussi par les fourbis généraux à la Quillen-Thomason, sans se battre avec un formalisme de chemins …) Les 'isomorphismes canoniques' qu'on peut établir entre une 'théorie C ' et une 'théorie C' ' (via deux constructions différentes, p. ex., de catégories de chemins, ou via l'approche de Quillen-Thomason, sans chemins) doivent toujours revenir à ceux

qu'on peu définir via la comparaison de la 'théorie C ' d'une part, la 'théorie C' ' de l'autre, avec ce que nous donne directement la flèche composée (∗) (p. 204). Finalement, il me semble que *l'existence* d'une théorie des 'fibres homotopiques' (ou 'W-homotopiques') d'une flèche $X \longrightarrow S$ dans Cat, résulte de l'équivalence (∗∗) (p. 204), et cette théorie pourrait sans

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doute dans une certaine mesure se développer à partir de là, sans choix d'un foncteur C ni d'une structure de catégorie de modèles au sens de Quillen.

Commentary · Examples [ms. 206]

A homotopy fibre which an ordinary kernel misses

The replacement construction is already visible in elementary homological algebra, where its comparison maps and the information lost by taking a raw fibre can both be written down. This gives a different example from the groupoid calculations in the commentary on earlier chapters. Let C, D be cochain complexes of abelian groups, with differential of degree 1, and let $f : C \to D$ be a cochain map. Take W to be the quasi-isomorphisms. Degreewise surjections preserve quasi-isomorphisms under pullback: pulling such a surjection $E \to D$ back along a map $A \to B$ over D gives a map between the short exact sequences with common kernel $\ker(E \to D)$ and quotient map $A \to B$. The long exact cohomology sequences prove the assertion. Thus these surjections belong to the class of W-fibrations used below.

Define a complex

$$E(f)^n = C^n \oplus D^{n-1} \oplus D^n, \qquad d(c, h, y) = (d_C c,\ y - f(c) - d_D h,\ d_D y).$$

The equation $d^2 = 0$ follows from $d_D f = f d_C$. The maps

$$j(c) = (c, 0, f(c)), \qquad p(c, h, y) = y, \qquad r(c, h, y) = c$$

are cochain maps, $pj = f$, and $rj = \mathrm{id}_C$. The degree -1 map

$$H(c, h, y) = (0, 0, h)$$

satisfies $dH + Hd = \mathrm{id}_{E(f)} - jr$. Consequently j is a cochain-homotopy equivalence, in particular a quasi-isomorphism, while p is degreewise surjective. We have obtained the required factorization $C \xrightarrow{j} E(f) \xrightarrow{p} D$ with its comparison map explicit.

The strict fibre of p over the zero map $0 \to D$ is

$$F(f)^n = C^n \oplus D^{n-1}, \qquad d(c, h) = (d_C c, -f(c) - d_D h).$$

With the convention $\mathrm{Cone}(f)^n = D^n \oplus C^{n+1}$ and differential $(y, c) \mapsto (d_D y + f(c), -d_C c)$, this is canonically isomorphic to $\mathrm{Cone}(f)[-1]$ after reordering the summands. The sign is forced by the shift; it is not an additional convention about homotopy fibres. The short exact sequence $0 \to D[-1] \to F(f) \to C \to 0$ gives the familiar exact sequence relating the cohomology of the fibre, C, and D.

For a concrete calculation, let $C = D = \mathbf{Z}[0]$ and let f be multiplication by an integer $m \geq 2$. The ordinary kernel is zero. The homotopy fibre is the two-term cochain complex

$$\mathbf{Z} \xrightarrow{-m} \mathbf{Z}$$

in degrees 0 and 1. It has $H^0 = 0$ and $H^1 = \mathbf{Z}/m\mathbf{Z}$. The replacement has retained the failure of surjectivity which the ordinary kernel discards. This example is classical: Brown lists complexes in an abelian category, with quasi-isomorphisms and surjections, among his categories of fibrant objects (Brown, 1973, Part I, §1, Example 3, p. 421). The calculation explains why Grothendieck insists on replacing the structural map before taking its fibre.

4 Axiomatisation des 'fibres W-homotopiques'.

La reflexion de hier me fait revenir sur l'axiomatique dans des 'catégories de modèles'

$$(\mathcal{M}, W)$$

([119]) plus générales que Cat, dans l'esprit de VI (cf. théorème-scholie 2, p. 108-113). Pour le moment, on va supposer seulement $\mathcal{M}$ stable par produits fibrés (on n'a pas besoin d'objet final). On introduit l'ensemble

$$\mathrm{Fib}_W \subseteq \mathrm{Fl}(\mathcal{M})$$

des W-*fibrations* (sous-entendu 'ultrafortes'!) par la condition

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$f : X \longrightarrow Y$ est dans Fib_W si et seulement si le foncteur changement de base par X,

$$\begin{array}{rcl} Y' & \longmapsto & X' = X \times_Y Y' \\ \mathcal{M}/Y & \longrightarrow & \mathcal{M}/X \end{array}$$

ed. 135

transforme W_X dans W_Y, où $W_X \subseteq \underline{\mathrm{Fl}}(\mathcal{M}/X)$ `[est]` défini par
$u \in W_X \iff \sigma(U) \in W$ ($\sigma : \mathcal{M}/X \longrightarrow \mathcal{M}$ le foncteur 'source', ou 'oubli de X').

([120]). Posons, pour tout S dans $\mathcal{M}$,

$$\mathrm{Ho}^!_W(S) = (\mathcal{M}/S)W_S^{-1},$$

qui joue le rôle d'une catégorie de W-types d'homotopie relatifs *localement constants* sur S. On aimerait décrire une loi contravariante en S pour $\mathrm{Ho}^!_W(S)$, i.e. pour

$$f : S' \longrightarrow S$$

flèche quelconque dans $\mathcal{M}$, définir

$$(*) \qquad f^*_W : \mathrm{Ho}^!_W(S) \overset{?}{\to} \mathrm{Ho}^!_W(S').$$

La difficulté est, bien sûr, que le foncteur changement de base

$$f^* : \mathcal{M}/S \longrightarrow \mathcal{M}/S'$$

n'est *pas* en général compatible avec les localiseurs W_S, $W_{S'}$. La notion de W-fibration sert donc en premier lieu à dégager la condition la plus évidente sur une flèche $f : S' \longrightarrow S$ dans $\mathcal{M}$, pour que le foncteur changement de base soit compatible avec les localiseurs W_S, $W_{S'}$ (c'est *ça* la définition), donc induit $(*)$ sur les localisées.

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Commentary · Later developments [ms. 208]

The intrinsic fibrations are sharp maps

The purpose of the new definition is to extract from $(\mathcal{M}, W)$ the largest class of maps along which ordinary pullback is homotopical. For $f : X \to Y$ the correctly typed condition is

$$f^* : (\mathcal{M}/Y, W_Y) \longrightarrow (\mathcal{M}/X, W_X).$$

The edition reverses W_X and W_Y in the sentence following the displayed pullback functor; the direction of that functor fixes the intended reading. Unpacked, the definition says that whenever $a : A \to B$ is in W and is a map over Y, the map

$$X \times_Y A \longrightarrow X \times_Y B$$

is again in W. The quantification over all objects over Y is essential. Testing only pullbacks of weak equivalences whose target is exactly Y would omit part of the requirement.

119. W satisfait l'axiome de saturation faible : les isomorphismes `[sont]` dans W, et si deux parmi f, g, gf `[sont]` dans W, le troisième aussi.

120. **NB** Donner ici la proposition: Fib_W *stable par composition et par changement de base, et contient les isomorphismes, stable par facteurs directs.*

In a model category, this is precisely the condition called *sharp* by Rezk (1998, §2). His Proposition 2.2 says that all model fibrations are sharp exactly when the model category is right proper. His Proposition 2.7 further identifies sharp maps, in a right proper model category, with those maps all of whose ordinary pullback squares are homotopy cartesian. These hypotheses distinguish three statements which can otherwise look indistinguishable: a map can be a fibration for a chosen model structure, preserve weak equivalences by pullback, or compute all homotopy pullbacks strictly. The first is extra structure, the second depends only on $(\mathcal{M}, W)$, and the equivalence with the third needs the stated properness hypothesis. Rezk attributes the sharp-map idea to Hopkins; we are comparing equivalent conditions, not asserting a historical line of transmission from this manuscript.

The basic stability properties require little machinery. If $f : X \to Y$ and $g : Y \to Z$ preserve weak equivalences by pullback, so does gf, because $(gf)^*$ is canonically f^*g^*. To see stability under a pullback $f' : X \times_Y T \to T$, regard a weak equivalence over T also as a map over Y; its pullback by f' is the same underlying map as its pullback by f. Isomorphisms plainly have the property. The statement about retracts in footnote 120 is different: it needs retract closure of W. Under that hypothesis, pulling back a retract diagram makes the map to be tested a retract of a map in W, proving the result. Under only the weak saturation explicitly listed in footnote 119, the retract assertion fails, as the following finite example shows.

Le *'problème des fibres homotopiques'* est justement de définir $(*)$ pour *toute* flèche dans $\mathcal{M}$, pas seulement pour les W-fibrations, de tellle façon que les $\mathrm{Ho}^!_W(S)$ forment, pour S variable, les fibres d'une catégorie fibrée remarquable sur $\mathcal{M}$. Quand $\mathcal{M}$ admet un objet final, de sorte que l'on peut regarder les flèches

$$p : e \longrightarrow S \qquad (p \text{ comme 'point'})$$

dans $\mathcal{M}$ comme des 'points' de S, la 'fibre W-homotopique' d'un objet quelconque X/S de $\mathcal{M}/S$ en ledit point sera par définition l'élément de

$$\mathrm{Ho}_W = \mathrm{Ho}_W(\mathcal{M}) \simeq \mathrm{Ho}^!_W(e)$$

défini par

$$p^*(\mathrm{ho}^!_{W/S}(X/S)),$$

ed. 136 où ms. page 210

$$\mathrm{ho}^!_{W/S} : \mathcal{M}/S \longrightarrow \mathrm{Ho}^!_W(S) \overset{\text{déf}}{=} (\mathcal{M}/S)(W_S)^{-1}$$

est le foncteur de localisation canonique.

Commentary · Development (commentators') [ms. 209]

The retract condition needs a hypothesis

The minimal axioms are being separated here so that the fibre theory will apply beyond familiar model categories. A small counterexample helps to keep that economy honest. We use the usual meaning of a direct factor of a morphism, namely a retract in the arrow category. The observation below addresses only the retract clause of footnote 120; composition, pullback, and isomorphism stability remain valid under the assumptions stated in the text.

Finite-set counterexample to the retract clause. [ours] Let $\mathcal{M}$ be the category of finite sets, and let W consist of all maps $A \to B$ for which $|A| \equiv |B| \pmod 2$. Then $\mathcal{M}$ has pullbacks and W contains all isomorphisms and satisfies two-out-of-three. Nevertheless the class Fib_W defined on manuscript p. 208 is not closed under retracts.

Proof. The axioms for W follow by applying equality in $\mathbf{Z}/2\mathbf{Z}$ to the cardinalities of the three objects in a composable pair. Set $S = \{0, 1\}$. Let E have one element over 0 and three over 1, and write $p : E \to S$ for its structural map. For every finite set $T \to S$,

$$|E \times_S T| = |T_0| + 3|T_1| \equiv |T| \pmod 2.$$

Thus if $A \to B$ over S belongs to W, its pullback along p also belongs to W. This proves $p \in \mathrm{Fib}_W$.

Let E' have one element over 0 and two over 1, and let $p' : E' \to S$. Include E' into E by choosing two of the three elements over 1. Define a retraction $E \to E'$ which is the identity on the chosen elements and sends the remaining element to either of the two elements over 1. Together with the identity of S, these maps exhibit p' as a retract of p in the arrow category. But the map $\varnothing \to S$, considered as a map from the empty object to the identity object of $\mathcal{M}/S$, lies in W, since both cardinalities are even. Its pullback along p' is $\varnothing \to E'$, whose target has cardinality 3. This map is not in W. Hence $p' \notin \mathrm{Fib}_W$. □

This counterexample also makes clear what repairs the statement. One may assume W is retract closed, as in a model category; strong saturation is more than sufficient. Alternatively, one can omit the retract clause, which is not needed for the formal base-change construction on the following pages. We checked the axioms in Brown's Part I, Cisinski's §1, and Rezk's sharp-map discussion; those sources do not provide this particular example. The argument is supplied here as a local clarification of the manuscript, without a claim of priority.

Pour tout S dans $\mathcal{M}$, soit

$$\underline{\mathrm{Fib}}_W S \subseteq \mathcal{M}/S$$

la sous-catégorie pleine formée des objets X/S de $\mathcal{M}/S$ tels que $X \longrightarrow S$ soit W-fibrant. On va définir une variante

$$\mathrm{Ho}^{\mathrm{lc}}_W(S)$$

de $\mathrm{Ho}^{!}_W(S)$, comme

$$\mathrm{Ho}^{\mathrm{lc}}_W(S) \overset{\text{déf}}{=} (\underline{\mathrm{Fib}}_W S)(W^{\mathrm{u}}_S)^{-1},$$

où

> $W^{\mathrm{u}}_S \subseteq \underline{\mathrm{Fl}}(\underline{\mathrm{Fib}}_W S)$ est le localiseur formé des flèches $u : X \longrightarrow Y$ qui sont 'universellement sur S dans W', i.e. telles que pour toute flèche $S' \longrightarrow S$ de $\mathcal{M}$ de but S, la flèche déduite par changement de base,
>
> $$u' : X' = X \times_S S' \longrightarrow Y' = Y \times_S S'$$
>
> est dans $W_{S'}$.

On a 'fait ce qu'il fallait' pour que $\mathrm{Ho}^{\mathrm{lc}}_W(S)$ soit contravariant en S pour des flèches *quelconques* $f : S' \longrightarrow S$ de $\mathcal{M}$, elles définissent

$$(**) \qquad f^{*\,\mathrm{lc}}_W : \mathrm{Ho}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{Ho}^{\mathrm{lc}}_W(S'),$$

puisque les W-fibrations sont stables par changement de base, et que par suite $f^* : \mathcal{M}/S \longrightarrow \mathcal{M}/S'$ induit

$$f^*_{\underline{\mathrm{Fib}}_W} : \underline{\mathrm{Fib}}_W S \longrightarrow \underline{\mathrm{Fib}}_W S'$$ ms. page 211

qui, par définition de W^{u}_S et $W^{\mathrm{u}}_{S'}$, est compatible avec les localiseurs

$$f^*_{\underline{\mathrm{Fib}}_W} : (\underline{\mathrm{Fib}}_W S, W^{\mathrm{u}}_S) \longrightarrow (\underline{\mathrm{Fib}}_W S', W^{\mathrm{u}}_{S'}),$$

et passe donc aux catégories des fractions en une flèche $(*)$.

On peut se demander pourquoi ne pas être plus généreux, et définir $\mathrm{Ho}^{\mathrm{lc}}_W(S)$ comme $(\mathcal{M}/S)(W^{\mathrm{u}}_S)^{-1}$, en définissant W^{u}_S dans $\mathcal{M}/S$ de la même façon que tantôt dans $\underline{\mathrm{Fib}}_W S$ - ce qui ne coûte pas plus cher. Une première réponse est d'expérience le localiseur W^{u}_S dans $\mathcal{M}/S$ tout entier s'avère pratiquement inutilisable, d'une part parce que l'appartenance à W^{u}_S (pour une flèche entre objets de $\mathcal{M}/S$ qui ne sont pas très particuliers, p.ex. W-fibrés sur S) est invérifiable (sauf pour des isomorphismes et des choses comme ça …), et que d'autre part la ed. 137 catégorie de fractions est prohibitivement grosse et ne peut servir (semble-t-il) à rien. Une autre réponse, c'est qu'on veut que la catégorie obtenue réponde bien à l'intuition de types d'homotopie relatifs *localement constants* sur S - et on veut donc que pour un changement de base quelconque $f : S' \longrightarrow S$ par un ms. page 212

Commentary · Exposition [ms. 211]

Two localizations with different immediate advantages

Grothendieck keeps two constructions in view because each makes one half of the desired theory immediate. Write

$$\mathcal{H}_S = (\mathcal{M}/S)[W_S^{-1}], \qquad \mathcal{L}_S = (\mathrm{Fib}_W S)[(W_S^{\mathrm{u}})^{-1}].$$

The first accepts every object over S. Its ordinary inverse-image functor is initially available only for a base map in Fib_W. The second accepts only objects whose structural map is a W-fibration, but its inverse images exist for every base map. The universal quantifier defining W_S^{u} is precisely what makes this second assertion formal.

For $f : T \to S$, ordinary pullback sends $\mathrm{Fib}_W S$ into $\mathrm{Fib}_W T$, as just proved. If $u \in W_S^{\mathrm{u}}$ and $g : U \to T$, the iterated pullback $g^* f^* u$ is isomorphic to $(fg)^* u$, hence belongs to W. Therefore $f^* u \in W_T^{\mathrm{u}}$. The universal property of localization supplies a functor

$$P_f : \mathcal{L}_S \longrightarrow \mathcal{L}_T.$$

The canonical isomorphisms between iterated ordinary pullbacks survive localization and give $P_g P_f \simeq P_{fg}$. Their associativity follows from the universal property of the pullback: both routes identify an iterated pullback with the same object, carrying the same projections. Thus the result is naturally a pseudofunctor, or equivalently a fibred category. No strictly functorial global choice of pullbacks is needed.

One can also see exactly what must be checked to define the fibre at a generalized point. Fix $X \to S$ in $\mathrm{Fib}_W S$. Sending $T \to S$ to the absolute homotopy type of $X \times_S T$ is a covariant functor on $\mathcal{M}/S$; a map $T' \to T$ induces the corresponding map between total pullbacks. If that map of generalized points is in W_S, its image is in W by the defining property of $X \to S$. The functor consequently inverts weak equivalences between generalized points. This assertion is compatible with replacing X by a morphism in W_S^{u}, since evaluation at every T then gives a weak equivalence. It therefore makes sense for all $\xi \in \mathcal{L}_S$, not only for a chosen representative.

This explains the two separate roles of the subcategory and the localizer. Restricting to $\mathrm{Fib}_W S$ controls variation of the generalized point by a weak equivalence. Restricting the arrows to W_S^{u} controls variation of the representative by a weak equivalence after any base change. Omitting either condition loses one of these immediate guarantees. The manuscript's remark that the unrestricted universal localization is too large is a methodological judgment; it is not a theorem saying that every such localization is mathematically useless. What has been proved is the precise pair of invariance properties above.

S' dans $\mathcal{M}/S$, d'où un foncteur composé

$$\mathcal{F}_S(W_S^{\mathrm{u}})^{-1} \xrightarrow{f_W^{*\,\mathrm{lc}}} \mathcal{F}_{S'}(W_{S'}^{\mathrm{u}})^{-1} \longrightarrow \mathrm{Ho}_W$$
$$\mathcal{F}_S(W_S^{\mathrm{u}})^{-1} \xrightarrow[\varphi_{S'}]{} \mathrm{Ho}_W$$

([121]) déduit des foncteurs de catégories avec localiseurs

$$(\mathcal{F}_S, W_S^{\mathrm{u}}) \xrightarrow{f^*} (\mathcal{F}_{S'}, W_{S'}^{\mathrm{u}}) \xrightarrow{\text{oubli de } S'} (\mathcal{M}, W)$$

(foncteur composé $\varphi_{S'}$ qui peut être vu comme le foncteur *'fibre homotopique en l'objet S' sur S'*, cet objet étant visualisé comme un 'point' de S 'à valeurs dans S' ' suivant le yoga fonctoriel général …), le foncteur

$$\begin{aligned} S' &\longmapsto \varphi_{S'}(\xi) \qquad \text{pour } \xi \text{ dans } \mathcal{F}_S(W_S^{\mathrm{u}})^{-1} \\ \mathcal{M}/S &\longrightarrow \mathrm{Ho}_W \end{aligned}$$

transforme flèches dans W_S en isomorphismes : la 'fibre W-homotopique' en S'/S d'un objet de $\mathcal{F}_S(W_S^{\mathrm{u}})^{-1}$ (vu comme W-type d'homotopie *localement constant* sur S) doit être 'la même', si S' remplacé par S'' avec $S'' \longrightarrow S'$ donné et dans W_S. C'est pour cela qu'il est raisonnable ms. page 213

de prendre $\mathcal{F}_S$ 'assez petit' (savoir $\underline{\mathrm{Fib}}_W S$, qui est exactement ajusté pour cela), plutôt que de prendre $\mathcal{M}/S$ tout entier.

On peut donc dire que la catégorie $\underline{\mathrm{Fib}}_W S$, et le localiseur W_S^{u} ont été construits ad-hoc, le plus simplement possible, pour donner lieu à un localisé $\mathrm{Ho}_W^{\mathrm{lc}}(S)$ qui réponde à l'intuition de W-types d'homotopie relatifs localement constants des deux façons suivantes :

1°) Il y a spontanément une loi contravariante en S, pour des flèches quelconques $S' \longrightarrow S$ dans Cat (on a défini W_S^{u} juste exprès en conséquence).

2°) que la notion de 'fibre W-homotopique en S'/S' d'un objet ξ de $\mathrm{Ho}_W^{\mathrm{lc}}$ satisfasse à la condition précédente, i.e. une flèche $S'' \longrightarrow S'$ dans $\mathcal{M}/S$ qui est dans W_S définisse un *isomorphisme* (dans Ho_W) des fibres homotopiques

$$\underbrace{\xi_{S''}}_{\varphi_{S''}(\xi)} \longrightarrow \underbrace{\xi_{S'}}_{\varphi_{S'}(\xi)}$$

(et on a choisi $\mathcal{F}_S = \underline{\mathrm{Fib}}_W(S)$ juste exprès pour ça).

ed. 138 On a donc une excellente notion de fibres W-homotopiques pour des X W-fibrés sur S - cette fois excellente pour des changements de base *quelconque* - et il n'y a aucun mérite à ça puisqu'on a tout défini exprès pour ça!

Pour le dire autrement, on a défini *deux* notions de W-types d'homotopie relatifs sur S, ms. page 214
l'une de façon à donner un tel type d'homotopie $\mathrm{ho}_{W,S}^{!}(X)$ pour *tout* objet de $\mathcal{M}/S$, mais il n'y a a priori une bonne notion de changement de base que pour les changements de base $S' \longrightarrow S$ qui sont W-fibrés, l'autre, qui correspond plus directement à l'intuition d'un W-type d'homotopie localement constant sur S, n'associe un tel type $\mathrm{ho}_{W,S}^{\mathrm{lc}}(X)$ qu'aux X de $\mathcal{M}/S$ qui sont W-fibrés sur S, en revanche cette notion admet une bonne notion d'image inverse et de 'fibre W-homotopique' pour des changements de base quelconques.

Une 'bonne' théorie des fibres W-homotopiques doit réunir en une seule les deux notions précédentes, de sorte que les avantages des deux s'ajoutent, et les désavantages s'éliminent! Elle s'obtient à l'aide d'un axiome draconien sur la situation (alors que jusqu'à maintenant on n'a rien utilisé sur $W \subseteq \underline{\mathrm{Fl}}(\mathcal{M})$, si ce n'est tout au plus la saturation faible, en plus de l'existence des produits fibrés) :

ⓐ (*Axiome de la fibre homotopique.*) Pour tout objet S dans $\mathcal{M}$, le foncteur canonique ms. page 215

$$\mathrm{Ho}_W^{\mathrm{lc}}(S) \longrightarrow \mathrm{Ho}_W^{!}(S)$$

déduit de l'inclusion

$$(\underline{\mathrm{Fib}}_W(S), W_S^{\mathrm{u}}) \longrightarrow (\mathcal{M}/S, W_S)$$

est une *équivalence de catégories.*

Cet axiome implique que dans $\underline{\mathrm{Fib}}_W S$, W_S^{u} et le localiseur induit par W_S (que je désigne encore par W_S par abus de notation) ont même saturé fort. En fait, il est naturel de compléter l'axiome précédent par le suivant.

ⓑ (Axiome adjoint de l'axiome de la fibre homotopique.) Sur $\underline{\mathrm{Fib}}_W S$, les localiseurs W_S^{u} et W_S coïncident, i.e. on a ceci :

> Soient X, Y dans $\underline{\mathrm{Fib}}_W S$, $X \xrightarrow{u} Y$ un S-morphisme tel que $u \in W_S$, alors $u \in W_S^{\mathrm{u}}$ (i.e. le morphisme $u' : X' \longrightarrow Y'$ déduit par tout changement de base $S' \longrightarrow S$ est encore dans W).

Je présume que a) sera rarement satisfait sans que b) le soit (et inversement d'ailleurs), et ms. page 216
que la démonstration de a) et b) se fera pratiquement en une seule haleine.

121. $\mathcal{F}$ désigne ici une sous-catégorie fibrée de $\mathcal{M}/\mathcal{M} = \underline{\underline{\mathrm{Fl}}}(\mathcal{M})$ sur $\mathcal{M}$, substitut éventuel de celle formée de $\underline{\mathrm{Fib}}_W S$ ou de $\mathcal{M}/S$ tout entier.

Commentary · Exposition [ms. 215]

Transporting base change and separating the two axioms

The comparison axiom now combines the two advantages. Let

$$J_S : \mathcal{L}_S \longrightarrow \mathcal{H}_S$$

be induced by inclusion. Axiom a says that J_S is an equivalence. Choose an inverse R_S together with the unit and counit of an adjoint equivalence. For an arbitrary $f : T \to S$, define derived inverse image by

$$f^!_W = J_T P_f R_S : \mathcal{H}_S \longrightarrow \mathcal{H}_T.$$

This is the operation denoted f^*_W in the manuscript; the temporary exclamation mark here only distinguishes it from raw pullback. For composable $U \xrightarrow{g} T \xrightarrow{f} S$, the comparison

$$J_U P_g R_T J_T P_f R_S \xrightarrow{\sim} J_U P_g P_f R_S \xrightarrow{\sim} J_U P_{fg} R_S$$

uses $R_T J_T \simeq \mathrm{id}$ and the ordinary pullback comparison. The triangle identities of the chosen adjoint equivalences and the associativity of pullback give the unit and pentagon identities. Equivalently, these identities can be checked after the fully faithful equivalences J_S, where they reduce to those same identities. A different choice of inverses gives the unique normalized comparison described above, so the resulting fibred theory is independent of that choice up to the corresponding coherent equivalence.

Axiom a also has a useful immediate consequence which is weaker than axiom b. If u between W-fibrant objects belongs to W_S, then J_S sends its image in $\mathcal{L}_S$ to an isomorphism. An equivalence reflects isomorphisms, so u belongs to the strong saturation of W^{u}_S on $\mathrm{Fib}_W S$. Since $W^{\mathrm{u}}_S \subseteq W_S$, the two classes have the same strong saturation there. This is exactly the intermediate observation in the text. It does not, under bare two-out-of-three, identify the two original classes of arrows.

There is a clean case in which axiom b follows. Suppose W itself is strongly saturated: a morphism belongs to W exactly when it is inverted by $\mathcal{M} \to \mathcal{M}[W^{-1}]$. Then W^{u}_S is strongly saturated in $\mathrm{Fib}_W S$. Indeed, if u becomes invertible after localization at W^{u}_S, every evaluation functor

$$X \longmapsto X \times_S T \longmapsto \mathcal{M}[W^{-1}]$$

for $T \to S$ sends u to an isomorphism. Strong saturation gives $u_T \in W$ for every T, which is the definition of $u \in W^{\mathrm{u}}_S$. Thus axiom a implies axiom b under this extra hypothesis. Without it, the statement in the manuscript should retain its saturation formulation; the author's expectation that the two axioms will usually be verified together is not a formal converse.

The construction gives a derived base-change theory, but it does not by itself assert all the axioms of a derivator on arbitrary diagram categories. Those require further control of diagrams, pointwise detection, and homotopy Kan extensions. The distinction preserves the considerable content of the comparison axiom without crediting it with statements that are proved elsewhere.

Si $\mathcal{M} = \mathrm{Cat}$, W un localiseur fondamental, alors la notion adaptée ici de W-fibration (lire : 'ultraforte', dans la terminologie spécifique développée dans Cat) est à tel point forte, qu'elle
ed. 139 implique déjà pratiquement b) (moyennant W(4a′), il est vrai, cf. p. 53), et a) n'est guère vérifiable que si on a W(4a) ([122]) (loc. cit.). Donc ici c'est 'l'axiome des fibrations' sur W, sous sa forme forte W(4a), qui est la clef de voûte d'une bonne théorie des fibres homotopiques.

Une telle théorie (revenant au cas général) revient à dire, pour l'essentiel, qu'il y a 'assez' de W-fibrations ('ultrafortes') pour pouvoir décrire *tout* objet d'un $\mathcal{M}/S$, à W_S-équivalence près, comme un objet W-fibrant sur S. Dans les cas à ma connaissance, on a de plus ceci : on a pour tout S dans $\mathcal{M}$ une sous-catégorie $\mathcal{F}_{0\,S}$ de $\underline{\mathrm{Fib}}_W S$ (je pense par exemple à $\underline{\mathrm{Parf}}_W S$, ms. page 217
voire $\underline{\mathrm{Parf}}_{W_\omega} S$, dans le cas où $\mathcal{M} = \mathrm{Cat}$), telle que pour toute sous-catégorie pleine $\mathcal{F}$ de $\mathcal{M}/S$ telle que

$$\mathcal{F}_{0\,S} \subseteq \mathcal{F} \subseteq \mathcal{M}/S$$

(a fortiori, pour une $\mathcal{F}$ pleine telle que $\underline{\mathrm{Fib}}_W S \subseteq \mathcal{F} \subseteq \mathcal{M}/S$), les inclusions induisent des

122. Non, il me semble que W(4b) doit suffire …

équivalences de catégories

$$\mathcal{F}_{0\,S}(W_S)^{-1} \xrightarrow{\sim} \mathcal{F}(W_S)^{-1} \xrightarrow{\sim} \underbrace{(\mathcal{M}/S)(W_S)^{-1}}_{\overset{\text{déf}}{=}\mathrm{Hot}^{!}_{W}(S)}.$$

Le choix de $\mathcal{F}_{0\,S}$ ne semble pas pouvoir se fixer de façon 'la meilleure possible', une fois pour toutes dans le contexte axiomatique adopté ici. C'est un élément de structure supplémentaire dans $\mathcal{M}$ (en plus du localiseur W), au même titre que la donnée des classes de Q-fibrations et Q-cofibrations, à la Quillen. J'ignore si les axiomes a), b) ci-dessus (axiome de la fibre homotopique) impliquent déjà la variante de ⓐ :

ms. page 218

Commentary · Later developments [ms. 216]

What the weaker model formalisms establish

The new axioms are close to Brown's economy of assumptions, but they organize it differently. Brown begins with specified fibrations and weak equivalences on a category whose objects are fibrant, and assumes the path-object and pullback axioms needed for a calculus of homotopy. Here the fibrations are extracted from W itself by asking for homotopical base change; the existence of enough suitable representatives is imposed afterwards. One should therefore compare the actual hypotheses before using a theorem about one formalism in the other.

Cisinski's derivable categories give a precise later setting for this programme. Definition 1.1 separates weak equivalences, fibrations, and factorization toward fibrant objects. Proposition 1.8 proves that restricting to fibrant objects does not change the homotopy category. Theorem 2.21 then constructs a weak left derivator on finite direct categories, with the additional diagram axiom stated there (Cisinski, 2010). This is a useful progression for reading the present section: comparison of a single localization comes first; coherent diagram calculus requires further work. None of these results says that an arbitrary category equipped only with a two-out-of-three class automatically has homotopy fibres.

For ordinary spaces or simplicial sets with weak homotopy equivalences, a right proper model presentation supplies a familiar way to meet the requirements. Model fibrations are sharp, and ordinary pullback of a sharp map computes homotopy pullback. Weak equivalences between such representatives remain weak equivalences after derived base change. These are the roles of properness and replacement in the usual theory, rather than unexplained special properties of a path formula. In the generality of the manuscript, the corresponding assertions are the explicit axioms a and b, unless a chosen factorization theorem proves them.

The fibrewise ordinary diagram $s \mapsto \mathrm{hot}_W(X_s)$ still forgets coherence. In manuscript p. 200 the functor through ΠS° is being used for the locally constant subcategory: the missing superscript lc on the displayed $\mathrm{HOT}_W(S)$ should be understood in that context. It is not a factorization of every varying relative type through the fundamental groupoid. The commentary on earlier chapters already treats the difference between an ordinary diagram in a homotopy category and a coherent family (notably on Chapter VI, manuscript pp. 7 and 74). The equivalence of localized slice categories here concerns the coherent family; evaluation in ordinary homotopy types is one invariant of it.

ⓒ Pour toute sous-catégorie pleine $\mathcal{F}$ de $\mathcal{M}/S$ telle que

$$\underline{\mathrm{Fib}}_W S \subseteq \mathcal{F} \subseteq \mathcal{M}/S$$

les inclusions induisent des équivalences

$$\underbrace{(\underline{\mathrm{Fib}}_W S)(W_S)^{-1}}_{\substack{=\,\mathrm{Ho}^{\mathrm{lc}}_W(S)\\ \text{moyennant a)}}} \xrightarrow{\sim} \mathcal{F}(W_S)^{-1} \xrightarrow{\sim} \underbrace{(\mathcal{M}/S)(W_S)^{-1}}_{=\mathrm{Ho}^{!}_W(S)}.$$

Mais je présume que pratiquement, quand on pourra vérifier ⓐ, on pourra vérifier ⓒ également ([123]).

ed. 140

123. Cf. page 240 ff. Il faudrait ici développer un peu plus le formalisme de $\mathrm{Ho}^{\mathrm{lc}}_W(S)$ et les fibres homotopiques, sous l'hypothèse ⓐⓑ plus haut.

5 Axiomatique des W-types d'homotopie relatifs non localement constants $\mathrm{Ho}^{\mathcal{L}}_W(S)$.

On n'a rien dit jusqu'à ce point d'éventuels W-types d'homotopie relatifs (sur un objet S de $\mathcal{M}$) qui ne seraient *pas* "localement constants". Il en était question pourtant dans loc. cit. (VI, théorème-scholie 2, partie II du théorème pages 109-113.) Je profite de l'occasion pour revenir là dessus. Il faut introduire, dans chaque $\mathcal{M}/S$, un deuxième localiseur

$$W^0_S \subseteq \mathrm{Fl}(\mathcal{M}/S), \qquad W^0_S \subseteq W_S$$

(plus fin que W_S), ce qui permettra de définir ms. page 219

$$\mathrm{Ho}_W(S) \overset{\text{déf}}{=} (\mathcal{M}/S)(W^0_S)^{-1}.$$

Qu'il n'y ait pas de choix *canonique* qui s'impose ici, déduit de la seule structure $(\mathcal{M}, W)$, est claire sur l'exemple de (Cat, W), auquel cas on trouve *deux* choix symétriques, transformés l'un en l'autre par l'involution

$$X \longmapsto X^\circ : (\mathrm{Cat}, W) \xrightarrow{\sim} (\mathrm{Cat}, W),$$

à savoir W^{g}_S et W^{d}_S. Le choix de l'un ou de l'autre, disons (pour fixer les idées) de W^{g}_S, est associé à une notion d'*isomorphisme local* (ou de 'coisomorphisme local' dans le cas de W^{d}_S) que je voudrais axiomatiser. Il semble bien que pour les principales catégories de modèles pour Hot (et pour ses variantes), on dispose de façon assez naturelle d'une telle notion d'isomorphisme local (sans l'embarras, le plus souvent, d'un automorphisme involutif fournissant une notion symétrique …).

Commentary · Present state [ms. 218]

Intermediate categories and the extra information in a replacement

The last question of this section asks for more than essential surjectivity. It asks whether one may admit any collection of additional representatives without changing the localized category. A functorial replacement proves exactly this strengthened assertion, and it is useful to isolate the short proof. Let $\mathcal{F}_0 \subseteq \mathcal{A}$ be full, and suppose a coaugmented replacement $C : \mathcal{A} \to \mathcal{F}_0$, $i : \mathrm{id} \to C$, has components in W. For every full intermediate category

$$\mathcal{F}_0 \subseteq \mathcal{E} \subseteq \mathcal{A},$$

restricting C and i to $\mathcal{E}$ gives a replacement into $\mathcal{F}_0$. The localization proof on p. 205 then applies both to $\mathcal{F}_0 \subseteq \mathcal{E}$ and to $\mathcal{F}_0 \subseteq \mathcal{A}$. Consequently

$$\mathcal{F}_0[W^{-1}] \xrightarrow{\sim} \mathcal{E}[W^{-1}] \xrightarrow{\sim} \mathcal{A}[W^{-1}].$$

Fullness ensures that the components of the restricted coaugmentation really are arrows of $\mathcal{E}$; no closure of $\mathcal{E}$ under limits or factorizations is needed. When $\mathcal{F}_0 = \mathrm{Fib}_W S$ and axiom b identifies the weak-equivalence classes, this is axiom c in the text. If the replacement lands in a smaller full category of perfect objects, the stronger assertion preceding c follows as well.

The point is that a replacement supplies comparisons inside every such intermediate category. An abstract equivalence after localization has forgotten where its representing zigzags live. One therefore cannot justify c merely by factoring an equivalence of categories through the intermediate localization: a composite of two functors can be an equivalence without either factor being an equivalence. The source recognizes precisely this issue when it distinguishes axioms a and b from its question about c.

Our conclusion about the source's unanswered implication is deliberately narrower than a general claim of openness. The implication from the bare axioms a and b to c is not proved in these pages, and the literature checked here does not settle that exact formulation. The argument above establishes c under the explicit functorial-replacement hypothesis already available in the categorical

construction. Grothendieck returns to the issue at manuscript p. 240 and following in this same chapter, with factorization hypotheses and a qualification on p. 241 that every object should be W-fibrant over the terminal object; that later discussion is commented with those pages. Brown's and Cisinski's theorems provide other sufficient structures, not a proof that the extra structure follows from a and b alone. Thus the practical expectation on p. 218 has a complete proof in the replacement setting, while its unrestricted logical formulation is kept separate.

On supposera donc, comme dans loc. cit. p. 94-96, donné une classe

$$\mathcal{L} \subseteq \mathrm{Fl}(\mathcal{M})$$

de morphismes, jounant le rôle de 'flèches de localisation' ou 'd'isomorphismes locaux', satisfaisant les conditions suivantes. ms. page 220

Commentary · Exposition [ms. 219]

What the finer relative theory remembers

Grothendieck now wants a family to retain the direction in which its fibres vary. The preceding fibre theory has deliberately made transport invertible. A general coefficient system on a category need not have this property: a specialization can create a component, identify two components, or alter a higher homotopy group. Localizing the total category over the base at all total-space equivalences loses such information. The additional class W_S^0 is introduced to preserve it.

There are consequently two distinct localizations to keep in view. The category $(\mathrm{Cat}/S)[W_S^{-1}]$ describes a homotopy type equipped with a map to the homotopy type of S. In the ordinary homotopy theory, its derived fibres form a locally constant coherent family over $|NS|$. A directional diagram, by contrast, is a functor from NS, or from NS^{op} according to the chosen variance, to spaces. Its transition maps need not be equivalences. It is the localization of the whole diagram category at objectwise equivalences that records this datum; a diagram in the ordinary category of homotopy types need not retain its coherence. The distinction is the one already made in the commentary on Chapter VI, pp. 64–74, and not a new objection to the fibre construction.

The notation W_S^{g} has a concrete test inherited from Chapter VI, manuscript p. 41: a map $u : X \to Y$ over S belongs to this class precisely when

$$X/s \longrightarrow Y/s \quad \text{belongs to } W \qquad (s \in \mathrm{Ob}\, S).$$

Here $X/s = X \times_S (S/s)$. Replacing these over-categories by the under-categories $s\backslash X$ gives W_S^{d}. Taking opposite categories exchanges the two tests, together with the base S and its opposite. It does not give an equivalence between the two theories over a fixed, unmodified S. The phrase that neither choice is canonically imposed should be read as a statement about the absence of a preferred orientation in the specified data, not as a theorem that no invariant construction of any kind is possible (Grothendieck, 1990b, Chapter VI, manuscript pp. 41–42).

The axioms on $\mathcal{L}$ make the test intrinsic to a chosen class of base changes. Since id_S is an admissible test, membership in $W_S^{\mathcal{L}}$ implies membership in W_S. Testing against every base change is stronger, giving $W_S^{\mathrm{u}} \subseteq W_S^{\mathcal{L}}$. All three agree on the good locally constant representatives by the preceding fibre axiom. Their agreement there says nothing about their agreement on the whole slice: the whole purpose of the finer theory is to retain objects outside that subcategory.

(∗) $\mathcal{L}$ contient les isomorphismes, est stable par composition, et par changement de base.

On pose, pour S dans $\mathcal{M}$,

$$\mathcal{L}_S \subseteq \mathcal{M}/S \qquad \text{sous-catégorie pleine des } S'/S \text{ telle que } S' \longrightarrow S \texttt{ [soit]} \text{ dans } \mathcal{L}.$$

On définit alors pour tout S dans $\mathcal{M}$

$$W_S^{\mathcal{L}} \subseteq \mathrm{Fl}(\mathcal{M}/S)$$
$$(X \xrightarrow{u} Y) \in W_S^{\mathcal{L}} \iff \forall\, S' \in \mathcal{L}_S,\ (u_{S'} : X \times_S S' \longrightarrow Y \times_S S') \text{ est dans } W.$$

ed. 141 C'est donc une variante du localiseur W_S^{u}, où maintenant on ne prend comme changements de base que les flèches '$\mathcal{L}$-localisantes', i.e. $\in \mathcal{L}$. On a donc

$$W_S^{\mathrm{u}} \subseteq W_S^{\mathcal{L}} \subseteq W_S$$

en tant que parties de $\mathrm{Fl}(\mathcal{M}/S)$, et de même dans toute sous-catégorie pleine $\mathcal{F}$ de $\mathcal{M}/S$. Dans le cas de $\underline{\mathrm{Fib}}_W S$, moyennant l'axiome (b) ci-dessus ($W_S^{\mathrm{u}} = W_S$ dans $\underline{\mathrm{Fib}}_W S$), les trois localiseurs coïncident.

On pose

$$\mathrm{Ho}_W^{\mathcal{L}}(S) \text{ (ou } \mathrm{Ho}_W(S)) \overset{\text{déf}}{=} (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1},$$

et on a donné dans loc. cit. (partie II du théorème-scholie) un résultat intéressant sur cet invariant, pour S fixé. Mon propos présent est de voir ce qui se passe pour S variable. ms. page 221
Visiblement, une notion de W-type d'homotopie relatif sur un objet quelconque S de $\mathcal{M}$ n'est satisfaisante, que si elle permet des images inverses par des flèches *quelconques*

$$f : S' \longrightarrow S$$

dans $\mathcal{M}$, qui doivent définir absolument

$$f_W^{*\mathcal{L}} \text{ ou } f_W^* : \mathrm{Ho}_W^{\mathcal{L}}(S) \longrightarrow \mathrm{Ho}_W^{\mathcal{L}}(S').$$

On est donc confronté à la même perplexité que tantôt - et ce n'est *pas,* cette fois, l'introduction de $\underline{\mathrm{Fib}}_W(S)$ qui va sauver la situation, car

$$(\underline{\mathrm{Fib}}_W S)(\underbrace{W_S^{\mathcal{L}}}_{\substack{=W_S \\ =W_S^{\mathrm{u}}}})^{-1} \longrightarrow (\mathcal{M}/S)(W_S^{\mathcal{L}})^{-1}$$

n'a aucune envie, cette fois, d'être une équivalence de catégories, mais bien plutôt une inclusion (pleinement fidèle) des W-types d'homotopie relatifs constants sur S, dans les W-types d'homotopie quelconques. Il faut visiblement un axiome qui remplace l'axiome 'de la fibre homotopique'.

ms. page 222

Commentary · Examples [ms. 221]

An endomorphism which becomes invertible only after changing the fibre

A single irreversible transition already explains why the new theory is needed. Consider the category $B\mathbf{N}$ with one object and endomorphism monoid $(\mathbf{N}, +)$. Let a covariant diagram of spaces assign a circle to its object and the degree-two map $d : S^1 \to S^1$ to its generating endomorphism. This diagram cannot be locally constant: multiplication by 2 on $\pi_1(S^1) = \mathbf{Z}$ is not invertible. Objectwise weak equivalences of diagrams preserve this obstruction.

Its homotopy colimit is the mapping torus

$$T_d = (S^1 \times [0,1])/((z,1) \sim (d(z),0)).$$

One can obtain this model directly from the bar construction for the free monoid on one generator: the higher simplices express compositions of that generator, and collapse onto the mapping-cylinder description. Van Kampen, using the circle in the fibre and the loop round the parameter as generators, gives

$$\pi_1(T_d) = \langle a, t \mid tat^{-1} = a^2 \rangle.$$

The direction chosen for t fixes the displayed convention. This group is nonabelian: the transformations $a(x) = x + 1$ and $t(x) = 2x$ of the real line satisfy the relation and do not commute. The cellular chain calculation for a mapping torus is the cone of $1 - d_*$ on the chains of the fibre. On $H_1(S^1)$ this is multiplication by -1, and on $H_0(S^1)$ it is zero. The resulting exact sequence yields

$$H_0(T_d; \mathbf{Z}) = \mathbf{Z}, \qquad H_1(T_d; \mathbf{Z}) = \mathbf{Z}, \qquad H_q(T_d; \mathbf{Z}) = 0 \quad (q \geq 2).$$

Thus the total space has the homology of the base circle although it retains a substantial nonabelian fundamental group.

Now regard the projection $T_d \to S^1$ solely as a map of homotopy types. Its homotopy fibre is the infinite cyclic cover of the mapping torus. Cutting that cover at the integral parameter values exhibits it as the telescope of successive degree-two maps. Its fundamental group is therefore

$$\operatorname{colim}(\mathbf{Z} \xrightarrow{2} \mathbf{Z} \xrightarrow{2} \cdots) = \mathbf{Z}[1/2],$$

and its higher homotopy groups vanish: every map from a sphere has compact image in a finite portion of the telescope, which retracts onto a circle. Transport around the base now acts invertibly, by multiplication by 2 or its inverse according to orientation. The passage to locally constant relative homotopy types has changed the fibre from $K(\mathbf{Z}, 1)$ to $K(\mathbf{Z}[1/2], 1)$.

This example can be made entirely categorical. Use the one-object groupoid $B\mathbf{Z}$ and the functor induced by $m \mapsto 2m$, then form its Grothendieck construction over $B\mathbf{N}$. Its nerve has the indicated homotopy-colimit type by the homotopy-colimit theorem (Thomason, 1979, Theorem 1.2); the precise statement and natural comparison are reproduced by Rubin (2020, §2.3, Theorem 2.6). The example extends the earlier walking-arrow tests: it shows, through an explicit group calculation, the information that is lost when all transport is forced to be invertible.

Dans le cas de Cat, la classe $\mathcal{L}$ des isomorphismes locaux (disons) permet de définir, comme ci-dessus, le localiseur $W_S^{\mathcal{L}} = W_S^{\mathrm{g}}$ dans les Cat/S, comme formé des $u : X \longrightarrow Y$ sur S qui sont dans W et le restent par tout changement de base $S' \longrightarrow S$ qui est dans $\mathcal{L}$. Mais il y a lieu de *saturer* $\mathcal{L}$, en regardant la classe $\mathcal{L}^\wedge$ des flèches $f : S \longrightarrow S'$ de Cat telles que le changement de base

$$f^* : \mathrm{Cat}/S \longrightarrow \mathrm{Cat}/S'$$

envoie W_S^{g} dans $W_{S'}^{\mathrm{g}}$. Ce $\mathcal{L}^\wedge$ est tautologiquement stable par composition, et contient les
ed. 142 isomorphismes (mieux, contient $\mathcal{L}$ = ensemble des isomorphismes locaux). Il n'est pas clair, par la définition donnée ici, qu'il soit stable par changement de base - on va renforcer la définition en conséquence, en exigeant, pour que $f : S' \longrightarrow S$ soit dans $\mathcal{L}^\wedge$, que pour tout T sur S, considérons $f_T : T' = S' \times_S T \longrightarrow T$, $f_T^* : \mathrm{Cat}/T \longrightarrow \mathrm{Cat}/T'$ soit compatible avec les localiseurs W_T^{g}, $W_{T'}^{\mathrm{g}}$. Alors $\mathcal{L}^\wedge$ est (tout comme $\mathcal{L}$) stable par composition et par changement de base, et de plus il contient $\mathcal{L}$. De plus, pour tout S, on a $W_S^{\mathcal{L}} = W_S^{\mathcal{L}^\wedge}$. Ce petit sorite pourrait se faire bien sûr pour tout modéliseur $(\mathcal{M}, W)$. Mais la chose intéressante ici, c'est que $\mathcal{L}^\wedge$ contient l'ensemble Liss_W des flèches W-lisses. (J'ignore s'il est égal.) On aurait donc pu aussi bien partir de l'ensemble de flèches Liss_W, pour définir les localiseurs W_S^{g} comme ms. page 223

$$W_S^{\mathrm{g}} = W_S^{\mathrm{Liss}_W} \qquad \text{au lieu de } W_S^{\mathrm{g}} = W_S^{\mathrm{Isoloc}},$$

ce qui est équivalent.

Commentary · Development (commentators') [ms. 222]

The maximal class of harmless base changes

Grothendieck strengthens his first definition of $\mathcal{L}^\wedge$ because a useful class of changes of base must remain useful after it is pulled back. The strengthened definition does more than supply that stability. It identifies a largest class of tests which leaves the relative weak equivalences unchanged.

Proposition (universal saturation of the tests). [ours] Let $\mathcal{M}$ have pullbacks, let W be a class invariant under isomorphisms of arrows, and let $\mathcal{L}$ contain all isomorphisms and be closed under composition and pullback. Write $E_S = W_S^{\mathcal{L}}$. Define $\widehat{\mathcal{L}}$ to contain $f : S' \to S$ when, for every $T \to S$, pullback along $f_T : S' \times_S T \to T$ sends E_T into $E_{S' \times_S T}$. Then $\widehat{\mathcal{L}}$ contains $\mathcal{L}$, is closed under composition and pullback, and

$$W_S^{\widehat{\mathcal{L}}} = E_S, \qquad \widehat{\widehat{\mathcal{L}}} = \widehat{\mathcal{L}}.$$

Moreover, if $\mathcal{K}$ has the same closure properties and $W_S^{\mathcal{K}} = E_S$ for every S, then $\mathcal{K} \subseteq \widehat{\mathcal{L}}$.

Proof. First take f in $\mathcal{L}$ and u in E_T. To test $f_T^* u$ at $g : R \to S' \times_S T$ in $\mathcal{L}$, pull back u along the composite $f_T g$. Both factors belong to $\mathcal{L}$, so that composite is an admissible test and its pullback of u belongs to W. Thus $\mathcal{L} \subseteq \widehat{\mathcal{L}}$.

For the pullback of a member f of $\widehat{\mathcal{L}}$, every subsequent test is already among the base changes in its definition. This proves pullback stability. For composable members $f : S' \to S$ and $g : S'' \to S'$, each base change of fg factors as the corresponding base change of f followed by one of g. Their pullback functors preserve the respective E classes in succession. This proves composition stability; identities satisfy the same condition immediately.

Since $\mathcal{L} \subseteq \widehat{\mathcal{L}}$, testing against $\widehat{\mathcal{L}}$ can only shrink the resulting class, so $W_S^{\widehat{\mathcal{L}}} \subseteq E_S$. Conversely, for $u \in E_S$ and $f : S' \to S$ in $\widehat{\mathcal{L}}$, the defining condition with $T = S$ gives $f^* u \in E_{S'} \subseteq W_{S'}$. Hence u passes every $\widehat{\mathcal{L}}$ test. This proves equality. Repeating the construction consequently uses exactly the same family E_S, so gives exactly the same class $\widehat{\mathcal{L}}$.

Finally, take $f \in \mathcal{K}$ and any $T \to S$. If $u \in E_T = W_T^{\mathcal{K}}$, its pullback along f_T passes every $\mathcal{K}$ test because such a test followed by f_T is again in $\mathcal{K}$. The pulled-back map therefore belongs to $W^{\mathcal{K}}_{S' \times_S T} = E_{S' \times_S T}$. This is precisely $f \in \widehat{\mathcal{L}}$. □

The proof does not use a model structure, a factorization axiom, or even two-out-of-three for W. Invariance under arrow isomorphisms only ensures independence of the choices of pullbacks and their canonical associativity isomorphisms. If W is weakly saturated, each E_S is weakly saturated because it is an intersection of inverse images of W under functors. We give the maximality and idempotence explicitly as a development of the source's sorite; we did not locate this exact formulation in the checked accounts of smooth functors and derived base change. The label makes no claim of priority. The variance of the first displayed f on manuscript p. 222 is reversed in the edition; the strengthened definition, and the argument above, use $f : S' \to S$.

D'autre part, en travaillant avec les localisés $\mathrm{HOT}^{\circ}_W(S) = (\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}$, le rôle de $\underline{\mathrm{Fib}}_W S$ pour la théorie des $\mathrm{HOT}^{\mathrm{lc}}_W(S)$ est joué ici par $\underline{\mathrm{Prop}}_W(S)$. On voit donc que dans la théorie des W-types d'homotopie relatifs non localement constants, mode HOT°_W, les ensembles de flèches Liss_W et Prop_W jouent un rôle en quelque sorte complémentaire : les flèches $f : S' \longrightarrow S$ de Liss_W sont (pratiquement) celles qui permettent directement de définir

$$f^{*\circ}_W : \mathrm{HOT}^{\circ}_W(S) \longrightarrow \mathrm{HOT}^{\circ}_W(S')$$

comme induit par

$$f^* : (\mathrm{Cat}/S, W_S^{\mathrm{g}}) \longrightarrow (\mathrm{Cat}/S', W_{S'}^{\mathrm{g}}),$$

tandis que les flèches de Liss_W permettent de définir pour tout S dans Cat la sous-catégorie pleine

$$\underline{\mathrm{Prop}}_W S \subseteq \mathrm{Cat}/S,$$

donnant lieu à une équivalence de catégories

$$\begin{cases} (\underline{\mathrm{Prop}}_W S)(W_S^{\mathrm{u}})^{-1} \xrightarrow{\sim} \underbrace{(\mathrm{Cat}/S)(W_S^{\mathrm{g}})^{-1}}_{=\mathrm{HOT}^{0}_W(S)} \\ \text{avec} \\ W_S^{\mathrm{u}} = W_S^{\mathrm{g}} \text{ sur } \underline{\mathrm{Prop}}_W S \end{cases}$$

(ce qui visiblement est le substitut annoncé de "l'axiome de la fibre W-homotopique"). Cette équivalence, qui permet de décrire $\mathrm{Hot}^{\circ}_W(S)$ en termes de $\underline{\mathrm{Prop}}_W S \subseteq \mathrm{Cat}/S$ et d'un localiseur W_S^{u} *stable par* **tout** *changement de base,* permet d'autre part de définir la loi contravariante attendue pour $\mathrm{HOT}^{\circ}_W(S)$. ms. page 224

Commentary · Later developments [ms. 223]

The saturation question has an affirmative answer in Cat

The question in parentheses on p. 222 can be answered using the later characterization of smooth functors. With Grothendieck's localizer W_S^{g} and the universal definition of $\mathcal{L}^\wedge$, one has

$$\mathcal{L}^\wedge = \mathrm{Liss}_W.$$

This is an implication of an established theorem, not a new smoothness criterion claimed by the commentators. A functor is W-smooth exactly when all its base changes preserve W-aspherical functors (Maltsiniotis, 2005, Theorem 1.2(c)).

Here is the precise translation. A functor $v : A \to T$, regarded as a morphism from A/T to the terminal object T/T of the slice, belongs to W_T^{g} exactly when $A/t \to T/t$ belongs to W for every t. Since T/t has a terminal object, this says exactly that every A/t is W-aspherical. Thus it says that v is a W-aspherical functor. If $f : S' \to S$ belongs to $\mathcal{L}^\wedge$, apply the preservation condition after every $T \to S$ to these particular arrows $v : A \to T$. Their pullbacks are again W-aspherical. The cited characterization implies that f is W-smooth. The opposite inclusion is the smooth base-change property used in the source. This proves the equality, including its universal quantifier.

There is an editorial distinction on this page that should not obscure the argument. The sentence introducing the full subcategory $\underline{\mathrm{Prop}}_W S$ repeats "Liss" where its role requires "Prop". The underbrace subsequently prints $\mathrm{HOT}^0_W(S)$, with a zero, although the theory throughout the paragraph is $\mathrm{HOT}^\circ_W(S)$. Both readings are retained and recorded; neither denotes a third relative theory.

ed. 143 D'autre part, quand on échange le rôle de Liss_W et Prop_W, on trouve la théorie complémentaire du $\mathrm{HOT}_W(S)$,

$$\begin{cases} \mathrm{HOT}_W(S) \overset{\text{déf}}{=} (\mathrm{Cat}/S)(W_S^{\mathrm{d}})^{-1} \overset{\sim}{\leftarrow} (\underline{\mathrm{Liss}}_W S)(W_S^{\mathrm{u}})^{-1}, \\ \text{sur } \underline{\mathrm{Liss}}_W S,\ W_S^{\mathrm{d}} = W_S^{\mathrm{u}}, \end{cases}$$

la description de $\mathrm{HOT}_W(S)$ en termes de $\underline{\mathrm{Liss}}_W S$ et de W_S^{u} permet d'en définir le caractère contravariant en S, pour *toute* flèche $f : S' \longrightarrow S$ dans Cat. Tandis que si on veut pouvoir passer aux catégories de fractions sur Cat/S et Cat/S' tout entier (via W_S^{d}, $W_{S'}^{\mathrm{d}}$), il faut se limiter aux changements de base W-propres.

Du coup, me vient la perplexité si cette complémentaire, voire cette dualité, entre deux ensembles de flèches, les 'lisses' et les 'propres', ne serait pas un fait général qui se retrouvait tout au moins dans les catégories de modèles proche de la description de HOT ou d'un Hot_W.

Il se pourrait bien que dans $\mathbf{\Delta}^\wedge$ on ait une dualité parfaite entre ces deux espèces de notions (il faudra que j'examine la chose), peut-être même dans toute catégorie $\mathcal{A}^\wedge$ (mais dans $\mathbf{\Delta}^\wedge$ il y a un automorphisme involutif $K \longmapsto K^\circ$, correspondant au $X \longmapsto X^\circ$ dans Cat …). Et après tout, même dans le contexte des espaces topologiques, les notions de propreté d'une part, d'asphéricité locale d'autre part, pour une flèche $X \longrightarrow Y$, sont bien définies, et pourraient bien s'ajuster comme le font les notions correspondantes dans Cat, et (qui sait) donner lieu à deux versions complémentaires, ou 'duales' l'une de l'autre, d'une notion de type d'homotopie relatif. ms. page 225

N'en ayant pas le cœur net, je suis un peu indécis comment présenter une description axiomatique de la théorie des $\mathrm{Ho}^{\mathcal{L}}_W(S)$, pour une catégorie de modèles générale $(\mathcal{M}, W)$. Dois-je me donner d'emblée

$$\mathrm{Liss}, \mathrm{Prop} \subseteq \mathrm{Fl}(\mathcal{M}),$$

stables par composition, par changement de base, contenant les isomorphismes, et tels que (peut-être)

$$\mathrm{Liss} \cap \mathrm{Prop} \subseteq \mathrm{Fib}_W,$$

mais surtout ils permettent de définir, pour S de $\mathcal{M}$ ms. page 226

$$\begin{cases} W_S^{\mathrm{g}} \overset{\text{déf}}{=} W_S^{\mathrm{Liss}} \subseteq \mathcal{F}(\mathcal{M}/S) \\ W_S^{\mathrm{d}} \overset{\text{déf}}{=} W_S^{\mathrm{Prop}} \subseteq \mathcal{F}(\mathcal{M}/S) \end{cases}$$

avec les propriétés suivantes :

Commentary · Context [ms. 225]

Smoothness and properness depend on a coefficient theory

Grothendieck's proposed pair of classes is meant to encode two ways in which a square admits base change. The geometric names come from proper and smooth base-change theorems in étale cohomology, but their categorical definitions concern the invertibility of specific comparison transformations. This is why exchanging the two classes in the text also exchanges the direction of the relative theory. The comparison with geometry is a source of the formalism; it does not identify a categorical proper functor with a proper continuous map.

Once a coherent indexed category $\mathscr{E} : S \mapsto \mathscr{E}(S)$ is given, an adjunction $f_! \dashv f^*$ produces a base-change mate for each pullback square, and an adjunction $f^* \dashv f_*$ produces its other version. Requiring the relevant mate to be invertible after all further base changes defines a stable class. Thus a meaningful duality has three pieces of data: the coefficient theory, its two adjunctions where they exist, and the chosen variance. Merely knowing which arrows of an ambient category are weak equivalences does not supply all these pieces.

A modern formulation makes precisely this dependence explicit. Anel and Weinberger (2024, Definition 9 and Proposition 10) associate smooth and proper classes to an indexed category over a base with finite limits; these are the largest classes along which the corresponding indexed sums and products exist with stable Beck–Chevalley compatibility. Their examples in Sections 3.2–3.3 include categorical and topological versions. This gives a general setting for the complementarity sought here. It is not a proof that every proposed $(\mathcal{M}, W, \mathcal{L})$ satisfies the localization equivalences of p. 226.

The involution on simplicial sets mentioned by Grothendieck is entirely concrete. Reverse the order on each finite ordinal: for $\alpha : [m] \to [n]$, set $\alpha^{\mathrm{rev}}(i) = n - \alpha(m-i)$. These maps define an involutive automorphism of Δ, hence, by precomposition, an involution $K \mapsto K^{\mathrm{op}}$ of simplicial sets. The identity $N(C^{\mathrm{op}}) = (NC)^{\mathrm{op}}$ is immediate from a string of composable arrows. The corresponding assertion for a general presheaf category requires an appropriate duality of its indexing category; none is part of the present hypotheses. Consequently the speculation about all presheaf categories should remain a programme with specified additional structure, rather than an unqualified theorem or a single well-posed open conjecture.

ⓐ Si $f : S' \longrightarrow S$ est dans Prop (resp. Liss), alors $f^* : \mathcal{M}/S \longrightarrow \mathcal{M}/S'$ transforme W_S^{d} dans $W_{S'}^{\mathrm{d}}$ (resp. W_S^{g} dans $W_{S'}^{\mathrm{g}}$),

ed. 144 ce qui est une pure tautologie, et ne concerne pas encore les relations de Prop avec Liss, mais par contre :

ⓑ
$$\begin{cases} (\underline{\mathrm{Liss}}S)(W_S^{\mathrm{u}})^{-1} \xrightarrow{\sim} (\mathcal{M}/S)(W_S^{\mathrm{d}})^{-1} \texttt{ [est]} \text{ une équivalence} \\ \text{et } W_S^{\mathrm{u}} = W_S^{\mathrm{d}} \text{ sur } \underline{\mathrm{Liss}}S, \end{cases}$$

et dualement

$$\begin{cases} (\underline{\mathrm{Prop}}S)(W_S^{\mathrm{u}})^{-1} \xrightarrow{\sim} (\mathcal{M}/S)(W_S^{\mathrm{g}})^{-1} \texttt{ [est]} \text{ une équivalence} \\ \text{et } W_S^{\mathrm{u}} = W_S^{\mathrm{g}} \text{ sur } \underline{\mathrm{Prop}}S. \end{cases}$$

De plus, il y a lieu d'énoncer ⓒ deux propriétés d'adjonction (duales l'une de l'autre), pour lesquelles je renvoie à VI, théorème-scholie 2, p. 108-113. Il se pose ici la question subsidiaire de simplifier les conditions de validité de la formule d'adjonction, un peu hétéroclites dans loc. cit.

Il se pose d'autre part la question de faire la théorie des lois covariantes $f_!$ et f_* pour $\mathrm{Ho}_W^{\mathrm{lc}}(S)$ et pour $\mathrm{Ho}_W^{\mathrm{Prop}}(S)$, $\mathrm{Ho}_W^{\mathrm{Liss}}(S)$.

ms. page 227

Commentary · Development (commentators') [ms. 226]

Postcomposition already gives a direct image

Grothendieck postpones the direct images because the hard part of his construction has been the inverse image. One part of his expectation follows from the elementary structure of the test classes alone. It is useful to isolate it before bringing in a theory of derived adjunctions.

Proposition (descent of postcomposition). [ours] Under the hypotheses of the universal-saturation proposition, every $f : S' \to S$ induces a functor

$$f_{\#} : (\mathcal{M}/S')[(W^{\mathcal{L}}_{S'})^{-1}] \longrightarrow (\mathcal{M}/S)[(W^{\mathcal{L}}_{S})^{-1}]$$

by postcomposition with f. These functors compose canonically. If f belongs to $\widehat{\mathcal{L}}$, ordinary pullback also descends, and the ordinary adjunction $f_{\#} \dashv f^*$ descends with its unit and counit.

Proof. Take $u : X \to Y$ over S' in $W^{\mathcal{L}}_{S'}$. For a test $T \to S$ in $\mathcal{L}$, its pullback $T' = S' \times_S T \to S'$ again belongs to $\mathcal{L}$. The underlying arrow obtained by pulling back $f_{\#}u$ to T is canonically the underlying arrow obtained by pulling back u to T'. It belongs to W. Thus postcomposition sends the required weak equivalences to weak equivalences and factors through localization. The equalities of ordinary postcomposition give the composition isomorphisms, with their associativity.

When $f \in \widehat{\mathcal{L}}$, its pullback functor is homotopical by definition. Both functors in the ordinary slice adjunction therefore descend. A natural transformation between relative functors descends as well: its naturality equation for an inverted arrow remains valid after that arrow is inverted. Apply this observation to the unit and counit. Their two triangle identities are equalities before localization and remain equalities afterwards, proving the adjunction. □

This argument establishes the direct-image functor for every f, without a functorial replacement. It does not by itself establish an adjunction to an inverse image obtained indirectly from proper or smooth representatives when raw f^* is not homotopical. In that case one must compare the replacement with the ordinary slice adjunction. The source's proposed axioms are designed to make this possible; equality of the three localizers on a full subcategory alone is not the comparison. The earlier Chapter VI commentary, at manuscript pp. 137–145, supplies that comparison in its categorical setting. The proposition above records the additional generality that follows directly from the present definition, without claiming a new priority for descent of adjunctions.

J'ai bon espoir que le foncteur $f_!$ peut être décrit, et son adjonction avec f^* établie, à aussi bon compte dans ce contexte général que dans Cat. Mais je n'ai pas envie de m'y arrêter à présent.

6 Cofibre W-homotopique dans Cat.

Par contre, j'ai envie de regarder ici sous quelles conditions sur le localiseur fondamental W, il existe dans (Cat, W) une théorie duale de celle de la fibre W-homotopique.

La question-clef est, bien sûr, celle d'avoir une bonne prise sur les W-cofibrations. Or je ne connais jusqu'à présent *que* les immersions ouvertes de Dwyer, les immersions fermées de Dwyer, et ce qui s'en déduit par composition etc. Le travail de Thomason peut motiver l'espoir que la connaissance de ces immersions-là, qui semblent ici jouer un rôle similaire (dual) à celui des morphismes W-parfaits, suffira pour faire une bonne théorie des 'cofibres W-homotopiques'.

||| Je suppose dans toute cette section que W satisfait W(7 bis) (axiome des carrés cocartésiens), pour avoir des bonnes propriétés des immersions ouvertes de Dwyer ([124]), et plus généralement, pour les W-cofibrations.

ed. 145 Il se pose donc la question de la factorisation duale d'une flèche $f : X \longrightarrow Y$ de Cat en ms. page 228

$$X \xrightarrow[i_f]{W\text{-cofibration}} \tilde{Y} \xrightarrow[p_f]{\in W} Y.$$

124.

a) Ce sont des W-cofibrations (résulte de b) et c)).

b) Si

Commentary · Exposition [ms. 227]

Why a longer categorical interval is needed

The passage from fibres to cofibres begins with a manageable class of maps along which strict gluing has the correct homotopical effect. For an inclusion $A \hookrightarrow B$, the Dwyer condition provides exactly such control. In its original form it says that A is a full sieve, and that its inclusion in the cosieve $Z \subseteq B$ generated by A has a right adjoint $r : Z \to A$ restricting to the identity of A. The counit $ir \to \mathrm{id}_Z$ is a categorical deformation of the neighbourhood Z onto A. This accounts for the usefulness of Dwyer maps in gluing (Thomason, 1980, Definition 4.1 and Proposition 4.3).

For $I = \Phi_n$, the endpoints 0 and $2n$ form a sieve. Their generated cosieve consists of the four vertices $0, 1, 2n-1, 2n$ when $n \geq 2$. It is the disjoint union of two arrows, $0 \to 1$ and $2n \to 2n-1$. Send the first arrow to the endpoint 0 and the second to $2n$. This is the required right adjoint. For $n = 1$, the generated cosieve is the whole cospan $0 \to 1 \leftarrow 2$. A right adjoint to the inclusion of the two endpoints would have to assign to 1 an endpoint receiving all arrows from either endpoint, which is impossible. The lower bound $n \geq 2$ therefore has an exact purpose: the two deformation neighbourhoods must not meet.

Taking a product with X preserves the sieve and cosieve conditions and the adjunction, now with right adjoint $\mathrm{id}_X \times r$. The nerve of Φ_n realizes to a subdivided interval, so $X \times I \to X$ is an ordinary weak equivalence. The stronger description as W_ω-perfect records the corresponding universal statement in the categorical theory. Thus the factorization of the codiagonal has both the disjoint endpoint inclusion needed for the mapping cylinder and the weakly contractible cylinder direction needed for its retraction.

The more general fibred base change asserted in the source has a similarly explicit proof. Pull a Dwyer inclusion $A \hookrightarrow B$ back along a Grothendieck fibration $E \to B$. Over its generated cosieve Z, lift each counit arrow $r(b) \to b$ cartesianly to an arrow $r'(e) \to e$. Choose identity lifts over A. The cartesian universal property defines r' on morphisms, with functoriality following from uniqueness. For an object a' over A, an arrow $a' \to e$ factors uniquely through that cartesian lift: its image in B first factors uniquely through the counit by the original adjunction. Hence r' is right adjoint to the pulled-back inclusion. Moreover every object over Z receives such a lifted arrow from an object over A, so the pulled-back cosieve is exactly the generated one. This proves the asserted stability and explains why an arbitrary base change, without a lifting condition, was not claimed.

The terminology must remain precise. Grothendieck's W-cofibration is a map whose cobase changes preserve the relevant W-equivalences. It is a homotopical gluing condition, not the definition by a lifting property in Thomason's model structure. Dwyer maps supply examples; one should not identify the two classes. Nor should one assert that every Thomason cofibration is a Dwyer map: Cisinski (1999, pp. 227–231) gave a counterexample to closure of the original Dwyer maps under retracts and repaired the properness argument. That correction leaves the explicit mapping-cylinder construction below intact.

Les résultats de Thomason assurent que ça existe, avec i une immersion ouverte de Dwyer, et p une W-fibration - du moins dans le cas $W = W_\infty$. Je vais essayer de le voir directement, en calquant dualement la construction de Cartan-Serre.

Commentary · Exposition [ms. 228]

An interval with room at both ends

Grothendieck wants to replace every map by an inclusion along which gluing respects the chosen weak equivalences. This will make a homotopy cofibre accessible to the same relative methods that have already produced homotopy fibres. The interval is therefore being chosen for a relative property of its two endpoints. Its contractibility supplies only the other half of the construction.

We recall precisely the relevant meaning of cofibration. A map $i : A \to B$ is a W-cofibration when the pushout functor

$$A\backslash\mathcal{M} \longrightarrow B\backslash\mathcal{M}, \qquad (A \to T) \longmapsto (B \to B \amalg_A T),$$

preserves morphisms belonging to W. This is the definition on manuscript pp. 109–110 (Grothendieck, 1990–1991d, Definition 2(c)). It includes every subsequent change of the object under A. Associativity of pushouts immediately makes this class stable under pushout and composition. A Quillen cofibration is defined by a lifting property; the relation between the two notions requires a properness or gluing theorem. In the present argument, the required gluing property is part of the definition of Cof_W.

The geometry of a Dwyer map provides a tractable supply of such inclusions. Write an open immersion as a full sieve $A \subseteq B$: an arrow of B whose target belongs to A has its source in A. It is Dwyer when the inclusion of A into its generated cosieve $Z_B A$ has a right adjoint r with $r|_A = 1_A$. The counit gives arrows $r(b) \to b$, coherently in b. Thus objects immediately outside A have canonical points of attachment in A; this is the categorical structure used when the inclusion is pushed out (Thomason, 1980, Definition 4.1).

For the fence $I = \Phi_n$, the cosieve generated by its endpoints is

$$\{0 \longrightarrow 1\} \amalg \{2n \longrightarrow 2n-1\}, \qquad n \geq 2.$$

The right adjoint sends the first component to 0 and the second to $2n$. The unit is the identity, and the two displayed arrows give its counit. Taking its product with X gives the required adjoint for $X \amalg X \subseteq X \times I$. The same argument also explains the lower bound on n. For $n = 1$, the vertex 1 receives arrows from both 0 and 2. If a right adjoint existed, its value at 1 would have to represent the functor on the discrete two-point category which takes both objects to a singleton. Neither object represents that functor. The two endpoint neighbourhoods need a vertex between them before the Dwyer condition can hold.

The notation W_ω belongs to the earlier construction of a minimal localizer from saturation, the relative Theorem A condition, and the terminal-object axiom (Grothendieck, 1990–1991f, manuscript p. 14). The elementary contraction of this finite fence already belongs to that small class. One can remove the outermost endpoint, then its newly terminal neighbour, and repeat; each removal is witnessed by an adjunction between the remaining posets. Products preserve these adjunctions. Consequently $X \times I \to X$ is a homotopy equivalence for every X, and remains one after an arbitrary change of its base X.

Il s'agit donc d'abord de factoriser

$$X \amalg X \xrightarrow{\text{codiag}_X} X,$$

et pour ceci je choisis un intervalle fixe

$$I \;=\; \Phi_n \;=\; (0 \longrightarrow 1 \longleftarrow 2 \longrightarrow \cdots \longrightarrow 2n-1 \longleftarrow 2n)$$

avec

$$n \geq 2$$

de sorte que l'inclusion

$$\varepsilon = \text{catégorie discrète } \{0,1\} \xrightarrow{\partial} I, \qquad \partial(0) = 0, \quad \partial(1) = 2n,$$

soit une immersion ouverte de Dwyer. Cette propriété est stable par changement de base Cat-fibrant, en particulier en prenant un produit

$$\underbrace{X \times \varepsilon}_{\simeq X \coprod X} \xrightarrow{i_X} X \times I,$$

ed. 146 d'autre part

$$\begin{array}{ccc} U & \xrightarrow{\;i\;} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U' & \xrightarrow{\;i'\;} & X' \end{array}$$

ms. page 229

avec i immersion ouverte de Dwyer, alors

$$\begin{aligned} i \in W \quad &\Longleftrightarrow \quad i' \in W \\ p \in W \quad &\Longleftrightarrow \quad q \in W, \end{aligned}$$

j'ai donc besoin de W(7 bis) *et* même de W(7) pour la deuxième implication.

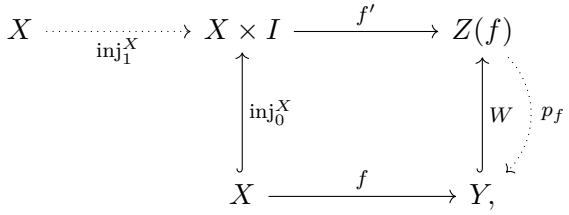
c) Les immersions ouvertes de Dwyer [sont] stables par cochangement de base et par composition, contiennent les isomorphismes.

$$X \times I \xrightarrow{\mathrm{pr}_1} X$$

est W_ω-parfait, d'où la factoriastion cherchée

$$X \amalg X \xrightarrow{i_X} X \times I \xrightarrow{p_X = \mathrm{pr}_1^X} X$$

- il n'y a pas plus simple! Comment l'utiliser pour construire la factorisation d'une flèche $f : X \longrightarrow Y$ quelconque, en dualisant les constructions relatives aux espaces de chemins? On définit $Z(f)$ par le carré cocartésien dans Cat



on a

$$Z(f) \xrightarrow{\alpha} Y \qquad \text{induit par } \mathrm{pr}_1 : X \times I \longrightarrow X,$$

et la factorisation cherchée de f est

$$X \xrightarrow{i_f = f' \circ \mathrm{inj}_1^X} Z(f) \xrightarrow{p_f} Y.$$

Commentary · Development (commentators') [ms. 228]

Conservative gluing is an additional hypothesis

The continued footnote 124 is stronger than the gluing theorem usually used in homotopy theory. It prints two biconditionals for a Dwyer pushout. Its stated hypotheses include the conservative axioms W(7 bis), and for the second assertion W(7), inherited from earlier pages (Grothendieck, 1990–1991d, Chapter XII, manuscript pp. 76–78). The conditional assertion should not be silently replaced by a theorem about W_∞. In fact the reverse implications fail in the ordinary homotopy theory, even for finite posets and an open cover.

Example (a conservative converse fails). [ours] There is a pushout square of finite posets

$$\begin{array}{ccc} A & \xhookrightarrow{i} & B \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ C & \xhookrightarrow[i']{} & D \end{array}$$

in which i and p are Dwyer maps, i' and q belong to W_∞, and neither i nor p belongs to W_∞.

Construction and proof. Start with a connected finite acyclic complex K with nontrivial fundamental group. For completeness, the presentation complex with two generators and two relations

$$\langle a, b \mid a^5 b^{-3},\ b^3 (ab)^{-2} \rangle$$

has cellular boundary matrix $\left(\begin{smallmatrix} 5 & -2 \\ -3 & 1 \end{smallmatrix}\right)$, of determinant -1, so its reduced integral homology vanishes. Sending a to (12345) and b to (253) defines a nontrivial homomorphism of its fundamental group to A_5: $a^5 = b^3 = (ab)^2 = 1$, since $ab = (12)(45)$. Hence it is not contractible. This is the acyclic complex of Hatcher (2002, Example 2.38). Choose a finite simplicial model of the same homotopy type. Such a model can be obtained by replacing the two attaching maps by simplicial maps between subdivided finite graphs and using their finite triangulated mapping cylinders before adjoining the two discs.

Let $P(L)$ be the poset of nonempty simplices of a finite simplicial complex L, ordered by inclusion. Set

$$A = P(K), \quad B = P(C_+ K), \quad C = P(C_- K), \quad D = P(\Sigma K).$$

The two cones are subcomplexes of the suspension and meet in K. Their face posets are sieves, their union is D, and there are no arrows between the two complements. Their ordinary pushout is therefore D. For $A \hookrightarrow B$, the cosieve generated by A consists of all simplices except the one-vertex apex simplex. On this cosieve, deleting the apex defines a right adjoint to the inclusion of A: a simplex of K is a face of σ if and only if it is a face of σ with the apex removed. This verifies the Dwyer condition. The proof for $A \hookrightarrow C$ is identical.

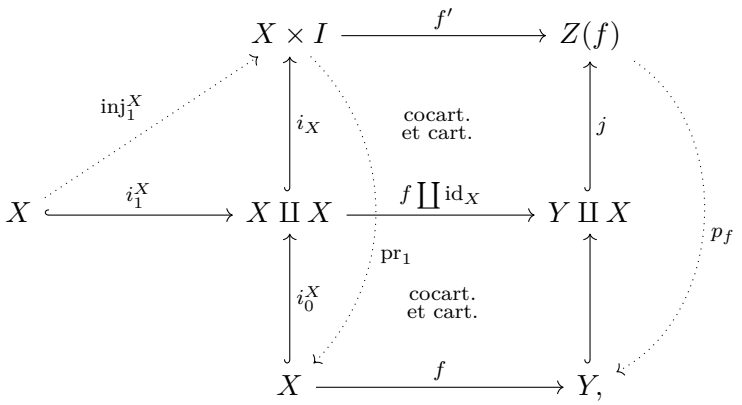
The realization of $NP(L)$ is the barycentric subdivision of L. Both B and C are therefore contractible. The suspension of a connected space is simply connected, and its reduced homology is the shifted reduced homology of the space. Thus ΣK is simply connected and acyclic. If it had a first nonzero homotopy group, the Hurewicz theorem would identify that group with a nonzero homology group. It has none, and Whitehead's theorem makes ΣK contractible. Consequently i' and q are weak equivalences, whereas i and p cannot be, because $|NA|$ has nontrivial π_1. □

This application of a classical acyclic example is included to expose the strength of the assumed axiom, not as a claim of novelty. We checked the original Dwyer pushout theorem, its retract correction, and the later categorical refinement; they assert preservation, not these converse implications. The earlier W(7 bis), formulated for a union of open sieves, already fails on the displayed union. The claim in Chapter XII, manuscript p. 78, that W_∞ satisfies the stronger W(7), cannot supply it. The present proof of the mapping cylinder and Proposition 16 uses only the forward gluing implications, which are valid for W_∞. Keeping this distinction allows the construction to survive with its natural hypotheses.

Je dis que

$$\begin{cases} i_f \text{ est immersion ouverte de Dwyer, donc une } W\text{-cofibration.} \\ p_f \in W. \end{cases}$$

On voit sur le diagramme plus complet ms. page 230



ed. 147

$$i_f = f' \circ \mathrm{inj}_1^X = f' i_X i_1^X = j \underbrace{(f \amalg \mathrm{id}_X) i_1^X}_{\substack{\text{inclusion canonique} \\ \text{de } X \text{ dans} \\ Y \coprod X}} ,$$

c'est [une] immersion ouverte de Dwyer, comme composé de $Y \longrightarrow Y \amalg X$ et $j : Y \amalg X \longrightarrow Z(f)$ qui le sont (j obtenu par cochangement de base à partir de i_X, qui est immersion ouverte de Dwyer).

D'autre part, $X \xrightarrow{\mathrm{inj}_0^X} X \times I$ est [une] immersion ouverte de Dwyer et dans W, donc coüniversellement dans W, donc $Y \longrightarrow Z(f)$ est elle aussi dans W, donc aussi la rétraction p_f.

Commentary · Examples [ms. 230]

A cylinder and a circle in finite categories

The mapping cylinder puts the preceding two properties to work simultaneously. Its construction is worth following through a small example, because the example also shows what a homotopy cofibre retains that a strict categorical quotient can lose. First write its two structural maps without the changing names in the display:

$$Z(f) = (X \times I) \amalg_{X\times\{0\}} Y, \qquad s_f : Y \longrightarrow Z(f), \qquad p_f : Z(f) \longrightarrow Y.$$

The map p_f is induced by $f \operatorname{pr}_1$ on $X \times I$ and by 1_Y on Y. Hence $p_f s_f = 1_Y$, while the free endpoint defines $i_f : X \to Z(f)$ with $p_f i_f = f$. These equations also show that the construction is functorial in a commutative square of functors: the two maps on the pieces agree on the identified endpoint and therefore induce a unique map on their pushout.

To see that the free endpoint is still a Dwyer inclusion, separate both endpoints before making the attachment. The pushout of $X \amalg X \hookrightarrow X \times I$ along $f \amalg 1_X$ is

$$j : Y \amalg X \longrightarrow Z(f).$$

The desired inclusion is j composed with the X-summand inclusion. Both are Dwyer maps. The printed sentence has $Y \to Y \amalg X$ in this position; its preceding formula and its source X specify the intended summand. The other endpoint inclusion $X \to X \times I$ is a weak equivalence, and its pushout s_f is one by the forward gluing property. Two-out-of-three applied to $p_f s_f = 1_Y$ proves $p_f \in W$. No fibration property of p_f is required for this finite construction.

Now take $X = \{a, b\}$ discrete, $Y = \{y\}$, and let f be the unique functor. Choose $I = \Phi_2$. The category $Z(f)$ has the nine objects

$$y, \quad a_1, a_2, a_3, a_4, \quad b_1, b_2, b_3, b_4.$$

Its nonidentity arrows are exactly, for $x = a, b$,

$$y \longrightarrow x_1 \longleftarrow x_2 \longrightarrow x_3 \longleftarrow x_4.$$

There are no composable pairs of nonidentity arrows. Thus its nerve has precisely these vertices and eight nondegenerate edges, and no higher nondegenerate simplices. Its realization is a tree with two arms. The contraction to y verifies directly that p_f is a weak homotopy equivalence. The free endpoints a_4, b_4 form a Dwyer subcategory: their generated cosieve consists of the two separate arrows $a_4 \to a_3$ and $b_4 \to b_3$.

We may consequently form the homotopy cofibre by the strict pushout

$$C(f) = Z(f) \amalg_{\{a_4,b_4\}} \{t\}.$$

It has eight objects and eight nonidentity arrows. Its nerve is the graph obtained from the tree by identifying its two free endpoints. This graph is an eight-edge circle. In particular,

$$H_1(NC(f); \mathbb{Z}) = \mathbb{Z}, \qquad \pi_1(NC(f), t) \cong \mathbb{Z}.$$

The direct pushout $\{y\} \amalg_{\{a,b\}} \{t\}$ in Cat is instead the terminal category: both maps identify the two objects at once. The cylinder has retained the two paths by which the gluing takes place. This is the finite categorical model of the familiar homotopy pushout $* \amalg^h_{S^0} * \simeq S^1$, and it is the reason the question on p. 232 requires a proof.

Commentary · Later developments [ms. 230]

What the appeal to Thomason supplies

Grothendieck's appeal to Thomason supplies a stronger factorization in the ordinary spatial case. The cellular construction in the proof of the model-category theorem factors a functor as a Dwyer inclusion followed by a trivial fibration for the Thomason model structure (Thomason, 1980, proof of CM5 and Theorem 4.9). The cylinder just constructed gives a finite and transparent substitute for the weaker factorization needed here.

A later correction matters when using that historical reference. Cisinski (1999) exhibited a retract of a Dwyer map which is a Thomason cofibration but is not itself Dwyer, and repaired the properness argument. Thomason's assertion that all cofibrations are Dwyer therefore cannot be used to justify the first paragraph. The particular cellular factorization survives, as do the pushout and composition properties used in this passage. Grothendieck's explicit cylinder is also independent of closure under retracts. The distinction protects exactly the argument being made, without changing its conclusion.

D'autre part, la proposition duale de

$$\underline{\mathrm{Fib}}_W S \cap W_S = W_{S\,\mathrm{univ}}^{\mathrm{u}}$$

est valable pour les W-cofibrations :

Proposition 16 ([125]). *Sur la sous-catégorie pleine* $\underline{\mathrm{Cofib}}_W(S)$ *de* $S\backslash\mathrm{Cat}$*, le localiseur* ${}_SW$ *est égal au* $({}_SW)^{\mathrm{univ}}$ *(*W*-équivalences couniverselles), i.e. on a ceci : Soient* ms. page 231

$$S \longrightarrow X, \qquad S \longrightarrow Y$$

des W*-cofibrations, et soit* $f : X \longrightarrow Y$ *un morphisme de* $S\backslash\mathrm{Cat}$ *tel que* $f \in W$*. Alors pour tout cochangement de base* $S \longrightarrow S'$*, la flèche correspondante*

$$f' : X' = X \amalg_S S' \longrightarrow Y' = Y \amalg_S S'$$

est dans W*.*

Commentary · Exposition [ms. 230]

The mapping cylinder, with the two endpoints distinguished

The construction now turns the endpoint geometry into a factorization of an arbitrary functor. Let $a_0, a_1 : X \to X \times I$ denote the two endpoint inclusions and form

$$Z(f) = Y \amalg_{X,a_0} (X \times I).$$

Write $k : Y \to Z(f)$ and $f' : X \times I \to Z(f)$ for the coprojections. The maps $f\,\mathrm{pr}_X$ and id_Y agree on the identified copy of X, hence induce a unique $p_f : Z(f) \to Y$. Put $i_f = f'a_1$. Then $p_f i_f = f$ and $p_f k = \mathrm{id}_Y$. The displayed α in the source is this same p_f; the projection used to define it is followed by f on the cylinder summand.

The endpoint a_0 is a Dwyer map and a weak equivalence. Its pushout k is consequently a weak equivalence by the forward gluing property. Two-out-of-three applied to $p_f k = \mathrm{id}_Y$ gives $p_f \in W$. To prove the other assertion, use both endpoints at once. The map

$$j : Y \amalg X \longrightarrow Z(f)$$

is the pushout of $a_0 \amalg a_1 : X \amalg X \to X \times I$ along $f \amalg \mathrm{id}_X$, hence is Dwyer. The inclusion of the second summand $X \hookrightarrow Y \amalg X$ is Dwyer, and its composite with j is i_f. Closure under composition proves the required assertion. The edition says $Y \to Y \amalg X$ in this sentence; the formula immediately preceding it requires the second summand X.

The construction is functorial in f. A commutative square $bf = f'a$ gives maps b on the Y summand and $a \times \mathrm{id}_I$ on the cylinder summand, compatible on the attaching copy of X. The universal property gives the induced map of mapping cylinders, and uniqueness proves compatibility with composition and identities. This supplies an actual replacement functor on the arrow category, not merely objectwise existence of a factorization. Nothing in this argument says that the displayed p_f is a Thomason fibration; the source only claims $p_f \in W$ for this explicit factorization. The stronger model-category factorization mentioned earlier is a different choice.

A small computation shows what the construction buys. Let X be the discrete two-object category and let $f : X \to e$ be the unique functor. With $I = \Phi_2$, $Z(f)$ is made of two four-edge intervals sharing their attached endpoint y. Form the further pushout $e \amalg_X Z(f)$, identifying their

125. N'a rien à voir avec Cat - une chose générale quand il y a assez de W-cofibrations.

two free endpoints to a second point z. The resulting poset has eight vertices: y, z, and three interior vertices on each arm. Every nonidentity arrow runs from an even vertex to an adjacent odd vertex, so there are no nondegenerate simplices of its nerve in dimensions above one. Its realization is the graph of two four-edge paths from y to z, hence a circle. Its cellular differential has rank seven, giving $H_1 = \mathbf{Z}$ and $H_q = 0$ for $q > 1$; a spanning tree leaves one generator of π_1. This is the homotopy cofiber of the collapse $S^0 \to *$, obtained by an ordinary categorical pushout only after the mapping-cylinder replacement.

Démonstration. On factorise $S \longrightarrow S'$ en

$$S \xrightarrow{i} S'_0 \xrightarrow{j} S',$$

avec i immersion ouverte de Dwyer (donc W-cofibration), et $j \in W$. On a donc le diagramme à carrés cocartésiens

$$\begin{array}{ccccc} S & \longrightarrow & X & \xrightarrow{f} & Y \\ {\scriptstyle i}\downarrow & & \downarrow & & \downarrow \\ S'_0 & \longrightarrow & X'_0 & \xrightarrow[W]{f'_0} & Y'_0 \\ {\scriptstyle \in W}\downarrow & & \downarrow{\scriptstyle W} & & \downarrow{\scriptstyle W} \\ S' & \longrightarrow & X' & \xrightarrow{f'} & Y'. \end{array}$$

Comme i est une W-cofibration et $f \in W$, on a $f'_0 \in W$. D'autre part, comme $S \longrightarrow X$, donc
ed. 148 $S'_0 \longrightarrow X'_0$ est une W-cofibration, $X'_0 \longrightarrow X'$ est dans W, et pour la même raison, $Y'_0 \longrightarrow Y$ ms. page 232
l'est aussi. Donc f' l'est aussi.

NB Ici, c'est un argument formel, valable (sous forme duale) dans tout $(\mathcal{M}, W)$ où on a un énoncé de factorisation pour une flèche quelconque $f : X \longrightarrow Y$ en $X \xrightarrow{i_f} \bar{X} \xrightarrow{\alpha} Y$, avec $i_f \in W$ et α une W-fibration. Alors, si X, Y sont W-fibrés sur S, toute S-flèche de X dans Y qui est dans W, est dans W^{u}_S. Et dualement, si toute flèche $X \longrightarrow Y$ se factorise en $X \longrightarrow \tilde{Y} \longrightarrow Y$, avec $X \longrightarrow \tilde{Y}$ une W-cofibration et $\tilde{Y} \longrightarrow Y$ dans W.

Commentary · Exposition [ms. 232]

Weak equivalences between objects under a fixed base

Proposition 16 provides the independence of replacement that a relative homotopy construction needs. A map under S can be a weak equivalence while its image after a badly behaved attachment is not. The proposition locates a class of objects under S for which every attachment behaves correctly: both structure maps from S must be W-cofibrations. It does not require the map between those objects to be a cofibration.

The argument can be expressed in any category with the pushouts in question. Let $a : S \to X$ and $b : S \to Y$ be W-cofibrations, and let $f : X \to Y$ satisfy $fa = b$ and $f \in W$. To change S into an arbitrary T, factor the change of base as

$$S \xrightarrow{k} T_0 \xrightarrow{q} T, \qquad k \in \mathrm{Cof}_W, \quad q \in W.$$

Set $X_0 = X \amalg_S T_0$, $Y_0 = Y \amalg_S T_0$, and set $X_T = X \amalg_S T$, $Y_T = Y \amalg_S T$. The map $f_0 : X_0 \to Y_0$ is in W because k is a W-cofibration and it pushes out the map f in $S\backslash\mathcal{M}$. On the other hand, $T_0 \to X_0$ and $T_0 \to Y_0$ are pushouts of a and b, so they are W-cofibrations. Pushing out q along them gives weak equivalences

$$u : X_0 \longrightarrow X_T, \qquad v : Y_0 \longrightarrow Y_T.$$

The equality $f_T u = v f_0$ now proves $f_T \in W$ by two-out-of-three. These are three separate applications of the cofibration condition, each with its own fixed base. They account for every marked W in the displayed proof.

There is a useful immediate consequence. If $a : S \to X$ is both a W-cofibration and a weak equivalence, apply the proposition to the two objects $1_S : S \to S$ and $a : S \to X$ of $S\backslash\mathcal{M}$. Every

pushout of a belongs to W. Thus, once these factorizations exist, a trivial W-cofibration is exactly a couniversal weak equivalence. This is the abstract version of the special Dwyer fact used to construct the cylinder. The order of proof is sound: the special Dwyer statement constructs enough cofibrations first, and Proposition 16 then gives the general statement.

The dual formulation on p. 232 is obtained by taking opposite categories throughout. A weak equivalence between two W-fibrations over S survives every pullback along $T \to S$. The symmetry is a symmetry of the relative arguments. It does not assert that every object of a slice or coslice is already suitable for calculating a derived change of base.

Il me vient ici une (petite) inquiétude : ai-je au moins prouvé que si

$$S \longrightarrow X$$

est une W-cofibration, alors pour toute flèche $S' \longrightarrow S$ dans Cat, le carré cocartésien dans Cat

$$\begin{array}{ccc} S & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & X' \end{array}$$

est aussi W-cocartésien ([126])? Autrement, à quoi ce carré-là serait-il bon?! Pour le prouver, factorisons encore $S \longrightarrow S'$ en

ms. page 233

$$\underset{\substack{\text{immersion}\\ \text{ouverte de}\\ \text{Dwyer}}}{S \hookrightarrow} \; S'_0 \underset{\in W}{\longrightarrow} S',$$

d'où le diagramme de 2 carrés cocartésiens

$$\begin{array}{ccc} S & \xrightarrow{\mathrm{Cof}_W} & X \\ {\scriptstyle D_W}\downarrow & & \downarrow{\scriptstyle D_W} \\ S'_0 & \xrightarrow{\mathrm{Cof}_W} & X'_0 \\ {\scriptstyle \in W}\downarrow & & \downarrow{\scriptstyle \in W} \\ S' & \xrightarrow{\mathrm{Cof}_W} & X', \end{array}$$

ed. 149 où $X'_0 \longrightarrow X'$ est $\in W$ d'après l'hypothèse sur X sous S. Il faut donc vérifier :

a) Un carré cocartésien construit sur une flèche qui est `[une]` immersion de Dwyer, est W-cocartésien.

b) Un carré commutatif dans lequel deux côtés opposés sont dans W, est W-cocartésien.

c) Le composé de deux carrés W-cocartésiens est W-cocartésien.

Commentary · Exposition [ms. 232]

Why Proposition 16 is a formal gluing lemma

Grothendieck needs weak equivalences between objects under S to survive an arbitrary change of the attaching object. The hypothesis that both objects are W-cofibrant under S is exactly what controls the two comparisons introduced by a factorization of that change. It cannot be replaced by a statement about the underlying objects alone.

Assume an ambient category has pushouts, a weakly saturated class W, and a class of W-cofibrations stable under pushout, with the property that pushout along such a map preserves W-equivalences. Suppose every map factors as a W-cofibration followed by a member of W. Let

126. dans le sens ancien, spécial à Cat (déf. 3, p. 131). Pour le comparer avec la théorie générale, cf. prop. 20, p. 262.

$S \to X$ and $S \to Y$ be W-cofibrations, and let $f : X \to Y$ be a W-equivalence under S. Factor the chosen $S \to S'$ as

$$S \xrightarrow{i} S'_0 \xrightarrow{j} S', \qquad i \in \mathrm{Cof}_W, \quad j \in W.$$

Put $X'_0 = X \amalg_S S'_0$ and $Y'_0 = Y \amalg_S S'_0$. The map $X \to X'_0$ is a pushout of i, hence a W-cofibration. Pushing out f along this map gives $f'_0 : X'_0 \to Y'_0$ in W; its target is $Y \amalg_X X'_0 \cong Y \amalg_S S'_0$. The maps $S'_0 \to X'_0$ and $S'_0 \to Y'_0$ are pushouts of the two original cofibrations. Pushing out j along them gives weak equivalences $v_X : X'_0 \to X'$ and $v_Y : Y'_0 \to Y'$. In the square

$$f' v_X = v_Y f'_0,$$

the right-hand composite belongs to W, and v_X belongs to W. Two-out-of-three proves $f' \in W$.

This is the whole argument. It uses neither a lifting axiom nor the conservative converses of footnote 124. It also identifies the dual hypotheses precisely: pullbacks, fibrations whose pullback functors preserve weak equivalences, stability under pullback, and a factorization as a weak equivalence followed by a fibration. Under those hypotheses a weak equivalence between fibrations over the same base is universally a weak equivalence over that base. Thus the parallel with the preceding fibre theory is mathematical, not just terminological.

The small slips in the final prose are harmless once the diagram is read: the comparison is $Y'_0 \to Y'$, not $Y'_0 \to Y$, and the subsequent cobase-change arrow is $S \to S'$, not $S' \to S$. Both are printed slips and remain in the source text. They are not gaps in Proposition 16.

Commentary · Exposition [ms. 233]

Comparing the two meanings of homotopy pushout

Grothendieck now asks whether the replacement method agrees with the earlier construction special to small categories. The earlier definition starts with a span $X_1 \leftarrow X_0 \to X_2$, integrates that diagram into a category $\mathcal{X}$, and calls a cocone to Y homotopy cocartesian when its comparison $\mathcal{X} \to Y$ belongs to W. The new method starts with an ordinary pushout and asks whether a W-cofibration makes that pushout homotopically correct. The two descriptions have different starting data; their agreement is the content of the corollary on p. 239, and later of Proposition 20.

The proof has three stages, and keeping their dependence visible avoids a misleading impression of circularity. A pushout along a Dwyer inclusion is first reduced to a pushout whose two incoming maps are Dwyer inclusions, followed by a square with two opposite sides in W. In the first square, both categories embed as open subcategories of their union, so the earlier open-cover calculation applies. In the second, the definition of W-cofibration supplies the second weak equivalence. The only remaining operation is composition of homotopy-cocartesian squares.

That composition rule is then proved independently, by replacing the comparison maps with inclusions between categories of elements. Its proof uses restrictions of a single diagram over a six-object poset and a union of closed subcategories. It therefore does not presuppose that all Dwyer pushouts have already been identified as homotopy pushouts. Finally, an arbitrary W-cofibration is handled by factoring the other incoming map through a Dwyer inclusion and a weak equivalence. The two resulting squares fall under the established cases.

The assertion about two opposite weak equivalences has an especially short explanation. In the source's notation, if $i_2 : X_0 \to X_2$ is in W, the other leg $j_1 : X_1 \to \mathcal{X}$ of the integrated span is in W. The cocone satisfies $\varphi j_1 = f_1$. If f_1 also belongs to W, two-out-of-three gives $\varphi \in W$. This argument concerns the actual coherent square, including its commutativity. It will reappear in the general formalism as the rule that a homotopy pushout of an equivalence is an equivalence.

Admettons provisoirement b) et c), et prouvons a), pour un carré cocartésien

$$\begin{array}{ccc} U & \xhookrightarrow{\ i\ } & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U' & \xhookrightarrow{\ i'\ } & X', \end{array}$$

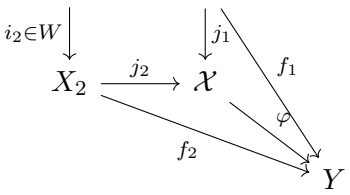
où i est immersion ouverte de Dwyer. On procède encore comme tantôt, en factorisant p en $U \xhookrightarrow{j} U'_0 \xrightarrow{p'} U'$, où j est [une] immersion ouverte de Dwyer, $p' \in W$. On a donc un ms. page 234

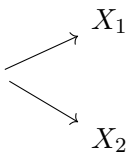
diagramme de deux carrés cocartésiens accolés

$$\begin{array}{ccc} U & \xrightarrow{i\in D_W} & X \\ {\scriptstyle D_W}\downarrow & & \downarrow \\ U'_0 & \xrightarrow{D_W} & X'_0 \\ {\scriptstyle p'\in W}\downarrow & & \downarrow{\scriptstyle q'\in W} \\ U' & \longrightarrow & X', \end{array}$$

où le premier carré est formé d'immersions ouverts. On sait alors qu'il est W-cocartésien. Dans le deuxième, on a $p' \in W$, donc $q' \in W$ puisque i est une immersion de Dwyer, donc une W-cofibration. Donc il est également W-cocartésien. On conclut alors grâce à c) ci-dessus.

Prouvons b), pour $\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1, \\ {\scriptstyle i_2\in W}\downarrow & & \downarrow{\scriptstyle f_1\in W} \\ X_2 & \xrightarrow{f_2} & Y \end{array}$ en l'insérant dans $\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1, \\ {\scriptstyle i_2\in W}\downarrow & & \downarrow{\scriptstyle j_1} \\ X_2 & \xrightarrow{j_2} & \mathcal{X} \end{array}$ où



$\mathcal{X} = \int X_0$  .

Comme $i_2 \in W$, on sait qu'il en résulte $j_1 \in W$, et comme $f_1 \in W$, il en résulte que $\varphi \in W$, q.e.d.

Commentary · Exposition [ms. 234]

Completing the pasting step

At the end of manuscript page 234, the proof that a pushout along a W-cofibration is W-cocartesian still invokes the composition of homotopy pushout squares. This is the deferred item c), not a defect in item b), whose proof has just been given. The composition statement is a standard consequence of coherent homotopy colimits; in derivator language it is the composition part of Groth (2013, Proposition 3.13, published version). We spell out the argument to make clear which compatibility is required.

Consider a coherent commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \\ \downarrow & & \downarrow \\ E & \longrightarrow & F. \end{array}$$

Suppose each small square is a homotopy pushout. Its two universal properties give

$$D \simeq B \amalg^h_A C, \qquad F \simeq D \amalg^h_C E.$$

For any test object T, write $M_U = \mathrm{Map}(U, T)$. These mapping spaces are derived mapping spaces, so that homotopy colimits become homotopy limits. Then

$$\begin{aligned} \mathrm{Map}(F, T) &\simeq (M_B \times^h_{M_A} M_C) \times^h_{M_C} M_E \\ &\simeq M_B \times^h_{M_A} M_E. \end{aligned}$$

The second equivalence is the associative law for homotopy pullbacks: both sides are limits of the same diagram after eliminating the repeated M_C. The natural equivalences identify the maps to M_B

and M_E and their common homotopy over M_A. The mapping-space universal property therefore identifies F with $B \amalg^h_A E$, including its cocone. This proves that the outside rectangle is a homotopy pushout.

For Grothendieck's W-cocartesian comparison, we give the categorical calculation explicitly. Suppose W is a fundamental localizer. First, integrating an objectwise W-equivalence $\theta : H \to H'$ of covariant diagrams on J gives a W-equivalence. Indeed, the over-category $(\int_J H)/j$ of the projection to J has an evaluation functor

$$(i, x, \alpha : i \to j) \longmapsto H(\alpha)x.$$

It is left adjoint to $y \mapsto (j, y, \mathrm{id}_j)$, with identity counit. Its nerve is therefore a homotopy equivalence. The analogous evaluation for H' commutes with θ; two-out-of-three shows that the induced maps of over-categories belong to W. The relative Theorem A gives $\int \theta \in W$. Here and below an ordinary weak equivalence belongs to W by Cisinski (2004, Theorem 2.2.11).

Write $H_1 = \int(B \leftarrow A \to C)$ and $H_2 = \int(D \leftarrow C \to E)$. The two small squares being W-cocartesian means that their canonical maps $H_1 \to D$ and $H_2 \to F$ lie in W. Objectwise invariance now shows that

$$H_{12} = \int (H_1 \leftarrow C \to E) \longrightarrow H_2 \longrightarrow F$$

belongs to W. To compare this iterated integral with the outside span, introduce the five-object poset K whose only nonidentity arrows are

$$a \to b, \quad a \to c_1, \quad c_0 \to c_1, \quad c_0 \to e.$$

Assign A, B, C, C, E to a, b, c_1, c_0, e, with the identity on the arrow between the two copies of C. Flattening the two Grothendieck constructions identifies its integral with H_{12}: the outer copy of C is c_0, and the copy inside H_1 is c_1.

Let K' have objects a, b, c, e and arrows generated by $a \to b$ and $a \to c \to e$. The functor $r : K \to K'$ identifies c_0, c_1 with c. Let $G : K' \to \mathrm{Cat}$ be the original diagram $A \to B$, $A \to C \to E$. Thus the flattened diagram is r^*G. Let $L \subset K'$ be the full subcategory on a, b, e; its integral is the outside-span integral $H_0 = \int(B \leftarrow A \to E)$.

Both r and $L \hookrightarrow K'$ are homotopy final. This can be checked without an abstract pasting theorem. For r, the comma category a/r is the tree-shaped poset K; c/r is the cospan $c_1 \leftarrow c_0 \to e$; and $b/r, e/r$ are singletons. All have contractible nerves. For the inclusion of L, the comma categories at a, b, c, e are respectively a span with initial object a, a singleton, the singleton e, and a singleton. Homotopy cofinality for homotopy colimits, together with the nerve comparison for integration (Thomason, 1979, Theorem 1.2), therefore gives ordinary weak equivalences

$$H_{12} \longrightarrow \int_{K'} G \longleftarrow H_0.$$

All these functors respect the cocone to F. Since $H_{12} \to F$ lies in W, two-out-of-three first gives $\int_{K'} G \to F$ in W and then $H_0 \to F$ in W. This proves the required pasting statement for every fundamental localizer. It supplies the comparison and its compatibility with the cocone, rather than merely an equality of unstructured homotopy types.

Item b) has an equally transparent interpretation. If $A \to C$ is a weak equivalence, then $B \to B \amalg^h_A C$ is a weak equivalence. If also $B \to D$ is a weak equivalence, the comparison from that homotopy pushout to D is one by two-out-of-three. The final integral diagram in the source is a categorical model for exactly this proof. For a general W, use the objectwise invariance of integration just proved instead of appealing to ordinary spaces.

These two facts complete the reduction. Factor the cobase-change map through a Dwyer inclusion followed by a weak equivalence. The first pushout is W-cocartesian by the Dwyer gluing theorem, and the second has the two opposite weak equivalences supplied by the W-cofibration hypothesis. Item b) handles the second square; pasting handles their composite. This proof establishes the intended statement without adding a converse detection axiom for individual edges.

The Dwyer condition ultimately controls more than the weak homotopy type of the nerve. Hackney et al. (2024, Theorems 1.5 and 1.6; arXiv version 3) prove that pushouts along Dwyer maps are preserved when ordinary categories are viewed as $(\infty, 1)$-categories. Equivalently, the comparison from the pushout of nerves to the nerve of the ordinary pushout is a categorical equivalence, a stronger assertion than Thomason's weak homotopy equivalence. Their formulation also covers suitable generalized Dwyer maps.

This later theorem concerns the fidelity of the gluing construction. It supplies no converse of the kind excluded by the acyclic example above. It is compatible with Grothendieck's strategy of finding strict representatives on which homotopy operations become ordinary operations, while explaining why the coherence of the original diagram must be retained. In the pointed derivator formulation, a cofiber is correspondingly functorial on the category of coherent arrows, rather than determined by an arrow in the ordinary homotopy category alone (Groth, 2013, Definition 3.18).

Overview · XIIc-a

These pages begin by asking what the categorical loop constructions actually detect, and end by constructing their dual, a categorical mapping cylinder (Grothendieck, 1990–1991h, manuscript pp. 171–234). Between these two constructions Grothendieck makes a decisive change of viewpoint. A path category is first a concrete device for computing homotopy groups. It then becomes one possible presentation of an intrinsically defined operation on localized categories. The problem is to retain enough of the family of relative homotopy types that changing the presentation does not change the operation or its compatibility with a further change of base.

The first section tests this programme against a classical comparison between cohomology and homotopy. For a small category X, write $BX = |NX|$. Constant coefficients detect connected components, while locally constant coefficients detect transport around the fundamental groupoid. Grothendieck's several formulations of W_0 and W_1 are therefore exact descriptions of the information that a zero-type and a one-type retain. The nonabelian degree-one condition matters: an abelian cohomology test alone can leave a perfect fundamental group undetected. Our commentary spells out the small group-theoretic argument showing why surjectivity on constant nonabelian H^1, together with the local-system H^0 test, already determines the fundamental groupoid.

The attempted extension to higher degrees is more delicate. With his notation, a_n means that every homotopy fibre is n-connected; b_n means that the map induces isomorphisms on homotopy groups through degree n, including components; and c_n means that it induces the corresponding cohomology isomorphisms through degree n for all permitted locally constant coefficients. The range of i is missing from the printed statement of c_n, but is fixed by the proof. The corrected theorem establishes

$$a_n \implies b_n \implies c_n \implies a_{n-1}.$$

The homotopy exact sequence explains the first loss of one degree. The next implication uses the cohomological spectral sequence of the homotopy fibre, with its monodromy retained. The last implication tests the first possible nonzero fibre group against itself as a coefficient module. These arguments are sound once the stated verifications concerning twisted coefficients and the degree-one fibre group are supplied.

The withdrawn equivalence $b_n \Leftrightarrow c_n$ really does fail, but the footnote's suggested degree-two failure does not occur. The inclusion $S^2 \to \mathbf{CP}^\infty = K(\mathbf{Z}, 2)$ is a cohomology isomorphism through degree three with every abelian coefficient group and has trivial fundamental-group data; it kills $\pi_3(S^2)$. In degree two, by contrast, after identifying the fundamental groups, the coinduced local system

$$\mathrm{Hom}_{\mathbf{Z}}(\mathbf{Z}[\pi], \mathbf{Q}/\mathbf{Z})$$

dually detects the second homology of the universal cover. The natural Hurewicz isomorphism then detects π_2. The commentary gives the complete argument and distinguishes it from a claim of historical priority. It also completes the universal-cover spectral-sequence approach that Grothendieck leaves aside: a finite-range truncation argument for derived invariants supplies the needed comparison without assuming an isomorphism in the next degree.

There is a second correction at the end of the section. The class a_n is not a localizer, because it fails two-out-of-three. For a nonzero abelian group A, the composite

$$* \longrightarrow K(A, n+1) \longrightarrow *$$

is the identity, and the second map has n-connected fibre, whereas the first does not. The class b_n, the equivalences detected by Postnikov n-truncation, supplies the intended weak equivalences. There is nevertheless a precise sense in which a_n suffices: its saturation generates

the same localization. The two conditions already agree for maps to a point, so this correction leaves Theorem 2 intact. That theorem recognizes weak contractibility by connectedness of all iterated loop categories. At the degree-two endpoint its reverse induction needs the abelianness of $\pi_1\Omega X$; we supply the Eckmann–Hilton verification. The modern identification of the minimal fundamental localizer with W_∞ and the later loop-based characterization are established by Cisinski (2004, Theorems 2.2.11 and 2.3.6).

The question on p. 191 is consequently no longer whether the path objects work, but in what sense their answers are canonical. Fix a base S. A replacement consists of an endofunctor C of Cat/S, a natural map $i : 1 \to C$ whose components belong to W, and the requirement that $C(X)$ be W-fibred over S. Two replacements compare through their composite. The compatibility of this comparison with the units is exactly naturality, and two-out-of-three shows that the comparison maps are weak equivalences. These elementary observations already contain the solution to the coherence problem posed a few pages later.

Indeed, fixing one replacement C_0, composition gives natural transformations from the identity functor on the category of replacements to $C \mapsto CC_0$, and from the constant functor at C_0 to the same functor. Passing to nerves contracts the identity to a constant map. Thus this category is contractible whenever it is nonempty, and its fundamental groupoid is equivalent to the terminal groupoid. The manuscript's assertion of filteredness is not justified by its common-target construction: equalizing parallel arrows is a separate requirement. Contractibility provides exactly the comparison that is needed without that assertion. The localization argument on pp. 204–207 gives a second formulation: the inclusion of suitable fibred representatives induces an equivalence of localized categories, and the unit of replacement supplies the specified comparison to its quasi-inverse.

This concern with operations independent of their computation is continuous with the programme of Grothendieck (1983, §69, printed pp. 1–3). There the derived categories of diagrams, together with their change-of-index functors and adjoints, replace the information lost on keeping only one derived category. The letter to Thomason makes the distinction between the relative category $(\mathcal{M}, W)$ and supplementary choices of fibrations and cofibrations explicit (Grothendieck, 1991a, points 1–2). The present pages examine that distinction in a concrete and demanding case. An ordinary functor $\Pi S \to \mathrm{Hot}$ records the homotopy types of fibres and their first transport, but it does not by itself retain all higher coherence of a locally constant family. The later theories of derived diagrams and homotopy Kan extensions preserve the additional structure rather than identifying these two levels of information.

The fourth section asks how much of this picture can be stated for a category with pullbacks and weak equivalences. Its intrinsic W-fibrations are maps whose pullback functors preserve weak equivalences. These are the maps later called sharp in a model-category setting; they should not be identified indiscriminately with an independently chosen class of Quillen fibrations. Grothendieck compares the localization of the entire slice at underlying weak equivalences with the localization of the subcategory of such fibrations at universally preserved weak equivalences. The fibre axiom says that these localizations agree. The adjoining axiom says that the two weak-equivalence classes already agree on fibrant representatives, before localization.

Some care with the hypotheses is essential here. The stated two-out-of-three assumption does not imply closure under retracts. Our finite-set example shows that the corresponding claimed retract closure of intrinsic fibrations also fails at precisely that level of generality; imposing retract closure on W repairs it. On the other hand, a functorial fibrant replacement with a weak-equivalence unit gives the desired localization comparison for every full intermediate subcategory containing its image. This proves the intermediate-subcategory assertion in the situations for which the manuscript actually has a replacement construction. The bare logical implication from the two fibre axioms to that assertion is not settled here; this restricted conclusion is recorded as such, not presented as a newly identified open problem.

The subsequent manuscript discussion at pp. 240–241 adds a factorization hypothesis and its own caution about fibrancy over a terminal object.

Non-locally-constant relative types require more structure than the preceding slice localization. Grothendieck introduces a pullback-stable class $\mathcal{L}$ of allowed tests and defines $W_S^{\mathcal{L}}$ by preservation under these tests. In Cat the two comma-category variances give W_S^{g} and W_S^{d}. Smooth and proper functors provide complementary ways of changing the base and choosing representatives. Our commentary proves the formal universal-saturation construction and identifies the saturation asked about on p. 222 with W-smooth functors. The key is to test a W-aspherical map as an arrow to the terminal object of its slice and use the universal pullback characterization in Maltsiniotis (2005, Theorem 1.2(c)). A calculation with a degree-two self-map of a circle then shows why a genuinely varying diagram and a locally constant family are different objects. The abstract existence of both adjoints to restriction requires further hypotheses; in the combinatorial model-category setting the resulting strong derivator is established by Groth (2013, Proposition 1.30).

Finally the interval Φ_n, with $n \geq 2$, gives a Dwyer inclusion of its two endpoints. Multiplying by X produces a cylinder, and attaching its first endpoint to Y along $f : X \to Y$ gives a factorization of f into a Dwyer inclusion and a weak equivalence. The commentary identifies the required adjoints explicitly, explains why the shorter interval fails for the two-endpoint inclusion, and verifies the induced pushout and comparison maps. Proposition 16 then follows by factoring a change of cobase and applying preservation of weak equivalences along cofibrations. The final pasting step is completed with its hypotheses stated; its later derivator formulation is Groth (2013, Proposition 3.13).

The two biconditionals in footnote 124 need a different reading. They use the manuscript's stronger conservative pushout axioms, whose converse implications are not properties of ordinary weak homotopy equivalences. A finite acyclic, noncontractible complex and the two cones in its suspension give a counterexample by face posets, even with Dwyer inclusions. This does not invalidate the mapping-cylinder construction or Proposition 16: their proofs use forward preservation, which is the part available for W_∞. Keeping that distinction visible lets the successful construction retain its force while preventing an extra axiom from silently becoming a claim about classical homotopy theory.

Reste à prouver c). Je vais le déduire d'une p. 150

Proposition 17. *Soit un carré commutatif*

$$Q \quad = \quad \begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\downarrow & & \downarrow{\scriptstyle f_1} \\ X_2 & \xrightarrow{f_2} & Y \end{array}$$

dans Cat*, et $\mathscr{X}_Q$ la catégorie fibrée sur*

$$\tilde{\Phi} \quad (\simeq \mathbf{\Delta}^1 \times \mathbf{\Delta}^1) \quad = \quad \begin{array}{ccc} 0 & \longleftarrow & 1 \\ \uparrow & & \uparrow \\ 2 & \longleftarrow & 3 \end{array}$$

qui l'intègre. Soit $\mathscr{X}$ l'image inverse du fermé

$$\begin{array}{ccc} 0 & \longleftarrow & 1 \\ \uparrow & & \\ 2 & & \end{array}$$

dans $\mathscr{X}_Q$, qui intègre le diagramme [page 235]

$$\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$$

. Pour que Q soit W-cocartésien, il faut et il suffit que l'inclusion

$$i : \mathscr{X} \longrightarrow \mathscr{X}_Q$$

soit dans W.

Comme $\mathscr{X}_Q$ est lisse sur $\tilde{\Phi}$, et 3 est objet initial de $\tilde{\Phi}$, il s'ensuit que l'inclusion de la fibre de $\mathscr{X}_Q$ en le sommet 3 de $\tilde{\Phi}$ est dans W,

$$Y \xrightarrow{i_Y} \mathscr{X}_Q \qquad\qquad \in W.$$

Considérons alors le diagramme

$$\begin{array}{ccc} \mathscr{X} & \xrightarrow{\varphi_Q} & Y \\ & {\scriptstyle i}\searrow \quad \swarrow{\scriptstyle i_Y \in W} & \\ & \mathscr{X}_Q. & \end{array} \qquad \alpha$$

Ce diagramme n'est pas commutatif, mais il y a une flèche de commutativité p. 151

$$\alpha : i_Y \varphi_Q \longrightarrow i.$$

Commentary · Exposition [ms. 235]

The integral criterion with its natural transformation

Proposition 17 turns the comparison with a homotopy pushout into a test on one inclusion. This is what makes the pasting proof manageable: several comparison maps can then be placed in the same category and related by literal inclusions. The natural transformation in the triangle is essential to that change of viewpoint.

Here is an explicit model, fixing the variance. Regard the square as a contravariant diagram $F : \widetilde{\Phi}^{\mathrm{op}} \to \mathrm{Cat}$, with $F(0) = X_0$, $F(1) = X_1$, $F(2) = X_2$, and $F(3) = Y$. The fibred Grothendieck construction $E = \int F$ has objects (b, x) for $x \in F(b)$. A morphism $(b, x) \to (c, y)$ consists of an arrow $u : b \to c$ in the base together with an arrow $x \to F(u)(y)$ in $F(b)$. Composition uses the contravariance of F.

The vertex 3 is initial in $\widetilde{\Phi}$. Its unique arrows $3 \to b$ therefore define a functor

$$r : E \longrightarrow Y, \qquad r(b, x) = F(3 \to b)(x).$$

For $y \in Y$ there is a natural bijection

$$\mathrm{Hom}_E\big((3, y), (b, x)\big) = \mathrm{Hom}_Y\big(y, r(b, x)\big).$$

Thus the fibre inclusion $i_Y : Y \to E$ is left adjoint to r, with $r i_Y = 1_Y$. The counit $i_Y r \Rightarrow 1_E$ is the arrow over $3 \to b$ represented by the identity of $F(3 \to b)x$. This displays the homotopy equivalence of Y with E directly.

Now restrict to $\mathcal{X} = E|_{\{0,1,2\}}$. The restricted functor $r|_{\mathcal{X}}$ is exactly the source's comparison $\varphi_Q : \mathcal{X} \to Y$. Restricting the counit gives

$$\alpha : i_Y \varphi_Q \Longrightarrow i, \qquad i : \mathcal{X} \hookrightarrow E.$$

This is an arrow between functors with source $\mathcal{X}$ and target E. In the edition it is drawn inside the triangle. The transcription had instead attached it to the vertices Y and $\mathcal{X}$; the correction register restores the interior arrow.

To finish the criterion, a natural transformation $g \Rightarrow h$ determines a functor $\mathcal{X} \times [1] \to E$. The projection $\mathcal{X} \times [1] \to \mathcal{X}$ and its two sections belong to the localizer. Consequently g belongs to W exactly when h does. Applying this to α, and then using $i_Y \in W$, gives

$$i \in W \iff i_Y \varphi_Q \in W \iff \varphi_Q \in W.$$

The triangle need not commute strictly. Its specified transformation is precisely enough because the localizer already identifies naturally homotopic functors.

On sait que ceci implique que

$$i \in W \qquad \Longleftrightarrow \qquad (i_Y \varphi_Q) \in W$$

(cf. lemme p. 2), et comme $i_Y \in W$,

$$(i_Y \varphi_Q) \in W \qquad \Longleftrightarrow \qquad \varphi_Q \in W,$$

d'où

$$i \in W \qquad \Longleftrightarrow \qquad \varphi_Q \in W,$$

q.e.d.

Nous pouvons maintenant prouver

Proposition 18 ([127]). (Moyennant W(7 bis).) *Considérons un diagramme commutatif* [page 236]

$$(*) \qquad \begin{array}{ccccc} X_0 & \longrightarrow & X_1 & \longrightarrow & X_2 \\ \downarrow & 1 & \downarrow & 2 & \downarrow \\ X'_0 & \longrightarrow & X'_1 & \longrightarrow & X'_2 \end{array}$$

127. devrait être vrai dans toute catégorie de modèles ayant assez de cofibrations.

d'où un carré composé

$$\begin{array}{ccc} X_0 & \longrightarrow & X_2 \\ \downarrow & 3 & \downarrow \\ X'_0 & \longrightarrow & X'_2. \end{array}$$

Supposons que le carré 1 *soit W-cocartésien. Alors le carré* 2 *est W-cocartésien si et seulement si le carré* 3 *l'est.*

Démonstration.

`[Maintenant,` X_i `et` X'_i `échangent leurs rôles.]`

Soit p. 152

$$I \quad = \quad \begin{pmatrix} 0 & \longleftarrow & 1 & \longleftarrow & 2 \\ \downarrow & & \downarrow & & \downarrow \\ 0' & \longleftarrow & 1' & \longleftarrow & 2' \end{pmatrix}$$

et soit $\mathscr{X}$ l'intégrale du diagramme $(*)$, catégorie fibrée sur $\mathscr{X}$. On désigne par $\mathscr{X}_{010'}$ la sous-catégorie pleine de $\mathscr{X}$ image inverse de la sous-catégorie pleine

$$\underbrace{I_{010'}}_{=\{0,1,0'\}} \quad \subseteq I,$$

et de même pour toute autre partie A de $\operatorname{Ob} I = \{0, 1, 2, 0', 1', 2'\}$, donnant une sous-catégorie pleine $\mathscr{X}_A$ de $\mathscr{X}$. On pose

$$U \quad = \quad \begin{array}{ccccc} 0 & \longleftarrow & 1 & \longleftarrow & 2 \\ \downarrow & & & & \\ 0' & & & & \end{array} \quad , \qquad V \quad = \quad \begin{array}{ccccc} 0 & \longleftarrow & 1 & \longleftarrow & 2 \\ \downarrow & & \downarrow & & \\ 0' & \longleftarrow & 1' & & \end{array} \quad ,$$

d'où

$$\mathscr{X}_U = \mathscr{X}_{0120'}, \qquad \mathscr{X}_V = \mathscr{X}_{0120'1'}.$$

Les flèches entre les $\mathscr{X}_A$ sont les inclusions

déduites d'inclusions $A \hookrightarrow A'$ dans $\mathfrak{P}(\operatorname{Ob} I)$. Ainsi, le caractère W-cocartésien des carrés 1, 2, 3 s'exprime respectivement par la condition que les inclusions [page 237]

$$\begin{array}{ccc} \mathscr{X}_{010'} & \xrightarrow{\varphi_1} & \mathscr{X}_{010'1'} \\ \mathscr{X}_{121'} & \xrightarrow{\varphi_2} & \mathscr{X}_{121'2'} \\ \mathscr{X}_{020'} & \xrightarrow{\varphi_1} & \mathscr{X}_{020'2'} \end{array}$$

soient dans W. Considérons alors le diagramme de flèches d'inclusion p. 153

$$\begin{array}{ccc} \mathscr{X}_{020'} & \xrightarrow{\varphi_3} & \mathscr{X}_{020'2'} \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\ \mathscr{X}_U & \xrightarrow{\psi_3} & \mathscr{X} \\ {\scriptstyle\psi_1}\downarrow & & \downarrow{\scriptstyle\mathrm{id}} \\ \mathscr{X}_V & \xrightarrow{\psi_2} & \mathscr{X} \\ {\scriptstyle\alpha'}\uparrow & & \uparrow{\scriptstyle\beta'} \\ \mathscr{X}_{121'} & \xrightarrow{\varphi_2} & \mathscr{X}_{121'2'}. \end{array}$$

Je dis que dans ce diagramme on a

$$(*) \qquad \alpha, \beta, \alpha', \beta' \in W$$

et

$$(**) \qquad \varphi_1 \in W \ \text{ (i.e. carré 1 } W\text{-cocartésien)} \qquad \Longleftrightarrow \qquad \psi_1 \in W.$$

Les relations $(*)$ impliquent déja que

$$\begin{aligned} \varphi_2 \in W \quad &\Longleftrightarrow \quad \psi_2 \in W \\ \varphi_3 \in W \quad &\Longleftrightarrow \quad \psi_3 \in W, \end{aligned}$$

et sous l'hypothèse W-cocartésienne $(**)$, impliquant $\psi_1 \in W$,
on trouve [page 238]

$$\psi_2 \in W \qquad \Longleftrightarrow \qquad \psi_3 \in W,$$

d'où

$$\varphi_2 \in W \qquad \Longleftrightarrow \qquad \varphi_3 \in W,$$

ce qu'il fallait prouver.

Il reste à établir $(*)$ et $(**)$. Les relations $(*)$ proviennent de la remarque que si $i : A \longrightarrow A'$ est une inclusion de sous-catégories pleines de I, telle que i soit W-*asphérique* (i.e. A/j W-asphérique pour $j \in \mathrm{Ob}\, A' \backslash \mathrm{Ob}\, A$), alors $\mathscr{X}_A \longrightarrow \mathscr{X}_{A'}$ l'est aussi (changement de base lisse), donc est `[dans]` W. On doit vérifier la W-asphéricité pour les quatre inclusions

$$\begin{aligned} \{0,2,0'\} &\longrightarrow U = \{0,1,2,0'\} && \mathbf{\Delta}^0 \\ \{0,2,0',2'\} &\longrightarrow I && \mathbf{\Delta}^0/\mathbf{\Delta}^1 \\ \{1,2,1'\} &\longrightarrow V = \{0,1,2,0',1'\} && \mathbf{\Delta}^1,\ \Phi \\ \{1,2,1',2'\} &\longrightarrow I && \mathbf{\Delta}^1,\ \mathbf{\Delta}^1 \times \mathbf{\Delta}^1. \end{aligned}$$

Commentary · Examples [ms. 238]

The seven slices behind the pasting theorem

The six-object poset is the bookkeeping device that puts the left square, right square and outer rectangle in one ambient diagram. Grothendieck reduces the substantial part of pasting to seven very small slice categories. Computing them explicitly both explains the proof and corrects one harmless shape label in its margin.

Use the arrows of I exactly as printed: the top row is $2 \to 1 \to 0$, the bottom row is $2' \to 1' \to 0'$, and the vertical arrows are $k \to k'$. A slice A/j has as objects the elements of A admitting an arrow to j. Since I is a poset, its morphisms are just the induced order. For the inclusions underlying $\alpha, \beta, \alpha', \beta'$ we obtain

inclusion	j	A/j
α	1	$\{2\}$
β	1	$\{2\}$
β	$1'$	$(2 \longrightarrow 2')$
α'	0	$(2 \longrightarrow 1)$
α'	$0'$	$(2 \longrightarrow 1 \longrightarrow 1')$
β'	0	$(2 \longrightarrow 1)$
β'	$0'$	$\{1,2,1',2'\} \cong [1] \times [1]$.

Each displayed category has initial object 2. For an object j already in A, the slice A/j has terminal object 1_j, so no separate check is necessary. In the fifth line the composite $2 \to 1'$ exists: the inclusion is full, and the defining rectangle is commutative. The slice is consequently $[2] = \Delta^2$. The edition prints Φ in this place. The asphericity argument uses the initial object and is unaffected by this label.

We can also make the passage from base slices to total-category slices explicit. Let $F : I^{\mathrm{op}} \to \mathrm{Cat}$ be the integrated diagram and let $e = (j, y)$ be an object of its total category. There is a projection

$$(\mathcal{X}_A)/e \longrightarrow A/j.$$

An object above $u : a \to j$ is an arrow $x \to F(u)y$ in $F(a)$. The choice $x = F(u)y$, with the identity arrow, defines a section. For an object z of the total slice, every arrow from its image in A/j to u has a unique lift from z to this chosen object over u. Hence the projection is left adjoint to the section. The two categories are homotopy equivalent. Asphericity of A/j therefore proves asphericity of $(\mathcal{X}_A)/e$; the localizer's Theorem A condition gives $\mathcal{X}_A \to \mathcal{X}_{A'} \in W$. This is the "smooth change of base" step in concrete form.

The last ingredient is gluing. The two subsets

$$U = \{0, 1, 2, 0'\}, \qquad U' = \{0, 1, 0', 1'\}$$

are cosieves in $V = \{0, 1, 2, 0', 1'\}$: following an arrow from either subset never leaves that subset. Their total categories are therefore closed subcategories covering $\mathcal{X}_V$, with intersection $\mathcal{X}_{010'}$. The forward part of the closed-cover axiom sends $\varphi_1 \in W$ to $\psi_1 \in W$. As $\psi_3 = \psi_2\psi_1$, two-out-of-three then says that $\psi_2 \in W$ exactly when $\psi_3 \in W$. The four slice equivalences transport this assertion back to φ_2 and φ_3.

The proof thus needs only the forward part of W(7 bis). The source writes the stronger equivalence $\varphi_1 \in W \iff \psi_1 \in W$, which its full hypothesis permits, but its reverse direction is not used. Cancellation of the right square against the outer rectangle is produced by a known weak equivalence ψ_1 and two-out-of-three. It should be kept separate from the stronger conservative gluing condition examined in the appendix.

p. 154

(NB J'ai marqué en marge les catégories qu'on obtient comme A/j, à isomorphisme près, dans les quatre cas envisagés - elles ont toutes un objet initial, donc sont W-asphériques.)

Reste à prouver (∗∗). Soit

$$U' = \{0, 1, 0', 1'\}, \qquad \text{donc } V = U \cup U', \quad U \cap U' = \{0, 1, 0'\},$$

où U, U' sont des *fermés* de V qui recouvrent V. Donc on aura de même

$$\mathscr{X}_V = \mathscr{X}_U \cup \mathscr{X}_{U'}, \qquad \mathscr{X}_U \cap \mathscr{X}_{U'} = \mathscr{X}_{010'},$$

[page 239]

$$\begin{array}{ccc} \mathscr{X}_{U\cap U'} & \longrightarrow & \mathscr{X}_U \\ \downarrow{\scriptstyle \varphi_1} & & \downarrow{\scriptstyle \psi_1} \\ \mathscr{X}_{U'} & \longrightarrow & \mathscr{X}_V. \end{array}$$

Donc par W(7 bis) (cas dual des immersions fermées), on a

$$\varphi_1 \in W \qquad \Longleftrightarrow \qquad \psi_1 \in W,$$

q.e.d.

Notons pour mémoire le corollaire qui nous avait motivé.

Corollaire. *Supposons* W(7 bis) (comme dans toute cette section). *Soit un carré cocartésien dans* Cat

$$\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3, \end{array}$$

avec i_1 (ou i_2) une W-cofibration (p.ex. une immersion de Dwyer). Alors ce carré est W-cocartésien.

Commentary · Later developments [ms. 239]

From categorical gluing to coherent cocartesian squares

The corollary finishes the comparison that motivated this detour. Factor the second leg of a strict pushout as a Dwyer map followed by a weak equivalence. The first resulting square is homotopy

cocartesian by the Dwyer case; the second is so because its opposite sides are weak equivalences, using the first leg's W-cofibration property. Proposition 18 identifies their composite as homotopy cocartesian. The categorical integral and the replacement construction therefore give the same answer on these pushouts.

In the modern derivator language, put $\square = [1] \times [1]$ and let $\ulcorner \subset \square$ omit $(1,1)$. A coherent square $Q \in \mathbb{D}(\square)$ is cocartesian when the counit

$$\iota_! \iota^* Q \longrightarrow Q, \qquad \iota : \ulcorner \hookrightarrow \square,$$

is an isomorphism. The composition and cancellation rule for such squares is Groth (2013, Proposition 3.13). It requires neither stability nor a zero object. This identifies the modern form of the assertion in footnote 127: the coherence belongs to an object of $\mathbb{D}([2] \times [1])$, and the left square controls equivalence of the two remaining cocartesianness tests.

For Dwyer maps the later result is stronger still. Hackney et al. (2024, Theorems 1.5–1.6, arXiv version 3) show that if $D = B \amalg_A C$ and $A \to B$ is Dwyer, then

$$NB \amalg_{NA} NC \longrightarrow ND$$

is a weak categorical equivalence in the Joyal model structure. Thus the ordinary pushout represents the pushout of the corresponding $(\infty, 1)$-categories, retaining categorical information before passage to the homotopy types of their classifying spaces. The hypotheses concern the attaching map, so the result applies directly to the finite cylinders above.

The source's final sentence on cofibres should be read at this finite level. It has exhibited functorial replacements, proved the relative invariance needed to compare them, and established the basic gluing calculus. Those constructions supply the material for the abstract homotopy-square theory in the next section. A claim that all homotopy Kan extensions exist for an arbitrary relative category would require further hypotheses; no such claim is established by this closing factorization alone.

Cette digression (p. 232 ff.) n'est pas nécessaire d'ailleurs pour formuler dès maintenant la conclusion de nos cogitations sur les cofibres W-homotopiques : *Il y a une telle théorie dans* Cat, pourvu seulement que W satisfasse à W(7 bis). En effet, tout résulte du fait que toute flèche $f : X \longrightarrow Y$ dans Cat se factorise en

$$X \xrightarrow{i_f} \tilde{Y} \xrightarrow{p_f} Y,$$

avec i_f une W-cofibration (et même une immersion ouverte de Dwyer), et $p_f \in W$.

p. 155
[page 240]

7 Retour sur l'axiomatique de la fibre W-homotopique.

Je reviens sur la réflexion du n° 4. Finalement, pour

$$(\mathcal{M}, W \subseteq \mathrm{Fl}(\mathcal{M}))$$

donnés avec W faiblement saturé, il apparaît qu'une excellente théorie des fibres W-homotopiques, satisfaisant aux conditions ⓐⓑⓒ du n° 4 (pages 215-218), existe dès que l'axiome d'apparence anodine suivant est satisfait (en plus de l'existence de l'objet final et des produits fibrés ([128])).

Factorisation gauche : ([129], [130]). Toute flèche $f : X \longrightarrow Y$ de $\mathcal{M}$ se factorise en

$$X \xrightarrow{i_f} \bar{X} \xrightarrow{p_f} Y,$$

128. et encore! il me semble qu'il suffit de supposer que les $X \longrightarrow Y$ qui sont W-fibrants sont 'quarrables'.

129. Sous ces conditions, $\mathcal{M}$ muni de (W, F_W) est une catégorie à cofibrations de K.S. Brown-Anderson, et 'tout est O.K.', mais il faudrait vérifier a) b) c) ci-dessus (p. 215, 218), je ne suis pas sûr de l'avoir fait.

130. **NB** L'axiome de factorisation gauche n'est suffisant pour l'existence d'une bonne théorie des fibres homotopiques (semble-t-il) que dans le cas où tout X dans $\mathcal{M}$ est W-fibrant, i.e. est dans $\underline{\mathrm{Fib}}_W e$.

avec

$$i_f \in W, \qquad p_f \in \mathrm{Fib}_W.$$

L'axiome dual donne lieu à une bonne théorie des cofibres W-homotopiques.

Factorisation droite : Toute flèche $f : X \longrightarrow Y$ dans $\mathcal{M}$ se factorise en

$$X \xrightarrow{j_f} \tilde{Y} \xrightarrow{q_f} Y,$$

avec

$$q_f \in W, \qquad j_f \in \mathrm{Cofib}_W.$$

Je voudrais à présent développer quelque peu le formalisme des fibres W-homotopiques, sous l'hypothèse (disons) que W satisfait l'axiome de factorisation gauche. [page 241]

Commentary · Context [ms. 240]

The factorization axiom and Brown's hypothesis

Grothendieck now wants to detach the theory of homotopy fibres from the particular geometry of categories. The interval and mapping-cylinder arguments of the preceding pages have supplied a mechanism; he asks which part of that mechanism belongs to an arbitrary category with weak equivalences. His phrase "catégorie de modèles" here means a category equipped with a class W and the properties under discussion. It does not silently import all the axioms of a Quillen model category.

The relevant meaning of W-fibration is likewise intrinsic to W. Precisely, for $p : X \to Y$, the pullback functor $p^* : \mathcal{M}/Y \to \mathcal{M}/X$ must preserve the arrows whose underlying maps belong to W. This also implies the same assertion after another base change. This is the property used below, together with the stability of these maps under composition and pullback. In particular, one must not replace Fib_W by the fibrations of an arbitrary model structure without checking the corresponding properness statement. Here "weakly saturated" means that W contains the isomorphisms and satisfies two-out-of-three; it does not mean that every arrow inverted by localization already belongs to W. A factorization $f = pi$ with $i \in W$ and $p \in \mathrm{Fib}_W$ is sufficient for the replacement operations that are actually written down; it is not, by its form alone, a pair of lifting axioms.

The two footnotes beneath the axiom record a real issue in its scope. Brown's category of fibrant objects requires every map to the terminal object to be a fibration. Brown also requires pullbacks of fibrations and of trivial fibrations to exist and to remain of the same kind, and a path-object factorization of every diagonal. His factorization lemma produces a factorization of every map into a weak equivalence followed by a fibration; conversely, applying such a factorization to the diagonal supplies the path-object axiom (Brown, 1973, Part I, §1, pp. 420–422). Thus the all-object-fibrancy qualification in footnote 130 gives the clean connection with Brown. The word "cofibrations" in footnote 129 is a slip for the fibration version of this theory.

Here is why a weakened existence condition on pullbacks is reasonable. If S is fixed and $A \to S$, $B \to S$ are W-fibrations, their product over S is formed by pulling back one of those fibrations. The projection to the other factor is again a fibration, and the composite to S is a fibration. The same observation constructs a product with any finite number of factors; the empty product is $S \to S$. A factorization of a diagonal then provides the path objects required in this relative category. Arbitrary pullbacks in $\mathcal{M}$ are consequently stronger than what these constructions use. Footnote 128 is a proposed economy of axioms, rather than an obstruction to the argument.

The distinction between Brown's category and a category merely admitting fibrant replacement matters for the footnote's present status. In the language of left derivable categories, the written factorization and pullback hypotheses already suffice for the finite diagram theorem. We verify this below, including the required stability of trivial fibrations; no global-fibrancy hypothesis has to be added. Functorial factorization gives a shorter separate proof of the two replacement lemmas, but is also unnecessary in the derivable-category argument. Thus the source's cautious necessity suggestion is not part of the modern theorem.

Considérons un diagramme cartésien

$$\begin{array}{ccc} X_1 & \xleftarrow{q_2} & Y \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle q_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$$

dans $\mathcal{M}$. Si p_1 est dans Fib_W, alors Y sur X_2 peut s'interpréter comme la fibre W-homotopique de X_1/S en X_2/S. Plus précisément, il en est ainsi de la classe $\mathrm{ho}^{\mathrm{lc}}_{W,X_2}(Y)$, p. 156

$$(*) \qquad \mathrm{ho}^{\mathrm{lc}}_{W,X_2}(Y) \simeq p_2^*(\mathrm{ho}^{\mathrm{lc}}_{W,S}(X_1/S)).$$

Mais en tant qu'objet sur X_1, Y peut s'interpréter également comme fibre W-homotopique de X_2/S en X_1/S :

$$(**) \qquad \mathrm{ho}^{\mathrm{lc}}_{W,X_1}(Y/X_1) \simeq p_1^*(\mathrm{ho}^{\mathrm{lc}}_{W,S}(X_2/S)).$$

À dire vrai, il faudrait le prouver. Pour ceci, considérons une factorisation

$$X_2 \xrightarrow{i_2} \bar{X}_2 \xrightarrow{\bar{p}_2} S \qquad i_2 \in W \; p_2 \in \mathrm{Fib}_W,$$

de sorte qu'on trouve un diagramme de carrés cartésiens

$$\begin{array}{ccccc} X_1 & \xleftarrow{\bar{q}_2} & \bar{Y} & \xleftarrow[W]{j_2} & Y \\ \downarrow & & \downarrow & & \downarrow \\ S & \xleftarrow{\bar{q}_2} & \bar{X}_2 & \xleftarrow[W]{i_2} & X_2. \end{array}$$

Par définition de p_1^*, on a [page 242]

$$p_1^*(\mathrm{ho}^{\mathrm{lc}}_{W,S}(X_2/S)) \simeq \mathrm{ho}^{\mathrm{lc}}_{W,X_1}(\bar{Y}/X_1),$$

d'autre part, comme $X_1 \longrightarrow S$ est dans Fib_W, et $i_2 \in W$, on a $j_2 \in W$, donc

$$\mathrm{ho}^{\mathrm{lc}}_{W,X_1}(Y/X_1) \xrightarrow{\sim} \mathrm{ho}^{\mathrm{lc}}_{W,X_1}(\bar{Y}/X_1),$$

d'où la formule $(**)$.

Ainsi, l'objet Y a deux interprétations homotopiques différentes, suivant qu'on le regarde comme étant sur X_1, ou sur X_2. On aimerait alors une interprétation qui englobe les deux à la fois. On va considérer le diagramme

$$\begin{array}{ccc} X_1 & & \\ \downarrow & & \\ S & \longleftarrow & X_2 \end{array}$$

de type Ψ°, ou

$$\Psi \quad = \quad \begin{array}{ccc} 1 & & \\ \uparrow & & \\ 0 & \longrightarrow & 2 \end{array} \qquad \text{ou} \qquad \begin{array}{ccc} & & 1 \\ & \nearrow & \\ 0 & & \\ & \searrow & \\ & & 2 \end{array} \quad ,$$

dont Y se déduit par produit fibré, en complétant ce diagramme en un diagramme de type p. 157 [page 243]

$\tilde{\Psi}^\circ$, où

$$\tilde{\Psi} = \begin{array}{ccccc} & & 1 & & \\ & \nearrow & & \searrow & \\ 0 & & & & 3 \\ & \searrow & & \nearrow & \\ & & 2 & & \end{array} \qquad (\simeq \mathbf{\Delta}^1 \times \mathbf{\Delta}^1).$$

Pour toute catégorie I, soit

$$\mathrm{Ho}_{\mathcal{M},W}(I) \;\text{ ou }\; \mathrm{Ho}_W(I) \overset{\text{déf}}{=} \mathcal{M}(I)W(I)^{-1},$$

où

$$\begin{cases} \mathcal{M}(I) & = & \underline{\mathrm{Hom}}(I^\circ, \mathcal{M}) \\ W(I) & \subseteq & \mathrm{Fl}(\mathcal{M}(I)) \text{ est formé des } u : F \longrightarrow G \text{ telles que} \\ & & u_i : F(i) \longrightarrow G(i) \text{ dans } W \text{ pour tout } i \text{ dans } \mathrm{Ob}\, I. \end{cases}$$

Commentary · Exposition [ms. 242]

One pullback with two relative meanings

The symmetry that Grothendieck is seeking concerns the whole square. A pullback $P = A \times_S B$ has a map to A and a map to B; a construction that remembers only its isomorphism class in a homotopy category has forgotten part of the answer. Suppose $A \to S$ is a W-fibration and factor $B \to S$ as $B \xrightarrow{w} \bar{B} \to S$, with $w \in W$ and $\bar{B} \to S$ a W-fibration. Then

$$A \times_S B \longrightarrow A \times_S \bar{B}$$

is the pullback of w along the base change of $A \to S$. It belongs to W. It is also a morphism over A, so it identifies the class of the original pullback over A with the result of deriving the base-change operation on B. This proves the second interpretation in the text, while the first interpretation is already computed by the good representative $A \to S$. No second fibration hypothesis on $B \to S$ has been introduced.

The natural domain of this simultaneous construction is therefore the category of cospans before localization. With the manuscript's contravariant convention,

$$\Psi = (1 \leftarrow 0 \rightarrow 2), \qquad \mathcal{M}(\Psi) = \mathrm{Fun}(\Psi^{\mathrm{op}}, \mathcal{M}),$$

so an object of $\mathcal{M}(\Psi)$ is $A \to S \leftarrow B$. The additional vertex 3 of $\tilde{\Psi}$ points, after taking the opposite, to both 1 and 2. A right Kan extension along $\Psi^{\mathrm{op}} \hookrightarrow \tilde{\Psi}^{\mathrm{op}}$ consequently puts a limit at vertex 3. This orientation explains the star in i_*: it is a right adjoint to restriction.

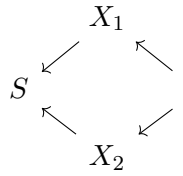
One should retain the distinction already emphasized in the commentary on Chapter VI, manuscript p. 87. The category

$$\mathrm{Fun}(\Psi^{\mathrm{op}}, \mathcal{M})[W(\Psi)^{-1}]$$

retains how the maps and their weak-equivalence comparisons occur together. It is not being replaced by $\mathrm{Fun}(\Psi^{\mathrm{op}}, \mathcal{M}[W^{-1}])$. In the latter, a commutativity equation is only an equality of homotopy classes. The former is the diagram category on which a canonical homotopy completion can be defined. This is precisely the reason that the later derivator formalism forms diagram categories first and localizes afterwards (Groth, 2013, pp. 313–315).

The coming proof accordingly has two tasks. It must show that the chosen good cospans present the localization, and then show that taking their strict pullbacks respects weak equivalences. The first task concerns all morphisms and relations in a localized category; replacing one object at a time does not settle it.

Ainsi, le diagramme $\begin{array}{ccc} & & X_1 \\ & \swarrow & \\ S & & \\ & \nwarrow & \\ & & X_2 \end{array}$ définit un objet ξ de $\mathcal{M}(\Psi)$, donc de $\mathrm{Ho}_W(\Psi)$, et le

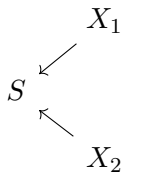
diagramme  peut de même s'interpréter comme un objet $\tilde{\xi}$ de $\mathrm{Ho}_W(\tilde{\Psi})$, dont la restriction à Ψ est ξ.

Ceci posé, j'aimerais définir un foncteur

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi})$$

(où

$$i : \Psi \longrightarrow \tilde{\Psi}$$

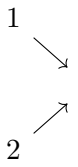
est l'inclusion, qui est une immersion ouverte), lequel doit correspondre, dans le cas d'un objet  de $\mathrm{Ho}_W(\Psi)$ avec $p_1 \in \mathrm{Fib}_W$ [: $X_1 \longrightarrow S$], à l'opération de produit fibré ([131]).

La construction de ce foncteur i_* devrait être essentiellement sorital, avec ce dont nous disposons. Nous allons définir d'abord, pour S *fixé,* un foncteur canonique p. 158 [page 244]

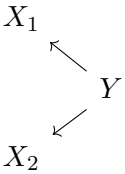
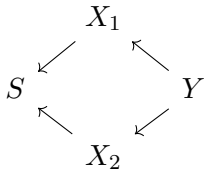
$$(*) \qquad \mathrm{Ho}^{\mathrm{lc}}_W(S) \times \mathrm{Ho}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{Ho}_{W,S}(\Psi),$$

où le deuxième membre est défini, par analogie avec les $\mathrm{Ho}_W(I)$, par

$$\mathrm{Ho}_{W,S}(I) \overset{\text{déf}}{=} ((\mathcal{M}/S)(I))(W_S(I))^{-1}$$

- i.e. c'est la même définition, sauf que $(\mathcal{M}, W)$ est remplacé par $(\mathcal{M}/S, W_S)$. Quant à $\Phi =$  , on le considère comme une sous-catégorie fermée de $\tilde{\Psi}$, qui s'en déduit en lui rajoutant un objet initial 0 (tout comme $\tilde{\Psi}$ se déduit de Ψ en lui rajoutant un objet final 3). D'autre part, on a un foncteur tautologique

$$(*') \qquad \mathrm{Ho}_{W,S}(\Phi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}),$$

qui sur les objet consiste à compléter un diagramme  de $\mathcal{M}/S$ en le diagramme carré  de $\mathcal{M}$.

[page 245]

131. **NB** Cette construction, qui se fait ici assez lourdement, devient immédiate, et aussi ses principales propriétés, si on admet que la catégorie à fibrations $(\mathcal{M}, W, \mathrm{Fib}_W)$ définit un *dérivateur* sur le domaine des ensembles ordonnés finis, ce qui se prouve en effet assez simplement (avec pas plus de mal que je ne me donne ici).

Le composé de $(*)$ et de $(*')$ donnera donc un foncteur

$$(**) \qquad \mathrm{Ho}^{\mathrm{lc}}_W(S) \times \mathrm{Ho}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}),$$

très proche déjà du foncteur i_* à construire. Pour construire $(*)$, on interprète (à équivalence de catégories près) le premier membre comme

$$(\underline{\mathrm{Fib}}_W S \times \underline{\mathrm{Fib}}_W S)(W_S \times W_S)^{-1},$$

donc il faut construire un foncteur

$$\underline{\mathrm{Fib}}_W S \times \underline{\mathrm{Fib}}_W S \longrightarrow \mathrm{Ho}_{W,S}(\Phi),$$

qui transforme flèches dans $W_S \times W_S$ en isomorphismes. On prendra le foncteur

$$(X_1, X_2) \longmapsto \begin{pmatrix} X_1 & \longleftarrow & X_1 \times_S X_2 \\ & & \downarrow \\ & & X_2 \end{pmatrix},$$

et il faut seulement vérifier que si

p. 159

$$X_1 \xrightarrow{\alpha_1} X_1', \qquad X_2 \xrightarrow{\alpha_2} X_2'$$

sont deux flèches dans $\underline{\mathrm{Fib}}_W S$, d'où

$$X_1 \times_S X_2 \xrightarrow{\alpha = \alpha_1 \times_S \alpha_2} X_1' \times_S X_2',$$

alors

$$\alpha_1 \in W,\ \alpha_2 \in W \qquad \Longrightarrow \qquad \alpha \in W,$$

ce qui est immédiat sur la factorisation

$$X_1 \times_S X_2 \underbrace{\longrightarrow}_{\substack{\in W \\ \text{car } X_2 \in \underline{\mathrm{Fib}}_W S}} X_1' \times_S X_2 \underbrace{\longrightarrow}_{\substack{\in W \\ \text{car } X_1' \in \underline{\mathrm{Fib}}_W S}} X_1' \times_S X_2'.$$

On a prouvé la

Proposition 18. *Soit S dans $\mathcal{M}$. Il existe, à isomorphisme unique près, un foncteur et un seul* [page 246]

$$\mathrm{Ho}^{\mathrm{lc}}_W(S) \times \mathrm{Ho}^{\mathrm{lc}}_W(S) \xrightarrow{\pi} \mathrm{Ho}_{W,S}(\Phi)$$

donnant lieu à un carré commutatif à isomorphisme près

$$\begin{array}{ccc} & X_1 & \\ & \nwarrow & \\ (X_1, X_2) \longmapsto & X_1 \times_S X_2 & \\ & \swarrow & \\ & X_2 & \end{array}$$

$$\begin{array}{ccc} \underline{\mathrm{Fib}}_W S \times \underline{\mathrm{Fib}}_W S & \longrightarrow & (\mathcal{M}/S)(\Phi) \\ \downarrow & & \downarrow \\ \mathrm{Ho}^{\mathrm{lc}}_W(S) \times \mathrm{Ho}^{\mathrm{lc}}_W(S) & \longrightarrow & \mathrm{Ho}_{W,S}(\Phi) \\ (= (\mathcal{M}/S)(W_S)^{-1} \times (\mathcal{M}/S)(W_S)^{-1}) & & (= \mathcal{M}/S(\Phi)(W_S(\Phi))^{-1}). \end{array}$$

Pour X_1, X_2 dans $\mathcal{M}/S$, le diagramme

$$\pi(X_1, X_2) \overset{\text{déf}}{=} \pi(\mathrm{ho}^{\mathrm{lc}}_{W,S}(X_1), \mathrm{ho}^{\mathrm{lc}}_{W,S}(X_2))$$

se 'calcule' ainsi, à isomorphisme canonique près : on choisit

$$X_1 \xrightarrow{\alpha_1} \bar{X}_1, \qquad X_2 \xrightarrow{\alpha_2} \bar{X}_2 \qquad \textit{flèches dans } \mathcal{M}/S$$

avec

$$\alpha_1, \alpha_2 \in W_S, \qquad \bar{X}_1, \bar{X}_2 \in \operatorname{Ob} \underline{\operatorname{Fib}}_W S,$$

ce qui définit

$$\pi(\alpha_1, \alpha_2) : \pi(X_1, X_2) \xrightarrow{\sim} \pi(\bar{X}_1, \bar{X}_2) = \operatorname{ho}_{S,W,\Phi} \begin{pmatrix} \bar{X}_1 & \longleftarrow & \bar{X}_1 \times_S \bar{X}_2 \\ & & \downarrow \\ & & \bar{X}_2 \end{pmatrix}.$$

Commentary · Exposition [ms. 246] p. 160 [page 247]

A product together with its projections

The intermediate construction over a fixed base is more than a calculation of the object $A \times_S B$. Grothendieck keeps both projections in the value of π. Thus π lands in the localized category of Φ^{op}-diagrams in $\mathcal{M}/S$, where $\Phi = (1 \to 3 \leftarrow 2)$. The occurrence of Ψ in the first display of ms. 244 is an edition slip: the next functor, and Proposition 18 itself, require Φ.

Write $\mathcal{C}_S$ for the full subcategory of $\mathcal{M}/S$ on the W-fibrations. If $a : A \to A'$ and $b : B \to B'$ are weak equivalences in this category, the comparison of products has the factorization

$$A \times_S B \xrightarrow{a \times_S 1} A' \times_S B \xrightarrow{1 \times_S b} A' \times_S B'.$$

The first arrow is a pullback of a along a base change of $B \to S$; the second is a pullback of b along a base change of $A' \to S$. Both belong to W. The same morphism of spans has a and b at its other vertices, so it is an objectwise weak equivalence. This proves exactly the compatibility needed to descend the entire span functor.

There is also a small categorical point behind the source's product of localizations. For relative categories $(\mathcal{C}, U)$ and $(\mathcal{D}, V)$, with identities in U, V, there is a canonical equivalence

$$(\mathcal{C} \times \mathcal{D})[(U \times V)^{-1}] \simeq \mathcal{C}[U^{-1}] \times \mathcal{D}[V^{-1}].$$

Indeed a functor out of $\mathcal{C} \times \mathcal{D}$ inverts $U \times V$ precisely when it inverts $(u, 1)$ and $(1, v)$ separately. Localizing successively in the two variables gives the displayed universal property, including natural transformations. Applied to $\mathcal{C}_S$ twice, this justifies the domain used in Proposition 18. The localization equivalence between $\mathcal{C}_S$ and the relative homotopy category is an earlier input to this paragraph, not a consequence of the displayed product calculation.

The phrase "unique up to unique isomorphism" is understood relative to the specified comparison with the functor on good representatives. Without that comparison, a functor can have automorphisms and the uniqueness assertion would be too strong. With it, full faithfulness of restriction along a localization supplies the required uniqueness. A pair of replacement arrows $A \to \bar{A}$, $B \to \bar{B}$ therefore gives the canonical comparison written by Grothendieck; it does not require a preferred strict map between every two independently chosen replacements.

This proposition is numbered 18 a second time in the edition. We distinguish it from the pasting proposition on ms. 236 by its page and by the name "fixed-base product proposition". Its content will also be used when the base itself varies, where a further argument is needed.

Corollaire. *Description du foncteur*

$$(\xi_1, \xi_2) \longmapsto \xi_1 \times_S \xi_2 : \mathrm{Ho}^{\mathrm{lc}}_W(S) \times \mathrm{Ho}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{Ho}^{\mathrm{lc}}_W(S).$$

Ce foncteur est bel et bien un produit cartésien dans $\mathrm{Ho}^{\mathrm{lc}}_W(S)$ ([132]), *où les produits finis existent* ([133]).

132. mais c'est un cas particulier des cas de Ho_W (si $\mathcal{M}$ a des produits binaires), en remplaçant $\mathcal{M}$ par $\mathcal{M}/S$.
133. Dire aussi que les foncteurs $f^* : \mathrm{Ho}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{Ho}^{\mathrm{lc}}_W(S')$ (pour $f : S' \longrightarrow S$) commutent aux produits finis.

Pour prouver ce dernier point, posant

$$\mathcal{H} = \mathrm{Ho}_W^{\mathrm{lc}}(S),$$

il faut montrer que les foncteurs

$$\mathcal{H} \underset{\pi}{\overset{\delta}{\rightleftarrows}} \mathcal{H} \times \mathcal{H}$$

sont adjoints, avec

$$\begin{array}{rcccccc} \delta & = & \mathrm{diag}_{\mathcal{H}} & : & \xi & \longmapsto & (\xi, \xi) \\ \pi & = & \text{'produit'} & : & (\xi_1, \xi_2) & \longmapsto & \xi_1 \times_S \xi_2. \end{array}$$

On va définir les homomorphismes d'adjonction

$$\begin{array}{rcccccc} \mathrm{id}_{\mathcal{H}} & \overset{\alpha}{\to} & \pi\delta & \text{i.e.} & \xi & \overset{\alpha_\xi}{\longrightarrow} & \xi \times_S \xi \\ \mathrm{id}_{\mathcal{H}\times\mathcal{H}} & \overset{\beta}{\leftarrow} & \delta\pi & \text{i.e.} & (\xi_1, \xi_2) & \overset{\beta_{\xi_1,\xi_2}}{\longleftarrow} & (\xi_1 \times_S \xi_2, \xi_1 \times_S \xi_2). \end{array}$$

Pour ceci, on décrit $\mathcal{H}$ comme

$$\mathcal{H} = (\underline{\mathrm{Fib}}_W S)(W_S)^{-1}$$

et on note que δ, π proviennent de foncteurs adjoints au niveau de $\mathcal{H}_0 = \underline{\mathrm{Fib}}_W S$

$$\mathcal{H}_0 \underset{\pi_0}{\overset{\delta_0}{\rightleftarrows}} \mathcal{H}_0 \times \mathcal{H}_0 \ ,$$

lesquels donnent lieu à α_0, β_0

$$\begin{array}{rcl} \mathrm{id}_{\mathcal{H}_0} & \overset{\alpha_0}{\longrightarrow} & \pi_0\delta_0 \\ \mathrm{id}_{\mathcal{H}_0\times\mathcal{H}_0} & \overset{\beta_0}{\longleftarrow} & \delta_0\pi_0. \end{array}$$

On définit α, β par passage aux catégories de fractions pour α_0, β_0. Les 2 compatibilités, étant vérifiées pour α_0, β_0, le sont aussi pour α, β. Q.e.d. p. 161 [page 248]

Je dégage ici le

Lemme. *Soit* $\mathcal{X} \underset{g}{\overset{f}{\rightleftarrows}} \mathcal{Y}$ *un couple de foncteurs adjoint, et soient* Σ, Σ' *des localiseurs dans* $\mathcal{X}$, $\mathcal{Y}$ *tels que* $f(\Sigma) \subseteq \Sigma'$, $g(\Sigma') \subseteq \Sigma$, *de sorte qu'on a un couple de foncteurs*

$$\mathcal{X}\Sigma^{-1} \underset{\bar{g}}{\overset{\bar{f}}{\rightleftarrows}} \mathcal{Y}\Sigma'^{-1} \ .$$

Alors $\bar{f}$, $\bar{g}$ *sont adjoints, les flèches d'adjonction étant déduites de celles pour* f, g.

Corollaire. *Soit* $(\mathcal{M}, W)$ *une catégorie de modèles, supposons que dans* $\mathcal{M}$ *les produits binaires (resp. les produits finis) existent, et que pour* $u_1 : X_1 \longrightarrow X_1'$ *et* $u_2 : X_2 \longrightarrow X_2'$, *si* $u_1, u_2 \in W$, *alors* $u_1 \times u_2 : X_1 \times X_2 \longrightarrow X_1' \times X_2'$ *est dans* W. *Alors dans* $\mathcal{M}W^{-1}$ *les produits binaires (resp. produits finis) existent, et le foncteur localisation* $\mathcal{M} \longrightarrow \mathcal{M}W^{-1}$ *y commute.*

On applique le lemme à

$$\mathcal{M} \underset{\pi}{\overset{\delta}{\rightleftarrows}} \mathcal{M} \times \mathcal{M} \qquad \text{et } W,\ W \times W,$$

et pour l'objet final,

$$\mathcal{M} \underset{\bullet}{\overset{e}{\rightleftarrows}} e \qquad \text{et } W,\ W_0 = \{\mathrm{id}_e\}.$$

[page 249]

Commentary · Exposition [ms. 248]

Why the product remains a categorical product

The assertion about products in $\mathrm{Ho}^{\mathrm{lc}}_W(S)$ is stronger than the existence of a homotopically well-defined binary operation. Grothendieck proves the universal property by descending an adjunction. This is an efficient place to make the small localization argument completely explicit.

Suppose $F : \mathcal{C} \rightleftarrows \mathcal{D} : G$ is an adjunction, with F left adjoint to G, and both functors preserve the chosen weak equivalences. Its unit $\eta : 1 \to GF$ and counit $\epsilon : FG \to 1$ have images in the localizations. Their naturality for an inverted weak equivalence follows from their naturality for that weak equivalence: multiply the naturality square by the inverse image of the equivalence. Naturality then holds for any composite of ordinary arrows and formal inverses, which are the morphisms of the localization. The identities

$$\epsilon_{FX} F(\eta_X) = 1_{FX}, \qquad G(\epsilon_Y)\eta_{GY} = 1_{GY}$$

are equations already true before localization, hence remain true afterwards. The descended unit and counit establish $\bar{F} \dashv \bar{G}$. No lifting property and no fullness assertion about the localization functor are needed.

Apply this argument to the diagonal functor on $\mathcal{C}_S$ and its right adjoint, the ordinary product over S. The preceding block proves that this right adjoint preserves weak equivalences. The unit is the diagonal $A \to A \times_S A$; the counit consists of the two projections. Consequently the operation constructed from good representatives is the categorical product in the localized relative category. Applying the same argument to the unique functor $\mathcal{C}_S \to e$ and the functor selecting $S \to S$ proves that this object remains terminal. This supplies all finite products.

The statement in footnote 133 has an equally concrete proof. Pullback along $f : S' \to S$ sends good objects to good objects and, by the preceding relative invariance theorem, sends weak equivalences between them to weak equivalences. The canonical strict isomorphism

$$f^*(A \times_S B) \cong f^*A \times_{S'} f^*B$$

is compatible with both projections; also $f^*(S \to S) = (S' \to S')$. Their images after localization are therefore the comparison isomorphisms for a functor preserving finite products. This proof concerns the product in the relative homotopy category. It does not assert that an arbitrary strict pullback in $\mathcal{M}$ is already a homotopy pullback.

Je vais maintenant construire le foncteur

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}).$$

Il me faut d'abord prouver le

Lemme 1. *Soit $\mathcal{F}$ la sous-catégorie pleine de $\mathcal{M}(\Psi)$, catégorie des diagrammes*

$$\begin{array}{ccc} & & X_1 \\ & {\scriptstyle p_1}\swarrow & \\ S & & \\ & {\scriptstyle p_2}\nwarrow & \\ & & X_2 \end{array}$$

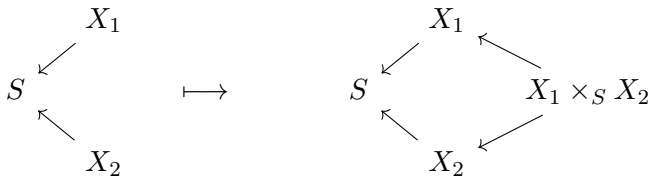
dans S [plutôt dans $\mathcal{M}$, sur S], formée de ceux pour lesquels p_1, p_2 sont dans Fib_W. Soit $W_{\mathcal{F}}$ le localiseur induit sur $\mathcal{F}$ par $W(\Psi)$ (W-équivalences terme à terme). Alors

$$\mathcal{F}W_{\mathcal{F}}^{-1} \xrightarrow{\sim} \mathcal{M}(\Psi)W(\Psi)^{-1} \ (\overset{déf}{=} \mathrm{Ho}_W(\Psi)).$$

Nous allons prouver ce lemme plus bas. En l'admettant, pour construire i_*, il suffit de construire p. 162

$$\mathcal{F} \longrightarrow \mathrm{Ho}_W(\tilde{\Psi})$$

qui applique $W_{\mathcal{F}}$ en des isomorphismes. Je prendrai, bien sûr,



$$\mathcal{F} \quad \longrightarrow \quad \mathrm{Ho}_W(\Psi).$$

Vérifions qu'il transforme flèches de $W_{\mathcal{F}}$ en isomorphismes.

[page 250]

Commentary · Exposition [ms. 249]

The deferred lemmas with functorial factorization

Grothendieck has deliberately separated the existence of useful representatives from the assertion that they compute the whole localization. Lemmas 1 and 2 assert the latter. Their proofs are deferred, and the promise on ms. 257 does not complete them in the pages supplied here. There is a particularly short completion when the factorization axiom is available functorially. We state that extra hypothesis explicitly, since the axiom on ms. 240 does not include it.

Let $D = \mathcal{M}(\Psi)$, and suppose that the assignment

$$(A \xrightarrow{p} S) \longmapsto (A \xrightarrow{j_p} R(p) \xrightarrow{r_p} S)$$

is a functor on the arrow category, with $j_p \in W$, $r_p \in \mathrm{Fib}_W$, and $r_p j_p = p$. For $d = (A \xrightarrow{p} S \xleftarrow{q} B)$, set

$$R_D d = (R(p) \xrightarrow{r_p} S \xleftarrow{r_q} R(q)).$$

This gives a functor $R_D : D \to \mathcal{F}$ and a natural transformation $j : 1_D \to JR_D$, where $J : \mathcal{F} \hookrightarrow D$ is inclusion. Every component of j is an objectwise weak equivalence. If $a : d \to d'$ is in $W(\Psi)$, naturality gives

$$(JR_D a) j_d = j_{d'} a.$$

At each vertex the two maps $j_d, j_{d'}$ and the map a belong to W; two-out-of-three therefore implies that $R_D a$ is in $W_{\mathcal{F}}$. The functor R_D consequently descends to localization. Restricted to $\mathcal{F}$, the same natural transformation has both source and target in $\mathcal{F}$. After localization it supplies an isomorphism $1 \cong \bar{R}_D \bar{J}$, while its unrestricted version supplies $1 \cong \bar{J}\bar{R}_D$. These are precisely the two isomorphisms that show $\bar{J}$ is an equivalence. This proves Lemma 1 with the additional hypothesis. Over a fixed base S, the same argument applies to every full intermediate category between $\mathrm{Fib}_W(S)$ and $\mathcal{M}/S$: the replacement always lands in $\mathrm{Fib}_W(S)$ and the natural comparison stays in the intermediate category. It therefore proves the strengthened replacement assertion (c) recalled on ms. 240 in this functorial setting.

For Lemma 2 use the identical construction on the three vertices of a square belonging to Ψ. Keep its fourth vertex T unchanged, and replace its maps to A and B by their composites with j_p and j_q. Their composites to S still agree, so this is a strictly commutative square. It defines a functor $R_E : \mathcal{M}(\tilde{\Psi}) \to \mathcal{G}$ with a natural objectwise weak equivalence $1 \to KR_E$, whose component at T is the identity. The same two-out-of-three argument makes R_E homotopical, and its restriction to $\mathcal{G}$ gives the other inverse comparison. Hence $\mathcal{G}[W_{\mathcal{G}}^{-1}] \to \mathrm{Ho}_W(\tilde{\Psi})$ is an equivalence.

This argument also specifies what the arbitrary choice of a factorization would fail to provide: not the value on an object, but the functor and the natural transformation used in the proof. It uses neither an assumption that every object is fibrant nor a saturation theorem identifying W with all maps inverted by localization. The required input is a functorial factorization, which is a separate sufficient hypothesis.

Commentary · Later developments [ms. 249]

A nonfunctorial repair by derivability

The passage to diagrams can be made without functorial choices, and the later theory also removes the need to add global fibrancy to the written hypotheses. The appropriate structure is a

left derivable category in the sense of Cisinski (2010, Definition 1.1). We check its one potentially hidden axiom. Let $p : X \to Y$ belong to $W \cap \mathrm{Fib}_W$, and let $u : Y' \to Y$ be arbitrary. Factor u as $Y' \xrightarrow{j} Z \xrightarrow{r} Y$, where $j \in W$ and $r \in \mathrm{Fib}_W$. Since r^* preserves W, the map $X \times_Y Z \to Z$ is in W. The map $X \times_Y Y' \to X \times_Y Z$ is the pullback of j along a base change of p, hence is also in W. The commutative square with bottom map j and two-out-of-three now give $X \times_Y Y' \to Y' \in W$. Thus trivial fibrations remain trivial under arbitrary pullback.

The other derivability axioms are immediate from the standing assumptions: a terminal object exists, weak equivalences contain isomorphisms and satisfy two-out-of-three, fibrations compose and are stable under the available pullbacks, and the required factorization is the axiom of ms. 240. These properties are stronger than the restricted pullback and factorization requirements of Definition 1.1. The precise later results are the finite-diagram structures and localization comparison of Theorem 1.30 and Corollaries 1.31–1.32, followed by the right-Kan-extension theorem, Proposition 2.6; Theorem 2.21 packages the construction as a one-sided derivator on finite direct categories.

This also settles the strengthened fixed-base condition (c) recalled from ms. 218. The slice $\mathcal{M}/S$, with the inherited fibrations and weak equivalences, is left derivable and has $\mathrm{Fib}_W(S)$ as its full subcategory of fibrant objects. Every full intermediate category $\mathcal{E}$ containing these objects is left derivable with the same fibrant objects: the pullbacks demanded by the axioms take place among fibrant objects, and the middle object in a factorization with fibrant target is again fibrant. Proposition 1.8 identifies the localization of each such category with that of its fibrant objects. It follows that both inclusions in condition (c) are localization equivalences. Condition (b), the equality of ordinary and universal weak equivalences between good objects, is the relative invariance argument dual to Proposition 16; together they give (a).

For clarity, the particular inclusions $\mathcal{F} \subset D$ and $\mathcal{G} \subset E$ can be checked by that paper's approximation theorem, Theorem 3.12. We give the verification, since $\mathcal{G}$ imposes no fibration condition on its fourth vertex. On $\mathcal{F}$ take a morphism (a_0, a_1, a_2) to be a fibration when the three maps

$$S \to S', \qquad A \to S \times_{S'} A', \qquad B \to S \times_{S'} B'$$

are fibrations. On $\mathcal{G}$ use these same conditions and additionally require the map $T \to T'$ between fourth vertices to be a fibration. Take weak equivalences objectwise in both cases. An object of $\mathcal{F}$ is fibrant precisely when its base S is fibrant in $\mathcal{M}$. An object of $\mathcal{G}$ is fibrant precisely when both S and its fourth vertex T are fibrant. Its remaining vertices are then fibrant because their maps to S are fibrations.

Here is the factorization needed both for these structures and for approximation. Given a map from an arbitrary cospan to a fibrant good cospan $(A' \to S' \leftarrow B')$, first factor its base map as

$$S \xrightarrow{\sim} S_1 \twoheadrightarrow S'.$$

Factor the resulting maps $A \to S_1 \times_{S'} A'$ and $B \to S_1 \times_{S'} B'$ as weak equivalences followed by fibrations. Call the middle objects A_1, B_1. Their maps to S_1 are fibrations, and the resulting cospan maps to the target by a fibration of the displayed kind. This constructs an objectwise weak equivalence from the original cospan to an object of $\mathcal{F}$, compatibly with the given map to the target.

For a map of squares with fibrant target in $\mathcal{G}$, make those three factorizations and put

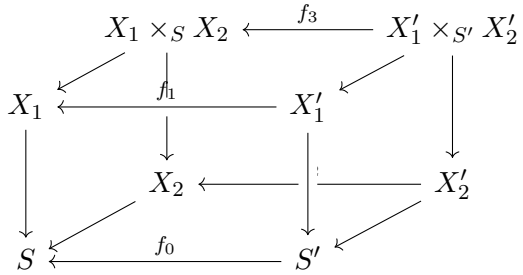
$$P_1 = A_1 \times_{S_1} B_1, \qquad P' = A' \times_{S'} B'.$$

The map $P_1 \to P'$ is a fibration. Indeed it factors through $S_1 \times_{S'} P'$; the first map is a composite of pullbacks of the two relative leg fibrations, and the second is a pullback of $S_1 \to S'$. The source corner T maps canonically to $P_1 \times_{P'} T'$. Factor this map into $T \xrightarrow{\sim} T_1$ followed by a fibration. Since $P_1 \times_{P'} T' \to T'$ is itself a fibration, so is $T_1 \to T'$. The middle square lies in $\mathcal{G}$ and is fibrant: its base maps fibrantly to the fibrant target base, and T_1 maps fibrantly to the fibrant object T'. Thus the factorization has all the required properties.

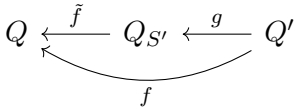
The remaining derivability axioms can be checked on these same relative maps. Composition is a composite of a relative map and a pullback of the next one. A pullback of a fibration is computed vertex by vertex; at either leg its relative map is the corresponding pullback of the original relative map, so the resulting object remains good. For a trivial fibration, the relative leg maps are also weak equivalences, by two-out-of-three applied to the weak base map and the good target leg. Their pullbacks remain weak equivalences, as do the fourth-vertex maps. This verifies the trivial-fibration axiom. Factoring a diagonal of a fibrant object by the construction above supplies its path object. The ambient categories of cospans and squares carry the objectwise fibration structures

of Cisinski's Corollary 1.31; the nontrivial factorization axiom there is supplied by the finite inverse diagram construction. The inclusions preserve fibrations, trivial fibrations, terminal objects and their pullbacks; they are exact and reflect weak equivalences.

Finally, Cisinski's approximation condition AP2 asks, for a map from an ambient fibrant object to an included fibrant object, for a weak-equivalence comparison with an object of the smaller category over the same target. The explicit factorizations just constructed supply it, with the preliminary weak equivalence of the ambient source taken to be its identity. AP1 is reflection of weak equivalences. Theorem 3.12 now proves both localization equivalences. This proves the two deferred lemmas under the written factorization and pullback hypotheses. The derivable-category argument controls the morphisms without either global fibrancy or functorial choices of replacements.



Il faut prouver que si $f_0, f_1, f_2 \in W$, alors $f_3 \in W$. Mais considérons l'image inverse du carré cartésien Q par $f_0 : S' \longrightarrow S$, d'où un carré cartésien $Q_{S'}$ et une factorisation



de $Q' \longrightarrow Q$. Comme $f_0 \in W$, et que X_1, X_2, $X_1 \times_S X_2$ sont dans $\underline{\mathrm{Fib}}_W S$, les flèches horizontales réalisant $Q_{S'} \longrightarrow Q$ sont toutes dans W. Ceci, plus le fait que $f_1, f_2 \in W$, impliquent que dans le morphisme g, les flèches g_1, g_2 sont dans W. Il en résulte que la flèche

$$g_3 : X_1' \times_{S'} X_2' \longrightarrow (X_1 \times_S X_2) \times_S S'$$

l'est aussi (théorie du produit dans $\underline{\mathrm{Fib}}_W S'$ et $\mathrm{Ho}^{\mathrm{lc}}_W(S)$), donc $f_3 = \tilde{f}_3 g_3 \in W$, q.e.d.

[page 251]

Commentary · Exposition [ms. 250]

The base is allowed to move

Having obtained good representatives, Grothendieck must show that their pullbacks vary homotopically in the base as well as in the two inputs. This is the extra content of the argument on ms. 250. The fixed-base product proposition cannot by itself be applied to a morphism of cospans whose middle objects differ.

Suppose a morphism of good cospans has components

$$a : A' \to A, \qquad b : B' \to B, \qquad s : S' \to S$$

in W. Pull the target cospan back along s, obtaining

$$\tilde{A} = S' \times_S A, \quad \tilde{B} = S' \times_S B, \quad \tilde{P} = S' \times_S (A \times_S B).$$

The projections $\tilde{A} \to A$ and $\tilde{B} \to B$ are weak equivalences because $A \to S$ and $B \to S$ are W-fibrations. The map $A \times_S B \to S$ is a composite of W-fibrations, so $\tilde{P} \to A \times_S B$ is also in W. The given maps factor through $\tilde{A}$ and $\tilde{B}$; two-out-of-three shows that $A' \to \tilde{A}$ and $B' \to \tilde{B}$ lie in W. These are now maps over the same base S'. The product calculation gives

$$A' \times_{S'} B' \longrightarrow \tilde{A} \times_{S'} \tilde{B} \cong \tilde{P} \quad \text{in } W.$$

Composition proves that the original map of pullbacks is a weak equivalence. All four components of the completed square have therefore been checked. The final parenthesis of the source should refer to $\mathrm{Ho}^{\mathrm{lc}}_W(S')$, since this intermediate product is taken over S'; the printed S is retained as an edition slip.

The calculation explains why one can descend the strict completion functor on $\mathcal{F}$. It also shows that its homotopicality is a theorem about a whole diagram, not an assertion that the pullback functor preserves arbitrary weak equivalences in each uncontrolled variable. The good-map assumptions are used at the three projections above and at the two fixed-base product maps. In Brown's setting the corresponding pullback invariance is already the content of Brown (1973, Part I, §4, Lemmas 1–2, pp. 428–429).

Notons qu'on a un isomorphisme canonique

$$(*) \qquad \alpha : i^* i_* \xrightarrow{\sim} \mathrm{id}_{\mathrm{Ho}_W(\Phi)}.$$

J'ai envie de prouver que les foncteurs

$$\mathrm{Ho}_W(\Phi) \underset{i_*}{\overset{i^*}{\leftrightarrows}} \mathrm{Ho}_W(\tilde{\Phi})$$

sont adjoints, et que $(*)$ est une des flèches d'adjonction. Il faut définir l'autre flèche d'adjonction p. 163

$$(**) \qquad \beta : i_* i^* \longleftarrow \mathrm{id}_{\mathrm{Ho}_W(\tilde{\Phi})}.$$

Mais pour ceci, je dois utiliser le

Lemme 2. Soit $\mathcal{G}$ la sous-catégorie pleine de $\mathcal{M}(\tilde{\Phi})$, formée des $\begin{array}{ccc} X_1 & \xleftarrow{q_2} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle q_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ tels que $p_1, p_2 \in \mathrm{Fib}_W$ et soit $W_{\mathcal{G}}$ le localiseur induit par $W(\tilde{\Phi})$ (W-équivalences terme à terme). Alors l'inclusion $\mathcal{G} \longrightarrow \mathcal{M}(\tilde{\Phi})$ induit une équivalence

$$\mathcal{G}W_{\mathcal{G}}^{-1} \xrightarrow{\sim} \mathcal{M}(\tilde{\Phi})W(\tilde{\Phi})^{-1} \overset{\text{déf}}{=} \mathrm{Ho}_W(\tilde{\Phi}).$$

Je vais pour le moment admettre ce

lemme, tout comme le lemme 1 (p. 249). Je vais considérer i^*, i_* comme des flèches entre les categories $\mathcal{F}W_{\mathcal{F}}^{-1}$, $\mathcal{G}W_{\mathcal{G}}^{-1}$. Elles sont induites par les flèches [page 252]

$$\mathcal{F} \underset{i_{0*}}{\overset{i_0^*}{\leftrightarrows}} \mathcal{G}$$

entre $\mathcal{F}$ et $\mathcal{G}$ eux mêmes,

$$i_0^* : \quad \mathcal{G} \quad \longrightarrow \quad \mathcal{F} : \qquad \begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2 \end{array} \qquad \longmapsto \qquad \begin{array}{c} X_1 \\ \downarrow \\ S \end{array} \begin{array}{c} \\ \longleftarrow \\ \\ \end{array} \begin{array}{c} X_2 \\ \downarrow \\ X_2 \end{array}$$

$$i_{0*} : \quad \mathcal{F} \quad \longrightarrow \quad \mathcal{G} : \qquad \begin{array}{c} X_1 \\ \downarrow \\ S \end{array} \begin{array}{c} \\ \longleftarrow \\ \\ \end{array} \begin{array}{c} X_2 \\ \\ X_2 \end{array} \qquad \longmapsto \qquad \begin{array}{ccc} X_1 & \longleftarrow & X_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2. \end{array}$$

Ces deux foncteurs sont adjoints l'un de l'autre, et compatibles aux localiseurs $W_{\mathcal{F}}$, $W_{\mathcal{G}}$. Donc les foncteurs déduits de i_0^*, i_{0*} par passage aux catégories de fractions sont également adjoints (cf. lemme p. 248).

Ainsi, modulo la démonstration des lemmes 1 et 2, on a prouvé ceci :

Théorème. *Soit $(\mathcal{M}, W)$ une catégorie de modèles ($W \subseteq Fl\,\mathcal{M}$) satisfaisant l'axiome de factorisation gauche (p. 240), donnant donc lieu à une bonne théorie des fibres W-homotopiques. Soient* p. 164 [page 253]

$$\begin{array}{ccc} & & 1 \\ \Psi = 0 & & \\ & & 2 \end{array} \qquad \xrightarrow{i} \qquad \begin{array}{ccccc} & & 1 & & \\ \tilde{\Psi} = 0 & & & & 3 \\ & & 2 & & \end{array}$$

(immersions ouvertes dans Cat*), et considérons le foncteur restriction*

$$i^* : \mathrm{Ho}_W(\tilde{\Psi}) \longrightarrow \mathrm{Ho}_W(\Psi).$$

Ce foncteur admet un adjoint à droite

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}).$$

Ce foncteur i_* se 'calcule' ainsi : sa valeur en $\begin{array}{ccc} & & X_1 \\ & & \downarrow{\scriptstyle p_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ *est canoniquement isomorphe (dans* $\mathrm{Ho}_W(\tilde{\Psi})$*) à* $\begin{array}{ccc} X_1 & \longleftarrow & X_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2 \end{array}$ *, si on suppose que p_1 ou p_2 est une W-fibration.*

Dans le cas général il se calcule en factorisant p_1 en

$$X_1 \xrightarrow{i_1} \bar{X}_1 \xrightarrow{p_1'} S, \qquad i_1 \in W,\ p_1' \in Fib_W,$$

et en prenant $\begin{array}{ccc} \bar{X}_1 & \longleftarrow & \bar{X}_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2 \end{array}$ *, ou en*

procédant de façon symétrique, ou en factorisant simultanement p_1 et p_2, et prenant [page 254]

$$\begin{array}{ccc} \bar{X}_1 & \longleftarrow & \bar{X}_1 \times_S \bar{X}_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & \bar{X}_2. \end{array}$$

Commentary · Examples [ms. 254]

The missing corner can be an entire group

A groupoid calculation makes the difference between a strict pullback and its homotopy replacement visible without any topological machinery. Work in groupoids with equivalences as W, and let G be a group. Write BG for the one-object groupoid with automorphism group G. Consider the cospan

$$* \longrightarrow BG \longleftarrow *.$$

Its strict pullback is the terminal groupoid: there is one pair of objects, and the only pair of morphisms is the pair of identities. The homotopy pullback will instead be the discrete groupoid on the set G.

To compute it, let EG be the groupoid whose objects are the elements of G and with exactly one morphism from each object to every other object. Define $p : EG \to BG$ by sending the unique morphism $g \to h$ to hg^{-1}. The composition formula

$$(kh^{-1})(hg^{-1}) = kg^{-1}$$

shows that this is a functor. The inclusion $* \to EG$ selecting the identity element is fully faithful and essentially surjective, hence is an equivalence; its composite with p is the specified map $* \to BG$.

The functor p is an isofibration. Given an object g and an arrow a in BG, the arrow $g \to ag$ lies above a. Isofibrations of groupoids have the required W-fibration property. For completeness, pull back an equivalence $u : C \to D$ over a base along an isofibration $E \to S$. The induced functor on pullbacks is fully faithful because u is. Given an object (d, e) of the target pullback, choose an isomorphism $u(c) \cong d$; its image in S lifts through the isofibration, and the lift changes e to an object lying strictly over the image of c. This produces an object in the source pullback isomorphic to (d, e), proving essential surjectivity. The same reasoning applies after any base change.

We can therefore replace the left leg by $* \xrightarrow{\sim} EG \xrightarrow{p} BG$. In the strict pullback $EG \times_{BG} *$, an object is an element $g \in G$. A morphism from g to h exists precisely when its image hg^{-1} in BG is the identity, which is equivalent to $g = h$. Thus

$$EG \times_{BG} * \cong G_{\mathrm{disc}}.$$

For $G = \mathbb{Z}/2$, the strict pullback has one connected component while the homotopy pullback has two. For a general group, its elements record the possible isomorphisms between the two images of the chosen object of $*$. The derived completion preserves those comparisons.

This is the cospan version of the groupoid computations already used in the commentary on Chapter VI, manuscript p. 107. Its purpose here is narrower: it exhibits the actual fourth vertex produced by the theorem and explains why the maps $* \to BG$ must be replaced. The whole completed square, including its two projections and its map to the original cospan after localization, is the value of i_*.

p. 165

La loi fonctorielle s'obtient à l'aide du lemme 1 p. 249, qui permet de se borner aux arguments $\begin{array}{ccc} X_1 & & \\ {\scriptstyle p_1}\downarrow & & \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ *tels que* $p_1, p_2 \in Fib_W$*, de sorte que l'on trouve un foncteur*

$$\begin{array}{ccc} X_1 & & \\ \downarrow & & \\ S & \longleftarrow & X_2 \end{array} \quad \longmapsto \quad \begin{array}{ccc} X_1 & \longleftarrow & X_1 \times_S X_2 \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2, \end{array}$$

compatible aux localiseurs, et passant aux catégories de fractions $\mathrm{Ho}_W(\Psi)$ *et* $\mathrm{Ho}_W(\tilde{\Psi})$ *respectivement.*

NB En procédant de façon similaire, plus simple (sans besoin d'un recours au lemme 1), on trouve l'existence d'un *adjoint à gauche* $i_!$ de $i^* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi})$, sous réserve de l'existence d'un objet initial φ dans $\mathcal{M}$. Cet foncteur $i_!$ est donné par

[page 255]

$$i_! \left(\begin{array}{ccc} X_1 & & \\ \downarrow & & \\ S & \longleftarrow & X_2 \end{array} \right) \quad = \quad \left(\begin{array}{ccc} X_1 & \longleftarrow & \varphi \\ \downarrow & & \downarrow \\ S & \longleftarrow & X_2 \end{array} \right),$$

la loi fonctorielle étant évidente.

Commentary · Exposition [ms. 254]

The adjunction determines the completion

The adjunction is what makes Grothendieck's construction a canonical completion of a cospan. On the good-diagram categories, restriction $r : \mathcal{G} \to \mathcal{F}$ has right adjoint c, where c supplies the strict pullback. A morphism from a square Q to $c(d)$ is uniquely determined by its three components on the cospan $r(Q)$: the fourth component exists uniquely by the pullback universal property. This gives the adjunction bijection before localization.

The counit $rc \to 1_{\mathcal{F}}$ is the identity on the original three vertices. The unit at a square with fourth vertex T is the map to its completion, whose fourth component is

$$T \longrightarrow A \times_S B.$$

At a completed square this map is the identity, and restricting the unit to its three original vertices also gives the identity. These observations verify both triangle identities directly. Restriction preserves objectwise weak equivalences; the preceding base-change calculation proves that completion does too. The localization lemma on ms. 248 therefore descends the adjunction. Once Lemmas 1 and 2 identify the localized good-diagram categories with the required diagram categories, it is exactly $i^* \dashv i_*$, with invertible counit $i^* i_* \to 1$.

The formula using only one fibrant leg follows by an explicit comparison, not by a symmetry convention. If $A \to S$ is already a W-fibration, replace only B by $\bar{B}$. The map

$$A \times_S B \longrightarrow A \times_S \bar{B}$$

is in W, as in the block at ms. 242, and forms an objectwise weak equivalence between the completed square and one with two good legs. The latter computes i_*. For an arbitrary cospan, first replace A; the same argument compares the one-leg and two-leg formulas. Interchanging A and B gives the other formula. These comparisons concern objects of $\mathrm{Ho}_W(\tilde{\Psi})$ and retain all the projection maps.

There is also the left adjoint mentioned in the concluding NB. If $\mathcal{M}$ has an initial object φ, extending a cospan by putting φ at vertex 3 is left adjoint to restriction already on ordinary diagram categories. The fourth component of any extension of a map of cospans is the unique map $\varphi \to T$. This extension preserves objectwise weak equivalences, since its fourth component on every morphism is 1_φ, so the adjunction descends by the same lemma. It needs no fibrant replacement. The displayed type of i^* in that NB is reversed in the edition; its correct type remains $\mathrm{Ho}_W(\tilde{\Psi}) \to \mathrm{Ho}_W(\Psi)$. Likewise the $\Phi, \tilde{\Phi}$ used on ms. 251–252 denote the $\Psi, \tilde{\Psi}$ of the theorem, not a second pair of diagram shapes.

Commentary · Exposition [ms. 255]

Two extensions of the same cospan

Grothendieck has now reached the point at which a homotopy fibre can be characterized by its relation to all diagrams, rather than by a particular construction of paths. The two extensions of the three-vertex diagram make this distinction visible. Restriction has the direction

$$i^* : \mathrm{Ho}_W(\tilde{\Psi}) \longrightarrow \mathrm{Ho}_W(\Psi).$$

The reversed direction in the preceding NB is an edition slip. Its left adjoint, discussed in the preceding block, fills the fourth vertex with an initial object; its right adjoint fills that vertex with a homotopy pullback. These operations answer different universal problems on the same cospan.

For the right adjoint, the fourth vertex must instead receive a compatible pair of maps. Once the two legs to S are W-fibrations, that universal object is $X_1 \times_S X_2$. The extra work in Lemmas 1 and 2 is what allows this calculation on suitably replaced diagrams to determine a functor and an adjunction on the original localization. The homotopy limit of the cospan is then obtained by evaluating this completed square at vertex 3.

The indexing convention explains the apparently reversed endpoint in the following lemma. An object of $\mathcal{M}(I)$ is a functor $I^{\mathrm{op}} \to \mathcal{M}$. If t is terminal in I, each arrow $i \to t$ induces $F(t) \to F(i)$. A natural transformation from the constant diagram on A to F is consequently determined by its component $A \to F(t)$, and every such component gives a natural transformation. Thus

$$\mathrm{Hom}_{\mathcal{M}(I)}(p^* A, F) \cong \mathrm{Hom}_{\mathcal{M}}(A, F(t)).$$

Evaluation at t is a right adjoint. The displayed formula survives localization because both functors preserve the designated weak equivalences. In particular, the object constructed here is universal for maps of coherent cospans. There is no assertion that taking an ordinary pullback in $\mathcal{M}[W^{-1}]$ would recover it.

Ce n'est pas encore la fin du sorite! Il faut encore interpréter le foncteur composé

$$\mathrm{Ho}_W(\Psi) \xrightarrow{i_*} \mathrm{Ho}_W(\tilde{\Psi}) \xrightarrow{\varepsilon_3^*} \mathrm{Ho}_W(e) = \mathrm{Ho}_W,$$

où

$$\varepsilon_3 : e \xrightarrow{\sim} \{3\} \qquad \longrightarrow \qquad \begin{array}{ccccc} & & 1 & & \\ & \swarrow & & \nwarrow & \\ 3 & & & & 0 \\ & \nwarrow & & \swarrow & \\ & & 2 & & \end{array}$$

est l'inclusion du sommet 3 dans $\tilde{\Psi}$. Si $\mathrm{Ho}_W(\Psi)$ est interprété, conformément au lemme 1, comme $\mathcal{F}W_{\mathcal{F}}^{-1}$, ce foncteur composé n'est autre que

$$\begin{array}{ccc} X_1 & & \\ \downarrow & \longmapsto & X_1 \times_S X_2 \\ S \longleftarrow & X_2 & \end{array}$$

$$\text{(pour } \begin{array}{ccc} X_1 & & \\ {\scriptstyle p_1}\downarrow & & \\ S & \xleftarrow[p_2]{} & X_2 \end{array} \in \mathrm{Ob}\,\mathcal{F}\text{, i.e. } p_1, p_2 \in \mathrm{Fib}_W.)$$

On veut interpréter ce foncteur comme un $\varprojlim$ W-homotopique, i.e. vu comme un 'produit fibré W-homotopique'. De façon précise, on a ceci : p. 166

Corollaire 1. *Considérons $p_\Psi : \Psi \longrightarrow e$, et le foncteur* [page 256]

$$p_\Psi^* : \mathrm{Ho}_W(e) = \mathrm{Ho}_W \longrightarrow \mathrm{Ho}_W(\Psi),$$

déduit par passage aux catégories de fractions du foncteur

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & \mathcal{M}(\Psi) \\ X & \longrightarrow & \begin{array}{ccc} X & & \\ \downarrow & & \\ X & \longleftarrow & X \end{array} \end{array}.$$

Ce foncteur admet le foncteur $\varepsilon_3^* i_*$ précédent comme adjoint à droite, i.e.

$$p_{\Psi *} = \varepsilon_3^* i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W.$$

Démonstration. On peut regarder p_ψ comme le composé

$$\Psi \xrightarrow{i} \tilde{\Psi} \xrightarrow{p_{\tilde{\Psi}}} e,$$

et on a donc

$$p_\Psi^* \simeq i^*(p_{\tilde{\Psi}})^*.$$

Donc l'existence et la détermination d'un adjoint à droite pour p_Ψ^* résultera de l'existence et de la détermination d'adjoints à droite i_*, $(p_{\tilde{\Psi}})_*$ de i^*, $(p_{\tilde{\Psi}})^*$, via [page 257]

$$(p_\Psi)_* \simeq (p_{\tilde{\Psi}})_* i_*. \qquad \text{p. 167}$$

Donc le corollaire 1 résulte de la formule

$$p_{\tilde{\Psi}\,*} = \varepsilon_3^*,$$

qui elle-même est un cas particulier du

Lemme. *Soit I dans* Cat*, avec objet final e, considérons l'inclusion*

$$\varepsilon : e \longrightarrow I, \qquad \text{et } p : I \longrightarrow e.$$

Soit $(\mathcal{M}, W)$ une catégorie de modèles, et considérons

$$\mathrm{Ho}_W(I) \xleftarrow{p^*} \mathrm{Ho}_W.$$

Ce foncteur admet un adjoint à gauche `[plutôt à droite]` *p_*, lequel est isomorphe à ε^*.*

Démonstration. Considérons le couple de foncteurs adjoints

$$\mathcal{M}(I) \underset{p_*=\varprojlim_I}{\overset{p^*}{\leftleftarrows}} \mathcal{M} \quad ;$$

comme I admet l'objet final e, le foncteur p_* est isomorphe à ε^*. Ces foncteurs p^*, ε^* sont compatibles aux localiseurs $W(I)$, W, donc on conclut par le lemme page 248.

Il ne reste plus qu'à prouver les lemmes 1 et 2. Je le ferai demain.

Aujourd'hui, j'ai envie encore de donner les sorites des carrés W-cartésiens dans une [page 258]

Proposition 19 : *Soit* $\begin{array}{ccc} X_1 & \xleftarrow{f_1} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle f_2} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$ *un carré commutatif dans $\mathcal{M}$. Considérons des factorisations de p_1, p_2 en*

$$X_1 \xrightarrow{i_1} \bar{X}_1 \xrightarrow{\bar{p}_1} S, \qquad X_2 \xrightarrow{i_2} \bar{X}_2 \xrightarrow{\bar{p}_2} S,$$

$i_1, i_2 \in W$, $\bar{p}_1, \bar{p}_2 \in F_W$ `[= Fib`$_W$`]`, et considérons les flèches induites par $X_3 \longrightarrow X_1 \times_S X_2$ ($\bar{X}_1 \times_S X_2$ déduite de Q) via $X_1 \times_S X_2 \longrightarrow \begin{array}{c} \nearrow \bar{X}_1 \times_S X_2 \\ \longrightarrow\ \bar{X}_1 \times_S \bar{X}_2 \\ \searrow X_1 \times_S \bar{X}_2 \end{array}$ *, de sorte qu'on trouve un diagramme commutatif*

$$\begin{array}{ccc} & & \bar{X}_1 \times_S X_2 \\ & \overset{\pi_1}{\nearrow} & {\scriptstyle j_2}\downarrow \\ X_3 & \underset{\pi_2}{\overset{\pi_0}{\longrightarrow}} & \bar{X}_1 \times_S \bar{X}_2 \\ & \underset{\pi_2}{\searrow} & {\scriptstyle j_1}\uparrow \\ & & X_1 \times_S \bar{X}_2, \end{array}$$

où $j_1 = i_1 \times_S \mathrm{id}_{\bar{X}_2}$ et $j_2 = \mathrm{id}_{\bar{X}_1} \times i_2$ sont dans W (car i_1 et i_2 le sont, et $\bar{X}_1$ et $\bar{X}_2$ sont dans $\underline{\mathrm{Fib}}_W S$). On a donc

$$(\pi_1 \in W) \qquad \Longleftrightarrow \qquad (\pi_2 \in W) \qquad\qquad (\Longleftrightarrow \ \pi_0 \in W),$$

ce qui prouve que la condition $\pi_1 \in W$, qui semble dépendre du choix de la factorisation par i_1, n'en dépend pas, et idem pour $\pi_2 \in W$, et pour $\pi_0 \in W$ (qui semble dépendre à la fois de i_1, et de i_2).

Commentary · Exposition [ms. 258]

Independence of replacements and invariance of the test

Grothendieck wants the property of a square to be independent of the auxiliary factorization used to recognize it. Proposition 19 achieves this with a particularly economical comparison: one changes the first leg while holding the second fixed, then exchanges their roles. We spell out the argument because it also supplies the invariance used in the next pasting proof.

Write a square as $D \to A$, $D \to B$ over the cospan $A \to S \leftarrow B$. Choose $A \xrightarrow{a} \bar{A} \to S$ and $B \xrightarrow{b} \bar{B} \to S$ as in the proposition. The two arrows

$$\bar{A} \times_S B \longrightarrow \bar{A} \times_S \bar{B}, \qquad A \times_S \bar{B} \longrightarrow \bar{A} \times_S \bar{B}$$

are in W, since each pulls back a member of W along a W-fibration. Thus the three comparison maps from D are simultaneously in W. Now choose a second factorization of $A \to S$, retaining the chosen factorization of $B \to S$. Both first-leg tests are equivalent to the very same test $D \to A \times_S \bar{B}$. They therefore agree. The argument with A and B interchanged proves independence of the other choice. No map between the two factorizations of A has been assumed.

For the pasting argument we also need the following verification. Suppose a map of squares $Q \to Q'$ is in W at all four vertices. Denote the cospans by $A \to S \leftarrow B$ and $A' \to S' \leftarrow B'$, and choose a factorization

$$A' \xrightarrow{a'} \widehat{A}' \xrightarrow{q'} S', \qquad a' \in W, \quad q' \in \mathrm{Fib}_W.$$

Set $T = S \times_{S'} \widehat{A}'$. The projection $T \to \widehat{A}'$ is in W, being the pullback of $S \to S'$ along q'. The induced $A \to T$ is in W by two-out-of-three, since its composite to $\widehat{A}'$ is $A \to A' \xrightarrow{a'} \widehat{A}'$. Moreover $T \to S$ is a W-fibration. This is an admissible replacement of the first leg of Q.

With these choices the comparison of the two pullback objects is

$$T \times_S B \cong \widehat{A}' \times_{S'} B \longrightarrow \widehat{A}' \times_{S'} B'.$$

It is in W, since it is obtained by pulling back $B \to B'$ along the pullback of q'. The square relating these objects to $D \to D'$ commutes. Its vertical arrows are in W, so two-out-of-three shows that the comparison for Q is in W if and only if that for Q' is. This proves invariance under objectwise weak equivalences, and hence under zigzags of them, using only the factorization hypothesis and the intrinsic definition of Fib_W. Strong saturation, functorial choices of factorization, and the deferred adjunction lemmas are unnecessary for this particular assertion.

Définition. *Quand ces conditions sont satisfaites, on dit que le carré Q est W-cartésien.* p. 168

Corollaire 1. *Si Q est W-cartésien, alors l'image de Q dans $\mathrm{Ho}_W(\tilde{\Psi})$ est dans l'image essentielle de* [page 259]

$$i_* : \mathrm{Ho}_W(\Psi) \longrightarrow \mathrm{Ho}_W(\tilde{\Psi}),$$

i.e. la flèche canonique dans $\mathrm{Ho}_W(\tilde{\Psi})$

$$Q \longrightarrow i_* i^* Q$$

est [un] *isomorphisme. La réciproque est vraie si W est fortement saturé.*

Commentary · Exposition [ms. 259]

The unit and the role of strong saturation

Grothendieck now compares his concrete test with the intrinsic language of homotopy Kan extension. This comparison separates two tasks which coincide in the familiar examples: constructing an isomorphism after localization, and recognizing that a specified original arrow belongs to W.

Let Q^r be obtained by replacing $X_1 \to S$ and $X_2 \to S$ by $\bar{X}_1 \to S$ and $\bar{X}_2 \to S$, while keeping X_3 and S. There is an objectwise weak equivalence $Q \to Q^r$. The unit on Q^r is represented by the map to the completed pullback square which is the identity on $\bar{X}_1$, $\bar{X}_2$, and S, and is

$$\pi_0 : X_3 \longrightarrow \bar{X}_1 \times_S \bar{X}_2$$

at the fourth vertex. If $\pi_0 \in W$, this map of squares becomes an isomorphism. Naturality of the unit then gives the assertion for Q.

Conversely, if the unit on Q is invertible, the unit on Q^r is invertible. Evaluation at vertex 3 shows that π_0 is inverted by $\mathcal{M} \to \mathcal{M}[W^{-1}]$. Strong saturation says exactly that this forces $\pi_0 \in W$. Two-out-of-three by itself makes no such assertion about every map inverted by localization. This is the reason for the qualification in Corollary 1, rather than an incidental technical precaution.

Since $i^* i_* \cong \mathrm{id}$, the right adjoint is fully faithful, and the usual adjunction argument identifies its essential image with the squares whose unit is invertible. In modern derivator terminology this essential-image condition defines cartesian squares; the concrete W test agrees with it under the saturation hypothesis just explained. See Groth (2013, Proposition 1.20 and Lemma 1.21) for the fully faithful Kan-extension formulation. That formulation presupposes the construction of the adjunction, so it does not by itself discharge the two deferred localization lemmas in this manuscript.

Corollaire 2. *Considérons un diagramme commutatif dans* $\mathcal{M}$

$$\text{(D)} \qquad \begin{array}{ccccc} X'_0 & \longleftarrow & X'_1 & \longleftarrow & X'_2 \\ \downarrow & 1 & \downarrow & 2 & \downarrow \\ X_0 & \longleftarrow & X_1 & \longleftarrow & X_2, \end{array}$$

d'où deux carrés commutatifs 1, 2, et un carré composé

$$\begin{array}{ccc} X'_0 & \longleftarrow & X'_2 \\ \downarrow & 3 & \downarrow \\ X_0 & \longleftarrow & X_2. \end{array}$$

Supposons le carré 1 W-cartésien. Alors le carré 2 est W-cartésien si et seulement si le carré 3 l'est.

Démonstration. Considérons une factorisation de $\begin{array}{c} X'_0 \\ \downarrow \\ X_0 \end{array}$ en

$$\begin{array}{ccc} \bar{X}'_0 & \longleftarrow & X'_0 \\ & \searrow & \downarrow \\ & & X_0 \end{array}$$

avec $(X'_0 \longrightarrow \bar{X}'_0) \in W$, et $(\bar{X}'_0 \longrightarrow X_0) \in \mathrm{Fib}_W$. On peut alors remplacer dans (D) X'_0 par $\bar{X}'_0$, obtenant un diagramme $(\bar{D})$, et on voit aussitôt que chacun des carrés 1(D), 2(D), 3(D) est W-cartésien si et seulement si le carré correspondant $1(\bar{D})$, $2(\bar{D})$ ou $3(\bar{D})$ l'est. Cela montre qu'on peut supposer $X_0 \longrightarrow S$ dans $\mathrm{Fib}_W S$. Considérons alors le diagramme commutatif déduit de (D) p. 169 [page 260]

$$\begin{array}{ccccccc} X'_0 & \longleftarrow & \tilde{X}'_1 \overset{\text{déf}}{=} X'_0 \times_{X_0} X_1 & \overset{\alpha}{\longleftarrow} & X'_1 & \longleftarrow & X'_2 \\ \downarrow & & \downarrow & & & & \downarrow \\ X_0 & \longleftarrow & X_1 & \longleftarrow & & & X_2, \end{array}$$

où $\alpha \in W$ en vertu de l'hypothèse W-cartésienne sur le carré 1. On voit encore que le carré 2 est W-cartésien si et seulement si le carré correspondant avec X'_1 remplacé par $\tilde{X}'_1$ l'est, quant aux carrés 1, 3, ils ne changent pas par cette modification. Cela montre qu'on peut supposer le carré 1 *cartésien*. Considérons alors le diagramme

$$\begin{array}{ccccccc} X'_0 & \longleftarrow & X'_1 & \longleftarrow & \tilde{X}'_2 \overset{\text{déf}}{=} X'_1 \times_{X_1} X_2 & \overset{\beta}{\longleftarrow} & X'_2 \\ {\scriptstyle \mathrm{Fib}_W}\downarrow & \text{cart.} & \downarrow{\scriptstyle \mathrm{Fib}_W} & \text{cart.} & \downarrow{\scriptstyle \mathrm{Fib}_W} & & \\ X_0 & \longleftarrow & X_1 & \longleftarrow & X_2. & & \end{array}$$

Alors la proposition 19 et la définition montrent que le carré 2 est W-cartésien si et seulenment si $\beta \in W$, et de même pour le carré 3. Ce qui achève la démonstration.

Commentary · Exposition [ms. 260]

Pasting reduces two questions to the same arrow

Grothendieck's proof makes pasting effective by arranging that the first square is an actual pullback along a W-fibration. The apparently immediate replacements in the proof are justified by the objectwise invariance established in our preceding block. The printed phrase $X_0 \to S \in \mathrm{Fib}_W S$ should be read as $X_0' \to X_0 \in \mathrm{Fib}_W$; it is the latter arrow that has just been factored.

Write $E \to X_0$ for a fibrant replacement of $X_0' \to X_0$. Replacing X_0' by E changes neither relevant cartesianness test. Because the first square is W-cartesian, its comparison

$$X_1' \longrightarrow E \times_{X_0} X_1$$

is in W. We may therefore replace X_1' by $E_1 = E \times_{X_0} X_1$, preserving the test for the second square. The first square is now strictly cartesian, and $E_1 \to X_1$ is a W-fibration.

Both remaining conditions are consequently tested by the one arrow

$$X_2' \longrightarrow E_1 \times_{X_1} X_2 \cong E \times_{X_0} X_2.$$

Using the expression on the left tests the second square; using the expression on the right tests the outer rectangle. The two conditions are identical. This proves both the composition implication and its indicated cancellation, without assuming that arbitrary base change preserves W.

The result became a basic part of the calculus of derivators (Groth, 2013, Proposition 3.13). Its orientation matters: in the manuscript the horizontal arrows point towards the left, so its left-hand square is the square adjacent to the common target. One cannot replace the hypothesis by either small square indiscriminately in an unstable theory.

Corollaire 2. *Soit*
$$\begin{array}{ccc} X_1 & \xleftarrow{f_2} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle f_1} \\ S & \xleftarrow[p_2]{} & X_2 \end{array}$$
un carré W-cartésien. Alors

$$\begin{aligned} p_1 \in W \quad &\Longrightarrow \quad f_1 \in W \\ p_2 \in W \quad &\Longrightarrow \quad f_2 \in W, \end{aligned}$$

plus précisément, si $p_i \in W$, alors Q est W-cartésien si et seulement si $f_i \in W$.

Supposons $p_1 \in W$. Alors dans la factorisation de p_1 en $X_1 \longrightarrow \bar{X}_1 \longrightarrow S$, on peut prendre $\bar{X}_1 = S$. Donc le carré est W-cartésien si et seulement si [page 261]

$$(X_3 \longrightarrow \bar{X}_1 \times_S X_2 = X_2) \in W,$$

i.e. $f_1 \in W$, q.e.d.

Bien sûr, on ne peut en général affirmer les implications inverses : $f_i \in W$ n'implique nullement $p_i \in W$.

Commentary · Examples [ms. 261]

A nonempty counterexample to the converse

Grothendieck's final caution about cartesian squares concerns a loss of information under restriction to part of the base. A finite example already shows the issue, without empty objects or any higher homotopy calculation.

Take $\mathcal{M} = \mathrm{Set}$ and let W be the bijections. Every map is an intrinsic W-fibration, and the identity followed by the given map provides the required factorization. Put

$$S = \{0,1\}, \quad X_1 = \{a,b,c\}, \quad X_2 = \{0\}, \qquad p_1(a) = 0, \quad p_1(b) = p_1(c) = 1,$$

and let p_2 be the inclusion of $\{0\}$ in S. The pullback is $X_3 = \{a\}$. Its projection $f_1 : X_3 \to X_2$ is a bijection, whereas $p_1 : X_1 \to S$ is not. The square is strictly cartesian and hence W-cartesian.

Only the fibre over 0 has been tested. Nothing in the upper projection records the two elements over 1. The forward assertion in the source has no such problem: if $p_1 \in W$, use the factorization $X_1 \xrightarrow{p_1} S \xrightarrow{\mathrm{id}} S$, and the cartesianness comparison is precisely f_1. The later appendix will ask for a converse to the analogous pushout statement. That converse is an additional conservativity condition, rather than a formal consequence of the existence of homotopy pushouts.

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Tous ces résultats, et la notion de carré W-cartésien, se dualisent (moyennant l'axiome idoine de factorisation à droite, cf. page 240) en une théorie des sommes amalgamées W-homotopiques, et des carrés W-cocartésiens. Cette fois on travaille avec la catégorie

$$\Phi = \begin{pmatrix} & & 1 \\ & \swarrow & \\ 0 & & \\ & \nwarrow & \\ & & 2 \end{pmatrix} \qquad \overset{j}{\hookrightarrow} \qquad \tilde{\Phi} = \begin{pmatrix} & & 1 & & \\ & \swarrow & & \nwarrow & \\ 0 & & & & 3 \\ & \nwarrow & & \swarrow & \\ & & 2 & & \end{pmatrix},$$

j étant une immersion fermée. On a un foncteur restriction

$$j^* : \mathrm{Ho}_W(\tilde{\Phi}) \longrightarrow \mathrm{Ho}_W(\Phi),$$

dont on construit un adjoint à gauche

$$j_! : \mathrm{Ho}_W(\Phi) \longrightarrow \mathrm{Ho}_W(\tilde{\Phi}).$$

Et on trouve une notion naturelle de carré W-cocartésien, qui paraît irréprochable à tout points de vue, et d'une simplicité parfaite. [page 262]

Cela amène à revoir la notion en question dans Cat, où elle avait l'air très astucieuse, profonde et un peu obscure, avec des intégrales et tout ça. C'est le moment de se rendre compte que 'ce n'est que ça'. Donc de prouver la

Proposition 20. *Supposons $\mathcal{M} = \mathrm{Cat}$, W un localiseur fondamental satisfaisant W(7 bis). Considérons un carré commutatif dans* Cat

$$Q = \begin{pmatrix} S & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3 \end{pmatrix}.$$

Pour qu'il soit W-cartésien [plutôt W-cocartésien] *au sens de la théorie développée ici (pour des catégories de modèles générales), il faut et il suffit que*

$$\varphi_Q : \quad \mathcal{X}_Q = \begin{pmatrix} & & X_1 \\ & \swarrow & \\ \int S & & \\ & \nwarrow & \\ & & X_2 \end{pmatrix} \qquad \longrightarrow \qquad X_3$$

[plutôt

$$\varphi_Q : \quad \mathcal{X}_Q = \begin{pmatrix} & & X_1 \\ & \nearrow & \\ \int S & & \\ & \searrow & \\ & & X_2 \end{pmatrix} \qquad \longrightarrow \qquad X_3]$$

p. 171

soit dans W. (NB L'intérêt de ce critère est qu'il est 'intrinsèque' à Q, ne faisant pas appel à une factorisation de β_1 ou de β_2 [où $\beta_i : S \longrightarrow X_i$] en $S \xrightarrow{\in \mathrm{Cof}_W} \tilde{X}_i \xrightarrow{\in W} X_i$. Cf. application intéressante dans section V 3.4., p. 146 ff., th. 4 et corollaires.)

Commentary · Later developments [ms. 262]

The integral as a categorical homotopy pushout

Grothendieck wants to identify the abstract operation just constructed with the categorical integral used earlier in the chapter. Proposition 20 says that the older construction was already providing a model for the same homotopy pushout. Its advantage is that the comparison can be read directly from the original square.

Here Φ has arrows $1 \to 0 \leftarrow 2$, and the diagram of categories is contravariant: its data are $X_1 \leftarrow S \to X_2$. The associated fibred Grothendieck construction has objects (i, x), with x an object of the fibre at i. For $\alpha : i \to j$, a morphism over α is a map $x \to F(\alpha)(y)$ in the fibre at i. A compatible cocone to X_3 gives the displayed functor φ_Q.

For ordinary weak homotopy equivalences, this interpretation is the content of Thomason's homotopy-colimit theorem: the classifying space of the integral models the homotopy colimit of the classifying spaces of the diagram (Thomason, 1979, Theorem 1.2). We checked its statement and a later proof in Chachólski and Scherer (2001, Theorem 26.8 and Corollary 26.9); the verification note distinguishes this from inspection of the original 1979 proof. In the present shape it gives

$$|N\mathcal{X}_Q| \simeq |NX_1| \amalg^h_{|NS|} |NX_2|.$$

The fibred version follows from the usual covariant theorem by applying it to the opposite fibre categories on Φ^{op} and then taking the opposite total category. Taking opposites leaves classifying spaces canonically homeomorphic. The equivalence explains the notation, but the manuscript's argument addresses a general W under its stated axioms, rather than only W_∞.

Its proof has one directional slip worth resolving before following the reduction. The transformation $\alpha : \tilde{Q} \to Q$ induces

$$\lambda : \mathcal{X}_{\tilde{Q}} \longrightarrow \mathcal{X}_Q, \qquad \varphi_{\tilde{Q}} = \varphi_Q \lambda.$$

The edition reverses the horizontal arrow in this triangle. With the correct direction, the fibrewise weak-equivalence assertion gives $\lambda \in W$, and two-out-of-three gives the required equivalence of tests. Footnote 134 expressly identifies the strong localization axiom used for this step; it should not be silently dropped when W is left arbitrary.

After both legs have been replaced by Dwyer immersions, put $P = X_1 \amalg_S X_2$. The earlier result on pushouts of these immersions supplies $\mathcal{X}_Q \to P$ in W. The map to X_3 factors as $\mathcal{X}_Q \to P \xrightarrow{\pi} X_3$, so $\varphi_Q \in W$ if and only if $\pi \in W$. The integral and the abstract replacement criterion therefore agree. None of this requires the reverse assertion that a homotopy pushout reflects weak equivalences in an attached leg; that separate condition enters the appendix.

Démonstration. Considérons une factorisation de $S \longrightarrow X_1$ en [page 263]

$$S \xrightarrow{i_1} \tilde{X}_1 \xrightarrow{p_1} X,$$

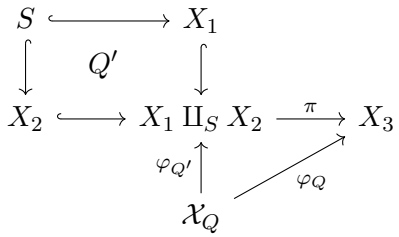
avec i_1 immersion ouverte de Dwyer, $p_1 \in W$. Donc on a un carré commutatif

$$\tilde{Q} = \begin{pmatrix} S & \xhookrightarrow{\mathrm{Dw}} & \tilde{X}_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3 \end{pmatrix}$$

et un homomorphisme $\tilde{Q} \xrightarrow{\alpha} Q$ qui est l'identité sur les trois sommets distincts de $\tilde{X}_1$, et qui est p_1 en $\tilde{X}_1$. On en déduit un diagramme commutatif

$$\begin{array}{ccc} \mathcal{X}_Q & \xrightarrow{\mathcal{X}_\alpha = \lambda} & \mathcal{X}_{\tilde{Q}} \\ & {\scriptstyle \varphi_Q} \searrow \quad \swarrow {\scriptstyle \varphi_{\tilde{Q}}} & \\ & X_3, & \end{array}$$

et on a $\lambda \in W$ (car c'est une Φ-flèche de catégories fibrées sur Φ, qui induit une W-équivalence sur chacune de trois fibres.) Donc $\varphi_Q \in W \iff \varphi_{\tilde{Q}} \in W$, d'autre part il est clair que Q est W-cocartésien si et seulement si $\tilde{Q}$ l'est. Donc on peut supposer que $S \hookrightarrow X_1$ est [une] immersion ouverte (de Dwyer) ([134]). Appliquant ceci en changeant les rôles de X_1 et X_2, on [page 264] voit qu'on peut supposer de plus que $S \longrightarrow X_2$ est également [une] immersion ouverte de Dwyer. Considérons alors



On a $\varphi_Q = \pi\varphi_{Q'}$, d'autre part on sait que $\varphi_{Q'} \in W$ (cor. 3, p. 149). Donc p. 172

$$\pi \in W \qquad \Longleftrightarrow \qquad \varphi_Q \in W,$$

où $\pi \in W$ signifie que Q est W-cocartésien, q.e.d.

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Commentary · Examples [ms. 264]

A four-object circle behind a strict pushout

Return to the two-point span whose cylinder cofiber was calculated at ms. 230. The integral gives a smaller categorical model of the same circle, and explains why the word "homotopy" cannot be omitted from Proposition 20. Let S be the discrete category on two objects a, b, and take $X_1 = X_2 = e$, both maps $S \to e$ being the unique functor. The ordinary pushout in Cat is e.

The fibred integral of this span has four objects, which we denote u, v, a, b. Its only nonidentity arrows are

$$u \longrightarrow a, \qquad u \longrightarrow b, \qquad v \longrightarrow a, \qquad v \longrightarrow b.$$

There are no composable pairs of nonidentity arrows. Thus its nerve has four vertices, four nondegenerate edges, and no nondegenerate simplices of dimension at least two. Its realization is the four-edge cycle, homeomorphic to a circle. Equivalently, the cellular boundary map has rank three, so its first homology is $\mathbb{Z}$.

The comparison $\mathcal{X}_Q \to e$ therefore fails to be in W_∞. The strict pushout square is not a homotopy pushout, and its actual homotopy pushout has the homotopy type of S^1. This four-edge circle and the earlier eight-edge cylinder cofiber both model the suspension of the two-point space. Neither original leg $S \to e$ is a Dwyer immersion: both identify the two objects. Replacing these legs before taking the ordinary pushout is precisely what preserves the missing circle. The example also fixes the variance of the integral without relying on a pictorial convention for its arrows.

8 Appendice : Retour sur les axiomes W(7), W(7 bis) (pages 75, 76).

([135]). On a vu (prop. 6, p. 78) que W(7 bis) implique W(7). Montrons une réciproque partielle.

134. On utilise ici W(3) (forme forte), i.e. Loc(3 bis) avec les notations de XVI. Mais on peut s'en passer, cf. plus bas.

135. Reflexion provisoire, supplanté par XVI.

Commentary · Exposition [ms. 265]

What the two converse gluing assertions require

Grothendieck wants to recognize a weak equivalence from the pieces of a decomposition. The familiar direction starts with equivalences on the common part and the two pieces and concludes that their homotopy pushouts are equivalent. His appendix asks for two converses: can the common part be recovered from the pieces and their union, and can the pieces be recovered from the common part and their union? These are substantive additions to homotopy-colimit invariance.

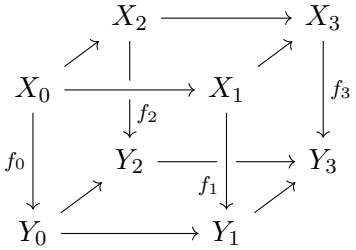
Write a square as $A \to B$, $A \to C$, with fourth vertex P, and write

$$C_e(u) = e \amalg^{\mathrm{h}}_{\mathrm{dom}(u)} \mathrm{cod}(u)$$

for the homotopy cofiber formed using the terminal object e. Whenever the square is a homotopy pushout, an equivalence $A \to B$ gives an equivalence $C \to P$. Grothendieck's conservative axiom asks for the converse as well. Applying that converse to the defining square of $C_e(u)$ says that a terminal homotopy cofiber detects an equivalence. This detection property is the extra ingredient in the proposition, through the full hypothesis W(7 bis).

It is useful to keep the three levels separate as the proof unfolds. Invariance of a homotopy pushout is formal once a coherent homotopy-colimit theory exists. The fact that an equivalence has terminal homotopy cofiber is equally formal. The converse, that terminal cofiber forces equivalence, is generally false for spaces; a complete example appears below. Thus the proposition under W(7 bis) and the later theorem for a general category of models must be read with different hypotheses. The last theorem drops an assumption that its third assertion still needs.

Proposition. *Supposons* W(7 bis) (axiome du carré cocartésien) *satisfait, et plaçons nous sous les conditions de l'énoncé de* W(7), *et plus généralement, dans le cas d'un diagramme cubique*



où les deux carrés X et Y (face supérieure et inférieure du cube) sont W-cocartésiens. Considérons les conditions suivantes sur ce diagramme

a) $f_0 \in W$.

b) f_1 *et* $f_2 \in W$.

c) $f_3 \in W$.

Alors on a les implications

$$a \ \& \ b \Longrightarrow c, \qquad b \ \& \ c \Longrightarrow a.$$

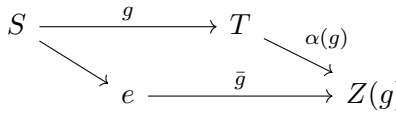
De plus, si W est stable par facteurs directs (p. ex. W fortement saturé), alors on a aussi

$$a \ \& \ c \Longrightarrow b,$$

donc la conjonction [de] *deux quelconques parmi les conditions a) b) c) implique la troisième.*

Démonstration. Utilisant la factorisation fonctorielle d'une flèche quelconque (ici les f_i) en pi, avec i immersion ouverte de Dwyer et $p \in W$ (p. 229), on est remené au cas où les f_i sont des immersions ouvertes de Dwyer. Mais pour toute flèche $g : S \longrightarrow T$ qui est dans Cof_W, soient $Z(g)$ son mapping-cone

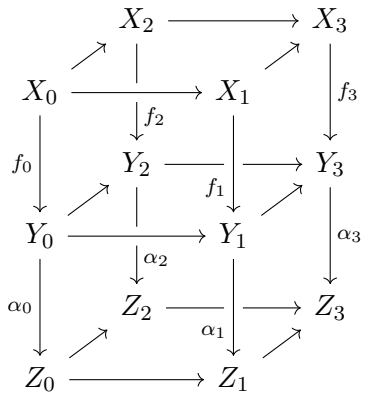
p. 173
[page 266]



et

$$\alpha(g) : T = \mathrm{but}(g) \longrightarrow Z(g)$$

la flèche canonique ([136]). Celle-ci est fonctorielle en g, de sorte que le diagramme cubique donné se prolonge en un double cube



avec

$$Z_i = Z(f_i),\ \alpha_i = \alpha(f_i) \qquad \text{pour } i \in \{0,1,2,3\}.$$

Notons que W(7 bis) implique que si $g \in \mathrm{Cof}_W$, on a

[page 267]

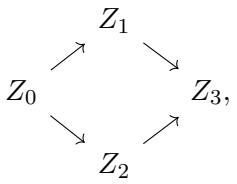
$$g \in W \qquad \Longleftrightarrow \qquad \bar{g} \in W, \qquad \text{i.e. } Z(g) \text{ } W\text{-asphérique},$$

en particulier

$$f_i \in W \qquad \Longleftrightarrow \qquad Z_i \text{ } W\text{-asphérique}.$$

Ainsi, l'énoncé à prouver sur le cube initial se ramène à un énoncé sur le carré

p. 174



savoir que les trois conditions

a′) Z_0 W-asphérique.

b′) Z_1 et Z_2 W-asphériques.

c′) Z_3 W-asphérique.

donnant lieu aux implications

$$\mathrm{a}' \ \&\ \mathrm{b}' \Longrightarrow \mathrm{c}', \qquad \mathrm{b}' \ \&\ \mathrm{c}' \Longrightarrow \mathrm{a}',$$

et que si W est stable par facteurs directs, alors

$$\mathrm{a}' \ \&\ \mathrm{c}' \Longrightarrow \mathrm{b}'.$$

C'est à peu de choses près le corollaire de la page 76, complété à la page 77 (corollaire), à cela près qu'on y suppose les flèches des immersions ouvertes, et le carré cartésien (donc aussi cocartésien) dans Cat, i.e. $Z_3 = Z_1 \cup Z_2$, $Z_0 = Z_1 \cap Z_2$. Mais ce qui importe pour la validité de cet énoncé, c'est (on s'en doute) que Z_* soit W-cocartésien. Donc il faut d'abord prouver ceci.

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136. C'est un peu vif, car il n'est pas clair que $X_1' \amalg_{X_0} X_2' \longrightarrow X_1 \amalg_{X_0} X_2$ soit `[dans]` W, sans utiliser déjà un des trois ingrédients de W(7) (savoir $f_0, f_1, f_2 \in W \implies f_3 \in W$). Il semble donc que notre argument ne marche que moyennant une hypothèse supplémentaire de cet ordre …Mais il y a le *lemme de Brown,* cf. XVI §3, notamment p. 20.

Commentary · Context [ms. 267]

The gluing step that Brown supplies

The factorization into a Dwyer immersion and a weak equivalence gives Grothendieck an explicit replacement of every vertical edge. To use the replacement on an entire cube, however, one must know that the induced replacement of a pushout is again a weak equivalence. Footnote 136 identifies this dependence. It is a gap in the chosen proof, not a counterexample to the conditional proposition.

The later discussion is unusually precise about the dependency. In Chapter XVI, Commentaire 3.2, Grothendieck observes that the appendix had used the fact that Dwyer immersions are W-cofibrations at a point where his earlier proof of that fact already used a gluing assertion. Lemma 3.2.1 and Corollary 3.2.2 isolate the required Brown lemma; Proposition 3.3 then records the intended equivalence of the strong axioms. The chapter identifies the repair but also says explicitly that its author does not have a proof of that lemma there (Grothendieck, 1991c, §3, ms. pp. 18–23).

Here is the precise external result that closes this dependency. Suppose a class C contains the isomorphisms and all maps from an initial object, is stable under composition and pushout, and every map factors as a member of C followed by a weak equivalence. Suppose also that pushout preserves $C \cap W$. These are the relevant axioms of a category of cofibrant objects. Brown's gluing lemma implies that pushing any weak equivalence out along a member of C preserves weak equivalences. In Brown's original formulation this is the dual of Part I, §4, Lemma 2; the proof uses his factorization lemma and the invariance result in Lemma 1 (Brown, 1973, pp. 428–429). With C the Dwyer immersions, this yields $C \subseteq \mathrm{Cof}_W$ and legitimizes the replacement of the cube. No converse detection property is supplied by Brown's lemma; that remains the role of W(7 bis).

Lemme. *Si le diagramme cubique p. 265 est tel que les faces X, Y soient W-cocartésiennes, et les f_i ($i \in \{0,1,2,3\}$) des W-cofibrations, alors le carré Z des cofibres des f_i est lui aussi W-cocartésien.*

Ce lemme résulte de la théorie du 'déploiement des diagrammes cubiques', relatif à une catégorie de modèles $(\mathcal{M}, W)$ admettant une théorie des cofibres W-homotopiques, qu'il me faudra bien écrire! (Mais l'existence pour (Cat, W) d'une telle théorie des cofibres W-homotopiques, et par suite de celle des déploiements des diagrammes dans $\mathcal{M}$ de type $(\mathbf{\Delta}^1)^I$, I ensemble d'indices fini, ne dépend que de W(7 bis) - on n'a pas à faire appel à W(7).)

Commentary · Exposition [ms. 268]

Taking cofibers commutes with the pushout

The useful part of Grothendieck's large diagram is the compatibility of two successive homotopy pushouts. We can prove the lemma on p. 268 directly from this compatibility, without using the claim that a terminal cofiber detects an equivalence.

Let $J = (1 \leftarrow 0 \rightarrow 2)$, and let $X \to Y$ be a coherent morphism of homotopy-pushout squares in a homotopy theory with a terminal object and finite homotopy colimits. Put $Z_j = C_e(X_j \to Y_j)$. Computing the colimit in the two orders gives

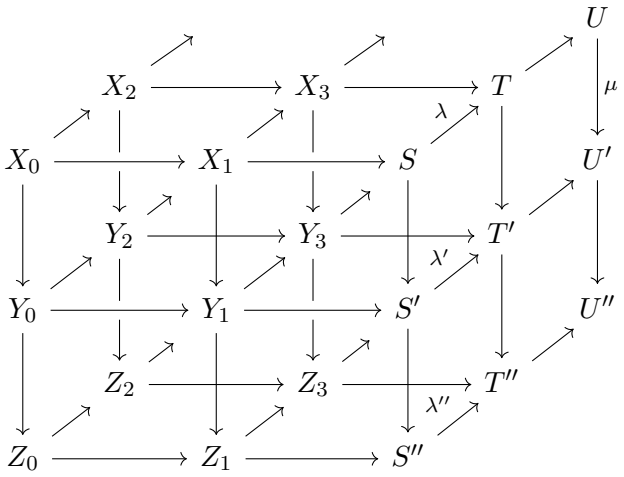
$$\begin{aligned} \operatorname*{hocolim}_{j\in J} Z_j &\simeq \left(\operatorname*{hocolim}_{j\in J} Y_j\right) \amalg^{\mathrm{h}}_{\operatorname{hocolim}_{j\in J} X_j} \left(\operatorname*{hocolim}_{j\in J} e\right) \\ &\simeq Y_3 \amalg^{\mathrm{h}}_{X_3} e = Z_3. \end{aligned}$$

The first equivalence is the interchange of colimits of the single diagram indexed by J and the cofiber span. The constant span $e \leftarrow e \rightarrow e$ has homotopy pushout e, which proves the second equivalence. Under these identifications the comparison map is the one in the given square Z, so this proves the lemma, including its compatibility with the prescribed edges. In derivator language the same calculation is the composition of left Kan extensions, with the parameter retained; the pointwise criterion is Groth (2013, Corollary 3.14).

This proof also locates the excessive assertion on p. 269. From $u \in W$ one may conclude $C_e(u) \simeq e$ in any such theory. Going back from $C_e(u) \simeq e$ to $u \in W$ uses conservativity. The full W(7 bis) hypothesis permits that return, but the statement that the whole reduction used only its forward half is false. The pointwise-cofiber lemma survives without the reverse half; the reduction of all three cube implications does not.

Ce 'deploiement' est un diagramme infini dont toutes les suites infinies, dans les trois directions des axes de coordonnées, sont des 'suites de suspension'. Je reproduis la partie utile du déploiement.

p. 175



La condition que X soit W-cocartésien s'exprime par $\lambda \in W$, ou encore par $U \in \mathrm{Asph}_W$ (car U s'identifie à la cofibre homotopique $Z(\lambda)$ de λ …), de même Y W-cocartésien s'exprime par $U' \in \mathrm{Asph}_W$, et Z W-cocartésien par $U'' \in \mathrm{Asph}_W$. Or $U'' \sim Z(\mu)$, où $\mu : U \longrightarrow U'$ est dans W si U, U' sont W-asphériques, donc $U'' = Z(\mu)$ est alors W-asphérique. Cela prouve le lemme.

Cela montre que, moyennant W(7 bis) (en fait, on a utilisé seulement la partie 'universelle' de W(7 bis)), savoir que dans $\begin{array}{ccc} T_0 & \longrightarrow & T_2 \\ {\scriptstyle i}\searrow & & \searrow{\scriptstyle i'} \\ & T_1 & \longrightarrow & T_3 \end{array}$ diagramme cocartésien d'immersions ouvertes, $i \in W \Longrightarrow i' \in W$ (au lieu de $\Longleftrightarrow$),

[page 270]

la validité de W(7) se ramène au cas où le carré Y est le carré trivial $\begin{array}{ccc} & e & \\ e & & e \\ & e & \end{array}$, e

l'objet final de Cat.

On peut maintenant conclure : On a (moyennant la partie directe de W(7 bis))

$$\mathrm{a}' \;\&\; \mathrm{b}' \Longrightarrow \mathrm{c}',$$

car si Z_0, Z_1 sont $\in \mathrm{Asph}_W$, alors $i_1 \in W$, donc $j_1 \in W$, et si de plus $Z_2 \in \mathrm{Asph}_W$, on a donc $Z_3 \in \mathrm{Asph}_W$.

$$\begin{array}{ccccc} & & Z_1 & & \\ & {\scriptstyle i_1}\nearrow & & \searrow{\scriptstyle j_2} & \\ Z_0 & & & & Z_3 \\ & {\scriptstyle i_2}\searrow & & \nearrow{\scriptstyle j_1} & \\ & & Z_2 & & \end{array}$$

Si on a aussi la partie inverse ('inhabituelle') de W(7 bis), alors

$$\mathrm{b}' \;\&\; \mathrm{c}' \Longrightarrow \mathrm{a}',$$

car si Z_1 et Z_3 sont W-asphériques, alors $j_2 \in W$, donc $i_2 \in W$, et si de plus Z_2 est W-asphérique, on conclut Z_0 W-asphérique.

Enfin, supposons que W soit *stable par facteurs directs,* et W(7 bis) satisfait sous forme directe. Je dis qu'alors

p. 176

$$\mathrm{c}' \;\&\; \mathrm{a}' \Longrightarrow \mathrm{b}',$$

i.e.

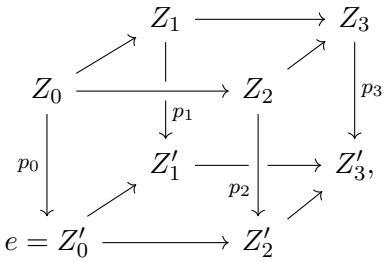
$$Z_0 \text{ et } Z_3 \ \in \mathrm{Asph}_W \qquad \Longrightarrow \qquad Z_1 \text{ et } Z_2 \ \in \mathrm{Asph}_W.$$

Factorisant i_1 et i_2 en

$$Z_0 \xrightarrow{i'_1} Z'_1 \xrightarrow{\alpha_1} Z_1,$$
$$Z_0 \xrightarrow{i'_2} Z'_2 \xrightarrow{\alpha_2} Z_2,$$

avec i'_1, i'_2 des immersions ouvertes de Dwyer, et $\alpha_1, \alpha_2 \in W$, on est ramené au cas où

i_1, i_2 sont des *immersions ouvertes de Dwyer,* et où le carré est cocartésien (ce qui implique [page 271]
qu'il est W-cocartésien). Faisant alors le cochangement de base par $Z_0 \xrightarrow{p_0} Z'_0 = e$, on trouve un homomorphisme de carrés cocartésiens



où toutes les faces du cube sont W-cocartésiennes, et comme $p_0 \in W$ (Z_0 étant W-asphérique), les autres flèches verticales p_1, p_2, p_3 sont `[dans]` W, donc le carré Z'_* satisfait aussi Z'_0 et Z'_3 `[`W`-]`asphériques (et de plus $Z'_0 = e$), et il faut prouver Z'_1, Z'_2 W-asphériques (d'où itou pour Z_1, Z_2). Donc, on est ramené au cas particulier où $Z_0 = e$.

Mais dans ce cas il existe une rétraction (triviale!) de Z_1 sur Z_0, donc aussi de Z_3 sur Z_2. Ainsi, Z_2 est un facteur

direct d'un objet W-asphérique, ce qui implique, par l'hypothèse sur W, que Z_2 est [page 272]
W-asphérique. Par symétrie, Z_1 l'est aussi. On a gagné!

Commentary · Exposition [ms. 272]

The retract argument is valid at the level of objects

Grothendieck's reduction to a common part equal to e is sound and deserves to be retained. If

$$Z_3 \simeq Z_1 \amalg^{\mathrm{h}}_{Z_0} Z_2, \qquad Z_0 \simeq e,$$

then homotopy-colimit invariance identifies Z_3 with the homotopy coproduct of two objects under e. Each summand is a retract of that coproduct: the retraction onto Z_1 is induced by the identity of Z_1 and the composite $Z_2 \to e \to Z_1$. Both agree on e, so the universal property supplies the retraction; its composite with the summand inclusion is the identity. The same construction applies to Z_2. Consequently, if Z_3 is W-aspherical and W-aspherical objects are closed under retracts, both summands are W-aspherical. This proves the object statement without any conservative axiom.

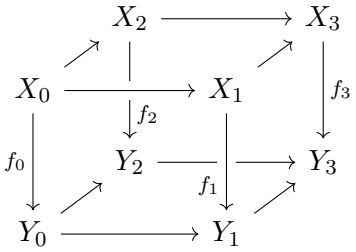
For the source's weaker hypothesis of closure under strict retracts, realize this argument using cofibration representatives $e \to Z_i^c$ in the category under e and their ordinary pushout $P^c = Z_1^c \amalg_e Z_2^c$. These representatives compute the displayed homotopy pushout. The identity of Z_1^c and the composite $Z_2^c \to e \to Z_1^c$ agree strictly on e, so the pushout retraction and its splitting are literal maps, with composite the identity. The comparison with Z_3 makes P^c aspherical; strict retract closure applies to Z_i^c, and the replacement weak equivalences give the assertion for Z_i.

The phrase *facteur direct* here means a retract, as the explicit construction shows. Strong saturation of W also suffices because localization carries a retract diagram to a retract diagram, and a retract of an isomorphism is an isomorphism.

What this argument does not supply is the return from the objects $Z_i = C_e(f_i)$ to the maps f_i. Terminal cofibers need not detect equivalences. The object statement just proved therefore cannot by itself justify assertion 3° of the following general theorem, or the equivalence asserted in Corollary 1. Under the full conservative hypothesis that return is valid, and the final Corollary 2 survives.

NB En fait, ces arguments sont généraux, et donnent ceci :

Théorème. *Soit $(\mathcal{M}, W)$ une catégorie des modèles, admettant une théorie de la cofibre W-homotopique ([137]) (i.e. on a une factorisation de toute flèche $f : X \longrightarrow Y$ en pi, avec $i \in \mathrm{Cof}_W$, $p \in W$.) On a donc une notion de carré W-cocartésien dans $\mathcal{M}$ ([138]). Soit alors $f : X \longrightarrow Y$* p. 177 *un homomorphisme d'un carré W-cocartésien X dans un autre Y, $f = (f_0, f_1, f_2, f_3)$.*



Considérons les conditions sur f

a) $f_0 \in W$.

b) f_1 et $f_2 \in W$.

c) $f_3 \in W$.

Alors

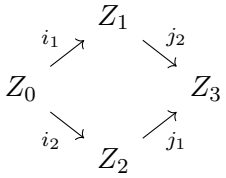
1°) a & b $\Longrightarrow$ c.

2°) Supposons que $\mathcal{M}$ ait un objet final e. Alors l'implication [page 273]

$$b \ \& \ c \Longrightarrow a$$

pour tout *homomorphisme $f : X \longrightarrow Y$ de carrés W-cocartésiens équivaut à l'axiome suivant sur W :*

Axiome des carrés W-cocartésiens conservatifs ([139]) : *Si*



est W-cocartésien, alors $j_1 \in W \Longrightarrow i_1 \in W$ (donc en fait $i_1 \in W \Longleftrightarrow j_1 \in W$ et $i_2 \in W \Longleftrightarrow j_2 \in W$).

Cet axiome est équivalent à la conjonction des deux axiomes suivants.

Commentary · Exposition [ms. 273]

A direct test for conservative homotopy pushouts

Grothendieck now isolates the property needed by the preceding reduction. Once homotopy pushouts are available coherently, there is a short proof of the test that also clarifies Cons 1 and Cons 2. We work in a homotopy theory with a terminal object and finite homotopy colimits, and use "equivalence" for a morphism detected as an isomorphism by its localization.

137. Mais on ne sait vérifier l'existence d'une telle théorie dans (Cat, W) que si on admet la forme faible de W(7), i.e. $f_0, f_1, f_2 \in W \implies f_3 \in W$, ou quelque cas particulier de celle-ci. Mais le *lemme de Brown* sauve la situation, cf. XV §3.

138. à vérifier que la factorisation suffit, même si tous les objets ne sont pas fibrants sur e. Mais on peut alors remplacer $\mathcal{M}$ par $\mathcal{M}_f$, et développer ce formalisme des fibres homotopiques en conséquence …

139. cf. argument page 78 qui montre que l'implication b & c $\Longrightarrow$ a entraine l'axiome conservatif.

For every homotopy-pushout square

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ \downarrow & & \downarrow \\ C & \xrightarrow[v]{} & P \end{array}$$

associativity gives a canonical equivalence

$$C_e(v) = e \amalg^{\mathrm{h}}_C P \simeq e \amalg^{\mathrm{h}}_C (C \amalg^{\mathrm{h}}_A B) \simeq e \amalg^{\mathrm{h}}_A B = C_e(u).$$

Thus, if terminal homotopy cofiber detects equivalences, v being an equivalence forces u to be one. Conversely, apply the conservative-pushout axiom to the defining cofiber square: if $C_e(u) \simeq e$, its lower edge $e \to C_e(u)$ is an equivalence, and the axiom forces u to be an equivalence. This proves the equivalence of conservativity and Cons 1, with the latter interpreted for a cofibration replacement of every map. The intended cocartesian Cons 2 follows: if B, C, P are terminal up to equivalence, the edge $B \to P$ is an equivalence, hence so is $A \to C$; consequently $A \simeq e$. It is therefore redundant under this coherent formulation.

These observations complete the proof of the two converse cube implications under the appropriate hypothesis. Given a morphism $f : X \to Y$ of homotopy-pushout squares, its pointwise cofibers $Z_i = C_e(f_i)$ form a homotopy-pushout square by the preceding lemma. If f_1, f_2, f_3 are equivalences, then $Z_1, Z_2, Z_3 \simeq e$; conservativity gives $Z_0 \simeq e$, and Cons 1 gives $f_0 \in W$. Conversely, the implication $b \,\&\, c \Rightarrow a$ applied to the map from a square (A, B, C, P) to (B, B, P, P) with components $(u, 1_B, v, 1_P)$ proves conservativity immediately.

If instead f_0, f_3 are equivalences and aspherical objects are closed under retracts, the object argument on pp. 271–272 gives $Z_1, Z_2 \simeq e$; Cons 1 then gives $f_1, f_2 \in W$. Together with ordinary pushout invariance this proves the final Corollary 2. It also identifies exactly what must be added to the printed assertion 3 °: a cofiber-detection hypothesis, equivalently conservative homotopy pushouts, in addition to the retract condition.

Cons 1. *Si on a un carré cocartésien* p. 178

$$\begin{array}{ccc} S & \longrightarrow & T \\ & \searrow \quad \searrow & \\ & e \longrightarrow & U, \end{array}$$

avec $f \in \mathrm{Cof}_W$*, alors* $U \in \mathrm{Asph}_W \Longrightarrow f \in W$ *(donc* $U \in \mathrm{Asph}_W \Longleftrightarrow f \in W$*).*

Cons 2. *Si on a un carré cartésien,*

$$\begin{array}{ccccc} & & Z_1 & & \\ & \overset{i_1}{\nearrow} & & \searrow & \\ Z_0 & & & & Z_3, \\ & \underset{i_2}{\searrow} & & \nearrow & \\ & & Z_2 & & \end{array}$$

avec $i_1, i_2 \in \mathrm{Cof}_W$*, alors : Si* Z_1*,* Z_2*,* Z_3 *sont* W*-asphériques,* Z_0 *l'est aussi.*

Commentary · Examples [ms. 273]

Why Cons 2 must concern pushouts

The edition prints *cartésien* in Cons 2. Taken literally, this gives a different and false consequence of conservativity. The issue can be seen inside a stable homotopy theory, where conservative homotopy pushouts do hold.

Let k be a field and use homologically graded chain complexes. Put $A = ka$ in degree zero, and let

$$B = (kb \xrightarrow{d} ka), \qquad C = (kc \xrightarrow{d} ka), \qquad d(b) = d(c) = a.$$

Embed both in the complex

$$D = (kd \longrightarrow kb \oplus kc \longrightarrow ka), \qquad \partial d = b - c, \quad \partial b = \partial c = a.$$

Here d as a basis symbol lies in degree two. Both B and C are acyclic. In D the first differential is injective, its image is the kernel of the second, and the second is surjective, so D is acyclic too. Their ordinary pullback over D is the intersection A, whose zeroth homology is k. All four inclusions are degreewise monomorphisms. Such a map is a W-cofibration for quasi-isomorphisms: pushout leaves its cokernel unchanged, and the long exact homology sequences show that a weak equivalence of sources induces one of the pushouts. Over a field these maps are also cofibrations in the usual model structure.

Thus a strict cartesian square with the printed cofibration hypotheses need not satisfy Cons 2, even when the surrounding derived theory is stable. Replacing *cartésien* by *cocartésien* restores the statement used in the argument. The source wording is retained in the text and recorded as an edition slip.

[page 274]

3°) Supposons que tout facteur direct d'un objet W-asphérique soit W-asphérique, ce qui est le cas notamment si W est lui-même stable par facteurs directs, a fortiori si W est fortement saturé. Supposons de plus que $\mathcal{M}$ ait `[un]` *objet final. Alors on a (sous les conditions générales du début)*

$$c \ \& \ a \Longrightarrow b,$$

i.e. si f_0, f_3 sont `[dans]` *W, de même f_1, f_2.*

Cette dernière partie peut se déployer ainsi :

Commentary · Development (commentators') [ms. 274]

An explicit cube testing the generalization

Grothendieck's retraction proof invites a test of the stronger assertion about arbitrary maps of squares. We give the test in full, since it distinguishes a missing hypothesis from a missing step in a proof. The acyclic complex used below is classical; the assembly into the two diagnostic cubes is our application to this passage. We checked the source's later Chapter XVI discussion, Brown's gluing lemmas, and the cofiber and Mayer–Vietoris treatments cited here; we did not find this particular counterexample to the printed generalization in those sources. No priority is claimed.

A cube with a and c, but not b. [ours] In simplicial sets with weak homotopy equivalences, there is a map between homotopy-pushout squares whose zeroth, second and third components are equivalences and whose first component is not.

Proof. Begin with the finite CW complex A having one vertex, two edges a, b, and two two-cells attached by a^5b^{-3} and $b^3(ab)^{-2}$ (Hatcher, 2002, Example 2.38). Its cellular boundary in positive dimensions is

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\left(\begin{smallmatrix} 5 & -2 \\ -3 & 1 \end{smallmatrix}\right)} \mathbb{Z}^2 \longrightarrow 0.$$

The determinant is -1, so A is acyclic. Its fundamental group is nontrivial: sending a to (12345) and b to (153) respects the relations, since their product is (23)(45), and gives a nontrivial permutation representation. The suspension ΣA is simply connected by van Kampen and acyclic by the suspension isomorphism in reduced homology. Hurewicz and Whitehead therefore make it contractible (Hatcher, 2002, Theorem 4.32 and Corollary 4.33).

Choose a simplicial set K of the homotopy type of A, let $CK = K \star \Delta^0$, and write $e = \Delta^0$. The inclusion $K \hookrightarrow CK$ is a cofibration, CK is contractible, and $CK \amalg_K e$ has the homotopy type of ΣA, hence is weakly contractible. Consider

$$X = \begin{pmatrix} K & = & K \\ \downarrow & & \downarrow \\ CK & = & CK \end{pmatrix}, \qquad Y = \begin{pmatrix} K & \longrightarrow & e \\ \downarrow & & \downarrow \\ CK & \longrightarrow & CK \amalg_K e \end{pmatrix}.$$

Both strict pushouts compute homotopy pushouts, since the common vertical leg is a cofibration. There is a strict map $X \to Y$ with components

$$(f_0, f_1, f_2, f_3) = (1_K,\ K \to e,\ 1_{CK},\ CK \to CK \amalg_K e).$$

Its zeroth and second components are identities; its third connects weakly contractible objects; its first is not a weak equivalence because $\pi_1(K) \neq 1$. This proves the assertion. All simplicial sets

are cofibrant, factorizations exist, and weak equivalences are strongly saturated and closed under retracts. Thus the failure is present under the printed hypotheses of assertion 3°, not a failure of replacement machinery.

The same K tests the other converse. Map the homotopy-pushout square with vertices $(K, CK, CK, CK \amalg_K CK)$ to the constant square at e. Its two cone vertices and its suspension vertex are weakly contractible, while K is not. Hence $b \,\&\, c \Rightarrow a$ fails for spaces, as the conservative axiom predicts. Finally, the object statement in Corollary 1 remains valid for spaces by the retract proof above. Its proposed equivalence with $a \,\&\, c \Rightarrow b$ therefore fails as printed. Adding cofiber detection repairs both assertions.

Corollaire 1. *Sous l'hypothèse supplémentaire* 3°) *sur W, la validité de c & $a \Longrightarrow b$ (pour toute flèche f entre carrés W-cocartésiens) équivaut à chacun des deux autres axiomes suivants sur W (qui sont donc équivalents) :*

1) Si on a un carré W-cocartésien

$$\begin{array}{ccc} & Z_1 & \\ \nearrow & & \searrow \\ Z_0 & & Z_3, \\ \searrow & & \nearrow \\ & Z_2 & \end{array}$$

alors Z_0 et Z_3 W-asphériques $\Longrightarrow$ Z_1 et Z_2 W-asphériques.

1′) Comme 1), avec $Z_0 = e$ (dans le cas où on suppose que $\mathcal{M}$ admet `[un]` *objet final e).*

Corollaire 2. *Supposons que W satisfasse l'axiome des carrés W-cocartésiens conservatifs, que tout facteur direct d'un objet W-asphérique soit W-asphérique (p. ex. W stable par facteurs directs), enfin que $\mathcal{M}$ admette un objet final. Alors pout tout homomorphisme $f : X \longrightarrow Y$ de carrés W-cocartésiens, la conjonction de deux parmi les conditions a) b) c) de p. 272 entraîne la troisième.* [page 275]

Commentary · Later developments [ms. 275]

The stable form of the cube calculus

The last corollary has a particularly transparent home in a stable derivator. Coherent cocartesian squares then are also cartesian, and their geometry gives a Mayer–Vietoris triangle

$$X_0 \xrightarrow{(i_1,-i_2)} X_1 \oplus X_2 \xrightarrow{(j_2,j_1)} X_3 \longrightarrow \Sigma X_0.$$

The existence of this triangle is Groth et al. (2014a, Theorem 6.1, arXiv v2); their Theorem 7.1 compares stability with the invertibility of suspension and of the cofiber/fiber adjunction. This gives an established realization of the suspension calculus suggested by the large diagram on p. 269.

For a morphism of two such squares, the induced map of triangles makes the three implications immediate but still instructive. Equivalences on the first and middle terms give one on the third; equivalences on the middle and third give one on the first. Equivalences on the first and third give an equivalence $f_1 \oplus f_2$. Each f_i is a retract of that direct-sum map in the arrow category, so each is an equivalence. This is the additive form of the retraction argument, with cofiber detection now supplied by stability.

Conservativity should nevertheless not be identified with stability. The former is the particular detection property isolated on p. 273; stability also requires a pointed theory and the agreement of cartesian and cocartesian squares. The source's general constructions of coherent squares and iterated cofibers remain useful in unstable theories, while the converse arguments require their stated additional hypotheses.

Overview · XIIc-b

From replacements to a calculus of coherent squares. The pages gathered here form a single mathematical movement. Grothendieck first constructs categorical cylinders and uses them to establish the gluing rules for homotopy pushouts. He then asks which part of this geometry can be recovered from a category $\mathcal{M}$ and a class W of weak equivalences alone. The answer takes the form of a homotopy completion of a cospan, characterized by an adjunction. Finally he returns to pushouts and investigates how far weak equivalences can be recovered from the pieces of a decomposition. The first two movements build the calculus; the appendix tests a considerably stronger conservativity property.

These pages are manuscript pages 228–275, printed pages 145–178 of the edition (Grothendieck, 1990–1991h): the second part of section VI of Chapter XII, that is, the end of its section 6 followed by sections 7 and 8. The audit covers that entire range, including the fragment of a footnote continued at the beginning. Page labels in the margins reproduce the two numbering systems of the transcription; citations in the commentary use manuscript pages unless another system is specified.

Why the diagrams must precede localization. The motivation is older than the technical apparatus of these pages. In section 69 of *Pursuing Stacks*, Grothendieck traces the need for derivators to the trace formalism of SGA 5: a single derived category did not retain the diagrammatic information needed for the desired compatibilities (Grothendieck, 1983, §69, pp. 1–2). His letter to Thomason of 2 April 1991 returns to this history and to the system of localized diagram categories (Grothendieck, 1991a, §2, pp. 5–6). These are historical statements checked in those texts; we do not infer that every later definition was already present there in its final form.

Here this motivation becomes a concrete distinction. With the manuscript's contravariant convention, the relevant category is

$$\mathbb{D}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})[W(I)^{-1}],$$

where $W(I)$ consists of the objectwise weak equivalences. It is not obtained by first passing to $\mathcal{M}[W^{-1}]$ and then forming its ordinary diagram category. A cospan remembers two maps to a common base together; its homotopy completion must preserve that relationship and its naturality in all three vertices. This explains both the attention given to the whole completed square and the two localization lemmas whose proofs the manuscript postpones. The modern introduction in Groth (2013, pp. 313–315) makes the same structural distinction.

The constructive part and its later justification. The fence interval Φ_n, with $n \geq 2$, lets both endpoints enter by a Dwyer immersion. The associated mapping cylinder gives a functorial factorization into a W-cofibration followed by a weak equivalence. The commentary proves the endpoint condition and follows the construction on the map from two discrete objects to a point. Taking its categorical cofiber produces an eight-edge circle. Later, the Grothendieck integral of the same span gives a four-edge circle. These are two explicit models of the homotopy pushout; the ordinary categorical pushout is only a point.

The appeal to Thomason is valid in the needed cellular form. It must not be strengthened to the assertion that every Thomason cofibration is a Dwyer map: the retract assertion used to obtain that conclusion is false, and Cisinski (1999) gives the counterexample and repair. The recent comparison of Dwyer pushouts with $(\infty, 1)$-categorical pushouts is stronger in a different direction, identifying what the nerve preserves rather than adding the appendix's conservative axiom (Hackney et al., 2024, Theorems 1.5–1.6, arXiv v3).

In section 7, a W-fibration means a map whose pullback functor preserves the designated weak equivalences. The factorization assumption says that every arrow is a weak equivalence followed by such a fibration. It is essential to retain this intrinsic meaning: the manuscript has not assumed a Quillen model structure with arbitrary lifting classes. Under its stated

terminal-object and pullback hypotheses, these data form a left derivable category. The commentary verifies the nontrivial axiom, stability of trivial fibrations under arbitrary base change, by a two-step factorization and two-out-of-three argument.

This observation gives a complete modern repair of the deferred Lemmas 1 and 2. We construct the appropriate fibration structures on the good cospans and good squares, check the factorization and pullback conditions, and verify approximation for their inclusions in the ambient diagram categories. Cisinski's Proposition 1.8 also supplies the strengthened replacement assertion for every full intermediate subcategory of a slice containing its fibrant objects; his Theorem 3.12 establishes the two diagram-localization equivalences (Cisinski, 2010). No functorial factorization or global assumption that all objects are fibrant is required. A separate, shorter deformation argument explains what functorial factorization would give. The audit still labels the source proofs as deferred or sketched: the later completion does not change what is written on these pages.

With those equivalences available, restriction from squares to cospans has a fully faithful right adjoint. On a cospan $A \to S \leftarrow B$ with one good leg, its missing corner is represented by $A \times_S B$. The groupoid example $* \to BG \leftarrow *$ gives the discrete groupoid G at that corner, whereas the strict pullback without replacement is a point. The adjunction retains the two projections and their compatibility, as well as the corner's homotopy type. Proposition 19 then gives an intrinsic test for cartesianness, and we prove its independence of choices, invariance under objectwise weak equivalences, and pasting directly from the factorization axiom. Strong saturation enters precisely when an isomorphism after localization must be recognized as an original member of W.

What the appendix adds, and where its generalization fails. The ordinary gluing implication says that equivalences on the common part and the two pieces induce an equivalence on their homotopy pushouts. The appendix asks for converses. Its full W(7 bis) hypothesis is substantially stronger: a homotopy pushout must also reflect weak equivalences in an attached leg. In a coherent homotopy-pushout theory this is equivalent to detection by the terminal homotopy cofiber

$$C_e(u) = e \amalg^{\mathrm{h}}_{\mathrm{dom}(u)} \mathrm{cod}(u).$$

The forward implication $u \in W \Rightarrow C_e(u) \simeq e$ is formal. The reverse implication is the additional detection property. We prove the equivalence with conservativity by pasting cofiber squares.

This distinction exposes an actual error in the general theorem on manuscript page 274. Retract closure of aspherical objects alone does not imply that equivalences at the initial and final vertices of a map of homotopy-pushout squares force equivalences at the two middle vertices. Our counterexample uses Hatcher's finite acyclic noncontractible complex A. Its suspension is contractible; the map from the square (A, A, CA, CA) to $(A, *, CA, CA/A)$ has three weak-equivalence components and one component $A \to *$ that is not a weak equivalence. Both squares are homotopy pushouts. We give the presentation of A, its cellular homology calculation, a nontrivial permutation representation of its fundamental group, and the suspension argument (Hatcher, 2002, Example 2.38). The related equivalence asserted in Corollary 1 therefore fails as printed. Adding cofiber detection repairs both statements, and the final Corollary 2 already has the required conservative hypothesis.

There is also a source slip in Cons 2: the edition prints *cartésien* where its argument requires *cocartésien*. A finite chain-complex example shows that the literal cartesian assertion fails even in a stable theory. The proof of the intended criterion shows that its cocartesian version follows from Cons 1 under the coherent formulation. In stable derivators the entire corrected cube calculus has a familiar expression through the Mayer–Vietoris triangle (Groth et al., 2014a, Theorems 6.1 and 7.1, arXiv v2).

The appendix's earlier circularity is a different issue. Chapter XVI identifies the dependence on the Dwyer-cofibration gluing lemma but explicitly does not give its proof there

(Grothendieck, 1991c, §3, ms. pp. 18–23). Brown's published gluing lemma, dualized, supplies the missing result (Brown, 1973, Part I, §4, Lemma 2). Thus the initial conditional proposition survives, while the later hypothesis-free generalization does not. These are distinct diagnoses, with distinct repairs.

Editorial and verification limits. The text preserves slips printed in the edition. Its one mathematical transcription correction restores the interior natural transformation α on manuscript page 235, which the transcription had drawn as an arrow between the wrong objects. The separate registers record the source's proof status, the present repair, and the evidence for each reading. Bibliography notes identify exactly which statements, proofs, or metadata were checked; in particular, the original proof of Thomason's 1979 theorem was not inspected, although its statement and a later proof were checked.

Appendice au chapitre XII

XIIapp · Chapter XII, appendix · manuscript pages 1–18

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Appendice au Chapitre XII

ed. 1

Proposition (?) ([1]). *Soit X dans* Cat*, avec b, c dans X tels que* [ms. 1]

a) $b \times c$ *est dans* X*.*

b) $\forall\ x$ *dans* X*,* $b(x)$ $(\overset{déf}{=} \mathrm{Hom}_X(x,b))$ *et* $c(x)$ *sont* $\neq \emptyset$*, i.e.* $X/b \longrightarrow X$ *et* $X/c \longrightarrow X$ *sont surjectifs sur les objets.*

Alors X *est* $\mathbf{W}_\infty$*-asphérique.* ([2])

Le fait que b soit un plus grand élément de ObX assure déjà la 0-connexité.

Commentary · Context [ms. 1]

What an ordinary pushout forgets

Grothendieck is looking for categorical gluings that preserve a homotopy type. The little span that appears throughout this appendix is a particularly severe test. Its indexing category

$$\Psi = (b_0 \longleftarrow a_0 \longrightarrow c_0)$$

has an initial object, hence a contractible nerve. Nevertheless, taking the ordinary colimit of its representable presheaves can introduce higher homotopy into the category of elements. The purpose of the examples is to locate this discrepancy precisely enough to decide which inclusions can be used in a homotopy theory on Cat.

We write $BC = |NC|$, and interpret $\mathbf{W}_\infty$ as the functors inducing weak homotopy equivalences on these classifying spaces. In this terminology "aspherical" means weakly contractible, not merely a space with vanishing higher homotopy groups. For a presheaf $F : C^{\mathrm{op}} \to \mathrm{Ens}$, the notation C/F denotes its category of elements: its objects are (c, x), and an arrow $(c, x) \to (c', x')$ is an arrow $u : c \to c'$ with $F(u)(x') = x$. Thus a represented presheaf has contractible category of elements C/c. The central phenomenon is that a colimit of represented presheaves need not retain that property.

This is already the tension described in Grothendieck (1983, §69): categorical integration gives homotopy colimits, whereas actual amalgamated sums require additional care. The letter to Thomason of 2 April 1991 explains why the two adjoints to restriction, together with their exactness properties, should be treated as the essential structure (Grothendieck, 1991a, pp. 8–9). The appendix examines the small categorical configurations behind that program. Its final gluing theorem belongs to the theory of homotopy colimits; its earlier counterexamples concern ordinary pushouts in Cat. Keeping the two operations distinct makes the apparent reversal between the examples and the closing conjecture intelligible.

There are also two different uses of asphericity for an inclusion. A functor $U \to X$ is aspherical in the usual sense if every comma category U/x is weakly contractible. An open immersion in these pages is a full sieve, and the local condition used for it is imposed on $U \to \overline{U}$, where $\overline{U}$ is its cosieve closure. Objects outside $\overline{U}$ have empty comma categories. This distinction will matter for the boundary of a subdivided square: that boundary has the homotopy type of a circle, even though the indicated local condition holds. We refer to the precise conventions of Chapter XII, §V.2 (Grothendieck, 1990–1991d), and to the commentary on Chapter II, manuscript p. 36 ("The quantifier in total asphericity") for the universal quantifier in that second notion.

Commentary · Exposition [ms. 1]

What the product actually contracts

Grothendieck begins by asking how much of a terminal object can be replaced by the existence of maps. The objects b and c are *weakly terminal*: every object admits a map to each of them, without any assertion of uniqueness. Their product $a = b \times c$ is weakly terminal as well. The argument establishes the exact conclusion needed at the first stage: the nerve NX is connected and simply connected. Here and below $BX = |NX|$, and "aspherical" in the terminology of these pages means weakly contractible, not merely a $K(\pi, 1)$.

The cancellation argument can be compressed without losing its content. For two maps $s, t : x \to a$, insert the mixed pair $(p_b s, p_c t)$. Cancellation against the now invertible projections

1. Cf. pages 11 ff. pour les contrexemples-clef.
2. Faux, cf. plus bas. X est 1-connexe, mais pas nécessairement 2-connexe.

shows that all three maps have the same image in the groupoid completion ΠX. Choose one such image $\alpha_x : x \to a$ for each object, taking $\alpha_a = 1_a$. For every arrow $u : x \to y$, both $\alpha_y u$ and α_x come from arrows $x \to a$ and therefore agree. The same identity then holds for inverse arrows and their composites. Every morphism $x \to a$ in ΠX is consequently α_x. Thus a is terminal in this groupoid, which is equivalent to the terminal groupoid. The longer zigzag induction in the text is a complete proof of this fact; its undefined a_0 and u_0 are printing slips, not a missing argument.

There is a significant distinction between this proposition and the later four-object example. In that example $b \amalg_a c = 1$ in the presheaf category, but a is *not* $b \times c$: at the added object there are two maps to a and only one pair of maps to (b, c). It cannot by itself refute the opening product proposition. The finite two-object category below does refute that proposition, although its first obstruction occurs in degree three. We give the calculation because the provisional degree-two calculation in the manuscript does not survive verification.

Pour la 1-connexité, prouvons que dans le groupoïde fondamental $\Pi(X)$, $a \overset{\text{déf}}{=} b \times c$ devient un élément final.

$$\begin{array}{ccc} b \times c & \longrightarrow & b \\ \downarrow & & \\ c & & \end{array}$$

Pour ceci, prouvons d'abord que $\forall$ x dans X, tous les éléments de $a(x) = b(x) \times c(x)$ sont égaux dans $\mathrm{Fl}\Pi(X)$. En effet, comme $b \times c \longrightarrow b$ est `[un]` isomorphisme, donc `[un]` monomorphisme dans $\Pi(X)$, cela montre déjà que si $\beta, \beta' \in b(x)$, $\gamma, \gamma' \in c(x)$, d'où (β, γ), (β, γ'), $(\beta', \gamma') \in b(x) \times c(x)$, on a $(\beta, \gamma) = (\beta, \gamma')$ dans $\Pi(X)$ (puisque $\mathrm{pr}_1(\beta, \gamma) = \mathrm{pr}_1(\beta, \gamma')$, et de même (en utilisant que pr_2 est `[un]` isomorphisme, donc `[un]` monomorphisme) $(\beta, \gamma') = (\beta', \gamma')$). Donc tous les $(\beta, \gamma) \in b(x) \times c(x) = a(x)$ sont égaux dans $\mathrm{Fl}\Pi(X)$. Pour achever de prouver que a est objet final de $\Pi(X)$, il suffit de voir que toute flèche *dans* $\Pi(X)$ d'un $x \in \mathrm{Ob}X$ dans a, provient d'une flèche $x \longrightarrow a$ dans X. Mais une flèche dans $\Pi(X)$ est obtenue par un chemin dans X

$$\begin{array}{ccccccccc} x = x_0 & \xrightarrow{u_1} & x_1 & \xleftarrow{u_2} & x_2 & \longrightarrow & \cdots & \xleftarrow{u_{2N}} & x_{2N} = a \\ & \searrow^{\alpha_0} & \downarrow{\scriptstyle \alpha_1} & \swarrow_{\alpha_2} & & & & & \\ & & a. & & & & & & \end{array}$$

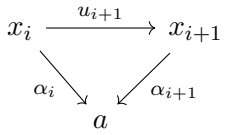
Prenons une flèche $\alpha_0 : x \longrightarrow a_0$ dans X, je dis que la classe d'homotopie du chemin c (i.e. son image dans $\mathrm{Fl}(\Pi(X))$ est égale à celle de u_0. Pour le voir, choisissons pour tout $i \in [0, 2N]$ une flèche $\alpha_i : x_i \longrightarrow a$. (Ça existe grâce à l'hypothèse b)). Je dis que pour chaque $i \in [0, 2N]$, le triangle [ms. 2] [ed. 2]

$$\begin{array}{ccc} x_0 & \xrightarrow{c_i} & x_i \\ & \searrow^{\alpha_0} \quad \swarrow_{\alpha_i} & \\ & a & \end{array}$$

dans $\Pi(X)$ est commutatif, où c_i désigne la classe d'homotopie du chemin de x_0 à x_i définie par les u_i ($i \in [1, i]$) `[plutôt` ($i \in [1, 2N]$)`]`. Nous le faisons par récurrence sur i, l'assertion est triviale si $i = 0$ (c_0 représente l'application identitique). Le diagramme

$$\begin{array}{ccccc} & & c_{i+1} & & \\ & \overset{}{\frown} & & \overset{}{\frown} & \\ x_0 & \xrightarrow{c_i} & x_i & \xrightarrow{u_{i+1}} & x_{i+1} \\ & \searrow^{\alpha_0} \quad \swarrow_{\alpha_i} & & \swarrow_{\alpha_{i+1}} & \\ & & a & & \end{array}$$

montre qu'il suffit de prouver que pour $i \in [0, 2N-1]$ le triangle



est commutatif dans $\Pi(X)$. (**NB** u_{i+1} est en sens directe ou rétrograde, suivant que i est paire ou impaire.) Mais cela résulte du fait que l'image de $\mathrm{Hom}_X(x_i, a)$ (resp. $\mathrm{Hom}_X(x_{i+1}, a)$) dans $\mathrm{Fl}\Pi(X)$ est réduite à un seul élément. [ms. 3]

Ainsi on trouve que X *est* 1-*connexe.* Il y a donc de l'espoir! Il reste à prouver

$$H^i(X, M) = 0, \qquad i \geq 2, \quad M \text{ coefficient constant.}$$

J'ai regardé le cas lorsqu'on a ceci : $b = c$, $\mathrm{Ob}X = \{b, b \times b\}$, $\mathrm{Hom}_X(b \times b)$ [plutôt $\mathrm{Hom}(b \times b, b)$] = { réunion disjointe des ensembles $u \circ \mathrm{pr}_1$ et $u \circ \mathrm{pr}_2$, avec $u \in \mathrm{Hom}_X(b, b) = G$ un monoïde quelconque }. Alors X° est fibrée, via $I \longmapsto b^I$, sur la sous-catégorie pleine S de Ens formée de $\{0\}$ et $\{0, 1\}$, et donc X est cofibrée (en particulier propre) sur $S^\circ \overset{\text{déf}}{=} T$. (NB $S^\circ = T$ a un objet initial, mais non un objet final. Sa cohomologie à valeurs dans un préfaisceau quelconque n'est pas toujours triviale, cf. ci-dessous.)

On trouve [`On trouve' barré]

$$H^n(X, M) \Longleftarrow H^p(T, \underbrace{R^q f_*(M_X)}_{=\mathcal{H}^q}), \qquad \text{où } f : X \longrightarrow S^\circ$$

[$n = p + q$]. J'admet ici (provisoirement) ceci :

$$\begin{aligned} &H^p(T, F) = 0 \ , && \text{pour } p \geq 2, \text{ tout } F \in T^\wedge \ , \\ &H^0(T, F) = \mathrm{Ker}\delta \ , \\ &H^2(T, F) = \mathrm{Coker}\delta \ , \end{aligned}$$

ed. 3

où $\delta : F_0 \longrightarrow F_1$ est défini comme la différence $\delta = p_1^* - p_2^*$, où p_1, p_2 sont les deux projections de $\{0, 1\} \simeq \{0\}^2$ dans $T = S^\circ$. On a de plus

$$(\mathcal{H}^q)_0 = H^q(G, M), \qquad (\mathcal{H}^q)_1 = H^q(G \times G, M),$$

et

$$p_1^*, \ p_2^* \ : \ H^q(G, M) \rightrightarrows H^q(G \times G, M) \qquad \text{sont les transposés de } \mathrm{pr}_1, \ \mathrm{pr}_2 : G \times G \longrightarrow G.$$

Commentary · Exposition [ms. 3]

The finite-power category and the missing symmetries

Grothendieck now makes the product condition explicit in a category small enough to calculate. Write X_G for the category with objects $1, 2$, representing b, b^2, and

$$\mathrm{Hom}_{X_G}(m, n) = \big(\{1, \ldots, m\} \times G\big)^n.$$

A morphism lists, for each output coordinate, one input coordinate and one element of the monoid G; composition substitutes these expressions. Thus $(i, g) : b^m \to b$ means $g \circ p_i$. This convention fixes multiplication even when G is not commutative. The evident maps $2 \to 1$ exhibit 2 as the product of 1 with itself. Both objects map to 1, so the hypotheses of the opening proposition hold. For a group G, the opposite category is the full category of free G-sets of ranks one and two.

Let S be the full subcategory of finite sets on $\mathbf{1}, \mathbf{2}$, and put $T = S^{\mathrm{op}}$. The functor $X_G \to T$ forgets the monoid labels. A map $v : m \to n$ of T, equivalently a function $n \to m$, induces the

coordinate-selection homomorphism $G^m \to G^n$. The arrows with all labels equal to 1_G are chosen cocartesian lifts, so X_G is the Grothendieck construction of $n \mapsto BG^n$, where here BG^n denotes the one-object category. This proves the asserted opfibration and justifies the cohomological spectral sequence, with constant abelian coefficients M:

$$E_2^{p,q} = R^p \varprojlim_{S} F_q \implies H^{p+q}(BX_G; M), \qquad F_q(n) = H^q(BG^n; M).$$

The functor F_q is covariant on S. A modern treatment of this spectral sequence for Grothendieck constructions is Cegarra (2020, Corollaries 3–4); the underlying homotopy-colimit comparison is Thomason (1979, Theorem 1.2).

The provisional formula is already contradicted by a constant diagram $F = M$. The category T has an initial object, hence contractible nerve, and $H^{>0}(T, M) = 0$. Yet the proposed differential is $1_M - 1_M = 0$, whose cokernel is M. Changing the printed H^2 to the evidently intended H^1 does not repair this. The full category S contains two inclusions, their common retraction, and the transposition of $\mathbf{2}$; retaining only a parallel pair loses relations and higher derived limits. On the other hand the formula for H^0 is correct: a compatible pair is determined by its value at $\mathbf{1}$, and the equality of the two induced values at $\mathbf{2}$ implies compatibility with the retraction and transposition.

Commentary · Development (commentators') [ms. 3]

A small resolution for the actual base category

The point of repairing the base calculation is to retain Grothendieck's finite example, while locating its obstruction in the correct degree. The following elementary computation supplies all the derived limits used below. We did not find this calculation for the appendix's two-object category in the literature checked; the argument is included in full, without a claim of priority.

Derived-limit calculation. [ours] Let $F : S \to \mathrm{Ab}$, let $i_1, i_2 : \mathbf{1} \to \mathbf{2}$ be the inclusions, let $r : \mathbf{2} \to \mathbf{1}$ be the unique map, and let τ transpose $\mathbf{2}$. Put

$$A = F(\mathbf{1}), \quad B = F(\mathbf{2}), \quad L = \ker(F(r) : B \to A), \quad L^- = \{v \in L : F(\tau)v = -v\}.$$

Then

$$\begin{aligned} R^0 \varprojlim_{S} F &= \ker(F(i_1) - F(i_2) : A \to B), \\ R^1 \varprojlim_{S} F &= L^- / (F(i_1) - F(i_2))A, \\ R^p \varprojlim_{S} F &= \mathrm{Ext}^{p-1}_{\mathbb{Z}[C_2]}(\mathbb{Z}_{\mathrm{sgn}}, L) \qquad (p \geq 2). \end{aligned}$$

The action of C_2 on L is induced by $F(\tau)$.

Proof. Work in the abelian category of covariant functors on S. The functors $P_j = \mathbb{Z}[S(\mathbf{j}, -)]$ are projective, since $\mathrm{Hom}(P_j, F) = F(\mathbf{j})$ and evaluation is exact. The augmentation $P_1 \to \underline{\mathbb{Z}}$ gives

$$0 \longrightarrow K \longrightarrow P_1 \longrightarrow \underline{\mathbb{Z}} \longrightarrow 0.$$

Here $K(\mathbf{1}) = 0$, and $K(\mathbf{2}) = \mathbb{Z}(i_1 - i_2)$ is the sign representation; the two constant endomorphisms act as zero. This immediately gives the formula for degree zero and identifies the degree-one group as the stated cokernel, because a map $K \to F$ is precisely an element of L^-.

For completeness the higher Ext groups can be computed inside the same tiny category. Precomposition with r gives a monomorphism $P_1 \to P_2$. It has the natural retraction induced by i_1, because $ri_1 = 1$. Consequently $Q = P_2/P_1$ is projective. It is zero at $\mathbf{1}$ and is the regular representation $\mathbb{Z}[C_2]$ at $\mathbf{2}$, with basis the classes of 1 and τ; both constant maps act as zero. Any free $\mathbb{Z}[C_2]$-resolution of the sign representation therefore gives a projective resolution of K, by replacing the regular module with Q. Applying $\mathrm{Hom}(-, F)$ gives exactly the complex computing $\mathrm{Ext}^*_{\mathbb{Z}[C_2]}(\mathbb{Z}_{\mathrm{sgn}}, L)$: maps out of Q are elements annihilated by $F(r)$. Finally dimension shifting through the displayed short exact sequence proves the formula for $p \geq 2$. This also proves that no omitted higher relation is being assumed away. □

For example, take $F(\mathbf{1}) = 0$, $F(\mathbf{2}) = \mathbb{Z}$ with τ acting as -1. The formula gives $R^3 \varprojlim F \cong \mathbb{Z}/2$, from the usual alternating resolution of C_2. Thus even the claimed vanishing in all degrees at least two is false for the full base category.

On vérifie aisément que [ms. 4]

$$H^0(T, \mathcal{H}^q) = 0 \qquad \text{si } q \geq 1,$$

et il reste

$$H^n(X, M) \simeq H^1(S, \mathcal{H}^{n-1}) = \mathrm{Coker}(\delta : H^{n-1}(G) \longrightarrow H^{n-1}(G \times G)).$$

Cela donne, pour $n = 2$,

$$H^2(X, M) \simeq \mathrm{Coker}\Big(\delta : \mathrm{Hom}_{\mathrm{mon}}(G, M) \longrightarrow \underbrace{Hom_{\mathrm{mon}}(G \times G, M)}_{\mathrm{Hom}_{\mathrm{mon}}(G,M) \times \mathrm{Hom}_{\mathrm{mon}}(G,M)} \Big)$$
$$\simeq \mathrm{Coker}\Big(u \longmapsto (u, -u) : H \longrightarrow H \times H\Big),$$

où H est le groupe abélien $\mathrm{Hom}_{\mathrm{mon}}(G, M)$.

D'où

$$H^2(X, M) \;\simeq\; H = \mathrm{Hom}_{\mathrm{mon}}(G, M).$$

Donc

> X est 2-connexe si et seulement si le groupe *abélien* enveloppant du monoïde G est nul.

Ainsi, si G est un groupe abélien $\neq 0$, X *n'est pas* 2-connexe.

Commentary · Development (commentators') [ms. 4]

The counterexample survives, one degree later

Grothendieck's proposed category is indeed nonaspherical for $G = C_2$. Its degree-two cohomology, however, is zero. The distinction matters twice: it changes the audit of the first four pages, and it supplies a sound finite example for the pushouts that follow.

The actual low-degree calculation. [ours] For every monoid G, the space BX_G is 2-connected. For $G = C_2$ one moreover has

$$H^3(BX_{C_2}; \mathbb{F}_2) \cong \mathbb{F}_2.$$

In particular the opening weak-contractibility assertion is false under its exact product hypotheses.

Proof. The groupoid argument already proves simple connectivity. Use the spectral sequence and notation of the preceding two comments. The $q = 0$ row is the constant diagram M, so its terms vanish for $p > 0$. For $q > 0$ the two coordinate pullbacks

$$H^q(BG; M) \rightrightarrows H^q(BG^2; M)$$

have zero equalizer: restriction along $g \mapsto (g, 1)$ sends the first pullback to the identity and the second to zero. Thus $E_2^{0,q} = 0$ for $q > 0$.

Set $A = H^1(BG; M) = \mathrm{Hom}_{\mathrm{mon}}(G, M)$, with M regarded as an additive group. A normalized degree-one cocycle on the monoid is exactly an additive homomorphism, which proves this identification directly. A homomorphism $G^2 \to M$ is uniquely the sum of its restrictions to the two factors, so the degree-one diagram is

$$F_1(\mathbf{1}) = A, \qquad F_1(\mathbf{2}) = A \oplus A,$$

with i_1, i_2 inducing the two inclusions, r inducing addition, and τ interchanging the factors. Hence $L = \{(a, -a) : a \in A\}$, $L^- = L$, and the map $A \to L$ in the degree-one formula is an isomorphism. Thus $E_2^{1,1} = 0$. All three entries of total degree two vanish; consequently $H^2(BX_G; M) = 0$ for every abelian M. Simple connectivity and the universal coefficient theorem give $H^2(BX_G; M) = \mathrm{Hom}(H_2(BX_G; \mathbb{Z}), M)$. Taking $M = H_2(BX_G; \mathbb{Z})$ forces this group to be zero. The Hurewicz theorem now gives $\pi_2(BX_G) = 0$; see Hatcher (2002, Theorems 3.2 and 4.32).

Now put $G = C_2$ and $M = \mathbb{F}_2 = k$. The same degree-one diagram has $L = k$ with trivial C_2-action. The base calculation gives

$$E_2^{2,1} = \mathrm{Ext}^1_{\mathbb{Z}[C_2]}(\mathbb{Z}_{\mathrm{sgn}}, k) = k.$$

Indeed the free resolution of the sign module has differentials alternating between $1+\tau$ and $1-\tau$, both zero after applying $\mathrm{Hom}(-,k)$. No differential can kill this term: its sole possible incoming differential is from $E_2^{0,2}=0$ and its sole possible outgoing differential lands in $E_2^{4,0}=0$.

To identify the whole degree-three group, the only remaining possible entry is $E_2^{1,2}$. Write $H^*(BC_2;k)=k[t]$ and $H^*(B(C_2^2);k)=k[x,y]$, with all three generators in degree one; these are the usual projective-space and Künneth calculations (Hatcher, 2002, Theorems 3.15 and 3.19). In degree two, $F_2(r)$ sends each of x^2, xy, y^2 to t^2. The τ-fixed part of its kernel is therefore $k(x^2+y^2)$, exactly the image of $F_2(i_1)-F_2(i_2)$. The degree-one formula gives $E_2^{1,2}=0$. All other entries of total degree three vanish, proving the displayed isomorphism. □

This proof also explains the discrepancy in the manuscript. The abstract cokernel of $a\mapsto(a,-a)$ is indeed A, but it is not the derived limit being computed. The transposition and the common retraction replace that cokernel by the preceding resolution. For $G=C_2$ the category has two objects and $2+4+4+16=26$ arrows. It supplies the finite counterexample promised in the text, with a degree-three obstruction. The stronger assertion in footnote 2 that the general product hypotheses need not imply 2-connectivity requires a different example. The next comment verifies it with an actual product and a two-sphere.

Commentary · Development (commentators') [ms. 4]

A product whose category is a two-sphere

The correction of the monoid calculation leaves one assertion in footnote 2 to check: the exact product hypotheses may fail to force 2-connectivity. That assertion is true. One extra object supplies a smaller example in object count, and reveals geometrically the second homotopy group that the proposed monoid calculation did not produce. The construction below is our synthesis of the product condition and the elementary nerve-of-a-cone calculation; we did not locate this particular finite example in the literature searched, and make no claim of priority.

A family realizing wedges of two-spheres. [ours] For every integer $r\geq 1$ there is a category Y_r with three objects d,b,a, where b is weakly terminal and $a=b\times b$, such that

$$BY_r \simeq \bigvee^{(r-1)(r-2)/2} S^2.$$

The wedge with zero summands means a point. In particular Y_3 has 21 arrows, satisfies the opening hypotheses with $c=b$, and has the homotopy type of S^2.

Proof. Let D be an r-element set. On the two objects $b=b^1$ and $a=b^2$ use the category $T=S^{\mathrm{op}}$ already encountered, with

$$\mathrm{Hom}_T(b^m,b^n)=\{1,\dots,m\}^{\{1,\dots,n\}},\qquad m,n\in\{1,2\}.$$

These are coordinate-selection maps, composed by substitution. Add the object d, its identity, the maps

$$\mathrm{Hom}(d,b)=D,\qquad \mathrm{Hom}(d,a)=D^2,$$

and no maps from a or b to d. Composition with old arrows again selects coordinates. Substitution is associative, so this defines a category. The product bijection

$$\mathrm{Hom}(z,a)\xrightarrow{(p_1,p_2)}\mathrm{Hom}(z,b)\times\mathrm{Hom}(z,b)$$

holds for $z=a,b$ by the definition of T, and for $z=d$ it is the identity on D^2. Hence $a=b\times b$ in the enlarged category. As D is nonempty, all three objects map to b.

To see the homotopy type, consider $E=d\backslash T$. An object of E is a list of length one or two with entries in D; a morphism selects coordinates to obtain its target list. Let P be the poset of nonempty subsets of D of size at most two, ordered by reverse inclusion. Sending a list to its support defines $s:E\to P$. Choose one ordering of D and let $j:P\to E$ list each subset without repetition in that order. Coordinate selection makes j a functor and $sj=1_P$. There is a natural transformation $js\to 1_E$: from the distinct list of a support to any given list, repeat the indicated coordinates. This map is unique because each label occurs once in its source. Its uniqueness also proves naturality for every coordinate-selection arrow. Therefore $BE\simeq BP$.

For clarity, the last gluing step can be read directly on the nerve. Every nondegenerate simplex of NY_r is either wholly in NT, or starts at d and continues in T. A simplex of the second kind is precisely the cone on a simplex of $N(d\backslash T)$. Its face opposite d is obtained by forgetting the initial arrow from d. Thus NY_r is the simplicial mapping cone of $NE \to NT$. After realization it is the mapping cone of $BE \to BT$. The category T has initial object b, so BT is contractible and

$$BY_r \simeq \Sigma BE \simeq \Sigma BP.$$

The space BP is the barycentric subdivision of the complete graph on the r vertices of D. For $r \geq 1$ that graph is connected and has $\binom{r}{2} - r + 1 = (r-1)(r-2)/2$ independent cycles: collapse a spanning tree to obtain the wedge of that many circles. Suspending gives the asserted wedge of two-spheres. Finally T has 8 arrows; adding d contributes $1 + r + r^2$ arrows. For $r = 3$ the total is 21. □

This example verifies footnote 2 under the actual product assumptions, including its failure of 2-connectivity. It also keeps the two repairs distinct. The two-object category X_{C_2} first detects nonasphericity in degree three; the three-object category Y_3 detects it in degree two. The four-object category considered next in the manuscript has the latter homotopy type but satisfies only the terminal-presheaf-span condition.

Corollaire. $\Psi = \begin{array}{ccc} & \nearrow & b_0 \\ a_0 & & \\ & \searrow & c_0 \end{array}$ *n'est* pas *totalement* $\mathbf{W}_\infty$*-asphérique; i.e. si on a un dia-*

gramme $\begin{array}{ccc} & \nearrow & b \\ a & & \\ & \searrow & c \end{array}$ *dans une catégorie* $X \in \mathrm{ObCat}$*, tel que sa somme amalgamée dans* $X^\wedge$ *soit égale à* e_X*, il ne s'ensuit pas que* X *soit* $\mathbf{W}_\infty$*-asphérique.* (Mais peut-être voudra bien être 1-connexe?) ed. 4

Question. *Y-a-t-il d'autres objets totalement* $\mathbf{W}_\infty$*-asphériques de* Cat*, que les catégories ayant un objet final?* Je suspecte à présent que non – donc que l'on ne peut améliorer substantiellement sur la notion déjà connue d'immersion ouverte de Dwyer. [ms. 5]

Commentary · Present state [ms. 5]

Total asphericity without a terminal object

The question concerns how far the useful gluing condition extends beyond representability. In the sense used here, a category A is totally $\mathbf{W}_\infty$-aspherical when, for every functor $f : A \to C$, the presheaf $f_!1_A$ has weakly contractible category of elements. This is the universal left-Kan-extension condition of Chapter XII, ms. 88–90; the editors explicitly distinguish it from the total asphericity of *Pursuing Stacks*. The suspicion that a terminal object is necessary is false. Filtered categories provide counterexamples; indeed Grothendieck records this correction himself in Chapter XII, ms. 109, footnote 78 (Grothendieck, 1990–1991d). Chapter II, Corollaries 2.17–2.18 (ms. 34–35), gives the underlying filtered argument (Grothendieck, 1990–1991b). We recall its mechanism because it answers the question on this page, while the commentary on Chapter II, manuscript pp. 35–36, already discusses its broader setting.

For any small A, the canonical presentation of the terminal presheaf gives

$$f_!1_A \cong \operatorname*{colim}_{a \in A} C(-, f(a)).$$

Suppose that A is filtered, and denote this presheaf by F. We verify directly that C/F is filtered. It is nonempty, since the identity of any $f(a)$ supplies an object. Given (c, x) and (d, y), choose representatives $u : c \to f(a)$ and $v : d \to f(b)$. A common successor k of a, b produces arrows from both objects to $(f(k), [\mathrm{id}_{f(k)}])$. Finally, let $r, s : (c, x) \rightrightarrows (d, y)$ be parallel arrows. Represent y by $v : d \to f(b)$. The equality $F(r)y = F(s)y$ says that vr and vs become equal at some later stage of the filtered colimit. Thus there is an arrow $b \to k$ such that its image, composed with v, equalizes r and s. This is an equalizer after postcomposition inside C/F.

The nerve of a filtered category is weakly contractible. One way to see this is to take the finitely many objects and arrows occurring in a simplicial representative of a homotopy class, choose a common target, and then equalize the finitely many competing composites. The resulting compatible cone gives a null homotopy. Equivalently, the usual finite-diagram characterization of filteredness cones off every finite simplicial diagram. Consequently C/F is weakly contractible for every f, as required. The elementary eventual-equality fact used above is the standard filtered-colimit argument (The Stacks Project Authors, 2026d, §4.19).

For a concrete infinite example take $A = (\mathbb{N}, \leq)$. It has no greatest element and hence no terminal object. The phenomenon is not a consequence of infinitude: the one-object category with endomorphism monoid $\{1, e\}$, $e^2 = e$, is filtered, since postcomposition by e equalizes any pair of endomorphisms. Its unique object is not terminal, because it has two endomorphisms. For a functor out of this category, $f_!1_A$ is the image of the induced idempotent on the represented presheaf $C(-, f(*))$. Taking $C = A$ and $f = \mathrm{id}_A$ makes this image the terminal presheaf, with one element at the unique object, whereas the sole representable has two elements there. Thus the gluing weight need not be representable. These examples settle the question as asked; they do not assert an exhaustive classification of totally aspherical categories.

Dans le diagramme

$$(*) \qquad \begin{array}{ccccc} U & \xrightarrow{\ i\ } & X & \xleftarrow{\ j\ } & T \\ {\scriptstyle p}\downarrow & \text{cocart. et cart.} & {\scriptstyle q}\downarrow & & \vdots\ \text{iso} \\ U' & \xrightarrow{\ i'\ } & X' & \xleftarrow{\ j'\ } & T', \end{array}$$

prenons

$$\begin{aligned} U = \Psi, \quad & T = \{t\} = e \text{ catégorie finale,} \\ & t \text{ objet final de } X, \\ & \text{i.e. } H_t = (u \longmapsto \mathrm{Hom}_X(u,t)) = e_U \in \mathrm{Ob}U^\wedge \ ({}^3). \end{aligned}$$

Prenons pour U' la catégorie dont il est question dans la 'proposition' hypothétique (et fausse) p. 1 avec $b = c$, plus précisément une catégorie avec deux objets b et $b \times b$, un groupe (voire un monoïde) G d'automorphismes `[(voire d'endomorphismes)]` de b, et les autres flèches celles qui s'en déduisent, en exigeant que $b \times b$ soit bel et bien le carré cartésien de b. On trouve une catégorie cofibrée (donc propre) sur la catégorie T opposée de celle des ensembles finis de la forme $\{0\} \cup \{0, 1\}$ (deux objets) avec comme fibres B_G et $B_{G\times G}$ respectivement (B_π = catégorie avec un seul objet, et monoïde d'endomorphismes π). Le foncteur

$$U = \Psi \longrightarrow U'$$

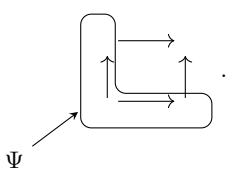
est celui donné par le diagramme

[ms. 6]

$$\begin{array}{ccccc} & & & \xrightarrow{\ \mathrm{pr}_1\ } & b \\ b \times b & & & & \\ & & & \xrightarrow{\ \mathrm{pr}_2\ } & b \end{array}$$

ed. 5

3. i.e. X déduit de $U = \Psi$ en lui rajoutant un objet final t – donc $X \simeq \mathbf{\Delta}^1 \times \mathbf{\Delta}^1$,



dans U', et les hypothèses a), b) de la page 1 assurent que la somme amalgamée $\varinjlim$ de ce diagramme dans $U'^{\wedge}$ est $e_{U'}$, i.e. que

$$p_!(e_U) = e_{U'}.$$

Il s'ensuit que dans la somme amalgamée, X' envisagée dans $(*)$, l'unique élément t' de T' est objet final de X'. On a vu que U' est 1-connexe, mais non 2-connexe si le groupe abélien enveloppant de G (i.e. $G/$(groupe des commutateurs) dans le cas où G est un groupe) n'est pas réduit à 1. L'exemple le plus économique s'obtient en prenant $G = \{\pm 1\}$ le groupe à deux éléments. C'est donc un cas où dans $(*)$ on a

$$i \in W \quad \text{(et même } i \text{ asphérique)}, \qquad \text{mais } i' \notin W,$$

et qui a l'avantage sur l'exemple plus sophistiqué de la page 103 (prop. 10) de ne mettre en jeu que des catégories finies – et même fort petites !

Commentary · Examples [ms. 6]

Adjoining a terminal object after the pushout

This construction uses the failure of asphericity to test whether an ordinary pushout preserves weak equivalences. Its mechanism is transparent on representables. For any object x of a category satisfying the opening hypotheses, put $B_x = \mathrm{Hom}(x, b)$ and $C_x = \mathrm{Hom}(x, c)$. Both sets are nonempty, and

$$B_x \amalg_{B_x \times C_x} C_x = \{*\}.$$

Every element of B_x is identified with every element of C_x, so there is exactly one equivalence class. As presheaf colimits are computed objectwise, the span $b \leftarrow b \times c \rightarrow c$ has terminal colimit. This verifies $p_! 1_\Psi = 1_{U'}$ without a homotopy assertion.

Write $\Psi^{\triangleright}$ for Ψ with a new terminal object. In the pushout $U' \amalg_\Psi \Psi^{\triangleright}$, a map from $x \in U'$ to the new object is represented by a map from x to an object of the span, modulo its two identifications. Thus the presheaf of maps to that object is precisely $p_! 1_\Psi$, and the pushout is $(U')^{\triangleright}$. The original inclusion $\Psi \rightarrow \Psi^{\triangleright}$ is aspherical: the comma category over the new vertex is Ψ, with initial object, and the comma categories over old vertices have terminal objects. After the pushout, $U' \rightarrow (U')^{\triangleright}$ is a weak equivalence exactly when BU' is weakly contractible.

Taking $U' = X_{C_2}$ therefore validates the announced failure. Both the original span and its cone are contractible, whereas $H^3(BU'; \mathbb{F}_2) = \mathbb{F}_2$. The source's asserted obstruction in π_2 must be replaced by the degree-three calculation above; the categorical pushout and its terminal object require no alteration. This is also the point where the distinction from a homotopy pushout becomes visible: a homotopy pushout along the equivalence $\Psi \rightarrow \Psi^{\triangleright}$ would retain the homotopy type of U', while the ordinary categorical pushout has become a cone.

Procédons maintenant comme dans la proposition 11, en désignant par U'' la catégorie déduite de U' en lui ajoutant un objet final, et soit [ms. 7]

$$p' : U' \longrightarrow U''$$

l'inclusion, de sorte que

$$p'_!(e_{U'}) \quad = \quad (p'p)_!(e_U) = U' \qquad \text{en tant qu'objet dans } U''^{\wedge}$$
$$\text{(sous-préfaisceau du préfaisceau final } e_{U''}).$$

Considérons

$$\begin{array}{ccccc} U & \xrightarrow{\ i\ } & X & \xleftarrow{\ j\ } & T \\ {\scriptstyle p''=p'p}\downarrow & \text{cocart. et cart.} & {\scriptstyle q''}\downarrow & & \downarrow{\scriptstyle \text{iso}} \\ U'' & \xrightarrow{\ i''\ } & X'' & \xleftarrow{\ j''\ } & T''. \end{array}$$

Cette fois U, X *et* U'' sont $\mathbf{W}_\infty$-asphériques (et même contractiles toutes trois), donc i, p'' sont dans W, pourtant X'' n'est pas $\mathbf{W}_\infty$-asphérique (donc $i'', q'' \notin \mathbf{W}_\infty$), car pour qu'elle le soit, i.e.

pour que $i'' \in \mathbf{W}_\infty$, il faudrait (prop. 7, cor. page 85 ([4])) que i'' soit $\mathbf{W}_\infty$-asphérique, i.e. que U''/t'' soit $\mathbf{W}_\infty$-asphérique. Mais par construction $U''/t'' \simeq U'$, qui n'est *pas* $\mathbf{W}_\infty$-asphérique.

Commentary · Exposition [ms. 7]

Two cone vertices and a suspension

Grothendieck strengthens the example by making the other map out of Ψ a weak equivalence as well. Let $C = U'$, let e be the terminal object adjoined to form $U'' = C^{\triangleright}$, and retain t for the cone vertex coming from $X = \Psi^{\triangleright}$. The left Kan extension $p'_!1_C$ is the presheaf with value one on the old objects of C and value zero at e: no map runs from the new terminal object back to C. This is the subterminal presheaf called U' in the displayed formula.

Consequently the pushout X'' consists of C with two incomparable new terminal vertices e, t, and one map from each old object to each vertex. There are no arrows between e and t. Every nondegenerate nerve simplex lies in one of the two cones; their intersection is exactly NC. Hence

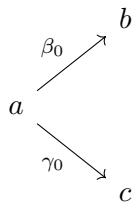
$$BX'' = C(BC) \cup_{BC} C(BC) \simeq \Sigma BC.$$

This gives the omitted nonasphericity argument directly. For the finite choice $C = X_{C_2}$, suspension in reduced cohomology yields

$$\widetilde{H}^4(BX''; \mathbb{F}_2) \cong \widetilde{H}^3(BC; \mathbb{F}_2) \cong \mathbb{F}_2.$$

Thus X'' is not weakly contractible, although Ψ, X and U'' are contractible. Both maps to X'' from the latter two categories fail to be weak equivalences. No converse to Quillen's Theorem A is required. In general a noncontractible space may have contractible suspension if it is connected and acyclic, so a bare appeal to the nonasphericity of a comma category would not suffice. Here the explicit surviving cohomology class proves exactly what is needed.

Voilà un exemple encore plus simple d'un foncteur $f : \Psi \longrightarrow X$ tel que $f_!(e_\Psi)$ ne soit pas $\mathbf{W}_\infty$-asphérique, i.e. d'un diagramme



ed. 6

dans une catégorie V, telle que la limite inductive $L = b \amalg_a c$ *dans* $V^\wedge$ soit un objet non [ms. 8] asphérique de $V^\wedge$, i.e. V/L soit non asphérique. Je sais qu'il faut, pour un tel contre-exemple, que $a \longrightarrow b$ et $a \longrightarrow c$ ne soient pas des monomorphismes. L'idée, c'est de prendre un exemple universel où ces flèches ne soient pas `[des]` monomorphismes, en prenant [ms. 9]

$$V = \left(\begin{array}{ccccc} & & & & b \\ & & & \overset{\beta}{\nearrow} & \\ 0 & \underset{\mu}{\overset{\lambda}{\rightrightarrows}} & a & & \\ & & & \underset{\gamma}{\searrow} & \\ & & & & c \end{array} \right)$$

avec

$$\beta\lambda = \beta\mu \qquad \text{et} \qquad \gamma\lambda = \gamma\mu$$

les deux seules flèches non identiques, en plus des quatre flèches marquées β, γ, λ, μ.

4. pas encore démontré !

Commentary · Exposition [ms. 8]

A presheaf pushout remembers only components

Grothendieck now seeks a finite obstruction that can be inserted into a much larger category. The decisive distinction is between the ordinary pushout of three representable presheaves and the homotopy carried by its category of elements. We make that distinction explicit before using the example. Write $y(a) = \mathrm{Hom}_C(-, a)$, and let

$$L = y(b) \amalg_{y(a)} y(c)$$

for a span $b \xleftarrow{\beta} a \xrightarrow{\gamma} c$ in C. For $x \in C$, form the bipartite multigraph K_x with vertices $b(x) \amalg c(x)$ and one edge for every $u \in a(x)$, joining βu to γu. Then $L(x) = \pi_0 K_x$. An ordinary pushout has forgotten the cycles in this graph.

This description proves the assertion about monomorphisms, including the case where L is not terminal. Let E have objects (v, x, u), with $v \in \Psi$ and $u : x \to f(v)$ for the given span $f : \Psi \to C$. A morphism to (v', x', u') consists of $h : v \to v'$ and $r : x \to x'$ satisfying $f(h)u = u'r$. The projection $E \to \Psi$ has right adjoint $v \mapsto (v, f(v), 1)$, so BE is contractible. There is a canonical functor

$$P : E \longrightarrow C/L, \qquad (v, x, u) \longmapsto (x, [u]).$$

Here C/L denotes the category of elements of the presheaf L. For $(x, l) \in C/L$, the comma category $(x, l)\backslash E$ retracts up to a natural transformation onto the incidence category of the component l of K_x. Indeed an object of this comma category includes $r : x \to x'$; send it to $(v, x, u'r)$ and use the morphism given by r for the comparison with the original object. The incidence category has one object for each vertex and edge, and two arrows from each edge to its endpoints.

If β is monic, every vertex in $b(x)$ has degree at most one. Every connected component of K_x is consequently a star, possibly a single vertex, and its incidence category is contractible. Quillen's Theorem A, in its comma-category form, gives $BP : BE \simeq B(C/L)$. Thus L is aspherical. The same argument applies with b and c interchanged. This supplies the proof promised in the source; the only external ingredient is the comma-category criterion (Quillen, 1973, §1, Theorem A).

There is also a useful conclusion without a monomorphism hypothesis. If $L = e_C$, every K_x is nonempty and connected, and C is simply connected. To see this directly, work in ΠC. Put $d = [\beta][\gamma]^{-1} : c \to b$. Send a vertex $v \in b(x)$ to $[v] : x \to b$ and a vertex $w \in c(x)$ to $d[w]$. The two endpoints of an edge $u \in a(x)$ have the same image, since $d[\gamma u] = [\beta u]$. Connectedness therefore gives a single arrow $s_x : x \to b$. For every $r : x \to y$, precomposition of a vertex of K_y proves $s_y[r] = s_x$. These identities extend to all zigzags and force every arrow from x to y in ΠC to equal $s_y^{-1} s_x$. Thus ΠC is equivalent to the terminal groupoid. The correction on manuscript p. 10 is justified: the first possible obstruction is in degree two.

On trouve que

$$b \amalg_a c \xrightarrow{\sim} e_X \qquad \text{(somme amalgamée dans } V^\wedge\text{)}.$$

Pourtant V n'est pas asphérique, ni même 1-connexe : il est homotope à la sous-catégorie ouverte $V_0 = (0 \underset{\mu}{\overset{\lambda}{\rightrightarrows}} a)$, car on a une rétraction $V \xrightarrow{r} V_0$ (envoyant b et c sur a), et une flèche $ir \longrightarrow \mathrm{id}_V$ (où $i : X_0 \hookrightarrow X$ est l'inclusion). Donc V a le même type d'homotopie *que* V_0, qui est un cercle. Donc on a intérêt à prendre ce V, où $\tilde{V}$ déduit de V en rajoutant un élément final e,

$$\tilde{V} = \left(\begin{array}{ccccc} & & & & b & & \\ & & & \nearrow & & \searrow & \\ 0 & \rightrightarrows & a & & & & e \\ & & & \searrow & & \nearrow & \\ & & & & c & & \end{array} \right),$$

comme U', pour les contre-exemples pages 6 et 7.

Commentary · Examples [ms. 9]

The missing retraction and the two-sphere

The four-object category is useful because its relations can be seen as attaching two disks to the same circle. We use the notation Z of the following pages, suppressing subscripts: its objects are $0, a, b, c$, its generating arrows are $\lambda, \mu : 0 \rightrightarrows a$, $\beta : a \to b$, and $\gamma : a \to c$, and its relations identify the two composites to each of b, c. The commentary on Chapter II, manuscript p. 36 ("The retracted proposition and a visible sphere") already records the general obstruction to total asphericity. Here we need the precise attaching maps, because the next pushout will suspend this particular sphere once more.

Let $D = (0 \rightrightarrows a)$. A retraction $Z \to D$ would fix a, so it must send b to a and β to 1_a. Applying it to $\beta\lambda = \beta\mu$ would give $\lambda = \mu$, a contradiction. This is the exact point at which the proposed retraction fails. The full subcategories on $\{0, a, b\}$ and $\{0, a, c\}$ are instead the two terminal cones on D. Every composable string in Z belongs to one of them, and their intersection is D. Consequently

$$NZ = N(D^{\triangleright_b}) \amalg_{ND} N(D^{\triangleright_c}), \qquad BZ \cong \Sigma BD \cong S^2.$$

The first identity is an actual identity of simplicial sets. Realization preserves this union, and BD is the circle with two vertices and two edges. This proves the correction already supplied by footnote 7, rather than replacing one guessed homotopy type by another.

The calculation is also visible in normalized chains. There are four vertices, six nonidentity arrows, four nondegenerate 2-simplices, and no higher nondegenerate simplices. With $[u, v]$ denoting a composable pair, a fundamental cycle is

$$[\lambda, \beta] - [\mu, \beta] - [\lambda, \gamma] + [\mu, \gamma].$$

Its boundary vanishes by the two defining relations; it generates $H_2(BZ; \mathbf{Z}) \cong \mathbf{Z}$. In particular $\pi_1(BZ) = 0$ and $\pi_2(BZ) \cong \mathbf{Z}$.

The presheaf pushout nevertheless is terminal. Its pointwise calculation is

x	0	a	b	c
$\lvert a(x)\rvert$	2	1	0	0
$\lvert b(x)\rvert$	1	1	1	0
$\lvert c(x)\rvert$	1	1	0	1
$\lvert L(x)\rvert$	1	1	1	1

and all restriction maps of L are the unique maps between singletons. Thus $Z/L \cong Z$, while $L = e_Z$. The same data establish that Ψ is not totally $\mathbf{W}_\infty$-aspherical. Both arrows out of a fail to be monic, and the same pair λ, μ witnesses both failures. The word "universal" refers to the free category with these specified generators and relations; arbitrary failures of the two monomorphisms need not have a common witnessing pair.

[ms. 10]

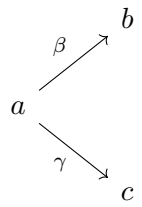
2.6. Contre-exemples pour la totale asphéricité.

Je me suis finalement convaincu que

$$\Psi = \left\{ \begin{array}{ccc} & & b_0 \\ & \nearrow & \\ a_0 & & \\ & \searrow & \\ & & c_0 \end{array} \right\}$$

n'est *pas* totalement asphérique (pour le localiseur $\mathbf{W}_\infty$ – dans le présent numéro, on supposera $W = \mathbf{W}_\infty$), ou ce qui revient au même, qu'on peut avoir un diagramme de type Ψ

$(*)$



dans une catégorie $Z \in \mathrm{Ob}\,\mathrm{Cat}$, dont la limite inductive $b \amalg_a c$ dans $Z^\wedge$ soit le préfaisceau final e_Z, i.e. tel que l'on ait

$$b(z) \amalg_{a(z)} c(z) \xrightarrow{\sim} \text{ensemble ponctuel}$$

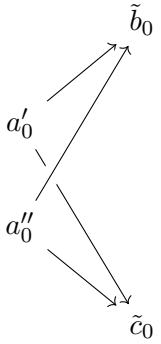
pour tout objet z de Z (où $a(z) = \mathrm{Hom}_Z(a, z)$ `[plutôt` $\mathrm{Hom}_Z(z, a)$`]`) sans que pour autant Z soit asphérique, ni même 1-connexe ([5]). L'exemple le plus simple que j'ai trouvé, est l'exemple 'universel' d'un diagramme $(*)$ où ni l'une ni l'autre des deux flèches $a \longrightarrow b$, $a \longrightarrow c$ n'est `[un]` monomorphisme (car si elle l'est, alors on peut prouver que Z est bel et bien asphérique, comme nous verrons par la suite). C'est la catégorie

$$Z = \left\{ \begin{array}{ccccc} & & & & b_0 \\ & & & \overset{\beta_0}{\nearrow} & \\ 0_0 & \underset{\mu_0}{\overset{\lambda_0}{\rightrightarrows}} & a_0 & & \\ & & & \underset{\gamma_0}{\searrow} & \\ & & & & c_0 \end{array} \right\}$$

déduite de Ψ en lui rajoutant un objet 0_0, relié à a_0 par deux flèches distinctes λ_0, μ_0, telles que [ms. 11]

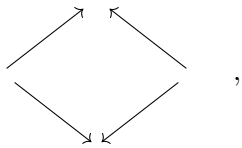
$$\beta_0\lambda_0 = \beta_0\mu_0 \qquad \text{et} \qquad \gamma_0\lambda_0 = \gamma_0\mu_0,$$

ces deux flèches composées étant, avec les quatre flèches β_0, γ_0, λ_0, μ_0 marquées sur le diagramme, les seules (six) flèches non identiques de Z. Ψ immergée dans Z comme un fermé, de complémentaire la catégorie ponctuelle réduite à 0_0. (NB L'immersion fermée $\Psi \longrightarrow Z$ n'est pas immersion de Dwyer, i.e. (comme Z est le plus petit ouvert de Z contenant Ψ) n'est pas coasphérique, car la catégorie $0\backslash\Psi$, cofibrée sur Ψ avec fibres discrètes, de cardinal 2 en a_0 et 1 en b_0 et c_0, est la catégorie



5. faux – il est 1-connexe forcément ([6]), mais non 2-connexe.
6. cf. commentaire XII, p. 101.

i.e.



,

qui a le type d'homotopie du cercle.) Ni l'immersion ouverte $\{0_0\} \longrightarrow Z$, car $\{0_0\}/a_0$ est la catégorie discrète de cardinalité 2.

Commentary · Exposition [ms. 11]

The link detects the failure of Dwyer's condition

Grothendieck is locating the obstruction where it will survive a change of ambient category. For the closed inclusion $\Psi \hookrightarrow Z$, the relevant link is $0\backslash\Psi$. Its four objects are the two arrows λ, μ to a and the unique arrows to b, c. Each of the first two objects maps to each of the last two, and there are no other nonidentity arrows. Its nerve is the square graph, hence its classifying space is S^1. Projection to Ψ is the discrete opfibration classified by the covariant functor $\mathrm{Hom}_Z(0,-)|_\Psi$, with fibre cardinalities $2, 1, 1$.

This link has neither an initial object nor a contractible nerve. The inclusion is therefore neither a closed Dwyer inclusion nor coaspherical. These are separate conclusions: Dwyer's condition asks for a left adjoint to the inclusion into the smallest open neighbourhood, equivalently for initial objects in the relevant comma categories; weak contractibility alone would be a weaker requirement. For the complementary open inclusion $\{0\} \hookrightarrow Z$, the slice over a is the discrete two-object category, so this inclusion also fails Dwyer's condition. The larger category can be simply connected while these local links retain nontrivial information. That local information, rather than the nonexistent retraction, will drive the finite pushout counterexample.

L'inclusion $i : Z_0 \longrightarrow Z$ de

$$Z_0 \;=\; (\; 0_0 \overset{\lambda_0}{\underset{\mu_0}{\rightrightarrows}} a_0 \;)$$

dans Z est un homotopisme ([7]) (car il existe une rétraction évidente r de Z sur Z_0 ([7]), et une flèche $ir \longrightarrow \mathrm{id}_Z$), donc Z est homotope à Z_0, qui a le type d'homotopie du cercle, donc n'est pas 1-connexe.

D'autre part, la vérification de [ms. 12]

$$b_0 \amalg_{a_0} c_0 \;=\; e_Z \qquad \text{dans } Z^\wedge$$

est immédiate.

On va utiliser cette flèche

$$\Psi \longrightarrow Z$$

pour construire des exemples intéressants (et déchirants) de sommes amalgamées

$$\begin{array}{ccc} U & \overset{i}{\hookrightarrow} & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle q} \\ U' & \overset{i'}{\hookrightarrow} & X', \end{array}$$

avec i immersion ouverte, telle qu'on ait même

$$i :\; U \longrightarrow \overline{U} \qquad \text{est asphérique, donc dans } W,$$

alors que pourtant on aura

$$i' :\; U' \longrightarrow \overline{U'} \qquad \notin W,$$

7. faux, il n'y a pas de rétraction du tout! En fait, Z est la *suspension* de la $\rightrightarrows$, qui a le type d'homotopie (topologique) du cercle, et il a donc le type d'homotopie de la 2-sphère.

et ce qui mieux (ou pis) est,

$$p \in W \qquad \text{mais} \qquad q \notin W$$

ed. 9

(et éventuellement, en même temps, $i \in W$ mais $i' \notin W$). De plus (contrairement aux constructions des propositions 10, 11 (p. 101-107)), les catégories envisagées pourront toutes être finies. Je suis particulièrement intéressé par le cas où X est un 'carré'

$$X = \Phi_n \times \Phi_m \qquad (n, m \geq 2),$$

et où U est son 'bord'

$$U = \dot{X} \overset{\text{déf}}{=} (\dot{\Phi}_n \times \Phi_m) \cup (\Phi_n \times \dot{\Phi}_m).$$

Commentary · Exposition [ms. 12]

The boundary and its categorical closure

Grothendieck wants to know whether ordinary subdivisions of a square provide attaching maps compatible with weak equivalences. The notation carries two geometries: the realization of the nerve is a genuine square, while the open and closed subsets of the category are determined by the directions of its arrows. Write

$$\Phi_n = (0 \longrightarrow 1 \longleftarrow 2 \longrightarrow \cdots \longrightarrow 2n-1 \longleftarrow 2n), \qquad \dot{\Phi}_n = \{0, 2n\}.$$

This is the convention in Chapter XII, manuscript p. 114 (Grothendieck, 1990–1991d); the commentary on directed intervals in Chapter VII, manuscript pp. 1–19, explains its relation to the underlying unoriented interval. The boundary U consists of the pairs with at least one coordinate an endpoint. It is a sieve, and

$$BX \cong [0,1]^2, \qquad BU \cong \partial[0,1]^2 \cong S^1.$$

In particular $U \to X$ is not a weak equivalence. The expression "immersion ouverte W-asphérique" in the proposition must be read in the local sense $U \to \overline{U}$: it is this induced functor to the categorical closure that is aspherical.

The slice calculation makes the distinction exact. Put

$$E_n = \{0, 2n\}, \qquad E_n^+ = \{0, 1, 2n-1, 2n\}.$$

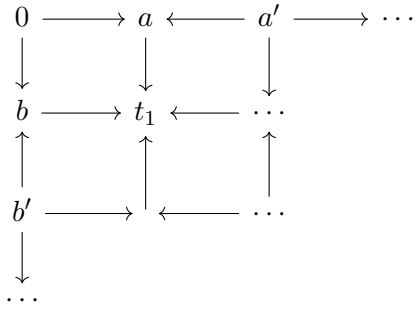
The cosieve generated by U is

$$\overline{U} = (E_n^+ \times \Phi_m) \cup (\Phi_n \times E_m^+).$$

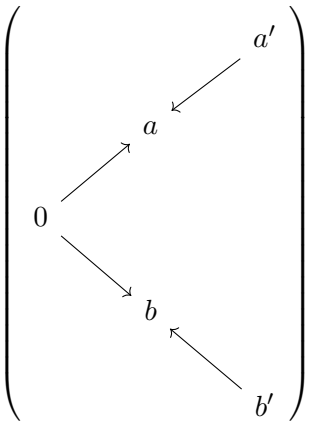
For $t \notin \overline{U}$, the slice U/t is empty. For $t \in U$ it has terminal object t. Suppose that $t \in \overline{U} \setminus U$. If exactly one coordinate is adjacent to an endpoint and the other is an interior even vertex, then U/t is a point. If that other coordinate is an odd vertex not adjacent to an endpoint, U/t is the three-object interval with a terminal middle vertex. Finally, when both coordinates are adjacent to endpoints, U/t is the five-object category $\widetilde{\Psi}$ drawn in the source. There are exactly four such t, because $n, m \geq 2$.

The nerve of $\widetilde{\Psi}$ is the path with successive vertices $a', a, 0, b, b'$. It is contractible, although neither a nor b receives a map from the other. Thus every slice over $\overline{U}$ is contractible, proving the claimed local asphericity, while the four exceptional slices have no terminal object. For a sieve of posets, Dwyer's condition is precisely the existence of these terminal objects over its generated cosieve: they assemble into the right adjoint to the inclusion. This proves both that $U \hookrightarrow X$ is not a Dwyer inclusion and that deleting the four exceptional vertices makes it one. Contractible slices suffice for a weak equivalence to the closure; an adjoint supplies the stronger structure that behaves well under arbitrary pushout.

[ms. 13]



Proposition. *Soient $n, m \geq 2$, et considérons l'inclusion $U = \partial X \overset{i}{\hookrightarrow} X$ du bord du 'carré' $X = \Phi_n \times \Phi_m$ dans X. C'est une immersion ouverte W-asphérique, de façon précise, les U/t ($t \in \dot{U} = \bar{U}\backslash U$) sont isomorphes à e (catégorie ponctuelle) ou à $\Phi =$*  *(catégorie avec élément final), sauf pour les quatre éléments t_i des quatre 'coins' de $\dot{U}$, où c'est la catégorie*

$\tilde{\Psi} =$ 

n'ayant pas d'élément final. Donc $U \overset{i}{\longrightarrow} X$ n'est pas immersion ouverte de Dwyer (mais $U \longrightarrow X\backslash\{t_1, t_2, t_3, t_4\}$ l'est …). Ceci posé, je dis que $i : U \longrightarrow X$ n'est pas une W-cofibration. ed. 10

DÉMONSTRATION. Soit

$$F \hookrightarrow U$$

l'inclusion dans $U = \partial X$ de la catégorie-coin $[V =] \begin{matrix} 0 \longrightarrow a \\ \downarrow \\ b \end{matrix}$ (cf. figure), isomorphe à Ψ. On voit aussitôt que c'est une immersion fermée de Dwyer (le plus petit ouvert de U contenant Ψ est justement la catégorie $\tilde{\Psi} = U/t_1$ de tantôt), donc une W-cofibration. Si $U \longrightarrow X$ était une W-cofibration, il en serait de même du composé $V \longrightarrow X$. On va voir que ce n'est pas le cas.

Soit $\overline{V} = \begin{matrix} 0 \longrightarrow a \\ \downarrow \quad\quad \downarrow \\ b \longrightarrow t_1 \end{matrix}$ l'adhérence de V dans X, et considérons l'inclusion [ms. 14]

$$\overline{V} \longrightarrow X,$$

je dis que [c]'est une immersion fermée de Dwyer, i.e. que pour tout $s \in W\backslash\overline{V}$, où W est le plus petit ouvert de X contenant $\overline{V}$, i.e.

$$W = \begin{matrix} 0 \longrightarrow a \longleftarrow a' \\ \downarrow \quad\quad \downarrow \quad\quad \downarrow \\ b \longrightarrow \downarrow \longleftarrow a'' \\ \uparrow \quad\quad \uparrow \quad\quad \uparrow \\ b' \longrightarrow b'' \longleftarrow c, \end{matrix}$$

la catégorie $s\backslash\overline{V}$ a un objet initial. (Les catégories $s\backslash\overline{V}$ qu'on obtient, à partir des cinq choix possibles pour s, sont $\mathbf{\Delta}^1$ (si $s = a'$ ou b') ou e (pour $s = a''$, b'' ou c)). Donc $\overline{V} \longrightarrow X$ est une W-cofibration.

Ceci posé, on sait qu'il existe, pour [?] l'inclusion (immersion ouverte)

$$V \hookrightarrow \overline{V} \qquad \text{i.e. } \Psi \hookrightarrow \mathbf{\Delta}^1 \times \mathbf{\Delta}^1,$$

un foncteur $V \longrightarrow V'$, donnant lieu à un carré cocartésien

$$(*) \qquad \begin{array}{ccc} V & \xrightarrow{i} & \overline{V} \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \overline{p}} \\ V' & \xrightarrow{i'} & \overline{V'} \end{array}$$

avec

$$i, p \in W \text{ (car } V, V', \overline{V} \text{ } W\text{-asphérique), mais } i', \overline{p} \notin W \text{ (i.e. } \overline{V'} \textit{ pas } W\text{-asphérique).}$$

ed. 11

Je rappelle qu'on prend d'abord

$$V'_0 = \left(0' \underset{\mu}{\overset{\lambda}{\rightrightarrows}} 0 \begin{array}{l} \overset{\alpha}{\nearrow} a \\ \underset{\beta}{\searrow} b \end{array} \right) \qquad \text{avec} \quad \begin{cases} \alpha\lambda = \alpha\mu \\ \beta\lambda = \beta\mu \end{cases}$$

et l'immersion fermée 'identique' de V dans V' [plutôt V'_0]

$$p_0 : V \longrightarrow V'_0$$

(qui est telle que [ms. 15]

$$p_{0!}(e_V) = e_{V'_0} \quad),$$

puis on prend pour V' la catégorie obtenue en ajoutant un objet final à V'_0

$$V' = \left(0' \underset{\mu}{\overset{\lambda}{\rightrightarrows}} 0 \begin{array}{ccc} & \overset{\alpha}{\nearrow} a & \overset{\rho}{\searrow} \\ & & e \\ & \underset{\beta}{\searrow} b & \underset{\sigma}{\nearrow} \end{array} \right) \qquad \text{avec} \quad \begin{cases} \alpha\lambda = \alpha\mu \\ \beta\lambda = \beta\mu \\ \rho\alpha = \sigma\beta \end{cases}$$

et pour p le foncteur composé

$$V \xrightarrow{p_0} V'_0 \xrightarrow{p_1} V', \qquad p$$

où p_1 est l'inclusion (ouverte). On trouve alors le carré $(*)$, avec $\overline{V'}$ non W-asphérique, ou ce qui revient au même, $i' \notin W$. Car comme $\overline{V'}\backslash V' \simeq \overline{V}\backslash V$ est la catégorie ponctuelle, $i' \in W$ équivaut à V'/t W-asphérique ([8]) (cf. cor. 4, p. 149 dans XII). Or par construction $V'/t \simeq V'_0$, qui n'est pas W-asphérique (dans le cas, disons, où $W \subseteq W_1$, car $\pi_1(V'_0) \simeq \mathbf{Z}$).

8. moyennant (W 7 bis).

Commentary · Examples [ms. 15]

The pushout is a three-sphere

The purpose of adjoining the new terminal object e is to make the replacement V' contractible without removing the obstruction in the link of the other cone point t. It is possible to calculate the entire pushout, so the argument need not depend on the converse excision principle mentioned in footnote 8. Rename V_0' as Z and the original three-object corner as V. Thus Z is the four-object category just analysed, $V' = Z^{\triangleright_e}$, and $\overline{V} = V^{\triangleright_t}$.

Let $Q = V' \amalg_V \overline{V}$. Its objects are the four objects of Z and two further objects e, t. It contains Z as a full subcategory. For each $z \in Z$ there is exactly one arrow $z \to e$ and exactly one arrow $z \to t$; there are no arrows between e and t. To check the only nontrivial assertion, the cocone over V supplies the maps from its three objects to t. The two composites from the extra object $0'$ to t agree, because the relation $\alpha\lambda = \alpha\mu$ already identifies their composites through a. Hence there is a unique arrow $0' \to t$. This description has the pushout universal property: a functor out of Q is exactly a functor out of Z together with one cocone over Z with vertex the image of e and one with vertex the image of t; the latter is uniquely determined by its restriction to V.

Every simplex of NQ lies in one of the two cones on NZ, and their intersection is NZ. Therefore

$$BQ \cong \Sigma BZ \cong S^3, \qquad \widetilde{H}_k(BQ; \mathbf{Z}) = \begin{cases} \mathbf{Z}, & k = 3, \\ 0, & k \neq 3. \end{cases}$$

Since BV, $B\overline{V}$, and BV' are contractible, the maps i and p are weak equivalences, but i' and $\overline{p}$ are not. This is a completely finite example: Q has six objects, fourteen nonidentity arrows, sixteen nondegenerate 2-simplices, and eight nondegenerate 3-simplices. Its nerve has no higher nondegenerate simplices.

The sentence $\pi_1(V_0') \cong \mathbf{Z}$ is present in the edition and must remain in the historical text. Its correct replacement in the argument is

$$\pi_1(V_0') = 0, \qquad \pi_2(V_0') \cong \mathbf{Z}, \qquad \pi_3(Q) \cong \mathbf{Z}.$$

The example proves the required assertion for $W = \mathbf{W}_\infty$. It does not prove the printed extension to every $W \subseteq W_1$: both $V' \to Q$ and $\overline{V} \to Q$ are equivalences on 2-types. Here W_k means equivalence on homotopy groups in degrees at most k and on components.

There is a second, logically independent issue. For an arbitrary open inclusion with one-point complement, membership in $\mathbf{W}_\infty$ does not force its comma category over that point to be contractible. The referenced Chapter XII corollary explicitly assumes the inverse part of $W(7\,\text{bis})$. The acyclic-link construction in the commentary on manuscript pp. 17–18 exhibits the failure of that inverse implication for spaces. In the present example both the link and the pushout are known spheres, so the conclusion follows from the displayed nerve calculation with no appeal to this extra axiom.

Considérons alors le diagramme de carrés cocartésiens [ms. 16]

$$\begin{array}{ccccc} V & \xhookrightarrow{i} & \overline{V} & \hookrightarrow & X \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle \overline{p}} & & \downarrow{\scriptstyle q} \\ V' & \xhookrightarrow{i'} & \overline{V'} & \hookrightarrow & X' \end{array} \quad .$$

ed. 12

Comme $\overline{V} \hookrightarrow X$ est [une] immersion fermée de Dwyer, le carré de droite est W-cocartésien, donc $(\overline{p} \in W) \iff (q \in W)$, donc (comme $\overline{p} \notin W$) on a

$$p \in W, \qquad \text{mais} \qquad q \notin W,$$

ce qui montre bien que $V \longrightarrow X$ n'est pas une W-cofibration. Factorisant $V \longrightarrow X$ par U (au lieu de $\overline{V}$), on trouve

$$\begin{array}{ccccc} V & \longrightarrow & U & \longrightarrow & X \\ {\scriptstyle p\in W}\downarrow & & \downarrow{\scriptstyle q_U \in W} & & \downarrow{\scriptstyle q \notin W} \\ V' & \longrightarrow & U' & \longrightarrow & X' \end{array} \quad ,$$

où $q_U \in W$ puisque $p \in W$ et $V \longrightarrow U$ est [une] immersion de Dwyer. Cela rend explicite que $U \longrightarrow X$ n'est pas une W-cofibration, puisque

$$q_U \in W, \qquad \text{mais} \qquad q \notin W,$$

q.e.d.

Commentary · Exposition [ms. 16]

Carrying the local obstruction to the boundary

The last step places the six-object defect inside the subdivided rectangle. Grothendieck has chosen the corner so that the two surrounding inclusions are closed Dwyer inclusions. We record their adjoints, because they explain exactly which pushouts preserve homotopy and complete the logical step in the proof.

Use coordinates $0 = (0,0)$, $a = (1,0)$, $b = (0,1)$, and $t_1 = (1,1)$. The smallest open neighbourhood of $V = \{0, a, b\}$ in U consists of these three points and $a' = (2,0), b' = (0,2)$. Its inclusion of V has the left adjoint fixing V and sending a' to a and b' to b. Indeed these images are the least points of V above the corresponding objects. The inclusion $V \hookrightarrow U$ is a cosieve, so it is a closed Dwyer inclusion.

Similarly $\overline{V} = \{0,1\} \times \{0,1\}$ is a cosieve of X. Its smallest open neighbourhood is $\{0,1,2\}^2$. Define $r_0(0) = 0$, $r_0(1) = r_0(2) = 1$; then $r_0 \times r_0$ is left adjoint to the inclusion of $\overline{V}$ into this neighbourhood. Thus $\overline{V} \hookrightarrow X$ is also a closed Dwyer inclusion. This gives the source's five comma categories explicitly: $a'\backslash\overline{V}$ and $b'\backslash\overline{V}$ are two-object chains, whereas the other three are points.

Thomason's pushout theorem, applied after taking opposite categories, now identifies the nerves of the relevant categorical pushouts with homotopy pushouts (Thomason, 1980, Proposition 4.3 and Corollary 4.4). One additional observation is needed for the reverse implication written in the source: $\overline{V} \to X$ is itself a weak equivalence, since both classifying spaces are contractible. Consequently its pushout $Q \to X'$ is a weak equivalence. In the commutative square

$$\begin{array}{ccc} \overline{V} & \xrightarrow{\sim} & X \\ {\scriptstyle \overline{p}}\downarrow & & \downarrow{\scriptstyle q} \\ Q & \xrightarrow{\sim} & X' \end{array}$$

two-out-of-three gives $\overline{p} \in \mathbf{W}_\infty$ if and only if $q \in \mathbf{W}_\infty$. Since $BQ \simeq S^3$, we obtain $BX' \simeq S^3$ and $q \notin \mathbf{W}_\infty$. A homotopy pushout by itself would not justify the converse: the two horizontal equivalences are the missing reason in this particular square.

Finally let $U' = U \amalg_V V'$. The other closed Dwyer inclusion, $V \to U$, ensures that $q_U : U \to U'$ is a weak equivalence because $p : V \to V'$ is one. Associativity of ordinary pushout identifies $X \amalg_U U'$ with the same X'. Thus the cobase-change functor along $U \to X$ takes the weak equivalence q_U to the map q which is not a weak equivalence. This proves exactly the definition of failure of a W-cofibration used in Chapter XII: the functor $U\backslash\mathrm{Cat} \to X\backslash\mathrm{Cat}$ fails to preserve weak equivalences. In modern terminology, this inclusion is not an h-cofibration. The target replacement still satisfies $BU' \simeq S^1$; the change in the ambient category is detected by its new third homology group.

Commentary · Later developments [ms. 16]

Dwyer maps, homotopy pushouts, and products

Grothendieck's distinction between contractible links and adjoint neighbourhoods remains useful in the modern theory. Thomason's original Dwyer condition guarantees that the comparison from a pushout of nerves to the nerve of a categorical pushout is a weak homotopy equivalence. It therefore controls the entire change-of-base operation, rather than merely the homotopy type of the initial inclusion. One should not identify Dwyer maps with all cofibrations in the Thomason model structure: Cisinski (1999, Lemma 4 and Proposition 2) corrected the claim that the original Dwyer class is closed under retracts. His larger class of pseudo-Dwyer maps retains the required pushout property.

The later result of Hackney et al. (2024, Theorems 1.5–1.6) strengthens the comparison to an equivalence of $(\infty,1)$-categories. Thus pushouts along Dwyer maps preserve more than the

classifying-space homotopy type. Our two uses of closed Dwyer maps are obtained by passing to opposites; the reversal changes the direction of the neighbourhood adjunction but does not change the classifying-space conclusion.

The negative product phenomenon also has a precise later form. Pavlov (2024, Definition 2.6) uses h-cofibration for the preservation condition Grothendieck calls W-cofibration. In the same paper, Proposition 4.3 proves that no model structure on Cat with Thomason weak equivalences can be cartesian. Its proof uses a different finite cube construction, credited in Proposition 4.2 to Maltsiniotis. The present appendix supplies a particularly transparent earlier obstruction: the boundary of a product of alternating intervals can have contractible links in its closure and still fail the preservation condition. These are results about strict categorical pushouts; homotopy pushouts of spaces retain their usual invariance.

Commentary · Development (commentators') [ms. 16]

The same defect on higher cubical boundaries

The rectangle argument is local enough to survive in every higher dimension. This gives a concrete continuation of Grothendieck's construction: the defect need not grow with the dimension of the ambient cube. The literature checked for this extension comprised Thomason's pushout theorem, Cisinski's correction, the Dwyer pushout comparison of Hackney–Ozornova–Riehl–Rovelli, and Pavlov's distinct cube obstruction. We did not locate the following precise family there; the proposition is an elementary extension of the appendix, without a claim of priority.

Proposition (one fixed obstruction in every dimension). [ours] Let $d \geq 2$, let $n_j \geq 2$, and put

$$X_d = \prod_{j=1}^{d} \Phi_{n_j}, \qquad U_d = \bigcup_{j=1}^{d} \Phi_{n_1} \times \cdots \times \dot{\Phi}_{n_j} \times \cdots \times \Phi_{n_d}.$$

There is a finite weak equivalence $U_d \to U'_d$ whose pushout along $U_d \hookrightarrow X_d$ has target X'_d with $BX'_d \simeq S^3$. In particular every inclusion $U_d \hookrightarrow X_d$ fails to be a $\mathbf{W}_\infty$-cofibration.

Proof. Fix the sink 1 in each of the last $d-2$ factors. Let V be the three-object full subposet with first two coordinates $(0,0), (1,0), (0,1)$ and all other coordinates 1. Let S add the point $(1,1,1,\ldots,1)$. Thus $V \cong \Psi$ and $S \cong [1] \times [1]$. Every point of V lies in U_d, while the fourth point of S does not. Since each coordinate 1 is maximal in its alternating interval, all successors of S remain in S; all successors of V that remain in U_d remain in V. Hence $S \hookrightarrow X_d$ and $V \hookrightarrow U_d$ are closed inclusions.

Their Dwyer structures can be given without induction. The smallest open neighbourhood of S is $\{0,1,2\}^d$. Map its first two coordinates by $r_0(0) = 0$, $r_0(1) = r_0(2) = 1$, and map every remaining coordinate to 1. This sends an object to the least element of S above it, and is therefore left adjoint to the inclusion of S. The smallest open neighbourhood of V in U_d is

$$\big(\{0,1,2\} \times \{0\} \;\cup\; \{0\} \times \{0,1,2\}\big) \times \{0,1,2\}^{d-2}.$$

The same map lands in V and is its left adjoint. Both inclusions are closed Dwyer inclusions.

Use the finite weak equivalence $p : V \to V' = Z^{\triangleright_e}$ constructed above. Put

$$U'_d = U_d \amalg_V V', \quad Q = S \amalg_V V', \quad X'_d = X_d \amalg_V V'.$$

The first Dwyer pushout gives $U_d \simeq U'_d$. Both BS and BX_d are contractible, so $S \to X_d$ is a weak equivalence as well as a closed Dwyer inclusion. Its pushout gives a weak equivalence $Q \to X'_d$. The calculation on manuscript p. 15 identifies $BQ \simeq S^3$, so $BX'_d \simeq S^3$. Finally associativity of pushouts gives $X'_d = X_d \amalg_{U_d} U'_d$. The resulting map from the contractible BX_d to BX'_d is not a weak equivalence, proving the assertion. All categories involved are finite, since the displayed pushouts are obtained by the same finite gluing constructions. □

Recollement dans Cat **d'un ouvert et d'un fermé complémentaire.** [ms. 17]

`[Supposons donné]`

$$X_0 \underbrace{\hookrightarrow}_{\text{ouvert}} X \underbrace{\hookleftarrow}_{\substack{\text{fermé}\\ \text{complé-}\\ \text{mentaire}}} X_1$$

dans Cat, [et soit]

$$\tilde{X}_{01} \subseteq \mathrm{Fl}(X)$$

[la] sous-catégorie pleine des flèches de source dans X_0, but dans X_1.

Commentary · Exposition [ms. 17]

The arrows which join the two pieces

Grothendieck now turns from unsuccessful pushouts to a decomposition which retains the attaching data. An open subcategory $A = X_0$ is a full sieve; its complementary closed subcategory $B = X_1$ is a full cosieve. Thus every object belongs to exactly one piece, and there are no arrows from B to A. The additional information in X is the functor

$$M : A^{\mathrm{op}} \times B \longrightarrow \mathrm{Ens}, \qquad M(a,b) = \mathrm{Hom}_X(a,b).$$

In the language of distributors, X is the collage of M; this description, and its simplicial counterpart, are given in Joyal (2008, §§7.1–7.2, pp. 309–311). The direction of these attaching arrows is the feature that permits the argument below.

Write $E = \widetilde{X}_{01}$. Its objects are arrows $f : a \to b$ with $a \in A$ and $b \in B$; a morphism $f \to f'$ is a commuting square

$$(u,v) : f \longrightarrow f', \qquad u : a \to a', \quad v : b \to b', \quad vf = f'u.$$

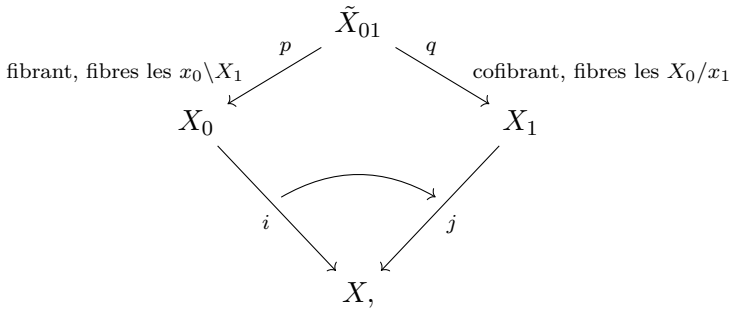
Both vertical sides point in their ordinary directions. In particular E is the ordinary crossing-arrow category, not a twisted-arrow category. The projection $p : E \to A$ is a split Grothendieck fibration: pullback along $u : a \to a'$ sends $f' : a' \to b$ to $f'u : a \to b$. Its fibre is $a\backslash B$. Dually, $q : E \to B$ is a split opfibration, with fibre A/b and transport given by postcomposition. The arrows f themselves constitute a natural transformation $ip \Rightarrow jq$.

After realization this natural transformation provides a specified homotopy between the two maps $BE \rightrightarrows BX$, where $BC = |NC|$. Consequently it defines a map from the double mapping cylinder of $BA \leftarrow BE \to BB$ to BX. This is the comparison whose weak equivalence is at issue. The ordinary square does not commute strictly, and replacing its specified homotopy by an arbitrary one would change the question.

The easy cases stated in the manuscript have a useful explanation. If i is aspherical, every A/b is contractible, hence q is a weak equivalence; the homotopy pushout then has the homotopy type of BA, which maps equivalently to BX. If j is coaspherical, the dual argument uses the fibres $a\backslash B$ of p. The general argument must retain these fibres rather than assume their contractibility.

Question. *Ce diagramme*

(∗)



ed. 13

est-il W-cocartésien?

OK si i W-asphérique ou j W-coasphérique, i.e. $X \longrightarrow \mathbf{\Delta}^1$ lisse ou propre – c'est un cas où on a soit $X_0 \xrightarrow{W} X$, soit $X_1 \xrightarrow{W} X$. Mais même en supposant i ou j dans W, et de plus X asphérique (donc X_0 et X asphériques, ou X_1 et X asphériques), et $W = \mathbf{W}_\infty$, je n'ai pu prouver que le diagramme est cocartésien, i.e. $q \in W$ resp. $p \in W$ ([9]). (Pas non plus si

9. Je suppose que W satisfait Loc (6).

$X_0 \simeq e$, X, X_1 W-asphérique $\Longrightarrow$ $e\backslash X$ W-asphérique, i.e. $X_1 \longrightarrow X$ W-coasphérique ...)
Mais j'ai prouvé du moins, dans tous les cas, que

$$(**) \qquad \mathcal{X} \overset{\text{déf}}{=} \int \tilde{X}_{01} \begin{array}{c} \nearrow X_0 \\ \searrow X_1 \end{array} \xrightarrow{\varphi} X$$

est dans $\mathbf{W}_1$.

Chose *fausse* (sauf si i ou j dans $W = \mathbf{W}_\infty$) si on remplace $\tilde{X}_{01}$ par $\tilde{X}'_{01}$, formée des triples (x_0, x_1, c), avec $x_0 \in \mathrm{Ob}\, X_0$, $x_1 \in \mathrm{Ob}\, X_1$, $c : x_0 \dashrightarrow x_1$ un *chemin* de x_0 à x_1 dans X! On a donc une situation 'topologique' tout à fait spéciale à la catégorie de modèles Cat, qui [ms. 18]
ne semble pas avoir d'analogue dans des catégories telles que Esp ou $\mathbf{\Delta}^\wedge$, où pourtant les notions d'ouvert et de fermé complémentaire (avec grain de sel dans $\mathbf{\Delta}^\wedge$) ont un sens.

Commentary · Development (commentators') [ms. 17]

Acyclic links and a false converse

The proposed $\mathbf{W}_\infty$ gluing formula and the opposite-side reflection invoked by the manuscript's "i.e." have different strengths. The formula is true. Footnote 9 adds Loc (6), a numbered hypothesis which requires qualification. If it uses the full reversible convention of Chapter XVI, Corollary 3.7, then cocartesianness together with $i \in W$ does imply $q \in W$, and dually for j, p: this is the additional condition Loc (6C) (Grothendieck, 1991c, §3, Corollary 3.7, ms. 26). Under that interpretation the conditional inference is valid, but its specialization to $\mathbf{W}_\infty$ is incompatible with the axiom. The following construction proves the failure for ordinary weak homotopy equivalences.

Proposition (a contractible union with a noncontractible link). [ours] There is a finite poset X, a one-object sieve $A = \{a\}$, and its complementary cosieve B, such that BA, BB, and BX are contractible, whereas $B\widetilde{X}_{01}$ is not contractible. In particular neither p nor q need be a weak equivalence, even when both i and j are weak equivalences. Moreover, $\mathbf{W}_\infty$ fails Loc (6b) in the convention of Chapter XVI.

Proof. We first choose a connected acyclic finite complex P with nontrivial fundamental group. A standard source of such complexes is Hatcher (2002, Example 2.38). For explicitness, take the presentation complex

$$\langle s, t \mid s^3(st)^{-2},\ t^5(st)^{-2} \rangle.$$

It has one vertex, two edges and two two-cells. Its cellular differential in positive degrees is

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\left(\begin{smallmatrix} 1 & -2 \\ -2 & 3 \end{smallmatrix}\right)} \mathbb{Z}^2 \longrightarrow 0.$$

The determinant is -1, so its reduced integral homology vanishes. The assignments $s \mapsto (1\,2\,3)$ and $t \mapsto (1\,4\,3\,2\,5)$ define a nontrivial homomorphism to A_5: their product is $(1\,4)(2\,5)$, and both relations hold. Thus P is not simply connected. By the finite form of the simplicial replacement theorem (Hatcher, 2002, Theorem 2C.5), choose a finite simplicial complex $L \simeq P$ and let K be its poset of nonempty faces. Its order complex is the barycentric subdivision of L, so $BK \simeq P$.

Adjoin two incomparable new elements a, b to K, each smaller than every element of K, and call the resulting poset X. Set $A = \{a\}$ and $B = \{b\} \cup K$. Then A is a sieve and B is its complementary cosieve; b is initial in B. The nerve of X is the union of two cones on NK, along NK itself. Hence

$$BX \cong \Sigma BK.$$

This suspension is simply connected because BK is connected, and its reduced homology vanishes by the suspension isomorphism. The simply connected homology form of Whitehead's theorem therefore makes it contractible (Hatcher, 2002, Corollary 4.33).

The crossing arrows are exactly the unique arrows $a \to k$, for $k \in K$. Their commuting squares are exactly the order relations of K. Thus $E \cong K$, $p : K \to \{a\}$, and $q : K \hookrightarrow \{b\} \cup K$. Neither map is a weak equivalence, since $\pi_1(BK)$ is nontrivial.

For the last assertion, consider the full sieves

$$U = (\{a\} \cup K)^{\mathrm{op}}, \qquad V = (\{b\} \cup K)^{\mathrm{op}} \quad \text{in } X^{\mathrm{op}}.$$

They cover X^{op} and intersect in K^{op}. Map their Mayer–Vietoris square to the square whose four vertices are the terminal category. The component maps from U, V, X^{op} are weak equivalences: their classifying spaces are two cones and the contractible suspension. The component from K^{op} is not. This violates precisely the inverse cube implication Loc (6b); the full axiom includes that implication (Grothendieck, 1991c, §3, ms. 16′–17′, footnote 13). □

For $\mathbf{W}_\infty$, this also answers negatively the subsidiary question with $X_0 \simeq e$ and X, X_1 aspherical. Its intended comma category is $a\backslash X_1 = K$; the printed $a\backslash X$ would contain the initial object id_a and would be automatically contractible. The example does not contradict the gluing conjecture: its homotopy pushout is exactly the contractible suspension ΣBK. It disproves opposite-side reflection for $\mathbf{W}_\infty$, while leaving the stronger conditional principle intact.

The axiom labels cannot be identified solely by their numbers. Chapter XV, §9, ms. 117, calls a fibration axiom Loc (6) (Grothendieck, 1990–1991k) and refers to Chapter XII, §IV, ms. 53–59; there, variant a′ is a pullback condition for W_ω-perfect functors (Grothendieck, 1990–1991f). Chapter XVI instead writes the forward Mayer–Vietoris condition as Loc (6′a), whereas the appendix's last line prints Loc (6a′) (Grothendieck, 1991c, §3, ms. 17′). We retain these distinct labels and cannot certify that footnote 9 invokes the later reversible convention. We checked the standard acyclic-complex construction and the homotopy-colimit sources cited below; we did not locate this precise finite-poset application there. The marker records our worked application, without a claim of priority.

L'argument catégorique quasi-trivial qui établit que $(**)$ est dans $\mathbf{W}_1$, i.e. induit une équivalence de catégories entre $\mathrm{Rev}(X)$ et

$$\mathrm{Rev}(\mathcal{X}) \;\simeq\; \mathrm{Rev}(X_0) \overset{2}{\times}_{\mathrm{Rev}(\tilde{X}_{01})} \mathrm{Rev}(X_1),$$

me semble devoir s'étendre aux dimensions supérieures, à coups de n-gerbes sur X – mais les fondements sur la théorie des n-gerbes manquent. (Et je n'ai pas eu le courage d'expliciter l'argument même pour $n = 2$ (où pourtant Giraud fournit ce qu'il faut) ([10]).) Aussi je conjecture que φ est tout au moins dans $\mathbf{W}_\infty$, i.e. que le diagramme $(*)$ est W-cocartésien si $W \supseteq \mathbf{W}_\infty$ (et satisfait Loc (6 a′)).

Commentary · Development (commentators') [ms. 18]

The open–closed gluing formula

The final conjecture $\varphi \in \mathbf{W}_\infty$ has an affirmative answer. Its proof comes from the fact that a simplex in NX crosses from A to B at most once. We give the precise specialization of Thomason's homotopy-colimit comparison that settles this question. We did not locate this exact formulation in the checked sources; the result below is presented as a consequence of an established theorem, without a claim of priority.

Proposition (directed gluing). [ours] For every small category X with a full sieve A and complementary full cosieve B, let E be its crossing-arrow category. The natural transformation $ip \Rightarrow jq$ induces a weak homotopy equivalence

$$\mathrm{hocolim}(BA \longleftarrow BE \longrightarrow BB) \longrightarrow BX.$$

Consequently the comparison $\varphi : \mathcal{X} \to X$ of the manuscript belongs to $\mathbf{W}_\infty$.

Proof. To keep track of the unique crossing, introduce the bisimplicial set

$$S_{r,s} = \left\{ a_0 \longrightarrow \cdots \longrightarrow a_r \xrightarrow{f} b_0 \longrightarrow \cdots \longrightarrow b_s \;\middle|\; a_i \in A,\ b_j \in B \right\}, \qquad r, s \geq 0.$$

Its faces compose or omit arrows on the indicated side, and its degeneracies insert identities. The last horizontal face precomposes f; the first vertical face postcomposes f. Thus S is the simplicial replacement of the contravariant functor

$$A^{\mathrm{op}} \longrightarrow \mathrm{Cat}, \qquad a \longmapsto a\backslash B.$$

10. Mais j'ai fait quand même une esquisse de vérification pour $n = 2$, qui me convainc qu'on a bien $\varphi \in \mathbf{W}_2$. Et je n'ai pas guère de doute qu'on a même $\varphi \in \mathbf{W}_\infty$.

Its Grothendieck construction, with cartesian variance, is exactly E: a morphism over $u : a \to a'$ is a map $f \to f'u$ in $a\backslash B$, namely a v with $vf = f'u$.

There is a comparison $\theta : \operatorname{diag} S \to NE$. On an n-simplex it gives the chain of crossing arrows

$$a_i \longrightarrow a_n \xrightarrow{f} b_0 \longrightarrow b_i, \qquad 0 \le i \le n,$$

whose successive morphisms have sides $a_i \to a_{i+1}$ and $b_i \to b_{i+1}$. This is Thomason's comparison for the preceding contravariant functor. It is a weak equivalence; the precise variance and this formula are verified in Cegarra (2020, §3.1, equation (9), and Theorem 3), which gives an independent proof of the comparison originally due to Thomason (1979, Theorem 1.2). Both endpoint maps commute with θ: they simply forget the b-chain or the a-chain.

It remains to identify what the mixed simplices assemble to. Let $R = \|S\|$ be the double geometric realization. It has maps $r_A : R \to BA$ and $r_B : R \to BB$ obtained by forgetting one side. The natural diagonal-to-double-realization homeomorphism identifies R with $|\operatorname{diag} S|$. Form

$$D(R) = \big(BA \amalg (R \times [0,1]) \amalg BB\big)/\big((z,0) \sim r_A(z),\ (z,1) \sim r_B(z)\big).$$

For a mixed simplex as above, send $(u, v, t) \in \Delta^r \times \Delta^s \times [0,1]$ to the point of Δ^{r+s+1} with coordinates

$$((1-t)u_0, \dots, (1-t)u_r, tv_0, \dots, tv_s).$$

This respects faces and degeneracies, and respects the endpoint identifications. It supplies a homeomorphism $D(R) \cong BX$. Indeed, in a point with positive total weight on both pieces, t is the total weight on the B vertices; dividing the two sets of coordinates by $1 - t$ and t recovers u, v. At $t = 0, 1$ precisely the identifications defining $D(R)$ occur. This description is compatible with the realization quotient topology because it is the usual simplex-join homeomorphism on every simplex. Pure simplices give the two endpoint spaces.

The double mapping cylinder computes a homotopy pushout. Replacing its middle space R by the weakly equivalent CW complex BE therefore preserves its homotopy type, by homotopy invariance of this construction. The resulting map to BX is the canonical one: the homotopy obtained from $ip \Rightarrow jq$, after composition with θ, is homotopic relative to its endpoints to the displayed barycentric homotopy. On each prism, both lie in the same mixed simplex, and linear interpolation supplies compatible homotopies on its faces. Thus the canonical comparison, and not merely an abstract map between the same homotopy types, is a weak equivalence.

Finally, let J be the span category with arrows $* \to 0$ and $* \to 1$, and take its covariant diagram with values E, A, B and structure maps p, q. Its Grothendieck construction models $\mathcal{X}$. A comparison to X sends its A- and B-objects to themselves and its E-object $f : a \to b$ to a; the arrows towards the B-part use f. Thomason's theorem identifies its classifying space with the preceding homotopy pushout, compatibly with this cocone. Hence $B\varphi$ is a weak equivalence. □

The cases $A = \varnothing$ or $B = \varnothing$ are included: then E is empty and the assertion reduces to the other piece. For every localizer W containing $\mathbf{W}_\infty$, the comparison just proved also lies in W. If the span Grothendieck construction represents the W-homotopy pushout, this proves W-cocartesianness as well. For a fundamental localizer, Chapter XVI explicitly supplies that interpretation under its forward axiom Loc (6′a) (Grothendieck, 1991c, §3, ms. 24, footnote 15). This comparison hypothesis requires no opposite-side reflection. The manuscript cites Loc (6a′) for its general-W interpretation; because of the numbering ambiguity just documented, we do not silently identify that label with the distinct forward or full axioms of Chapter XVI. The theorem for $\mathbf{W}_\infty$ is independent of that identification.

The proof also clarifies the preceding observation about a phenomenon peculiar to categories. Its realization is an explicit topological double mapping cylinder, and the mixed-simplex construction has a directed simplicial formulation. What a bare pair of complementary topological subsets does not supply is the specified category of attaching arrows.

Commentary · Examples [ms. 18]

Why replacing arrows by all paths changes the answer

Grothendieck's caution about paths concerns the attaching information, rather than an insufficiency of path spaces as models of homotopy types. A bridge of directed arrows records how the

two pieces were assembled. A bridge of all paths in the already assembled space usually records a homotopy pullback instead. These two bridges can differ even in the smallest example with a loop.

Let X have objects a, b and precisely two nonidentity arrows $\lambda, \mu : a \rightrightarrows b$. Take $A = \{a\}$ and $B = \{b\}$. There are no nonidentity commuting squares between its two crossing arrows, so E is the discrete two-point category. The nerve of X has two vertices and two edges and no nondegenerate higher simplices. Thus

$$BX \cong S^1, \qquad \operatorname{hocolim}(* \longleftarrow \{\lambda, \mu\} \longrightarrow *) \cong S^1.$$

Now replace the bridge by a model of the space of all paths from a to b in BX. Choosing one reference path identifies this space up to homotopy with ΩS^1, whose components are contractible and indexed by $\mathbb{Z}$. The new homotopy pushout is therefore the graph with two vertices and one edge e_k for each $k \in \mathbb{Z}$. Collapsing e_0 gives

$$\bigvee_{k \in \mathbb{Z} \setminus \{0\}} S^1.$$

The comparison to S^1 sends the loop represented by $e_k e_0^{-1}$ to the integer k. Its map on fundamental groups is the homomorphism from the free group on these generators to $\mathbb{Z}$ which sends the generator indexed by k to k. It is surjective but not injective; for example the word consisting of the generator indexed by 2 followed by the inverse square of the generator indexed by 1 lies in its kernel and is a nonempty reduced word. Thus even the $\mathbf{W}_1$ conclusion fails.

More generally, when a path category models $BA \times^h_{BX} BB$, the exception stated in the manuscript is valid: if $BA \to BX$ is a weak equivalence, its projection to BB is a weak equivalence, so the associated homotopy pushout maps equivalently to BX; the other case is dual. This is a sufficient condition, not a characterization of every exceptional case. The manuscript specifies the objects of $\widetilde{X}'_{01}$ but does not spell out its morphisms. Our computation concerns the usual homotopically correct all-paths interpretation; it does not assert an equivalence for an unspecified raw category of paths.

Commentary · Later developments [ms. 18]

From covering descent to higher local systems

The last paragraph identifies the right source of confidence in the conjecture: a local system can be reconstructed from its restrictions to the two pieces and its transport across every attaching arrow. We can make the argument for coverings completely explicit before passing to higher coefficients.

Use covariant transport to identify $\mathrm{Rev}(C)$ with the category of functors $C \to \mathrm{Ens}$ which send every arrow to a bijection. An object of

$$\mathrm{Rev}(A) \underset{\mathrm{Rev}(E)}{\overset{2}{\times}} \mathrm{Rev}(B)$$

is a pair L_A, L_B with an isomorphism $\alpha : p^* L_A \xrightarrow{\sim} q^* L_B$. Define L on the two full subcategories by L_A, L_B, and on a crossing arrow $f : a \to b$ by $L(f) = \alpha_f$. To check functoriality, there are only two mixed cases. The squares $(u, \mathrm{id}) : fu \to f$ and $(\mathrm{id}, v) : f \to vf$ in E give

$$\alpha_{fu} = \alpha_f L_A(u), \qquad \alpha_{vf} = L_B(v) \alpha_f.$$

They are exactly the two composition laws needed in X, since a composable chain cannot cross from A to B twice. All transports are bijective. A compatible pair of morphisms between such descent data gives a natural transformation of the reconstructed functors, and restriction is inverse to this construction. This proves the equivalence of covering categories in the manuscript; applying the same gluing description to the categorical mapping cylinder gives its displayed formula for $\mathrm{Rev}(\mathcal{X})$.

This argument also identifies the fundamental groupoid directly. Its generators are the arrows internal to the two pieces and the crossing arrows, now formally invertible; the equations just verified supply all mixed composition relations. Consequently the comparison φ induces an equivalence of fundamental groupoids. The source offers a sketch of this $\mathbf{W}_1$ assertion and reports a further sketch for $\mathbf{W}_2$; the preceding mixed-simplex proof establishes all degrees at once.

The contemporary interpretation of the proposed higher-gerbe argument uses local systems valued in spaces or in n-groupoids. If $\mathcal{S}$ denotes the ∞-category of spaces, define $\mathrm{Loc}(C) = \mathrm{Fun}(BC, \mathcal{S})$, viewing BC as its homotopy type. The proved pushout formula, by the universal property of a colimit, yields

$$\mathrm{Loc}(X) \simeq \mathrm{Loc}(A) \times^{h}_{\mathrm{Loc}(E)} \mathrm{Loc}(B).$$

One can restrict the values to n-truncated spaces to obtain every finite stage. Straightening gives the precise realization of space-valued coefficients, and higher descent formalizes such pullbacks of coefficient categories; see Lurie (2009, Theorem 2.2.1.2, Corollary 3.3.4.6, and Theorem 6.1.3.9). Giraud's theory supplied the stack and gerbe language for the first nonabelian stages (Giraud, 1971); the claim here is about locally constant higher coefficients, not an identification of all stacks with local systems. The categorical question of the appendix is therefore settled without requiring a separate construction and comparison of n-gerbes for each n.

Overview · XIIapp

Grothendieck's appendix to Chapter XII examines a difficulty that becomes visible as soon as one tries to use small categories as models for spaces: an ordinary categorical pushout can discard the homotopy information that a homotopy pushout must retain. The investigation begins with a proposed contractibility criterion, passes through explicit finite counterexamples, and ends with a conjectural gluing theorem. Its exploratory order matters. An unsuccessful cohomology calculation and an impossible retraction are part of the route to the examples; correcting them changes some intermediate claims without destroying the main obstruction. The electronic edition already marks several of these revisions (Grothendieck, 1990–1991g).

Throughout these pages $BC = |NC|$ denotes the classifying space of a small category. The localizer $\mathbf{W}_\infty$ consists of the functors inducing weak homotopy equivalences. The word "aspherical" means weakly contractible in this usage. A presheaf $F : C^{\mathrm{op}} \to \mathrm{Ens}$ has a homotopy type through its category of elements C/F. In particular, a representable presheaf is aspherical, since its category of elements is a slice C/c with a terminal object. These conventions connect the appendix to the integration of homotopy types described in *Pursuing Stacks*, §69, and to the adjoints-to-restriction viewpoint of the 1991 letter to Thomason (Grothendieck, 1983, 1991a).

The opening hypothesis asks for two weakly terminal objects b, c, meaning that every object admits a map to each, and for their categorical product $a = b \times c$. The source proves that BC is simply connected. Its proof is sound: in the groupoid completion, cancellation against the two projections identifies all maps from an object to a; those unique classes are compatible with every arrow and its inverse. The completion is therefore equivalent to the terminal groupoid. The proposed stronger conclusion, weak contractibility, is false, but the first example in the manuscript needs a different calculation from the one printed.

That example is the category X_G of the first two formal powers of an object with endomorphism monoid G:

$$\mathrm{Hom}_{X_G}(b^m, b^n) = \big(\{1, \dots, m\} \times G\big)^n, \qquad m, n \in \{1, 2\}.$$

It is an opfibration over the opposite of the full finite-set category on a singleton and a two-element set, with fibres the one-object categories of G and G^2. The resulting cohomological spectral sequence is legitimate. The provisional derived-limit formula used to evaluate it is not: the transposition of the two-element set and the common retraction of its two inclusions supply relations that the proposed two-term complex has omitted. Even a constant coefficient diagram contradicts the displayed cokernel formula.

The commentary gives an explicit projective resolution for the actual indexing category. It proves that BX_G is 2-connected for every monoid G, while

$$H^3(BX_{C_2}; \mathbb{F}_2) \cong \mathbb{F}_2.$$

Thus the two-object, twenty-six-arrow category remains a finite counterexample to the opening proposition. Its obstruction occurs one degree later than the manuscript asserts. The repaired computation is supplied as a commentators' development, with complete proofs and a separate normalized-nerve verification. The standard spectral-sequence and homotopy-colimit ingredients are identified explicitly (Cegarra, 2020; Thomason, 1979).

A second development supplies the degree-two counterexample promised by footnote 2 under the exact product hypotheses. For an integer $r \geq 1$, adjoin a new source d to the category of the first two powers for the trivial monoid, with r maps $d \to b$ and the corresponding r^2 maps $d \to b^2$. Composition selects coordinates. The product remains an actual categorical product, and its new source link retracts up to homotopy onto the poset of nonempty subsets of an r-element set of cardinality at most two. Its nerve is the subdivided complete graph.

The resulting three-object category Y_r satisfies

$$BY_r \simeq \bigvee^{(r-1)(r-2)/2} S^2.$$

For $r = 3$ this gives a twenty-one-arrow model of S^2. It verifies the footnote without confusing the opening product condition with the weaker presheaf condition used subsequently.

The persistent indexing shape is the span $\Psi = (b_0 \leftarrow a_0 \rightarrow c_0)$. For a span of representables in $C^\wedge$, its value at x is the set of components of a bipartite graph: the vertices are the maps $x \to b$ and $x \to c$, and the edges are the maps $x \to a$. A terminal ordinary pushout says that all these graphs are connected. It does not say that their cycles vanish. The commentary proves both useful consequences stated in the source: a terminal pushout forces simple connectivity of BC, and if either arrow out of a is monic, the presheaf pushout is aspherical. The latter follows because the graph components are then stars, and Quillen's Theorem A applies to their incidence categories (Quillen, 1973).

The four-object example Z makes the missing cycles visible. It has two parallel arrows $0 \rightrightarrows a$, followed by arrows $a \to b, c$ which equalize them. Its presheaf pushout $b \amalg_a c$ is terminal. Its nerve, however, is two cones on the parallel-arrow circle, and hence $BZ \cong S^2$. The proposed retraction onto the circle cannot be a functor: applying it to either equalizing relation would identify the two distinct arrows. Footnote 7 corrects the homotopy type, but the later statement $\pi_1(Z) \cong \mathbb{Z}$ repeats the earlier error. We retain both printed readings and explain their mathematical status.

Adding two incomparable cone vertices to Z gives the six-object category Q with $BQ \cong S^3$. This is the reliable finite obstruction in the later pushout square: its three input categories are contractible, while its ordinary categorical pushout is not. The assertion is established directly from the nerve, without an unrestricted converse to a comma-category theorem. Such a converse would be false: a connected acyclic noncontractible link can have contractible suspension. The appendix's extension to arbitrary $W \subseteq W_1$ therefore does not follow from this example; the obstruction for this pushout lies in degree three.

Grothendieck then inserts this local defect into the boundary of a product $\Phi_n \times \Phi_m$ of alternating directed intervals. The boundary has classifying space S^1, while the product is contractible. Its slices over its categorical closure are contractible, but four of them have no terminal object. Accordingly the boundary inclusion satisfies the local asphericity condition of Chapter XII and fails the stronger Dwyer neighbourhood-adjoint condition. Two surrounding closed Dwyer inclusions allow the six-object defect to survive the enlargement. The necessary pushout comparison is Thomason's; Cisinski's later correction distinguishes Dwyer maps from their retracts (Thomason, 1980; Cisinski, 1999). The resulting inclusion fails to preserve weak equivalences under cobase change. In contemporary terminology it is not an h-cofibration.

The same construction extends, with proof here, to the boundary of every product of at least two such intervals: fixing interior sinks in the remaining coordinates carries the same S^3 defect into higher dimensions. Later theory gives two complementary perspectives. Pushouts along Dwyer maps are preserved even as $(\infty, 1)$-categorical pushouts (Hackney et al., 2024); no model structure on Cat with Thomason weak equivalences can be cartesian (Pavlov, 2024, Proposition 4.3). These results explain the continuing significance of the appendix's distinction between contractible links and actual adjoint neighbourhoods.

The question whether total asphericity forces a terminal object has a simpler answer. Here "total" quantifies over every left Kan extension $f_!1_A$, a convention expressly distinguished in the edition from the total asphericity of *Pursuing Stacks*. Filtered categories satisfy this condition, because the category of elements of a filtered colimit of representables is filtered. Grothendieck already records the correction elsewhere in the manuscript (Grothendieck, 1990–1991b, Chapter II, Corollaries 2.17–2.18). Both $(\mathbb{N}, \leq)$ and the one-object category of an unsplit idempotent give examples without a terminal object. The surrounding classification

is discussed in the commentary on Chapter II, manuscript pages 35–36, and is not repeated here.

The closing conjecture concerns a different, homotopically correct gluing. If A is a full sieve of X, B its complementary full cosieve, and E the category of arrows from A to B with ordinary commuting squares, then

$$BX \simeq \operatorname{hocolim}(BA \longleftarrow BE \longrightarrow BB).$$

The comparison is defined by the natural transformation supplied by the crossing arrows. We prove the formula using mixed simplices of the nerve: each crosses from A to B once, and its barycentric coordinates split into a point on each side and an interval coordinate. Thomason's comparison identifies the resulting bisimplicial bridge with BE. This settles the $\mathbf{W}_\infty$ conjecture and supplies the whole hierarchy of local-system descent statements at once. The exact specialization is worked out here; the general comparison theorem predates the manuscript, and no claim about priority for this formulation is made.

Two qualifications are mathematically necessary. First, for $\mathbf{W}_\infty$, the true gluing formula does not imply the subsidiary assertion that one of its bridge projections must be a weak equivalence whenever a boundary inclusion is. The acyclic-link example disproves that implication even for finite posets with contractible pieces and union. A stronger reversible localizer axiom would supply the implication, but $\mathbf{W}_\infty$ does not satisfy that axiom. The numbered references in the appendix require caution: Chapters XV and XVI use different Loc (6) conventions, and the positions of the primes in their variants differ. The detailed commentary records the checked formulations without certifying which convention the appendix intended (Grothendieck, 1990–1991k, 1991c). Second, replacing crossing arrows by all paths in the completed space changes the bridge: for the parallel-arrow circle, two crossing arrows give S^1, whereas an all-paths bridge gives a countable wedge of circles. The source's covering-descent argument is fully justified by gluing locally constant set-valued functors across commuting squares; higher local systems give its contemporary extension (Lurie, 2009). The explicit directed attaching data are what make the final formula work.

Chapter XIII

Catégories de modèles (1)

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p. –

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XIII

Catégories de modèles (1)

p. –

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède. Les index, la liste des axiomes, le glossaire, et les références bibliographiques à la fin du chapitre ont été compilés par les éditeurs. Les numéros des pages dans les index sont ceux de l'original.
Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :
[e-mail address removed]

G. Maltsiniotis

1 Couples de Quillen (Φ, Ψ)

p. 1
[page 1]

Je veux d'abord donner une variante affaiblie de la notion de triple de Quillen ("model category" de Quillen). Ce ne sera pas une notion tout-à-fait autoduale – ainsi il y aura *deux* variantes affaiblies, duales l'une de l'autre.

Commentary · Exposition [ms. 1]

Two calculi behind a model structure

Grothendieck begins by separating the machinery that constructs replacements from the equivalences those replacements are meant to respect. This separation will make it possible to ask whether the homotopy theory belongs to the pair $(\mathcal{M}, W)$ itself. The first section therefore contains no weak equivalences: its subject is the elementary calculus of lifting and factorization.

For arrows $i : A \to B$ and $p : X \to Y$, write $i \,\square\, p$ when every commutative square with these vertical sides admits a diagonal $h : B \to X$, satisfying $hi = a$ and $ph = b$, where $a : A \to X$ and $b : B \to Y$ are its horizontal sides. No uniqueness is required. In current notation Grothendieck's Φ_* is $\Phi^{\square}$, and Ψ^* is ${}^{\square}\Psi$. Consequently

$$\Phi \subseteq {}^{\square}\Psi \quad \Longleftrightarrow \quad \Psi \subseteq \Phi^{\square}.$$

This is an order-reversing Galois connection between classes of arrows. The arrows in the two displayed boxes operate on classes, rather than on individual morphisms; the editors' correction is mathematically essential, while the missing parenthesis in that correction is merely typographical. For a large category we work relative to universes, or speak of classes instead of forming an unrestricted set of all its subclasses.

There are already two different closure notions in view. A class is *left lifting closed* when it has the form ${}^{\square}\Psi$. Later, a class will be enlarged by isomorphisms, pushouts, transfinite composites and retracts; this is its *weak saturation* in current model-category language. These operations agree for the closure generated by a set in a locally presentable category, by the small object argument. Their identification requires that theorem; it is not a consequence of the Galois connection alone. Conversely, Grothendieck's phrase "W faiblement saturée" on manuscript p. 7 means two-out-of-three together with the isomorphisms. Keeping these uses separate prevents a genuine ambiguity in the conjectural argument below.

Mais avant encore, un sorite sur les LLP (left lifting property) et RLP (right lifting property) ([1]). Soit $\mathcal{M}$ une catégorie quelconque (le plus souvent, $\mathcal{M}$ sera grande). On considère des parties

$$\Phi,\ \Psi\ \subset\ \mathrm{Fl}(\mathcal{M})$$

et on définit des applications

$$\boxed{\begin{array}{ccc} \Phi & \longmapsto & \Phi_* \\ \Psi & \longmapsto & \Psi^* \end{array}} \qquad \mathrm{Fl}(\mathcal{M}) \longrightarrow \mathrm{Fl}(\mathcal{M})$$

[plutôt $\mathfrak{P}(\mathrm{Fl}(\mathcal{M}) \longrightarrow \mathfrak{P}(\mathrm{Fl}(\mathcal{M}))$,
où pour tout ensemble E, $\mathfrak{P}(E)$ désigne l'ensemble des parties de E] par

$$\begin{aligned} \Phi_* &= \{\text{ensemble des } p \in \mathrm{Fl}(\mathcal{M}) \text{ qui ont la RLP par rapport à } \Phi\} \\ \Psi^* &= \{\text{ensemble des } i \in \mathrm{Fl}(\mathcal{M}) \text{ qui ont la LLP par rapport à } \Psi\}, \end{aligned}$$

et ensuite

$$\boxed{\begin{array}{ccc} \Phi & \longmapsto & \widetilde{\Phi} \\ \Psi & \longmapsto & \overline{\Psi} \end{array}} \qquad \mathrm{Fl}(\mathcal{M}) \longrightarrow \mathrm{Fl}(\mathcal{M})$$

1. donner définition en forme, avec renvoi à Quillen.

[plutôt $\mathfrak{P}(\mathrm{Fl}(\mathcal{M}) \longrightarrow \mathfrak{P}(\mathrm{Fl}(\mathcal{M}))$] par

$$\begin{aligned} \widetilde{\Phi} &= (\Phi_*)^* \\ \overline{\Psi} &= (\Psi^*)_* \ . \end{aligned}$$

Je résume les propriétés les plus immédiates de ces opérations dans une [page 2]
Proposition 1.

a) $\Phi \longmapsto \Phi_*$, $\Psi \longmapsto \Psi^*$ *sont des applications décroissantes de* $\mathrm{Fl}(\mathcal{M})$ [plutôt $\mathfrak{P}(\mathrm{Fl}(\mathcal{M}))$] *dans lui-même, pour la relation d'ordre* $\subset$, *et* $\Phi \longmapsto \widetilde{\Phi}$, $\Psi \longmapsto \overline{\Psi}$ *des applications croissantes.*

p. 2

b) *On a*

$$\Phi \ \subset \ \widetilde{\Phi},$$

et pour qu'on ait $\Phi = \widetilde{\Phi}$, *il faut et il suffit que* Φ *soit de la forme* Ψ^*. *On dit alors que* Φ *est* clos à gauche ([2]). *L'ensemble des parties de* $\mathrm{Fl}(\mathcal{M})$ *closes à gauche est stable par intersections quelconques* ([3]), *et* $\widetilde{\Phi}$ *est la plus petite partie close de* $\mathrm{Fl}(\mathcal{M})$ *qui contienne* Φ *et la plus grande partie* Φ' *de* $\mathrm{Fl}(\mathcal{M})$ *telle que* $\Phi'_* = \Phi_*$ *(ou telle que* $\Phi'_* \supset \Phi_*$*).*
Énoncés symétriques pour Ψ *: on a*

$$\Psi \ \subset \ \overline{\Psi},$$

on a $\Psi = \overline{\Psi}$ *si et seulement si* Ψ [est] *de la forme* Φ_*, *on dit alors que* Ψ *est* clos à droite. *L'ensemble des parties de* $\mathrm{Fl}(\mathcal{M})$ *closes à droite est stable par intersections quelconques* ([4]), *et* $\overline{\Psi}$ *est la plus petite partie close à droite contenant* Ψ, *et aussi la plus grande partie* Ψ' *de* $\mathrm{Fl}(\mathcal{M})$ *telle que* $\Psi'^* = \Psi^*$ *(ou telle que* $\Psi'^* \supset \Psi^*$*).*

[page 3]

c) *Les applications*

$$\Phi \ \longmapsto \ \Phi_*, \qquad \Psi \ \longmapsto \ \Psi^*$$

établissent des bijections inverses l'une de l'autre entre l'ensemble des parties Φ *closes à gauche de* $\mathrm{Fl}(\mathcal{M})$, *et l'ensemble des parties* Ψ *closes à droite. C'est un anti-isomorphisme d'ensembles ordonnés. On a alors*

$$(\Phi \cap \Phi')_* \ = \ \begin{array}{l} \textit{partie close à droite de } \mathrm{Fl}(\mathcal{M}) \textit{ “engendrée par } \Phi_*,\ \Phi'_* \textit{ ”,} \\ \text{i.e. [la] } \textit{plus petite partie close à droite contenant } \Phi_*, \\ \Phi'_*, \text{ i.e. } \overline{\Phi_* \cup \Phi'_*} \end{array}$$

[pour Φ, Φ' parties closes à gauche], *plus généralement*

$$(\bigcap_i \Phi_i)_* \ = \ \overline{\bigcup_i (\Phi_{i\,*})}$$

pour toute famille $(\Phi_i)_{i\in I}$ *de parties closes à gauche de* $\mathrm{Fl}(\mathcal{M})$. *Formules duales* [pour des parties closes à droite]

$$\begin{cases} (\Psi \cap \Psi')^* \ = \ \widetilde{\Psi^* \cup \Psi'^*} \\ (\bigcap_j \Psi_j)^* \ = \ \widetilde{\bigcup_j (\Psi_j^*)} \ . \end{cases}$$

d) *Si* $\Phi, \Psi \subset \mathrm{Fl}(\mathcal{M})$, *alors on a*

$$\Phi \subset \Psi^* \iff \Psi \subset \Phi_*, \qquad \textit{on écrit alors } \Phi \longleftrightarrow \Psi,$$

p. 3

cela signifie que Φ *a la LLP pour* Ψ, *ou que* Ψ *a la RLP pour* Φ. [page 4]

Commentary · Exposition [ms. 3]

Closure, transfinite lifting and the retract argument

Grothendieck wants the formal part of Quillen's method to be available before choosing any examples. Proposition 1 provides exactly this preparation. To see its content directly, let $L(R) = {}^{\square}R$ and $R(L) = L^{\square}$. Antitonicity follows by adding lifting problems. Every arrow of L lifts against every arrow of $R(L)$, so $L \subseteq L(R(L))$. Applying antitonicity gives $R(L(R(L))) \subseteq R(L)$, while the reverse inclusion is the same elementary observation with the two roles exchanged. Thus $RLR = R$ and $LRL = L$. These identities prove idempotence, describe the fixed points and show that the two fixed-point posets are oppositely isomorphic. For instance,

$$R\left(\bigcap_i L_i\right) = \overline{\bigcup_i R(L_i)}$$

when each L_i is left lifting closed. The hypothesis matters: the right side is the join in the lattice of right lifting closed classes, not their ordinary union.

The stability assertions in Proposition 2 express how to transport a chosen solution. Isomorphisms lift by inversion. A lifting problem for a composite is solved successively. A lifting problem for a pushout is equivalent, by its universal property, to one for the original map. For a continuous ordinal chain, begin with the prescribed map on its initial object, extend at each successor, and take the unique induced map at each limit. Transfinite recursion then supplies the diagonal against the composite. The existence of the colimits appearing in the chain is the only colimit hypothesis used here; no presentability has yet entered.

The retract assertion deserves an explicit formulation, since it is the bridge from a constructed factorization to a lifting characterization. Suppose $f : A \to B$ factors as $A \xrightarrow{i} E \xrightarrow{p} B$, and suppose $f \,\square\, p$. Apply this lifting property to the square with top i and bottom 1_B. The resulting $s : B \to E$ satisfies $sf = i$ and $ps = 1_B$. In the arrow category, the morphisms $(1_A, s) : f \to i$ and $(1_A, p) : i \to f$ have composite 1_f. Hence f is a retract of i. In particular, a factorizing pair (L, R) with $L \,\square\, R$ becomes a weak factorization system as soon as both classes are retract closed. This is the content of the standard retract argument; compare Riehl (2014, Lemmas 11.1.4 and 11.2.3). It is this system of two classes of arrows that the manuscript calls a *couple de Quillen*. A modern *Quillen adjunction* is a different notion, involving two model categories and an adjoint pair of functors.

Proposition 2. *Soit* Φ *une partie close à gauche de* $\mathrm{Fl}(\mathcal{M})$. *Alors* Φ *satisfait les conditions* a) b) c) *ci-dessous :*

a) Φ *contient les isomorphismes, et est stable par composition, et par cochangement de base (*i.e. *si on a* $\begin{array}{ccc} A & \xrightarrow{i} & B \\ \downarrow & & \downarrow \\ A' & \xrightarrow{i'} & B' \end{array}$ *carré cocartésien dans* $\mathcal{M}$, *avec* $i \in \Phi$, *on a* $i' \in \Phi$*).*

b) Φ *stable par facteurs directs (en tant que partie de l'ensemble des objets de* $\underline{\mathrm{Fl}}(\mathcal{M})$*).*

c) Φ *"stable par limites inductives ordinales", dans le sens suivant : Soit* I *un ensemble bien ordonné,* $(A_i)_{i\in I}$ *un système inductif dans* $\mathcal{M}$, *tel que les* $u_{ji} : A_i \longrightarrow A_j$ *soient dans* Φ, *et* $A_{i_0} \xleftarrow{\sim} \varinjlim_{i<i_0} A_i$ *si* $i_0 \in I$ *est un élément limite. Supposons que* $A = \varinjlim_I A_i$ *existe. Alors* $\forall\ i_0 \in I$, *la flèche canonique* $A_{i_0} \longrightarrow A$ *est dans* Φ.

Il y a un énoncé dual pour une partie Ψ close à droite; elle contient les isomorphismes, est stable par composition et par changement de base, par facteurs directs, et enfin stable par limites projectives ordinales.

On aura une réciproque partielle dans les catégories accessibles. C'est pour l'énoncé de [page 5] cette réciproque que j'ai mis à part la condition *b*) de stabilité par facteurs directs, au lieu de la bloquer avec *a*).

2. Donner définition en forme.
3. de façon précise, $\bigcap_i(\Psi_i^*) = (\bigcup_i \Psi_i)^*$
4. de façon précise, on a $\bigcap_i(\Phi_{i*}) = (\bigcup_i \Phi_i)_*$

Définition 1.

a) (Pour mémoire.) RLP, LLP, la relation $\Phi \longleftrightarrow \Psi$ entre parties de $\mathrm{Fl}(\mathcal{M})$ (relation non symétrique!).

b) (Pour mémoire.) Partie de $\mathrm{Fl}(\mathcal{M})$ close à gauche, close à droite.

c) Soit $\Phi \longleftrightarrow \Psi$. On dit que le couple (Φ, Ψ) a la *propriété de factorisation de Quillen*, si *toute* flèche f de $\mathcal{M}$ se factorise en

$$f = pi, \qquad i \in \Phi,\ p \in \Psi.$$

On dit que (Φ, Ψ) est un *couple de Quillen* si c'est un couple associé par `[les]` propriétés LLP et RLP ($\Phi \longleftrightarrow \Psi$), et `[si]` elles `[les parties` Φ `et` Ψ`]` sont closes (*i.e.* $\Phi = \Psi^*$, $\Psi = \Phi_*$), et si elles ont la propriété de factorisation de Quillen.

J'énonce dès maintenant le résultat technique essentiel, assurant l'existence de beaucoup de couples de Quillen :

Commentary · Examples [ms. 5]

A weak factorization system in sets

The elementary example of injections and surjections makes the preceding abstraction concrete. We work in sets with the axiom of choice. Every injection $i : A \hookrightarrow B$ lifts against every surjection $p : X \twoheadrightarrow Y$: retain the given map on A, and choose a preimage in X of the prescribed image of each point of $B \setminus A$. Conversely, lifting against $\varnothing \to \{*\}$ characterizes surjections. If a map i has the left lifting property against all surjections, factor it as

$$A \xrightarrow{\iota} A \amalg B \xrightarrow{[i,1_B]} B.$$

The second arrow is surjective, so the retract argument makes i a retract of the injection ι, hence an injection. Thus $(\mathrm{Mono}, \mathrm{Surj})$ is a weak factorization system, generated by the single arrow $\varnothing \to \{*\}$.

The factorization displayed above is functorial: a square $(u, v) : i \to i'$ acts on its middle object by $u \amalg v$. The diagonal in an arbitrary lifting problem need not be unique or functorially chosen. This is why functorial factorization and a specified coherent family of fillers are additional kinds of structure, even though both are useful in later developments. Nor should this factorization be confused with the familiar image factorization, whose left class is surjections and whose right class is injections. The same two classes occur in opposite order, with a different lifting relation. The asymmetry noted in Definition 1(a) is already visible in this example.

Théorème 1. *Supposons que* $\mathcal{M}$ *soit* accessible, *et stable par (petites) limites inductives. (Il doit suffire stabilité par sommes amalgamées, et par* $\varinjlim$ *filtrantes, voire ordinales …). Soit* Φ_0 *une* petite *partie de* $\mathrm{Fl}(\mathcal{M})$, $\Psi = (\Phi_0)_*$, *et* Φ *une partie de* $\mathrm{Fl}(\mathcal{M})$ *telle que* p. 4 [page 6]

$$\Phi_0 \ \subset\ \Phi\ \subset \Psi^*\ (= \widetilde{\Phi}_0).$$

Alors :

a) *Pour que* Φ *soit close à gauche,* i.e. $\Phi = \widetilde{\Phi}_0$, *il faut et il suffit qu'elle satisfasse les conditions* a) b) c) *de la proposition 2. Quand il en est ainsi, le couple* (Φ, Ψ) *est un couple de Quillen.*

b) *Supposons seulement que* Φ *satisfasse aux conditions* a), c) *de la proposition 2 (à l'exclusion de la condition* b) *des facteurs directs) – donc on ne peut affirmer que* Φ *soit close à gauche. Néanmoins,* $(\Phi, \underbrace{\Psi}_{=\Phi_*})$ *satisfait à la propriété de factorisation de Quillen.*

Commentary · Exposition [ms. 6]

The small object theorem and its precise input

Theorem 1 is sound under its stated cocompleteness and accessibility hypotheses, understood in the usual locally small sense: an accessible cocomplete category is locally presentable. Its proof explains why the small generating family and the separate retract condition have been singled out. Let $I = \Phi_0$, choose a regular cardinal κ such that the domains of all arrows in I are κ-presentable, and fix $f : X \to Y$. Start with $X_0 = X$. At stage α, take every square

$$\begin{array}{ccc} A & \longrightarrow & X_\alpha \\ {\scriptstyle j}\downarrow & & \downarrow \\ B & \longrightarrow & Y \end{array} \qquad (j \in I)$$

and attach one copy of B along its specified map from A. This is a set of squares because I is small and the category is locally small. Their coproduct and pushout give $X_\alpha \to X_{\alpha+1}$ over Y. At limit stages take colimits. After κ stages put $E = \operatorname{colim}_{\alpha<\kappa} X_\alpha$.

Every map $A \to E$ with A a chosen domain factors through some X_α, since the ordinal κ is κ-filtered. A square $j \to (E \to Y)$ is therefore one of the squares attached at the next stage; its attaching cell supplies a filler into E. Hence $E \to Y$ belongs to $I^{\square} = \Psi$. The arrow $X \to E$ is an I-cell complex. The construction is functorial if every square is attached, without making choices of representatives depending on f.

For the theorem's slightly unusual formulation, coproducts of attaching maps cause no problem. A simultaneous attachment indexed by a set can be replaced, after well ordering that set, by successive pushouts. Thus closure under pushouts and continuous transfinite composition already puts every I-cell complex in Φ. This proves part (b). Moreover

$$I \subseteq \Phi \subseteq {}^{\square}(I^{\square}) \quad \Longrightarrow \quad \Phi^{\square} = I^{\square}.$$

If Φ is also retract closed, any $g \in {}^{\square}(I^{\square})$ is a retract of the left factor of its cellular factorization. Consequently $g \in \Phi$ and $\Phi = {}^{\square}(I^{\square})$, proving part (a). This proof supplies the theorem only stated here; a modern account is Riehl (2014, Theorem 12.2.2 and Corollary 12.2.4).

The parenthetical weakening of cocompleteness can also be made precise. For this construction it suffices to have the needed pushouts, ordinal colimits, local smallness and one regular cardinal giving the required smallness of the generating domains relative to the cellular chains. Pushouts together with ordinal colimits construct each set of attachments successively from the existing object X_α, so a separately postulated empty coproduct is unnecessary. What cannot be dropped is the smallness condition guaranteeing that a lifting problem has appeared before the construction ends.

Remarque. Dans la démonstration, l'existence de $\Phi_0 \subset \Phi$, qui joue le rôle d'un petit ensemble de "générateurs" (à gauche) pour Φ, joue un rôle essentiel. Il y aurait lieu de développer la notion de *parties accessibles* de $\mathrm{Fl}(\mathcal{M})$, qui impliquerait l'existence d'un petit [page 7] ensemble $\Phi_0 \subset \Phi$ tel que $\Phi \subset \widetilde{\Phi}_0$, et qui aurait la propriété que

$$\begin{aligned} \Phi \text{ accessible} \quad &\Longrightarrow \quad \Phi_* \text{ accessible} \\ \Psi \text{ accessible} \quad &\Longrightarrow \quad \Psi^* \text{ accessible.} \end{aligned}$$

Tout couple (Φ, Ψ) de parties closes associées ($\Phi = \Psi^*$, $\Psi = \Phi_*$) et accessible, serait donc un couple de Quillen.

Commentary · Examples [ms. 7]

Accessibility does not supply generating arrows

Grothendieck's proposed notion of an accessible class is meant to make the small object theorem reusable without displaying its generators. Modern terminology separates three properties: accessibility of a full subcategory of the arrow category, accessibility of a chosen factorization functor, and generation of its right lifting class by a set of arrows. The following known example shows why these cannot be substituted for one another; see Rosický (2021, Introduction and Examples 3.7).

Let $\mathcal{M} = \mathrm{Pos}$, and let L consist of the order embeddings. This is a left lifting closed class. Indeed, a monotone map from a subposet $A \subseteq B$ to the two-element chain $2 = \{0 < 1\}$ extends to B by taking the upward closure of the inverse image of 1. Thus every order embedding lifts against $2 \to 1$. Conversely, if $u : A \to B$ is not an order embedding, there exist $a, a' \in A$ with $u(a) \leq u(a')$ but $a \not\leq a'$. This also detects failure of injectivity by choosing the appropriate ordered pair of distinct points with the same image. The characteristic function of the principal upper set $\{x \mid a \leq x\}$ takes a to 1 and a' to 0, and cannot extend over u. Hence

$$L = {}^{\square}\{2 \to 1\}.$$

One can even construct the associated factorization. Embed A into the complete lattice $2^{\mathrm{Up}(A)}$ by recording membership in all its upper sets, and factor $f : A \to B$ through its graph in $B \times 2^{\mathrm{Up}(A)}$. Projection to B lifts against order embeddings, because a monotone map into a complete lattice extends by taking joins of the images of elements below the new point. Thus $(L, L^{\square})$ is a weak factorization system.

As a full subcategory of $\mathrm{Pos}^{[1]}$, L is accessible and accessibly embedded. Filtered colimits preserve order reflection: any relation between images is witnessed at a finite stage, where it already reflects. Every embedding is a filtered union of its finite subembeddings; such finite arrows are finitely presentable in the arrow category and remain so in this full subcategory.

Nevertheless no set $I \subseteq L$ satisfies $I^{\square} = L^{\square}$. Choose an infinite regular κ larger than the cardinalities of all domains and codomains in I, and let P be the poset of subsets of κ of cardinality less than κ. A map $h : A \to P$ extends along every $i : A \hookrightarrow B$ in I by

$$\bar{h}(b) = \bigcup_{i(a) \leq b} h(a).$$

Regularity makes this union an element of P, and order reflection gives $\bar{h}(i(a)) = h(a)$. Therefore $P \to 1$ belongs to $I^{\square}$. Now embed the discrete poset on κ into the poset obtained by adjoining a top element. The map $a \mapsto \{a\}$ cannot extend to that top, since its image would have to contain all of κ. Thus $P \to 1 \notin L^{\square}$. The proposed generating-set conclusion fails even for an accessible, left lifting closed class having factorizations. This does not refute the possibility of defining the stronger notion Grothendieck wants; it identifies what such a definition would have to add.

2 Catégories de Quillen faibles, et construction d'icelles (Théorème-scholie conjectural)

Dans toute la suite, on se donne une *catégorie de modèles*

$$\begin{cases} (\mathcal{M}, W), \text{ où } W \subset \mathrm{Fl}(\mathcal{M}) \\ \\ W \textit{ faiblement saturée } ({}^{5}) : \begin{array}{ll} a) & W \text{ contient les isomorphismes.} \\ b) & \text{Si deux parmi } f,\ g,\ gf \texttt{ [sont] dans } W, \\ & \text{le troisième aussi.} \end{array} \end{cases}$$

On veut décrire des données supplémentaires un peu plus faibles que celles d'une catégorie de modèles de Quillen. On suppose données des parties p. 5

$$C,\ F,\ TC,\ TF\ \subset\ \mathrm{Fl}(\mathcal{M}),$$

avec les propriétés suivantes. Comme axiome préliminaire, je pose

M 0 Stabilité de $\mathcal{M}$ par $\varprojlim$ finies et $\varprojlim$ [plutôt $\varinjlim$] finies.

M 1 Propriété de relèvement-prolongement

$$C \longleftrightarrow TF, \qquad TC \longleftrightarrow F. \qquad \text{[page 8]}$$

5. C'est l'axiome M 5 de Quillen.

M 2 Propriété de factorisation pour (C, TF), (TC, F).

M 3 Stabilité de F par composition et par changement de base, de C par composition et cochangement de base, F et C contiennent les isomorphismes.

M 4 Stabilité de TF par composition et changement de base, de TC par composition et cochangement de base, et TF et TC contiennent les isomorphismes.

M′ $TC \subset C \cap W$, $TF \subset F \cap W$.

M″ $C \subset \underbrace{\mathrm{Cof}_W}_{W\text{-cofibrations}}$, $F \subset \underbrace{\mathrm{Fib}_W}_{W\text{-fibrations}}$.

[Un morphisme $p : X \longrightarrow Y$ de $\mathcal{M}$ est une W-fibration si pour tout diagramme de carrés cartésiens de la forme :

$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow{t} & X'' \\ {\scriptstyle p}\downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \xleftarrow[s]{} & Y'' , \end{array}$$

si $s \in W$, alors $t \in W$. La notion de W-cofibration est la notion duale.]

Dans le cas de Quillen (Q-*catégories de modèles*), M′ est renforcé en

$$TC = C \cap W, \qquad TF = F \cap W,$$

et il n'y a pas d'axiome M″. L'axiome M″ équivaut à la propriété de "propriety", correspondant aux Q-catégories de modèles *propres.*

Commentary · Exposition [ms. 8]

Relative equivalences and properness

The additional four classes serve two purposes: they furnish replacements and make ordinary base change behave homotopically. In modern language $(\mathcal{M}, W)$ is first a relative category with two-out-of-three. It is not yet a Quillen model category. Axiom M 2 supplies both factorizations, and M 1 supplies the corresponding lifting relations; without retract closure these pairs need not yet be weak factorization systems. The strict inclusions allowed in M′ deliberately leave open whether the chosen trivial cofibrations exhaust $C \cap W$, and similarly for fibrations.

The intrinsic class Fib_W has a stronger universal quantifier than the assertion that one pullback preserves weak equivalences. A map $p : X \to Y$ belongs to it precisely when, after every base change $Y' \to Y$, the functor

$$(p')^* : \mathcal{M}/Y' \longrightarrow \mathcal{M}/(X \times_Y Y')$$

takes weak equivalences to weak equivalences. Its dual is Cof_W. For an ordinary model structure, pullbacks of fibrations are again fibrations. Therefore right properness says exactly that $F \subseteq \mathrm{Fib}_W$; left properness similarly gives $C \subseteq \mathrm{Cof}_W$. The identification in the manuscript is correct, with the customary completeness and retract conventions supplied when translating to a modern closed model category.

There is a useful unconditional inclusion

$$W_{\mathrm{univ}} \subseteq \mathrm{Fib}_W \cap W, \qquad W^{\mathrm{univ}} \subseteq \mathrm{Cof}_W \cap W.$$

For the first, put a universally weak equivalence vertically in a cartesian square. Both vertical sides are in W. If the bottom horizontal arrow is in W, two-out-of-three puts the top arrow in W; apply the same argument after any further base change. The reverse inclusions require the factorization argument of Proposition 4. This logical dependence explains why Proposition 3 first invokes the displayed equality and then proves the result which justifies it. There is no circularity: Proposition 4 uses factorization and the intrinsic fibrations only, not Proposition 3's saturation construction.

Proposition 3. *Supposons les conditions* M 0 *(pour mémoire),* M 1, M 2, M′, M″ *satisfaites. Alors en remplaçant C, TC par les saturés pour les propriétés de stabilité de contenir les* p. 6 [page 9]

isomorphismes, d'être stable par composition et par cochangement de base, et F, TF par celles de contenir les isomorphismes et d'être stable par composition et par changement de base, on trouve C', TC', F', TF' satisfaisant à tous les axiomes M 1 à M 4, M′, M″.

Démonstration. La proposition 2 nous assure que M 1 continue à être satisfait, et pour M 2 c'est trivial. M 3 et M 4 seront valables par construction. D'autre part, M″ reste valable, car Cof_W a les stabilités requises pour C', et Fib_W celles requises pour F'. Reste à prouver M′, *i.e.*

$$TC' \subset C' \cap W, \qquad TF' \subset F' \cap W,$$

et pour ceci il suffit de voir que $C' \cap W$ resp. $F' \cap W$ a les stabilités requises pour TC' resp. TF'. Cela résulte du

Lemme. *Soit C une partie de Cof_W. Si C contient les isomorphismes, il en est de même de $C \cap W$. Si de plus C est stable par composition resp. par cochangement de base, il en est de même* [de] *$C \cap W$. Énoncé dual pour une partie F de Fib_W.*

Cela provient du fait que

$$\boxed{\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}\ (^6)} \overset{\text{déf}}{=} \begin{array}{l}\text{ensemble des flèches } A \longrightarrow B \text{ telles que pour tout} \\ A \longrightarrow A',\ A' \longrightarrow B' = A' \vee_A B \ \text{ est dans } W.\end{array}$$

est lui-même stable par composition et par cochangement de base, et que l'on a bien sûr, si $C \subset \mathrm{Cof}_W$,

$$C \cap W \ = \ C \cap (\mathrm{Cof}_W \cap W) \ = \ C \cap W^{\mathrm{univ}}.$$

Donc il reste à prouver le [page 10]

Corollaire 1. *Supposons qu'il existe des parties C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfaisant* M 2, M′ (7), M″. *Alors*

$$\begin{aligned}\mathrm{Cof}_W \cap W \ &= \ W^{\mathrm{univ}} \\ \mathrm{Fib}_W \cap W \ &= \ W_{\mathrm{univ}} \ \ (^8).\end{aligned}$$

Pour voir ceci, montrons p. ex. le deuxième (le premier s'en déduira par dualité).

$$\begin{array}{ccc} X & \longleftarrow & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \longleftarrow & Y' \end{array} \qquad\qquad f \in \mathrm{Fib}_W,\ f \in W \Longrightarrow f' \in W.$$

On a mieux :

Proposition 4. *Supposons que toute flèche de $\mathcal{M}$ se factorise en pi, avec $i \in W$ et* p. 7 *$p \in \mathrm{Fib}_W$. Alors, si X, Y sont dans $\underline{\mathrm{Fib}}_W S \subset \mathcal{M}/S$* [sous-catégorie pleine de $\mathcal{M}/S$ formée des $r : Z \longrightarrow S$, où $r \in \mathrm{Fib}_W$], *alors toute $f : X \longrightarrow Y$ dans $\mathcal{M}/S$ qui est dans W_S* [formé des flèches au-dessus de S qui sont dans W] *est dans W_S^{u},* i.e. *elle reste dans W après tout changement de base $S' \longrightarrow S$.*

$$\begin{array}{ccc} X & \xrightarrow{\ f\ } & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array} \qquad\qquad p, q \in \mathrm{Fib}_W$$

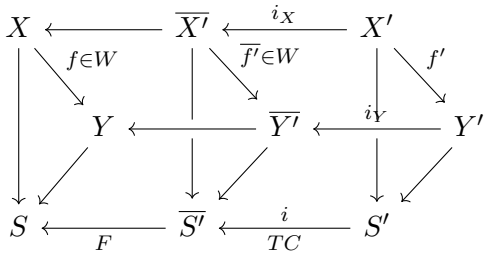
Énoncé dual pour les flèches de $S\backslash\underline{\mathrm{Cof}}_W \subset S\backslash\mathcal{M}$ [sous-catégorie pleine de $S\backslash\mathcal{M}$ formée des $i : S \longrightarrow Z$, où $i \in \mathrm{Cof}_W$], *une telle flèche qui est dans W est S-coüniversellement dans W, si toute flèche de $\mathcal{M}$ se factorise en qj, avec $j \in \mathrm{Cof}_W$ et $q \in W$.*

6. flèches *coüniversellement* dans W, ou *W-équivalences coüniverselles*.

7. On n'a pas besoin de tout M′, seulement de $TC \subset W$, $TF \subset W$. [page 11]

8. Flèches $f : X \longrightarrow Y$ *universellement* dans W, *i.e.* telles que pour tout $Y' \longrightarrow Y$, $f' : X' = X \times_Y Y' \longrightarrow Y'$ soit dans W.

Démonstration.



On factorise $S' \longrightarrow S$ en $S' \xrightarrow{i} \overline{S'} \xrightarrow{p} S$, avec $i \in W$, $p \in \mathrm{Fib}_W$, par M 2. Donc $f \in W \Longrightarrow \overline{f'} \in W$. Comme $X, Y \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_W S$, on a $\overline{X'}, \overline{Y'} \in \mathrm{Ob}\,\underline{\mathrm{Fib}}_W \overline{S'}$, et comme $i : S' \longrightarrow \overline{S'}$ est dans W, donc aussi i_X, i_Y, on a $f' \in W$ par M 5 (saturation faible de W).

Commentary · Exposition [ms. 11]

Why weak equivalences between fibrations survive base change

Proposition 4 turns the factorization machinery into a statement about W alone. Its proof is short, but each use of the intrinsic fibration condition has a different role. Given a weak equivalence $f : X \to Y$ over S, with $X \to S$ and $Y \to S$ in Fib_W, factor a proposed base change as

$$S' \xrightarrow{i} \overline{S} \xrightarrow{q} S, \qquad i \in W, \quad q \in \mathrm{Fib}_W.$$

Pulling f back along q gives $\bar{f} \in W$: it is a pullback of the weak equivalence f along the base change $Y \times_S \overline{S} \to Y$ of q. Next, the maps $X \times_S \overline{S} \to \overline{S}$ and $Y \times_S \overline{S} \to \overline{S}$ remain intrinsic fibrations. Pulling i back along them gives weak equivalences

$$X \times_S S' \longrightarrow X \times_S \overline{S}, \qquad Y \times_S S' \longrightarrow Y \times_S \overline{S}.$$

The commutative square with these sides and $\bar{f}$ then places $f \times_S S'$ in W by two-out-of-three. No lifting property, functorial choice of factorization, or retract argument is used.

For Corollary 1 apply this assertion in the slice over the codomain of $f \in \mathrm{Fib}_W \cap W$, comparing f with the identity fibration. The converse inclusion was proved above. Duality gives the cofibration equality. Now Proposition 3 can be completed: adjoining the specified composition and base-change operations preserves lifting by Proposition 2 and preserves available factorizations because the four classes only grow. The intrinsic classes have the same finite closure properties. Finally $C' \cap W = C' \cap W^{\mathrm{univ}}$ is stable under pushout and composition, and therefore contains the closure of TC; the fibration side is dual. The strengthening by retracts on manuscript p. 11 is valid if W is retract closed, since pullback and pushout transport retract diagrams to retract diagrams.

Commentary · Development (commentators') [ms. 11]

The factorization hypothesis seen on three objects

Grothendieck's intrinsic classes make it possible to isolate the role of factorization without any topology. The following small calculation shows exactly what would go wrong if Corollary 1 were read as a formal property of two-out-of-three alone. We checked the comparisons with relative-category and factorization-system treatments in Riehl (2014); Rosický (2021); we did not find this particular three-object calculation there. Its purpose is to clarify the hypothesis, not to claim a new general phenomenon.

A weak equivalence which is an intrinsic fibration but is not universal. [ours] Let $\mathcal{M}$ be the chain $0 < 1 < 2$, regarded as a category, and let W consist of the identities and the arrow $w : 0 \to 2$. Then W has two-out-of-three, $w \in \mathrm{Fib}_W$, but $w \notin W_{\mathrm{univ}}$.

Proof. The only composite of two nonidentity arrows is $0 \to 1 \to 2$. Only its composite is in W, so two-out-of-three imposes no further weak equivalence; composites involving identities satisfy the condition. Pullback in this chain is minimum. Every base change of w is $0 \to y$ for $y \in \{0, 1, 2\}$. Below $y = 0$ or 1, the only weak equivalences are identities. Below 2, the sole additional test is

$0 \to 2$, whose pullback along $0 \to 2$ is 1_0. Thus all base changes of w preserve weak equivalences and $w \in \mathrm{Fib}_W$. Its pullback along $1 \to 2$ is $0 \to 1$, which is not in W. This proves the assertion.

The missing factorization can also be located. The map $1 \to 2$ is not in Fib_W, since it pulls w back to $0 \to 1$. A factorization of $1 \to 2$ with first factor in W must have first factor 1_1, leaving precisely this nonfibration as second factor. Thus the hypothesis of Proposition 4 fails for a specified arrow. This finite category has all small limits and colimits; filtered diagrams of arrows have colimits attained in their finite directed image. Hence W is even stable under filtered colimits. Those size and closure properties do not replace factorization in the corollary. □

Commentaire sur Proposition 3. On peut être encore plus exigeant, en demandant dans M 3, M 4 également la stabilité par facteurs directs, et (quand seuls M 1, M2, M′, M″ sont satisfaits) en saturant C, TC, F, TF également pour cette stabilité supplémentaire. La proposition 2 assure encore que M 1 continue à être vérifié (et M 2, M 3, M 4 trivialement), on voit qu'il en est ainsi encore de M′, M″ à condition qu'on renforce l'axiome de saturation faible M 5 sur W par

stabilité de W par facteurs directs,

ce qui implique en effet que Cof_W, Fib_W sont stables par facteurs directs, ainsi que W^{univ} et W_{univ}.

Il ne semble pas que les axiomes M 0 à M 4, plus M′ et M″ suffisent à développer les suites [page 12]
exactes de suspension et de cosuspension. Il faut un petit chouia de plus :

Définition 2. J'appelle *pré-Q-catégorie de modèles à gauche,* la donnée de $\mathcal{M}$ et de W, p. 8
C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$, satisfaisant les axiomes M 0 à M 5, M′, M″, plus la forme renforcée de la première inclusion dans M′

$$\mathrm{M}'_0\ (\mathrm{g}) \qquad\qquad TC = C \cap W \qquad\qquad (\text{non seulement } TC \subset C \cap W).$$

De même, on définit la *pré-Q-catégorie de modèles à droite,* par la condition

$$\mathrm{M}'_0\ (\mathrm{d}) \qquad\qquad TF = F \cap W.$$

NB Quand les deux conditions sont satisfaites à la fois, alors la donnée de TC, TF résulte déjà de celle de W, C, F, et on a simplement une Q-catégorie de modèles "propre" (au sens de Quillen).

Commentary · Exposition [ms. 12]

The purpose of the one-sided equalities

The one-sided conditions are intended to recover exactness of homotopy fibres or cofibres while leaving the auxiliary choices flexible. They should not be read as a claim that every relative category has such a structure, or that a left structure and a right structure must use the same four classes. Grothendieck explicitly allows different choices. The assertion about a proper Quillen category when both equalities hold uses Quillen's original conventions; for a modern closed model category one also requires the retract conditions, or replaces the classes by their retract closures when the stated hypotheses allow it.

The historical motivation is coherent with the earlier reduction to categories of fibrant objects. Brown's formulation develops homotopy using weak equivalences, fibrations and path objects, without specifying a full model structure; see Brown (1973, §1, pp. 420–421). Grothendieck is pursuing a further independence statement: the resulting homotopy operations should be attached to W, with (C, TC, F, TF) used to prove that they exist and have exactness properties. Later in these pages, the same issue reappears for all diagram categories, where objectwise replacement alone does not automatically construct a coherent derived Kan extension.

The claims on manuscript p. 13 about exactness are explicitly provisional: the source says that they still require checking. They are therefore recorded as conjectural assertions in the audit. The completion proved in the commentary on manuscript p. 32 settles their conclusion under the printed hypotheses: either one-sided equality implies exact factorization of weak equivalences, after which the two designated trivial classes can be enlarged to $C \cap W$ and $F \cap W$. The resulting structure

is biexact with the same W, C, F, though it need not be closed. Thus one-sided structures retain flexibility in the designated trivial maps, but their existence already yields a possibly nonclosed Quillen structure. This is a later verification of the assertion, not a proof supplied on p. 13. The cross-reference to "proposition 3" on p. 13 points in fact to Proposition 4: the property cited there is the universal base-change property just proved, together with Corollary 1.

Scholie. Pour moi, l'introduction de C, TC, F, TF est purement auxiliaire – ce qui m'importe, ce sont les propriétés de W. Ainsi, l'existence de C, TC, F, TF satisfaisant aux axiomes faibles M 0 à M 4, M′, M″, du fait qu'elle donne les factorisations dont il est [page 13]
question dans [la] proposition 3, entraîne la propriété remarquable de W énoncée dans cette proposition, et dans le corollaire qui la précède. De même, ces propriétés de factorisation ([9]) assurent l'existence d'une bonne théorie des fibres et cofibres W-homotopiques sans besoin de référence dans cette théorie à C, TC, F, TF (dont l'introduction donne tout au plus des petits compléments …). Mais je ne vois pas comment, à partir des seules hypothèses de factorisation, prouver la propriété cruciale d'exactitude pour les carrés W-cartésiens, et W-cocartésiens, nécessaire pour pouvoir développer la suite *exacte* de cosuspension et de suspension. C'est pour avoir cette exactitude que l'introduction auxiliaire, et transitoire, de l'attirail (C, TC, F, TF) est utile, l'axiome supplémentaire à gauche $TC = C \cap W$ assurant (sauf erreur) l'exactitude des carrés W-cartésiens, et l'axiome à droite $TF = F \cap W$ celle des carrés W-cocartésiens. (Chose à vérifier avec plus de soin que je ne l'ai fait à présent.) Une fois ceci obtenu, et éventuellement itou pour les $(\mathcal{M}(I), W(I))$ pour toute catégorie-diagramme I, on peut totalement oublier ces ensembles auxiliaires – j'ai envie en tous cas de les oublier. [page 14]

Commentary · Context [ms. 14]

The auxiliary status of the Quillen classes

Grothendieck wants the eventual constructions to belong to the pair $(\mathcal{M}, W)$, even when their existence is established by choosing cofibrations and fibrations. His wish to forget those choices is therefore a mathematical requirement: two suitable presentations should give the same homotopy theory of diagrams, including its Kan extensions and their comparison maps. It does not assert that the choices are dispensable in a proof.

The letter to Thomason of 2 April 1991 places this passage in its immediate setting. Grothendieck begins with the accessibility theory he had been developing from SGA 4, then distinguishes the intrinsic localizer from the additional classes used to calculate with it. Concerning closed Quillen triples he writes, "J'avais cru pouvoir m'en passer, mais finalement n'y suis pas parvenu" (Grothendieck, 1991a, pp. 3–4). The contrast he draws there with Brown's one-sided methods explains why the present chapter keeps trying weaker combinations of axioms. The asymmetry is deliberate: the existence of homotopy fibres and that of homotopy cofibres need not initially be established by the same auxiliary structure.

There are two distinct tasks in such a programme. One must first construct the localized categories of diagrams, with sufficient control over their morphisms to form derived operations. One must then prove that the operations satisfy comparison laws as the indexing category varies. An equivalence between two ordinary homotopy categories settles neither task by itself. This is why the apparently elementary question whether a lifting problem can be solved naturally in a diagram will become decisive on manuscript page 41. A family of componentwise solutions need not be a natural transformation.

The later literature realizes these two tasks in several settings, with hypotheses that must be retained. Cisinski's construction of homotopy right Kan extensions starts from a closed Quillen model category with small limits; it does not require an injective model structure on every diagram category (Cisinski, 2003, Theorem 6.11). His subsequent theory of derivable categories provides a further framework for one-sided and exact-category examples (Cisinski, 2010, Introduction and §1). These results give a precise continuation of the programme, but they should not be substituted for a proof of the particular conjectural recognition criterion now being formulated. Here the outstanding issue is the smallness of the classes from which the factorization systems are to be built.

9. sous une forme légèrement renforcée il est vrai – il faut une factorisation *fonctorielle* pour f …

Mon espoir, c'est que pour une W donnée, même si on n'arrive pas à trouver une structure de triple de Quillen (W, C, F) associée (p. ex. parce qu'il n'y en aurait pas), on peut trouver à la fois des quintuplets (W, C, F, TC, TF) qui soient des pré-Quillen à gauche, et des quintuplets (d'*autres* quintuplets!) qui soient des pré-Quillen à droite.

Voici le genre d'énoncé en forme, pour le moment conjectural voire totalement heuristique, auquel je songe :

Théorème-scholie conjectural : *On suppose que $\mathcal{M}$ est stable par $\varinjlim$ "quelconques", et par $\varprojlim$ finies, et que W est stable par $\varinjlim$ filtrantes (en plus de la saturation faible).*

① *Alors* Cof_W *et* W^{univ} *ont les propriétés de stabilité* a) *et* c) *de la proposition* 2 (p. 4), *et aussi la propriété de stabilité* b) *par facteurs directs, pourvu que W lui-même l'a.* p. 9

② *Supposons* $\mathcal{M}$ accessible, *de sorte qu'on dispose du théorème* 1. *Conditions équivalentes sur W :* [page 15]

a) *Il existe une partie petite C_0 de* Cof_W, *et TC_0 de* W^{univ}, *telles que*

$$TF \overset{\text{déf}}{=} C_{0*} \subset W, \qquad (TC_0)_* \overset{\text{déf}}{=} F \subset \mathrm{Fib}_W.$$

a′) *Il existe un quadruplet (C, TC, F, TF) de parties de* $\mathrm{Fl}(\mathcal{M})$, *satisfaisant les conditions* M 1 *à* M 4, M′, M″ *de la page* 6 ([10]).

a″) *Comme* a′), *en supposant* M′ *sous forme renforcée avec* M'_0 (g), i.e. *on a une structure de pré-Q-catégorie de modèles à gauche ($TC = C \cap W$).*

b) *On a*

$$(\mathrm{Cof}_W)_* \subset W, \qquad (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W$$

(donc $(\mathrm{Cof}_W)_* \subset \mathrm{Fib}_W \cap W$ *puisque* $(\mathrm{Cof}_W)_* \subset (W^{\mathrm{univ}})_*$ ([11])*).*

b′) *On a*

$$(\mathrm{Cof}_W)_* \subset W_{\mathrm{univ}}, \qquad (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W.$$

③ *Pour qu'il existe un quadruplet qui, avec W, définisse une pré-Q-catégorie de modèles à droite ($TF = F \cap W$)* ([12]), *il faut et il suffit qu'on ait* b′) *ci-dessus, ainsi que*

$$(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \quad \textit{est clos à droite.}$$

④ *Les hypothèses préliminaires sur $(\mathcal{M}, W)$, y inclus l'accessibilité, ainsi que les conditions équivalentes dans* ②, *sont stables par passage de $(\mathcal{M}, W)$ à $(\mathcal{M}(I), W(I))$. Mais qu'en est-il de la condition dans* ③ *? Est-elle conséquence des conditions dans* ② (i.e. *leurs est-elle équivalente) ?* [page 16]

Je vais esquisser la démonstration du théorème-scholie, pour cerner le travail technique qui reste à faire.

Commentary · Exposition [ms. 16]

The closure part of the conjectural theorem

The theorem-scholie combines a correct closure calculation with a much more substantial assertion that small generating data exist. It helps to finish the closure calculation before attempting the existence argument. Assume the colimit and weak-equivalence hypotheses stated on manuscript p. 14. Then W^{univ} and Cof_W contain isomorphisms and are stable under pushout and composition, by pasting pushout squares and using two-out-of-three where appropriate.

For transfinite composition, let $(A_\alpha)_{\alpha \leq \lambda}$ be continuous. Suppose first that every successor map is a universal weak equivalence. Push the chain out along an arbitrary $A_0 \to T$, obtaining

10. avec C, TC clos à gauche, F, TF clos à droite.
11. $W^{\mathrm{univ}} \subset \mathrm{Cof}_W$.
12. On suppose toujours $\mathcal{M}$ accessible dans 3° et 4°, avec en plus F, TF (et C, TC) closes à droite (resp. à gauche).

$T_\alpha = T \amalg_{A_0} A_\alpha$. Each successor is in W. Transfinite induction shows that $T \to T_\beta$ is in W: the successor step uses composition, while a limit step is the filtered colimit of the arrows $T \to T_\alpha$. Thus $A_0 \to A_\lambda$ lies in W^{univ}.

For Cof_W, start with a weak equivalence $T \to T'$ under A_0. The two pushed-out chains satisfy $T_0 \to T'_0 \in W$. The successor step preserves this relation because the corresponding base change of $A_\alpha \to A_{\alpha+1}$ is a W-cofibration; at a limit step use filtered-colimit closure of W. It follows that pushing out along $A_0 \to A_\lambda$ preserves weak equivalences. Repeating after arbitrary cobase change establishes the universal quantifier in the definition of Cof_W.

Retract closure is available here without a further assumption on W. If f is a retract of $g \in W$ in the arrow category, write the retract maps as $s : f \to g$ and $r : g \to f$, with $rs = 1_f$. Then f is the colimit of

$$g \xrightarrow{sr} g \xrightarrow{sr} g \xrightarrow{sr} \cdots .$$

Indeed a compatible cocone $h : g \to z$ satisfies $h(sr) = h$ and factors uniquely as $hs : f \to z$ followed by r. Since this is a filtered diagram of arrows in W, its colimit f belongs to W. Pullback and pushout preserve retract diagrams, so the intrinsic and universal classes inherit retract closure. This is the elementary mechanism behind Beke (2000, Proposition 1.19); the stronger accessibility hypothesis is not needed when filtered-colimit closure is already assumed.

These arguments establish item ①. They do not establish that either whole class is defined by a lifting property, nor do they produce a small generating family. The distinction becomes decisive in the next paragraph of the manuscript.

1°) Est de la vérification de routine.

2°) Voici les implication ± tautologiques entre les conditions envisagées :

$$(*) \qquad a'' \Longrightarrow a' \overset{?}{\Longrightarrow} a \Longrightarrow b \Longleftarrow b',$$

mais je vois que $a' \Longrightarrow a$ n'est nullement tautologique, et peut-être même faux après tout ! p. 10
([13]). Il faudrait supposer que dans `[le]` quadruplet (C, F, TC, TF), F et TF sont clos à droite, et plus précisément, que $F = (TC)_*$, $TF = C_*$, ce qui sera le cas p. ex. si on a une structure de Q-catégorie de modèles "close" ("closed model category"), au sens de Quillen. Les autres trois implications dans $(*)$ sont bel et bien tautologiques.

$a \Longrightarrow a'$. On peut dans a) supposer

$$TC_0 \ \subset \ C_0$$

(quitte à remplacer C_0 par $C_0 \cup TC_0$), puis, grâce à 1°) (W^{univ} et Cof_W sont clos à gauche), [page 17]
posant

$$\overline{TC_0} = TC, \qquad \overline{C_0} = C, \qquad \texttt{[plutôt } \widetilde{TC_0} = TC, \qquad \widetilde{C_0} = C, \texttt{]}$$

on aura $TC \subset W^{\mathrm{univ}}$, $C \subset \mathrm{Cof}_W$. Passant de TC_0 à TC, de C_0 à C, on a perdu seulement que ces parties de $\mathrm{Fl}(\mathcal{M})$ soient petites. Mais par le théorème 1, posant

$$F \ = \ (TC)_*, \qquad TF \ = \ C_*,$$

les couples de parties closes (à gauche – à droite)

$$(C,\, TF), \qquad (TC,\, F)$$

satisfont à la condition de factorisation de Quillen. On aura alors les propriétés M 1 à M 4, plus les inclusions $TC \ \subset \ C \cap W$ (car $TC \ \subset \ W^{\mathrm{univ}} \ \subset \ W$, et $TC \ \subset \ C$ par construction), et $C \subset \mathrm{Cof}_W$ de M′ et M″. Il reste à vérifier $TF \subset F \cap W$, $F \subset \mathrm{Fib}_W$, mais $F \ (= (TC)_*)$ $\subset \mathrm{Fib}_W$ par hypothèse, et $TF \ (= C_*) \ \subset W$ aussi ; d'autre part $TF \subset F$ résulte de $TC \subset C$.

13. C'est pourquoi je l'ai modifié, en renforçant a') par `[la condition de la note 10]`.

Commentary · Exposition [ms. 17]

From generators to the two factorizations

The implication $(a) \Rightarrow (a')$ can be proved completely without the unsupported assertion that Cof_W and W^{univ} themselves are left lifting closed. Replace C_0 by $C_0 \cup TC_0$; this preserves the required inclusions and makes $TC_0 \subseteq C_0$. Put

$$C = \mathrm{cof}(C_0), \qquad TC = \mathrm{cof}(TC_0), \qquad TF = C_0^{\square}, \qquad F = TC_0^{\square},$$

where cof means pushouts, transfinite composites and retracts. The small object theorem identifies these left classes with the corresponding double lifting closures and supplies both factorizations. The closure calculation in the preceding block gives

$$C \subseteq \mathrm{Cof}_W, \qquad TC \subseteq W^{\mathrm{univ}}, \qquad TC \subseteq C.$$

Condition (a) supplies $TF \subseteq W$ and $F \subseteq \mathrm{Fib}_W$; antitonicity gives $TF \subseteq F$. All of M 1–M 4, M′ and M″ follow.

This repairs the proof, not by assuming the desired global closure statement, but by applying the theorem to the actual small sets that condition (a) provides. The reverse implication remains a separate issue: a right lifting closed class need not be generated by a set on the left. The order-embedding example above explains why the footnote adding lifting closedness to (a') does not, by itself, supply that missing smallness. Thus $(a'') \Rightarrow (a')$, $(a) \Rightarrow (b)$ and $(b') \Rightarrow (b)$ are immediate, and $(a) \Rightarrow (a')$ is now proved; the remaining arrows in the proposed equivalence require additional arguments.

$a' \Longrightarrow a''$. On veut donc, en plus des conditions M 1 à M 4 et M′, M″, la condition

$$TC \;=\; C \cap W \;(= C \cap W^{\mathrm{univ}}).$$

L'idée, c'est de remplacer le couple [page 18]

$$C,\ TC \qquad (C \subset \mathrm{Cof}_W,\ TC \subset C \cap W),$$

donnant naissance à

$$F = (TC)_*, \qquad TF = C_*,$$

par

$$C,\ C \cap W\ (\supset TC),$$

ce qui ne change pas le couple (C, TF), mais bien le couple (TC, F), remplacé par p. 11

$$\underbrace{(TC)'}_{\supset TC} \;=\; \underbrace{C \cap W}_{=C \cap W^{\mathrm{univ}}}, \qquad F' \;=\; \underbrace{(TC')_*}_{\subset F}$$

(on a remplacé F par un F' plus petit) ([14]). Le seul problème, c'est si (TC', F') satisfait encore à la condition de factorisation de Quillen. Ce sont des parties closes (à gauche pour TC', à droite pour F') "duales" l'une de l'autre,

$$TC' = F'^{*}, \qquad F' = TC'_{*},$$

mais comme on ne sait pas si TC' est de la forme $\widetilde{TC'_0}$, avec TC'_0 petite, on n'est pas sûr que la conclusion du théorème 1 s'applique. Mais si on suppose dans a') que C est une partie *accessible* de $\mathrm{Fl}(\mathcal{M})$ (*i.e.*, étant déjà close à gauche, que $C = \widetilde{C_0}$, C_0 petit), alors je présume qu'on pourra prouver qu'il en est de même de $C \cap W$ (W elle-même étant accessible, grâce à la condition de stabilité par petites $\varinjlim$ filtrantes), donc le théorème 1 s'applique bel et bien à $TC' = C \cap W$. Mais si on garde a') telle quelle, alors il faudrait sans doute prouver que a') *implique* a') sous la forme plus forte, avec C *accessible.* Mais comme on a $a' \Longrightarrow a$, et qu'on [page 19]

14. NB TC' sera encore clos à gauche comme intersection de deux parties closes à gauche.

vient d'établir que $a \Longrightarrow a'$ sous cette forme plus forte, on gagne (modulo le fait conjectural admis sur les parties admissibles [plutôt accessibles] de $\mathrm{Fl}(\mathcal{M})$).

Commentary · Later developments [ms. 19]

The solution-set repair and Smith's theorem

Grothendieck is trying to enlarge TC to $C \cap W$ while keeping a factorization theorem. The successful modern recognition theorem places the smallness assumption on the weak equivalences, where it can be checked independently. Smith's theorem, published with proof by Beke (2000, Theorem 1.7), assumes a locally presentable category, a set I of arrows, two-out-of-three and retract closure for W, the inclusion $I^{\square} \subseteq W$, closure of $\mathrm{cof}(I) \cap W$ under pushouts and transfinite composition, and a solution-set condition for maps from each $i \in I$ into W in the arrow category. It then gives a cofibrantly generated model structure with cofibrations $\mathrm{cof}(I)$ and fibrations $(\mathrm{cof}(I) \cap W)^{\square}$. Accessibility and accessible embeddedness of the full weak-equivalence subcategory imply the required solution-set condition; this is Beke's Proposition 1.15.

Here is the exact consequence for these pages. Retain their hypotheses and condition (a), and assume in addition that the full subcategory of weak equivalences is accessible and accessibly embedded in $\mathcal{M}^{[1]}$. Use $I = C_0 \cup TC_0$ and the two factorizations constructed in the preceding block. Corollary 1 gives $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$. Hence

$$C \cap W = C \cap W^{\mathrm{univ}}$$

is closed under pushouts, transfinite composites and retracts. Smith's hypotheses hold, so $C \cap W$ is generated by a set and the desired replacement of TC has factorizations. In fact the result is stronger than the proposed left pre-Q structure: it is a full combinatorial model structure. It is proper because $C \subseteq \mathrm{Cof}_W$ and

$$(C \cap W)^{\square} \subseteq TC_0^{\square} \subseteq \mathrm{Fib}_W.$$

The first inclusion uses $TC_0 \subseteq C \cap W$.

This application is a repair under an explicit additional hypothesis, not a proof of the original theorem-scholie. Closure under filtered colimits says that colimits of diagrams already in W stay in W; accessibility additionally supplies a set of sufficiently presentable arrows from which all of W is built. Nor is the manuscript's parenthetical definition $C = \widetilde{C}_0$ with C_0 small equivalent to accessibility of the full arrow subcategory. The example in Pos separates those notions even when C is left lifting closed.

Il reste donc à prouver

$$b \implies a, \qquad b \implies b',$$

et il suffit de prouver la première implication, car on sait que $a) \Longrightarrow a')$, et que $a')$ implique que

$$\mathrm{Fib}_W \cap W \;=\; W_{\mathrm{univ}}$$

(cor. 1, page 10), donc $(\mathrm{Cof}_W)_* \subset W$ et $(W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W$ (*i.e.* b)) impliquent, comme $(\mathrm{Cof}_W)_* \subset (W^{\mathrm{univ}})_*$ (car $W^{\mathrm{univ}} \subset \mathrm{Cof}_W$), $(\mathrm{Cof}_W)_* \subset W \cap \mathrm{Fib}_W = W_{\mathrm{univ}}$, ce qui est la partie à établir de b') (l'autre $(W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W$ étant tautologiquement contenue dans b).

Notons maintenant que b) n'est autre que la variante affaiblie de a), quand on omet de [page 20] spécifier que C_0, TC_0 doivent être petits. Donc le seul problème, pour $b \Longrightarrow a$, c'est de voir ceci :

Lemme conjectural ([15]).

① *Si on a une partie close à gauche* Φ *de* $\mathrm{Fl}(\mathcal{M})$*, telle que* $\Phi_* \subset \mathrm{Fib}_W$ *(resp.* $\Phi_* \subset W$*), alors il existe une partie* petite $\Phi_0 \subset \Phi$*, telle que* $\Phi_{0*} \subset \mathrm{Fib}_W$ *(resp.* $\Phi_0 \subset W$*).*

② *Si* C*,* C' *sont deux parties closes à gauche de* $\mathrm{Fl}(\mathcal{M})$*, si* C *et* C' *sont accessibles, il en est de même de* $C \cap C'$*.*

p. 12

15. Ce lemme a l'air un peu faux, mais ça doit pouvoir s'arranger …

Je crois que c'est bel et bien vrai sous les hypothèses faites sur $\mathcal{M}$, W – qui doivent impliquer que dans un sens convenable, W est une partie accessible de l'ensemble $\mathrm{Fl}(\mathcal{M})$ des flèches de la catégorie accessible $\mathcal{M}$. Il doit en résulter qu'une partie de $\mathcal{M}$ construite explicitement en termes de W, telle Fib_W, est elle-même accessible, et le lemme doit sortir de là.

Ainsi, 2° est prouvé, modulo ce lemme d'accessibilité, qu'il me faudra prouver ± en même temps que le théorème 1 (que je n'ai fait ici qu'énoncer).

[page 21]

Commentary · Later developments [ms. 20]

The intersection lemma has a later affirmative answer

The second part of the conjectural lemma has a precise modern resolution when its word "accessible" is read in the sense specified on manuscript p. 18, namely set-generated under left lifting closure. In a locally presentable category, if

$$C = \mathrm{cof}(I), \qquad C' = \mathrm{cof}(J)$$

for sets I, J, then $C \cap C'$ is again $\mathrm{cof}(K)$ for a set K. This follows from Makkai and Rosický (2014a, Theorem 3.2 and Remark 3.8): combinatorial cellular structures are stable under the relevant pseudopullbacks. Apply their result to the two identity functors from $(\mathcal{M}, C)$ and $(\mathcal{M}, C')$ into $\mathcal{M}$ with every arrow cellular. The resulting structure has exactly the intersection as its cellular class. This resolves the intersection assertion; it does not imply that the entire intrinsic class W^{univ} has such generators.

The first part of the lemma asks for a different kind of compactness: from $\Phi^{\square} \subseteq T$, where T is W or Fib_W, it seeks a set $\Phi_0 \subseteq \Phi$ with $\Phi_0^{\square} \subseteq T$. Neither the Galois connection nor the preceding intersection theorem provides this reduction. The final parenthesis in the edition has $\Phi_0 \subseteq W$; its intended expression is $\Phi_{0*} \subseteq W$, as the surrounding reduction from (b) to (a) requires. The printed slip is retained and recorded separately.

We found the applicable positive results in the small object theorem, Smith's recognition theorem, and the cellular-category theorem just cited. They do not prove this small-subfamily assertion under precisely the preliminary assumptions of the manuscript. We therefore leave that assertion unresolved in this audit rather than attach a claim of current openness to a loosely specified accessibility principle. In a small category the selection is immediate by taking $\Phi_0 = \Phi$; its entire content here is the passage from a proper class to a set. Modern accessible weak factorization systems also require care: accessibility refers to a factorization functor, and does not guarantee that its full left or right class is an accessible subcategory. The corrected statements and the distinction from set generation are recorded in Rosický (2021, §3, especially Remarks 3.2, 3.5 and Theorem 3.9).

Prouvons 3°. Supposons d'abord qu'il existe un quadruplet "clos" (C, TC, F, TF), s'insérant dans un "pré-quintuplet de Quillen clos" qui soit une Q-catégorie de modèles à droite, *i.e.*

$$TF \;=\; F \cap W \;=\; F \cap W_{\mathrm{univ}}.$$

Prouvons alors `[que]`

(∗) $\underbrace{(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}}_{A_W}$ est clos à droite.

En effet, on a

$$TC \;\subset\; W^{\mathrm{univ}},$$

d'où

$$(W^{\mathrm{univ}})_* \;\subset\; (TC)_* \;=\; F,$$

donc

$$A = (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \;\subset\; F \cap W_{\mathrm{univ}} \overset{(\text{par hyp.})}{=} TF,$$

or TF est clos à droite et contenu dans W_{univ}, d'où

$$\overline{A} \ \subset \ TF \ \subset \ W_{\mathrm{univ}},$$

et comme $(W^{\mathrm{univ}})_*$ est clos à droite, on a aussi

$$\overline{A} \ \subset \ (W^{\mathrm{univ}})_*,$$

d'où $\overline{A} \subset W_{\mathrm{univ}} \cap (W^{\mathrm{univ}})_* = A$, donc A est clos à droite, q.e.d.

Prouvons la réciproque, *i.e.* supposons satisfaites les conditions de 2°, *plus* la condition $(*)$ ci-dessus. On doit construire le quadruplet.

[page 22]

Commentary · Exposition [ms. 21]

A closure obstruction before the existence problem

Grothendieck is trying to extract a property of the weak equivalences which is independent of the auxiliary Quillen structure. The class

$$A_W = (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}$$

is designed for this purpose: its first condition is a lifting condition, its second says that every pullback is a weak equivalence. There is a useful separation here between a necessary condition, which is established without an accessibility argument, and a construction of factorizations, which still needs one.

Suppose that a closed weak structure is right exact. Since $TC \subset W^{\mathrm{univ}}$, we have $(W^{\mathrm{univ}})_* \subset F$. Consequently

$$A_W = (W^{\mathrm{univ}})_* \cap TF.$$

Indeed, the forward inclusion uses $TF = F \cap W_{\mathrm{univ}}$, while the reverse inclusion uses $TF \subset W_{\mathrm{univ}}$. This writes A_W as an intersection of two right lifting classes. It is therefore right closed. This short proof is equivalent to the closure computation in the manuscript and shows exactly why no generation hypothesis enters the necessary condition.

The converse is more delicate. Set $L = W^{\mathrm{univ}}$, $F = L_*$, $T = F \cap W_{\mathrm{univ}}$ and $C = T^*$. If T is right closed, then $C_* = T$, and $L \subset C$. These are statements about orthogonality; they do not factor an arbitrary arrow. To obtain the proposed structure, one must additionally know that (L, L_*) and (T^*, T) admit factorizations and that $T^* \subset \mathrm{Cof}_W$. If also L is left closed, these conditions give the two weak factorization systems. Since $L \subset C \cap W$, $T \subset F \cap W$, and $F \subset \mathrm{Fib}_W$, they give a closed weak structure; the right exactness follows from $F \cap W = F \cap W_{\mathrm{univ}}$, using the universal-equivalence criterion already proved on manuscript pp. 10–11.

The sentence on the next page which attributes left lifting closure of Cof_W and W^{univ} to part 1 of the conjectural theorem passes over precisely this distinction. Part 1 establishes stability under certain constructions. Left lifting closure is a stronger assertion unless a generation theorem applies. The earlier block *From generators to the two factorizations* supplies the appropriate repair when generators have been specified. We read the diagrams on pp. 22–24 as the formal organization of a proposed construction, with the remaining factorization hypotheses still visible; they are not an additional proof of those hypotheses.

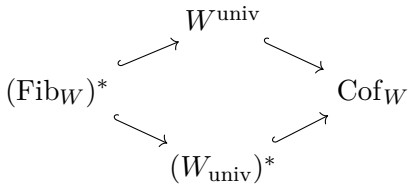
Notons d'abord, pour nous y reconnaître, que les relations b') de 2°), et les relations transposées (compte tenu que Cof_W et W^{univ} sont clos à gauche, par 1°), savoir

$$\left\{ \begin{array}{lcl l lcl} (\mathrm{Cof}_W)_* & \subset & W_{\mathrm{univ}} & , & (W^{\mathrm{univ}})_* & \subset & \mathrm{Fib}_W \\ (W_{\mathrm{univ}})^* & \subset & \mathrm{Cof}_W & , & (\mathrm{Fib}_W)^* & \subset & W^{\mathrm{univ}}, \end{array} \right.$$

donnant ([16]) les deux diagrammes carrés d'inclusions p. 13

16. compte tenu de $W^{\mathrm{univ}} \subset \mathrm{Cof}_W$, $W_{\mathrm{univ}} \subset \mathrm{Fib}_W$ et des inclusions transposées.

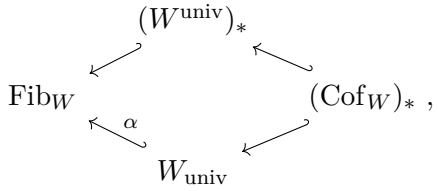
(I)



([17])

et

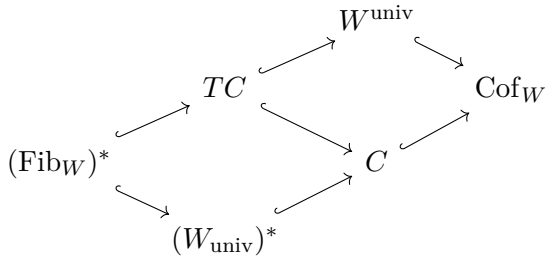
(II)



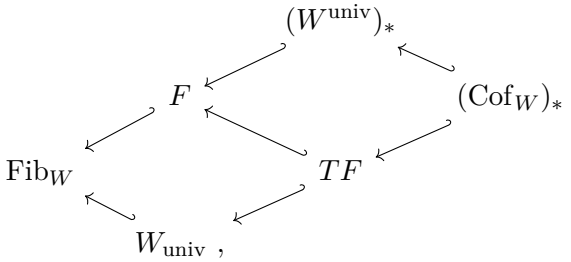
dont le premier est transposé (par $\Phi \longmapsto \Phi^*$) du second, et où les 8 ensembles de flèches envisagés, à l'exception de Fib_W et W_{univ} (reliés par α dans le carré II) sont clos (les quatre ensembles de I clos à gauche, et les deux ensembles de II non intéressés par α, clos à droite).

Les "pré-quintuplets de Quillen clos", associés à W, correspondent au choix de parties closes à gauche TC et C, s'insérant dans le diagramme (I) comme montré dans I′ ci-dessous, et donnent lieu par transposition à F et TF comme dans II′ ci-dessous, [page 23]

(I′)



(II′)



le "hic" étant donc de choisir TC et C assez grandes pour qu'on ait

$$\underbrace{(TC)_*}_{=F} \subset \mathrm{Fib}_W, \qquad \underbrace{C_*}_{=TF} \subset W_{\mathrm{univ}} \quad -$$

et les inclusions stipulées dans 2°, b') assurent qu'on peut bel et bien trouver TC et C, quitte à les prendre les plus grandes possibles, $TC = W^{\mathrm{univ}}$, $C = \mathrm{Cof}_W$ – auquel cas, d'ailleurs, on aura p. 14 [page 24]

$$TC \;=\; C \cap W^{\mathrm{univ}},$$

i.e. on a une pré-Q-catégorie de modèles à gauche. Mais il s'agit d'assurer maintenant, par contre, que

$$TF \;=\; F \cap W_{\mathrm{univ}},$$

17. **NB** On a ces carrés d'inclusions, chaque fois qu'on a un système de Quillen (W, C, F, TC, TF) *clos.*

ce qui implique, on vient de le voir, (en notant $(W^{\mathrm{univ}})_* \subset F$) que

$$(*) \qquad \overline{(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}} \ \subset \ \overline{F \cap W_{\mathrm{univ}}} = \overline{TF} = TF \subset W_{\mathrm{univ}}.$$

Inversement, si $(*)$ est satisfait, prenons

$$F = (W^{\mathrm{univ}})_*$$

le plus petit possible, *i.e.* $C = W^{\mathrm{univ}}$ `[plutôt` $TC = W^{\mathrm{univ}}$`]` le plus grand possible. On doit donc prendre

$$TF \ = \ F \cap W_{\mathrm{univ}} \ = \ (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}},$$

et cet ensemble est bien clos à droite par hypothèse.

Corollaire. *Supposons la condition* b′) *du théorème-scholie satisfaite. Les conditions suivantes* a), b) *sur W sont équivalentes (et impliquent la condition de* 3° *du théorème-scholie).*

[page 25]

ⓐ $(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} = (\mathrm{Cof}_W)_*$.

ⓑ $\widetilde{W^{\mathrm{univ}} \cup (W_{\mathrm{univ}})}^{\,*} = \mathrm{Cof}_W$, i.e. Cof_W *est la partie close à gauche engendrée par* W^{univ} *et* $(W_{\mathrm{univ}})^*$,

et ces conditions impliquent

ⓒ *qu'il existe une structure de* triple clos de Quillen (W, C, F) *incluant W (donc avec $TC = W \cap C$, $TF = W \cap F$) (lequel sera clos au sens de Quillen si et seulement si W* `[est]` *stable par facteurs directs – car C et F sont déjà stables spontanément).*

Il suffit en effet de prendre alors

$$TC \ = \ W^{\mathrm{univ}}, \qquad C \ = \ \mathrm{Cof}_W.$$

Mais je doute que l'existence d'un triple clos de Quillen (W, C, F) suffise à impliquer les conditions *a*), *b*) ci-dessus, ni même la condition plus faible

$$(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \ = \ (\mathrm{Cof}_W)_*$$

qui signifie *exactement* que $C = \mathrm{Cof}_W$, $TC = W^{\mathrm{univ}}$, et leurs transposés $TF = (\mathrm{Cof}_W)_*$, $F = (W^{\mathrm{univ}})_*$ forment une catégorie de Quillen close.

p. 15
[page 26]

Conditions sur $(\mathcal{M}, W)$ avec

a) $\mathcal{M}$ stable par $\varprojlim$ finies et $\varinjlim$ (petites),

b) W stable par $\varinjlim$ filtrantes,

c) $\mathcal{M}$ accessible.

Commentary · Exposition [ms. 25]

The two closures in the maximal construction

The corollary seeks a particularly economical structure: take every universal weak cofibration as a trivial cofibration and every W-cofibration as a cofibration. Its strength comes from this maximal choice, and the two different closures in the formulas measure its cost.

For arbitrary classes L, R of arrows, the lifting calculus gives

$$\big(L \cup R^*\big)_* = L_* \cap (R^*)_* = L_* \cap \overline{R}.$$

Thus the right orthogonal of $W^{\mathrm{univ}} \cup (W_{\mathrm{univ}})^*$ is $(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}}$, with a bar on the second factor. It is not, without another hypothesis, $(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}$. If Cof_W is left closed, applying the Galois correspondence proves the equivalence

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} = (\mathrm{Cof}_W)_* \quad \Longleftrightarrow \quad \widetilde{W^{\mathrm{univ}} \cup (W_{\mathrm{univ}})^*} = \mathrm{Cof}_W.$$

The superscript star must belong to the second summand inside the tilde. The printed formula on manuscript p. 25 places it outside the visible tilde; the accompanying prose and the summary on p. 26 identify the intended expression. We preserve the edition's formula and record the discrepancy separately.

Under condition b' one has $(\mathrm{Cof}_W)_* \subset W_{\mathrm{univ}}$. Hence the stronger displayed equality implies

$$(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} = (\mathrm{Cof}_W)_*.$$

For the inclusion from right to left, use $W^{\mathrm{univ}} \subset \mathrm{Cof}_W$ and b'; for the reverse inclusion, enlarge W_{univ} to its right closure and apply the stronger equality. The latter equality is exactly the compatibility needed by the maximal two pairs, once their factorizations exist. It does not itself supply factorizations.

This explains the hierarchy I–IV on p. 26. At the level of formal implications, IV imposes a closure identity stronger than the identity in III, and a maximal structure in III is a special instance of a structure in II. The alleged equivalences inside I and the construction clauses inside II retain the accessibility and factorization problems from pp. 17–20. The hierarchy is a research programme with several proved implications, not four unconditional recognition theorems. In particular, the correct formula in II(c) is $C = (TF)^*$; the printed subscript star is a variance slip.

I
- ⓐ existence d'un système de Quillen faible (W, C, TC, F, TF) (clos si on y tient)
- ⇕
- ⓑ existence d'un système de Quillen faible exact à gauche ($TC = C \cap W$).
- ⇕
- ⓒ $\underbrace{(\mathrm{Cof}_W)_* \subset W,\ (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W}_{(*)}$ $(\Longleftrightarrow \quad (\mathrm{Cof}_W)_* \quad \subset \quad W_{\mathrm{univ}},$ $(W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W)$.

$\Uparrow$

II $\begin{cases} \text{ⓐ} & \text{existence d'un système de Quillen faible exact à droite } (TF = F \cap W) \\ \Updownarrow & \\ \text{ⓓ} & \text{On a } (*), \text{ et de plus } (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \text{ clos à droite.} \\ \Updownarrow & \\ \text{ⓑ} & \text{existence d'un système de Quillen clos (closed model category), } TC = C \cap W,\ TF = F \cap W,\ F = (TC)_*,\ TF = C_*,\ TC = F^*,\ C = (TF)^*. \\ \Updownarrow & \\ \text{ⓒ} & \text{Posant } TC = W^{\mathrm{univ}},\ F = (TC)_*,\ TF = F \cap W_{\mathrm{univ}},\ C = (TF)_*, \text{ on trouve avec } W \text{ un système de Quillen clos.} \end{cases}$

$\Uparrow$

III $\begin{cases} C = \mathrm{Cof}_W \text{ et } TC = W^{\mathrm{univ}} \text{ font partie (avec } W) \text{ d'un système de Quillen clos (donc } F = (W^{\mathrm{univ}})_*,\ TF = (\mathrm{Cof}_W)_*,\ W = TF \circ TC). \\ \Updownarrow \\ \texttt{[On a]}\ (*), \text{ et de plus } (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} = (\mathrm{Cof}_W)_*. \end{cases}$

$\Uparrow$

IV $\begin{cases} \text{On a } (*) \text{ et } \mathrm{Cof}_W = \widetilde{(W_{\mathrm{univ}})^* \cup W^{\mathrm{univ}}}\ . \\ \Updownarrow \\ \text{On a } (*) \text{ et } (W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} = (\mathrm{Cof}_W)_*. \end{cases}$

Cas $\mathcal{M} = \mathrm{Cat}$, $W = \mathbf{W}_\infty$. On sait que II est satisfait, par Thomason. Donc on a p. 16

$$(\mathrm{Cof}_W)_* \subset W_{\mathrm{univ}}, \qquad (W^{\mathrm{univ}})_* \subset \mathrm{Fib}_W, \qquad (W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}} \text{ clos à droite.}$$

J'ai pu prouver directement les deux premières inclusions (*i.e.* I), mais pas que $(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}$ soit clos à droite. (Je devrais au moins essayer!) ([18]). On n'a pas IV, *i.e.* on a

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} \neq (\mathrm{Cof}_W)_*$$

car

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} \not\subset W_{\mathrm{univ}}$$

(contient les $X \longrightarrow e$ avec X discrète), donc on a

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} \supsetneq \underbrace{(W^{\mathrm{univ}})_* \cap W_{\mathrm{univ}}}_{\text{n'est } \textit{pas} \text{ l'orthogonal de } W^{\mathrm{univ}} \cup (W_{\mathrm{univ}})^*} .$$

[page 27]

Commentary · Development (commentators') [ms. 26]

Discrete categories detect the extra closure

Grothendieck's example deserves a proof because it distinguishes two conditions that otherwise look nearly identical. We take W in Cat to be the functors inducing weak homotopy equivalences of nerves. Thomason's theorem gives a proper model structure with this W; its existence does not imply the maximal closure condition IV (Thomason, 1980, Theorem 4.9 and Corollary 5.5).

A discrete obstruction to condition IV. [ours] For every nonempty discrete category D, its terminal map belongs to

$$(W^{\mathrm{univ}})_* \cap \overline{W_{\mathrm{univ}}} \cap \mathrm{Fib}_W.$$

If D has at least two objects, this map does not belong to W_{univ}.

Proof. We first examine a functor $i : A \to B$ having the left lifting property against every universally weakly equivalent functor. We claim that $\pi_0(i) : \pi_0 A \to \pi_0 B$ is injective. Suppose that two distinct components of A have images connected in B by a zigzag of length m. Choose a finite poset J whose Hasse diagram is an alternating linear zigzag with more than m edges. Its nerve is an interval subdivided into edges, so $J \to e$ is a weak equivalence. It is universally so: its pullback over any category E is the projection $J \times E \to E$, and the nerve preserves products; a product with this finite contractible simplicial set is a weak homotopy equivalence.

Send the first chosen component of A to one endpoint of J, the second to the other endpoint, and every other component to either endpoint. This defines a functor $a : A \to J$, constant on components. A lift against $J \to e$ would extend a to $B \to J$. The zigzag of length m in B would then connect the two endpoints of J by at most m edges. Every nonidentity arrow of the alternating poset joins adjacent vertices, so this is impossible. This proves the claim.

Now a functor $A \to D$ is just a function $\pi_0 A \to D$. If $i \in (W_{\mathrm{univ}})^*$, injectivity of $\pi_0(i)$ lets us extend this function to $\pi_0 B$: use its prescribed values on the image and one fixed object of D on all remaining components. Thus $D \to e$ has the right lifting property against $(W_{\mathrm{univ}})^*$, which is exactly membership in $\overline{W_{\mathrm{univ}}}$.

Every functor in W^{univ} is in W, so induces a bijection on components. The same extension argument, now with a unique possible value on every component, gives $D \to e \in (W^{\mathrm{univ}})_*$. Finally, any base change of $D \to e$ is a coproduct of copies of an identity map. Pulling a weak equivalence back along it gives the corresponding coproduct of that weak equivalence; the nerve and geometric realization verify that the result is again a weak equivalence. Therefore $D \to e \in \mathrm{Fib}_W$. If $|D| \geq 2$, the map fails already to induce a bijection on components, so is not in W, and a fortiori not in W_{univ}. □

This also explains the qualification required in the footnote: a nonempty discrete category supplies the displayed membership, but a counterexample requires at least two components. The empty category is not included in the lifting argument because extending a function to a newly

18. Il suffirait de savoir que $\overline{W_{\mathrm{univ}}} \cap \mathrm{Fib}_W = W_{\mathrm{univ}}$, ce qui impliquerait que III et IV sont équivalents ([19]).
19. Mais c'est faux, p. ex. si X `[est une]` catégorie discrète non vide, $X \longrightarrow e$ est dans $\overline{W_{\mathrm{univ}}} \cap \mathrm{Fib}_W$.

> appearing component would then be impossible. The proof expands the manuscript's example; we checked Thomason's construction and the cited modern treatments of lifting and localization, without finding this precise zigzag proof there. The marker records a proof supplied for this commentary, not a claim of priority.

3 Catégorie de Quillen (deuxième mouture)

On se donne toujours un couple

$$(\mathcal{M}, W), \qquad W \subset \mathrm{Fl}(\mathcal{M}), \qquad W \text{ faiblement saturé},$$

$\mathcal{M}$ satisfait M 0 (stabilité par $\varprojlim$ finies et $\varinjlim$ finies).
Définition 1 : Une *structure de Quillen faible* associée à la catégorie de modèles $(\mathcal{M}, W)$ est la donnée de quatre parties (C, TC, F, TF) de $\mathrm{Fl}(\mathcal{M})$, satisfaisant les conditions suivantes.

- **M 1** $C \longleftrightarrow TF$, $TC \longleftrightarrow F$ (*i.e.* $TF \subset C_*$, $F \subset (TC)_*$).
- **M 2** Propriété de factorisation de Quillen pour (C, TF) et (TC, F).
- **M 3** Stabilité de C par composition et par cochangement de base, de F par composition et changement de base – C et F contiennent les isomorphismes.
- **M 4** Stabilité similaire pour TC et TF.
- **M′** $TC \subset C \cap W$, $TF \subset F \cap W$.
- **M″** $C \subset \mathrm{Cof}_W$, $F \subset \mathrm{Fib}_W$.

On dit que la structure de Quillen est *exacte* si elle satisfait à l'axiome supplémentaire

M 2 bis $W = TF \circ TC$. p. 17

On dit qu'elle est *exacte à gauche* si elle satisfait l'axiome supplémentaire

M$'_0$ (g) $TC = C \cap W$ (au lieu de $TC \subset C \cap W$ seulement),

exacte à droite si elle satisfait

M$'_0$ (d) $TF = F \cap W$ (au lieu de $TF \subset F \cap W$ seulement) ([20]).

Remarque. [page 28]

a) Une structure de Quillen exacte à gauche ou exacte à droite est exacte.

b) Si on a une donnée (C, TC, F, TF) satisfaisant M 1, M 2, M$'$, M$''$, quand on sature C, TC à gauche et F, TF à droite, pour les stabilités précisées dans M 2, M 3, on trouve une structure de Quillen. (Tous les axiomes sont évidents, sauf M$'$, pour lequel je renvoie à p. 9, 10, notamment prop. 4.) Si on avait M 2 bis, cela sera encore vérifié (a fortiori) après saturation, *i.e.* on a une structure de Quillen exacte.

Commentary · Exposition [ms. 27]

Exactness belongs to the chosen classes

The second definition rearranges the earlier axioms around the task they have to perform. The inclusions $TC \subset C \cap W$ and $TF \subset F \cap W$ distinguish designated trivial maps from all weak equivalences among the designated cofibrations and fibrations. This distinction is essential before retract closure has been imposed. In modern terminology, the closed case consists of two nested weak factorization systems; Grothendieck's M$''$ adds properness relative to the specified weak equivalences.

There are three different assertions. Exactness says that every weak equivalence has a factorization using the two designated trivial classes. Left exactness identifies the designated trivial cofibrations with $C \cap W$; right exactness identifies the designated trivial fibrations with $F \cap W$. Either identification implies exactness. For example, if $w = pi$ with $i \in C$ and $p \in TF$, then $w, p \in W$ imply $i \in W$, and left exactness gives $i \in TC$. The dual argument uses the other factorization.

Neither this argument nor the two-out-of-three axiom proves that the original chosen class TF is retract closed. Footnote 20 tries to infer too much from retract closure of C, F, W. Its conclusion becomes correct after replacing the missing trivial class by its retract closure, but the chosen class need not already equal that closure. The next example makes the distinction concrete. For closed structures, by contrast, exactness alone forces both equalities by the retract argument developed later in the commentary on manuscript p. 57.

Définition 1 bis. La structure de Quillen faible est dite *structure de Quillen* (tout court) si elle est exacte à gauche et à droite, ou (comme on dit) "biexacte", alors (C, F) determine déjà (C, TC, F, TF). (Ça correspond exactement aux "model categories" de Quillen `[propres]`.) On dit que la structure de Quillen faible est *close* si on a

$$C \leftarrow\!+\!\rightarrow TF\ , \qquad TC \leftarrow\!+\!\rightarrow F\ ,$$

i.e. $TF = C_*$, $C = (TF)^*$ et $F = (TC)_*$, $TC = F^*$ ([21]). Une structure de Quillen faible à la fois biexacte (*i.e.* pas faible du tout!) et close est appelée structure de Quillen *parfaite*. (Ça correspond exactement à la "closed model category" `[propre]` de Quillen.)

Remarque. Pour qu'une structure de Quillen faible soit parfaite, il suffit qu'elle soit biexacte et qu'on ait [page 29]

$$C = (TF)^*, \qquad F = (TC)_*$$

(ce qui implique déjà que $TC = F^*$, $TF = C_*$, *i.e.* que TC est clos à gauche et TF clos à droite). C'est prouvé dans Quillen. Ça implique aussi que W est fortement saturé – c'est le pays de cocagne!

20. **NB** Si C, F, W `[sont]` stables par facteurs directs (donc aussi $C \cap W$ et $F \cap W$), alors l'exactitude à gauche $TC = C \cap W$ *équivaut* à l'exactitude à droite $TF = F \cap W$, et implique que C, TC, F, TF sont *clos* (à gauche pour C, TC, à droite pour F, TF).

21. Cela signifie aussi que C, TF, TC, F sont stables par facteurs directs. On dit que la structure de Quillen faible est *close à gauche* (resp. *à droite*) si C, TC (resp. F, TF) sont clos à gauche (resp. à droite).

Commentary · Development (commentators') [ms. 28]

A retract condition missing from footnote 20

Grothendieck wants the ordinary model-category identities to reappear as soon as retracts behave well. They do reappear after completing the designated classes, but not necessarily before. The following example isolates the missing step without involving any topological pathology.

Left exactness without right exactness. [ours] Assume the axiom of choice. In Set put

$$W = \mathrm{Fl}(\mathrm{Set}), \qquad C = TC = \{\text{injections}\}, \qquad F = \{\text{surjections}\},$$

and let TF consist of those surjections all of whose fibres are either singletons or infinite sets. These classes satisfy all the weak Quillen axioms on manuscript p. 27. The classes C, F, W are retract closed, and the structure is left exact, but it is not right exact.

Proof. Injections have the left lifting property against surjections: on the complement of the injected subset, choose a preimage in each relevant nonempty fibre. Both required lifting properties follow. For an arbitrary map $f : A \to B$, use

$$A \longrightarrow A \amalg (B \times \mathbb{N}) \xrightarrow{p} B, \qquad p|_A = f, \quad p(b, n) = b.$$

The first map is injective and every fibre of the second is infinite; when B is empty, necessarily A is empty and the assertion is vacuous. This one factorization serves both required pairs.

Injections contain the isomorphisms and are stable under composition and pushout. Surjections have the corresponding pullback and composition properties. For the special class TF, pullback preserves each fibre up to bijection. A fibre of a composite is a disjoint union of nonempty fibres of the first map indexed by a fibre of the second. If the indexing fibre is a singleton, the result is a permitted fibre of the first map; if it is infinite, the disjoint union is infinite. Thus TF also has the required stability. The inclusions $TC \subset C \cap W$ and $TF \subset F \cap W$ hold. Since all maps are weak equivalences, Cof_W and Fib_W are both the class of all maps, so M'' holds as well.

The usual elementwise arguments show that injections and surjections are retract closed in the arrow category, while W trivially is. We have $TC = C \cap W$. However, the map from a two-element set to a singleton belongs to $F \cap W$ and does not belong to TF. It is a retract of $\mathbb{N} \to \{*\}$, which does belong to TF. This proves both the failure of right exactness and the failure of retract closure of the designated class. □

The repaired assertion can be stated precisely. Suppose C, F, W are retract closed and $TC = C \cap W$. Then TC is retract closed. The factorization and lifting axioms give $TC = {}^{\perp}F$ and $F = TC^{\perp}$. If $p \in F \cap W$, factor $p = qi$ with $i \in C$ and $q \in TF$. Two-out-of-three gives $i \in TC$, and lifting i against p exhibits p as a retract of q. Conversely every retract of a map in TF lies in $F \cap W$. Hence

$$\mathrm{Retr}(TF) = F \cap W.$$

Replacing TF by this class preserves lifting, factorization and properness and produces a perfect structure. The standard retract argument also gives $C = {}^{\perp}(F \cap W)$ and $C^{\perp} = F \cap W$. The dual repair applies to a right exact structure. We checked the standard model-category retract arguments and Cisinski's treatments; the particular fibre-cardinality example is supplied here to settle this exact footnote, without a priority claim.

Scholie : Le couple $TC \longleftrightarrow F$, satisfaisant à la condition de factorisation de Quillen, et satisfaisant $TC \subset W$, $F \subset \mathrm{Fib}_W$, permet déjà de développer une théorie satisfaisante des fibres W-homotopiques dans $\mathcal{M}$ – mais sans propriété d'exactitude pour les carrés W-cartésiens. Dualement, la donnée de $C \longleftrightarrow TF$, avec factorisation, et $TF \subset W$, $C \subset \mathrm{Cofib}_W$, donne une bonne théorie des cofibres W-homotopiques, mais sans propriété d'exactitude pour les carrés W-cocartésiens. La structure de Quillen faible réunit ces deux données duales l'une de l'autre, avec en plus le lien supplémentaire entre ces deux donnés, par $TC \subset C$, $TF \subset F$ (stipulé dans M'). Cette donnée assure déjà que p. 18

$$\underbrace{\mathcal{M}_{\mathrm{cf}}}_{\substack{\text{sous-catégorie pleine de } \mathcal{M} \text{ formée des objets fibrants et cofibrants à la fois}}} \qquad W_{\mathrm{cf}}^{-1} \xrightarrow{\sim} \mathcal{M}W^{-1} = \mathrm{Ho}_W$$

est une équivalence, mais non pas que le foncteur localisation [page 30]

$$\mathcal{M}_{\mathrm{cf}} \longrightarrow \mathcal{M}_{\mathrm{cf}} W_{\mathrm{cf}}^{-1}$$

est un simple passage au quotient sur les flèches, *i.e.* que si X, Y sont dans $\mathcal{M}_{\mathrm{cf}}$, alors

$$(*) \qquad \mathrm{Hom}_{\mathcal{M}}(X,Y) \longrightarrow \underbrace{\mathrm{Hom}_W(X,Y)}_{\overset{\text{déf}}{=} \mathrm{Hom}_{\mathrm{Ho}_W}(X,Y)}$$

soit surjectif. Ceci est assuré par l'axiome d'exactitude M 2 bis,

$$W = TF \circ TC.$$

Avec ce seul axiome, on n'a pas de caractérisation sympathique de la relation d'équivalence dans $\mathrm{Hom}_{\mathcal{M}}(X,Y)$ définie par la flèche $(*)$ (pour X, Y dans $\mathcal{M}_{\mathrm{cf}}$).

Commentary · Exposition [ms. 29]

Replacement controls arrows as well as objects

The passage to fibrant-cofibrant objects is the first point where the lifting axioms recover information which would be lost by merely choosing a weakly equivalent object. Grothendieck's assertion of an equivalence of localizations is correct under the weak axioms. The mechanism is worth making explicit because exactness will be needed only for the next assertion, that ordinary arrows represent all localized arrows.

Choose $q_X : QX \to X$ in TF with QX cofibrant. Given $f : X \to Y$, lift against q_Y to obtain $Qf : QX \to QY$ with $q_Y Qf = f q_X$. This lift need not be unique as an arrow. It is unique in the localization of the cofibrant objects. Indeed, for two lifts $a, b : QX \to QY$, form $P = QY \times_Y QY$. Its projections belong to TF by pullback stability. Choose a cofibrant replacement $q_P : QP \to P$ and lift (a,b) through it. The diagonal $QY \to P$ likewise lifts through q_P because QY is cofibrant. Call this lift $d : QY \to QP$. The two composites

$$QY \xrightarrow{d} QP \xrightarrow{\mathrm{pr}_j\, q_P} QY$$

are identities. Each $\mathrm{pr}_j\, q_P$ belongs to TF, so $d \in W$. After localization both projections are therefore the inverse of d, and $a = b$ there. This uniqueness proves compatibility with identities and composition. Two-out-of-three shows that a weak equivalence f gives a weak equivalence Qf. Thus Q defines an inverse, after localization, to the inclusion of the cofibrant objects.

For the second step choose $r_X : X \to RX$ in TC, with RX fibrant, when X is cofibrant. Then RX remains cofibrant. An arrow $X \to Y$ extends to $RX \to RY$ by lifting r_X against $RY \to e$. To compare two extensions, form $P = RX \amalg_X RX$. The two inclusions $RX \to P$ are in TC. Choose $P \to RP$ in TC with RP fibrant. The fold $P \to RX$ extends to $d : RP \to RX$ because RX is fibrant. If $v_0, v_1 : RX \to RP$ are the two induced maps, then $dv_0 = dv_1 = 1$, and d, v_0, v_1 are weak equivalences. The two extensions consequently agree after localization, by first extending their

common pushout map $P \to RY$ to $RP \to RY$. This gives the inverse equivalence for the inclusion of fibrant-cofibrant objects into cofibrant objects.

No equality $TC = C \cap W$ or $TF = F \cap W$ entered these two arguments. What is indispensable is the ability to compare choices inside the relevant full subcategory. This is the same distinction already emphasized in the commentary on Chapter VI, manuscript p. 87. For the usual model-category statement, see Cisinski (2019a, Theorem 2.2.15 and Corollary 2.2.18); the argument above records the weaker hypotheses actually used here.

Si on a exactitude à gauche,

$$TC = C \cap W \qquad (\Longleftrightarrow\ TC = C \cap W^{\mathrm{univ}}),$$

alors cette relation d'équivalence est celle engendrée par la relation de *Q-homotopie à droite* ([22], [23]) (impliquée par la Q-homotopie à gauche), et la catégorie quotient ainsi définie

$$\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{cf}} \;=\; \pi\mathcal{M}_{\mathrm{cf}} \;\simeq\; \mathrm{Ho}_W,$$

s'envoie dans $\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}}$,

$$\pi\mathcal{M}_{\mathrm{cf}} \longrightarrow \pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}},$$

et ce foncteur peut être interprété (sauf erreur)

Commentary · Examples [ms. 30]

Two path conventions and a nontransitive homotopy

Footnote 22 requires a distinction which the text does not explicitly make. A diagonal factorization

$$Y \xrightarrow{s} PY \xrightarrow{p=(d_0,d_1)} Y \times Y$$

may be required to have $s \in TC$ and $p \in F$, or only $s \in W$ and $p \in F$. Call these, respectively, a *designated* path object and a *broad* path object. In standard model-category terminology the first is a very good path object and the second a good path object. The raw relation associated with a fixed path object declares $f, g : X \to Y$ related when $(f, g) = pH$ for some $H : X \to PY$. Taking the union of these relations over all broad path objects is another operation; it must not be silently identified with the relation from a designated one.

Every designated path object gives a reflexive and symmetric relation, for every source. Reflexivity uses sf. For symmetry, lift s against p, with top map s and bottom map τp, where τ interchanges the two factors of Y^2. The filler $r : PY \to PY$ satisfies $pr = \tau p$, and rH reverses a homotopy. Moreover, if $Y \xrightarrow{s'} P'Y \xrightarrow{p'} Y^2$ is another designated path object, the square with vertical maps s, p' and horizontal maps s', p has a filler $PY \to P'Y$ preserving the endpoints. Interchanging the objects gives the converse comparison. Thus *one fixed designated path object and all designated path objects define the same raw relation*, without fibrancy, cofibrancy or retract-closure assumptions. Transitivity is the additional issue.

A counterexample with every object fibrant and cofibrant. The following is a dual application of the graph example of Kurz and Rosický (2005, Definition 3.1, Observation 3.3 and Example 3.6). Let $\mathcal{G}$ be the category of reflexive directed multigraphs. An object consists of sets of vertices and edges, maps $s, t : E \rightrightarrows V$, and a distinguished loop $e : V \to E$ satisfying $se = te = 1_V$. Morphisms preserve these data. This is a functor category, hence has all small limits and colimits. Let

$$L = \mathrm{Mono}(\mathcal{G}), \qquad R = L^{\square}.$$

Here monomorphisms are injective on both vertices and edges. For precision, R is defined by the literal lifting property, not by an abbreviated assertion about whether two image vertices are adjacent: in a multigraph a lifting problem specifies the target edge as well as its endpoints.

22. Il semblerait que la relation de Q-homotopie à droite est une relation d'équivalence, dès que Y [est] dans $\mathcal{M}_{\mathrm{f}}$ et sans supposer la structure de Quillen faible exacte à droite ou à gauche.

23. **NB** D'après Baues, il semblerait que sur $\mathcal{M}_{\mathrm{cf}}$, cette relation d'homotopie à droite équivaut à la relation d'homotopie à gauche, et que celle-ci ne dépend pas du cylindre choisi …

The pair (L, R) is a weak factorization system. One generating map adds a vertex with its distinguished loop to the empty graph. The other adds a directed edge between the two vertices of a discrete two-vertex reflexive graph. In a pushout the two endpoints may coincide, so the latter generator also adds a nondistinguished loop. Every subgraph inclusion is a composite of pushouts of these maps: first adjoin its missing vertices, then its missing edges. Conversely such pushouts, their composites and their retracts are monomorphisms. The finite generating graphs are finitely presentable, so the small object theorem proved earlier gives exactly $(L, L^{\square})$.

Pass to $\mathcal{M} = \mathcal{G}^{\mathrm{op}}$ and put

$$W = C = \mathrm{Mor}(\mathcal{M}), \qquad TF = \mathrm{Iso}(\mathcal{M}), \qquad TC = R^{\mathrm{op}}, \qquad F = L^{\mathrm{op}}.$$

Both (C, TF) and (TC, F) are weak factorization systems. In particular their classes contain the isomorphisms and have the required composition, pushout or pullback stability, and even retract stability. The inclusions $TC \subset C \cap W$ and $TF \subset F \cap W$ hold. Since every arrow is weak, the intrinsic classes Cof_W and Fib_W also contain every arrow. Thus all the printed weak axioms, including M″, hold.

Every object is cofibrant because C consists of all arrows. Every object is fibrant because its map to the terminal object of $\mathcal{M}$ is opposite to an inclusion of the empty graph, hence belongs to L^{op}. These observations make the example independent of any failure to replace its source or target.

Let V denote the graph with one vertex and its single distinguished loop. Let D have two vertices $0, 1$ and exactly one directed edge for each ordered pair of vertices, with the two diagonal edges distinguished. The inclusion $j : V \amalg V \to D$ belongs to L. The unique map $q : D \to V$ belongs to R: a graph map from a subgraph to D extends by sending each additional vertex to 0, and then sending each edge to the unique edge with the prescribed endpoints. This extends every lifting problem against a monomorphism. The equality $qj = \nabla_V$ therefore becomes a designated path factorization in the opposite category,

$$Y \xrightarrow{q^{\mathrm{op}}} D \xrightarrow{j^{\mathrm{op}}} Y \times Y, \qquad Y = V \quad \text{viewed in } \mathcal{M}.$$

Now let X be the reflexive graph with vertices a, b, c, distinguished loops, and four further edges

$$a \rightleftarrows b \rightleftarrows c,$$

with no edge from a to c or from c to a, again viewed as an object of $\mathcal{M}$. The three vertex maps $V \to X$ in $\mathcal{G}$ are three arrows $f_a, f_b, f_c : X \to Y$ in $\mathcal{M}$. A right homotopy through the displayed path object is precisely a graph map $D \to X$ with the specified vertex images. Such maps exist for (a, b) and (b, c), whereas none exists for (a, c). Consequently

$$f_a \sim_r f_b, \qquad f_b \sim_r f_c, \qquad f_a \not\sim_r f_c.$$

The comparison argument above shows that allowing every designated path object does not change this calculation. Raw designated right homotopy is not transitive even though every object is fibrant and cofibrant and all four lifting classes are retract closed.

This weak structure is not exact. Indeed $TF \circ TC = TC$, and the opposite of a vertex inclusion $V \to X$ is not in TC: the vertex inclusion fails the right lifting property against the map adding a vertex. Thus $TC \neq W$. There is also a useful exact variant: replace C by TC and TF by F, retaining W, TC, F. This is a perfect model structure with all arrows weak. The same path calculation still fails transitivity for the same source, which is now noncofibrant. To see the latter directly, $X \to V$ in $\mathcal{G}$ does not lift the edge-adding generator with endpoints mapped to a, c. Hence even exactness does not make the raw relation from very good paths transitive for unrestricted sources.

The broad relation for a fibrant target. Under the printed weak axioms, the relation defined by existence of a homotopy through *some* broad path object is an equivalence relation whenever Y is fibrant, with arbitrary source X. Here is the direct argument. Reflexivity uses the constant section of any designated, hence broad, path object. Symmetry uses the same broad object with its endpoints interchanged. For transitivity choose broad path objects

$$Y \xrightarrow{s_P} P \xrightarrow{(d_0, d_1)} Y^2, \qquad Y \xrightarrow{s_Q} Q \xrightarrow{(e_0, e_1)} Y^2$$

and homotopies from f to g through P and from g to h through Q. Form

$$K = P \times_Y Q \quad \text{using } d_1 \text{ and } e_0, \qquad t = (s_P, s_Q) : Y \longrightarrow K.$$

Each endpoint map d_i, e_i belongs to F, since it is the composite of a path fibration with a projection $Y^2 \to Y$, and the latter is a pullback of $Y \to e$. Each endpoint is also in W by two-out-of-three applied to its composite with the constant section. The projection $K \to P$ is the pullback of $e_0 \in W$ along $d_1 \in F \subset \mathrm{Fib}_W$, so it is in W. Its composite with t is $s_P \in W$, and hence $t \in W$.

The map $K \to Y^3$ recording the two outer endpoints and their common middle endpoint is the pullback of

$$P \times Q \longrightarrow Y^2 \times Y^2$$

along $(y_0, y_1, y_2) \mapsto (y_0, y_1, y_1, y_2)$. Products and pullbacks of F-maps are in F, so this map is in F. The projection $Y^3 \to Y^2$ retaining the outer coordinates is again a pullback of $Y \to e$. Their composite $K \to Y^2$ is therefore in F. Thus $Y \xrightarrow{t} K \to Y^2$ is a broad path object. The two given homotopies induce $X \to K$, supplying the homotopy from f to h. No exactness or retract closure has entered.

The missing implication for designated paths is now visible: this argument proves $t \in W$, but does not prove $t \in TC$. The graph example shows that the omission cannot be repaired from the weak axioms alone. In that example the broad path $Y \xrightarrow{\Delta} Y^2 \xrightarrow{1} Y^2$ makes every pair related, whereas every designated path retains the nontransitive relation just calculated. Consequently footnote 22 has a positive answer for the existential broad convention and a negative answer for the designated convention used in the replacement constructions. Its status is a question of these precise conventions, not an unresolved general homotopy theorem.

p. 19
[page 31]

comme un *adjoint à droite* (pleinement fidèle) du foncteur de localisation

$$\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}} \longrightarrow (\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}})W_{\mathrm{c}}^{-1} \;\simeq\; \mathrm{Ho}_W,$$

et Ho_W se déduit de $\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}}$ par un *calcul de fractions à gauche.* Dualement, si on a exactitude à droite, *i.e.* $TF = F \cap W$ ($\Longleftrightarrow TF = F \cap W_{\mathrm{univ}}$), alors la relation d'équivalence pertinente sur $\mathrm{Hom}_{\mathcal{M}}(X, Y)$, pour X, Y dans $\mathcal{M}_{\mathrm{cf}}$, est la Q-homotopie à gauche (impliquée par la Q-homotopie à droite), et cette fois la flèche naturelle de $\pi_{\mathrm{g}}\mathcal{M}_{\mathrm{cf}} = \pi\mathcal{M}_{\mathrm{cf}} \simeq \mathrm{Ho}_W$ dans $\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{f}}$ [plutôt $\pi_{\mathrm{g}}\mathcal{M}_{\mathrm{f}}$],

$$\underbrace{\pi\mathcal{M}_{\mathrm{cf}}}_{\simeq \mathrm{Ho}_W} \longrightarrow \pi_{\mathrm{d}}\mathcal{M}_{\mathrm{f}} \qquad [\text{plutôt } \pi_{\mathrm{g}}\mathcal{M}_{\mathrm{f}}]$$

est un *adjoint à gauche* (pleinement fidèle) du foncteur de localisation

$$\pi_{\mathrm{d}}\mathcal{M}_f \longrightarrow \mathcal{M}_{\mathrm{f}}W_{\mathrm{f}}^{-1} \;\simeq\; \mathrm{Ho}_W \qquad [\text{plutôt } \pi_{\mathrm{g}}\mathcal{M}_f \longrightarrow \mathcal{M}_{\mathrm{f}}W_{\mathrm{f}}^{-1} \;\simeq\; \mathrm{Ho}_W],$$

et la localisée s'explicite par un calcul de fractions à droite.

Commentary · Development (commentators') [ms. 30]

Exactness and the congruence generated by both homotopies

Grothendieck has now isolated the hypothesis which allows a weak equivalence between good objects to be inverted by an ordinary arrow. His observation admits a precise completion even before one-sided exactness identifies a preferred homotopy relation.

A chosen cylinder of X is a factorization

$$X \amalg X \xrightarrow{(j_0, j_1)} IX \xrightarrow{q} X, \qquad (j_0, j_1) \in C, \quad q \in TF.$$

A chosen path object of Y is a factorization

$$Y \xrightarrow{s} PY \xrightarrow{(p_0, p_1)} Y \times Y, \qquad s \in TC, \quad (p_0, p_1) \in F.$$

Left and right Q-homotopies are maps $IX \to Y$ and $X \to PY$ with the prescribed endpoints. We allow all such factorizations in these definitions.

A quotient description under exactness alone. [ours] Assume the weak Quillen axioms and $W = TF \circ TC$. Let $\equiv$ be the smallest congruence on $\mathcal{M}_{\mathrm{cf}}$ containing both left and right Q-homotopy. Then

$$\mathcal{M}_{\mathrm{cf}}/\!\equiv \;\simeq\; \mathcal{M}[W^{-1}].$$

In particular every morphism between fibrant-cofibrant objects in the localization is represented by an ordinary morphism.

Proof. Both kinds of homotopy become equalities after localization: for a cylinder, $qj_0 = qj_1 = 1$ and $q \in W$; for a path object, $p_0 s = p_1 s = 1$ and $s \in W$. It therefore suffices to prove that the quotient inverts every weak equivalence between fibrant-cofibrant objects.

Let $w : X \to Y$ be such a weak equivalence. Exactness gives $w = pi$ with $i : X \to Z$ in TC and $p : Z \to Y$ in TF. The intermediate object Z is cofibrant because $TC \subset C$ and X is cofibrant; it is fibrant because $TF \subset F$ and Y is fibrant. Lifting i against $X \to e$ gives $r : Z \to X$ with $ri = 1_X$. Lifting $\varnothing \to Y$ against p gives $t : Y \to Z$ with $pt = 1_Y$.

To compare ir with 1_Z, choose a path object $Z \xrightarrow{s} PZ \xrightarrow{v} Z \times Z$. The square with left edge i, right edge v, top edge si and bottom edge $(ir, 1_Z)$ commutes. The lifting axiom $TC \perp F$ supplies a right homotopy from ir to 1_Z. To compare tp with 1_Z, choose a cylinder $Z \amalg Z \xrightarrow{j} IZ \xrightarrow{q} Z$. The square with left edge j, right edge p, top edge $(tp, 1_Z)$ and bottom edge pq commutes. The lifting axiom $C \perp TF$ supplies a left homotopy from tp to 1_Z.

Consequently i and p are invertible in the quotient, with inverses r and t, and w has inverse rt. The universal property of localization now identifies the quotient with $\mathcal{M}_{\mathrm{cf}}[W_{\mathrm{cf}}^{-1}]$. The replacement argument in the preceding block identifies that category with $\mathcal{M}[W^{-1}]$. □

The congruence formulation deliberately keeps the two homotopies together. Under left exactness a left homotopy implies right homotopy: for a cofibrant source, either cylinder endpoint is in $C \cap W = TC$, so the familiar endpoint lifting square against a path fibration produces the right homotopy. Dually, right exactness converts a right homotopy into a left homotopy. Thus the proposition also proves the manuscript's description by the relation *generated* by right homotopy under left exactness, and its dual. We have checked Cisinski's model-category and derivable-category treatments; their standard statements identify a single homotopy relation under their own hypotheses. The joint-congruence formulation is supplied here for the exact weak axioms on these pages, without claiming that it is new.

Dans le cas biexacte (*i.e.* celui des véritables structures de Quillen), les deux relations de Q-homotopie dans $\mathcal{M}_{\mathrm{cf}}$ (plus généralement dans $\mathrm{Hom}(X, Y)$, si X dans $\mathcal{M}_{\mathrm{c}}$ et Y dans $\mathcal{M}_{\mathrm{f}}$) coïncident, et deviennent même particulièrement simples. (Car si $f, g : X \rightrightarrows Y$, avec X cofibrant et Y fibrant, alors $f \sim g$ peut se tester sur *n'importe quel* "objet cylindre" pour X, ou sur *n'importe quel* "objet chemins" pour Y.) [page 32]

Commentary · Exposition [ms. 31]

Why the localization has a fully faithful adjoint

The proposed adjoint is a way to recognize fibrant replacement from a universal property in the homotopy quotient. Assume left exactness, and write $\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}}$ for the category of cofibrant objects modulo the equivalence relation generated by right Q-homotopies. Right homotopy is compatible with both compositions: precomposition is immediate, while postcomposition follows by extending a map between the constant paths across $s : Y \to PY$ using $TC \perp F$. Thus taking its generated equivalence relation really defines a category.

Choose $r_X : X \to RX$ in TC with RX fibrant. For any fibrant-cofibrant Y, restriction along r_X gives a bijection

$$\mathrm{Hom}_{\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}}}(RX, Y) \xrightarrow{\sim} \mathrm{Hom}_{\pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}}}(X, Y).$$

Surjectivity follows by lifting r_X against $Y \to e$. For injectivity, a right homotopy between ur_X and vr_X extends across r_X by lifting against the path fibration $PY \to Y^2$, giving a right homotopy between u and v. For a chain of homotopies, choose extensions of all intermediate maps and apply this argument successively. The same proof works for chains with either orientation.

The previous block identifies the homotopy quotient on fibrant-cofibrant objects with Ho_W. Moreover its inclusion in the larger quotient is fully faithful: any relation in the larger quotient gives equality after localization, and on fibrant-cofibrant objects such equality is already the generated right-homotopy relation. The displayed bijection therefore says exactly that the localization

$$L : \pi_{\mathrm{d}}\mathcal{M}_{\mathrm{c}} \longrightarrow \mathrm{Ho}_W$$

is left adjoint to the inclusion of the fibrant-cofibrant homotopy category. The unit is represented by r_X. This proves the suggested fully faithful right adjoint without any claim that a chosen lift is unique before homotopy.

It also proves the stated calculus of fractions. Write S for the images of the arrows of W in this quotient; identities and compositions belong to S. Precomposition by $s : X \to X'$ in S is bijective on maps into any fibrant-cofibrant object, since the displayed bijection identifies these Hom sets with those in the localization. Given $f : X \to Y$ and $s : X \to X'$ in S, choose the denominator $r_Y : Y \to RY$. That bijectivity supplies a map $g : X' \to RY$ with

$$gs = r_Y f,$$

which is the common-denominator condition. For cancellation, suppose $f, g : X \to Y$ and $s : X' \to X$ in S satisfy $fs = gs$. The same bijectivity, now with target RY, gives $r_Y f = r_Y g$. Thus the denominator r_Y also supplies the cancellation condition. Every localized arrow consequently has a representative $X \to RY \leftarrow Y$, with the right-hand arrow in S; this specifies the manuscript's left-fraction convention.

Applying the whole argument to the opposite category proves the right-exact statement and the fully faithful left adjoint on the quotient of fibrant objects. The editors' changes from π_{d} to π_{g} on this page are therefore mathematically required.

For the cofibrant sources and fibrant targets considered here, the biexact case gives a single homotopy relation, computable using any cylinder or path object (Cisinski, 2019a, Lemma 2.2.12). Footnote 22 omits the cofibrancy assumption on the source and leaves the path convention implicit. The block *Two path conventions and a nontransitive homotopy* resolves that distinction. For designated TC–F paths, raw right homotopy can fail transitivity even when every object is fibrant and cofibrant if the weak structure is not exact; it can also fail in a perfect structure for a noncofibrant source. Both failures come from the graph example of Kurz and Rosický (2005, Example 3.6). In contrast, the existential relation obtained by allowing all W–F path objects is an equivalence relation for a fibrant target and an arbitrary source, by direct concatenation. The generated relation used in the adjoint argument above is therefore a deliberate and sufficient formulation.

Dans le cas où la structure de Quillen faible est exacte à gauche, *i.e.* où la relation d'équivalence pertinente dans $\mathrm{Fl}\,\mathcal{M}_{\mathrm{cf}}$ est la Q-homotopie à droite, alors l'axiome d'exactitude est vérifié pour les carrés W-cartésiens, mais je doute qu'il en soit de même pour les carrés W-cocartésiens ([24]). (Si on ne fait une hypothèse supplémentaire, p. ex. l'exactitude à droite de la structure de Quillen.) Si on n'a que l'exactitude (mais ni à gauche ni à droite), il semble qu'on ne peut en conclure l'exactitude ni des carrés W-cartésiens, ni des carrés W-cocartésiens, donc pas celle des suites de cosuspension (loop sequences), ni des suites de suspension. On pourrait donc penser que la condition d'exactitude, à elle seule, ne sert pas à grand chose pour [page 33] le formalisme homotopique dans les catégories de fractions Ho_W etc. Mais on a pourtant ceci :

soit $\Psi = \begin{array}{ccc} & & 1 \\ 0 & \nearrow & \\ & \searrow & \\ & & 2 \end{array}$, et supposons que $(\mathcal{M}, W)$ *et* $(\mathcal{M}(\Psi), W(\Psi))$ [où pour toute petite catégorie I, $\mathcal{M}(I)$ désigne la catégorie des préfaisceaux sur I à valeurs dans $\mathcal{M}$, et $W(I)$ est la partie de $\mathrm{Fl}(\mathcal{M}(I))$ formée des morphismes de préfaisceaux p. 20 qui sont dans W "argument par argument"] admettent une structure de Quillen faible exacte. Alors dans $\mathcal{M}$ les carrés W-cartésiens sont W-exactes, donc on a dans $\mathcal{M}$ des suites exactes longues de cosuspension (exact loop sequences) ([25]). (L'hypothèse sur $(\mathcal{M}(\Psi), W(\Psi))$ sert uniquement à assurer que toute flèche de $\mathrm{Ho}_{W(\Psi)} = \mathcal{M}(\Psi)W(\Psi)^{-1}$ est isomorphe à une flèche provenant d'une flèche de $\mathcal{M}(\Psi)$ lui-même. Mais c'est là une propriété apparemment très délicate, et je ne connais d'autre façon de l'obtenir que via l'existence d'une structure de Quillen faible exacte.) Dualement, si $(\mathcal{M}, W)$ et $(\mathcal{M}(\Phi), W(\Phi))$ admettent des structures de Quillen faibles exactes, alors les carrés W-cocartésiens dans $\mathcal{M}$ sont W-exacts, et on trouve (si $\mathcal{M}$ est ponctué) des suites exactes longues de suspension.

24. Il semble pourtant que c'est le cas, d'après Baues, cf. annotation marginale correspondante page 30.
25. il semble que je me suis canulé ici, et je doute que le résultat évoqué soit exact.

Commentary · Development (commentators') [ms. 32]

Exactness permits completion of the trivial classes

Grothendieck's hesitation about exactness can be resolved by changing the designated trivial classes while keeping the weak equivalences, cofibrations and fibrations. This is a different operation from asserting that the original designated classes were already complete. It is therefore compatible with the counterexample to footnote 20.

Completion under exactness. [ours] Suppose (C, TC, F, TF) is an exact weak Quillen structure. Then

$$\big(C,\ C\cap W,\ F,\ F\cap W\big)$$

is a biexact Quillen structure, with the same C, F, W. No retract-closure hypothesis on these classes is needed for this assertion. If all four original lifting classes were closed, the completion makes no change.

Proof. The only new lifting problems involve $C\cap W$ against F, and C against $F\cap W$. Let $j\in C\cap W$. Exactness gives $j=qi$ with $i\in TC$ and $q\in TF$. Since $j\in C$ and $q\in TF$, the lifting axiom exhibits j as a retract of i in the arrow category. Lifting properties are preserved by retracts, so j has the left lifting property against every map in F. Dually, if $p\in F\cap W$, factor $p=qi$ with $i\in TC$ and $q\in TF$. Lifting i against p exhibits p as a retract of q, so it has the right lifting property against C.

The original factorizations still serve, because $TC\subset C\cap W$ and $TF\subset F\cap W$. The classes C and F have not changed. The classes $C\cap W$ and $F\cap W$ have the required pushout or pullback stability by M″ and the universal-equivalence criterion of manuscript pp. 10–11; composition and identities follow from two-out-of-three. Thus every axiom holds, and both exactness equalities hold by definition. When TC and TF are retract closed, the two retract arguments also show $C\cap W\subset TC$ and $F\cap W\subset TF$, giving equality with the original classes. □

This completion is a possibly nonclosed model structure in Quillen's original terminology. It suffices for the ordinary homotopy theory of cofibrant and fibrant objects: the standard endpoint arguments now use $C\cap W\perp F$ and $C\perp F\cap W$. In particular both homotopies classify the localized maps from a cofibrant object to a fibrant object, and any good chosen cylinder or path object from the original weak structure remains available. The existence of different designated classes therefore does not obstruct the usual homotopy calculation once exactness holds. This also explains the applicability of the nonclosed Quillen case in Cisinski (2010, Example 5.2).

We checked the quoted Quillen and derivable-category frameworks and did not locate this completion stated using the exact four-class axioms of the manuscript. The proof is a direct application of the standard retract argument; its purpose is to settle the hypotheses on these pages, not to claim a new model-category theorem.

[page 34]

Ceci montre que si, pour une catégorie de modèles $(\mathcal{M}, W)$ donnée, on s'intéresse à "faire de l'algèbre homotopique" pour tous les couples $(\mathcal{M}(I), W(I))$ à la fois, avec I dans Cat, il suffit (si on veut avoir une bonne théorie des fibres et cofibres W-homotopiques dans ces catégories, y inclus la propriété de W-exactitude pour les carrés W-cartésiens et pour les carrés W-cocartésiens) de disposer pour chaque $(\mathcal{M}(I), W(I))$ d'une structure de Quillen faible exacte sans plus.

Commentary · Exposition [ms. 33]

Exactness of squares after completing the classes

Grothendieck introduces $\mathcal{M}(\Psi)$ to recover relations among maps while representatives remain available. Exactness in that diagram category indeed says that every localized arrow is isomorphic, in the arrow category, to an actual natural transformation: replace its source and target by fibrant-cofibrant objects and apply the quotient description above. His footnote correctly warns that this representability argument has not, by itself, established compatibility with the chosen homotopy pullback.

There is a more direct repair. The completion just proved turns an exact weak structure on $\mathcal{M}$ into a possibly nonclosed proper Quillen structure with the same W, C, F. Both homotopy-cartesian and homotopy-cocartesian exactness follow from that structure. For this conclusion the additional exact structure on $\mathcal{M}(\Psi)$ is unnecessary. Thus the asserted conclusion on p. 33 is valid, while the proposed route through arrow representability leaves a step unproved.

Here is the local lifting calculation behind the cartesian assertion. After taking the usual fibrant representatives, write the homotopy pullback as the ordinary pullback $P = E \times_S Y$, where $p : E \to S$ is a fibration and E, Y, S are fibrant. Let T be cofibrant. Suppose maps $a : T \to E$ and $b : T \to Y$ have homotopic composites to S. Represent the homotopy by $H : IT \to S$, with endpoints pa and qb. The endpoint inclusion $j_0 : T \to IT$ is in $C \cap W$: it is a cofibration because T is cofibrant, and a weak equivalence because the cylinder projection is one. The completed lifting property against p gives a lift $\widetilde{H} : IT \to E$. Its other endpoint $a' = \widetilde{H} j_1$ is homotopic to a and satisfies $pa' = qb$. The pair (a', b) therefore gives a map $T \to P$. This proves the required surjectivity onto pairs of homotopy classes with equal composites. The dual argument gives cocartesian exactness and the corresponding suspension sequences in the pointed case.

The geometry of the calculation is visible in the cospan $* \to S^1 \leftarrow *$. Its homotopy pullback is ΩS^1, not the one-point strict pullback. Lifting a based loop to $\mathbb{R}$ with initial point 0 assigns its endpoint in $\mathbb{Z}$; two loops are homotopic relative endpoints precisely when those integers agree. Hence $\pi_0 \Omega S^1 = \mathbb{Z}$. The path between the two composites is part of the homotopy-pullback data. The lifting argument preserves that datum while making the square strictly commute. This is the compatibility that mere representability of an arrow does not explain.

Ceci nous montre donc l'intérêt de développer des critères, pour une $(\mathcal{M}, W)$, assurant qu'il en est bien ainsi. À ce sujet, je note que je ne sais comment vérifier l'exactitude d'une structure de Quillen *que si* je dispose de l'exactitude à gauche, ou à droite. C'est dire que je dois dégager en fait des conditions sur $(\mathcal{M}, W)$ qui assurent (disons) que les $(\mathcal{M}(I), W(I))$ admettent des structures de Quillen faibles exactes à gauche.

Les structures de Quillen faibles pour les $(\mathcal{M}(I), W(I))$ seront déduites d'une structure de Quillen faible pour $(\mathcal{M}, W)$, de la façon suivante : elles seront toutes *"closes à droite", i.e.* [page 35]

$$F(I) \;=\; (TC(I))_*, \qquad TF(I) \;=\; (C(I))_*, \tag{1}$$

en particulier (pour $I = e$)

$$F \;=\; (TC)_*, \qquad TF \;=\; C_*,$$

ce qui implique déjà (puisque $C \subset \mathrm{Cof}_W$, $TC \subset W^{\mathrm{univ}}$)

$$(\mathrm{Cof}_W)_* \;\subset\; W, \qquad (W^{\mathrm{univ}})_* \;\subset\; \mathrm{Fib}_W \tag{2}$$

(d'où $(\mathrm{Cof}_W)_* \subset W_{\mathrm{univ}}$). En vertu de (1), pour définir les structures de Quillen faibles pour les $(\mathcal{M}(I), W(I))$, il suffit de définir $C(I)$, $TC(I)$, et même, comme on veut l'exactitude à gauche, *i.e.*

$$TC(I) \;=\; C(I) \cap W(I) \tag{3}$$

(en particulier

$$TC \;=\; C \cap W \quad), \tag{4}$$

p. 21

il reste à définir C. On définira $C(I)$ "argument par argument", *i.e.* pour $f \in \mathrm{Fl}(\mathcal{M}(I))$

$$f \in C(I) \qquad \Longleftrightarrow \qquad \alpha_s^*(f) \in C \;\text{ pour tout } s \in \mathrm{Ob}\, I, \tag{5}$$

où

$$\alpha_s : e \longrightarrow I \qquad \texttt{[est]} \text{ défini par } \alpha_s(e) = s.$$

Il en résulte d'ailleurs aussitôt, par (3),

$$f \in TC(I) \qquad \Longleftrightarrow \qquad \alpha_s^*(f) \in TC \;\text{ pour tout } s \in \mathrm{Ob}\, I. \tag{6}$$

On suppose donc que $(\mathcal{M}, W)$ admet une structure de Quillen faible exacte à gauche et close à droite (C, TC, F, TF), définie en termes de

[page 36]

$$C \;\subset\; \mathrm{Fl}(\mathcal{M}),$$

par

$$TC \;=\; C \cap W, \qquad F \;=\; (TC)_*, \qquad TF = C_*.$$

Pour que cela donne bel et bien une structure de Quillen faible (forcément exacte à gauche et close à droite), il faut simplement vérifier les conditions suivantes.

Commentary · Exposition [ms. 35]

The injective choice of diagram structure

Grothendieck now makes a choice which determines the variance of the whole construction. Cofibrations and weak equivalences will be tested at each object of the indexing category. Fibrations will be characterized by lifting against these objectwise classes. In modern terminology this is the injective choice. The asymmetry is productive: every restriction of diagrams preserves the classes specified objectwise, so its right adjoint becomes a candidate for a right Quillen functor.

Keep the variance explicit: $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$. If $C(I)$ is the objectwise class associated with C, then

$$TC(I) = C(I) \cap W(I) = (C \cap W)(I)$$

is an identity of classes, not an existence theorem. Once $F(I) = TC(I)_*$ and $TF(I) = C(I)_*$ are defined, lifting is automatic and the right classes have their required composition and pullback stability. Objectwise pushouts show the stability of $C(I)$; the corresponding statement for $TC(I)$ uses the universal weak-equivalence property already obtained in the base category.

The remaining two tasks have different characters. The inclusions $TF(I) \subset W(I)$ and $F(I) \subset \mathrm{Fib}_{W(I)}$ can be checked by evaluation and its adjoints; pp. 37–39 give that argument. Factorization in the diagram category requires a construction of coherent fillers and cannot be inferred merely by factoring every component. Even functorial componentwise factorizations do not make both lifting classes objectwise. The walking-arrow example below shows the distinction in the smallest nontrivial indexing category.

ⓘ Factorisation pour $(C, TF = C_*)$ et $(TC = C \cap W, F = (TC)_*)$ – on reviendra sur ce point.

ⓘⓘ Stabilité de C par composition et cochangement de base, et C contient les isomorphismes (d'où résultera la stabilité similaire de $TC = C \cap W$, compte tenu que $C \subset \mathrm{Cof}_W$) (NB Les stabilités idoines pour F, TF sont triviales par définition $F = (TC)_*$, $TF = C_*$.)

ⓘⓘⓘ $C \subset \mathrm{Cof}_W$, $\underbrace{C_*}_{=TF} \subset W$, $(C \cap W)_* \subset \mathrm{Fib}_W$.

Pour vérifier, de même, que $(C(I), TC(I), F(I), TF(I))$ soit une structure de Quillen faible pour $W(I)$, il faut vérifier les conditions similaires à ⓘⓘⓘ . Or (*ii*) pour $C(I)$ résulte aussitôt de la condition (*ii*) pour C. Il reste à examiner (*iii*) et (*i*).

Les inclusions de (*iii*).

L'inclusion p. 22 [page 37]

$$C \;\subset\; \mathrm{Cof}_W$$

implique que toute $f \in C(I)$ est dans Cof_W argument par argument. Cela implique que f est dans $\mathrm{Cof}_{W(I)}$, comme on voit aussitôt sur la définition – je l'explicite en le

Lemme 1 : *Soit $f : X_\bullet \longrightarrow Y_\bullet$ une flèche de $\mathcal{M}(I)$. Pour que f soit dans* $\mathrm{Cof}_{W(I)}$ (resp. *dans* $\mathrm{Fib}_{W(I)}$), *il* suffit *que pour tout s dans* $\mathrm{Ob}\, I$, $\alpha_s^*(f)$ *soit dans* Cof_W (resp. *dans* Fib_W). *Itou (pour mémoire) pour une condition pour que $f \in W(I)$ – là il faut et il suffit que $\alpha_s^*(f) \in W \;\forall\, s \in \mathrm{Ob}\, I$.*

Nous allons appliquer ce même lemme pour établir les inclusions

$$C(I)_* \;\subset\; W(I), \qquad (C(I) \cap W(I))_* \;\subset\; \mathrm{Fib}_{W(I)}.$$

Donc il suffit de voir que

$$\begin{aligned} \alpha_s^*(C(I)_*) \;&\subset\; C_* && (\subset\; W) \\ \alpha_s^*(\underbrace{(C(I) \cap W(I))}_{(C \cap W)(I)}{}_*) \;&\subset\; (C \cap W)_* && (\subset\; \mathrm{Fib}_W), \end{aligned}$$

et plus généralement

$$\alpha_s^*(\Phi(I)_*) \ \subset \ \Phi_* \tag{7}$$

pour toute (?) partie Φ de $\mathrm{Fl}(\mathcal{M})$, définissant donc $\Phi(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ terme à terme.

En explicitant la chose, on trouve la condition [page 38]

$$\alpha_{s\,!}(\Phi) \subset \Phi(I) \qquad \text{pour tout } s \in \mathrm{Ob}\, I,$$

pour que ça marche, *i.e.* on veut que

$$i \in \Phi \qquad \Longrightarrow \qquad \alpha_{s\,!}(i) \in \Phi(I), \quad \textit{i.e.} \ \ \alpha_t^*(\alpha_{s\,!}(i)) \in \Phi \ \ \forall\, t \in \mathrm{Ob}\, I.$$

Or le foncteur $\alpha_t^*\alpha_{s\,!}$ s'explicite comme

$$A \longmapsto \underbrace{\mathrm{Hom}_I(t,s) \times A}_{\substack{\text{somme de } \mathrm{Hom}_I(t,s) \\ \text{copies de } A \text{ dans } \mathcal{M}}},$$

donc il faut supposer Φ *stable par sommes directes* (du type envisagé $\mathrm{Hom}_I(t,s) \times A$). C'est bien le cas pour $\Phi = C$ et $\Phi = C \cap W$ et pour des sommes directes *finies,* grâce à (*ii*). On trouve donc le

p. 23

Lemme 2.

a) *Soit I dans* Cat *telle que les sommes directes de type* $\mathrm{Hom}_I(t,s)$ *existent dans* $\mathcal{M}$. *(P. ex. on suppose que I est* finie, *plus généralement que les* $\mathrm{Hom}_I(t,s)$ *sont des ensembles finis ; ou que les "petites" sommes directes existent dans* $\mathcal{M}$.*) Soit* Φ *une partie de* $\mathrm{Fl}(\mathcal{M})$, *stable par ce type de sommes directes, et soit* $\Phi(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ *la partie correspondante. Alors pour tout* $s \in \mathrm{Ob}\, I$, *on a*

[page 39]

$$\alpha_s^*(\Phi(I)_*) \ \subset \ \Phi_*.$$

b) *Supposons que les* $\mathrm{Hom}_I(t,s)$ *soient finis (p. ex. I finie). Alors l'hypothèse de* a) *s'applique pour* $\Phi = C$ *et pour* $\Phi = C \cap W$, *donc la condition* (iii) *p.* 36 *est satisfaite pour la structure* $(W(I), C(I), TC(I), F(I), TF(I))$ *dans* $\mathcal{M}(I)$. *Même conclusion sans hypothèse de finitude sur les* $\mathrm{Hom}_I(t,s)$, *quand on suppose* $\mathcal{M}$ *stable par "petites" sommes directes, et C et W stables également par de telles sommes directes.*

Commentary · Exposition [ms. 38]

Evaluation transports the lifting problem

The copower formula is the essential calculation in the two lemmas. Let s be an object of I. The left adjoint to evaluation at s is

$$(\alpha_{s!}A)_t = \coprod_{\mathrm{Hom}_I(t,s)} A.$$

For an arrow $t' \to t$, precomposition sends $\mathrm{Hom}_I(t,s)$ to $\mathrm{Hom}_I(t',s)$ and defines the presheaf restriction map. A natural transformation $\alpha_{s!}A \to X$ is uniquely determined by the map from the summand indexed by 1_s to X_s; all other summands are obtained by restriction. This proves the adjunction and fixes the direction of the Hom set.

Suppose Φ is stable under the copowers appearing in this formula. Then $\alpha_{s!}(i) \in \Phi(I)$ for every $i \in \Phi$. For $p \in \Phi(I)_*$, a lifting square from i to p_s transposes to a square from $\alpha_{s!}i$ to p. Its diagonal transposes back to the required lift. Hence

$$\alpha_s^*(\Phi(I)_*) \subset \Phi_*.$$

The statement on p. 37 for an arbitrary class, accompanied by a question mark, therefore needs exactly the additional closure used in Lemma 2. We are not assuming that evaluation itself preserves right orthogonals without an adjoint test.

Finite copowers require no new hypothesis for C or TC. For maps $i : A \to B$ and $j : A' \to B'$, the coproduct map factors as

$$A \amalg A' \longrightarrow B \amalg A' \longrightarrow B \amalg B',$$

where each factor is a pushout of one of the original maps. The empty copower is an identity on the initial object. Induction proves the finite statement. For arbitrary copowers one must either assume their stability or obtain it from transfinite composition by adjoining the summands in a well-order. The required colimits must exist.

Finally, finite limits and colimits of presheaves are computed objectwise. A double pushout diagram testing a $W(I)$-cofibration evaluates to the corresponding test in $\mathcal{M}$, and an objectwise weak equivalence remains one after all these tests. This proves Lemma 1 and its dual. Combining it with the lifting argument proves all the inclusions on p. 36 under the finite-Hom hypothesis, while deliberately leaving factorization for the next stage.

NB Vu le scholie plus haut, il est parfaitement raisonnable de se borner au besoin aux catégories I *finies* – vu que la catégorie de ces I est stable par les produits $I \longmapsto I \times \Psi$, $I \longmapsto I \times \Phi$.

Il reste, pour être heureux, à vérifier la condition de factorisation ⓘ pour (C, TF), (TC, F), mais aussi et surtout pour les $(C(I), TF(I), TC(I), F(I))$. C'est dire que de partir même d'une structure de Quillen close (pays de cocagne) ne nous avance guère à lui tout seul, faute de [page 40] savoir que l'axiome de factorisation se conserve, en passant de $(\mathcal{M}, W, \ldots)$ à $(\mathcal{M}(I), W(I), \ldots)$.

Commentary · Examples [ms. 39]

A complete calculation for the walking arrow

The formulas become especially transparent when $I = [1]$, with its arrow $0 \to 1$. A presheaf is a map $X_1 \to X_0$. Let Φ contain identities and be stable under finite coproducts, and put $R = \Phi_*$. Write $x : X_1 \to X_0$ and $y : Y_1 \to Y_0$ for the structural maps. For a natural transformation $p : X \to Y$, consider the matching map

$$m_p : X_1 \longrightarrow Y_1 \times_{Y_0} X_0.$$

Then p has the right lifting property against every objectwise Φ-map if and only if

$$p_0 : X_0 \to Y_0 \ \in R, \qquad m_p \in R.$$

To prove sufficiency, take a square from an objectwise Φ-map $A \to B$ to p. First lift its component at 0 against p_0, obtaining $B_0 \to X_0$. The component at 1, together with this already chosen map composed with $B_1 \to B_0$, gives a lifting square against m_p. A lift provides $B_1 \to X_1$ and enforces precisely the naturality relation. Thus the two component lifts fit together.

For necessity, evaluation at 0 and its left adjoint show $p_0 \in R$. To test m_p against $i : A \to B$ in Φ, write a square as maps $a : A \to X_1$, $b_1 : B \to Y_1$, and $b_0 : B \to X_0$ with

$$p_1 a = b_1 i, \qquad xa = b_0 i, \qquad p_0 b_0 = y b_1.$$

Use the map of arrow diagrams

$$(A \xrightarrow{i} B) \longrightarrow (B \xrightarrow{1} B),$$

whose components are i at 1 and 1_B at 0. This is objectwise in Φ. A lift of its induced square against p is exactly a solution to the original square against m_p. This proves necessity.

Apply the calculation in sets with Φ the injections and R the surjections, assuming choice. Let $E_1 = \{a\}$, $E_0 = \{0, 1\}$, and let $E_1 \to E_0$ send a to 0. The transformation from E to the terminal arrow diagram is surjective at both objects. Its matching map is the non-surjective map $\{a\} \to \{0, 1\}$, so it is not in $\Phi(I)_*$. Explicitly, a proposed lift choosing the element 1 at vertex 0 cannot be extended to vertex 1. This is a failure of naturality, despite the existence of lifts at each object separately.

For the injective model structure, taking $\Phi = C$ gives the characterization of trivial fibrations, and taking $\Phi = TC$ gives the characterization of fibrations. It follows in particular that an injectively fibrant arrow has fibrant target and a fibration as its structural arrow. The general admissible-edge result later in the manuscript is an extension of this relative lifting calculation, rather than a consequence of pointwise fibrancy alone.

C'est ici que le théorème de factorisation du no. 1 doit s'avérer providentiel. Mais pour pouvoir l'appliquer aux couples $(C(I), C(I)_*)$, $(TC(I), TC(I)_*)$, il faut (en plus de l'accessibilité de $\mathcal{M}$, ce qui est une condition fort anodine) des conditions de stabilité très fortes pour $C(I)$, $TC(I)$ ([26]). À vrai dire, il reste la stabilité de C et de $C \cap W$ par limites inductives ordinales (ce qui impliquera la même stabilité pour les $C(I)$, $(C \cap W)(I) = C(I) \cap W(I)$).

À dire vrai, je vois peu de chances de trouver une structure de Quillen faible (C, TC, F, TF) associé à W, et satisfaisant toutes les conditions dites, y compris celle de stabilité de C et de $C \cap W$ par $\varinjlim$ ordinales, sans supposer que W elle-même ne soit déjà stable par (petites) limites inductives filtrantes.

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On peut songer à utiliser une factorisation *fonctorielle* des flèches dans $\mathcal{M}$ en

$$\begin{aligned} f = pi \quad \text{avec} \quad & i \in TC, \quad p \in F \\ \text{ou} \quad & i \in C, \quad\ \ p \in TF, \end{aligned}$$

Commentary · Later developments [ms. 40]

The factorization problem remains a coherence problem

The final paragraph should be read as a choice of a sufficient strategy, not as a necessity theorem about weak equivalences. Transfinite closure of cofibrations and trivial cofibrations is part of the small-object method; closure of every weak equivalence under every filtered colimit is a convenient, stronger condition from which Grothendieck hopes to obtain it. Nothing on this page proves that such closure of W is necessary for diagram model structures.

The later injective existence theorem in the combinatorial setting answers the construction problem with explicit generation hypotheses; its precise application belongs to the discussion of manuscript pp. 42–44. There is also a different route through weaker diagram-compatible structures: Cisinski (2010, Theorem 6.22) proves stability under presheaf categories for homotopically complete model categories in his sense; the theorem also covers homotopically complete and cocomplete weak model categories. These are two ways to obtain the coherence which the walking-arrow calculation detects. Neither licenses treating the pointwise right class as the required right lifting class.

p. 24

pour déduire une factorisation similaire pour les flèches f de $\mathcal{M}(I)$, en

$$\begin{aligned} f = pi \quad \text{avec} \quad & i \in TC(I), \quad p \in F(I) \\ \text{ou} \quad & i \in C(I), \quad\ \ p \in TF(I), \end{aligned}$$

où $C(I)$, $TC(I)$, $F(I)$, $TF(I)$ sont ici définis via C, TC, F, TF "argument par argument". Mais l'ennui, c'est que en procédant de cette façon naïve, on n'aura plus nécessairement $C(I) \longleftrightarrow TF(I)$, $TC(I) \longleftrightarrow F(I)$. Il faut considérablement rapetisser $TF(I)$ et $F(I)$, pour avoir une propriété RLP par rapport à $C(I)$ resp. $TC(I)$.

Donc je ne vois guère d'autre moyen de s'en tirer, dans ce contexte général, qu'en supposant la stabilité de W par $\varinjlim$ filtrantes (ce qui n'est après tout pas très méchant).

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Commentary · Exposition [ms. 41]

Natural factorizations and natural fillers

26. Plus les *petits* systèmes de générateurs pour C, TC.

Grothendieck has reached the obstruction which makes a theory of diagrams more than a repetition of the theory of objects. A functorial factorization in $\mathcal{M}$ does produce a factorization of every natural transformation. Its middle objects and their transition maps fit together because the original factorization is functorial on the arrow category. But a choice of a filler for each component of a lifting problem need not commute with the transition maps. The missing datum is a compatible family of fillers, and the factorization functor supplies no such family.

The walking-arrow calculation in the commentary at manuscript pp. 38–39 makes this failure completely explicit. For diagrams $X_1 \to X_0$ and $Y_1 \to Y_0$, right lifting against all objectwise maps in a left class L requires both

$$X_0 \longrightarrow Y_0 \in R, \qquad X_1 \longrightarrow Y_1 \times_{Y_0} X_0 \in R, \qquad R = L_*.$$

The second condition remembers the compatibility of the lift with the arrow. Requiring only $X_1 \to Y_1 \in R$ forgets it. This is the first matching-object condition of the later Reedy formalism; the general condition appears in Hovey (1999a, Definition 5.2.2 and Theorem 5.2.5).

There are consequently two different ways to organize the model structure of a diagram category. In the injective structure the cofibrations and weak equivalences are objectwise, and fibrations are defined by a global right lifting property. In the projective structure the fibrations and weak equivalences are objectwise, and the cofibrations carry the compatibility condition. The manuscript is pursuing the injective structure. Its decision to keep $C(I)$ and $TC(I)$ objectwise therefore forces the replacement of the naive classes of fibrations by the smaller orthogonal classes $(TC(I))_*$ and $C(I)_*$.

This choice has a purpose beyond obtaining factorizations. Restriction along any functor preserves the two objectwise left classes. Its right adjoint will therefore preserve the two right classes, which is exactly the input needed to construct homotopy right Kan extensions. Thus the apparent asymmetry of the construction already prepares the three variances of the next section.

Bon gré, mal gré, on revient à la situation envisagée dans le paragraphe 2. On va réénoncer l'essentiel de ce qui a été pressenti alors :

Théorème 2 (encore un peu conjectural). *Soit $(\mathcal{M}, W)$ une catégorie de modèles, avec W (comme toujours) faiblement saturée, mais de plus on suppose $\mathcal{M}$ stable par petites $\varinjlim$ et par $\varprojlim$ finies, et W stable par $\varinjlim$ filtrantes, enfin $\mathcal{M}$ accessible. Considérons les conditions suivantes :*

a) *Il existe une structure de Quillen faible pour W, exacte à gauche, close à droite, (C, TC, F, TF).*

a′) *Comme* a)*, mais avec de plus C, TC clos à gauche (*i.e. *on a une structure de Quillen faible close), et C, TC de la forme $\widetilde{C_0}$, $\widetilde{TC_0}$, où C_0 et TC_0 sont petits.*

a″) *Il existe $C_0 \subset \mathrm{Cof}_W$, $TC_0 \subset W^{\mathrm{univ}}$, parties petites, telles que $C_{0*} \subset W$, $(TC_0)_* \subset \mathrm{Fib}_W$.*

b) *On a*

$$(\mathrm{Cof}_W)_* \ \subset \ W_{\mathrm{univ}}, \qquad (W^{\mathrm{univ}})_* \ \subset \ \mathrm{Fib}_W$$

(d'où résulte, Cof_W et W^{univ} étant clos à gauche, que

$$(W_{\mathrm{univ}})^* \subset \mathrm{Cof}_W, \qquad (\mathrm{Fib}_W)^* \subset W^{\mathrm{univ}} \qquad).$$

[page 43]

Ces conditions impliquent que pour toute I dans Cat*, il existe une structure de Quillen faible exacte à gauche et close pour $W(I)$ dans $\mathcal{M}(I)$. Une telle structure se définit, en termes d'une structure similaire C, $TC = C \cap W$, $F = (TC)_*$, $TF = C_*$ pour W, par*

$$C(I) \ = \ \Big\{ f \in \mathrm{Fl}(\mathcal{M}(I)) \ \Big| \ \alpha_s^*(f) \in C \ \ \forall\, s \in \mathrm{Ob}I \Big\},$$

puis

$$\begin{aligned} TC(I) \ &= \ C(I) \cap W(I) \ = \ \Big\{ f \in \mathrm{Fl}(\mathcal{M}(I)) \ \Big| \ \alpha_s^*(f) \in TC \ \ \forall\, s \in \mathrm{Ob}I \Big\} \\ F(I) \ &= \ (TC(I))_* \\ TF(I) \ &= \ C(I)_* \,. \end{aligned}$$

p. 25

Corollaire-Scholie. *Ces conditions impliquent que pour tout I dans* Cat, $(\mathcal{M}(I), W(I))$ *donne lieu à une bonne théorie des fibres et cofibres $W(I)$-homotopiques, avec carrés $W(I)$-cartésiens $W(I)$-exacts, et carrés $W(I)$-cocartésiens $W(I)$-exacts – donc avec des suites* exactes *longues de cosuspension et de suspension, dans le cas où $\mathcal{M}$, donc aussi $\mathcal{M}(I)$, est une catégorie avec objet neutre.*

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Commentary · Later developments [ms. 43]

The injective existence theorem and its precise hypothesis

Grothendieck wants the factorization theorem to work uniformly after adding any small indexing category. One part of this proposal is now a standard existence theorem: a combinatorial model category has injective and projective model structures on each small diagram category. In the terminology of the present pages, a closed structure exact on the left is already an ordinary closed model structure, and condition a' supplies the two sets of generators. Since an accessible cocomplete category is locally presentable, a' places us in the combinatorial situation.

For the injective construction, the set of generators on the diagram category is the substantive point. One cannot simply use the free diagrams on the generators at each object: those give the projective left class. A theorem of Makkai and Rosický (2014a, Corollary 3.4, author preprint of 21 November 2013) instead says that the pointwise left class associated with a set-generated, retract-closed cellular structure on a locally presentable category is again set-generated. Apply it separately to C and TC. The two resulting small-object factorizations have left classes $C(I)$ and $TC(I)$ and right classes $C(I)_*$ and $(TC(I))_*$.

Here is why the compatibility of these two factorizations survives. Evaluation takes either right class into its corresponding right class in $\mathcal{M}$: its left adjoint is the copower by the representable presheaf, and copowers preserve C and TC. Consequently $C(I)_* \subset W(I)$, while an element of $(TC(I))_*$ which belongs to $W(I)$ is objectwise in TF. To obtain the reverse lifting assertion without confusing componentwise and natural fillers, factor such a map f using $(C(I), C(I)_*)$, say $f = pi$. The maps p and f are weak equivalences, so $i \in C(I) \cap W(I) = TC(I)$. The lifting property of f against i exhibits f as a retract of p. Hence $f \in C(I)_*$. This proves the second trivial-class equality, and the model axioms follow.

Properness also passes to these structures. Pushouts and pullbacks are objectwise; injective cofibrations are objectwise cofibrations, and injective fibrations are objectwise fibrations by the preceding evaluation argument. The base properness axioms therefore apply at every object. These conclusions are recorded in Barwick (2007, Theorems 1.17 and 1.19, Propositions 1.21–1.22); the numbering here is that of the 2007 preprint.

This settles the diagram construction under a'. It does not establish every proposed recognition implication involving a, a'' or b. Closure under filtered colimits alone is not the solution-set condition in a model-structure recognition theorem. The standard Smith theorem requires an accessible, accessibly embedded class of weak equivalences in the arrow category, as well as a specified set of generating cofibrations; see Barwick (2007, Proposition 1.7). No assertion that the whole conjectural theorem has been proved is justified by citing the injective existence theorem. Conversely, once a combinatorial model structure is given, its diagram structures do not require preservation of *every* filtered colimit by the weak equivalences. That stronger hypothesis belongs to Grothendieck's proposed route to recognition.

Je m'occuperai de la démonstration du théorème précédent, ainsi que du théorème de factorisation du paragraphe 1, quand j'aurai mes notes sur les catégories et foncteurs accessibles et filtrations cardinales. En attendant, j'ai envie déjà d'expliciter un peu plus le formalisme auquel on parvient moyennant les hypothèses faites ci-dessus. (Je suis à peu près sûr tout au moins que la condition a'') suffit à impliquer toutes les autres. Et c'est bien sous la forme a'') que ces conditions vont se vérifier en pratique, je présume, notamment dans le cas où $\mathcal{M} =$ Cat.)

Commentary · Exposition [ms. 44]

The variance is fixed by presheaves

Grothendieck now turns the diagram model structures into operations. Throughout these pages

$$\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M}), \qquad u : I' \longrightarrow I.$$

This convention determines the comma categories in both pointwise Kan formulas. If $X' : (I')^{\mathrm{op}} \to \mathcal{M}$, then

$$(u_! X')(i) = \operatorname*{colim}_{(i \downarrow u)^{\mathrm{op}}} X'(i'),$$
$$(u_* X')(i) = \lim_{(u \downarrow i)^{\mathrm{op}}} X'(i').$$

The manuscript writes $i\backslash I'$ and I'/i, with the contravariant convention understood. Making the opposites explicit explains the otherwise surprising occurrence of an *initial* object in footnote 27: it becomes terminal in the indexing category of the colimit. It also prevents interchanging the two comma categories when passing to modern covariant notation.

The lifting argument behind Proposition 2 is a precise form of the principle that adjunction transfers extension problems. Suppose $L : \mathcal{A} \rightleftarrows \mathcal{B} : R$ and let $i : A \to A'$ and $p : B \to B'$. By the adjunction, the two horizontal maps in a square from Li to p correspond to the two horizontal maps in a square from i to Rp. Naturality of the adjunction bijection identifies the commutativity conditions. A diagonal $LA' \to B$ corresponds to a diagonal $A' \to RB$, and each of the two triangle identities required of a filler is carried to the corresponding identity. The sets of fillers are therefore naturally bijective. In particular

$$Li \perp p \quad \Longleftrightarrow \quad i \perp Rp.$$

Applying this first to $u^* \dashv u_*$ proves the preservation of F and TF by u_*. Applying it to $u_! \dashv u^*$ proves the two equivalences in (3). This is the same elementary mechanism as Hovey (1999a, Lemma 1.3.4).

For a useful special case, suppose every component of $i \downarrow u$ has an initial object e_α. The left Kan formula reduces to

$$(u_! X')(i) \cong \coprod_{\alpha} X'(e_\alpha).$$

A left orthogonal class is closed under coproducts: a square out of a coproduct is a family of squares, and a family of fillers assembles to a filler. Thus $u_!$ preserves C and TC under this hypothesis, and u^* preserves F and TF. No claim about preservation of general colimits by the weak equivalences is involved. The comma-category hypothesis has replaced that issue by a coproduct calculation.

4 Les variances u^*, u_*, $u_!$ pour $(\mathcal{M}(I), W(I))$ (I variable)

Je suppose les conditions du paragraphe précédent vérifiées pour $(\mathcal{M}, W)$ (cf. théorème p. 42), et choisis une structure de Quillen faible close et exacte à gauche (C, TC, F, TF) pour W, d'où pour tout I dans Cat

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une structure de Quillen faible close et exacte à gauche $(C(I), TC(I), F(I), TF(I))$ pour $W(I)$, dans $\mathcal{M}(I)$, décrite page 43. Soit maintenant

$$u \ : \ I' \longrightarrow I$$

une flèche dans Cat, d'où une flèche

$$u^* \ : \ (\mathcal{M}(I), W(I)) \ \longrightarrow \ (\mathcal{M}(I'), W(I')).$$

On sait, vu que $\mathcal{M}$ est stable par $\varinjlim$ quelconques et aussi (étant de plus accessible) par $\varprojlim$ quelconques, que u^* admet des foncteurs adjoints à gauche et à droite,

$$u_!, \ u_* \ : \ \mathcal{M}(I') \ \longrightarrow \ \mathcal{M}(I),$$

p. 26

et je me propose ici de résumer les propriétés les plus immédiates de u^* et de $u_!$, u_* par rapport aux structures de Quillen faibles envisagées.

Proposition 1 : *Avec les notations précédentes, on a les relations :*

$$u^*(C(I)) \subset C(I'), \qquad u^*(TC(I)) \subset TC(I'). \tag{1}$$

$$u_*(F(I')) \subset F(I), \qquad u_*(TF(I')) \subset TF(I). \tag{2}$$

$$\begin{aligned} u^*(F(I)) \subset F(I') &\iff u_!(TC(I')) \subset TC(I) \\ u^*(TF(I)) \subset TF(I') &\iff u_!(C(I')) \subset C(I). \end{aligned} \tag{3}$$

D'autre part, je ne sais rien dire de positif pour les inclusions

$$\begin{aligned} u_*(C(I')) \ &\subset \ C(I), \ \ u_*(TC(I')) \ \subset \ TC(I) \\ u_!(F(I')) \ &\subset \ F(I), \ \ u_!(TF(I')) \ \subset \ TF(I). \end{aligned}$$

Si ce n'est que pour un u tant soit peu général (u pas un isomorphisme, disons), u_* n'a aucune envie

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d'être compatible avec les structures C, TC (propriété d'exactitude homotopique à droite, moralement, vu que u_* est exact à gauche et en général pas du tout à droite), ni $u_!$ avec les structures F, TF (commentaire dual à donner, $u_!$ est exacte à *droite*, et n'a pas envie d'être "homotopiquement exact à gauche").

Démonstration de (1), (2), (3). Les relations (1) sont évidentes par définition. Les relations (2), (3) le sont à peine moins, vu la propriété soritale générale suivante, dont la place serait au paragraphe 1 :

Proposition 2. *Soit*

$$C' \underset{u^*}{\overset{u_!}{\rightleftarrows}} C$$

un couple de foncteurs adjoints $(u_!, u^*)$*, et soit* $\Phi' \subset \mathrm{Fl}(C')$*,* $\Psi \subset \mathrm{Fl}(C)$*. Alors*

$$(u_!(\Phi') \longleftrightarrow \Psi) \qquad \Longleftrightarrow \qquad (\Phi' \longleftrightarrow u^*(\Psi)).$$

Corollaire. (C'est le même énoncé, avec seulement des notations changées.) *Supposons que* $u^* : C \longrightarrow C'$ *admette un adjoint à* droite u_* *(pas forcément un adjoint à gauche)*

$$C' \underset{u_*}{\overset{u^*}{\rightleftarrows}} C,$$

soit $\Phi \subset \mathrm{Fl}(C)$*,* $\Psi' \subset \mathrm{Fl}(C')$*, alors* p. 27

$$(u^*(\Phi) \longleftrightarrow \Psi') \qquad \Longleftrightarrow \qquad (\Phi \longleftrightarrow u_*(\Psi')).$$

C'est la proposition 2 pour $C'_0 \underset{u_0^*}{\overset{u_{0!}}{\rightleftarrows}} C_0$, avec $C'_0 = C$, $C_0 = C'$, $(u_{0!} : C \longrightarrow C') = u^*$, et $(u_0^* : C' \longrightarrow C) = u_*$.

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Pour les inclusions (2), on utilise le corollaire de la proposition 2 pour les couples $\Phi = TC(I)$, $\Psi' = F(I')$, resp. $\Phi = C(I)$, $\Psi' = F(I')$ `[plutôt` $\Psi' = TF(I')$`]`, et on applique (1) qui donne $u^*(\Phi) \longleftrightarrow \Psi'$, d'où $\Phi \longleftrightarrow u_*(\Psi')$, ce qui n'est autre que (2). Pour les équivalences (3), on utilise la proposition 2, en faisant $\Psi = F(I)$, $\Phi' = TC(I')$ resp. $\Psi = TF(I)$, $\Phi' = C(I')$.

Corollaire 1 (de la proposition 1) : *Supposons $u : I' \longrightarrow I$ telle que pour tout $i \in \mathrm{Ob}I$, la catégorie $i\backslash I'$ soit discrète* ([27]) *(p. ex. I' discrète). Alors on a*

$$\begin{cases} u_!(C(I')) \subset C(I), \;\; u_!(TC(I')) \subset TC(I) \\ u^*(F(I)) \subset F(I'), \;\; u^*(TF(I)) \subset TF(I'). \end{cases}$$

En effet, les relations de la deuxième ligne équivalent (à cause de (3)) à celles de la première. Il faut prouver que si f dans $\mathrm{Fl}\mathcal{M}(I')$ est dans $C(I')$ resp. $TC(I')$, $u_!(f)$ est dans $C(I)$ resp. $TC(I)$, ce qui signifie aussi que $\alpha_s^*(u_!(f))$ est dans C resp. TC, pour tout $s \in \mathrm{Ob}I$. Or le foncteur

$$\alpha_s^* u_! \; : \; \mathcal{M}(I') \longrightarrow \mathcal{M}$$

s'explicite par la formule bien connue

$$\alpha_s^*(u_!(X')) \overset{\text{déf}}{=} u_!(X')(s) \simeq \varinjlim_{i' \in s\backslash I'} X'(i').$$

Comme

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$s\backslash I'$ est discrète, on a donc l'isomorphisme fonctoriel en X' dans $\mathcal{M}(I')$

$$\alpha_s^*(u_!(X')) \simeq \coprod_{(s', x : s \longrightarrow u(s')) \text{ dans } \mathrm{Ob}(s\backslash I')} X'(s'),$$

donc si $f : X' \longrightarrow Y'$, $\alpha_s^*(u_!(f))$ s'explicite comme une somme directe dans $\underline{\mathrm{Fl}}(\mathcal{M})$ de flèches de la forme

$$f_{s'} \; : \; X'(s') \longrightarrow Y'(s'),$$

lesquelles sont dans C resp. TC puisque f est dans $C(I')$ resp. $TC(I')$. Comme C, TC sont p. 28
stables par sommes directes quelconques (C, TC étant clos à gauche), on gagne.

Corollaire 2. *Soit I dans* Cat, *et $f \in \mathrm{Fl}(\mathcal{M}(I))$. Pour que f soit dans $F(I)$* (resp. *dans $TF(I)$), il* faut *que $\forall\, s \in \mathrm{Ob}I$, $f_s : X(s) \longrightarrow Y(s)$ le soit.* (Mais cette condition n'a pas du tout l'air d'être suffisante !) *En particulier, soit $X \in \mathrm{Ob}\mathcal{M}(I)$, alors $X \in \mathrm{Ob}\mathcal{M}(I)_f \Longrightarrow X(s) \in \mathrm{Ob}\mathcal{M}_f \;\; \forall\, s \in \mathrm{Ob}I$.*

Commentary · Examples [ms. 48]

Ordinary limits and the two homotopical exactness properties

Grothendieck's hesitation on pp. 49–50 identifies a necessary distinction. A right Kan extension can preserve fibrations and trivial fibrations without preserving weak equivalences between arbitrary diagrams. Its derived functor is computed on fibrant diagrams, and it is this functor which preserves homotopy limits. The distinction already matters for a finite poset.

Work in simplicial sets, and let K be the nerve of the groupoid with two objects and exactly one morphism between every ordered pair. The simplicial set K is Kan and contractible. Consider the cospan

$$* \xrightarrow{0} K \xleftarrow{1} *,$$

where the two maps select distinct vertices. Its strict pullback is empty. Collapsing K to a point gives an objectwise weak equivalence from this cospan to the constant one-point cospan, whose strict pullback is a point. Hence the ordinary limit functor does not preserve objectwise weak equivalences, even though every value of both diagrams is fibrant. The square formed by this weak equivalence over the identity of the terminal diagram is homotopy cartesian; its image under the strict limit is not, since it would require the map from the empty space to the point to be a weak equivalence.

27. Plus généralement, il suffit que toute composante connexe Z_α de $i\backslash I'$ ait un objet initial e_α, de sorte que $\varinjlim_{i\backslash I'} X'(i') \simeq \coprod_{\alpha \in \pi_0(i\backslash I')} X'(e_\alpha)$. C'est le cas notamment si $I' \longrightarrow I$ est un isomorphisme local, p. ex. une immersion ouverte.

Fibrancy of values has not imposed the matching condition. The homotopy pullback of the first cospan is the space of paths from 0 to 1 in K, which is contractible, as it must be.

There is an equally concrete colimit example. Form the simplicial span

$$\Delta^1 \longleftarrow \partial\Delta^1 \longrightarrow \Delta^1$$

using the two boundary inclusions. Its pushout is a simplicial circle. Collapsing both intervals gives an objectwise weak equivalence to $* \leftarrow \partial\Delta^1 \rightarrow *$, whose strict pushout is a point. The first span consists of cofibrations and calculates the homotopy pushout; the second does not. Thus objectwise cofibrancy of all simplicial sets does not make every span a cofibrant diagram for computing colimits.

Even after deriving the right Kan extension, its preservation of homotopy pushouts is a different assertion and is generally false. Take u from the discrete two-object category to the terminal category and work with pointed spaces. Then $\mathbf{R}u_*$ is the product. In the category of pairs, the coproduct of $(S^1, *)$ and $(*, S^1)$ is (S^1, S^1). Applying the product gives the torus $S^1 \times S^1$, whereas taking the coproduct after applying the product gives $S^1 \vee S^1$. The comparison induces

$$\mathbb{Z} * \mathbb{Z} \longrightarrow \mathbb{Z} \times \mathbb{Z}$$

on fundamental groups and is not an isomorphism. Since a pointed homotopy coproduct is a homotopy pushout over the zero object, this supplies the finite-poset example requested in the manuscript.

Restriction has a different behavior. Once the diagram localizations form a derivator, restrictions between shifted derivators preserve homotopy Kan extensions in the independent diagram variable, and hence both cartesian and cocartesian squares; this is Groth (2013, Proposition 2.5). The two claims about restriction are therefore compatible. The asymmetry belongs to a general left or right Kan extension, and disappears in stable situations for *finite* homotopy limits and colimits, where the two notions coincide.

On applique le corollaire 1 à $\alpha_s : e \longrightarrow I$.

Scholie. La proposition 1 (relation (1)) montre que le foncteur u^* transforme carrés W-cocartésiens en itou, suites exactes longues de suspension en itou – c'est "l'exactitude à droite homotopique" de u^*. On s'attend à ce qu'il en soit de même pour les carrés cartésiens et les suites exactes

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longues de cosuspension, mais pour le prouver il faudra un travail, qu'on fera dans la suite.

Les relations (2) montrent que le foncteur u_* transforme carrés W-cartésiens en itou, donc suites exactes longues de cosuspension en itou. Je ne m'attends *pas* à ce qu'il en soit de même en général pour les carrés W-cocartésiens – *i.e.* que u_* soit "homotopiquement exact à droite", et non seulement à gauche. Mais il faudrait au moins que je fasse des exemples (avec I', I des catégories finies ordonnées).

Quant à $u_!$, la proposition ne dit rien de bien palpable à son sujet. Néanmoins on s'attend à ce que $u_!$ transforme carrés W-cocartésiens en itou – c'est la moindre des choses ! Ou peut-être je fais erreur, en confondant les foncteurs u_*, $u_!$ envisagés ici, avec les foncteurs similaires

$$\mathrm{HO}_W(I') \underset{u_*^W}{\overset{u_!^W}{\rightrightarrows}} \mathrm{HO}_W(I),$$

adjoints à gauche resp. à droite de

$$\mathrm{HO}_W(I') \xleftarrow{u_W^*} \mathrm{HO}_W(I),$$

induit par

$$u^* \ : \ (\mathcal{M}(I), W(I)) \ \longrightarrow \ (\mathcal{M}(I'), W(I')).$$

Mais en général u_*, $u_!$ ne sont pas compatibles

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avec les localiseurs $W(I')$, $W(I)$, donc ils ne doivent pas être confondus avec les foncteurs $u_!^W$, u_*^W. Donc j'ai été peut-être hâtif, dans ce que j'ai dit au sujet de u_* et de $u_!$. Je serai obligé d'étudier la situation de très près, quand j'essayerai de construire les foncteurs u_*^W et $u_!^W$. Pour le moment, je n'ai encore aucune assurance que les conditions envisagées pour $(\mathcal{M}, W)$ suffisent à assurer l'existence de u_*^W, $u_!^W$ – pas même pour I, I' des ensembles ordonnés finis !

Corollaire 1. *Soit $X \in \mathcal{M}(I)_{\mathrm{f}}$, alors pour toute $\alpha : s \longrightarrow t$ dans I qui est* [un] monomorphisme, *$X(\alpha) : X(t) \longrightarrow X(s)$ est dans F.*

(On applique [le] corollaire 1 (p. 47) à $\tilde{\alpha} : \mathbf{\Delta}_1 \longrightarrow I$.) p. 29

Proposition 3. ([28]) *Soit X dans $\mathcal{M}(\mathbf{\Delta}_1)$, $X = (X_0 \xleftarrow{\alpha} X_1)$. Alors X est dans $\mathcal{M}(\mathbf{\Delta}_1)_{\mathrm{f}}$ si et seulement si X_0, X_1* [sont] *dans $\mathcal{M}_{\mathrm{f}}$ et $\alpha \in F$.*

Corollaire 2. *Soit $I' \overset{u}{\hookrightarrow} I$* [une] immersion ouverte. *Alors pour $X \in \mathrm{Ob}\,\mathcal{M}(I)_{\mathrm{f}}$, $\varprojlim_I X(i) \longrightarrow \varprojlim_{I'} X(i)$ est dans F.*

5 Catégories de modèles quillénisables

Je reviens au cas général d'une catégorie de modèles $(\mathcal{M}, W)$.

Commentary · Exposition [ms. 50]

The two finite lifting arguments

Grothendieck isolates consequences of diagram fibrancy which can be proved before any large existence theorem. The walking-arrow criterion of Proposition 3 is the special case $Y = *$ of the matching-map calculation discussed above: $X = (X_1 \to X_0)$ is injectively fibrant precisely when X_0 is fibrant and $X_1 \to X_0$ is a fibration. The fibrancy of X_1 then follows by composition. This proves the proposition without an appeal to the later lemma cited in footnote 28.

The monomorphism hypothesis in Corollary 1 has an exact combinatorial role. Let $\alpha : s \to t$ be monic and let $a : [1] \to I$ classify it. For any object i, the category $i \downarrow a$ has objects given by maps $i \to s$ or $i \to t$. A map of the first kind is joined to its composite with α; a map of the second kind has at most one such predecessor because α is monic. Thus every component is either a singleton or a single arrow, and has an initial object. Footnote 27 shows that a^* preserves injective fibrancy. Applying Proposition 3 to a^*X gives $X(t) \to X(s) \in F$. No assumption that I is itself a poset is needed.

For the restriction of limits, let $j : J \hookrightarrow I$ be a sieve, the presheaf convention for the manuscript's open immersion, and let X be injectively fibrant. We prove directly that

$$\lim_I X \longrightarrow \lim_J j^* X$$

has the right lifting property against TC. Take a square against $a : A \to B \in TC$. The upper map is a cone from A to X over I, while the lower map is a cone from B over J extending it. Since j is a sieve, left extension $j_!$ is extension by the initial object outside J. The two cones combine to a map into X from

$$T = (\mathrm{const}_I\, A) \amalg_{j_! \,\mathrm{const}_J\, A} j_! \,\mathrm{const}_J\, B.$$

The canonical map $T \to \mathrm{const}_I\, B$ is the identity at objects of J and is a elsewhere. It belongs to $TC(I)$. The fibrancy of X supplies a lift $\mathrm{const}_I\, B \to X$, which is exactly the missing cone. This proves the corollary whenever the displayed limits and the required finite colimits exist. The construction exhibits the compatibility information in the lifting problem instead of suppressing it in a pointwise calculation.

Définition 1. On dit que $(\mathcal{M}, W)$ est *quillénisable* s'il existe une structure de Quillen faible *exacte* (C, TC, F, TF) pour $\mathcal{M}$.

C'est là une condition autoduale, mais que je ne sais vérifier en pratique que moyennant l'exactitude soit à gauche, soit à droite – donc par une condition plus forte qui détruit la

28. Comparer avec le lemme p. 63 et sa démonstration (plus exactement, le corollaire du lemme).

symétrie. C'est pour la sauvegarder dans la mesure du possible que j'introduis la définition précédente, ainsi que la

Définition 2. Soit, en plus de $(\mathcal{M}, W)$, Diag $\subset$ Cat une sous-catégorie pleine de Cat. On dit que W est *quillénisable sur* Diag, si pour tout I dans Diag, $(\mathcal{M}(I), W(I))$ est quillénisable. [page 51]

En pratique, j'utiliserai cette définition dans le cas où Diag = Cat, ou alors, qu'il satisfasse au moins les propriétés de stabilité utiles pour le domaine d'un dérivateur : stable par $\varprojlim$ finies, par sommes finies – eventuellement par $\varinjlim$ finies, etc ([29]).

Commentary · Exposition [ms. 51]

What quillenizability promises

Grothendieck wants a condition on the relative category $(\mathcal{M}, W)$, independent of the auxiliary choices of cofibrations and fibrations. Quillenizability says that some exact weak structure exists. Quillenizability on a domain of diagrams repeats this existence statement for every I in the domain. It does not itself choose compatible structures. The exact-completion argument in the commentary to pp. 32–34 does allow one to enlarge the two trivial classes to $C \cap W$ and $F \cap W$, recovering the original, possibly nonclosed Quillen axioms without changing C, F or W. Thus exactness has real force even before closure is imposed. The adjective "weak" must nevertheless retain the definition from p. 27. It is not a synonym for the later notions of a semi-model category or of a weak model category.

There are three separate questions. First, can one form the localization $\mathcal{M}(I)[W(I)^{-1}]$? Second, does it carry the fibre and cofibre operations constructed by replacement? Third, do restriction and Kan extension relate these categories with the exactness and pointwise formulas required of a derivator? Existence of localizations answers the first question, an exact weak structure addresses the second, and the uniform constructions aim at the third. Treating these three tasks separately explains why the author returns to examples from ordinary homological algebra rather than simply assuming arbitrary colimits from the outset.

His quotation marks around "triangulée" are also essential. In a pointed unstable theory there are suspension and loop functors, and long exact sequences obtained from fibres or cofibres. The suspension need not be an equivalence, the homotopy category need not be additive, and these data are not a triangulated category in Verdier's sense. Stable derivators do yield canonical triangulations, with exact restriction and Kan extension functors; the stability hypothesis is the additional mathematical content in Groth (2013, Theorem 4.16). For complexes, shifting is invertible on the unbounded derived category, whereas nonnegative chain complexes provide a pointed but generally unstable example. This is precisely the distinction the next example needs.

Commentary · Exposition [ms. 51]

Quillenisability and the strength of its qualifiers

Grothendieck is looking for hypotheses on a category with weak equivalences which survive passage to diagrams. The point is not merely to construct another presentation of its homotopy category. A derivator must retain enough of the original category to restrict a diagram, replace it, and eventually take its homotopy limit. His succession of adjectives records different parts of this programme; reading them all as the modern expression "model category" would erase the question he is asking.

Throughout these pages we use his variance:

$$\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M}), \qquad W(I) = \{f \mid f_i \in W \text{ for every } i\}.$$

Thus the ordered category $0 \to 1$ indexes an arrow $X_1 \to X_0$ in $\mathcal{M}$. A terminal object of I computes the ordinary limit of such a diagram. This explains several formulas which would have the opposite orientation with covariant diagrams.

The glossary supplied with the chapter makes the terminology precise. A weak Quillen structure has the two factorizations (C, TF) and (TC, F), their lifting properties, and $TC \subset C \cap W$, $TF \subset F \cap W$. The axiom called exactness is

$$W = TF \circ TC.$$

29. Je supposerai que Diag contient tout au moins les catégories provenant d'ensembles ordonnés finis.

Exactness on the left is the stronger-looking equality $TC = C \cap W$; exactness on the right is $TF = F \cap W$. Closedness says that the four classes are determined by their respective lifting properties. The additional axiom $\mathbf{M}''$ concerns preservation of weak equivalences under base change along fibrations and cobase change along cofibrations. In a usual model structure these are the properness requirements. Finally, **M**0 asks only for finite limits and finite colimits. Small completeness must therefore be supplied when an infinite Kan extension is used.

There is a useful simplification precisely when *closedness and exactness are both present*, as they will be in §6. It is the retract argument, not an extra smallness theorem. Suppose $f \in C \cap W$. Exactness supplies $f = pi$ with $i \in TC$ and $p \in TF$. The left lifting property of f against p makes f a retract of i in the arrow category. A class defined by a left lifting property is closed under retracts, so $f \in TC$. For $f \in F \cap W$, use the same factorization and lift i against f; this exhibits f as a retract of p, and hence puts it in TF. Consequently

$$TC = C \cap W, \qquad TF = F \cap W$$

under the closed exact hypotheses. This application of the familiar retract argument is particularly consequential here: see Hovey (1999a, Lemma 1.1.9). The proof uses neither arbitrary products nor a prior assertion that W is closed under retracts.

Quillenisability by itself, in Definition 1, does not include closedness. We therefore do not substitute the stronger hypothesis into §5. Once a uniform closed exact quillenisability has been assumed in §6, however, the preceding equalities are available in every diagram category. They will allow the apparently separate questions about finite posets, products, towers and group actions to be treated by one argument. Keeping this distinction visible preserves both the breadth of the definition and the actual force of the theorem's hypotheses.

Exemple 1 : Celui des paragraphes 3, 4, où on a dégagé des conditions sur $(\mathcal{M}, W)$, avec $\mathcal{M}$ une "grosse" catégorie et $\mathcal{M}$ accessible, pour que W soit quillénisable sur Cat tout entier. C'est le cas qui me paraît à présent le plus intéressant.

Exemple 2 : Soient $\mathcal{A}$ une catégorie abélienne, et $\mathcal{M} = K^\bullet(A)$ la catégorie des complexes de cochaînes à valeurs dans $\mathcal{A}$, $W = W_\mathcal{A}$ l'ensemble des quasi-isomorphismes, donc

$$\mathcal{M}W^{-1} \;=\; K^\bullet(\mathcal{A})W_\mathcal{A}^{-1} \;\simeq\; D^\bullet(\mathcal{A}), \qquad \text{catégorie dérivée de } \mathcal{A}.$$

On définit aussi $\mathcal{M}^+$, $\mathcal{M}^-$, $\mathcal{M}^\mathrm{b}$, avec les localiseurs induits $W_\mathcal{A}^+$, $W_\mathcal{A}^-$, $W_\mathcal{A}^\mathrm{b}$, et les catégories de fractions $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, $D^\mathrm{b}(\mathcal{A})$. Ou encore, pour I dans Cat, [page 52]

$$\mathcal{M}(I) \;=\; K^\bullet(\mathcal{A}(I))$$

$$W(I) \;=\; \underbrace{W_{\mathcal{A}(I)}}_{\text{quasi-isomorphismes}} \;\subset\; \mathrm{Fl}(K^\bullet(\mathcal{A}(I))), \qquad \text{p. 30}$$

et de même pour $\mathcal{M}^\alpha(I)$, $W^\alpha(I)$ ($\alpha \in \{+, -, \mathrm{b}\}$), du moins *si I est finie.* Il y a aussi les sous-catégories $K^{\geq 0}(\mathcal{A})$ et $K^{\leq 0}(\mathcal{A})$ des complexes de cochaînes resp. de chaînes au sens strict (avec $K^i = 0$ si $i \geq 0$, resp. si $i \leq 0$ `[plutôt si` $i < 0$`, resp. si` $i > 0$`]`). On aura pour I quelconque

$$\begin{cases} K^{\geq 0}(\mathcal{A})(I) \;\simeq\; K^{\geq 0}(\mathcal{A}(I)) \\ W_\mathcal{A}^{\geq 0}(I) \;=\; W_{\mathcal{A}(I)}^{\geq 0}, \end{cases}$$

et de même pour $K^{\leq 0}$, $W^{\leq 0}$.

D'après Quillen page 1.2, $(K^+(\mathcal{A}), W_\mathcal{A}^+)$ est quillénisable si $\mathcal{A}$ a assez d'injectifs, $(K^-(\mathcal{A}), W_\mathcal{A}^-)$ l'est si $\mathcal{A}$ a assez de projectifs. Il ne dit rien pour $(K^\bullet(\mathcal{A}), W_\mathcal{A})$ sans plus – pourtant on fait de l'algèbre cohomologique avec $D^\bullet(\mathcal{A})$ sans plus, et même sans faire d'hypothèse d'existence d'injectifs ou de projectifs. J'ai un peu oublié ce que Verdier a fait à ce sujet – je devrais quand même regarder sa thèse (!), ou bien l'exposé de Hartshorne.

Commentary · Examples [ms. 52]

Two meanings of a bounded diagram of complexes

Grothendieck's qualification "du moins si I est finie" protects an important distinction. We use $\mathrm{Ch}(\mathcal{A})$ for his $K^\bullet(\mathcal{A})$, which here denotes complexes and chain maps, before quotienting by chain homotopy. For unbounded complexes there is a genuine identification

$$\mathrm{Fun}(I^{\mathrm{op}}, \mathrm{Ch}(\mathcal{A})) \cong \mathrm{Ch}(\mathrm{Fun}(I^{\mathrm{op}}, \mathcal{A})).$$

Both sides specify objects X_i^n, differentials, and structure maps satisfying the same commutation relations. Quasi-isomorphisms are objectwise because kernels, cokernels, and hence cohomology in an abelian functor category are computed objectwise. The same argument works for a fixed half-line of degrees, such as $n \geq 0$.

For Ch^+ there are two possible bounds. An object of $\mathrm{Fun}(I^{\mathrm{op}}, \mathrm{Ch}^+(\mathcal{A}))$ has a lower bound depending on i. An object of $\mathrm{Ch}^+(\mathcal{A}(I))$ has one lower bound valid for every i. A finite object set allows us to take the minimum of the finitely many lower bounds, so the two categories agree. An infinite object set need not. For $I = \mathbb{N}$ discrete and a nonzero abelian group A, the family $A[n]$, $n \geq 0$, has one nonzero term in degree $-n$ and gives an immediate counterexample. There are analogous upper-bound and two-sided-bound distinctions.

This also explains footnote 31. In $\mathrm{Ch}^+(\mathbf{Ab})$ take the filtered system

$$X_m = \bigoplus_{0 \leq n \leq m} \mathbb{Z}[n], \qquad X_m \longrightarrow X_{m+1}$$

with the evident inclusions and zero differentials. Its colimit in unbounded complexes has a copy of $\mathbb{Z}$ in every nonpositive degree. It does not belong to Ch^+. Thus the assertion "$K^\bullet(\mathcal{A})$ etc. have the same properties" cannot include bounded-below or bounded-above complexes without a common bound; the manuscript itself flags this point.

More seriously, the difference can destroy an adjoint. Suppose the right adjoint to the constant-diagram functor

$$D^+(\mathbf{Ab}) \longrightarrow \prod_{n \geq 0} D^+(\mathbf{Ab})$$

existed at the family $(\mathbb{Z}[n])_{n \geq 0}$, and call its value P. Applying the adjunction to $\mathbb{Z}[k]$ gives

$$\begin{aligned} H^{-k}(P) &\cong \mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}[k], P) \\ &\cong \prod_{n \geq 0} \mathrm{Hom}_{D(\mathbf{Ab})}(\mathbb{Z}[k], \mathbb{Z}[n]) \\ &\cong \mathbb{Z} \qquad (k \geq 0). \end{aligned}$$

The last equality uses the projectivity of $\mathbb{Z}$: only the factor $n = k$ is nonzero. No bounded-below complex can have these cohomology groups. Therefore this right adjoint does not exist. For a discrete indexing category, the localization of the objectwise bounded complexes is indeed the product of the bounded derived categories: choose a bounded-below injective resolution separately in each component and represent morphisms by chain maps between these resolutions.

Accordingly, the familiar $\mathbf{R}u_* : D^+(\mathcal{A}(I')) \to D^+(\mathcal{A}(I))$ refers to a *uniformly* bounded-below input. It must not be used without comment for the generally larger localization of $\mathrm{Fun}((I')^{\mathrm{op}}, \mathrm{Ch}^+(\mathcal{A}))$. This failure concerns the requested adjoint, not a proof that the latter relative category has no possible weak quillenization. The commentary on Chapter VI, manuscript p. 131, already records a uniform-bound obstruction; here the obstruction is located directly in the proposed domain of a derived Kan extension.

[page 53]

Toujours est-il qu'on a là des exemples de type tout différent de ceux envisagés dans 1°. Ainsi, il n'y a pas à supposer que $\mathcal{M}$ soit stable par petites $\varinjlim$ – $\mathcal{M}$ peut même être une petite catégorie, elle l'est si et seulement si $\mathcal{A}$ elle même l'est. Si $\mathcal{A}$ est stable par (petites) $\varinjlim$ filtrantes, et est accessible (p. ex. a une petite famille génératrice, et les $\varinjlim$ filtrantes sont exactes), alors $K^\bullet(\mathcal{A})$ etc. ([30]) ont les mêmes propriétés, et $(K^+(\mathcal{A}), W_{\mathcal{A}}^+)$ est quillénisable sur

30. !!

Cat tout entier ([31]). (C'est le cas où on sait que $\mathcal{A}$ a assez d'injectifs, et l'algèbre cohomologique se fait surtout à coups de résolutions injectives …) Dans ce cas on sait que les

$$u_* \ : \ D^+(I', \mathcal{A}) \longrightarrow D^+(I, \mathcal{A})$$

existent comme adjoints à droite des

$$u^* \ : \ D^+(I, \mathcal{A}) \longrightarrow D^+(I', \mathcal{A}),$$

mais je ne suis plus trop sûr si on sait définir u_* aussi sur $D(I', \mathcal{A})$ tout entier, et je doute que $u_!$ existe sans hypothèse sur u (genre : dimension cohomologique finie). Sans doute on doit toujours avoir

$$u_! \ : \ D^-(I', \mathcal{A}) \longrightarrow D^-(I, \mathcal{A}),$$

adjoint à gauche de

$$u^* \ : \ D^-(I, \mathcal{A}) \longrightarrow D^-(I', \mathcal{A}). \qquad \text{[page 54]}$$

Commentary · Examples [ms. 53]

A bound on each component is not a bound on the diagram

Grothendieck turns to complexes because they offer a familiar theory of derived coefficients without the large-colimit hypotheses of the preceding sections. His notation $K^\bullet(\mathcal{A})$ denotes complexes and chain maps here. To avoid conflict with the now customary use of $K(\mathcal{A})$ for the chain-homotopy category, we write $\mathrm{Ch}(\mathcal{A})$ for that category of complexes. Localization at quasi-isomorphisms gives the derived category in either convention.

For unbounded complexes the exchange of the two diagram directions is literal:

$$\mathrm{Fun}(I^{\mathrm{op}}, \mathrm{Ch}(\mathcal{A})) \simeq \mathrm{Ch}(\mathrm{Fun}(I^{\mathrm{op}}, \mathcal{A})).$$

Both sides are the same collection of objects, differentials, transition maps and commutativities. Quasi-isomorphisms agree because kernels, cokernels and cohomology in the functor category are computed at each object. The fixed truncation conditions $X^n = 0$ for $n < 0$, or for $n > 0$, also commute with arbitrary diagram categories. The editor's correction to the reversed inequalities on manuscript p. 52 is essential.

The movable bound in Ch^+ behaves differently. A functor into $\mathrm{Ch}^+(\mathcal{A})$ gives a lower bound b_i for each component. An object of $\mathrm{Ch}^+(\mathcal{A}(I))$ requires one bound valid for all components. These conditions agree for a finite set of objects, since one can take $\min_i b_i$, but need not agree otherwise. The same distinction applies to upper bounds and to bounded complexes. This is why the words "si I est finie" belong to the argument, rather than merely to its notation.

Here is the obstruction in its simplest form. Let k be a field and put $X_n = k[n]$, with our cochain convention $X[n]^j = X^{j+n}$. Thus X_n has a single nonzero term in degree $-n$. Each X_n is bounded below. The degreewise product

$$\prod_{n \geq 0} k[n]$$

has nonzero cohomology in every degree $0, -1, -2, \ldots$ and is not bounded below. This is not merely a defect of that chosen complex. If a product P of the family existed in $D^+(k)$, the defining universal property, tested on $k[m]$, would give

$$H^{-m}(P) = \mathrm{Hom}_{D(k)}(k[m], P) \cong \prod_{n \geq 0} \mathrm{Hom}_{D(k)}(k[m], k[n]) \cong k$$

for every $m \geq 0$. No object of $D^+(k)$ has this property. Thus the projection from an infinite discrete category does not have a right adjoint on the theory of *objectwise* bounded-below diagrams with values in $D^+(k)$.

The filtered-colimit warning in footnote 31 has an equally concrete witness. Let $P_r = \bigoplus_{0 \leq n \leq r} k[n]$ and include P_r into P_{r+1}. Each P_r is bounded, whereas their degreewise filtered

31. attention, $K^+(\mathcal{A})$ n'est *pas* stable par $\varinjlim$ filtrantes infinies – donc on ne peut appliquer le tapis des paragraphes 2, 3 directement à K^+.

colimit has nonzero terms arbitrarily far to the left. Hence Ch^{+}, and likewise Ch^{b}, is not closed under arbitrary filtered colimits in Ch. The dual example uses $k[-n]$ to treat Ch^{-}. The unbounded category and the two fixed half-line categories do not have this problem.

Enough injectives do give the familiar bounded-below resolutions, and enough projectives give bounded-above resolutions. What the argument on these pages does not establish is quillenisability on every infinite diagram shape for the category allowing a different bound at every object. The accessibility theorem quoted from the preceding sections cannot be applied directly to that category. We record this as a missing justification, not as a counterexample to quillenisability itself: the product calculation concerns a Kan extension, which a finite-limit model structure is not obliged to possess. Passing to unbounded complexes is the clean way to obtain the unrestricted diagram theory.

p. 31

Mais je ne sais plus trop s'il faut s'attendre à pouvoir définir $u_!$ sur $D(I', \mathcal{A})$ tout entier, en dehors d'hypothèses genre dimension cohomologique finie. La triste vérité, c'est que j'ai ± oublié la théorie des foncteurs dérivés ! (Il faudrait que je demande encore à Jean Malgoire de m'envoyer des textes canoniques.)

Commentary · Later developments [ms. 54]

Unbounded resolutions remove the dimensional restriction

Grothendieck's uncertainty about unbounded derived functors has a precise modern answer for Grothendieck abelian categories. An unbounded complex needs a resolution which is injective as a complex, in the homotopical sense, rather than merely injective in each degree. A complex J is K-injective when $\mathrm{Hom}^{\bullet}(A, J)$ is acyclic for every acyclic complex A. This condition makes homotopy classes of maps into J invariant under quasi-isomorphisms in the source.

Spaltenstein (1988, Theorems C and D) constructed unbounded resolutions for modules and for sheaves of modules. For an arbitrary Grothendieck abelian category, the general existence result is stated and proved in The Stacks Project Authors (2026h, Theorem 19.12.6, tag 079P). One may also equip $\mathrm{Ch}(\mathcal{A})$ with the proper injective model structure whose weak equivalences are quasi-isomorphisms and whose cofibrations are monomorphisms; see Cisinski and Déglise (2009, Theorem 1.2), where the existence theorem is attributed to Beke.

For a small category I, the abelian category $\mathcal{A}(I)$ is again Grothendieck. Exactness of filtered colimits is objectwise. A generating family is obtained by applying the left adjoints to evaluation to a generating family of $\mathcal{A}$; these left adjoints are coproducts indexed by hom-sets. Applying the injective and projective diagram constructions to unbounded complexes consequently gives

$$\mathbf{L}u_! \dashv u^* \dashv \mathbf{R}u_* \quad \text{on} \quad D(\mathcal{A}(I')) \text{ and } D(\mathcal{A}(I))$$

for every functor of small indexing categories. Here "projective diagram structure" refers to the diagram model structure and does not assert that $\mathcal{A}$ has enough projective objects. No finite cohomological dimension of u is required for existence on the unbounded categories.

Finite dimensionality becomes relevant when one asks these functors to preserve a chosen bounded subcategory. This can be seen without any sophisticated example. Let G be cyclic of prime order p, let $k = \mathbb{F}_p$, and consider constant coefficients k over BG. Write $\epsilon = g - 1$ in $k[G] \cong k[\epsilon]/(\epsilon^p)$. The periodic free resolution of the trivial module has differentials alternating multiplication by ϵ and ϵ^{p-1}:

$$\cdots \xrightarrow{\epsilon} k[G] \xrightarrow{\epsilon^{p-1}} k[G] \xrightarrow{\epsilon} k[G] \longrightarrow k \longrightarrow 0.$$

It is exact because the kernel of multiplication by ϵ is the image of multiplication by ϵ^{p-1}, and conversely. After tensoring with k, or applying $\mathrm{Hom}_{k[G]}(-, k)$, every differential is zero. Thus

$$H^{-n}(\mathbf{L}\operatorname{colim}_{BG} k) \cong k, \qquad H^{n}(\mathbf{R}\lim_{BG} k) \cong k \quad (n \geq 0).$$

The left derived colimit of this degree-zero input is not bounded below, and the right derived limit is not bounded above. Both nevertheless exist as unbounded complexes. The source's worries about existence and about boundedness therefore have different resolutions. For an arbitrary abelian category, existence of the Verdier localization alone still does not supply either unbounded resolutions or all these Kan extensions.

Commentary · Later developments [ms. 54]

From a presentation to a homotopy theory of diagrams

The uncertainty about derived direct images brings Grothendieck back to the reason for keeping all indexing categories at once. A construction on one localized category can conceal the choices needed to make it functorial. Passing first to diagrams and then localizing preserves the information on which the construction depends. In particular, a coherent square is an object of the localization of the category of squares; it is not merely a commutative square in the ordinary homotopy category.

Renaudin makes the comparison with Quillen presentations precise for combinatorial model categories. In the published version, the bicategorical localization of left proper combinatorial model categories at Quillen equivalences maps locally equivalently to the 2-category of derivators with adjunctions as morphisms; restricting to derivators of small presentation gives a biequivalence (Renaudin, 2009, Theorems 3.3.2 and 3.4.4). The choice of morphisms is part of this statement. It is not an assertion that every prederivator, or every transformation of prederivators, comes from an arbitrary relative category in this way.

The analogous construction for an infinity-category $\mathcal{C}$ has the particularly transparent form

$$I \longmapsto \mathrm{Ho}\,\mathrm{Fun}(NI^{\mathrm{op}}, \mathcal{C}).$$

Here N is the nerve, and the opposite records the presheaf convention of this chapter. Limits and colimits in $\mathcal{C}$ give the homotopy Kan extensions of this system. Riehl and Verity express the relevant comparison law as a Beck–Chevalley theorem for comma squares, and explain the resulting derivator of a complete and cocomplete quasi-category (Riehl and Verity, 2017, Proposition 5.3.9 and Remark 5.3.10).

This description also clarifies the role of the classes C and F in the preceding calculations. They supply replacements that compute these intrinsic operations. The right replacement for a presheaf is generally more than a list of fibrant values: the transition maps carry compatibility conditions. Grothendieck's arrow example on manuscript page 50 already exhibits this fact. On the other hand, forgetting to derive an ordinary Kan extension destroys precisely the homotopy invariance that the intrinsic description requires. The withdrawal on manuscript pages 49–50 corrects the construction while preserving its objective.

Scholie. Si $(\mathcal{M}, W)$ est quillénisable sur Diag $\subset$ Cat (et même sans qu'il soit quillénisable), il définit un prédérivateur $\mathbf{D}_{\mathcal{M},W}$ sur Diag. On posera parfois

$$\mathrm{HO}_{\mathcal{M},W}(I) \text{ ou } \mathrm{HO}_W(I) = \mathbf{D}_{\mathcal{M},W}(I).$$

L'hypothèse quillénisable implique, sur chacune de ces catégories homotopiques, la structure (relativement faible) "triangulée", provenant des suites *exactes* de suspension et de cosuspension ([32]). (Le caractère quillénisable assure l'exactitude de ces suites …) Les foncteurs $u^* : \mathrm{HO}_W(I) \longrightarrow \mathrm{HO}_W(I')$ sont "homotopiquement exacts" dans le sens des ces structures "triangulées" (non commutatives), *i.e.* transforment suites exactes de suspension resp. de cosuspension en itou. [page 55]

Commentary · Context [ms. 54]

An answer already present in Joyal's letter

The questions about complexes and uniform diagram structures have a particularly close antecedent in the correspondence surrounding *Pursuing Stacks*. In the letter begun on 11 April 1984, Joyal describes Heller's two model structures on simplicial presheaves and explicitly connects them with derived right and left Kan extensions for an arbitrary functor of indexing categories. In the continuation dated 15 April, Theorem 2 states the injective model structure on unbounded complexes in an arbitrary Grothendieck abelian category, with monomorphisms as cofibrations and homology isomorphisms as weak equivalences (Joyal, 1984, pp. 3 and 10).

This locates the present hesitation accurately. The manuscript records a return to an earlier set of questions, not the historical first appearance of the unrestricted complex example. The later

32. Dans le cas où $\mathcal{M}$ est ponctuée par `[un]` objet neutre. Sinon, il y a la structure associée aux carrés cartésiens et cocartésiens. Il serait assez intéressant d'axiomatiser une telle espèce de structure.

proofs by Beke and the constructions of K-injective resolutions make those statements accessible in systematic published forms. The letter is also unusually direct evidence for the role of Heller in the diagram-category approach; it supports that precise connection without requiring us to infer a chronology from similarities between later axiomatizations.

Commentary · Later developments [ms. 54]

Unbounded resolutions answer the derived-image question

Grothendieck's hesitation about unbounded derived images concerns the existence of resolutions on which an ordinary functor can safely be evaluated. A complex of injective objects is not automatically suitable when it is unbounded: what is needed is a condition on the whole complex. A complex J is K-injective when $\mathrm{Hom}^\bullet(A, J)$ is acyclic for every acyclic complex A. Equivalently, maps into J in the homotopy category already compute maps into its image in the derived category.

Spaltenstein's work was already available when these manuscript pages were written. His Theorem C treats unbounded complexes of modules, Theorem D treats sheaves of modules, and Theorem 4.5 gives the latter existence construction; these results should not be retrospectively attributed to the present text (Spaltenstein, 1988). For an arbitrary Grothendieck abelian category the general result is associated with Serpé (2003). A detailed modern statement, including a functorial monomorphic quasi-isomorphism into a termwise injective K-injective complex, is Theorem 19.12.6, Tag 079P, of the Stacks Project (The Stacks Project Authors, 2026g).

The application to right Kan extensions is short once this existence theorem has been supplied. Let $u : I' \to I$ be a functor of small categories and let $\mathcal{A}$ be Grothendieck abelian. The category $\mathcal{A}(I')$ is again Grothendieck: exactness of filtered colimits is checked componentwise, and free diagrams on a set of generators give a set of generators. Restriction u^* is exact, and its ordinary right adjoint u_* exists. If J is K-injective in $\mathrm{Ch}(\mathcal{A}(I'))$, then for any acyclic A in $\mathrm{Ch}(\mathcal{A}(I))$ the ordinary adjunction gives

$$\mathrm{Hom}^\bullet(A, u_*J) \cong \mathrm{Hom}^\bullet(u^*A, J).$$

The right-hand side is acyclic because u^*A is acyclic. Thus u_*J is K-injective. Replacing X' by such a J defines $\mathbf{R}u_*X' = u_*J$ on the entire unbounded derived category, and the same Hom identity proves $u^* \dashv \mathbf{R}u_*$ there.

The corresponding unbounded left Kan extensions also exist for Grothendieck coefficients. The precise model-theoretic input is Beke (2000, Proposition 3.13): unbounded complexes have a cofibrantly generated model structure with monomorphisms as cofibrations and quasi-isomorphisms as weak equivalences. The underlying category is locally presentable, so this is a combinatorial model; its projective diagram structures and the Brown argument supply all left Kan extensions. Neither construction requires a finite cohomological dimension for the indexing functor. Such dimension restrictions instead govern preservation of bounded ranges. Since restriction is exact for the standard degreewise t-structures, its right adjoint preserves lower bounds and its left adjoint preserves upper bounds. Thus the usual statements on globally bounded-below and globally bounded-above diagrams remain meaningful, while a bounded object can acquire cohomology in arbitrarily many degrees on the other side.

For this reason the letter $u_!$ here must be read as the left adjoint to restriction of diagrams. It is not a direct image with proper supports on a space. Those functors have their own geometric hypotheses; importing them would introduce difficulties unrelated to the question asked on pp. 53–54. The example of a finite group below will display the genuine boundedness issue without any such ambiguity.

Quels sont les axiomes des dérivateurs qui sont vérifiés sous la seule hypothèse quillénisable ? **Der** 1 `[voir chapitre I, p. 43]` l'est sans hypothèse aucune sur la catégorie de modèles $(\mathcal{M}, W)$:

$$\mathrm{HO}_W(\coprod_\alpha I_\alpha) \;\simeq\; \prod_\alpha \mathrm{HO}_W(I_\alpha)$$

pour une famille *finie* d'objets I_α dans Diag. L'axiome de localisation **Der** 2 `[voir chapitre I, p. 43]` l'est sous l'hypothèse que W soit *fortement saturé,* et quillénisable. On utilise le fait que toute flèche de $\mathrm{HO}_W(I)$ provient, à isomorphisme près, d'une flèche dans $\mathcal{M}(I)$. (C'était

là un des points qui restait en suspens dans VI, dans la théorie de HOT_W, $W \subset \mathrm{Fl}(\mathrm{Cat})$ un localiseur fondamental …)

Commentary · Exposition [ms. 55]

The two elementary derivator axioms

Grothendieck is now testing how much of the derivator is already present before constructing homotopy Kan extensions. The two assertions about finite coproducts of shapes and detection of isomorphisms have different proofs, and the latter uses a homotopical input which cannot be replaced by saturation alone.

For a finite family of shapes there is an isomorphism of relative categories

$$\mathcal{M}\Big(\coprod_{r=1}^{n} I_r\Big) \cong \prod_{r=1}^{n} \mathcal{M}(I_r)$$

with componentwise weak equivalences. Localization commutes with this finite product. One way to see the point is to separate each arrow $(f_1, \dots, f_n)$ into the n coordinate arrows, inserting identities in the other positions. Coordinate arrows commute, and this commutation remains true after inverting weak equivalences. Finite zigzags in the individual coordinates can therefore be padded with identities and assembled into a finite zigzag in the product. The same operation on the elementary relations between zigzags gives full faithfulness as well as essential surjectivity. The empty family gives the terminal category. Infinite products require an additional argument: one cannot pad zigzags of unbounded length to a single finite length. This is why the statement on the page is deliberately finite.

For detection of isomorphisms, use the representation property of the exact weak structure proved in the commentary to pp. 29–34: every arrow ξ in $\mathcal{M}(I)[W(I)^{-1}]$ is isomorphic in the arrow category to the localization of an actual map $f : X \to Y$. If every evaluation of ξ is invertible, the same is true of every f_i in $\mathcal{M}[W^{-1}]$. Strong saturation of W then gives $f_i \in W$ for all i, hence $f \in W(I)$, and ξ is invertible. The role of representability is to put all components into one honest natural transformation before applying saturation. The commentary on Chapter VI, manuscript p. 130, already explains why saturation by itself is insufficient.

These arguments establish the two stated axioms with their stated qualifications. They do not produce the pointwise Kan-extension formula. In the modern model-category treatment, that remaining structure is available: Groth (2013, Proposition 1.30) proves the strong derivator theorem in the combinatorial case, and his Theorem 1.32 records Cisinski's theorem for model categories in general. The word "model" in that latter statement has the ordinary model-category meaning. The exact-completion lemma above first recovers Quillen's original axioms from the weak structure; closedness, size conventions and the required small limits and colimits must still be tracked when invoking a modern formulation.

L'axiome **Der** 3, nouvelle mouture `[voir chapitre I, pp. 111-112]`, implique un localiseur fondamental $\mathbf{W} \subset \mathrm{Fl}(\mathrm{Cat})$ (pas forcément $\mathbf{W}_\infty$) ([33]). La forme la plus forte serait celle-ci : si

$$u \ : \ I' \longrightarrow I$$

est dans $\mathbf{W}$, on veut que le foncteur u^* induise une *équivalence* [page 56]

$$u^*_{\mathrm{lc}} \ : \ \mathrm{HO}^{\mathrm{lc}}_W(I) \xrightarrow{\sim} \mathrm{HO}^{\mathrm{lc}}_W(I').$$

Mais je doute qu'on soit armé pour vérifier un axiome aussi fort, sans avoir au moins une théorie des opérations $u_!$ ou u_*. On pourrait à tout hasard poser cet axiome en prenant $\mathbf{W} = \mathbf{W}_\alpha \overset{\text{déf}}{=}$ le plus petit localiseur fondamental satisfaisant W (1 à 7) (le plus grand de ces localiseurs étant $\mathbf{W}_\infty$). Pour l'instant, la façon de vérifier cet axiome (en termes de la donnée $(\mathcal{M}, W)$) est pour moi un mystère ± total. (Mais je sais le vérifier pour HOT_W, quand $W \subset \mathrm{Fl}(\mathrm{Cat})$ satisfait W (1, 2, 3, 4, 5, …).) p. 32

33. Il est naturel ici de supposer que $\mathbf{W}$ satisfait à tous les "bons axiomes" sur les localiseurs fondamentaux W (1, 2, 3, 4, 5, 6, 7) `[voir chapitre XII]`.

Commentary · Exposition [ms. 55]

Localization axioms, stability, and locally constant coefficients

Grothendieck now asks which parts of the derivator formalism are already forced by a good calculus of weak equivalences. Two separate gains should be distinguished. Localizing categories of diagrams preserves their coherent origin; obtaining homotopy Kan extensions requires additional functors between these localized categories. The prederivator exists before the second gain has been made.

Finite disjoint unions are formal. A diagram on $I_1 \amalg \cdots \amalg I_r$ is a tuple of diagrams, and weak equivalences are componentwise. Localization commutes with this finite product: finite zigzags may be padded by identities, and the relations may be checked in one coordinate at a time. This explains the absence of a quillenisability hypothesis in the stated finite version of **Der**1. The same argument is not an argument for arbitrary products of unrelated relative categories, since there may be no uniform finite zigzag length. No such infinite assertion is needed here.

For **Der**2, the point is to represent a localized arrow by a genuine arrow after replacing its endpoints. Let f be an arrow of $\mathrm{HO}_W(I)$ whose evaluations are all invertible. Under the stated homotopy calculus, choose a representative $a : A \to B$ in $\mathcal{M}(I)$, with the endpoint identifications weak equivalences. Every a_i is invertible after localization in $\mathcal{M}$. Strong saturation of W implies $a_i \in W$, so $a \in W(I)$ and f is invertible. The source is correctly using both ingredients: componentwise detection in the original relative category and representability of arrows in the localized diagram category. Strong saturation alone would not supply the latter.

The word "triangulée" has a deliberately broader meaning in this passage than the Verdier definition. In a pointed homotopy theory one has cofiber sequences, fiber sequences, suspension and loop objects. Suspension need not be invertible, and there is no reason for the homotopy category to be additive. Pointed spaces give the basic example. An actual triangulated category emerges in the stable case, where suspension and loop are inverse equivalences and cocartesian squares agree with cartesian squares. For strong stable derivators, the canonical triangulations and exactness of the induced functors are established in Groth (2013, Theorem 4.16 and Proposition 4.18). This is a later completion of the stable part of the programme, not an assumption hidden in the quoted scholium.

The locally constant condition on p. 56 is likewise a condition on a coherent diagram. Its transition arrows become isomorphisms after localization, but the object still belongs to $\mathrm{HO}_W(I)$, not merely to $\mathrm{Fun}(I^{\mathrm{op}}, \mathrm{HO}_W(e))$. The higher coherence carries information. For ordinary model categories, the precise later answer is Cisinski (2009, Theorem 1.3): a functor inducing a weak equivalence of nerves induces an equivalence between the corresponding categories of locally constant diagrams. One can interpret this as dependence only on the infinity-groupoid obtained by inverting all arrows of the indexing category. It does not assert the result for every relative category merely satisfying the definition of quillenisability in §5; the necessary homotopy theory must still be available.

The axiom numbers are historical. The present **Der**3 concerns a localizer and locally constant objects, whereas **Der**4 and **Der**5 below concern existence and pointwise computation of right images. Modern texts commonly number the existence and pointwise axioms as Der3 and Der4, with strength as an additional Der5. A reference to a modern numbered axiom must therefore be translated by its content.

Commentary · Later developments [ms. 56]

Locally constant diagrams depend on the homotopy type

Grothendieck wants a homotopy-invariance theorem for coefficients, stronger than invariance of a single homotopy limit. A locally constant diagram sends every arrow of the indexing category to a weak equivalence. Its transition maps consequently remember transport along paths, together with the higher compatibility data carried by the original diagram before localization. This is why the whole homotopy type of the indexing category, rather than only its set of connected components, enters.

For an ordinary model category $\mathcal{V}$, Cisinski (2009, Theorem 1.3) proves the precise statement anticipated here: if $N(u)$ is a weak homotopy equivalence, restriction induces an equivalence between the homotopy categories of locally constant $\mathcal{V}$-valued diagrams. Corollary 1.28 gives the converse when all coefficient model categories are allowed; simplicial sets already suffice. Apply the theorem to the opposite indexing categories to match the presheaf convention of these pages. This is an

established answer for the usual localizer of nerve weak equivalences, not a conjecture awaiting the construction of $u_!$ or u_*.

In the language of higher categories, the reason is that a locally constant diagram factors through the groupoid completion of the index. That completion is the homotopy type represented by its nerve. An equivalence of those homotopy types induces an equivalence of their categories of functors into any fixed homotopy theory. The model-category theorem makes this explanation precise while retaining strict diagrams and their objectwise weak equivalences.

Two qualifications keep this comparison faithful to the text. The source's "Der 3, nouvelle mouture" is an axiom about local systems and a chosen fundamental localizer; it is not the modern axiom usually numbered (Der3), which asserts existence of homotopy Kan extensions. Also, a fixed coefficient theory may detect fewer distinctions between shapes than all coefficient theories together. Cisinski's converse uses the latter quantifier. We refer to the commentary on Chapter VI, manuscript pp. 161–165, for the detailed comparison of those two forms of detection, rather than repeating it here.

Je vais m'en occuper, une fois que j'aurai une maîtrise raisonnable sur le formalisme de u_*, $u_!$ relatif aux $u : I' \longrightarrow I$ dans Diag. L'hypothèse quillénisable seule semble trop grossière pour assurer l'existence de $u_!$ ou de $u_\star$ dans des cas intéressants.

[page 57]

6 Catégorie de modèles quillénisables, et foncteurs $u_!$, u_*

Définition 1 :

ⓐ On dit que $(\mathcal{M}, W)$ est *quillénisable uniformément à gauche* sur Diag, si on peut trouver une structure de Quillen faible (C, TC, F, TF) pour $\mathcal{M}$ qui soit *close,* et qui de plus ait la propriété suivante : pour I dans Cat, soient $C(I)$, $TC(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ définis par

$$(u : X \longrightarrow Y) \in C(I) \text{ (resp. } TC(I)) \text{ si et seulement si } \forall\, s \in \mathrm{Ob}\, I,$$
$$u_s : X_s \longrightarrow Y_s \text{ est dans } C(I) \text{ (resp. } TC(I)).$$

Soit de plus

$$F(I) \;=\; TC(I)_*, \qquad TF(I) \;=\; C(I)_*.$$

La condition envisagée, c'est que pour tout I *dans* Diag, $(C(I), TC(I), F(I), TF(I))$ est une structure de Quillen faible *close exacte* pour $W(I)$ ([34]).

ⓑ On dit que $(\mathcal{M}, W)$ est *quillénisable sur Diag uniformément et exactement* `[à]` *gauche*, si on a de plus $TC = C \cap W$, *i.e.* si (C, TC, F, TF) est exacte à gauche, donc défini par C seul, via

$$TC = C \cap W, \qquad F = (TC)_*, \qquad TF = C_* \qquad ([35])$$

On a vu au paragraphe 3 que si $\mathcal{M}$ est stable par petites $\varinjlim$ et accessible, et W stable par petites $\varinjlim$ filtrantes, enfin si W admet une structure de Quillen faible (C, TC, F, TF) qui soit close et exacte à gauche – alors les conditions de la définition 1 sont satisfaites. p. 33 [page 58]

Commentary · Exposition [ms. 57]

Why uniform left structures produce right images

Grothendieck's word "uniform" puts the same replacement mechanism on every diagram category. With cofibrations and trivial cofibrations defined objectwise, restriction preserves both classes for the elementary reason that it merely selects components. This makes restriction the left-hand member

34. (C, TC, F, TF) est appelé alors une *quillénisation uniforme à gauche* de $(\mathcal{M}, W)$ sur Diag.

35. On dit que $(C, \ldots)$, ou simplement C, définit une *quillénisation uniforme et exacte à gauche* de $(\mathcal{M}, W)$ sur Diag.

of the relevant Quillen adjunction, even though its geometric name is "inverse image". Its right adjoint is therefore the functor on which fibrant replacement should be performed.

Write $u : I' \to I$ and assume that the ordinary right Kan extension exists. The lifting adjunction immediately gives

$$u_* F(I') \subset F(I), \qquad u_* TF(I') \subset TF(I).$$

For example, lifting a map of $\mathcal{M}(I)$ in $C(I)$ against $u_* p$ is equivalent to lifting its restriction, which lies in $C(I')$, against $p \in TF(I')$. The argument for F uses TC instead. Since u_* preserves the terminal object, it also preserves fibrant objects. These are the facts summarized before the theorem-scholie; they do not yet appeal to a formula for a homotopy limit.

The pointwise formula determines which ordinary limits must exist:

$$(u_* X')(s) = \varprojlim_{(i',\, u(i') \to s) \in I'/s} X'(i').$$

Here the displayed limit is the limit of the contravariant diagram on I'/s. A finite category has finitely many morphisms as well as finitely many objects, so these are finite limits when I and I' are finite. The phrase "$\varinjlim$ finies" on p. 59 is a source slip: the relevant half of **M**0 is finite *limits*. For an infinite comma category, neither **M**0 nor the word quillenisable creates the missing limit.

There are two ways to continue. The manuscript isolates preservation of weak equivalences between fibrant diagrams as a new condition called admissibility, then proves it by finite gluing. This is a sound and instructive construction of finite homotopy limits. The closed exact hypotheses also permit the general Brown argument given below, in the block "The Brown argument makes admissibility automatic". It shows that admissibility is automatic whenever the ordinary right Kan extension exists. Consequently the remaining substantive issue is existence of appropriate diagram structures and ordinary limits, rather than an independent homotopy-invariance property of their right adjoints.

For comparison, a combinatorial model category has both the injective and projective model structures on every small diagram category. The injective structure is the modern counterpart of the objectwise-cofibration construction used here, and the projective structure of its dual. The existence statement and its derivator application are respectively Theorem 1.28 and Proposition 1.30 of Groth (2013). The original finite-limit formulation is still useful when arbitrary small limits do not exist, as in bounded complexes; it then yields only the Kan extensions for which the underlying limit problem can be solved.

D'autre part, notons que la proposition 1 page 45 et ses corollaires, y inclus la proposition 3 (page 50) et ses corollaires, sont valables dès lors que les quillénisations des $(\mathcal{M}(I), W(I))$ envisagés sont définies "uniformément à gauche" par une quillénisation uniforme à gauche (pas nécessairement exacte à gauche) de $(\mathcal{M}, W)$ sur Diag.

Commentary · Exposition [ms. 58]

Closure and exactness already imply both trivial-class equalities

Grothendieck introduces uniform quillenization to remove the dependence of the cofibration classes on the shape. The exact definition on p. 57 has a useful consequence which shortens the later discussion. In each diagram category it asks for a structure which is both *closed* and *exact*. Together these hypotheses force the two equalities which define biexactness.

Indeed, suppose $i \in C \cap W$. Exactness factors this very map as

$$i = pj, \qquad j \in TC, \quad p \in TF.$$

Since $i \in C$ and $p \in TF$, the square with top map j and bottom map the identity has a lift. The usual retract argument makes i a retract of j in the arrow category. Closedness of TC gives $i \in TC$. Thus $C \cap W = TC$. Dually, factoring $p \in F \cap W$ as qj with $j \in TC$, $q \in TF$ makes p a retract of q, so $F \cap W = TF$. These are applications of the ordinary retract argument, as in Hovey (1999a, Lemma 1.1.9); no transfinite construction is involved.

Consequently a closed exact weak structure is perfect in the manuscript's terminology. In particular, if the domain contains the terminal category, as stipulated on p. 51, Definition 1(a) already implies the additional equality in Definition 1(b). The advertised distinction between

uniform quillenizability and uniform *exact* left quillenizability is therefore redundant under the printed hypotheses. This is a logical consequence of those hypotheses, not a proposed emendation of the French text. If "exact" were removed from Definition 1(a), there would be a different and genuinely weaker notion, but that is not the definition supplied here.

The formula in Definition 1(a) has a separate typographical error: a component $u_s : X_s \to Y_s$ belongs to C or TC, not to $C(I)$ or $TC(I)$. The latter classes contain natural transformations, so the printed condition is not well-typed. The correction ledger records the error as a slip of the edition and preserves the printed text. After reading the component condition correctly, the uniform structures are precisely injective model structures with the same underlying cofibration classes, subject here also to the properness axiom M″ and to the available finite or small limits.

La quillénisation uniforme gauche nous servira à établir l'existence des images directes homotopiques u_*, pour certains u. La quillénisation uniforme droite, dualement, aura comme conséquence principale l'existence des $u_!$ homotopiques pour certains $u : I' \longrightarrow I$ dans Diag.

Je vais résumer ce que j'ai pu établir jusqu'à présent dans ce sens, dans le

Théorème-scholie : *Soit* (C, TC, F, TF) *une quillénisation uniforme gauche de* W *sur* Diag $\subset$ Cat. [page 59]

1) *Soit* $u : I' \longrightarrow I$ *une flèche dans* Diag. *On suppose* Diag *accordé à* $\mathcal{M}$ *de façon que pour* u*, flèche dans* Diag, *les* $\varprojlim$ *dans* $\mathcal{M}$ *(de type* I'/i*) requises pour assurer l'existence de*

$$u_* \ : \ \mathcal{M}(I') \longrightarrow \mathcal{M}(I)$$

existent dans $\mathcal{M}$*. (C'est O.K. si* I*,* I' `[sont]` *finies,* $\mathcal{M}$ *étant supposée stable par* $\varinjlim$ *finies de toutes façons.) On sait* (prop. 1, page 45) *que* u_* *induit*

$$(*) \qquad u_*^{\mathrm{f}} \ : \ \mathcal{M}(I')_{\mathrm{f}} \longrightarrow \mathcal{M}(I)_{\mathrm{f}} \, .$$

On dit que u *est* admissible (à gauche) pour la quillénisation uniforme gauche donnée de W sur Diag, *si* u_*^{f} *est compatible avec les localiseurs* $W(I')_{\mathrm{f}}$, $W(I)_{\mathrm{f}}$, i.e. *si* $f' : X' \longrightarrow Y'$ *dans* $\mathcal{M}(I')$*,* X'*,* Y' *fibrants,* $f' \in W(I')$ *implique* $u_*(f') \in W(I)$*. On dit que* Z *dans* Diag *est* admissible (pour la quillénisation uniforme gauche donnée de W sur Diag …) *si le foncteur* $Z \longrightarrow e$ *est admissible. Ceci posé :*

a) *Pour que* u *soit admissible, il suffit que les* I'/s *pour* $s \in \mathrm{Ob}\, I$*, le soient. NB il y a une réciproque, quand* I *est ordonné, donc les* $I'/s \longrightarrow I'$ *des immersions ouvertes.*

b) *Tout ensemble ordonné fini est admissible. Plus généralement (sauf erreur) tout ensemble ordonné* I *tel que l'espace topologique associé à* I° *soit noethérien (*i.e. *toute suite croissante d'ouverts de* I° *est stationnaire, ou encore toute suite croissante de fermées de* I *est stationnaire), et si de plus toute partie fermée irréductible de* I° *admet un point générique,* i.e. I° *est (non seulement un espace noethérien, mais aussi) un espace topologique* sobre. *(Pour* I*, cela signifie que toute partie ouverte filtrante croissante de* I *admet un plus grand élément.)* [page 60]

Commentary · Later developments [ms. 59]

The Brown argument makes admissibility automatic

The purpose of admissibility is to ensure that a limit of a suitably fibrant diagram depends only on its homotopy class. Under the closed exact hypotheses actually imposed in Definition 1, this is already a consequence of the lifting and factorization axioms. We give the argument explicitly, since it both strengthens the theorem-scholie and settles the later questions without assuming the existence of limits which the source has not provided.

First recall the consequence of closedness and exactness proved above: $F \cap W = TF$. Let $T : \mathcal{C} \to \mathcal{E}$ be a functor, where $\mathcal{C}$ has the source's closed exact structure and $\mathcal{E}$ has a class V containing the identity maps and satisfying two-out-of-three. Assume that T carries trivial fibrations

between fibrant objects into V. If $f : X \to Y$ is a weak equivalence between fibrant objects, factor its graph as

$$X \xrightarrow{i} Z \xrightarrow{p} X \times Y, \qquad i \in TC, \quad p \in F, \quad pi = (1_X, f).$$

Put $p_X = \mathrm{pr}_X\, p$ and $p_Y = \mathrm{pr}_Y\, p$. The two projections are fibrations because X and Y are fibrant. Consequently Z is fibrant and $p_X, p_Y \in F$. The identities $p_X i = 1_X$ and $p_Y i = f$, together with $i, f \in W$, imply $p_X, p_Y \in W$. Both maps therefore lie in TF. Their images under T belong to V. Since $T(p_X)T(i) = 1$, two-out-of-three first gives $T(i) \in V$ and then gives

$$T(f) = T(p_Y)T(i) \in V.$$

This is the dual form of Ken Brown's lemma (Hovey, 1999a, Lemma 1.1.12). We have displayed its proof to identify the exact hypotheses used: finite products, factorization, composition and pullback stability of F, two-out-of-three, and the equality $TF = F \cap W$. There is no appeal to smallness, accessibility, or filtered colimits.

Apply this to $T = u_*$. Uniformity makes u^* preserve $C(I)$, so the lifting adjunction makes u_* preserve $TF(I')$. It also preserves fibrant objects. The lemma gives precisely

$$f \in W(I'), \quad X', Y' \in \mathcal{M}(I')_{\mathrm{f}} \quad \Longrightarrow \quad u_* f \in W(I).$$

Thus every functor u for which the ordinary right Kan extension exists is admissible under the stated uniform closed exact assumptions. The finite-poset theorem remains correct, but its restriction on shapes is unnecessary for this particular homotopy-invariance assertion.

The conclusion should not be enlarged in a different direction. It does not manufacture an ordinary right adjoint in a category lacking the required limits. Nor does it, by itself, prove the pointwise homotopy formula: that formula additionally needs fibrant restriction to the comma categories, as discussed at p. 72. These are genuine existence and compatibility questions. The question of preserving weak equivalences *after the relevant fibrant replacements exist* is answered by the short argument above. We attribute it to the established Brown lemma, rather than present the application as a new result.

Commentary · Later developments [ms. 60]

Every available right Kan extension is admissible

Grothendieck's admissibility condition asks for the exact homotopical property needed to derive u_*. Under the uniform hypotheses actually printed on p. 57, it holds for every functor for which the ordinary right Kan extension exists. There is no restriction to finite posets. The reason is the dual of Ken Brown's lemma, whose standard statement is Hovey (1999a, Lemma 1.1.12).

We give the short proof in the present notation. By the preceding retract argument, each uniform structure is perfect. Proposition 1 says that u_* preserves F and TF; as a right adjoint it preserves the terminal object, and hence fibrant objects. Let $f : X \to Y$ be a weak equivalence between fibrant diagrams. Factor its graph as

$$X \xrightarrow{j} Z \xrightarrow{q} X \times Y, \qquad j \in TC, \quad q \in F.$$

The composites $q_X : Z \to X$ and $q_Y : Z \to Y$ are fibrations because X and Y are fibrant. Their identities $q_X j = 1_X$ and $q_Y j = f$, together with $j, f \in W$, show that both belong to W. Biexactness gives $q_X, q_Y \in TF$. Their images under u_* therefore lie in W. The equality $(u_* q_X)(u_* j) = 1$ and two-out-of-three show $u_* j \in W$; hence $u_* f = (u_* q_Y)(u_* j) \in W$. This is precisely admissibility.

Choosing a fibrant replacement $X \to RX$ now defines

$$\mathbf{R}u_*[X] = [u_* RX].$$

Brown's lemma ensures independence of the replacement in the localized category. The adjunction with the derived left functor of u^* is the usual derived Quillen adjunction; since restriction already preserves all objectwise weak equivalences, that left derived functor is u^* itself. Thus $u^* \dashv \mathbf{R}u_*$. The dual argument with uniformly right-defined structures gives $\mathbf{L}u_! \dashv u^*$. This is exactly the construction developed in Hovey (1999a, Definition 1.3.6 and Lemma 1.3.10).

The pointwise criterion in part (a) remains useful for calculations. In the right Kan formula, the projection $I'/s \to I'$ is a discrete fibration. When the requisite fibre coproducts exist, in

particular for finite indexing categories, the left Kan extension of presheaves along such a projection is computed by those coproducts: each relevant comma component has an initial object obtained by lifting the arrow to the fibre. It therefore preserves TC, so restriction preserves fibrant diagrams. The admissibility of the slice then makes the pointwise limit preserve weak equivalences. If I is a poset, the slice projection identifies its domain with a sieve, as the author says. Under the full uniform hypotheses, the global Brown argument has already established both sides of the proposed converse, without requiring these auxiliary coproducts.

This proof uses finite products and factorizations in the diagram categories, plus the existence of the indicated right adjoint. The parenthesis on p. 59 should invoke finite *limits*; it prints finite colimits. Both are included in M 0, so this slip does not invalidate the finite-index assertion. Up to manuscript p. 60 the theorem is only stated; the proof above supplies its admissibility conclusion independently of the continuation.

Commentary · Exposition [ms. 60]

The noetherian sober poset condition is finite

Grothendieck tries to enlarge the finite-poset case by replacing finite induction with topological noetherian induction. For the usual Alexandrov topology attached to a poset, however, the two hypotheses he writes down already force finiteness. The observation concerns this topology, in which arbitrary intersections of open sets are open; it would not be true for arbitrary noetherian sober spaces.

To make the order convention immaterial, let P be the specialization poset and take its Alexandrov opens to be the upper sets. Closed sets are lower sets. An irreducible nonempty closed set is then exactly a directed lower set. For one direction, if D is directed and meets two upper open sets, a common upper bound in D lies in their intersection. Conversely, if a lower set D is irreducible and $x, y \in D$, the open sets $\uparrow x$ and $\uparrow y$ both meet D, so their intersection meets D; that is a common upper bound in D. Sobriety says that each such directed lower set is $\downarrow z$ for a unique z.

It follows that P has no strictly increasing infinite sequence. Otherwise, from $x_0 < x_1 < \cdots$ form the directed lower set $D = \bigcup_n \downarrow x_n$. If $D = \downarrow z$, then $z \leq x_N$ for some N, whereas $x_{N+1} \leq z$, a contradiction. This is the sobriety criterion of Picado and Pultr (2012, Chapter II, Proposition 6.4.4).

Every nonempty subset of P consequently has a maximal element: starting from an element with no maximal element above it would produce such an increasing sequence. A maximal element m of a subset S is isolated in S, since $(\uparrow m) \cap S = \{m\}$. The Alexandrov space is therefore scattered, meaning that every nonempty subspace has an isolated point.

Now use noetherianity only once. Among the finite open subsets choose a maximal one, U; it exists because a strictly increasing sequence of finite open subsets is forbidden. If the complement $P \setminus U$ were nonempty, scatteredness would give an isolated point x in that subspace. There would be an open V of P with $V \cap (P \setminus U) = \{x\}$. Then $U \cup V = U \cup \{x\}$ would be a strictly larger finite open set, contradicting maximality. Hence $P = U$ is finite.

Applied to I^{op}, the proof shows that the class on p. 60 consists exactly of finite posets. Both chain conditions are used: the order condition controls sobriety, and topological noetherianity controls opens. The sequence arguments use classical choice.

Overview · XIII-a

Chapter XIII, first part: building the machinery that can later be forgotten.

These sixty manuscript pages turn from the intrinsic calculus of a localizer to the auxiliary structures that make it calculable. Grothendieck calls a pair $(\mathcal{M}, W)$ a category of models when W contains the isomorphisms and satisfies two-out-of-three. This is much less than what "model category" ordinarily means today. He seeks classes of cofibrations and fibrations that provide replacements, homotopy fibres and cofibres, and eventually the two derived Kan extensions. Their purpose is to establish properties of the localized diagram categories

$$\mathrm{HO}_W(I) = \mathcal{M}^{I^{\mathrm{op}}}[W(I)^{-1}],$$

where $W(I)$ is defined objectwise. Once these constructions and their universal properties have been established, the auxiliary classes should cease to be part of the intrinsic answer (Grothendieck, 1990–1991e, manuscript pp. 12–14).

The chapter opens with the algebra of lifting properties. In current notation, Grothendieck's Φ_* is $\Phi^{\square}$, his Ψ^* is ${}^{\square}\Psi$, and his "couple de Quillen" is a weak factorization system. It is not what is now called a Quillen adjunction. The Galois connection explains why left classes are stable under pushouts, retracts, and continuous transfinite compositions, and why right classes have the dual properties. The small-object theorem then supplies factorizations from a set of generating arrows in a locally presentable category. Our commentary proves the construction with its cardinal bound and separates the cellular class from its retract closure. This distinction matters because factorization can exist before the two classes are closed under retracts.

The intrinsic classes Cof_W and Fib_W express preservation of weak equivalences under cobase change and base change. They are not defined by a lifting property. Their relation to the universally trivial arrows is established using the factorization hypotheses, not just two-out-of-three. The base-change argument on manuscript pages 10–11 is particularly economical: factor the change of base, apply the fibration condition along its weak-equivalence part, and use two-out-of-three. The commentary expands this proof and gives a three-element-poset example showing why the factorization assumption cannot simply be omitted.

The conjectural theorem of manuscript pages 14–26 asks whether the intrinsic classes contain small collections of tests sufficient to build the desired structures. Here three different forms of smallness enter: closure under filtered colimits, accessibility of a full subcategory of the arrow category, and generation of a lifting class by a set. They do not mean the same thing. In particular, even an accessible left lifting class can fail to be generated by a set in the ordinary small-object sense; the order-embedding example is worked out in the commentary. Conversely, one specific conjecture in these pages has a precise later answer: intersections of two set-generated left classes in a locally presentable category are again set-generated, by the limit theorem for combinatorial cellular categories (Makkai and Rosický, 2014a, Theorem 3.2 and Remark 3.8).

Smith's recognition theorem gives a useful repair of the larger programme. One fixes a set of generating cofibrations, assumes that its right orthogonal consists of weak equivalences, requires the trivial cofibrations to be stable under pushouts and transfinite composition, and imposes a solution-set condition on W. Under these hypotheses one obtains an ordinary cofibrantly generated model structure (Beke, 2000, Theorem 1.7). With the intrinsic inclusions envisaged here, the resulting structure is proper. This is a theorem with explicit extra hypotheses, not a proof that the manuscript's filtered-colimit assumption alone yields every assertion in its conjectural equivalence. The audit records the individual implications and their dependencies rather than assigning one status to the whole discussion.

The second formulation, beginning on manuscript page 27, isolates a different issue. A weak Quillen structure is called exact when $W = TF \circ TC$; it is left exact when $TC = C \cap W$

and right exact when $TF = F \cap W$. These distinctions are meaningful for classes that have not been closed under retracts. The marginal claim that retract closure of C, F, W alone makes the two one-sided exactness conditions equivalent is false. Our example in sets takes $C = TC$ to be the monomorphisms, F the surjections, W all maps, and TF the surjections whose fibres are singletons or infinite. It satisfies the weak axioms and left exactness, but excludes the map from a two-element set to a singleton from TF. The missing retract condition is visible directly.

Exactness nevertheless permits a useful completion without any retract closure assumption. If $j \in C \cap W$, factor $j = qi$ with $i \in TC$ and $q \in TF$. The lifting axiom makes j a retract of i, so j has the left lifting property against every fibration. The dual argument applies to $F \cap W$. Replacing the two designated trivial classes by $C \cap W$ and $F \cap W$ therefore gives a biexact Quillen structure with the same C, F, W. Properness supplies the required stability under base and cobase change. This is a completion of the designated classes; the counterexample shows why it cannot be replaced by an assertion that they were equal already. When all four original classes are closed, the completion makes no change, and a *closed exact* weak structure is already perfect. We use this observation again at the end of these pages to remove an unnecessary admissibility restriction on right Quillen functors.

The discussion of homotopy on manuscript pages 29–34 concerns the actual morphisms in the localization. Replacing objects by objects that are both cofibrant and fibrant is not yet the same as proving that all localized morphisms are represented by single arrows. Exactness supplies such representatives. We first prove a description by the congruence generated jointly by left and right homotopies. The completion then identifies each of the usual homotopies with equality after localization for a cofibrant source and a fibrant target. It also resolves the doubts on manuscript pages 32–34: homotopy-cartesian and homotopy-cocartesian squares are exact under the weak exact axioms themselves. An endpoint-lifting argument makes the first assertion explicit; the extra hypothesis about the diagram category used in the source's proposed proof is unnecessary for this conclusion. Pointedness gives the suspension and loop sequences. Stability is the additional condition needed for a triangulated structure. The separate suggestion about raw right homotopy depends on the path-object convention. A graph example, adapted from Kurz and Rosický (2005, Example 3.6), shows that paths supplied by the designated factorization system need not give a transitive relation without exactness, even on fibrant-cofibrant objects. Allowing some weak-equivalence–fibration path object gives an equivalence relation for a fibrant target by the concatenation argument proved in the commentary.

The passage to presheaves supplies the chapter's central technical test. Objectwise cofibrations lead to an injective diagram structure. Its fibrations are defined by lifting against all objectwise trivial cofibrations, and need not be merely objectwise fibrations. The calculation

$$(\alpha_{s!}A)(t) = \coprod_{\mathrm{Hom}_I(t,s)} A$$

explains the coproduct hypothesis in the evaluation lemma. The subsequent failure of componentwise factorization to solve natural lifting problems is exhibited by explicit examples. Modern existence results for injective structures apply in the combinatorial setting, while the cellular-category theorem also accounts for the set-generation of the pointwise left classes (Makkai and Rosický, 2014a, Corollary 3.4).

The variances then become transparent. Restriction preserves the objectwise cofibrations and trivial cofibrations, so its right adjoint is right Quillen. Its left adjoint is not automatically left Quillen for these same injective structures. Above all, an ordinary Kan extension and its derived functor are different constructions. Grothendieck explicitly withdraws the premature identification on manuscript pages 49–50. The commentary supplies counterexamples and the appropriate derived statements. Under the uniform closed exact hypotheses finally adopted on

manuscript page 57, every existing ordinary right Kan extension preserves weak equivalences between injectively fibrant diagrams by Brown's lemma. Consequently the final admissibility conclusion holds for every such indexing functor, beyond the finite posets singled out in the text.

Complexes give a complementary warning about size. For an infinite indexing category, a diagram of individually bounded-below complexes need not have a common lower bound. It therefore need not be a bounded-below complex in the abelian category of diagrams. A family with nonzero cohomology in successively lower degrees makes the resulting failure of a product in $D^{+}(\mathbf{Ab})$ explicit. Unbounded derived operations in Grothendieck abelian categories are instead available through suitable resolutions and model structures; the boundedness issue concerns the chosen subcategory and cannot be removed by renaming it.

The later derivator constructions realize the intended independence from presentation. Cisinski supplies homotopy right Kan extensions for general closed model categories with small limits (Cisinski, 2003, Theorem 6.11); Renaudin compares the localized 2-category of suitable combinatorial presentations with derivators of small presentation (Renaudin, 2009, Theorems 3.3.2 and 3.4.4); and the infinity-categorical treatment recovers the same pointwise comparison laws from Kan extensions in the underlying infinity-category (Riehl and Verity, 2017, Proposition 5.3.9 and Remark 5.3.10). These results give precise answers in their respective settings. The final order-topological extension also becomes precise: a noetherian sober Alexandrov T_0 space is finite, so under the ordinary poset topology the apparently broader class of shapes on manuscript page 60 adds no new cases. For the weakest recognition assertions of the manuscript, what was established here is kept apart from what was not located in the checked literature. An unlocated proof is not certified as an open problem.

p. 34

c) *En particulier, si I', I sont des ensembles ordonnés finis, alors tout $u : I' \longrightarrow I$ est admissible. Il en est de même, plus généralement, si I, I' sont des ensembles ordonnés et I'° est un espace noethérien sobre.*

2) *Supposons $u : I' \longrightarrow I$ admissible. Considérons*

$$u_*^W \ : \ \underbrace{\mathcal{M}(I')_{\mathrm{f}} W(I')_{\mathrm{f}}^{-1}}_{\simeq \mathrm{HO}_W(I')} \ \longrightarrow \ \underbrace{\mathcal{M}(I)_{\mathrm{f}} W(I)_{\mathrm{f}}^{-1}}_{\mathrm{HO}_W(I)},$$

induit par u_*. Ce foncteur est adjoint à droite de $u_W^* : \mathrm{HO}_W(I) \longrightarrow \mathrm{HO}_W(I')$.
De plus, l'homomorphisme canonique (pour $s \in \mathrm{Ob}I$, $X' \in \mathrm{Ob}\mathrm{HO}_W(I')$) [page 61]

$$u_*^W(X')(s) \ \simeq \ p_{s*}(X'|I'/s) \ \overset{\text{déf}}{=} \ \underset{\longleftarrow I'/s}{\mathrm{holim}}_W(X'|I'/s)$$

(où $p_s : I'/s \longrightarrow e$) est un isomorphisme, si on suppose que les I'/s sont admissibles (on dira alors que u est strictement *admissible).*

Corollaire. *Sur le domaine formé des ensembles ordonnés finis, le prédérivateur* HO_W *satisfait (en plus des axiomes* **Der** 1, **Der** 2*) aux axiomes* **Der** 4 *et* **Der** 5 *(existence des images directes u_*, adjoints à droite des foncteurs u^*, et calcul standard des fibres d'une image directe).*

Démonstration du théorème-scholie.

1°) *a*) est immédiat. Pour *b*), on procède par récurrence sur le cardinal n de l'ensemble ordonné fini I, les cas $n \in \{0, 1\}$ étant triviaux. Supposons donc $n \geq 2$, et le théorème prouvé pour les $n' < n$. Je distingue deux cas : si I admet un plus grand élément a, alors, désignant par $\alpha : e \longrightarrow I$ le foncteur de valeur a, par $p : I \longrightarrow e$ la projection, on sait que l'on a

$$p_* \ \simeq \ \alpha^*, \qquad i.e. \ \varprojlim_I X_i \simeq X_a,$$

d'où le fait que sur $\mathcal{M}(I)$ tout entier (pas seulement sur $\mathcal{M}(I)_{\mathrm{f}}$), p_* est compatible aux localiseurs $W(I)$, W. [page 62]

Commentary · Exposition [ms. 61]

Finite limits assembled from a two-piece cover

Grothendieck's finite-poset induction makes the geometry of a homotopy limit visible. Instead of treating a large compatibility diagram at once, he cuts the indexing poset into two lower sets and glues the two resulting limits over their overlap. This argument remains useful even though the Brown lemma gives a shorter proof of admissibility under the stronger standing hypotheses.

Let $I = I_1 \cup I_2$ with I_1, I_2 lower subsets, and put $I_0 = I_1 \cap I_2$. Define $v : I \to \Psi = (1 \leftarrow 0 \to 2)$ by sending the overlap to 0, the points belonging only to I_1 to 1, and those belonging only to I_2 to 2. The lower-set condition makes this assignment order preserving: an arrow cannot travel from one exclusive part into the other. Its comma categories over $0, 1, 2$ are precisely I_0, I_1, I_2. Right Kan extension along v therefore produces the span

$$\varprojlim_{I_1} X \longrightarrow \varprojlim_{I_0} X \longleftarrow \varprojlim_{I_2} X.$$

Taking its pullback gives $\varprojlim_I X$. Indeed, a cone to X is exactly a pair of cones to its restrictions whose two cones on I_0 agree. This proves the ordinary limit identity by its universal property, including the cases in which the overlap is empty.

For a finite nonempty poset without a greatest element, partition its maximal elements into two nonempty sets M_1, M_2 and take $I_j = \bigcup_{m \in M_j} \{i : i \leq m\}$. Every point lies below a maximal

element, so the lower sets cover I, and neither is all of I. Their overlap is also smaller. The inductive argument is therefore well founded. If I has a greatest element a, the limit is X_a in the source's contravariant convention; the empty poset has the terminal object as its limit.

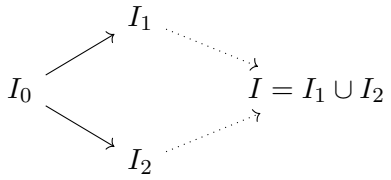
The homotopical step is to ensure that the span just constructed is fibrant. Right Kan extension preserves fibrancy, and a fibrant Ψ-diagram consists of fibrant objects A_0, A_1, A_2 with both arrows $A_j \to A_0$ fibrations. Its ordinary pullback then represents its homotopy pullback. The induction replaces "take an arbitrary pullback" by "take a pullback whose legs satisfy a lifting condition". The latter is the operation stable under weak equivalences of the input span.

Composition of admissible functors is also transparent at this point. If u and v preserve weak equivalences between fibrant diagrams after right Kan extension, then u_* first yields another fibrant diagram, to which v_* can be applied. The canonical identification $(vu)_* \cong v_* u_*$ proves admissibility of the composite. No new comparison of replacements is required. This explains why the elementary shape Ψ controls the whole finite induction.

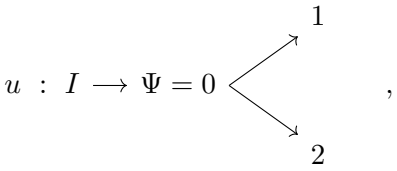
Si I n'a pas de plus grand élément, alors l'ensemble M des ses éléments maximaux est de cardinal ≥ 2, on peut donc l'écrire $M = M_1 \amalg M_2$, et on aura

$$I \;=\; I_1 \cup I_2, \qquad \text{où } I_1 = I/M_1,\ I_2 = I/M_2$$

sont deux ouverts de I, différents de I, donc de cardinaux n_1, $n_2 \;<\; n$. Considérons aussi $I_0 = I_1 \cap I_2$, c'est un ouvert de cardinal $< n$. Le diagramme



correspond à une flèche p. 35

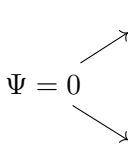


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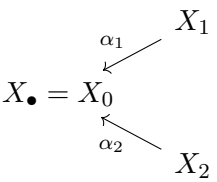
avec

$$I_1 \;=\; I/1, \qquad I_2 \;=\; I/2, \qquad I_0 \;=\; I/0.$$

L'hypothèse de récurrence et *a*) impliquent que u est admissible. Comme un composé de foncteurs admissibles est admissible (tautologiquement, via $(vu)_* \simeq v_* u_*$), on est réduit finalement au [page 63]

Lemme. *La catégorie*  *est admissible.*

Ceci va résulter du

Corollaire ([36]). *Soit*  *dans $\mathcal{M}(\Psi)$. Pour que $X_\bullet$ soit fibrant, il faut*

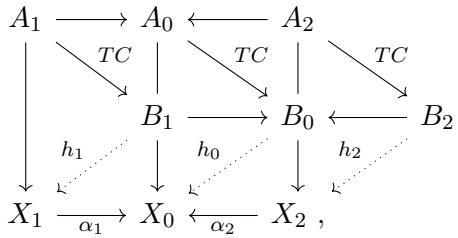
et il suffit que l'on ait

36. **NB** Il y a une variante évidente, caractérisant les X tels que $X \longrightarrow e$ soit dans $TF(\Psi)$ comme ceux pour lesquels X_0, X_1, X_2 sont fibrants *triviaux,* et α_1, $\alpha_2 \in TF$.

a) X_0, X_1, X_2 *sont fibrants, et*

b) α_1, $\alpha_2 \in F$.

La nécessité de *a*) est cas particulier du corollaire 2, page 48, celle de *b*) est cas particulier du corollaire 1 page 50. Pour la suffisance (dont on n'aura pas besoin pour le théorème), on sait montrer que dans le diagramme commutatif à traits pleins



avec trois flèches marquées TC, on peut trouver h_1, h_0, h_2 laissant le diagramme commutatif (trois triangles et deux carrés commutatifs à respecter). Pour ceci, on commence par choisir h_0 de façon à rendre le triangle vertical médian commutatif – c'est possible car X_0 est fibrant, $A_0 \longrightarrow B_0$ dans TC.

p. 36
[page 64]

Commentary · Examples [ms. 63]

Matching objects explain the lifting diagram

Grothendieck's three lifts express a recursive compatibility condition. The lift at X_0 must be chosen first, because the two remaining lifts are required to induce that same map after composition with $X_1 \to X_0$ and $X_2 \to X_0$. Choosing three unrelated lifts from objectwise fibrancy would not enforce the squares. The fibrations on the two legs are exactly what turns each final choice into a lifting problem with a prescribed target.

The calculation is the smallest instance of the matching-object description for an inverse diagram. Let I be a finite poset, still indexing contravariant diagrams. For $i \in I$ put

$$M_i X = \varprojlim_{j<i} X_j.$$

An injectively fibrant diagram is characterized by the requirement that each natural map $X_i \to M_i X$ be a fibration. When i has no smaller element, the matching object is terminal, so this says that X_i is fibrant. For $I = \Psi$, the matching objects are $M_0 X = e$ and $M_1 X = M_2 X = X_0$, giving exactly the corollary. This is the inverse-category case of the familiar Reedy description (Hovey, 1999a, Theorem 5.1.3).

The lifting criterion can be proved without invoking the terminology. Choose a linear ordering of the finite set of vertices in which smaller vertices occur first. Given an objectwise trivial cofibration $A \to B$ and a map $A \to X$, suppose the lift $B_j \to X_j$ has been chosen for every $j < i$. These maps induce $B_i \to M_i X$. The desired map $B_i \to X_i$ is now a lift in the square whose left side is $A_i \to B_i$ and whose right side is $X_i \to M_i X$. Proceeding through the vertices gives a natural transformation $B \to X$.

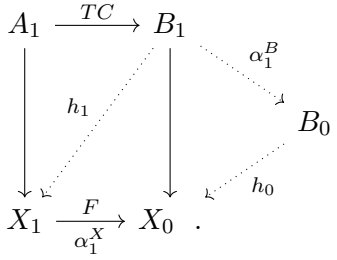
For necessity, start with a lifting problem $A \to X_i$, $B \to M_i X$, where $A \to B$ is a trivial cofibration. Form two diagrams which equal X_j for $j < i$, take the values A and B at i, respectively, and equal the initial object outside the lower subset $\{j : j \le i\}$. The given maps to the matching object specify their transition maps. Extension by the initial object is consistent because this subset is lower. Their comparison is objectwise a trivial cofibration, and the first diagram maps to X by the identity below i and the given map at i. A lift against $X \to e$ supplies the required $B \to X_i$. This proves the converse. The same finite recursion, factoring $X_i \to M_i(RX)$ at each stage as a trivial cofibration followed by a fibration, constructs a fibrant replacement $X \to RX$.

The condition becomes more informative at a branching vertex. Consider the diamond $0 < 1 < 3$, $0 < 2 < 3$. Its top matching object is

$$M_3 X = X_1 \times_{X_0} X_2.$$

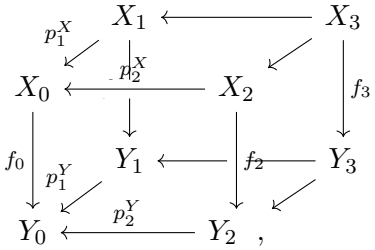
> In cochain complexes of vector spaces over a field k, take $X_0 = 0$ and $X_1 = X_2 = X_3 = k[0]$, with both maps from X_3 equal to the identity. All objects are fibrant in the usual model structure, and every nonidentity arrow of this diagram is a fibration: the maps are degreewise surjective. Yet
>
> $$X_3 \longrightarrow M_3X = k[0] \oplus k[0]$$
>
> is the diagonal and is not surjective. The diagram is not injectively fibrant. Thus the two separate legs happen to suffice for Ψ, whereas a general poset requires compatibility with the entire lower matching diagram. This distinction is why replacements of diagrams carry more information than separate replacements of their components.

Ensuite, on voit séparément pour h_1, h_2, qu'on peut les trouver en satisfaisant chacune aux deux contraintes de commutativité (1 triangle vertical et un carré penché). Pour h_1 p. ex. l'existence se voit sur le diagramme carré auxiliaire (traits pleins)



NB C'est la démonstration de la proposition 3, page 50.

Le corollaire (partie "il faut") étant vu, le lemme en résulte, car on sait que si on a un homomorphisme f de carrés cartésiens X, Y



avec

$$p_1^X,\ p_2^X,\ p_1^Y,\ p_2^Y \ \text{ fibrants } \texttt{[appartenant à } F\texttt{]} \text{ (il suffit même } p_1^X \text{ et } p_1^Y \text{ fibrant)},$$

et

$$f_0,\ f_1,\ f_2 \ \in\ W,$$

alors $f_3 \in W$. (Ça a été vu dans la théorie des produits fibrés W-homotopiques `[voir chapitre XII, p. 253]`, et la vérification directe est immédiate.) De l'hypothèse initiale [page 65]

pertinente que $\begin{matrix} & X_1 & \\ X_0 & & \\ & X_2 & \end{matrix}$ et $\begin{matrix} & Y_1 & \\ Y_0 & & \\ & Y_2 & \end{matrix}$ sont fibrants, on n'a besoin d'utiliser qu'un petit

bout, p. ex. que p_1^X et p_1^Y sont fibrants, qui assure que les deux carrés X, Y sont W-homotopiquement cartésiens, ce qui suffit à impliquer alors que $f_0,\ f_1,\ f_2 \in W \Longrightarrow f_3 \in W$.

Commentary · Examples [ms. 64]

A homotopy pullback computed as a complex

Grothendieck uses the pullback lemma to make finite gluing invariant under weak equivalences. A concrete calculation shows what fibrant replacement adds to an ordinary pullback. Work with cochain complexes over a field and consider maps $a : X_1 \to X_0$ and $b : X_2 \to X_0$. A model for their homotopy pullback is

$$H = \mathrm{Cone}(a - b : X_1 \oplus X_2 \to X_0)[-1].$$

With coordinates chosen so that

$$H^n = X_1^n \oplus X_2^n \oplus X_0^{n-1}, \quad d(x_1, x_2, h) = (dx_1, dx_2, -dh + a(x_1) - b(x_2)),$$

the equation $d^2 = 0$ follows from the fact that a and b are chain maps. The third coordinate records the homotopy witnessing equality of the first two images. A strict pullback asks those images to be equal without this additional coordinate.

To verify that this complex computes the derived pullback, replace a by the surjective map $\widetilde{X}_1 \to X_0$, where

$$\widetilde{X}_1^n = X_1^n \oplus X_0^n \oplus X_0^{n-1}, \quad d(x, z, h) = (dx, dz, -dh + a(x) - z),$$

and the map to X_0 is $(x, z, h) \mapsto z$. The inclusion $x \mapsto (x, a(x), 0)$ is a quasi-isomorphism: the projection back to X_1 has kernel the contractible complex with differential $(z, h) \mapsto (dz, -dh - z)$, contracted by $(z, h) \mapsto (-h, 0)$. The new map is degreewise surjective and hence a fibration. Its ordinary pullback along b imposes $z = b(x_2)$ and is exactly the displayed complex H.

There is a short exact sequence of complexes

$$0 \longrightarrow X_0[-1] \longrightarrow H \longrightarrow X_1 \oplus X_2 \longrightarrow 0.$$

Its connecting map on cohomology is induced by $a - b$. This gives the usual exact sequence for a homotopy pullback and also proves invariance directly: a componentwise quasi-isomorphism between two such spans induces an isomorphism on the two outside terms of the associated long exact sequences, and hence on the middle term. If $a - b$ is degreewise surjective, its ordinary kernel maps quasi-isomorphically to H. In particular this occurs when either a or b is degreewise surjective, exactly the fibration condition relevant to this model.

For the span $0 \to k[0] \leftarrow 0$, the ordinary pullback is zero, while $H = k[-1]$ has $H^1(H) = k$. A fibration replacement of one leg restores this missing class. This is the algebraic version of the familiar fact that the homotopy pullback of the basepoint maps $* \to Y \leftarrow *$ is the loop object of Y. It explains both why the manuscript requires fibrant legs and why its gluing step cannot be replaced by taking limits in the homotopy category with no coherent diagram behind them.

In the source's general framework, the corresponding invariance statement uses the W-fibration property included in $\mathbf{M}''$, or the equivalent derived-functor argument under the closed exact hypotheses. No failure of properness is being overlooked in the proof on pp. 64–65: the relevant base-change compatibility is already among its assumptions. The statement that one fibrant leg suffices concerns this particular pullback computation, not a claim that all transition maps in an arbitrary diagram can be ignored.

Je vais essayer de généraliser ce raisonnement, pour traiter le cas d'un ensemble ordonné I qui peut être infini, mais tel que I° soit noethérien et sobre, donc toute suite croissante de parties fermées de I, et toute suite décroissante de parties ouvertes, est stationnaire, et de plus toute partie ouverte filtrante croissante a un plus grand élément.

On veut prouver que I est admissible, et par "récurrence noethérienne" on est ramené p. 37
à prouver ceci : $(*)$ Si I est tel que pour tout ouvert $I' \subset I$, avec $I' \neq I$, I' soit admissible, alors I aussi.

$(*)$ admis, prouvons que $I = I_0$ est admissible.

S'il ne l'était, par $(*)$ il existerait $I_1 \subset I_0$, $I_1 \neq I_0$, tel que I_1 non admissible. De même, [page 66]
il y aurait $I_2 \subset I_1$, $I_2 \neq I_1$, non admissible etc. – on aurait construit une suite strictement décroissante d'ouverts I_n de I, contrairement au caractère noethérien de I°.

Prouvons $(*)$, et distinguons encore deux cas.

a) I° est irréductible, donc admet un point générique, *i.e.* un plus petit élément, *i.e.* I admet un plus grand élément. Ce cas a déjà été traité tantôt – toute catégorie avec objet final est admissible.

b) I° est réductible, donc réunion de deux fermés I_1°, I_2° distincts de I°, donc I est réunion de deux ouverts I_1, I_2 distincts de I. Alors par l'hypothèse dans $(*)$, I_1, I_2 et $I_0 = I_1 \cap I_2$ sont admissibles. On termine alors comme tantôt, étant ramené à l'admissibilité

de $\Psi = 0 \begin{cases} \nearrow 1 \\ \searrow 2 \end{cases}$!

Prouvons la partie *c*) du théorème-scholie, partie 1°). Cela résulte de *a*), une fois vu [page 67] que pour $s \in I$, I'/s satisfait aux mêmes hypothèses que I'. Or I étant ordonné, I/s est une sous-catégorie ouverte de I, donc I'/s une sous-catégorie ouverte de I', dont $(I'/s)^\circ$ un sous-ensemble ordonné fermé de I'°. Ce dernier étant noethérien et sobre, il en est de même de tout fermé, d'où la conclusion.

Commentary · Exposition [ms. 66]

The proposed noetherian enlargement is actually finite

Grothendieck's noetherian induction is patterned on the decomposition of a closed space into irreducible pieces. A generic point makes one piece have a greatest indexing object, and a reducible piece can be glued from two smaller ones. The induction is valid. For the specific Alexandrov topology of a poset, however, the combination "noetherian and sober" is stronger than it first appears: it forces the poset to be finite. Thus this paragraph does not in fact add infinite diagram shapes.

We verify the assertion in the precise order convention used here. Opens of I are lower subsets, so opens of I^{op} are upper subsets of I. Sobriety of I^{op} says that every nonempty directed lower subset of I has a greatest element. If $x_0 < x_1 < \cdots$ were a strictly increasing sequence, its downward closure would be directed. It could not have a greatest element: any alleged greatest element belongs below some x_n, whereas x_{n+1} is strictly larger. Consequently every nonempty subset of I has a maximal element; otherwise dependent choice would construct such a sequence. This is the order-theoretic sobriety criterion, with the order reversed, in Lindenhovius (2014, Proposition B.5).

Let X denote I with its upper topology, that is, the topology of I^{op}. Every nonempty subspace $S \subset X$ has an isolated point: take a maximal element m of S, so that $S \cap \{x : m \leq x\} = \{m\}$. We now use noetherianity. The union U of all finite open subsets of X is open. Every open subset of a noetherian space is quasi-compact, so the cover of U by its finite open subsets has a finite subcover. Hence U itself is finite. If $X \setminus U$ were nonempty, choose an isolated point x of that subspace. There would be an open $V \subset X$ with $V \cap (X \setminus U) = \{x\}$. Then $U \cup V = U \cup \{x\}$ would be a finite open subset not contained in U, contradicting the definition of U. Therefore $X = U$ is finite.

Conversely, every finite poset has a noetherian sober Alexandrov topology. Noetherianity is immediate. A nonempty irreducible closed subset has finitely many minimal elements in the lower-topology convention, and irreducibility forces exactly one of them; the closed subset is its closure. Thus the two hypotheses in the proposed enlargement are equivalent to finiteness for posets.

This observation should not be confused with the false assertion that every noetherian sober topological space is finite. Spectra supply abundant infinite examples. The essential extra hypothesis is the Alexandrov topology, in which arbitrary intersections of opens remain open. Nor is it necessary to insert a defect into the manuscript's induction: its irreducible and reducible cases work. The correction is to its claimed extra scope. Infinite towers and group actions on the following pages require another argument, which the Brown lemma supplies whenever the necessary diagram structures and limits exist.

Prouvons la partie 2°) du théorème-scholie. Considérons

$$\begin{array}{ccccc} \mathcal{M}(I')_{\mathrm{c}} & \hookrightarrow & \mathcal{M}(I') & \hookleftarrow & \mathcal{M}(I')_{\mathrm{f}} \\ \uparrow{\scriptstyle u_{\mathrm{c}}^*} & & \uparrow{\scriptstyle u^*} & & \downarrow{\scriptstyle u_*^{\mathrm{f}}} \\ \mathcal{M}(I)_{\mathrm{c}} & \hookrightarrow & \mathcal{M}(I) & \hookleftarrow & \mathcal{M}(I)_{\mathrm{f}}\ , \end{array}$$

où u^* et a fortiori u_{c}^* sont compatibles avec $W(I)$, $W(I')$, et par hypothèse u_* est compatible avec $W(I')_{\mathrm{f}}$, $W(I)_{\mathrm{f}}$. En passant aux catégories de fractions, on trouve

p. 38
[page 68]

$$\begin{array}{ccccc} \mathcal{M}(I')_{\mathrm{c}}W(I')_{\mathrm{c}}^{-1} & \xrightarrow{\sim} & \mathrm{HO}_W(I') & \xleftarrow{\sim} & \mathcal{M}(I')_{\mathrm{f}} \\ \uparrow{\scriptstyle u_W^*} & & & & \downarrow{\scriptstyle u_*^W} \\ \mathcal{M}(I)_{\mathrm{c}}W(I)_{\mathrm{c}}^{-1} & \xrightarrow{\sim} & \mathrm{HO}_W(I) & \xleftarrow{\sim} & \mathcal{M}(I)_{\mathrm{f}} \end{array}$$

`[plutôt` $\mathcal{M}(I')_{\mathrm{f}}W(I')_{\mathrm{f}}^{-1}$ `(resp.` $\mathcal{M}(I)_{\mathrm{f}}W(I)_{\mathrm{f}}^{-1}$`) au lieu de` $\mathcal{M}(I')_{\mathrm{f}}$ `(resp.` $\mathcal{M}(I)_{\mathrm{f}}$`)]` et on veut montrer que u_W^* et u_*^W sont adjoints, de façon précise trouver un isomorphisme bifonctoriel, pour F *dans* $\mathcal{M}(I)_{\mathrm{c}}$ et G' *dans* $\mathcal{M}(I')_{\mathrm{f}}$

$$(*) \qquad \mathrm{Hom}_W(u^*F, G') \xrightarrow{\sim} \mathrm{Hom}_W(F, u_*G')$$

(où Hom_W désigne le Hom dans $\mathrm{HO}_W(I')$ resp. $\mathrm{HO}_W(I)$). Comme F est dans $\mathcal{M}(I)_{\mathrm{c}}$, u^*F est dans $\mathcal{M}(I')_{\mathrm{c}}$, et comme G' est dans $\mathcal{M}(I')_{\mathrm{f}}$, u_*G' est dans $\mathcal{M}(I)_{\mathrm{f}}$. Il s'ensuit que les deux membres de $(*)$ sont des quotients des deux membres dans la bijection canonique

$$(**) \qquad \mathrm{Hom}_{\mathcal{M}(I')}(u^*F, G') \xrightarrow[\sim]{\alpha} \mathrm{Hom}_{\mathcal{M}(I)}(F, u_*G').$$

Il faut simplement montrer que cette dernière transforme la relation d'équivalence pertinente R' du premier membre, en celle R du second. [page 69]

Commentary · Exposition [ms. 68]

The adjunction after localization, with both inverses

Grothendieck wants the limit functor to retain its universal property after replacement. Knowing that it sends weak equivalences to weak equivalences makes it descend to the homotopy category; it does not alone identify its adjoint. The argument on pp. 68–71 checks that the ordinary adjunction identifies the two homotopy relations. We can make its two directions explicit without choosing a globally functorial replacement.

Choose F cofibrant in $\mathcal{M}(I)$ and G fibrant in $\mathcal{M}(I')$. Then u^*F is cofibrant and u_*G is fibrant. Every morphism between these objects in the respective homotopy categories is represented by an ordinary map. Restriction followed by the ordinary counit defines

$$\beta : \mathrm{Hom}_{\mathrm{HO}(I)}(F, u_*G) \longrightarrow \mathrm{Hom}_{\mathrm{HO}(I')}(u^*F, G).$$

It is well defined because restriction preserves all weak equivalences. It is surjective because every ordinary map $u^*F \to G$ has a strict adjoint $F \to u_*G$. This is the one direction available before admissibility has been used.

To construct the inverse, choose a trivial cofibration $i' : u^*F \to F'$ with F' fibrant, and let $i : F \to u_*F'$ be its strict adjoint. The object F' is also cofibrant. Composition with i' induces a bijection

$$\mathrm{Hom}_{\mathrm{HO}(I')}(F', G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{HO}(I')}(u^*F, G).$$

For a class g on the right, let a be its unique inverse image and define

$$\gamma(g) = u_*^W(a) \circ [i].$$

This is where admissibility enters: u_*^W is a functor on the localization of fibrant objects, so it acts on the class a, independently of its representative. Naturality of the ordinary counit, together with the fact that i' is adjoint to i, gives $\beta\gamma(g) = a[i'] = g$.

For the other composite, represent a class $b : F \to u_*G$ by an ordinary map and let $g : u^*F \to G$ be its ordinary adjoint. Since i' is a trivial cofibration and G is fibrant, g extends to an ordinary map $a : F' \to G$ satisfying $ai' = g$. Its class is the inverse image used in defining γ. The ordinary adjunction then gives $u_*a \circ i = b$, proving $\gamma\beta([b]) = [b]$. Thus β is bijective. Its naturality, already built into restriction and the counit, gives the required bifunctorial adjunction.

This is the derived-adjunction construction; compare Hovey (1999a, Lemma 1.3.10). Independence of the chosen F' follows from uniqueness of the inverse of β, rather than from an additional comparison of arbitrary replacements. It also explains the manuscript's remark that functoriality in G appears a posteriori. The adjoint obtained is

$$u_W^* \dashv u_*^W.$$

The two occurrences of "adjoint à gauche" on p. 71 are slips in the source and must read "adjoint à droite". The Hom formula itself already fixes the orientation, so the proof is unaffected. Likewise the target containing u^*F' on p. 70 is ill typed: F' is already an I'-diagram, and the intended object is u^*F.

Je dis que l'on a de toute façon

$$R \;\leq\; \alpha(R')$$

sans hypothèse d'admissibilité sur u, *i.e.* que la bijection invese de $(**)$ donne par passage aux quotients un homomorphisme

$$(***) \qquad \mathrm{Hom}_W(u^*F, G') \longleftarrow \mathrm{Hom}_W(F, u_*G'),$$

forcément surjectif, et de plus fonctoriel en F `[, G' barré]`, variables dans $\mathrm{HO}_W(I)$ et $\mathrm{HO}_W(I')$ respectivement. En effet, on définit la flèche ($*$ bis) `[plutôt (***)]` comme composé

$$\mathrm{Hom}_W(F, u_*G') \longrightarrow \mathrm{Hom}_W(u^*F, u^*u_*G') \longrightarrow \mathrm{Hom}_W(u^*F, G').$$

car u^* est compatible avec $W(I)$, $W(I')$ sur $\mathcal{M}(I)$ tout entier, et on interprète les deux membres comme des Hom dans $\mathcal{M}(I)W(I)^{-1}$ et $\mathcal{M}(I')W(I')^{-1}$.

composition avec l'homomorphisme d'adjonction $u^*u_*G' \longrightarrow G'$

Cela implique donc que le foncteur sur $\mathrm{HO}_W(I)$ $F \longmapsto \mathrm{Hom}_{\mathrm{HO}_W(I')}(u^*F, G')$ est, sinon représentable, du moins quotient terme à terme du foncteur représenté par u_*G' ([37]). p. 39

Mais supposons maintenant que u soit admissible. Alors je vais définir la flèche en sens initial ($*$) (p. 68) de la façon suivante : Je choisis une flèche [page 70]

$$i' \;:\; u^*F \longrightarrow F' \qquad \text{avec } i' \in TC \subset C \cap W,\; F' \in \mathrm{Ob}\mathcal{M}(I')_{\mathrm{f}}$$

(donc en fait $F' \in \mathrm{Ob}\mathcal{M}(I')_{\mathrm{cf}}$, car u^*F est cofibrant), et par adjonction la flèche i' correspond à

$$i \;:\; F \longrightarrow u_*F'.$$

37. Mais il n'est pas clair, sans hypothèse supplémentaire sur la situation, que ce foncteur représentable puisse être choisi en dépendance fonctorielle de G', envisagé dans $\mathrm{HO}_W(I')$.

Je considère le diagramme (commutatif!)

$$\begin{array}{ccc} \mathrm{Hom}(F', G') & \xrightarrow{\ \alpha\ } & \mathrm{Hom}(u_*F', u_*G') \\ \Big\downarrow{\scriptstyle \lambda : f' \longmapsto f' \circ i'} & & \Big\downarrow{\scriptstyle \mu : f \longmapsto f \circ i} \\ \mathrm{Hom}(u^*F, G') & \xrightarrow{\ \text{adj.}\ } & \mathrm{Hom}(F, u_*G')\ . \end{array}$$

On note que l'on a encore F' cofibrant (car u^*F l'est et $i' \in TC \subset C$), donc $\mathrm{Hom}_W(F', G')$ est un quotient de $\mathrm{Hom}(F', G')$, et λ induit un isomorphisme

$$\mathrm{Hom}_W(F', G') \xrightarrow{\ \sim\ } \mathrm{Hom}_W(u^*F', G')$$

(car $i' \in TC \subset W$). Comme F', $G' \in \mathrm{Ob}\mathcal{M}(I')_{\mathrm{f}}$ et que u_*^{f} est compatible avec $W(I')_{\mathrm{f}}$, $W(I)_{\mathrm{f}}$, il s'ensuit que la flèche horizontale α est compatible avec les relations d'équivalence définies par [page 71]

$$\mathrm{Hom}(F', G') \longrightarrow \mathrm{Hom}_{W(I')}(F', G') \quad \text{resp.} \quad \mathrm{Hom}(u_*F', u_*G') \longrightarrow \mathrm{Hom}_{W(I)}(u_*F', u_*G')\ .$$

Et il en est de même de la flèche $\mu : f \longmapsto f \circ i$, trivialement. Donc la flèche d'adjonction $\mathrm{Hom}(u^*F, G') \longrightarrow \mathrm{Hom}(F, u_*G')$ est compatible avec les relations R, R', ce qui achève de prouver que (∗∗), page 68, induit une bijection (∗), page 68.

Ceci montre donc

1°) que pour un objet fixé G' dans $\mathrm{HO}_W(I')$ (qu'on peut toujours supposer, à isomorphisme près, provenir de $\mathcal{M}(I')_{\mathrm{f}}$), le foncteur

$$F \longmapsto \mathrm{Hom}_{\mathrm{HO}_W(I')}(u_W^*F, G')$$

est représentable – *i.e.* le foncteur u_W^* admet un adjoint à gauche, et

2°) que cet adjoint à gauche se calcule comme $u_*(G')$ (quand l'argument G' est pris fibrant) – vu que l'homomorphisme (∗∗∗), page 69, est *fonctoriel en* F. (A posteriori il s'ensuit qu'il est aussi fonctoriel en G' …)

Commentary · Exposition [ms. 71] p. 40

Pointwise right images require fibrant restriction [page 72]

Grothendieck now wants more than an abstract right adjoint: he wants to compute its value at an object by a homotopy limit over a comma category. The comparison is canonical, but it becomes an isomorphism only after the same replacement is known to be suitable on both sides. This is the content of the invocation of local isomorphisms on p. 72.

Fix $s \in I$, let $q : I'/s \to I'$ be the projection, and write $p_s : I'/s \to e$. If $X' \to RX'$ is fibrant replacement and q^* preserves fibrant objects, then q^*RX' is a fibrant replacement of q^*X', because restriction always preserves the objectwise weak equivalences. The ordinary pointwise formula therefore identifies

$$\left(\mathbf{R}u_*X'\right)_s = \left(u_*RX'\right)_s \cong \varprojlim_{I'/s} q^*RX' \cong \mathbf{R}(p_s)_*q^*X'.$$

The last step uses admissibility of p_s. Thus the comparison is computed on one common representative, rather than by comparing two unrelated fibrant replacements.

For clarity, there is a simple sufficient condition for the fibrancy assertion. The functor q is a discrete fibration: an arrow $j \to j'$ in I' and an object $(j', u(j') \to s)$ have a unique lift with source $(j, u(j) \to u(j') \to s)$. In our contravariant convention, when the necessary coproducts exist, its left Kan extension has the formula

$$(q_!A)_j \cong \coprod_{x \in q^{-1}(j)} A_x.$$

Coproducts of trivial cofibrations have the left lifting property against every fibration, so $q_!$ preserves the objectwise trivial cofibrations. Its right adjoint q^* consequently preserves fibrations and fibrant

objects. This is the contravariant version of the discrete-opfibration argument in Groth (2013, Lemma 1.31).

If I is a poset, each fiber has at most one element, so only the initial object is needed for this coproduct formula. In the large cocomplete model examples, all small fibers are allowed. In a setting with only finite limits and finite colimits, an arbitrary infinite fiber requires an additional existence hypothesis; it must not be supplied by notation. This is why the general admissibility result and the pointwise formula have been stated separately in the commentary. The finite-poset corollary satisfies the needed conditions, and the usual combinatorial model setting satisfies them for every small functor.

Modern derivator theory packages this calculation as the pointwise Kan-extension axiom. The square of shapes contains the canonical natural transformation $uq \Rightarrow sp_s$; its mate is exactly the displayed comparison. Thinking in terms of that mate explains its compatibility with composition and base change, while the fibrant-representative calculation proves that it is invertible in the present models.

The same proof gives a useful class of base-change squares, with a condition that can be checked directly. Let $v : I_1 \to I$ be a discrete fibration, form the strict pullback $I'_1 = I' \times_I I_1$, and denote its projections by $v' : I'_1 \to I'$ and $u_1 : I'_1 \to I_1$. Assume that the required right Kan extensions and the coproducts over the fibers exist. For each $t \in I_1$ the natural functor

$$I'_1/t \longrightarrow I'/v(t)$$

is an isomorphism. Indeed, an object of the target is an arrow $u(i') \to v(t)$. Its unique lift along v is an arrow $t' \to t$ with $v(t') = u(i')$, and therefore supplies the object $((i', t'), t' \to t)$ of the source. The uniqueness of lifts gives the inverse also on morphisms. This proves the ordinary base-change identity by the pointwise limit formula.

The pullback v' is again a discrete fibration, so the preceding coproduct argument shows that $(v')^*$ preserves fibrant objects. Evaluate both sides of the ordinary identity on a fibrant replacement of X'. They then compute the two derived functors, proving the canonical isomorphism

$$v^*\mathbf{R}u_*X' \cong \mathbf{R}(u_1)_*(v')^*X'.$$

This is the same discrete-fibration mechanism used in the comma calculation, applied to a family of objects at once; it is part of the standard pointwise derivator calculus cited above.

A strict pullback square without this hypothesis need not have such an isomorphism. Take $I = (0 \to 1)$, let $I' = \{0\}$ and $I_1 = \{1\}$ be the two endpoint inclusions, and use complexes over a field. Their pullback is empty. For $X' = k[0]$, the left side evaluated at 1 is $k[0]$, because $\{0\}/1$ is terminal. The right side is zero, the limit of the empty diagram in complexes. Thus the base-change map cannot be an isomorphism. The pointwise axiom guarantees the designated comma squares; additional squares require the corresponding exactness argument.

Le dernier énoncé dans 2° du théorème-scholie résulte du calcul de u_*^W, compte tenu que $I'/s \longrightarrow I'$ est un isomorphisme local, donc le foncteur image inverse envoie

$$\mathcal{M}(I')_{\mathrm{f}} \longrightarrow \mathcal{M}(I'/s)_{\mathrm{f}},$$

donc si G' est dans $\mathcal{M}(I')_{\mathrm{f}}$,

$$u_*(G')(s) \;\simeq\; p_{s\,*}(G'|I'/s)$$

est un calcul de $p_{s\,*}^W(G'|I'/s)$.

Cela achève la démonstration du théorème-scholie sur les images directes.

Commentaires.

1) Ce théorème me paraît loin d'être satisfaisant, c'est un pis-aller. J'avais espéré, avec les hypothèses du paragraphe 3, qu'on aurait l'existence des u_*^W pour *toute* flèche $u : I' \longrightarrow I$ dans Cat. Je doute à présent que ce soit le cas, mais n'ai pas encore réussi de faire un contre-exemple. Les cas test les plus importants pour la question d'admissibilité d'une catégorie I me paraissent les suivants :

a) I catégorie discrète infinie (disons dénombrable). La question revient ici à celle-ci : soient $f_i : X_i \longrightarrow Y_i$ dans W, avec X_i, Y_i fibrants, *i.e.* dans F `[i.e.` $X_i \longrightarrow e$`,` $Y_i \longrightarrow e$ `dans` F`]`. Alors $f = \prod f_i : \prod X_i \longrightarrow \prod Y_i$ est-il dans W ?

[page 73]

Commentary · Present state [ms. 72]

Products and towers have a positive answer

Grothendieck's test cases isolate two ways in which an infinite limit could lose a weak equivalence. The first takes infinitely many independent factors; the second imposes infinitely many compatibility equations. Under the closed model hypotheses he explicitly mentions, both questions have affirmative answers. Their difficulty is the availability of suitable replacements, not a failure of weak equivalence to survive the limit of a fibrant diagram.

For a set S, equip $\mathcal{M}^S$ with the product structure. The diagonal functor preserves cofibrations and trivial cofibrations componentwise. Its right adjoint, whenever it exists, is $\prod_S$. The Brown argument therefore shows that a product of weak equivalences between fibrant objects is a weak equivalence. This remains true for uncountable S whenever the required products exist. No assumption about exact filtered colimits is used. The derived product is consequently computed by the product of fibrant replacements; the same conclusion appears explicitly in Hovey (1999a, Example 1.3.11).

For the tower shape, use the matching-object description already seen in finite posets. A tower

$$X_0 \longleftarrow X_1 \longleftarrow X_2 \longleftarrow \cdots$$

is fibrant precisely when X_0 is fibrant and every bonding map is a fibration; all other X_n are then fibrant by composition. The recursive lifting proof has one step for each natural number and defines a compatible lift on the whole tower. Taking the limit is right adjoint to the constant-tower functor, which preserves the objectwise cofibrations and trivial cofibrations. Brown's lemma again gives invariance under levelwise weak equivalences between these towers. In the ordinary complete model setting this is also the inverse Reedy construction.

The manuscript correctly observes that the tower assertion implies the countable product assertion. For a countable family (X_n) of fibrant objects, put $P_r = \prod_{0 \le n \le r} X_n$. The projection $P_{r+1} \to P_r$ is a pullback of $X_{r+1} \to e$, hence a fibration. A family of weak equivalences induces weak equivalences between the finite products, by the finite case; the inverse limit of the resulting map of towers is the original map of infinite products. This reduction proves the stated implication, although the direct product adjunction is shorter once the general lemma is recognized.

The affirmative answer does not mean that ordinary inverse limits preserve all levelwise weak equivalences. The following example shows exactly where the bonding-map condition matters. It is also a useful safeguard when replacing an abstract existence theorem by an explicit tower of complexes.

b) I catégorie

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots,$$

i.e. la catégorie ordonnée associée à l'ensemble ordonné $\mathbf{N}$, pour la relation d'ordre ordinaire $m \le n$. Donc $\mathcal{M}(I)$ est formée des systèmes *projectifs* dans $\mathcal{M}$, indexés par $\mathbf{N}$. On prouve facilement (en procédant comme dans le corollaire du lemme p. 63) que $\mathcal{M}(I)_{\mathrm{f}}$ est formée des systèmes projectifs

$$X_0 \longleftarrow X_1 \longleftarrow X_2 \longleftarrow \cdots \longleftarrow X_n \xleftarrow{p_n} X_{n+1} \longleftarrow \cdots$$

tels que les X_i soient dans $\mathcal{M}_{\mathrm{f}}$, et les p_n dans F. La question est donc : si $X_\bullet$, $Y_\bullet$ sont fibrants et $f_\bullet : X_\bullet \longrightarrow Y_\bullet$ est telle que $f_i \in W$ pour $i \in \mathbf{N}$, alors a-t-on $\varprojlim f_i : X_\infty \longrightarrow Y_\infty$ dans W ?

Si la reponse est affirmative pour $\mathbf{N}$ (pour un W et une quillénisation donnée), elle l'est aussi pour toute catégorie discrète dénombrable, comme on le voit aussitôt. Mais je ne vois pas comment déduire même *a*) des axiomes, disons pour un triplet de Quillen clos

(closed model category) tel que $\mathcal{M}$ soit de plus stable par $\varinjlim$ filtrantes, que celles-ci soient exactes, et qu'il y ait un petit ensemble de générateurs dans $\mathcal{M}$ (donc $\mathcal{M}$ est accessible), que W soit stable par $\varinjlim$ filtrantes. Je devrais demander à Thomason s'il connait un contre-exemple. p. 41 [page 74]

Commentary · Examples [ms. 73]

An inverse limit of acyclic complexes can have cohomology

The insistence on fibrant towers is mathematically necessary, even in the most familiar algebraic model. Fix a prime p and, for $n \geq 1$, form the cochain complex

$$A_n = \left[\mathbb{Z} \xrightarrow{p^n} \mathbb{Z} \xrightarrow{\bmod p^n} \mathbb{Z}/p^n\right]$$

in degrees $0, 1, 2$. It is acyclic: the first map is injective, the image of that map is the kernel of the second, and the second is surjective. Define $A_{n+1} \to A_n$ by multiplication by p in degree 0, the identity in degree 1, and reduction modulo p^n in degree 2. These maps commute with both differentials. The maps of towers $A_n \to 0$ are therefore quasi-isomorphisms at every level.

Their ordinary inverse limit is readily computed. A compatible sequence in degree 0 satisfies $a_n = pa_{n+1}$. Its first coordinate is divisible by every power of p, and hence vanishes; then every coordinate vanishes. The limit in degree 1 is $\mathbb{Z}$, and the limit in degree 2 is $\mathbb{Z}_p$. Thus

$$\varprojlim_n A_n = \left[0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}_p\right].$$

The map $\mathbb{Z} \to \mathbb{Z}_p$ is injective but not surjective. For example, $\mathbb{Z}_p$ is uncountable, since a sequence of digits in $\{0, \ldots, p-1\}$ uniquely determines a p-adic integer, while $\mathbb{Z}$ is countable. Consequently

$$H^2\left(\varprojlim_n A_n\right) = \mathbb{Z}_p/\mathbb{Z} \neq 0.$$

The inverse limit of the levelwise quasi-isomorphism $A_\bullet \to 0$ is not a quasi-isomorphism.

There is no conflict with the positive result on the preceding page. In the projective model structure on unbounded complexes of abelian groups every object is fibrant, but a fibration is degreewise surjective. The bonding map of this tower is multiplication by p in degree 0, so it is not a fibration. Individual fibrancy has not made the *diagram* fibrant. A fibrant tower replacement would have an acyclic inverse limit by the Brown argument.

This example separates three statements which can otherwise look deceptively similar: ordinary inverse limit is not exact; inverse limit need not preserve levelwise quasi-isomorphisms of arbitrary towers; inverse limit does preserve weak equivalences between fibrant towers in the model structure under discussion. Only the third statement is the admissibility problem posed in the manuscript. Its positive answer is compatible with the failure exhibited here.

c) Les deux exemples précédents étaient des catégories ordonnées. Comme exemple typique de catégorie non-ordonnée, où je ne vois pas ce qui se passe, je signale $I = B_G$, G un groupe (donc un seul objet, avec un groupe d'automorphismes G). Aussi le cas de $\mathbf{\Delta}[1] = \{\mathbf{\Delta}_0, \mathbf{\Delta}_1\}$ (sous-catégorie pleine de $\mathbf{\Delta}$) n'est pas clair pour moi.

Commentary · Examples [ms. 74]

A group action detects the difference between fixed and homotopy fixed points

Grothendieck introduces B_G because automorphisms are absent from the finite-poset argument. The limit of a G-diagram is a fixed-point object. Its derived version measures the coherence of invariance under the whole action, and a single fibrant value does not guarantee that the action diagram is injectively fibrant. The algebraic example below makes the difference explicit and at the same time answers the boundedness question from §5.

Let $G = C_p = \langle g \mid g^p = 1\rangle$, let k be a field of characteristic p, and give k the trivial action. Using inversion to change a right action into a left action if necessary, diagrams of complexes on B_G

are complexes of $R = k[G]$-modules. With $t = g - 1$, one has $R \cong k[t]/(t^p)$. The trivial module has the free resolution

$$\cdots \xrightarrow{t} R \xrightarrow{t^{p-1}} R \xrightarrow{t} R \longrightarrow k \longrightarrow 0.$$

Its exactness follows immediately from

$$\ker(t : R \to R) = (t^{p-1}), \qquad \ker(t^{p-1} : R \to R) = (t).$$

Applying $\mathrm{Hom}_R(-, k)$ makes every differential zero, since t acts by zero on k. Tensoring with k does the same. Therefore

$$\mathrm{Ext}^n_R(k, k) \cong k, \qquad \mathrm{Tor}^R_n(k, k) \cong k \qquad (n \geq 0).$$

For the projection $p : B_G \to e$ this gives

$$\mathbf{R}p_* k = \mathbf{R}\,\mathrm{Hom}_{k[G]}(k, k), \qquad \mathbf{L}p_! k = k \otimes^{\mathbf{L}}_{k[G]} k.$$

The first object has nonzero cohomology in every nonnegative degree, and the second in every nonpositive degree. Both exist in $D(k)$. Neither belongs to $D^{\mathrm{b}}(k)$, although the input is concentrated in degree zero. Finite cohomological dimension is therefore a boundedness condition in this example, not a prerequisite for the unbounded operations.

Ordinary invariants of the original diagram give only $k[0]$. Moreover, over a field its underlying complex is already fibrant. The additional cohomology arises because injective fibrancy in the diagram category is a condition on the $k[G]$-complex, not merely on the underlying k-complex. Replacing that diagram by a K-injective $k[G]$-complex and taking ordinary invariants computes $\mathbf{R}p_*$. The Brown lemma proves invariance between such fibrant diagrams, so the presence of a nontrivial automorphism group creates no exception to admissibility.

Exactly the same general argument applies to the full subcategory of $\mathbf{\Delta}$ on [0] and [1] mentioned next in the manuscript. It is not the walking arrow: it contains all maps between those two ordinals, including the two sections and the retraction. It has two objects and seven morphisms: the four Hom sets have respectively $1, 2, 1, 3$ elements. Thus its ordinary limit uses only the finite limits of $\mathbf{M}0$. Its relations matter when constructing a coherent diagram, but the assumed uniform structure and the lifting-adjunction proof already give admissibility; no further infinite-completeness assumption is needed for this test.

2) On aimerait pouvoir traiter de façon duale l'existence des $u^W_!$, à défaut de mieux. Dans l'hypothèse de départ du paragraphe 4 sur $(\mathcal{M}, W)$, cela amène donc à changer de tactique. Cette fois, on devra supposer l'existence d'une quillénisation close *exacte à droite* pour W (ce qui est plus exigeant, semble-t-il, qu'une telle structure exacte à gauche, dans le cas envisagé), donc

$$TF \;=\; F \cap C,$$

et de plus encore C, TC tels qu'on ait [page 75]

$$C \;=\; \widetilde{C_0}, \qquad TC \;=\; \widetilde{TC_0}$$

avec C_0, TC_0 petits ([38]). On posera maintenant

$$\begin{aligned} F(I) \;&=\; \{f \in \mathrm{Fl}\,\mathcal{M}(I) \;\mid\; f_s \in F \quad \forall\, s \in \mathrm{Ob}I\} \\ TF(I) \;&=\; \{f \in \mathrm{Fl}\,\mathcal{M}(I) \;\mid\; f_s \in TF \;\; \forall\, s \in \mathrm{Ob}I\}, \end{aligned}$$

puis

$$\begin{aligned} C(I) \;&=\; TF(I)^* \\ TC(I) \;&=\; F(I)^* \;. \end{aligned}$$

L'espoir (que je devrai vérifier ultérieurement), c'est qu'on obtient ainsi une quillénisation uniforme à droite de W, sur Cat tout entier. Le seul problème, c'est la validité des axiomes de factorisation de $f \in \mathrm{Fl}\,\mathcal{M}(I)$ en $f = pi$, avec $i \in TC$, $f \in F$, et aussi avec $i \in C$, $f \in TF$.

38. On dit alors que C, TC (en tant que parties closes à droite `[plutôt à gauche]`) sont "accessibles".

J'ai bon espoir que c'est vrai – cela se réduit essentiellement à voir que $(C(I), TF(I))$ et $(TC(I), F(I))$, soit $(\Phi(I), \Psi(I))$, satisfont en plus de $\Phi(I) = \Psi(I)^*$, à $\Psi(I) = \Phi(I)_*$ (*i.e.* $\Psi(I)$ est clos à droite), et que $\Phi(I)$ est accessible (tout comme $\Phi = \Phi(e)$). Si ça marche, on aura donc l'existence de $u_!$ pour u un foncteur entre catégories ordonnées finies, et sous des conditions légèrement plus générales. (P. ex. $u : I' \longrightarrow I$ avec I ordonnée, I' ordonnée et à espace sous-jacent noethérien et sobre.) Peut-être que cette fois, l'hypothèse que W est stable par $\varinjlim$ filtrantes va quand même servir! Ainsi, le cas des catégories discrètes, et des catégories filtrantes à gauche comme $\mathbf{N}^\circ$, ne va plus poser de problèmes à cause de cela. Il y a donc de l'espoir! Tout ensemble ordonné qui est $\varinjlim$ filtrante de sous-ensembles ordonnés *fermés* noethériens et sobres, est alors W-admissible à droite. Mais qu'en est-il de catégories de type $\Gamma = B_G$ p. ex. ? ? ? p. 42 [page 76]

Commentary · Later developments [ms. 75]

The projective construction supplies the dual theory

Grothendieck's change of tactic is the right one for left images. To derive a colimit, fibrations should be defined objectwise, so restriction becomes a right Quillen functor and its left adjoint acts on cofibrant diagrams. The extra small generating sets are needed to construct these projective diagram structures. Once they have been constructed, the dual Brown argument supplies homotopy invariance for every existing left Kan extension, rather than only for finite posets.

Here is the construction with its size hypothesis visible. Suppose that $\mathcal{M}$ is a locally presentable, cofibrantly generated model category, with generating cofibrations C_0 and generating trivial cofibrations T_0. For $i \in I$, evaluation has a left adjoint L_i, given in the contravariant convention by

$$(L_i A)_j = \coprod_{I(j,i)} A.$$

Use the sets $\{L_i c : i \in I, c \in C_0\}$ and $\{L_i t : i \in I, t \in T_0\}$. By the evaluation adjunction, their right lifting classes are exactly the objectwise trivial fibrations and the objectwise fibrations, respectively. The sets really are small: I and the two generating collections are small. For a sufficiently large regular cardinal the domains are presentable, because evaluation preserves filtered colimits and

$$\mathrm{Hom}_{\mathcal{M}^{(I)}}(L_i A, X) \cong \mathrm{Hom}_{\mathcal{M}}(A, X_i).$$

The small object argument therefore gives both desired factorizations.

One also checks the weak equivalences in the transferred structure. A relative cell complex built from the $L_i t$ is componentwise a composite of pushouts of coproducts of trivial cofibrations. The base trivial cofibrations form a left lifting class and are closed under these operations, so the resulting map is an objectwise trivial cofibration and hence an objectwise weak equivalence. Taking retracts gives the same conclusion for every map in $TC(I)$. Moreover $TC(I) \subset C(I)$, because the objectwise trivial fibrations are objectwise fibrations. If $f \in C(I) \cap W(I)$, factor it as $f = pj$ with $j \in TC(I)$ and $p \in F(I)$. Two-out-of-three gives $p \in W(I)$, so $p \in TF(I)$ by the objectwise definitions. The lifting property of f against p makes f a retract of j, hence $f \in TC(I)$. This identifies the required acyclic class and completes the model-structure verification. It is the projective transfer argument sought in the paragraph following the two formulas on p. 75. See Groth (2013, Theorem 1.28 and Lemma 1.29); the small object argument itself is recorded in Hovey (1999a, Theorem 2.1.14).

For $u : I' \to I$, restriction now preserves the objectwise fibrations and trivial fibrations. Adjointness makes $u_!$ preserve cofibrations and trivial cofibrations. The dual Brown lemma shows that it preserves weak equivalences between cofibrant diagrams, and the derived adjunction is

$$\mathbf{L}u_! \dashv u^*.$$

The hypotheses of local presentability and cofibrant generation make all small diagram shapes available, including B_G. An objectwise construction of fibrations does not mean that cofibrant diagrams are objectwise cofibrant with no further condition; the latter would miss precisely the relations that the free diagrams $L_i A$ impose.

Two slips in the source obscure this otherwise correct programme. Exactness on the right is $TF = F \cap W$, not $F \cap C$. In the factorizations $f = pi$, the membership assertions must refer to $p \in F$ or $p \in TF$, not to f. They are retained as printed and explained in the correction register.

The role of filtered colimits also deserves precision: in the projective transfer just described, closure of trivial cofibrations under transfinite composites follows from their lifting property. A separate assertion that all weak equivalences are closed under filtered colimits is not what proves that step.

[page 77]

Commentary · Context [ms. 76]

Why the homotopy problem returns to SGA 4

Grothendieck turns to cardinal filtrations because a construction on every diagram category needs a controlled supply of small pieces. The issue is already visible in the proposed right uniform quillenisation: a lifting condition must be generated by a set of maps, and the factorisations built from that set must remain available after changing the indexing category. The return to SGA 4, I, §9 is therefore part of the homotopical argument. It supplies a way of bringing a large category within reach of set-indexed constructions.

The later letter to Thomason makes this connection explicit. In its opening pages Grothendieck describes his renewed work on accessible categories and accessible parts as a rebuilding of the provisional account in SGA 4. In the next section of the letter, he distinguishes the relative category $(\mathcal{M}, W)$, which specifies the homotopy theory, from the cofibrations and fibrations used to construct its operations (Grothendieck, 1991a, pp. 2–4 of the electronic edition). The present chapter places these two concerns next to one another: first find enough replacements to derive direct images; then explain the smallness that makes such replacements possible. This is also why the size of a *diagram*, rather than just the sizes of its values, becomes decisive near the end of §7.

There are two historical cautions. The word "accessible" here follows the cardinal conventions of SGA 4: "grand devant π" bounds every subset of size at most π, and thus corresponds to π^+-directedness in the regular-cardinal terminology now customary. Also, the closing definition tests membership in a part against *every* perfectly adapted presentation. It is stronger than merely exhibiting the part as a category of sufficiently directed colimits of small objects. The distinction is mathematically effective: the counterexamples below separate those conditions even in a powerset ordered by inclusion. The later development of accessibility does not license us to identify formulations whose quantifiers differ.

Commentary · Present state [ms. 76]

What has been settled before the cardinal detour

The final questions of §6 can now be separated by the task they perform. For the uniform closed exact structures assumed in the theorem-scholie, the Brown lemma settles admissibility of every functor with the requisite ordinary right adjoint; its dual settles left admissibility. Products, towers, group actions and the truncated simplex category are all included. The finite-poset proof remains a direct construction of the finite calculus, while the noetherian sober Alexandrov condition adds no infinite posets.

The next question is existence of those uniform structures. In the usual combinatorial setting, injective and projective diagram model structures, derived adjunctions, and the pointwise comma-category formulas furnish a strong derivator on all small categories. This is the later theorem corresponding most directly to Grothendieck's programme. In smaller categories such as bounded complexes, one retains finite operations and must check each infinite ordinary limit or colimit separately. The bounded-product example above prevents a blanket completeness assertion.

Section 7 addresses the remaining constructive burden: how a large lifting class or a large collection of objects can be controlled by a small family at a sufficiently large cardinal. That question is substantial even though the preceding homotopy-invariance doubts have a formal answer. It is also where the older SGA 4 language of accessibility and the later language of presentability need to be compared with particular care. The change of subject is therefore motivated by the production of factorizations, rather than by a need to repair Brown's argument.

7 Filtrations cardinales et parties accessibles d'une grosse catégorie

Commentary · Exposition [ms. 77]

A size filtration and a presentability rank

Grothendieck returns to cardinal estimates because the constructions of model structures need a controlled supply of small objects. A replacement construction may be performed object by object and still fail to be controlled uniformly over a category. The filtration is intended to provide that control: a large object is assembled from bounded pieces, while bounded constructions on bounded pieces remain bounded. The categorical purpose of the estimates should remain visible through the set theory.

There is a one-cardinal shift in the language. Throughout these pages, "grand devant π" means that every subset of cardinal at most π has an upper bound. In modern terminology such a nonempty poset is π^+-directed, where π^+ is the successor cardinal. Write $\kappa = \pi^+$ when translating the assertions. An object called π-accessible in the sense of SGA 4 is correspondingly κ-presentable: its covariant Hom functor preserves these directed colimits. This meaning of "accessible object" must also be kept distinct from the new meaning of "accessible part" introduced on manuscript page 86. The latter concerns a full subcategory and its position inside the ambient category.

The three filtration conditions do not say that every object of $\mathrm{Filt}^\pi(E)$ is π^+-presentable. They say that this stage is essentially small, is closed under the specified bounded colimits, and generates E under sufficiently directed colimits. Preservation of such colimits by Hom is an additional hypothesis. This explains why the hypothesis $\mathrm{Filt}^\pi(E) \subset E_\pi$, initially left tacit, becomes necessary on pages 90–94. Under suitable accessibility hypotheses, SGA 4 supplies useful cardinals at which size and presentability do coincide; see Artin et al. (1972, I, 9.16–9.18). It does not identify the two notions at every index of an arbitrary filtration.

The universe convention is also doing real work. "Essentially small" means that a skeleton belongs to the fixed universe; it does not mean that there are finitely or countably many objects. Once a skeleton is chosen, the relevant comma categories are small because E is a $\mathfrak{U}$-category. All diagrams below remain small relative to this universe. No assertion that the whole large category has a set of objects in that same universe is needed.

Finally, the filtration is exhaustive even though exhaustiveness is not listed separately. Choose the presentation from condition (c) at π_0, and choose $\rho \geq \pi_0$ larger than the cardinality of its indexing poset. Its terms lie in $\mathrm{Filt}^\rho(E)$, and condition (b) places its colimit in the same stage. Thus every object eventually enters the filtration. This short argument is the content of footnote 39.

Je me réfère à la théorie des foncteurs et catégories accessibles de SGA 4, I, §9 (p. 138-178). Je me donne une grosse catégorie (toujours une $\mathfrak{U}$-catégorie) E avec une *filtration cardinale* (croissante)

$$\Big(\mathrm{Filt}^\pi(E)\Big)_{\pi \geq \pi_0}, \qquad \pi_0 \text{ un cardinal infini fixé,}$$

par les sous-catégories strictement pleines $\mathrm{Filt}^\pi(E)$. On a `[SGA 4, I, 9.12]`

a) $\mathrm{Filt}^\pi(E)$ essentiellement petite, *i.e.* équivalente à une petite catégorie (*i.e.* à une catégorie appartenant à $\mathfrak{U}$).

b) E satisfait L_{π_0} (*i.e.* E est stable par $\varinjlim$ de type I, quand I est un ensemble ordonné grand devant π_0), et $\mathrm{Filt}^\pi(E)$ est stable par les limites inductives filtrantes indexées par I grand devant π_0, et tel que $\mathrm{card}\, I \leq \pi$.

c) Pour tout $\pi \geq \pi_0$, et pour tout $X \in \mathrm{Ob}E$, on a

$$X \simeq \varinjlim_{i \in I} X_i,$$

où les X_i `[sont]` dans $\mathrm{Filt}^\pi(E)$, et où I est un ensemble ordonné grand devant π. De plus, si $X \in \mathrm{ObFilt}^{\pi'}(E)$, avec $\pi' > \pi$, on peut prendre I tel que $\mathrm{card}\, I \leq \pi'^\pi$ (donc

si $\pi' = 2^{c'}$ avec $c' \geq \pi$, d'où $(\pi')^\pi = (2^{c'})^\pi = 2^{c'\pi} = 2^{c'} = \pi'$, on aura $\operatorname{card} I \leq \pi'$). (**NB** Si on admet l'hypothèse du continu généralisée, on aura toujours $\pi'^\pi = \pi'$, donc $\operatorname{card} I \leq \pi'$.) ([39])

Commentary · Examples [ms. 77]

The useful exponent bound and the singular-cardinal exception

The bound $(\pi')^\pi$ measures how many small pieces may be needed to describe an object of size π'. It should be retained until an actual cardinal identity permits its replacement by π'. The calculation for a power-set cardinal is correct: if $\pi' = 2^{c'}$ and $c' \geq \pi$ are infinite, then

$$(\pi')^\pi = (2^{c'})^\pi = 2^{c'\cdot\pi} = 2^{c'} = \pi'.$$

Thus arbitrarily large useful choices are available without a continuum hypothesis.

The subsequent claim about the generalized continuum hypothesis is false as written. If μ is singular and $\operatorname{cf}(\mu) \leq \pi < \mu$, then $\mu^\pi > \mu$, independently of GCH. Here is the elementary obstruction. Choose increasing infinite cardinals $\mu_\alpha < \mu$, for $\alpha < \delta = \operatorname{cf}(\mu)$, with supremum μ. König's diagonal argument gives

$$\mu = \sum_{\alpha<\delta} \mu_\alpha < \prod_{\alpha<\delta} \mu_\alpha^+ \leq \mu^\delta \leq \mu^\pi.$$

For completeness, the strict inequality follows by partitioning a purported list of all elements of the product into pieces S_α of size μ_α. In coordinate α, choose a value outside the α-coordinates of S_α; the resulting element is absent from every piece of the list. Taking $\mu = \aleph_\omega$ and $\pi = \aleph_0$ already disproves the printed assertion.

Under GCH the conclusion does hold when $\pi < \operatorname{cf}(\pi')$. In that case each map $\pi \to \pi'$ has bounded image. For every infinite $\nu < \pi'$, GCH gives

$$\nu^\pi \leq (2^\nu)^\pi = 2^{\max(\nu,\pi)} = \max(\nu,\pi)^+ \leq \pi',$$

and the union of π' such sets has size at most π'. The constant functions give the reverse inequality. The missing cofinality condition matters precisely at singular cardinals. Nothing later in the construction requires the erroneous unconditional simplification: the bound $(\pi')^\pi$, and the unconditional power-set choices, suffice.

Considérons la représentation de c), p. 43 [page 78 barrée] [page 79]

$$X \;=\; \varinjlim_{I} X_i \qquad \begin{cases} X_i \in \operatorname{Filt}^\pi, \\ I \text{ grand devant } \pi. \end{cases}$$

Soit

$$\tilde{I} \;=\; \{I' \subset I \mid \operatorname{card} I' \leq \pi, \quad I' \text{ grand devant } \pi_0\},$$

muni de la relation d'ordre d'inclusion. Pour

$$I' \;\in\; \tilde{I}, \qquad \text{soit} \;\; X_{I'} = \varinjlim_{I'} X_i \;,$$

cette limite existe par L_{π_0} (hypothèse b), elle est dans Filt^π par b), donc on a un système inductif

$$(X_{I'})_{I'\in\tilde{I}}, \qquad X_{I'} \in \operatorname{Filt}^\pi,$$

avec

$$\operatorname{card} \tilde{I} \;\leq\; (\operatorname{card} I)^\pi,$$

et si $X \in \operatorname{Filt}^{\pi'}$ $(\pi' > \pi)$, prenant

$$\operatorname{card} I \;\leq\; \pi'^\pi,$$

39. NB b) et c) impliquent que $E = \bigcup_{\pi\geq\pi_0} \operatorname{Filt}^\pi(E)$.

on aura $\operatorname{card} \tilde{I} \leq (\pi'^{\pi})^{\pi} = \pi'^{\pi^2} = \pi'^{\pi}$, donc

$$\operatorname{card} \tilde{I} \ \leq \ \pi'^{\pi}.$$

Je dis que $\tilde{I}$ est lui aussi *grand devant* π, tout comme I. En effet, soit [page 80]

$$(I_\alpha)_{\alpha \in A}, \qquad I_\alpha \in \tilde{I}, \qquad \operatorname{card} A \ \leq \ \pi,$$

prouvons qu'il existe $I' \in \tilde{I}$ qui majore les I_α, *i.e.* qui contient $I'_0 = \bigcup_{\alpha \in A} I_\alpha$. On aura $\operatorname{card} I'_0 \leq \pi^2 = \pi$ (car $\operatorname{card} A \leq \pi$ et $\operatorname{card} I_\alpha \leq \pi$ pour tout α), et il suffit de voir ceci :

[En fait, l'assertion est évidente : comme $\operatorname{card} I'_0 \leq \pi$, et comme I est grand devant π, I'_0 admet un majorant i'_0 dans I, et on peut prendre $I' = I'_0 \cup \{i'_0\}$.]

Lemme 1 : *Soient π_0, π deux cardinaux, avec $\pi \geq \pi_0$, I un ensemble ordonné grand devant π, et I'_0 une partie de I de cardinal $\leq \pi$. Alors il existe une partie I' de I satisfaisant les conditions :* ([40])

a) $I' \supset I'_0$.

b) $\operatorname{card} I' \leq \pi$.

c) I' *grand devant* π_0. p. 44

[Le lemme tel qu'il est énoncé est évident pour les mêmes raisons que ci-dessus (même sans l'hypothèse de la note de bas de page). La démonstration qui suit prouve en fait un résultat plus fin :

LEMME 1′ : *Soient π_0, π deux cardinaux, avec $\pi_0 < \pi$ et $\pi^{\pi_0} = \pi$, I un ensemble ordonné grand devant π_0, et I'_0 une partie de I de cardinal $\leq \pi$. Alors il existe une partie I' de I satisfaisant les conditions :*

a) $I' \supset I'_0$.

b) $\operatorname{card} I' \leq \pi$.

c) I' *grand devant* π_0.]

On va essayer de construire I' par récurrence transfinie, en définissant les $I'_\alpha \subset I$ ainsi :

$$I'_0 = I'_0 \,,$$
$$I'_\alpha = \textstyle\bigcup_{\beta<\alpha} I'_\beta \quad \text{si } \alpha \text{ est un ordinal limite,}$$
$$I'_{\alpha+1} = \begin{cases} \text{déduit de } I'_\alpha \text{ ainsi : pour toute partie } J \text{ de } I'_\alpha \text{ telle que} \\ \operatorname{card} J \leq \pi_0, \text{ qui n'est pas majorée dans } I'_\alpha, \text{ on choisit} \\ \text{un majorant } i_J \text{ dans } I \text{ ([qui] existe, car } I \text{ [est] grand} \\ \text{devant } \pi, \text{ donc devant } \pi_0) \text{ et } I'_{\alpha+1} \text{ s'obtient en adjoignant} \\ \text{ces } i_J \text{ à } I'_\alpha. \end{cases}$$

Soit alors α_0 le premier ordinal qui soit de cardinal $> \pi_0$, et considérons

$$I' \ = \ I'_{\alpha_0} \ = \ \bigcup_{\substack{\alpha<\alpha_0 \\ i.e.\ \operatorname{card}\alpha\leq\pi_0}} I'_\alpha. \qquad \text{[page 81]}$$

Alors I' contient I'_0, il est grand devant π_0, mais quel est son cardinal ? Prouvons par récurrence transfinie que

$$\operatorname{card} I'_\alpha \ \leq \pi \qquad \text{si } \alpha < \alpha_0, \quad i.e. \ \operatorname{card}\alpha \leq \pi_0.$$

C'est O.K. pour $\alpha = 0$ par hypothèse sur I'_0. Si c'est vrai pour I'_α, est-ce vrai pour $I'_{\alpha+1}$? On adjoint à I'_α un ensemble d'éléments i_J qui est de cardinal $\leq \operatorname{card} \mathfrak{P}_{\leq\pi_0}(I'_\alpha)$, donc de cardinal $\leq \operatorname{card} (I'_\alpha)^{\pi_0} \leq \pi^{\pi_0}$, et si on suppose

$$\pi^{\pi_0} \ = \ \pi,$$

40. Pourvu qu'on suppose $\pi > \pi_0$, et de plus $\pi^{\pi_0} = \pi$ (p. ex. $\pi = 2^c$ avec $c \geq \pi_0$).

on gagne. Supposons enfin que α soit un ordinal limite (de cardinal $\leq \pi_0$). Donc on a

$$I'_\alpha \;=\; \bigcup_{\beta<\alpha} I'_\beta,$$

donc p. 45

$$\operatorname{card} I'_\alpha \;\leq\; \underbrace{(\operatorname{card}\alpha)\pi}_{\leq\pi\pi=\pi^2=\pi} \qquad \text{puisque les } \operatorname{card} I'_\beta \leq \pi, \qquad \text{[page 82]}$$

on gagne encore.

On trouve enfin, par le même argument,

$$\operatorname{card} I'_{\alpha_0} \;\leq\; \underbrace{(\operatorname{card}\alpha_0)\pi \leq \pi^2}_{\substack{\text{si } \operatorname{card}\alpha_0\leq\pi,\\ \textit{i.e.}\ \text{si } \pi_0<\pi}} = \pi,$$

donc on gagne encore.

Commentary · Exposition [ms. 82]

Why the transfinite closure has to run far enough

The editors distinguish two different closure problems. If I is already grand devant π and $S \subset I$ has size at most π, a single upper bound s gives the required subposet $S \cup \{s\}$. It has a largest element, so it is grand devant every cardinal. This proves the stated Lemma 1 without any exponent assumption. The longer proof is useful when the ambient poset is only grand devant π_0; that is the corrected Lemma 1′.

Put $\lambda = \pi_0^+$. Starting with $S_0 = S$, adjoin at each successor stage an upper bound for every subset of S_α having cardinal at most π_0, and take unions at limit stages. Suppose $\pi > \pi_0$ and $\pi^{\pi_0} = \pi$. Each successor step adds at most $\pi^{\pi_0} = \pi$ elements. A union of at most π sets of size at most π again has size at most π, so every S_α, including S_λ, satisfies the desired bound.

The omitted verification is that S_λ is grand devant π_0. Given $A \subset S_\lambda$ with $|A| \leq \pi_0$, choose a stage containing each of its elements. Since $\lambda = \pi_0^+$ is regular, these at most π_0 stages are bounded below λ. Hence $A \subset S_\alpha$ for some $\alpha < \lambda$, and the next stage contains an upper bound for A. This is why merely iterating the operation finitely many times, or through a sequence of cofinality at most π_0, would not prove the assertion.

The exponent hypothesis is sufficient rather than intrinsic. For example, when the desired bound is exactly π_0^+, enumerate the initial subset in a sequence of length π_0^+ and recursively choose an upper bound for the elements enumerated so far and the previously chosen bounds. At each proper stage there are at most π_0 requirements. Adjoining the resulting increasing sequence to the initial subset produces a subposet grand devant π_0, of cardinal at most π_0^+, without assuming $(\pi_0^+)^{\pi_0} = \pi_0^+$. We will use this elementary successor-cardinal improvement when returning to the unfinished final lemma.

These are two applications of the same method: control the number of requirements at a successor step, then choose a length whose cofinality catches every small family of requirements. Modern accessibility arguments still use this mechanism. The explicit stronger size assumption in Lurie (2009, Lemma 5.4.2.8) serves exactly this purpose in the setting of ∞-categories.

Ainsi, on a trouvé un deuxième système inductif

$$(X_{\tilde{\imath}})_{\tilde{\imath}\in\tilde{I}}$$

du même type que le système initial, ayant comme limite inductive X (qu'il ait même limite inductive est immédiat). Mais qu'a-t-on gagné ? Ceci : dans $\tilde{I}$ toute partie de cardinal $\leq \pi$ et grande devant π_0 est non seulement *majorée* dans $\tilde{I}$, mais même y admet une borne supérieure. Cela résulte du

Lemme 2 : *Soit I grand devant $\pi > \pi_0$, avec $\pi^{\pi_0} = \pi$* `[il suffit $\pi \geq \pi_0$]` *(cf. lemme 1). Soit $(I'_\alpha)_{\alpha\in A}$ un ensemble de parties de I qui sont dans $\tilde{I}$,* i.e. [page 83]

a) *de cardinal $\leq \pi$, et*

b) *grandes devant* π_0,

et supposons que $\operatorname{card} A \leq \pi$ *et que* A, *ordonné par inclusion, soit grand devant* π_0. *Alors* $I' = \bigcup_{\alpha \in A} I'_\alpha$ *est encore dans* $\tilde{I}$, i.e. I' *est de cardinal* $\leq \pi$ *(car de cardinal* $\leq \pi^2 = \pi$*), et de plus, grande devant* π_0. (Vérification immédiate.)

Ainsi, on a prouvé ceci :

Proposition 1. *Soit* E `[une]` *grosse catégorie, avec filtration cardinale* $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$. *Soit* π *un cardinal avec*

$$\pi > \pi_0, \qquad \pi^{\pi_0} = \pi \qquad \textit{(p. ex. } \pi = 2^c\textit{, avec } c \geq \pi_0\textit{)} \quad \texttt{[}\pi \geq \pi_0 \texttt{ suffit]}.$$

Alors pour tout X *dans* E, *on peut trouver un système inductif* $(X_i)_{i \in I}$, *de limite* X, *avec*

a)
$$\begin{cases} X_i \in \mathrm{Filt}^\pi \ \textit{ pour tout } i \in I, \\ I \textit{ grand devant } \pi, \end{cases}$$

satisfaisant aux conditions supplémentaires suivantes :

b) *Toute partie* J *de* I *de cardinal* $\leq \pi$, *et grande devant* π_0, *admet une borne supérieure* i_J *dans* I.

c) *On a, sous ces conditions*

$$X_{i_J} \xleftarrow{\sim} \varinjlim_{i \in J} X_i. \qquad \text{p. 46 [page 84]}$$

De plus, si $X \in \mathrm{Filt}^{\pi'}$ *avec* $\pi' > \pi$, *on peut prendre* I *de cardinal* $\leq \pi'^{\pi}$ *(donc de cardinal* $\leq \pi'$ *si* $\pi'^{\pi} = \pi'$, *p. ex. si* $\pi' = 2^{c'}$, *avec* $c' \geq \pi$*).*

Définition. On appelle un système inductif $(X_i)_{i \in I}$, satisfaisant les conditions *a*), *b*), *c*) ci-dessus, un *système inductif bien adapté* (à la filtration cardinale donnée, pour les cardinaux π_0, π donnés), et la représentation correspondante

$$X \xleftarrow{\sim} \varinjlim X_i$$

une *représentation bien adaptée* de X en termes d'objets de Filt^π.

Commentary · Exposition [ms. 84]

A presentation that is continuous on its small pieces

Grothendieck wants the indexing set to remember small colimits already performed. An arbitrary directed presentation has upper bounds, but its value at an upper bound need not equal the colimit of the stages below it. Replacing indices by small directed subsets builds this continuity into the presentation.

Take $X = \varinjlim_{i \in I} X_i$ as in condition (c), and let P consist of the subposets $S \subset I$ of size at most π which are grand devant π_0. Order P by inclusion and put $Z_S = \varinjlim_{s \in S} X_s$. The ambient hypothesis on I gives an upper bound for the union of at most π members of P; adjoining that element produces another member of P. Thus P is grand devant π. Condition (b) gives $Z_S \in \mathrm{Filt}^\pi(E)$, while the elementary bound on subsets gives $|P| \leq |I|^\pi$.

The equality of the two colimits can be verified directly, without a cofinality convention. A compatible family of maps $Z_S \to Y$ gives maps $X_i \to Y$ by using the singleton subset $\{i\}$. For $i \leq j$, the two-element subset $\{i, j\}$, which has largest element j, ensures compatibility. Conversely a cocone on the X_i induces one on every Z_S, and these two operations are inverse. Hence $\varinjlim_P Z_S \simeq X$.

Now let $A \subset P$ be grand devant π_0 and have cardinal at most π. Its union $S = \bigcup_{T \in A} T$ has size at most π. For a subset $B \subset S$ of size at most π_0, choose one $T_b \in A$ containing each b. An upper bound $T \in A$ for these members contains B, and an upper bound for B inside T belongs to S. Thus $S \in P$, and S is the supremum of A. A cocone on the diagrams over all $T \in A$ is exactly a cocone over their union, so

$$Z_{\sup A} = Z_S \simeq \varinjlim_{T \in A} Z_T.$$

This proves the continuity clause in Proposition 1, not merely the existence of suprema in its indexing poset.

When all filtered colimits exist and the filtration is strict in the bounded sense of footnote 41, the same proof uses ordinary directed subposets S and ordinary directed families A. A union of a directed family of directed subposets is directed. A subset of I of size at most π can be enlarged to a directed subset of that same size by adding upper bounds for finite subsets through countably many stages. This yields Proposition 1 bis. The word "perfectly" records the stronger continuity, with no change to the cardinality bound on the partial diagrams. The inverse-limit sign in the preceding display on manuscript page 85 is a source slip: the construction uses a colimit.

NB Quand E est stable par petites $\varinjlim$ filtrantes, la situation se simplifie considérablement. On peut alors prendre

$$\tilde{I} \;=\; \{I' \subset I \mid \operatorname{card} I' \leq \pi,\ I' \text{ filtrant}\}$$

(sans s'encombrer de la condition I' grand devant π_0) et définir encore le système inductif $(X_{I'})_{I' \in \tilde{I}}$ par

$$X_{I'} \;=\; \varinjlim_{i \in I'} X_i \ .$$

On a encore [page 85]

$$\operatorname{card} \tilde{I} \;\leq\; (\operatorname{card} I)^{\pi} \;\leq\; \pi'^{\pi} \qquad \text{si} \ \ \operatorname{card} I \leq \pi'^{\pi}$$

(notamment, si $X \in \mathrm{Filt}^{\pi'}$, et qu'on prenne I comme indiqué dans la condition c des filtrations cardinales). De plus, cette fois, le fait que $\tilde{I}$ soit grand devant π est trivial, on a même mieux, à part que I soit filtrante :

(∗) *Toute* partie $(I'_\alpha)_{\alpha \in A}$ de $\tilde{I}$ filtrante et de cardinal $\leq \pi$ admet une borne supérieure $\tilde{\imath}_A$,

savoir la réunion (qui est bien filtrante). De plus, le système inductif $(X_{\tilde{\imath}})_{\tilde{\imath} \in \tilde{I}}$ a la propriété plus forte que

$$X_{\tilde{\imath}_A} \xleftarrow{\sim} \varprojlim_{\tilde{\imath} \in A} X_{\tilde{\imath}} \ .$$

Ainsi on a la

Proposition 1 bis. *Supposons E stable par petites $\varinjlim$ filtrantes, et Filt^π stable par celles-ci, où $\pi \geq \pi_0$ est donné. Alors pour tout X dans E, on peut trouver un système inductif $(X_i)_{i \in I}$ de limite inductive X, avec*

a) *$X_i \in \mathrm{Filt}^\pi$ pour tout $i \in I$, I ensemble ordonné grand devant π.*

b) *Toute partie J de I filtrante de cardinal $\leq \pi$ admet une borne supérieure i_J dans I.*

c) *Sous ces conditions*

p. 47
[page 86]

$$X_{i_J} \;=\; \varinjlim_{i \in J} X_i \ .$$

De tels systèmes inductifs seront dits *"parfaitement adaptés"* (à la filtration cardinale donnée, pour le cardinal π donné – π_0 n'y figure plus, si ce n'est comme début de la filtration cardinale sans plus).

Commentary · Examples [ms. 86]

Sets and modules as perfectly adapted presentations

The basic example makes every condition concrete. In $E = \mathrm{Set}$ put $\mathrm{Filt}^\pi(E) = \{S : |S| \leq \pi\}$. A skeleton is a set of cardinals, so the stage is essentially small. A colimit of a diagram with at most π objects, each of size at most π, is a quotient of a disjoint union of size at most $\pi \cdot \pi = \pi$. Thus the required bounded closure holds, even for arbitrary colimits of that size.

For a set X consider

$$I = [X]^{\leq \pi}, \qquad X_A = A \quad (A \in I),$$

with inclusion maps. The union of at most π members has size at most π, so I is grand devant π. Every nonempty directed family $J \subset I$ of size at most π has supremum $\bigcup J$, and its colimit in sets is this same union. The colimit of the entire diagram is X, since the singletons cover it. If $|X| \leq \pi'$, then $|I| \leq (\pi')^{\pi}$ for infinite X; finite X causes no difficulty. This is a perfectly adapted representation with the precise bound appearing in the source.

A set A of size at most π is π^+-presentable. To factor a map $A \to \varinjlim_I Y_i$ through one stage, choose representatives for its at most π values, and then one upper bound for their indices. If two such factorizations induce the same map, each of their at most π component equalities holds at some later stage; another upper bound makes the maps equal. Both surjectivity and injectivity of the Hom comparison are used. They are the categorical counterparts of choosing generators and then checking relations.

For modules over a ring R, fix an infinite $\pi \geq |R|$. Use modules whose underlying sets have size at most π. They form an essentially small class, and bounded filtered colimits preserve the size bound. Every module X is the union of its submodules of size at most π. Any set of at most π elements generates such a submodule: a linear combination uses finitely many coefficients, so there are at most π combinations. The poset of these submodules is grand devant π, and directed unions of at most π of them again have size at most π. This gives the same perfectly adapted presentation.

The module presentability assertion also includes relations. A module of size at most π has a surjection from a free module on at most π generators. That free module and its kernel have size at most π, so the kernel is generated by at most π elements. Factoring a homomorphism through a π^+-directed colimit requires at most π choices for generators and at most π equations for relations; a common later stage handles them all. Equality of two factorizations is checked on the generators. Thus the chosen stage consists of π^+-presentable modules.

These examples also locate an omission in the glossary. There, "strict filtration" is stated as closure of each stage under all filtered colimits, with no bound on the indexing cardinal. Together with condition (c), that statement would force $E = \mathrm{Filt}^{\pi}(E)$ for every π, hence would make E essentially small. The definition actually used here is the bounded one in footnote 41: closure under filtered diagrams of cardinal at most π. For instance, the union of π^+ many successively larger subsets of a set of cardinal π^+ shows why the bound cannot be discarded.

Comme pratiquement, on aura à manier surtout des E stables par petites $\varinjlim$ filtrantes, et des filtrations cardinales stables en conséquence (renforcement de la condition b) des filtrations cardinales), c'est le cas, et la notion de *représentation parfaitement adaptée* de X comme $\varinjlim_I X_i$ ($X_i \in \mathrm{Filt}^{\pi}$), que je mettrai en avant. Le cas "général", où on suppose seulement `[que]` E satisfait L_{π_0}, est plus alambiqué et se traite moins joliment ([41]).

Définition 2. Soit E munie d'une filtration cardinale stricte $(\underline{\mathrm{Filt}}^{\pi})_{\pi \geq \pi_0}$. Soit M une partie de ObE, $\underline{M}$ la sous-catégorie pleine de E qu'elle définit, et soit π un cardinal $\geq \pi_0$. On dit que M est π-*accessible,* où que $\underline{M}$ est π-*accessible*, si les conditions suivantes sont satisfaites. Pour tout système inductif [page 87]

$$(X_i)_{i \in I}$$

parfaitement adapté à la filtration cardinale pour π (donc I ensemble ordonné filtrant tel que toute partie filtrante J de cardinal $\leq \pi$ ait une borne supérieure i_J, et $X_{i_J} \xleftarrow{\sim} \varinjlim_{j \in J} X_j$) ; soit I_M l'ensemble des $i \in I$ tels que $X_i \in M$. Ceci posé, on veut ceci : pour que X $(= \varinjlim X_i)$ soit dans M, il faut et il suffit que I_M soit cofinal dans I (de sorte que X apparaît comme $\varinjlim$ filtrante, avec ensemble d'indices I_M grand devant π, d'éléments de $M \cap \underline{\mathrm{Filt}}^{\pi} \overset{\text{déf}}{=} M^{\pi}$).

J'ai envie de reprendre la définition de la π-accessibilité de M, pour une filtration cardinale donnée, à coups de conditions équivalentes. Pour tout système inductif $(X_i)_{i \in I}$ dans M `[plutôt dans` E`]`, je garde la notation [page 88]

$$I_M \;=\; \{i \in I \mid X_i \in M\}.$$

41. On parlera de filtration cardinale *stricte,* si les $\underline{\mathrm{Filt}}^{\pi}$ sont stables par $\varinjlim$ filtrantes de cardinal $\leq \pi$ (et que ces limites filtrantes petites existent dans E).

Je pose, pour tout cardinal $\pi \geq \pi_0$,

$$M^\pi \;=\; M \cap \mathrm{Filt}^\pi.$$

(a$_\pi$) M est l'ensemble des objets de la forme

$$\varinjlim_{i\in I} X_i \ ,$$

où les X_i sont dans M^π, et I est grand devant π.

(b$_\pi$) Soit $(X_i)_{i\in I}$ un système inductif dans M [plutôt dans E] parfaitement adapté à la filtration cardinale pour π, *i.e.* I filtrante et toute partie filtrante J de I de cardinal $\leq \pi$ admet une borne supérieure i_J, et $X_{i_J} \xleftarrow{\sim} \varinjlim_J X_i$, et enfin $X_i \in \mathrm{Filt}^\pi$ pour tout $i \in I$. p. 48

Soit $X = \varinjlim X_i$. Pour que l'on ait $X \in M$, il faut et il suffit que I_M soit cofinal dans I.

(c$_\pi$) $\underline{M}^\pi = \underline{M} \cap \underline{\mathrm{Filt}}^\pi$ est une sous-catégorie de $\underline{M}$ génératrice par épimorphismes stricts, $\underline{M}$ est stable par $\varinjlim$ grandes devant π, enfin pour tout $X \in M$, $\underline{M}^\pi/X$ est une catégorie filtrante.

Je présume que ces conditions sont équivalentes.

[page 89]

Commentary · Exposition [ms. 89]

Three different ways to recognize a part

The purpose of the new definition is to recognize membership in a class M from small approximations. This would be particularly useful when E is an arrow category and M is a class of weak equivalences. Grothendieck proposes three tests, but their logical strengths differ.

The first, a_π, says that M is exactly the class of π^+-directed colimits of objects of $M^\pi = M \cap \mathrm{Filt}^\pi(E)$. It includes a closure assertion as well as an existence assertion. The second, b_π, says that *every* perfectly adapted presentation detects membership by a cofinal set of M-stages. It therefore imposes a condition on the individual stages of a chosen presentation, beyond the existence of some good presentation. The third, c_π, combines strict generation, closure under π^+-directed colimits, and ordinary filteredness of M^π/X.

At a useful cardinal, with M^π consisting of π^+-presentable objects, a_π is the familiar description by an Ind_{π^+}-completion. Indeed the canonical colimit functor out of that completion is fully faithful; the Hom calculation is the one in Artin et al. (1972, I, 9.18(a)). It follows in particular that the resulting full subcategory is closed under π^+-directed colimits. This explains the modern accessibility of the inclusion, but it does not make b_π automatic.

There are two distinct issues in the proposed equivalence. Ordinary filteredness controls finite diagrams, whereas a_π asks for compatibility of up to π constraints at once. Separately, two cofinal collections of small approximations need not meet: one may pick only successor stages and another only limit stages. Continuity of the second presentation does not force membership to pass to its countable colimits. The following examples isolate these two failures while keeping all objects in the small stage genuinely presentable.

Commentary · Development (commentators') [ms. 89]

Two counterexamples to the proposed equivalence

The distinction just made can be checked in a category with no nontrivial parallel arrows. This removes any possible ambiguity about coherence and leaves the cardinal issue exposed. We give the examples as our own verification of the conjecture on page 89. We did not find these particular examples in the relevant portions inspected of SGA 4 I, §9, *Les Dérivateurs* XVIII, §4.13 and Lemma 4.16.7.11, and the accessible-category references listed below; no priority is asserted.

Proposition: the implication $a_\pi \Rightarrow b_\pi$ fails at the same cardinal. [ours] Take $\pi = \aleph_0$ and let E be the poset $\mathcal{P}(\omega_1)$, viewed as a category under inclusion. Give it the strict filtration by cardinality, $\mathrm{Filt}^\rho(E) = \{S \subset \omega_1 : |S| \leq \rho\}$. Let

$$M = \{\alpha + 1 : \alpha < \omega_1\} \cup \{\omega_1\},$$

where ordinals denote their sets of predecessors. Then a_π and c_π hold, while b_π fails. Every object of $\mathrm{Filt}^\pi(E)$ is π^+-presentable in E.

Proof. Colimits in E are unions. Each filtration stage is small, since it is a subset of the universe-small set $\mathcal{P}(\omega_1)$. A union of at most ρ sets of size at most ρ again has size at most ρ, proving the bounded closure axiom for every infinite ρ. Each $S \subset \omega_1$ is the union of its subsets of size at most ρ; their poset is ρ^+-directed because a union of at most ρ such subsets is another one. If $|S| \le \rho'$ and $\rho' > \rho$, the number of indices is at most $(\rho')^\rho$; the finite cases satisfy this bound as well. This verifies the full filtration axiom (c), including its size estimate. Finally, in a π^+-directed union every countable subset of the union is contained in one stage. Since Hom sets in E have at most one element, this factorization assertion proves presentability of all the countable objects.

The objects of M^π are exactly the countable successor ordinals. Each is a constant directed colimit of such objects, and ω_1 is the union of all countable successor ordinals; their ordered set is countably directed. Conversely, consider a countably directed family of countable successor ordinals. If its union β is countable and is not attained, choose a countable cofinal subfamily. A single member bounding this subfamily would contain its union, contradicting nonattainment. Therefore a countable union arising this way is an attained successor ordinal. An uncountable union is ω_1. This proves a_π and closure of M under countably directed colimits.

For $X \in M$, the slice M^π/X is a nonempty linearly ordered set, hence is filtered. Its union is X. In a poset this says exactly that the canonical family of arrows from M^π is strictly epimorphic: any common upper bound of that family in M is an upper bound of X, and the required factorization is unique. Thus c_π holds as well.

Now index a diagram by the nonzero countable limit ordinals $\delta < \omega_1$, and set $X_\delta = \delta$. The index is countably directed: the supremum of any countable family is still countable, and a larger countable limit ordinal is an upper bound. Every nonempty directed subset of this index of cardinal at most $\aleph_0$ has its supremum in the index; a supremum of nonzero limit ordinals is again a nonzero limit ordinal. At that supremum the diagram value is the union of the earlier values. The presentation is therefore perfectly adapted, and its colimit is $\omega_1 \in M$. Nevertheless no stage belongs to M. Its set I_M is empty. This disproves b_π. □

Proposition: the implication $c_\pi \Rightarrow a_\pi$ also fails. [ours] In the same filtered category E, replace M by

$$N = \{S \subset \omega_1 : S \text{ is finite}\} \cup \{\omega_1\}.$$

Then c_π holds for N, but a_π does not.

Proof. Here N^π consists of the finite subsets. For a finite X, the slice N^π/X has largest element X; for $X = \omega_1$, it is the directed set of finite subsets, whose union is ω_1. These unions prove strict generation, as in the first example. A countably directed family of finite subsets has finite union. Otherwise choose countably many distinct elements of the union and a stage containing each; an upper bound for these countably many stages would be a finite set containing infinitely many elements. A countably directed diagram in N that contains ω_1 has union ω_1. Thus N is closed under the required colimits, proving c_π.

But no countably directed colimit of objects of N^π can equal ω_1, by the same finite-union argument. Hence a_π fails. □

The examples are legitimate $\mathfrak{U}$-categories. If one requires the ambient category itself to be non-small, take $E \times \mathrm{Set}$ with

$$\mathrm{Filt}^\rho(E \times \mathrm{Set}) = \{(S,T) : |S| \le \rho,\ |T| \le \rho\}.$$

A small skeleton is obtained by retaining the actual subsets $S \subset \omega_1$ and replacing T by a cardinal. Colimits are computed in each coordinate, so bounded closure follows from the union calculation above and the set-colimit calculation in the block "Sets and modules as perfectly adapted presentations." For condition (c), use the product poset $[S]^{\le\rho} \times [T]^{\le\rho}$. It is ρ^+-directed and has colimit (S,T). If both coordinates have size at most $\rho' > \rho$, its cardinal is at most $(\rho')^\rho \cdot (\rho')^\rho = (\rho')^\rho$. Countable objects are $\aleph_1$-presentable by the same simultaneous choice of representatives and equations in the two coordinates. Restrict the second coordinate of M or N to the empty set. All presentations, strict generating families and counterexample diagrams used above then have empty second coordinate, so every preceding verification remains valid. The repair to c_π is substantial: one needs a π^+-filtered canonical slice with the asserted canonical colimit, rather than merely an ordinary filtered slice. The first example shows that even this repaired generation condition does not recover the universal-presentation condition b_π at the same rank.

$b_\pi \Longrightarrow a_\pi$. On suppose b) vérifié, on veut prouver a). D'abord que tout $X \in M$ est de la forme dite $\varinjlim_I X_i$, avec les $X_i \in M^\pi$ et I grand devant π. Par [la] proposition 1, on sait qu'on a une présentation parfaite adaptée pour π, $X = \varinjlim_{i\in I} X_i$. Par b) "il faut", la partie I_M de I est cofinale dans I, donc comme I est grand devant π, I_M aussi, et on aura (posant $J = I_M$)

$$X \;=\; \varinjlim_{i\in J} X_i, \qquad\qquad X_i \in M^\pi, \quad J \text{ grand devant } \pi,$$

O.K. Inversement, prouvons que si X se met sous la forme dite dans a), il est dans M. Cela va résulter du

Lemme 1 : *Si la condition* b) (*ou* a)) *est satisfaite, M est stable par limites inductives grandes devant π.*

Il faut prouver que si [page 90]

$$X \;=\; \varinjlim_{i\in I} X_i \;,$$

avec I grand devant π, les X_i dans M, alors $X \in M$. La condition b) "il suffit" l'assure quand on suppose de plus les X_i dans Filt^π (donc dans M^π), de sorte que l'on aura $I_M = I$, donc I_M cofinal, donc $X \in M$ ([42]). Si on ne suppose pas les X_i dans Filt^π, il faut travailler un peu plus.

Passer tout de suite à la partie de la démonstration ci-dessous, page 96, qui prouve $b) \Longrightarrow a$). Le reste de la démonstration va prouver le

Lemme 2. *Si tout élément de Filt^π est π-accessible, alors* b_π) *(et même* a_π*) implique que M est stable par $\varinjlim$ grandes devant π.*

Soit p. 49

$$\mathcal{C} \;=\; \underline{M}^\pi/\mathcal{X} \qquad \text{[page 91]}$$

la catégorie suivante. Ses objets sont les triples

$$(Z, i, \alpha) \quad \text{tels que} \quad Z \in M^\pi,\ i \in I,\ \alpha : Z \longrightarrow X_i \;,$$

les flèches

$$(Z, i, \alpha) \;\longrightarrow\; (Z', i', \alpha')$$

sont les couples (u, λ) avec

$$u : Z \longrightarrow Z', \qquad \lambda : i \longrightarrow i', \qquad (\textit{i.e.} \text{ on a } i \leq i' \text{ et } \lambda \text{ est l'unique flèche de } i \text{ dans } i'),$$

tels que le diagramme

$$\begin{array}{ccc} Z & \xrightarrow{\ \alpha\ } & X_i \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle \text{transition } \lambda_{i',i}} \\ Z' & \xrightarrow{\ \alpha'\ } & X_{i'} \end{array}$$

commute. On peut donc définir $\mathcal{C}$ comme le produit fibré dans le carré cartésien

$$\begin{array}{ccccc} & & \mathcal{C} & \xrightarrow{\text{cofibré}} & I \\ & {\scriptstyle \mathcal{Z}}\swarrow & \downarrow & & \downarrow{\scriptstyle \mathcal{X}:X\longmapsto X_i} \\ \underline{M}^\pi & \xleftarrow{\ \text{source}\ } & \underline{M}^\pi/E & \xrightarrow[\text{cofibré}]{\text{but}} & E. \end{array}$$

La fibre $\mathcal{C}_i$ de $\mathcal{C}$ en $i \in I$ est Filt^π/X_i [plutôt M^π/X_i]. On a, comme $b) \Longrightarrow$ première moitié de a), que $\underline{M}^\pi$ est un système de générateurs stricts de $\underline{M}$ [cela n'est pas clair sans

42. Pas tout à fait – on y suppose plus, sur le système inductif. On ne peut se dispenser, même dans le cas $X_i \in M^\pi\ \forall\, i \in I$, faire le raisonnement des pages suivantes. [page 92]

`autres hypothèses sur` $\underline{M}$`]`, et $X_i \in \mathrm{Ob}\,\underline{M}$ par hypothèse,

$$X_i \;=\; \varinjlim_{Z \in \mathrm{Ob}\,\underline{M}^\pi/X_i} Z \;=\; \varinjlim_{\mathcal{C}_i} \mathcal{Z}|\mathcal{C}_i,$$

donc le système inductif donné $(X_i)_{i\in I}$ sur I à valeurs dans X, se déduit du système inductif (non ordonné en général) sur $\mathcal{C}$, à valeurs dans $\underline{M}^\pi$, en "intégrant sur les fibres" de $\mathcal{C} \longrightarrow I$. D'ailleurs, comme $\underline{M}^\pi/E \longrightarrow E$ est cofibrée, donc propre, il en est de même de $\mathcal{C} \longrightarrow I$, donc les $\varinjlim$ de $\mathcal{Z}$ sur les fibres $\mathcal{C}_i$ $(= \underline{M}^\pi/X_i)$ forment bien un foncteur covariant $I \longrightarrow E$, dont la limite inductive est celle de $\mathcal{Z}$ ([43]).

Soit, pour tout $i \in I$, $\tilde{\mathcal{C}}_i$ l'ensemble ordonné

p. 50
[page 93]

$$\tilde{\mathcal{C}}_i = \{\text{sous-catégories } \mathcal{C}'_i \text{ de } \mathcal{C}_i = \underline{M}^\pi/X_i \mid \mathrm{card}\,\mathcal{Z}(\mathcal{C}'_i) \leq \pi, \;\; \mathcal{C}'_i \text{ filtrante}\}.$$

J'admets pour l'instant le

Lemme ([44]). *Pour tout $X \in M$, la catégorie $\underline{M}^\pi/X$ est filtrante. (Sous l'hypothèse* b), *ou que tout objet de M est de la forme dite dans* a).*)*

Commentary · Exposition [ms. 93]

Presentability supplies the strict generators

Grothendieck now tries to replace a diagram of large M-objects by one diagram of small M-objects. The comma category is the right construction: it records a small object together with the map by which it participates in a particular large object. The editors' reservation on page 92 concerns the claim that a chosen colimit presentation already makes *all* maps from small objects into a canonical colimit presentation. Presentability supplies the missing comparison.

Here is the precise assertion. Let $\kappa = \pi^+$, let $A \subset E$ be an essentially small full subcategory of κ-presentable objects, and suppose

$$X \simeq \varinjlim_{j\in J} Y_j, \qquad J \text{ } \kappa\text{-directed}, \qquad Y_j \in A.$$

Then A/X is κ-filtered, and its canonical diagram has colimit X. This is the canonical-diagram form of the fully faithful Ind_κ construction in Artin et al. (1972, I, 9.18(a)).

To prove the assertion, take any diagram of fewer than κ morphisms in A/X. Each of its objects is a map $T \to X$ with T κ-presentable, so it factors through a Y_j. Choose one common upper bound for the indices used. For every arrow in the given diagram, the two resulting maps into this common stage agree after passage to X. Presentability of its source makes that equality hold at some later stage. There are fewer than κ equalities, so another common upper bound makes them all hold together. The resulting object $Y_{j'} \to X$ is a cocone in A/X. This proves κ-filteredness, including the case of the empty diagram.

To identify the colimit, let a compatible family $T \to Z$ be given for all objects $(T \to X)$ of A/X. Restrict it to the chosen presentation $Y_j \to X$, producing a unique map $h : X \to Z$. Any $T \to X$ factors through a Y_j, and compatibility with that factorization says that the prescribed map $T \to Z$ is h composed with $T \to X$. Thus h induces the whole family, uniquely. This proves the universal property of the canonical colimit. In particular its canonical family is strictly epimorphic in the ambient category, hence also in any full subcategory containing X and A.

Apply this with $A = M^\pi$. The first half of a_π, already obtained from b_π, supplies the required presentations. The tacit hypothesis in footnote 44 supplies presentability. Under those hypotheses the lemma on page 93 is correct, and in fact its slice is π^+-filtered, a stronger conclusion than printed. The precise missing step is factorization and eventual equality for *arbitrary* maps from M^π, not an additional assumption about epimorphisms in M.

The same argument explains why this construction is stable under a sufficiently directed change of the large object. For a κ-directed diagram (X_i) in M, form the category C of triples $(T, i, T \to X_i)$.

43. **NB** Si on admet la validité de *a*), alors il en résulte déjà que $\varinjlim_{i\in I} X_i \simeq \varinjlim_{\alpha\in C} Z_\alpha$ est dans M. (Stabilité de M par $\varinjlim$ grandes devant π).

44. Utilise l'hypothèse tacite que les éléments de Filt^π sont π-accessibles.

A diagram of fewer than κ arrows in C can first be moved into one fibre over I, and then given a cocone in that fibre by the preceding paragraph. Thus C is κ-filtered. Its colimit is $\varinjlim_I X_i$, because giving a cocone on C is giving compatible cocones on all the fibres. After the usual cofinal directed-poset replacement, this is a presentation by objects of M^π. If a_π holds, it proves closure of M under κ-directed colimits. The directed-poset replacement is the ordinary-category case of Lurie (2009, Proposition 5.3.1.18); it changes the index, not the colimit.

Il en résulte, comme $X_i \in M$, que $\mathcal{C}_i$ est filtrante, donc la réunion des $\mathcal{C}'_i \in \tilde{\mathcal{C}}_i$ est $\mathcal{C}_i$; l'ensemble ordonné $\tilde{\mathcal{C}}_i$ est filtrant, et toute partie $(\mathcal{C}'_{i_\alpha})_{\alpha\in A}$ filtrante et de cardinal $\leq \pi$ de $\tilde{\mathcal{C}}_i$ y admet une borne supérieure $\mathcal{C}'_{i_A}$, savoir la réunion des $\mathcal{C}'_{i_\alpha}$. $\tilde{\mathcal{C}}_i$ est donc un ensemble ordonné π-adapté.

On peut considérer les ensembles ordonnés $\tilde{\mathcal{C}}_i$, pour $i \in I$ variable, comme les fibres d'une application croissante d'ensembles ordonnés

$$p \,:\, \tilde{\mathcal{C}} \longrightarrow \mathcal{C},$$

en écrivant

$$\begin{cases} (\mathcal{C}'_i \subset \mathcal{C}_i) \;\leq\; (\mathcal{C}'_j \subset \mathcal{C}_j) \\ \text{si et seulement si } i \leq j, \\ \text{et le foncteur canonique de transition } \mathcal{C}_i \longrightarrow \mathcal{C}_j \text{ applique } \mathcal{C}'_i \text{ dans } \mathcal{C}'_j. \end{cases}$$

On voit donc que $\tilde{\mathcal{C}}$ est lui aussi filtrant et grand devant π.

On définit un système inductif sur $\tilde{\mathcal{C}}$, à valeurs dans $\underline{\mathrm{Filt}}^\pi$, en posant [page 94]

$$Z_{\mathcal{C}'_i} \;=\; \varinjlim_{\alpha\,\in\,\mathcal{C}'_i\ (\subset \mathcal{C}_i = \underline{M}^\pi/X_i)} Z_\alpha$$

(où un $\alpha \in \mathcal{C}_i$ est formé, je le rappelle, de $Z_\alpha \xrightarrow{u_\alpha} X_i$, avec $Z_\alpha \in M^\pi \subset \underline{\mathrm{Filt}}^\pi$). Ces $Z_{\mathcal{C}'_i}$ ne sont plus forcément dans M^π, mais seulement dans Filt^π. (Alors à quoi ça a servi de définir les $\mathcal{C}_i$ eux-mêmes comme $\underline{M}^\pi/X_i$, au lieu de $\underline{\mathrm{Filt}}^\pi/X_i$?)

Le système inductif

$$(Z_{\mathcal{C}'_i})_{\mathcal{C}'_i\in\tilde{\mathcal{C}}_i}$$

est parfaitement bien adapté à la filtration cardinale pour le cardinal π, et il est immédiat que sa $\varinjlim$ est isomorphe à $\varinjlim_{\alpha\in \mathrm{Ob}\,\mathcal{C}_i=\underline{M}^\pi/X_i} Z_\alpha = X_i \in M$. Donc par b), l'ensemble $\tilde{\mathcal{C}}_M$ des $\mathcal{C}'_i \in \tilde{\mathcal{C}}_i$ tels que $Z_{\mathcal{C}'_i} \in M$ (donc $Z_{\mathcal{C}'_i} \in M^\pi$) est cofinal dans $\tilde{\mathcal{C}}_i$. Il s'ensuit donc que l'ensemble $\tilde{\mathcal{C}}_M$ est lui aussi cofinal dans $\tilde{\mathcal{C}}$.

On vérifie aussi immédiatement que p. 51 [page 95]

$$\varinjlim_{\alpha\in\tilde{\mathcal{C}}} Z_\alpha \;=\; \varinjlim X_i \;=\; X,$$

donc, $\tilde{\mathcal{C}}_M$ étant cofinal,

$$\varinjlim_{\alpha\in\tilde{\mathcal{C}}_M} Z_\alpha \;=\; X.$$

Au total, on est arrivé à représenter X sous la forme

$$X \;=\; \varinjlim_{\alpha\in A} X_\alpha,$$

où cette fois les X_α sont *dans* M^π (pas seulement dans M), et A est grand devant π. Mais on ne sait pas que toute partie filtrante B de cardinal $\leq \pi$ de A admet une borne supérieure

α_B dans A, et encore moins que $X_{\alpha_B} = \varinjlim_{\alpha\in B} X_\alpha$ ([45]). Donc on ne peut encore appliquer directement b). S'il ne s'agissait que de prouver la deuxième moitié de a), où on suppose déjà les $X_i \in M^\pi$ et I grand devant π, on n'a rien gagné du tout par nos labeurs !

Commentary · Development (commentators') [ms. 95]

The size of an indexing category cannot be replaced by the size of its image

The partial-colimit construction is meant to keep all intermediate objects within Filt^π. For that purpose it must bound the indexing category. The literal condition on page 93, $\operatorname{card}\mathcal{Z}(C'_i) \le \pi$, bounds only the image of the diagram. That is insufficient, even if its cardinal counts image morphisms as well as image objects.

A filtered counterexample to the image bound. [ours] Let $E = M = \mathrm{Set}$ with the usual cardinal filtration, let $\pi = \aleph_0$, and choose an uncountable set X. Inside M^π/X, take the category C' whose objects are injections $a : n \hookrightarrow X$ from the standard finite sets $n = \{0, \dots, n-1\}$. A morphism from $a : n \hookrightarrow X$ to $b : m \hookrightarrow X$ is a map $u : n \to m$ satisfying $bu = a$.

This category is filtered. Two objects map into an injection whose image is the union of their two finite images. Parallel arrows are equal, since their composites with the injective target map are equal. The source functor $\mathcal{Z} : C' \to \mathrm{Set}$ has image inside the countable category of standard finite sets and maps; hence $\operatorname{card}\mathcal{Z}(C') \le \aleph_0$. Yet

$$\varinjlim_{C'} \mathcal{Z} \simeq X.$$

Indeed every element of X occurs through a one-element injection, and two representatives determine the same element precisely when they become equal in a finite union of their images. Equivalently the compatible injections give the usual directed union of the finite subsets of X. This colimit is not in $\mathrm{Filt}^{\aleph_0}(E)$. □

The corrected construction uses filtered *indexing categories* $D \subset C_i$ with at most π morphisms, or directed indexing posets of that size mapping to C_i. This correction does not require a bound on every Hom set of C_i. The subcategories need not be full. Start with at most π specified objects and arrows; close under identities and composition, adjoin cones for finite diagrams, and repeat countably often. Each step adds at most π data, since π is infinite. The resulting filtered subcategory still has at most π morphisms. In particular the poset of these small filtered subcategories is grand devant π: perform this construction on the union of any at most π of them.

A directed union of at most π such subcategories is again a filtered subcategory of size at most π and is their supremum. Colimits over this union are the colimits of the partial colimits. To apply the source's filtration axiom, which is phrased for posets, a filtered category with at most π arrows can be replaced cofinally by a directed poset of size at most π: the usual finite-diagram construction adjoins finitely many compatibility data at each of countably many stages. Thus the corrected partial colimit lies in Filt^π, and its presentation over the poset of small subcategories is perfectly adapted.

This also repairs the rest of the argument on pages 93–95. The map from the total poset of pairs $(i, D \subset C_i)$ has target I, as its displayed order relation specifies; the printed target C is a slip. To bound at most π such pairs, move their indices to one index of I and enlarge the union of their images to a small filtered subcategory of the corresponding fibre. Cofinality of the good partial colimits in every fibre then gives their cofinality in the total poset. The final diagram so obtained need not itself be continuous at all small directed suprema, exactly as Grothendieck observes on page 95. The reindexing on page 96 is therefore still needed.

The counterexample concerns this particular image-size formulation. The corrected small-subcategory construction is the familiar filtered-diagram argument, already used in SGA 4 I, 9.13.1. We did not find the specific counterexample above in the literature checked; it records our verification of the printed hypothesis, not a claim of priority.

Mais revenons à ce cas [page 96]

$$X = \varinjlim X_i \;\Big|\; \begin{array}{l} X_i \in M^\pi \\ I \text{ grand devant } \pi, \end{array}$$

45. Si on suppose (a_π), cet argument prouve du moins que M est stable par $\varinjlim$ grandes devant π.

et introduisons à nouveau

$$\tilde{I} \;=\; \{I' \subset I \mid I' \text{ filtrant, de cardinal } \leq \pi\},$$

donc cette fois $\tilde{I}$ est bien adapté à π (filtrant, et toute partie filtrante de cardinal $\leq \pi$ a une borne supérieure), et si on pose

$$X_{\tilde{\imath}} \;=\; \varinjlim_{i \in I'_{\tilde{\imath}}} X_i,$$

on voit que le système inductif

$$(X_{\tilde{\imath}})_{\tilde{\imath} \in \tilde{I}}$$

est parfaitement adapté à π et la filtration cardinale. De plus, je dis que l'ensemble $\tilde{I}_M$ est *cofinal* dans $\tilde{I}$. En effet, $\tilde{I}_M$ contient le sous-ensemble $\tilde{I}_0$ formé des $I' \in \tilde{I}$ qui ont un plus grand élément – et $\tilde{I}_0$ est bien cofinal ! Maintenant, on peut appliquer le "il suffit" dans b), p. 52
qui conclue que $X = \varinjlim X_i \in M$, q.e.d. [page 97]

On a prouvé $b \Longrightarrow a$, sans utiliser l'hypothèse que tout objet de Filt^π est π-accessible. Mais on a utilisé cette hypothèse pour prouver que b (ou a) implique que M est stable par $\varinjlim$ grandes devant π (lemme 2, p. 90).

Commentary · Exposition [ms. 97]

The implication that does not need presentable objects

The short argument on pages 96–97 is more robust than the intervening attempt to prove closure under large colimits. It proves $b_\pi \Rightarrow a_\pi$ directly, using only the strict filtration and the existence of perfectly adapted presentations. In particular it should not inherit the tacit presentability hypothesis needed for the preceding comma-category construction.

First suppose $X \in M$. Choose a perfectly adapted presentation $X = \varinjlim_I X_i$ at π. By b_π, the subset I_M is cofinal. A cofinal subset of a π^+-directed poset is again π^+-directed: take an upper bound in the original poset for any at most π elements, and then move above it into the cofinal subset. Restricting to I_M therefore presents X as a π^+-directed colimit of objects of M^π.

For the other implication in a_π, suppose $X = \varinjlim_{i \in I} Y_i$ with $Y_i \in M^\pi$ and I grand devant π. Let P be the poset of nonempty directed subsets $S \subset I$ of cardinal at most π, and put $Z_S = \varinjlim_S Y_i$. Strictness places each Z_S in Filt^π. The union construction makes $(Z_S)_{S \in P}$ perfectly adapted, with colimit X.

The key cofinal family consists of the subsets having a largest element. For any $S \in P$, choose an upper bound s of all of S in I and put $S^+ = S \cup \{s\}$. Then $S^+ \in P$, it contains S, and

$$Z_{S^+} \simeq Y_s \in M^\pi.$$

Hence P_M is cofinal in P. The sufficiency direction in b_π gives $X \in M$. This completes both directions of a_π.

The earlier inference on page 90 applied b_π directly to an arbitrary directed presentation and omitted its continuity hypothesis; footnote 42 explicitly retracts that inference. The argument just given is the source's own repair. After it is established, the additional hypothesis that M^π consists of π^+-presentable objects permits the closure argument through canonical comma categories. These are two different uses of the hypotheses, and retaining that separation prevents an artificial circularity in the proof.

$a \overset{?}{\Longrightarrow} b$. Le "il suffit" dans b) est conséquence immédiate de a). Il reste à prouver que si, dans le système inductif π-adapté $(X_i)_{i \in I}$ de limite X, on a $X \in M$, alors I_M est cofinal dans I. Mais par hypothèse a), on aura $X = \varinjlim_J Y_j$, avec J grand devant π, les Y_j dans M^π. Admettons que $\mathrm{Filt}^\pi \subset E_\pi$, *i.e.* tout élément de Filt^π, en particulier les X_i, sont π-accessibles.

Je pense que je n'arriverai pas à prouver $a_\pi \Longrightarrow b_\pi$, mais seulement ceci : Si π' est un cardinal $> \pi$ (disons le successeur de π dans la suite ordinale des cardinaux), alors $a_\pi \Longrightarrow b_\pi$.

Ceci va résulter du [page 98]

Lemme 3. *Soient* $X \simeq \varinjlim_{I} X_i$, $X \simeq \varinjlim_{J} Y_j$ *deux représentations de* X *sous forme des* $\varinjlim$ *filtrantes, avec* I *grand devant* π', J *grand devant* π, *avec* π, π' *deux cardinaux tels que* $\pi' > \pi$. *On suppose* $(X_i)_{i\in I}$ π'*-adapté,* i.e. *pour toute partie filtrante* I' *de* I *de cardinal* $\leq \pi'$, I' *a une borne supérieure* $i_{I'}$, *et* $X_{i_{I'}} = \varinjlim_{i\in I'} X_i$. *Alors pour tout* $i \in I$, *on peut trouver une partie* I' *de* I, *contenant* i, grande devant π, *de cardinal* $\leq \pi'$, *et une partie* J' *de* J, *enfin un isomorphisme* $I' \xrightarrow[\sim]{\lambda} J'$ *et une application strictement croissante* $\mu : J' \longrightarrow I'$ *et des homomorphismes* `[fin du chapitre]`

Commentary · Exposition [ms. 98]

Where the manuscript ends

Grothendieck has reached the comparison of two presentations when the manuscript breaks off. The final words announce homomorphisms without specifying them. We therefore have a programme for a comparison lemma, not a completed statement whose omitted proof can simply be supplied. The separate larger-cardinal repair is a theorem with its own stated hypotheses; it does not turn this fragment retrospectively into a finished argument.

The material that follows belongs to the edition's apparatus. Its notation and terminology indices refer to manuscript page numbers. Thus the reference "77" beside cardinal filtrations points to manuscript page 77, on printed page 42 of this edition. The glossary then restates definitions from the whole chapter, including sections outside these pages. Its axiom list is particularly useful here: it records finite limits and colimits in **M 0**, two factorizations in **M 2**, the properness conditions in **M′**, and the separate requirement $W = TF \circ TC$ in **M 2 bis**. These distinctions explain why the names of the various Quillen structures cannot be replaced indiscriminately by the single modern phrase "model category".

The glossary is a guide to those definitions, and should be checked against their explicit formulation in the manuscript. In its account of a strict cardinal filtration, the qualification "of cardinality at most π" has disappeared; manuscript page 86, footnote 41, supplies it. Its final paragraph also says "well adapted" where the immediately preceding definition requires "perfectly adapted". The audit registers these two discrepancies separately, retaining the printed wording. The bound matters: by the representation axiom every object of E is a filtered colimit of objects in any fixed stage, so unrestricted filtered-colimit closure would identify that stage with all of E. The printed glossary would then express a substantially different definition.

Finally, the bibliography after the glossary is the edition's bibliography. It includes the 1996 publication of Verdier's thesis; its presence is not evidence that Grothendieck had consulted that publication while writing these pages in 1990–1991. Our later references and our repairs are placed in the commentary and its own bibliography, where the date and the responsibility for each statement remain visible.

Commentary · Development (commentators') [ms. 98]

A complete comparison after one successor step

The chapter stops while Grothendieck is constructing a comparison between two presentations. The final Lemma 3 has no completed conclusion or proof, and we do not supply words to its unfinished sentence. His preceding proposal does, however, suggest a precise statement which can be proved under explicit presentability assumptions. The printed implication on page 97 repeats b_π; the surrounding reference to a larger cardinal indicates $b_{\pi'}$ as the intended target.

Successor comparison. [ours] Let E carry a strict cardinal filtration and let $\pi \geq \pi_0$ be infinite. Put $\kappa = \pi^+$ and $\mu = \kappa^+$. Assume that objects of $\mathrm{Filt}^\pi(E)$ are κ-presentable and that objects of $\mathrm{Filt}^\kappa(E)$ are μ-presentable. Let M be a replete full subcategory satisfying a_π. Then M satisfies b_κ, that is, b_{π^+} in the manuscript's notation.

We prove the statement rather than infer it from a general assertion that increasing a cardinal preserves every accessibility property. The literature checked includes SGA 4 I, 9.18, Grothendieck's later Chapter XVIII, §4.13 and Lemma 4.16.7.11, and Lieberman et al. (2020, §8). The argument below is a formulation for the particular continuous presentations used here; its use of alternating factorizations belongs to the standard comparison method for filtrations. We make no priority claim.

Proof. The canonical-slice argument above shows first that M is closed under κ-directed colimits. Indeed every object of M has a presentation by objects of M^{π}, these objects are κ-presentable, and a κ-directed diagram of such presentations can be combined into one κ-filtered diagram of their small terms. The closure half of a_{π} then applies.

We next obtain a more highly directed presentation of each $X \in M$, while keeping its terms in M. Write $X = \varinjlim_{j \in J} Y_j$ as in a_{π}. Any subset $S \subset J$ of cardinal at most κ lies in a κ-directed subposet $K \subset J$ of size at most κ. To see this, first adjoin any element of the nonempty poset J if S is empty, then enumerate S in length at most κ, repeating elements if necessary. At stage $\alpha < \kappa$, choose an upper bound in J for the current enumerated element and all previously chosen bounds. There are at most π of them. The increasing sequence of bounds, together with S, is κ-directed: any fewer than κ of its elements is accounted for before some stage, since κ is regular.

Let H be the poset of all these κ-directed subposets of J of size at most κ. It is μ-directed. The union of at most κ of its members has size at most κ, and the construction just given places that union in another member of H. Put

$$Z_K = \varinjlim_{j \in K} Y_j \qquad (K \in H).$$

The closure already proved gives $Z_K \in M$, and strictness of the filtration gives $Z_K \in \mathrm{Filt}^{\kappa}(E)$. Thus Z_K is μ-presentable by hypothesis. The union-of-subdiagrams argument gives $X \simeq \varinjlim_{K \in H} Z_K$.

Now take a perfectly adapted presentation $X = \varinjlim_{i \in I} X_i$ at cardinal κ. Its index I is μ-directed and its objects are μ-presentable. Fix $i_0 \in I$. We will construct increasing sequences $(i_{\alpha})_{\alpha<\kappa}$ in I and $(K_{\alpha})_{\alpha<\kappa}$ in H, together with maps over X

$$X_{i_{\alpha}} \xrightarrow{u_{\alpha}} Z_{K_{\alpha}} \xrightarrow{v_{\alpha}} X_{i_{\alpha+1}},$$

such that

$$v_{\alpha} u_{\alpha} = X(i_{\alpha} \le i_{\alpha+1}),$$

and, for $\alpha < \beta < \kappa$,

$$u_{\beta} X(i_{\alpha+1} \le i_{\beta}) v_{\alpha} = Z(K_{\alpha} \subset K_{\beta}). \tag{SC}$$

The first equality makes a trip through the other presentation agree with a transition in the X-diagram; the second does the same for the Z-diagram. They will give inverse maps on colimits.

Here are all the inductive choices. At a limit stage $\beta < \kappa$, choose i_{β} above the earlier i_{α} and $i_{\alpha+1}$; at a successor it has already been chosen. Since $X_{i_{\beta}}$ is μ-presentable, its map to $X = \varinjlim_{H} Z_K$ factors through some Z_K. Enlarge K above all the previously chosen K_{α}. For each $\alpha < \beta$, the two maps from $Z_{K_{\alpha}}$ to this stage occurring in (SC) agree after passage to X. Presentability of $Z_{K_{\alpha}}$ makes that equality hold at some later H-stage. There are fewer than κ, hence fewer than μ, such equalities. The μ-directedness of H permits one enlargement making all of them hold. Call the resulting index K_{β} and the factorization u_{β}.

Next factor $Z_{K_{\beta}} \to X$ through an X-stage, using its μ-presentability. Enlarge that stage above i_{β}. The composite with u_{β} and the transition out of $X_{i_{\beta}}$ have equal images in X; presentability of $X_{i_{\beta}}$ allows a further enlargement on which they are equal. This gives $i_{\beta+1}$ and v_{β}. The same process starts at $\beta = 0$, when there are no equations (SC) from earlier stages. Every choice takes place in a small diagram, and the regularity and directedness just used justify every limit stage.

Equations (SC) imply that the u_{α} define a map between the two chain colimits: compose (SC) with u_{α} and use $v_{\alpha} u_{\alpha} = X(i_{\alpha} \le i_{\alpha+1})$. They also imply that the v_{α}, followed by the structural maps to the X-chain colimit, are compatible. The two resulting maps are inverse, by the same displayed equalities. Consequently

$$\varinjlim_{\alpha<\kappa} X_{i_{\alpha}} \simeq \varinjlim_{\alpha<\kappa} Z_{K_{\alpha}} \in M.$$

The membership assertion uses closure of M under κ-directed colimits, since the regular ordinal κ is a κ-directed poset.

The perfectly adapted X-diagram has a supremum $i_* = \sup_{\alpha<\kappa} i_{\alpha}$ and is continuous there: the image of this chain is a directed subset of I with at most κ elements. Therefore $X_{i_*} \simeq \varinjlim_{\alpha<\kappa} X_{i_{\alpha}}$ belongs to M, and $i_* \ge i_0$. Thus I_M is cofinal whenever $X \in M$. Conversely, if I_M is cofinal then it is μ-directed, hence κ-directed, and closure of M gives $X = \varinjlim_{I_M} X_i \in M$. This proves b_{κ}. $\square$

The cardinal shift has a definite function. It permits a comparison chain of length π^+ whose final colimit remains a single stage of the larger adapted presentation. At the original cardinal, that chain is too large for the continuity clause. The counterexample with successor and limit ordinals shows that this obstruction can be real. No continuum hypothesis entered the proof, but presentability at both filtration levels did; the filtration axioms alone do not provide it.

Commentary · Later developments [ms. 98]

Canonical slices, later accessibility, and weak equivalences

Grothendieck returned to these questions at much greater length in Chapter XVIII. Definition 4.13.1 there asks that a full subcategory be replete, accessible, and accessibly included in the ambient accessible category. Lemma 4.16.7.11 tests membership by cofinality of the small canonical slice inside the ambient small-object slice. That is an intrinsic form of the generation condition, with presentability built into the indexing category. It does not assert that every continuous presentation already has cofinally many stages in the subcategory at the original rank. The two powerset examples above therefore leave the later criterion intact. The edition consulted is dated 17 December 2025; the cited lemma lies on manuscript pages 488–490 (Grothendieck, 1990–1991n).

In present terminology, accessibility and accessible inclusion are distinct conditions, and a rank belongs to the statement. The definition and the description as a completion under directed colimits are given by Adámek and Rosický (1994, Definition 2.1 and Theorem 2.26). The earlier monograph of Makkai and Paré (1989) situates these constructions within categorical model theory. For the higher-categorical formulation, Lurie (2009, Proposition 5.4.2.2) characterizes accessible ∞-categories by small generating subcategories of κ-compact objects for a suitable regular κ. Its nearby rank-enlargement results retain explicit cardinal conditions; a bare change from one regular cardinal to a larger one is not a general theorem.

The modern theory of filtrations also distinguishes a directed presentation from a continuous chain. Lieberman et al. (2020, §8, especially Lemma 8.4 and the following paragraph) discusses continuous filtrations and their agreement on a closed unbounded set of indices when the size has the appropriate regularity. This helps explain the example above: the successor-ordinal presentation of ω_1 is not continuous at its countable limit stages. The comparison theorem for two smooth filtrations therefore cannot turn its stages into stages of the continuous limit-ordinal presentation. We checked that discussion and Theorem 8.8 in the authors' 5 June 2019 version; we do not identify its notion of filtration with Grothendieck's cardinal filtration of the whole category.

For homotopical algebra, the relevant application is to a full subcategory of the arrow category. Raptis and Rosický (2015, Proposition 2.2 in author version 2; Proposition 2.3 in the journal) gives a controlled accessibility rank for weak equivalences in a combinatorial model category after the replacement functors preserve the specified presentable objects. Their Theorem 4.2 supplies better bounds under additional hypotheses, and Corollary 5.1 proves finite accessibility for weak equivalences of simplicial sets. These are genuine refinements of the small-generation problem motivating this chapter. They do not show that weak equivalences in every combinatorial model category are finitely accessible, nor do they establish the source's universal-presentation condition for an arbitrary filtration. The present status of the local questions is thus precise: $b_\pi \Rightarrow a_\pi$ is proved in the manuscript; the three-way equivalence as stated is false; the unfinished comparison has the explicit conditional completion proved above.

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Liste des axiomes

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$\mathcal{M}$ catégorie
$W \subset \mathrm{Fl}(\mathcal{M})$, $C, F, TC, TF \subset \mathrm{Fl}(\mathcal{M})$

M 0 La catégorie $\mathcal{M}$ admet des limites projectives finies et des limites inductives finies.

M 1 Les flèches appartenant à C ont la propriété de relèvement à gauche relativement à celles appartenant à TF, et les flèches appartenant à F ont la propriété de relèvement à droite relativement à celles appartenant à TC.

M 2 Toute flèche de $\mathcal{M}$ se décompose en une flèche de C suivie d'une flèche de TF, et en une flèche de TC suivie d'une flèche de F.

M 3 La partie F de $\mathrm{Fl}(\mathcal{M})$ est stable par composition et changement de base, C est stable par composition et cochangement de base, F et C contiennent les isomorphismes.

M 4 La partie TF de $\mathrm{Fl}(\mathcal{M})$ est stable par composition et changement de base, TC est stable par composition et cochangement de base, TF et TC contiennent les isomorphismes.

M 5 Si dans un triangle commutatif de $\mathcal{M}$ deux des trois flèches sont dans W, il en est de même de la troisième ; W contient les isomorphismes.

M′ $TC \subset C \cap W, TF \subset F \cap W$.

M″ Les flèches appartenant à C sont des W-cofibrations, et les flèches appartenant à F sont des W-fibrations.

M′$_0$(g) $TC = C \cap W$.

M′$_0$(d) $TF = F \cap W$.

M 2 bis $W = TF \circ TC$.

Glossaire

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Section 1

— $\Phi \subset \mathrm{Fl}(\mathcal{M})$
Φ est *clos à gauche* : il existe $\Psi \subset \mathrm{Fl}(\mathcal{M})$ tel que Φ soit formé des flèches ayant la propriété de relèvement à gauche relativement aux flèches de Ψ.

— $\Psi \subset \mathrm{Fl}(\mathcal{M})$
Ψ est *clos à droite* : il existe $\Phi \subset \mathrm{Fl}(\mathcal{M})$ tel que Ψ soit formé des flèches ayant la propriété de relèvement à droite relativement aux flèches de Φ.

— $\Phi, \Psi \subset \mathrm{Fl}(\mathcal{M})$
(Φ, Ψ) a la *propriété de factorisation de Quillen* :
a) les flèches de Φ ont la propriété de relèvement à gauche relativement aux flèches de Ψ ;
b) toute flèche f de $\mathcal{M}$ se factorise en

$$f = pi \quad , \qquad i \in \Phi \ , \quad p \in \Psi \ .$$

— (Φ, Ψ) *couple de Quillen* :
a) (Φ, Ψ) a la propriété de factorisation de Quillen ;
b) Φ est clos à gauche et Ψ clos à droite.

Section 2

— $W \subset \mathrm{Fl}(\mathcal{M})$
W *faiblement saturé* :
a) W contient les isomorphismes ;
b) si deux parmi les trois flèches d'un triangle commutatif sont dans W, la troisième aussi.

— $W \subset \mathrm{Fl}(\mathcal{M})$
Un morphisme $p : X \longrightarrow Y$ de $\mathcal{M}$ est une *W-fibration* ($p \in \mathrm{Fib}_W$) si pour tout diagramme de carrés cartésiens

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$$\begin{array}{ccccc} X & \longleftarrow & X' & \xleftarrow{t} & X'' \\ {\scriptstyle p}\downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \xleftarrow[s]{} & Y'' \end{array} \quad ,$$

si $s \in W$, alors $t \in W$.

- $W \subset \mathrm{Fl}(\mathcal{M})$
Un morphisme $i : A \longrightarrow B$ de $\mathcal{M}$ est une *W-cofibration* ($i \in \mathrm{Cof}_W$) si pour tout diagramme de carrés cocartésiens

$$\begin{array}{ccccc} A'' & \xleftarrow{s} & A' & \longleftarrow & A \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle i} \\ B'' & \xleftarrow[t]{} & B' & \longleftarrow & B \end{array} \quad ,$$

si $s \in W$, alors $t \in W$.

- $W \subset \mathrm{Fl}(\mathcal{M})$
  Un morphisme $p : X \longrightarrow Y$ de $\mathcal{M}$ est *universellement dans* W, ou une W*-équivalence universelle* ($p \in W_{\mathrm{univ}}$) si pour tout carré cartésien

$$\begin{array}{ccc} X & \longleftarrow & X' \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle p'} \\ Y & \longleftarrow & Y' \end{array} \quad ,$$

  la flèche p' appartient à W.
- $W \subset \mathrm{Fl}(\mathcal{M})$
  Un morphisme $i : A \longrightarrow B$ de $\mathcal{M}$ est *coüniversellement dans* W, ou une W*-équivalence coüniverselle* ($i \in W^{\mathrm{univ}}$) si pour tout carré cocartésien

$$\begin{array}{ccc} A' & \longleftarrow & A \\ {\scriptstyle i'}\downarrow & & \downarrow{\scriptstyle i} \\ B' & \longleftarrow & B \end{array} \quad ,$$

  la flèche i' appartient à W.

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- *Pré-Q-catégorie de modèles à gauche* (resp. *à droite*) :
  $\mathcal{M}$ et W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfaisant aux axiomes **M 0** à **M 5**, **M′**, **M″** et $\mathbf{M'_0(g)}$ (resp. $\mathbf{M'_0(d)}$).
- *Q-catégorie de modèles*, *triple de Quillen* :
  catégorie de modèles au sens de Quillen.
- *Q-catégorie de modèles "close"*, *triple clos de Quillen*, *catégorie de Quillen close*, *système de Quillen clos* :
  catégorie de modèles fermée au sens de Quillen.
- *Pré-Quillen à gauche* (resp. *à droite*) :
  pré-Q-catégorie de modèles à gauche (resp. à droite).
- *Pré-quintuplet de Quillen clos* :
  W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfaisant aux axiomes **M 0** à **M 5**, **M′**, **M″**, C, TC étant clos à gauche et F, TF clos à droite.
- *Système de Quillen faible* :
  W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfaisant aux axiomes **M 0** à **M 5**, **M′**, **M″**.
- *Système de Quillen faible exact à gauche* (resp. *à droite*) :
  W, C, F, TC, $TF \subset \mathrm{Fl}(\mathcal{M})$ satisfaisant aux axiomes **M 0** à **M 5**, **M′**, **M″** et $\mathbf{M'_0(g)}$ (resp. $\mathbf{M'_0(d)}$).

Section 3

- $(\mathcal{M}, W)$ *catégorie de modèles* :
  $\mathcal{M}$ catégorie satisfaisant à l'axiome **M 0** ;
  $W \subset \mathrm{Fl}(\mathcal{M})$ faiblement saturé.
- *Structure de Quillen faible associée à la catégorie de modèles* $(\mathcal{M}, W)$:
  quadruplet (C, F, TC, TF) de parties de $\mathrm{Fl}(\mathcal{M})$ satisfaisant aux axiomes **M 1** à **M 5**, **M′**, **M″**.

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- *Structure de Quillen faible exacte* :
  structure de Quillen faible satisfaisant en outre à l'axiome **M 2 bis**.
- *Structure de Quillen faible exacte à gauche* (resp. *à droite*) :
  structure de Quillen faible satisfaisant en outre à l'axiome $\mathbf{M_0'(g)}$ (resp. $\mathbf{M_0'(d)}$).
- *Structure de Quillen faible close à gauche* (resp. *à droite*) :
  structure de Quillen faible (C, F, TC, TF) telle que C, TC (resp. F, TF) soient clos à gauche (resp. à droite).
- *Structure de Quillen faible close* :
  structure de Quillen faible à la fois close à gauche et à droite.
- *Structure de Quillen* :
  structure de Quillen faible "biexacte" :
  structure de Quillen faible à la fois exacte à gauche et à droite
  = catégorie de modèles de Quillen propre.
- *Structure de Quillen parfaite* :
  structure de Quillen faible "biexacte" et close
  = catégorie de modèles fermée de Quillen propre.

Section 5

- Catégorie de modèles $(\mathcal{M}, W)$ *quillénisable* :
  il existe une structure de Quillen faible *exacte* (C, F, TC, TF) associée à $(\mathcal{M}, W)$.
- $(\mathcal{M}, W)$ *quillénisable sur* Diag $\subset$ Cat :
  pour tout I dans Diag, la catégorie des modèles $(\mathcal{M}(I), W(I))$ est quillénisable.

Section 6

- Catégorie de modèles $(\mathcal{M}, W)$ *quillénisable uniformément à gauche sur* Diag :
  il existe une structure de Quillen faible *close* (C, F, TC, TF) pour $(\mathcal{M}, W)$ telle que pour tout I dans Diag, si l'on définit $C(I)$ et $TC(I)$ "argument par argument" (*), et $F(I)$ et $TF(I)$ par la propriété de relèvement à droite relativement à $TC(I)$ et $C(I)$ respectivement (†), le quadruplet $(C(I), TC(I), F(I), TF(I))$ soit une structure de Quillen faible *close exacte* pour $(\mathcal{M}(I), W(I))$. On dit alors que (C, F, TC, TF) est une *quillénisation uniforme à gauche* de $(\mathcal{M}, W)$ sur Diag).

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- Catégorie de modèles $(\mathcal{M}, W)$ *quillénisable sur* Diag *uniformément et exactement à gauche* :
  $(\mathcal{M}, W)$ admet une quillénisation uniforme à gauche sur Diag (C, F, TC, TF) telle que $TC = C \cap W$ (autrement dit, telle que (C, F, TC, TF) soit une strucure de Quillen faible *exacte à gauche*). On dit alors que (C, F, TC, TF) est une *quillénisation uniforme et exacte à gauche* de $(\mathcal{M}, W)$ sur Diag.

∗. Un morphisme de foncteurs $u : X \longrightarrow Y$ de I° vers $\mathcal{M}$ étant dans $C(I)$ (resp. $TC(I)$) si et seulement si pour tout objet i de I, u_i est dans C (resp. TC).

†. Un morphisme de $\mathcal{M}(I)$ étant dans $F(I)$ (resp. $TF(I)$) si et seulement si il a la propriété de relèvement à droite relativement aux flèches appartenant à $TC(I)$ (resp. $C(I)$).

Section 7

- I ensemble ordonné filtrant, π cardinal
  I *grand devant* π :
  toute partie de I de cardinal inférieur ou égal à π admet un majorant dans I.
- E catégorie
  Une *filtration cardinale* de E est une filtration croissante

$$\Big(\mathrm{Filt}^{\pi}(E)\Big)_{\pi\geq\pi_0} \quad , \qquad \pi_0 \text{ un cardinal infini fixé,}$$

  de E par des sous-catégories strictement pleines $\mathrm{Filt}^{\pi}(E)$, indexée par les cardinaux $\pi \geq \pi_0$ appartenant à un univers donné, et satisfaisant aux conditions suivantes :
  a) pour tout $\pi \geq \pi_0$, $\mathrm{Filt}^{\pi}(E)$ est équivalente à une petite catégorie ;
  b) la catégorie E admet des limites inductives filtrantes indexées par des ensembles ordonnés I grands devant π_0, et pour tout $\pi \geq \pi_0$, $\mathrm{Filt}^{\pi}(E)$ est stable par les dites limites si $\mathrm{card}\, I \leq \pi$;
  c) Pour tout $\pi \geq \pi_0$, et pour tout $X \in \mathrm{Ob}\, E$, il existe un isomorphisme

$$X \;\simeq\; \varinjlim_{i\in I} X_i \ ,$$

  où $(X_i)_{i\in I}$ est un système inductif filtrant dans E, indexé par un ensemble ordonné I grand devant π, et où les X_i sont des objets de $\mathrm{Filt}^{\pi}(E)$. De plus, si $X \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}(E)$, $\pi' \geq \pi$, on peut prendre I tel que $\mathrm{card}\, I \leq \pi'^{\pi}$.
- E catégorie, $(\mathrm{Filt}^{\pi}(E))_{\pi\geq\pi_0}$ filtration cardinale de E, $\pi \geq \pi_0$ cardinal
  Un *système inductif bien adapté* (à la filtration cardinale donnée, pour les cardinaux π_0, π donnés) est un système inductif filtrant $(X_i)_{i\in I}$ satisfaisant aux conditions suivantes :
  a) I est grand devant π, et pour tout $i \in I$, $X_i \in \mathrm{Filt}^{\pi}(E)$;
  b) Toute partie J de I de cardinal $\leq \pi$, et grande devant π_0, admet une borne supérieure i_J, et on a

$$X_{i_J} \;\xleftarrow{\sim}\; \varinjlim_{i\in J} X_i \ .$$

  Si X est un objet de E, une *représentation bien adaptée* de X, en termes d'objets de $\mathrm{Filt}^{\pi}(E)$, est la donnée d'un système inductif bien adapté $(X_i)_{i\in I}$, à la filtration cardinale donnée, pour les cardinaux π_0, π, et d'un isomorphisme p. 63

$$X \;\xleftarrow{\sim}\; \varinjlim_{i\in I} X_i \ .$$

- E catégorie admettant des petites limites inductives filtrantes
  Une *filtration cardinale stricte* de E est une filtration cardinale $(\mathrm{Filt}^{\pi}(E))_{\pi\geq\pi_0}$ telle que pour tout cardinal $\pi \geq \pi_0$, $\mathrm{Filt}^{\pi}(E)$ soit stable par limites inductives filtrantes.
- E catégorie admettant des petites limites inductives filtrantes
  $(\mathrm{Filt}^{\pi}(E))_{\pi\geq\pi_0}$ filtration cardinale stricte de E, $\pi \geq \pi_0$ cardinal
  Un *système inductif parfaitement adapté* (à la filtration cardinale donnée, pour le cardinal π) est un système inductif filtrant $(X_i)_{i\in I}$ satisfaisant aux conditions suivantes :
  a) I est grand devant π, et pour tout $i \in I$, $X_i \in \mathrm{Filt}^{\pi}(E)$;
  b) Toute partie J de I filtrante de cardinal $\leq \pi$, admet une borne supérieure i_J, et on a

$$X_{i_J} \;\xleftarrow{\sim}\; \varinjlim_{i\in J} X_i \ .$$

Si X est un objet de E, une *représentation parfaitement adaptée* de X, en termes d'objets de $\mathrm{Filt}^{\pi}(E)$, est la donnée d'un système inductif bien adapté $(X_i)_{i\in I}$, à la filtration cardinale donnée, pour le cardinal π, et d'un isomorphisme

$$X \xleftarrow{\sim} \varinjlim_{i\in I} X_i \ .$$

p. 64

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Overview · XIII-b

The second part of Chapter XIII joins two constructions that Grothendieck needs for the same purpose. He wants a homotopy theory to provide a coherent calculus of diagrams and their direct images. He also wants the replacements used in that calculus to be obtainable from a controlled set of small data. Sections 5 and 6 address the first problem through quillenisation; section 7 returns to the cardinal filtrations of SGA 4 to address the second. The manuscript ends during the statement of its final comparison lemma. The break is part of the surviving source, and the commentary does not supply a purported continuation of Grothendieck's text (Grothendieck, 1990–1991q).

Throughout these pages, $\mathcal{M}(I)$ means the category of contravariant diagrams $I^{\mathrm{op}} \to \mathcal{M}$, and $W(I)$ consists of objectwise weak equivalences. This variance explains both the right Kan extension formula over I'/s and the fact that a greatest element of a poset computes the ordinary limit by evaluation. Grothendieck's phrase "catégorie de modèles" initially specifies much less than a modern Quillen model category: it is a category with finite limits and colimits and a two-out-of-three class of morphisms. Cofibrations, fibrations, closedness, and exactness are additional data or properties. Keeping those qualifications visible is necessary both to understand the proofs and to avoid attributing modern theorems to a weaker hypothesis.

A closed exact structure has more force than the subsequent doubts suggest. Exactness says that every weak equivalence factors as a trivial cofibration followed by a trivial fibration. Closedness supplies closure under retracts for the lifting classes. The elementary retract argument therefore gives $TC = C \cap W$ and $TF = F \cap W$. In a uniformly left quillenised diagram theory, restriction preserves the objectwise cofibrations and trivial cofibrations, so its ordinary right adjoint preserves trivial fibrations. The graph-factorisation proof of Ken Brown's lemma then shows that this right adjoint preserves weak equivalences between fibrant diagrams. Consequently every u for which the ordinary u_* exists is admissible under those uniform closed exact hypotheses. This answers the product, tower, and group-indexing questions in the scope where the required ordinary operations exist. The argument uses only finite products, the specified factorizations, lifting, and two-out-of-three; it does not infer the existence of missing infinite limits (Hovey, 1999a, Lemmas 1.1.9 and 1.1.12).

Grothendieck nevertheless gives a useful geometric proof for finite posets. A downward-closed cover reduces a limit to a pullback, and the fibrancy of the cospan makes that pullback homotopy invariant. The construction exposes the local geometry behind pointwise derived direct images. His subsequent noetherian induction is correct, but its proposed larger class has an unexpected limitation: a sober noetherian Alexandrov space associated to a poset is finite. The commentary proves the assertion directly. Pointwise computation for a general comma category also deserves its own hypothesis: restriction to that comma category must preserve the chosen fibrant diagrams. This holds for the finite-poset situation and in the usual cocomplete diagram-model setting, where the left adjoint to restriction along a discrete fibration is computed by fibrewise coproducts. Admissibility and the pointwise comparison are thus separately justified.

The derived-category examples illuminate a different boundary. For a finite shape, a diagram of bounded-below complexes has a common lower bound. For an infinite discrete shape, its components may be bounded below at degrees that tend to minus infinity. Accordingly, $\mathrm{Ch}^+(\mathcal{A}(I))$ and $\mathrm{Fun}(I^{\mathrm{op}}, \mathrm{Ch}^+(\mathcal{A}))$ need not agree, and products need not stay in $D^+(\mathcal{A})$. An explicit family of shifted copies of a field k makes the obstruction visible. Unbounded resolutions answer the substantive existence question in the standard module and sheaf settings: K-injective resolutions allow right derived functors to be defined on the full unbounded derived category. Spaltenstein's work predates the present manuscript, and the general Grothendieck-category theorem is recorded precisely in the references. Beke's construction of the combinatorial injective model structure on unbounded complexes, combined with the diagram-model theorem, gives the all-small-category derivator for any Grothendieck abelian

category (Beke, 2000, Proposition 3.13). As a direct later algebraic treatment under the additional assumption of enough projectives, Alonso Tarrío, Álvarez Díaz, and Jeremías López prove the all-small-category strong derivator theorem for a Grothendieck category with enough projectives (Alonso Tarrío et al., 2023, Theorem 3.4, arXiv version 3).

The modern comparison also separates pointedness from stability. Fibre and cofiber sequences are available in pointed homotopy theories without making their homotopy categories additive or their suspension functors invertible. The manuscript's deliberately broad "triangulée" language must be read accordingly. Strong stable derivators give actual canonical triangulations, with exact diagram operations, as proved by Groth (2013, Theorem 4.16 and Proposition 4.18). His Theorem 1.28 and Proposition 1.30 provide the combinatorial model-category construction of the full derivator. These results explain the successful modern form of the programme without replacing the original finite-limit framework by an unstated cocompleteness assumption.

Section 7 changes the technical language but keeps the construction problem in view. A cardinal filtration $(\mathrm{Filt}^{\pi}(E))$ consists of essentially small stages, bounded colimit-closure conditions, and presentations of every object by sufficiently directed diagrams in each stage. "Grand devant π" means that every subset of cardinality at most π has an upper bound, hence means π^{+}-directed in the customary regular-cardinal convention. Membership in $\mathrm{Filt}^{\pi}(E)$ is a size condition; preservation of these colimits by $\mathrm{Hom}(A,-)$ is a presentability condition. The filtration axioms do not identify the two. SGA 4's comparison results explain why suitable cardinals permit one to use both, and why the extra hypothesis becomes visible in the discussion of canonical slice categories (Artin et al., 1972, I, 9.12 and 9.16–9.18).

The adapted-presentation construction itself survives the audit well. Replace an indexing poset by its small sufficiently directed subposets and take the colimit over each piece. The new index admits the relevant small suprema, and the new diagram preserves them. The editors correctly observe that the first enlargement lemma is immediate under its stated hypotheses: one may simply adjoin an upper bound to the given small subset. The transfinite proof does something more useful when only the weaker directedness of the ambient poset is assumed. Its cardinal equation controls the number of newly imposed bounds. The commentary proves both versions and checks their different assumptions. The separate assertion that generalized continuum hypothesis always makes $(\pi')^{\pi} = \pi'$ is false at singular π' with small cofinality; the displayed power-of-two case remains valid without that hypothesis.

The proposed equivalence of the three conditions on a part M needs a more substantial correction. Condition a_{π} describes M as the objects obtainable by π^{+}-directed colimits from $M \cap \mathrm{Filt}^{\pi}(E)$. Condition b_{π} asks, for every perfectly adapted presentation, that its colimit belong to M if and only if its M-stages are cofinal. Condition c_{π} combines strict generation, closure under sufficiently directed colimits, and ordinary filteredness of a canonical slice. These quantify over different data. The commentary gives two fully checked counterexamples in $E = \mathcal{P}(\omega_1)$ with $\pi = \aleph_0$. Taking M to be the countable successor initial segments together with ω_1 satisfies a_{π}, but a continuous diagram of countable limit initial segments violates b_{π}. Taking M to be the finite subsets together with ω_1 satisfies c_{π} but violates a_{π}. Both examples use the ordinary cardinality filtration, whose stages have the required presentability. They therefore cannot be removed by adding only the compactness hypothesis mentioned in the manuscript.

What does remain is substantial. Grothendieck's proof of $b_{\pi} \Rightarrow a_{\pi}$ is completed without any presentability assumption, using a cofinal family of small subposets with greatest elements. With presentable small stages, the canonical slice construction gives the stronger directedness needed to prove colimit closure. Near page 94 there is a distinct size error: bounding the image of an indexing category does not bound that category itself. A filtered diagram of finite sets can have an uncountable colimit even though only countably many finite sets and maps occur as values. The repair bounds the indexing morphisms. Finally, the enlarged-cardinal

idea is developed as a separate conditional result, with every object of $\mathrm{Filt}^{\pi}(E)$ assumed π^{+}-presentable and every object of $\mathrm{Filt}^{\pi^{+}}(E)$ assumed π^{++}-presentable, and with a complete alternating-chain proof. It is not presented as the missing end of Lemma 3. The powerset counterexamples and this conditional completion are marked as the commentators' work, with no claim of priority.

The accompanying registers preserve the distinction between a transcription mistake, a slip already present in the edition, and a mathematical claim requiring a proof or a correction. Every doubtful reading and every `[sic]` marker of the transcription is accounted for. The editorial indices, axiom list, and glossary have also been checked, including the missing cardinal bound in the glossary's definition of strict filtration. References report the portions actually verified. The purpose throughout is to let the reader follow the mathematical argument while retaining an exact account of what this fragment establishes, what later theory supplies, and what the present commentary proves separately.

Chapter XIV

Carrés h-cartésiens et h-cocartésiens

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XIV

Carrés h-cartésiens et h-cocartésiens

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Pour les quelques corrections évidentes, ou rares commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Un point d'interrogation entre crochets signifie que l'on n'est pas sûr du mot qui précède.

Ce chapitre est inachevé. Dans la première section, Grothendieck introduit une notion de "catégorie quadrangulée" à gauche (resp. à droite). Le texte s'interrompt brutalement au début de la deuxième section, consacrée aux exemples, quand Grothendieck réalise, sans doute, que l'exemple principal qu'il a en vue (la catégorie homotopique d'une catégorie de modèles munie des carrés homotopiquement cartésiens (resp. cocartésiens)) ne satisfait *pas* à l'axiomatique des catégories quadrangulées. Cela est brièvement expliqué dans une postface des éditeurs.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[e-mail address removed]

G. Maltsiniotis

p. 1 [page 1]

1 Axiomes de structures quadrangulées (une tentative avortée)

Soit $(\mathcal{M}, W)$ une catégorie de modèles admettant une théorie de la fibre homotopique. Alors on a dans $\mathrm{Ho}_W = \mathcal{M}W^{-1}$ une notion de *carré W-cartésien* (ou W-homotopiquement cartésien), par quoi on entend un diagramme carré

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

dans Ho_W provenant, à isomorphisme près, d'un carré W-cartésien dans $\mathcal{M}$ – donc isomorphe, dans $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathrm{Ho}_W)$, a un carré *cartésien* `[dans` $\mathcal{M}$`]`

$$\begin{array}{ccc} Y_1 & \longleftarrow & Y_3 \ (\simeq Y_1 \times_{Y_0} Y_2) \\ {\scriptstyle p_1}\downarrow & & \downarrow \\ Y_0 & \underset{p_2}{\longleftarrow} & Y_2 \end{array} \quad ,$$

avec $p_1,\ p_2 \in \mathrm{Fib}_W$. (En fait, il suffit de p_1 ou p_2 dans Fib_W pour assurer le caractère W-cartésien du carré cartésien envisagé.) Je voudrais axiomatiser cette situation, histoire surtout de faire la liste des propriétés serviables essentielles de cette notion, et de la notion duale de carré W-cocartésien (ou W-homotopiquement cocartésien), dans le cas où $(\mathcal{M}, W)$ admet une théorie de la cofibre homotopique. L'espèce de structure, savoir celle des "catégories quadrangulées" (à gauche, à droite, bilatères) est une variante, dans un cas non additif, de [page 2]
la notion de catégorie triangulée dégagée par Verdier. Comme celle-ci, cette notion est en fait trop faible – ce n'est *pas* "la bonne" – mais elle a un intérêt pratique provisoire. La "bonne" notion (s'il en est une à part celle de dérivateur) devrait travailler avec une notion plus fine de carré et de carré W-cartésien, en travaillant systématiquement dans $\mathrm{HO}_W(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1) = \mathcal{M}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1)W(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1)^{-1}$ (où vivent les "vrais" carrés W-cartésiens !), et catégories similaires, telles que $\mathrm{HO}_W(\Psi)$, $\mathrm{HO}_W(\Phi)$. Mais les inclure dans une axiomatique ne faisant pas mention d'une catégorie de modèles $(\mathcal{M}, W)$, revient manifestement à mettre le doigt dans l'engrenage de la notion de dérivateur, et ce n'est pas mon propos pour l'instant.

Commentary · Exposition [ms. 1]

What a square must remember

Grothendieck wants a usable calculus of homotopy pullbacks on the category of homotopy types. The attraction is clear: four objects and four arrows are much easier to manipulate than a model category. The difficulty is already identified in his next paragraph. The comparison

$$\mathrm{dia}_{\square} : \mathcal{M}^{\square}(W^{\square})^{-1} \longrightarrow (\mathcal{M}W^{-1})^{\square}$$

forgets part of a diagram. Here $\square$ has arrows $3 \to 1 \to 0$ and $3 \to 2 \to 0$, with the two composites equal; $\Phi \subset \square$ omits 3. We use covariant diagrams, following the arrows actually drawn. An object on the left is a *coherent square*; on the right it is an ordinary commutative square in the localized category. Calling an object coherent does not mean that its chosen representative has stopped commuting strictly. It means that the whole diagram was localized together.

Write $\mathrm{Cart}_{\mathrm{im}}$ for the replete image, on the right, of the homotopy-cartesian objects on the left. Membership says that *some* coherent realization is cartesian. Proposition 1 will assert, incorrectly, that *every* strict realization is then cartesian. An isomorphism of underlying squares need not lift to an isomorphism of coherent squares. Likewise, four arrows forming a cube after localization need not form a morphism of the chosen coherent squares. These are the two losses that the postface makes explicit.

The opening fibration formula belongs to the assumed theory of homotopy fibres. For an arbitrary Quillen model category, a safe version is that $Y_1 \times_{Y_0} Y_2$ computes the homotopy pullback when $Y_1 \to Y_0$ is a fibration and Y_0, Y_2 are fibrant; right properness permits the familiar version without these fibrancy assumptions (Hovey, 1999b, Theorem 5.2.5 and the mixed Reedy argument of Lemma 5.2.6, dualized). Thus the opening paragraph should not be detached from its hypotheses or used to diagnose a false formula in the transcription. The editors state the precise Quillen version below.

The commentary on Chapter I, manuscript p. 7 ("Coherent squares and what the axioms remember"), and on Chapter IV, manuscript p. 75 ("Constructing the loop square without choosing homotopies"), supplies the general background. The new issue here is reflection of a universal property through $\mathrm{dia}_{\square}$, and the chapter itself supplies a counterexample.

Définition 1. J'appelle *"catégorie à carrés h-cartésiens",* ou *"catégorie quadrangulée à gauche",* une catégorie $\mathcal{H}$, munie d'un sous-ensemble Cart de l'ensemble $\mathrm{Hom}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathcal{H})$
p. 2 des diagrammes carrés `[par carré on entend carré commutatif]`

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

dans $\mathcal{H}$, carrés appelés *"h-cartésiens"* (h comme "homotopiquement"), cette donnée étant [page 3]
soumise aux axiomes qui suivent, Cart (1) à Cart (…).

Commentary · Context [ms. 2]

The earlier reason for keeping diagram categories

The retreat to diagram categories on manuscript page 2 is part of a long-standing programme. In *Pursuing Stacks*, §69, Grothendieck relates it to the difficulty of formulating trace additivity and determinant multiplicativity using only a derived category. His proposed object retains the derived categories of diagram categories, including filtered complexes of varying lengths (Grothendieck, 1983, pp. 1–2 of the edited excerpt). The letter to Thomason of 2 April 1991 returns to the mapping-cone problem and the need for this whole family (Grothendieck, 1991a, §2, pp. 5–6). These passages explain why an apparently modest square calculus leads back to derivators.

Heller's *Homotopy Theories* already organizes homotopy Kan extensions on diagram categories (Heller, 1988, Introduction, pp. 1–2). This is a historical parallel, not a warrant for all the proposed square rules. The point of the present fragment is precisely to test how much of that structure can survive on the single category $\mathcal{H}$.

Two different judgments in the source must therefore be distinguished. The ordinary square is too impoverished to carry the desired coherent universal property; nevertheless, some of the axioms imposed on such squares are too strong for their intended examples. There is no contradiction between these diagnoses. Nor should the editors' suggestion that Grothendieck stopped when he noticed the failure be converted into a documented account of his decision: the editors' postface proves the mathematical obstruction, not that biographical explanation.

Comme je ne suis pas sûr de connaître d'avance toutes les propriétés essentielles, le nombre de ces axiomes reste pour le moment ouvert.

Cart (1) (Symétrie). Si un carré

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

est h-cartésien, de même le carré symétrique (obtenu en échangeant le rôle de X_1 et de X_2).

Cart (2) (Carrés triviaux). Soit

$$\begin{array}{ccc} X_1 & \xleftarrow{q_2} & X_3 \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle q_1} \\ X_0 & \xleftarrow[p_2]{} & X_2 \end{array}$$

un carré dans $\mathcal{H}$, avec p_1 un isomorphisme. Alors le carré est h-cartésien si et seulement si q_1 est un isomorphisme.

Cart (3) (Carrés composés). Soit un diagramme commutatif

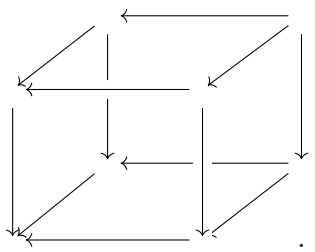
$$\begin{array}{ccccc} X'_0 & \longleftarrow & X'_1 & \longleftarrow & X'_2 \\ \downarrow & \text{I} & \downarrow & \text{II} & \downarrow \\ X_0 & \longleftarrow & X_1 & \longleftarrow & X_2 \end{array} \quad ,$$

d'où les carrés I, II et III :

$$\begin{array}{ccc} X'_0 & \longleftarrow & X'_2 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array}$$

p. 3 (qu'on appelle *carré composé* ("horizontal") des carrés I et II). Supposons I h-cartésien. Alors II est h-cartésien si et seulement si III l'est `[voir postface des éditeurs]`.

Corollaire. *Soit un diagramme commutatif cubique, qu'on peut considérer comme homomorphisme d'un carré (face supérieure du cube) dans un autre (face inférieure) :*



Si toutes les faces du cube[1] *sauf la face supérieure sont des carrés h-cartésiens (savoir la face inférieure et les quatre faces latérales), il en est de même de la face supérieure.*[2] [page 4]

Commentary · Exposition [ms. 3]

Pasting, cancellation, and the cube

The first three axioms try to isolate the part of pullback calculus that can be used without constructing a pullback. Cart (2) has exactly the usual orientation: if the left vertical arrow is invertible, a square is distinguished precisely when the right vertical arrow is invertible. Cart (1) supplies the transposed assertion. Cart (3) then includes both composition and cancellation, with the left square held cartesian. Its failure for $\mathrm{Cart}_{\mathrm{im}}$ will be a failure of cancellation; the postface should not be read as disproving each implication separately.

We can fill in the cube argument entirely formally. Regard the cube as two horizontal squares S_X, S_Y and name the opposite lateral faces

$$B = \begin{pmatrix} X_1 & \leftarrow & X_3 \\ \downarrow & & \downarrow \\ Y_1 & \leftarrow & Y_3 \end{pmatrix}, \qquad F = \begin{pmatrix} X_0 & \leftarrow & X_2 \\ \downarrow & & \downarrow \\ Y_0 & \leftarrow & Y_2 \end{pmatrix}.$$

[1] En fait, il suffit de la face inférieure, et des deux faces parallèles latérales (p. ex. celle de devant et celle de derrière).

[2] **Corollaire** (de Cart (3) et Cart (2)) : *tout carré isomorphe à un carré h-cartésien est h-cartésien.*

If B and S_Y are distinguished, their vertical composite R is distinguished. The same rectangle is obtained by composing S_X with F. If F is distinguished, vertical cancellation implies that S_X is distinguished. Vertical pasting is horizontal pasting after Cart (1). This proves exactly the stronger three-face version in the first footnote; the other two lateral faces are unnecessary. An isomorphism of squares makes B, F distinguished by Cart (2), so the second footnote follows as well.

There is also an economy in the later list: Cart (5)(b) is already a consequence of Cart (1)–(3). For a cube between distinguished S_X, S_Y, suppose p_0, p_1, p_2 are isomorphisms. Then F is distinguished. Composing S_X and F makes R distinguished; cancelling S_Y makes B distinguished. Applying Cart (2) to B and p_1 forces p_3 to be an isomorphism. This deduction concerns the abstract axioms. The editors' two counterexamples will show why the intended class cannot satisfy them.

Ces axiomes sont dans la nature d'axiomes de stabilité, nous donnant des critères affirmant que certains carrés sont h-cartésiens, si telles conditions sont remplies (p. ex. que certains autres carrés sont déjà h-cartésiens). J'ai l'impression d'avoir épuisé ici les propriétés pertinentes de ce type (*i.e.* que les autres devraient pouvoir s'en déduire). À présent, je m'occupe de propriétés d'existence et d'unicité de carrés h-cartésien.

Cart (4) (Axiome du produit fibré). Tout diagramme

$$\begin{array}{ccc} X_1 & & \\ \downarrow & & \\ X_0 & \longleftarrow & X_2 \end{array}$$

dans $\mathcal{H}$ peut se compléter en un carré h-cartésien

$$\begin{array}{ccc} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{array} \quad .$$

L'axiome suivant (*cf.* son corollaire) assurera entre autres que le carré h-cartésien en question est *unique à isomorphisme près* – mais on fera attention qu'il n'est en général pas unique "à isomorphisme unique près".

p. 4 **Cart (5) (Axiome d'exactitude)** Considérons deux carrés h-cartésiens

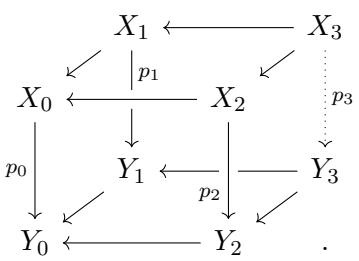
$$X = \begin{pmatrix} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{pmatrix} \qquad \text{et} \qquad Y = \begin{pmatrix} Y_1 & \longleftarrow & Y_3 \\ \downarrow & & \downarrow \\ Y_0 & \longleftarrow & Y_2 \end{pmatrix}$$

et un homomorphisme $p = (p_0, p_1, p_2)$ entre les diagrammes de type

$$\Phi \;=\; \begin{array}{ccc} 1 & & \\ \downarrow & & \\ 0 & \longleftarrow & 2 \end{array}$$

associés (*cf.* diagramme en traits pleins ci-dessous) :

[page 5]



.

Alors :

a) Il existe un homomorphisme $\bar{p} : X \longrightarrow Y$ induisant p, *i.e.* il existe une flèche p_3 de façon à compléter le diagramme cubique commutatif (en respectant donc la contrainte de commutativité pour les deux carrés adjacents à l'arête p_3).

b) Si de plus p_0, p_1, p_2 sont des isomorphismes, il en est de même de p_3 `[voir postface des éditeurs]`.

Prenant le cas où p_0, p_1, p_2 sont des identités, on trouve :

Corollaire 1. *Sous les conditions de* Cart (4)*, si on a* deux *carrés h-cartésiens X, X' qui complètent*

$$\begin{array}{ccc} & & X_1 \\ & \swarrow & \\ X_0 & & \\ & \nwarrow & \\ & & X_2 \end{array} \quad ,$$

il existe un homomorphisme de X dans X' qui induise l'identité sur les sommets d'indices 0, 1, 2*, et un tel homomorphisme est toujours un isomorphisme.*

Mais cet isomorphisme n'est pas forcément unique, en d'autres termes, un automorphisme d'un carré h-cartésien peut induire l'identité sur les sommets X_0, X_1, X_2, sans être l'identité.

Commentary · Exposition [ms. 5]

Existence of a comparison is not a chosen comparison

Cart (4) supplies an object over each cospan. Cart (5)(a) asks that a map of cospans extend to some map of these objects; it does not specify a choice compatible with composition. Cart (5)(b), with its universal quantifier on the already chosen p_3, says that *every* such extension over an isomorphism of cospans is invertible. Replacing "every" by "there exists an invertible extension" would be a different axiom.

For a fixed cospan $A \to C \leftarrow B$, consider the category of cones whose objects are triples $(T, a : T \to A, b : T \to B)$ satisfying $fa = gb$, and whose arrows preserve a, b. The first corollary says that the full subcategory consisting of distinguished completions is nonempty, that any two of its objects admit a comparison, and that every comparison is invertible. This is a connected groupoid, not a proof that it is a terminal category. Existence follows by Cart (5)(a) with three identities; invertibility follows by (b). The source's nonuniqueness remark records the distinction between these conclusions. It supplies no separate example satisfying all five axioms with a nontrivial automorphism group.

For the intended homotopy class, the distinction is quite visible: the square with vertices $\Omega X, 0, 0, X$ and all edges zero permits every automorphism of ΩX over the fixed cospan. It also permits every *noninvertible* endomorphism, including zero. Thus this example illustrates nonuniqueness but simultaneously invalidates (b); it cannot validate the literal axiom system. In an additive category satisfying the whole system, the development below proves that no such nonuniqueness is possible.

In the coherent repair the relevant base is itself a category of coherent cospans. Restriction from cartesian coherent squares to that category is an equivalence. This gives an honest functorial construction there, while forgetting coherence can again identify distinct maps. Neither a noncanonical choice of apex nor an arbitrary commuting cube restores the information that was forgotten.

p. 5 **Corollaire 2.** *Considérons un carré commutatif* [page 6]

$$Y = \begin{pmatrix} Y_1 & \longleftarrow & Y_3 \\ \downarrow & & \downarrow \\ Y_0 & \longleftarrow & Y_2 \end{pmatrix} \quad ,$$

et un objet T de $\mathcal{H}$, d'où (posant $\mathrm{Hom}(T,X) = X(T)$*) un carré commutatif dans* (Ens)

$$\begin{array}{ccc} Y_1(T) & \longleftarrow & Y_3(T) \\ \downarrow & & \downarrow \\ Y_0(T) & \longleftarrow & Y_2(T) \end{array} \quad ,$$

et par suite

$$(*) \qquad Y_3(T) \longrightarrow Y_1(T) \times_{Y_0(T)} Y_2(T) \quad .$$

Si Y est h-cartésien, cette application $(*)$ *est* surjective *quel que soit T.*

On applique Cart (5) à Y et au carré constant X de valeur T

$$X = \begin{array}{ccc} T & \xleftarrow{\mathrm{id}} & T \\ {\scriptstyle \mathrm{id}}\downarrow & & \downarrow{\scriptstyle \mathrm{id}} \\ T & \xleftarrow{\mathrm{id}} & T \end{array} \quad ,$$

en notant que la donnée de $p = (p_0, p_1, p_2)$ de

$$\begin{array}{ccc} T & & Y_1 \\ \downarrow & \text{dans} & \downarrow \\ T & \longleftarrow T & Y_0 \longleftarrow Y_2 \end{array}$$

revient à la donnée d'un élément du deuxième membre de $(*)$,

$$\begin{array}{ccc} Y_1 & \xleftarrow{p_1} & T \\ \downarrow & \swarrow^{p_0} & \downarrow{\scriptstyle p_2} \\ Y_0 & \longleftarrow & Y_2 \end{array} \quad ,$$

et que les éléments de $Y_3(T)$ qui s'envoient (par $(*)$) dans cet élément ne sont autres que les p_3 dont l'axiome Cart (5) affirme l'existence.

N.B. On notera que l'application surjective $(*)$ n'est en général *pas* bijective, *i.e.* le carré h-cartésien n'est en général pas cartésien.

Commentary · Exposition [ms. 6]

Weak universality and the information in a homotopy

The second corollary identifies the positive content of the existence axiom. A distinguished square is a *weak pullback*: every compatible pair of maps from T factors through its apex, without uniqueness. The constant square on T is distinguished by Cart (2), and applying Cart (5)(a) to it gives exactly the surjection displayed in the text. Conversely, if every distinguished square is a weak pullback, apply its weak property to $T = X_3$ and the two composites $X_3 \to X_i \to Y_i$. This supplies Cart (5)(a). Thus, given the constant squares, the Hom-set assertion and that part of Cart (5) are equivalent.

For an actual homotopy pullback the surjection has a geometric explanation. In an enhanced homotopy theory, put

$$M_i = \mathrm{Map}(T, Y_i). \qquad \mathrm{Map}(T, Y_3) \simeq M_1 \times^h_{M_0} M_2.$$

A point of the homotopy pullback consists of two maps and a path between their composites. Equality in $\pi_0 M_0$ ensures that such a path exists, which proves surjectivity after taking components. Different choices of path can give different components. For the loop square, taking $T = S^0$ in pointed spaces and $X = S^1$ gives the map $\pi_0 \Omega S^1 \cong \mathbb{Z} \to \{*\}$, a fully explicit failure of injectivity.

This statement is about the intended homotopy class, not an example of all five literal axioms. Mapping spaces supply the universal property before π_0 is taken; compare Lurie (2009, §1.2.13).

There is a useful rigidity argument whenever an ordinary pullback $P = A \times_C B$ already exists in $\mathcal{H}$. Let Q be a distinguished completion. Its canonical map $q : Q \to P$ has a section $s : P \to Q$ by weak universality. The endomorphism sq of Q is over the identity cospan, so Cart (5)(b) makes it invertible. But $(sq)^2 = sq$; an invertible idempotent is the identity. Hence q is an isomorphism, and Q is an ordinary pullback. Conversely, Cart (4) and invariance under isomorphisms show that every existing ordinary pullback square is distinguished. Consequently, if $\mathcal{H}$ has all ordinary pullbacks, the proposed class is forced to be precisely that class. This is the familiar minimality argument for weak universal objects; its weak-kernel form is explicitly recorded by Krause (2016, §1, "Weak kernels", p. 6 of the 2014 manuscript).

p. 6 **N.B.** Il y a une notion duale de *catégorie à carrés h-cocartésiens,* ou *catégorie quadrangulée à droite.* Le plus souvent, les deux structures sont présentes simultanément. Il y a lieu de dégager des propriétés qui relient les deux structures duales sur une même catégorie. Ainsi, sauf erreur, dans le cas $\mathcal{H}$ additive, les deux espèces de carrés, les h-cartésiens et les h-cocartésiens, devraient *coïncider.* (Il me faudra revenir là-dessus par la suite.) Dans le cas $\mathcal{H}$ "ponctuée", il y a l'adjonction des foncteurs Σ, Ω liés l'un à la structure quadrangulée droite, l'autre à la structure quadrangulée gauche. Et il y a, quand il y a des $\underline{\mathrm{Hom}}$ internes ou externes, des propriétés éventuelles de $\underline{\mathrm{Hom}}(X, Y)$ par rapport aux quadrangles h-cocartésiens en X, h-cartésiens en Y – mais c'est à peine un lien entre les deux structures … [page 7]

Commentary · Later developments [ms. 7]

Additivity, stability, and the two variables of Hom

Grothendieck is looking for a common structure behind fibres and cofibres. Additivity alone cannot supply it. Take $\mathcal{H} = \mathrm{Vect}_k$, with ordinary pullback squares on the left and ordinary pushout squares on the right. Each structure satisfies the corresponding five axioms. For $V \neq 0$, the square

$$\begin{array}{ccc} 0 & \longleftarrow & 0 \\ \downarrow & & \downarrow \\ V & \longleftarrow & 0 \end{array}$$

is a pullback, since $0 \times_V 0 = 0$, but is not a pushout, since $0 \amalg_0 0 = 0$. This refutes the additive speculation within the literal framework, without using any defective homotopy example.

For coherent diagrams, the corrected hypothesis is *stability.* A pointed derivator is stable when its cartesian and cocartesian coherent squares coincide. The suspension–loop adjunction exists already in the pointed case; the equivalence of that adjunction characterizes stability (Groth, 2013, Proposition 3.17; Groth et al., 2014a, Theorem 7.1, arXiv v2). The corresponding higher-categorical statement appears in Lurie (2017, Proposition 1.1.3.4). These theorems concern coherent squares, and do not rescue Cart (3) on the ordinary homotopy category.

The Hom remark also has a precise scope. For fixed T, a derived mapping-space functor sends homotopy limits in its second variable to limits of spaces. In the first variable, mapping out of a homotopy colimit produces a homotopy limit. In a pointed enhancement this gives

$$\mathrm{Map}(\Sigma X, Y) \simeq \Omega\,\mathrm{Map}(X, Y) \simeq \mathrm{Map}(X, \Omega Y),$$

where loops in the mapping space are based at the zero map; taking components gives the adjunction. Cisinski constructs the external derived Hom for model-derived derivators and proves its continuity (Cisinski, 2003, Proposition 6.13, pp. 239–240). Internal Hom requires a specified closed monoidal structure and the appropriate derived adjunctions. Nothing on this page supplies such structure to an arbitrary category $\mathcal{H}$.

Commentary · Development (commentators') [ms. 7]

Binary products force the literal structure to be ordinary

The proposed axioms are meant to retain a weak universal property while allowing nonunique comparisons. The preceding rigidity argument shows that this freedom disappears wherever an ordinary pullback already exists. Pasting makes the restriction stronger: binary products alone force all the missing pullbacks to exist. This applies in particular to additive categories, without any assumption of kernels or of an abelian structure.

We assume Cart (1)–(5) literally throughout this block. The weak-kernel minimality argument has the antecedent cited above. We checked the editors' postface, the coherent-square discussion of Groth (2013, §3), and Krause (2016, §1), together with targeted literature searches. We did not find the following deductions from these particular axioms there. The marking identifies proofs supplied by us, not a claim of priority.

Pullback along a monomorphism. [ours] A distinguished square with a monomorphism among its two base arrows is an ordinary pullback.

Proof. Write the square as the left square below, with f monic. To test uniqueness of its lifting property, apply Cart (4) to $P \xrightarrow{q} B \xleftarrow{q} P$ and obtain the right square:

$$\begin{array}{ccccc} A & \xleftarrow{p} & P & \xleftarrow{r} & R \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle q} & & \downarrow{\scriptstyle s} \\ C & \xleftarrow[g]{} & B & \xleftarrow[q]{} & P. \end{array}$$

Cart (3) makes the outer rectangle distinguished. Its base cospan is $A \xrightarrow{f} C \xleftarrow{fp} P$, because $gq = fp$. Since f is monic, this cospan has ordinary pullback P with projections p and 1_P. The rigidity argument proved in the preceding exposition makes the canonical comparison $s : R \to P$ an isomorphism.

Weak universality of the right square supplies a diagonal $d : P \to R$ with $rd = sd = 1_P$. Thus $d = s^{-1}$ and $r = s$. If $u, v : T \to P$ satisfy $qu = qv$, the same weak property gives $w : T \to R$ with $rw = u$ and $sw = v$; hence $u = v$. Therefore q is monic. The original square is a weak pullback, and its factorization maps are unique because their composites with q agree. It is an ordinary pullback. Cart (1) treats the other base arrow. □

Collapse under binary products. [ours] If $\mathcal{H}$ has binary products and satisfies Cart (1)–(5), it has all pullbacks, and its distinguished squares are exactly its ordinary pullback squares.

Proof. Given $A \xrightarrow{f} C \xleftarrow{g} B$, form the cospan

$$C \xrightarrow{\Delta} C \times C \xleftarrow{f \times g} A \times B.$$

The diagonal is monic. A distinguished completion exists by Cart (4), and the preceding result makes it an ordinary pullback. Its apex is $A \times_C B$: a pair $a : T \to A$, $b : T \to B$ with $fa = gb$ gives the unique map (a, b) to $A \times B$ and the compatible map fa to C, so the displayed pullback property gives a unique map to the apex. All pullbacks therefore exist. Rigidity shows that each distinguished square is ordinary; Cart (4) and isomorphism invariance show that every ordinary pullback square is distinguished. □

No terminal object was used. Additivity supplies the required binary products as biproducts, so the theorem is stronger than an additive statement. It does not classify categories lacking binary products, and we draw no general nonexistence conclusion in that setting.

Pointed coincidence obstruction. [ours] If a pointed category carries both literal structures and their distinguished classes coincide, every object is isomorphic to zero.

Proof. For any X, the cospan $0 \to X \leftarrow 0$ has ordinary pullback 0. Cart (4), rigidity, and isomorphism invariance designate the square with base X and its other three vertices zero. Coincidence makes it distinguished cocartesian. Its upper span $0 \leftarrow 0 \to 0$ has ordinary pushout 0; dual rigidity therefore gives $X \cong 0$. □

Even without coincidence, the literal loop construction is always zero, by the same pullback argument, and the literal suspension is zero if the right structure is present. Their adjunction would be the vacuous zero-functor adjunction. The meaningful suspension and loop functors in homotopy theory belong to the coherent setting anticipated on manuscript page 2.

2 Exemples

Un premier exemple trivial, est celui où $\mathcal{H}$ est stable par produits fibrés, et où les carrés h-cartésiens sont définis comme les carrés cartésiens. Les axiomes Cart (1) à Cart (5) sont alors des propriétés bien connues, ± tautologiques de la notion de carré cartésien. Et en un [page 8] sens, le cas général doit se ramener à ce cas particulier banal.

Le seul type d'autre cas que je connaisse, est celui où $\mathcal{H} = \mathcal{M}W^{-1}$, où W est un localiseur admettant une théorie de la fibre W-homotopique (*cf.* chapitre XII). Dans ce cas, on l'a dit, on prend comme carrés h-cartésiens dans $\mathcal{H}$ ceux qui sont isomorphes, dans $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathcal{H})$, à un carré W-cartésien dans $\mathcal{M}$. Encore faut-il vérifier les axiomes Cart (1) à Cart (5). Je pense que je n'ai guère de chance d'y arriver que moyennant une

Proposition 1. *Moyennant des hypothèses sur $(\mathcal{M}, W)$ qui vont être dégagées ci-dessous, vérifiées en tous cas si $(\mathcal{M}, W)$ est quillénisable, sur la catégorie des ensembles ordonnés finis* (*cf.* XIII, p. 51)*, on a ceci : soit*

$$X = \begin{pmatrix} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{pmatrix}$$

un carré dans $\mathcal{M}$, et soit $\mathrm{ho}_W(X)$ *son image dans* $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1, \mathcal{H})$*. Pour que* $\mathrm{ho}_W(X)$ *soit h-cartésien, il* faut (*et il suffit, bien sûr*) *que X soit W-cartésien* `[voir postface des éditeurs]`*.*

Commentary · Exposition [ms. 8]

The precise assertion that fails in Proposition 1

Grothendieck hopes to verify the axioms by transporting a homotopy-cartesian criterion through localization. The sufficient direction of Proposition 1 is the definition of $\mathrm{Cart}_{\mathrm{im}}$: a homotopy-cartesian strict square has a distinguished underlying image. The necessary direction would make homotopy cartesianness a property of the ordinary underlying square alone. The editors disprove precisely this reflection statement.

It matters that the obstruction occurs in Quillen model categories with entirely familiar diagram model structures. It cannot be removed by the promised availability of models for finite ordered indexing categories. The walking square and the walking cube are already finite posets; the example below lives in complexes over a field, where all their diagram model structures are available. An additional hypothesis could restrict the homotopy theory drastically, but Quillen presentability of diagrams does not establish the proposition.

The preliminary example, ordinary pullbacks in an ordinary category, remains valid. Symmetry, the isomorphism test, and pasting follow directly from the pullback universal property; every cospan has a completion by hypothesis; a map of cospans induces a unique map of pullbacks; an invertible base map induces an invertible comparison. These statements also prove the stronger uniqueness omitted from the proposed homotopical axioms. The preceding development gives a precise circumstance in which the general axiom system really does reduce to this example: the existence of binary products. It does not vindicate the intended model-category example.

Postface des éditeurs

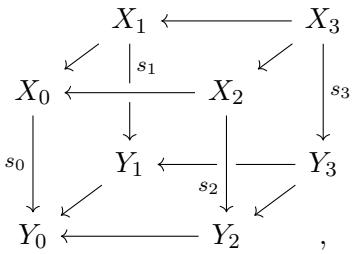
La proposition ci-dessus est fausse. On rappelle que si $\mathcal{M}$ est une catégorie de modèles fermée de Quillen de classe d'équivalences faibles W, un carré commutatif

$$X = \begin{pmatrix} X_1 & \longleftarrow & X_3 \\ \downarrow & & \downarrow \\ X_0 & \longleftarrow & X_2 \end{pmatrix}$$

de $\mathcal{M}$ est W-cartésien si et seulement si il existe un carré *cartésien*

$$Y = \begin{pmatrix} Y_1 & \longleftarrow & Y_3 \\ \downarrow & & \downarrow \\ Y_0 & \longleftarrow & Y_2 \end{pmatrix}$$

avec $Y_1 \longrightarrow Y_0$ une fibration, et Y_0 et Y_2 fibrants (cette dernière condition étant superflue si la catégorie de modèles $\mathcal{M}$ est propre à droite), et un morphisme de carrés $s : X \longrightarrow Y$, $s = (s_0, s_1, s_2, s_3)$, *i.e.* un cube commutatif



tel que s soit une équivalence faible argument par argument, *i.e.* $s_0, s_1, s_2, s_3 \in W$. Il est équivalent de demander que pour *tout* diagramme commutatif à flèches verticales des équivalences faibles

$$\begin{array}{ccccc} X_1 & \longrightarrow & X_0 & \longleftarrow & X_2 \\ \downarrow & & \downarrow & & \downarrow \\ Y_1 & \longrightarrow & Y_0 & \longleftarrow & Y_2 \end{array}$$

tel que $Y_1 \longrightarrow Y_0$ soit une fibration, et Y_0, Y_2 des objets fibrants, la flèche canonique $X_3 \longrightarrow Y_1 \times_{Y_0} Y_2$ soit une équivalence faible.

Commentary · Exposition [ms. 8]

Postface: why one strict comparison cube suffices

Editors' postface, printed p. 7. The editors replace the cospan before taking its limit. This makes the missing comparison map canonical, and explains the difference between a weak equivalence of whole squares and a mere isomorphism after applying $\mathrm{dia}_\square$.

Here is a verification with exactly their fibrancy hypotheses. Give $\Phi = (1 \to 0 \leftarrow 2)$ the Reedy degrees $\deg(2) = 0$, $\deg(0) = 1$, $\deg(1) = 2$, with $2 \to 0$ direct and $1 \to 0$ inverse. A diagram Y is Reedy fibrant precisely when Y_0, Y_2 are fibrant and $Y_1 \to Y_0$ is a fibration. The constant-diagram functor is left Quillen: for a (trivial) cofibration its relative latching maps at $1, 2$ are that map, and at 0 the identity. Hence its right adjoint, the pullback functor, is right Quillen. This is the mixed Reedy method of Hovey (1999b, Theorem 5.2.5 and the proof of Lemma 5.2.6), dualized.

For a cospan K, choose a Reedy trivial cofibration $K \to RK$ with fibrant target. Any objectwise weak equivalence $K \to Y$ to a cospan with the printed fibrancy conditions extends to a map $RK \to Y$ by lifting against $Y \to *$. It is a weak equivalence by two-out-of-three. Applying the right Quillen pullback functor preserves it, since both diagrams are fibrant. Consequently the two comparison

maps from the original fourth vertex are weak equivalences simultaneously. This proves "some" if and only if "every". Taking $Y_3 = \lim Y$ makes the fourth component of the strict cube the unique induced cone map, so it also proves the single-cube criterion.

For completeness, right properness removes both objectwise fibrancy conditions. If $K_1 \to K_0$ is a fibration and $K \to R$ is a fibrant replacement of the cospan, put $K_1' = R_1 \times_{R_0} K_0$. Right properness and two-out-of-three make $K_1 \to K_1'$ a weak equivalence between fibrant objects of the slice $\mathcal{M}/K_0$. Pullback to K_2 is right Quillen, so

$$K_1 \times_{K_0} K_2 \longrightarrow R_1 \times_{R_0} K_2 \longrightarrow R_1 \times_{R_0} R_2$$

are weak equivalences: the first by that slice argument, the second by right properness. This verifies the parenthesis in the source, without adding a properness assumption to the fibrant version.

Supposons maintenant que la catégorie de modèles $\mathcal{M}$ soit ponctuée, autrement dit qu'elle admette un objet $*$ à la fois initial et final, et soit X un objet fibrant de $\mathcal{M}$. Choisissons une décomposition de la flèche $* \longrightarrow X$, en une équivalence faible suivie d'une fibration

$$* \longrightarrow C \longrightarrow X \quad ,$$

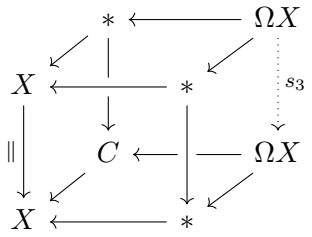
p. 8 et formons le carré cartésien

$$\begin{array}{ccc} C & \longleftarrow & \Omega X \\ \downarrow & & \downarrow \\ X & \longleftarrow & * \end{array} \quad ,$$

où par définition $\Omega X = C \times_X *$ est l'*espace des lacets* de X. Ce carré est W-cartésien. Par ailleurs, il est isomorphe dans $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1,\ \mathcal{M}W^{-1})$ au carré trivial

$$\begin{array}{ccc} * & \longleftarrow & \Omega X \\ \downarrow & & \downarrow \\ X & \longleftarrow & * \end{array} \quad .$$

Néanmoins, ce dernier n'est *pas* en général W-cartésien. En effet, s'il l'était, l'unique flèche $s_3 : \Omega X \longrightarrow \Omega X$ de $\mathcal{M}$ rendant commutatif le cube



serait une équivalence faible. Or, comme la flèche s_3 se factorise par $* \times_X * \simeq *$, cela impliquerait que $\Omega X \to *$ est une équivalence faible. Cette condition est très restrictive. Si, par exemple, $\mathcal{M}$ est la catégorie de modèles des espaces topologiques pointés, et X un CW-complexe pointé, elle équivaut à la contractibilité de la composante connexe de X contenant le point base. De même, si $\mathcal{M}$ est la catégorie de modèles des complexes d'une catégorie abélienne de Grothendieck et X un complexe, elle signifie que X est acyclique.

Commentary · Examples [ms. 8]

Postface: a two-term complex and the missing filling

Editors' postface, printed p. 8. The counterexample is especially transparent when every map can be written as a scalar. Let k be a nonzero field and use unbounded cochain complexes, with quasi-isomorphisms as weak equivalences and degreewise surjections as fibrations (Hovey, 1999b,

Theorem 2.3.11, with degrees reversed). Our shift convention is $(V[r])^n = V^{n+r}$, with differential $(-1)^r d_V$.

Take $X = k[0]$ and the acyclic complex $C = (k \xrightarrow{1} k)$ in degrees $0, 1$. Let $p : C \to X$ be the identity in degree 0 and zero in degree 1. This is a surjective chain map. Its kernel P is $k[-1]$, concentrated in degree 1. Thus $0 \xrightarrow{\sim} C \xrightarrow{p} X$ is the editors' factorization and

$$\begin{array}{ccc} C & \xleftarrow{\ i\ } & P \\ {\scriptstyle p}\downarrow & & \downarrow \\ X & \longleftarrow & 0 \end{array}$$

is both an ordinary and a homotopy pullback. The contraction $s^1 : C^1 \to C^0$ is the identity, and $d_C s + s d_C = 1_C$. In $D(k)$, therefore, $C \cong 0$, and the underlying square is isomorphic to the square with apex P, base X, and all edges zero.

That latter square, taken *strictly* in complexes, has comparison $s_3 = 0_P$ to the displayed fibrational replacement: commutativity requires $i s_3 = 0$, and i is injective. The comparison induces zero on $H^1(P) = k$, so it is not a quasi-isomorphism. This computes the dotted arrow in the postface and disproves the necessary direction of Proposition 1.

To locate the information lost, write down a homotopy-pullback model for $f : A \to Z \leftarrow B : g$:

$$\begin{aligned} H(f,g)^n &= A^n \oplus B^n \oplus Z^{n-1}, \\ d(a,b,z) &= (d_A a, d_B b, fa - gb - d_Z z). \end{aligned}$$

The chain-map identities imply $d^2 = 0$. This is a shifted mapping cone of $f - g$, with the last coordinate negated relative to one common convention. A cone with maps $u : T \to A$, $v : T \to B$ and a specified degree -1 homotopy h satisfying

$$d_Z h + h d_T = fu - gv$$

has comparison $c_h = (u, v, h) : T \to H(f,g)$. The displayed identity is exactly its chain-map condition. The coherent square is cartesian precisely when this comparison is a quasi-isomorphism.

For $A = B = 0$, $Z = X = k[0]$, and $T = P = k[-1]$, the four edges are zero and $H(0,0) = P$. Every scalar $\lambda \in k$ specifies a filling $h^1_\lambda = \lambda$, and c_{h_λ} is multiplication by λ on P. It is a quasi-isomorphism exactly when $\lambda \neq 0$. Its mapping cone is the two-term complex

$$k \xrightarrow{\lambda} k \quad \text{in degrees } 0, 1,$$

which is acyclic for $\lambda \neq 0$ and has k in both cohomology degrees for $\lambda = 0$. Thus identical vertices and edges support both a cartesian and a noncartesian coherent filling.

We can check the transport sign as well. Put $t = ps$, so $d_X t + t d_C = p$. The map

$$H(p,0) \longrightarrow X[-1], \qquad (c, z) \longmapsto z - tc$$

is a chain map with acyclic kernel, the diagonal copy of C. Composing with $P \to H(p,0)$, $a \mapsto (i(a), 0)$, gives -1_P. The strict pullback transported through this contraction has filling h_{-1}, whereas the strict zero-edge square has filling h_0. The sign depends on the displayed cone convention; its being nonzero does not. With the boundary and its identifications fixed, the classes of fillings are indexed by k, because $\underline{\mathrm{Hom}}(P, X)$ is concentrated in degree -1 and has no degree -2 boundaries. Allowing scalar automorphisms of the apex identifies all nonzero fillings, while the zero filling remains separate.

The editors' other interpretations agree with this computation. In pointed spaces, $X = S^1$ gives $\pi_0 \Omega X = \mathbb{Z}$, so the zero self-map cannot be a weak equivalence. For a CW complex in general, the homotopy groups of its loop space detect all positive homotopy groups of the base component; Whitehead's theorem then makes exactly that component contractible (Hatcher, 2002, Theorem 4.5 and §4.3). Other components are unrestricted. For unbounded complexes in a Grothendieck abelian category, an acyclic middle term in $0 \to P \to C \to X \to 0$ gives $H^n(P) \cong H^{n-1}(X)$, so P is acyclic exactly when X is. The standard injective model structure supplies this setting (Hovey, 2001, Theorem 2.2 and Proposition 2.12 of the preprint); no assertion that acyclic complexes are chain-contractible is needed.

Des considérations analogues montrent que les carrés h-cartésiens de $\mathcal{M}W^{-1}$, définis dans la section 2 comme étant les carrés commutatifs de $\mathcal{M}W^{-1}$ qui sont isomorphes dans $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1,\ \mathcal{M}W^{-1})$ à un carré W-cartésien, ne satisfont pas à toutes les conditions de Cart (3) et Cart (5) de la section 1. En effet, supposons toujours que $\mathcal{M}$ soit ponctuée, et gardons les notations ci-dessus. Considérons le diagramme commutatif dans $\mathcal{M}W^{-1}$

$$\begin{array}{ccccc} * & \longleftarrow & \Omega X & \xleftarrow{*_{\Omega X}} & \Omega X \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & * & \longleftarrow & * \end{array} \quad ,$$

où $*_{\Omega X}$ désigne l'unique endomorphisme de ΩX se factorisant par $*$. Il résulte de ce qui précède que le carré de gauche et le carré composé sont h-cartésiens. En revanche, le carré de droite ne l'est pas en général. En effet, si ce carré était isomorphe dans $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1 \times \mathbf{\Delta}_1,\ \mathcal{M}W^{-1})$ à un carré W-cartésien, la flèche $*_{\Omega X}$ serait un isomorphisme

p. 9 de $\mathcal{M}W^{-1}$, donc une équivalence faible de $\mathcal{M}$, ce qui impliquerait les mêmes conditions très restrictives que ci-dessus. Ainsi l'une des implications de l'axiome Cart 3 n'est pas satisfaite.

De même le cube commutatif dans $\mathcal{M}W^{-1}$

$$\begin{array}{ccccc} & & * & \longleftarrow & \Omega X \\ & \swarrow & \downarrow & \swarrow & \downarrow{\scriptstyle *_{\Omega X}} \\ X & \longleftarrow & * & & \\ \downarrow{\scriptstyle =} & & \downarrow & & \\ & & * & \longleftarrow & \Omega X \\ & \swarrow & & \swarrow & \\ X & \longleftarrow & * & & \end{array} \quad ,$$

dont les faces horizontales sont h-cartésiennes, montre que la condition (b) de l'axiome Cart (5) n'est pas satisfaite en général.

Commentary · Present state [ms. 8]

Postface: the two failed quantifiers

Editors' postface, printed pp. 8–9. The two final counterexamples use the same object $P = \Omega X$ and the same underlying zero-edge square Q. They apply whenever P is not isomorphic to zero, a condition realized by both worked examples above. In the horizontal pair, the left square and outer rectangle are Q, hence belong to $\mathrm{Cart}_{\mathrm{im}}$. In the right square the bottom edge is 1_0 and the top edge is 0_P. A homotopy pullback of an equivalence has an equivalence as opposite edge, so this right square cannot belong to $\mathrm{Cart}_{\mathrm{im}}$. This is exactly the implication "I and III cartesian imply II cartesian" that fails.

For Cart (5)(b), use the endomorphism of Q that is the identity at the three cospan vertices and zero at P. It is a commuting cube in the ordinary homotopy category, but its fourth component is not invertible. The positive existence statement in (a), and the failure of uniqueness already noticed in Corollary 2, should not be conflated with this false assertion about *every* fourth component. As the formal deduction above shows, (b) would have followed from the full pasting axiom anyway.

The implication $0_P \in W \Rightarrow P \to 0 \in W$ used in the postface needs no cofibrancy assumption. Writing $r : P \to 0$ and $s : 0 \to P$, the arrow r is a retract, in the arrow category, of $sr = 0_P$, using components $(1_P, s)$ and $(1_P, r)$. Weak equivalences are closed under retracts. The converse follows from two-out-of-three. Equivalently, an invertible zero endomorphism makes the object zero in the homotopy category. These are settled counterexamples, not conjectures awaiting a stronger model-category theorem.

Commentary · Exposition [ms. 8]

Why composition nevertheless survives

The failure of cancellation leaves a useful positive statement: for every Quillen model category, $\mathrm{Cart}_{\mathrm{im}}$ is closed under composition. Two adjacent cartesian witnesses can be made compatible along their shared edge. A left square and an outer rectangle, as in the counterexample, are different data. We give the rectification argument to keep this distinction precise.

For any model category $\mathcal{A}$, the arrow comparison

$$U_{\mathcal{A}} : \mathrm{Ho}(\mathcal{A}^{[1]}) \longrightarrow \mathrm{Ho}(\mathcal{A})^{[1]}$$

is full, essentially surjective, and conservative. The first two properties are the arrow-shape lifting property used in derivator strength (Groth, 2013, Lemma 1.9). To verify fullness directly, take a Reedy-cofibrant source arrow $f : X_0 \to X_1$ and a Reedy-fibrant target arrow $g : Y_0 \to Y_1$. Thus X_0 is cofibrant, f is a cofibration, and Y_0, Y_1 are fibrant. Represent a square in the homotopy category by maps a_0, a_1 and a right homotopy between $a_1 f$ and $g a_0$. The endpoint map $\mathrm{ev}_0 : \mathrm{Path}(Y_1) \to Y_1$ is a trivial fibration. Lifting against f extends that homotopy from X_0 to X_1, replacing a_1 by a homotopic a_1' with $a_1' f = g a_0$ strictly. This gives the required coherent-arrow morphism. Any arrow in the homotopy category has an actual representative between cofibrant-fibrant objects, proving essential surjectivity. Finally, pointwise weak equivalences and saturation imply conservativity. In particular, an isomorphism of underlying arrows lifts to an isomorphism of coherent arrows.

Now choose cartesian strict lifts of the two adjacent squares. Their common columns may differ, but the given underlying identifications specify an isomorphism between these columns in $\mathrm{Ho}(\mathcal{M})^{[1]}$. Apply the preceding result for $\mathcal{A} = \mathcal{M}$ to lift it. The two squares can then be regarded as composable morphisms in $\mathrm{Ho}(\mathcal{M}^{[1]})$. Choose bifibrant representatives for the three column objects in the arrow model category, with a single shared middle representative, and represent the two morphisms by actual maps. They form one strict rectangle in $\mathcal{M}$.

Apply the arrow comparison a second time, now to $\mathcal{A} = \mathcal{M}^{[1]}$. Each new small square has an underlying arrow in $\mathrm{Ho}(\mathcal{A})$ isomorphic to that of its original witness. Fullness lifts this isomorphism and conservativity makes the lift invertible in $\mathrm{Ho}(\mathcal{M}^{\square})$. Thus the two new squares are cartesian as coherent squares. Coherent pasting makes their outer rectangle cartesian, and its underlying square is isomorphic to the required composite. This proves the assertion without a combinatorial or stability assumption, and without lifting an arbitrary incoherent cube.

In the stable setting, Christensen and Frankland (2022, Proposition 6.11, pp. 17–18 of the 7 June 2021 author version) prove the corresponding pasting statement even for homotopy-cartesian squares in an arbitrary triangulated category. The valid remnant of the proposed calculus is therefore substantial: symmetry, the isomorphism test, completion of cospans, weak universality, and positive composition survive for the homotopy-image class. Cancellation and Cart (5)(b) do not.

Commentary · Later developments [ms. 8]

The coherent replacement of the square axioms

After the editors' postface. The construction Grothendieck names on manuscript page 2 gives the appropriate replacement. Let $\mathbb{D}$ be a derivator and $j : \Phi \hookrightarrow \square$ the full inclusion omitting the prospective pullback vertex. A coherent square $Q \in \mathbb{D}(\square)$ is cartesian when the adjunction unit

$$Q \longrightarrow j_* j^* Q$$

is invertible. Right Kan extension along j is fully faithful; consequently

$$j^* : \mathbb{D}(\square)_{\mathrm{cart}} \simeq \mathbb{D}(\Phi)$$

is an equivalence, with inverse j_* (Groth, 2013, Proposition 1.20 and Definition 3.9). A coherent cospan therefore has a cartesian completion, and a morphism of coherent cospans has a unique extension between these completions once their identifications are fixed. If its three components are isomorphisms, pointwise conservativity makes it an isomorphism in $\mathbb{D}(\Phi)$; the equivalence makes the extended square morphism invertible. This proves the correct fourth-vertex assertion. Its input is a morphism in the coherent category, not an arbitrary cube in $\mathbb{D}(e)$.

The full pasting-and-cancellation law is a statement about one coherent rectangle. To see the mechanism in a model or higher-category enhancement, write it as

$$\begin{array}{ccccc} A & \longleftarrow & B & \longleftarrow & C \\ \downarrow & & \downarrow & & \downarrow \\ D & \longleftarrow & E & \longleftarrow & F. \end{array}$$

If the left square is cartesian, $B \simeq A \times^h_D E$. Associativity of the homotopy limit gives

$$B \times^h_E F \simeq (A \times^h_D E) \times^h_E F \simeq A \times^h_D F.$$

Coherence identifies the comparison from C to the first expression with its comparison to the last. One is an equivalence exactly when the other is. For derivators this is the right-Kan-extension pasting statement of Groth (2013, Proposition 3.13); the higher-categorical proof is given, dually, by Lurie (2009, Lemma 4.4.2.1). The editors' rectangle loses the compatibility needed to make those two comparison maps agree.

In a stable enhancement there is a useful translation into the underlying triangulated category. A coherent cartesian square with projections $u : P \to A$, $v : P \to B$ and base maps $f : A \to C$, $g : B \to C$ supplies a distinguished triangle

$$P \xrightarrow{(u,-v)} A \oplus B \xrightarrow{(f,g)} C \xrightarrow{\delta} \Sigma P$$

(Groth et al., 2014a, Theorem 6.1, arXiv v2). The connecting morphism is part of this conclusion. In the zero-edge loop example it is an isomorphism $X \to \Sigma P$, with the sign dictated by conventions. The cube with fourth component 0_P and base component 1_X cannot preserve that morphism, since it would require $\delta = (\Sigma 0_P)\delta = 0$. The triangle therefore diagnoses exactly why that cube is incoherent, even though its four visible faces commute in the homotopy category.

The relation with higher categories is a comparison of enhanced diagram theories. For a suitable model presentation, localizing diagram categories gives the homotopy categories of coherent diagrams; precise comparisons appear in Lurie (2017, Propositions 1.3.4.24–1.3.4.25). It is not necessary here to identify every abstract derivator with an infinity category. The correct conclusion is narrower and sufficient: the familiar homotopy theories retain the diagram information on which the failed inference depended.

Overview · XIV

Chapter XIV is a short experiment in reducing homotopy theory to a calculus of squares in a single category. Its eight manuscript pages introduce five proposed axioms and begin the examples. The editors' postface, which has no manuscript pagination, explains why the intended model-category example fails. The fragment is especially instructive because Grothendieck identifies the necessary finer setting before attempting the reduction: square diagrams must be localized as diagrams, together with their maps, rather than reconstructed from four arrows in the localized category.

The mathematical object under discussion can be described without ambiguity. For $\mathcal{H} = \mathcal{M}W^{-1}$, let $\mathrm{Cart}_{\mathrm{im}}$ consist of those commutative squares in $\mathcal{H}$ that are isomorphic to the underlying images of homotopy-cartesian strict squares in $\mathcal{M}$. With $\square$ denoting the walking commutative square, the relevant functor is

$$\mathrm{dia}_{\square} : \mathcal{M}^{\square}(W^{\square})^{-1} \longrightarrow \mathcal{H}^{\square}.$$

The domain retains coherent diagrams. The target retains ordinary commutativity after localization. Membership in $\mathrm{Cart}_{\mathrm{im}}$ is an existential statement about a coherent realization. Proposition 1 would promote it to a statement about every strict realization, and this is the false inference.

There is a specific historical reason for wanting the larger diagram theory. In *Pursuing Stacks*, §69, Grothendieck connects it with the trace and determinant problems that motivated him in the period of SGA 5; the letter to Thomason of 2 April 1991 again stresses derived diagram categories and mapping cones (Grothendieck, 1983, 1991a). The present chapter tests whether one can retain a useful part of their calculus in a simpler language. The editors' preliminary notice suggests that recognition of the failed example explains the abrupt ending. That suggestion is conjectural; the postface establishes the mathematical failure, not the author's reason for interrupting the manuscript.

The formal axioms are consistent: any category with pullbacks, equipped with its ordinary pullback squares, satisfies them. Cart (1) gives symmetry, Cart (2) tests trivial squares against isomorphisms, and Cart (3) combines pasting with cancellation. Cart (4) requires a completion of every cospan. Cart (5)(a) extends a map between the three-vertex boundaries of two distinguished squares; (b) asserts that every such extension is invertible whenever the boundary map is invertible. The commentary supplies the cube proof, including the stronger three-face hypothesis in the footnote, and proves that (b) already follows from Cart (1)–(3).

The source's second corollary isolates the weak universal property that does survive in homotopy theory. For a homotopy pullback P of $A \to C \leftarrow B$, there is a surjection

$$[T, P] \longrightarrow [T, A] \times_{[T,C]} [T, B].$$

A compatible pair in the target admits a homotopy between its composites; a lift remembers a choice of that homotopy. Passing to components forgets the choice, so the map need not be injective. In the loop square for S^1, with $T = S^0$ in pointed spaces, it is the map $\mathbb{Z} \to \{*\}$. The surjection is equivalent to Cart (5)(a) for a class containing the constant squares. The much stronger assertion in (b) does not follow from this weak property alone.

Positive composition does hold for the homotopy-image class of every Quillen model category. The commentary proves this by applying fullness and conservativity of the arrow comparison twice, first to the model category and then to its arrow category, so that two cartesian witnesses acquire a single shared middle column. This is compatible with the postface: its counterexample concerns cancellation.

The editors' example distinguishes the underlying square from its realization. Factor the basepoint of a fibrant object as

$$0 \xrightarrow{\sim} C \twoheadrightarrow X, \qquad P = C \times_X 0.$$

The resulting strict pullback is homotopy cartesian. Its image in $\mathcal{H}^{\square}$ is isomorphic to the square with apex P, base X, the other vertices zero, and all edges zero. But that zero-edge square, regarded strictly in $\mathcal{M}$, has comparison $0_P : P \to P$ to the fibrational replacement. It is homotopy cartesian exactly when P is weakly zero. This proves the failure of Proposition 1 whenever the loop object is nontrivial. The same underlying square gives the two final counterexamples: a horizontal rectangle with left and outer squares distinguished but right square not distinguished, and a cube whose first three components are identities while the fourth is zero.

The worked algebraic example takes $X = k[0]$ and $C = (k \xrightarrow{1} k)$ in degrees $0, 1$. Then $P = k[-1]$. A specified filling of the fixed zero-edge square is a scalar λ; the homotopy-pullback comparison is multiplication by λ on P. Its cone is $k \xrightarrow{\lambda} k$, so the coherent square is cartesian precisely for $\lambda \neq 0$. The commentary computes the transport through a contraction and its sign explicitly. It also checks the editors' topological qualification: for a CW complex, weak triviality of the loop object means contractibility of the base component, with no restriction on other components. For unbounded complexes in a Grothendieck abelian category it means acyclicity.

The literal axioms have a stronger limitation than the displayed counterexample alone reveals. Our development proves that, if $\mathcal{H}$ has binary products, the axioms force all ordinary pullbacks to exist and force the distinguished class to be exactly their class. The proof first compares a distinguished weak pullback with an existing ordinary one: a section and the invertibility of a cone-preserving idempotent make the comparison an isomorphism. This is the minimality argument familiar for weak kernels (Krause, 2016, §1, p. 6 of the 2014 manuscript). Pasting a distinguished square with a weak kernel pair then shows that pullbacks along monomorphisms are ordinary. Finally one pulls back the diagonal $C \to C \times C$ along $A \times B \to C \times C$. The complete argument is given in the marked development block; its application to these axioms was not located in the bounded literature search, and no priority is claimed.

Additive categories satisfy the binary-products hypothesis. Moreover, even without products, the literal loop object in a pointed category must be zero, because $0 \to X \leftarrow 0$ already has ordinary pullback zero. If the dual structure also exists, its suspension is zero. If the two classes coincide, the square with base X and three zero vertices makes X itself zero. Thus Grothendieck's additive coincidence speculation is false for the literal structures: ordinary pullbacks and pushouts in vector spaces give an immediate counterexample.

The modern positive statement belongs to coherent squares. For the inclusion $j : \Phi \hookrightarrow \square$ of the cospan, cartesian coherent squares are the essential image of the fully faithful right Kan extension j_*. Restriction consequently identifies their category with $\mathbb{D}(\Phi)$. This makes the completion functorial and proves the correct fourth-vertex assertion for coherent morphisms. Coherent rectangles admit pasting and cancellation. In a pointed derivator the suspension–loop adjunction exists; stability, rather than additivity, is the condition under which cartesian and cocartesian coherent squares agree, equivalently that adjunction is an equivalence (Groth, 2013, Proposition 3.17; Groth et al., 2014a, Theorem 7.1, arXiv v2). External derived Hom has the expected continuity, with the usual enrichment hypotheses for an internal version (Cisinski, 2003, Proposition 6.13). None of these assertions makes the forgetful functor reflect the property disproved in the postface.

The proposed main example and Proposition 1 are therefore settled negatively, while the coherent replacements are established theorems. The manuscript's broad request for further compatibility axioms does not by itself specify a present open problem. This commentary does not classify all literal quadrangulated categories without binary products or produce a nonordinary example in that generality. Its audit keeps that limit separate from the explicit counterexamples and the proved structural deductions.

Chapter XV

Théorèmes de factorisation (Modèles (2))

p. —

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XV

Théorèmes de factorisation
(Modèles (2))

p. —

Ce texte a été déchiffré et transcrit en LaTeX 2$_\varepsilon$ par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Seules quelques corrections mineures évidentes ont été effectuées. Pour les rares ajouts ou commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Construction fondamentale : la suite transfinie de factorisation d'une flèche f.

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[page 1]

Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie, stable par "petites" (*i.e.* $\mathfrak{U}$-petites) $\varinjlim$ filtrantes. Soit

$$C_0 \subset \mathrm{Fl}(\mathcal{M}) \tag{1.1}$$

un *petit* ensemble de flèches, on se propose de construire pour toute flèche f de $\mathcal{M}$, une factorisation

$$f = p\,i\,, \qquad i \in \widetilde{C_0} = ((C_0)_*)^*\,,\ p \in (C_0)_*\,. \tag{1.2}$$

Pour ceci, on va construire une suite ordinale de foncteurs

$$\left\{\begin{array}{rcl} f & \longmapsto & X_\alpha(f) = X_\alpha(C_0, f) \\ X_\alpha : \underline{\mathrm{Fl}}(\mathcal{M}) & \longrightarrow & \mathcal{M} \end{array}\right. \tag{1.3}$$

et des homomorphismes fonctoriels

$$i_\alpha(f) : \mathrm{source}\, f \longrightarrow X_\alpha(f)\,, \tag{1.4}$$

$$p_\alpha(f) : X_\alpha(f) \longrightarrow \mathrm{but}\, f\,, \tag{1.5}$$

de sorte qu'on aura le diagramme (où $X = \mathrm{source}\, f$, $Y = \mathrm{but}\, f$)

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle i_\alpha(f)}\searrow \quad \nearrow{\scriptstyle p_\alpha(f)} & \\ & X_\alpha(f) & \end{array}\,, \tag{1.6}$$

ce diagramme étant commutatif

$$f = p_\alpha(f) i_\alpha(f)\,. \tag{1.7}$$

On peut dire aussi qu'on a une section du foncteur canonique de "composition", transposé de $\mathbf{\Delta}_1 \longrightarrow \mathbf{\Delta}_2$, donné par $0 \longmapsto 0$, $1 \longmapsto 2$:

$$\left\{\begin{array}{l} \underline{\mathrm{Nerf}}_2(\mathcal{M}) \underset{\Phi_\alpha}{\overset{\gamma}{\rightleftarrows}} \underline{\mathrm{Nerf}}_1(\mathcal{M}) = \underline{\mathrm{Fl}}(\mathcal{M})\,, \\ \\ \end{array}\right. \tag{1.8}$$

Φ comme "factorisation", où

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[page 2]

$$\Phi_\alpha(f) = \{\, X \xrightarrow{i_\alpha(f)} X_\alpha(f) \xrightarrow{p_\alpha(f)} Y \,\} \tag{1.9}$$

avec $p_\alpha(f) i_\alpha(f) = f$. La fonctorialité de 1.4 et 1.5, ou celle de Φ_α, *i.e.* de 1.9, s'exprime par le fait que tout carré commutatif

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ {\scriptstyle u_X}\downarrow & & \downarrow{\scriptstyle u_Y} \\ X' & \xrightarrow{\quad f' \quad} & Y' \end{array} \tag{1.10}$$

dans $\mathcal{M}$, interprété comme flèche $u = (u_X, u_Y)$ de f dans f' dans $\underline{\mathrm{Fl}}(\mathcal{M})$, donne naissance au double carré commutatif

$$\begin{array}{ccccc} X & \xrightarrow{i_\alpha(f)} & X_\alpha(f) & \xrightarrow{p_\alpha(f)} & Y \\ \downarrow{\scriptstyle u_X} & & \downarrow{\scriptstyle X_\alpha(u)} & & \downarrow{\scriptstyle u_Y} \\ X' & \xrightarrow{i_\alpha(f')} & X_\alpha(f') & \xrightarrow{p_\alpha(f')} & Y' \,. \end{array} \tag{1.11}$$

N.B. Quand on voudra expliciter la dépendance de ces constructions par rapport à C_0, on écrira

$$X_\alpha(C_0, f)\,, \qquad i_\alpha(C_0, f)\,, \qquad p_\alpha(C_0, f)\,, \qquad \Phi_\alpha(C_0, f)\,,$$

ou

$$X_\alpha(C_0, -) \text{ ou } X_\alpha^{C_0}\,, \qquad i_\alpha(C_0, -) \text{ ou } i_\alpha^{C_0} \qquad \textit{etc.}\,,$$

au lieu de $X_\alpha(f)$, $i_\alpha(f)$, …

Commentary · Context [ms. 1]

Why this construction belongs in a theory of derivators

Grothendieck needs factorizations which can be used uniformly across a category of coefficients. A construction for isolated arrows would not supply the functorial replacements needed when those coefficients occur in diagrams. The emphasis on an endofunctor of the arrow category, and then on an entire transfinite sequence of such endofunctors, should be read with that purpose in view.

Two strands of the chapter have different origins. Its language of lifting and factorization is that of homotopical algebra; its treatment of size continues the cardinal filtrations and accessibility developed in (Artin et al., 1972, Exposé I, §9). A filtration does more than assign a cardinal label to an object: it specifies presentations, closure under controlled colimits, and the bounds within which represented functors commute with those colimits. Here that apparatus will turn an unbounded repetition of a local operation into a uniform existence theorem.

The commentary on Chapter VI already discusses the two forms of the replacement axiom, by a functorial replacement and by a factorization “à la Quillen”, at manuscript p. 108, and the distinction between replacements at manuscript p. 180. The present pages address the additional existence problem: how to manufacture the replacements from a set of maps. They should therefore be read as preparation for the diagram theory, rather than as another formulation of its axioms. The modern account of the same small object mechanism in (Riehl, 2014, Chapter 12) makes this separation of construction and use especially clear.

Commentary · Exposition [ms. 1]

Factorization by adjoining solutions

Grothendieck wants a construction which can be used uniformly for every arrow and subsequently compared as its generators vary. The object to be constructed is therefore more than an isolated factorization. It is a section of the composition functor

$$\mathcal{M}^{[2]} \longrightarrow \mathcal{M}^{[1]}, \qquad (X \longrightarrow E \longrightarrow Y) \longmapsto (X \longrightarrow Y),$$

where $[n] = \{0 < \cdots < n\}$. This explains the apparently leisurely passage through the arrow category: naturality in a commutative square will eventually supply natural replacements and comparisons between them.

Write $I = C_0$. In modern lifting notation the two desired classes are

$$R = I^{\perp}, \qquad L = {}^{\perp}R.$$

Here $j \perp p$ means that every commutative square with j on the left and p on the right admits a diagonal filler; uniqueness is not required. Thus Grothendieck's $(C_0)_*$ is R, and $\widetilde{C_0}$ is L. The

construction will first produce a much more concrete subclass of L, consisting of maps built by attaching copies of the generators. Retracts enter only when one identifies all of L.

There are two independent tasks. Adjoining a cell for every lifting problem makes all problems present at one stage solvable at the next. Smallness of the domains then ensures that a problem appearing at a sufficiently long limit was already present earlier. Neither task asserts that one application solves every problem in its newly enlarged domain. This distinction is the reason for the transfinite sequence.

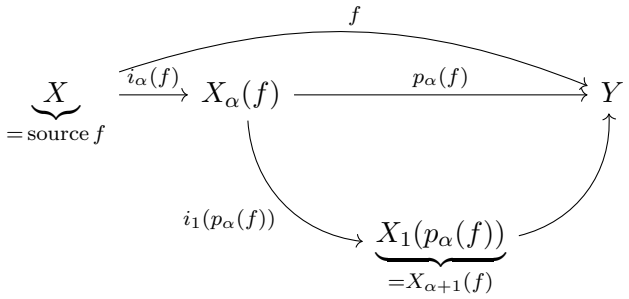
The resulting argument is the classical small object argument, whose modern formulation and retract conclusion are given in Riehl (2014, Theorem 12.2.2 and Corollary 12.2.4). Grothendieck isolates a useful economy in its hypotheses: arbitrary coproducts in $\mathcal{M}$ are unnecessary. Pushouts of the generating arrows, together with filtered colimits, suffice to form all the amalgamated sums under the fixed source. The word "accessible" concerns objects as tested by their covariant representable functors; in the proof below, only factorization of maps from the generating domains through stages is used. The categorical accessibility assumptions later in this chapter are a way of obtaining a uniform bound for that property.

Voici les propriétés essentielles du système inductif ordinal des $X_\alpha(f)$ ou $\Phi_\alpha(f)$.

1. $X_0(f) = X$, $i_0(f) = \mathrm{id}_X$, $p_0(f) = f$ (*i.e.* la factorisation $\Phi_0(f)$ de f est la factorisation triviale $X \xrightarrow{\mathrm{id}_X} X \xrightarrow{f} Y$). [page 3]

2.

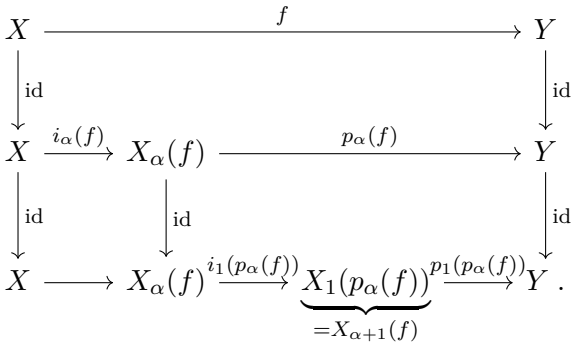
$$\begin{aligned} X_{\alpha+1}(f) &= X_1(p_\alpha(f)) \\ i_{\alpha+1}(f) &= i_1(p_\alpha(f)) \circ i_\alpha(f) \\ p_{\alpha+1}(f) &= p_1(p_\alpha(f)) \end{aligned}$$



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ou mieux

(1.12)



Le morphisme de transition $X_\alpha(f) \longrightarrow X_{\alpha+1}(f)$ est

$$X_\alpha(f) \xrightarrow{\ i_1(p_\alpha(f))\ } X_1(p_\alpha(f)) \;=\; X_{\alpha+1}(f)\,.$$

3. Si α est un ordinal limite, on a

$$\begin{cases} X_\alpha(f) &= \varinjlim_{\alpha'<\alpha} X_{\alpha'}(f). \\ i_\alpha(f) &= \text{morphisme canonique de } X = X_0(f) \text{ dans la limite inductive.} \\ p_\alpha(f) &= \text{limite projective des } p_{\alpha'}(f) : X_{\alpha'}(f) \longrightarrow Y. \end{cases}$$

(On peut considérer la limite inductive définissant $X_\alpha(f)$ comme une limite dans $X\backslash\mathcal{M}/Y\ldots$)

Les morphismes de transition

$$X_{\alpha'}(f) \longrightarrow X_\alpha(f) \qquad (\,\alpha' < \alpha\,)$$

sont les morphismes canoniques relatifs au système inductif des $X_{\alpha'}(f)$.

On peut donc dire que la suite ordinale des factorisations fonctorielles $\Phi_\alpha(f)$ de f variable, [page 4] se déduit de la factorisation fonctorielle $\Phi_1(f)$ de f en

$$X \xrightarrow{\ i_1(f)\ } X_1(f) \xrightarrow{\ p_1(f)\ } Y \qquad (f : X \to Y)$$

par "itération transfinie à droite", *i.e.* en regardant les opérations Φ_α comme des foncteurs

$$\begin{cases} p_\alpha : \underline{\mathrm{Fl}}(\mathcal{M}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{M}) \\ f \longmapsto p_\alpha(f)\,, \end{cases}$$ [1]

munis d'homomorphismes fonctoriels (au-dessus de $\mathcal{M}$)

$$\mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha\,.$$

Ainsi, si on considère

p. 4

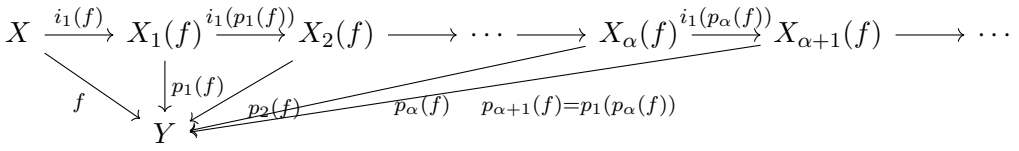
$$\begin{cases} \underline{\mathrm{Fl}}(\mathcal{M}) & \text{comme objet de CAT}/\mathcal{M} \text{ par le foncteur } f \longmapsto \text{but } f \\ \text{où} & \text{CAT est la catégorie des catégories } \textit{éléments de } \mathfrak{V}\text{, univers} \\ & \text{tel que } \mathfrak{U} \in \mathfrak{V}\text{, d'où Cat} \overset{(\text{déf})}{=} \mathrm{Cat}^{\mathfrak{U}},\ \mathcal{M},\ \underline{\mathrm{Fl}}(\mathcal{M}) \in \mathrm{CAT}, \end{cases}$$

1. **N.B.** p_α respecte les structures de catégories *sur* $\mathcal{M}$, via les foncteurs but : $\underline{\mathrm{Fl}}(\mathcal{M}) \longrightarrow \mathcal{M}$, *i.e.* $\underline{\mathrm{Fl}}(\mathcal{M})$ est considérée comme objet de CAT$/\mathcal{M}$.

on a un endofoncteur p_1 de $\underline{\mathrm{Fl}}(\mathcal{M})$ sur $\mathcal{M}$, et un homomorphisme fonctoriel

(1.13) $$i_1 \ : \ \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})/\mathcal{M}} \longrightarrow p_1 \ , \qquad \textit{i.e.} \qquad \begin{array}{ccccc} X & & \xrightarrow{\quad f \quad} & & Y \\ & {\scriptstyle i_1(f)}\searrow & & \nearrow{\scriptstyle p_1(f)} & \\ & & \underbrace{X_1(f)}_{=\,\mathrm{source}\,p_1(f)} & & \end{array} \ ,$$

et la description de la suite ordinale des endofoncteurs p_α de $\underline{\mathrm{Fl}}\,\mathcal{M}/\mathcal{M}$ résulte de cette seule donnée, compte tenu que les petites limites filtrantes existent dans les fibres de $\underline{\mathrm{Fl}}\,\mathcal{M} \xrightarrow{\text{but}} \mathcal{M}$. [page 5]

(1.14)


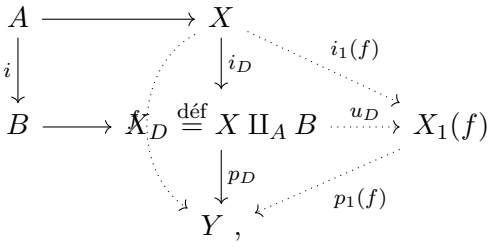
Donc, pour achever de définir la suite ordinale des $X_\alpha(f)$ (munie des $i_\alpha(f)$, $p_\alpha(f)$), ou des $p_\alpha(f)$ (munie des $i_\alpha(f)$), il reste à décrire $X_1(f)$, $i_1(f)$, $p_1(f)$.

La propriété essentielle de ce triple dont nous aurons besoin est la suivante :

(1.15) **Fact**(C_0) (propriété de factorisation par rapport à C_0).
Pour tout diagramme commutatif

$$D : \qquad \begin{array}{ccc} A & \longrightarrow & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \longrightarrow & Y \end{array} \qquad \text{avec } i \in C_0 \ ,$$

d'où un diagramme traits pleins



il existe un homomorphisme de la factorisation $f = p_D i_D$ de f dans la factorisation $f = p_1(f)i_1(f)$. p. 5

Pour formuler cette condition, j'ai utilisé l'existence des sommes amalgamées X_D, *i.e.* la condition sur C_0 : [page 6]

Commentary · Exposition [ms. 5]

A single stage and the colimits it needs

The first stage is arranged to solve every prescribed extension problem without choosing one problem in preference to another. For $f : X \to Y$, its indexing set can be written explicitly as

$$\Delta(I, f) = \coprod_{(j:A\to B)\in I} \{(a, b) \in \mathrm{Hom}(A, X) \times \mathrm{Hom}(B, Y) \mid fa = bj\}.$$

This is a small set: I is small and the category is locally small in the chosen universe. Each triple $D = (j, a, b)$ gives the pushout $X_D = X \amalg_A B$, together with its induced map to Y. In the category

of objects under X, take their coproduct. This gives exactly (1.23), and the inclusion of the Dth cell supplies the map $B \to X_1(f)$ required by (1.17).

The existence argument deserves to be explicit because the stated assumptions do not include all small colimits. A pushout of a pushout of j is again a pushout of j: this is the pasting property of pushout squares. Consequently every map $X \to X_D$ is itself pushoutable along every map out of X. Any finite family of these objects therefore has an amalgamated sum under X, built by repeated binary pushouts. For a finite subset $F \subseteq \Delta(I, f)$, write this sum as X_F, and set $X_\varnothing = X$. Chosen colimit cocones give transition maps for inclusions $F \subseteq G$. The poset of finite subsets is filtered, and

$$X_1(f) = \operatorname*{colim}_{F \subseteq \Delta(I,f),\ F \text{ finite}} X_F.$$

To verify its universal property, a map from this colimit to an object under X is a compatible family of maps from the finite amalgamations. Restriction to singleton subsets identifies such a family with precisely one compatible map from each X_D. This proves the required coproduct property in the undercategory, including the case $\Delta(I, f) = \varnothing$.

If ordinary coproducts also exist, the same object is the pushout

$$\begin{array}{ccc} \coprod_{D \in \Delta(I,f)} A_D & \longrightarrow & X \\ \downarrow & & \downarrow{\scriptstyle i_1(f)} \\ \coprod_{D \in \Delta(I,f)} B_D & \longrightarrow & X_1(f). \end{array}$$

This familiar presentation is convenient for calculations; the preceding finite-subset construction explains why it is not an additional assumption in Theorem 1. The construction of a stage uses the codomains B_D, but the eventual lifting argument will require smallness only of the domains A_D.

(1.16) Les flèches $i \in C_0$ sont coquarrables, *i.e.* pour un diagramme

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \\ B & & \end{array}$$

avec $i \in C_0$, la somme amalgamée existe.

Mais on peut formuler 1.15 aussi ainsi, indépendamment de l'hypothèse (1.16) :

(1.17) Pour tout carré commutatif

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow{v} & Y \end{array} \quad ,$$

avec $i \in C_0$, il existe un carré commutatif traits pleins

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle i_1(f)} \\ B & \xrightarrow{v_1} & X_1(f) \\ & {\scriptstyle v}\searrow & \downarrow{\scriptstyle p_1(f)} \\ & & Y \end{array}$$

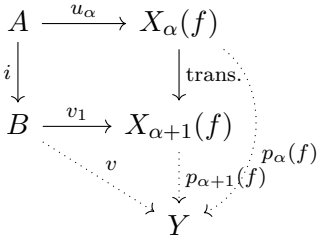
tel que $v = p_1(f) v_1$.

Ceci a pour conséquence le

Corollaire 1 (de 1.17). *Soit $f \in \mathrm{Fl}(\mathcal{M})$, et α un ordinal ($\in \mathfrak{U}$). Pour tout carré commutatif*

$$\begin{array}{ccc} A & \xrightarrow{u_\alpha} & X_\alpha(f) \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle p_\alpha(f)} \\ B & \xrightarrow{v} & Y \end{array}$$

il existe un carré commutatif p. 6



tel que

$$v \;=\; p_{\alpha+1}(f)\; v_1 \;.$$

Corollaire 2. *Soit π un cardinal ($\in \mathfrak{U}$) tel que pour toute flèche $i : A \longrightarrow B$ dans C_0, la source A soit π-accessible. Soit α un ordinal grand devant π* (i.e. *tel que l'ensemble ordonné I_α des ordinaux $\alpha' < \alpha$ soit grand devant π*) *– p. ex. α le premier ordinal dont le cardinal soit $> \pi$. Alors pour toute flèche $f \in \mathrm{Fl}(\mathcal{M})$, on a $p_\alpha(f) \in (C_0)_*$.* [page 7]

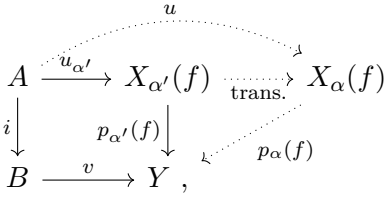
DÉMONSTRATION. Cela signifie que pour tout carré commutatif

$$(*) \qquad \begin{array}{ccc} A & \xrightarrow{u} & X_\alpha(f) \\ {\scriptstyle i}\downarrow & \overset{w}{\nearrow} & \downarrow{\scriptstyle p_\alpha(f)} \\ B & \xrightarrow{v} & Y \end{array},$$

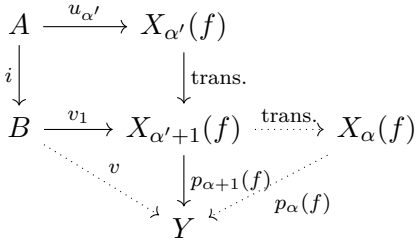
avec $i \in C_0$, la flèche pointillée w (avec $wi = u$, $p_\alpha(f)w = v$) existe. Or

$$X_\alpha(f) = \varinjlim_{\alpha' \in I_\alpha} X_{\alpha'}(f) \;,$$

donc A étant π-accessible et I_α grand devant π, u provient d'une flèche $u_{\alpha'} : A \longrightarrow X_{\alpha'}(f)$, et (quitte à remplacer α' par $\alpha'' > \alpha'$ et $u_{\alpha'}$ par le composé $u_{\alpha''} : A \longrightarrow X_{\alpha'}(f) \longrightarrow X_{\alpha''}(f)$) on peut supposer $p_{\alpha'}(f)u_{\alpha'} = vi$, *i.e.* que $(*)$ provient d'un carré commutatif



et appliquant le corollaire 1, on trouve p. 7



[page 8]

avec $v = p_{\alpha+1}(f)v_1$, et on prendra

$$w \;=\; \mathrm{trans}_{\alpha,\alpha'+1} \circ v_1 \ .$$

Commentary · Exposition [ms. 7]

Why the cofinality bound solves every problem

Grothendieck's Corollary 2 supplies the decisive passage from a succession of partial solutions to an actual right lifting property. The statement needs the additional word *limit*. For a limit ordinal, the condition that every subset of cardinal at most π has an upper bound is equivalent to cofinality greater than π. More invariantly, choose a regular cardinal κ for which every generating domain is κ-presentable, and use a limit ordinal λ with $\mathrm{cf}(\lambda) \geq \kappa$. The ordered set of stages $\beta < \lambda$ is then κ-filtered.

Consider a square $j : A \to B$ against $p_\lambda : X_\lambda \to Y$, with top map $a : A \to X_\lambda$ and bottom map $b : B \to Y$. Since

$$X_\lambda = \operatorname*{colim}_{\beta<\lambda} X_\beta,$$

smallness gives $\beta < \lambda$ and $a_\beta : A \to X_\beta$ representing a. The compatibility of the maps to Y gives

$$p_\beta a_\beta = p_\lambda a = bj.$$

Thus the square already commutes at this stage. The parenthetical suggestion to enlarge β is harmless but unnecessary: the target Y has remained fixed throughout. The corresponding cell at stage $\beta + 1$ provides $h : B \to X_{\beta+1}$ with $hj = t_{\beta+1,\beta}a_\beta$ and $p_{\beta+1}h = b$. Because λ is a limit, $\beta + 1 < \lambda$, and $t_{\lambda,\beta+1}h$ is the desired filler.

No smallness assertion about B was used. In fact the proof needs only the surjectivity of the map from the colimit of the relevant Hom sets, rather than its injectivity. This is why the usual relative-smallness hypothesis, restricted to the cellular chains actually constructed, is sufficient as well. The bound must be common to all generating domains, and the stopping ordinal must be common to all input arrows if one wants a single factorization functor. Since the generators form a set, individual accessibility bounds can be replaced by a common larger one.

The omitted limit condition is a real, although easily repaired, boundary error. In the definition used in Artin et al. (1972, Exposé I, Definition 9.1), a poset with a maximum is "grand devant π" for every π. Thus $I_1 = \{0\}$ satisfies the printed hypothesis. Take sets, generators $\varnothing \to 1$ and the fold $2 \to 1$, and $f = 1_1$. The first generator adds one point; the square for the fold identifies no distinct old points. Hence $X_1(f) = 2$ and $p_1(f) : 2 \to 1$ is not injective. Since the right class for these two generators consists of bijections, Corollary 2 would be false at $\alpha = 1$. Its proof itself shows the repair: choose a limit ordinal with the indicated cofinality, as in the successor-cardinal example given in the text. The same qualification is required in (1.22) and Theorem 1(b).

The two printed labels $p_{\alpha+1}$ at the end of the proof, on printed page 7, omit the prime on α. Their intended value is $p_{\alpha'+1}$. The displayed source of that arrow determines the correction; the edition's readings are retained in the French text and registered in the corrections table.

Mais on veut une factorisation de f, flèche quelconque dans $\mathcal{M}$, en

$$f \;=\; p\ i\ , \qquad \text{avec}\ \ p \in (C_0)_*\ \ \textit{et}\ \ i \in \widetilde{C_0}\ ,$$

donc il faut encore énoncer une propriété, assurant que les $i_\alpha(f)$ sont dans $\widetilde{C_0}$. On aura même quelque chose de plus précis. Introduisons

(1.18) $C_0' \;=\;$ plus petit sous-ensemble de $\mathrm{Fl}(\mathcal{M})$ contenant C_0, stable par cochangement de base (chaque fois qu'il existe).

$C_0'' \;=\;$ ensemble des flèches qui sont soit dans C_0', soit des sommes amalgamées d'une famille *non vide* de telles flèches $A \xrightarrow{i_\alpha} B_\alpha$ de même source A, $i_\alpha \in C_0'$; la flèche $A \longrightarrow B = \coprod_\alpha {}_{(A)}B_\alpha$ est dans C_0''.

(La condition 1.16 implique que les éléments de C_0' sont coquarrables, et cela suffit, avec la condition de stabilité de $\mathcal{M}$ par $\varinjlim$ filtrantes, à assurer que les sommes amalgamées infinies envisagées existent dans $\mathcal{M}$.)

N.B. C_0'' est *égal* au plus petit ensemble de flèches contenant C_0 et stable par cochangement de base et par petites sommes amalgamées. `[Si les petites sommes existent dans` $\mathcal{M}$`, alors` C_0'' `est aussi l'ensemble des flèches obtenues par cochangement de base à partir de sommes de flèches de` C_0`.]`

(1.19) $C_0''' =$ plus petit sous-ensemble de $\mathrm{Fl}(\mathcal{M})$ contenant C_0'', et stable de plus par limites inductives ordinales strictes, *i.e.* pour un système inductif ordinal $(A_\alpha)_{\alpha<\alpha_0}$, tel que pour α ordinal limite on ait $A_\alpha \xleftarrow{\sim} \varinjlim_{\alpha'\in I_\alpha} A_{\alpha'}$, si les morphismes de transition $i_{\beta,\alpha} : A_\alpha \longrightarrow A_\beta$ sont dans C_0''', de même les morphismes canoniques

$$A_\alpha \xrightarrow{i_\alpha} \varinjlim_{\beta\in I_{\alpha_0}} A_\beta \ .^{2}$$

Commentary · Examples [ms. 8] p. 8 [page 9]

Empty families and a complete calculation in sets

Grothendieck has now separated attachment from lifting, and the simplest generator makes both operations visible. Work in sets and let $I = \{\varnothing \to \{*\}\}$. For $f : X \to Y$, a lifting problem is exactly an element of Y. Hence

$$X_1(f) = X \amalg Y, \qquad i_1(f) = \mathrm{in}_X, \qquad p_1(f) = [f, 1_Y].$$

The right class is the class of surjections, because a lift against $\varnothing \to \{*\}$ is a preimage of one specified element. With the usual axiom of choice, the left class is the class of injections. An injection lifts against a surjection by extending the prescribed map over its complement. Conversely, if $g : A \to B$ is not injective, choose distinct a, a' with $g(a) = g(a')$, map A to $\{0, 1\}$ with different values at these points, and use the surjection $\{0, 1\} \to \{*\}$. A filler would factor that map through g, which is impossible.

The whole transfinite construction can be calculated, with the obvious compatible choices of coproducts:

$$X_\alpha(f) = X \amalg (Y \times \alpha), \qquad p_\alpha(x) = f(x), \qquad p_\alpha(y, \beta) = y.$$

At a successor one adds another copy of Y; at a limit one takes the union of the earlier copies. For every $\alpha > 0$, p_α is already a surjection, but if $Y \neq \varnothing$ every transition to the next stage adds elements. Success in the lifting problem therefore does not mean that the unmodified construction has stabilized.

There is also a small literal omission in (1.18)–(1.20). The definition of C_0'' expressly allows only *nonempty* amalgamated families. In this example C_0' consists of the injections which add one element, and C_0'' consists of the injections with nonempty complement. For $f : \varnothing \to \varnothing$, however, $\Delta(I, f)$ is empty and (1.23) gives $i_1(f) = 1_\varnothing$. Thus (1.20), read with the printed definition, is false in this boundary case. The minimal repair is to permit the empty amalgamated family, whose structural map is the identity, or explicitly to adjoin identities to C_0''. Under the printed convention the always valid assertion is $i_1(f) \in C_0'' \cup \mathrm{Iso}(\mathcal{M})$. This is a convention repair in the source, not a transcription correction, and it does not affect the resulting weak factorization theorem. Zero-length cellular compositions must in any event include identities.

La proposition 2, p. 4 de XIII assure que l'on a

$$C_0''' \ \subset \ \widetilde{C_0} \ .$$

Ceci posé, voici la deuxième propriété essentielle de la factorisation $f = p_1(f)i_1(f)$ (la première étant 1.17) :

2. $C_0 \subset C_0' \subset C_0'' \subset C_0'''$

(1.20) Pour toute flèche $f \in \mathrm{Fl}(\mathcal{M})$, on a $i_1(f) \in C_0''$.

Il en résulte aussitôt, par récurrence ordinale :

Commentary · Exposition [ms. 9]

The cellular closure and its successive steps

The finer class C_0''' records how a map was constructed, before retracts erase that information. Its intended modern description is the class of relative I-cell maps: continuous transfinite composites in which every successor map is a pushout of an amalgamated family of copies of generators. Identities are allowed, and all diagrams are small in the chosen universe. The separate stages C_0' and C_0'' distinguish one attachment from simultaneous attachment of a family.

The assertion that these maps lie in ${}^{\perp}(I^{\perp})$ has a direct proof. Fix $q : U \to V$ in $I^{\perp}$. A lifting problem for a pushout of $j \in I$ reduces, by the pushout property, to one for j. For an amalgamated family of such maps, choose a filler for each member with the common prescribed value on the source; the resulting maps agree there and therefore induce a filler on the amalgamation. For a continuous chain $(Z_\beta)_{\beta \le \lambda}$, construct the lift on Z_β by transfinite recursion. At a successor extend the lift already constructed using the lifting property of $Z_\beta \to Z_{\beta+1}$. At a limit the compatible lifts determine a map from the colimit. This proves the lifting property for $Z_0 \to Z_\lambda$. It uses existence of fillers, together with the ordinary choice principle, and no uniqueness.

The word "successor" is essential in this formulation. In the explanatory sentence of (1.19), the displayed symbols $i_{\beta,\alpha}$ can be read as demanding that *every* transition already belong to the class. That condition by itself would not imply closure under finite composition. For example, in the poset category $0 < 1 < 2$, take $I = \{0 \to 1, 1 \to 2\}$. Pushouts and amalgamations give these two arrows and identities, but not $0 \to 2$. A chain all of whose transitions lie in this class cannot contain both 0 and 2; its colimit introduces no missing arrow. Nevertheless two successive cell attachments factor $0 \to 2$ through 1. Thus the induction in (1.21) requires the usual successorwise interpretation of "limites inductives ordinales strictes", or an explicit additional closure under composition. We adopt that intended interpretation throughout; the claim about the lifting class is valid independently of this notational compression.

(1.21) Pour toute flèche $f \in \mathrm{Fl}(\mathcal{M})$ et tout ordinal $\alpha \in \mathfrak{U}$, on a $i_\alpha(f) \in C_0'''$, a fortiori $i_\alpha(f) \in \widetilde{C_0}$.

Donc si α est grand devant π (*cf.* cor. 2, page 7), la factorisation $f = p_\alpha(f) i_\alpha(f)$ satisfait

(1.22) $\left\{ \begin{array}{l} f = p_\alpha(f) i_\alpha(f), \text{ avec } i_\alpha(f) \in C_0''' \subset \widetilde{C_0} = \Phi \texttt{ [à ne pas} \\ \texttt{confondre avec } \Phi_\alpha\texttt{]}, \ p_\alpha(f) \in C_{0*} = \Psi, \text{ lorsque } \alpha \text{ est} \\ \text{grand devant } \pi, \text{ les sources des } i \in C_0 \ \pi\text{-accessibles.} \end{array} \right.$

Il reste à construire le foncteur $f \longmapsto X_1(f)$, avec $i_1(f)$, $p_1(f)$, satisfaisant les conditions (1.17) et (1.20), sous l'hypothèse (1.16) que les $i \in C_0$ sont coquarrables. On considère alors, pour $f \in \mathrm{Fl}(\mathcal{M})$ fixé, l'ensemble $\Delta(f)$ (ou $\Delta(C_0, f)$) de *tous* les diagrammes D de la forme (1.15), avec $i \in C_0$, et on pose

(1.23) $$X_1(f) \ = \ \coprod_{D \in \Delta(f)} {}_{(X)}X_D$$

(somme amalgamée sous X, *i.e.* somme indexée par $\Delta(f)$ dans $X \backslash \mathcal{M}$), le morphisme $X \xrightarrow{i_1(f)}$ [page 10] $X_1(f)$ étant le morphisme costructural, et le morphisme $X_1(f) \xrightarrow{p_1(f)} Y$ étant celui ayant pour composantes les p_D. En somme, on fait "au plus économique" (et au plus naturel) pour assurer la validité de (1.17), sous la forme équivalente (1.15). D'autre part, par définition de C_0', C_0'', les i_D sont dans C_0', et $i_1(f)$ est dans C_0'', ce qui est (1.20) [3].

3. Il faudrait quand même expliciter la dépendance fonctorielle de $X_1(f)$ par rapport à f !!!

Commentary · Exposition [ms. 10]

The omitted functoriality in the input arrow

The footnote asks for the functorial dependence which makes Theorem 1 usable. Its proof is short once the indexing squares are retained as data. A morphism $(u, v) : f \to f'$ in the arrow category consists of $u : X \to X'$ and $v : Y \to Y'$ with $vf = f'u$. It sends

$$(j, a, b) \longmapsto (j, ua, vb)$$

to a well-defined member of $\Delta(I, f')$. For this image D', the pair of maps $X \xrightarrow{u} X'$ and 1_B induces a map

$$X_D = X \amalg_A B \longrightarrow X' \amalg_A B = X'_{D'}.$$

These maps respect the structural maps from X and the maps to the target. Composing them with the inclusions into $X_1(f')$ gives a cocone over the family defining $X_1(f)$, hence a unique map $X_1(u, v) : X_1(f) \to X_1(f')$.

Identity and composition laws follow by restricting to X and to each copy of B: there the asserted equalities are the identity and composition laws for u and v, and the universal property makes their extensions unique. The same argument proves the naturality of i_1 and p_1. At successor ordinals one applies this functor to the already constructed square between $p_\alpha(f)$ and $p_\alpha(f')$; at limits one takes the map of colimits. Transfinite induction gives all of (1.10)–(1.14), including compatibility with every transition map.

One must fix the relevant colimit cocones to obtain literal functors. Another choice gives canonically isomorphic functors by the same universal properties. There is no need to choose fillers in arbitrary lifting problems: the attached cell itself supplies the particular solution required at each stage. This distinction explains why the factorization is functorial even when the fillers subsequently used to characterize the left or right class are highly nonunique.

Pour résumer :

Théorème 1. *Soient $\mathcal{M}$ une $\mathfrak{U}$-catégorie stable par $\mathfrak{U}$-limites inductives filtrantes, $C_0 \subset \mathrm{Fl}(\mathcal{M})$ un petit ensemble de flèches de $\mathcal{M}$. On suppose les flèches $i \in C_0$ coquarrables. Cela assure la définition d'un endofoncteur $f \longmapsto p_1(f)$ de $\underline{\mathrm{Fl}}(\mathcal{M})$, et d'un homomorphisme fonctoriel $i_1 : \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_1$,* i.e. *$i_1(f) : f \longrightarrow p_1(f)$, induisant l'identité sur les objets buts de f et de $p_1(f)$,*

$$\begin{array}{ccc} X & \xrightarrow{i_1(f)} & X_1(f) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle p_1(f)} & \\ & Y\,, & \end{array}$$

d'où par itération un système inductif ordinal d'endofoncteurs $f \longmapsto p_\alpha(f)$ de $\underline{\mathrm{Fl}}(\mathcal{M})$, et p. 9
d'homomorphismes $i_\alpha : \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha$, i.e.

$$\begin{array}{ccc} X & \xrightarrow{i_\alpha(f)} & X_\alpha(f) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle p_\alpha(f)} & \\ & Y\,, & \end{array}$$

formant un système inductif ordinal de factorisations de f fonctorielles en f.

Ceci posé, on a : [page 11]

a) *Pour tout ordinal α, on a $i_\alpha(f) \in \Phi$.*

b) *Supposons que les sources des $i \in C_0$ soient accessibles et soit alors π un cardinal dans $\mathfrak{U}$ tel qu'elles soient π-accessibles. Alors pour tout nombre ordinal α grand devant π, on a $p_\alpha(f) \in C_{0*}$, donc*

$$f \;=\; p_\alpha(f)\, i_\alpha(f) \qquad \textit{avec} \qquad i_\alpha(f) \in \Phi\,, \qquad p_\alpha(f) \in \Psi\,.$$

On a un résultat légèrement plus précis, que voici :

Corollaire 1. *Soit* Φ_0 *une partie de* $\mathrm{Fl}(\mathcal{M})$ *telle que*

$$C_0''' \subset \Phi_0 \subset \Phi \qquad (d'où\ C_0 \subset \Phi_0 \subset \Phi)\,, \tag{1.24}$$

où $C_0''' \subset \Phi$ *est définie dans* 1.19, *p.* 8. *Par exemple, si* Φ_0 *est une partie de* $\mathrm{Fl}(\mathcal{M})$ *satisfaisant*

$$C_0 \subset \Phi_0 \subset \Phi\,,$$

et stable par cochangement de base, par petites sommes amalgamées (*cf.* (1.18), p. 8), *par* $\varinjlim$ *ordinales strictes* (*cf.* (1.19), p. 8), *elle satisfait* (1.24). (**N.B.** *Si* $\mathcal{M}$ *admet des sommes amalgamées simples, de tels* Φ_0 *existent trivialement, il suffit de prendre* $\Phi_0 = \Phi$.)

Ceci posé, pour toute flèche f *dans* $\mathcal{M}$, *et tout ordinal* α, *on a* [page 12]

$$i_\alpha(f) \in \Phi_0\,.$$

Corollaire 2. *Supposons que les sources des flèches* $i \in C_0$ *soient accessibles, et soit* Φ_0 *comme dans le corollaire* 1 (p. ex. $\Phi_0 = C_0'''$). *Alors le couple* (Φ_0, Ψ) *est un précouple de Quillen,* i.e.

a) $\Phi_0 \longleftrightarrow \Psi$, i.e. $\Psi \subset \Phi_{0*}$ (*ou ce qui revient au même,* $\Phi_0 \subset \Psi^*$), *et*

b) *on a la propriété de factorisation de Quillen.*

Cela implique donc que Φ $(= \widetilde{C_0} = \widetilde{\Phi_0})$ *est formé des flèches qui sont facteurs directs de flèches de* Φ_0. *Donc si* Φ_0 *est stable par facteurs directs, on a* $\Phi_0 = \Phi$, i.e. Φ_0 *est clos à gauche.*

Commentary · Exposition [ms. 12]

Why the retract conclusion is exact

The corollary identifies the amount of closure required after the cellular construction. Put $R = I^{\perp}$ and $L = {}^{\perp}R$. If $I \subseteq \Phi_0 \subseteq L$, then $\Phi_0^{\perp} = R$: the inclusion $I \subseteq \Phi_0$ gives one containment, and $\Phi_0 \subseteq {}^{\perp}R$ gives the other. Assume, as in the corollary, that every map factors as a map in Φ_0 followed by a map in R.

Take $g : A \to B$ in L and choose a factorization

$$A \xrightarrow{j} E \xrightarrow{q} B, \qquad j \in \Phi_0, \quad q \in R, \quad qj = g.$$

The square with left map g, top map j, right map q and bottom map 1_B admits a filler $s : B \to E$. It satisfies $sg = j$ and $qs = 1_B$. In the arrow category the maps

$$g \xrightarrow{(1_A, s)} j \xrightarrow{(1_A, q)} g$$

compose to the identity. Thus g is a retract of a member of Φ_0, with the source held fixed. Conversely a retract of a map with the left lifting property has that property: insert the retract maps into a lifting square, solve the resulting square, and compose back. Therefore L is exactly the retract closure of Φ_0.

The cellular class and its retract closure consequently have different jobs. Cellular maps are the output of the explicit construction; the retract closure is the class characterized purely by lifting. In particular, if Φ_0 is retract closed then it is already L. The stronger description using C_0''' follows from the transfinite attachment construction, with identities included as explained above. This is the content, rather than an additional hypothesis, of the assertion that (Φ_0, R) is a Quillen precouple. Section 4 will isolate the same retract mechanism without any reference to the particular generating set.

Commentary · Later developments [ms. 12]

From a pointed endofunctor to algebraic factorization

Grothendieck's emphasis on $1 \to p_1$ makes the construction especially close to the later algebraic treatment. It should not, however, be read as already providing a monad: a pointed endofunctor comes with a unit but has no specified multiplication or associativity law. The set example above exhibits the difference concretely, since the ordinary iteration keeps adding solutions to problems which have already been solved.

Garner (2009, Theorem 4.4) constructs an algebraically free natural weak factorization system on a small category of generating arrows under specified cocompleteness and smallness hypotheses, in particular for a locally presentable category. Its right factor carries a monad and its left factor a comonad. For a discrete generating category, the algebras on a right map encode chosen solutions of the generating lifting problems; morphisms preserve those choices. The added algebraic structure gives a universal property to the construction. It is stronger than the existence of the functorial factorization obtained here.

This places the chapter's two concerns in their subsequent setting. The transfinite argument proves that the required lifting classes form a weak factorization system, while the algebraic refinement controls the choices of solutions and their compatibility. A functorial section of composition, an algebraic weak factorization system, and an orthogonal factorization system with unique fillers are three distinct structures. Nothing in Theorem 1 requires unique fillers.

p. 10

2 Variation des factorisations $f = p_\alpha i_\alpha$ avec la partie C_0 génératrice

Je dis que le système inductif ordinal de foncteurs $p_\alpha = p_\alpha^{C_0} : \underline{\mathrm{Fl}}(\mathcal{M}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{M})$, muni [page 13]
des flèches

$$i_\alpha^{C_0} \ : \ \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha^{C_0}\,,$$

dépend fonctoriellement de C_0, quand on considère C_0 comme un élément de l'ensemble ordonné des *petites* parties de $\mathrm{Fl}(\mathcal{M})$, formées de flèches coquarrables (donc de l'ensemble ordonné

$$\mathfrak{P}_{\mathfrak{U}\text{-petites}}(\mathrm{Fl}_{\mathrm{coquar.}}(\mathcal{M}))\quad).$$

Si on a une inclusion

$$C_0 \ \hookrightarrow \ C_1\,, \tag{2.1}$$

on en déduit pour tout ordinal α un homomorphisme fonctoriel

$$u_\alpha \ = \ u_\alpha^{C_1,C_0} \ : \ X_\alpha^{C_0} \ \longrightarrow \ X_\alpha^{C_1}\,, \tag{2.2}$$

rendant commutatifs les carrés déduits de $f : X \longrightarrow Y$ dans $\mathcal{M}$,

$$\begin{array}{ccccc} X & \xrightarrow{i_\alpha^{C_0}(f)} & X_\alpha^{C_0}(f) & \xrightarrow{p_\alpha^{C_0}(f)} & Y \\ {\scriptstyle \mathrm{id}}\downarrow & & \downarrow{\scriptstyle u_\alpha(f)} & & \downarrow{\scriptstyle \mathrm{id}} \\ X & \xrightarrow{i_\alpha^{C_1}(f)} & X_\alpha^{C_1}(f) & \xrightarrow{p_\alpha^{C_1}(f)} & Y\,, \end{array} \tag{2.3}$$

et compatible avec les morphismes de transition pour α variable,

$$\begin{array}{ccc} X_\alpha^{C_0} & \xrightarrow{\mathrm{trans}_{\beta,\alpha}^{C_0}} & X_\beta^{C_0} \\ {\scriptstyle u_\alpha}\downarrow & & \downarrow{\scriptstyle u_\beta} \\ X_\alpha^{C_1} & \xrightarrow{\mathrm{trans}_{\beta,\alpha}^{C_1}} & X_\beta^{C_1}\,. \end{array} \tag{2.4}$$

[page 14]

Pour définir les u_α, satisfaisant aux conditions précédentes, il suffit de les définir pour $\alpha = 1$ "et d'itérer" :

(2.5)
$$\begin{array}{ccccc} & & X_1^{C_0}(f) & & \\ & {\scriptstyle i_1^{C_0}(f)}\nearrow & \downarrow & \searrow{\scriptstyle p_1^{C_0}(f)} & \\ X & & {\scriptstyle u_1(f)} & & Y \\ & {\scriptstyle i_1^{C_1}(f)}\searrow & \downarrow & \nearrow{\scriptstyle p_1^{C_1}(f)} & \\ & & X_1^{C_1}(f) & & \end{array} .$$

Or $X_1^{C_0}(f)$ est défini comme une somme directe dans $X\backslash\mathcal{M}/Y$, relative à un ensemble d'indices $\Delta(C_0, f)$ *contenu* dans $\Delta(C_1, f)$ – et la famille d'objets dont on prend la somme directe étant induite d'une famille (non petite) dont l'ensemble d'indices serait $\Delta(\mathrm{Fl}_{\mathrm{coquar.}}(\mathcal{M}), f)$, savoir formé de *tous* les diagrammes carrés p. 11

$$D : \quad \begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow{v} & Y \end{array} \qquad \text{avec } i \text{ coquarrable sans plus ,}$$

la définition des X_D, i_D, p_D associés à D ne dépendant que de D, et non de l'ensemble C_0 ou C_1 de flèches coquarrables qui contient i. Ceci dit, la flèche $u_1(f)$ dans (2.5) n'est autre que l'inclusion canonique d'une somme amalgamée partielle, indexée par $\Delta(C_0, f)$, dans la somme amalgamée correspondante à l'ensemble d'indices plus grand. La fonctorialité de cette flèche $u_1(f)$, pour f variable, "est immédiate", elle résulte de la commutativité du carré, associé à un morphisme $v = (v_X, v_Y)$ de flèches [page 15]

(2.6)
$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ {\scriptstyle v_X}\downarrow & & \downarrow{\scriptstyle v_Y} \\ X' & \xrightarrow{f'} & Y' , \end{array}$$

savoir le diagramme d'applications entre ensembles d'indices

(2.7)
$$\begin{array}{ccc} \Delta(C_0, f) & \xrightarrow{u_1(f)} & \Delta(C_1, f) \\ {\scriptstyle \text{dépendance fonctorielle en } f \text{ de } \Delta(C_0,f)}\downarrow & & \downarrow{\scriptstyle \text{dépendance fonctorielle en } f \text{ de } \Delta(C_1,f)} \\ \Delta(C_0, f') & \xrightarrow{u_1(f')} & \Delta(C_1, f') \end{array}$$

(commutativité qui revient à dire que $\Delta(C_0, f)$ peut être considéré comme un foncteur en le couple (C_0, f) comme objet de la catégorie produit $\mathfrak{P}_{\mathrm{pet.}}(\mathrm{Fl}_{\mathrm{coquar.}}(\mathcal{M})) \times \underline{\mathrm{Fl}}(\mathcal{M})$ …). Le carré commutatif (2.7) est la partie "indices" d'un carré commutatif entre familles d'objets $\mathscr{X}(C_0, f)$, $\mathscr{X}(C_0, f')$, $\mathscr{X}(C_1, f)$, $\mathscr{X}(C_1, f')$, indexés par ces ensembles, savoir les $X_D^{C_0}$ et les $X_D^{C_1}$ relatifs aux diagrammes D impliquant, soit la flèche f, soit la flèche f' … [page 16]

Commentary · Exposition [ms. 15]

Functoriality in the generators, including limit stages

Grothendieck next keeps track of what happens when more lifting problems are declared relevant. The one-stage comparison is induced by the inclusion of indexing sets, but at later stages the input to the attachment functor has also changed. Writing the recursive formula makes the joint naturality precise.

Let $I \subseteq J$, put $P_I = p_1^I$ and $P_J = p_1^J$, and let $\tau : P_I \to P_J$ be the natural transformation constructed from the inclusion of squares. Both endofunctors lie over the codomain functor, and

their units satisfy $\tau\eta_I = \eta_J$. Let $u_\alpha : P_I^\alpha \to P_J^\alpha$ denote the comparison at stage α, where the superscript means the continuous iteration used in the text, not ordinary iteration at a limit. Then

$$u_0 = 1, \qquad u_{\alpha+1} = P_J(u_\alpha)\,\tau_{P_I^\alpha}, \qquad u_\lambda = \operatorname*{colim}_{\alpha<\lambda} u_\alpha \quad (\lambda \text{ a limit}).$$

The successor formula starts with $P_I(P_I^\alpha f)$, changes the outer attachment functor to P_J, and then changes its input to $P_J^\alpha f$. Naturality of τ allows these two operations in the opposite order as well.

Here is the compatibility with a successor transition, which is the point needed to make the limit formula meaningful. Naturality of η_J and the equation $\tau\eta_I = \eta_J$ give

$$\begin{aligned} u_{\alpha+1}\eta_{I,P_I^\alpha} &= P_J(u_\alpha)\eta_{J,P_I^\alpha} \\ &= \eta_{J,P_J^\alpha} u_\alpha. \end{aligned}$$

Compatibility with all transitions follows by composition and the defining colimit property at limits. Each u_α is natural in f by the same induction. For $I \subseteq J \subseteq K$, the equation $u_\alpha^{K,I} = u_\alpha^{K,J} u_\alpha^{J,I}$ holds at stage one because both maps restrict to the same inclusions of cells. At successors it follows by naturality of the one-stage transformations, and at limits by equality on every colimit component. Identity inclusions are handled identically. This proves the functoriality on the product of the poset of generating sets with the arrow category that the final paragraph of Section 2 leaves unexpanded.

The term "inclusion" describes the indexing sets. It does not by itself assert that the resulting morphism of objects is a monomorphism in an arbitrary category. The following elementary refinement explains when the same word can also be used for the actual stages.

Or la construction canonique du système inductif ordinal p_α, avec les $i_\alpha : \mathrm{id}_{\underline{\mathrm{Fl}}(\mathcal{M})} \longrightarrow p_\alpha$, est fonctorielle par rapport à la donnée initiale p_1, i_1 – je n'explicite pas le détail de la fonctorialité, bien que je devrais …

Commentary · Development (commentators') [ms. 16]

Comparisons which are monomorphisms

The parameter dependence in Section 2 is useful when a family of generators is enlarged gradually. For presheaves, monomorphic generators give a stronger conclusion: an enlargement can be seen stage by stage inside the resulting objects. We record the precise two-variable statement, with its proof, for this construction. We checked the small object construction in Riehl (2014, Section 12.2) and the algebraic comparison in Garner (2009, Sections 3–4); we did not find this particular formulation there. The statement below is an elementary elaboration of the displayed construction, with no claim of priority.

Proposition (monomorphic comparison). [ours] Let $\mathcal{E} = \mathbf{Set}^{\mathcal{A}^{\mathrm{op}}}$ for a small category $\mathcal{A}$. Let $I \subseteq J$ be small sets of monomorphisms in $\mathcal{E}$. Suppose $(u, v) : f \to f'$ is a commutative square whose source map $u : X \to X'$ and target map $v : Y \to Y'$ are both monomorphisms. For every ordinal α for which the construction is performed, the jointly natural comparison

$$X_\alpha^I(f) \longrightarrow X_\alpha^J(f')$$

is a monomorphism. In particular, at a fixed input f, enlargement of a set of monomorphic generators gives monomorphisms at every stage.

Proof. First consider one stage. The map

$$\Delta(I, f) \longrightarrow \Delta(J, f'), \qquad (j, a, b) \longmapsto (j, ua, vb)$$

is injective. Indeed equal images have the same generator j, and cancellation of the monomorphisms u and v gives equality of both maps a and b. Evaluate the pushout formula at an object $c \in \mathcal{A}$. Evaluation preserves colimits, and every $j : A_j \to B_j$ is pointwise injective. Thus, as sets,

$$X_1^I(f)(c) \cong X(c) \amalg \coprod_{(j,a,b)\in\Delta(I,f)} \big(B_j(c) \setminus j(A_j(c))\big).$$

This decomposition is only used at the level of sets; the complements need not form subpresheaves. On the first summand the comparison is $u(c)$, which is injective. On each remaining summand it is

the identity on the same complement, placed in the summand with image index. Injectivity of the indexing map shows that distinct summands remain distinct. Hence the one-stage comparison is pointwise injective.

At stage zero the assertion is the hypothesis on u. If it holds at stage α, apply the one-stage result to the square between $p^I_\alpha(f)$ and $p^J_\alpha(f')$; its source map is monic by induction and its target map is still v. This proves the successor step. At a limit λ, evaluate at c and use the two compatible diagrams of sets. If two representatives in the first colimit have equal images in the second, they become equal at a common later stage of the second diagram. The injectivity of the comparison at that stage makes their images in the first diagram equal there as well. They therefore represent the same element of the first colimit. This proves the limit step and the proposition. □

The assumption on the target map carries real information. With $I = J = \{\varnothing \to \{*\}\}$, take $f : \varnothing \to \{0,1\}$ and $f' : \varnothing \to \{*\}$, the identity on the empty source and the constant map on targets. The induced map between first stages is $\{0,1\} \to \{*\}$, so it is not monic. The construction distinguishes lifting problems by their target component as well as their boundary map.

3 Majoration des cardinaux

Je suppose à présent que $\mathcal{M}$ est muni d'une *filtration cardinale*

$$(\mathrm{Filt}^\pi)_{\pi \geq \pi_0} \qquad (\pi_0,\ \pi \text{ cardinaux dans } \mathfrak{U}) \tag{3.1}$$

(SGA 4 I 9.12, p. 149), les Filt^π stables par sommes amalgamées simples (chaque fois qu'elles existent), donc aussi par sommes amalgamées infinies indexées par des ensembles d'indices de cardinal $\leq \pi$. On suppose de plus que tout objet de Filt^{π_0} soit accessible, et on désigne par π_1 le plus petit cardinal tel que $\pi_1 \geq \pi_0$ et que les objets de Filt^{π_0} soient π_1-accessibles. Désignant, pour tout cardinal, par $\mathcal{M}_\pi$ le sous-ensemble de $\mathrm{Ob}\,\mathcal{M}$ formé des objets π-accessibles, on aura alors pour tout $\pi \geq \pi_1$ (*loc. cit.* 9.16)

p. 12

$$\mathcal{M}_\pi \ \subset\ \mathrm{Filt}^\pi \ \subset\ \mathcal{M}_{(\pi^{\pi_0})} \qquad (\text{si } \pi \geq \pi_1)\,, \tag{3.2}$$

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Commentary · Exposition [ms. 16]

A filtration measures the cost of a construction

Grothendieck now asks for more than the existence of a factorization. He wants to perform the construction inside a bounded part of the category, uniformly for every arrow in that part. This is what will eventually let a small amount of data control a model structure. The three quantities to keep separate are the size of the domains and codomains of the generators, the number of generators actually used, and the length of the construction. Isomorphic generators have the same lifting property, but the unmodified construction attaches a cell for each of them.

The terminology comes from Artin et al. (1972, Exposé I, Definitions 9.1–9.3 and 9.12). A partially ordered set is *large relative to* an infinite cardinal π if each of its subsets of cardinality at most π has an upper bound. Thus the modern directedness index is π^+. An object called π-accessible here has a representable functor preserving such colimits; in the usual setting this is π^+-presentability, not presentability at the cardinal π. The filtration is additional structure: its levels are essentially small, are increasing, admit bounded presentations of objects at larger levels, and have specified closure properties. Membership in a filtration level should therefore not be replaced by an informal assertion that an object has at most π elements.

For the calculation ahead we need the following concrete consequences. Put $\mathcal{M}^\pi = \mathrm{Filt}^\pi$. If $A \in \mathcal{M}^r$, it has a filtered presentation by objects of $\mathcal{M}^{\pi_0}$ with at most r^{π_0} indices. If $X \in \mathcal{M}^s$ and $s \geq \pi_1$, it has a presentation by objects of $\mathcal{M}^{\pi_1}$ over a π_1^+-directed poset with at most s^{π_1} indices. These are the presentations used on ms. 20. The stronger closure condition (3.13), which is explicitly assumed below, includes *all* filtered diagrams of cardinality at most the level, not only those large relative to π_0.

This stronger closure also supplies the retract argument in (3.2). If X is a retract of $Y \in \mathcal{M}^\pi$, with idempotent $e : Y \to Y$, the countable chain

$$Y \xrightarrow{e} Y \xrightarrow{e} Y \xrightarrow{e} \cdots$$

has colimit X, so $X \in \mathcal{M}^\pi$. If X is π-accessible, factor its identity through a stage of its π^+-directed presentation by $\mathcal{M}^\pi$-objects. This makes it such a retract. In the other direction, a presentation with at most π^{π_0} arrows expresses $\mathrm{Hom}(X, -)$ as a limit of that many representables. Directed colimits of sets with index larger than this bound commute with those limits. This proves the inclusions in (3.2) under the closure hypotheses used here and explains why their equality at suitable cardinals is useful.

donc si $\pi^{\pi_0} = \pi$, p. ex. si $\pi = 2^c$ avec $c \geq \pi_0$, on aura

$$\mathrm{Filt}^\pi \;=\; \mathcal{M}_\pi\,. \tag{3.2 bis}$$

(Si on admet l'hypothèse du continu généralisée, on a $\pi^{\pi_0} = \pi$ pour tout $\pi > \pi_0$, donc $\mathrm{Filt}^\pi = \mathcal{M}_\pi$ si $\pi \geq \pi_1$ et $\pi > \pi_0$.)

Je pose, pour tout cardinal $\pi \geq \pi_0$,

$$\mathrm{Fl}^\pi(\mathcal{M}) \;=\; \{\; f \in \mathrm{Fl}(\mathcal{M}) \mid \text{source et but de } f \text{ sont dans } \mathrm{Filt}^\pi \;\} \tag{3.3}$$

– ce qui sans doute donne une filtration cardinale sur $\underline{\mathrm{Fl}}(\mathcal{M})$ (mais ce n'est pas là notre souci …). Mon but à présent, c'est de dégager des conditions sur $C_0 \subset \mathrm{Fl}_{\mathrm{coquar.}}(\mathcal{M})$ (petite partie de $\mathrm{Fl}(\mathcal{M})$) et sur le cardinal π, et l'ordinal α, pour que l'on ait

$$\begin{aligned} f \in \mathrm{Fl}^\pi(\mathcal{M}) \;&\overset{?}{\Longrightarrow}\; X_\alpha(f) \in \mathrm{Filt}^\pi \\ &\quad (\textit{i.e. } p_\alpha(f) \in \mathrm{Fl}^\pi(\mathcal{M}),\ \text{ou encore } i_\alpha(f) \in \mathrm{Fl}^\pi(\mathcal{M}))\,. \end{aligned} \tag{3.4}$$

Plus généralement, je me propose de donner une estimation de la "cardinalité" de $X_\alpha(f)$, pour C_0 fixé, en fonction de C_0, de f, de α, plus exactement, en fonction des trois cardinaux $\pi(C_0)$, $\pi(f)$, $\pi(\alpha)$ définis comme

$$\left\{\begin{array}{rcl} \pi(C_0) &=& \text{plus petit cardinal tel que } C_0 \subset \mathrm{Fl}^\pi(\mathcal{M}) \\ \pi(f) &=& \pi(\{f\}) \;=\; \text{plus petit cardinal tel que } f \in \mathrm{Fl}^\pi(\mathcal{M}) \\ \pi(\alpha) &=& \mathrm{card}\,\alpha\,. \end{array}\right. \tag{3.5}$$

Donc je cherche une fonction cardinale de variables cardinales φ, telle que l'on ait

$$X_\alpha(f) \;\in\; \mathrm{Filt}^\pi \qquad \text{si} \qquad \pi \;=\; \varphi(\underbrace{\pi(C_0)}_{r}, \underbrace{\pi(f)}_{s}, \underbrace{\pi(\alpha)}_{t})\,. \tag{3.6}$$

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L'estimation cruciale concernera sûrement le cas $\alpha = 1$. Notons que $X_1^{C_0}(f) = X_1(f)$ est défini comme une somme amalgamée (en général infinie) d'objets X_D, pour un ensemble $\Delta(C_0, f) = \Delta(f)$ de diagrammes, savoir les diagrammes commutatifs

$$\begin{array}{ccc} A & \longrightarrow & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \longrightarrow & Y \end{array}$$

avec $i \in C_0$. Pour un i fixé, et un f fixé, le cardinal de l'ensemble des diagrammes de ce type est majoré par celui de $\mathrm{Hom}(A, X) \times \mathrm{Hom}(B, Y)$. On a $A, B \in \mathrm{Filt}^r$, $X, Y \in \mathrm{Filt}^s$, donc il nous faut une majoration de

$$\mathrm{card}\,\mathrm{Hom}_{\mathcal{M}}(A, X)\,, \qquad A \;\in\; \mathrm{Filt}^r\,, \qquad X \;\in\; \mathrm{Filt}^s\,.$$

p. 13

Commentary · Exposition [ms. 17]

The cofinality qualification in the cardinal arithmetic

Grothendieck looks for cardinals at which the estimate stops growing. The unconditional choice $\pi = 2^c$, $c \geq \pi_0$, works because

$$(2^c)^{\pi_0} = 2^{c\pi_0} = 2^c.$$

The parenthetical appeal to the generalized continuum hypothesis has a different status. Even under GCH, $\pi > \pi_0$ does not imply $\pi^{\pi_0} = \pi$. For example, $\pi = \aleph_\omega$ and $\pi_0 = \aleph_0$ give $\pi^{\pi_0} > \pi$. More generally, König's theorem gives $\pi^{\operatorname{cf}\pi} > \pi$ for every infinite π. A direct proof of the case just used is instructive: if $(f_\xi)_{\xi<\aleph_\omega}$ enumerated all maps $\mathbb{N} \to \aleph_\omega$, choose $g(n)$ outside the set $\{f_\xi(n) : \xi < \aleph_n\}$. For every ξ some n satisfies $\xi < \aleph_n$, and then $g(n) \neq f_\xi(n)$, a contradiction.

Under GCH the additional condition $\pi_0 < \operatorname{cf}\pi$ is sufficient. The general exponentiation rule and its cofinality qualification are recorded in Marks (2026, Theorems 10.12 and 11.7, Exercise 11.8). Nothing in the factorization theorem requires GCH: the fixed exponent identity above is all that is needed. The same correction applies to the repeated GCH assertion on ms. 19.

The sentence about the arrow category immediately before (3.4) should also be read with its size threshold in view. Pointwise filtered colimits and essential smallness of the bounded arrow categories are immediate. Uniformly bounded presentations of arrows require control of the Hom sets between small objects and simultaneous realization of the arrow in two presentations. At a sufficiently enlarged initial level this is the usual accessibility theorem for a finite diagram category. The text does not supply that extra argument at the original π_0, and does not use the assertion in its proof.

[page 19]

Voici les majorations essentielles dans ce sens :

Proposition 3.1. *Soit π_0 le cardinal initial de la filtration cardinale $(\mathrm{Filt}^\pi)_{\pi\geq\pi_0}$ envisagé, π_1 le plus petit cardinal $\geq \pi_0$ tel que les objets de Filt^{π_0} soient π_1-accessibles, $C_1 = \operatorname{card} \underbrace{\widetilde{\underline{\mathrm{Fl}}^{\pi_1}(\mathcal{M})}}_{=\underline{\mathrm{Fl}}(\mathcal{M}^{\pi_1})}$[4]. Pour tout cardinal $\pi \geq \pi_0$, je désigne par $\mathcal{M}^\pi$ la sous-catégorie strictement pleine de $\mathcal{M}$ dont l'ensemble des objets est Filt^π. Enfin, si $\mathcal{C}$ est une catégorie, je désigne par $\widetilde{\mathcal{C}}$ l'ensemble des classes d'isomorphie d'objets de $\mathcal{C}$.*[5]

1°) *Si $r \geq \sup(\pi_1, C_1)$, on a*

$$\operatorname{card}\big(\underbrace{\widetilde{\underline{\mathrm{Fl}}^r(\mathcal{M})}}_{=\underline{\mathrm{Fl}}(\mathcal{M}^r)}\big) \leq 2^{(r^{\pi_0})} \quad (= 2^r, \text{ si } r = 2^c, \text{ avec } c \geq \pi_0)\,. \tag{3.7}$$

2°) *Si $r, s \geq \sup(\pi_1, C_1)$, et si $A \in \mathrm{Filt}^r$, $X \in \mathrm{Filt}^s$, on a*

$$\operatorname{card}\operatorname{Hom}_{\mathcal{M}}(A, X) \leq s^{(r^{\pi_0})} \tag{3.8}$$

$$\operatorname{card}\underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X}) \leq s^{(r^{\pi_0})} \tag{3.9}$$

$$\begin{cases} = s^r & \text{si } r = 2^c, \text{ avec } c \geq \pi_0\,, \\ = s & \text{si } s = 2^{c'}, \text{ avec } c' \geq r^{\pi_0}\,. \end{cases}$$

N.B. Si on admet l'hypothèse du continu généralisée, qui implique que $\pi^{\pi'} = \pi$ si $\pi > \pi'$ (π cardinal infini, $\pi' \geq 1$), alors prenant r tel que $r > \pi_0$, les formules précédentes donnent (avec de plus $r \geq \sup(\pi_1, C_1)$) :

$$\operatorname{card}\widetilde{\underline{\mathrm{Fl}}^r(\mathcal{M})} \leq 2^r\,, \tag{3.7 bis}$$

$$\operatorname{card}\operatorname{Hom}_{\mathcal{M}}(A, X) \leq s^r \qquad (\text{si } A \in \mathrm{Filt}^r,\ X \in \mathrm{Filt}^s)\,, \tag{3.8 bis}$$

$$\operatorname{card}\underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X}) \leq s^r \qquad (\text{si } X \in \mathrm{Filt}^s)\,. \tag{3.9 bis}$$

4. *cf.* plus bas la notation $\widetilde{\mathcal{C}}$.

5. Signalons qu'on a toujours $\operatorname{card}\widetilde{\underline{\mathrm{Fl}}(\mathcal{C})} \geq \operatorname{card}\widetilde{\mathcal{C}}$ car $\widetilde{\mathcal{C}} \hookrightarrow \widetilde{\underline{\mathrm{Fl}}(\mathcal{C})}$ via $\mathrm{cl}(X) \longmapsto \mathrm{cl}(\mathrm{id}_X)$.

Sans hypothèse du continu, ces formules sont aussi valables pour r de la forme 2^c, avec $c \geq \pi_0$. Ce sont ces dernières formules qui on l'air les plus naturelles. Pour (3.8 bis) et (3.9 bis), je signale que le deuxième membre s^r est le cardinal de l'ensemble $\mathrm{Hom}_{\mathrm{Ens}}(A, X)$, où A, X sont deux ensembles tels que $\mathrm{card}\, A = r$, $\mathrm{card}\, X = s$. On ne peut donc espérer améliorer (3.8 bis) ! Pour la

première formule (3.7 bis), notons que l'on a égalité lorsque $\mathcal{M}$ est la catégorie des couples (A, B), A un ensemble et B une partie de A, avec les flèches évidentes, et qu'on définit $\mathcal{M}^\pi$ (pour $\pi \geq \pi_0$) comme la sous-catégorie strictement pleine formée des (A, B) tels que $\mathrm{card}\, A \geq \pi$. p. 14 [page 20]

Commentary · Examples [ms. 19]

An orbit count is not a bound on a Hom set

The purpose of C_1 is to pay for every choice of a map between the small objects. As printed, however, it counts isomorphism classes of objects of the arrow category. These two numbers can be very different. This is a mathematical issue in the edition, not an error introduced by the transcription.

Let G be any small group and let BG have one object $*$, with $\mathrm{Hom}(*, *) = G$. Give BG the constant filtration $\mathcal{M}^\pi = BG$, with $\pi_0 = \pi_1 = \aleph_0$. Every filtered diagram in BG is isomorphic to the constant diagram: transport from a fixed index to another through a common upper bound; a further common upper bound shows that transport does not depend on the choices. Consequently filtered colimits exist, and $\mathrm{Hom}(*, -)$ preserves them. Pushouts of spans of isomorphisms exist as well. The constant filtration has all the presentation and closure properties needed here.

Every two arrows g, h are isomorphic as objects of $\mathrm{Fl}(BG)$: take the square with vertical maps 1 and hg^{-1}. Thus the printed constant is $C_1 = 1$, whereas $|\mathrm{Hom}(*, *)| = |G|$. Take $r = s = \aleph_0$ and a group of cardinality greater than $\aleph_0^{(\aleph_0^{\aleph_0})}$. Formula (3.8), with the printed C_1, is false. The example also identifies precisely the invalid step on ms. 20: the indicated bound $|\mathrm{Hom}(A_i, X_j)| \leq C_1$ does not follow from the definition of C_1.

There is a straightforward repair. Choose a small full skeleton K_1 of $\mathcal{M}^{\pi_1}$, choosing its representatives compatibly with those at level π_0, and use

$$\chi = \max\{\aleph_0, \pi_1, |\mathrm{Mor}(K_1)|\}$$

as the threshold for r and s. It counts actual maps between representatives, including automorphisms. This change is also the natural reading of the lowercase c_1 in Theorem 2: the entire strictly full level need not be a small category, so its raw collection of arrows cannot serve as a small cardinal. The choice of skeleton affects a numerical bound, not the factorization or the lifting classes.

Démonstration de 3.1. Prouvons d'abord la formule (3.8),

$$\mathrm{card}\,\mathrm{Hom}(A, X) \;\leq\; s^{(r^{\pi_0})} \qquad \begin{cases} \text{si } A \in \mathrm{Filt}^r,\ X \in \mathrm{Filt}^s, \\ r \geq \pi_1,\ s \geq \sup(\pi_1, C_1). \end{cases}$$

(**N.B.** Je n'utiliserai pas $r \geq C_1$, postulé dans 3.1 1°.) J'écris `[<<J'écris>> rayé]`

$$A \;=\; \varinjlim_I A_i\ , \quad \mathrm{card}\, I \;\leq\; r^{\pi_0}\ , \quad A_i \;\in\; \mathrm{Filt}^{\pi_0}\ ,$$
$$X \;=\; \varinjlim_J X_j\ , \quad \mathrm{card}\, J \;\leq\; s^{\pi_1}\ , \quad X_j \;\in\; \mathrm{Filt}^{\pi_1}\ , \qquad\qquad J \text{ grand devant } \pi_1\ .$$

On aura

$$\mathrm{Hom}(A, X) \;\simeq\; \varprojlim_{i \in I} \underbrace{\mathrm{Hom}(A_i, \varinjlim_J X_j)}_{\overset{(6)}{\simeq} \varinjlim_{j \in J} \underbrace{\mathrm{Hom}(A_i, X_j)}_{\text{de cardinal } \leq C_1}}\ ,$$

donc pour tout $i \in I$

$$\mathrm{Hom}(A_i, X) \;\leq\; \mathrm{card}(J) \times C_1 \;\leq\; s^{\pi_1} \times C_1 \;=\; s^{\pi_1} \;,$$

car $s \geq C_1$, donc $s^{\pi_1} \geq C_1$, d'où

$$\mathrm{card}(\varprojlim_{j \in I} \mathrm{Hom}(A_i, X)) \;\leq\; (s^{\pi_1})^{\mathrm{card}\, I} \;\leq\; \underbrace{(s^{\pi_1})^{(r^{\pi_0})}}_{s^{\pi_1 r^{\pi_0}}} \;.$$

Comme $r \geq \pi_1$, on a $r^{\pi_0} \geq \pi_1$, donc $r^{\pi_0}\pi_1 = r^{\pi_0}$, d'où

$$\mathrm{card}\,\mathrm{Hom}(A, X) \;\leq\; s^{(r^{\pi_0})} \;,$$

c'est bien la formule (3.8). Pour en tirer une estimation pour $\mathcal{M}^r/X$ (avec $X \in \mathrm{Filt}^s$), je vais d'abord majorer le cardinal des classes d'isomorphie d'objets de $\mathcal{M}^r/X$, *i.e.* des flèches à isomorphisme près (dans $\mathcal{M}^r$) [page 21]

$$A \;\longrightarrow\; X \;, \qquad X \text{ fixé dans } \mathrm{Filt}^s,\; A \text{ variable dans } \mathcal{M}^r \;.$$

On aura

$$A \;=\; \varinjlim_{i \in I} A_i \qquad \text{avec } A_i \in \mathrm{Filt}^{\pi_0},\; I \text{ ensemble ordonné de cardinal} \leq r^{\pi_0} \;.$$

Pour I fixé, le cardinal de l'ensemble des systèmes inductifs indexés par I, dans $\mathcal{M}^{\pi_0}/X$, à isomorphisme près, peut se majorer ainsi : pour tout $i \in I$, il faut choisir A_i dans $\mathcal{M}^{\pi_0}$ p. 15
(mais seule sa classe dans $\widetilde{\mathcal{M}^{\pi_0}}$ importe), donc c'est majoré par $\mathrm{card}\,\widetilde{\mathcal{M}^{\pi_0}} \leq C_0$ [(7)], et pour tous les i, par $\mathrm{card}\, I \times C_0 \leq r^{\pi_0} \times C_0$, et si $r \geq C_0$, donc $r^{\pi_0} \geq C_0$, on trouve une majoration des choix possibles du système des *objets* A_i dans $\mathcal{M}^{\pi}$, à isomorphisme près, par r^{π_0}. Si on se donne de plus les morphismes de transition $A_i \longrightarrow A_j$ pour $i < j$, cela fait pour chaque paire comparable, élément de $I \times I$, un choix majoré par C_0, donc encore une majoration par $\underbrace{\mathrm{card}(I \times I)}_{=(\mathrm{card}\, I)^2 = \mathrm{card}\, I} \cdot C_0 \leq r^{\pi_0} \cdot C_0 = r^{\pi_0}$. Donc l'ensemble des systèmes inductifs en question, à isomorphisme près, pour I fixé, est de cardinal $\leq r^{\pi_0} \cdot r^{\pi_0} = r^{\pi_0}$. Enfin, les *homomorphismes* d'un système inductif donné dans X sont majorés par $\mathrm{card}\, I \times s^{(r^{\pi_0})}$ (par (3.8) déjà prouvé), donc par

$$r^{\pi_0} \cdot s^{(r^{\pi_0})} \;=\; s^{(r^{\pi_0})} \qquad (\text{puisque } r^{\pi_0} \leq s^{(r^{\pi_0})}) \;.$$

Au total, pour I fixé, l'ensemble des systèmes inductifs à isomorphisme près, indexés par I, à valeurs dans $\mathcal{M}^r/X$, est de cardinal majoré par $\underbrace{r^{\pi_0}}_{\substack{\text{nombre des}\\ \text{systèmes inductifs}\\ \text{dans } \mathcal{M}^{\pi}, \text{ à}\\ \text{isomorphisme près}}} \cdot \underbrace{s^{(r^{\pi_0})}}_{\substack{\text{nombre des}\\ \text{flèches d'un}\\ \text{tel système inductif}\\ \text{dans } X}} =$

$s^{(r^{\pi_0})}$. [page 22]

Mais il faut maintenant faire varier I, à isomorphisme près, dans la catégorie des ensembles ordonnés (filtrants *etc.*) de cardinal $\leq \lambda = r^{\pi_0}$. On voit de suite que cet ensemble de classes d'isomorphie (qui peut s'interpréter comme un ensemble quotient d'un certain ensemble de parties d'un ensemble fixe I_0 de cardinal λ) est de cardinal $\leq 2^{\lambda}$. On trouve donc finalement

$$\mathrm{card}\,\widetilde{\mathcal{M}^r/X} \;\leq\; s^{(r^{\pi_0})} \cdot 2^{(r^{\pi_0})} \;=\; s^{(r^{\pi_0})} \;,$$

sous condition que

$$r \;\geq\; \sup(\pi_1, C_0) \;, \qquad s \;\geq\; \sup(\pi_1, C_1) \;.$$

6. car les $A \in \mathrm{Filt}^{\pi_0}$ sont π_1-accessibles.

7. On a posé ici $C_0 = \mathrm{card}\,\widetilde{\mathrm{Fl}^{\pi_0}}$, donc $C_0 \leq C_1$.

Il faut prouver l'inégalité similaire pour card $\widetilde{\underline{\mathrm{Fl}}^r\,\mathcal{M}}$ (3.9). Or, comme pour A, B dans $\mathcal{M}^r$, on a, si $r \geq \sup(\pi_1, C_1)$,

$$\operatorname{card} \operatorname{Hom}_{\mathcal{M}}(A,B) \;\leq\; r^{(r^{\pi_0})}$$

en vertu de (3.8) où on fait $s = r$, on trouve que

$$\begin{aligned} \operatorname{card} \underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X}) \;&\leq\; \Big(\operatorname{card} \widetilde{\mathcal{M}^r/X}\Big)^2 \cdot r^{(r^{\pi_0})} \\ &\leq\; s^{(r^{\pi_0})} \cdot r^{(r^{\pi_0})} \\ &=\; s^{(r^{\pi_0})} \qquad \text{si } s \geq r\ . \end{aligned}$$

(On va regarder plus bas les cas $s \leq r$ [8]). C'est bien la formule (3.9).

Venons-en à la catégorie $\mathcal{M}^r$ elle-même. L'argument de la page 21 montre que l'ensemble des classes d'isomorphie d'objets de $\mathcal{M}^r$ indexés par un I fixé, de cardinal $\leq r^{\pi_0}$, comme $A = \varinjlim_i A_i$ avec $A_i \in \mathrm{Filt}^{\pi_0}$, est de cardinal majoré par r^{π_0} (si $r \geq C_0$). Si on fait varier I, on trouve une majoration par $r^{\pi_0} \cdot 2^{(r^{\pi_0})} = 2^{(r^{\pi_0})}$. On en conclut comme tantôt que l'ensemble $\widetilde{\mathrm{Fl}\,\mathcal{M}^r}$ est de cardinal majoré par

p. 16 [page 23]

$$(2^{(r^{\pi_0})})^2 \times r^{(r^{\pi_0})} \qquad (\text{si } r \geq \sup(\pi_1, C_1))\ ,$$

donc par $r^{(r^{\pi_0})}$. Je dis qu'on a

$$r^{(r^{\pi_0})} \;=\; 2^{(r^{\pi_0})}\ ,$$

en effet, on a $\geq$, et pour prouver $\leq$, on note que $r < 2^r$, donc $r^{(r^{\pi_0})} \leq (2^r)^{(r^{\pi_0})} = 2^{r \cdot r^{\pi_0}} = 2^{(r^{\pi_0})}$, O.K.! Cela établit donc 3.7. Cet argument prouve en fait le

Commentary · Exposition [ms. 22]

The estimates survive the corrected counting argument

Grothendieck's desired bound has enough room to absorb the choices that were omitted in the displayed counting. We give the calculation with the corrected threshold $r, s \geq \chi$ of the preceding block. Write $\mu = r^{\pi_0}$. For the presentations on ms. 20, accessibility gives

$$\operatorname{Hom}(A_i, X) \cong \varinjlim_j \operatorname{Hom}(A_i, X_j), \qquad |\operatorname{Hom}(A_i, X)| \leq s^{\pi_1}.$$

Taking a limit over the presentation of A then yields

$$|\operatorname{Hom}(A, X)| \leq (s^{\pi_1})^{|I|} \leq s^{\pi_1 \mu} = s^{\mu}.$$

This proves the repaired (3.8). The last equality uses $\pi_1 \leq r \leq \mu$ and infinite cardinal arithmetic.

The choices of all the objects A_i must be counted by a product, not by $|I|$ times the number of possibilities at one index. The same applies to the transition arrows. A poset with at most μ objects has at most μ arrows; a diagram on it with values in a fixed small skeleton at level π_0 can therefore be specified in at most $\chi^{\mu} \leq 2^{\mu}$ ways. The requirements of identity and composition merely reduce this number. There are at most 2^{μ} isomorphism types of posets of size at most μ, since both their underlying sets and their order relations can be coded inside a fixed set of size μ and its square. Taking colimits gives

$$|\operatorname{IsoOb}(\mathcal{M}^r)| \leq 2^{\mu}.$$

This replaces the unsupported fixed-index bound r^{π_0} on ms. 21 and 23 by 2^{μ}; the eventual bound is unchanged.

For a fixed $X \in \mathcal{M}^s$, choose one representative of each object at level r. There are at most s^{μ} arrows from any such representative to X, hence at most $2^{\mu} s^{\mu} = s^{\mu}$ objects of $\mathcal{M}^r/X$ up to isomorphism. Between two of them there are at most $r^{\mu} = 2^{\mu}$ morphisms, because a morphism in the slice is a particular underlying morphism in $\mathcal{M}$. Consequently

$$|\operatorname{IsoOb}(\mathrm{Fl}(\mathcal{M}^r/X))| \leq (s^{\mu})^2 2^{\mu} = s^{\mu}.$$

8. *cf.* cor. 2, page 23.

This is (3.9), including the case $s < r$. Finally, applying the same argument to the arrow category without a fixed target gives (3.7). In particular the appearance of products where exponential counts are required is a false intermediate inference, but it does not force a worse final estimate once the small skeleton has been counted correctly.

The elementary identity used here is $a^b = 2^b$ for infinite b and $2 \leq a \leq b$: one has $2^b \leq a^b \leq (2^b)^b = 2^b$. These are the hypotheses under which Lemma 3.2 is used. Its unqualified wording should not be extended to arbitrary finite cardinals, for which, for example, $3^3 \neq 2^3$.

Lemme 3.2. *Soient r un cardinal, r' un cardinal $\geq r$. Alors on a*

$$r^{r'} \;=\; 2^{r'} \qquad (= s^{r'} \text{ pour tout } 2 \leq s \leq r)\ .$$

Corollaire 1. *Soient $r \geq \sup(\pi_1, C_1)$, et $A, X \in \mathrm{Filt}^{\pi}$. Alors*

$$\operatorname{card} \mathrm{Hom}_{\mathcal{M}}(A, X) \;\leq\; 2^{(r^{\pi_0})} \qquad (= 2^r \text{ si } r = 2^c,\ \text{avec } c \geq \pi_0)\ . \tag{3.10}$$

On applique (3.8), qui donne la majoration $r^{(r^{\pi_0})}$, et le lemme 3.2.

On notera qu'on trouve la même majoration pour le cardinal de $\underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r})$ tout entier (3.7)! De même, (3.9) donne :

Corollaire 2. *Soit $r \geq \sup(\pi_1, C_1)$, et $X \in \mathrm{Ob}\, \mathcal{M}^r$. Alors*

$$\operatorname{card} \underline{\mathrm{Fl}}(\widetilde{\mathcal{M}^r/X}) \;\leq\; 2^{(r^{\pi_0})} \qquad (= 2^r \text{ si } r = 2^c,\ \text{avec } c \geq \pi_0)\ . \tag{3.11}$$

c'est donc la formule (3.9) pour $s \leq r$, laissée en suspens.

Revenons à la majoration cardinale des $X_\alpha(f)$, où $f : X \longrightarrow Y$ est dans $\mathrm{Fl}^s(\mathcal{M})$, *i.e.* $X, Y \in \mathrm{Filt}^s$. On suppose de plus $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$, [page 24]

$$\begin{aligned} &C_0 \;\subset\; \mathrm{Fl}^r(\mathcal{M}) \;=\; \mathrm{Fl}(\mathcal{M}^r)\ , \qquad (f : X \longrightarrow Y) \in \mathrm{Fl}^s(\mathcal{M})\ , \quad \textit{i.e. } X, Y \in \mathrm{Filt}^s\ , \\ &\text{avec } s \geq r\ . \end{aligned} \tag{3.10}$$

Commentary · Examples [ms. 23]

What the proposed sharpness example does show

Grothendieck rightly observes that the Hom bound cannot improve on the elementary set-theoretic exponent s^r: sets of cardinalities r and s have exactly that many functions. The claim that the category of pairs (A, B), $B \subseteq A$, attains the arrow-isomorphism bound 2^r at every allowed r needs a separate check. First, the printed inequality $|A| \geq \pi$ on ms. 20 must be $|A| \leq \pi$ to describe the intended increasing bounded filtration; the original sign remains in the source text and is recorded in the corrections register.

Even after that change, equality does not hold at every cardinal. Assume GCH and take $r = \aleph_1$, $\pi_0 = \pi_1 = \aleph_0$. A map of such pairs is a function between sets with two colours, required to preserve the distinguished colour. Its isomorphism type is determined by the colour of a target element, the two cardinalities of its source fibre, and the multiplicity of each such fibre type. Among cardinalities at most $\aleph_1$ there are only countably many choices. Thus there are countably many fibre types, and at most

$$\aleph_1^{\aleph_0} = \aleph_1$$

possible multiplicity lists. The count of arrow-isomorphism types is therefore at most $\aleph_1$, strictly less than $2^{\aleph_1} = \aleph_2$. The threshold involving the printed C_1 is met: arrows between countable coloured sets have at most $2^{\aleph_0} = \aleph_1$ isomorphism types.

The same example does attain $2^{\aleph_0}$ at the countable level. For each subset S of the positive integers, take one fibre of size n for each $n \in S$, and add a countably infinite fibre so that the source remains countable. Distinct S give nonisomorphic arrows. This is an orbit-count example at a level below the proposition's threshold $C_1 = 2^{\aleph_0}$, rather than an admissible equality case for that proposition. These calculations retain the useful point of the example while separating a genuine sharp Hom estimate from a stronger, false, universal sharpness assertion about arrow orbits.

[Deux fois (3.10).] Il y a une difficulté idiote, c'est que $\mathcal{M}^r$ est équivalente à une petite catégorie, mais n'est pas elle-même petite. Même si C_0 ne contenait qu'une seule classe d'isomorphie de flèches de $\mathcal{M}^r$, son cardinal pourrait être (pour r fixé) arbitrairement élevé. Aussi, dans ce qui suit, je dois supposer la condition de "non pléthoricité cardinale" pour C_0 :

(3.11) $\forall\, i \in C_0$, désignant par $C_0(i)$ l'ensemble des $i' \in C_0$ telles que $i \simeq i'$ dans $\mathrm{Fl}(\mathcal{M})$, j'impose que $\operatorname{card} C_0(i) \leq 2^{(r^{\pi_0})}$ (ce qui est aussi la valeur de la majoration pour le cardinal de $\widetilde{\underline{\mathrm{Fl}}^r(\mathcal{M})} = \widetilde{\underline{\mathrm{Fl}}(\mathcal{M}^r)}$).

[Deux fois (3.11).] Cela implique donc que l'on a $\operatorname{card} C_0 \leq (2^{(r^{\pi_0})})^2 = 2^{(r^{\pi_0})}$, p. 17

$$\operatorname{card} C_0 \;\leq\; 2^{(r^{\pi_0})}\ , \tag{3.12}$$

qui est équivalente à (3.11) (moyennant $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$, postulé dans (3.10)).

Il s'ensuit par (3.8) que l'on a

$$\operatorname{card} \Delta(C_0, f) \;\leq\; \underbrace{\operatorname{card}(C_0)}_{\leq 2^{(r^{\pi_0})}} \cdot s^{(r^{\pi_0})} \cdot s^{(r^{\pi_0})} \;\leq\; s^{(r^{\pi_0})}\ .$$

D'autre part, les X_D dont on prend la somme amalgamée sous X, pour obtenir $X_1(f)$, sont dans $\mathcal{M}^s$, si on fait l'hypothèse suivante. [page 25]

(3.13) Les $\mathcal{M}^\pi = \underline{\mathrm{Filt}}^\pi$ sont stables par sommes amalgamées simples, et par $\varinjlim_{\pi_0}$ filtrantes de cardinal $\leq \pi$ (pas nécessairement grandes devant π_0) ,

compte tenu que $r \leq s$, donc $\mathcal{M}^r \subset \mathcal{M}^s$, donc A, B, X dans $X_D = B \amalg_A X$ sont dans $\mathcal{M}^s$. Les sommes amalgamées partielles *finies* des X_D sont donc aussi dans $\mathcal{M}^s$, par récurrence sur le cardinal de l'ensemble d'indices. La somme amalgamée infinie apparaît ainsi comme $\varinjlim$ filtrante d'objets de $\mathcal{M}^s$, l'ensemble d'indices étant l'ensemble $\mathfrak{P}_f^*\Delta(C_0, f)$ des parties finies non vides de $\Delta(C_0, f)$. Cet ensemble est de cardinal majoré par s', si s' est un cardinal *infini* tel que $\operatorname{card} \Delta(C_0, f) \leq s'$. On peut donc prendre $s' = s^{(r^{\pi_0})}$. On a donc prouvé, compte tenu de (3.13) appliqué à $\mathcal{M}^{s'}$, le

Lemme 3.3. *On suppose* (3.10), i.e. $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$, $f \in \mathrm{Fl}^s(\mathcal{M})$, *avec* $s \geq r \geq \sup(\pi_1, C_1)$, *et de plus que* C_0 *satisfait la condition de non-pléthoricité* (3.11) (*ce qu'on peut toujours supposer, quitte à réduire* C_0 *sans changer* $\widetilde{C_0} = \Phi$ *et* $C_{0*} = \Psi$), *et que la filtration cardinale satisfait la condition de stabilité (anodine!)* (3.13). *Posons*

$$s' \;=\; s^{(r^{\pi_0})}\ . \tag{3.14}$$

Alors $X_1(C_0, f)$ *est dans* $\mathcal{M}^{s'}$. *Plus généralement et plus joliment, pour* C_0 *et* r *fixés* (*avec* $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$ *et satisfaisant la condition de non-pléthoricité* (3.11) *relativement à* r), *pour tout cardinal de la forme* [page 26]

$$s' \;=\; s^t\ , \qquad \text{avec } t \geq r^{\pi_0},\ s \geq 2 \tag{3.15}$$

(d'où $s' \geq 2^{(r^{\pi_0})} > r^{\pi_0} \geq r$), $\underline{\mathrm{Fl}}^{s'}(\mathcal{M})$ *est stable sous le foncteur* $p_1^{C_0} : \underline{\mathrm{Fl}}(\mathcal{M}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{M})$.

En effet, si on applique la première partie du lemme au couple (r, s'), on trouve que $X_1(f)$ est dans $\mathcal{M}^{s''}$, avec $s'' = s'^{(r^{\pi_0})}$. Mais

$$s'' \;=\; s'^{(r^{\pi_0})} \;=\; (s^t)^{(r^{\pi_0})} \;=\; s^{tr^{\pi_0}} \underset{\substack{| \\ \text{car } t \geq r^{\pi_0}, \\ \text{donc } tr^{\pi_0} = t}}{=} s^t \;=\; s'\ ,$$

donc on voit que $f \in \mathrm{Fl}^{s'}$ implique $X_1(f) \in \mathcal{M}^{s'}$, ou ce qui revient au même, $p_1(f) \in \mathrm{Fl}^{s'}$. p. 18

Corollaire. *Soient C_0, r, $s' = s^t$ ($t \geq r^{\pi_0}$, $s \geq 2$) comme dessus, et soit α un nombre ordinal tel que*

$$\operatorname{card} \alpha \ \leq \ s' \ = \ s^t \ .$$

Alors $\underline{\mathrm{Fl}}^{s'}(\mathcal{M})$ est stable par le foncteur $p_\alpha = p_\alpha^{C_0}$.

On le voit aussitôt par récurrence transfinie, compte tenu du lemme 3.3, et de l'hypothèse de stabilité (3.13).

Pour résumer :

Théorème 2. *Soit $\mathcal{M}$ une catégorie stable par petites $\varinjlim$ filtrantes, et munie d'une filtration cardinale $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$, telle que les Filt^π soient stables par $\varinjlim$ filtrantes de cardinal $\leq \pi$ (pas forcément grandes devant π_0), et par les sommes amalgamées simples qui existent dans $\mathcal{M}$. On suppose que les éléments de Filt^{π_0} soient accessibles, et on désigne par π_1 le plus petit cardinal $\geq \pi_0$, tel que les objets de Filt^{π_0} soient π_1-accessibles. On pose* [page 27]

$$c_1 \ = \ \operatorname{card} \mathrm{Fl}(\mathcal{M}^{\pi_1}) \ = \ \operatorname{card} \mathrm{Fl}^{\pi_1}(\mathcal{M})$$

(où pour tout $\pi \geq \pi_0$, $\mathcal{M}^\pi$ désigne la sous-catégorie strictement pleine de $\mathcal{M}$ définie par Filt^π, et $\mathrm{Fl}^\pi(\mathcal{M}) = \mathrm{Fl}(\mathcal{M}^\pi)$).

Commentary · Exposition [ms. 26]

One fixed exponent controls every stage

The geometric calculation has now become very simple. If $C_0 \subseteq \mathrm{Fl}(\mathcal{M}^r)$ and there are at most 2^μ generators, where $\mu = r^{\pi_0}$ and $r \geq \chi$, then for an arrow with endpoints in $\mathcal{M}^s$, $s \geq r$, the set of attaching squares has cardinality at most s^μ. Each individual pushout stays in $\mathcal{M}^s$. Their common-source amalgam can be obtained as a filtered colimit of finite partial amalgams, using at most s^μ indices. This proves Lemma 3.3 with the corrected counting constant. If there are no squares, the amalgam is simply the original source; this case is included by allowing the empty finite subset, or by treating it separately.

What matters for iteration is an exponent fixed point. If $S = c^t$ with $c \geq 2$ and $t \geq \mu$, then $S^\mu = c^{t\mu} = c^t = S$. Thus the successor construction preserves $\mathcal{M}^S$. A limit stage indexed by an ordinal of cardinality at most S stays there by (3.13). Transfinite induction proves that every stage up to any ordinal α with $|\alpha| \leq S$ has its middle object in $\mathcal{M}^S$.

To obtain the lifting property at the final stage, the ordinal must also be a limit of cofinality greater than μ. The initial ordinal of μ^+ is a uniform choice: $\mu^+ \leq 2^\mu \leq S$. The assertion in Theorem 2 that the initial ordinal of *any* cardinal $r' > \mu$ will do is too broad as a justification of the cofinality condition; a singular r' may have cofinality at most μ. Choosing the successor cardinal, as the theorem itself suggests, repairs this point. The size estimate and the cofinality test perform different jobs and must both be retained.

Soit C_0 une petite partie de $\mathrm{Fl}(\mathcal{M})$, formée de flèches coquarrables, et posons

$$\Phi \ = \ \widetilde{C_0} \ = \ \Psi^* \ , \qquad \text{où} \ \ \Psi = C_{0*} = \Phi_* \ .$$

*Soit r un cardinal $\geq \sup(\pi_1, c_1)$, et tel que l'on ait $C_0 \subset \mathrm{Fl}^r(\mathcal{M})$. On suppose que C_0 est irrédondant (*i.e. *deux objets de C_0 isomorphes sont égaux), ou plus généralement, que C_0 satisfait la très anodine condition de non-pléthoricité* (3.11) *relativement à r. On considère la suite transfinie des foncteurs $p_\alpha = p_\alpha^{C_0}$ sur $\underline{\mathrm{Fl}}(\mathcal{M})$ (cf.* th. 1*), donnant lieu aux factorisations fonctorielles d'une $f \in \underline{\mathrm{Fl}}(\mathcal{M})$, $f : X \longrightarrow Y$, en*

$$X \ \xrightarrow{i_\alpha(f)} \ X_\alpha(f) \ \xrightarrow{p_\alpha(f)} \ Y \ . \tag{3.16}$$

On a alors ce qui suit :

1°) *On a (pour mémoire) $i_\alpha(f) \in \Phi$ pour tout α, et de plus,* si α est grand devant r^{π_0}, *on a aussi $p_\alpha(f) \in \Psi$. Si r' est un cardinal $> r^{\pi_0}$ (p. ex. le cardinal successeur de r^{π_0}), on peut prendre pour α le premier ordinal dont le cardinal soit égal à r'.*

2°) *Pour tout cardinal s de la forme*

$$s \;=\; c^t \qquad \textit{avec}\;\; t \geq r^{\pi_0},\; c \geq 2$$

(donc $s = 2^t$ si on prend $c \leq t$, cf. *lemme 3.2),* $\underline{\mathrm{Fl}}^s(\mathcal{M})$ *est stable sous les p_α tels que* $\operatorname{card}\alpha \leq s$. *Si on prend p. ex. pour α le premier ordinal dont le cardinal soit $> r^{\pi_0}$, alors* $\underline{\mathrm{Fl}}^s(\mathcal{M})$ *est stable par p_α, qui par 1°) réalise donc une factorisation fonctorielle* (3.16) *avec $i_\alpha(f) \in \Phi$, $p_\alpha(f) \in \Psi$.*

Démonstration. Tout a été prouvé, si on se rappelle que les objets de $\mathcal{M}^r$ sont r^{π_0}-accessibles (*cf.* (3.2)). p. 19 [page 28]

Commentary · Development (commentators') [ms. 27]

Keeping the multiplicity of generators in the estimate

The non-plethoricity hypothesis is convenient when one is free to choose a generating set. In a family of constructions, however, discarding isomorphic copies can obscure the given maps between generating data. We can keep those copies and make their cost explicit. The following is a quantitative completion of the calculation, with no claim of priority. We checked the cardinal-filtration argument in SGA 4 and the accessibility bounds for functorial factorizations in Low (2016, Theorem 4.4); neither is cited as asserting this exact formula with the raw multiplicity parameter.

A uniform bound with multiplicities. [ours] Assume the filtered-colimit, pushout and cardinal-filtration hypotheses of Section 3, including (3.13). Choose χ by counting the actual arrows of a small full skeleton of $\mathcal{M}^{\pi_1}$. Let C_0 be any small set of coquarrable arrows whose endpoints lie in $\mathcal{M}^r$, where $r \geq \chi$. Let f have endpoints in $\mathcal{M}^s$, and let α be any small ordinal. Put

$$\mu = r^{\pi_0}, \qquad q = \max\{2, r, s, |C_0|, |\alpha|\}, \qquad B = q^\mu.$$

Then $X_\beta(C_0, f) \in \mathcal{M}^B$ for every $\beta \leq \alpha$. If α is a limit with $\mathrm{cf}(\alpha) > \mu$, the terminal factor $p_\alpha(f)$ has the right lifting property against C_0.

Proof. The cardinal μ is infinite, $B \geq q \geq r, s$, and

$$B^\mu = q^{\mu\mu} = q^\mu = B.$$

At stage zero the assertion follows from monotonicity of the filtration. Suppose it is true at β. The repaired Hom estimate applies to each endpoint of a generator and to each endpoint of $p_\beta(f)$, and gives at most $B^\mu = B$ possibilities for either side of an attaching square. There are therefore at most $|C_0|B^2 \leq B$ squares. Each of their pushouts belongs to $\mathcal{M}^B$. A finite partial amalgam also belongs to $\mathcal{M}^B$, by induction on the number of its terms; the empty one is X_β. The poset of finite subsets of the set of squares is filtered and has cardinality at most B. Condition (3.13) puts its colimit, which is $X_{\beta+1}$, in the same level.

If $0 < \lambda \leq \alpha$ is a limit, its diagram has cardinality $|\lambda| \leq |\alpha| \leq B$. Again (3.13) puts $X_\lambda = \varinjlim_{\beta<\lambda} X_\beta$ in $\mathcal{M}^B$. This completes the induction. Finally, a generator domain is μ-accessible by (3.2), so any map from it to X_α factors through some X_β, $\beta < \alpha$. The prescribed map from the generator codomain to the fixed target then determines a square at that stage, solved at $\beta + 1 < \alpha$. Its image in X_α is the required lift. □

In particular take α to be the initial ordinal of μ^+. The formula gives an explicit answer to the size question (3.6) for the unaltered generating set. The extra parameter $|C_0|$ is necessary in this formulation: in Set, arbitrarily many distinct copies of the isomorphism class $\varnothing \to \{*\}$ attach arbitrarily many points to $\varnothing \to \{*\}$ at the first stage, while the filtration levels of the generator endpoints remain fixed.

Commentary · Later developments [ms. 28]

From bounded factorizations to small model structures

Theorem 2 addresses a question that remains useful long after the existence of the small object argument is understood: can a chosen factorization be performed inside a controlled small part of the ambient category? In the locally presentable setting, Low (2016, Theorem 4.4) gives a functorial weak factorization system whose functors preserve specified presentable objects under explicit bounds

on the generators, their Hom sets and two regular cardinals. His Theorem 5.9 then tests when two such systems form a model structure by restricting them to an appropriate small full subcategory; Proposition 5.12 gives the corresponding statement for an arbitrary combinatorial model category.

These results use regular presentability indices and the relation of one cardinal being sharply below another. The numerical dictionary is therefore not obtained by replacing every Filt^{π} with the modern class of π-presentable objects. What is common is the argument's architecture: control one stage, choose a cardinal stable under its operations, and keep the entire factorization within that bound. This is the immediate background for the model-structure criteria that Grothendieck develops in Section 4.

4 Compléments sur les structures de Quillen

Définition 4.1. Soit $\mathcal{M}$ une catégorie. On appelle *précouple de Quillen* dans $\mathcal{M}$ un couple (Φ, Ψ) de parties de $\mathrm{Fl}(\mathcal{M})$, satisfaisant les deux conditions suivantes.

a) On a $\Phi \longleftrightarrow \Psi$, *i.e.* $\Psi \subset \Phi_*$, ou ce qui revient au même, $\Phi \subset \Psi^*$ – soit encore : pour tout carré commutatif dans $\mathcal{M}$

(4.1.1)
$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle i}\downarrow & \overset{w}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow{v} & Y\ , \end{array}$$

avec $i \in \Phi$, $p \in \Psi$, il existe w telle que $wi = u$, $pw = v$.

b) Toute $f \in \mathrm{Fl}(\mathcal{M})$ se factorise en

$$f \;=\; p\,i\ , \qquad \text{avec } i \in \Phi,\ p \in \Psi\ .$$

On écrit

(4.1.2)
$$\Phi \;\leftrightarrows\; \Psi$$

pour indiquer que (Φ, Ψ) est un précouple de Quillen.

On dit que le précouple de Quillen est *clos*, si Φ est clos à gauche, Ψ clos à droit. On va voir plus bas (cor. 2, page 31) que cela implique (donc équivaut à)

(4.1.3)
$$\Phi \;=\; \Psi^*\ , \qquad \Psi \;=\; \Phi_*\ .$$

Un précouple de Quillen clos est aussi appelé un couple de Quillen. [page 29]

Commentary · Exposition [ms. 28]

From a factorization procedure to its two lifting classes

Grothendieck has constructed factorizations from a set of generators. He now asks which parts of that construction survive when the generators and their cardinal estimates are forgotten. This change of viewpoint explains the apparently elementary opening of Section 4: the next existence argument will need to recognize a model structure from two factorization procedures, and a factorization procedure does not by itself determine the classes of maps which ought to be called cofibrations and fibrations.

Write $L^{\square}$ for the maps having the right lifting property against every member of L, and ${}^{\square}R$ for the corresponding left lifting class. Grothendieck's notation is $L_* = L^{\square}$ and $R^* = {}^{\square}R$. A *précouple* supplies factorizations and the inclusion $L \subset {}^{\square}R$. A *couple* supplies the equalities

$$L = {}^{\square}R, \qquad R = L^{\square}.$$

Thus a couple is what is now called a weak factorization system. Neither uniqueness of the diagonal filler nor a functorial choice of factorization belongs to this definition. The constructions of Section 1 provide functoriality under their additional smallness hypotheses. Modern definitions and the retract argument can be compared directly in Riehl (2014, Definition 11.2.1 and Lemma 11.2.3).

The word *clos* concerns equality with a lifting class. It must be distinguished from closure under the operations listed later in (4.4.3). Those operations preserve a left lifting class, but the converse requires an existence theorem. Section 1 furnishes that converse for classes generated by a small set of sufficiently small maps. This distinction will become decisive when the class $C \cap W$ is known to have the expected stability properties but no generating set has yet been found.

The expression "facteur direct" also has a categorical meaning here. It is a retract in the arrow category, without any additive direct-sum decomposition. For arrows $j : A \to B$ and $i : A' \to B'$, it means that there are morphisms of arrows $j \to i \to j$ whose composite is the identity of j. Proposition 4.2 shows that a factorization of j lets us keep its source fixed. That stronger form is why one diagonal filler is enough to turn a lifting condition into a retract statement.

Proposition 4.2. *Soit* (Φ, Ψ) *un précouple de Quillen dans* $\mathcal{M}$*, et soit* $j \in \mathrm{Fl}(\mathcal{M})$ [9]*. On a alors les conditions équivalentes sur* j *:*

a) $j \in \widetilde{\Phi}$.

b) $j \in \Psi^*$.

c) j *est facteur direct d'un* $i \in \Phi$.

c′) j *est facteur direct d'un* $i \in \Phi$ *ayant même source que* j*, plus précisément, il existe un diagramme*

(4.2.1)
$$\begin{array}{ccc} A & \xrightarrow{\ j\ } & B \\ {\scriptstyle \mathrm{id}_A}\downarrow\uparrow{\scriptstyle \mathrm{id}_A} & & {\scriptstyle q}\downarrow\uparrow{\scriptstyle p} \\ A & \xrightarrow{\ i\ } & B' \end{array},$$

avec $i \in \Phi$*, et* p. 20

(4.2.2)
$$qj \;=\; i\,, \qquad pi \;=\; j\,, \qquad pq \;=\; \mathrm{id}_B\,,$$

de sorte que j *est un facteur direct de* $i \in \Phi$*, par* (id_A, q)*, admettant la rétraction* (id_A, p).

d) *Pour tout diagramme commutatif traits pleins*

(4.2.3)
$$\begin{array}{ccc} A & \xrightarrow{\ j\ } & B \\ & {\scriptstyle i}\searrow \quad \nearrow{\scriptstyle p} \quad \curvearrowleft{\scriptstyle q} & \\ & B' & \end{array},$$

i.e. *toute factorisation de* j *en* pi*, si on a* $p \in \Psi$*, il existe une section* q *de* B' *sur* B *dans* $A\backslash\mathcal{M}$, i.e. *une flèche* $q : B \longrightarrow B'$ *telle que*

$$pq \;=\; \mathrm{id}_B \qquad et \qquad qj \;=\; i\,.$$

Démonstration de la proposition. Elle se fait par la suite d'implications suivante, toutes (±) tautologiques :

$$(c') \implies (c) \implies (a) \implies (b) \implies (d) \implies (c')\,,$$

dont les quatre premières utilisent seulement l'hypothèse $\Phi \longleftrightarrow \Psi$ (de (a) de la définition [page 30]

9. **N.B.** On utilise seulement que $\Phi \longleftrightarrow \Psi$, et que j peut se mettre sous la forme pi, avec $i \in \Phi$, $p \in \Psi$.

4.1), et la dernière seulement, $(d) \Longrightarrow (c')$, fait appel à l'hypothèse de factorisation de Quillen ((b) dans 4.1) ; et encore il suffit de la supposer pour j seulement !

$(c') \Longrightarrow (c)$ tautologie pure, $(c) \Longrightarrow (a)$ car on sait que $\widetilde{\Phi}$ est stable par facteurs directs (et contient Φ), $(a) \Longrightarrow (b)$ car $\widetilde{\Phi} \subset \Psi^*$ ($\widetilde{\Phi}$ étant par définition la plus petite partie close à gauche de $\mathrm{Fl}(\mathcal{M})$ contenant Φ, et Ψ^* étant close à gauche), $(b) \Longrightarrow (d)$ car on écrit le diagramme donné (4.2.2) (traits pleins) comme

$(*)$

$$\begin{array}{ccc} A & \xrightarrow{u} & X \\ {\scriptstyle j\in\Psi^*}\downarrow & \overset{q}{\nearrow} & \downarrow{\scriptstyle p\in\Psi} \\ B & \xrightarrow{v} & Y\,, \end{array}$$

en y faisant $Y = B$, $v = \mathrm{id}_B$, et en rebaptisant X par B', et u par i : l'hypothèse (b), $j \in \Psi^*$, signifie que dans un diagramme traits pleins $(*)$, il existe toujours q telle que $qj = u$, $pq = v$, et en l'exprimant dans le cas particulier où $v = \mathrm{id}_B$, on trouve (d).

Reste à prouver $(d) \Longrightarrow (c')$, en utilisant cette fois l'hypothèse de factorisation sur j, *i.e.* qu'on peut écrire j sous la forme pi :

$$\begin{array}{ccc} A & \xrightarrow{j} & B \\ & {\scriptstyle i}\searrow \quad \overset{p}{\nearrow} & \\ & B' \quad \overset{q}{\leftarrow} & \end{array}\,,$$

avec $i \in \Phi$, $p \in \Psi$. En vertu de (d), comme $p \in \Psi$, il existe donc $q : B \longrightarrow B'$ satisfaisant

p. 21
[page 31]

$$pq \;=\; \mathrm{id}_B\,, \qquad qj \;=\; i\,,$$

ce qui prouve (c'), puisque $i \in \Phi$.

Commentary · Exposition [ms. 30]

The retraction which completes both classes

The role of Proposition 4.2 is to identify exactly what can be missing from a précouple. Suppose $j : A \to B$ lifts against R, and choose a factorization $j = pi$ with $i : A \to Z$ in L and $p : Z \to B$ in R. Apply the lifting property to

$$\begin{array}{ccc} A & \xrightarrow{i} & Z \\ {\scriptstyle j}\downarrow & \overset{q}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow[\mathrm{id}_B]{} & B. \end{array}$$

The equations are $qj = i$ and $pq = \mathrm{id}_B$. Hence $(\mathrm{id}_A, q) : j \to i$ and $(\mathrm{id}_A, p) : i \to j$ exhibit j as a retract of i. Conversely, a lifting property passes to retracts: extend a proposed lifting square along the section, solve it for the larger arrow, and compose the resulting filler with the retraction. These two observations prove

$$^{\square}R = \mathrm{Retr}(L), \qquad L^{\square} = \mathrm{Retr}(R)$$

whenever (L, R) is a précouple. The dual assertion uses a factorization with target fixed.

This proof requires a factorization only of the particular arrow j under consideration, as footnote 9 correctly emphasizes. There is no induction hidden in the argument and no completeness hypothesis on the category. Once all arrows have factorizations, closing the two classes under retracts therefore produces a weak factorization system without changing any chosen factorization. This is the precise content of the two corollaries on manuscript page 31.

It also explains a useful economy in model-category arguments. One first constructs maps in classes with convenient explicit descriptions, such as relative cell complexes. One then enlarges those classes by retracts. The enlargement does not require repeating the construction: lifting supplies the necessary retraction. The enlargement is nevertheless real, and omitting it from a recognition statement can change that statement's truth value.

Corollaire 1. *Soit* (Φ, Ψ) *un précouple de Quillen. Alors on a*

$$\widetilde{\Phi} \;=\; \Psi^* \;=\; \text{ensemble des objets de } \underline{\mathrm{Fl}}(\mathcal{M}) \text{ qui sont facteurs directs d'éléments de } \Phi\ , \tag{4.3}$$

et dualement

$$\overline{\Psi} \;=\; \Phi_* \;=\; \text{ensemble des objets de } \underline{\mathrm{Fl}}(\mathcal{M}) \text{ qui sont facteurs directs d'éléments de } \Psi\ . \tag{4.4}$$

Corollaire 2. *Les conditions suivantes sur le précouple de Quillen* (Φ, Ψ) *sont équivalentes.*

a) *Le précouple est* clos, i.e. $\Phi = \widetilde{\Phi}$, $\Psi = \overline{\Psi}$.

b) $\Phi = \Psi^*$, $\Psi = \Phi_*$.

c) Φ *et* Ψ *sont stables par facteurs directs.*

Soit maintenant

$$W \;\subset\; \mathrm{Fl}(\mathcal{M}) \tag{4.5}$$

un localiseur, satisfaisant les conditions habituelles

$$\begin{cases} a) & W \text{ contient les isomorphismes.} \\ b) & \text{Si } X \xrightarrow{f} Y \xrightarrow{g} Z \text{ dans } \mathcal{M}, \text{ si deux parmi } f,\ g,\ gf \text{ sont dans } W, \text{ alors le troisième aussi.} \end{cases} \tag{4.6}$$

(W modérément saturé). Supposons de plus données quatre autres parties

$$C\ ,\ TC\ , \qquad F\ ,\ TF \tag{4.7}$$

de $\mathrm{Fl}(\mathcal{M})$, satisfaisant les conditions suivantes.

$$\begin{cases} a) & (C, TF) \text{ et } (TC, F) \text{ sont des précouples de Quillen.} \\ b) & TC \subset C \cap W,\ TF \subset F \cap W. \end{cases} \tag{4.8}$$

Ces propriétés s'explicitent ainsi : [page 32]

M 1 Propriétés de relèvement-prolongement, $C \longleftrightarrow TF$ et $TC \longleftrightarrow F$.

M 2 Propriété de factorisation de toute flèche f de $\mathcal{M}$ en $f = pi$, avec, au choix, $i \in C$, $p \in TF$, ou $i \in TC$, $p \in F$.

M′ $TC \subset C \cap W$, $TF \subset F \cap W$.

C'est donc une partie des axiomes d'une structure de Quillen faible pour W (XIII, p. 27, déf. 1) – il manque les axiomes de stabilité M 3, M 4, et les axiomes de "propreté" (propriety) M″, savoir $F \subset \mathrm{Fib}_W$, $C \subset \mathrm{Cof}_W$. p. 22

Proposition 4.3. *Soient* $\mathcal{M}$ *une catégorie,* (C, TF) *et* (TC, F) *deux précouples de Quillen dans* $\mathcal{M}$*, satisfaisant*

$$TC \;\subset\; C\ , \qquad TF \;\subset\; F\ , \tag{4.3.1}$$

et soit W *une partie de* $\mathrm{Fl}(\mathcal{M})$ *dont on ne suppose pour l'instant que* 4.6, (b) [10]*, et*

$$W \;\supset\; TC\ ,\ TF\ , \tag{4.3.2}$$

10. On utilise seulement que si $X \xrightarrow{f} Y \xrightarrow{g} Z$, et si $gf \in W$, alors $f \in W \iff g \in W$. On n'a pas besoin que W contienne les isomorphismes, ni qu'il soit stable par composition.

donc on a

$$TC \subset C \cap W , \qquad TF \subset F \cap W . \tag{4.3.3}$$

Considérons les conditions de prolongement-relèvement suivantes (a), (a′)*, et les conditions de factorisation* (b), (b′), (c), (d) (*en se rappelant que* $\widetilde{TC} = F^*$, $\overline{TF} = C_*$*, en vertu de la proposition* 4.2 (*cor.* 1) *appliquée aux précouples de Quillen* (TC, F), (C, TF)) :

a) $C \cap W \subset \widetilde{TC}$ (*donc* $TC \subset C \cap W \subset \widetilde{TC}$).

a′) $F \cap W \subset \overline{TF}$ (*donc* $TF \subset F \cap W \subset \overline{TF}$).

b) $W \subset TF \circ \widetilde{TC}$.

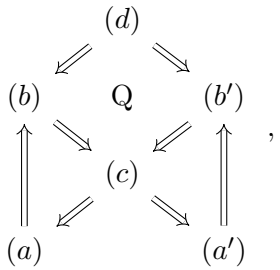
b′) $W \subset \overline{TF} \circ TC$.

c) $W \subset \overline{TF} \circ \widetilde{TC}$.

d) $W \subset TF \circ TC$ (*donc, par* (4.3.2), $W = TF \circ TC$ *si* W *est stable par composition*). [11]

Alors toutes les conditions sauf la dernière sont équivalentes (*et elles sont donc impliquées par cette dernière*)*. Donc lorsque* TC *ou* TF *sont stables par facteurs directs* (*p. ex. si* TC *est clos à gauche, ou* TF *clos à droite*)*, toutes les conditions envisagées sont équivalentes* (*puisque par le corollaire* 2 *à la proposition* 4.2*, on a alors* $TC = \widetilde{TC}$ *resp.* $TF = \overline{TF}$*, donc* (b′) *resp.* (b) *équivaut alors à* (d)). [page 33]

DÉMONSTRATION. Elle se fait par le diagramme d'implications



où les quatre implications du carré supérieur Q sont tautologiques. Il suffit donc d'établir les implications p. 23

$$(c) \implies (a) \implies (b) ,$$

les implications $(c) \Longrightarrow (a') \Longrightarrow (b')$ en résultent par dualité.

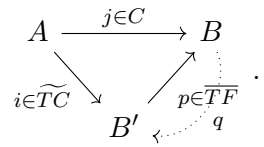
Supposons donc (c), *i.e.*

$$W \subset \overline{TF} \circ \widetilde{TC} , \tag{$*$}$$

et prouvons (a), *i.e.* que

$$C \cap W \subset \widetilde{TC} .$$

Mais si $j \in C \cap W$, pour prouver que $j \in \widetilde{TC}$, écrivons $j = pi$, $i \in \widetilde{TC}$, $p \in \overline{TF}$, ce qui est possible par l'hypothèse $(*)$ et $j \in W$.



[page 34]

Utilisons maintenant l'hypothèse $j \in C$ et $C \longleftrightarrow \overline{TF}$, on trouve donc (prop. 3.2, $(a) \Longrightarrow (d)$) une flèche $q : B \longrightarrow B'$ telle que $pq = \mathrm{id}_B$, $qj = i$, donc j apparaît comme facteur direct de i. Comme $i \in \widetilde{TC}$ et que $\widetilde{TC}$ est stable par facteurs directs, on en conclut $j \in \widetilde{TC}$, *q.e.d.*

11. **N.B.** La condition (a) signifie aussi que $(C \cap W, F)$ est encore un précouple de Quillen, resp. (b) signifie que $(C, F \cap W)$ en est un – et alors cette paire de précouples de Quillen satisfait cette fois à (d), *i.e.* $W = TF' \circ TC'$.

$(a) \Longrightarrow (b)$. On suppose donc

$$(*) \qquad C \cap W \ \subset \ \widetilde{TC} \ ,$$

et on veut prouver (b), *i.e.*

$$W \ \subset \ TF \circ \widetilde{TC} \ .$$

Soit donc $f \in W$, et factorisons f en pi, avec $i \in C$, $p \in TF$. (O.K. à cause du précouple de Quillen (C, TF)). Comme $TF \subset W$, on a $p \in W$, et comme $pi \in W$, on en conclut $i \in W$ (saturation modérée de W – on doit quand même l'utiliser !). On a donc $i \in C \cap W$, donc $i \in \widetilde{TC}$ par $(*)$, d'où $f \in TF \circ \widetilde{TC}$, *q.e.d.*

Corollaire 1. *Supposons TC stable par facteurs directs. Alors toutes les conditions de la proposition* 4.3 *sont équivalentes, et équivalent à la relation*

$$(4.3.4) \qquad C \cap W \ = \ TC \ .$$

Dualement, si TF est stable par facteurs directs, les conditions de la proposition 4.3 *sont toutes équivalentes et équivalent à*

$$(4.3.4') \qquad F \cap W \ = \ TF \ .$$

En effet, par le corollaire 2 de la proposition 4.2, l'hypothèse sur TC implique que $TC = \widetilde{TC}$, donc comme $TC \subset C \cap W$, l'hypothèse $C \cap W \subset \widetilde{TC}$ $(= TC)$ équivaut à $C \cap W = TC$.

p. 24
[page 35]

Commentary · Exposition [ms. 34]

Recognition by two compatible factorizations

Grothendieck wants to describe weak equivalences through their two possible factorizations. The names TC and TF are provisional at this stage: the assumptions only give $TC \subset C \cap W$ and $TF \subset F \cap W$. Proposition 4.3 measures what must be added before those inclusions become the desired equalities.

The proof has two different mechanisms. A factorization

$$j = pi, \qquad j \in C \cap W, \quad i \in \widetilde{TC}, \quad p \in \overline{TF}$$

exhibits j as a retract of i, because C lifts against $\overline{TF} = C_*$. This gives $C \cap W \subset \widetilde{TC}$ without using a property of W beyond membership of j in it. In the other direction, factor an arbitrary $w \in W$ using (C, TF). If $w = pi$, then $p \in TF \subset W$; cancellation in W forces $i \in C \cap W$. This is the sole use of the two-out-of-three axiom in that half of the argument. The dual proof exchanges the two factorizations.

Consequently, the equivalences among (a), (a′), (b), (b′), and (c) need only the restricted cancellation condition stated in footnote 10: whenever $gf \in W$, membership of f in W is equivalent to membership of g in W. A displayed equality $W = TF \circ TC$ additionally uses closure of W under composition for the reverse inclusion. The source itself makes this qualification beside (d). There is no need to strengthen the main proposition to a saturation condition involving localization.

The subsequent corollaries must be read with their respective closure hypotheses. If TC is retract closed, then $\widetilde{TC} = TC$, so the recognition condition becomes $C \cap W = TC$. The analogous statement for TF needs retract closure of TF. If C and W are retract closed, there is an even shorter proof of Corollary 2 than the printed one: $TC \subset C \cap W$ gives

$$\widetilde{TC} = \mathrm{Retr}(TC) \subset C \cap W,$$

and condition (a) supplies the reverse inclusion. The printed relation $C \cap W \longleftrightarrow TC$ in this proof is a slip. The lifting relation available here is $C \cap W \longleftrightarrow F$, and the relevant précouple is $(C \cap W, F)$.

Corollaire 2. *Supposons C et W stables par facteurs directs. Alors les conditions équivalentes de la proposition* 4.3 *équivalent à*

$$(4.3.5) \qquad C \cap W \ = \ \widetilde{TC} \ .$$

Dualement, si F, W sont stables par facteurs directs, les conditions équivalentes de la proposition 4.3 *équivalent à*

$$F \cap W \;=\; \overline{TF}\ . \tag{4.3.5$'$}$$

En effet, si C et W sont stables par facteurs directs, il en est de même de $C \cap W$, et dès lors si $C \cap W \subset \widetilde{TC}$, comme $C \cap W$ contient TC, on en conclut $\widetilde{C \cap W} = \widetilde{TC}$. Mais comme $C \cap W \longleftrightarrow TC$ et $C \cap W$ est stable par facteurs directs, $C \cap W$ est clos à gauche par corollaire 2 de la proposition 4.2, *i.e.* $C \cap W = \widetilde{C \cap W}$, d'où $C \cap W = \widetilde{TC}$, *q.e.d.*

Corollaire 3. *Supposons les conditions équivalentes* (a) *à* (c) *de* 4.3 *satisfaites, et de plus* W stable par facteurs directs[12]. *Alors les conditions préliminaires* (4.3.1) *et* (4.3.2) *restent satisfaites quand on remplace* (C, TF) *et* (TC, F) *par leurs clôtures* $(C', TF') = (\widetilde{C}, \overline{TF})$ *et* $(TC', F') = (\widetilde{TC}, \bar{F})$, *et les conditions équivalentes de la proposition* 4.3 *sont encore satisfaites pour ces couples, sous les formes fortes*

$$\begin{cases} C' \cap W \;=\; TC' & (= \widetilde{TC'}) \\ F' \cap W \;=\; TF' & (= \overline{TF'}) \\ \qquad W \;=\; TF' \circ TC'\ . \end{cases} \tag{4.3.6}$$

Le triple (W, C', F') *est un triple de Quillen clos* (*correspondant à une "closed model category" de Quillen, du moins si on suppose* $\mathcal{M}$ *stable par* $\varinjlim$ *finies et* $\varprojlim$ *finies, et que* W *est modérément saturé*).[13] [page 36]

Démonstration. Les relations (4.3.1) impliquent

$$\widetilde{TC} \;\subset\; \widetilde{C} \qquad \text{et} \qquad \overline{TF} \;\subset\; \bar{F}\ ,$$

i.e. p. 25

$$TC' \;\subset\; C'\ , \qquad TF' \;\subset\; F'\ ,$$

i.e. (4.3.1) pour les nouveaux couples de Quillen. La relation 4.3.2, $W \supset TC, TF$, implique

$$W \;\supset\; \widetilde{TC}\ ,\ \overline{TF}\ , \qquad \textit{i.e.} \qquad W \;\supset\; TC'\ ,\ TF'\ ,$$

en vertu du corollaire 2 à la proposition 4.2 et du fait que W est stable par facteurs directs ; donc on a (4.3.2) pour les nouveaux couples. Comme par hypothèse on a

$$W \;=\; \overline{TF} \circ \widetilde{TC}\ ,$$

ceci s'écrit

$$W \;=\; TF' \circ TC'\ ,$$

donc les nouveaux couples satisfont aux mêmes conditions, qui cette fois, les couples étant clos, prennent la forme forte (4.3.6).

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12. En fait, il suffit de supposer que l'on a $W \supset \widetilde{TC}$, $\overline{TF}$, *i.e.* que tout facteur direct d'un élément de W qui est soit dans C, soit dans F, est encore dans W. Alors $(\widetilde{C}, \widetilde{TC}, \bar{F}, \overline{TF})$ définissent une structure de Quillen close, et on sait que cela implique qu'en fait, W lui aussi est stable par facteurs directs.

13. Il en résulte le

Corollaire 4. *Soient W un localiseur modérément saturé, et (C, TF), (TC, F) deux précouples de Quillen tels que $TC \subset C \cap W$, $TF \subset F \cap W$. Alors les conditions suivantes sont équivalentes.*

a) $TC = C \cap W$.

b) $TF = F \cap W$.

c) $W = TF \circ TC$.

d) (W, C, F) *est un triple de Quillen clos.*

Prouvons que le triple (W, C', F') satisfait aux axiomes de Quillen d'un triple de Quillen clos. On a bien, par (4.3.6),

$$\begin{cases} C' \;=\; (TF')^* \;=\; (F' \cap W)^* \;, & F' \;=\; (TC')_* \;=\; (C' \cap W)_* \;, \\ W \;=\; TF' \circ TC' \;=\; C'_* \circ F'^* \;, \end{cases}$$

qui correspond à la définition des catégories de modèles de Quillen *clos* : les axiomes M 1, M 2, M 3, M 4 de Quillen sont réalisés de façon triviale (M 3 et M 4 parce que C et F, TC et TF sont clos à gauche resp. à droite). L'axiome M 5 est la saturation modérée de W. [page 37]

Commentary · Development (commentators') [ms. 36]

The missing closure hypothesis in the footnote

Grothendieck's main closure result, Corollary 3, deliberately replaces the four provisional classes by their retract closures. Footnote 13 then proposes a shorter Corollary 4 in which that replacement disappears. As printed, its equivalence with a *closed* Quillen structure is false. The following example isolates the defect while retaining invariance of the classes under isomorphism; it is not an objection based merely on a choice of representatives.

A précouple whose two classes have not both been closed. [ours] In the category of sets, with choice, let W be all maps. Let L consist of the injections $i : A \hookrightarrow B$ for which $B \setminus i(A)$ is empty or infinite, and let R consist of all surjections. Set

$$C = TC = L, \qquad F = TF = R.$$

Then both required pairs are précouples, and conditions (a), (b), and (c) of the printed Corollary 4 hold, but (W, C, F) is not closed.

To verify the lifting condition, consider an injection $i : A \hookrightarrow B$ and a surjection $p : X \twoheadrightarrow Y$. A commutative square specifies the filler on $i(A)$; on each remaining element of B, choose an inverse image under p of its prescribed image in Y. This proves $L \longleftrightarrow R$. For factorization, take any $f : A \to B$ and form

$$A \;\longrightarrow\; A \amalg (B \times \mathbb{N}) \;\xrightarrow{[f, \mathrm{pr}_B]}\; B.$$

The first map has empty complement when $B = \varnothing$ and infinite complement otherwise; the second is surjective. Thus it is an (L, R)-factorization. Since W contains every map, the equalities $TC = C \cap W$ and $TF = F \cap W$ hold, and the factorization just given proves $W = TF \circ TC$.

On the other hand, $\varnothing \to \{0\}$ is a retract of $\varnothing \to \mathbb{N}$: use the inclusion selecting 0 and the unique map $\mathbb{N} \to \{0\}$. The larger map belongs to L, whereas the smaller one does not. Therefore L is not retract closed and cannot equal $^{\square}R$. This proves the failure of (d).

The appropriate repair of the footnote is to require the two input pairs to be *couples*, or equivalently to require all four classes to be retract closed. Proposition 4.3 then gives the equivalence of (a), (b), and (c), and the resulting two weak factorization systems give (d), subject to the required existence of limits and colimits. Another valid conclusion under the weaker hypotheses is the one actually stated in Corollary 3: take retract closures first. The general modern formulation, including automatic retract closure of W once the two weak factorization systems exist, is in Riehl (2014, Definition 11.3.1 and Lemma 11.3.3).

We checked those general criteria and the recognition theorems cited below. We did not find this particular isomorphism invariant example in those sources; the claim here is the explicit counterexample and its verification, not a claim of priority for the retract obstruction.

4.4. Voici l'idée pour construire plus ou moins "toutes" les structures de Quillen closes (C, TC, F, TF) associées à un localiseur donné $W \subset \mathcal{M}$. On suppose que $\mathcal{M}$ satisfait aux conditions du paragraphe précédent, concernant l'existence de $\varinjlim$ filtrantes, d'une filtration cardinale, l'accessibilité dans $\mathcal{M}$ (celle des éléments de Filt^{π_0}). On part d'une partie petite

$$C_0 \;\subset\; \mathrm{Fl}(\mathcal{M})$$

telle que

$$C_{0*} \;\subset\; W \;, \tag{4.4.1}$$

et on pose

$$(4.4.2)\qquad \begin{cases} C \;=\; \widetilde{C_0}\;, & TC \;=\; C\cap W\;,\\ TF \;=\; C_* \;=\; C_{0*}\;, & F \;=\; (TC)_*\;. \end{cases}$$

Il y a un problème seulement, vu que l'on sait déjà que (C, TF) est un couple de Quillen par le théorème de factorisation dans §1 (p. 10) : il faut la factorisation pour le couple $(TC, F \;=\; (TC)_*)$. À ce moment là, les conditions préliminaires (4.3.1), (4.3.2) de la proposition 4.3 sont satisfaites :

$$TC \;\subset C\;,\qquad TF \;\subset F\;,\qquad W \;\supset\; TC\;,\, TF\;,$$

ainsi que la condition (a), $C\cap W \;\subset\; \widetilde{TC}$, puisque $C\cap W \;=\; TC$. Donc on trouve, par l'équivalence des conditions (a), (a'), (c), et vu que TC et TF sont clos,

$$TF \;=\; F\cap W\;,\qquad W \;=\; TF\circ TC\;,$$

d'autre part l'hypothèse sur W implique que W est stable par facteurs directs, C l'est aussi, étant close à gauche, donc $TC = C\cap W$ aussi, donc par proposition 4.1, corollaire 2, on a $TC \;=\; \widetilde{TC}$, TC est close à gauche. On gagne donc, par la dernière assertion du corollaire 3, proposition 4.3 (p. 35). p. 26 [page 38]

En fait, indépendamment des conditions particulières sur $\mathcal{M}$, W, on vient de prouver ceci :

Lemme 4.4.1 (corollaire de la proposition 4.3.). *Soient $\mathcal{M}$ une catégorie, W un localiseur dans $\mathcal{M}$ modérément saturé et stable par facteurs directs. Alors les structures de Quillen closes (C, F) associées à W s'obtiennent ainsi : on prend une partie C_0 de* $\mathrm{Fl}(\mathcal{M})$ *telle que*

$$TF \;\overset{\text{déf}}{=}\; C_{0*} \;\subset\; W\;,$$

on pose

$$C \;=\; \widetilde{C_0} \;=\; TF^*\;,\qquad TC \;=\; C\cap W\;,\qquad F \;=\; (TC)_*\;,$$

et on doit se débrouiller pour que (C, TF) et (TC, F) satisfassent à la condition de factorisation de Quillen, i.e. *soient des précouples de Quillen.* (En fait, C, TF et F étant clos par construction, et $TC = C\cap W$ étant stable par facteurs directs (C et W l'étant), ce seront même des *couples* de Quillen.)

Les hypothèses supplémentaires faites tantôt sur $\mathcal{M}$ servaient à assurer que si je prends C_0 petite, alors la factorisation marche pour (C, TF) automatiquement. Il reste le problème pour TC, F. [page 39]

Notons que si elle marche, alors $TC = \widetilde{TC} = F^*$, donc TC est clos à gauche, et *a fortiori* satisfait les conditions de stabilité correspondantes :

$$(4.4.3)\qquad \begin{cases} a) & \text{Stable par cochangement de base, par composition, par}\\ & \text{facteurs directs (ces deux dernières sont assurées auto-}\\ & \text{matiquement, car } C,\ W \text{ sont stables).}\\ b) & \text{Stable par } \varinjlim \text{ ordinales strictes.} \end{cases}$$

L'hypothèse de stabilité sur W sert (entre autres) à garantir la stabilité (b) pour TC (en fait, il suffirait pour cela seulement que W soit stable par $\varinjlim$ ordinales strictes). Il reste la stabilité de $TC = C\cap W$ par cochangement de base, qui est problématique car W n'est en général *pas* stable par cochangement de base. C'est pour assurer cette stabilité que je me vois pratiquement contraint ici à imposer

$$(4.4.4)\qquad C_0 \;\subset\; \mathrm{Cof}_W\;,$$

p. 27 [page 40]

je dis qu'avec l'hypothèse de stabilité faite sur W, cela implique

$$C \overset{\text{déf}}{=} \widetilde{C_0} \subset \mathrm{Cof}_W \ . \tag{4.4.5}$$

En effet, on a :

Commentary · Exposition [ms. 39]

Why preservation under pushout is the right condition

The second factorization cannot be obtained merely by declaring $TC = C \cap W$. Its left class must survive the operations used in the small object construction. Composition and retracts are already available, but an arbitrary weak equivalence need not remain a weak equivalence after pushout. Grothendieck's choice $C_0 \subset \mathrm{Cof}_W$ makes the missing stability a property of the cofibrations along which one pushes out.

With the definition recalled in Chapter XIII, manuscript page 8, a map $i : A \to B$ lies in Cof_W when its pushout functor

$$i_! : A\backslash\mathcal{M} \longrightarrow B\backslash\mathcal{M}, \qquad (A \to X) \longmapsto (B \to B \amalg_A X),$$

preserves weak equivalences. All pushouts mentioned here are assumed to exist. This is often called the condition of being an h-cofibration relative to W. The property automatically passes to cobase changes, by the pasting law for pushout squares. It is defined from W, before a model structure has been chosen.

The distinction between preserving weak equivalences under pushout *along* i and having every pushout *of* i belong to W is essential. The latter condition defines W^{univ}. Two-out-of-three gives $W^{\mathrm{univ}} \subset \mathrm{Cof}_W \cap W$: in the square comparing the cobase changes of i to X and to Y, the two cobase changes and the given map $X \to Y$ are weak equivalences. The reverse inclusion needs a factorization hypothesis. Grothendieck had explicitly used that hypothesis in Chapter XIII, Corollary 1 and Proposition 4, manuscript pages 10–11; it has been dropped from Corollary 4.4.3 here. The next block records the resulting counterexample and restores the hypothesis exactly where it is needed.

The filtered-colimit assertion of Proposition 4.4.2 itself is sound and has a short explicit proof. Let $i_t : A_t \to B_t$ be a filtered diagram in Cof_W, with colimit $i : A \to B$. Given a weak equivalence $u : X \to Y$ under A, restrict it under each A_t. Each map

$$B_t \amalg_{A_t} X \longrightarrow B_t \amalg_{A_t} Y$$

lies in W. Their filtered colimit is $B \amalg_A X \to B \amalg_A Y$, since colimits commute with colimits. Closure of W under filtered colimits proves the required preservation. It also gives retract closure of W: a retract is the colimit of the countable sequence which repeats the associated idempotent. Thus a retract of a map in Cof_W again preserves W, because its pushout comparisons are retracts of the corresponding comparisons for the larger map. Together with pushout and composition, these facts prove the asserted stability under continuous transfinite composition by induction on the ordinal.

Proposition 4.4.2. *Soit $\mathcal{M}$ stable par petites $\varinjlim$ inductives filtrantes, et $W \subset \mathrm{Fl}(\mathcal{M})$ un localiseur* (*modérément saturé*) stable par lesdites limites. *Alors* Cof_W *satisfait les conditions de stabilité* (4.4.3) ("*stabilité de Quillen gauche*"), *et est stable aussi par $\varinjlim$ filtrantes quelconques.*

En effet, la condition (a) est satisfaite sans condition supplémentaire sur W. D'autre part, il est immédiat que cette condition par rapport aux $\varinjlim$ filtrantes implique que Cof_W *est stable par limites inductives filtrantes* dans $\underline{\mathrm{Fl}}(\mathcal{M})$.

Corollaire 4.4.3. *Sous les conditions de la proposition* 4.4.2, $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ (*ensemble des W-équivalences co-universelles dans $\mathcal{M}$*) *est stable par limites inductives filtrantes, et donc "stable à gauche" en sens de* (4.4.3).

Corollaire 4.4.4. *Sous les conditions de la proposition* 4.4.2, *supposons de plus que les sommes amalgamées simples existent dans $\mathcal{M}$, et soit C_0 une petite partie de* Cof_W *dont les éléments soient accessibles, et considérons*

$$C \ = \ \widetilde{C_0} \ , \qquad TC \overset{\text{déf}}{=} C \cap W \ .$$

Alors on a

$$C \;=\; \mathrm{Cof}_W\ , \qquad TC \;=\; C \cap W^{\mathrm{univ}} \;\subset\; W^{\mathrm{univ}}\ ,$$

et C et TC sont Q-stables à gauche. [page 41]

Commentary · Development (commentators') [ms. 40]

A missing hypothesis in the couniversal-equivalence assertion

The intended application will have a useful factorization available before the second small object argument starts. Remembering that factorization repairs Corollary 4.4.3 and preserves the eventual recognition theorem. Forgetting it makes the intermediate assertion false even in a finite lattice.

A finite counterexample. [ours] Let $\mathcal{M}$ be the diamond lattice, viewed as a category, with $0 < a < 1$ and $0 < b < 1$, and with a, b incomparable. Let W consist of the identity maps and $i : 0 \to a$. Then W satisfies two-out-of-three and is closed under all filtered colimits; all objects of $\mathcal{M}$ are finitely presentable. Nevertheless,

$$i \in \mathrm{Cof}_W \cap W, \qquad i \notin W^{\mathrm{univ}}.$$

Indeed, a filtered diagram in a finite poset has a greatest element in its image, and that element is its colimit. The same observation applies to the finite arrow poset, proving closure of W under filtered colimits. It also shows directly that each representable preserves filtered colimits. To check two-out-of-three, the only nonidentity composable pairs are the two chains from 0 to 1 through a and b; neither chain has two of its three maps in W.

Pushout along i is the operation $x \mapsto a \vee x$. Among the arrows under 0, the only nonidentity weak equivalence is $0 \to a$, and it is sent to id_a. Hence $i \in \mathrm{Cof}_W$. But its pushout along $0 \to b$ is $b \to 1$, which does not belong to W. This verifies every claimed property and disproves the equality in Corollary 4.4.3 under its printed hypotheses. If $C_0 = \{i\}$, then $C = \widetilde{C_0}$ contains i and its pushout $b \to 1$. Thus $TC = C \cap W$ contains i but not that pushout, disproving also the asserted pushout stability of TC in Corollary 4.4.4. The other identities involving W^{univ} in that corollary require the same repair.

The factorization that repairs the argument. Assume that every map factors as a member of Cof_W followed by a member of W. Then $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$. This is the dual form of Grothendieck (1990–1991q, Chapter XIII, Proposition 4, ms. pp. 10–11), and we give the complete argument in the notation used here.

Take $i : A \to B$ in $\mathrm{Cof}_W \cap W$ and any $u : A \to X$. Factor $u = qj$ with $j : A \to Z$ in Cof_W and $q : Z \to X$ in W. Form $P = B \amalg_A Z$ and $Q = B \amalg_A X$. The map $k : Z \to P$ is weak because it is the pushout of the weak equivalence i along the W-cofibration j. It is also a W-cofibration, being a pushout of i. Therefore the induced map $q' : P \to Q$, the pushout of q along k, is weak. The equation $q'k = i'q$, where $i' : X \to Q$ is the desired cobase change, and two-out-of-three imply $i' \in W$. The reverse inclusion was proved in the preceding block.

Proposition 4.4.6 supplies exactly this hypothesis: its first factorization has left map in $C \subset \mathrm{Cof}_W$ and right map in $TF \subset W$. Once that first factorization is constructed, the proof above establishes the pushout stability of TC, and all subsequent uses of Corollary 4.4.4 are justified. The false intermediate generalization therefore does not invalidate the final recognition proposition. The class W^{univ} itself remains closed under filtered colimits and the stated left stability operations even without the extra factorization.

Our literature check covered the earlier manuscript argument and the modern comparison in Batanin and Berger (2017, Lemma 1.10), which distinguishes couniversal weak equivalences from trivial h-cofibrations and gives equality in left proper model categories. We did not find this finite lattice example in those sources.

En effet, par le théorème 1 du paragraphe 1 (p. 10) et son corollaire, $\widetilde{C_0}$ est la plus petite partie de $\mathrm{Fl}(\mathcal{M})$ contenant C_0 et Q-stable à gauche. Comme Cof_W est Q-stable à gauche et contient C_0, on a donc $C = \widetilde{C_0} \subset \mathrm{Cof}_W$. Donc $TC \overset{\text{déf}}{=} C \cap W \subset \mathrm{Cof}_W \cap W \overset{\text{déf}}{=} W^{\mathrm{univ}}$, donc $TC = C \cap W^{\mathrm{univ}}$. Comme W^{univ} est lui aussi Q-stable à gauche (corollaire précédent), il en est de même de $C \cap W$, *q.e.d.*

Corollaire 4.4.5. *Supposons que tout objet de $\mathcal{M}$ soit accessible. Sous les conditions du*

corollaire précédent, posant

$$TF \;=\; C_{0*} \;=\; C_* \;, \qquad F \;=\; (TC)_* \;,$$

le couple (C, TF) *est un couple de Quillen. Pour qu'il en soit de même du couple* (TC, F)*, il suffit qu'il existe une* petite *partie* TC_0 *de* $C \cap W$*, telle que* $\widetilde{TC_0} = TC$ $(\overset{déf}{=} C \cap W)$.

La première assertion est, on l'a vu tantôt, le théorème 1 du paragraphe 1. La deuxième résulte de ce même théorème. On peut préciser dans ce sens :

Commentary · Examples [ms. 41]

An inclusion which cannot be an equality

Grothendieck has established enough closure to apply the small object theorem to a chosen set $C_0 \subset \mathrm{Cof}_W$. The conclusion is that the class it generates stays inside Cof_W. The display in Corollary 4.4.4 prints $C = \mathrm{Cof}_W$, although the proof on the following manuscript page proves $C \subset \mathrm{Cof}_W$. This is a slip in the edition, not in the transcription.

A small example shows why the stronger assertion cannot follow. Take $\mathcal{M} = \mathrm{Set}$, let W consist of the bijections, and take $C_0 = \{\mathrm{id}_{\{0\}}\}$. Every map belongs to Cof_W, since every pushout functor preserves isomorphisms. Every map also has the right lifting property against $\mathrm{id}_{\{0\}}$, so

$$C_0^{\square} = \mathrm{Mor}(\mathrm{Set}), \qquad {}^{\square}(C_0^{\square}) = \mathrm{Iso}(\mathrm{Set}).$$

For the second equality, a map which lifts against every map lifts against itself in the square with identity horizontal maps, and the resulting filler is a two-sided inverse. Thus C consists of the bijections, whereas Cof_W contains, for example, $\varnothing \to \{0\}$. All objects of Set are presentable for some regular cardinal, and the required colimits exist, so the example meets the hypotheses of the corollary.

Under the extra factorization hypothesis explained in the preceding block, the rest of the corollary has its intended force:

$$TC = C \cap W = C \cap W^{\mathrm{univ}}.$$

Both factors in the last intersection have the required stability under pushout, retracts, and continuous transfinite composition. Hence so does TC. This establishes the necessary closure of the proposed trivial cofibrations; the existence of a small set generating them remains the next problem.

Proposition 4.4.6. *Soit* $\mathcal{M}$ *une catégorie stable par petites* $\varinjlim$ *filtrantes, par sommes amalgamées, et dont tous les objets sont accessibles,* W *un localiseur modérément saturé dans* $\mathcal{M}$*, stable par* $\varinjlim$ *filtrantes,* C_0 *une* petite *partie de* $\mathrm{Fl}(\mathcal{M})$*, telle que* [page 42]

$$C_0 \;\subset\; \mathrm{Cof}_W \;, \qquad \underbrace{C_{0*}}_{(=TF)} \;\subset\; W \;,$$

posons p. 28

$$C \;=\; \widetilde{C_0} \qquad TC \;=\; C \cap W \;, \qquad TF \;=\; C_* \;, \qquad F \;=\; (TC)_* \;,$$

et soit TC_0 *une petite partie de* $TC = C \cap W$*. On a alors*

$$(4.4.6) \qquad \begin{cases} C \;\subset\; \mathrm{Cof}_W \;, & TC \;=\; C \cap W^{\mathrm{univ}} \quad (\textit{pour mémoire}) \\ \widetilde{TC_0} \;\subset\; TC \;, & F_0 \supset F \quad (\textit{ce dernier trivial}) \;, \end{cases}$$

et les conditions suivantes sont équivalentes :

a) $\widetilde{TC_0} = TC$ *(on a* $\subset$*)*;

a′) $F_0 = F$ *(on a* $\supset$*)*;

b) $TF = F_0 \cap W$ *(on a* $\subset$*)* [14];

14. **N.B.** On voit facilement que $\boxed{TF = F \cap W,}$ mais ça ne suffit pas pour avoir $TF = F_0 \cap W$ pour TC_0 convenable.

c) $W = TF \circ \widetilde{TC_0}$ $\quad (on\ a \supset)$ [15] ;

et impliquent que (W, C, F) est un triple de Quillen clos.

Démonstration. L'inclusion $\widetilde{TC_0} \subset TC$ résulte de la Q-stabilité à gauche de TC (cor. 4.4.4) et du fait que $TC_0 \subset TC$ est petite (on applique alors le théorème 1 tout comme pour $C \subset \mathrm{Cof}_W \ldots$). L'équivalence de (a), (b), (c) n'est autre que la proposition 4.3 appliquée à W et aux deux couples de Quillen (C, TF), $(\widetilde{TC_0}, F_0)$ (compte tenu du théorème 1 du paragraphe 1). Qu'on a alors un triple de Quillen clos a été vu dans le corollaire 3 de la proposition 4.3 (p. 35). Il reste à prouver l'équivalence de (a) et de (a'). On a $(a) \Longrightarrow (a')$, puisque $(TC_0)_* = (\widetilde{TC_0})_* = F_0$. Inversement, (a') implique $\widetilde{TC_0} = \widetilde{TC}$, donc le théorème de factorisation pour (TC, F) en vertu du théorème de factorisation, corollaire 1 (p. 11) appliqué à $TC_0 \subset TC \subset \widetilde{TC_0}$ et à $F_0 = F = (TC_0)_*$.

Commentary · Exposition [ms. 42]

The two identities already available before the second factorization

Grothendieck's two footnotes on this page separate a formal recognition problem from a problem of size. Put $C = \widetilde{C_0}$, $TF = C^{\square} \subset W$, $TC = C \cap W$, and $F = TC^{\square}$. The first weak factorization system (C, TF) already gives

$$W = TF \circ TC, \qquad TF = F \cap W,$$

without assuming a factorization with left map in TC for every arrow. The missing quantifier concerns arbitrary arrows, not only weak equivalences.

Here are complete proofs. Given $w \in W$, factor it as $w = pi$ using (C, TF). Since $p \in TF \subset W$, two-out-of-three gives $i \in C \cap W = TC$. Conversely, a composite of a member of TC and a member of TF belongs to W. For the second identity, $TF \subset F \cap W$ follows from $TC \subset C$. If $f \in F \cap W$, factor $f = pi$ with $i \in C$ and $p \in TF$. Again $i \in TC$. The square with left arrow i, right arrow f, top arrow the identity of the source of f, and bottom arrow p has a diagonal s satisfying $si = \mathrm{id}$ and $fs = p$. Consequently f is a retract of p in the arrow category. The right lifting class TF is retract closed, so $f \in TF$.

For a candidate set $TC_0 \subset TC$, the notation left implicit in the statement of Proposition 4.4.6 is $F_0 = TC_0^{\square}$. Smallness produces a weak factorization system $(\widetilde{TC_0}, F_0)$, but only the inclusions

$$\widetilde{TC_0} \subset TC, \qquad F \subset F_0$$

are automatic. Proposition 4.4.6 says that the candidate succeeds precisely when $F_0 \cap W = TF$, equivalently when every weak equivalence has the more restrictive factorization with left map in $\widetilde{TC_0}$. If $F_0 = F$, then $\widetilde{TC_0} = {}^{\square}F = \widetilde{TC}$; combined with $\widetilde{TC_0} \subset TC \subset \widetilde{TC}$, this gives equality immediately. Thus the final implication does not assume the desired second factorization in disguise.

To see the distinction concretely, take $\mathcal{M} = \mathrm{Set}$ and W all maps. Let C_0 contain $\varnothing \to \{0\}$ and the fold $\{0,1\} \to \{0\}$. Right lifting against the first map is surjectivity, and right lifting against the second is injectivity. Hence TF consists of bijections and C consists of all maps. Then TC is all maps and F consists of bijections. If the candidate TC_0 contains only $\varnothing \to \{0\}$, its right class F_0 consists of surjections. The projection $\{0,1\} \to \{0\}$ belongs to $F_0 \cap W$ but not to TF. Both identities of the preceding paragraph still hold; the candidate has simply omitted a necessary generating map.

[page 43]

Ainsi, le seul problème, une fois trouvé C_0 (petite) satisfaisant $C_0 \subset \mathrm{Cof}_W$, $C_{0*} \subset W$, c'est de trouver une *petite* partie $TC_0 \subset TC$ qui "engendre $TC = C \cap W$ ", *i.e.* qui satisfasse aux conditions équivalentes (a), (a'), (b), (c) de la proposition précédente. J'ai passé des jours à essayer de prouver l'existence de TC_0, en supposant $\mathcal{M}$ munie d'une filtration cardinale, et W une partie accessible de $\mathcal{M}$, sans y être arrivé. Bien sûr, tout ce qui nous importe vraiment, c'est que (TC, F) satisfasse à la condition de factorisation. Mais je ne vois aucun moyen de le prouver, si ce n'est en "engendrant" à gauche TC par un *petit* sous-ensemble TC_0.

15. **N.B.** On sait que $\boxed{W = TF \circ TC,}$ ce qui n'est nullement suffisant pour impliquer (c) !

Commentary · Later developments [ms. 43]

The generating set supplied by Smith's theorem

The question at the end of Section 4 became a standard recognition theorem for combinatorial model categories. In the locally presentable setting, Smith's theorem, published with proof by Beke (2000, Theorem 1.7), gives the missing set of generating trivial cofibrations. Its hypotheses are precisely organized around the difficulty seen here: W has two-out-of-three and is retract closed, $C_0^{\square} \subset W$, $C \cap W$ is stable under pushout and transfinite composition, and W satisfies a solution set condition at each map of C_0. Accessibility of the full subcategory $W \subset \mathcal{M}^{[1]}$ supplies the last condition by Beke (2000, Proposition 1.15).

The solution set condition says that, for each $i \in C_0$, a set of maps $i \to w_s$ with $w_s \in W$ is weakly initial among all maps from i to members of W in the arrow category. This is the categorical form of the bounded choices Grothendieck is seeking. Smith's argument converts these choices into a set $J \subset C \cap W$ with $\mathrm{cof}(J) = C \cap W$; Beke's Lemmas 1.8 and 1.9 supply the construction and proof.

This answers the existence question under the stated modern hypotheses. A cardinal filtration on its own should not silently be substituted for local presentability, and the claim does not cover every category allowed by the more flexible assumptions on the preceding pages. Nor does the theorem require C itself to be stable under all filtered colimits, the additional route Grothendieck begins to explore in Section 5.

Commentary · Present state [ms. 43]

The form closest to the proposed localizer construction

Grothendieck's use of Cof_W also survives in a recognition theorem whose hypotheses closely follow these pages. In the updated online text of Lurie (2009, Proposition A.2.6.15), a locally presentable category, a perfect class W, and a set of generating cofibrations satisfying the pushout preservation condition and $C_0^{\square} \subset W$ produce a left proper combinatorial model structure. Here "perfect" means that W contains the isomorphisms, satisfies two-out-of-three, is closed under filtered colimits, and is generated under those colimits by a set. The proof first establishes the required stability of $C \cap W$, then applies the small-generation result of Lemma A.2.6.9 and the general recognition Proposition A.2.6.10. These are the numbers in the online revision consulted, not the earlier printed numbering.

The later theorem makes precise both parts of the intended conclusion: the second factorization exists, and the chosen cofibrations preserve weak equivalences under pushout. The latter is left properness. It remains separate from the formal two-factorization axioms studied at the beginning of Section 4; an arbitrary model structure need not have it.

Commentary · Later developments [ms. 43]

The passage from models to diagram homotopy theories

The point of obtaining a model structure is that it can support coherent operations on diagrams. In the combinatorial setting, the later passage is precise: Groth (2013, Theorem 1.28 and Proposition 1.30) constructs projective and injective model structures on diagram categories and proves that $J \mapsto \mathrm{Ho}(\mathcal{M}^J)$ is a strong derivator. The two diagram model structures supply the appropriate derived Kan extensions.

The existence of these diagram model structures is a convenient sufficient route. The broader construction of homotopy right Kan extensions for model categories with small limits is treated by Cisinski (2003, Theorem 6.11), in that paper's left-derivator convention. This later theorem explains why the goal is larger than a particular generating-set argument. Chapter XV builds and controls models; the derivator records their homotopy theories of diagrams and the functors between them. Neither the cardinal estimates nor the two lifting systems alone constitute the passage between the two.

5 Résultats techniques préliminaires sur les parties accessibles de $\mathrm{Ob}\,\mathcal{M}$, stables par limites inductives filtrantes

Je crois pourtant que je peux arriver à quelque chose d'utilisable, en travaillant systématiquement avec des C ($\subset \mathrm{Cof}_W$, Q-closes à gauche, Q-engendrées à gauche par une *petite* partie C_0 de C) qui sont stables par toutes limites inductives filtrantes.

Je suppose à nouveau sur $\mathcal{M}$ les conditions suivantes : p. 29 [page 44]

a) $\mathcal{M}$ stable par petites limites inductives (et pas seulement filtrantes).

b) $\mathcal{M}$ muni d'une filtration cardinale $(\mathrm{Filt}^\pi)_{\pi\geq\pi_0}$.

c) Les objets de $\mathcal{M}$ sont accessibles (**N.B.** Il suffit de le supposer pour les objets de Filt^{π_0}).

La condition (*b*) implique que $\mathcal{M}$ contient une petite famille génératrice par épimorphismes stricts (p. ex. une sous-catégorie $\mathcal{C}$ petite de $\underline{\mathrm{Filt}}^{\pi_0} = \mathcal{M}^{\pi_0}$ telle que l'inclusion $\mathcal{C} \longrightarrow \underline{\mathrm{Filt}}^{\pi_0}$ soit une équivalence de catégories). Par SGA 4 I 8.12.7, il en résulte que $\mathcal{M}$ est stable par petites limites projectives. Il s'ensuit le

Lemme 5.1 (sous les conditions (*a*), (*b*)). *Pour tout objet X de $\mathcal{M}$, l'ensemble* $\mathrm{Ss}(X)$ *des sous-objets de X, muni de la relation d'ordre naturelle "d'inclusion", admet des* inf *et des* sup *quelconques.*

Des inf, car ceux-ci s'expriment comme des limites projectives dans $\mathcal{M}$. Or l'existence des inf implique celle des sup, puisque $\sup_J = \inf_{J'}$, où J' est l'ensemble des majorants de J dans $\mathrm{Ss}(X)$.

Corollaire 5.2. *Pour toute flèche $f : Z \longrightarrow X$ dans $\mathcal{M}$, il existe, parmi les sous-objets X' de X par lesquelles f se factorise, un plus petit (qu'on notera* $\mathrm{Im}\, f$*). Le morphisme $Z \longrightarrow \mathrm{Im}\, f$* [page 45] *est un épimorphisme.*

Commentary · Exposition [ms. 44]

Images and the meaning of a small subobject

Grothendieck now wants the size of an object to be visible inside the object itself. A presentation by a filtered diagram can repeat objects, identify elements, and carry many choices. The ordered collection of its small subobjects is a more geometric presentation: its transition maps are inclusions, and its small directed suprema can be calculated in the ambient category. This change of presentation is the preparation for recognizing a class of objects from its bounded part.

We keep the three conditions on the filtration separate. Essential smallness says that a bounded collection has a set of isomorphism classes; generation says that every object has an appropriately directed presentation by bounded objects; preservation of such colimits by a represented functor is an additional condition on the representing object. In the terminology of these pages, "grand devant π" means that every subset of cardinal at most π has an upper bound. For infinite π, this is the usual π^+-directedness. A π-accessible object is consequently a π^+-presentable object in current terminology. It is not an object with exactly π elements, and the bound is not assumed regular.

The set-sized assertion in Lemma 5.1 also has content. Let G run through a small set of representatives of the base level. A subobject $U \hookrightarrow X$ determines the subsets of $\mathrm{Hom}(G, X)$ consisting of the maps which factor through U. These subsets determine U: a generating presentation of its domain lifts uniquely to any other subobject having the same subsets, and these unique lifts assemble. Thus there is a set of subobjects of X.

The same base level is a conservative generating family. Suppose $f : X \to Y$ induces bijections on maps from its objects. Write $Y = \varinjlim_i G_i$ using the filtration. Each structural map $G_i \to Y$ lifts uniquely to X; uniqueness makes the lifts compatible, so they induce $g : Y \to X$ with $fg = \mathrm{id}_Y$. For a presentation $X = \varinjlim_j H_j$, injectivity of $\mathrm{Hom}(H_j, f)$ gives $gfh_j = h_j$ at every structural map h_j, hence $gf = \mathrm{id}_X$. No accessibility rank is needed. The dual of (Artin et al., 1972, Exposé I, 8.12.7) now supplies completeness: the functor of cones $Z \mapsto \varprojlim_j \mathrm{Hom}(Z, X_j)$ takes colimits to limits and is represented by the desired limit. This argument uses conservative generation, without requiring the stronger density assertion in the source.

Once intersections exist, the image used here is the intersection of *all* subobjects through which the map factors. If $Z \to \mathrm{Im}(f)$ were not epic, two different maps out of its codomain would agree after composition; their equalizer would give a smaller subobject through which f factors. This

proves Corollary 5.2. In fact the first map in this factorization is extremal epic: any factorization through a monomorphism would contradict the minimality of the image unless that monomorphism were invertible.

In Lemma 5.3 the adjective *strict* has the categorical meaning of SGA 4, not merely "different from the whole object"; the latter requirement is written separately. A strict monomorphism which is also epic is invertible. Conversely, failure to be epic is detected by a proper equalizer, which is a strict monomorphism. This proves the stated criterion, after replacing the accidental name g by f. The distinction matters, for example, for the epimorphism $\mathbb{Z} \to \mathbb{Q}$ in commutative rings: it factors through a proper monomorphism, namely itself, but not through a proper strict monomorphism. The terminology is fixed in (Artin et al., 1972, Exposé I, 10.2–10.5).

Cette dernière assertion provient de celle-ci :

Lemme 5.3. *Soient $\mathcal{M}$ une catégorie où les noyaux des doubles flèches existent. Soit $f : Z \longrightarrow Y$ une flèche de $\mathcal{M}$. Alors g est un épimorphisme si et seulement si g ne se factorise pas par un sous-objet* strict Y_0 *de Y différent de Y.*

On va faire une hypothèse supplémentaire (pratiquement anodine) sur les Filt^π :

d) Filt^π est stable par limites inductives indexées par des catégories d'indices I telles que $\operatorname{card} \mathrm{Fl}(I) \leq \pi$ [16]. (Il suffit de supposer la stabilité par objet initial, sommes amalgamées simples, et $\varinjlim$ filtrantes indexées par un ensemble ordonné I de cardinal $\leq \pi$.) De plus, si $f : Z \longrightarrow X$ est une flèche de $\mathcal{M}$ telle que Z ou X est dans Filt^π, alors $\operatorname{Im} f \in \mathrm{Filt}^\pi$.

Cette dernière condition sur $\operatorname{Im} f$ se décompose en deux : [page 46]

d_1) Tout sous-objet d'un objet de Filt^π est dans Filt^π.

d_2) Si Z est dans Filt^π, alors tout quotient Z' de Z tel que $\operatorname{Im} p = Z'$ (où $p : Z \longrightarrow Z'$ est le morphisme canonique) est dans Filt^π. [17]

Bien sûr, pratiquement il sera toujours vrai que Filt^π est stable par passage aux sous-objets et aux quotients d'un objet (pas seulement les quotients particuliers dont il est question dans (d_2). D'ailleurs, lorsque tout bimorphisme (*i.e.* monomorphisme *et* épimorphisme) de $\mathcal{M}$ est un isomorphisme, alors (d_2) équivaut bien à la stabilité de Filt^π par quotients d'objets. p. 30

Proposition 5.3. `[Deux fois 5.3.]` *Supposons les conditions* (a), (b), (d) *vérifiées pour $\mathcal{M}$. Soient π un cardinal $\geq \pi_0$, et X un objet de $\mathcal{M}$. Soit $\mathrm{Ss}_\pi(X)$ l'ensemble des sous-objets de X qui sont dans* Filt^π*. Cet ensemble, ordonné par la relation d'ordre induite par $\mathrm{Ss}(X)$, est grand devant π (et en particulier filtrant). Si de plus, les limites inductives filtrantes de monomorphismes dans $\mathcal{M}$ sont des monomorphismes* [18] *(p. ex. si les limites inductives filtrantes sont exactes à gauche), ou si les objets de* Filt^π *sont π-accessibles, on a $\varinjlim_{\alpha \in \mathrm{Ss}_\pi(X)} Z_\alpha \xrightarrow{\sim} X$* [19].

Démonstration. Soit $(Z_\alpha)_{\alpha \in A}$ une famille de sous-objets de X tels que $Z_\alpha \in \mathrm{Filt}^\pi$, et $\operatorname{card} A \leq \pi$. Soit $Z = \coprod_{\alpha \in A} Z_\alpha$, on a $Z \in \mathrm{Filt}^\pi$ par (c). Le plus petit sous-objet de X majorant les Z_α n'est autre que $\operatorname{Im}(Z \longrightarrow X)$, qui est dans Filt^π par (d_1). Prouvons que $\varinjlim_{\alpha \in \mathrm{Ss}_\pi(X)} Z_\alpha \xrightarrow{\sim} X$ est un isomorphisme, quand on sait que c'est un monomorphisme. Or [page 47]

$$X = \varinjlim_{I} X_i \ ,$$

où $(X_i)_{i \in I}$ est un système inductif filtrant, I ensemble ordonné grand devant π, les X_i dans Filt^π. Soit, pour tout $i \in I$, X'_i l'image de X_i dans X. On a donc $X'_i \in \mathrm{Filt}^\pi$ par (d_2). Utilisant

16. Cette condition sera nommée (d_0) par la suite.

17. Il semblerait qu'on n'aura pas besoin de (d_1), seulement de (d_2). Mais en fait, (d_1) doit être conséquence de (d_2), et (d_2) est conséquence de la première partie de (d) (stabilité par $\varinjlim$ finies) si tout épimorphisme dans $\mathcal{M}$ est strict, donc effectif (si les Filt^π sont stables par $\varprojlim$ finies aussi …).

18. C'est une *condition d'exactitude* de $\mathcal{M}$, que je vais noter comme :

© Les monomorphismes dans $\mathcal{M}$ sont stables par $\varinjlim$ filtrantes dans $\underline{\mathrm{Fl}}(\mathcal{M})$.

19. Dire aussi que $\mathrm{Ss}_\pi(X)$ est π-adapté, et le système inductif $(Z_\alpha)_\alpha$ aussi.

le fait que les $X_i \longrightarrow X'_i$ sont des épimorphismes, on voit de suite que $X = \varinjlim_I X'_i$. Considérons l'application

$$\begin{array}{ccc} I & \longrightarrow & \mathrm{Ss}_\pi(X) \\ i & \longmapsto & X'_i \ . \end{array}$$

C'est une application croissante, soit J son image, partie filtrante grande devant π de $\mathrm{Ss}_\pi(X)$, telle que

$$\varinjlim_{\alpha\in J} Z_\alpha \ \xrightarrow{\sim}\ X \ .$$

On a donc [page 48]

$$\underbrace{\varinjlim_{\alpha\in J} Z_\alpha}_{\simeq X} \ \longrightarrow \ \varinjlim_{\alpha\in \mathrm{Ss}_\pi X} Z_\alpha \ \xrightarrow{\varphi}\ X \ ,$$

le composé étant l'identité. Si on sait que φ est un monomorphisme, il en résulte que c'est un isomorphisme (monomorphisme ayant une section). Or φ sera un monomorphisme si toute $\varinjlim$ filtrante de monomorphismes est un monomorphisme, et alors on gagne.

Commentary · Development (commentators') [ms. 46]

Quotient bounds do not bound subobjects

The two image conditions serve different purposes. The second controls the image of a bounded source; the first controls an arbitrary subobject of a bounded target. The suggestion in footnote 17 that the second should imply the first is false even in a module category with exact filtered colimits. Giving the example explicitly explains why the two conditions must be distinguished, even in arguments where only one of them is used.

A filtration separating (d_1) and (d_2). [ours] Fix an infinite cardinal π, a field k, and a set S of cardinal greater than π. Put

$$R = k[x_s \mid s \in S], \qquad \mathrm{Filt}^\kappa(R\text{-Mod}) = \{N \mid N \text{ is generated by at most } \kappa \text{ elements}\}, \quad \kappa \geq \aleph_0.$$

This is a cardinal filtration satisfying (a), (b), (c), (d_0), (d_2) and (e), but (d_1) fails at π.

Proof. Modules generated by at most κ elements are quotients of $R^{(\kappa)}$, so they have a set of isomorphism classes. A colimit of a diagram with at most κ arrows and κ-generated values is a quotient of the sum of at most κ such modules; it is again κ-generated. This proves (d_0) and the required closure part of the filtration, and quotient closure proves (d_2). Every module is presentable for some sufficiently large regular cardinal: choose a presentation, and bound both its generators and relations; factorization and eventual equality in a sufficiently directed colimit are then checked on those sets. This proves (c). Filtered colimits of modules are exact, giving (e).

For the quantitative generation clause, let N have a chosen generating set T of size at most κ', where $\kappa' \geq \kappa$. The submodules generated by subsets of T of size at most κ form a κ^+-directed diagram with colimit N. Its indexing set has cardinal at most $(\kappa')^\kappa$. This is precisely the bound required of a cardinal filtration, including when the ring itself is much larger than κ.

Finally R is cyclic, whereas its ideal $\mathfrak{a} = (x_s \mid s \in S)$ is not π-generated. Indeed $R/\mathfrak{a} = k$ and

$$\mathfrak{a}/\mathfrak{a}^2 \cong \bigoplus_{s\in S} k\,\overline{x_s}.$$

Any generating set of $\mathfrak{a}$ maps to a spanning set of this k-vector space and therefore has cardinal at least $|S|$. Thus $\mathfrak{a} \hookrightarrow R$ disproves (d_1). □

The example does not challenge the propositions which follow: they assume (d), hence both image conditions. It also explains why increasing the cardinal can remove the difficulty in familiar concrete categories. If $\kappa \geq |R|$, every submodule of a κ-generated module has at most κ underlying elements and is itself κ-generated. We checked the filtration definition and the relevant size arguments in (Artin et al., 1972, Exposé I, 9.12–9.16); the particular counterexample above is our explicit verification of the remark's failure, with no claim of priority.

Si on suppose que les éléments de Filt^π sont π-accessibles, on gagne aussi, car on trouve : p. 31

Corollaire 5.4. *Si les objets de* Filt^π *sont* π*-accessibles (p. ex. si* $\pi \geq \pi_1$ *et* $\pi = \pi^{\pi_0}$*, p. ex.* $\pi \geq \pi_1$ *et* $\pi = 2^c$*, avec* $c \geq \pi_0$*,* cf. (3.2) *p. 16), alors pour toute partie filtrante* J *de* $\mathrm{Ss}(X)$ *grande devant* π*, telle que* $\varinjlim_{\alpha\in J} Z_\alpha \xrightarrow{\sim} X$*,* J *est cofinale dans* $\mathrm{Ss}_\pi(X)$ *(donc* $\varinjlim_{\alpha\in J} Z_\alpha \xrightarrow{\sim} \varinjlim_{\alpha\in \mathrm{Ss}(X)} Z_\alpha$*).*

C'est immédiat.

[page 49]

Scholie 5.5. Quand on fait sur $\mathcal{M}$ l'hypothèse d'exactitude assez anodine que les $\varinjlim$ filtrantes de monomorphismes sont des monomorphismes, *ou* qu'on se borne aux cardinaux π tels que les objets de Filt^π soient π-accessibles, on peut *canoniser* la représentation d'un objet X comme $\varinjlim_{i\in I} X_i$, avec $X_i \in \mathrm{Filt}^\pi$ et I grand devant π, en prenant $I = \mathrm{Ss}_\pi(X)$, et le système inductif correspondant. On a de plus l'agréable propriété supplémentaire que voici : toute partie filtrante J de I, de cardinal $\leq \pi$, admet une borne supérieure i_J, et (si on suppose que $\mathcal{M}$ satisfait la condition (e), page 46, note 18), on aura

$$X_{i_J} \;=\; \varinjlim_{i\in J} X_i \;;$$

en d'autres termes, on a une représentation de X par un système inductif (parfaitement) π*-adapté* (*cf.* XIII, p. 85, 86).

[page 50]

Soit maintenant

$$M \;\subset\; \mathrm{Ob}\,\mathcal{M}$$

une partie de $\mathrm{Ob}\,\mathcal{M}$, stable par la relation d'isomorphie. Si M est π-accessible (suivant la condition b_π de XIII, p. 88), alors en posant

$$M^\pi \;=\; M \cap \mathrm{Filt}^\pi \;,$$

M^π détermine M par la condition suivante sur $X \in \mathrm{Ob}\,\mathcal{M}$.

(5.1) $X \in M \iff$ le sous-ensemble $\mathrm{Ss}_\pi^M(X)$ de $\mathrm{Ss}_\pi(X)$ formé des $X_\alpha \in \mathrm{Ss}_\pi(X)$ tels que $X_\alpha \in M$ ($\iff X_\alpha \in M^\pi$) est cofinal dans $\mathrm{Ss}_\pi(X)$.

Notons que lorsque $X \in \mathrm{Filt}^\pi$, ce critère est trivialement valable, puisque alors $\mathrm{Ss}_\pi(X)$ admet X comme plus grand élément, et les sous-ensembles cofinaux sont ceux qui contiennent cet objet final. Donc (5.1) est intéressant seulement pour les $X \in \mathrm{Ob}\,\mathcal{M}\backslash\mathrm{Filt}^\pi$. Notons que pour *toute* partie M^π de Filt^π, on peut définir une partie M de $\mathrm{Ob}\,\mathcal{M}$ par la condition (5.1), et on aura trivialement

(5.2) $$M^\pi \;=\; M \cap \mathrm{Filt}^\pi \;.$$

Mais sans doute, bien que M satisfasse (5.1), il n'est pas pour autant nécessairement π-accessible. En effet, pour qu'il le soit, il faut qu'il soit stable par $\varinjlim$ grandes devant π d'éléments de M^π, ce qui implique que tout facteur direct d'un élément de M^π est dans M, donc aussi dans M^π, d'après la condition de stabilité (d) sur Filt^π. C'est là une condition particulière sur la partie M^π de Filt^π. Aussi, on devra avoir une condition sur $\mathcal{M}$ plus forte que la validité de (5.1), savoir :

p. 32

[page 51]

(5.3) Pour tout $X \in \mathrm{Ob}\,\mathcal{M}$ et pour toute partie J de $\mathrm{Ss}_\pi(X)$, telle que
- *a)* J est cofinale dans $\mathrm{Ss}_\pi(X)$ et
- *b)* Si $J' \subset J$ est une partie filtrante de J, de cardinal $\leq \pi$, alors sa borne supérieure $\alpha_{J'}$ dans $\mathrm{Ss}_\pi(X)$ appartient à J,

on a $X \in M$ si et seulement si $J^M \overset{\text{déf}}{=} J \cap \mathrm{Ss}_\pi^M(X)$ est cofinal dans J (ou ce qui revient au même, dans $\mathrm{Ss}_\pi(X)$)[20].

Commentary · Exposition [ms. 49]

The canonical directed union

The gain in Proposition 5.3 is a canonical presentation whose indexing order has actual suprema. Write $\mathrm{Sub}_\pi(X)$ for Grothendieck's $\mathrm{Ss}_\pi(X)$. Given at most π members $U_a \hookrightarrow X$, form their coproduct and then its image in X. The coproduct belongs to Filt^π by (d_0), and its image belongs to Filt^π by (d_2). It is their least upper bound in the full subobject lattice. These are the two assumptions used in the proof; the printed references to (c) and (d_1) on manuscript page 47 are misplaced.

Here is the colimit argument in a form that also explains the role of epimorphisms. Choose a generating presentation $X = \varinjlim_i X_i$ by Filt^π-objects, and factor each map to X as

$$X_i \twoheadrightarrow U_i \hookrightarrow X.$$

A compatible family $U_i \to T$ gives, by composition, a compatible family $X_i \to T$ and hence a unique map $X \to T$. Its restriction to U_i is the given map because $X_i \to U_i$ is epic. Therefore $\varinjlim_i U_i = X$. Passing to the image of the indexing order in $\mathrm{Sub}_\pi(X)$ does not change this colimit: above any chosen U_i, the original directed order supplies a common upper bound for every finite family of representatives.

There is consequently a section of the canonical map

$$q : \varinjlim_{U \in \mathrm{Sub}_\pi(X)} U \longrightarrow X.$$

Under (e) this map is monic, being the filtered colimit of the monomorphisms $U \to X$; a monomorphism with a section is invertible. Alternatively, if every Filt^π-object is π-accessible, any small subobject $V \hookrightarrow X$ factors through some U_i, and the factorization is a map of subobjects. The U_i are then cofinal in $\mathrm{Sub}_\pi(X)$, proving the same result without (e).

For the stronger continuity assertion of Scholie 5.5, (e) is used again. If $K \subset \mathrm{Sub}_\pi(X)$ is directed and $|K| \le \pi$, its colimit belongs to Filt^π by (d_0), and its map to X is monic by (e). It is the supremum of K. Thus the canonical diagram carries small directed suprema to colimits. This is the precise content of a *perfectly π-adapted* representation. It is stronger than the mere requirement that its index be π^+-directed.

In a presheaf category on a small category A, this construction has a concrete form once π bounds $|\mathrm{Fl}(A)|$. Use the filtration by presheaves whose total set of elements has cardinal at most π. Closing at most π chosen elements under all restriction maps still gives at most π elements. The poset of such subpresheaves is π^+-directed, its small directed suprema are unions, and their union is the whole presheaf. This familiar example is the geometric picture underlying the abstract image argument.

Si on admet déjà le "il suffit" de (5.1), alors il suffit dans (5.3) de supposer $X \in M$ et exiger qu'alors J^M soit cofinal dans J.

Mais notons la [page 52]

Commentary · Exposition [ms. 51]

Accessibility tests every adapted presentation

The word "accessible" now applies to a *class of objects*. Its meaning must be recovered from the definition to which Grothendieck refers, since it is more specific than simply saying that the corresponding full subcategory is accessible in the modern sense. The relevant condition b_π in (Grothendieck, 1990–1991q, Chapter XIII, manuscript pp. 88–89) is this: for every perfectly π-adapted presentation

$$X = \varinjlim_{i \in I} X_i, \qquad X_i \in \mathrm{Filt}^\pi,$$

20. Au lieu de J cofinale dans $\mathrm{Ss}_\pi(X)$, il suffit même que l'on ait $\varinjlim_{\alpha \in J} X_\alpha \xrightarrow{\sim} X$ – la condition correspondante sur M devient donc plus exigeante. Mais si les objets de Filt^π sont π-accessibles, cette condition est en fait équivalente à la condition plus faible.

one has $X \in M$ if and only if $\{i \mid X_i \in M\}$ is cofinal in I. Thus the test concerns all such presentations, including presentations whose maps identify elements.

Formula (5.1) applies this test to the canonical subobject presentation. It records a necessary condition, and gives a way to reconstruct a candidate class from any chosen bounded part M^π. On bounded objects the recovery formula is automatic: X itself is the largest element of $\mathrm{Sub}_\pi(X)$, so a cofinal subset must contain it. This explains (5.2), but supplies no invariance under changing the presentation.

Condition (5.3) asks for that invariance among the continuous cofinal subdiagrams of the canonical diagram. The closure requirement on J says that it contains the supremum of every directed subset of itself of cardinal at most π. It makes the induced diagram perfectly adapted. It does not turn a nonmonic presentation into a subobject presentation, and this is where the first version of Proposition 5.6 overreaches. The source itself identifies the obstruction and inserts the correction in footnote 21.

The repair on manuscript page 55 asks that M^π be closed under filtered colimits of diagrams of cardinal at most π. Then the good indices in a perfectly adapted diagram are themselves closed under the same directed suprema. They form another perfectly adapted presentation whenever they are cofinal. Lemma 5.7 can therefore be applied to that presentation, reducing its colimit to a continuous cofinal family of monomorphic stages. This is why the extra closure condition repairs the converse: it allows the comparison lemma to be used a second time, on the good part of the diagram.

Proposition 5.6.[21] *Supposons π choisi de telle façon que les objets de* Filt^π *soient π-accessibles (p. ex. $\pi = 2^c$, avec $c \geq \pi_1$,* cf. (3.2)*, p. 16). Alors, pour une partie M de* $\mathrm{Ob}\,\mathcal{M}$*, M est π-accessible si et seulement si elle satisfait au critère* (5.3).

On a vu que c'est nécessaire (tautologiquement, compte tenu de la proposition 5.3 et du scholie 5.5, bas de la page 49). Prouvons que c'est suffisant, en établissant d'abord le

Lemme 5.7. *Sous les conditions de* 5.6 *sur π, et la condition* (e) *sur $\mathcal{M}$* (p. 46)*, pour toute représentation π-adaptée $X = \varinjlim_{i\in I} X_i$ d'un objet X de $\mathcal{M}$, le sous-ensemble I' des $i \in I$ tels que $X_i \longrightarrow X$ soit un monomorphisme, est cofinal dans I. De plus, I' est stable par bornes supérieures dans I de parties filtrantes J de I' de cardinal $\leq \pi$.*

Montrons comment ce lemme 5.7 implique le "il suffit" de la proposition 5.6. Soit donc $X = \varinjlim_{i\in I} X_i$ une représentation π-adaptée de X ($X_i \in \mathrm{Filt}^\pi$, I grand devant π, *etc.*). Il faut prouver que, si 5.3 est valable, alors $X \in M$ si et seulement si l'ensemble I_M est cofinal dans I.

[page 53]

Supposons $X \in M$. Considérons le sous-ensemble $I' \subset I$ décrit dans le lemme. On a une application $I' \longrightarrow \mathrm{Ss}_\pi(X)$, soit $i \longmapsto X'_i = \mathrm{Im}\,(X_i \hookrightarrow X)$, et on a

$$X \simeq \varinjlim_{i\in I} X_i \;\simeq\; \varinjlim_{i\in I'} X_i \;\simeq\; \varinjlim_{i\in I'} X'_i \;\simeq\; \varinjlim_{\alpha\in J} Z_\alpha\ ,$$

où J est l'image de I' dans $\mathrm{Ss}_\pi(X)$. Comme les objets de Filt^π sont π-accessibles, on en conclut que J est cofinal dans $\mathrm{Ss}_\pi(X)$. Par 5.3, il s'ensuit que l'ensemble J_M des $\alpha \in J$ tels p. 33
que $Z_\alpha \in M$ est cofinal dans J, donc $\neq \emptyset$. Donc I'_M, qui est l'image inverse de J_M dans I', est en tous cas non vide. Appliquant le même résultat en remplaçant I' par le sous-ensemble $I'_{\geq i'_0}$, on voit que I'_M est cofinal dans I', donc aussi dans I, puisque I' est cofinal dans I ; donc $I_M \supset I'_M$ est cofinal dans I, *q.e.d.*

Inversement, supposons que I_M soit cofinal dans I, et prouvons $X \in M$. La difficulté ici provient du fait que la représentation $X = \varinjlim_{i\in I_M} X_i$ n'est plus adaptée à $\pi - I_M$ n'est pas nécessairement stable dans I par bornes supérieures (dans I) de parties filtrantes dans I_M, I' de cardinal $\leq \pi$. En somme, il faut prouver que M est stable par $\varinjlim$ grandes devant π sans [page 54]

21. Faux tel quel, vrai si on suppose que M^π est stable par $\varinjlim$ filtrantes.

plus, et je doute maintenant que ça résulte des seules hypothèses faites ici, qui ne concernent que les systèmes inductifs où les morphismes de transition sont des monomorphismes.

Voici, je crois, un contre-exemple. Prenons

(5.4) M = ensemble des objets de $\mathcal{M}$ isomorphes à un objet fixe $X_0 \in \mathrm{Filt}^\pi$, tel que X_0 n'ait pas d'endomorphisme qui soit un monomorphisme autre que l'identité.

Alors la condition 5.3 est satisfaite. Car :

a) Si $X \in M$, *i.e.* $X \simeq X_0$, alors $X \in \mathrm{Filt}^\pi$, donc $\mathrm{Ss}_\pi(X)$ admet X comme objet final, et J, étant cofinal, contient cet élément, donc J_M (qui le contient aussi) est cofinal.

Et :

b) Si J_M est cofinal, l'hypothèse (5.4) implique que l'ensemble des morphismes de transition entre les Z_α ($\alpha \in J_M$) est formé d'isomorphismes, donc la limite inductive X des Z_α en question est isomorphe à chacun d'eux, donc $X \simeq X_0$, *i.e.* $X \in M$.

Pourtant, M n'est π-accessible que s'il est stable par $\varinjlim$ grandes devant π, ce qui implique qu'il est stable par *facteurs directs.* Mais l'hypothèse (5.4) n'exclut nullement que X_0 admette [page 55]
un projecteur p ($p^2 = p$) non trivial. Celui-ci aura une image X' dans $\mathcal{M}$, qui sera facteur direct de X_0 non isomorphe à X_0, *i.e.* non dans M.

Ces difficultés disparaissent si on suppose que M^π est stable par $\varinjlim$ filtrantes indexées par des ensembles ordonnés I de cardinal $\leq \pi$. En effet, dans ce cas, dans la démonstration précédente, le sous-ensemble I_M est stable par bornes supérieures (dans I) de parties J de I_M filtrantes et de cardinal $\leq \pi$. Donc le système inductif $(X_i)_{i \in I_M}$ est encore π-adapté, et on peut lui appliquer le lemme 5.7, pour trouver un sous-ensemble cofinal $I'_M \subset I_M$, tel que les $X_i \longrightarrow X$ pour $i \in I'_M$ soient des monomorphismes. L'image J de I'_M dans $\mathrm{Ss}_\pi(X)$ est alors une partie de $\mathrm{Ss}_\pi(X)$ telle que $\varinjlim_{\alpha \in J} Z_\alpha = X$, donc (par l'hypothèse faite sur π) cofinale dans $\mathrm{Ss}_\pi(X)$ (5.4), avec $Z_\alpha \in M^\pi$ pour tout $\alpha \in J$. Donc par 5.3 on a $X \in M$, *q.e.d.*

Il reste à prouver le lemme 5.7. Avec les hypothèses et notations du lemme, soit donc [page 56]
$i_0 \in I$, il faut trouver $i' \in I'$ tel que $i' \geq i_0$. Considérons l'image X'_i de X_i dans X, et

Commentary · Development (commentators') [ms. 55]

A two-point model of the retract obstruction

Grothendieck's proposed counterexample asks for an object which has no proper monic self-map, but does have a nontrivial retract. The walking arrow provides such an object in an elementary ambient category, and also makes the failure visible in a perfectly adapted presentation. This gives a concrete completion of the mechanism proposed in (5.4).

A class satisfying (5.3) but failing b_π. [ours] Let $\mathcal{M} = \mathrm{Pos}$ be the category of small partially ordered sets and monotone maps. For an infinite cardinal π, use the filtration by underlying cardinality at most π, and let M consist of the objects isomorphic to the two-element chain $P = \{0 < 1\}$. Then M satisfies (5.3) but is not π-accessible.

Proof. Monomorphisms in Pos are the injective monotone maps, so every monic endomorphism of P is the identity. The constant map $P \to P$ with value 0 is nevertheless an idempotent whose image is the one-point poset 1. The ambient hypotheses are satisfied: a colimit is obtained from a disjoint union by generating the order relations and quotienting by the resulting equivalence relation, so a diagram of cardinal at most π with π-small values has a π-small colimit. Subobjects and image quotients have the same cardinal bound. Filtered colimits have underlying set the filtered colimit of the underlying sets, and an order relation between two representatives holds exactly when it holds at some later common stage. This description proves (*e*). A poset of cardinal at most π has at most π order relations; choosing representatives for its elements and making those relations hold at one common stage proves its π-accessibility. The eventual-equality part follows by the same argument on the at most π elements.

To check (5.3), first suppose $X \cong P$. Every cofinal subset of $\mathrm{Sub}_\pi(X)$ contains its largest member X, so its good members are cofinal. Conversely, suppose J^M is cofinal in a continuous cofinal subset J of $\mathrm{Sub}_\pi(X)$. Then X is the filtered colimit of the subobjects indexed by J^M. Each

has type P, and every transition between them is an isomorphism. The index is nonempty and directed, so its colimit is again P. This establishes both directions of (5.3).

For a direct failure of the actual definition b_π, put $\lambda = \pi^+$ and index a diagram by the ordinals $\alpha < \lambda$. Take $X_\alpha = P$ at 0 and at every successor ordinal, and $X_\alpha = 1$ at every nonzero limit ordinal. Give each X_α its distinguished least element. For $\alpha < \beta$, let the transition map be the constant map to that element of X_β. These maps compose as required. The colimit is 1, since every element is identified with a distinguished element at a later stage.

This diagram is perfectly π-adapted. The ordinal λ is π^+-directed. Every nonempty directed subset of cardinal at most π has a supremum below λ; if it has a maximum, the colimit is its final stage, and otherwise its supremum is a nonzero limit ordinal and both the colimit and that stage are 1. Yet the good indices, namely 0 and the successor ordinals, are cofinal, while the colimit is not in M. Thus b_π fails. □

The same phenomenon can be described as splitting an idempotent by a long filtered diagram. The continuous version above shows why replacing arbitrary presentations by subobject presentations loses precisely the needed information. We checked the definition in Chapter XIII, manuscript pp. 85–97, and the treatment of idempotents in (Artin et al., 1972, Exposé I, 8.6.5 and 10.6). We did not find this specified two-point diagram in those sources; it is an elaboration of the manuscript's counterexample, not a claim of priority.

p. 34

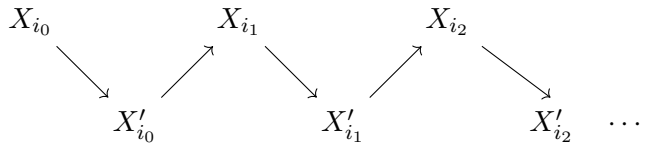
l'inclusion

$$\underbrace{X'_{i_0}}_{\in \mathrm{Filt}^\pi} \longrightarrow X \simeq \varinjlim X_i \,.$$

Comme $X'_{i_0} \in \mathrm{Filt}^\pi$, donc est π-accessible, il existe un $i_1 > i_0$ tel que $X'_{i_0} \longrightarrow X$ se factorise par

$$X'_{i_0} \longrightarrow X_{i_1} \,.$$

Appliquant ce même résultat à X_{i_1}, X'_{i_1} *etc.*, on trouve une suite infinie de morphismes



telle que les morphismes composés de deux en deux soient, sur la ligne supérieure, les transitions

$$X_{i_m} \longrightarrow X_{i_n} \qquad (\, i_m < i_n \,) \,,$$

et sur la ligne inférieure, les inclusions

$$X'_{i_m} \hookrightarrow X'_{i_n} \qquad (\hookrightarrow X) \,.$$

Soit $i_\omega = \sup_n i_n$ dans I, donc

$$X_{i_\omega} = \varinjlim_n X_{i_n} \simeq \varinjlim_n X'_{i_n} \simeq X'_{i_\omega} \,,$$

donc $X_{i_\omega} \longrightarrow X$ est un monomorphisme.

La dernière assertion de 5.7 résulte aussitôt du fait qu'une limite inductive filtrante de monomorphismes dans $\mathcal{M}$ est un monomorphisme (condition (e), p. 46). [page 57]

On peut préciser la proposition 5.6 obtenue ainsi :

Proposition 5.7. `[Deux fois 5.7.]` *Soient $\mathcal{M}$ une catégorie satisfaisant les conditions* (a), (b), (c), (d) (p. 44, 45) et (e) (p. 46). *Soit $\mathcal{M}$ une partie de* $\mathrm{Ob}\,\mathcal{M}$, *stable par isomorphie, π un cardinal tel que les objets de* Filt^π *soient π-accessibles. Les conditions suivantes sont équivalentes :*

a) *M est π-accessible, et stable par $\varinjlim$ filtrantes.*

b) *M est π-accessible, et $M \cap \mathrm{Filt}^\pi$ est stable par $\varinjlim$ filtrante de cardinal $\leq \pi$.*

c) *Il existe une partie M^π de Filt^π, stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$ et telle que pour tout $X \in \mathrm{Ob}\,\mathcal{M}$, on ait $X \in M$ si et seulement si l'ensemble J_M des $Z_\alpha \in J$ tels que $Z_\alpha \in M^\pi$ est cofinal dans J.*

(*De plus, sous la condition* (c), *on a* $M^\pi = M \cap \mathrm{Filt}^\pi$.) p. 35

c′) *Il existe une partie M_0 de Filt^π telle que M soit la plus petite partie de $\mathcal{M}$ stable par limites inductives filtrantes.* (*De plus, on peut alors prendre M_0* petite.)[22]. [page 58]

Commentary · Exposition [ms. 57]

Making the image factorization a transition

Lemma 5.7 is the comparison that makes the repaired recognition theorem work. Its diagonal construction is sound once one supplies an eventual-equality step. That step uses the accessibility of X_i as well as the accessibility of its image; factorization of the image alone does not identify the resulting composite with a transition of the original diagram.

Write $q_i : X_i \twoheadrightarrow U_i$ for the image factorization of the structural map to X. Given i_0, accessibility of $U_{i_0} \in \mathrm{Filt}^\pi$ gives $j \geq i_0$ and a factorization

$$U_{i_0} \xrightarrow{a} X_j \longrightarrow X.$$

The two maps aq_{i_0} and $X(i_0 \leq j)$ from X_{i_0} to X_j have the same composite to X. Accessibility of X_{i_0} therefore gives a further stage $i_1 \geq j$ at which these maps agree. After replacing a by its composite to X_{i_1}, the upper composite in Grothendieck's zigzag really is the transition $X_{i_0} \to X_{i_1}$. Its lower composite is the inclusion $U_{i_0} \hookrightarrow U_{i_1}$, since both maps are over X and the latter image maps monomorphically to X.

Repeating gives an interleaving of the chains (X_{i_n}) and (U_{i_n}). A cocone on either chain extends uniquely to a cocone on the other, so their colimits agree. Let $i_\omega = \sup_n i_n$. Perfect adaptation identifies X_{i_ω} with that colimit, while (e) makes the map $\varinjlim_n U_{i_n} \to X$ monic. Thus i_ω is a monomorphic stage above i_0. The same use of (e) and continuity proves that the monomorphic stages are closed under directed suprema of cardinal at most π. The choices may be non-strictly increasing; the proof does not require that the index have no maximal element.

There is a second detail in applying this lemma on manuscript page 53. The image of a continuous cofinal map of directed posets need not itself be closed under directed suprema, as the example below shows. Consequently, the image J of the monomorphic stage indices cannot simply be inserted into (5.3) without checking its closure requirement.

The correction already adopted on manuscript page 55 supplies the repair. If $X \in M$, the good small subobjects form a cofinal subset closed under small directed unions, because M^π has that closure. Starting with any monomorphic stage, choose a good small subobject containing it, then a later monomorphic stage containing that subobject, and continue alternately. At the countable supremum the stage and the union of the good subobjects agree, so the resulting stage is good. Hence good stages are cofinal. In the converse direction, if good stages are cofinal, their closure under small directed suprema makes their restricted diagram perfectly adapted. Applying Lemma 5.7 to it produces cofinally many good monomorphic stages; their subobjects are cofinal in $\mathrm{Sub}_\pi(X)$, so (5.1) gives $X \in M$. This proves the corrected assertion without imposing a closure property on the raw image of an indexing map.

22. **N.B.** On a

$$\begin{array}{ccc} (a) & \Longleftrightarrow & (b) \\ \Downarrow & & \Downarrow \\ (c') & \overset{?}{\Longrightarrow} & (c) \end{array},$$

sans conditions sur π (sauf $\pi \geq \pi_0$).

Commentary · Development (commentators') [ms. 57]

A continuous image need not be closed

The distinction just used is easy to miss because the indexing map preserves every supremum which can be formed upstairs. The difficulty is that a directed family of image values need not have coherently ordered representatives. The following example isolates that point in sets.

A perfectly adapted diagram with nonclosed image. [ours] Let I be the countable subsets of the successor ordinals below ω_1, ordered by inclusion. Choose $e \notin \omega_1$, put $X = \omega_1 \cup \{e\}$, and define subobjects of X by

$$X_S = \bigcup_{\alpha \in S} \alpha \ \cup \ \{e \mid |S| \geq 2\}.$$

With inclusion maps this is a perfectly $\aleph_0$-adapted presentation of X, and its image is cofinal among the countable subobjects of X, but that image is not closed under countable directed suprema.

Proof. The union of countably many countable subsets is countable, so I is $\aleph_1$-directed, and countable directed suprema in I are unions. If $\mathcal{K} \subset I$ is directed, its union contains two distinct ordinals exactly when some member of $\mathcal{K}$ contains both: direct the two witnesses to a common member. It follows that

$$X_{\bigcup \mathcal{K}} = \bigcup_{S \in \mathcal{K}} X_S.$$

Every X_S is countable, and the union of the whole diagram is X, proving perfect adaptation. Any countable subset of X is contained in X_S for a two-element S containing a successor ordinal above its ordinal elements; this proves cofinality. Finally every positive finite ordinal n is $X_{\{n\}}$. Their directed supremum is ω without e. No S with two or more elements gives this value, because its value contains e; a singleton gives a successor ordinal, and the empty set gives the empty set. Thus ω is not in the image. □

The literature check here was bounded to the adapted-presentation discussion in Chapter XIII, manuscript pp. 85–97, and SGA 4, Exposé I, 9.1–9.3 and 9.12–9.16. This explicit example supplies a missing verification in the present argument; we make no assertion that the observation is new.

Corollaire 5.8. *Les applications*

$$M \ \longmapsto \ M^\pi = M \cap \mathrm{Filt}^\pi \ ,$$

$$M^\pi \quad \longmapsto \quad \begin{array}{l} \textit{l'ensemble des } X \in \mathrm{Ob}\,\mathcal{M} \textit{ tels que l'ensemble} \\ \mathrm{Ss}_\pi^{M^\pi}(X) \textit{ des } Z_\alpha \in \mathrm{Ss}_\pi(X) \textit{ qui sont dans } M^\pi \\ \textit{soit cofinal } (= \textit{plus petit sous-ensemble de} \\ \mathrm{Ob}\,\mathcal{M} \textit{ stable par } \varinjlim \textit{ filtrantes}) \end{array}$$

établissent des bijections, inverses l'une de l'autre, entre l'ensemble des $M \subset \mathrm{Ob}\,\mathcal{M}$ satisfaisant les conditions équivalentes de la proposition 5.7*, et l'ensemble des parties M^π de* Filt^π *stables par* $\varinjlim$ *filtrantes de cardinal $\leq \pi$.*

Démonstration de 5.7. L'implication $(b) \Longrightarrow (c)$ est tautologique, compte tenu de la définition de la π-accessibilité de M, et de la proposition 5.3. Et $(a) \Longrightarrow (b)$ provient de la stabilité de Filt^π par $\varinjlim$ filtrantes de cardinal $\leq \pi$ (condition (d)).

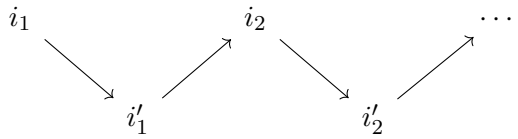
Il reste donc à prouver $(c) \Longrightarrow (a)$. (C'est ici qu'on doit utiliser l'hypothèse particulière sur π.) Une fois ceci vu, on saura l'équivalence de (a), (b), (c), et d'autre part, il est immédiat que $(a) \Longrightarrow (c')$, en prenant $M_0 = M \cap \mathrm{Filt}^\pi$. D'autre part, $(c') \Longrightarrow (c)$ (sans hypothèses sur π ?), car on peut supposer que la partie M_0 de Filt^π est stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$. Mais alors, connaissant l'équivalence de (a), (b) (c), d'où aussitôt le corollaire, il s'ensuit que l'enveloppe stable par $\varinjlim$ filtrantes de M_0 dans $\mathcal{M}$ n'est autre que la partie accessible déduite de M_0 dont il est question dans (c), et on gagne. [page 59]

Donc tout revient à prouver que, si tous les objets de Filt^π sont accessibles, alors (c) implique (a). Montrons d'abord que (c) implique le critère d'appartenance (5.3) (p. 51), donc par la proposition 5.6 "revue et corrigée" dans la note 21, il s'ensuit déjà que $(c) \Longrightarrow (b)$, et il restera à établir $(b) \Longrightarrow (a)$, *i.e.* que (b) implique la stabilité de M par $\varinjlim$ filtrantes quelconques (pas seulement grandes devant π).

Que (c) implique (5.3) va résulter du lemme suivant, appliqué à $I = \mathrm{Ss}_\pi(X)$:

Lemme 5.8. [Deux fois 5.8.] *Soit I un ensemble ordonné* π-adapté, i.e. *filtrant et tel que toute partie filtrante K de I, de cardinal $\leq \pi$, ait une borne supérieure i_K dans I. Je dis qu'une partie J de I est* π-cofinale, *si elle est cofinale, et si pour toute partie filtrante K de J de cardinal $\leq \pi$, sa borne supérieure i_K dans I est dans J. (Il s'ensuit donc que J est lui aussi π-adapté, pour la structure d'ordre induite.) Ceci dit, l'intersection de deux parties π-cofinales de I est encore π-cofinale.* p. 36 [page 60]

Il suffit de voir qu'elle est cofinale, car sa stabilité par bornes supérieures dans I de parties filtrantes de cardinal $\leq \pi$ est tautologique (si elle est vraie pour des J_α, elle est vraie pour $\bigcap_\alpha J_\alpha$). Soit $i_0 \in I$, et construisons un élément de $J \cap J'$ qui majore i_0. On construit de proche en proche une double suite



avec les i_α dans J, les i'_α dans J', et $i_1 \geq i_0$. On a donc

$$\sup i_\alpha \;=\; \sup i'_\alpha \,,$$

et si i est la valeur commune des deux membres, on a $i \in J \cap J'$.

Ceci vu, pour prouver la validité de (5.3), on note que la partie $J' = \mathrm{Ss}_\pi^{(M^\pi)}(X)$ de $I = \mathrm{Ss}_\pi(X)$, formée des $Z_\alpha \subset X$ tels que $Z_\alpha \in M^\pi$, est stable par bornes supérieures de parties filtrantes de cardinal $\leq \pi$, vu que M^π est stable par $\varinjlim$ de cardinal $\leq \pi$. Donc si J est une partie π-cofinale de $\mathrm{Ss}_\pi(X)$, $J \cap J'$ est cofinale dans J, *i.e.* dans I, si et seulement si J' est cofinale dans I (*i.e.* si et seulement si $X \in M$, par la condition (c)). Car $J \cap J'$ cofinale $\Longrightarrow$ J' cofinale tautologiquement, et J' cofinale implique J' π-cofinale, donc $J \cap J'$ π-cofinale par le lemme 5.8, donc cofinale. [page 61]

Commentary · Exposition [ms. 60]

Closed cofinal families and the four descriptions

The alternating construction in Lemma 5.8 is a small version of the closed-unbounded argument familiar in set theory. Its purpose here is representation independence. Two canonical-looking systems of small subobjects need not meet cofinally merely because each is cofinal. Closure under countable directed suprema is what allows an alternating sequence to finish in both systems.

For example, the even and odd natural numbers are cofinal in $\mathbb{N}$ and disjoint. That order lacks the required countable suprema. In a π-adapted order, with π infinite, start above any prescribed element and alternate choices in two closed cofinal subsets. The supremum of the even subsequence equals that of the odd subsequence, and belongs to both. The intersection is therefore cofinal; closure of the intersection is immediate. This gives exactly the assertion used in the manuscript, without any assumption that π itself is regular.

We can now organize the corrected Proposition 5.7. Let $S \subset \mathrm{Filt}^\pi$ be invariant under isomorphism and closed under filtered colimits with at most π arrows. Define M_S by cofinality of its S-subobjects in $\mathrm{Sub}_\pi(X)$. Under (e) these good subobjects are closed under small directed suprema. Their intersection with any other closed cofinal family is cofinal by the preceding argument. Thus M_S satisfies (5.3), and the corrected Proposition 5.6 identifies it as a π-accessible class when the Filt^π-objects are π-accessible. Proposition 5.9, in this same range of cardinals, supplies closure under all filtered colimits.

Conversely, a π-accessible class closed under all filtered colimits has bounded part $S = M \cap \mathrm{Filt}^\pi$ with the required small-colimit closure, by (d_0), and (5.1) reconstructs M. These two operations are inverse because on a bounded object cofinality is equivalent to containing the largest subobject, namely the object itself. This is Corollary 5.8.

The description by an arbitrary generating set $M_0 \subset \mathrm{Filt}^\pi$ follows by first closing M_0 under the filtered colimits of cardinal at most π which exist inside the bounded level, and under isomorphism.

Call the resulting class S. It remains essentially small because it is contained in Filt^π. The class M_S is filtered-colimit closed and contains M_0. Any filtered-colimit closed class containing M_0 contains S, and then contains M_S, since every object of M_S is a filtered colimit of its good small subobjects. Hence M_S is precisely the filtered-colimit closure of M_0.

The printed condition (c'), and the matching parenthesis in Corollary 5.8, omit the words saying that the smallest closed class must *contain* the proposed generators. Those words are indispensable: without them the smallest class closed under nonempty filtered colimits is the empty class. The intended statement and its proof are unambiguous from the surrounding discussion. The stronger question in footnote 22 concerns retaining the same rank without the special assumption on the bounded objects; it is not resolved merely by the proof at the admissible ranks used in the proposition.

Reste à prouver seulement que $(b) \Longrightarrow (a)$, *i.e.* la stabilité de M par limites inductives filtrantes quelconques, moyennant (b). Ici à nouveau, cela ne doit pas utiliser l'hypothèse particulière sur π, ni même l'hypothèse d'exactitude (e) sur $\mathcal{M}$, mais être une histoire générale d'accessibilité :

Proposition 5.9. *Soit $\mathcal{M}$ une catégorie satisfaisant* (a), (b), (c), (d_0) (pages 44, 45), *$\pi \geq \pi_0$ un cardinal, M une partie π-accessible de* $\mathrm{Ob}\,\mathcal{M}$, i.e. *satisfaisant la condition b_π de* XIII, p. 88. *Alors M est stable par $\varinjlim$ filtrantes si et seulement si $M^\pi \overset{déf}{=} M \cap \mathrm{Filt}^\pi$ est stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$.*

Je n'ai pas envie d'expliciter ici la démonstration, ayant l'impression pénible de refaire à perpette la même démonstration (ici et dans XIII). Je devrai revenir là-dessus dans une situation plus décantée.

Je vais quand même esquisser la démonstration. Soit $X = \varinjlim_I X_i$, I filtrant, les X_i dans [page 62]
M, il faut prouver que $X \in M$ (moyennant l'hypothèse de stabilité sur $M^\pi = M \cap \mathrm{Filt}^\pi$). p. 37

$$\begin{array}{ccc} \mathcal{C} & \longrightarrow & I \\ \downarrow & \text{cart.} & \downarrow \varphi \\ \underline{M}^\pi \text{ “source”} & \underline{M}^\pi/\underline{M} \xrightarrow[\text{“but”}]{} & \underline{M} \end{array}$$

$\mathcal{C}$ est une catégorie cofibrée sur I, dont les fibres sont

$$\mathcal{C}_i \simeq M^\pi/X_i\ ,$$

or comme $X_i \in M$, on voit, à cause de la condition de π-accessibilité, que M^π/X_i est *filtrante.* On va le dégager en

Lemme 5.10. *Soit M une partie π-accessible de $\mathcal{M}$. Alors pour tout $Y \in M$, M^π/Y est une catégorie* filtrante, *et $\mathrm{Ss}_\pi(Y)$ y est cofinale.*

Considérons le foncteur canonique [page 63]

$$I \;=\; \mathrm{Ss}_\pi(Y) \xrightarrow{\varphi} \underline{M}^\pi/Y \;=\; I'\ ,$$

où la catégorie source I est filtrante. Par SGA 4 I 8.1.3, ce foncteur est cofinal si et seulement s'il satisfait deux conditions F 1, F 2, et alors I' est également filtrante. La condition F 1 est la condition habituelle que tout $i' \in \mathrm{Ob}\,I'$ est “majoré” par un $\varphi(i)$, *i.e.* tout $Z \longrightarrow Y$, avec $Z \in M^\pi$, est “majoré” par un *monomorphisme* $Z_\alpha \hookrightarrow Y$, avec $Z_\alpha \in M^\pi$, ce qui résulte du fait que c'est majoré par $Z_\alpha \hookrightarrow Y$ avec $Z_\alpha \in \mathrm{Filt}^\pi$, *i.e.* $Z_\alpha \in \mathrm{Ss}_\pi(Y)$, et que parmi ces Z_α ceux qui sont dans M^π sont cofinaux (vu que $Y \in M$). L'autre condition [23] est qu'une double flèche $i' \overset{f'}{\underset{g'}{\rightrightarrows}} \varphi(i)$ peut s'égaliser par une flèche $\varphi(h)$, avec $h : i \longrightarrow j$. Mais en fait, pour un $i' \in \mathrm{Ob}\,I'$, [page 64]
i.e. un $Z \longrightarrow Y$ donné ($Z \in M^\pi$), et un $i \in \mathrm{Ob}\,I$, *i.e.* un $Z_\alpha \hookrightarrow Y$ donné ($Z_\alpha \in M^\pi$), il existe

23. Comme I est filtrante.

au plus *une* flèche dans $I' = \underline{M}^\pi/Y$ de $Z \longrightarrow Y$ dans $Z_\alpha \hookrightarrow Y$. Donc la condition F 2 est trivialement satisfaite (on prend $h = \mathrm{id}_i$).

Corollaire 5.11. *Si $(X_i)_{i\in I}$ est un système inductif filtrant, avec les X_i dans M^π, alors la catégorie $\mathcal{C} = \underline{M}^\pi/\underline{M} \times_{\underline{M}} (I, \varphi)$, construite page* 62, *est filtrante, et le foncteur canonique (fibrant) $\mathcal{C} \longrightarrow I$ est cofinal.*

Que $\mathcal{C}$ soit filtrante et $\mathcal{C} \longrightarrow I$ cofinal résulte formellement du fait que I l'est, que les fibres de $\mathcal{C}$ le sont, et que $\mathcal{C}$ est cofibrée sur I. (Mérite un lemme, à toutes fins utiles.)

N.B. On peut remplacer $\mathcal{C}$ par $\mathcal{C}_0$ dont les fibres sont les

$$\mathrm{Ss}_\pi^{M^\pi}(X_i) \ \subset \ \underline{M}^\pi \ ; \qquad \text{cofinale dans } \mathcal{C}_i = \underline{M}^\pi/X_i \ ,$$

on trouve une sous-catégorie de $\mathcal{C}$ cofinale dans $\mathcal{C}$, et qui est cette fois ordonnée tout comme I. [page 65]

Commentary · Exposition [ms. 64]

The slice category that must be integrated

The final argument returns from subobjects to maps, because maps assemble functorially when the target moves. Its fibre over $Y \in M$ is M^π/Y, the category of maps $A \to Y$ with $A \in M^\pi$. The poset naturally included in this fibre is

$$\mathrm{Sub}_\pi^M(Y) = \{U \hookrightarrow Y \mid U \in M \cap \mathrm{Filt}^\pi\}.$$

The unrestricted $\mathrm{Ss}_\pi(Y)$ printed in Lemma 5.10 and its first display must be restricted in this way. An arbitrary small subobject of an object of M need not itself belong to M, so the printed unrestricted functor is not defined in general.

Under the image and continuity hypotheses used earlier in this section, the corrected assertion follows directly from (5.1). Given $A \to Y$, its image is small by (d_2), and is contained in a good small subobject by cofinality. Hence every object of M^π/Y maps to an object of $\mathrm{Sub}_\pi^M(Y)$. There is at most one map over Y into any fixed subobject. The good subobjects form a directed poset, and their inclusion in the slice is final: the relevant comma category is nonempty, and two of its objects have a common upper bound. This also proves that M^π/Y is filtered. Its source diagram has colimit Y, because the final subobject diagram does.

For a filtered diagram $i \mapsto Y_i$ in M, the total category $\mathcal{E}$ has objects (i, A, a), where $A \in M^\pi$ and $a : A \to Y_i$. A morphism to (j, B, b) is a pair (u, v) with $u : i \to j$, $v : A \to B$, and $bv = Y(u)a$. Postcomposition gives an opfibration $\mathcal{E} \to I$. The adjective "fibrant" in Corollary 5.11 has the wrong variance for this construction; no pullback closure of M^π is assumed.

The total category is filtered. Two objects can first be moved to a common base index and then given a common successor in that fibre. For two parallel arrows, first equalize their base arrows in I, then equalize the resulting two vertical arrows in the target fibre. The same argument shows that $\mathcal{E} \to I$ is final. More particularly, a cocone on the source objects A of $\mathcal{E}$ is exactly a compatible family of cocones on the fibres. Taking their colimits gives

$$\varinjlim_{(i,A,a)\in\mathcal{E}} A \cong \varinjlim_{i\in I} \left(\varinjlim_{(A\to Y_i)\in M^\pi/Y_i} A \right) \cong \varinjlim_i Y_i.$$

This is an ordinary colimit calculation; no homotopy-cofinality theorem is being used.

Corollary 5.11 prints $Y_i \in M^\pi$, a stronger hypothesis than the argument needs. The same proof works for $Y_i \in M$, and this is the form required on manuscript page 62. If the base index is a poset, the full subcategory consisting of good subobjects in each fibre is again a poset and is final in $\mathcal{E}$. Its projection need not remain an opfibration: the image of a good subobject after postcomposition need not be good. The proof requires only finality of this subcategory, which the preceding factorization through good subobjects establishes.

p. 38

On a

$$X_i \ = \ \varinjlim_{Z_\alpha \in \mathcal{C}_i} Z_\alpha \ ,$$

donc

$$X \;=\; \varinjlim_{\alpha\in\mathcal{C}} Z_\alpha \;,$$

au total on est arrivé à représenter X comme $\varinjlim$ filtrante d'éléments Z_α qui sont (non seulement dans M, mais même) dans M^π.

Changeons la notation, en prenant

$$X \;=\; \varinjlim_{i\in I} X_i \;, \qquad I \text{ ensemble ordonné filtrant, } X_i \in M^\pi \;\; \forall\, i \in I \;,$$

soit I' l'ensemble des parties *filtrantes* $J \subset I$ telles que $\operatorname{card} J \leq \pi$, I' ordonné par inclusion. Alors I' est grand devant π, et on a un système inductif

$$(X_J)_{J\in I'} \;, \qquad X_J \;=\; \varinjlim_{i\in J} X_i \;.$$

Par l'hypothèse de stabilité de M^π (qui n'a pas encore servi), les X_J sont dans M^π. On a donc

$$X \;=\; \varinjlim_{J\in I'} X_J \;, \qquad \text{les } X_J \text{ dans } M^\pi,\ I' \text{ grande devant } \pi \;,$$

donc $X \in M$ puisque M est π-admissible, *q.e.d.*

[page 66]

Commentary · Exposition [ms. 65]

Completing the filtered-colimit closure argument

The last reindexing in Proposition 5.9 is the right way to pass from bounded filtered colimits to arbitrary ones. There is, however, a scope issue in the preceding reduction. The proposition drops the image hypotheses $(d_1), (d_2)$ and the exactness hypothesis (e), whereas the proof of Lemma 5.10 as printed uses the earlier subobject presentation. We give a complete proof under the additional assumption that every Filt^π-object is π-accessible. This is the assumption already present in Proposition 5.7 and suffices for every application to the four equivalent descriptions above. The printed general assertion for an arbitrary rank is not certified by the image proof alone.

Put $A = M \cap \mathrm{Filt}^\pi$, and suppose that A is closed under filtered colimits indexed by categories with at most π arrows. When forming slices, replace this essentially small class by a small full set of representatives; this does not change their colimits. The necessary direction of Proposition 5.9 is immediate from (d_0), so consider a filtered diagram $Y_i \in M$ with colimit Y. For each Y_i, choose a perfectly π-adapted presentation by objects of Filt^π. Such presentations exist under $(b), (d_0)$: apply the reindexing by small filtered subcategories given below to any bounded generating presentation. Condition b_π makes its A-stages cofinal. Thus Y_i is a π^+-directed colimit of objects of A.

Smallness of the domains now supplies the slice argument without images. Any finite diagram in A/Y_i factors through one stage of the chosen A-presentation: factor the finitely many structural maps, choose a common stage, and then pass further until its finitely many commutativity equations hold. The same reasoning works for a diagram with at most π arrows. Hence A/Y_i is π^+-filtered. The chosen presentation is final in it, since every map from an A-object factors through a stage and two factorizations become equal at a later stage. In particular

$$Y_i \cong \varinjlim_{A/Y_i} A.$$

The opfibration argument in the preceding block then expresses Y as a filtered colimit of A-objects, say $Y = \varinjlim_{e\in E} A_e$ with E small and filtered.

To make this presentation perfectly adapted, let P be the poset of filtered subcategories $J \subset E$ with at most π arrows. Every set of at most π arrows of E is contained in such a subcategory: repeatedly adjoin identities, composites, cocones for finite diagrams and equalizers of parallel pairs. There are at most π choices at each step, and countably many repetitions suffice, since π is infinite. This also shows that P is π^+-directed. A directed family of at most π members of P has its union as supremum; that union is still filtered and has at most π arrows.

Define $A_J = \varinjlim_{e\in J} A_e$. By the assumed bounded closure, $A_J \in A$. Colimits commute with this directed union of indexing categories: a cocone on the union is exactly a compatible family of cocones on the pieces. Consequently

$$Y \cong \varinjlim_{J\in P} A_J, \qquad A_{\bigcup K} \cong \varinjlim_{J\in K} A_J \quad (K \subset P \text{ directed},\ |K| \le \pi).$$

This is a perfectly π-adapted presentation with every stage in M. Applying b_π gives $Y \in M$, which completes the proof in the stated range.

The proof also makes the remaining issue precise. To remove the extra presentability assumption one needs, under the proposition's weaker hypotheses, a replacement for the slice factorization and density argument. The statement on manuscript page 44 about generation by strict epimorphisms would provide such density if established at the required bounded level, but it should not be inferred merely from a choice of colimit presentations. Density and generation under colimits are distinct notions. The earlier discussion already flags this difficulty: Chapter XIII, manuscript p. 92, questions the asserted strict generation, and footnote 44 on manuscript p. 93 explicitly identifies the tacit hypothesis that the bounded objects are π-accessible (Grothendieck, 1990–1991q). We have not settled that unrestricted fixed-rank step here; this is a limit of the present verification, not a claim that it is an open problem in the literature.

Définition 5.12.

1) Soit $\mathcal{M}$ une catégorie satisfaisant les conditions (a), (b), (c), (d) (p. 44-46). Soit $\pi \ge \pi_0$ un cardinal. Une partie M de $\operatorname{Ob}\mathcal{M}$ est dite *L-π-accessible,* si elle est π-accessible et stable par petites $\varinjlim$ filtrantes[24], ou ce qui revient au même (5.9), si elle est π-accessible et si $M^\pi = M \cap \operatorname{Filt}^\pi$ est stable par $\varinjlim$ filtrantes de cardinal $\le \pi$. (**N.B.** Si les objets de Filt^π sont π-accessibles, 5.8 nous dit que cela signifie aussi qu'il suffit pour cela qu'il existe une partie M_0 de Filt^π telle que M soit la L-enveloppe (enveloppe stable par $\varinjlim$ filtrantes) de M_0 – condition qui est de toutes façons toujours nécessaire, en prenant $M_0 = M^\pi$.)
2) Soit M une partie de $\operatorname{Ob}\mathcal{M}$. On dit qu'elle est *L-accessible,* si elle satisfait à l'une des quatre conditions équivalentes suivantes :
   - a) Il existe un cardinal π tel que M soit L-π-accessible.
   - a') M est accessible et L-stable.
   - b) Il existe une petite partie M_0 de $\operatorname{Ob}\mathcal{M}$, telle que M soit la L-enveloppe de M_0[25].
   - b') Il existe une partie M_0 de $\mathcal{M}$ contenue dans une partie Filt^π (ou ce qui revient au même, telle que la sous-catégorie pleine $\underline{M}_0$ de $\mathcal{M}$ soit essentiellement petite, *i.e.* équivalente à une petite catégorie, telle que M soit la *L*-enveloppe de M_0).

Commentary · Later developments [ms. 66]

Filtered-colimit classes and modern accessible categories

Definition 5.12 isolates the feature needed for the intended application: a class is generated by bounded objects and is closed under *all* filtered colimits. At the good cardinals used in Proposition 5.7, its full subcategory is π^+-accessible in the current sense, and its inclusion into $\mathcal{M}$ preserves filtered colimits. Indeed its bounded objects are π^+-presentable even in the ambient category, every object has a π^+-directed presentation by them, and the required colimits exist in the full subcategory because they are computed in $\mathcal{M}$. This is the familiar small-generation description of accessible categories; see (Adámek and Rosický, 1994, Definition 2.1 and Theorem 2.26). The manuscript's b_π also controls every continuous bounded presentation, so one should not identify its fixed-rank definition with the modern definition without the comparison just proved.

A simple example shows that L-accessibility does not mean local presentability of the full subcategory. In Set, let M be the nonempty sets. Its bounded part consists of the nonempty sets of

24. On dit alors que M est *L-stable.*
25. Cette condition ne dépend pas de la filtration cardinale choisie de $\mathcal{M}$.

cardinal at most π and is closed under filtered colimits of cardinal at most π. For a nonempty X, every small subset can be enlarged to a nonempty small subset; for $X = \varnothing$ there is no good subobject at all. Thus the small-subobject criterion recovers exactly M. Filtered colimits of nonempty sets are nonempty, so this is an L-accessible class. Its full subcategory has no initial object: a singleton has more than one map to a two-element set, and a larger set does not have a unique map either. It is therefore accessible but not cocomplete.

Under the hypotheses of Proposition 5.7, the statement independent of a chosen filtration is the existence of an essentially small family whose filtered-colimit closure is the given class. To pass from a small family to a bounded one, bound the filtration indices of its representatives. To pass back, take representatives of the bounded level. At a sufficiently large cardinal satisfying the accessibility requirement, Proposition 5.7 identifies this closure with the class detected by cofinally many good small subobjects. The cardinal filtration is then a means of making the construction uniform; the resulting notion of L-accessibility does not retain that choice.

There is a final scope convention to keep visible. Definition 5.12 restates $(a), (b), (c), (d)$ but omits (e). The definition of L-π-accessibility itself makes sense with those hypotheses. Its cited classification through Proposition 5.7, however, was proved using (e), and our small-subobject argument retains that hypothesis. We have not supplied a proof removing it from the classification, just as we have not removed the rank restriction from Proposition 5.9. These qualifications concern the claimed equivalences under weaker hypotheses, not the definition or the preceding example.

This is the part of the chapter which is closest to the later theory of accessible and accessibly embedded subcategories. It supplies a method for obtaining a small amount of generating data, while the earlier factorization argument turns small generating *maps* into lifting factorizations. These are different passages: existence of a small generating family under filtered colimits does not by itself assert that the same maps generate a weak factorization system. The additional closure and lifting arguments remain necessary.

Overview · XV-a

Grothendieck devotes these pages to the passage from a small amount of generating data to a factorization theory on a large category. The point is not merely to factor an individual map. The construction must vary with that map, remain compatible with enlargement of the generators, and admit a bound which is independent of the particular object being treated. Only then can it contribute to the construction of categories of models and, through them, to the homotopy theories of diagrams which the manuscript seeks to organize.

The chapter brings together two kinds of control. Lifting properties say what a factorization must accomplish; cardinal filtrations say when the construction has accomplished enough and how large its intermediate objects can become. The former belongs to the language of model categories, while the latter continues the systematic treatment of generation and accessibility in SGA 4, Exposé I. Their interaction is the mathematical subject of this part of Chapter XV. The later discussion of accessible classes is therefore not a digression. It addresses the problem of replacing a class of prospective generators by a set to which the preceding construction can be applied.

Attaching solutions and taking a sufficiently long limit. For a set C_0 of arrows and a map $f : X \to Y$, the first stage attaches, simultaneously, one solution for every square from an element of C_0 to f. Grothendieck realizes this simultaneous attachment as the filtered colimit of the finite amalgamations. This permits him to work with the relevant pushouts and filtered colimits, without assuming arbitrary coproducts at the outset. The residual map to Y is then subjected to the same operation, and limit stages are formed by filtered colimits. The result is the classical small object construction, expressed as a transfinite sequence of functorial factorizations; compare Riehl (2014, Chapter 12).

The decisive step is a factorization of a map from the domain of a generator through an earlier stage. If these domains are π-accessible in the manuscript's convention, a limit ordinal of cofinality greater than π suffices. A square reaching the limit then already occurs at some stage and receives its solution at the next. The word *limit* is necessary: an ordinal with a last predecessor can satisfy the order-theoretic definition of "grand devant π" while providing no further stage in which to attach the required solution. The printed statements omit this qualification. Their usual successor-cardinal choice of stopping ordinal has the required property and repairs them.

The cellular left class is compared with the left lifting class by the familiar retract argument. Factoring a map which already has the left lifting property exhibits it as a retract of its cellular factor. The commentary also supplies the functoriality arguments left to the reader, including compatibility with inclusions of generating sets. One should not confuse completion of the lifting task with stabilization of the sequence: for the sole generator $\varnothing \to 1$ in sets, the stages have the form $X \amalg (Y \times \alpha)$. They continue growing after the residual map has become surjective. Garner's algebraic refinement treats the additional structure of chosen fillers and its composition laws (Garner, 2009); that structure is stronger than the functoriality established here.

The arithmetic behind a uniform factorization. A cardinal filtration presents a large object by sufficiently directed diagrams of smaller ones. The two base cardinals have distinct roles: π_0 bounds the complexity of these presentations, while π_1 provides a uniform accessibility bound for the first bounded level. The estimates then depend on a small category of representatives and on its actual arrows. Counting only isomorphism classes in the arrow category loses the automorphisms and cannot bound individual Hom sets. The one-object category of an arbitrarily large small group gives a particularly short counterexample: its arrows have one isomorphism class, although its unique Hom set has the cardinality of the group.

Choose a small full skeleton K_1 of the π_1-level and put

$$\chi = \max(\aleph_0, \pi_1, |\operatorname{Mor}(K_1)|), \qquad \mu = r^{\pi_0}, \qquad r, s \geq \chi.$$

The corrected counting argument gives the intended forms of the bounds: at most 2^μ isomorphism classes of objects and arrows at level r, and at most s^μ maps from an r-bounded object to an s-bounded one. The number of choices of a diagram must be counted by exponentiation, not by the printed products. With that change the final estimates still follow. The asserted universal sharpness of the two-coloured-set example does not: under GCH its arrow-orbit count at $r = \aleph_1$ is smaller than 2^r.

The fixed-point device is robust. Taking a sufficiently large $\rho = 2^c$ makes $\rho^{\pi_0} = \rho$ and absorbs the repeated attachment bounds. The manuscript's alternative GCH shortcut needs a cofinality condition: GCH does not imply $\rho^{\pi_0} = \rho$ at singular cardinals of cofinality at most π_0. The commentary separates these two choices and proves a bound retaining the actual number of generators. If their endpoints lie at level r, the input endpoints at level s, and the construction is followed through α, then

$$B = \left(\max(2, r, s, |C_0|, |\alpha|)\right)^{r^{\pi_0}}$$

bounds every stage, under the stated closure hypotheses. Thus the principal uniform-factorization conclusion survives the correction of its numerical threshold. Low's later analysis identifies regular-cardinal conditions under which factorization functors preserve presentable objects and a small subcategory detects a combinatorial model structure (Low, 2016, Theorem 4.4, Theorem 5.9, Proposition 5.12).

Recognizing a category of models. Section 4 separates the elementary logic of lifting from the smallness needed to produce the factorizations. A precouple supplies lifting and factorization; closing its two classes under retracts gives the corresponding weak factorization system. Two compatible systems, together with the cancellation properties of W, provide the model-category identities. This distinction explains both the strength of Proposition 4.2 and the defect in the later footnote which treats unclosed precouples as though they were already closed. A set-theoretic counterexample in the commentary satisfies the displayed equalities but fails retract closure.

The class Cof_W records preservation of weak equivalences by pushout. An arrow in $W \cap \mathrm{Cof}_W$ need not, under the hypotheses printed at that point, remain in W after every cobase change. A four-object diamond lattice makes the difference explicit. The omitted sufficient hypothesis is already present in Chapter XIII: every map must factor as a W-cofibration followed by a weak equivalence. The commentary proves the resulting identification with couniversal weak equivalences. The eventual recognition theorem, Proposition 4.4.6, supplies this factorization through its first weak factorization system, so its principal conclusion is preserved.

For the existence of the second small generating family, modern accessibility hypotheses give a precise answer. Smith's theorem as published by Beke (2000, Theorem 1.7 and Proposition 1.15) supplies the relevant solution-set argument in a locally presentable category. The formulation in Lurie (2009, Proposition A.2.6.15, updated online version) is especially close to the proposed construction from W and Cof_W. These results resolve the problem in their stated setting; they do not justify deleting hypotheses from an arbitrary cardinally filtered category. Once such a combinatorial model category is obtained, its diagram homotopy categories form a strong derivator (Groth, 2013, Proposition 1.30). This identifies the later destination of the machinery without treating this portion of the manuscript as a proof of every derivator axiom.

Recovering a class from its bounded part. Section 5 replaces presentations by maps with presentations by small subobjects. Images, closure of bounded levels under suitable

colimits, and exactness of filtered colimits make the poset of small subobjects into a continuous, sufficiently directed presentation of the object. The hypotheses on images are independent in the direction questioned by the source: a cyclic module may contain a submodule needing arbitrarily many generators. The commentary gives the complete polynomial-ring example, and distinguishes a strict monomorphism from an arbitrary proper monomorphism in the preceding image lemmas.

The definition of an accessible class used here tests *every* perfectly adapted presentation. Recognizing membership from cofinally many good small subobjects is therefore initially only a necessary condition. The first version of Proposition 5.6 misses the information carried by nonmonic presentations; Grothendieck himself corrects it in footnote 21. A continuous diagram of two-point and one-point posets makes the retract obstruction explicit. Closure of the bounded part under small filtered colimits repairs the assertion. The remaining comparison is a closed-cofinal-set argument: two cofinal families meet cofinally when both contain the relevant directed suprema. The supplied proofs include the eventual-equality step in Lemma 5.7 and avoid assuming that the image of a continuous cofinal indexing map is itself closed.

At cardinals where the bounded objects are themselves π-accessible, these arguments give the four descriptions in the corrected Proposition 5.7, the recovery bijection of Corollary 5.8, and filtered-colimit closure. Proposition 5.9 is printed in wider generality. Its proof invokes a slice argument whose supplied justification uses hypotheses no longer retained. We prove the proposition at the cardinals required for the preceding recognition theorem, but do not certify the unrestricted fixed-rank assertion. This is a stated limit of the present verification, not a designation of an open problem. In the verified range, the resulting full subcategory is accessible and accessibly embedded; it need not be cocomplete, as the nonempty sets show (Adámek and Rosický, 1994, Definition 2.1).

The commentary keeps these distinctions visible in its apparatus. The edition remains the authority for the historical text; genuine transcription corrections are applied, while slips visible in the edition are retained and explained. The claim register records the source's proof status separately from the repair. Results developed here carry the marker [ours] in their commentary blocks and no claim of priority.

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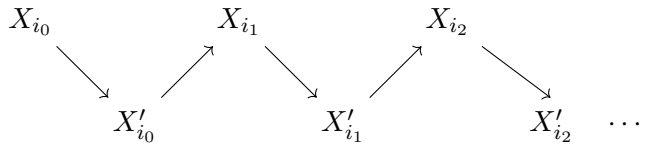
l'inclusion

$$\underbrace{X'_{i_0}}_{\in \mathrm{Filt}^\pi} \longrightarrow X \simeq \varinjlim X_i \,.$$

Comme $X'_{i_0} \in \mathrm{Filt}^\pi$, donc est π-accessible, il existe un $i_1 > i_0$ tel que $X'_{i_0} \longrightarrow X$ se factorise par

$$X'_{i_0} \longrightarrow X_{i_1} \,.$$

Appliquant ce même résultat à X_{i_1}, X'_{i_1} *etc.*, on trouve une suite infinie de morphismes



telle que les morphismes composés de deux en deux soient, sur la ligne supérieure, les transitions

$$X_{i_m} \longrightarrow X_{i_n} \qquad (\, i_m < i_n \,) \,,$$

et sur la ligne inférieure, les inclusions

$$X'_{i_m} \hookrightarrow X'_{i_n} \qquad (\hookrightarrow X) \,.$$

Soit $i_\omega = \sup_n i_n$ dans I, donc

$$X_{i_\omega} = \varinjlim_n X_{i_n} \simeq \varinjlim_n X'_{i_n} \simeq X'_{i_\omega} \,,$$

donc $X_{i_\omega} \longrightarrow X$ est un monomorphisme.

La dernière assertion de 5.7 résulte aussitôt du fait qu'une limite inductive filtrante de monomorphismes dans $\mathcal{M}$ est un monomorphisme (condition (e), p. 46). [page 57]

On peut préciser la proposition 5.6 obtenue ainsi :

Commentary · Exposition [ms. 57]

Small pieces and recognition of a class

Grothendieck wants a supply of small generators which remains useful after one imposes several conditions on an object or on a morphism. The immediate example will be a class of cofibrations intersected with a class of weak equivalences. Before asking whether such an intersection generates a factorization system, he separates the more elementary question: can membership be recognized on sufficiently many small pieces? This is the purpose of the end of §5. Its positive results survive the failure of the proposed factorization theorem in §6.

Here and below all indexing categories are small in the fixed universe, and π is infinite. Put $\mathcal{K}_\pi = \underline{\mathrm{Filt}}^\pi(\mathcal{M})$ and write $\mathrm{Ss}_\pi(X)$ for the poset of subobjects of X whose domains belong to $\mathcal{K}_\pi$. A poset is "large with respect to π" when every subset of cardinal at most π has an upper bound; in current terminology it is π^+-directed. A π-adapted presentation has the additional continuity property

$$X_{\sup J} \simeq \varinjlim_{j \in J} X_j \quad \text{for every directed } J \text{ with } |J| \leq \pi, \qquad X_i \in \mathcal{K}_\pi.$$

The last requirement is part of the definition. It will be precisely the requirement lost on manuscript page 83. The term "π-accessible object" means that its covariant hom functor preserves these large filtered colimits, hence means π^+-presentable in the modern cardinal convention. Accessibility of a *class* in these pages is the stronger, explicitly embedded recognition condition b_π: in every adapted presentation of X, the object X belongs to M exactly when the indices with $X_i \in M$ are cofinal. It must not be replaced without argument by the assertion that the full category on M is accessible. The definition is recorded explicitly in Chapter XIII, manuscript pages 86–89 (Grothendieck, 1990–1991e).

The standing assumptions from the preceding pages have concrete roles. Cocompleteness supplies the colimits. The cardinal filtration makes each $\mathcal{K}_\pi$ essentially small and supplies presentations. Condition (d) on manuscript pages 45–46 includes closure under colimits of diagrams with at most π arrows, under subobjects, and under images of maps with small source. Condition (e) says that filtered colimits in the arrow category preserve monomorphisms. Consequently the union of small subobjects is computed by their filtered colimit, and $\mathrm{Ss}_\pi(X)$ itself gives an adapted presentation. These hypotheses concern ordinary colimits, before homotopy enters the discussion.

There are two independent meanings of generation to keep in view. The L-envelope of M_0 is its closure under arbitrary small filtered colimits. The class $\widetilde{C}$ of arrows is a lifting closure, namely ${}^{\square}(C^{\square})$. A filtered presentation of a morphism need not be a sequence of pushout attachments. Nothing in Proposition 5.7 identifies these two closure operations.

The proposition compares three descriptions of an L-accessible class: an intrinsic recognition condition; closure of its bounded part under bounded filtered colimits; and a cofinality test on small subobjects. In condition (c), J is $\mathrm{Ss}_\pi(X)$. In condition (c′), the phrase "containing M_0" is absent from the edition and has to be understood; otherwise the smallest filtered-colimit-closed class is the empty class. The same omission occurs in the parenthesis of Corollary 5.8. We retain these printed readings and record their intended meaning in the audit. The duplicate numbers 5.7 and 5.8 likewise belong to the edition: the proposition and corollary here are distinct from the lemmas with those numbers.

Proposition 5.7. [Deux fois 5.7.] *Soient $\mathcal{M}$ une catégorie satisfaisant les conditions* (a), (b), (c), (d) (p. 44, 45) et (e) (p. 46). *Soit $\mathcal{M}$ une partie de* $\mathrm{Ob}\mathcal{M}$*, stable par isomorphie, π un cardinal tel que les objets de* Filt^π *soient π-accessibles. Les conditions suivantes sont équivalentes :*

a) *M est π-accessible, et stable par $\varinjlim$ filtrantes.*

b) *M est π-accessible, et $M \cap \mathrm{Filt}^\pi$ est stable par $\varinjlim$ filtrante de cardinal $\leq \pi$.*

c) *Il existe une partie M^π de* Filt^π*, stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$ et telle que pour tout $X \in \mathrm{Ob}\mathcal{M}$, on ait $X \in M$ si et seulement si l'ensemble J_M des $Z_\alpha \in J$ tels que $Z_\alpha \in M^\pi$ est cofinal dans J.*

(*De plus, sous la condition* (c), *on a* $M^\pi = M \cap \mathrm{Filt}^\pi$.) p. 35

c′) *Il existe une partie M_0 de* Filt^π *telle que M soit la plus petite partie de $\mathcal{M}$ stable par limites inductives filtrantes.* (*De plus, on peut alors prendre M_0* petite.) [22]. [page 58]

Corollaire 5.8. *Les applications*

$$M \longmapsto M^\pi = M \cap \mathrm{Filt}^\pi ,$$

$$M^\pi \longmapsto \begin{array}{l} \text{l'ensemble des } X \in \mathrm{Ob}\mathcal{M} \text{ tels que l'ensemble} \\ \mathrm{Ss}_\pi^{M^\pi}(X) \text{ des } Z_\alpha \in \mathrm{Ss}_\pi(X) \text{ qui sont dans } M^\pi \\ \text{soit cofinal } (= \text{plus petit sous-ensemble de } \mathrm{Ob}\mathcal{M} \\ \text{stable par } \varinjlim \text{ filtrantes}) \end{array}$$

établissent des bijections, inverses l'une de l'autre, entre l'ensemble des $M \subset \mathrm{Ob}\mathcal{M}$ satisfaisant les conditions équivalentes de la proposition 5.7*, et l'ensemble des parties M^π de* Filt^π *stables par $\varinjlim$ filtrantes de cardinal $\leq \pi$.*

Démonstration de 5.7. L'implication $(b) \Longrightarrow (c)$ est tautologique, compte tenu de la définition de la π-accessibilité de M, et de la proposition 5.3. Et $(a) \Longrightarrow (b)$ provient de la stabilité de Filt^π par $\varinjlim$ filtrantes de cardinal $\leq \pi$ (condition (d)).

22. **N.B.** On a

$$\begin{array}{ccc} (a) & \Longleftrightarrow & (b) \\ \Downarrow & & \Downarrow \\ (c') & \overset{?}{\Longrightarrow} & (c) \end{array} ,$$

sans conditions sur π (sauf $\pi \geq \pi_0$).

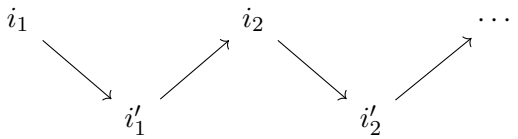
Il reste donc à prouver $(c) \Longrightarrow (a)$. (C'est ici qu'on doit utiliser l'hypothèse particulière sur π.) Une fois ceci vu, on saura l'équivalence de (a), (b), (c), et d'autre part, il est immédiat que $(a) \Longrightarrow (c')$, en prenant $M_0 = M \cap \mathrm{Filt}^\pi$. D'autre part, $(c') \Longrightarrow (c)$ (sans hypothèses sur π ?), car on peut supposer que la partie M_0 de Filt^π est stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$. Mais alors, connaissant l'équivalence de (a), (b) (c), d'où aussitôt le corollaire, il s'ensuit que l'enveloppe stable par $\varinjlim$ filtrantes de M_0 dans $\mathcal{M}$ n'est autre que la partie accessible déduite de M_0 dont il est question dans (c), et on gagne. [page 59]

Donc tout revient à prouver que, si tous les objets de Filt^π sont accessibles, alors (c) implique (a). Montrons d'abord que (c) implique le critère d'appartenance (5.3) (p. 51), donc par la proposition 5.6 "revue et corrigée" dans la note 21, il s'ensuit déjà que $(c) \Longrightarrow (b)$, et il restera à établir $(b) \Longrightarrow (a)$, *i.e.* que (b) implique la stabilité de M par $\varinjlim$ filtrantes quelconques (pas seulement grandes devant π).

Que (c) implique (5.3) va résulter du lemme suivant, appliqué à $I = \mathrm{Ss}_\pi(X)$:

Lemme 5.8. `[Deux fois 5.8.]` *Soit I un ensemble ordonné* π-adapté, i.e. *filtrant et tel que toute partie filtrante K de I, de cardinal $\leq \pi$, ait une borne supérieure i_K dans I. Je dis qu'une partie J de I est* π-cofinale, *si elle est cofinale, et si pour toute partie filtrante K de J de cardinal $\leq \pi$, sa borne supérieure i_K dans I est dans J. (Il s'ensuit donc que J est lui aussi π-adapté, pour la structure d'ordre induite.) Ceci dit, l'intersection de deux parties π-cofinales de I est encore π-cofinale.* p. 36 [page 60]

Il suffit de voir qu'elle est cofinale, car sa stabilité par bornes supérieures dans I de parties filtrantes de cardinal $\leq \pi$ est tautologique (si elle est vraie pour des J_α, elle est vraie pour $\bigcap_\alpha J_\alpha$). Soit $i_0 \in I$, et construisons un élément de $J \cap J'$ qui majore i_0. On construit de proche en proche une double suite



avec les i_α dans J, les i'_α dans J', et $i_1 \geq i_0$. On a donc

$$\sup i_\alpha \;=\; \sup i'_\alpha \;,$$

et si i est la valeur commune des deux membres, on a $i \in J \cap J'$.

Ceci vu, pour prouver la validité de (5.3), on note que la partie $J' = \mathrm{Ss}_\pi^{(M^\pi)}(X)$ de $I = \mathrm{Ss}_\pi(X)$, formée des $Z_\alpha \subset X$ tels que $Z_\alpha \in M^\pi$, est stable par bornes supérieures de parties filtrantes de cardinal $\leq \pi$, vu que M^π est stable par $\varinjlim$ de cardinal $\leq \pi$. Donc si J est une partie π-cofinale de $\mathrm{Ss}_\pi(X)$, $J \cap J'$ est cofinale dans J, *i.e.* dans I, si et seulement si J' est cofinale dans I (*i.e.* si et seulement si $X \in M$, par la condition (c)). Car $J \cap J'$ cofinale $\Longrightarrow J'$ cofinale tautologiquement, et J' cofinale implique J' π-cofinale, donc $J \cap J'$ π-cofinale par le lemme 5.8, donc cofinale. [page 61]

Commentary · Exposition [ms. 60]

Why a closed cofinal set is the right index

Grothendieck strengthens cofinality because it is not preserved by intersection alone. Two cofinal subsets of a directed set can be disjoint: the even and odd natural numbers already show this. His extra condition is continuity under bounded directed suprema. It ensures that alternating between two sets eventually produces a point belonging to both.

For clarity, the indexing hypotheses imply more than finite directedness. If $S \subset I$ has cardinal at most π, start with S, choose upper bounds for all pairs, repeat this operation countably many times, and call the union K. Since π is infinite, $|K| \leq \pi$; the set K is directed. Its supremum exists by π-adaptedness and bounds S. Thus I is π^+-directed. If $J \subset I$ is cofinal and closed under the specified suprema, the same construction, with each new upper bound subsequently enlarged into J, shows that J has the corresponding properties.

Now let J, J' be closed and cofinal, and fix $i_0 \in I$. Choose

$$i_0 \le j_0 \le j_0' \le j_1 \le j_1' \le \cdots, \qquad j_n \in J, \quad j_n' \in J'.$$

Both subsequences have the same supremum. Closure places that supremum in both sets, and it still lies above i_0. This proves cofinality of $J \cap J'$; closure is immediate. The argument uses only a countable supremum, so no regularity assumption on π is needed. The ambient convention $\pi \ge \aleph_0$ matters: the result, isolated from that convention and read for finite π, would fail for the even and odd subsets of $\mathbb{N}$.

Apply this observation to the small subobjects of an object X. If $N \subset \mathrm{Filt}^\pi$ is closed under filtered colimits of cardinal at most π, then

$$J_N(X) = \{Z \in \mathrm{Ss}_\pi(X) : Z \in N\}$$

is closed under the corresponding suprema. Therefore, whenever $J \subset \mathrm{Ss}_\pi(X)$ is closed and cofinal, $J_N(X)$ is cofinal if and only if $J \cap J_N(X)$ is cofinal. Membership is independent of the chosen continuous cofinal subsystem. This is the precise reason for the closure assumption on M^π in the corrected version of Proposition 5.6.

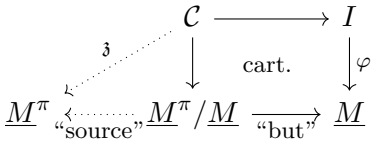
The opening zigzag of these pages explains the remaining comparison with an arbitrary adapted presentation. At a cardinal for which the X_i and their images are π-accessible, a map from $\mathrm{Im}(X_i \to X)$ into the colimit factors through a later stage. One must also enlarge that stage so that the composite from X_i equals the transition map, rather than merely having the same composite to X. Preservation of the filtered colimit by $\mathrm{Hom}(X_i, -)$ supplies this equality. Repeating the construction gives the interlacing diagram printed on p. 34. At its countable supremum the two colimits agree; the image-side colimit maps monomorphically to X by condition (e). Hence there are cofinally many monomorphic stages. This fills the implicit equality step without changing Grothendieck's argument.

Reste à prouver seulement que $(b) \Longrightarrow (a)$, *i.e.* la stabilité de M par limites inductives filtrantes quelconques, moyennant (b). Ici à nouveau, cela ne doit pas utiliser l'hypothèse particulière sur π, ni même l'hypothèse d'exactitude (e) sur $\mathcal{M}$, mais être une histoire générale d'accessibilité :

Proposition 5.9. *Soit $\mathcal{M}$ une catégorie satisfaisant* (a), (b), (c), (d_0) (pages 44, 45), *$\pi \ge \pi_0$ un cardinal, M une partie π-accessible de* Ob$\mathcal{M}$, i.e. *satisfaisant la condition b_π de* XIII, p. 88. *Alors M est stable par $\varinjlim$ filtrantes si et seulement si $M^\pi \overset{déf}{=} M \cap \mathrm{Filt}^\pi$ est stable par $\varinjlim$ filtrantes de cardinal $\le \pi$.*

Je n'ai pas envie d'expliciter ici la démonstration, ayant l'impression pénible de refaire à perpette la même démonstration (ici et dans XIII). Je devrai revenir là-dessus dans une situation plus décantée.

Je vais quand même esquisser la démonstration. Soit $X = \varinjlim_I X_i$, I filtrant, les X_i dans [page 62]
M, il faut prouver que $X \in M$ (moyennant l'hypothèse de stabilité sur $M^\pi = M \cap \mathrm{Filt}^\pi$). p. 37



$\mathcal{C}$ est une catégorie cofibrée sur I, dont les fibres sont

$$\mathcal{C}_i \simeq M^\pi/X_i\,,$$

or comme $X_i \in M$, on voit, à cause de la condition de π-accessibilité, que M^π/X_i est *filtrante.* On va le dégager en

Lemme 5.10. *Soit M une partie π-accessible de $\mathcal{M}$. Alors pour tout $Y \in M$, M^π/Y est une catégorie* filtrante, *et* $\mathrm{Ss}_\pi(Y)$ *y est cofinale.* [page 63]

Considérons le foncteur canonique

$$I \;=\; \mathrm{Ss}_\pi(Y) \xrightarrow{\varphi} \underline{M}^\pi/Y \;=\; I' \;,$$

où la catégorie source I est filtrante. Par SGA 4 I 8.1.3, ce foncteur est cofinal si et seulement s'il satisfait deux conditions F 1, F 2, et alors I' est également filtrante. La condition F 1 est la condition habituelle que tout $i' \in \mathrm{Ob}I'$ est "majoré" par un $\varphi(i)$, *i.e.* tout $Z \longrightarrow Y$, avec $Z \in M^\pi$, est "majoré" par un *monomorphisme* $Z_\alpha \hookrightarrow Y$, avec $Z_\alpha \in M^\pi$, ce qui résulte du fait que c'est majoré par $Z_\alpha \hookrightarrow Y$ avec $Z_\alpha \in \mathrm{Filt}^\pi$, *i.e.* $Z_\alpha \in \mathrm{Ss}_\pi(Y)$, et que parmi ces Z_α ceux qui sont dans M^π sont cofinaux (vu que $Y \in M$). L'autre condition [23] est qu'une double flèche $i' \overset{f'}{\underset{g'}{\rightrightarrows}} \varphi(i)$ peut s'égaliser par une flèche $\varphi(h)$, avec $h : i \longrightarrow j$. Mais en fait, pour un $i' \in \mathrm{Ob}I'$, [page 64]
i.e. un $Z \longrightarrow Y$ donné ($Z \in M^\pi$), et un $i \in \mathrm{Ob}I$, *i.e.* un $Z_\alpha \hookrightarrow Y$ donné ($Z_\alpha \in M^\pi$), il existe au plus *une* flèche dans $I' = \underline{M}^\pi/Y$ de $Z \longrightarrow Y$ dans $Z_\alpha \hookrightarrow Y$. Donc la condition F 2 est trivialement satisfaite (on prend $h = \mathrm{id}_i$).

Corollaire 5.11. *Si $(X_i)_{i\in I}$ est un système inductif filtrant, avec les X_i dans M^π, alors la catégorie $\mathcal{C} = \underline{M}^\pi/\underline{M} \times_{\underline{M}} (I, \varphi)$, construite page* 62, *est filtrante, et le foncteur canonique (fibrant) $\mathcal{C} \longrightarrow I$ est cofinal.*

Que $\mathcal{C}$ soit filtrante et $\mathcal{C} \longrightarrow I$ cofinal résulte formellement du fait que I l'est, que les fibres de $\mathcal{C}$ le sont, et que $\mathcal{C}$ est cofibrée sur I. (Mérite un lemme, à toutes fins utiles.)

N.B. On peut remplacer $\mathcal{C}$ par $\mathcal{C}_0$ dont les fibres sont les

$$\mathrm{Ss}_\pi^{M^\pi}(X_i) \;\subset\; \underline{M}^\pi \;; \qquad \text{cofinale dans } \mathcal{C}_i = \underline{M}^\pi/X_i \;,$$

on trouve une sous-catégorie de $\mathcal{C}$ cofinale dans $\mathcal{C}$, et qui est cette fois ordonnée tout comme [page 65]
I.

Commentary · Exposition [ms. 64]

Integrating small presentations

Grothendieck now wants to flatten a filtered diagram of objects which themselves admit small presentations. His category $\mathcal{C}$ remembers both the outer index and a small object mapping into that stage. Keeping these maps, rather than choosing presentations separately and hoping they are compatible, makes the construction functorial.

The notation in Lemma 5.10 needs a restriction. The subobjects which form a subcategory of $\underline{M}^\pi/Y$ are

$$\mathrm{Ss}_\pi^{M^\pi}(Y) = \{Z \hookrightarrow Y : Z \in M^\pi\},$$

not all of $\mathrm{Ss}_\pi(Y)$. For example, in sets let M be the class of nonempty sets and let Y be nonempty. The empty subobject of Y belongs to $\mathrm{Ss}_\pi(Y)$ but is not an object of $\underline{M}^\pi/Y$. The proof and the subsequent displayed formula use the restricted collection, so the repair is unambiguous.

Under the standing subobject and image hypotheses the intended lemma has a short proof. Any $Z \to Y$ with $Z \in M^\pi$ factors through its image, a small subobject by condition (d). Since $Y \in M$, the recognition criterion enlarges this image to a subobject $T \hookrightarrow Y$ with $T \in M^\pi$. Such subobjects form a directed poset: first bound finitely many small subobjects in $\mathrm{Ss}_\pi(Y)$ and then enlarge the bound into the cofinal subset. Finally two maps $Z \rightrightarrows T$ over Y are equal because $T \to Y$ is monic. Thus every finite diagram in the slice has a cone through this poset, and the inclusion is final. Its colimit is Y because the small subobjects are cofinal among all small subobjects.

The formal lemma requested after Corollary 5.11 can also be made explicit. Let $p : E \to I$ be an opfibration, with I filtered and every fibre nonempty and filtered. Given finitely many objects of E, move them by opcartesian arrows to one common fibre and take a cone there. Given parallel arrows of E, first equalize their images by moving farther in I, then use filteredness of the new fibre to equalize the resulting vertical arrows. Hence E is filtered. For every $i \in I$, the category $(i \downarrow p)$ is nonempty, and the same argument makes it filtered; therefore p is final. These are ordinary finality statements and do not require a homotopical version of Theorem A.

23. Comme I est filtrante.

For the category used here, objects are triples $(i, Z, u : Z \to X_i)$; a morphism to (j, Z', u') consists of $a : i \to j$ and $v : Z \to Z'$ satisfying $u'v = X(a)u$. Postcomposition gives the opcartesian transport. Its tautological source functor is the dotted $\mathfrak{z}$ of the diagram. A cocone from this source functor to T is exactly a compatible family of cocones on the fibres. Consequently

$$\varinjlim_{(i,Z,u)\in\mathcal{C}} Z \simeq \varinjlim_{i\in I}\left(\varinjlim_{Z\to X_i} Z\right) \simeq \varinjlim_{i\in I} X_i.$$

This proves the interchange actually needed on page 65. Corollary 5.11 is printed with $X_i \in M^\pi$; the argument works for $X_i \in M$, which is the case required for Proposition 5.9. Its word "fibrant" should be "cofibrant", in the older terminology for an opfibration.

p. 38

On a

$$X_i \;=\; \varinjlim_{Z_\alpha\in\mathcal{C}_i} Z_\alpha \;,$$

donc

$$X \;=\; \varinjlim_{\alpha\in\mathcal{C}} Z_\alpha \;,$$

au total on est arrivé à représenter X comme $\varinjlim$ filtrante d'éléments Z_α qui sont (non seulement dans M, mais même) dans M^π.

Changeons la notation, en prenant

$$X \;=\; \varinjlim_{i\in I} X_i \;, \qquad I \text{ ensemble ordonné filtrant, } X_i \in M^\pi \;\; \forall\, i \in I \;,$$

soit I' l'ensemble des parties *filtrantes* $J \subset I$ telles que $\operatorname{card} J \leq \pi$, I' ordonné par inclusion. Alors I' est grand devant π, et on a un système inductif

$$(X_J)_{J\in I'} \;, \qquad X_J \;=\; \varinjlim_{i\in J} X_i \;.$$

Par l'hypothèse de stabilité de M^π (qui n'a pas encore servi), les X_J sont dans M^π. On a donc

$$X \;=\; \varinjlim_{J\in I'} X_J \;, \qquad \text{les } X_J \text{ dans } M^\pi,\ I' \text{ grande devant } \pi \;,$$

donc $X \in M$ puisque M est π-admissible, *q.e.d.*

Commentary · Exposition [ms. 65] [page 66]

The cardinal regrouping in Proposition 5.9

The last step of the proof turns an unrestricted filtered colimit into an adapted colimit whose stages remain bounded. This is a useful construction independently of the subobject description. Suppose

$$X = \varinjlim_{i\in I} Z_i, \qquad Z_i \in M^\pi,$$

where I is a directed poset. Let $P_\pi(I)$ consist of its nonempty directed subsets of cardinal at most π. Every subset of I of cardinal at most π lies in such a directed subset: adjoin chosen upper bounds of all pairs repeatedly and take the countable union. It follows that $P_\pi(I)$ is π^+-directed. The supremum of a directed family of at most π members is its union, again a member of $P_\pi(I)$.

Set $Z_J = \varinjlim_{j\in J} Z_j$. Bounded filtered-colimit closure puts Z_J in M^π. If $\mathcal{J} \subset P_\pi(I)$ is directed with $|\mathcal{J}| \leq \pi$, the universal property of a cocone gives

$$\varinjlim_{J\in\mathcal{J}} Z_J \simeq \varinjlim_{j\in\bigcup\mathcal{J}} Z_j = Z_{\bigcup\mathcal{J}}.$$

Moreover $\varinjlim_{J\in P_\pi(I)} Z_J \simeq X$, because every object and every comparison in the original diagram occurs in some J. Thus the regrouped presentation is adapted and every one of its stages belongs

to M. Condition b_π now implies $X \in M$. This supplies the continuity check implicit in the last displayed formula of page 65.

There is a genuine distinction between the hypotheses of Proposition 5.9 as printed and those used by its subsequent proof. The proposition lists only (a), (b), (c), and (d_0), whereas Lemma 5.10 is proved using images and the canonical subobject presentation from the stronger standing assumptions. The regrouping argument itself uses neither images nor exactness. Its preceding flattening step does. One complete version of the stated weakening is obtained by requiring that the objects of Filt^π be π-accessible. Then, for $Y \in M$, start with an adapted presentation of Y. The good indices are cofinal, and bounded filtered-colimit closure makes them closed under the required suprema. Maps from objects of M^π into Y factor through this subsystem, and equalities of maps hold at a sufficiently late stage, by presentability. These two properties show directly that M^π/Y is filtered, that the chosen subsystem is final in it, and that its colimit is Y. The preceding block and the regrouping then prove Proposition 5.9 using (d_0) and this presentability assumption, without (d_1), (d_2), or (e).

Alternatively, under the full standing assumptions (a)–(e), the subobject proof in the preceding block works at the given cardinal. The assertion that both sets of additional hypotheses may be removed simultaneously is not established by the written sketch. This is a gap in its claimed generality, not a counterexample to its conclusion. Chapter XIII, manuscript pages 90–97, explicitly retains the corresponding presentability condition when flattening slices (Grothendieck, 1990–1991e). The operative form of Proposition 5.7 uses good cardinals, so its proof is unaffected by this distinction.

To finish that proposition, start with a class $M_0 \subset \mathrm{Filt}^\pi$ and close it under bounded filtered colimits inside Filt^π, obtaining N. Define M_N by cofinality of $\mathrm{Ss}^N_\pi(X)$. The preceding arguments show that M_N is L-stable and, at a good cardinal, π-accessible. It contains M_0, while each of its objects is a filtered colimit of objects of N, themselves in the L-envelope of M_0. Therefore M_N is exactly that envelope. On a bounded object X, the poset $\mathrm{Ss}_\pi(X)$ has largest element X, so the recognition condition gives $M_N \cap \mathrm{Filt}^\pi = N$. This proves the inverse correspondence of Corollary 5.8, including its uniqueness assertion.

Définition 5.12.

1) Soit $\mathcal{M}$ une catégorie satisfaisant les conditions (a), (b), (c), (d) (p. 44-46). Soit $\pi \geq \pi_0$ un cardinal. Une partie M de $\mathrm{Ob}\mathcal{M}$ est dite *L-π-accessible,* si elle est π-accessible et stable par petites $\varinjlim$ filtrantes [24], ou ce qui revient au même (5.9), si elle est π-accessible et si $M^\pi = M \cap \mathrm{Filt}^\pi$ est stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$. (**N.B.** Si les objets de Filt^π sont π-accessibles, 5.8 nous dit que cela signifie aussi qu'il suffit pour cela qu'il existe une partie M_0 de Filt^π telle que M soit la L-enveloppe (enveloppe stable par $\varinjlim$ filtrantes) de M_0 – condition qui est de toutes façons toujours nécessaire, en prenant $M_0 = M^\pi$.)
2) Soit M une partie de $\mathrm{Ob}\mathcal{M}$. On dit qu'elle est *L-accessible,* si elle satisfait à l'une des quatre conditions équivalentes suivantes :
   - a) Il existe un cardinal π tel que M soit L-π-accessible.
   - a') M est accessible et L-stable.
   - b) Il existe une petite partie M_0 de $\mathrm{Ob}\mathcal{M}$, telle que M soit la L-enveloppe de M_0 [25].
   - b') Il existe une partie M_0 de $\mathcal{M}$ contenue dans une partie Filt^π (ou ce qui revient au même, telle que la sous-catégorie pleine $\underline{M}_0$ de $\mathcal{M}$ soit essentiellement petite, *i.e.* équivalente à une petite catégorie, telle que M soit la L-enveloppe de M_0).

L'équivalence de (a) et (a'), ou de (b) et (b') est plus ou moins tautologique. L'équivalence de (a) et (b) résulte aussitôt du N.B. inclus dans 1°, compte tenu que les cardinaux π tels que les objets de Filt^π soient π-accessibles sont cofinaux dans l'ensemble des cardinaux dans $\mathfrak{U}$. (*Cf.* 3.2, page 16). p. 39 [page 67]

Proposition 5.13. *Toute réunion finie de parties L-π-accessibles de* $\mathrm{Ob}\mathcal{M}$ *est L-π-accessible. L'intersection d'une famille* $(M_\alpha)_{\alpha\in A}$ *de parties L-π-accessibles, avec* $\operatorname{card} A \leq \pi$*, est L-π-accessible.*

24. On dit alors que M est *L-stable.*
25. Cette condition ne dépend pas de la filtration cardinale choisie de $\mathcal{M}$.

Démonstration. La première assertion provient du fait qu'une réunion *finie* de parties L-stables de M est L-stable, et qu'on a itou pour la notion de π-accessibilité. Les deux choses se prouvent d'ailleurs pratiquement de la même façon, en utilisant le

Lemme 5.14. *Soit I un ensemble ordonné filtrant, $(I_\alpha)_{\alpha\in A}$ une famille* finie *non vide de parties de I, telle que $\bigcup I_\alpha = I$. Alors il existe $\alpha \in A$ tel que I_α soit cofinal.*

Démonstration par l'absurde. Si pour aucun α, I_α n'était cofinal, on en conclurait [page 68] que pour tout α il existe un i_α tel que $I/i_\alpha \cap I_\alpha = \emptyset$. Soit i majorant les i_α, on aura donc $I/i \cap I_\alpha = \emptyset \ \ \forall\, \alpha \in A$, contrairement à l'hypothèse que $\bigcup I_\alpha = I$.

Commentary · Exposition [ms. 67]

Two closure operations on accessible classes

Grothendieck's letter L records a deliberate strengthening of accessibility: the class is closed under *all* small filtered colimits, including ordinary sequential colimits. Increasing the accessibility cardinal does not remove that requirement. Conversely, once a class is generated under filtered colimits by a small family, one may choose a cardinal bounding that family and then a good cardinal for the ambient filtration. Proposition 5.7 supplies the recognition condition at that cardinal. This is why the absolute notion in Definition 5.12 does not depend on the particular cardinal filtration.

The equivalence between a small generating family and a bounded essentially small family also has a precise size content. Each filtration stage is essentially small, so its members have a small set of representatives. Conversely, choose one representative of every isomorphism class in a given essentially small family. Each representative belongs to some stage, and the small set of those cardinal bounds has a common bound in the universe. Replacing the family by its isomorphism closure changes none of its colimits. These are the size reductions used in clauses (b) and (b′), rather than an assertion that a potentially large collection of literally equal objects is a set.

The finite-union statement may at first seem surprising, since diagrams can contain objects from different classes. Its explanation is entirely about the index. If a directed set is covered by finitely many subsets, one subset is cofinal. Otherwise choose, for each subset, a point above which it never occurs, and then bound those finitely many points. No subset can contain that upper bound, a contradiction. A cofinal subset of a directed poset is itself directed, so a filtered diagram whose objects lie in $M_1 \cup \cdots \cup M_n$ has a cofinal subdiagram lying in one M_r. If every M_r is L-stable, its colimit lies in that M_r and hence in the union.

The same finite-colour argument proves the recognition property in an adapted presentation. If the union of the good-index subsets is cofinal, apply the lemma inside that union; one good-index subset is then cofinal in the union and hence in the original index. The individual good-index subsets need not all be filtered before one knows that they are cofinal. The parenthesis "filtrantes" on page 72 is therefore unnecessary and, if read as an unconditional assertion, incorrect. For example a diagram of sets indexed by the fork $a, b \le t$, with two singleton stages mapping to the two different elements of a two-element terminal stage, has two singleton indices without a common singleton upper bound. The finite-cover lemma does not require that additional assertion.

For intersections, a different argument is needed. If X belongs to every M_α, each corresponding good-index subset is both cofinal and closed under bounded directed suprema, because M_α^π is closed under the matching colimits. The required common presentation is therefore obtained by intersecting closed cofinal subsets. This is the role of Lemma 5.15.

Le cas d'une *intersection* de parties L-π-accessibles : elle est bien sûr L-stable, donc il reste à prouver qu'elle est π-accessible, *i.e.* satisfait la condition b_π de *loc. cit.*

Soit donc [page 69]

$$X \;=\; \varinjlim_{i\in I} X_i\ , \qquad\qquad X_i \in \mathrm{Filt}^\pi\ ,$$

une représentation π-adaptée de X, avec I un ensemble ordonné π-adapté. Donc on a

$$(*) \qquad X \in M_\alpha \iff \begin{array}{c}\text{l'ensemble } I_{M_\alpha} \text{ des } i \in I \text{ tels que } X_i \in M_\alpha \\ \text{est cofinal dans } I.\end{array}$$

D'ailleurs, comme $M_\alpha^\pi = M_\alpha \cap \mathrm{Filt}^\pi$ est stable par $\varinjlim$ filtrante de cardinal $\le \pi$, on voit tout de suite que la partie I_{M_α} de I est stable par bornes supérieures dans I de parties J de I_{M_α}

filtrantes et de cardinal $\leq \pi$. Donc I_{M_α} cofinal équivaut à I_{M_α} π-*cofinal* (*cf.* 5.8, page 59). Soit $M = \bigcap_{\alpha\in A} M_\alpha$, je veux prouver que

$(**)$ $\qquad X \in M \iff$ l'ensemble I_M $(= \bigcap I_{M_\alpha})$ est cofinal dans I.

Or $(*)$ implique que $X \in M$ $(= \bigcap M_\alpha)$ équivaut à I_{M_α} π-cofinal pour tout α. Donc au total on veut prouver

$(***)$ $\qquad I_{M_\alpha}$ cofinal dans I pour tout $\alpha \iff \bigcap I_{M_\alpha}$ est cofinal dans I.

On a bien sûr $\Longleftarrow$, donc il reste à prouver $\Longrightarrow$. Si l'ensemble d'indices A est fini, cela résulte du lemme 5.8, compte tenu que la cofinalité des I_{M_α} implique leur π-cofinalité. On doit donc simplement améliorer 5.8, par le [page 70]

Lemme 5.15. *Soient π un cardinal, I un ensemble ordonné filtrant π-adapté, et soit $(I_\alpha)_{\alpha\in A}$ une famille de parties I_α π-cofinales* (cf. 5.8), *telle que* $\operatorname{card} A \leq \pi$. *Alors $\bigcap I_\alpha$ est π-cofinale.* p. 40

Il suffit de prouver qu'elle est cofinale, car la propriété de stabilité par bornes supérieures de parties filtrantes de cardinal $\leq \pi$ est évidente. Quitte à remplacer I par I/i_0, et les I_α par leurs traces sur I/i_0, la question est simplement de prouver que $\bigcap_\alpha I_\alpha \neq \emptyset$. On va construire un élément s de $\bigcap I_\alpha$ par induction transfinie. Prenons un bon ordre sur A, et prouvons par récurrence transfinie que pour tout $\alpha \in A$, l'ensemble [page 71]

$$J_\alpha \;=\; \bigcap_{\beta<\alpha} I_\beta$$

est cofinal. Si on sait que J_α est cofinal, il en est de même de $J_{\alpha+1} = J_\alpha \cap I_\alpha$, par 5.8. Donc on est ramené à prouver que J_α est cofinal si α est un ordinal limite, donc à prouver le lemme 5.15 pour une suite transfinie *décroissante* de parties π-cofinales J_β $(\beta < \alpha)$ quand $\operatorname{card}\alpha \leq \pi$, telle que $J_\beta = \bigcap_{\beta'<\beta} J_{\beta'}$ quand β est limite. À prouver que l'intersection est π-cofinale – et on est réduit à prouver qu'elle est non vide. On va construire par récurrence transfinie une suite *croissante* d'éléments

$$i_\beta \;\in\; J_\beta \qquad (\,\beta \;\leq\; \alpha\,)\,.$$

Si i_β est construit, comme $J_{\beta+1}$ est cofinal, il existe $i_{\beta+1} \geq i_\beta$ dans $J_{\beta+1}$, O.K. Si β est un ordinal limite, on prend

$$i_\beta \;=\; \sup_{\beta'<\beta} i_{\beta'}\;,$$

i_β est dans chaque I_{β_0} pour $\beta_0 < \beta$, puisque les $i_{\beta'}$ pour $\beta_0 \leq \beta' < \beta$ sont tous dans $J_{\beta_0} \supset J_{\beta'}$, donc aussi leur limite, donc [page 72]

$$i_\beta \;\in\; \bigcap_{\beta_0<\beta} J_{\beta_0} \;=\; J_\beta\;.$$

Donc la récurrence transfinie marche, et nous donne donc un élément s_α de J_α !

Commentary · Exposition [ms. 71]

The transfinite intersection argument and its bound

Grothendieck is arranging one approximation which satisfies every condition at once. The induction works because a limit of approximations eventually satisfying a fixed condition still satisfies that condition. Here is the proof with its terminal index included, so that the last intersection is obtained as well as all its proper initial intersections.

Write the indexing set as an ordinal λ with $|\lambda| \leq \pi$, and put

$$K_\alpha = \bigcap_{\beta<\alpha} I_\beta \qquad (\alpha \leq \lambda).$$

We prove by transfinite induction that every K_α is closed and cofinal. For $\alpha = 0$ it is I. Closure holds for every intersection. At a successor ordinal it follows from the two-set lemma that $K_{\alpha+1} = K_\alpha \cap I_\alpha$

is cofinal. Suppose now that $\delta \leq \lambda$ is a nonzero limit and that all the earlier intersections are cofinal. Given $i \in I$, construct an increasing family $(x_\beta)_{\beta<\delta}$ with $x_0 \geq i$ and $x_\beta \in K_\beta$. At a successor stage enlarge x_β into $K_{\beta+1}$. At a limit stage $\eta < \delta$, take the supremum of the previous terms. This supremum exists since the chain has cardinal at most π. For each $\gamma < \eta$, its tail from γ onward lies in K_γ; closure therefore places the supremum in every K_γ, hence in K_η.

Finally take $x = \sup_{\beta<\delta} x_\beta$. The same tail argument puts x in K_γ for every $\gamma < \delta$, and hence in K_δ. It also lies above i. This proves the limit step and gives cofinality of K_λ, the desired full intersection. In the manuscript's version the line referring to I_{β_0} is naturally read with J_{β_0}, as in the following line; alternatively one starts the tail after β_0. Neither adjustment changes the proof. What matters is to run the induction through the order type of the entire indexing set, not merely through its elements.

The cardinal restriction is exact at this level of generality. Let T be a set of cardinal strictly larger than π, and let I be the poset of subsets of T of cardinal at most π. It is directed, and a directed family of at most π such subsets has its union as supremum. For $t \in T$, the set

$$I_t = \{S \in I : t \in S\}$$

is cofinal and closed under these suprema. Yet $\bigcap_{t\in T} I_t$ is empty, because no member of I contains all of T. Thus one cannot intersect an arbitrarily large family at a fixed cardinal by this argument.

Corollary 5.16 is nevertheless an absolute small-intersection theorem. Given a small family of L-accessible classes, choose a common sufficiently large good cardinal witnessing their accessibility and bounding the size of the family. Apply Proposition 5.13 there. The witness cardinal may change; the absolute class does not. This is the flexibility that will allow several properties of arrows to be imposed simultaneously.

Je vais prouver aussi en forme l'assertion pour $\bigcup_{\alpha\in A} M_\alpha$, quand A est fini. Posant cette fois $M = \bigcup_\alpha M_\alpha$, il faut prouver, moyennant les relations $(*)$ (p. 68), qu'on a

$$(**) \qquad \begin{array}{lcl} X \in M & \iff & \text{l'ensemble } I_M \ (= \bigcup_\alpha I_{M_\alpha}) \\ (\textit{i.e. } \exists\ \alpha \text{ avec } X \in M_\alpha) & & \text{est cofinal dans } I\ , \end{array}$$

et moyennant $(*)$ le premier terme de cette équivalence à établir signifie qu'il existe α avec I_{M_α} cofinal dans I. Bien sûr, si un des I_{M_α} est cofinal dans I, a fortiori leur réunion. Mais inversement, si la réunion est cofinale, elle est filtrante et réunion des parties (filtrantes) I_{M_α}, dont l'une doit donc être cofinale dans elle par le lemme 5.14, donc aussi dans I, *q.e.d.*

Corollaire 5.16 (de la proposition 5.13). *Toute réunion finie de parties L-accessibles de $\mathcal{M}$ est L-accessible. Toute intersection $\bigcap M_\alpha$ d'une famille petite de parties L-accessibles de $\mathcal{M}$, est L-accessible.*

[page 73]

p. 41

6 Essai avorté pour un critère d'engendrement des parties L-accessibles de $\mathrm{Fl}(\mathcal{M})$.

Soit $\mathcal{M}$ une catégorie munie d'une filtration cardinale $(\mathrm{Filt}^\pi)_{\pi\geq\pi_0}$, et soit S un objet de Cat, considérons

$$\mathcal{M}(S) \ = \ \underline{\mathrm{Hom}}(S, \mathcal{M})\ ,$$

et munissons $\mathcal{M}(S)$ de la filtration par les

$$\mathcal{M}^\pi(S) \ = \ \underline{\mathrm{Hom}}(S, \mathcal{M}^\pi = \underline{\mathrm{Filt}}^\pi(\mathcal{M}))\ .$$

Je dis que c'est là une filtration cardinale de $\mathcal{M}(S)$. Il faut vérifier les conditions (a), (b), (c) de SGA 4 I 9.12. Les conditions (a) (Filt^π essentiellement petite) et (b) (condition L_{π_0}, et stabilité de Filt^π par $\varinjlim_I$, I filtrante grande devant π_0 et de cardinal $\leq \pi$) passent aussitôt de $\mathcal{M}$ à $\mathcal{M}(S)$. La condition (c) est plus délicate (représentation d'un objet comme $\varinjlim$ filtrante

[page 74]

grande devant π d'objets de Filt^{π}). Je ne vais pas refaire le travail, fait dans un cas beaucoup plus général et plus délicat dans SGA 4 I §9, savoir dans le corollaire 9.24 du théorème 9.22 (p. 177). La seule restriction, c'est qu'on y suppose les objets de $\mathcal{M}$ accessibles, et désignant par π_1 le plus petit cardinal tel que $\pi_1 \geq \pi_0$ et que les objets de Filt^{π_0} soient π_1-accessibles, on se borne pour la filtration cardinale de $\underline{\mathrm{Hom}}(S, \mathcal{M})$ aux cardinaux $\boxed{\pi \geq c = 2^{c_0},}$ où

$$c_0 \;=\; \sup(\pi_1, \operatorname{card} \mathrm{Fl}(S)) \;.$$

C'est une restriction raisonnable et anodine.

Commentary · Examples [ms. 73]

Why finite unions are the natural statement

Grothendieck's finite-union theorem has a useful algebraic test. Fix a prime p and work in abelian groups with the cardinal filtration by underlying sets of cardinal at most π, where π is infinite. For $n \geq 1$, let

$$M_n = \{A : p^n A = 0\}.$$

The class is closed under filtered colimits: every element of a filtered colimit is represented by an element of a stage, and multiplication by p^n is zero at that stage. It is also recognized by small subgroups. If $p^n A = 0$, all subgroups satisfy the same equation. Conversely, if small subgroups annihilated by p^n are cofinal, every element of A belongs to one of them, so it is annihilated by p^n. These good subgroups are closed under bounded directed unions. At this cardinal the bounded groups are π^+-presentable: a group with at most π elements has a presentation with at most π generators and relations, and a map or an equality out of such a presentation is settled at one stage of a π^+-filtered diagram. Thus Proposition 5.7 applies and each M_n is L-π-accessible.

The increasing union $\bigcup_n M_n$ is the class of groups annihilated by some finite power of p, with a bound allowed to depend on the whole group. It is not L-stable. Consider

$$\mathbb{Z}/p \longrightarrow \mathbb{Z}/p^2 \longrightarrow \mathbb{Z}/p^3 \longrightarrow \cdots, \qquad [a] \longmapsto [pa].$$

All arrows are injective. Sending $[a] \in \mathbb{Z}/p^n$ to $a/p^n + \mathbb{Z}$ identifies its colimit with $\mathbb{Z}[1/p]/\mathbb{Z}$. For every n this group has an element of order p^{n+1}; consequently no power p^n kills the whole colimit. Every stage is in the union and the colimit is not. This computes the failure of an infinite-union analogue without any size pathology.

The example also explains the distinction between a global bound and a local condition. The class of *all* p-primary torsion groups is L-stable: each element is annihilated by some power of p, and an element in a filtered colimit comes from a stage. Its small subgroups give the same cofinality test. Passing from bounded exponent to elementwise finite exponent changes the class, and is exactly what happens in the displayed colimit. Accessibility alone does not authorize exchanging these two quantifiers.

Pour l'instant, je suis intéressé surtout au cas où $S = \mathbf{\Delta}_1$, donc à

$$\underline{\mathrm{Hom}}(\mathbf{\Delta}_1, \mathcal{M}) \;=\; \underline{\mathrm{Fl}}(\mathcal{M}) \;,$$

auquel cas on a $c_0 = \pi_1$, donc il suffit de se borner aux cardinaux $\pi \geq 2^{\pi_1}$ pour avoir une bonne filtration cardinale sur $\underline{\mathrm{Fl}}(\mathcal{M})$.

C'est donc l'ensemble des deux conditions (b), (c) sur $\mathcal{M}$ (p. 44) qui est stable par passage aux $\underline{\mathrm{Hom}}(S, \mathcal{M}) = \mathcal{M}(S)$. Il en est de même de la condition (a) (stabilité par petites limites inductives), et aussi de la condition (d) (p. 45). De même la condition d'exactitude (e) (p. 46) [page 75]
sur les $\varinjlim$ filtrantes (elles doivent transformer les monomorphismes en monomorphismes).

Commentary · Context [ms. 74]

The passage to diagrams comes from SGA 4

Grothendieck needs the preceding theory in an arrow category, since weak equivalences and cofibrations are classes of arrows. His citation is precise: SGA 4, Exposé I, Corollary 9.24 gives

the objectwise cardinal filtration on categories of sections, under the stated accessibility and filtered-colimit hypotheses, after the bound $\pi \geq 2^{\aleph_0}$ (Artin et al., 1972, Exposé I, 9.22–9.24). Ordinary functor categories are the case of a constant fibred category. The enlarged bound pays for simultaneously choosing the objects, the structure maps, and the equations making them a diagram; small approximations of individual values alone do not provide those equations.

The other transferred properties can be checked directly. Colimits in $\mathcal{M}^S$ are computed at each object of S. A sufficiently small colimit of diagrams whose values lie in Filt^π again has its values there. Essential smallness follows by choosing a small skeleton of Filt^π: a functor assigns one of its objects to each object of S and one of its arrows to each arrow of S, subject to equations. Since S is small, the resulting set of choices is small. Images and subobjects are also computed pointwise under the standing hypotheses. For images, the arrow induced from $\mathrm{Im}(F(s) \to G(s))$ to $\mathrm{Im}(F(t) \to G(t))$ is forced by the image factorization and is unique after the monomorphic inclusion into $G(t)$; these unique arrows satisfy functoriality. This checks the part of condition (d) that is easy to overlook.

A natural transformation is monic exactly when each component is monic in this setting. One direction follows directly from the definition. For the other, evaluation at s has a left adjoint sending A to the diagram $t \mapsto \coprod_{S(s,t)} A$; testing monicity against these diagrams detects monicity of the s-component. Filtered colimits therefore preserve monomorphisms objectwise whenever they do so in $\mathcal{M}$. The same pointwise argument transfers commutation of filtered colimits with finite limits when that stronger exactness condition is available.

For $S = [1]$, a bounded object of the arrow category is a map *whose source and target are both bounded*. This concrete description should remain visible throughout §6. A map from a small object into a fixed large object is not a bounded object of the arrow category. Nor does a family of such maps become a bounded presentation merely because it is continuous and its indexing poset has the correct suprema.

The relationship with modern accessible categories is easiest at good cardinals: the bounded diagram objects can then be compared with presentable objects, and the categorical smallness of all their structure maps is absorbed by increasing the regular cardinal. The historical filtration language retains this choice explicitly. It gives uniform control over several classes in the same ambient category, which is exactly the control required for $C \cap W$.

Pour être tranquille, je vais faire sur $\mathcal{M}$ les hypothèses (a) à (e) inclus (bien que (e) me semble un peu artificielle, on devrait pouvoir l'éliminer, dans un second souffle !). Ces conditions ont donc la propriété de stabilité souhaitable par passage aux $\mathcal{M}(S)$, et en particulier par passage à $\mathrm{Fl}(\mathcal{M})$. Quitte à augmenter π_0, on peut donc supposer que les $\mathrm{Fl}^\pi(\mathcal{M}) = \mathrm{Hom}(\mathbf{\Delta}_1, \mathcal{M}^\pi)$ $(\pi \geq \pi_0)$ forment aussi une filtration cardinale de $\underline{\mathrm{Fl}}(\mathcal{M})$.

On dispose donc des notions de parties π-accessibles, accessibles, L-π-accessibles, L-accessibles de $\mathrm{Fl}(\mathcal{M}) = \mathrm{Ob}\underline{\mathrm{Fl}}(\mathcal{M})$. Ces notions sont stables par réunions et intersections finies, et les notions absolues (sans π) stables par intersections "quelconques" (*i.e.* pour des *petites* familles de parties). Ce sera là une chose particulièrement cruciale, le cas d'une intersection $C \cap W$! p. 42 [page 76]

Le résultat principal, pour lequel j'ai fait les gammes du paragraphe précédent, est le suivant :

Théorème 6.1 [26]. *Soit $\mathcal{M}$ une catégorie satisfaisant les conditions dites :*

a) *Stabilité par petites $\varinjlim$ (pas nécessairement filtrantes).*

b) *$\mathcal{M}$ munie d'une filtration cardinale* $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$.

c) *Les objets de $\mathcal{M}$ sont accessibles.*

d) *Les* Filt^π *sont stables par $\varinjlim$ indexées par des catégories I telles que* $\operatorname{card}\mathrm{Fl}(I) \leq \pi$, *et si $Z \in \mathrm{Filt}^\pi$, tout quotient Z' de Z (tel que $Z \longrightarrow Z'$ ne se factorise pas par un sous-objet Z'_0 de Z', différent de Z') est dans* Filt^π.

e) *Les limites inductives filtrantes dans $\mathcal{M}$ transforment les monomorphismes en monomorphismes.*

26. Démonstration canulée, énoncé peut-être *faux* !!

Soit C une partie L-accessible de $\mathrm{Fl}(\mathcal{M}) = \mathrm{Ob}\underline{\mathrm{Fl}}(\mathcal{M})$, i.e. (5.12) *une partie stable par* $\varinjlim$ *filtrantes, et accessible. Supposons de plus C formée de monomorphismes*[27]*, et que C contient les* isomorphismes. *Alors il existe une partie petite $C_0 \subset C$, telle que $C \subset \widetilde{C_0}$. Plus précisément, si π est un cardinal tel que C soit π-accessible, et si on pose $C^\pi = C \cap \mathrm{Fl}^\pi(\mathcal{M})$, alors on a* [page 77]

$$C \subset \widetilde{C^\pi} = \widetilde{C_0}\,, \tag{6.1.1}$$

où C_0 est n'importe quelle petite partie de C^π telle que tout objet de C^π soit isomorphe à un objet de C_0.

Corollaire 6.2. *On a $C_* = C_{0*}$, $\widetilde{C} = \widetilde{C_0}$.*

Corollaire 6.3. *Conditions équivalentes sur la partie L-accessible C de* $\mathrm{Fl}(\mathcal{M})_{\mathrm{mono}}$*, contenant les isomorphismes :*

a) *$C = \widetilde{C}$.*

b) *C est stable par cochangement de base et par composition.*

b′) *C est stable par cochangement de base, et par sommes amalgamées simples,* i.e. *si $A \longrightarrow B_1$, $A \longrightarrow B_2$ sont dans C, alors $A \longrightarrow B_1 \amalg_A B_2$ aussi.*

c) *C^π est stable dans $\mathrm{Fl}^\pi(\mathcal{M}) = \mathrm{Fl}(\mathcal{M}^\pi)$ par cochangement de base et par composition.*

c′) *C^π est stable dans $\mathrm{Fl}(\mathcal{M}^\pi)$ par cochangement de base et par sommes amalgamées* (cf. b′).

Quand ces conditions sont satisfaites, alors (C, C_*) est un couple de Quillen, justiciable du théorème 1 *du paragraphe* 1 (p. 10).

La dernière assertion résulte du théorème 6.1 et du corollaire au théorème 1 du paragraphe 1. On a d'autre part tautologiquement p. 43 [page 78]

$$\begin{array}{ccccc} (a) & \Longrightarrow & (b) & \Longrightarrow & (b') \\ & & \Downarrow & & \Downarrow \\ & & (c) & \Longrightarrow & (c')\,, \end{array}$$

et il reste à prouver $(c') \Longrightarrow (a)$, ce qu'on fera plus bas.

Commentary · Exposition [ms. 77]

What the proposed generation theorem would have supplied

Grothendieck wants the accessibility calculation to finish the existence problem for factorizations. If every arrow in C lay in the lifting closure of a small bounded part C_0, the small object construction from §1 would provide factorizations controlled by C. This is a substantially stronger assertion than saying that C is a filtered-colimit closure of C_0.

Recall the order-reversing operations

$$C^\square = \{p : p \text{ has the right lifting property against } C\},$$
$$^\square R = \{i : i \text{ has the left lifting property against } R\}.$$

In the manuscript $C_* = C^\square$ and $\widetilde{C} = {}^\square(C^\square)$. From $C_0 \subset C \subset \widetilde{C_0}$ one obtains

$$C^\square \subset C_0^\square = (\widetilde{C_0})^\square \subset C^\square.$$

Thus Corollary 6.2 is a correct formal consequence of the proposed inclusion. It supplies no independent evidence for that inclusion. The source itself marks the proof of Theorem 6.1 as withdrawn.

The distinction between lifting closure and closure under elementary constructions is also relevant to Corollary 6.3. Every left lifting class contains the isomorphisms and is closed under pushouts,

27. Finalement, on se débarrasse de cette hypothèse.

composition, transfinite composition, and retracts. If a class is already closed under all filtered colimits, its closure under retracts can be proved without a separate hypothesis: an idempotent $e : f \to f$ in the arrow category has its image as the colimit of the sequence

$$f \xrightarrow{e} f \xrightarrow{e} f \xrightarrow{e} \cdots .$$

A splitting of the idempotent identifies that image with the desired retract. Closure under filtered colimits and finite composition likewise supplies the limit stages of transfinite composition. These observations explain why the hypotheses in (b) look close to being sufficient. They do not produce a *set* of lifting generators. The implication promised from (c′) to (a), and the assertion of a Quillen factorization obtained from §1, cannot be certified by the cancelled proof of 6.1.

For comparison, modern "accessible weak factorization system" refers to accessibility of a chosen functorial factorization, not to an arbitrary class satisfying the manuscript's recognition condition. Rosický's example of split monomorphisms and split epimorphisms in abelian groups gives an accessible weak factorization system which is not generated by a set (Rosický, 2017, Example 2.8). It is not by itself a counterexample to Theorem 6.1: split monomorphisms are not closed under all filtered colimits. An argument about this theorem must verify the additional L-condition and the embedded recognition condition.

The later counterexample discussed at manuscript page 84 does verify those conditions: pure monomorphisms in a finite presheaf diagram category provide the required separation between filtered generation and generation by lifting. Theorem 6.1 is therefore false in its proposed generality, rather than merely lacking the particular proof attempted here. The many valid cardinal and cofinality calculations of §5 remain available; what fails is their conversion into uniformly small relative attachments.

Prouvons donc 6.1. Dans (6.1.1) l'égalité $\widetilde{C_0} = \widetilde{C^\pi}$ est tautologique. Donc tout revient à prouver

$$C \subset \widetilde{C^\pi} .$$

Une difficulté ici, c'est qu'on ne sait *pas* que $\widetilde{C^\pi}$ soit stable par $\varinjlim$ filtrantes – on le sait seulement pour C. Soit donc

$$A \hookrightarrow B$$

une flèche dans C, prouvons qu'elle est dans $\widetilde{C^\pi}$. Pour ceci, on va construire, par induction transfinie, une suite transfinie *strictement* croissante de sous-objets A_α de B contenant $A = A_0$,

$$\underbrace{A}_{=A_0} \xhookrightarrow{i_\alpha} A_\alpha \xhookrightarrow{i_{\beta,\alpha}} A_\beta \xrightarrow{j_\beta} B ,$$

telle que l'on ait

$$i_\alpha \in \widetilde{C^\pi} , \qquad \text{plus généralement les } i_{\beta,\alpha} \in \widetilde{C^\pi} , \qquad j_\alpha \in C ,$$

et que l'on ait

$$A_\alpha = \varinjlim_{\beta<\alpha} A_\beta \qquad \text{si } \alpha \text{ ordinal limite} .$$

Cela précise déjà la définition récurrente pour $\alpha = \alpha_0$ ordinal limite. Notons que le A_{α_0} construit en termes des A_β ($\beta < \alpha_0$) satisfait bien aux conditions que [page 79]

a) les $i_{\alpha_0,\beta} : A_\beta \longrightarrow A_{\alpha_0}$ sont dans $\widetilde{C^\pi}$, car $\widetilde{C^\pi}$ est stable par "limites inductives ordinales strictes" (*cf.* 1.19, p. 8),

b) $j_{\alpha_0} : A_{\alpha_0} \longrightarrow B$ est dans C, car c'est la limite inductive filtrante du système des $A_\beta \longrightarrow B$ ($\beta < \alpha_0$) qui sont dans C, et C est stable par ces limites.

Il reste donc à préciser comment on passe de A_α à $A_{\alpha+1}$. Cela revient à prouver le

Lemme 6.4. *Soit $f : A \hookrightarrow B$ dans C, telle que f ne soit pas un isomorphisme. Alors il existe un sous-objet A_1 de B,* distinct de A, *tel que $A \longrightarrow A_1$ soit dans $\widetilde{C^\pi}$, et $A_1 \longrightarrow B$ dans C.*

Si ce lemme est admis, cela signifie que la construction transfinie se continue aussi longtemps que l'on trouve des A_α tels que $A_\alpha \neq B$, *i.e.* $A_\alpha \hookrightarrow B$ ne soit pas un isomorphisme. Mais comme les A_α sont tous différents, si π est l'ensemble des sous-objets de B (on sait que c'est un

ensemble petit, à cause des hypothèses faites sur $\mathcal{M}$), les ordinaux α pour lesquels A_α existe satisfont $\operatorname{card} A_\alpha \leq \pi$. Donc il doit exister un α tel que $A_\alpha = B$. Mais alors $i_\alpha : A \longrightarrow A_\alpha$ est égal à f, donc $f \in \widetilde{C^\pi}$, *q.e.d.* p. 44 [page 80]

Il reste donc à prouver 6.4. Pour ceci, soit I l'ensemble des sous-objets de B qui sont dans Filt^π, ordonné par inclusion. (Si B est dans Filt^π, A aussi, donc f est déjà dans $C^\pi \subset \widetilde{C^\pi}$, et il n'y a rien à prouver.) C'est un ensemble filtrant grand devant π (et même π-*adapté*), dont la $\varinjlim$ est B (5.3, p. 46). Considérons les

$$A_i \;=\; B_i \cap A\;,$$

ils sont dans Filt^π, comme sous-objets d'objets de Filt^π (condition (d) sur la filtration cardinale). Ainsi, la flèche f apparaît comme limite inductive filtrante des flèches

$$f_i \;:\; A_i \longrightarrow B_i\;.$$

Il est trivial que le système des A_i est cofinal dans $\mathrm{Ss}_\pi(A)$, donc $A = \varinjlim A_i$, d'où $f = \varinjlim f_i$.

D'autre part, je dis que le système inductif des f_i dans $\underline{\mathrm{Fl}}(\mathcal{M})$ est *bien adapté à π*. On sait déjà que I est bien adapté à π, le sup dans I d'une partie filtrante J de cardinal $\leq \pi$ étant le sup dans l'ensemble ordonné $\mathrm{Ss}(B)$ (lequel sup se trouve dans $\mathrm{Ss}_\pi(B)$). Il est essentiel, pour notre argument, de savoir que si $i_J = \sup J$, alors non seulement $B_{i_J} = \varinjlim_{i\in J} B_i$, mais aussi $A_{i_J} = \varinjlim_{i\in J} A_i$, de sorte que le système inductif des f_i est π-adapté dans $\underline{\mathrm{Fl}}(\mathcal{M})$. Il semble qu'il faille donc une condition d'exactitude sur $\mathcal{M}$ voisine de (e), savoir : [page 81] ‖

(e′) Les foncteurs $\varinjlim$ filtrantes commutent au changement de base.

(Il le suffit ici dans le cas très particulier d'un changement de base par un monomorphisme $A \longrightarrow B$, et d'une limite inductive filtrante de sous-objets de B pour des morphismes d'inclusion comme transitions.) Notons que (e), (e′) se trouvent généralement comme conséquence de la condition

(e_0) Les $\varinjlim$ filtrantes dans I sont exactes à gauche.

Ceci posé, on voit donc, C étant π-accessible et I grand devant π, que l'ensemble I_C des $i \in I$ tels que $f_i \in C$ est *cofinal* dans I. Mais comme $f_i \in \mathrm{Filt}^\pi$, $f_i \in C$ équivaut à $f_i \in C^\pi$. [page 82] Comme $A \longrightarrow B$ n'est pas un isomorphisme, il existe un i_0 tel que $B_{i_0} \not\subset A$, puis un $i \geq i_0$ tel que $f_i \in C^\pi$, et aussi $B_i \not\subset A$ (puisque $B_i \supset B_{i_0}$). Ainsi on a prouvé ceci :

Commentary · Examples [ms. 81]

Exact filtered unions do not make every amalgam a subobject

Grothendieck identifies two different requirements while trying to construct the next subobject of B. First, the approximating intersections $A \cap B_i$ must commute with bounded filtered unions. Second, after attaching B_i to A along that intersection, the pushout must still map monomorphically into B. The first requirement concerns a limit followed by a filtered colimit; the second concerns a pushout followed by an image comparison. They have different answers even in familiar algebraic categories.

Under condition (e′), the identity

$$A \times_B \left(\varinjlim_i B_i \right) \simeq \varinjlim_i (A \times_B B_i)$$

provides the desired continuity. Only this pullback along the fixed monomorphism $A \to B$ and filtered diagrams of subobjects of B are needed. Commutation of filtered colimits with finite limits is

sufficient. It also implies condition (e): a map is monic exactly when its diagonal to its self-pullback is an isomorphism, and filtered colimits then preserve that characterization. The phrase "dans I" in the printed condition (e_0) should be read as "dans $\mathcal{M}$".

For the second requirement consider groups. Let $B = \mathbb{Z} \times \mathbb{Z}$, let $A = \mathbb{Z} \times 0$, and let $B' = 0 \times \mathbb{Z}$. Both maps into B are monomorphisms and $A \cap B'$ is the trivial group. Their pushout in groups is

$$A * B' \cong F(a, b),$$

the free group on two generators. The canonical map to B sends a and b to its two coordinate generators. It sends the commutator $aba^{-1}b^{-1}$ to zero, but that reduced word is nontrivial in the free group. Hence the canonical comparison $A * B' \to B$ is not monic. Meanwhile filtered colimits of groups commute with finite limits: both are formed on underlying sets, and a finite list of equations or a finite tuple of representatives is settled in one sufficiently late filtered stage. Thus even condition (e_0) does not repair this pushout problem.

In sets and presheaves of sets the corresponding pushout of two subobjects over their intersection is their union and does embed into B. The proof is componentwise: two elements in the disjoint union are identified precisely when they represent the same element in the intersection. This favourable geometry removes the obstruction just exhibited for groups. It still does not force the residual inclusion $A \cup B' \to B$ to remain in an arbitrarily chosen accessible class C. The pure-monomorphism counterexample later in the commentary occurs in such a favourable presheaf category, and therefore isolates that remaining obstruction.

A further hypothesis has slipped between the statement and this proof. The version of condition (d) restated in Theorem 6.1 omits closure under subobjects, although page 80 uses it to conclude $A \cap B_i \in \mathrm{Filt}^\pi$. The original condition on manuscript pages 45–46 includes that closure. Retaining it makes this approximation step valid; it does not mend the final inference on page 84.

(∗) Sous les hypothèses de (6.4) et moyennant l'hypothèse (e′) sur $\mathcal{M}$, il existe un sous-objet B' de B satisfaisant les conditions suivantes.

a) $B' \in \mathrm{Filt}^\pi$.

b) $B' \not\subset A$.

c) Posant $A' = B' \cap A$, l'inclusion

$$i' \ : \ A' \ \longrightarrow \ B'$$

est dans C^π.

De plus, l'ensemble des $J \subset \mathrm{Ss}_\pi(B)$ formé des B' en question est cofinal dans $\mathrm{Ss}_\pi(B)$, et même π-cofinal. p. 45

Il faut prouver encore ce dernier point – que J (dont on sait qu'il est cofinal) est même π-cofinal, *i.e.* stable par sup (dans $\mathrm{Ss}_\pi(B)$) des parties filtrantes $J' \subset J$ telles que $\mathrm{card}\, J' \leq \pi$. Mais cela résulte aussitôt du fait que C étant stable par

limites inductives filtrantes, $C^\pi = C \cap \mathrm{Filt}^\pi$ est stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$, [page 83] en vertu de la condition (*d*) sur Filt^π. Considérons maintenant le carré cocartésien

$$\begin{array}{ccc} A' & \xhookrightarrow{\ i'\ } & B' \\ \downarrow & & \downarrow \\ A & \xrightarrow{\ i\ } & C' = A \amalg_{A'} B' \end{array} \xrightarrow{\ j\ } B \ .$$

Comme $i' \in C^\pi$, on a $i \in \widetilde{C^\pi}$ (car $\widetilde{C^\pi}$ stable par cochangement de base). D'autre part, j'ai besoin du fait que j soit dans C, et je ne sais pas même qu'il est un monomorphisme ! Aussi il faut repasser aux notations indicielles

$$\begin{array}{ccc} A_i & \xhookrightarrow{\ \alpha_i\ } & B_i \\ \downarrow & & \downarrow \\ A & \xrightarrow{\ \beta_i\ } & C_i \end{array} \xrightarrow{\ \gamma_i\ } B \ .$$

Comme la formation des sommes amalgamées commute aux limites inductives filtrantes, on voit que $\varinjlim C_i \simeq A \amalg_{\varinjlim A_i} \varinjlim B_i = A \amalg_A B = B$, donc la limite inductive des γ_i est un isomorphisme (essentiellement id_B). D'autre part, le même argument montre que le système inductif des C_i ($i \in J$) dans $\mathcal{M}$ est π-adapté, donc aussi celui des γ_i [28] dans $\underline{\mathrm{Fl}}(\mathcal{M})$. Comme [page 84]
$\varinjlim \gamma_i$ est un isomorphisme, il est dans C. Comme C est π-accessible, il s'ensuit que l'ensemble J' formé des $i \in J$ tels que $\gamma_i \in C$ est lui aussi cofinal [29]. Donc on trouve i avec $\alpha_i \in C^\pi$, $\gamma_i \in C$, ce qui établit le lemme 6.4, et achève la démonstration du théorème 6.1.

Commentary · Development (commentators') [ms. 83]

A continuous diagram outside the cardinal bound

Grothendieck's footnote 28 locates the decisive obstruction accurately: the arrows $\gamma_i : C_i \to B$ are not known to be bounded objects of the arrow category. The following explicit test shows that the omitted bound cannot be recovered from continuity and an isomorphic colimit. We give the construction to isolate the inference, without making it serve as a counterexample to the entire theorem.

An unbounded counterdiagram. [ours] Let π be an infinite cardinal, let B be a set with $|B| > \pi$, and let $I = [B]^{\leq \pi}$, ordered by inclusion. In the arrow category of sets consider

$$\gamma_S : S \hookrightarrow B \qquad (S \in I),$$

with transition squares given by inclusion on sources and the identity on B. The poset I is π-adapted, the diagram preserves suprema of nonempty directed subsets of cardinal at most π, and its colimit is the identity of B. Nevertheless no γ_S is an isomorphism.

Proof. A family of at most π subsets of cardinal at most π has union of cardinal at most π. Thus I is π^+-directed and directed unions of that size are its suprema. Colimits of the source sets in such a directed family are their unions. The target is the constant diagram at B, whose colimit is B because the indexing category is nonempty and connected. Colimits in the arrow category are pointwise, proving the asserted continuity. Taking the union of all $S \in I$ gives B, and the colimit arrow is consequently id_B. Finally every S is a proper subset of B, so every stage fails to be an isomorphism. □

For completeness, the class of isomorphisms has exactly the accessibility properties relevant here. It is closed under filtered colimits in the arrow category, since a compatible family of inverse maps induces the inverse of the colimit map. Its bounded part is closed under bounded filtered colimits. For an isomorphism $f : A \to D$, a small subarrow $A_0 \to D_0$ can be enlarged to the isomorphism $f^{-1}(D_0) \to D_0$, still small; hence bounded isomorphic subarrows are cofinal. Conversely, if isomorphic subarrows are cofinal, their filtered colimit is f, so f is an isomorphism. The standard cardinal filtration of the arrow category of sets has bounded objects π^+-presentable, and Proposition 5.7 therefore makes this class L-π-accessible in the manuscript's sense.

The displayed diagram does not contradict that recognition property: its target B is too large, so none of its arrows belongs to $\mathrm{Fl}^\pi(\mathbf{Set})$. It does contradict the proposed inference after deleting precisely that hypothesis. Even the strongest possible conclusion about the colimit, namely that it is an identity, says nothing about cofinally many isomorphic stages in this situation.

This is an elementary diagnostic construction supplied here. We checked the bounded-presentation requirements in Chapter XIII, manuscript pages 86–89, and the diagram-filtration statement of SGA 4 I 9.24; neither licenses the deleted hypothesis. We do not claim priority for this example. Its purpose is to make the cancelled inference visible before discussing a counterexample to the generation theorem itself.

Commentary · Development (commentators') [ms. 84]

A presheaf counterexample to the abandoned generation theorem

Grothendieck has isolated the precise defect of his argument, but he still asks whether the theorem might survive it. There is now a negative answer in a particularly elementary presheaf

28. *Sauf* qu'on ne sait pas que les C_i soient dans Filt^π – en fait, ils ne seront pas en général, puisqu'on ne suppose pas que $A, B \in \mathrm{Filt}^\pi$.

29. *Canulé, cf.* annotation précédente.

category. The modern obstruction is a theorem about pure monomorphisms. To apply it here, however, we must verify the manuscript's bounded cofinality condition; replacing that condition by the modern word "accessible" would leave the essential comparison undone.

Let P be the three-object category with its two nonidentity arrows

$$1 \longleftarrow 0 \longrightarrow 2, \qquad \mathcal{E} = \mathrm{Set}^{P}.$$

Thus an object is a span of sets $E_1 \leftarrow E_0 \rightarrow E_2$. This is the presheaf category on P^{op}. Give $\mathcal{E}$ the filtration by the total cardinality of its three sets. For every infinite π, the objects of size at most π are π^+-presentable: a natural transformation out of one of them is specified by at most π elements and equations, all of which can be represented simultaneously at one stage of a π^+-filtered diagram. Subobjects, quotients, and colimits with at most π arrows preserve this bound. All colimits and finite limits are computed on the three underlying sets. In particular, this example satisfies even the strengthened exactness assumptions introduced after Theorem 6.1.

An inclusion $U \hookrightarrow V$ in $\mathcal{E}$ is *pure* if every finite system of equations in the three sorts, with parameters in U, that has a solution in V already has a solution in U. Equivalently, for every commutative square

$$\begin{array}{ccc} K & \xrightarrow{u} & U \\ {\scriptstyle a}\downarrow & & \downarrow \\ L & \xrightarrow[v]{} & V \end{array}$$

with K, L finite, there is $t : L \to U$ with $ta = u$; the other triangle is not required to commute. Indeed, the finitely many elements and structure equations of L, together with $t(a(k)) = u(k)$, give exactly such a system. Conversely, a finite system and its parameters can be encoded by a morphism between finite spans. Write $\mathcal{P}$ for the class of pure monomorphisms.

Proposition (comparison with the manuscript). [ours] *The class $\mathcal{P}$ is L-π-accessible for every infinite cardinal π in this filtration. It contains the isomorphisms and is stable under pushouts, composition, transfinite composition, and retracts. Nevertheless there is no set $S \subset \mathcal{P}$ with $\mathcal{P} \subset \widetilde{S}$. Consequently the assertion of Theorem 6.1 is false, even for a presheaf category.*

We give the comparison and closure arguments, and use the published non-generation theorem only at the last step. First, filtered colimits preserve purity. Given a finite system with parameters in the source of a filtered colimit arrow and a solution in its target, represent its parameters and solution at a common stage. The finitely many equations hold at a further common stage. Purity there supplies a solution in that stage's source, which gives the required solution in the colimit. The colimit arrow is a monomorphism because filtered colimits of sets preserve monomorphisms. This argument permits arbitrary transition squares; it is not restricted to diagrams of inclusions.

Next fix a pure inclusion $A \subset B$ and a subarrow $U_0 \subset V_0$ of size at most π. Thus $U_0 \subset A$, $V_0 \subset B$, and $U_0 \subset V_0$, but there is no requirement that $U_0 = A \cap V_0$. We construct increasing subarrows $U_n \subset V_n$. For every finite equation system with parameters in U_n that is solvable in V_n, choose a solution in A, using the purity of $A \subset B$. There are at most π systems to consider. Adjoin all their chosen solutions to U_n and close under the two structure maps of the span, obtaining U_{n+1} of size at most π. Let V_{n+1} be the subspan of B generated by $V_n \cup U_{n+1}$. Its size is still at most π. At the countable union,

$$U = \bigcup_{n<\omega} U_n, \qquad V = \bigcup_{n<\omega} V_n,$$

the inclusion $U \subset V$ is pure. A finite list of parameters and a finite solution in V both occur at some stage, so its required solution was added at the next stage. Hence the pure subarrows of size at most π are cofinal among all subarrows of that size. They are also closed under filtered unions with at most π members, by the preceding paragraph. Conversely, an arrow whose bounded pure subarrows are cofinal is their filtered colimit and is pure. This is precisely the cofinal small-subobject recognition condition of Proposition 5.7(c), applied to $\underline{\mathrm{Fl}}(\mathcal{E})$. That proposition therefore gives the manuscript's L-π-accessibility, with its original, stronger meaning.

For completeness, pushout stability can be proved without a further smallness assumption. Every pure arrow is a filtered colimit of split monomorphisms between finite spans. To see this, the category of finite arrows mapping to a fixed arrow $f : A \to B$ is filtered, and its colimit is f: finite sets of elements and relations can be put into one finite subarrow. If $a : K \to L$ is such an arrow, purity gives $t : L \to A$ extending its map $K \to A$. Form

$$D = L \amalg_K L, \qquad j_0, j_1 : L \longrightarrow D.$$

The arrow j_0 is split by the fold $D \to L$. There is a morphism $a \to j_0$ in the arrow category with source component a and target component j_1. There is also a morphism $j_0 \to f$: its source component is t, and its target component is induced by ft on the first copy of L and the original $L \to B$ on the second. These two maps agree on K. Thus every finite arrow over f maps to a finite split monomorphism over f. The full subcategory of such split monomorphisms is filtered: any finite diagram first has a cone in the category of finite arrows over f, and the construction just given replaces that cone by a split one. For any fixed finite arrow $a \to f$, the same construction, keeping the map from a as part of the diagram, makes its comma category of split targets nonempty and filtered. This proves finality, so the colimit of the split subcategory is f.

Now push out this filtered presentation of f along any $A \to A'$. The resulting arrow is a filtered colimit of pushouts of split monomorphisms, each of which is split. It is pure by filtered-colimit stability. Composition of pure inclusions follows by reflecting a finite system twice. Retract stability follows by composing a finite square with the section into the larger arrow, taking its pure extension, and composing with the retraction. Transfinite composition follows by induction, using composition at successor stages and filtered-colimit stability at limit stages. These arguments establish every closure property asserted above.

The external ingredient is now exact and short. Cox et al. (2026b, Theorem 5.1) prove that pure monomorphisms in Set^D are cofibrantly generated if and only if every coslice d/D is linearly preordered. For our P, the two objects $0 \to 1$ and $0 \to 2$ of $0/P$ admit no morphism in either direction. Thus $\mathcal{P}$ is not cofibrantly generated. If Theorem 6.1 held, it would give a set $S \subset \mathcal{P}$ with $\mathcal{P} \subset \widetilde{S}$. The small object argument in $\mathcal{E}$ identifies $\widetilde{S}$ with the closure of S under pushouts, transfinite composition, and retracts. The closure properties just proved give the reverse inclusion $\widetilde{S} \subset \mathcal{P}$. This would make $\mathcal{P}$ cofibrantly generated, a contradiction. □

This also explains why the successful theorem of the next section has genuinely more structure. Purity has cancellation in the direction "gf pure implies f pure". Grothendieck's construction in Section 7 needs the other cancellation: from f and gf in W, it must recover $g \in W$. That step keeps the residual inclusion in the desired class after a small attachment. It is precisely what the preceding argument cannot obtain for a general L-accessible class.

The non-generation theorem and its presheaf classification belong to Cox, Feigert, Kamsma, Mazari-Armida, and Rosický; no priority claim is made here. Our contribution is the explicit check of the manuscript's bounded recognition condition and the resulting application to Theorem 6.1. We checked Theorem 5.1 and its reduction to the preceding results, Definition 2.7, Remark 2.8, Fact 2.9, and Example 5.3 in the 26-page third arXiv version, dated 29 April 2026. We did not find this specific application to *Les Dérivateurs* stated there.

7 Construction de triples de Quillen clos avec $C = \mathbf{Mon} \subset \mathbf{Cof}_W$

Finalement, tous les beaux développements des paragraphes 5, 6 ne me donnent pas le "sweeping result" souhaité. Peut-être est-il faux – peut-être la propriété d'engendrement "~" p. 46
par une petite partie (pour les ensembles accessibles, ou L-accessibles) est-elle beaucoup plus rarement satisfaite, que je ne l'aurais imaginé. Pour bien faire, il faudrait que je traite jusqu'au bout quelques exemples, p. ex. celui de Cof_W, et de W^{univ} dans Cat, pour un localiseur [page 85]
fondamental plus ou moins général, et surtout dans le cas $W = \mathbf{W}_\infty$. Sûrement Cof_W et W^{univ} sont accessibles (il faudrait que je le vérifie pourtant), et si le "théorème" hypothétique du paragraphe 6 est faux pour eux, du moins en aurai-je le cœur net. Mais il reste pourtant la question : si $C = \widetilde{C}$ est stable par $\varinjlim$ filtrantes, et de la forme $\widetilde{C_0}$, C_0 petit, et si W est L-accessible également, alors $C \cap W$ est-il aussi de la forme $\widetilde{TC_0}$, avec TC_0 petit ? Je viens encore d'essayer de le prouver, en supposant de plus $C \subset$ mono, et ne vois aucun moyen d'y parvenir. Il faudrait donc un contre-exemple pour dissiper ces perplexités ! Et le mieux serait dans Cat, tant qu'à faire.

Il surnage pourtant un résultat tangible positif, que je vais maintenant essayer de bien cerner et prouver : [page 86]

Théorème 7.1. *Soit $\mathcal{M}$ une catégorie stable par petites $\varinjlim$ filtrantes, $M_0 \subset \mathrm{Ob}\,\mathcal{M}$ un sous-ensemble de $\mathrm{Ob}\,\mathcal{M}$ générateur par monomorphismes stricts, plus précisément on suppose ceci :*

Commentary · Exposition [ms. 85]

What the generation argument must preserve

Grothendieck now changes the question in a decisive way. The unsuccessful argument of §6 tried to recover a lifting class from accessibility alone. The useful extra information is that an arrow in the desired class can be enlarged a little while its remaining part stays in the same class. This is a geometric condition on a factorization, and it is stronger than the assertion that the arrow is a filtered colimit of small arrows. A filtered presentation allows both its source and its target to vary. A cellular construction keeps the original source fixed and attaches successive pieces to it. The discrepancy between those two operations was precisely what invalidated the preceding proof.

We use modern lifting notation to make the goal explicit. For a class S of arrows, $S^{\square}$ consists of arrows having the right lifting property against every member of S, and ${}^{\square}T$ denotes the corresponding left lifting class. Grothendieck's S_* is $S^{\square}$, and his left Q-closure $\widetilde{S}$ is ${}^{\square}(S^{\square})$. When S is a set and the small object argument applies, this is the closure of S under pushouts, transfinite composition and retracts. A proof that $C \subset \widetilde{C_0}$ must therefore account for this particular closure; being generated by filtered colimits is a different assertion.

The property in Theorem 7.1 supplies the missing control. Suppose $A \hookrightarrow B$ is not yet an isomorphism. A small subobject $B' \hookrightarrow B$ detects something outside A, and the union

$$D = A \amalg_{A\cap B'} B'$$

still embeds in B. Thus $A \subsetneq D \subset B$ is a genuine enlargement, obtained by one permitted attachment. For trivial cofibrations one must additionally keep $D \hookrightarrow B$ a weak equivalence. This is where cancellation in W enters: once $A \to D$ and $A \to B$ lie in W, so does $D \to B$. The hypotheses that look separate in the statement are arranged to secure these two facts, geometric containment and homotopical cancellation.

The question on manuscript page 85 has a precise later answer in the setting of model recognition. If $\mathcal{M}$ is locally presentable, $C = \mathrm{cof}(I)$ for a set I, W is an accessible full subcategory of the arrow category with an accessible inclusion, W has two-out-of-three and is closed under retracts, $I^{\square} \subset W$, and $C \cap W$ is closed under pushouts and transfinite compositions, then $C \cap W$ is generated by a set. This is Smith's theorem, published with proof by Beke (2000, Theorem 1.7 and Lemmas 1.8–1.9). Its solution-set hypothesis is supplied by accessibility. It does not assert that the intersection of an arbitrary accessible class with a lifting class is automatically a lifting class: the closure and compatibility hypotheses remain part of the theorem. Consequently it resolves the model-structure problem which motivates this passage, without turning the tentative claim of §6 into a general theorem.

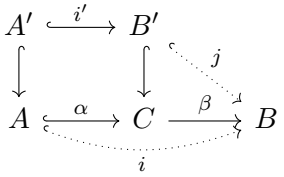
Pour tout monomorphisme $i : A \hookrightarrow B$ dans $\mathcal{M}$ qui n'est pas un isomorphisme, il existe un monomorphisme $B' \overset{j}{\hookrightarrow} B$ ayant les propriétés suivantes :

a) *$B' \in M_0$.*

b) *j ne se factorise pas par i.*

c) *$A' = A \times_B B'$ existe dans $\mathcal{M}$, ainsi que $C = A \amalg_{A'} B'$, A' est dans M_0, et dans le diagramme canonique*

(7.1.1)



la flèche β est un monomorphisme.

(**N.B.** Ces conditions sont satisfaites si on suppose, en plus de la condition que M_0 soit générateur pour les monomorphismes, que M_0 contient, avec tout objet B', et pour tout objet A' qui est soit un quotient, soit un sous-objet de B', un objet A'_0 isomorphe à A' ; et si de plus $\mathcal{M}$ satisfait aux conditions d'exactitude suivantes :

1°) Toute flèche f de $\mathcal{M}$ se factorise en ip, avec i monomorphisme, p épimorphisme, et :

2°) Si A_1, A_2 sont deux sous-objets d'un objet B de $\mathcal{M}$, alors l'intersection $A_0 = A_1 \times_B A_2$ existe, ainsi que la somme amalgamée $C = A_1 \amalg_{A_0} A_2$, et le morphisme canonique $C \longrightarrow B$ est un monomorphisme.) p. 47 [page 87]

Supposons de plus que $\mathcal{M}$ satisfait la condition ©, p. 76, i.e. *que toute* $\varinjlim$ *filtrante de monomorphismes est un monomorphisme.*

1°) *Soit* $\mathrm{Mon} \subset \mathrm{Fl}(\mathcal{M})$ *l'ensemble des monomorphismes, et* Mon_0 *le sous-ensemble de* Mon *formé des* $i \in \mathrm{Mon}$ *tels que source et but appartiennent à* M_0. *On a alors*

$$\text{(7.1.2)} \qquad \mathrm{Mon} \subset \widetilde{\mathrm{Mon}_0} \quad , \qquad d'où \quad \begin{cases} \widetilde{\mathrm{Mon}} & = & \widetilde{\mathrm{Mon}_0} \\ (\mathrm{Mon})_* & = & (\mathrm{Mon}_0)_* \end{cases} .^{30}$$

2°) *Si* M_0 *est* petit *et si les objets de* M_0 *sont accessibles* (p. ex. $\mathcal{M}$ satisfait ©, p. 76), *et si* Mon *est stable par cochangement de base, alors il existe une factorisation fonctorielle de toute flèche* f *de* $\mathcal{M}$ *en*

$$f = p\,i\,,$$

avec $i \in \mathrm{Mon}$, $p \in (\mathrm{Mon})_*$. (**N.B.** Une flèche $p : X \longrightarrow Y$ de $\mathcal{M}$ est dans $(\mathrm{Mon})_*$ si et seulement si elle fait de X un objet *injectif* de $\mathcal{M}/Y$ – donc l'assertion de factorisation assure que dans les catégories induites $\mathcal{M}/Y$ "il y a assez d'injectifs".) *De plus, on a*

$$\text{(7.1.3)} \qquad \mathrm{Mon} = \widetilde{\mathrm{Mon}_0} \quad .$$

3°) *Soit* W *une partie de* $\mathrm{Fl}(\mathcal{M})$ *qui est* L-accessible (i.e. *accessible, et stable par limites inductives filtrantes*). (*On suppose à présent que* $\mathcal{M}$ *satisfait les conditions* (a), (b), (c), (d), (e) *de la page 76, d'où une bonne théorie des parties L-accessibles de* $\mathrm{Ob}\,(\mathcal{M})$ *ou de* $\mathrm{Fl}(\mathcal{M})$.) *On suppose, plus précisément, que* W *satisfait les conditions :* [page 88]

$$\text{(7.1.4)} \qquad \begin{cases} \text{a)} & W \textit{ est L-accessible } (\textit{i.e.} \text{ stable par } \varinjlim \text{ filtrantes, et accessible}). \\ \text{b)} & \textit{Soient } \alpha, \beta \textit{ composables, avec } \beta\alpha \textit{ et } \alpha \textit{ dans } W, \textit{ alors } \beta \in W. \text{ (\textbf{N.B.} Cette condition sera satisfaite si } W \text{ est un localiseur modérément saturé, et c'est à cette situation que nous allons appliquer notre énoncé.)} \\ \text{c)} & \mathrm{Mon} \cap W \textit{ est stable par cochangement de base.} \end{cases}$$

et que $\mathcal{M}$ *satisfait les conditions supplémentaires*

$$\text{(7.1.5)} \qquad \begin{cases} \text{e}') & \textit{Les } \varinjlim \textit{ filtrantes commutent au changement de base.} \\ \text{f)} & \textit{La condition d'exactitude } (7.2.1) \textit{ ci-dessous.} \end{cases}$$

Soit π *un cardinal tel que* W *soit* π*-accessible, et* $M_0 \subset \mathrm{Filt}^\pi$ (*donc* M_0 *essentiellement petite*). *Alors on a*

$$\text{(7.1.6)} \qquad W \cap \mathrm{Mon} \subset \widetilde{(W \cap \mathrm{Mon})^\pi}$$

(*où pour toute partie* C *de* $\mathrm{Fl}(\mathcal{M})$, *on pose* $C^\pi = C \cap \mathrm{Fl}^\pi(\mathcal{M})$), *donc*

30. **N.B.** On a là un critère d'injectivité d'objets dans les catégories $\mathcal{M}/Y$ – il suffit de tester la condition d'injectivité par des monomorphismes i qui sont dans Mon_0. p. 48

(7.1.7) $$\widetilde{W \cap \mathrm{Mon}} \;=\; (\widetilde{W \cap \mathrm{Mon}})^{\pi}\ ,$$

donc $\widetilde{W \cap \mathrm{Mon}}$ *est Q-engendré par une* petite *partie de* $W \cap \mathrm{Mon}$.

Commentary · Examples [ms. 87]

Small attachments in presheaves and in group actions

Grothendieck's generator condition has an especially concrete meaning for presheaves. Let A be small and choose an infinite cardinal $\kappa \geq |\operatorname{Mor} A|$. For $F \in \widehat{A}$, put $|F| = \sum_{a \in \operatorname{Ob} A} |F(a)|$. Up to isomorphism the presheaves of cardinal at most κ form a set. Let M_0 be a skeleton of this class, and let I contain all monomorphisms between its objects.

Given a proper inclusion $U \subset V$, choose a and $v \in V(a) \setminus U(a)$. The associated map $h_a \to V$ has image B', a subpresheaf of cardinal at most κ, and B' is not contained in U. Its intersection $A' = U \cap B'$ again has cardinal at most κ. Since unions and intersections are computed objectwise, the canonical map

$$U \amalg_{A'} B' \longrightarrow V$$

is a monomorphism, with image $U \cup B'$. This verifies every part of (7.1.1). The image is essential: $h_a \to V$ need not itself be a monomorphism, so directly attaching h_a need not produce a subobject of V. Passing to its quotient image is exactly the reason for allowing quotients among the small pieces.

There is a more explicit calculation when $\widehat{A}$ is the category of left G-sets. Every monomorphism $U \hookrightarrow V$ identifies U with a union of G-orbits in V, and its complement is a disjoint union of orbits G/H. Thus the set

$$I_G = \{\varnothing \longrightarrow G/H \mid H \leq G\}$$

generates all monomorphisms by coproducts and transfinite attachments; retracts are unnecessary for this particular generation. A map $p : X \to Y$ belongs to $I_G^{\square}$ precisely when, for every subgroup $H \leq G$, the map on fixed points

$$X^H \longrightarrow Y^H$$

is surjective. Indeed, an equivariant map $G/H \to Y$ is determined by an H-fixed element y, and a lift is exactly an H-fixed x with $p(x) = y$. Conversely the fixed-point condition solves every orbit attachment and hence every monomorphism lifting problem.

This example explains the word "injective" in the source. It refers to an injective object of the slice, tested against all monomorphisms in that slice. Ordinary surjectivity need not suffice. If G is nontrivial, the equivariant map $G \to *$ is surjective as a map of sets, but G^G is empty, so it fails the test for $H = G$. The small object factorization adds precisely the missing equivariant extensions; it does not merely repair the underlying map of sets.

Corollaire 7.2. *Soit* $\mathcal{M}$ *satisfaisant* (a), (b), (c), (d), (e) *et aux conditions supplémentaires :*

(7.2.1)
- e′) *Les* $\varinjlim$ *filtrantes commutent au changement de base.*
- f) *Si* A_1, A_2 *sont deux sous-objets d'un objet* B *de* $\mathcal{M}$, *et* $A_0 = A_1 \cap A_2$, *alors* $A_1 \amalg_{A_0} A_2 \hookrightarrow B$ *est un monomorphisme*[31].

31. Une autre façon de le dire : si $C = \sup_{\mathrm{Ss}(B)}(A_1, A_2)$, le carré cartésien

$$\begin{array}{ccc} A_0 & \longrightarrow & A_1 \\ \downarrow & & \downarrow \\ A_2 & \longrightarrow & C \end{array}$$

est aussi cocartésien.

Soit W une partie de $\mathrm{Fl}(\mathcal{M})$ *satisfaisant les conditions suivantes.*

(7.2.3)
- a) *W est un localiseur modérément saturé.*
- b) *W est L-accessible,* i.e. *il est stable par* $\varinjlim$ *filtrantes, et est accessible.*
- c) $\mathrm{Mon} \cap W$ *est stable par cochangement de base,* i.e. *si on a un diagramme* cocartésien

$$\begin{array}{ccc} A & \xrightarrow{i} & B \\ {\scriptstyle\alpha}\downarrow & \text{cocart.} & \downarrow{\scriptstyle\beta} \\ A' & \xrightarrow{i'} & B' \end{array}$$

avec i monomorphisme appartenant à W, alors i' est un monomorphisme appartenant à W.[32]

[page 89]

Sous ces conditions, on a

$$\underbrace{\mathrm{Mon}}_{C} \cap W \;=\; \widetilde{W \cap \mathrm{Mon}} \quad , \qquad \text{i.e. } \mathrm{Mon} \cap W \text{ est Q-clos à gauche} \, , \tag{7.2.4}$$

et p. 49

$$\text{il existe une petite partie } TC_0 \subset \mathrm{Mon}, \text{ telle que } \mathrm{Mon} \cap W = \widetilde{TC_0} \, . \tag{7.2.5}$$

(De façon plus précise, si $\pi \geq \pi_0$ est un cardinal tel que W soit π-accessible, on peut prendre $TC_0 = (\mathrm{Mon} \cap W)^{\pi}$ (ou une partie petite *de $(\mathrm{Mon} \cap W)^{\pi}$ ayant les mêmes classes d'isomorphie).) Donc* (th. 1 de §1) *le couple $(TC = \mathrm{Mon} \cap W,\ F = (TC)_*)$ est un couple de Quillen clos, et il y a même une factorisation* fonctorielle *de toute flèche f en*

$$f \;=\; p\,i \, ; \qquad i \;\in\; TC \, , \qquad p \;\in\; F \, . \tag{7.2.4}$$

Si on pose

$$C \;=\; \mathrm{Mon} \, , \qquad TC \;=\; \mathrm{Mon} \cap W \, , \qquad F \;=\; (TC)_* \, , \qquad TF \;=\; C_* \, , \tag{7.2.5}$$

le triple (W, C, F) est un triple de Quillen clos, si et seulement si on a les conditions supplémentaires suivantes.

- (7.2.6) ⓖ *(sur $\mathcal{M}$ seulement)* Mon *stable par cochangement de base.*
- (7.2.7) $\underbrace{TF}_{\overset{\text{déf}}{=} (\mathrm{Mon})_*} \;\subset\; W$ i.e. *si* d) *toute $f : X \longrightarrow Y$ qui fait de X un objet injectif de $\mathcal{M}/Y$, est dans W (donc aussi dans W_{univ}).*

Démonstration de 7.1. On peut donner un énoncé un peu plus délicat, en supposant d'emblée donnée, non une partie M_0 de $\mathrm{Ob}\mathcal{M}$, mais une partie [page 90]

$$\mathrm{Mon}_0 \;\subset\; \mathrm{Mon} \, , \tag{7.1}$$

satisfaisant la condition suivante

32. *Cf.* plus bas pour une quatrième condition (*d*) sur W, qui assure que W s'insère dans un triple de Quillen.

Pour tout monomorphisme $i : A \longrightarrow B$ dans $\mathcal{M}$, il existe $i' \in \mathrm{Mon}_0$, $i' : A' \longrightarrow B'$, et une flèche $i' \longrightarrow i$ dans $\underline{\mathrm{Fl}}(\mathcal{M})$,

(7.2)
$$\begin{array}{ccc} A' & \xrightarrow{i'} & B' \\ \downarrow & & \downarrow \\ A & \xrightarrow{i} & B\ , \end{array}$$

telle que la flèche correspondante

$$C \;=\; A \amalg_{A'} B' \;\longrightarrow\; B$$

soit un monomorphisme, et que $B' \longrightarrow B$ ne se factorise pas par A.

C'est cela qui était visé par la condition (7.1.1) sur M_0, qui implique (7.2) sur Mon_0 défini en termes de M_0 comme dit dans 1°.

Dans ce qui suit, on suppose donnée une partie $\mathrm{Mon}_0 \subset \mathrm{Fl}(\mathcal{M})$ satisfaisant (7.2). En pratique, elle sera souvent déduite d'une partie M_0 de $\mathrm{Ob}\mathcal{M}$, génératrice pour les monomorphismes, comme explicité dans 1°, mais on n'a pas besoin du fait que $i' \longrightarrow i$ soit un monomorphisme (*i.e.* (7.1.1) est plus forte que ce qu'il nous faut). P. ex. si $B' \in M_0$ et $B' \longrightarrow B$ est donné, ne se factorisant pas par A, on pourra prendre $A' = B' \times_B A$ et $i' : A' \longrightarrow B'$ (pour assurer $i' \in \mathrm{Mon}_0$, il faudra donc supposer M_0 stable par passage aux sous-objets), et il y a des chances p. 50

que $C \longrightarrow B$ sera alors bel et bien un monomorphisme … [page 91]

1°) Il faut prouver

$$\mathrm{Mon} \;\subset\; \widetilde{\mathrm{Mon}_0}$$

moyennant (7.2) (et la stabilité de $\mathcal{M}$ par $\varinjlim$ filtrantes). Cela va résulter directement du

Lemme 7.3. *Soient $\mathcal{M}$ une catégorie stable par limites inductives filtrantes, et C_0, C des parties de* $\mathrm{Fl}(\mathcal{M})$. *On supposera*

ⓐ *C formée de monomorphismes, et*

ⓑ *que la condition suivante (généralisant* (7.2) *ci-dessus) soit vérifiée.*

Pour toute flèche $i : A \hookrightarrow B$ dans C telle que i ne soit pas un isomorphisme, il existe $i' : A' \longrightarrow B'$ dans $\widetilde{C_0}$, et une flèche $i' \longrightarrow i$ dans $\underline{\mathrm{Fl}}(\mathcal{M})$, telle que dans le diagramme

(7.3.1)
$$\begin{array}{ccccc} A' & \xrightarrow{i'} & B' & & \\ \downarrow & & \downarrow & \searrow^{\beta} & \\ A & \xrightarrow{\alpha} & D \;=\; A \amalg_{A'} B' & \xrightarrow{\beta} & B\ , \\ & \searrow_{i} & & \nearrow & \end{array}$$

la flèche β soit dans C, et que $B' \longrightarrow B$ ne se factorise pas par A (ou, ce qui revient au même, que le sous-objet de B défini par D et β ne soit pas égal à celui défini par A et i).

On suppose de plus

ⓒ *C stable par $\varinjlim$ filtrantes, et*

ⓓ *que pour tout $B \in \mathrm{Ob}\mathcal{M}$, l'ensemble des sous-objets de B est petit.* (Condition des plus anodines! C'est une condition *sur* $\mathcal{M}$, n'impliquant pas C). [page 92]

Alors on a

$$C \subset \widetilde{C_0}\ , \qquad \textit{donc } \widetilde{C} \subset \widetilde{C_0}\ , \tag{7.3.2}$$

donc, si $C_0 \subset C$, on conclut

$$\widetilde{C} = \widetilde{C_0}\ . \tag{7.3.3}$$

(Ce lemme implique aussitôt la première assertion de 7.1, 1°.) Prouvons ce lemme, *i.e.* que toute $i \in C$ est dans $\widetilde{C_0}$, en construisant par récurrence transfinie une suite strictement croissante de sous-objets de B, p. 51

$$A_\alpha \overset{j_\alpha}{\hookrightarrow} B\ , \qquad \text{avec les } j_\alpha \in C\ ,$$

et avec :

① $A_0 = A$. On prouvera que $A \xrightarrow{i_\alpha = i_{\alpha,0}} A_\alpha$ dans $\widetilde{C_0}$, et il y aura un ordinal final α tel que $A_\alpha = B$; on en conclura donc que $i = i_\alpha$ est dans $\widetilde{C_0}$.

Si i est un isomorphisme, on sait que $i \in \widetilde{C_0}$, il n'y a rien à prouver (la construction transfinie s'arrête au cran 0). Les deux pas de la construction, après le pas initial 1°, sont :

② Soit A_α construit. Si $A_\alpha = B$, la construction est arrêtée. Si $A_\alpha \xrightarrow{j_\alpha} B$ n'est pas un isomorphisme, on applique (7.3.1) à $j_\alpha \in C$, pour trouver un $D \overset{j}{\hookrightarrow} B$, donc $j \in C$ et $D \supsetneqq A_\alpha$. De plus,

$$i_{\alpha+1,\alpha} : A_\alpha \longrightarrow A_{\alpha+1} \overset{\text{déf}}{=} D \qquad \text{est dans } \widetilde{C_0}\ , \tag{7.3.4}$$

car déduit de $i' \in \widetilde{C_0}$ par cochangement de base.

③ Si α est ordinal limite, on pose

$$A_\alpha = \varinjlim_{\beta<\alpha} A_\beta\ , \tag{7.3.5}$$

et la flèche canonique [page 93]

$$A_\alpha \xrightarrow{j_\alpha} B$$

est dans C, car les j_β le sont et C est stable par limites inductives filtrantes.

On a ainsi achevé la description des systèmes transfinis. Pour tout α, on a

$$\operatorname{card}\alpha \leq \operatorname{card}\operatorname{Ss}(B)\ ,$$

puisque l'application $\beta \longmapsto A_\beta$ de l'ensemble $I_\alpha = \{\beta \mid \beta < \alpha\}$ dans $\operatorname{Ss}(B)$ est *strictement* croissante. Donc il existe un cardinal α tel que $\operatorname{card}\alpha \leq \operatorname{card}\operatorname{Ss}(B)$, tel que $A_\alpha = B$. Mais alors $i \in \widetilde{C_0}$, comme limite inductive (ou composé) transfinie de flèches appartenant à $\widetilde{C_0}$. (Il suffit de la condition $\forall\ \beta$, $i_{\beta+1,\beta} \in \widetilde{C_0}$ pour impliquer que tous les morphismes de transition $i_{\beta',\beta}$ sont dans $\widetilde{C_0}$.)

Revenons à 7.1, 2°. Il reste à prouver que si Mon est stable par cochangement de base, alors on a

$$\operatorname{Mon} = \widetilde{\operatorname{Mon}_0}\ .$$

Mais comme Mon est déjà stable par composition, et par limites inductives filtrantes quelconques (donc aussi par facteurs directs), il s'ensuit par le corollaire 2 du théorème 1, §1 (p. 11), que $\operatorname{Mon} = \widetilde{\operatorname{Mon}_0}$. L'énoncé de factorisation est le théorème 1 du §1.

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[page 94]

Commentary · Exposition [ms. 93]

The transfinite argument and its stopping condition

The strength of Lemma 7.3 is that its proof never guesses a universal set of cells. It assumes that a permissible next cell can be found whenever a remainder is nontrivial, and then proves that the process reaches the target. Here is the full argument, including its size condition.

Fix $i : A \hookrightarrow B$ in C, and let $\mathcal{S}(B)$ be the set of subobjects of B. Choose an ordinal θ whose cardinal is strictly greater than $|\mathcal{S}(B)|$. Starting with $A_0 = A$, apply condition (b) whenever $A_\alpha \hookrightarrow B$ is not invertible, obtaining a pushout attachment $A_\alpha \to A_{\alpha+1}$ in $\widetilde{C_0}$ and a proper inclusion $A_\alpha \subsetneq A_{\alpha+1} \subset B$ whose remainder lies in C. At a nonzero limit ordinal λ, form $A_\lambda = \varinjlim_{\alpha<\lambda} A_\alpha$. The arrows $A_\alpha \to B$ form a filtered diagram in the arrow category with constant target; condition (c) puts its colimit $A_\lambda \to B$ in C, and hence makes it monic. This identifies A_λ with the union of the preceding subobjects.

If the construction had not stopped before θ, it would give more distinct subobjects of B than $\mathcal{S}(B)$ contains. Therefore $A_\lambda = B$ at some stage. Every successor attachment is in $\widetilde{C_0}$, so transfinite closure gives $A \to B \in \widetilde{C_0}$. This also proves the claimed equality of closures when $C_0 \subset C$.

The printed phrase asserting the existence of a "cardinal α" is an ordinal/cardinal slip. The argument produces a stopping *ordinal* λ with $|\lambda| \le |\mathcal{S}(B)|$, not necessarily a stopping stage which is itself an initial ordinal. The distinction changes no conclusion. More substantively, well-poweredness is explicitly assumed in Lemma 7.3, but is absent from the short list at the start of Theorem 7.1(1). Its proof needs this hypothesis, or another smallness argument that forces termination. It holds in all the presheaf applications here. We retain it when stating the general result; we have not obtained a counterexample to the theorem with that hypothesis omitted.

A second small point explains the retract assertions used subsequently. If f is a retract of g in the arrow category, with section s and retraction r, then $e = sr$ is an idempotent endomorphism of g. The sequential diagram $g \xrightarrow{e} g \xrightarrow{e} \cdots$ has colimit f: a cocone is exactly a map out of g invariant under e, and these are exactly maps out of its splitting f. Thus closure under filtered colimits implies closure under retracts. This observation justifies the source's parenthesis without importing a model structure that has not yet been constructed.

Prouvons 7.1, 3°, *i.e.* la formule (7.1.6),

$$W \cap \mathrm{Mon} \;\subset\; \widetilde{(W \cap \mathrm{Mon})^\pi} \ .$$

Pour ceci, on applique encore le lemme 7.3 à

$$C_0 \;=\; (W \cap \mathrm{Mon})^\pi \ , \qquad C \;=\; W \cap \mathrm{Mon} \ .$$

Les conditions (a), (b), (c), (d) du lemme sont vérifiées. C'est clair pour toutes, sauf pour (b), qui demande une vérification non tautologique. Soit donc

$$f \;:\; A \hookrightarrow B \ , \qquad f \in \mathrm{Mon} \cap W,\ f \text{ non isomorphisme} \ .$$

On considère B comme limite inductive [33]

$$B \;=\; \varinjlim_{I} B_i \ , \qquad I \;=\; \mathrm{Ss}_\pi(B) \ .$$

Pour tout i, on pose

$$A_i \;=\; A \cap B_i \;=\; A \times_B B_i \ ,$$

et

$$f_i \;:\; A_i \;\hookrightarrow\; B_i \ ,$$

donc

$$f \;=\; \varinjlim_{i \in I} f_i \ .$$

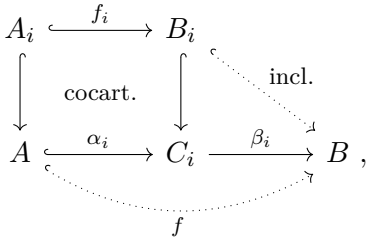
[page 95]

On a besoin de savoir que le système inductif des flèches f_i est π-adapté, ce qui nous amène à postuler la condition d'exactitude supplémentaire suivante sur $\mathcal{M}$:

(7.4) e') les $\varinjlim$ filtrantes dans $\mathcal{M}$ commutent au changement de base ,

il suffit même de le supposer dans le cas très particulier qu'on a envisagé ici.

Comme W est π-accessible, et $(f_i)_{i\in I}$ est π-adapté, comme $\varinjlim f_i \in W$, il s'ensuit que l'ensemble I_W des $i \in I$ tels que $f_i \in W$ est cofinal. Comme $A \overset{f}{\hookrightarrow} B$ n'est pas un isomorphisme, il existe donc un $i \in I_W$ tel que $B_i \hookrightarrow B$ ne se factorise pas par A. On a alors le diagramme



avec $f_i \in (C \cap W)^\pi = C_0$, donc $\alpha_i \in \widetilde{C_0}$. On a besoin de savoir que $\alpha_i \in W$, donc une condition supplémentaire sur W : p. 53

Si on a un carré cocartésien

(7.5)
$$\begin{array}{ccc} A & \xrightarrow{\ i\ } & B \\ \downarrow & & \downarrow \\ A' & \xrightarrow{\ i'\ } & B' \ , \end{array}$$

avec $i \in \mathrm{Mon} \cap W$, alors $i' \in W$ [34].

(j'ai renforcé (7.1.4) en conséquence).

Comme par hypothèse $f = \beta_i\alpha_i$ est dans W, et $\alpha_i \in W$, il s'ensuit par (7.1.4), (a) que $\beta_i \in W$. Il faudra que nous sachions de plus que β_i est un monomorphisme, donc $\beta_i \in \mathrm{Mon} \cap W = C$, ce qui achèvera d'établir la condition manquante (b) de 7.3. C'est la condition d'exactitude (7.2.1), qu'il me faut visiblement rajouter aux hypothèses dans 7.1, 3°. Ayant fait cela, le lemme 7.3 s'applique et établit 7.1, 3°. [page 96]

Démonstration de 7.2. Posons

$$C \ = \ \mathrm{Mon} \ , \qquad TC \ = \ C \cap W \ , \qquad TC_0 \ = \ TC^\pi \ ,$$

donc par 7.1, 3°, on a

$$\underbrace{TC_0}_{\substack{\text{essentiellement}\\ \text{petite}}} \quad \subset \ TC \ \subset \ \widetilde{TC_0} \ = \ \widetilde{TC} \ ,$$

et par le théorème 1, §1 (p. 10-12), cor. 2, pour vérifier que $TC = \widetilde{TC_0}$ (*i.e.* TC est Q-clos à gauche), il revient au même de voir qu'il est Q-stable à gauche, *i.e.* stable par composition, par cochangement de base, par $\varinjlim$ ordinales strictes, par facteurs directs. Or les deux derniers sont impliqués par le fait que $\widetilde{TC} = C \cap W$ est stable par $\varinjlim$ filtrantes, vu que W l'est, et C aussi par la condition ⓔ (p. 76) sur $\mathcal{M}$. D'autre part, $C = \mathrm{Mon}$ et W étant stables par composition, itou pour $C \cap W$. Donc il reste le cochangement de [page 97]

33. *D'un système inductif π-adapté.*

34. **N.B.** Cette condition est vérifiée.

Commentary · Exposition [ms. 96]

Keeping the remainder a weak equivalence

Theorem 7.1(3) succeeds because it repairs the precise defect of §6. Let $f : A \hookrightarrow B$ lie in W. Use the full filtration hypothesis (d) from manuscript pages 45–46, including closure under subobjects; its abbreviated restatement in 6.1 omitted this clause. Write B as the π-adapted union of its π-small subobjects B_i, and put $A_i = A \cap B_i$. Subobject closure gives $A_i \in \mathrm{Filt}^{\pi}$, and preservation of filtered colimits by pullback gives

$$f = \varinjlim_i (A_i \hookrightarrow B_i).$$

The same preservation proves continuity at the small filtered suprema in the index poset. Thus this is a π-adapted presentation in the arrow category, rather than merely an arbitrary filtered presentation.

Accessibility now has a legitimate input: it says that the indices for which $f_i : A_i \hookrightarrow B_i$ belongs to W are cofinal. If $A \neq B$, some B_i meets the complement of A; cofinality permits choosing such an index with $f_i \in W$. Form

$$A \xrightarrow{\alpha_i} D_i = A \amalg_{A_i} B_i \xrightarrow{\beta_i} B.$$

The first arrow is a pushout of a small trivial monomorphism, so it lies both in $\widetilde{(W \cap \mathrm{Mon})}^{\pi}$ and in W. Effective unions, condition (f), say that β_i is monic. Since $\beta_i \alpha_i = f \in W$, the cancellation clause (7.1.4)(b) gives $\beta_i \in W$. The attachment is proper because B_i was not contained in A. These facts verify precisely the successor-step hypothesis of Lemma 7.3 for $C = W \cap \mathrm{Mon}$.

No assertion that the large remainder $D_i \to B$ is itself π-small has been used. In §6 that assertion was needed to apply accessibility to a second diagram, and it was unavailable. Here the remainder is put in W by cancellation instead. This is the mathematical reason why the new theorem survives the failure of the earlier one.

Once $TC = W \cap \mathrm{Mon}$ is generated by its small members, the additional closure properties make it equal to its left Q-closure. Composition follows from composition in W and in Mon; pushout stability is assumed; transfinite composition and retract closure follow from filtered-colimit stability, with identities supplying the degenerate cases. The small object argument then constructs the weak factorization system $(TC, TC^{\square})$. Its ordinal length must be chosen large enough for the domains of the chosen generating set to be small relative to the construction. Merely saying that the generating set is small is not itself this domain-smallness assertion; the accessibility hypotheses supply it.

base, et c'est assuré par 7.2.3, (c) (dont on avait déjà besoin dans 7.1, 3°, pour établir $C \cap W \subset \widetilde{TC_0}$). (**N.B.** les hypothèses de 7.2 sont pratiquement celles de 7.1, 3°, sauf qu'on suppose que W est un localiseur (sous-entendu : modérément saturé), de façon précise on n'a utilisé, jusqu'à présent, comme hypothèse supplémentaire, que la stabilité de W par composition.)

Il reste à établir que $(W, C = \mathrm{Mon}, F = (TC)_*)$ est un triple de Quillen clos, si et seulement si les conditions (7.2.6) et (7.2.7), *i.e.*

$$\begin{cases} C = \mathrm{Mon} \ \text{ stable par cochangement de base,} \\ TF = C_* = (\mathrm{Mon})_* \subset W, \end{cases}$$

sont satisfaites. La nécessité est évidente, par définition des triples de Quillen clos, et pour la suffisance, on note qu'il résulte du théorème 7.1, 2°, et de la partie déjà prouvée de 7.2, que p. 54
(C, TF), (TC, F) sont des couples de Quillen clos, et on a de plus

$$TC \ = \ C \cap W \ , \qquad TF \ \subset \ F \cap W$$

(puisque $TF \subset W$ par (7.2.7), et $TF \subset F$ puisque $TC \subset C$). On sait (paragraphe 4.3, corollaire 4, page 36) qu'il en résulte que (W, C, F) est un triple de Quillen clos. Cela achève la démonstration de 7.1 et 7.2. [page 98]

Résumons l'essentiel :

Corollaire 7.4.[35] *Soit $\mathcal{M}$ une catégorie satisfaisant les conditions* (a), (b), (c), (d), (e) (p. 76) *et* (e′), (f) ((7.2.1), p. 88), *enfin* (g) (7.2.6), i.e. *stabilité de l'ensemble* Mon *des*

35. **N.B.** C'est finalement ce corollaire qui est le résultat principal de la présente section.

monomorphismes de $\mathcal{M}$ par cochangement de base. Soit $W \subset \mathrm{Fl}(\mathcal{M})$ un localiseur dans $\mathcal{M}$, qui est stable par $\varinjlim$ filtrantes *et de plus* (*condition plus subtile, mais pratiquement anodine*) accessible. *On pose*

$$C = \mathrm{Mon}\,, \qquad TC = C \cap W\,, \qquad TF = C_*\,, \qquad F = (TC)_*\,.$$

Alors :

1°) (C, TF) *est un couple de Quillen* (*énoncé indépendant de la donnée de W*), *et même un couple de Quillen à factorisation fonctorielle.*

2°) *Pour que (TC, F) soit un couple de Quillen, il faut et il suffit que TC soit stable par cochangement de base.* (*Et alors il y a factorisation fonctorielle.*) [36].

3°) *Pour que (W, C, F) soit un triple de Quillen clos, il faut et il suffit que les conditions suivantes soient satisfaites.*

(7.4.1)
$$\begin{cases} \text{a)} \quad TC \overset{\text{déf}}{=} W \cap \mathrm{Mon} \text{ stable par cochangement de base.} \\ \text{b)} \quad TF \overset{\text{déf}}{=} (\mathrm{Mon})_* = \text{applications de type injectif} \subset W. \end{cases}$$

4°) *Pour que (W, C, F) soit un triple de Quillen clos et propre à gauche* (i.e. $C \subset \mathrm{Cof}_W$), *il faut et il suffit qu'on ait la condition* (7.4.1), (b) (i.e. *W contient les morphismes de type injectif*), *et de plus* [page 99]

$$C \subset \mathrm{Cof}_W, \text{ ou encore : pour tout carré cocartésien}$$

(7.4.2)
$$\begin{array}{ccc} A & \xrightarrow{\ i\ } & B \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\ A' & \xrightarrow{\ i'\ } & B' \end{array},$$

avec i un monomorphisme, alors $\alpha \in W \Longrightarrow \beta \in W$.

Commentary · Exposition [ms. 98]

Why the two factorization systems fit together

The remaining issue in Corollary 7.4 is compatibility. Two weak factorization systems on the same category do not by themselves make a model structure. Here the required compatibility is concentrated in the inclusion $TF \subset W$, and the retract argument explains why this one inclusion is enough.

Assume that (C, TF) and (TC, F) are weak factorization systems, that $TC = C \cap W$, that $TF \subset W$, and that W has two-out-of-three and is closed under retracts. Because $TC \subset C$, every map right orthogonal to C is right orthogonal to TC; hence $TF \subset F \cap W$. Conversely, let $p : X \to Y$ lie in $F \cap W$. Factor it using the first system as

$$X \xrightarrow{i} Z \xrightarrow{q} Y, \qquad i \in C, \quad q \in TF.$$

Since $q, p \in W$, two-out-of-three gives $i \in W$, hence $i \in TC$. Lifting i against p gives $r : Z \to X$ with $ri = 1_X$ and $pr = q$. In the arrow category the maps $(i, 1_Y)$ and $(r, 1_Y)$ exhibit p as a retract of q. Right lifting classes are retract-closed, so $p \in TF$. Consequently

$$TF = F \cap W, \qquad TC = C \cap W,$$

and the two factorizations and lifting properties are exactly the model-category axioms. Completeness and cocompleteness are supplied by the ambient hypotheses in the applications. This proves the sufficiency left implicit in the reference to §4.3.

36. **N.B.** Dans les cas 1° et 2°, C, $TC = C \cap W$ sont Q-engendrés à gauche par des *petites* parties.

For $C = \mathrm{Mon}$ the class TF is independent of W. It is already fixed by the ordinary category, and condition (7.2.7) asks the proposed homotopy theory to regard all those maps as equivalences. This is a substantial condition. In Set, for example, $\mathrm{Mon}^{\square}$ is the class of surjections, assuming the usual axiom of choice: testing $\varnothing \to \{*\}$ gives surjectivity, and choices of preimages solve the general lifting problem. If W consists only of bijections, $TF \subset W$ fails. Thus the category of sets with weak equivalences the bijections cannot have all injections as cofibrations and the factorization systems prescribed here.

Left properness has a separate role. It asks that a pushout of any weak equivalence along a cofibration be a weak equivalence. At this stage the source uses its previously established identity $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$, where the superscript denotes universal stability under *cobase* change, to obtain pushout stability of TC. In the presheaf applications, once the model structure is constructed, there is also the general explanation that every object is cofibrant: initial maps are monomorphisms, and these are the cofibrations. Homotopy invariance of pushouts of cofibrant objects then gives left properness. The assertion about initial maps is a property of presheaves, not of arbitrary categories. None of these arguments gives right properness automatically. The symbols W^{univ} and W_{univ} must therefore retain their opposite variances throughout this passage.

p. 55

5°) *Pour que* (W, C, F) *soit un triple de Quillen clos et* propre, *il faut et il suffit que les deux conditions de 4° soient satisfaites, ainsi que l'on ait :*

$$F \subset \mathrm{Fib}_W, \ \textit{ou encore : pour tout carré cartésien}$$

$$\begin{array}{ccc} X & \xrightarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ Y & \xrightarrow[p]{} & Y' \end{array} \qquad (7.4.3)$$

$$\textit{si } f \in F \ (\text{i.e. } f \in (C \cap W)_*) \textit{ et } p \in W, \ \textit{on a } q \in W.$$

Démonstration de 7.4. Les parties 1°, 2°, 3° ne sont qu'un résumé de ce qui a été établi. Pour 4°, il faut voir que les conditions dites impliquent (7.4.1), (a), or on a $C \subset \mathrm{Cof}_W$, donc $C \cap W \subset \mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ donc $C \cap W = C \cap W^{\mathrm{univ}}$, qui est stable par cochangement de base, puisque C et W^{univ} le sont. Quant à 5°, c'est une tautologie, compte tenu de 4°. [page 100]

Commentary · Examples [ms. 99]

Which vertical map is the fibration

Grothendieck wants the last clause of Corollary 7.4 to record right properness. In the displayed cartesian square the fibration hypothesis belongs to $f' : X' \to Y'$, the arrow *along which* $p : Y \to Y'$ is pulled back. The edition instead prints $f \in F$, referring to its pullback. This is a source slip, not a transcription error. The correction is necessary even in the most familiar proper model category.

In simplicial sets with the Kan–Quillen structure, take $Y' = \Delta[1]$, let $Y = \{0\}$, and let $X' = \{1\}$. Both maps to Y' are the indicated endpoint inclusions. Their pullback is empty, so the square is

$$\begin{array}{ccc} \varnothing & \longrightarrow & \{1\} \\ \downarrow & & \downarrow \\ \{0\} & \longrightarrow & \Delta[1]. \end{array}$$

The bottom map is a weak equivalence. The left map $\varnothing \to \{0\}$ is a Kan fibration: every horn is nonempty, so there are no lifting problems with a map from a horn into its empty domain. The top map is not a weak equivalence, since it changes the set of connected components from empty to a singleton. The right endpoint inclusion is not a Kan fibration; a one-dimensional horn asking for the missing endpoint has no lift. Thus the erroneous condition on the left-hand vertical map does not express right properness. Replacing it by $f' \in F$ restores the intended statement.

8 Application aux catégories de modèles $(A^\wedge, W_A)$

Soit A dans Cat, et considérons dans $A^\wedge$ le localiseur W_A, image inverse par

$$\text{(8.1)} \qquad \begin{cases} i_A : A^\wedge & \longrightarrow & \mathrm{Cat} \\ F & \longmapsto & A/F \end{cases}$$

d'un localiseur fondamental donné W, satisfaisant les conditions

(8.2) $\mathrm{W}(\underbrace{1}_{\text{sat.}}, \underbrace{2}_{\text{loc.}}, \underbrace{3}_{\text{objet fin.}}, \underbrace{5}_{\text{conn. } \pi_0}, \underbrace{6}_{\varinjlim}, \underbrace{7\text{ bis}}_{\substack{\text{carrés coc.} \\ \text{ou "M.V."}}}, \underbrace{8}_{\text{acc.}})$, où $\mathrm{W}(8)$ est l'accessibilité de W (donc W est L-accessible s'il satisfait aussi $\mathrm{W}(6)$).

Pour pouvoir cependant appliquer les développements sur les parties accessibles d'une grosse catégorie du §5, il faut vérifier les conditions (a), (b), (c), (d), (e) du §5, tant dans Cat, que (surtout) dans $A^\wedge$.

a) Stabilité par petites $\varinjlim$ filtrantes, O.K.

b) Filtration cardinale. Bien connu pour $A^\wedge = \underline{\mathrm{Hom}}(A, \mathrm{Ens})$, à partir de celle de Ens par les cardinaux (*cf.* SGA 4 I, 9.24, p. 177). Pour Cat, on définit une filtration cardinale

$$\text{(8.3)} \qquad (\mathrm{Cat}^\pi)_{\pi \geq \pi_0}\ , \qquad \text{où } \pi_0 = \text{cardinal des ensembles dénombrables}\ ,$$

par la condition, pour A dans Cat,

p. 56 [page 101]

$$\text{(8.4)} \qquad A \in \mathrm{Filt}^\pi(\mathrm{Cat}) \ = \ \mathrm{Cat}^\pi \qquad \Longleftrightarrow \qquad \mathrm{card}\,\mathrm{Fl}(A) \ \leq \ \pi\ .$$

Il faut vérifier les conditions (a), (b), (c) de *loc. cit.* 9.12. La condition (a) est évidente, (b) est contenue dans la condition plus forte de stabilité de Cat^π par $\varinjlim_I$, pour I une catégorie telle que $\mathrm{card}\,\mathrm{Fl}(I) \leq \pi$ (c'est la condition (d_0) de ces notes, p. 45-46). Notons que Cat^π est également stable par $\varprojlim$ finies (et si π est de la forme 2^c, il est stable même par limites projectives $\varprojlim_I$, avec $\mathrm{card}\,\mathrm{Fl}(I) \leq c$ ...). La condition (c) se vérifie aisément par la méthode du §5, en notant que pour X dans Cat, l'ensemble

$$\mathrm{Ss}_\pi(X)$$

des sous-catégories de X qui sont dans Cat^π est filtrant croissant et grand devant π, et que X en est la limite inductive. D'autre part, il est clair aussi que si $X \in \mathrm{Cat}^{\pi'}$ avec $\pi' > \pi$, alors

$$\mathrm{card}\,\mathrm{Ss}_\pi(X) \ \leq \ \pi'^\pi\ ,$$

ce qui achève d'établir la condition (c).

De plus, la condition (d) de p. 45 se vérifie immédiatement pour les Cat^π, et aussi pour les $A^{\wedge\pi}$ (pour $\pi \geq \mathrm{card}\,\mathrm{Fl}(A)$).

Aussi que les monomorphismes sont stables par $\varinjlim$ filtrantes (condition (e), page 46).

Passons aux conditions supplémentaires (e'), (f), (g) exigées dans le corollaire 7.4. La condition (e') (commutativité des $\varinjlim$ filtrantes au changement de base) est satisfaite dans $A^\wedge$ et dans Cat. La condition (f) dit que pour deux sous-objets A_1, A_2 d'un objet B, si $A_0 = A_1 \cap A_2$, alors [page 102]

$$A_1 \amalg_{A_0} A_2 \longrightarrow B$$

est un monomorphisme. Je doute que c'est vrai dans Cat, et ce n'est pas grave, car c'est à $A^\wedge$, non à Cat, qu'on voudra appliquer 7.4 – et c'est évidemment vrai dans $A^\wedge$, et dans tout topos, l'étant dans (Ens). Enfin, la stabilité des monomorphismes par cochangement de base est sûrement fausse dans Cat, mais vraie dans $A^\wedge$ – donc c'est O.K.

Comme W est stable par $\varinjlim$ filtrantes (W (6)), et que i_A y commute, W_A est également stable. De même, W_A est L-accessible, en vertu du

Lemme 8.1. *Soient $\mathcal{M}$, $\mathcal{M}'$ deux catégories satisfaisant* (a), (b), (c), (d), (e), *et $\varphi : \mathcal{M} \longrightarrow \mathcal{M}'$ un foncteur commutant aux $\varinjlim$ filtrantes. Alors pour toute partie L-accessible M' de $\mathcal{M}'$, $\varphi^{-1}(M')$ est L-accessible.*

De façon précise, soit π un cardinal tel que M' soit π-accessible et que φ applique $\mathcal{M}^\pi$ dans $\mathcal{M}'^\pi$ (par exemple, pour cette dernière condition, il suffit $\pi = 2^c$ pour c assez grand, cf. loc. cit. 9.14, p. 153). *Alors $M = \varphi^{-1}(M')$ est π-accessible.* [page 103]

Commentary · Development (commentators') [ms. 102]

Two finite categorical obstructions to the exactness hypotheses

Grothendieck separates Cat from a presheaf topos because the generation argument needs effective unions and stable monomorphisms, not only filtered-colimit exactness. His doubts about these two properties in Cat admit very small witnesses. We give them explicitly, so that the restriction to presheaves can be seen in the actual constructions. We checked the model-recognition discussions of Beke (2000, Section 1) and Cisinski (2006a, Sections 1.2 and 4.2); neither is being cited as the source of the particular finite presentations below. The examples establish familiar failures and carry no claim of priority.

Failure of effective unions. [ours] Let B be the four-element diamond poset, with objects $0, 1, 2, 3$ and nonidentity relations

$$0 < 1 < 3, \qquad 0 < 2 < 3.$$

Let A_1 contain all four objects and just the nonidentity arrows $0 \to 1$ and $2 \to 3$, and let A_2 contain all four objects and just $0 \to 2$ and $1 \to 3$. Each is a subcategory of B, and $A_0 = A_1 \cap A_2$ is discrete. Their pushout $P = A_1 \amalg_{A_0} A_2$ is the free category on the directed diamond: a functor from P is exactly the data of functors from A_1 and A_2 agreeing on objects, with no imposed relation between the two length-two paths. Thus P has two distinct arrows $0 \to 3$, whereas B has one. The comparison $P \to B$ is not faithful and therefore is not a monomorphism in Cat. This disproves condition (f) of (7.2.1).

Failure of stability of monomorphisms under pushout. [ours] Let $A = [1]$, with arrow $a : 0 \to 1$, and let B be the category obtained by making a invertible. The functor $A \to B$ is injective on objects and faithful, hence monic. Let C have objects $0, 1, 2$, arrows

$$a : 0 \to 1, \quad b, c : 1 \to 2, \quad d : 0 \to 2, \qquad ba = d = ca,$$

with $b \neq c$ and no further nonidentity arrows or relations. This is a finite category, and $A \to C$ is the evident inclusion. In the pushout $D = B \amalg_A C$, the image of a is invertible. Therefore the equality $ba = ca$ forces $b = c$ in D. The induced functor $C \to D$ is not faithful. It is the pushout of the monomorphism $A \to B$, so condition (g) also fails.

Both failures concern creation and identification of composites. In a presheaf category the corresponding constructions are computed in sets at each object; no new paths have to be composed, and there is no cancellation forced by an inverse newly adjoined in another piece. For the finite-union example the precise defect is an additional relation on two newly composable paths. For the pushout example it is a cancellation relation imposed by a new inverse. These are different obstructions, which is why conditions (f) and (g) were listed separately.

Vérification immédiate.

On a donc le

Lemme 8.2. *Soit A dans* Cat. *Alors $(A^\wedge, W_A)$ satisfait aux conditions préliminaires du corollaire* 7.4, *ainsi qu'aux conditions suivantes.* p. 57

(8.2.1) $$\underbrace{\mathrm{Mon}(A^\wedge)}_{\substack{\text{ensemble des}\\ \text{monomorphismes}\\ \text{dans } A^\wedge}} \quad \subset \ \mathrm{Cof}_{W_A} ,$$

(8.2.2) $$\mathrm{Mon} \cap W_A \qquad \text{stable par cochangement de base} ,$$

en d'autres termes (*compte tenu que* Mon *est stable par cochangement de base*), *pour un carré cocartésien*

$$\begin{array}{ccc} A & \xrightarrow{\ i\ } & B \\ {\scriptstyle\alpha}\searrow & & \searrow{\scriptstyle\beta} \\ & A' \xrightarrow{\ i'\ } & B' \end{array} \tag{8.2.3}$$

avec i un monomorphisme, si $\alpha \in W_A$ (resp. $i \in W_A$), on a $\beta \in W_A$ (resp. $i' \in W_A$). En fait, on a mieux :

$$i \in W_A \iff i' \in W_A\ , \qquad \alpha \in W_A \iff \beta \in W_A\ . \tag{8.2.4}$$

Il reste à vérifier (8.2.4), qui a été vu dans XII … [page 104]

Commentary · Exposition [ms. 103]

The filtration and the inverse-image lemma

Grothendieck needs the class W_A to inherit the smallness properties of W before the generation theorem can be applied. The cardinal bookkeeping is straightforward once the correct presentation is fixed. For an infinite π, a category with at most π arrows has at most π objects. A colimit of a diagram with at most π arrows, whose values have at most π arrows, is a quotient of the free category on a graph of size at most π. Finite strings in such a graph still form a set of size at most π, so the same cardinal bound holds for the colimit. Finite limits are subsets of finite products of object and arrow sets, with composition inherited componentwise. The estimate for larger limits when $\pi = 2^c$ uses $(2^c)^c = 2^c$, for infinite c.

Every set of at most π arrows of an arbitrary category is contained in a subcategory with at most π arrows, obtained by adding identities and finite composites. The poset of these small subcategories is π-directed in the source's cardinal convention, admits the required small directed unions, and has colimit the whole category. If the ambient arrow set has cardinal π', there are at most $(\pi')^\pi$ such subcategories. For presheaves, restriction maps replace finite composites in the same calculation, and the bound $\pi \geq |\operatorname{Mor} A|$ controls the closure under all restrictions.

Lemma 8.1 then amounts to transporting an adapted presentation. Choose π for which M' is π-accessible and $\varphi(\mathcal{M}^\pi) \subset \mathcal{M}'^\pi$. For a π-adapted presentation $X = \varinjlim_i X_i$, preservation of filtered colimits gives an adapted presentation $\varphi X = \varinjlim_i \varphi X_i$. Membership of φX in M' is equivalent to cofinality of the indices with $\varphi X_i \in M'$. This is exactly the criterion for $X \in \varphi^{-1} M'$. Closure of the inverse image under arbitrary filtered colimits follows separately from the same preservation property. The proof establishes the source's L-accessibility; it does not identify this notion merely by terminology with modern accessibility of a full arrow subcategory.

For $i_A : F \mapsto A/F$, the objects are pairs (a, x) and an arrow $(a, x) \to (b, y)$ is an arrow $u : a \to b$ with $x = F(u)(y)$. Filtered colimits of these finite data are computed as expected, so i_A preserves them. Its cardinal bound follows from the number of choices of u and y. On enlarging the initial cardinal, the inverse-image lemma therefore applies to $W_A = i_A^{-1} W$.

On conclut donc de 7.4 1°, 2°, 3° :

Corollaire 8.3. *Posons*

$$\begin{cases} C \ = \ \operatorname{Mon}(A^\wedge)\ , & TC \ = \ C \cap W\ , \\ F \ = \ (TC)_*\ , & TF \ = \ C_*\ . \end{cases} \tag{8.3.1}$$

Alors (C, TF) et (TC, F) sont des couples de Quillen à factorisation fonctorielle. (**N.B.** *C et TC étant de plus Q-engendrés à gauche par des petites parties de* $\operatorname{Fl}(A^\wedge)$.) *Pour que (W_A, C, F) soit un triple de Quillen clos, il faut et il suffit qu'on ait*

$$\underbrace{TF}_{\overset{\text{déf}}{=}(\operatorname{Mon})_*} \ \subset \ W_A\ , \tag{8.3.2}$$

i.e. *que toute flèche* $f : X \longrightarrow Y$ *de type injectif dans* $A^\wedge$ (i.e. *qui fait de* X *un objet injectif de* $A^\wedge/Y$) *soit dans* W_A.

Quand cette condition est satisfaite, le triple de Quillen (W_A, C, F) *est propre à gauche, par* 8.2.1.

Il n'est pas clair qu'il soit forcément propre-tout-court, *i.e.* qu'on ait

$$F \ \subset \mathrm{Fib}_W \qquad (?)\ . \tag{8.3.3}$$

Nous y reviendrons par la suite. Pour l'instant, la chose essentielle, c'est d'examiner la condition (8.3.2) sur A.

Théorème 8.4. *Soit* W *un localiseur fondamental dans* Cat *satisfaisant* W (1, 2, 3, 6, 7, 8), p. 58
notations de XII, (i.e. *tous les axiomes envisagés jusqu'à présent, sauf peut-être celui* W (4) [page 105]
des fibrations `[et celui W(5) de connexité]`), *supposons de plus* W *stable par facteurs directs* (*renforcement* W (1 bis) *de l'axiome de saturation modérée* W (1)).[37] *Soit* A *un objet de* Cat, *et* $W_A = i_A^{-1}(W)$ *le localiseur de* $A^\wedge$ *déduit de* W. *Avec les notations de* 8.3 :

1°) *Les conditions suivantes sont équivalentes.*

a) *On a* (8.3.2),

$$TF \ \subset \ W_A\ ,$$

i.e. *toute flèche de type injectif dans* $A^\wedge$ *est dans* W_A.

b) *Soit*

$$L \ = \ L_A \tag{8.4.1}$$

l'élément de Lawvere dans $A^\wedge$, *donc le préfaisceau*

$$a \quad \longmapsto \quad \begin{array}{l} \text{ensemble des sous-préfaisceaux de } a \\ (\text{i.e. ensemble des ouverts de } A/a, \\ \text{i.e. des cribles de } A/a)\ , \end{array}$$

et considérons la flèche $L \longrightarrow e_A$ *(objet final de* $A^\wedge$*). On veut que cette flèche soit* W_A*-asphérique,* i.e. *soit universellement dans* W_A *– ou ce qui revient au même, que pour tout* $a \in A$, $L_A \times a$ $(= L_{A/a})$ *soit un objet* W_A*-asphérique de* $A^\wedge$ *(ou, ce qui revient au même, de* $(A/a)^\wedge$*).* [page 106]

c) *Il existe un objet* I *de* $A^\wedge$ *tel que* $I \longrightarrow e_A$ *soit* W_A*-asphérique, avec deux sections* δ_0, δ_1 *de* I *sur* e_A, *avec* $\mathrm{Ker}(\delta_0, \delta_1) = \emptyset_A$ *(préfaisceau vide).*

d) *Les catégories localisées* $A/a = B$ *sont des* catégories-test *pour* W (cf. *Pursuing Stacks, §65, notamment pages 172-173), ce qui revient à dire que l'on a ceci, pour chacune des catégories* $B = A/a$ *:*

Les foncteurs

$$\begin{cases} i_B : \ B^\wedge \ \longrightarrow \ \mathrm{Cat} \ \ (\ F \ \longmapsto \ B/F\ ) \\ j_B : \ \mathrm{Cat} \ \longrightarrow \ B^\wedge \ \ (\ C \ \longmapsto \ (b \longmapsto \mathrm{Hom}(B/b, C))\ ) \end{cases} \tag{8.4.2}$$

(*où* j_B *est l'adjoint à droite de* i_B) *sont tous les deux "model preserving"*, i.e.

$$\begin{cases} W_B \ = \ i_B^{-1}(W) \qquad \textit{(par définition de } W_B\textit{, en fait)} \\ W \ = \ j_B^{-1}(W_B)\ , \end{cases} \tag{8.4.3}$$

37. **N.B.** Pour la validité de 1°, 3° on n'a pas besoin de W (6) ($\varinjlim$) ni de W (8) (accessibilité), ni même de W (7) (axiome du carré cocartésien). Ce dernier est nécessaire déjà pour assurer la partie (*a*) du corollaire 8.5. Mais pour la partie 2° du théorème 8.4, on a besoin de *tous* les axiomes du localiseur fondamental W, sauf seulement W (4) `[et W(5)]`.

Commentary · Present state [ms. 104]

The forward gluing theorem and the extra converses

Grothendieck's immediate objective is a model structure, for which the forward implications in (8.2.4) are sufficient: pushouts of weak equivalences along monomorphisms remain weak equivalences, and pushouts of trivial monomorphisms remain trivial. The two converse implications printed in that formula impose a stronger requirement. They belong to the strengthened excision framework invoked here and must not be silently identified with the ordinary gluing theorem for homotopy types.

This distinction has already been analysed in the commentary on Chapter XII, manuscript p. 78 ("A finite counterexample to the asserted reverse excision") and p. 126 ("Why converse excision is an additional axiom"). We refer to those blocks rather than reproduce their examples. In particular, an application to W_∞ must retain the forward statements and use the actual presheaf gluing theorem, rather than assume that the stronger biconditionals hold for ordinary weak homotopy equivalences.

The surviving construction is substantial. Under the forward stability, the two weak factorization systems in Corollary 8.3 exist, and the sole remaining model-recognition condition is that maps right orthogonal to all monomorphisms belong to W_A. Theorem 8.4 turns that apparently large condition into an interval condition. Later, Cisinski (2006a, Corollary 4.2.18) proves the model-structure theorem for every accessible fundamental localizer and every corresponding local test category. This theorem supplies the intended application without the reverse-excision assertion.

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et les foncteurs localisés

$$\mathrm{Hot}_{B,W} \underset{\overline{j_B}}{\overset{\overline{i_B}}{\rightleftarrows}} \mathrm{Hot}_W$$

sont des équivalences de catégories. De plus, ces équivalences de catégories sont quasi-inverses l'une de l'autre, les morphismes d'adjonction étant déduits des morphismes d'adjonction pour le couple de foncteurs adjoints (i_B, j_B), ces morphismes d'adjonction étant de plus dans W resp. dans W_B.[38]

[page 107]

Commentary · Exposition [ms. 106]

The Lawvere object and the local interval test

Grothendieck is looking for an intrinsic cylinder in $\widehat{A}$. The subobject classifier $\Omega = L_A$ is available before any homotopy theory is chosen, and its two sections classify the empty and the full subobject. They are disjoint because a sieve cannot be both empty and maximal. Thus Ω is always a separated interval as a piece of topos theory; the additional homotopical question is whether its projection to the terminal object is universally a weak equivalence.

Injectivity of Ω has a direct proof. Given a monomorphism $U \hookrightarrow V$ and a map $U \to \Omega$, let $S \hookrightarrow U$ be its classified subobject. The composite $S \hookrightarrow V$ is a subobject of V, so its characteristic map $V \to \Omega$ extends the original map: pulling $S \hookrightarrow V$ back to U recovers $S \hookrightarrow U$. Consequently $\Omega \to 1$ has the right lifting property against all monomorphisms. The same is true after every base change, since right lifting properties are stable under pullback. This proves the implication (a)⇒(b) in 8.4, including its universal quantifier.

Checking universal asphericity on representables is the relevant local criterion. For $a \in A$, the slice topos $\widehat{A}/h_a$ identifies with $\widehat{A/a}$. Under that identification $\Omega \times h_a \to h_a$ is its subobject classifier. More generally, a separated interval I with $I \times h_a \to h_a$ a weak equivalence for every a gives a cylinder after passage to every slice. The localizer's localization axiom then extends the test to arbitrary bases. The equality $L_A \times a = L_{A/a}$ in the source is understood in this slice equivalence; the two objects are not literally presheaves on the same category before that identification.

The category-of-elements functor also explains the right adjoint appearing in (8.4.2). A map $F \to j_A(C)$ assigns to every element $x \in F(a)$ a functor $A/a \to C$, compatibly with restriction along

38. **N.B.** La condition (d), d'apparence très forte, est équivalente à la condition d'apparence plus anodine :
d') le foncteur $j_A : \mathrm{Cat} \longrightarrow A^\wedge$ transforme éléments de W en flèches W_A-asphériques ;
et il suffit même de l'exiger pour $C \longrightarrow e$ dans Cat, où C est un objet de Cat ayant objet final.
[La condition (d') ci-dessus est trop forte et n'est pas équivalente à la condition (d) ; en revanche la condition plus faible envisagée ensuite est bien équivalente à (d)].

arrows of A. This is equivalent to a functor $A/F \to C$: evaluate the assigned functor at 1_a, and use the compatibility to recover its action on arrows. Conversely, restrict a functor $A/F \to C$ along the canonical functor $A/a \to A/F$ attached to x. These constructions are inverse. Hence $i_A \dashv j_A$ is an ordinary adjunction, and the stronger test-category assertion says that its unit and counit become weak equivalences on all relevant slices.

The interval characterization is already present in Grothendieck (1983, §65, Proposition 3 and Corollary 1); in the 2015 transcription the proposition spans printed pages 171–172, corresponding to the manuscript pagination cited here. It is the local condition, followed by the additional asphericity of A, which distinguishes a local test category from a test category.

2°) *Si les conditions équivalentes de* 1° *sont satisfaites, alors le triple*

$$(W_A\,,\ C = \mathrm{Mon}_{A^\wedge}\,,\ F = (C \cap W_A)_*)$$

est un triple de Quillen (clos) propre à gauche.

3°) *Si les conditions équivalentes de* 1° *sont satisfaites, et si de plus A est W-asphérique, alors (et alors seulement) A elle-même est une* catégorie-test *pour W, ce qui revient à dire que les conditions sur la catégorie B dans* (d) *sont satisfaites quand B est* soit *A, soit une catégorie localisée A/a.*

Corollaire 8.5. *Les conditions équivalentes de* 1° *et de* 3° *sont satisfaites dans chacun des cas suivants (et ceci quel que soit W, satisfaisant aux hypothèses dites).*

a) *$A = \mathbf{\Delta}$, $\mathbf{\Delta}^\sim$, $\square$, $\square^\sim$ (catégories des simplexes combinatoires standard, ordonnés ou non, et des cubes combinatoires standard, ordonnés ou non …)*[39].

b) *A est stable par produits finis*[40]*, et contient un objet I tel que I ait deux sections δ_0, δ_1 sur l'objet final, avec δ_0, δ_1 disjointes,* i.e. $\mathrm{Ker}(\delta_0, \delta_1) = \emptyset_{A^\wedge}$ *(sous-préfaisceau vide du préfaisceau final).*

[page 108]

Commentary · Examples [ms. 107]

A terminal category fails and a separated interval succeeds

Theorem 8.4 is not saying that every category with a terminal object is a test category. For $A = e$, presheaves are sets and i_A regards a set as a discrete category. For $W = W_\infty$, the induced class W_A is therefore the bijections. The Lawvere object is the two-element set, whose projection to a singleton is not a weak equivalence. Equivalently, there cannot be a separated interval in sets which is weakly equivalent to a point. The terminal object of A supplies asphericity of the indexing category, but cannot supply the missing interval.

The positive criterion in Corollary 8.5(b) fills exactly this gap. Suppose A has a terminal object and an object I with disjoint endpoint sections, and that $a \times I$ exists for every a. The representable presheaf h_I has disjoint sections, and

$$h_a \times h_I \simeq h_{a \times I}.$$

Its category of elements is $A/(a \times I)$, which has a terminal object. Thus the required local asphericity follows from the terminal-object axiom. Since A itself has a terminal object, the additional global asphericity holds as well. This proves the weakened finite-product hypothesis in footnote 40; no products unrelated to I were used.

For the ordinary simplex category, the interval is $\Delta[1]$. The product $\Delta[n] \times \Delta[1]$ is the nerve of the product poset $[n] \times [1]$, whose largest element contracts its nerve. The two endpoint simplicial subsets are disjoint. This supplies the concrete homotopy behind the local criterion, while the terminal object $[0]$ makes Δ aspherical. The names of the cubical categories in the source require

39. $\square^\sim$ est essentiellement la catégorie A "universelle", stable par produits finis, et contenant un "intervalle séparant" (I, δ_0, δ_1) comme dans (b).

40. En fait, il suffit que $I \longrightarrow e$ soit quarrable, *i.e.* que A ait un objet final, et que les produits $I \times a$ ($a \in \mathrm{Ob}\, A$) existent dans A.

their prescribed choices of maps; one should not transfer the statement to every category called "cubical", since diagonals, symmetries and connections change the indexing category.

Footnote 38 distinguishes a genuinely stronger, false reformulation. In the simplicial case let $u : e \to [1]$ select one endpoint. Both categories have terminal objects, so $u \in W_\infty$. The two endpoint maps $j_A(e) \to j_A([1])$ are disjoint: at every a they are the distinct constant functors $A/a \to [1]$. Pulling one endpoint back along the other gives $\varnothing \to j_A(e) = 1$. For $A = \Delta$ this is not a weak equivalence. Hence $j_A(u)$ is not universally a weak equivalence, although A is a test category. The weaker condition involving only projections $C \to e$ with C terminal-pointed avoids this counterexample, exactly as the editors' bracketed correction says.

Le cas (b) du corollaire est justiciable de la condition (c) dans 1°. (De plus, A ayant un objet final, est W-asphérique, donc on est dans les conditions de 3°.) Les cas $\mathbf{\Delta}$, $\mathbf{\Delta}^\sim$, $\square$ (et $\square^\sim$, qui tombe en fait dans le cas (b)) sont traités dans Pursuing Stacks §34 (p. 47-50) `[voir aussi §36]`.

Démonstration du théorème 8.4. La partie essentielle, techniquement, est bien sûr 1° [41]. On a les implications ± tautologiques

$$(a) \implies (b) \implies (c)\ ,$$

où $(a) \Longrightarrow (b)$ provient du fait que L est évidemment un objet injectif de $A^\wedge$. (L'objet de Lawvere d'un topos quelconque est un objet injectif.) Il reste à établir

$$(c) \implies (d) \implies (a)\ .$$

Que l'on ait $(c) \Longrightarrow (d)$ (et $(c) \Longrightarrow (b)$) est établi dans Pursuing Stacks, voir le résumé des notions W-test dans Pursuing Stacks, §65, p. 168-179; l'énoncé pertinent ici est la proposition 3, page 173. p. 60

Il reste à établir $(d) \Longrightarrow (a)$ – mais à présent on joue sur du velours! Je note ici le lemme suivant, dont la place serait plutôt à la fin du §4 des généralités sur les triples de Quillen : [page 109]

Lemme 8.6. *Soit $\mathcal{M}$ une catégorie, (C, TF) un précouple de Quillen dans $\mathcal{M}$, et $W \subset \mathrm{Fl}(\mathcal{M})$ une partie de $\mathrm{Fl}(\mathcal{M})$ stable par facteurs directs. Conditions équivalentes :*

a) $TF \subset W$.

b) *Toute flèche f dans $\mathcal{M}$ se factorise en $f = pi$, avec $i \in C$, $p \in W$.*

b′) *Comme* (b), *avec f dans TF.*

Démonstration. Les implications $(a) \Longrightarrow (b) \Longrightarrow (b')$ sont tautologiques et ne font pas appel à l'hypothèse de stabilité sur W. Mais pour un $f \in TF$ donné, pour voir qu'il est dans W, si W est stable par facteurs directs, il suffit que cet-f-là se factorise en pi, avec $i \in C$, $p \in W$. En effet, on voit aussitôt qu'alors f est facteur direct de p (car $i \in C$), et comme $p \in W$, on a $i \in W$, *q.e.d.*

Revenons à 8.4, démonstration de $(d) \Longrightarrow (a)$.

`[Suit un passage rayé, en bas de la page 109 et en haut de la page 110.]`

[page 110]

Commentary · Exposition [ms. 109]

The retract in Lemma 8.6

Grothendieck isolates a useful recognition principle: to prove that a right lifting class consists of weak equivalences, it can be enough to factor its arrows through weak equivalences. Write $f : X \to Y$ in TF as $f = pi$, where $i : X \to Z$ is in C and $p : Z \to Y$ is in W. The lifting square with top map 1_X, left edge i, bottom map p, and right edge f yields $r : Z \to X$ satisfying

$$ri = 1_X, \qquad fr = p.$$

41. La partie 2° en résulte, grâce au corollaire 8.3.

Therefore $(i, 1_Y) : f \to p$ and $(r, 1_Y) : p \to f$ exhibit f as a retract of p in the arrow category. Retract closure of W gives $f \in W$. The final $i \in W$ printed in the proof is a slip: the assertion proved and needed is $f \in W$.

The other direction uses the factorization already included in the precouple hypothesis: factor any arrow as a C-map followed by a TF-map, and then use $TF \subset W$. Thus one must retain both ingredients of the precouple, factorization and the lifting relation; a bare orthogonality inclusion would not prove (a)⇒(b). No two-out-of-three property is needed for this lemma. Its economy is what makes it attractive here, where the interval construction is intended to provide the weaker factorization C followed by W before the model structure is known.

Finalement, mon projet de démonstration se modifie, car bien que (d) ait l'air très, très fort comme condition, telle quelle je ne vois pas, finalement, comment en déduire (a). Mais en fait on peut considérer que l'équivalence de (b), (c), (d) est mise ici pour mémoire, elle est établie dans *loc. cit.*, donc je n'ai à me préoccuper que de l'implication $(c) \Longrightarrow (a)$, qui va résulter du lemme 8.6, qui nous ramène à prouver que toute flèche $f : X \longrightarrow Y$ dans $A^\wedge$ [page 111] se factorise en pi, avec i monomorphisme et $p \in W_A$. On fait l'argument du mapping-cone `[plutôt mapping-cylinder]`, en partant de

$$\begin{array}{ccc} e_A \amalg e_A & \xrightarrow[\text{mono}]{(\delta_0,\delta_1)} & I \\ & \searrow \quad \swarrow_{(W_A)_{\text{univ}}} & \\ & e_A\ , & \end{array}$$

d'où pour tout X dans $A^\wedge$, en prenant le produit par X,

$$\begin{array}{ccc} X \amalg X & \xrightarrow[\text{mono}]{(\delta_0^X,\delta_1^X)} & X \times I \\ & \searrow_{(\text{id},\text{id})} \quad \swarrow_{\text{pr}_1^X \in W_A} & \\ & X\ , & \end{array}$$

et on construit le mapping-cone `[plutôt mapping-cylinder]` de f par le diagramme cocartésien p. 61

(8.6.1)

$$\begin{array}{ccccc} X & & \xrightarrow{\quad f \quad} & & Y \\ & {\scriptstyle \delta_0^X}\big\uparrow & \text{pr}_1^X \text{ cocart.} & {\scriptstyle \beta}\big\uparrow & \big\uparrow{\scriptstyle p_f} \\ X & \xrightarrow{\delta_1^X} & X \times I & \xrightarrow{\quad f' \quad} & Z(f)\ , \\ & \searrow & & \nearrow & \\ & & {\scriptstyle i_f} & & \end{array}$$

p_f déduit de $\text{pr}_1^X : X \times I \to X$

la factorisation cherchée de f est

$$f \;=\; p_f\, i_f\ , \qquad \text{où} \quad i_f = f' \delta_1^X\ .$$

Ici, comme $\text{pr}_1^X \in W_A$, on en conclut que δ_0^X, δ_1^X (qui en sont des sections) sont dans [page 112] W_A, comme δ_0^X est un monomorphisme, on a donc $\delta_0^X \in TC$, donc (TC étant stable par cochangement de base) $\beta \in W_A$, donc enfin p_f (qui est une rétraction de $Z(f)$ sur Y) est aussi dans W_A. Il reste à voir que i_f est un monomorphisme, ce qui se voit sur le diagramme plus complet (*cf.* XII, p. 230) que j'ai sous les yeux : i_f apparaît comme composé de deux monomorphismes $X \longrightarrow Y \amalg X$ et $Y \amalg X \longrightarrow Z(f)$ … Le théorème 8.4 est démontré.

Je vais quand même isoler l'argument formel sempiternel (que j'ai utilisé surtout sous forme duale) dans un

Lemme 8.7. *Soit $\mathcal{M}$ une catégorie, W un localiseur modérément saturé dans $\mathcal{M}$, C une partie stable par composition et par cochangement de base. On se propose, pour X fixé dans $\mathcal{M}$, de trouver une factorisation fonctorielle de toute flèche $f : X \longrightarrow Y$* (i.e. *du morphisme costructural d'un Y dans $X\backslash\mathcal{M}$) en $p_f i_f$, avec $i_f \in C$, $p_f \in W$. On suppose pour ceci qu'on a une factorisation de ce type pour $X \amalg X \xrightarrow{(\mathrm{id},\mathrm{id})} X$,* i.e. *un objet I_X et une factorisation* [page 113]

$$X \amalg X \xrightarrow{i_X=(\delta_0^X,\delta_1^X)} I_X \xrightarrow{\quad p_X \quad} X \ ,$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \searrow \quad \nearrow$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (\mathrm{id}_X,\mathrm{id}_X)$$

avec $i_X \in C$, $p_X \in W$. On associe alors à toute f le mapping cone [plutôt cylinder] *$Z(f)$ comme dans le diagramme* (8.6.1) *(où il faut lire I_X au lieu de $X \times I$), et on a alors $p_f \in W$, $i_f \in C$ comme voulu. Mais il faut supposer, pour savoir que $i_f \in C$, que $X \in \mathcal{M}_C$ (objets C-cofibrants),* i.e. *que pour tout Y, $X \longrightarrow Y \amalg X$ est dans C. (De plus, on a supposé tacitement $\mathcal{M}$ stable par $\varinjlim$ finies.)*

Commentary · Exposition [ms. 112]

What the mapping cylinder proves

The mapping cylinder converts a homotopy between two copies of the source into a factorization of an arbitrary map. For $f : X \to Y$ in a presheaf category, form

$$Z(f) = Y \amalg_{X,\delta_0} (X \times I), \qquad i_f(x) = [(x, \delta_1)],$$

and let $p_f : Z(f) \to Y$ equal 1_Y on Y and $f\,\mathrm{pr}_X$ on $X \times I$. The compatibility on the identified copy of X makes p_f well defined, and $p_f i_f = f$.

There are two independent verifications. First, the map

$$Y \amalg X \longrightarrow Z(f)$$

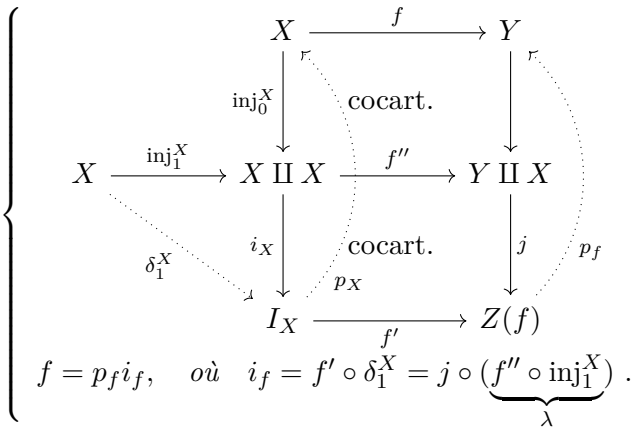
is the pushout of the boundary inclusion $X \amalg X \hookrightarrow X \times I$. It is monic, as are coproduct inclusions in a presheaf category. Their composite $i_f : X \to Y \amalg X \to Z(f)$ is therefore monic. This proves the assertion about the left factor and explains why disjoint endpoints are needed.

Second, $\mathrm{pr}_X \in W_A$, and its sections δ_0^X, δ_1^X lie in W_A by two-out-of-three. If trivial monomorphisms are stable under pushout, then the map $\beta : Y \to Z(f)$ is in W_A. Since $p_f\beta = 1_Y$, two-out-of-three gives $p_f \in W_A$. This proves the assertion about the right factor. The two arguments use different hypotheses: ordinary monomorphism stability for the first, and *weak-equivalence* stability for the second.

The construction is functorial in maps $Y \to Y'$ under the fixed source X, because the universal property of the pushout supplies $Z(f) \to Z(f')$ and these maps respect identities and composition. A varying-source functorial factorization additionally requires a functorial choice of the cylinders I_X and their structure maps. In the present application $I_X = X \times I$, so this further functoriality is available. Compare the commentary on Chapter XII, manuscript p. 230 ("The mapping cylinder, with the two endpoints distinguished"); the present discussion addresses the additional issue of which hypotheses remain when the argument is extracted as Lemma 8.7.

Le diagramme complet est

(8.7.1)



Commentary · Development (commentators') [ms. 113] [page 114]

The omitted hypothesis in the abstract cylinder lemma

Grothendieck has established the mapping-cylinder argument in a presheaf category where trivial monomorphisms are stable under pushout. Lemma 8.7 then extracts an abstract statement but retains only stability of C under composition and pushout. That suffices to prove $i_f \in C$ under the stated coproduct condition; it does not suffice to prove $p_f \in W$. The following example separates the two assertions completely. We checked the cylinder criterion of Cisinski (2006a, Lemma 1.4.13) and the model-recognition argument of Beke (2000, Section 1); we did not find this particular counterexample in those sources.

Counterexample to Lemma 8.7 as stated. [ours] Take $\mathcal{M} = \mathrm{Grp}$, let C be all group homomorphisms, and let W consist of the homomorphisms $u : G \to H$ such that

$$\mathrm{Hom}(C_2, G) \longrightarrow \mathrm{Hom}(C_2, H)$$

is a bijection. The class W is the inverse image of the isomorphisms of sets under a functor. It therefore contains all isomorphisms, has two-out-of-three, and is closed under retracts; it also satisfies the usual saturation condition involving two opposite composites. The category has finite colimits, and C satisfies every stability and coproduct condition in the lemma.

Let $X = 1$, the trivial group, and choose $I_X = \mathbb{Z}$. There is a factorization of the fold map

$$X * X = 1 \longrightarrow \mathbb{Z} \longrightarrow 1 = X.$$

Its left factor is in C, and its right factor is in W, since $\mathbb{Z}$ has no nontrivial element of order two. Both endpoint maps are the unique map $1 \to \mathbb{Z}$; the abstract lemma does not require them to be disjoint, and its condition on i_X is satisfied.

For $f : 1 \to C_2$, the prescribed mapping cylinder is the free product

$$Z(f) = C_2 * \mathbb{Z}, \qquad p_f : C_2 * \mathbb{Z} \longrightarrow C_2,$$

where p_f is the identity on C_2 and kills the generator z of $\mathbb{Z}$. If t is the nonidentity element of C_2, both t and ztz^{-1} are involutions in the free product. They are distinct by reduced-word normal form, but p_f maps both to t. Thus the induced map on $\mathrm{Hom}(C_2, -)$ is not injective, and $p_f \notin W$. Every explicit hypothesis of Lemma 8.7 holds, while its right-factor conclusion fails.

The repair can be stated at the exact level used by the proof. Assume that $\delta_0 : X \to I_X$ belongs to W and that its pushout along each map $f : X \to Y$ under consideration belongs to W. Then $\beta : Y \to Z(f)$ is in W, and $p_f \beta = 1_Y$ gives $p_f \in W$. A uniform sufficient condition is $C \cap W$ stable under pushout, together with $\delta_0 \in C$. The latter follows if $X \to X \amalg X$ and $i_X : X \amalg X \to I_X$ are in C. For the left factor one also needs $X \to Y \amalg X$ in C, precisely the explicit condition used in the printed proof. In the usual convention this is automatic when Y is cofibrant, and it holds for every X, Y in a presheaf category with $C = \mathrm{Mon}$. Calling it simply "X cofibrant" reverses which added summand supplies the condition.

With these hypotheses the proof on the preceding page is complete: $j : Y \amalg X \to Z(f)$ is a pushout of i_X, so $i_f = j \circ (X \to Y \amalg X)$ lies in C; the separate argument with β proves $p_f \in W$. The failure of the abstract lemma therefore does not invalidate the presheaf application under its original gluing assumptions.

On a $i_X \in C$ par hypothèse, donc $j \in C$ (stabilité de C par cochangement de base), on a $\lambda : X \longrightarrow Y \amalg X$ dans C car X dans $\mathcal{M}_C$, donc le composé $i_f = j \circ \lambda$ est dans C (stabilité de C par composition) – on gagne!

Commentaires. Pour l'équivalence de (a), (b), (c) dans le théorème 8.4, 1°, on n'a donc pas eu à faire appel aux développements de Pursuing Stacks, qui servent seulement à établir l'équivalence des trois conditions en question avec la condition "modélisante" (d). Pour résumer :

1°) Les implications

$$(a) \implies ((b) \iff (c) \iff (d))$$

sont valables sous les seules conditions

$$\mathrm{W}\,(1,\ 2\,\text{bis},\ 3)\ ,$$

et pour l'équivalence de (a) aux autres conditions, il suffit de

$$\mathrm{W}\,(1\,\text{bis},\ 2\,\text{bis},\ 3)$$

(où W (1 bis) renforce W (1) par la condition des facteurs directs, tandis que W (2 bis) affaiblit W (2), en exigeant seulement que les morphismes W-asphériques soient dans W). Les conclusions de 8.4, 3°, relatives au cas où A est W-asphérique, sont valables également sous les conditions W (1, 2 bis, 3) sans plus.

[page 115]

Commentary · Exposition [ms. 114]

The interval proof that avoids pushouts

Grothendieck's final reduction of the hypotheses in 8.4 asks for an argument that no longer uses the pushout axiom. Such an argument exists: the map under examination already has a lifting property, so the interval can be used to construct a homotopy directly. This is the proof later given in Cisinski (2006a, Lemma 1.4.13, printed pp. 56–57). We spell it out with the hypotheses of the present text.

Suppose $\widehat{A}$ has a separated interval I such that $X \times I \to X$ belongs to W_A for every X, and suppose W_A has two-out-of-three and is closed under retracts. Let $p : X \to Y$ have the right lifting property against all monomorphisms. Lifting against $\varnothing \hookrightarrow Y$ gives a section $s : Y \to X$, with $ps = 1_Y$. The endpoint inclusion

$$X \amalg X \hookrightarrow X \times I$$

is monic. Map its two summands to X by 1_X and sp, and map $X \times I$ to Y by $p\,\mathrm{pr}_X$. The equalities $p1_X = p$ and $psp = p$ make this a commutative lifting square. A lift $H : X \times I \to X$ therefore satisfies

$$H\delta_0 = 1_X, \qquad H\delta_1 = sp.$$

Both endpoint maps are in W_A, because they are sections of $\mathrm{pr}_X \in W_A$. From $H\delta_0 = 1_X$, two-out-of-three gives $H \in W_A$, and hence $sp = H\delta_1 \in W_A$.

Finally p is a retract of sp as an arrow: the maps are $(1_X, s) : p \to sp$ and $(1_X, p) : sp \to p$, and their composite on p is $(1_X, ps) = 1_p$. Retract closure gives $p \in W_A$. This proves (c)⇒(a) without a single pushout, without accessibility, and without a model structure. If one uses the weak-saturation axiom saying that ps and sp in W_A force $p, s \in W_A$, the final retract step may be replaced by that axiom.

The reverse implications remain the elementary ones: the subobject classifier is injective, and its empty/full sections form a separated interval. Thus the equivalence of (a), (b) and (c) really has the weaker character claimed at the end of §8. The argument corrects a proof dependency, not the theorem's conclusion. Accessibility is needed later to produce factorizations; it is not needed to decide whether this interval test recognizes the intended weak equivalences.

2°) Le corollaire 8.5, se référant aux conditions (b), (c), (d) de 1° (à l'exclusion de (a), qui suppose W stable par facteurs directs), est valable pour un W satisfaisant

$$\mathrm{W}\,(1,\ 2\,\mathrm{bis},\ 3,\ \underbrace{7\,\mathrm{bis}}_{\substack{\text{carré}\\ \text{cocartésien}}}\ )\ .$$

p. 63

3°) Pour la validité de 8.4, 2°, *i.e.* pour avoir un triple de Quillen (W_A, C, F), il nous faut (presque) tout l'éventail des conditions sur W,

$$\mathrm{W}\,(1\,\mathrm{bis},\ 2\,\mathrm{bis},\ 3,\ \underbrace{6}_{\text{limites}},\ \underbrace{7\,\mathrm{bis}}_{\substack{\text{carré}\\ \text{cocartésien}}},\ \underbrace{8}_{\text{accessibilité}}\ )\ ^{42},$$

à l'exclusion de $\mathrm{W}\,(4)$, et en se contentant pour $\mathrm{W}\,(2)$ de la forme affaiblie $\mathrm{W}\,(2\,\mathrm{bis})$ sans plus. (Même l'axiome de connexité $\mathrm{W}\,(5)$ n'a pas servi, mais je l'avais d'abord inclus, car il y a peu de chances qu'on aura jamais à travailler avec des W qui n'y satisfont pas! Il se pourrait d'ailleurs, en présence des autres axiomes, ou seulement de $\mathrm{W}\,(1\,\mathrm{bis}, 2\,\mathrm{bis}, 3, 7\,\mathrm{bis})$, que si $\mathrm{W}\,(5)$ n'est *pas* satisfait, on ait $W = \mathrm{Fl}\,\mathrm{Cat}$ tout entier, donc $W_A = \mathrm{Fl}\,A^{\wedge}$!)

[page 116]

Commentary · Later developments [ms. 115]

What later model recognition establishes

The construction sought here became a theorem with fewer auxiliary hypotheses. For a fundamental localizer W generated by a set and a W-local test category A, Cisinski (2006a, Corollary 4.2.18) constructs the cofibrantly generated model structure with weak equivalences W_A and cofibrations all monomorphisms. The interval characterization is Theorem 4.1.19. The source's question about right properness is a separate question: Cisinski's Theorem 4.3.24 makes properness of the fundamental localizer independent of the chosen test category, and the minimal localizer is proper (Example 4.3.22).

There are three smallness statements to keep distinct. Grothendieck's L-accessibility includes closure under *every* filtered colimit together with his adapted-presentation criterion. Modern accessibility of a full subcategory of the arrow category concerns a sufficiently large regular cardinal and an accessible inclusion. Cisinski calls a localizer accessible when it is generated by a *set as a localizer*; for presheaf localizers, his Scholie 1.4.6 compares this with accessibility of the arrow class. These comparisons are theorems, not interchangeable definitions. In particular, the model-recognition theorem for a general accessible presheaf localizer should not be reported as assuming closure under all filtered colimits merely because the present chapter does so.

The logical order of the achievement also matters. A separated interval first proves that injective-type maps are weak equivalences; a bounded generation argument then supplies the weak factorization systems; their compatibility yields the model structure. The associated derivator can subsequently be formed from diagram categories and their weak equivalences. This order explains why the elementary topos-theoretic object L_A is central to a chapter whose ultimate purpose is the construction of homotopy theories.

9 Révision des notations, pour les axiomes d'un localiseur fondamental [43]

Loc (1) (Saturation).

a) Les isomorphismes sont dans W.

b) Si f, g sont composables, et deux parmi f, g, gf sont dans W, le troisième aussi.

42. **N.B.** On devrait renuméroter les axiomes $\mathrm{W}\,(\ldots)$.

43. *Cf.* XVII `[plutôt XVI]` pour une version définitive (?).

c) Si $X \underset{g}{\overset{f}{\rightleftarrows}} Y$, si gf et fg sont dans W, alors f et g aussi.

Variante renforcée :

Loc (1 bis) Loc (1) plus

d) W est stable par facteurs directs.

Loc (2) (Axiome de l'objet final). Si X dans Cat a un objet final, alors X est W-asphérique, *i.e.* $X \longrightarrow e$ est dans W.

Loc (3) (Axiome de localisation, forme faible). Soit $f : X \longrightarrow Y$ tel que pour tout $y \in Y$, X/y soit W-asphérique (*i.e.* le morphisme $f/y : X/y \longrightarrow Y/y$ déduit de f par changement de base $Y/y \longrightarrow Y$ est dans W). Alors $f \in W$.

Variante renforcée de Loc (3) :

Loc (3 bis) (Axiome de localisation, forme forte). Soit $f : X \longrightarrow Y$ un morphisme dans Cat/S, tel que $\forall\ s \in S$, $f/s : X/s \longrightarrow Y/s$ (déduit de f par changement de base $S/s \longrightarrow S$) soit dans W. Alors $f \in W$. (**N.B.** Sauf erreur, il suffit le l'exiger si X, Y sont Cat-fibrés scindés sur S, et f S-cartésien compatible avec le scindage – en supposant Loc (1, 2)).

p. 64
[page 117]

Commentary · Exposition [ms. 116]

The catalogue distinguishes several notions of localizer

Grothendieck is reorganizing the hypotheses so that each construction can carry only the assumptions it needs. The list is a catalogue of possible conditions, not a declaration that all of them are in force. This matters particularly for localization: in these pages a "localiseur fondamental" without qualification satisfies Loc (1,2,3), whereas the subsequent standard definition uses the relative condition Loc (3 bis).

The distinction can be expressed without terminology. The weak condition recognizes a functor $u : A \longrightarrow B$ when every comma category A/b has the homotopy type of a point, relative to W. The relative condition starts with a map of diagrams over a third category S and recognizes u from the maps $A/s \longrightarrow B/s$. Here A/s has objects $(a, p(a) \longrightarrow s)$; it is not generally the strict fibre of $p : A \longrightarrow S$. The equivalence between the two formulations of Loc (3) uses the terminal object $(b, 1_b)$ of B/b and two-out-of-three. It does not make the relative condition redundant.

There is a similarly necessary distinction within "saturation". Loc (1 c) is the two-sided inverse test: if gf and fg belong to W, then so do f and g. Loc (1 bis) adds closure under retracts in the arrow category. Strong saturation, later called Loc (1 ter), says that an arrow belongs to W precisely when its image in Cat$[W^{-1}]$ is invertible. The last condition implies the preceding ones, but its proof requires work; it is not part of the three displayed clauses of Loc (1).

The modern vocabulary records these distinctions. The definitions of weak and ordinary fundamental localizers in Maltsiniotis (2020, §§1.1 and 2.1) use respectively the absolute and relative forms of Theorem A. The theorem that every weak fundamental localizer contains the nerve weak equivalences is Cisinski (2006a, Theorem 6.1.18). Containment of the usual weak equivalences does not, by itself, supply every stability assertion concerning the additional arrows of W.

We should also keep the strength of Loc (4) visible. Its reverse implication is an extra excision condition, not a general property of homotopy pushouts. The commentary on Chapter XII already gives the relevant warnings: manuscript p. 126 ("Why converse excision is an additional axiom") and, in the appendix, manuscript p. 17 ("Acyclic links and a false converse"). Those examples are not repeated here. In this chapter Loc (4) may legitimately be assumed; it should not be silently assigned to every localizer obtained from a familiar homotopy theory. Finally, the parenthetical reduction of Loc (3 bis) to split fibred categories is proposed rather than demonstrated. A proof must specify its fibration convention and compare the comma resolutions with the original categories; replacing comma categories by strict fibres without that comparison would lose the hypothesis being tested.

Loc (4) (Axiome du carré cocartésien). Soit X dans Cat, X_1, X_2 deux ouverts de réunion X, $X_0 = X_1 \cap X_2$, d'où un diagramme cocartésien dans Cat,

$$\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\downarrow & & \downarrow{\scriptstyle j_2} \\ X_2 & \xrightarrow{j_1} & X\ . \end{array}$$

On a alors $i_1 \in W \iff j_1 \in W$ (et symétriquement $i_2 \in W \iff j_2 \in W$).

Loc (6) [44] (Axiome des W-fibrations). Il y a des variantes ⓐ, (a′), ⓑ, (b′), (b″), (c), (c′), *cf.* XII, p. 53-59. La forme la plus forte ⓐ est celle-ci :
Soit $f : X \longrightarrow Y$ dans Cat une W-fibration au sens ancien, *i.e.* telle que pour toute $Y'' \longrightarrow Y'$ dans Cat$/Y$, avec Y', Y'' contractiles, la flèche $X'' \longrightarrow X'$ déduite par le changement de base $X \longrightarrow Y$ soit dans W. Alors f est dans Fib$_W$, *i.e.* le foncteur de changement de base par f, Cat$/Y \longrightarrow$ Cat$/X$, transforme flèches dans W en flèches dans W.

Loc (7) (Axiome des limites inductives). W est stable par limites inductives filtrantes.

Forme renforcée de Loc (7) :

Loc (7 bis) (L-accessibilité). W est stable par $\varinjlim$ filtrantes *et* accessible, *i.e.* il existe une petite partie W_π de Fl(Cat), telle que W soit la plus petite partie de Fl(Cat) stable par $\varinjlim$ filtrantes, contenant W_π.

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Commentary · Context [ms. 117]

Which fibration hypothesis is being invoked

The variants imported from chapter XII concern the old notion of W-fibration. Write F_{old} for that class, reserving Fib$_W$ for the intrinsic condition that pullback preserves every W-map. The reference is Grothendieck (1990–1991f, ms. 53); after the renumbering, its W (4) becomes Loc (6). The seven tests are as follows.

a	$F_{\text{old}} \subseteq \text{Fib}_W$;
a'	the same test restricted to Parf$_\omega$;
b	$f : X \to Y$ over S, $p, q \in F_{\text{old}}$, $f \in W \implies f_s \in W$ for all s;
b'	the preceding test for split fibred objects and split cartesian f;
b''	the preceding test with p, q both W-perfect;
c	$p \in F_{\text{old}} \cap W \implies p^{-1}(s)$ is W-aspherical;
c'	the preceding test restricted to Parf$_\omega \cap W$.

Here $p : X \to S$, $q : Y \to S$, and Parf$_\omega$ refers to the least weak fundamental localizer. The restricted classes are part of the hypotheses. In particular, the proof on ms. 121 invokes c' after writing only $p \in \text{Parf}_W$. It needs a replacement in the restricted perfect class, together with componentwise weak equivalences for its first factor. These are dependencies of that short argument. The modern theorem identifying the least weak fundamental localizer with W_∞ clarifies the subscript, but does not supply an unmentioned replacement construction merely by changing the notation.

Loc (8) [45] (Axiome de connexité). $f \in W \Longrightarrow \pi_0(f)$ bijectif.

N.B. Tous les axiomes, sauf le dernier (et un chouïa de l'avant dernier) sont des axiomes de stabilité – ils disent tous que W est "grand" : il contient certaines flèches, et s'il contient telles flèches, il contient telles autres. Loc (8), et dans une certaine mesure Loc (7 bis), sont des axiomes "limitatifs" sur la taille de W : Loc (7) nous dit que $W \subset \mathbf{W}_0$ (= ensemble des

44. Il y aura un Loc (5) plus "basique" plus bas.

45. J'ai changé la numérotation, pour tenir compte d'un Loc (5) plus bas, qui est encore un axiome de stabilité [initialement les axiomes Loc (6) à Loc (8) étaient notés Loc (5) à Loc (7)].

$f \in \mathrm{Fl}\,\mathrm{Cat}$ telles que $\pi_0(f)$ soit bijectif), Loc (7 bis) nous dit l'existence d'un cardinal tel que W soit égal à la L-enveloppe de W^π (donc $W \subset \text{L-env}(W^\pi)$).

J'ai essayé de ranger les axiomes par ordre d'importance pratique. J'ai fini par rajouter l'axiome de connexité à la fin, car dans les développements théorique sur les localiseurs généraux, il n'intervient pratiquement jamais. Mais je rappelle (XII, p. 20, th. 2) que si W satisfait Loc (1, 2, 3, 6), et si Loc (8) n'est *pas* satisfait, alors $W = \mathrm{Fl}(\mathrm{Cat})$; donc en présence de Loc (1, 2, 3, 6), Loc (8) *équivaut* à $W \neq \mathrm{Fl}(\mathrm{Cat})$ tout entier.

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Commentary · Context [ms. 118]

A size condition with a particular closure operation

The accessibility hypothesis is doing a different job from the homotopical axioms. It is intended to turn a proper class of potential attachments into a set from which a factorization can be constructed. In Loc (7 bis), the closure operation is specifically closure under all small filtered colimits in the arrow category of Cat. Its relationship with the cardinal filtration was developed in §5. Thus "L-accessible" contains both a bounded generation assertion and a closure assertion.

Later authors also call a fundamental localizer accessible when a set generates it *as a fundamental localizer*; this is Cisinski (2006a, Definition 4.2.10). The latter closure includes the localizer axioms. These two formulations must not be identified simply because they share the word "accessible". A comparison requires a theorem connecting the two closure operations. The theorem on filtered colimits in the mature theory is recorded separately in Maltsiniotis (2020, Proposition 2.4.12 and Corollary 2.4.14). The size and closure requirements in Grothendieck's stated theorems are retained in the audit rather than replaced by an unqualified modern accessibility assumption.

Les axiomes qui ont servi pour 8.4, 1°, 3° et 8.5, (b) sont

$$\mathrm{Loc}\,(1\,\mathrm{bis},\ 2,\ 3)\ .$$

Ceux qui sont nécessaires pour 8.5, (a) sont p. 65

$$\mathrm{Loc}\,(1\,\mathrm{bis},\ 2,\ 3,\ 4)\ .$$

Enfin, ceux nécessaires pour 8.4, 2° sont

$$\mathrm{Loc}\,(1\,\mathrm{bis},\ 2,\ 3,\ 4,\ 7\,\mathrm{bis})\ ,$$

et ils impliquent la *saturation forte de* W (*cf.* paragraphe suivant).

Il me faut réparer un oubli, ayant traité un peu par le mépris (!) les questions de composantes connexes, jusqu'à présent.

Proposition 9.1. *Supposons que* W *satisfait* Loc (1, 2, 3, 4) *sans plus. Soit* $f_i : X_i \longrightarrow Y_i$ $(i \in I)$ *une famille* finie *de flèches de* Cat. *Pour que* $f = \coprod f_i : \coprod X_i \longrightarrow \coprod Y_i$ *soit dans* W, *il suffit que* $\forall\ i \in I$, *on ait* $f_i \in W$. *Si* W *satisfait de plus à* Loc (7) (*axiome des limites inductives*), *alors cette conclusion reste valable sans condition de finitude sur* I.

Démonstration. Le cas I fini se traite par récurrence sur $n = \mathrm{card}\, I$, on est ramené aussitôt au cas où $n = 2$ [46]. On voit d'abord le

Corollaire 9.2. *Sous l'hypothèse* Loc (1, 4), *soit* $f : X \longrightarrow Y$ *une flèche de* Cat, *et* Z *un objet de* Cat. *Alors* $f \in W$ *si et seulement si* $f \amalg \mathrm{id}_Z : X \amalg Z \longrightarrow Y \amalg Z$ *est dans* W. [page 120]

Immédiat sur le carré cocartésien, où inj_0^X est une immersion ouverte de Dwyer,

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ {\scriptstyle \mathrm{inj}_0^X}\downarrow & & \downarrow{\scriptstyle \mathrm{inj}_0^Y} \\ X \amalg Z & \xrightarrow[f' = f \amalg \mathrm{id}_Z]{} & Y \amalg Z\ , \end{array}$$

46. Le cas infini s'y ramène sous l'hypothèse Loc (7), par passage à la limite.

qui montre que $f \in W \iff f' \in W$.

En itérant ce résultat, on trouve, si

$$f_1 \ : \ X_1 \longrightarrow Y_1 \ , \qquad f_2 \ : \ X_2 \longrightarrow Y_2$$

sont donnés, que

$$f_1 \ \in \ W \ , \ f_2 \ \in \ W \quad \Longrightarrow \quad f_1 \amalg f_2 \ \in \ W \ ,$$

en regardant le diagramme de deux carrés cocartésiens :

$$\begin{array}{ccc} X_1 & \xrightarrow{\quad f \quad} & Y_1 \\ \downarrow & & \downarrow \\ X_1 \amalg X_2 & \xrightarrow[f_1'=f_1 \amalg \mathrm{id}_{X_2}]{} & Y_1 \amalg X_2 \\ {\scriptstyle f_2'=\mathrm{id}_{X_1} \amalg f_2}\downarrow & & \downarrow{\scriptstyle g_2=\mathrm{id}_{Y_1} \amalg f_2} \\ X_1 \amalg Y_2 & \xrightarrow[g_1=f_1 \amalg \mathrm{id}_{Y_2}]{} & Y_1 \amalg Y_2 \ . \end{array}$$

Si f_1 et f_2 sont dans W, il en est de même de f_1', f_2', g_1, g_2, par 9.2, donc aussi de $f_1 \amalg f_2 = g_2 f_1'$.

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Commentary · Exposition [ms. 120] [page 121]

Adding a summand and recovering a summand

The point of this discussion is to make weak equivalence compatible with the decomposition of a category into connected components. There are two operations to control. One adds an unrelated component to an equivalence. The other extracts a component from an equivalence already given. Proposition 9.4 separates them cleanly, and its proof is entirely formal once those operations are stated.

Write P for the condition that $f \in W$ implies $f \amalg 1_Z \in W$, and R for the condition that restriction to a summand of the target preserves W. Assuming identities and composition belong to the saturation rules, P implies closure under binary sums because

$$f_1 \amalg f_2 = (1_{Y_1} \amalg f_2)(f_1 \amalg 1_{X_2}).$$

Conversely, binary closure implies P. If $f_1 \amalg f_2 \in W$, condition R, applied to each of the two target summands, recovers f_1 and f_2. If binary sums are both preserved and reflected, any $f : X \longrightarrow Y_0 \amalg Y_1$ decomposes canonically as $f_0 \amalg f_1$, since a functor cannot carry an arrow between two different components of the target. Reflection then proves R.

The same argument works for an arrow $X \longrightarrow X'$ over a category Y and a summand $Y_0 \subset Y$: decompose X and X' by inverse image of Y_0 and its complement. This proves that R is exactly the assertion that the inclusion $Y_0 \hookrightarrow Y$ belongs to Fib_W. The target of the restricted arrow in Corollary 9.3 must therefore be X_0', not Y_0. The latter is the base of the restricted diagram, not its target.

Reflection for an arbitrary family already follows from binary reflection. For a fixed index i, group all other indices into one summand. No limiting argument is needed for this direction. Preservation for an infinite family requires an additional step: the coproduct arrow is the filtered colimit of its finite subcoproducts, indexed by the finite subsets of the index set. Thus Loc (7) upgrades finite preservation to arbitrary preservation. This explains why Loc (5 bis) only adds the infinite version of the forward implication. It also proves the two equivalent componentwise formulations of Corollary 9.5.

The proof of Corollary 9.2 deserves a small qualification. Its displayed square is a pushout along a summand inclusion, but the horizontal arrow f is arbitrary. Literally, Loc (4) on ms. 117 is phrased for a union of two open subcategories. To apply an excision theorem to this square one must use the corresponding extension to pushouts along Dwyer inclusions, or first perform the mapping-cylinder reduction established in the preceding discussion. The word “immédiat” suppresses that extension. Under the stronger pushout formulation the claimed equivalence is immediate; the displayed square alone is not an instance of the literal two-open-cover formulation. We do not infer a counterexample to the conclusion from this mismatch of formulations.

Mais je ne vois aucun moyen de prouver la réciproque, que $f_1 \amalg f_2 \in W \Longrightarrow f_1, f_2 \in W$. Grâce à ce qui a été prouvé, on est ramené au cas où f_1, f_2 sont des immersions ouvertes de Dwyer, mais il ne semble pas que cela nous avance d'un cheveu ! En somme, il faut savoir que les applications canoniques $Y_i \hookrightarrow \coprod Y_i$ sont dans Fib_W. Comme Y_i est un sommand direct, on est ramené au cas d'une immersion canonique $Y_1 \hookrightarrow Y_1 \amalg Y_2 = Y$: si $f : X \longrightarrow Y$ est dans W, en est-il de même de $X_1 = X \times_Y Y_1 \longrightarrow Y_1$? C'est une propriété qui, bien sûr, serait conséquence de l'ancien W (4 a) (XII, p. 53), ou seulement W (4 a′), donc de l'actuel Loc (6 a′). Il suffit même de la forme W (4 c′) = Loc (6 c′) la plus faible de toutes, comme on voit en décomposant $X_1 \longrightarrow Y$ en $X \xrightarrow{i} X' \xrightarrow{p} Y$, avec $i \in W$ (plus précisément, $i_1 : X_1 \longrightarrow X'_1$ et $i_2 : X_2 \longrightarrow X'_2$ dans W !), et $p \in \mathrm{Parf}_W$. Par Loc (6 c′), $f \in W$, donc $p \in W$ implique que les fibres de p sont W-asphériques, ce qui implique que p est dans W_{univ}, donc $p_1 : X'_1 \longrightarrow Y_1$ est dans W.

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Donc notons par acquis de conscience :

Corollaire 9.3. *Si W satisfait* Loc (1, 2, 3, 6 c′) (*où* Loc (6 c′) *est la variante la plus faible de toutes de l'axiome des fibrations*), *alors pour toute famille $(f_i : X_i \longrightarrow Y_i)_{i\in I}$ de flèches dans* Cat, $f = \coprod f_i \in W \Longrightarrow$ *les f_i sont dans W $\forall\, i \in I$* [47]. *Et si $i : Y_0 \hookrightarrow Y$ est une flèche qui est à la fois immersion ouverte et immersion fermée* (i.e. *un isomorphisme avec un sommand direct de Y*), *alors i est dans* Fib_W, i.e. *si $f : X \longrightarrow X'$ est une flèche dans* Cat$/Y$, *alors $f \in W$ implique $(f_0 : X_0 \longrightarrow Y_0) \in W$, où $X_0 = X|Y_0$, $X'_0 = X'|Y_0$.*

Mais il est assez scandaleux d'avoir à utiliser un axiome aussi fort, pour en conclure un résultat aussi anodin ! Je note ceci :

Proposition 9.4. *Soit $W \subset$* Fl(Cat), *satisfaisant* Loc (1) *sans plus. Les conditions suivantes sont équivalentes* (*et impliquées par* Loc (1, 2, 3, 4, 6 c′)).

a) *Si $f_1 : X_1 \longrightarrow Y_1$, $f_2 : X_2 \longrightarrow Y_2$ sont deux flèches de* Cat, *alors*

$$f_1 \amalg f_2 \in W \iff f_1 \in W,\ f_2 \in W\,.$$

b) *Les deux conditions suivantes sur W sont satisfaites :*

[page 123]

1°) *Si $f : X \longrightarrow Y$ est dans W, alors il l'est coüniversellement sous $\emptyset$,* i.e. *pour tout Z dans* Cat, $f \amalg \mathrm{id}_Z : X \amalg Z \longrightarrow Y \amalg Z$ *est dans W.* (**N.B.** *Ceci équivaut à : toute somme finie de flèches $f_i \in W$ est dans W – et itou même pour sommes infinies, si* Loc (7), *axiome des limites, est satisfait.*)

2°) *Soit $Y_0 \hookrightarrow Y$ une inclusion d'un sommand direct d'un objet de* Cat. *Alors si $X \xrightarrow{f} Y$ est dans W, de même $X_0 = X|Y_0 \longrightarrow Y_0$.* (**N.B.** *Ceci équivaut à la condition que l'inclusion d'un sommand direct est toujours une W-fibration.*)

De plus, si $\pi_0(Y)$ est fini, ou si W satisfait l'axiome des limites Loc (7), *il suffit d'exiger 2° pour Y_0 une composante connexe de Y.*

p. 67

c) *Pour toute famille* finie $(f_i)_{i\in I}$ *de flèches $f_i : X_i \longrightarrow Y_i$ dans* Cat,

$$\coprod f_i \in W \iff f_i \in W \quad \textit{pour tout } i \in I.$$

De plus, si W satisfait Loc (7), *alors il est inutile dans* (c) *de se borner à I fini.* (*Et en tous cas, les conditions équivalentes* (a), (b), (c) *impliquent pour* toute *famille de flèches $(f_i)_{i\in I}$, $\coprod f_i \in W \Longrightarrow f_i \in W\ \forall\, i \in I$.*)

[page 124]

D'après 9.3, ces conditions sont satisfaites moyennant Loc (1, 2, 3, 4, 6 c′), mais Loc (6 c′) est un marteau-pilon pour une mouche ! On aurait envie plutôt d'ajouter un axiome aux axiomes Loc (1) à Loc (4) et Loc (6) à Loc (8), qui avec Loc (8) serait l'axiome le plus anodin de tous.

47. En fait, je ne sais prouver l'implication en sens inverse, moyennant seulement Loc (1, 2, 3, 6 c′).

Loc (5) (Axiome de décomposition, forme faible). Si $(f_i)_{i\in I}$ est une famille finie de flèches de Cat, alors $f = \coprod f_i$ est dans W si et seulement si les f_i le sont. (Ce sont les conditions équivalentes de 9.4.) [48]

Loc (5 bis) (Axiome de décomposition, forme forte). Comme Loc (5), mais sans supposer I finie. [49]

Ainsi, moyennant l'axiome des limites Loc (7), les deux axiomes précédents sont équivalents. De plus, notons ceci :

Corollaire 9.5 (de la proposition 9.4). *L'axiome* Loc (5 bis) *équivaut à chacune des propriétés suivantes :*

ⓐ *Pour tout Y de* Cat, *décomposé en ses composantes connexes Y_i ($i \in I = \pi_0(Y)$), et pour toute flèche $f : X \longrightarrow Y$ dans* Cat, *on a $f \in W$ si et seulement si les $f_i : X_i = X|Y_i \longrightarrow Y_i$ sont dans W.*

ⓑ *Pour Y, Y_i comme dans* ⓐ, *pour toute flèche $f : X \longrightarrow X'$ dans* Cat$/Y$, *on a $f \in W$ si et seulement si $\forall\ i \in I$, le morphisme $f_i : X_i \longrightarrow X'_i$ au-dessus de Y_i est dans W.* [page 125]

Corollaire 9.6. *Si l'axiome* Loc (5) *est satisfait, alors l'axiome de connexité* Loc (8) *équivaut à ceci : pour toute flèche $f : X \longrightarrow Y$ dans W, avec Y 0-connexe, X est 0-connexe.*

N.B. Les axiomes Loc (8) (connexité) et Loc (5 bis) (de décomposition) ont l'air de nature voisine, étant liés l'un et l'autre à la décomposition des objets de Cat en composantes connexes. (Néanmoins, je pense qu'ils sont indépendants.) Je présume qu'on n'aura jamais à travailler vraiment avec des W qui ne satisfont à ces deux axiomes ! (En plus de Loc (1, 2, 3), qui eux constituent le "minimum vital".) Mais apparemment il n'y en a pas besoin dans les développements théoriques généraux jusqu'ici. (Mais l'axiome de connexité Loc (8) pourrait jouer un rôle, quand on parle de la détermination réciproque des structures de contractibilité, et de W-asphéricité, *cf.* Pursuing Stacks. p. 68

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Commentary · Development (commentators') [ms. 125]

Connectivity supplies the missing retraction

Grothendieck asks whether the decomposition axiom and the component axiom are independent. The question changes once closure under retracts is available. The following elementary observation makes the relation precise, and accounts for why the modern situation is more rigid than the tentative list suggests. We checked the localizer discussions in Maltsiniotis (2020, §2.1 and §2.4) and Cisinski (2006a, §9.3); the particular retract argument below is supplied here, without a claim of priority.

Summand reflection from connectivity. [ours] Let W be a class of functors containing the identities and closed under retracts in $\underline{\mathrm{Fl}}(\mathrm{Cat})$. Suppose every W-map induces a bijection on connected components. Then, for every small family $(f_i : X_i \longrightarrow Y_i)_{i\in I}$,

$$\coprod_{i\in I} f_i \in W \quad \Longrightarrow \quad f_i \in W \quad \text{for every } i.$$

Consequently, if W is also closed under arbitrary coproducts, it satisfies Loc (5 bis).

Proof. Fix i. If X_i is empty, then Y_i is empty: otherwise the component of any object of Y_i would not be in the image of $\pi_0(\coprod f_j)$, since all the other source summands map into the other target

48. Il peut être commode aussi d'introduire la condition encore plus faible :
Loc (5 a) Si f_1, f_2 sont dans W, alors $f_1 \amalg f_2 \in W$.
C'est aussi la condition (b), 1° de la proposition 9.4.
Loc (5 b) Si $f_1 \amalg f_2$ est dans W, alors f_1 et f_2 aussi.
$\iff$ Si $Y_0 \subset Y$ est un sommand direct, alors $Y_0 \hookrightarrow Y$ est dans Fib_W.

49. `[Réécrivons :]`
Loc (5) a) Si $f_1, f_2 \in W$, alors $f_1 \amalg f_2 \in W$.
b) Si $f_1 \amalg f_2 \in W$, alors $f_1, f_2 \in W$
($\iff b'$) tout $Y_0 \hookrightarrow Y$ sommand direct est dans Fib_W).
Loc (5 bis) Comme Loc (5), mais (a) remplacé par (a bis) une famille (pas nécessairement finie).

summands. In this case f_i is an identity. If X_i is nonempty, choose an object x_i. The inclusion of f_i into $\coprod f_j$ is an arrow-category map. Define a retraction on sources to be the identity on X_i and, on each X_j with $j \neq i$, the constant functor at x_i. Define the retraction on targets to be the identity on Y_i and constant at $f_i(x_i)$ on the other Y_j. These two functors form a commutative square: on the selected summand this is tautological, and on each other summand both composites are the same constant functor. Their composite with the inclusion is the identity of f_i. Thus f_i is a retract of $\coprod f_j$ and belongs to W. □

This proves a sharper statement than the finite reflection needed in Proposition 9.4. It uses no lifting property and no properness theorem. The empty-source case is the only point where preservation of components is needed; the chosen point supplies the retraction in every other case.

Corollary 9.6 can now be read directly at the level of components, even without this retract argument. Under Loc (5), an arrow in W restricts to an arrow in W over each connected component of its target. If every such arrow with connected nonempty target has connected nonempty source, these restrictions each have exactly one source component; therefore the original map induces a bijection on components. The converse follows by applying a component-preserving map to a target having one component. Likewise, the equivalence between (a) and (c) of Proposition 9.11 is formal: a component bijection identifies the coproduct decompositions, and Loc (5 bis) applies to that coproduct. Conversely, criterion (c) implies component preservation and reconstructs both preservation and reflection of arbitrary coproducts by inspecting their component maps. Only the comparison with nondegeneracy in (b) invokes the collapse theorem discussed below.

For a fundamental localizer in the modern, relative sense, coproduct closure and retract closure are available. Hence Loc (8) implies Loc (5 bis) in that setting. The reverse implication fails: the class of all functors satisfies decomposition and does not preserve components. Thus the proposed mutual independence does not survive in the modern setting. For the literal weaker assumptions Loc (1,2,3) of the present manuscript, the cited modern theorems do not eliminate the distinction between weak and relative localization; we do not claim that they settle that separate implication.

There is also a direct explanation of footnote 57. In any category with sequential colimits, a retract a of an object b, with inclusion i and retraction r, is the colimit of the constant-object sequence

$$b \xrightarrow{ir} b \xrightarrow{ir} b \xrightarrow{ir} \cdots .$$

Indeed, a compatible cocone is one arrow $h : b \longrightarrow z$ satisfying $h = hir$, and it factors uniquely through r as $(hi)r$. Apply this observation in the arrow category. A class closed under filtered colimits and isomorphism is therefore closed under retracts. Loc (7) supplies the extra saturation condition Loc (1 bis); it does not by this argument alone establish strong saturation.

Notons ceci :

Proposition 9.7.[50] *Supposons que W satisfait* Loc (1, 2, 3, 5), i.e. *c'est un localiseur fondamental satisfaisant la condition de décomposition* (*forme faible*). *Si l'axiome* Loc (8) *de connexité n'est* pas *satisfait, alors la sous-catégorie* Hot_W^* *des* $X \in \mathrm{Cat}$ *non vides est équivalente à la catégorie ponctuelle* (*donc* Hot_W *est équivalente soit à la catégorie ponctuelle, si* $\emptyset \longrightarrow e$ *est un isomorphisme dans* Hot_W, *soit à* $\mathbf{\Delta}_1$ *dans le cas contraire*[51].)

Ceci est une variante du th. 2 de XII (p. 20), où on faisait l'hypothèse Loc (6 c′), plus forte que Loc (5) et même que Loc (5 bis), mais avec la conclusion plus forte que $W = \mathrm{Fl}(\mathrm{Cat})$.

Démonstration. Soit

$$f \ : \ X \longrightarrow Y \ , \qquad f \ \in \ W \ ,$$

telle que $\pi_0(f)$ ne soit pas *surjective.* Si Y_0 est une composante connexe qui n'est pas dans $\mathrm{Im}\pi_0(f)$, alors $X_0 = X|Y_0$ est vide, et par Loc (7), $\emptyset \longrightarrow Y_0$ est dans W. Donc $\forall\ Z$ dans Cat, $\underbrace{\emptyset \times Z}_{=\emptyset} \longrightarrow Y_0 \times Z$ est dans W, et pour toute flèche $g : Z \longrightarrow Z'$, $\mathrm{id}_{Y_0} \times g : Y_0 \times Z \longrightarrow Y_0 \times Z'$ est dans W. Prenant g telle que $\pi_0(g)$ ne soit pas injective, p. ex. $e \amalg e \longrightarrow e$, on trouve une flèche $\varphi = \mathrm{id}_{Y_0} \times g$ dans W telle que $\pi_0(\varphi)$ ne soit pas injective.

50. *Cf.* version plus forte dans 9.9 plus bas (débarrassée de l'hypothèse Loc (5).) [page 127]

51. Le premier cas est aussi celui où $\exists\ f \in W$ avec $\pi_0(f)$ *non surjectif,* ou encore $\emptyset \longrightarrow X$ dans W, avec X non vide.

Soit donc $\varphi : I \longrightarrow J$ dans W, avec $\pi_0(\varphi)$ non injective. Procédant comme tantôt en utilisant Loc (5), on est ramené au cas où de plus J est 0-connexe. (**N.B.** Dans les deux cas, on aurait pu aussi invoquer 9.6.) Choisissons deux éléments i_0, i_1 dans I dans deux composantes connexes distinctes, soient j_0, j_1 leurs images dans J. On va traiter (I, i_0, i_1) et (J, j_0, j_1) comme des "intervalles homotopiques". Le fait que i_0, i_1 appartiennent à des composantes distinctes de I implique que deux flèches $f_0, f_1 : X \rightrightarrows Y$ sont "homotopes par rapport à (I, i_0, i_1)", *i.e.* il existe $h : X \times I \longrightarrow Y$ dans Cat telle que $f_0 = h\partial_0^X$, $f_1 = h\partial_1^X$, où

$$\partial_0^X \ , \ \partial_1^X \ : \ X \ \longrightarrow \ I \times X$$

sont données par $x \longmapsto (i_\alpha, x)$ $(\alpha \in \{0,1\})$. Pour trouver h, on décompose I en

$$I \ = \ I_0 \amalg I_1 \ , \qquad \text{avec} \quad i_0 \in I_0, \ i_1 \in I_1 \ ,$$

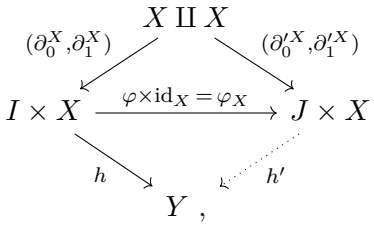
d'où $I \times X \ \simeq \ (I_0 \times X) \amalg (I_1 \times X)$, et on définit h par ses restrictions h_0, h_1 aux deux morceaux $I_0 \times X$, $I_1 \times X$, savoir

$$h_0 \ = \ f_0 \circ \mathrm{pr}_2^{I_0} \ , \qquad h_1 \ = \ f_1 \circ \mathrm{pr}_2^{I_1} \ ,$$

où p. 69

$$\mathrm{pr}_2^{I_\alpha} \ : \ I_\alpha \times X \longrightarrow X \ .$$

Considérons maintenant le diagramme (traits pleins) [page 128]



où $\partial_0'^X$, $\partial_1'^X$ sont définis comme ∂_0^X, ∂_1^X, mais utilisant $j_0, j_1 \in \mathrm{Ob}J$ au lieu de $i_0, i_1 \in \mathrm{Ob}I$. S'il existait dans Cat une flèche h' (pointillée) telle que $h' \circ \varphi_X = h$, il s'ensuivrait que h' est une homotopie de f_0 et f_1 suivant (J, j_0, j_1). Comme J est 0-connexe, il s'ensuivrait aussitôt que f_0, f_1 sont homotopes au sens habituel dans Cat, donc que f_0 et f_1 ont même image dans Hot_W. Mais si on ne peut trouver une flèche h' dans Cat, du moins peut-on la trouver dans Hot_W, puisque $\varphi_X \in W$, donc un isomorphisme dans Hot_W (et h' dans $\mathrm{Hom}_W(J \times X, Y)$ sera même uniquement déterminé). Donc dans Hot_W, on a

$$f_0 \ = \ h'\partial_0'^X \ , \qquad f_1 \ = \ h'\partial_1'^X \ .$$

Pour prouver que [page 129]

$$f_0 \ = \ f_1 \qquad \text{dans} \quad \mathrm{Hom}_W(X, Y) \ ,$$

il suffit de voir que

$$\partial_0'^X \ = \ \partial_1'^X \qquad \text{dans} \quad \mathrm{Hom}_W(X, J \times X) \ .$$

Or en fait $\partial_0'^X$ et $\partial_1'^X$ sont homotopes, car $e \overset{j_0}{\underset{j_1}{\rightrightarrows}} J$ sont homotopes (j_0 et j_1 appartenant à la même composante connexe de J), donc aussi leur produit par id_X.

On voit donc que pour deux objets quelconques X, Y de Cat, toutes les flèches de X dans Y sont égales dans Hot. L'argument fait prouve même que toutes les flèches *dans* Hot_W de X dans Y sont égales. Cela signifie que la catégorie Hot_W^* (où pour deux objets quelconques X, Y, $\mathrm{Hom}_{\mathrm{Cat}}(X, Y)$, et a fortiori $\mathrm{Hom}_W(X, Y)$ est $\neq \emptyset$) est équivalente à la catégorie ponctuelle. Quant à Hot_W elle-même, dont l'ensemble des objets est $\{\emptyset\} \amalg \mathrm{Ob}\mathrm{Hot}_W^*$, avec $\emptyset$ objet initial, [page 130]

elle sera donc équivalente à la catégorie ponctuelle, ou bien à $\mathbf{\Delta}_1$, selon que $\emptyset \longrightarrow e$ est un isomorphisme dans Hot_W, ou non.

Commentaire 9.8. Si les axiomes Loc (1, 2, 3, 5) [52] sont satisfaits, alors l'axiome de connexité Loc (8) *équivaut* à chacun des suivants :

Commentary · Examples [ms. 129]

The two degenerate homotopy categories

Grothendieck's conclusion allows two outcomes because the empty category can retain information after every nonempty category has become indistinguishable. Both outcomes occur, and computing them helps separate a statement about a localization from a statement about its specified class of denominators.

Let W_{-1} consist of all functors whose source and target are both nonempty, together with the identity of the empty category. Equivalently, these are precisely the functors which preserve and reflect emptiness. This class satisfies two-out-of-three and the inverse test by applying the functor

$$E : \mathrm{Cat} \longrightarrow [1], \qquad E(A) = \begin{cases} 0 & A = \emptyset, \\ 1 & A \neq \emptyset. \end{cases}$$

It satisfies the terminal-object axiom. It even satisfies the relative Theorem A: if $A/s \longrightarrow B/s$ preserves emptiness for every $s \in S$ and B is nonempty, choose $b \in B$ and $s = p(b)$. Then B/s contains $(b, 1_s)$, so A/s is nonempty and therefore A is nonempty. The reverse nonemptiness implication is automatic for a functor. Thus the comparison itself lies in W_{-1}.

The localization at W_{-1} is [1]. To check the universal property, let $F : \mathrm{Cat} \longrightarrow \mathcal{H}$ invert W_{-1}. For nonempty A, the arrow $A \longrightarrow e$ becomes invertible. For any $u : A \longrightarrow B$ between nonempty categories, the equation $p_B u = p_A$ forces

$$F(u) = F(p_B)^{-1} F(p_A).$$

Consequently every arrow between nonempty categories has a uniquely prescribed image, and the unique arrow $\emptyset \longrightarrow e$ supplies the remaining arrow of [1]. This is exactly the required factorization, with its natural-isomorphism version as well. Localizing at all functors additionally makes $\emptyset \longrightarrow e$ invertible and gives the terminal category.

The class W_{-1} fails component preservation: the fold $e \amalg e \longrightarrow e$ belongs to it. It also fails summand reflection, since

$$(\emptyset \longrightarrow e) \amalg (1_e : e \longrightarrow e)$$

has nonempty source and target, while its first summand is not in W_{-1}. This is the precise obstruction behind the concern on ms. 121. A weak equivalence of a disjoint union need not restrict to a weak equivalence merely because both unions are nonempty.

By contrast, for $W_0 = \{u \mid \pi_0(u) \text{ is bijective}\}$, the localization is Set. The component functor has a section sending a set to its discrete category, and the canonical functor $A \longrightarrow \pi_0(A)$ belongs to W_0. Its universal property identifies the localization with Set. This supplies infinitely many isomorphism classes, including the finite discrete categories of every cardinality. Thus the alternatives of Proposition 9.12 already occur in these elementary examples; the point of its argument is to show that loss of components leaves no intermediate nonempty homotopy theory under the necessary tensor hypothesis.

p. 70

a) Hot_W n'est équivalente ni à e, ni à $\mathbf{\Delta}_1$.

b) Hot_W a une infinité de classes d'isomorphie.

c) $f_0 : X_0 \longrightarrow Y_0$ (flèche donnée d'avance dans Cat, telle que $\pi_0(f_0)$ n'est pas un isomorphisme, et $X_0 \neq \emptyset$) n'est *pas* un isomorphisme de Hot_W.

d) X_0 et Y_0 ne sont pas isomorphes dans Hot_W (où X_0, Y_0 sont deux objets donnés d'avance dans Cat, avec X_0, Y_0 non vides et $\mathrm{card}\,\pi_0(X_0) \neq \mathrm{card}\,\pi_0(Y_0)$).

52. En fait, ce commentaire vaut même en l'absence de l'hypothèse Loc (5), *cf.* 9.9 ci-dessous.

On peut donc dire que l'axiome de connexité joue le rôle d'un *axiome de non trivialité* pour la catégorie Hot_W localisée de Cat. Il sera satisfait automatiquement pour tous les localiseurs fondamentaux W qu'on sera amené à construire ou à introduire. Mais on n'aura pratiquement jamais à l'invoquer comme une hypothèse, en vue de certaines conclusions.

Je m'aperçois finalement que dans la proposition 9.7 l'hypothèse Loc (5) est inutile :

Proposition 9.9.[53] *Soit W un localiseur fondamental dans* Cat (i.e. *on a* Loc (1, 2, 3) *sans plus*), *plus la condition que la* somme *de $f_1, f_2 \in W$ est dans W*[54]. *Si l'axiome de connexité n'est pas satisfait, alors* Hot_W *est équivalente à la catégorie ponctuelle* (*si $\emptyset \longrightarrow e$ est un isomorphisme dans* Hot_W[55]) *ou à $\mathbf{\Delta}_1$* (*dans le cas contraire*), *et* Hot_W^* ($\overset{\text{déf}}{=} \mathrm{Hot}_W \backslash \{\emptyset\}$) *est équivalente à la catégorie ponctuelle.*

La condition sur les sommes assure que dans Hot_W la somme de deux objets existe, et [page 131] que $\mathrm{Cat} \longrightarrow \mathrm{Hot}_W$ commute aux sommes de deux objets. Ceci dit, soit

$$(*) \qquad f \ : \ X \longrightarrow Y$$

une flèche telle que $f \in W$, mais $\pi_0(f)$ non surjectif. Supposons d'abord $X \neq \emptyset$ (cas le moins évident). Si Y_0 est un sommand non vide de Y tel que f se factorise par $Y_1 = Y \backslash Y_0$, alors f est somme de

$$f_1 \ : \ X \ = \ X_1 \longrightarrow Y_1 \qquad \text{et} \qquad f_0 \ : \ X_0 \ = \ \emptyset \longrightarrow Y_0$$

dans Cat, donc aussi dans Hot_W. Donc on a une somme $f \ = \ f_0 \amalg f_1$ dans Hot_W qui est un isomorphisme, donc pour tout Z dans Hot_W,

$$\mathrm{Hom}_W(Y_0, Z) \times \mathrm{Hom}_W(Y_1, Z) \ \xrightarrow{\sim} \ \underbrace{\mathrm{Hom}_W(X_0, Z)}_{=e} \times \mathrm{Hom}_W(X_1, Z)$$

(**N.B.** $\emptyset$ est objet initial de Hot_W, l'étant de Cat), *i.e.* le composé

$$Z(Y_0) \times Z(Y_1) \ \xrightarrow{\mathrm{pr}_2} \ Z(Y_1) \ \xrightarrow{Z(f_1)} \ Z(X_1)$$

est bijectif. Comme $X \ = \ X_1 \neq \emptyset$, donc aussi $Y_1 \neq \emptyset$, et $Y_0 \neq \emptyset$, les ensembles $Z(Y_0)$, $Z(Y_1)$, $Z(X_1)$ sont $\neq \emptyset$ si $Z \neq \emptyset_{\mathrm{Cat}}$. Le composé étant bijectif, la première flèche pr_2 est injective, ce qui implique que $Z(Y_0)$ est réduit à un point (les deux facteurs $Z(Y_0)$, $Z(Y_1)$ étant $\neq \emptyset$). Cela p. 71 signifie que Y_0 est un objet initial dans Hot_W^*. Il s'ensuit notamment que $Y_0 \amalg Y_0$ est aussi un [page 132] objet initial, et l'application canonique $Y_0 \amalg Y_0 \longrightarrow Y_0$ est donc un isomorphisme dans Hot_W, alors que l'application induite sur les π_0 n'est pas injective. (C'est $\pi_0(Y_0) \amalg \pi_0(Y_0) \longrightarrow \pi_0(Y_0)$, avec $\pi_0(Y_0)$ non vide.) Ce cas va être traité ci-dessous.

Si dans $(*)$ X est vide, Y non vide (puisque $\pi_0(f)$ non surjectif), alors Y est objet initial de Hot_W, et on voit encore que $Y \amalg Y \longrightarrow Y$ est un isomorphisme dans Hot_W, bien que n'induisant pas un isomorphisme des π_0. La proposition 9.9 va donc résulter du

Lemme 9.10. *Soit W localiseur fondamental sans plus* (*axiomes* Loc (1, 2, 3)). *Supposons qu'il existe $f : I \longrightarrow J$ dans* Cat *tel que* $\mathrm{hot}_W(f)$ *soit un isomorphisme, mais que $\pi_0(f)$* ne soit pas injectif[56]. *Alors* Hot_W^* *est équivalente à la catégorie ponctuelle e, et* Hot_W *est équivalente à e* (*si $\emptyset \longrightarrow e$ est un isomorphisme dans* Hot_W) *ou à $\mathbf{\Delta}_1$* (*dans le cas contraire*).

La démonstration est celle donnée pour 9.7 : on prouve que si X, Y sont deux objets de [page 133] Cat, *non vides*, alors deux flèches $f_0, f_1 : X \rightrightarrows Y$ sont égales dans Hot_W, par un argument

53. *Cf.* meilleur énoncé 9.12 (p. 133 bis).

54. Cette condition Loc (5 a) est vérifiée notamment si W satisfait de plus l'un des deux axiomes Loc (4) (*cf.* 9.1), ou Loc (6 c′) (*cf.* 9.3).

55. Cela signifie aussi qu'il existe $Z \neq \emptyset$ dans Cat, tel que $\emptyset \longrightarrow Z$ soit un isomorphisme dans Hot_W.

56. Ou bien tel que I soit 0-connexe, et J non 0-connexe (*i.e.* $\mathrm{card}\, \pi_0(I) \ = \ 1$, $\mathrm{card}\, \pi_0(J) \ \geq \ 2$). Ce deuxième cas se traite par la même méthode essentiellement.

d'homotopie, en choisissant deux points i_0, i_1 de I qui sont dans deux composantes connexes distinctes, et tels que leurs images j_0, j_1 dans J soient dans la même composante.

Proposition 9.11. *Soit W un localiseur fondamental* (i.e. *on a* Loc (1, 2, 3)). *Alors les conditions suivantes sont équivalentes.*

a) *On a* Loc (5 bis) (*axiome de décomposition, forme forte*) *et* Loc (8) (*axiome de connexité*).

b) *On a* Loc (5 bis), *et* Hot_W *n'est pas équivalente à e ou à* $\mathbf{\Delta}_1$ (cf. *commentaire* 9.8).

c) *Pour toute $f : X \longrightarrow Y$, on a $f \in W$ si et seulement si $\pi_0(f)$ est un isomorphisme, et pour toute composante connexe X_i de X, si Y_i désigne la composante connexe correspondante de Y, la flèche induite $f_i : X_i \longrightarrow Y_i$ est dans W.*

La raison pour laquelle je n'isole pas cette propriété en un des axiomes Loc (…), c'est que ce n'est *pas* un axiome de stabilité, mais un axiome de nature mixte.

Proposition 9.12. *Soit W un localiseur fondamental* (*sans plus*), i.e. *satisfaisant* Loc (1, 2, 3). *Les conditions suivantes sont équivalentes.* [page 133 bis]

a) *W satisfait l'axiome de connexité* Loc (8), i.e. *$f \in W \Longrightarrow \pi_0(f)$ bijectif.*

b) *Si $f, g : X \rightrightarrows Y$ dans* Cat *sont telles que* $\mathrm{hot}_W(f) = \mathrm{hot}_W(g)$, *alors* $\pi_0(f) = \pi_0(g)$.

c) *Il existe X, Y dans* Cat *tel que l'application* $\mathrm{hot}_W^{X,Y} : \mathrm{Hom}_{\mathrm{Cat}}(X, Y) \longrightarrow \mathrm{Hom}_W(X, Y)$ *ne soit pas constante.*

c′) *La sous-catégorie* Hot_W^* *de* Hot_W, *formée des objets non vides, n'est pas équivalente à la catégorie ponctuelle.*

c″) *La catégorie* Hot_W *n'est pas équivalente à la catégorie e ni à la catégorie* $\mathbf{\Delta}_1$.

d) *L'ensemble des classes d'isomorphie d'objets de* Hot_W *est infini.*

Démonstration. On a $(a) \Longrightarrow (b)$, puisque (a) équivaut à dire que le foncteur $\pi_0 :$ Cat $\longrightarrow$ Ens se factorise par Hot_W. D'ailleurs le foncteur correspondant [p. 72]

$$\mathrm{Hot}_W \longrightarrow (\mathrm{Ens})$$

est essentiellement surjectif – il y a un inverse à isomorphisme près, $I \longmapsto \mathrm{hot}_W(I)$. Cela prouve que (a) implique (d), et on a $(d) \Longrightarrow (c')$, $(d) \Longrightarrow (c'')$ tautologiquement. Un argument plus ou moins tautologique montre d'ailleurs que si Hot_W^* est équivalente à e, alors Hot_W est équivalente à e ou à $\mathbf{\Delta}_1$ (car $\emptyset$ est objet initial de Hot_W) – donc $(c'') \Longrightarrow (c')$. Donc on a plus ou moins tautologiquement

$$\begin{array}{ccccccc} (\mathrm{a}) & \Longrightarrow & (\mathrm{d}) & \Longrightarrow & (\mathrm{c}'') & \Longrightarrow & (\mathrm{c}') \\ \Downarrow & & & & & \nearrow\!\nearrow & \\ (\mathrm{b}) & & \Longrightarrow & & (\mathrm{c}) & & \end{array}$$

où les implications $(b) \Longrightarrow (c) \Longrightarrow (c')$ sont elles aussi triviales [c'est plutôt l'implication (c′) ⟹ (c) qui est triviale ; l'implication (c) ⟹ (c′) est démontrée après l'énoncé du corollaire 9.13, avant la preuve de ce dernier]. Donc tout revient à prouver que $(c) \Longrightarrow (a)$, ou encore : [page 133 ter]

Corollaire 9.13. *Si l'axiome de connexité n'est pas satisfait, alors si X, Y sont deux objets de* Cat, *toutes les flèches $X \longrightarrow Y$ dans* Cat *sont égales dans* Hot_W, i.e. *l'application* $\mathrm{Hom}_{\mathrm{Cat}}(X, Y) \longrightarrow \mathrm{Hom}_W(X, Y)$ *est constante.*

Cela implique donc que tous les objets de Cat*, *i.e.* tous les objets non vides de Cat, sont isomorphes dans Hot_W, et on trouve un isomorphisme canonique entre elles, avec transitivité. Soit alors $\mathcal{H}$ la catégorie quotient de Cat, ayant mêmes objets que Cat, et ayant exactement une flèche entre deux objets non vides de Cat et exactement une flèche de $\emptyset$ dans tout autre

objet de Cat (et pas de flèche d'un objet non vide de Cat dans $\emptyset$). C'est donc une catégorie équivalente à $\mathbf{\Delta}_1$. On a vu que $\mathrm{Cat} \longrightarrow \mathrm{Hot}_W$ se factorise en

$$\mathrm{Cat} \xrightarrow{\text{can.}} \mathcal{H} \longrightarrow \mathrm{Hot}_W \ .$$

Si $\mathrm{Cat} \longrightarrow \mathcal{H}$ transforme toute flèche appartenant à W en isomorphisme, alors il est immédiat que $\mathcal{H}$ est solution au problème universel définissant Hot_W, donc que $\mathcal{H} \longrightarrow \mathrm{Hot}_W$ est un isomorphisme. Cela signifie aussi que W ne contient pas de flèche $\emptyset \longrightarrow X$ avec X non vide (auquel cas W contient toutes les flèches). Donc dans ce cas $\mathrm{Hot}_W \simeq \mathbf{\Delta}_1$. Dans le cas contraire, soit $\mathcal{H}'$ la catégorie déduite de $\mathcal{H}$ en inversant les flèches $\emptyset \longrightarrow X$, *i.e.* le groupoïde 1-connexe sur l'ensemble d'objets ObCat. Ce n'est plus une catégorie quotient de Cat, mais on a encore une factorisation

$$\mathrm{Cat} \xrightarrow{\text{épi}} \mathcal{H} \xrightarrow{\text{loc.}} \mathcal{H}' \longrightarrow \mathrm{Hot}_W \ ,$$

et cette fois c'est $\mathcal{H}'$ qui est solution du problème universel ($\mathrm{Cat} \longrightarrow \mathcal{H}'$ transformant flèches de W en isomorphismes). Donc dans ce cas, $\mathcal{H}' \longrightarrow \mathrm{Hot}_W$ est un isomorphisme, et Hot_W est équivalente à e. Dans les deux cas, Hot_W^* est équivalente à e. p. 73 [page 133 quater]

Donc, la proposition 9.12 résulte du corollaire 9.13, qu'il s'agit à présent de démontrer.

Commentary · Exposition [ms. 133]

The precise hypothesis behind the collapse argument

Grothendieck has found a remarkably economical way to detect nontriviality: two points must remain distinguishable. The argument is correct once product with a category carries weak equivalences to weak equivalences. This is exactly the operation used when an equality of two points is promoted to an equality of two maps with arbitrary source. We give the full argument with that hypothesis visible, rather than silently proving a statement under stronger axioms than those printed.

Assume Loc (1,2,3), and in addition

(T) $$u \in W \quad \Longrightarrow \quad u \times 1_X \in W \quad \text{for every small category } X.$$

Write $L : \mathrm{Cat} \longrightarrow \mathcal{H} = \mathrm{Cat}[W^{-1}]$. Natural transformations already identify functors in $\mathcal{H}$ under the weak axioms. To see this directly, the projection $[1] \times X \longrightarrow X$ has comma category $[1] \times (X/x)$ over x, which has a terminal object. Loc (2,3) therefore puts the projection in W. Its two sections have the same image, being inverses to the same isomorphism. A natural transformation $F \Rightarrow G$ is a functor $[1] \times X \longrightarrow Y$, and composition with the sections gives $L(F) = L(G)$. A finite zigzag of natural transformations gives the same conclusion.

Suppose first that $u : I \longrightarrow J$ belongs to W and identifies two different components of I. Choose i_0, i_1 in those components, with images j_0, j_1 in the same component of J. Given $f_0, f_1 : X \longrightarrow Y$, split $I = I_0 \amalg I_1$ with the chosen objects in different summands, and define $h : I \times X \longrightarrow Y$ by f_0 on the first summand and f_1 on the second. By (T), $L(u \times 1_X)$ is invertible. The two inclusions $X \longrightarrow J \times X$ at j_0, j_1 are joined by a zigzag of natural transformations, hence equal in $\mathcal{H}$. Their inverse images through $L(u \times 1_X)$ are equal, and composing with $L(h)$ proves $L(f_0) = L(f_1)$.

Suppose instead that u misses a component of J. Let ε be the discrete category $\{0, 1\}$. Define $g_0 : J \longrightarrow \varepsilon$ to be constantly zero, and g_1 to be zero on every component met by u and one on a chosen complementary nonempty union. The equation $g_0 u = g_1 u$ and invertibility of $L(u)$ imply $L(g_0) = L(g_1)$. Evaluation at an object of the complementary part gives equality of the two arrows $e \longrightarrow \varepsilon$ in $\mathcal{H}$. By (T), product with X induces an endofunctor of $\mathcal{H}$, so the two coproduct inclusions $X \longrightarrow X \amalg X$ have equal images. Composing with $(f_0, f_1) : X \amalg X \longrightarrow Y$ again gives $L(f_0) = L(f_1)$. Notice the target Y: the printed X in the final display is a slip.

It remains to justify why equality of ordinary functors determines all arrows of the localization. Let $\mathcal{Q}$ have the objects of Cat, one arrow between any two nonempty objects, one arrow from the empty object to every object, and no arrow from a nonempty object to the empty one. Equality of all parallel ordinary functors gives a factorization of L through $\mathcal{Q}$. If W contains no arrow with empty source and nonempty target, the functor $\mathrm{Cat} \longrightarrow \mathcal{Q}$ inverts W and has the same universal property as L. Thus $\mathcal{H} \simeq [1]$. Otherwise invert the unique missing type of arrow in $\mathcal{Q}$; the resulting category is equivalent to e and has the universal property. In either case the full subcategory on

the nonempty categories is equivalent to e. This argument includes arbitrary zigzags and does not assume a calculus of fractions.

If, on the other hand, every element of W preserves components, π_0 factors through $\mathcal{H}$. The discrete finite categories then remain pairwise nonisomorphic, and the two points of ε remain distinct. These observations prove all six equivalences of Proposition 9.12 under (T). In particular, (c)⇒(c′) is immediate: a nonconstant map on a hom-set exhibits two distinct arrows between nonempty objects. The argument through $\mathcal{Q}$ proves the contrapositive of (c′)⇒(c). The editorial bracket on ms. 133 bis interchanges these two roles; the original direction called trivial there was correct.

The additional hypothesis is available for relative fundamental localizers; see Maltsiniotis (2020, Propositions 2.1.3 and 2.1.9). The modern classification of the two exceptional localizers is Cisinski (2006a, Proposition 9.3.2). Under the manuscript's literal weak hypotheses, the displayed proof has a gap, already flagged by the edition on ms. 133 quinquies. We found no proof in the checked sources that removes it. Finally, the conclusion $\mathcal{H} \simeq e$ should not be substituted for $W = \mathrm{Fl}(\mathrm{Cat})$ without a saturation argument. The ambiguous parenthesis on ms. 133 ter is read with this distinction maintained.

Soit $f : I \longrightarrow J$ dans Cat telle que $\pi_0(f)$ ne soit pas injective. On a vu dans la démonstration de 9.7 comment on en déduit (en choisissant deux éléments i_0, i_1 dans deux composantes connexes distinctes de I, telles que $j_0 = f(i_0)$ et $j_1 = f(i_1)$ soient dans une même composante connexe de J) que pour X, Y dans Cat, deux flèches $f_0, f_1 : X \rightrightarrows Y$ sont toujours égales dans Hot_W (*cf.* lemme 9.10).

Soit maintenant

$$f \ : \ I \longrightarrow J$$

dans Cat telle que $\pi_0(f)$ ne soit pas surjective, donc $J = J_0 \amalg J_1$, avec f se factorisant par J_0, et $J_1 \neq \emptyset$. Considérons alors la catégorie discrète à deux objets,

$$\varepsilon \ = \ \{0, 1\} \ ,$$

et deux flèches

$$g_0, \ g_1 \ : \ J \ \rightrightarrows \ \varepsilon \ ,$$

g_0 étant constante de valeur 0, et g_1 étant égale à 0 sur J_0, à 1 sur J_1. On a

$$g_0 f \ = \ g_1 f \ ,$$

donc, comme $f \in W$,

$$g_0 \ = \ g_1 \qquad \text{dans} \quad \mathrm{Hot}_W \ .$$

On en conclut, en prenant un point j_1 dans J_1, et l'inclusion $\alpha_{j_1} : e \longrightarrow J$ correspondante, que

$$g_0 \alpha_{j_1} \ = \ g_1 \alpha_{j_1} \ ,$$

i.e. les deux sections de ε sur e sont égales dans Hot_W. Comme le foncteur Cat $\longrightarrow \mathrm{Hot}_W$ commute aux produits finis [cette propriété n'est pas établie sous les seules conditions Loc(1,2,3) ; elle est démontrée dans Pursuing Stacks, section 64, en remplaçant Loc(3) par l'axiome plus fort Loc(3bis)], il s'ensuit que pour X dans Cat, les deux flèches [page 133 quinquies]

$$X \ \overset{\partial_0^X}{\underset{\partial_1^X}{\rightrightarrows}} \ X \times \varepsilon \ = \ X \amalg X$$

sont égales dans Hot_W. Si alors

$$f_0, \ f_1 \ : \ X \ \rightrightarrows \ Y$$

sont deux flèches dans Cat, elles sont de la forme

$$f_0 \ = \ h\partial_0^X \ , \qquad f_1 \ = \ h\partial_1^X \ ,$$

avec

$$h \ = \ (f_0, f_1) \ : \ X \amalg X \ \longrightarrow \ X \ ,$$

et comme $\partial_0^X = \partial_1^X$ dans Hot_W, on en conclut $f_0 = f_1$ dans Hot_W, *q.e.d.* p. 74 [page 134]

10 Application au dérivateur HOT_W

Théorème 10.1. *Soit W un localiseur dans* Cat *satisfaisant les conditions* Loc (1, 2, 3, 4, 7 bis) (*axiomes de saturation forme forte* (*facteurs directs*), *de l'objet final, de localisation forme faible, du carré cocartésien, de la L-accessibilité* [57]). *Soit* HOT_W *le prédérivateur sur* (Cat) *associé à* (Cat, W) *de la façon habituelle,*

$$I \longmapsto \mathrm{Cat}(I)\, W(I)^{-1} \;=\; \underline{\mathrm{Hom}}(I^\circ, \mathrm{Cat})\, W(I)^{-1}\,. \tag{10.1.1}$$

On a alors ce qui suit.

1 °) *Pour toute flèche $u : I \longrightarrow J$ dans* Cat, *le foncteur $u_!$* (*adjoint à gauche de u^**) *existe* (**N.B.** On n'a pas besoin pour ceci de Loc (4) ni de Loc (7 bis), il suffit que W soit un localiseur fondamental `[on a néanmoins besoin de l'axiome de localisation forme forte Loc (3 bis)]`), *et il se calcule de la façon standard* (*axiomes* Der 4 bis *et* Der 5 bis). *Le foncteur u_*, adjoint à droite de u^*, existe* [58] [*si u est de la forme $v \times \mathrm{id}_S$, où $v : I_0 \longrightarrow J_0$ est une flèche entre catégories ordonnées finies, et S un objet quelconque de* Cat], *et ce foncteur u_* se calcule de façon standard* (*axiomes* Der 4 *et* Der 5). *Donc on a une théorie des carrés W-cartésiens et des carrés W-cocartésiens dans les catégories* $\mathrm{HOT}_W(S)$, *et des suites longues* [page 135] *de suspension et de cosuspension dans les catégories* $\mathrm{HOT}^\bullet_W(S)$. (*Plus exactement, ce sont des théories relatives aux dérivateurs induits* $\mathrm{HOT}_{W,S}$, $\mathrm{HOT}^\bullet_{W,S}$ *sur S, $I \longmapsto \mathrm{HOT}_W(S \times I)$ resp. $I \longmapsto \mathrm{HOT}^\bullet_W(S \times I)$ …*). *De plus, les carrés cocartésiens et les carrés cartésiens relatifs aux dérivateurs induits* $\mathrm{HOT}_{W,S}$ *sont* exacts, i.e. *les suites longues de suspension et de cosuspension sont exactes* (*axiomes d'exactitude à droite et à gauche* Der 6 bis *et* Der 6).

Commentary · Exposition [ms. 134]

What the model structure supplies to the derivator

Grothendieck is returning to the purpose of the factorization theorems: they should make homotopy Kan extensions available for diagrams of every shape. The final paragraph says that the proof will be given next, but the announced paragraph is absent. The theorem must therefore be read as a collected statement with several recalled ingredients, not as a theorem proved in these pages.

There is now a convenient independent route to its main structural conclusions. For an accessible fundamental localizer in the relative modern sense, Cisinski (2006a, Theorem 5.2.15) constructs a combinatorial model structure on Cat with weak equivalences W. The associated diagram theory is a strong derivator by Groth (2013, Proposition 1.30). These are sufficient modern hypotheses, not a verbatim replacement of the manuscript's L-accessibility condition or a proof that weak Loc (3) implies relative Loc (3 bis).

Here is how this route accounts for the assertions. The projective and injective model structures on diagram categories have the same objectwise weak equivalences. Projective cofibrant replacements compute left derived Kan extensions; injective fibrant replacements compute right derived Kan extensions. Derived adjunctions give both adjoints to restriction. The pointwise formulas retain the variance in (10.1.1). If $X : I^{\mathrm{op}} \longrightarrow \mathrm{Cat}$ and $u : I \longrightarrow J$, they have the form

$$(u_! X)_j \simeq \mathrm{hocolim}_{(j/u)^{\mathrm{op}}} X_i, \qquad (u_* X)_j \simeq \mathrm{holim}_{(u/j)^{\mathrm{op}}} X_i.$$

In the first comma category an object is $j \longrightarrow u(i)$, and in the second it is $u(i) \longrightarrow j$. The opposite categories occur because the coefficient diagrams are contravariant. They should not be omitted by importing a formula written for covariant diagrams.

The model of a diagram also explains evaluation conservativity. Represent a morphism of the diagram homotopy category by a map from a cofibrant replacement to a fibrant replacement. If every evaluated morphism is invertible in the underlying homotopy category, every component of this representing map is a weak equivalence. Hence the map is an objectwise weak equivalence and is invertible in the diagram homotopy category. This is the content needed for Der 2; it is not a formal

57. **N.B.** Loc (1 bis), *i.e.* $W = W^\natural$, est déjà impliqué par la condition que W soit stable par $\varinjlim$ filtrantes.
58. *Toujours ! Cf.* XIX, p. 29.

consequence of strong saturation of W alone. Compare the commentary on Chapter VI, manuscript p. 130.

Finite decomposition requires much less. A diagram on $I \amalg J$ is a pair of diagrams, and localization at the corresponding product class gives the product of the two localizations when identities belong to that class. This is the finite-product localization argument already discussed in the commentary on Chapter X, manuscript p. 5 ("Finite products and sums already descend"). It does not require the coproduct axiom Loc (5) for functors in Cat: a coproduct of *index categories* and a coproduct of *coefficient objects* are different operations.

Finally, shifts $I \mapsto \mathbf{D}(S \times I)$ are derivators, by Groth (2013, Theorem 1.25). Pointed versions therefore have the coherent fibre, cofibre, loop and suspension constructions needed for the sequences mentioned here. No assertion of stability is involved: for spaces, suspension and looping are not inverse equivalences, and cocartesian squares need not be cartesian. The older exactness axioms should be read in their pointed, unstable sense. The limitation to certain finite shapes for u_* is superseded under the modern model hypotheses; footnote 58 itself records that Grothendieck later removed it.

2°) *Le prédérivateur* HOT_W *satisfait aussi l'axiome* Der 1 *de décomposition finie* [59] (*pour* $I = \coprod_\alpha I_\alpha$), *et l'axiome* Der 2 *de localisation. De plus,* W *est fortement saturé,* i.e. $f \in W \iff \mathrm{hot}_W(f)$ *isomorphisme.*

3°) *Concernant l'axiome* Der 3 *et ses variantes* (*cf.* I, pages 111-112, 125), *on a ceci :*

a) [60]. *Soit* $f : S' \longrightarrow S$ *une flèche dans* Cat. *Pour que ce soit une* HOT_W*-équivalence,* i.e. *pour que le foncteur* $f^* : \mathrm{HOT}_W(S) \longrightarrow \mathrm{HOT}_W(S')$ *induise une* équivalence *entre les sous-catégories pleines formées des objets* constants *sur* S, S', *il faut et il suffit que* f *soit dans* W. *L'énoncé correspondant dans les dérivateurs induits* $\mathrm{HOT}_{W,I}$ *est également valable, pour* $I \neq \emptyset$. *Donc a fortiori,* HOT_W *et* $\mathrm{HOT}_{W,I}$ *satisfont l'axiome de l'objet final,* i.e. *si* S *dans* Cat *a un objet final, alors* $S \longrightarrow e$ *est une* HOT_W*-équivalence, et même une* $\mathrm{HOT}_{W,I}$*-équivalence pour toute* I *dans* Cat. [page 136] p. 75

Commentary · Examples [ms. 135]

Constant objects retain coherent maps

The detection statement in 10.1(3a) becomes natural when we ask what the morphisms between constant diagrams actually are. The full subcategory of constant objects has all the morphisms inherited from the homotopy category of diagrams. These morphisms remember the indexing homotopy type, even though the displayed objects are constant. This is the same issue introduced in the commentary on Chapter I, manuscript pp. 61–62 ("Equivalence measured by constant coefficients" and "How much constant coefficients can detect"); the following computation makes its role in the present theorem explicit.

For the ordinary homotopy theory of spaces, write $BS = |N(S)|$ and let K, L be spaces. The derived space of maps between their constant diagrams on S is the homotopy limit of the constant mapping-space diagram, hence

$$\mathrm{Map}_{\mathrm{Fun}(S^{\mathrm{op}},\mathcal{S})}(\mathrm{const}\, K, \mathrm{const}\, L) \simeq \mathrm{Map}(BS, \mathrm{Map}(K, L)).$$

On connected components this gives

$$(*) \qquad \mathrm{Hom}_{\mathbf{D}(S)}(K_S, L_S) \cong [BS \times K, L].$$

One can obtain the same formula by the adjunction between homotopy colimit and the constant diagram functor: the homotopy colimit of the constant diagram K is $BS \times K$. This derivation also shows that restriction along $f : S' \longrightarrow S$ corresponds to precomposition with $Bf \times 1_K$.

If Bf is a weak equivalence, every such map of hom-sets is a bijection. Conversely, take $K = *$ and let L vary. Full faithfulness on constant objects says that

$$[BS, L] \longrightarrow [BS', L]$$

59. Sans aucun axiome sur W ! `[On a besoin simplement que les identités soient dans` W`.]`

60. On n'a besoin ici que de Loc (1 ter, 2, 3) (Loc (1 ter) est la saturation forte, *i.e.* $f \in W \iff \mathrm{hot}_W(f)$ isomorphisme).

is a bijection for every space L. The contravariant Yoneda lemma in the homotopy category then says that Bf is invertible there, hence is a weak equivalence. Essential surjectivity on the full constant subcategories is automatic: the constant object with value L on S' is the restriction of the constant object with value L on S. Thus full faithfulness and equivalence give the same test here.

For a concrete calculation let $S = B\mathbb{Z}$, the one-object category of the group $\mathbb{Z}$, and let $K = *$, $L = S^1$. Its classifying space is S^1. Formula $(*)$ gives

$$\mathrm{Hom}_{\mathbf{D}(B\mathbb{Z})}(*_{B\mathbb{Z}}, (S^1)_{B\mathbb{Z}}) \cong [S^1, S^1] \cong \mathbb{Z}.$$

The integer is the degree of the corresponding map. On restriction along the object $e \longrightarrow B\mathbb{Z}$, all these morphisms map to the unique element of $[*, S^1]$. The distinction between degrees has been forgotten. Consequently this restriction is not fully faithful on constant objects, exactly as the theorem predicts. Looking only at strict natural transformations between the original constant diagrams would miss these derived morphisms.

For a general W, the analogous proof uses the homotopy-colimit formula for constant families, product stability, invariance under passage to opposite categories, and strong saturation. These ingredients are available in the relative accessible setting of the preceding block. The smaller hypothesis list in footnote 60 should be read with the earlier construction it invokes, not as a new proof of those ingredients under weak Loc (3). In particular, strong saturation is what changes "the comparison is invertible in Hot_W" into "$f \in W$".

b) *Supposons que W satisfasse* Loc (1 ter, 2, 3, 5 b) (*donc l'axiome* Loc (5 b) *des W-fibrations, variante* b). *Alors pour toute $f : S' \longrightarrow S$ dans W,* $\mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S')$ *est une* équivalence de catégories (*la réciproque étant clair par* (a)). *Itou pour*

$$\mathrm{HOT}^{\mathrm{lc}}_{W,I}(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_{W,I}(S') \qquad (\text{i.e. } \textit{pour } \mathrm{HOT}^{\mathrm{lc}/I}_W(S \times I) \longrightarrow \mathrm{HOT}^{\mathrm{lc}/I}_W(S' \times I)\ ).$$

En d'autres termes, pour le dérivateur HOT_W *et pour tous les* $\mathrm{HOT}_{W,I}$ (**D** *étant un des ces dérivateurs*) [61], *toute* **D***-équivalence est une* **D***-équivalence* 2*-complète,* i.e. *pour $f : S' \longrightarrow S$, pour que $f^* : \mathbf{D}(S) \longrightarrow \mathbf{D}(S')$ induise une équivalence entre les sous-catégories $\mathbf{D}^{\mathrm{lc}}(S) \longrightarrow \mathbf{D}^{\mathrm{lc}}(S')$, il suffit déjà qu'elle induise un foncteur pleinement fidèle sur les sous-catégories pleines formées des objets constants, et ceci équivaut (si $I \neq \emptyset$) à la condition que $f \in W$.*

Commentary · Development (commentators') [ms. 136] [page 137]

A finite monodromy test for the displayed hypotheses

The extra hypothesis in 10.1(3b) is meant to control locally constant families, whose fibres may be transported nontrivially around loops. There is a consequential numbering slip: the printed Loc (5 b) is described as the fibration axiom, but in the new list Loc (5 b) means reflection of coproducts. The intended reference is Loc (6 b), corresponding to W (4 b) in chapter XII. The distinction can be tested by a finite group.

Failure under the literal reduced list. [ours] Let $W_{\mathbb{Q}}$ be the functors inducing isomorphisms on rational homology of nerves. It satisfies strong saturation, the terminal-object axiom, relative Theorem A and coproduct reflection. Nevertheless there is $f \in W_{\mathbb{Q}}$ for which restriction on locally constant objects is not essentially surjective.

Proof. Strong saturation follows directly from the definition: rational homology factors through the localization, so an arrow made invertible by localization must induce an isomorphism on rational homology. The terminal-object assertion follows from contractibility of the nerve. Rational homology of a coproduct is the direct sum of the homologies of its summands. A block-diagonal direct sum map is invertible precisely when each component map is invertible, which proves coproduct preservation and reflection. That $W_{\mathbb{Q}}$ is a relative fundamental localizer follows from Cisinski (2006a, Corollary 9.4.3 and §9.4.4); equivalently, it is the nerve pullback of the homological localization of simplicial sets.

Choose a nontrivial finite group G, for example C_2. The map $p : BG \longrightarrow e$ belongs to $W_{\mathbb{Q}}$. Indeed, the augmentation $\mathbb{Q}[G] \longrightarrow \mathbb{Q}$ splits as a map of $\mathbb{Q}[G]$-modules by

$$1 \longmapsto \frac{1}{|G|} \sum_{g \in G} g.$$

61. À vérifier avec soin dans le cas $\mathbf{D} = \mathrm{HOT}_{W,I}$.

Thus the trivial module $\mathbb{Q}$ is projective, and

$$H_n(BG;\mathbb{Q}) = \operatorname{Tor}_n^{\mathbb{Q}[G]}(\mathbb{Q},\mathbb{Q}) = 0 \quad (n>0), \qquad H_0(BG;\mathbb{Q}) = \mathbb{Q}.$$

Let $A : (BG)^{\mathrm{op}} \longrightarrow \mathrm{Cat}$ be the discrete category on the regular right G-set, with its action. Every structure map is an isomorphism, so A is locally constant.

The component functor sends every $W_{\mathbb{Q}}$-map to a bijection. To verify this last point without a finiteness assumption, $H_0(NC;\mathbb{Q})$ is the vector space with basis $\pi_0(C)$, and a map induced by a function on those bases is invertible only when that function is bijective: two identified basis vectors give a nonzero kernel element, while an omitted basis vector cannot lie in the image. Applied objectwise, π_0 therefore descends to a functor

$$\mathrm{Cat}^{(BG)^{\mathrm{op}}}[W_{\mathbb{Q}}(BG)^{-1}] \longrightarrow \mathrm{Set}^{(BG)^{\mathrm{op}}}.$$

An object in the image of p^* has trivial G-action after this functor is applied. The regular G-set is nonempty and has no point fixed by all of G, so it is not equivariantly isomorphic to any nonempty trivial G-set. Hence A cannot be isomorphic to a constant object in the localized diagram category. This proves the failure of essential surjectivity. □

The example also satisfies the opening excision and size hypotheses of 10.1. For Loc (4), a simplex in the nerve of a category covered by two open subcategories lies in one of the two subcategories: for sieves, its last vertex determines this, and for the opposite convention its first vertex does. Consequently

$$C_*(NX, NX_2;\mathbb{Q}) \cong C_*(NX_1, NX_0;\mathbb{Q}).$$

The two inclusions $X_0 \longrightarrow X_1$ and $X_2 \longrightarrow X$ are rational homology equivalences precisely when these isomorphic relative chain complexes are acyclic. This proves both directions of Loc (4). Unlike the corresponding assertion for ordinary weak homotopy equivalences, this reverse implication is valid for the homological invariant used here.

For completeness, L-accessibility can be checked in the manuscript's own sense. Fix an infinite cardinal π and a rational homology equivalence $f : X \longrightarrow Y$. Start with a subarrow $f_0 : X_0 \longrightarrow Y_0$ having at most π arrows in total. The rational mapping cone $C_*(Nf_0;\mathbb{Q})$ has at most π elements. For every cycle in it choose a bounding chain in the acyclic cone $C_*(Nf;\mathbb{Q})$. Each bounding chain uses finitely many simplices. Adjoin all their objects and arrows, their images under f, and all necessary composites. This gives a subarrow f_1 still of size at most π. Repeat countably often. In the union $\bar{f}$, every cycle has finite support and therefore occurs at one stage; its chosen bounding chain occurs at the next. Thus $\bar{f} \in W_{\mathbb{Q}}$ and has size at most π.

It follows that the $W_{\mathbb{Q}}$-subarrows of size at most π are cofinal among all subarrows of that size and form a filtered family whose union is f. Rational chain complexes of nerves commute with filtered colimits, and filtered colimits of vector spaces preserve exactness, so $W_{\mathbb{Q}}$ is closed under all filtered colimits. There is a set of representatives of the arrows of size at most π. They generate $W_{\mathbb{Q}}$ under this closure. This is precisely the condition displayed in Loc (7 bis). It also gives the stronger bounded-subobject criterion used in the second Proposition 5.7: categories and functors of size at most π have presentations with at most π generators and relations, hence maps out of them factor through a stage of a π^+-filtered diagram, with equalities detected at a later stage. Thus the comparison with manuscript L-accessibility is explicit here; it is not inferred from the modern terminology.

The example therefore refutes the assertion with current Loc (5 b) read literally, even together with the opening assumptions. It does not refute the corrected assertion with the intended fibration axiom. The exact issue is that a test which preserves components of disjoint unions need not eliminate monodromy.

The regular action makes the missing fibration hypothesis visible as well. Its translation groupoid EG has one arrow between any two objects and is contractible. The projection $EG \longrightarrow BG$ has discrete fibre G. It is an old-style W-fibration: on a contractible base the discrete covering is trivializable, and a map between two such bases induces weak equivalences on the corresponding sheets. Compare it over BG with the identity fibration $BG \longrightarrow BG$. The map of total categories $EG \longrightarrow BG$ is a rational homology equivalence, while its fibre map $G \longrightarrow e$ is not. This directly violates Loc (6 b). Pullback along $e \longrightarrow BG$ also gives $G \longrightarrow EG$, which is not a rational homology equivalence because $|G| > 1$, illustrating failure of the base-change condition Loc (6 a). We checked the manuscript's earlier fibration variants at Grothendieck (1990–1991f, chapter XII, ms. 53–56) and the rational-localizer example in Cisinski's §9.4; we did not find this precise diagnostic of the

renumbered assertion stated there. The proof is offered as a check of the manuscript, not as a new claim about homological localization.

Commentary · Development (commentators') [ms. 136]

Why a nonempty shift still detects the base test

Footnote 61 asks that the shifted assertion be checked with care. One half is formal and explains exactly why the exclusion of the empty category is necessary. The other half needs a theorem about coherent diagrams with the shifted coefficient category; checking each value separately does not by itself prove full faithfulness of the whole diagram functor.

Reflection by a nonempty shift. [ours] Let $\mathbf{D}$ be a prederivator, let I have an object i, and put $\mathbf{D}^I(S) = \mathbf{D}(I \times S)$. If restriction along $f : S' \longrightarrow S$ is fully faithful on the full subcategories of constant objects for $\mathbf{D}^I$, then it is fully faithful on the corresponding full subcategories for $\mathbf{D}$.

Proof. The projection $I \times S \longrightarrow S$ and the inclusion $s \mapsto (i, s)$ induce functors

$$c_S : \mathbf{D}(S) \longrightarrow \mathbf{D}(I \times S), \qquad e_S : \mathbf{D}(I \times S) \longrightarrow \mathbf{D}(S)$$

with $e_S c_S = 1$, up to the canonical coherence isomorphism if pseudofunctors are used. These functors commute with restriction along f. If $A, B \in \mathbf{D}(e)$, the objects $c_S(A_S), c_S(B_S)$ are constant in the S-direction, with values $c_e(A), c_e(B) \in \mathbf{D}(I)$. Therefore the map

$$\mathrm{Hom}_{\mathbf{D}(S)}(A_S, B_S) \longrightarrow \mathrm{Hom}_{\mathbf{D}(S')}(A_{S'}, B_{S'})$$

is a retract, in the arrow category of sets, of the corresponding map between the hom-sets of the shifted constant objects. The latter is bijective by hypothesis. A retract of a bijection is bijective: if q is the large bijection, its inverse on the retract is obtained by inclusion, then q^{-1}, then retraction. The two commutative squares and the retraction identities verify both inverse equations. This proves full faithfulness. □

Every constant object on S' is the restriction of the constant object with the same coefficient on S, so the test is also reflection of equivalence between the full constant subcategories. The argument uses an object of I, not an initial or terminal object, and it does not require I to be connected. If $I = \emptyset$ and $\mathbf{D}$ satisfies the usual empty-index axiom, every category $\mathbf{D}^I(S)$ is terminal; every f then passes the test, so detection of W is impossible.

For the ordinary localizer W_∞, the forward statement and its locally constant refinement hold for every shift: apply Cisinski (2009, Theorem 1.3) to a model category of I-diagrams. This coefficient category retains coherent diagrams; it is not merely the ordinary functor category with values in Hot. The theorem just cited concerns nerve weak equivalences. It is not a theorem that every element of an arbitrary larger localizer preserves all locally constant coefficients; the preceding finite-group example explains why such a reformulation would be false. We checked this formal shifted argument against Groth (2013, Theorem 1.25) and the derivator discussion in Cisinski's §3, and supply the retract proof to settle the specific reflection point left implicit here.

Commentaire. Dans cet énoncé récapitulatif de ce que je sais prouver à présent pour HOT_W, j'ai un peu mélangé les provenances diverses – ce qui est mis "pour mémoire" de VI (et qui ne dépend pas de Loc (4, 7 ou 7 bis)), *i.e.* ce qui dépend pour l'essentiel de la construction de $u_!$ donnée dans VI (donc ce qui concerne $u_!$ dans 1°, *moins* l'exactitude à droite, et tout ce qui concerne 3°, (a), (b)).

La démonstration de 10.1 résultera des développements du paragraphe suivant `[qui n'existe pas]`, compte tenu du résultat principal du paragraphe 8 (th. 8.4), et du paragraphe 7 (cor. 7.4), qui sont les ingrédients "durs" du théorème 10.1.

Commentary · Later developments [ms. 137]

The later theorem and the scope of the unfinished proof

The principal construction announced here has a full modern realization through model categories and their diagram homotopy theories. Locally constant coefficients likewise have a precise invariant formulation: Cisinski (2009, Theorem 1.3 and Corollaries 1.28 and 3.8) proves invariance under nerve

weak equivalences, and conversely characterizes those equivalences by testing all model-category or derivator coefficients. In the published paper these are the numbers; the shorter preprint has different numbering. In higher-category language the underlying object is the functor category out of the groupoid completion of the indexing category, which is the perspective described in that paper's introduction.

The unresolved matters in these pages are therefore specific. The missing next paragraph does not provide a proof of 10.1 under each of its reduced hypothesis lists. The weak-versus-relative localization issue remains visible in 9.12 and in the claimed construction of $u_!$. The fibration reference in 10.1(3b) must be read after correcting its old numbering. None of these points is a reason to withhold the established constructions under their verified hypotheses, and none is repaired simply by citing a theorem about the ordinary localizer while leaving an arbitrary W in the statement.

Overview · XV-b

The end of Chapter XV asks how a homotopy theory can be constructed from a class of weak equivalences and a supply of small pieces. Grothendieck approaches that question in three stages. He first develops a calculus for classes recognized on bounded approximations. He then tries to turn that calculus into a general factorization theorem, discovers that the proof has lost its cardinal bound, and recovers a more structured theorem for monomorphisms. Finally, he applies the result to presheaves and returns to the axioms for a fundamental localizer and its associated derivator. The progression is mathematically instructive because the abandoned argument and the successful one use nearly the same constructions. What distinguishes them is control of the morphism left over after an attachment (Grothendieck, 1990–1991k).

The accessibility language must be read with its original conventions. A cardinal π is an infinite bound on sizes, and a filtered indexing category "large with respect to π" has the directedness now associated with the regular cardinal π^+. At a good cardinal, an object called π-accessible is therefore π^+-presentable in current terminology. The accessibility of a *class*, however, is not merely accessibility of its full subcategory. It is the embedded recognition condition b_π: in every continuous, suitably directed presentation by objects of the chosen bounded filtration, the colimit belongs to the class if and only if cofinally many stages do. The letter L adds closure under all small filtered colimits. These distinctions are essential both to the positive results of §5 and to the defect found in §6.

Under the standing subobject, image, and exactness assumptions, an object has a canonical presentation by its small subobjects. Proposition 5.7, its corollary, and the ensuing lemmas explain when a class can be recovered from its bounded part. The basic index is a directed poset closed under bounded directed suprema. A subset that is both cofinal and closed under those suprema can be intersected with another such subset without losing cofinality: alternately enlarge an approximation into each subset and take a countable supremum. A transfinite version gives intersections of at most π such subsets. The same idea proves closure of L-π-accessible classes under suitably bounded intersections. Finite unions use a different fact: among finitely many subsets covering a directed set, one is cofinal. Thus finite unions and arbitrary small intersections are available for the absolute L-accessibility notion, after increasing the witness cardinal when necessary.

The flattening argument in Proposition 5.9 deserves to be read alongside these cofinality calculations. A filtered family of objects, each described by bounded pieces, is replaced by the category of all those pieces mapping into the various stages. This category is an opfibration over the original index, with filtered fibres. Its total category is filtered, and the tautological source functor has the desired colimit. Regrouping the resulting diagram by its directed subdiagrams of cardinal at most π produces an adapted presentation. The commentary supplies the intervening finality and continuity arguments. It also records the precise limit of the written proof: the proposition advertises the weakened assumption (d_0), but its subobject proof uses stronger hypotheses. A complete proof is given under the full standing assumptions, and another under (d_0) at a good cardinal. We do not identify the remaining weakening with a disproved assertion.

The passage from objects to arrows comes from the theory of categories of sections in SGA 4, Exposé I, 9.22–9.24 (Artin et al., 1972). It requires a common cardinal bound large enough to accommodate the diagram's objects, structure maps, and equations. In the arrow category a bounded object has *both* bounded source and bounded target. Theorem 6.1 attempts to show that any L-accessible class of monomorphisms containing the isomorphisms is generated, in the lifting sense, by a small subcollection. Grothendieck builds a transfinite succession of attachments inside a fixed target. At the crucial step he invokes accessibility on a continuous family whose targets remain that fixed, possibly large object. His footnotes correctly withdraw the inference: this is no longer a presentation by bounded arrows.

The distinction can be seen without homotopy theory. For a set B larger than π, the arrows $S \hookrightarrow B$, indexed by the subsets S of size at most π, form a continuous, highly directed diagram with colimit id_B. None of its stages is an isomorphism, although the isomorphisms form an L-π-accessible class. This diagnoses the failed inference. To decide the proposed theorem itself requires more. A recent classification provides that decision: pure monomorphisms in the category of spans of sets are not cofibrantly generated (Cox et al., 2026b, Theorem 5.1). The development at manuscript page 84 proves that this class satisfies the manuscript's stronger recognition condition, as well as closure under pushouts, transfinite composition, and retracts. It is therefore an actual presheaf counterexample to Theorem 6.1. The cited authors prove the non-generation theorem; the commentary supplies its comparison with Grothendieck's cardinal filtration.

Section 7 succeeds by imposing structure which an arbitrary class does not have. In a presheaf category, two subobjects glued over their intersection form their union, still a subobject of the ambient object. Monomorphisms are stable under pushout and filtered colimits. Small quotient images of representables supply enough bounded pieces. If the class being constructed is $\mathrm{Mon} \cap W$, pushing out a bounded trivial monomorphism gives another weak equivalence, and cancellation in W proves that the residual inclusion is again a weak equivalence. The induction can therefore continue. Well-poweredness makes it terminate, by preventing a strictly increasing chain of subobjects from having arbitrarily large length.

Once a set of generators has been obtained, the small object argument gives two weak factorization systems. The remaining recognition condition is that every map right orthogonal to all monomorphisms belongs to W. A retract argument then identifies these maps with the trivial fibrations, and the two factorizations fit together into a model structure. The general recognition theorem later associated with Smith, in the form proved by Beke (2000, Theorem 1.7), places this reasoning in locally presentable categories. Its closure and solution-set hypotheses explain what survives of the question on manuscript page 85. They do not validate the unrestricted theorem of §6.

Section 8 turns the remaining recognition condition into geometry. For a small category A, the category-of-elements functor $i_A : \widehat{A} \to \mathrm{Cat}$ transports a fundamental localizer to a class W_A of presheaf maps. The subobject classifier, called the Lawvere object here, is injective with respect to monomorphisms. Its two disjoint sections provide the universal candidate for an interval. Theorem 8.4 relates universal asphericity of such an interval, triviality of maps right orthogonal to all monomorphisms, and the test-category property of every slice A/a. An additional asphericity condition on A itself gives the global test-category conclusion. These are statements about modelling homotopy types, not automatically about preserving their cartesian products.

The mapping-cylinder proof makes the interval criterion concrete. In its presheaf application, the endpoint inclusion is a trivial monomorphism and remains so after pushout. The resulting projection is a weak equivalence by two-out-of-three. The abstract Lemma 8.7 omits that pushout condition. The commentary gives an explicit counterexample in groups, then restores the needed hypothesis. It also presents the direct interval-lifting proof, available when the morphism being tested has the right lifting property against every monomorphism; this proof validates the reduced hypothesis formulation of the interval criterion without invoking the faulty abstract lemma. The corresponding later argument is Cisinski's Lemma 1.4.13 (Cisinski, 2006a).

Two other distinctions govern the application. First, the right properness criterion in (7.4.3) puts the fibration condition on the wrong vertical side of the pullback square; a small simplicial example detects the error. Second, the two converses in (8.2.4) belong to the manuscript's stronger excision axiom. They are not consequences of ordinary left properness and do not hold for arbitrary weak homotopy equivalences. Earlier commentary blocks indexed in DER XIIb discuss that obstruction; they are referred to here without reproducing them.

Cisinski's later model theorem for accessible fundamental localizers and local test categories supplies the intended presheaf construction using the appropriate gluing results (Cisinski, 2006a, Corollary 4.2.18). His term "accessible localizer" means generation by a set *as a localizer*; its relationship with accessibility of an arrow class is a theorem, rather than a change of spelling for L-accessibility.

The final two sections return from models to the intrinsic theory. Section 9 reorganizes the localizer axioms and studies how coproducts and connected components behave after localization. The distinction between weak Theorem A, Loc(3), and its relative version over an arbitrary base, Loc(3 bis), remains significant. The argument that a failure to preserve connected components collapses the localization to a point or to the walking arrow uses compatibility with products. The editors explicitly observe that this compatibility has not been obtained from the weak axiom alone. The commentary retains that qualification and proves the degeneration mechanism under the needed product hypothesis. Later treatments make the relative hypothesis and its consequences systematic; they should not be silently substituted for the weaker definition printed on these pages.

The decomposition problem also admits a useful direct argument. If a retract-closed class preserves connected components and a coproduct of maps belongs to it, each nonempty summand is an arrow-category retract of the whole map, by collapsing the other summands to a chosen point. An empty source summand has empty target because the total map preserves components. Thus reflection of weak equivalences by summands follows under these hypotheses. This specifies one setting in which the proposed independence of decomposition and connectivity cannot hold, while keeping the manuscript's weaker axiom system separate.

Theorem 10.1 gathers the desired conclusions for HOT_W: homotopy Kan extensions, pointwise detection of isomorphisms, strong saturation, and invariance of locally constant diagrams. Its promised next section is absent. Modern model-category and derivator results provide the appropriate existence statements under their stated hypotheses; the commentary identifies those hypotheses and the exact published references. Locally constant diagrams require particular care, because they retain monodromy. The printed reference to Loc(5 b) in part 3(b) is stale: the prose means the fibration axiom, now numbered Loc(6 b). A rational-homology equivalence $BC_2 \to e$, together with the nontrivial action of C_2 on a two-element set, shows why summand reflection alone cannot replace that fibration condition. The shifted statement likewise requires the nonempty indexing category indicated in the manuscript; an empty shift cannot detect a weak equivalence.

Chapter XVI

Les localiseurs fondamentaux

XVI · Chapter XVI · manuscript pages 1–124

p. [non numérotée]

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XVI
Les localiseurs fondamentaux

Ce texte a été déchiffré et transcrit en LaTeX 2ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Seules quelques corrections mineures évidentes ont été effectuées. Pour les rares ajouts ou commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés.

p. [non numérotée]

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Axiomes Loc(1) à Loc(3) (et variantes) : localiseurs fondamentaux

p. 1
[page 1]

J'ai envie de reprendre l'exposé des axiomes utiles pour un localiseur fondamental

$$W \subset \mathrm{Fl}(\mathrm{Cat}),$$

en me guidant cette fois plus sur la signification géométrique des axiomes, que sur les alias ± techniques qu'on rencontre (ou ne rencontre pas) pour les vérifier. Je révise donc dans cet esprit la présentation de XV §9.

Loc (1) (*saturation*).

a) Iso $\subset W$.

b) Si deux parmi f, g, gf sont dans W, aussi le troisième.

c) Si gf et fg sont dans W, f et g aussi.

Les conditions (*a*) et (*b*) toutes seules, je vais les appeler (pour un $W \subset \mathrm{Fl}\,\mathcal{M}$ général) *saturation faible* du localiseur. Les conditions (*a*), (*b*), (*c*), *saturation modérée* ("mild saturation"). Autres variantes :

Loc (1 bis) Les conditions (*a*), (*b*), (*c*) du haut, plus

d) W stable par facteurs directs.

C'est bien sûr impliqué par :

Loc (1 ter) W est *fortement saturé, i.e.* si $f \in \mathrm{Cat}$ est telle que $\mathrm{hot}_W(f)$ est un isomorphisme, alors $f \in W$.

Loc (2) (*Objet final*). Si X dans Cat a un objet final, alors X est W-asphérique, *i.e.* $X \longrightarrow e$ est dans W.

Parfois, cet axiome est conséquence de la variante affaiblie suivante.

Loc (2_0) (Axiome du segment). $\mathbf{\Delta}^1$ est W-asphérique.

Loc (3) (*localisation,* forme faible). Soit $f : X \longrightarrow S$ une flèche dans Cat *W-asphérique, i.e.* telle que $\forall\ s \in \mathrm{Ob}\,S$, la catégorie localisée X/s sur S en s, $X/s = X \times_S S/s$, soit W-asphérique. Alors $f \in W$. [1]

[page 2]

Rappel 1. Loc (1, 2, 3) impliquent que l'on a

$$W = W^\circ \qquad \textit{i.e. } f \in W \iff f^\circ \in W,$$

où on dénote par $X \longmapsto X^\circ$, $f \longmapsto f^\circ$ le foncteur de passage à la catégorie opposée, foncteur

$$X \longmapsto X^\circ : \mathrm{Cat} \xrightarrow{\sim} \mathrm{Cat}$$

p. 2

qui est un *automorphisme* de Cat (donc commute à toutes $\varinjlim$ et $\varprojlim$). Il s'ensuit que l'ensemble d'axiomes Loc (1, 2, 3) équivaut à Loc (1, 2^{opp}, 3^{opp}), où Loc (2^{opp}) est l'axiome "dual" de Loc (2), savoir que tout objet de Cat ayant un objet initial est W-asphérique, et de même Loc (3^{opp}) l'axiome que toute $f : X \longrightarrow S$ *W-coasphérique, i.e.* telle que les catégories colocalisées $s\backslash X$ de X sur S soient W-asphériques, est dans W.

Définition 1.1. On appelle *localiseur fondamental* une partie W de Fl(Cat) satisfaisant les axiomes Loc (1, 2, 3) (*i.e.* saturation modérée, axiome de l'objet final, axiome de localisation).

1. Dans une version finale, il faudrait intervertir les notations pour Loc (3) et pour Loc (3 bis), car ce dernier m'apparaît comme le plus important des deux - il est satisfait pour tous les localiseurs qui ont été utilisées (à ma connaissance), et est valable pour tout localiseur de la forme $W_{\mathbf{D}}$, où $\mathbf{D}$ est un dérivateur (ou seulement un prédérivateur satisfaisant Der (4,5) `[il faut aussi Der (2) ; la numérotation des axiomes est celle du chapitre I, p. 68]`).

[page 3]

Commentary · Context [ms. 2]

Axioms for equivalences before a choice of model

Commentary and mathematical audit AI-produced by GPT-6 Astra (OpenAI, ChatGPT Pro mode), checked and integrated by Claude (Anthropic).

Grothendieck starts with a class of functors, rather than with cofibrations and fibrations. The distinction matters throughout this chapter: the latter classes are to be recovered, when possible, from stability properties of the former. His letter to Thomason of 2 April 1991, §1, describes precisely this separation between the equivalences carrying the homotopy theory and the auxiliary devices used to calculate it (Grothendieck, 1991a). Here the basic datum is a class W in Cat; the localization $L : \mathrm{Cat} \to \mathrm{Hot}_W$ exists in the chosen larger universe and need not come with a calculus of single fractions.

Three meanings of saturation must consequently remain distinct. The two-out-of-three property controls a composable pair. The extra condition (1c) controls a pair $f : A \to B$, $g : B \to A$ when both composites are already equivalences. It is just what is needed to recognize a homotopy inverse before one knows that every arrow inverted by L belongs to W. Closure under retracts is an additional requirement here. Strong saturation means $W = L^{-1}(\mathrm{Iso})$; it implies retract closure because an arrow which is a retract of an isomorphism is an isomorphism. One cannot substitute strong saturation silently in arguments made under (1).

There is also a consequential change of terminology in the later literature. Definition 1.1 calls the class satisfying (1,2,3) a fundamental localizer. Later authors generally call this a *weak* fundamental localizer and reserve the unqualified name for the relative axiom (3 bis). The distinction concerns the hypothesis of Theorem A, not merely a word. In (3), the comma categories X/s are individually aspherical. In (3 bis), the two comma categories may each have substantial homotopy, provided that the comparison between them is an equivalence. The latter implication does not follow by applying the former separately to $X \to S$ and $Y \to S$.

The editor's addition of Der (2) in footnote 1 is therefore mathematically substantive. A pointwise comparison of diagram objects gives an isomorphism only when the evaluations detect isomorphisms. The derivator axioms numbered here are those of Chapter I; importing a modern numbering without translating it would conceal the missing conservativity assumption. This is the same point already isolated in the commentary to XIIa on constructing localizers from derivators.

Rappel 1.2. Si W est un localiseur fondamental, alors tout *homotopisme* $f : X \longrightarrow Y$ de Cat est dans W (c'est pour cela essentiellement que l'on a imposé (c) dans la saturation modérée Loc (1)). Si $f, g : X \rightrightarrows Y$ sont homotopes (*i.e.* appartiennent à une même composante connexe de $\underline{\mathrm{Hom}}(X, Y)$), alors $\mathrm{hot}_W(f) = \mathrm{hot}_W(g)$ et $f \in W \iff g \in W$.

Voici la forme forte de Loc (3).

Loc (3 bis) (*Localisation,* forme forte). Soit S dans Cat et $f : X \longrightarrow Y$ un morphisme dans Cat/S tel que "f soit une W-équivalence localement sur S", *i.e.* pour tout $s \in \mathrm{Ob}\, S$, le morphisme induit

$$f/s : X/s \longrightarrow Y/s$$

pour les catégories localisées de X, Y sur S en s, soit dans W. Alors $f \in W$.

L'axiome Loc (3) n'est autre que le cas particulier de Loc (3 bis) relatif au cas où $Y = S$.

Proposition 1.3. *Soit $W \subset \mathrm{Fl}(\mathcal{M})$. Alors on a les équivalences suivantes*

$$\mathrm{Loc}\ (1,\ 2,\ 3) \iff \mathrm{Loc}\ (1,\ 2_0,\ 3')$$
$$\mathrm{Loc}\ (1,\ 2,\ 3\ \mathrm{bis}) \iff \mathrm{Loc}\ (1,\ 2_0,\ 3'\ \mathrm{bis}),$$

où les axiomes Loc $(3')$ *et* Loc $(3'$ bis$)$ *se formulent ainsi :*

Loc (3′) ⓐ (Axiome des Cat-fibrations, forme directe faible). Soient S dans Cat, X une catégorie fibrée scindée sur S, telle que les fibres X_s $(s \in \mathrm{Ob}\, S)$ soient W-asphériques. Alors $X \longrightarrow S$ est dans W. [page 4]

ⓑ Soit X dans Cat/S, alors $i_X : X \longrightarrow \Phi_S(X)$ est dans W [2].

2. (b) concerne la *comparaison* d'une X quelconque sur S, avec la catégorie fibrée scindée $\Phi_S(X)$ sur S.

Loc (3′ bis) ⓐ (Axiome des Cat-fibrations, forme directe forte). Soient S dans Cat, X, Y deux catégories fibrées scindées sur S, $f : X \longrightarrow Y$ un S-foncteur cartésien compatible aux scindages, tel que $\forall\ s \in \mathrm{Ob}\, S$, le foncteur induit sur les fibres p. 3

$$f_s : X_s \longrightarrow Y_s$$

soit dans W. Alors $f \in W$.

ⓑ Comme Loc (3′) (b).

N.B. En fait, moyennant Loc (1, 2), les énoncés Loc (3′), Loc (3′ bis) impliquent les énoncés similaires sans scindages, et sans même supposer f cartésien dans Loc (3′ bis), *cf.* démonstration ci-dessous.

Démonstration. a) Loc (1, 2, 3) $\Longrightarrow$ Loc (1, 2_0, 3′). Il suffit de prouver Loc (3′). Prouvons Loc (3′ a). Pour ceci, sous les conditions de Loc (3′ a), prouvons que $X \longrightarrow S$ est W-coasphérique, *i.e.* que les catégories

$$s\backslash X = X \times_S s\backslash S$$

sont W-asphériques. Il en résultera bien que $f \in W$ en vertu de Loc (3) appliqué à f° et de $W = W^\circ$. Mais considérons l'inclusion canonique [page 5]

$$i_s : X_s \hookrightarrow s\backslash X,$$

il résulte du fait que $p : X \longrightarrow S$ est Cat-fibrant que ce foncteur est W-asphérique (il a même un adjoint à droite, *i.e.* les catégories X_s/u, où $u \in \mathrm{Ob}\,(s\backslash X)$, ont un objet final). Donc par Loc (3) on a $i_s \in W$, et comme X_s est W-asphérique par hypothèse, de même $s\backslash X$, q.e.d.

Commentary · Exposition [ms. 4]

Comma categories replace a map by a fibration

The construction $\Phi_S(X)$ makes Proposition 1.3 transparent. For $p : X \to S$, its objects can be written

$$(s, x, \alpha : s \longrightarrow p(x)).$$

A morphism to (t, y, β) is a pair $(h : s \to t, k : x \to y)$ satisfying $p(k)\alpha = \beta h$. Projection to s is a split Grothendieck fibration; reindexing along $h : s \to t$ replaces (t, y, β) by $(s, y, \beta h)$. Thus its fibre over s is $s\backslash X$. The inclusion

$$i_X(x) = (p(x), x, \mathrm{id}_{p(x)})$$

has left adjoint $q(s, x, \alpha) = x$: a map from (s, x, α) to $i_X(y)$ is determined uniquely by $k : x \to y$. In particular, $qi_X = \mathrm{id}_X$, and the unit supplies the other homotopy. This explains the comparison axiom (3′b), including the direction of its adjoint.

If $p : X \to S$ is itself a fibration, the inclusion $X_s \to s\backslash X$ has a right adjoint: choose the cartesian pullback $\alpha^* x$ of x along $\alpha : s \to p(x)$. The adjunction follows from the defining universal property of a cartesian arrow. Its comma categories have terminal objects, so (2) and (3) put the inclusion in W. Consequently the fibrewise hypothesis can be transferred to the comma categories. This is the sole place where the fibration condition is doing work; a general functor's strict fibres need not compute the comma categories.

For example, take $S = (0 \to 1)$ and let $X = e$ map to 1. The strict fibre over 0 is empty, although $0\backslash X = e$. Replacing X by $\Phi_S(X)$ produces S itself. A condition on strict fibres of an arbitrary map would incorrectly reject this contractible example; the replacement repairs exactly that defect.

The reverse implication in the proposition first gets $\Delta^1 \times X \to X$ from the fibre axiom and the segment axiom. The two endpoint inclusions become equal inverses after localization. A natural transformation therefore makes its endpoint functors equal in Hot_W, and two-out-of-three gives the corresponding equivalence of membership in W. For a homotopy equivalence, (1c) then recognizes the two inverse functors as members of W. A category with a final or initial object is contractible by these natural transformations, which recovers (2). Finally the displayed square involving $\Phi_S(f)$ transfers the relative fibre condition back to f. This proof also explains why a chosen splitting and

preservation of its cartesian arrows are unnecessary in the final fibrewise criterion: the comparison with the comma categories works for every functor over S between fibrations. For the later systematic separation of fibrations and asphericity structures, see Maltsiniotis (2005, §1).

Two slips in the printed proof should not change the argument. The last conclusion on ms. 5 is $i_X \in W$, rather than its printed repetition $i_X^\circ \in W^\circ$. On ms. 7 the fibres $s\backslash X$ are *aspherical categories*; coasphericity properly describes the original map $X \to S$.

Prouvons Loc (3′ *b*) : On note que $i_X : X \longrightarrow \Phi_S(X)$ est W-*coasphérique* (il a même un adjoint à gauche), donc i_X° est W-asphérique, donc dans W, donc $i_X^\circ \in W^\circ$ car $W = W^\circ$.

a bis) Loc (1, 2, 3 bis) $\Longrightarrow$ Loc (1, 2_0, 3′ bis). Cette fois il faut prouver seulement Loc (3′ bis). Pour (*b*), c'est déjà fait. Reste à prouver (*a*). Si f est comme dans l'énoncé Loc (3′ bis *a*), prouvons que pour tout s dans S, le morphisme induit

$$s\backslash f : s\backslash X \longrightarrow s\backslash Y$$

est dans W. Comme $W = W^\circ$, on aura donc $(s\backslash f)^\circ \in W$, et appliquant Loc (3 bis) à f°, on trouve $f^\circ \in W$, donc $f \in W$, et on aura fini. Mais $s\backslash f$ s'insère dans le carré commutatif [page 6]

$$\begin{array}{ccc} X_s & \xrightarrow{f_s} & Y_s \\ {\scriptstyle i_s^X}\downarrow & & \downarrow{\scriptstyle i_s^Y} \\ s\backslash X & \xrightarrow{s\backslash f} & s\backslash Y, \end{array}$$

où, on l'a vu dans (*a*), i_s^X et i_s^Y sont dans W. Par hypothèse on a $f_s \in W$, donc $s\backslash f \in W$, p. 4
q.e.d.

b) Loc (1, 2_0, 3′) $\Longrightarrow$ Loc (1, 2, 3). Prouvons d'abord Loc (2). Appliquant Loc (3′ *a*) à la projection

$$\mathbf{\Delta}^1 \times X \longrightarrow X \qquad (\text{où } X \in \mathrm{Ob}\,\mathrm{Cat}),$$

qui est fibrée scindée à fibres isomorphes à $\mathbf{\Delta}^1$, donc W-asphériques par Loc (2_0), on voit que cette projection est dans W. Cela permet d'appliquer les arguments standard d'homotopie (*cf.* XII, p. 1 …) pour conclure qu'un homotopisme est dans W, donc une catégorie contractile est W-asphérique, a fortiori une catégorie avec objet final (ou avec objet initial) est W-asphérique, *i.e.* on a Loc (2).

Prouvons Loc (3). Comme Loc (1, 2, 3) $\Longleftrightarrow$ Loc (1, 2°, 3°), il suffit de prouver Loc (3°), vu qu'on a prouvé Loc (2°). Soit

$$f : X \longrightarrow S \qquad W\text{-}co\text{asphérique},$$

prouvons $f \in W$. Introduisons la catégorie *fibrée* sur S dont les fibres sont les $s\backslash X$, soit $\Phi_S(X)$, et considérons l'inclusion canonique

$$i_X : X \longrightarrow \Phi_S(X).$$

Par Loc (3′ *b*) elle est dans W. D'autre part, les fibres $s\backslash X$ de $\Phi_S(X)$ sont W-coasphériques [page 7]
par hypothèse, donc par Loc (3′ *a*) $\Phi_S(X) \longrightarrow S$ est dans W. Comme $i_X \in W$, il s'ensuit $(f : X \longrightarrow S) \in W$, q.e.d.

b bis) Loc (1, 2_0, 3′ bis) $\Longrightarrow$ Loc (1, 2, 3 bis). On a déjà prouvé Loc (1, 2, 3) (vu que Loc (3′ bis) contient Loc (3′) comme cas particulier). Prouvons donc Loc (3 bis) en admettant déjà tout ça. Il suffira encore de le prouver sous forme duale, puisque

$$\text{Loc (1, 2, 3 bis)} \iff \text{Loc(1, 2°, 3 bis°)},$$

et que Loc (2°) est connu par Loc (1, 2, 3) $\Longleftrightarrow$ Loc (1, 2°, 3°). Il reste donc à prouver que si [page 8]

$$f : X \longrightarrow Y \qquad \text{dans } \mathrm{Cat}/S$$

est telle que

$$s\backslash f : s\backslash X \longrightarrow s\backslash Y$$

est dans W pour tout $s \in \mathrm{Ob}\, S$, alors $f \in W$. Considérons alors le carré commutatif

$$\begin{array}{ccc} X & \xrightarrow{i_X} & \Phi_S(X) \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle \Phi_S(f)} \\ Y & \xrightarrow{i_Y} & \Phi_S(Y), \end{array}$$

où $i_X, i_Y \in W$ par Loc (3′ bis b). L'hypothèse sur f signifie que $\Phi_S(f)$ induit des W-équivalences sur les fibres, donc par Loc (3′ bis) $\Phi_S(f) \in W$, donc $f \in W$ par Loc (1), q.e.d.

p. 5

2 Axiomes Loc (4) et Loc (5) (des sommes directes, de connexité) [3]

Loc (4) (Axiome des *sommes directes*). Pour une famille $(f_i)_{i\in I}$ de flèches dans Cat, si $f = \coprod f_i$, on a $f \in W$ si et seulement si $f_i \in W$ pour tout $i \in I$.

Cet axiome se décompose en deux, en le "si" et le "seulement si", que je vais formuler sous la forme la plus économique :

Proposition 2.1. *L'axiome* Loc (4) *des sommes équivaut à la conjonction des deux axiomes suivants :*

[page 9]

Loc (4 a) Soit $(f_i)_{i\in I}$ une famille de flèches dans Cat, si les $f_i \in W$, alors $f = \coprod f_i \in W$.

Loc (4 b) Soit S dans Cat, et $S_0 \hookrightarrow S$ un sommand direct (sous-catégorie à la fois ouverte et fermée). Soit X dans Cat/S. Si $X \longrightarrow S$ est dans W, alors $X_0 = X|S_0 \longrightarrow S_0$ aussi. (En d'autres termes, l'inclusion $S_0 \hookrightarrow S$ est une "W-fibration", *i.e.* est dans Fib_W.)

Commentary · Later developments [ms. 8]

The weak minimum and the stronger axiom

There is a definitive later answer to one foundational issue that must be distinguished from the stronger-axiom problem. Let

$$W_\infty = \{f \mid |Nf| \text{ is a weak homotopy equivalence}\}.$$

Theorem 6.1.18 of Cisinski (2006a, pp. 242–243) proves that W_∞ is the *smallest weak fundamental localizer*. The cited pagination is that of the published Astérisque volume; an earlier author version places the same theorem at pp. 213–214. Definition 6.1.12 and Proposition 6.1.16, pp. 240–241, are indispensable for checking that this theorem really concerns the weak axiom used here. His split-retraction saturation condition follows from (1c), while W_∞ itself satisfies (1c), so the minimum comparison applies to the precise class of localizers in this manuscript. Thus every W of Definition 1.1 contains W_∞, and the intersection of all such W equals W_∞. It is not necessary to assume (3 bis) for this conclusion.

This does not prove that every weak localizer satisfies (3 bis). A theorem identifying the least objects in two partially ordered classes does not identify the two classes. Chiche explicitly records the latter problem as open in his 2014 thesis, Remark 2.1.15, and Cisinski still reported no example distinguishing the classes in his dated 2019 discussion (Chiche, 2014; Cisinski, 2019b). We found no

3. Il s'agit ici d'axiomes qu'on peut considérer comme étant de nature plus ou moins "triviale". Dans tous les cas à ma connaissance, ils sont bel et bien trivialement vérifiés.

resolution in the targeted literature search for this commentary. These are dated statements of the problem's documented status, not a claim that a search can certify the absence of every subsequent result.

There is a further useful distinction. Strong fundamental localizers in the later sense are strongly saturated, by Cisinski (2006a, Proposition 4.2.4). That theorem may be used after (3 bis) has been obtained. It cannot supply strong saturation for an arbitrary W satisfying just Definition 1.1. In particular, neither the finite-product arguments below nor the speculative reverse-excision axioms should acquire extra hypotheses invisibly through a change in terminology.

Corollaire 2.2. *Considérons la variante de* Loc (4 a) *obtenue en se bornant aux familles finies.* (*Elle équivaut à* Loc (4) *lorsque W est stable par* $\varinjlim$ *filtrantes,* cf. *axiome* Loc (8) *plus bas*). *Cette variante équivaut à*

Loc (4 a$_0$) Tout objet X de Cat est "W-cofibrant", *i.e.* $\emptyset \longrightarrow X$ est dans Cof_W, *i.e.* pour toute $f : Y \longrightarrow Y'$ qui est dans W, le morphisme correspondant $\mathrm{id}_X \amalg f : X \amalg Y \longrightarrow X \amalg Y'$ est dans W.

Rappel 2.3. De toutes façons, si W est un localiseur fondamental (ce que nous supposerons désormais, sauf mention du contraire), l'analogue *fini* de Loc (4 a) pour les produits est vrai, *i.e.* si on a une famille *finie* de flèches $(f_i)_{i\in I}$ dans Cat, si les f_i sont dans W, alors $f = \prod f_i \in W$. `[Pour cela, il faut supposer que` W `satisfait à l'axiome de localisation, forme` *`forte`*`, Loc 3 bis.]` L'analogue pour une famille infinie n'est vérifié pratiquement pour aucun localiseur fondamental.

Proposition 2.4. *Soit W un localiseur fondamental. Alors les produits finis existent dans* Hot_W*, et le foncteur* $\mathrm{hot}_W : \mathrm{Cat} \longrightarrow \mathrm{Hot}_W \overset{déf}{=} (\mathrm{Cat})W^{-1}$ *y commute. Si W satisfait à* Loc (4 a) (resp. Loc (4 a$_0$)), *alors les sommes quelconques* (resp. *finies*) *existent dans* Hot_W*, et le foncteur* hot_W *y commute. De plus, dans ce cas les sommes sont distributives par rapport aux produits (vu qu'il en est ainsi dans* Cat). [page 10]

Commentary · Exposition [ms. 9]

Sums and products require different hypotheses

The two halves of the sum axiom do different jobs. Closure under sums builds a global equivalence from component equivalences. Pullback along a summand recovers the component equivalences from the global one. Indeed, in a coproduct map $\coprod X_i \to \coprod Y_i$, the inverse image of the summand Y_i is exactly X_i. This proves Proposition 2.1. For finite sums, the factorization

$$f \amalg g = (\mathrm{id} \amalg g)(f \amalg \mathrm{id})$$

proves that closure is equivalent to the condition that each $\emptyset \to X$ is an intrinsic W-cofibration. Empty and one-term sums cause no problem because identities belong to W.

The parenthesis in Corollary 2.2 should say Loc (4*a*), rather than Loc (4). A coproduct is the filtered colimit of its finite partial coproducts, so filtered-colimit stability promotes finite *closure* to arbitrary closure. This argument gives no summand-reflection statement (4b). The later summary on ms. 15′ correctly supplies a separate fibration axiom for that half. For an explicit counterexample, let W_{ne} consist of all functors between nonempty categories, together with $\mathrm{id}_\emptyset$. This is even a strong localizer: a relative comma comparison cannot belong to it when its source is empty and its target has a chosen object. It is closed under arbitrary coproducts and filtered colimits, since both constructions preserve the test of nonemptiness. However, $e \to e \amalg e$ is in W_{ne}, while its pullback over the other summand is $\emptyset \to e$, which is not. Thus (4b) really fails. The printed parenthesis is retained as an edition slip.

For products the editors' qualification is equally necessary. With (3 bis), consider $f \times \mathrm{id}_C : A \times C \to B \times C$ over C. Over c the comparison is $f \times \mathrm{id}_{C/c}$. Since C/c has a final object, the projection $A \times (C/c) \to A$ has a right adjoint and belongs to W, and likewise for B. Two-out-of-three makes the local comparison a member of W; relative Theorem A then gives $f \times \mathrm{id}_C \in W$. Factoring a finite product into maps changing one coordinate proves finite-product stability. This proof specifically uses the relative axiom at its last step.

The source's phrase that infinite stability holds for practically no localizer should be read as an informal warning, not a quantified theorem. There are immediate degenerate exceptions, such as the class of all functors. Ordinary weak equivalences do not in general enjoy the asserted strict-product stability for arbitrary families of categories: Thomason (1980, pp. 306–308) exhibits the discrepancy using products of a category with two parallel arrows. The lesson needed here is simply that the finite argument must not be extended to infinitely many coordinate changes.

La démonstration résulte simplement du résultat général :

Lemme 2.5. *Soit* $(\mathcal{M}, W)$ *une catégorie de modèles (*W *partie quelconque de* $\mathrm{Fl}\,\mathcal{M}$*). Supposons que les produits* (resp. *les produits finis,* resp. *les produits binaires*) *existent dans* $\mathcal{M}$*, et que de plus pour toute famille* $(f_i)_{i\in I}$ *de flèches de* $\mathcal{M}$ (resp. *toute famille finie,* resp. *famille binaire*) $f_i \in W$ *pour tout* $i \in I$ *implique* $f = \prod f_i \in W$*. Alors les produits du type envisagé existent dans* $\mathrm{Ho}_W = \mathcal{M}W^{-1}$*, et le foncteur* $\mathrm{ho}_W : \mathcal{M} \longrightarrow \mathcal{M}W^{-1}$ *y commute. Enfin, si* $\mathcal{M}$ *admet un objet final* e*, alors* e *est aussi objet final de* Ho_W*.* p. 6

[Il n'est pas clair que l'assertion soit vraie dans cette généralité pour les produits *infinis* ; elle est vraie si W est la classe des équivalences faibles d'une *catégorie de modèles de Quillen.*]

On en conclut les énoncés duaux concernant les sommes directes et les objets initiaux. [page 11]

Commentary · Development (commentators') [ms. 10]

A missing hypothesis in the product lemma

A finite counterexample to the literal generality of Lemma 2.5. [ours] The qualification "W an arbitrary subset" already causes a problem for binary products, independently of the editors' infinite-product warning. Let $\mathcal{M}$ be the four-element Boolean lattice

$$0 < a < 1, \qquad 0 < b < 1,$$

with a and b incomparable, regarded as a category, and put

$$W = \{\mathrm{id}_1,\ a \longrightarrow 1\}.$$

All small products exist and are meets. A product of arrows in W is id_1 if all factors are identities, and is $a \to 1$ otherwise; this includes the empty product. Thus even the strongest product-closure hypothesis printed in the lemma holds.

After localization a and 1 are isomorphic, but 0 and b are not. To verify the latter assertion without guessing the localized category, use the functor to the chain $0 < b < 1$ sending a to 1 and fixing $0, b, 1$. It inverts W, hence factors through the localization, and distinguishes 0 from b. Since $a \times b = 0$ in $\mathcal{M}$, preservation of binary products would make the image of 0 a product of the images of 1 and b. The image of 1 remains terminal, so this product is the image of b, a contradiction. This elementary counterexample is included to repair the wording, with no claim of historical priority.

The appropriate finite version assumes that W contains all identities, as explicitly in Maltsiniotis (2020, Lemma 2.1.7 and Proposition 2.1.8, pp. 114–115). For finite I, localization then commutes with the product of relative categories:

$$(\mathcal{M}^I)[(W^I)^{-1}] \simeq (\mathcal{M}[W^{-1}])^I.$$

One can prove this by changing one coordinate at a time. Every formal inverse in one coordinate lifts to an inverse of a tuple with identities in the other coordinates; coordinate changes commute, and finitely many finite words combine into one finite word. If products preserve W, the diagonal/product adjunction therefore descends and proves the corrected finite lemma. The same reasoning applies to coproducts. Every localizer under consideration in the chapter contains identities, so the counterexample does not affect those finite applications.

For infinite I, coordinatewise finite words need not have a common finite length; the displayed comparison is no longer a formal consequence of the localization universal property. We do not claim a general resolution of the infinite variant under just its printed assumptions. In a Quillen model category the editors' repair works: the product of fibrant replacements computes the derived product, and assumed closure of W under these products identifies it with the ordinary product. Proposition 2.4 should accordingly be read with the editors' strong-localization hypothesis for finite products and with a separate justification whenever arbitrary coproducts are asserted; the finite adjunction argument alone is insufficient for an infinite family.

La démonstration est un cas particulier du résultat bien utile suivant, appliqué au couple de foncteurs adjoints

$$\mathcal{M} \underset{\text{produits dans } \mathcal{M} \text{ indexés par } I}{\overset{\text{diag}}{\rightleftarrows}} \mathcal{M}^I \qquad :$$

Lemme 2.6 [4]. *Soient $(\mathcal{M}, W)$, $(\mathcal{M}', W')$ deux catégories de modèles, et*

$$\mathcal{M} \underset{\psi}{\overset{\varphi}{\rightleftarrows}} \mathcal{M}'$$

un couple de foncteurs adjoints compatibles aux localiseurs, i.e. *tels que*

$$\varphi(W) \subset W', \qquad \psi(W') \subset W.$$

Alors les foncteurs localisés

$$\mathcal{M}W^{-1} \underset{\bar{\psi}}{\overset{\bar{\varphi}}{\rightleftarrows}} \mathcal{M}'W'^{-1}$$

sont également adjoints, les flèches d'adjonction pour $\bar{\varphi}\bar{\psi}$ et $\bar{\psi}\bar{\varphi}$ étant déduites de celles pour $\varphi\psi$ et $\psi\varphi$ par composition avec les foncteurs de localisation $\mathcal{M} \longrightarrow \mathcal{M}W^{-1}$, $\mathcal{M}' \longrightarrow \mathcal{M}'W'^{-1}$.

Loc (5) (Axiome de *connexité* ou de *non-trivialité*).

[page 12]

Commentary · Exposition [ms. 11]

What descends from an adjunction

Lemma 2.6 itself has no product-size defect. Suppose $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ is an adjunction and both functors send the designated equivalences into equivalences. Its unit and counit give natural transformations after localization. To check naturality on a formally inverted arrow, invert the corresponding naturality square; thus naturality on ordinary arrows suffices. The two triangle identities remain identities after applying the localization functors. They prove $\overline{F} \dashv \overline{G}$ directly, without a calculus of fractions or any claim that localization commutes with an infinite product of categories.

This also proves the final assertion of Lemma 2.5 for an arbitrary class W: a terminal object remains terminal. Use the adjunction between the unique functor $\mathcal{M} \to e$ and the selection of its terminal object, allowing the image of designated arrows to be merely inverted, as the universal property permits. Dually an initial object remains initial. Strictness is another matter; localization may create arrows into an initial object by reversing arrows out of it. The hypotheses of Lemma 2.14 are designed precisely to prevent that phenomenon. Keeping existence, strictness, and preservation of an entire family of products separate prevents several otherwise tempting shortcuts in the next pages.

Soit $f \in W$, alors $\pi_0(f)$ est bijectif.

Cette propriété de W est d'une utilisation constante en pratique, quand on travaille avec W au lieu de se borner à construire des W et faire joujou avec les axiomes et leur stabilité par passage à des $\underline{\mathrm{Hom}}(I, M)$ (où ici $\mathcal{M} = \mathrm{Cat}$). D'autre part, on peut aussi le regarder comme un axiome de *non-trivialité* pour Hot_W. Je rappelle à ce sujet :

Proposition 2.7. *Soit W un localiseur fondamental. Conditions équivalentes :*

a) *W satisfait l'axiome de connexité* Loc (5).

b) *Le foncteur $\pi_0 : \mathrm{Cat} \longrightarrow \mathrm{Ens}$ se factorise par un foncteur*

$$\mathrm{Cat} \dashrightarrow \mathrm{Hot}_W \xrightarrow{\pi_0} \mathrm{Ens}. \qquad \text{(} \pi_0 \text{ : Hot}_W \to \text{Ens)}$$

p. 7

4. *cf.* XII, lemme page 248.

c) *La catégorie* $\mathrm{Hot}_W^* = \mathrm{Hot}_W \backslash \{\emptyset\}$ *n'est pas équivalente à la catégorie ponctuelle.*

c′) *La catégorie* Hot_W *n'est pas équivalente à e ni à* $\mathbf{\Delta}^1$.[5]

Pour la démonstration, et pour d'autres formes équivalentes de l'axiome de connexité, je renvoie à XV 9.12 (p. 133 bis). Pour le nom "axiome de connexité", je peux offrir l'explication suivante 2.8 et 2.9. [page 13]

Commentary · Present state [ms. 13]

Connectivity and a hypothesis hidden in the recalled proof

The equivalence of (2.7a) and (2.7b) is exactly the universal property of localization: π_0 factors if and only if it inverts W. The remaining equivalences require more care than the reference to XV might suggest. In the proof of XV, Proposition 9.12, ms. 133 quinquies, the editors explicitly flag the use of finite-product preservation as unproved under the weak axioms (1,2,3); it is available under (3 bis) (Grothendieck, 1990–1991k). Thus XVI, Proposition 2.7 inherits a missing hypothesis in the cited proof. We do not turn this gap into either a counterexample or an assertion that the weak formulation is an established theorem.

Here is the complete argument once finite-product preservation is supplied, in particular under (3 bis). Suppose $f : A \to B$ belongs to W but does not induce a bijection on components. If two different components of A are identified, choose points in them and a map $A \to \{0, 1\}$ separating them. Their images in B are connected, hence equal as points after localization, so invertibility of f makes the two sections of $\{0, 1\} \to e$ equal in Hot_W. If a component of B is missed, compare the constant map $B \to \{0, 1\}$ with a map which is 1 on that component and 0 on the image of A. Their composites with f agree. Inverting f and evaluating at a point in the missed component gives the same equality of sections.

Product preservation now makes the two inclusions $X \to X \amalg X$ equal for every X. Composing with a map specified by any two functors $X \rightrightarrows Y$ shows that their localized images are equal. For nonempty X, a chosen point $e \to X$ is therefore inverse to $X \to e$. The localization is equivalent to Δ^1 if no arrow from the empty category to a nonempty category is inverted, and to e otherwise. Conversely, if π_0 factors, the discrete categories with different numbers of objects remain distinguishable, so neither degeneration occurs.

Corollary 2.8's forward implication has no such gap: $X \to e$ in W immediately gives $\pi_0 X = \{*\}$. Its converse, as argued through the recalled classification and strong saturation, needs the same repair to that classification. The test of Corollary 2.9 has a direct proof independent of it. Pull a W-map back to each component of its target using (4b); the stated connected-target condition says that every inverse image is nonempty and connected, exactly the bijectivity of π_0. The rest of this section explicitly assumes (5), so the concrete description of points below does not depend on the unresolved weak-hypothesis step.

Corollaire 2.8. *Si l'axiome de connexité* Loc (5) *est satisfait, alors les objets W-asphériques sont* 0*-connexes* (*tautologie*), *et la réciproque est vraie si W est fortement saturé.* (*Ou aussi, si W satisfait l'axiome des fibrations* Loc (7) *ci-dessous, puisque par* XII, th. 2, p. 20, *dans ce cas,* Loc (5) *équivaut à* $W \neq \mathrm{Fl}(\mathcal{M})$ …).

Car si l'axiome de connexité n'est pas satisfait, alors pour toute X non vide dans Cat, $X \longrightarrow e$ est un isomorphisme, donc (W étant fortement saturé) $X \longrightarrow e$ est dans W, *i.e.* X est W-asphérique, donc il y a des objets W-asphériques non 0-connexe. Notons aussi :

Corollaire 2.9. *Supposons que W satisfait à l'axiome* Loc (4) *des sommes directes* (*ou seulement à* Loc (4 b)). *Pour que W satisfasse l'axiome de connexité, il faut et il suffit que pour toute flèche* $f : X \longrightarrow Y$ *avec Y* 0*-connexe, si f est dans W, alors X est* 0*-connexe.* (*C'est nécessaire bien sûr sans supposer* Loc (4 b) *ni rien du tout.*)

`[Les numéros 2.7, 2.8 et 2.9 apparaissent deux fois.]` [page 14]

5. **NB** Si W est fortement saturée (condition Loc (1 ter), alors les conditions ci-contre (*i.e.* Loc (5)) équivalent à ceci :

d) *W n'est égal ni à Fl*(Cat) *tout entier, ni à Fl*(Cat^*) $\cup$ $\{\mathrm{id}_\emptyset\}$ (i.e. *Fl*(Cat)$\backslash W$ *n'est ni vide, ni égal à l'ensemble des flèches* $\emptyset \longrightarrow X$ *avec* $X \neq \emptyset$).

Si W satisfait Loc (7 D_{W_ω}), alors W satisfait Loc (5) si et seulement si $W \neq \mathrm{Fl}(\mathrm{Cat})$ (*cf.* XII, th. 2, p. 20).

2.7. Considérons maintenant le foncteur composé

$$\text{(2.7.1)} \qquad \text{Ens} \xrightarrow{I \longmapsto I_{\text{Cat}}} \text{Cat} \xrightarrow{\text{hot}_W} \text{Hot}_W, \qquad I \longmapsto I_{\text{Hot}_W} \text{ ou } I_W,$$

où le premier foncteur est celui qui associe à l'ensemble I la catégorie discrète I_{Cat} ayant I comme ensemble d'objets. (Le plus souvent, on identifiera I_{Cat} avec I dans les notations, tout comme pour un ensemble ordonné J, on identifie J et la "catégorie ordonné" définie par J. Ici I peut être considéré comme muni de l'ordre discret.) On a d'ailleurs

$$\text{(2.7.2)} \qquad I_{\text{Cat}} \simeq I \times e \quad (\overset{\text{déf}}{=} \text{somme de } I \text{ copies de } e),$$

donc quand Loc (4 a) est satisfait, donc hot_W commute aux sommes directes, on aura un isomorphisme canonique

$$\text{(2.7.3)} \qquad I_W \simeq I \times e_W \quad \begin{array}{l}(\text{où } e_W \text{ est l'objet final de Hot}_W,\\ \textit{i.e. } e \text{ considéré comme objet de Hot}_W).\end{array}$$

Proposition 2.8. *Supposons l'axiome de connexité* Loc (5) *satisfait, d'où un foncteur* p. 8

$$\text{(2.8.1)} \qquad \pi_0 : \text{Hot}_W \longrightarrow \text{Ens}.$$

Ce foncteur est adjoint à gauche *du foncteur* $I \longmapsto I_W$ (2.7.1), i.e. *on a une bijection fonctorielle en* (X, I) *dans* $\text{Hot}_W \times \text{Ens}$:

[page 15]

$$\text{(2.8.2)} \qquad \text{Hom}_{\text{Ens}}(\pi_0(X), I) \simeq \text{Hom}_W(X, I_W).$$

Le foncteur π_0 *est également* adjoint à droite *du* même *foncteur* $I \longmapsto I_W$, i.e. *on a un bijection fonctorielle*

$$\text{(2.8.3)} \qquad \text{Hom}_{\text{Ens}}(I, \pi_0(X)) \simeq \text{Hom}_W(I_W, X).$$

Prenant $I = e$ (ensemble ponctuel), on trouve le :

Corollaire 2.9. *On a l'isomorphisme fonctoriel en* X *dans* Hot_W

$$\text{(2.8.4)} \qquad \boxed{\text{Hom}_W(e_W, X) \simeq \pi_0(X).}$$

Commentaire 2.10. La formule d'adjonction (2.8.2) est d'ailleurs familière, chaque fois qu'on a une catégorie $\mathcal{H}$ (telle Cat, où Hot_W *etc.*) munie d'un foncteur π_0, $\pi_0(X)$ correspondant à l'intuition d'une décomposition d'un objet en "composantes connexes". (Je ne vais pas expliciter ici les conditions générales sous lesquelles on définit un tel foncteur.) Par contre, l'adjonction en sens inverse de ces deux mêmes foncteurs est un phénomène tout à fait spécial à des catégories du type Hot_W. Si on tient compte de la description de I_W comme $I \times e_W$, la formule d'adjonction inhabituelle (2.8.3) équivaut, bien sûr, à la formule (2.8.4) [page 16]
- elle s'en déduit en "élevant à la puissance I". On peut renverser la vapeur, et dans une catégorie générale $\mathcal{H}$, *définir* un foncteur $\pi_0 : \mathcal{H} \longrightarrow \text{Ens}$ par le formule (2.8.4),

$$\pi_0(X) \overset{\text{déf}}{=} \text{Hom}(e_{\mathcal{H}}, X) \qquad (= \Gamma_{\mathcal{H}}(X), \text{ avec une notation familière}),$$

où $e_{\mathcal{H}}$ est un objet final de $\mathcal{H}$. On aura donc, en posant $I_{\mathcal{H}} = I \times e_{\mathcal{H}}$, la formule

$$\text{Hom}(I_{\mathcal{H}}, X) \xrightarrow{\sim} \text{Hom}(I, \pi_0(X))$$

(*i.e.* la formule (2.8.3)), disant que l'on a un couple $(I_{\mathcal{H}}, \pi_0)$ de foncteurs adjoints. Supposons d'autre part que pour tout ensemble I, l'application canonique d'adjonction

$$I \longrightarrow \pi_0(I_{\mathcal{H}}) = \text{Hom}_{\mathcal{H}}(e_{\mathcal{H}}, e_{\mathcal{H}} \times I)$$

soit *bijective*[6], *i.e.* que $I \longmapsto I_{\mathcal{H}}$ soit *pleinement fidèle.* On définit alors, pour tout I, une application canonique

Commentary · Exposition [ms. 15]

The double adjunction without an infinite-coproduct shortcut

Write $d(I) = I_W$ for the image of the discrete category on a set I. The first adjunction $\pi_0 \dashv d$ follows by localizing the familiar adjunction on Cat: every functor to a discrete category is constant on components. The second,

$$d \dashv \pi_0, \qquad \mathrm{Hom}_{\mathrm{Hot}_W}(dI, X) = \mathrm{Set}(I, \pi_0 X),$$

is special to the localized situation. In the printed proof it is reduced to $I = e$ by treating dI as the coproduct of I copies of e. For arbitrary I that reduction calls on the earlier infinite-coproduct assertion. The following direct proof avoids it and works with just (1,2,3,5).

First suppose two functors $u, v : I \to X$ choose points in the same component for every i. For each i choose a finite zigzag joining $u(i)$ to $v(i)$, and let J_i be its walking zigzag category. Its nerve is contractible, so $J_i \to e$ is in $W_\infty \subset W$; alternatively this follows by successive elementary contractions. Put $J = \coprod_i J_i$. The projection $J \to I$ has comma category J_i over i, and (3) therefore puts it in W directly, without (4). Its two endpoint sections are both inverse to it in the localization. Composing with the chosen functor $J \to X$ proves $L(u) = L(v)$. The lengths of the individual zigzags need not be bounded.

Next let $w : X \to Y$ belong to W and let $u : I \to Y$. Surjectivity of $\pi_0(w)$ allows a choice $s : I \to X$ such that ws and u lie in the same component pointwise. The previous paragraph gives $L(w)L(s) = L(u)$. Any localization morphism from dI is represented by a finite word of ordinary arrows and inverses of arrows in W. Whenever its first reversed arrow occurs, this lifting replaces the preceding part of the word by an ordinary map from I. Induction eliminates all reversed arrows. Thus every morphism $dI \to X$ is represented by an actual functor $I \to X$; two representatives agree exactly when their component functions agree, since the descended π_0 detects that distinction. This proves the desired natural bijection, including $I = \emptyset$.

Taking $I = e$ proves Corollaries 2.9 and 2.12. Taking $X = dJ$ shows that d is fully faithful. It also proves, rather than assumes, that dI is the coproduct of I copies of e in Hot_W, since its maps to X are the I-indexed families of points of X. Here $I \times e$ denotes a *copower*, as the source says; it is not an assertion that an external set is already an object participating in a categorical product.

p. 9

$$\begin{array}{ccc} \mathrm{Hom}(X, I_{\mathcal{H}}) & \longrightarrow & \mathrm{Hom}(\pi_0(X), I) \\ & {\scriptstyle \pi_0^{X, I_{\mathcal{H}}}} \searrow \quad \nearrow {\scriptstyle \sim} & \\ & \mathrm{Hom}(\pi_0(X), \pi_0(I_{\mathcal{H}})). & \end{array}$$

La "bi-adjonction" des foncteurs π_0 et $I \longmapsto I_{\mathcal{H}}$ revient à dire que cette flèche est bijective, ou encore, que la flèche $\pi_0^{X, I_{\mathcal{H}}}$: [17]

$$\mathrm{Hom}(X, I_{\mathcal{H}}) \longrightarrow \mathrm{Hom}(\Gamma_{\mathcal{H}}(X), \Gamma_{\mathcal{H}}(I_{\mathcal{H}}) \simeq I)$$

soit bijective.

Corollaire 2.11. *Le foncteur $I \longmapsto I_W$ de* Ens *dans* Hot_W *est pleinement fidèle,* i.e. *l'homomorphisme d'adjonction*

$$\pi_0(I_W) \longrightarrow I$$

est bijectif. (*Où encore, π_0 est un foncteur le localisation.*)

En effet, cette bijectivité est évidente quand on travaille dans Cat, et s'en déduit aussitôt dans Hot_W.

6. ça a tendance à être ultrafaux si $\mathcal{H}$ est additive. Par contre, vrai si $\mathcal{H}$ est une catégorie à sommes distributives par rapport aux produits (voire universelles), dont l'objet final est *connexe* …

Commentary · Examples [ms. 17]

Why the two adjunctions are exceptional

For a category $\mathcal{H}$ with terminal object e and the required coproducts, the copower functor $dI = \coprod_I e$ is automatically left adjoint to $\Gamma = \mathrm{Hom}(e, -)$. The unit $I \to \Gamma(dI)$ is bijective precisely when d is fully faithful. Neither this condition nor the existence of coproducts automatically supplies the other adjunction: that further assertion is that every function $\Gamma(X) \to I$ comes from exactly one morphism $X \to dI$.

The examples distinguish the conditions. In sets both adjunctions hold with $d = \Gamma = \mathrm{id}$. In Cat, d is fully faithful and $\pi_0 \dashv d$, but $\Gamma(X) = \mathrm{Ob}\,(X)$ is usually different from $\pi_0 X$; two objects joined by an arrow still define different functors $e \to X$. Localization makes those two functors equal. In an additive category the terminal object is the zero object, so every copower of it is again zero and $\Gamma(X)$ is a singleton. Consequently $I \to \Gamma(dI)$ is bijective only for singleton I. This explains the warning in footnote 6 without treating additivity as a mild exceptional case of the asserted unit condition.

In the triangular display the edition prints the horizontal top arrow as well as the two dotted downward arrows, the right one labelled $\sim$; the transcription keeps all three. The top arrow is the composite of the left arrow with the inverse of the right arrow, exactly as the surrounding prose requires.

Démonstration de 2.8. La formule d'adjonction (2.8.2) résulte de la formule correspondante pour Cat, évidente, et du lemme 2.6 sur les foncteurs adjoints entre catégories de modèles. L'adjonction dans l'autre sens se réduit à la formule (2.8.4). On a des flèches canoniques

$$\mathrm{Hom}_W(e_W, X) \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \pi_0(X),$$

où α est $\pi_0^{e_W, X}$, compte tenu que $\pi_0(e_W) = \pi_0(e_{\mathrm{Cat}}) = e_{\mathrm{Ens}}$, et $\mathrm{Hom}(e_{\mathrm{Ens}}, \pi_0(X)) \simeq \pi_0(X)$. La flèche β est la flèche composée

$$\mathrm{Ob}\, X \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Cat}}(e_{\mathrm{Cat}}, X) \xrightarrow{\mathrm{hot}_W} \mathrm{Hom}_W(e_W, X)$$

en notant que cette flèche passe au quotient en $\pi_0(X) \longrightarrow \mathrm{Hom}_W(e_W, X)$, vu que deux [18]
applications homotopes de e_{Cat} dans X sont égales dans Hot_W. Je dis que

$$\beta\alpha = \mathrm{id}\ , \qquad \alpha\beta = \mathrm{id}\ .$$

La relation $\alpha\beta = \mathrm{id}$ revient à dire que la flèche $i_x : e_{\mathrm{Cat}} \longrightarrow X$ définie par $x \in \mathrm{Ob}\, X$, donne $\pi_0(i_x) : e_{\mathrm{Ens}} \longrightarrow \pi_0(X)$ dont la valeur est la classe de x dans $\pi_0(X)$, c'est tautologique. Pour prouver que $\beta\alpha = \mathrm{id}$, il suffit alors de prouver que β est surjective, *i.e.* le

Corollaire 2.12. *Pour tout X dans* Cat, *l'application*

$$\underbrace{\mathrm{Hom}_{\mathrm{Cat}}(e_{\mathrm{Cat}}, X)}_{\simeq \mathrm{Ob}\, X} \longrightarrow \mathrm{Hom}_W(e_W, X)$$

est surjective (*moyennant l'axiome de connexité* Loc (5)). p. 10

Démonstration de 2.12. Si $X = \emptyset$, l'assertion revient à dire que $\mathrm{Hom}_W(e_W, \emptyset) = \emptyset$, mais plus généralement, on a le

Corollaire 2.13. (Moyennant Loc (5).) *Soit X non vide dans* Cat, *alors* $\mathrm{Hom}_W(X, \emptyset) = \emptyset$. (*Donc $\emptyset$ est un objet initial* strict *de* Hot_W.)

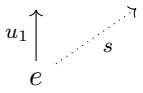
En effet, si on avait $X \longrightarrow \emptyset$ dans Hot_W, `[il y a un décalage dans la numérotation]` [page 13']
on en déduirait $\pi_0(X) \longrightarrow \emptyset$, ou $\pi_0(X) = \emptyset$, absurde !

Reste le cas où X est non vide. Cela résulte du fait général :

Lemme 2.14. *Soit $\mathcal{M}$ une catégorie ayant un objet initial* strict (*si* $\mathrm{Hom}_{\mathcal{M}}(X, \emptyset_{\mathcal{M}}) \neq \emptyset$, *on a* $X = \emptyset_{\mathcal{M}}$), *un objet final e, et telle que pour tout $X \neq \emptyset_{\mathcal{M}}$, $X \longrightarrow e$ ait une section,* i.e. $\mathrm{Hom}(e, X) \neq \emptyset$. *Soit* $W \subset \mathrm{Fl}\, \mathcal{M}$

1°) *ne contenant pas de flèches* $\emptyset_{\mathcal{M}} \longrightarrow X$ *avec* $X \neq \emptyset_{\mathcal{M}}$, *et*

2°) *tel que pour tout diagramme* $X_1 \xleftarrow{u_2} X_2$ *avec* $u_2 \in W$ $\exists$ s dans $\mathcal{M}$ *tel que* $u_2 s = u_1$



dans $\mathcal{M}W^{-1}$.

Alors

a) $\emptyset_{\mathcal{M}}$ *est un objet initial* strict *de* $\mathcal{M}W^{-1}$.

b) *Pour tout* $X \neq \emptyset_{\mathcal{M}}$, $\mathrm{Hom}_{\mathcal{M}}(e, X) \longrightarrow \mathrm{Hom}_W(e, X)$ *est surjectif.*

Démonstration. Que $\emptyset_{\mathcal{M}}$ soit objet initial de $\mathcal{M}W^{-1}$ a été vu (dernière assertion du lemme 2.5, forme duale), prouvons qu'il est strict, *i.e.* que si $\mathrm{Hom}_W(X, \emptyset_{\mathcal{M}}) \neq \emptyset$, on a $X = \emptyset_{\mathcal{M}}$. En effet, si on avait $u : X \longrightarrow \emptyset_{\mathcal{M}}$ dans $\mathcal{M}W^{-1}$, u serait représenté par

$$X = X_0 \xleftarrow{W} X_1 \longrightarrow X_2 \xleftarrow{W} \cdots X_{2n-2} \longrightarrow X_{2n-1} \xleftarrow{W} X_{2n} = \emptyset_{\mathcal{M}}.$$

On procède par récurrence sur la demi-longueur de cette chaîne pour voir que $X = \emptyset_{\mathcal{M}}$. En effet, l'hypothèse sur W implique que $X_{2n-1} = \emptyset_{\mathcal{M}}$, puis celle que $\emptyset_{\mathcal{M}}$ est strict implique $X_{2n-1} = \emptyset_{\mathcal{M}}$, d'où la conclusion.

Prouvons la surjectivité de $\mathrm{Hom}_{\mathcal{M}}(e, X) \longrightarrow \mathrm{Hom}_W(e, X)$ pour $X \neq \emptyset_{\mathcal{M}}$, en montrant que tout chemin réduit de longueur 2n+1 est "homotope" à un chemin réduit de longueur $2n - 1$: [page 14']

$$e = X_0 \longrightarrow X_1 \xleftarrow{W} X_2 \longrightarrow X_3 \longleftarrow \cdots \longleftarrow X_{2n} \longrightarrow X_{2n+1} = X.$$

En effet, on peut remplacer la section

$$e = X_0 \xrightarrow[u_1]{} X_1 \xleftarrow[u_2]{W} X_2 \xrightarrow[u_3]{} X_3 \xleftarrow{W} X_4$$

(s : $X_0 \longrightarrow X_2$)

par [p. 11]

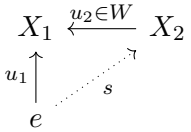
$$e \xrightarrow[u_3 s]{} X_3 \xleftarrow{W} \cdots,$$

où s est une section de X_2 sur e telle que

$$u_2 s = u_1 \qquad \textit{dans} \qquad \mathcal{M}W^{-1}\ .$$

Cela prouve donc par récurrence sur n que le chemin est homotope à un chemin de longueur 1, $e \longrightarrow X$, q.e.d.

Il faut donc, dans le cas actuel, vérifier 1°) (c'est essentiellement le corollaire 2.12.), et 2°)



.

Mais comme $u_2 \in W$, on a $\pi_0(u_2)$ isomorphisme, donc surjectif, donc $\exists$ s tel que $u_2 s$ soit homotope à u_1 dans Cat, ce qui implique $u_2 s = u_1$ dans Hot_W, q.e.d. [page 15']

Commentaire final. L'axiome de connexité Loc (5) diffère de tous les autres axiomes Loc (1) à Loc (8) (mais à l'exception de Loc (8 bis) d'accessibilité) par le fait que ce n'est pas un axiome de stabilité. Les autres axiomes Loc (1) à Loc (8) (et aussi Loc (8 bis) d'accessibilité et même les axiomes ultérieures d'exactitude[7]) sont satisfaits si $W = \mathrm{Fl}\,\mathcal{M}$ - l'axiome de connexité est donc le seul qui s'oppose à la *trivialité* de la théorie homotopique définie par W.

7. il n'y en aura en fait pas besoin, grâce à la providentielle théorie de K. S. Brown !

C'est dire qu'il ne peut être conséquence de l'un des autres. Par contre, l'axiome Loc (4) des sommes directes n'est pas indépendant des autres. Signalons à ce propos (rappel) :

Loc (4 a_0)	est conséquence de l'axiome Loc (6 a) du carré cocartésien (forme directe).
Loc (4 a)	est conséquence de Loc (6 a) précédent + l'axiome des limites inductives filtrantes Loc (8).
Loc (4 b)	est conséquence de l'axiome Loc (7) des fibrations. Donc
Loc (4)	est conséquence de Loc (6 a, 7, 8).

p. 12

Commentary · Exposition [ms. 15'] [page 16']

Shortening localization words and preserving emptiness

Lemma 2.14 is a useful statement about localization words, rather than a calculus-of-fractions theorem. A morphism is a finite zigzag; it need not have a single denominator. If a zigzag ends at a strict initial object 0, the prohibition on W-arrows $0 \to X$ with $X \neq 0$ forces the adjacent object to be 0. Strictness then forces the preceding object to be 0, and induction runs through the zigzag. This proves strictness after localization. The repeated subscript $2n-1$ in the printed proof is an edition slip: the second conclusion is $X_{2n-2} = 0$.

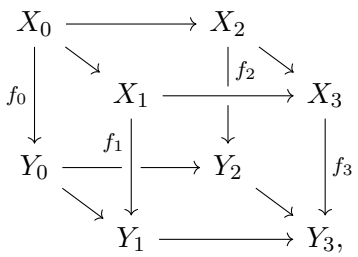
For a word beginning at e, the lifting condition replaces $e \to X_1 \leftarrow X_2$ by a genuine map $e \to X_2$ having the same localized composite. Composing with the next arrow shortens the word by two. In Cat, (5) supplies this lifting because every component of X_1 has a representative in X_2, while a zigzag inside a component becomes equality of points after localization. The citation of Corollary 2.12 when checking condition (1) is circular as printed; that condition follows directly from (5), or from the empty-target argument of Corollary 2.13. Neither slip affects the corrected induction.

There is a useful logical division in the final commentary. The class of all functors satisfies every purely closure-type axiom under discussion, so no conjunction of those axioms can force (5). Conversely, sum closure need not be an independent axiom once a gluing theory is available: a coproduct is the union of two disjoint open subcategories, giving finite closure from (6a), and its finite partial unions give arbitrary closure after adding filtered-colimit stability. Reflection on a summand still comes from the separate base-change requirement. This is why (6a,7,8), rather than just (6a,8), appears in the correct final implication to (4).

3 Axiome Loc (6) du carré cocartésien (ou de "Mayer-Vietoris", ou des W-cofibrations) [8]

Je vois plusieurs variantes, et n'ai pas encore l'esprit totalement net sur les implications mutuelles. Je commence par la variante plus compliquée, mais dont on aura besoin en tous cas :

Loc 6 (Axiome des W-cofibrations, ou lemme des 5, ou axiome des homomorphismes de carrés cocartésiens). Considérons dans Cat un diagramme commutatif cubique



8. C'est le nom "*axiome des W-cofibrations*" qui m'apparaît à présent comme le meilleur, – c'est ce nom qui me paraît le mieux saisir la signification conceptuelle et non technique de l'axiome.

homomorphisme $f = (f_0, f_1, f_2, f_3)$ d'un diagramme carré X dans un autre Y. On suppose que X provient d'un objet X_3 de Cat de deux ouverts X_1, X_2 de X_3 *recouvrant* X_3, et de leur intersection $X_0 = X_1 \cap X_2$, et de même pour Y. (Appelons ces diagrammes dans Cat des diagrammes MV ("*ouverts*") - "*Mayer-Vietoris*".) On a alors ce qui suit [9] :

a) $f_0, f_1, f_2 \in W \Longrightarrow f_3 \in W$ [10].

b) $f_1, f_2, f_3 \in W \Longrightarrow f_0 \in W$.

c) $f_0, f_3 \in W \Longrightarrow f_1, f_2 \in W$ [11].

[12].

[13]. p. 13

[page 17']

Commentary · Exposition [ms. 16']

The three cube rules measure different phenomena

An open subcategory here is a sieve: whenever an arrow has target in it, its source belongs to it as well. If $X = U \cup V$ is a cover by two sieves and $S = U \cap V$, every nerve simplex lies wholly in U or wholly in V, by looking at its last vertex. Therefore

$$NX = NU \amalg_{NS} NV.$$

Both inclusions are simplicial monomorphisms, so this strict simplicial pushout also represents the homotopy pushout of spaces. For cosieves the same reasoning uses the first vertex. This elementary geometry underlies the forward cube rule (6a).

The reverse rule (6b) is a different assertion: equivalences on the two pieces and the union must detect an equivalence on the intersection. A homotopy pushout does not ordinarily have this property. The third rule (6c) asks for detection on both pieces from the intersection and union. It is consequently misleading to call all three rules merely different forms of the ordinary gluing lemma. Their implications require arguments and, for the converse involving (6c), the stated retract hypothesis.

The manuscript's evolving terminology should be followed literally. Footnotes 11 and 13 remove (6c) from the name Loc (6), but Loc (6) still means *both* (6a) and (6b). Only (6a) is the familiar forward gluing property of ordinary weak homotopy equivalences. The commentary to XIIb-a, ms. 78–79, and XII Appendix, ms. 6–17, already separates these rules and gives the acyclic-link obstruction to reverse excision for W_∞; those examples need not be duplicated here. The finite-dimensional examples below settle the particular prediction made on the present page.

The small cubes in Proposition 3.1 can be checked by recording their vertical tuples. For Q the tuple is $(\mathrm{id}, i_1, \mathrm{id}, j_1)$; for Q' it is $(i_1, \mathrm{id}, j_1, \mathrm{id})$. Substitution immediately gives all three implications. The printed second occurrence of f_1 in the explanation of Q' means f_2. The doubtful id label belongs to the front vertical edge $X_1 \to X_1$, as the diagram and the edition both confirm.

N.B. Je distingue ces conditions comme Loc (6 a), Loc (6 b), Loc (6 c). Je présume qu'on n'aura jamais à faire usage à Loc (6 b), sans supposer également Loc (6 a), mais l'inverse n'est sûrement pas le cas. Ainsi, $W_0 = \{f \in \mathrm{Fl}\,\mathrm{Cat} \mid \pi_0(f) \text{ bijectif}\}$ satisfait a), mais non

9. Il est bon d'introduire des propriétés du diagramme

1°) $f_0 \in W$.

2°) f_1 et f_2 dans W.

3°) f_3 dans W.

et de formuler les parties a) b) c) de Loc (6) ainsi

a) $1° + 2° \Longrightarrow 3°$.

b) $2° + 3° \Longrightarrow 1°$.

c) $1° + 3° \Longrightarrow 2°$.

10. N.B. il semble que dans Loc (6 a), il suffit de l'énoncé en supposant f_0, f_2 des *isomorphismes, cf.* p. 22.

11. Il ne faudrait pas inclure c dans Loc 6 *cf.* p. 26.

12. N.B. On verra plus bas (3.3) que Loc 6 c implique déjà Loc (6 a, b), *i.e.* Loc (6) tout entier. Et si W satisfait Loc (1 bis), *i.e.* stable par facteurs directs, alors Loc (6 a, b) $\Longrightarrow$ Loc (6 c).

13. Je vais désigner par Loc (6) l'axiome Loc (6 a, b), sans y inclure c.

b). Je présume (sans avoir vérifié) que c'est pareil pour les W_n, $n \in \mathbf{N}$, de la "*n-équivalence homotopique*".

Il se pourrait que Loc (6 a) et Loc (6 a, b) soient équivalentes à la forme plus faible, et plus simple, suivante. (Si oui, c'est sous la forme la plus simple, bien sûr, qu'il convient de les énoncer.)

Loc 6′ (Axiome des carrés cocartésiens, ou de Mayer-Vietoris). Soit

$$\begin{array}{ccc} X_0 & \xrightarrow{i_2} & X_2 \\ {\scriptstyle i_1}\searrow & & \searrow{\scriptstyle j_1} \\ & X_1 & \xrightarrow{j_2} X_3 \end{array}$$

un diagramme MV dans Cat. Alors

a) Si $i_1 \in W$, $j_1 \in W$.

b) Si $j_1 \in W$, $i_1 \in W$.

Proposition 3.1. [page 18']

$$\begin{aligned} \text{Loc (6 a)} &\implies \text{Loc (6′ a)}, \\ \text{Loc (6 b)} &\implies \text{Loc (6′ b)}, \\ \text{Loc (6 c)} &\implies \text{Loc (6′ a, b)}\,[14]. \end{aligned}$$

Commentary · Examples [ms. 17']

A reverse-excision counterexample in every finite degree

Let W_n consist of functors inducing a bijection on components and isomorphisms on homotopy groups through degree n, for every choice of base point. Grothendieck's prediction is correct for every finite $n \geq 0$: W_n satisfies forward gluing, but fails (6b).

Here is one uniform finite example. Let K be the boundary of an $(n+1)$-simplex, so $|K| = S^n$, and take its two simplicial cones $a * K$ and $b * K$. For a simplicial complex L, let $F(L)$ be the poset of its nonempty faces, ordered by inclusion. Its nerve is the barycentric subdivision of L. The inclusions

$$F(K) \subset F(a * K), \qquad F(K) \subset F(b * K)$$

are sieves, and the two cone face posets cover

$$F((a * K) \cup_K (b * K)).$$

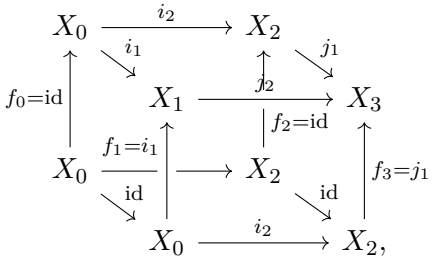
Thus they form an MV square. Map this entire square to the square all of whose vertices are e. The two piece maps are ordinary weak equivalences because cones are contractible. The union has homotopy type S^{n+1}, so its map to e belongs to W_n. The intersection has homotopy type S^n, so its map to e does not belong to W_n: for $n = 0$ it has two components, and for $n > 0$ its nth homotopy group is $\mathbb{Z}$. This is exactly a failure of (6b), using finite posets throughout.

For completeness, the forward assertion can be seen from the universal property of n-truncation. A map is an n-equivalence exactly when it induces equivalences of derived mapping spaces into all n-types. Mapping from a homotopy pushout into an n-type gives a homotopy pullback of mapping spaces. Equivalences on its three entries therefore give an equivalence on the result. Since an MV square has the homotopy-pushout realization just proved, (6a) follows. This proves the source's conjectural example rather than merely restating it. Notice that passage from S^n to its suspension is the mechanism of failure: the union moves the first detectable homotopy group one degree beyond the range being tested.

14. N.B. on a même Loc (6 c) $\implies \underbrace{\text{Loc 6 (a, b)}}_{\text{Loc (6)}}$.

Ça se voit sur le diagramme cubique

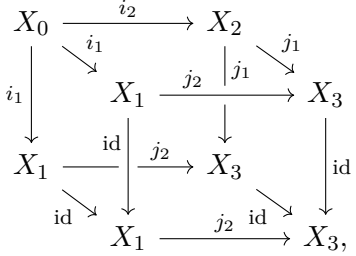
(Q)



où les f_0, f_1, f_2, f_3 de l'énoncé de Loc (6) deviennent id_{X_0}, i_1, id_{X_2}, j_1. Loc (6 a) donne donc $i_1 \in W \Longrightarrow j_1 \in W$, *i.e.* Loc (6′ a), et Loc (6 c) $j_1 \in W \Longrightarrow i_1 \in W$, *i.e.* Loc (6′ b). Pour Loc (6 b) $\Longrightarrow$ Loc (6′ b), on utilise (*cf.* XII, p. 78) le cube

p. 14

(Q′)



où $f_0 = i_1$, $f_1 = \mathrm{id}_{X_1}$, $f_1 = j_1$, $f_3 = \mathrm{id}_{X_3}$, donc

$$j_1 \in W \Longrightarrow i_1 \in W, \quad \textit{i.e.}\ \text{Loc (6 b)} \Longrightarrow \text{Loc (6′ b)},$$

et si on a Loc (6 c), on voit que $i_1 \in W \Longrightarrow j_1 \in W$, *i.e.* Loc (6′ a).

Commentaire 3.2. Loc (6 a) m'apparaît comme le "minimum vital" pour établir l'existence d'une théorie des cofibres W-homotopique dans Cat. Pour cela, il faut en effet disposer d'un critère serviable assurant qu'une flèche $i : A \longrightarrow B$ dans Cat est dans Cof_W.

On trouve, quand on dispose de Loc (6 a) (sous forme duale, *i.e.* pour les *fermés,* c'est kif-kif), que les *immersions ouvertes de Dwyer* sont dans Cof_W, ce qui suffit pour avoir *une bonne théorie des cofibres W-homotopiques, dans laquelle les carrés MV sont W-cocartésiens* (chose a laquelle on tient absolument). [page 19]

Quant à Loc (6 b), si je dispose à la place de Loc (6′ b) + Loc (6 a), je dois m'en tirer pour prouver Loc (6). Dans XII Appendice (pages 265 ff.) je prouve, paraît-il, que (avec les notations actuelles) Loc (6′) implique Loc (6), et même Loc (6 c) si W est stable par facteurs directs (condition Loc (1 bis)). Mais la démonstration est incomplète, puisque j'admets tacitement que les immersions de Dwyer sont dans Cof_W, chose que je n'avais établie que moyennant l'axiome W (7), en fait moyennant Loc (6 a) (ce qui est une partie de W (7)).

Mais ici me vient l'idée du lemme de K. S. Brown, cité par Baues, que je vais redonner ici : [page 20]

Commentary · Exposition [ms. 19]

The Dwyer neighborhood and the missing gluing step

A Dwyer inclusion $A \hookrightarrow B$ is a sieve whose inclusion into its generated cosieve U has a right adjoint $r : U \to A$ restricting to the identity on A. Thus $A \to U$ is a categorical homotopy equivalence. Put $V = B \setminus A$ and $T = U \cap V$. Both U and V are cosieves, and $B = U \cup_T V$. On pushing $A \to B$ out along $A \to A'$, one obtains the corresponding cover

$$B' = U' \cup_T V, \qquad U' = U \amalg_A A'.$$

The inclusion $A' \to U'$ again has a right-adjoint retraction, induced by r, so is in every W. This is the neighborhood calculation in Thomason (1980, Definition 4.1 and Proposition 4.3, pp. 313–317);

its proof does not require the claim elsewhere in that paper that all retracts of Dwyer maps are Dwyer maps, disproved in Cisinski (1999).

It explains both uses of the excision axioms here. If $A \to A'$ is in W, then $U \to U'$ is in W by the two homotopy-equivalent neighborhoods. The cube from the first closed cover to the second has its intersection and V maps equal to identities. The restricted case of (6a) therefore gives $B \to B' \in W$. This proves that Dwyer inclusions are intrinsic cofibrations, since their pushouts are again Dwyer inclusions.

Alternatively, suppose $A \to B$ itself is in W and assume (6′a,b). Then $U \to B$ is in W. The reverse single-square rule, applied to $B = U \cup_T V$, gives $T \to V \in W$. The forward rule in the new cover gives $U' \to B' \in W$, hence $A' \to B' \in W$. Thus trivial Dwyer inclusions are stable under every pushout. Opposite invariance justifies using closed covers in both arguments. This is precisely condition (4) in Brown's lemma, and makes explicit where both directions of (6′) enter the second proof.

The other factorization hypothesis also has a concrete explanation. Let $J = (0 \to 1 \leftarrow 2)$ and form

$$M(f) = (A \times J) \amalg_{A \times \{2\}} B$$

for $f : A \to B$. The inclusion of $A \times \{0\}$ is Dwyer: its cosieve neighborhood is $A \times (0 \to 1)$, with right-adjoint projection to A. The projection $M(f) \to B$ is a categorical homotopy equivalence. Indeed, the endomorphism of J sending 0 to 1 and fixing $1, 2$ receives natural transformations from both id_J and the constant functor at 2, stationary at 2; these descend after the pushout and contract $M(f)$ onto B. Hence every functor factors as a Dwyer inclusion followed by a member of W. No transfinite construction or functorial factorization axiom is needed for this particular factorization.

Lemme 3.2.1. (K. S. Brown) *Soit $(\mathcal{M}, W)$ un localiseur (W modérément saturé), muni d'un ensemble $C \subset \mathrm{Fl}\,\mathcal{M}$ satisfaisant les conditions :*

1) *C stable par composition et par cochangement de base* (N.B. *on n'a à supposer l'existence des sommes amalgamées* $\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\searrow & & \searrow \\ & X_2 & \longrightarrow X_3 \end{array}$ *que si $i_1 \in C$), contient les isomorphismes.*
2) *$\mathcal{M}$ contient un objet initial, $\emptyset_{\mathcal{M}}$, et pour tout objet X de $\mathcal{M}$, $\emptyset_{\mathcal{M}} \longrightarrow X$ est dans C.* (N.B. (1) *et* (2) *impliquent que $\mathcal{M}$ est stable par sommes finies.*)
3) *Toute flèche f de $\mathcal{M}$ se factorise en pi, avec $i \in C$, $p \in W$.*
4) *$C \cap W$ est stable par cochangement de base,* i.e. *si*

(∗)
$$\begin{array}{ccc} X_0 & \xrightarrow{i_1} & X_1 \\ {\scriptstyle i_2}\searrow & & \searrow \\ & X_2 & \xrightarrow[j_1]{} X_3 \end{array}$$

est cocartésien et $i_1 \in C \cap W$, alors $j_1 \in C \cap W$. (N.B. *$j_1 \in C$ est déjà contenu dans* (1), *il reste à exiger $j_1 \in W$.*) p. 15

Sous ces conditions, on a $C \subset \mathrm{Cof}_W$.

On peut aussi énoncer le lemme ainsi

Corollaire 3.2.2. *Supposons que C satisfait* (1), (2), (3). *Alors les conditions suivantes sur C sont équivalentes.* [page 21]

a) *Condition* (4) *ci-dessus,* i.e. *stabilité de $C \cap W$ par cochangement de base.*
a′) *$C \cap W \subset W^{\mathrm{univ}}$.*
b) *$C \subset \mathrm{Cof}_W$.*
b′) *Pour tout carré cocartésien*

$$\begin{array}{ccc} X_0 & \xrightarrow{i} & X_1 \\ {\scriptstyle p}\searrow & \textit{coc.} & \searrow{\scriptstyle q} \\ & X_0' & \xrightarrow{j} X_1' \end{array}$$

avec $i \in C$, $p \in W$, on a $q \in W$.

Démonstration. On a

$$\begin{aligned} \mathrm{b} &\iff \mathrm{b}' \\ \mathrm{a} &\iff \mathrm{a}' \end{aligned}$$

car C est stable par cochangement de base. On a $\mathrm{b} \Longrightarrow \mathrm{a}'$ car $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ grâce à l'hypothèse 3°). On a $\mathrm{a} \Longrightarrow \mathrm{b}'$ par le lemme de Brown.

On va appliquer ceci au cas $\mathcal{M} = \mathrm{Cat}$, $C =$ immersions ouvertes de Dwyer. Les conditions 1), 2) ne dépendent pas de W, elles sont vérifiées, et 3) aussi dès que W est un localiseur fondamental. Quant à 4), c'est une conséquence de l'ancien W (7 bis), *i.e.* Loc (6′) (on a besoin apparemment de Loc (6′ a *et* b).) On en conclut $C \subset \mathrm{Cof}_W$ - c'est merveilleux! (Mais je ne sais pas prouver le lemme …) [page 22]

Commentary · Exposition [ms. 21]

A complete proof of the Brown lemma used here

The relevant historical result is the dual of the properness argument for categories of fibrant objects, not just the isolated factorization lemma: see Brown (1973, Part I, §1, pp. 420–422, and §4, Lemmas 1–2, pp. 428–429). A modern one-sided formulation appears in Rădulescu-Banu (2009, §§1.3–1.4). Here is a proof directly under the four displayed hypotheses, filling the gap Grothendieck explicitly acknowledges on ms. 22.

First, Brown's functor criterion. A functor F from such a category to a category with a two-out-of-three class of equivalences preserves all W-maps if it preserves $C \cap W$. For $w : X \to Y$ in W, factor

$$(w, \mathrm{id}_Y) : X \amalg Y \xrightarrow{k} Z \xrightarrow{p} Y, \qquad k \in C,\ p \in W.$$

The restrictions k_X, k_Y belong to C, because coproduct inclusions do, and both are in W by two-out-of-three. Their images under F are equivalences. Since $pk_Y = \mathrm{id}_Y$, $F(p)$ is an equivalence, and so is $F(w) = F(p)F(k_X)$.

Second, change the initial object. For each A, consider the category $\mathcal{C}_A$ of arrows $A \to X$ belonging to C, with all maps under A. Take equivalences and cofibrations to be those whose underlying maps are in W and C. It satisfies the same hypotheses: its initial object is id_A, pushouts and factorizations are inherited, and composing a factorization's first map with $A \to X$ keeps the new object in $\mathcal{C}_A$. For any $u : A \to A'$, the functor

$$u_! : \mathcal{C}_A \longrightarrow \mathcal{C}_{A'}, \qquad (A \to X) \longmapsto (A' \to A' \amalg_A X)$$

preserves trivial cofibrations by condition (4). The functor criterion shows that it preserves all equivalences. This assertion concerns maps *between* cofibrant objects under A, and is not yet the desired properness statement; the distinction prevents a circular proof.

Third, prove properness. We must show that the pushout of $w : A \to A'$ in W along $i : A \to B$ in C is a W-map $B \to P$. Applying the factorization of the first paragraph to $(w, \mathrm{id}_{A'})$ writes $w = pj$, where $j \in C \cap W$ and p has a section $s \in C \cap W$. Pushing out j is covered by (4). We are reduced to the case $ws = \mathrm{id}_{A'}$, with $s : A' \to A$ a trivial cofibration.

Form $B_1 = A \amalg_{A'} B$, using $s : A' \to A$ and $is : A' \to B$. The map $q : B \to B_1$ is a pushout of s, hence a trivial cofibration. There is a retraction $g : B_1 \to B$ induced by (i, id_B), so $g \in W$. The structural maps $A \to B_1$ and $A \to B$ belong to C: the former is a pushout of $is \in C$. Consequently g is an equivalence in $\mathcal{C}_A$. Apply $w_!$. Since $ws = \mathrm{id}_{A'}$, associativity of pushouts gives

$$A' \amalg_A B_1 \simeq B, \qquad A' \amalg_A B = P.$$

Under these identifications, $w_!(g)$ is precisely $B \to P$, which is therefore in W. Stability of C under pushouts makes the same argument apply to every cobase change of i, proving $C \subset \mathrm{Cof}_W$.

For the converse in Corollary 3.2.2, suppose $C \subset \mathrm{Cof}_W$, let $i : A \to B$ be in $C \cap W$, and push it out along an arbitrary $f : A \to A'$. Factor $f = pj$ with $j \in C$ and $p \in W$. Pushing i out along j gives a W-map because j is an intrinsic cofibration. Pushing p out along that new C-map gives another W-map. Two-out-of-three, applied to their composite and p, proves that the final pushout of i is in W. This verifies all four equivalent conditions without assuming the equality $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ as an unexplained extra theorem. The argument only uses identities and two-out-of-three in W; retract closure and functorial factorizations are unnecessary here.

Cela légitime la démonstration faite dans XII *loc. cit.*, et on trouve la

Proposition 3.3. *La condition* Loc (6′) (*axiome du carré cocartésien, forme forte* (a) *et* (b)) équivaut *à* Loc (6 a, b) (*axiome des homomorphismes des carrés cocartésiens*) *et est* impliqué *par* Loc (6 c). *Si W satisfait à* Loc (1 bis), i.e. *est stable par facteurs directs, alors* Loc (6′) équivaut *à* Loc (6 c) *et à* Loc (6 a b c).

Commentaire 3.4. Ainsi, on trouve une théorie des cofibres W-homotopiques *soit* sous l'hypothèse Loc (6′) (équivalente à Loc (6) plus compliquée), soit sous l'hypothèse plus faible (et plus compliquée) Loc (6 a).

Dans le premier cas, c'est une théorie de nature très particulière, puisque les carrés W-cocartésiens sont "réversibles" (*i.e.* on a non seulement $i_1 \in W \Longrightarrow j_1 \in W$, mais aussi l'implication inverse).

Il reste donc la question si Loc (6′ a) suffit pour une théorie des cofibres W-homotopiques, ou ce qui revient sûrement au même, si ça implique déjà Loc (6 a) (donc lui est équivalent). C'est entièrement suspendu à la *question* :

p. 16 [page 23] ? ‖

l'axiome Loc (6′ a) (axiome du carré cocartésien, forme directe) implique-t-il que l'ensemble des immersions ouvertes de Dwyer qui sont dans W (immersions de Dwyer W-triviales) est stable par cochangement de base ? En fait, on voit qu'il suffirait pour cela de vérifier l'axiome du cube (ou des homomorphismes de carrés cocartésiens), forme directe Loc (6 a), dans le cas où f_0 et f_2 sont des identités, donc à prouver qu'alors $f_1 \in W \Longrightarrow f_3 \in W$ (ce qui donnerait directement $\mathrm{Dw} \subset \mathrm{Cof}_W$). Mais je ne vois aucun moyen de le prouver …

Supposons maintenant que la condition Loc (6 a) est vérifiée, donc qu'on dispose d'une "bonne théorie des cofibres W-homotopiques", donc aussi d'une théorie des carrés W-cocartésiens (*cf.* XII, p. 258-262). Il y a deux questions qui se posent, concernant la validité, avec ces hypothèses plus faibles, des développements de XII :

1°) Soit un carré commutatif

$$\begin{array}{ccc} X_0 & \longrightarrow & X_2 \\ \searrow & & \searrow \\ & X_1 & \longrightarrow & X_3 \end{array}$$

dans Cat, est-il vrai que ce carré est W-cocartésien si et seulement si la flèche

$$\text{canonique } \int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array} \longrightarrow X_3 \text{ est dans } W \text{ ?}$$ [15]

[page 24]

2°) Est-il encore vrai que si A est dans Cat, et $u : F \hookrightarrow G$ un homomorphisme dans $A^\wedge$, alors $u \in \mathrm{Cof}_{W_A}$, et si $u \in W$, $u \in W_A^{\mathrm{univ}}$, donc a-t-on

$$\mathrm{Mono}_{A^\wedge} \subset \mathrm{Cof}_{W_A}, \qquad \mathrm{Mono}_{A^\wedge} \cap W_A \subset W_A^{\mathrm{univ}} \text{ ?}$$ [16]

---

15. Mieux, dans tous les cas, quand on a un diagramme $\begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array}$, son intégrale $\int$ est une somme amalgamée W-homotopique dudit diagramme (moyennant Loc (6 a) sans plus, et même seulement Loc (6′ a)).

16. N.B. Loc (6′ a) suffit au lieu de (Loc (6 a)).

Mais 1°) est prouvé dans XII, prop. 20, p. 262 - je viens de vérifier que la démonstration est valable dans les hypothèses actuelles [17], pourvu que je vérifie que le corollaire 3, p. 149 dans XII est valable, *i.e.*

1° bis) Si $\begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X \end{array}$ est un carré MV dans Cat, alors $\int \begin{array}{ccc} & & X_1 \\ X_0 & \nearrow & \\ & \searrow & \\ & & X_2 \end{array} \longrightarrow X$ est dans W (donc, moyennant 1°), le carré est W-cocartésien).

Pour le vérifier, je suis amené à vérifier que le théorème 4 dans loc. cit. (p. 146) est encore valable. p. 17

Je viens de le faire, et ai annoté en conséquence le théorème 4 et ses corollaires (certains résultats (cor. 4 et cor. 5) utilisant W (7 bis) dans toute sa force, *i.e.* Loc (6′), *i.e.* Loc (6 a *et* b). Mais le corollaire 3 du théorème 4 est O.K.

Je viens aussi de vérifier la validité de 2°), en vérifiant la démonstration du corollaire du théorème 5 (XII, p. 168) et en l'annotant en conséquence.

3.5. Résumons : on a [page 25]

$$\text{Loc (6 c)} \implies \text{Loc (6)} \iff \text{Loc (6}') \tag{3.5.1}$$

et

$$\text{Si } W = W^{\natural}, \text{ on a Loc (6 c)} \iff \text{Loc (6)} \iff \text{Loc (6}'). \tag{3.5.2}$$

D'autre part

$$\text{Loc (6 a)} \implies \text{Loc (6}' \text{ a)}, \tag{3.5.3}$$

et j'ignore si l'implication inverse

$$\text{Loc (6}' \text{ a)} \overset{?}{\implies} \text{Loc (6 a)}$$

est valable [18]. Pour établir Loc (6 a), il suffit de le faire en tous cas quand f_0 et f_2 sont des isomorphismes.

L'axiome Loc (6 a) semble le "minimum vital" pour que W donne lieu à une "bonne théorie des cofibres W-homotopiques", donc aussi des carrés W-cocartésiens. De façon plus précise, j'ai envie de prouver ceci :

Proposition 3.6. *L'axiome* Loc (6 a) *équivaut à la conjonction des deux propriétés suivantes.*

Loc (6 A) Cat a "assez de W-cofibrations", *i.e.* toute flèche f de Cat se factorise en $f = pi$, avec $i \in \mathrm{Cof}_W$ et $p \in W$. (Cela signifie aussi que Cat muni de (W, Cof_W) est une catégorie à cofibrations de K. S. BROWN - ANDERSON, et tous les résultats de la théorie de BROWN-QUILLEN sont applicables. En particulier, on a une bonne théorie des carrés W-cocartésiens.) p. 18 [page 26]

Loc (6 B) Si X dans Cat est réunion de deux ouverts U, V, d'intersection S, alors le carré

$$\begin{array}{ccc} S & \longrightarrow & U \\ \downarrow & & \downarrow \\ V & \longrightarrow & X \end{array}$$

17. Mais elle utilise de plus Loc (3 bis). Mais avec une autre démonstration plus simple que je viens de rajouter (p 264), on peut s'en passer (et on obtient une vison plus complète de la situation).

18. C'est le cas en présence de Loc (8, 9).

est W-cocartésien [19].

D'autre part, la condition Loc (6 A) peut être remplacée par la condition (en apparence plus faible)

Loc (6 A′) Tout morphisme de Dwyer est dans Cof_W.

Corollaire 3.7. *La condition* Loc (6), i.e. Loc (6 a, b) (*ou encore la condition équivalente* Loc (6′)) *équivaut à la conjonction des conditions* Loc (6 A) (*ou* Loc (6 A′)) *et* Loc (6 B) *ci-dessus*), *et de la condition*

Loc (6 C) Pour tout carré W-cocartésien

$$\begin{array}{ccc} X_0 & \xrightarrow{i} & X_1 \\ \downarrow & & \downarrow \\ X_2 & \xrightarrow{j} & X \end{array}$$

dans Cat, $j \in W \Longrightarrow i \in W$.

D'autre part, je soupçonne que Loc (6 c) n'est guère vérifiée que si $W = W^{\natural}$ (cas qui est le plus commun bien sûr), auquel cas Loc (6 c) *équivaut* à Loc (6). Il n'y a sans doute pas lieu de l'inclure dans Loc (6), pas plus que $W = W^{\natural}$ dans Loc (1). [page 27]

Commentary · Present state [ms. 25]

What the excision reformulations establish

The new contribution of the present argument, relative to the earlier cube calculations in XIIb-a and XII Appendix, is the missing cofibration criterion. The Dwyer-neighborhood proof and Brown's lemma establish it from (6′a,b), so the earlier passage from single-square reversibility to the two cube rules can now be used without assuming the desired cofibration theory in advance. The source's equivalence

$$(6\mathrm{a}, 6\mathrm{b}) \Longleftrightarrow (6'\mathrm{a}, 6'\mathrm{b})$$

should be read with that repair. Closure of W under retracts is a separate premise for the converse implication to (6c); it concerns W, not closure of the class of Dwyer inclusions under retracts. The latter closure is false (Cisinski, 1999, pp. 228–229).

There is also a short direct check of the cover comparison in question (1° bis). Let G be the Grothendieck construction on the span $S \to U$, $S \to V$, and $q : G \to X$ its canonical functor. For $x \in U$, the sieve condition gives $U/x = X/x$ and $V/x = S/x$. Thus G/x is the integral of the span $S/x \to X/x$, $S/x \xrightarrow{\mathrm{id}} S/x$. The redundant branch contracts by a zigzag of natural transformations, leaving X/x, which has terminal object id_x. The same argument applies if $x \in V$. Hence every G/x is contractible and $q \in W$ by (3). This proves the cover comparison without (3 bis), in agreement with the revised proof mentioned in footnote 17. For general spans, identifying their integral with the intrinsic homotopy pushout still uses the cofibration theory; the earlier commentary to XIIb-b, ms. 120–149, discusses that distinction in detail.

The statements about monomorphisms of presheaves on ms. 24 refer to W_A as defined using the category of elements. They are claims about the induced relative category, not a claim that every fully faithful functor in Cat is a cofibration. Nor does "enough intrinsic cofibrations" by itself provide both lifting systems of a Quillen model structure. The conclusion licensed at this stage is Brown's one-sided cofibration theory, with its homotopy pushouts. Proposition 3.6 and footnote 19 further reduce the issue to enough such cofibrations; the promised implication (6A)⇒(6B) is addressed at ms. 66–68, not proved at its first announcement here.

The bare implication (6′a)⇒(6a) is explicitly left as a question in this passage; footnote 18 supplies the additional assumptions (8,9), not an unconditional answer. Likewise the final suggestion that (6c) might force retract closure is a conjectural observation. For strong localizers retract closure follows independently from later saturation theory, but that does not answer the question for the

19. En fait, Loc (6 B) est conséquence de Loc (6 A), laquelle donc est *équivalente* à Loc (6 a). (*Cf.* esquisse de démonstration dans 4.4.15, pages 66-68.)

weak framework of Definition 1.1. No result settling these precise weakest-hypothesis formulations was found in the sources checked. The next two blocks instead extract a definite, and rather restrictive, consequence of the reverse cube axiom itself.

Commentary · Development (commentators') [ms. 27]

Integral homology supplies a reversible localizer

Homological Theorem A and reversible excision. [ours] Let $C_*(A) = C_*(NA;\mathbb{Z})$ be the integral nerve chain complex and define

$$W_{H\mathbb{Z}} = \{f : A \to B \mid H_k(NA;\mathbb{Z}) \xrightarrow{\sim} H_k(NB;\mathbb{Z}) \text{ for all } k \geq 0\}.$$

This is a strongly saturated fundamental localizer satisfying (3 bis), (5), and all three cube rules (6a,b,c). We include the proof because the ensuing minimality statement needs the actual Theorem A hypothesis, not just the observation that homology has a Mayer–Vietoris sequence. The ingredients are classical; the formulation and proof assembled here are offered as a development of this passage, with no priority claim.

The local-to-global assertion. For $u : A \to S$, use the first-quadrant double complex $D(u)_{p,q}$ freely generated by strings

$$a_0 \longrightarrow \cdots \longrightarrow a_q, \qquad u(a_q) \longrightarrow s_0 \longrightarrow \cdots \longrightarrow s_p.$$

The vertical faces compose or omit arrows in the A-string, including composition with the connecting arrow when a_q is omitted. The horizontal faces do the same for the S-string. Use unnormalized chains for this description and total differential $d_v + (-1)^q d_h$. There is an augmentation

$$r_A : \operatorname{Tot} D(u) \longrightarrow C_*(A)$$

which forgets s_0 in horizontal degree zero and is zero in positive horizontal degrees. For a fixed A-string, the horizontal complex is that of $u(a_q)\backslash S$. Its initial object is $\mathrm{id}_{u(a_q)}$, so its augmented chain complex is contractible. Filtering the total complex therefore proves that r_A is a quasi-isomorphism for every u.

Now take a triangle $A \xrightarrow{f} B \to S$. It induces $D(A \to S) \to D(B \to S)$. For a fixed string $s_0 \to \cdots \to s_p$, the corresponding vertical complexes are precisely $C_*(A/s_0)$ and $C_*(B/s_0)$. If all the comma comparisons are integral homology equivalences, this map is a quasi-isomorphism in every such column. The other first-quadrant filtration consequently makes the map on totals a quasi-isomorphism. There is no convergence issue: in each total degree only finitely many bidegrees occur. The naturality square for the two augmentations r_A, r_B identifies the resulting map with $C_*(f)$, proving (3 bis). Taking $B = S$ and using the terminal objects of S/s also proves the weak axiom (3).

For readers wishing to check the comparison with the ordinary homological Theorem A directly, there is a second augmentation $r_S : \operatorname{Tot} D(u) \to C_*(S)$, forgetting the A-string in vertical degree zero. The signed concatenation

$$h(a_\bullet, u(a_q) \to s_\bullet) = (-1)^q \big[u(a_0) \to \cdots \to u(a_q) \to s_0 \to \cdots \to s_p\big]$$

satisfies $dh + hd = r_S - u_* r_A$. The internal faces cancel in pairs; deleting the sole A-vertex when $q = 0$ yields r_S, and deleting the sole S-vertex when $p = 0$ yields $-u_* r_A$. If each A/s has the homology of a point, the vertical filtration makes r_S a quasi-isomorphism as well. This explicit homotopy verifies that the isomorphism produced by the two filtrations is induced by the given functor, rather than merely an abstract isomorphism between homology groups.

The other localizer axioms. A terminal object contracts the nerve, giving (2). Identities, two-out-of-three, mild saturation, and retract closure follow by applying integral homology. Strong saturation follows too: homology factors through $\mathrm{Cat}[W_{H\mathbb{Z}}^{-1}]$, so any arrow inverted in that localization already induces a homology isomorphism. In degree zero, a functor induces the free abelian map on its function of components; such a map is an isomorphism exactly when the function is a bijection. This proves (5).

The three cube rules. For an open MV square, the nerve decomposition gives a natural short exact sequence of normalized chain complexes

$$0 \longrightarrow C_*(S) \xrightarrow{(i_*,-j_*)} C_*(U) \oplus C_*(V) \longrightarrow C_*(X) \longrightarrow 0.$$

A map of MV squares gives a map of these short exact sequences. If two of the three vertical maps are quasi-isomorphisms, so is the third, by their long exact homology sequences. This yields (6a) and (6b). For (6c), the two outer quasi-isomorphisms force the middle map to be a quasi-isomorphism. That middle map is block diagonal, so each of its two summand maps is a quasi-isomorphism separately. These arguments apply to disconnected and empty categories as written. The same proof works for closed covers, or one may pass to opposite categories. Thus the homological class is an actual example of the reversible theory sought here, rather than an analogy with an exact sequence.

Commentary · Development (commentators') [ms. 27]

The least localizer satisfying reverse cube excision

Minimality forced by one suspension. [ours] Among weak fundamental localizers in the sense of Definition 1.1 satisfying (6b), the least one is $W_{H\mathbb{Z}}$. In particular, the least such localizer satisfying both directions of Loc (6), or satisfying all three cube rules, is again $W_{H\mathbb{Z}}$. The previous block proves the required upper bound. We now prove that every such W contains every integral homology equivalence. This gives a precise consequence of Grothendieck's reversibility requirement; it does not claim that every larger reversible weak localizer is strong or saturated.

By Cisinski (2006a, Theorem 6.1.18), $W_\infty \subset W$ even under the weak axioms. Let $f : A \to B$ be an integral homology equivalence. Its map on components is a bijection, and its restriction to each matched pair of components is a homology equivalence: homology is the direct sum over components and the induced map respects this decomposition. Write these maps as $f_i : A_i \to B_i$, with each category nonempty and connected. If there are no components, f is $\mathrm{id}_\emptyset$ and the assertion is immediate.

For a nonempty category C, form C_- and C_+ by adjoining one new terminal object in each case, and let ΣC be the category obtained by adjoining both objects, with no morphisms between them. Then

$$\begin{array}{ccc} C & \longrightarrow & C_- \\ \downarrow & & \downarrow \\ C_+ & \longrightarrow & \Sigma C \end{array}$$

is an open MV square: both cones are sieves, their intersection is C, and their union is ΣC. The nerve realizes the union of the two ordinary cones on $|NC|$, hence the unreduced suspension of $|NC|$. Each cone is contractible. Because C is connected, its suspension is simply connected, by van Kampen, and the suspension isomorphism identifies its reduced integral homology with that of C, shifted one degree.

Apply this construction to each f_i. The map on either cone is in W_∞. The map $|N\Sigma A_i| \to |N\Sigma B_i|$ is a homology equivalence between simply connected CW complexes. The homological Whitehead theorem makes it a weak homotopy equivalence (Hatcher, 2002, Corollary 4.33, pp. 367–368). Thus the three maps on the pieces and union belong to $W_\infty \subset W$. Reverse cube excision (6b) forces $f_i \in W$.

One must not finish by taking a sum of these conclusions unless sum closure has been assumed. Instead take the disjoint union of the entire componentwise cube diagrams *before* applying (6b). This is a single open MV cube whose intersection map is the original f. The two cone maps and the union map are ordinary weak equivalences globally: their component functions are the same bijection, and their restrictions are the weak equivalences already established. Therefore they belong to W without any use of (4), and (6b) directly gives $f \in W$. This proves the lower bound and the theorem.

The conclusion also locates the obstruction to identifying reversible excision with ordinary homotopy theory. Noncontractible acyclic spaces, such as the example recalled in Hatcher (2002, discussion following Corollary 4.33, p. 368), have the homology of a point but a nontrivial fundamental group; categorical models give maps in $W_{H\mathbb{Z}} \setminus W_\infty$. Such examples were already used in the commentary on Chapter XII to disprove reverse excision for W_∞. The new content here is the

universal lower bound: reverse cube excision forces *every* integral homology equivalence, not just the particular acyclic examples. We did not find this precise least-localizer formulation in the primary sources examined; its proof is included in full so that the assertion can be assessed independently of any priority question.

p. 19

4 Axiome Loc (7) des W-fibrations

4.1. Rappelons la notion de *W-fibration faible* (XII p. 51, 52) : la version la plus forte des 8 conditions équivalentes pour que $f : X \longrightarrow S$ soit une fibration faible est celle-ci.

(Ff) $\forall\ S'' \longrightarrow S'$ dans Cat/S, avec S', S'' *contractiles,* le morphisme correspondant $X'' = X \times_S S'' \longrightarrow X' = X \times_S S'$ est dans W.

Quand W est fortement saturé, alors ceci implique la condition, en apparence plus forte :

(Ff′) $\forall\ S'' \longrightarrow S'$ dans Cat/S, telle que $S'' \longrightarrow S'$ soit un homotopisme, $X'' \longrightarrow X'$ est dans W.

Je désigne par Fibf_W l'ensemble des fibrations faibles. Je résume dans un diagramme d'inclusions les principales classes de flèches dans Cat que nous avons rencontrées, s'apparentant à des classes de "fibrations" (pour W, dans un sens convenable) :

[page 28]

$$\begin{array}{ccccccccc} & & & & \mathrm{Fib}_W & & & & \\ & & & & \downarrow & & & & \\ \mathrm{Parf}_\omega & \longrightarrow & \mathrm{Parf}_W & \longrightarrow & \mathrm{Fibf}_W & \longleftarrow & \underbrace{\mathrm{Catfib}^W}_{\mathcal{F}_W} & \longleftarrow & \underbrace{\mathrm{Catfib}^\omega}_{\mathcal{F}_\omega}\ , \end{array}$$

où Fib_W désigne les W-fibrations ($f : S' \longrightarrow S$ est dans Fib_W si et seulement si le foncteur changement de base $X \longmapsto X' = X \times_S S'$ de Cat/S dans Cat/S' est compatible avec les localiseurs W_S et $W_{S'}$), Catfib les Cat-fibrations, Catfib^W (resp. Catfib^ω) les Cat-fibrations telles que les foncteurs changement de base soient dans W (resp. soient dans W_ω, *cf.* déf. plus bas) [20], Parf_W les morphismes W-parfaits, *i.e.* W-lisses et W-propres, enfin Parf_ω les morphismes W_ω-parfaits, où W_ω est le plus petit localiseur satisfaisant Loc (1, 2, 3). Parmi les espèces de "fibrations" envisagées, c'est cette dernière seulement, avec Catfib^ω, qui est indépendante de W.

À ce titre, elle me paraît jouer un rôle similaire, dual, à celui des morphismes de Dwyer, et je présume qu'on peut leur faire jouer un rôle similaire. En tous cas, on sait que toute flèche [page 29]
f de Cat se factorise en pi, avec $i \in W_\omega$ et $p \in \mathrm{Parf}_\omega$, donc $i \in W$ pour tout W satisfaisant Loc (1, 2, 3). On sait aussi que Catfib^ω satisfait l'énoncé de factorisation forte, f se factorise en $f = pi$, $i \in W_\omega$, $p \in \mathrm{Catfib}_0^\omega$. [21]

Commentary · Exposition [ms. 28]
Three meanings of fibration

Grothendieck is looking for maps along which an ordinary pullback already performs the correct homotopical operation. This explains the change of terminology at the beginning of the section: the class now called Fib_W is defined by the operation it makes possible, whereas the former fibrations, now Fibf_W, are recognized by tests over contractible categories. The difference is the central issue of the next forty manuscript pages.

20. N.B. $\mathrm{Catfib}_0^W = \mathrm{Catfib} \cap \mathrm{Fibf}_W$, $\mathrm{Catfib}_0^\omega = \mathrm{Catfib} \cap \mathrm{Fibf}_{W_\omega}$

21. Donc la propriété de factorisation est valable pour toutes ces classes, sauf peut-être Fib_W.

For clarity, write W_S for the maps over S whose underlying functor belongs to W, and write W_S^{u} for maps which remain in W after every change of base $T \to S$. If $p : X \to S$, its intrinsic fibration property says that every W-equivalence $T' \to T$ over S induces a W-equivalence

$$X \times_S T' \longrightarrow X \times_S T.$$

Both T and T' vary. Testing only maps $T' \to S$ misses the required stability after a first change of base. By contrast, weak fibrancy makes the same demand only when T and T' are contractible in the categorical homotopy sense used in the preceding chapters. A category with contractible nerve need not admit a contraction through a finite chain of natural transformations; these two meanings should not be interchanged in reading the test.

The third notion, a Grothendieck fibration, concerns Cartesian lifts of arrows of S. A cleavage gives inverse-image functors $\alpha^* : X_t \to X_s$ for $\alpha : s \to t$. Membership in $\mathcal{F}_W$ means that these functors belong to W. Thus $\mathcal{F}_W$ combines strict categorical transport with homotopical local constancy. It supplies convenient representatives of families, but its definition does not say that pullback preserves every equivalence of bases.

Perfect maps enter from a different direction. Smoothness controls one comma-category comparison and properness the opposite comparison; perfection imposes both. The comparison with strict bifibrations was already discussed in the commentary on Chapter VI, manuscript p. 186. Here perfection matters because it supplies replacements, and because the class of perfect maps is closed under composition. Those facts make Brown's argument available. No lifting-property definition of a Quillen fibration is being introduced.

The inclusion diagram is consequently asymmetric in a useful way. The geometric classes $\mathcal{F}_W$ and Parf_W sit inside the class detected on contractible bases; the intrinsic class Fib_W also sits there, but its inclusion in the other direction is a substantive assertion about W. Proving that reverse inclusion is the purpose of Loc (7).

Notons aussi que Parf_ω et Parf_W, ainsi que Fib_W, sont stables non seulement par changement de base (c'est le cas des 5 classes envisagées, qui de plus contiennent tous les isomorphismes), mais aussi par composition. (De plus, Fib_W est stable par facteurs directs, et je présume qu'il en est de même de Liss_W et de Prop_W, donc de Parf_W, en particulier de Parf_{W_ω}, mais je n'ai pas pris la peine de le vérifier.) Par contre, a priori Fibf_W n'est pas en général stable par composition, et Catfib^W, Catfib^ω n'ont pas de raison non plus d'être p. 20
stables par composition [22].

4.2. Finalement, il paraît plus "éclairant" d'introduire, au lieu du localiseur fondamental W_ω bien déterminée, un localiseur fondamental

$$W_0 \subset \mathrm{Fl}(\mathrm{Cat}),$$

qui jouera le rôle d'une sorte de "localiseur de référence" pour formuler certaines propriétés de W. On supposera toujours que

$$W \supset W_0.$$

Suivant les cas, W_0 pourra être W_ω, ou quelque autre localiseur fondamental défini comme [page 30]
le "plus petit" satisfaisant à des axiomes de stabilité donnés. Ou ça pourra être $\mathbf{W}_\infty$ (les équivalences faibles habituelles), quand on s'intéresse aux localiseurs qui contiennent $\mathbf{W}_\infty$ (les seuls, apparemment, avec lesquels on ait travaillé jusqu'à présent en homotopie). Ça pourra être aussi W, et les énoncés obtenus en spécialisant les résultats obtenus au cas où $W = W_0$ ne sont nullement tautologiques. Le diagramme de la page 28 se remplace alors par le diagramme

22. il faudrait expliciter par des exemples.

d'inclusions suivant, plus éclairant pour notre propos :

(4.2.1)
$$\begin{array}{ccccc} & & \mathrm{Fib}_W & & \\ & & \downarrow & & \\ \mathrm{Parf}_W & \longrightarrow & \mathrm{Fibf}_W & \longleftarrow & \mathcal{F}_W \\ \uparrow & & \uparrow & & \uparrow \\ \mathrm{Parf}_{W_0} & \longrightarrow & \mathrm{Fibf}_{W_0} & \longleftarrow & \mathcal{F}_{W_0} \\ & & \uparrow & & \\ & & \mathrm{Fib}_{W_0}, & & \end{array}$$

où pour tout localiseur fondamental W, je pose pour abréger

$$\mathcal{F}_W = \mathrm{Catfib}^W,$$

ce qui définit donc également $\mathcal{F}_{W_0}$. La classe Fib_W doit être regardée comme le point de mire de notre attention ; la plupart des propriétés de W que nous allons passer en revue ici, s'exprimeront par une inclusion de la forme [page 31]

$$\mathcal{F} \subset \mathrm{Fib}_W,$$

où $\mathcal{F}$ est une des sept autres classes du diagramme (4.2.1) ; à l'exception tout au plus du cas $\mathcal{F} = \mathrm{Fib}_{W_0}$, *i.e.* de l'inclusion $\mathrm{Fib}_{W_0} \subset \mathrm{Fib}_W$. Comme propriété sur W, cette inclusion paraît très faible et pour cela pas utile à grand chose, sauf bien sûr dans le cas où $\mathrm{Fib}_{W_0} \longrightarrow \mathrm{Fibf}_{W_0}$ est une égalité, de sorte que l'inclusion envisagée équivaut à $\mathrm{Fibf}_{W_0} \subset \mathrm{Fib}_W$ (et implique que Parf_{W_0} et $\mathcal{F}_{W_0}$ sont eux aussi contenus dans Fib_W). C'est là par contre, comme nous allons voir, une propriété assez forte et fort utile de W, impliquant notamment le fait que $(\mathrm{Cat}, W, \mathrm{Fib}_W)$ est une catégorie à fibrations.

p. 21
[page 32]

Commentary · Exposition [ms. 31]

Why a reference localizer is retained

The second localizer allows Grothendieck to separate the construction of a family from the equivalences which that family must preserve. A map may have transport belonging to a smaller class W_0, while the desired base-change theorem concerns the larger class W. This distinction is valuable precisely when changing the homotopy theory makes more maps into equivalences: additional equivalences of bases need not remain harmless for an existing family.

From $W_0 \subset W$ one immediately obtains

$$\mathcal{F}_{W_0} \subset \mathcal{F}_W, \qquad \mathrm{Fibf}_{W_0} \subset \mathrm{Fibf}_W, \qquad \mathrm{Parf}_{W_0} \subset \mathrm{Parf}_W.$$

Each defining condition on the left asks specified comparison functors to belong to the smaller class. There is no corresponding formal inclusion $\mathrm{Fib}_{W_0} \subset \mathrm{Fib}_W$. Here the quantified class of equivalences of bases has also enlarged. This is why the intrinsic fibrations appear at opposite ends of the diagram rather than in its vertical chain.

The pointwise pullback test at ms. 36 follows from this same distinction. For a class $\mathcal{F}$ stable under arbitrary base change, it suffices to check the outermost square with a map in W at its bottom. Indeed, to test $p \in \mathcal{F}$ against a map $T' \to T$ over S, first pull p back to T; the resulting map still belongs to $\mathcal{F}$, so the proposed single-square test applies. This proves the full intrinsic fibration property, with no assumption about the saturation of W beyond the ordinary equivalence calculus.

The observation about retracts at ms. 29 deserves the same care as in the commentary on Chapter XII, manuscript p. 209. If W is closed under retracts, then Fib_W is closed under retracts: every pullback comparison for a retract of p is a retract of the corresponding comparison for p. Without that hypothesis this argument does not establish the assertion. We retain the source's claim in the audit, with this dependency explicit, rather than silently replacing mild saturation by retract closure. It has no role in the composition and replacement arguments below.

Aux fins de référence ultérieure, je vais formuler maintenant les divers aspects principaux de *l'axiome des W-fibrations* Loc (7). La formulation à laquelle j'étais parvenu (*cf.* I, VI, XII) utilisait la notion que j'appelle à présent "W-fibrations *faible*" (l'ex W-fibration, désormais détrônée !), et sous sa forme la plus forte s'écrivait comme l'inclusion

$$\mathrm{Fibf}_W \subset \mathrm{Fib}_W.$$

Cette notion de W-fibration faible, tout comme celle de morphisme W-parfait, si utiles soient-elles, m'apparaît cependant un peu technique, et je préfère donner une formulation qui ne fasse appel qu'à des notions plus familières et d'apparence plus élémentaires. C'est pourquoi je vais remplacer Fibf_W par la sous-classe $\mathcal{F}_W = \mathrm{Catfib}^W$, ce qui (on le verra) revient en fait au même.

Cette classe $\mathcal{F}_W$ est extrêmement naturelle, et s'impose même dans le yoga des dérivateurs. [page 33] En effet, dans la construction de $\mathrm{HOT}_W(S)$ pour S dans Cat, on passe à une catégorie de fractions de la catégorie

$$\mathrm{Cat}(S) = \underline{\mathrm{Hom}}(S, \mathrm{Cat}),$$

par un localiseur $W(S)$ (que nous avions généralement noté W_S^{g} ou W_S^{u} …). Cette catégorie $\mathrm{Cat}(S)$ est d'autre part isomorphe à la catégorie

$$\mathrm{Fibsc}\, S$$

des catégories fibrées et scindées sur S, avec comme flèches les S-foncteurs qui respectent le scindage (et en particulier cartésiens) (*cf.* VI, et notamment le diagramme p. 175). On a vu (*loc. cit.*) que la catégorie localisée $\mathrm{HOT}_W(S)$ est équivalente à la catégorie localisée de la catégorie très voisine

$$\mathrm{Fib}\, S \qquad \text{(que je préférerais maintenant noter } \mathrm{Catfib}\, S \text{ ou } \mathrm{Catfib}_S\text{)},$$

formé des catégories *fibrées* sur S, avec comme flèches les seuls foncteurs cartésiens. (On [page 34] fera attention cependant que le foncteur canonique $\mathrm{Fibsc}\, S \longrightarrow \mathrm{Fib}\, S$ est fidèle mais non pleinement, néanmoins il induit une équivalence sur les catégories localisées, en définissant de façon évidente, par la W-équivalence fibre par fibre, *i.e.* comme W_S^{u}, le localiseur pertinent sur $\mathrm{Fib}\, S$.) Donc pour cette raison de principe, et d'ailleurs tout au long du développement de la théorie de HOT_W, la catégorie $\mathrm{Fib}\, S$ et par là, les foncteurs fibrants, ont joué un rôle essentiel. La classe de flèches que nous notons ici $\mathcal{F}_W$ ou Catfib^W (on pourrait aussi la noter Fib^W, si ce n'était pour le risque de confusion avec Fib_W) s'interprète alors très naturellement à partir de là. Une flèche $f : X \longrightarrow S$ est dans $\mathcal{F}_W$ si et seulement si elle est a) fibrante, et par là définit de façon particulièrement directe un objet de $\mathrm{HOT}_W(S)$, puisque à S-équivalence près, X/S provient d'un objet de $\mathrm{Fibsc}\, S$ ou encore de $\mathrm{Cat}(S)$, et b) l'objet dans $\mathrm{HOT}_W(S)$ défini [page 35] par f est dans $\mathrm{HOT}_W^{\mathrm{lc}}(S)$, *i.e.* dans l'image essentielle de la sous-catégorie strictement pleine $\mathrm{Fibsc}_0^W S$ de $\mathrm{Fibsc}\, S$ (par le foncteur canonique $\mathrm{Fibsc} \longrightarrow \mathrm{HOT}_W(S)$), ou ce qui revient au [p. 22] même (du moins si W est fortement saturée) que le foncteur

$$\begin{array}{ccc} S^\circ & \longrightarrow & \mathrm{Hot}_W \\ s & \longmapsto & \mathrm{hot}_W(X_s) \end{array}$$

soit un foncteur "localement constant". Ainsi, on considérera les $f \in \mathcal{F}_W$ comme les flèches les plus économiques et les plus naturelles dans Cat, qui définissent des W-types d'homotopie relative localement constants.

Notons aussi que toutes les classes de flèches (4.2.1) sont stables par changement de base (et elles contiennent les isomorphismes). Les catégories Fib_W, Parf_W et leurs variantes W_0 sont aussi stables par composition, par contre, ce n'est pas le cas pour Fibf_W et $\mathcal{F}_W$ et leurs variantes W_0, sauf axiomes particuliers sur W ou sur W_0). Il en résulte aussitôt que si $\mathcal{F}$ est [page 36]

une de ces classes, la condition

$$\mathcal{F} \subset \mathrm{Fib}_W$$

équivaut à celle ci :

Commentary · Exposition [ms. 35]

Local constancy belongs to the coherent family

Grothendieck's preferred representatives carry their coherence with them. A split Grothendieck fibration is described, in the variance used here, by a strict functor $S^{\mathrm{op}} \to \mathrm{Cat}$; forgetting the splitting leads to a pseudofunctor with canonically coherent transport. The display at ms. 33 prints $\underline{\mathrm{Hom}}(S, \mathrm{Cat})$, whereas the displayed fibre functor at ms. 35 has the required opposite. We use the latter convention. The change is recorded as an edition slip and is not silently applied to his text.

The equivalence after localization between split and unsplit presentations does not mean that the forgetful functor is full before localization. A Cartesian functor may preserve chosen lifts only up to the unique vertical isomorphisms supplied by their universal property. Splitting replacements account for this discrepancy. The commentary on Chapter VI, manuscript pp. 26 and 53–59, treats the comparison itself; its role here is to justify using either presentation when formulating the axiom.

There is also a distinction between saying that a coherent family is locally constant and retaining only its diagram in the ordinary homotopy category. The former is the object of $\mathrm{HOT}_W(S)$ together with the assertion that its transition maps are equivalences. Passing to $S^{\mathrm{op}} \to \mathrm{Hot}_W$ is a test for this property when W is strongly saturated. It is not a reconstruction theorem for the family: higher compatibility data can disappear in the ordinary homotopy category. The commentary on Chapter VI, manuscript pp. 7 and 73–74, gives the relevant examples; we do not repeat them here.

The modern comparison is especially transparent for ordinary homotopy types. Cisinski (2009, Definitions 1.1–1.2, Theorem 1.3, and Corollary 1.28) characterize weak homotopy equivalences of indexing categories by their effect on locally constant homotopy-coherent diagrams. This is the later form of the geometric intuition behind Grothendieck's use of $\mathrm{HOT}^{\mathrm{lc}}_W$. The theorem concerns that coherent localization, with its coefficient category allowed to vary, and does not identify it with the category of ordinary functors into Hot.

In the present argument the practical benefit is more elementary. The free split-fibration replacement $X \to \Phi_S(X)$ is natural in X, and for a weakly W_0-fibred $X \to S$ its comparison is universally a W_0-equivalence over S. Naturality will later let one replace not merely two objects but also the arrow between them. That extra feature is exactly what transfers condition B.

Pour tout diagramme cartésien

(4.2.2)
$$\begin{array}{ccc} X & \xleftarrow{q} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow[p]{} & S', \end{array}$$

avec $f \in \mathcal{F}$ et $p \in W$, on a $q \in W$.

(N.B. On a aussi $f' \in \mathcal{F}$, par la stabilité rappelée plus haut.)

4.3. Voici donc (*cf.* plus bas) l'axiome pertinent, Loc (7) avec ses trois aspects a, b, c [23]. On a vu (et on reverra) que l'on a les implications

$$\text{a} \quad \Longrightarrow \quad \text{b} \quad \Longrightarrow \quad \text{c} \quad (\Longrightarrow \quad \text{F}).$$

Quand on référera à Loc (7) sans autre précision, il sera entendu qu'il s'agit de la forme la plus forte Loc (7 a), qui implique donc les deux autres. Il n'en faut pas moins pour être totalement à l'aise. Mais je rappelle que la variante affaiblie Loc (7b) est suffisante pour faire la théorie des foncteurs $f_!$ pour les catégories $\mathrm{HOT}_W(S)$ [24] (*cf.* VI).

23. J'y ai rajouté finalement un quatrième F (comme "fibration").

24. Ici ma mémoire me joue un tour - on n'a nullement besoin pour cela de l'axiome déjà très fort Loc (7b), mais seulement de Loc (3 bis), dont il semble bien qu'il soit vérifiée pour tous les localiseurs fondamentaux "raisonnables". [page 37]

Commentary · Later developments [ms. 36]

Intrinsic fibrations and sharp maps

The definition chosen here has a later name which makes its independence of auxiliary lifting classes particularly clear. It is the definition of a sharp map in Rezk (1998, Section 2): pullback along the map, after an arbitrary preliminary base change, preserves weak equivalences. In a right proper model category Rezk's Proposition 2.7 identifies these maps with those whose ordinary pullback squares are homotopy Cartesian. His Proposition 2.2 says that all model fibrations are sharp exactly when the model category is right proper.

These statements explain the intended geometry of Fib_W, but they do not identify it with the fibrations of an arbitrary chosen model structure. Nor do they turn a weakly W-fibred category into an intrinsic fibration merely by changing terminology. The latter conclusion is Loc (7), whose failure for connected-component equivalences is demonstrated below.

A useful consequence can already be read directly from the definition. Intrinsic fibrations are stable under composition: if $X \to Y \to S$ have this property and $T' \to T$ is a W-equivalence over S, first pull it back to Y, and then to X. Both comparison maps lie in W. They are also stable under every base change, because a test over the new base is already a test over the original one. These two elementary operations are the structural resources which the weaker class must acquire if it is to replace Fib_W.

C'est d'ailleurs une chose surprenante, puisque via le yoga général des dérivateurs associés aux catégories de modèles, on s'attendrait plutôt que pour avoir des adjoints à gauche $f_!$ des foncteurs f^* d'image inverse, il faut faire sur W des hypothèses du type *W-cofibration* (essentiellement, Loc (6 a)). Il me semble peu probable que Loc (7 b), ni même Loc (7), p. 23
implique Loc (6 a), ni même sa variante plus faible Loc (6′ a) (axiome du carré cocartésien, ou de Mayer-Vietoris, forme faible). Si d'autre part ces axiomes ne sont pas *vérifiés,* alors que Loc (7) l'est, il y a des chances que cela donnera un exemple où il y a un formalisme des $f_!$, sans que l'axiome d'exactitude à gauche soit vérifié.

Quant à la forme Loc (7 c), plus faible encore, elle suffit pourtant (comme nous le verrons plus bas) à assurer que Cat admet suffisamment de W-fibrations (résultat qui n'a rien d'évident, [page 38]
et ne pouvait m'apparaître qu'après avoir pris connaissance du substantiel lemme de K. S. Brown, dans son beau travail où il introduit les catégories à fibrations et à cofibrations).

Nous donnerons aussi plus bas des cas fort simples, où la condition Loc (7 b) entraîne déjà Loc (7) (W fortement saturé, *i.e.* Loc (1 bis)), voire même où Loc (7 c) entraîne Loc (7) (cas où W satisfait à l'axiome Loc (8) des limites inductives). Je signale en passant que je ne connais pas de localiseur fondamental satisfaisant l'axiome Loc (8) (qui implique déjà $W = W^{\natural}$, *i.e.* Loc (1 bis)), sans que W soit fortement saturé (*i.e.* en satisfasse même Loc (1 ter)).

Loc (7)

a) On a $\mathcal{F}_W \subset \mathrm{Fib}_W$, *i.e.* pour tout diagramme cartésien (4.2.2) dans Cat, avec $f \in \mathcal{F}_W$ (*i.e.* f Cat-fibrant et faiblement W-fibrant) et $p \in W$, on a $q \in W$.

b) Pour tout diagramme commutatif

$$\begin{array}{ccc} X & \xrightarrow{\ f\ } & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$$

avec $p, q \in \mathcal{F}_W$ et $f \in W$, on a $f \in W_S^{\mathrm{u}}$ (ou, ce qui revient au même comme X, Y sont W-lisses sur S, $f_s \in W$ pour tout $s \in \mathrm{Ob}\, S$) [25]. [page 39]

c) On a

$$\mathcal{F}_W \cap W \subset W^{\mathrm{u}},$$

25. **N.B.** La démonstration de $B(\mathcal{F}_W) \Longrightarrow B(\mathrm{Fibf}_W)$ montre en fait qu'il suffit de postuler b) lorsque f est cartésien, et même lorsque X, Y sont scindées sur S et f compatible avec le scindage.

en d'autres termes, pour toute flèche $f : X \longrightarrow S$ dans Cat telle que $f \in \mathcal{F}_W$ (on pourrait appeler ces flèches des Cat-W-fibrations, étant entendu qu'elles sont aussi des Cat-fibrations, mais non nécessairement des W-fibrations), si $f \in W$, alors $f \in W_S^{\mathrm{u}}$ (ou, ce qui revient au même, les fibres X_s de f ($s \in \mathrm{Ob}\, S$) sont W-asphériques).

p. 24

F) [26] Cat a suffisamment de W-fibrations, *i.e.* Cat muni de (W, Fib_W) est une catégorie à cofibrations (propre, et même de Brown, *i.e.* $X \longrightarrow e$ est dans Fib_W pour tout X dans Cat.

Il est clair que c est le cas particulier de b obtenu en prenant $Y = S$, $q = \mathrm{id}_S$, donc b $\Longrightarrow$ c. L'implication a $\Longrightarrow$ b est moins évidente, elle est établie d'abord dans XII, et nous y reviendrons ci-dessous (4.4.2 ci-dessous). Le reste du présent paragraphe est consacré à étudier diverses variantes des conditions a) b) c) précédentes et les implications entre celles-ci.

[page 40]

Commentary · Exposition [ms. 39]

The three tests measure different operations

Grothendieck separates three assertions which are easily conflated if one thinks only of familiar topological fibrations. Condition A concerns changing the base of an arbitrary W-equivalence. Condition B concerns changing the base of a W-equivalence between two suitable objects already lying over the same base. Condition C concerns one suitable map which is itself a W-equivalence. The implication from B to C uses the identity family $S \to S$ as the second object.

For a Grothendieck fibration in $\mathcal{F}_W$, the strict fibres are good representatives of the appropriate relative homotopy fibres. This accounts for the fibrewise reformulations of B and C. It does not remove their force: a total-space equivalence between locally constant families need not induce an equivalence on each strict fibre for a coarser homotopy theory. The covering obstruction at the end of the section will make this failure explicit in degree zero.

The added assertion F says only that every functor admits a factorization

$$X \xrightarrow{i} \bar{X} \xrightarrow{p} S, \qquad i \in W, \quad p \in \mathrm{Fib}_W.$$

The word "cofibrations" in its printed definition is a slip for "fibrations". A separate dependency concerns the assertion that every object is intrinsically fibrant over the final category: this is exactly stability of W under the functors $X \times -$. The editorial qualification to Rappel 2.3 must therefore be carried here; that product stability is established under Loc (3 bis), not by the cited argument from the weak axioms alone. Factorizations by themselves should not silently be identified with an all-objects-fibrant Brown structure without this additional fact. There is no resulting gap in the implication from D: every $X \to e$ is perfect directly from its comma-category definition, so $\mathrm{Parf}_{W_0} \subset \mathrm{Fib}_W$ independently supplies all the required terminal fibrations, and therefore product stability. F still does not say that the already available class $\mathcal{F}_W$ consists of intrinsic fibrations.

The footnote correcting the recollection about $f_!$ is mathematically significant. Construction of the left homotopy Kan extensions is controlled here by relative Theorem A, Loc (3 bis); it should not be attributed to the much stronger B merely because B had accompanied earlier arguments. This is consistent with the later exposition of Maltsiniotis (2005, pp. 3–4). Smooth and proper base-change formulas, locally constant families, and existence of Kan extensions remain distinct pieces of the formalism.

The speculative independence from the cofibration axioms at ms. 37 is therefore a question about the scope of an abstract presentation, not a reason to distrust the familiar homotopy theory of spaces. Its present status must take account of the minimum-localizer theorem, discussed at ms. 65 below.

4.4. Soit, plus généralement,

$$\mathcal{F} \subset \mathrm{Fl}(\mathrm{Cat})$$

un ensemble de flèches dans Cat, en pensant plus particulièrement au cas où $\mathcal{F}$ est un des sept ensembles du diagramme (4.2.1) (p. 30) autres que Fib_W. Nous pouvons formuler trois

26. rajouté après coup.

propriétés similaires pour $\mathcal{F}$ à celles formulées dans Loc (7 a b c) dans le cas de $\mathcal{F}_W$, notons ces propriétés

$$\mathrm{A}(\mathcal{F})\,,\ \mathrm{B}(\mathcal{F})\,,\ \mathrm{C}(\mathcal{F})\,,\ ^{27}$$

donc

$$(4.4.1)\qquad \begin{cases} \mathrm{A}(\mathcal{F}) & : \ \mathcal{F} \subset \mathrm{Fib}_W. \\ \mathrm{B}(\mathcal{F}) & : \ \mathrm{Fl}(\mathcal{F}_S) \cap W_S \subset W_S^{\mathrm{u}} \\ & \quad (\text{où } \mathcal{F}_S \text{ désigne la sous-catégorie pleine de } \mathrm{Cat}/S \\ & \quad \text{formée des } f : X \longrightarrow S \text{ avec } f \in \mathcal{F}). \\ \mathrm{C}(\mathcal{F}) & : \ \mathcal{F} \cap W \subset W^{\mathrm{u}}. \end{cases}$$

Il est clair qu'on a $\mathrm{B}(\mathcal{F}) \Longrightarrow \mathrm{C}(\mathcal{F})$, et je vais donner une condition sur $\mathcal{F}$ qui implique qu'on a aussi $\mathrm{A}(\mathcal{F}) \Longrightarrow \mathrm{B}(\mathcal{F})$.

(Nous l'appliquerons dans les cas suivants : [page 41]

$$(4.4.2)\qquad \mathcal{F} \ = \ \mathcal{F}_W,\ \mathrm{Fibf}_W,\ \mathrm{Parf}_W,\ \mathcal{F}_{W_0},\ \mathrm{Fibf}_{W_0}.$$

Je signale d'ailleurs que l'on verra plus bas (cor. 4.4.10) que

$$(4.4.3)\qquad \mathrm{A}(\mathcal{F}_W) \iff \mathrm{A}(\mathrm{Fibf}_W), \qquad \mathrm{A}(\mathcal{F}_{W_0}) \iff \mathrm{A}(\mathrm{Fibf}_{W_0}),$$

et de même pour les variantes B, C, de sorte que pour l'implication que nous voulons établir, il n'y a pas lieu de distinguer les cas de $\mathcal{F}_W$ et Fibf_W, de $\mathcal{F}_{W_0}$ et Fibf_{W_0}. D'ailleurs il suffira de regarder les ensembles $\mathcal{F}_{W_0}$, Fibf_{W_0}, Parf_{W_0}, puisque le cas des ensembles relatifs à W en sont un cas particulier (en prenant $W_0 = W$). [28]

Lemme 4.4.1. [29] *Supposons que $\mathcal{F}$ satisfasse la condition suivante : toute flèche f de* Cat *se factorise en $f = pi$, avec $i \in W$, $f \in \mathcal{F}$. Alors la condition* $\mathrm{A}(\mathcal{F})$ *sur $\mathcal{F}$ implique que dans* Cat *il y a "assez de W-fibrations", et par suite (par un argument formel bien connu), que l'on a* $\mathrm{B}(\mathrm{Fib}_W)$, *et a fortiori (comme par hypothèse $\mathcal{F} \subset \mathrm{Fib}_W$)* $\mathrm{B}(\mathcal{F})$. p. 25 [page 42]

Commentary · Examples [ms. 41]

An identity needed by the formal scheme

Grothendieck now abstracts the three tests so that one proof can serve several classes of maps. The six classes to which he applies the scheme all contain the identities. That small hypothesis is needed for the assertion, made for an arbitrary $\mathcal{F}$, that B implies C.

Here is a complete test of the literal statement without it. Take $W = \mathbf{W}_\infty$ and let $\mathcal{F}$ consist only of the inclusion $j : e \to [1]$ selecting 0. The full subcategory $\mathcal{F}_{[1]}$ has one object and only its identity endomorphism; for other bases it is empty. Thus every weak equivalence between its objects remains a weak equivalence after every base change, and $\mathrm{B}(\mathcal{F})$ holds. Both e and $[1]$ are contractible, so $j \in W$. Pulling j back along the inclusion of 1, however, gives $\varnothing \to e$, which is not a weak homotopy equivalence. Hence $\mathrm{C}(\mathcal{F})$ fails.

The repair is simply to require $\mathrm{id}_S \in \mathcal{F}$ for every S. Apply B to the arrow $p : (X \to S) \to (S \to S)$ whenever $p \in \mathcal{F} \cap W$; the resulting universal equivalence is exactly C. All subsequent applications satisfy this hypothesis. This is a missing condition in the unrestricted formal sentence, not a defect in the six geometric cases.

A second notational repair keeps the scheme readable. In Lemma 4.4.1 the right factor is $p \in \mathcal{F}$, not the printed $f \in \mathcal{F}$. The literal latter requirement would demand that every map already belong to $\mathcal{F}$, which is neither used nor intended. The same distinction between the original arrow and its right factor recurs at ms. 50.

27. **N.B.** Ces conditions ont un sens si $\mathcal{F} \subset \mathrm{Fl}(\mathcal{M})$, si $(\mathcal{M}, W)$ est une catégorie de modèles quelconque.

28. Cela fera donc a priori $3 \times 6 = 18$ conditions différentes, mais à cause de (4.4.3) il n'y en a (à équivalence près) que $4 \times 3 = 12$, et en fait deux parmi ces quatre triples de conditions se réduisent à une seule (pour $\mathcal{F} = \mathcal{F}_W$ et $\mathcal{F} = \mathcal{F}_{W_0}$), de sorte qu'il ne reste au total que *huit* conditions différentes, que nous résumons dans le diagramme récapitulatif (4.4.7.1) de la page 48. Il y en a même seulement *quatre,* A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0}, comme fonctions de W_0, les quatre autres A, B, C, D s'obtiennent en spécialisant au cas où $W_0 = W$.

29. Vaut dans une catégorie de modèles quelconque.

N.B. Il est clair en effet que si on a $\mathcal{F} \subset \mathcal{F}'$, alors

$$(*) \qquad \mathrm{A}(\mathcal{F}') \Longrightarrow \mathrm{A}(\mathcal{F}) \qquad \text{et de même pour B, C.}$$

Or la condition préliminaire sur $\mathcal{F}$ est bien satisfaite dans le cas d'un ensemble de l'une des formes $\mathcal{F} = \mathcal{F}_W$, Fibf_W, Parf_W. Il suffit de le voir dans le cas des sous-ensembles $\mathcal{F}_W$, Parf_W de Fibf_W, *i.e.* que toute flèche f de Cat se factorise en

$$f = pi,$$

avec $i \in W$, et p étant, au choix, dans Parf_W, ou dans $\mathcal{F}_W$. Or dans le cas de Parf_W, c'est ce qui est établie dans VII (Catégories de chemins (2)) grâce au formalisme des $\underline{\mathrm{Ch}}(S) \longrightarrow S \times S$. Pour trouver une factorisation avec $f = qj$, $j \in W$, $q \in \mathcal{F}_W$, il suffit de voir que p se factorise ainsi en $p = q_0 j_0$, on aura ‖

$$f = \underbrace{q_0}_{\in \mathcal{F}_{W_0}} \underbrace{(j_0 i)}_{\in W},$$

donc on peut supposer $f \in \mathrm{Parf}_W$. Mais si X est W-parfait sur S, donc un W-fibré faible sur S, alors dans la factorisation [page 43]

$$\begin{array}{ccc} X & \xrightarrow{\ i\ } & \Phi_S(X) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle p} & \\ & S & \end{array}$$

avec $i \in W$ et $p \in \mathrm{Catfib}$ (*cf.* VI), on sait que p est aussi une W-fibration faible, *i.e.* $p \in \mathcal{F}_W$, d'où la conclusion. Ainsi on trouve le

Corollaire 4.4.2. *Supposons que $\mathcal{F}$ soit l'un des six ensembles* (4.4.2) (i.e. *l'un des ensembles du diagramme* (4.2.1), *à l'exclusion de Fib_W et Fib_{W_0}). Alors on a*

$$\mathrm{A}(\mathcal{F}) \implies \mathrm{B}(\mathcal{F}) \implies \mathrm{C}(\mathcal{F})\,.$$

Lemme 4.4.3. *Soit $(\mathcal{M}, W)$ une catégorie de modèles, et soit $\mathcal{F} \subset \mathrm{Fl}(\mathcal{M})$ satisfaisant les conditions suivantes*

1) *$\mathcal{F}$ stable par changement de base, composition, contient les isomorphismes.*
2) *$\mathcal{M}$ a un objet final, et pour tout $X \in \mathrm{Ob}\,\mathcal{M}$, $X \longrightarrow e$ est dans $\mathcal{F}$.*
3) *Toute flèche f de $\mathcal{M}$ se factorise en $f = pi$, avec $i \in W$, $p \in \mathcal{F}$ (c'est la condition du lemme* 4.4.1).

Alors les conditions $\mathrm{A}(\mathcal{F})$, $\mathrm{B}(\mathcal{F})$, $\mathrm{C}(\mathcal{F})$ *sur $\mathcal{F}$ sont équivalentes.* p. 26

En effet, en vertu de 4.4.1, il reste à prouver l'implication

$$\mathrm{C}(\mathcal{F}) \implies \mathrm{A}(\mathcal{F}),$$

[page 44]

Commentary · Exposition [ms. 43]

Why enough intrinsic fibrations gives B

The first implication has a short proof which clarifies why factorization is relevant. Suppose every map admits a factorization as a W-equivalence followed by an intrinsic W-fibration. Let $u : X \to Y$ be a W-equivalence over S, where both structural maps $X \to S$ and $Y \to S$ are intrinsically fibrant. We must show that u remains a W-equivalence after an arbitrary base change $t : T \to S$.

Factor that base change, rather than u, as

$$T \xrightarrow{a} Z \xrightarrow{b} S, \qquad a \in W, \quad b \in \mathrm{Fib}_W.$$

Since b is intrinsically fibrant, its pullback preserves u, so $u_Z : X \times_S Z \to Y \times_S Z$ lies in W. The structural maps of these two pullbacks over Z are again intrinsically fibrant. Consequently the maps

$$X \times_S T \longrightarrow X \times_S Z, \qquad Y \times_S T \longrightarrow Y \times_S Z$$

induced by a both belong to W. In the commutative square formed by these maps and u_Z, two-out-of-three gives $u_T \in W$. The map t was arbitrary, which proves $u \in W_S^{\mathrm{u}}$.

If $\mathcal{F} \subset \mathrm{Fib}_W$ and every map factors through $\mathcal{F}$ after a W-equivalence, the assumed intrinsic factorizations exist. The argument therefore proves $\mathrm{A}(\mathcal{F}) \Rightarrow \mathrm{B}(\mathrm{Fib}_W)$, and restriction gives $\mathrm{B}(\mathcal{F})$. It also explains why the statement concerns the full subcategory of maps over S, rather than merely Cartesian arrows between fibred categories: no Cartesian condition on u was used.

The remaining input in Corollary 4.4.2 is the availability of the six factorizations. Perfect replacement is inherited from the path-category construction in VII. Applying Φ_S to a weakly fibred representative gives the locally constant Grothendieck-fibration replacement. For the latter argument, the underbrace $q_0 \in \mathcal{F}_{W_0}$ at ms. 42 should be read with W in place of W_0, unless the entire argument has first been carried out relative to W_0. Both versions are available; their parameters should not be mixed within one factorization.

This completes the "formal well-known argument" suppressed in the source. It uses base change, two-out-of-three, and the specified factorizations. No claim about full faithfulness of a localization functor, and no strong saturation assumption, is needed.

i.e. ceci : si $\mathcal{F}$ satisfait aussi la condition

4) $\mathcal{F} \cap W \subset W^{\mathrm{u}}$ (ou ce qui revient au même, comme $\mathcal{F}$ est stable par changement de base, que $\mathcal{F} \cap W$ est stable par changement de base),

alors on en conclut

$$\mathcal{F} \subset \mathrm{Fib}_W.$$

Or ceci n'est autre que le lemme de K. S. Brown, sous la forme duale de celle donnée plus haut (3.2.1, page 20).

Parmi les six ensembles de flèches envisagés dans (4.4.2), les hypothèses 1) 2) 3) sont satisfaites pour Parf_W et Parf_{W_0}, à l'exclusion des quatre autres. (Les autres satisfont à toutes les conditions, à la seule exception de la stabilité par composition [30]). On trouve donc :

Corollaire 4.4.4. *Les conditions* $\mathrm{A}(\mathrm{Parf}_W)$, $\mathrm{B}(\mathrm{Parf}_W)$, $\mathrm{C}(\mathrm{Parf}_W)$ *sont équivalentes* (nous désignons cette condition sur W par D). *De même les conditions* $\mathrm{A}(\mathrm{Parf}_{W_0})$, $\mathrm{B}(\mathrm{Parf}_{W_0})$, $\mathrm{C}(\mathrm{Parf}_{W_0})$ *sont équivalentes.*

(On désignera la condition obtenue sur le couple $W \supset W_0$ de localiseurs fondamentaux par D_{W_0}.) [page 45]

Ainsi on a (comme $\mathrm{Parf}_{W_0} \subset \mathrm{Parf}_W$)

$$\mathrm{D} = \mathrm{D}_W \implies \mathrm{D}_{W_0}\,, \tag{4.4.4}$$

et si $W_0' \subset W_0$, on aura

$$\mathrm{D}_{W_0} \implies \mathrm{D}_{W_0'}\,. \tag{4.4.5}$$

Corollaire 4.4.5. *Désignons par* A, B, C *respectivement les conditions* Loc (7 a), Loc (7 b), Loc (7 c). *On a alors les implications*

$$\mathrm{A} \implies \mathrm{B} \implies \mathrm{C}\,. \tag{4.4.5.1}$$

De plus, si on désigne (*pour un ensemble* $\mathcal{F}$ *de flèches de* Cat) *par* $\mathrm{Comp}(\mathcal{F})$ *la condition sur* $\mathcal{F}$ *que* $\mathcal{F}$ *soit stable par composition, on a les équivalences*

$$\mathrm{A} \iff (\mathrm{C} + \mathrm{Comp}(\mathcal{F}_W)) \iff (\mathrm{C} + \mathrm{Comp}(\mathrm{Fibf}_W)). \tag{4.4.5.2}$$

Démonstration. La première assertion est contenue dans le corollaire 4.4.2, appliqué au cas $\mathcal{F} = \mathcal{F}_W$. D'autre part, si $\mathcal{F}_W$ satisfait à Comp($\mathcal{F}_W$), il satisfait aux conditions 1) 2) 3) du lemme de Brown 4.4.3, donc joint à C, il implique A. Il reste, pour la première équivalence, à prouver que

$$A \implies \mathrm{Comp}(\mathcal{F}_W).$$

[page 46]

Commentary · Context [ms. 45]

The precise contribution of Brown's lemma

Grothendieck's enthusiasm comes from a reversal of the expected direction of verification. To prove that a candidate fibration class preserves all weak equivalences under pullback, one may first verify that its trivial members remain trivial after pullback. If the class is closed under composition and base change, contains the maps to the final object, and supplies weak-equivalence–fibration factorizations, Brown's theory promotes this apparently smaller assertion to the full base-change theorem.

In the original paper, the factorization lemma is in Part I, §1, pp. 421–422, and the relevant pullback theorem is Part I, §4, Lemma 2, pp. 428–429 (Brown, 1973). Lemma 1 of that section first proves invariance of base change between fibrant objects over a fixed base; Lemma 2 then treats a weak equivalence of bases. The full dual argument has been supplied in the commentary to XVI, ms. 20–22. Here it yields $\mathrm{C}(\mathcal{F}) \Rightarrow \mathrm{A}(\mathcal{F})$ under the hypotheses of Lemma 4.4.3.

Perfect maps satisfy the composition requirement before Loc (7) is imposed. The locally constant Grothendieck fibrations need not. This single difference explains why each triple for Parf_W or Parf_{W_0} immediately collapses to the condition denoted D, while the corresponding triples for $\mathcal{F}_W$ and $\mathcal{F}_{W_0}$ remain separated for the moment.

Footnote 28 has these classes interchanged: its unconditional collapse concerns perfect maps. It is not evidence for an unproved unconditional use of Brown's lemma on $\mathcal{F}_W$. The body of the argument, especially Corollaries 4.4.4–4.4.6, keeps the missing composition condition visible.

Mais on a par définition

$$(*) \qquad \mathcal{F}_W = \mathrm{Catfib} \cap \mathrm{Fibf}_W$$

et on sait que Catfib est stable par composition. Donc $\mathcal{F}_W$ l'est aussi, pourvu que le composé gf de deux flèches dans $\mathcal{F}_W$ soit dans Fibf_W. Or par hypothèse A, qui signifie $\mathcal{F}_W \subset \mathrm{Fib}_W$, on a $f, g \in \mathrm{Fib}_W$, donc $gf \in \mathrm{Fib}_W$, puisque Fib_W est stable par composition, a fortiori $gf \in \mathrm{Fibf}_W$.

D'autre part, la formule précédente $(*)$ montre que $\mathrm{Comp}(\mathrm{Fibf}_W) \Longrightarrow \mathrm{Comp}(\mathcal{F}_W)$, donc on a gratis l'implication $\Longleftarrow$ dans la deuxième équivalence (4.4.5.2). L'implication en sens inverse revient alors à

$$\mathrm{A} \implies \mathrm{Comp}(\mathrm{Fibf}_W),$$

ce qui sera évident grâce au résultat (donné plus bas) [page 47]

$$\mathrm{A} = \mathrm{A}(\mathcal{F}_W) \iff \mathrm{A}(\mathrm{Fibf}_W) \qquad \text{et de même pour B, C,}$$

qui signifie donc que

$$\mathcal{F}_W \subset \mathrm{Fib}_W \qquad \text{implique} \qquad \mathrm{Fibf}_W \subset \mathrm{Fib}_W \qquad (\text{donc } \mathrm{Fibf}_W = \mathrm{Fib}_W),$$

or Fib_W étant stable par composition, il en est de même pour Fibf_W, q.e.d.

Corollaire 4.4.6. *Supposons que $\mathcal{F}_{W_0}$ soit stable par composition, ce qui est le cas notamment (par* 4.4.5.2 *appliqué à W_0 au lieu de W) si W_0 satisfait* Loc (7). *Alors les conditions* $\mathrm{A}(\mathcal{F}_{W_0})$, $\mathrm{B}(\mathcal{F}_{W_0})$, $\mathrm{C}(\mathcal{F}_{W_0})$ *(que nous notons aussi simplement* A_{W_0}, B_{W_0}, C_{W_0})

30. **N.B.** Pour Fib_{W_0}, ce sont les conditions 1) 2) qui sont vérifiées, et par contre, la condition 3) qui fait problème. Notons là $\mathrm{F}(W_0)$. Si elle est satisfaite (*i.e.* si Cat a assez de W_0-fibrations), alors on a aussi équivalence des conditions $\mathrm{A}(\mathrm{Fib}_{W_0})$, $\mathrm{B}(\mathrm{Fib}_{W_0})$, $\mathrm{C}(\mathrm{Fib}_{W_0})$.

sont équivalentes. (***N.B.*** *Si on ne fait pas l'hypothèse précédente sur* W_0*, on aura seulement* $\mathrm{A}_{W_0} \Longrightarrow \mathrm{B}_{W_0} \Longrightarrow \mathrm{C}_{W_0}$*, en vertu de* 4.4.2.)

4.4.7. On a fait le tour à présent de toutes les conditions de type "fibration" sur W, ou sur le couple (W, W_0), qui m'ont parues utiles. Je vais les déployer ici dans un diagramme (plus ou moins) récapitulatif :

[page 48]

Commentary · Examples [ms. 47]

A finite failure of composition

The composition condition in Corollary 4.4.5 can be tested on a very small family. The commentary on Chapter I, manuscript pp. 118 and 122, already examined failures of composition. The following presentation singles out the precise class $\mathcal{F}_W$ used here and needs only connected components.

Put $W = \mathbf{W}_0$, the class of functors inducing bijections on π_0, and let G be a nontrivial finite group. Define a category B with two objects b_0, b_1 by

$$\mathrm{Hom}_B(b_0,b_0) = G, \quad \mathrm{Hom}_B(b_1,b_1) = \{1\}, \quad \mathrm{Hom}_B(b_0,b_1) = G, \quad \mathrm{Hom}_B(b_1,b_0) = \varnothing.$$

Write $a_h : b_0 \to b_1$ for the arrow labelled by h and prescribe $a_h g = a_{hg}$. There is a Grothendieck fibration $q : B \to [1]$ with fibres BG and e. Its inverse-image functor along $0 \to 1$ is $e \to BG$. Since both categories are connected, $q \in \mathcal{F}_{\mathbf{W}_0}$.

Let E be a transitive G-set with more than one element. The functor $r : B \to BG$ sending g and a_g to g pulls the corresponding locally constant discrete family on BG back to a discrete fibration $p : A \to B$. Every transport map of this family is a bijection of sets. Thus p lies in $\mathcal{F}_{\mathbf{W}_0}$, and in fact its transports are isomorphisms of discrete categories.

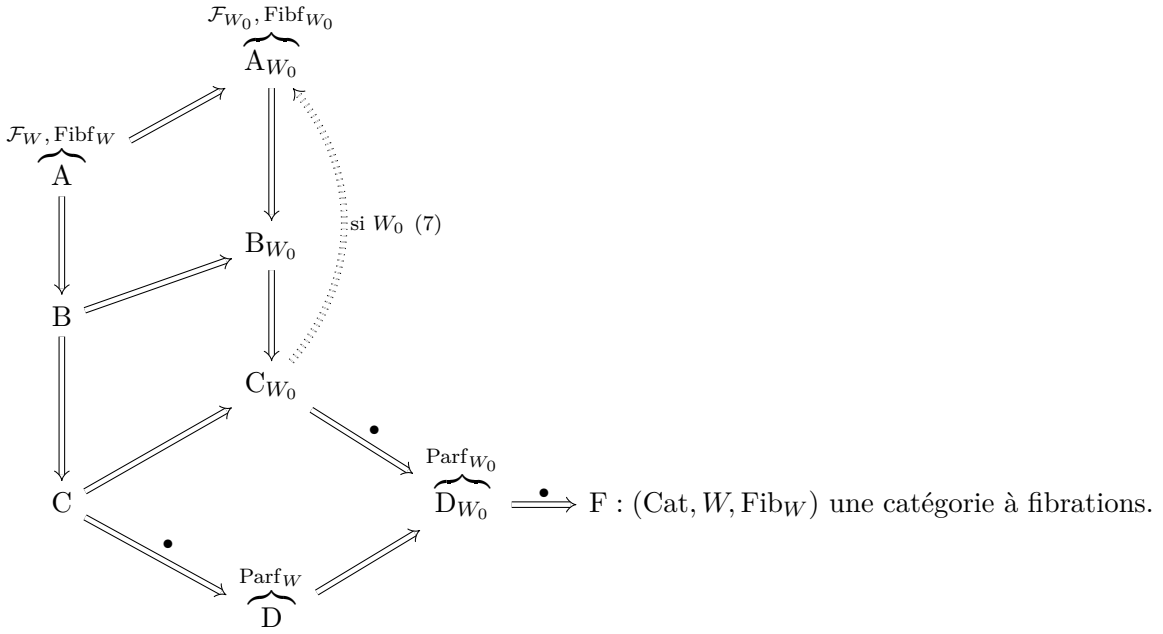
The composite qp is again a Grothendieck fibration. Its fibre over 0 is the action groupoid $E /\!\!/ G$, which is connected by transitivity; its fibre over 1 is the discrete category E. The inverse-image functor between these fibres is the inclusion of the objects of the action groupoid. On connected components it is the map

$$E \longrightarrow E/G = \{*\},$$

which is not bijective. Consequently $qp \notin \mathcal{F}_{\mathbf{W}_0}$, although both p and q belong to that class. Taking $G = C_2$ with its regular action gives a base B with two objects and five arrows, and A with four objects and ten arrows; all categories in the example are finite.

The mechanism is the loss of monodromy information when a transport functor is required to preserve only connected components. A second locally constant family can detect precisely the information the first equivalence discarded. Corollary 4.4.5 says that C together with composition closure prevents this phenomenon. It does not say that C can be used without checking composition.

p. 28



$\overset{\bullet}{\Longrightarrow}$ F : $(\mathrm{Cat}, W, \mathrm{Fib}_W)$ une catégorie à fibrations.

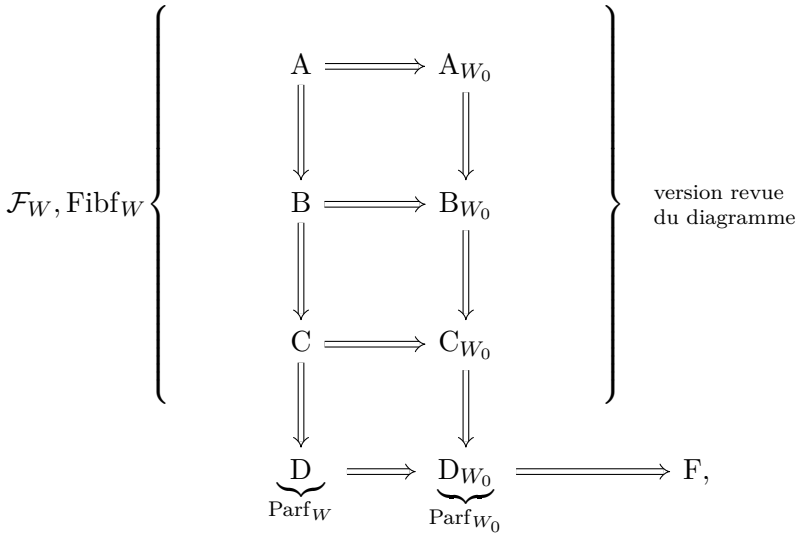


diagramme que je vais commenter. Toutes les implications en traits pleins du diagramme ont été déjà obtenues ou sont tautologiques, à l'exception des trois $\overset{\bullet}{\Longrightarrow}$ sur lesquels je vais revenir. L'implication en pointillés $\mathrm{C}_{W_0} \dashrightarrow \mathrm{A}_{W_0}$ est une implication conditionnelle, soumise à la condition marquée sur le diagramme comme W_0 (7), ce qui signifie : W_0 satisfait à Loc (7). Cette implication, *i.e.* l'équivalence de A_{W_0}, B_{W_0}, C_{W_0} sous la condition W_0 (7), a été vue dans 4.4.6. J'ai marqué, au dessus des deux groupes (A, B, C) et (A_{W_0}, B_{W_0}, C_{W_0}) ainsi qu'au dessus de D et de D′ (qui correspondent chacun à un tel groupe de conditions $\mathrm{A}(\mathcal{F})$, $\mathrm{B}(\mathcal{F})$, $\mathrm{C}(\mathcal{F})$ toutes équivalentes, avec $\mathcal{F} = \mathrm{Parf}_W$ ou $\mathcal{F} = \mathrm{Parf}_{W_0}$) les ensembles de flèches [page 49]
qui leur donnent naissance respectivement : on notera qu'il y en a deux pour les pour les deux triplets nommés en premier, A, B, C pouvant se définir via $\mathcal{F}_W$ ou Fib_W indifféremment, et de même A_{W_0}, B_{W_0}, C_{W_0} via $\mathcal{F}_{W_0}$ ou Fib_{W_0}, comme déjà énoncé (4.4.3) et comme nous le verrons plus bas (cor. 4.4.10, pages 54-55). Par contre, les triplets réduits D et D_{W_0} sont définis chacun par un $\mathcal{F}$ seulement, savoir par Parf_W et

Parf_{W_0} respectivement. (En tout, cela fait intervenir six ensembles $\mathcal{F}$ distincts, à savoir p. 29
les six ensembles du diagramme (4.2.1) autres que Fib_W et Fib_{W_0}.)

L'implication non tautologique C $\overset{\bullet}{\Longrightarrow}$ D est un cas particulier de l'implication $\mathrm{C}_{W_0} \overset{\bullet}{\Longrightarrow}$ D_{W_0}, laquelle est contenue dans ce qui a été annoncé à l'instant, à savoir $\mathrm{C}_{W_0}(\mathcal{F}_{W_0}) \Longrightarrow$ $\mathrm{C}_{W_0}(\mathrm{Fib}_{W_0})$, compte tenu que $\mathrm{Parf}_{W_0} \subset \mathrm{Fibf}_{W_0}$ (*cf.* (∗), p. 42). Faisant $W_0 = W$, on trouve [page 50]

donc l'implication A $\Longrightarrow$ D, qui est cependant contenue dans les autres implications du diagramme, comme composée dans la chaîne d'implications

$$\mathrm{A} \Longrightarrow \mathrm{B} \Longrightarrow \mathrm{C} \Longrightarrow \mathrm{D} \Longrightarrow \mathrm{D}_{W_0} \overset{\bullet}{\Longrightarrow} \mathrm{F}.$$

Il reste, dans ces commentaires généraux, à dire un mot de cette dernière implication

$$\mathrm{D}_{W_0} \overset{\bullet}{\Longrightarrow} \mathrm{F},$$

i.e. du fait que la plus faible de toutes les conditions A à D_{W_0} envisagées ici implique déjà que Cat a assez de W-fibrations, *i.e.* est une catégorie à cofibrations pour (W, Fib_W). C'est clair quand on prend D_{W_0} sous la forme $\mathrm{A}(\mathrm{Parf}_{W_0})$, *i.e.*

$$\mathrm{Parf}_{W_0} \subset W.$$

Comme toute flèche f de Cat se factorise en $f = pi$, avec

$$i \in W_0 \subset W \quad \text{et} \quad f \in \mathrm{Parf}_{W_0} \subset \mathrm{Fib}_W\,,$$

on trouve la factorisation avec $i \in W$, $f \in \mathrm{Fib}_W$.

4.4.8. Pour terminer d'établir les implications du diagramme p. 48, il reste seulement à établir les équivalences [page 51]

$$\text{(4.4.8.1)} \qquad \begin{cases} \mathrm{A}(\mathcal{F}_{W_0}) & \Longleftrightarrow & \mathrm{A}(\mathrm{Fibf}_{W_0}) \\ \mathrm{B}(\mathcal{F}_{W_0}) & \Longleftrightarrow & \mathrm{B}(\mathrm{Fibf}_{W_0}) \\ \mathrm{C}(\mathcal{F}_{W_0}) & \Longleftrightarrow & \mathrm{C}(\mathrm{Fibf}_{W_0}), \end{cases}$$

qui donneront les équivalences similaires pour W, en prenant $W_0 = W$.

Commentary · Exposition [ms. 50]

How to read the rectangle of conditions

The large diagram records a small number of operations. Moving down a column first specializes from arbitrary base equivalences to equivalences between families, then to a family equivalent to its base, and finally replaces weakly fibred families by perfect ones. Moving to the right restricts the admissible families from W to W_0. The first three downward arrows ultimately depend on the natural comparison with Φ_S, whereas the perfect-map triple has already been handled by Brown's lemma.

The last arrow, $\mathrm{D}_{W_0} \Rightarrow \mathrm{F}$, has an especially direct meaning. Given $f : X \to S$, take its path factorization

$$X \xrightarrow{i} \bar{X} \xrightarrow{p} S, \qquad i \in W_0 \subset W, \quad p \in \mathrm{Parf}_{W_0}.$$

The condition D_{W_0}, in its A-form, says that p is an intrinsic W-fibration. This is the required factorization for F. The isolated display $\mathrm{Parf}_{W_0} \subset W$ in the source is therefore an edition slip for $\mathrm{Parf}_{W_0} \subset \mathrm{Fib}_W$; it is not an additional theorem asserting that all perfect maps are equivalences. The subsequent occurrences of f as the right factor should likewise be p.

This interpretation also explains the role of the smallest reference localizer. It supplies the largest scope for the existence argument, because asking its perfect maps to preserve W is the weakest of the displayed perfect-map conditions. The existence of replacements remains separate from the claim that every weakly W-fibred family is already a suitable replacement. The connected-component example will satisfy the first assertion and fail the second.

None of the solid arrows in this rectangle relies on the conditional arrows which return upwards. Those additional arguments enter only after the natural replacement lemma is established. Reading the diagram in that order avoids a circular use of the equivalence between $\mathcal{F}_{W_0}$ and Fibf_{W_0}.

La première équivalence (4.4.8.1) sera conséquence du lemme (± tautologique) suivant :

Lemme 4.4.9. *Soient* $(\mathcal{M}, W)$ *une catégorie de modèles,* $\mathcal{F}, \mathcal{G} \subset \mathrm{Fl}(\mathcal{M})$, *tels que tout* $f \in \mathcal{F}$, $f : X \longrightarrow S$, *s'écrive* $f = pi$, *avec* $i \in W_S^u$, $p \in \mathcal{G}$. *Alors* $\mathrm{A}(\mathcal{G}) \Longrightarrow \mathrm{A}(\mathcal{F})$, i.e.

$$\mathcal{G} \subset \mathrm{Fib}_W \implies \mathcal{F} \subset \mathrm{Fib}_W, \tag{4.4.9.1}$$

et de même $\mathrm{C}(\mathcal{G}) \Longrightarrow \mathrm{C}(\mathcal{F})$, i.e.

$$\mathcal{G} \cap W \subset W^u \implies \mathcal{F} \cap W \subset W^u. \tag{4.4.9.2}$$

Démonstration. A) En effet, si $\mathcal{G} \subset \mathrm{Fib}_W$, la factorisation $f = pi$ de f est telle que $p \in \mathrm{Fib}_W$. Or un composé pi, avec $p \in \mathrm{Fib}_W$ et $i \in W_S^{\mathrm{u}}$, est dans Fib_W, comme on vérifie aussitôt sur le diagramme de carrés cartésiens p. 30

$$\begin{array}{ccccc}
X & \longleftarrow & X' & \longleftarrow & X'' \\
{\scriptstyle W_S^{\mathrm{u}}}\downarrow & & {\scriptstyle (W)}\downarrow & & {\scriptstyle (W)}\downarrow \\
\overline{X} & \longleftarrow & \overline{X'} & \xleftarrow{(W)} & \overline{X''} \\
{\scriptstyle \mathrm{Fib}_W}\downarrow & & \downarrow & & \downarrow \\
S & \longleftarrow & S' & \xleftarrow{W} & S''
\end{array}$$

sur lequel on lit que $X'' \longrightarrow X'$ est dans W, q.e.d.

Ce lemme s'applique notamment au cas où $\mathcal{F} = \mathrm{Fibf}_{W_0}$, $\mathcal{G} = \mathcal{F}_{W_0}$, car f se factorise en [page 52]

$$\begin{array}{ccc}
X & & \\
{\scriptstyle f}\downarrow & \searrow^{i} & \\
 & & \Phi_S(X) \\
 & \swarrow_{p} & \\
S, & &
\end{array}$$

et on sait que $i \in W_{0S}^{\mathrm{u}} \subset W_S^{\mathrm{u}}$, et $p \in \mathcal{F}_{W_0} \subset \mathcal{F}_W$, ce qui sont les conditions voulues.

C) Démonstration similaire : Soit $f \in \mathcal{F} \cap W$, factorisant en $f = pi$, avec $i \in W_S^{\mathrm{u}}$, $g \in \mathcal{G}$, on aura alors $g \in W$, donc $g \in \mathcal{G} \cap W \subset W^{\mathrm{u}}$ (par hypothèse $\mathrm{C}(\mathcal{G})$), a fortiori on a $g \in W_S^{\mathrm{u}}$, donc $f \in W_S^{\mathrm{u}}$ comme composé de deux flèches de W_S^{u}, et comme le but de f est S lui-même, cela signifie $f \in W^{\mathrm{u}}$, q.e.d.

$$\begin{array}{ccc}
X & & \\
 & \searrow^{i \in W_S^{\mathrm{u}}} & \\
{\scriptstyle \mathcal{F} \cap W}\downarrow & & \bar{X} \\
 & \swarrow_{g} & \\
S & &
\end{array}$$

Je ne vois pas de façon pour prouver également

$$\mathrm{B}(\mathcal{G}) \qquad \Longrightarrow \qquad \mathrm{B}(\mathcal{F}),$$

avec les seules hypothèses du lemme précédent sur le couple $\mathcal{F}$, $\mathcal{G}$. Il faudrait que pour tout S dans $\mathcal{M}$, et toute flèche $u : X \longrightarrow Y$ dans $\mathcal{M}/S$, on puisse insérer le diagramme obtenu dans [page 53] $\mathcal{M}$

$$\begin{array}{ccc}
X & \xrightarrow{u} & Y \\
 & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\
 & S &
\end{array}$$

p. 31

dans un diagramme commutatif ci-dessous

(4.4.9.3)
$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ \bar{X} & \xrightarrow{\bar{u}} & \bar{Y} \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & S & \end{array}$$

avec

$$f = pi, \qquad g = qj, \qquad i, j \in W_S^{\mathrm{u}}, \qquad p, q \in \mathcal{G}.$$

Ce sera le cas en tous cas si pour tout S, et un objet variable X de $\mathcal{M}/S$, on peut trouver une factorisation *fonctorielle en* X de son morphisme structural f_X en $p_X i_X$, avec $i_X \in W_S^{\mathrm{u}}$ et $p_X \in \mathcal{G}$, *i.e.* si i, p se déduisent d'un foncteur

$$\varphi : \mathcal{F}_S \longrightarrow \mathcal{M}/S \ ^{31},$$

et d'un morphisme de foncteurs

$$i : \underbrace{\mathrm{inc}_{\mathcal{F}_S}}_{\substack{\text{foncteur} \\ \text{d'inclusion}}} \longrightarrow \varphi.$$

Admettant l'existence d'un diagramme ayant les propriétés dites, pour toute flèche u dans $\mathcal{F}_S$, ou seulement toute flèche u qui soit aussi dans W_S, nous pouvons prouver maintenant que $u \in W_S \Longrightarrow u \in W_S^{\mathrm{u}}$, pourvu que la propriété similaire soit vrai pour $\mathcal{G}$ à la place de $\mathcal{F}$.

En effet, l'hypothèse sur $u \in W$ implique $\bar{u} \in W$, et l'hypothèse sur $\mathcal{G}$ implique dès lors $\bar{u} \in W_S^{\mathrm{u}}$. Comme W_S^{u} est un localiseur dans $\mathcal{M}/S$, on en conclut bien $u \in W_S^{\mathrm{u}}$, q.e.d. [page 54]

Ainsi on a obtenu

Corollaire 4.4.10. *Supposons, sous les conditions du lemme* 4.4.9, *que tout diagramme commutatif triangulaire* $\begin{array}{ccc} X & \xrightarrow{u} & Y \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & S & \end{array}$ *s'insère dans un diagramme du type* (4.4.9.3) *comme ci-dessus* (*avec les relations explicites sur les flèches qui y figurent*). *Alors*

$$\mathrm{B}(\mathcal{G}) \implies \mathrm{B}(\mathcal{F}).$$

Donc si de plus $\mathcal{G} \subset \mathcal{F}$, *de sorte que* $\mathrm{A}(\mathcal{F}) \Longrightarrow \mathrm{A}(\mathcal{G})$ *tautologiquement, et de même pour* B *et pour* C, *on trouve*

$$\begin{aligned} \mathrm{A}(\mathcal{F}) &\iff \mathrm{A}(\mathcal{G}) \\ \mathrm{B}(\mathcal{F}) &\iff \mathrm{B}(\mathcal{G}) \\ \mathrm{C}(\mathcal{F}) &\iff \mathrm{C}(\mathcal{G}). \end{aligned}$$

Nous pouvons appliquer ce corollaire dans le cas qui nous occupe ici, avec

$$\mathcal{G} \overset{\text{déf}}{=} \mathcal{F}_{W_0} \subset \mathcal{F} \overset{\text{déf}}{=} \mathrm{Fibf}_{W_0},$$

vu que dans ce cas on a bien une factorisation *fonctorielle* d'un $f \in \mathrm{Fibf}_{W_0}$, $f : X \longrightarrow S$, en $f = pi$, $i \in W_{0S}^{\mathrm{u}} \subset W_S^{\mathrm{u}}$, $p \in \mathcal{F}_{W_0} = \mathcal{G}$, savoir celle rappelée page 43 : p. 32 [page 55]

$$\begin{array}{ccc} X & \xrightarrow{i_X} & \bar{X} = \Phi_S(X) \\ & {\scriptstyle f=f_X}\searrow \quad \swarrow{\scriptstyle p_X} & \\ & S. & \end{array}$$

31. $\mathcal{F}_S$ est la sous-catégorie pleine de $\mathcal{M}/S$ formée des X tels que $f_X \in \mathcal{F}$.

Commentary · Exposition [ms. 54]

Why universal comparison maps transfer the tests

Grothendieck's replacement lemma isolates the exact information needed from a comparison map. A global weak equivalence is insufficient because the tests concern all base changes. Suppose that every structural map $f : X \to S$ in $\mathcal{F}$ has a factorization

$$X \xrightarrow{i} \bar{X} \xrightarrow{p} S, \qquad i \in W_S^{\mathrm{u}}, \quad p \in \mathcal{G}.$$

The notation W_S^{u} concerns changes of the base S, not arbitrary changes of the codomain $\bar{X}$. That relative condition is exactly sufficient.

For A, let $T' \to T$ be a W-equivalence over S. The two comparison maps $i_{T'}$ and i_T lie in W. If A holds for $\mathcal{G}$, the map $\bar{X}_{T'} \to \bar{X}_T$ also lies in W. Commutativity and two-out-of-three then put $X_{T'} \to X_T$ in W. This proves A for $\mathcal{F}$.

For C, suppose f itself lies in W. Since $i \in W$, two-out-of-three gives $p \in W$. If C holds for $\mathcal{G}$, every map $p_T : \bar{X}_T \to T$ lies in W. Composing it with i_T gives $f_T \in W$ for every $T \to S$. Thus f is universally a W-equivalence. The change from p to g in the printed proof is only a change of name, but should not be mistaken for a second factorization.

B involves a further datum: an arrow between two families. Suppose a W-equivalence $u : X \to Y$ over S fits into a commuting square

$$\begin{array}{ccc} X & \xrightarrow{u} & Y \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle j} \\ \bar{X} & \xrightarrow[\bar{u}]{} & \bar{Y} \end{array} \qquad i, j \in W_S^{\mathrm{u}}, \quad \bar{X}, \bar{Y} \in \mathcal{G}_S.$$

The square first implies $\bar{u} \in W$. Assuming B for $\mathcal{G}$, every pullback $\bar{u}_T$ is in W. The pulled-back square, with $i_T, j_T \in W$, then gives $u_T \in W$. Notice that the square is needed only for those u which are in W; a replacement functor defined on all arrows is a convenient sufficient construction, not a logically necessary extra assumption.

The free split-fibration construction supplies such a functor. For weakly W_0-fibred objects, its unit lies in $W_{0,S}^{\mathrm{u}} \subset W_S^{\mathrm{u}}$, and the resulting structural map belongs to $\mathcal{F}_{W_0}$. Applying the three arguments with $\mathcal{G} = \mathcal{F}_{W_0}$ and $\mathcal{F} = \mathrm{Fibf}_{W_0}$ gives every announced equivalence. The reverse implications are restrictions along $\mathcal{G} \subset \mathcal{F}$.

This proof explains why it is legitimate to postpone the equivalences earlier in the section: they depend on a particular natural replacement, not on any assertion that all weakly fibred families are intrinsically fibrant. It also states the genuine limit of Lemma 4.4.9. Its objectwise hypotheses prove A and C; the text does not establish the B-transfer under those hypotheses alone. We leave that unrestricted assertion unclaimed.

Cela prouve donc les équivalences annoncées (4.4.8.1) et achève d'établir la validité des implications du diagramme de la page 48.

Ces implications, à la seule exception de l'implication en pointillés $\mathrm{C}_{W_0} \dashrightarrow \mathrm{A}_{W_0}$ (qui suppose qu'on ait W_0 (7)), ne font appel à aucune hypothèse particulière sur les localiseurs fondamentaux W et W_0 (à part l'inclusion $W \supset W_0$). Mais nous pouvons, en utilisant le lemme 4.4.10, donner des implications conditionnelles remarquables, sous la seule condition que W, ou W_0, satisfassent à l'anodin *axiome des limites* Loc (8), et même seulement sous la forme faible de stabilité des localiseurs par $\varinjlim$ de *suites*. Donnons-les d'abord sous forme technique, avant de déployer les conséquences dans un théorème scholie : [page 56]

Corollaire 4.4.11. *Supposons que W_0 satisfasse à l'axiome* Loc (8), *ou seulement* Loc (8 bis) (*stabilité par $\varinjlim$ de suites*). *Alors* $\mathrm{D}_{W_0} \Longrightarrow \mathrm{A}_{W_0}$ (*donc les conditions* A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0} *sont équivalentes*). *En particulier* (*faisant $W_0 = W$*), *si W satisfait à* Loc (8 bis), *alors on a* $\mathrm{D}_W \Longrightarrow \mathrm{A}_W$ (*donc les conditions* A_W, B_W, C_W, D_W *sont équivalentes*).

En effet, l'implication à prouver s'écrit aussi

$$\mathrm{A}(\mathrm{Parf}_{W_0}) \iff \mathrm{A}(\mathrm{Fibf}_{W_0}).$$

L'implication non tautologique $\Longrightarrow$ est justiciable du lemme 4.4.9. Il faut vérifier que

moyennant Loc (8 bis) pour W_0, tout $f \in \mathrm{Fib}_{W_0}$, $f : X \longrightarrow S$, se factorise en $f = pi$, avec $i \in W_0^{\mathrm{u}}$, $p \in \mathrm{Parf}_{W_0}$. Or on sait [32] (VI, p. 193 ff.) que l'axiome Loc (8 bis) implique que [page 57]
$f \in \mathrm{Fibf}_{W_0}$ se factorise en

$$\begin{array}{ccc} X & \xrightarrow{i_\infty} & \bar{X} = \Phi_\infty(X) \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle p_\infty} & \\ & S, & \end{array}$$

(catégorie de chemins infinis d'un côté, relativisée sur S, *i.e.* des couples (x, c), avec $x \in \mathrm{Ob}\, X$ et $c : f(x) \longrightarrow s$ un chemin infini dans S, de source $f(x)$.)

avec $p_\infty \in \mathrm{Parf}_{W_0}$ (comme on veut), et $i_\infty \in W_{0S}^{\mathrm{u}}$, et comme $W_{0S}^{\mathrm{u}} \subset W_S^{\mathrm{u}}$, on a bien $i_\infty \in W_S^{\mathrm{u}}$.

Corollaire 4.4.12. *Supposons que* W et W_0 *satisfassent à* Loc (8 bis). *Alors on a* $\mathrm{D}_{W_0} \Longrightarrow \mathrm{A}$ (*donc les huit conditions* A, B, C, D, A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0} *sont équivalentes*).

Cela signifie en effet que p. 33

$$\mathrm{A}(\mathrm{Parf}_{W_0}) \implies \mathrm{A}(\mathrm{Fibf}_W),$$

et c'est encore justiciable du lemme 4.4.9. En effet, la factorisation ci-dessus d'une flèche f ne dépend pas du choix d'un localiseur, et p_∞ est W-parfait pour *tout* localiseur satisfaisant Loc (8 bis), donc en particulier pour W_0. Et la flèche i_∞ est dans W_S^{u}, pour tout localiseur satisfaisant Loc (8 bis), donc pour W_0.

Je vais récapituler les résultats obtenus dans cette section dans un théorème, qui constituera [page 58]
le résultat principal du présent paragraphe sur les axiomes de W-fibrations.

Commentary · Examples [ms. 57]

The two localizers have different jobs

The infinite-path argument is designed to improve a family until both comma-category comparisons are controlled. Alternating the two one-sided constructions gives

$$X \longrightarrow \Phi(X) \longrightarrow \Psi\Phi(X) \longrightarrow \Phi\Psi\Phi(X) \longrightarrow \cdots .$$

The odd and even subsequences are cofinal. Under sequential-colimit stability they show that the resulting object is both smooth and proper. If the input is weakly W-fibred, the successive comparison maps are universally W-equivalent over the base, and so is their colimit. This is the argument of VI, ms. 193–195 (Grothendieck, 1990b); the commentary on Chapter VI, manuscript p. 194, treats the alternation in detail.

Corollary 4.4.12 uses two independent conclusions from that construction. To transfer A from Parf_{W_0} to Fibf_W, one needs a right factor in Parf_{W_0} and a comparison unit in W_S^{u}. The underlying construction does not depend on a localizer. Its perfection uses Loc (8 bis) for W_0, while the unit's universal W-equivalence uses Loc (8 bis) for W and weak W-fibrancy of the input. Lemma 4.4.9 then applies with exactly the required parameters.

The last sentence of the printed proof goes further, asserting the universal unit conclusion for W_0 as well. That conclusion is not available for an arbitrary weakly W-fibred input. The qualification is explicit in VI: for a general input the unit is a global equivalence, while the stronger relative conclusion requires weak fibrancy for the same localizer.

One can see the obstruction without a complicated path calculation. Let $G \neq 1$ and take the Grothendieck fibration over [1] classified by

$$F(0) = e, \qquad F(1) = BG, \qquad F(1) \longrightarrow F(0) \text{ the unique functor.}$$

Its total category has an initial object, so its nerve is contractible. The map is weakly $\mathbf{W}_0$-fibred, since both fibres are connected, but it is not weakly $\mathbf{W}_\infty$-fibred. A perfect ordinary-homotopy replacement has contractible homotopy fibres over the contractible base: its total space is weakly equivalent to the original contractible total space. Consequently its fibre over 1 is contractible,

32. et c'est là une trouvaille tout à fait inattendue, un peu comme le lemme de K. S. Brown …

whereas the original strict fibre is BG. The unit on that fibre cannot be a weak homotopy equivalence. This rules out the extra $W_0 = \mathbf{W}_\infty$ assertion while leaving the $W = \mathbf{W}_0$ assertion intact.

Thus the theorem is repaired by using the weaker unit statement it actually needs. No additional axiom and no change to the conclusion of Corollary 4.4.12 is necessary.

Théorème 4.4.13. *Soit W un localiseur fondamental dans* Cat. *Pour tout localiseur W_0 contenu dans W, on a explicité un ensemble de quatre conditions sur la paire W_0, W, qui pour W fixé, dépendent de W_0 seul, notées*

(4.4.13.1) $$\mathrm{A}_{W_0},\ \mathrm{B}_{W_0},\ \mathrm{C}_{W_0},\ \mathrm{D}_{W_0},$$

et définies respectivement comme

$$\mathrm{A}(\mathcal{F}_{W_0}),\ \mathrm{B}(\mathcal{F}_{W_0}),\ \mathrm{C}(\mathcal{F}_{W_0}), \qquad \textit{et}$$

$$\mathrm{A}(\mathrm{Parf}_{W_0})\Big(\iff \mathrm{B}(\mathrm{Parf}_{W_0}) \iff \mathrm{C}(\mathrm{Parf}_{W_0})\Big),$$

avec les notations A(·), B(·), C(·) *introduites dans* (4.4.1). *Faisant $W_0 = W$, on trouve les conditions sur W*

(4.4.13.2) $$\mathrm{A},\ \mathrm{B},\ \mathrm{C},\ \mathrm{D}$$

avec $\mathrm{A} = \mathrm{A}_W$, $\mathrm{B} = \mathrm{B}_W$, $\mathrm{C} = \mathrm{C}_W$, $\mathrm{D} = \mathrm{D}_W$, *dont les trois premières ne sont autres que* Loc (7 a), Loc (7 b), Loc (7 c) (cf. 4.3, *page* 38).

① *On a pour ces conditions les implications suivantes.*

a) *On a les implications*

(4.4.13.3) $$\mathrm{A}_{W_0} \implies \mathrm{B}_{W_0} \implies \mathrm{C}_{W_0} \implies \mathrm{D}_{W_0},$$

(*d'où en particulier*

[page 59]

(4.4.13.4) $$\mathrm{A}_W \implies \mathrm{B}_W \implies \mathrm{C}_W \implies \mathrm{D}_W\ ).$$

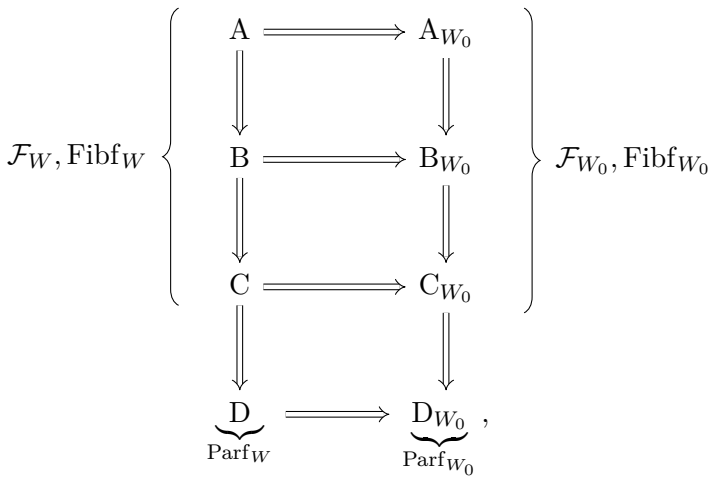
b) *Si $W_0' \subset W_0$ est un localiseur fondamental contenu dans W_0, on a* (*tautologiquement*)

$$\mathrm{A}_{W_0} \implies \mathrm{A}_{W_0'}, \qquad \mathrm{B}_{W_0} \implies \mathrm{B}_{W_0'}, \qquad \mathrm{C}_{W_0} \implies \mathrm{C}_{W_0'}.$$

On trouve donc, par (a) *et* (b) (*y faisant pour le couple (W, W_0) comme (W_0, W_0')*), *le rectangle d'implications*

p. 34

(4.4.13.5)



où on a rappelé, à côté des accolades, quels sont l'ensemble, ou les ensembles $\mathcal{F} \subset \mathrm{Fl}(\mathrm{Cat})$ *qui définissent les conditions envisagées (comme l'une des conditions* $\mathrm{A}(\mathcal{F})$, $\mathrm{B}(\mathcal{F})$, $\mathrm{C}(\mathcal{F})$ *ou* $\mathrm{D}(\mathcal{F}) = \mathrm{A}(\mathcal{F})$ *dans le cas où les trois conditions sont équivalentes). La plus faible des huit conditions envisagées est* D_{W_0}*, la plus forte* A. (*Et pour* W_0 *variable, on trouve la condition la plus faible de toutes en prenant* D_{W_ω}*, où* W_ω *est le plus petit des localiseurs fondamentaux dans* Cat.)

c) *On a de plus, pour la condition la plus faible* D_{W_0}*, l'implication*

$$\mathrm{D}_{W_0} \implies \mathrm{F},$$

où F = Loc (7 F) *est la condition que* Cat *possède suffisamment de* W*-fibrations,* i.e. [page 60] *que c'est une catégorie de fibrations pour* (W, Fib_W).

② *Si* W *et* W_0 *satisfont à* Loc (8 bis) *(axiome des limites dénombrables), alors* $\mathrm{D}_{W_0} \Longrightarrow \mathrm{A}$, i.e. *les 8 conditions envisagées sont équivalentes. Donc si* W *satisfait* Loc (8 bis)*, considérant le plus petit localiseur fondamental* W_L *satisfaisant à* Loc (8 bis) *(de sorte que* $W \supset W_L$*), on voit que les huit conditions*

$$\mathrm{A},\ \mathrm{B},\ \mathrm{C},\ \mathrm{D},\ \mathrm{A}_{W_L},\ \mathrm{B}_{W_L},\ \mathrm{C}_{W_L},\ \mathrm{D}_{W_L}$$

sont équivalentes[33]*. En particulier, sous la même hypothèse (fort anodine) sur* W*, les conditions* Loc (7) = Loc (7 a), Loc (7 b) *et* Loc (7 c) *sont équivalentes, et elles équivalent aussi à la condition* $\mathrm{D} = \mathrm{A}(\mathrm{Parf}_W)$, i.e. *à*

$$\mathrm{Parf}_W \subset \mathrm{Fib}_W \qquad (\textit{cas particulier de Loc (7 a)}),$$

et à ses deux variantes d'apparence plus faible $\mathrm{B}(\mathrm{Parf}_W)$, $\mathrm{C}(\mathrm{Parf}_W)$ *(cas particulier de* Loc (7 b) et de Loc (7 c) *respectivement).*

③ *Supposons que* W_0 *satisfasse à* Loc (8 bis) *(sans hypothèse particulière sur* W*). Alors les* p. 35 *quatre conditions* A_{W_0}, B_{W_0}, C_{W_0}, D_{W_0} *sont équivalentes, et le diagramme d'implications de* [page 61] 1°) *(p.* 59*) se réduit donc au diagramme linéaire*

$$\mathrm{A} \implies \mathrm{B} \implies \mathrm{C} \implies \mathrm{D} \implies \mathrm{D}_{W_0}\ (\Longleftrightarrow \mathrm{A}_{W_0} \ldots),$$

dont le premier terme est Loc (7 a) = Loc (7)*, et le dernier équivaut à l'inclusion*

$$\mathcal{F}_{W_0} \subset \mathrm{Fib}_W,$$

cas particulier de A = Loc (7 a)*, ou encore à l'inclusion (en apparence plus forte)*

$$\mathrm{Fibf}_{W_0} \subset \mathrm{Fib}_W,$$

(cas particulier de la forme équivalente

$$\mathrm{Fibf}_W \subset \mathrm{Fib}_W,$$

de l'axiome Loc (6 a)).

④ *Enfin, on a les implications plus techniques conditionnelles suivantes.*

33. N.B. La forme la plus faible, D_{W_L}, de ces conditions, est la relation $\boxed{\mathrm{Parf}_{W_L} \cap W \subset W^{\mathrm{u}}.}$

a) *Si W_0 satisfait à Loc (7 F), alors les conditions* A_{W_0}, B_{W_0}, C_{W_0} *sont équivalentes, donc le diagramme (4.4.13.5) se réduit au diagramme*

$$\begin{array}{ccc} \mathrm{A} & & \\ \Downarrow & & \\ \mathrm{B} & & \\ \Downarrow & & \\ \mathrm{C} & \Longrightarrow & \mathrm{C}_{W_0} \\ \Downarrow & & \Downarrow \\ \mathrm{D} & \Longrightarrow & \mathrm{D}_{W_0}\ . \end{array}$$

b) *Si W est fortement saturé* (*axiome* Loc (1 ter), *qui est satisfait dans la grande majorité des cas, et est même impératif pour construire un prédérivateur satisfaisant à l'axiome de localisation* Der 2) *et satisfait à* Loc (3 bis) (*axiome de localisation fort,* cf. §1), *alors on a* $\mathrm{B} \Longrightarrow \mathrm{A}$, *donc* [page 62]

$$\mathrm{A} \iff \mathrm{B},$$

i.e. Loc (7 a) = Loc (7) *équivaut à* Loc (7 b).

Commentary · Exposition [ms. 61]

The summary theorem and its printed hypotheses

The theorem gathers a substantial amount of reasoning, and its hypotheses should be read from the proved statements rather than from the abbreviated labels alone. With no extra hypotheses, there are the two downward chains, the restrictions to the reference localizer, and the implication from the weakest perfect-map condition to enough intrinsic fibrations. Sequential-colimit stability of W_0 collapses its column; stability for both localizers identifies the two columns. These are precisely Corollaries 4.4.11 and 4.4.12.

Two local errors in the summary can otherwise change its meaning. At the end of part 3, $\mathrm{Fibf}_W \subset \mathrm{Fib}_W$ is called Loc (6 a); it is Loc (7 a). In part 4(a), the printed assumption is Loc (7 F) for W_0. The proof previously supplied in Corollary 4.4.6 assumes Loc (7), which ensures that $\mathcal{F}_{W_0}$ is closed under composition. The diagram at ms. 63 also labels the upward arrow by $W_0(7)$. Footnote 30 does discuss the weaker F, but there the class is Fib_{W_0}, not $\mathcal{F}_{W_0}$. Substituting the former class for the latter requires an argument which that footnote does not provide.

Accordingly, the direct proof in this section establishes part 4(a) under Loc (7). This observation is a dependency correction, not a counterexample to the printed conclusion. A later argument using minimality settles the relevant comparison without that extra hypothesis, as explained next. It is better to distinguish these two proofs than to pretend that the summary's F follows from the preceding paragraphs.

Part 4(b) has its own independent mechanism. The previous proof in XII, Proposition 1, ms. 54–56, interprets a pullback comparison as the map on homology with locally constant coefficients. The equivalence of the locally constant coefficient categories supplies an isomorphism in Hot_W; strong saturation then returns membership in W. Loc (3 bis) supplies the homotopy Kan extensions used for the interpretation (Grothendieck, 1990–1991f). This explains both hypotheses, and why strong saturation cannot simply be replaced in that proof by closure under retracts.

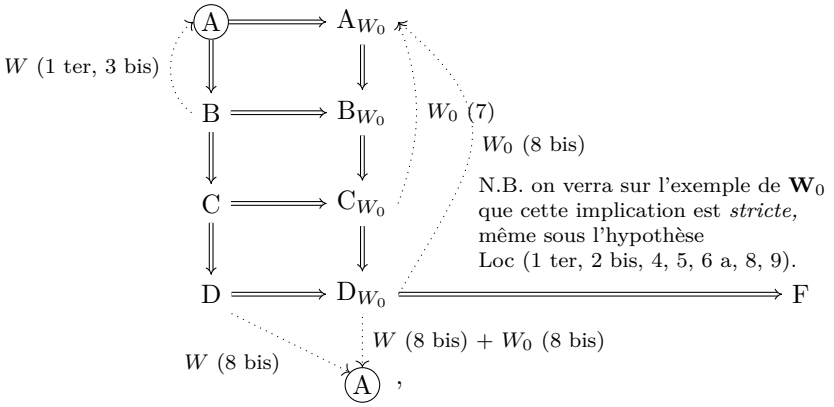
Démonstration. Tout ce qui est énoncé a été vu dans les pages précédentes, à la seule exception de 4°, b. Mais cette implication $\mathrm{B} \Longrightarrow \mathrm{A}$ pour W fortement saturé est prouvé dans XII (prop. 1, p. 54). Je rappelle que la démonstration utilise de façon essentielle le formalisme p. 36
du prédérivateur HOT_W (construit et étudié surtout dans VI), et la théorie des foncteurs

$f_!$, et notamment le résultat (valable si W satisfait à Loc (3 bis, 7 b)) que pour une flèche $f : S' \longrightarrow S$ dans W, la flèche correspondante

$$f^*_{\mathrm{lc}} : \mathrm{HOT}^{\mathrm{lc}}_W(S) \longrightarrow \mathrm{HOT}^{\mathrm{lc}}_W(S')$$

est une équivalence de catégories. Donc l'implication B $\Longrightarrow$ A, elle aussi, est un résultat qui n'a rien d'évident.

4.4.14. Je résume les implications obtenues jusqu'à présent, y compris les implications conditionnelles, dans le diagramme récapitulatif suivant : [page 63]



où j'ai cerclé la condition la plus forte et la plus importante de toutes, A = Loc (7), qui apparaît en deux extrémités opposées du diagramme.

Je voudrais encore faire un commentaire sur les ingrédients des résultats prouvés ici, et notamment de l'équivalence des huit conditions A à D_{W_0}, quand W et W_0 satisfont Loc (8 bis) : [page 64]

1) La théorie des *catégories de chemins* $\underline{\mathrm{Ch}}(X)$ (VII), s'appuyant sur celle des *morphismes lisses, propres, parfaits.*
2) Le lemme de K. S. Brown (3.2.1, page 20).
3) Pour l'implication B $\Longrightarrow$ A, *i.e.* Loc (7 b) $\Longrightarrow$ Loc (7) (sous l'hypothèse W (1 ter, 3 bis)), j'utilise la théorie du prédérivateur Hot_W (VI), et plus particulièrement celle des foncteurs $f_!$ (définis sous la seule hypothèse W (3 bis), renforçant la condition automatique W (3) (faisant partie de la définition d'un localiseur fondamental)).

Il est clair que je ne serais pas arrivé à y voir clair, sans le travail fondamental de K. S. Brown. Il reste pourtant encore des questions ouvertes. Ainsi, si on ne suppose pas que W satisfasse à Loc (8 bis), mais au besoin à Loc (3 ter) (et je n'ai pas eu l'occasion encore de travailler avec un localiseur qui n'y satisfasse …), est-il vrai néanmoins que l'on a encore l'implication p. 37

$$\mathrm{D} \quad \Longrightarrow \quad \mathrm{A},$$

de façon que les conditions A, B, C, D soient équivalentes (et en particulier Loc (7 c) implique déjà Loc (7) dans toute sa force) ? Il est clair dans tout cela que je ne me suis pas suffisamment attaché à développer des exemples et des contre-exemples. Sans compter qu'il me semble que tous les localiseurs fondamentaux sur Cat qu'on rencontre "en pratique" satisfont déjà à Loc (8) et a fortiori à Loc (8 bis), donc pour eux on a bien l'implication escomptée. La même remarque s'applique à l'implication hypothétique [page 65]

$$\mathrm{D}_{W_0} \overset{?}{\Longrightarrow} \mathrm{A},$$

sous hypothèse Loc (8 bis) sur W_0 et sur W. Un candidat évident pour un contre-exemple déjà à la première question serait W_ω = plus petit localiseur fondamental sur Cat, un qu'il

est presque évident qu'il ne satisfait *pas* à Loc (8) ni même Loc (8 bis). Mais il n'y a aucune raison qu'il satisfasse non plus à F, de telle façon que (s'il n'y satisfait pas, donc non plus à D) l'implication D $\Longrightarrow$ A est bel et bien valable. Cela nous amène à la question, plus substantielle, si on n'aurait pas l'implication [page 66]

$$\mathrm{F} \overset{?}{\Longrightarrow} \mathrm{A}\ ^{34},$$

du moins sous des conditions convenables assez anodines sur W, genre Loc (1 ter, 3 bis, 8 bis).

Commentary · Later developments [ms. 64]

Why ordinary homotopy types satisfy A

The ordinary localizer provides a useful verification of the axioms, as well as the eventual answer to the minimum question. Start with a Grothendieck fibration $p : X \to S$ all of whose inverse-image functors are weak homotopy equivalences. For $s \in S$, the inclusion of the strict fibre X_s into the comma category $s\backslash X$ has a right adjoint: it sends $(x, a : s \to p(x))$ to a Cartesian inverse image a^*x. The counit is the chosen Cartesian arrow. Hence this inclusion induces a homotopy equivalence on nerves. The transition functors between these comma categories induce, through these comparisons, the given inverse-image functors on strict fibres.

Quillen's theorem B, applied to the opposite categories, therefore identifies the strict fibres with the homotopy fibres of Np. A precise published formulation is Cisinski (2004, paragraph 2.3.1 and Theorem 2.3.2, p. 136); the comma-category variance is why taking opposites is convenient here. If p also belongs to $\mathbf{W}_\infty$, its homotopy fibres are weakly contractible. Thus every strict fibre X_s is weakly contractible.

After any change of base $T \to S$, the resulting Grothendieck fibration has these same strict fibres. The dual theorem-A criterion now makes its projection a weak homotopy equivalence: the appropriate comma categories retract by adjunction onto the fibres, so they too are weakly contractible. We have proved C for $\mathbf{W}_\infty$, including its universal conclusion, without first assuming A. Empty bases cause no exception: a category mapping to the empty category is itself empty.

To recover A from this verification, the already proved Corollary 4.4.11 applies because $\mathbf{W}_\infty$ is stable under sequential filtered colimits. This stability can be seen without an additional axiom. The nerve commutes with filtered colimits: an n-simplex is a finite string of composable arrows, and the equations defining such a string are finite. The simplicial fibrant replacement Ex^∞ also commutes with filtered colimits, since each simplex of $\mathrm{Ex}^k K$ is specified by a map from the finite simplicial set $\mathrm{Sd}^k \Delta^n$, and its sequential union commutes with filtered colimits. A filtered colimit of Kan complexes is Kan because each horn has only finitely many simplices. Its components, its based homotopy classes of spheres, and the identifications given by homotopies are consequently represented at finite stages. A filtered diagram of weak equivalences therefore induces a bijection on components and isomorphisms on all homotopy groups in its colimit. Applying this to the nerves proves the required closure.

This gives the converse needed in the following status discussion: ordinary weak homotopy equivalences satisfy C, hence D, and then A by the infinite-path comparison. The use of filtered-colimit stability here proves a property of the ordinary localizer; it does not assert that an arbitrary weak fundamental localizer has that stability. Nor does the argument identify every categorical functor with a fibration. It identifies the particular locally constant Grothendieck fibrations with maps whose pullbacks perform the correct homotopy operation, and the natural replacement lemma extends that conclusion to the chapter's weak fibrations.

Commentary · Present state [ms. 65]

The minimum settles the implication questions

Grothendieck is testing whether sequential limits are genuinely needed or merely belong to the replacement method he has available. The proposed candidate W_ω turns out to answer that question in the opposite direction from his prediction. Cisinski (2006a, Theorem 6.1.18, pp. 242–243) proves that ordinary weak homotopy equivalences form the smallest *weak* fundamental localizer.

34. C'est faux, *cf.* plus bas, l'implication Loc (7 D_{W_ω}) $\Longrightarrow$ Loc (7 F) est *stricte.*

The definition used there is given in paragraph 6.1.12, p. 240, and uses the nonrelative theorem-A condition. Thus the theorem applies to the minimum of this chapter:

$$W_\omega = \mathbf{W}_\infty.$$

The distinction from the earlier minimum theorem for the stronger relative axiom is essential. The assertion that W_ω almost certainly fails Loc (8 bis) is therefore false: ordinary weak homotopy equivalences are stable under filtered colimits, as verified above.

Combining weak minimality with the fibre-detection theorem in §6 settles the whole rectangle, without imposing Loc (8 bis) on the two arbitrary localizers. For weak fundamental localizers $W_0 \subset W$, all eight conditions are equivalent, and they hold precisely in the following two cases:

$$W = \mathrm{Fl}(\mathrm{Cat}) \qquad \text{or} \qquad W = W_0 = \mathbf{W}_\infty.$$

Here is the deduction, with the role of connectedness made explicit. Each of the eight conditions implies D_{W_0}, which implies D_{W_ω} by monotonicity of the perfect classes. If W is all arrows, every test is automatic. Assume otherwise. Then D_{W_ω} itself forces Loc (5); one need not assume the potentially problematic converse from mere nondegeneracy to Loc (5).

Indeed, an empty W-aspherical category would put $\varnothing \to e$ in W. Theorem A would then put $\varnothing \to X$ in W for every X, and two-out-of-three would give all arrows. If a nonempty W-aspherical A were disconnected, choose $a : e \to A$ and factor it as $e \to E \xrightarrow{p} A$, with the first map in W_ω and p W_ω-perfect. The category E is connected because $e \to E$ lies in the tautological subclass $W_\omega \subset \mathbf{W}_\infty$. The map p belongs to W and is universally in W by D_{W_ω}. Its strict fibre over a different component of A is empty, again giving the contradiction $\varnothing \to e \in W$. Thus every W-aspherical category is connected and nonempty.

Now factor any $f \in W$ as a W_ω-equivalence followed by a W_ω-perfect p. Every strict fibre of p is W-aspherical by D, hence connected and nonempty. The inclusion $W_\omega \subset \mathbf{W}_\infty$ makes p ordinarily perfect, so it compares these fibres with the corresponding comma categories by ordinary weak equivalences. The connected-component version of Theorem A therefore makes $\pi_0(p)$ bijective. The left factor also induces a bijection on components, so $\pi_0(f)$ is bijective. This proves Loc (5) directly. The commentary at ms. 85 gives the full fibre-detection argument and this nondegenerate reduction.

Theorem 6.1 now gives $W \subset \mathbf{W}_\infty$; weak minimality gives the reverse containment. Finally $\mathbf{W}_\infty \subset W_0 \subset W$ forces $W_0 = \mathbf{W}_\infty$. Conversely ordinary weak homotopy equivalences satisfy A by the preceding theorem-B argument, so all eight tests hold in the second case too.

Consequently both questions D ⇒ A and D_{W_0} ⇒ A have affirmative answers, without sequential-colimit assumptions. The suggested example at ms. 37 satisfying Loc (7) while failing ordinary excision cannot occur: the ordinary and the all-arrow localizers both satisfy the excision axiom. This is a deduction from later weak minimality and the present chapter's perfect-replacement arguments, not the proof Grothendieck had in hand. F remains strictly weaker, as the connected-component example shows. The proper coarse localizer remembering only emptiness is not an exception to the result: it fails D, and the argument does not treat nontriviality alone as connectedness.

C'est sans doute là la question la plus substantielle et la plus intéressante qui est soulevée par le diagramme de la page 63, et qui ne fait plus appel à l'introduction d'un localiseur fondamental W_0 auxiliaire. Vu le lemme de Brown, il ne me semble pas absolument exclu qu'un tel miracle ait bel et bien lieu !

4.4.15 (Digression sur l'axiome Loc (6) des cofibrations.) Cela attire mon attention sur la question en quelque sorte duale : Si Cat a assez de W-*co*fibrations, *i.e.* si muni de (W, Cof_W) c'est une catégorie à cofibrations, cela implique-t-il déjà l'axiome Loc (6 a) (qui avait été dégagé, essentiellement, pour assurer qu'il en soit bien ainsi). La proposition 3.6. (p. 25) nous dit qu'il revient au même de prouver que, si W satisfait Cof, alors les carrés de M. V. ouverts sont W-cocartésien. Mais il me semble à présent que nous pouvons le prouver, sous la seule [page 67]
condition supplémentaire Loc (3 bis) [35], laquelle assure l'existence des foncteurs $f_!$, et en p. 38

35. et peut-être, de plus, Loc (1 ter), *i.e.* W fortement saturée.

particulier la construction de Holim d'un diagramme $\begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$ comme $\int \begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array}$.

Donc dire qu'un diagramme de $\mathcal{M}$ ouvert $\begin{array}{ccccc} & & X_1 & & \\ & \nearrow & & \searrow & \\ X_0 & & & & X \\ & \searrow & & \nearrow & \\ & & X_2 & & \end{array}$ est W-cocartésien, doit revenir à dire que la flèche canonique

$$\int \begin{array}{ccc} & & X_1 \\ & \nearrow & \\ X_0 & & \\ & \searrow & \\ & & X_2 \end{array} \longrightarrow X$$

est dans W ; du moins si W est supposé fortement saturé. Or cela a été bel et bien vu dans XII 3.4 (théorème 4 et son *corollaire* 3, p. 149). Dans loc. cit. j'énonce les conclusions sous l'hypothèse que j'appelais W (7 bis) (l'actuel Loc (6)), mais je viens de vérifier que pour sa partie *directe,* donc aussi pour ses corollaires 1 à 3, on n'a pas besoin de cette hypothèse (comme je l'indique en annotations marginales). [page 68]

Commentary · Exposition [ms. 67]

Why an open cover gives the needed integral comparison

The brief digression uses a geometric fact already isolated in XII: an actual open cover has a homotopically faithful categorical integral. The commentary on Chapter XII, manuscript p. 149, treats the cover criterion separately from the stronger converse-excision assertions. That separation matters here, because Grothendieck is trying to avoid assuming the very cofibration axiom he hopes to recover.

Let $X = U_1 \cup U_2$ be a cover by full open subcategories, in the convention that each U_i is closed under predecessors, and put $U_0 = U_1 \cap U_2$. Let J have arrows $0 \to 1$ and $0 \to 2$, and form the Grothendieck construction E of the diagram $U_0 \to U_1$, $U_0 \to U_2$. There is a canonical augmentation $q : E \to X$. We can verify directly that it is in every weak fundamental localizer.

For an object $x \in X$, consider the comma category E/x. If $x \in U_1 \setminus U_2$, every arrow $u \to x$ starts in U_1, since U_1 is open. In particular an occurrence labelled by 2 must have $u \in U_0$. Forgetting the label defines $T : E/x \to U_1/x$, with section I selecting label 1. Define $H : E/x \to E/x$ by changing label 2 to label 0 and keeping labels $0, 1$. This is well-defined because every label-2 occurrence now lies in the intersection; it sends an arrow from label 0 to label 2 to the corresponding arrow within label 0. The arrows of the span give natural transformations $H \to \mathrm{id}_{E/x}$ and $H \to IT$, while $TI = \mathrm{id}$. Thus T is a categorical homotopy equivalence. Since U_1/x has terminal object x, the category E/x is W-aspherical. The case $x \in U_2 \setminus U_1$ is symmetric.

If $x \in U_0$, all objects occurring in E/x lie in U_0, and the construction is the product of the span index J with U_0/x. The first factor has initial object 0 and the second has terminal object x. Their categorical contractions give a contraction of the product by a finite chain of natural transformations; no general product-stability assertion about W is needed. Hence it is aspherical. The theorem-A axiom now gives $q \in W$.

The homotopy-colimit interpretation identifies the class of E with the homotopy pushout of the span; relative Theorem A supplies the needed Kan-extension formalism. Strong saturation permits one to read an isomorphism in the localized category back as membership in W. Combined with the comparison criterion of Proposition 3.6, this is the intended route from enough intrinsic cofibrations to Loc (6 a). The source explicitly labels its verification provisional. The open-cover comparison itself is secure; extending its argument to arbitrary pushout squares without the indicated hypotheses would reintroduce the invalid converse-excision principle already audited in XII.

4.16. Le résultat que je viens de prouver (sans vérifier avec soin la démonstration) me paraît aussi délicat, qu'il est satisfaisant. Cela rend la "conjecture" hésitante de 4.4.14, savoir

que

$$\mathrm{F} \overset{?}{\Longrightarrow} \mathrm{A} \qquad \text{donc} \qquad \mathrm{F} \Longleftrightarrow \mathrm{A}$$

en supposant que W satisfait Loc (1 ter, 3 bis, 8 bis), nettement plus engageante. Car on commence à s'habituer aux miracles les plus époustouflants …

Pourtant, je viens de vérifier que cette conjecture est fausse, sur l'exemple de

$$\mathbf{W}_0 = \{f \in \mathrm{Fl}(\mathrm{Cat}) \mid \pi_0(f) \text{ iso}\},$$

qui satisfait à F, mais pas à A, ni même à D_{W_ω}. Ainsi l'implication

$$\mathrm{D}_{W_\omega} \implies \mathrm{F}$$

et a fortiori les implications

$$\begin{cases} \text{Loc (7 a)} \implies \mathrm{F},\ \text{Loc (7 b)} \implies \mathrm{F},\ \text{Loc (7 c)} \implies \mathrm{F} \\ \mathrm{D}_{W_0} \implies \mathrm{F} \end{cases}$$

sont *strictes,* même en se limitant à des localiseurs fondamentaux W satisfaisant (comme $\mathbf{W}_0$) à

$$\text{Loc (1 ter, 2, 3 bis, 4, 5, 6 a, 8, 9)} \ ^{36}.$$

p. 39 [page 69]

Que $\mathbf{W}_0$ satisfasse à cet ensemble d'axiomes se vérifie par AQT `[âne qui trotte]`. (**N.B.** Qu'il satisfasse à Loc (1 ter, 2, 3 bis) est déjà conséquence du fait que $\mathbf{W}_0$ est de la forme $W_{\mathbf{D}}$, où $\mathbf{D}$ (= $\mathbf{D}_{\mathbf{W}_0}$ en l'occurrence) est un prédérivateur satisfaisant Der (3, 4, 5), ce qui est bien le cas pour $\mathbf{D}_{\mathbf{W}_0}$, qui est un vrai *dérivateur.*) Il faut vérifier que tout f dans Cat se factorise en $f = pi$, avec $i \in \mathbf{W}_0$, p une $\mathbf{W}_0$-fibration. Or si $f : X \longrightarrow Y$, $I = \pi_0(X)$, on prend $\bar{X} = \coprod_{\alpha \in I} Y_{f(\alpha)}$, d'où une factorisation $f = pi$, où $\pi_0(i)$ est un isomorphisme, *i.e.* $i \in \mathbf{W}_0$, et où $p : \bar{X} \longrightarrow Y$ est, au dessus de la composante connexe Y_j de Y, donné par le produit $\bar{X}|Y_j \simeq Y_j \times \underbrace{\pi_0(f)^{-1}(j)}_{\text{catégorie discrète}}$. Le morphisme p est bien une $\mathbf{W}_0$-fibration, car grâce à Loc (4), il suffit de le vérifier sur chaque Y_j, et là c'est déduit d'une flèche $I_j \longrightarrow e$ (une $\mathbf{W}_0$-fibration) par changement de base $Y_j \longrightarrow e$.

D'autre part, que $\mathbf{W}_0$ ne satisfait *pas* même à D_{W_ω} a déjà été vu (I, p. 87), en réalisant l'application du n-revêtement $f : S^1 \longrightarrow S^1$ dans Cat par un morphisme W_ω-parfait (en fait, par une Cat-fibration localement triviale), chose très facile [37]. [page 70]

Commentary · Examples [ms. 69]

Enough fibrations can be computed from components

Grothendieck's counterexample is effective because its replacement can be calculated completely. Put $W = \mathbf{W}_0$, and let $f : X \to Y$ be any functor. Write $I = \pi_0 X$, $J = \pi_0 Y$, and $\phi = \pi_0 f$. Define

$$\bar{X} = \coprod_{\alpha \in I} Y_{\phi(\alpha)},$$

where Y_j is the full connected component of Y indexed by j. Send an object x of X to the copy of $f(x)$ indexed by the component of x; arrows remain in that copy. This defines $i : X \to \bar{X}$, and the projections of the copies define $p : \bar{X} \to Y$ with $f = pi$.

Each summand of $\bar{X}$ is connected and nonempty, so the summands are exactly its connected components. Thus $\pi_0(i)$ is the identity on I, and $i \in \mathbf{W}_0$. To verify the stronger property of p, let

36. **N.B.** $\mathbf{W}_0$ ne satisfait *pas* Loc (6 b), ni Loc (7 D_{W_ω}) (*cf.* plus bas), par contre Loc (7 F).

37. Autre façon de le voir : on *sait* (XII) que W (5, 7 D_W) $\Longrightarrow W \subset \mathbf{W}_\infty$, or on a $\mathbf{W}_\infty \subset \mathbf{W}_0$, et l'inclusion est *stricte !*

$T \to Y$ be arbitrary. The pullback is a disjoint union of the components of T, with a copy indexed by each α whose target component agrees with theirs. Consequently there is a natural identification

$$\pi_0(T \times_Y \bar{X}) \cong \pi_0 T \times_J I.$$

If $T' \to T$ is a $\mathbf{W}_0$-equivalence over Y, it gives a bijection on the first factor, compatible with the maps to J. Its pullback therefore gives a bijection on the displayed fibre products. Hence $p \in \mathrm{Fib}_{\mathbf{W}_0}$, proving F for all functors. The formula includes empty categories: if I is empty then both the replacement and every pullback are empty.

This factorization keeps precisely the component data of the source and fills each selected target component with a whole copy of that component. It does not try to preserve a nontrivial local system already present in $X \to Y$. That explains why enough intrinsic fibrations can coexist with failure of A, B, C, and even D_{W_ω}.

For the latter failure, take a transitive action of a group G on a finite set E with $|E| > 1$. The action-groupoid projection $E /\!\!/ G \to BG$ has connected source and target, and so belongs to $\mathbf{W}_0$. It is locally trivial with discrete fibre E, hence perfect for every fundamental localizer. Pullback to the unique object of BG gives $E \to e$, which is not a $\mathbf{W}_0$-equivalence. Thus $\mathrm{C}(\mathrm{Parf}_{W_\omega})$ fails. This is the covering obstruction discussed in the commentary on Chapter VI, manuscript p. 43, expressed in a form suited to the present factorization.

Je pense qu'à présent l'ensemble des différentes variantes de l'axiome des W-fibrations est à peu près compris, à cela près qu'il n'est toujours pas clair, en dehors de l'axiome Loc (8 bis), si les implications du diagramme

$$\begin{array}{ccc}
\mathrm{A} & \Longrightarrow & \mathrm{A}_{W_0} \\
\Downarrow & & \Downarrow \\
\mathrm{B} & \Longrightarrow & \mathrm{B}_{W_0} \\
\Downarrow & & \Downarrow \\
\mathrm{C} & \Longrightarrow & \mathrm{C}_{W_0} \\
\Downarrow & & \Downarrow \\
\mathrm{D} & \Longrightarrow & \mathrm{D}_{W_0}
\end{array}$$

sont strictes ou non. Mais il me semble que tous les localiseurs fondamentaux intéressants rencontrés jusqu'à présent satisfont Loc (8), et même Loc (1 ter, 2, 3 bis, 4, 5, 6 a, 8, 9) (peut-être même Loc (7 F) - il faudrait que je teste sur les $\mathbf{W}_n$), donc la perplexité non résolu ? me paraît sans grande conséquence.

[page 71]

Commentary · Development (commentators') [ms. 70]

The component-monodromy criterion

Grothendieck's counterexample suggests a more precise question: which locally constant categorical families really are intrinsic fibrations for connected-component equivalences? A complete answer can be given inside the class of Grothendieck fibrations. It explains both the covering obstruction and the unusually simple replacement just constructed.

Component-monodromy criterion. [ours] Let $p : X \to S$ be a Grothendieck fibration whose transport functors induce bijections on connected components. Let

$$L : S^{\mathrm{op}} \longrightarrow \mathrm{Ens}, \qquad L(s) = \pi_0(X_s),$$

be its component functor. For each connected component C of S, put $I_C = \mathrm{colim}_{C^{\mathrm{op}}} L$. Then the following conditions are equivalent: $p \in \mathrm{Fib}_{\mathbf{W}_0}$; every canonical map $L(s) \to I_C$, for $s \in C$, is a bijection; the monodromy action of the fundamental groupoid of C on L is trivial. Here trivial

means that every loop acts identically, not that a chosen identification of the different fibres was part of the original data.

We first justify the component calculation on which the proof depends. For any Grothendieck fibration $Z \to T$, not necessarily locally constant, there is a canonical bijection

$$\pi_0 Z \cong \operatorname{colim}_{T^{\mathrm{op}}} \pi_0(Z_t).$$

A connected component on the left is generated by vertical arrows in the fibres and by arrows lying over arrows of T. Cartesian factorization writes every arrow as a vertical arrow followed by a Cartesian arrow. Its relation on components is therefore exactly the relation identifying a component of Z_t with its inverse image in Z_s along $s \to t$. These are precisely the relations in the set colimit. Conversely each such relation is represented by a Cartesian arrow. This proves both surjectivity and injectivity of the canonical map. The construction is natural in changes of base and in functors over the base.

Assume first that p is intrinsically $\mathbf{W}_0$-fibrant. For any $s \in C$, the functor $e_s \to C$ is a $\mathbf{W}_0$-equivalence. Pull it back along $X|_C \to C$. The resulting functor $X_s \to X|_C$ induces a bijection on components. By the calculation above this is exactly $L(s) \to I_C$. Thus the second condition is necessary.

Conversely suppose all these canonical maps are bijections. They are natural in s, so they identify L on each C with the constant functor of value I_C. For any $t : T \to S$, the pullback fibration has component functor $L \circ t^{\mathrm{op}}$. Applying the component calculation component by component gives the natural formula

$$\pi_0(X \times_S T) \cong \coprod_{D \in \pi_0 T} I_{\pi_0(t)(D)}.$$

A $\mathbf{W}_0$-equivalence $u : T' \to T$ over S bijects the indexing components and preserves their image in $\pi_0 S$. On the corresponding summands its map is the identity of I_C. Hence it induces a bijection on the displayed component sets. This is the full intrinsic fibration test, proving the first condition.

It remains to identify the second condition with monodromy. All arrows of L are bijections, so L extends uniquely to the groupoid obtained from S by formally inverting its arrows. Within a connected component, choose an object s. Every element of the colimit can be transported to $L(s)$ along a zigzag, and two elements of $L(s)$ have the same image precisely when they lie in the same orbit of the loop group at s. Thus $L(s) \to I_C$ is the orbit map. It is bijective precisely when every orbit has one element, equivalently when every loop acts trivially. This completes the proof, including components with empty fibres.

For $S = BG$, the criterion says exactly that a locally constant family with fibre component set E is intrinsically $\mathbf{W}_0$-fibrant if and only if the G-action on E is trivial. A transitive nontrivial action fails; a constant discrete family succeeds. The replacement at ms. 69 succeeds because its component functor is already the constant family indexed by the copies of each target component.

This is our explicit development of the source's example, not a priority claim. We checked the general smooth-functor treatment in Maltsiniotis (2005, Theorem 1.2), the sharp-map account in Rezk (1998, Section 2), and the component and covering discussions in the commentary on the earlier chapters. Those sources supply the surrounding notions; we did not find this exact component-monodromy formulation there. The proof above is independent of any unverified identification with a particular model structure.

p. 40

5 Axiome des limites Loc (8), et l'axiome Loc (9) d'accessibilité

C'est l'axiome :

Loc (8) W est stable par $\varinjlim$ filtrantes.

Une variante affaiblie utile est la suivante :

Loc (8 bis) W est stable par $\varinjlim$ dénombrables.

Explicitement, si on a deux systèmes inductifs filtrants dans Cat

$$\underline{X} = (X_i)_{i \in I}$$
$$\underline{Y} = (Y_i)_{i \in I},$$

et une flèche

$$\underline{f} = (f_i)_{i\in I} : \underline{X} \longrightarrow \underline{Y},$$

donnée par des $f_i : X_i \longrightarrow Y_i$, posant $X = \varinjlim X_i$, $Y = \varinjlim Y_i$, $f = \varinjlim f_i : X \longrightarrow Y$, Loc (8) nous dit que si les $f_i \in W$, alors $f \in W$. Ici I est une (petite) catégorie filtrante quelconque, [page 72] et quitte à la remplacer par une catégorie cofinale, on peut supposer I un ensemble ordonné. L'axiome Loc (8 bis) correspond au cas où $I = \mathbf{N}$.

Commentary · Exposition [ms. 71]

Two different ways of controlling size

Grothendieck now asks whether a homotopy theory can be recovered from its small pieces. The two conditions he separates answer different parts of that question. Filtered-colimit stability says that a compatible family of already known equivalences remains an equivalence upon passage to the union. Accessibility says that equivalences have witnesses within a bounded range of sizes. Neither condition should be read as the assertion that a particular finite stage of an arbitrary convergent diagram is already an equivalence.

For example, let A be a countable connected category. Its finite subcategories need not be connected, and a finite collection of arrows need not even generate a finite category: one endomorphism can generate infinitely many powers. An infinite cardinal bound behaves better. Closing at most κ arrows under identities and finite composition still gives at most κ arrows. This is why the canonical presentations below use an infinite cardinal and allow enlargement of a stage.

There is a terminological change between these pages and contemporary accessible-category theory. Here "$\aleph_0$-accessible" means generated, in the specified filtered-colimit sense, by categories or functors with at most countably many arrows. It does not mean generated by finitely presentable objects. In modern regular-cardinal terminology, a category with at most κ arrows is κ^+-presentable. Indeed, a functor out of it is specified by at most κ choices of objects and arrows, subject to at most κ equations. In a κ^+-filtered colimit those choices and equations can be realized at a common stage. The same argument applies to commutative squares between functors, hence to the arrow category. This elementary verification is enough to translate the size estimates used here without silently shifting the cardinal index.

The path replacement serves a different purpose. It converts the question about an arbitrary arrow into a question about the fibres of one specially controlled arrow. The argument needs the replacement functor to commute with filtered colimits. It does not require its two output classes to be closed under those colimits. Grothendieck's parenthetical prediction on manuscript page 73 nevertheless contains an error: W_ω is in fact $\mathbf{W}_\infty$, and is filtered-colimit stable. We return to the precise minimality theorem after the proof of 6.1.

Notons que grâce à la construction de la catégorie des chemins, nous avons une factorisation, fonctorielle en f, de toute flèche $f : X \longrightarrow Y$ de Cat en $f = pi$:

$$X \xrightarrow{i=i_f} \bar{X}(f) = \bar{X} \xrightarrow{p=p_f} Y,$$

avec $i \in W_\omega$, $p \in \mathrm{Parf}_{W_\omega}$, a fortiori $i \in W$, $p \in \mathrm{Parf}_W$. Ce foncteur $f \longmapsto (\bar{X}(f), p_f)$ commute aux limites inductives. Ainsi, si on applique ceci à chacun des $f_\alpha : X_\alpha \longrightarrow Y_\alpha$, le factorisant en

$$X_\alpha \xrightarrow{i_\alpha} \bar{X}_\alpha \xrightarrow{p_\alpha} Y_\alpha,$$

on trouve un système inductif de diagrammes de type $\mathbf{\Delta}^2$ (factorisant les f_i), et en passant à la $\varinjlim$ on trouve

$$X \xrightarrow{i} \bar{X} \xrightarrow{p} Y,$$

qui n'est autre que la factorisation canonique de $f = \varinjlim f_i$ en pi, avec $i \in W_\omega$, $p \in \mathrm{Parf}_{W_\omega}$ [page 73] (N.B. Alors que ni W_ω, ni Parf_{W_ω} ne sont stables par $\varinjlim$ filtrantes, à coup sûr !). On a donc $i \in W$, et la condition $f \in W$, *i.e.* $pi \in W$, équivaut donc à $p \in W$. Cela nous ramène, pour vérifier l'axiome Loc (8) (resp. Loc (8 bis)) au cas où les f_i sont dans Parf_{W_ω}.

Supposons maintenant que W satisfasse à D_{W_ω} (la plus faible de toutes les variantes envisagées ici de l'axiome des W-fibrations Loc (7), à la seule exception de F). Alors on a

$$p \in W \qquad \Longleftrightarrow \qquad \text{les fibres de } p \text{ sont } W\text{-asphériques},$$

l'implication $\Longrightarrow$ résultant de l'hypothèse D_{W_ω} $\mathrm{Parf}_{W_\omega} \cap W \subset W^{\mathrm{u}}$, et $\Longleftarrow$ résultant du sorite des morphismes W-lisses *etc.*, via Loc (3). De même, l'hypothèse que les $f_i \in W$ équivaut à celle que les fibres des p_i sont W-asphériques. D'autre part, il est immédiat que toute fibre de p est isomorphe à la $\varinjlim_I$ des fibres correspondantes des p_α. De cette façon, on trouve : p. 41 [page 74]

Proposition 5.1. *Supposons que W satisfasse l'axiome des W-fibrations* Loc (7 D_{W_ω}) (*ce qui est donc le cas s'il satisfait un des axiomes plus forts* Loc (7 a *ou* b *ou* c)). *Pour que W satisfasse à* Loc (8) (resp. *à* Loc (8 bis)), *il faut et il suffit que l'ensemble des objets dans* Cat *qui sont W-asphériques soit stable par* $\varinjlim$ *filtrante* (resp. *par* $\varinjlim$ *de suites dénombrables*).

5.2. Notons que la catégorie Cat est un paratopos, a fortiori elle est *accessible* (*cf.* XVIII, §8), et il en est donc de même de $\underline{\mathrm{Fl}}(\mathrm{Cat})$. Ainsi on a dans $\mathrm{Fl}(\mathrm{Cat}) = \mathrm{Ob}\,\underline{\mathrm{Fl}}(\mathrm{Cat})$ la notion de *partie accessible.* On dira donc qu'une partie W de $\mathrm{Fl}\,\mathrm{Cat}$ (plus généralement d'une catégorie $\mathrm{Fl}(\mathcal{M})$, où $\mathcal{M}$ est une catégorie accessible) est *accessible,* si elle l'est par rapport à la catégorie ambiante $\underline{\mathrm{Fl}}\,\mathrm{Cat}$. Ceci posé, l'axiome Loc (9) est le suivant. [page 75]

Commentary · Exposition [ms. 74]

Taking a fibre after a filtered colimit

Grothendieck wants to replace a closure condition on arrows by a closure condition on aspherical categories. Proposition 5.1 achieves exactly this, provided the path fibration is an intrinsic W-fibration. One small indexing choice makes the proof literal.

Write $p_i : E_i \to B_i$ for the perfect replacements and $p : E \to B$ for their filtered colimit. Choose $b \in B$. After replacing the indexing category by a cofinal filtered poset, choose a stage i_0 and $b_{i_0} \in B_{i_0}$ representing b. For $i \geq i_0$ let b_i be its image. There is then a canonical isomorphism

$$E_b \cong \operatorname*{colim}_{i \geq i_0} (E_i)_{b_i}.$$

To see surjectivity on objects, represent an object of E_b at a stage. The equality of its image with b holds at a later stage, since equality in a filtered colimit of sets is eventually witnessed. The same argument applies to arrows, including the requirement that their image be the identity of b_i. Eventual equality also gives injectivity. Composition is inherited from a common later stage. This proves the displayed isomorphism directly; equivalently, filtered colimits commute with the finite-limit data defining categories and their strict fibres.

Suppose now that W-aspherical categories are closed under the relevant colimits. Since $f_i = p_i i_i$ and $i_i \in W_\omega \subset W$, each $f_i \in W$ implies $p_i \in W$. The hypothesis D_{W_ω} makes each p_i universally a W-equivalence, so every fibre $(E_i)_{b_i}$ is W-aspherical. The displayed formula makes E_b aspherical. The colimit replacement p is again W_ω-perfect because it is the canonical replacement of colim f_i, not because arbitrary filtered colimits of perfect maps have been shown perfect. The aspherical-fibre criterion gives $p \in W$, and then $f \in W$ by two-out-of-three. Conversely, apply closure of W to the maps $X_i \to e$. The proof for sequences is identical, using a tail of the sequence.

This is also the correction needed when a basepoint is chosen on manuscript page 82. A point of a colimit has a representative at some stage; it does not have an image in every earlier stage. Once a representative is fixed, the later images form the based diagram needed for iterated loops.

Loc (9) W est une partie *accessible* de $\mathrm{Fl}(\mathrm{Cat}) = \mathrm{Ob}\,\underline{\mathrm{Fl}}(\mathrm{Cat})$.

5.3. Rappelons que cette notion un peu technique se simplifie considérablement, si on suppose que W satisfait l'axiome de limites Loc (8). Alors W est accessible si et seulement si il existe une *petite* partie W_0 de $\mathrm{Fl}(\mathrm{Cat})$, telle que W soit la "L_1-enveloppe" de W_0, *i.e.* la plus petite partie de $\mathrm{Fl}(\mathrm{Cat})$ contenant W_0 et L_1-stable (*i.e.* stable par $\varinjlim$ filtrantes). S'il en est ainsi, il est clair qu'on peut prendre W_0 de la forme

$$W_\pi = W \cap \mathrm{Fl}_\pi(\mathrm{Cat}),$$

où π est un cardinal infini convenable, et où $\mathrm{Fl}_\pi(\mathrm{Cat}) \overset{\text{déf}}{=} \mathrm{Fl}(\underline{\mathrm{Cat}}_\pi)$, $\underline{\mathrm{Cat}}_\pi$ désignant la sous-catégorie strictement pleine de Cat formée des objets X tels que $\mathrm{card}\, X \leq \pi$ (où par définition, $\mathrm{card}\, X = \mathrm{card}\, \mathrm{Fl}\, X$).

Notons que la filtration qu'on vient de décrire sur Cat (indexée par les petits cardinaux infinis) n'est autre que sa filtration canonique, *i.e.* $\underline{\mathrm{Cat}}_\pi$ est la sous-catégorie pleine de Cat formée des objets π-accessibles de Cat, et de même dans $\underline{\mathrm{Fl}}(\mathrm{Cat})$. On trouve ainsi, grâce aux développements de XVIII, §13, la caractérisation suivante, que je rappelle ici pour mémoire (en me bornant pour simplifier au cas où W satisfait à Loc (8)) : [page 76]

Proposition 5.3.1. [38] *Soit W une partie de* $\mathrm{Fl}(\mathrm{Cat})$ *stable par* $\varinjlim$ *filtrantes, et soient pour tout cardinal infini π, $W_\pi = W \cap \mathrm{Fl}_\pi(\mathrm{Cat})$, et $\underline{W}$, $\underline{W}_\pi$ les sous-catégories pleines correspondantes de* $\underline{\mathrm{Fl}}(\mathrm{Cat})$. *On dit que W est* π-accessible *si W est la L-enveloppe de W_π (donc W est accessible si et seulement s'il existe un cardinal infini π tel que W soit π-accessible). Ceci posé, les conditions suivantes sont équivalentes :* p. 42

a) *W est π-accessible.*

b) *Pour tout système inductif $(f_i)_{i\in I}$, L_1-π-adapté dans* $\underline{\mathrm{Fl}}\,\mathrm{Cat}$ (cf. *ci-dessous*), *de limite inductive f, on a $f \in W$ si et seulement si la sous-catégorie pleine I_W des $i \in Ob I$ tels que $f_i \in W$ est cofinale dans I.* [page 77]

b′) (*Quand on a choisi, pour tout $f \in \mathrm{Fl}(\mathrm{Cat})$, une représentation L_1-π-adaptée de f comme* $\varinjlim$) : *Si $f \in W$, alors la sous-catégorie pleine I_W définie dans* b) *est cofinale dans I.*

Rappelons ici la notion de système inductif L_1-π-adapté. Il s'agit des conditions :

1°) I est grand devant π, et stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$.

2°) Le foncteur $i \longmapsto x_i : I \longrightarrow \underline{\mathrm{Fl}}(\mathrm{Cat})$ commute aux $\varinjlim$ en question.

3°) Les x_i sont dans $\underline{\mathrm{Fl}}(\mathrm{Cat})_\pi \overset{\text{déf}}{=} \underline{\mathrm{Fl}}_\pi(\mathrm{Cat})$.

Notons que dans le cas d'un "paratopos parabélien" tel que Cat ou $\underline{\mathrm{Fl}}\,\mathrm{Cat}$, il y a une façon canonique, π étant donné, de représenter un objet f comme $\varinjlim$ d'un système inductif L_1-π-adapté. Ici,

$$f : X \longrightarrow Y$$

étant donné, on considère l'ensemble I des couples $(X_i \subset X, Y_i \subset Y)$ de sous-catégories de X et de Y, telles que l'on ait $X_i, Y_i \in \mathrm{Ob}\,\mathrm{Cat}_\pi$ (*i.e.* $\mathrm{card}\,(X_i), \mathrm{card}\,(Y_i) \leq \pi$) et que $f(X_i) \subset Y_i$. Il est immédiat que I est filtrant, pour la relation d'ordre évidente [page 78]

$$(X_i, Y_i) \subset (X_j, Y_j) \qquad \Longleftrightarrow \qquad X_i \subset Y_i \ \text{ et } \ X_j \subset Y_j,$$

et même grand devant π, que I est stable par $\varinjlim$ filtrantes de cardinal $\leq \pi$, et que le foncteur

$$i \longmapsto (f_i : X_i \longrightarrow Y_i) : I \longrightarrow \underline{\mathrm{Fl}}\,\mathrm{Cat}$$

commute auxdites $\varinjlim$, enfin les f_i sont dans $\underline{\mathrm{Fl}}(\mathrm{Cat})_\pi$ par construction. Ainsi on a un système inductif L_1-π-adapté, et il est clair que sa limite inductive est canoniquement isomorphe à f. Ceci vu :

Corollaire 5.3.2. *La condition que W soit π-accessible signifie donc que pour tout $f : X \longrightarrow Y \in W$, et pour toute sous-catégorie X_i de X et Y_i de Y, de cardinal $\leq \pi$ (avec $f(X_i) \subset Y_i$), mais en fait c'est inutile, quitte à agrandir Y_i), il existe des sous-catégories $X_j \supset X_i$ et $Y_j \supset Y_i$ de cardinal $\leq \pi$, telles que f induise $f_j : X_j \longrightarrow Y_j$ et que l'on ait $f_j \in W$.* [page 79]

38. *cf. loc. cit.* 13.36.

Commentary · Examples [ms. 78]

Enlarging a countable subfunctor

The order on pairs of subcategories is meant to describe simultaneous approximation of a functor, rather than separate approximation of its source and target. In the displayed formula on this page the right-hand side must be read as $X_i \subset X_j$ and $Y_i \subset Y_j$. The printed comparisons $X_i \subset Y_i$ and $X_j \subset Y_j$ do not even compare two stages. We retain that edition slip in the transcription.

Fix an infinite cardinal κ and a functor $f : X \to Y$. Any collection of at most κ pairs (A_i, B_i) has an upper bound: take the subcategory of X generated by all arrows of the A_i, and the subcategory of Y generated by all arrows of the B_i together with their necessary images under f. Each still has at most κ arrows. A filtered union of at most κ such pairs is itself a pair of the permitted size. Every arrow of X or Y occurs at some stage. These observations verify, without an appeal to abstract accessibility, the canonical adapted presentation used in Corollary 5.3.2.

Here is the admitted Lemma 5.6 in that presentation. Suppose $\pi_0(f)$ is bijective and begin with a countable subfunctor $f_0 : A_0 \to B_0$. We construct increasing countable subfunctors $f_m : A_m \to B_m$. At the next stage perform two operations. For every object $b \in B_m$, choose an object $a \in X$ whose image lies in the connected component of b, and choose a finite zigzag in Y from $f(a)$ to b. Adjoin a, this zigzag, and their images to the relevant categories. Next, for every pair $a, a' \in A_m$ whose images are connected in B_m, choose a finite zigzag between a and a' in X. Such a zigzag exists by injectivity of $\pi_0(f)$. Adjoin it and its image. There are countably many choices to make, each with finite support. Closing under identities and composition preserves countability.

Let $A_\infty = \bigcup_m A_m$ and $B_\infty = \bigcup_m B_m$. Every object of B_∞ occurs at some stage, so the first operation makes $\pi_0(A_\infty) \to \pi_0(B_\infty)$ surjective. If two objects of A_∞ have connected images in B_∞, the finite zigzag witnessing this lies at some stage containing both objects. The second operation at the next stage makes them connected in A_∞. Thus the map is injective. We have found a countable enlargement belonging to $\mathbf{W}_0$.

Finally, π_0 preserves colimits because it is left adjoint to the discrete-category functor. A filtered colimit of bijections is a bijection, so $\mathbf{W}_0$ satisfies Loc (8). The enlargement criterion now proves its asserted countable accessibility. Taking the target to be the terminal category gives the analogous result for connected categories. A small initial subcategory can therefore be enlarged to a connected one; it need not have been connected at the outset.

La notion d'accessibilité de W me semble aussi bien comprise que celle de limites, je n'y vois plus aucun mystère. D'autre part, il me semble que tous les localiseurs fondamentaux utilisés à ce jour y satisfont (tout comme à l'axiome Loc (8)). Ici, je vais me borner à établir :

Théorème 5.4.[39] *Le localiseur fondamental* $\mathbf{W}_\infty$ *(équivalences faibles "habituelles") satisfait à l'axiome* Loc (9) (*en fait, il satisfait à* tous *les axiomes vu jusqu'à présent, savoir* Loc (1 ter, 2, 3 bis, 4, 5, 6, 7, 8, 9)). *Plus précisément,* $\mathbf{W}_\infty$ *est* $\aleph_0$*-accessible (où* $\aleph_0$*, ou* π_0*, est le premier cardinal infini).* p. 43

Je ne reviens pas ici sur la vérification des autres axiomes, et me borne à prouver Loc (9), l'accessibilité, en utilisant ce qui peut m'être utile. Je me borne à indiquer la démonstration (assez technique) dans ses grandes lignes. Pour le moment, W est à nouveau un localiseur fondamental quelconque, sauf qu'il satisfait à Loc (8). [page 80]

Commentary · Exposition [ms. 79]

A direct completion of the accessibility argument

Grothendieck's theorem concerns a concrete size bound, and one can prove it without completing the unfinished general statement of Lemma 5.5(b). We give that proof because it identifies exactly which witnesses must be added. The accessibility assertion of Theorem 5.4 is correct. Its parenthetical list of all axioms must, however, be qualified: the reverse cube condition Loc (6b), if included in the bare label Loc (6) as in Section 3, is not valid for ordinary weak homotopy equivalences. The valid direct excision assertion is Loc (6a).

39. Mérite bien le nom de *théorème,* surtout si on y inclut toute la liste des axiomes Loc mentionnés ci-contre ! Ce théorème n'a rien d'évident, à partir de la définition cohomologique de $\mathbf{W}_\infty$ - c'est un résultat ("bien connu") qui me paraît profond.

Put $R(A) = \mathrm{Ex}^{\infty} N(A)$. This is a Kan complex with the weak homotopy type of the nerve; the Kan construction and its proof are recalled in Cisinski (2006a, Proposition 2.1.39). Two elementary finiteness facts drive the argument. First, if A has countably many arrows, then $R(A)$ has countably many simplices. In each degree, Ex is the set of maps from the finite simplicial set $\mathrm{Sd}\,\Delta^n$, and countably many finite choices remain countable. Iteration and the sequential union preserve that bound. Second, every finite simplicial configuration in $R(A)$ has support in a finitely generated subcategory of A. It appears at some finite Ex stage and involves finitely many original simplices; their finitely many arrows generate a countable subcategory. The same statement holds for a configuration together with a finite simplicial homotopy. These facts also show that R commutes with filtered colimits.

Let $f : A \to B$ be in $\mathbf{W}_{\infty}$ and let $A_0 \to B_0$ be a countable subfunctor. We build a sequence of countable enlargements. At stage m, first carry out the connected-component operations of the previous block. For each object $a \in A_m$, each $n \geq 1$, and each element of

$$\pi_n(R(B_m), f(a)),$$

choose a preimage in $\pi_n(R(A), a)$. Choose simplicial representatives and a homotopy exhibiting that the image of the chosen representative equals the prescribed class in $R(B)$. Adjoin the support of these choices. Next, for every element of $\pi_n(R(A_m), a)$ whose image is zero in $\pi_n(R(B_m), f(a))$, choose a null homotopy in $R(A)$ and adjoin its support. These choices exist because $R(f)$ is a weak equivalence. A countable Kan complex has countably many homotopy classes in every degree, so there are only countably many requirements over all a and n. After adjoining images under f and closing under composition, the next stage remains countable.

At the union, the connected-component argument still applies. To prove surjectivity of a positive homotopy group at an object a, represent a target class by a finite simplicial sphere in $R(B_{\infty})$. Its support and the object a occur at some stage. The next stage supplies the selected preimage and the required comparison homotopy. For injectivity, represent a class in $R(A_{\infty})$ and a null homotopy of its image in $R(B_{\infty})$. Both are finite configurations, so they occur together at some stage. The next stage supplies a null homotopy in the source. Consequently $A_{\infty} \to B_{\infty}$ is a weak equivalence. This proves the enlargement condition of Corollary 5.3.2 for $\kappa = \aleph_0$.

The remaining closure condition is equally finite. Given a filtered diagram of weak equivalences, a sphere, a null homotopy, or an equality of homotopy classes in its Kan replacement is represented at a stage. Bijectivity on components and homotopy groups therefore passes to the colimit. This proves Loc (8) for $\mathbf{W}_{\infty}$ and completes the accessibility theorem. The construction is a direct proof of the known assertion of 5.4, rather than a claim of a new result.

The final reformulation of Lemma 5.5(b) also needs a qualification before it is used outside this example. The literal inclusion $\mathrm{Asph}_W \subset \mathrm{Cat}_{\mathrm{connected}}$ follows from Loc (5); without it, the total localizer makes even the empty category aspherical. The unfinished extra hypothesis in the manuscript cannot be reconstructed uniquely, but this particular missing condition is visible.

Lemme 5.5. [40]

a) *Pour que le critère* 5.3.1 b) *de π-accessibilité de W soit satisfait, il faut et il suffit qu'il le soit lorsque les f_i sont W_ω-parfaits.*

b) *Supposons que $Parf_{W_\omega} \cap W \subset W^u$,* i.e. *que W satisfait à* Loc (7 D_{W_ω}) (*la plus faible des variantes de l'axiome des W-fibrations, à l'exception seulement de* Loc (7 F)). *Supposons de plus (pour être tranquille) que W satisfait* `[phrase non terminée]. [?Alors]` *W est π-accessible si et seulement si l'ensemble* $\mathrm{Asph}_W \subset \mathrm{Ob}\,\mathrm{Cat}$ *des objets W-asphériques de* Cat *est π-accessible. Ou encore, que la partie* $\mathrm{Asph}_W \subset \mathrm{Ob}\,\mathrm{Cat}_{0\text{-}connexe}$ *soit π-accessible, où* $\mathrm{Cat}_{0\text{-}connexe}$ *désigne la sous-catégorie pleine de* Cat *formée des objets* 0*-connexes.*

Pour l'essentiel, la démonstration est la même que celle de 5.1, en un peu plus technique. L'outil essentiel est toujours la factorisation fonctorielle $f = pi$ d'une flèche de Cat, avec $i \in W_\omega$, $p \in \mathrm{Parf}_{W_\omega}$, et la propriété de commutation aux $\varinjlim$ filtrantes de ladite factorisation. [page 81]

Revenons maintenant au cas de $W = \mathbf{W}_{\infty}$. On doit prouver que $\mathrm{Asph}_{\mathbf{W}_\infty} \subset \mathrm{Ob}\,\mathrm{Cat}_{0\text{-connexe}}$ est π_0-accessible. Donc à prouver le critère correspondant à 5.3.1 b), en se plaçant maintenant

40. On suppose que W satisfasse à Loc (8).

dans $\mathrm{Cat}_{0\text{-connexe}}$ au lieu de $\mathrm{Fl}\,\mathrm{Cat}$. On a donc

$$X = \varinjlim_{i\in I} X_i,$$

représentation L_1-π-admissible de X $\mathbf{W}_\infty$-asphérique, à prouver que pour tout i dans I, $\exists j$ dans I et $i \longrightarrow j$, avec X_j $\mathbf{W}_\infty$-asphérique.

J'admets ici le lemme AQT `[âne qui trotte]` :

Lemme 5.6. *Soit* $\mathbf{W}_0 = \{f \in \mathrm{Fl}\,\mathrm{Cat} \mid \pi_0(f)$ *isomorphisme*$\}$. *Alors* $\mathbf{W}_0$ *est* π_0*-accessible (et* L_1*-stable). A fortiori, l'ensemble des objets* 0*-connexes de* Cat *est* π_0*-accessible.*

Ceci nous ramène, quitte à remplacer I par une catégorie cofinale, à supposer que les X_i sont 0-connexes. (Mais j'ai oublié que cette réduction est déjà faite ! Donc il faut voir 5.6 comme ingrédient

de la démonstration de 5.5 b).) p. 44

Soit x un objet (point) de X, x_i son image dans X_i. Considérons (via ma construction standard des catégories de chemins) les $\Omega^1(X,x)$, $\Omega^1(X_i,x_i)$. On trouve [page 82]

$$\Omega^1(X,x) \simeq \varinjlim_{I} \Omega^1(X_i,x_i),$$

et on vérifie que c'est là encore un système inductif L_1-π_0-adapté. En itérant, on trouve pour $n \in \mathbf{N}^*$

$$\Omega^n(X,x) \simeq \varinjlim_{I} \Omega^n(X_i,x_i),$$

limite inductive L_1-π_0-adaptée. Utilisant 5.6, on en conclut que pour tout n, le sous-ensemble (supposons I un ensemble ordonné, que diable ça suffit !)

$$I_n = \{i \in I \mid \Omega^n(X_i,x_i) \ \text{ est 0-connexe}\}$$

est cofinal dans I, de plus il est stable par $\varinjlim$ filtrantes de cardinal $\leq \pi_0$. Donc

$$I_\infty \overset{\text{déf}}{=} \bigcap I_n$$

est aussi cofinal dans I (*cf.* XVIII, 13.10.2, p. 302).

Or $i \in I_\infty$ signifie que tous les $\Omega^n(X_i,x_i)$ sont 0-connexes, ce qui signifie aussi que [page 83]
X_i est $\mathbf{W}_\infty$-asphérique (puisqu'il est déjà 0-connexe) (*cf.* XII, cor. p. 190). Cela achève la démonstration du théorème 5.4.

Commentary · Exposition [ms. 82]

Why the countable intersection is cofinal

Grothendieck's loop argument eliminates the homotopy groups one at a time. The final passage from each degree separately to all degrees together is a small diagonal argument, and its closure hypothesis is essential.

Let I be a directed poset in which every increasing countable sequence has a supremum. Suppose each $J_n \subset I$ is cofinal and closed under suprema of increasing countable sequences contained in J_n. Choose a sequence $n(0), n(1), \ldots$ in which every positive integer occurs infinitely often. Starting above an arbitrary $i \in I$, choose successively

$$i \leq i_0 \leq i_1 \leq \cdots, \qquad i_k \in J_{n(k)}.$$

Cofinality makes every choice possible. Put $j = \sup_k i_k$. For a fixed n, the subsequence with $n(k) = n$ is cofinal in this increasing sequence. It has the same supremum j, and all its terms belong to J_n. Hence $j \in J_n$. This holds for every n, so $j \in \bigcap_n J_n$. Since $j \geq i$, the intersection is cofinal.

Applied to the I_n of the text, the argument finds a stage at which all iterated loop categories are connected. Connectedness of the original stage is already part of the reduction; together these conditions say that all homotopy groups vanish. The loop-detection comparison itself was treated in

the commentary on Chapter XII, manuscript p. 190. Here the new issue is making countably many conditions hold at one stage.

Two apparent defects should therefore be distinguished. The point x must first be represented at a stage, as explained above. By contrast, the cardinal π_0 in the bound $\leq \pi_0$ is not a misprint: Theorem 5.4 has expressly used π_0 as another notation for $\aleph_0$. Replacing it is unnecessary and would erase the author's declared convention.

6 La condition Loc (5, 7 D_{W_ω}) implique $W \subset \mathbf{W}_\infty$

Je vais redonner ici, pour mémoire, la démonstration du théorème 1 de XII :

Théorème 6.1. *Soit W un localiseur fondamental satisfaisant à* D_{W_ω} *(où W_ω est le plus petit de tous les localiseurs fondamentaux). Alors on a $W = \mathrm{Fl}\,\mathrm{Cat}$,* ou *$W \subset \mathbf{W}_\infty$ (suivant que W ne satisfait pas à l'axiome de connexité* Loc (5), *ou qu'il y satisfait.)*

Je vais faire la démonstration en supposant que W satisfait à Loc (5). Pour la démonstration du fait que dans le cas contraire, on a $W = \mathrm{Fl}(\mathrm{Cat})$, je renvoie à XII, th. 2, p. 20. (Voir aussi le résultat très voisin, plus général et plus faible, rappelé dans prop. 2.7, page 12.)

Lemme 6.2. *Soient W, W' deux localiseurs fondamentaux, W satisfaisant à* Loc (7 D_{W_ω}). *Pour qu'on ait $W \subset W'$, il faut et il suffit qu'on ait* $\mathrm{Asph}_W \subset \mathrm{Asph}_{W'}$.

Il suffit bien sûr de prouver "il suffit", donc supposons que tout objet W-asphérique est [page 84]
W'-asphérique, et prouvons que $W \subset W'$. Soit donc $f : X \longrightarrow Y$ dans W, prouvons $f \in W'$. Pour ceci, j'utilise la factorisation canonique $f = pi$,

$$X \xrightarrow{i \in W_\omega} \bar{X} \xrightarrow{p \in \mathrm{Parf}_{W_\omega}} Y.$$

On a donc aussi $i \in W$, et comme $f = pi \in W$, on a $p \in W$, donc $p \in \mathrm{Parf}_{W_\omega} \cap W$. Or D_{W_ω} p. 45
signifie que cet ensemble est contenu dans W^{u}, donc on a $p \in W^{\mathrm{u}}$, et en particulier les fibres de p sont W-asphériques. Elles sont donc W'-asphériques, et comme p est W_ω-parfait et a fortiori W'-parfait, il s'ensuit que $p \in W'$. On a aussi $i \in W_\omega \subset W'$, donc $i \in W'$, donc aussi $f = pi \in W'$, q.e.d.

Revenons alors à la démonstration de 6.1. En vertu du lemme, il suffit de prouver que si X dans Cat est W-asphérique, il est $\mathbf{W}_\infty$-asphérique. Mais par hypothèse Loc (5) on sait déjà que X est 0-connexe, et choisissant un point x dans X, qu'il soit $\mathbf{W}_\infty$-asphérique s'exprime par la condition que les $\Omega^n(X, x)$ (pour $n \geq 1$) sont 0-connexes (*cf.* XII, cor. p. 190 déjà cité). [page 85]
Pour ceci, à cause de Loc (5) il suffit de prouver qu'ils sont W-asphériques, et par induction il suffit de prouver que $\Omega^1(X, x)$ l'est. Or par définition de $\Omega^1(X, x)$, c'est là la fibre en x d'une flèche dans Cat

$$E = \underline{\mathrm{Ch}}(X, x, -) \xrightarrow{p \in \mathrm{Parf}_{W_\omega}} X,$$

où la source E est W_ω-asphérique, a fortiori W-asphérique. Comme X l'est également, on a donc $p \in W$, donc $p \in \mathrm{Parf}_{W_\omega} \cap W \subset W^{\mathrm{u}}$, donc p est à fibres W-asphériques, et en particulier $\Omega^1(X, x)$ est W-asphérique, q.e.d.

Corollaire 6.3. (Caractérisation de $\mathbf{W}_\infty$.) $\mathbf{W}_\infty$ *est le plus grand localiseur fondamental satisfaisant à* Loc (5, 7 D_{W_ω}) *- donc aussi le plus grand localiseur fondamental satisfaisant à* Loc (5, 7).

Je pense que nous trouverons plus bas une caractérisation "interne" de $\mathbf{W}_\infty$, via un "axiome de descente" (très fort) Loc (10), comme l'unique localiseur fondamental satisfaisant Loc (5, 7 D_{W_ω}, 10) (ou encore Loc (5, 7, 10)). [page 86]

Commentary · Exposition [ms. 85]

The fibration axiom recovers every homotopy group

Grothendieck's comparison theorem explains why the seemingly modest axiom D_{W_ω} is so restrictive. Its role is not merely to furnish some factorization. It makes the fixed categorical path

replacement valid for the chosen equivalences W, and that replacement retains all ordinary homotopy groups.

The proof has two logically separate steps. First, under D_{W_ω} a class W is detected by its aspherical objects. If every W-aspherical category is W'-aspherical and $f \in W$, factor $f = pi$ with $i \in W_\omega$ and p W_ω-perfect. Then $p \in W$. The axiom makes all its fibres W-aspherical, hence W'-aspherical. Perfectness for W_ω implies perfectness for W', and the aspherical-fibre criterion gives $p \in W'$. Since $i \in W'$, we have $f \in W'$. Only the source localizer W needs the fibration axiom in this argument.

Second, let X be W-aspherical and assume Loc (5). It is connected and nonempty, so choose $x \in X$. The categorical based-path construction gives a map $E_xX \to X$ with E_xX W_ω-aspherical and with fibre the categorical loop object. Both total space and base are W-aspherical, so the map is in W. Since it is also W_ω-perfect, its fibre is W-aspherical. Repeat at the constant loop. Loc (5) says that every resulting loop category is connected. Thus all ordinary positive homotopy groups of NX vanish. The first step, with $W' = \mathbf{W}_\infty$, now proves $W \subset \mathbf{W}_\infty$.

The connectedness hypothesis supplies the basepoint and turns asphericity into vanishing of components at every loop stage. The conclusion $W \subset \mathbf{W}_\infty$ cannot hold for the total localizer $W = \mathrm{Fl}(\mathrm{Cat})$, which satisfies the universal-fibration condition but contains all functors. The shorter recollection on manuscript page 102 must therefore retain that exception or the explicit Loc (5) hypothesis.

The total-localizer branch, cited rather than proved in the source, admits the following direct completion. Suppose that W satisfies D_{W_ω} and that a W-aspherical category A is disconnected. Choose an object a in one component and an object b in another. Since $A \to e$ is in W, two-out-of-three makes $a : e \to A$ a W-equivalence. Factor it canonically as

$$e \xrightarrow{i} E \xrightarrow{p} A, \qquad i \in W_\omega, \quad p \in \mathrm{Parf}_{W_\omega}.$$

The tautological inclusion $W_\omega \subset \mathbf{W}_\infty$ makes E weakly contractible, in particular nonempty and connected. This inclusion only uses that $\mathbf{W}_\infty$ is a fundamental localizer; it does not use the later equality with W_ω. The image of p is consequently contained in the component of a. Its strict fibre over b is empty. On the other hand, $p \in W$ by two-out-of-three, so D_{W_ω} makes its empty fibre W-aspherical. Thus $\varnothing \to e$ belongs to W.

The same conclusion holds immediately if the original aspherical category is empty. Once $\varnothing$ is aspherical, Loc (3) applies to $\varnothing \to B$ for every category B: each of its comma categories is empty and hence aspherical. Therefore $\varnothing \to B$ is in W for all B. For any $f : B \to C$, two-out-of-three applied to $\varnothing \to B \to C$ gives $f \in W$. Hence W is total.

Consequently, if W is not total, every W-aspherical category is nonempty and connected. The loop argument above uses exactly this property: it applies to X and successively to all its W-aspherical loop categories. It therefore proves $W \subset \mathbf{W}_\infty$ in the nontotal case, and in particular implies Loc (5). This completes the whole dichotomy without a product-preservation or strong-saturation assumption. It also avoids the separate dependency issue with Proposition 2.7 discussed in the Section 2 audit; that general issue is not being used to withhold the stronger conclusion available under D_{W_ω}.

6.4. *Commentaire sur les axiomes des W-fibrations du types* Loc (7 …), *et notamment sur* Loc (7 D_{W_ω}). *Relation conjecturale avec les* "types d'homotopie simples" *de* Whitehead.

a) Il me semble que *tous* les localiseurs fondamentaux avec lesquels on a travaillé tacitement jusqu'à présent (les $\mathbf{W}_n$ ($n \in \mathbf{N}^*$), équivalence homologique et cohomologique, équivalence rationnelle …) correspondent à des localiseurs $W \supset \mathbf{W}_\infty$. Par le théorème 6.1, il s'ensuit donc qu'*aucun* d'eux, à part $\mathbf{W}_\infty$ lui-même, ne satisfait à la plus faible version D_{W_ω} de toutes de l'axiome Loc (7 …), à la seule exception de Loc (7 F). Ainsi, les localiseurs W qui y satisfont sont à l'exception de D_{W_ω} lui-même, d'un type tout nouveau - des localiseurs qui sont strictement contenus dans $\mathbf{W}_\infty$.

b) Ceci soulève, bien sûr, la question de l'existence de localiseurs fondamentaux contenus dans $\mathbf{W}_\infty$ et distincts de $\mathbf{W}_\infty$. Je n'ai pas même prouvé que $W_\omega \neq \mathbf{W}_\infty$, alors que je n'ai pas [page 87]
le moindre doute que tel est bien le cas. `[Cisinski a démontré que` $W_\omega = \mathbf{W}_\infty$`.]` A fortiori, j'ignore si le plus petit localiseur fondamental satisfaisant Loc (7 D_{W_ω}) est différent de $\mathbf{W}_\infty$.

Je serais tout particulièrement intéressé par le plus petit localiseur $\tilde{\mathbf{W}}_\infty \subset \mathbf{W}_\infty$ satisfaisant tous les axiomes de stabilité Loc (1 ter, 2, 3 bis, 4, 6, 7, 8) (**N.B.** il satisfait nécessairement Loc (5) car $\tilde{\mathbf{W}}_\infty \subset \mathbf{W}_\infty \subset \mathbf{W}_0$, de plus Loc (7) équivaut, en présence de Loc (8), à la forme très faible Loc (7 D_{W_ω}) : $\mathrm{Parf}_{W_\omega} \cap W \subset W^{\mathrm{u}}$. Enfin, on montre que Loc (4) est conséquence des autres axiomes, *cf.* plus bas.) Je conjecture à présent que $\tilde{\mathbf{W}}_\infty$ (qui donne lieu à un formalisme homotopique tout aussi complet que la sempiternelle "équivalence faible") est bel et bien différent de $\mathbf{W}_\infty$ `[non par Cisinski]`. Et je suspecte que la catégorie $\mathrm{Hot}_{\tilde{\mathbf{W}}_\infty}$ mérite d'être appelée catégorie des *"types d'homotopie simples"* (à la Whitehead) [41]. Un premier test décisif, c'est de vérifier que l'invariant cohomologique principal de Whitehead d'un type d'homotopie simple, savoir un élément dans un $K(\mathbf{Z}[\pi_1])$ (si je me rappelle bien [42]) - ou quelque chose d'approchant, peut se définir déjà en termes des objets dans $\mathrm{Hot}_{\bullet\,\tilde{\mathbf{W}}_\infty}$ (le Hot pointé). Et deuxio, si on associe à une "maquette simpliciale" finie (décrivant combinatoirement une triangulation finie) l'ensemble ordonné de ses simplexes, vu comme un objet de Cat, et l'objet correspondant dans $\mathrm{Hot}_{\tilde{\mathbf{W}}_\infty}$, celui-ci ne change pas [43], à isomorphisme canonique près, quand on effectue sur ce "polyèdre" envisagé une opération élémentaire à la Whitehead. Cela suggère d'ailleurs d'essayer de définir de telles opérations élémentaires dans Cat tout entier, ou tout au moins dans la sous-catégorie des (petits) ensembles ordonnés, dans l'espoir de parvenir (peut-être) à une construction plus directe du localiseur pertinent pour le type d'homotopie simple.

p. 46

[page 88]

[page 89]

Commentary · Later developments [ms. 87]

The weak minimality theorem settles the finer-localizer hopes

Grothendieck is looking for a homotopy theory finer than weak homotopy equivalence while retaining the elementary local axioms. The subsequent minimality theorem rules out that particular route. The exact result needed here is stronger than minimality among localizers satisfying the relative Theorem A axiom.

Cisinski (2006a, Theorem 6.1.18, pp. 242–243) proves that the weak homotopy equivalences of nerves form the least *weak* fundamental localizer. His Definition 6.1.12 uses the ordinary aspherical-functor axiom, corresponding to this chapter's Loc (3). Thus it applies to Definition 1.1 here, and gives

$$W_\omega = \mathbf{W}_\infty.$$

The earlier strong-localizer theorem alone would not justify this substitution. The weak theorem does. The bracketed correction on manuscript page 87 is consequently valid in the precise axiom system of these pages.

Combining this equality with Theorem 6.1 gives a particularly transparent consequence. If W satisfies Loc (5) and D_{W_ω}, then

$$\mathbf{W}_\infty = W_\omega \subset W \subset \mathbf{W}_\infty,$$

so $W = \mathbf{W}_\infty$. There is no additional size or countable-colimit assumption in this deduction. With the completed dichotomy in the preceding block, D_{W_ω} alone allows exactly the two possibilities $W = \mathrm{Fl}(\mathrm{Cat})$ and $W = \mathbf{W}_\infty$. In particular, the theory sought here cannot retain more information than ordinary homotopy types by choosing a smaller fundamental localizer.

The proposed $\widetilde{\mathbf{W}}_\infty$ requires a further reading decision. With Loc (6) understood as the intended direct condition Loc (6a), $\mathbf{W}_\infty$ satisfies the defining list, and the same sandwich makes the intersection equal to $\mathbf{W}_\infty$. Literally, however, Section 3 defines Loc (6) to include the reverse condition (6b). No fundamental localizer contained in $\mathbf{W}_\infty$ can satisfy that full list: weak minimality would make it $\mathbf{W}_\infty$ itself, which fails (6b). Equivalently, Loc (5) together with Loc (7) forces $W = \mathbf{W}_\infty$, contradicting reverse excision. Thus the literal requested nondegenerate family is empty. One cannot call its purported minimum $\mathbf{W}_\infty$ without making the direct-excision qualification explicit. This is the same distinction required in the parenthesis of Theorem 5.4.

41. Canulé (?), *cf.* commentaire page 92.

42. non, c'est un invariant plus fin …

43. C'est évident même pour W_ω, puisque le type d'homotopie *stricte ne change pas !* - Non, il y a ici confusion entre homotopie topologique et homotopie combinatoire, *cf.* p. 92 …

The theorem does not say that every weak fundamental localizer satisfies relative Loc (3 bis). Minimality is an inclusion statement about the least class; relative locality is an additional property of an arbitrary larger class. Keeping these separate is necessary when reading the disputed recovery argument in 7.7.2.

c) Je termine par un commentaire sur le sens de l'axiome Loc (7 D_{W_ω}), par opposition avec l'axiome plus faible Loc (7 F) qui assure seulement l'existence d'une "structure à fibrations" (W, Fib_W) sur Cat. Je signale aussi en passant que Loc (7 F), tout comme Loc (5) et comme Loc (9), et à la différence de tous les autres axiomes passés en revue jusqu'à présent, n'est *pas* un axiome de stabilité : il affirme, non que certaines flèches (sous des hypothèses convenables) sont dans W, mais l'existence d'une factorisation. Aussi il n'est nullement clair que l'intersection [page 90]
de deux localiseurs fondamentaux satisfaisant à Loc (7 F) y satisfasse.

Loc (7 D_{W_ω}) est exactement ce qu'il faut pour assurer que la factorisation canonique d'une flèche $f : X \longrightarrow Y$ de Cat en $f = pi$, avec $i \in W_\omega$, $p \in \mathrm{Parf}_{W_\omega}$, est aussi une factorisation du type stipulé par l'axiome de factorisation Loc (7 F) : on a (bien sûr) $i \in W$, mais aussi $p \in \mathrm{Fib}_W$ (car D_{W_ω} équivaut à sa forme forte $\mathrm{Parf}_{W_\omega} \subset \mathrm{Fib}_W$). Cela signifie aussi que la factorisation canonique de f peut être utilisée pour "calculer" ses fibres homotopiques, non seulement en les points de Y, mais aussi par rapport à un morphisme de changement de base $Y' \longrightarrow Y$.

En l'absence de cet axiome, admettant seulement Loc (7 F), on sait qu'il existe une [page 91]
théorie des fibres W-homotopiques, mais on n'a a priori aucun moyen pratique pour "calculer" p. 47
lesdites fibres, faute de disposer d'un critère maniable suffisant pour assurer $p \in \mathrm{Fib}_W$. J'ai donc l'impression que l'axiome Loc (7 F) va être d'un maigre usage. (Il est vrai que l'exemple de $\mathbf{W}_0$ devrait me rendre prudent ! Il me faudrait regarder le cas des $\mathbf{W}_n$, s'ils satisfont Loc (7 F) …). À mon sens, l'axiome Loc (7 D_{W_ω}) est l'exact "minimum vital" pour être à l'aise dans le maniement des fibres W-homotopiques. Il est remarquable que ceci implique qu'on doive avoir $W \subset \mathbf{W}_\infty$, et que cela exclut du même coup *toutes* les variantes considérées à ce jour de la notion de type d'homotopie !

Commentary · Examples [ms. 90]

Why ordinary paths fail after truncation

Grothendieck distinguishes a usable prescribed replacement from the bare existence asserted by Loc (7F). A familiar example shows why that distinction survives even when a localized homotopy theory is completely well behaved.

Use the convention that $\mathbf{W}_n$ detects components and homotopy groups through degree n, and choose a connected based space $X = K(G, n+1)$ with $G \neq 0$; for $n = 0$ any nontrivial group is allowed, and for $n \geq 1$ take an abelian group. The based path space $P_x X$ is contractible, so

$$p : P_x X \longrightarrow X$$

is a $\mathbf{W}_n$-equivalence. Its ordinary fibre is ΩX, whose homotopy group in degree n is G (whose components are G if $n = 0$). Hence $\Omega X \to e$ is not a $\mathbf{W}_n$-equivalence. Pulling p back along the basepoint therefore destroys the equivalence.

The same calculation can be made with categorical models and the categorical path construction used in these pages: its nerves represent the indicated path space and loop space. It disproves universality of this particular $\mathbf{W}_n$-equivalence. It does not show that the localized theory has no homotopy fibre. Indeed, after n-truncation, both $P_x X$ and X become terminal, so the homotopy fibre of the induced map in the localized theory is terminal. The nontrivial ordinary loop object was the wrong replacement for that calculation.

This explains the force of D_{W_ω}. It insists that one path construction, which knows all homotopy groups, continue to calculate fibres after changing the equivalences. Truncation deliberately forgets a homotopy group of the base that becomes a visible homotopy group of the ordinary fibre. The obstruction is therefore structural. For $\mathbf{W}_0$, the explicit factorization at manuscript page 69 gives a useful replacement by copies of connected components; it avoids this obstruction and proves the weaker Loc (7F). For every finite n, the proper localized Thomason models discussed at page 124

likewise prove Loc (7F). Thus the source's caution about the usefulness of F should be read as a concern about calculations, not as a theorem of nonexistence.

[page 92]

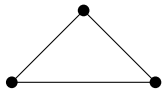
6.5. Finalement, la relation entrevue avec les "types d'homotopie simples" de Whitehead s'évanouit [44] : en effet, ces derniers sont strictement plus fin que les types d'homotopie (*stricts*) ordinaires, et l'invariant de Whitehead n'est *pas* un invariant du type d'homotopie ordinaire. Il est vrai que lorsqu'on considère le squelette simplicial d'un polyèdre triangulé comme objet de Cat, l'homotopie (continue) ordinaire entre deux tels polyèdres s'interprète, dans Cat, non pas (me semble-t-il) par l'existence d'un homotopisme entre les objets correspondants de Cat, mais par leur isomorphie dans $\mathrm{Hot}_{\mathbf{W}_\infty} \overset{\text{déf}}{=} \mathrm{Cat}(\mathbf{W}_\infty)^{-1}$. Mais le procédé d'approximations simpliciales de Brouwer montre que si les polyèdres X et Y sont homotopes, alors il existe des subdivisions simpliciales X' et Y' de X et de Y qui sont "combinatoirement homotopes" (dans un sens convenable). Donc je présume que les maquettes simpliciales de X' et de Y' sont alors homotopes dans Cat. Il s'ensuivrait que X et Y, j'entends leurs maquettes, sont [page 93] bien homotopes dans Cat, s'il est vrai qu'une subdivision simpliciale (il suffirait la subdivision barycentrique, en itérant …) X' de X a une maquette dans Cat qui lui est homotope. La question soulevée ici me paraît intéressante, il faut visiblement en trouver la réponse. Elle se pose déjà pour tout ensemble ordonnée I, qu'on va supposer fini pour simplifier, en prenant l'ensemble (ordonné par inclusion) $I' = \mathrm{sd}(I)$ des parties totalement ordonnées de I. On sait que les réalisations topologiques de I et de $\mathrm{sd}(I)$ sont canoniquement isomorphes, mais I et $\mathrm{sd}(I)$ sont-ils homotopes dans Cat ? On a bien une application croissante canonique

$$f : \mathrm{sd}(I) \longrightarrow I,$$

qui associe à toute partie totalement ordonnée de I son plus grand élément. Mais existe-t-il une application croissante en sens inverse, qui soit un homotopisme inverse du précédent à homotopie près ? Je viens de vérifier dans le cas

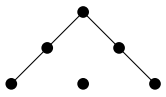
manque [page 94]

du bord du 2-simplexe [page 95]


ou

,

subdivision barycentrique p. 48


ou


(hexagone), que pour toute application croissante $g : I \longrightarrow \mathrm{sd}(I)$, sa réalisation topologique est homotope à une application constante, donc g ne peut être un homotopisme dans Cat (car cela impliquerait que sa réalisation topologique est un homotopisme).

Ceci montre donc que la notion d'homotopisme dans Cat est *strictement plus fine* que celle pour les réalisations topologiques, ce qui redonne à nouveau substance à l'espoir que l'invariant de Whitehead pourrait se définir comme invariant d'homotopie sur Cat, ou du moins sur une sous-catégorie pleine convenable, telle celle des ensembles ordonnés (éventuellement restreints à être finis), ou des maquettes simpliciales. Et même, que l'ensemble des Wh-types d'homotopie (types d'homotopie simples) peut se plonger canoniquement dans un Hot_W [page 96] convenable, où W serait un localiseur fondamental strictement plus fin que $\mathbf{W}_\infty$, p.ex. (qui sait ?) $\tilde{\mathbf{W}}_\infty$. Le premier travail qu'on aurait envie de faire, c'est de construire un tel W *ad hoc*, pour que l'invariant de Whitehead puisse s'écrire comme un foncteur sur $\mathrm{Hot}_{\bullet W}$. Pour

44. Mais peut-être quand même pas - la question reste ouverte.

commencer, il faudrait déjà s'assurer que cet invariant peut être vu comme un invariant du type d'homotopie, *i.e.* qu'il s'interprète comme un foncteur sur $\mathrm{Hot}_\bullet\mathrm{Cat} \simeq \mathrm{Cat}(\underbrace{\text{homotop}}_{\substack{\text{ensemble des}\\ \text{homotopismes}}})^{-1}$.

Ce travail suggérera peut-être aussi la définition des "invariants de Whitehead supérieurs" (impliquant les π_i, non seulement le π_1), lesquels pourraient peut-être servir (avec celui de dimension 1) à décrire le localiseur hypothétique W ("de Whitehead").

Commentary · Development (commentators') [ms. 95]

A sharp degree bound for the polygon examples

Grothendieck's six-point example is decisive because subdivision doubles the number of edges without giving a categorical homotopy inverse. We can make its obstruction quantitative, and thereby see that the phenomenon persists through every iterated subdivision.

For $m \geq 3$, let C_m be the height-one poset with minimal elements v_i and maximal elements e_i, indexed by $i \in \mathbf{Z}/m$, and with the only strict relations

$$v_i < e_i, \qquad v_{i+1} < e_i.$$

It is the face poset of a polygonal circle. Its order complex is a circle with $2m$ edges. A choice of the cyclic ordering fixes the sign of the degree of a map of realizations.

Degree spectrum for crowns. [ours] For $m, n \geq 3$, the degrees of the maps $|NC_m| \to |NC_n|$ induced by monotone maps $C_m \to C_n$ are exactly the integers d satisfying $|d|n \leq m$.

Proof. A monotone map induces a simplicial map of the order complexes. Each of the $2m$ domain edges maps either to a single target edge or to a vertex. Give every target edge length one, and lift the image of the oriented domain circle to the universal covering line of the target circle, starting at a chosen lift of its initial vertex. Each traversed domain edge changes the lifted coordinate by at most one in absolute value. After all $2m$ edges the displacement is $2nd$, where d is the degree. Therefore $2n|d| \leq 2m$.

For the converse, choose a closed walk of m steps in the set of polygon vertices of the target. Its first $n|d|$ steps go around the polygon $|d|$ times, in the sign prescribed by d; all remaining steps are stationary at the initial vertex. Send the ith minimal element of C_m to the ith vertex of this walk. If a step traverses a target edge, send the corresponding maximal element to that edge. If the step is stationary, send the maximal element to the same target vertex. In either case the images of both adjacent minimal elements are less than or equal to the image of the maximal element. This defines a monotone map. Its realized walk has degree d, as required. □

The nonempty-chain poset $\mathrm{sd}(C_m)$ is isomorphic to C_{2m}: its minimal elements are the singleton chains, and its maximal elements are the two-element chains, arranged in the subdivided cycle. Induction gives $\mathrm{sd}^r(C_m) \cong C_{2^r m}$. For every $r \geq 1$ the proposition forces every monotone map

$$C_m \longrightarrow \mathrm{sd}^r(C_m)$$

to have degree zero. Maps between circles of degree zero are null-homotopic, whereas the maximum map in the other direction has degree one. No categorical homotopy inverse can exist. Taking $m = 3$ and $r = 1$ proves the claim on manuscript page 95 in full.

There is also a direct categorical rigidity check. If an endomap $h : C_m \to C_m$ satisfies $h \geq \mathrm{id}$, it fixes the maximal elements. The image of v_i must then lie below both e_{i-1} and e_i and above v_i, forcing $h(v_i) = v_i$. Thus $h = \mathrm{id}$. Dually $h \leq \mathrm{id}$ also forces equality. Any finite zigzag of natural transformations starting at the identity is therefore constant. A categorical homotopy equivalence between two crowns would have strictly inverse composites and hence be an isomorphism; crowns of different sizes cannot be categorically homotopy equivalent.

The general distinction between finite-space homotopy and simple homotopy is established in the literature discussed next. We did not find this explicit degree-spectrum calculation in the consulted finite-space papers. We include it as an elementary development of Grothendieck's example, without a priority claim.

Commentary · Later developments [ms. 96]

The finite-poset version of simple homotopy

Grothendieck's attempted localizer is ruled out by minimality, but his suggestion to work directly with finite ordered sets has a substantial later realization. These are two different claims, and the failure of the first does not diminish the second.

For a finite T_0 space, regarded as a finite poset, Barmak and Minian define elementary simplifications by deleting a weak point: a point whose strict lower set or strict upper set is contractible. Their Theorem 3.10 identifies the resulting simple-equivalence classes with simple homotopy types of finite simplicial complexes, through the order-complex and face-poset constructions. Their Theorem 4.13 gives the corresponding statement for maps (Barmak and Minian, 2008). This is a precise version of the finite combinatorial program suggested here. It is not a fundamental localizer on all of Cat finer than $\mathbf{W}_\infty$.

The crown calculation explains why strict categorical homotopy equivalence is too fine for the desired comparison. A polygon and its subdivision have the same simple homotopy type, while their face posets have different strict categorical homotopy types. The elementary finite-space operations therefore must allow more than deletion of a point with a genuine order-theoretic deformation retraction. Conversely, passing all the way to weak homotopy equivalence forgets the simple structure. The successful theory sits between these two equivalence relations on finite models.

The invariant mentioned on manuscript page 88 also needs its classical input. For a homotopy equivalence $f : K \to L$ of connected finite CW complexes, Whitehead torsion is an element

$$\tau(f) \in \mathrm{Wh}(\pi_1 L) = K_1(\mathbf{Z}[\pi_1 L])/\langle \pm g \mid g \in \pi_1 L \rangle.$$

It vanishes exactly when the equivalence is simple, with the usual finite CW structures. It is attached to an equivalence and its simple structures, rather than being a canonical element attached to an unmarked homotopy type (Milnor, 1966, Section 7). Thus a proposed functorial invariant must specify its objects and morphisms before one asks whether it factors through a localization. The manuscript's suggestion of higher invariants does not specify enough data to identify it with a particular later torsion theory.

The page marked 94 is explicitly missing in the edition. The surrounding argument still supplies the finite counterexample and its purpose; we do not reconstruct an absent manuscript page from those conclusions.

7 Commentaires sur les axiomes

7.1. La principale raison d'être d'un localiseur fondamental W, c'est de définir une théorie d'homotopie', et plus précisément un prédérivateur HOT_W, dont on espère que ce sera même un dérivateur, ou qu'il satisfasse du moins à certains des axiomes des dérivateurs. Les axiomes Loc (1) à Loc (9) sont en majeure partie orientés vers cet objectif, à cela près que Loc (1, 2, 3) constitue un "minimum vital" absolu, pour pouvoir commencer à travailler avec W. Les principales questions qui se posent me paraissent les suivantes :

1) Quand Cat munie de (W, Cof_W), resp. de (W, Fib_W), est-elle une catégorie à cofibrations resp. à fibrations, de façon que les résultats de K. S. Brown soient applicables ?
2) Sous quelles conditions, sur W et sur f, $f_!$ et f_* existent-ils ? Et quand a-t-on, de plus, exactitude des carrés cocartésiens resp. cartésiens ?
3) Plus généralement, en termes des axiomes imposés à W, quelles sont les axiomes Der (…) que sont vérifiés pour le prédérivateur HOT_W ? (Ce qui pourrait aussi dépendre du choix du domaine Diag $\subset$ Cat. Le prédérivateur est défini a priori sur Cat tout entier). En particulier, [page 98]
4) quand HOT_W est-il bel et bien un dérivateur sur Cat tout entier, *i.e.* quand satisfait-il à tous les axiomes ? p. 49

Une question qui me paraît moins importante (notamment que 1°) est la suivante :

1 bis) Quand W s'insère-t-il dans un triple de Quillen clos (W, C, F) (propre de préférence).

Je vais passer en revue ces questions une à une.

7.2. On a vu (mais sans que j'aie fait une vérification soigneuse pour la réciproque) que (W, Cof_W) définit une structure à cofibrations si l'axiome *Loc (6 a)* est satisfait, et que cette condition est aussi nécessaire si Loc (3 bis) est satisfait [45].

Alors les immersions ouvertes de Dwyer (ou au choix les immersions fermées de Dwyer) [page 99] sont dans Cof_W, et la construction standard du mapping-cone définit, pour toute flèche f dans Cat, une factorisation *fonctorielle*

$$f = pi \quad \begin{array}{l} i \text{ immersion ouverte de Dwyer (donc dans } \mathrm{Cof}_W), \\ p \in W. \end{array}$$

Cela, plus la construction du foncteur $f_!$, et en particulier des W-sommes amalgamées $\int X_0 \begin{array}{c} \nearrow X_1 \\ \searrow X_2 \end{array}$, données dans VI (théorie de HOT_W) et XII, nous donne une maîtrise parfaite (il me semble) sur la "structure homotopique gauche" définie par W, sous le seul axiome Loc (6 a) (Axiome du cube, forme directe). Je rappelle que les foncteurs $f_!$ sont construits dans VI sous la seule hypothèse *Loc (3 bis)* (f flèche quelconque de Cat), mais sauf erreur, la construction et l'interprétation W-homotopique de $\int X_0 \begin{array}{c} \nearrow X_1 \\ \searrow X_2 \end{array}$ donnée dans XII ne [page 100] dépend pas même de cette hypothèse [46] - mais je ne peux le garantir. Il me semble que XII (et notamment le long §5 sur les sommes amalgamées) est écrit sous l'hypothèse que W (1, 2, 3), *i.e.* Loc (1, 2, 3bis), sont satisfaits.

7.3. C'est une chose assez extraordinaire que les foncteurs $f_!$ pour HOT_W existent, sous la seule hypothèse *Loc (3 bis)* (axiome de localisation, forme forte) - et ils se "calculent" d'une façon féeriquement simple ! Je suis à peu près sûr que sous cette seule hypothèse, *l'axiome d'exactitude pour les carrés cocartésiens* n'est *pas* satisfait [47]. Je ne vois aucun moyen de le vérifier, sauf via la théorie de Brown, sous l'hypothèse *Loc (6 a)* (*cf.* 7.2).

Mais sous la seule condition Loc (3 bis) le calcul standard des fibres d'un $f_!(\xi)$ est valable, p. 50 en d'autre termes le prédérivateur HOT_W satisfait Der (4*, 5*) - et de plus Der (6*) (exactitude [page 101] "à gauche") si W satisfait Loc (6 a).

7.4. Venons-en à la question si Cat muni de (W, Fib_W) est une catégorie à fibrations. C'est l'axiome *Loc (7 F),* mais je ne connais guère de moyen de le vérifier (en dehors du cas très, très particulier $W = \mathbf{W}_0$) que moyennant tout au moins l'axiome *Loc (7 D_{W_ω}),* ou l'une ou l'autre des variantes plus fortes de Loc (7). De Loc (7 D_{W_ω}) il a été abondamment question précédemment, *cf.* notamment 6.4 (pages 89-91). En mon sens, cet axiome assure une maîtrise comparable sur la structure W-homotopique droite (fibres W-homotopiques, suites exactes de cosuspension) que Loc (6 a) pour la structure gauche. En particulier, on a la factorisation [page 102] *fonctorielle* de toute flèche f de Cat en

$$f = pi, \qquad i \in W_\omega \subset W,\ p \in \mathrm{Parf}_{W_\omega} \subset \mathrm{Fib}_W.$$

45. Vérifier quand même si Loc (3 bis) n'est pas déjà utilisé dans la partie directe ! **O.K.**
46. *cf.* XII, page 264
47. Il faudrait un exemple !

Mais *l'axiome Loc (7 D_{W_ω}) me paraît de nature considérablement plus forte que Loc (6 a),* lequel (si je ne m'abuse) est vérifié pour tous les localiseurs qui ont été envisagés jusqu'à présent. (Mais j'avoue n'avoir pas regardé le cas des $\mathbf{W}_n$, $n \in \mathbf{N}$, sauf $\mathbf{W}_0$.) Par contre, l'axiome Loc (7 D_{W_ω}) implique $W \subset \mathbf{W}_\infty$ (théorème 6.1), relation vraiment draconienne (*cf.* commentaire 6.4 a).

Par ailleurs, le seul axiome Loc (7 D_{W_ω}), pour autant que je sache, n'implique pas l'existence de f_* pour *toute* flèche dans Cat, mais seulement pour des flèches de la forme [page 103]
$f \times \mathrm{id}_X : S \times X \longrightarrow T \times X$, où $f : S \longrightarrow T$ est un morphisme entre ensembles ordonnés finis. Ou plus généralement, entre ensembles ordonnés S, T tels que S° soit artinien, *i.e.* S noethérien, et que $\mathrm{Esp}(S^\circ)$ soit noethérien (cette dernière hypothèse étant inutile sous l'axiome Loc (8) des limites, *cf.* XVII, th. 3.1, p. 50). Donc ce n'est pas le résultat rêvé.

7.5. Le seul cas que je connaisse où f_* existe pour *toute* flèche dans Cat, est le cas où W satisfait à

(∗) *Loc (6 a, 8, 9)* [48] [49],

c'est à dire l'axiome des W-cofibrations faibles, déjà abondamment mentionné et enfin ceux des limites et d'accessibilité, *i.e.* W est L_1-accessible dans $\underline{\mathrm{Fl}}(\mathrm{Cat})$. Le résultat-clef pertinent est donné dans XV 8.4 (p. 105-107), qu'il faut joindre à XV 7.4 (p. 98-99) pour en faire bon usage, et en déduire le théorème 10.1 (XV, p. 134-135). Mieux que l'existence des f_*, [page 104]
l'ensemble d'axiomes étonnamment faible que je viens d'expliciter suffit à assurer que HOT_W est un *dérivateur* sur Cat tout entier, en plein sens du terme, *i.e.* qu'il satisfait à Der (1, 2, 3, 4, 5, 6, 4*, 5*, 6*). Ceci est obtenu en remplaçant (grâce à XV 8.4) Cat par $A^\wedge$, où A est p. 51
une catégorie-test convenable, W par $W_A \subset \mathrm{Fl}\, A^\wedge$, et en prouvant que W_A s'insère dans un triple de Quillen clos (W, C, F), avec $C = \mathrm{Mono}(A^\wedge)$, ce qui détermine F comme $(C \cap W_A)_*$. On applique alors XV 7.4 à cette situation (résultats résumés dans XIX, th. 2.3 (p. 19), 2.6 (p. 25), 2.9 (p. 28). On retrouve l'existence des $f_!$ (avec un mode de calcul bien différent de celui dans Cat !), et en plus on trouve l'existence de f_* (plus l'exactitude dans cor. [?] 5 des [page 105]
carrés W-cartésiens) - ce qui pour moi, pendant un bon moment, était *le* gros problème pour la théorie de HOT_W !).

Commentary · Exposition [ms. 104]

A presentation supplies operations at every diagram shape

The shift from Cat to a presheaf category is a change of computational model. Its purpose is to make the homotopy theory of diagrams available uniformly. A homotopy category alone contains too little information for this task: an object of $\mathrm{Ho}(\mathcal{M})^S$ is a diagram whose relations commute only after localization, whereas an object of $\mathrm{Ho}(\mathcal{M}^S)$ retains a coherent realization of those relations. A derivator keeps the latter categories, their restrictions, and their homotopy Kan extensions. This distinction is why the existence of f_* for every shape is stronger than the ability to write down a fibre for a single arrow.

Modern statements require some translation before they can be applied here. In Groth (2013, Definition 1.5), Der (1) concerns arbitrary small coproducts of shapes, Der (2) says that evaluations jointly detect isomorphisms, Der (3) gives both Kan adjoints, and Der (4) calculates them pointwise. Grothendieck's original numbering separates the existence, calculation, and exactness assertions into Der (4)–(6) and their starred partners; his Der (3) is a contractibility condition. Moreover, the convention here uses presheaves $\mathcal{M}(S) = \mathcal{M}^{S^{\mathrm{op}}}$. Passing to opposite shapes translates this convention into the covariant diagrams used by many modern authors. Numbers and the words "left" and "right" cannot safely be transferred without that translation.

For a combinatorial model category the modern realization theorem is Groth (2013, Proposition 1.30). The proof uses diagram model structures to derive restriction and ordinary Kan extension. It supplies the uniform operations sought in these pages. A complementary comparison is Lenz

49. Vérifier s'il ne faut pas aussi la condition Loc (3 bis) en plus de Loc (6 a, 8, 9). Vérification faite, c'est O.K. (sans Loc (3 bis)).

50. **N.B.** Après nouvelle réflexion, il apparaît qu'il suffit même Loc (6′ a, 8, 9).

(2018, Theorem 3.33): the homotopy prederivator of a cofibration category agrees with that of its quasicategory of frames. Its Corollary 4.1 then obtains the corresponding derivators from complete or cocomplete quasicategories. These are precise realization and comparison results. They do not say that every prederivator is representable by a model category, or that the original class W can automatically be recovered from any equivalent presentation. The latter is a separate recognition argument, undertaken on manuscript pages 111–118.

Je trouve époustouflant qu'on parvienne à un résultat aussi fort avec des axiomes aussi ridiculement faibles. Plus particulièrement, qu'on ait toutes les propriétés désirables de la structure homotopique droite, sans avoir à faire sur W aucune hypothèse du type W-fibration ! Ces axiomes n'impliquent *pas* l'axiome Loc (7) (équivalent à Loc (7 D_{W_ω}) en présence de Loc (8)), comme on a vu sur l'exemple de $\mathbf{W}_0$. Impliquent-ils tout au moins Loc (7 F), *i.e.* l'existence d'assez de W-fibrations ? Je ne hasarderais aucune conjecture à ce sujet ! Mais il faut bien se dire qu'en l'absence de Loc (7) (et même en admettant Loc (7 F)), tout en [page 106] sachant que la structure homotopique droite associée à W est parfaite à tous égards, on n'a a priori aucune maîtrise sur celle-ci, notamment pour le calcul des fibres W-homotopiques et la construction des carrés W-cartésiens. Même en ramenant toutes ces constructions à $A^\wedge$ (prenant, si on y tient, $A = \mathbf{\Delta}$, *i.e.* se ramenant au contexte semisimplicial), quoi qu'on sache qu'il y a une factorisation *fonctorielle* $f = pi$ avec $i \in \mathrm{TC} \subset W_A$, $p \in F \subset \mathrm{Fib}_{W_A}$, je ne vois aucune façon concrète de l'expliciter. À moins que la considération des chemins infinis, qui (en l'absence de Loc (7)) s'avère inopérante dans Cat sauve la mise dans $\mathbf{\Delta}^\wedge$? ?

Question qui devrait être facile à élucider, mais sur laquelle je ne veux pas m'attarder à [page 107] présent.

Notons en passant que le résultat de saturation forte de Quillen, pour les triples de Quillen clos, montre que l'ensemble d'axiomes $(*)$ implique l'axiome Loc (1 ter), *i.e.* que W est *fortement saturé.*

Notons aussi que contrairement à ce qui a lieu pour les $f_!$, je ne sais prouver l'existence de f_*, et notamment des suites de cosuspension, que sous des conditions (soit Loc (7 D_{W_ω}), soit Loc (6a, 8, 9)) qui assurent déjà que les carrés W-cartésiens sont exacts, et par suite les suites de cosuspension exactes.

Je pense que dans les pages précédentes, j'ai dit tout ce que je sais dire de pertinent, concernant les questions 1, 2, 4 soulevées dans 7.1. Ces réponses me paraissent satisfaisantes [page 108] - plus satisfaisantes que je ne l'aurais osé espérer pendant les premiers deux mois de mes cogitations ! Il reste les questions 3 et 1 bis, que je vais à présent examiner.

7.6. Sur les différents axiomes des dérivateurs : sous quelles conditions sur W un de ces p. 52 axiomes est-il satisfait pour HOT_W [51] ? Pour ce qui est des axiomes Der (4, 5, 6, 4^*, 5^*, 6^*), je crois avoir dit tout ce que je sais en dire, et avoir des critères ce validité parfaitement maniables, dans les pages précédentes. Résumons pourtant :

structure à cofibrations	:	Loc (6) (+ Loc (3 bis) ?)
structure à fibrations	:	Loc (7 D_{W_ω})
Der (4^*, 5^*)	:	Loc (3 bis)
Der (4^*, 5^*, 6^*)	:	Loc (3 bis, 6a) ou Loc ($6'$ a, 8, 9)
Der (4, 5, 6) (sur Cat) (et tout le reste !)	:	Loc ($6'$ a, 8, 9)
Der (4, 5, 6) (sur Diag = ens. ordonnés finis)	:	Loc (7 D_{W_ω}) (ou seulement Loc (7 F)).

Il reste les axiomes Der (1) (décomposition), Der (2) (localisation), Der (3) (objet unité), que je vais passer à présent en revue. On vérifie de façon quasi-tautologique que l'axiome Der [page 109]

51. Pour les axiomes Der (1) à Der (5^*), *cf.* I, p. 67-68. Les axiomes Der (6, 6^*) sont les axiomes d'exactitude, dont la nécessité m'est apparue seulement ultérieurement.

1 est satisfait sans condition sur W (pas même Loc (2, 3) !) - en fait, il est satisfait pour tout prédérivateur $\mathbf{D}_{(\mathcal{M},W)}$ défini par une catégorie de modèles.

L'axiome de localisation Der (2) est par contre de nature délicate. Il est valable, plus généralement, pour une catégorie de modèles $(\mathcal{M}, W)$ (définissant un prédérivateur $\mathbf{D} = \mathbf{D}_{(\mathcal{M},W)}$) et un objet S de Cat, lorsque $(\mathcal{M}(S), W(S))$ est associé à une structure à fibrations ou à cofibrations, ce qui par K. S. Brown implique que toute flèche de $\mathrm{HO}_W(S) = \mathcal{M}(S)W(S)^{-1}$ se "réalise" par une flèche de $\mathcal{M}(S)$ - et alors le résultat est évident. La condition que je viens de dire est satisfaite lorsque $(\mathcal{M}, W)$ lui-même est associé à une structure de cofibrations ou de fibrations, et si de plus on suppose que l'on est dans l'un de deux cas suivants : [page 110]

a) S est un ensemble ordonné fini (ce qui nous amène à nous restreindre, à défaut de mieux, à Diag = sous-catégorie de Cat formée des ensembles ordonnés finis).

b) La factorisation stipulée dans l'axiome de factorisation $f = pi$ (avec $i \in \mathrm{Cof}$, $p \in W$ (cas de cofibrations) ou $i \in W$, $p \in \mathrm{Fib}$ (cas de fibrations)) peut être choisie fonctorielle en f.

Or, dans le cas de Cat, dans les cas où on sait prouver que W est associé à une structure à cofibrations (axiome *Loc (6 a)*) où à fibrations (axiome *Loc (7 D_{W_ω})*), on trouve bel et bien une factorisation fonctorielle. Donc chacun de ces axiomes implique Der (2).

Commentary · Examples [ms. 109]

Finite coproducts of shapes and the length of a zigzag

The assertion called quasi-tautological is correct in its original scope. Chapter I, manuscript page 67, footnote 3, defines Der (1) for a *finite* sum of shapes. Let $(\mathcal{C}, U)$ and $(\mathcal{E}, V)$ be relative categories, with identities among their weak equivalences. There is a canonical equivalence

$$(\mathcal{C} \times \mathcal{E})[(U \times V)^{-1}] \simeq \mathcal{C}[U^{-1}] \times \mathcal{E}[V^{-1}].$$

Here is the mechanism. Every ordinary arrow (a, b) factors as $(a, 1)$ followed by $(1, b)$. A formal inverse of (u, v) decomposes in the same fashion, using the inverses of $(u, 1)$ and $(1, v)$. Horizontal and vertical arrows commute in every rectangle. The same commutation holds when one or both arrows are inverted, by multiplying the commutative square by the relevant inverses. Thus a word in the localized product can be rearranged into a word in the first coordinate followed by a word in the second. Conversely, two localized coordinate words define such a composite; the defining composition, identity, and inverse relations in either coordinate remain valid in the product localization. These operations are mutually inverse on hom-sets and preserve composition. This proves the equivalence, including its faithfulness.

Induction gives finite products. Since

$$\mathcal{M}^{(S \amalg T)^{\mathrm{op}}} \cong \mathcal{M}^{S^{\mathrm{op}}} \times \mathcal{M}^{T^{\mathrm{op}}},$$

and objectwise weak equivalences correspond under this isomorphism, localizing proves finite Der (1). The empty shape gives the terminal category as well. No localizer axiom is needed.

An explicit example explains why the same reasoning must not be extended to infinitely many factors without hypotheses. For $n \geq 1$, let J_n have objects $0, \ldots, n$, with precisely the nonidentity arrows between adjacent vertices directed from the even vertex to the odd vertex. Thus every nonidentity arrow has length one, and no two nonidentity arrows compose. Let $\mathcal{M} = \coprod_{n \geq 1} J_n$ and declare every arrow a weak equivalence. In $\mathcal{M}^{\mathbb{N}}$ take objects x and y whose nth coordinates are the vertices 0 and n in J_n. Each coordinate has a finite zigzag joining these vertices, so their images are isomorphic in $\prod_n \mathrm{Ho}(\mathcal{M})$.

There is nevertheless no arrow from x to y in $\mathrm{Ho}(\mathcal{M}^{\mathbb{N}})$. Such an arrow would be represented by a finite word of, say, k ordinary arrows and formal inverses. In coordinate n, each letter changes the vertex by at most one. Hence the word can travel at most k edges, whereas the endpoints in J_n are n edges apart. Taking $n > k$ gives a contradiction. The comparison to the infinite product is therefore not full. The obstruction is the absence of a uniform finite zigzag, not a defect in Grothendieck's finite axiom. Modern arbitrary-coproduct Der (1) requires the additional homotopical structure supplied by the realization theorems. A related bounded-length obstruction already occurs in Thomason (1980, p. 307), where the fundamental group of an infinite product of elementary categorical circles gives bounded integer families rather than all integer families.

Reste l'axiome de contractibilité : si S dans Cat est contractile, est-il HOT_W-asphérique, *i.e.* le foncteur [page 111]

$$\mathrm{HOT}_W(e) = \mathrm{Hot}_W \longrightarrow \mathrm{HOT}_W(S)$$

est-il pleinement fidèle ? Quand on dispose de la théorie de $f_!$ "version Cat" (donc sous le seul axiome *Loc (3 bis)*), on trouve ceci (noté dans VI, p. 202) [52] : p. 53

> Soit $f : S \longrightarrow T$ une flèche dans Cat. Pour que ce soit une HOT_W-équivalence, *i.e.* pour que $f^* : \mathrm{HOT}_W(T) \longrightarrow \mathrm{HOT}_W(S)$ induise un foncteur *pleinement fidèle* sur la sous-catégorie des objets constants de $\mathrm{HOT}_W(T)$, il faut et il suffit que $\mathrm{hot}_W(f) : \mathrm{hot}_W(S) \longrightarrow \mathrm{hot}_W(T)$ soit un isomorphisme. Donc il *suffit* que $f \in W$, et cette condition est aussi nécessaire si W est fortement saturé (axiome Loc (1 ter)).

Par acquit de conscience, j'explicite ici la démonstration. L'existence des foncteurs $f_!$, en particulier des objets d'homologie $H_\bullet^{\mathrm{HOT}_W}(S, \xi)$ (où simplement $H_\bullet^W(S, \xi)$) pour $\xi \in \mathrm{HOT}_W(e) = \mathrm{Hot}_W$, permet d'interpréter homologiquement la condition de HOT_W-équivalence, comme signifiant que pour tout $\xi = \mathrm{hot}_W(X)$ dans Hot_W, la flèche canonique dans Hot_W [page 112]

$$(*) \qquad H_\bullet^W(S, \xi) \longrightarrow H_\bullet^W(T, \xi)$$

est un isomorphisme. Or le calcul des $f_!$ (dans le cas de $p : S \longrightarrow e$ et de $T \longrightarrow e$) nous montre que les deux membres de $(*)$ ne sont autres (à isomorphisme près) que $\mathrm{hot}_W(X \times S)$ et $\mathrm{hot}_W(X \times T)$, la flèche étant induite par

$$(*) \qquad \mathrm{id}_X \times f : X \times S \longrightarrow X \times T.$$

Donc la condition envisagée signifie que cette flèche est un isomorphisme dans Hot_W pour tout X dans Cat. Prenant $X = e$, on voit que cela implique que $\mathrm{hot}_W(f)$ est un isomorphisme, et cela suffit, vu que le foncteur $\mathrm{hot}_W : \mathrm{Cat} \longrightarrow \mathrm{Hot}_W$ commute aux produits finis (il suffit de Loc (1, 2, 3)) `[plutôt Loc (1, 2, 3 bis)]` et que la flèche $(*)$ dans Hot_W s'identifie à [page 113]

$$\mathrm{id}_{\mathrm{hot}_W(X)} \times \mathrm{hot}_W(f)\,.$$

Cela montre donc que si S est un objet W-asphérique, *i.e.* tel que $S \longrightarrow e$ soit dans W (et a fortiori, si S est contractile), alors $S \longrightarrow e$ est HOT_W-asphérique.

Commentary · Exposition [ms. 110]

What it means to realize a localized arrow

The word "realize" in the proof of Der (2) permits replacement of the source and target by weakly equivalent objects. It does not say that the localization functor is full on the original, fixed pair of objects. Even a formal inverse of a weak equivalence need not be represented by an ordinary arrow in the reverse direction between those same objects.

In a model category, the precise calculation uses a cofibrant replacement $QX \to X$ and a fibrant replacement $Y \to RY$. Every arrow $X \to Y$ in the homotopy category is represented by an ordinary map $QX \to RY$. Applied in a suitable diagram model category, this gives a map of replacement diagrams. If every component of the original localized arrow is an isomorphism, every component of the representing map is an isomorphism in the base homotopy category. Strong saturation of model weak equivalences makes those components weak equivalences. The representing map is therefore an objectwise weak equivalence, and becomes an isomorphism in the localized diagram category. The weak-equivalence replacements show that the original arrow is also invertible. This proves the pointwise conservativity required by Der (2).

Grothendieck invokes the analogous Brown calculation for a cofibration or fibration category. In the finite-poset case, one constructs the necessary replacement diagram inductively, placing an object after its already treated predecessors and factoring the resulting latching map, or dually using

52. moyennant Loc (3 bis).

matching maps. A functorial factorization can instead be applied directly to natural transformations; functoriality makes the chosen middle objects into a diagram. This is the specific advantage of the functorial constructions recalled on pages 99 and 102.

For all small shapes, a modern cofibration-category theorem carries more structure than a finite Brown theorem. Lenz (2018, Definition 1.5 and Warning 1.7) explicitly includes coproduct and sequential-colimit conditions in the ABC framework. The distinction matters: a proof that works by finite induction cannot simply be reindexed by an arbitrary small category. Conversely, a theorem proved for ABC cofibration categories should not be used to certify an unqualified claim about every finite Brown category. Lenz's Example 1.49 also exhibits a general relative category whose homotopy prederivator fails even weak strength. A class of denominators alone supplies a prederivator; the homotopical factorization and gluing conditions do substantive work.

Pour résumer, on trouve le tableau

$$\begin{cases} \text{Der (1)} & : & \text{sans hypothèse sur } W \\ \text{Der (2)} & : & \text{Loc (6 a) } ou \text{ Loc (7 D}_{W_\omega}\text{)} \\ \text{Der (3)} & : & \text{Loc (3 bis) } ou \text{ Loc (6}' \text{ a, 8, 9)} \ ^{53}. \end{cases}$$

7.7. Il faut seulement justifier encore que Loc (6′ a, 8, 9) implique Der (3). Mais alors, on l'a vu, HOT_W peut être défini par un triple de Quillen très spécial. Mais même pour un triple de Quillen spécial sans plus $(\mathcal{M}, W)$, le prédérivateur **D** qu'il définit satisfait (en plus de Der (4*, 5*, 6*)) aux conditions Der (1) (trivialement), Der (2)

(*cf.* p. 110, cas b)) et *de plus* Der (3). (Donc dans le cas d'un triple de Quillen *très* spécial, le prédérivateur obtenu est bel et bien un dérivateur.) Je vais expliciter ici la démonstration. p. 54 [page 114]

Commentary · Exposition [ms. 113]

Recovering the equivalences from constant diagrams

The recognition calculation is a particularly useful part of this survey. It also records exactly where the later circular argument will enter. Let $\mathbb{D} = \mathrm{HOT}_W$, let $p_S : S \to e$, and write $c_S = p_S^*$. Assuming the left Kan extensions constructed under Loc (3 bis), set

$$H(S, \xi) = p_{S!} c_S \xi.$$

Adjunction identifies

$$\mathrm{Hom}_{\mathbb{D}(S)}(c_S\xi, c_S\eta) \cong \mathrm{Hom}_{\mathbb{D}(e)}(H(S,\xi), \eta).$$

For $f : S \to T$, restriction on constant objects corresponds under this identification to precomposition with the canonical map

$$H(S, \xi) \longrightarrow H(T, \xi).$$

Consequently restriction is fully faithful on constants precisely when these maps are isomorphisms for every ξ. For the nontrivial direction, bijectivity of the displayed hom-set maps for every η implies the assertion by the ordinary Yoneda criterion for an isomorphism. This is the exact reason that a condition on all morphisms between constants can be tested by homology objects.

Write $q : \mathrm{Cat} \to \mathrm{Hot}_W$ for localization. Every object ξ of this homotopy category is represented by a category X. The categorical Kan-extension calculation gives

$$H(S, qX) \cong q(X \times S), \qquad H(T, qX) \cong q(X \times T),$$

with the comparison induced by $1_X \times f$. Taking $X = e$ shows that full faithfulness on constants forces $q(f)$ to be an isomorphism. Conversely, the product calculation identifies the comparison with $1_{qX} \times q(f)$, so invertibility of $q(f)$ implies all the required isomorphisms. The argument thus identifies the class of $\mathbb{D}$-equivalences with

$$\{f \mid q(f) \text{ is an isomorphism}\}.$$

53. **N.B.** On verra plus bas que Loc (6 a, 8, 9) implique déjà Loc (3 bis).

It identifies that class with W itself only when W is strongly saturated. This qualification is expressly present on page 111.

There are two different localization facts in this proof. The preceding block used finite products of *categories being localized* to prove finite Der (1). Here one needs the cartesian product operation on Cat to descend to its localization and to agree with the products used in the homology formula. That uses the localizer's stability properties and the preceding Kan-extension theory. Grothendieck's bracketed correction from Loc (1,2,3) to Loc (1,2,3 bis) is therefore retained in the audited reading; a formal finite-product localization lemma cannot remove the hypothesis from the whole recognition argument.

For $T = e$, a W-aspherical S has $q(S) \cong e$, so the comparison is invertible for every X. Thus c_S is fully faithful. This verifies the contractibility axiom in this setting without identifying a contractible diagram shape with an arbitrary diagram category: constant diagrams embed fully faithfully, while nonconstant diagrams can still contain substantial information.

Je vais désigner par $(\mathcal{M}, \Sigma)$ la catégorie de modèles donnée ($\Sigma \subset \mathrm{Fl}\,\mathcal{M}$), par $\mathbf{D} = \mathbf{D}_{\mathcal{M},W}$ le prédérivateur correspondant, et par

$$W = W_D$$

le localiseur dans Cat associé, formé des "$\mathbf{D}$-équivalences". On voit tout de suite que W est fortement saturé (vrai pour tout $W_{\mathbf{D}}$, $\mathbf{D}$ un prédérivateur), *i.e.* satisfait Loc (1 ter), a fortiori Loc (1), d'autre part il satisfait à Loc (3 bis*), comme conséquence du fait que $\mathbf{D}$ satisfait à Loc (4*, 5*). A fortiori il satisfait à Loc (3*), et pour voir que tout X contractile dans Cat est W-asphérique, *i.e.* $X \longrightarrow e$ dans W, on sait (*cf.* §1) qu'il suffit de prouver qu'il en est ainsi si [page 115]
X a un objet initial (axiome Loc (2*) dual de l'axiome Loc (2)). À ce moment, on saura que W est un localiseur fondamental, et de plus il satisfait Loc (3* bis), donc aussi Loc (3 bis) (forme forte de l'axiome de localisation).

Donc *on peut supposer* que si S a un objet initial, alors S est W-asphérique, *i.e.* $S \longrightarrow e$ est dans W, *i.e.* $p^* : \mathbf{D}(e) \longrightarrow \mathbf{D}(S)$ est pleinement fidèle. Il revient au même de dire que la flèche d'adjonction

$$p_! p^* \xi \longrightarrow \xi$$

est un isomorphisme, pour tout ξ dans $\mathbf{D}(e)$. Or le calcul de $p_!(\eta)$ (pour $\eta \in \mathrm{Ob}\,\mathbf{D}(S)$, *i.e.* $\eta \in \mathrm{Ob}\,\mathcal{M}(S)$) se fait ainsi : on choisit une flèche

$$\eta' \xrightarrow{\Sigma_S} \eta$$

avec η' "cofibrant" (pour la structure à cofibrations pertinente sur $\mathcal{M}(S)$), et on pose[54] [page 116]

$$p_!(\mathrm{ho}_{W,S}(\eta)) = \mathrm{ho}_W(p_!(\eta')).$$

Quand $\eta = p^*(\xi)$, la flèche d'adjonction pertinente dans $\mathrm{Ho}_\Sigma = \mathbf{D}(e)$ s'obtient par composition

$$(*) \qquad p_!(\eta') \longrightarrow p_!(p^*(\xi)) \xrightarrow{\text{adj.}} \xi,$$

où ici $p_!$ est encore pris dans le sens ordinaire ($\mathcal{M}(S) \longrightarrow \mathcal{M}(e) = \mathcal{M}$). Or S ayant un objet initial 0, on a

$$\begin{aligned} p_!(\eta') &\simeq \eta'(0) \\ p_!(p^*(\xi)) &\simeq p^*(\xi)(0) \simeq \xi, \end{aligned}$$

et la flèche d'adjonction dans $(*)$ est l'identité, la première flèche est la flèche $\eta'(0) \longrightarrow \eta(0)$ induite par $\eta' \longrightarrow \eta$, elle est dans Σ, donc $(*)$ est dans Σ, donc c'est un isomorphisme dans $\mathrm{Ho}_\Sigma = \mathbf{D}(e)$, q.e.d.

Le même argument marche d'ailleurs si $\mathcal{M}$ est seulement une catégorie à cofibrations ou à p. 55
fibrations, mais en se bornant alors au cas où S est un ensemble ordonné fini. [page 117]

Je vais dégager à part ce qu'on vient d'obtenir :

Commentary · Exposition [ms. 116]

Why the initial object computes this left adjoint

There is no initial-versus-terminal mistake in the displayed calculation. Its direction records the presheaf convention. An object of $\mathcal{M}(S)$ is a functor $S^{\mathrm{op}} \to \mathcal{M}$. If 0 is initial in S, it is terminal in S^{op}, and hence

$$p_!\eta = \operatorname*{colim}_{S^{\mathrm{op}}} \eta \cong \eta(0).$$

For completeness, the cocone is given by the unique arrows $s \to 0$ in S^{op}. A cocone from η to an object z is uniquely determined by its component $\eta(0) \to z$, since compatibility forces every other component to be its composite with $\eta(s) \to \eta(0)$. This proves the universal property and the natural isomorphism.

Now take an object $\xi \in \mathcal{M}$, let $c\xi = p^*\xi$, and choose a cofibrant replacement of this diagram,

$$q : Q(c\xi) \longrightarrow c\xi,$$

whose components belong to Σ. The derived counit of $\mathbf{L}p_! \dashv p^*$ is represented by

$$p_!Q(c\xi) \longrightarrow p_!c\xi \longrightarrow \xi.$$

Under the displayed evaluation isomorphisms this is simply $q_0 : Q(c\xi)(0) \to \xi$, because the second arrow is the identity. It is a weak equivalence, so the derived counit is an isomorphism. A right adjoint is fully faithful exactly when its counit is an isomorphism. Therefore the constant-diagram functor is fully faithful, which is the required asphericity conclusion for this shape.

The argument proves a property of the derivator produced by the model structure. It has no need to identify its associated fundamental localizer with a previously given W. That is why Proposition 7.7.1 can be used to verify the original Der (3) independently of the disputed recognition statement. The symbols W and Σ in the surrounding text should still be distinguished: Σ denotes weak equivalences in the chosen model category, whereas $W_{\mathbb{D}}$ is a class of functors between small categories. Several printed subscripts switch them inadvertently; the correction ledger records those slips without rewriting the edition.

Proposition 7.7.1. *Soit* $(\mathcal{M}, W)$ *une catégorie de modèles, associée à un triple de Quillen clos* spécial (W, C, F). *Alors le prédérivateur* $\mathbf{D} = \mathbf{D}_{\mathcal{M},W}$ *sur* Cat *satisfait aux axiomes* Der (1, 2, 3, 4*, 5*, 6*). *Si de plus le triple de Quillen est* très *spécial* (*donc en particulier* $C = \mathrm{Mono}(\mathcal{M})$), *alors il satisfait de plus à* Der (4, 5, 6), i.e. *c'est un dérivateur au plein sens du terme. D'autre part, le localiseur* $W = W_{\mathbf{D}}$ *dans* Cat *associé à* $\mathbf{D}$ (*dans le cas d'un triple de Quillen spécial sans plus*) *est un* localiseur fondamental, *satisfaisant de plus* Loc (1 ter, 3 bis).

N.B. Ce qui est dit ici pour le cas d'un triple de Quillen très spécial (qui ne fait pas appel à la théorie des localiseurs fondamentaux dans Cat) montre à nouveau que si W localiseur fondamental dans Cat satisfait à Loc ($6'$ a, 8, 9), alors HOT_W est un vrai dérivateur. On a vu de plus dans 7.6 (page 111) que l'on a alors $W = W_{\mathrm{HOT}_W}$ [55], donc il s'ensuit : [page 118]

Corollaire 7.7.2 : *On a l'implication*

$$\text{Loc } (6' \text{ a, 8, 9}) \implies \text{Loc(1 ter, 3 bis)} \text{ [56]}.$$

Notons aussi, pour mémoire :

Corollaire 7.7.3. *Soit* $(\mathcal{M}, W)$ *une catégorie de modèles, associée à une catégorie à cofibrations. Alors sur le domaine* Diag *formé des ensembles ordonnés finis, le prédérivateur* $\mathbf{D} = \mathbf{D}_{\mathcal{M},W}$ *satisfait aux axiomes* Der (1, 2, 3, 4*, 5*, 6*), *et dualement. Donc si* W *est associé à la fois à une structure de cofibrations et de fibrations,* $\mathbf{D}$ *est un dérivateur sur* Diag. [page 119]

54. opération $p_!$ au niveau de $\mathcal{M}(S)$.
55. Mais c'était sous l'hypothèse Loc (3 bis) ! Gare au cercle vicieux.
56. Loc (3 bis) pas prouvé, probablement faux.

[Les six lignes qui suivent semblent hors contexte.]

Commentary · Present state [ms. 118]

The circular inference and the separate recognition problem

The two authorial footnotes are mathematically decisive. The proof that a suitable presheaf presentation yields a derivator concerns the prederivator HOT_W. Proposition 7.7.1 then associates a strongly saturated fundamental localizer $W_{\mathbb{D}}$ satisfying the relative locality axiom to that derivator. To conclude the same relative locality assertion for the original W, one still needs

$$W = W_{\mathrm{HOT}_W}.$$

The proof on pages 111–113 establishes this equality under strong saturation *and* Loc (3 bis). Applying it to prove Loc (3 bis) therefore assumes the very conclusion under consideration. The derivator existence theorem does not remove that logical dependence. This is the circle explicitly marked by footnote 55 and acknowledged in footnote 56.

Strong saturation has a different route. If the presentation identifies the original weak equivalences with the inverse image of the model weak equivalences, the model category's saturation theorem supplies Loc (1 ter). That statement does not require the invalid recognition inference. Thus the two conclusions in Corollary 7.7.2 should not receive a single undifferentiated verdict: the saturation argument has already been given on page 107, while the proposed proof of relative locality fails. The claim register records the latter argument as a sketch with a circularity gap. A circular proof does not by itself refute its conclusion, and no counterexample to that exact implication is asserted here.

Nor does weak minimality solve this separate problem. Theorem 6.1.18 of Cisinski (2006a) identifies the *least* weak fundamental localizer. It does not say that every weak fundamental localizer satisfies the relative axiom. The distinction remains necessary throughout this commentary. The verified modern model-existence theorem near the end of the chapter assumes a strong basic localizer, so it cannot be used to supply the missing implication by changing terminology mid-proof.

There is also a broader recognition question: how much of a homotopy theory survives passage to all its diagram homotopy categories? A precise positive result is Arlin (2020, Theorem 37), read in the February 2020 arXiv revision. A map of quasicategories whose induced prederivator morphism is an equivalence is itself an equivalence. The proof recovers equivalences on mapping spaces from diagram categories. This detects a given map; it does not assert that every abstract prederivator has a quasicategorical or model presentation. Together with Lenz's comparison theorem, it explains the strength of the derivator perspective while leaving the hypotheses of the particular equality $W = W_{\mathrm{HOT}_W}$ visible.

alors la flèche de changement de base

$$p^* f_! \longrightarrow f'_! q^*$$

entre foncteurs $\mathrm{HOT}_W(X) \longrightarrow \mathrm{HOT}_W(S')$ *est un isomorphisme.*

C'est pratiquement trivial, compte tenu, dans le cas f W-lisse, que si Z sur X est W-lisse sur X, il l'est sur S ; et dans le cas p W-propre, en utilisant le dernier énoncé ("oubli") dans ⑥, p. 199.

7.8. Je termine en mettant en vedette ce qui, à présent, me paraît la substance principale de tous ces "rappels" et comme la quintessence de la théorie des localiseurs fondamentaux développée jusqu'à présent : p. 56

$$\begin{cases} \text{Loc (3 bis)} & \Longrightarrow & \text{Der (4*, 5*)} \\ \text{Loc (3 bis, 6 a)} & \Longrightarrow & \text{Der (1, 2, 3, 4*, 5*, 6*)} \\ \text{Loc (6' a, 8, 9)} & \Longrightarrow & \text{tous les axiomes des dérivateurs (+ Loc (1 ter)).} \end{cases}$$

Je considère les axiomes Loc (3 bis), Loc (8), Loc (9) comme des axiomes "anodins" - tous les localiseurs qu'on définit par voie cohomologique ou homotopique ordinaire, y satisfont. Il reste l'axiome Loc (6 a) ou Loc (6′ a) des W-cofibrations (forme faible), qui visiblement joue un rôle crucial dans la théorie de HOT_W. Par contre, les axiomes Loc (4) (décomposition), [page 120]

Loc (5) (connexité) et Loc (7) et ses variantes (et bien sûr aussi Loc (6 b)) n'interviennent absolument pas ! C'est particulièrement impressionnant, en ce qui concerne l'axiome des W-fibrations Loc (7) et ses variantes (toutes équivalentes à Loc (7) en présence de Loc (8), à la seule exception de Loc (7 F)). Comme on a vu, Loc (7) n'est nullement impliqué par Loc (6 a, 8, 9) ($\mathbf{W}_0$ n'y satisfait pas), tout au plus se pourrait-il que Loc (7 F) le soit. Il en est de même de l'axiome de connexité Loc (5), comme on voit sur l'exemple de $W = \mathrm{Fl}(\mathrm{Cat})$, qui satisfait à tous les axiomes vus jusqu'à présent, à la seule exception de Loc (5). Quant à [page 121] la question de l'indépendance de l'axiome Loc (4) par rapport à Loc (6 a, 8, 9), nous allons examiner ce genre de questions au paragraphe suivant.

Commentary · Exposition [ms. 119]

The detached base-change assertion

The edition expressly marks six lines as out of context. Their symbols determine the square they refer to, but do not recover all of its missing hypotheses. The evident intended arrangement is

$$\begin{array}{ccc} X' & \xrightarrow{q} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow[p]{} & S. \end{array}$$

The printed comparison is $p^* f_! \to f'_! q^*$, with the variance of these pages. The following explanation distinguishes two sufficient geometric situations: f is W-smooth, or p is W-proper. It invokes stability of smoothness under composition in the first case and the previously established "forgetting" statement on page 199 in the second. It would be unsafe to turn this fragment into a theorem for arbitrary commutative squares.

The structural meaning is clear. A base-change assertion says that restricting a homotopy Kan extension and extending the restricted diagram agree. It requires a square for which the two pointwise indexing constructions are homotopically equivalent. For comparison, Groth (2013, Proposition 1.24) gives an exact modern statement: a pullback square of categories is homotopy exact if its right vertical map is a Grothendieck fibration or its bottom horizontal map is a Grothendieck opfibration, in that article's covariant-diagram convention. The proof reduces the pointwise calculation to actual fibres using adjunctions. This illustrates the mechanism; it does not identify all W-smooth or W-proper functors with Grothendieck fibrations, nor does it reconstruct the absent surrounding sentence. The fragment and its editorial warning therefore remain in the text.

L'ensemble d'axiomes Loc (6′ a, 8, 9) me paraît à présent si important, que j'ai envie de lui donner un nom, et d'appeler *localiseur* (fondamental) *spécial* dans Cat [57], un localiseur fondamental qui y satisfait. Ce sont donc aussi ceux pour lesquels nous savons que HOT_W est un vrai *dérivateur*. Si on ne peut écarter comme "pathologique" une structure aussi riche et délicate qu'un dérivateur, il s'ensuit aussitôt que l'on ne peut non plus écarter les localiseurs spéciaux dans Cat (même si on affecte de regarder de haut les localiseurs fondamentaux [page 122] généraux !). En effet, W se récupère de façon unique à l'aide de $\mathbf{D} = \mathrm{HOT}_W$, ou de tout dérivateur équivalent à celui-ci, comme l'ensemble des $\mathbf{D}$-équivalences dans Cat [58] : c'est le localiseur fondamental (correspondant à une notion d'équivalence "universelle", *i.e.* dans la catégorie Cat de tous les "espaces" élémentaires) canoniquement associé au dérivateur. Ce n'est donc nullement un localiseur de référence fixe assez fin, tel $\mathbf{W}_\infty$, qui est apte à jouer un rôle !

Cela nous amène à nouveau à une question du genre déjà soulevé : si $W \subset \mathbf{W}_\infty$ est le plus petit localiseur qui satisfait à Loc (6′ a, 8) (ou Loc (3 bis, 6′ a, 8)), peut-on dire a) que W est même *spécial*, *i.e.* satisfait également Loc (9) (accessibilité) ?, et de plus b) a-t-on $W \neq \mathbf{W}_\infty$? [page 123] Je suis persuadé que c'est le cas (il y a moins de mérite pour ce W-là d'être différent de $\mathbf{W}_\infty$, que pour le $\tilde{\mathbf{W}}_\infty$ envisagé dans 6.4 (p. 87)). p. 57

57. J'hésite si je ne dois pas rajouter l'axiome Loc (3 bis), qui assure que $W = W_{\mathbf{D}}$, $\mathbf{D} = \mathrm{HOT}_W$.
58. Mais ceci n'est prouvé que moyennant Loc (3 bis).

7.9. Oubli : la question 1 bis (p. 98), sur l'existence d'une structure de Quillen close sur Cat, correspondant à W. Le seul résultat que je connaisse dans ce sens est celui de Thomason, construisant explicitement une telle structure de Quillen (de plus propre), dans le cas où $W = \mathbf{W}_\infty$. Je n'ai pas vraiment regardé la démonstration de Thomason, et ignore si celle-ci peut s'étendre à des localiseurs fondamentaux plus généraux, par exemple à ceux satisfaisant Loc (6′ a, 8, 9). S'il en était ainsi, cela prouverait d'ailleurs que ces localiseurs ont assez de [page 124]
cofibrations et assez de fibrations, donc qu'ils satisfont Loc (6 a) et Loc (7 F). (Pour ce qui est de Loc (6 a), il semble qu'on puisse le prouver directement très simplement, *cf.* le paragraphe suivant. Mais ici encore, il faudrait regarder quelques exemples (équivalences homologiques ou cohomologiques, $\mathbf{W}_n$, *i.e.* les n-équivalences d'homotopie), et regarder déjà si Loc (7 F) est satisfait.)

Commentary · Later developments [ms. 123]

Both proposed minima give ordinary homotopy types

The two minima asked for on pages 122–123 have an unequivocal answer. Let $W^{(a)}$ be the least weak fundamental localizer satisfying Loc (6′a,8), and let $W^{(b)}$ be the least one satisfying Loc (3 bis,6′a,8). The ordinary class $\mathbf{W}_\infty$ satisfies both lists: here the cube condition is explicitly the direct, weak form, so the reverse-excision problem of page 87 does not arise. It is therefore a member of each class being intersected. Conversely, weak minimality puts $\mathbf{W}_\infty$ in every weak fundamental localizer. Hence

$$\mathbf{W}_\infty \subset W^{(a)} \subset \mathbf{W}_\infty, \qquad \mathbf{W}_\infty \subset W^{(b)} \subset \mathbf{W}_\infty.$$

Both equalities follow. By the accessibility theorem proved above, both minima satisfy Loc (9). Thus part (a), accessibility, is affirmative, whereas part (b), strict inequality with ordinary weak homotopy equivalences, is negative. These answers use Cisinski (2006a, Theorem 6.1.18), not merely a theorem asserting minimality among already strong localizers.

The conclusion does not make other localizers irrelevant. Coarser localizers can distinguish different localizations of ordinary homotopy theory, and their derivators can retain different homotopy-limit and homotopy-colimit operations. What fails is this specific expectation of a new theory obtained by imposing the stated axioms below $\mathbf{W}_\infty$. The separate proposal to encode Whitehead torsion needs additional finite or relative structure, as in the finite-space construction discussed at page 96.

Il ne me semble pas que la théorie de Thomason ait été jusqu'à présent `[mots illisibles]` être très utile, techniquement parlant. Psychologiquement, je crois qu'il a été important, pour donner confiance que Cat est vraiment une excellente catégorie de modèles pour Hot. J'ai été vraiment enchanté et stimulé d'apprendre ce théorème. Mais la notion d'application de Dwyer (que j'ai apprise en 1983 dans l'article de Thomason) a été, techniquement, beaucoup plus importante pour moi : c'était exactement ce qui manquait à ma panoplie de notions et d'intuitions "homotopiques" dans Cat …

Commentary · Later developments [ms. 124]

Generalized Thomason models and the role of properness

The model-existence question has a substantial positive answer with precise hypotheses. Cisinski (2006a, Theorem 5.2.15, p. 225) constructs, for every accessible *strong* basic localizer W, a cofibrantly generated model structure on Cat with weak equivalences W and Thomason cofibrations. It is left proper. It is right proper exactly when the localizer is proper in the sense used there. The reference is the published Astérisque version; its pagination differs from the shorter author preprint. This theorem extends the classical structure recalled in the preceding Theorem 5.2.12. Its adjunction uses $\mathrm{cat}\,\mathrm{Sd}^2$ and $\mathrm{Ex}^2 N$; the two subdivisions are part of the construction, rather than an identification of every categorical homotopy with a simplicial one.

This answers the generalized Thomason question in a large specified class. It does not prove that Grothendieck's weak axioms Loc (6′a,8,9), without relative locality, imply that class of hypotheses. His phrase about extending Thomason's structure can naturally include its explicitly mentioned

properness. With that reading, the ensuing conclusion about enough intrinsic cofibrations and fibrations is justified. If one only knows existence of an arbitrary Quillen structure, however, identifying its fibrations with the intrinsic class Fib_W requires an additional argument. The properness qualification explains exactly what makes the intended inference work.

The implication supplied by properness is elementary and worth making explicit. Suppose (W, C, F) is a right proper model structure on a category with pullbacks, and let $p : E \to B$ belong to F. For any $B' \to B$, the pullback $p' : E' \to B'$ is again a model fibration. If $a : A \to A'$ is a weak equivalence over B', pull back p' first to A'. Right properness applied to that fibration shows

$$E' \times_{B'} A \longrightarrow E' \times_{B'} A'$$

is a weak equivalence. Thus every base change of p preserves weak equivalences under pullback: p is an intrinsic W-fibration. The model factorization of f as a trivial cofibration followed by a fibration is therefore a factorization as a W-equivalence followed by an intrinsic W-fibration. This gives enough such fibrations, namely the content needed for Loc (7F). Cisinski records the corresponding comparison in Example 6.1.14.

Dually, in a left proper model structure, a model cofibration remains one after cobase change, and pushout along it preserves weak equivalences. Hence it is an intrinsic W-cofibration. Factoring any map as a cofibration followed by a trivial fibration gives the required cofibration factorization. In the generalized Thomason theorem this conclusion is available because left properness is part of the theorem. The fibration conclusion requires the additional right properness hypothesis. One should therefore separate the affirmative realization theorem from the intrinsic-fibration question for an arbitrary localizer.

The finite-truncation examples specifically requested here do satisfy the stronger hypothesis. In Cisinski (2006a, Section 9.2.1, p. 364), $\mathbf{W}_n$ is generated by making the simplex category $\Delta/\partial\Delta^{n+2}$ aspherical. It is accessible and has aspherical generators; Corollary 4.3.28 proves such a localizer proper. Corollary 9.2.15, p. 369, identifies its simplicial weak equivalences with the maps inducing a bijection on components and isomorphisms on homotopy groups through degree n, so this is the class meant in the manuscript, not merely a similarly named localization. Theorem 5.2.15 therefore gives a proper model on Cat for every finite $n \geq 0$. Applying the preceding pullback argument proves that $\mathbf{W}_n$ has enough intrinsic fibrations. This answers the particular Loc (7F) question on pages 91 and 124 affirmatively, even though D_{W_ω} fails for each of these strict truncations.

This distinction also resolves the apparent tension with the earlier truncated examples. A theory can possess valid derived right Kan extensions through a model presentation even though the fixed ordinary path replacement is invalid for its weak equivalences. Derived constructions use the fibrant replacements and lifting conditions of the localized model. They need not be the ordinary loop categories that retain every homotopy group. The explicit $\mathbf{W}_0$ factorization at manuscript page 69 supplies a concrete instance; the $K(G, n+1)$ example at page 90 explains why the unmodified ordinary path construction cannot provide a universal replacement for all truncations.

Overview · XVI

Chapter XVI, manuscript pages 1–124: equivalences before models.

Grothendieck returns to the axioms because he wants to understand which part of a homotopy theory belongs to the equivalences themselves. A useful factorization supplies a way of calculating a homotopy fibre or a homotopy pushout; it should not determine what that construction means. This conviction explains both the long inventory in these pages and its recurrent changes of emphasis. The object being studied is a class W of functors between small categories. From it one wants to recover the localization Hot_W, categories of coherent coefficients over every index category, and the operations relating those categories. The chapter asks how much geometry is forced by a few closure properties of W, and how much must still be supplied by an axiom of descent, a factorization, or a smallness condition.

The letter to Thomason of 2 April 1991 states this position particularly clearly: a category with weak equivalences is the underlying datum, while cofibrations and fibrations are additional means of constructing its homotopy theory. The same letter recalls the observation that a derived category "ne vient jamais seule": diagrams of complexes have their own derived categories, whose relationships carry information lost on passing immediately to diagrams in a single derived category (Grothendieck, 1991a, §§1–2). Chapter XVI belongs to that project. Its geometric language of fibres, open and closed subcategories, and integration is a language for constructing coherent diagram theories. The connection with the theory of coefficients is structural, rather than an analogy added after the constructions have been made.

The first distinction: absolute and relative localization.

The starting axioms are deliberately modest. They impose the specified saturation conditions, declare a category with a terminal object to be aspherical, and require an aspherical functor to be an equivalence. Here asphericity of $u : A \to B$ is tested on the comma categories A/b. The stronger axiom Loc (3 bis) concerns a map $A \to B$ over an arbitrary third category S, and tests the maps $A/s \to B/s$. This difference governs much of the chapter. The first condition controls individual objects and functors with contractible fibres in the comma sense. The second permits the comparison of two varying families. In later terminology the first system defines a *weak* fundamental localizer; the usual definition of a fundamental localizer incorporates the relative condition.

The comparison with split Grothendieck fibrations in §1 makes this distinction geometric. Replacing a category over S by its free split fibration turns comma categories into actual fibres. The comparison with the original category has an adjoint and is therefore already harmless for the weak axioms. What remains is exactly the question whether fibrewise equivalences of two families are equivalences of their total categories. The commentary spells out the variance in this replacement: Grothendieck fibrations are described by contravariant category-valued functors, and their fibres are compared with undercategories over the base. Interchanging these conventions silently would change the argument.

The connectedness axiom of §2 rules out the degenerate theories in which all nonempty categories become indistinguishable. It allows π_0 to descend through localization and explains the special role of discrete categories. The converse characterization by nontriviality invokes an earlier product argument whose editors explicitly require (3 bis); its weak-axiom generality is therefore not established by that reference. This section also contains a useful warning about localization arguments. Adjunctions that preserve the designated equivalences descend by their units and counits. An identification between the localization of an infinite product of relative categories and the product of their localizations requires additional justification. The finite and infinite cases must therefore be separated. Moreover, the literal formulation of Lemma 2.5 with an arbitrary class of arrows omits an identity hypothesis needed even in the finite case. The audit isolates this defect without disturbing the intended application to localizers, which already contain the identities.

Excision and the two meanings of fibration.

The next two sections study complementary ways of making constructions invariant. For pushouts, the intrinsic class Cof_W consists of maps along which cobase change preserves W-equivalences. The forward Mayer–Vietoris condition is an excision principle for a cover by two sieves, or dually by two cosieves. Dwyer inclusions supply enough explicit maps with which to work. Brown's lemma is the decisive bridge from stability of trivial cofibrations to stability of arbitrary equivalences under these pushouts (Brown, 1973). The commentary gives the missing argument, retaining the distinction between Grothendieck's sketch, the cited general result, and its application in Cat.

Forward excision must be distinguished from the reverse implication, which tries to recover an equivalence on the intersection from equivalences on the two pieces and their union. The latter is substantially stronger. It fails for ordinary weak homotopy equivalences, and the blanket assertion in Theorem 5.4 that the ordinary localizer satisfies Loc (6) must be read with this defect exposed: its forward part is valid, its reverse part is not. Cone and suspension constructions explain why this is a mathematical issue rather than a matter of terminology. They also give a useful further development: reverse cube excision forces the inversion of integral homology equivalences. The detailed proof appears at the end of the commentary on §3, with its precise scope and the literature checked.

For pullbacks, Fib_W is defined intrinsically by preservation of W-equivalences under base change. It is not the class of fibrations in an arbitrarily chosen Quillen model structure. Nor is it the class of Grothendieck fibrations with W-equivalent transition functors. The latter is an explicit source of candidate maps, and §4 asks when those candidates belong to the intrinsic class. Universal equivalences, weak fibrations, perfect maps, and the auxiliary localizer $W_0 \subseteq W$ provide progressively different tests. The many implications labelled A, B, C, D and F record which tests are sufficient for which calculations. Their hypotheses matter: a statement about maps over a fixed base does not automatically assert stability under every change of target, and the existence of enough intrinsic fibrations does not identify a prescribed geometric class with those fibrations.

These pages are exploratory, but their main mechanism is robust. A factorization by a sufficiently well behaved class reduces preservation of all equivalences to preservation of trivial maps in that class. Repeated path constructions then turn local constancy into the stronger comparison needed for arbitrary base change. The commentary follows this mechanism through the diagram of implications, provides the missing intermediate verifications, and distinguishes genuine proof gaps from incorrectly copied indices and from slips already present in the edition. A worked analysis of component monodromy gives a concrete test for the difference between locally constant Grothendieck fibrations and intrinsic fibrations for π_0-equivalences.

What the minimality theorem changes.

The most consequential later result for these pages is stronger than the minimality theorem in its familiar formulation. Cisinski proves that ordinary weak homotopy equivalences form the least *weak* fundamental localizer (Cisinski, 2006a, Theorem 6.1.18, pp. 242–243). Consequently Grothendieck's W_ω, defined here using only the weak axioms, equals $\mathbf{W}_\infty$. This settles the suggestions that W_ω might fail the filtered-colimit axiom, and that a fundamental localizer strictly contained in $\mathbf{W}_\infty$ might retain simple homotopy information. It is essential to cite this stronger theorem: the relative theorem alone would not justify its application to the definition used at the beginning of Chapter XVI.

Theorem 6.1 gives a second consequence. Its fibration hypothesis, together with connectedness, forces $W \subseteq \mathbf{W}_\infty$. Minimality supplies the opposite inclusion. The other branch can be proved directly: a disconnected W-aspherical category, after perfect replacement of a chosen point, produces an empty fibre; universal base change then forces every functor into W. This supplies the missing dependency without invoking the general nontriviality criterion. Consequently D_{W_ω} permits exactly the ordinary localizer and the total localizer; Loc (5),

or non-totality under this fibration hypothesis, selects the former. The eight conditions in the main rectangle are therefore equivalent, while F remains strictly weaker. This explains why the stronger fibre calculations exclude the truncated and homological examples that otherwise belong naturally to the theory. The finite truncation localizers do have enough intrinsic fibrations: their proper generalized Thomason models answer the question about F positively (Cisinski, 2006a, §9.2.1, Corollary 9.2.15 and Theorem 5.2.15).

One qualification is necessary for the list on ms. 87: its bare Loc (6), if read as including reverse excision, is incompatible with the simultaneous connectedness and fibration requirements. Equality with $\mathbf{W}_\infty$ describes the intended forward-excision version, not that inconsistent literal list.

The finite-poset calculation in §6 nevertheless retains its independent interest. Ordinary homotopy of realizations is weaker than homotopy through natural transformations of functors. Subdivision preserves the realization but need not provide a categorical homotopy inverse. The commentary proves the cycle example explicitly and distinguishes categorical homotopy, weak homotopy type, and simple homotopy type. Minimality prevents the proposed refinement from being encoded by a fundamental localizer on all small categories; it does not identify these three notions or erase the finer finite combinatorics.

Smallness and the passage to derivators.

Accessibility in §5 supplies the finiteness principle needed to construct models. The relevant smallness is a bounded-subcategory property: a prescribed small part of an equivalence can be enlarged, without exceeding the chosen cardinal bound, to a small equivalence. The argument for $\mathbf{W}_\infty$ uses a countable succession of homotopy witnesses. The notation π_0 in the cardinal estimate is expressly used for $\aleph_0$ and is not a transcription error. Translating this passage into modern presentability language requires care about successor cardinals; one must not identify a countable-category bound with finite presentability. The commentary gives an explicit closure construction and supplies the cofinal-tail choice needed to represent a basepoint in a filtered system.

The final section returns to the goal of the axioms. It asks when HOT_W has homotopy Kan extensions, detects isomorphisms pointwise, and satisfies the required exactness properties. The source's numbering of the derivator axioms is kept separate from modern numbering. In particular its decomposition axiom concerns finite sums. Its assertion for arbitrary relative categories is therefore not refuted by a failure involving infinite products. Likewise, the calculation of a colimit at an initial object in §7.7 is correct with the manuscript's contravariant diagram convention.

Two later theorems give firm landmarks. Accessible fundamental localizers in the relative sense admit the generalized Thomason model structures constructed by Cisinski; left properness is part of the theorem, whereas right properness requires the corresponding properness condition (Cisinski, 2006a, Theorem 5.2.15). Suitable model categories produce derivators, with homotopy Kan extensions and their pointwise formulas (Cisinski, 2003; Groth, 2013). These results answer the model-existence problem under explicit hypotheses. They do not validate the circular inference in Corollary 7.7.2, where the identification of W with the localizer detected by HOT_W uses the very relative axiom it is meant to establish. The edition preserves the warning about that circle, and so does the audit. For the remaining weak-axiom variants, the results found are kept apart from the implications that the sources checked do not settle.

Chapter XVII

Catégories à fibrations et à cofibrations (Modèles (3))

p. –

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XVII

Catégories à fibrations et à cofibrations (Modèles (3))

With commentary

Commentary and audit AI-produced by GPT-6 Astra (OpenAI, ChatGPT Pro mode), checked and integrated by Claude (Anthropic)

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Seules quelques corrections mineures évidentes ont été effectuées. Pour les rares ajouts ou commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Préliminaires [1]

p. 1
[page 1]

Proposition 1. *Soit $\mathcal{M}$ une catégorie stable par $\varinjlim$ quelconques, $C \subset \mathrm{Fl}(\mathcal{M})$ contenant les isomorphismes, stable par cochangement de base et par composition* transfinie. *Soit, pour tout petit ensemble ordonné I, $C_I \subset \mathrm{Fl}(\mathcal{M}(I^\circ))$ l'ensemble des flèches $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ telles que $\forall\ i_0 \in I$, dans le diagramme*

$$(*) \qquad \begin{array}{ccc} X'_{i_0} = \varinjlim_{i<i_0} X_i & \longrightarrow & X_{i_0} \\ {\scriptstyle \alpha_{i_0} = \varinjlim_{i<i_0} u_i}\downarrow & \text{cocart.} & \downarrow{\scriptstyle \beta_{i_0}} \\ Y'_{i_0} = \varinjlim_{i<i_0} Y_i & \longrightarrow & \tilde{Y}_{i_0} \\ & & \downarrow{\scriptstyle \tilde{u}_{i_0}} \\ & & Y_{i_0}\ , \end{array} \qquad u_{i_0} : \tilde{Y}_{i_0} \to Y_{i_0}$$

(défini comme somme amalgamée)

la flèche $\tilde{u}_{i_0}$ soit dans C [2]*. Je dis que si I est* bien ordonné, *alors $u_\bullet \in C_I$ implique*

a) *que $\varinjlim u_i : \varinjlim X_i \longrightarrow \varinjlim Y_i$ est dans C, et*

b) *$\forall\ i \in I$, $(u_i : X_i \longrightarrow Y_i) \in C$* [3]*.*

.

Corollaire. *Soit $W \subset \mathcal{M}$ localiseur modérément saturé, tel que $TC \overset{déf}{=} C \cap W$ soit stable par cochangement de base, et par composition* transfinie. *Soit $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ dans C_I. Alors $u_\bullet \in W_{I^\circ}$ (i.e. $\forall\ i \in I$, on a $u_i \in W$) si et seulement si $u_\bullet \in (TC)_I$, i.e. si et seulement si pour tout $i \in I$, la flèche $\tilde{u}_i$ dans $(*)$ est dans TC, ou ce qui revient au même, dans W (puisque par hypothèse elle est déjà dans C). Si cette condition est satisfaite, alors $\varinjlim u_i : \varinjlim X_i \longrightarrow \varinjlim Y_i$ est dans W.*

[page 2 n'existe pas]
p. 2
[page 3]

Commentary · Exposition [ms. 1]

The relative latching map measures a new attachment

Grothendieck wants a condition on a map of diagrams which survives taking their colimits. The condition must record how the new value at an index is attached to everything already constructed. Merely asking that each component be a cofibration loses precisely this information. The pushout in $(*)$ isolates the additional attachment, and the rest of the preliminaries explains how to assemble these attachments without losing control of the resulting map.

We use the order convention of the manuscript: there is a unique arrow $i \to j$ when $i \leq j$, and an open subset is a lower ideal. Thus $\mathcal{M}(I^\circ)$ here means the category of covariant functors $I \to \mathcal{M}$; the circle belongs to Grothendieck's indexing convention, not to a reversal of the arrows in the diagrams being considered. Put

$$L_iX = \operatorname*{colim}_{j<i} X_j, \qquad \rho_i(u) : X_i \amalg_{L_iX} L_iY \longrightarrow Y_i.$$

1. **N.B.** Ces préliminaires seront complètement repris et dépassés dans le paragraphe 2. À ne pas lire !
2. Condition stable par passage de I à un sous-ensemble ordonné *ouvert* I' de I.
3. Dire aussi que l'on a

b′) *Si $i' \leq i$, alors dans le diagramme similaire à $(*)$,*

$$\begin{array}{ccc} X'_{i'} & \longrightarrow & X'_i \\ \downarrow & \text{cocart.} & \downarrow \\ Y'_{i'} & \longrightarrow & \tilde{Y}'_i \\ & & \downarrow{\scriptstyle u_{i',i}} \\ & & Y'_i\ , \end{array}$$

on a $u_{i',i} \in C$ (on retrouve $(*)$ en faisant $i' = i_0$, $i = i_0 + 1$). Le cas b) s'obtient en faisant $i' = 0$, i quelconque, d'où $(X'_i \longrightarrow Y'_i) \in C$. L'appliquant à $i + 1$ au lieu de i, on trouve $(u_i : X_i \longrightarrow Y_i) \in C$.

The map $\rho_i(u)$ is the relative latching map. In section 1, membership in C_I means exactly that all these maps belong to C. When i is minimal, both latching objects are initial and $\rho_i(u) = u_i$. At a nonminimal index, the latching object includes every incoming relation, rather than just a chosen list of immediate predecessors. This matters even for finite posets: two paths from an earlier object have to impose the same identification in the colimit.

The hypotheses on C say that attachments may be transported by pushout and composed, including through a continuous ordinal sequence. They do not require lifting properties, retract closure, or a choice of generating cofibrations. Accordingly, the first proposition is a statement about colimits and a class of morphisms; weak equivalences enter only in the corollary. Throughout these paragraphs a transfinite composite means the composite along a small ordinal-indexed chain whose value at a limit stage is the colimit of its preceding values.

For a lower ideal $J \subset I$, the incoming category at any $j \in J$ is unchanged by restriction: every $i < j$ also lies in J. Consequently $L_j(X|_J) = L_j X$, and the relative latching test restricts to J. This proves the assertion in footnote 2 and explains why lower ideals, rather than arbitrary full subposets, organize the argument. The same observation will allow us to apply a colimit statement to all predecessors of a single index and recover a statement about its component.

There is a change of definition ahead. In section 2, C_I will be defined by all open subsets of each principal lower ideal, even for a poset that is not artinian. The present one-latching test and that stronger definition agree for artinian posets under the stated closure hypotheses. They need not agree for an arbitrary poset. Keeping track of this change prevents the later counterexample from appearing to contradict Proposition 1.

Démonstration du corollaire. Supposons $u_\bullet \in (TC)_I$, alors par la proposition, $\varinjlim_{i\in I} u_i$ est dans TC, ou ce qui revient au même, dans W. D'autre part, appliquant ce même résultat aux systèmes inductifs induits sur $I_{<i_0}$, on trouve que dans $(*)$, la flèche α_{i_0} est dans TC, donc aussi β_{i_0} par cochangement de base, donc aussi $u_{i_0} = \tilde{u}_{i_0}\beta_{i_0}$. Donc il reste à prouver qu'inversement, si on suppose les $u_{i_0} \in W$, alors les $\tilde{u}_{i_0}$ sont eux aussi dans W (i.e. dans $TC = C \cap W$), ou ce qui revient au même, que les β_{i_0} sont dans W, ou encore dans TC. Mais comme TC est stable par cochangement de base, il suffit pour cela de prouver que $\alpha_{i_0} \in TC$, ou encore $\alpha_{i_0} \in C$. Donc on est ramené à prouver que

c) $u \in W_{I^\circ} \cap C_I \Longrightarrow (\varinjlim_{i<i_0} u_i) \in W$ pour tout $i_0 \in I$,

donc, changeant de notation, à prouver `[phrase incomplète]`. Nous prouverons ensemble a) b) c) par induction transfinie sur le nombre ordinal α tel que $I \simeq I_\alpha = \{\text{ensemble des ordinaux } i < \alpha\}$.

1°) Cas $\alpha = 0$, i.e. $I = \emptyset$: il est trivial.

[page 4]

2°) Si c'est ok pour I_α, aussi pour $I_{\alpha+1} = \{\beta \mid \beta \leq \alpha\}$. Ici, $I_{\alpha+1}$ ayant un plus grand élément α, a) est contenu dans b) (pour $i = \alpha$, plus grand élément de $I_{\alpha+1}$.) Pour prouver b), il reste à prouver que $u_\alpha \in C$. On regarde le diagramme $(*)$ pour $i_0 = \alpha$. Par l'hypothèse de récurrence (forme a), appliquée à I_α, on a $\alpha_{i_0} \in C$, donc par cochangement de base $\beta_{i_0} \in C$, donc par composition $u_{i_0} = \tilde{u}_{i_0}\beta_{i_0} \in C$. Pour prouver c), il reste à prouver que $\varinjlim_{i<\alpha} u_i \in W$. Mais par l'hypothèse de récurrence appliquée à $I_\alpha = \{i \mid i < \alpha\}$, on sait déjà que $(u_i)_{i<\alpha} \in (TC)_{I_\alpha}$, donc par la partie prouvée du corollaire ou de la proposition, $\varinjlim_{i<\alpha} u_i \in TC$.

3°) Si α est un ordinal limite, et si a) b) c) sont vrais pour I_β avec $\beta < \alpha$, ils le sont aussi pour I_α. Cette fois, b) est contenu dans l'hypothèse de récurrence, il faut prouver a), i.e. que

[page 5]

$$u_\alpha \overset{\text{déf}}{=} \varinjlim_{i<\alpha} u_i \ : \ \underbrace{X_\alpha}_{\overset{\text{déf}}{=}\varinjlim_{i<\alpha} X_i} \longrightarrow \underbrace{Y_\alpha}_{\overset{\text{déf}}{=}\varinjlim_{i<\alpha} Y_i}$$

est dans C. Quant à c), il est déjà contenu dans la hypothèse de récurrence. Donc il reste finalement à prouver

$$u_\alpha \ \in \ C$$

quand on suppose que la proposition est prouvée pour les ordinaux $\beta < \alpha$.

On va construire un troisième système ordinal sur I_α, en introduisant pour tout $0 \leq i < \alpha$, i.e. $i \in I_\alpha$, Z_i par le carré cocartésien p. 3

$$\begin{array}{ccc} X_i & \longrightarrow & X_\alpha \\ \downarrow & \text{cocart.} & \downarrow \\ Y_i & \longrightarrow & Z_i \ . \end{array}$$

Les flèches de transition entre les Z_i sont évidentes, et on aura $\varinjlim_i Z_i \xrightarrow{\sim} Y_\alpha$. Ainsi, $u_\alpha : X_\alpha (= Z_{-1}) \longrightarrow Y_\alpha$ est décomposé en composé transfini de morphismes [page 6]

$$(*) \qquad Z_i \xrightarrow{v_{i+1}} Z_{i+1} \qquad (-1 \leq i < \alpha)$$

et

$$(**) \qquad \varinjlim_{i<i_0} Z_i \xrightarrow{v_{i_0}} Z_{i_0} \qquad i_0 \text{ ordinal limite } < \alpha.$$

(**N.B.** Le cas où i_0 n'est *pas* ordinal limite n'est autre que $(*)$ pour $i = \beta_0 - 1$.) Tout revient à prouver que ces flèches sont dans C (donc leur composé transfini aussi).

Cas $(*)$.

4

$$\begin{array}{ccccc} X_i & \longrightarrow & X_{i+1} & \longrightarrow & X_\alpha \\ {\scriptstyle u_i}\downarrow & \text{cocart.} & {\scriptstyle \beta_{i+1}}\downarrow & \text{cocart.} & \downarrow \\ Y_i & \longrightarrow & \tilde{Y}_{i+1} & \longrightarrow & Z_i \\ & & {\scriptstyle \tilde{u}_{i+1}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle v_{i+1}} \\ & & Y_{i+1} & \longrightarrow & Z_{i+1} \ . \end{array}$$

Comme $\tilde{u}_{i+1} \in C$, $v_{i+1} \in C$ par cochangement de base, OK. Il faut cependant traiter à part le cas $i = 0$, i.e. la flèche $X_\alpha \longrightarrow Z_0$, qui est incluse dans

$$\begin{array}{ccc} X_0 & \longrightarrow & X_\alpha = Z_{-1} \\ {\scriptstyle u_0}\downarrow & \text{cocart.} & \downarrow{\scriptstyle v_0} \\ Y_0 & \longrightarrow & Z_0 \ , \end{array}$$

où $u_0 \in C$, donc $v_0 \in C$, OK.

Cas $(**)$. Si $i_0 < \alpha$ est ordinal limite, inspectons $(**)$. On a par passage à la limite

$$Z'_{i_0} \overset{\text{déf}}{=} \varinjlim_{i<i_0} Z_i \ \simeq \ Y'_{i_0} \amalg_{X'_{i_0}} X_\alpha$$

(cf. diagramme $(*)$ pour la définition de X'_{i_0}, Y'_{i_0}), s'insérant dans le diagramme suivant greffé sur $(*)$

$$\begin{array}{ccccc} X'_{i_0} & \longrightarrow & X_{i_0} & \longrightarrow & X_\alpha \\ {\scriptstyle \alpha_{i_0}}\downarrow & \text{cocart.} & {\scriptstyle \beta_{i_0}}\downarrow & \text{cocart.} & \downarrow \\ Y'_{i_0} & \longrightarrow & \tilde{Y}_{i_0} & \longrightarrow & Z'_{i_0} \\ & & {\scriptstyle \tilde{u}_{i_0}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle v_{i_0}} \\ & & Y_{i_0} & \longrightarrow & Z_{i_0} \ . \end{array}$$

p. 4
[page 7]

4. Pour se donner confiance – on pourrait traiter toute de suite le cas général.

On a $\tilde{u}_{i_0} \in C$ par hypothèse, donc $v_{i_0} \in C$, ce qui achève la démonstration.

Commentary · Exposition [ms. 7]

A continuous filtration of the colimit map

The auxiliary objects Z_i put all stages over the fixed object X_α, so that their successive changes are pushouts of the relative latching maps. This is the right idea. Their indexing becomes more transparent if the value immediately before an attachment is distinguished from the value immediately after it. In particular, the original diagram X need not itself be continuous at a limit ordinal.

Here is a continuous version of the argument, including the marginal strengthening. Let the indexing ordinal be λ, and write

$$X_{<\beta} = \operatorname*{colim}_{i<\beta} X_i, \quad Y_{<\beta} = \operatorname*{colim}_{i<\beta} Y_i, \quad X = X_{<\lambda}, \quad Y = Y_{<\lambda}.$$

For $0 \le \beta \le \lambda$, set

$$P_\beta = X \amalg_{X_{<\beta}} Y_{<\beta}.$$

Then $P_0 = X$ and $P_\lambda = Y$. If $\delta \le \lambda$ is a nonzero limit ordinal, the universal property of colimits gives

$$\operatorname*{colim}_{\beta<\delta} P_\beta \simeq X \amalg_{X_{<\delta}} Y_{<\delta} = P_\delta.$$

Indeed, a compatible cocone from the left side consists of a map from X and compatible maps from the $Y_{<\beta}$ agreeing on every $X_{<\beta}$. These are exactly the data defining a map from the displayed pushout. This verifies continuity without an extra exactness assumption on $\mathcal{M}$.

At a successor stage the square

$$\begin{array}{ccc} X_\beta \amalg_{L_\beta X} L_\beta Y & \xrightarrow{\rho_\beta(u)} & Y_\beta \\ \downarrow & & \downarrow \\ P_\beta & \longrightarrow & P_{\beta+1} \end{array}$$

is cocartesian. To check this, a map from its pushout amounts to a map from X and a map from Y_β whose restrictions along every X_i, $i \le \beta$, agree. Since β is terminal in the subposet $i \le \beta$, the resulting object is $X \amalg_{X_\beta} Y_\beta = P_{\beta+1}$. Thus each successor arrow belongs to C, and transfinite closure proves $X \to Y \in C$. Restricting the entire argument to $i \le i_0$ proves $u_{i_0} \in C$.

For the marginal assertion start the same construction at any $\gamma \le \lambda$. It proves

$$X_{<\lambda} \amalg_{X_{<\gamma}} Y_{<\gamma} \longrightarrow Y_{<\lambda} \quad \in C.$$

The proof uses only the cells with indices $\gamma \le i < \lambda$. In particular, it does not need the map $X_{<\gamma} \to Y_{<\gamma}$ to belong to C. The source later checks this map in order to invoke Proposition 1 as already stated; the relative filtration itself is slightly more economical.

This also locates the two harmless indexing slips on manuscript page 6. The predecessor in the parenthesis is $i_0 - 1$, not the printed $\beta_0 - 1$. The special arrow $X_\alpha = Z_{-1} \to Z_0$ has source index -1, although the following sentence calls it the case $i = 0$. Both printed readings are retained; neither affects the pushout argument. The continuous P-filtration avoids counting a nontrivial attachment at a limit as though the original Z-diagram were already continuous there.

Proposition 2. *$(\mathcal{M}, C)$ et le cas échéant W, comme dans proposition 1 et son corollaire, I ensemble ordonné localement fini, i.e. tel que $\forall\ i_0 \in I$, I/i_0 soit fini. Soit*

$$u_\bullet : X_\bullet \longrightarrow Y_\bullet$$

une flèche dans[5] *$\mathcal{M}(I^\circ)_c$ = sous-catégorie pleine de systèmes inductifs sur I, à valeurs dans*

5. en fait, l'hypothèse que $X_\bullet$, $Y_\bullet$ dans $\mathcal{M}(I^\circ)_c$ est inutile, tant dans la proposition 2 que dans ses deux corollaires.

$\mathcal{M}$, tels que $\underbrace{\emptyset_{\mathcal{M}(I^\circ)}}_{\substack{\text{objet}\\ \text{initial}\\ \text{de } \mathcal{M}(I^\circ)}} \longrightarrow Z_\bullet$ *soit dans C_I (définie dans prop. 1), i.e. tel que $\forall\ i_0 \in I$, la flèche*

$$Z'_{i_0} \;=\; \varinjlim_{i<i_0} Z_i \;\longrightarrow\; Z_{i_0}$$

soit dans C, et supposons $u_\bullet \in C_I$. Alors

a) $\varinjlim_{i\in I} u_\bullet : X_I \overset{\text{déf}}{=} \varinjlim_{i\in I} X_i \longrightarrow Y_I \overset{\text{déf}}{=} \varinjlim_{i\in I} Y_i$ *est dans C,* [page 8]
et

b) $\forall\ i_0 \in I$, *on a $\alpha_{i_0}, \beta_{i_0} \in C$ dans le diagramme $(*)$ p. 1.*

Commentary · Later developments [ms. 7]

The later Reedy formulation

The relative latching condition became a standard language for diagrams built in increasing degree. For a finite or locally finite poset, one may assign to an object the maximum length of a strictly increasing chain ending there. Every nonidentity arrow raises this natural-number degree. With inverse arrows restricted to identities, this is a direct Reedy category. The cells in the preceding proof are its relative latching maps.

The general cellular decomposition of a natural transformation is developed in Riehl and Verity (2014, Proposition 6.3); their Lemma 7.1 identifies a Reedy cofibration which is an objectwise weak equivalence by the triviality of all its relative latching maps. These are the two mechanisms used here. The comparison does not import a full model structure into Grothendieck's first propositions: their proofs already work with his stated closure properties.

For cofibration categories, Rădulescu-Banu (2009, Theorem 9.4.1(1)(a,b)) gives the later relative-colimit statement for an open embedding of direct categories. It includes both the transition map for a cofibrant diagram and the relative map associated with a Reedy cofibration. His cofibrancy conventions and existence assumptions are more restrictive than the cocomplete ambient category used in these preliminaries, so the formulations should not be identified without that qualification.

An arbitrary artinian poset need not admit a natural-number degree. Its rank can be a larger ordinal. The transfinite version of the degree convention is treated explicitly, in the dual orientation, by Shulman (2015, Definition 4.6 and Theorem 4.7). The ordinal formulation is the one appropriate to the unification Grothendieck reaches on manuscript pages 23–25.

Corollaire. *De plus (supposant donné W comme dans prop. 1, cor. 1), les conditions suivantes sur une $u_\bullet$ entre $X_\bullet$, $Y_\bullet$ C-cofibrants sont équivalentes.*

1°) $u_\bullet \in (TC)_I$, *i.e. les $\tilde{u}_{i_0} \in TC$ ($\forall\ i_0 \in I$).*

2°) $u_\bullet \in T_I \cap W_{I^\circ}$, *i.e. $\forall\ i_0 \in I$, $\tilde{u}_{i_0} \in C$, $u_{i_0} \in W$.*

3°) *Pour tout $i_0 \in I$, on a*

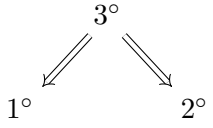
$$\alpha_{i_0},\ \beta_{i_0},\ \tilde{u}_{i_0},\ u_{i_0} \;\in\; TC$$

(i.e. $\alpha_{i_0}, \tilde{u}_{i_0} \in TC$, ce qui implique $\beta_{i_0} \in TC$, puis $u_{i_0} \in TC$ par cochangement de base et par composition).

Et ces conditions impliquent

4°) *que $u_I : X_I \longrightarrow Y_I$ est dans TC.*

Voyons comment ou dans quelle mesure le corollaire se déduit de la proposition. On a bien sur p. 5



tautologiquement. D'ailleurs, la proposition appliquée à TC au lieu de X montre qu'on a $1° \Longrightarrow 3°$ & $4°$, car en l'appliquant aux ensembles $I_{<i_0}$ on trouve que $\alpha_{i_0} \in TC \ \forall\ i_0 \in I$. Donc la proposition 2 implique

$$(1° \Longleftrightarrow 3°) \Longrightarrow 2° \ (\&\ 4°).$$

[page 9]

Commentary · Exposition [ms. 8]

Why objectwise equivalences become trivial attachments

Grothendieck next wants the same colimit construction to preserve weak equivalences. The essential hypothesis is that $TC = C \cap W$ has the same attachment stability as C. It permits the cellular proof to be used a second time, with TC in place of C. No assertion that arbitrary colimits preserve W is being made.

The reverse implication in the corollary can be written without simultaneous induction on three differently phrased statements. Assume the $\rho_i(u)$ belong to C and the u_i belong to W. Induct through the well-founded indices, assuming that $\rho_j(u) \in TC$ for every $j < i$. Applied to the predecessor diagram, the already established cellular argument for the class TC gives

$$L_iX \longrightarrow L_iY \quad \in TC.$$

Its pushout $X_i \to X_i \amalg_{L_iX} L_iY$ is therefore in TC. Its composite with $\rho_i(u)$ is $u_i \in W$, so the two-out-of-three property gives $\rho_i(u) \in W$. Since the relative latching condition already places it in C, it belongs to TC. The induction starts at a minimal object, where $\rho_i(u) = u_i$.

Conversely, if all $\rho_i(u)$ lie in TC, the same cellular argument shows that every component and the total colimit map belong to TC. Restriction to lower ideals also gives $L_iX \to L_iY \in TC$, and pushout gives the corresponding $\beta_i \in TC$. These are exactly the four maps listed in condition $3°$ of the second corollary. The assertion is an equivalence between two descriptions of trivial diagram cofibrations.

There is a genuine but local misprint in the argument on manuscript page 3. After reducing to $\alpha_{i_0} \in TC$, the text says "ou encore $\alpha_{i_0} \in C$". The proposition already supplies membership in C; what remains is membership in W. The next displayed condition (c) states precisely that remaining assertion. The intended proof therefore survives after replacing the final C in that sentence by W. Similarly the T_I printed in condition $2°$ on manuscript page 8 is C_I in the intended statement, as its immediately following explanation confirms.

Donc pour prouver $2° \Longrightarrow 3°$, il resterait à prouver que $2°$ suffit à impliquer $4°$ (et ceci quel que soit I ensemble ordonné localement fini) – car en appliquant ceci aux $I_{<i_0}$, on trouve que $2°$ implique $\alpha_{i_0} \in TC = C \cap W$ (non seulement $\alpha_{i_0} \in C$, selon la proposition 2), donc $\beta_{i_0} \in TC \subset W$, donc $\tilde{u}_{i_0} \in W$ par saturation, donc $\tilde{u}_{i_0} \in TC$, d'où $3°$.

Notons d'autre part que le b) dans la proposition 2 est conséquence de a), appliqué à la restriction de $u_\bullet$ à $I_{<i_0}$. Au total, il y a donc à prouver seulement les deux choses

$$(*) \qquad \begin{cases} \text{a)} \quad u_\bullet \in C_I \implies u_I \in C \\ \text{b)} \quad u_\bullet \in C_I \cap W \implies u_I \in C \cap W \ (\overset{\text{déf}}{=} TC) \\ \qquad\qquad\qquad\qquad\quad (\text{et } u_\bullet \in (TC)_I) \end{cases}$$

Nous le prouverons par réduction au cas particulier de la proposition 1, déjà prouvée.

J'ai d'ailleurs aussi envie de prouver le

Corollaire 2. *$\mathcal{M}$, C (et éventuellement W). Les hypothèses sur $X_\bullet, Y_\bullet \in \mathcal{M}(I°)_{\mathrm{c}}$ et sur $u_\bullet \in C_I$, enfin sur I localement fini, sont celles de la proposition 2. Pour toute partie J de I, soit*

$$u_J \overset{\text{déf}}{=} \varinjlim_{i \in J} u_i \ : \ X_J \overset{\text{déf}}{=} \varinjlim_{i \in J} X_i \ \longrightarrow \ Y_J \overset{\text{déf}}{=} \varinjlim_{i \in J} Y_i \ .$$

Soient $I' \subset I''$ deux parties ouvertes *de I, et considérons le diagramme*

[page 10]

(∗)
$$\begin{array}{ccc} X_{I'} & \xrightarrow{\tau_X} & X_{I''} \\ {\scriptstyle \alpha=u_{I'}}\downarrow & \text{cocart.} & {\scriptstyle \beta}\downarrow \\ Y_{I'} & \xrightarrow{\tilde{\tau}_Y} & \tilde{Y}_{I''} \\ & {\scriptstyle \tau_Y}\searrow & \downarrow{\scriptstyle \tilde{u}} \\ & & Y_{I''}\ . \end{array} \qquad u_{I''}$$
(défini comme somme amalgamée)

Alors on a p. 6

$$\begin{cases} u_{I'}, u_{I''} \in C \text{ (donc aussi } \beta \in C\text{) (i.e. pour tout } J \text{ ouvert dans } I, \\ u_J : X_J \longrightarrow Y_J \text{ est dans } C)^6, \text{ et } \tau_X, \tau_Y \in C \text{ (donc aussi } \tilde{\tau}_Y \in C\text{)} \\ \text{(i.e. pour } I' \subset I'' \text{ inclusion d'ouverts, et } X_\bullet \text{ } C\text{-cofibrant dans } \mathcal{M}(I^\circ), \\ \text{on a } X_{I'} \longrightarrow X_{I''} \text{ dans } C)^7, \text{ et enfin, même} \\ \qquad\qquad \boxed{\tilde{u} \in C.} \end{cases}$$

N.B. Le cas de la proposition 2 sous la forme (∗) a), page 9, (i.e. aussi le fait que les $u_{I'}$, $u_{I''}$, … sont dans C) est obtenu en prenant $I'' = I$, $I' = \emptyset$, le diagramme devient

$$\begin{array}{ccc} \emptyset & \longrightarrow & X_I \\ {\scriptstyle \mathrm{id}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle \mathrm{id}} \\ \emptyset & \longrightarrow & X_I \\ & & \downarrow{\scriptstyle \tilde{u}=u_I} \\ & & Y_I\ , \end{array}$$

et la conclusion est donc $u_I \in C$ [8]. La *définition* de C_I, i.e. la condition $u \in C_I$, se retrouve en exigeant $\tilde{u} \in C$ dans (∗) ci-dessus, dans le cas $I'' = I/i_0$, $I' = I_{<i_0}$. Il est remarquable que cette définition de C_I donne une conclusion aussi forte que $\tilde{u} \in C$ pour *tout* couple $I' \subset I''$ de sous-ensembles ouverts de I. [page 11]

Commentary · Examples [ms. 10]

Which conclusions need a cofibrant source diagram

The relative comparison map is the main invariant of Corollary 2. Its source combines the whole old X-diagram with the part of Y already constructed, so its being a cofibration does not require either diagram to start cofibrantly. By contrast, the transition map $X_{I'} \to X_{I''}$ concerns the attachment of X itself and does require a condition on X.

This distinction is visible in the two footnotes. Footnote 5 says that the assumptions on the objects are unnecessary in Proposition 2 and both its corollaries. Read literally, this is too broad.

6. C'est bien sûr contenu dans prop. 2, appliquée à $u_\bullet|_J$.

7. C'est pour cette assertion $\tau_X \in C$ (d'où $\tilde{\tau}_Y, \tau_Y \in C$, moyennant $\tilde{u} \in C$) qu'on a besoin de l'hypothèse $X_\bullet \in \mathcal{M}(I^\circ)_{\mathrm{c}}$, $\tilde{u} \in C$ vrai sans cette hypothèse.

8. Le cas $(Y_{I'} \xrightarrow{\tau_Y} Y_{I''}) \in C$ est obtenu en prenant $X_\bullet = \emptyset_\bullet$, on aura

$$\begin{array}{ccc} \emptyset & \xrightarrow[\sim]{\tau_X} & \emptyset \\ \downarrow & \text{cocart.} & \downarrow \\ Y_{I'} & \xrightarrow{\mathrm{id}} & \tilde{Y}_{I''} = Y_{I'} \\ & {\scriptstyle \tau_Y}\searrow & \downarrow{\scriptstyle \tilde{u}=\tau_Y} \\ & & Y_{I''}\ . \end{array}$$

Footnote 7 gives the correct qualification for the transition maps: X must be a cofibrant diagram to deduce $\tau_X \in C$. The relative conclusion $u_{I',I''} \in C$ holds without that hypothesis.

A complete counterexample to the blanket reading takes $\mathcal{M} = \mathrm{Set}$, C the isomorphisms, and $I = \{0 < 1\}$. Let $X = Y$ be the diagram

$$\varnothing \longrightarrow \{*\}, \qquad u = \mathrm{id}_X.$$

At 0, the relative latching map is $\mathrm{id}_\varnothing$. At 1, its source is

$$X_1 \amalg_{X_0} Y_0 = \{*\} \amalg_\varnothing \varnothing = \{*\},$$

and its target is $Y_1 = \{*\}$; the map is again an identity. Thus $u \in C_I$. For the open subsets $I' = \{0\}$ and $I'' = I$, however,

$$\tau_X : X_{I'} = \varnothing \longrightarrow X_{I''} = \{*\}$$

is not an isomorphism. This example satisfies every closure hypothesis on C, including transfinite closure. The failure therefore concerns the missing hypothesis, not the colimit construction.

Under the stated cofibrancy hypothesis the transition conclusions follow formally from the relative one. Apply it first to $\varnothing_I \to X$ to obtain $X_{I'} \to X_{I''} \in C$. Its pushout along $u_{I'}$ is $Y_{I'} \to X_{I''} \amalg_{X_{I'}} Y_{I'}$, hence also lies in C. Composing with $u_{I',I''} \in C$ gives $Y_{I'} \to Y_{I''} \in C$. Thus cofibrancy of Y need not be assumed separately once X is cofibrant and u satisfies the relative latching condition. The useful division is between a condition on the map and a condition on its initial diagram.

C'est finalement sous cette forme qu'on va prouver la proposition 2 – c'est *ça* à présent qui me semble *le* bon énoncé dans cette direction. Avec le complément, bien sûr, que si de plus $u_\bullet \in W_{I^\circ}$, i.e. les u_i sont dans W, alors $u_\bullet$ est même dans $(TC)_I$, ou encore simplement que les $u_{I'}$ *sont dans* TC (d'où il résultera donc que $\tilde{u}$ dans $(*)$, p. 10, est aussi dans TC). p. 7

On va procéder par construction transfinie. On peut supposer bien sûr que $I'' = I$. On va quand même d'abord prouver le cas particulier où $I' = \emptyset$, i.e. prouver au moins que $u_I \in C$. On construit par récurrence transfinie une suite strictement croissante d'ouverts

$$I_\alpha \ \subset \ I,$$

aboutissant à

$$I_{\alpha_0} \ = \ I,$$

en posant

$$\begin{cases} I_0 \ = \ \emptyset\,, \\ I_\alpha \ = \ \bigcup_{\beta<\alpha} I_\beta \qquad \text{si } \alpha \text{ ordinal limite}\,, \\ I_{\alpha+1} \ = \ I_\alpha \cup (I/s_\alpha) \quad \text{si } I_\alpha \neq I, \text{ et en choisissant } s_\alpha \in I \backslash I_\alpha\,. \end{cases}$$

(On finit par avoir $I_{\alpha_0} = I$, bien sûr.) On pose [page 12]

$$\begin{aligned} X_\alpha \ = \ X_{I_\alpha} \ &= \ \varinjlim_{i \in I_\alpha} X_i \\ Y_\alpha \ = \ Y_{I_\alpha} \ &= \ \varinjlim_{i \in I_\alpha} Y_i \\ u_\alpha \ = \ u_{I_\alpha} \ &: \ X_\alpha \longrightarrow Y_\alpha\,, \end{aligned}$$

homomorphisme de systèmes inductifs transfinis. On aura

$$X_{\alpha_0} \ = \ X_I, \qquad Y_{\alpha_0} \ = \ Y_I, \qquad u_{\alpha_0} \ = \ u_I,$$

et on doit donc prouver que $u_{\alpha_0} \in C$. Pour ceci, on applique la proposition 1, il suffit donc de prouver que $(u_\alpha) \in C_{I_{\alpha_0+1}}$ (où $I_{\alpha_0+1} = \{\text{ordinaux } i \mid i < \alpha_0 + 1, \text{ i.e. } i \leq \alpha_0\}$). Donc il faut

prouver que les flèches $\tilde{u}_{i_0}$ dans $(*)$ p. 1 (où cette fois les i, i_0 sont des ordinaux, $i_0 \leq \alpha_0$) sont dans C. Je vais distinguer les deux cas où $\alpha = i_0$ est ordinal limite, qui est trivial car alors $X_{i_0} \xleftarrow{\sim} \varinjlim_{i<i_0} X_i$, $Y_{i_0} \xleftarrow{\sim} \varinjlim_{i<i_0} Y_i$, et $\tilde{u}_{i_0} = \mathrm{id}$, et le cas α non limite, i.e.

$$\begin{array}{ccc} X_{I_{\alpha-1}} & \longrightarrow & X_{I_\alpha} \\ {\scriptstyle u_{\alpha-1}}\downarrow & & {\scriptstyle \beta_\alpha}\downarrow \\ Y_{I_{\alpha-1}} & \longrightarrow & \tilde{Y}_{I_\alpha} \\ & & {\scriptstyle \tilde{u}_\alpha}\downarrow \\ & & Y_{I_\alpha}\ , \end{array} \qquad u_\alpha$$

à prouver que $\tilde{u}_\alpha \in C$. Lorsqu'on suppose de plus $u_0 \in W_{I_0}$, i.e. les $u_i \in W$, il faut de plus p. 8
prouver que $\tilde{u}_\alpha \in TC = C \cap W$. La proposition 2 et son corollaire 1 résulteront de là. Pour [page 13]
prouver $\tilde{u}_\alpha \in C$ resp. $\tilde{u}_\alpha \in TC$, on peut supposer $I_\alpha = I$ (quitte à changer de notation). On pose donc

$$\begin{aligned} I_\alpha &= I, \qquad I_{\alpha-1} = I' \\ I/s_{\alpha-1} &= I_0, \quad I_0 \cap I' = I'_0\ , \end{aligned}$$

de sorte qu'on a un diagramme commutatif d'inclusions ouvertes

$$\begin{array}{ccc} I_0 \cap I' = I'_0 & \hookrightarrow & I_0 \\ \downarrow & & \downarrow \\ I' & \hookrightarrow & I = I_0 \cup I'. \end{array}$$

On utilisera seulement le fait que I_0, donc I'_0, est fini, i.e. que dans le couple $(I' \subset I)$, I peut s'écrire sous la forme $I = I' \cup I_0$, avec I_0 un ouvert *fini,* pour en conclure que dans le diagramme

$$\begin{array}{ccc} X_{I'} & \xrightarrow{\tau^X_{I',I}} & X_I \\ {\scriptstyle u_{I'}}\downarrow & \text{cocart.} & {\scriptstyle u_{I',I}}\downarrow \\ Y_{I'} & \longrightarrow & \tilde{Y}_I \\ & {\scriptstyle \tau^Y_{I',I}}\searrow & {\scriptstyle \tilde{u}=u_{I',I}}\downarrow \\ & & Y_I \end{array} \qquad u_I$$

on a $\tilde{u} \in C$ resp. $\tilde{u} \in TC = C \cap W$.

[page 14]

Commentary · Exposition [ms. 13]

Excision for a cover by lower ideals

Grothendieck reduces an attachment over a large open subset to an attachment over one new principal open subset. The reduction is an ordinary colimit identity. Stating that identity before the larger diagrams makes clear which part of the argument uses the order and which part uses the class C.

Let A, B be lower ideals of a poset, put $H = A \cap B$ and $T = A \cup B$, and let $X : T \to \mathcal{M}$ be any diagram. Then

$$X_T \simeq X_A \amalg_{X_H} X_B.$$

To prove this, maps from the right side into an object Q are cocones on $X|_A$ and $X|_B$ that agree on H. Every arrow of T belongs to A or to B: if its target lies in one of these lower ideals, its source lies there too. The two cocones therefore combine uniquely into a cocone on all of X. This establishes the displayed universal property. The same argument proves that the square of full subcategories with vertices H, A, B, T is a pushout in Cat.

Now let $u : X \to Y$, and define the relative domain

$$R_{A,T} = Y_A \amalg_{X_A} X_T, \qquad u_{A,T} : R_{A,T} \longrightarrow Y_T.$$

We claim that $u_{A,T}$ is a pushout of $u_{H,B}$. Indeed,

$$\begin{aligned} R_{A,T} &\simeq Y_A \amalg_{X_H} X_B \\ &\simeq Y_A \amalg_{Y_H} (Y_H \amalg_{X_H} X_B) = Y_A \amalg_{Y_H} R_{H,B}. \end{aligned}$$

Pushing out $R_{H,B} \to Y_B$ consequently gives

$$R_{A,T} \amalg_{R_{H,B}} Y_B \simeq Y_A \amalg_{Y_H} Y_B \simeq Y_T.$$

These canonical isomorphisms identify the induced comparison map with $u_{A,T}$, proving the claim. If $u_{H,B} \in C$, or in TC, its pushout belongs to the same class. No cofibrancy assumption on X or Y appears in the proof.

For the application in the text, take $A = I'$, $B = I_0$ and $H = I'_0$. Local finiteness ensures that $B = I/s$ is finite; hence the relative problem on $H \subset B$ can be solved by a finite induction before the transfinite assembly of the large index set is attempted. This is the precise role of local finiteness in Proposition 2. It does not imply that the full index set is finite, and here it means that every principal lower ideal is finite, not merely that every interval $[i, j]$ is finite.

The reduction also explains why the square is a cover by opens. For arbitrary subposets A, B, there can be arrows with one endpoint in each complement. Compatible cocones on the two pieces would then fail to include the conditions imposed by those arrows, and their pushout would not compute X_T.

Il faut des notations générales, d'une part pour une inclusion $J \subset K$ d'ouverts de I, donnant lieu au diagramme

(1)
$$\begin{array}{ccc} X_J & \xrightarrow{\tau^X_{J,K}} & X_K \\ {\scriptstyle u_J}\downarrow & \text{cocart} & \downarrow{\scriptstyle \tilde{u}_{J,K}} \\ Y_J & \longrightarrow & Y_{J,K} \\ & {\scriptstyle \tau^Y_{J,K}}\searrow & \downarrow{\scriptstyle u_{J,K}} \\ & & Y_K \end{array} \qquad \begin{array}{l} \\ \\ {\scriptstyle u_K}\ \text{(anciennement } \tilde{Y}_K\text{)} \\ \\ \\ , \end{array}$$

et pour des inclusions de trois ouverts emboîtés $J \subset K \subset L$ de I,

p. 9

(2)
$$\begin{array}{ccccc} X_J & \longrightarrow & X_K & \longrightarrow & X_L \\ {\scriptstyle u_J}\downarrow & \text{cocart} & \downarrow{\scriptstyle \tilde{u}_{J,K}} & \text{cocart} & \downarrow{\scriptstyle \tilde{u}_{J,L}} \\ Y_J & \longrightarrow & Y_{J,K} & \longrightarrow & Y_{J,L} \\ & & {\scriptstyle u_{J,K}}\downarrow & \text{cocart} & \downarrow{\scriptstyle u^L_{J,K}} \\ & & Y_K & \longrightarrow & Y_{K,L} \\ & & & & {\scriptstyle u_{K,L}}\downarrow \\ & & & & Y_L \end{array} \qquad \begin{array}{l} \\ \\ {\scriptstyle \tilde{u}_{K,L}} \\ \\ {\scriptstyle u_{J,L}} \\ \\ , \end{array}$$

donnant lieu à une nouvelle flèche $u^L_{J,K}$, déduite de $u_{J,K}$ par le cochangement de base $Y_{J,K} \longrightarrow Y_{J,L}$. Nous utiliserons ce diagramme dans les deux cas où $(J \subset K \subset L)$ est l'un des deux triplets [page 15]

$$\begin{cases} I'_0 \hookrightarrow I_0 \hookrightarrow I \\ I'_0 \hookrightarrow I' \hookrightarrow I, \end{cases}$$

issus du carré cocartésien (dans Cat) (carré MV)

(3)
$$\begin{array}{ccc} I'_0 & \hookrightarrow & I_0 \\ \downarrow & & \downarrow \\ I' & \hookrightarrow & I\ . \end{array}$$

Il faut insérer les deux diagrammes correspondants du type (∗) [de la page 10] dans un même diagramme. On notera à ce propos que les carrés dans $\mathcal{M}$ déduits de (3) en lui appliquant $J \longmapsto X_J$ et $J \longmapsto Y_J$

(4_X) (4_Y)
$$\left.\begin{array}{ccccc} X_{I'_0} & \longrightarrow & X_{I_0} & & \\ \downarrow{\scriptstyle u_{I'_0}} & \searrow & \downarrow & \searrow & \\ X_{I'} & & X_{I'} & \longrightarrow & X_I \\ \downarrow & & \downarrow & & \downarrow{\scriptstyle u_I} \\ Y_{I'_0} & \longrightarrow & Y_{I_0} & \searrow & \\ \downarrow & \searrow & \downarrow & & \\ Y_{I'} & & Y_{I'} & \longrightarrow & Y_I \end{array}\right\} (4)$$

sont cocartésien dans $\mathcal{M}$. (Transitivité des $\varinjlim$ …).

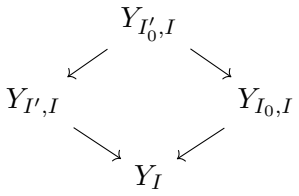
On a un homomorphisme $(u_{I'_0}, u_{I_0}, u_{I'}, u_I)$ déduit de u du carré (4_X) dans les carré (4_Y), [page 16] et cet homomorphisme lui aussi s'insère dans un diagramme, incluant les deux avatars déjà signalé de (2). Voici le diagramme pertinent : p. 10

(5 [9])
$$\begin{array}{ccccc} & & X_{I_0} & \longrightarrow & X_I \\ & \nearrow & \downarrow & \nearrow & \downarrow \\ X_{I'_0} & \longrightarrow & X_{I'} & & \\ \downarrow & & \downarrow & & \\ & & Y_{I'_0,I_0} & \longrightarrow & Y_{I'_0,I} \\ & \nearrow & \downarrow & \nearrow & \downarrow{\scriptstyle u^I_{I'_0,I_0}} \\ Y_{I'_0} & \xrightarrow{u_{I'_0,I_0}} & Y_{I'_0,I'} & & \\ \downarrow & \searrow & \downarrow & & \\ & & Y_{I_0} & \longrightarrow & Y_{I_0,I} \\ & & & \searrow & \downarrow{\scriptstyle u_{I_0,I}} \\ & & & & Y_{I'I} \\ & & \downarrow & \searrow & \downarrow{\scriptstyle u_{I',I}} \\ & & Y_{I'} & & Y_I\ . \end{array}$$

Le cube supérieur est le cube dans $\mathcal{M}$ à 6 faces cocartésiens déduit des trois flèches de $X_{I'_0}$ dans X_{I_0}, $X_{I'}$, $Y_{I'_0}$. On aura besoin de savoir que la *base* du cube est un carré cocartésien. La partie antérieure du diagramme est l'avatar de (2) correspondant aux inclusions $I'_0 \subset I' \subset I$, la partie derrière l'avatar pour l'inclusion $I'_0 \subset I_0 \subset I$. Donc les deux autres carrés verticaux, en dessous du cube supérieur, sont également cocartésiens, ainsi que le carré oblique en pointillés, lequel n'est autre que la base (4_Y) du diagramme (4). De ces caractères du diagramme (5) résulte aisément (quoique j'ai mis une bonne journée de gribouillis en bonheur-la-chance, [page 17]

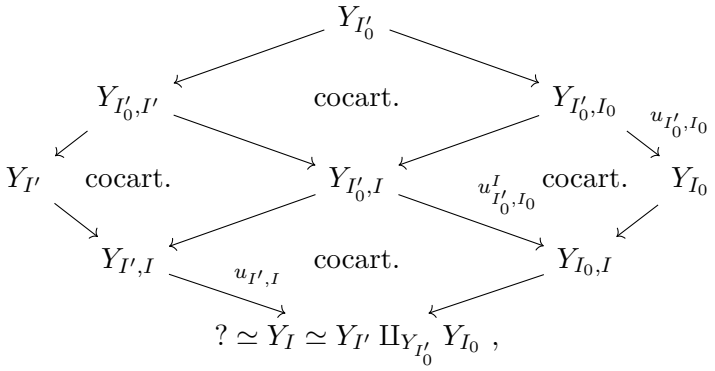
9. diagramme imbitable, à remplacer par diagramme (6) ci-dessus.

avant de m'en rendre compte enfin …) que le carré en losange à traits pleins



est également cocartésien. On le voit trivialement sur le diagramme déployé p. 11

(6)



qui est finalement le diagramme pertinent (le diagramme (5) ne servait à rien, c'est un mauvais bout par lequel j'ai pris les choses). On voit sur le diagramme (6) que pour prouver $u_{I',I} \in C$ resp. TC, il suffit de prouver $u_{I'_0,I_0} \in C$ resp. TC. Mais c'est là une assertion relative à l'inclusion d'ensembles ordonnés $I'_0 \subset I_0$ *finis.* On est donc réduit à prouver le lemme suivant, où I_0 devient I : [page 18]

Lemme. *Soit I un ensemble ordonné fini, $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ une flèche dans $\mathcal{M}(I^\circ)$, où $\mathcal{M}$ est une catégorie munie d'un ensemble $C \subset \mathrm{Fl}(\mathcal{M})$ de flèches tel que C soit stable par cochangement de base et par composition, et contienne les isomorphismes. On suppose $u_\bullet \in C_I$, i.e. que pour tout $i \in I$, la flèche $u_{I_{<i},I/i}$ est dans C. Alors pour toute partie ouverte I' de I, la flèche $u_{I',I}$ est dans C* [10].

Commentary · Exposition [ms. 18]

The diamond follows from compatible cocones

The long cube is intended to prove one fact: the relative attachment across $I' \subset I$ is inherited from the smaller attachment across $I'_0 \subset I_0$. The preceding excision identity proves it directly. We can also verify the diamond on its own terms, which explains why Grothendieck's second diagram succeeds where the first obscures the calculation.

Continue with $H = A \cap B$, $T = A \cup B$, and write

$$R_{H,T} = Y_H \amalg_{X_H} X_T, \quad R_{A,T} = Y_A \amalg_{X_A} X_T, \quad R_{B,T} = Y_B \amalg_{X_B} X_T.$$

Consider compatible maps $R_{A,T} \to Q$ and $R_{B,T} \to Q$ which agree on $R_{H,T}$. They determine maps $a : Y_A \to Q$ and $b : Y_B \to Q$ agreeing on Y_H, together with one common map $x : X_T \to Q$. The defining pushout conditions say that $x|_{X_A} = a\,u_A$ and $x|_{X_B} = b\,u_B$. Because $X_T = X_A \amalg_{X_H} X_B$, these conditions uniquely determine x from a, b. There is therefore no extra datum beyond the compatible pair a, b. Such pairs are precisely maps $Y_T = Y_A \amalg_{Y_H} Y_B \to Q$. Naturality in Q proves

$$R_{A,T} \amalg_{R_{H,T}} R_{B,T} \simeq Y_T.$$

This is the lower diamond of diagram (6).

10. **N.B.** Bien sûr, $u_{J,K} \in C$ si $J \subset K$ sont deux ouverts de I, comme on voit en appliquant le lemme à K au lieu de I.

The same calculation explains transitivity for three nested lower ideals $J \subset K \subset L$. The map $u_{J,L}$ factors through $R_{K,L}$, and the first factor

$$R_{J,L} \longrightarrow R_{K,L}$$

is the pushout of $u_{J,K} : R_{J,K} \to Y_K$ along the canonical map $R_{J,K} \to R_{J,L}$. Thus relative maps belonging to C compose after one pushout. This is exactly the operation needed when one adds finitely many elements or finitely many principal opens.

Everything here is an identity of ordinary pushouts. The commentary on earlier chapters discussed homotopy pushouts and failures of converse excision; the present calculation requires none of those stronger hypotheses. In particular, the proof does not infer an unknown face of an arbitrary cube from a collection of homotopy-cartesian or homotopy-cocartesian faces. It identifies all the relevant objects by their actual universal properties. For continuity with the earlier commentary, compare the discussion of the cube principle in the commentary on Chapter XII, manuscript pages 78–79; the issue there is different because the asserted faces are homotopical rather than specified ordinary colimits.

Corollaire. *Si de plus* $W \subset \mathrm{Fl}(\mathcal{M})$ *est un localiseur modérément saturé, tel que* $TC \overset{\text{déf}}{=} W \cap C$ *soit stable par cochangement de base (et l'est de toutes façons par composition, et contient les isomorphismes), alors on a*

$$C_I \cap W_{I^\circ} \;=\; (TC)_I,$$

donc si $u_\bullet \in C_I$ *(cf. ci-dessus) et les* $u_i \in W$*, alors* $u_\bullet \in (TC)_I$*, donc en appliquant ce qui est dit pour* C *à* TC*, on a* $u_{I',I} \in TC$ *pour toute partie ouverte* I' *de* I*.* [page 19]

Commentary · Exposition [ms. 18]

A finite induction with the last attachment exposed

The finite lemma supplies the local input for the transfinite construction. It is useful to write its induction so that the exceptional terminal-object case is unavoidable: this is the case in which the latching condition itself must be used, rather than another induction hypothesis.

Fix a finite poset I and a lower ideal K. Enumerate $I \setminus K$ by repeatedly selecting an element minimal among those remaining, and adjoin these elements one at a time. Every intermediate subset is a lower ideal. The transitivity calculation above reduces the problem to $I = K \cup \{i\}$, with all predecessors of i in K. Let $B = I/i$ and $H = B \cap K = I_{<i}$. Excision identifies $u_{K,I}$ as a pushout of $u_{H,B}$.

If $B \neq I$, the ambient poset B has smaller cardinality, so the induction hypothesis applies. If $B = I$, the element i is terminal and $K = I_{<i}$; the map $u_{K,I}$ is exactly $\rho_i(u)$. This proves the assertion for C. The printed condition on manuscript page 19 is $I'_0 \neq I$, but I'_0 excludes i_0 and always differs from I. The condition needed in this dichotomy is $I_0 \neq I$. The following sentence, "Sinon, i_0 est un objet final", confirms that reading.

For the assertion involving W, assume in addition that every component of u is in W. In the terminal case, the induction hypothesis puts u_K in TC. Hence its pushout $X_i \to X_i \amalg_{L_iX} L_iY$ lies in TC. The composite with $\rho_i(u)$ is $u_i \in W$, and two-out-of-three gives $\rho_i(u) \in W$. Together with the already established membership in C, this yields $\rho_i(u) \in TC$. The smaller-poset case and the successive attachments use only pushouts and ordinary composition in TC.

At cardinality zero every comparison is an identity of the initial object, so both inductions have the stated base case. Conversely, if all relative latching maps lie in TC, apply the proved lemma with TC replacing C and $K = \varnothing$; this gives the colimit map in TC. Restriction to each principal ideal gives every component in TC. We have therefore proved the full equality

$$C_I \cap W_{I^\circ} = (TC)_I$$

for finite I, including both implications and all relative maps $u_{K,I}$. Only finite colimits and ordinary composition have been used. The later appeal to transfinite closure is needed to assemble infinitely many such finite pieces; it is not hidden inside the finite lemma.

On prouvera en même temps le lemme et son corollaire, en procédant par récurrence sur $\operatorname{card} I = n$. Si $n = 0$, i.e. $I = \emptyset$, les énoncés sont triviaux. Par transitivité, en utilisant les diagrammes du type (2), on est ramené au cas où $I = I' \cup \{i_0\}$, où $i_0 \in I \backslash I'$. On pose encore $I_0 = I/i_0$, $I'_0 = I_0 \cap I'$, et le raisonnement qu'on vient de faire nous montre qu'il suffit de prouver que $u_{I'_0,I_0} \in C$ resp. $\in TC$. Si $I'_0 \neq I$, cela résulte de l'hypothèse de récurrence. Sinon, i_0 est un objet final de I, et $I' = I_{<i_0}$, et alors $u_{I',I} \in C$ est inclus dans la définition de C_I. Reste à prouver alors que $u_{I',I} \in TC$ si les $u_i \in W$. On a le diagramme p. 12

$$\begin{array}{ccc} X_{I'} & \longrightarrow & X_I \\ {\scriptstyle u_{I'}}\downarrow & \text{cocart.} & \downarrow{\scriptstyle \tilde{u}_{I',I}} \\ Y_{I'} & \longrightarrow & Y_{I',I} \\ & & {\scriptstyle u_{I',I}}\downarrow \\ & & Y_I \end{array} \qquad u_I$$

et l'hypothèse de récurrence implique $u_{I'} \in TC$, donc $\tilde{u}_{I',I} \in TC$ par cochangement de base. [page 20] D'autre part, la flèche $u_I : X_I \longrightarrow Y_I$ s'identifie à $X_{i_0} \longrightarrow Y_{i_0}$ (i_0 étant élément final de I), elle est donc dans W. Donc par saturation, $u_{I',I} \in W$, donc $u_{I',I} \in TC$, q.e.d.

Ainsi, la proposition 2 est prouvée, ainsi que son corollaire 1 – et on voit que si dans la proposition 2 et son corollaire, on suppose I *fini,* alors il est inutile dans les hypothèses pour C, resp. $TC = C \cap W$, de supposer la stabilité par composition *transfinie,* la composition ordinaire suffit. Mais il reste à prouver le corollaire 2, que je viens de prouver dans le cas I *fini,* sans avoir à faire appel à une hypothèse "transfinie". Il faut maintenant prouver que c'est encore OK dans le cas localement fini, en supposant C stable par composition transfinie. (Il n'y a plus à revenir sur le cas de W, car on sait déjà maintenant que $C_I \cap W_{I^\circ} = (TC)_I$, par le corollaire 1 déjà prouvé.)

La démonstration est essentiellement la même que pour le cas particulier déjà traité $I' = \emptyset$, [page 21] $I'' = I$, et on aurait pu aussi bien rédiger la démonstration dans ce cadre, à l'aide d'une suite strictement croissante transfinie d'ouverts dans I, I_α, avec $\boxed{I_0 = I',}$ $I_{\alpha_0} = I''$ $(= I)$, et

$$\begin{array}{lll} I_\alpha & = & I_{\alpha-1} \cup I/s_{\alpha-1} \quad \text{si } \alpha \text{ non limite, } I_{\alpha-1} \neq I, \text{ en choisissant } s_{\alpha-1} \in I \backslash I_{\alpha-1}, \\ I_\alpha & = & \bigcup_{\beta<\alpha} I_\beta \qquad\qquad \text{si } \alpha \text{ limite,} \end{array}$$

et en posant encore

$$X_\alpha = X_{I_\alpha}, \qquad Y_\alpha = Y_{I_\alpha}, \qquad u_\alpha = u_{I_\alpha} : X_\alpha \longrightarrow Y_\alpha.$$

Soit α_0 tel que $I_{\alpha_0} = I$ (indice terminal de la suite), il faut prouver que dans

$$\begin{array}{ccc} X_0 & \longrightarrow & X_{\alpha_0} \\ \downarrow & & \downarrow \\ Y_0 & \longrightarrow & Y_{0,\alpha_0} \\ & & {\scriptstyle u_{0,\alpha_0}}\downarrow \\ & & Y_{\alpha_0} \end{array}$$

on a p. 13

$$u_{0,\alpha_0} \in C. \qquad \text{[page 22]}$$

Commentary · Examples [ms. 20]

A fork whose ordinary colimit loses a circle

The colimit theorem is meant to control the identifications made when diagrams are glued. A small example shows why its latching condition cannot be replaced by objectwise information. Work in simplicial sets, with monomorphisms as cofibrations and weak homotopy equivalences as weak equivalences, and let I have the two arrows $a \leftarrow s \rightarrow b$.

Define diagrams X, Y by

$$X_s = \partial\Delta^1, \qquad X_a = X_b = \Delta^1, \qquad Y_s = \partial\Delta^1, \qquad Y_a = Y_b = \Delta^0.$$

The two structure maps of X are the boundary inclusions. Those of Y collapse the two vertices to a point. Let u_s be the identity and let u_a, u_b be the collapse maps $\Delta^1 \to \Delta^0$. Each component of u is a weak equivalence: the first is an isomorphism and the other two identify contractible realizations. The diagram X is cofibrant, since its latching maps are $\varnothing \to \partial\Delta^1$ and the two inclusions $\partial\Delta^1 \hookrightarrow \Delta^1$. The diagram Y fails the condition at both outer vertices.

The two ordinary colimits are

$$X_I = \Delta^1 \amalg_{\partial\Delta^1} \Delta^1, \qquad Y_I = \Delta^0 \amalg_{\partial\Delta^1} \Delta^0 = \Delta^0.$$

The first simplicial set has two vertices and two nondegenerate edges, with the same endpoints, and no nondegenerate simplices of higher dimension. Its realization is a circle. One can see the obstruction directly in the normalized chain complex:

$$C_1 = \mathbb{Z}e_a \oplus \mathbb{Z}e_b, \quad C_0 = \mathbb{Z}v_0 \oplus \mathbb{Z}v_1, \quad \partial e_a = \partial e_b = v_1 - v_0.$$

Thus $H_1(X_I) = \mathbb{Z}(e_a - e_b)$, whereas $H_1(Y_I) = 0$. The colimit of u is not a weak equivalence.

Its relative latching maps identify the defect exactly. At a and b they are

$$\Delta^1 \amalg_{\partial\Delta^1} \partial\Delta^1 = \Delta^1 \longrightarrow \Delta^0,$$

which are not monomorphisms. Consequently $u \notin C_I$, and the corollary makes no false prediction. This example is also a warning about the future derived colimit: replacing the two collapse maps in Y by the two boundary inclusions produces the cofibrant diagram X with its comparison to Y. The homotopical information is carried by the manner in which the common boundary is attached, even though every individual value has already been compared by a weak equivalence.

Compte tenu du complément marginal à la proposition 1 (qui se prouve de la même façon …), on est ramené à prouver que l'homomorphisme des systèmes ordinaux envisagés est dans $C_{I_{\alpha_0+1}}$ (où I_{α_0+1} est l'ensemble des ordinaux $\leq \alpha_0$). Or qu'il en soit ainsi se prouve exactement comme tantôt, à cela près qu'on doit d'abord faire une vérification de $(*)$ p. 1 ($\tilde{u}_\bullet \in C$) pour $i_0 = 0$. Alors le diagramme $(*)$ p. 1 est

$$\begin{array}{ccc} \emptyset & \longrightarrow & X_0 = X_{I'} \\ \downarrow{\scriptstyle \text{iso}} & & {\scriptstyle \text{iso}}\downarrow{\scriptstyle \text{id}} \\ \emptyset & \longrightarrow & X_0 = X_{I'} \\ & & \downarrow{\scriptstyle u_0 = u_{I'}} \\ & & Y_0 = Y_{I'} \ , \end{array}$$

et il faut donc vérifier $u_0 = u_{I'} \in C$. Or ceci est contenu dans la proposition 2 elle-même, déjà prouvé, qu'on doit donc utiliser. Le reste se prouve comme avant. Ainsi, le corollaire 2 de la proposition 2 est également prouvé.

Commentary · Exposition [ms. 22]

Starting a filtration at an arbitrary open subset

Grothendieck returns to Corollary 2 because the relative statement is the one needed for later Kan extensions. The proof with $I_0 = I'$ can be made completely independent of the behavior of the

map over I' itself. This also supplies the general gluing step that is abbreviated in the lemma on manuscript page 24.

Let $(A_\beta)_{\beta\le\lambda}$ be a continuous increasing chain of lower ideals, with $A_0 = K$ and $A_\lambda = J$. Assume that every relative map $u_{A_\beta,A_{\beta+1}}$ belongs to C. Fix the final X_J and form

$$P_\beta = X_J \amalg_{X_{A_\beta}} Y_{A_\beta}.$$

The initial object of this chain is the domain of $u_{K,J}$ and its final object is Y_J. At a successor stage, the map $P_\beta \to P_{\beta+1}$ is the pushout of $u_{A_\beta,A_{\beta+1}}$ by transitivity of pushouts. At a limit stage it is the colimit of the preceding stages, by the same cocone calculation as in the ordinal proof. It follows that $u_{K,J} \in C$. This relative filtration does not require $u_K \in C$. The source's check at index zero is valid, but belongs to its chosen reduction to the absolute Proposition 1.

For later use, suppose now that J is any small poset and that, for every $j \in J$ and every lower ideal $H \subset J/j$, one knows $u_{H,J/j} \in C$. Choose a well-order of the elements of $J \setminus K$. Starting at K, adjoin the principal lower ideals of these elements in that order, taking unions at limit stages. Repetitions do no harm. If the current ideal is A and the next principal ideal is $B = J/j$, the successor relative map is a pushout of $u_{A\cap B,B}$, by excision. It therefore belongs to C, and the preceding filtration proves $u_{K,J} \in C$.

This argument uses no well-foundedness of the order on J: the stronger hypothesis over every open subset of every principal ideal replaces it. In the locally finite case these local comparisons were supplied by the finite lemma. On manuscript page 24 they are supplied by the definition of the set P. The same construction is the main mechanism behind Theorem 2.1(1), so the preliminary proof is best read as anticipating that theorem's organization.

Commentaires : Il y a un malaise, car les proposition 1 et proposition 2 (ainsi que leurs corollaires) ont essentiellement la même conclusion, avec des hypothèses sur I qui sont différentes : bien ordonné dans un cas, localement fini dans un autre. L'idée naturelle, c'est de donner un énoncé commun, qui coiffe les deux à la fois, avec I artinien, i.e. toute suite décroissante est stationnaire. Ce qu'il faut, c'est une hypothèse sur I d'une part, sur $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ dans $\mathcal{M}(I^\circ)$ d'autre part, qui assure ceci : [page 23]

(∗) $\forall\, i_0 \in I$, et toute partie ouverte J' de $J = I/i_0$, la flèche $u_{J',J}$ dans le diagramme

$$\begin{array}{ccc} X_{J'} & \longrightarrow & X_J \\ {\scriptstyle u_{J'}}\downarrow & \text{cocart} & \downarrow{\scriptstyle \tilde{u}_{J',J}} \\ Y_{J'} & \longrightarrow & Y_{J',J} \\ & & {\scriptstyle u_{J',J}}\downarrow \\ & & Y_J \end{array} \qquad u_J$$

est dans C.

Question : est-il exacte qu'il suffit, pour la validité de (∗), de supposer

a) $u_\bullet \in C_I$, i.e. $u_{J',J} \in C$ quand $J = I/i_0$, $J' = J\setminus\{i_0\}$, et

b) *I localement fini ?*

Commentary · Exposition [ms. 23] p. 14 [page 24]

The artinian induction and the wording of the question

The two successful constructions suggest a common hypothesis: attachments should have no indefinitely descending dependence. For a poset this is well-foundedness. In the usual set-theoretic setting with choice, it is equivalent to the manuscript's artinian condition that every descending sequence eventually stabilizes. An ordinal is well founded, and a poset with finite principal lower ideals is well founded, so this hypothesis includes both of the cases already treated.

One can construct a compatible ordinal filtration directly. Define the rank recursively by

$$r(i) = \sup_{j<i}\big(r(j)+1\big).$$

Choose a well-order within each rank and concatenate the ranks. Every predecessor of an element then occurs earlier. For a specified lower ideal $K \subset J$, apply this construction to $J \setminus K$ and place K first. Adding one element at a time gives lower ideals A_β with $A_0 = K$ and union J. When $A_{\beta+1} = A_\beta \cup \{i\}$, all strict predecessors of i lie in A_β. Excision identifies the successor relative map with a pushout of $\rho_i(u)$. The relative filtration therefore proves $u_{K,J} \in C$ for every $K \subset J$. This is a complete proof of the proposed common generalization.

The literal question on manuscript page 23 asks whether the original latching condition together with local finiteness suffices. It does, and that case has just been proved in Corollary 2. The surrounding discussion, and the immediately following artinian corollary, show that the useful new question is whether local finiteness can be weakened to the artinian condition. We record both the printed wording and its mathematical context, without silently changing the word in the source.

The lemma on manuscript page 24 can also be completed in its stated generality. Suppose every $i < i_0$ belongs to P, set $J = I/i_0$ and $J' = J \setminus \{i_0\}$, and take a lower ideal $K \subset J$. If $K = J$ the relative map is an isomorphism. Otherwise $K \subset J'$. For every $j \in J'$, the assumption $j \in P$ supplies all the local relative comparisons over I/j. The principal-open gluing argument of the preceding block therefore gives $u_{K,J'} \in C$, even if J' is infinite or not well founded. The map $u_{J',J} = \rho_{i_0}(u)$ is in C by hypothesis, so transitivity gives $u_{K,J} \in C$.

Thus P is progressive. Well-founded induction gives $P = I$. The displayed progressive condition in the proof must use $I_{<i_0} \subset P$, not $I/i_0 \subset P$, since the latter already contains i_0 and is tautological. Likewise the element chosen below $x_0 \notin P$ must be $x_1 \notin P$. These printed slips and the repeated case $K = J$ are recorded in the correction register. The missing gluing paragraph is a proof omission repaired by the argument above; the artinian conclusion itself is valid.

Lemme. *Soit $P \subset I$ l'ensemble des $i_0 \in I$ tels que pour tout ouvert K de $J = I/i_0$, la flèche $u_{K,J}$ est dans C. Ceci dit, soit $i_0 \in I$ tel que tout $i < i_0$ soit dans P. Alors $i_0 \in P$.* (On suppose $u_\bullet \in C_I$, mais I quelconque.)

En effet, soit $K \subset J = I/i_0$ un ouvert. Si $K = J$, il est trivial que $u_{K,J} \in C$ (c'est un isomorphisme). Si $K = J$, on a

$$K \ \subset \ J' \overset{\text{déf}}{=} J\backslash\{i_0\} \ \subset \ J,$$

et pour prouver que $u_{K,J} \in C$, il suffit de prouver que $u_{K,J'}, u_{J',J} \in C$. Or $u_{J',J} \in C$ par hypothèse $u_\bullet \in C_I$. Et $u_{K,J'} \in C$ par ce que a été prouvé pour le cas de J', compte tenu que par hypothèse sur i_0, tout $i \in J'$ (i.e. tout $i < i_0$) est dans P.

Corollaire. *Si I est artinien, $P = I$.*

Car toute partie P de I ayant la propriété qu'on vient de dire ($I/i_0 \subset P \Longrightarrow i_0 \in P$) est égale à I. Si en effet il y avait $x_0 \notin P$, il existerait $x_1 < x_0$ avec $x_0 \notin P$, et par récurrence on construirait une suite strictement décroissante dans I, absurde.

[page 25]

Commentary · Development (commentators') [ms. 24]

Well-foundedness detected by maps of set diagrams

Grothendieck's artinian hypothesis is more than a convenient way to arrange the induction. One can characterize it by the validity of the latching test, using only diagrams of sets. The following formulation makes the obstruction independent of any model category. We checked the cellular descriptions in Riehl and Verity (2014, Proposition 6.3 and Corollary 6.8), the ordinal degree formulation in Shulman (2015, Section 4), and the relative-colimit theorem of Rădulescu-Banu (2009, Section 9.4). We did not find this universal converse in those inspected passages. We give our formulation and proof, without a claim of priority.

Recognition by the latching test. [ours] Work with ordinary choice, and let I be a small poset. The following conditions are equivalent: I is well founded; every map of set-valued I-diagrams whose relative latching maps are injective has all its component maps injective; every such map has an injective map on colimits.

Proof. If I is well founded, the preceding relative filtration applies with C the injections in Set. Pushouts of injections are injections: in the disjoint-union description of a pushout, the imposed equivalence relation cannot identify two different elements on the opposite side, because an element in the subobject has at most one representative there. Continuous transfinite composites of injections

are injections, since equality of two representatives in the colimit is witnessed at a common later stage. The filtration consequently proves the colimit assertion, and restriction to I/i proves the component assertion.

Conversely, suppose I is not well founded. By dependent choice there is a strictly descending chain

$$i_0 > i_1 > i_2 > \cdots .$$

Let U be its upward closure:

$$U = \{j \in I \mid i_n \leq j \text{ for some } n\}.$$

For a set A, define a diagram $E(A)$ by $E(A)_j = A$ when $j \in U$ and $E(A)_j = \varnothing$ otherwise. All arrows within U act by the identity of A; all other arrows have empty source. This is a functor because U is upward closed.

For $j \in U$, the poset $U \cap I_{<j}$ is nonempty and connected. To prove both assertions, choose n with $i_n \leq j$; then $i_{n+1} < j$ belongs to it. Given any two of its objects a, b, choose n_a, n_b with $i_{n_a} \leq a$ and $i_{n_b} \leq b$. For $m > \max(n_a, n_b)$, the element i_m lies below both a, b and still lies strictly below j. This supplies the zigzag $a \leftarrow i_m \rightarrow b$. A constant diagram with identity transition maps over a nonempty connected category has colimit A: its cocones are precisely one map $A \rightarrow Q$. The empty values outside U impose no further condition. Therefore

$$L_j E(A) = A \quad (j \in U), \qquad L_j E(A) = \varnothing \quad (j \notin U),$$

and the latching map is an identity in both cases.

Choose the noninjective map $q : \{0, 1\} \rightarrow \{*\}$ and let $u = E(q)$. At $j \in U$ its relative latching map is

$$\{0, 1\} \amalg_{\{0,1\}} \{*\} \xrightarrow{\mathrm{id}} \{*\},$$

an isomorphism; outside U it is $\mathrm{id}_\varnothing$. Nevertheless $u_j = q$ at every $j \in U$, so the component condition fails. The same common-lower-bound argument shows that U itself is nonempty and connected. Hence $\operatorname{colim}_I E(A) = A$, and the colimit of u is again q, so the colimit condition fails as well. This proves the equivalence. □

The obstruction is especially sharp: every relative latching map in the counterexample is an isomorphism. An endless predecessor chain can make the local condition see no new attachment anywhere, although a nontrivial map is present throughout the chain. This is exactly the failure that a well-founded order excludes. The counterexample on manuscript pages 37–38 exhibits a particular instance; the recognition statement explains why the phenomenon occurs for every non-well-founded poset, after extending the descending-chain example by empty values outside its upward closure.

2 Description de $C_I \subset \mathrm{Fl}(\mathcal{M}(I^\circ))$, pour I ensemble ordonné

Soit $\mathcal{M}$ une catégorie, stable par (petites) $\varinjlim$, $C \subset \mathrm{Fl}(\mathcal{M})$ (“cofibrations”). On suppose

(2.1)
$$\begin{cases} \text{a)} \quad C & \text{contient les isomorphismes et est stable} \\ & \text{par cochangement de base.} \\ \text{b)} \quad C & \text{stable par composition transfinie.} \end{cases}$$

Soit I un (petit) ensemble ordonné, et considérons

$$\mathcal{M}(I^\circ) \;=\; \underline{\mathrm{Hom}}(I, \mathcal{M}),$$

on va définir une partie

$$C_I \;\subset\; \mathrm{Fl}(\mathcal{M}(I^\circ))$$

ainsi, pour $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ flèche dans $\mathcal{M}(I^\circ)$:

(2.2)
$$u_\bullet \in C_I \quad \Longleftrightarrow \quad \begin{array}{l} \forall\, i_0 \in I \text{ et pour tout ouvert } J \subset I/i_0 = K, \\ \text{la flèche } u_{J,K} \text{ est dans } C. \end{array}$$

Commentary · Exposition [ms. 25]

A local definition that already sees every open part

Grothendieck restarts the discussion because he wants the useful conclusion of the preliminary arguments to become the definition. The change is substantial. In §1, one tested the single boundary $I_{<i} \subset I/i$ at each object. In §2, one tests every lower open $K \subset I/i$. We shall call these respectively the latching condition and the strong open condition, solely to distinguish the two definitions during this transition. The notation C_I is reused in the source; the meanings agree under the artinian hypothesis, but they must not be identified before that hypothesis has been used.

We write $\downarrow i = \{j \mid j \leq i\}$ and $X_K = \operatorname{colim}_{k\in K} X_k$. Here an open means a lower set, and $\mathcal{M}(I^\circ)$ means the category of covariant diagrams $I \to \mathcal{M}$. For a natural transformation $f : X \to Y$, put

$$Q_f(K,J) = X_J \amalg_{X_K} Y_K, \qquad d_f(K,J) : Q_f(K,J) \longrightarrow Y_J \quad (K \subset J).$$

Thus $Q_f(K,J)$ is Grothendieck's $Y_{K,J}$ and $d_f(K,J)$ is his $u_{K,J}$. The object $Q_f(K,J)$ contains the whole source diagram over J, together with the part of the target already installed over K. The comparison $d_f(K,J)$ attaches the remaining target data. This interpretation explains why its membership in C is the correct relative condition.

Three immediate specializations should be kept available. For $K = \varnothing$ the comparison is $X_J \to Y_J$. For $K = J$ it is an isomorphism. For $J = \downarrow i$ and $K = I_{<i}$ it becomes

$$X_i \amalg_{L_iX} L_iY \longrightarrow Y_i, \qquad L_iX = \operatorname{colim}_{j<i} X_j.$$

This is the relative latching map. On an artinian poset it supplies all the strong open tests; on a general poset it need not even imply that the individual f_i belong to C. The reverse infinite chain on page 38 will isolate precisely this failure.

Theorem 2.1(1) is stronger in a different direction: the strong open tests imply $d_f(K,J) \in C$ for arbitrary lower opens $K \subset J$, without any artinian hypothesis. Its proof is an ordinary cellular argument with principal lower opens as the pieces. All colimits in this assertion are ordinary colimits in $\mathcal{M}$. Neither a homotopy colimit nor a model structure has yet been used.

Voici la définition de $u_{J,K}$. Pour tout ouvert J de I, on pose

$$(2.3) \qquad \begin{cases} X_J = \varinjlim_{i\in J} X_i, & Y_J = \varinjlim_{i\in J} Y_i, \\ u_J : X_J \longrightarrow Y_J, & u_J = \varinjlim_{i\in J} u_i, \end{cases}$$

et on considère, pour $K \subset J$, le diagramme

p. 15
[page 26]

$$\begin{array}{ccccc} X_K & \xrightarrow{\tau^X_{K,J}} & X_J & & \\ {\scriptstyle u_K}\downarrow & \text{cocart} & \downarrow{\scriptstyle u_{K,J}} & \searrow & \\ Y_K & \xrightarrow{\tau^Y_{K,J}} & Y_{K,J} & \xrightarrow{u} & Y_K \amalg_{X_K} X_J \\ & & {\scriptstyle u_{K,J}}\downarrow & \swarrow & \\ & & Y_J\ . & & \end{array}$$

Théorème 2.1. *Soit $\mathcal{M}$ $\mathfrak{U}$-catégorie stable par petites $\varinjlim$, $C \subset \mathrm{Fl}(\mathcal{M})$, satisfaisant les conditions de stabilité* (2.1) *(par cochangement de base et par composition transfinie), I un petit ensemble ordonné, $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ une flèche dans $\mathcal{M}(I^\circ)$.*

1°) *On suppose $u_\bullet \in C_I$ (cf.* (2.2)*). Soient $K \subset J$ deux ouverts de I. Alors $u_{K,J}$ (défini dans* (2.4)*) est dans C.*

2°) *Supposons I artinien. Pour qu'on ait $u_\bullet \in C_I$, i.e. pour que $u_{K,J}$ soit dans C pour tout couple $(K \subset J)$ d'ouverts tel que J ait un plus grand élément, i.e. soit de la forme I/i_0 (i.e. aussi, par 1°), pour* tout *couple d'ouverts $J \subset K$), (il faut et) il suffit qu'il en soit ainsi dans le cas particulier où $J = I/i_0$, $J' = I'/i_0 \overset{déf}{=} I/i_0 - \{i_0\} = \{i \in I \mid i < i_0\}$.* [page 27]

3°) *Supposons encore I artinien*[11]*. Soit $W \subset \mathrm{Fl}(\mathcal{M})$ un localiseur modérément saturé, tel que $TC \overset{déf}{=} C \cap W$ satisfasse également aux conditions de stabilité* (2.1) *(par cochangement de base, par composition transfinie). Les conditions suivantes sur une flèche $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ dans $\mathcal{M}(I^\circ)$ sont équivalentes.*

a) *$u_\bullet \in (TC)_I$, i.e. pour tout $i_0 \in I$ et K ouvert de $J = I/i_0$, la flèche $u_{K,J}$ est dans TC.*

b) *$u_\bullet \in C_I \cap W_{I^\circ}$, i.e. pour tout $i_0 \in I$, on a $u_{i_0} \in W$, et pour tout ouvert K de $I/i_0 = J$, on a $u_{K,J} \in C$.*

Donnons tout de suite les corollaires principaux.

Corollaire 2.2. *Supposons $(u_\bullet : X_\bullet \longrightarrow Y_\bullet) \in C_I$. Alors*

a) *$\forall\ i \in I$, $u_i : X_i \longrightarrow Y_i$ est dans C.*

b) *La flèche*

$$u_I \ : \ X_I \longrightarrow Y_I \tag{2.2.1}$$

est dans C. Plus généralement, pour toute partie ouverte J de I, la flèche

$$u_J \ : \ X_J \longrightarrow Y_J \tag{2.2.2}$$

est dans C.

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Commentary · Exposition [ms. 27] [page 28]

The three assertions of Theorem 2.1 have different hypotheses

The theorem separates a gluing result from two induction results. Its first part uses the strong open condition and closure of C under pushout and continuous transfinite composition. Its second part identifies that condition with the latching condition when I is artinian. Its third part adds a class W satisfying two out of three and assumes the same closure properties for $TC = C \cap W$; under the artinian hypothesis it identifies trivial diagram cofibrations by the formula

$$(TC)_I = C_I \cap W_{I^\circ}.$$

The corollaries concerning weak equivalences inherit this artinian hypothesis. In particular, Corollary 2.3 is not an additional assertion that this equality holds for every poset. The question on page 38 would make no sense under that reading.

The mechanism of the third assertion is short enough to record independently. Assume, by well-founded induction, that the restrictions below i already satisfy the strong TC condition. If $K \subsetneq\downarrow i$ is lower, then $K \subset I_{<i}$. Applying part (1) with TC to the restricted transformation gives $f_K \in TC$. In the factorization

$$X_i \xrightarrow{a} X_i \amalg_{X_K} Y_K \xrightarrow{d_f(K,\downarrow i)} Y_i,$$

a is a pushout of f_K, hence belongs to TC. The composite is $f_i \in W$. Two out of three gives $d_f(K, \downarrow i) \in W$, while the strong C condition gives membership in C. This proves the induction step. The reverse inclusion follows immediately by taking $K = \varnothing$.

This argument shows exactly why no closure of all of W under pushouts is required. Only the pushout of the already established trivial cofibration f_K is formed. Nor is an arbitrary colimit of objectwise weak equivalences being declared a weak equivalence: the relative cofibration conditions are what allow part (1) to be applied. These distinctions are essential when the theorem is later used to put a cofibration structure on a diagram category.

Corollary 2.4 is the corresponding absolute statement. Taking X to be the initial diagram says that a cofibrant diagram has cofibration maps $Y_K \to Y_J$ for every inclusion of lower opens. This includes the individual structure maps but imposes more than their being cofibrations. At a branching object the entire latching colimit must embed by a cofibration, including the compatibility among different incoming branches.

11. J'ignore si cette hypothèse est nécessaire, cf. commentaire p. 39.

Pour a), on applique 1°) (ou simplement la *définition* de C_I) dans le cas de l'inclusion d'ouverts $\emptyset \hookrightarrow I/i$, vu qu'on a

$$u_{\emptyset,I/i} \simeq u_i .$$

Pour b), on applique 1°) à l'inclusion $\emptyset \longrightarrow I$, vu que

$$u_{\emptyset,I} \simeq u_I ,$$

et plus généralement

$$u_{\emptyset,J} \simeq u_J .$$

Corollaire 2.3. *Supposons $u_\bullet \in C_I \cap W_{I^\circ}$. Alors pour toute inclusion d'ouverts $K \subset J$ de I, on a $u_{K,J} \in TC$, et en particulier (prenant $K = \emptyset$), on a $u_J \in TC$. Enfin, pour tout $i \in I$, on a $u_i \in TC$.*

Corollaire 2.4. *Soit $X_\bullet$ dans $\mathcal{M}(I^\circ)_{\mathrm{c}}$ un système inductif C-cofibrant, i.e. tel que $\emptyset_I \longrightarrow X_\bullet$ soit dans C_I, i.e. tel que pour tout $i_0 \in I$ et tout ouvert K de $J = I/i_0$, la flèche de transition $\tau_{K,J} : X_K = \varinjlim_{i\in K} X_i \longrightarrow X_J \simeq X_{i_0}$ soit dans C. Alors pour* tout *couple d'ouverts $K \subset J$, la flèche*

$$\tau_{K,J} \ : \ \varinjlim_{i\in K} X_i \longrightarrow \varinjlim_{i\in J} X_i$$

est dans C[12]*. En particulier (faisant $K = I/k$, $J = I/j$ pour $k \leq j$), si $k \leq j$ sont dans I, le morphisme de transition*

$$\tau_{k,j} \ : \ X_k \longrightarrow X_j$$

est dans C. Enfin, si $W \subset \mathrm{Fl}(\mathcal{M})$ est comme dans (2.1. 3°)*, et si $\emptyset_I \longrightarrow X$ est dans W_{I°, i.e.* [page 29]
les X_i sont W-coasphériques, alors $X_\bullet$ est dans $\mathcal{M}(I^\circ)_{TC}$ (où $TC = C \cap W$), et pour tout ouvert J de I, $X_J = \varinjlim_{i\in J} X_i$ est dans $\mathcal{M}_{TC}$, donc X_J est W-coasphérique, et pour tout couple d'ouverts $K \subset J$, la flèche de transition $\tau_{K,J}$ est dans TC.

Signalons enfin le complément suivant au théorème 2.1, qui résultera de sa démonstration :

Corollaire 2.5. *Supposons que l'espace topologique associé à I (dont les ouverts sont les ouverts de I, et les I/i forment une base d'ouverts) soit* noethérien*, i.e. toute suite croissante d'ouverts est stationnaire. Alors toutes les conclusions dans le théorème* 2.1 *et ses corollaires restent valables, en supposant seulement la stabilité de C ou de $TC = C \cap W$ par composition simple (au lieu de composition transfinie), en plus des deux autres stabilités.*

p. 17
[page 30]

Commentary · Examples [ms. 29]

A branching diagram and the finite meaning of the noetherian case

Grothendieck's open-set formulation makes a small example particularly instructive. Let I be the square poset with elements a, b, c, d and relations $a < b < d$, $a < c < d$, with b, c incomparable. Take $\mathcal{M} = \mathrm{Set}$ and let C be the injections. A diagram is cofibrant precisely when the maps

$$\varnothing \longrightarrow X_a, \quad X_a \longrightarrow X_b, \quad X_a \longrightarrow X_c, \quad X_b \amalg_{X_a} X_c \longrightarrow X_d$$

are injections. The last condition controls the intersection of the two branches. When all maps have been realized as inclusions into X_d, it says $X_b \cap X_c = X_a$. Individual injectivity of every arrow does not imply this equality.

For a complete failure, take $X_a = \varnothing$ and $X_b = X_c = X_d = \{*\}$, with the maps into X_d the identity. Every structure map is injective. The latching map at d is the fold $\{b, c\} \to \{*\}$, which is not. The open $K = \{a, b, c\}$ has colimit a two-element set, and its map to X_d identifies the two branches. Thus the extra open-set tests detect the precise identification that ordinary pointwise cofibrancy misses.

The topological adjective in Corollary 2.5 also needs its order convention. $\mathrm{Esp}(I)$ has the lower sets as its opens. Noetherianity means that no strictly increasing infinite sequence of lower sets

12. et réciproquement, caractérise les objets C-cofibrants.

exists. The process on page 30 adds a principal lower set whenever the current open is proper; in a noetherian space this process must stop after finitely many additions. The argument therefore uses only finitely many pushouts and compositions, even when the chosen principal opens themselves are infinite. This removes the transfinite *closure* assumption; it does not by itself remove the need for the colimits occurring inside those opens.

There is a useful simplification when noetherianity is combined with artinianity. Under the usual choice principles, an artinian poset with noetherian lower-set topology is finite. Indeed, if I were infinite, choose a minimal element of I, then a minimal element of what remains, and continue. After n choices the chosen finite set J_n is lower: when an element was chosen, every strict predecessor had already been chosen. The sequence $J_0 \subsetneq J_1 \subsetneq \cdots$ contradicts noetherianity. Conversely every finite poset has both properties. This observation explains why finite colimits suffice in the first alternative of Theorem 3.1. Noetherianity alone allows infinite examples, such as $\mathbf{N}^{\mathrm{opp}}$; those examples are not artinian.

Démonstration du théorème 2.1 et du corollaire 2.5.

2.1. 1°) On peut supposer $J = I$, $K \subset I$. On construit une suite transfinie strictement croissante d'ouverts I_α, avec

$$\begin{cases} I_0 &= K, \\ I_{\alpha+1} &= I_\alpha \cup I/s_\alpha \quad \text{où } s_\alpha \in I \backslash I_\alpha \text{ (si } I_\alpha \neq I \text{, sinon, la construction transfinie s'arrête)}, \\ I_\alpha &= \bigcup_{\beta<\alpha} I_\beta \quad \text{si } \alpha \text{ ordinal limite.} \end{cases}$$

Il existe un α_0 tel qu'on ait $I_{\alpha_0} = I$. On voit alors comme au paragraphe 1 que pour tout α, on a [page 31]

$$(*) \qquad u_{I_\alpha, I_{\alpha+1}} \in C,$$

car il est déduit de $u_{(I/s_\alpha)\cap I_\alpha, I/s_\alpha}$ par cochangement de base. Cf. diagramme (6), p. 17, qu'il faudrait dégager en lemme, débarrassé des particularités techniques de la situation, i.e. un lemme sur les homomorphismes de carrés cartésiens :

Commentary · Exposition [ms. 31]

The cube reduces to one universal property

Grothendieck extracts Lemma 2.6 to make the long preliminary cube reusable. The useful invariant of a commutative square is its comparison from the pushout of the first three vertices to the fourth. The lemma says that a suitable pushout of squares transports this comparison by pushout. A calculation with universal properties gives the assertion without relying on the perspective of a three-dimensional drawing.

In the notation of the lemma, suppose the two horizontal squares are pushouts,

$$X_1' = X_1 \amalg_{X_0} X_0', \qquad Y_1' = Y_1 \amalg_{Y_0} Y_0'.$$

Let $Q_0 = Y_0 \amalg_{X_0} X_0'$ and $Q_1 = Y_1 \amalg_{X_1} X_1'$. There is a canonical identification

$$Q_1 \cong Y_1 \amalg_{X_0} X_0', \qquad Q_1 \amalg_{Q_0} Y_0' \cong Y_1 \amalg_{Y_0} Y_0' = Y_1'.$$

To check the second identification, a map from either side to an object T is a pair of maps $Y_1 \to T$, $Y_0' \to T$ agreeing on Y_0. The extra map from X_0' is already supplied through Y_0', and the relations imposed by Q_0 are exactly the stated agreement. Hence the square $Q_0 \to Y_0'$, $Q_1 \to Y_1'$ is cocartesian. This is the claimed pushout of defect maps.

For two lower opens $A, B \subset I$, a cocone over $A \cup B$ is exactly a pair of cocones over A and B agreeing over $A \cap B$. A mixed relation introduces no additional condition: if $a \leq b$ and $b \in B$, lower closure puts a in B. Consequently

$$X_{A\cup B} \cong X_A \amalg_{X_{A\cap B}} X_B,$$

and likewise for Y. Applying the lemma with $A = I_\alpha$ and $B = {\downarrow} s_\alpha$ shows that the attachment from I_α to $I_\alpha \cup {\downarrow} s_\alpha$ is a pushout of the strong local comparison over B. This is the precise connection between the abstract cube and Theorem 2.1.

The sentence introducing the lemma says "carrés cartésiens", although the lemma concerns cocartesian squares. The same missing prefix occurs in the final sentence of its proof in the edition. We retain the edition's wording in the source and use the pushout calculation above for its mathematical reading. The diagram's f'_0 is the full map $X'_0 \to Y'_0$, not either horizontal map between intermediate pushouts.

The description of the intermediate cube as "totalement cocartésien" is stronger than the hypotheses justify if it means that all six faces are pushouts. Take all the X-vertices empty, $Y_0 = Y'_0 = \varnothing$, and $Y_1 = Y'_1 = \{*\}$. Both original horizontal squares are pushouts, but the back face with vertices X_0, X_1, Y_0, Y_1 is not. The bottom face of the intermediate cube is nevertheless a pushout, as the calculation proves, and this is the only extra face needed. The lemma is correct; its proof should invoke that face rather than an assertion about every face.

Lemme 2.6. *Soit un homomorphisme de carrés cocartésiens $X_\bullet$, $Y_\bullet$, dans une catégorie $\mathcal{M}$ avec sommes amalgamées (**N.B.** il suffit que les sommes amalgamées intervenant ci-dessus existent) :*

(2.6.1)
$$\begin{array}{ccccc} X_0 & \longrightarrow & X_1 & & \\ \downarrow{\scriptstyle f_0} & \searrow & \downarrow & \searrow & \\ Y_0 & \longrightarrow & X'_0 & \longrightarrow & X'_1 \\ & \searrow & \downarrow{\scriptstyle f'_0} & & \downarrow{\scriptstyle f_1} \\ & & Y_0 & \longrightarrow & Y_1 \\ & & & \searrow & \downarrow{\scriptstyle f'_1} \\ & & & & Y'_0 \longrightarrow Y'_1 \ . \end{array}$$

Soient Z_0, Z_1 les deux carrés latéraux et considérons, pour tout carré commutatif $Z = \begin{array}{ccc} X & \longrightarrow & X' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y' \end{array}$, la flèche u_Z du diagramme

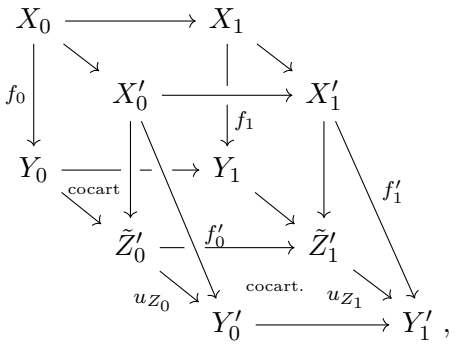
$$\begin{array}{ccccc} X & \longrightarrow & X' & & \\ \downarrow & \text{cocart.} & \downarrow & & \\ Y & \longrightarrow & \tilde{Z} & \xrightarrow[u_Z]{} & Y' \ . \end{array}$$

Alors la flèche u_{Z_1} se déduit de u_{Z_0} par cochangement de base, de façon plus précise, p. 18 [page 32]
on a un carré cocartésien, où $T = \begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ Y_0 & \longrightarrow & Y_1 \end{array}$ et $T' = \begin{array}{ccc} X'_0 & \longrightarrow & X'_1 \\ \downarrow & & \downarrow \\ Y'_0 & \longrightarrow & Y'_1 \end{array}$ désignent la face postérieure et antérieure du cube (2.6.1) :

(∗)
$$\begin{array}{ccc} \tilde{Z}_0 & \xrightarrow{u_{Z_0}} & Y'_0 \\ \downarrow & \text{cocart.} & \downarrow \\ \tilde{Z}_1 & \xrightarrow{u_{Z_1}} & Y'_1 \ , \end{array}$$

qu'on voit sur la base du diagramme

(∗∗)



dans lequel le cube est un cube totalement cocartésien, déduit du carré cocartésien X en l'"exponentiant" par la flèche $f_0 : X_0 \longrightarrow Y_0$ `[plutôt, par la flèche` (f_0, f_1) `de` $\underline{\mathrm{Fl}}(\mathcal{M})$`]`.

La base du cube, et le carré composé de celui-ci et de (∗) (base du diagramme polyédral (∗∗)) étant cartésien, il en est de même de (∗). [page 33]

Lemme 2.7. *Soit $(I_\alpha)_{\alpha\leq\alpha_0}$ une suite ordinale croissante d'ouverts de I. Alors $u_{I_0,I_{\alpha_0}} : Y_{I_0,I_{\alpha_0}} \longrightarrow Y_{I_{\alpha_0}}$ est composé transfini, indexé par l'ensemble des ordinaux $-1 \leq \alpha < \alpha_0$, d'une suite de flèches v_α, où chaque v_α est déduit de $u_{I_\alpha,I_{\alpha+1}}$ par cochangement de base (et de $u_{I_0} : X_{I_0} \longrightarrow Y_{I_0}$ pour $\alpha = 0$).*

Commentary · Exposition [ms. 33]

The relative transfinite composition starts after the initial attachment

Lemma 2.7 provides the second ingredient in the global argument: once the successive local changes have been understood, they must be assembled into the desired relative map. Its printed statement combines two different starting points and does not state continuity at limit ordinals. Both points can be repaired without changing the filtration actually used in Theorem 2.1.

Fix a continuous increasing filtration of lower opens $(I_\alpha)_{\alpha\leq\lambda}$ with $I_0 = K$ and $I_\lambda = J$, where $I_\delta = \bigcup_{\alpha<\delta} I_\alpha$ at every nonzero limit. Set

$$Z_\alpha = X_J \amalg_{X_{I_\alpha}} Y_{I_\alpha}.$$

Then $Z_0 = Q_f(K, J)$ and $Z_\lambda \cong Y_J$. The universal property of a cocone over a union gives $X_{I_\delta} = \operatorname{colim}_{\alpha<\delta} X_{I_\alpha}$ and the analogous identity for Y. Since colimits commute with colimits, it follows that $Z_\delta = \operatorname{colim}_{\alpha<\delta} Z_\alpha$. At a successor stage the square

$$\begin{array}{ccc} Q_f(I_\alpha, I_{\alpha+1}) & \longrightarrow & Z_\alpha \\ {\scriptstyle d_f(I_\alpha,I_{\alpha+1})}\downarrow & & \downarrow \\ Y_{I_{\alpha+1}} & \longrightarrow & Z_{\alpha+1} \end{array}$$

is a pushout, as follows by substituting the definitions. Thus $d_f(K, J) : Z_0 \to Z_\lambda$ is the continuous transfinite composite of these successor pushouts. There is no additional initial factor involving f_K.

If one wants instead the absolute map $f_J : X_J \to Y_J$, one must first prepend $X_J \to Z_0$, the pushout of f_K, and then take the relative composite. This is the legitimate role of an extra index -1. It cannot be included while the asserted composite still has domain $Q_f(K, J)$. Likewise, the displayed identification $X_J = Z_0$ on page 34 is valid only in the special case $K = \varnothing$, not for the relative assertion as stated.

Continuity cannot be discarded from the lemma. In Set let I have one object, let X be empty, let Y be a singleton, and take $I_n = \varnothing$ for every finite n, but $I_\omega = I$. Every successor attachment below ω is an identity of the empty set, whereas the requested final map is $\varnothing \to \{*\}$. It is therefore not their transfinite composite. The filtration constructed on page 30 is continuous, so this defect of the isolated statement does not invalidate the theorem. The modern terminology makes continuity explicit in the definition of a transfinite composite (Riehl and Verity, 2014, Definition 5.3).

Pour le voir, on introduit la suite transfinie d'objets $Z_\alpha = Y_{I_\alpha,I_{\alpha_0}}$, définis par le diagramme p. 19

$$\begin{array}{ccc} X_{I_\alpha} & \longrightarrow & X_{I_{\alpha_0}} \\ \downarrow & & \downarrow \\ Y_{I_\alpha} & \longrightarrow & Y_{I_\alpha, I_{\alpha_0}} = Z_\alpha \\ & & \downarrow \\ & & Y_{I_{\alpha_0}} \ , \end{array}$$

on a $Z_0 = Y_{I_\alpha, I_{\alpha_0}}$, $Z_{\alpha_0} = Y_{I_\alpha, I_{\alpha_0}}$, $Z_\alpha = \varinjlim_{\beta<\alpha} Z_\beta$ si α ordinal limite. Donc il reste à déterminer [34]
les $v_{\alpha,\alpha+1} : Z_\alpha \longrightarrow Z_{\alpha+1}$ (pour $\alpha < \alpha_0$), ce qu'on voit sur le diagramme

$$\begin{array}{ccccc} X_{I_\alpha} & \longrightarrow & X_{I_{\alpha+1}} & \longrightarrow & X_{I_{\alpha_0}} = Z_0 \\ \downarrow & & \downarrow & & \downarrow \\ Y_{I_\alpha} & \longrightarrow & Y_{I_\alpha, I_{\alpha+1}} & \longrightarrow & Y_{I_\alpha, I_{\alpha_0}} = Z_\alpha \\ & & \downarrow{\scriptstyle u_{I_\alpha, I_{\alpha+1}}} & & \downarrow{\scriptstyle v_{\alpha,\alpha+1}} \\ & & Y_{I_{\alpha+1}} & \longrightarrow & Y_{I_{\alpha+1}, I_{\alpha_0}} = Z_{\alpha+1} \ . \end{array}$$

Reprenons la démonstration de 2.1. On va prouver 1° par les deux lemmes précédents. Notons que lorsque l'espace sous-jacent à I est noethérien, alors la construction transfinie est forcément finie, donc on n'a pas besoin de la stabilité de C par composés transfinis infinis.

Prouvons 2°. On suppose donc que pour tout $i_0 \in I$, on ait $u_{I_{<\alpha_0}, I/i_0} \in C$, on veut en [35]
déduire que $u_\bullet \in C_I$, i.e. $u_{J,I/i_0} \in C$ pour *tout* ouvert.

Lemme 2.8. (Sous hypothèse artinienne sur I.) *Soit $P \subset I$ l'ensemble des $i_0 \in I$ tels que pour $u_\bullet | I/i_0$ soit dans C_{I/i_0}, i.e. que pour tout $i \leq i_0$, et tout ouvert K de I/i, on ait $u_{K,I/i} \in C$. Soit $i_0 \in I$ tel que $I_{<i_0} \subset P$. Alors $i_0 \in P$.*

Démonstration. Il reste à prouver que si $K \subset I/i_0 = J$ est un ouvert, alors $u_{K,J} \in C$. On peut supposer $K \neq I/i_0$, i.e. $K \subset I_{<i_0} \overset{\text{déf}}{=} J'$. Mais alors par hypothèse initiale sur $u_\bullet$, on a $u_{J',J} \in C$, et d'autre part, par hypothèse sur i_0, $u_\bullet | J'$ est dans $C_{J'}$. Donc par le théorème 2.1. 1°, on a $u_{K,J'} \in C$. Donc par transitivité $u_{K,J} \in C$, q.e.d.

Corollaire 2.9. *Si I est artinien, alors $P = I$, i.e. $u_\bullet \in C_I$.*

Cela prouve donc 2.1. 2°. Notons que $\mathrm{Esp}(I)$ est noethérien, il en est de même de tout ouvert (tel J' ci-dessus), donc pour la conclusion du lemme, inutile de supposer C stable par [36]
composition infinie.

Commentary · Exposition [ms. 35]

Well-founded induction turns boundaries into all open tests

Grothendieck now uses artinianity to recover the economical definition with which the chapter began. For a poset, the relevant hypothesis is well-foundedness of the strict relation: every nonempty subset has a minimal element. Under dependent choice this is equivalent to the stated absence of an infinite strictly descending sequence. It is not a finiteness assertion about the predecessor set of an object.

Assume every relative latching map lies in C. Let P consist of those i for which the transformation on $\downarrow i$ satisfies the strong open condition. Suppose all strict predecessors of i lie in P. For a proper lower open $K \subset \downarrow i$, put $J' = I_{<i}$. The restricted transformation on J' has the strong open condition, because this property is tested on the principal lower sets of its objects. Theorem 2.1(1), already proved without artinianity, gives $d_f(K, J') \in C$. By hypothesis $d_f(J', \downarrow i) \in C$. The comparison $d_f(K, \downarrow i)$ factors as a pushout of the first of these maps followed by the second; hence it belongs to C. The case $K = \downarrow i$ is an identity. Well-founded induction proves $P = I$.

The proof is logically independent of Theorem 2.1(3), and uses part (1) only on a smaller lower open. This matters because the induction on the trivial class will repeat the argument with an extra

use of two out of three. The duplicate label "2.9" for the corollary here and the lemma on page 37 should be read by their content, not as a logical back-reference to a later statement.

Modern Reedy theory packages the same boundary calculation as a factorization by degree. For a finite or locally finite artinian poset one can use a degree in $\mathbf{N}$, such as the maximal length of a chain ending at an object. A general artinian poset can require an ordinal rank

$$r(i) = \sup_{j<i}\big(r(j)+1\big).$$

For example $\omega+1$ is artinian but its last element has arbitrarily long finite chains below it, so no strictly increasing $\mathbf{N}$-valued degree exists. Grothendieck's transfinite argument therefore exceeds the finite-degree convention used in many introductions. The relative latching maps themselves are the familiar ones; it is the allowable height and the required colimits that change. The published Reedy account identifies these maps and their cellular presentations explicitly (Riehl and Verity, 2014, Definition 4.14 and Proposition 6.3).

Prouvons 2.1. 3°. On a a) $\Longrightarrow$ b), car par 2.1. 1°, $u_\bullet \in (TC)_I$ implique que pour $i \in I$, $u_i : X_i \longrightarrow Y_i$ est dans TC (cf. cor. 2.1 a). Donc il reste à prouver b) $\Longrightarrow$ a), i.e. si $u_\bullet \in C \cap W_{I^\circ}$, alors $u_\bullet \in (TC)_I$, i.e. les $u_{K,I/i_0}$ sont dans TC. On sait déjà que $u_{K,I/i_0} \in C$ par hypothèse $u_\bullet \in C_I$, donc il faut prouver que c'est dans W. On est ramené à prouver que $u_K \in W$, i.e. $u_K \in TC$, car par le diagramme (où $J = I/i_0$) p. 20

$$\begin{array}{ccc} X_K & \longrightarrow & X_J \\ {\scriptstyle u_K}\downarrow & \text{cocart} & \downarrow{\scriptstyle \tilde{u}_{K,J}} \\ Y_K & \longrightarrow & Y_{K,J} \\ & & {\scriptstyle u_{K,J}}\downarrow \\ & & Y_J \end{array} \qquad u_J \text{ (curved arrow } X_J \to Y_J),$$

$u_K \in TC$ implique $\tilde{u}_{K,J} \in TC$, et comme $u_J \simeq u_{i_0} \in W$ par hypothèse, il en résulte bien $u_{K,J} \in W$ par saturation. Mais on va s'y prendre autrement :

Lemme 2.9. *I ensemble ordonné quelconque, C, W comme dans th.* 2.1, 1° *et* 3°, *$u_\bullet : X_\bullet \longrightarrow Y_\bullet$ dans $C_I \cap W_{I^\circ}$. Soit P l'ensemble des $i_0 \in I$ tels que $u_\bullet | I/i_0$ soit dans $(TC)_{I/i_0}$. Si $i_0 \in I$ est tel que $I_{<i_0} \subset P$, alors $i_0 \in P$.* [37]

En effet, il reste à prouver que pour tout ouvert K de $J = I/i_0$, $K \neq J$, on a $u_{K,J} \in TC$. Mais pour ceci, il suffit de voir que $u_K, u_J \in W$ (comme on venait de le voir). Or $u_J \simeq u_{i_0} \in W$ par hypothèse. D'autre part $u_\bullet | K \in (TC_K)$ puisque $K \subset I_{<i_0} \subset P$, donc par le théorème 2.1, 1°, on conclut que $u_K \in TC$.

Si maintenant on suppose I artinien, il résulte de 3.9 que $P = I$, donc $u_\bullet \in (TC)_I$, q.e.d.

Cela achève la démonstration du théorème 2.1, et aussi du corollaire 2.5, vu que pour la démonstration du lemme 2.9 également, les hypothèses infinies sur C, $C \cap W$ sont inutiles si $\mathrm{Top}(I)$ est noethérien, donc ses ouverts sont aussi noethériens.

Remarques.

1) Dans 2.1. 2°, l'hypothèse I artinien n'est par superflue, comme on voit en prenant $I = \mathbf{N}^{\mathrm{opp}} \simeq \mathbf{Z}^{\leq 0}$, et pour une flèche [38]

$$\emptyset_{\mathcal{M}(I)} \longrightarrow Y_\bullet \ .$$

Dire que cette flèche est dans C_I signifie que les morphismes de transition $Y_i \xrightarrow{u_i} Y_{i+1}$ sont dans C, *et* que les Y_i sont C-cofibrants, i.e. $\emptyset_{\mathcal{M}} \longrightarrow Y_i$ est dans C, pour tout i. Mais la condition envisagée dans 2.1. 1° est seulement que les u_i sont dans C. Donc en prenant pour $Y_\bullet$ un système inductif constant de valeur Y, avec Y non C-cofibrant, i.e. $\emptyset_{\mathcal{M}} \longrightarrow Y$ non dans C, on a un contre-exemple.

Avec le même I, on voit que $u_\bullet : X_\bullet \longrightarrow Y_\bullet$ est dans C si et seulement si

a) les $u_i \in C$, et

b) les $u_{i,i+1} : Y_{i,i+1} \longrightarrow Y_{i+1}$ sont dans C.

Si de plus les u_i sont dans W, donc dans TC, on voit comme ci-dessus que $u_{i,i+1} \in TC$ pour tout i, donc $u_\bullet \in (TC)_I$. Donc ici on a encore p. 21

$$(TC)_I \;=\; C_I \cap W_{I^\circ}\ ,$$

quoique I ne soit pas artinien. J'ignore si l'hypothèse artinienne dans 2.1. 3° est superflue ou non.

2) Soit $u_\bullet \in \mathrm{Fl}(\mathcal{M}(I^\circ))$ [13], donc $u_\bullet$ est un objet de $\underline{\mathrm{Fl}}(\mathcal{M}(I^\circ)) \simeq \mathcal{M}(I^\circ \times \mathbf{\Delta}^1) = \mathcal{M}((I \times (\mathbf{\Delta}^1)^{\mathrm{opp}})^{\mathrm{opp}})$. On vérifie que pour que cet objet soit C-cofibrant, il faut et il suffit que

1°) $X_\bullet$ soit C-cofibrant, et

2°) $u_\bullet \in C_I$ (ce qui implique que $Y_\bullet$ est aussi cofibrant).

[39]

Commentary · Examples [ms. 38]

What the reverse chain does and does not refute

Grothendieck's counterexample separates the local latching tests from cofibrancy of the values. Write $I = \mathbf{Z}^{\leq 0}$ with its usual order. Every strict predecessor set of i has final object $i-1$, so $L_iY = Y_{i-1}$. The latching test for $\varnothing \to Y$ therefore asks only that $Y_{i-1} \to Y_i$ belong to C. By contrast, every nonempty lower open of $\downarrow i$ has a largest element; the strong condition asks for all these transition maps and also for $\varnothing \to Y_i$.

Choose, for instance, $\mathcal{M} = \mathrm{Set}$ and C the isomorphisms. This class contains the isomorphisms and is stable under pushout and transfinite composition. Let Y be the constant singleton diagram. Every latching map is the identity of a singleton, while $\varnothing \to Y_i$ is not an isomorphism. The latching criterion therefore fails to imply the strong criterion. This makes the counterexample completely concrete; it does not require a model category or a choice of W.

For an arbitrary transformation $f : X \to Y$ on this chain, the strong condition is equivalent to $f_i \in C$ and the adjacent relative maps $d_f(\downarrow i, \downarrow (i+1)) \in C$. A larger interval is obtained by finitely many such attachments, and the empty lower open supplies the pointwise condition. If in addition every $f_i \in W$, then every $f_i \in TC$. In the adjacent factorization of f_{i+1}, the first map is a pushout of f_i, hence is in TC, and two out of three puts the adjacent relative map in TC. Finite composition gives all remaining tests. Thus the equality involving TC survives in this example even though the latching characterization does not.

The two issues on page 38 are consequently distinct. The displayed constant diagram disproves omission of artinianity from Theorem 2.1(2). It is not a counterexample to Theorem 2.1(3) without artinianity. The latter is the question Grothendieck leaves undecided here.

Commentary · Development (commentators') [ms. 38]

The trivial-cofibration equality on chains without a first element

Grothendieck's reverse-chain calculation suggests that a missing first element need not obstruct the trivial-cofibration equality. We can extend his example by using cofinality in place of well-foundedness. The statement concerns the strong open definition of C_I throughout. We checked the direct-diagram definitions in Rădulescu-Banu (2009, Sections 9.1–9.2) and the relative latching and cellular presentations in Riehl and Verity (2014, Definition 4.14 and Proposition 6.3); these specified passages do not state the following extension. We record the argument as a continuation of the source, without a claim of priority or a claim to settle arbitrary posets.

Proposition. [ours] Assume $\mathcal{M}$ is cocomplete, C contains isomorphisms and is closed under pushout and small transfinite composition, W contains isomorphisms and satisfies two out of three, and $TC = C \cap W$ has the same closure properties as C. Suppose every principal lower set $\downarrow i$ of I is linearly ordered. Then

$$(TC)_I = C_I \cap W_{I^\circ},$$

13. $I^\circ = I^{\mathrm{opp}}$.

without an artinian hypothesis on I.

Proof. First let K be a nonempty linearly ordered set. It admits a strictly increasing well-ordered cofinal subset. To construct one, choose $s_0 \in K$. At a stage where the previously chosen elements are not cofinal, choose an element strictly greater than all of them; such an element exists because failure of cofinality gives a k above which no earlier choice lies, and linearity then puts all earlier choices below k. Distinct stages give distinct elements of the set K, so the construction stops before an ordinal of cardinality larger than K. At its stopping stage the chosen subset $S = \{s_\alpha \mid \alpha < \lambda\}$ is cofinal. A largest element, if present, simply ends the construction.

Now let $f \in C_I \cap W_{I^\circ}$ and fix a lower open $K \subset\!\downarrow i$. If $K = \varnothing$, then f_K is an identity. Otherwise choose such a cofinal chain $S \subset K$. For each $\alpha < \lambda$ put

$$U_\alpha = \bigcup_{\beta<\alpha} \downarrow s_\beta.$$

The earlier chosen elements are cofinal in U_α, so their X- and Y-colimits are X_{U_α} and Y_{U_α}. This identification follows directly from cocones: every object of U_α maps to an earlier chosen element, and a common later choice compares any two such maps. The relative latching map of $f|_S$ at α is therefore

$$X_{s_\alpha} \amalg_{X_{U_\alpha}} Y_{U_\alpha} \longrightarrow Y_{s_\alpha},$$

namely $d_f(U_\alpha, \downarrow s_\alpha)$, which lies in C by the strong condition. The pointwise maps of $f|_S$ are in W. Proposition 1 and its corollary, applied to the well-ordered set S, give $\operatorname{colim}_S f \in TC$. Cofinality identifies this map with f_K, so $f_K \in TC$.

The composite $X_i \to X_i \amalg_{X_K} Y_K \to Y_i$ is f_i. Its first factor is a pushout of $f_K \in TC$, and its second factor belongs to C. Two out of three makes that second factor a weak equivalence. These are all the strong TC tests at i. The reverse inclusion follows by taking $K = \varnothing$ in the definition. This proves the proposition.

In particular the result covers every linearly ordered small set, including dense chains and chains with no first element. It also covers branching posets in which branching occurs only upwards, so that each individual object's predecessors remain linearly ordered. The proof uses the full transfinite hypotheses; it does not assert that sequential closure alone suffices at arbitrary cofinalities.

Commentary · Exposition [ms. 38]

An arrow is a cofibrant two-layer diagram

The second remark turns a relative assertion into an absolute one. Label the two layers so that the arrow $f : X \to Y$ is the diagram on $I \times [1]$ with X at level 0 and Y at level 1. This fixes the endpoint convention independently of the opposite-category notation in the source. A lower open in this product is determined by lower opens $B \subset A \subset I$, and its colimit is

$$X_A \amalg_{X_B} Y_B.$$

The formula follows because the arrows from level 0 to level 1 impose precisely the relations already imposed over B.

Suppose X is strongly cofibrant and $f \in C_I$. At an object $(i, 0)$ the absolute cofibration test is a test for X. At $(i, 1)$ an arbitrary lower-open test has the form

$$X_A \amalg_{X_B} Y_B \longrightarrow Y_i, \qquad B \subset A \subset\!\downarrow i.$$

It factors through $X_i \amalg_{X_B} Y_B$. The first map is a pushout of $X_A \to X_i$, which belongs to C by cofibrancy of X; the second is the strong comparison for f. Thus the two-layer diagram is cofibrant. Conversely, testing its level-0 objects gives cofibrancy of X, and taking $A =\!\downarrow i$ in the level-1 tests gives all the strong tests for f. Taking the whole level-1 principal lower set also proves cofibrancy of Y.

This calculation explains why the relative construction is natural under iteration. A map of diagrams is itself a diagram with one additional direction, and its relative latching object is the boundary contribution in that direction. The observation is valid for the strong definition without assuming artinianity. Artinianity is needed only when one replaces all these tests by the single latching test at each object.

3 Catégories de modèles à cofibrations

Soit $\mathcal{M}$ une catégorie munie de deux ensembles de flèches

$$W,\ C\ \subset\ \mathrm{Fl}(\mathcal{M})\ .$$

On suppose que $\mathcal{M}$ contient un objet initial, et :

Cof$_0$ W est modérément saturé, i.e. contient les isomorphismes et si f, g sont composables, et si deux parmi f, g, gf sont dans W, le troisième aussi.

Cof$_1$ C contient les isomorphismes, est stable par composition et par cochangement de base.

Cof$_2$ $TC \overset{\text{déf}}{=} C \cap W$ est stable par cochangement de base.

Cof$_3$ Toute flèche f dans $\mathcal{M}$ s'écrit

$$f\ =\ pi$$

avec $i \in C$, $p \in W$.

Ce sont les axiomes de K. S. Brown (forme duale), qui suppose de plus que les objets de $\mathcal{M}$ sont C-cofibrants, i.e. $\emptyset_{\mathcal{M}} \longrightarrow X$ est dans C. Je ne ferai pas cette hypothèse. La sous-catégorie $\mathcal{M}_c$ de $\mathcal{M}$ formée des objets C-cofibrants satisfait aux axiomes de Brown, de plus on prouve [40] que

$$\mathrm{Ho}_W(\mathcal{M}_c) \xrightarrow{\sim} \mathrm{Ho}_W(\mathcal{M})$$

est une équivalence. Ainsi les résultats de Brown relatifs à $\mathcal{M}_c$, muni des ensembles de flèches induits par W, C, sont applicables à $\mathrm{Ho}_W(\mathcal{M})$ et à ses avatars (tels $\mathrm{Ho}_W(S\backslash\mathcal{M})$, $\mathrm{Ho}_W(\mathcal{M}/T)$ et $\mathrm{Ho}_W(S\underset{u_0}{\backslash}\mathcal{M}/T)$, où $u_0 : S \longrightarrow T$ est donné). Il se trouve que pour l'essentiel, les résultats homotopiques standard (notamment suite exacte de cofibrations relatifs aux opérations homotopiques "à droite" (ou faut-il dire à gauche ? il s'agit pourtant de questions d'exactitude à droite, non à gauche)) sont valables. Il me faudra revenir dessus de façon plus circonstanciée par la suite. Notamment, il y a une bonne théorie de carrés W-cocartésiens (et *dans* $\mathcal{M}_c$, il y a assez de W-cofibrations [14], …), et de plus, ceux-ci sont *exacts*. La catégorie p. 22 $\mathrm{Ho}_W(\mathcal{M}) \simeq \mathrm{Ho}_W(\mathcal{M}_c)$ "se calcule" par un calcul de fractions à gauche dans $\pi\mathcal{M}_c$, où $\pi\mathcal{M}_c$ [41] est la catégorie déduite de $\mathcal{M}_c$ en y remplaçant les $\mathrm{Hom}_{\mathcal{M}_c}(X,Y) = \mathrm{Hom}_{\mathcal{M}}(X,Y)$ par le quotient par une relation d'homotopie convenable :

$f, g : X \rightrightarrows Y$ sont QB-homotopes (au sens de *Quillen-Brown*) s'il existe un $X' \xrightarrow{\alpha} X$ dans W (et on peut supposer $X' \in \mathrm{Ob}\,\mathcal{M}_c$ également) tel que $f\alpha \overset{1}{\sim} g\alpha$ au sens de Quillen, i.e. il existe un objet cylindre $\begin{array}{ccc} I_{X'} & \xrightarrow{\in W} & X' \\ {\scriptstyle \in C}\uparrow & \nearrow & \\ X' \vee X' & & \end{array}$ pour X', tel que $(f\alpha, g\alpha)\ :\ X' \vee X' \longrightarrow Y$ se factorise par un $h' : I_{X'} \longrightarrow Y$.

Commentary · Context [ms. 39]

Brown's axioms and the role of noncofibrant objects

Grothendieck introduces cofibrations as auxiliary data for a fixed localization. The weak equivalences are required to satisfy two out of three; the cofibrations supply the pushouts and factorizations needed to construct homotopy colimits. No lifting relation with a class of fibrations is imposed. The historical reference is to the dual of Brown's theory of fibrant objects (Brown, 1973). The broad comparison with Quillen's axioms was treated in the commentary on Chapter VI, manuscript p. 108; the new point here is the systematic inclusion of noncofibrant objects.

The existence convention in "stable par cochangement de base" matters. The Brown interpretation includes existence of the required pushouts of cofibrations. If the words were read as closure only

14. Plus précisément, par K. S. Brown, *dans* $\mathcal{M}_c$, les objets de C sont des W_c-cofibrations.

for pushouts that happened to exist, Cof_0–Cof_3 together with an initial object would not provide the finite constructions used below. The explicit finite-colimit hypothesis of Theorem 3.1 removes this issue there. With the Brown existence convention, the cofibrant objects form a category of cofibrant objects: a pushout of a cofibration between cofibrant objects remains cofibrant, a factorization of a map with cofibrant source has cofibrant middle object, and finite coproducts are pushouts of the cofibrations from the initial object.

In particular, the cylinder needed later is obtained by factoring the fold map

$$X \amalg X \longrightarrow X \quad \text{as} \quad X \amalg X \hookrightarrow IX \xrightarrow{\sim} X.$$

The hook here denotes membership in C, not literal injectivity in an arbitrary category. Each endpoint inclusion $X \to IX$ is a trivial cofibration: it is a composite of cofibrations, and its composite with $IX \to X$ is the identity. Thus cylinders are constructed from Cof_3 rather than introduced as an independent choice of extra structure.

Allowing noncofibrant objects does not enlarge the localized theory. Factoring $\varnothing \to X$ gives a cofibrant replacement $QX \xrightarrow{\sim} X$. This proves essential surjectivity of $\mathrm{Ho}_W(\mathcal{M}_c) \to \mathrm{Ho}_W(\mathcal{M})$ immediately; full faithfulness requires the approximation argument, not just the existence of these replacements. The source returns to it in Lemma 3.4. A precise modern treatment is Rădulescu-Banu (2009, Theorem 6.1.6). We do not replace this argument by a tacit assumption that replacement choices are functorial.

The same construction applies to slices and coslices with their induced weak equivalences and cofibrations. In a coslice $S\backslash\mathcal{M}$, cofibrancy means that the structure map $S \to X$ is a cofibration; it does not mean that X was cofibrant in $\mathcal{M}$. This relative interpretation is what makes the later discussion of exact homotopy pushout squares possible.

C'est de ce critère que résulte l'exactitude (à gauche) des carrés W-cocartésiens.

Je veux voir comment une structure de cofibrations sur $\mathcal{M}$ s'étend aux catégories $\mathcal{M}(I^\circ) = \underline{\mathrm{Hom}}(I, M)$. On a dans $\mathcal{M}(I^\circ)$ l'ensemble W_{I°, et C_I (il vaut mieux sans doute le noter aussi C_{I°) défini au paragraphe précédent. Il s'agit de vérifier les axiomes. L'axiome Cof_0 est immédiat, et bien connu pour I une catégorie quelconque. [42]

Commentary · Exposition [ms. 41]

Fractions, homotopy, and weak pushout exactness

Grothendieck is preparing a calculus of representatives for morphisms in the localization. His QB relation first allows a weak equivalence $\alpha : X' \to X$ in the source, then asks for an ordinary cylinder homotopy between $f\alpha$ and $g\alpha$. That is the formula printed in the edition. It should not be silently replaced by the formula one expects from the dual of a category of fibrant objects.

Every relation of the displayed kind implies equality of f and g in $\mathrm{Ho}_W(\mathcal{M})$. Indeed, for the two endpoint maps $e_0, e_1 : X' \to IX'$ and the cylinder projection q, the equations $qe_0 = qe_1 = 1$ show that $e_0 = e_1$ after localization. A map $h : IX' \to Y$ gives $f\alpha = he_0 = he_1 = g\alpha$ there, and invertibility of α gives $f = g$. This verifies the direction used whenever such an explicit homotopy is supplied.

A useful, separately verified equality criterion is the Brown formulation with enlargement of the target: for cofibrant X, Y, equality in the localization is equivalent to the existence of a trivial cofibration $b : Y \to Y'$ and a cylinder homotopy from bf to bg. This is the convention in Rădulescu-Banu (2009, Section 6.3, diagram (6.1), and Theorem 6.3.1). Its formal cancellation is in the target, whereas the displayed QB definition cancels in the source. The source does not explicitly assert that its quotient $\pi\mathcal{M}_c$ is already the full category of morphisms in the localization: it still localizes by fractions. Therefore the difference between these formulas is not, by itself, a counterexample to his assertion about fractions.

For the applications here one may use this target-enlargement criterion and Brown's homotopy calculus. A morphism is represented by $X \to Z \xleftarrow{\sim} Y$; the denominator may be chosen to be a trivial cofibration (Rădulescu-Banu, 2009, Theorem 6.4.5(1)(a)). This makes both the direction of the denominator and the role of the homotopy quotient explicit.

The square-exactness requirement has a concrete consequence which is useful throughout the construction. Write $\mathcal{H} = \mathrm{Ho}_W(\mathcal{M})$. For a homotopy-cocartesian square completing a span

$B \leftarrow A \to C$ to P, and any test object T, the comparison

$$\mathrm{Hom}_{\mathcal{H}}(P,T) \longrightarrow \mathrm{Hom}_{\mathcal{H}}(B,T) \times_{\mathrm{Hom}_{\mathcal{H}}(A,T)} \mathrm{Hom}_{\mathcal{H}}(C,T)$$

is surjective. Grothendieck explicitly gives the dual Hom-set formulation in Chapter XIV, manuscript page 6, Corollary 2, and introduces its cocartesian dual on page 7 (Grothendieck, 1990–1991r). XVII invokes Brown's calculus for square exactness here and labels the corresponding requirement Dér 6 in Corollary 3.9. We use the verified Hom-surjectivity statement; we do not identify it with every clause of the stronger quadrangulated-category axioms proposed in XIV.

Here is the proof from that calculus. Represent the homotopy pushout by a span of cofibrant objects whose two legs are cofibrations, with ordinary pushout P, and replace T by a cofibrant representative. Suppose classes $b : B \to T$ and $c : C \to T$ agree on A. The fraction theorem represents them by actual maps $b' : B \to T'$ and $c' : C \to T'$ after one common trivial-cofibration enlargement $T \to T'$: push out their two denominators over T to obtain such a common target. The equality criterion gives, after a further trivial-cofibration enlargement if necessary, a cylinder $A \amalg A \to IA \xrightarrow{q} A$ and $h : IA \to T'$ with endpoint values $b'|_A$ and $c'|_A$.

To make the agreement strict, let $d_0, d_1 : A \to IA$ be the endpoints and put

$$B' = B \amalg_{A,d_0} IA, \qquad P' = B' \amalg_{A,d_1} C.$$

The map $B \to B'$ is a trivial cofibration, being a pushout of d_0. The cylinder projection defines a retraction $r : B' \to B$, which is therefore a weak equivalence. The other endpoint $A \to B'$ is a cofibration: the map $B \amalg A \to B'$ is a pushout of $A \amalg A \to IA$, and $A \to B \amalg A$ is a cofibration because B is cofibrant. The maps b' and h glue on B', and their value on the other endpoint agrees with c'. They consequently give $P' \to T'$. The gluing lemma applied to the objectwise weak equivalence of cofibrant spans $(A, B', C) \to (A, B, C)$ gives a weak equivalence $P' \to P$. Thus

$$P \xleftarrow{\sim} P' \longrightarrow T' \xleftarrow{\sim} T$$

defines the desired morphism in $\mathcal{H}$. Its restrictions are b and c because the construction on B' restricts to b' on B, the map $B \to B'$ becomes the inverse of r, and the construction on C is c'.

This is a weak pushout property, with no uniqueness assertion. In spaces the homotopy pushout of $* \leftarrow S^0 \to *$ is S^1. Testing against S^1 gives the comparison $\mathbb{Z} \to \{*\}$: self-maps of the circle have all integer degrees, whereas the two point maps and their restrictions have a single unbased homotopy class. Replacing Hom-surjectivity by an ordinary pushout universal property would therefore be false. The proof above applies in a coefficient category as soon as its required cofibration structure has been established.

Cof_1 : C_{I° contient les isomorphismes, c'est clair, parce que C les contient. Stable par composition $X_\bullet \xrightarrow{u} Y_\bullet \xrightarrow{v} Z_\bullet$, regardons $K \subset J$, posons $w = v \circ u$,

$$\begin{array}{ccccccc}
X_K & \longrightarrow & X_J & & & & \\
\downarrow & & \downarrow & & & & \\
Y_K & \longrightarrow & Y_{K,J} & \xrightarrow{u_{K,J}} & Y_J & & \\
\downarrow & & \downarrow & & \downarrow & & \\
Z_K & \longrightarrow & \bullet & \xrightarrow{u'_{K,J}} & Z_{K,J} & \xrightarrow{v_{K,J}} & Z_J \,,
\end{array}$$

with $w_{K,J}$ spanning $\bullet \to Z_J$.

$w_{K,J}$ est composé de $v_{K,J}$, qui est dans C, et $u'_{K,J}$, qui l'est car déduit de $u_{K,J}$ par cochangement de base.

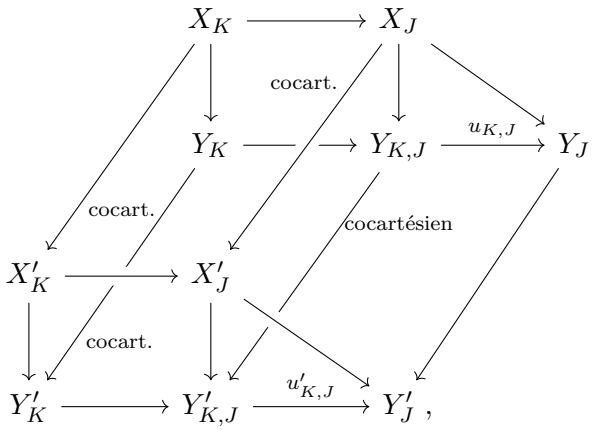
(**N.B.** On a déjà fait le raisonnement avec un seul $u_\bullet$, et *trois* ouverts $J \subset K \subset L$. Il y aurait lieu d'introduire la notion de C-carré $\begin{array}{ccc} X_0 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ X_2 & \longrightarrow & X_3 \end{array}$, comme un carré tel que [43]

$X_1 \amalg_{X_0} X_2 \longrightarrow X_3$ soit dans C (**N.B.** notion symétrique pour les foncteurs de symétrie des carrés), et dire que le composé de deux C-carrés est un C-carré.) p. 23

Cochangement de base

$$\begin{array}{ccc} X_\bullet & \xrightarrow{u} & Y_\bullet \\ \downarrow & & \downarrow \\ X'_\bullet & \xrightarrow{u'} & Y'_\bullet \,. \end{array}$$

Considérons $K \subset J$, on a



où les trois carrés latéraux marqués "cocart." sont cocartésiens, ainsi que le carré en plan incliné. Il en résulte, d'abord que le quatrième carré latéral du cube est lui aussi cocartésien, puis que le carré ayant deux côtés $u_{K,J}$, $u'_{K,J}$ est cocartésien. (**N.B.** comparer le diagramme similaire p. 32 – il y aurait un lemme commun à dégager, distinguant deux cas …) [44]

Commentary · Exposition [ms. 43]

C-squares and the two gluing calculations

Grothendieck's proposed C-square records the same relative defect used throughout the chapter. For a commutative square with vertices A, B, D, E, in that order, its defect is $B \amalg_A D \to E$. Requiring this map to belong to C is symmetric under interchange of the two coordinate directions. It does not require any of the four boundary arrows separately to belong to C.

The composition statement can be proved once in a form that covers both uses in the text. Consider a commutative rectangle with top row $A \to B \to C'$ and bottom row $D \to E \to F$. Its outer defect factors as

$$C' \amalg_A D \longrightarrow C' \amalg_B E \longrightarrow F.$$

The first factor is the pushout of $B \amalg_A D \to E$ along the canonical map to $C' \amalg_A D$; the second is the defect of the right-hand square. If both inner squares are C-squares, pushout closure and composition closure show that the rectangle is one. Transposing gives the corresponding vertical-pasting statement.

There are then two applications. If one holds $K \subset J$ fixed and composes transformations $X \to Y \to Z$, this formula proves closure of C_I under composition. If one holds a transformation fixed and composes inclusions $K \subset J \subset L$, it proves the relative transitivity used in the induction of Theorem 2.1. The pictures differ only in which direction of a diagram is regarded as the variable.

For cobase change, form a pointwise pushout $Y' = Y \amalg_X X'$ of diagrams. Colimits over an open commute with this pushout, so

$$Q_{f'}(K, J) \amalg_{Q_f(K,J)} Y_J \cong Y'_J.$$

The comparison $d_{f'}(K, J)$ is consequently a pushout of $d_f(K, J)$. This is the assertion that the polyhedral diagram on pages 43–44 is designed to show. The proof uses only the universal properties of the displayed colimits and therefore applies equally to C and TC.

These are familiar Leibniz or pushout-product calculations in later terminology. The relative latching map of a composite is explicitly treated in Riehl and Verity (2014, Observation 5.1 and

Example 5.2). The reference identifies a later general organization of the calculation; no lifting axiom, monoidal model structure, or retract closure is needed for the elementary identities just proved.

Cof_2. Pour prouver que $C_{I^\circ} \cap W_{I^\circ}$ est stable par cochangement de base, je ne vois pas de façon de le faire, si je ne sais déjà qu'on a

$$(*) \qquad C_{I^\circ} \cap W_{I^\circ} \;=\; (TC)_{I^\circ} \;,$$

et alors on applique le résultat précédent (où W n'entrait pas) à TC au lieu de C. Mais pour avoir $(*)$, il faut que je suppose qu'on est dans l'un des deux cas suivants (de façon à pouvoir appliquer le théorème 2.1 et son corollaire 2.5).

a) I fini, ou plus généralement I artinien *et* $\mathrm{Top}(I)$ est noethérien.

b) I artinien, et de plus C et TC sont stables par composition transfinie.

Cof_3. L'axiome de facorisation me paraît a priori le plus délicat [15]. Mais cela sort finalement sans trop de mal. Soit donné

$$f \;:\; X_\bullet \longrightarrow Y_\bullet$$

dans $\mathcal{M}(I^\circ)$, considérons les triples p. 24 [45]

$$(J,\; Z,\; i,\; p) \;,$$

où $J \subset I$ est un ouvert, Z un objet de $\mathcal{M}(J^\circ)$, et i, p sont des flèches

$$\begin{cases} X|J \xrightarrow{\;i\;} Z \xleftarrow{\;p\;} Y|J \\ \text{avec } i \in C_{J^\circ},\; p \in W_{J^\circ}. \end{cases}$$

On ordonne les quadruples par prolongement. Si on a un ensemble totalement ordonné, de $(J_\alpha, Z_\alpha, i_\alpha, p_\alpha)$, posant $J = \bigcup I_\alpha$, on trouve de Z, i, p uniques sur J qui prolongent les Z_α, i_α, p_α donnés, de plus on aura $i \in C_{J^\circ}$, $p \in W_{J^\circ}$, car l'appartenance à C_{J° resp. W_{J° est question locale sur J. Donc par Zorn, on est ramené à prouver que si (J, Z, i, p) est maximal, on a $J = I$ – d'ou la factorisation voulue $f = pi$. Donc à prouver que si $J \neq I$, le quadruple n'est pas maximal. Considérons le fermé non vide $I \backslash J$. I est supposé artinien, il y a un *élément minimal* i_0 de $I \backslash J$. Posons [46]

$$J' \;=\; J \cup \{i_0\} \;=\; J \cup I/i_0,$$

et montrons qu'on peut prolonger le triplet Z, i, p en Z', j', p' sur J'. Posant $K' = I/i_0$, $K = J \cap K = I_{<i_0}$, on a donc

$$\begin{array}{ccc} K = J \cap K' & \hookrightarrow & K' = I/i_0 \\ \downarrow & & \downarrow \\ J & \hookrightarrow & J' = J \cup K'. \end{array}$$

On voit alors que les prolongements de (Z, i, p) de J à J' correspondent à ceux de $(Z|K', i|K', p|K')$ à K. Pour ceci, on note que pour une flèche φ d'objets de $\mathcal{M}(J'^\circ)$, cette flèche est dans $C_{J'^\circ}$ resp. $W_{J'^\circ}$, si et seulement si ses restrictions à J et à K' le sont.

15. L'argument qui suit s'applique à d'autres situations de factorisation, par exemple pour un précouple de Quillen donné (Φ, Ψ) dans $\mathcal{M}$.

Commentary · Exposition [ms. 46]

Factorization by adjoining one minimal missing object

Grothendieck's construction is the relative latching factorization, written without a degree function. Its purpose is to preserve all the factorization data already chosen on a lower open. The only new choice at a stage is a factorization in $\mathcal{M}$ of one map from a pushout.

Suppose $f : X \to Y$ has been factored on a lower open J as $X|J \xrightarrow{i} Z \xrightarrow{p} Y|J$, with $i \in C_J$ and p objectwise in W. Choose a minimal element s of $I \setminus J$. Every strict predecessor of s lies in J, so $J \cup \{s\}$ is lower. Put $K = I_{<s}$ and form

$$P_s = X_s \amalg_{X_K} Z_K.$$

The original map f_s and the map $Z_K \xrightarrow{p_K} Y_K \to Y_s$ agree on X_K, hence define $P_s \to Y_s$. Factor it by Cof_3 as

$$P_s \xrightarrow{j_s} Z_s \xrightarrow{p_s} Y_s, \qquad j_s \in C, \quad p_s \in W.$$

The maps from the predecessor values Z_t to Z_s are the composites through Z_K and P_s. They are compatible because they come from a cocone. Define i_s by $X_s \to P_s \to Z_s$. The equality $p_s i_s = f_s$ and the naturality equations follow from the defining pushout. The new relative latching map for i is exactly j_s; all older latching maps are unchanged. Theorem 2.1(2) gives the strong cofibration condition on the enlarged lower open, and p remains a pointwise weak equivalence.

This proves the required extension and also explains the direction of p in the source: it must run from Z to $Y|J$. The backward arrow on page 45 is printed in the edition and is retained as an edition slip. On page 46 the restriction to K must be extended to K', rather than the other way around as the sentence there says.

One may finish with a well-ordered recursion instead of applying Zorn to a possibly large class of partial factorizations. Fix a well-order of the set of objects of I, choose at each successor stage the first minimal missing object, and take the union of the strictly compatible data at a limit stage. This union is an actual diagram: each object, each arrow, and each required naturality equation already occurs at some previous stage. Cofibrancy is local on principal lower sets, so it persists at the union. The recursion stops before more distinct objects have been chosen than I contains. This avoids a size ambiguity in the Zorn wording while preserving the argument. The local factorization choices can be made in the ambient universe; they need not be functorial.

The argument requires the latching colimits X_K, Z_K to exist. In the artinian noetherian alternative the indexing poset is finite, as proved above. In the transfinite alternative suitable small colimits must actually be available. Closure of a class of morphisms under transfinite composition cannot supply a colimit that the ambient category does not have.

On est donc ramené au cas standard d'un ensemble ordonné avec objet final e_0, ensemble ordonné qu'on va noter I, et l'ensemble ordonné I' des objets $i < e_0$, i.e. $I' = I\backslash\{i_0\}$. On a $f : X_\bullet \longrightarrow Y_\bullet$, flèche dans $\mathcal{M}(I^\circ)$, une factorisation de $f|I'$ en $p' \circ i'$ avec $i' \in C_{I'^\circ}$, $p' \in W_{I^\circ}$, et il s'agit de la prolonger en une factorisation similaire de f. On note que la donnée d'un [47]
objet $X_\bullet$ de $\mathcal{M}(I^\circ)$ équivaut à celle d'un objet $X'_\bullet$ de $\mathcal{M}(I'^\circ)$ (la restriction de X à I') et d'une flèche

$$X'_{I'} \xrightarrow{\alpha_X} X_{e_0} = X_0$$

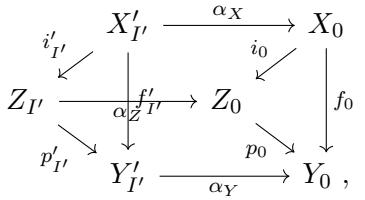
dans $\mathcal{M}$, où $X'_{I'} = \varinjlim_{i' \in I'} X'_{i'}$. Donc la donnée de la flèche $f : X_\bullet \longrightarrow Y_\bullet$ revient à celle de
$f' = f|I' : X'_\bullet \longrightarrow Y'_\bullet$, et d'une flèche $f_0 : X_0 \longrightarrow Y_0$ rendant commutatif le carré p. 25

$$\begin{array}{ccc} X'_{I'} & \xrightarrow{\alpha_X} & X_0 \\ {\scriptstyle f'_{I'}}\downarrow & & \downarrow{\scriptstyle f_0} \\ Y'_{I'} & \xrightarrow{\alpha_Y} & Y_0 \ . \end{array}$$

De même, trouver une factorisation de f en pi, qui prolonge une factorisation donnée

$f' = p'i'$ sur J', revient à la donnée d'un diagramme commutatif

(∗)



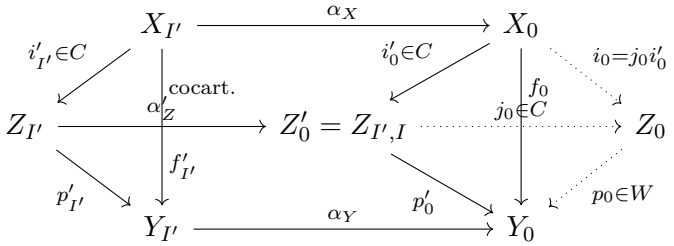
où l'objet Z_0, et les trois flèches α_Z, i_0, p_0 sont à déterminer. Si on a $i' \in C_{I'\circ}$, $p \in W_{I'\circ}$, alors on conclut par théorème 2.1. 1° (cor. 2.2) que [48 vide] [49]

$$i'_{I'} \ \in \ C \ ,$$

et si on veut que $i \in C_{I\circ}$, $p \in W_{I\circ}$, il reste seulement à exiger

$$p_0 \in W, \qquad (i'_{I',I} : \underbrace{Z_{I'} \amalg_{X_{I'}} X_0}_{Z'_0} \longrightarrow Z_0) \in C,$$

d'après le théorème 2.1. 2°. On va alors construire le diagramme (∗), à partir du diagramme similaire ci-dessous



(avec Z_0 de (∗) remplacé par $Z'_0 = Z_{I',I}$) en factorisant la flèche canonique $p'_0 : Z'_0 \longrightarrow Y_0$ en

$$p'_0 \ = \ p_0 j_0 \ , \qquad \text{avec } p_0 \in W, \ j_0 \in C \ ,$$

et en posant

$$i_0 \ = \ j_0 i'_0 \ ,$$

ce qui achève la construction.

Commentary · Examples [ms. 49]

A factorization at a branching vertex in chain complexes

The last construction is easiest to see when one can write its new object explicitly. Let $\mathcal{M}$ be chain complexes of vector spaces over a field k, with homological differentials, degreewise injections as cofibrations, and chain homotopy equivalences as weak equivalences. The injections split degree by degree. We use the square poset from the earlier example, with $X = 0$ and with the target diagram $Y_a = 0$, $Y_b = Y_c = Y_d = k[0]$. Both maps into Y_d are the identity. The initial map to this diagram is pointwise a cofibration but fails the diagram cofibration condition at d, where its latching map is the fold $k \oplus k \to k$.

Leave the factorization unchanged at a, b, c. At d the object P_d in the preceding construction is $A = k^2[0]$ and its map to $B = Y_d = k[0]$ is $q(a_1, a_2) = a_1 + a_2$. Define a complex Z_d by

$$(Z_d)_1 = k^2, \qquad (Z_d)_0 = k \oplus k^2, \qquad d(x_1, x_2) = (x_1 + x_2, -x_1, -x_2),$$

and zero in other degrees. Let $j : A \to Z_d$ be $j(a_1, a_2) = (0, a_1, a_2)$ in degree zero, and let $p : Z_d \to B$ be

$$p(b, a_1, a_2) = b + a_1 + a_2$$

in degree zero, with $p = 0$ in degree one. Then $pj = q$, p is a chain map, and j is degreewise injective. The maps from b and c are the two summand inclusions followed by j, so the extended data form a commutative square.

The map p is a homotopy equivalence, not merely a quasi-isomorphism asserted without calculation. Its section sends b to $(b, 0, 0)$. Its kernel in degree zero consists of $(-a_1 - a_2, a_1, a_2)$ and is identified with k^2. Under this identification the differential from degree one is $-1 : k^2 \to k^2$. The homotopy inverse to this differential, with the same minus sign, contracts the kernel. Thus Z_d is the direct sum of B and a contractible complex. The resulting map of diagrams $Z \to Y$ is pointwise in W, and $0 \to Z$ is a diagram cofibration because the new latching map is exactly j.

The construction preserves the homotopy type at the branching vertex while replacing an identification of two branches by an explicit cofibration. Its quotient complex $Z_d/j(A)$ is $k^2 \xrightarrow{(1,1)} k$ in degrees one and zero. Its degree-one homology is $\{(t, -t) \mid t \in k\} \cong k$. This records the relation joining the two branches. The example makes visible why a diagram factorization has more information than choosing independent factorizations of the component maps.

On a ainsi prouvé le

Théorème 3.1. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations (cf. page 39), $\mathcal{M}$ stable par $\varinjlim$ finies. Pour tout ensemble odonné I, considérons le tuple $(\mathcal{M}(I^\circ), W_{I^\circ}, C_{I^\circ})$ correspondant, où C_{I° est défini au paragraphe 2 (cf. à ce sujet le théorème 2.1). Supposons que I soit un ensemble ordonné* artinien *(p.ex. I localement fini, i.e. les I/i finis). Supposons de plus qu'on soit dans l'un des deux cas suivants :* p. 26 [50]

a) *L'espace* $\mathrm{Esp}(I)$ *est noethérien, i.e. toute suite croissante d'ouverts de I est stationnaire (p.ex. I fini).*

b) *C et $TC = C \cap W$ sont stables, non seulement par composition ordinaire, mais aussi par composition transfinie. Dans ce cas, on suppose aussi $\mathcal{M}$ stable par petites $\varinjlim$ filtrantes (donc par petites $\varinjlim$ quelconques).*

Sous ces conditions, $(\mathcal{M}(I^\circ), W_{I^\circ}, C_{I^\circ})$ est également une catégorie à cofibrations.

Commentary · Exposition [ms. 50]

The diagram structure and the size of its indexing category

Grothendieck has now reached the purpose of the preceding combinatorics: a diagram admits a replacement whose ordinary colimit can be trusted homotopically. Write $\mathcal{M}^I = \mathrm{Fun}(I, \mathcal{M})$, remembering that this is his $\mathcal{M}(I^\circ)$. For a well-founded poset the cofibrations of Theorem 3.1 are exactly the maps whose relative latching maps

$$X_i \amalg_{\mathrm{colim}_{j<i} X_j} \mathrm{colim}_{j<i} Y_j \longrightarrow Y_i$$

belong to C. Weak equivalences remain objectwise. The construction at pages 45–49 therefore gives the familiar inductive replacement: after the values below i have been chosen, factor the single map from the relative latching object to the prescribed value at i. The boundary has already encoded every naturality equation involving an incoming arrow. No lifting property between two distinguished classes is needed.

The alternatives in the theorem concern existence of these boundaries as well as preservation of the relevant classes. In case (b), finite colimits and small filtered colimits give all small colimits. Indeed, an arbitrary coproduct is the filtered colimit of its finite subcoproducts, and a general colimit is the coequalizer of the two familiar coproducts indexed by objects and arrows. Transfinite closure of C and $C \cap W$ then supplies precisely the closure needed in the boundary calculations. Merely having ordinary finite composites would not justify a limit stage.

There is a useful simplification of case (a). With the usual choice principles, a well-founded poset whose lower Alexandrov topology is noetherian is finite. If it were infinite, choose successively a minimal element of the remaining poset. After n choices the chosen elements form a lower set: every predecessor of the next chosen element was already chosen. These finite lower sets give an infinite strictly increasing sequence of opens, a contradiction. Consequently the apparent generalization beyond finite I in this branch introduces no additional posets. This observation also explains

why finite colimits suffice here; one must not diagnose an absent infinite-colimit hypothesis in this particular assertion.

The theorem supplies more than an objectwise cofibrant replacement. On the span $a \leftarrow b \rightarrow c$, an objectwise cofibrant diagram can still have a noncofibrant attaching map out of b. A replacement cofibrant for C_I controls those attachments, so its pushout is the model for the homotopy pushout. The distinction is the reason for carrying the relative, rather than just the absolute, latching construction throughout the chapter.

Proposition 3.2. *Soit $u : I \longrightarrow I'$ une application croissante entre ensembles ordonnés, d'où un couple de foncteurs adjoints*

$$\mathcal{M}(I^\circ) \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{M}(I'^\circ) \quad . \tag{3.2.1}$$

a) *Le foncteur u^* est compatible avec les localiseurs $W_{I'^\circ}$, W_{I°. Si I est discrète, ou u étale, alors u^* applique $C_{I'^\circ}$ dans C_{I°. Il en est encore de même quand u est localement cofiltrant, i.e. si pour tout $i' \in \mathrm{Ob}\, I'$, la catégorie $i'\backslash I$ est localement filtrante, i.e. ses composantes connexes sont cofiltrantes (condition* $\mathrm{PS_g}$ 1 *sur $i'\backslash I$)* [16].* [51]

b) *Dans tous les cas, $u_!$ transforme C_{I° en $C_{I'^\circ}$, TC_{I° en $TC_{I'^\circ}$, enfin $\mathcal{M}(I^\circ)_c$ en $\mathcal{M}(I'^\circ)_c$.*

Commentary · Exposition [ms. 51]

Why restriction needs a condition on the indexing map

Grothendieck wants to know which changes of indices preserve the replacements just constructed. Preservation of weak equivalences is immediate for every restriction functor: the component of u^*f at i is $f_{u(i)}$. Preservation of cofibrations is different, because restriction can change the shape and the connected components of a boundary. Proposition 3.2(a) identifies a condition which prevents this change from changing the relevant colimit.

Here is the poset argument behind the reference to Chapter II. Fix $i \in I$, let $U \subseteq I/i$ be a lower set, and let U' be the lower set of $I'/u(i)$ generated by $u(U)$. For $b \in U'$, the comma category $(b \downarrow u|_U)$ is the poset

$$S_b = \{x \in U \mid b \leq u(x)\}.$$

It is nonempty by the definition of U'. Every object of S_b maps to i in $(b \downarrow u)$, since $x \leq i$ and $b \leq u(i)$. Thus all these objects belong to one connected component of that comma category. If its connected components are cofiltered, any two objects x, y of S_b have a common predecessor z there. Since U is lower, $z \in U$, and hence $z \in S_b$. A poset has no nontrivial parallel-arrow equalization condition, so S_b itself is cofiltered. In particular its underlying undirected graph is connected.

The ordinary final-functor criterion now gives

$$\mathrm{colim}_U\, u^*X' \;\simeq\; \mathrm{colim}_{U'}\, X'.$$

The same holds for Y', and the values at i and $u(i)$ are identical. Consequently the two relative boundary maps in the displayed cube differ only by isomorphisms. This proves the asserted preservation of C_I without introducing any homotopy cofinality theorem. The localizer does not enter this step.

For an étale map of posets the same conclusion is even more direct: the map $I/i \to I'/u(i)$ is an isomorphism. For a discrete source, each boundary is empty, so only the component maps are tested. These are two special ways of making the boundary comparison harmless. The printed phrase "localement filtrante" on page 51 conflicts with the immediately following definition; the proof uses cofiltered connected components, as the statement's initial terminology already indicates. We retain the printed wording and record the slip.

Démonstration. Prouvons a), dans le cas le plus gênant, où on suppose seulement que u est localement faiblement cofiltrant. On applique II 7.4 (p. 92), qui nous dit que $\forall i \in \mathrm{Ob}\, I$, considérant le morphisme induit par u

16. inutile de parler de "faiblement" localement cofiltrant, car il s'agit d'ensembles ordonnés où il n'y a pas à distinguer entre notions forte et faible. [52]

$$I/i \longrightarrow I'/i' \qquad (\text{où } i' = u(i)),$$

celui-ci est *(globalement)* faiblement cofiltrant, et que pour tout ouvert U de I/i, désignant par U' l'ouvert engendré par son image, le morphisme induit $v : U \longrightarrow U'$ est encore faiblement cofiltrant, et a fortiori *cofinal.* Soit alors $f' : X' \longrightarrow Y'$ une flèche dans $C_{I'\circ}$, prouvons que

$$\underbrace{u^*(f')}_{f} \;:\; \underbrace{u^*(X')}_{X} \longrightarrow \underbrace{u^*(Y')}_{Y}$$

est dans $C_{I\circ}$, i.e. que pour tout $i \in I$, et U ouvert de I/i, dans le diagramme canonique p. 27

$$(*) \qquad \begin{array}{ccc} X_U & \longrightarrow & X(i) \\ \downarrow & \text{cocart.} & \downarrow \\ Y_U & \longrightarrow & Y_{U,i} \\ & & \downarrow{\scriptstyle v_{U,i}} \\ & & Y(i)\ , \end{array}$$

la flèche $v_{U,i}$ est dans C. Le diagramme $(*)$ s'envoie dans le diagramme similaire pour [53] $f' : X' \longrightarrow Y'$, $i' = u(i)$, et l'ouvert $U' \subset I'/i'$ engendré par $u(U)$:

$$(**) \qquad \begin{array}{ccccc} X_U & \xrightarrow{\quad} & X(i) & & \\ {\scriptstyle c_U^{X'}}\downarrow & \searrow^{f_U} & \text{cocart.}\downarrow & \searrow & \\ & Y_U & \longrightarrow & Y_{U,i} & \xrightarrow{v_{U,i}} \\ & {\scriptstyle c_U^{Y'}}\downarrow & \text{id}\downarrow & {\scriptstyle c^{f'}_{U,i}}\downarrow & \\ X'_{U'} & \xrightarrow[\text{cocart.}]{} & X'(i') & & Y(i) \\ & \searrow^{f'_{U'}} & \downarrow & \searrow & \downarrow{\scriptstyle \text{id}} \\ & & Y'_{U'} & \longrightarrow & Y'_{U',i'} \\ & & & \searrow^{v_{U',i'}} & Y'(i')\ . \end{array}$$

Par cofinalité, les deux flèches verticales $c_U^{X'}$ et $c_U^{Y'}$ sont des isomorphismes, en plus de $X(i) \xrightarrow{id} X'(i')$, donc la quatrième flèche du cube, $c^{f'}_{U,i}$, est un isomorphisme. Comme $v_{U',i'}$ est dans C, $v_{U,i}$ aussi, on gagne.

Commentary · Examples [ms. 53]

A constant point which is not a cofibrant diagram

The condition on restriction has a visible effect even in sets. Take $\mathcal{M} = \mathsf{Set}$, let C be the injections, and let every map be a weak equivalence. This satisfies the four axioms on page 39: injections are stable under pushout and composition, and any map factors as its source identity followed by a weak equivalence. All objects are cofibrant. The example concerns the cofibration structure; its deliberately indiscriminate weak equivalences play no role in the computation.

Let I have three objects a, b, t, with $a < t$ and $b < t$, and no relation between a and b. For the unique map $q : I \to e$, restrict the one-point set $*$ to the constant diagram. At either minimal object the latching map is $\varnothing \to *$, an injection. At t it is

$$* \amalg * \longrightarrow *,$$

the fold map, which is not injective. Thus q^* does not preserve cofibrant objects. The comma category $(* \downarrow q) = I$ is connected but not cofiltered: a and b have no common predecessor. This is exactly the obstruction detected by the hypothesis of Proposition 3.2(a).

A cofibrant replacement can be written without any choices. Put $Q_a = Q_b = *$, $Q_t = \{a, b\}$, and use the two inclusions as transition maps. The relative boundary at t is the identity of the

two-point set. The obvious map $Q \to \mathrm{const}(*)$ is objectwise a weak equivalence. Its colimit is Q_t, whereas the colimit of the constant diagram is $*$. Those two objects are equivalent for the chosen W, but the two strict diagrams carry different attaching information. The calculation explains why the proof of Theorem 3.7 must work even when ordinary restriction fails to preserve cofibrancy.

For comparison, if $I = [1]$, the boundary at its final vertex is a single copy of $*$. Its map to the final value is the identity, and the constant point is cofibrant. The passage from one predecessor to two disconnected predecessors is already sufficient to destroy preservation. Neither the cardinality of the object values nor a complicated homotopy theory is responsible.

Prouvons b). Soit donc

$$f \ : \ X \longrightarrow Y$$

une flèche dans C_I, et soit

$$f' \ = \ u_!(f) \ : \ u_!(X) = X' \longrightarrow u_!(Y) = Y',$$

prouvons que c'est dans $C_{I'^\circ}$, donc que pour deux ouverts $U' \subset V'$ de I' (où on peut supposer [54]
que $V' = I'/i'$), dans le diagramme-base de $(**)$, $v_{U',V'}$ soit dans C. Soit $U = u^{-1}(U')$, $V = u^{-1}(V')$, et considérons le diagramme similaire à $(**)$, défini par les deux inclusions d'ouverts $U' \subset V'$, $U \subset V$. Cette fois, comme la formation de $u_!$ commute aux restrictions aux ouverts, par transitivité des $\varinjlim$, les trois flèches vertivales $c^X_{U'}$, $c^Y_{U'}$, $c^X_{V'}$ sont des isomorphismes, donc aussi $c^f_{U',V'}$. De même, la flèche $Y_V \longrightarrow Y'_{V'}$ (à la place de $Y(i) \longrightarrow Y'(i')$) est un isomorphisme. Comme $v_{U,V} \in C$, il en est de même de $v_{U',V'}$, ce qui prouve que $f' \in C_{I'^\circ}$. On procède de même pour TC_I. Enfin, l'assertion que $u_!$ applique $\mathcal{M}(I^\circ)_\mathrm{c}$ dans $\mathcal{M}(I'^\circ)_\mathrm{c}$ résulte des définitions et de $u_!(C_{I^\circ}) \subset C_{I'^\circ}$.

Commentary · Exposition [ms. 54]

Left Kan extension transports the whole relative boundary

The asymmetry between the two parts of Proposition 3.2 is deliberate. Restriction changes a boundary, but left Kan extension is designed to preserve the colimit describing that boundary. For any lower set $V' \subseteq I'$ and any diagram $X : I \to \mathcal{M}$, there is a canonical isomorphism

$$\mathrm{colim}_{V'}\, u_!X \ \simeq \ \mathrm{colim}_{u^{-1}V'}\, X,$$

whenever the indicated colimits exist. One can see it directly from cocones. An object contributing to $(u_!X)(b)$ is a pair $(i, u(i) \leq b)$ with $b \in V'$. Since V' is lower, $u(i) \in V'$. Conversely every $i \in u^{-1}V'$ contributes via $b = u(i)$ and the identity arrow. These identity contributions determine every cocone, and their transition relations are exactly those of the restricted diagram on $u^{-1}V'$.

Apply this twice, to $U' \subseteq V'$ and to X, Y. The relative boundary for $u_!f$ is thereby identified with

$$Y_{u^{-1}U'} \amalg_{X_{u^{-1}U'}} X_{u^{-1}V'} \longrightarrow Y_{u^{-1}V'}.$$

Theorem 2.1(1) puts this map in C. Applying the same reasoning to $TC = C \cap W$ puts it in TC if $f \in (TC)_I$. Taking X to be the initial diagram proves preservation of cofibrant objects. This is why the strong definition involving every open subset was convenient before well-foundedness was imposed: the inverse images of the target opens are arbitrary opens of the source, not necessarily single principal ideals.

There is a small logical distinction in the next corollary. The source may be assumed well-founded while the target is not. The equality

$$(TC)_{I'} = C_{I'} \cap W_{I'^\circ}$$

cannot then simply be imported from the artinian part of Theorem 2.1. The proof needs only the inclusion from left to right. This inclusion is automatic for the strong definition: take the empty lower set inside I'/i' to see that each component lies in TC. Thus the unnecessary equality in the proof of Corollary 3.3 can be replaced by this inclusion, with no change to its conclusion. Existence of the ordinary Kan extension remains a separate hypothesis when one leaves the cocomplete setting.

Corollaire 3.3 (de prop. 3.2 et th. 3.1). *Soit $u : I \longrightarrow I'$ comme dans* 3.2. *Supposons de plus I artinien, et qu'on soit pour $\mathcal{M}$, W, C, I dans l'une des conditions* a), b) *du théorème* 3.1. *Alors $u_!$ induit une flèche* p. 28 [55]

$$u_{!\,\mathrm{cof}} \ : \ \mathcal{M}(I^\circ)_\mathrm{c} \longrightarrow \mathcal{M}(I'^\circ)_\mathrm{c} \ \subset \ \mathcal{M}(I'^\circ)$$

qui transforme flèches de $W(I^\circ)$ en flèches de $W(I'^\circ)$ (i.e. compatible avec les localiseurs). Il induit donc un foncteur sur les localisés, d'où un

$$u_!^W \ : \ \mathrm{Ho}_W(I^\circ) \longrightarrow \mathrm{Ho}_W(I'^\circ) \qquad (17). \tag{3.3.1}$$

Démonstration. Par le théorème 3.1, on sait que $\mathcal{M}(I^\circ)$, $W(I^\circ)$, $C(I^\circ)$ est une structure de catégorie de modèles à cofibrations. On sait par ailleurs que $TC_{I^\circ} = C_{I^\circ} \cap W(I^\circ)$, par th. 2.1. 3° (p. 27). Donc par 3.2, $u_!$ applique $C_{I^\circ} \cap W(I^\circ)$ dans $TC_{I'^\circ} = C_{I'^\circ} \cap W(I'^\circ)$, et a fortiori, en des flèches dans $W(I'^\circ)$. Par ailleurs, par la théorie des catégories à cofibrations, on trouve que toute flèche f entre objets cofibrants est composée $f = pi$, où $i \in C_{I^\circ}$, et où p est dans $W(I^\circ)$, et de plus *inverse à gauche* d'une flèche dans $C_{I^\circ} \cap W(I^\circ)$, i.e. d'une flèche de $(TC)_{I^\circ}$. Quand $f \in W(I^\circ)$, alors $i \in W(I^\circ)$, donc $i \in W(I^\circ) \cap C_{I^\circ} = (TC)_{I^\circ}$. Au total, toute [56] flèche $f \in W(I^\circ)_\mathrm{c}$ est composée pi dans $\mathcal{M}(I^\circ)_\mathrm{c}$, où $i \in (TC)_{I^\circ}$, et p est inverse à gauche d'une $j \in (TC)_{I^\circ}$. Comme $u_!$ transforme $(TC)_{I^\circ}$ en $(TC)_{I'^\circ}$, il transforme f en $f' = p'i'$, avec $i' \in (TC)_{I'^\circ} \subset W_{I'^\circ}$, et p' inverse à gauche d'une $j' = u_!(j)$ qui est également dans $W_{I'^\circ}$. Mais alors $i' \in W(I'^\circ)$, donc aussi $f' = p'i' \in W_{I'^\circ}$, ce qui prouve que $u_!(W(I^\circ)_\mathrm{c}) \subset W(I'^\circ)$. La dernière assertion provient du lemme suivant, appliqué à $(\mathcal{M}(I^\circ), W(I^\circ), C_{I^\circ})$ au lieu de $(\mathcal{M}, W, C)$:

Lemme 3.4. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations. Alors l'inclusion $\mathcal{M}_\mathrm{c} \subset \mathcal{M}$ induit une équivalence*

$$\mathcal{M}_\mathrm{c} W_\mathrm{c}^{-1} \xrightarrow{\sim} \mathcal{M} W^{-1} = \mathrm{Ho}_W \ .$$

Nous donnerons la démonstration (par AQT `[âne qui trotte]`) ultérieurement.

Cela achève la démonstration de 3.3.

Commentary · Exposition [ms. 56]

The mapping-cylinder argument behind Corollary 3.3

Grothendieck must now pass from preservation of trivial cofibrations to preservation of all weak equivalences between cofibrant diagrams. The useful fact is a one-sided factorization, stronger than the bare factorization axiom: a map between cofibrant objects factors as a cofibration followed by a weak equivalence which has a trivial-cofibration section. Here is a complete construction in the notation of page 39.

For a cofibrant object A, factor its fold map as

$$A \amalg A \xrightarrow{(d_0,d_1)} IA \xrightarrow{r} A, \qquad (d_0, d_1) \in C, \quad r \in W.$$

Each inclusion $A \to A \amalg A$ is a pushout of $\varnothing \to A$, hence belongs to C. Thus $d_0, d_1 \in C$. The equations $rd_0 = rd_1 = \mathrm{id}_A$ and two-out-of-three show $d_0, d_1 \in TC$. Given $f : A \to B$ with B cofibrant, form

$$M_f = B \amalg_{A,d_0} IA.$$

The inclusion $j : B \to M_f$ is a pushout of d_0, so $j \in TC$. There is a map $p : M_f \to B$ whose restrictions are id_B and fr, and $pj = \mathrm{id}_B$ implies $p \in W$. The other endpoint defines $i : A \to M_f$ with $pi = f$.

To check that i is a cofibration, first observe that $B \amalg A \to M_f$ is the pushout of $(d_0, d_1) : A \amalg A \to IA$ along (f, id_A). The map $A \to B \amalg A$ is a cofibration because B is cofibrant. Their composite is

17. **N.B.** Cette flèche est définie, même sans supposer I' ordonné, car $u_{!\,\mathrm{cof}}$ transforme encore flèches de $W(I^\circ)$ en flèches de $W(I'^\circ)$. Cf. commentaire p. 86.

i. Therefore $f = pi$ has all the claimed properties. If $f \in W$, then $i \in W$ by two-out-of-three, so both i and the section j are trivial cofibrations.

Now let F be any functor carrying trivial cofibrations between cofibrant objects to weak equivalences. Applied to this factorization, $F(j)$ is a weak equivalence and $F(p)F(j) = \mathrm{id}$, so $F(p)$ is one also. If $f \in W$, then $F(i) \in W$, and hence $F(f) = F(p)F(i) \in W$. Taking $F = u_!$ gives the assertion used here. The printed last inference on page 56 repeats $i' \in W$ where the new information is $p' \in W$; the section equation supplies the missing inference. This is a correct argument obscured by a repeated letter, not a failure of Corollary 3.3.

Commentary · Later developments [ms. 56]

What is required of cofibrant replacement

The deferred Lemma 3.4 ensures that the candidate just constructed is defined on the intended homotopy category. Essential surjectivity is immediate: factor $\varnothing \to X$ as $\varnothing \to QX \xrightarrow{q} X$, with QX cofibrant and $q \in W$. Full faithfulness requires more than one such choice for each object. A morphism in a localization is represented by a finite zigzag, and its relations must remain valid after replacing the objects appearing in that zigzag.

The factorization argument of pages 45–49 supplies these compatible replacements on finite direct diagrams. In particular it replaces a span or a composable pair together with its structure maps, rather than replacing its vertices independently. The mapping-cylinder argument in the preceding block compares different choices after passing to homotopy. These are the ingredients of the usual cofibrant-approximation proof. A precise later statement is Rădulescu-Banu (2009, Theorem 6.1.6(1)); its proof passes through the approximation theorem and the comparison lemmas in §6.1. It applies already to precofibration categories, so no infinite-composition hypothesis is needed for Lemma 3.4.

This also clarifies the meaning of the "canonical" functor in (3.3.1). If

$$j_I : \mathcal{M}^I_{\mathrm{c}}[W^{-1}] \longrightarrow \mathcal{M}^I[W^{-1}]$$

is the equivalence and $\overline{u_!}$ is the functor induced on the cofibrant subcategory, then the candidate is $\overline{u_!}\, j_I^{-1}$. Its uniqueness is understood relative to the specified comparison on cofibrant objects. One does not obtain a preferred strict functor before making choices, and one does not obtain an adjunction merely by writing this composite. The next pages are devoted to that second issue.

3.5. Il se pose la question si le foncteur $u_!^W$ de (3.3.1) est bien un adjoint à gauche du foncteur en sens inverse u_W^*, induit par u^*. Notons que c'est le cas si u^* transforme $\mathcal{M}(I'^{\circ})_{\mathrm{c}}$ en $\mathcal{M}(I^{\circ})_{\mathrm{c}}$, et quand on suppose que I' est également artinien et satisfait également à l'une des hypothèses a) b) du théorème 3.1. Car alors $(\mathcal{M}(I^{\circ}), W(I^{\circ}), C_{I^{\circ}})$ est une catégorie à cofibrations par le théorème 3.1, on peut lui appliquer le lemme précédent, donc u_W^* et $u_!^W$ se réalisent à partir d'un couple de foncteurs [57]

$$\mathcal{M}(I^{\circ})_{\mathrm{c}} \underset{u_{\mathrm{c}}^*}{\overset{u_!^{\mathrm{c}}}{\rightleftarrows}} \mathcal{M}(I'^{\circ})_{\mathrm{c}} \quad ,$$

lesquels sont visiblement adjoint l'un de l'autre (puisque induits par un couple de foncteurs adjoints $u_!$, u^*), et de plus compatibles aux localiseurs. Donc on peut appliquer le lemme des foncteurs adjoints pour les localisés. Pour résumer : p. 29

Corollaire 3.5. *Supposons que I, I' soient artiniens, et qu'on soit dans l'un des deux cas suivants :*

a) *Les espaces* $\mathrm{Esp}(I)$, $\mathrm{Esp}(I')$ *sont noethériens (p.ex. I, I' finis).* [page 58]

b) *C et $TC = C \cap W$ sont stables par composition transfinie.*

Supposons de plus que u^* applique $\mathcal{M}(I'^{\circ})_{\mathrm{c}}$ dans $\mathcal{M}(I^{\circ})_{\mathrm{c}}$, ce qui est le cas notamment si u est localement cofiltrant, i.e. les ensembles ordonnés localisés $i'\backslash I$ sont localement cofiltrants. Alors le foncteur $u_!^W$ construit dans (3.3.1) est adjoint à gauche de u_W^*.

Remarque 3.6. Quand on a un couple de foncteurs adjoints

$$\mathcal{C} \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{C}' \quad ,$$

avec les localiseurs W, W' de $\mathcal{C}$, $\mathcal{C}'$, u^* compatible aux localiseurs, et si de plus on a une sous-catégorie pleine $\mathcal{C}_0$ de $\mathcal{C}$ telle que

a) $u_!|\mathcal{C}_0$ transforme flèches de W en flèches de W', et

b) l'inclusion $\mathcal{C}_0 \hookrightarrow \mathcal{C}$ induit une *équivalence* $\mathcal{C}_0 W_0^{-1} \xrightarrow{\sim} \mathcal{C}W^{-1}$ (où $W_0 = W \cap \mathrm{Fl}(\mathcal{C}_0)$),

on trouve deux foncteurs canoniques

$$\mathcal{C}W^{-1} \underset{u_W^*}{\overset{u_!^W}{\rightleftarrows}} \mathcal{C}'W'^{-1} \quad ,$$

où $u_!^W$ est défini seulement à isomorphisme canonique près.

Contrairement à ce que prétend Anderson, il n'est par vrai en général que ces deux foncteurs soient adjoints, sans hypothèses supplémentaires. Voici un contrexemple : on prend $\mathcal{C} = \mathcal{C}'$, $u_! = u^* = \mathrm{id}_{\mathcal{C}}$, $W' = \mathrm{Is}(\mathcal{C})$, $W = \mathrm{Fl}(\mathcal{C})$, donc u_W^* est le foncteur canonique [page 59]

$$\underbrace{\Pi\mathcal{C}}_{\text{groupoïde fondamental}} \longleftarrow \mathcal{C} \ .$$

Supposons que $\mathcal{C}$ soit 1-connexe, i.e. $\Pi\mathcal{C}$ soit équivalente à e. Soit $e \in \mathrm{Ob}\,\mathcal{C}$ tel que $\mathrm{End}_{\mathcal{C}}(e)$ soit le groupe unité, et prenons $\mathcal{C}_0 = \{e\}$. Alors l'hypothèse sur $\mathcal{C}_0$ est satisfaite. Mais si $u_!^W$ était adjoint à gauche de u_W^*, e serait objet initial de $\mathcal{C}$. Par exemple si $\mathcal{C}$ a un objet final e (ce qui implique que $\mathcal{C}$ est 1-connexe et e rigide), cet objet final devrait être initial si Anderson avait raison avec "it is clear that …".

Commentary · Examples [ms. 59]

The two-object test for an alleged derived adjunction

Grothendieck's counterexample isolates a genuine logical error: an equivalence after localization of a chosen subcategory does not by itself make that subcategory a suitable domain for deriving an adjunction. Everything can be computed in the walking arrow $\mathcal{C} = [1] = \{0 \to 1\}$. Put $\mathcal{C}' = \mathcal{C}$, let both ordinary functors be the identity, take W to be all arrows, and take W' to be the isomorphisms. The right-hand identity functor is compatible with these localizers, because it sends an isomorphism to an arrow of W.

Localizing $\mathcal{C}$ at W gives the groupoid with two isomorphic objects and a unique map between any ordered pair; it is equivalent to e. Localizing $\mathcal{C}'$ at W' leaves $[1]$ unchanged. Choose $\mathcal{C}_0 = \{1\}$. Its localization is e, and its inclusion into $\mathcal{C}$ induces an equivalence after localization. The restricted identity functor sends its only arrow to an isomorphism. Thus both hypotheses of Remark 3.6 hold.

After identifying $\mathcal{C}[W^{-1}]$ with e, the constructed candidate $L : e \to [1]$ selects 1, and the localized restriction $R : [1] \to e$ is the unique functor. If $L \dashv R$, then

$$\mathrm{Hom}_{[1]}(1, 0) \simeq \mathrm{Hom}_e(*, *).$$

The left side is empty and the right side has one element. Hence there is no such adjunction. In fact L is right adjoint to R, since 1 is terminal. The defect is the direction in which the chosen representative can map to other objects.

There is a second computation, needed again on page 68. Choose instead $\mathcal{C}_0 = \{0\}$. The candidate now selects the initial object and is indeed left adjoint to R. Nevertheless R is not fully faithful: the map

$$\mathrm{Hom}_{[1]}(1, 0) = \varnothing \longrightarrow \mathrm{Hom}_e(*, *) = \{*\}$$

is not surjective. Before localization the right adjoint was the identity, hence fully faithful; afterwards it has acquired an extra morphism because the two relative structures invert different arrows. This

second version establishes exactly the warning about full faithfulness, without confusing it with the first version's failure of adjunction.

The mathematical counterexample is complete and independent of the historical attribution. We verified the metadata of Anderson (1978), but did not obtain a readable copy of the original argument containing the phrase criticized here. We therefore attribute the criticism to this manuscript and do not attach it to an unverified theorem number in Anderson's publication.

Commentary · Later developments [ms. 59]

The comparison map is part of a derived functor

What the two-point example lacks is the universal comparison carried by a derived functor. If $F : \mathcal{C} \to \mathcal{C}'$ and γ, γ' are the localization functors, a total left derived functor comes with

$$\lambda : (\mathbf{L}F)\gamma \longrightarrow \gamma' F,$$

and a universal property for that transformation. A functor obtained by transporting $F|\mathcal{C}_0$ across an equivalence of localizations is not automatically equipped with this universal property. The distinction is substantive even though both constructions are often described informally as "apply F to a replacement."

For an adjunction $F \dashv G$, absolute total derived functors $\mathbf{L}F$ and $\mathbf{R}G$ form an adjunction. The adjective "absolute" means that the defining Kan-extension property survives postcomposition by every functor. Maltsiniotis (2007) proves the adjunction theorem formally from these properties, including compatibility with the original unit and counit; the proof does not require a particular model-category factorization. This is a later precise repair of the general inference under discussion.

In Grothendieck's terminal-object example there cannot even be a comparison λ of the indicated direction. Its component at 0 would have to be a map $1 \to 0$ in [1]. Thus the false assertion is not a counterexample to the modern derived-adjunction theorem. It shows why the comparison and its universal property cannot be suppressed when passing from a homotopically dense subcategory to a derived functor. The following mapping-cone proof constructs the needed adjunction in the geometric situation at hand instead of relying on that suppressed inference.

[page 60]

Néanmoins :

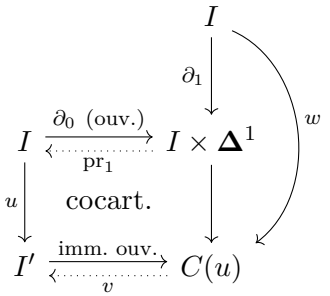
Théorème 3.7. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations stable par $\varinjlim$ finies, et $u : I \longrightarrow I'$ une application croissante entre ensembles ordonnés artiniens. On suppose qu'on est sous l'une des hypothèses suivantes.*

a) $\mathrm{Esp}(I)$, $\mathrm{Esp}(I')$ *sont noethériens, p.ex. I, I' finis.*

b) *C et $TC \overset{déf}{=} C \cap W$ sont stables par composition transfinie, et $\mathcal{M}$ stable par petites $\varinjlim$ filtrantes.* p. 30

Alors le foncteur $u^W_! : \mathrm{Ho}_W(I^\circ) \longrightarrow \mathrm{Ho}_W(I'^\circ)$ construit dans (3.3.1) *est adjoint à gauche du foncteur u^*_W en sens opposé.*

Démonstration. On utilise le diagramme bien connu



pour factoriser u en

$$u \;=\; vw\ ,$$

où w est une *immersion fermée,* et v est déduit de $\mathrm{pr}_1 : I \times \mathbf{\Delta}^1 \longrightarrow I$ par push-out. C'est une rétraction du mapping-cone C(u) sur l'ouvert I', dont le complémentaire fermé s'identifie à I via w.

[page 61]

Le foncteur de recollement entre I', I

$$H \ : \ I'^{\circ} \times I \ \longrightarrow \ \mathrm{Ens}$$

est donné par

$$(*) \qquad \underbrace{H(i',i)}_{=\mathrm{Hom}_{C(u)}(i',i)} \ = \ \mathrm{Hom}_{I'}(i', u(i)) \ .$$

La rétraction v de $C(u)$ dans I' applique i dans $u(i)$, et une flèche $i' \longrightarrow i$ correspondant à $i' \longrightarrow u(i)$ dans I' via $(*)$, est transformé par v en cette même flèche $i' \longrightarrow u(i)$. Calculons, pour i'_0 donné, la catégorie $i'_0\backslash\Big(C(u),\ v : C(u) \longrightarrow I'\Big)$, des couple d'un objet ξ de $C(u)$ et d'une flèche $i'_0 \longrightarrow v(\xi)$ dans I'. On voit que $i'_0\backslash C(u)$ admet l'ouvert $i'_0\backslash I'$, avec le complémentaire fermé $i'_0\backslash I$,

$$\begin{array}{ccccc} & & & & \text{fermé} \\ & & & & \text{complémentaire} \\ & & & & \text{de } I' \\ I' & \underset{\text{ouvert}}{\hookrightarrow} & C(u) & \hookleftarrow & I \\ & & \downarrow & & \\ & & I' \ , & & \end{array}$$

ces deux catégories étant recollées par le bifoncteur

$$\begin{array}{rcl} (i'_0\backslash I')^{\circ} \times (i'_0\backslash I) & \longrightarrow & \mathrm{Ens} \\ \Big((\alpha : i'_0 \longrightarrow i'), (\beta : i'_0 \longrightarrow u(i), i)\Big) & \longmapsto & \mathrm{Hom}_{i'_0\backslash I'}(i', u(i)) \end{array}$$

induit par $(*)$; i.e. les flèches de $(\alpha : i'_0 \longrightarrow i') \in \mathrm{Ob}\,(i'_0\backslash I')$ dans $(\beta : i'_0 \longrightarrow u(i)) \in i'_0\backslash I$ sont les flèches $i' \xrightarrow{\gamma} u(i)$ telles que le triangle

p. 31

$$\begin{array}{ccc} & i'_0 & \\ {\scriptstyle\alpha}\swarrow & & \searrow{\scriptstyle\beta} \\ i' & \xrightarrow{\ \gamma\ } & u(i) \end{array}$$

commute.

[page 62]

Si $i' = i'_0$, $\alpha = \mathrm{id}_{i'_0}$, alors cette commutativité revient à $\gamma = \beta$, ce qui signifie qu'il y a une flèche et une seule de α dans (i, β). Cela montre donc que $(i'_0, \mathrm{id}_{i'_0})$ est un objet initial de $i'_0\backslash C(u)$. On a donc prouvé le lemme :

Lemme 3.8. *Soit $u : I \longrightarrow I'$ une flèche dans* (Cat)*, factorisons-le en*

$$I \xrightarrow{w} C(u) \xrightarrow{v} I' \ ,$$

où $C(u)$ est le mapping-cone défini page 60*, w une immersion fermée de complémentaire I', v la rétraction canonique de $C(u)$ sur I'. Ceci posé, pour tout $i'_0 \in \mathrm{Ob}\, I'$, la catégorie $i'_0\backslash(C(u), v)$ admet un objet initial – donc $v : C(u) \longrightarrow I'$ est (tout ce qu'il y a de) cofiltrante.*

Commentary · Exposition [ms. 62]

The mapping cone as a category of two pieces

Grothendieck's factorization separates two operations for which the adjunction can be controlled. It is useful to describe the category $C(u)$ before reading its comma categories. Its objects are a disjoint copy of the objects of I' and a copy of those of I. Within each copy the old arrows remain. There are no arrows from the I-copy to the I'-copy, and the mixed arrows are

$$\mathrm{Hom}_{C(u)}(i', i) = \mathrm{Hom}_{I'}(i', u(i)).$$

Composition of a mixed arrow with arrows on either side is induced by composition in I' and functoriality of u. This is the collage of the displayed profunctor H. The open–closed language and its profunctor description were developed earlier; the commentary on Chapter XII, manuscript pp. 59–68, explains that construction in its general form.

Let $j : I' \hookrightarrow C(u)$ denote the open inclusion, $w : I \hookrightarrow C(u)$ the closed inclusion, and let $v : C(u) \to I'$ equal the identity on I' and u on I. The mixed-hom formula gives, naturally in $b \in I'$ and $c \in C(u)$,

$$\mathrm{Hom}_{C(u)}(j(b), c) \simeq \mathrm{Hom}_{I'}(b, v(c)).$$

Thus $j \dashv v$, with $vj = \mathrm{id}$. This elementary adjunction is another way to recognize the initial object of each $(b \downarrow v)$: it is $(j(b), \mathrm{id}_b)$. For an object $(c, \beta : b \to v(c))$, the unique arrow from that initial object is the mixed arrow corresponding to β, or the corresponding arrow within I' when $c \in I'$. The same hom-set description proves its uniqueness, including compatibility with every triangle in the comma category.

If I and I' are posets, the collage is a poset: each mixed hom-set is empty or a singleton, and a mixed arrow cannot be followed by an arrow back to the first piece. If both posets are well-founded, so is the collage. A decreasing sequence can cross from the closed piece into the open piece at most once; any infinite decreasing tail would then lie entirely in one of the two original posets. In the finite branch the collage is finite. More generally an increasing chain of lower sets in the collage stabilizes as soon as its intersections with both pieces stabilize, which proves the noetherian-topology assertion directly.

The comma calculation therefore has exactly the strength used in Corollary 3.5. It proves that v is locally cofiltered, so v^* preserves cofibrant diagrams. It is not an invocation of an unspecified homotopy equivalence of the two indexing categories. The original adjunction $j \dashv v$ already identifies the relevant initial objects on the level of categories.

Nous pouvons maintenant prouver 3.7. Tout d'abord, les hypothèses faites sur I, I' se transportent à $C(u)$ (hypothèse artinienne, et éventuellement hypothèse noethérienne sur $\mathrm{Esp}(C(u))$). Donc on peut appliquer les résultats précédents, notamment 3.5, aux flèches v, [page 63]
w. Le lemme précédent montre bien que v satisfait aux conditions de 3.5, donc le foncteur $v_!^W$ construit dans 3.3.1 est adjoint à gauche de v_W^*. Considérons d'autre part le foncteur $w_!^W$ construit dans 3.3.1, je dis que la même conclusion s'applique. Cela provient du fait que w est une immersion fermée. En effet, dans ce cas il est immédiat que

$$w_! \ : \ \mathcal{M}(I^\circ) \ \longrightarrow \ \mathcal{M}(I'^\circ), \qquad \text{où } I = C(u),$$

est compatible avec les localiseurs, puisque l'on a

$$\begin{cases} w_!(X)(j) \ \simeq \ X(j) & \text{si } j \in I, \\ w_!(X)(j) \ \simeq \ \emptyset_{\mathcal{M}} & \text{si } j \in J \backslash I, \end{cases}$$

par le calcul des fibres

$$w_!(X)(j) \ \simeq \ \varinjlim_{i \in I/j} X(i) \ .$$

Donc la définition de $w_!^W$ donnée dans 3.3.1 revient en fait à passer directement aux catégories localisées pour $w_!$, sans avoir à se restreindre à $\mathcal{M}(I^\circ)_{\mathrm{c}}$, et le lemme des foncteurs adjoints p. 32
nous dit que ce foncteur est adjoint à gauche de w_W^*. [page 64]

Comme on a évidemment

$$u_!^W \;\simeq\; v_!^W \circ w_!^W \;, \qquad\qquad u_W^* \;\simeq\; w_W^* \circ v_W^* \;,$$

on en conclut que $u_!^W$ est bien adjoint à gauche de u_W^*, q.e.d.

Commentary · Exposition [ms. 64]

The two adjunctions and their composite

The closed inclusion w supplies the other half of Theorem 3.7 for a particularly transparent reason. A closed full subposet is upward closed. Hence, for $c \notin I$, the comma category $(w \downarrow c)$ is empty: an arrow from an object of the closed piece to c would force c into that piece. For $c \in I$, the same comma category has terminal object (c, id). Therefore

$$(w_! X)(c) = \begin{cases} X(c), & c \in I, \\ \varnothing_{\mathcal{M}}, & c \notin I. \end{cases}$$

This is extension by the initial object, with the unique maps out of that object on mixed arrows. It preserves every objectwise weak equivalence; on the complement its components are identities. Restriction w^* preserves them as well.

Whenever both functors of an ordinary adjunction preserve the relevant weak equivalences, the unit and counit descend to localization. Their naturality continues to hold for formally inverted arrows, and their triangle identities remain identities after applying localization. This proves the localized adjunction in the closed-inclusion case. It also shows that its left adjoint agrees with the candidate defined using cofibrant replacement: $w_!$ already sends the replacement comparison itself to a weak equivalence.

For v, Proposition 3.2 and the preceding comma calculation put both functors on the full cofibrant subcategories. There they remain adjoint, since fullness leaves the old hom-set bijections unchanged. Both are homotopical on these subcategories, so their adjunction descends. Lemma 3.4 transports it to the homotopy categories of all diagrams. This is the restricted adjunction of Corollary 3.5, with its hypotheses now verified.

Finally, ordinary Kan extensions give $(vw)_! \simeq v_! w_!$. On a cofibrant I-diagram, $w_!$ again produces a cofibrant diagram, so no intermediate replacement changes the derived composite. Consequently the candidate $u_!^W$ is canonically isomorphic to $v_!^W w_!^W$. The right adjoints satisfy $u_W^* = w_W^* v_W^*$ simply by restriction. Composing the two established adjunctions gives Theorem 3.7. There is no use of the conclusion to establish this comparison: preservation of cofibrancy supplies the comparison before the final adjunction is deduced.

The symbols in the printed extension-by-initial-object formula require a local repair. Its target is $C(u)$, so the parenthetical "$I = C(u)$" should identify the target, and the subsequent J denotes that same target. The categorical construction above makes both readings unambiguous while leaving the edition's text intact.

Corollaire 3.9. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations. Alors le prédérivateur $\mathbf{D}_W$, restreint à la catégorie* Diag_0 *formée des catégories I* ordonnées *artiniennes et telles que* $\mathrm{Esp}(I)$ *soit noethérien, satisfait aux axiomes* Dér 1 (décomposition), Dér 2 (localisation), Dér 3 (objet final), Dér 4 bis (existence de $u_!$), Dér 5 bis (mode de calcul des $u_!$), Dér 6 (exactitude des carrés W-cocartésiens). *Si C, $TC = C \cap W$ sont stables par composition transfinie, on a le même résultat sur la catégorie* Diag *formée des catégories I ordonnées artiniennes sans plus.*

Commentary · Later developments [ms. 64]

The derivator conclusion and its precise scope

The point of Corollary 3.9 is that homotopy colimits have acquired the two properties that make them usable: an adjunction characterizes them, and a comma-category formula computes their values. In the covariant notation used here the latter is

$$(\mathbf{L}u_! X)_{i'} \simeq \mathrm{hocolim}_{(u \downarrow i')} X.$$

The category $(u \downarrow i')$ records arrows $u(i) \to i'$. For posets it is the lower subposet $u^{-1}(I'/i')$ of I. Restricting a cofibrant diagram to that lower subposet preserves its latching maps. Thus ordinary pointwise Kan extension applied to a cofibrant replacement computes the right-hand side, and the comparison is compatible with the adjunction already proved.

The convention in this formula must be kept visible when the corollary names the historical prederivator. Put

$$H(I) = \mathrm{Fun}(I, \mathcal{M})[W^{-1}], \qquad \mathbf{D}_W(J) = \mathrm{Fun}(J^{\mathrm{op}}, \mathcal{M})[W^{-1}].$$

Then $H(I) = \mathbf{D}_W(I^{\mathrm{op}})$. The preceding induction and the displayed comma formula directly concern H. Expressed for $\mathbf{D}_W$, they concern the opposite indexing categories. This makes no difference to the finite-poset domain, which is closed under opposites. In the infinite branch, the opposite of an artinian poset need not be artinian. Thus the induction as written directly proves the presheaf statement on opposites of artinian posets; it must not simply be relabelled as the same induction on artinian presheaf indices. Under the stated cocomplete and transfinite hypotheses, the later ABC theorem discussed below supplies all small shapes and hence also the asserted artinian domain. This is a qualification of the route of proof, not a counterexample to the corollary. The switch back to presheaves becomes explicit in paragraph 3.11 and determines the final comma square.

The statement retains the assumptions of Theorem 3.7. In particular, the infinite branch needs the indicated small colimits in $\mathcal{M}$ as well as transfinite closure. The abbreviated sentence in Corollary 3.9 must not be read as deriving arbitrary small colimits from the four finite cofibration axioms. The historical list Dér 1–6 also needs its own numbering: Chapter I, manuscript pages 43 and 67–68, keeps Dér 2 as conservativity of evaluations and inserts Dér 3 before moving Kan adjunctions to Dér 4 and their pointwise formulas to Dér 5 (Grothendieck, 1990–1991a).

The “objet final” version of Dér 3 requires constants to be fully faithful when the indexing category has a final object. The final object is in the indexing category. This part has a direct proof for the diagram localizations under discussion. In the source’s presheaf convention, if t is final in J, the constant functor $\mathcal{M} \to \mathrm{Fun}(J^{\mathrm{op}}, \mathcal{M})$ is left adjoint to evaluation at t: a map from a constant diagram to X is uniquely determined by its component into X_t. Its unit is the identity. Both functors preserve the respective weak equivalences, so this adjunction and its identity unit descend to localization. The localized constant functor is therefore fully faithful. This proves precisely the final-object clause independently of the construction of general Kan extensions. The cocartesian exactness clause in Dér 6 invokes the Brown homotopy calculus discussed on pages 40–41; its Hom-surjectivity formulation is proved there by a cylinder-and-gluing construction.

For the finite theory, Cisinski (2010, Corollary 2.24) gives a weak right derivator in his convention from a right derivable category, on finite direct categories, satisfying his additional axiom Der 5. His terminology calls the homotopy-colimit side “right.” For the small-diagram theory, Rădulescu-Banu (2009, Theorems 9.6.3, 9.6.5, and 10.3.2) gives the adjunction, pointwise formula, and strong Heller derivator for ABC cofibration categories. The latter axioms explicitly include the required coproduct and sequential-composition properties. These are precise later realizations of the program, with their hypotheses visible.

The extra ABC closure properties do follow from the stronger transfinite assumptions imposed here. To see the coproduct assertion directly, well-order a small family of maps $f_\alpha : A_\alpha \to B_\alpha$ in C. Start with $\coprod_\alpha A_\alpha$, and replace A_α by B_α one summand at a time. At stage β the object is

$$\coprod_{\alpha<\beta} B_\alpha \amalg \coprod_{\alpha\geq\beta} A_\alpha.$$

Each successor map is a pushout of f_β. At a limit stage the displayed object is the colimit of the preceding stages, as is verified by giving a map from each summand into an arbitrary test object. Transfinite closure therefore puts $\coprod f_\alpha$ in C. The same argument with TC proves the trivial-cofibration assertion. Applied to $\varnothing \to A_\alpha$, it also proves cofibrancy of the coproduct when the summands are cofibrant. The sequential-composition axiom is the countable instance of the assumed transfinite closure. Thus the later theorem applies to the actual cocomplete branch, rather than only to a vaguely similar axiom system. The dual of Cisinski (2010, Corollary 6.21) gives another published arbitrary-small-category result under the corresponding homotopical cocompleteness conditions of his Definition 6.1.

For an ABC cofibration category with all objects cofibrant, Lenz (2018, Theorem 3.33) identifies its homotopy prederivator with that of its quasicategory of frames. The all-cofibrant ABC convention

is explicit in his Definition 1.6 and Warning 1.7; it includes infinite-colimit axioms absent from the four finite axioms here. Corollary 1.41 passes from a general ABC category to its cofibrant subcategory at the level of diagram localizations, and Corollaries 4.1 and 4.5 provide the derivator and strength conclusions. Thus this comparison applies within the preceding ABC framework. It preserves coherent diagrams; it does not identify $\mathcal{M}^I[W^{-1}]$ with $\mathrm{Fun}(I, \mathcal{M}[W^{-1}])$.

3.10. Ainsi, pour avoir un dérivateur au plein sens du terme, soit sur Diag_0, soit sur Diag, il ne manque que Dér 4 (existence des u_*), Dér 5 (calcul des u_*) et Dér 6 (exactitude des carrés W-cartésiens). On va les obtenir en principe à l'aide d'une structure de *catégorie à fibrations* $(\mathcal{M}, F, W)$ associé à $(\mathcal{M}, W)$, i.e. l'espèce de structure duale de celle intro- duite page 39. Ces deux structures n'ont pas à être reliées entre elles d'une façon autre que par le fait qu'elles correspondent au même W. Elles sont considérablement plus faciles à construire que des structures de Quillen (W, C, F), où C, F doivent être reliés de façon extrêmement stricte, avec les propriétés $C \longleftrightarrow \underbrace{TF}_{=F\cap W}$, $\underbrace{TC}_{=C\cap W} \longleftrightarrow F$ (dont il n'est pas question dans l'axiomatique de K. S. Brown), et factorisation pour (C, TF) et (TC, F). [page 65]

Commentary · Context [ms. 65]

Two auxiliary structures over one relative category

Grothendieck's observation on page 65 concerns the economy of the axioms. Once the weak equivalences have been fixed, a cofibration structure may organize homotopy colimits and a fibration structure may organize homotopy limits. Their common object is the relative category $(\mathcal{M}, W)$. A Quillen structure imposes further relations between the two distinguished classes: lifting properties and the two complementary factorizations. None of those relations was used in the mapping-cylinder proof above.

This proposal has a precise later form. Cisinski (2010, Definition 2.25 and Corollary 2.28) defines a derivable category by placing compatible left and right derivability structures over the same weak equivalences, and obtains a derivator on finite direct categories. Compatibility here means the common relative category and the stated axioms; it does not insert Quillen's orthogonality relations. The finite result is the appropriate comparison with Grothendieck's paragraph.

There is an indexing issue in the infinite case which the wording "en principe" leaves to the subsequent development. A proof by well-founded latching induction is not automatically a proof by matching induction on the same infinite poset. The opposite of a well-founded poset need not be well-founded: $\mathbb{N}$ is the basic example. Additional fibration hypotheses, existence of the required limits, and a construction on the relevant opposite shapes must therefore be supplied before a two-sided conclusion on all the stated shapes is justified. We record page 65 as a program, not as an already completed proof of the remaining axioms. The later ABC theory treats the two sides with their respective completeness hypotheses.

Mais revenant à la construction des $u_!^W$, il reste le problème de les construire pour une flèche $u : I' \longrightarrow I$ *quelconque* dans Cat. [page 66]

À vrai dire, la construction du foncteur

$$u_!^W \ : \ \mathrm{Ho}_W(I^\circ) \ \longrightarrow \ \mathrm{Ho}_W(I'^\circ) \tag{3.3.1}$$

est valable sans supposer que I' soit une catégorie ordonnée – ça peut être une catégorie quelconque. Cela provient du fait que le foncteur induit par $u_!$,

$$u_{!\,\mathrm{cof}} \ : \ \mathcal{M}(I^\circ)_{\mathrm{c}} \ \longrightarrow \ \mathcal{M}(I'^\circ)$$

est encore compatible avec les localiseurs, comme on voit par le calcul des fibres

$$u_!(X)(i) \ \simeq \ \varinjlim_{i\in I/i'} X(i) \ .$$

Ainsi, on est ramené aux cas où $I' = e$, quitte à remplacer I par I/i', et en notant que $(I/i' \longrightarrow I$ étant étale) l'image inverse de X sur I/i' est cofibrant (cf. 3.2.a).

Commentary · Development (commentators') [ms. 66]

A finite source can demand an infinite Kan extension

Grothendieck next removes the assumption that the target category is a poset. In the cocomplete branch the pointwise argument survives: the comma category $(u \downarrow i')$ is still a well-founded poset when I is one. Indeed there is at most one arrow between two of its objects because the projection to I is faithful, and a nonidentity decreasing chain projects to a nonidentity decreasing chain in I. The projection is étale in the relevant sense, so restriction preserves cofibrancy. What does not survive automatically is the finite-colimit branch. A finite source can have infinitely many different arrows to a fixed target object.

A finite-colimit obstruction. [ours] There is a category satisfying all four axioms on page 39 and possessing finite colimits, together with a functor from a one-object poset to a small category, for which the desired left adjoint does not exist.

Proof. Take $\mathcal{M} = \mathsf{FinSet}$, W the bijections, and C all maps. Cof_0 is immediate, Cof_1 follows from finite cocompleteness, and Cof_2 says that pushouts of bijections are bijections. For Cof_3, factor $f : A \to B$ as f followed by id_B. All objects are cofibrant. Let $u : e \to B\mathbb{Z}$ be the unique functor to the one-object category of the infinite cyclic group. Restriction is the forgetful functor from finite $\mathbb{Z}$-sets, equivalently finite sets with a permutation, to finite sets. Since W consists of isomorphisms, the localized categories and functors are the same categories and functors.

If this forgetful functor had a left adjoint, let P be its value on the singleton. The adjunction would require

$$\mathrm{Hom}_{\mathbb{Z}}(P, \mathbb{Z}/n) \simeq \mathrm{Hom}_{\mathsf{FinSet}}(*, \mathbb{Z}/n)$$

for every $n \geq 1$. If $P = \varnothing$, the left side has one element, contradicting $n = 2$. Otherwise decompose P into its finitely many nonempty cyclic orbits, of lengths $m_1, \ldots, m_r$. Choose a prime $p > \max_j m_j$. An equivariant map from an orbit of length m_j to the transitive p-cycle would imply $p \mid m_j$: applying the generator m_j times fixes the source point and hence its image. No such map exists. Thus the left side for $n = p$ is empty while the right side has p elements. This contradiction proves the assertion.□

Here $(u \downarrow *)$ is the discrete set $\mathbb{Z}$, so the unavailable pointwise value is the infinite copower $\mathbb{Z} \times *$. The failure occurs before adjunction descent and cannot be repaired by choosing another cofibrant replacement. The example uses $\mathbb{Z}$ specifically; an arbitrary infinite group need not give the same finite-action obstruction.

The corrected scope is to retain the cocomplete and transfinite assumptions, or to require the relevant comma colimits and their homotopical preservation separately. For the construction of the candidate in the finite branch, it suffices, for example, that every $(u \downarrow i')$ be finite: each such poset then admits the finite latching argument. This addresses existence and homotopicality; the adjunction is still a separate assertion. We checked the scope of the arbitrary-index results in Rădulescu-Banu (2009, §9.6) and Cisinski (2010, §2); we did not find this particular FinSet test stated there. The displayed result is supplied as an explicit application of the elementary free-action obstruction, without a claim of priority. The manuscript's footnote on page 55 refers forward to page 86, outside these pages; the audit concerns the assumptions visible here and does not claim that the later manuscript never revisits them.

[page 67]

Mais il se présente à nouveau la perplexité si le foncteur (3.3.1) ainsi construit est bien adjoint à gauche de u_W^*. On peut songer à utiliser encore la même factorisation en $u = vw$, en utilisant le mapping-cone. La difficulté, cependant, c'est qu'à présent $J = C(u)$ n'est plus une p. 33
catégorie ordonnée, donc on n'a pas une définition toute prête de $C_{J^\circ} \subset \mathrm{Fl}(\mathcal{M}(J^\circ))$, donnant lieu à une sous-catégorie $\mathcal{M}(J^\circ)_{\mathrm{c}}$ etc. En fait, il semblerait qu'on soit pratiquement paralysé, tant qu'on n'arrive à construire sur $\mathcal{M}(J^\circ)$ également une structure de catégorie à cofibrations, se comportant bien par rapport au foncteurs $v_!$. La "démonstration" d'Anderson de l'existence de $u_!$ pour *toute* flèche dans Cat, foire totalement (même pour $I \longrightarrow e$, I ensemble ordonné fini – mais là j'ai pu rétablir la situation), car il se ramène au cas où I est ordonné localement fini, et il admet sans plus que *dans ce cas* $u_!^W$ est adjoint à gauche de u_W^*, et *de plus,* que si u [page 68]
est $\mathbf{W}_0$-connexe, donc $u^* : \mathcal{M}(I'^\circ) \longrightarrow \mathcal{M}(I^\circ)$ pleinement fidèle, qu'il en soit de même de u_W^* déduit par passage aux localisés. Or même si on admet que $u_!^W$ est adjoint à gauche de u_*^W, faute de disposer d'un morphisme d'adjonction tout près "en bas" (dans $\mathrm{Ho}_W(I'^\circ)$), déduit du morphisme d'adjonction dans $\mathcal{M}(I^\circ)$, il n'est absolument pas clair que u_W^* soit pleinement

fidèle, i.e. $u_!^W$ un foncteur de localisation. Le contrexemple 3.6 montre bien que, même si on a adjonction (i.e. si e est un objet initial de $\mathcal{C} = \mathcal{C}'$), le foncteur $u^* = \mathrm{id}_{\mathcal{C}}$ a beau être pleinement fidèle (et même une équivalence de catégories), le foncteur localisé n'a aucune raison de l'être : cela pouverait que tout objet de Cat ayant un objet initial est équivalent à e !

Commentary · Exposition [ms. 68]

Full faithfulness is controlled by the derived counit

The remaining warning on page 68 concerns the passage from a left adjoint to a localization functor. For any adjunction $L \dashv R$, the right adjoint is fully faithful exactly when the counit $LR \to \mathrm{id}$ is an isomorphism. To recall the mechanism, an invertible counit identifies the adjunction bijection on $\mathrm{Hom}(RB, RC)$ with the desired bijection $\mathrm{Hom}(B, C)$. Conversely, when R is fully faithful, the counit represents the identity under the same bijections and is invertible. Thus one must control the counit after deriving the functors, not merely the original counit before replacement.

In the second walking-arrow example above, $L : e \to [1]$ selects 0 and R is the projection. The counit at 1 is $0 \to 1$, which is not an isomorphism in the target localization, since that localization inverts only isomorphisms. This locates the entire defect in one component. The original identity adjunction had an identity counit, but its restriction to the chosen replacement subcategory changed the candidate left adjoint. The missing comparison with replacement is precisely what would have been needed to transport full faithfulness.

For actual diagram categories, ordinary finality is likewise weaker than homotopy finality. The strict colimit comparison tests connectedness of comma categories, whereas homotopy colimits can detect their higher homotopy. Here is a finite computation which exhibits the counit itself. Let P be the face poset of the boundary of a triangle: three vertices, three edges, and the six incidence relations from a vertex to an edge. The unique functor $q : P \to e$ is ordinarily final because P is nonempty and connected. Its constant-diagram functor is fully faithful even before localization: naturality along the incidences forces all components of a transformation between constant diagrams to be the same map.

Use simplicial sets with their usual weak homotopy equivalences and monomorphisms. A cofibrant replacement Q of the constant point on P is explicit. Assign Δ^0 to each vertex and Δ^1 to each edge, with the two incident vertices mapping to its distinct endpoints. The latching maps at vertices are $\varnothing \to \Delta^0$; those at edges are $\partial\Delta^1 \to \Delta^1$. All are monomorphisms, so Q is a cofibrant diagram. The unique transformation $Q \to \mathrm{const}(*)$ is an objectwise weak equivalence because every value is contractible. Its ordinary colimit glues three intervals cyclically along their endpoints. Its realization is the triangular circle, and its first integral homology is $\mathbb{Z}$: its cellular boundary sends each oriented edge to the difference of its endpoints, so its kernel is generated by the sum of the three consistently oriented edges, and there are no two-cells.

Consequently the derived counit at the point is represented by this circle mapping to $*$. Integral homology sends weak homotopy equivalences of simplicial sets to isomorphisms, so factors through their homotopy category; its value shows that this counit is not invertible. Thus q^* fails to be fully faithful after localization, despite being fully faithful as an ordinary functor. This calculation uses only the finite cofibrant replacement machinery established here; no appeal to an uncomputed homotopy colimit is hidden in the example.

The objection does not contradict Theorem 3.7. That theorem constructs an adjunction by two controlled factorizations. Full faithfulness of a particular restriction requires the additional counit computation appropriate to that restriction, or a homotopy finality theorem with its actual hypotheses.

[page 69]

3.11. Disons qu'on a une *catégorie à cofibrations stricte,* si dans l'axiome Cof_3 de factorisation (cf. p. 39), la factorisation peut être prise fonctoriellement en f. Il revient ± au même de dire que l'axiome de factorisation reste valable, quand on pense à une catégorie $\mathcal{M}(I)$ pour I dans Cat, en y définissant $W(I)$ et $C(I)$ (à ne pas confondre avec C_I) fibre par fibre. Cela revient au même tout au moins si $\mathcal{M}$ est équivalente à une petite catégorie. Sinon, il faut supposer fonctorialité pour toute *petite* sous-catégorie pleine de $\underline{\mathrm{Fl}}(\mathcal{M})$. Il vaut mieux définir la catégorie à cofibrations stricte par cette propriété de stabilité.

Commentary · Exposition [ms. 69]

Functorial factorization supplies arbitrary parameters

Grothendieck introduces strictness to permit an arbitrary diagram category to serve as the coefficient category in the finite or well-founded construction. A functorial factorization assigns to every arrow $f : X \to Y$ an object $E(f)$ and maps

$$X \xrightarrow{i_f} E(f) \xrightarrow{p_f} Y, \qquad i_f \in C, \quad p_f \in W,$$

and assigns to every commutative square a compatible arrow between the middle objects. Given a natural transformation $f : X \to Y$ of covariant Z-diagrams, apply this assignment to its components. The arrows $E(f_z) \to E(f_{z'})$ coming from the naturality squares satisfy the identity and composition laws because E is a functor on the arrow category. Hence they form a middle Z-diagram and give the required objectwise factorization. Since Z is arbitrary, the same assertion applies to presheaves by replacing Z with Z^{op}.

The converse explains the size qualification. If $\mathcal{M}$ is small, use the covariant index $Z = \mathrm{Fl}(\mathcal{M})$ and the universal natural transformation from its source functor to its target functor. A factorization in $\mathcal{M}^Z$ is exactly a functorial factorization of every arrow of $\mathcal{M}$; in presheaf notation the index is $\mathrm{Fl}(\mathcal{M})^{\mathrm{op}}$. If $\mathcal{M}$ is only essentially small, transport the construction along a small skeleton. For a large category, an assertion quantifying only over small Z yields factorizations on every small diagram of arrows; it does not automatically choose one globally coherent factorization on a proper class. Grothendieck's final preference for stability under small diagram categories keeps that quantification explicit.

Paragraph 3.11 explicitly returns to $\mathbf{D} = \mathbf{D}_{\mathcal{M}}$ and $\mathcal{M}(I)$, hence to the historical presheaf convention. For its parameter category Z, the coefficient category is therefore

$$\mathcal{N} = \mathcal{M}(Z) = \mathrm{Fun}(Z^{\mathrm{op}}, \mathcal{M}), \qquad \mathbf{D}_Z(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{N})[W^{-1}].$$

Equip $\mathcal{N}$ with the pointwise classes $C(Z)$ and $W(Z)$. The first three cofibration axioms hold objectwise, while strictness supplies the fourth. Colimits and the relevant closure properties also pass objectwise. Apply Theorem 3.7 in this coefficient category to $u^{\mathrm{op}} : I^{\mathrm{op}} \to I'^{\mathrm{op}}$. The identification

$$\mathrm{Fun}(I^{\mathrm{op}}, \mathcal{N}) \simeq \mathrm{Fun}((I \times Z)^{\mathrm{op}}, \mathcal{M})$$

identifies its restriction with $(u \times \mathrm{id}_Z)^*$ on the historical $\mathbf{D}$. This proves the parameter assertion directly for finite indices, and in the well-founded branch when I^{op} and I'^{op} are artinian. The infinite-domain qualification explained at Corollary 3.9 applies here as well.

The two cofibration notations must stay distinct. $C(Z)$ tests each arrow component separately. The relative latching class in this application is $C_{I^{\mathrm{op}}}$ inside the coefficient category $\mathcal{N}$: it controls the attachments in the contravariant I direction. Replacing it by pointwise cofibrancy would discard the boundary information that the chapter has established.

Under ABC hypotheses, stability under arbitrary small diagram categories was later established without assuming a globally chosen functorial factorization; see Lenz (2018, Theorem 1.24). That result uses infinite colimits. It therefore completes this program in the stronger setting without making the finite hypotheses sufficient by themselves.

On voit alors que le prédérivateur $\mathbf{D} = \mathbf{D}_{\mathcal{M}}$ sur Cat est un dérivateur *relativement à* $\mathcal{D} = \mathrm{Diag}_0$ défini plus haut (catégories ordonnées I artiniennes, telles que $\mathrm{Esp}(I)$ soit noethérien), ou même relativement à $\mathcal{D} = \mathrm{Diag}$ (catégories ordonnées I artiniennes sans plus) lorsque C et TC sont stables par composition transfinie. J'entends par là que $u_!$ existe quand u est une flèche de $\mathcal{D}$, et que ça reste valable quand on passe à un dérivateur induit $\mathbf{D}_Z$ (Z dans Cat quelconque) ; en d'autre termes, que $v_!$ existe quand v est de la forme $\mathrm{id}_Z \times u$, où $u : I \longrightarrow I'$ est une flèche de $\mathcal{D}$, et que le calcul se fait *au dessus de* I' selon la règle Dér 5 bis. Cela signifie donc que la flèche de changement de base pour les $f_!$ est un isomorphismes dans [page 70]

le cas suivant :

$$\begin{array}{ccc} I \times Z & \longleftarrow & (i'\backslash I) \times Z \\ \downarrow{\scriptstyle u\times \mathrm{id}_Z} & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ I' \times Z & \longleftarrow & \{i'\} \times Z \end{array} \quad ,$$

où i' est un objet quelconque de I'.

Commentary · Examples [ms. 70]

The comma category after returning to presheaves

The final square is correct in the presheaf convention to which paragraph 3.11 has returned. Its slash differs from the earlier covariant formula because the underlying functor is now $I^{\mathrm{op}} \to \mathcal{M}$. For $u : I \to I'$ set

$$A = (i' \downarrow u) = i'\backslash I, \qquad p : A \longrightarrow I.$$

An object of A is a pair $(i, \alpha : i' \to u(i))$. The usual pointwise formula for presheaves derives left Kan extension along u^{op}, and the relevant comma category is canonically

$$(u^{\mathrm{op}} \downarrow i') \simeq A^{\mathrm{op}}.$$

Indeed its objects are the same pairs. An arrow from (i, α) to (j, β) in this opposite comma category is an arrow $j \to i$ in I satisfying $u(j \to i)\beta = \alpha$, exactly the opposite of an arrow in A. Thus for an ordinary presheaf X the formula is

$$(\mathrm{Lan}_{u^{\mathrm{op}}} X)(i') \simeq \mathrm{colim}_{A^{\mathrm{op}}} X p^{\mathrm{op}}.$$

The notation $\mathbf{D}_W(A)$ already supplies this opposite in the index of the coefficient diagram. Replacing A by $u \downarrow i'$ would alter the formula.

The source's parameterwise assertion is therefore

$$(i' \times \mathrm{id}_Z)^* \, \mathbf{L}(u \times \mathrm{id}_Z)_! X \;\simeq\; \mathbf{L}(\mathrm{pr}_Z)_! \, (p \times \mathrm{id}_Z)^* X, \qquad p : (i' \downarrow u) \longrightarrow I,$$

as an isomorphism in $\mathbf{D}_W(Z)$. The arrows $i' \to u(i)$ define the natural transformation from the constant functor at i' to up; after passing to presheaves, its mate is the displayed comparison. The 2-cell is part of the meaning of this comma square.

Here is a direct proof in the range supplied by the preceding construction. Regard X as an I^{op}-diagram in $\mathcal{N} = \mathrm{Fun}(Z^{\mathrm{op}}, \mathcal{M})$ and choose a cofibrant replacement in that coefficient category. If I and I' are posets, A is the upper subposet of elements i for which $i' \leq u(i)$. Consequently A^{op} is a lower subposet of I^{op}, and restriction preserves its latching data. The ordinary pointwise Kan formula then computes both sides from the same restricted cofibrant diagram. Its colimit isomorphism gives the required derived comparison. This proves the finite case, and the well-founded case with I^{op} artinian and the stated coefficient colimits and closure. The later arbitrary-shape ABC theorem removes that indexing restriction under its stronger hypotheses.

A one-arrow computation makes the change of convention visible. Let $u : e \to [1] = \{0 \to 1\}$ select 0, take $i' = 1$, and let $X = *$ in sets with weak equivalences the bijections and cofibrations all maps. For covariant diagrams, $u \downarrow 1$ is a singleton, so $(\mathrm{Lan}_u X)(1) = *$. For presheaves, however, $1 \downarrow u$ is empty. Hence $(\mathrm{Lan}_{u^{\mathrm{op}}} X)(1) = \varnothing$, just as the printed square asserts. At 0 both extensions have value $*$. In particular the presheaf extension is the diagram with values $*$ at 0 and $\varnothing$ at 1, with its unique restriction map $\varnothing \to *$. The covariant extension is the constant point. These are different variance conventions for the same indexing map, not contradictory values of one functor.

Thus the printed $i'\backslash I$ requires no textual correction. The earlier covariant formula remains valid for $H(I) = \mathbf{D}_W(I^{\mathrm{op}})$; the final formula is its correctly reindexed presheaf counterpart.

Overview · XVII-a

Chapter XVII, manuscript pages 1–70: attachments, diagrams, and derived left Kan extensions

Grothendieck is looking for a way to construct the operations of a derivator from a category endowed with weak equivalences and cofibrations (Grothendieck, 1990–1991m). The principal operation in these pages is a left adjoint to restriction of diagrams. Its construction has two logically separate parts. First one must find diagrams on which an ordinary colimit or ordinary left Kan extension respects weak equivalences. Then one must show that the functor obtained after localization has the required adjunction. Much of the force of the chapter lies in its refusal to identify these two tasks. The counterexample in Remark 3.6 makes the distinction precise, and the mapping-cone argument in Theorem 3.7 supplies the missing adjunction for the indexing posets under consideration.

The first twenty-four manuscript pages are a preliminary investigation which the author asks the reader to skip once section 2 has been written. They nevertheless disclose the mathematical problem that determines the stronger definition. Throughout, an open subset of an ordered set is a lower ideal, and $\mathcal{M}(I^\circ)$ denotes covariant I-diagrams. For a map $u : X \to Y$, the elementary test is that

$$X_i \amalg_{\operatorname{colim}_{j<i} X_j} \operatorname*{colim}_{j<i} Y_j \longrightarrow Y_i$$

be a cofibration. This relative latching map measures the attachment at i after all its predecessors have been incorporated. If the indexing order is well founded, one can perform these attachments in a compatible ordinal order. Pushout stability and transfinite-composition stability then put both every component of u and its colimit map in C. Applying the same argument to $TC = C \cap W$, and using two-out-of-three at the next attachment, identifies trivial diagram cofibrations with the diagram cofibrations that are objectwise weak equivalences. The commentary supplies a continuous filtration so that this argument does not tacitly assume continuity of the original diagrams.

The relative statement is more useful than the absolute one. For lower ideals $K \subset J$, write $X_K = \operatorname{colim}_K X$ and similarly for Y, and put

$$r_{K,J}(u) : Y_K \amalg_{X_K} X_J \longrightarrow Y_J.$$

The essential excision identity says that when a new lower ideal B is adjoined to A, the relative map over $A \subset A \cup B$ is a pushout of the relative map over $A \cap B \subset B$. A succession of such comparisons is controlled by a continuous filtration with terms $X_J \amalg_{X_{A_\alpha}} Y_{A_\alpha}$. These two elementary colimit calculations account for the large diagrams of the preliminary section and for Lemmas 2.6–2.7. They also explain the distinction between the relative assertion, which does not require X to be cofibrant, and the assertion $X_K \to X_J \in C$, which does. Footnote 5 overstates the dispensability of this latter hypothesis; footnote 7 already gives the necessary qualification.

Section 2 makes all these relative comparisons part of the definition. A map belongs to the new C_I when $r_{K,I/i}(u) \in C$ for every i and every lower ideal $K \subset I/i$. Theorem 2.1(1) extends that condition to every inclusion of lower ideals, without assuming that I is artinian. Part (2) recovers the single relative-latching test when I is artinian. Part (3) identifies $(TC)_I$ with $C_I \cap W_{I^\circ}$ under the same well-foundedness and closure hypotheses. This change of definition is indispensable: on a poset with an endless descending chain, all relative latching maps can be isomorphisms while the components of a map of diagrams fail even to be injective. The development attached to manuscript page 24 proves that this obstruction detects every failure of well-foundedness, using diagrams of sets extended from the upward closure of a descending chain.

The question left on manuscript pages 38–39 concerns part (3), not part (2): can the artinian hypothesis be removed from the equality involving TC when the stronger definition of C_I is retained? Grothendieck verifies one descending-chain example. The present commentary extends that argument to linearly ordered sets and, more generally, to posets whose principal lower ideals are linearly ordered. A well-ordered cofinal subset of a lower chain reduces its colimit to the established ordinal case. We did not find a resolution of the unrestricted question under exactly the assumptions of Theorem 2.1 in the literature inspected. This is a statement about the scope of our verification, not a claim that the problem is known to remain open. The later theorems on diagram cofibration structures use their own definitions and additional hypotheses, and therefore do not by themselves settle this precise question.

Several corrections concern the formulation of an argument whose conclusion remains valid. Lemma 2.7 must use a chain continuous at limit ordinals; an arbitrary increasing chain can acquire new objects at a limit stage, so its successor comparisons cannot account for the entire relative map. The printed lemma also confuses its initial relative domain with the initial object of an absolute filtration. Both points are repaired explicitly. The description of the intermediate cube in Lemma 2.6 as totally cocartesian is too strong if it means that all six faces are pushouts; pushout associativity proves the needed face without that assertion. These repairs are recorded separately from transcription corrections. The edition's printed slips remain in the source text, while the commentary states and proves the mathematically intended claims.

Section 3 turns the attachment calculus into homotopical algebra. The axioms Cof_0–Cof_3 require two-out-of-three for W, composition and pushout stability for C, pushout stability for TC, and a factorization of every map as a cofibration followed by a weak equivalence. They do not require every object to be cofibrant. The full subcategory of cofibrant objects has the same localization, so Brown's methods can be applied there. The factorization of a diagram is then constructed one index at a time: a factorization already chosen on the predecessors of i produces the map

$$X_i \amalg_{L_iX} L_iZ \longrightarrow Y_i,$$

to which the ambient factorization axiom is applied. This is the direct-diagram form of the Reedy construction. Its later cellular formulation is made explicit in Riehl and Verity (2014, Proposition 6.3 and Lemma 7.1). The source works also with ordinal ranks, which exceed the natural-number degree convention used for many familiar Reedy categories.

The two finiteness branches in Theorem 3.1 deserve to be read literally. In the first, I is artinian and its lower-Alexandrov topology is noetherian. Under the ordinary choice convention used in the argument, their conjunction forces I to be finite: otherwise successive choices of a minimal element of the remaining poset give a strictly increasing sequence of finite lower ideals. The second branch allows arbitrary small artinian posets, requires the necessary small colimits, and imposes transfinite stability on both C and TC. Thus the theorem has a finite construction and a construction with explicit infinitary hypotheses. The commentary keeps these alternatives separate when it discusses the available Kan extensions.

Proposition 3.2 proves that left Kan extension preserves diagram cofibrations, and that restriction preserves them under a suitable local cofilteredness condition. Brown's cylinder factorization then shows that left Kan extension carries weak equivalences between cofibrant diagrams to weak equivalences. Together with the localization equivalence for cofibrant objects, this yields the candidate $u_!^W$. The warning in Remark 3.6 is categorical and complete: choosing a subcategory which computes a localization need not make its inclusion a derived replacement suitable for an adjunction. With the identity adjunction on the walking arrow, different choices of weak equivalences give the stated failure. The related assertion that full faithfulness survives localization also fails without control of the localized counit. These are failures of general inferences, not counterexamples to the properly constructed derived adjunction.

Theorem 3.7 restores the adjunction by factoring $u : I \to I'$ through its categorical mapping cone. One factor is a closed inclusion, whose left Kan extension is extension by the initial object. The other is a retraction whose relevant comma categories have initial objects. Each factor consequently falls under a case where the adjunction can be checked, and their left adjoints compose. The discussion of arbitrary target categories that follows needs a further size qualification in the finite branch. For $u : e \to B\mathbb{Z}$ and $\mathcal{M} = \mathrm{FinSet}$, a left adjoint to the forgetful functor from finite $\mathbb{Z}$-sets would have to provide a free finite $\mathbb{Z}$-set on one generator; finite cycles of unbounded order show that no such object exists. A finite source does not make its comma categories finite. This obstruction disappears when the required small coproducts and colimits are supplied, but that hypothesis must be stated.

The chapter belongs to the program described in *Pursuing Stacks*, section 69, and in the letter to Thomason of 2 April 1991: diagram localizations retain operations that a single homotopy category does not specify, while cofibrations and fibrations are auxiliary means of constructing those operations (Grothendieck, 1983, 1991a). Later work establishes the desired diagram formalism with precise hypotheses. Cisinski (2010, Corollary 2.24) obtains a one-sided derivator on finite direct categories: in his convention, a weak right derivator satisfying the additional axiom Der 5. For ABC cofibration categories, Rădulescu-Banu (2009, Theorems 9.6.3, 9.6.5, and 10.3.2) gives arbitrary small homotopy left Kan extensions, their adjunctions, pointwise formulas, and the resulting derivator. Lenz (2018, Theorem 3.33) identifies the homotopy prederivator of an ABC cofibration category in which every object is cofibrant with that of its quasicategory of frames. These comparisons place the construction in modern homotopy theory while preserving the distinctions in assumptions that matter in this chapter.

The last paragraph introduces strictness in order to apply the same construction to categories of diagrams with an arbitrary additional parameter. Functorial factorization gives objectwise diagram factorizations; the universal arrow diagram gives the converse when the relevant arrow category is small. Here the source returns explicitly to its historical presheaf convention, $\mathbf{D}_W(J) = \mathrm{Fun}(J^{\mathrm{op}}, \mathcal{M})[W^{-1}]$. The preceding covariant construction concerns $H(I) = \mathbf{D}_W(I^{\mathrm{op}})$, so its immediate infinite range must be translated by opposition; the finite range is unchanged. In this presheaf convention the final printed comma category $i'\backslash I = (i' \downarrow u)$ is correct: the colimit computing a left Kan value runs over its opposite. The covariant formula instead uses $(u \downarrow i') = I/i'$. A walking-arrow calculation in both conventions explains the difference and prevents a change of variance from being misdiagnosed as an edition slip. The later ABC theorem supplies the arbitrary-shape extension under the full infinitary hypotheses. The manuscript's own further construction and forward references lie beyond manuscript page 70.

lesquels sont visiblement adjoint l'un de l'autre (puisque induits par un couple de foncteurs adjoints $u_!$, u^*), et de plus compatibles aux localiseurs. Donc on peut appliquer le lemme des foncteurs adjoints pour les localisés. Pour résumer : p. 29

Corollaire 3.5. *Supposons que I, I' soient artiniens, et qu'on soit dans l'un des deux cas suivants :*

a) *Les espaces* $\mathrm{Esp}(I)$, $\mathrm{Esp}(I')$ *sont noethériens (p.ex. I, I' finis).* [page 58]

b) *C et $TC = C \cap W$ sont stables par composition transfinie.*

Supposons de plus que u^* applique $\mathcal{M}(I'^{\circ})_{\mathrm{c}}$ dans $\mathcal{M}(I^{\circ})_{\mathrm{c}}$, ce qui est le cas notamment si u est localement cofiltrant, i.e. les ensembles ordonnés localisés $i'\backslash I$ sont localement cofiltrants. Alors le foncteur $u_!^W$ construit dans (3.3.1) est adjoint à gauche de u_W^*.

Remarque 3.6. Quand on a un couple de foncteurs adjoints

$$\mathcal{C} \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{C}' \quad ,$$

avec les localiseurs W, W' de $\mathcal{C}$, $\mathcal{C}'$, u^* compatible aux localiseurs, et si de plus on a une sous-catégorie pleine $\mathcal{C}_0$ de $\mathcal{C}$ telle que

a) $u_!|\mathcal{C}_0$ transforme flèches de W en flèches de W', et

b) l'inclusion $\mathcal{C}_0 \hookrightarrow \mathcal{C}$ induit une *équivalence* $\mathcal{C}_0 W_0^{-1} \xrightarrow{\sim} \mathcal{C}W^{-1}$ (où $W_0 = W \cap \mathrm{Fl}(\mathcal{C}_0)$),

on trouve deux foncteurs canoniques

$$\mathcal{C}W^{-1} \underset{u_W^*}{\overset{u_!^W}{\rightleftarrows}} \mathcal{C}'W'^{-1} \quad ,$$

où $u_!^W$ est défini seulement à isomorphisme canonique près.

Commentary · Exposition [ms. 58]

The adjunction must survive the choice of resolutions

Grothendieck has constructed the candidate for a left homotopy Kan extension and now asks for the property that makes this candidate useful: maps out of it must correspond to maps into the restricted diagram. The distinction matters because an equivalence between two localized categories remembers neither a preferred direction of replacement nor the comparison maps with the original functor. Corollary 3.5 has enough compatibility to descend an actual adjunction. Remark 3.6 removes that compatibility and exposes what is lost.

Write $F \dashv G$, let $j : \mathcal{C}_0 \hookrightarrow \mathcal{C}$ be the chosen full subcategory, and write γ and δ for localization. If $\mathrm{Ho}(j)$ is an equivalence and Fj preserves weak equivalences, the composite

$$\overline{F} = \mathrm{Ho}(Fj)\,\mathrm{Ho}(j)^{-1}$$

is certainly a functor. Its construction has not supplied a natural transformation $\overline{F}\gamma \to \delta F$. Such a comparison is part of the data of a left derived functor, with a universal property; it cannot be reconstructed from the objectwise choice alone.

The smallest example makes the distinction quite concrete. Let $\mathcal{C}$ be the walking arrow $0 \to 1$, use all arrows as W, only isomorphisms as W', and take $F = G = 1_{\mathcal{C}}$. Inverting the single nonidentity arrow makes $\mathcal{C}[W^{-1}]$ equivalent to the terminal category. Both subcategories $\{0\}$ and $\{1\}$ therefore give localization equivalences. Choosing $\mathcal{C}_0 = \{1\}$ makes $\overline{F}$ select 1, while $\overline{G} : \mathcal{C} \to 1$ is the unique functor. An adjunction would require

$$\mathrm{Hom}_{\mathcal{C}}(1, 0) \simeq \mathrm{Hom}_1(*, *).$$

The first set is empty and the second is a singleton. Choosing $\{0\}$ instead gives an adjunction, since 0 is initial. Even then $\overline{G}$ is not fully faithful: the empty hom-set from 1 to 0 again maps to

a singleton. This also proves the second warning on manuscript p. 68. The two failures concern the formal inferences; they do not disprove a correctly constructed homotopy Kan extension or a correctly justified homotopy-cofinality theorem.

A useful sufficient repair is a functor $Q : \mathcal{C} \to \mathcal{C}_0$ and a natural weak equivalence $q : jQ \to 1_{\mathcal{C}}$, with F homotopical on $\mathcal{C}_0$ and G homotopical on all of its domain. The comparison is then $\delta F q$, rather than an unspecified isomorphism after localization. This is the left-deformation construction of Riehl (2014, Theorems 2.2.8 and 2.2.11). A more intrinsic sufficient condition is that the total derived functors are absolute Kan extensions; their adjunction is the unnumbered theorem of Maltsiniotis (2007).

The direction of q explains the example. The constant choice $Q(X) = 0$ admits $q_X : 0 \to X$; the constant choice $Q(X) = 1$ admits no component $1 \to 0$. Thus the obstruction is visible before any hom-set calculation in the localization. The passage to the categorical mapping cylinder in the next proof is a way of recovering compatible adjunctions without first choosing a functorial resolution of every diagram.

Contrairement à ce que prétend Anderson, il n'est par vrai en général que ces deux foncteurs soient adjoints, sans hypothèses supplémentaires. Voici un contrexemple : on prend $\mathcal{C} = \mathcal{C}'$, $u_! = u^* = \mathrm{id}_{\mathcal{C}}$, $W' = \mathrm{Is}(\mathcal{C})$, $W = \mathrm{Fl}(\mathcal{C})$, donc u^*_W est le foncteur canonique [page 59]

$$\underbrace{\Pi\mathcal{C}}_{\text{groupoïde fondamental}} \qquad\qquad \longleftarrow \mathcal{C}\ .$$

Supposons que $\mathcal{C}$ soit 1-connexe, i.e. $\Pi\mathcal{C}$ soit équivalente à e. Soit $e \in \mathrm{Ob}\,\mathcal{C}$ tel que $\mathrm{End}_{\mathcal{C}}(e)$ soit le groupe unité, et prenons $\mathcal{C}_0 = \{e\}$. Alors l'hypothèse sur $\mathcal{C}_0$ est satisfaite. Mais si $u^W_!$ était adjoint à gauche de u^*_W, e serait objet initial de $\mathcal{C}$. Par exemple si $\mathcal{C}$ a un objet final e (ce qui implique que $\mathcal{C}$ est 1-connexe et e rigide), cet objet final devrait être initial si Anderson avait raison avec "it is clear that …".

Néanmoins :

[page 60]

Commentary · Exposition [ms. 59]

Why the mapping cylinder repairs the argument

Grothendieck replaces the given map of indexing posets by two maps for which the adjunction can be controlled separately. The construction called a mapping-cone here is a categorical mapping cylinder. It is also the collage of the bifunctor $\mathrm{Hom}_{I'}(-, u-)$, the gluing construction already encountered in the commentary on Chapter XII, manuscript pp. 59–61. Here its particular purpose is to make one factor preserve cofibrancy and the other preserve weak equivalences without deriving it.

Keep the convention of Theorem 3.7 explicit: $\mathcal{M}(I^\circ) = \mathrm{Fun}(I, \mathcal{M})$ consists of covariant diagrams. The objects of $C(u)$ are the disjoint union of those of I' and I. Within either piece the original arrows remain; a mixed arrow from $j \in I'$ to $i \in I$ is an arrow $j \to u(i)$ in I'; there are no mixed arrows in the reverse direction. Composition is induced by composition in I' and by u. Thus the inclusion $w : I \hookrightarrow C(u)$ is a cosieve, and the inclusion $s : I' \hookrightarrow C(u)$ is a sieve. The retraction v is the identity on I' and equals u on I.

The essential feature of v can be read directly from its hom-sets:

$$\mathrm{Hom}_{C(u)}(sj, x) \simeq \mathrm{Hom}_{I'}(j, vx).$$

Thus $s \dashv v$, with $vs = 1$. In particular $(j, 1_j)$ is initial in $(j \downarrow v)$. For an object $(x, \beta : j \to vx)$ of this comma category, its unique arrow from $(j, 1_j)$ is the arrow represented by β itself. This proves Lemma 3.8, including uniqueness, for arbitrary categories. An initial object gives a common predecessor for every finite diagram and equalizes each parallel pair; hence the required cofilteredness follows. The result is stronger than a statement that the comma category has contractible nerve.

The other factor is controlled by extension by the initial object. If $j \notin I$, the category $(w \downarrow j)$ is empty, because a cosieve admits no arrow from its objects to an object outside it. If $j \in I$, that category has terminal object $(j, 1_j)$. Consequently

$$(w_! X)_j = \begin{cases} X_j, & j \in I, \\ \varnothing_{\mathcal{M}}, & j \in I'. \end{cases}$$

An objectwise weak equivalence is sent to the same weak equivalence on I and to an identity on I'. Both $w_!$ and w^* are therefore homotopical, and their unit and counit descend. No inference of the kind disproved in Remark 3.6 is needed.

There are two small inheritance checks behind the application of the earlier cofibration theorem. A chain in $C(u)$ can cross from one piece to the other at most once, so artinianity passes from the two pieces to their collage. An increasing chain of open subsets stabilizes once its intersections with the two pieces stabilize, which proves the stated noetherianity assertion. Corollary 3.5 therefore applies to v, while the preceding extension argument applies to w. Compatibility of the already constructed left Kan extensions with composition identifies $u_!^W$ with $v_!^W w_!^W$. Composing the two actual adjunctions proves Theorem 3.7.

For a small worked instance, take I to be two isolated objects and $I' = e$. The cylinder has an initial vertex $*$ and two arrows $* \to a$, $* \to b$. Extending the pair (A, B) along w gives $(\varnothing \to A, \varnothing \to B)$; taking its colimit along v gives $A \amalg B$. Its cocones to Y are exactly the pairs of arrows $A \to Y$, $B \to Y$. With cofibrant replacements $QA \to A$ and $QB \to B$, the same construction gives the derived coproduct $QA \amalg QB$. The proof above, rather than the mere existence of these replacements, supplies its adjunction in the homotopy categories.

Théorème 3.7. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations stable par $\varinjlim$ finies, et $u : I \longrightarrow I'$ une application croissante entre ensembles ordonnés artiniens. On suppose qu'on est sous l'une des hypothèses suivantes.*

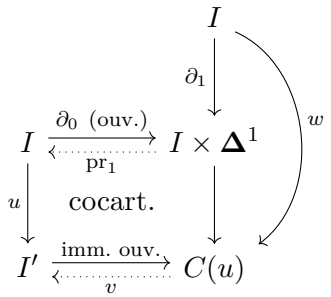
a) $\mathrm{Esp}(I)$, $\mathrm{Esp}(I')$ *sont noethériens, p.ex. I, I' finis.*

b) *C et $TC \overset{déf}{=} C \cap W$ sont stables par composition transfinie, et $\mathcal{M}$ stable par petites $\varinjlim$ filtrantes.*

p. 30

Alors le foncteur $u_!^W : \mathrm{Ho}_W(I^\circ) \longrightarrow \mathrm{Ho}_W(I'^\circ)$ construit dans (3.3.1) *est adjoint à gauche du foncteur u_W^* en sens opposé.*

Démonstration. On utilise le diagramme bien connu



pour factoriser u en

$$u \;=\; vw \;,$$

où w est une *immersion fermée,* et v est déduit de $\mathrm{pr}_1 : I \times \mathbf{\Delta}^1 \longrightarrow I$ par push-out. C'est une rétraction du mapping-cone C(u) sur l'ouvert I', dont le complémentaire fermé s'identifie à I via w.

Le foncteur de recollement entre I', I [page 61]

$$H \;:\; I'^\circ \times I \;\longrightarrow\; \mathrm{Ens}$$

est donné par

$$(*) \qquad \underbrace{H(i', i)}_{= \mathrm{Hom}_{C(u)}(i', i)} \;=\; \mathrm{Hom}_{I'}(i', u(i)) \;.$$

La rétraction v de $C(u)$ dans I' applique i dans $u(i)$, et une flèche $i' \longrightarrow i$ correspondant à $i' \longrightarrow u(i)$ dans I' via $(*)$, est transformé par v en cette même flèche $i' \longrightarrow u(i)$. Calculons,

pour i'_0 donné, la catégorie $i'_0\backslash\Big(C(u),\ v : C(u) \longrightarrow I'\Big)$, des couple d'un objet ξ de $C(u)$ et d'une flèche $i'_0 \longrightarrow v(\xi)$ dans I'. On voit que $i'_0\backslash C(u)$ admet l'ouvert $i'_0\backslash I'$, avec le complémentaire fermé $i'_0\backslash I$,

$$\begin{array}{ccccc} & & & & \text{fermé} \\ & & & & \text{complémentaire} \\ & & & & \text{de } I' \\ I' & \xhookrightarrow[\text{ouvert}]{} & C(u) & \hookleftarrow & I \\ & & \Big\downarrow & & \\ & & I' & & \end{array},$$

ces deux catégories étant recollées par le bifoncteur

$$\begin{array}{rcl} (i'_0\backslash I')^\circ \times (i'_0\backslash I) & \longrightarrow & \text{Ens} \\ \Big((\alpha : i'_0 \longrightarrow i'), (\beta : i'_0 \longrightarrow u(i), i)\Big) & \longmapsto & \mathrm{Hom}_{i'_0\backslash I'}(i', u(i)) \end{array}$$

induit par $(*)$; i.e. les flèches de $(\alpha : i'_0 \longrightarrow i') \in \mathrm{Ob}\,(i'_0\backslash I')$ dans $(\beta : i'_0 \longrightarrow u(i)) \in i'_0\backslash I$ sont les flèches $i' \xrightarrow{\gamma} u(i)$ telles que le triangle p. 31

$$\begin{array}{ccc} & i'_0 & \\ {\scriptstyle\alpha}\swarrow & & \searrow{\scriptstyle\beta} \\ i' & \xrightarrow{\ \gamma\ } & u(i) \end{array}$$

commute.

Si $i' = i'_0$, $\alpha = \mathrm{id}_{i'_0}$, alors cette commutativité revient à $\gamma = \beta$, ce qui signifie qu'il y a une flèche et une seule de α dans (i, β). Cela montre donc que $(i'_0, \mathrm{id}_{i'_0})$ est un objet initial de $i'_0\backslash C(u)$. On a donc prouvé le lemme : [page 62]

Lemme 3.8. *Soit $u : I \longrightarrow I'$ une flèche dans* (Cat)*, factorisons-le en*

$$I \xrightarrow{w} C(u) \xrightarrow{v} I' ,$$

où $C(u)$ est le mapping-cone défini page 60*, w une immersion fermée de complémentaire I', v la rétraction canonique de $C(u)$ sur I'. Ceci posé, pour tout $i'_0 \in \mathrm{Ob}\, I'$, la catégorie $i'_0\backslash(C(u), v)$ admet un objet initial – donc $v : C(u) \longrightarrow I'$ est (tout ce qu'il y a de) cofiltrante.*

Nous pouvons maintenant prouver 3.7. Tout d'abord, les hypothèses faites sur I, I' se transportent à $C(u)$ (hypothèse artinienne, et éventuellement hypothèse noethérienne sur $\mathrm{Esp}(C(u))$). Donc on peut appliquer les résultats précédents, notamment 3.5, aux flèches v, w. Le lemme précédent montre bien que v satisfait aux conditions de 3.5, donc le foncteur $v_!^W$ construit dans 3.3.1 est adjoint à gauche de v_W^*. Considérons d'autre part le foncteur $w_!^W$ construit dans 3.3.1, je dis que la même conclusion s'applique. Cela provient du fait que w est une immersion fermée. En effet, dans ce cas il est immédiat que [page 63]

$$w_! \ :\ \mathcal{M}(I^\circ) \ \longrightarrow\ \mathcal{M}(I'^\circ), \qquad \text{où } I = C(u),$$

est compatible avec les localiseurs, puisque l'on a

$$\begin{cases} w_!(X)(j) \ \simeq\ X(j) & \text{si } j \in I, \\ w_!(X)(j) \ \simeq\ \emptyset_{\mathcal{M}} & \text{si } j \in J\backslash I, \end{cases}$$

par le calcul des fibres p. 32

$$w_!(X)(j) \ \simeq\ \varinjlim_{i \in I/j} X(i)\ . \qquad \text{[page 64]}$$

Donc la définition de $w_!^W$ donnée dans 3.3.1 revient en fait à passer directement aux catégories localisées pour $w_!$, sans avoir à se restreindre à $\mathcal{M}(I^\circ)_c$, et le lemme des foncteurs adjoints nous dit que ce foncteur est adjoint à gauche de w_W^*.

Comme on a évidemment

$$u_!^W \simeq v_!^W \circ w_!^W , \qquad u_W^* \simeq w_W^* \circ v_W^* ,$$

on en conclut que $u_!^W$ est bien adjoint à gauche de u_W^*, q.e.d.

Corollaire 3.9. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations. Alors le prédérivateur $\mathbf{D}_W$, restreint à la catégorie* Diag_0 *formée des catégories I* ordonnées *artiniennes et telles que* $\mathrm{Esp}(I)$ *soit noethérien, satisfait aux axiomes* Dér 1 (décomposition), Dér 2 (localisation), Dér 3 (objet final), Dér 4 bis (existence de $u_!$), Dér 5 bis (mode de calcul des $u_!$), Dér 6 (exactitude des carrés W-cocartésiens). *Si C, $TC = C \cap W$ sont stables par composition transfinie, on a le même résultat sur la catégorie* Diag *formée des catégories I ordonnées artiniennes sans plus.*

Commentary · Context [ms. 64]

The invariant theory and the auxiliary structures

The point of Corollary 3.9 is that a one-sided structure on the coefficient category already constructs the corresponding half of the theory of diagrams. Cofibrations organize the calculation of left homotopy Kan extensions. A dual structure of fibrations organizes right homotopy Kan extensions. Grothendieck does not require the two chosen classes to determine one another by lifting properties before recognizing the resulting operations as part of the same homotopy theory.

This viewpoint is stated particularly clearly in the letter to Thomason of 2 April 1991: weak equivalences specify the essential data, while fibrations and cofibrations serve to construct operations and establish their existence (Grothendieck, 1991a, point 1, printed pp. 3–4). Brown's categories of fibrant objects provided a one-sided precedent (Brown, 1973, Section 1); the text here works dually and allows noncofibrant objects before passing to the cofibrant subcategory. A Quillen structure imposes the further agreement of the pairs $(C, F \cap W)$ and $(C \cap W, F)$ through lifting and factorization. That agreement is a powerful construction device, but it is a stronger condition than having two separate ways to calculate the operations attached to W.

Two conventions need attention when reading the corollary. First, its axiom numbers are Grothendieck's. Dér 2 is the localization or evaluation-conservativity assertion. Dér 3 concerns the full faithfulness of the constant-diagram functor when the indexing category has a final object; it does not assert that $\mathbf{D}(e)$ has a final object. Dér 4 bis and Dér 5 bis are the existence and pointwise-computation assertions for left Kan extensions. Dér 6 includes the corresponding exactness of base change on the specified homotopy-cocartesian squares. This older list should not be substituted term by term for a modern list numbered Der1–Der4.

Second, the preceding theorem is about covariant diagrams $\mathcal{M}(I^\circ)$, whereas the usual presheaf notation is $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$. Passing to the presheaf prederivator reverses the admissible indexing orders. The artinian condition of Theorem 3.7 becomes a noetherian condition on I; the extra topological condition becomes noetherianity of $\mathrm{Esp}(I^\circ)$. Corollary 3.9 and the corresponding sentence of 3.11 omit this reversal in their wording. Grothendieck gives the corrected presheaf formulation explicitly in Chapter XIX, Proposition 3.2 and the discussion in 3.4 (Grothendieck, 1990–1991j). The finite examples are unaffected, but the infinite indexing convention should be fixed before applying the result.

This also explains why the comma symbols change. For a covariant diagram $X : I \to \mathcal{M}$ the value of $u_!X$ at j is a colimit over $(u \downarrow j) = I/j$. For a presheaf $X : I^{\mathrm{op}} \to \mathcal{M}$ it is a colimit over $(j \downarrow u)^{\mathrm{op}}$. The printed $i'\backslash I$ in the square on manuscript p. 70 is therefore correct in the presheaf convention of 3.11. The square carries its canonical comma natural transformation; it is not meant as an equality of its two composites in the underlying ordinary category of categories.

3.10. Ainsi, pour avoir un dérivateur au plein sens du terme, soit sur Diag_0, soit sur Diag, il ne manque que Dér 4 (existence des u_*), Dér 5 (calcul des u_*) et Dér 6 (exactitude des carrés W-cartésiens). On va les obtenir en principe à l'aide d'une structure de *catégorie à fibrations* $(\mathcal{M}, F, W)$ associé à $(\mathcal{M}, W)$, i.e. l'espèce de structure duale de celle introduite page 39. Ces deux structures n'ont pas à être reliées entre elles d'une façon autre que par le [page 65]

fait qu'elles correspondent au même W. Elles sont considérablement plus faciles à construire que des structures de Quillen (W, C, F), où C, F doivent être reliés de façon extrêmement stricte, avec les propriétés $C \longleftrightarrow \underbrace{TF}_{=F\cap W}$, $\underbrace{TC}_{=C\cap W} \longleftrightarrow F$ (dont il n'est pas question dans l'axiomatique de K. S. Brown), et factorisation pour (C, TF) et (TC, F).

Mais revenant à la construction des $u_!^W$, il reste le problème de les construire pour une flèche $u : I' \longrightarrow I$ *quelconque* dans Cat. [page 66]

À vrai dire, la construction du foncteur

$$u_!^W \ : \ \mathrm{Ho}_W(I^\circ) \ \longrightarrow \ \mathrm{Ho}_W(I'^\circ) \tag{3.3.1}$$

est valable sans supposer que I' soit une catégorie ordonnée – ça peut être une catégorie quelconque. Cela provient du fait que le foncteur induit par $u_!$,

$$u_{!\,\mathrm{cof}} \ : \ \mathcal{M}(I^\circ)_\mathrm{c} \ \longrightarrow \ \mathcal{M}(I'^\circ)$$

est encore compatible avec les localiseurs, comme on voit par le calcul des fibres

$$u_!(X)(i) \ \simeq \ \varinjlim_{i\in I/i'} X(i) \ .$$

Ainsi, on est ramené aux cas où $I' = e$, quitte à remplacer I par I/i', et en notant que ($I/i' \longrightarrow I$ étant étale) l'image inverse de X sur I/i' est cofibrant (cf. 3.2.a).

Commentary · Development (commentators') [ms. 66]

A finite source can require an infinite supply of coefficients

Grothendieck observes that the target of a Kan extension need not itself be ordered: the calculation can be made over comma categories. This is correct in the transfinite setting, but it does not extend the finite-colimit alternative without a further hypothesis. The new comma categories may be much larger than the source. We make the obstruction precise in a case where weak equivalences introduce no additional difficulty.

Take $\mathcal{M} = \mathrm{FinSet}$, let W be the isomorphisms, and let every map be a cofibration. All the finite cofibration axioms hold: finite colimits exist, pushouts preserve isomorphisms, and $f = 1 \circ f$ is a cofibration–weak-equivalence factorization. Every object is cofibrant. For the functor $e \to BG$, covariant restriction is the forgetful functor

$$U : \mathrm{FinSet}^G \longrightarrow \mathrm{FinSet}.$$

The comma category $(e \downarrow *)$ is the discrete set G, so the proposed pointwise formula on a singleton requires $\coprod_G 1$. If G is infinite this colimit is absent from FinSet. There is also a sharper test for whether any left adjoint exists.

Finite-action criterion. [ours]

For a group G, set

$$N = \bigcap_{[G:H]<\infty} H.$$

The forgetful functor U has a left adjoint if and only if $[G : N]$ is finite. When it exists, the left adjoint sends a finite set S to $(G/N) \times S$, with G acting on the first factor.

Proof. Suppose $F \dashv U$, put $P = F(1)$, and let $p \in P$ be the point specified by the unit. Equivariance and the adjunction give a natural bijection

$$\mathrm{Hom}_G(P, A) \longrightarrow U(A), \qquad f \longmapsto f(p).$$

In particular P is nonempty. For the trivial two-element G-set, an equivariant map is precisely a function on the orbit set of P. If P has k orbits, the left side has 2^k elements and the right side has two. Thus $k = 1$ and P is transitive. Write $H = \mathrm{Stab}_G(p)$, so $P \simeq G/H$ and H has finite index.

For every finite-index subgroup $K \subset G$, apply the bijection to $A = G/K$ and the distinguished point K. The required map taking p to K forces $H \subset K$. Hence $H \subset N$. The reverse inclusion follows because H itself occurs in the intersection, so $H = N$ and $[G : N] < \infty$.

Conversely, the intersection defining N is normal, since conjugation permutes the finite-index subgroups. For every finite G-set A and every $a \in A$, its stabilizer has finite index, so N fixes a. All finite actions therefore factor through G/N. If G/N is finite, $(G/N) \times S$ is a finite G-set, and an equivariant map from it to A is uniquely specified by the arbitrary function $s \mapsto f(N, s)$. This is the required adjunction, naturally in both variables. □

For $G = \mathbb{Z}$, the intersection of the subgroups $n\mathbb{Z}$ is zero and has infinite index. Thus U has no left adjoint, even though the source of $e \to B\mathbb{Z}$ has one object and no nonidentity arrows. Equivalently, a hypothetical universal point in a finite $\mathbb{Z}$-set would have a finite orbit of length r, but could not map to the distinguished point in the cycle of length $r + 1$. This disproves the unrestricted finite-colimit reading of the sentence on manuscript p. 66.

The criterion also separates existence of a left adjoint from the pointwise colimit formula. An infinite group with no nontrivial finite quotients has $N = G$, so all its finite actions are trivial and U is an equivalence. Its left adjoint exists although the infinite discrete coproduct appearing in the proposed formula does not. In the general theory, Dér 5 bis is therefore substantive even when Dér 4 bis has been obtained.

We did not find this explicit finite-action criterion in the derived-functor and ABC sources checked here, nor in a targeted search for adjoints of the finite-action forgetful functor. The elementary argument is supplied as a diagnostic continuation of the text, with no claim of priority. The repair needed for Grothendieck's construction is to retain the transfinite colimit hypotheses, or to require that each comma category and its colimit remain within the range of the finite theorem.

Mais il se présente à nouveau la perplexité si le foncteur (3.3.1) ainsi construit est bien [page 67]
adjoint à gauche de u_W^*. On peut songer à utiliser encore la même factorisation en $u = vw$, en
utilisant le mapping-cone. La difficulté, cependant, c'est qu'à présent $J = C(u)$ n'est plus une p. 33
catégorie ordonnée, donc on n'a pas une définition toute prête de $C_{J^\circ} \subset \mathrm{Fl}(\mathcal{M}(J^\circ))$, donnant lieu à une sous-catégorie $\mathcal{M}(J^\circ)_{\mathrm{c}}$ etc. En fait, il semblerait qu'on soit pratiquement paralysé, tant qu'on n'arrive à construire sur $\mathcal{M}(J^\circ)$ également une structure de catégorie à cofibrations, se comportant bien par rapport au foncteurs $v_!$. La "démonstration" d'Anderson de l'existence de $u_!$ pour *toute* flèche dans Cat, foire totalement (même pour $I \longrightarrow e$, I ensemble ordonné fini – mais là j'ai pu rétablir la situation), car il se ramène au cas où I est ordonné localement fini, et il admet sans plus que *dans ce cas* $u_!^W$ est adjoint à gauche de u_W^*, et *de plus,* que si u [page 68]
est $\mathbf{W}_0$-connexe, donc $u^* : \mathcal{M}(I'^\circ) \longrightarrow \mathcal{M}(I^\circ)$ pleinement fidèle, qu'il en soit de même de u_W^* déduit par passage aux localisés. Or même si on admet que $u_!^W$ est adjoint à gauche de u_*^W, faute de disposer d'un morphisme d'adjonction tout près "en bas" (dans $\mathrm{Ho}_W(I'^\circ)$), déduit du morphisme d'adjonction dans $\mathcal{M}(I^\circ)$, il n'est absolument pas clair que u_W^* soit pleinement fidèle, i.e. $u_!^W$ un foncteur de localisation. Le contrexemple 3.6 montre bien que, même si on a adjonction (i.e. si e est un objet initial de $\mathcal{C} = \mathcal{C}'$), le foncteur $u^* = \mathrm{id}_{\mathcal{C}}$ a beau être pleinement fidèle (et même une équivalence de catégories), le foncteur localisé n'a aucune raison de l'être : cela pouverait que tout objet de Cat ayant un objet initial est équivalent à e !

3.11. Disons qu'on a une *catégorie à cofibrations stricte,* si dans l'axiome Cof_3 de factori- [page 69]
sation (cf. p. 39), la factorisation peut être prise fonctoriellement en f. Il revient ± au même de dire que l'axiome de factorisation reste valable, quand on pense à une catégorie $\mathcal{M}(I)$ pour I dans Cat, en y définissant $W(I)$ et $C(I)$ (à ne pas confondre avec C_I) fibre par fibre. Cela revient au même tout au moins si $\mathcal{M}$ est équivalente à une petite catégorie. Sinon, il faut supposer fonctorialité pour toute *petite* sous-catégorie pleine de $\underline{\mathrm{Fl}}(\mathcal{M})$. Il vaut mieux définir la catégorie à cofibrations stricte par cette propriété de stabilité.

On voit alors que le prédérivateur $\mathbf{D} = \mathbf{D}_{\mathcal{M}}$ sur Cat est un dérivateur *relativement à* $\mathcal{D} = \mathrm{Diag}_0$ défini plus haut (catégories ordonnées I artiniennes, telles que $\mathrm{Esp}(I)$ soit noethérien), ou même relativement à $\mathcal{D} = \mathrm{Diag}$ (catégories ordonnées I artiniennes sans plus) lorsque C et TC sont stables par composition transfinie. J'entends par là que $u_!$ existe quand u est une flèche de $\mathcal{D}$, et que ça reste valable quand on passe à un dérivateur induit $\mathbf{D}_Z$ (Z [page 70]
dans Cat quelconque) ; en d'autre termes, que $v_!$ existe quand v est de la forme $\mathrm{id}_Z \times u$, où $u : I \longrightarrow I'$ est une flèche de $\mathcal{D}$, et que le calcul se fait *au dessus de* I' selon la règle Dér 5 bis. Cela signifie donc que la flèche de changement de base pour les $f_!$ est un isomorphismes dans

le cas suivant :

$$\begin{array}{ccc} I \times Z & \longleftarrow & (i'\backslash I) \times Z \\ \downarrow{\scriptstyle u\times \mathrm{id}_Z} & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ I' \times Z & \longleftarrow & \{i'\} \times Z \end{array} ,$$

où i' est un objet quelconque de I'.

L'hypothèse *stricte* pour une catégorie à cofibrations me paraît assez anodine, dans la plupart des cas à ma connaissance, elle est satisfaite. Mais même moyennant cette hypothèse, je ne vois aucun moyen de prouver l'existence de $u_!$ pour *toute* flèche de Cat, et c'est une situation bien sûr très gênante. On voudrait absolument dégager des conditions qui assurent la validité de Dér 4 bis et de Dér 5 bis (existence de $u_!$ pour *toute* flèche de Cat, et mode de calcul des fibres …). p. 34 [page 71]

Commentary · Later developments [ms. 70]

Strict factorization and arbitrary diagram shapes

Grothendieck introduces strictness to make the theorem uniform in an extra parameter. A natural transformation of Z-diagrams is itself a functor $Z \to \mathrm{Arr}(\mathcal{M})$. A functorial factorization on the arrow category factors all its components together and supplies the transition maps between the intermediate objects. Thus the coefficient category $\mathcal{M}^Z$, with weak equivalences and cofibrations tested componentwise, again has the required factorization axiom. Colimits and the stated closure properties are also tested componentwise. Applying the earlier theorem to these new coefficients constructs the extension along $u \times 1_Z$.

There is a converse behind the qualification about size. If $\mathcal{M}$ is small, factor the universal arrow diagram indexed by $\mathrm{Arr}(\mathcal{M})$; its two endpoint functors and their universal natural transformation recover a functorial factorization on all arrows. For an essentially small category, first choose a small equivalent one. In a large category the all-small-diagrams assertion only makes this choice on every small full subcategory of the arrow category, which is the formulation Grothendieck finally adopts. No global choice on a proper class is needed for the application to a particular small Z.

The arbitrary-shape question at the end of 3.11 was subsequently answered under hypotheses appropriate to all small homotopy colimits. In Rădulescu-Banu (2009, Theorems 9.6.3 and 9.6.5), the existence of the left adjoint and its pointwise comma-category formula are proved separately for ABC cofibration categories; Theorem 10.3.2 identifies the resulting strong left Heller derivator. Here we use the numbering of arXiv:math/0610009v4. Its axioms CF5 and CF6 provide small coproduct and sequential-composition control. The theorem therefore does not assert that the finite axioms alone suffice. The construction proceeds through direct simplicial resolutions of arbitrary categories, rather than requiring a projective Quillen structure on every original diagram category.

Cisinski (2010, Theorem 6.11 and Corollary 6.21) gives the corresponding presheaf and derivator statements in a fibration-oriented convention, with the required homotopical completeness assumptions; the dual statements concern cofibrations. These references establish the later affirmative results without turning the source's uncertainty into a present-day open problem. They also explain why strictness and specialness should remain distinct: strictness is a coherent factorization condition, while the next section adds a set of generators and transfinite closure to obtain a particular transfer construction.

4 Catégories à cofibrations spéciales

4.1. Je suppose à présent que

ⓐ $\mathcal{M}$ est stable par $\varinjlim$ filtrantes, et

ⓑ que ses objets sont accessibles,

de sorte qu'on est sous les conditions d'application du théorème de factorisation (XV, §1). On suppose :

ⓒ donné $W, C \subset \mathrm{Fl}(\mathcal{M})$, satisfaisant les conditions Cof_1, Cof_2 de page 39. Pour Cof_3, cf. plus bas.

ⓓ C, $TC \overset{\text{déf}}{=} C \cap W$ *stables par composition transfinie* (c'est le cas si la structure provient d'une structure de Quillen close (W, C, F)).

$$TF \overset{\text{déf}}{=} C_* \ \subset \ \mathrm{Fl}(\mathcal{M}) \ , \tag{4.1}$$

il résulte du théorème de factorisations que toute flèche f de $\mathcal{M}$ se factorise en

$$f \ = \ pi \ , \qquad p \in TF, \ i \in C \ ,$$

pourvu qu'on fasse l'hypothèse

ⓔ

$$\exists\, C_0 \subset C, \ \textit{petite} \text{ partie de } C, \ \text{telle que } C \subset \tilde{C}_0, \ \text{i.e. } TF = (C_0)_*. \tag{4.3}$$

Sous ces conditions, si on suppose

ⓕ

$$TF \subset W \ , \tag{4.4}$$

alors l'axiome de factorisations Cof_3 est satisfait. Inversement, si Cof_3 est satisfait, et si W est stable par facteurs directs, alors on a 4.4. Donc si W est stable par facteurs directs, Cof_3 équivaut à (4.4). [page 72]

Commentary · Exposition [ms. 71]

What the small set of cofibrations contributes

Grothendieck now changes the method. The induction over a well-founded indexing order is replaced by an induction that attaches solutions to lifting problems. A small set of maps can test all the required right lifting properties at once. This is why the set C_0 enters the argument precisely when the indexing category is allowed to have arbitrary arrows and endomorphisms.

For a class S of maps, write

$$S^{\perp} = \{p : s \perp p \text{ for every } s \in S\}, \qquad \widetilde{S} = {}^{\perp}(S^{\perp}).$$

The tilde denotes closure for the lifting relation, not closure under weak equivalence. If $C_0 \subset C \subset \widetilde{C_0}$, then $C^{\perp} = C_0^{\perp}$ by the two inclusions in the lifting relation. Under the smallness hypotheses, $\widetilde{C_0}$ is also the class of retracts of relative C_0-cell maps. These are the two descriptions connected by the small-object argument; they should not be identified without its hypotheses.

Here is the factorization mechanism in the form needed in 4.1. Given $f : X \to Y$, begin with $X_0 = X$. At a successor stage, form the coproduct of all maps $a : A \to B$ in C_0, indexed further by all commutative squares from a to $X_\alpha \to Y$. Push this coproduct out along its maps to X_α, obtaining $X_\alpha \to X_{\alpha+1}$ and an induced map $X_{\alpha+1} \to Y$. At a limit stage take the colimit of the preceding objects. The set of generators and local smallness ensure that each attaching family is a set. Choose a regular cardinal large enough for the domains of all the generators, and carry the construction to a limit ordinal of at least that cofinality. Every map from a generating domain to the final object then factors through some stage, so its associated lifting square is solved at the next stage. The resulting map $p : X_\lambda \to Y$ lies in $C_0^{\perp} = C^{\perp}$.

The left factor $i : X \to X_\lambda$ lies in C. Indeed, a coproduct of cofibrations can be attached one summand at a time: well-order the indexing set, replace one source summand by its target at each step, and take colimits at limits. Each step is a pushout of a map in C, and transfinite closure finishes the argument. The same closure applies to the successive attaching maps and then to their composite. This gives $f = pi$ with $i \in C$ and $p \in C_*$; the extra assumption $C_* \subset W$ supplies the required weak equivalence. For the standard modern formulation, see Riehl (2014, Theorem 12.2.2 and Corollary 12.2.4).

The converse on manuscript p. 72 is a short retract argument. Factor $p \in C_*$ as $p = qi$ with $i \in C$ and $q \in W$. In the square having top map $1_{\mathrm{dom}\, p}$, left map i, right map p, and bottom map q, a lift h satisfies $hi = 1$ and $ph = q$. Therefore p is a retract of q in the arrow category. Retract closure of W gives $p \in W$. This is exactly where that additional hypothesis is used.

The accessibility assumption should also be read at its stated strength. It provides a smallness cardinal for each relevant object, and the set C_0 permits a common bound for the domains occurring in the construction. It does not, by itself, state that the whole category has a set of presentable generators. Thus a special cofibration category is not automatically being identified here with a combinatorial model category. A second lifting/factorization pair and its compatibility with W are still to be constructed.

4.2. Il me semble que ces conditions, en pratique, ne seront guère jamais remplies, que si W (qui est donné en tout premier lieu, satisfaisant Cof_0 [18]), n'est pas stable par $\varinjlim$ filtrantes. À ce moment, pour construire la structure à cofibrations, on part de C_0 petite p. 35
partie de $\mathrm{Fl}(\mathcal{M})$, prise telle que l'on ait

$$(C_0)_* \subset W \ , \tag{4.5}$$

on prend pour C au choix le saturé C_0' de C_0 par cochangement de base et composition transfinie, ou $C = \tilde{C}_0 =$ ensemble des facteurs directs de C_0'. Les conditions Cof_1 (stabilités de C), Cof_3 (factorisation) sont satisfaites automatiquement, et il reste à s'occuper de Cof_2, qui revient à la stabilité de $C \cap W$ par cochangement de base (la stabilité par composition et que ça contienne les isomorphismes, étant immédiates). C'est une condition délicate, et la seule façon standard que je vois pour l'assurer, c'est en supposant

$$C_0 \subset \mathrm{Cof}_W \ ; \tag{4.6}$$

la stabilité de W par $\varinjlim$ filtrantes (donc aussi par facteurs directs) assure alors qu'on a également [page 73]

$$C \subset \mathrm{Cof}_W \ , \tag{4.7}$$

et par suite

$$\begin{cases} C \cap W \subset \mathrm{Cof}_W \cap W = W^{\mathrm{univ}}, \\ \text{d'où } \ TC \overset{\text{déf}}{=} C \cap W = C \cap W^{\mathrm{univ}} \ , \end{cases} \tag{4.8}$$

ce qui assure la stabilité de $C \cap W$ par cochangement de base. La stabilité de TC par composition transfinie résulte de celle de C, et de la stabilité de W par composition transfinie.

4.3. Je reviens aux conditions générales ⓐ à ⓕ de 4.1, sans supposer nécessairement W stable par $\varinjlim$ filtrantes. Je veux à présent étendre la structure à cofibrations aux $\mathcal{M}(I)$, I quelconque dans Cat. Nous n'avons pas pour ceci (je crois) à utiliser les développements des paragraphes 2, 3, mais nous ferons le lien avec ceux-ci par la suite.

Désignons par I_0 la catégorie discrète ayant mêmes objets que I, d'où un foncteur canonique

$$v = v_I \ : \ I_0 \longrightarrow I \ . \tag{4.9}$$

L'idée est de définir la structure de $\mathcal{M}(I)$ via la structure évidente de $\mathcal{M}(I_0) \simeq \mathcal{M}^{I_0}$, en jouant sur le couple de foncteurs adjoints

$$\mathcal{M}(I_0) \underset{\psi = v^*}{\overset{\varphi = v_!}{\rightleftarrows}} \mathcal{M}(I) \quad .$$

Nous allons présenter les choses de façon un peu plus générale.

[page 74]

18. i.e. saturation faible.

Commentary · Exposition [ms. 73]

Why intrinsic cofibrations control the weak equivalences

The difficulty in 4.2 is not producing a large class of cofibrations. Pushouts and transfinite compositions already do that. The problem is keeping those maps that are also weak equivalences stable under arbitrary cobase change. Grothendieck's intrinsic class Cof_W addresses precisely this issue: a map belongs to it when pushout along it preserves weak equivalences between objects under its source. This class is determined by W, while C_0 is a chosen supply of maps with which to construct factorizations.

Suppose W is stable under filtered colimits of diagrams of arrows. Pushout stability of Cof_W follows from the associativity of pushouts. For a transfinite composite of intrinsic cofibrations, push a weak equivalence through the successive stages. At a successor stage it remains a weak equivalence by the defining property; at a limit stage it remains one by filtered-colimit stability. This proves the transfinite closure needed to pass from C_0 to its cellular closure.

The same hypothesis on W also gives retract closure, by a useful elementary argument. If an arrow f is a retract of an arrow $g \in W$, let e be the corresponding idempotent endomorphism of g in the arrow category. The sequence

$$g \xrightarrow{e} g \xrightarrow{e} g \xrightarrow{e} \cdots$$

has colimit f. To see this, compose the retraction and section maps with the defining cocones; the identities for a split idempotent establish both universal properties. All objects of this sequence lie in W, so its colimit does as well. Retracts of intrinsic cofibrations are then intrinsic cofibrations: any pushout comparison for the retract is a retract of the corresponding comparison for the original arrow. This proves the passage from the cellular closure to $\widetilde{C_0}$.

The equality $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$ used in the text relies here on the factorization already obtained, and admits a direct verification. Let $a : A \to B$ be both intrinsic and weakly invertible, and take any map $A \to A'$. Factor the latter as $A \xrightarrow{c} Z \xrightarrow{q} A'$ with $c \in C \subset \mathrm{Cof}_W$ and $q \in W$. Pushing a along c gives a weak equivalence $Z \to B \amalg_A Z$. Since the pushout $Z \to B \amalg_A Z$ of a is itself intrinsic, pushing the weak equivalence q along it gives a weak equivalence

$$B \amalg_A Z \longrightarrow B \amalg_A A'.$$

The commutative square with these two maps and q shows, by two-out-of-three, that $A' \to B \amalg_A A'$ is a weak equivalence. Conversely, if all pushouts of a are weak equivalences, the two horizontal pushouts of a in any cobase-change comparison are in W; two-out-of-three shows that the comparison is in W whenever its input is. Thus a is intrinsic.

Consequently $TC = C \cap W^{\mathrm{univ}}$ is pushout-stable. Its transfinite stability follows from that of C and the limit-stage hypothesis on W. The negative wording "n'est pas stable" at the end of manuscript p. 72 is printed in the edition but conflicts with the positive stability hypothesis used throughout this argument. We retain the wording in the source and record the intended sign in the correction register.

Définition 4.4. J'appelle structure de cofibrations *spéciale* une structure $(\mathcal{M}, W, C)$ satisfaisant les conditions a) à f) de 4.1, que je reformule :

ⓐ $\mathcal{M}$ stable par petites limites inductives, et les objets de $\mathcal{M}$ sont accessibles. p. 36

ⓑ W est faiblement saturé (i.e. Cof_0 de p. 39), C stable par cochangement de base et par composition, $TC \overset{\text{déf}}{=} C \cap W$ stable par cochangement de base (et par composition automatiquement).

ⓑ′ C et TC sont stables même par composition *transfinie.*

ⓒ $\exists$ petite partie $C_0 \subset C$ telle que $C \subset \tilde{C}_0$ ($\Longrightarrow \tilde{C} = \tilde{C}_0$).

ⓓ $TF \overset{\text{déf}}{=} C_* \subset W$ (ce qui implique, joint àⓐ àⓒ, l'axiome de factorisation Cof_3 de page 39).

4.5. Nous allons donner une recette générale d'extension d'une telle structure, quand on

dispose d'un couple de foncteurs adjoints

$$\mathcal{M} \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}' \ ,$$

$\mathcal{M}$ étant muni d'une structure de cofibrations spéciale (W, C), et $\mathcal{M}'$ satisfaisant la condition liminaire ⓐ. On définira alors les parties W', C' de $\mathrm{Fl}(\mathcal{M}')$ par [page 75]

$$(4.5.1) \qquad \begin{cases} W' \ = \ (v^*)^{-1}(W) \\ C' \ = \ \text{clôture stable par cochangement de base} \\ \qquad\quad \text{et composition transfinie de } v_!(C), \end{cases}$$

donc

$$(4.5.2) \qquad v_!(C) \ \subset \ C' \ \subset \ (v_!(C))^{\sim} \ = \ \tilde{C}' \ .$$

Je dis que (W', C'), et au choix $(W', \tilde{C}')$, est une structure de cofibrations spéciale sur $\mathcal{M}'$. On va vérifier les différentes conditions de 4.4. La condition a) est déjà acquise.

Commentary · Exposition [ms. 75]

The transfer proof and its exact hypotheses

The adjunction in 4.5 is meant to reconstruct a diagram from its values at objects. Grothendieck first proves the transfer abstractly so that the construction does not depend on a particular indexing category. Write $L = v_!$ and $U = v^*$. The proposed weak equivalences are detected by U, while the proposed cofibrations are built from the images under L of the old cofibrations. These are different parts of the construction, and the proof must show that their intersection behaves correctly.

Let $C' = \mathrm{cell}(LC)$ denote closure under pushouts and transfinite composition, including isomorphisms, and let $W' = U^{-1}W$. The two substantive assumptions are

$$U \text{ preserves small colimits}, \qquad UL(C) \subset C.$$

Every operation forming C' is then sent to the corresponding operation in the source category. Induction through a cellular construction gives $U(C') \subset C$. Therefore

$$U(C' \cap W') \subset C \cap W = TC.$$

If $a' \in C' \cap W'$ is pushed out in the target, its image under U is a pushout of a trivial cofibration. It remains in TC, so the target pushout lies in W' as well as in C'. For a transfinite composite of maps in $C' \cap W'$, preservation of the relevant colimits gives a transfinite composite in TC; the same conclusion follows. This proves the two closure statements whose compatibility was initially in doubt.

The set of generators is transferred using the left adjoint, not the right one. If $C_0 \subset C \subset \widetilde{C_0}$, then L preserves pushouts and transfinite composites, and every functor preserves retract diagrams. Hence

$$L(C_0) \subset C' \subset \widetilde{L(C_0)}.$$

The target smallness hypothesis permits the small-object argument for this new set. Notice that the assertion is about a generating set for the lifting closure; it does not claim that every map of C' is a cell complex on $L(C_0)$ without taking retracts.

Finally, right lifting properties pass across an adjunction exactly. A commutative square from $L(a)$ to p' corresponds to a square from a to $U(p')$, and diagonal fillers correspond under the same bijection. Since left lifting properties survive pushouts, transfinite compositions and retracts,

$$(C')^{\perp} = (LC)^{\perp} = U^{-1}(C^{\perp}) = U^{-1}(TF) \subset U^{-1}(W) = W'.$$

The small-object factorization therefore has a weakly invertible right factor. These arguments prove Proposition 4.6 without assuming that U preserves all weak equivalences by some separate theorem: that property is part of the definition of W'.

For the optional replacement of C' by $\widetilde{C'}$, the proof needs C to be retract-closed. Then $U(\widetilde{C'}) \subset C$, and the pushout and transfinite arguments apply again to $\widetilde{C'} \cap W'$. This does not require silently adding retract closure of W'. Footnote 20 offers a different route, developed in Proposition 5.1, using additional stability assumptions on W. The announcement in 4.5 should be read with the hypotheses collected in the actual proposition, not as a theorem preceding them.

In the diagram application, U is evaluation on the discrete object set and genuinely does preserve colimits. The composite UL is a coproduct of the old objects indexed by arrows of the indexing category. The stability of C under such coproducts, proved by attaching summands as above, is what makes the abstract assumption $UL(C) \subset C$ usable there. The next question concerns trivial cofibrations and is not answered merely by this cofibration calculation.

b) [19]. La saturation faible de W' résulte de celle de W, via (4.5.1). La stabilité de C' par cochangement de base et par composition (même transfinie) est acquise par définition de C', et itou pour $\tilde{C}'$ (étant Q-close à gauche). Stabilité de $TC' \overset{\text{déf}}{=} C' \cap W'$ par cochangement de base ? Il suffit de voir que si on a un carré cocartésien

$$\begin{array}{ccc} X'_0 & \xrightarrow{i'} & X'_1 \\ \downarrow & \text{cocart.} & \downarrow \\ X'_2 & \xrightarrow{j'} & X'_3 \end{array}$$

dans $\mathcal{M}'$, avec i' dans $TC' = C' \cap W'$, alors $j' \in W'$ (car on sait déjà $j' \in C'$), i.e. $v^*(j') \in W'$. ||
Nous allons supposer que v^* *commute aux sommes amalgamées,* et on est donc ramené à prouver que $v^*(i') \in TC$. Donc on doit prouver le p. 37
[page 76]

Lemme 4.5.2. *Sous les conditions* (4.5.3) *et* (4.5.4) *ci-dessous, on a*

$$v^*(C') \ \subset \ C \ , \qquad v^*(TC') \ \subset \ TC \ ,$$

et si $C = \tilde{C}$, *on a* $v^*(\tilde{C}') \subset C$, $v^*(T\tilde{C}') \subset TC$.

Pour $v^*(C') \subset C$, il faut d'abord prouver

$$v^*(v_!(C)) \ \subset \ C \ , \tag{4.5.3}$$

et je ne vois aucun moyen – il faut le poser comme hypothèse. De plus, que v^* commute aux $\varinjlim$ quelconques (cochangement de base déjà admis – à présent on a besoin des limites inductives filtrantes)

$$v^* \text{ commute aux petites } \varinjlim \text{ quelconques} \tag{4.5.4}$$

pour conclure de (4.5.3) que $v^*(C' = \text{clôture de } v_!(C)$ pour cochangement de base et composition transfinie) est également dans C. Passant à $v^*(TC') \subset TC$, par ce qui précède, $v^*(C') \subset C$, il suffit de prouver $v^*(TC') \subset W$, mais $v^*(TC') \subset v^*(W') \subset W$ par définition de W'.

Supposons $C = \tilde{C}$, i.e. C stable par facteurs directs, et prouvons $v^*(\tilde{C}') \subset C$, $v^*(T\tilde{C}') \subset TC$. Par c) plus bas, $\tilde{C}'$ est formé des facteurs directs d'éléments de C', donc $v^*(C') \subset C$ implique $v^*(\tilde{C}') \subset C$ puisque C stable par facteurs directs. [page 77]

Prouvons enfin $v^*(T\tilde{C}') \subset TC \overset{\text{déf}}{=} C \cap W$. Comme $T\tilde{C}' = \tilde{C}' \cap W'$, cela résulte de $v^*(\tilde{C}') \subset C$, et de $v^*(W') \subset W$.

Prouvons b′). La stabilité de C' et de $\tilde{C}'$ par composition transfinie est acquise, il reste à prouver la stabilité de $TC' = C' \cap W'$ et de $T\tilde{C}' = \tilde{C}' \cap W'$. Il suffit de voir que le composé transfinie f en question est dans W', donc que $v^*(f) \in W$, mais cela résulte du fait que v^* commute à la composition transfinie par (4.5.4), et que v^* transforme TC' et $T\tilde{C}'$ dans TC.

19. **N.B.** Il vaut mieux commencer par prouver c).

Prouvons c) pour C' (donc aussi pour $\tilde{C}'$). Soit C_0 une petite partie de C telle que $C \subset \tilde{C}_0$, i.e. C est contenu dans la clôture de C_0 par les trois opérations de cochangement de base, de composition transfinie, des facteurs directs. On voit alors que $v_!(C)$ est dans la clôture de $v_!(C_0)$ pour les opérations, donc aussi C', et on peut alors prendre $C'_0 = v_!(C_0)$.

Prouvons enfin d), c'est la même assertion pour C' et pour $\tilde{C}'$, puisque

$$C'_* \ = \ \tilde{C}'_* \ (\overset{\text{déf}}{=} \ TF' \ ) \ ,$$

il faut donc prouver [page 78]

$$TF' \overset{\text{déf}}{=} C'_* \ \subset \ W' \ .$$

Cela va résulter du p. 38

Lemme 4.5.3. *On a* $TF' = (v^*)^{-1}(TF)$.

C'est formel par définition de TF' comme

$$TF' \ = \ (v_!(C))_* \ ,$$

de TF comme

$$TF \ = \ C_* \ ,$$

et de l'adjonction de $v_!$, v^*. Comme $TF \subset W$, on en conclut

$$TF' \ = \ (v^*)^{-1}(TF) \ \subset \ (v^*)^{-1}(W) = W' \ ,$$

q.e.d.

Donc on a prouvé ceci :

Proposition 4.6. *Considérons un couple de foncteurs adjoints*

$$\mathcal{M} \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}' \ ,$$

où $\mathcal{M}$ *est muni d'une structure de cofibrations* spéciale (W, C). *On suppose de plus les conditions suivantes satisfaites.*

(4.6.1)
- 1°) $\mathcal{M}'$ *stable par petites* $\varinjlim$ *, ses éléments sont accessibles.*
- 2°) v^* *commute aux petites* $\varinjlim$.
- 3°) $v^*(v_!(C)) \subset C$.

Considérons les parties W', C' *de* $\mathrm{Fl}(\mathcal{M}')$ *définies par* [page 79]

(4.6.2)
$$\begin{cases} W' \ = \ (v^*)^{-1}(W), \\ C' \ = \ \textit{clôture de } v_!(C) \textit{ par cochangement de base et par composition} \\ \qquad \textit{transfinie.} \end{cases}$$

Alors $(\mathcal{M}', W', C')$ *est une structure à cofibrations spéciale. Si de plus* $C = \tilde{C}$, *i.e.* C *est stable par facteurs directs, alors* $(\mathcal{M}', W', \tilde{C}')$ *est également une structure à cofibrations spéciale* [(20)].

4.7. J'aimerais, en vue d'énoncés du type 3.7 (établissant l'existence de foncteurs du type $u_!^W$), avoir une prise sur $TC' = C' \cap W'$, et sur $T\tilde{C}' = \tilde{C}' \cap W'$, de façon précise, j'aimerais avoir la relation

$$v_!(TC) \ \subset TC' \subset v_!(TC)^\sim \ , \tag{4.7.1}$$

voire la relation légèrement plus forte

$$\underbrace{v_!(TC) \ \subset TC'}_{} \ \overset{\text{tautol.}}{\subset} \ \underbrace{T\tilde{C}' \ \subset \ (v_!(TC))^\sim}_{} \ , \tag{4.7.2}$$

p. 39

qui se décompose en deux

$$\left\{\begin{array}{rcl} v_!(TC) & \subset & TC' \\ T\tilde{C}' & \subset & (v_!(TC))^\sim . \end{array}\right. \tag{4.7.3}$$

La moins délicate des deux est la première, $v_!(TC) \subset TC'$, qui revient, en vertu des définitions, impliquant [page 80]

$$v_!(TC) \subset v_!(C) \subset C' ,$$

à

$$v_!(TC) \ \subset \ W' ,$$

i.e. par définition de W', à

$$v^* v_!(TC) \ \subset \ TC , \tag{4.7.4}$$

ce qui est une hypothèse de type (4.6.1) 3°), à joindre aux hypothèses (4.6.1).

Commentary · Exposition [ms. 79]

The remaining comparison is a comparison of lifting classes

Grothendieck has enough structure to factor arrows in the new category, but he needs more to derive arbitrary left Kan extensions in a controlled way. The further issue is whether the transferred trivial cofibrations are generated, for lifting purposes, by the images of the original trivial cofibrations. It is this question, rather than the already established existence of a special cofibration structure, that occupies 4.7.

The forward inclusion is transparent once the hypotheses are separated. Since $L(TC) \subset L(C) \subset C'$, one has

$$L(TC) \subset TC' \quad \Longleftrightarrow \quad UL(TC) \subset W \quad \Longleftrightarrow \quad UL(TC) \subset TC.$$

The last equivalence uses $UL(C) \subset C$. Thus equation (4.7.4) is an additional preservation condition, not a consequence of Proposition 4.6 alone.

Assume it from now on. The inclusion $LC \supset L(TC)$ does not control the reverse comparison $TC' \subset \widetilde{L(TC)}$: a cofibration built from LC may happen to be weakly invertible without presenting itself as an attachment of maps from $L(TC)$. Passing to right lifting classes isolates the exact missing assertion. With $F = (TC)^\perp$ and $F' = (TC')^\perp$, adjunction gives

$$(L(TC))^\perp = U^{-1}(F), \qquad F' \subset U^{-1}(F).$$

The reverse inclusion is equivalent to $TC' \subset {}^\perp((L(TC))^\perp)$, which is precisely the desired Quillen-closure inclusion. The equivalence is a formal Galois calculation and does not assume an unproved identification of a lifting closure with a cellular closure.

The same argument applies after retract completion, replacing TC' by $\widetilde{C'} \cap W'$. Proposition 5.1 will compare these two trivial classes; Proposition 5.8 will provide the stronger transfer theorem with the required factorization hypotheses. The question on manuscript p. 80 should therefore be read as a problem that directs the next section. The later propositions answer it under further hypotheses; they do not assert that every special cofibration structure already satisfies this stronger comparison.

La relation hypothétique

$$(*) \qquad \underbrace{T\tilde{C}'}_{=\tilde{C}'\cap W'} \ \subset \ (v_!(TC))^\sim , \quad \text{ou seulement} \quad TC' \subset (v_!(TC))^\sim ,$$

m'apparaît de nature nettement plus délicate. Comme $T\tilde{C}' \supset TC' \supset v_!(TC)$, elle équivaut à l'*égalité* de $(T\tilde{C}')_*$ resp. $(TC')_*$ et $(v_!(TC))_* = (v^*)^{-1}(TC)_*$. Posons

$$\left\{\begin{array}{rcl} (TC)_* & \overset{\text{déf}}{=} & F \\ (TC')_* & \overset{\text{déf}}{=} & F' \ \supset \ (T\tilde{C}')_* \ = \ F'_\sim . \end{array}\right. \tag{4.7.5}$$

20. **N.B.** C'est là un fait général pour les catégories à cofibrations spéciales, et devrait être explicité plus bas (cf. 5.1). L'hypothèse sur C est inutile ici (ou plutôt, il suffit au lieu de cela que W soit stable par facteurs directs *et* par composition transfinie, ce qui n'est pas une hypothèse plus réjouissante …).

On a donc, en vertu de l'inclusion

$$v_!(TC) \ \subset \ TC' \ (\subset \ T\tilde{C}' \ ) \ ,$$

la relation

$$F'_\sim \ \subset \ F' \ \subset \ (v^*)^{-1}(F) \ , \tag{4.7.6}$$

et la question est donc si on a égalité $F' = (v^*)^{-1}(F)$, voire $F'_\sim = (v^*)^{-1}(F)$ (d'où $F'_\sim = F'$).

En fait, on montre assez facilement qu'on a $F'_\sim = F'$, i.e. $T\tilde{C}' \overset{\text{déf}}{=} \tilde{C}' \cap W' \subset (C' \cap W')^\sim$ – c'est une histoire générale de catégories à cofibrations spéciales [21]. Mais je ne vois guère comment prouver (sous des hypothèses convenables) que $F' = (v^*)^{-1}(F)$. Il me faudra revenir là-dessus au besoin [22]. Pour l'instant, je me bornerai à récolter le fruit de 4.6 dans un cas particulier : [page 81]

Commentary · Exposition [ms. 81]

Free diagrams and the missing coproduct lemma

Grothendieck now has a way to give a diagram its cofibrations without choosing an order in which to construct its objects. The price of this freedom is to begin with a diagram having no relations, then attach relations by pushouts. This is the purpose of the adjunction attached to the inclusion of the objects of I. Write $L = v_!$ and $R = v^*$. Since $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$, the free diagram on a family (X_j) is

$$(LX)(i) = \coprod_{j \in \mathrm{Ob}\, I} \mathrm{Hom}_I(i,j) \cdot X_j.$$

Here $S \cdot X = \coprod_{s \in S} X$ is a copower. The exponent in the source denotes this coproduct, not a product or an internal Hom. An arrow $i' \to i$ acts by precomposition on the indexing sets $\mathrm{Hom}_I(i,j)$. Thus an arbitrary collection of maps $X_j \to Y(j)$ extends uniquely to a natural transformation $LX \to Y$: on the summand indexed by $i \to j$, use the corresponding structure map of the contravariant diagram Y. This also verifies the variance of the formula directly.

The assertion that the product category has a special cofibration structure contains a smallness argument worth making explicit. If C_0 is a generating set for the right lifting tests in $\mathcal{M}$, put a map of C_0 in one coordinate and the identity of the initial object in every other coordinate. The union of these sets over $\mathrm{Ob}\, I$ is still a set. A family of maps has the right lifting property against it exactly when every component lies in C_*. This gives the required generating set for the product. Colimits and the stated closure properties are computed componentwise. Likewise, a diagram of accessible objects is accessible: choose a regular cardinal larger than the cardinality of the arrows of I and larger than the smallness bounds for all its components. The naturality equations express its Hom functor as an equalizer of two products of component Hom functors. At this cardinal, the relevant filtered colimits of sets commute with those products and the equalizer. The same argument works with sufficiently long ordinal chains in the small-object formulation of accessibility. No common small generating family for the whole category $\mathcal{M}$ is required by this argument.

The coproduct assertion singled out in footnote 25 follows from precisely the two closure operations already assumed. Suppose $\mathcal{A}$ contains the isomorphisms and is stable under pushout and continuous transfinite composition. Given a set of maps $a_s : A_s \to B_s$ in $\mathcal{A}$, well-order its index set as an ordinal λ and form

$$X_\alpha = \left(\coprod_{s<\alpha} B_s \right) \amalg \left(\coprod_{\alpha \le s < \lambda} A_s \right) \qquad (\alpha \le \lambda).$$

The arrow $X_\alpha \to X_{\alpha+1}$ is the pushout of a_α along its coproduct inclusion. At a limit ordinal the displayed object is the colimit of the previous ones, as follows by specifying a map out of each summand. Therefore $X_0 \to X_\lambda$, which is $\coprod_s a_s$, lies in $\mathcal{A}$. The empty family gives the identity of the initial object. Apply this argument separately to C and to TC. It proves that RL preserves both

21. cf. prop. 5.1 ci-dessous.
22. voir cependant prop. 5.8 ci-dessous.

classes and supplies the lemma requested in the footnote. This is the familiar cellular coproduct argument; compare Hovey (1999a, Lemma 2.1.13).

The outcome of Proposition 4.8 is consequently quite concrete. A freely attached diagram cell is a family of copies of a coefficient cell, one for each incoming arrow at an object. Closing under pushouts and transfinite attachment produces diagram cofibrations; the right lifting tests are pointwise. The first formula (4.8.1) has $TC(I)$ where both occurrences must be $TC(I_0)$, since RL is an endofunctor of the category of families. The proof itself uses the correctly typed assertion.

Proposition 4.8. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations spéciale* (déf. 4.4, page 94). *Soit I objet de* Cat*, I_0 la catégorie discrète associée, considérons $v : I_0 \longrightarrow I$ et* p. 40

$$\mathcal{M}(I_0) \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}(I) \quad ,$$

enfin munissons $\mathcal{M}(I_0) = \mathcal{M}^{I_0}$ de la structure de catégorie à cofibrations, produit *de celles des facteurs, i.e. définissons $W(I_0)$, $C(I_0)$ "fibre à fibre". (Cela coïncide avec la définition générale donnée au paragraphe* 2*, pour I_0 muni de la structure d'ordre discret, et fait de $\mathcal{M}(I_0)$ une catégorie à cofibrations spéciale.) Alors le couple $(v_!, v^*)$ satisfait aux conditions de la* [page 82] *proposition* 4.6*, ainsi qu'à la condition supplémentaire à* (4.6.1)*,*

$$4^\circ) \qquad v^*(v_!(TC(I))) \ \subset \ TC(I) \ . \tag{4.8.1}$$

Donc on trouve une structure de catégorie à cofibrations spéciale sur $\mathcal{M}(I)$, en posant

$$W(I) \ = \ (v^*)^{-1}(W(I_0)) \tag{4.8.2}$$

(c'est en fait équivalent à la définition habituelle de $W(I)$) et

$$C(I) \quad = \quad \text{clôture de } v_!(C(I_0)) \text{ pour les opérations de cochangement de base et composition transfinie.} \tag{4.8.3}$$

On a de plus sous ces conditions :

$$\underbrace{TF(I)}_{\overset{\text{déf}}{=}(C(I))_*} \ = \ (v^*)^{-1}\Big(\ \underbrace{TF(I_0)}_{\substack{\overset{\text{déf}}{=}(C_{I_0})_*, \\ \text{se définit aussi} \\ \text{fibre par fibre}}} \ \Big) \ , \tag{4.8.8}$$

$$\begin{cases} v_!\big(\overbrace{(TC)(I_0)}^{C(I_0)\cap W(I_0)} \big) & \subset \quad TC(I) \\ F(I) \overset{\text{déf}}{=} (TC(I))_* & \subset \quad (v^*)^{-1}(F(I_0)). \end{cases} \tag{4.8.9}$$

(23)

Démonstration. On sait déjà que $\mathcal{M}(I)$ est stable par petites $\varinjlim$, et que tout élément est accessible, comme il résulte aussitôt des propriétés idoines de $\overrightarrow{\mathcal{M}}$. Il est clair aussi que v^* commute aux petites $\varinjlim$. Il reste à prouver (4.6.1. 3°), et 4° dans (4.8.1), i.e. que [page 83] $v^*v_! : \mathcal{M}(I_0) \longrightarrow \mathcal{M}(I_0)$ applique C dans C, TC dans TC. Or on a

$$v_!(F_0)(i) \ = \ \varinjlim_{(\alpha,j)\in i\backslash I_0} F_0(j) \tag{24}$$

et $i\backslash I_0$ est la catégorie discrète au dessus de I_0, dont la fibre en $j \in \mathrm{Ob}\, I_0 \ = \ \mathrm{Ob}\, I$ est $\mathrm{Hom}_I(i,j)$:

$$i\backslash I_0 \ = \ \coprod_{j\in \mathrm{Ob}\, I} (\text{catégorie discrète associée à } \mathrm{Hom}_I(i,j)) \ , \qquad \text{p. 41}$$

donc

$$v_!(F_0)(i) \;=\; \coprod_{j\in \mathrm{Ob}\, I=\mathrm{Ob}\, I_0} \underbrace{F_0(j)^{\mathrm{Hom}_I(i,j)}}_{\substack{\text{somme directe}\\ \text{de } \mathrm{Hom}_I(i,j)\\ \text{copies de } F_0(j)}},$$

donc on trouve

$$v^* v_! F_0(\underbrace{\quad i \quad}_{\in \mathrm{Ob}\, I_0 = \mathrm{Ob}\, I}) \;=\; v_!(F_0)(i) \;=\; \curvearrowright \quad .$$

Comme C et TC sont stables par sommes directes quelconques (l'étant par sommes directes finies, à cause des conditions de stabilité finies sur C et TC, et aussi par somme infinies à cause de l'axiome de stabilité transfinie …[25]), il s'ensuit bien que $v^*v_!$ applique $C(I_0)$ et $TC(I_0)$ en eux-mêmes.

Ainsi, la première assertion de 4.8 résulte de 4.6. De plus, (4.8.8) et (4.8.9) ont été vus dans 4.7, dans les conditions générales de 4.6 et en supposant de plus le 4° de (4.7.4).

Commentary · Examples [ms. 83]

A free cell does not make every cofibration free

The two hoped-for equalities in footnote 23 have different meanings. The equality concerning fibrations is established under the stronger assumptions of Proposition 5.8. The literal equality between $v_!(TC(I_0))$ and $TC(I)$ cannot hold in that generality: the class on the left consists of free maps, whereas the class on the right is allowed to attach them to diagrams that already have relations. A small equivariant example makes the distinction visible without any homotopical calculation.

Give **Set** the model structure in which every map is a weak equivalence, cofibrations are injections, and fibrations are surjections. We work with the usual axiom of choice. Injections lift against surjections by choosing a preimage for each new element. Conversely, lifting against $\varnothing \to \{*\}$ characterizes surjections, and a map lifting against all surjections is injective, by the retract argument after factoring it through the injection

$$A \longrightarrow A \amalg B \longrightarrow B.$$

Here the second map is f on A and the identity on B. Thus both weak factorization systems are the injection–surjection system. It is generated by the single map $\varnothing \to \{*\}$, so this is a special closed triple in Grothendieck's sense.

Let $I = BG$ for the group $G = \mathbb{Z}/2$, viewed as a one-object category. Contravariant diagrams are right G-sets, R forgets the action, and $L(S) = S \times G$ is free. Consider the equivariant inclusion

$$k : \{*\} \longrightarrow \{*\} \amalg G,$$

where $*$ is fixed and G acts regularly on the second summand. This is the pushout of $L(\varnothing \to \{*\})$, so it belongs to $TC(I)$; all maps are weak equivalences in the transferred structure. It also has the requisite lifting property directly. Given a lifting square against an equivariant map whose underlying function is surjective, the image of $*$ is already specified. Choose a preimage of the image of the identity element of the free orbit, and extend that choice equivariantly to G. This defines the diagonal and verifies both triangle identities.

On the other hand, k is not isomorphic in the arrow category to $L(a)$ for any injection $a : S \to T$. The domain of $L(a)$ is the free G-set $S \times G$. A nonempty free G-set has no fixed points, whereas the domain of k has one. Even the identity of the fixed point would already disprove the literal equality. The nonidentity map k shows why the failure belongs to the geometry of attachment rather than to a convention about including identities.

The correct conclusion under the hypotheses of Proposition 5.8 is

$$TC(I) = \widetilde{v_!(TC(I_0))}\,, \qquad F(I) = (v^*)^{-1}(F(I_0)).$$

23. Ces inclusions devraient être des égalités, mais je ne sais pas le prouver sans faire des hypothèses plus fortes, du type *catégorie de Quillen spéciale.*

24. où $j \in \mathrm{Ob}\, I_0 = \mathrm{Ob}\, I$, et $\alpha : i \longrightarrow v(j) = j$.

25. mérite un lemme général …

The tilde in the first equality is indispensable. Theorem 6.1(b) gives exactly this statement. Modern descriptions by representable cells have the same distinction; see Riehl (2014, Theorem 12.3.2). We therefore read the first hope in footnote 23 as needing this closure, and the second as the precise question that the next section resolves.

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5 Relation des catégories à cofibrations spéciales avec les catégories de Quillen spéciales

Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations spéciale. J'introduis

$$TF = C_*, \qquad TC = C \cap W, \qquad F = (TC)_* , \tag{5.1}$$

donc on a

$$\begin{array}{ccc} TC & \hookrightarrow & C \\ \downarrow & & \downarrow \\ F & \hookrightarrow & TF \end{array} , \tag{5.2}$$

(C, TF) étant un précouple de Quillen clos à droite. On sait alors (XV 4.2 cor. 1, page 31) que $\tilde{C}$ est formé des facteurs directs d'éléments de C. Posons

$$T\tilde{C} = \tilde{C} \cap W , \tag{5.3}$$

d'où

$$TC \subset T\tilde{C} . \tag{5.4}$$

Proposition 5.1 *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations spéciale. On a $T\tilde{C} \subset (TC)^\sim$ (et même $T\tilde{C} \subset (TC)^\natural$ [26]), ou ce qui revient au même en vertu de (5.4), TC et $T\tilde{C}$ ont même Q-clôture à gauche :*

$$T\tilde{C} = (T\tilde{C})^\sim = (TC)^\sim \qquad \textit{(et même } T\tilde{C} \subset (TC)^\natural\textit{)}, \tag{5.5}$$

d'où p. 42

$$(\tilde{C})_* = C_* (= TF), \qquad (T\tilde{C})_* = (TC)_* (= F) \qquad [27]. \tag{5.6}$$

De plus, si W stable par facteurs directs (ou si seulement tout facteur direct d'une flèche de TC est dans W), et si de plus W est stable par composition transfinie, alors $(\mathcal{M}, W, \tilde{C})$ est une catégorie à cofibrations spéciale. De plus, $T\tilde{C}$ est Q-saturé à gauche, donc (par 5.5)

$$T\tilde{C} = (TC)^\natural . \tag{5.7}$$

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Commentary · Exposition [ms. 84]

What the retract argument actually identifies

Grothendieck introduces fibrations here in order to measure the remaining obstruction to constructing a model structure. The cofibration factorization is already available. His task is to decide what happens when the cofibrations are enlarged by retracts and whether the trivial

26. **N.B.** Pour une partie A de $\mathrm{Fl}(\mathcal{M})$, $A^\natural$ désigne l'ensemble des facteurs directs d'éléments de A.
27. Ces relations ne dépendent pas de l'hypothèse W stable par facteurs directs.

cofibrations admit a second factorization. These are separate questions, and the calculation of Proposition 5.1 answers the first without assuming an answer to the second.

For a class A of arrows write $A^{\perp}$ for its right lifting class and ${}^{\perp}A$ for its left lifting class. In this notation Grothendieck's Q-closure is

$$\widetilde{A} = {}^{\perp}(A^{\perp}).$$

It is a closure defined by lifting. By contrast, closure under pushouts, transfinite compositions and retracts is the class now usually called weak saturation. The two coincide for a set of generating maps when the small object argument applies. They must not be identified merely because a class is stable under those three operations. A left lifting class has all three stability properties, but the converse requires an additional argument.

Let $f : A \to B$ lie in $\widetilde{C} \cap W$. Use the available factorization $f = pi$, with $i : A \to E$ in C and $p : E \to B$ in $TF = C^{\perp}$. Since $TF \subset W$, the two-out-of-three property gives $i \in C \cap W = TC$. The definition $f \in \widetilde{C} = {}^{\perp}TF$ supplies a map $s : B \to E$ with

$$ps = \mathrm{id}_B, \qquad sf = i.$$

In the arrow category under A, the maps (id_A, s) and (id_A, p) exhibit f as a retract of i. This proves the strong, correctly qualified conclusion

$$TC \subset \widetilde{C} \cap W \subset \mathrm{Retr}(TC) \subset \widetilde{TC}.$$

Taking Q-closures gives

$$\widetilde{\widetilde{C} \cap W} = \widetilde{TC}, \qquad (\widetilde{C} \cap W)^{\perp} = (TC)^{\perp}.$$

This is all that is needed to identify $F'_{\sim}$ with F' on manuscript p. 81. In the source's proof, the occurrence $f \in C$ and the corresponding diagram label should be read as $f \in \widetilde{C}$. The lifting argument remains valid with exactly that hypothesis. Its modern general form is the retract argument of Riehl (2014, Lemma 11.2.3).

The first equality in (5.5), identifying $\widetilde{C} \cap W$ with its own Q-closure, goes beyond this argument. The preceding prose asserts equality of the two closures, and that is the formulation proved here. Likewise, the equality with $\mathrm{Retr}(TC)$ printed at the start of the proof must at that point be read as the inclusion just established. If retracts of trivial cofibrations belong to W, the reverse inclusion follows: a retract of a map in TC belongs to $\widetilde{C}$ and, by this additional hypothesis, to W. Hence

$$\widetilde{C} \cap W = \mathrm{Retr}(TC).$$

We retain these source displays in the transcription and use their justified forms in the audit. No counterexample to the extra first equality of (5.5) under the full stated assumptions is asserted here.

There is a reason for keeping the retract under its fixed source. Push out the retract diagram in $A \backslash \mathcal{M}$ along $A \to A'$. Functoriality of pushout preserves its two splitting identities, so the new map is a retract of a pushout of $i \in TC$. That pushout still lies in TC. Retract stability of W therefore proves the required weak-equivalence assertion after pushout. Transfinite stability of the enlarged trivial class follows under the further hypothesis used by Grothendieck, namely transfinite stability of W itself. The other conditions for a special cofibration structure pass to $\widetilde{C}$ because its right lifting class and its generating set have not changed. This proves his stated sufficient result without silently adding the missing second factorization.

Démonstration. Prouvons

$$T\tilde{C} \ \subset \ (TC)^{\sim},$$

et plus précisément, que tout élément de $T\tilde{C}$ est facteur direct d'un élément de TC :

$$T\tilde{C} = (TC)^{\natural}.$$

Soit donc $f \in T\tilde{C} \overset{\text{déf}}{=} \tilde{C} \cap W$, considérons leur factorisation de f en pi, avec $i \in C$, $p \in TF$. Comme $TF \subset W$, on a $p \in W$, donc $i \in W$, donc $i \in C \cap W \overset{\text{déf}}{=} TC$. Comme $f \in C$, $p \in TF$,

il existe $q : Y \longrightarrow X'$ tel que $pq = \mathrm{id}_Y$, $qf = i$,

$$\begin{array}{ccc} X & \xrightarrow{f \in C \cap W} & Y \\ & {\scriptstyle i \in C} \searrow \quad \overset{q}{\vdots} \quad \nearrow {\scriptstyle p \in TF} & \\ & X' & \end{array}$$

donc f apparaît comme facteur direct de $i \in TC$, q.e.d.

Prouvons que $(\mathcal{M}, W, \tilde{C})$ est une catégorie à cofibrations spéciale (déf. 4.4, page 74). Condition a) sur $\mathcal{M}$ acquise, condition b), la partie sur W, aussi. La stabilité de $\tilde{C}$ par cochangement de base et par composition OK, aussi (pour b′) la stabilité par composition transfinie. Stabilité de $T\tilde{C} = \tilde{C} \cap W$ par cochangement de base : il reste à prouver que si on a un carré cocartésien

$$\begin{array}{ccc} X & \xrightarrow{\ i\ } & Y \\ \downarrow & \text{cocart.} & \downarrow \\ X' & \xrightarrow{\ i'\ } & Y' \end{array},$$

si $i \in \tilde{C} \cap W$, alors $i' \in W$ (car on sait déjà que $i' \in \tilde{C}$). Mais la démonstration précédente montre que c'est facteur direct *dans* $X \backslash \mathcal{M}$ d'une flèche dans TC, donc i' est facteur direct d'une flèche de $TC \subset W$. Donc si W est stable par facteurs directs, $i' \in W$.

Cela achève de prouver b). Pour b′), il reste à voir que $T\tilde{C}$ est stable par composition transfinie. Je ne vois pas d'autre façon de le voir, qu'en supposant que W lui-même est stable par composition transfinie (ce qui paraît à peine moins exigeant que la stabilité de W par $\varinjlim$ `[filtrantes]`). Cela achève de prouver b, b′). La condition c) sur C implique la même condition sur $\tilde{C}$. Enfin, comme $\tilde{C}_* = C_*$, la condition d) est la même pour C et $\tilde{C}$. Cela achève de prouver que $(\mathcal{M}, W, \tilde{C})$ est une catégorie à cofibrations spéciale. (**N.B.** on avait besoin de deux hypothèses supplémentaires sur W, stabilité par facteurs directs et par composition transfinie, pour établir que $T\tilde{C}$ est stable par cochangement de base et par composition transfinie). Le dernière assertion de 5.1 et la relation (5.7) qui en résulte, revient à dire que $T\tilde{C} = \tilde{C} \cap W$ est stable par facteurs directs, ce qui est évident par la même propriété de $\tilde{C}$, et l'hypothèse faite sur W de stabilité par facteurs directs (ou seulement que les facteurs directs d'éléments de $T\tilde{C}$ sont dans W, donc dans $T\tilde{C}$).

p. 43
[page 86]

Proposition 5.2. *Soit* $(\mathcal{M}, W, C)$ *une catégorie à cofibrations spéciale. Alors avec les notations de* (5.1)*, on a*

[page 87]

$$W = TF \circ TC, \tag{5.8}$$

$$TF = F \cap W. \tag{5.9}$$

Démonstration. Soit $f \in W$, écrivons $f = pi$, avec $i \in C$, $p \in TF$, donc $p \in W$, donc $i \in W$, donc $i \in C \cap W = TC$, d'où (5.8).

Prouvons (5.9). On a $TF \subset F \cap W$, prouvons inversement qu'une $f \in F \cap W$ est dans TF. Écrivons $f = pi$, avec $p \in TF$, $i \in TC$.

$$\begin{array}{ccc} X & \xrightarrow{f \in F \cap W} & Y \\ & {\scriptstyle i \in TC} \searrow \quad \nearrow {\scriptstyle p \in TF} & \\ & X' & \end{array}$$

Comme $f \in F$, $i \in TC$, et $TC \longleftrightarrow F$ par définition de F, $\exists\, r : X' \longrightarrow X$ tel que $ri = \mathrm{id}_X$, $fr = p$. Donc f est facteur direct de $p \in TF$, et comme $TF = C_*$ est stable par facteurs directs, on conclut que $f \in TF$.

[page 88]

Corollaire 5.3. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations spéciale. Pour que (W, C, F), $(F \overset{déf}{=} (TC)_*)$ soit un triple de Quillen dans $\mathcal{M}$, il faut et il suffit que le couple (TC, F) satisfasse à l'axiome de factorisation (i.e. que ce soit un précouple de Quillen). Pour que cette structure de Quillen soit clos, il faut et il suffit que C soit stable par facteurs directs.*

Démonstration. La nécessité de l'axiome de factorisation pour (TC, F), pour avoir un triple de Quillen, est tautologique. Pour la suffisance, ayant déjà les deux précouples de Quillen (C, TF), (TC, F) avec

p. 44

$$\begin{array}{ccc} TC & \hookrightarrow & C \\ \updownarrow & & \updownarrow \\ F & \hookleftarrow & TF \end{array} , \tag{5.2}$$

et $TC = C \cap W$, le fait que (W, C, F) est un triple de Quillen est le corollaire 4 p. 36 de XV : Les conditions $TC = C \cap W$, $TF = F \cap W$, $W = TF \circ TW$ sont équivalentes, et équivalentes au fait que (W, C, F) soit un triple de Quillen.

Par la définition de Quillen des triples *clos,* par les propriétés

$$\underbrace{W = TF \circ TC}_{\substack{\text{vraie pour tous} \\ \text{le triples} \\ \text{de Quillen}}}, \qquad C = (TF)^*, \qquad \underbrace{F = (TC)_*}_{\substack{\text{vraie dans} \\ \text{le cas actuel}}} ,$$

il reste à exprimer que $C = (TF)^*$, ce qui équivaut à C stable par facteurs directs (cf. loc. cit. cor. 1, p. 31), q.e.d.

Remarque 5.4. Les énoncés 5.2 et 5.3, et la relation $T\tilde{C} \subset (TC)^\natural$ dans proposition 5.1, ne font pas appel à la totalité de l'hypothèse que $(\mathcal{M}, W, C)$ soit une catégorie à cofibrations spéciale, mais seulement que l'on ait une catégorie à cofibrations, avec $C \longleftrightarrow \underbrace{TF}_{\overset{\text{déf}}{=} C_*} \subset W$, i.e. une catégorie avec W, C satisfaisant Cof_0, Cof_1, Cof_2 de la page 39, plus $TF \overset{\text{déf}}{=} C_* \subset W$, plus l'axiome de factorisation pour (C, TF) (lequel implique Cof_3).

[page 89]

Commentary · Exposition [ms. 88]

The second factorization is the remaining requirement

Grothendieck's comparison with Quillen theory becomes short once the role of the two factorizations is isolated. The factorization associated with C already determines which fibrations must be trivial, even before a general fibration factorization has been constructed. Proposition 5.2 is the calculation that makes this assertion exact.

Set $F = (TC)^\perp$. If $f \in W$, factor $f = pi$ with $i \in C$ and $p \in TF$. Since $p \in W$, two-out-of-three gives $i \in TC$; conversely, a composite of an arrow in TC and an arrow in TF belongs to W. Thus $W = TF \circ TC$. To identify TF, take $f : A \to B$ in $F \cap W$ and use this factorization again. The lifting property of f against i gives $r : E \to A$ satisfying

$$ri = \mathrm{id}_A, \qquad fr = p.$$

The maps (i, id_B) and (r, id_B) exhibit f as a retract of p. A right lifting class is retract-stable, so $f \in TF$. The reverse inclusion is immediate from $TC \subset C$ and $TF \subset W$. We have proved $TF = F \cap W$ without needing W itself to be retract-stable.

It follows that the new requirement for a Quillen triple is precisely a factorization by (TC, F). If $TC_0 \subset TC$ is a set with $(TC_0)^\perp = (TC)^\perp$, the small object argument factors every map as a relative TC_0-cell map followed by an arrow of F. The left map belongs to TC because this class is stable under pushouts and transfinite compositions; coproducts were accounted for above. This proves Corollary 5.5. The source's formula $W = TF \circ TW$ on p. 88 has TC as its intended second factor.

Once the second factorization exists, W is in fact retract-stable. Here is a proof that does not presuppose this conclusion or retract stability of TC. Let $f : A \to B$ be a retract of $w : C \to D \in W$,

with splitting pairs $(a,b) : f \to w$ and $(r,s) : w \to f$, so $ra = \mathrm{id}_A$ and $sb = \mathrm{id}_B$. First suppose $f \in F$. Factor $w = vu$, with $u : C \to E \in TC$ and $v : E \to D \in TF$. Lifting u against f in the square with top r and bottom sv gives $\ell : E \to A$ such that $\ell u = r$ and $f\ell = sv$. The pairs (ua, b) and (ℓ, s) now exhibit f as a retract of v. Thus $f \in TF \subset W$, because $TF = C^{\perp}$ is retract-stable.

For arbitrary f, use the second factorization $f = hg$, with $g : A \to Z \in TC$ and $h : Z \to B \in F$. Push out g along a, writing $E = C \amalg_A Z$ and $j : C \to E$, $t : Z \to E$ for the structure maps. Pushout stability gives $j \in TC$. The compatible maps w and bh induce $k : E \to D$ with $kj = w$ and $kt = bh$, so two-out-of-three gives $k \in W$. The compatible maps gr and id_Z induce $q : E \to Z$ satisfying

$$qj = gr, \qquad qt = \mathrm{id}_Z, \qquad hq = sk.$$

The last identity can be checked on the two pushout summands. Therefore (t, b) and (q, s) exhibit h as a retract of k. The special case already proved gives $h \in W$, and hence $f = hg \in W$. This is the argument of Riehl (2014, Lemma 11.3.3), observed here to require only the indicated factorization, lifting, and pushout properties.

If C is retract-stable, so is $TC = C \cap W$. Factoring a map of ${}^{\perp}F$ by (TC, F) and retracting its left factor therefore proves $TC = {}^{\perp}F$. The first factorization similarly gives $C = {}^{\perp}TF$. Thus the source's criterion for a closed triple supplies the full modern lifting characterizations as well. The second factorization is essential in this explanation; this argument cannot be inserted earlier into Proposition 5.1 to justify its extra equality.

The terminology "special closed triple" thus points toward cofibrant generation: there is one set of tests for trivial fibrations and another for fibrations. Its smallness assumption is imposed on all coefficient objects. Modern treatments generally impose only the smallness needed for the chosen generators; see Hovey (1999a, Theorem 2.1.14 and Definition 2.1.17). If the underlying category is locally presentable, the resulting model category is combinatorial in the now customary terminology. Grothendieck's condition that each object is accessible does not itself state local presentability, which also requires an appropriate set of objects generating the category by filtered colimits. Corollary 5.7 should therefore not be translated by replacing the word "special" with "combinatorial" without that extra hypothesis.

Corollaire 5.5. *Soit $(\mathcal{M}, W, C)$ une catégorie à cofibrations spéciale, telle qu'il existe une* petite *partie $TC_0 \subset TC = C \cap W$, avec*

$$TC \ \subset \ TC_0^{\sim} \ , \tag{5.5.1}$$

ou ce qui revient au même, $(TC)^{\sim} = (TC_0)^{\sim}$, i.e.

$$\underbrace{F}_{\overset{\text{déf}}{=}(TC)_*} \ = \ (TC_0)_* \ . \tag{5.5.2}$$

Alors (W, C, F) est un triple de Quillen. Ce triple est clos si et seulement si C est stable par facteurs directs.

En effet, le théorème de factorisation dans XV §1 implique que moyennant l'hypothèse faite sur $TC_0 \subset TC$, le couple (TC, F) satisfait à la condition de factorisation. En effet, par les conditions b, b′ de la définition 4.4, TC a les propriétés de stabilité voulues (contient les isomorphismes, est stable par cochangement de base et par composition transfinie).

Définition 5.6. Les triples de Quillen déduits comme dans 5.5 d'une catégorie à cofibrations spéciale sont dits *spéciaux.* Ainsi un triple de Quillen (W, C, F) dans $\mathcal{M}$ est spécial s'il satisfait les conditions a) à d) dans la définition (4.4), concernant (W, C) exclusivement, plus les deux conditions suivantes. [page 90]

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(c′) $\exists$ petite partie TC_0 de TC, telle que $TC \subset (TC_0)^{\sim}$.

(e) $F = (TC)_*$.

Corollaire 5.7. *Soit (W, C, F) un triple de Quillen* clos *dans $\mathcal{M}$. Pour que ce triple soit spécial, il faut et il suffit qu'il satisfasse les conditions suivantes.*

ⓐ $\mathcal{M}$ *stable par petites* $\varinjlim$*, et ses éléments sont accessibles.*

(c, c′) *Il existe une petite partie* C_0 *de* C*,* TC_0 *de* TC*, telles que l'on ait*

$$C = C_0^{\sim}\ , \qquad TC = (TC_0)^{\sim}\ . \tag{5.7.1}$$

En effet, toutes les autres conditions a) à e), y inclus c′) de 5.6, sont automatiquement satisfaites.

Proposition 5.8. *Considérons un couple de foncteurs adjoints* [page 91]

$$\mathcal{M} \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}'\ , \tag{5.8.1}$$

où $\mathcal{M}$ *est munie d'un triple de Quillen spécial clos* $(W, C, F = (TC)_*)$*, avec* $TC = C \cap W$ (cf. définition 5.6). *On suppose satisfaites les conditions suivantes* (comparer prop. 4.6, relations (4.6.1), et (4.7.4)) :

$$\begin{cases} \text{a)} & \mathcal{M}' \text{ stable par petites } \varinjlim \text{ , ses éléments sont accessibles.} \\ \text{b)} & v^* \text{ commute aux petites } \varinjlim\ . \\ \text{c)} & v^*(v_!(C)) \subset C,\ v^*(v_!(TC)) \subset TC. \end{cases} \tag{5.8.2}$$

Définissons (W', C', F') *dans* $\mathcal{M}'$ (comparer prop. 4.6, notamment 4.6.2) *par*

$$\begin{cases} W' &= (v^*)^{-1}(W) \\ C' &= v_!(C)^{\sim} \\ F' &= (TC')_*\ , \qquad \text{où } TC' \overset{\text{déf}}{=} C' \cap W'\ . \end{cases} \tag{5.8.3}$$

Alors (W', C', F') *est un triple de Quillen clos spécial dans* $\mathcal{M}'$*, et on a de plus*

$$F' = (v^*)^{-1}(F)\ , \qquad TF' = (v^*)^{-1}(TF)\ , \tag{5.8.4}$$

$$TC' = (v_!(TC))^{\sim}\ . \tag{5.8.5}$$

Commentary · Exposition [ms. 91]

The two factorizations meet in the transfer theorem

Grothendieck now returns to the equality of fibrations that had resisted the argument of §4. The extra structure supplied by the second generating set does exactly what is needed: it produces a candidate trivial cofibration factorization in the target, and the previously constructed cofibrations then force that candidate to be the correct one. The proof is a transfer argument, but its decisive step is again a retract under a fixed source.

Write $L = v_!$, $R = v^*$, and choose generating sets $I_0 \subset C$ and $J_0 \subset TC$ in the coefficient category. To avoid the ambiguity between the raw image of a class and its closure, put

$$C' = \mathrm{cof}(LI_0), \qquad K' = \mathrm{cof}(LJ_0), \qquad W' = R^{-1}(W),$$

where cof denotes the class of retracts of relative cell maps. The smallness assumptions ensure that these are also the corresponding left lifting classes. Adjointness transports lifting squares and their diagonals, giving

$$(LI_0)^{\perp} = R^{-1}(TF), \qquad (LJ_0)^{\perp} = R^{-1}(F).$$

The equality for trivial fibrations is therefore formal. The equality for fibrations will follow only after K' has been identified with $C' \cap W'$.

First, $K' \subset C'$ because $J_0 \subset C$. It is also contained in W'. Indeed, $RL(TC) \subset TC$ by hypothesis, while R preserves pushouts and transfinite colimits. It follows by induction along an attachment

that R carries every relative LJ_0-cell map into TC. Functors preserve retract diagrams, and TC is retract-stable in the closed coefficient model structure, so $R(K') \subset TC \subset W$. This proves $K' \subset C' \cap W'$. Notice that preservation of weak equivalences alone would not prove this step: it is the cellular construction and the stronger assertion about $RL(TC)$ that control the entire attachment.

For the converse, let $f : A \to B$ belong to $C' \cap W'$. Factor it by the small object argument for LJ_0 as

$$A \xrightarrow{j} E \xrightarrow{p} B, \qquad j \in K', \quad Rp \in F.$$

The first part of the proof gives $j \in W'$. As $f \in W'$, two-out-of-three gives $p \in W'$, hence $Rp \in F \cap W = TF$. Thus $p \in R^{-1}(TF) = (C')^{\perp}$. The lifting property of $f \in C'$ supplies a section $s : B \to E$ with $ps = \mathrm{id}_B$ and $sf = j$. Consequently f is a retract of j, and K' is retract-stable. This proves

$$C' \cap W' = K', \qquad (C' \cap W')^{\perp} = R^{-1}(F).$$

Both target factorizations, both lifting properties, and the equality of the target weak equivalences with the prescribed inverse image are now established. The generating sets are LI_0 and LJ_0.

This presentation also explains the status of Remark 5.9. If the source triple is not closed, use the pushout–transfinite closure C' of $L(C)$ and the relative LJ_0-cell class for the left factor. Its image under R still lies in TC. The same argument shows that every map in $C' \cap W'$ is a retract of one of these cell maps, which suffices to identify the right lifting class. The cell maps themselves lie in $C' \cap W'$, so their small-object factorization supplies the required factorization for that larger class. One need not first assert that either of these classes is closed under retracts.

In the modern transfer theorem, the condition that the right adjoint preserve all colimits can be replaced by a direct acyclicity condition on the transferred trivial cells, together with the smallness needed to form the two factorizations. This is the form of Hirschhorn (2003, Theorem 11.3.2). Grothendieck's hypotheses give a transparent sufficient test for that acyclicity condition. Their main application here is exceptionally well suited to them: restriction from diagrams to their objects computes colimits pointwise, and the free diagram functor produces coproducts of coefficient cells.

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Démonstration. Par 4.6, on sait déjà que $(\mathcal{M}', W', C')$ est une catégorie à cofibrations spéciale. (**N.B.** Ce qui est noté C' ici, était noté $\tilde{C}'$ dans la proposition 4.5. On doit utiliser la dernière assertion de 4.6, où on suppose C stable par facteurs directs, OK car (W, C, F) supposé clos.) Pour prouver que $(W', C', F' \overset{\text{déf}}{=} (TC')_*)$ est un triple de Quillen spécial, il reste, par 5.3 et définition 5.6, à prouver qu'il existe une petite partie TC'_0 de TC', telle que p. 46

$$(\ TC'_0\ \subset\ )\ TC' \ \subset\ (TC'_0)^{\sim}\ . \tag{5.8.6}$$

(Ce triple de Quillen sera de plus clos, puisque C' est Q-clos à gauche par définition.) Soit donc TC_0 une petite partie de TC telle que

$$TC \ \subset\ (TC_0)^{\sim}\ , \tag{5.8.7}$$

qui existe par hypothèse sur (W, C, F). On prendra

$$TC'_0 \ \overset{\text{déf}}{=}\ v_!(TC_0)\ . \tag{5.8.8}$$

On a vu dans 4.7 qu'on a

$$v_!(TC) \ \subset\ TC' \tag{5.8.9}$$

(résulte de l'hypothèse $v^*(v_!(TC)) \subset TC$ dans (5.8.2. c)), a fortiori

$$TC'_0 \ =\ v_!(TC_0) \ \subset\ TC'\ ,$$

il reste à prouver

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$$TC' \ \overset{?}{\subset}\ (TC'_0)^{\sim}\ , \tag{5.8.10}$$

ce qui équivaut à $(TC')^{\sim} \subset (TC'_0)^{\sim}$, ou encore à

$$(TC'_0)_* \overset{?}{\subset} (TC')_* \overset{\text{déf}}{=} F' , \tag{5.8.10 bis}$$

alors qu'on a a priori l'inclusion inverse (déduite de $TC'_0 \subset TC'$)

$$F' \subset (TC'_0)_* . \tag{5.8.11}$$

Or par adjonction, on a

$$\begin{aligned} (\underbrace{TC'_0}_{v_!(TC_0)})_* &= (v_!(TC_0))_* \\ &= (v^*)^{-1}(\underbrace{(TC_0)_*}_{=(TC)_* \overset{\text{déf}}{=} F}) \\ &= (v^*)^{-1}(F) , \end{aligned}$$

donc (5.8.11) s'écrit

$$F' \subset (v^*)^{-1}(F) , \qquad \text{i.e. } v^*(F') \subset F \tag{5.8.12}$$

(comparer (4.7.6)), et il faut montrer que cette inclusion est une égalité (sur quoi j'avais buté p. 80, 81 …), ce qui n'est autre que la première des deux relations (5.8.4). Mais c'est sous la forme initiale (5.8.10) (transposée de (5.8.4)) que nous allons prouver la chose, finalement. Notons que l'on a

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$$(\underbrace{v_!(TC_0)}_{TC'_0})^{\sim} = (v_!(TC))^{\sim} ,$$

puisque TC est la clôture de TC_0 pour les opérations de cochangement de base, de composition transfinie, et de facteurs directs, et que $v_!$ commute à ces opérations. Donc on doit montrer simplement

$$TC' \subset (v_!(TC))^{\sim} \tag{5.8.12 bis}$$

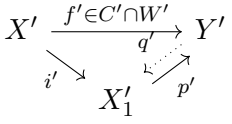
(où il n'est plus question de TC_0). Mais le théorème de factorisation s'applique au couple

$$\Big((v_!(TC))^{\sim},\ (v_!(TC))_* = (v^*)^{-1}(F)\Big) ,$$

puisque $(v_!(TC))^{\sim} = (TC'_0)^{\sim}$, TC'_0 petit. Si donc on a $f' \in TC' = C' \cap W'$, factorisons f' en $p'i'$, avec

$$i' \in (v_!(TC))^{\sim} , \qquad p' \in (v^*)^{-1}(F) . \tag{5.8.13}$$

(∗)



Notons qu'on a

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$$(v_!(TC))^{\sim} \subset TC' \ (\overset{\text{déf}}{=} C' \cap W') \tag{5.8.14}$$

car $v_!(TC) \subset TC' \subset C'$, donc $(v_!(TC))^{\sim} \subset \tilde{C}' = C'$, et il reste à prouver que $v_!(TC)^{\sim} \subset W'$, i.e. $v^*((v_!(TC))^{\sim}) \subset W$. Or je dis qu'on a même

$$v^*((v_!(TC))^{\sim}) \subset TC ,$$

i.e.

$$v_!(TC)^\sim \ \subset \ (v^*)^{-1}(TC)\ . \tag{5.8.15}$$

Cela résulte du fait que TC est Q-clos, donc Q-saturé à gauche, ce qui implique, comme v^* commute aux petites $\varinjlim$, que $(v^*)^{-1}(T)$ l'est également. D'autre part, il contient $v_!(TC)$, donc aussi le Q-saturé à gauche de $v_!(TC)$. Mais ce dernier est *égal* à la Q-clôture $(v_!(TC))^\sim$, par le théorème de factorisation, vu que $TC'_0 \subset v_!(TC) \subset (TC'_0)^\sim$, avec TC'_0 petit.

Ainsi, (5.8.13) et (5.8.14) impliquent $i' \in TC' = C' \cap W'$, donc $i' \in W'$, d'où $p' \in W'$, i.e. $v^*(p') \in W$, et comme par hypothèse (5.8.13) on a $v^*(p') \in F$, on a $v^*(p') \in TF$, i.e. p. 48 [page 96]

$$p' \ \in \ (v^*)^{-1}(TF) \ = \ TF' \ (\overset{\text{déf}}{=} \ C' \ )\ ,$$

l'égalité $(v^*)^{-1}(TF) = TF'$, i.e.

$$(v^*)^{-1}(C_*) \ = \ (v_!(C))_*\ ,$$

résultant de l'adjonction de $v_!$, v^*. Au total, on a donc

$$p' \ \in \ TF' \ (\ = \ C'_*\ )\ , \tag{5.8.16}$$

et comme $f' \in C'$, le diagramme $(*)$, page 114, nous montre qu'il existe

$$q' \ : \ Y' \longrightarrow X'_1\ ,$$

avec $p'q' = \mathrm{id}$, $q'f' = i'$. Donc f' est facteur direct de $i' \in v_!(TC)^\sim$, donc on a $f' \in v_!(TC)^\sim$, d'où la relation

$$TC' \ \subset \ v_!(TC)^\sim\ ,$$

ce qui achève de prouver que (W', C', F') est un triple de Quillen spécial et clos.

Les deux relations (5.8.4) ont été prouvées chemin faisant (la deuxième étant essentiellement triviale, et la première représentant la difficulté principale dans la démonstration). Reste à établir (5.8.5). Mais on sait déjà (5.8.9) que

$$v_!(TC) \ \subset \ TC'\ ,$$

d'où [page 97]

$$v_!(TC)^\sim \ \subset \ TC',$$

puisque TC' est Q-clos à gauche (ou parce que ça a été prouvé dans (5.8.14)), on a déjà prouvé l'inclusion en sens inverse (5.8.12 bis), assertion équivalente à (5.8.4) (et qui représentait la principale difficulté).

Remarque 5.9. Si on ne suppose pas que la structure sur $\mathcal{M}$ soit close, alors la démonstration donnée de 5.8 montre qu'on trouve pourtant sur $\mathcal{M}'$ une structure de Quillen spéciale, en définissant cette fois C' comme dans 4.6. Mais je doute qu'on aura jamais à travailler avec des triples de Quillen spéciaux qui ne soient aussi clos. De toute façon, si W est stable par petites $\varinjlim$ filtrantes [28] (et je doute qu'on trouve des triples de Quillen spéciaux en dehors de ce cas), alors si (W, C, F) est un triple de Quillen spécial, il en est de même de $(W, \tilde{C} = C^\natural, F)$, et je présume qu'on a tout intérêt à remplacer le triple initial par celui-ci.

28. ce qui implique la stabilité par facteurs directs, et implique aussi la même propriété pour $W' \overset{\text{déf}}{=} (v^*)^{-1}(W)$.

Commentary · Development (commentators') [ms. 97]

Special triples need not preserve ordinary filtered colimits

Grothendieck closes the section by expressing doubt that special triples occur outside the case where weak equivalences are stable under filtered colimits. The smallness conditions do explain why sufficiently filtered colimits behave well. They do not force the bound to be finite. A countably presented coefficient gives a concrete separation, already in abelian groups.

A special closed triple with a failing sequential colimit. [ours]

Let $A = \mathbb{Z}[1/2]$ and, in **Ab**, define

$$W_A = \{f \mid \mathrm{Hom}(A, f) \text{ is bijective}\},$$
$$I_A = \{0 \longrightarrow A, \quad A \oplus A \xrightarrow{\nabla} A\},$$
$$C_A = {}^{\perp}W_A, \qquad F_A = \mathrm{Mor}(\mathbf{Ab}).$$

Then (W_A, C_A, F_A) is a special closed Quillen triple. Its weak equivalences are not closed under sequential colimits in the arrow category.

Proof. We first identify the right lifting tests. Lifting against $0 \to A$ says that every map $A \to Y$ lifts through $f : X \to Y$; equivalently, $\mathrm{Hom}(A, f)$ is surjective. A lifting square from the fold ∇ to f is a pair of homomorphisms $a, b : A \to X$ with $fa = fb$. A diagonal exists exactly when $a = b$: it must be the common map $A \to X$. The right lifting property against the fold is therefore injectivity of $\mathrm{Hom}(A, f)$. We obtain

$$I_A^{\perp} = W_A.$$

The domains of I_A are small. More explicitly, A has the countable presentation with generators x_n, $n \geq 0$, and relations $2x_{n+1} = x_n$. For a map from this presentation into an $\aleph_1$-filtered colimit, choose representatives of the countably many images of the generators and stages witnessing the countably many relations. Filteredness at $\aleph_1$ provides a common stage at which they define a homomorphism. The same argument applied to the countably many generator equalities proves eventual equality of two such homomorphisms. Thus A, and likewise $A \oplus A$, is $\aleph_1$-presentable. Every abelian group has a presentation of some set cardinality, so the analogous argument supplies the accessibility condition for every object of **Ab**.

The small object argument now gives the weak factorization system (C_A, W_A), with $C_A = \mathrm{cof}(I_A)$. The standard theorem being used is Hovey (1999a, Theorem 2.1.14); all its smallness conditions have just been checked. Since W_A is the inverse image of the isomorphisms of abelian groups under the functor $\mathrm{Hom}(A, -)$, it contains the isomorphisms, satisfies two-out-of-three, and is closed under retracts.

If $f : X \to Y$ lies in both C_A and W_A, apply its left lifting property to the square having f on both vertical sides and identities horizontally. Its diagonal $h : Y \to X$ satisfies $hf = \mathrm{id}_X$ and $fh = \mathrm{id}_Y$. Therefore

$$C_A \cap W_A = \mathrm{Is}(\mathbf{Ab}).$$

The second weak factorization system is consequently $(\mathrm{Is}(\mathbf{Ab}), \mathrm{Mor}(\mathbf{Ab}))$: factor a map as its identity followed by itself, and invert the left map to solve every lifting problem. This verifies all model axioms. Both left classes are defined by lifting and hence are stable under pushout, transfinite composition and retracts. The generating cofibrations are I_A; the generating trivial cofibrations form the empty set, whose right lifting class is all arrows and whose left lifting class is the isomorphisms. Together with cocompleteness and accessibility of the coefficient objects, this verifies every condition of a special closed triple.

To see the failure, consider the sequence in the arrow category whose nth arrow is $0 \to \mathbb{Z}$ and whose transition square is $(\mathrm{id}_0, \times 2)$. Every homomorphism $A \to \mathbb{Z}$ is zero, because the image of each x_n must be divisible by arbitrarily high powers of 2. Thus all these arrows lie in W_A. Their colimit is

$$0 \longrightarrow \mathrm{colim}(\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{2} \cdots) \cong \mathbb{Z}[1/2] = A.$$

The map $\mathrm{Hom}(A, 0) \to \mathrm{Hom}(A, A)$ is not surjective, since its target contains the nonzero identity. The colimit arrow is not in W_A, as required. □

The example does preserve $\aleph_1$-filtered colimits of weak equivalences, by the presentability calculation in the proof. The general distinction is formalized in Raptis and Rosický (2015, Proposition 4.1): generating cofibrations between κ-presentable objects give stability of weak equivalences under κ-filtered colimits in a combinatorial model category. We checked that result, the standard

small-object and transfer constructions, and searches for this particular $\mathrm{Hom}(\mathbb{Z}[1/2], -)$ example; we did not find this precise illustration in the literature. The construction is a specialization of established methods, and the mark [ours] records the argument assembled here, with no claim of priority.

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6 Application à la construction des dérivateurs (associés aux triples de Quillen spéciaux)

Théorème 6.1.[29] *Soit (W, C, F) un triple de Quillen clos spécial sur la catégorie $\mathcal{M}$* (cf. déf. 5.6, p. 90, et cor. 5.7). *Pour tout I dans* Cat*, considérons la catégorie*

$$\mathcal{M}(I) \;=\; \underline{\mathrm{Hom}}(I^\circ, \mathcal{M})\;, \tag{6.1.1}$$

et considérons

$$W(I),\; F(I) \;\subset\; \mathrm{Fl}(\mathcal{M}(I))\;,$$

définis par

$$\begin{array}{l} u : X \longrightarrow Y \text{ dans } \mathcal{M}(I) \text{ est dans } W(I) \text{ (resp. } F(I)\text{) si et seulement} \\ \text{si } \forall\, i \in \mathrm{Ob}\, I,\; u(i) : X(i) \longrightarrow Y(i) \text{ est dans } W \text{ (resp. } F\text{).} \end{array} \tag{6.1.2}$$

a) *Il existe une structure de Quillen close (nécessairement unique) dans $\mathcal{M}(I)$, ayant $W(I)$ et $F(I)$ comme ensembles des quasi-isomorphismes et de fibrations respectivement. On a donc aussi*

$$\begin{cases} TF(I) \;=\; F(I) \cap W(I) \;=\; \{u \in \mathrm{Fl}(\mathcal{M}(I)) \mid u(i) \in TF \quad \forall\, i \in \mathrm{Ob}\, I\}, \\ C(I) \;=\; TF(I)^*. \end{cases} \tag{6.1.3}$$

Cette structure de Quillen est spéciale.

b) *Soit I_0 la catégorie discrète ayant même ensemble d'objets que I, et* [page 99]

$$v_I \quad ou \quad v \;:\; I_0 \longrightarrow I \tag{6.1.4}$$

le foncteur canonique. Désignons par

$$C(I) \;=\; (TF(I))^*, \qquad TC(I) \;=\; C(I) \cap W(I) \;=\; F(I)^* \tag{6.1.5}$$

les ensembles de cofibrations resp. de cofibrations triviales dans $\mathcal{M}(I)$, et utilisons les mêmes notations $C(I_0)$, $TC(I_0)$ pour I_0. On a alors

$$\begin{array}{l} u : X_0 \longrightarrow Y_0 \text{ dans } \mathcal{M}(I_0) \text{ est dans } C_0 \text{ (resp. } TC_0\text{) si et seulement si} \\ \text{pour tout } i \text{ dans } \mathrm{Ob}\, I,\; u(i) : X_0(i) \longrightarrow Y_0(i) \text{ est dans } C \text{ (resp. } TC\text{).} \end{array} \tag{6.1.6}$$

Enfin on a

$$\begin{cases} C(I) \;=\; (v_!(C(I_0)))^\sim \\ TC(I) \;=\; (v_!(TC(I_0)))^\sim\,. \end{cases} \tag{6.1.7}$$

c) *Soit*

$$u \;:\; I \longrightarrow I'$$

une flèche dans Cat*, considérons le couple des foncteurs adjoints*

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$$\mathcal{M}(I) \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{M}(I') \quad . \tag{6.1.8}$$

Ce couple satisfait aux conditions de Quillen pour les couples de foncteurs adjoints entre catégories de Quillen (Hom. Alg. Chp. I, §4, th. 3, p. 4.5), *de sorte qu'ils définissent par passage aux localisés un couple des foncteurs adjoints*

$$\mathrm{Ho}_W(I) \underset{u_W^*}{\overset{u_!^W}{\rightleftarrows}} \mathrm{Ho}_W(I') \quad . \tag{6.1.9}$$

On a, plus précisément, les relations

$$\begin{cases} u_!(C(I)) \ \subset \ C(I') \\ u_!(TC(I)) \ \subset \ TC(I') \ , \end{cases} \tag{6.1.10}$$

d'où

$$(u_!(\mathcal{M}(I)_{\mathrm{c}})) \ \subset \ \mathcal{M}(I')_{\mathrm{c}} \ , \tag{6.1.11}$$

et de plus

$$\begin{array}{l} \textit{pour toute flèche } f \textit{ dans } \mathcal{M}(I)_{\mathrm{c}}\textit{, telle que } f \in W(I)\textit{, on a} \\ u_!(f) \in W(I')\textit{, i.e. } (u_!)_{\mathrm{c}} : \mathcal{M}(I)_{\mathrm{c}} \longrightarrow \mathcal{M}(I')_{\mathrm{c}} \textit{ est compatible} \\ \textit{avec les localiseurs.} \end{array} \tag{6.1.12}$$

D'autre part on a

$$\begin{cases} u^*(F(I')) \ \subset \ F(I) \\ u^*(TF(I')) \ \subset \ TF(I) \\ u^*(W(I')) \ \subset \ W(I) \ , \end{cases} \tag{6.1.13}$$

donc

$$\begin{cases} u^*(\mathcal{M}(I')_{\mathrm{f}}) \ \subset \ \mathcal{M}(I)_{\mathrm{f}} \\ u_{\mathrm{f}}^* : \mathcal{M}(I')_{\mathrm{f}} \longrightarrow \mathcal{M}(I)_{\mathrm{f}} \textit{ est compatible avec les localiseurs.} \end{cases} \tag{6.1.14}$$

Commentary · Exposition [ms. 99]

The projective structure and its free diagrams

Grothendieck has now supplied the structure that was missing from the mapping-cone argument. A diagram can be replaced without discarding its transition maps, and the replacement is good for every left Kan extension. The weak equivalences and fibrations are still tested at individual objects; it is the cofibrations that encode the compatibility needed to make these tests into a theory of diagrams.

We fix the variance before using the theorem. Here

$$\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M}), \qquad (u^*X)(i) = X(u(i)).$$

For $j \in I$, evaluation has a left adjoint F_j, given by

$$(F_jA)(i) = \coprod_{I(i,j)} A.$$

A square from F_jf to a diagram map p is exactly a square from f to $p(j)$; the same correspondence identifies their fillers. Thus, if S and T generate the two lifting systems in $\mathcal{M}$, the sets

$$\{F_js : j \in I,\ s \in S\}, \qquad \{F_jt : j \in I,\ t \in T\}$$

29. C'est le résultat principal dans cette partie XVII des notes.

generate the projective cofibrations and projective trivial cofibrations. The smallness conditions in the preceding sections justify the required factorizations. Formula (6.1.7) is the version using all objectwise cofibrations at once instead of selected generators. This is the modern projective model structure; the construction and its acyclicity check appear in Riehl (2014, Theorem 12.3.2).

The Quillen property of $u_! \dashv u^*$ can also be seen without repeating the cellular construction. If $f \in C(I)$ and $p \in TF(I')$, transpose a lifting square from $u_! f$ to p across the adjunction. Its right side is u^*p, which is still an objectwise trivial fibration. A filler exists by the definition of $C(I)$ and transposes back. Replacing (C, TF) by (TC, F) gives the second preservation statement. In particular, the conclusion does not depend on u preserving the order properties used earlier to construct cofibrant diagrams.

There is a useful economy in (6.1.9). Restriction preserves every weak equivalence, so its functor on homotopy categories is already defined. Only the left adjoint needs to be derived: for a projectively cofibrant replacement $QX \to X$,

$$\mathbf{L}u_!(X) = u_!(QX).$$

The adjunction asserts a natural bijection of morphisms in the localized categories, not merely the existence of this formula on objects. This is the precise additional conclusion that the earlier counterexample made necessary. The elementary lifting criterion for a Quillen adjunction and the preservation of weak equivalences between cofibrant objects are recorded in Riehl (2014, Lemmas 11.3.11 and 11.3.14).

[page 101]

d) *Supposons que I soit une catégorie ordonnée, et considérons l'ensemble $C_I \subset \mathrm{Fl}(\mathcal{M}(I))$ défini dans §2*[30] (cf. page 25, cf. aussi th. 2.1, p. 26, et ses corollaires). *On a alors*

$$\begin{cases} C_I \supset C(I) \\ TC_I \supset TC(I)\,, \end{cases} \tag{6.1.15}$$

et on a égalité quand I est un ensemble ordonné artinien.

Commentary · Development (commentators') [ms. 101]

The order hypothesis follows the variance

The comparison with the earlier order-theoretic cofibrations is meant to show that the new construction recovers the old one. At this point a change of notation matters: the earlier argument used covariant diagrams on an artinian poset, whereas (6.1.1) now uses contravariant diagrams on I. Accordingly, the sufficient condition is that I^{op} be artinian, or that I be noetherian. Corollary 6.3 states exactly this condition. The word *artinien* in 6.1(d), repeated in the sentence preceding 6.3, is printed in the edition and is retained in the source text.

An inverse-sequence counterexample[ours]. The condition printed in 6.1(d) is false in general. We work in sets with the axiom of choice and take

$$W = \mathrm{Mor}(\mathrm{Set}), \qquad C = \{\text{injections}\}, \qquad F = \{\text{surjections}\}.$$

These classes give a closed special Quillen structure. Indeed, injections and surjections are mutually determined by lifting, every map $A \to B$ factors as

$$A \longrightarrow A \amalg B \longrightarrow B,$$

and the single map $\varnothing \to \{*\}$ generates the lifting system. The other specialness conditions hold because sets have all small colimits, each set is small for a sufficiently large regular cardinal, and injections are stable under pushout and transfinite composition. Since every map is in W, the two required factorizations are the same one.

Let $I = \mathbb{N}$ with its usual order. It is artinian but not noetherian. In $\mathcal{M}(I)$ consider $f : \varnothing \to \mathbf{1}$, where $\mathbf{1}$ is the constant singleton inverse sequence. Every nonempty upward closed subset of I is a

30. Au paragraphe 2, c'était noté C_{I° au lieu de C_I.

tail $[n,\infty)$. The colimit of $\mathbf{1}$ over its opposite is a singleton, while the colimit over the empty subset is empty. Therefore every relative-colimit map associated with closed subsets $U \subseteq V$ is one of

$$\varnothing \longrightarrow \varnothing, \qquad \varnothing \longrightarrow \{*\}, \qquad \{*\} \xrightarrow{1} \{*\}.$$

All are injections. In particular $f \in C_I$, whether one uses the defining principal closed-set tests or the equivalent tests for arbitrary closed pairs.

Nevertheless, f is not projectively cofibrant. Define a presheaf E by

$$E(n) = \{k \in \mathbb{N} : k \geq n\}, \qquad E(n+1) \longrightarrow E(n) \quad \text{the inclusion.}$$

The map $E \to \mathbf{1}$ is an objectwise surjection, hence a projective trivial fibration. A filler for the square from $\varnothing \to \mathbf{1}$ to $E \to \mathbf{1}$ would choose compatible elements $k_n \in E(n)$. Compatibility forces all k_n to be the same integer, which would have to belong to every tail. There is no such integer. Thus $C(I) \neq C_I$.

This example isolates the missing hypothesis: the partial-lift argument must start at objects with no predecessors in the direction of the diagram. For inverse sequences indexed by $\mathbb{N}$, that direction has no starting object. We checked the standard projective and Reedy constructions in Riehl (2014, §§12.3 and 14.2) and Riehl and Verity (2014, Observation 4.16); the example is our explicit diagnostic for the printed variance error, with no claim of priority.

Démonstration. L'assertion a) est un cas particulier de 5.8, appliqué au couple de foncteurs adjoints

$$\mathcal{M}(I_0) \underset{v^*}{\overset{v_!}{\rightleftarrows}} \mathcal{M}(I) \quad .$$

Les conditions (5.8.2) sont satisfaites, cela a été vu dans (4.8). La proposition 5.8 nous dit alors l'existence d'une structure de triple de Quillen close sur $\mathcal{M}(I)$, dont l'ensemble des quasi-isomorphismes $W(I)$ et l'ensemble des fibrations $F(I)$, sont donnés par la première formule (5.8.3) et par la première formule (5.8.4), ce qui équivaut aux définitions (6.1.2). (**N.B.** Le fait que $\mathcal{M}(I_0)$, avec la structure $W(I_0)$, $C(I_0)$, $F(I_0)$ évidente (produit), est bien une catégorie de Quillen spéciale close, est ± tautologique.) p. 51 [page 102]

Prouvons b). La relation (6.1.6) est tautologique. Les relations (6.1.7) sont contenues dans 5.8 (cf. deuxième relation dans (5.8.3) pour $C' = v_!(C)^\sim$, i.e. $C(I) = (v_!(C(I_0)))^\sim$, et (5.8.5) pour $TC(I) = (v_!(TC(I_0)))^\sim$).

Prouvons c). La première assertion est, par Quillen, un cas particulier des relations (6.1.10) à (6.1.14), c'est elles qu'il s'agit d'établir. Les relations (6.1.13) et par suite (6.1.14) sont tautologiques en vertu des définitions, donc il s'agit d'établir *(6.1.10)* et (6.1.12).

Pour les formules (6.1.10), j'utilise le diagramme commutatif

$$\begin{array}{ccc} I & \xleftarrow{\ v\ } & I_0 \\ {\scriptstyle u}\downarrow & & \downarrow{\scriptstyle u_0} \\ I' & \xleftarrow{\ v'\ } & I'_0\ , \end{array}$$

et je note que les foncteurs $u_!^{\mathcal{M}}$, $u_{0!}^{\mathcal{M}}$, commutant aux petites limites inductives, se comportent bien pour l'opération [page 103]

$$\Phi \longmapsto \Phi^\sim\ ,$$

quand Φ est une partie de $\mathrm{Fl}(\mathcal{M}(I))$ resp. de $\mathrm{Fl}(\mathcal{M}(I_0))$ qui est Q-accessible, i.e. telle qu'il existe une *petite* partie $\Phi_0 \subset \Phi$ avec $\Phi \subset \tilde{\Phi}_0$, plus précisément, on a

$$u_!(\Phi^\sim) \ \subset\ (u_!(\Phi))^\sim\ .$$

Ceci noté, on a par b)

$$C(I) \ =\ (v_!(C(I_0)))^\sim\ ,$$

d'où

$$u_!(C(I)) \ \subset \ \underbrace{(u_!(v_!(C(I_0))))}_{(uv)_!(C(I_0))}{}^{\sim} ,$$

et le deuxième membre s'écrit aussi

$$(v'_! u_{0\,!}(C(I_0)))^{\sim} \ \overset{?}{\subset} \ (v'_!(I'_0))^{\sim} \ = \ C(I') \ .$$

Donc tout revient à montrer que

$$u_{0\,!}(C(I_0)) \ \subset \ C(I'_0) \ , \tag{6.1.16}$$

ce qui résulte aussitôt du calcul de $u_{0\,!}$ à coups de sommes directes, et du fait que C est stable par sommes directes quelconques. Cela prouve donc $u_!(C(I)) \subset C(I')$, et on prouve de même $u_!(TC(I)) \subset TC(I')$, qui se ramène de même au cas particulier

$$u_{0\,!}(TC(I_0)) \ \subset \ TC(I'_0) \ . \tag{6.1.17}$$

Il reste à prouver (6.1.12), et on procède comme pour 3.3 (p. 54), en utilisant le fait qu'une flèche f dans $\mathcal{M}(I)_{\mathrm{c}}$ qui est dans $W(I)$, s'écrit dans $\mathcal{M}(I)_{\mathrm{c}}$ sous la forme $f = pi$, où $i \in TC(I)$, et où p est inverse à gauche (rétraction) d'une flèche $j \in TC(I)$. On aura donc $u_!(f) = u_!(p)u_!(i)$, et $u_!(i) \in TC(I')$ par (6.1.10), et $u_!(p)$ est inverse à gauche de $u_!(j) \in TC(I')$. On a alors $u_!(i), u_!(j), u_!(p) \in W(I')$, d'où $u_!(f) \in W(I')$, q.e.d. p. 52 [page 104]

Prouvons enfin d). Compte tenu de la définition de C_I, pour établir $C(I) \subset C_I$, il suffit de voir ceci :

Commentary · Exposition [ms. 104]

Relative colimits measure the new attachment

Grothendieck wants a cofibration of diagrams to remain a cofibration when part of the indexing category has already been assembled. The map to inspect is therefore not just the map between two colimits. It is the remaining attachment after the source diagram and the chosen part of the target have been put together:

$$X_V \amalg_{X_U} Y_U \longrightarrow Y_V .$$

For a cosieve $U \subseteq V$ in I, the actual diagram direction is V^{op}, and U^{op} is a sieve. These are exactly the subsets on which a partial covariant diagram can be constructed before its complement. In these formulas X_U means $\mathrm{colim}_{U^{\mathrm{op}}} X$, and similarly for V; the repeated U in the edition's definition of X_V is a slip.

The reduction to an arrow is useful because it separates the categorical bookkeeping from the lifting argument. After restricting to V, take the characteristic map $V \to [1]$ whose fibre at 1 is U. Left Kan extension turns the pair of colimits into the arrow $X_U \to X_V$, and turns a map of diagrams into its square of colimits. The relative map of that square is then the displayed attachment. Only the preservation of projective cofibrations by left Kan extension and the arrow-category calculation remain to be checked.

The restriction step is explicitly deferred in the source. It is justified by 6.4(c) and 6.5: a cosieve inclusion is a discrete opfibration, and its right Kan extension is given objectwise by either evaluation or an empty product. Thus restriction preserves the two projective left classes by adjunction. The temporary assumption in the proof of 6.2 is discharged within these pages.

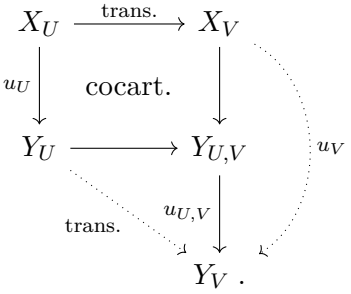
Corollaire 6.2.[31] *Soit I quelconque dans* Cat*, soient $U \subset V$ deux fermés dans I (i.e. des ouverts dans I°), et soit $f : X_\bullet \longrightarrow Y_\bullet$ une flèche dans $C(I)$ (resp. dans $TC(I)$). Posons encore* [page 105]

$$X_U \ = \ \varinjlim_{i \in U} X(i), \qquad X_V \ = \ \varinjlim_{i \in U} X(i) \ ,$$

31. **N.B.** Ce résultat se généralise, sous les conditions générales dans 6.2.5.

et de même pour Y, et considérons le diagramme

(6.2.1)



Alors la flèche $u_{U,V}$ est dans C (resp. TC).

Démonstration de 6.2. Admettons que la restriction d'une flèche de $C(I)$ resp. $TC(I)$ à un fermé V de I est dans $C(V)$ resp. $TC(V)$ (cf. … ci-dessous). Cela nous permet de supposer $V = I$. La donnée de U, qui est un ouvert de I°, équivaut à la donnée d'une flèche $v : I^\circ \longrightarrow \mathbf{\Delta}^1$ (en prenant la fibre en 0), ou encore à la donnée d'une flèche $u : I \longrightarrow (\mathbf{\Delta}^1)^\circ \simeq \mathbf{\Delta}^1$ (en prenant le fibre en 1). Or

$$u \,:\, I \longrightarrow \mathbf{\Delta}^1\,, \qquad U \;=\; I_1 \overset{\text{déf}}{=} u^{-1}(\{1\})\,. \tag{6.2.2}$$

On a alors [page 106]

$$X_U \;=\; u_!(X)(1), \quad X_I \;=\; u_!(X)(0)\,,$$

et le morphisme de transition $X_U \longrightarrow X_I$ est déduit de la flèche $0 \xrightarrow{\partial} 1$ dans $\mathbf{\Delta}^1$, et de même pour Y. Soit

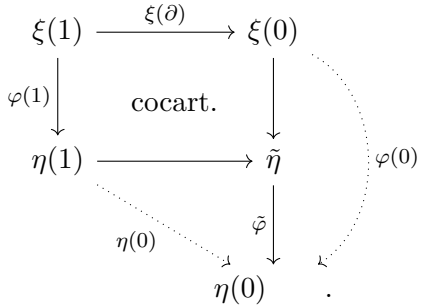
$$\xi \;=\; u_!(X), \qquad \eta \;=\; u_!(Y)\,,$$

donc le diagramme (6.2.1) s'identifie au diagramme déduit de p. 53

$$\varphi \,:\, \xi \longrightarrow \eta\,, \qquad \varphi \;=\; u_!(f)\,,$$

comme

(6.2.3)



Or par 6.1.10 appliqué à $u : I \longrightarrow \mathbf{\Delta}^1$, la relation $f \in C(I)$ resp. $f \in TC(I)$ implique $\varphi \in C(\mathbf{\Delta}^1)$ resp. $\varphi \in TC(\mathbf{\Delta}^1)$. Donc pour établir 6.1, on est ramené au cas particulier où $I = \mathbf{\Delta}^1$, $U = \{1\}$, $V = I$, donc à prouver le

Lemme 6.2.4. *Soit $\varphi : \xi \longrightarrow \eta$ une flèche dans $\mathcal{M}(\mathbf{\Delta}^1)$. Si $\varphi \in C(\mathbf{\Delta}^1)$ (resp. $\varphi \in TC(\mathbf{\Delta}^1)$), alors la flèche $\tilde{\varphi}$ dans le diagramme* (6.2.3) *est dans C (resp. TC).* [page 107]

Plus généralement, et plus précisément :

Lemme 6.2.5. *Soit $\mathcal{M}$ une catégorie stable par sommes amalgamées, et Φ une partie Q-close à gauche* (on prendra $\Phi = C$, et $\Phi = TC$, dans le cas du lemme 6.2.4). *Soit $\Psi = \Phi_*$, et considérons, pour toute I dans* Cat, *$\Phi(I)$, $\Psi\langle I\rangle$ dans $\mathcal{M}(I)$, définis par :*

$$\begin{aligned} &u : X \longrightarrow Y \text{ dans } \mathcal{M}(I) \text{ est dans } \Psi\langle I\rangle \text{ si et seulement si } \forall\, i \in \operatorname{Ob} I,\\ &u(i) : X(i) \longrightarrow Y(i) \text{ est dans } \Psi, \end{aligned} \tag{6.2.5.1}$$

$$\Phi(I) \;=\; \Psi\langle I\rangle^{*}. \tag{6.2.5.2}$$

Prenons le cas $I = \mathbf{\Delta}^1 = \{0 \xrightarrow{\partial} 1\}$, et soit $u : X \longrightarrow Y$ une flèche dans $\mathcal{M}(\mathbf{\Delta}^1)$, correspondant à un carré commutatif

$$\begin{array}{ccc} X(1) & \xrightarrow{X(\partial)} & X(0) \\ {\scriptstyle u(1)}\downarrow & & \downarrow{\scriptstyle u(0)} \\ Y(1) & \xrightarrow{Y(\partial)} & Y(0)\,. \end{array} \tag{6.2.5.3}$$

Considérons le diagramme correspondant

p. 54
[page 108]

$$\begin{array}{ccccc} X(1) & \xrightarrow{X(\partial)} & X(0) & & \\ {\scriptstyle u(1)}\downarrow & \text{cocart.} & \downarrow & \searrow & \\ Y(1) & \longrightarrow & \tilde{Y} & & u(0) \\ & {\scriptstyle Y(\partial)}\searrow & \downarrow{\scriptstyle \tilde{u}} & \swarrow & \\ & & Y(0) & & \end{array} \qquad . \tag{6.2.5.4}$$

Pour qu'on ait $u \in \Phi(\mathbf{\Delta}^1)$, il faut et il suffit que l'on ait

$$u(0),\; u(1),\; \tilde{u} \;\in\; \Phi\;. \tag{6.2.5.5}$$

J'ai déjà vérifié la chose (sous forme duale), mais ne retrouve plus l'endroit (peut-être n'ai-je pas pris des notes). Je n'ai pas envie de rechercher la démonstration ici.

Commentary · Exposition [ms. 108]

The omitted arrow-category calculation

The short lemma that Grothendieck leaves unwritten is the local lifting step used throughout the remainder of the proof. We give it with precisely the pushout hypothesis he states. This also explains why the three maps in (6.2.5.5) are slightly more than one needs to test.

Let $\Phi = {}^{\perp}\Psi$ and $\Psi = \Phi^{\perp}$. Write a map of arrows as

$$\begin{array}{ccc} A_1 & \longrightarrow & A_0 \\ {\scriptstyle f_1}\downarrow & & \downarrow{\scriptstyle f_0} \\ B_1 & \longrightarrow & B_0, \end{array} \qquad P = A_0 \amalg_{A_1} B_1, \qquad t : P \longrightarrow B_0.$$

Then $f \in {}^{\perp}(\Psi\langle[1]\rangle)$ if and only if $f_1, t \in \Phi$. When these conditions hold, $f_0 \in \Phi$ as well.

For sufficiency, take a square from f to a map of arrows $p : (E_1 \to E_0) \to (F_1 \to F_0)$ whose two components belong to Ψ. First choose a filler $h_1 : B_1 \to E_1$ against f_1 and p_1. The given map $A_0 \to E_0$ and the composite $B_1 \xrightarrow{h_1} E_1 \to E_0$ agree on A_1. They define $P \to E_0$. Together with $B_0 \to F_0$, this is a square from t to p_0. A filler $h_0 : B_0 \to E_0$ solves it. Its restriction to B_1 is the prescribed composite, so (h_1, h_0) is a natural filler of the original square.

For necessity of the relative condition, take any square from t to $q : E \to F$ in Ψ, with upper map $a : P \to E$ and lower map $b : B_0 \to F$. Consider the map of arrows

$$(B_1 \xrightarrow{a|_{B_1}} E) \longrightarrow (B_1 \xrightarrow{b|_{B_1}} F)$$

with components $(1_{B_1}, q)$. The maps $(f_1, a|_{A_0})$ and $(1_{B_1}, b)$ define a square from f to this map. A filler has first component 1_{B_1}; its second component is exactly a filler for the prescribed square from t to q. Thus $t \in \Phi$.

To test f_1, one must make an arbitrary square at the first vertex into a square of arrows. Suppose that square has maps $a : A_1 \to E$, $b : B_1 \to F$ and right side $q : E \to F$ in Ψ. Form

$$K = F \amalg_{B_1} B_0.$$

There is a map of arrows $(E \to K) \to (F \to K)$ with components $(q, 1_K)$. The given maps at vertex 1, together with $A_0 \to B_0 \to K$ and $B_0 \to K$ at vertex 0, form a square from f. A filler supplies the desired map $B_1 \to E$. Hence $f_1 \in \Phi$. This construction uses a pushout in place of any terminal-object argument.

Finally, $A_0 \to P$ is the pushout of f_1, and f_0 is its composite with t. A left lifting class is closed under pushout and composition, so $f_0 \in \Phi$. This proves the printed criterion, including its redundant component condition. The same two-stage construction is the direct-category case of the relative-latching lifting argument of Riehl and Verity (2014, Observation 4.16); no model structure or factorization axiom is used here.

Pour achever de prouver 6.1. d), i.e. l'égalité $C(I) = C_I$ pour I artinien, j'ai envie de prouver encore quelque chose de plus générale que voici :

Corollaire 6.3. *Soient* $(\mathcal{M}, \Phi, \Psi)$ *et* $I \longmapsto \Phi(I) \subset \mathrm{Fl}(\mathcal{M}(I))$ *comme dans* 6.2.5. *Soit* I *un ensemble ordonné, et considérons aussi* Φ_I *défini (dans la situation duale) page* 25 (2.2) *(mais avec des notations duales …). Je dis que l'on a*

$$\Phi_I \supset \Phi(I) , \tag{6.3.1}$$

et on a égalité si I *est noethérien, i.e.* I° *est artinien.* [page 109]

Démonstration. La démonstration de $\Phi(I) \subset \Phi_I$ est celle qu'on vient de donner. Il reste à prouver que si I est noethérien, on a $\Phi_I \subset \Phi(I)$. Étant donnée $i : A \longrightarrow B$ dans Φ_I, il faut prouver que $u \in \Phi(I)$, i.e. que pour un diagramme commutatif dans $\mathcal{M}(I)$,

$$\begin{array}{ccc} A & \xrightarrow{\alpha} & X \\ {\scriptstyle i}\downarrow & \overset{h}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow[\beta]{} & Y \end{array} \tag{6.3.2}$$

avec $p \in \Psi\langle I\rangle$, i.e. $p(i) : X(i) \longrightarrow Y(i)$ dans Ψ pour tout i, $\exists\, h : B \longrightarrow X$ laissant le diagramme commutatif. On va revenir aux notations duales, en travaillant dans $\mathcal{M}(I^\circ)$, avec I artinien. On considère l'ensemble $\mathcal{H}$ des couple (U, h_U) d'un ouvert U de I et d'un $h_U : B|U \longrightarrow X|U$, rendant commutatif la restriction à U de (6.3.2). On ordonne $\mathcal{H}$ par prolongement, on note que toute partie totalement ordonnée (U_α, h_{U_α}) dans $\mathcal{H}$ a une borne supérieure, savoir $U = \bigcup_\alpha U_\alpha$, h_U défini par les h_{U_α} sur les U_α.

Commentary · Exposition [ms. 109]

The lift is built in the direction of the arrows

The long cube calculation has a simple purpose: when a new vertex is added, its lift must extend the lifts already chosen on all incoming arrows. The relative latching map incorporates that compatibility in a single lifting problem. We formulate the argument in a fixed covariant convention so that its order hypothesis is visible throughout.

Let J be a small artinian poset, let $A, B, X, Y : J \to \mathcal{M}$, and suppose $p : X \to Y$ is objectwise in Ψ. Assume that the relevant colimits exist. For $j \in J$, put

$$L_j A = \mathrm{colim}_{k<j} A_k, \qquad L_j B = \mathrm{colim}_{k<j} B_k, \qquad \ell_j(f) : A_j \amalg_{L_j A} L_j B \longrightarrow B_j.$$

If every $\ell_j(f)$ belongs to Φ, any square from $f : A \to B$ to p has a natural filler.

Indeed, order partial fillers on downward closed subsets by extension. The empty subset supports a filler, and a chain has an upper bound obtained by taking the union of its domains and its compatible component maps. Zorn's lemma gives a maximal partial filler h_U. If $U \neq J$, choose

a minimal element j of $J \setminus U$. Every $k < j$ lies in U, so the partial filler gives a map $L_jB \to X_j$. Together with $A_j \to X_j$ it induces

$$A_j \amalg_{L_jA} L_jB \longrightarrow X_j.$$

The original square and the naturality of the maps make this a square with lower map $B_j \to Y_j$ and right side p_j. The lifting relation $\Phi \perp \Psi$ supplies $h_j : B_j \to X_j$. Its restriction along $L_jB \to B_j$ is the map already prescribed by h_U, hence it is compatible with every incoming arrow. There are no arrows from j to an old object of U, since U is downward closed. The result is a filler on $U \cup \{j\}$, contradicting maximality. Therefore $U = J$.

This proves the difficult inclusion in 6.3 without shortening any of its mathematical content. Grothendieck's $\widetilde{B}$, $\widetilde{i}$ and $\widetilde{h}_0$ are the pushout, relative latching map and upper arrow just displayed. The identities on pp. 113–116 verify that the proposed upper arrow and lower arrow make the required square commute; they hold by the pushout universal property. The sentence at the end must use the given hypothesis $i \in \Phi_I$, not the printed $i \in \Phi(I)$, which is the conclusion.

The passage from J back to the source's contravariant notation is $J = I^{\mathrm{op}}$. Thus I must be noetherian. The colimit assumption is inherited from the use of the classes defined in §2; pushouts alone, the hypothesis of the arrow lemma, do not ensure that arbitrary L_jA exist. No small-object argument enters this proof. Its induction is legitimate for any small well-founded poset, including those whose ranks are transfinite.

(**N.B.** L'ensemble $\mathcal{H}$ n'est pas vide, car $(\emptyset, h_\emptyset)$ est dans $\mathcal{H}$ – mais c'est contenu dans l'énoncé précédent que toute partie totalement ordonnée de $\mathcal{H}$ y a une borne supérieure …). Par Zorn, il existe un élément maximal (U, h_U). À prouver que $U = I$, et on aura terminé. Ou encore que si (U, h_U) tel que $U \neq I$, ce n'est *pas* maximal. Soit $F = I \backslash U \neq \emptyset$ le fermé complémentaire. Par l'hypothèse artinienne sur I, F a un élément minimal i_0. Soit

p. 55
[page 110]

$$U' \;=\; U \cup \overbrace{I/i_0}^{V} \;=\; U \cup \{i_0\}\ ,$$

on va prolonger h_U en $h_{U'}$ sur U', on aura terminé. Pour ceci, soit $V_0 = V \cap U = V \backslash \{i_0\}$, on est ramené à prolonger la restriction h_{V_0} de h_U à V_0, de V_0 à V. Au total, on peut supposer $I = I/i_0$ muni d'un élément final i_0, $U = I \backslash \{i_0\}$, et montrer qu'une $(U, h_U) \in \mathcal{H}$ se prolonge à I. Soient A', B', X', Y' les restrictions de A, B, X, Y à U (qu'on pourra aussi noter I'). La donné de A équivaut à celle de A', celle de $A_0 = A(i_0) \in \mathrm{Ob}\,\mathcal{M}$, et d'une flèche de

$$A'_{I'} \overset{\text{déf}}{=} \varinjlim_{i' \in I'} A'(i') \xrightarrow{\lambda_A} A_0 \ (\simeq A_I)\ ,$$

et de même pour B, X, Y. Plus précisément, la catégorie $\mathcal{M}(I^\circ)$ s'interprète comme celle des triples $X = (X', X_0, \lambda_X)$, où $X' \in \mathcal{M}(I'^\circ)$, $X_0 \in \mathcal{M}$, et $\lambda_X : X'_{I'} \longrightarrow X_0$. À ce compte, la donnée du diagramme (6.3.2) équivaut à celle d'un diagramme carré similaire [page 111]

(6.3.3)
$$\begin{array}{ccc} A' & \xrightarrow{\alpha'} & X' \\ {\scriptstyle i'}\downarrow & \overset{h'}{\nearrow} & \downarrow{\scriptstyle p'} \\ B' & \xrightarrow[\beta']{} & Y' \end{array}$$

dans $\mathcal{M}(I'^\circ)$, d'où un diagramme dans $\mathcal{M}$

(6.3.4)
$$\begin{array}{ccc} A'_{I'} & \xrightarrow{\alpha'_{I'}} & X'_{I'} \\ {\scriptstyle i'_{I'}}\downarrow & \overset{h'_{I'}}{\nearrow} & \downarrow{\scriptstyle p'_{I'}} \\ B'_{I'} & \xrightarrow[\beta'_{I'}]{} & Y'_{I'}\ , \end{array}$$

d'un autre diagramme carré *dans* $\mathcal{M}$

$$\begin{array}{ccc} A_0 & \xrightarrow{\alpha_0} & X_0 \\ {\scriptstyle i_0}\downarrow & & \downarrow{\scriptstyle p_0} \\ B_0 & \xrightarrow[\beta_0]{} & Y_0\ , \end{array} \tag{6.3.5}$$

et enfin d'un homomorphisme $\lambda = (\lambda_A, \lambda_B, \lambda_X, \lambda_Y)$ du carré (6.3.4) dans le carré (6.3.5) : p. 56 [page 112]

$$\begin{array}{ccccc} A'_{I'} & & \xrightarrow{\alpha'_{I'}} & & X'_{I'} \\ & \searrow^{i'_{I'}} & \nearrow_{h'_{I'}} & & \downarrow{\scriptstyle \lambda_X} \quad \searrow^{p'_{I'}} \\ \downarrow{\scriptstyle \lambda_A} & & B'_{I'} & \xrightarrow{\beta'_{I'}} & Y'_{I'} \\ & & \downarrow{\scriptstyle \lambda_B} & & \downarrow \\ A_0 & \xrightarrow{\alpha_0} & X_0 & & \downarrow{\scriptstyle \lambda_Y} \\ & \searrow^{i_0} & \nearrow_{h_0} & \searrow^{p_0} & \\ & & B_0 & \xrightarrow[\beta_0]{} & Y_0 \end{array} \tag{6.3.6}$$

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D'autre part, la question de prolonger $(I', h') \in \mathcal{H}$ en (I, h), revient à celle de trouver $h_0 : B_0 \longrightarrow X_0$ dans $\mathcal{M}$, de façon qu'avec $h'_{I'}$ cela fasse encore un diagramme commutatif de (6.3.6), ce qui revient à dire que d'une part les deux triangles formés dans le carré de base sont commutatifs :

$$p_0 h_0 \ = \ \beta_0, \qquad h_0 i_0 \ = \ \alpha_0\ , \tag{6.3.7}$$

et d'autre part le carré diagonal défini par $h'_{I'}$, h_0 soit commutatif (ce qui exprime qu'on a une flèche h de B dans X) :

$$\lambda_X h'_{X'} \ = \ h_0 \lambda_B\ . \tag{6.3.8}$$

Mais le diagramme (6.3.6), où la flèche en traits pointillés denses $h'_{I'}$ est déjà acquise, et h_0 en pointillés espacés reste à trouver, donne naissance au diagramme, dans lequel est mis en évidence la flèche $\tilde{i} \in \Phi$ qui exprime l'hypothèse $i \in C(i)$, compte tenu de $i' \in C(I')$: [page 113]

$$\begin{array}{ccccc} A'_{I'} & & \xrightarrow{\alpha'_{I'}} & & X'_{I'} \\ & \searrow^{i'_{I'}} & \nearrow_{h'_{I'}} & & \downarrow{\scriptstyle \lambda_X} \quad \searrow^{p'_{I'}} \\ \downarrow{\scriptstyle \lambda_A} & & B'_{I'} & \xrightarrow{\beta'_{I'}} & Y'_{I'} \\ & \text{cocart.} & \downarrow{\scriptstyle s} & & \downarrow \\ A_0 & \xrightarrow{\alpha_0} & & & X_0 \quad \downarrow{\scriptstyle \lambda_Y} \\ & \searrow^{r} & \nearrow_{\tilde{h}_0} & & \searrow^{p_0 \in \Psi} \\ & & \tilde{B} & \xrightarrow{\tilde{\beta}_0} & Y_0 \\ & \searrow^{i_0} & \downarrow{\scriptstyle \tilde{i} \in \Phi} & \nearrow_{h_0} & \\ & & B_0\ , & \xrightarrow[\beta_0]{} & \end{array} \tag{6.3.9}$$

où la flèche $\tilde{h}_0$ en traits pointillés denses de $\tilde{B} \overset{\text{déf}}{=} A_0 \amalg_{A'_{I'}} B'_{I'}$ dans X_0, est définie par les p. 57

32. **N.B.** La condition $p \in \Psi\langle I \rangle$ s'exprime par $p' \in \Psi\langle I' \rangle$ et $\boxed{p_0 \in \Psi.}$

conditions de commutativité

$$\tilde{h}_0 r \;=\; \alpha_0, \qquad \tilde{h}_0 s \;=\; \lambda_X h'_{I'} \;, \tag{6.3.10}$$

ce qui est licite, vu que

$$\begin{aligned} \alpha_0\lambda_A &= (\lambda_X h'_{I'})i'_{I'} \\ &= \lambda_X(h'_{I'}i'_{I'}) \\ &= \lambda_X\alpha'_{I'}. \end{aligned}$$

Quant à la flèche $\tilde{\beta}_0 : \tilde{B} \longrightarrow Y_0$, elle est donnée par la condition de commutativité [page 114]

$$\tilde{\beta}_0 r \;=\; p_0\alpha_0, \qquad \tilde{\beta}_0 s \;=\; \lambda_Y \beta'_{I'} \;, \tag{6.3.11}$$

ce qui est encore licite, puisque

$$(p_0\alpha_0)\lambda_A \;=\; (\lambda_Y\beta'_{I'})i'_{I'}$$

en vertu de la commutativité du cube (6.3.6). Pour établir la commutativité totale du diagramme (6.3.9) (traits pleins, et traits pointillés denses pour $h'_{I'}$, $\tilde{h}_0$ et $\tilde{\beta}_0$ – à l'exclusion de h_0 pas encore construit), il faut encore vérifier d'une part les relation

$$\begin{cases} \tilde{h}_0 r \;=\; \alpha_0, \quad p_0\tilde{h}_0 \;=\; \tilde{\beta}_0, \\ \lambda_X h'_{I'} \;=\; \tilde{h}_0 s \end{cases} \qquad {}^{33}, \tag{6.3.12}$$

établissant la commutativité du cube supérieur avec le carré diagonal (diagramme du type (6.3.6), la flèche pointillée h_0 y incluse), d'autre part les relations

$$\begin{cases} \tilde{i} r \;=\; i_0 \qquad (\text{et } \tilde{i}s = \lambda_B \text{ pour mémoire}) \\ \beta_0\tilde{i} \;=\; \tilde{\beta}_0 \;, \end{cases} \tag{6.3.13}$$

où la première ligne est déjà connue par définition de $\tilde{B}$ et de $\tilde{i}$. La deuxième se ramène aux deux relations [page 115]

$$\begin{aligned} (\beta_0\tilde{i})r &= \beta_0(\tilde{i}r) \\ &= \beta_0 i_0 \\ &= p_0\alpha_0 \\ &= \tilde{\beta}_0 r \;, \end{aligned}$$

OK par commutativité du carré de base de (6.3.5),

$$\begin{aligned} (\beta_0\tilde{i})s &= \beta_0(\tilde{i}s) \\ &= \beta_0\lambda_B \\ &= \lambda_Y\beta'_{I'} \\ &= \tilde{\beta}_0 s \;, \end{aligned}$$

OK par commutativité du carré antérieur dans (6.3.6). p. 58

Commentary · Examples [ms. 113]

A diamond and the compatibility at its last vertex

Grothendieck's cube records information that an objectwise cofibration test would miss. A finite example makes the attachment visible. Work in simplicial sets, whose cofibrations are the monomorphisms (Riehl, 2014, Example 11.3.5), and let J be the poset with objects a, b, c, d and

33. **N.B.** la première et la troisième de ces relations sont déjà connues, cf. (6.3.10).

relations $a < b < d$ and $a < c < d$. We use covariant diagrams, or equivalently the source's notation $\mathcal{M}(J^{\mathrm{op}})$.

Define a diagram B by

$$B_a = \Delta^0, \qquad B_b = B_c = \Delta^1, \qquad B_d = \Delta^1 \amalg_{\Delta^0} \Delta^1,$$

where the maps out of B_a choose the initial endpoint of each interval, and the maps into B_d are the two inclusions. The latching objects are

$$L_a B = \varnothing, \qquad L_b B = L_c B = \Delta^0, \qquad L_d B = \Delta^1 \amalg_{\Delta^0} \Delta^1.$$

Every latching map is a simplicial monomorphism: at d it is the identity. Consequently B is projectively cofibrant by the lifting argument just given. Its colimit is B_d, since d is terminal in J; its realization is two intervals joined at an endpoint. A compatible map out of B_b and B_c is exactly a pair of paths with the same value at their initial endpoints. The pushout in the latching object expresses precisely this requirement.

Now retain the first three values but replace B_d by Δ^0, with both maps from the intervals collapsing them. All four values remain cofibrant simplicial sets. However, the last latching map becomes

$$\Delta^1 \amalg_{\Delta^0} \Delta^1 \longrightarrow \Delta^0,$$

which is not a monomorphism, since it identifies distinct vertices. The diagram is therefore not projectively cofibrant. A choice of lifts at its four objects need not be compatible along the two paths through the diamond. This is why the proof asks for $\widetilde{i} \in \Phi$, rather than merely the component map $i_0 \in \Phi$.

Donc il reste à vérifier la deuxième relation (6.3.12),

$$p_0 \tilde{h}_0 \;=\; \tilde{\beta}_0 \ ,$$

ce qui se ramène à

$$\begin{aligned} (p_0 \tilde{h}_0) r \;&=\; p_0 (\tilde{h}_0 r) \\ &=\; p_0 \alpha_0 \\ &=\; \tilde{\beta}_0 r \ , \end{aligned}$$

OK par (6.3.11),

$$\begin{aligned} (p_0 \tilde{h}_0) s \;&=\; p_0 (\tilde{h}_0 s) \\ &=\; p_0 (\lambda_X h'_{I'}) \\ &=\; (p_0 \lambda_X) h'_{I'} \\ &=\; (\lambda_Y p'_{I'}) h'_{I'} \\ &=\; \lambda_Y (p'_{I'} h'_{I'}) \\ &=\; \lambda_Y \beta'_{I'} \\ &=\; \tilde{\beta}_0 s \ , \end{aligned}$$

OK.

Ceci posé, je dis que les contraintes (6.3.7), (6.3.8) sur h_0 reviennent à la commutativité des deux triangles dans les carré

(6.3.14)
$$\begin{array}{ccc} \tilde{B} & \xrightarrow{\tilde{h}_0} & X_0 \\ {\scriptstyle \tilde{i} \in \Phi}\downarrow & \overset{h_0}{\nearrow} & \downarrow{\scriptstyle p_0 \in \Psi} \\ B_0 & \xrightarrow[\beta_0]{} & Y_0 \ , \end{array}$$

i.e. aux deux relations

[page 116]

(6.3.15)
$$h_0 \tilde{i} \;=\; \tilde{h}_0 \ , \qquad p_0 h_0 \;=\; \beta_0 \ .$$

La deuxième de ces relations n'est autre que la première relation (6.3.7), et il reste à voir que la première relation (6.3.15) équivaut à l'ensemble des deux relations restantes dans (6.3.7), (6.3.8) :

$$h_0 i_0 \;=\; \alpha_0\ , \qquad \lambda_X h'_{I'} \;=\; h_0 \underbrace{\lambda_B}_{\tilde{i}s} \tag{6.3.16}$$

Or la relation en question $h_0\tilde{i} = \tilde{h}_0$ équivaut à l'ensemble des deux relations p. 59

$$\begin{aligned}(h_0\tilde{i})r &= h_0(\tilde{i}r)\\ &= h_0 i_0\\ &\overset{?}{=} \alpha_0\\ &= \tilde{h}_0 r,\end{aligned}$$

équivalent à $h_0 i_0 = \alpha_0$,

$$\begin{aligned}(h_0\tilde{i})s &= h_0(\tilde{i}s)\\ &= h_0\lambda_B\\ &\overset{?}{=} \lambda_X h'_{I'}\\ &= \tilde{h}_0 s,\end{aligned}$$

équivalent à $h_0\lambda_B = \lambda_X h'_{I'}$, c'est OK.

Ceci posé, dans le carré (6.3.14), les hypothèses $i \in \Phi(I)$, $p \in \Psi\langle I\rangle$ impliquent $\tilde{i}_0 \in \Phi$, $p_o \in \Psi$. Donc la flèche cherchée h_0 existe, q.e.d.

Cela achève la démonstration du corollaire 6.3, et par là aussi, celle de la dernière partie, d), de 6.1. On est heureux !

[page 117]

Commentary · Later developments [ms. 116]

The direct part of Reedy theory

The proof just completed has become the direct part of Reedy theory. For a category with degrees and unique direct–inverse factorizations, a new vertex has both incoming and outgoing constraints. Relative latching maps encode the former and relative matching maps the latter. In a direct category the inverse part has only identities, so matching objects are terminal and Reedy fibrations are objectwise fibrations. Whenever the projective structure exists, the two model structures then agree because their fibrations and weak equivalences agree.

Riehl and Verity (2014, Definition 4.14, Observation 4.16 and Theorem 4.18) give the relative maps, the lifting construction and the resulting model structure; their Lemma 7.1 also identifies trivial Reedy cofibrations by trivial relative latching maps. The familiar natural-number degree convention in that treatment is sufficient for finite posets. Grothendieck's well-founded-poset argument allows transfinite rank directly, without requiring a uniform finite bound on the lengths of chains entering a vertex. The equality in 6.3 concerns this direct situation, after the variance has been fixed; it is not an equality of the projective and arbitrary Reedy structures on every diagram category.

Proposition 6.4.[34] *Soit $\mathcal{M}$ une catégorie, et $\Phi \longleftrightarrow \Psi$ un couple Q-clos en Q-dualité ($\Phi = \Psi^*$, $\Psi = \Phi_*$). Soit $u : I \longrightarrow I'$ une flèche dans* Cat*, d'où une suite de foncteurs adjoints*

$$\mathcal{M}(I) \;\underset{u_*}{\overset{u_!}{\underset{\longrightarrow}{\overset{\longrightarrow}{\underset{u^*}{\longleftarrow}}}}}\; \mathcal{M}(I')\quad .$$

On définit $\Phi(I)$, $\Psi\langle I\rangle$, $\Phi(I')$, $\Psi\langle I'\rangle$ comme dans 6.2.5*. On a alors :*

34. Sous réserve d'existence des $\varinjlim$ et des $\varprojlim$ pertinentes dans $\mathcal{M}$. On peut supposer pour simplifier $\mathcal{M}$ stable par $\varinjlim$ et $\varprojlim$.

ⓐ

$$\begin{cases} u_!(\Phi(I)) \ \subset \ \Phi(I') \ , \\ u^*(\Psi\langle I'\rangle) \ \subset \ \Psi\langle I\rangle \ . \end{cases} \tag{6.4.1}$$

ⓑ *On a l'équivalence*

$$u^*(\Phi(I')) \subset \Phi(I) \qquad \Longleftrightarrow \qquad u_*(\Psi\langle I\rangle) \subset \Psi\langle I'\rangle \ . \tag{6.4.2}$$

ⓒ *Les relations équivalentes* (6.4.2) *ont lieu dans chacun des trois cas suivants :*

1°) *u est coétale.*

2°) *I est discrète.*

3°) *Pour tout $i' \in \mathrm{Ob}\, I'$, les composantes connexes de I/i' admettent un objet final.*

Corollaire 6.5. *Soit (W, C, F) un triple de Quillen clos spécial dans $\mathcal{M}$, et définissons, pour toute I dans* Cat*, $C(I)$, $TC(I)$ et $F(I)$, $TF(I)$ comme dans le théorème* 6.1. a)*. On a alors* p. 60

les équivalences [page 118]

$$\Big(u^*(C(I')) \ \subset \ C(I)\Big) \quad \Longleftrightarrow \quad \Big(u_*(TF) \ \subset \ TF'\Big)\ , \tag{6.5.1}$$

$$\Big(u^*(TC(I')) \ \subset \ TC(I)\Big) \quad \Longleftrightarrow \quad \Big(u_*(F) \ \subset \ F'\Big)\ . \tag{6.5.2}$$

Les conditions équivalentes (6.5.1) *ainsi que* (6.5.2) *sont satisfaites quand on est dans l'un des cas* 1°) 2°) 3°) *de* 6.4. c).

Commentary · Exposition [ms. 118]

Why terminal comma components preserve cofibrations

The final proposition asks when restriction can serve on the left of a Quillen adjunction as well as on the right. The distinction is useful in applying the construction to a subdiagram. Restriction always preserves the objectwise right classes. To preserve the left classes as well, its right adjoint must preserve the corresponding objectwise right classes. The commentary on Chapter XII, manuscript p. 114 ("A terminal fibre controls all the ordinary direct images"), discusses terminal fibres and ordinary direct images; the additional point here is the product stability of right lifting classes.

Here is the formal detail behind the equivalence in 6.4(b). With all required copowers available, one has

$$\Psi\langle I\rangle = \Phi(I)^{\perp}.$$

One inclusion follows from the definition $\Phi(I) = {}^{\perp}(\Psi\langle I\rangle)$. For the other, use the free evaluation diagrams $F_i A = I(-, i) \cdot A$. If $a \in \Phi$, then $F_i a \in \Phi(I)$ by adjunction. A map right-orthogonal to $\Phi(I)$ therefore has each component right-orthogonal to Φ, hence in Ψ. Transposing lifting problems across $u^* \dashv u_*$ now proves both directions of (6.4.2). Factorizations are not needed.

The geometric hypothesis makes right Kan extension particularly simple. Its actual pointwise formula for presheaves is

$$(u_*X)(i') = \lim_{(I/i')^{\mathrm{op}}} X.$$

If each connected component of I/i' has a terminal object e_α, that object is initial in the opposite indexing category. A cone on the restricted diagram is then determined by its map into $X(e_\alpha)$. Splitting the index into its components gives

$$(u_*X)(i') \cong \prod_{\alpha} X(e_\alpha).$$

For a morphism of diagrams the same formula gives a product of its components. Right lifting classes are closed under products: solve a square against the product by projecting to each factor, choosing the factorwise fillers, and using the universal property of the product. The empty product gives an identity map of a terminal object. Thus the displayed formula proves the required preservation.

For completeness, if u is a discrete opfibration and $(i, \gamma : u(i) \to i')$ is an object of I/i', lift γ uniquely to $i \to \gamma_! i$. Its target lies over i', and (i, γ) has a unique map to $(\gamma_! i, 1_{i'})$. A morphism in the comma category leaves this target unchanged by uniqueness of lifting. Consequently the comma category is a disjoint union of components, each with one of these terminal objects. If I is discrete, the comma category itself is discrete and the assertion is immediate. These facts prove the two implications that the source abbreviates as $1° \Rightarrow 3°$ and $2° \Rightarrow 3°$.

Démonstration de 6.4. Les assertions ⓐ , ⓑ sont ± tautologiques, par adjonction, comparer XIII, prop. 2, et cor., page 46. Pour prouver c), il suffit de traiter le cas 3°), car 1° $\Longrightarrow$ 3°, 2° $\Longrightarrow$ 3°. Mais le calcul de

$$u_*(X)(i') \ \simeq \ \varprojlim_{i \in I/i'} X(i)$$

montre que si les composantes connexes $(I/i')_\alpha$ de I/i' ont chacune un objet final e_α, alors

$$u_*(X)(i') \;\simeq\; \prod_\alpha X(e_\alpha)\ .$$

Or Ψ, étant clos à droite, est stable par produits quelconques, donc $X \in \Psi\langle I\rangle$ implique que les $u_*(X)(i')$ sont dans Ψ, donc u_* est dans $\Psi\langle I'\rangle$, q.e.d.

Commentary · Development (commentators') [ms. 118]

A converse when the source is a poset

Grothendieck's terminal-component condition is sufficient for every Q-dual pair. The elementary model of sets used above shows that, for a poset source, it is also a necessary condition if one wants such preservation even in this one test theory. This identifies exactly how much of the comma category the assertion detects.

Proposition [ours]. Let $u : I \to I'$ be a functor between small categories, with I a poset. Give sets the model structure with all maps weak equivalences, injections cofibrations and surjections fibrations, and give presheaf categories their projective structures. The following conditions are equivalent:

1. Every connected component of every I/i' has a terminal object.
2. u^* preserves projective cofibrations.
3. u^* takes every representable presheaf on I' to a projective presheaf on I.

Here projective means having the lifting property against objectwise surjections, and the axiom of choice is understood.

We first establish the elementary description of projectives needed for the converse. If P is a presheaf on a small poset I, form the canonical objectwise surjection

$$q : \coprod_{(i,x),\ x\in P(i)} I(-,i) \longrightarrow P.$$

It is surjective at j because the summand indexed by (j,z) sends 1_j to every given $z \in P(j)$. If P is projective, choose a section s of q. The induced map of categories of elements takes each connected component of $\int P$ into a single representable summand, say the summand indexed by (i,x). For any element (j,z) in that component, $qs = 1$ says that the arrow selected by s has the form $j \to i$ and sends x to z. Hence (j,z) has an arrow to (i,x) in $\int P$, so (i,x) belongs to the same component. Because I is a poset there is at most one such arrow. Thus (i,x) is terminal in that component.

Conversely, if every component of $\int P$ has a terminal element (i_α, x_α), the maps represented by these elements give an isomorphism

$$\coprod_\alpha I(-,i_\alpha) \xrightarrow{\ \sim\ } P.$$

At j, existence and uniqueness of the arrow to the appropriate terminal element give respectively surjectivity and injectivity. A representable is projective by evaluation, and a coproduct of representables is projective by choosing a lift in each summand. We have proved that projective presheaves on a poset are exactly coproducts of representables, equivalently presheaves whose categories of elements have terminal connected components. The general background is the description of indecomposable projectives as retracts of representables and its Cauchy-complete refinement in van Oosten (2021, Proposition 0.7 and Theorem 0.8); the poset case eliminates nonidentity idempotents.

Condition 1 implies condition 2 by 6.4 and 6.5. Condition 2 implies condition 3 because $\varnothing \to I'(-,i')$ is a projective cofibration: any lifting problem against an objectwise surjection is a lifting problem for an element at i'. Restriction preserves the initial presheaf. Finally,

$$\int \big(u^* I'(-,i')\big) = I/i',$$

including both its objects and its morphisms. The description of projectives just proved therefore makes condition 3 imply condition 1, completing the proof.

The poset hypothesis has a precise role. In a general category a projective presheaf may be a proper retract of a representable, controlled by a nontrivial idempotent; a terminal object in its

category of elements need not exist. We checked the projective-presheaf results just cited and the projective diagram construction in Riehl (2014, §12.3). We did not locate this exact formulation of the converse to 6.4(c) in those sources. The proposition is an elementary application of those established results, developed here to clarify the scope of Grothendieck's criterion, not a claim of priority.

Commentary · Later developments [ms. 118]

The pointwise formula and the derivator theorem

Grothendieck has now constructed enough compatible diagram structures to return to the purpose of the chapter: computing a homotopy Kan extension at one object. The last restriction criterion is the bridge. We spell out this application because the printed chapter stops immediately after the criterion, before drawing the conclusion in these terms.

We keep the convention of §6, so that $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$. Given $u : I \to I'$ and $i' \in I'$, put $S = i'\backslash I$, with objects $(i, \alpha : i' \to u(i))$, and let $q : S \to I$ and $p : S \to e$ be the projections. An arrow of S over $a : i \to j$ is forced by the equation $\beta = u(a)\alpha$. Thus q is a discrete opfibration. More explicitly, for each $j \in I$, the category S/j splits according to the composite $i' \to u(i) \to u(j)$; the part indexed by $\gamma : i' \to u(j)$ has terminal object $((j, \gamma), \mathrm{id}_j)$. This verifies exactly condition 3 of Proposition 6.4, with no claim that arbitrary restrictions preserve projective cofibrations.

There is also a direct check of the required preservation of cofibrations which uses only coproducts, and therefore does not import the right Kan extension existence proviso in Proposition 6.4. Write $F_i(A)(j) = \coprod_{I(j,i)} A$ for the free presheaf at i. Unique outgoing lifts along q give

$$q^* F_i(A) \cong \coprod_{s \in q^{-1}(i)} F_s(A).$$

Restriction preserves colimits and retracts. This formula and the generating descriptions in Theorem 6.1(b) show that it preserves projective cofibrations and trivial cofibrations. Choose a projectively cofibrant replacement $QX \to X$; then q^*QX is projectively cofibrant on S. Ordinary left Kan extension gives

$$(u_! QX)(i') \cong p_! q^* QX = \operatorname*{colim}_{S^{\mathrm{op}}} q^* QX.$$

The first expression computes the value of $\mathbf{L}u_! X$, and the second computes $\mathbf{L}p_!(q^*X)$. Evaluation and restriction preserve all objectwise weak equivalences, so the replacement maps identify this comparison independently of the chosen cofibrant representative. We obtain the canonical pointwise formula

$$i'^* \mathbf{L}u_! X \cong \operatorname*{hocolim}_{(i'\backslash I)^{\mathrm{op}}} q^* X.$$

Here the opposite on the comma category is essential. In the covariant convention of §3 the corresponding indexing category is I/i', without this opposite. The formula is the content of the left Kan extension axiom, not merely an objectwise recipe unrelated to the adjunction.

The other elementary properties can also be read directly from the diagram model structures. Diagrams on a disjoint union form a product of diagram categories; cofibrant and fibrant representatives show that their homotopy categories decompose accordingly. Evaluations jointly detect isomorphisms: represent a homotopy-category morphism between cofibrant–fibrant objects, and use saturation of weak equivalences in a closed model category in each component. If I has a terminal object t, the constant-presheaf functor is left adjoint to evaluation at t, with identity unit; both functors preserve weak equivalences, so constant diagrams remain fully faithful after localization. These are the precise elementary assertions behind the older axiom names.

The full, two-sided existence theorem is now available without the chapter's special-generation hypotheses, for a model category admitting all small limits and colimits. Cisinski (2003, Theorem 6.11 and its dual) establishes the homotopy Kan extension and pointwise assertions; his "left" terminology there is attached to the right Kan extension convention. For combinatorial model categories, Groth (2013, Theorem 1.28, Proposition 1.30, and Theorem 1.32) presents the projective–injective argument and records the general theorem. The present chapter provides the projective, colimit side; existence of all ordinary right Kan extensions is not part of its cocompleteness assumption.

Commentary · Later developments [ms. 118]

What the coherent diagram remembers

Grothendieck's decision to localize the category of whole diagrams has a precise counterpart in the language of higher categories. Suppose, for a direct comparison, that $\mathcal{M}$ is a combinatorial simplicial model category, and let $\mathcal{C}_\infty$ be the coherent nerve of its simplicial subcategory of cofibrant–fibrant objects, with universes fixed as usual. Rectification identifies

$$\mathrm{Ho}\big(\mathcal{M}^{I^{\mathrm{op}}}\big) \simeq h\,\mathrm{Fun}\big(N(I^{\mathrm{op}}), \mathcal{C}_\infty\big).$$

One precise reference is Lurie (2009, Proposition 4.2.4.4, Remark 4.2.4.5, and Corollary 4.2.4.8); the statement is made there with size-controlled subcategories, which give the displayed comparison after choosing an ambient universe. Projective and injective presentations have the same weak equivalences. The homotopy Kan extensions above present the corresponding higher categorical Kan extensions.

The comparison is with the homotopy category of a coherent diagram category. It does not identify that category with $\mathrm{Fun}(I^{\mathrm{op}}, h\mathcal{C}_\infty)$. Passing to the homotopy category before forming diagrams discards the compatibility of homotopies. The latching maps in the proof of Corollary 6.3 provide a concrete model-level account of precisely this compatibility: each new lift must extend the lifts already chosen, and its domain is a pushout recording those earlier choices.

Overview · XVII-b

From cofibrant diagrams to a calculus of homotopy Kan extensions

The second half of Chapter XVII supplies a construction that the preceding discussions have repeatedly needed: a homotopy theory of diagrams in a category of coefficients, compatible with changing the category of indices. Grothendieck begins with the economical axioms of a category with cofibrations and ends with projective model structures on arbitrary small diagram categories. The intervening work explains why the latter are useful. They provide a uniform source of derived adjunctions, while the earlier constructions on ordered sets explain how the resulting cofibrations can be recognized locally. The text therefore moves between two kinds of argument: a formal calculus of lifting and adjunction, and an inductive construction in which the part of a diagram already built controls the next attachment.

This is one concrete realization of the programme described in Grothendieck (1983, §69, opening discussion). A single derived category does not retain enough information about compatible constructions on complexes or spaces. One must also retain the localized categories of diagrams, together with restriction and the operations adjoint to restriction. The letter to Thomason of 2 April 1991 explains the role of the auxiliary model structures in this programme: the weak equivalences specify the homotopy theory; cofibrations and fibrations make its operations constructible (Grothendieck, 1991a, point 1, printed pp. 3–4). Chapter XVII shows both the reach of this principle and the difficulty of implementing it with a minimal axiom system.

A convention must remain visible throughout. In §3 the diagrams under construction are covariant, written $\mathcal{M}(I^\circ) = \mathrm{Fun}(I, \mathcal{M})$. In §§4–6 the notation returns to $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$. A left Kan extension therefore uses I/i' in the first convention and $(i'\backslash I)^{\mathrm{op}}$ in the second. Likewise, well-founded induction concerns an artinian poset in the covariant convention and a noetherian poset in the presheaf convention. The corrections and commentary preserve the printed text where it slips between these formulations, but do not treat a legitimate change of convention as a transcription error.

The adjunction must survive the construction

The first obstruction is already present in ordinary localization. An adjunction $L \dashv R$ does not automatically yield an adjunction between a functor obtained by restricting L to a convenient subcategory and the localization of R. An equivalence between the localized subcategory and the whole localization is insufficient. Grothendieck's counterexample in Remark 3.6 is decisive: localizing all arrows in a contractible category can identify it with the terminal category, while selecting a terminal object of the original category does not make that object initial. A left adjoint from the terminal category would have to select an initial object. The defect concerns the universal property of the constructed functor, not its existence as a functor.

Theorem 3.7 repairs the argument for the allowed ordered index categories. A functor is factored through its categorical mapping cylinder, or collage. One part is a closed inclusion; its left extension is extension by the initial object. The other part is a retraction with an initial object in every relevant comma category. Each part admits the localized adjunction already justified by the earlier cofibration arguments. Their composite therefore has the desired adjoint property. This is a proof, not an appeal to a general principle that localization preserves adjunctions without hypotheses.

The subsequent attempt to allow arbitrary targets exposes a separate size issue. Finite colimits do not provide the coproduct indexed by every hom-set of an arbitrary target. For example, finite sets admit the finite-branch cofibration axioms with every map a cofibration and only isomorphisms weak equivalences, but restriction from finite $\mathbb{Z}$-sets to finite sets has no free-action left adjoint. The commentary gives the argument and states precisely which additional colimits repair the construction. Later results for cocomplete cofibration categories

require corresponding infinite coproduct and sequential-colimit axioms; they should not be read back into the finite axioms of these pages.

Small generation and transfer

Sections 4 and 5 organize a second route. A special cofibration structure has enough colimits, individually accessible objects, a small family generating its cofibrations in the lifting sense, and the inclusion $C^{\square} \subseteq W$. The accessibility condition is used to run the small object argument. It does not by itself assert that the category has a small generating family of presentable objects; identifying it without qualification with local presentability would strengthen the source's hypotheses.

Given $L \dashv R$, the candidate weak equivalences in the target are detected by R, while the candidate cofibrations are built from $L(C)$ by pushouts and transfinite compositions. The central calculation is the lifting identity

$$(L(C))^{\square} = R^{-1}(C^{\square}).$$

To make this into a cofibration structure, the source requires R to preserve the relevant colimits and RL to preserve cofibrations. Preservation of trivial cofibrations is a distinct condition. This distinction explains the difficulty of identifying the transferred fibrations: they are initially defined by lifting against the transferred trivial cofibrations, whose description is the very point at issue.

The free-diagram adjunction is particularly favourable. If I_0 is the discrete category of objects of I, the left adjoint to restriction along $I_0 \to I$ sends a family (A_j) to

$$i \longmapsto \coprod_{j \in I} I(i,j) \cdot A_j.$$

For a map of families, every component of its free-diagram image is a coproduct of coefficient maps. Pushout stability and transfinite closure give the needed coproduct stability. The commentary supplies both the cardinal argument for accessibility of an entire diagram and the elementary transfinite construction of these coproducts.

Section 5 then isolates the remaining model-category problem. Put $TC = C \cap W$, $TF = C^{\square}$ and $F = (TC)^{\square}$. The established factorization gives $W = TF \circ TC$; a retract argument gives $TF = F \cap W$. To obtain a Quillen triple one needs the second factorization, into a trivial cofibration followed by a fibration. A small generating family for TC supplies it. With this second factorization, a pushout and retract argument also proves retract stability of W, justifying the later comparison with closed model categories. Under the corresponding preservation hypotheses on RL, Proposition 5.8 identifies transferred trivial cofibrations as the lifting closure of $L(TC)$ and transferred fibrations as $R^{-1}(F)$.

The distinction between a generating image, its cellular closure, its retract closure, and its double-orthogonal closure matters here. The equality suggested in footnote 23 cannot be true for the raw image of the free-diagram functor; a worked equivariant example shows why pushouts are indispensable. Proposition 5.1 also contains a displayed equality stronger than the argument preceding the additional hypotheses establishes. The audit records the precise inclusions proved and separates this unresolved strengthening from the later, correctly proved transfer theorem. We do not infer falsity merely from an incomplete justification.

The local structure of a projective cofibration

Theorem 6.1 constructs, for every small I, the model structure on $\mathcal{M}(I)$ whose weak equivalences and fibrations are objectwise. Thus every left Kan extension is a left Quillen functor and its derived functor is left adjoint to restriction. Its cofibrations are generally more restrictive than objectwise cofibrations. The rest of the chapter explains what their compatibility condition says.

For a map of arrows, the decisive map is the relative latching map

$$X_0 \amalg_{X_1} Y_1 \longrightarrow Y_0.$$

Along with the cofibration at the source of the arrow, it characterizes the projective lifting condition. Lemma 6.2.5 states this criterion but leaves its proof unwritten. The commentary proves both directions under exactly the stated pushout hypothesis. No terminal object or model-category factorization is needed for that lemma.

For a well-founded diagram, the same pushout occurs every time one extends an already constructed lift to a new object. Grothendieck uses Zorn's lemma to organize the extensions and writes out the final compatibility cube in detail. The mathematical content of that cube is that one lift from a relative latching object enforces both the pointwise lifting equation and naturality with all previous components. Corollary 6.3 correctly places the noetherian hypothesis on I in the presheaf convention. The contrary word in Theorem 6.1(d) is a genuine printed variance slip: the inverse sequence of nonempty tails of $\mathbb{N}$, with empty inverse limit, gives a counterexample to the literal artinian version. Chapter XIX, Proposition 3.2 and §3.4, subsequently uses the corrected noetherian formulation (Grothendieck, 1990–1991j).

The final proposition identifies restrictions that preserve projective cofibrations. When the connected components of each I/i' have terminal objects, right Kan extension is a product of evaluations. Right lifting classes are closed under products, and adjunction gives the preservation assertion. The commentary also develops a converse test with set-valued presheaves on a poset, making precise how much the terminal-component condition can detect.

What is settled, and what the commentary adds

The arbitrary-shape derivator construction is no longer an open problem for model categories with small limits and colimits. Cisinski proves the required homotopy Kan extension theorem without the special-generation restriction (Cisinski, 2003, Theorem 6.11 and its dual); Groth gives a projective–injective treatment in the combinatorial case (Groth, 2013, Proposition 1.30 and Theorem 1.32). For a combinatorial simplicial presentation, rectification identifies the localized diagram categories with $h\,\mathrm{Fun}(N(I^{\mathrm{op}}), \mathcal{C}_\infty)$ (Lurie, 2009, Proposition 4.2.4.4). These results concern coherent diagrams, retaining the compatibility which the latching construction makes explicit.

The developments marked [ours] are proved in the accompanying blocks and make no claim of priority. In particular, a special closed model structure on abelian groups detected by $\mathrm{Hom}(\mathbb{Z}[1/2], -)$ shows that weak equivalences need not be closed under all filtered colimits. The counterexample addresses the expectation expressed on manuscript p. 97 while respecting the chapter's generation and accessibility hypotheses. Together with the poset restriction criterion and the complete missing arrow-category proof, it turns the commentary into a mathematical continuation of the arguments at the points where a reader needs one.

Chapter XVIII

Catégories et ensembles accessibles

XVIII-a · Chapter XVIII, part 1 of 5 · manuscript pages 50–138

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XVIII

Catégories et ensembles accessibles

Ce texte a été déchiffré et transcrit en LaTeX 2_ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Seules quelques corrections mineures évidentes ont été effectuées. Pour les rares ajouts ou commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés. Quelques notes de bas de page ajoutées par les éditeurs sont signalées par « N. Éd. ». La numérotation des pages de ce chapitre commence à la page 50, car à l'origine ce chapitre était la suite du chapitre XIX qui se termine au milieu de la page 50.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

4 Paratopos et l'opération $\mathcal{M} \boxtimes \mathcal{N}$

p. 1
[page 50]

4.1 Pseudo-topos. Problème du produit.

J'appelle *catégorie pseudo-$\mathfrak{U}$-topos*, ou simplement *pseudo-topos*, une $\mathfrak{U}$-catégorie $\mathcal{M}$ stable par $\mathfrak{U}$-limites inductives, et qui admet une *petite* sous-catégorie $\mathcal{M}_0$ *strictement génératrice* (i.e. génératrice par épimorphismes stricts). Alors $\mathcal{M}$ admet aussi des $\mathfrak{U}$-limites projectives (SGA 4 I 8.12.7). (Mais l'exemple de (Gr), cf. loc. cit. 8.12.9, montre que $\mathcal{M}^\circ$ n'est par forcément, pour autant, un pseudo-$\mathfrak{U}$-topos, faute à $\mathcal{M}$ d'admettre une petite sous-catégorie strictement *co*génératrice. Mais le théorème 2.6 `[du chapitre XIX]` donne des conditions sur $\mathcal{M}$ (surtout du type exactitude) pour que $\mathcal{M}$ admette assez d'injectifs – ce qui impliquera alors [page 51] l'existence d'une petite sous-catégorie strictement cogénératrice, donc que $\mathcal{M}^\circ$ est également un pseudo-topos ([1]).

Commentary · Context [ms. 50]

Why the theory of coefficients leads to cardinal arithmetic

Grothendieck is looking for categories of coefficients in which constructions on diagrams can be made uniformly. The first sentence already contains the two ingredients that will govern the chapter: arbitrary small colimits, which make extensions possible, and a small family of objects, which makes those extensions controllable. His geometrical language is deliberate. Calling a category a category of sheaves encourages us to regard a colimit-preserving functor as an inverse image, even when no topology and no left exactness have been supplied. The subsequent tensor problem will test how much of this geometrical picture survives.

There is unusually direct historical evidence for the place of this digression. In the opening paragraph of the letter of 2 April 1991 to Thomason, Grothendieck describes work on accessible categories as a return to SGA 4, I, 9, needed for his broader homotopical project; the editor identifies Chapter XVIII explicitly (Grothendieck, 1991a, opening paragraph, printed p. 2). Thus the relevance of the present pages does not have to be inferred from a modern theory of derivators. The connection to factorization methods is also internal to the manuscript: Chapter XIX, Theorem 2.6, uses accessibility, cardinal filtrations, and exactness assumptions to construct the required lifting factorizations (Grothendieck, 1990–1991j, ms. pp. 25–26).

We keep the universe fixed. "Small" means small relative to $\mathfrak{U}$, and a $\mathfrak{U}$-category has $\mathfrak{U}$-small hom-sets. The whole category need not be small. A small full subcategory $S \subset \mathcal{M}$ is strictly generating in the sense used here when its canonical families of maps reconstruct objects and their morphisms; equivalently in this cocomplete setting, the restricted Yoneda functor

$$\rho_S : \mathcal{M} \longrightarrow [S^{\mathrm{op}}, \mathrm{Ens}], \qquad x \longmapsto \mathcal{M}(-, x)|_S$$

is fully faithful. Such a subcategory is called dense in current terminology. In particular,

$$x \simeq \operatorname*{colim}_{(s \to x) \in S/x} s.$$

The category S/x is small: its objects form a small union of hom-sets and its arrows a small union of subsets of hom-sets of S. This observation is where the size of the generating family first does mathematical work.

A pseudo-topos category here is consequently a cocomplete category with a small dense subcategory. It is not assumed accessible in the modern sense, and the adjective does not assert the exactness axioms of a topos. Later, the paratopos condition adds a uniform presentability bound on a small generating family; the resulting categories are the locally presentable categories. We establish that identification in the commentary to Section 4.3, with the cardinal convention made explicit. It should not be read back into the first definition.

For the diagram-theoretic motivation, the commentary on Chapter I, manuscript p. 4 ("From derived coefficients to diagram categories"), and on Chapter XII, manuscript p. 225 ("Smoothness and properness depend on a coefficient theory"), provides the appropriate continuity. Here we develop the additional size arguments, rather than repeating the construction of a derivator from a relative category.

1. C'est le cas notamment si $\mathcal{M}$ est un topos.

Je visualise les catégories pseudo-topos comme étant la catégorie des "faisceaux" sur un objet $\mathcal{X}$, visualisé comme une sorte d'"espace" que je distingue de la catégorie $\mathcal{M} = \mathrm{Fais}(\mathcal{X})$ des faisceaux sur $\mathcal{X}$. C'est $\mathcal{X}$ qui sera pour ceci le "pseudo-topos". Les pseudo-topos forment une 2-catégorie, la variance dans $\mathcal{X}$ étant opposée de celle dans $\mathrm{Fais}(\mathcal{X}) = \mathcal{M}$. De façon précise, si $\mathcal{X}$ correspond à $\mathcal{M}$, $\mathcal{Y}$ à $\mathcal{N}$, un morphisme f de $\mathcal{X}$ dans $\mathcal{Y}$ sera donné par un foncteur

$$f^* \ : \ \mathcal{N} \longrightarrow \mathcal{M} \tag{4.1.1}$$

en sens inverse, commutant aux $\varinjlim$ inductives quelconques ([2]). (**N.B.** si $\mathcal{X}$, $\mathcal{Y}$ sont même des topos, i.e. $\mathcal{M}$, $\mathcal{N}$ des catégories-topos, et si on veut que f soit un morphisme de topos, on exigera de plus que f^* soit de plus *exact à gauche*. On dira qu'un *morphisme de pseudo-topos est exact* si de plus f^* est exact à gauche.)

En vertu du dual de loc. cit. 8.12.7, cela équivaut au fait que f^* admette un adjoint à droite f_*, [page 52]

$$f_* \ : \ \mathcal{M} \longrightarrow \mathcal{N} \ , \tag{4.1.2}$$

lequel commute donc aux petites $\varprojlim$. La donnée de f_* satisfaisant à cette propriété détermine donc f^* à isomorphisme unique près. Mais en général, il ne suffit pas de se donner un tel f_*, pour pouvoir affirmer qu'il corresponde à un f^*, i.e. qu'il admette un adjoint à droite ; pour ceci, il faudrait qu'on sache que les foncteurs $\mathcal{M} \longrightarrow \mathrm{Ens}$ qui commutent aux petites $\varprojlim$ sont représentables, ce qui sera le cas si $\mathcal{M}$ admet une petite sous-catégorie *co*génératrice, i.e. si $\mathcal{M}^\circ$ est également une catégorie pseudo-topos.

2. Dire ici que les objets de $\mathcal{M}$ sont visualisés comme des "ouverts" généralisés de $\mathcal{X}$. On a $\mathcal{M} \xrightarrow{\sim} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathrm{Ens})$, i.e. $\mathcal{M}$ s'interprète bel et bien comme les "faisceaux" sur cette catégorie $\mathcal{M}$ d'ouverts …

Il est sans doute judicieux de procéder comme dans le cas des topos, et d'appeler *morphisme de pseudo-topos* $f : \mathcal{X} \longrightarrow \mathcal{Y}$ un quadruple p. 2

$$(f^*, f_*, \alpha, \beta) \tag{4.1.3}$$

de deux foncteurs adjoints (f^*, f_*), et des morphismes d'adjonction. Quand f^* est une équivalence, ce qui équivaut à f_* une équivalence, ou α et β des isomorphismes, on dira qu'on a une *équivalence de pseudo-topos*, et à toutes fins pratiques, on pourra alors regarder $\mathcal{X}$ et $\mathcal{Y}$ comme des objets essentiellement identiques. [page 53]

Commentary · Exposition [ms. 52]

Adjoints, representability, and the direction of a morphism

Grothendieck wants to recover the direct image from the inverse image, just as for a geometric morphism of topoi. The relevant existence theorem is the dual special adjoint functor theorem. In the present situation it says that a colimit-preserving functor out of a pseudo-topos category has a right adjoint. The representability statement used is SGA 4, I, Corollaries 8.12.7–8.12.8, applied to the opposite category (Artin et al., 1972).

Here is the application with its variance exposed. If $L : \mathcal{N} \to \mathcal{M}$ preserves small colimits and $m \in \mathcal{M}$, then

$$n \longmapsto \mathcal{M}(Ln, m) : \mathcal{N}^{\mathrm{op}} \longrightarrow \mathrm{Ens}$$

sends colimits in $\mathcal{N}$ to limits of sets. Representability provides an object Rm and bijections

$$\mathcal{M}(Ln, m) \simeq \mathcal{N}(n, Rm).$$

Yoneda makes these choices functorial in m and gives $L \dashv R$. Applying the same argument to the diagonal $\mathcal{M} \to \mathcal{M}^I$, which preserves colimits, supplies all small limits in $\mathcal{M}$. No assertion that $\mathcal{M}^{\mathrm{op}}$ is another pseudo-topos is needed for either step.

The converse problem starts with R, not with L. Preservation of limits alone does not give a left adjoint on an arbitrary complete category. The small cogenerating hypothesis on $\mathcal{M}$ is a sufficient condition; it permits the undualized representability theorem to be applied to $m \mapsto \mathcal{N}(n, Rm)$. The printed phrase "un adjoint à droite" at the end of manuscript p. 52 has the wrong direction: the missing adjoint to f_* is the left adjoint f^*. This is an edition slip, retained in the source and registered separately.

The asymmetry can already be seen in groups. There are arbitrarily large simple groups, for example the finitary alternating groups on sufficiently large infinite sets. Given a small family of groups (G_i), choose a simple group H whose cardinal exceeds every $|G_i|$. Every homomorphism $H \to G_i$ is trivial, since a nontrivial map from a simple group is injective. Such maps cannot separate the identity endomorphism of H from the trivial endomorphism. Thus no small family cogenerates groups, although finite free groups give a small generating supply. This independently explains the obstruction invoked in SGA 4, I, 8.12.9; it is not an objection to the existence of right adjoints to colimit-preserving functors out of groups.

Finally, an adjunction includes its unit and counit. The equivalence between descriptions by left and right adjoints reverses natural transformations: a transformation $L \Rightarrow L'$ corresponds to a transformation $R' \Rightarrow R$. This is the reason for the opposite category in (4.1.5). Invertible unit and counit characterize an adjoint equivalence. Requiring f^* also to preserve finite limits gives the usual extra condition for a geometric morphism when the coefficient categories are topoi; it is an additional requirement, not a consequence of having an adjoint.

Pour deux pseudo-topos $\mathcal{X}$, $\mathcal{Y}$, j'ai envie d'introduire (comme dans le cas des topos) un pseudo-topos produit

$$\mathcal{X} \times \mathcal{Y}\,, \tag{4.1.4}$$

qui ait les propriétés d'un 2-produit dans la 2-catégorie des pseudo-topos ([3]). Je devrais

3. Il apparaît finalement que ce qu'on construit n'est *pas* un 2-produit dans la 2-catégorie envisagée, donc il vaut mieux le noter par un signe tel que $\mathcal{X} \boxtimes \mathcal{Y}$. Mais si $\mathcal{X}$, $\mathcal{Y}$ sont deux topos, $\mathcal{X} \boxtimes \mathcal{Y}$ est un 2-produit dans la 2-catégorie des topos.

préciser que par définition

$$\begin{aligned}\underline{\mathrm{Hom}}_{\mathrm{pstop}}(\mathcal{X},\mathcal{Y}) &\simeq \underline{\mathrm{Hom}}_{!}(\mathcal{F}(\mathcal{Y}),\mathcal{F}(\mathcal{X}))^{\circ}\\ &\simeq \underline{\mathrm{Hom}}^{!'}(\mathcal{F}(\mathcal{X}),\mathcal{F}(\mathcal{Y})),\end{aligned} \tag{4.1.5}$$

où l'indice ! (resp. l'exposant !) indique la sous-catégorie pleine dans la catégorie $\underline{\mathrm{Hom}}$ envisagée formée des foncteurs qui commutent aux petites $\varinjlim$ (resp. aux petites $\varprojlim$), et l'accent dans le $\underline{\mathrm{Hom}}^{!'}$ dans le dernier membre indique qu'on exige de plus que le foncteur en question ait un adjoint à gauche. [page 54]

Soient donc

$$\mathcal{M} = \mathcal{F}(\mathcal{X}), \qquad \mathcal{N} = \mathcal{F}(\mathcal{Y}), \tag{4.1.6}$$

et soit $\mathcal{Z}$ un troisième pseudo-topos, décrit par

$$\mathcal{P} = \mathcal{F}(\mathcal{Z}). \tag{4.1.7}$$

On veut expliciter, à équivalence près, la catégorie produit $\underline{\mathrm{Hom}}_{\mathrm{pstop}}(\mathcal{Z},\mathcal{X}) \times \underline{\mathrm{Hom}}_{\mathrm{pstop}}(\mathcal{Z},\mathcal{Y})$, i.e.

$$\underline{\mathrm{Hom}}_{!}(\mathcal{M},\mathcal{P})^{\circ} \times \underline{\mathrm{Hom}}_{!}(\mathcal{N},\mathcal{Q})^{\circ}. \tag{4.1.8}$$

Je vais introduire pour cela une catégorie p. 3 [page 55]

$$\mathcal{M}\boxtimes\mathcal{N} \tag{4.1.9}$$

par la formule

$$\mathcal{M}\boxtimes\mathcal{N} \simeq \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^{\circ}\times\mathcal{N}^{\circ},\mathrm{Ens}), \tag{4.1.10}$$

où le signe $\underline{\mathrm{Hom}}^{!!}$ désigne la sous-catégorie pleine de la catégorie $\underline{\mathrm{Hom}}$, formée des bi-contrafoncteurs

$$F : \mathcal{M}^{\circ}\times\mathcal{N}^{\circ} \longrightarrow \mathrm{Ens}, \tag{4.1.10.1}$$

tels que pour tout x dans $\mathcal{M}$, le foncteur

$$y \longmapsto F(x,y) : \mathcal{N}^{\circ} \longrightarrow \mathrm{Ens} \tag{4.1.10.2}$$

soit dans $\underline{\mathrm{Hom}}^{!}(\mathcal{N}^{\circ},\mathrm{Ens})$, i.e. transforme petites $\varinjlim$ en $\varprojlim$, et que, symétriquement, pour tout y dans $\mathcal{N}$, le foncteur

$$x \longmapsto F(x,y) : \mathcal{M}^{\circ} \longrightarrow \mathrm{Ens} \tag{4.1.10.3}$$

soit dans $\underline{\mathrm{Hom}}^{!}(\mathcal{M}^{\circ},\mathrm{Ens})$. Mais cela signifie aussi que ces foncteurs sont *représentables*; donc on trouve aussi

$$\mathcal{M}\boxtimes\mathcal{N} \xleftarrow{\sim} \underline{\mathrm{Hom}}^{!}(\mathcal{N}^{\circ},\mathcal{M}) \tag{4.1.11}$$

et

$$\mathcal{M}\boxtimes\mathcal{N} \xleftarrow{\sim} \underline{\mathrm{Hom}}^{!}(\mathcal{M}^{\circ},\mathcal{N}) \tag{4.1.12}$$

en utilisant (dans chacune de ces formules) dans deux côtés les équivalences de catégories [page 56]

$$\begin{cases}\mathcal{M} & \xrightarrow{\sim} \underline{\mathrm{Hom}}^{!}(\mathcal{M}^{\circ},\mathrm{Ens})\\ \mathcal{N} & \xrightarrow{\sim} \underline{\mathrm{Hom}}^{!}(\mathcal{N}^{\circ},\mathrm{Ens}).\end{cases} \tag{4.1.13}$$

On peut dire aussi que la catégorie $\mathcal{M} \boxtimes \mathcal{N}$ est équivalente à celle des *couples* de foncteurs

$$\begin{cases} \varphi & : & \mathcal{M}^\circ & \longrightarrow & \mathcal{N} \\ \psi & : & \mathcal{N}^\circ & \longrightarrow & \mathcal{M}\,, \end{cases} \tag{4.1.14}$$

munis de deux morphismes d'adjonction

$$\begin{cases} \psi\varphi^\circ & \overset{\alpha}{\longleftarrow} & \mathrm{id}_{\mathcal{M}} \\ \varphi\psi^\circ & \overset{\beta}{\longleftarrow} & \mathrm{id}_{\mathcal{N}} \end{cases} \tag{4.1.15}$$

satisfaisant les deux compatibilités habituelles exprimant l'adjonction de φ° et de ψ (ou, de façon équivalente, de ψ° et de φ). Le foncteur

$$(\varphi, \psi) \longmapsto \varphi \qquad (\text{resp. } (\varphi, \psi) \longmapsto \psi)$$

donne une équivalence de la catégorie de ces couples avec $\underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathcal{N})$ (resp. avec $\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ, \mathcal{M})$), de sorte qu'on a un carré commutatif d'équivalences canoniques de foncteurs :

Commentary · Exposition [ms. 54]

The actual product and the proposed tensor

The attempted product is useful because it makes Grothendieck identify the universal property he really needs. Two inverse-image functors into a coefficient category and one functor preserving colimits separately in two variables carry different information. The former already have a simple representing category.

Let $\mathcal{M}, \mathcal{N}, \mathcal{P}$ be pseudo-topos categories, and denote by $\mathrm{Cocont}(\mathcal{A}, \mathcal{B})$ the category of colimit-preserving functors and natural transformations. There is an equivalence

$$\mathrm{Cocont}(\mathcal{M} \times \mathcal{N}, \mathcal{P}) \simeq \mathrm{Cocont}(\mathcal{M}, \mathcal{P}) \times \mathrm{Cocont}(\mathcal{N}, \mathcal{P}).$$

To prove it, write 0 for initial objects and use the two inclusions $m \mapsto (m, 0)$ and $n \mapsto (0, n)$. Every pair (m, n) is canonically the coproduct of $(m, 0)$ and $(0, n)$. A cocontinuous F therefore satisfies

$$F(m, n) \simeq F(m, 0) \amalg F(0, n).$$

Conversely, two cocontinuous functors G, H determine $F(m, n) = G(m) \amalg H(n)$; interchange of colimits shows that F is cocontinuous. A natural transformation between two such functors is uniquely the coproduct of its restrictions. These constructions are mutually inverse up to the indicated canonical isomorphisms. Moreover, $\mathcal{M} \times \mathcal{N}$ is a pseudo-topos category: colimits are coordinatewise, and the union of the two small strictly generating families $(s, 0)$ and $(0, t)$ is strictly generating.

After reversing the variance and the two-cells as in (4.1.5), this is precisely the bicategorical product of pseudo-topoi. The tensor category which the text is about to define solves a different universal problem, once its existence has been established:

$$\mathrm{Cocont}(\mathcal{M} \boxtimes \mathcal{N}, \mathcal{P}) \simeq \mathrm{Cocont}_2(\mathcal{M} \times \mathcal{N}, \mathcal{P}),$$

where the subscript 2 means preservation of colimits in each variable separately. We will use this formula only in the accessible setting or in the explicit examples where it is proved.

The difference is visible without large categories of coefficients. For sets, separate continuity of the contravariant bifunctor gives

$$F(X, Y) \simeq F(1, 1)^{X \times Y},$$

since X and Y are coproducts of singleton sets. Hence $\mathrm{Ens} \boxtimes \mathrm{Ens} \simeq \mathrm{Ens}$, whereas the category representing the first displayed universal property is $\mathrm{Ens} \times \mathrm{Ens}$. These categories are not equivalent: the terminal object of Ens has two subobjects, while the terminal object of $\mathrm{Ens} \times \mathrm{Ens}$ has four. The footnote on manuscript p. 53 correctly retracts the product interpretation. Within the smaller bicategory of topoi and geometric morphisms, imposing left exactness changes the problem and recovers the geometrical product; the two bicategories must not be interchanged. p. 4 [page 57]

Commentary · Exposition [ms. 56]

Birepresentability and contravariant adjunctions

Grothendieck's definition packages a candidate coefficient object by all of its pairings with objects in the two factors. This is why a bifunctor, rather than a pair of functors, is the natural starting point. Fix $F : \mathcal{M}^{\mathrm{op}} \times \mathcal{N}^{\mathrm{op}} \to \mathrm{Ens}$ preserving limits separately. Representability in the first variable gives objects $\psi(y) \in \mathcal{M}$ such that

$$F(x, y) \simeq \mathcal{M}(x, \psi(y)).$$

An arrow $y' \to y$ in $\mathcal{N}$ induces a transformation $F(-, y) \to F(-, y')$ and therefore an arrow $\psi(y) \to \psi(y')$. This determines a functor $\psi : \mathcal{N}^{\mathrm{op}} \to \mathcal{M}$. It preserves limits because each representable $\mathcal{M}(x, -)$ does, and the family of all representables detects a limiting cone. Representing in the other variable gives $\varphi : \mathcal{M}^{\mathrm{op}} \to \mathcal{N}$ and the natural bijection

$$\mathcal{N}(y, \varphi(x)) \simeq F(x, y) \simeq \mathcal{M}(x, \psi(y)).$$

Thus $\varphi^{\mathrm{op}} : \mathcal{M} \to \mathcal{N}^{\mathrm{op}}$ is left adjoint to ψ. The apparent two units in (4.1.15) are the unit and the opposite of the counit of this ordinary adjunction. Their triangular identities, not just the existence of two comparison arrows, are required.

The argument also works on morphisms. A transformation $F \to F'$ gives compatible transformations $\psi \to \psi'$ and $\varphi \to \varphi'$ by Yoneda. Conversely, either transformation determines the bifunctor transformation. This proves the equivalences (4.1.11)–(4.1.16), including their two-cell content. An object of the tensor is therefore a contravariant adjunction, not a cocontinuous functor from $\mathcal{M}$ to $\mathcal{N}$.

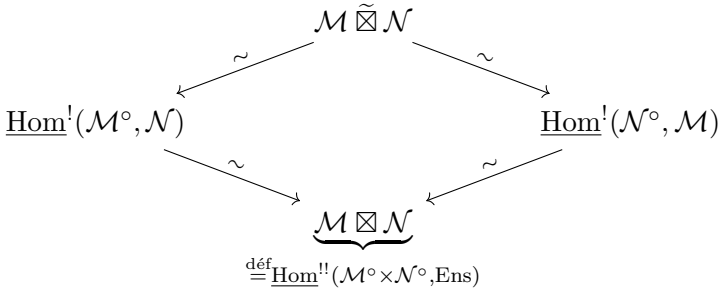
Small limits of such bifunctors are calculated pointwise. Indeed, if (F_i) is a diagram and (x_j) a colimit diagram in $\mathcal{M}$, then

$$\left(\lim_i F_i\right)(\mathrm{colim}_j x_j, y) \simeq \lim_i \lim_j F_i(x_j, y) \simeq \lim_j \lim_i F_i(x_j, y).$$

The interchange is the universal property of limits in sets, with no filteredness assumption. The same argument treats the other variable. This establishes completeness. It does not construct coproducts or a left adjoint to the forthcoming inclusion in a presheaf category. The sentence on p. 57 announcing that the tensor is a pseudo-topos is a proposed conclusion; the questions immediately following it and the later footnote to Lemma 4.2.3 show that it is not yet proved.

There is also a precise size benefit to restricting both variables to small dense subcategories. Once Corollary 4.2.2 is established, transformations between two candidate tensors form a subset of a product indexed by $\mathrm{Ob}(S \times T)$ of small sets. This proves local smallness of the candidate tensor before cocompleteness is known. The restriction theorem addresses the size of hom-sets; accessibility will address the size of constructions.

(4.1.16)



[où $\mathcal{M} \widetilde{\boxtimes} \mathcal{N}$ est la catégorie des couples définie dans la page précédente.]

Je dis que $\mathcal{M} \boxtimes \mathcal{N}$ *est une catégorie pseudo-topos*. Il est clair, [en] outre, qu'elle admet des petites $\varprojlim$, qui se calculent argument par argument dans $\underline{\mathrm{Hom}}(\mathcal{M}^\circ \times \mathcal{N}^\circ, \mathrm{Ens})$, ou au choix dans $\underline{\mathrm{Hom}}^!(\mathcal{M}^\circ, \mathcal{N})$, ou $\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ, \mathcal{M})$. Comment construire des petites $\varinjlim$? Il n'est pas question d'invoquer un argument de cogénérateurs. Il faut d'ailleurs trouver aussi une

petite sous-catégorie strictement génératrice. Visiblement, il y a quelque chose de non trivial à démontrer !

4.2 Cafouillages sur $\mathcal{M} \boxtimes \mathcal{N}$

Revenons à la situation générale d'un pseudo-topos $\mathcal{X}$, décrit par la catégorie pseudo-topos $\mathcal{M}$. Si $\mathcal{N}$ est une catégorie quelconque, on définit la catégorie [page 58]

$$\mathrm{Fais}(\mathcal{X},\mathcal{N}) \qquad \text{ou} \qquad \mathcal{F}(\mathcal{X},\mathcal{N})$$

des faisceaux sur $\mathcal{X}$, à valeurs dans $\mathcal{N}$, par

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \ . \tag{4.2.1}$$

On a donc un foncteur pleinement fidèle

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \hookrightarrow \mathrm{Hom}^!(\mathcal{N}^\circ,\overline{\mathcal{M}}) \ , \tag{4.2.2}$$

où $\overline{\mathcal{M}}$ désigne la sous-catégorie de $\mathcal{M}^\wedge$ (**N.B.** $\mathcal{M}^\wedge$ n'est pas en général une $\mathcal{U}$-catégorie qu'à cela ne tienne) formée des foncteurs *représentables* $\mathcal{M}^\circ \longrightarrow \mathrm{Ens}$ (laquelle est une $\mathfrak{U}$-catégorie, équivalente à $\mathcal{M}$ par l'inclusion canonique $\mathcal{M} \hookrightarrow \overline{\mathcal{M}}$). Donc on a à isomorphisme unique près une factorisation de (4.2.2) en

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{pl. fid.}}{\hookrightarrow} \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathcal{M}) \ , \tag{4.2.3}$$

l'image essentielle étant formée des foncteurs

$$\psi \ : \ \mathcal{N}^\circ \longrightarrow \mathcal{M}$$

tels que pour tout x dans $\mathcal{M}$,

$$y \longmapsto \mathrm{Hom}(y,\psi(x)) \ : \ \mathcal{N}^\circ \longrightarrow \mathrm{Ens}$$

soit représentable, i.e. des foncteurs qui admettent un *adjoint à gauche*. Quand $\mathcal{N}$ lui-même est une catégorie pseudo-topos, cette condition est automatique (elle résulte de l'hypothèse p. 5
que ψ commute aux petites $\varprojlim$). Donc dans ce cas, (4.2.3) est une équivalence de catégories. [page 59]

Soit maintenant S (S comme "site") une *petite* sous-catégorie de $\mathcal{M}$, génératrice par épimorphismes stricts. On sait que cela signifie que le foncteur composé

$$\mathcal{M} \overset{\sim}{\longrightarrow} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathrm{Ens}) \xrightarrow{\text{restriction à } S} S^\wedge \ (\overset{\text{déf}}{=} \underline{\mathrm{Hom}}(S^\circ,\mathrm{Ens})) \tag{4.2.4}$$

est pleinement fidèle, et ceci implique plus généralement, que pour toute catégorie $\mathcal{N}$, le foncteur restriction

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \overset{\text{pl. fid.}}{\hookrightarrow} \underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) \tag{4.2.5}$$

est pleinement fidèle. En effet, le deuxième membre, via le plongement pleinement fidèle

$$\mathcal{N} \overset{\text{pl. fid.}}{\hookrightarrow} \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens}) \hookrightarrow \mathcal{N}^\wedge \ ,$$

se plonge dans

$$\underline{\mathrm{Hom}}(S^\circ,\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens})) \simeq \mathrm{Hom}^!(\mathcal{N}^\circ,S^\wedge) \ ,$$

et la pleine fidélité de (4.2.5) équivaut donc à celle du foncteur composé

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,S^\wedge) \simeq \underline{\mathrm{Hom}}(S^\circ,\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens})) \ . \tag{4.2.6}$$ [page 60]

Or on a commutativité dans le carré

$$\begin{array}{ccc} \mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) & \xrightarrow{(4.2.5)} & \underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) \\ {\scriptstyle (4.2.3)}\big\downarrow \text{pleinement fidèle} & & \big\downarrow \text{pleinement fidèle} \\ \underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathcal{M}) & \xrightarrow[\text{pleinement fidèle}]{(4.2.4)} & \underbrace{\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,S^\wedge)}_{\simeq \underline{\mathrm{Hom}}(S^\circ,\underline{\mathrm{Hom}}^!(\mathcal{N}^\circ,\mathrm{Ens}))} \end{array} \tag{4.2.7}$$

(où les flèches verticales sont des équivalences, si $\mathcal{N}$ est lui-même une catégorie pseudo-topos). Notons d'autre part que l'on a une équivalence

$$\underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) \xleftarrow{\sim} \underline{\mathrm{Hom}}^!(S^{\wedge\circ},\mathcal{N}) \overset{\text{déf}}{=} \mathcal{F}(\mathrm{Top}(S),\mathcal{N}) , \tag{4.2.8}$$

où $\mathrm{Top}(S)$ est le pseudo-topos (en fait, un topos) défini par la catégorie pseudo-topos $S^\wedge$. Pour ceci, notons qu'on a

$$\underline{\mathrm{Hom}}^!(S^{\wedge\circ},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_!(S^\wedge,\mathcal{N}^\circ)^\circ ,$$

et d'autre part le foncteur restriction à $S \hookrightarrow S^\wedge$ donne

$$\underline{\mathrm{Hom}}_!(S^\wedge,\mathcal{N}^\circ) \xrightarrow{\sim} \underline{\mathrm{Hom}}(S,\mathcal{N}^\circ) ,$$

d'où

$$\underline{\mathrm{Hom}}(S^{\wedge\circ},\mathcal{N}) \xrightarrow{\sim} (\underline{\mathrm{Hom}}(S,\mathcal{N}^\circ))^\circ \xrightarrow{\sim} \underline{\mathrm{Hom}}(S^\circ,\mathcal{N}) ,$$

p. 6
[page 61]

ce qui n'est autre que (4.2.8). Ainsi :

Commentary · Exposition [ms. 59]

Density reconstructs a continuous functor

The move to a small site is the first effective reduction in the tensor problem. Grothendieck needs to remember a limit-preserving functor on a small collection of test objects, and density supplies exactly that reconstruction. It is helpful to prove full faithfulness directly, so that no completeness of an arbitrary target is silently inserted.

Let $S \subset \mathcal{M}$ be small and dense, and let $F, G : \mathcal{M}^{\mathrm{op}} \to \mathcal{N}$ preserve all small limits of $\mathcal{M}^{\mathrm{op}}$. This formulation includes the requirement that the relevant image cones have limits in $\mathcal{N}$. For each x, density gives $x = \mathrm{colim}_{S/x} s$, hence

$$F(x) \simeq \lim_{(S/x)^{\mathrm{op}}} F(s), \qquad G(x) \simeq \lim_{(S/x)^{\mathrm{op}}} G(s).$$

Given a transformation $\eta : F|_{S^{\mathrm{op}}} \to G|_{S^{\mathrm{op}}}$, compose the projections from $F(x)$ with the maps η_s. They form a cone to the diagram defining $G(x)$, and therefore give a unique map $\eta_x : F(x) \to G(x)$. If $u : x \to x'$ is an arrow in $\mathcal{M}$, both composites from $F(x')$ to $G(x)$ have the same projections to every $G(s)$ indexed by $s \to x$; they are equal. Thus η_x is natural in x. Uniqueness follows from the same limiting projections. This proves (4.2.5) for any $\mathcal{N}$ in which the functors under consideration exist.

The symmetry construction has an equally transparent formula. Given F, represent in $\mathcal{M}$ the functor

$$x \longmapsto \mathcal{N}(y, Fx),$$

for each $y \in \mathcal{N}$. The resulting $\psi : \mathcal{N}^{\mathrm{op}} \to \mathcal{M}$ lies in the image exactly when, for every x, the presheaf

$$y \longmapsto \mathcal{M}(x, \psi(y))$$

is representable in $\mathcal{N}$. This is the well-typed formula intended on manuscript p. 58. The printed $\mathrm{Hom}(y, \psi(x))$ interchanges the variables and cannot be evaluated with the stated types. In terms of adjunctions, ψ must have a left adjoint $\mathcal{M} \to \mathcal{N}^{\mathrm{op}}$.

Now take both coefficient categories to be pseudo-topos categories. Restrict first along $S \subset \mathcal{M}$ and then along $T \subset \mathcal{N}$. The argument just given proves

$$\mathcal{M} \boxtimes \mathcal{N} \longhookrightarrow [(S \times T)^{\mathrm{op}}, \mathrm{Ens}]$$

fully faithful. It also explains why the construction respects natural transformations, an issue that cannot be settled merely by asserting that every object is a colimit of generators. The functors being reconstructed must carry those presentations to the appropriate limits. This qualification will reappear in the proof of Theorem 4.4.16.

Proposition 4.2.1. *Soient $\mathcal{M} = \mathcal{F}(\mathcal{X})$ une catégorie pseudo-topos, munie d'une sous-catégorie pleine strictement génératrice S, $\mathcal{N}$ une $\mathfrak{U}$-catégorie quelconque, alors le foncteur "restriction à S" suivant est pleinement fidèle.*

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^{!}(\mathcal{M}^{\circ},\mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}(S^{\circ},\mathcal{N}) \xleftarrow{\sim} \underbrace{\underline{\mathrm{Hom}}(S^{\wedge\circ},\mathcal{N})}_{\overset{\text{déf}}{=}\mathcal{F}(\mathrm{Top}(S),\mathcal{N})} , \tag{4.2.1.1}$$

et il s'insère dans le carré commutatif (4.2.7) de foncteurs pleinement fidèles.

Corollaire 4.2.2. *Supposons que $\mathcal{N} = \mathcal{F}(\mathcal{Y})$ soit également une catégorie pseudo-topos, munie d'une sous-catégorie pleine T strictement génératrice. Alors*

$$\mathcal{F}(\mathcal{X},\mathcal{N}) \simeq \mathcal{F}(\mathcal{Y},\mathcal{M}) \simeq \underbrace{\underline{\mathrm{Hom}}^{!!}(\mathcal{M}^{\circ} \times \mathcal{N}^{\circ}, \mathrm{Ens})}_{=\mathcal{M}\boxtimes\mathcal{N}} ,$$

et le foncteur restriction

$$\begin{aligned} \mathcal{M} \boxtimes \mathcal{N} \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^{\circ} \times \mathcal{N}^{\circ}, \mathrm{Ens}) &\longrightarrow \underline{\mathrm{Hom}}(S^{\circ} \times T^{\circ}, \mathrm{Ens}) \\ &\simeq (S \times T)^{\wedge} \\ &\overset{\text{déf}}{=} \mathcal{F}(\mathrm{Top}(S \times T)) \end{aligned} \tag{4.2.2.1}$$

est pleinement fidèle.

La première assertion a été vue dans 4.1. Pour la pleine fidélité de (4.2.2.1), on note que ce foncteur s'interprète, à équivalence de catégories près, comme le composé du foncteur pleinement fidèle (4.2.1.1), et du foncteur similaire relatif à $\mathcal{N}$ (donc lui aussi pleinement fidèle)

[page 62]

$$\mathcal{F}(\mathcal{Y}, S^{\wedge}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^{!}(\mathcal{N}^{\wedge}, S^{\wedge}) \xrightarrow{\text{restr. à } T} \underbrace{\underline{\mathrm{Hom}}(T^{\circ}, S^{\wedge})}_{\underline{\mathrm{Hom}}(S^{\circ}\times T^{\circ},\mathrm{Ens})} , \tag{4.2.2.2}$$

compte tenu de l'équivalence de symétrie

$$\mathcal{F}(\mathcal{Y}, S^{\wedge}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}^{!}(\mathcal{N}^{\circ}, S^{\wedge}) \simeq \underline{\mathrm{Hom}}^{!}(S^{\wedge\circ}, \mathcal{N}) . \tag{4.2.2.3}$$

Commentary · Examples [ms. 61]

The missing completeness hypothesis in the presheaf equivalence

The restriction theorem is sound, but the displayed equivalence (4.2.8) asks for more: every functor $S^{\mathrm{op}} \to \mathcal{N}$ is supposed to extend continuously to $\widehat{S}^{\mathrm{op}}$. For an arbitrary $\mathcal{N}$ this is false. The free-cocompletion argument in the text is being applied with target $\mathcal{N}^{\mathrm{op}}$, which needs the colimits used by that argument.

Take S to be the category with one object and only its identity, so that $\widehat{S} = \mathrm{Ens}$, and take $\mathcal{N} = \mathrm{FinSet}$. A functor $S^{\mathrm{op}} \to \mathcal{N}$ can select the two-element set 2. If it extended to a continuous $F : \mathrm{Ens}^{\mathrm{op}} \to \mathrm{FinSet}$, the countable coproduct decomposition $\mathbb{N} = \coprod_{n\in\mathbb{N}} 1$ would force

$$F(\mathbb{N}) \simeq \prod_{n\in\mathbb{N}} 2.$$

No finite set represents this product in FinSet: testing the product universal property on 1 would give a bijection from a finite set to the infinite set $2^{\mathbb{N}}$. Thus restriction is not essentially surjective. This counterexample concerns the claimed equivalence, not the already proved full faithfulness of restriction.

Here is the sufficient hypothesis and its proof. Suppose $\mathcal{N}$ has all small limits and let $A : S^{\mathrm{op}} \to \mathcal{N}$. For a presheaf P, define

$$\overline{A}(P) = \lim_{(S/P)^{\mathrm{op}}} A(s).$$

The category of elements S/P is small. For $n \in \mathcal{N}$, the universal property gives

$$\mathcal{N}(n, \overline{A}(P)) \simeq \mathrm{Nat}\big(P, \mathcal{N}(n, A(-))\big).$$

The right-hand side carries colimits in P to limits of sets. Since representables detect limits, $\overline{A}$ carries colimits of presheaves to limits in $\mathcal{N}$. For $P = h_s$, Yoneda identifies $\overline{A}(h_s)$ with $A(s)$. The density argument of the preceding block gives uniqueness on objects and on transformations. Therefore

$$[S^{\mathrm{op}}, \mathcal{N}] \simeq \mathrm{Cont}(\widehat{S}^{\mathrm{op}}, \mathcal{N})$$

under the completeness hypothesis.

There are also two missing superscripts in the printed formulas on p. 61: the initial equivalence and the final term of (4.2.1.1) use unrestricted $\underline{\mathrm{Hom}}(\widehat{S}^{\mathrm{op}}, \mathcal{N})$ where the continuous functors $\underline{\mathrm{Hom}}^!$ are intended. Adding these superscripts repairs the notation but does not repair the missing hypothesis on $\mathcal{N}$. Corollary 4.2.2 takes $\mathcal{N}$ to be a pseudo-topos category, which is complete, so the corollary survives intact.

Au total, quand $\mathcal{M}$, $\mathcal{N}$ sont tous deux des catégories pseudo-topos, munies de sous-catégories strictement génératrices S resp. T, on a un plongement pleinement fidèle canonique

$$\mathcal{M} \boxtimes \mathcal{N} \xhookrightarrow{\ \rho\ } (S \times T)^{\wedge} , \tag{4.2.9}$$

qu'on interprète par la suite comme $(f_S^{\mathcal{M}} \times f_T^{\mathcal{N}})_*$, où

$$\begin{cases} f_S^{\mathcal{M}} \ : \ \mathcal{X} \longrightarrow \mathrm{Top}S \\ f_T^{\mathcal{N}} \ : \ \mathcal{Y} \longrightarrow \mathrm{Top}T \end{cases} \tag{4.2.10}$$

p. 7

sont les morphismes de pseudo-topos, donnés par les plongements pleinement fidèles (foncteurs image directe)

$$\begin{cases} (f_S^{\mathcal{M}})_* \ : \ \mathcal{M} \longrightarrow S^{\wedge} \\ (f_T^{\mathcal{N}})_* \ : \ \mathcal{N} \longrightarrow T^{\wedge} , \end{cases} \tag{4.2.11}$$

lesquels admettent bel et bien des adjoints à gauche

$$\begin{cases} (f_S^{\mathcal{M}})^* \ : \ S^{\wedge} \longrightarrow \mathcal{M} \\ (f_T^{\mathcal{N}})^* \ : \ T^{\wedge} \longrightarrow \mathcal{N} , \end{cases} \tag{4.2.12}$$

[page 63]

à savoir les prolongements canoniques (par commutativité aux petites $\varinjlim$) des foncteurs d'inclusion $S \hookrightarrow \mathcal{M}$, $T \hookrightarrow \mathcal{N}$.

La question essentielle, à présent, c'est visiblement de construire un adjoint à gauche pour le foncteur (4.2.9). Il revient au même, pour (s,t) dans $S \times T$, de définir un objet

$$s \boxtimes t \ \in \ \mathrm{Ob}\, \mathcal{M} \boxtimes \mathcal{N} , \tag{4.2.13}$$

tel que l'on ait, pour F variable dans $\mathcal{M} \boxtimes \mathcal{N}$, un isomorphisme fonctoriel en F,

$$\underbrace{\rho(F)(s,t)}_{\overset{\text{déf}}{=} F(s,t)} \ \simeq \ \mathrm{Hom}(s \boxtimes t, F) .$$

Donc il faut prouver ceci :

Lemme 4.2.3 ([4]). *Soient $\mathcal{M}$, $\mathcal{N}$ deux catégories pseudo-topos, et $s \in \mathrm{Ob}\,\mathcal{M}$, $t \in \mathrm{Ob}\,\mathcal{N}$,*

4. Pas prouvé ; peut-être faux !

et considérons sur $\mathcal{M}\boxtimes\mathcal{N} \overset{déf}{=} \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^\circ\times\mathcal{N}^\circ, \mathrm{Ens})$ *le foncteur* $F \longmapsto F(s,t)$*. Ce foncteur est* représentable *(par un objet que je note* $s\boxtimes t$*).*

L'idée naturelle serait de définir $s\boxtimes t$ par

$$(s\boxtimes t)(x,y) \;=\; s(x)\times t(y) \;=\; \mathrm{Hom}_{\mathcal{M}}(x,s)\times\mathrm{Hom}_{\mathcal{N}}(y,t)\ ,$$

ce qui définit bien un foncteur [page 64]

$$\mathcal{M}^\circ\times\mathcal{N}^\circ \longrightarrow \mathrm{Ens}\ .$$

Malheureusement, pour un des arguments fixé, disons pour y fixé, ce foncteur n'est *pas* compatible aux limites projectives, p.ex. il ne transforme pas l'objet final de $\mathcal{M}^\circ$ correspondant à l'objet initial de $\mathcal{M}$) en objet final de Ens, mais en $t(y)=\mathrm{Hom}_{\mathcal{N}}(y,t)$. Néanmoins, je pense que $s\boxtimes t$ doit bel et bien se définir comme produit de deux objets de $\mathcal{M}\boxtimes\mathcal{N}$, dépendant fonctoriellement de s resp. de t ([5]). On est donc amené dès à présent, qu'on le veuille ou non, à essayer de définir des foncteurs canoniques, commutant aux petites $\varinjlim$, p. 8

$$\mathcal{M}\longrightarrow\mathcal{M}\boxtimes\mathcal{N}\longleftarrow\mathcal{N} \tag{4.2.13}$$

(lesquels seront les foncteurs image inverse pour les projections canoniques $\mathcal{X}\times\mathcal{Y}\to\mathcal{X}$, $\mathcal{X}\times\mathcal{Y}\to\mathcal{Y}$)

([6]).

Commentary · Exposition [ms. 63]

The reflector is the actual existence problem

At this point the tensor has a faithful realization as a category of presheaves. Grothendieck wants to build colimits in it by first building ordinary presheaf colimits and then reflecting them. The individual coefficient categories already illustrate this procedure without any accessibility assumption.

For the restricted nerve $\rho_S:\mathcal{M}\to\widehat{S}$, define

$$L_S(P)=\operatorname*{colim}_{(s,a)\in S/P} s.$$

For each $x\in\mathcal{M}$, a map from this colimit to x is a compatible family of maps $s\to x$, one for each element $a\in P(s)$. Equivalently, it is a natural transformation $P\to\rho_S x$. Thus $L_S\dashv\rho_S$. Density says that the counit $L_S\rho_S x\to x$ is invertible. This constructs the morphisms (4.2.10)–(4.2.12) and shows exactly where cocompleteness of $\mathcal{M}$ enters.

For $\rho:\mathcal{E}=\mathcal{M}\boxtimes\mathcal{N}\hookrightarrow\widehat{S\times T}$ the same formula would require the colimits one is trying to prove exist. If a reflector L exists, then

$$s\boxtimes t=L(h_{(s,t)}),\qquad \mathcal{E}(s\boxtimes t,F)\simeq F(s,t),$$

and arbitrary colimits in $\mathcal{E}$ are L applied to the corresponding colimits of nerves. This is a consequence of the adjunction, not a prior recipe for constructing L.

The restriction to generator pairs in Lemma 4.2.3 is therefore natural. Their evaluation functors are the coordinate functions of the embedding. If $\mathcal{E}$ were already cocomplete and these evaluations representable, their representing objects could be extended from $S\times T$ to all presheaves by the displayed colimit construction. At the present stage cocompleteness is not available. The wording "Pas prouvé; peut-être faux!" should be retained as a genuine record of the state of the argument. The accessible case will have a positive answer; the general case needs a separate existence theorem and is not settled by the formal Yoneda calculations.

5. également faux, cf. plus bas.

6. Le cas où $\mathcal{M}$, $\mathcal{N}$ sont des catégories-topos, nous montre que ces foncteurs devraient être décrits comme $x\longmapsto x\boxtimes e_{\mathcal{N}}$, $y\longmapsto e_{\mathcal{M}}\boxtimes y$. Mais il n'est pas clair qu'on ait en dehors du cas des topos, et même en admettant l'existence du $x\boxtimes y$, la formule (valable pour un topos) $x\boxtimes y\simeq(x\boxtimes e_{\mathcal{N}})\times(e_{\mathcal{M}}\boxtimes y)$.

4.3 L'opération $\mathcal{M} \boxtimes \mathcal{N}$ et l'axiome d'accessibilité

J'ai passé la journée hier, essayant d'établir l'existence des $x \boxtimes y$ dans $\mathcal{M} \boxtimes \mathcal{N}$, sans y parvenir – du moins pas sans faire une hypothèse supplémentaire d'accessibilité, dont il devient évident à présent qu'il faut la faire de toutes façons. Je vais quand même énoncer le positif de mes perplexités. [page 65]

Commentary · Development (commentators') [ms. 64]

Why the naive formula fails, and the abelian calculation

Grothendieck has reached a useful obstruction: the external product of two representable presheaves need not be a separately continuous bifunctor. We can sharpen the obstruction to an exact criterion and then calculate an example in which the proposed product decomposition is impossible. These are elementary consequences of the definitions, developed here to locate the failed geometrical intuition. We checked the tensor constructions in Kelly (1982, Section 6.5) and the algebraic examples in Freyd (1966, p. 95); the latter already contains the tensor calculus for module theories. We make no priority claim for the following formulations or proofs.

The external-product criterion. [ours] Let $\mathcal{M}, \mathcal{N}$ be pseudo-topos categories. The bifunctor

$$H_{s,t}(x, y) = \mathcal{M}(x, s) \times \mathcal{N}(y, t)$$

belongs to $\mathcal{M} \boxtimes \mathcal{N}$ if and only if s and t are terminal in their respective categories.

Proof. If $H_{s,t}$ preserves the empty limit in the first variable, evaluate at the initial object $0_{\mathcal{M}}$. For every y we obtain

$$1 \simeq H_{s,t}(0_{\mathcal{M}}, y) \simeq \mathcal{N}(y, t).$$

This says exactly that t is terminal. Interchanging the variables proves that s is terminal. Conversely, if both are terminal, $H_{s,t}$ is the constant singleton bifunctor and preserves every limit separately. This proves both directions, including the degenerate case of a one-object coefficient category. □

For the tensor itself, there is a particularly instructive classical calculation:

$$\mathrm{Ab} \boxtimes \mathrm{Ab} \simeq \mathrm{Ab}, \qquad A \boxtimes B \longleftrightarrow A \otimes_{\mathbb{Z}} B.$$

We give the full argument because it identifies the universal elements, rather than only the ambient category. Under (4.1.11), an object is a continuous $\psi : \mathrm{Ab}^{\mathrm{op}} \to \mathrm{Ab}$. Composing with the forgetful functor to sets gives a continuous contravariant set-valued functor, hence $U\psi(B) \simeq \mathrm{Ab}(B, C)$ for some C. The group structure on these sets is natural in B. By Yoneda, its multiplication is a morphism $\mu : C \times C \to C$ in Ab, with an internal unit. That unit is necessarily zero; since μ is a group homomorphism and has zero as its unit,

$$\mu(c, d) = \mu(c, 0) + \mu(0, d) = c + d.$$

The inverse is consequently the usual negative. Thus the internal abelian group structure is forced, and

$$\psi(B) = \mathrm{Ab}(B, C)$$

with its ordinary pointwise group structure. Natural transformations between two such functors are exactly homomorphisms between their representing groups; they automatically preserve these structures. Under the tensor description, the associated bifunctor is

$$F_C(A, B) = \mathrm{Ab}\big(A, \mathrm{Ab}(B, C)\big) \simeq \mathrm{Ab}(A \otimes_{\mathbb{Z}} B, C).$$

The last bijection follows directly by sending a homomorphism $A \to \mathrm{Ab}(B, C)$ to its bilinear evaluation map $(a, b) \mapsto f(a)(b)$, and then using the defining generators and bilinear relations of $A \otimes_{\mathbb{Z}} B$. It proves that evaluation at (A, B) is represented by $A \otimes_{\mathbb{Z}} B$.

Now the terminal object of Ab is 0. The proposed formula involving terminal coefficients would give

$$(A \boxtimes 0) \times (0 \boxtimes B) = 0,$$

whereas $\mathbb{Z} \boxtimes \mathbb{Z}$ corresponds to $\mathbb{Z}$. This is a complete counterexample to the proposed decomposition outside topoi. The failure has no size-theoretic cause: abelian groups are locally finitely presentable and their tensor exists. What fails is the identification of a bilinear operation with a cartesian product operation.

Commentary · Later developments [ms. 64]

The modern tensor comparison and its precise scope

The construction becomes a well-behaved tensor on locally presentable categories. Its universal property involves functors preserving colimits separately; its objects can be realized by the separately continuous bifunctors used in (4.1.10). Kelly's construction of tensors for small categories with prescribed colimits supplies an earlier categorical framework for this operation (Kelly, 1982, Section 6.5, equation (6.24)). Thus this part of the comparison predates the manuscript. The accessible presentation developed below connects the small presentation with the large category of coefficients.

The ordinary topos assertion also has a precise predecessor: Pitts (1985, Theorem 2.3, pp. 50–52) proves that the product of Grothendieck topoi is their tensor as cocomplete categories, in a more general indexed setting. The construction uses the topology on a product of sites generated by covers in each variable. Specializing the base to sets gives the assertion in Grothendieck's footnote on p. 53 and the external-product formula in the footnote on p. 64. Pitts's Definition 2.2 explicitly separates this theorem from an unrestricted existence claim for arbitrary locally small cocomplete categories.

The higher analogue is especially close in its formulas. For presentable ∞-categories, Lurie (2017, Proposition 4.8.1.15, Lemma 4.8.1.16, and Proposition 4.8.1.17) constructs the tensor and identifies it with the category of limit-preserving functors $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Examples 4.8.1.19–4.8.1.20 identify its restriction to ∞-topoi with their product and identify spaces as the tensor unit. These are structural analogues of the present formulas, with mapping spaces in place of hom-sets.

There is no implication here that replacing sets by spaces converts every assertion in these pages into a correct higher assertion. Ordinary filtered colimits, strict categories of elements, and cardinal estimates on actual hom-sets all have to be replaced by their appropriate higher versions. Conversely, the derivator formalism by itself does not establish an accessibility bound on an arbitrary coefficient category. The present chapter is supplying such bounds before the homotopical constructions are made. The unconditional pseudo-topos tensor problem and the positive accessible tensor theorem must therefore remain separate entries in the audit.

Commentary · Exposition [ms. 65]

The representability problem behind the tensor

Grothendieck has found the right category of bilinear coefficients; what is missing is the ability to make objects in it. A separately continuous bifunctor is easy to recognize, and limits of such bifunctors remain separately continuous. Colimits are different: their pointwise values generally destroy the continuity conditions. The proposed object $x \boxtimes y$ would solve this problem at its smallest visible scale, by representing the operation which reads a coefficient at (x, y).

Write $\mathcal{E} = \mathcal{M} \boxtimes \mathcal{N}$ and $P = (S \times T)^{\wedge}$, and let $\alpha : \mathcal{E} \hookrightarrow P$ be restriction. If a reflection $L \dashv \alpha$ exists, then

$$\mathcal{E}\big(Lh_{(s,t)}, F\big) \cong P(h_{(s,t)}, \alpha F) \cong F(s,t).$$

Thus $s \boxtimes t = Lh_{(s,t)}$. The same adjunction supplies every small colimit:

$$\operatorname*{\overset{\mathcal{E}}{colim}}_{i \in I} F_i \cong L\left(\operatorname*{\overset{P}{colim}}_{i \in I} \alpha F_i\right).$$

The images of representables generate $\mathcal{E}$ densely, because every presheaf is the colimit of its representables and L preserves that colimit. These are the sound implications $c \Rightarrow b'$ and $c \Rightarrow a$ in Proposition 4.3.1. No left exactness of L is needed.

There is a genuine difficulty in the reverse implication as written. Lemma 4.3.2 assumes that $\mathcal{E}$ has small *limits*, although its proof constructs a *colimit* of representing objects. The printed variance $\mathcal{E}^{\circ} \to$ Ens is also reversed: $\xi \mapsto \alpha(\xi)(x)$ is covariant. These readings occur in the edition, not only in the transcription. The valid elementary lemma says that if $\mathcal{E}$ is cocomplete and each evaluation is corepresented by $r(x)$, then

$$LF = \operatorname*{colim}_{(x,u) \in X/F} r(x) \quad \text{satisfies} \quad \mathcal{E}(LF, e) \cong P(F, \alpha e).$$

Indeed, applying $\mathcal{E}(-, e)$ turns this colimit into the limit of the evaluations $\alpha(e)(x)$ over the category of elements of F, which is precisely the set of natural transformations $F \to \alpha(e)$. The representing

objects form a functor $r : X \to \mathcal{E}$ by Yoneda, so the displayed colimit has the required coherent transition maps.

Completeness alone cannot replace cocompleteness in that elementary lemma. Let $\mathcal{E}$ be the opposite of the ordered category of ordinals belonging to the fixed universe. It has every small limit, since a small family of ordinals has a supremum, but it has no initial object. Take $X = \{*\}$ and $\alpha(e) = \mathcal{E}(0, e)$. Its only evaluation is corepresentable. A left adjoint would nevertheless have to send the empty set to an initial object of $\mathcal{E}$, which does not exist. This example concerns the lemma's generality; it is not a counterexample to the tensor construction, where α has additional properties. The written argument for $b' \Rightarrow c$ therefore needs an existence theorem beyond this lemma. The accessibility hypothesis introduced next provides one, as we prove below.

Proposition-perplexité 4.3.1. *Soient $\mathcal{M}$, $\mathcal{N}$ deux catégories pseudo-topos, d'où $\mathcal{M} \boxtimes \mathcal{N}$ comme dans 4.1. Soient $S \subset \mathcal{M}$, $T \subset \mathcal{N}$ deux petites sous-catégories strictement génératrices, et considérons l'inclusion pleinement fidèle correspondante :*

$$\underbrace{\mathcal{M} \boxtimes \mathcal{N}}_{\overset{\text{déf}}{=} \underline{\mathrm{Hom}}^{!!}(\mathcal{M}^\circ \times \mathcal{N}^\circ, \mathrm{Ens})} \overset{\varphi\,?}{\hookleftarrow} \xrightarrow[\alpha]{} S^\wedge \boxtimes T^\wedge \;\simeq\; (S \times T)^\wedge . \tag{4.3.1.1}$$

Considérons les conditions suivantes :

a) $\mathcal{M} \boxtimes \mathcal{N}$ est un pseudo-topos.

b) Pour $x \in \mathrm{Ob}\,\mathcal{M}$, $y \in \mathrm{Ob}\,\mathcal{N}$, $x \boxtimes y \in \mathrm{Ob}\,\mathcal{M} \boxtimes \mathcal{N}$ existe, i.e. le foncteur (commutant aux petites $\varprojlim$)

$$F \longmapsto F(x, y)\ , \qquad \mathcal{M} \boxtimes \mathcal{N} \longrightarrow \mathrm{Ens} \tag{4.3.1.2}$$

est représentable.

b′) Comme b), avec $x \in \mathrm{Ob}\,S$, $y \in \mathrm{Ob}\,T$.

c) Le foncteur α (lequel commute aux petites $\varprojlim$) admet un adjoint à gauche

$$\varphi \;:\; (S \times T)^\wedge \longrightarrow \mathcal{M} \boxtimes \mathcal{N}\ .$$

On a alors les implications

$$b \implies b' \iff c \implies a\ . \tag{4.3.1.3}$$

De plus, si c est satisfait, on a ceci : pour $s \in S$, $t \in T$, considérons p. 9

$$(s, t) \;\in\; (S \times T) \;\subset\; (S \times T)^\wedge \;\simeq\; S^\wedge \boxtimes T^\wedge\ ,$$

qui s'explicite aussi comme

$$\begin{cases} (s,t) \;=\; \mathrm{pr}_1^*(s) \times \mathrm{pr}_2^*(t) \;\simeq\; s \overset{S^\wedge, T^\wedge}{\boxtimes} t \in \mathrm{Ob}\ (S \times T)^\wedge\ , \\ \textit{i.e. } (s,t)(s',t') \;=\; s(s') \times t(t') \;=\; \mathrm{Hom}_S(s', s) \times \mathrm{Hom}_T(t', t) \\ \qquad\qquad\qquad\qquad\quad \;=\; \mathrm{Hom}_{S \times T}((s', t'), (s, t))\ , \end{cases} \tag{4.3.1.4}$$

$$\begin{array}{ccc} S \times T & \xrightarrow{\mathrm{pr}_2} & T \\ \downarrow{\scriptstyle \mathrm{pr}_1} & & \\ S\ , & & \end{array}$$

on a [page 66]

$$s \overset{\mathcal{M},\mathcal{N}}{\boxtimes} t \simeq \varphi(s,t)\ , \tag{4.3.1.4}$$

et $\varphi(S\times T)=\{s\boxtimes t \mid s\in S,\ t\in T\}$ est une partie strictement génératrice de $\mathcal{M}\boxtimes\mathcal{N}$ ([7]).
N.B. Je ne connais pas d'exemple où les conditions envisagées ne soient satisfaites, mais doute pourtant à présent que même la plus faible le soit toujours, i.e. que $\mathcal{M}\boxtimes\mathcal{N}$ soit un pseudo-topos dès que $\mathcal{M}$ et $\mathcal{N}$ le sont.
Démonstration de 4.3.1. L'implication b $\Longrightarrow$ b′ est tautologique. L'implication b′ $\Longleftrightarrow$ c résulte du
Lemme 4.3.2. *Soient $\mathcal{E}$, X des $\mathfrak{U}$-catégories avec X petite, $\mathcal{E}$ stable par petites $\varinjlim$,*

$$\alpha\ :\ \mathcal{E}\longrightarrow X^\wedge$$

un foncteur. Pour que α admette un adjoint à gauche φ, il faut et il suffit que pour tout $x\in \mathrm{Ob}\,X$, le foncteur

$$\begin{cases} \xi & \longmapsto & \mathrm{Hom}_{X^\wedge}(x,\alpha(\xi)) \ =\ \alpha(\xi)(x) \\ \mathcal{E}^\circ & \longrightarrow & \mathrm{Ens} \end{cases}$$

soit représentable.
C'est évidemment nécessaire. La suffisance résulte du fait que $X\subset X^\wedge$ est strictement génératrice dans $X^\wedge$, i.e. que tout objet F de $X^\wedge$ est limite inductive $F=\varinjlim_I F_i$, les F_i dans X (on prend $I=X/F$), d'où p. 10 [page 67]

$$\begin{aligned}\mathrm{Hom}(F,\alpha(\xi)) &\simeq \varprojlim_I \underbrace{\mathrm{Hom}(F_i,\alpha(\xi))}_{\mathrm{Hom}(\rho(F_i),\xi)} \\ &\simeq \mathrm{Hom}(\varinjlim_I(\rho(F_i),\xi))\ ,\end{aligned}$$

donc le foncteur $\xi\longmapsto\mathrm{Hom}(F,\alpha(\xi))$ est représenté par $\rho(F)\overset{\text{déf}}{=}\varinjlim_I \rho(F_i)$.
Prouvons c $\Longrightarrow$ a. Pour l'existence des petites $\varinjlim$ dans $\mathcal{M}\boxtimes\mathcal{N}$, et l'existence d'une partie strictement génératrice, on invoque le lemme général :
Lemme 4.3.3. *Soit $\alpha:\mathcal{E}\hookrightarrow\mathcal{F}$ un foncteur pleinement fidèle admettant un adjoint à gauche ρ.*

ⓐ *Si les $\varinjlim$ d'un type donné I existent dans $\mathcal{F}$, elles existent dans $\mathcal{E}$, et se calculent comme*

$$\varinjlim_I{}_{\mathcal{E}}\,\xi_i \ =\ \rho(\varinjlim_I{}_{\mathcal{F}}\,\alpha(\xi_i))\ . \tag{4.3.3.1}$$

ⓑ *Si U est une petite partie strictement génératrice de $\mathcal{F}$, $\rho(U)$ est une petite partie strictement génératrice dans $\mathcal{F}$.*

En effet, si ξ désigne le deuxième membre [de l'égalité 4.3.3.1], et η un objet variable de $\mathcal{E}$, on aura

$$\begin{aligned}\mathrm{Hom}_{\mathcal{E}}(\xi,\eta) &\overset{\text{adj.}}{\simeq} \mathrm{Hom}_{\mathcal{F}}(\varinjlim_I \alpha(\xi_i),\alpha(\eta)) \\ &\simeq \varprojlim_I \underbrace{\mathrm{Hom}_{\mathcal{F}}(\alpha(\xi_i),\alpha(\eta))}_{\simeq \mathrm{Hom}_{\mathcal{E}}(\xi_i,\eta)\ \text{ car } \alpha \text{ pleinement fidèle}}\ . \\ &\simeq \varprojlim_I \mathrm{Hom}_{\mathcal{E}}(\xi_i,\eta)\ ,\end{aligned}$$

[page 68]

7. **N.B.** On a bien sûr $(s,t)=s\overset{S^\wedge,T^\wedge}{\boxtimes}t$.

ce qui prouve que ξ est une limite inductive des ξ_i dans $\mathcal{E}$ ([8]).
Ceci prouve que (moyennant c)) $\mathcal{M} \boxtimes \mathcal{N}$ est stable par petites $\varinjlim$, donc il reste à prouver que $\mathcal{M} \boxtimes \mathcal{N}$ admet une petite famille strictement génératrice. Cela résultera du b) dans le lemme, puisque $S \times T$ engendre strictement $(S \times T)^\wedge$. Cela prouve en même temps la dernière assertion de la proposition, sous réserve de prouver la formule (4.1.3.4). Il faut donc prouver un isomorphisme fonctoriel en $F \in \mathrm{Ob}\, \mathcal{M} \boxtimes \mathcal{N}$,

$$\mathrm{Hom}(\rho(s,t), F) \;\simeq\; F(s,t)\ ,$$

mais par adjonction on a

$$\mathrm{Hom}(\rho(s,t), F) \;\simeq\; \mathrm{Hom}((s,t), \alpha(F)) \;=\; \alpha(F)(s,t) \;=\; F(s,t)\ ,$$

OK.

Proposition 4.3.4. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie. Les conditions suivantes sont équivalentes.* p. 11 [page 69]

a) $\mathcal{M}$ est une catégorie $\mathfrak{U}$-pseudo-topos, i.e. elle est stable par $\mathfrak{U}$-limites inductives, et admet une $\mathfrak{U}$-petite sous-catégorie S strictement génératrice.

b) Il existe une $\mathfrak{U}$-petite catégorie S, et un foncteur $i : \mathcal{M} \hookrightarrow S^\wedge$ pleinement fidèle, *ayant un adjoint à gauche.*

Démonstration. On a a) $\Longrightarrow$ b), car il suffit de prendre pour S dans b) celui spécifié dans a), l'hypothèse sur S implique que le foncteur canonique

$$\mathcal{M} \;\longrightarrow\; S^\wedge\ , \qquad \text{composé de } \mathcal{M} \hookrightarrow\; \mathcal{M}^\wedge \;\xrightarrow{\text{restriction}}\; S^\wedge$$

est pleinement fidèle. Qu'il admette un adjoint à gauche résulte du lemme 4.3.2. L'implication b) $\Longrightarrow$ a) résulte du lemme 4.3.3.

Commentary · Exposition [ms. 69]

A reflective presentation and its generators

Grothendieck now moves from the particular tensor problem to the presentation of its possible factors. The useful feature of a pseudo-topos is that its objects can be reconstructed from a fixed small category of probes. This is the precise content of the restricted Yoneda embedding; it is stronger than asking the probes merely to distinguish parallel arrows.

For a small full strictly generating subcategory $S \subset \mathcal{M}$, put

$$N(x)(s) = \mathcal{M}(s,x), \qquad L(F) = \operatorname*{colim}_{(s,u)\in S/F} s.$$

The colimit exists because $\mathcal{M}$ is cocomplete. The universal property of a colimit and Yoneda give $\mathcal{M}(LF, x) \cong \mathrm{Nat}(F, Nx)$. Strict generation says that the canonical diagram $S/x \to \mathcal{M}$ has colimit x; equivalently, the counit $LNx \to x$ is an isomorphism. Hence N is fully faithful. This proves Proposition 4.3.4 directly, without the incorrect completeness hypothesis of Lemma 4.3.2. The density statement is the one formulated in Artin et al. (1972, Exposé I, Proposition 7.2(i)).

Conversely, suppose $L : P(S) \rightleftarrows \mathcal{M} : N$ is a reflection. If $D : I \to \mathcal{M}$ is a small diagram, then $L\,\mathrm{colim}_I ND$ represents its cocones, by adjunction and full faithfulness. Thus $\mathcal{M}$ is cocomplete. For $x \in \mathcal{M}$, apply L to the category-of-elements presentation of Nx:

$$x \cong LNx \cong \operatorname*{colim}_{(s,u)\in S/Nx} Lh_s.$$

This presentation is natural in x and exhibits the family Lh_s as dense. The target of this family is $\mathcal{M}$; correspondingly, the last $\mathcal{F}$ in the printed Lemma 4.3.3(b) must be read mathematically as $\mathcal{E}$. The source wording is retained in the text and recorded in the corrections register.

8. Pour prouver b), on note que la pleine fidélité de α implique que ρ est essentiellement surjectif ; comme ρ commute aux petites $\varinjlim$ et que tout objet de $\mathcal{F}$ est de la forme $\varprojlim_I x_i$, les x_i dans U, on aura $\rho(x) = \varinjlim_I \rho(x_i)$, les $\rho(x_i)$ dans $\rho(U)$, on gagne.

The adjunction also explains the accessibility assertion in Corollary 4.3.5. Suppose all Lh_s preserve sufficiently filtered colimits under their Hom functors. Since S is small, choose one regular cardinal κ which works for every s. Colimits in a presheaf category are pointwise, so

$$N \operatorname{colim}_I x_i \cong \operatorname{colim}_I Nx_i \quad \text{for every } \kappa\text{-filtered } I$$

is equivalent to the same assertion after evaluating at each s, hence to κ-presentability of all Lh_s. Every x is a small colimit of these objects. Choose a larger regular cardinal λ than κ and than the number of arrows in one such presentation of x. In sets, λ-filtered colimits commute with limits of fewer than λ arrows. Applying this to the Hom formula for that presentation proves that x is λ-presentable. Thus each object has a rank, although the ranks of all objects need not share one bound.

Corollaire 4.3.5. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie. Pour que $\mathcal{M}$ soit un $\mathfrak{U}$-pseudo-topos* accessible *(i.e. dont tous les objets sont accessibles), il faut et il suffit qu'il existe S dans* Cat *et un foncteur pleinement fidèle et* accessible $\mathcal{M} \overset{\alpha}{\hookrightarrow} S^\wedge$*, admettant un adjoint à gauche ρ.*

Démonstration. Il suffit de noter que si S, α sont comme dans 4.3.4 b), alors α est accessible si et seulement si les $\rho(S)$ sont formées d'objets accessibles dans $\mathcal{M}$; comme cette partie est strictement génératrice, on sait que cela implique que *tout* objet de $\mathcal{M}$ est accessible. (Cf. SGA 4 I 9.9.) [page 70]

Commentary · Exposition [ms. 70]

From accessible probes to an ind-presentation

Theorem 4.3.6 seeks an intrinsic description of the categories for which the tensor construction should work. In contemporary terminology its main conditions describe a locally presentable category. The indexing convention needs care: throughout these pages, "grand devant π" permits simultaneous solutions to diagrams with at most π arrows. With π infinite, this is modern κ-filteredness for $\kappa = \pi^+$. An object called π-accessible here is therefore π^+-presentable, not generally π-presentable. The implication from a cocomplete category with a small strong generator of presentable objects to local presentability is also stated in Adámek and Rosický (1994, Theorem 1.20). For the dense generator used here, a direct proof clarifies the role of the ranks.

Assume $\mathcal{M}$ is cocomplete and has a small full dense subcategory S of accessible objects. Choose an infinite regular κ such that every object of S is κ-presentable. Enlarge S to a small full subcategory C closed, up to isomorphism, under colimits of diagrams with fewer than κ arrows. Here is a construction which also proves the required smallness. Start with S; at each successor stage choose colimits of every such diagram in the preceding stage, and take the full subcategory on the resulting set of objects. At a limit stage take the union of the preceding object sets. Carry out κ stages. All sets used remain in the universe. By regularity, a diagram with fewer than κ arrows in the union appears at some earlier stage, so its chosen colimit also belongs to the union.

All objects of C remain κ-presentable. For a diagram $D : K \to \mathcal{M}$ of that size, the formula

$$\mathcal{M}(\operatorname{colim}_K D, -) \cong \lim_{K^{\mathrm{op}}} \mathcal{M}(D(-), -)$$

and the corresponding interchange theorem in sets prove closure of presentable objects under these colimits. At every stage this applies to the objects just constructed.

For any $x \in \mathcal{M}$, the category C/x is κ-filtered. Given a diagram of fewer than κ arrows in C/x, take the colimit of its underlying diagram in C. Its structure maps to x induce a map from this colimit to x, giving a cocone in C/x. This includes the empty diagram, because C contains an initial object. Moreover

$$x \cong \operatorname*{colim}_{(c,u)\in C/x} c.$$

To see density after enlarging S, a cocone from C/x to y restricts to one from S/x, hence to a unique map $x \to y$. For each $c \to x$, its agreement with the given map $c \to y$ can be tested after every map $s \to c$, and density of S gives that agreement. Thus the restriction has neither lost nor added cocones.

The functor which takes a κ-filtered diagram of objects of C to its colimit in $\mathcal{M}$ is now essentially surjective. It is fully faithful because, for diagrams $(c_i)_{i\in I}$ and $(d_j)_{j\in J}$, presentability gives

$$\mathcal{M}(\operatorname{colim}_I c_i, \operatorname{colim}_J d_j) \cong \lim_{i\in I^{\mathrm{op}}} \operatorname*{colim}_{j\in J} \mathcal{M}(c_i, d_j),$$

the defining Hom formula for ind-objects. Consequently $\mathcal{M} \simeq \mathrm{Ind}_\kappa C$, with the subscript here in the modern convention. When $\kappa = \pi^+$ this is Grothendieck's $\mathrm{Ind}_\pi C$.

This argument proves the structural part of the theorem without asserting that the particular slice-cardinality filtration chosen on page 72 already satisfies all the axioms of a cardinal filtration. That stronger numerical statement is exactly where the subsequent attempted proof encounters difficulty. The converse structural implication uses the cocompleteness of $\mathrm{Ind}_\kappa C$ when C has the indicated small colimits; Section 4.4 supplies it. No inaccessible cardinal beyond the standing universe is required for this passage.

Commentary · Exposition [ms. 70]

Why accessibility supplies the tensor reflector

The accessible presentation gives a concrete answer to the existence problem which opened Section 4.3. It replaces a class of continuity requirements by a set of requirements on a small category. We give the construction, because that change of size is the mathematical reason the added axiom works.

Take a common regular cardinal κ and small full subcategories $C \subset \mathcal{M}$, $D \subset \mathcal{N}$ of κ-presentable objects, closed under colimits of fewer than κ arrows, with $\mathcal{M} \simeq \mathrm{Ind}_\kappa C$ and $\mathcal{N} \simeq \mathrm{Ind}_\kappa D$. Write $P = (C \times D)^\wedge$. In P consider the set Σ of maps

$$\operatorname*{colim}_{i \in K}{}^{P} h_{(c_i,d)} \longrightarrow h_{(\operatorname{colim}_K^C c_i,d)}$$

for all diagrams in C of fewer than κ arrows and all $d \in D$, together with the analogous maps in the second variable. There is a set of such maps, since the categories are small and one may choose a set of representatives of all indexing shapes of the stated size. A presheaf F is orthogonal to Σ exactly when it takes those colimits to limits separately: orthogonality means that precomposition with each displayed map induces a bijection into F.

The orthogonal subcategory $\Sigma^\perp \subset P$ is reflective. To see the size control explicitly, choose a regular μ such that all domains and codomains in Σ are μ-presentable presheaves. This is possible by writing each as a small colimit of representables and taking one bound for the set Σ. Starting with $X_0 \in P$, perform the following two operations at every successor stage. First push out the coproduct of all maps $s : A \to B$ in Σ along all maps $A \to X_\alpha$, thus adjoining every required extension. Then coequalize every pair of maps $B \rightrightarrows Y$ into that new object Y which agree on A, for every $s \in \Sigma$. At limit stages take the colimit. Continue for μ stages.

For an orthogonal object Z, each operation induces a bijection from maps out of the new stage into Z to maps out of the preceding stage into Z. The pushout operation does so because extensions into Z exist uniquely; the coequalizer operation does so because those extensions agree whenever they agree on A. The same holds at limits. Thus $X_0 \to X_\mu$ is already universal for maps into orthogonal objects. The endpoint is itself orthogonal: a map $A \to X_\mu$ factors through one stage by presentability and is extended at the next; two maps $B \rightrightarrows X_\mu$ agreeing on A factor through a common stage, their equality on A holds at a later stage, and the coequalizer step then makes them equal. The ordinal μ is μ-filtered, which justifies each factorization. Hence this construction is the reflector $L : P \to \Sigma^\perp$.

To identify this category with Grothendieck's tensor, first recall a useful description of the restricted ind-completion. A presheaf H on a small κ-cocomplete category C belongs to $\mathrm{Ind}_\kappa C$ if and only if it carries the indicated colimits in C to limits of sets. In one direction, representables have that property, and κ-filtered colimits commute with those limits. In the other, a diagram of fewer than κ arrows in C/H has a cocone: take its underlying colimit in C and use the compatible elements to obtain the element of H at the colimit. The canonical category-of-elements presentation then proves the assertion.

Each factor is complete, and its reflective presheaf embedding computes limits. Indeed, objects in a full reflective subcategory are precisely those orthogonal to every reflection unit. A limit of such objects remains orthogonal to every unit, since Hom into a limit is a limit; orthogonality to its own unit then makes that unit invertible. Thus the presheaf limit belongs to the reflective subcategory.

It follows that a separately continuous $F \in \Sigma^\perp$ gives a functor $D^{\mathrm{op}} \to \mathcal{M}$ preserving limits of fewer than κ arrows. The dual functor $D \to \mathcal{M}^{\mathrm{op}}$ therefore preserves the corresponding colimits.

By the ind-completion universal property proved in Section 4.4, it extends uniquely to a colimit-preserving functor $\mathcal{N} \to \mathcal{M}^{\mathrm{op}}$. Dualizing and applying the Hom pairing recovers a separately continuous bifunctor on $\mathcal{M}^{\mathrm{op}} \times \mathcal{N}^{\mathrm{op}}$. Conversely, restriction of such a bifunctor lies in $\Sigma^{\perp}$, and its values on general objects are determined by their ind-presentations. These constructions are inverse, including natural transformations.

Thus $\mathcal{M} \boxtimes \mathcal{N} \simeq \Sigma^{\perp}$ is a reflective subcategory of P. It is cocomplete, with colimits obtained by reflection, and has the small dense family $Lh_{(c,d)}$. Its inclusion preserves μ-filtered colimits, since orthogonality to maps with μ-presentable endpoints is preserved by those colimits. Accordingly the reflected representables are μ-presentable, and the tensor is again locally presentable. Finally every evaluation $F \mapsto F(x, y)$ has a representing object: it preserves all limits and is accessible, or, equivalently, it is obtained from the small evaluation objects using the cocontinuity in x and y and the now-established colimits. For the latter explicit construction, choose presentations $x = \mathrm{colim}_{C/x}\, c$, $y = \mathrm{colim}_{D/y}\, d$ and take

$$x \boxtimes y = \operatorname*{colim}_{(c \to x, d \to y) \in (C/x) \times (D/y)} Lh_{(c,d)}.$$

Applying $\mathrm{Hom}(-, F)$ verifies the representing property directly. This proves all the existence assertions of Proposition 4.3.1 under the accessibility hypothesis, without using the disputed general implication $b' \Rightarrow c$.

Je vais essayer de donner un énoncé plus complet :

Théorème 4.3.6. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie. Conditions équivalentes :*

a) $\mathcal{M}$ stable par petites $\varinjlim$, et elle admet une petite sous-catégorie pleine S, strictement génératrice, et formée d'objets accessibles de $\mathcal{M}$.

a′) $\mathcal{M}$ est une catégorie $\mathfrak{U}$-pseudo-topos faiblement accessible, i.e. satisfait a), avec en plus tout *élément de $\mathcal{M}$ (pas seulement ceux de S) accessible dans $\mathcal{M}$.*

b) Il existe une petite catégorie S, et un foncteur $\alpha : \mathcal{M} \hookrightarrow S^{\wedge}$ pleinement fidèle et accessible, *tel que α admette un adjoint à gauche.*

c) $\mathcal{M}$ est stable par petites $\varinjlim$, est accessible, et admet une filtration cardinale [page 71] $(\mathrm{Filt}^{\pi}\, \mathcal{M})_{\pi \geqslant \pi_0}$.

d) Il existe un cardinal π_1, une petite catégorie $\mathcal{M}_{\pi_1}$, stable par limites inductives de cardinal $\leqslant \pi_1$, tels qu'on ait une équivalence de catégories

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_{\pi_1}\, \mathcal{M}_{\pi_1}\ , \tag{4.3.6.1}$$

où le deuxième membre désigne la sous-catégorie pleine de $\mathrm{Ind}(\mathcal{M}_{\pi_1})$ *formée des ind-objets $(X_i)_{i \in I}$ de $\mathcal{M}_{\pi_1}$ tels que I soit grand devant π_1.*

Démonstration de 4.3.6 ([9]). On a déjà vu a′ $\Longrightarrow$ a $\Longrightarrow$ b $\Longrightarrow$ a′, i.e. l'équivalence de a, a′, b, soit PTA cette condition. Nous allons prouver les implications

$$\mathrm{PTA} \implies \mathrm{c} \implies \mathrm{d} \implies \mathrm{PTA}\ ,$$

ce qui établira le théorème.

Prouvons PTA $\Longrightarrow$ c. Soit S une petite sous-catégorie pleine strictement génératrice. Ainsi, p. 12 pour tout x dans $\mathcal{M}$ on a [page 72]

$$x \;=\; \varinjlim_{s \in S/x} s \tag{4.3.6.2}$$

(cf. SGA 4 I 7.2 (i)). Soit $\pi_0 = \mathrm{card}\, \mathrm{Fl}(S)$, et pour tout cardinal $\pi \geqslant \pi_0$, soit

$$\mathrm{Filt}^{\pi}\, \mathcal{M} \;=\; \text{sous-catégorie pleine de } \mathcal{M} \text{ formée des objets } x \text{ tels que } \mathrm{card}\,(\mathrm{Fl}(S/x)) \leqslant \pi\ . \tag{4.3.6.3}$$

9. Pour une démonstration complète (celle donnée ici canule à partir d'un certain moment), cf. §4.6 (p. 139-148).

Je veux prouver que l'on trouve ainsi une filtration cardinale sur $\mathcal{M}$, i.e. que les conditions a) b) c) de SGA 4 I 9.12 sont satisfaites. C'est prouvé dans loc. cit. 9.13 sous des hypothèses légèrement différentes, la condition d'accessibilité de $\mathcal{M}$ étant remplacée par celle (beaucoup moins sympathique!) que les familles de flèches épimorphiques strictes dans $\mathcal{M}$ sont épimorphiques strictes universelles. Cette condition n'est utilisée que dans la démonstration de la condition b), à savoir la stabilité de $\mathrm{Filt}^{\pi}\mathcal{M}$ par $\varinjlim_{I}$ de cardinal $\leqslant \pi$ (sans avoir d'ailleurs à supposer I filtrante, ni a fortiori grand devant π_0). Il faut d'ailleurs dire que la définition de $\mathrm{Filt}^{\pi}\mathcal{M}$ donnée dans loc. cit. n'est pas tout à fait la même que (4.3.6.3), mais c'est [page 73]

(4.3.6.3′) $\mathrm{Filt}'^{\pi}\mathcal{M}$ = sous-catégorie pleine de $\mathcal{M}$ formée des objets x tels qu'il existe une famille épimorphique stricte $x_i \xrightarrow{\alpha_i} x$ de cardinal $\leqslant \pi$, avec $x_i \in S \ \forall\ i \in I$.

Ainsi on a

$$\mathrm{Filt}^{\pi}\mathcal{M} \ \subset\ \mathrm{Filt}'^{\pi}\mathcal{M}\ ,$$

mais dans loc. cit. on prouve que pour x dans $\mathrm{Filt}'^{\pi}\mathcal{M}$, on a

$$\operatorname{card}\mathrm{Fl}(S/x)\ \leqslant\ \pi^{\pi_0}\ ,$$

d'où

$$\mathrm{Filt}^{\pi}\mathcal{M}\ \subset\ \mathrm{Filt}'^{\pi}\mathcal{M}\ \subset\ \mathrm{Filt}^{\pi^{\pi_0}}\mathcal{M}\ ,$$

donc si $\pi = \pi^{\pi_0}$, p.ex. $\pi = 2^c$ avec $c \geqslant \pi_0$, alors on trouve

$$\mathrm{Filt}^{\pi}\mathcal{M}\ =\ \mathrm{Filt}'^{\pi}\mathcal{M}\ .$$

Donc les deux filtrations on l'air ± équivalentes. Mais la démonstration donnée de la stabilité de Filt'^{π} semble vraiment faire appel de façon essentielle à la condition d'universalité des familles épimorphiques strictes (donc épimorphiques effectives, avec les hypothèses faites impliquant l'existence de petites $\varprojlim$). Donc il me faut bon gré, mal gré, refaire le travail!

Condition a) : Filt^{π} équivalente à une petite catégorie. C'est immédiat, puisque par (4.3.6.2), Filt^{π} est formée de $\varprojlim_{I}$ dans $\mathcal{M}$ d'objets de S (petite catégorie), avec I telle que $\operatorname{card}\mathrm{Fl}\,I \leqslant \pi$. p. 13 [page 74]

Condition b) : Filt^{π} stable par $\varprojlim_{I}$, quand $\operatorname{card}\mathrm{Fl}\,I \leqslant \pi$ ([10]). Soit donc $(x_i)_{i\in I}$, où

$$\begin{array}{ccc} I & \longrightarrow & \mathcal{M} \\ i & \longmapsto & x_i \end{array}$$

un système inductif dans $\mathcal{M}$, avec des x_i dans Filt^{π}. Considérons le produit fibré

(4.3.6.4)
$$\begin{array}{ccccc} I & \xleftarrow{\text{cofibré}} & S/I = J & & \\ {\scriptstyle\varphi}\downarrow & & \downarrow & \searrow^{\psi} & \\ \mathcal{M} & \xleftarrow[\text{Cat-cofibré}]{\text{but}} & S/\mathcal{M} & \xrightarrow{\text{source}} & S \hookrightarrow \mathcal{M}\ , \end{array}$$

où J est Cat-cofibré sur I, sa fibre en $i \in \mathrm{Ob}\,I$ étant isomorphe à S/x_i, donc $\operatorname{card}\mathrm{Fl}\,J_i \leqslant \pi$. Comme par hypothèse $\operatorname{card}\mathrm{Fl}\,I \leqslant \pi$, on en conclut que $\operatorname{card}\mathrm{Fl}\,J \leqslant \pi^3 = \pi$. (**N.B.** L'ensemble des flèches $v : j_0 \longrightarrow j_1$ de J au dessus d'une flèche $u : i_0 \longrightarrow i_1$ de I est "contenu" dans l'ensemble produit $\mathrm{Ob}\,J_{i_0} \times \mathrm{Fl}\,J_{i_1}$, par l'application

$$v \longmapsto (\operatorname{source} v, v' : u_*(j_0) \longrightarrow j_1)\ ,$$

donc de cardinal majoré par π^2.) On a alors

10. quand I filtrante et grande devant π_0 – je ne le prouve que dans ce cas, hélas! [page 75]

$$x \overset{\text{déf}}{=} \varinjlim_{I} x_i = \varinjlim_{J} \underbrace{\xi_j}_{=\psi(j)} \qquad ([11]),$$

où cette fois la limite est une limite d'objets de S, prise dans $\mathcal{M}$. Il s'ensuit que $x \in \mathrm{Filt}'^{\pi}$, ce n'est pas tout à fait $x \in \mathrm{Filt}^{\pi}$. (Ça implique $x \in \mathrm{Filt}^{\pi^{\pi_0}}$, donc $x \in \mathrm{Filt}^{\pi}$ si $\pi = \pi^{\pi_0}$, p.ex. $\pi = 2^c$ avec $c \geqslant \pi_0$.) Mais c'est le moment d'utiliser le fait que les objets de S sont accessibles, et on prendra π_0 tel que les objets de S soient π_0-accessibles, et cette fois on prendra quand même I filtrante et grande devant π_0 (pour nous borner à ce qui est exigé dans la condition b) de loc. cit.). On doit prouver que $\mathrm{card}\,(\mathrm{Fl}(S/x)) \leqslant \pi$ (ce qui signifie $x \in \mathrm{Filt}^{\pi}$). Mais comme les objets de S sont π_0-accessibles et I grande devant π_0, on voit aussitôt que

$$S/x \simeq \varinjlim_{i} S/x_i ,$$

d'où p. 14

$$\mathrm{Fl}\ S/x \simeq \varinjlim_{i} \mathrm{Fl}\ S/x_i ,$$

donc

$$\mathrm{card}\,\mathrm{Fl}(S/x) \leqslant \underbrace{(\mathrm{card}\, I)}_{\leqslant \pi}\ \underbrace{(\sup_i \mathrm{card}\,\mathrm{Fl}(S/x_i))}_{\leqslant \pi} \leqslant \pi^2 = \pi ,$$

q.e.d.

Prouvons c) ([12]), i.e. que pour tout $\pi' > \pi \geqslant \pi_0$ et tout $x \in \mathrm{Ob}\ \mathrm{Filt}^{\pi'}$, on peut écrire [page 76]

(4.3.6.5) $\qquad x = \varinjlim_{I} x_i \qquad I$ grande devant π, les x_i dans Filt^{π}, $\mathrm{card}\, I \leqslant \pi'^{\pi}$.

La démonstration se fait comme dans loc. cit. (p. 151-152), en changeant légèrement la définition de I, comme désignant l'ensemble des sous-catégories pleines i de S/x qui sont filtrantes, de cardinal $\leqslant \pi$ *et de plus grandes devant* π_0 ([13]). (On pourra supposer S/x filtrante en augmentant S de façon à ce que S soit stable dans $\mathcal{M}$ par petites $\varinjlim$ finies, donc les catégories S/x aussi, ce qui implique qu'elles sont filtrantes. Il est `[phrase incomplète]`)

On a donc prouvé PTA $\Longrightarrow$ c).

Commentary · Exposition [ms. 76]

Why the first filtration argument stops

Grothendieck wants a presentation which controls not only the existence of small pieces but also their number. The two proposed measurements record different information. The condition $|\mathrm{Fl}(S/x)| \leq \pi$ counts *all* maps from probes and their relations. The primed condition asks for one strictly epimorphic family from at most π probes. The first implies the second, but a family which covers x need not furnish lifts of every map from a probe into x.

This distinction matters already on page 73. The estimate $|\mathrm{Fl}(S/x)| \leq \pi^{\pi_0}$ cited from SGA 4 is proved there for a universally epimorphic, pullbackable covering family; see Artin et al. (1972, Exposé I, Proposition 7.6 and Proposition 9.13). That hypothesis allows a map into x to be pulled back to a covering family and then encoded by maps among the probes. Accessibility alone is not the hypothesis of that estimate. Its use in the present argument is therefore not justified by the

11. On utilise la propriété de distributivité des $\varinjlim$, relative à un foncteur Cat-cofibrant

$$\begin{array}{ccc} J & \longrightarrow & I \\ & \searrow & \\ & & \mathcal{M} \end{array} ,$$

explicitée dessus.

12. *Démonstration canulée*, ne marche que si $\pi^{\pi_0} = \pi$!!

13. Il n'est pas sûr que I soit grande devant π, sauf si on suppose $\pi^{\pi_0} = \pi$!

cited result. This identifies a missing inference, rather than a counterexample to every possible version of the estimate.

Several parts of the argument do work. There are only a small set of presheaves on S whose categories of elements have at most π arrows, up to isomorphism, so the unprimed layer is essentially small. If I is κ-filtered, the objects of S are κ-presentable, $|\operatorname{Fl} I| \le \pi$, and $|\operatorname{Fl}(S/x_i)| \le \pi$, then

$$|\operatorname{Ob}(S/\operatorname{colim}_I x_i)| \le \sum_{i\in I} |\operatorname{Ob}(S/x_i)| \le \pi.$$

For a fixed arrow $s \to t$ in S, the arrows over it are determined by a map $t \to \operatorname{colim}_I x_i$. Thus the number of all arrows is at most $|\operatorname{Fl} S| \cdot \pi \le \pi$. This is a precise version of the calculation on pages 74–75. It uses filtered colimits of the relevant size, not arbitrary colimits. The two projective-limit signs on page 74 are source slips; the argument concerns colimits.

The proposed proof of the decomposition axiom on page 76 requires more. Starting with at most π arrows of S/x, one must close the resulting subcategory under simultaneous cocones for at most π_0 constraints. The number of constraints can be π^{π_0}, so closure need not stay within π. A fixed-point hypothesis such as $\pi^{\pi_0} = \pi$ permits the familiar closure construction: adjoin chosen cocones for all these constraints, iterate through an ordinal of cofinality greater than π_0, and keep cardinality π. Without a bound of this kind, the ordered collection of proposed small subcategories has not been shown to be large enough in the required sense. The manuscript explicitly retracts this step in its footnotes; the sentence claiming that PTA implies condition (c) is not supported by the preceding proof.

The failure can be seen without uncertain cardinal arithmetic. Take $\mathcal{M} = \mathbf{Set}$, let S be a skeleton of finite sets, and put $\pi_0 = \pi = \aleph_0$. Let x be an infinite set. The proposed poset I consists of countable full subcategories of S/x which are $\aleph_1$-filtered. For each $t \in x$, the singleton subcategory on the map $1 \to x$ with value t is such a subcategory. If countably many distinct such singleton subcategories had an upper bound $K \in I$, $\aleph_1$-filteredness would give a common cocone for their countable family of objects. Its vertex would be a finite set mapping to x whose image contains all those distinct values, which is impossible. Thus this particular I is not grand devant $\aleph_0$. Sets still have the desired decompositions, for instance by their countable subsets; the failed part is the proposed indexing construction.

The repair should preserve that distinction. The direct construction in the preceding block proves an ind-presentation from accessible probes. It does not identify the provisional filtration with the later intrinsic filtration by presentability ranks. The latter comparison is the subject of Section 4.5, and the manuscript itself points forward to Section 4.6 for a corrected proof of the theorem. An audit can therefore retain the theorem's structural content while recording that this particular numerical proof is abandoned.

Prouvons c) $\Longrightarrow$ d). Soit

$$\pi_1 \;=\; 2^c \qquad \text{avec} \qquad c \;\geqslant\; \pi_0\ .$$

On sait alors que

$$\operatorname{Filt}^{\pi_1}\mathcal{M} \;=\; \mathcal{M}_{\pi_1} \overset{\text{déf}}{=} \text{ sous-catégorie pleine de } \mathcal{M} \text{ des objets } \pi_1\text{-accessibles de } \mathcal{M}$$

(loc. cit. 9.16), ce qui implique notamment que $\mathcal{M}_{\pi_1}$ est stable par $\varinjlim$ de cardinal $\leqslant \pi_1$, en vertu de loc. cit. 9.9 (page 145). De plus, par loc. cit. 9.18 b) (p. 156), on a

[page 77]

$$\operatorname{Ind}_{\pi_1}(\mathcal{M}_{\pi_1}) \xrightarrow{\sim} \mathcal{M}\ ,$$

ce qui prouve d).

Prouvons d) $\Longrightarrow$ PTA. Par loc. cit. 9.18 a) (p. 156), on voit que les objets de $\mathcal{M}_{\pi_1}$ sont π_1-accessibles dans $\operatorname{Ind}_{\pi_1}(\mathcal{M}_{\pi_1})$, d'autre part il est clair que $\mathcal{M}_{\pi_1}$ est une sous-catégorie pleine strictement génératrice de $\operatorname{Ind}_{\pi_1}(\mathcal{M}_{\pi_1})$, donc il reste à voir que $\operatorname{Ind}_{\pi_1}(\mathcal{M}_{\pi_1})$ est stable par petites $\varinjlim$, i.e. le

Lemme 4.3.7. *Soit $\mathcal{M}_\pi$ une $\mathfrak{U}$-catégorie stable par limites inductives de cardinal $\leqslant \pi$, où π est un cardinal infini donné. Alors $\operatorname{Ind}_\pi \mathcal{M}_\pi$ est stable par petites limites inductives. De* p. 15

plus, $\mathcal{M} = \mathrm{Ind}_\pi \mathcal{M}_\pi$, munie de $\mathcal{M}_\pi \longrightarrow \mathcal{M}$, est solution d'un problème 2-universel, à savoir que pour toute catégorie $\mathcal{N}$ stable par petites limites inductives, le foncteur restriction induit une équivalence

$$\underline{\mathrm{Hom}}_!(\mathcal{M},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{!\,\pi}(\mathcal{M}_\pi,\mathcal{N}) \ , \tag{4.3.7.1}$$

où le signe ! (resp. $!\,\pi$) désigne la sous-catégorie pleine de la catégorie Hom*, formée des foncteurs qui commutent aux petites* $\varinjlim$ *(resp. aux limites inductives de cardinal $\leqslant \pi$).*

Prouvons d'abord que les $\varinjlim$ de cardinal $\leqslant \pi$ *dans* $\mathcal{M}_\pi$ sont aussi des $\varinjlim$ *dans* $\mathcal{M} = \mathrm{Ind}_\pi \mathcal{M}_\pi$ ([14]). On a donc [page 78]

$$x \xleftarrow{\sim} \varinjlim_I x_i \qquad \textit{dans } \mathcal{M}_\pi \ ,$$

i.e.

$$(*) \qquad \mathrm{Hom}(x,y) \xrightarrow{\sim} \varprojlim_I \mathrm{Hom}(x_i,y) \qquad \text{si } y \in \mathrm{Ob}\,\mathcal{M}_\pi \ ,$$

prouvons que cela reste vrai pour y dans $\mathcal{M}$, i.e. y de la forme

$$y \ = \ \varinjlim_J y_j \ , \qquad J \text{ grande devant } \pi, \text{ les } y_j \text{ dans } \mathcal{M}_\pi.$$

On a en effet

$$\begin{aligned} \mathrm{Hom}(x,y) \ &\simeq \ \varinjlim_J \mathrm{Hom}(x,y_j) \\ &\overset{(*)}{\simeq} \ \varinjlim_J \varprojlim_I \mathrm{Hom}(x_i,y_j) \\ &\overset{\text{SGA 4 I 9.8}}{\simeq} \ \varprojlim_I \varinjlim_J \mathrm{Hom}(x_i,y_j) \\ &\simeq \ \varprojlim_I \mathrm{Hom}(x_i,y) \ , \end{aligned}$$

au total

$$\mathrm{Hom}(x,y) \ \simeq \ \varprojlim_I \mathrm{Hom}(x_i,y),$$

q.e.d.

Pour prouver que les petites $\varinjlim$ sont représentables dans $\mathcal{M} = \mathrm{Ind}_\pi(\mathcal{M}_\pi)$, comme elles le sont dans $\mathrm{Ind}(\mathcal{M}_\pi)$ (SGA 4 I 8.9.5 b, on utilise l'hypothèse que les $\varinjlim$ *finies* sont représentables dans $\mathcal{M}_\pi$), il reste à voir que $\mathrm{Ind}_\pi(\mathcal{M}_\pi)$ est stable dans $\mathcal{M}_\pi$ par ces limites.

Commentary · Examples [ms. 78]

Which colimits the ind-completion remembers

Grothendieck's warning on page 78 is essential to the next section. An ind-completion always remembers the original category fully faithfully, but it remembers a colimit *as a colimit* only when the objects allowed as targets impose the appropriate continuity condition. Here the large filtering condition supplies precisely that condition.

Let $\kappa = \pi^+$, suppose C has colimits of diagrams with fewer than κ arrows, and let $x = \mathrm{colim}_{i\in I}\, x_i$ be such a colimit in C. If $Y = \mathrm{colim}_{j\in J}\, h_{y_j}$ is a κ-filtered colimit of representables, then

$$\begin{aligned} \mathrm{Hom}(h_x, Y) &\cong \mathrm{colim}_{j\in J}\, C(x,y_j) \\ &\cong \mathrm{colim}_{j\in J}\, \lim_{i\in I^{\mathrm{op}}} C(x_i,y_j) \\ &\cong \lim_{i\in I^{\mathrm{op}}} \mathrm{colim}_{j\in J}\, C(x_i,y_j) \cong \lim_{i\in I^{\mathrm{op}}} \mathrm{Hom}(h_{x_i}, Y). \end{aligned}$$

The middle interchange is justified because there are fewer than κ simultaneous representatives and compatibility equations to choose. First choose a common upper stage for the representatives; then

14. Mais *attention*, $\mathcal{M}_\pi \longrightarrow \mathrm{Ind}(\mathcal{M}_\pi)$ ne commute *pas* à ces $\varinjlim$!

choose a common upper stage where the equations hold. Injectivity follows by doing the same for the equations expressing equality. Thus $C \to \mathrm{Ind}_\kappa C$ preserves the indicated colimit. Notice that the colimit need not be the pointwise colimit in $C^\wedge$.

One explicit example separates the two completions. Let C be a skeleton of the category of countable sets, let $\pi = \aleph_0$, and put $\kappa = \aleph_1$. Countable colimits of countable sets are countable. Every set X is the $\aleph_1$-filtered union of its countable subsets, since a countable collection of countable subsets has countable union. A map from a countable set into such a union factors through one stage, and two maps which become equal agree at a later stage. It follows that the restricted Yoneda functor identifies

$$\mathrm{Ind}_{\aleph_1} C \simeq \mathbf{Set}.$$

In particular, the countable coproduct of the singleton objects is represented by the countable set $\mathbb{N}$.

In the ordinary filtered completion $\mathrm{Ind}_{\aleph_0} C$, however, consider the filtered diagram of finite subsets $A \subset \mathbb{N}$. Its colimit is the presheaf

$$F(c) = \mathrm{colim}_{A \subset \mathbb{N} \text{ finite}} C(c, A) = \{f : c \to \mathbb{N} \mid f(c) \text{ is finite}\}.$$

This is not $h_{\mathbb{N}}$: evaluating at $c = \mathbb{N}$ misses the identity map. Since finite coproducts of representables agree with those in C, this F is the countable coproduct of the singleton in the ordinary ind-completion. The inclusion of C therefore does not preserve this countable coproduct there.

The proposed final sentence of page 78 cannot establish closure of $\mathrm{Ind}_\pi C$ under colimits *inside* ordinary $\mathrm{Ind}\, C$ either. In this example its countable coproduct would be F, which is not $\aleph_1$-continuous on C°: $F(\mathbb{N})$ consists only of finite-image maps, whereas the product of countably many copies of $F(1) = \mathbb{N}$ contains every sequence. Section 4.4 correctly turns instead to constructing colimits intrinsically in the restricted completion.

[page 79]

4.4 Étude de $\mathrm{Ind}_\pi(C)$ (pour C petite catégorie stable par $\varinjlim$ de cardinal $\leqslant \pi$)

4.4.1. Finalement, la démonstration du lemme 4.3.7 n'est pas tellement évidente que je l'anticipais, et il me faut lui consacrer des préliminaires techniques, que je vais réunir ici. Dans toute la suite, C représente une $\mathfrak{U}$-catégorie équivalente à une petite catégorie.

Proposition 4.4.2. *Soit $F \in C^\wedge$ un foncteur ind-représentable, i.e. tel que C/F soit filtrante. Soit d'autre part $i \longmapsto x_i$ un foncteur $I \longrightarrow C$, où I est une petite catégorie filtrante, et soit* p. 16

$$\alpha \;:\; x \;=\; \varinjlim_I {}^{C^\wedge} x_i \longrightarrow F \qquad (4.4.2.1)$$

[15]

*un système projectif de flèches $x_i \longrightarrow F$ (**N.B.** la limite inductive x est prise dans $C^\wedge$, et est calculée argument par argument), ce qui équivaut à une factorisation $\varphi_\alpha : I \longrightarrow C/F$ de $I \longrightarrow C$ en*

$$I \xrightarrow{\varphi_\alpha} C/F \underset{\psi}{\xleftarrow{can.}} C\ .$$

Ceci dit, pour que α (4.4.2.1) soit un isomorphisme dans $\mathrm{Ind}'(C) \subset C^\wedge$ (sous-catégorie pleine des foncteurs ind-représentables sur C), il faut et il suffit que φ_α soit cofinal*, ou ce qui revient au même, I étant filtrante, qu'il satisfasse les deux conditions habituelles :*

F 1) *$\forall\, \xi \in \mathrm{Ob}\, C/F$, $\exists\, i \in \mathrm{Ob}\, I$ et une flèche $\xi \longrightarrow \varphi_\alpha(i)$.*

F 2) *$\forall\, \xi \in \mathrm{Ob}\, C/F$, et $i \in \mathrm{Ob}\, I$, et toute double-flèche*

$$\xi \rightrightarrows \varphi_\alpha(i)\ ,$$

[page 80]

il existe une flèche $i \xrightarrow{u} j$ dans I telle que $\varphi_\alpha(u)$ égalise la double-flèche.

15. La donnée de α équivaut aussi à celle d'un homomorphisme d'ind-objets $(x_i)_{i \in I} \longrightarrow F$, F étant considéré comme ind-objet (généralisé) indexé par C/F.

Démonstration. Si φ_α est cofinal, on en conclut par définition que $\varinjlim \varphi_\alpha \longrightarrow \varinjlim \psi\varphi_\alpha$ est un isomorphisme, or cette flèche n'est autre que α. Inversement, supposons que cette flèche soit un isomorphisme, donc $C/F \xleftarrow{\sim} C/\varinjlim x_i$, et il reste à voir que le foncteur canonique

$$I \longrightarrow C/\varinjlim_{C^\wedge} x_i$$

est cofinal, i.e. satisfait F 1) et F 2). La vérification est ± tautologique, en utilisant le fait que pour tout y dans C, on a

$$\mathrm{Hom}(y, \varinjlim_{C^\wedge} x_i) \xleftarrow{\sim} \varinjlim_{\mathrm{Ens}} \mathrm{Hom}(y, x_i),$$

et du "calcul" des limites inductives filtrantes dans Ens.

Commentary · Exposition [ms. 80]

Cofinality as an elementwise presentation criterion

The point of Proposition 4.4.2 is to replace a statement about equality of ind-objects by a statement about the category indexing their elements. That change makes later simultaneous constructions possible: one can arrange factorization and equality at stages, rather than manipulate abstract maps out of a completed category.

Let $F : C^{\mathrm{op}} \to \mathbf{Set}$, and write C/F for its category of elements. An object is a pair (c, u) with $u \in F(c)$; a morphism $(c, u) \to (d, v)$ is a map $a : c \to d$ satisfying $F(a)(v) = u$. There is always a canonical presentation

$$F \cong \operatorname*{colim}_{(c,u)\in C/F} h_c \qquad \text{in } C^\wedge.$$

A map $\alpha : \mathrm{colim}_{i\in I} h_{x_i} \to F$ is consequently the same as compatible choices $u_i \in F(x_i)$, hence a functor $q : I \to C/F$ over C.

If q is final, replacing C/F by I in the canonical presentation gives α as an isomorphism. Conversely, suppose I is filtered and α is an isomorphism. For any $(c, u) \in C/F$, the identity

$$F(c) \cong \mathrm{colim}_{i\in I} C(c, x_i)$$

represents u by some arrow $c \to x_i$. This is F1. If two arrows $(c, u) \rightrightarrows q(i)$ are given, their underlying maps $c \rightrightarrows x_i$ have the same image in that colimit of sets. Filteredness supplies a later arrow $i \to j$ at which the two composites agree. This is F2. These conditions make every comma category $(c, u) \downarrow q$ nonempty and connected: two objects can first be moved to a common index of I, and their remaining discrepancy can then be equalized. Thus q is final.

For $\kappa = \pi^+$, the same observation yields the clean formulation of Corollary 4.4.5:

$$F \in \mathrm{Ind}_\kappa C \quad \Longleftrightarrow \quad C/F \text{ is } \kappa\text{-filtered.}$$

One implication follows by the presentation criterion and the stability of κ-filteredness under a final functor from a κ-filtered category. For the other, the canonical presentation itself is a κ-filtered presentation. If ind-objects are required to use ordered indexing sets, the replacement by a final κ-directed poset is justified in the next block. What is not required is that each bounded collection of arrows of C/F lie inside a subcategory with an actual terminal object. The distinction is exactly the one exposed by the editors' idempotent example.

Corollaire 4.4.3. *Soit π un ordinal infini. Pour que F soit dans $\mathrm{Ind}'_\pi(C)$ (image essentielle de $\mathrm{Ind}_\pi(C)$ dans $C^\wedge$), il faut et il suffit qu'il existe une catégorie ordonnée grande devant π I, et un foncteur cofinal $I \longrightarrow C/F$.*

Il nous faut donc donner des conditions, sur une catégorie J $(= C/F)$, pour qu'il existe une catégorie ordonnée I grande devant π, et un foncteur cofinal $I \longrightarrow J$.

Une telle catégorie s'appellera *grande devant π*. (Il est immédiat que dans le cas où J est elle-même ordonnée, cette terminologie est cohérente avec celle déjà adoptée pour les ensembles ordonnés, i.e. que s'il existe une application croissante cofinale $\varphi : I \longrightarrow J$, avec I grand devant π, alors J est lui-même grand devant π.) Une telle catégorie est automatiquement [page 81] p. 17

filtrante (SGA 4 I 8.1.3 b)). Notons aussi que si on a un foncteur cofinal $J \longrightarrow J'$, si J est grande devant π, J' l'est aussi (par transitivité de la cofinalité).

Proposition 4.4.4. *Soient J une $\mathfrak{U}$-catégorie, π un cardinal infini. Conditions équivalentes :*

a) J est grande devant π, i.e. il existe un ensemble ordonné grand devant π et un foncteur cofinal $\varphi : I \longrightarrow J$.

b) J satisfait les deux conditions suivantes :

PF$_\pi$ 1 *$\operatorname{Ord} J$ est grande devant π, i.e. pour toute famille $(y_\alpha)_{\alpha\in A}$ d'objets de J, avec $\operatorname{card} A \leqslant \pi$, il existe un majorant y dans J (i.e. un y et des flèches $y_\alpha \xrightarrow{v_\alpha} y$).*

PF$_\pi$ 2 *Pour y_0, y_1 dans J et tout ensemble $(u_\alpha)_{\alpha\in A}$ de flèches $y_0 \xrightarrow{u_\alpha} y_1$ avec $\operatorname{card} A \leqslant \pi$, il existe une flèche $y_1 \xrightarrow{u'} y_2$ qui les égalise, i.e. telle que $u'u_\alpha = u'u_\beta \;\forall\, \alpha, \beta \in A$. (Il suffit de l'exiger quand $\operatorname{card} A \geqslant 2$.)*

c) Pour toute partie A de $\operatorname{Fl} J$, avec $\operatorname{card} A \leqslant \pi$, existe une sous-catégorie J' de J, avec $\operatorname{card} \operatorname{Fl} J' \leqslant \pi$, telle que $A \subset \operatorname{Fl} J'$ et que J' admette un objet final.

d) Soit I l'ensemble des sous-catégories J' de J telles que $\operatorname{card} J' \leqslant \pi$ et ayant un objet final

et ordonnons I par inclusion. Alors I est grand devant π, de plus le foncteur canonique [page 82]
$I \longrightarrow J$ ($J' \longmapsto e_{J'}$ = objet final de J') est cofinal ([16], [17], [18]).

([19])

Démonstration.

a $\Longrightarrow$ b vérification immédiate, en utilisant le critère F 1, F 2 de la cofinalité pour I filtrante.

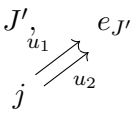
b $\Longrightarrow$ c par AQT [âne qui trotte].

c $\Longrightarrow$ d par AQT [âne qui trotte].

d $\Longrightarrow$ a Considérons l'ensemble ordonné grand devant π construit dans l'énoncé de d), et le *foncteur* p. 18

$$I \longrightarrow J$$

donné par $J' \longmapsto e_{J'}$ = objet final de J'. (Vérification que c'est un foncteur immédiate.) Ce foncteur est surjectif, donc satisfait la condition F 1 de cofinalité. La condition F 2 résulte aussi de d) en regardant les trois objets J', $j \xrightarrow{u_1} e_{J'}$, $j \xrightarrow{u_2} e_{J'}$ de I,



16. Introduire $I(J)$ avant l'énoncé de la proposition. **N.B.** La condition de cofinalité $I(J) \longrightarrow J$ n'est pas automatique, comme on voit en prenant $J = B_G$, $G = \{\pm 1\}$.

17. **N.B.** L'hypothèse de cofinalité est superflue si J est rigide, i.e. tout endomorphisme d'un objet de J est identique.

18. N. Éd. Dans cette condition (d), le "*de plus le foncteur … est cofinal*" est ajouté par Grothendieck *a posteriori*, ce qui explique sa vaine tentative de l'établir dans l'implication d $\Longrightarrow$ a ci-dessous, implication qui devient évidente après cet ajout. Par ailleurs, les conditions (c) et (d) ne sont *pas* impliquées par les conditions équivalentes (a) et (b). Par exemple, si J est la catégorie ayant un seul objet et un seul morphisme non identique p tel que $p^2 = p$, alors J satisfait aux conditions (a) et (b), mais *pas* aux conditions (c) et (d).

19. Autres conditions :

e) Pour tout ensemble A avec $\operatorname{card} A \leqslant \pi$, $\operatorname{diag}_J : J \longrightarrow J^A$ est cofinal.

e′) (Si A_0 est un ensemble donné, de cardinal $= \pi$.) $J \longrightarrow J^{A_0}$ est cofinal.

Voir aussi 4.9.6.9.1 et 4.9.6.9.2, pages 237, 238.

et en prenant un J'' qui les majore tous trois, alors $e_{J'} \longrightarrow e_{J''}$ égalise u_1, u_2 ([20]).

Corollaire 4.4.5. *Soit $F \in \mathrm{Ob}\, C^\wedge$. Pour que F appartienne à $\mathrm{Ind}'_\pi(C)$, il faut et il suffit que C/F satisfasse à l'ensemble des quatre conditions équivalentes de 4.4.4.*

Corollaire 4.4.6. *Soit J une catégorie stable par $\varinjlim$ de cardinal $\leqslant \pi$. Alors J est grande devant π.*

En effet, les conditions $\mathrm{PF}_\pi 1$, $\mathrm{PF}_\pi 2$ sont visiblement vérifiées.

Commentary · Development (commentators') [ms. 82]

A corrected terminal-subcategory criterion

Grothendieck is trying to turn simultaneous solvability of equations into a particularly concrete indexing poset: its elements would be small subcategories already possessing a final object. The editors explain why this construction is too restrictive. We first give the correct general replacement, then show precisely when his proposed one works. The filtered-category argument is classical; the second statement isolates and proves the repair suggested by his rigidity footnote. We checked the formulation in SGA 4 I, Section 8.1, the present edition's note 18, and the later product criterion cited below; we did not find this exact bounded rigid formulation there. The label on the second result records our proof of that formulation, not a claim of priority.

For a small category J and an infinite cardinal π, put $\kappa = \pi^+$. Conditions $\mathrm{PF}_\pi 1$ and $\mathrm{PF}_\pi 2$ are equivalent to saying that every diagram with fewer than κ arrows has a cocone. To prove the nontrivial direction, first choose a common target for all objects in the diagram. For each arrow $a : i \to j$ there are then two maps from its source to that target, namely the chosen map at i and the composite through j. Use $\mathrm{PF}_\pi 2$ to choose a map from the common target to an object z_a equalizing that individual pair, then choose a common target z of all the z_a. The resulting maps from the original common target to z need not coincide; apply $\mathrm{PF}_\pi 2$ to this family. Their common composite equalizes every required pair. If there are no compatibility pairs, the first common target suffices. The empty diagram is included in $\mathrm{PF}_\pi 1$.

Such a small J admits a final functor from a κ-directed poset. Here is a construction which does not presume the false terminal-subcategory condition. Begin with a discrete poset P_0 having a vertex over each object of J. Inductively maintain a functor $q_\alpha : P_\alpha \to J$ such that every principal downset in P_α has fewer than κ elements. At the next stage, for every subset $A \subset P_\alpha$ of size less than κ, adjoin a new vertex above its downward closure. That closure has size less than κ by regularity, so a chosen cocone of its image defines the new vertex and all arrows into it. Also, for every $p \in P_\alpha$ and every parallel pair $u, v : j \rightrightarrows q_\alpha(p)$, adjoin a vertex above the downset of p whose transition map equalizes u, v. This is possible by applying the cocone condition to that downset with a new formal source and those two arrows. Distinct new vertices are left incomparable. At limit stages take unions.

After κ stages the union P is a set, all its principal downsets remain of size less than κ, and any subset of P of size less than κ occurs at one earlier stage and hence has an upper bound. Thus P is κ-directed. The functor $q : P \to J$ satisfies F1 because of P_0, and F2 because every parallel-pair requirement is addressed at a successor stage. The preceding presentation criterion proves that q is final. This supplies the implication $b \Rightarrow a$ of Proposition 4.4.4 in the small case needed for $J = C/F$.

Proposition (the rigid repair). [ours] Suppose J is small and every endomorphism of every object of J is an identity. Then J is κ-filtered if and only if every set of at most π arrows of J is contained in a subcategory with at most π arrows and a terminal object. In that event, the poset of these terminal subcategories is κ-directed, and the functor choosing their terminal objects is final.

Proof. Let A be the prescribed set of arrows. Close it under identities and finite composition to obtain a subcategory K with at most π arrows. Choose a cocone $\eta : K \to j$ in J. Adjoin j, the arrows $\eta_k : k \to j$, and the identity of j. If j was not in K, the generated subcategory has no nonidentity arrow leaving j, and every path ending at j equals the appropriate η_k by naturality. Hence j is terminal. If j already belongs to K, rigidity forces $\eta_j = 1_j$. For an old arrow $a : k \to j$, naturality gives $a = \eta_j a = \eta_k$. All newly generated paths ending at j reduce to the same arrow, again by naturality. Thus j is terminal in this case too. The enlarged subcategory still has at most π arrows.

20. Un peu bref, le cas où $e_{J'} = j$ échappe à cet argument, et demande qu'on `[phrase incomplète]`

Conversely, the arrow set of a diagram of at most π arrows is contained in such a terminal subcategory, whose terminal object gives a cocone. For directedness, take at most π terminal subcategories, form the union of their arrow sets, and apply the proved condition. Their union has at most π arrows. To prove finality of the terminal-object functor, every object of J occurs in its singleton subcategory, giving F1. Given a terminal subcategory K and two maps $j \rightrightarrows e_K$, enlarge the arrow set of K by this pair and choose a terminal subcategory containing it. The unique map from e_K to the new terminal object equalizes the pair, giving F2. This completes the proof.

The rigidity hypothesis addresses the precise obstruction. In the one-object category with arrows $1, p$ and $p^2 = p \neq 1$, every family of parallel arrows is equalized by postcomposition with p, so the category is κ-filtered for every infinite κ. A subcategory containing p cannot have a terminal object: its only object has two endomorphisms. Its only terminal subcategory is therefore the identity-only subcategory. The resulting functor from the singleton indexing poset is not final, since the relevant comma category has two disconnected objects. This proves the editors' failure of both (c) and (d), and explains why Corollary 4.4.5 must refer to (a) or (b), rather than to all four printed conditions.

Commentary · Later developments [ms. 82]

The infinite-product criterion

The diagonal conditions in note 19 point toward a valid alternative. For a small J, finality of $J \to J^A$ for every $|A| \leq \pi$, including $A = \varnothing$, means that J-colimits of sets commute with these products. Since π is infinite, Adámek et al. (2000, Theorem 3.1) identifies this condition with π^+-filteredness. The restriction to infinite π matters: finite-product commutation alone is the weaker condition of siftedness.

The single-power condition (e′) needs a nonemptiness clause. For $J = \varnothing$ and a nonempty set A_0, both J and J^{A_0} are empty, so the diagonal is final vacuously. For nonempty J, finality of the indicated diagonal first forces connectedness. Padding a smaller family with constant singleton diagrams then reduces its product comparison to the A_0 comparison. Thus (e′) recovers (e) with that clause added.

[page 83]

4.4.6. Je me donne un foncteur

$$J \xrightarrow{\varphi} \operatorname{Ind}'_\pi C\ , \tag{4.4.6.1}$$

J une petite catégorie, et on cherche une $\varinjlim$ dans $\operatorname{Ind}'_\pi C$, en postulant au besoin l'existence des $\varinjlim$ de cardinal $\leqslant \pi$ dans C, et l'hypothèse $\operatorname{card} \operatorname{Fl} J \leqslant \pi$. Le procédé le plus naturel pour ramener la question de l'existence des $\varinjlim$ dans $\operatorname{Ind}'_\pi C$, à la question similaire dans C, est explicité dans SGA 4 I 8.8.3. On considère l'inclusion pleinement fidèle

$$C \xhookrightarrow{\alpha} \operatorname{Ind}'_\pi C\ , \tag{4.4.6.2}$$

laquelle commute aux $\varinjlim$ de cardinal $\leqslant \pi$ (celles qui sont représentables dans C, cf. p. 78 et référence à SGA 4 I 9.8), d'où

$$\underline{\operatorname{Hom}}(J, C) \xhookrightarrow{\alpha} \underline{\operatorname{Hom}}(J, \operatorname{Ind}'_\pi C) \tag{4.4.6.3}$$

également pleinement fidèle, et commutant aux $\varinjlim$ de cardinal $\leqslant \pi$ (représentables dans le premier membre). Cette flèche s'insère dans le carré commutatif relatif à la donnée d'un $j \in \operatorname{Ob} J$, et des foncteurs d'inclusion

$$e \xrightarrow{\varepsilon_j} J\ , \qquad e_j(e) \ = \ j\ ; \tag{4.4.6.4}$$

p. 19
[page 84]

$$\begin{array}{ccc} \underline{\operatorname{Hom}}(J, C) & \xrightarrow{\alpha^J} & \underline{\operatorname{Hom}}(J, \operatorname{Ind}'_\pi C) \\ \downarrow{\scriptstyle \varepsilon_j^*} & & \downarrow{\scriptstyle \varepsilon_j^{*\prime}} \\ C & \xrightarrow{\alpha} & \operatorname{Ind}'_\pi(C) \end{array} \qquad \begin{array}{c} \ni \varphi \\ \downarrow \\ \ni \varphi(j)\ , \end{array} \tag{4.4.6.2}$$

([21])

lequel induit un foncteur

$$\underbrace{\underline{\mathrm{Hom}}(J/C)/\varphi}_{\overset{\text{déf}}{=} I} \xrightarrow{\varepsilon_j^{\varphi}} C/\varphi(j)\ . \tag{4.4.6.3}$$

Si C est stable par $\varinjlim$ relatives à un ensemble K du type de conditions $\leqslant \pi$ (p.ex. $\varinjlim$ finies), il en est de même de $\underline{\mathrm{Hom}}(J,C)$, et aussi de $\underline{\mathrm{Hom}}(J,C)/\varphi$, et de $C/\varphi(j)$, en vertu du fait que α^J et α commutent aux $\varinjlim$ du type envisagé, de plus (4.4.6.3) commute aux limites en question. Supposons donc C stable tout au moins par $\varinjlim$ finies, donc les catégories source et but dans (4.4.6.3) sont filtrantes. (Le fait que le foncteur commute aux $\varinjlim$ finies ne semble pas utile dans le contexte présent.) Cela nous servira pour avoir le critère commode de cofinalité SGA 4 I 8.1.3 b) par F 1, F 2.

Commentary · Exposition [ms. 84]

One index category for an entire diagram

The next difficulty is coherence. An individual ind-object has many filtered presentations, but presentations chosen separately at the vertices of a diagram need not carry its arrows and relations. Grothendieck uses the category of all coherent approximations at once:

$$I = \mathrm{Fun}(J, C)/\varphi.$$

An object of I consists of a C-valued diagram D and a natural transformation $\alpha D \to \varphi$, where $\alpha : C \to \mathrm{Ind}_\pi C$. A morphism is a natural transformation over φ. Evaluating at j gives $\varepsilon_j^{\varphi} : I \to C/\varphi(j)$. In general I is a category, not an ordered set; the phrase calling it an ordered set on page 85 is not needed by the argument.

The purpose of Lemma 4.4.9 is to show that each evaluation is final. We can prove its stronger assertion directly and avoid the temporary dependency on coproducts of arbitrary ind-objects. Assume that C admits the copowers indexed by the sets $J(j, j')$, all of cardinality at most π. Define

$$(L_j c)(j') = \coprod_{a \in J(j,j')} c.$$

An arrow $b : j' \to j''$ carries the summand indexed by a to the summand indexed by ba. This makes $L_j c$ a diagram in C. A natural transformation $L_j c \to D$ is uniquely determined by the map from the identity summand $c \to D(j)$; its component on the a-summand must be $D(a)$ composed with that map. Hence L_j is left adjoint to evaluation on C-valued diagrams.

Page 78 proves that α preserves these particular coproducts. Therefore, for any $\varphi : J \to \mathrm{Ind}_\pi C$, the same calculation gives

$$\mathrm{Nat}(\alpha L_j c, \varphi) \cong \mathrm{Hom}(\alpha c, \varphi(j)).$$

This calculation uses coproducts of copies of the representable object c only. It does not require the still-pending existence of coproducts of arbitrary objects of $\mathrm{Ind}_\pi C$.

For an object $(c, f : \alpha c \to \varphi(j))$ of $C/\varphi(j)$, let $\widetilde{f} : \alpha L_j c \to \varphi$ be the corresponding natural transformation. Set

$$R_j(c, f) = (L_j c, \widetilde{f}) \in I.$$

A map from $R_j(c, f)$ to (D, η) in I is a transformation $\theta : L_j c \to D$ with $\eta\alpha\theta = \widetilde{f}$. By the preceding calculation, this is exactly a map $g : c \to D(j)$ with $\eta_j \alpha(g) = f$, which is a map $(c, f) \to \varepsilon_j^{\varphi}(D, \eta)$ in the slice. These bijections are natural, proving $R_j \dashv \varepsilon_j^{\varphi}$. Every comma category $(c, f) \downarrow \varepsilon_j^{\varphi}$ has the resulting universal object as an initial object, so evaluation is final.

If C has all colimits of cardinality at most π, then so does I: take a colimit of approximating diagrams pointwise in C, use preservation by α, and form the induced map to φ. Thus I is π^+-filtered. Finality of each evaluation gives

$$\varphi(j) \cong \operatorname*{colim}_{i \in I} \alpha D_i(j).$$

21. Les foncteurs ε_j^*, $\varepsilon_j^{*\prime}$ sont les foncteurs évaluation en j.

If $|\operatorname{Fl} J| \leq \pi$, form $c_i = \operatorname{colim}_{j\in J} D_i(j)$ in C. Testing against any target Z and interchanging two limits of Hom sets gives

$$\operatorname{Hom}(\operatorname{colim}_{i\in I} \alpha c_i, Z) \cong \lim_{j\in J^{\mathrm{op}}} \operatorname{Hom}(\varphi(j), Z).$$

The left colimit exists by its π^+-filtered presentation. It therefore supplies the desired J-colimit in $\operatorname{Ind}_\pi C$. This proves the content of Lemma 4.4.7 and Corollary 4.4.8 after their explicitly stated cofinality reservation has been discharged.

[page 85]

Rappelons que la cofinalité des foncteurs (4.4.6.3), quand elle a lieu (?), implique le fait que φ est une limite inductive du foncteur composé

(4.4.6.4)
$$\begin{array}{ccccc} I & \xrightarrow{\psi_J} & \underline{\mathrm{Hom}}(J,C) & \xrightarrow{\alpha^J} & \underline{\mathrm{Hom}}(J,\mathrm{Ind}'_\pi(C)) \\ \| & & \vdots{\scriptstyle \varepsilon_j^*} & & \vdots{\scriptstyle \varepsilon_j^{*\prime}} \\ \underline{\mathrm{Hom}}(J,C)/\varphi & & & & \\ {\scriptstyle \varepsilon_j^\varphi}\downarrow & & & & \\ C/\varphi(j) & \xrightarrow{\psi_j} & C & \xrightarrow{\alpha} & \mathrm{Ind}'_\pi(C)\ , \end{array}$$

suivant l'ensemble ordonné filtrant I, plus précisément que φ est la $\varinjlim$ "argument par argument en j", ce qui signifie que pour tout $j \in \operatorname{Ob} J$, $\varphi(j) = \varepsilon_j^*(\varphi)$ est limite inductive de $\varepsilon_j^*(\alpha^J\psi) = (\alpha\psi_j)\varepsilon_j^\varphi$. (Or on sait en effet que $\varphi(j)$ est la limite de $\alpha\psi$, puisque $\varphi(j) \in \operatorname{Ob} C^\wedge$ est ind-représentable, donc aussi celle de $(\alpha\psi_j)\varepsilon_j^\varphi$ si ε_j^φ est cofinal.)

Ainsi, admettant la cofinalité de (4.4.6.3) et interprétant ψ_J dans (4.4.6.4)

[page 86]

$$\psi_J \ :\ I \longrightarrow \underline{\mathrm{Hom}}(J,C)$$

comme un foncteur

$$I \times J \xrightarrow{\Psi} C\ ,$$

on trouve la "représentation indicielle" de φ

$$\varphi(j) \ = \ \underbrace{\varinjlim_{i\in I} \psi(i,j)}_{\substack{\text{limite inductive}\\ \text{prise dans } \mathrm{Ind}'_\pi(C)\\ \text{d'objets de } C\hookrightarrow \mathrm{Ind}'_\pi(C)}} \quad ,$$

d'où

p. 20

$$\begin{aligned} \underbrace{\varinjlim_{j\in J} {}_{\mathrm{Ind}'_\pi C}\varphi(j)}_{\text{sous réserve d'existence}} &= \varinjlim_{j\in J} \varinjlim_{i\in I} \psi(i,j) \\ &= \varinjlim_{i\in I} \varinjlim_{j\in J} \psi(i,j)\ , \end{aligned}$$

toutes les limites inductives étant prises dans $\mathrm{Ind}'_\pi(C)$. Mais si les $\varinjlim$ de type J sont représentables dans C, et $\operatorname{card} J \leqslant \pi$, on sait que la $\varinjlim_{j\in J} \psi(i,j)$ *dans* C est aussi leur $\varinjlim$ dans $\mathrm{Ind}'_\pi C$, et on trouve donc que $\varinjlim_{j\in J} \varphi(j)$ existe dans $\mathrm{Ind}'_\pi(C)$ si et seulement si $\varinjlim_{i\in I} \Psi(i)$ existe dans $\mathrm{Ind}'_\pi(C)$, où pour $i \in I$, on pose

$$\Psi(i) \ = \ \varinjlim_{j\in J} \psi(i,j) \qquad \textit{dans } C.$$

Résumons ce que nous avons obtenu :

[page 87]

Lemme 4.4.7. *Soit C une petite catégorie, π un cardinal infini, J une catégorie telle que* $\operatorname{card} \operatorname{Fl} J \leqslant \pi$, *$\varphi : J \longrightarrow \mathrm{Ind}'_\pi(C)$ un foncteur. Supposons les $\varinjlim$ de type J existant dans C, et*

que les foncteurs ε_j^{φ} ($j \in \mathrm{Ob}\, J$) de (4.4.6.3) soient cofinaux *(ce qui s'explicite de façon simple surtout si source et but sont filtrants, chose acquise si C est stable par $\varinjlim$ finies). Alors φ a une "représentation indicielle canonique"*

$$\psi \ : \ I \times J \longrightarrow C \tag{4.4.7.1}$$

avec

$$I \ = \ \underline{\mathrm{Hom}}(J, C)/\varphi \ , \tag{4.4.7.2}$$

i.e. on a l'isomorphisme fonctoriel en $j \in \mathrm{Ob}\, J$

$$\varphi(j) \ \simeq \ \varinjlim_{I} \psi(i, j) \qquad ([22]) \ . \tag{4.4.7.3}$$

Posons pour tout $i \in \mathrm{Ob}\, I$

$$\Psi(i) \ = \ \varinjlim_{J} \psi(i, j) \qquad ([23]) \ . \tag{4.4.7.4}$$

Alors $\varinjlim_{j \in J} \varphi(j)$ (dans $\mathrm{Ind}'_\pi\, C$) existe si et seulement si la $\varinjlim_{i \in I} \Psi(i)$ existe (dans $\mathrm{Ind}'_\pi\, C$ également), et les deux sont égales. [page 88]

Corollaire 4.4.8. *Supposons que C soit stable par $\varinjlim$ de cardinal $\leqslant \pi$ (pas seulement par celles de type J). Alors $\underline{\mathrm{Hom}}(J, C)$ et $I = \underline{\mathrm{Hom}}(J, C)/\varphi$ ont la même propriété, I est grande devant π, et $\varinjlim_{j \in J} \varphi(j) \ = \ \varinjlim_{i \in I} \Psi(i)$ existe dans $\mathrm{Ind}'_\pi(C)$, sous réserve de cofinalité des foncteurs (4.4.6.3).*

Il suffit en effet de vérifier ce

Lemme 4.4.8.1. *Soit I une catégorie grande devant π, et $\Psi : I \longrightarrow C$ un foncteur. Alors $\varinjlim_i \Psi(i)$ existe dans $\mathrm{Ind}'_\pi(C)$, et c'est la limite inductive dans $C^\wedge$, ou encore "c'est" l'ind-objet grand devant π représenté par Ψ.* (En d'autres termes ([24]), $\mathrm{Ind}'_\pi(C) \subset C^\wedge$ est une sous-catégorie strictement pleine *stable* par $\varinjlim$ grandes devant π.) p. 21

Ce lemme est une tautologie : si F désigne la limite inductive dans $C^\wedge$, il est clair par définition (en prenant un foncteur $I_0 \longrightarrow I$ cofinal, avec I_0 *ordonnée* grande devant π, de sorte que la $\varinjlim$ peut être prise suivant I_0) que F est dans $\mathrm{Ind}'_\pi(C)$, et alors c'est a fortiori une $\varinjlim$ dans $\mathrm{Ind}'_\pi\, C$. [page 89]

Pour vérifier l'existence des $\varinjlim$ de cardinal $\leqslant \pi$ dans $\mathrm{Ind}'_\pi\, C$, dans le cas où C lui-même est stable par lesdites limites, il suffit donc, grâce au corollaire 4.4.8, de prouver la cofinalité des foncteurs (4.4.6.3). Cela va résulter du

Lemme 4.4.9. *Soient C une petite catégorie, π un cardinal infini, J une petite catégorie telle que pour tous $j, j' \in \mathrm{Ob}\, J$, on ait* $\mathrm{card}\, \mathrm{Hom}(j, j') \leqslant \pi$. *Sous ces conditions, pour tout $j \in \mathrm{Ob}\, J$, le foncteur (4.4.6.3)* admet un adjoint à gauche, *donc a fortiori il est cofinal (car les catégories colocalisées au dessus de $C/\varphi(j)$ ont un objet initial, a fortiori elles sont* 0*-connexes).*

Démonstration de 4.4.8. Revenons au diagramme (4.4.6.2), dont on déduit (4.4.6.3), nous allons le réécrire de façon simplifiée : [page 90]

$$\begin{array}{ccc} B & \xrightarrow{\beta} & B' \qquad \ni y'_0 \\ {\scriptstyle\lambda}\uparrow\downarrow{\scriptstyle\varepsilon} & & {\scriptstyle\lambda'}\uparrow\downarrow{\scriptstyle\varepsilon'} \qquad \downarrow \\ C & \xrightarrow{\gamma} & C' \qquad \ni z'_0 \ . \end{array} \tag{4.4.9.1}$$

22. limite inductive dans Ind'_π d'objets de C.
23. limite inductive prise dans C, mais valable aussi dans $\mathrm{Ind}'_\pi\, C$.
24. C'est abusif, cette stabilité est un énoncé nettement plus fort! Je le prouve page 99.

Je dis que ε, ε' ont *des adjoints à gauche* λ resp. λ', et que l'homomorphisme canonique de "changement de base"

$$\lambda'\gamma \longrightarrow \beta\lambda \tag{4.4.9.2}$$

est un *isomorphisme.* En effet, l'existence des adjoints de ε, ε', à savoir des foncteurs $\varepsilon_{j\,!}$ et $\varepsilon'_{j\,!}$ relatifs à l'inclusion

$$\varepsilon_j \ : \ e \longrightarrow J$$

et aux catégories de foncteurs $\underline{\mathrm{Hom}}(J, C)$ resp. $\underline{\mathrm{Hom}}(J, \mathrm{Ind}'_\pi C)$, résultant de l'existence des sommes de type $\mathrm{Hom}(j, j')$ ($j' \in \mathrm{Ob}\, J$) dans C (déjà acquise), *et dans* $\mathrm{Ind}'_\pi C$ *;* et le fait que le foncteur $C \longrightarrow \mathrm{Ind}'_\pi C$ commute auxdites sommes (chose également déjà acquise, puisque card $(\mathrm{Hom}(j, j')) \leqslant \pi$) implique alors que (4.4.4.2) est un isomorphisme. Moyennant la vérification en suspens (l'existence des sommes de cardinal $\leqslant \pi$ dans $\mathrm{Ind}'_\pi(C)$), le lemme 4.4.9 résultera du résultat plus général : [page 91] p. 22

Lemme 4.4.10. *Considérons un carré commutatif (4.4.9.1) (traits pleins) de catégories, où ε admet un adjoint à gauche λ, ε' un adjoint à gauche λ', et la flèche de changement de base (4.4.9.2) est un isomorphisme. Alors pour tout y'_0 dans B', posant $\varepsilon'(y'_0) = z'_0$, le foncteur induit par $(\varepsilon, \varepsilon')$,*

$$\overline{B} \ = \ B/y'_0 \xrightarrow{\bar{\varepsilon}} \overline{C} = C/z'_0 \ ,$$

admet un adjoint à gauche $\overline{\lambda}$, donné par

$$\overline{\lambda}(\gamma(z) \xrightarrow{v} z'_0) \quad = \quad \Big(\lambda'\gamma(z) \simeq \beta(\lambda(z)) \xrightarrow{\lambda'(v)} \lambda'(z'_0) \Big\| \ \lambda'\varepsilon'(y'_0) \Big\downarrow \mathrm{adj.} \ y'_0\Big) \ .$$

($\overline{\lambda}(v) = $ composé adj$\circ\lambda'(v)$)

La démonstration se fait par un AQT `[âne qui trotte]` fastidieux (j'avoue ne pas avoir poussé jusqu'au dernier détail …). [page 92]

Commentary · Exposition [ms. 92]

The adjunction on categories of presentations

Grothendieck needs to present an entire diagram by objects of C, while keeping the prescribed value of that diagram at a chosen vertex. The passage to comma categories in Lemma 4.4.10 is what makes these two requirements compatible. Its proof is indeed formal, but the formal statement includes the invertibility of the specified mate, not merely the existence of four functors making a square commute.

Write the square as $\varepsilon'\beta = \gamma\varepsilon$, and write η, ϵ and η', ϵ' for the units and counits of $\lambda \dashv \varepsilon$ and $\lambda' \dashv \varepsilon'$. Let

$$\chi : \lambda'\gamma \longrightarrow \beta\lambda$$

be the invertible mate used in the source. Fix $y' \in B'$. An object of $(\gamma \downarrow \varepsilon' y')$ is a pair (c, v) with $v : \gamma c \to \varepsilon' y'$. Define

$$q_v = \epsilon'_{y'}\, \lambda'(v)\, \chi_c^{-1} : \beta\lambda c \longrightarrow y'.$$

Thus the proposed left adjoint sends (c, v) to $(\lambda c, q_v)$. On a morphism $a : (c, v) \to (d, w)$, the equation $w\gamma a = v$ and naturality of χ show that λa is a morphism of the required comma category. This already supplies the functor whose existence the displayed formula asserts.

We check the adjunction by retaining the equation defining a morphism in the comma category. Given (b, u) with $u : \beta b \to y'$, an arrow from $(\lambda c, q_v)$ to (b, u) is an arrow $f : \lambda c \to b$ such that $u\beta f = q_v$. Its transpose under $\lambda \dashv \varepsilon$ is

$$k = \varepsilon(f)\eta_c : c \longrightarrow \varepsilon b.$$

The definition of the mate gives

$$\varepsilon'(\chi_c)\eta'_{\gamma c} = \gamma(\eta_c).$$

Consequently the transpose, under $\lambda' \dashv \varepsilon'$, of $u\beta f\chi_c$ is $\varepsilon'(u)\gamma(k)$. The transpose of $q_v\chi_c$ is v. Since transposition is a bijection, the required equation on f is equivalent to

$$\varepsilon'(u)\gamma(k) = v.$$

This is exactly the condition that k be a morphism from (c, v) to the image of (b, u) in $(\gamma \downarrow \varepsilon' y')$. Both transposition bijections are natural in the objects, so this proves the comma-category adjunction, including its action on morphisms.

In the application, ε is evaluation at j. Its left adjoint sends c to the diagram $j' \mapsto \coprod_{J(j,j')} c$. The comparison χ is invertible because the embedding of C preserves these coproducts. Lemmas 4.4.11 and 4.4.12 establish their existence in the completion without presupposing general cocompleteness. A functor with a left adjoint is final: for each object in its target, the unit exhibits an initial object in the corresponding comma category. This explains why the adjunction provides precisely the cofinality needed for the simultaneous presentation.

Pour terminer la démonstration de 4.4.9, il reste à prouver un dernier (?)

Lemme 4.4.11. *Soit C une petite catégorie, stable par sommes de cardinal $\leqslant \pi$. Alors $\mathrm{Ind}'_\pi C$ est stable par sommes de cardinal $\leqslant \pi$ (et, pour mémoire, le foncteur $\alpha : C \longrightarrow \mathrm{Ind}'_\pi C$ y commute).*

Appliquant le corollaire 4.4.8 au lemme 4.4.7, on est ramené à prouver la cofinalité de (4.4.6.3) quand J est discrète, plus le fait que $\underline{\mathrm{Hom}}(J, C)/\varphi$ est grande devant π, dans ce cas. Mais la donnée de φ revient à celle d'une famille $(\varphi_j)_{j\in J}$ d'objets de $\mathrm{Ind}'_\pi(C)$, et le foncteur (4.4.6.3) s'explicite comme la projection

$$\prod_{j\in J}(C/\varphi_j) \longrightarrow C/\varphi_j$$

du produit total sur un des facteurs. Donc on est ramené à ceci :

Lemme 4.4.12. *Soit $(C_j)_{j\in J}$ une famille de petites catégories grandes devant π, indexée par un petit ensemble J. Alors $\prod C_j$ est grande devant π, et les projections (pour $j_0 \in J$) $\prod C_j \longrightarrow C_{j_0}$ sont cofinales.* [page 93]

Prouvons d'abord la cofinalité, qui est le plus important pour nous. C'est évident si $J = \{j_0\}$, sinon on a $J = J' \cup \{j_0\}$, et $\prod_{j\in J} C_j \simeq C' \times C_{j_0}$, où $C' = \prod_{j\in J'} C_j$, donc il faut voir que $C' \times C_{j_0} \longrightarrow C_{j_0}$ est cofinal, i.e. que toute catégorie $\xi\backslash(C' \times C_{j_0})$ (pour $\xi \in \mathrm{Ob}\, C_{j_0}$) est 0-connexe, or elle est isomorphe à $C' \times (\xi\backslash C_{j_0})$, et $\xi\backslash C_{j_0}$ étant 0-connexe, il reste à voir que C' est 0-connexe. On fera attention à cet égard qu'un produit *infini* de catégories 0-connexes n'est pas forcément 0-connexe, donc la cofinalité n'est pas totalement tautologique! Mais si on prouve que la catégorie $\prod_{j\in J} C_j$ est grande devant π, donc filtrante, cela implique qu'elle est 0-connexe, et on peut appliquer ce même résultat à $C' = \prod_{j\in J'} C_j$, et on gagne. p. 23 [page 94]

Donc on est ramené à prouver que $\prod C_j$ est grande devant π. Ceci se voit soit sur le critère $\mathrm{PF}_\pi\, 1$, $\mathrm{PF}_\pi\, 2$ de 4.4.4 b, soit en se ramenant au cas où les C_j sont ordonnées, mais cela a l'air plus long. Notons que l'on prouve de même qu'un (petit) produit de catégories filtrantes est encore une catégorie filtrante.

Commentary · Exposition [ms. 94]

Products of indexing categories and the first coproducts

The coproduct argument isolates the part of the construction that can be done before arbitrary colimits have been established. Put $\kappa = \pi^+$. A category called "grande devant π" here is κ-filtered. If $(D_j)_{j\in J}$ is any small family of such categories, a diagram with fewer than κ arrows in $\prod_j D_j$ projects to such a diagram in every factor. Choose a cocone in each factor and take their product. This proves that the product is κ-filtered, including its nonemptiness; the usual axiom of choice is being used in making these simultaneous choices.

The projection to D_{j_0} is final for a similarly concrete reason. For $d \in D_{j_0}$ its comma category is

$$(d \downarrow \mathrm{pr}_{j_0}) \simeq (d \downarrow D_{j_0}) \times \prod_{j \neq j_0} D_j.$$

The first factor is filtered and nonempty, and the remaining product is filtered and nonempty by the argument just given. Their product is therefore connected. The finality criterion follows. This proof also explains the warning about products of connected categories. Connectedness alone supplies zigzags of possibly unbounded length; filteredness supplies compatible cocones of one fixed shape.

For a literal example of that distinction, let P_n be the poset on $0, 1, \ldots, n$, with each even vertex below its adjacent odd vertices and no further nonidentity comparisons. Every P_n is connected. In $\prod_{n \geq 1} P_n$, however, the tuples $(0)_n$ and $(n)_n$ cannot be joined by a finite zigzag. A zigzag of length r projects to one of length r in every factor, whereas the distance from 0 to n in the underlying undirected graph of P_n is n. Choosing $n > r$ gives a contradiction. Thus the source's caution is a real one, even for a product of finite posets.

Now let $|J| \leq \pi$ and $F_j \in \mathrm{Ind}'_\pi C$, assuming only that C has coproducts with at most π summands. Set $I = \prod_j (C/F_j)$. For $i \in I$, write $c_{i,j} \to F_j$ for its jth component and put $s_i = \coprod_j c_{i,j}$ in C. The object

$$H = \varinjlim_{i \in I} y(s_i)$$

belongs to $\mathrm{Ind}'_\pi C$, because I is κ-filtered. Here y denotes the Yoneda embedding, and this colimit is taken in the presheaf category. To identify H, take $G \in \mathrm{Ind}'_\pi C$. Such a G sends the given coproducts in C to products of sets: it is a κ-filtered colimit of representables, and these colimits commute with products having fewer than κ factors. Hence

$$\begin{aligned} \mathrm{Hom}(H, G) &\simeq \varprojlim_i G(s_i) \simeq \varprojlim_i \prod_j G(c_{i,j}) \\ &\simeq \prod_j \varprojlim_i G(c_{i,j}) \simeq \prod_j \mathrm{Hom}(F_j, G). \end{aligned}$$

The last isomorphism uses the finality of the projections just proved. All the maps are induced by the evident coproduct cocone, so this proves the required universal property. For an empty family, I is terminal and s_i is the initial object of C; the same calculation includes the empty coproduct. There is therefore no circular use of the cocompleteness theorem in Lemma 4.4.11.

Remarques.

1. Cet argument ne montre nullement (même si C est stable par $\varinjlim$ de cardinal $\leqslant \pi$) que pour tout $\varphi \in \underline{\mathrm{Hom}}(J, \mathrm{Ind}(C))$, le foncteur similaire à (4.4.6.3)

$$\underline{\mathrm{Hom}}(J, C)/\varphi \longrightarrow C/\varphi(j)$$

soit cofinal, et je doute qu'il le soit toujours. La difficulté provient du fait que $C \longrightarrow \mathrm{Ind}\, C$ ne commute pas aux sommes de cardinal $\leqslant \pi$, seulement aux sommes finies, donc le lemme [page 95] 4.4.10 ne peut être invoqué.

2. Nous venons de prouver, via 4.4.8 et 4.4.9 (dont la démonstration est achevée avec 4.4.12), que si C est une petite catégorie stable par $\varinjlim$ de cardinal $\leqslant \pi$, alors $\mathrm{Ind}_\pi C$ est également stable par ces limites. De plus, les systèmes inductifs de ce type J dans $\mathrm{Ind}_\pi C$ admettent une représentation indicielle

$$\psi \ : \ I \times J \longrightarrow C \ ,$$

avec I grande devant π. Et on peut toujours supposer, quitte à prendre $I' \longrightarrow I$ cofinal avec I' ensemble ordonné grand devant π, que I est de plus un ensemble ordonné.

Commentary · Examples [ms. 94]

An ordinary ind-object whose action has no stage

The doubt on page 94 concerns the loss of a cardinal bound when $\mathrm{Ind}_\pi(C)$ is replaced by the ordinary ind-completion. It has a concrete negative answer. Fix a small skeleton C of the category of at most countable sets. This category has all countable colimits, so it satisfies the hypothesis under discussion with $\pi = \aleph_0$. In $\widehat{C}$, define

$$F(c) = \{f : c \longrightarrow \mathbb{Z} \mid f(c) \text{ is finite}\}.$$

The presheaf F is an ordinary ind-object. Indeed, the finite subsets $T \subset \mathbb{Z}$ form a filtered poset, and

$$F = \varinjlim_{T \subset \mathbb{Z},\ T \text{ finite}} y(T)$$

pointwise: a map lies in the displayed colimit exactly when its image is contained in one finite T. Translation on $\mathbb{Z}$ defines an action on F, hence a diagram $\varphi : B\mathbb{Z} \to \mathrm{Ind}'(C)$.

An object of $C^{B\mathbb{Z}}/\varphi$ consists of a countable $\mathbb{Z}$-set c and an equivariant map of presheaves $y(c) \to F$. By Yoneda, the map corresponds to a function $f : c \to \mathbb{Z}$ with finite image. Equivariance says

$$f(n \cdot x) = n + f(x) \qquad (n \in \mathbb{Z}).$$

If c were nonempty, the image of f would therefore be a nonempty finite subset of $\mathbb{Z}$ invariant under every translation. Taking any element in that image produces all its translates, a contradiction. Thus every such c is empty.

Evaluation at the unique object of $B\mathbb{Z}$ consequently has image only the initial presentation $y(\varnothing) \to F$. In C/F, the element $0 \in F(1)$ defines a presentation $y(1) \to F$, and there is no morphism from this presentation to $y(\varnothing) \to F$. The corresponding comma category for evaluation is empty. Evaluation is therefore not final, exactly as Grothendieck suspected.

The example also locates the obstruction. At $c = \mathbb{N}$, the identity-shaped enumeration of an infinite set cannot be represented at a finite stage. Accordingly, F does not send a countable coproduct of singletons to the corresponding product: its value there consists of finite-image sequences. It is not an object of $\mathrm{Ind}'_{\aleph_0} C$. The stronger filteredness in the positive theorem is what prevents this failure of simultaneous realization. The distinction between a formal ind-object and the colimit of its values already appeared in the commentary on Chapter VII, manuscript p. 123 ("A colimit need not be the same ind-object"); here it obstructs a specific comparison of diagram categories.

Commentary · Later developments [ms. 94]

The modern comparison for diagram categories

Grothendieck's simultaneous presentation is the essential-surjectivity part of a familiar modern comparison. With $\kappa = \pi^+$ and J having fewer than κ arrows, it gives

$$\mathrm{Ind}_\kappa(C^J) \longrightarrow \mathrm{Ind}_\kappa(C)^J$$

when C has κ-small colimits. Here Ind_κ uses the modern regular-cardinal convention, so it is Grothendieck's Ind_π, not his ordinary Ind.

Henry (2025a, Theorem 1.2 and Corollary 2.2) proves that this comparison is an equivalence under precisely these hypotheses, and characterizes the diagram shapes for which the statement holds uniformly in C. The paper also shows why removing the colimit hypothesis on C requires additional restrictions on J; its Theorem 1.3 treats the corresponding well-founded shapes. These are results about the chosen presentability rank and about realizing diagrams by stages. They should be distinguished from the mere existence of some accessibility rank for a functor category. We use the theorem numbering of the checked arXiv version 2, dated 9 October 2024; the published article appeared in 2025 under the title cited in the bibliography.

C'est là le moyen donc, pour la démonstration du

Théorème 4.4.13. *Soit C une catégorie équivalente à une petite catégorie, et stable par $\varinjlim$ de cardinal $\leqslant \pi$, où π est un cardinal infini donné. Alors :* [page 96]

a) (Pour mémoire.) Le foncteur canonique

$$C \longrightarrow \mathrm{Ind}_\pi(C) \xrightarrow{\sim} \mathrm{Ind}'_\pi(C)$$

commute aux $\varinjlim$ *de cardinal* $\leqslant \pi$.

b) (Pour mémoire.) Les systèmes inductifs de cardinal $\leqslant \pi$ *dans* $\mathrm{Ind}_\pi(C)$ *(i.e. les foncteurs* $\varphi : J \longrightarrow \mathrm{Ind}_\pi(C)$*, avec* $\mathrm{card}\,\mathrm{Fl}(J) \leqslant \pi$*) admettent une représentation indicielle*

$$\psi \ : \ I \times J \longrightarrow C \ ,$$

avec I *ensemble ordonné grand devant* π*, et admettent une limite inductive dans* $\mathrm{Ind}_\pi C$ *qui se calculent par*

$$\underbrace{\varinjlim_{j\in J} \varphi(j)}_{\text{dans } \mathrm{Ind}_\pi(C)} \quad \simeq \quad \text{“} \varinjlim_{i\in I} \text{”}\Psi(i)$$

avec

$$\Psi(i) \ = \ \varinjlim_{j\in J}{}_C \, \psi(i,j) \ .$$

p. 24

c) $\mathrm{Ind}_\pi C$ *est stable par petites limites inductives* ([21]).

Il reste à établir ce dernier point. Comme on a déjà les $\varinjlim$ finies par b, il reste à prouver l'existence des petites $\varinjlim$ filtrantes, ou au choix des sommes infinies, ce qui sera plus joli en fait. Mais on pourra aussi bien traiter le cas général d'une $\varinjlim$ de type J quelconque (J petite). Pour ceci, soit [page 97]

$$I \ = \ I(J) \ = \ \text{ensemble des sous-catégories } J' \text{ de } J \text{ telles que } \mathrm{card}\,\mathrm{Fl}(J') \leqslant \pi,$$

ordonnons I par inclusion. Il est évident que $I \neq \emptyset$ (car $\emptyset \in I$), et même que I *est grand devant* π, car si $(J'_\alpha)_{\alpha\in A}$ avec $\mathrm{card}(A) \leqslant \pi$ est une famille d'éléments de I, alors la sous-catégorie de J engendrée par les J'_α est encore dans I (vérification immédiate), et majore les J'_α. En fait, la considération de $I(J)$ associée à J va nous permettre de prouver ceci, qui généralise 4.4.13 c), compte tenu de b) :

Corollaire 4.4.14 ([22]). *Soit* $\mathcal{M}$ *une catégorie,* π *un cardinal infini tel que*

a) $\mathcal{M}$ *est stable par* $\varinjlim$ *de cardinal* $\leqslant \pi$*, et*

b) $\mathcal{M}$ *est stable par* $\varinjlim$ *sur les ensembles ordonnés* I *grands devant* π. [page 98]

Alors $\mathcal{M}$ *est stable par petites* $\varinjlim$.

([23]).

En effet, si $\varphi : J \longrightarrow \mathcal{M}$ est donné, on en construit

$$\psi \ : \ I \ = \ I(J) \ \longrightarrow \ \mathcal{M}$$

par

$$\psi(J') \ = \ \varinjlim_{J'} \varphi(J') \ ,$$

qui existe par hypothèse sur $\mathcal{M}$. Il est alors immédiat que $\varinjlim_J \varphi$ existe si et seulement si $\varinjlim_I \psi$ existe, or cette existence est assurée du fait que I est grand devant π, et l'hypothèse sur $\mathcal{M}$.

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21. rajouter aussi, en d), le corollaire 4.4.16, page 101.

22. Comparer aussi 4.11.3, page 270.

23. **Corollaire 4.4.14.1.** $\mathcal{M}$ étant comme dans 4.4.14, et $f : \mathcal{M} \longrightarrow \mathcal{E}$ étant un foncteur qui commute aux $\varinjlim$ de cardinal $\leqslant \pi$, et aux petites $\varinjlim$ grandes devant π, alors f commute aux petites $\varinjlim$.

Commentary · Exposition [ms. 98]

Why bounded colimits suffice for cocompleteness

The passage from bounded colimits to arbitrary small colimits is the structural conclusion of this section. Its proof uses a directed family of portions of the indexing category, so it concerns the diagram and its relations together. A family of subsets of objects would not retain the equations imposed by its arrows.

For a small category J, let $\mathcal{I}_\pi(J)$ be the poset of its subcategories with at most π arrows, with the empty subcategory included. A union of at most π such arrow sets still has size at most π. Closing it under identities and finite compositions does not increase this bound, because π is infinite. The resulting subcategory is an upper bound. Thus $\mathcal{I}_\pi(J)$ is π^+-directed.

Given $X : J \to \mathcal{M}$, suppose that the colimit $X_K = \varinjlim_K X$ exists for every $K \in \mathcal{I}_\pi(J)$. These objects form a diagram indexed by $\mathcal{I}_\pi(J)$. We verify directly that its colimit is the colimit of X. A compatible family of maps $X_K \to Z$ gives a map $X(j) \to Z$ by using the subcategory containing only the identity of j. For an arrow $a : j \to j'$, choose a subcategory containing a and its two objects. Its cocone forces the required equality between the two maps to Z. Conversely, a cocone on X restricts to every K, hence factors uniquely through every X_K, and the resulting maps are compatible with inclusions. These two constructions are inverse and natural in Z. Therefore

$$\varinjlim_J X \simeq \varinjlim_{K \in \mathcal{I}_\pi(J)} \left(\varinjlim_K X \right)$$

whenever the right side exists. This proves Corollary 4.4.14 without an interchange of limits whose existence has not yet been established.

The same cocone argument proves the preservation statement in footnote 27. If a functor preserves the inner colimits and the outer directed colimit, it preserves the colimit of X. No hypothesis about exactness, generators or representability is used. This makes the result useful beyond the particular ind-completion being constructed.

There is a size distinction that becomes consequential on page 110. If $|\operatorname{Fl} J| \leq \rho$, then the set $\mathcal{I}_\pi(J)$ can have size ρ^π, rather than ρ. Thus the same decomposition proves a bound-preserving assertion only if this extra exponent has been controlled, or if a different argument supplies a smaller final indexing family. The unbounded cocompleteness theorem is sound; it does not by itself establish the later assertion that a pointwise cardinal bound survives every bounded colimit.

Pour achever de prouver le théorème 4.4.13, on est donc ramené au cas particulier des limites inductives $\varinjlim_J \varphi(j)$, quand $\varphi : J \longrightarrow \operatorname{Ind}'_\pi C$ est donné, avec J ensemble ordonné grand devant π. Mais ceci va résulter du résultat plus précis : p. 25 [page 99]

Corollaire 4.4.15 ([24]). *Soit C une petite catégorie. Alors la sous-catégorie strictement pleine $\operatorname{Ind}'_\pi C$ de $C^\wedge$ est stable dans $C^\wedge$ par $\varinjlim$ grandes devant π, en d'autres termes, ces $\varinjlim$ existent dans $\operatorname{Ind}'_\pi(C)$, et le foncteur d'inclusion y commute.*

Démonstration. Soit I ensemble ordonné, grand devant π, et $i \longmapsto F_i$ un foncteur $I \longrightarrow \operatorname{Ind}'_\pi(C)$, $F = \varinjlim_I F_i$ dans $C^\wedge$, prouvons que $F \in \operatorname{Ob} \operatorname{Ind}'_\pi(C)$, i.e. (4.4.3) que C/F est grande devant π. J'utilise le critère b) de 4.4.4, il faut vérifier $\mathrm{PF}_\pi 1$, $\mathrm{PF}_\pi 2$. Soient

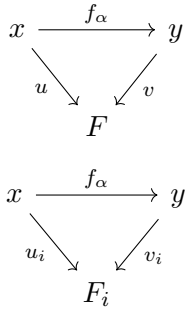
$$x_\alpha \xrightarrow{u_\alpha} F = \varinjlim F_i$$

des objets de C/F indexés par A avec $\operatorname{card} A \leqslant \pi$. Chaque u_α provient de $x_\alpha \longrightarrow F_{i_\alpha}$, et en prenant un majorant commun des i_α (I grand devant π), on trouve un $i \in I$ et des $v_\alpha : x_\alpha \longrightarrow F_i$. Comme F_i est dans $\operatorname{Ind}'_\pi(C)$, i.e. C/F_i grande devant π, considérant les v_α comme des objets de C/F_i, ils sont majorés par un objet $x \longrightarrow F_i$ de C/F_i. On a donc

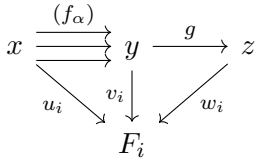
$$\begin{array}{ccccc} x_\alpha & \longrightarrow & F_{i_\alpha} & & \\ \downarrow & & \downarrow & \searrow & \\ x & \longrightarrow & F_i & \longrightarrow & F\,, \end{array}$$

24. comparer à 4.7.3.1, qu'il y aurait lieu d'expliciter ici.

qui montre que les $x_\alpha \longrightarrow F$ sont majorés par $x \longrightarrow F$, ce qui prouve $\mathrm{PF}_\pi\, 1$. Prouvons $\mathrm{PF}_\pi\, 2$, [page 100]
i.e. soient $x \xrightarrow{u} F$, $y \xrightarrow{v} F$ deux objets de C/F, et $(f_\alpha)_{\alpha\in A}$ une famille de cardinal $\leqslant \pi$ de flèches de u dans v. On sait que u, v proviennent de $u_i : x \longrightarrow F_i$, $v_i : y \longrightarrow F_i$ pour i assez grand.



Pour tout α, on peut choisir $i_\alpha \geqslant i$ tel que $v_i f_\alpha$ et u_i deviennent égaux en composant avec $F_i \longrightarrow F_{i_\alpha}$, et comme $\operatorname{card} A \leqslant \pi$ et I grand devant π, il existe un majorant i' des i_α. Au total, on peut supposer (remplaçant i par i') que $v_i f_\alpha = u_i$ pour tout α. Alors u_i, v_i étant considérés comme objets de C/F_i, les f_α sont des flèches de C/F_i de u_i dans v_i. Comme C/F_i est grand devant π (par l'hypothèse $F_i \in \mathrm{Ob}\, \mathrm{Ind}'_\pi(C)$), on trouve qu'il existe un égalisateur $(y, v_i) \xrightarrow{g} (z, w_i)$. p. 26



Alors l'image de g dans C/F (via $F_i \longrightarrow F$) égalise les flèches f_α, en tant que flèches de C/F, ce qui prouve $\mathrm{PF}_\pi\, 2$ et achève la démonstration de 4.4.14, et par là aussi celle du théorème 4.4.13, et du lemme 4.3. Par là est aussi achevée la démonstration de la première assertion du lemme 4.3.7 (p. 77), ce qui achève la démonstration du théorème 4.3.6 (p. 70-71). [page 101]
Il reste à établir la propriété universelle énoncée dans 4.3.7, que je vais rappeler :

Commentary · Exposition [ms. 100]

Filtered closure and its exact strength

Grothendieck now proves the closure property that footnote 24 had correctly distinguished from the definition of an ind-object. A single κ-filtered colimit of representables belongs to $\mathrm{Ind}_\kappa C$ by definition. The closure assertion says that a κ-filtered colimit of objects already in this category belongs to it again. It is this second assertion that allows arbitrary cocompleteness to be deduced from the preceding decomposition.

Here is the argument in a form that displays the two uses of the same cardinal. Let $\kappa = \pi^+$, let I be κ-filtered, and let $F = \varinjlim_i F_i$ in $\widehat{C}$, with every category of elements C/F_i κ-filtered. Replace I by a final κ-directed poset if desired. Given fewer than κ objects of C/F, choose representatives of their defining elements at stages of I. A common upper bound places all of them in one C/F_i. Filteredness of that category supplies a common target, whose image is the desired target in C/F.

For the second filteredness condition, start with fewer than κ parallel arrows between two objects of C/F. Place the two defining elements in one F_i. Each equality saying that an arrow is over F becomes true at some later stage, because equality in a filtered colimit of sets is eventual equality. A common upper bound makes all these equations true together. The corresponding arrows now lie in $C/F_{i'}$, where they can be simultaneously coequalized. Their images coequalize the original arrows. This proves both filteredness conditions and hence $F \in \mathrm{Ind}_\kappa C$.

The word "*égalisateur*" on page 100 denotes a morphism that makes the given arrows equal after composition. No universal coequalizer is asserted or needed. Similarly, no colimits in C are used in this closure proof. The hypothesis that C has κ-small colimits belongs to the cocompleteness theorem, not to Corollary 4.4.15 itself.

The inclusion into $\widehat{C}$ creates these κ-filtered colimits, but generally does not preserve all ordinary filtered colimits. For instance, let C again consist of countable sets and take $\pi = \aleph_0$. The completion

$\mathrm{Ind}'_\pi C$ identifies with sets via $X \mapsto (c \mapsto X^c)$: every set is the $\aleph_1$-filtered union of its countable subsets, and a map out of a countable set factors through one such subset. The increasing union of finite subsets of $\mathbb{N}$ has colimit $\mathbb{N}$ in this completion. Its colimit in $\widehat{C}$, however, contains only finite-image maps $c \to \mathbb{N}$. At $c = \mathbb{N}$ it omits the identity map. This is the precise failure used in the preceding example and the reason the proof of Corollary 4.4.16 must retain the larger filteredness threshold.

Corollaire 4.4.16 ([25]). *Soient C une catégorie équivalente à une petite catégorie, stable par $\varinjlim$ de cardinal $\leqslant \pi$, où π est un cardinal infini donné, et considérons le foncteur d'inclusion canonique*

$$C \xrightarrow{\alpha} \mathrm{Ind}_\pi(C)\ . \tag{4.4.16.1}$$

Ce foncteur est 2-universel pour les foncteurs $f : C \longrightarrow \mathcal{E}$, où $\mathcal{E}$ est une $\mathcal{U}$-catégorie stable par petites $\varinjlim$, et f un foncteur commutant à ces limites inductives. En d'autres termes, on a

1°) α possède les propriétés précédentes, i.e. $\mathrm{Ind}_\pi(C)$ est stable par $\varinjlim$, et le foncteur α commute aux $\varinjlim$ de cardinal $\leqslant \pi$.

2°) Pour toute $\mathfrak{U}$-catégorie $\mathcal{E}$ stable par petites limites inductives, le foncteur

$$\begin{array}{ccc} g & \longmapsto & g \circ \alpha \\ \underline{\mathrm{Hom}}_!(\mathrm{Ind}_\pi C, \mathcal{E}) & \longrightarrow & \underline{\mathrm{Hom}}_{!\,\pi}(C, \mathcal{E}) \end{array} \tag{4.4.16.2}$$

est une équivalence de catégories, où l'indice $!$ (resp. $!\,\pi$) indique la sous-catégorie strictement pleine de la catégorie $\underline{\mathrm{Hom}}$ envisagée, formée des foncteurs qui commutent aux petites $\varinjlim$ (resp. à celles de cardinal $\leqslant \pi$). [page 102]

On peut donc dire aussi que $\mathrm{Ind}_\pi(C)$ joue le rôle d'une "enveloppe" stable par petites $\varinjlim$ de la catégorie C, quand les $\varinjlim$ de cardinal $\leqslant \pi$, lesquelles existent déjà dans C, sont respectées par les plongement de C dans son enveloppe. C'est un résultat nullement tautologique qu'il suffit pour cela d'ajouter à C des $\varinjlim$ *grandes devant* π d'objets de C, et que plus précisément, la catégorie enveloppe est réalisée par $\mathrm{Ind}_\pi C$ (ou, au choix et moins pléthoriquement, par $\mathrm{Ind}'_\pi C \subset C^\wedge$).

Commentary · Exposition [ms. 102]

Density determines the continuous extensions

The universal property is the main result of the construction: the passage from C to $\mathrm{Ind}_\pi C$ freely supplies all small colimits while preserving the colimits of size at most π that were already present. There is a local error in the proof that should be separated from this correct theorem. The unrestricted restriction functor displayed across pages 102–103 is not fully faithful. What is fully faithful is its restriction to functors preserving the canonical presentations by objects of C.

For the required version, write $\mathcal{M} = \mathrm{Ind}'_\pi C$ and use a small skeleton of C. Each $F \in \mathcal{M}$ has its canonical presentation

$$F \simeq \varinjlim_{(c,u) \in C/F} y(c).$$

If $g, h : \mathcal{M} \to \mathcal{E}$ preserve these colimits, a natural transformation $\theta : gy \to hy$ extends at F by taking the colimit of its components along C/F. Naturality for $F \to F'$ follows by postcomposition $C/F \to C/F'$ and the defining cocones. The extension is unique because maps out of the displayed colimit are determined by their composites with its structure maps. On representables it restricts to θ, since $C/y(c)$ has its terminal object (c, id). This proves full faithfulness on the required functor categories. It is the precise density statement of SGA 4, Exposé I, Proposition 7.8 (Artin et al., 1972). Cocontinuous functors satisfy its hypothesis; so do functors preserving the relevant π^+-filtered presentations.

25. Pourrait être incorporé en partie d) au théorème 4.4.13, car c'est un complément important, et pas tellement évident finalement.

The failure without the preservation hypothesis is visible in sets. Let C be a small skeleton of sets of cardinal at most an infinite π, so that $\mathrm{Ind}'_\pi C \simeq \mathrm{Set}$. Define a functor $Q : \mathrm{Set} \to \mathrm{Set}$ by taking $\mathcal{P}(X)$ and identifying all subsets of cardinal at most π to a single distinguished element $*$. Direct image under a function is well-defined on this quotient: the image of a subset of cardinal at most π still has cardinal at most π. For every $X \in C$, $Q(X)$ is a singleton. The identity transformation of Q and the transformation that sends every value to $*$ therefore have identical restrictions to C. They are different on any X with $|X| > \pi$, since the class of X is then different from $*$. Restriction is not even faithful on arbitrary functors.

This example explains why strict generation cannot justify the unrestricted assertion. A generating family detects morphisms of objects; density expresses objects as particular colimits; a functor must respect those colimits for its values and transformations to be reconstructed from the generators. The source uses the correct restricted conclusion throughout the rest of the proof. Replacing the one unrestricted assertion by the density argument above repairs the proof without changing Corollary 4.4.16.

Démonstration. Le 1°) est contenu dans 4.4.13 a) et c). Prouvons donc 2°). Il est évident que α est un foncteur pleinement fidèle dont l'image essentielle engendre strictement $\mathcal{M} = \mathrm{Ind}_\pi C$, d'où résulte que le foncteur (4.4.16.2) est pleinement fidèle, étant induit par le foncteur pleinement fidèle p. 27

$$\alpha^*_{\mathcal{E}} \ : \ \underline{\mathrm{Hom}}(\mathcal{M}, \mathcal{E}) \ \longrightarrow \ \underline{\mathrm{Hom}}(C, \mathcal{E})$$

sur les sous-catégorie pleines envisagées. Il reste donc à montrer que (4.4.16.2) est essentiellement surjectif. Soit donc $f : C \longrightarrow \mathcal{E}$ dans $\underline{\mathrm{Hom}}_{!\,\pi}(C, \mathcal{E})$, considérons son extension canonique [page 103]

$$\text{(4.4.16.3)} \qquad \begin{cases} Lf \ : \ \mathrm{Ind}(C) \ \longrightarrow \ \mathcal{E} \\ Lf((X_i)_{i\in I}) \ = \ \varinjlim_{i\in I} f(X_i) \end{cases}$$

(cf. SGA 4 I 8.7.3), et soit

$$\text{(4.4.16.4)} \qquad \psi(f) \ = \ Lf|\,\mathrm{Ind}_\pi(C) \ .$$

Rappelons que Lf commute aux petites $\varinjlim$ filtrantes, mais comme le foncteur d'inclusion

$$\text{(4.4.16.5)} \qquad \mathrm{Ind}_\pi C \ \longrightarrow \ \mathrm{Ind}(C)$$

ne commute pas à ces $\varinjlim$, il n'est pas clair a priori que $\psi(f)$ commute auxdites limites, ni a fortiori qu'il commute aux petites $\varinjlim$ sans plus. Mais on a vu que (4.4.16.5) commute aux $\varinjlim$ *grandes devant* π, ce qui est contenu dans le corollaire 4.4.15 (vu que les limites inductives *filtrantes* dans $\mathrm{Ind}(C)$ sont les limites *dans* $C^\wedge$) : dire que $\mathrm{Ind}'_\pi(C) \longrightarrow \mathrm{Ind}'(C)$ commute à un certain type de $\varinjlim$ filtrantes, revient à dire qu'il en est ainsi du composé avec $\mathrm{Ind}'_\pi(C) \hookrightarrow C^\wedge$, ce qui est établi par 4.4.15. De ceci résulte que $\psi(f)$ *commute aux* $\varinjlim$ *grandes devant* π. Pour établir que $g = \psi(f)$ commute aux petites $\varinjlim$, il reste donc à voir qu'il commute aux $\varinjlim$ de cardinal $\leqslant \pi$, en vertu de 4.4.14.1 (marge page 98). Pour l'établir, on utilise le théorème 4.4.13 b, en représentant un foncteur $\varphi : J \longrightarrow \mathrm{Ind}_\pi(C)$ (où $\mathrm{card}\,\mathrm{Fl}\,J \leqslant \pi$) sous forme indicielle par [page 104]

$$\text{(4.4.16.6)} \qquad \Phi \ : \ I \times J \ \longrightarrow \ C \ ,$$

I étant un ensemble ordonné grand devant π. On a donc

$$\varphi(j) \ = \ \varinjlim_{i\in I} {}_{\mathrm{Ind}_\pi C}\, \Phi(i, j) \ ,$$

et comme $g = \psi(f)$ commute aux $\varinjlim$ de type I par ce qu'on a vu, on aura

$$g(\varphi(j)) \ = \ \varinjlim_{i\in I} {}_{(\mathcal{E})}\, g(\Phi(i, j)) \ ,$$

d'où

$$(*) \qquad \varinjlim_{j\in J}{}_{(\mathcal{E})}\, g(\varphi(j)) \;=\; \varinjlim_{j\in J} \varinjlim_{i\in I} \underbrace{g(\Phi(i,j))}_{=f(\Phi(i,j))} \qquad \text{(limites dans } \mathcal{E})\,.$$

p. 28

D'autre part, on a

$$\begin{aligned} \varinjlim_{j\in J} \operatorname{Ind}_\pi \varphi(j) \;&\simeq\; \varinjlim_{j\in J} \varinjlim_{i\in I} \Phi(i,j) \\ &\simeq\; \varinjlim_{i\in I} \underbrace{\varinjlim_{j\in J} \Phi(i,j)}_{\substack{\text{peut être calculé}\\ \text{dans } C\text{, soit}\\ \Psi(i)\in \operatorname{Ob} C}}\,, \end{aligned}$$

donc

$$(**) \qquad \varinjlim_{j\in J} \operatorname{Ind}_\pi \varphi(j) \simeq \varinjlim_{i\in I} \Psi(i)\,,$$

d'où en appliquant g, qui commute aux $\varinjlim$ de type I, et se rappelant maintenant que $g|C \simeq f$, [page 105]

$$(***) \qquad g(\varinjlim_{J} \operatorname{Ind}_{\pi\, C} \varphi(j)) \;\simeq\; \varinjlim_{i\in I} f(\Psi(i))\,.$$

On a d'autre part

$$\Psi(i) \overset{\text{déf}}{=} \varinjlim_{j\in J} \Phi(i,j)\,, \qquad\qquad \varinjlim \text{ dans } C\,,$$

et comme f commute aux $\varinjlim$ de type J (car $\operatorname{card}\operatorname{Fl} J \leqslant \pi$), on en conclut

$$f(\Psi(i)) \;\simeq\; \varinjlim_{j\in J} f(\Phi(i,j))\,,$$

donc par $(***)$

$$(****) \qquad g(\varinjlim_{j\in J} \varphi(j)) \;\simeq\; \varinjlim_{i\in I} \varinjlim_{j\in J} f(\Phi(i,j))\,.$$

En composant avec $(*)$ et utilisant à nouveau la commutativité de $\varinjlim_I$ à $\varinjlim_J$, on trouve donc

$$g(\varinjlim \varphi(j)) \;\xleftarrow{\sim}\; \varinjlim g(\varphi(j))\,,$$

ce qui prouve que g commute aux limites inductives de type J, et achève de prouver que g est [dans] $\underline{\operatorname{Hom}}_!(\operatorname{Ind}_\pi(C),\mathcal{E})$. Comme $g|C \simeq f$, on en conclut que le foncteur (4.4.16.2) est essentiellement surjectif, ce qui achève de prouver 4.4.16. De plus, on a construit un foncteur quasi-inverse explicite, ψ, s'insérant dans le diagramme commutatif [page 106]

(4.4.16.6)
$$\begin{array}{ccccc} \underline{\operatorname{Hom}}_{!\,\pi}(C,\mathcal{E}) & \xrightarrow[(\sim)]{\psi} & \underline{\operatorname{Hom}}_!(\operatorname{Ind}_\pi(C),\mathcal{E}) & \longrightarrow & \underline{\operatorname{Hom}}_{!\,\pi}(\operatorname{Ind}_\pi(C),\mathcal{E}) \\ \Big\downarrow{\scriptstyle \text{incl. pl. fid.}} & & \Big\vdots{\scriptstyle \text{pl. fid.}} & \nearrow{\scriptstyle R\ \text{(restriction)}} & \\ \underset{\text{(cf. SGA 4 I 8.7.3)}}{\underline{\operatorname{Hom}}(C,\mathcal{E})} & \xrightarrow[\sim]{L} & \underline{\operatorname{Hom}}_{!\,\operatorname{filt}}(\operatorname{Ind}(C),\mathcal{E}) & & \end{array}\,.$$

Commentary · Exposition [ms. 106]

The extension and the codomain of restriction

The explicit inverse constructed here makes the universal property usable. For a functor $f : C \to \mathcal{E}$ preserving colimits of cardinal at most π, put

$$\overline{f}(F) = \varinjlim_{(c,u)\in C/F} f(c).$$

This is the restriction of the usual extension Lf on the ordinary ind-completion. The latter is characterized by preservation of filtered colimits, as in SGA 4, Exposé I, Proposition 8.7.3 (Artin et al., 1972). It follows from Corollary 4.4.15 that $\overline{f}$ preserves π^+-filtered colimits. The proof on pages 104–105 supplies the remaining preservation property rather than assuming it from the inclusion into $\mathrm{Ind}(C)$.

To see the mechanism, present a diagram $X : J \to \mathrm{Ind}_\pi C$, with $|\operatorname{Fl} J| \le \pi$, by $c : I \times J \to C$, where I is π^+-filtered. Put $s_i = \varinjlim_{j\in J} c(i,j)$ in C. Then all the indicated colimits exist, and the two comparisons are

$$\overline{f}\left(\varinjlim_j X_j\right) \simeq \varinjlim_i f(s_i) \simeq \varinjlim_i \varinjlim_j f(c(i,j)),$$
$$\varinjlim_j \overline{f}(X_j) \simeq \varinjlim_j \varinjlim_i f(c(i,j)).$$

Both are the colimit in $\mathcal{E}$ of the same diagram indexed by $I \times J$. The isomorphism between them is therefore the actual comparison induced by the colimit cocone, not merely an unspecified isomorphism between two objects. Corollary 4.4.14.1 then gives preservation of every small colimit. Natural transformations extend by the same construction, so this proves the asserted equivalence of categories, including its two-dimensional content.

There is a second, independent typing error in the final diagram. Its arrow R, on the whole bottom-middle category, need not land in $\underline{\mathrm{Hom}}_{!\,\pi}(\mathrm{Ind}_\pi C, \mathcal{E})$. A constant singleton functor to sets preserves all nonempty filtered colimits, but it does not preserve the initial object. It is therefore an object of the stated source of R whose restriction is not in the stated target. One may repair the diagram by replacing that target with the full functor category, or by restricting the source of R to the extensions of functors that preserve the specified colimits in C. The dotted downward arrow remains meaningful: it is the fully faithful functor $g \mapsto L(g|_C)$ supplied by the two universal properties. These qualifications preserve the explicit inverse ψ while making every arrow of the diagram well-typed.

p. 29

4.5 Filtration cardinale canonique d'une catégorie $\mathfrak{U}$-paratopos

Définition 4.5.1. On appelle *catégorie* $\mathfrak{U}$*-paratopos* une catégorie $\mathcal{M}$ $\mathfrak{U}$-pseudo-topos et accessible, i.e. qui satisfait aux conditions équivalentes du théorème 4.3.6 (p. 70). On dit que c'est une *catégorie* π-$\mathfrak{U}$*-paratopos*, si elle est π-accessible, i.e. si elle admet une petite sous-catégorie génératrice dont les objets soient π-accessible, ou ce qui revient au même, si $\mathcal{M}$ est équivalente à une catégorie de la forme $\mathrm{Ind}'_\pi(\mathcal{M}_\pi)$, où $\mathcal{M}_\pi$ est une petite catégorie (ou : une catégorie équivalente à une petite catégorie), et stable par $\varinjlim$ de cardinal $\leqslant \pi$. (Et on peut prendre pour $\mathcal{M}_\pi$ la sous-catégorie pleine de $\mathcal{M}$ formée des objets π-accessibles). [page 107]

Dans la suite, nous allons, plus généralement, considérer une catégorie C équivalente à une petite catégorie, stable par $\varinjlim$ de cardinal $\leqslant \pi_0$ ([30]), et la catégorie

$$\mathcal{M} \overset{\text{déf}}{=} \mathrm{Ind}'_{\pi_0} C \ \subset\ C^\wedge\ , \tag{4.5.1}$$

où π_0 est un cardinal infini donné. Je me propose d'introduire sur $\mathcal{M}$ une filtration cardinale canonique :

30. Il n'y a pas lieu de le supposer tout de suite.

Proposition 4.5.2 ([31]). *Soit $F \in \mathrm{Ob}\,\mathcal{M} \subset \mathrm{Ob}\,C^\wedge$, et soit π un cardinal $\geqslant \pi_1 = \sup(\pi_0, \mathrm{card}\,\mathrm{Fl}\,C_0)$, où C_0 est une petite sous-catégorie pleine de C telle que $C_0 \overset{\sim}{\hookrightarrow} C$. Les conditions suivantes sont équivalentes :*

a) $\mathrm{card}\,\mathrm{Fl}(C_0/F) \leqslant \pi$.

a′) $\mathrm{card}\,\mathrm{Ob}\ (C_0/F) \leqslant \pi$.

a″) Pour tout x dans C_0, $\mathrm{card}\,F(x) \leqslant \pi$.

b) Il existe une petite catégorie filtrante I (pas nécessairement ordonnée), grande devant *π_0, avec* $\mathrm{card}\,I \leqslant \pi$, *et un système inductif $(x_i)_{i\in I}$ dans C, tel que l'on ait*

$$F \ \simeq\ \varinjlim_{i\in I} x_i\ , \tag{4.5.2.1}$$

où la $\varinjlim$ est prise dans $\mathcal{M}$, ou ce qui revient au même (en vertu de 4.4.15, I étant grande devant π_0) dans $C^\wedge$, i.e. F est isomorphe au foncteur ind-représentable défini par l'ind-objet $\underline{x} = (x_i)_{i\in I}$ de C.

c) F est π-accessible dans $\mathcal{M}$.

c′) F est π-accessible dans $C^\wedge$.

Démonstration. On a les implications ± tautologiques [page 108]

(∗)
$$\begin{array}{ccccc} & & \mathrm{a}' & \Longrightarrow & \mathrm{a}'' \\ \mathrm{a} & \Longrightarrow & & & \\ & & \mathrm{b} & \Longrightarrow & \mathrm{c}' \Longrightarrow \mathrm{c}\ , \end{array}$$

où $\mathrm{a}' \Longrightarrow \mathrm{a}''$ provient de l'inclusion canonique $F(x) = \mathrm{Hom}_{\mathcal{M}}(x, F) \hookrightarrow \mathrm{Ob}\,C/F$, $\mathrm{a} \Longrightarrow \mathrm{b}$ p. 30 s'obtient en prenant $I = C_0/F$ et notant que I est stable par limites inductives de cardinal $\leqslant \pi_0$ (C l'étant), donc est grande devant π_0 ([32]). On a $\mathrm{b} \Longrightarrow \mathrm{c}'$ en vertu du fait que la sous-catégorie strictement pleine $C^\wedge_\pi$ de $C^\wedge$ définie par

$$\left\{ \begin{array}{lcl} C^\wedge_\pi & = & \text{sous-catégorie strictement pleine de } C^\wedge \text{ formée des} \\ & & F \in \mathcal{M} \text{ qui sont } \pi\text{-accessibles.} \end{array} \right. \tag{4.5.2.2}$$

est stable par $\varinjlim$ de cardinal $\leqslant \pi$ (SGA 4 I 9.9). Enfin $\mathrm{c}' \Longrightarrow \mathrm{c}$ provient du fait que le foncteur d'inclusion $\mathcal{M} \hookrightarrow C^\wedge$ est pleinement fidèle et π_0-accessible, i.e. commute aux limites inductives grandes devant π_0, et a fortiori à celles grandes devant π. Par les implications du diagramme ci-dessus, il suffit donc de prouver qu'on a

$$\mathrm{c} \overset{(33)}{\Longrightarrow} a \qquad \text{et} \qquad \mathrm{a}'' \Longrightarrow \mathrm{a}\ .$$

Commentary · Exposition [ms. 108]

Two meanings of the size of a presheaf

The new section asks whether the intrinsic presentability rank of an object can be read from the size of its canonical presentation. This would turn the generators used to construct a paratopos into an effective cardinal filtration. The potential discrepancy is already visible in Proposition 4.5.2: presentability in $\mathcal{M}$ and presentability in the ambient presheaf category test different colimits.

31. N'est prouvé (et n'est vrai) que si $\pi^{\pi_0} = \pi$, voir rectification 4.5.6, page 116.

32. On n'a pas besoin de cette stabilité, de toutes façons C_0/F est grande devant π_0, en vertu de 4.4.3, p. 80.

33. n'est prouvé (et n'est vrai) que si $\pi^{\pi_0} = \pi$.

Replace C by the small category C_0 in the statement, and put $\theta = \pi_0$, $\mathcal{M} = \mathrm{Ind}_{\theta^+} C$, and $\kappa = \pi^+$, with $\pi \geq \max(\theta, |\operatorname{Fl} C|)$. The following equivalences require no exponentiation hypothesis:

$$|\operatorname{Fl}(C/F)| \leq \pi \iff |\operatorname{Ob}(C/F)| \leq \pi \iff \forall c \in C, |F(c)| \leq \pi,$$
$$\iff F \text{ has a } \theta^+\text{-filtered presentation with at most } \pi \text{ arrows},$$
$$\iff F \text{ is } \kappa\text{-presentable in } \widehat{C}.$$

We give the details, since they isolate the part of the proposition which survives unchanged.

The objects of C/F are pairs (c, a) with $a \in F(c)$, so their number is bounded by π exactly when all the values are bounded by π. A morphism of C/F is determined by a morphism $a : c \to d$ of C and an element of $F(d)$; its source element is then forced. Consequently

$$|\operatorname{Fl}(C/F)| = \sum_{a:c\to d} |F(d)| \leq \pi \cdot \pi = \pi.$$

The reverse bound follows from identity arrows. If these bounds hold, the category C/F itself is the required θ^+-filtered presentation. Conversely, a presentation with at most π arrows has values bounded by π: every representable has that bound, and a colimit of at most π sets of that size again has that size.

Such a presentation also makes F κ-presentable in $\widehat{C}$. Indeed, a colimit of at most π representables represents a limit of at most π evaluation functors, and κ-filtered colimits of sets commute with those limits. For the converse, write any presheaf as the colimit of its partial presentations indexed by the subcategories of C/F with at most π arrows. The preceding small-subcategory argument makes this indexing poset κ-directed. Each partial presentation is pointwise bounded by π. If F is κ-presentable, its identity factors through one of them, making F a retract of a pointwise bounded presheaf. Every value of F then has cardinal at most π. This proves the final equivalence.

The class needed in (4.5.2.2) is the class of all κ-presentable objects of $\widehat{C}$. The printed condition $F \in \mathcal{M}$ in that definition is too restrictive for the assertion of closure under colimits taken in $\widehat{C}$. For example, if C is the terminal category, then $\mathcal{M}$ contains only the singleton presheaf, whereas its empty colimit in $\widehat{C} = \mathrm{Set}$ is empty. The proof is repaired by using the ambient class, as above.

Finally, ambient κ-presentability implies presentability in $\mathcal{M}$, since the inclusion creates κ-filtered colimits. The converse is the unsettled implication $c \Rightarrow a$ in the source. Colimits in $\mathcal{M}$ may enlarge the values of a presheaf; preservation of the colimits that happen to be computed pointwise does not control this enlargement. The condition $\pi^{\pi_0} = \pi$ introduced on page 116 supplies the missing uniform estimate. It is not needed for the five equivalences just proved.

Prouvons a″ $\Longrightarrow$ a. Considérons l'application [page 109]

$$(*) \qquad \operatorname{Fl} C/F \longrightarrow \operatorname{Fl} C$$

induit par le foncteur étale $C' = C/F \longrightarrow C$, alors l'ensemble des flèches au-dessus d'une flèche $\alpha : x \longrightarrow y$ de C est en correspondance biunivoque avec $\operatorname{Ob} C'_y = F(y)$, puisqu'une telle flèche α' est connue quand on connaît son but, qui est un objet quelconque de $F(y)$ ($C' \longrightarrow C$ étant étale, i.e. fibrant à fibres discrètes). Donc le fibre de $(*)$ en α est de cardinal $= \operatorname{card} F(y) \leqslant \pi$, donc on a

$$\operatorname{card} \operatorname{Fl} C/F \leqslant \pi \underbrace{\operatorname{card} \operatorname{Fl} C}_{\leqslant \pi_1} \leqslant \pi \underbrace{\pi_1}_{\leqslant \pi} \leqslant \pi^2 = \pi ,$$

ce qui prouve a.

Prouvons enfin c $\Longrightarrow$ a. Pour ceci, nous allons utiliser la première inclusion $\mathcal{M}_\pi \subset \mathrm{Filt}^\pi \mathcal{M}$ de SGA 4 I 9.16, qui s'applique dès lors que nous prouvons le

Corollaire 4.5.3 ([34]). *Sous les conditions de 4.5.2, soit, pour tout cardinal $\pi \geqslant \pi_1$, $\mathrm{Filt}^\pi(\mathcal{M})$ la sous-catégorie strictement pleine de $\mathcal{M}$ formée des $F \in \operatorname{Ob} \mathcal{M}$ qui satisfont la condition a) de 4.5.2. Alors la famille des $(\mathrm{Filt}^\pi)_{\pi \geqslant \pi_1}$ est une filtration cardinale de $\mathcal{M}$.* [page 110]

34. Pas prouvé, la condition c) des filtrations cardinales peut-être pas satisfaite. Sans doute *faux* en général.

La condition a) de loc. cit. 9.12 (les $\underline{\mathrm{Filt}}^\pi$ sont équivalentes à des petites catégories) est évidente. Prouvons b), i.e. la stabilité de $\underline{\mathrm{Filt}}^\pi$ par $\varinjlim$ filtrantes $F = \varinjlim F_i$ *grandes devant* π_0, de cardinal $\leqslant \pi$. Comme $\mathcal{M} \hookrightarrow C^\wedge$ commute à ces limites, pour x dans C

$$F(x) \;=\; \varinjlim_i F_i(x) \;,$$

d'où $\operatorname{card} F(x) \leqslant \operatorname{card} \operatorname{Ob} I \cdot \pi \leqslant \pi \cdot \pi = \pi$, ce qui (par l'équivalence a $\iff$ a″) implique $F \in \operatorname{Ob} \mathrm{Filt}^\pi$. Prouvons la condition c) de loc. cit. Pour ceci, nous aurons besoin du fait, plus précis que le b), qu'on vient d'établir : p. 31

Corollaire 4.5.4 ([35]). *Pour tout $\pi \geqslant \pi_1$, Filt^π est stable par $\varinjlim$ (dans $\mathcal{M}$) de cardinal $\leqslant \pi$ (pas nécessairement filtrantes, a fortiori pas nécessairement grandes devant π_0).*

Il suffit, en vertu de ce qui est déjà prouvé de du lemme 4.4.14 ([36]), de montrer que Filt^π est stable par $\varinjlim$ *de cardinal* $\leqslant \pi_0$. Si $\varphi : J \longrightarrow \mathrm{Filt}^\pi$ est un foncteur avec $\operatorname{card} J \leqslant \pi_0$, on le met sous forme indicielle

$$\psi \;:\; I \times J \;\longrightarrow\; C \;,$$

grâce à 4.14.3.b, avec I grande devant π_0.

Je rappelle qu'on pouvait prendre [page 111]

$$I \;=\; \underline{\mathrm{Hom}}(J, C)/\varphi \;,$$

où $\underline{\mathrm{Hom}}(J, C) \hookrightarrow \underline{\mathrm{Hom}}(J, \mathcal{M} = \mathrm{Ind}'_\pi C)$, quitte à ne pas exiger que I soit ordonnée. On a envie de prouver que $\operatorname{card} \mathrm{Fl}\, I \leqslant \pi$. L'ennui, c'est qu'on arrive seulement à la majoration

$$\operatorname{card} \mathrm{Fl}\, I \;\leqslant\; \pi^{\pi_0} \;.$$

(Même si J est discrète, on n'a pas mieux, puisqu'alors

$$I \;\simeq\; \prod_{j \in J} C/\varphi(j) \;, \qquad \text{d'où} \quad \mathrm{Fl}\, I = \prod_{j \in J} \mathrm{Fl}\, C/\varphi(j) \;,$$

et si $\operatorname{card} \mathrm{Fl}\, C/\varphi(j) = \pi$ pour tout j, on trouve $\operatorname{card} \mathrm{Fl}\, I = \pi^{\pi_0}$ – il n'y a pas moyen de faire mieux !)

Donc je n'arrive pas à prouver directement 4.5.4, ce qui m'aurait arrangé pour prouver la condition c) de loc. cit. Mais à vrai dire, on a fait ici le travail de la démonstration du théorème 4.3.6, partie b $\Longrightarrow$ c – on y a déjà prouvé que la filtration qu'on vient de construire (par la condition a) de 4.5.2) est une filtration cardinale. La condition c) de SGA 4 I 9.12 est prouvée à la page 76. Mais je vais refaire la démonstration plus sérieusement. [page 112]

Commentary · Exposition [ms. 112]

Counting simultaneous presentations

Grothendieck's obstruction on page 111 is not a failure of the existence theorem for colimits. It concerns the amount of data needed to describe a compatible family of presentations at one time. This distinction is what the initial argument for a cardinal filtration had left uncontrolled.

Suppose $|\mathrm{Fl}\, C| \leq \pi$, $|\mathrm{Fl}\, J| \leq \theta$, and every value $\varphi(j)(c)$ has cardinal at most π. An object of C^J/φ consists of a choice of objects and arrows of C for the diagram, together with at most θ elements defining a compatible transformation to φ. There are at most π^θ such choices. An arrow in this comma category adds at most θ morphisms of C, subject to equations; imposing equations cannot increase the bound. Hence

$$|\mathrm{Fl}(C^J/\varphi)| \leq \pi^\theta.$$

For a discrete J of size exactly θ, the category is $\prod_{j \in J}(C/\varphi(j))$. If each factor has π arrows, its arrow set has size π^θ. Thus the counting argument itself cannot improve this estimate to π. The

35. Prouvé seulement si $\pi^{\pi_0} = \pi$! Sans doute *faux* en général.

36. Il faut une variante du lemme 4.4.14, pour les $\varinjlim$ de cardinal $\leqslant \pi'$, qui se prouve de la même façon.

exact equality in the source's parenthesis tacitly takes $|J| = \pi_0$; for a smaller discrete J, the exponent is $|J|$.

There is a useful positive conclusion that does not require closing the entire argument prematurely. The pointwise bounded objects form an essentially small class: choose standard sets of cardinal at most π at the set of objects of C, and then choose all structure maps satisfying the functor equations. They are also closed under θ^+-filtered colimits indexed by categories with at most π arrows, because those colimits are computed pointwise. These prove the first two estimates used on page 110. They do not prove that every object can be assembled from bounded ones through a π^+-filtered diagram.

To obtain the third property by the method on page 112, one must take a prescribed set of at most π arrows in C/F and enlarge it to a θ^+-filtered subcategory without exceeding π arrows. That is the content of Lemma 4.5.5. It is a genuinely stronger demand than adjoining a cocone to a single small diagram. Closing under all the required cocones may require π^θ simultaneous choices. The original unqualified Corollaries 4.5.3 and 4.5.4 should consequently be read with their retraction notes, not used as completed lemmas while proving Proposition 4.5.2.

Soit donc $F \in \mathrm{Ob}\,\mathcal{M}$, considérons

$$J \;=\; C/F\ ,$$

c'est une catégorie stable par $\varinjlim$ de cardinal $\leqslant \pi_0$, donc grande devant π_0, et on a

$$F \;\simeq\; \varinjlim_{j\in J} x_j \qquad (\varinjlim \text{ dans } \mathcal{M})$$

en désignant par $j \longmapsto x_j$ le foncteur canonique $C/F \longrightarrow C$. Soit alors I l'ensemble des sous-catégories J' de J qui sont *de cardinal $\leqslant \pi$ et grandes devant π_0*, I ordonné par inclusion, et pour tout $J' \in I$, posons p. 32

$$X_{J'} \;=\; \varinjlim_{j\in J'} x_j\ .$$

Comme J' est grande devant π_0, la limite inductive dans $\mathcal{M}$ est aussi une limite dans $C^\wedge$, d'où résulte aussitôt, comme $\mathrm{card}\,\mathrm{Ob}\,J' \leqslant \mathrm{card}\,\mathrm{Fl}\,J' \leqslant \pi$, que $X_{J'}$ est dans Filt^π. Alors

$$J' \;\longmapsto\; X_{J'} \;\in\; \mathrm{Ob}\ \mathrm{Filt}^\pi$$

est un foncteur

$$I \;\longrightarrow\; \mathrm{Filt}^\pi\ ,$$

dont la limite inductive est celle des x_j, donc isomorphe à F. D'autre part, il est immédiat que I est grand devant π, i.e. que si $(J'_\alpha)_{\alpha\in A}$ est une famille d'éléments de I, celle-ci est majorée [page 113] par un $J' \in I$. Ceci résulte du

Lemme 4.5.5 ([37]). *Soient π, π_0, avec $\pi > \pi_0$, des cardinaux infinis, et soit J une catégorie grande devant π_0. Alors pour toute partie Φ de $\mathrm{Fl}\,J$, telle que $\mathrm{card}\,\Phi \leqslant \pi$, il existe une sous-catégorie J' de J* grande devant π_0, *telle que* $\mathrm{card}\,J' \leqslant \pi$ ([38]), *et* $\mathrm{Fl}\,J' \supset \Phi$.

Ce lemme étant admis, on a donc représenté un objet quelconque F de $\mathcal{M}$ comme une $\varinjlim_{i\in I} X_i$, avec I ensemble ordonné grand devant π, les X_i dans Filt^π. Il reste à prouver que si $\pi' > \pi$ et $F \in \mathrm{Filt}^{\pi'}$, alors on peut prendre I tel que $\mathrm{card}\,I \leqslant \pi'^\pi$. Mais $F \in \mathrm{Filt}^{\pi'}$ signifie que $J = C/F$ satisfait $\mathrm{card}\,\mathrm{Fl}\,J \leqslant \pi'$. Alors l'ensemble des parties de cardinal $\leqslant \pi$ de $\mathrm{Fl}\,J$ a un cardinal majoré par π'^π, a fortiori $\mathrm{card}\,I \leqslant \pi'^\pi$, ce qui achève de prouver que $(\mathrm{Filt}^\pi)_{\pi\geqslant\pi_1}$ est une filtration cardinale sur $\mathcal{M}$.

Pour prouver le lemme, on commence par saturer Φ par composition finie, on trouve une [page 114] sous-catégorie J'_0 de J telle que $\mathrm{card}\,\mathrm{Fl}\,J'_0 \leqslant \pi_0$ … Mais je m'aperçois que pour prouver le

37. N'est prouvé *que* si $\pi^{\pi_0} = \pi$! Sans doute faux en général.
38. C'est OK si on exige seulement $\mathrm{card}\,J' \leqslant \pi^{\pi_0}$.

lemme, j'ai encore besoin de $\pi^{\pi_0} = \pi$ – une enveloppe de J' grande devant π_0 va avoir un cardinal majoré (au mieux) par π^{π_0}.

Prenons p.ex.

$$J \;=\; \mathfrak{P}(S) \qquad \text{(ensemble ordonné)}$$

avec $\operatorname{card} S = \pi > \pi_0$, donc J est grand devant π_0. Considérons l'application

$$S \;\longrightarrow\; J = \mathfrak{P}(S)\ , \qquad s \longmapsto \{s\}\ ,$$

et prenons pour $\Phi \subset \operatorname{Fl} J$ l'ensemble des $\mathrm{id}_{\{s\}}$. Donc on cherche une partie J' de J, contenant S, et grande devant π_0 pour l'ordre induit par J. On voit tout de suite que la plus petite partie J' faisant l'affaire est l'ensemble $\mathfrak{P}_{\pi_0}(S)$ des parties de S de cardinal $\leqslant \pi_0$. Mais on trouve

$$\operatorname{card} J' \cdot \pi_0! \;=\; \pi^{\pi_0}\ ,$$

où

$$\begin{aligned} \pi_0! \;&=\; \operatorname{card} \operatorname{Aut}(E_0) \qquad (E_0 \text{ ensemble de cardinal } \pi_0) \\ &=\; \pi_0^{\pi_0} \\ &=\; 2^{\pi_0}\ , \end{aligned}$$

p. 33 [page 115]

donc

$$(*) \qquad \operatorname{card} J' \cdot 2^{\pi_0} \;=\; \pi^{\pi_0}\ .$$

D'autre part, il est immédiat que $\operatorname{card} J' \geqslant \operatorname{card} \mathfrak{P}(E_0) = 2^{\pi_0}$ (où E_0 est une partie de S de cardinal π_0), donc le premier membre de $(*)$ est égal à $\operatorname{card} J'$, donc on trouve

$$\operatorname{card} J' \;=\; \pi^{\pi_0}\ ,$$

sorry !

Finalement, je me suis convaincu que la proposition 4.5.2, et les corollaires et lemmes précédents par lesquels j'ai essayé de l'établir, n'est vrai que pour ces cardinaux $\pi > \pi_0$ tels que

$$\pi^{\pi_0} \;=\; \pi\ ,$$

ce qui, moyennant l'hypothèse du continu généralisée, équivaut à

$$\pi^{\pi_0} \;<\; 2^{\pi}\ (\;=\; \pi^{\pi}\;)$$

(ou encore à $\pi^{\pi_0} \neq 2^{\pi}\ (= \pi^{\pi})$). Je vais donner ici un énoncé rectifié. [page 116]

Commentary · Development (commentators') [ms. 115]

A corrected obstruction to a small directed hull

The purpose of the example on pages 114–115 is to show that the cardinal cost of making an indexing category sufficiently filtered cannot always be avoided. Its chosen ambient poset, however, makes the proposed counterexample fail. In $J = \mathcal{P}(S)$, the family consisting of all singletons together with S has a greatest element. It is therefore directed at every cardinal threshold, has cardinal $|S|$, and contains every prescribed singleton. The family $[S]^{\leq \pi_0}$ is not the smallest directed family with that property. Its cardinality, correctly calculated in the source, is not a lower bound for every possible directed hull.

The required obstruction is a statement about cofinality. We give a direct repair of this passage. We checked the relevant arguments in SGA 4, Exposé I, sections 8–9, and the diagram-completion comparison in Henry (2025a); neither supplies a correction to this particular example. The following elementary application of diagonalization is offered as a repair, without a claim of priority.

Proposition (a singular-cardinal obstruction). [ours] Let $\mu > \aleph_0$ be a singular cardinal of countable cofinality. The poset

$$J = [\mu]^{\leq \aleph_0},$$

ordered by inclusion, is $\aleph_1$-directed. There is no $\aleph_1$-directed subposet $J' \subseteq J$ of cardinal at most μ containing every singleton.

Proof. A countable union of countable sets is countable, so J is $\aleph_1$-directed. If J' were as stated, then for every countable $A \subseteq \mu$, the family of singletons $\{\{a\} : a \in A\}$ would have an upper bound in J'. Thus J' would be cofinal in J.

Choose an increasing sequence of infinite cardinals $\mu_n < \mu$ with supremum μ. If $|J'| \leq \mu$, we can write $J' = \bigcup_{n<\omega} \mathcal{H}_n$ with $|\mathcal{H}_n| \leq \mu_n$. Each union $U_n = \bigcup \mathcal{H}_n$ has cardinal at most $\mu_n \cdot \aleph_0 < \mu$. Choose $a_n \in \mu \setminus U_n$, and put $A = \{a_n : n < \omega\}$. For any $H \in J'$, some n has $H \in \mathcal{H}_n$, and then $a_n \notin H$. Consequently no member of J' contains A, contradicting cofinality. This proves the proposition. □

Taking $\pi_0 = \aleph_0$, $\pi = \mu$, and the prescribed arrows to be the identities of all singletons gives an actual counterexample to Lemma 4.5.5 as originally stated. One may take $\mu = \beth_\omega$, which also lies above the elementary size bounds that occur earlier in the section. The argument establishes failure in ZFC and does not assume the generalized continuum hypothesis.

For comparison, the corrected upper estimate can be proved constructively at the level of categories. Let J be θ^+-filtered, let $\mu > \theta$ satisfy $\mu^\theta = \mu$, and start with at most μ specified arrows. Close them under identities and finite compositions. At each successor stage, choose cocones in J for all diagrams with at most θ arrows in the current subcategory, adjoin their vertices and legs, and again close under composition. There are at most $\mu^\theta = \mu$ such diagrams, since both their shapes and their values can be encoded by at most θ pieces of data from sets of size at most μ. Each stage therefore has at most μ arrows.

Continue this construction for θ^+ stages and take the union. Its cardinal remains at most μ, since $\theta^+ \leq \mu$. Every diagram with at most θ arrows in the union occurs at one stage, by regularity of θ^+, and acquires a cocone at the next stage. The union is therefore θ^+-filtered and gives the desired hull. Without a fixed-point hypothesis, the same construction with the bound $\nu = \mu^\theta$ works because $\nu^\theta = \nu$. This proves the enlarged bound recorded in footnote 38. The two arguments distinguish the valid upper estimate from the invalid minimality assertion and provide both the positive and negative parts needed at this point of the manuscript.

Proposition 4.5.6.

(1°) *Sous les hypothèses générales de 4.5.2 sur C, π_0, π_1, π, les conditions a, a′, a″, b, c′ de cette proposition sont équivalentes (moyennant $\pi \geqslant \pi_1$) ([39]), et impliquent la condition c (F est π-accessible). Quand $\pi^{\pi_0} = \pi$ (notamment quand $\pi = 2^c$, avec $c \geqslant \pi_0$), alors toutes les conditions a à c′ de 4.5.2 sont équivalentes.*

(2°) *Plus généralement et plus précisément, pour $\pi \geqslant \pi_1$ donné, considérons les conditions sur π :*

1) $\pi^{\pi_0} = \pi$.

2) $\mathrm{Filt}^\pi(\mathcal{M})$ (définie dans 4.5.3) est stable par $\varinjlim$ de cardinal $\leqslant \pi$.

3) $\forall$ F dans $\mathcal{M}$, on a $F \simeq \varinjlim_I X_i$, avec I ensemble ordonné grand devant π, les X_i dans $\mathrm{Filt}^\pi(\mathcal{M})$. De plus, si $\pi' > \pi$ et $F \in \mathrm{Filt}^{\pi'}(\mathcal{M})$, on peut prendre I de cardinal $\leqslant \pi'^\pi$.

4) Comme 3), sans la majoration de cardinaux, et en supposant seulement I catégorie grande devant π (pas nécessairement catégorie ordonné).

5) La condition c) de 4.5.2 implique les autres conditions (elles sont équivalentes, par 1°), en d'autres termes

$$\mathcal{M}_\pi \subset \mathrm{Filt}^\pi \qquad \text{(donc } \mathcal{M}_\pi = \mathrm{Filt}^\pi \text{, puisqu'on sait que } \mathrm{Filt}^\pi \subset \mathcal{M}_\pi\text{)} ,$$

p. 34
[page 117]

39. **N.B.** Pour cette équivalence, l'hypothèse que C soit stable par $\varinjlim$ de cardinal $\leqslant \pi$ est inutile, et aussi pour que c′) implique c), et lui soit équivalente si $\pi = \pi^{\pi_0} \geqslant \pi_1$, cf. 4.7.1, p. 161.

Commentary · Exposition [ms. 116]

Two measurements of size

Grothendieck is separating the size of an object from the size of the collection of all maps into it from his chosen test objects. This distinction is the reason for the correction of Proposition 4.5.2. We retain his infinite cardinals $\pi_0 \leq \pi$, and write $\lambda = \pi_0^+$ and $\mu = \pi^+$. Thus "π-accessible object" means μ-presentable object, and a category "large relative to π" is μ-filtered. All assertions here concern ordinary categories.

Let C_0 be a small skeleton of C, with at most π_1 arrows, and put $\mathcal{M} = \mathrm{Ind}'_{\pi_0}(C)$. Its embedding in $C^\wedge$ identifies an object X with the presheaf $c \mapsto \mathrm{Hom}(c, X)$. For $\pi \geq \pi_1$, the following three bounds are equivalent:

$$|\,\mathrm{Fl}(C_0/X)| \leq \pi, \qquad |\mathrm{Ob}\,(C_0/X)| \leq \pi, \qquad |\mathrm{Hom}(c, X)| \leq \pi \quad (c \in C_0).$$

Indeed, an object of C_0/X is a pair (c, u), while an arrow over $a : c \to d$ is determined by its target $v : d \to X$, its source being va. Counting first objects and then arrows proves the equivalence. Such a presheaf is μ-presentable in the presheaf category: its category-of-elements presentation uses at most π representables. Conversely, evaluation at c preserves presentability once $|\mathrm{Hom}_C(c, d)| \leq \pi$; the precise adjunction is explained in the next block. This proves the equivalence with condition c′ without any cardinal exponent assumption.

The distinction becomes concrete in sets. Take C_0 to represent sets of cardinality at most π_0. An object X of Ens is μ-presentable exactly when $|X| \leq \pi$. The forward implication follows by writing X as the μ-filtered union of its subsets of size at most π and factoring id_X through one of them. The converse follows because a function from at most π elements into a μ-filtered colimit can be lifted at one stage, and equality of two such functions can be witnessed at one later stage. On the other hand,

$$X \in \mathrm{Filt}^\pi \quad \Longleftrightarrow \quad |X|^{\pi_0} \leq \pi.$$

For $|X| \geq 2$, the exponent on the right bounds all maps from the objects of C_0, and is itself attained by maps from a set of size π_0; the empty and singleton cases satisfy the same conclusion directly. Consequently equality $\mathrm{Filt}^\pi = (\mathrm{Ens})_\pi$ is equivalent to $\pi^{\pi_0} = \pi$. This proves a genuine necessity result, independently of the unsuccessful power-set argument on manuscript page 114.

The corrected estimate

$$\mathrm{Filt}^\pi \subseteq \mathcal{M}_\pi \subseteq \mathrm{Filt}^{\pi^{\pi_0}}$$

retains useful information at every cardinal. Its second inclusion follows by replacing π by $\rho = \pi^{\pi_0}$: since $(\pi^{\pi_0})^{\pi_0} = \pi^{\pi_0}$, the corrected equivalence applies at ρ, and $\mathcal{M}_\pi \subseteq \mathcal{M}_\rho$. There is no assumption of the generalized continuum hypothesis in this argument.

There is one local repair in the proof on pages 117–118. A fixed Filt^π is not closed under arbitrarily large λ-filtered colimits: arbitrarily large unions of small sets already contradict this. What is needed there is only closure under retracts. It holds immediately, since $\mathrm{Hom}(c, X)$ is a retract of $\mathrm{Hom}(c, Y)$ whenever X is a retract of Y. The implication 4)⇒5) is therefore correct, with this replacement of the printed justification.

où $\mathcal{M}_\pi$ désigne la sous-catégorie strictement pleine de $\mathcal{M}$ formée des objets π-accessibles.

Ceci posé, on a les implications

$$1) \implies 2) \implies 3) \implies 4) \implies 5)\ ,$$

et dans le cas particulier où $\mathcal{M} = (\mathrm{Ens})$, C = sous-catégorie pleine formée des ensembles de cardinal $\leqslant \pi_0$, on a aussi 5) $\implies$ 1), i.e. toutes ces conditions sont équivalentes.

Démonstration.

1) $\implies$ 2). Cf. démonstration de 4.5.4.

2) $\implies$ 3). La catégorie $J = \mathrm{Filt}^\pi / F$ est stable par $\varinjlim$ de cardinal $\leqslant \pi$, Filt^π l'étant, donc J est grande devant π, d'ailleurs $F = \varinjlim_{j \in J} X_j$, où $j \longmapsto X_j$ est le foncteur source

$\mathrm{Filt}^\pi/F \longrightarrow \mathrm{Filt}^\pi$. D'où la représentation voulue de F, quitte à remplacer J par un ensemble ordonné cofinal I, p.ex. en prenant pour I l'ensemble des sous-catégories J de I telles que $\operatorname{card}\mathrm{Fl}\,J \leqslant \pi$ [cette deuxième partie de la phrase "p.ex. ..." demande correction]. La majoration $\operatorname{card} I \leqslant \pi'^{\pi}$, pour $F \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}$, se fait par AQT [âne qui trotte].

L'implication 3) $\Longrightarrow$ 4) est tautologique.

4) $\Longrightarrow$ 5) ([40]), car si $F \in \mathrm{Ob}\,\mathcal{M}_\pi$, on trouve par 4) que F est facteur direct d'un objet de Filt^π, et Filt^π étant stable par $\varinjlim$ grandes devant π_0, cela implique que Filt^π est stable par facteurs directs, d'où $F \in \mathrm{Ob}\,\mathrm{Filt}^\pi$. [page 118]

Enfin, si $\mathcal{M} = \mathrm{Ens}$, $C = \mathrm{Ens}(\pi_0)$. On note que

$$\mathrm{Filt}^\pi \;=\; \{\; X \in \mathrm{Ob}\,\mathrm{Ens} \mid \operatorname{card}(X)^{\pi_0} \;\leqslant\; \pi \;\}\;.$$

D'autre part, on trouve facilement

$$\begin{aligned}(\mathrm{Ens})_\pi \quad &\overset{\text{déf}}{=} \quad \text{sous-catégorie pleine de (Ens) formée}\\ &\qquad\quad \text{des objets } \pi\text{-accessibles}\\ &= \quad \{X \in \mathrm{Ob}\,\mathrm{Ens} \mid \operatorname{card} X \leqslant \pi\}\;.\end{aligned}$$

Donc on a $(\mathrm{Ens})_\pi \subset \mathrm{Filt}^\pi$ si et seulement si $\pi^{\pi_0} \leqslant \pi$, i.e. $\pi^{\pi_0} = \pi$, q.e.d. ([41]).

Corollaire 4.5.7. *Pour tout cardinal $\pi \geqslant \pi_1$, on a*

$$\mathrm{Filt}^\pi \;\subset\; \mathcal{M}_\pi \;\subset\; \mathrm{Filt}^{\pi^{\pi_0}}\;. \tag{4.5.7.1}$$

L'inclusion $\mathrm{Filt}^\pi \subset \mathcal{M}_\pi$ est l'implication a $\Longrightarrow$ c de la partie 1°) de 4.5.6. Pour prouver $\mathcal{M}_\pi \subset \mathrm{Filt}^{\pi^{\pi_0}}$, on pose $\pi' = \pi^{\pi_0}$ et on note que $\pi'^{\pi_0} = \pi^{\pi_0\cdot\pi_0} = \pi^{\pi_0} = \pi'$. Donc on peut appliquer 1° à π' au lieu de π, et on trouve que $\mathcal{M}_{\pi'} = \mathrm{Filt}^{\pi'}$. Comme $\mathcal{M}_\pi \subset \mathcal{M}_{\pi'}$, on en conclut $\mathcal{M}_\pi \subset \mathrm{Filt}^{\pi'}$, q.e.d.

J'ai oublié de démontrer le 1°) dans 4.5.6. Vu le diagramme d'implications (∗), p. 108, et l'implication a'' $\Longrightarrow$ a déjà prouvé (p. 109), il reste à prouver seulement p. 35

$$\mathrm{c}' \;\Longrightarrow\; \mathrm{a}''\;, \qquad \text{[page 119]}$$

i.e. que F π-accessible *dans* $C^\wedge$ implique $\operatorname{card} F(x) \leqslant \pi \;\forall\; x \in C$. Ceci va résulter du résultat un peu plus général :

Lemme 4.5.8.

1°) Dans la catégorie $\mathcal{E} = (\mathrm{Ens})$, les objets π-accessibles (π un cardinal infini) sont les ensembles X tels que $\operatorname{card} X \leqslant \pi$.

2°) ([42]). Soit C une catégorie équivalente à une petite catégorie, telle que $\forall\; x, y \in \mathrm{Ob}\,C$, on ait $\operatorname{card}\mathrm{Hom}(x,y) \leqslant \pi$. *Soit $\mathcal{E}$ une $\mathfrak{U}$-catégorie stable par $\varinjlim$ grandes devant π, et faiblement π-accessible (i.e. admettant une sous-catégorie génératrice formée d'objets π-accessibles), et stable par produits de cardinal $\leqslant \pi$. Alors pour qu'un objet F de $\underline{\mathrm{Hom}}(C,\mathcal{E})$ soit π-accessible, il* faut *que $\forall\; x \in \mathrm{Ob}\,C$, $F(x)$ soit un objet π-accessible de $\mathcal{E}$, et cette condition est aussi suffisante si* $\operatorname{card}\text{réd}\,\mathrm{Fl}(C) \leqslant \pi$ ([43]).

Démonstration. 1°) $\operatorname{card} X \leqslant \pi \Longrightarrow X$ π-accessible dans $\mathcal{E} = (\mathrm{Ens})$, car $X = \coprod_{x\in X}\{x\}$, et on applique SGA 4 I 9.9. Inversement, supposons X π-accessible, et écrivons $X \simeq \varinjlim_{i\in I} X_i$, où I est l'ensemble des parties de X qui sont de cardinal $\leqslant \pi$. Comme I est grande devant π, l'application identique $X \xrightarrow{\mathrm{id}} X$ se factorise par un ensemble X_i, donc $X_i = X$, donc $\operatorname{card} X = \operatorname{card} X_i \leqslant \pi$, q.e.d.

40. comparer le lemme 4.5.14, page 133.
41. Cf. démonstration de la partie 1°) de 4.5.6, plus bas (bas de page et page suivante).
42. Cf. généralisation 4.14.18, p. 391 b) et c). [page 120]
43. même *sans* condition de stabilité de $\mathcal{E}$ par produits de cardinal $\leqslant \pi$.

2°) Considérons, pour $x \in \mathrm{Ob}\, C$, le foncteur

$$e \xrightarrow{\varepsilon_x} C \ ,$$

d'où un foncteur

$$\varepsilon_x^* \ : \ \underline{\mathrm{Hom}}(C, \mathcal{E}) \longrightarrow \mathcal{E} \ , \qquad F \longmapsto F(x) \ ,$$

admettant un adjoint à droite

$$\varepsilon_{x\,*} \ : \ \mathcal{E} \ \longrightarrow \ \underline{\mathrm{Hom}}(C, \mathcal{E}) \ ,$$

donné par

$$\varepsilon_{x\,*}(E)(y) \ = \ E^{\mathrm{Hom}(x,y)}$$

(sous réserve d'existence des produits de cardinal $\leqslant \pi$). Ce foncteur $\varepsilon_{x\,*}$ est π-accessible en vertu de loc. cit. 9.8.

Ceci posé, soit F un objet π-accessible de $\underline{\mathrm{Hom}}(C, \mathcal{E})$, on voit que $\varepsilon_x^*(F)$ est π-accessible, ε_x^* ayant un adjoint à droite π-accessible :

$$E \ \longmapsto \ \mathrm{Hom}_{\mathcal{E}}(\underbrace{F(x)}_{\varepsilon_x^*(F)}, E) \ \simeq \ \mathrm{Hom}_{\underline{\mathrm{Hom}}(C,\mathcal{E})}(F, \varepsilon_{x\,*}(E))$$

est composé du foncteur π-accessible $\varepsilon_{x\,*}$, et du foncteur covariant π-accessible représenté par F dans $\underline{\mathrm{Hom}}(C, \mathcal{E})$. p. 36

Inversement, si les $F(x)$ sont π-accessibles, prouvons que F l'est, moyennant $\mathrm{card}\, \mathrm{Fl}(C) \leqslant \pi$. Se fait par AQT `[âne qui trotte]`, je ne vais pas l'expliciter ici.

[page 121]

Commentary · Exposition [ms. 120]

The presentability of a diagram

Grothendieck wants to recover the sizes of the values of a presheaf from the size of the presheaf as an object. Evaluation supplies the necessary bridge, but its adjoint must have the correct variance. For a covariant functor $F : C \to \mathcal{E}$, the right adjoint to evaluation at c is

$$R_c(E)(d) = \prod_{a : d \to c} E.$$

The edition writes $E^{\mathrm{Hom}(c,d)}$. That expression has the opposite variance in d and is not the asserted right adjoint. To verify the corrected formula directly, a map $t : F(c) \to E$ determines the component $F(d) \to R_c(E)(d)$ whose projection at $a : d \to c$ is $tF(a)$. Naturality recovers every component from the projection at id_c, establishing the adjunction. This is the pointwise right Kan extension along $\{c\} \to C$; compare Riehl (n.d., Proposition 6.1.5 and Theorem 6.2.1).

Suppose $\mathcal{E}$ has the products and μ-filtered colimits used here, and has a conservative family of representables preserving those colimits. This is the meaning of the generating-family hypothesis inherited from SGA 4, rather than merely faithfulness of those representables. Products indexed by at most π elements commute with μ-filtered colimits in $\mathcal{E}$: applying a conservative representable reduces the assertion to sets. In sets, lift the at most π coordinates at a common stage to obtain surjectivity, and witness their at most π equalities at a common later stage to obtain injectivity. Thus R_c preserves μ-filtered colimits when every $\mathrm{Hom}(d, c)$ has cardinality at most π. If F is μ-presentable, then

$$\mathrm{Hom}_{\mathcal{E}}(F(c), -) \cong \mathrm{Hom}_{[C,\mathcal{E}]}(F, R_c(-))$$

preserves these colimits, so $F(c)$ is μ-presentable.

For the converse, assume a skeleton of C has at most π arrows. The missing AQT argument is an equalizer computation. For any $G : C \to \mathcal{E}$,

$$\mathrm{Nat}(F, G) = \mathrm{Eq}\left(\prod_{c \in \mathrm{Ob}\, C} \mathrm{Hom}(F(c), G(c)) \rightrightarrows \prod_{a : c \to d} \mathrm{Hom}(F(c), G(d)) \right).$$

The two maps send a family (t_c) to $(G(a)t_c)$ and $(t_dF(a))$. If all $F(c)$ are μ-presentable, every factor preserves μ-filtered colimits in G, and so does the displayed equalizer: the total number of coordinates and compatibility conditions is at most π. Since diagram-category colimits are pointwise, this proves presentability of F. No products in $\mathcal{E}$ are required for this direction; only products of hom-sets are used. The source's parenthetical qualification is therefore exact.

This calculation also explains why the bound on the number of arrows cannot be silently replaced by a bound on the number of objects. One must make every naturality equation hold at the same stage. A one-object category may carry arbitrarily many such equations through its endomorphisms.

Proposition 4.5.9. *Soit C une catégorie équivalente à une petite catégorie, et soit $\mathcal{M} = \mathrm{Ind}'_{\pi_0}(C)$, π_0 un cardinal infini donné. On suppose C stable par $\varinjlim$ de cardinal $\leqslant \pi_0$. Pour tout cardinal $\pi \geqslant \pi_0$, soit $\mathcal{M}_\pi \supset C$ la sous-catégorie pleine de $\mathcal{M}$ formée des objets π-accessibles de $\mathcal{M}$. Soit d'autre part $\pi_1 = \sup(\pi_0, \operatorname{card} \mathrm{Fl}\, C)$. Alors $(\mathcal{M}_\pi)_{\pi \geqslant \pi_1}$ est une filtration cardinale de $\mathcal{M}$ ([44]), et de plus, pour $\pi \geqslant \pi_0$, $\mathcal{M}_\pi$ est stable dans $\mathcal{M}$ par $\varinjlim$ de cardinal $\leqslant \pi$.*

Démonstration. La stabilité de $\mathcal{M}_\pi$ est un rappel, cf. SGA 4 I 9.9. Comme on a

$$\mathcal{M}_\pi \ \subset\ \mathrm{Filt}^{\pi^{\pi_0}}$$

pour $\pi \geqslant \pi_1$ (cf. 4.5.7), on voit donc que les $\mathcal{M}_\pi$ sont équivalentes à des petites catégories (condition a) des filtrations cardinales). La condition b) des filtrations cardinales est contenue dans la propriété de stabilité des $\mathcal{M}_\pi$ qu'on vient de rappeler. Prouvons enfin la condition c). Soit donc $\pi \geqslant \pi_0$, et soit x dans $\mathcal{M}$, considérons la catégorie

$$I \ =\ \mathcal{M}_\pi/x\ ,$$

elle est stable par $\varinjlim$ de cardinal $\leqslant \pi$, d'après la stabilité similaire de $\mathcal{M}_\pi$, donc I est grande devant π et équivalente à une petite catégorie, d'autre part, on a [page 122]

$$x \ =\ \varinjlim_{i \in I} x_i\ ,$$

les x_i dans $\mathcal{M}_\pi$ ($\varinjlim$ dans $\mathcal{M}$ des foncteurs $I = \mathcal{M}_\pi/x \longrightarrow \mathcal{M}_\pi$). Quitte à remplacer I par une catégorie ordonnée cofinale, cela prouve la partie qualitative de c. Supposons enfin $x \in \mathrm{Ob}\, \mathcal{M}_{\pi'}$, où $\pi' > \pi$, montrons que l'on a alors

$$\operatorname{card} I_0,\ \operatorname{card} I' \ \leqslant\ \pi'^{\pi} \qquad ([45], [46])\ , \tag{4.5.9.1}$$

ce qui achèvera de prouver que l'on a une filtration cardinale. Il nous faut une majoration des cardinaux.

Lemme 4.5.10 ([47]). *Soit π un cardinal $\geqslant \pi_1$. Alors on a*

$$\operatorname{card} \mathrm{Fl}((\mathrm{Filt}^\pi)_0) \ \leqslant\ 2^\pi\ . \tag{4.5.10.1}$$

On a en effet, en désignant par $C^\wedge(\pi)$ la sous-catégorie strictement pleine de $C^\wedge$ formée des $F : C^\circ \longrightarrow \mathrm{Ens}$ avec $\operatorname{card} F(x) \leqslant \pi$ pour tout x dans C (i.e. F se factorise par $\mathrm{Ens}(\pi)$), et par $C^\wedge(\pi)_0$ une sous-catégorie réduite exhaustive,

$$\operatorname{card} C^\wedge(\pi)_0 \ \leqslant\ 2^\pi\ , \tag{4.5.10.2}$$

p. 37
[page 123]

d'où résulte (4.5.10.1), puisque pour $\pi \geqslant \pi'$, Filt^π est une sous-catégorie pleine de $C^\wedge(\pi')$. Pour établir (4.5.10.2), on note que l'ensemble des classes à isomorphie près de $\mathrm{Ens}(\pi')$ s'identifie à

44. **N.B.** Toutes les conditions des filtrations cardinales sont vérifiées à l'exclusion, quand $\pi \geqslant \pi_0$, de celle de la majoration de cardinaux $\operatorname{card} I \leqslant \pi'^{\pi}$ condition c).

45. où I_0, I'_0 sont des sous-catégories réduites de I, I' telles que $I_0 \hookrightarrow I$, $I'_0 \hookrightarrow I'$ soient des équivalences `[Grothendieck a biffé l'indice 0 de I' dans 4.5.9.1]`.

46. On ne le prouve que si $\pi \geqslant \pi_1$.

47. Donner énonce commun avec 4.5.11, et prouver d'abord 4.5.11.

l'ensemble des cardinaux $\leqslant \pi'$, lequel ensemble est de cardinal $\leqslant \pi'$. Remplaçant $\mathrm{Ens}(\pi)$ par $\mathrm{Ens}(\pi')_0$, C par C_0, on doit majorer $\operatorname{card}\mathrm{Fl}(\underline{\mathrm{Hom}}(C_0, \mathrm{Ens}(\pi')_0))$. Or $\mathrm{Ob}\,\underline{\mathrm{Hom}}(C_0, \mathrm{Ens}(\pi')_0)$ est contenu dans l'ensemble des applications de $\mathrm{Fl}\,C_0$ dans $\mathrm{Fl}(\mathrm{Ens}(\pi')_0)$, le premier est majoré par π_1, le deuxième par $\pi^\pi \times \pi \times \pi = \pi^{2\pi} = \pi^\pi = 2^\pi$ (**N.B.** $\pi < 2^\pi$, d'où $\pi^\pi \leqslant (2^\pi)^\pi = 2^{\pi^2} = 2^\pi$), donc $\mathrm{Ob}\,\underline{\mathrm{Hom}}$ majoré par $(2^\pi)^{\pi_1} = 2^{\pi\pi_1} = 2^\pi$. D'autre part, si F, G sont dans $\underline{\mathrm{Hom}}(C_0, \mathrm{Ens}(\pi_0))$, $\mathrm{Hom}(F, G)$ est majoré par $(\pi^\pi)^{\pi_1} = \pi^\pi = 2^\pi$, donc $\mathrm{Fl}\,\underline{\mathrm{Hom}}$ est majoré par $2^\pi \times 2^\pi \times 2^\pi$, q.e.d.

Corollaire 4.5.11. *Soit $F \in \mathrm{Ob}\ \mathrm{Filt}^\pi$, $F' \in \mathrm{Ob}\ \mathrm{Filt}^{\pi'}$, avec $\pi' \geqslant \pi \geqslant \pi_1$. Alors*

$$\operatorname{card}\mathrm{Hom}(F, F') \leqslant \pi'^\pi \ . \tag{4.5.11.1}$$

C'est essentiellement la même démonstration, légèrement plus simple. (On aurait dû commencer par là.) [page 124]

Corollaire 4.5.12. *Si $\pi' \geqslant \pi \geqslant \pi_1$, on a pour $x \in \mathrm{Ob}\,\mathcal{M}_{\pi'}$*

$$\operatorname{card}(\mathcal{M}_\pi/x)_0 \leqslant \pi'^{(\pi^{\pi_0})} \ . \tag{4.5.12}$$

En effet,

$$\operatorname{card}\mathrm{Ob}\ (\mathcal{M}_\pi/x)_0 \overset{\text{par 4.5.11}}{\leqslant} \underbrace{\underbrace{\operatorname{card}\mathrm{Ob}\ (\mathcal{M}_\pi)_0}_{\leqslant 2^{(\pi^{\pi_0})}\ \text{par 4.5.10}} \times (\pi'^{\pi_0})^{\pi^{\pi_0}}}_{=\pi'^{(\pi^{\pi_0})}} \ ,$$

i.e.

$$\operatorname{card}\mathrm{Ob}\ (\mathcal{M}_\pi/x)_0 \leqslant \pi'^{(\pi^{\pi_0})} \ ,$$

d'où

$$\operatorname{card}\mathrm{Fl}(\mathcal{M}_\pi/x)_0 \leqslant \pi'^{(\pi^{\pi_0})} \times \pi'^{(\pi^{\pi_0})} \times (\pi'^{\pi_0})^{(\pi^{\pi_0})} \leqslant \pi'^{(\pi^{\pi_0})} \ .$$

Cela donne donc (revenant à 4.5.9.1)

$$\operatorname{card} I_0 \leqslant \pi'^{(\pi^{\pi_0})} \ .$$

D'autre part, I' peut se définir en termes de I_0 comme l'ensemble des sous-catégories de I_0 qui sont de cardinal $\leqslant \pi$, donc leur nombre est majoré par $(\operatorname{card} I_0)^\pi \leqslant (\pi'^{\pi^{\pi_0}})^\pi = \pi'^{\pi\pi^{\pi_0}} = \pi'^{(\pi^{\pi_0})}$.

Mais ce n'est pas tout à fait la majoration exigée, puisque

$$\pi'^\pi \leqslant \pi'^{(\pi^{\pi_0})} \qquad ([48]),$$

et qu'il est possible, a priori, que l'inégalité soit stricte ! Apparemment, les "majorations" p. 38 [page 125]

$$\mathcal{M}_\pi \subset \mathrm{Filt}^{\pi^{\pi_0}} \ , \qquad \mathcal{M}_{\pi'} \subset \mathrm{Filt}^{\pi'^{\pi_0}}$$

sont trop grossières. Aussi il s'agit de mieux appréhender les catégories $\mathcal{M}_\pi$, pour $\pi \geqslant \pi_0$ ([49]). Je vais hasarder une

48. **N.B.** On a égalité si on a $\pi' = 2^{c'}$ avec $c' \geqslant \pi^{\pi_0}$, soit $\pi = 2^c$ avec $c \geqslant \pi_0$.

49. Mais on peut utiliser 4.5.10, 4.5.11 pour conclure que pour $x \in \mathrm{Ob}\,\mathcal{M}_{\pi'} \subset \mathrm{Ob}\ \mathrm{Filt}^{\pi'^{\pi_0}}$, on a $\operatorname{card}\mathrm{Fl}(C_0/x) \leqslant (\pi'^{\pi_0})^{\pi_0} = \pi'^{\pi_0}$, on aura donc $x = \varinjlim_J x_j$ ($x_j \in \mathrm{Ob}\,C$, $\varinjlim$ filtrante, $\operatorname{card}\mathrm{Fl}\,J \leqslant \pi'^{\pi_0}$), et en introduisant l'ensemble I des sous-catégories $J' \subset J$ qui sont de cardinal $\leqslant \pi$, et posant $x_{J'} = \varinjlim_{j \in J'} x_j$, on trouve $x = \varinjlim_{J' \in I} x_{J'}$, où cette fois I est un *ensemble ordonné* grand devant π, les $x_{J'} \in \mathcal{M}_\pi$, et $\operatorname{card} I \leqslant (\operatorname{card} J)^\pi \leqslant (\pi'^{\pi_0})^\pi = \pi'^{\pi\pi_0} = \pi'^\pi$, c'est la bonne majoration !

Commentary · Exposition [ms. 124]

The canonical filtration and its cardinal bound

Grothendieck's preferred filtration is now the intrinsic one: $\mathcal{M}_\pi$ consists of the π^+-presentable objects. The strength of Proposition 4.5.9 is that it supplies the required presentations at every sufficiently large π, although the more concrete test $|\mathrm{Hom}(c, X)| \leq \pi$ may fail. The footnote on manuscript page 125 supplies the right construction; counting the whole slice too early obscures it.

First, $\mathcal{M}_\pi$ is closed under colimits with at most π arrows. If $A = \mathrm{colim}_{j\in J} A_j$ and $|\operatorname{Fl} J| \leq \pi$, then $\mathrm{Hom}(A, -) = \lim_{J^{\mathrm{op}}} \mathrm{Hom}(A_j, -)$, and a limit of at most π set-valued functors preserving π^+-filtered colimits again preserves them. This is the elementary closure result used throughout these pages; the historical reference is Artin et al. (1972, Exposé I, 9.8–9.9). Essential smallness follows either from the presheaf bound $\mathcal{M}_\pi \subseteq \mathrm{Filt}^{\pi^{\pi_0}}$ or from the presentation theorem proved below.

For a direct construction, write $X = \mathrm{colim}_{j\in J} c_j$ using $J = C_0/X$. Let P be the poset of subcategories $K \subseteq J$ with at most π arrows, and set

$$X_K = \mathrm{colim}_{k\in K} c_k \quad \text{in } \mathcal{M}.$$

The category P is π^+-directed. A family of at most π such subcategories has a union of at most π arrows; closing the union under identities and finite composition still adds at most π arrows. Every X_K belongs to $\mathcal{M}_\pi$. Moreover $X = \mathrm{colim}_{K\in P} X_K$: a compatible family of maps from the X_K is exactly a cocone on the original diagram, since every object and arrow of J occurs in some K. This proof does not assert that the subcategories K are themselves π_0^+-filtered.

If $X \in \mathcal{M}_{\pi'}$, with $\pi' \geq \pi \geq \pi_1$, then $|\operatorname{Fl}(C_0/X)| \leq (\pi')^{\pi_0}$ by Corollary 4.5.7. The number of choices of K is bounded by

$$|P| \leq \left((\pi')^{\pi_0}\right)^{\pi} = (\pi')^{\pi}.$$

The number of order relations of this infinite poset has the same bound. This proves the quantitative clause of the cardinal filtration without assuming $\pi^{\pi_0} = \pi$.

The numerical lemmas on pages 122–124 are correct after normalizing the stray primes in the proof of 4.5.10. For presheaves with at most π values at each object of a category with at most π arrows, one chooses representative sets among the cardinals at most π, and then structure maps between them. There are at most $(2^\pi)^\pi = 2^\pi$ resulting functors and at most 2^π natural transformations between each pair. Similarly, if the source values are bounded by π and target values by π', then

$$|\operatorname{Nat}(F, G)| \leq \prod_{c\in \mathrm{Ob}\, C_0} |G(c)|^{|F(c)|} \leq (\pi')^{\pi}.$$

The weaker exponent in Corollary 4.5.12 is a valid intermediate estimate. It is not a counterexample to the sharper presentation bound: that bound counts the subdiagrams used above rather than every map from every presentable object at once.

Proposition 4.5.13. *Les hypothèses et notations sont celles de 4.5.9. Soit π un cardinal $\geqslant \pi_0$, et soit $x \in \mathrm{Ob}\,\mathcal{M}$. Considérons les conditions :*

a) Il existe une catégorie I, avec $\operatorname{card}\operatorname{Fl} I \leqslant \pi$, *et un foncteur $i \longmapsto x_i : I \longrightarrow C$, tel que x soit isomorphe à $\varinjlim x_i$.*

a′) Comme a), avec I filtrante.

a″) Comme a), avec I ordonnée filtrante.

b) Il existe un foncteur cofinal $I \longrightarrow C/x$, *avec* $\operatorname{card}\operatorname{Fl} I \leqslant \pi$.

b′) La catégorie filtrante C/x est "essentiellement de cardinal $\leqslant \pi$", i.e. il existe une catégorie ordonnée `[non, cf. b″)]` filtrante I, $\operatorname{card}\operatorname{Fl} I \leqslant \pi$, *et un foncteur cofinal $I \longrightarrow C/x$.*

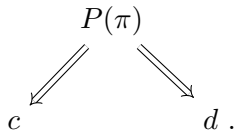
b″) Comme b′), avec de plus I ordonnée.

c) Il existe une famille de flèches $(u_\alpha)_{\alpha\in A}$, $x_\alpha \longrightarrow x$, épimorphique stricte, indexée par un ensemble A tel que $\operatorname{card} A \leqslant \pi$.

d) On a $x \in \mathrm{Ob}\,\mathcal{M}_\pi$, i.e. x est π-accessible dans $\mathcal{M}$. [page 126]

Ceci posé :

(1°) *Les six premières conditions a, a′, a″, b, b′, b″ sont équivalentes, soit $P(\pi)$, et impliquent les deux dernières,*



Si $\pi \geqslant \pi_1$, alors $P(\pi)$ est équivalente à c.

(2°) *Soit* $\mathrm{Filt}'^{\pi} \subset \mathcal{M}$ *la sous-catégorie strictement pleine de $\mathcal{M}$ formée des objets qui satisfont la condition $P(\pi)$ ci-dessus, et* Filt^{π} *la sous-catégorie strictement pleine définie dans 4.5.3 (p. 109). On a alors*

$$\mathrm{Filt}^{\pi} \ \subset \ \mathrm{Filt}'^{\pi} \ \subset \ \mathcal{M}_{\pi} \ , \tag{4.5.13.1}$$

et pour qu'on ait égalité des trois membres, il suffit qu'on ait $\pi^{\pi_0} = \pi$. (Cf. prop. 4.5.6, page 116). De plus, Filt'^{π} *est stable dans $\mathcal{M}$ par* sommes *de cardinal $\leqslant \pi$.*

(3°) *Les conditions suivantes sur C, π_0, π sont équivalentes.*

p. 39

1) Filt'^{π} *est stable dans $\mathcal{M}$ par $\varinjlim$ de cardinal $\leqslant \pi$.*

1′) Filt'^{π} *est stable par conoyaux de double-flèches.*

2) Tout objet x de $\mathcal{M}$ est de la forme $\varinjlim_{I} x_i$, où x_i dans Filt'^{π}*, I grande devant π.*

3) La condition d) de 1°) implique $P(\pi)$ (donc lui est équivalente), i.e. on a

$$\mathcal{M}_{\pi} \ = \ \mathrm{Filt}'^{\pi} \, .$$

Ces conditions sont satisfaites dans les trois cas suivants : [page 127]

A) $\pi^{\pi_0} = \pi$ (pour mémoire, c'est contenu dans 2°).

B) Les familles épimorphiques strictes dans $\mathcal{M}$ sont épimorphiques strictes universelles.

C) Les limites inductives filtrantes dans $\mathcal{M}$ commutent au changement de base (ce qui est le cas notamment si les $\varinjlim$ filtrantes sont exactes à gauche) ([50]).

Commentary · Development (commentators') [ms. 126]

A small colimit need not be cofinal in the canonical presentation

Grothendieck is looking for a presentation whose size can be read directly from the object. There are two different requirements in Proposition 4.5.13: being the colimit of a small diagram of objects of C, and possessing a small cofinal diagram in C/X. The proof on page 129 establishes the first requirement with a filtered indexing poset; it does not establish cofinality in C/X. The distinction persists even in sets, where all the exactness hypotheses later imposed are satisfied.

A counterexample to a)⇒b). [ours] Take $\pi_0 = \aleph_0$, let C be a skeleton of the countable sets, and let X have cardinality $\pi = \beth_\omega$. Here $\pi \geq \pi_1$, C has countable colimits, and $\mathcal{M} = \mathrm{Ens}$. The set X is the coproduct of its π singleton subsets. It is also the filtered colimit of its finite subsets, indexed by a poset with π arrows. Thus a), a′), and a″) all hold.

Suppose a category I with at most π arrows admitted a cofinal functor $I \to C/X$. Write its objects as maps $u_i : c_i \to X$. For every countable subset $A \subseteq X$, nonemptiness of the comma category $(A \hookrightarrow X) \downarrow I$ would give some i with $A \subseteq u_i(c_i)$. The at most π countable subsets $u_i(c_i)$ would therefore be cofinal among the countable subsets of X.

No such family exists. Indeed, write any family $\mathcal{H}$ of at most π countable subsets as an increasing union $\bigcup_{n<\omega} \mathcal{H}_n$ with $|\mathcal{H}_n| < \pi$. Each union $\bigcup \mathcal{H}_n$ has cardinality less than π. Choose $s_n \in X \setminus \bigcup \mathcal{H}_n$. The countable set $\{s_n : n < \omega\}$ is contained in no member of $\mathcal{H}$: a member belonging to $\mathcal{H}_n$ omits

50. Je suppose que cette condition revient à la même condition sur C, pour les $\varinjlim$ filtrantes de cardinal $\leqslant \pi$.

s_n. This contradiction proves failure of b), hence also of b′) and b″). The choice $\beth_\omega$ avoids any continuum hypothesis; its countable cofinality and its being larger than $2^{\aleph_0}$ are sufficient.

For the corrected positive assertion, let $X = \operatorname{colim}_I c_i$ with $|\operatorname{Fl} I| \leq \pi$. Every finite set of arrows of I gives a finite incidence diagram: introduce one vertex for each object used and one extra vertex for the source of each selected arrow, with its identity arrow to the source vertex and the given arrow to the target vertex. Its colimit belongs to C, because C has finite colimits. Ordering finite sets of arrows by inclusion gives a filtered poset with at most π arrows, and the colimit of these finite presentations is X. A cocone on all of them is exactly a cocone on I. This establishes a)⇒a″), including categories with loops: finite subsets need not generate finite subcategories, which is why the incidence-diagram description is preferable.

Thus a), a′), and a″) are equivalent. Each of b), b′), and b″) implies the corresponding colimit condition, but the two triples cannot be identified. For the rest of our commentary, Filt'^{π} denotes the one-step colimit class defined by a), the class used in the subsequent presentation arguments. This resolves an ambiguity created by the false equivalence in the printed definition; it is not a change to the source text.

Remarque. La filtration de $\mathcal{M}$ pour les Filt'^{π} n'est autre que celle envisagée dans SGA 4 I 9.13, où on fait de plus l'hypothèse B) sur $\mathcal{M}$, qui après-coup n'était plus tellement de mon goût. Toujours est-il que d'après (4.5.13.1), cette filtration se rapproche plus de la "bonne" filtration par les $\mathcal{M}_\pi$, que celle que j'avais eu encore naguère tendance à lui préférer, celle par les Filt^{π}. Elle a d'ailleurs sur cette dernière l'avantage, dans le cas envisagé B), d'être une filtration cardinale (ce qui n'est pas nécessairement le cas pour Filt^{π}, du moins s'il existe des cardinaux infinis $\pi \geqslant \pi_0$ avec $\pi^{\pi_0} \neq \pi$). Je soupçonne maintenant qu'en général (et s'il existe des cardinaux infinis $\pi > \pi_0$ tels que $\pi^{\pi_0} \neq \pi$), en dehors des conditions A) B) C), la filtration par les (Filt'^{π}) n'est *pas* une filtration cardinale. En tous cas, 4.5.6 et la présente proposition, [page 128]
3°, montrent que si soit $(\mathrm{Filt}^{\pi})_{\pi\geqslant\pi_2}$, soit $(\mathrm{Filt}'^{\pi})_{\pi\geqslant\pi_2}$ est une filtration cardinale (π_2 étant un cardinal donné $\geqslant \pi_1$), alors elle coïncide avec la filtration par les $\mathcal{M}_\pi$. Donc apparemment, dans tous les cas où je sais construire une filtration cardinale, sur un paratopos du moins (i.e. sur une catégorie $\mathcal{M}$ stable par petites $\varinjlim$, et dont tous les objets sont accessibles – l'existence d'une filtration cardinale impliquant de plus celle d'une petite sous-catégorie strictement génératrice …), celle-ci n'est autre que celle par les $\mathcal{M}_\pi$ – chose dont je ne m'étais par aperçu en rédigeant SGA 4 I 9.

Commentary · Development (commentators') [ms. 127]

A precise reduction of universality to the generators

Footnote 50 suggests testing condition C) inside the small generating category. As printed, that suggestion needs a qualification: C is only assumed to have colimits with at most π_0 arrows, and it need not have pullbacks. With the latter hypothesis added, the reduction is valid, and the bound π_0 itself suffices. The proof below makes explicit which data must be approximated together.

A bounded test for universal filtered colimits. [ours] Let κ be infinite, let C be essentially small with pullbacks and all colimits indexed by categories with at most κ arrows, and put

$$\lambda = \kappa^+, \qquad \mathcal{M} = \mathrm{Ind}'_\kappa(C).$$

Then ordinary filtered colimits in $\mathcal{M}$ are universal if and only if filtered colimits in C indexed by categories with at most κ arrows are universal in C. Here universality in C means stability under every base change that is a morphism of C. Thus the additional hypothesis is existence of pullbacks in C; no terminal object or colimits of the larger bound π are needed. The marker identifies the present formulation and proof, without a priority claim.

We first recall the presheaf facts that control the proof. The category $\mathcal{M}$ consists of the presheaves on C that carry colimits with at most κ arrows in C to limits of sets. This description follows from the canonical category-of-elements presentation: the preservation condition makes that category λ-filtered, and conversely a λ-filtered colimit of representables has the preservation condition. Limits of such presheaves, and λ-filtered colimits of them, are computed pointwise. Indeed, limits commute with limits, and λ-filtered set colimits commute with limits having at most κ arrows. Consequently λ-filtered colimits in $\mathcal{M}$ commute with finite limits. The embedding $j : C \to \mathcal{M}$ preserves pullbacks,

by Yoneda and the pointwise description of limits. It also preserves the prescribed colimits: for any such colimit $c = \operatorname{colim}_i c_i$ in C and any $F \in \mathcal{M}$,

$$\operatorname{Hom}_{\mathcal{M}}(j(c), F) = F(c) = \lim_i F(c_i) = \lim_i \operatorname{Hom}_{\mathcal{M}}(j(c_i), F).$$

We need a simultaneous diagram approximation. If J has at most κ arrows, every J-diagram in $\mathcal{M}$ is a λ-filtered colimit of C-valued J-diagrams. Here is a direct verification of this instance of the diagram construction used earlier. Let E be the full subcategory of $\mathcal{M}^J$ consisting of the C-valued diagrams, up to isomorphism. It is essentially small and closed under colimits with at most κ arrows, since those colimits are computed pointwise and preserved by j. For a diagram D, the category E/D is therefore λ-filtered: take the colimit of any diagram of at most κ arrows in that slice. The empty diagram is handled by its initial object.

To see that the canonical diagram $E/D \to \mathcal{M}^J$ has colimit D, use the free diagrams

$$L_a(c)(b) = \coprod_{J(a,b)} c \qquad (a, b \in J,\ c \in C).$$

They belong to E, since every indexing set $J(a,b)$ has at most κ elements, and L_a is left adjoint to evaluation at a. A cocone from E/D to a diagram G therefore assigns compatible maps $c \to G(a)$ to all maps $c \to D(a)$ from C. The canonical presentation of $D(a)$ gives a unique map $t_a : D(a) \to G(a)$. For an arrow $a \to b$ in J, the induced maps $L_b(c) \to L_a(c)$ show that the t_a satisfy naturality. Finally, for any $A \to D$ in E/D, testing with all maps $L_a(c) \to A$ shows that its given cocone map is $A \to D \xrightarrow{t} G$. Those tests suffice because the maps from C to every $A(a)$ are jointly epimorphic. This proves the universal property and hence the simultaneous approximation. A small skeleton of E/D supplies a small filtered index.

Assume now the bounded universality hypothesis in C. First consider a filtered category I with at most κ arrows, a diagram $D : I \to \mathcal{M}$ with a cocone to Z, and a map $T \to Z$. We will prove the more flexible comparison

$$(*) \qquad \operatorname{colim}_{i \in I}(T \times_Z D_i) \longrightarrow T \times_Z \operatorname{colim}_{i \in I} D_i$$

is an isomorphism. Adjoin to I the cocone vertex z, all arrows $i \to z$, and one further vertex t with an arrow $t \to z$, imposing the cocone relations. The resulting shape still has at most κ arrows. Apply the simultaneous approximation to this entire shape. We obtain a λ-filtered diagram, indexed by p, of data

$$D_p : I \to C, \qquad D_p \longrightarrow Z_p, \qquad T_p \longrightarrow Z_p,$$

whose colimit is exactly the original data. In particular, one does not assume that Z_p is the colimit of D_p.

Put $Q_p = \operatorname{colim}_I D_p$ in C. Applying the assumed universality to the base change

$$T_p \times_{Z_p} Q_p \longrightarrow Q_p$$

gives, in C and hence in $\mathcal{M}$,

$$\operatorname{colim}_{i \in I}(T_p \times_{Z_p} D_{p,i}) \simeq T_p \times_{Z_p} Q_p.$$

All objects and pullbacks in this equation lie in C, and the preceding facts ensure that its colimit and pullbacks are preserved by j. Take the λ-filtered colimit over p. On the left, interchange the two colimits and commute the p-colimit with each pullback. On the right, commute it with the displayed pullback and use

$$\operatorname{colim}_p Q_p = \operatorname{colim}_p \operatorname{colim}_i D_{p,i} \simeq \operatorname{colim}_i D_i.$$

The resulting isomorphism is exactly (*). Setting $Z = \operatorname{colim}_I D_i$ proves universality for every bounded filtered diagram in $\mathcal{M}$.

It remains to remove the bound on I. Let I be any small filtered category, and let P be the poset of its filtered subcategories with at most κ arrows. Every set of at most κ arrows of I lies in such a subcategory. Start by closing the selected arrows under identities and finite composition. Choose in I a cocone for every finite diagram in the resulting subcategory, add its objects and arrows, and close under composition again. Repeat this operation countably many times. At each step there

are at most κ finite diagrams and at most κ new arrows; the final union still has at most κ arrows. Every finite diagram in the union occurs at some stage and receives a cone at the next, so the union is filtered. If the initial selection is empty, first choose one object of I.

The same construction applied to the union of at most κ members of P proves that P is λ-directed. Every object and arrow of I occurs in some member of P, so for any $D : I \to \mathcal{M}$ the cocone universal property gives

$$\operatorname{colim}_{K\in P} \operatorname{colim}_{i\in K} D_i \simeq \operatorname{colim}_{i\in I} D_i.$$

Apply the already proved comparison (*) to each restricted diagram $D|_K$, with the fixed cocone to Z and the fixed arrow $T \to Z$. Taking the λ-filtered colimit over P, and again commuting it with pullback, proves (*) for arbitrary filtered I. This is condition C).

Conversely, if condition C) holds in $\mathcal{M}$, apply it to a bounded filtered diagram in C and a base-change arrow in C. The embedding j preserves its colimit and pullbacks and reflects isomorphisms, so the corresponding comparison in C is an isomorphism. This proves the converse and the theorem.

With $\kappa = \pi_0$, this gives a precise affirmative form of footnote 50. Without the pullback hypothesis the phrase "the same condition on C" needs a relative formulation using pullbacks calculated in $\mathcal{M}$; it is not the intrinsic statement just proved. The argument addresses the proposed reduction of C), and makes no assertion about the separate implication B)⇒C).

Démonstration de 4.5.13.

Prouvons 1°. On a les implications tautologiques

$$\begin{array}{ccccc} \mathrm{b}'' & \Longrightarrow & \mathrm{b}' & \Longrightarrow & \mathrm{b} \\ \Downarrow & & \Downarrow & & \Downarrow \\ \mathrm{a}'' & \Longrightarrow & \mathrm{a}' & \Longrightarrow & \mathrm{a}\ , \end{array} \qquad (^{51})\ ,$$

et pour prouver l'équivalence de ces six conditions, il suffit de prouver

$$\mathrm{a} \implies \mathrm{b}''\ .$$

Si on a

$$x \simeq \varinjlim_{I} x_i\ , \qquad \operatorname{card} \mathrm{Fl}\, I \leqslant \pi\ ,$$

p. 40 [page 129]

soit I' l'ensemble des *sous-diagrammes* $J \subset I$ finis de I, ordonné par inclusion, donc I' est un ensemble ordonné filtrant, et du fait que C est stable par $\varinjlim$ finies, on trouve un foncteur

$$J \longmapsto x_J = \varinjlim_{i\in J} x_i \ :\ I' \longrightarrow C\ ,$$

et on aura

$$x \simeq \varinjlim_{J\in I'} x_J\ .$$

D'autre part, on a

$$\begin{aligned} \operatorname{card} I' &= \operatorname{card} \mathfrak{P}_f(\ \underbrace{\mathrm{Fl}(I)}_{\text{de card} \leqslant \pi}\ ) \\ &\leqslant \pi + \pi^2 + \ \cdots\ + \pi^n + \ \ldots \\ &\leqslant \pi\aleph_0 = \pi\ , \end{aligned}$$

q.e.d. Pour établir 1°, il reste à établir

$$\begin{array}{ccc} & P(\pi) & \\ \swarrow & & \searrow \\ \mathrm{c} & & \mathrm{d}\ , \end{array}$$

51. compte tenu que C est strictement génératrice dans $\mathcal{M}$, donc $x \simeq \varinjlim_{i\in C/x} x_i$.

et l'implication inverse c $\Longrightarrow P(\pi)$ quand $\pi \geqslant \pi_1$. Mais $P(\pi) \Longrightarrow$ c est une tautologie, et $P(\pi) \Longrightarrow$ d puisque $\mathcal{M}_\pi$ est stable par $\varinjlim$ de cardinal $\leqslant \pi$. Reste à prouver que c $\Longrightarrow P(\pi)$ si $\pi \geqslant \pi_1$.

Commentary · Development (commentators') [ms. 128]

One step of colimits already gives all presentable objects

The useful question remaining after the preceding counterexample is whether the one-step colimit class Filt'^{π} is smaller than $\mathcal{M}_\pi$. In fact the two classes coincide for every $\pi \geq \pi_0$. The general presentation theorem is known; it is discussed as Remark 1.30 of Adámek–Rosický in the primary proof of Barton (2019). We give a direct proof in Grothendieck's setting. The marker below identifies our formulation and proof, rather than a claim that the theorem is new.

A bounded presentation theorem, with an explicit flattening proof. [ours] Suppose C is essentially small and closed under colimits with at most π_0 arrows, and $\mathcal{M} = \mathrm{Ind}'_{\pi_0}(C)$. Let D_π be the full subcategory of objects expressible as a colimit in $\mathcal{M}$ of a C-valued diagram with at most π arrows. Then

$$D_\pi = \mathcal{M}_\pi \qquad (\pi \geq \pi_0).$$

In particular D_π is closed under all colimits with at most π arrows, without the extra hypotheses A), B), or C) of Proposition 4.5.13.

The point requiring proof is that a map into a presentation can itself be recorded using only boundedly many of its data. Let $Y = \mathrm{colim}_{j\in J}\, d_j$, with $d_j \in C$, and let $P_0(J)$ be the poset of subcategories of J with at most π_0 arrows. This poset is π_0^+-directed: take unions of at most π_0 arrow sets and close under finite composition. For $K \in P_0(J)$, choose $d_K = \mathrm{colim}_K\, d_j$ in C. The embedding $C \to \mathcal{M}$ preserves these colimits. Their universal properties give compatible transition maps, and

$$Y = \mathrm{colim}_{K\in P_0(J)}\, d_K.$$

This equality follows by identifying cocones, since every object and arrow of J appears in some K. Every $c \in C$ is π_0^+-presentable in $\mathcal{M}$. Hence a map $c \to Y$ factors through some d_K. Two maps can be factored through the same d_K by passing to a common upper bound. Notice that K itself need not be filtered.

We now prove closure under coequalizers. Let $f, g : X \rightrightarrows Y$ be maps between objects of D_π, and choose presentations

$$X = \mathrm{colim}_{i\in I}\, c_i, \qquad Y = \mathrm{colim}_{j\in J}\, d_j, \qquad |\operatorname{Fl} I|, |\operatorname{Fl} J| \leq \pi.$$

Writing $s_i : c_i \to X$ for the first cocone, choose, for every i, one $K_i \in P_0(J)$ and two arrows

$$u_i, v_i : c_i \rightrightarrows d_{K_i}$$

whose composites into Y are fs_i and gs_i. Construct a graph of objects of C as follows. Retain the objects and arrows of the J-diagram. For each i, add a fresh vertex with value d_{K_i} and the complete colimit cone $d_j \to d_{K_i}$ for $j \in K_i$. Add a further fresh vertex with value c_i, and the two arrows u_i, v_i into that d_{K_i}. The free category on this graph, equipped with the assigned composites in C, has at most π arrows. Indeed, there are at most π choices of i, each adds at most π_0 edges, and taking finite paths preserves the infinite bound π. One may also impose the original diagram relations; the bound is unchanged.

A cocone on the original J-diagram with target Z is a map $h : Y \to Z$. Its extension to every new vertex d_{K_i} is forced uniquely, because the displayed cone on K_i is a genuine colimit cone in C and in $\mathcal{M}$. Its extension to the new copy of c_i exists precisely when $hfs_i = hgs_i$, and is then unique. These equalities for all i are equivalent to $hf = hg$, since the presentation maps s_i are jointly epimorphic. Thus cocones on the augmented diagram are exactly maps out of $\mathrm{coeq}(f, g)$. This proves that this coequalizer belongs to D_π. No compatibility between the choices of the K_i is needed: only equations on the jointly epimorphic presentation maps are being imposed.

The class D_π is also closed under coproducts of at most π objects, by taking disjoint unions of their indexing categories. The standard expression of a colimit as a coequalizer of two coproducts now proves closure under every diagram with at most π arrows. The inclusion $D_\pi \subseteq \mathcal{M}_\pi$ follows from closure of presentable objects under these colimits.

For the converse, express an arbitrary X as a colimit of a small C-valued diagram. Its subcategories of at most π arrows give a π^+-filtered presentation of X by objects of D_π, exactly as in the

preceding block. If $X \in \mathcal{M}_\pi$, its identity factors through one term, so X is a retract of an object $Y \in D_\pi$. A retract $X \xrightarrow{i} Y \xrightarrow{r} X$, $ri = 1$, is the coequalizer of 1_Y and ir. Closure under coequalizers gives $X \in D_\pi$, completing the proof.

This supplies the positive answer sought on pages 127–128 and 138. Iterated closure and one-step closure agree here; the bounded cofinal class and the pointwise presheaf bound can still be smaller. The three notions differ because only the first permits the additional relations introduced by the flattening construction.

Supposons donc qu'on ait une famille épimorphique stricte [page 130]

$$x_\alpha \longrightarrow x \qquad (\alpha \in A,\ \operatorname{card} A \leqslant \pi)\ ,$$

cela signifie donc que x est limite inductive d'un système inductif indexé par la catégorie B ayant comme ensemble d'objets $A \amalg (A \times A)$, les flèches non identiques étant représentées par les flèches canoniques

$$\begin{array}{ccc} & x_\alpha \times_x x_\beta & \\ {\scriptstyle p_{\alpha,\beta}}\swarrow & & \searrow{\scriptstyle q_{\alpha,\beta}} \\ x_\alpha & & x_\beta\ . \end{array}$$

(**N.B.** B n'est par une catégorie ordonnée, car pour l'objet (α, α), représenté par $x_\alpha \times_x x_\alpha$, il y a *deux* flèches dans l'objet α, représenté par x_α. Mais on a évidemment $\operatorname{card} \mathrm{Fl}(B) \leqslant \pi$.) Si on savait que C est stable par produits binaires et par passage à des sous-objets (ici les $x_\alpha \times_x x_\beta \ \subset\ x_\alpha \times x_\beta$), on aurait prouvé la condition a). À défaut, utilisons encore le fait que C est génératrice par épimorphismes (même stricts) et choisissons pour tout couple (α, β) un ensemble épimorphique de flèches :

$$\underbrace{x_{\alpha,\beta,\gamma}}_{\in \mathrm{Ob}\, C_0} \xrightarrow{u_{\alpha,\beta,\gamma}} x_\alpha \times_x x_\beta\ ,$$

où C_0 est une sous-catégorie réduite exhaustive de C. La flèche $u_{\alpha,\beta,\gamma}$ est connue quand on connaît les composés p. 41 [page 131]

$$\begin{array}{ccc} & & x_\alpha \\ & {\scriptstyle v_{\alpha,\beta,\gamma}}\nearrow & \\ x_{\alpha,\beta,\gamma} & & \\ & {\scriptstyle w_{\alpha,\beta,\gamma}}\searrow & \\ & & x_\beta\ , \end{array}$$

donc pour $x_{\alpha,\beta,\gamma} \in \mathrm{Ob}\, C_0$ fixé, les choix possibles pour $u_{\alpha,\beta,\gamma}$ sont de cardinal majoré par $\pi_1\pi_1 = \pi_1$, de plus on a $\operatorname{card} \mathrm{Ob}\, C_0 \leqslant \pi_1$, donc le nombre total des flèches $v_{\alpha,\beta,\gamma}$, $w_{\alpha,\beta,\gamma}$ (pour α, β fixés) est majoré par $\pi_1 \cdot \pi_1 = \pi_1$. Donc le nombre total des flèches $u_{\alpha,\beta,\gamma}$, pour α, β, γ variables, est majoré par $\pi \cdot \pi_1 = \pi$ (si $\pi \geqslant \pi_1$). Donc x est représenté par une catégorie I ayant comme objets l'ensemble somme de A et de l'ensemble des indices triples α, β, γ, et pour chaque tel indice, exactement deux flèches $v_{\alpha,\beta,\gamma}$, $w_{\alpha,\beta,\gamma}$ non identiques de source cet objet, et allant respectivement dans x_α, x_β. On a donc $\operatorname{card} \mathrm{Fl}\, I \leqslant \pi$, et on a un foncteur $I \longrightarrow C$, $\alpha \longmapsto x_\alpha$, $(\alpha, \beta, \gamma) \longmapsto x_{\alpha,\beta,\gamma}$, tel que $x = \varinjlim_{i \in I} x_i$. Cela achève la démonstration de 1°).

Prouvons 2°. La première inclusion (4.5.13.1) est tautologique, la deuxième ne fait qu'exprimer $P(\pi) \Longrightarrow$ d. Le fait que $\pi^{\pi_0} = \pi$ implique l'égalité des membres extrêmes est un rappel de 4.5.6. Reste à prouver que Filt'^{π} est stable dans $\mathcal{M}$ par sommes de cardinal $\leqslant \pi$. Mais si $(x_\alpha)_{\alpha \in A}$ est une famille d'objets de Filt'^{π}, avec $\operatorname{card} A \leqslant \pi$, et si $x = \coprod_{\alpha \in A} x_\alpha$, prenant pour tout x_α un foncteur $\varphi_\alpha : I_\alpha \longrightarrow C$, avec $\operatorname{card} \mathrm{Fl}\, I_\alpha \leqslant \pi$, tel que x_α soit la $\varinjlim$ dans $\mathcal{M}$ de ce foncteur, on trouve que x est $\varinjlim$ du foncteur $\varphi : I \longrightarrow C$ défini par les φ_α, où $I = \coprod_\alpha I_\alpha$ (somme dans Cat). Il est évident que $\operatorname{card} \mathrm{Fl}(I) \leqslant \pi \cdot \pi = \pi$, ce qui achève de prouver 2°. [page 132]

Commentary · Exposition [ms. 131]

Strict epimorphic presentations and the number of relations

Grothendieck's counting argument on pages 130–131 gives a useful concrete form of the one-step presentation theorem. It should be read with the hypothesis $x_\alpha \in C$ in condition c); this is used in every hom-set estimate, although it is absent from the printed statement. Without it the identity of an arbitrary X would satisfy c), making the proposed characterization vacuous.

Suppose $(c_\alpha \to X)_{\alpha\in A}$ is a strict epimorphic family, $|A| \le \pi$, with all $c_\alpha \in C$. Because $\mathcal{M}$ has pullbacks, strictness means that X is the colimit of the family together with all intersections $c_\alpha \times_X c_\beta$ and their two projections. An intersection need not lie in C. To replace it, use all maps $c \to c_\alpha \times_X c_\beta$ with $c \in C_0$. Such a map is determined by its two composites $c \to c_\alpha, c \to c_\beta$. For $\pi \ge \pi_1$, there are at most π_1 choices for each fixed pair (α, β), and at most $\pi^2\pi_1 = \pi$ choices altogether.

Form a category with one vertex for each c_α and one vertex for each selected c, with two arrows from the latter vertex to its two targets. Its colimit is X. To verify this, a cocone gives maps $c_\alpha \to Z$ agreeing after every selected map into an intersection. The family of selected maps is jointly epimorphic on that intersection, so the two maps from the intersection to Z agree. Strictness of the original family then gives one and only one map $X \to Z$. This proves condition a) directly and justifies the arrow count. The two projections for $\alpha = \beta$ must both be retained; the indexing category is not generally a poset.

Conversely, the legs of any colimit diagram form a strict epimorphic family: a family compatible with all equalities in the common target is in particular compatible with every arrow of the diagram, and therefore descends uniquely by the colimit property. Thus, for $\pi \ge \pi_1$, condition c), with its missing domain requirement restored, characterizes $D_\pi = \mathcal{M}_\pi$.

The presentation theorem also sharpens the counts left unfinished around (4.5.9.1). Throughout this count assume $\pi' \ge \pi \ge \pi_1$. When $\pi \ge \pi_1$, there are at most 2^π isomorphism classes of diagrams of at most π arrows with values in C_0, and hence at most 2^π objects in a skeleton of $\mathcal{M}_\pi$. If $A \in \mathcal{M}_\pi$ and $X \in \mathcal{M}_{\pi'}$, then a one-step presentation of A gives

$$|\mathrm{Hom}(A, X)| \le \prod_{i\in \mathrm{Ob}\, I} |\mathrm{Hom}(c_i, X)| \le \big((\pi')^{\pi_0}\big)^{\pi} = (\pi')^{\pi}.$$

Taking $\pi' = \pi$ in the hom estimate also bounds the arrows in a skeleton of $\mathcal{M}_\pi$ by 2^π. An arrow of the slice is determined by such an arrow $A \to B$ and a map $B \to X$. Consequently a skeleton of $\mathcal{M}_\pi/X$ has at most $(\pi')^\pi$ arrows. The sharper bound is therefore available even for the canonical slice, once the bounded presentation theorem has been established.

Prouvons 3°. On a les implications tautologiques $1 \Longrightarrow 1'$, et $1' \Longrightarrow 1$, puisque Filt'^{π} est stable par sommes de cardinal $\leqslant \pi$. Donc $1 \Longleftrightarrow 1'$, et implique que pour tout x dans $\mathcal{M}$, $I = \mathrm{Filt}'^{\pi}/x$ est stable par $\varinjlim$ de cardinal $\leqslant \pi$, donc est grand devant π. Comme $x \simeq \varinjlim_{i\in I} x_i$, [page 133]
cela prouve que $1 \Longrightarrow 2$. L'implication $2 \Longrightarrow 3$ est standard, et résulte de $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$, je vais expliciter dans un

Lemme 4.5.14. *Soient $\mathcal{M}$ une catégorie stable par petites $\varinjlim$, π un cardinal infini, $\mathcal{M}_0$ une sous-catégorie strictement pleine de $\mathcal{M}$. Considérons les conditions*

a) $\mathcal{M}_0$ est une sous-catégorie strictement génératrice, et stable dans $\mathcal{M}$ par $\varinjlim$ de cardinal $\leqslant \pi$.

b) Pour tout objet de $\mathcal{M}$, on a un isomorphisme

$$x \simeq \varinjlim_{i\in I} x_i\ ,$$

les x_i dans $\mathcal{M}_0$, I grande devant π.

b′) Comme b), avec I catégorie ordonnée. p. 42

c) $\mathcal{M}_\pi \subset \mathcal{M}_0$ (où $\mathcal{M}_\pi$ est la sous-catégorie strictement pleine de $\mathcal{M}$ formée des objets π-accessibles).

On a donc les implications

$$a \implies b \iff b' \implies c\,,$$

et si on suppose $\mathcal{M}_0 \subset \mathcal{M}_\pi$, et $\mathcal{M}_\pi$ sous-catégorie strictement génératrice de $\mathcal{M}$, alors les quatre conditions a, b, b′, c sont équivalentes, et équivalent à $\mathcal{M}_0 = \mathcal{M}_\pi$ ([52]).

Enfin (revenant à la démonstration de 4.5.13, 3°), l'implication 3) $\Longrightarrow$ 1) provient du fait [page 134] que $\mathcal{M}_\pi$ est stable par $\varinjlim$ de cardinal $\leqslant \pi$.

Reste à prouver que ces conditions, i.e. la stabilité de Filt'^{π} par conoyaux de double-flèches, est satisfaite dans les cas A) B) C), le cas A) étant d'ailleurs déjà connu. Le cas B) résulte immédiatement de la transitivité des familles épimorphiques strictes universelles, et c'est d'ailleurs aussi contenu explicitement (stabilité de Filt'^{π} par $\varinjlim$ de cardinal $\leqslant \pi$) dans SGA 4 I 9.13 (sinon dans l'énoncé, du moins c'est signalé dans la démonstration). En rédigeant SGA 4 I 9, je savais donc qu'on a $\mathcal{M}_\pi \subset \mathrm{Filt}'^{\pi}$ pour tout π, mais l'inclusion, plus évidente pourtant dans le cas d'espèce, $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$ (quand on suppose les objets de C π_1-accessibles) m'avait apparemment échappé ?

Commentary · Development (commentators') [ms. 134]

The retract missing from the abstract lemma

Grothendieck isolates Lemma 4.5.14 to make the passage from presentations to presentability reusable. Its essential mechanism is correct: presentability factors the identity through one stage. What this produces is a retract, and a strictly full subcategory need not contain all retracts of its objects.

A counterexample to b)⇒c) in Lemma 4.5.14. [ours] Fix an infinite cardinal π, take $\mathcal{M} = \mathrm{Ens}$, and let $\mathcal{M}_0$ be the full subcategory of sets of cardinality at most π except the singleton sets. Every object of $\mathcal{M}_0$ is π-accessible, and $\mathcal{M}_0$ is essentially small and strictly generating: its two-element sets already map jointly surjectively to every nonempty set. Also $\mathcal{M}_\pi$ is strictly generating, so the additional hypotheses in the last sentence of the printed lemma are satisfied.

For a set X with at least two elements, use the poset of its subsets A with $2 \leq |A| \leq \pi$. This is π^+-directed, because a union of at most π such subsets still has at most π elements, and its colimit is X. The empty set has its constant presentation. For a singleton, index a diagram by the ordinal π^+, put the two-element set $\{0,1\}$ at every stage, and use the constant map with value 0 for every strict transition. Identities remain identities, and composition is respected because the constant map is idempotent. The ordinal is π^+-directed, and the colimit of this diagram is a singleton. Thus b′), and hence b), hold for every X. Condition c) fails, since a singleton is π-accessible but is not in $\mathcal{M}_0$.

The corrected implication is

$$\text{b} \quad \Longrightarrow \quad \mathcal{M}_\pi \subseteq \{\text{retracts of objects of } \mathcal{M}_0\}.$$

Its proof is exactly the identity-factorization argument. Adding closure of $\mathcal{M}_0$ under retracts restores b)⇒c), and restores the final equivalence when the other hypotheses of the lemma hold. In the applications, essential smallness of the relevant slices has already been established; an abstract formulation should retain an appropriate small-presentation hypothesis when using $\mathcal{M}_0/X$ as an indexing category.

Under the small-presentation hypothesis just specified, the implication a)⇒c), which is the one used in Proposition 4.5.17, survives. Closure under colimits of size at most π includes coequalizers, hence splitting of idempotents. Starting from the canonical dense presentation, the same closure makes $\mathcal{M}_0/X$ π^+-filtered; the resulting retract therefore belongs to $\mathcal{M}_0$. Likewise our proof that $D_\pi = \mathcal{M}_\pi$ established coequalizer closure before invoking identity factorization. The repair does not weaken the final canonical filtration theorem.

Reste à traiter le cas C). (J'ai l'impression qu'il est un peu plus général que le cas B), i.e. que la condition B) sur $\mathcal{M}$ implique C), mais non inversement. Ainsi, la condition C) est vérifiée dès que $\mathcal{M}$ correspond à une espèce de structure algébrique définie en termes de $\varprojlim$ [page 135] *finies*, i.e. quand $\mathcal{M}$ est de la forme $\mathrm{Ind}(C)$, C stable par $\varinjlim$ *finies*, alors que je doute que

52. cf. démonstration de proposition 4.5.6, 4 $\Longrightarrow$ 5, page 117.

la condition B) soit toujours satisfaite pour ces catégories. Par exemple dans la catégorie Cat (qui est un paratopos de cette espèce), les $\varinjlim$ filtrantes sont exactes à gauche, mais je ne crois pas que les familles épimorphiques strictes soient universelles. Prenons p.ex. deux homomorphismes de groupes $G_0 \xrightarrow{u_0} G$, $G_1 \xrightarrow{u_1} G$ tels que $G_0 * G_1 \longrightarrow G$ soit surjectif, alors cette famille à deux éléments est épimorphique stricte dans Gr et dans Cat ([53]), mais n'est pas forcément épimorphique stricte universelle, même si p.ex. $G_0 = G_1 = H$, $G = H \times H$, et u_0, u_1 sont des injections $H \rightrightarrows H \times H$, données par $u_0(h) = (h, 1)$, et $u_1(h) = (1, h)$. Si je prends l'application diagonale $H \longrightarrow H \times H$, les images inverses de u_0, u_1 sont le même objet de $\mathcal{M}/H$, savoir $e \longrightarrow H$, et cette famille n'est épimorphique stricte que si H est le groupe unité.)

Le cas C) va résulter du résultat plus précis :

Corollaire 4.5.15. *Sous les conditions générales de 4.5.13, supposons que dans $\mathcal{M}$ les* $\varinjlim$ filtrantes *commutent au changement de base (p.ex. les* $\varinjlim$ *filtrantes sont exactes). Alors pour tout $\pi \geqslant \pi_1$,* Filt'^{π} *est formée des $x \in \mathrm{Ob}\,\mathcal{M}$ satisfaisant la condition suivante.* [page 136]

Commentary · Examples [ms. 135]

A free product repairs the group example

Grothendieck wants to show that universality of filtered colimits is compatible with failure of universality for strict epimorphic families. The intended distinction is sound, but the two axes of $H \times H$ do not give the promised example. For nontrivial H, their common intersection is the trivial group. Maps from the two copies of H to $H * H$ agree on that intersection, but do not extend to $H \times H$: the images of the two factors would have to commute. The normal form in a free product shows that nonidentity elements in the two factors do not commute. Thus the axes are not a strict epimorphic family, contrary to the assertion retained in footnote 53.

A nearby example proves exactly what is needed. Put

$$G = \langle a, b \mid a^2 = b^2 = 1 \rangle = C_2 * C_2, \qquad A = \langle a \rangle, \quad B = \langle b \rangle.$$

The inclusions $A \to G$ and $B \to G$ form a strict epimorphic family in groups. Any compatible pair of maps out of A and B extends uniquely to the free product; their intersection is trivial, so compatibility imposes only agreement on the unit. The element ab has infinite order, since every nonzero power has a nonempty reduced word. Hence $K = \langle ab \rangle$ is infinite cyclic, and reduced-word normal form gives

$$K \cap A = 1 = K \cap B.$$

Pulling the family back along $K \hookrightarrow G$ therefore gives two copies of $1 \to K$. This family is not even jointly epimorphic: the trivial homomorphism $K \to C_2$ and the homomorphism sending ab to the nonidentity element agree on both copies of 1.

The same construction works in Cat by viewing groups as one-object categories. A compatible pair of functors $BA, BB \to \mathcal{D}$ must select the same object of $\mathcal{D}$, because their restrictions to the common terminal category agree. Their homomorphisms into that object's automorphism group extend uniquely to $A * B$. This proves strict epimorphy in Cat itself, with arbitrary target categories, rather than merely in its subcategory of groups. Pullback along $BK \to BG$ gives the same failure.

Finally, ordinary filtered colimits in Cat commute with finite limits. One direct verification uses the nerve. For every finite ordinal $[n]$, a functor $[n] \to \mathrm{colim}_i \mathcal{A}_i$ is specified by finitely many objects, arrows, and composition equalities, so it occurs at one stage of a filtered diagram; equality between two such functors is likewise witnessed at a later stage. Thus the nerve preserves filtered colimits. It also preserves limits and reflects isomorphisms. In simplicial sets the required commutation is pointwise the familiar set-theoretic fact, so it holds in categories. This settles the direction relevant to Grothendieck's example: condition C) does not imply condition B). The converse suggestion B)⇒C) is a separate assertion, not established by this example.

c′) Il existe une famille de flèches $(u_\alpha)_{\alpha \in A}$, $u_\alpha : x_\alpha \longrightarrow x$, dans $\mathcal{M}$, qui est épimorphique stricte universelle *(i.e. couvrante pour la topologie canonique de $\mathcal{M}$), avec $x_\alpha \in \mathrm{Ob}\,C$ $\forall\ \alpha \in A$, et* $\mathrm{card}\,A \leqslant \pi$. p. 43

53. !, c'est *faux* en général, mais vrai dans le cas particulier plus bas.

Démonstration : il faut prouver que cette condition équivaut à la condition c) de 4.5.13 1°). Comme on a c′ ⟹ c trivialement, il reste à montrer que c ⟹ c′. Mais on a vu dans 4.5.13 1°) que c implique a″, i.e. qu'on a un isomorphisme

$$x = \varinjlim_{i\in I} x_i ,$$

les x_i dans C, et I ensemble *ordonné filtrant*, avec $\operatorname{card} I \leqslant \pi$. Il reste à voir que si x dans $\mathcal{M}$ est une limite inductive *filtrante* d'objets x_i, alors l'ensemble des flèches $x_i \longrightarrow x$ est épimorphique stricte *universelle.* Or cela est immédiat, car pour $x' \longrightarrow x$, on a $x' = \varinjlim_i x'_i$ (par hypothèse C), où $x'_i = x' \times_x x_i$, donc la famille $x'_i \longrightarrow x'$ est épimorphique stricte, q.e.d.

Prouvons maintenant que dans le cas C, la condition de 3°, 1 est satisfaite, i.e. que Filt'^{π} est stable par $\varinjlim$ de cardinal $\leqslant \pi$. Cela résulte aussitôt du [page 137]

Corollaire 4.5.16. (de 4.5.15). *Sous les conditions de 4.5.15 (i.e. la condition C) de 4.5.13, 3°), soit x un objet de $\mathcal{M}$, et supposons qu'il existe une famille* épimorphique universelle *$(x_\alpha \longrightarrow x)_{\alpha\in A}$ de flèches de $\mathcal{M}$, avec $x_\alpha \in \mathrm{Ob}\ \mathrm{Filt}'^{\pi}$ pour tout $\alpha \in A$, et $\operatorname{card} A \leqslant \pi$. Alors $x \in \mathrm{Ob}\ \mathrm{Filt}'^{\pi}$.*

En effet, l'hypothèse sur x_α équivaut, en vertu de 4.5.15, à l'existence d'une famille épimorphique stricte universelle de flèches $(x_{\alpha\beta} \longrightarrow x_\alpha)_{\beta\in B_\alpha}$, avec $\operatorname{card} B_\alpha \leqslant \pi$ et les $x_{\alpha\beta}$ dans C. Soit $B = \coprod_{\alpha\in A} B_\alpha$, on a alors $\operatorname{card} B \leqslant \pi^2 = \pi$, et d'autre part, la famille des flèches composées $x_{\alpha\beta} \longrightarrow x_\alpha \longrightarrow x$ indexée par B est épimorphique stricte universelle (par transitivité de ce type de familles), donc $x \in \mathrm{Ob}\ \mathrm{Filt}'^{\pi}$, q.e.d.

Avec tout ça, on n'a toujours pas donné de caractérisation de $\mathcal{M}_\pi$ lui-même, qui est meilleure que Filt^{π} et Filt'^{π} du point de vue catégorique, car seuls les $\mathcal{M}_\pi$ définissent une filtration cardinale, et de plus ils ont les propriétés de stabilité désirables dans ce contexte, [page 138] pour les $\varinjlim$ de cardinal $\leqslant \pi$. Le seul critère non tautologique que j'arrive à dégager dans le cas général (pour tout $\pi \geqslant \pi_0$, et sans hypothèse d'exactitude supplémentaire sur $\mathcal{M}$) est le suivant, beaucoup moins "explicite" ou "calculatoire" que les critères très concrets pour Filt^{π} et pour Filt'^{π} :

Commentary · Development (commentators') [ms. 137]

Universal epimorphisms can conceal too many relations

Grothendieck's condition C) gives a useful covering interpretation of a small presentation. If $X = \operatorname{colim}_{i\in I} X_i$ is filtered and such colimits commute with pullback, then every pullback of its cocone is again a colimit cocone, hence is strictly epimorphic. Combined with the finite-diagram replacement above, this proves Corollary 4.5.15: for $\pi \geq \pi_1$, the one-step colimit class consists exactly of objects with a universal strict epimorphic family of at most π maps from C.

Corollary 4.5.16 replaces "strict epimorphic" by "epimorphic" for the outer family, although its proof still uses strict descent. The weaker hypothesis does not suffice. The following example isolates the missing information: a surjection can cover every element while supplying none of the relations.

A counterexample to Corollary 4.5.16 as printed. [ours] Let $\mathcal{R}$ be the category whose objects are pairs (S, R) with $R \subseteq S^{\mathbb{N}}$, and whose arrows are functions preserving this one countably-ary relation. We first verify the ambient hypotheses, so that the example does not rely on a category outside the scope of the proposition.

Limits have the usual underlying set-theoretic limits, with a tuple declared related precisely when every projection is related. A colimit has the underlying set-theoretic colimit, and its relation is the union of the images of the relations in the diagram. These descriptions follow directly by testing cones and cocones. All these colimits are universal. Indeed, for a colimit $(S, R) = \operatorname{colim}_i(S_i, R_i)$ and an arrow $(T, Q) \to (S, R)$, the underlying set of the pulled-back colimit is T, by universality of set colimits. If $q \in Q$, its image in R comes from some $r_i \in R_i$, by the definition of the colimit relation. Pairing the coordinates of q and r_i gives a related tuple in $(T, Q) \times_{(S,R)} (S_i, R_i)$ mapping to q. Thus its colimit relation is all of Q, proving the assertion, including condition C).

Let C be a skeleton of objects with countable underlying set and countable relation. It is closed under countable colimits by the preceding formula. Its objects are $\aleph_1$-presentable. To lift a map

from such an object into an $\aleph_1$-filtered colimit, lift its countably many elements, choose witnesses for its countably many related tuples, and then witness the resulting countably many coordinate equalities at one common stage. This produces a relation-preserving lift; the same common-stage argument proves eventual equality of two lifts. Every object is an $\aleph_1$-filtered union of its countable partial substructures: take countable sets of elements and countable sets of related tuples, including the countably many coordinates of the chosen tuples. These choices form a countably directed poset. Therefore $\mathcal{R} \simeq \mathrm{Ind}'_{\aleph_0}(C)$, with $\pi_0 = \aleph_0$. There are at most $2^{\aleph_0}$ arrows in a skeleton of C: code countable carriers by subsets of $\mathbb{N}$, countable relations by countable lists of sequences, and maps by functions between countable sets.

Epimorphisms in $\mathcal{R}$ are exactly the maps surjective on underlying sets. Surjectivity suffices because the forgetful functor is faithful. If a map misses an element, two functions to the two-element set equipped with its full relation can distinguish that element while agreeing on the image; both are morphisms. Such epimorphisms are universal, since pullbacks are set-theoretic on carriers.

Choose $\pi = \beth_\omega$ and a set S of cardinality π. The arrow

$$q : (S, \varnothing) \longrightarrow (S, S^{\mathbb{N}}), \qquad q = \mathrm{id}_S,$$

is universally epimorphic. Its source is π^+-presentable: its hom-functor is simply $\mathrm{Hom}_{\mathrm{Ens}}(S, U(-))$, where U preserves colimits. Its target is not π^+-presentable. Indeed, $|S^{\mathbb{N}}| > \pi$. To see this directly, partition any purported list of at most π sequences into an increasing union of sublists L_n of cardinality less than π, and choose the nth coordinate outside the coordinates appearing there. The resulting sequence differs from every listed sequence.

Now write $(S, S^{\mathbb{N}})$ as the colimit of (S, R_0) over all subsets $R_0 \subseteq S^{\mathbb{N}}$ with $|R_0| \leq \pi$. This index is π^+-directed, and the transition maps are identities on carriers. If the target were presentable, its identity would factor through some (S, R_0). The factor map would necessarily be the identity on S, which preserves the full relation only if $S^{\mathbb{N}} \subseteq R_0$, an impossibility. The source belongs to D_π already as a coproduct of π singleton structures with empty relation, whereas the target does not, since $D_\pi \subseteq \mathcal{M}_\pi$. The single arrow q therefore violates precisely the conclusion of Corollary 4.5.16, with $\pi \geq \pi_1$ and all its other hypotheses satisfied.

The same category even satisfies B). A family is strictly epimorphic precisely when it is jointly surjective on carriers and every relation tuple of its target is the image of a relation tuple in one member. For necessity, strictness implies joint epimorphy, and the preceding two-point test proves joint surjectivity. Factor the family through the structure on the same carrier with only the image relation. The factor maps are compatible with the original family, so strictness supplies a map back to that structure. Joint surjectivity forces its underlying function to be the identity, hence forces every target tuple to lie in the image relation. Conversely, compatible maps out of the family agree on each underlying pairwise intersection, since the intersection structure has the set-theoretic pullback as carrier. They therefore descend to a unique function on the target carrier; the tuple condition makes this function a morphism. Both conditions are preserved by pullback, by the coordinate-pairing argument above. Thus the counterexample is entirely about the omission of strictness, not a failure of either exactness condition.

The positive statement is restored by requiring the outer family to be universally strict epimorphic. Choose the universal strict covers of its members furnished by Corollary 4.5.15. Their composites form a universal strict cover of the target by transitivity, with at most $\pi \cdot \pi = \pi$ terms, so 4.5.15 applies again. This is the covering-composition principle for the canonical topology (Artin et al., 1972, Exposé II, 2.5–2.6). We found no published treatment of this particular counterexample to the manuscript assertion; the proof is supplied here without a claim of priority.

Proposition 4.5.17. *Soit π un cardinal $\geqslant \pi_0$. Alors $\mathcal{M}_\pi$ est égal à la plus petite sous-catégorie strictement pleine de $\mathcal{M}$, contenant C, et stable par $\varinjlim$ de cardinal $\leqslant \pi$.* (Il est important de noter qu'il *ne suffit pas* de dire : stable par $\varinjlim_I$, avec I *grande devant* π_0, et $\operatorname{card} I \leqslant \pi$. On trouve alors Filt^π, qui risque d'être une catégorie strictement plus petite que $\mathcal{M}_\pi$. Ce n'est pas non plus la catégorie des $\varinjlim_I x_i$, avec $x_i \in C\ \forall\ i \in \mathrm{Ob}\, I$, et $\operatorname{card} \mathrm{Fl}\, I \leqslant \pi$ – puisque cette catégorie-là est Filt'^π.)

Démonstration. Soit $\mathcal{M}'$ la catégorie de l'énoncé, on a donc $\mathcal{M}' \subset \mathcal{M}_\pi$, puisque $\mathcal{M}_\pi$

contient C et est stable par les $\varinjlim$ de cardinal $\leqslant \pi$. Inversement, le lemme 4.5.14, a $\Longrightarrow$ c.

Commentary · Present state [ms. 138]

The closure description and the later presentation theorem

Grothendieck ends this portion with a characterization that is intrinsic to the category: start with its chosen small collection of π_0^+-presentable objects and close under colimits having at most π arrows. Proposition 4.5.17 is correct. The generating collection lies in $\mathcal{M}_\pi$, and that class is closed under the prescribed colimits. Conversely, the canonical filtered presentation and the splitting of its resulting retract give the reverse inclusion. Only a)⇒c) of the corrected Lemma 4.5.14 is needed.

The additional warning that a single colimit of this size cannot suffice should not be retained as a present-day conclusion. Our explicit flattening proof shows that such one-step colimits already form $\mathcal{M}_\pi$. The general theorem is the assertion attributed to Adámek–Rosický, Remark 1.30, in Barton (2019); Barton's authored proof removes the retract using good diagrams and the fat small object argument. That proof also records why replacing arbitrary small colimits by colimits that are π_0^+-filtered is a stronger demand. The arithmetic obstruction exposed by sets remains real for the latter demand.

There are therefore two distinct outcomes to the exploratory discussion. The intrinsic filtration exists at every rank $\pi \geq \pi_1$, with the stated quantitative bound, and its one-step colimit description needs no extra exactness condition. The full canonical presheaf presentation can still have too many maps, and a small presentation need not be cofinal in it. The unrepaired implication in Lemma 4.5.14 and the group example do not invalidate these positive results; their corrections explain which arguments actually prove them.

Overview · XVIII-a

The opening of Chapter XVIII asks what a category of coefficients must retain if the formal operations of sheaf theory are to survive outside a topos. Grothendieck begins with colimits and a small dense collection of test objects. The proposed tensor then forces him to ask whether continuity conditions define a reflective category, and that question leads to accessibility. The long investigation of cardinal filtrations is therefore part of the construction of coefficients. His letter to Thomason of 2 April 1991 explicitly describes this chapter as a return to the accessibility ideas of SGA 4, Exposé I, Section 9 (Grothendieck, 1991a; Artin et al., 1972).

These pages run from manuscript page 50 to page 138, corresponding to printed pages 1–43 of the edition. They contain Sections 4.1–4.5: pseudo-topoi and the tensor problem, its small-presheaf formulation, the proposed accessibility criterion, the restricted ind-completion, and the intrinsic cardinal filtration. The text often records an unsuccessful proof and then retains the objection which caused its abandonment. A faithful reading has to preserve this exploratory structure. An assertion subsequently withdrawn is not evidence that the intended theorem is false; conversely, the truth of the final theorem does not validate an intermediate equivalence.

We work in a fixed Grothendieck universe and use smallness relative to that universe. One translation governs the entire discussion. If π is infinite, a category called *grande devant* π is filtered with respect to diagrams having at most π arrows. In the modern regular-cardinal convention it is π^+-filtered. Grothendieck's π-accessible objects are consequently π^+-presentable objects. His cocomplete categories with accessible small generators are locally presentable categories; the broader modern notion of an accessible category does not include cocompleteness. The cardinal attached to an object and one uniform cardinal for a chosen small generating family must also be distinguished.

A pseudo-topos has enough density for the dual special adjoint functor theorem. Colimit-preserving functors out of it have right adjoints, and a contravariant set-valued functor taking colimits to limits is representable. These facts make the opening construction plausible. They also expose a distinction in the motivating terminology. If morphisms of pseudo-topoi are described by their inverse-image functors, then the actual bicategorical product has underlying category $\mathcal{M} \times \mathcal{N}$: a cocontinuous functor out of that category is determined by its restrictions to $(m, 0)$ and $(0, n)$. The proposed $\mathcal{M} \boxtimes \mathcal{N}$ represents functors cocontinuous in each variable separately. It is a tensor, with **Set** as unit, rather than this Cartesian product. For Grothendieck topoi the tensor nevertheless agrees with the product in the more restrictive category of topoi and geometric morphisms; Pitts's Theorem 2.3 proves the ordinary comparison (Pitts, 1985).

The tensor is first described as the category of bifunctors

$$\mathcal{M}^{\mathrm{op}} \times \mathcal{N}^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

which take colimits to limits separately. Density embeds it in a presheaf category on a small product of test categories. This proves local smallness and gives a useful formula for its continuous extensions, but it does not by itself provide the reflector. Nor does the naive external product of two representable presheaves usually satisfy the continuity condition: the empty colimit already forces both representing objects to be terminal. The example $\mathbf{Ab} \boxtimes \mathbf{Ab} \simeq \mathbf{Ab}$ makes the failure concrete. Its external tensor is the ordinary tensor product over $\mathbb{Z}$, and it cannot be recovered from the Cartesian product formula suggested in the manuscript.

Accessibility supplies the missing reflector. Choose a common regular cardinal and small full subcategories of presentable objects closed under colimits below that cardinal. Separate continuity can now be imposed by orthogonality to a *set* of maps between small presheaves.

The commentary constructs the reflection by alternately adjoining extensions and identifying extensions which agree on their domains. Presentability controls the length of the construction. The resulting category is again locally presentable, and reflection of representables supplies its elementary tensors. This gives a complete existence proof in the setting that the chapter is trying to isolate. Kelly's treatment of categories with prescribed colimits supplies the ordinary categorical background (Kelly, 1982, Section 6.5). Lurie's tensor on presentable ∞-categories is a later comparison with the same universal pattern, not a substitute for the ordinary cardinal estimates here (Lurie, 2017, Section 4.8.1).

The structural assertion behind Theorem 4.3.6 survives the failure of its first numerical proof. Starting from a small dense family of presentable objects, close that family under suitably small colimits. Every slice of the resulting category over an object is filtered at the required rank, and its canonical colimit is that object. The ind-object Hom formula proves full faithfulness. Thus the category is a restricted ind-completion. This argument does not require the provisional filtration by sizes of categories of elements to have the properties claimed for it on pages 72–76. The latter proof uses an unestablished directed-hull bound and imports a covering estimate from SGA 4 without its universality hypothesis. The manuscript's own corrective footnotes are justified.

Section 4.4 gives the complementary construction. When C has colimits of diagrams with at most π arrows, its π^{+}-ind-completion is cocomplete and is the free cocomplete category which preserves those existing colimits. The proof passes through cofinality, simultaneous presentations of diagrams, bounded coproducts, and the expression of an arbitrary indexing category as a directed union of small subcategories. We supply the omitted comma-adjunction argument and separate the existence of individual ind-presentations from closure under filtered colimits. The unrestricted claim that restriction on all functor categories is fully faithful is false; the needed restriction to functors preserving the density presentations is fully faithful and repairs the universal-property proof. Henry's recent theorem gives an exact modern comparison for these diagram presentations, with the hypotheses on the diagram shape and on C stated explicitly (Henry, 2025a).

The difference between ordinary and more highly filtered ind-objects is essential. Let C consist of countable sets. Its $\aleph_1$-ind-completion is sets, but the ordinary ind-object obtained from finite subsets of $\mathbb{Z}$ has, at c, only the finite-image maps $c \to \mathbb{Z}$. Translation gives it a $\mathbb{Z}$-action which cannot occur at any nonempty equivariant representable stage. This answers negatively the specific cofinality question on manuscript page 94. It also explains why the restricted completion need not be closed under ordinary filtered colimits calculated in the presheaf category.

Section 4.5 measures an object in two ways. Write $\mathcal{M}_\pi$ for its intrinsic presentability layer, and Filt^π for the layer whose restricted nerve has at most π maps from each test object, once the chosen small category itself is bounded by π. In sets, with tests of size at most π_0, these conditions read respectively

$$|X| \le \pi, \qquad |X|^{\pi_0} \le \pi.$$

They need not coincide. The corrected comparison is

$$\mathrm{Filt}^\pi \subseteq \mathcal{M}_\pi \subseteq \mathrm{Filt}^{\pi^{\pi_0}}.$$

The fixed-point condition $\pi^{\pi_0} = \pi$ restores equality. No continuum hypothesis is needed for the comparison itself. The counterexample sketched in the manuscript by a supposedly smallest directed family of subsets is not valid; the commentary replaces it with a diagonal argument at a singular cardinal of countable cofinality.

There is a second distinction within the proposed primed filtration. An object may be the colimit of a diagram of at most π objects and arrows from C without admitting a cofinal diagram of that size in C/X. For C the countable sets and $|X| = \beth_\omega$, the coproduct of the singleton subsets gives the first presentation. A cofinal presentation would give at most $\beth_\omega$

countable subsets cofinal among all countable subsets of X, which the same diagonal argument excludes. Thus the two triples of conditions in Proposition 4.5.13 cannot be identified.

The positive outcome is stronger than the closing warning on page 138 suggests. The class of *one-step* colimits of C-diagrams with at most π arrows is already $\mathcal{M}_\pi$, for every $\pi \geq \pi_0$. The commentary proves closure under coequalizers by recording, for each generator in a presentation of the source, only the bounded portion of a target presentation through which its two images factor. This keeps the enlarged diagram within the original cardinal bound. Coproducts and splitting of idempotents finish the proof. The general presentation theorem is known; Barton's authored proof discusses its earlier attribution and the difficulty with the originally cited justification (Barton, 2019). Our argument is included as a direct repair in Grothendieck's setting, without a claim of priority. It also yields the sharper bounds on the number of objects and arrows which the chapter seeks.

Two later corrections prevent the descent arguments from obscuring that conclusion. Presentability factors an identity through a stage as a *retract*; Lemma 4.5.14 must retain closure under retracts before replacing this by membership in the chosen subcategory. And a universal epimorphism need not transmit a bound on the number of relations. A category of sets with one countably-ary relation gives a counterexample to Corollary 4.5.16 even when all colimits are universal. Requiring the outer family to be universally *strict* epimorphic restores the corollary. Both corrections leave the intrinsic filtration theorem intact.

Footnote 50 suggests testing universality of filtered colimits in the generating category. We prove a precise version: if C also has pullbacks, then ordinary filtered colimits in $\mathrm{Ind}'_{\pi_0}(C)$ are universal exactly when the filtered colimits in C with at most π_0 arrows are universal. The proof approximates the entire cocone and its base-change arrow together, and then expresses any filtered indexing category as a π_0^{+}-directed union of filtered subcategories with at most π_0 arrows. The additional pullback hypothesis makes the test intrinsic to C.

The remaining verification limits concern the disputed implication in Proposition 4.3.1 for arbitrary pseudo-topoi and the suggestion that universality of strict epimorphic families implies universality of ordinary filtered colimits. We do not describe these as literature-established open problems. The relevant older discussions of density, ind-objects, and the distinction between formal and realized colimits are cross-referenced where they are used.

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1) Filt'^{π} *est stable dans* $\mathcal{M}$ *par* $\varinjlim$ *de cardinal* $\leqslant \pi$.

1′) Filt'^{π} *est stable par conoyaux de double-flèches.*

2) Tout objet x *de* $\mathcal{M}$ *est de la forme* $\varinjlim_{I} x_i$, *où* x_i *dans* Filt'^{π}, I *grande devant* π.

3) La condition d) de 1°) implique $P(\pi)$ *(donc lui est équivalente), i.e. on a*

$$\mathcal{M}_\pi \;=\; \mathrm{Filt}'^{\pi}\,.$$

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Ces conditions sont satisfaites dans les trois cas suivants :

A) $\pi^{\pi_0} = \pi$ *(pour mémoire, c'est contenu dans 2°).*

B) Les familles épimorphiques strictes dans $\mathcal{M}$ *sont épimorphiques strictes universelles.*

C) Les limites inductives filtrantes dans $\mathcal{M}$ *commutent au changement de base (ce qui est le cas notamment si les* $\varinjlim$ *filtrantes sont exactes à gauche)* ([50]).

Remarque. La filtration de $\mathcal{M}$ pour les Filt'^{π} n'est autre que celle envisagée dans SGA 4 I 9.13, où on fait de plus l'hypothèse B) sur $\mathcal{M}$, qui après-coup n'était plus tellement de mon goût. Toujours est-il que d'après (4.5.13.1), cette filtration se rapproche plus de la "bonne" filtration par les $\mathcal{M}_\pi$, que celle que j'avais eu encore naguère tendance à lui préférer, celle par les Filt^{π}. Elle a d'ailleurs sur cette dernière l'avantage, dans le cas envisagé B), d'être une filtration cardinale (ce qui n'est pas nécessairement le cas pour Filt^{π}, du moins s'il existe des cardinaux infinis $\pi \geqslant \pi_0$ avec $\pi^{\pi_0} \neq \pi$). Je soupçonne maintenant qu'en général (et s'il existe des cardinaux infinis $\pi > \pi_0$ tels que $\pi^{\pi_0} \neq \pi$), en dehors des conditions A) B) C), la filtration par les (Filt'^{π}) n'est *pas* une filtration cardinale. En tous cas, 4.5.6 et la présente proposition, 3°, montrent que si soit $(\mathrm{Filt}^{\pi})_{\pi\geqslant\pi_2}$, soit $(\mathrm{Filt}'^{\pi})_{\pi\geqslant\pi_2}$ est une filtration cardinale (π_2 étant un cardinal donné $\geqslant \pi_1$), alors elle coïncide avec la filtration par les $\mathcal{M}_\pi$. Donc apparemment, dans tous les cas où je sais construire une filtration cardinale, sur un paratopos du moins (i.e. sur une catégorie $\mathcal{M}$ stable par petites $\varinjlim$, et dont tous les objets sont accessibles – l'existence d'une filtration cardinale impliquant de plus celle d'une petite sous-catégorie strictement génératrice …), celle-ci n'est autre que celle par les $\mathcal{M}_\pi$ – chose dont je ne m'étais par aperçu en rédigeant SGA 4 I 9.

[page 128]

Démonstration de 4.5.13.

Prouvons 1°. On a les implications tautologiques

$$\begin{array}{ccccc} \mathrm{b}'' & \Longrightarrow & \mathrm{b}' & \Longrightarrow & \mathrm{b} \\ \Downarrow & & \Downarrow & & \Downarrow \\ \mathrm{a}'' & \Longrightarrow & \mathrm{a}' & \Longrightarrow & \mathrm{a}\,, \end{array} \qquad ([51])\,,$$

et pour prouver l'équivalence de ces six conditions, il suffit de prouver

$$\mathrm{a} \;\Longrightarrow\; \mathrm{b}''\,.$$

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Si on a

$$x \;\simeq\; \varinjlim_{I} x_i\,, \qquad \mathrm{card}\,\mathrm{Fl}\,I \;\leqslant\; \pi\,,$$

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soit I' l'ensemble des *sous-diagrammes* $J \subset I$ finis de I, ordonné par inclusion, donc I' est un ensemble ordonné filtrant, et du fait que C est stable par $\varinjlim$ finies, on trouve un foncteur

$$J \longmapsto x_J \;=\; \varinjlim_{i\in J} x_i \;:\; I' \longrightarrow C\,,$$

50. Je suppose que cette condition revient à la même condition sur C, pour les $\varinjlim$ filtrantes de cardinal $\leqslant \pi$.

51. compte tenu que C est strictement génératrice dans $\mathcal{M}$, donc $x \simeq \varinjlim_{i\in C/x} x_i$.

et on aura

$$x \simeq \varinjlim_{J \in I'} x_J \ .$$

D'autre part, on a

$$\begin{aligned} \operatorname{card} I' &= \operatorname{card} \mathfrak{P}_f(\ \underbrace{\operatorname{Fl}(I)}_{\text{de card} \leqslant \pi}\ ) \\ &\leqslant \pi + \pi^2 + \ \cdots\ + \pi^n + \ \ldots \\ &\leqslant \pi \aleph_0 \ = \ \pi \ , \end{aligned}$$

q.e.d. Pour établir 1°, il reste à établir

$$\begin{array}{ccc} & P(\pi) & \\ \swarrow\!\!\swarrow & & \searrow\!\!\searrow \\ \mathrm{c} & & \mathrm{d} \ , \end{array}$$

et l'implication inverse $\mathrm{c} \Longrightarrow P(\pi)$ quand $\pi \geqslant \pi_1$. Mais $P(\pi) \Longrightarrow \mathrm{c}$ est une tautologie, et $P(\pi) \Longrightarrow \mathrm{d}$ puisque $\mathcal{M}_\pi$ est stable par $\varinjlim$ de cardinal $\leqslant \pi$. Reste à prouver que $\mathrm{c} \Longrightarrow P(\pi)$ si $\pi \geqslant \pi_1$.

Commentary · Exposition [ms. 128]

The intrinsic size of an object

Grothendieck is trying to replace a convenient measure of the size of a presentation by a measure belonging to the object itself. This is the reason for the repeated comparison of Filt^π, Filt'^π and $\mathcal{M}_\pi$. The first two remember how an object has been assembled from a chosen small category C; the third asks which diagrams a represented functor can see. A change of presentation should not alter the last answer.

There is a shift of convention to make before using modern references. In these pages "grand devant π" means that every diagram having at most π arrows admits a cocone. Thus, with $\lambda = \pi^+$, this is λ-filteredness, and an object called π-accessible here is λ-presentable in the usual terminology. No regularity of π is being assumed. The successor π^+ is regular, and it is this regular cardinal which enters the standard interchange theorem for sets. We shall retain the manuscript's indices while making this translation whenever a modern formulation is needed.

The essential calculation is

$$\operatorname{Hom}\left(\varinjlim_{j\in J} X_j, \varinjlim_{i\in I} Y_i\right) = \varprojlim_{j\in J} \operatorname{Hom}\left(X_j, \varinjlim_{i\in I} Y_i\right).$$

If $|\operatorname{Fl} J| \leq \pi$, I is π^+-filtered and every X_j is π-accessible, the right side becomes $\varprojlim_j \varinjlim_i \operatorname{Hom}(X_j, Y_i)$ and then $\varinjlim_i \varprojlim_j \operatorname{Hom}(X_j, Y_i)$. The last interchange holds in sets: at most π representatives can be put at one stage, and at most π required equalities can be made true at a later stage. This proves closure of $\mathcal{M}_\pi$ under the indicated colimits. It also explains why relations, and not merely generators, enter a size bound.

The historical comparison with Artin et al. (1972, Exposé I, §9) concerns this same passage from bounded presentations to bounded represented functors. The manuscript's provisional claim that one of the presentation filtrations is automatically the intrinsic filtration must therefore be read with its hypotheses. The following pages repeatedly repair precisely that identification. The relevant modern recognition theorem is the characterization by a small strong generator of presentable objects, recorded as Theorem 1.20 in Adámek and Rosický (1994); the word "strong" carries mathematical content.

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Supposons donc qu'on ait une famille épimorphique stricte

$$x_\alpha \longrightarrow x \qquad (\alpha \in A,\ \operatorname{card} A \leqslant \pi) \ ,$$

cela signifie donc que x est limite inductive d'un système inductif indexé par la catégorie B ayant comme ensemble d'objets $A \amalg (A \times A)$, les flèches non identiques étant représentées par les flèches canoniques

$$\begin{array}{ccc} & x_\alpha \times_x x_\beta & \\ {\scriptstyle p_{\alpha,\beta}} \swarrow & & \searrow {\scriptstyle q_{\alpha,\beta}} \\ x_\alpha & & x_\beta \ . \end{array}$$

Commentary · Exposition [ms. 129]

Finite pieces of a diagram

Grothendieck wants to make an arbitrary bounded colimit into an ordinary filtered colimit without enlarging its cardinal. His argument works if a finite subdiagram is a finite set of constraints. It would not work if it meant a finite subcategory closed under composition: one endomorphism can generate infinitely many arrows.

Here is a precise construction which also makes the counting transparent. For a finite set V of objects of I and a finite set E of arrows whose endpoints lie in V, put

$$X_{V,E} = \operatorname{coeq}\left(\coprod_{f \in E} X_{s(f)} \rightrightarrows \coprod_{v \in V} X_v \right).$$

The two maps use respectively the inclusion of the source summand and the given arrow followed by the inclusion of its target. These are finite colimits and therefore belong to C. Enlarging (V, E) induces a canonical map of coequalizers. The ordered set of these pairs is filtered, has cardinal at most π when $|\operatorname{Fl} I| \leq \pi$, and its colimit is $\varinjlim_I X_i$. Indeed, a compatible family of maps from all $X_{V,E}$ to an object Y is exactly a family of maps $X_i \to Y$ satisfying every arrow relation of I. This proves the asserted universal property, including diagrams with loops.

The calculation illustrates an advantage of the primed filtration. It allows arbitrary bounded presentations to be flattened into a single bounded filtered presentation when finite colimits in C are available. It does not replace ordinary filteredness by π_0^+-filteredness. The latter requires simultaneous solutions of much larger families of constraints and is the source of the cardinal exponent appearing later.

(**N.B.** B n'est par une catégorie ordonnée, car pour l'objet (α, α), représenté par $x_\alpha \times_x x_\alpha$, il y a *deux* flèches dans l'objet α, représenté par x_α. Mais on a évidemment $\operatorname{card} \operatorname{Fl}(B) \leqslant \pi$.) Si on savait que C est stable par produits binaires et par passage à des sous-objets (ici les $x_\alpha \times_x x_\beta \ \subset \ x_\alpha \times x_\beta$), on aurait prouvé la condition a). À défaut, utilisons encore le fait que C est génératrice par épimorphismes (même stricts) et choisissons pour tout couple (α, β) un ensemble épimorphique de flèches :

$$\underbrace{x_{\alpha,\beta,\gamma}}_{\in \operatorname{Ob} C_0} \xrightarrow{u_{\alpha,\beta,\gamma}} x_\alpha \times_x x_\beta \ ,$$

où C_0 est une sous-catégorie réduite exhaustive de C. La flèche $u_{\alpha,\beta,\gamma}$ est connue quand on connaît les composés p. 41 [page 131]

$$\begin{array}{ccc} & & x_\alpha \\ & {\scriptstyle v_{\alpha,\beta,\gamma}} \nearrow & \\ x_{\alpha,\beta,\gamma} & & \\ & {\scriptstyle w_{\alpha,\beta,\gamma}} \searrow & \\ & & x_\beta \ , \end{array}$$

donc pour $x_{\alpha,\beta,\gamma} \in \operatorname{Ob} C_0$ fixé, les choix possibles pour $u_{\alpha,\beta,\gamma}$ sont de cardinal majoré par $\pi_1 \pi_1 = \pi_1$, de plus on a $\operatorname{card} \operatorname{Ob} C_0 \leqslant \pi_1$, donc le nombre total des flèches $v_{\alpha,\beta,\gamma}$, $w_{\alpha,\beta,\gamma}$ (pour α, β fixés) est majoré par $\pi_1 \cdot \pi_1 = \pi_1$. Donc le nombre total des flèches $u_{\alpha,\beta,\gamma}$, pour α, β, γ

variables, est majoré par $\pi \cdot \pi_1 = \pi$ (si $\pi \geqslant \pi_1$). Donc x est représenté par une catégorie I ayant comme objets l'ensemble somme de A et de l'ensemble des indices triples α, β, γ, et pour chaque tel indice, exactement deux flèches $v_{\alpha,\beta,\gamma}$, $w_{\alpha,\beta,\gamma}$ non identiques de source cet objet, et allant respectivement dans x_α, x_β. On a donc $\operatorname{card} \mathrm{Fl}\, I \leqslant \pi$, et on a un foncteur $I \longrightarrow C$, $\alpha \longmapsto x_\alpha$, $(\alpha, \beta, \gamma) \longmapsto x_{\alpha,\beta,\gamma}$, tel que $x = \varinjlim_{i \in I} x_i$. Cela achève la démonstration de 1°).

Prouvons 2°. La première inclusion (4.5.13.1) est tautologique, la deuxième ne fait qu'ex- [page 132]
primer $P(\pi) \Longrightarrow \mathrm{d}$. Le fait que $\pi^{\pi_0} = \pi$ implique l'égalité des membres extrêmes est un rappel de 4.5.6. Reste à prouver que Filt'^{π} est stable dans $\mathcal{M}$ par sommes de cardinal $\leqslant \pi$. Mais si $(x_\alpha)_{\alpha \in A}$ est une famille d'objets de Filt'^{π}, avec $\operatorname{card} A \leqslant \pi$, et si $x = \coprod_{\alpha \in A} x_\alpha$, prenant pour tout x_α un foncteur $\varphi_\alpha : I_\alpha \longrightarrow C$, avec $\operatorname{card} \mathrm{Fl}\, I_\alpha \leqslant \pi$, tel que x_α soit la $\varinjlim$ dans $\mathcal{M}$ de ce foncteur, on trouve que x est $\varinjlim$ du foncteur $\varphi : I \longrightarrow C$ défini par les φ_α, où $I = \coprod_\alpha I_\alpha$ (somme dans Cat). Il est évident que $\operatorname{card} \mathrm{Fl}(I) \leqslant \pi \cdot \pi = \pi$, ce qui achève de prouver 2°.

Prouvons 3°. On a les implications tautologiques $1 \Longrightarrow 1'$, et $1' \Longrightarrow 1$, puisque Filt'^{π} est stable par sommes de cardinal $\leqslant \pi$. Donc $1 \Longleftrightarrow 1'$, et implique que pour tout x dans $\mathcal{M}$, $I = \mathrm{Filt}'^{\pi}/x$ est stable par $\varinjlim$ de cardinal $\leqslant \pi$, donc est grand devant π. Comme $x \simeq \varinjlim_{i \in I} x_i$, [page 133]
cela prouve que $1 \Longrightarrow 2$. L'implication $2 \Longrightarrow 3$ est standard, et résulte de $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$, je vais expliciter dans un

Lemme 4.5.14. *Soient $\mathcal{M}$ une catégorie stable par petites $\varinjlim$, π un cardinal infini, $\mathcal{M}_0$ une sous-catégorie strictement pleine de $\mathcal{M}$. Considérons les conditions*

a) $\mathcal{M}_0$ est une sous-catégorie strictement génératrice, et stable dans $\mathcal{M}$ par $\varinjlim$ de cardinal $\leqslant \pi$.

b) Pour tout objet de $\mathcal{M}$, on a un isomorphisme

$$x \simeq \varinjlim_{i \in I} x_i \ ,$$

les x_i dans $\mathcal{M}_0$, I grande devant π.

b′) Comme b), avec I catégorie ordonnée. p. 42

c) $\mathcal{M}_\pi \subset \mathcal{M}_0$ (où $\mathcal{M}_\pi$ est la sous-catégorie strictement pleine de $\mathcal{M}$ formée des objets π-accessibles).

On a donc les implications

$$a \implies b \iff b' \implies c \ ,$$

et si on suppose $\mathcal{M}_0 \subset \mathcal{M}_\pi$, et $\mathcal{M}_\pi$ sous-catégorie strictement génératrice de $\mathcal{M}$, alors les quatre conditions a, b, b′, c sont équivalentes, et équivalent à $\mathcal{M}_0 = \mathcal{M}_\pi$ ([52]).

Enfin (revenant à la démonstration de 4.5.13, 3°), l'implication 3) $\Longrightarrow$ 1) provient du fait [page 134]
que $\mathcal{M}_\pi$ est stable par $\varinjlim$ de cardinal $\leqslant \pi$.

Commentary · Development (commentators') [ms. 133]

The retract which a filtered presentation supplies

Grothendieck is isolating the argument which should identify the small stages with the intrinsically accessible objects. The argument does identify them when the stages are closed under the stated colimits. The standalone lemma goes further than its proof permits: presentability gives a retract, and strict fullness does not put that retract back into the subcategory.

Corrected implication. Suppose every object of $\mathcal{M}$ is a π^+-filtered colimit of objects of $\mathcal{M}_0$. Then every object of $\mathcal{M}_\pi$ is a retract of an object of $\mathcal{M}_0$. Consequently the printed implication $b \Rightarrow c$ holds if $\mathcal{M}_0$ is closed under retracts.

To prove this, write $X = \varinjlim_i X_i$ with $X_i \in \mathcal{M}_0$. If X is π-accessible, its identity is represented by some $s : X \to X_i$. Writing $r : X_i \to X$ for the colimit map gives $rs = 1_X$. Nothing in this argument says that X itself belongs to an arbitrary strictly full subcategory. If $\mathcal{M}_0$ is closed under

52. cf. démonstration de proposition 4.5.6, 4 $\Longrightarrow$ 5, page 117.

the colimits in condition a, the missing conclusion does follow: r is a coequalizer of sr and 1_{X_i}, so $X \in \mathcal{M}_0$. This is why the later use of $a \Rightarrow c$ in Proposition 4.5.17 remains sound.

Counterexample to the unrestricted lemma. [ours] In Ens, let $\mathcal{M}_0$ contain exactly the sets of cardinal at most π other than singletons. This is a strictly full subcategory of $\mathcal{M}_\pi$. Every set with at least two elements is the π^+-filtered union of its subsets of cardinal at most π whose cardinal is not one: the union of at most π such subsets is another such subset, and each element belongs to a two-element subset. The empty set has its constant presentation.

For the singleton, index by the ordinal $\lambda = \pi^+$, put $X_\alpha = \{0, 1\}$, and give every strictly increasing transition the idempotent map e with $e(0) = e(1) = 0$. The identity transitions are identities, and $e^2 = e$ makes this a functor. Its colimit is a singleton, since every element becomes 0 at a later stage. The ordinal is π^+-filtered, but the singleton is absent from $\mathcal{M}_0$. Thus even the additional assumptions $\mathcal{M}_0 \subset \mathcal{M}_\pi$ and density of $\mathcal{M}_\pi$ do not repair the printed equivalence.

The retract principle is standard in the theory of presentable objects; the same argument appears in the higher categorical formulation of Lurie (2009, Lemma 5.4.2.4). We include the explicit counterexample to audit this particular formulation; it is not a claim of priority. Our literature comparison concerns the valid retract statement, not an alleged later proof of the erroneous implication.

Reste à prouver que ces conditions, i.e. la stabilité de Filt'^{π} par conoyaux de double-flèches, est satisfaite dans les cas A) B) C), le cas A) étant d'ailleurs déjà connu. Le cas B) résulte immédiatement de la transitivité des familles épimorphiques strictes universelles, et c'est d'ailleurs aussi contenu explicitement (stabilité de Filt'^{π} par $\varinjlim$ de cardinal $\leqslant \pi$) dans SGA 4 I 9.13 (sinon dans l'énoncé, du moins c'est signalé dans la démonstration). En rédigeant SGA 4 I 9, je savais donc qu'on a $\mathcal{M}_\pi \subset \mathrm{Filt}'^{\pi}$ pour tout π, mais l'inclusion, plus évidente pourtant dans le cas d'espèce, $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$ (quand on suppose les objets de C π_1-accessibles) m'avait apparemment échappé ?

Reste à traiter le cas C). (J'ai l'impression qu'il est un peu plus général que le cas B), i.e. que la condition B) sur $\mathcal{M}$ implique C), mais non inversement. Ainsi, la condition C) est vérifiée dès que $\mathcal{M}$ correspond à une espèce de structure algébrique définie en termes de $\varprojlim$ *finies*, i.e. quand $\mathcal{M}$ est de la forme $\mathrm{Ind}(C)$, C stable par $\varinjlim$ *finies*, alors que je doute que la condition B) soit toujours satisfaite pour ces catégories. Par exemple dans la catégorie Cat (qui est un paratopos de cette espèce), les $\varinjlim$ filtrantes sont exactes à gauche, mais je ne crois pas que les familles épimorphiques strictes soient universelles. Prenons p.ex. deux homomorphismes de groupes $G_0 \xrightarrow{u_0} G$, $G_1 \xrightarrow{u_1} G$ tels que $G_0 * G_1 \longrightarrow G$ soit surjectif, alors cette famille à deux éléments est épimorphique stricte dans Gr et dans Cat ([53]), mais n'est pas forcément épimorphique stricte universelle, même si p.ex. $G_0 = G_1 = H$, $G = H \times H$, et u_0, u_1 sont des injections $H \rightrightarrows H \times H$, données par $u_0(h) = (h, 1)$, et $u_1(h) = (1, h)$. Si je prends l'application diagonale $H \longrightarrow H \times H$, les images inverses de u_0, u_1 sont le même objet de $\mathcal{M}/H$, savoir $e \longrightarrow H$, et cette famille n'est épimorphique stricte que si H est le groupe unité.) [page 135]

Le cas C) va résulter du résultat plus précis :

Corollaire 4.5.15. *Sous les conditions générales de 4.5.13, supposons que dans $\mathcal{M}$ les $\varinjlim$ filtrantes commutent au changement de base (p.ex. les $\varinjlim$ filtrantes sont exactes). Alors pour tout $\pi \geqslant \pi_1$, Filt'^{π} est formée des $x \in \mathrm{Ob}\,\mathcal{M}$ satisfaisant la condition suivante.* [page 136]

Commentary · Examples [ms. 135]

A strict covering family which loses strictness under pullback

Grothendieck wants an elementary reason that exactness of filtered colimits is weaker than universality of every strict epimorphic family. Groups do provide that reason, but the direct-product example printed here does not: its initial family is already non-strict. The corrective footnote withdraws the general assertion and mistakenly exempts this special case.

53. !, c'est *faux* en général, mais vrai dans le cas particulier plus bas.

Take $H = C_2$, and regard its two copies as the coordinate subgroups of $C_2 \times C_2$. Their intersection is trivial. Map the first generator to $(12) \in S_3$ and the second to $(23) \in S_3$. These homomorphisms are compatible on every overlap test: if two maps into the two coordinate subgroups have the same composite into the direct product, both composites are trivial. A common extension would send the two commuting coordinate generators to two noncommuting transpositions. Hence no extension exists. The family is not strictly epimorphic in groups. The same argument applies to one-object group categories, so it is not strictly epimorphic in Cat either.

Replace the direct product by

$$G = C_2 * C_2 = \langle a, b \mid a^2 = b^2 = 1 \rangle.$$

The two free-factor inclusions are strictly epimorphic in groups. Compatibility on the intersection imposes no relation, and the universal property of the free product gives the unique extension. They are strictly epimorphic as functors of one-object categories as well. Compatibility forces the chosen target objects to coincide; the two homomorphisms into the automorphism group of that object then extend uniquely to G.

Now let $K = \langle ab \rangle$. The reduced-word description shows that ab has infinite order, so $K \simeq \mathbb{Z}$, and that K meets either free factor trivially. Pullback along $K \hookrightarrow G$ gives the family consisting of two copies of $1 \to \mathbb{Z}$. This family is not even epimorphic: the zero endomorphism and the identity of $\mathbb{Z}$ agree after both maps. Thus strict epimorphic families in groups, and in categories, need not be universal. Filtered colimits in both examples are nevertheless left exact; their objects, arrows or algebraic operations are governed by finite equations in sets.

The repair preserves the intended distinction while changing the ambient group. It also warns against replacing a strict epimorphic family by the assertion that its images generate a target. Generation detects uniqueness of an extension; strictness requires existence for every compatible family.

p. 43

c′) Il existe une famille de flèches $(u_\alpha)_{\alpha \in A}$, $u_\alpha : x_\alpha \longrightarrow x$, *dans* $\mathcal{M}$, *qui est* épimorphique stricte universelle *(i.e. couvrante pour la topologie canonique de* $\mathcal{M}$*), avec* $x_\alpha \in \mathrm{Ob}\, C$ $\forall\ \alpha \in A$, *et* $\mathrm{card}\, A \leqslant \pi$.

Démonstration : il faut prouver que cette condition équivaut à la condition c) de 4.5.13 1°). Comme on a c′ $\Longrightarrow$ c trivialement, il reste à montrer que c $\Longrightarrow$ c′. Mais on a vu dans 4.5.13 1°) que c implique a″, i.e. qu'on a un isomorphisme

$$x \;=\; \varinjlim_{i \in I} x_i \ ,$$

les x_i dans C, et I ensemble *ordonné filtrant*, avec $\mathrm{card}\, I \leqslant \pi$. Il reste à voir que si x dans $\mathcal{M}$ est une limite inductive *filtrante* d'objets x_i, alors l'ensemble des flèches $x_i \longrightarrow x$ est épimorphique stricte *universelle.* Or cela est immédiat, car pour $x' \longrightarrow x$, on a $x' = \varinjlim_i x'_i$ (par hypothèse C), où $x'_i = x' \times_x x_i$, donc la famille $x'_i \longrightarrow x'$ est épimorphique stricte, q.e.d.

Prouvons maintenant que dans le cas C, la condition de 3°, 1 est satisfaite, i.e. que Filt'^{π} est stable par $\varinjlim$ de cardinal $\leqslant \pi$. Cela résulte aussitôt du [page 137]

Corollaire 4.5.16. (de 4.5.15). *Sous les conditions de 4.5.15 (i.e. la condition C) de 4.5.13, 3°), soit x un objet de* $\mathcal{M}$*, et supposons qu'il existe une famille* épimorphique universelle $(x_\alpha \longrightarrow x)_{\alpha \in A}$ *de flèches de* $\mathcal{M}$*, avec* $x_\alpha \in \mathrm{Ob}\, \mathrm{Filt}'^{\pi}$ *pour tout* $\alpha \in A$, *et* $\mathrm{card}\, A \leqslant \pi$. *Alors* $x \in \mathrm{Ob}\, \mathrm{Filt}'^{\pi}$.

En effet, l'hypothèse sur x_α équivaut, en vertu de 4.5.15, à l'existence d'une famille épimorphique stricte universelle de flèches $(x_{\alpha\beta} \longrightarrow x_\alpha)_{\beta \in B_\alpha}$, avec $\mathrm{card}\, B_\alpha \leqslant \pi$ et les $x_{\alpha\beta}$ dans C. Soit $B = \coprod_{\alpha \in A} B_\alpha$, on a alors $\mathrm{card}\, B \leqslant \pi^2 = \pi$, et d'autre part, la famille des flèches composées $x_{\alpha\beta} \longrightarrow x_\alpha \longrightarrow x$ indexée par B est épimorphique stricte universelle (par transitivité de ce type de familles), donc $x \in \mathrm{Ob}\, \mathrm{Filt}'^{\pi}$, q.e.d.

Avec tout ça, on n'a toujours pas donné de caractérisation de $\mathcal{M}_\pi$ lui-même, qui est meilleure que Filt^{π} et Filt'^{π} du point de vue catégorique, car seuls les $\mathcal{M}_\pi$ définissent une [page 138]

filtration cardinale, et de plus ils ont les propriétés de stabilité désirables dans ce contexte, pour les $\varinjlim$ de cardinal $\leqslant \pi$. Le seul critère non tautologique que j'arrive à dégager dans le cas général (pour tout $\pi \geqslant \pi_0$, et sans hypothèse d'exactitude supplémentaire sur $\mathcal{M}$) est le suivant, beaucoup moins "explicite" ou "calculatoire" que les critères très concrets pour Filt^{π} et pour Filt'^{π} :

Commentary · Present state [ms. 137]

What the covering argument actually proves

Grothendieck is using descent along a bounded covering family to keep a presentation bounded. Corollary 4.5.15 makes the necessary descent condition explicit: the families are universally *strictly* epimorphic. In Corollary 4.5.16 the word "strictement" disappears from the hypothesis, although the proof still invokes transitivity of precisely that stronger notion.

With the stronger hypothesis restored, the proof is complete. Replace each X_α by a universal strict cover with at most π sources in C. The composite is again a universal strict cover, and its index set has cardinal at most $\pi \cdot \pi = \pi$. Corollary 4.5.15 applies. The weaker printed hypothesis does not justify that invocation of transitivity. We have not established a counterexample satisfying all the standing hypotheses, in particular $\pi \geq \pi_1$, and therefore record a gap in this proof rather than declaring its conclusion false.

There is a second point in the preceding "cela résulte aussitôt". Stability under universal strict covers is not, by itself, stability under arbitrary colimits. An arbitrary colimit cocone is strictly epimorphic but need not remain epimorphic after pullback, as the free-product example shows even when filtered colimits are exact. Thus the proposed deduction of the full case C of Proposition 4.5.13 requires an additional argument. The audit keeps that issue separate from the valid universal-cover calculation and from the general intrinsic closure statement which follows.

Proposition 4.5.17. *Soit π un cardinal $\geqslant \pi_0$. Alors $\mathcal{M}_\pi$ est égal à la plus petite sous-catégorie strictement pleine de $\mathcal{M}$, contenant C, et stable par $\varinjlim$ de cardinal $\leqslant \pi$.* (Il est important de noter qu'il *ne suffit pas* de dire : stable par $\varinjlim_I$, avec I *grande devant* π_0, et $\operatorname{card} I \leqslant \pi$. On trouve alors Filt^{π}, qui risque d'être une catégorie strictement plus petite que $\mathcal{M}_\pi$. Ce n'est pas non plus la catégorie des $\varinjlim_I x_i$, avec $x_i \in C \ \forall\ i \in \operatorname{Ob} I$, et $\operatorname{card} \operatorname{Fl} I \leqslant \pi$ – puisque cette catégorie-là est Filt'^{π}.)

Démonstration. Soit $\mathcal{M}'$ la catégorie de l'énoncé, on a donc $\mathcal{M}' \subset \mathcal{M}_\pi$, puisque $\mathcal{M}_\pi$ contient C et est stable par les $\varinjlim$ de cardinal $\leqslant \pi$. Inversement, le lemme 4.5.14, a $\Longrightarrow$ c.

Commentary · Exposition [ms. 138] p. 44 [page 139]

Why closing under relations gives the right category

Grothendieck now replaces a single bounded presentation by a closure operation. This is a less explicit description, but it has the advantage of being intrinsic once the initial generating category has been fixed. The distinction is between allowing one layer of relations and allowing every further relation needed by the construction.

Let D be the smallest strictly full subcategory containing C and closed under colimits indexed by categories of cardinal at most π. The represented-functor calculation already gives $D \subset \mathcal{M}_\pi$. To obtain the reverse inclusion, use the canonical presentation of an object X by D/X. Density supplies its colimit, and bounded-colimit closure makes the indexing category π^+-filtered. If X is π-accessible, its identity factors through one stage. The resulting retract is a coequalizer of an idempotent and an identity, and therefore still lies in D. This is the complete version of the short appeal to Lemma 4.5.14.

For sets, the construction is particularly concrete. Starting with a singleton and closing under at most π-sized coproducts and coequalizers gives every set of cardinal at most π, and these are exactly the π^+-presentable sets. A map from such a set into a π^+-filtered union is represented at one stage because its at most π elements can be treated simultaneously; equality of two maps is treated in the same way. Conversely, a larger set is the π^+-filtered union of its subsets of cardinal at most π, and its identity cannot factor through a proper subset. This simple model explains the intended statement without identifying an arbitrary one-step presentation with the completed closure.

4.6 Caractérisation des catégories paratopos (démonstration de 4.3.6)

Il est temps de revenir à la démonstration du théorème 4.3.6 (p. 70-71), celle que j'avais donnée étant incomplète (et erronée – j'affirmais que $(\mathrm{Filt}^{\pi})_{\pi\geqslant\pi_0}$ est une filtration cardinale, ce qui est faux si on n'admet que $\pi^{\pi_0}=\pi$ pour tout $\pi\geqslant\pi_1$).

On avait vu l'équivalence des conditions a, a′, b, soit PTA. Je vais achever la démonstration par le schéma logique

$$(*)\qquad \text{PTA} \implies \text{d)} \implies \text{c)} \implies \text{PTA}\ .$$

Prouvons PTA $\Longrightarrow$ d), i.e. que si $\mathcal{M}$ est un pseudo-topos accessible, alors existe un cardinal infini π_0, et une petite catégorie C_0 stable par $\varinjlim$ de cardinal $\leqslant\pi_0$, et une équivalence de catégories

$$(4.6.1)\qquad \mathcal{M}\xrightarrow{\sim}\mathrm{Ind}'_{\pi_0}(C_0)\ .$$

Plus précisément, on a ceci :

Proposition 4.6.1. *Soit $\mathcal{M}$ un paratopos, i.e. une $\mathfrak{U}$-catégorie $\mathcal{M}$ stable par petites $\varinjlim$, admettant une sous-catégorie pleine génératrice S équivalent à une petite catégorie, formée d'objets accessibles de $\mathcal{M}$. Soit π_0 un cardinal tel que les objets de S soient π_0-accessibles (i.e. $S\subset\mathcal{M}_{\pi_0}$), et soit C la plus petite sous-catégorie strictement pleine de $\mathcal{M}$, contenant S et stable par $\varinjlim$ de cardinal $\leqslant\pi_0$. Alors*

1°) C est équivalente à une petite catégorie.

2°) $C=\mathcal{M}_{\pi_0}$, sous-catégorie strictement pleine de $\mathcal{M}$ formée des objets π_0-accessibles. [page 140]

3°) Le foncteur canonique

$$\mathrm{Ind}'_{\pi_0}(C)\xrightarrow[\sim]{[\varphi]}\mathcal{M}$$

(défini à isomorphisme unique près par la condition que son composé avec l'inclusion $C\hookrightarrow\mathrm{Ind}'_{\pi_0}(C)$ soit isomorphe à l'inclusion $C\longrightarrow\mathcal{M}$, et que φ commute aux petites $\varinjlim$, ou encore aux $\varinjlim$ filtrantes grandes devant π_0 – cf. 4.4.16, page 101) est une équivalence de catégories ([54]).

Démonstration. 1°) On va construire C par induction transfinie, en posant

$$\begin{aligned} C_0 &= \text{sous-catégorie strictement pleine de } \mathcal{M} \text{ engendrée par } S.\\ C_{\alpha+1} &= \text{sous-catégorie strictement pleine de } \mathcal{M} \text{ engendrée par les } \varinjlim_{I} \text{ dans}\\ &\quad\ \mathcal{M} \text{ d'objets de } C_\alpha, \text{ avec } I \text{ catégorie telle que } \operatorname{card}\mathrm{Fl}\, I\leqslant\pi_0.\\ C_\alpha &= \textstyle\bigcup_{\alpha'<\alpha} C_{\alpha'} \quad \text{si } \alpha \text{ est un ordinal limite.}\end{aligned}$$

Cela définit de façon unique les C_α (pour tout petit cardinal α). Par la définition de C, on voit par induction transfinie que les C_α sont contenues dans C. Soit α_0 le premier ordinal tel que $\operatorname{card}\alpha_0>\pi_0$, donc l'ensemble I_{α_0} des cardinaux $\alpha<\alpha_0$ est grand devant π_0. Je dis qu'on a $C_{\alpha_0}=C$. Pour ceci, il suffit de voir que C_{α_0} est stable par $\varinjlim$ de cardinal $\leqslant\pi$. Mais si I est une catégorie telle que $\operatorname{card}\mathrm{Fl}\, I\leqslant\pi_0$, et $\varphi: I\longrightarrow C_{\alpha_0}=\bigcup_{\alpha<\alpha_0}C_\alpha$ un foncteur, alors la définition de α_0 implique que φ se factorise en $\varphi_\alpha: I\longrightarrow C_\alpha$, avec $\alpha<\alpha_0$ ([55]). Donc $\varinjlim\varphi\in\mathrm{Ob}\, C_{\alpha+1}\subset\mathrm{Ob}\, C_{\alpha_0}$, ce qui montre la stabilité requise. p. 45 [page 141]

54. donc on a aussi $\mathrm{Ind}'_{\pi_0}(C_0)\xrightarrow{\sim}\mathcal{M}$, où C_0 est une *petite* catégorie équivalente à C.

55. Un peu court. Il faut invoquer le fait que I_{α_0} est grande devant π_0.

Commentary · Exposition [ms. 140]

The small category behind a cocomplete category

Grothendieck's purpose is to recover a small category which contains all the operations needed to reconstruct $\mathcal{M}$. The transfinite argument does not require a precise estimate for every Hom set. It requires only that each stage have a set of isomorphism classes, that its Hom sets be small, and that the closure operation stabilize at a specified regular ordinal.

Put $\lambda = \pi_0^+$. Choose a small skeleton at each stage. There is a set of categories with at most π_0 arrows, a set of functors from each of them to that skeleton, and hence a set of chosen colimits. Local smallness then makes the full category on those chosen objects small. At limit stages take the union of the chosen sets of objects and then the full category they span. All these operations remain in the fixed universe.

If a diagram with at most π_0 arrows takes values in C_λ, its objects occur at at most π_0 earlier stages. Regularity of λ bounds those stages by a single $\alpha < \lambda$. Its colimit therefore occurs in $C_{\alpha+1}$. This proves C_λ is closed under the required colimits, and proves stabilization. The source's references to "cardinaux" in the indexing sequence must here mean ordinals; it is their cofinality which the proof uses.

The remaining reconstruction has a different, categorical, content. For $X_i, Y_j \in C$ and the relevant filtered indices,

$$\mathrm{Hom}_{\mathcal{M}}(\varinjlim_i X_i, \varinjlim_j Y_j) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i, Y_j).$$

This is the Hom formula for the filtered completion. It proves full faithfulness. The category C/X has the bounded colimits inherited from C and is therefore λ-filtered; density makes its colimit X. This proves essential surjectivity. These two arguments establish Proposition 4.6.1 independently of the sharper numerical estimate criticized in the next block.

Pour voir que C est équivalente à une petite catégorie, il suffit de le voir pour les C_α, en procédant par induction transfinie. Cela résulte du lemme suivant, immédiat :

Lemme 4.6.1.1.

1°) La limite inductive C d'un système inductive filtrant $(C_i)_{i\in I}$ de catégories C_i équivalentes à des petites catégories est équivalente à une petite catégorie. Si $\operatorname{card}\operatorname{réd}(C_i) \leqslant \pi$ *pour tout i, et* $\operatorname{card} I \leqslant \pi$, *alors* $\operatorname{card}\operatorname{réd}(C) \leqslant \pi$ *(π étant un cardinal infini). Ici, pour toute catégorie $\mathcal{A}$, je pose* $\operatorname{card}\mathcal{A} = \operatorname{card}\operatorname{Fl}\mathcal{A}$, $\operatorname{card}\operatorname{réd}(\mathcal{A}) = \operatorname{card}(\mathcal{A}_0)$, *où $\mathcal{A}_0$ est une catégorie réduite équivalente à $\mathcal{A}$.* [page 142]

2°) Soit $C_0 \subset \mathcal{M}$ une sous-catégorie pleine d'une catégorie $\mathcal{M}$, π_0 un cardinal infini, et soit C_0' la sous-catégorie strictement pleine de $\mathcal{M}$ engendrée par les limites inductives dans $\mathcal{M}$ (représentables dans $\mathcal{M}$) de foncteurs $I \longrightarrow C_0$, avec I catégorie telle que $\operatorname{card} I \leqslant \pi_0$. *Si C_0 est équivalente à une petite catégorie, il en est de même de C_0'. Plus précisément, si* $\operatorname{card}\operatorname{réd}(C_0) \leqslant \pi_1$, *$\pi_1$ un cardinal infini, alors* $\operatorname{card}\operatorname{réd}(C_0') \leqslant \pi_1^{\pi_0}$ ([56]).

Corollaire 4.6.1.2. *$\mathcal{M}$ et C_0 étant comme dans 2°) ci-dessus, soit C la plus petite sous-catégorie strictement pleine de $\mathcal{M}$ contenant C_0, et stable par $\varinjlim$ de cardinal $\leqslant \pi_0$ (celles qui sont représentables dans $\mathcal{M}$). Si C_0 est équivalente à une petite catégorie, il en est de même de C. Plus précisément, si* $\operatorname{card}\operatorname{réd}(C_0) \leqslant \pi_1$, *où π_1 est un cardinal infini, alors on a* $\operatorname{card}\operatorname{réd}(C) \leqslant \pi_1^{\pi_0}$ ([57]). [page 143]

Le 3°) se prouve par la démonstration transfinie précédente, on prouve par récurrence transfinie que

$$\operatorname{card}\operatorname{réd} C_\alpha \;\leqslant\; \pi_1^{\pi_0}$$

pour tout α. C'est OK pour $\alpha = 0$ par hypothèse (puisque $\pi_1 \leqslant \pi_1^{\pi_0}$). Si c'est vrai pour C_α, alors 2°) assure que

$$\operatorname{card}\operatorname{réd} C_\alpha \;\leqslant\; (\pi_1^{\pi_0})^{\pi_0} \;=\; \pi_1^{\pi_0\cdot\pi_0} \;=\; \pi_1^{\pi_0}\ ,$$

56. Cette majoration n'est par prouvée et probablement fausse si on ne fait pas des hypothèses un peu plus strictes.

57. majoration non prouvée, probablement fausse, cf. annotation ci-dessus.

enfin, si α est un ordinal limite $< \alpha_0$, on aura $C_\alpha = \bigcup_{\alpha'<\alpha} C_{\alpha'}$, donc

$$\operatorname{card}\operatorname{réd} C_\alpha \;\longrightarrow\; \pi_0 \cdot \pi_1^{\pi_0} \;=\; \pi_1^{\pi_0}$$

(vu que $\pi_1^{\pi_0} \geqslant \pi_0$), en vertu du 1° ci-dessus, et compte tenu du fait que $\operatorname{card} I_\alpha \leqslant \pi_0$. Soit π_0' le cardinal successeur de π_0, alors la même majoration nous donne

$$\operatorname{card}\operatorname{réd} C_{\alpha_0} \;\leqslant\; \pi_0'\pi_1^{\pi_0}\ .$$

Mais on a $\pi_1^{\pi_0} \geqslant 2^{\pi_0} > \pi_0$, donc $\pi_1^{\pi_0} \geqslant \pi_0'$, donc $\pi_1^{\pi_0} \geqslant \pi_0'$, donc $\pi_0'\pi_1^{\pi_0} = \pi_1^{\pi_0}$, d'où p. 46

$$\operatorname{card}\operatorname{réd} C_{\alpha_0} \;\leqslant\; \pi_1^{\pi_0}\ ,$$

ce qui est la majoration annoncée.

[page 144]

Commentary · Development (commentators') [ms. 143]

Counting presentations does not count all new arrows

Grothendieck has correctly counted the possible small diagrams, but then uses that count as a bound on the morphisms between their colimits. The warning in footnotes 56 and 57 identifies the problem accurately. A colimit can acquire many maps *from* an old object, and its universal property controls maps out of the colimit, not these maps into it.

Counterexample to the stated cardinal estimate. [ours] Let $\kappa > 2^{\aleph_0}$, and consider algebras with a binary operation m_ξ for every $\xi < \kappa$, subject only to the equations

$$m_\xi(x, x) = x \qquad (\xi < \kappa).$$

This is a finitary equational category. Its free algebra $F(1)$ on one generator has one element: induction on terms reduces every term in that generator to the generator itself. Thus the full subcategory C_0 with sole object $F(1)$ has one arrow.

The binary coproduct $F(1) \amalg F(1)$ is $F(2)$. Its elements include the terms $m_\xi(x, y)$, and these are pairwise distinct. To verify the last assertion without assuming a normal-form theorem, fix $\xi \neq \eta$ and interpret the operations on $\{0, 1\}$, taking m_ξ to be the first projection and m_η the second projection. Assign either projection to every remaining operation. All the idempotence equations hold, but the two specified terms have different values at $(0, 1)$. They cannot therefore be equal in the free algebra. It follows that

$$|\operatorname{Hom}(F(1), F(2))| = |F(2)| \geq \kappa > 2^{\aleph_0}.$$

Take $\pi_0 = \pi_1 = \aleph_0$. The hypothesis $\operatorname{card}\operatorname{réd} C_0 \leq \pi_1$ holds, but the full closure under even one binary coproduct has more than $\pi_1^{\pi_0}$ arrows in a skeleton. This disproves the quantitative assertions of Lemma 4.6.1.1(2) and Corollary 4.6.1.2 as stated. Their qualitative essential-smallness assertions remain true by the preceding argument.

The construction is an application of the standard free-algebra method, used here to test this particular estimate. We checked the primary representation discussions in Adámek and Rosický (1994) and the manuscript's two warnings; we did not locate this exact counterexample there. The marking records our supplied argument, without a priority claim.

A separate edge case affects Corollary 4.6.2(1). In Ens the one-object full subcategory $S = \{1\}$ is dense, so the printed definition gives $\pi_1 = \operatorname{card}\operatorname{réd} S = 1$. The proposed upper bound is then $1^\pi = 1$, although $\mathcal{M}_\pi$ contains many nonisomorphic objects. One must first replace π_1 by an infinite bound, such as $\max(\aleph_0, \operatorname{card}\operatorname{réd} S)$. Moreover the equality $\pi_1^\pi = 2^\pi$ used there requires $2 \leq \pi_1 \leq \pi$, exactly as the following proof shows. These corrections remove the immediate numerical contradictions; they do not retroactively validate an estimate obtained from the false unrestricted lemma. We retain the small-category reconstruction and distinguish it from the unproved quantitative refinement.

Prouvons 4.6.1 2°, i.e. $C = \mathcal{M}_{\pi_0}$. On a $C \subset \mathcal{M}_{\pi_0}$, puisque par hypothèse $S \subset \mathcal{M}_{\pi_0}$, et on sait que $\mathcal{M}_{\pi_0}$ est stable par $\varinjlim$ de cardinal $\leqslant \pi_0$. D'autre part, on a $\mathcal{M}_{\pi_0} \subset C$, en vertu de 4.5.19 a $\Longrightarrow$ c appliqué à C, qui est strictement génératrice dans $\mathcal{M}$, S l'étant, et $C \supset S$, et stable par $\varinjlim$ de cardinal $\leqslant \pi_0$ par construction.

Prouvons enfin 3°. Cela équivaut à ceci :

Lemme 4.6.1.3. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie stable par* petites $\varinjlim$ *, π_0 un cardinal, $\mathcal{M}_0$ une sous-catégorie de $\mathcal{M}$* strictement génératrice, *stable par $\varinjlim$ de cardinal $\leqslant \pi_0$, et formée d'objets π_0-accessibles. (En vertu de 4.6.1 2° déjà prouvée, appliqué à $S = \mathcal{M}_0$, si $\mathcal{M}_0$ est strictement pleine, on a donc $\mathcal{M}_0 = \mathcal{M}_{\pi_0}$.) Alors le foncteur canonique*

$$\mathrm{Ind}'_{\pi_0}(\mathcal{M}_0) \longrightarrow \mathcal{M}$$

est une équivalence de catégories.

Le fait que ce facteur est pleinement fidèle résulte aussitôt de l'hypothèse que les objets de $\mathcal{M}_0$ soient π_0-accessibles :

$$\begin{aligned}\mathrm{Hom}_{\mathcal{M}}(\varinjlim_{i\in I} \underbrace{X_i}_{\in\mathcal{M}_0},\ \varinjlim_{j\in J} \underbrace{Y_j}_{\in\mathcal{M}_0}\ ) &\simeq \varprojlim_{i\in I} \underbrace{\mathrm{Hom}_{\mathcal{M}}(X_i,\ \varinjlim_{j\in J} Y_j)}_{\varinjlim_{j\in J}\mathrm{Hom}(X_i,Y_j)} \\ &\simeq \underline{\mathrm{Hom}}_{\mathrm{Ind}_{\pi_0}(\mathcal{M})}((X_i),(Y_j))\ .\end{aligned}$$

Reste à voir que ce foncteur est essentiellement surjectif. Comme $\mathcal{M}_0$ est strictement [page 145]
génératrice, il s'ensuit que tout $x \in \mathcal{M}$ s'écrit

$$x = \varinjlim_{i\in I} x_i\ ,$$

où $I = \mathcal{M}_0/x$, et les $x_i \in \mathcal{M}_0$. L'hypothèse que $\mathcal{M}_0$ est stable par $\varinjlim$ de cardinal $\leqslant \pi_0$ implique la même propriété pour $I = \mathcal{M}_0/x$, donc I est grande devant π_0. Donc $(x_i)_{i\in I}$ définit un objet de $\mathrm{Ind}_{\pi_0}\mathcal{M}_0$, dont l'image dans $\mathcal{M}$ est isomorphe à x, q.e.d. (Comparer SGA 4 I 9.18.)

Corollaire 4.6.2. *Sous les conditions de 4.6.1, prenons pour π_0* le plus petit *cardinal tel que les objets de S soient π_0-accessibles. Pour tout $\pi \geqslant \pi_0$, soit $\mathcal{M}_\pi$ la sous-catégorie strictement pleine de $\mathcal{M}$ formée des objets π-accessibles, qui est aussi (par 4.6.1 2°) l'enveloppe de C dans $\mathcal{M}$ stable par $\varinjlim$ de cardinal $\leqslant \pi$.*

1°) Soit $\pi_1 = \mathrm{card}\,\mathrm{réd}\,S$. On a alors

$$\mathrm{card}\,\mathrm{réd}\,\mathcal{M}_\pi \leqslant \pi_1^\pi \qquad (= 2^\pi \text{ si } \pi \geqslant \pi_0). \tag{4.6.2.1}$$

p. 47

2°) $(\mathcal{M}_\pi)_{\pi\geqslant\pi_1}$ est une filtration cardinale de $\mathcal{M}$, où $\pi_1 = \sup(\pi_0, \mathrm{card}\,S)$. [page 146]

Démonstration. La majoration (4.6.2.1) est contenue dans 4.6.1.2, où π_0 est devenu π. Si $\pi \geqslant \pi_1$, on a

$$(*) \qquad \pi_1^\pi = 2^\pi$$

car on a $\pi_1 \geqslant 2$, d'où $\pi_1^\pi \geqslant 2^\pi$, d'autre part, on a $\pi_1 \leqslant 2^{\pi_1}$, d'où $\pi_1^\pi \leqslant (2^{\pi_1})^\pi = 2^{\pi_1\pi} = 2^\pi$, puisque $\pi_1\pi = \pi$ pour $\pi \geqslant \pi_1$; d'où $(*)$.

L'assertion 2° est mise pour mémoire ; en vertu de $\mathcal{M} \simeq \mathrm{Ind}'_{\pi_0}\mathcal{M}_{\pi_0}$, on peut supposer que l'on a $\mathcal{M} = \mathrm{Ind}'_{\pi_0}(\mathcal{M}_{\pi_0})$, et on peut alors appliquer 4.5.9. On aurait d'ailleurs envie de dire que $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$ est une filtration cardinale, sans se borner aux $\pi \geqslant \pi_1$. C'est vrai sauf pour la majoration de cardinaux dans la condition c) de la définition des filtrations cardinales (SGA 4 I 9.12) :

$$\underbrace{x}_{\in \mathrm{Ob}\,\mathcal{M}_{\pi'}} = \varinjlim_{i\in I} \underbrace{x_i}_{\in \mathrm{Ob}\,\mathcal{M}_\pi}\ , \qquad I \text{ grande devant } \pi \text{ et } \mathrm{card}\,I \leqslant \pi'^\pi$$

[page 147]

pour $\pi' \geqslant \pi$. Il faut supposer, pour cette majoration, que $\pi \geqslant \pi_1$.

On peut à présent achever la démonstration de 4.3.6, en prouvant les deux dernières implications $(*)$ (p. 139). On a d $\Longrightarrow$ c, i.e. si $\mathcal{M} \simeq \mathrm{Ind}'_{\pi_0}(C)$, C stable par limites inductives de cardinal $\leqslant \pi_0$, et C équivalente à une petite catégorie, alors $\mathcal{M}$ est stable par petites $\varinjlim$ (4.4.13, p. 95-96), ses éléments sont accessibles, et elle admet une filtration cardinale (cf. 4.5.9). Mais il faut vérifier quand même que les éléments de C sont π_0-accessibles. C'est contenu dans SGA 4 I 9.18, qui affirme que la pleine fidélité du foncteur $\mathrm{Ind}_{\pi_0}(C) \longrightarrow \mathcal{M}$ *équivaut* au fait que les éléments de C sont π_0-accessibles dans $\mathcal{M}$. La démonstration dans loc. cit. n'était pas claire, le "il suffit" a été vu bas de la page 144. Le "il faut" ne me paraît pas clair, avec la généralité donnée dans loc. cit. – et à présent je vois que c'est même faux, comme on voit en posant pour C la catégorie unité e, donc $\mathrm{Ind}'_{\pi_0} C \simeq e$, et l'assertion affirmant que si e est immergée pleinement dans $\mathcal{M}$ (donc par un objet *rigide* de $\mathcal{M}$, i.e. un objet a tel que $\mathrm{End}\, a = \{\mathrm{id}_a\}$), alors a est $\aleph_0$-accessible. C'est faux dans la catégorie des anneaux, où $\mathbf{R}$ est un objet rigide non $\aleph_0$-accessible !

[page 148]

Commentary · Examples [ms. 147]

A rigid object need not be small

Grothendieck's objection to the converse in SGA 4 is decisive. Full faithfulness of a completion functor can be vacuous when the chosen subcategory generates very little of the ambient category. A rigid object makes the point in the smallest possible example.

In the category of unital rings, every endomorphism $f : \mathbb{R} \to \mathbb{R}$ is the identity. It is injective because its kernel is an ideal in a field and $f(1) = 1$. A positive real number is the square of a nonzero real number, so its image is positive. Thus f preserves the order, and it fixes $\mathbb{Q}$. If $f(x) \neq x$, a rational number strictly between them contradicts order preservation. Hence $\mathrm{End}(\mathbb{R}) = \{1\}$.

Nevertheless $\mathbb{R}$ is not $\aleph_0$-accessible in the manuscript's convention. Let I be the ordered set of its countable unital subrings. The subring generated by a countable union of countable subrings is countable, so I is $\aleph_1$-filtered. Its union is $\mathbb{R}$, since every individual real belongs to a countable subring. If $\mathrm{Hom}(\mathbb{R}, -)$ preserved this colimit, the identity would factor through one countable subring $A \subset \mathbb{R}$. The composite of that factorization with the inclusion would be the identity, forcing $A = \mathbb{R}$, a contradiction.

The full subcategory on $\mathbb{R}$ is nevertheless equivalent to the terminal category, and its filtered completion is again terminal. Its inclusion into rings is fully faithful. This proves that the necessity assertion of Artin et al. (1972, Exposé I, Corollary 9.18(a)), taken at the stated generality, is false. For the intended application $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$ there is no difficulty: evaluation at $c \in C$ computes $\mathrm{Hom}(c, -)$, and the relevant filtered colimits of presheaves are pointwise. It is this calculation, rather than full faithfulness alone, which gives presentability.

Dans le cas qui nous occupe, $\mathcal{M} = \mathrm{Ind}'_{\pi_0} C$, il faut visiblement utiliser le calcul explicite des $\varinjlim$ grandes devant π_0 dans $\mathrm{Ind}'_{\pi_0} C \subset C^\wedge$, elles se calculent en effet "argument par argument", cf. 4.4.15.

Cela achève la démonstration de c $\Longrightarrow$ d dans $(*)$, page 139. Enfin l'implication c $\Longrightarrow$ PTA est tautologique, l'existence d'une filtration cardinale impliquant celle d'une petite sous-catégorie strictement génératrice. Cela achève la démonstration de 4.3.6.

Je vais redonner une formulation légèrement plus précise de 4.3.6., en précisant les cardinaux, et en y rajoutant une condition supplémentaire, inspirée par SGA 4 I 9.11 (revue et corrigée, cf. une annotation à l'exemplaire `[de Grothendieck]` de SGA 4 I ([58])).

p. 48

Théorème 4.6.3. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie.*

(1°) *Les conditions suivantes sont équivalentes pour un cardinal infini π_0 :*

58. N. Éd. Cette annotation dit "cf. résultat plus simple ci-dessus, si C est *strictement* génératrice" et le résultat en question, également en annotation marginale sur l'exemplaire de Grothendieck, est :

Corollaire. *Soit F admettant* `[une]` *sous-catégorie génératrice par épimorphismes stricts, formée d'objets accessibles, alors* tout *objet de F est accessible (utiliser 7.2. (i) b)).*

a_{π_0}) *$\mathcal{M}$ est une catégorie $\mathfrak{U}$-π_0-paratopos, i.e. $\mathcal{M}$ est stable par petites $\varinjlim$, et contient une petite sous-catégorie pleine* [strictement] *génératrice S, formée d'objets π_0-accessibles de $\mathcal{M}$.*

b_{π_0}) *Il existe une petite catégorie S, et un foncteur pleinement fidèle et π_0-accessible $\mathcal{M} \xrightarrow{i} S^\wedge$, admettant un adjoint à gauche.*

[page 149]

c_{π_0}) *Il existe une petite catégorie C, stable par $\varinjlim$ de cardinal $\leqslant \pi_0$, telle que $\mathcal{M} \simeq \mathrm{Ind}'_{\pi_0}(C)$.*

c'_{π_0}) *Comme c), avec C seulement équivalente à une petite catégorie.*

2°) *Ces conditions impliquent que $\mathcal{M}$ est stable par petites $\varinjlim$ et petites $\varprojlim$, et la condition :*

e_{π_0}) *Pour toute catégorie I grande devant π_0, le foncteur*

$$L = \varinjlim_I : \underline{\mathrm{Hom}}(I, \mathcal{M}) \longrightarrow \mathcal{M}$$

est exact à gauche (donc exact).

Cette condition implique notamment ceci :

$\alpha(\pi_0)$ *Sur la catégorie des double-flèches de $\mathcal{M}$, le foncteur* Ker *(à valeurs dans $\mathcal{M}$) est π_0-accessible.*

$\beta(\pi_0)$ *Pour toute flèche $y \leqslant x$ dans $\mathcal{M}$, le foncteur changement de base $\mathcal{M}/x \longrightarrow \mathcal{M}/y$ est π_0-accessible.*

([59]).

[page 150]

Commentary · Later developments [ms. 149]

Paratopoi and locally presentable categories

Grothendieck is now giving two descriptions of the same cocomplete category: a presentation by small objects and a reflection from a presheaf category. In modern language his π_0-paratopos is a locally π_0^+-presentable category. The connection with the work of Gabriel and Ulmer (1971) is structural; the publisher's contents place presentable objects, generators, and locally presentable categories together in the central part of that monograph. The equivalences used here are recorded more explicitly as Theorems 1.20 and 1.46 in the publisher's chapter introduction to Adámek and Rosický (1994).

The adjunction explains the equivalence with the presheaf formulation. If $L : \widehat{S} \rightleftarrows \mathcal{M} : N$ is a reflection and N preserves π_0^+-filtered colimits, then

$$\mathrm{Hom}_{\mathcal{M}}(Lh_s, X) \simeq N(X)(s).$$

Thus each reflected representable Lh_s is π_0-accessible. Since a presheaf is the colimit of its representables and L preserves colimits, these objects generate $\mathcal{M}$. Conversely, restricted Yoneda on a suitable small category of presentables is fully faithful, and its left adjoint sends a presheaf to the corresponding weighted colimit in $\mathcal{M}$.

There is a missing hypothesis in the later variable-cardinal statement, part 3(c). It says only that $\mathcal{M} \simeq \mathrm{Ind}'_{\pi_0}(S)$ for small S, omitting that S admits colimits of cardinal at most π_0. The proof on manuscript page 152 restores the hypothesis. Without it, take S to be a discrete category on two objects. Every filtered diagram in S is constant, so its filtered completion is still S, which has no coproduct of its two distinct objects. The corrected statement is the cocomplete recognition theorem; allowing arbitrary S leads instead to the accessible categories of the next section.

3°) *Les conditions suivantes sont équivalentes.*

a) $\mathcal{M}$ est une catégorie $\mathfrak{U}$-paratopos, i.e. est stable par petites $\varinjlim$, et contient une petite sous-catégorie pleine S génératrice par épimorphismes strictes, et formée d'objets accessibles.

a′) Comme a), mais en exigeant que $\mathcal{M}$ soit faiblement accessible, i.e. que tout *objet de $\mathcal{M}$ soit accessible.*

b) Il existe une petite catégorie S et un foncteur pleinement fidèle et accessible $\mathcal{M} \hookrightarrow S^\wedge$ admettant un adjoint à gauche.

ⓒ *Il existe un cardinal infini π_0, une petite catégorie S, et une équivalence $\mathcal{M} \simeq \mathrm{Ind}'_{\pi_0}(S)$.*

c′) Comme c, en supposant seulement S équivalente à une petite catégorie.

d) $\mathcal{M}$ est stable par petites $\varinjlim$, est faiblement accessible, i.e. tous ses objets sont accessibles, et admet une filtration cardinale.

p. 49

ⓔ ([60]). *$\mathcal{M}$ est stable par petites $\varinjlim$, contient une petite sous-catégorie pleine strictement génératrice, et il existe un cardinal infini π_0 tel que la condition e_{π_0} de 2°) soit satisfaite.*

ⓔ′ *Comme e, mais en exigeant, au lieu de e_{π_0}, la condition plus faible conjonction de $\alpha(\pi_0)$ et $\beta(\pi_0)$ dans 2°.*

Démonstration. La partie 1° a déjà été démontrée. La première assertions dans 2°), [page 151]
savoir la stabilité de $\mathcal{M}$ par petites $\varprojlim$, est bien connue (résulte de stabilité par petites $\varprojlim$, et de l'existence d'une petite famille génératrice, cf. SGA 4 I 8.12.7, version duale de l'annotation marginale). La validité de la condition e_{π_0} résulte du fait qu'elle est valable dans $C^\wedge$ (où C est la catégorie mentionnée dans 1°, c_{π_0}), et du fait que les $\varprojlim$ dans $\mathcal{M}$, et les $\varinjlim$ grandes devant π_0 dans $\mathcal{M}$, se calculent dans $C^\wedge$. On trouve d'ailleurs de même :

Corollaire 4.6.4. *Si $\mathcal{M}$ est une catégorie $\mathfrak{U}$-π_0-paratopos, alors pour tout cardinal $\pi \geqslant \pi_0$, et tout catégorie J telle que* $\mathrm{card}\, J \leqslant \pi$, *le foncteur*

$$\underline{\mathrm{Hom}}(J, \mathcal{M}) \xrightarrow{\varprojlim_J} \mathcal{M}$$

est π-accessible, ou ce qui revient au même, pour toute petite catégorie I grande devant π, le foncteur

$$\underline{\mathrm{Hom}}(I, M) \xrightarrow{\varinjlim_I} \mathcal{M}$$

commute aux $\varprojlim$ de cardinal $\leqslant \pi$.

[page 152]

Commentary · Exposition [ms. 151]

Exactness comes from the set-valued model

Grothendieck wants the cardinal used for generation also to control the passage between limits and colimits. This is available because the small-category presentation gives a faithful calculation of the particular colimits in question, not because an arbitrary colimit-preserving reflection is left exact.

Let C admit colimits indexed by categories of cardinal at most π_0. The objects of $\mathrm{Ind}_{\pi_0}(C)$ may be identified with the presheaves which take these colimits to limits. One implication follows because representables have this property and π_0^+-filtered colimits of sets commute with such limits. Conversely, if F has the property, then its category of elements C/F is π_0^+-filtered: the colimit in C of any bounded diagram of elements receives the unique compatible element supplied by preservation. The empty diagram gives nonemptiness. The canonical Yoneda presentation of F is therefore a presentation of the required filtered kind.

Limits of these presheaves are pointwise, since limits commute with limits. Their π_0^+-filtered colimits are pointwise by the set-theoretic interchange theorem. Evaluating any proposed comparison

59. **Question.** Condition e_{π_0} (voire, le couple $\alpha(\pi_0) + \beta(\pi_0)$ plus l'existence d'une petite sous-catégorie strictement génératrice (et la stabilité de $\mathcal{M}$ par petites $\varinjlim$), implique-t-il que $\mathcal{M}$ est un π_0-paratopos ? Je présume que non.)

60. C'est là la caractérisation intrinsèque la plus jolie des paratopos par la modalité "économie", l'autre, l'énoncé structurel, est ⓒ .

map at $c \in C$ consequently reduces it to the corresponding comparison in sets. In particular, for $\pi \geq \pi_0$ and $|\operatorname{Fl} J| \leq \pi$, the map

$$\varinjlim_{i \in I} \varprojlim_{j \in J} F_{ij} \longrightarrow \varprojlim_{j \in J} \varinjlim_{i \in I} F_{ij}$$

is an isomorphism whenever I is π^+-filtered. This proves Corollary 4.6.4 and includes the equalizer and base-change assertions in Theorem 4.6.3(2).

The converse recognition argument in part 3 is stated by reference to a corrected use of Artin et al. (1972, Exposé I, Proposition 9.11). Its conclusion allows an increase of cardinal. The set of subobjects and the set of generating maps into each object enter that argument, so exactness at one prescribed rank does not itself put the generators at that rank. The question in footnote 59 asks precisely whether this increase can be avoided.

D'autre part, le fait que e_{π_0} implique $\alpha(\pi_0)$ et $\beta(\pi_0)$ est immédiat.

La partie 3°), en ce qui concerne l'équivalence des conditions a) à d), est contenue dans 1°, et l'équivalence avec d) a été vue également (la seule chose non tautologique étant que la condition c), savoir $\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(C)$ avec C petite et stable par $\varinjlim$ de cardinal $\leqslant \pi_0$, implique l'existence d'une filtration cardinale, ce qui a été vu dans 4.5.9, et réaffirmé dans 4.6.2 2° ci-dessus). D'autre part, que ces conditions a) à d) impliquent ⓔ , et que ⓔ implique e′, a été vu dans 2°. Donc il reste à voir que les conditions de e′ impliquent a), et pour ceci, il reste à établir que les éléments de $\mathcal{M}$ sont accessibles. Mais c'est ce qui est établi dans SGA 4 I 9.11, à cela près que l'énoncé de la condition c), deuxième cas, est canulé, et devrait être remplacé par la condition $\beta(\pi_0)$ (la condition $\alpha(\pi_0)$ figurant dans la condition a) de loc. cit.).

[page 153]

Commentary · Development (commentators') [ms. 152]

Exactness does not determine the generating cardinal

Grothendieck suspects that the fixed-cardinal converse in footnote 59 is false. A topological example gives a negative answer for every infinite cardinal in his convention. It makes the obstruction visible in open covers: colimits may be exact while no nonzero object can be detected at the proposed size.

Proposition. [ours] For every infinite π_0 there is a Grothendieck topos with a small dense subcategory and exact filtered colimits which is not a π_0-paratopos. In fact its only π_0^+-presentable object is the initial object.

Put $\lambda = \pi_0^+$, $\kappa = \lambda^+$, and choose a set X of cardinal κ. Give it the topology whose nonempty opens are the complements of subsets of cardinal less than κ. This is a topology: finite unions of the excluded small subsets remain small, and an arbitrary union of nonempty opens contains one such open. Every nonempty open has cardinal κ. Let $\mathcal{E} = \mathrm{Sh}(X)$.

We first describe an obstruction valid for any noninitial sheaf F. Its support U, the image of $F \to 1$ regarded as an open subset of X, is nonempty. Equivalently, U is the set of points where F has a nonempty stalk; it is open because a germ comes from a local section. Choose distinct points $x_\alpha \in U$ for $\alpha < \lambda$, and set

$$U_\beta = U \setminus \{x_\alpha : \beta \leq \alpha < \lambda\} \qquad (\beta < \lambda).$$

Each U_β is open, since the removed set has cardinal at most $\lambda < \kappa$. The opens increase, their union is U, and each is a proper subset of U. The ordinal λ is a λ-filtered indexing poset: every set of fewer than λ ordinals below it is bounded below it.

Write $F_\beta = F \times U_\beta$, where an open is identified with its subterminal sheaf. Then

$$F = \varinjlim_{\beta < \lambda} F_\beta.$$

This equality can be checked on stalks. At a point in U the point eventually belongs to U_β, and the stalk then equals F_x; outside U every stalk is empty. Each map $F_\beta \to F$ is a monomorphism. If F were λ-presentable, its identity would factor through one of these monomorphisms. A monomorphism with a right inverse is an isomorphism, forcing the supports U_β and U to be equal. This contradicts their construction. Thus no noninitial F is λ-presentable.

For completeness, $\mathcal{E}$ has the other properties in the question; the general sheaf-theoretic facts used here are in Artin et al. (1972, Exposé II, Theorem 4.1 and Corollary 4.1.1). The category of opens is small, and its representable sheaves give the usual dense generating family. All colimits exist by sheafifying presheaf colimits. Filtered colimits are left exact: they are computed by first taking a pointwise filtered colimit of presheaves and then applying the left exact associated-sheaf functor; finite limits of sheaves agree with finite limits of presheaves. Hence e_{π_0} holds, as do $\alpha(\pi_0)$ and $\beta(\pi_0)$. Since X is nonempty, the topos is nondegenerate and cannot be generated by its initial object alone. It is therefore not a π_0-paratopos.

The construction uses the standard sheaf-theoretic relation between support and open covers; it does not contradict the variable-cardinal theorem, since a sheaf topos is locally presentable at a sufficiently large rank. We checked the primary SGA 4 material on sheaves and accessibility and the representation descriptions in Adámek and Rosický (1994); searches concerning compact topologies and presentable sheaves did not locate this exact co-small formulation. The proof is supplied as a development of the manuscript's question, without a claim of priority.

Remarque 4.6.5. Les développements des sections 4.4, 4.5, 4.6 précisent et affirment considérablement, à bien des égards, la théorie des filtrations cardinales de SGA 4 I 9. Il en est ainsi, tout au moins, dans le cas des $\mathfrak{U}$-catégories $\mathcal{M}$ stables par petites $\varinjlim$. Dans loc. cit. on travaille avec des $\mathfrak{U}$-catégories légèrement plus générales, celles satisfaisant la *condition* p. 50
L : il existe un cardinal infini π_0 tel que $\mathcal{M}$ soit stable par petites $\varinjlim$ *grandes devant* π_0. Je présume cependant qu'on doit pouvoir améliorer de façon similaire la théorie des filtrations cardinales *accessibles*, en s'inspirant du corollaire 9.18 de loc. cit., établissant une équivalence de catégories

$$\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(C) ,$$

pour un cardinal π_0 et une petite catégorie C convenable. Ici, on ne peut plus supposer C stable par tel ou tel type de $\varinjlim$. Néanmoins, 4.4.15 nous dit que sans autre hypothèse sur C, $\mathcal{M} \overset{\text{déf}}{=} \mathrm{Ind}_{\pi_0}(C)$ satisfait bien à L_{π_0}, i.e. est stable (et même stable dans $C^\wedge$)
par petites $\varinjlim$ grandes devant π_0. [page 154]

Ceci vu, j'ai envie d'appeler *accessible* une catégorie $\mathcal{M}$ qui admet une filtration cardinal, et qui est de plus faiblement accessible `[i.e.]` dont tout objet est accessible. Et de hasarder une

Conjecture 4.6.6 ([61]). *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie. Conditions équivalentes :*

a) $\mathcal{M}$ satisfait la condition L, et admet une petite catégorie strictement génératrice formée d'objets accessibles.

⇑ *b) Il existe une petite catégorie C, et un cardinal infini π_0, tel que $\mathcal{M}$ soit équivalente à la catégorie $\mathrm{Ind}'_{\pi_0}(C)$.*

⇑ *c) $\mathcal{M}$ est "accessible", i.e. est faiblement accessible (tout objet accessible) et admet une filtration cardinale.*

On a b $\Longrightarrow$ a $\Longleftarrow$ c de façon tautologique, et c $\Longrightarrow$ b a été vu dans loc. cit. 9.18, donc on a la suite

$$\mathrm{c} \Longrightarrow \mathrm{b} \Longrightarrow \mathrm{a} ,$$

donc il resterait à établir a $\Longrightarrow$ c.

Le travail accompli (et la sueur versée) dans 4.5 nous préservera d'ailleurs, le moment [page 155]
venu, d'erreurs fort naturelles dans la description d'une filtration cardinale, à partir d'une sous-catégorie S strictement génératrice : ce n'est ni Filt^π, ni Filt'^π, mais quelque chose de plus subtil – peut-être faut-il prendre encore la catégorie $\mathcal{M}_\pi$, ou alors saturer S par les limites inductives de cardinal $\leqslant \pi$ qui veulent bien être représentables dans $\mathcal{M}$: on trouve alors une $\mathrm{Filt}''^\pi \subset \mathcal{M}_\pi$, et si ces Filt''^π donnent bien une filtration cardinale, on doit avoir forcément $\mathcal{M}_\pi \subset \mathrm{Filt}''^\pi$, donc finalement $\mathrm{Filt}''^\pi = \mathcal{M}_\pi$. Donc je m'attends à ce qu'en tout

61. Tout ceci sera considérablement précisé dans la théorie qui sera développé dans les sections 4.7, 4.16.

état de cause, la filtration cardinale (si elle existe) se fasse par les $\mathcal{M}_\pi$. La condition a) de loc. cit. 9.12 ($\mathcal{M}_\pi$ équivalente à une petite catégorie), et la majoration de cardinaux dans loc. cit. c), ne doit pas faire de difficultés, et la condition b) est OK. La difficulté

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Commentary · Context [ms. 155]

Removing cocompleteness changes the problem

Grothendieck is now testing which part of the reconstruction survives when many colimits do not exist. The small objects can still be intrinsically accessible, but the category $\mathcal{M}_\pi/X$ need not be filtered. A collection of small objects is therefore not yet a collection of small stages of every object.

The modern definition of an accessible category makes the missing presentation part explicit. For a regular λ, one asks for λ-filtered colimits and a small full category of λ-presentable objects from which every object is obtained by such a colimit. Equivalently, the category is a suitable filtered completion of a small category, after allowing retracts among its presentables. This is the structural alternative in Conjecture 4.6.6(b). Its relation to the source's proposed "cardinal filtration" requires the generalized filtration developed next; one should not substitute a modern definition into condition (a) and thereby turn the unproved implication into a tautology.

There is a useful chronological distinction. Gabriel and Ulmer (1971) had treated the cocomplete theory, while Makkai and Paré (1989) had already developed accessible categories and their categorical constructions before these manuscript pages. Adámek and Rosický (1994) subsequently provided a systematic account of both theories. These references place the present calculations in an established mathematical development; they do not establish that Grothendieck was using those treatments here. The source's immediate interlocutor is its own SGA 4 terminology, whose meaning of "accessible category" is reconsidered explicitly in §4.8.

What has been proved on these pages is enough for the cocomplete case. What remains to be proved for a general completion is control of the indices and their retracts. That is why the next section studies three related filtrations instead of assuming that a single bound on a presentation is already the correct invariant.

provient du fait que $\mathcal{M}_\pi$ n'étant pas même stable par sommes et conoyaux de double-flèches (quand ceux-ci ne sont pas représentés dans $\mathcal{M}$), la catégorie $\mathcal{M}_\pi/x$ (pour x dans $\mathcal{M}$) n'a aucune raison d'être filtrante, pour autant que je vois, ni a fortiori d'être grande devant π. p. 51
Donc il n'est pas trop clair comment dès lors représenter x comme $\varinjlim_I x_i$, les x_i dans $\mathcal{M}_\pi$ et I grande devant π.

Il faut bien dire que dans SGA 4 I 9, je ne donne aucun *exemple* d'une filtration cardinale sur une catégorie où les petites $\varinjlim$ quelconques ne seraient représentables (cf. loc. cit. 9.13). Peut-être après tout est-ce contre-nature de vouloir elles au delà ? ? Il faut peut-être attendre d'y être forcé, pour essayer de développer une théorie en dehors du cadre des paratopos.

4.6.7. Nous sommes à présent en mesure de reformuler les kyrielles de conditions un peu techniques a) à f) du théorème 2.6 (p. 25 dans XVIII) concernant le caractère "spécial à gauche" de l'ensemble $\mathrm{Mon}(\mathcal{M}) \subset \mathrm{Fl}(\mathcal{M})$: [page 157]

Proposition 4.6.7 ([62]). *Les conditions préliminaires sur la $\mathfrak{U}$-catégorie $\mathcal{M}$, assurant la validité du théorème XVIII 2.6, sous une forme très légèrement modifié (savoir en modifiant dans 2.6 d) la condition de stabilité de* Filt^π *par quotients, en une propriété d'exactitude d) ci-dessous), équivaut à l'ensemble de six conditions suivantes a) à f).*

$$\begin{cases} \text{a)} & \mathcal{M} \text{ est stable par petites } \varinjlim . \\ \text{b)} & \mathcal{M} \text{ admet une petite sous-catégorie strictement génératrice.} \\ \text{c)} & \text{Les limites inductives filtrantes dans } \mathcal{M} \text{ sont exactes.} \end{cases}$$

Ces conditions impliquent que $\mathcal{M}$ est une catégorie $\mathfrak{U}$-paratopos *(cf. 4.6.3, 3°), et en particulier que les objets de $\mathcal{M}$ sont* accessibles, *et que $\mathcal{M}$ admet une filtration cardinale $(\mathcal{M}_\pi)_{\pi \geqslant \pi_0}$*

62. Peut-être on pourrait appeler ces catégories des catégories "*paratopos parabéliens*" ? ?

telle que pour tout $\pi \geqslant \pi_0$, $\mathcal{M}_\pi$ soit stable par $\varinjlim$ de cardinal $\leqslant \pi$ (en fait, on aura $\mathcal{M}_\pi =$ sous-catégorie pleine des objets π-accessibles de $\overrightarrow{\mathcal{M}}$). Aussi, elles impliquent la stabilité de $\mathcal{M}$ par petites $\varprojlim$. D'autre part, les trois conditions suivantes sont des conditions de stabilité pour les monomorphismes.

d) Soit $f : X \longrightarrow Y$ une flèche dans $\mathcal{M}$ et $R = X \times_Y X \rightrightarrows X$, alors $X/R \longrightarrow Y$ est un monomorphisme – en d'autres termes, f se factorise en un composé ip, avec i monomorphe et p un épimorphisme effectif.

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e) Si A_1, A_2 sont deux sous-objets d'un objet B de $\mathcal{M}$, et $A_0 = A_1 \cap A_2 \overset{déf}{=} A_1 \times_B A_2$, alors $A_1 \amalg_{A_0} A_2 \longrightarrow B$ est un monomorphisme. (Noter la proche parenté des conditions d) et e).)

f) L'ensemble $\mathrm{Mon}(\mathcal{M})$ *des monomorphismes dans $\mathcal{M}$ est stable par cochangement de base.*

En fait, il n'est pas clair que ces conditions, et notamment d), suffisent à entraîner la stabilité des $\mathcal{M}_\pi$ par quotients (seulement par quotients effectifs), comme postulé dans 2.6 d). Mais dans l'énoncé et la démonstration donnée dans XV cor. 7.4 (p. 98-99), on se contentait d'une forme un peu plus faible de cette propriété de stabilité, en se bornant aux quotients Z' de $Z \in \mathrm{Ob}\,\mathcal{M}_\pi$ *tels* qu'il n'existe pas de sous-objet Z'_0 de Z', distinct de Z', et tel que $Z \longrightarrow Z'$ se factorise par Z'_0 ; on exigeait donc que $Z' \in \mathrm{Ob}\,\mathcal{M}_\pi$. Mais si la condition d) est satisfaite, il s'ensuit aussitôt qu'un quotient Z' de Z ayant cette propriété supplémentaire est nécessairement effectif, d'où résulte bien que $Z' \in \mathrm{Ob}\,\mathcal{M}_\pi$.

p. 52

Commentary · Examples [ms. 158] [page 159]

The three conditions on monomorphisms

Grothendieck returns to the purpose of the size theory: constructing and controlling monomorphisms without losing the boundedness needed by a transfinite argument. The three categorical conditions separate an image factorization, the formation of a union, and a pushout. They agree in familiar exact settings but express different demands.

In an abelian category, condition (d) is the equality of image and coimage. For subobjects $A_1, A_2 \subset B$, the kernel of $A_1 \oplus A_2 \to B$, $(a_1, a_2) \mapsto a_1 + a_2$, is the image of

$$A_1 \cap A_2 \longrightarrow A_1 \oplus A_2, \qquad t \longmapsto (t, -t).$$

The pushout over the intersection is the quotient by this kernel, so it embeds in B. This proves (e). If $A \hookrightarrow B$ is monic and $f : A \to C$, its pushout is $(B \oplus C)/\{(a, -f(a)) : a \in A\}$, with the evident categorical cokernel interpretation. The map from C is monic: an element $(0, c)$ in the displayed image forces $a = 0$ because $A \to B$ is monic, and then $c = 0$. The same statement follows from the corresponding kernel diagram in any abelian category. This proves (f).

The distinction is visible in groups. For the two coordinate subgroups of $C_2 \times C_2$, the pushout over their trivial intersection is $C_2 * C_2$; its map to the direct product kills the nontrivial commutator, so (e) fails. For (f), take the normal Klein four subgroup $V \subset A_4$ and a surjection $q : V \to C_2$ with kernel generated by one of its nonidentity elements. In the pushout of $V \hookrightarrow A_4$ along q, that element is killed. Its conjugates in A_4 are the other two nonidentity elements of V, so all of V is killed. An element of V mapping to the generator of C_2 is therefore killed too. The induced map from C_2 is not monic. For commutative unital rings, the pushout of $\mathbb{Z} \hookrightarrow \mathbb{Q}$ along $\mathbb{Z} \to \mathbb{F}_p$ is the zero ring, so $\mathbb{F}_p \to 0$ gives another failure of (f).

The last paragraph of the source uses an extremal quotient: a map which cannot factor through a proper subobject. Under (d), its effective-epi–mono factorization has an invertible mono, hence the quotient is effective. This reduction is correct. One must still keep track of the size of the kernel-pair presentation when asserting preservation of π-accessibility; finite-colimit closure alone does not bound a kernel pair of an arbitrary quotient.

Corollaire 4.6.8. *Les conditions d), e), f) de 4.6.7 sont satisfaites si $\mathcal{M}$ est une catégorie abélienne. Donc l'ensemble des conditions a) à f) est satisfait dans chacun des deux cas suivants :*

1) $\mathcal{M}$ est une catégorie abélienne où les limites inductives filtrantes sont représentables et exactes, et admettant une petite catégorie strictement génératrice ([63]).

2) $\mathcal{M}$ est un topos.

Remarque 4.6.9. Je ne suis par sûr de connaître d'autres exemples où les conditions de 4.6.7 sont vérifiées, que ceux signalés dans 4.6.8 ci-dessus ([64]). On notera que le cas 1°) qui y est mentionné est celui de Tohoku, où j'établis l'existence de "suffisamment d'objets injectifs" dans une telle catégorie abélienne. Dans Tohoku, je me borne à supposer l'existence d'une petite sous-catégorie, *génératrice pour les monomorphismes*, condition en apparence plus faible que celle énoncée dans 4.6.8 1°). Mais dans une catégorie abélienne, les deux notions (et leurs diverses variantes dans SGA 4 I 7.1) sont équivalentes en vertu de SGA 4 I 7.3 (i).

Par contre, il semble que les conditions e) f) ne sont pratiquement jamais vérifiées pour les [page 160]
catégories définies par les espèces de structure algébrique "définissables par produits finis" sans plus, tel les groupes, les anneaux (de toutes sortes), etc. Mais la catégorie des groupes *commutatifs*, étant abélienne, est une exception. (On peut d'ailleurs l'interpréter comme $\mathcal{M} \boxtimes \mathcal{M}$, où $\mathcal{M} = (\mathrm{Gr})$, et on peut se demander dès lors s'il y a quelque généralisations aux catégories paratopos de cette forme, $\mathcal{M}$ étant une catégorie paratopos …)

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Commentary · Examples [ms. 160]

Pointed sets and the return to homological algebra

Grothendieck asks for examples beyond abelian categories and topoi. Pointed sets give a small but useful additional example of all six conditions. It shows that the monomorphism properties do not by themselves force additivity or the topos axioms.

The category Ens_* has all colimits, obtained from disjoint unions and quotient identifications while identifying the specified basepoints. Finite pointed sets form an essentially small dense category, and every pointed set is the filtered union of its finite pointed subsets. Filtered colimits and finite limits are computed on underlying sets with their induced basepoints, so filtered colimits are exact. These verify (a), (b), and (c).

For (d), a pointed map factors through its image, with a surjection onto that image; the surjection is the coequalizer of its kernel pair. For (e), two pointed subsets of a pointed set have intersection containing the basepoint, and their pushout over that intersection is precisely their union. For (f), the pushout of an inclusion $A \subset B$ along a pointed map $A \to C$ is the quotient of $B \amalg C$ identifying each a with its image. Such identifications do not identify two distinct elements of C: any passage through B can return to C only through the same element of the embedded subset A. Thus C still embeds in the pushout.

This category is not abelian: the coproduct of two pointed two-element sets has three elements, whereas their product has four. It is not a nondegenerate topos either: its initial and terminal objects coincide, while a topos with $0 \simeq 1$ is degenerate. Thus the example answers the concrete request for further settings without suggesting a classification.

The reference to Tôhoku has a different force. Grothendieck (1957, Theorem 1.10.1) proves the existence of enough injectives under AB5 and a generator. The present conditions seek a categorical environment in which analogous constructions with monomorphisms can be made beyond the additive case. Finally, the familiar identification of group objects in groups with abelian groups follows from interchange: the internal multiplication and the ambient multiplication have the same unit and satisfy $(ab) * (cd) = (a * c)(b * d)$; substituting units shows that the two multiplications coincide and are commutative. This explains the concrete phenomenon behind the closing suggestion, but supplies no general theorem for the speculative construction $\mathcal{M} \boxtimes \mathcal{M}$.

4.7 $\mathrm{Ind}_\pi C$ (C quelconque) et filtrations cardinales généralisées.

Je sens le besoin de me lancer dans des développements à ce sujet, pour parvenir à une meilleure compréhension de la notion de *partie accessible* d'un paratopos, et du théorème de

63. C'est le cas étudié dans "Tohoku".

64. ou qui s'en déduisent trivialement (produits etc.)

stabilité pour les paratopos par certaines opérations-clef sur ceux-ci.

Dans la suite, C désigne une catégorie équivalente à une petite catégorie, C_0 une sous-catégorie réduite exhaustive, π_0 un cardinal infini. Nous posons

(4.7.1) $$\mathcal{M} \;=\; \mathrm{Ind}_{\pi_0} C \;\subset\; C^{\wedge} \qquad ([65])\,.$$

On se propose d'introduire sur $\mathcal{M}$ diverses filtrations ± "cardinales", par des propriétés correspondantes sur un objet F de $\mathcal{M} \subset C^{\wedge}$, que je vais à présent introduire, en m'inspirant de 4.5.2 (p. 107) et 4.5.6 (p. 116).

Proposition 4.7.1. *Posons $\pi_1 = \sup(\pi_0, \operatorname{card} C_0)$. Soit $\pi \geqslant \pi_0$ un cardinal, et considérons les conditions suivantes (dépendant de π) sur un objet F de $\mathcal{M}$:*

a) $\operatorname{card} C_0/F \leqslant \pi$ *(i.e.* $\operatorname{card} \mathrm{Fl}\, C_0/F \leqslant \pi$*).*

a′) $\operatorname{card} \mathrm{Ob}\, C_0/F \leqslant \pi$.

a″) $\operatorname{card} F(x) \leqslant \pi$ *pour tout $x \in \mathrm{Ob}\, C$ (**N.B.** il suffit de l'exiger pour x dans C_0).*

p. 53
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Commentary · Exposition [ms. 161]

Presentation size and intrinsic smallness

Grothendieck now removes the existence of small colimits from the hypotheses on C. This changes the task. A presheaf may still be assembled from objects of C by sufficiently filtered colimits, but a construction performed on such presentations need not be available inside C itself. He therefore keeps several measures of size in play until he can prove that they agree.

For this section put $\lambda = \pi_0^+$. A category "grande devant π_0" is a λ-filtered category: every diagram having at most π_0 arrows admits a cocone. The category $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$ is thus the ordinary λ-Ind completion, regarded as a full subcategory of $\widehat{C} = \mathrm{Ens}^{C^{\mathrm{op}}}$. Its λ-filtered colimits are computed pointwise. Grothendieck's $\mathcal{M}_\pi$ consists of the π^+-presentable objects, those F for which $\mathrm{Hom}(F, -)$ preserves π^+-filtered colimits. The successor is essential: his diagrams have cardinality $\leq \pi$, whereas modern regular-cardinal notation uses cardinality strictly below the filtering cardinal.

The three tests in Proposition 4.7.1 now have distinct meanings. Conditions (a)–(a″) measure a particular presheaf through its elements. Condition (b) bounds a presentation by representables while retaining λ-filteredness. Condition (c) measures the object intrinsically in $\mathcal{M}$. The intermediate (c′) is intrinsic in the larger category $\widehat{C}$ and can initially be stronger. Finally Filt'^{π} is a closure under repeated bounded filtered colimits, whereas Filt^{π} asks for one presentation. The prime has changed meaning from the preceding cocomplete discussion; here its definition is the one in (b′), not a presentation by arbitrary colimits in an ambient cocomplete category.

Two easy implications deserve to be made explicit. Representables are π^+-presentable in $\widehat{C}$, and a colimit of at most π such presentable objects is again π^+-presentable, because π^+-filtered colimits of sets commute with limits over categories of at most π arrows. This gives $\mathrm{Filt}'^{\pi} \subseteq (\widehat{C})_\pi$. Since $\mathcal{M} \hookrightarrow \widehat{C}$ preserves the filtered colimits in question, ambient presentability implies presentability in $\mathcal{M}$. Neither argument bounds the number of objects of C.

The cardinal π_1 enters when one replaces the total number of elements by a pointwise bound. If $|\mathrm{Fl}\, C_0| \leq \pi$ and every $|F(c)| \leq \pi$, then C_0/F has at most π arrows. Without the bound on C_0, this summation is unavailable. The printed unconditional arrow (a″)⇒(c′) must therefore be treated separately; the example below shows an actual failure, rather than merely an incomplete proof.

b) *On a $F = \varinjlim_{i \in I} x_i$, avec les $x_i \in \mathrm{Ob}\, C$, I une catégorie (pas nécessairement ordonnée) grande devant π_0, et telle que* $\operatorname{card} I \leqslant \pi$. *(**N.B.** en vertu de 4.4.2 (p.79), pour I donnée, ceci équivaut à l'existence d'un foncteur cofinal $I \longrightarrow C/F$, ou encore $I \longrightarrow C_0/F$.)* ([66]).

b′) *On a $F \in \mathrm{Ob}\, \mathrm{Filt}'^{\pi}$, où Filt'^{π} est la plus petite sous-catégorie strictement pleine de $\mathcal{M}$ contenant C, et stable par $\varinjlim$ grandes devant π_0 et de cardinal $\leqslant \pi$.*

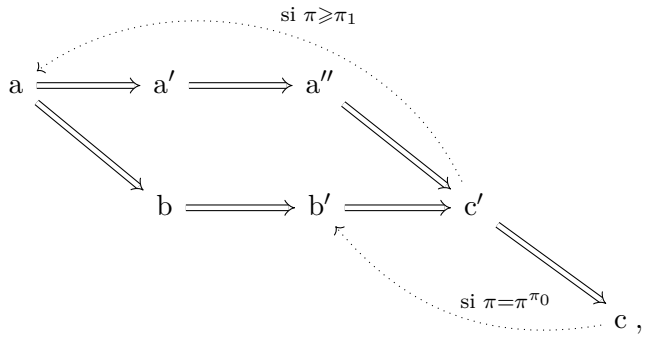
65. **N.B.** Je désigne maintenant par $\mathrm{Ind}C$, $\mathrm{Ind}_\pi C$ ce qui précédemment était noté $\mathrm{Ind}'C$, $\mathrm{Ind}'_\pi C$.

66. **N.B.** On désignera par $\mathrm{Filt}^{\pi} \subset \mathcal{M}$ la sous-catégorie strictement pleine de $\mathcal{M}$ formée des F satisfaisant cette condition.

c) *On a $F \in \mathrm{Ob}\,\mathcal{M}_\pi$, i.e. F est un objet π-accessible de $\mathcal{M}$.*

c′) *On a $F \in \mathrm{Ob}\,C^\wedge_\pi$, i.e. F est un objet π-accessible de $C^\wedge$.*

Ceci posé, on a le diagramme d'implications suivant.



où les deux implications en pointillés, $c' \Longrightarrow a$ et $c \Longrightarrow b$, sont conditionnelles soumises à la condition $\pi \geqslant \pi_1$, resp. $\pi = \pi^{\pi_0}$.

Corollaire 4.7.2. [page 163]

1°) Si $\pi \geqslant \pi_1$, les six conditions a, a′, a″, b, b′, c′ sont équivalentes, et la condition en question (qu'on peut écrire $F \in \mathrm{Ob}\,\mathrm{Filt}^\pi$ où $F \in \mathrm{Ob}\,\mathrm{Filt}'^\pi$ indifféremment) implique c :

$$\mathrm{Filt}^\pi \;=\; \mathrm{Filt}'^\pi \;\subset\; \mathcal{M}_\pi \qquad \textit{si } \pi \geqslant \pi_1 \tag{4.7.2.1}$$

(alors qu'en général, pour $\pi \geqslant \pi_0$, on a

$$\mathrm{Filt}^\pi \;\subset\; \mathrm{Filt}'^\pi \;(\subset C^\wedge_\pi \cap \mathcal{M}) \;\subset\; \mathcal{M}_\pi \qquad \textit{si } \pi \geqslant \pi_0). \tag{4.7.2.2}$$

2°) Quand $\pi = \pi^{\pi_0}$, p.ex. quand π est de la forme 2^c, avec $c \geqslant \pi_0$, alors les conditions b, b′, c, c′ sont équivalentes, i.e. on a

$$\mathrm{Filt}^\pi \;=\; \mathrm{Filt}'^\pi \;=\; (C^\wedge)_\pi \cap \mathcal{M} \;=\; \mathcal{M}_\pi\,. \tag{4.7.3.2}$$

3°) Quand $\pi \geqslant \pi_1$ et $\pi = \pi^{\pi_0}$, alors les sept conditions a), a′), a″), b) , b′), c) , c′) sont équivalentes, donc en particulier

$$\mathrm{Filt}^\pi \;=\; \mathrm{Filt}'^\pi \;=\; C^\wedge_\pi \cap \mathcal{M} \;=\; \mathcal{M}_\pi \qquad (\pi \geqslant \pi_1 \textit{ et } \pi = \pi^{\pi_0}). \tag{4.7.2.4}$$

Démonstration de 4.7.1. Les implications $\mathrm{a} \Longrightarrow \mathrm{a}' \Longrightarrow \mathrm{a}''$ sont tautologiques, ainsi que $\mathrm{b} \Longrightarrow \mathrm{b}'$. L'implication $\mathrm{b}' \Longrightarrow \mathrm{c}'$ provient de la stabilité de $C^\wedge_\pi$ par $\varinjlim$ de cardinal $\leqslant \pi$, et du fait que les $\varinjlim$ du type envisagé (en particulier, grandes devant π_0) *dans* $\mathcal{M}$ sont les mêmes que dans $C^\wedge$, i.e. se calculent "argument par argument" (cf. 4.4.15, p. 99). Ceci, plus le fait que l'inclusion $\mathcal{M} \hookrightarrow C^\wedge$ est pleinement fidèle, implique aussi $\mathrm{c}' \Longrightarrow \mathrm{c}$. p. 54

L'implication $\mathrm{a}'' \Longrightarrow \mathrm{c}'$ se voit comme dans 4.5.6 (p. 119) en appliquant le lemme 4.5.8 1° et 2°, au cas de $C^\wedge = \underline{\mathrm{Hom}}(C^\circ, \mathcal{E})$, où $\mathcal{E} = (\mathrm{Ens})$. L'implication conditionnelle inverse $\mathrm{c}' \Longrightarrow \mathrm{a}''$, pour $\pi \geqslant \pi_1$, se voit comme dans loc. cit., en appliquant ce lemme 4.5.8. Il reste donc à établir encore les trois implications [page 164]

$$(*) \qquad \begin{cases} \mathrm{a} & \Longrightarrow & \mathrm{b} & \\ \mathrm{a}'' & \Longrightarrow & \mathrm{a} & \text{si } \pi \geqslant \pi_1 \\ \mathrm{c} & \Longrightarrow & \mathrm{b} & \text{si } \pi = \pi^{\pi_0}. \end{cases}$$

Mais a $\Longrightarrow$ b est tautologique, en prenant $I = C_0/F$, puisqu'on sait que C_0/F est grande devant π_0 et que $F = \varinjlim_{i\in I} x_i$, pratiquement par définition de $\mathcal{M} = \underbrace{\mathrm{Ind}_\pi(C)}_{\simeq \mathrm{Ind}_\pi C_0} \subset C^\wedge$. Donc il reste à établir les deux implications conditionnelles de $(*)$. L'implication a$''$ $\Longrightarrow$ a pour $\pi \geqslant \pi_1$ s'établit comme dans 4.5.2 (cf. page 109), c'est essentiellement sorital. Reste à établir

$$(*) \qquad \text{c} \implies \text{b} \qquad \text{si } \pi = \pi^{\pi_0} .$$

Nous aurons besoin du résultat auxiliaire : [page 165]

Lemme 4.7.3. *Soit $\pi \geqslant \pi_0$ tel que $\pi^{\pi_0} = \pi$ (donc en fait $\pi > \pi_0$). Pour tout objet F de $\mathcal{M}$, on a*

$$F = \varinjlim_{i\in I} x_i \qquad (^{67}) . \tag{4.7.3.1}$$

Si de plus $\pi' > \pi$ et $F \in \mathrm{Ob}\,\mathrm{Filt}^{\pi'}$, alors on peut prendre I telle que

$$\mathrm{card}\, I \leqslant \pi'^{\pi} . \tag{4.7.3.2}$$

Commentary · Examples [ms. 164]

A pointwise singleton with a large presentation

Grothendieck wants the presheaf description to give a manageable test for accessibility. A thin category separates the pointwise size from the amount of data needed to present the presheaf, and makes the missing hypothesis in (a$''$)⇒(c$'$) visible without any algebraic complications.

Fix an infinite $\pi \geq \pi_0$, and let $\kappa > \pi$ be a regular cardinal. Take C to be the ordinal κ, with one arrow $\alpha \to \beta$ when $\alpha \leq \beta$. This is a small category in the working universe, and it is π^+-filtered: a family of at most π ordinals below κ has a supremum below κ. Write $h_\beta = C(-, \beta)$. Its value at α is a singleton if $\alpha \leq \beta$ and empty otherwise. Consequently

$$F = \varinjlim_{\beta<\kappa} h_\beta$$

is the terminal presheaf, with $F(\alpha) = \{*\}$ at every object. In particular $F \in \mathrm{Ind}_{\pi_0} C$, and condition (a$''$) holds even with the bound 1.

Yet there is no natural transformation $F \to h_\beta$: evaluation at $\beta + 1$ would require a map from a singleton to the empty set. Thus

$$\varinjlim_{\beta<\kappa} \mathrm{Hom}(F, h_\beta) = \varnothing, \qquad \mathrm{Hom}\left(F, \varinjlim_{\beta<\kappa} h_\beta\right) = \{1_F\}.$$

The same presentation witnesses failure of π^+-presentability both in $\widehat{C}$ and in $\mathrm{Ind}_{\pi_0} C$. Conditions (c$'$) and (c) fail. All arrows and colimits used here are ordinary categorical ones; there is no issue of a choice of model for the Ind completion.

The repair to Proposition 4.7.1 is to impose $\pi \geq \pi_1$ on the arrow (a$''$)⇒(c$'$), or to use condition (a), which bounds the whole category of elements. Under that size hypothesis, the six-way equivalence of Corollary 4.7.2(1) is unaffected. One should not extend that equivalence to small π merely because all the values of a presheaf happen to be small.

En effet, on a

$$F = \varinjlim_{j\in J} x_j \qquad \text{avec } J = C_0/F \qquad (^{68}) ,$$

avec J grande devant π_0. Soit alors

$$I = \begin{array}{l}\text{ensemble des sous-catégories } J' \text{ de } J \textit{ grandes devant } \pi_0,\\ \text{et telles que } \mathrm{card}\, J' \leqslant \pi.\end{array} \tag{4.7.3.3}$$

Je dis que I (ordonné par inclusion) *est grand devant* π ([69]). Cela résulte aussitôt du lemme 4.5.5, p. 113, (énoncé d'abord, de façon erronée, en supposant seulement $\pi > \pi_0$ et non $\pi^{\pi_0} = \pi$). Posons alors, pour tout $J' \in I$,

$$x_{J'} \;=\; \varinjlim_{j \in J'} x_j \;,$$

on a donc

$$x_{J'} \;\in\; \mathrm{Ob}\,\mathrm{Filt}^{\pi}$$

par définition de I et de Filt^{π}. D'autre part,

$$J' \;\longmapsto\; x_{J'} \;:\; I \;\longrightarrow\; \mathrm{Filt}^{\pi}$$

est un foncteur, donc [page 166]

$$\varinjlim_{J' \in I} x_{J'} \;\simeq\; \varinjlim_{j \in J} x_j \;\simeq\; x \;,$$

d'où la première assertion de 4.7.3. Reste la majoration de cardinaux pour $F \in \mathrm{Ob}\,\mathrm{Filt}^{\pi}$. Dans ce cas, quitte à remplacer C_0/F par une catégorie cofinale grande devant π_0, $\mathrm{card}\, J \leqslant \pi'$. On voit alors tout de suite que $\mathrm{card}\, I \leqslant \pi'^{\pi_0}$, ce qui est la majoration annoncée.

Nous pouvons maintenant prouver l'implication $(*)$, page 164, i.e.

$$\mathcal{M}_{\pi} \;\subset\; \mathrm{Filt}^{\pi} \qquad (\text{pour } \pi = \pi^{\pi_0}).$$

Soit donc F dans $\mathcal{M}_{\pi}$, i.e. π-accessible, écrivons F sous la forme (4.7.3.2), avec I grande devant π. Par définition de la π-accessibilité, appliqué à $F \xrightarrow{\mathrm{id}} F \simeq \varinjlim F_i$, on voit que F est un facteur direct d'un F_i, donc d'un objet de Filt^{π}. Faute de savoir que Filt^{π} est stable par facteurs directs, on ne peut en conclure que F est dans Filt^{π} (aussi je ne peux prouver c $\Longrightarrow$ b), mais seulement que F est dans Filt'^{π}, qui, lui, est stable par $\varinjlim$ grandes devant π_0, de cardinal π donné $> \pi_0$, donc aussi stable par facteurs directs. Cela achève la démonstration [page 167] de 4.7.1, et en même temps on obtient :

Commentary · Exposition [ms. 166]

Removing the retract without increasing the bound

Grothendieck's factorization of 1_F is the decisive accessibility argument, but it initially produces a retract of one stage. He explicitly stops short of the claimed implication (c)⇒(b). The missing step has a positive answer. It belongs to the retract-removal result of Makkai–Paré, Proposition 2.3.11; an accessible verification and a stronger closure statement are given by Chen (2021, Corollaries 4.13 and 4.15, including footnote 2). We give the elementary construction needed here, keeping track of the number of arrows.

Retract lemma. Let $\pi \geq \pi_0$ be infinite. If a presheaf H is a retract of

$$F = \varinjlim_{i \in I} h_{c_i}, \qquad I \text{ is } \pi_0^{+}\text{-filtered}, \qquad |\,\mathrm{Fl}\, I| \leq \pi,$$

then H has a presentation of the same form with an indexing category having at most π arrows. No closure property of C is needed.

Proof. Choose $s : H \to F$ and $r : F \to H$ with $rs = 1_H$, and put $e = sr$. Let $q_i : h_{c_i} \to F$ be the presentation maps. By Yoneda and pointwise computation of the colimit, for each i there are an object $t(i)$ of I and a morphism $v_i : c_i \to c_{t(i)}$ satisfying

$$q_{t(i)} h_{v_i} = e q_i.$$

67. les x_i dans Filt^{π}, I ensemble ordonné grand devant π.
68. où J peut aussi se remplacer par une catégorie cofinale dans [?] C_0/F grande devant π_0.
69. cf. démonstration 4.7.5, page 168-171.

Put $\alpha_i = rq_i$. Define J to have the same *labelled* objects as I. A morphism from label i to label j is an actual morphism $u : c_i \to c_j$ satisfying $\alpha_j h_u = \alpha_i$. Inside this category retain only the subcategory generated by the images of all arrows of I and by the arrows $v_i : i \to t(i)$. The latter are morphisms over H, since $re = r$. Finite compositions of an infinite set of at most π generators give at most π arrows, so $|\operatorname{Fl} J| \leq \pi$.

We check filteredness with all compatibility conditions included. Let $D : K \to J$ have at most π_0 arrows. For each vertex a of K, use the chosen map $v_{D(a)}$; choose an I-cocone on their targets and call its vertex k. This produces maps

$$w_a : c_{D(a)} \longrightarrow c_k, \qquad q_k h_{w_a} = s\alpha_{D(a)},$$

each a composite of a chosen v_i with an I-arrow. For an arrow $f : a \to b$ of K, the two maps $w_b D(f)$ and w_a have equal composites into F, because the α_i are a cocone over H. Equality in the filtered colimit

$$F(c_{D(a)}) = \varinjlim_{i \in I} C(c_{D(a)}, c_i)$$

is witnessed after an I-arrow out of k. There are at most π_0 such equalities. The π_0^+-filteredness of I gives a single further arrow $k \to l$ which makes all of them hold simultaneously: take a cocone on the diagram consisting of k, the chosen witnesses, and the arrows from k to them. The resulting maps $c_{D(a)} \to c_l$ are arrows of J and form the required cocone. This also handles the empty diagram, since I is nonempty. Thus J is π_0^+-filtered.

Finally the maps α_i induce $\varinjlim_J h_{c_i} \to H$. At an object c of C, every element of $H(c)$ lifts along s to $F(c)$ and therefore comes from some $C(c, c_i)$; the map is surjective. Suppose two such representatives $u : c \to c_i$ and $u' : c \to c_j$ have the same image in $H(c)$. After composition with v_i and v_j, their images in $F(c)$ are equal, since both are the image under s of that common element. Filteredness supplies a common I-stage where the two representatives agree. The arrows to that stage, and v_i, v_j, all belong to J, so the original representatives agree in the colimit over J. The map is injective at every c, hence an isomorphism. □

Apply the lemma to the retract produced on manuscript p. 166. At $\pi^{\pi_0} = \pi$ it proves the originally intended equality

$$\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi} = (\widehat{C})_{\pi} \cap \mathcal{M} = \mathcal{M}_{\pi}.$$

Thus the statement of Corollary 4.7.2(2) survives, while the proof actually written on these pages requires this additional argument. The construction uses the author's explicit permission that an indexing category need not be ordered; replacing it by a cofinal poset without a cardinal estimate would lose the point of the lemma.

Corollaire 4.7.4. *Si $\pi \geqslant \pi_0$ est tel que $\pi^{\pi_0} = \pi$, alors on a*

$$\mathrm{Filt}'^{\pi} \;=\; (\mathrm{Filt}^{\pi})^{\natural}\,, \tag{4.7.4.1}$$

i.e. Filt'^{π} *est formée des objets F de $\mathcal{M}$ qui sont facteurs directs d'un objet de* Filt^{π}.

En effet, comme $\mathrm{Filt}^{\pi} \subset \mathrm{Filt}'^{\pi}$, et que $(\mathrm{Filt}^{\pi})^{\natural} \subset \mathrm{Filt}'^{\pi}$, et que Filt'^{π} est stable par facteurs directs, on a $(\mathrm{Filt}^{\pi})^{\natural} \subset \mathrm{Filt}'^{\pi}$ pour tout $\pi \geqslant \pi_0$. Mais inversement, comme $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_{\pi}$, l'argument précédent montre que pour $\pi = \pi^{\pi_0}$, on a $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_{\pi} \subset (\mathrm{Filt}^{\pi})^{\natural}$, ce qui prouve 4.7.3.

Remarque 4.7.4.1 `[deux fois 4.7.4.1]`. Bien sûr, par 4.7.2, 1°, dès que $\pi \geqslant \pi_1$, on a même $\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi}$ (sans avoir à supposer $\pi = \pi^{\pi_0}$), donc *si* $\pi > \pi_0$ (ce qui sera la cas si $\pi_1 > \pi_0$, i.e. $\pi_1 \neq \pi_0$). Filt'^{π} étant stable par facteurs directs, Filt^{π} l'est aussi, (4.7.4.1) est satisfait trivialement. Mais si $\pi = \pi_1 = \pi_0$, on a $\mathrm{Filt}^{\pi} = \mathrm{Filt}^{\pi_0} = \mathrm{Filt}'^{\pi_0} =$ sous-catégorie des foncteurs *représentables* dans C (équivalent à C), et celle-ci n'est égale à $C^{\natural}$ dans $\mathcal{M}$ que si C est karoubienne, cf. … ci-dessous.

4.7.5. Pour que la démonstration de 4.7.1 soit complète, il me faudrait encore donner une démonstration de 4.5.5, p. 113, qui est le lemme technique clef pour établir le

p. 56
[page 168]

Commentary · Examples [ms. 167]

The initial level already contains idempotent splittings

Grothendieck's concern about retracts also reaches the endpoint $\pi = \pi_0$. Here the literal assertion that Filt^{π_0} consists only of representables is false under the definition just given. The obstruction is present in the smallest possible nontrivial idempotent category.

Let E have one object $*$ and endomorphism monoid $\{1, e\}$ with $e^2 = e$ and $e \neq 1$. Every diagram $K \to E$, of any size, admits a cocone whose components are e: for either possible arrow a of E, one has $ea = e$. Hence E is λ-filtered for every regular λ. It has no terminal object, since its unique object has two endomorphisms.

Take $C = E$. The representable h_* has value $\{1, e\}$. The natural endomorphism induced by e sends both elements to e, and its image is the singleton presheaf P. The colimit of the diagram $E \to \widehat{E}$ that sends the object to h_* and e to this endomorphism is P. It is a π_0^+-filtered presentation with just two indexing arrows, so $P \in \mathrm{Filt}^{\pi_0}$. It is not representable, since its value has one element rather than two.

This also clarifies the phrase "cofinale à e" in footnote 76. The unique functor $E \to e$ is final, as it is for any nonempty filtered category. This neither makes E equivalent to e nor supplies a final functor $e \to E$. In particular it does not say that the colimit of every diagram on E is one of its values. The relevant conclusion of the smallness assumption is a retract, as the next argument shows.

The corrected general statement is

$$\mathrm{Filt}^{\pi_0} = \mathrm{Filt}'^{\pi_0} = \mathcal{M}_{\pi_0} \simeq \mathrm{Kar}(C).$$

Indeed, if I is π_0^+-filtered and $|\mathrm{Fl}\, I| \leq \pi_0$, the identity diagram of I has a cocone $u_i : i \to i_*$. For any diagram $D : I \to \widehat{C}$, these arrows induce a map from its colimit L to $D(i_*)$. Its composite with $D(i_*) \to L$ is 1_L, by the colimit universal property. Thus a bounded presentation at the initial level gives a retract of a representable. Conversely every retract of a representable is the colimit of the two-arrow idempotent category just used. The same argument shows that this class is closed under further colimits of the prescribed size, proving the equality with Filt'^{π_0}.

For the equality with $\mathcal{M}_{\pi_0}$, write a π_0^+-presentable object as its defining π_0^+-filtered colimit of representables and factor its identity through one stage. Conversely representables are presentable and retracts preserve presentability. This is the ordinary categorical form of the idempotent-completion statement in Lurie (2009, Lemma 5.4.2.4). In particular the conclusion does not require C itself to be Karoubian. What Karoubian completeness provides is the further identification $\mathrm{Kar}(C) \simeq C$.

Lemme 4.7.5.1 (utilisé dans la démonstration de 4.7.3). *Soient π_0, π deux cardinaux infinis, avec $\pi \geqslant \pi_0$ et $\pi^{\pi_0} = \pi$ (donc $\pi > \pi_0$). Soit J une catégorie grande devant π_0. Alors l'ensemble I, ordonné par inclusion, des sous-catégories J' de J qui sont grandes devant π_0, et telles que* $\mathrm{card}\, J' \leqslant \pi$, *est grand devant π. (De plus, si* $\mathrm{card}\, J \leqslant \pi'$, *on a* $\mathrm{card}\, I \leqslant \pi'^{\pi}$.*)*

Il faut donc montrer le

Corollaire 4.7.5.2 ([70]) (c'est le lemme 4.5.5 non encore démontré). *Sous les conditions de 4.7.5, soit $K \subset \mathrm{Fl}J$ tel que* $\mathrm{card}\, K \leqslant \pi$. *Alors il existe $J' \in I$ tel que $K \subset J'$.*

On sait, π étant infini, que la catégorie K' engendrée par K est aussi de cardinal $\leqslant \pi$, donc on peut supposer que K "est" une sous-catégorie de J. On va construire une $J' \in I$ contenant K, en construisant par récurrence transfinie une suite croissante K_α de sous-catégories de J, avec $\mathrm{card}\, K_\alpha \leqslant \pi$, indexée par les ordinaux α tels que $\boxed{\mathrm{card}\, \alpha \leqslant \pi}$:

[page 169]

1°) $K_0 = K$.

2°) $K_\alpha = \bigcup_{\beta<\alpha} K_\beta$ ($= \varinjlim K_\beta$) si α est un ordinal limite (donc $\mathrm{card}\, K_\alpha \leqslant \pi$, puisque les $\mathrm{card}\, K_\beta \leqslant \pi$).

3°) $K_{\alpha+1}$ déduit de K_α de la façon suivante. Si K_α est grande devant π_0, on prend $K_{\alpha+1} = K_\alpha$ (donc, compte tenu de 2°, on aura $K_{\alpha'} = K_\alpha$ pour $\alpha' \geqslant \alpha$ – la construction transfinie s'arrête …). Sinon, $K_{\alpha+1}$ s'obtient ainsi :

70. Comparer avec la démonstration de 4.6.1.2.

a) pour toute partie A de cardinal $\leqslant \pi_0$ de $\mathrm{Ob}\, K_\alpha$, on choisit un majorant x_A dans J (on a envie de le prendre dans K_α, s'il y a un majorant dans K_α) ;

b) pour tout couple x, y d'objets de K_α, et toute partie de cardinal $\leqslant \pi_0$ de $\mathrm{Hom}_{K_\alpha}(x,y)$, on choisit une flèche $u_B : y \longrightarrow z_B$ dans J, qui égalise les $v_\beta \in B$ (et on prend cette flèche dans K_α, quand c'est possible) ;

c) on prend pour $K_{\alpha+1}$ la plus petite sous-catégorie contenant les id_{x_A} et les u_B.

Il faut vérifier que

$$\mathrm{card}\, K_{\alpha+1} \ \leqslant\ \pi\ .$$

Le nombre des choix possibles pour $A \in \mathfrak{P}_{\leqslant\pi_0}(\mathrm{Ob}\, J)$ est $\leqslant \pi^{\pi_0} = \pi$, le nombre des choix possibles pour $B \in \mathfrak{P}_{\pi_0}(\mathrm{Fl}\, J)$ est aussi $\leqslant \pi^{\pi_0} = \pi$, donc le nombre des flèches de la forme id_{x_A} ou u_B est $\leqslant \pi + \pi = \pi$. Mais alors la sous-catégorie $K_{\alpha+1}$ de J engendrée par cet ensemble de flèches est de cardinal $\leqslant 1$. Soit alors α_0 le plus petit ordinal tel que [page 170] p. 57

$$\mathrm{card}\, \alpha_0 \ >\ \pi_0\ ,$$

on a donc

$$\mathrm{card}\, \alpha_0 \ =\ \text{cardinal successeur de } \pi_0 \leqslant \pi\ ,$$

puisque $\pi^{\pi_0} = \pi$, d'où $\pi > \pi_0$ (car $\pi^{\pi_0} = \pi_0^{\pi_0} \geqslant 2^{\pi_0} > \pi_0$). Ainsi, α_0 fait partie des ordinaux α admissibles (i.e. tels que $\mathrm{card}\, \alpha \leqslant \pi$). Je dis que K_{α_0} *est grande devant* π_0 (ce qui prouvera 4.7.5.2, en prenant $J' = K_{\alpha_0}$). Cela résulte immédiatement du fait que

$$K_{\alpha_0} \ =\ \bigcup_{\alpha<\alpha_0} K_\alpha \qquad \text{(réunion } \textit{croissante}\text{)},$$

et que l'ensemble des $\alpha < \alpha_0$ est *grande devant* π_0 ([71]). Donc tout ensemble d'objets ou de flèches de K_{α_0}, de cardinal $\leqslant \pi_0$, est contenu dans un K_α avec $\alpha < \alpha_0$.

Cela implique, s'il s'agit d'un ensemble d'objets, qu'il y a un majorant commun dans $K_{\alpha+1}$, a fortiori dans K_{α_0}. S'il s'agit d'une partie d'un $\mathrm{Hom}_{K_\alpha}(x,y)$, qu'il y a un égalisateur $y \longrightarrow z$ dans $K_{\alpha+1}$, a fortiori dans K_{α_0}. Donc le critère PF_π 1,2 de 4.4.4 (page 89) est satisfait par K_{α_0}, ce qui achève la démonstration de 4.7.5.2. [page 171]

Commentary · Exposition [ms. 170]

Closing a small diagram under filteredness witnesses

Grothendieck is building a categorical version of a downward size argument: a prescribed small part of a filtered indexing category should lie inside a small filtered subcategory. The fixed-point assumption $\pi^{\pi_0} = \pi$ is exactly what pays for adjoining all witnesses of size at most π_0 at each step. The proof is sound after two pieces of construction data, implicit in the prose but absent from clause 3(c), are restored.

Let J be π_0^+-filtered and let K be a set of at most π arrows. First include their endpoints and identities and close under composition. This gives a subcategory K_0 with at most π arrows. At a successor stage retain *all* arrows of K_α. For each set $A \subseteq \mathrm{Ob}\, K_\alpha$ of cardinality at most π_0, choose an object z_A and arrows $a \to z_A$ for all $a \in A$, and adjoin the arrows as well as their target. Include the empty set here, so that nonemptiness is ensured even when K_0 is empty. For each pair of objects x, y and each family $B \subseteq K_\alpha(x,y)$ of cardinality at most π_0, adjoin an arrow $y \to z_B$ which coequalizes B. Then close under identities and finite composition. At a limit stage take the union.

There are at most $\pi^{\pi_0} = \pi$ choices of a small family of existing arrows or objects. Each contributes at most π_0 new arrows. Closing under composition still leaves at most π arrows. Thus all K_α with $|\alpha| \leq \pi$ have size at most π. The printed reference to subsets of $\mathrm{Ob}\, J$ and $\mathrm{Fl}\, J$ in the counting argument must refer to the current K_α; there is no bound on J itself. Likewise the printed final bound 1 is π.

71. expliciter ceci en un

Lemme. *Soit π_0 un cardinal, $\alpha_0 = \alpha_{\pi_0'}$ (π_0' le successeur de π_0) le plus petit ordinal tel que* $\mathrm{card}\, \alpha_0 > \pi_0$, *alors $I_{\alpha_0} = \{\alpha \mid \alpha < \alpha_0\}$ est grand devant π_0.*

Set $\delta = \pi_0^+$, regarded as its initial ordinal. It is regular, and $\delta \leq \pi$ because $\pi^{\pi_0} = \pi$ forces $\pi > \pi_0$. A family of at most π_0 arrows of K_δ is contained in some earlier K_α: take the supremum of their appearance stages, which is below δ. The next stage supplies the required common target or common coequalizer. These two properties imply the full cocone criterion. For a diagram of at most π_0 arrows, first send all vertices to one target; for each diagram arrow compare the two induced maps to that target. Coequalize each pair, send the resulting targets to one common target, and then coequalize the resulting family of parallel arrows out of the first common target. This produces one map which coequalizes every pair. Thus K_δ is π_0^+-filtered.

This proves the extension lemma. It also proves that the poset $\mathscr{I}$ of all such subcategories is π^+-directed: a family of at most π of them has union of arrow sets of cardinality at most π, and the extension construction supplies an upper bound. Every arrow of J lies in an object of $\mathscr{I}$. A compatible family of cocones on the restrictions to those subcategories is therefore exactly a cocone on all of J. Applying the universal property twice gives

$$\varinjlim_{J' \in \mathscr{I}} \varinjlim_{j \in J'} D(j) \;\simeq\; \varinjlim_{j \in J} D(j).$$

If $|\operatorname{Fl} J| \leq \pi'$, then $|\mathscr{I}| \leq (\pi')^\pi$, since every subcategory is determined by at most π arrows. This is the announced bound in Lemma 4.7.3; the exponent π_0 printed in its proof on p. 166 is not the bound established by this counting.

The same construction is the familiar mechanism for raising the index of accessibility; compare Vasey (2020, Theorem 3.4). In the higher categorical setting its poset form is used in Lurie (2009, Lemma 5.4.2.8 and Proposition 5.4.2.9). In each form the essential work is to retain enough witnesses while keeping their total size under control.

Corollaire 4.7.6 (de 4.7.1). *Pour tout cardinal $\pi \geqslant \pi_0$, on a*

$$(4.7.6.1) \qquad \underbrace{\mathrm{Filt}'^{\pi}}_{\substack{si\ \pi \geqslant \pi_1 \\ =\ \mathrm{Filt}^{\pi}}} \;\subset\; \mathcal{M}_\pi \;\subset\; \underbrace{\mathrm{Filt}^{\pi^{\pi_0}}}_{\substack{si\ \pi^{\pi_0} \geqslant \pi_1 \\ =\ \mathrm{Filt}^{\pi^{\pi_0}}}} \qquad (= (\mathrm{Filt}^{\pi^{\pi_0}})^\natural \text{ par cor. 4.7.4}),$$

donc si $\pi \geqslant \pi_1$, on a aussi

$$(4.7.6.2) \qquad \mathrm{Filt}^{\pi} \;\subset\; \mathcal{M}_\pi \;\subset\; \mathrm{Filt}^{\pi^{\pi_0}} \qquad (si\ \pi^{\pi_0} \geqslant \pi_1).$$

En effet, $\mathrm{Filt}'^{\pi} \subset \mathcal{M}_\pi$ n'est autre que l'implication b′ $\Longrightarrow$ c de 4.7.1, d'autre part, on a $\pi \leqslant \pi^{\pi_0}$, d'où

$$\mathcal{M}_\pi \;\subset\; \mathcal{M}_{\pi^{\pi_0}} \;=\; \mathrm{Filt}'^{\pi^{\pi_0}},$$

où la dernière égalité provient de $\mathcal{M}_{\pi'} = \mathrm{Filt}'^{\pi'}$ si $\pi'^{\pi_0} = \pi'$, i.e. de (4.7.2.3) (page 163), en faisant $\pi' = \pi^{\pi_0}$, compte tenu que $\pi'^{\pi_0} = (\pi^{\pi_0})^{\pi_0} = \pi^{\pi_0 \cdot \pi_0} = \pi^{\pi_0} = \pi$.

Proposition 4.7.7. *Considérons C, C_0, $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, $\pi_1 = \sup(\pi_0, \operatorname{card} \operatorname{Fl} C_0)$ comme précédemment. Pour tout cardinal $\pi \geqslant \pi_0$, soit $\mathcal{F}^\pi$ la sous-catégorie strictement pleine* Filt'^{π} *(resp. $(C^\wedge)_\pi \cap \mathcal{M}$, resp. $\mathcal{M}_\pi$) dans $\mathcal{M}$. Alors la filtration croissante $(\mathcal{F}^\pi)_{\pi \geqslant \pi_0}$ de $\mathcal{M}$ par les sous-catégories $\mathcal{F}^\pi$ satisfait les conditions suivantes.* [page 172]

a) Pour tout $\pi \geqslant \pi_0$, $\mathcal{F}^\pi$ est équivalente à une petite catégorie.

b) $\mathcal{M}$ satisfait à L_{π_0} *(i.e. est stable par $\varinjlim$ filtrantes grandes devant π_0), et pour tout $\pi \geqslant \pi_0$, $\mathcal{F}^\pi$ est stable par les limites filtrantes grandes devant π_0 et de cardinal $\leqslant \pi_0$ [?].* p. 58

c) Si $\pi \geqslant \pi_0$ est tel que $\boxed{\pi^{\pi_0} = \pi,}$ *alors pour tout F dans $\mathcal{M}$, on a un isomorphisme*

$$F \;=\; \varinjlim_{i \in I} x_i\,,$$

les x_i dans $\mathcal{F}^\pi$, et I catégorie ordonnée grande devant π_0. Si de plus $\pi' > \pi$ et $F \in \operatorname{Ob} F^{\pi'}$, on peut prendre I telle que

$$(4.7.7.1) \qquad \operatorname{card} I \;\leqslant\; \pi'^{\pi} \qquad \underline{si}\ \pi'^{\pi} \geqslant \pi_1. \qquad \text{[page 173]}$$

d) Si $\pi \geqslant \pi_0$, on a $\mathcal{F}^\pi \subset \mathcal{M}_\pi$, i.e. les objets de $\mathcal{F}^\pi$ sont π-accessibles.

N.B. Ce sont là exactement, à part d), les conditions a), b), c) des filtrations cardinales de SGA 4 I 9.7.12, à cela près seulement que dans c), on se restreint ici aux ordinaux π tels que $\pi^{\pi_0} = \pi$ ([72]). La condition supplémentaire d) est une "condition d'accessibilité", cf. plus bas.

Démonstration de 4.7.7. a) Comme on a

$$\mathrm{Filt}'^{\pi} \ \subset \ (C^\wedge)_\pi \cap \mathcal{M} \ \subset \ \mathcal{M}_\pi \ ,$$

on est réduit à prouver que $\mathcal{M}_\pi$ est équivalente à une petite catégorie. On est ramené au cas $\pi \geqslant \pi_1$, $\pi = \pi^{\pi_0}$, auquel cas on a $\mathcal{M}_\pi = \mathrm{Filt}^\pi$. En fait, on trouve :

Corollaire 4.7.7.1 `[deux fois 4.7.7.1]` ([73]). *Si $\pi \geqslant \pi_1$, on a*

$$\begin{cases} \operatorname{card} \text{réd}\, \mathrm{Filt}^\pi & \leqslant \ 2^\pi \\ \operatorname{card} \text{réd}\, \mathrm{Filt}'^{\pi} & \leqslant \ \operatorname{card} \text{réd}\, \mathcal{M}_\pi \ \leqslant \ 2^{(\pi^{\pi_0})} \qquad (= 2^\pi \text{ si } \pi^{\pi_0} = \pi). \end{cases}$$

La deuxième majoration résulte de la première et de

$$\mathcal{F}^\pi \ \subset \ \mathcal{M}_\pi \ \subset \ \mathrm{Filt}^{\pi^{\pi_0}} \qquad (\text{pour } \pi^{\pi_0} \geqslant \pi_1) \qquad ([74]).$$

Pour la première, on utilise la condition a$''$) de 4.7.1 (qui caractérise Filt^π pour $\pi \geqslant \pi_1$). Je passe sur les détails de la vérification.

Prouvons 4.7.2 b). La condition L_{π_0} est connue, reste la stabilité de $\mathcal{F}^\pi$. Dans le cas de Filt'^{π}, elle est acquise par définition. Dans les cas $C^\wedge_\pi \cap \mathcal{M}$ et $\mathcal{M}_\pi$, elle résulte de la stabilité de la notion de π-accessibilité par $\varinjlim$ de cardinal $\leqslant \pi$, soit dans $C^\wedge$, soit dans $\mathcal{M}$ (compte tenu que $C^\wedge$ et $\mathcal{M}$ ont une famille génératrice C formée d'objets π_0-accessibles, donc π-accessibles). [page 174]

Prouvons c) : c'est essentiellement le lemme 4.7.3, compte tenu que l'on a

$$\mathrm{Filt}^\pi \ \subset \ \mathcal{F}^\pi \ ,$$

sauf qu'il faut faire attention à la majoration de cardinaux, $\operatorname{card} I \leqslant \pi'^{\pi}$. Mais si $F \in \operatorname{Ob} \mathcal{F}^{\pi'}$, comme par (4.7.6.1) p. 59

$$\mathcal{F}^{\pi'} \ \subset \ \underbrace{\mathcal{M}_{\pi'}}_{\subset \mathcal{M}_{\pi'^\pi} \subset \mathrm{Filt}'^{\pi'^\pi}} \ \subset \ \mathrm{Filt}'^{\pi'^{\pi_0}},$$

donc a fortiori

$$\mathcal{F}^{\pi'} \ \subset \ \mathrm{Filt}^{\pi'^\pi} \ \underbrace{=}_{\text{si } \pi'^\pi \geqslant \pi_1} \ \mathrm{Filt}^{\pi'^\pi},$$

d'où, par 4.7.3, avec π' remplacé par $\pi'^\pi = \pi''$, la possibilité de prendre I avec

$$\operatorname{card} I \ \leqslant \ \pi''^\pi \ = \ (\pi'^\pi)^\pi \ = \ \pi'^{\pi \cdot \pi} \ = \ \pi'^\pi,$$

ce qui achève la démonstration.

Remarque 4.7.7.2. J'ai été obligé, dans l'énoncé de la condition c), de faire intervenir un deuxième cardinal π_1, et ceci faut de savoir si, pour π'' tel que $\pi'' = \pi''^{\pi_0}$, $\mathrm{Filt}^{\pi''}$ est stable par facteurs directs. (Il faudrait, pour bien faire, soit prouver qu'il est stable, donc qu'on a dans 4.7.1 c $\Longrightarrow$ b, i.e. $\mathcal{M}_\pi \subset \mathrm{Filt}^\pi$ (et non seulement c $\Longrightarrow$ b$'$, i.e. $\mathcal{M}_\pi \subset \mathrm{Filt}'^{\pi}$) si $\pi = \pi^{\pi_0}$, ou bien faire un contre-exemple!) Cet ingrédient un peu étrange, on sera hélas obligé de le traîner dans la définition des filtrations cardinales généralisées, si on ne veut s'interdire de regarder dans le cas présent $(\mathcal{F}^\pi)_{\pi \geqslant \pi_0}$ comme une telle filtration, et ne s'autoriser que de regarder $(\mathcal{F}^\pi)_{\pi \geqslant \pi_1}$ comme telle, ce qui me paraîtrait ridicule. Pratiquement (même si théoriquement [page 175]

72. et *de plus*, dans la majoration (4.7.7.1), aux cardinaux $\pi' > \pi$ tels que l'on ait de plus $\pi'^\pi \geqslant \pi_1$ (ce qui est automatique si on a $\pi \geqslant \pi_1$, ce qui est le cas, vu que $\pi = \pi^{\pi_0} \geqslant 2^{\pi_0}$, si on a $2^{\pi_0} \geqslant \pi_1$.)

73. Comparer avec 4.7.19.

74. Cf. (4.7.6.2).

l'introduction de π_1 dans la description d'une filtration cardinale généralisée était bel et bien nécessaire) on n'aura guère des cas où il faille faire mention de π_1 dans la condition c. En effet, $\pi \geqslant \pi_0$ et $\pi^{\pi_0} = \pi$ impliquent $\pi \geqslant 2^{\pi_0}$, et dès lors $\pi'^{\pi} \geqslant 2^{(2^{\pi_0})}$. Si donc $\pi_1 \leqslant 2^{(2^{\pi_0})}$, i.e. $\operatorname{card} C_0 \leqslant 2^{(2^{\pi_0})}$, alors la condition $\pi^{\pi'} \geqslant \pi_1$ dans c) est automatiquement satisfaite, et il n'y a donc pas à la mentionner. Or il est bien rare qu'on ait à faire à des petites catégories C_0, dont le cardinal soit $>$ puissance du continu c, et a fortiori, telle que $\operatorname{card} C_0 > 2^c$. Si par [page 176]
contre $\operatorname{card} C_0 \leqslant 2^c$, comme $2^c \leqslant 2^{(2^{\pi_0})}$ (puisque $\pi_0 \geqslant \aleph_0$), on aura bien $\pi_1 \leqslant 2^{(2^{\pi_0})}$, et on est tranquille.

Commentary · Later developments [ms. 175]

The bounded closure question has a positive answer

Grothendieck introduces the second threshold π_1 because he has not yet removed retracts or shown that a bounded iteration can be replaced by one bounded presentation. The general closure result now available answers precisely this difficulty. For infinite $\pi > \pi_0$, set $\lambda = \pi_0^+$ and $\mu = \pi^+$. Then Chen (2021, Corollary 4.13) identifies the closure of representables under μ-small λ-filtered colimits with the presheaves admitting a single such presentation. In the notation of these pages this says

$$\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi} \qquad (\pi > \pi_0).$$

The endpoint follows from the idempotent calculation above. The fixed-point presentability statement is the Makkai–Paré result recovered in Chen's Corollary 4.15. This answers the question on pp. 175–176 affirmatively; it is not an open problem of the current theory.

The distinction between the two arguments remains useful. Retract closure alone repairs the inference in the proof of 4.7.1 and suffices at $\pi^{\pi_0} = \pi$. The statement about arbitrary repeated colimits also resolves $\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi}$ when no fixed-point condition holds. One cannot obtain the latter merely by writing a formal double colimit: a diagram of already presented objects has transition maps which need not have been specified at chosen presentation stages. The bounded closure theorem supplies the necessary control of these maps.

These improvements do not make every cardinal useful for $\mathcal{M}$. Equality of the two *presentation* filtrations is distinct from equality with the *intrinsic* filtration $\mathcal{M}_\pi$. The latter compares the ability to lift morphisms out of an object with the size of a presentation inside the chosen Ind completion. The arithmetic condition in the small-subcategory construction still has work to do there.

Définition 4.7.8. Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie. On appelle $\mathfrak{U}$-*filtration cardinale généralisée* (ou simplement filtration cardinale généralisée, quand $\mathfrak{U}$ est sous-entendu) la donnée d'un cardinal infini π_0, et d'une filtration $(\mathcal{F}^\pi)_{\pi \geqslant \pi_0}$ de $\mathcal{M}$ par des sous-catégories strictement pleines $\mathcal{F}^\pi$, indexées par les $\mathfrak{U}$-cardinaux $\geqslant \pi_0$, et satisfaisant les conditions a) b) c) énoncées dans 4.7.7, où dans c) on pose

$$\pi_1 \;=\; \sup(\pi_0, \operatorname{card} \text{réd}\, \mathcal{F}^{\pi_0})\,. \tag{4.7.8.1}$$

(Donc si $\operatorname{card} \text{réd}\, \mathcal{F}^{\pi_0} \leqslant 2^{(2^{\pi_0})}$, il n'y a pas à tenir compte de la condition $\pi'^{\pi} \geqslant \pi_1$, cf. 4.7.7.1.) On dit que la filtration cardinale généralisée (fcg) est *accessible*, si elle satisfait de plus la condition d). On dit qu'elle est *canonique* si on a même

$$\mathcal{F}^\pi \;=\; \mathcal{M}_\pi \tag{4.7.8.2}$$

pour tout $\pi \geqslant \pi_0$.

Proposition 4.7.9. *Soient $(\mathcal{F}^\pi)_{\pi \geqslant \pi_0}$ une fcg sur la $\mathfrak{U}$-catégorie $\mathcal{M}$, et $\pi > \pi_0$ un cardinal tel que $\pi^{\pi_0} = \pi$. Alors on a* p. 60 [page 177]

$$\begin{cases} \mathcal{M}_\pi \subset \mathcal{F}^\pi \quad \textit{(si } \pi = \pi^{\pi_0}\textit{)}, \\ \textit{donc } \mathcal{M}_\pi = \mathcal{F}^\pi \textit{ si de plus la fcg est } \text{accessible}. \end{cases} \tag{4.7.9.1}$$

Commentary · Exposition [ms. 176]

Reading the generalized filtration axioms

Grothendieck's purpose is to retain only the properties of the preceding constructions that are needed to recover the category from its small objects. Two subscripts in the printed statement of 4.7.7 obscure that purpose. In (b), the closure bound is printed π_0 followed by the editors' question mark; the proof on p. 174 uses π. In (c), the presenting poset is printed large relative to π_0; Lemma 4.7.3 gives a poset large relative to π, and the factorization of a π-accessible object's identity on p. 177 requires this stronger assertion. We retain the edition's text, but the following discussion of its consequences uses those two repaired conditions explicitly.

In modern notation an accessible generalized filtration thus consists of increasing, essentially small, replete full subcategories $\mathcal{F}^\pi$, each consisting of π^+-presentable objects, with the specified bounded closure and generation properties. At a fixed point $\pi^{\pi_0} = \pi$, every object has a π^+-filtered presentation by objects of $\mathcal{F}^\pi$. An identity map out of a π^+-presentable object then factors through one stage, producing a retract. Closure under the permitted colimits supplies its splitting, and hence

$$\mathcal{F}^\pi = \mathcal{M}_\pi.$$

This is the content of 4.7.9. The hypothesis on the presenting index cannot be weakened to mere π_0^+-filteredness in this step: presentability at the larger cardinal says nothing about preservation of an arbitrary less-filtered colimit.

The cardinal estimates in 4.7.7.1 express essential smallness quantitatively. Write $\rho \geq \pi_1$. A presheaf with at most ρ elements at each object of a skeleton C_0 can be described by sets chosen inside ρ and functions assigned to the arrows of C_0. There are at most 2^ρ choices: $|\operatorname{Fl} C_0| \leq \rho$, there are at most 2^ρ functions between subsets of ρ, and $(2^\rho)^\rho = 2^\rho$. Between two such presheaves there are at most

$$(\rho^\rho)^{|\operatorname{Ob} C_0|} \leq 2^\rho$$

natural transformations. Consequently a skeleton of this full category has at most 2^ρ arrows. Taking $\rho = \pi$ gives the first estimate for Filt^π; taking $\rho = \pi^{\pi_0}$ gives the estimate for $\mathcal{M}_\pi$ after the inclusion of 4.7.6. This proves the details omitted on p. 173.

Finally, for any $\pi \geq \pi_0$, let $\rho = \pi^{\pi_0}$. Since $\rho^{\pi_0} = \rho$, monotonicity of presentability yields

$$\mathcal{F}^\pi \subseteq \mathcal{M}_\pi \subseteq \mathcal{M}_\rho = \mathcal{F}^\rho.$$

The equality $\mathcal{M}_\pi = \mathcal{M}_\rho$ printed in the proof of 4.7.10 must be an inclusion. Even in sets, increasing the cardinal bound admits genuinely larger objects. The conclusion (4.7.10.1) has the correct direction.

Démonstration ([75]). Appliquant la condition c) des fcg à un $F \in \operatorname{Ob}\mathcal{M}_\pi$, on trouve que F est facteur direct d'un $x_i \in \operatorname{Ob}\mathcal{F}^\pi$. Comme $\mathcal{F}^\pi$ est stable par $\varinjlim$ grandes devant π_0 et de cardinal $\leqslant \pi$, et que *de plus* $\pi > \pi_0$, on conclut que :

Lemme 4.7.9.1 `[deux fois 4.7.9.1]`. *Si $\pi > \pi_0$ ([76]), $\mathcal{F}^\pi$ est stable par facteurs directs.*

D'où résulte que $F \in \mathcal{F}^\pi$, d'où (4.7.9.1).

Corollaire 4.7.10. *Supposons la fcg accessible. Pour tout cardinal $\pi \geqslant \pi_0$, on a*

$$\mathcal{F}^\pi \ \subset\ \mathcal{M}_\pi \ \subset\ \mathcal{F}^{\pi^{\pi_0}}\ (=\ \mathcal{M}_{\pi^{\pi_0}})\,. \tag{4.7.10.1}$$

En effet, $\mathcal{F}^\pi \subset \mathcal{M}_\pi$ par définition de fcg accessibles, et $\mathcal{M}_\pi = \mathcal{M}_{\pi^{\pi_0}}$ (puisque $\pi \leqslant \pi^{\pi_0}$), or $\mathcal{M}_{\pi^{\pi_0}} = \mathcal{F}^{(\pi^{\pi_0})}$ en vertu de (4.7.9.1) appliqué à $\pi' = \pi^{\pi_0}$ au lieu de π (vu que $\pi'^{\pi_0} = \pi'$), d'où (4.7.10.1). On notera que cette inclusion généralise (4.7.9.1), elle donne $\mathcal{F}^\pi = \mathcal{M}_\pi$ si $\pi = \pi^{\pi_0}$.

4.7.11. Il me semble que les seules filtrations cardinales généralisées utiles sont celles qui sont *accessibles.* Par la suite, je vais me borner à ce cas. Comme alors $\mathcal{F}^\pi \subset \mathcal{M}_\pi$, on en conclut que le foncteur canonique [page 178]

75. `[Dans ce qui suit, $\mathcal{F}^\pi$ a parfois été écrit $\mathcal{F}_\pi$, souvent corrigé en $\mathcal{F}^\pi$ par l'auteur. Nous écrivons $\mathcal{F}^\pi$ partout.]`

76. Mais $\mathcal{F}^{\pi_0}$ n'est pas forcément stable par facteurs directs, comme on voit en prenant $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, C une petite catégorie non karoubienne, et la filtration cardinale pour les Filt'^π – on note que $\mathrm{Filt}'^{\pi_0} = \mathrm{Filt}^{\pi_0} \simeq e$, vu qu'une catégorie grande devant π_0 et de cardinal $\leqslant \pi_0$ est cofinale à e.

(4.7.11.1) $$\mathrm{Ind}_\pi(\mathcal{F}^\pi) \hookrightarrow \mathcal{M}$$

prolongeant l'inclusion $\mathcal{F}^\pi \subset \mathcal{M}$, de façon à commuter aux $\varinjlim$ grandes devant π, est *pleinement fidèle.* De plus, lorsque $\pi^{\pi_0} = \pi$, la condition c) de fcg nous dit entre autres que ce foncteur est essentiellement surjectif, donc que c'est une *équivalence de catégories* :

(4.7.11.2) $$\mathrm{Ind}_\pi(\mathcal{F}^\pi) \xrightarrow{\sim} \mathcal{M} \qquad (\text{si } \pi = \pi^{\pi_0}).$$

Proposition 4.7.12. *Soit $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$ une fcg accessible (fcga) sur la $\mathfrak{U}$-catégorie $\mathcal{M}$, et soit π un cardinal $\geqslant \pi_0$. Les conditions suivantes sont équivalentes.*

a) Le foncteur pleinement fidèle (4.7.11.1) est essentiellement surjectif, i.e. est une équivalence de catégories.

b) Pour tout $F \in \mathrm{Ob}\,\mathcal{M}$, $\mathcal{F}^\pi/F$ est une catégorie filtrante et grande devant π.

Quand ces conditions sont satisfaites, et si $\pi > \pi_0$, on a p. 61

(4.7.12.1) $$\mathcal{F}^\pi = \mathcal{M}_\pi .$$

Démonstration. L'équivalence de a) et b) résulte aussitôt de 4.4.2 et de la définition des catégories grandes devant π (p. 81). Que ces conditions impliquent $\mathcal{F}^\pi = \mathcal{M}_\pi$, i.e. $\mathcal{M}_\pi \subset \mathcal{F}^\pi$ [page 179] dans le cas où $\pi > \pi_0$, se voit comme (4.7.9.1) en utilisant le lemme 4.7.9.1.

Définition 4.7.13.

1°) Un cardinal $\pi \geqslant \pi_0$ est dit *cardinal utile* pour la fcga $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$, s'il satisfait les conditions équivalentes a) et b) de 4.7.12, et si de plus dans le cas où $\pi = \pi_0$, on a $\mathcal{F}^\pi = \mathcal{M}_\pi$. (Donc si π est un cardinal utile, on a toujours $\mathcal{F}^\pi = \mathcal{M}_\pi$.) Dans le cas contraire, on dit que π est un *cardinal superflu* (pour le fcga donnée).

2°) Un cardinal π est dit *régulier* pour la fcg donnée, s'il est utile, et si de plus la majoration de cardinaux énoncée dans (4.7.7.1) est valable.

4.7.14 Ainsi, tous les cardinaux $\pi \geqslant \pi_0$ tels que $\pi^{\pi_0} = \pi$ sont utiles, et même réguliers pour la filtration cardinale [générale**isée**] donnée. Notons que si tous les cardinaux $\pi \geqslant \pi_0$ sont utiles, alors on a $\mathcal{F}^\pi = \mathcal{M}_\pi$ pour tout $\pi \geqslant \pi_0$, donc la fcga est canonique. Notons aussi que dans tous les cas, si $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$ est une fcga, alors $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$ en est une aussi.
En effet, la condition a) des fcg (cf. 4.7.7) résulte de la même condition pour $(\mathcal{F}^\pi)$, et de [page 180] (4.7.10.1), $\mathcal{M}_\pi \subset \mathcal{F}^{\pi^{\pi_0}}$. La condition b) est une propriété générale de la π-accessibilité, compte tenu que $(\mathcal{F}^{\pi_0})_0$ est une petite sous-catégorie génératrice formée d'objets π_0-accessibles de $\mathcal{M}$. La condition d) est satisfaite tautologiquement, et il reste à vérifier la condition c). Mais on a vu que pour $\pi = \pi^{\pi_0}$, on a $\mathcal{M}_\pi = \mathcal{F}^\pi$ (4.7.9.1), donc la condition c) pour $(\mathcal{M}_\pi)$ résulte de la même condition pour $(\mathcal{F}^\pi)$. Ainsi, on trouve :

Commentary · Exposition [ms. 179]

Recovering the category at a useful cardinal

Grothendieck's useful cardinal is a cardinal at which the chosen level of the filtration actually presents the whole category. This is stronger than the assertion that its objects are small. The distinction explains both the comparison functor in 4.7.11 and the later preference for the canonical filtration.

Let $A \subseteq \mathcal{M}_\pi$ be an essentially small full subcategory, and assume $\mathcal{M}$ admits π^+-filtered colimits. Its extension

$$T : \mathrm{Ind}_\pi A \longrightarrow \mathcal{M}$$

is fully faithful. Indeed, for presentations $X = \varinjlim_i a_i$ and $Y = \varinjlim_j b_j$ over π^+-filtered categories,

$$\mathrm{Hom}_{\mathcal{M}}(TX, TY) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_A(a_i, b_j),$$

using the universal property in the first variable and presentability in the second. This is also the Hom formula in the Ind category. The extension has all of $\mathcal{M}$ as its essential image precisely when every object has such a presentation.

The equivalent condition in 4.7.12 uses the canonical presentation. If A/F is π^+-filtered and the restricted Yoneda functor determined by A is fully faithful, then the canonical map

$$\varinjlim_{(a\to F)\in A/F} a \longrightarrow F$$

is an isomorphism. To see this, apply $\mathrm{Hom}(a_0, -)$ for $a_0 \in A$: presentability moves it inside the colimit, and the category-of-elements calculation identifies the resulting set with $\mathrm{Hom}(a_0, F)$. Full faithfulness of restricted Yoneda reflects the isomorphism. If T is already an equivalence, the same category-of-elements description proves the converse immediately. However, density of a fixed-point level does not by itself imply density of every smaller level. The stated generalized-filtration axioms do not supply that missing hypothesis. The next block exhibits a counterexample to (b)⇒(a) in the proposition as printed; the repair is to require the level A to be dense, or to require its canonical cocones to be colimit cocones.

At a useful level every π^+-presentable object is a retract of a member of A. Taking $A = \mathcal{F}^\pi$ and using its retract closure gives $\mathcal{F}^\pi = \mathcal{M}_\pi$. This explains the equality in the endpoint clause of Definition 4.7.13, with usefulness understood through the realization equivalence rather than the faulty bare slice criterion; it is not an additional form of cardinal arithmetic. Nor is "regular for the filtration" the usual set-theoretic regularity of the cardinal π. It means useful together with a specified quantitative presentation bound. The ordinary regular filtering cardinal in all these statements is π^+, regardless of the cofinality of π itself.

Replacing $\mathcal{F}^\pi$ by $\mathcal{M}_\pi$ preserves the generalized filtration. Essential smallness follows from $\mathcal{M}_\pi \subseteq \mathcal{F}^{\pi^{\pi_0}}$. Bounded closure is the general fact that π^+-presentables are closed under colimits of diagrams having at most π arrows whenever those colimits exist. For the quantitative part, if $F \in \mathcal{M}_{\pi'}$, first use

$$F \in \mathcal{F}^{(\pi')^{\pi_0}}.$$

Then apply the old generation bound at the fixed-point level π. The new index has cardinal at most

$$\big((\pi')^{\pi_0}\big)^{\pi} = (\pi')^{\pi}.$$

This supplies the small substitution omitted from the proof of 4.7.15: equality of the two filtrations at the generating level π alone does not identify their arbitrary target levels π'.

Commentary · Development (commentators') [ms. 179]

A filtered slice need not be a presentation

Grothendieck wants to recognize a useful level by its slices, so that the Ind realization can be checked without choosing presentations. This recognition needs a density hypothesis. The ordinary category-of-elements theorem applies to a presheaf on the chosen level; identifying its realization with the original object is a separate assertion. The following example shows that the axioms stated in 4.7.7–4.7.8 do not imply it. We checked the general recognition criteria for Ind completions cited above; none dispenses with the generation requirement. The example is supplied as a local audit result, without a priority claim.

Counterexample to the slice criterion in 4.7.12 [ours]. For every infinite π_0, there is an accessible generalized filtration satisfying even the repaired strong forms of (b) and (c), such that $\mathcal{F}^{\pi_0}/X$ is π_0^+-filtered for every X, but $\mathrm{Ind}_{\pi_0}(\mathcal{F}^{\pi_0}) \to \mathcal{M}$ is not essentially surjective.

Proof. Let $\mathcal{M} = \mathrm{Ens}$, let $\theta = 2^{\pi_0}$, and define

$$\mathcal{F}^\pi = \begin{cases} \{\varnothing\}, & \pi_0 \le \pi < \theta, \\ \mathrm{Ens}_{\le\pi}, & \pi \ge \theta. \end{cases}$$

Take replete full subcategories; the first is equivalent to the one-object category. This is an increasing exhaustive filtration. Each level is essentially small. It consists of π^+-presentable sets. Sets have all the required filtered colimits. A filtered colimit of empty sets is empty, while a colimit indexed by a

category with at most π arrows of sets of size at most π again has size at most π. Thus condition (b), even with the intended bound π, holds.

If $\pi^{\pi_0} = \pi$, then necessarily $\pi \geq 2^{\pi_0} = \theta$. Every set X is the union of its subsets of cardinality at most π, indexed by the π^+-directed poset $[X]^{\leq\pi}$. If $X \in \mathcal{F}^{\pi'}$ with $\pi' > \pi$, the indexing poset has cardinal at most $(\pi')^\pi$. The threshold determined by $\mathcal{F}^{\pi_0}$ is $\pi_1 = \pi_0$, so its proviso is automatic. This verifies condition (c), with the stronger, intended π^+-filtered index. All accessible generalized-filtration axioms are satisfied.

For every set X there is exactly one map $\varnothing \to X$. Thus $\mathcal{F}^{\pi_0}/X$ is a one-object one-arrow category and is π_0^+-filtered. But a filtered colimit of copies of $\varnothing$ remains $\varnothing$. The realization $\mathrm{Ind}_{\pi_0}(\mathcal{F}^{\pi_0}) \to \mathrm{Ens}$ has only the empty set in its essential image, and is not an equivalence. □

The correction is substantive. A slice may be filtered because the chosen level sees almost nothing. Requiring the level to be dense restores the proof in the preceding block. Alternatively replace condition (b) by the conjunction that the slice is π^+-filtered and that its canonical diagram has colimit F. The later conclusions about useful cardinals should use the equivalence in condition (a); the counterexample does not affect the fixed-point levels, where the generation axiom supplies actual presentations.

Proposition 4.7.15. *Si $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$ est une fcga sur la $\mathfrak{U}$-catégorie $\mathcal{M}$, il en est de même de $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$. De plus, tout cardinal utile (resp. régulier) pour la première filtration, l'est pour la seconde (appelée la fcg* canonique associée *à la filtration cardinale* `[généralisée]` *donnée).*

La dernière assertion résulte du fait que si π est utile pour $(\mathcal{F}^\pi)$, alors on a $\mathcal{F}^\pi = \mathcal{M}_\pi$.

4.7.16. Il me semble qu'on peut dire que la fcg canonique associée à une fcga donnée est [page 181] toujours "meilleure" que celle-ci (si celle-ci n'est canonique, i.e. si les deux ne sont égales), à cause des deux faits suivants :

a) justement parce qu'elle est *canonique*, i.e. ne dépend d'aucun autre choix que celui du seul cardinal π_0, et

b) parce qu'il y a a priori plus des cardinaux utiles et de cardinaux réguliers pour la fcg canonique.

Donc au total, il me semble que les filtrations cardinales `[généralisées]` les seules utiles sont celles qui sont canoniques.

Notons qu'une filtration cardinale (au sens de SGA 4 I 9, que je conserve ici) qui est de plus p. 62 *accessible* (i.e. $\mathcal{F}^\pi \subset \mathcal{M}_\pi$ pour tout $\pi \geqslant \pi_0$), n'est autre qu'une fcga canonique, telle que *tous* les cardinaux sont réguliers (a fortiori utiles) pour ladite filtration. Je présume (sans avoir cependant construit d'exemple) que les filtrations cardinales `[généralisées]` canoniques des catégories de la forme $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, C une petite catégorie *karoubienne* (pour que π_0 soit un cardinal utile) ne sont pas forcément des fc (non généralisées, i.e. au sens SGA), i.e. je présume qu'il arrive qu'il y ait des cardinaux $\pi \geqslant \pi_0$ non réguliers, et même superflus.

Il me faut enfin étudier ce qui se passe quand on change le cardinal initial π_0 d'une fcga. [page 182]

Commentary · Present state [ms. 181]

Useful cardinals and the accessibility spectrum

Grothendieck suspects that an accessible category may have larger nonuseful cardinals. The relevant modern invariant is its *accessibility spectrum*. The sharp relation $\kappa \lhd \lambda$ between regular cardinals requires every set of size below λ to have a cofinal family of its subsets of size below κ, of cardinality below λ. It controls the general raising theorem. The equation $\pi^{\pi_0} = \pi$ implies $\pi_0^+ \lhd \pi^+$, since a set of size at most π has at most π subsets of size at most π_0. A mere inequality between cardinals does not give this control.

The example in Vasey (2020, Remark 3.7 and Example 3.8), attributed there to Adámek–Rosický, settles the existence question in 4.7.16. We give the details, including the explicit successor ranks and the unbounded choice needed for 4.8.6(4). The latter is an elementary verification of the cited criterion, not a priority claim.

Fix $\kappa = \aleph_1$, and let $\mathcal{D}$ be the category of nonempty κ-directed posets and monotone maps. For every regular $\lambda \geq \kappa$, its λ-filtered colimits are computed as filtered colimits of posets: representatives

and order relations can be compared at a common stage, and fewer than κ elements acquire an upper bound at one stage. We first verify that

$$P \text{ is } \lambda\text{-presentable in } \mathcal{D} \quad \Longleftrightarrow \quad |P| < \lambda.$$

For the forward implication, adjoin a new greatest element $\top$ to P. The resulting poset $P^{\top}$ is the λ-directed union of the subposets $S \cup \{\top\}$ for $S \subseteq P$, $|S| < \lambda$. Every stage belongs to $\mathcal{D}$, because it has a greatest element. If P is presentable, the inclusion $P \hookrightarrow P^{\top}$ factors through one stage. Its image then forces $P \subseteq S$, proving $|P| < \lambda$. Conversely, when $|P| < \lambda$, lift a map from P into a λ-filtered colimit by choosing representatives for its elements and witnesses for all its order relations. There are fewer than λ choices. A common stage makes them into a monotone map. Fewer than λ equality witnesses likewise prove essential uniqueness. This is exactly preservation of the colimit by $\mathrm{Hom}(P, -)$.

Every object P of $\mathcal{D}$ is a κ-directed union of its subposets having fewer than κ elements and a greatest element. Indeed, the union of fewer than κ such subposets has fewer than κ elements; adjoin an upper bound in P to obtain a larger subposet of the same kind. These subposets cover P. Their isomorphism classes form a set and they are κ-presentable by the criterion just proved. Thus $\mathcal{D}$ is $\aleph_1$-accessible. An idempotent on one of these presentables splits through its image, which is again κ-directed and has fewer than κ elements. Taking a skeleton C of the presentables therefore gives a small idempotent-complete category with

$$\mathcal{D} \simeq \mathrm{Ind}_{\aleph_0} C.$$

This meets the Karoubian-base requirement in 4.7.16.

Now let $\theta > \kappa$ be any singular cardinal of countable cofinality, and put $\lambda = \theta^{+}$. We claim that the inclusion poset $[\theta]^{<\kappa}$ has no cofinal subset of cardinality at most θ. Suppose a candidate family is $\mathscr{A} = \bigcup_{n<\omega} \mathscr{A}_n$, where $|\mathscr{A}_n| \le \theta_n < \theta$, $\theta_n \ge \kappa$, and $\sup_n \theta_n = \theta$. Since every member has cardinality below κ, the union of $\mathscr{A}_n$ has cardinality at most θ_n. Choose $a_n \in \theta$ outside that union. The countable set $\{a_n : n < \omega\}$ lies in $[\theta]^{<\kappa}$ but is contained in no member of $\mathscr{A}$, a contradiction.

The poset $Q = [\theta]^{<\kappa}$ itself belongs to $\mathcal{D}$. If it were a λ-filtered colimit of λ-presentables, each stage would have at most θ elements. For each $a < \theta$, choose a stage representing $\{a\} \in Q$. There are fewer than λ such choices, so one later stage has image $Y \subseteq Q$ containing all these singletons. A monotone image of a κ-directed poset is κ-directed. Therefore Y is cofinal in Q: any subset of θ of size below κ is contained in an upper bound of its singletons in Y. But $|Y| \le \theta$, contradicting the preceding paragraph. Thus $\mathcal{D}$ is not θ^{+}-accessible.

For a first explicit pair one may take $\theta = \aleph_\omega$. Singular cardinals of countable cofinality are unbounded, so this *one* category has arbitrarily large nonuseful manuscript cardinals $\pi = \theta$, although its initial cardinal $\pi_0 = \aleph_0$ is useful. Moreover its small dense subcategory C remains formed of θ^{+}-presentables and it has θ^{+}-filtered colimits. Hence it also gives the dense-presentable-generator separation requested in 4.8.6(1). Questions about the presentability ranks or internal sizes of individual objects are different; compare Lieberman et al. (2019, Definitions 2.5 and 2.10).

Commentary · Exposition [ms. 181]

Usefulness already gives the quantitative bound

Grothendieck's distinction between useful and regular cardinals reflects the unresolved presentation problem at this point of the manuscript. Once the retract-free presentation theorem is available, a useful cardinal is regular, and the proviso involving π_1 can be omitted. We include the argument because the requirement that the index be ordered deserves its own cardinal check.

Suppose $\mathrm{Ind}_{\pi}(A) \simeq \mathcal{M}$, where $A = \mathcal{F}^{\pi} = \mathcal{M}_{\pi}$, and let $F \in \mathcal{F}^{\pi'}$ with $\pi' > \pi$. Set $\mu = (\pi')^{\pi}$. Then $\mu^{\pi} = \mu$, and F is μ^{+}-presentable. Apply the fixed-point presentation theorem, now with initial cardinal π and generating category A: it gives a π^{+}-filtered category J of at most μ arrows and a presentation of F by objects of A. This estimate is independent of $|\operatorname{Fl} A|$.

To make the index ordered, put $\kappa = \pi^{+}$. We show that a κ-filtered category J with $|\operatorname{Fl} J| \le \mu = \mu^{\pi}$ has a cofinal functor from a κ-directed poset P with $|P| \le \mu$. Begin with a discrete poset labelled by the objects of J and its evident functor u to J. Build larger posets through κ stages. Every existing element will have fewer than κ predecessors. At a successor stage, for each subset S of the current poset of size below κ, take its downward closure and adjoin a new upper element.

This downward closure has size below κ; choose a cocone on its image in J to define the value of the new element and all new transition maps.

We also impose cofinality. For each pair p, q of existing elements and each fork $j \to u(p)$, $j \to u(q)$ in J, adjoin another upper element over the downward closure of $\{p, q\}$, choosing its image cocone to include this fork. The category of this augmented diagram still has fewer than κ arrows, so such a cone exists. In both constructions the predecessors of a new element are exactly the selected downward-closed set. Distinct new elements are initially incomparable; composition is defined by the chosen cocone, so the functor remains well defined. At limit stages take unions.

There are at most $\mu^{<\kappa} = \mu$ choices at each stage and $\kappa \leq \mu$ stages. Thus $|P| \leq \mu$. Regularity of κ ensures that any subset of P of size below κ is present at one earlier stage, and then receives an upper bound, so P is κ-directed. For each $j \in J$, the comma category $j\backslash P$ is nonempty because the initial stage includes j. Any two of its objects are joined to a common object by one of the added fork cones. The functor $P \to J$ is therefore cofinal. Reindexing the presentation of F along it retains the bound

$$|P| \leq \mu = (\pi')^{\pi}.$$

This is the full ordered-index bound in Definition 4.7.13, with no threshold proviso. Thus useful-but-nonregular cardinals do not occur under the repaired realization definition. Superfluous cardinals still occur, as the accessibility-spectrum discussion explains. The source's distinction is a faithful record of the stage reached in its argument, but should not be carried into a statement of the current theory.

Proposition 4.7.17. *Soit $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$ une fcga sur la $\mathfrak{U}$-catégorie $\mathcal{M}$, et soit π_0' un cardinal $\geqslant \pi_0$. Alors $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0'}$ est une fcga sur $\mathcal{M}$, qui est canonique si celle de départ est canonique. Pour un cardinal $\pi \geqslant \pi_0'$, celui-ci est utile pour $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0'}$ si et seulement s'il l'est pour $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$. Pour qu'il soit régulier pour $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0'}$, il* suffit *qu'il le soit pour $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$, et c'est aussi nécessaire quand on suppose de plus que $2^{\pi} \geqslant \pi_1' \overset{déf}{=} \sup(\pi_0', \operatorname{card} \operatorname{réd} \mathcal{F}^{\pi_0'})$, a fortiori quand $\pi \geqslant \pi_1'$.*

Démonstration. Les conditions a) b) sont trivialement satisfaites par la deuxième filtration, étant satisfaites par la première, ainsi que d) si elle l'est par la première. Il en est de même de la condition c) si $\pi = \pi^{\pi_0'}$, qui implique en effet $\pi^{\pi_0} = \pi$, puisqu'en général

$$\pi \leqslant \pi^{\pi_0} \leqslant \pi^{\pi_0'}.$$

Reste à examiner la majoration de cardinaux

$$\operatorname{card} I \longrightarrow \pi'^{\pi} \qquad \text{si } F \in \operatorname{Ob} \mathcal{F}^{\pi'} \text{ et } \pi'^{\pi} \geqslant \pi_1',$$

où cette fois $\pi_1' = \sup(\pi_0', \operatorname{card} \operatorname{réd} \mathcal{F}^{\pi_0'})$. En tout état de cause, on a $\pi_1' \geqslant \pi_1$, donc $\pi'^{\pi} \geqslant \pi_1'$ [page 183] implique $\pi'^{\pi} \geqslant \pi_1$, donc on peut appliquer la condition c) des fcg pour $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$, pour conclure qu'elle reste encore vérifiée. Donc en somme, *toutes* les conditions des fcg (resp. fcga) sont trivialement vérifiées pour la fcg *restreinte* de la première. De même, le fait qu'un cardinal $\pi' \geqslant \pi_0'$ est utile est trivialement indépendant de quelle des deux filtrations on considère, elle ne dépend que de la donnée du cardinal π' et de la sous-catégorie $\mathcal{F}^{\pi'}$ de $\mathcal{M}$. Reste la majoration de cardinaux, dans le cas où on s'intéresse à la *régularité* de π' pour les fcga envisagées. La majoration exigée est la même, mais la condition préalable de validité de cette majoration, savoir

$$\pi'^{\pi} \geqslant \pi_1' \qquad \text{resp.} \qquad \pi'^{\pi} \geqslant \pi_1,$$

n'est pas la même. Comme $\pi_1' \geqslant \pi_1$, on voit que la condition préalable est moins restrictive dans le cas de la première filtration, que de la seconde, d'où le "il suffit" dans la dernière assertion. Mais lorsque $2^{\pi} \geqslant \pi_1'$, même la plus restrictive est automatiquement vérifiée, puisque [page 184] $\pi'^{\pi} \geqslant 2^{\pi} \geqslant \pi_1'$, et alors la régularité de π pour la deuxième filtration implique la régularité pour le première, q.e.d.

Corollaire 4.7.18. *Supposons π_0 régulier pour la fcga ([77]). Si on restreint la fcga $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$* p. 63

aux cardinaux $\pi \geqslant \pi_0'$, où $\pi_0' \geqslant \pi_1$, alors on trouve une fcga où la majoration de cardinaux (pour $\pi = \pi^{\pi_0}$) est valable sans condition préalable $\pi'^{\pi} \geqslant \pi_1'$; plus précisément, on a alors

$$\pi_1' \leqslant 2^{(\pi_0'^{\pi_0})} \qquad \textit{(a fortiori } \pi_1' \leqslant 2^{(2^{\pi_0'})}\textit{)}, \tag{4.7.18.1}$$

donc la condition préalable $\pi'^{\pi} \geqslant \pi_1'$ est automatiquement vérifiée si $\pi \geqslant \pi_0$, $\pi = \pi^{\pi_0'}$, $\pi' > \pi$ (cf. remarque 4.7.7.2).

Ceci équivaut donc à la majoration de cardinaux suivante (appliquée au cas de $\pi = \pi_0'$).

Lemme 4.7.19. *Supposons π_0 régulier pour la fcga $(\mathcal{F}^{\pi})_{\pi \geqslant \pi_0}$. Alors pour tout cardinal $\pi \geqslant \pi_1$ ($\overset{déf}{=} \sup(\pi_0, \operatorname{card} \operatorname{réd} \mathcal{F}^{\pi_0})$), on a*

$$\operatorname{card} \operatorname{réd} \mathcal{F}^{\pi} \leqslant 2^{(\pi^{\pi_0})}, \tag{4.7.19.1}$$

et si $\pi' \geqslant \pi_1$ et $F \in \operatorname{Ob} \mathcal{F}^{\pi'}$, on a

$$\operatorname{card} \operatorname{réd} \mathcal{F}^{\pi}/F \leqslant \pi'^{(\pi^{\pi_0})} \qquad ([78]). \tag{4.7.19.2}$$

Démonstration du lemme. Comme π_0 est régulier, posant $C = \mathcal{F}^{\pi_0} = \mathcal{M}_{\pi_0}$, on trouve [page 185]
$\mathcal{M} \simeq \operatorname{Ind}_{\pi_0}(C)$, donc on peut supposer $\mathcal{M} = \operatorname{Ind}_{\pi_0}(C)$, et appliquer les résultats prouvés pour les catégories de cette forme. Comme $\mathcal{F}^{\pi} \subset \mathcal{M}_{\pi}$, $\mathcal{F}^{\pi'} \subset \mathcal{M}_{\pi'}$, les majorations (4.7.9.1) et (4.7.9.2) sont contenues dans

$$(*) \qquad \left\{ \begin{array}{rcll} \operatorname{card} \operatorname{réd} \mathcal{M}_{\pi} & \leqslant & 2^{(\pi^{\pi_0})} & \\ \operatorname{card} \operatorname{réd} \mathcal{M}_{\pi}/F & \leqslant & \pi'^{(\pi^{\pi_0})} & \text{si } F \in \operatorname{Ob} \mathcal{M}_{\pi'}, \text{ avec} \\ & & & \pi' \geqslant \pi \geqslant \pi_1. \end{array} \right\} \text{ pour } \pi \geqslant \pi_1 \qquad ([79]).$$

La première de ces majorations n'est autre que la deuxième majoration dans 4.7.7.1. Notons qu'on a bien

$$2^{(\pi^{\pi_0})} \leqslant 2^{(2^{\pi})},$$

ce qui résulte en effet de

$$\pi^{\pi_0} \leqslant 2^{\pi},$$

qu'on prouve en notant que $\pi < 2^{\pi}$, d'où $\pi^{\pi_0} \leqslant (2^{\pi})^{\pi_0} = 2^{\pi \cdot \pi_0} = 2^{\pi}$ puisque $\pi \geqslant \pi_0$. Il reste à établir la deuxième majoration

$$(*) \qquad \operatorname{card} \operatorname{réd} \mathcal{M}_{\pi}/F \leqslant \pi'^{(\pi^{\pi_0})} \qquad \text{si } F \in \operatorname{Ob} \mathcal{M}_{\pi'}, \text{ avec } \pi' \geqslant \pi \geqslant \pi_1.$$

(On n'en a pas besoin pour 4.7.18, mais ce sera sans doute utile par ailleurs.) On utilise $\mathcal{M}_{\pi'} \subset \operatorname{Filt}^{\pi'^{\pi_0}}$ si $\pi' \geqslant \pi_1$ (4.7.6.2), donc si $F \in \operatorname{Ob} \mathcal{M}_{\pi'}$, on a

Commentary · Exposition [ms. 184]

Changing the starting point and bounding a slice

Grothendieck wants the filtration to be intrinsic enough that deleting its first levels does not change the useful cardinals which remain. The only part of the definition that remembers the starting point quantitatively is the threshold π_1. Proposition 4.7.17 records this bookkeeping correctly once the repaired filtration axioms are used.

If $\pi_0' \geq \pi_0$ and $\pi^{\pi_0'} = \pi$, then

$$\pi \leq \pi^{\pi_0} \leq \pi^{\pi_0'} = \pi.$$

77. Il suffit pour cela que π_0 soit utile, i.e. $\mathcal{M} \xleftarrow{\sim} \operatorname{Ind}_{\pi_0}(\mathcal{F}^{\pi_0} = \mathcal{M}_{\pi_0})$, cela implique régulier, cf. 4.8.5.

78. **N.B.** J'ignore si on peut remplacer dans ces majorations π^{π_0} par π_0.

79. Comparer avec majorations $\operatorname{card} \operatorname{réd} \operatorname{Filt}^{\pi} \leqslant 2^{\pi}$, $\operatorname{card} \operatorname{réd} \operatorname{Filt}^{\pi}/F \leqslant \pi'^{\pi}$, si $F \in \operatorname{Ob} \operatorname{Filt}^{\pi'}$, avec $\pi' \geqslant \pi \geqslant \pi_1$, majorations plus agréables que celles de (4.7.19). Il faudrait quand même s'assurer sur un exemple si les majorations plus sympathiques ne seraient pas valables dans le cas général.

The old generation property therefore applies. The new threshold π_1' is at least π_1, so its size proviso is more restrictive and the old bound implies the new one. Usefulness concerns the same full subcategory $\mathcal{F}^\pi$ and the same realization functor, hence is unchanged. For regularity the converse needs the proviso to be automatic: $2^\pi \geq \pi_1'$ ensures $(\pi')^\pi \geq \pi_1'$ whenever $\pi' > \pi$. No additional category-theoretic argument is hidden here.

The slice estimate in 4.7.19 can be proved without the extra ordering $\pi' \geq \pi$ imposed midway through its proof. Assume only the stated $\pi, \pi' \geq \pi_1$ and set

$$a = \pi^{\pi_0}, \qquad b = (\pi')^{\pi_0}, \qquad \Gamma = (\pi')^a.$$

Choose a skeleton of C and of $\mathcal{M}_\pi$. Its object set and arrow set have cardinal at most 2^a. For $G \in \mathcal{M}_\pi$ and $F \in \mathcal{M}_{\pi'}$, the preceding bounds give $|G(c)| \leq a$ and $|F(c)| \leq b$. Hence

$$\begin{aligned} |\operatorname{Hom}(G,F)| &\leq \prod_{c \in \operatorname{Ob} C_0} |F(c)|^{|G(c)|} \\ &\leq (b^a)^{|\operatorname{Ob} C_0|} \leq \big((\pi')^{\pi_0}\big)^{a\pi_1} = (\pi')^a = \Gamma. \end{aligned}$$

The product, rather than a bare sum, is the direct count of all component maps of a natural transformation; naturality can only reduce the number. The infinite-cardinal identities use $\pi_0, \pi_1 \leq a$. Since $\pi' \geq 2$, also $2^a \leq \Gamma$. There are therefore at most Γ objects in the chosen representative slice, and at most

$$\Gamma^2\, 2^a = \Gamma$$

arrows. This proves $\operatorname{card} \operatorname{réd}(\mathcal{M}_\pi / F) \leq (\pi')^{\pi^{\pi_0}}$ in exactly the range of the statement. All counts are counts on skeletons; the literal set of objects of a replete full subcategory need not itself be small.

For 4.7.18 one needs only the first count, with $\pi = \pi_0'$. It gives $\pi_1' \leq 2^{(\pi_0')^{\pi_0}} \leq 2^{2^{\pi_0'}}$. If $\pi^{\pi_0'} = \pi$, then $\pi \geq 2^{\pi_0'}$, and $(\pi')^\pi \geq 2^\pi \geq \pi_1'$. This is the precise reason the threshold disappears after increasing the initial level.

p. 64

Commentary · Development (commentators') [ms. 185] [page 186]

Two different proposed improvements of the counting bounds

Grothendieck asks two different questions in the footnotes to 4.7.19. Footnote 78 literally asks whether the exponent π^{π_0} can be replaced by π_0. Footnote 79 asks about the more plausible bounds 2^π and $(\pi')^\pi$. These must be separated: the first has an elementary negative answer already for sets. We record the calculation because it prevents a small printed exponent from being mistaken for a still-open refinement. This is a local verification given here, not a claim of priority.

Failure of a uniform initial exponent [ours]. Fix an infinite π_0. In Ens, let $\mathcal{F}^\rho$ be the sets of cardinality at most ρ, and let $\pi = \pi' = 2^{\pi_0}$. Then

$$\operatorname{card} \operatorname{réd} \mathcal{F}^\pi > 2^{\pi_0}, \qquad \operatorname{card} \operatorname{réd}(\mathcal{F}^\pi / 1) > (\pi')^{\pi_0}.$$

Thus neither occurrence of π^{π_0} in 4.7.19 can uniformly be replaced by π_0.

Proof. We first verify the categorical setting. A set X with $|X| \leq \rho$ is ρ^+-presentable: a map from X to a ρ^+-filtered colimit has at most ρ elementwise witnesses, and all witnesses and equalities can be moved to a common stage. If $|X| > \rho$, its presentation as the union of subsets of cardinality at most ρ, indexed by the ρ^+-directed poset of those subsets, prevents the identity of X from factoring through a stage. Thus $\mathcal{F}^\rho = \mathcal{M}_\rho$. Every set is a filtered union of its subsets of cardinality at most π_0, so π_0 is useful for this canonical filtration. A skeleton C of $\mathcal{F}^{\pi_0}$ has at most 2^{π_0} arrows and at least that many endomorphisms of a π_0-element set; hence its reduced cardinal is exactly 2^{π_0} and $\pi_1 = \pi$.

Now choose X with $|X| = \pi$. It has exactly $\pi^\pi = 2^\pi$ endomorphisms. Any skeleton containing a representative of its isomorphism class has the same number of endomorphisms of that representative. Therefore $\operatorname{card} \operatorname{réd} \mathcal{F}^\pi \geq 2^\pi > \pi = 2^{\pi_0}$. In the slice over the singleton, the unique map $X \to 1$ has precisely the same endomorphism set. Hence

$$\operatorname{card} \operatorname{réd}(\mathcal{F}^\pi / 1) \geq 2^\pi > \pi = (2^{\pi_0})^{\pi_0} = (\pi')^{\pi_0}.$$

The strict inequalities are Cantor's theorem, and all hypotheses of 4.7.19 are met. □

In fact the elementary upper count also gives $\operatorname{card}\operatorname{réd}\mathcal{F}^\pi = 2^\pi$ in this example, so the first bound proposed in footnote 79 is sharp for sets. That observation does not prove its validity for arbitrary accessible categories. Our checks of the sources cited here, including the bounded-colimit theorem of Chen and the discussion of sizes by Lieberman–Rosický–Vasey, did not locate a proof or counterexample for the full pair of numerical inequalities in footnote 79 with these exact hypotheses. We leave that specific comparison unsettled; we do not label it a recognized open problem of the literature.

$$\operatorname{card} F(x) \longrightarrow \pi'^{\pi_0} \qquad \forall\, x \in \operatorname{Ob} C,$$

donc si $G \in \operatorname{Ob}\mathcal{M}_\pi \subset \operatorname{Ob}\operatorname{Filt}^{\pi^{\pi_0}}$, d'où

$$\operatorname{card} G(x) \leqslant \pi^{\pi_0} \qquad \forall\, x \in \operatorname{Ob} C,$$

on trouve

$$\operatorname{card}\operatorname{Hom}(G,F) \leqslant \underbrace{\underbrace{\operatorname{card}(\operatorname{Ob} C)}_{\leqslant \pi_1} \cdot (\pi'^{\pi_0})^{(\pi^{\pi_0})}}_{\pi'^{\pi_0}\cdot\pi^{\pi_0} = \pi'^{(\pi^{\pi_0})}},$$

d'où

$$\operatorname{card}\operatorname{Ob}\mathcal{M}_\pi/F \leqslant \underbrace{\underbrace{\operatorname{card}(\operatorname{Ob}\mathcal{M}_\pi)}_{\leqslant 2^{(\pi^{\pi_0})}} \cdot \pi'^{(\pi^{\pi_0})}}_{\gamma = \pi'^{\pi^{\pi_0}}},$$

et enfin

$$\operatorname{card}\operatorname{Fl}\mathcal{M}_\pi/F \leqslant \underbrace{\underbrace{\gamma\cdot\gamma}_{\gamma=\pi'^{(\pi^{\pi_0})}} \cdot \underbrace{\operatorname{card}(\operatorname{Fl}\mathcal{M}_\pi)}_{\leqslant 2^{(\pi^{\pi_0})}}}_{\pi'^{(\pi^{\pi_0})}},$$

ce qui est la majoration annoncée (la même que celle prouvée dans 4.5.12, la démonstration étant essentiellement la même).

4.7.20. Une fcg canonique est dite *maximale* (où *strictement canonique*) si elle n'est par déduite d'une fcg canonique strictement plus étendue $(\mathcal{F}^\pi)_{\pi\geqslant\pi_{00}}$ (donc avec $\pi_{00} < \pi_0$). Donc si $\mathcal{M}$ admet une fcga, alors elle admet même une fcg strictement canonique, et cette dernière est de plus unique. On va revenir là-dessus dans la section suivante. [page 187]

4.8 Catégories $\mathfrak{U}$-accessibles

Proposition 4.8.1. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie, π_0 un cardinal infini. Considérons les conditions suivantes :*

a) $\mathcal{M}$ satisfait à L_{π_0} (i.e. est stable par petites $\varinjlim$ grandes devant π_0), et contient une petite sous-catégorie pleine S strictement génératrice, *formée d'objets π_0-accessibles.*

b) Il existe une petite catégorie C, et une équivalence de catégories

$$\mathcal{M} \simeq \operatorname{Ind}_{\pi_0}(C).$$

c) $\mathcal{M}$ admet une filtration cardinale généralisée accessible $(\mathcal{F}^\pi)_{\pi\geqslant\pi_0}$, ayant π_0 comme cardinal initial, et telle que π_0 soit un cardinal utile pour ladite filtration.

c′) Comme c), avec π_0 cardinal régulier (et non seulement utile) pour la filtration cardinale généralisée accessible en question, et cette filtration étant supposée de plus canonique. p. 65 [page 188]

On a alors le diagramme d'implications

$$c' \iff c \iff b \implies a.$$

4.8.1.1. Il me faut ici préciser la terminologie, sur ce que j'entends par sous-catégorie (pleine) C *strictement génératrice* dans $\mathcal{M}$. Pour ceci, je me réfère à SGA 4 I 7.2, (i), aux conditions équivalentes b) et c) de loc. cit. :

ⓑ $\forall\, F \in \operatorname{Ob}\mathcal{M}$, la flèche canonique

$$\varinjlim_{i \in C/F} x_i \longrightarrow F$$

est un isomorphisme.

ⓒ Le foncteur canonique

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & C^\wedge \\ F & \longmapsto & (\underbrace{x}_{\in \operatorname{Ob} C} \longmapsto \underbrace{\operatorname{Hom}_{\mathcal{M}}(x,F)}_{\in \operatorname{Ob}(\mathrm{Ens})}) \end{array}$$

est pleinement fidèle.

Ces conditions impliquent :

a) C est génératrice dans $\mathcal{M}$ par épimorphismes stricts, i.e. pour tout F dans $\mathcal{M}$, la famille des flèches $x \longrightarrow F$ de source dans C, de but F, est épimorphique stricte.

Et lorsque $\mathcal{M}$ est stable par produits fibrés, cette condition a) est équivalente aux précédentes. Dans loc. cit. j'affirme

Commentary · Exposition [ms. 188]

Density records all relations [page 189]

Grothendieck now separates the existence of sufficiently filtered colimits from the assertion that a small family actually generates every object by such colimits. The distinction is indispensable once arbitrary colimits are unavailable. We continue to write $\lambda = \pi_0^+$: his π_0-accessible objects are the objects whose covariant Hom functors preserve λ-filtered colimits. His strictly generating functors are usually called *dense* functors. Thus the principal assertion of 4.8.1 is that a category of the form $\mathrm{Ind}_{\pi_0}(C)$ has a dense small subcategory of λ-presentable objects. The converse with that very same cardinal requires a separate argument; it is not included in the proposition's displayed implication chain. For an arbitrary generalized filtration, the term useful must retain the comparison-equivalence condition of 4.7.12: filteredness of its initial overcategories alone does not imply density, as the earlier audit explains. The canonical ind-completion examples satisfy the density hypothesis.

The density criterion can be proved without assuming that the ambient category has arbitrary colimits. Let $i : C \to \mathcal{M}$, and set $N_i X(c) = \mathrm{Hom}(i(c), X)$. A natural transformation $N_i X \to N_i Y$ is exactly a compatible family of maps $i(c) \to Y$ indexed by objects $(c, p : i(c) \to X)$ of C/X. Consequently

$$\mathrm{Nat}(N_i X, N_i Y) \cong \lim_{(c,p) \in (C/X)^{\mathrm{op}}} \mathrm{Hom}(i(c), Y).$$

Full faithfulness of N_i says precisely that the evident cocone with vertex X has the colimit universal property. The assertion supplies this particular colimit; it does not presuppose all colimits in $\mathcal{M}$. This is the content of the equivalence between b) and c) in 4.8.2, and explains the usefulness of the formulation inherited from SGA 4, I, 7.2 (Artin et al., 1972).

The passage through strict epimorphic families checks a different-looking compatibility condition. A family $(p_a : X_a \to X)$ is strict epimorphic when maps $u_a : X_a \to Y$ descend uniquely whenever $u_a v = u_b w$ for every pair $v : Z \to X_a$, $w : Z \to X_b$ with $p_a v = p_b w$. Density implies this property for the family of all maps from $i(C)$: its compatibility condition includes the arrows of C/X. Conversely, when i is fully faithful, the compatibility along C/X implies the required compatibility for arbitrary Z. Indeed, apply the joint epimorphicity of the family of all $i(c) \to Z$, which follows from the strict epimorphic hypothesis at Z, and use fullness to regard their composites with v, w as arrows of C. This explains exactly where the full faithfulness hypothesis enters. It also prevents us from replacing a dense subcategory by a mere collection of generators without recording its morphisms.

cette équivalence sans hypothèse restrictive sur $\mathcal{M}$ – mais en annotation marginale j'affirme que cette assertion est *fausse*, en dehors de l'hypothèse d'existence des produits fibrés dans $\mathcal{M}$. Pourtant, je viens de vérifier la démonstration donnée dans loc. cit. de l'implication a $\Longrightarrow$ b, et elle est correcte. Mais peut-être l'annotation résulte-t-elle d'une confusion entre la condition a), et la condition suivante, plus faible a priori, et qui lui ressemble comme un frère :

a′) Il existe une famille de flèches $u_i : x_i \longrightarrow F$, de source dans C, et de but F, qui est épimorphique stricte (cf. SGA 4 I 10.3).

On a envie de dire que quand on *agrandisse* une famille de flèches épimorphique stricte, en rajoutant d'autres flèches de même but, elle doit rester épimorphique stricte ("a fortiori") – ce qui donne l'impression que a′) doit impliquer a), donc lui être équivalent. Mais si on essaye de prouver cette propriété hypothétique de stabilité pour les familles épimorphiques strictes, p. 66
on voit qu'on n'y arrive que si on suppose que la famille initiale est de plus *épimorphique* [page 190]
universelle (si les produits fibrés existent dans $\mathcal{M}$), ou (sous la réserve précédente) si pour tout $x' \longrightarrow F$, le crible de $\mathcal{M}/F'$ image inverse du crible de $\mathcal{M}/F$ engendré par les $x_i \longrightarrow F$ est un "crible épimorphique". Cela fait entrevoir en effet qu'il doit y avoir des cas où la condition a′) est satisfaite, mais non la condition a) (équivalente à b et à c).

Mais nous sommes alors confrontés à la situation très déplaisante : si C est une sous-catégorie strictement génératrice de $\mathcal{M}$, il ne s'ensuit pas pour autant qu'une sous-catégorie pleine C' de $\mathcal{M}$ telle que $C' \supset C$ soit également strictement génératrice ! Or j'ai toujours utilisé précédemment, sans y réfléchir à deux fois, qui si C est strictement génératrice, il en est de même de $C' \supset C$. Donc il faudra que je sois à présent très attentif, et que je vérifie [page 191]
si cette erreur $\pm$ tacite rend inopérantes certaines démonstrations, voire même, rend faux certains énoncés. Aussi, il me faudrait donner un exemple où C est strictement génératrice, mais $C' \supset C$ ne l'est pas.

Finalement, réflexion faite, ces ennuis craints disparaissent. Je vais énoncer un résultat en forme :

Proposition 4.8.2. *Soient C, $\mathcal{M}$ deux $\mathfrak{U}$-catégories, $i : C \longrightarrow \mathcal{M}$ un foncteur, d'où un foncteur*

$$\begin{cases} \varphi : \mathcal{M} \longrightarrow C^\wedge \\ \varphi(X) = (x \longmapsto \mathrm{Hom}_{\mathcal{M}}(i(x), X)) \end{cases} \tag{4.8.2.1}$$

(foncteur composé $\mathcal{M} \xrightarrow{can.} \mathcal{M}^\wedge \underbrace{\xrightarrow{i^*}}_{\substack{\text{i.e. composition} \\ \text{avec } i}} C^\wedge$). Pour tout X dans $\mathcal{M}$, on considère la catégorie $C/X \simeq C/\varphi(X)$, et l'injection [?] canonique (dans $\mathcal{M}^\vee$)

$$\varinjlim_{x \in C/X} i(x) \longrightarrow X. \tag{4.8.2.2}$$

1°) Considérons les conditions suivantes.

- ⓐ *Pour tout $X \in \mathrm{Ob}\,\mathcal{M}$, la famille des flèches $i(x) \longrightarrow X$, indexée par $\mathrm{Ob}\,C/X$, est épimorphique stricte.*
- ⓑ *Pour tout $X \in \mathrm{Ob}\,\mathcal{M}$, la flèche canonique 4.8.2.2. est un isomorphisme, i.e. fait de X une $\varinjlim$ des foncteurs canoniques $C/X \longrightarrow \mathcal{M}$.* [page 192]
- ⓒ *Le foncteur φ (4.8.2.1) est pleinement fidèle.*

On a alors les implications

$$\mathrm{b} \Longleftrightarrow \mathrm{c} \Longrightarrow \mathrm{a},$$

et si on suppose i pleinement fidèle, alors les trois conditions a, b, c *sont équivalentes.* (**N.B.** On dit que le foncteur $i : C \longrightarrow \mathcal{M}$ est *strictement générateur*, si les conditions équivalentes b, c sont satisfaites. Si C est une sous-catégorie de $\mathcal{M}$ (pas nécessairement p. 67

pleine), on dit que C est *strictement génératrice* si le foncteur d'inclusion est strictement générateur, i.e. satisfait aux conditions équivalentes b), c). Pour ceci, il faut donc que la condition a) soit satisfaite, i.e. que pour tout X dans $\mathcal{M}$, la famille des flèches $x \longrightarrow X$ de source dans C, de but X, soit épimorphique stricte, et cette condition est aussi suffisante quand C est sous-catégorie pleine de $\mathcal{M}$.)

2°) ([80]). *Supposons que i soit strictement générateur* ([81]). *Alors pour tout objet X de $\mathcal{M}$, la famille des flèches $i(x) \longrightarrow X$, indexée par $\mathrm{Ob}\, C/X$, est non seulement épimorphique stricte (condition a) dans 1°), mais de plus, toute famille de flèches de but X, incluant la famille précédente, est également épimorphique stricte.*

3°) Considérons un diagramme commutatif de foncteurs de $\mathfrak{U}$-catégories :

[page 193]

$$\begin{array}{ccc} C & \xrightarrow{\alpha} & C' \\ & {\scriptstyle i}\searrow \quad \swarrow{\scriptstyle i'} & \\ & E & \end{array}$$

Si le foncteur i est strictement générateur ([82]), *et si de plus i' est pleinement fidèle, alors i' est strictement générateur.*

Corollaire 4.8.3. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie, C une sous-catégorie strictement génératrice, C' une sous-catégorie* pleine *de $\mathcal{M}$ qui contient C, alors C' est strictement génératrice.*

C'est en effet un cas particulier de 4.8.2 3°.

Démonstration de 4.8.2. La démonstration de 1°) est celle de SGA 4 I 7.2 (i). (Où on suppose que C est une sous-catégorie pleine de $\mathcal{M}$, mais la démonstration donnée prouve en fait l'énoncé donné ici.) Prouvons 2°). Rajoutons donc à la famille des [page 194]

$$i(x) \longrightarrow X \qquad (x \in \mathrm{Ob}\, C/X)$$

une famille

$$X_\alpha \xrightarrow{p_\alpha} X \qquad (\alpha \in A)$$

de nouvelles flèches, et prouvons que la famille totale ainsi obtenue est encore épimorphique stricte. On doit donc prouver que si Y est un objet de $\mathcal{M}$, et si on a une famille de flèches $u_x : i(x) \longrightarrow Y$, $u_\alpha : X_\alpha \longrightarrow Y$, satisfaisant la condition de compatibilité habituelle pour des couples de flèches :

(v_x, v_y)

$$\begin{array}{ccccc} & & i(x) & & \\ & {\scriptstyle v_x}\nearrow & & \searrow{\scriptstyle p_x} & \\ Z & & & & X\ , \\ & {\scriptstyle v_y}\searrow & & \nearrow{\scriptstyle p_y} & \\ & & i(y) & & \end{array}$$

ou p. 68

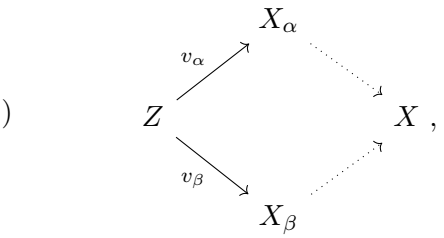
(v_x, v_α)

$$\begin{array}{ccccc} & & i(x) & & \\ & {\scriptstyle v_x}\nearrow & & \searrow{\scriptstyle p_x} & \\ Z & & & & X\ , \\ & {\scriptstyle v_\alpha}\searrow & & \nearrow{\scriptstyle p_\alpha} & \\ & & X_\alpha & & \end{array}$$

ou [page 195]

80. [2° se trouve sur la page 193.]
81. Il suffit même qu'il satisfasse la condition plus faible a) de 1°).
82. En fait, il suffit qu'il satisfasse la condition plus faible c) de 1°).

(v_α, v_β)



savoir les relations

$$(*) \qquad \begin{cases} u_x v_x = u_y v_y \\ u_x v_x = u_\alpha v_\alpha \\ u_\alpha v_\alpha = u_\beta v_\beta, \end{cases}$$

alors il existe une flèche

$$u : X \longrightarrow Y$$

(nécessairement unique, d'après la propriété c) d'épimorphicité stricte de la famille initiale indexée par C/X) telle qu'on ait

$$(*) \qquad \begin{cases} up_x = v_x & \text{pour tout } x \in \operatorname{Ob} C/X \\ up_\alpha = v_\alpha & \text{pour tout } \alpha \in A. \end{cases}$$

Or d'après la propriété a), il existe un unique $u : X \longrightarrow Y$, de telle façon que la première des relations $(*)$ soit satisfaite, pour tout $x \in \operatorname{Ob} C/X$. Donc il s'agit seulement de prouver que pour cet u, on a pour tout α la relation

[page 196]

$$up_\alpha = u_\alpha.$$

Or, en appliquant la propriété a) à X_α au lieu de X, on veut que la famille des flèches

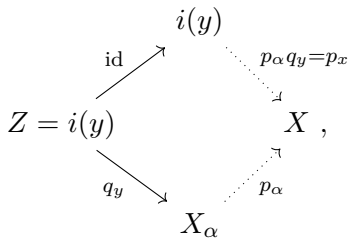
$$i(y) \xrightarrow{q_y} X_\alpha$$

(indexée par C/X_α) est épimorphique stricte, a fortiori épimorphique. Donc pour prouver $up_\alpha = u_\alpha$, il suffit de prouver qu'on a

$$(**) \qquad (up_\alpha)q_y = u_\alpha q_y$$

pour toute y dans C/X_α. Mais c'est là une conséquence de la compatibilité $(*)$ (deuxième formule), appliquée au cas suivant,

p. 69



on a donc, en posant $x = (y, p_\alpha q_y) \in \operatorname{Ob} C/X$,

$$\underbrace{u_x \circ \mathrm{id}_{i(y)}}_{u_x} = u_\alpha q_y ,$$

donc pour prouver $(**)$, il reste à vérifier

$$u_x = \underbrace{(up_\alpha)q_y}_{u(p_\alpha q_y) = up_x} ,$$

[page 197]

qui est en effet vérifiée par définition de u_x. Cela achève de prouver 2°.

Prouvons 3°. Comme i' est pleinement fidèle, en vertu de 1°, pour prouver que i' est strictement génératrice, il suffit d'établir que pour tout objet X de $\mathcal{M}$, la famille des flèches $i'(x') \longrightarrow X$, indexée par $\mathrm{Ob}\, C'/X$, est épimorphique stricte. Or cette famille contient la famille des $\underbrace{i'(\alpha(x))}_{=i(x)} \longrightarrow X$, indexée par $\mathrm{Ob}\, C/X$. Donc il résulte de l'hypothèse sur i, et de 2°, que cette famille est épimorphique stricte. On a gagné ! Ouf …

Démonstration de 4.8.1. On a

$$\mathrm{c}' \implies \mathrm{c} \implies \mathrm{b} \implies \mathrm{a}\,,$$

la première implication tautologique, la seconde résultant du fait que π_0 cardinal régulier pour la fcga envisagée, implique que

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_{\pi_0}(\mathcal{F}^{\pi_0})\ ,$$

$\mathcal{F}^{\pi_0}$ équivalente à une petite catégorie, d'où b).

Enfin b) $\implies$ a) résulte du fait que $\mathrm{Ind}_{\pi_0}(C)$ est stable par $\varinjlim$ grandes devant π_0 (4.4.15), et que C est une sous-catégorie strictement génératrice de $\overrightarrow{\mathrm{Ind}}_{\pi_0}(C)$, formée d'objets π_0-accessibles. Il nous faut expliciter le [page 198]

Commentary · Exposition [ms. 197]

The apparent difficulty with enlargement disappears

Grothendieck is checking that his earlier constructions survive enlargement of the generating category. The worry on manuscript pages 190–191 is withdrawn by Proposition 4.8.2; it is not an unresolved defect in the preceding theory. The useful fact is stronger than the permanence of density. Suppose that for every object Z, the family of all maps $i(c) \to Z$ is strict epimorphic. Adjoin arbitrary maps $q_a : Z_a \to X$ to the canonical family over X. Given compatible descent data on the enlarged family, its restriction supplies a unique $u : X \to Y$. To prove $uq_a = u_a$, precompose with every $i(c) \to Z_a$. Each resulting equality is already among the compatibility equations of the enlarged family. These maps are jointly epimorphic, so $uq_a = u_a$. Thus this particular strict epimorphic family remains strict after enlargement, with no pullback hypothesis.

There is also a short proof directly with natural transformations. If $C \subset C' \subset \mathcal{M}$, C' is full, and C is dense, a transformation $N_{C'}X \to N_{C'}Y$ restricts to a unique arrow $u : X \to Y$ on C. For $p : c' \to X$ and every $a : c \to c'$, naturality gives

$$\eta_{c'}(p)a = \eta_c(pa) = upa.$$

The maps from C to c' are jointly epimorphic, hence $\eta_{c'}(p) = up$. Faithfulness follows by restriction. Therefore C' is dense. This proof isolates the global hypothesis missing from the tempting assertion that *every* strict epimorphic family is preserved by arbitrary enlargement: the testing families are available at every newly adjoined domain.

Two local notational slips should not obscure this argument. In the display on manuscript page 195 the required equations are $up_x = u_x$ and $up_\alpha = u_\alpha$, not equations with the auxiliary comparison maps v_x, v_α. Footnote 82 should refer to condition a), as does footnote 81: condition c) is the density condition itself, so it is not a weaker assumption. Both readings are present in the edition and are retained in the transcription.

Lemme 4.8.4. *Soit C une $\mathfrak{U}$-catégorie, π_0 un cardinal infini. Alors C est strictement génératrice dans $\mathcal{M} = \mathrm{Ind}\, C$ et dans $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$.*

En effet, le foncteur canonique $\mathcal{M} \longrightarrow C^\wedge$ est pleinement fidèle, et on applique le critère c) de 4.8.2. [page 199]

Pour achever de prouver 4.8.1, il reste donc à établir l'implication

$$\mathrm{a} \implies \mathrm{c}'\,.$$

Supposons donc que $\mathcal{M}$ satisfasse à L_{π_0}, et admette une petite sous-catégorie strictement génératrice S, formée d'objets π_0-accessibles, donc un foncteur canonique

$$\mathrm{Ind}_{\pi_0} S \longrightarrow \mathcal{M} ,$$

prolongeant l'inclusion $S \hookrightarrow \mathcal{M}$, de façon à commuter aux $\varinjlim$ grandes devant π_0, et du fait que $S \subset \mathcal{M}_{\pi_0}$, ce foncteur est *pleinement fidèle.* Mais comment prouver qu'il est essentiellement surjectif ? Ce ne sera d'ailleurs pas vrai sans autre hypothèse sur S. L'idée naturelle est de saturer S pour les $\varinjlim$ de cardinal $\leqslant \pi_0$ représentables dans $\mathcal{M}$, et notamment par facteurs directs, mais même alors (et même en admettant que $\mathcal{M}_{\pi_0}$ lui-même est équivalent à une petite catégorie, et prenant $C = \mathcal{M}_{\pi_0}$), il n'est pourtant pas clair que le foncteur pleinement fidèle $\mathrm{Ind}_{\pi_0} C \longrightarrow \mathcal{M}$ soit essentiellement surjectif, ce qui équivaudrait aussi à dire que pour tout F dans $\mathcal{M}$, C/F est grande devant π_0. Il n'est en effet pas même clair que C/F soit filtrante. [page 200]

Donc finalement, je doute qu'on ait a) $\Longrightarrow$ c′) ou seulement a) $\Longrightarrow$ b), la situation n'est pas bien comprise visiblement.

Mais prouvons quand même

$$\mathrm{b} \implies \mathrm{c}' .$$

Pour ceci, on peut supposer $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$, C petite. On sait alors que la filtration de $\mathcal{M}$ par les $(\mathcal{M}_\pi)_{\pi \geqslant \pi_0}$ est une fcga (prop. 4.7.7 et définition 4.7.8), et même une fcg *canonique.* De plus, je dis que le cardinal π_0 est *utile* et même régulier pour cette filtration (ce qui achèvera de prouver c′). On a un résultat un peu plus précis :

Corollaire 4.8.5. *Soient C une catégorie équivalente à une petite catégorie, π_0 un cardinal infini, $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C) \subset C^\wedge$. Alors :* [page 201]

1°) π_0 est un cardinal utile *et même régulier pour la fcg canonique $(\mathcal{M}_\pi)_{\pi \geqslant \pi_0}$ de $\mathcal{M}$.*

2°) $\mathcal{M}_{\pi_0} = C^\natural$ (sous-catégorie pleine de $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$ formée des facteurs directs dans $\mathcal{M}$ d'objets de C), et $\mathcal{M}_{\pi_0}$ est une enveloppe de Karoubi de C.

([54], [55]).

Démonstration. p. 71

1°) On sait déjà que

$$(*) \qquad \mathrm{Ind}_{\pi_0} \mathcal{M}_{\pi_0} \longrightarrow \mathcal{M}$$

est pleinement fidèle, et pour voir que c'est essentiellement surjectif, on note que l'on a

$$\mathcal{M} \simeq \mathrm{Ind}_{\pi_0} C \hookrightarrow \mathrm{Ind}_{\pi_0} \mathcal{M}_{\pi_0} \longrightarrow \mathcal{M} ,$$

le composé étant isomorphe au foncteur identique, est essentiellement surjectif, donc aussi $(*)$. Cela prouve donc que π_0 est un cardinal utile pour la filtration envisagée, prouvons que ce cardinal est même régulier, donc que pour

$$F \in \mathrm{Ob}\, \mathcal{M}_\pi , \qquad \text{avec } \pi > \pi_0,\ \pi^{\pi_0} \geqslant \pi_1$$

54. **Corollaire 4.8.5.1.** *Si C est karoubienne, alors $\mathcal{M}_{\pi_0} \subset C^\wedge$ est la catégorie formée des foncteurs* représentables *dans $C \xrightarrow{\sim} \mathcal{M}_{\pi_0}$.*

55. **4.8.5.2.** Expliciter aussi ceci : Soit $u : C \longrightarrow C'$ un foncteur, avec C, C' équivalentes à des petites catégories. Conditions équivalentes :

a) u induit $\mathrm{Kar}\, C \xrightarrow{\sim} \mathrm{Kar}\, C'$.

a′) u pleinement fidèle, et tout objet de C' est isomorphe à un facteur direct d'un objet de l'image essentielle.

b) $\mathrm{Ind}(u) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$ est une équivalence.

c) $\mathrm{Ind}_{\pi_0}(u) : \mathrm{Ind}_{\pi_0}(C) \longrightarrow \mathrm{Ind}_{\pi_0}(C')$ est une équivalence (π_0 un cardinal infini donné).

d) $u_! : C^\wedge \longrightarrow C'^\wedge$ est une équivalence.

d′) $u_* : C'^\wedge \longrightarrow C^\wedge$ est une équivalence [plutôt $u_* : C^\wedge \longrightarrow C'^\wedge$].

d″) $u^* : C'^\wedge \longrightarrow C^\wedge$ est une équivalence.

on a

$$F \;=\; \varinjlim_{I} x_i \;, \qquad x_i \in \mathrm{Ob}\,\mathcal{M}_{\pi_0},\; I \text{ grande devant } \pi_0,\; \mathrm{card}\, I \leqslant \pi^{\pi_0}.$$

Pour ceci, on utilise (4.7.6.2), page 171, [page 202]

$$\mathcal{M}_\pi \;\subset\; \mathrm{Filt}^{\pi^{\pi_0}} \qquad (\text{si } \pi \geqslant \pi_0,\; \pi^{\pi_0} \geqslant \pi_1),$$

et la définition de $\mathrm{Filt}^{\pi'}$ (pour $\pi' \geqslant \pi_0$) nous montre que c'est là la conclusion cherchée.

Prouvons donc 2°). Comme $\mathcal{M}_{\pi_0}$ est stable par les $\varinjlim$ inductives finies qui sont représentables dans $\mathcal{M}$ (SGA 4 I 9.9), il est stable par facteurs directs, et comme il contient C, on a donc $C^\natural \subset \mathcal{M}_{\pi_0}$. L'inclusion inverse $\mathcal{M}_{\pi_0} \subset C^\natural$ se voit de la façon standard, en écrivant un objet F de $\mathcal{M}_{\pi_0}$ comme $\varinjlim_I x_i$, les x_i dans C, I grande devant π_0, alors F est facteur direct d'un des x_i.

Le fait que $C^\natural$ est une *enveloppe de Karoubi* de C dans $\mathcal{M} \subset C^\wedge$ provient du fait que $\mathcal{M}$ est stable par facteurs directs, et que C est une sous-catégorie pleine de $\mathcal{M}$, cf. SGA 4 I (? …).

4.8.6. Questions ouvertes. On a ainsi prouvé 4.8.1 revu et corrigé. Il reste la question :

1) Donner un exemple où 4.8.1 a) est vérifié, mais non b) etc. ([56]).
2) Dégager des conditions supplémentaires serviables qui, jointes à a), impliquent b), donc c, c′.

Signalons aussi pour mémoire les questions suivantes. [page 203]

3) Soit $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, et $\pi > \pi_0$. Alors Filt^π est-il stable par facteurs directs ? Cela est-il vrai du moins si $\pi = \pi^{\pi_0}$, ou ce qui revient au même (cf. 4.7.3), a-t-on dans ce cas $\mathrm{Filt}^\pi = \mathrm{Filt}'^\pi$? Si oui, il serait inutile dans la description 4.7.7 des filtrations cardinales, dans la majoration de cardinaux (4.7.7.1), $\mathrm{card}\, I \leqslant \pi'^\pi$, de mettre l'hypothèse restrictive $\pi'^\pi \geqslant \pi_1$.

Commentary · Exposition [ms. 202]

The compact part is an idempotent completion

Grothendieck identifies the part of an ind-completion that is already present at its initial cardinal. This is the first place where the completion by retracts must be treated as a structural operation. For $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, every representable is λ-presentable, and so is every retract of a representable. Conversely, write a λ-presentable F as a λ-filtered colimit $\varinjlim_i y(c_i)$. Preservation of that colimit by $\mathrm{Hom}(F, -)$ makes 1_F factor through one of its terms:

$$F \xrightarrow{s} y(c_i) \xrightarrow{r} F, \qquad rs = 1_F.$$

Hence $\mathcal{M}_{\pi_0}$ is exactly the retract closure of $y(C)$. Idempotents split in $\mathcal{M}$: a filtered presentation of the splitting is obtained by iterating the idempotent along an ordinal of cofinality greater than π_0, or by the finite idempotent indexing category used earlier in this commentary. Since idempotents split in the presheaf category, their splitting objects lie in the ind-completion. We have therefore proved

$$\mathcal{M}_{\pi_0} \simeq \mathrm{Kar}(C), \qquad \mathrm{Ind}_{\pi_0}(\mathcal{M}_{\pi_0}) \simeq \mathcal{M}.$$

The second equivalence follows either from the same presentations or by factoring the identity through the fully faithful comparison functor. It is the useful-cardinal assertion of 4.8.5; the separate quantitative notion called regularity in the manuscript additionally uses the presentation-size estimate from 4.7.6.

The list in footnote 84 is the corresponding Morita invariance statement. A fully faithful $u : C \to C'$ whose image contains every object of C' up to retract induces an equivalence on idempotent completions. Restriction then induces an equivalence of presheaf categories: any set-valued presheaf extends uniquely, up to unique isomorphism, across a split idempotent, because its value at that retract must be the image of the induced idempotent of a set. The left and right

56. Il suffit de trouver un cardinal non régulier, pour la fcg canonique d'une $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$.

Kan extensions are the inverse equivalences. They also preserve the indicated ind-subcategories. Conversely, an equivalence induced on Ind_{π_0} identifies its λ-presentable objects, and therefore identifies $\mathrm{Kar}(C)$ with $\mathrm{Kar}(C')$. The same argument at $\lambda = \aleph_0$ gives ordinary Ind. This proves the equivalences in the footnote; the variance of u_* must be $C^\wedge \to C'^\wedge$, as the editors already indicate.

The modern higher-categorical formulation identifies the compact objects of an ind-completion with the idempotent completion of its input (Lurie, 2009, Lemma 5.4.2.4). The ordinary proof above shows which part of that statement is already fully present here: the identity of a compact object factors through a single stage, and the resulting projector is the only extra object that completion has introduced at this cardinal.

Pour avoir une réponse affirmative à cette question, pour π donné, il suffirait (et probablement faudrait) d'avoir une réponse affirmative à celle-ci : Soient $\mathcal{D}$, $\mathcal{D}_0$ deux catégories grandes devant π_0, $\mathcal{D}_0$ facteur direct de $\mathcal{D}$, $\mathcal{D}$ admettant une catégorie cofinale I grande devant π_0, de cardinal $\leqslant \pi$. Alors en est-il de même de $\mathcal{D}_0$? Si p est une rétraction de $\mathcal{D}$ sur $\mathcal{D}_0$, la difficulté est que je ne sais pas si p est foncteur cofinal. En fait, je vois que ce n'est pas toujours le cas, même si $\mathcal{D}$ est finie. Prenons p.ex. pour $\mathcal{D}_0$ une catégorie ayant objet final e, et une flèche $x \longrightarrow e$ qui n'est pas un isomorphisme, et admettant une section (p.ex. $\mathcal{D}_0$ la sous-catégorie pleine de $\mathbf{\Delta}$ formée par $\mathbf{\Delta}_0$, $\mathbf{\Delta}_1$: $\mathbf{\Delta}_0 \underset{\partial_1}{\overset{\partial_0}{\rightrightarrows}} \mathbf{\Delta}_1$). Soit $\mathcal{D}$ déduit de $\mathcal{D}_0$ en ajoutant un objet final $\tilde{e}$. Alors $\mathcal{D}_0$, $\mathcal{D}$ sont filtrantes, et on obtient une rétraction de $\mathcal{D}$ sur $\mathcal{D}_0$, envoyant $\tilde{e}$ sur x. Ce foncteur n'est pas cofinal, car un foncteur cofinal doit envoyer $\tilde{e}$ en e. p. 72 [page 204]

Plus généralement, si $\mathcal{D}_0$ est une catégorie telle qu'il existe une flèche

$$(*) \qquad \alpha : e_{\mathcal{D}_0^\wedge} \longrightarrow x_0$$

dans $\mathcal{D}_0^\wedge$, avec $x_0 \in \mathrm{Ob}\,\mathcal{D}_0$, alors on voit que $\mathcal{D}_0$ est grande devant π_0 quel que soit le cardinal infini π_0, et que d'autre part, désignant encore par $\mathcal{D}$ la catégorie déduite de $\mathcal{D}_0$ en lui ajoutant un objet final, il y a une rétraction de $\mathcal{D}$ sur $\mathcal{D}_0$ qui envoie $e_{\mathcal{D}}$ dans x_0. Cette rétraction n'est cofinale que si x_0 est un objet final de $\mathcal{D}_0$, i.e. si et seulement si $(*)$ est un isomorphisme. Supposons qu'il n'en soit pas ainsi. Mais il y a un foncteur cofinal de la sous-catégorie I_0 de $\mathcal{D}_0$, telle que $\mathrm{Ob}\, I_0 = \{x_0\}$, $\mathrm{Fl}\, I_0 = \mathrm{End}_{I_0}(x_0) = \{p, \mathrm{id}_{x_0}\}$, où p est le composé $x_0 \xrightarrow{\text{can.}} e_{\mathcal{D}_0^\wedge} \xrightarrow{\alpha} x_0$. Donc pas de contre-exemple de cette façon !

4) Considérons $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, C petite, et considérons la filtration canonique $(\mathcal{M}_\pi)_{\pi \geqslant \pi_0}$. Donner des exemples où il y a des cardinaux non utiles, ou mieux, ou pour tout cardinal π, existe un cardinal $\pi' > \pi$ qui soit non utile. (Cela donnera l'exemple demandé dans 1°.) D'autre part, donner des exemples de cardinaux utiles qui ne soient réguliers – si possible, comme tantôt, de tels cardinaux aussi grands qu'on veut. [page 205]

Commentary · Examples [ms. 204]

A retraction can fail cofinality without losing a small presentation

Grothendieck's example separates two operations that are easy to confuse: retracting a category and choosing a final functor into it. Take the full subcategory $D_0 \subset \mathbf{\Delta}$ on [0] and [1]. Its terminal object is [0], while [1] → [0] has two sections. Choose one, $s : [0] \to [1]$. Adjoin a new terminal object t to D_0, obtaining D. The maps

$$a \longrightarrow [0] \xrightarrow{s} [1] \qquad (a \in D_0)$$

form a natural cocone, so they define a retraction $p : D \to D_0$ with $p(t) = [1]$ and $p|_{D_0} = 1$. This p is not final. If it were, then for every $a \in D_0$ the colimit of $\mathrm{Hom}(a, p(-))$ would be a singleton. Since D has terminal object t, that colimit is $\mathrm{Hom}(a, [1])$. Taking $a = [0]$ gives two elements.

Nevertheless a finite final indexing category remains. More generally a section $1 \to y(x)$ of the terminal presheaf map $y(x) \to 1$ determines an idempotent $e : x \to x$. For every arrow $v : a \to x$, naturality gives $ev = \alpha_a$, independent of v. Let E have one object and endomorphisms $1, e$, with

$e^2 = e$. The functor $E \to D_0$ picking x is final: in $(a \downarrow E)$ every object $v : a \to x$ maps, using e, to α_a. This comma category is nonempty and connected. The category E is itself large in front of every cardinal: the constant components e give a cocone on every diagram in E. Thus a failed retraction argument supplies no counterexample to bounded ind-presentations.

Question 4.8.6(3) is answered affirmatively by the retract-presentation argument in our commentary on §4.7. The conclusion holds for the manuscript's indexing convention, which allows non-ordered filtered categories. If one insists on a poset presentation, a finite indexing category with a nontrivial idempotent can no longer be used without changing its size; this is precisely why the indexing convention belongs in the statement. At fixed points $\pi^{\pi_0} = \pi$, the equality $\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi}$ then follows from 4.7.3. The example here explains why the proof should work on presentations themselves, rather than assuming that an arbitrary categorical retraction is final.

Définition 4.8.7. Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie, π_0 un cardinal infini. On dit que $\mathcal{M}$ est π_0-*accessible* si $\mathcal{M}$ est équivalente à une catégorie $\mathrm{Ind}_{\pi_0}(C)$, C une petite catégorie, i.e. si elle satisfait aux conditions équivalentes b, c, c′ de 4.8.1 (impliquant la condition a), sans doute strictement plus faible). On dit que $\mathcal{M}$ est accessible s'il existe un cardinal tel que $\mathcal{M}$ soit π_0-accessible (**N.B.** si π_0 est un tel cardinal, alors tout cardinal $\pi > \pi_0$ tel que $\pi^{\pi_0} = \pi$, notamment les cardinaux π'^{π_0}, avec $\pi' \geqslant 2$, ont la même propriété), ou ce qui revient au même [page 206]
d'après 4.8.1, si $\mathcal{M}$ admet une fcga, ou encore une fcg canonique.

Commentaire 4.8.8. Cette définition est en contradiction avec la terminologie introduite dans SGA 4 I 9.4 (p. 114) ; on désignera dorénavant sous les noms de "catégorie faiblement π_0-accessible", "catégorie faiblement accessible" les notions introduites dans 9.4 – i.e. les catégories admettant une sous-catégorie pleine génératrice (pas même strictement génératrice) dont les objets sont π_0-accessibles (resp. accessibles).

Notons que si $\mathcal{M}$ est π_0-accessible (resp. accessible), la condition de 4.8.1 nous dit que $\mathcal{M}$ est faiblement π_0-accessible (resp. faiblement accessible), et même qu'il existe une petite sous-catégorie pleine *strictement* génératrice, qui soit formée d'objets π_0-accessibles (resp. accessibles). Je doute fort que cette dernière condition nécessaire soit aussi suffisante pour que $\mathcal{M}$ soit π_0-accessible (resp. accessible). Mais en vertu de 4.6.3.3 (p. 150), il en est ainsi si p. 73
on suppose C stable par petites $\varinjlim$, ou ce qui revient au même, moyennant L_{π_0}, stable par limites inductives de cardinal $\leqslant \pi_0$.

Proposition 4.8.9. *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie accessible (4.8.7), π_0 un cardinal tel que $\mathcal{M}$ satisfait à L_{π_0}. Conditions équivalentes :* [page 207]

a) $\mathcal{M}$ est une catégorie paratopos (cf. 4.6.3, 3°, p. 150).

b) $\mathcal{M}$ est stable par petites limites inductives.

b′) $\mathcal{M}$ est stable par limites inductives de cardinal $\leqslant \pi_0$.

b″) (Si $\mathcal{M}$ est équivalente à une catégories $\mathrm{Ind}_{\pi_0} C$, avec C une petite catégorie karoubienne.*) La catégorie C est stable (en elle-même) par limites inductives de cardinal $\leqslant \pi_0$.*

c) $\mathcal{M}$ est stable par petites limites projectives.

c′) (Sous l'hypothèse $\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(C)$, et quand on se donne un cardinal π tel que $\pi = \pi^{\pi_0} \geqslant \pi_1 = \sup(\pi_0, \mathrm{card\,réd}\, C)$.) $\mathcal{M}_\pi$ est stable par $\varprojlim$ finies, et $\mathcal{M}$ est stable par petits produits infinis (ou encore, par $\varprojlim$ filtrantes).

c″) Il existe un cardinal infini π, et une petite catégorie $\mathcal{D}$ stable par $\varprojlim$ finies, telle que $\mathcal{M} \simeq \mathrm{Ind}_{\pi}(\mathcal{D})$, de plus $\mathcal{M}$ stable par produits infinis (ou encore par $\varprojlim$ filtrantes).

([57], [58]).

57. **N.B.** L'équivalence de c′, c″ avec les autres conditions n'est pas établie (ici, mais seulement dans 4.11 plus bas.)

58. **N.B.** Les critères c′, c″ me paraissent plus techniques et moins utiles `[c′, c″ ci-dessus sont mis en parenthèses]`.

Commentary · Present state [ms. 206]

Accessibility and the chosen cardinal

Grothendieck's change of terminology brings the chapter to the modern structural notion: an accessible category is generated by a small subcategory under sufficiently filtered colimits, with the generators presentable at the same cardinal. His earlier weak notion only asked for a small generating family of accessible objects. The cardinal is part of the assertion. In particular, the existence of a dense family of μ-presentable objects and of μ-filtered colimits does not by itself provide the filtered presentation required in the definition.

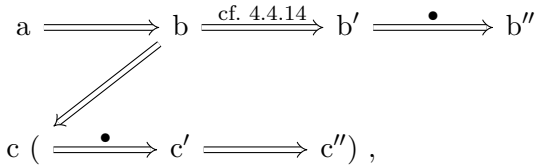
Questions 4.8.6(1) and the nonuseful-cardinal part of (4) have a common concrete answer. Let $\mathcal{K}$ be the category of nonempty $\aleph_1$-directed posets and monotone maps. It is $\aleph_1$-accessible: each object is the $\aleph_1$-directed union of its countable subposets with a greatest element. Its λ-presentable objects, for regular $\lambda \geq \aleph_1$, are precisely the posets of cardinal less than λ. Our commentary on §4.7 proves these facts and the resulting obstruction, following Vasey (2020, Example 3.8). For every singular $\theta > \aleph_1$ of countable cofinality, the poset $[\theta]^{<\aleph_1}$ has no cofinal subset of cardinal at most θ. If it were a θ^+-filtered colimit of θ^+-presentables, a single stage image would contain all θ singleton subsets. That image would be $\aleph_1$-directed and of cardinal at most θ, hence would be a forbidden cofinal subset. Thus $\mathcal{K}$ is not θ^+-accessible, although its original small dense family of countable presentables remains θ^+-presentable. It satisfies 4.8.1(a), but not (b), at manuscript cardinal θ. Taking arbitrarily large such θ answers the unbounded version of (4) for this one category.

The modern change-of-cardinal theorem explains the allowed upward steps through a sharp-inequality hypothesis; see Lurie (2009, Proposition 5.4.2.9 and Remark 5.4.2.10). Grothendieck's sufficient equation $\pi^{\pi_0} = \pi$ is a concrete way to ensure the requisite closure of small subsets under the previous stage's operations. The example shows why the cardinal condition has mathematical content.

For question (2), cocompleteness is a sufficient condition already established in 4.6.3. More economically, adjoining the existence of all colimits with at most π_0 arrows to L_{π_0} supplies cocompleteness by the earlier decomposition of a small diagram into bounded pieces. The closure of the chosen generators under those colimits then makes the overcategories filtered at the required cardinal. This is a repair by adding operations that create cones, rather than merely adding more generators.

The useful-but-nonregular alternative in question (4) is excluded by the repaired presentation argument of §4.7. If π is useful in the sense of the comparison equivalence, write $\mathcal{M} = \mathrm{Ind}_\pi(\mathcal{M}_\pi)$. For $\pi' > \pi$ put $\mu = (\pi')^\pi$, so that $\mu^\pi = \mu$. Given $F \in \mathcal{M}_{\pi'} \subseteq \mathcal{M}_\mu$, the fixed-point presentation theorem, with base category $\mathcal{M}_\pi$, supplies a π^+-filtered presentation of F of size at most μ. The bounded cofinal-poset construction in our commentary on §4.7 makes this index ordered without increasing its size. This is precisely the quantitative regularity required in 4.7.13, without an additional threshold π_1. The nonuseful cardinals supplied above therefore exhaust the possible kind of failure asked for in question (4).

Démonstration. Les implications suivantes sont tautologiques, ou déjà ± connues :



où seules les implications pointées $b' \Longrightarrow b''$ et $c \Longrightarrow c'$ demandent une démonstration. Pour [page 208]
$b' \Longrightarrow b''$, on note que l'hypothèse C karoubienne implique que $C \simeq \mathcal{M}_{\pi_0}$ (4.8.5, 2°), or on sait que $\mathcal{M}_{\pi_0}$ est stable dans $\mathcal{M}$ par limites inductives de cardinal $\leqslant \pi_0$ (et celles-ci sont représentables dans $\mathcal{M}$ en vertu de b′), donc $\mathcal{M}_{\pi_0}$ admet des $\varinjlim$ du type envisagé, donc C aussi.

$c \Longrightarrow c'$. Notons que si $\mathcal{M} = \mathrm{Ind}_{\pi_0}(C)$, alors les $\varprojlim$ représentables dans $\mathcal{M}$ se calculent dans $C^\wedge$. Il s'ensuit que si $\pi \geqslant \pi_1$, alors Filt^π est stable par $\varprojlim$ finies (Filt^π étant donné alors par la condition a″ de 4.7.1), d'autre part, si de plus $\pi^{\pi_0} = \pi$, π est un cardinal utile, donc $\mathcal{M} \simeq \mathrm{Ind}_\pi(\mathcal{D})$, où $\mathcal{D} = \mathrm{Filt}^\pi = \mathcal{M}_\pi$. Il suffit maintenant de voir que $\mathrm{Ind}_\pi(\mathcal{D})$ est alors stable par $\varprojlim$ finies, pour conclure qu'il en est de même de $\mathcal{M}$, donc que $\mathcal{M}$ est stable par limites

projectives quelconques, s'il l'est par petites produites ou par petites $\varprojlim$ filtrantes, d'où c′. Il reste donc à prouver le lemme suivant.

Lemme 4.8.10. *Soient C une petite catégorie, π_0 un cardinal infini, $\pi_1 = \sup(\pi_0, \operatorname{card} \text{réd}\, C)$.* p. 74
Supposons Ind_{π_0} stable par $\varprojlim$ finies (resp. et d'un certain type). Alors pour tout $\pi \geqslant \pi_1$, [page 209]
Filt^{π} est stable par lesdites $\varprojlim$. (Donc aussi $\mathcal{M}_\pi = \mathrm{Filt}^{\pi}$, si $\pi = \pi^{\pi_0}$.) Plus généralement, soit J une petite catégorie telle que $\mathrm{Ind}_{\pi_0} C$ soit stable par $\varprojlim$ de type J, et soit $c = \operatorname{card} \text{réd}\, J$, supposons que $\pi^c = \pi$ (p.ex. c fini). Alors Filt^{π} est stable dans $\mathcal{M}$ par $\varprojlim$ de type J. (Donc il en est de même de $\mathcal{M}_\pi = \mathrm{Filt}^{\pi}$, si de plus $\pi^{\pi_0} = \pi$.)

On utilise le critère a″ de 4.7.1, savoir $\operatorname{card} F(x) \leqslant \pi$ pour tout $x \in C$, pour l'appartenance à Filt^{π}. On peut supposer J réduite, donc $\operatorname{card} \mathrm{Ob}\, J \leqslant c$. Comme le foncteur d'inclusion $\mathrm{Ind}_{\pi_0}(C) \longrightarrow C^{\wedge}$ commute aux $\varprojlim$ quelconques, le calcul de $\varprojlim_{j \in J} x_j$ dans $C^{\wedge}$ fibre par fibre montre que dans $C^{\wedge}$, on a

$$X \overset{\text{déf}}{=} \varinjlim_{j \in J} x_j \;=\; \underbrace{\prod_{j \in \mathrm{Ob}\, J} x_j}_{\text{produit dans } C^{\wedge}} \quad ,$$

d'où

$$\operatorname{card} X(x) \;\leqslant\; \prod_{j \in \mathrm{Ob}\, J} \underbrace{\operatorname{card} x_j(x)}_{\leqslant \pi} \;\leqslant\; \pi^c \ ,$$

d'où (sans autre hypothèse sur π que $\pi \geqslant \pi_1$)

$$\varprojlim_{j \in J} x_j \;\in\; \mathrm{Ob}\, \mathrm{Filt}^{\pi^c} \ , \tag{4.8.10.1}$$

d'où la conclusion voulue si $\pi^c = \pi$, q.e.d.

[page 210]

Commentary · Exposition [ms. 209]

Limits are bounded by products

Grothendieck's cardinal estimate makes finite limits manageable at sufficiently large stages of the filtration. It uses the restricted Yoneda embedding, not a special property of the object being limited. An existing limit in $\mathcal{M}$ is carried to the pointwise limit of its restricted representable functors, because each $\mathrm{Hom}(c, -)$ preserves limits. If $X = \lim_{j \in J} X_j$, then

$$X(c) \hookrightarrow \prod_{j \in \mathrm{Ob}\, J} X_j(c), \qquad |X(c)| \leq \pi^{|\mathrm{Ob}\, J|}$$

whenever $|X_j(c)| \leq \pi$. The source display has two slips: an inductive limit in place of a projective one, and an equality with the product in place of an inclusion. A limit imposes the compatibility equations associated with the arrows of J; deleting those equations generally enlarges it. The next displayed inequality is correct and is all the proof needs.

For example, let J be the parallel-pair category and take two functions $f, g : A \rightrightarrows B$. Its limit is the equalizer $\{a \in A : f(a) = g(a)\}$, with the redundant B-coordinate determined by a. For $A = B = \{0, 1\}$, $f = 1$, and g the transposition, that limit is empty, while the product over the two objects of J has four elements. Thus the equality cannot be read literally even in sets. For a finite nonempty shape and infinite π, the estimate becomes $\pi^{|\mathrm{Ob}\, J|} = \pi$; the empty shape is the terminal presheaf and has singleton values. The latter should be treated separately when the notation $\mathrm{Filt}^{\pi^0} = \mathrm{Filt}^1$ would fall outside the filtration's declared range. This proves the intended finite-limit bound without changing any substantive conclusion of 4.8.10.

Nous allons prouver les implications supplémentaires

$$\mathrm{b}'' \implies \mathrm{b}\ , \qquad \mathrm{c} \implies \mathrm{b}\ , \qquad \mathrm{c}'' \overset{?}{\implies} \mathrm{c}\ ,$$

ce qui achève la démonstration.

L'implication b″ $\Longrightarrow$ a est contenue dans 4.6.3.

Pour l'implication c $\Longrightarrow$ b, on note que la stabilité de $\mathcal{M}$ par petites $\varinjlim$ équivaut à l'assertion que toute petite limite projective de foncteurs $\mathcal{M} \xrightarrow{\xi_i}$ Ens représentables, est représentable. Or les objets de $\mathcal{M}$ étant accessibles ($\mathcal{M}$ étant accessible), les foncteurs représentables ξ_i sont accessibles, donc aussi leur $\varprojlim$ (SGA 4 I 9.7). D'autre part, le foncteur ξ est exact à gauche, comme $\varprojlim$ de foncteurs représentables donc exactes à gauche. Donc le fait que ξ est représentable résulte du

Lemme 4.8.11. *Soit $\mathcal{M}$ une catégorie accessible, stable par $\varprojlim$ finies, $\mathcal{M} \longrightarrow$ Ens un foncteur. Pour que ξ soit représentable, il faut et il suffit qu'il soit exact à gauche et accessible. Donc (SGA 4 I 8.12.8.1) si $\mathcal{M}$ est stable par petites $\varprojlim$, ξ est représentable si et seulement s'il commute auxdites $\varprojlim$, et qu'il soit accessible.*

p. 75
[page 211]

Commentary · Examples [ms. 210]

Prorepresentability is the missing word

Grothendieck is trying to recover colimits from completeness by representing a limit of covariant representable functors. That strategy is sound, but the first sentence of Lemma 4.8.11, as printed, is false: accessibility and preservation of finite limits imply *prorepresentability*, not representability. This is exactly the conclusion of the cited SGA 4, I, 8.13.2(b). The following sentence, which adds all small limits, is the representability statement; SGA 4, I, 8.12.8.1 supplies the additional step (Artin et al., 1972).

An explicit example keeps the two conclusions apart. Put $A_n = \{0,1\}^n$, with the truncation surjections $A_{n+1} \to A_n$, and define a covariant functor on sets by

$$T(X) = \varinjlim_{n<\omega} X^{A_n}.$$

The transition maps repeat a function on the two extensions of each binary word. Thus $T(X)$ is the set of functions from $\{0,1\}^{\mathbb{N}}$ to X which depend on finitely many coordinates. It is equally the functor of continuous maps from the Cantor space to the discrete space X; compactness ensures that such a map factors through some A_n.

Each functor X^{A_n} preserves all limits and filtered colimits, since A_n is finite. Filtered colimits commute with finite limits in sets. It follows, by interchanging two colimits for the accessibility assertion, that T is finitary and preserves finite limits. However, the comparison

$$T(\{0,1\}^{\mathbb{N}}) \longrightarrow \prod_{k<\omega} T(\{0,1\})$$

does not contain the family of coordinate functions in its image. A preimage would have to be the identity function on $\{0,1\}^{\mathbb{N}}$, which depends on every coordinate. Therefore T fails to preserve this countable product and cannot be representable. Its defining inverse system (A_n) is its prorepresentation. This is a fully explicit counterexample to the printed first sentence, with $\mathcal{M} = \mathrm{Set}$ satisfying even stronger hypotheses than those stated.

The application $c \Rightarrow b$ in 4.8.9 survives. If $D : K \to \mathcal{M}$ is small and $\mathcal{M}$ is accessible and complete, the functor

$$H(X) = \varprojlim_{k \in K^{\mathrm{op}}} \mathrm{Hom}(Dk, X)$$

preserves all small limits. Choose a sufficiently large useful cardinal so that all Dk are presentable and K has smaller cardinality. Interchange of its defining small limit with sufficiently filtered colimits makes H accessible. The solution-set form of the adjoint functor theorem now represents H, and its representing object is $\varinjlim_K D$.

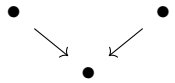
For completeness, the required solution set consists of all pairs (A, a) with A in a small presentable generating subcategory at a suitable cardinal and $a \in H(A)$. Every (X, x) receives a map from such a pair because H preserves the generating filtered colimits. The category of elements of H is complete, since H preserves all limits. A complete locally small category with a small weakly initial family has an initial object: take the product of the family to obtain a weakly initial object P, then equalize all endomorphisms of P. If $e : E \to P$ is that equalizer, choose $r : P \to E$ by weak initiality.

Since $ere = e$, monicity gives $re = 1_E$. For any $h : E \to E$, the endomorphism ehr of P satisfies $ehre = e$, hence $h = 1_E$. Equalizers of two maps out of E, together with weak initiality of E, then show uniqueness of every such map. Thus E is initial. This proves the representability step with the hypothesis actually needed.

Le "il faut" est remis pour mémoire, on vient de le voir comme conséquence du fait que tout objet de $\mathcal{M}$ est accessible. Le "il suffit" est essentiellement SGA 4 I 8.13.2.b), sauf que dans loc. cit. on suppose $\mathcal{M}$ munie d'une fc *stricte.* Mais la démonstration donnée dans loc. cit. montre aussi pour une fcga, en choisissant dans loc. cit. (p. 137) π tel que ξ soit π-accessible, et de plus π soit utile pour la filtration cardinale généralisée envisagée.

Reste à prouver $c'' \overset{?}{\Longrightarrow} c$, sous réserve que ce soit bien vrai. Ce serait une conséquence du lemme suivant (où $\mathcal{D}$, π sont remplacés par C, π_0).

Lemme 4.8.12 (?) ([88]). *Soient C_0 une petite catégorie, π_0 un cardinal infini. Si C est stable par $\varprojlim$ finies, alors $\mathrm{Ind}_{\pi_0} C$ également.*

L'assertion similaire pour $\mathrm{Ind}(C)$, au lieu de $\mathrm{Ind}_\pi C$, est établie dans SGA 4 I 8.9.1 c). Mais il n'est pas clair sans plus que $\mathrm{Ind}_\pi C$ soit stable dans $\mathrm{Ind}(C)$ par $\varprojlim$ finies, dans le cas envisagé. De façon précise, si $F = \varprojlim_{j\in J} F_j$, les F_j dans $\mathrm{Ind}_\pi C$, F dans $\varprojlim \mathrm{Ind}(C) \subset C^\wedge$, on sait [page 212]
que C/F est filtrant et il faut voir qu'il est filtrante grande devant π_0. On est ramené aux trois cas où $J = \emptyset$, $J = e \amalg e$, et $J =$  . Les deux premiers cas sont essentiellement triviaux (car e, et un produit de deux catégories grandes devant π_0, sont grandes devant π_0), le troisième cas par contre n'a pas l'air trivial a priori. C'est lié visiblement à la question des représentations indicielles de foncteurs $J \longrightarrow \mathrm{Ind}_{\pi_0}(C)$, pour $\mathrm{card\,réd}\, J \leqslant \pi_0$ – i.e. la validité de 4.4.13 b) en dehors de l'hypothèse que C soit stable par $\varinjlim$ de cardinal $\leqslant \pi_0$. Il nous faudra y revenir très prochainement, car cette question m'apparaît à présent une question-clef, pour les propriétés de stabilité de la notion de catégorie accessible (en dehors d'aucune hypothèse d'existence de $\varinjlim$ ou $\varprojlim$).

Proposition 4.8.13 ([89]). *Soit C une catégorie équivalente à une petite catégorie, et soit $\pi_0 \geqslant \mathrm{card}\, réd\, C$ un cardinal infini. Alors $\mathcal{M} = \mathrm{Ind}_{\pi_0} C = \mathcal{M}_{\pi_0}$ (i.e. tous les objets de $\mathrm{Ind}_{\pi_0} C$ sont π_0-accessibles), et $\mathcal{M}$ est une enveloppe de Karoubi de C.*

[page 213]

Commentary · Exposition [ms. 212]

Finite limits of a bounded ind-completion

Grothendieck isolates the real issue in Lemma 4.8.12: the pointwise limit must still admit a presentation with the stronger filteredness condition. The statement is true. Its ordinary ind-version is SGA 4, I, 8.9.1(c), and the later manuscript returns to it in 4.11.7 (Artin et al., 1972; Grothendieck, 1990–1991n). We give the bounded argument, since it needs only a span and does not assume an indexed representation theorem for arbitrary finite categories with endomorphisms.

Suppose C has finite limits, and put $\lambda = \pi_0^+$. The terminal presheaf is represented by the terminal object of C. For two λ-filtered presentations, the product indexing category is λ-filtered and filtered colimits commute with finite products in sets; hence their product is a λ-filtered colimit of the representing products. It remains to treat a pullback $F \times_H G$.

Choose small λ-filtered presentations $F = \varinjlim_i y(a_i)$, $G = \varinjlim_j y(b_j)$, and $H = \varinjlim_k y(c_k)$. Let L have objects (i, j, k, u, v), where $u : a_i \to c_k$ and $v : b_j \to c_k$ induce the given maps to H. Its arrows are triples of index arrows making the two squares commute. The category L is λ-filtered. To see this, begin with a diagram in L having fewer than λ arrows. Choose cones on its i-, j-, and k-components. The maps from the resulting a_i, b_j to H factor at some stages of the c-diagram, since representables are λ-presentable. Move them and the previously chosen k-cone to a common stage. Each desired compatibility equation holds in H, so it holds after a further transition. There are

88. Lemme non prouvé ici – cf. démonstration 4.11.7 (non achevée).

89. *Pas prouvé.* Mais OK si $\pi_0 > \mathrm{card\,réd}\, C$, cf. 4.15.10, p. 462.

fewer than λ such equations; λ-filteredness supplies one further stage satisfying all of them. The resulting object of L is the required cone. The same argument with an empty starting diagram gives nonemptiness.

There is a canonical comparison

$$\varinjlim_{(i,j,k,u,v)\in L} y(a_i \times_{c_k} b_j) \longrightarrow F \times_H G.$$

Evaluate at $d \in C$. A pair of elements on the right is represented by maps $d \to a_i$, $d \to b_j$ whose images agree in $H(d)$. Factor the two maps to H through a common c_k and enlarge k until their composites from d agree; the pair then factors through $a_i \times_{c_k} b_j$. This proves surjectivity. If two representatives have the same image, first move their i- and j-indices forward until the two coordinate maps agree; then choose a common k and impose the finitely many compatibility equations just used in the filteredness proof. The two representatives become equal in the left-hand colimit. This proves injectivity, naturally in d.

The displayed comparison is therefore an isomorphism. Its source is a λ-filtered colimit of representables, which proves the lemma. Together with arbitrary products it supplies all small limits. If instead all small cofiltered limits are assumed, an arbitrary product is the inverse limit of its finite subproducts, indexed by the opposite of the directed poset of finite subsets. Thus the alternative formulation using filtered projective limits in 4.8.9 is justified as well.

Signalons tout de suite les corollaires suivants.

Corollaire 4.8.14 ([90]). *Soit $\mathcal{M}$ une $\mathfrak{U}$-catégorie. Conditions équivalentes :*

a) $\mathcal{M}$ est accessible, et est équivalent à une petite catégorie.

a′) $\mathcal{M}$ est accessible, et il existe un cardinal infini π_0 tel que tous les objets de $\mathcal{M}$ soient π_0-accessibles (i.e. $\mathcal{M} = \mathcal{M}_{\pi_0}$).

b) $\mathcal{M}$ est équivalente à une petite catégorie, et est karoubienne, i.e. stable par facteurs directs ([91]).

L'équivalence de a) et a′) est claire, puisque $\mathcal{M}$ admet une filtration cardinale généralisée canonique $(\mathcal{M}_\pi)_{\pi\geqslant\pi_0}$, formée des catégories équivalentes à des petites catégories (condition a) des fcg), et telle que toute telle sous-catégorie soit contenue dans un des $\mathcal{M}_\pi$ (résulte de la condition b) des fcg). Comme $\mathcal{M}_{\pi_0}$ est équivalente à une petite catégorie, et est karoubienne (étant stable par $\varinjlim$ filtrantes grandes devant π_0), il est clair aussi que a′) $\Longrightarrow$ b). Reste à prouver b) $\Longrightarrow$ a′), or cela résulte aussitôt de 4.8.13 appliqué à $\mathcal{M}$, qui montre de plus qu'on peut prendre pour π_0 n'importe quel cardinal infini tel que $\pi_0 \geqslant \operatorname{card}\text{réd}\,\mathcal{M}$. p. 76

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Commentary · Examples [ms. 213]

An accessible category can have an accessible opposite

Grothendieck's proposed converse in footnote 91 would turn accessibility on both sides into a smallness criterion. Partial injections show why the existence of many filtered colimits is insufficient for this purpose. Let $\mathcal{P}$ have small sets as objects and partial injective functions as arrows. Thus an arrow $X \dashrightarrow Y$ consists of a subset $D \subseteq X$ and an injection $D \hookrightarrow Y$. Taking the inverse partial function defines an isomorphism $\mathcal{P}^{\mathrm{op}} \cong \mathcal{P}$. This category is finitely accessible; precise published treatments of the example include Heunen (2013, §3.2, Proposition 3.6, Lemma 3.8, Theorem 3.9). We give a direct argument, also keeping track of the larger cardinals used in this chapter.

Adjoining a basepoint identifies $\mathcal{P}$ with the category of pointed sets and pointed functions whose fibre over each nonbasepoint has at most one element. A partial function sends points outside its domain to the basepoint. Let $(X_i)_{i\in I}$ be a filtered diagram in $\mathcal{P}$, and form its colimit L_* in pointed sets. The canonical map $(X_i)_* \to L_*$ has at most one element over every nonbasepoint. Indeed, if two points of X_i have the same nonzero image, filteredness gives a stage at which they have the same nonzero image; the transition map is a partial injection, so the points were equal. Consequently the canonical maps define partial injections $X_i \dashrightarrow L$, where $L = L_* \setminus \{*\}$.

90. *Pas prouvé.* Si finalement, cf. 4.15.10, et 4.15.11, pages 462-463.

91. **N.B.** Ceci implique que $\mathcal{M}$ et $\mathcal{M}^\circ$ sont accessible, et je conjecture que la réciproque est vraie également.

The same argument proves the universal property within $\mathcal{P}$. A cocone $X_i \dashrightarrow Y$ gives a unique pointed function $L_* \to Y_*$. If two nonzero elements of L_* have the same nonzero image, represent them at a common stage. The cocone map at that stage is injective wherever it is defined; hence the representatives, and therefore the two colimit elements, coincide. The induced map is a partial injection. Thus filtered colimits exist in $\mathcal{P}$ and are created by this embedding into pointed sets.

Now let λ be an infinite regular cardinal. A set A of cardinal $< \lambda$ is λ-presentable in $\mathcal{P}$. To factor a partial injection $A \dashrightarrow L$ through a λ-filtered diagram, choose a representative for each of its fewer than λ defined values and move them to one stage. Distinct colimit values cannot have coincident representatives, so this gives a partial injection with exactly the prescribed domain. If two such factorizations have the same composite to L, the equality of their values, including any values that become the basepoint, is witnessed at later stages. There are fewer than λ witnesses; λ-filteredness puts them at one stage. This proves both surjectivity and injectivity of the presentability comparison map.

Conversely, X is the colimit of its subsets of cardinal $< \lambda$, ordered by inclusion. This poset is λ-filtered by regularity. If X were λ-presentable, its identity would factor through one of these inclusions $B \hookrightarrow X$. The composite being the total identity forces $X = B$, so $|X| < \lambda$. We have therefore proved the exact description

$$\mathcal{P}_\lambda^{\mathrm{pres}} = \{X : |X| < \lambda\}, \qquad \mathcal{P} \simeq \mathrm{Ind}_\lambda\big(\mathcal{P}_\lambda^{\mathrm{pres}}\big),$$

where Ind_λ in this display has its modern, regular-cardinal meaning. In the notation of the manuscript take $\lambda = \pi^+$: the π-accessible sets in $\mathcal{P}$ are exactly those of cardinal at most π. A skeleton of this subcategory is small, including its partial injections.

An isomorphism in $\mathcal{P}$ is an ordinary bijection: an inverse under partial composition forces both domains to be the whole sets. Thus $\mathcal{P}$ has objects of every small cardinality and is not essentially small relative to the chosen universe. Both it and its opposite are accessible. The conjectured converse is false; the equivalence in Corollary 4.8.14 between small accessible categories and small idempotent-complete categories is unaffected. The obstruction is visible without any higher category theory: a coproduct of two singleton sets does not exist in $\mathcal{P}$. The cocones in which exactly one singleton map is defined force both structural arrows to be everywhere defined and their images to be distinct. The cocone sending both singleton points to the same singleton then cannot factor through a partial injection. Completeness hypotheses cannot be dropped from the familiar duality restrictions for locally presentable categories.

4.8.15. Démonstration de 4.8.13. Nous allons prouver :

Lemme 4.8.15.1. *Sous les conditions de 4.8.13, soit $F \in \mathrm{Ob}\,\mathcal{M}$, $x \in \mathrm{Ob}\,C$, alors*

$$(*) \qquad \mathrm{card}\,F(x) \leqslant \pi_0 .$$

Ceci signifie donc, en vertu de 4.7.7, a$''$ $\Longrightarrow$ b pour $\pi \geqslant \pi_1$, compte tenu qu'ici $\pi_J = \pi_0$, $\pi = \pi_0$, qu'on a $F \in \mathrm{Filt}^{\pi_0}$, d'où a fortiori $F \in \mathcal{M}_{\pi_0}$, ce qui implique $\mathcal{M} = \mathcal{M}_{\pi_0}$, i.e. la proposition 4.8.13, compte tenu de 4.8.5. Donc tout revient à prouver $(*)$.

Écrivons

$$F = \varinjlim_{i \in I} x_i , \qquad \text{les } x_i \in \mathrm{Ob}\,C,\ I \text{ ensemble ordonné grand devant } \pi_0,$$

d'où

$$F(x) = \varinjlim_{i \in I} \mathrm{Hom}_C(x, x_i) .$$

On peut supposer

$$\mathrm{card}\,C = \pi_0 ,$$

quitte à remplacer C par une catégorie réduite équivalente.

Pour tout $\xi \in \mathrm{Ob}\,C$, soit

$$I_\xi = \{i \in I \mid x_i = \xi\} ,$$

alors

$$I \;=\; \bigcup_{\xi\in \mathrm{Ob}\, C} I_\xi \;,$$

réunion d'une famille de parties de I de cardinal $\mathrm{card}\,\mathrm{Ob}\,C \leqslant \pi_0$. Comme I est grand devant π_0, on voit que cela implique que l'ensemble des I_ξ est cofinal dans I. Quitte à remplacer I par I_ξ, on peut supposer $I_\xi = I$, i.e. que tous les x_i sont égaux à un même élément ξ. Quitte à remplacer C par la sous-catégorie pleine $C_0 = \{\xi\}$, on peut supposer C réduite à un seul élément. (Car si on prouve que F, en tant qu'objet de $\mathrm{Ind}_{\pi_0}(C_0)$, est facteur direct d'un objet de C_0 (i.e. de ξ), il en sera de même de F en tant qu'objet de $\mathrm{Ind}\,C$). Donc on peut supposer [page 215]

$$C \;=\; \{\xi\} \;,$$

donc C donné par le monoïde

$$M \;=\; \mathrm{End}_C(\xi) \;, \qquad \mathrm{card}\, M \;\leqslant\; \pi \;,$$

et la donnée du système inductif $(x_i)_{i\in I}$ revient à la donnée d'un système d'éléments

$$\alpha_{ji} \;\in\; M \qquad (i,j \in I,\; i \leqslant j)$$

satisfaisant p. 77

$$\alpha_{ii} \;=\; 1 \;, \qquad \alpha_{kj}\alpha_{ji} \;=\; \alpha_{ki} \;\text{ si } i \leqslant j \leqslant k.$$

On a

$$F(\xi) \;=\; \varinjlim_{i\in I} M_i \qquad \left\{ \begin{array}{l} \text{où } M_i = M \;\forall\, i \in I, \text{ et } M_i \longrightarrow M_j \\ \text{est donné par} \\ \qquad\qquad u \longmapsto \alpha_{ji} u \;. \end{array} \right.$$

J'ai pu me ramener, par des arguments similaires ([92]), au cas où le monoïde $M = \mathrm{End}(\xi)$ est un monoïde de monomorphismes, i.e. [page 216]

$$uv \;=\; uv' \;\;\Longrightarrow\;\; v \;=\; v' \qquad \text{si } u, v, v' \in \mathcal{M} \;,$$

ce qui implique que les flèches de transition dans le système inductif des $M_i = M$ sont des monomorphismes. Il s'ensuit que l'on ne peut avoir $\mathrm{card}\, F(\xi) \leqslant \pi_0$ (I étant grand devant π_0) que si α_{ji_0} est un isomorphisme pour $j \geqslant i_0$, et i_0 assez grand, i.e. si le système inductif induit pour $j \geqslant i_0$ est constant, a fortiori que le système inductif envisagé est essentiellement constant. Mais je viens de passer près de quatre heures pour essayer de prouver ce point, sans y être arrivé ! Cela semble plus délicat à prouver que je ne le prévoyais, mais je crois toujours que c'est bel et bien vrai.

[page 217]

Commentary · Present state [ms. 216]

The small-category theorem and its boundary

Grothendieck's intended conclusion is that enough filteredness forces a diagram in a small category to become constant up to splitting an idempotent. The reduction to a one-object category is valid. If a π_0^+-directed poset is the union of at most π_0 subsets, at least one subset is cofinal: otherwise choose one obstruction to cofinality for each subset and take a common upper bound, contradicting the union. The same subset, with its inherited order, remains π_0^+-directed. Applying this to the fibres of the object function gives a cofinal presentation whose values are one fixed object c; its remaining data lie in $\mathrm{End}_C(c)$.

92. Pour ces réductions, cf. 4.15.8 (p. 453-463). L'ultime réduction se trouve dans 4.15.8.9 (p. 461), et son corollaire 4.15.10.

The cancellation reduction and eventual-stationarity assertion on page 216 are not proved in these pages. Nor does the later annotation establish the exact endpoint printed in 4.8.13. The official continuation, 4.15.10–4.15.11, proves the conclusion under the strict inequality $\pi_0 > \operatorname{card}\operatorname{réd} C$ and explicitly describes it as a weaker form of 4.8.13 (Grothendieck, 1990–1991n). We use that strict-bound theorem for the applications here. It proves Corollary 4.8.14, because an essentially small category allows the cardinal to be enlarged. Thus an essentially small category is accessible exactly when it is idempotent complete; an accessible category with a common presentability bound on all objects is essentially small. The higher-categorical counterpart is Lurie (2009, Corollaries 5.4.3.5–5.4.3.6).

The equality case $\pi_0 = \operatorname{card}\operatorname{réd} C$ is deliberately kept separate in the audit. The checked forward reference and the general small-category result do not settle that fixed-cardinal assertion. We do not describe it as a currently open problem merely because its proof is absent here. What is established for this reading is the strict-bound result, the consequent small-category characterization, and the exact remaining endpoint in the printed proposition.

4.9 Le théorème de représentation indicielle pour $\mathrm{Ind}_\pi(C)$

4.9.1 ([93]). Soient C une catégorie équivalente à une petite catégorie, π_0 un cardinal infini,

$$\mathcal{M} \;=\; \mathrm{Ind}_{\pi_0}(C) \;\subset\; C^\wedge \;, \tag{4.9.1}$$

J une catégorie équivalente à une petite catégorie. Considérons l'inclusion pleinement fidèle $C \hookrightarrow \mathcal{M}$, et l'inclusion correspondante de $\mathfrak{U}$-catégories

$$\underline{\mathrm{Hom}}(J, C) \;\hookrightarrow\; \underline{\mathrm{Hom}}(J, \mathcal{M}) \qquad \text{(pleinement fidèle)}. \tag{4.9.2}$$

On sait, comme $C \subset \mathcal{M}_{\pi_0}$, que les objets de $\underline{\mathrm{Hom}}(J, C)$ donnent des objets π_0-*accessibles* de $\underline{\mathrm{Hom}}(J, \mathcal{M})$, du moins si on suppose

$$\operatorname{card}\operatorname{réd} J \;\leqslant\; \pi_0 \;, \tag{4.9.3}$$

ce que nous ferons désormais. (Cf. 4.5.8, 2°, p. 119.) Comme le deuxième membre de (4.9.2) satisfait L_{π_0} ($\mathcal{M}$ y satisfait), il s'ensuit que ce foncteur (4.9.2) se prolonge en

$$\mathrm{Ind}_{\pi_0}(\underline{\mathrm{Hom}}(J, C)) \;\hookrightarrow\; \underline{\mathrm{Hom}}(J, \underbrace{\mathcal{M}}_{=\mathrm{Ind}_{\pi_0}(C)}) \;, \tag{4.9.3}$$

et ce foncteur est *pleinement fidèle* grâce au résultat d'accessibilité qu'on vient d'énoncer. Je m'intéresse à établir des cas où ce foncteur est essentiellement surjectif, i.e. est une *équivalence* de catégories. Notons que pour que p. 78

$$\varphi \;:\; J \longrightarrow \mathcal{M}$$

appartienne à l'image essentielle, il faut et il suffit qu'il existe un foncteur

$$\Phi \;:\; I \;\longrightarrow\; \underline{\mathrm{Hom}}(J, C) \;,\; I \text{ grande devant } \pi_0 \text{ (et on peut supposer } I \text{ ordonnée)} \;, \tag{4.9.4}$$

tel qu'on ait

$$\varphi \;\xleftarrow{\sim}\; \varinjlim_{i \in I} \Phi(i) \;, \tag{4.9.5}$$

[page 218]

ou ce qui revient au même, tel qu'on ait

$$\varphi(j) \;\xleftarrow{\sim}\; \varinjlim_{i \in I} \Phi(i, j) \;, \qquad \text{pour } j \in J \;, \tag{4.9.6}$$

93. [Les numéros 4.9.1 à 4.9.6 sont utilisés deux fois, 4.9.3 trois fois.]

où on a posé

$$\Phi(i,j) \;=\; \Phi(i)(j)\ ,$$

interprétant donc Φ comme un foncteur

$$\Phi \;:\; I\times J \;\longrightarrow\; C\ ; \tag{4.9.7}$$

la loi fonctorielle de $\varphi(j)$ donnée par (4.9.6) étant évidente. En d'autres termes, les φ dans l'image essentielle de (4.4.3) sont ceux qui admettent une *représentation indicielle* par un foncteur Φ sous forme (4.9.4), ou sous forme (4.9.7), avec *de plus* I grand devant π_0. Je rappelle a ce sujet la

Proposition 4.9.2. *Sous les conditions précédentes sur C, π_0, J (notamment* $\operatorname{card}\operatorname{réd} J \leqslant \pi_0$*), si $\varphi : J \longrightarrow \mathcal{M} \overset{déf}{=} \operatorname{Ind}_{\pi_0}(C)$, pour que φ appartienne à l'image essentielle du foncteur pleinement fidèle (4.9.3), il faut et il suffit qu'elle satisfasse les deux conditions suivantes :*

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Commentary · Exposition [ms. 218]

A coherent diagram needs a common index

Grothendieck's aim is to present an entire diagram by diagrams in the small category, so that its arrows and relations are visible before taking the colimit. Write $[J,C]$ for $\underline{\mathrm{Hom}}(J,C)$ and assume $|\mathrm{Mor}\,J| \le \pi_0$. For $A : J \to C$ and a π_0^+-filtered diagram (X_i) in $[J,\mathcal{M}]$, the set of natural transformations is the equalizer

$$\mathrm{Nat}(A,X) = \mathrm{Eq}\left(\prod_{j\in \mathrm{Ob}\,J} \mathrm{Hom}(Aj, Xj) \rightrightarrows \prod_{u:j\to k} \mathrm{Hom}(Aj, Xk)\right).$$

Both products have at most π_0 factors. The defining commutation theorem in sets therefore proves that A is π_0^+-presentable. The usual Hom formula for ind-objects then gives full faithfulness of

$$\mathrm{Ind}_{\pi_0}([J,C]) \longrightarrow [J, \mathrm{Ind}_{\pi_0}(C)].$$

This argument proves uniqueness of morphisms between already indexed presentations. It does not produce the presentation of an arbitrary diagram.

For a diagram φ, let $K = [J,C]/\varphi$. If K is π_0^+-filtered and each evaluation $K \to C/\varphi(j)$ is final, the tautological diagram in $[J,C]$ has colimit φ, because colimits in the target functor category are computed objectwise. Conversely, suppose $\varphi = \varinjlim_{i\in I} A_i$ is already an indexed presentation. Any family of at most π_0 objects of K, together with at most π_0 relations, factors through one stage A_i by the presentability just proved. Any equality between those factorizations holds at a further common stage. Thus K is π_0^+-filtered and $I \to K$ is final. The same factorization-and-equality argument applied at a single j shows that $I \to C/\varphi(j)$ is final. Since its factorization through K has final first factor, evaluation from K is final as well. This proves the two directions of 4.9.2 with the size condition in its precise place.

a) $\underline{\mathrm{Hom}}(J,C)/\varphi$ *est grande devant* π_0.

b) Pour tout $j \in J$*, le foncteur*

$$\underline{\mathrm{Hom}}(J,C)/\varphi \;\longrightarrow\; C/\varphi(j) \qquad \begin{cases} \textit{foncteur entre catégories filtrantes grandes devant } \pi_0, \textit{ en vertu de a),} \end{cases} \tag{4.9.2.1}$$

induit par le foncteur

$$\begin{cases} \underline{\mathrm{Hom}}(J,C) \;\longrightarrow\; C \\ F \;\longmapsto\; F(j)\ , \end{cases} \tag{4.9.2.2}$$

est cofinal.

Le résultat principal dans la présente section est le suivant :

Théorème 4.9.3. *Sous les conditions générales de 4.9.1 pour C, π_0, J (notamment* $\operatorname{card}\operatorname{réd} J \leqslant \pi_0$*), le foncteur pleinement fidèle (4.9.3) est une équivalence de catégories dans chacun des deux cas suivants :*

a) C est stable par $\varinjlim$ filtrantes dénombrables.

*b) Il existe un cardinal $\pi_0 < \pi$, et un cardinal $\pi_0' > \pi_0$ (***N.B.** *On peut supposer que π_0' est le successeur de π_0, donc $\pi_0' \leqslant \pi$), tels que C soit stable par limites inductives grandes devant π_0, et de cardinal π_0'.*

([94]).

Remarque 4.9.4.

a) Ce théorème, cas a), contient le théorème 4.4.13, partie b) (p. 95-96), où on suppose que C est stable par limites inductives de cardinal $\leqslant \pi_0$, l'hypothèse a) de 4.9.3 étant considérablement plus faible.

b) Je présume qu'en dehors d'une hypothèse restrictive du type a) ou b) ci-dessus, le foncteur pleinement fidèle (4.9.3) n'est pas toujours essentiellement surjectif, même dans le cas particulier où J est une catégorie finie, p.ex. $J = B_G$, $G = \{\pm 1\}$. Pour un contre-exemple dans le cas très voisin où on prend $\mathcal{M} = \operatorname{Ind}(C)$ (au lieu de $\operatorname{Ind}_{\pi_0}(C)$), et où on considère le foncteur (pleinement fidèle si J est une catégorie équivalent à une catégorie *finie*)

$$\operatorname{Ind}(\underline{\operatorname{Hom}}(J, C)) \longrightarrow \underline{\operatorname{Hom}}(J, \operatorname{Ind}(C)),$$

[page 220]

je renvoie à l'exercice SGA 4 I 8.8.8.

L'application principale du théorème 4.9.3 consiste en le

Corollaire 4.9.5. *Soient $\mathcal{M}$ une catégorie* accessible *(4.8.7), J une catégorie équivalente à une petite catégorie. Alors la catégorie $\mathcal{M}(J^\circ) = \underline{\operatorname{Hom}}(J, \mathcal{M})$ est accessible. Plus précisément, soient π_0 un cardinal* utile *pour $\mathcal{M}$ (i.e. tel que $\operatorname{Ind}_{\pi_0}(\mathcal{M}_{\pi_0}) \xrightarrow{\sim} \mathcal{M}$, i.e. un cardinal tel que $\mathcal{M}$ soit π_0-accessible), soit π un cardinal infini tel que $\pi^{\pi_0} = \pi \geqslant \operatorname{card}\operatorname{réd} J$ ([95]), i.e. $\pi = c^{\pi_0}$, où c est n'importe quel cardinal $c \geqslant \sup(\pi_0, \operatorname{card}\operatorname{réd} J)$, alors $\underline{\operatorname{Hom}}(J, \mathcal{M})$ est π-accessible, plus précisément on a une équivalence de catégories*

$$\underline{\operatorname{Hom}}(J, \mathcal{M}) \xleftarrow{\sim} \operatorname{Ind}_\pi(\underline{\operatorname{Hom}}(J, \mathcal{M}_\pi)) \ . \tag{4.9.5.1}$$

On sait en effet, avec les hypothèses faites sur π_0, π, que π est utile pour $\mathcal{M}$, i.e. que

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$$\mathcal{M} \xleftarrow{\sim} \operatorname{Ind}_\pi(\mathcal{M}_\pi) \ ,$$

d'autre part, $\mathcal{M}_\pi$ est stable par $\varinjlim$ grandes devant π_0 et de cardinal $\leqslant \pi$. On peut donc appliquer 4.9.3 b), en prenant $C = \mathcal{M}_\pi$, π et π_0 comme ici (donc $\pi_0 < 2^{\pi_0} \leqslant \pi$), et $\pi_0' = \pi$.

Commentary · Development (commentators') [ms. 220]

An involution that no stage can carry

Grothendieck expects the indexed-representation theorem to fail for a finite category with a nontrivial automorphism. His reference to SGA 4, I, 8.8.8 gives the ordinary filtered example. The complete chapter also uses a countable signed-chain category in 4.9.6.8(2), on manuscript page 236. The following transfinite adaptation proves the failure of indexed representation for every infinite cardinal used here. We checked both antecedents and the manuscript's indexed-representation criteria; the cardinal-scaled argument below is supplied for this reading, without a claim of priority.

94. Cas commun : il existe un ordinal limite α, tel que 1) $\operatorname{card}\alpha \leqslant \pi_0$. 2) Les limites inductives [de] type $I_\alpha = \{\beta \mid \beta < \alpha\}$ sont représentables dans C.

95. On a donc $\pi = \pi^{\pi_0} \geqslant 2^{\pi_0} > \pi_0$.

A bounded-filtered involution obstruction [ours]. For every infinite cardinal π, there is a small category C and a diagram $B(\mathbb{Z}/2) \to \mathrm{Ind}_\pi(C)$ outside the essential image of

$$\mathrm{Ind}_\pi\big([B(\mathbb{Z}/2), C]\big) \longrightarrow [B(\mathbb{Z}/2), \mathrm{Ind}_\pi(C)].$$

Proof. Put $\lambda = \pi^+$ and let the objects of C be the ordinals $\alpha < \lambda$. There are no maps $\alpha \to \beta$ when $\alpha > \beta$, just the identity when $\alpha = \beta$, and two maps, labelled $+1, -1$, when $\alpha < \beta$. Composition multiplies the labels. This is a small category, and its only endomorphisms are identities. The positive maps define a functor $A : \lambda \to C$. The ordinal λ is λ-filtered, since every set of fewer than λ ordinals has supremum less than λ. Therefore

$$F = \varinjlim_{\alpha<\lambda} y(A\alpha)$$

belongs to $\mathrm{Ind}_\pi(C)$. For a fixed β, the tail $\alpha > \beta$ is cofinal, and $\mathrm{Hom}_C(\beta, \alpha) = \{+1, -1\}$ with identity transition maps on these two labels. Hence $F(\beta) = \{+1, -1\}$. For a morphism labelled ϵ, the presheaf action is multiplication by ϵ. Multiplication by -1 consequently defines a natural involution $\sigma : F \to F$, and it has no fixed point in any $F(\beta)$.

A functor $B(\mathbb{Z}/2) \to C$ must have trivial action, because every endomorphism in C is an identity. A natural map from such an object β to (F, σ) would be an element of $F(\beta)$ fixed by σ. No such element exists. Thus

$$[B(\mathbb{Z}/2), C]/(F, \sigma) = \varnothing.$$

In particular condition a) of 4.9.2 fails, and no indexed presentation can exist. Equivalently, every ind-colimit of the trivial actions still has trivial action, whereas $\sigma \neq 1_F$. This proves the assertion.□

At the ordinary filtered level the construction is the signed-chain version of the unequal two-interval example in SGA 4 (Artin et al., 1972, I, Exercise 8.8.8); the signed-chain category itself also occurs later in this chapter (Grothendieck, 1990–1991n, 4.9.6.8(2)). The transfinite chain is essential here: replacing it by the countable chain would not produce an object of $\mathrm{Ind}_\pi(C)$ when π is infinite. The obstruction is therefore a direct answer to the cardinal version posed on manuscript page 220, rather than an appeal to a counterexample at the wrong filteredness rank.

p. 80

4.9.6 ([96]). Le reste de la présente section 4.9 est consacré à la démonstration de 4.9.3, i.e. à la vérification, pour $J \longrightarrow \mathcal{M}$, des conditions a), b) de 4.9.2. Dans toute la suite, on se fixe donc une

$$\varphi \ : \ J \longrightarrow \mathcal{M} \ = \ \mathrm{Ind}_\pi(C) \ . \tag{4.9.6.1}$$

On peut supposer J réduite, donc petite et $\mathrm{card}\, J \leqslant \pi_0$. On va considérer le foncteur déduit du foncteur (4.9.2.1)

$$\underline{\mathrm{Hom}}(J, C)/\varphi \longrightarrow \prod_{j \in \mathrm{Ob}\, J} C/\varphi(j) \ , \tag{4.9.6.2}$$

dont les foncteurs en question se déduisent en composant avec les foncteurs projection

$$\prod C/\varphi(j) \longrightarrow C/\varphi(j_0) \ . \tag{4.9.6.3}$$

Comme les catégories facteurs $C/\varphi(j)$ sont filtrantes (et même grandes devant π), leur produit est également filtrant, et même grand devant π (immédiat), et de même pour les produits partiels. Il s'ensuit aussitôt que les projections (4.9.6.3) sont cofinales, en vertu du [page 222]

Commentary · Exposition [ms. 221]

Why colimits inside the small category repair coherence

Grothendieck's sufficient hypothesis makes the successive approximations to a diagram converge to an actual diagram in C. The countable case has a particularly transparent proof. Fix $\varphi : J \to$

96. Résultats préliminaires (à la démonstration de 4.9.3).

$\mathrm{Ind}_\pi(C)$, with $|\mathrm{Mor}\, J| \le \pi$, and assume that C has colimits of countable chains. Begin with arbitrary maps $X_0(j) \to \varphi(j)$ from objects of C. Since each $C/\varphi(j)$ is π^+-filtered, one can choose $X_1(j)$ and, for every arrow $u : j \to k$, a lift

$$d_0(u) : X_0(j) \longrightarrow X_1(k)$$

of its composite to $\varphi(k)$. The identity arrows give the transition maps $t_0(j) = d_0(1_j)$. Repeat this construction, imposing at the next stage all relations

$$d_{n+1}(v)d_n(u) = t_{n+1}(\ell)d_n(vu) \qquad (j \xrightarrow{u} k \xrightarrow{v} \ell).$$

Before refinement the two sides have the same image in $\varphi(\ell)$, so they become equal at a further stage of its category of elements. There are at most π equations and objects to dominate, and a common refinement imposes all of them. Taking $u = 1_j$ gives compatibility of the $d_n(v)$ with the transition maps. Thus their colimits define arrows of a functor $X : J \to C$, with $X(j) = \varinjlim_n X_n(j)$; the displayed relation proves composition and the identity relation proves the identity law.

There is a subtle but necessary final step. Yoneda need not preserve these colimits in C. Instead, every $\varphi(j)$, being a π^+-filtered colimit of representables, carries any existing colimit in C indexed by at most π arrows to a limit of sets. Indeed each representable has that property and π^+-filtered colimits of sets commute with those limits. Consequently the compatible elements of $\varphi(j)(X_n(j))$ define an element of $\varphi(j)(X(j))$, hence a map $X(j) \to \varphi(j)$. These maps form a natural transformation. This proves F 1 of page 226.

To prove F 3_π, start with $X_0(j) = G(j)$ and the given pointwise maps $u_\alpha(j) : F_\alpha(j) \to G(j)$. Besides the composition equations, impose the equations making $G \to X$ natural and making each composite $F_\alpha \to G \to X$ natural. For example the two maps $d_0(v)u_\alpha(j)$ and $t_0(k)u_\alpha(k)F_\alpha(v)$ already agree after mapping to $\varphi(k)$, so they can be equalized at the next refinement. Their number is still at most π. The resulting $G \to X$ and $F_\alpha \to X$ have exactly the compatibility requested by F 3_π. The faithful-functor lemma below now proves 4.9.3(a).

The same construction works along any infinite limit ordinal δ of cardinal at most π for which the relevant chain colimits exist in C. At a limit stage $\beta < \delta$, choose a cone in each $C/\varphi(j)$ on the preceding chain. At a successor stage, impose the compatibility of the new arrow lifts with every earlier transition and all composition equations. At most π data are involved, so the same filteredness argument applies. At the final colimit the equations make the arrow maps independent of the chosen stage, and the preceding preservation argument supplies the map to φ. This proves the intended two-cardinal case: if $\mu < \pi$ and C has μ^+-filtered colimits of cardinal $\mu^+ \le \pi$, use $\delta = \mu^+$. The theorem's printed b) reuses π_0 with a different role from its headline; writing the ind-rank as π and the smaller rank as μ removes that ambiguity.

For Corollary 4.9.5 take $C = \mathcal{M}_\pi$, with $\pi^{\pi_0} = \pi$ and $|\mathrm{Mor}\, J| \le \pi$. This C is closed under the existing π_0^+-filtered colimits of cardinal at most π, and the argument applies. Thus the theorem gives the claimed accessibility at the stated larger cardinal. The later statement that a small diagram category in an accessible category is accessible is therefore a continuation of this precise coherence argument; its higher-categorical counterpart is Lurie (2009, Proposition 5.4.4.3).

Lemme 4.9.6.1. *Soient I, I' deux catégories avec I' 0-connexe (p.ex. filtrante), alors le foncteur projection $pr_1 : I \times I' \longrightarrow I$ est cofinal.*

En effet, pour $i' \in \mathrm{Ob}\, I'$ `[plutôt` $i \in \mathrm{Ob}\, I$`]`, on a

$$i\backslash(I \times I') \;\simeq\; (i\backslash I) \times I' \;,$$

or $i\backslash I$ et I' étant 0-connexe, il en est de même de leur produit.

Lemme 4.9.6.2 ([97]). *Soit $(I_\alpha)_{\alpha\in A}$ une famille de catégories* filtrantes *(resp. grandes devant π). Alors $I_A \overset{déf}{=} \prod_{\alpha\in A} I_\alpha$ est filtrante (resp. grande devant π), et pour toute partie $B \subset A$, la projection canonique $I_A \longrightarrow I_B$ est cofinal.*

([98]). La stabilité de la propriété "filtrante" ou "grande devant π" par AQT `[âne qui trotte]`. On a, si $B' = A\backslash B$, $I_A \simeq I_B \times I_{B'} \longrightarrow I_B$, et on applique le lemme 4.9.6.1 pour conclure que ce foncteur est cofinal, $I_{B'}$ étant filtrante, donc 0-connexe.

[page 223]

97. Attention, il n'est pas vrai qu'un produit infini de catégories 0-connexes est 0-connexe !
98. Cf. démonstration élégante ci-dessous, 4.9.6.6.

Le foncteur (4.9.6.3) étant cofinal, pour établir que (4.9.2.1) l'est, il suffit de prouver qu'il en est ainsi du foncteur (4.9.6.2), ce que nous ferons. Notons de plus que ledit foncteur est *fidèle*, car il s'insère dans le carré commutatif

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(J,C)/\varphi & \longrightarrow & \prod C/\varphi(j) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Hom}}(J,C) & \longrightarrow & C^{\mathrm{Ob}\,J} \end{array} ,$$

où les flèches verticales et la deuxième flèche horizontale sont fidèles. En définitive, pour établir le théorème 4.9.3, nous allons étudier le foncteur *fidèle* (4.9.6.2) p. 81

$$(*) \qquad \begin{array}{l} f \ : \ \underline{\mathrm{Hom}}(J,C)/\Phi \ \longrightarrow \ \prod_{j\in J_0} C/\varphi(j) \\ (\text{où on a posé } J_0 = \mathrm{Ob}\,J,\ J_1 = \mathrm{Fl}\,J) \ , \end{array}$$

et nous allons prouver :

ⓐ Ce foncteur fidèle (∗) est *cofinal.*

ⓑ La catégorie source $\underline{\mathrm{Hom}}(J,C)/\varphi$ est grande devant π_0.

Nous allons pour ceci dégager un lemme auxiliaire sur la cofinalité des foncteurs *fidèles* :

Commentary · Development (commentators') [ms. 222]

Connected factors can require unbounded zigzags

Grothendieck's warning in footnote 97 explains why the product argument must use filteredness, not merely connectedness. We make the obstruction quantitative; this also gives a concrete test for an infinite-product argument elsewhere in the reader. We checked the product assertions in 4.9.6 and the standard filtered-category criteria in SGA 4, I, 8.1. The following exact component calculation is supplied here as a small development, without a priority claim.

Components of a product of fences[ours]. Let P_n have objects $0, 1, \dots, n$ and, besides identities, one arrow along each adjacent edge, always directed from the even vertex to the odd vertex. Then every P_n is connected, whereas two objects $a = (a_n)$ and $b = (b_n)$ of $\prod_{n\geq 1} P_n$ are connected if and only if

$$\sup_{n\geq 1} |a_n - b_n| < \infty.$$

Consequently the product has exactly $2^{\aleph_0}$ connected components.

Proof. There are no composable nonidentity arrows in P_n, so its underlying undirected graph is the path with n edges. A zigzag of length r in the product projects to a zigzag of length r in every factor; therefore $|a_n - b_n| \leq r$ for all n. Conversely, suppose all these distances are at most r. Choose the unique shortest path in each factor. Turn it into a zigzag with the fixed orientation pattern forward, backward, forward, backward, and so on, by inserting identity arrows when the next edge has the opposite direction. At most two positions are needed for each edge, and further identities pad every path to length $2r$. The resulting coordinatewise arrows are arrows of the product, so they give one finite zigzag from a to b.

There are at most continuum many objects in the product. For every real $t \in [0,1]$, take $a_n(t) = \lfloor tn \rfloor$. If $s \neq t$, the distances $|a_n(s) - a_n(t)|$ are unbounded, hence these objects lie in distinct components. This gives the matching lower bound. □

Filtered categories avoid this defect because any finite family of objects has a common target, and any parallel pair can be equalized. For a product of π^+-filtered categories, choose a cocone coordinatewise for each diagram of at most π arrows. Choice assembles these cones into a cocone in the product, including the empty-diagram case. Every projection to a subproduct is final: its comma categories are products of a category with an initial object and the complementary filtered product. This proves 4.9.6.1–4.9.6.2 while accounting for the warning.

Lemme 4.9.6.3 ([99]). *Soit $f : I \longrightarrow I'$ un foncteur, et considérons les conditions :* [page 224]

99. Ce lemme fait partie du sorite général sur la cofinalité et les catégories filtrantes.

a) I filtrante, f cofinal, i.e. (en présence de I filtrante) f satisfait les conditions familières (F 1), (F 2).

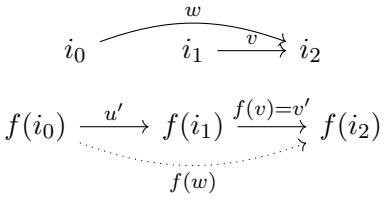
b) I′ est filtrante, et les i′\I (pour i′ ∈ Ob I′) sont filtrantes.

b′) I′ 0-connexe, satisfait PF (2) ([100]) (égalisation des double-flèches).

b″) I′ est filtrante, les i′\I pseudo-filtrantes, i.e. satisfont à la condition PF 1′.

c) I′ filtrante, et f satisfait à F 1, ainsi que la condition suivante.

(F 3) *Pour tout couple (i_0, i_1) d'objets de I et toute flèche $f(i_0) \xrightarrow{u'} f(i_1)$, il existe un objet i_2 de I, et une flèche $v' : f(i_1) \longrightarrow f(i_2)$, tels que v' et $v'u'$ soient relevables en des flèches $i_1 \xrightarrow{v} i_2$ et $i_0 \xrightarrow{w} i_2$ (i.e. il existe i_2 et $v : i_1 \longrightarrow i_2$, $w : i_0 \longrightarrow i_2$ tels qu'on ait $f(v)u' = f(w)$).*



1) On a les implications

$$a \iff b \iff b' \implies b'' \iff c\ ,$$

et si f est fidèle *(p.ex. I ordonnée), les cinq conditions a), b), b′), b″), c) sont équivalentes.*

p. 82
[page 225]

2) Soit π un cardinal infini, et considérons les conditions :

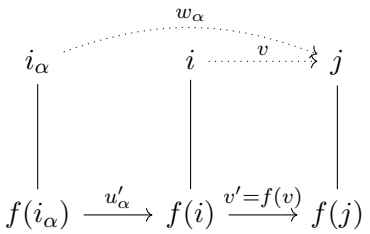
a_π) I grande devant π, f cofinal, i.e. F 1, F 2.

b_π) I′ grande devant π, les i′\I (i′ ∈ Ob I′) aussi.

b_π'') I′ grande devant π, les i′\I (i′ ∈ Ob I′) sont π-pseudo-filtrantes, i.e. satisfont PS $1_\pi{}'$.

c_π) I′ grande devant π, et f satisfait F 1 et

(F 3_π) ([101]). *Pour toute famille $(i_\alpha)_{\alpha\in A}$ d'objets de I, avec* card $A \leqslant \pi$, *et $i \in \mathrm{Ob}\, I$, et une famille de flèches $u'_\alpha : f(i_\alpha) \longrightarrow f(i)$, existe $j \in \mathrm{Ob}\, I$ et une flèche $v' : f(i) \longrightarrow f(j)$, telle que v'* et *les $v'u'_\alpha$ proviennent de flèches de I, i.e. il existe des flèches $v : i \longrightarrow j$, $w_\alpha : i_\alpha \longrightarrow j$ telles que $f(v)u'_\alpha = f(w_\alpha)\ \forall\ \alpha \in A$.*



Ceci posé, on a

$$a_\pi \iff b_\pi \implies c_\pi \implies b_\pi{}''\ ,$$

et si f est fidèle, ou I ordonné, les quatre conditions a_π, b_π, $b_\pi{}''$, c_π sont équivalentes.

([102]).

4.9.6.4. N.B. Dans le cas qui nous intéresse, où f est le foncteur $(*)$ (p. 223), on sait que I' est grande devant π. On voudrait établir que I l'est et que f est cofinal, i.e. la condition a_π. On le prouvera en prenant b_π (qui implique a_π puisque f est fidèle). Il reste donc à prouver, sur le foncteur f dans $(*)$, p. 223, les deux conditions suivantes : [page 226]

100. p.ex. PF 2 est satisfaite si I' est *ordonnée*, sans avoir à supposer d'avance I' filtrante, seulement 0-connexe.
101. **N.B.** La condition F 3_π est tautologiquement satisfaite si f est *pleinement fidèle.*
102. Démonstration de la proposition donnée dans 4.9.6.7, page 227.

F 1) Pour tout système de flèches dans $\mathcal{M}$

$$X(j) \xrightarrow{p(j)} \varphi(j)\ , \qquad j \in J_0 = \mathrm{Ob}\, J,\ \text{les } X(j) \text{ dans } \mathrm{Ob}\, C\ ,$$

il existe un foncteur $F : J \longrightarrow C$ et $F \xrightarrow{q} \varphi$, enfin un système de flèches $X(j) \xrightarrow{u(j)} F(j)$ dans les $C/\varphi(j)$, i.e. rendant commutatif les diagrammes

$$\begin{array}{ccc} X(j) & \xrightarrow{\quad u(j) \quad} & F(j) \\ & {\scriptstyle p(j)}\searrow \quad \swarrow{\scriptstyle q(j)} & \\ & \varphi(j)\ . & \end{array}$$

F 3_π) Pour tout système $(F_\alpha)_{\alpha\in A}$ ($\mathrm{card}\, A \leqslant \pi$) d'éléments dans $\underline{\mathrm{Hom}}(C, J)$, et des flèches

$$F_\alpha \xrightarrow{p_\alpha} \varphi \qquad \text{dans } \underline{\mathrm{Hom}}(J, \mathcal{M}),$$

et pour G dans $\underline{\mathrm{Hom}}(J, C)$ et $G \xrightarrow{q} \varphi$ dans $\underline{\mathrm{Hom}}(J, \mathcal{M})$, étant donné un système de flèches $u_\alpha(j)$, p. 83

$$\begin{array}{ccc} F_\alpha(j) & \xrightarrow[\text{comm.}]{u_\alpha(j)} & G(j) \\ & {\scriptstyle p_\alpha(j)}\searrow \quad \swarrow{\scriptstyle q(j)} & \\ & \varphi(j)\ , & \end{array} \qquad u_\alpha(j) \text{ flèche dans } C/\varphi(j)\ ,$$

il existe $H \xrightarrow{q} \varphi$ dans $\underline{\mathrm{Hom}}(J, C)$ et une flèche $G \xrightarrow{v} H$ dans $\underline{\mathrm{Hom}}(J, C)/\varphi$, telle que pour tout $\alpha \in A$, le système de flèches composés [page 227]

$$\begin{array}{ccccc} F_\alpha(j) & \xrightarrow[\text{comm.}]{u_\alpha(j)} & G(j) & \xrightarrow[\text{comm.}]{v(j)} & H(j) \\ & {\scriptstyle p_\alpha(j)}\searrow & \downarrow{\scriptstyle q(j)} & \swarrow{\scriptstyle q(j)} & \\ & & \varphi(j) & & \end{array}$$

(pour $j \in J_0 \in \mathrm{Ob}\, J$) provienne d'un *foncteur* $w_\alpha : F_\alpha \longrightarrow H$.

Pour prouver 4.9.3, nous sommes donc ramené à prouver les deux propositions F 1), F 3_π) précédentes, sous l'une des deux conditions a) ou b) de 4.9.3.

Commentary · Exposition [ms. 225]

The faithful cofinality test, with its hypotheses restored

Grothendieck reduces the construction of a coherent presentation to the ability to lift a bounded family of arrows after one common refinement. Faithfulness then turns equations seen downstairs into actual equations upstairs. This is the precise role of F 3_π; the evaluation functor into the product of the categories of elements is faithful because a natural transformation is determined by its components.

The printed condition b′ in 4.9.6.3 is incomplete. It only says that I' is connected and equalizes parallel pairs, so it cannot imply an assertion about I or f. The functor $\varnothing \to 1$ satisfies it and is not final. The proof in the continuation uses, in addition, that every $(i' \downarrow f)$ is filtered; inserting that clause gives the intended b′. The assertions involving b) and the faithful test c) are correct. Here is a proof of the version needed in these pages.

Suppose $f : I \to I'$ is faithful, I' is π^+-filtered, F 1 holds, and F 3_π holds. Given a diagram $D : K \to I$ with at most π arrows, choose a cone $u_k : fDk \to z$ in I', and then a map $z \to f(i)$ by

F 1. F 3_π gives $v : i \to h$ and $w_k : Dk \to h$ whose images are the composites of those cone maps with $f(v)$. For an arrow $a : k \to \ell$ of K we have

$$f(w_\ell D(a)) = f(w_k).$$

Faithfulness gives $w_\ell D(a) = w_k$, so I is π^+-filtered. Its nonemptiness follows from F 1 and nonemptiness of I'.

Apply the same construction to a bounded diagram of objects $(i_k, p_k : z \to f(i_k))$ in $(z \downarrow f)$. In I' take a cone on its image together with z and all p_k. After F 1 and F 3_π, the maps w_k obtained above satisfy

$$f(w_k)p_k = p$$

for one map $p : z \to f(h)$. They are therefore a cone in the comma category. The empty case is F 1. Every comma category is π^+-filtered, hence nonempty and connected, so f is final. This proves $\mathrm{c}_\pi \Rightarrow \mathrm{b}_\pi \Rightarrow \mathrm{a}_\pi$ under faithfulness, including all quantifiers used in 4.9.6.4.

Faithfulness cannot simply be discarded. For a nontrivial group G, the functor $BG \to 1$ satisfies F 1 and every F 3_π, since all equations in the target are automatic and arbitrary arrows upstairs can be chosen. But BG is not filtered: distinct automorphisms cannot be equalized. This example identifies the exact step that fails when equality of images is substituted for equality of arrows.

Corollaire 4.9.6.5 (de 4.9.6.3). *Soit $(f_\alpha : I_\alpha \longrightarrow I'_\alpha)$ une famille de foncteurs cofinaux, les catégories source étant filtrantes. Alors le foncteur produit $f = \prod f_\alpha : I = \prod_\alpha I_\alpha \longrightarrow I' = \prod_\alpha I'_\alpha$ est cofinal.*

En effet, il faut vérifier que les catégories $i'\backslash I$, pour $i' \in \mathrm{Ob}\, I'$, sont 0-connexes. Or si $i' = (i'_\alpha)_{\alpha \in A}$, on a

$$i'\backslash I \;\simeq\; \prod_\alpha (i'_\alpha \backslash I_\alpha)\ ,$$

et par 4.9.6.3, 1°) a $\Longrightarrow$ c, les $i'_\alpha\backslash I_\alpha$ sont filtrantes, donc leur produit est filtrant (cf. 4.4.6.2), donc 0-connexe, q.e.d. [page 228]

Corollaire 4.4.6.6. *Soit π un cardinal infini. Tout produit de catégories grandes devant π est grand devant π.*

Soit $I = \prod_{\alpha \in A} I_\alpha$, soit B un ensemble de cardinal $\leqslant \pi$, il faut prouver que $I \xrightarrow{\mathrm{diag}_I} I^B$ est cofinal. Or ce foncteur s'identifie au produit des foncteurs similaires $I_\alpha \longrightarrow I_\alpha^B$, lesquels sont cofinaux, les catégories source étant de plus filtrantes. Donc on peut conclure par le corollaire 4.9.6.5.

4.9.6.7. Démonstration de 4.9.6.3. (Partie 1°.)

a $\Longrightarrow$ b. On sait déjà que a) implique que I' est filtrante (SGA 4 I 8.1.3 b). Reste à prouver que pour tout $i' \in \mathrm{Ob}\, I'$, $i'\backslash I$ est filtrante ([103]). Deux conditions PF 1′ et PF 2 à vérifier (en plus du fait que $i'\backslash I \neq \emptyset$, qui est connu par cofinalité de f, impliquant que $i'\backslash I$ est 0-connexe).

1) Soient (i, u'), (j, v') deux éléments de $i'\backslash I$, il faut voir qu'ils sont majorés par un troisième élément. Or I étant filtrante, i, j sont majorés par k. On aurait fini donc si on savait que $f(u_1)u' = f(v_1)v'$, soit w', car alors (k, w') ferait l'affaire.

Commentary · Exposition [ms. 228]

Completing the argument of manuscript page 228

On manuscript page 228 Grothendieck's proof has reached the point just before the two candidate arrows from i' are equalized. The remaining step is short once finality is used in the correct comma category. If I is filtered and f is final, every $(i' \downarrow f)$ is connected. Given two of its objects, join them by a finite zigzag. Project that zigzag to I and take a cone there. Commutativity along each arrow of the zigzag makes the composites from i' agree at its vertex, so that vertex is an object receiving maps from the two original comma objects. Parallel arrows in the comma category can

103. Court-circuit : on sait déjà que $i'\backslash I$ est 0-connexe, d'autre part elle est *localement* filtrante, car coétale sur I qui est filtrante, donc elle est filtrante.

be equalized by filteredness of I; the resulting arrow is still over i'. Hence the comma category is filtered. This finishes the particular step begun in 4.9.6.7.

For the stronger cardinal statement, begin with a diagram of at most π arrows in $(i' \downarrow f)$. Choose one object and a finite connecting zigzag to each of the others. The union of their projections and of the original diagram still has at most π arrows, because π is infinite. A cone in I simultaneously imposes all relations and all equalities of maps from i'. Thus every comma category is π^+-filtered. Likewise a bounded diagram in I' can be lifted object by object using F 1; connect, inside the appropriate comma categories, the two choices associated with each arrow. A cone on the resulting bounded diagram in I yields a cone in I'. This proves $\mathrm{a}_\pi \Rightarrow \mathrm{b}_\pi$.

Conversely, suppose I' and all $(i' \downarrow f)$ are π^+-filtered. Finality is immediate. First derive F 3_π. Given $u'_\alpha : f(i_\alpha) \to f(i)$, filteredness of $(f(i_\alpha) \downarrow f)$ supplies arrows $w_\alpha : i_\alpha \to k_\alpha$ and $v_\alpha : i \to k_\alpha$ with $f(w_\alpha) = f(v_\alpha)u'_\alpha$. Regard all v_α as arrows from $(i, 1)$ in $(f(i) \downarrow f)$. A cone on this bounded diagram supplies maps $h_\alpha : k_\alpha \to h$ with all $h_\alpha v_\alpha$ equal to one $v : i \to h$. The arrows $h_\alpha w_\alpha$ prove F 3_π.

Now choose a cone on fD in I', follow its vertex by a map to some $f(i)$, and apply F 3_π. This gives arrows $w_k : Dk \to h$ which satisfy all the cone equations after applying f. For each arrow $a : k \to \ell$, the two arrows $w_\ell D(a), w_k$ are parallel arrows in $(fDk \downarrow f)$. Filteredness there gives $e_a : h \to h_a$ equalizing them. In $(f(h) \downarrow f)$ take a cone on the bounded diagram consisting of $(h, 1)$ and these arrows e_a. Its map $e : h \to H$ factors through every e_a, so the maps ew_k satisfy every cone equation in I. This proves π^+-filteredness of I without faithfulness, and establishes $\mathrm{b}_\pi \Rightarrow \mathrm{a}_\pi$.

Finally, for final functors $f_\alpha : I_\alpha \to I'_\alpha$ with filtered sources, the comma category of the product functor is

$$((i'_\alpha)_\alpha \downarrow \prod_\alpha f_\alpha) \cong \prod_\alpha (i'_\alpha \downarrow f_\alpha).$$

Each factor is filtered by the argument just given; their product is filtered and therefore connected. This proves 4.9.6.5. Applying it to the diagonal functors $I_\alpha \to I_\alpha^B$ gives the diagonal proof of the next corollary, whose heading is misnumbered 4.4.6.6 in the edition. The direct coordinatewise cone proof gives the same result and avoids treating infinite products of merely connected categories as connected.

Overview · XVIII-b

These pages belong to the ordinary categorical foundations of the theory of derivators. Grothendieck is asking how a large category, and then its categories of diagrams, can be recovered from controlled small data. The recurring difficulty is that a bound on the size of a presentation is not yet an invariant of the object presented. The passage from one to the other requires both a factorization argument and an account of the relations which that argument introduces. Retracts, categories of elements, and changes of indexing category are therefore the central constructions of this part of Chapter XVIII.

There is a cardinal convention to fix at the outset. A category called *grande devant* π admits cocones on diagrams with at most π arrows. In the customary notation for regular cardinals it is π^+-filtered. An object called π-accessible in the manuscript is consequently π^+-presentable. The parameter π itself need not be regular. We retain the manuscript's indices, including its later adjective "regular" for a useful cardinal satisfying a particular presentation bound. That adjective is a separate definition, not a statement about the set-theoretic cofinality of π.

The opening, which continues Proposition 4.5.13 from the preceding pages, compares two bounded-presentation filtrations with the intrinsic filtration $\mathcal{M}_\pi$. In a cocomplete category the basic calculation is straightforward: a colimit indexed by a category with at most π arrows, of π-accessible objects, is again π-accessible, because a limit of that size in sets commutes with π^+-filtered colimits. The converse requires generation. When every object is a sufficiently filtered colimit of objects of a small subcategory, the identity of a presentable object factors through one stage. What follows is a retract of a stage. Lemma 4.5.14 suppresses this last distinction. The commentary gives the corrected implication and an explicit counterexample in sets to its unrestricted wording. Its later application to a subcategory closed under bounded colimits remains valid, since such a subcategory also contains the required idempotent splittings.

The discussion of universal strict epimorphic families has a different purpose. It seeks hypotheses under which one bounded presentation already contains the relations needed for descent. The printed direct-product example in groups is defective even after the accompanying footnote: the two coordinate subgroups of $C_2 \times C_2$ do not form a strict epimorphic family. Replacing the product by the free product gives the intended example, including failure of epimorphy after pullback. We distinguish this correction from the unresolved proof transition in Corollary 4.5.16, where a universal epimorphic hypothesis is used as though it were universally strictly epimorphic. The strengthened version has a complete proof. The argument given in these pages does not establish the weaker printed version or the entire deduction of case C of Proposition 4.5.13.

Section 4.6 builds a small category closed under the colimits needed for reconstruction. The qualitative transfinite construction is sound: at each stage there is a set of diagrams and hence a set of chosen colimits; regularity of the stopping ordinal ensures that every bounded diagram appears early enough to be treated. The numerical estimate for the number of new morphisms is a separate assertion. It is false for an arbitrary small starting category, as the source's own later footnotes suspect. A finitary variety with many binary idempotent operations provides a direct counterexample. Thus the existence of a small reconstruction category must be retained while the claimed sharp bound requires another argument. The normalization of a finite initial cardinal in Corollary 4.6.2 repairs an additional elementary edge case, but does not repair that proof.

The resulting cocomplete recognition theorem is the locally presentable theory: a π_0-paratopos is, with the stated hypotheses restored, locally π_0^+-presentable. The presheaf reflection and the small-category presentation explain why suitable filtered colimits commute with bounded limits. This places the discussion alongside Gabriel–Ulmer duality and the systematic accounts of accessible categories, without asserting that Grothendieck was drawing on those accounts here (Gabriel and Ulmer, 1971; Makkai and Paré, 1989; Adámek and

Rosický, 1994). His immediate reference is SGA 4. One of its assertions, that full faithfulness of a filtered-completion functor forces the original objects to be accessible, fails at the stated generality; the rigid real ring supplies the manuscript's counterexample. The intended Ind application instead follows by evaluation of presheaves.

Exactness does not fix the generating cardinal. To answer footnote 59, we construct, for every prescribed infinite π_0, a sheaf topos whose only π_0^+-presentable object is initial, although filtered colimits are exact and a small dense category exists. The construction uses a set of cardinal π_0^{++} with the topology of complements of subsets of smaller cardinality. Every nonempty support has an increasing π_0^+-indexed cover by proper opens; restricting a sheaf to this cover rules out presentability at the proposed rank. This is a supplied development of the question, with a complete proof and no claim of priority. The closing examples of §4.6 also separate the conditions on images, unions and pushouts of monomorphisms. Pointed sets satisfy all six conditions under discussion and give an example beyond abelian categories and topoi.

Section 4.7 removes the assumption of cocompleteness. Now $\mathcal{M} = \mathrm{Ind}_{\pi_0} C$, and the primed filtration denotes closure under repeated bounded filtered colimits; its meaning must not be imported from the preceding section. The cardinal equation $\pi^{\pi_0} = \pi$ pays for adjoining all the small witnesses needed to make a subcategory of an indexing category sufficiently filtered. We spell out this closure operation, including the old arrows and cone legs omitted from the printed construction. The resulting comparison of presentation size and presentability is the ordinary version of the modern change-of-cardinal arguments (Vasey, 2020, §3).

The retract question which Grothendieck leaves at the centre of this comparison has a positive answer in the literature. Chen's bounded closure theorem, and his formulation of the Makkai–Paré retract-removal result, show that iterating the specified bounded filtered colimits does not enlarge the resulting class (Chen, 2021, Corollaries 4.13 and 4.15). We also give an elementary retract argument which preserves the number of arrows. Consequently $\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi}$ at every permitted cardinal. At the initial cardinal the common value is $\mathrm{Kar}(C)$, not necessarily C: a two-arrow idempotent category is sufficiently filtered at every rank and realizes a nonrepresentable retract. This endpoint explains several otherwise misleading assertions in the text.

The generalized filtration axioms and their useful cardinals require an additional distinction. A filtered slice is a presentation only when its canonical cocone has the correct colimit. Density supplies that condition, but the axioms printed in 4.7.7 do not force it at every level. We exhibit a filtration of sets whose initial slices are terminal and whose initial realization nevertheless misses every nonempty set. This disproves the bare converse in 4.7.12 and identifies the missing hypothesis. Once usefulness is understood through the realization equivalence, every useful cardinal satisfies the quantitative "regularity" requirement of 4.7.13. The commentary proves this with a size-controlled cofinal directed poset, so no example of a useful but nonregular cardinal exists under that interpretation. Conversely, the category of nonempty $\aleph_1$-directed posets has unboundedly many nonuseful manuscript cardinals: every singular $\theta > \aleph_1$ of countable cofinality gives failure of θ^+-accessibility. We provide the complete argument, refining the rank choices in Vasey (2020, Example 3.8). This concerns the accessibility spectrum, rather than the presentability rank of one object (Lieberman et al., 2019).

Section 4.8 returns to density and gives the conceptual recognition of accessible categories. The author's initial doubt about enlargement of a dense subcategory is eventually withdrawn, correctly: naturality against the smaller dense category determines the missing components. The characterization of the smallest presentable part as an idempotent completion is likewise valid. Two other assertions need more substantial correction. First, Lemma 4.8.11 omits "pro" from the conclusion supplied by SGA 4, I, 8.13.2(b). An accessible finite-limit-preserving functor need only be prorepresentable. Locally constant functions from the Cantor space

give an explicit counterexample to representability; preservation of all small limits gives the intended representability theorem. Second, the conjecture in footnote 91 that accessibility of a category and its opposite forces essential smallness is false. Sets and partial injections form a self-dual accessible category with objects of every small cardinality (Heunen, 2013, §3.2); we prove the precise presentability criterion directly.

The small-category theorem requires a more restrained verdict. The argument in these pages does not establish the equality endpoint in 4.8.13. The later manuscript proves a strict cardinal inequality, sufficient for Corollary 4.8.14, and explicitly calls it a weaker result. The continuity result checked in Tendas (2022) also has a strict threshold. We therefore retain the theorem on essentially small accessible categories while leaving that exact endpoint uncertified; this is not a claim that the endpoint is an established open problem today.

Section 4.9 explains why a diagram of ind-objects need not come from diagrams at individual stages. Coherence can impose a genuine obstruction, even for a one-object category with an involution. We extend the SGA 4 counterexample to every infinite filtering bound using a transfinite signed-chain category. The resulting ind-object has an involution which no stage admits. Under the appropriate colimit hypotheses on C, Grothendieck's common-index construction repairs the obstruction. The final cofinality lemmas express that construction through simultaneous factorization and equalization. We restore the missing hypotheses in their printed alternatives and complete the particular proof under way on manuscript page 228.

The later homotopical setting preserves this architecture while replacing sets of maps by mapping spaces and strict diagrams by coherent diagrams. Lurie's recognition theorem characterizes presentable infinity-categories through accessibility, cocompleteness and accessible presheaf localization; his diagram-category theorem supplies the corresponding stability under small diagrams (Lurie, 2009, Theorem 5.5.1.1 and Proposition 5.4.4.3). These are structural counterparts of the constructions here. They explain the relevance of this long ordinary-categorical preparation to the work on derivators without identifying strict diagrams in a homotopy category with coherent diagrams in a homotopy theory.

Le résultat principal dans la présente section est le suivant : p. 79

Théorème 4.9.3. *Sous les conditions générales de 4.9.1 pour C, π_0, J (notamment* $\operatorname{card}\operatorname{réd} J \leqslant \pi_0$*), le foncteur pleinement fidèle (4.9.3) est une équivalence de catégories dans chacun des deux cas suivants :*

a) C est stable par $\varinjlim$ filtrantes dénombrables.

b) Il existe un cardinal $\pi_0 < \pi$, et un cardinal $\pi_0' > \pi_0$ (**N.B.** *On peut supposer que π_0' est le successeur de π_0, donc $\pi_0' \leqslant \pi$), tels que C soit stable par limites inductives grandes devant π_0, et de cardinal π_0'.*

([94]).

Remarque 4.9.4.

a) Ce théorème, cas a), contient le théorème 4.4.13, partie b) (p. 95-96), où on suppose que C est stable par limites inductives de cardinal $\leqslant \pi_0$, l'hypothèse a) de 4.9.3 étant considérablement plus faible.

b) Je présume qu'en dehors d'une hypothèse restrictive du type a) ou b) ci-dessus, le foncteur pleinement fidèle (4.9.3) n'est pas toujours essentiellement surjectif, même dans le cas particulier où J est une catégorie finie, p.ex. $J = B_G$, $G = \{\pm 1\}$. Pour un contre-exemple dans le cas très voisin où on prend $\mathcal{M} = \operatorname{Ind}(C)$ (au lieu de $\operatorname{Ind}_{\pi_0}(C)$), et où on considère le foncteur (pleinement fidèle si J est une catégorie équivalent à une catégorie *finie*) [page 220]

$$\operatorname{Ind}(\underline{\operatorname{Hom}}(J, C)) \longrightarrow \underline{\operatorname{Hom}}(J, \operatorname{Ind}(C)),$$

je renvoie à l'exercice SGA 4 I 8.8.8.

L'application principale du théorème 4.9.3 consiste en le

Corollaire 4.9.5. *Soient $\mathcal{M}$ une catégorie* accessible *(4.8.7), J une catégorie équivalente à une petite catégorie. Alors la catégorie $\mathcal{M}(J^\circ) = \underline{\operatorname{Hom}}(J, \mathcal{M})$ est accessible. Plus précisément, soient π_0 un cardinal* utile *pour $\mathcal{M}$ (i.e. tel que $\operatorname{Ind}_{\pi_0}(\mathcal{M}_{\pi_0}) \xrightarrow{\sim} \mathcal{M}$, i.e. un cardinal tel que $\mathcal{M}$ soit π_0-accessible), soit π un cardinal infini tel que $\pi^{\pi_0} = \pi \geqslant \operatorname{card}\operatorname{réd} J$ ([95]), i.e. $\pi = c^{\pi_0}$, où c est n'importe quel cardinal $c \geqslant \sup(\pi_0, \operatorname{card}\operatorname{réd} J)$, alors $\underline{\operatorname{Hom}}(J, \mathcal{M})$ est π-accessible, plus précisément on a une équivalence de catégories*

$$\underline{\operatorname{Hom}}(J, \mathcal{M}) \xleftarrow{\sim} \operatorname{Ind}_\pi(\underline{\operatorname{Hom}}(J, \mathcal{M}_\pi)) \ . \tag{4.9.5.1}$$

On sait en effet, avec les hypothèses faites sur π_0, π, que π est utile pour $\mathcal{M}$, i.e. que [page 221]

$$\mathcal{M} \xleftarrow{\sim} \operatorname{Ind}_\pi(\mathcal{M}_\pi) \ ,$$

d'autre part, $\mathcal{M}_\pi$ est stable par $\varinjlim$ grandes devant π_0 et de cardinal $\leqslant \pi$. On peut donc appliquer 4.9.3 b), en prenant $C = \mathcal{M}_\pi$, π et π_0 comme ici (donc $\pi_0 < 2^{\pi_0} \leqslant \pi$), et $\pi_0' = \pi$.

Commentary · Exposition [ms. 220]

One indexing category for an entire diagram

Grothendieck wants an approximation of a diagram which remembers its arrows. Writing each value of $\varphi : J \to \operatorname{Ind}_\pi(C)$ as an ind-object separately does not provide this: an arrow can be represented only after enlarging the chosen indices, and a relation between arrows can require a further enlargement. The comparison functor therefore asks for one π-filtered system of genuine J-diagrams in C. Its essential surjectivity is a coherence assertion about presentations, not merely the existence of presentations of their values.

94. Cas commun : il existe un ordinal limite α, tel que 1) $\operatorname{card}\alpha \leqslant \pi_0$. 2) Les limites inductives [de] type $I_\alpha = \{\beta \mid \beta < \alpha\}$ sont représentables dans C.

95. On a donc $\pi = \pi^{\pi_0} \geqslant 2^{\pi_0} > \pi_0$.

We shall use a cardinal convention which removes a recurrent ambiguity in these pages. For an infinite cardinal ρ, "large before ρ" means that every diagram with at most ρ arrows has a cocone. This is modern ρ^+-filteredness. An object called ρ-accessible here is correspondingly ρ^+-presentable. We retain Ind_ρ for Grothendieck's completion, and reserve λ for a regular cardinal when using modern terminology. Thus $\mathrm{Ind}_\rho(C)$ is the completion under λ-filtered colimits with $\lambda = \rho^+$. This convention applies even when ρ itself is singular.

The precise sufficient condition established by the ensuing proof is as follows. Suppose C is essentially small, $|\operatorname{Mor} J| \le \rho$, and C admits colimits of chains of one nonzero limit ordinal α with $|\alpha| \le \rho$. Then

$$\mathrm{Ind}_\rho([J, C]) \longrightarrow [J, \mathrm{Ind}_\rho(C)]$$

is an equivalence. One may take $\alpha = \omega$ in case (a). For the intended case (b), take a smaller infinite cardinal $\theta < \rho$ and $\alpha = \theta^+$, assuming the stipulated θ^+-filtered colimits of size at most θ^+ exist in C. The printed statement moves between π_0 as the completion bound and π_0 as this smaller auxiliary cardinal. Corollary 4.9.5 and the formulation in 4.10 make the intended roles clear; the proof must keep the bound on $|\alpha|$ tied to the completion cardinal. Footnote 94 should be read with that same bound. This is a normalization of the statement, not an additional closure assumption on the intended theorem.

Full faithfulness has a short explanation which also identifies the cardinal restriction on J. For $A, B : J \to \mathcal{M}$,

$$\mathrm{Nat}(A, B) = \mathrm{Eq}\left(\prod_{j \in J} \mathrm{Hom}(A_j, B_j) \rightrightarrows \prod_{u : j \to k} \mathrm{Hom}(A_j, B_k) \right).$$

If every A_j is ρ^+-presentable, this expression commutes with ρ^+-filtered colimits: every product has at most ρ factors, and the equality conditions have the same bound. Consequently the objects of $[J, C]$ remain presentable in $[J, \mathrm{Ind}_\rho C]$, so extension under filtered colimits is fully faithful. The long construction which follows is needed only to produce enough such objects. Its antecedent is the indexed-presentation criterion of Artin et al. (1972, I, 8.8.2–8.8.3); the present cardinal accounting is what makes that criterion usable for accessible categories.

Commentary · Examples [ms. 220]

The involution obstruction at every cardinal

The reservation in Remark 4.9.4 has a concrete answer. The obstruction in SGA 4, I, Exercise 8.8.8 is an involution of an ind-object which cannot be reached equivariantly from any of the chosen approximating objects. The same construction works at every infinite completion bound; it is the case $I = B(\mathbf{Z}/2)$ of Henry (2025b, Example 1.4, Construction 3.1 and Section 3.2). We give the calculation because it distinguishes the theorem's actual hypothesis from the much weaker fact that $\mathrm{Ind}_\rho(C)$ is accessible.

Example (cardinal form of the involution obstruction). For every infinite cardinal ρ there is a small category C such that

$$\mathrm{Ind}_\rho([B(\mathbf{Z}/2), C]) \longrightarrow [B(\mathbf{Z}/2), \mathrm{Ind}_\rho(C)]$$

is not essentially surjective.

Proof. Put $\lambda = \rho^+$. The objects of C are the ordinals $\beta < \lambda$. There are no arrows $\beta \to \gamma$ when $\beta > \gamma$, there is only the identity when $\beta = \gamma$, and there are two arrows labelled $+1, -1$ when $\beta < \gamma$. Composition multiplies labels. These rules are associative and define a small category. Let $y : C \to \widehat{C}$ be Yoneda, and form

$$F = \operatorname*{colim}_{\beta < \lambda} y(\beta),$$

using the arrows labelled $+1$ as transitions. The ordinal λ is λ-filtered: a set of fewer than λ ordinals below λ has an upper bound below λ. Hence F belongs to $\mathrm{Ind}_\rho(C)$.

For a fixed $\gamma < \lambda$, evaluation gives

$$F(\gamma) = \operatorname*{colim}_{\beta < \lambda} C(\gamma, \beta) = \{+1, -1\}.$$

Indeed the terms for $\beta > \gamma$ are these two-element sets and all subsequent transition maps are identities. Precomposition with an arrow labelled ε multiplies by ε. Multiplication by -1 on every $F(\gamma)$ therefore defines a natural automorphism $s : F \to F$, with $s^2 = 1$. It defines an object (F, s) of the right-hand diagram category.

Every functor $B(\mathbf{Z}/2) \to C$ has trivial action, because every endomorphism in C is an identity. A map of actions $(y\gamma, 1) \to (F, s)$ would correspond by Yoneda to an element of $F(\gamma)$ fixed by multiplication by -1. There is no such element. If (F, s) were a filtered colimit of C-valued actions, each object of the nonempty indexing category would give exactly such an equivariant map to its colimit. This contradiction proves the assertion. □

For $\lambda = \omega$ the signed category is the abstract form of the asymmetric finite subsets used in SGA 4. The uncountable replacement supplies the stronger filteredness required here. Thus Grothendieck's question is settled affirmatively, and the same construction is now part of a general characterization of the indexing categories for which presentations always exist. The chosen category also explains the missing hypothesis: a strictly increasing countable chain has no colimit in C. Any cocone has an apex strictly above all its terms and one constant sign. Reversing that sign gives a second cocone to the same apex, which cannot factor through the first because the apex has only its identity endomorphism. This contradicts the universal property of a proposed colimit.

p. 80

4.9.6 ([96]). Le reste de la présente section 4.9 est consacré à la démonstration de 4.9.3, i.e. à la vérification, pour $J \longrightarrow \mathcal{M}$, des conditions a), b) de 4.9.2. Dans toute la suite, on se fixe donc une

$$\varphi \ : \ J \longrightarrow \mathcal{M} \ = \ \mathrm{Ind}_\pi(C) \ . \tag{4.9.6.1}$$

On peut supposer J réduite, donc petite et $\operatorname{card} J \leqslant \pi_0$. On va considérer le foncteur déduit du foncteur (4.9.2.1)

$$\underline{\mathrm{Hom}}(J, C)/\varphi \longrightarrow \prod_{j \in \mathrm{Ob}\, J} C/\varphi(j) \ , \tag{4.9.6.2}$$

dont les foncteurs en question se déduisent en composant avec les foncteurs projection

$$\prod C/\varphi(j) \longrightarrow C/\varphi(j_0) \ . \tag{4.9.6.3}$$

Comme les catégories facteurs $C/\varphi(j)$ sont filtrantes (et même grandes devant π), leur produit est également filtrant, et même grand devant π (immédiat), et de même pour les produits partiels. Il s'ensuit aussitôt que les projections (4.9.6.3) sont cofinales, en vertu du [page 222]

Commentary · Exposition [ms. 221]

Why the cardinal must grow with the diagram

Corollary 4.9.5 applies the representation theorem after enlarging the class of presentable objects. Its usefulness lies precisely in not claiming that an arbitrary diagram is assembled from the originally chosen presentability rank. The preceding involution example shows how that stronger assertion can fail. At a sufficiently large rank the subcategory $\mathcal{M}_\pi$ contains the intermediate colimits needed to enforce the diagram's equations.

The exponent condition has two separate jobs. If $\pi = c^{\pi_0}$ and $c \geq \max(\pi_0, |\operatorname{Mor} J|)$, then

$$\pi^{\pi_0} = (c^{\pi_0})^{\pi_0} = c^{\pi_0 \cdot \pi_0} = c^{\pi_0} = \pi, \qquad \pi \geq 2^{\pi_0} > \pi_0.$$

This gives enough room to carry out the accessibility enlargement established earlier in Chapter XVIII, and enough room to record all arrows and equations of J. Conversely, a cardinal satisfying the displayed fixed-point condition is of the indicated form by taking $c = \pi$. No generalized continuum hypothesis is being used.

For intuition take $\mathcal{M} = \mathbf{Set}$ and let π dominate the number of arrows of J. The sets of cardinal at most π are the π^+-presentable objects. Given at most π chosen elements of the values of a J-diagram,

96. Résultats préliminaires (à la démonstration de 4.9.3).

close them under its structure maps. At most $\pi \cdot \pi = \pi$ elements are added. The resulting subdiagram has pointwise cardinal at most π. The poset of these subdiagrams is π^+-filtered: the union of at most π of them still has that bound. Their union is the original diagram. Moreover the equalizer formula in the preceding block proves that every such bounded subdiagram is π^+-presentable in $[J, \mathbf{Set}]$. This is the elementary model for the corollary. In a general accessible category there are no underlying subsets on which to perform this closure; the comma categories and their cofinal functors provide the replacement.

Lemme 4.9.6.1. *Soient I, I' deux catégories avec I' 0-connexe (p.ex. filtrante), alors le foncteur projection $pr_1 : I \times I' \longrightarrow I$ est cofinal.*

En effet, pour $i' \in \mathrm{Ob}\, I'$ `[plutôt` $i \in \mathrm{Ob}\, I$`]`, on a

$$i\backslash(I \times I') \;\simeq\; (i\backslash I) \times I' \;,$$

or $i\backslash I$ et I' étant 0-connexe, il en est de même de leur produit.

Lemme 4.9.6.2 ([97]). *Soit $(I_\alpha)_{\alpha\in A}$ une famille de catégories* filtrantes *(resp. grandes devant π). Alors $I_A \overset{déf}{=} \prod_{\alpha\in A} I_\alpha$ est filtrante (resp. grande devant π), et pour toute partie $B \subset A$, la projection canonique $I_A \longrightarrow I_B$ est cofinal.*

([98]). La stabilité de la propriété "filtrante" ou "grande devant π" par AQT `[âne qui trotte]`. On a, si $B' = A\backslash B$, $I_A \simeq I_B \times I_{B'} \longrightarrow I_B$, et on applique le lemme 4.9.6.1 pour conclure que ce foncteur est cofinal, $I_{B'}$ étant filtrante, donc 0-connexe.

Le foncteur (4.9.6.3) étant cofinal, pour établir que (4.9.2.1) l'est, il suffit de prouver qu'il [page 223]
en est ainsi du foncteur (4.9.6.2), ce que nous ferons. Notons de plus que ledit foncteur est *fidèle*, car il s'insère dans le carré commutatif

$$\begin{array}{ccc} \underline{\mathrm{Hom}}(J,C)/\varphi & \longrightarrow & \prod C/\varphi(j) \\ \downarrow & & \downarrow \\ \underline{\mathrm{Hom}}(J,C) & \longrightarrow & C^{\mathrm{Ob}\, J} \end{array} ,$$

où les flèches verticales et la deuxième flèche horizontale sont fidèles. En définitive, pour p. 81
établir le théorème 4.9.3, nous allons étudier le foncteur *fidèle* (4.9.6.2)

$$(*) \qquad \begin{array}{l} f \;:\; \underline{\mathrm{Hom}}(J,C)/\Phi \;\longrightarrow\; \prod_{j\in J_0} C/\varphi(j) \\ (\text{où on a posé } J_0 = \mathrm{Ob}\, J,\; J_1 = \mathrm{Fl}\, J) \;, \end{array}$$

et nous allons prouver :

ⓐ Ce foncteur fidèle $(*)$ est *cofinal.*

ⓑ La catégorie source $\underline{\mathrm{Hom}}(J,C)/\varphi$ est grande devant π_0.

Nous allons pour ceci dégager un lemme auxiliaire sur la cofinalité des foncteurs *fidèles* :

Commentary · Examples [ms. 222]

Products need bounds on witnesses

The product trick is the first way Grothendieck makes independent choices simultaneous. If each I_a is λ-filtered and $D : K \to \prod_a I_a$ has fewer than λ arrows, choose a cocone on each projected diagram D_a. These coordinate cocones give a cocone on D. The empty diagram supplies nonemptiness. Thus arbitrary small products of λ-filtered categories are λ-filtered, using the usual axiom of choice. No bound on the number of factors is needed: the same small diagram is being solved in every coordinate.

97. Attention, il n'est pas vrai qu'un produit infini de catégories 0-connexes est 0-connexe !

98. Cf. démonstration élégante ci-dessous, 4.9.6.6.

The projection to a subproduct is final because its comma category at (x_b) is

$$\left(\prod_{b\in B}(x_b\backslash I_b)\right)\times\left(\prod_{a\notin B} I_a\right).$$

Both factors are nonempty and filtered. This calculation includes the empty subproduct, where the target is the terminal category. For the simpler projection $I\times I'\to I$, only connectedness and nonemptiness of I' are needed, since $(i\backslash I)\times I'$ has the same connected components as I'.

Footnote 97 correctly warns against replacing filteredness by connectedness in an infinite product. Here is a full example. Let P_n be the finite fence poset on vertices $0,1,\dots,n$, with the successive relations

$$0<1>2<3>4<\cdots.$$

Its underlying undirected comparability graph is a path, so P_n is connected, and any zigzag from 0 to n has at least n nonidentity steps. Consider the two objects $a=(0)_n$ and $b=(n)_n$ of $\prod_{n\geq 1}P_n$. A zigzag with m steps between them would project to a zigzag with at most m steps from 0 to n in every coordinate. Taking $n>m$ is impossible. Thus the product is disconnected. Filteredness avoids this failure because two objects always have a common target, giving a uniformly two-step zigzag in every coordinate. The issue is a uniform finite witness, not merely the existence of coordinatewise witnesses.

[page 224]

Lemme 4.9.6.3 ([99]). *Soit $f: I\longrightarrow I'$ un foncteur, et considérons les conditions :*

a) I filtrante, f cofinal, i.e. (en présence de I filtrante) f satisfait les conditions familières (F 1), (F 2).

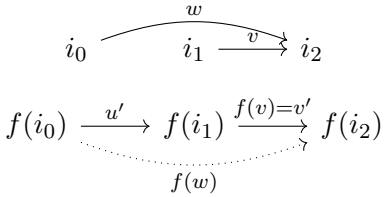
b) I' est filtrante, et les $i'\backslash I$ (pour $i'\in \mathrm{Ob}\, I'$) sont filtrantes.

b′) I' 0-connexe, satisfait PF (2) ([100]) (égalisation des double-flèches).

b″) I' est filtrante, les $i'\backslash I$ pseudo-filtrantes, i.e. satisfont à la condition PF $1'$.

c) I' filtrante, et f satisfait à F 1, ainsi que la condition suivante.

(F 3) *Pour tout couple (i_0,i_1) d'objets de I et toute flèche $f(i_0)\xrightarrow{u'}f(i_1)$, il existe un objet i_2 de I, et une flèche $v': f(i_1)\longrightarrow f(i_2)$, tels que v' et $v'u'$ soient relevables en des flèches $i_1\xrightarrow{v} i_2$ et $i_0\xrightarrow{w} i_2$ (i.e. il existe i_2 et $v: i_1\longrightarrow i_2$, $w: i_0\longrightarrow i_2$ tels qu'on ait $f(v)u'=f(w)$).*



1) On a les implications

$$a \iff b \iff b' \implies b'' \iff c\,,$$

et si f est fidèle *(p.ex. I ordonnée), les cinq conditions a), b), b′), b″), c) sont équivalentes.*

p. 82
[page 225]

2) Soit π un cardinal infini, et considérons les conditions :

a_π) I grande devant π, f cofinal, i.e. F 1, F 2.

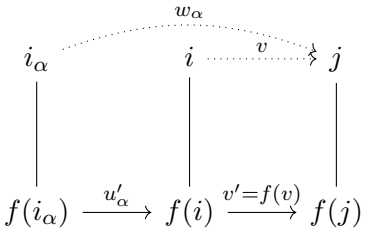
b_π) I' grande devant π, les $i'\backslash I$ ($i'\in \mathrm{Ob}\, I'$) aussi.

b_π'') I' grande devant π, les $i'\backslash I$ ($i'\in \mathrm{Ob}\, I'$) sont π-pseudo-filtrantes, i.e. satisfont PS $1_\pi{}'$.

c_π) I' grande devant π, et f satisfait F 1 et

99. Ce lemme fait partie du sorite général sur la cofinalité et les catégories filtrantes.
100. p.ex. PF 2 est satisfaite si I' est *ordonnée*, sans avoir à supposer d'avance I' filtrante, seulement 0-connexe.

F 3_π ([101]). *Pour toute famille $(i_\alpha)_{\alpha\in A}$ d'objets de I, avec* card $A \leqslant \pi$, *et $i \in \mathrm{Ob}\, I$, et une famille de flèches $u'_\alpha : f(i_\alpha) \longrightarrow f(i)$, existe $j \in \mathrm{Ob}\, I$ et une flèche $v' : f(i) \longrightarrow f(j)$, telle que v'* et *les $v'u'_\alpha$ proviennent de flèches de I, i.e. il existe des flèches $v : i \longrightarrow j$, $w_\alpha : i_\alpha \longrightarrow j$ telles que $f(v)u'_\alpha = f(w_\alpha)\ \forall\ \alpha \in A$.*



Ceci posé, on a

$$a_\pi \iff b_\pi \implies c_\pi \implies b_\pi{}'' ,$$

et si f est fidèle, ou I ordonné, les quatre conditions a_π, b_π, $b_\pi{}''$, c_π sont équivalentes.

([102]).

Commentary · Exposition [ms. 224]

What the cofinality lemma actually requires

Grothendieck has reduced a problem about natural transformations to a faithful functor into a product of comma categories. The new lemma is designed to exploit that faithfulness: one may arrange equality in the target and then reflect it back to the source. Keeping the two operations separate makes both the proof and its limitations transparent.

Write $D_d = (d \downarrow f)$ for $d \in I'$, and say explicitly that a category is *object-directed* if it is nonempty and any two objects admit maps to a common object. This is the particular property called "pseudo-filtrante" in this passage. It does not include equalization of parallel arrows and should not be confused with other uses of "pseudofiltered" in the literature. The coherent finite form of the lemma is

$$\begin{aligned} &I \text{ filtered and } f \text{ final} \iff I' \text{ filtered and every } D_d \text{ filtered,} \\ &I' \text{ filtered and every } D_d \text{ object-directed} \iff I' \text{ filtered, F1 and F3.} \end{aligned}$$

The first line implies the second. If f is faithful, the second also implies the first. F1 means that every d admits an arrow to some $f(i)$; F3 says that every arrow $f(i_0) \to f(i_1)$ becomes the image of an arrow out of i_0 after postcomposition with the image of an arrow out of i_1.

There is an actual omission in the printed condition b′: it requires only that I' be connected and satisfy PF2, without mentioning the comma categories. Taken literally it cannot be equivalent to a): the functor from a two-object discrete category to the terminal category is faithful and satisfies that printed b′, but its source is not filtered. The missing clause is "every D_d is filtered", as the proof and the explicit parenthesis on manuscript p. 236 confirm. With that clause restored, connectedness and PF2 for I' suffice; the argument below proves this slightly stronger formulation.

The nonempty requirement in b″ must likewise remain visible. If PF1′ is read only as a condition on pairs of objects, the empty category satisfies it vacuously, and $\varnothing \to 1$ disproves b″ ⇒c. Reading it as common upper bounds for all finite families, including the empty family, supplies precisely the required convention. The cardinal version explicitly quantifies over all families of cardinal at most π and therefore already includes the empty one. This is a clarification of the finite convention, not a counterexample to the intended criterion.

4.9.6.4. N.B. Dans le cas qui nous intéresse, où f est le foncteur $(*)$ (p. 223), on sait que I' est grande devant π. On voudrait établir que I l'est et que f est cofinal, i.e. la condition a_π. On le prouvera en prenant b_π (qui implique a_π puisque f est fidèle). Il reste donc à prouver, [page 226] sur le foncteur f dans $(*)$, p. 223, les deux conditions suivantes :

101. **N.B.** La condition F 3_π est tautologiquement satisfaite si f est *pleinement fidèle.*

102. Démonstration de la proposition donnée dans 4.9.6.7, page 227.

F 1) Pour tout système de flèches dans $\mathcal{M}$

$$X(j) \xrightarrow{p(j)} \varphi(j)\ , \qquad j \in J_0 = \mathrm{Ob}\, J,\ \text{les } X(j) \text{ dans } \mathrm{Ob}\, C\ ,$$

il existe un foncteur $F : J \longrightarrow C$ et $F \xrightarrow{q} \varphi$, enfin un système de flèches $X(j) \xrightarrow{u(j)} F(j)$ dans les $C/\varphi(j)$, i.e. rendant commutatif les diagrammes

$$\begin{array}{ccc} X(j) & \xrightarrow{\quad u(j) \quad} & F(j) \\ & {\scriptstyle p(j)}\searrow \quad \swarrow{\scriptstyle q(j)} & \\ & \varphi(j)\ . & \end{array}$$

F 3_π) Pour tout système $(F_\alpha)_{\alpha\in A}$ ($\mathrm{card}\, A \leqslant \pi$) d'éléments dans $\underline{\mathrm{Hom}}(C, J)$, et des flèches

$$F_\alpha \xrightarrow{p_\alpha} \varphi \qquad \text{dans } \underline{\mathrm{Hom}}(J, \mathcal{M}),$$

et pour G dans $\underline{\mathrm{Hom}}(J, C)$ et $G \xrightarrow{q} \varphi$ dans $\underline{\mathrm{Hom}}(J, \mathcal{M})$, étant donné un système de flèches $u_\alpha(j)$, p. 83

$$\begin{array}{ccc} F_\alpha(j) & \xrightarrow[\text{comm.}]{u_\alpha(j)} & G(j) \\ & {\scriptstyle p_\alpha(j)}\searrow \quad \swarrow{\scriptstyle q(j)} & \\ & \varphi(j)\ , & \end{array} \qquad u_\alpha(j) \text{ flèche dans } C/\varphi(j)\ ,$$

il existe $H \xrightarrow{q} \varphi$ dans $\underline{\mathrm{Hom}}(J, C)$ et une flèche $G \xrightarrow{v} H$ dans $\underline{\mathrm{Hom}}(J, C)/\varphi$, telle que pour tout $\alpha \in A$, le système de flèches composés [page 227]

$$\begin{array}{ccccc} F_\alpha(j) & \xrightarrow[\text{comm.}]{u_\alpha(j)} & G(j) & \xrightarrow[\text{comm.}]{v(j)} & H(j) \\ & {\scriptstyle p_\alpha(j)}\searrow & \downarrow{\scriptstyle q(j)} & \swarrow{\scriptstyle q(j)} & \\ & & \varphi(j) & & \end{array}$$

(pour $j \in J_0 \in \mathrm{Ob}\, J$) provienne d'un *foncteur* $w_\alpha : F_\alpha \longrightarrow H$.

Pour prouver 4.9.3, nous sommes donc ramené à prouver les deux propositions F 1), F 3_π) précédentes, sous l'une des deux conditions a) ou b) de 4.9.3.

Corollaire 4.9.6.5 (de 4.9.6.3). *Soit $(f_\alpha : I_\alpha \longrightarrow I'_\alpha)$ une famille de foncteurs cofinaux, les catégories source étant filtrantes. Alors le foncteur produit $f = \prod f_\alpha : I = \prod_\alpha I_\alpha \longrightarrow I' = \prod_\alpha I'_\alpha$ est cofinal.*

En effet, il faut vérifier que les catégories $i'\backslash I$, pour $i' \in \mathrm{Ob}\, I'$, sont 0-connexes. Or si $i' = (i'_\alpha)_{\alpha\in A}$, on a

$$i'\backslash I \ \simeq\ \prod_\alpha (i'_\alpha \backslash I_\alpha)\ ,$$

et par 4.9.6.3, 1°) a $\Longrightarrow$ c, les $i'_\alpha\backslash I_\alpha$ sont filtrantes, donc leur produit est filtrant (cf. 4.4.6.2), donc 0-connexe, q.e.d. [page 228]

Corollaire 4.4.6.6. *Soit π un cardinal infini. Tout produit de catégories grandes devant π est grand devant π.*

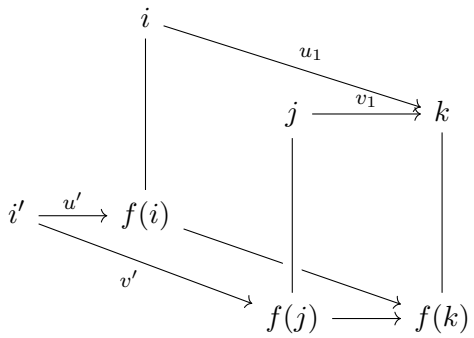
Soit $I = \prod_{\alpha\in A} I_\alpha$, soit B un ensemble de cardinal $\leqslant \pi$, il faut prouver que $I \xrightarrow{\mathrm{diag}_I} I^B$ est cofinal. Or ce foncteur s'identifie au produit des foncteurs similaires $I_\alpha \longrightarrow I_\alpha^B$, lesquels sont cofinaux, les catégories source étant de plus filtrantes. Donc on peut conclure par le corollaire 4.9.6.5.

4.9.6.7. Démonstration de 4.9.6.3. (Partie 1°.)

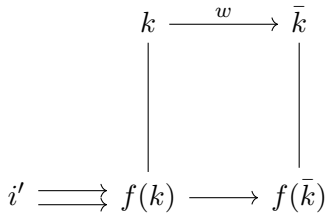
a $\Longrightarrow$ b. On sait déjà que a) implique que I' est filtrante (SGA 4 I 8.1.3 b). Reste à prouver que pour tout $i' \in \mathrm{Ob}\, I'$, $i'\backslash I$ est filtrante ([103]). Deux conditions PF 1′ et PF 2 à vérifier (en plus du fait que $i'\backslash I \neq \emptyset$, qui est connu par cofinalité de f, impliquant que $i'\backslash I$ est 0-connexe).

1) Soient (i, u'), (j, v') deux éléments de $i'\backslash I$, il faut voir qu'ils sont majorés par un troisième élément. Or I étant filtrante, i, j sont majorés par k. On aurait fini donc si on savait que $f(u_1)u' = f(v_1)v'$, soit w', car alors (k, w') ferait l'affaire.

p. 84



Mais on applique F 2 sur f à la double-flèche en question $i' \rightrightarrows f(k)$, on trouve $w : k \longrightarrow \bar{k}$, [page 229]



telle que

$$\underbrace{f(w)(f(u_1)u')}_{=f(wu_1)u'} = \underbrace{f(w)(f(v_1)v')}_{f(wv_1)v'},$$

et remplaçant k par $\bar{k}$, et u_1, v_1 par wu_1, wv_1, on trouve ce qu'on veut.

Commentary · Exposition [ms. 228]

The finite proof, with the lifting equation in its proper order

The proof is a calculus of eventual equality. It is useful to complete it once because the same operations reappear, with a cardinal bound, in the construction of sections. We use "final" for Grothendieck's right-cofinal functors.

Suppose first that I is filtered and f is final. A basic consequence is the following form of F2: if $r, s : d \rightrightarrows f(i)$, there is $w : i \to k$ such that $f(w)r = f(w)s$. Indeed (i, r) and (i, s) are joined by a finite zigzag in $(d \downarrow f)$. A cocone in I on the finite underlying graph makes the two composites from d equal. The two resulting maps from i to its apex may differ; equalizing them in I gives the single map w asserted. Together with F1, this proves that I' is filtered: two objects first map into the image of f and then to a common target; a parallel pair first maps to an object $f(i)$ and is then equalized by F2.

The same argument proves that every D_d is filtered. For two of its objects, choose a common target for their underlying objects of I, and then use F2 to make the two maps from d agree. For parallel arrows in D_d, equalize their underlying arrows in I. The resulting arrow remains in the comma category because its structure map is obtained by composition. This proves a)⇒b), including the target-filteredness which is invoked from SGA 4 in the text.

103. Court-circuit : on sait déjà que $i'\backslash I$ est 0-connexe, d'autre part elle est *localement* filtrante, car coétale sur I qui est filtrante, donc elle est filtrante.

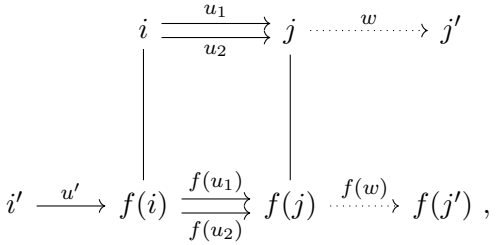
Now suppose only the corrected b′: I' is nonempty and connected, has right equalizers for parallel pairs, and each D_d is filtered. Finality follows immediately, hence I is nonempty and connected. First obtain F3 by taking a common target for $(i_0, 1)$ and (i_1, u') in $D_{f(i_0)}$. To equalize $a, b : i_0 \rightrightarrows i_1$, equalize $f(a), f(b)$ in I', use F1 to move the target into the image of f, and use F3 to replace that target arrow by an image $f(w)$. The resulting arrows wa, wb have the same image. They are parallel arrows in $D_{f(i_0)}$ from $(i_0, 1)$ to $(\operatorname{cod} w, f(wa))$, so filteredness of that comma category equalizes them in I.

For an outgoing span $i_1 \leftarrow i_0 \rightarrow i_2$, a common target in $D_{f(i_0)}$ supplies maps v_1, v_2 whose composites with the span have the same image. The equalization just proved makes the composites themselves equal. Thus every outgoing span completes to a commuting square. In a connected category this, together with parallel-pair equalization, gives filteredness: move along a finite zigzag between two objects and complete each successive outgoing span to keep a common target. This proves corrected b′)⇒a). The converse is immediate from b).

Finally suppose c). F1 makes each D_d nonempty. Given two objects of D_d, use filteredness of I' to form a common target for their maps from d, use F1 to put this target in the image of f, and apply F3 twice to lift both arrows after a further enlargement. Thus D_d is object-directed. Conversely, that property for $D_{f(i_0)}$ gives F3 from $(i_0, 1)$ and (i_1, u'), while nonemptiness gives F1. If f is faithful, a parallel pair in a comma category can be equalized in I', the equalizing arrow lifted by F3 after F1, and the resulting equality reflected by faithfulness. Hence b) follows.

The printed formula on p. 231, $f(w)v' = f(v)$, reverses the factors and is not typed: with $v : i_2 \to \bar{i}_2$, $w : i_1 \to \bar{i}_2$ and $v' : f(i_1) \to f(i_2)$ the correct equation is $f(v)v' = f(w)$. The proof above uses this order. This is an edition/source slip, not a failure of the argument. Its companion typo $\xi_1 = (i_0, v')$ must likewise be read as (i_1, v').

2) PF (2) pour $i'\backslash I$, i.e. on se donne un diagramme ci-contre `[ci-dessous]`,

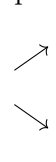


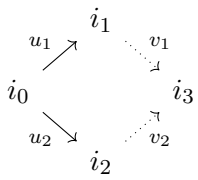
avec $f(u_1)u' = f(u_2)u'$, soit v' (double-flèche dans $i'\backslash I$ de (i, u') dans (j, v')), on veut l'égaliser à droite. Or I étant filtrant, il existe $w : j \longrightarrow j'$ telle que $wu_1 = wu_2$, or cette flèche définit une flèche $(j, v') \longrightarrow (j', f(w')v')$, qui égalise la double-flèche donnée.

b $\Longrightarrow$ b′ est clair.

b) $\Longrightarrow$ a). Comme les $i'\backslash I$ sont filtrantes, donc 0-connexes, f est cofinal, il reste à prouver que I est filtrant, donc à vérifier PF 0 (I 0-connexe), PF 1, PF 2 pour I. I est 0-connexe, car I' l'est, et f cofinal, i.e. $\mathbf{W}_0$-coasphérique, d'où PF 0.

[page 230]

Prouvons PF 1, i.e. que si  est donné, on peut compléter en un carré commutatif.



$$\begin{array}{ccccc} & & f(i_1) & & \\ & \overset{f(u_1)}{\nearrow} & & \overset{f(v_1)}{\searrow} & \\ f(i_0) & & & & f(i_3) \\ & \underset{f(u_2)}{\searrow} & & \underset{f(v_2)}{\nearrow} & \\ & & f(i_2) & & \end{array}$$

Mais considérons les objets $\xi_1 = (i_1, f(u_1))$, $\xi_2 = (i_2, f(u_2))$ de $f(i_0)\backslash I$, et utilisant PF 1′ pour cette catégorie, on trouve $\xi_3 = (i_3, w')$ dans ladite catégorie, et des flèches de ξ_1, ξ_2 dans ξ_3, i.e. $v_1 : i_1 \longrightarrow i_3$, $[v_2 : i_2 \longrightarrow i_3,]$ telles que $\underbrace{f(v_1)f(u_1)}_{f(v_1u_1)} = w'$, $\underbrace{f(v_2)f(u_2)}_{f(v_2u_2)} = w'$, donc

$$f(v_1u_1) \;=\; f(v_2u_2)\ .$$

Ainsi, on a une double-flèche $i_0 \overset{v_1u_1}{\underset{v_2u_2}{\rightrightarrows}} i_3$, égalisée par f. Si on prouve PF 2 sur I, on trouve $i_3 \xrightarrow{\alpha} \bar{i}_3$ telle que $\alpha(v_1u_1) = \alpha(v_2u_2)$, i.e. $(\alpha v_1)u_1 = (\alpha v_2)u_2$, et quitte à remplacer i_3 par $\bar{i}_3$, v_1, v_2 par αv_1, αv_2, on a gagné.

Reste à prouver PF 2 sur I, i.e. égaliser une double-flèche

$$i_0 \;\overset{v_1}{\underset{v_2}{\rightrightarrows}}\; i_1\ .$$

Or I' satisfaisant à PF 2, on peut égaliser la double-flèche $f(u_1)$, $f(u_2)$ par une flèche [page 231]
$f(i_1) \xrightarrow{v'} i'$ dans I', et comme $i'\backslash I$ non vide, on peut supposer que i' est de la forme $f(i_2)$.

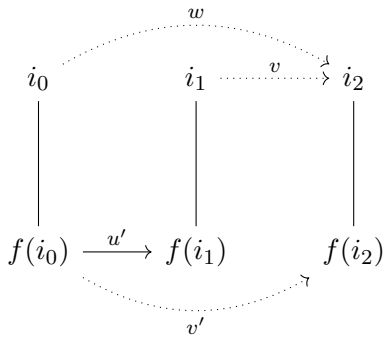
$$\begin{array}{ccccc} i_0 & \overset{u_1}{\underset{u_2}{\rightrightarrows}} & i_1 & \longrightarrow & i_2 \\ \Big| & & & & \\ f(i_0) & \overset{f(u_1)}{\underset{f(u_2)}{\rightrightarrows}} & f(i_1) & \xrightarrow{v'} & f(i_2) \end{array}$$

Admettons pour un instant la condition F 3 de c), que nous allons appliquer à (i_1, i_2, v'). Il existe donc $i_2 \xrightarrow{v} \bar{i}_2$, $w : i_1 \longrightarrow \bar{i}_2$ tels que l'on ait $f(w)v' = f(v)$. Il s'ensuit que $f(w)$ égalise $f(u_1)$ et $f(u_2)$, i.e. qu'on a $f(wu_1) = f(wu_2)$, i.e. la double-flèche initiale est remplacée par $(\bar{u}_1 = wu_1, \bar{u}_2 = wu_2)$, telle que $f(\bar{u}_1) = f(\bar{u}_2)$. En somme, on est ramené au cas d'une double-flèche (u_1, u_2) telle que $f(u_1) = f(u_2)$, soit v'. Mais alors soit $\xi_1 = (i_0, v')$, $\xi_0 = (i_0, \mathrm{id}_{f(i_0)})$, on a donc une double-flèche $(u_1, u_2) : \xi_0 \rightrightarrows \xi_1$ dans $f(i_0)\backslash I$. Cette catégorie est filtrante, donc [page 232]
cette double-flèche est égalisable à droite, ce qui implique aussitôt que la double-flèche (v_1, v_2) *dans* I est aussi égalisable à droite.

Donc il reste à prouver que b′) implique la condition F 3. Mais considérons les deux objets $\xi_0 = (i_0, \mathrm{id}_{f(i_0)})$, $\xi_1 = (i_1, u')$ de $f(i_0)\backslash I$, et utilisons la condition de filtration PF 1 sur cette catégorie, donc on trouve un objet $\xi_2 = (i_2, v')$, et deux flèches $\begin{array}{ccc} \xi_0 & \overset{w}{\searrow} & \\ & & \xi_2 \\ \xi_1 & \underset{v}{\nearrow} & \end{array}$, i.e. $i_0 \xrightarrow{w} i_2$ et

$i_1 \xrightarrow{v} i_2$ telles qu'on ait $f(w)\mathrm{id}_{i_0} = f(v)u' = v'$, d'où $f(w) = f(v)u'$, ce qui est la condition F p. 86

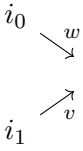
3.



Ainsi on a établi

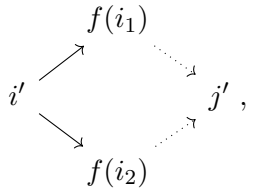
$$\text{a} \iff \text{b} \iff \text{b}' ,$$

prouvons maintenant

a $\Longrightarrow$ c. On sait déjà que a) implique I' filtrante, et il reste à établir F 3. Or I étant filtrante, il existe des flèches  . On n'a pas forcément $f(w) = f(v)u'$, on a là une double-flèche $f(i_0) = f(i_1)$ dans I'. Mais par la condition F 2 sur f, il existe $i_2 \xrightarrow{\alpha} \bar{i}_2$ dans I telle que $f(\alpha)$ égalise ladite double-flèche, donc $f(\underbrace{\alpha w}_{\bar{w}}) = f(\underbrace{\alpha v}_{\bar{v}})u'$, et quitte à remplacer i_2 par $\bar{i}_2$, et w, v par $\bar{w} = \alpha w$, $\bar{v} = \alpha v$, on gagne.

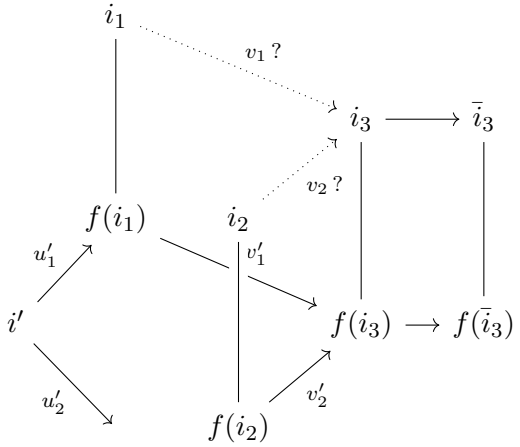
[page 233]

Reste à prouver (pour la partie 1°) que si f est *fidèle*, alors c) $\Longrightarrow$ b). Il reste donc à prouver que c) implique que pour i' dans I', $i'\backslash I$ est filtrante. On sait déjà par F 1 que $i'\backslash I$ est non vide, il reste à prouver les conditions PF 1′, PF 2 pour $i'\backslash I$. Comme I' filtrante, l'on peut compléter



et par F 1 on peut supposer j' de la forme $f(i_3)$.

p. 87



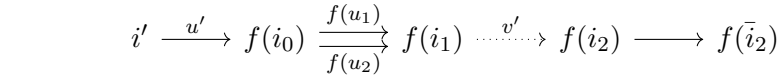
Appliquant F 3 à i_1, i_3, v'_1, on peut supposer (quitte à remplacer i_3 par $\bar{i}_3$) que v'_1 provient d'un $v_1 : i_1 \longrightarrow i_3$. Appliquant à nouveau F 3 à i_2, i_3, v'_2, on peut supposer que v'_2 également

[page 234]

provient d'un $v_2 : i_2 \longrightarrow i_3$. Alors $\xi_3 = (i_3, v_1'u_1' = v_2'u_2')$ est un majorant commun de $\xi_1 = (i_1, u_1')$, $\xi_2 = (i_2, u_2')$.

Prouvons enfin PF 2 pour $i'\backslash I$, donc on a un diagramme ci-contre [ci-dessous] avec $f(u_1)u' = f(u_2)u'$ (double-flèche dans $i'\backslash I$), on veut égaliser dans $i'\backslash I$.





Mais I' étant filtrante, on peut égaliser $(f(u_1), f(u_2))$ par $f(i_1) \longrightarrow j'$, et à cause de F 1 on peut supposer $j' = f(i_2)$. Appliquant encore F 3 à (i_1, i_2, v'), on trouve $i_1 \longrightarrow \bar{i}_2$ tel que $f(w)$ égalise $f(u_1)$, $f(u_2)$:

$$f(wu_1) \;=\; f(wu_2)\ .$$

Mais f étant *fidèle*, cela implique $wu_1 = wu_2$, et on a donc égalisé la double-flèche de $i'\backslash I$ par $w : (i_1, -) \longrightarrow (\bar{i}_2, -)$.

4.9.6.8. N.B.

1°) Si on ne suppose pas que f soit fidèle, on voit que c) implique pourtant que les $i'\backslash I$ satisfont PF 1′ (elles sont donc pseudo-filtrantes et 0-connexes). Donc f est cofinal.
On voit, par le même argument, que I satisfait PF 1′, i.e. est pseudo-filtrante. Ainsi, si on introduit la condition : [page 235]

(b″) I' est filtrante, les $i'\backslash I$ ($i \in \mathrm{Ob}\, I'$) sont pseudo-filtrantes (i.e. satisfont PF 1′) (donc f cofinal).

On trouve les implications

$$(\mathrm{a} \iff \mathrm{b} \iff \mathrm{b}') \implies (\mathrm{b}'' \iff \mathrm{c})\ ,$$

et les conditions équivalentes c), b″) impliquent que I est également pseudo-filtrante, et que pour toute double-flèche $\xi_0 \overset{u_1}{\underset{u_2}{\rightrightarrows}} \xi_1$ dans un $i'\backslash I$ ou dans I, il existe une flèche $\xi_1 \xrightarrow{v} \xi_2$ dans la même catégorie, telle que vu_1 et vu_2 : $\xi_0 \rightrightarrows \xi_2$ aient *même image dans* I'. Donc si f est fidèle, alors ces conditions c, c′ [plutôt b″] impliquent que I et les $i'\backslash I$ satisfont aussi la condition PF 2 de la double-flèche, donc sont filtrantes, donc dans ce cas les cinq conditions a) b) b′) c) c′) [plutôt b″)] sont équivalentes. Mais aussi lorsque I est ordonnée, on sait [?] que la condition de la double-flèche dans I est satisfaite automatiquement. p. 88

2°) Notons aussi que dans la condition b′), on ne peut supprimer l'hypothèse que I' satisfait PF 2 (en plus de I' 0-connexe et les $i'\backslash I$ filtrantes). Pour le voir, prenons $I' = I$, $f = \mathrm{id}$, donc l'hypothèse a ($\iff$ b $\iff$ b′) signifie que I est filtrante, f étant bien sûr cofinal. Mais b′ affaiblie signifie que I est 0-connexe et les $i\backslash I$ sont filtrantes, et on sait que ceci n'implique *pas* que I soit filtrante, même si on suppose I pseudo-filtrante. Un exemple simple est donné par I définie par [page 236]

$$\mathrm{Ob}\, I \;=\; \mathbf{N}\ , \qquad \mathrm{Hom}_I(i,j) \;=\; \begin{cases} \emptyset & \text{si } i > j \\ \{\mathrm{id}_i\} & \text{si } i = j \\ \{\pm 1\} & \text{si } i < j\ , \end{cases}$$

la composition des flèches se faisant par la multiplication des signes $\varepsilon_{k,j}\varepsilon'_{j,i} = (\varepsilon\varepsilon')_{k,i}$ (en posant $\mathrm{id}_i = (+1)_{i,i}$). Les catégories $i\backslash I$ sont isomorphes entre elles et sont *ordonnées* et pseudo-filtrantes, donc filtrantes, d'autre part, I est pseudo-filtrante, pourtant elle n'est pas filtrante, car ne satisfait pas PF 2.

3°) D'autre part, si f n'est pas fidèle, alors c) (équivalente à b'') n'implique pas a), comme on voit en prenant $I' = e$; alors b″) équivaut à I pseudo-filtrante, qui n'implique pas que I soit filtrante.

4.9.6.9. Démonstration de 4.9.6.3, partie 2°. [page 237]

Commentary · Examples [ms. 236]

Connected slices cannot replace equalization

Grothendieck's signed category measures exactly the part of filteredness lost when one retains only common upper bounds. Its objects are natural numbers, and for $n < m$ there are two arrows $n \to m$, labelled $+1$ and -1. Composition multiplies signs. Any two objects map to a larger integer, so the category is object-directed and connected. But two opposite-sign arrows can never be equalized on the right: multiplying both by the same sign leaves them opposite. It is therefore not filtered.

Fix n. The objects of $n\backslash I$ are 1_n and pairs (m, ε) for $m > n$. Between (m, ε) and (k, δ) with $m < k$ there is exactly one possible arrow, namely the arrow labelled $\delta\varepsilon^{-1}$. At equal levels the only arrows are identities between equal objects. Thus the slice is a poset, consisting of a bottom point followed by two points at each subsequent level, both below every point at a higher level. Two objects have an upper bound at any strictly larger level; hence the slice is filtered. Shifting levels shows that these slices are mutually isomorphic. Taking $f = 1_I$ gives exactly the counterexample promised on p. 236: connectedness of I' and filteredness of all D_d do not imply filteredness of I unless PF2 is retained for I'.

There is an even smaller illustration of the role of faithfulness. For a nontrivial group G, the functor $BG \to 1$ satisfies F1 and F3: there is only one target arrow, and one can choose both lifting arrows to be the identity. Its source is object-directed, but not filtered, since 1 and any $g \neq 1$ cannot be equalized by multiplying on the left by an invertible element. The functor is final because BG is connected. Finality alone is consequently weaker than having a filtered source. The same example works for every cardinal version of F3.

C'est essentiellement "la même" démonstration que pour la partie 1°).

$a_\pi) \Longrightarrow b_\pi)$. On peut court-circuiter la démonstration directe, en regardant, pour un ensemble A de cardinal $\leqslant \pi$, le diagramme

$$\begin{array}{ccc} I & \xrightarrow{\delta_A} & I^A \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f^A} \\ I' & \xrightarrow{\delta_{A'}} & I'^A \end{array}$$

[plutôt δ'_A]. Comme f est cofinal, I filtrante, f^A est aussi cofinal (4.9.8.5), d'autre part, I grande devant π signifie que δ_A est cofinal ($\forall$ A avec $\mathrm{card}\, A \leqslant \pi$). Donc $f^A\delta_A = \delta_{A'}f$ est cofinal, donc $\delta_{A'}f$ et f le sont, donc $\delta_{A'}$ aussi, ce qui signifie que I' est grande devant π.

Reste à prouver que pour i' dans I', $i'\backslash I$ est grande devant π. Cela résulte du fait que $i'\backslash I \xrightarrow{p} I$ est coétale, que $i\backslash I$ est 0-connexe, et du

Lemme 4.9.6.9.1. *Soit I une catégorie filtrante (resp. grande devant π). Alors pour toute J sur I coétale sur I et 0-connexe, J est filtrante (resp. grande devant π).*

Le cas filtrant est déjà connu (cf. II …). Pour le cas "grande devant π", on va utiliser le p. 89

Corollaire 4.9.6.9.2. *Soit J une catégorie. Pour que J soit grande devant π, il faut et il suffit qu'elle soit 0-connexe, et satisfasse PF_π 2 (cf. page 81), ainsi que la variante suivante de PF_π 1′.* [page 238]

PF_π 1) Pour tout objet x_0 de J, $x_0\backslash J$ satisfait PF_π 1′, i.e. pour toute famille de flèches $x_0 \xrightarrow{u_\alpha} y_\alpha$ ($\alpha \in A$, $\operatorname{card} A \leqslant \pi$), existe une flèche $x_0 \xrightarrow{v} z$ et des flèches $v_\alpha : y_\alpha \longrightarrow z$, avec $v_\alpha u_\alpha = v \ \forall\, \alpha \in A$.

Commentary · Exposition [ms. 237]

A connected category of elements inherits filteredness

The shortcut through a coétale functor expresses a useful structural fact. In modern language the functor $p : E \to I$ is a discrete opfibration: for every $e \in E$ and arrow $p(e) \to i$ there is a unique lift with source e. Equivalently, E is the category of elements of a covariant set-valued functor on I. The projection $(d \downarrow f) \to I$ is of this form, associated with $i \mapsto I'(d, f(i))$.

Assume I is λ-filtered and E is nonempty and connected. We give a direct proof that E is λ-filtered. Any finite zigzag in E has a common target: choose a cocone on its projected finite diagram in I and lift its legs. The lifted endpoint objects coincide along each edge by uniqueness of lifting, and hence along the entire zigzag. This proves that any two objects of E have a common target. A parallel pair in E is equalized by lifting an equalizing arrow in I; uniqueness of lifting identifies the resulting composites. Thus E is filtered.

For a family (e_a) of fewer than λ objects, choose a base object e_0 and, using the finite case, a common target z_a of e_0 and e_a. The projected arrows $p(e_0) \to p(z_a)$ have a common compatible target in I by λ-filteredness. Lift these arrows from the z_a. Their targets are all the same object, because the composites starting at e_0 are unique lifts of the same base arrow. It receives arrows from every e_a. A family of fewer than λ parallel arrows is equalized by the same lifting argument as before. Common upper bounds and simultaneous equalization now give cocones on arbitrary diagrams with fewer than λ arrows. This proves Lemma 4.9.6.9.1 completely.

The proof also explains Corollary 4.9.6.9.2. Connectedness plus the common-target condition for every family issuing from one object first gives the finite common-target property by following zigzags. For a larger family, join each member to a fixed base object and then apply the stipulated condition to those arrows from the base. Together with simultaneous equalization, this is exactly λ-filteredness. Connectedness cannot be discarded: a disjoint union of two terminal categories satisfies every condition localized at an individual object but has no cocone on the diagram consisting of one object in each component.

Démonstration. Si J est grande devant π, alors PF_π 1) résulte aussitôt de la conjonction de PF_π 1′ et PF_π 2, d'où le "il faut". Inversement, supposons que J satisfasse PF_π 1, PF_π 2, et soit 0-connexe, il faut prouver qu'elle satisfait PF_π 1′. Mais on sait déjà que les conditions en question, qui impliquent PF 1, PF 2, 0-connexe, impliquent que J est filtrante. Donc si on a une famille $(x_\alpha)_{\alpha\in A}$, $\operatorname{card} A \leqslant \pi$, choisissant un $x_0 \in \operatorname{Ob} J$, on peut pour tout α trouver y_α qui majore x_0 et x_α, d'où des flèches $\begin{array}{ccc} & & x_\alpha \\ & \searrow & \\ & & y_\alpha \\ & \nearrow & \\ x_0 & & \end{array}$, on peut leurs appliquer PF_π 1, d'où des flèches $y_\alpha \longrightarrow z$, et en composant des $x_\alpha \longrightarrow z$, q.e.d.

Prouvons maintenant le lemme. Il suffit (par le corollaire précédent) de prouver que J satisfait à PF_π 1 et PF_π 2. Or PF_π 1 est immédiat, puisque $\forall\, x_0$ dans J, $x_0\backslash J \xrightarrow{\sim} y_0\backslash I$, où y_0 est l'image de x_0 dans I. Quant à PF_π 2, si on a des $x_0 \overset{u_\alpha}{\rightrightarrows} x_1$ dans J, d'où des [page 239]
$g(x_0) \overset{g(u_\alpha)}{\rightrightarrows} g(x_1)$, on égalise dans I par $g(x_1) \xrightarrow{w} y$, mais comme g est coétale, $\exists\, x_1 \xrightarrow{v} x_2$ dans J telle que $g(v) = w$, donc on a $g(vu_\alpha) = g(vu_\beta)$ pour $\alpha, \beta \in A$, et comme g est fidèle, on en conclut $vu_\alpha = vu_\beta$, donc v égalise $(u_\alpha)_{\alpha\in A}$, q.e.d.

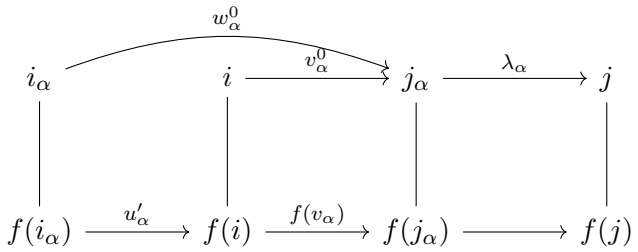
Prouvons $\mathrm{b}_\pi) \Longrightarrow \mathrm{a}_\pi)$: Si I' et les $i'\backslash I$ sont grandes devant π, alors I l'est aussi (et f est cofinal, les $i'\backslash I$ étant 0-connexes, car filtrantes). Comme on sait déjà que I est 0-connexe (I' l'étant et f cofinal), il reste, par 4.9.6.9.2, à vérifier PF_π 1 et PF_π 2.

La démonstration donnée dans 1° pour le cas similaire filtrant (implication $\mathrm{b}' \Longrightarrow \mathrm{a}$, pages [page 240]
231-232), s'applique mutatis mutandis. (La propriété F 3 a déjà été établie dans la partie 1°), comme conséquence de b, et peut donc être utilisée librement.)

Donc on a prouvé $a_\pi \iff b_\pi$. Prouvons $a_\pi \implies c_\pi$. On sait déjà que a_π implique que I' est grande devant π, et bien sûr la condition F 1, reste à établir la validité de la condition F 3_π. Par F 3, $\forall\, \alpha \in A$ on peut trouver j_α dans I et $w^0_\alpha : i_\alpha \longrightarrow j_\alpha$, $v^0_\alpha : i \longrightarrow j$ tels que

$$f(v_\alpha)u'_\alpha \;=\; f(w_\alpha)\ .$$

p. 90



Comme I satisfait PF 1_π (cf. 4.9.6.9.2), il existe h dans I et des $\lambda_\alpha : j_\alpha \longrightarrow j$, tels que $\lambda_\alpha v^0_\alpha = \lambda$ soit indépendant de α, alors on a pour tout $\alpha \in A$

$$f(\lambda)u'_\alpha \;\Big(\;=\; f(\lambda_\alpha)\underbrace{f(v_\alpha)u'_\alpha}_{f(w_\alpha)}\;\Big) \;=\; f(\underbrace{\lambda_\alpha w^0_\alpha}_{w_\alpha})\ ,$$

donc $\lambda : i \longrightarrow j$ et les $w_\alpha = \lambda_\alpha w^0_\alpha : i_\alpha \longrightarrow j$ satisfont aux conditions voulues.

$c_\pi \implies b''_\pi$. Il faut prouver que si c_π est satisfait, alors les $i'\backslash I$ satisfont PF $1'_\pi$.

Soit donc $\xi_\alpha = (i_\alpha, u'_\alpha)$ ($\alpha \in A$, $\operatorname{card} A \leqslant \pi$) une famille d'éléments de $i'\backslash I$, comme I' satisfait à PF 1_π (cf. 4.9.6.9.2), il existe j' dans I' et des flèches $u''_\alpha : f(i_\alpha) \longrightarrow j'$, telles que $u''_\alpha u'_\alpha = u'$ indépendant de α. Comme f satisfait F 1, on peut supposer que j' est de la forme $f(j)$. Appliquant F 3_π aux i_α, j, u''_α, on trouve $v : j \longrightarrow k$ et des $w_\alpha : i_\alpha \longrightarrow k$, tels que [page 241]

$$f(v)u''_\alpha \;=\; f(w_\alpha) \qquad \forall\, \alpha \in A\ ,$$

d'où

$$f(w_\alpha)u'_\alpha \;=\; f(v)\,\underbrace{u''_\alpha u'_\alpha}_{u'} \;=\; f(v)u' \qquad \text{indépendant de } \alpha,$$

soit w', donc les w_α définissent une flèche des ξ_α dans $\xi = (k, w')$, q.e.d.

Commentary · Exposition [ms. 240]

Simultaneous lifting at the cardinal bound

The cardinal proof requires one simultaneous enlargement. Repeating F3 separately for an infinite family would give unrelated targets, so the common-domain form of filteredness is essential. Put $\lambda = \pi^+$. For a functor $f : I \to I'$, with I' λ-filtered, let $F3_\pi$ have exactly the meaning stated on p. 225. Then F1 and $F3_\pi$ are equivalent to every $(d \downarrow f)$ being nonempty and admitting common targets for all families of at most π objects. The reverse implication, not displayed in the printed implication chain, follows by applying this property to $(i, 1)$ and the objects (i_a, u'_a) of $(f(i) \downarrow f)$. There are at most π of these objects because π is infinite.

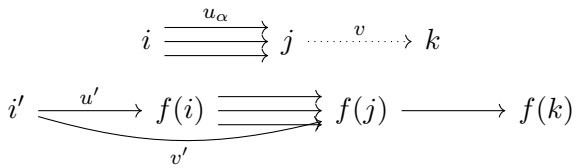
For the forward implication, a family $(i_a, r_a : d \to f(i_a))$ first acquires a compatible target in I'. By F1 its target may be replaced by $f(j)$. Apply $F3_\pi$ to all arrows $f(i_a) \to f(j)$ at once. The resulting maps $i_a \to k$ make the composites from d equal, so they are a common target in the comma category. If f is faithful, a family of at most π parallel comma arrows is equalized by first equalizing their images in I', moving the target into the image of f, and lifting the resulting arrow by the ordinary F3. Faithfulness reflects the common image. The comma categories are therefore λ-filtered.

We also fill in the cardinal descent which the source leaves to "mutatis mutandis". Suppose I' and every $(d \downarrow f)$ are λ-filtered. For a diagram $D : K \to I$ with at most π arrows, choose a cocone $fD \to d$ in I' and then a map $d \to f(i)$ by F1. The resulting family $fD(k) \to f(i)$ lifts simultaneously by $F3_\pi$ to maps $w_k : D(k) \to j$. Their naturality equations hold after applying f. For each arrow $a : k \to l$ of K, the pair $w_k, w_l D(a)$ is therefore a parallel pair in $(fD(k) \downarrow f)$ with

target $(j, f(w_k))$. Each such pair can be equalized in I by filteredness of that comma category. There are at most π choices of an equalizing map out of j; filteredness of $(f(j) \downarrow f)$ gives a compatible common successor for them. Composing w_k with its map from j makes every naturality equation hold. This is the required cocone on D.

Finally, if I is λ-filtered and f final, its comma categories are connected discrete opfibrations over I, hence λ-filtered by the previous block. The target I' inherits λ-filteredness by choosing image targets for all objects of a small diagram, and then using the filtered comma categories to impose its at most π relations; equivalently, the diagonal argument printed on p. 237 applies. These arguments prove all intended cardinal implications, with no need to assume π regular. The only regular cardinal used in the modern formulation is π^+.

$\mathrm{b}''_\pi \Longrightarrow \mathrm{b}_\pi$ si f est fidèle. On est ramené à prouver que si b''_π est satisfait, f fidèle, alors les catégories $i'\backslash I$, dont on sait déjà qu'elles satisfont PF 1_π, satisfont également PF 2_π. Soient donc $\xi = (i, u')$, $\eta = (j, v')$ deux objets de $i'\backslash I$, et $(u_\alpha)_{\alpha\in A}$ une famille de flèches $\xi \longrightarrow \eta$ (avec $\operatorname{card} A \leqslant \pi$), donc des flèches dans I telles que $f(u_\alpha)u' = v' \ \forall\, \alpha \in A$.



Comme I' est grande devant π, on peut égaliser par $f(j) \longrightarrow k'$, et comme $k'\backslash I \neq \emptyset$, on peut supposer k' de la forme $f(k)$. Or d'après l'implication $\mathrm{b}'' \Longrightarrow \mathrm{a} \Longrightarrow \mathrm{c}$ de la partie 1°) (cf. page 235) et comme $\mathrm{b}''_\pi \Longrightarrow \mathrm{b}''$, on sait déjà que f satisfait à F 3. En l'appliquant à j, k et $f(j) \longrightarrow f(k)$, on peut supposer que $f(j) \longrightarrow f(k)$ provient de $v : j \longrightarrow k$. On aura donc $f(v)f(u_\alpha) = f(v)f(u_\beta)\ \forall\, \alpha, \beta \in A$, donc $f(vu_\alpha) = f(vu_\beta)$. Comme f est fidèle, cela implique que $vu_\alpha = vu_\beta$, i.e. v égalise les u_α. D'ailleurs, v peut être considérée comme une flèche *dans* $i'\backslash I$, de (j, v') dans $(k, f(v)v')$. Q.e.d. [page 242] p. 91

4.9.6.10. Quand I est *ordonné*, alors les quatre conditions a_π, b_π, c_π, b''_π sont équivalentes, car les $i'\backslash I$ sont ordonnés, et il est claire qu'alors $\mathrm{b}''_\pi \Longleftrightarrow \mathrm{b}_\pi$.

Commentary · Development (commentators') [ms. 242] [page 243]

Replacing faithfulness by eventual equality in its fibres

The proof isolates a weaker hypothesis than faithfulness. In approximation problems, two different arrows may become equal after applying the comparison functor, yet already become equal at a later stage of the source. That is sufficient for the same conclusion.

Proposition (relative eventual equalization). [ours] Let π be infinite, I' be π^+-filtered, and $f : I \to I'$ satisfy F1 and $\mathrm{F3}_\pi$. Then the following are equivalent: I is π^+-filtered; and every family $(a_t : x \to y)_{t\in T}$ with $|T| \leq \pi$ and with $f(a_t)$ independent of t is equalized by some arrow $y \to z$ in I. Under these conditions f is final and each $(d \downarrow f)$ is π^+-filtered.

Proof. Necessity follows immediately from filteredness. Conversely, the preceding simultaneous-lifting argument shows that every comma category is nonempty and has common targets for at most π objects. For a family of at most π parallel arrows in such a comma category, equalize their images in I', use F1 to move the target into the image of f, and apply F3 to replace the resulting arrow by an image $f(w)$. The composites wa_t now have identical images. The asserted eventual-equality condition gives a further arrow h equalizing them in I. It is also an arrow of the comma category when the new structure map is defined by composition. Thus the comma category is π^+-filtered. The cardinal descent proved above gives π^+-filteredness of I, and nonempty connected comma categories give finality. □

Faithfulness implies the extra condition with $y \to z$ the identity. A functor from an ordered category has the same property for the still simpler reason that any parallel arrows are already equal. The proposition describes exactly what the printed proof uses when neither simplification is available. We checked the cofinality criteria in SGA 4, I, 8.1 and the filtered-colimit criteria in The Stacks Project Authors (2026d, Section 4.19); we did not find this particular cardinal formulation with the relative kernel condition stated there. The argument is a rearrangement of Grothendieck's proof, offered as a usable extension rather than a priority claim.

4.10 Le théorème de représentation indicielle : démonstration

On va d'abord reformuler le théorème 4.9.3 sous forme plus générale, en termes de catégories *cofibrées*. Soit

$$\mathcal{D} \longrightarrow J \tag{4.10.1}$$

un foncteur cofibrant, où J est équivalent à une petite catégorie, et de même pour les catégories fibres $\mathcal{D}_j$ ($j \in \mathrm{Ob}\, J$). On se donne un cardinal infini π. Soit

$$\mathcal{M}_j \;=\; \mathrm{Ind}_\pi(\mathcal{D}_j) \qquad (j \in \mathrm{Ob}\, J)\ , \tag{4.10.2}$$

alors les $\mathcal{M}_j$ sont les fibres d'une catégorie cofibrée $\mathcal{N}$ sur J, que je désigne aussi par $\mathrm{Ind}_\pi(\mathcal{D}/J)$, donc on pose

$$\mathcal{N} \;=\; \mathrm{Ind}_\pi(\mathcal{D}/J)\ . \tag{4.10.3}$$

On a donc une inclusion pleinement fidèle de catégories cofibrées sur J

$$\mathcal{D} \hookrightarrow \mathcal{N}\ ,$$

d'où une inclusion

$$\underline{\Gamma}(\mathcal{D}/J) \longrightarrow \underline{\Gamma}(\mathcal{N}/J)\ . \tag{4.10.4}$$

Dans le cas où $\mathcal{D}$ est de la forme $C \times J$, alors $\mathcal{N}$ est de la forme $\mathcal{M} \times J$, avec $\mathcal{M} = \mathrm{Ind}_\pi(C)$, [page 244]
et l'inclusion (4.10.3) n'est autre (à isomorphisme près) que

$$\underline{\mathrm{Hom}}(J, C) \longrightarrow \underline{\mathrm{Hom}}(J, \mathcal{M})\ .$$

Revenant au cas général, on voit encore que :

1°) Le deuxième membre de (4.10.3) est stable par $\varinjlim$ grandes devant π, lesquelles se calculent fibre à fibre.

2°) Le premier membre de (4.10.3) est formé d'objets π-accessibles du second.

D'où un foncteur *pleinement* fidèle

$$\mathrm{Ind}_\pi(\underline{\Gamma}(\mathcal{D}/J)) \hookrightarrow \underline{\Gamma}(\mathcal{N}/J)\ . \tag{4.10.4}$$

([104]).

Commentary · Exposition [ms. 243]

Sections turn many approximation problems into one

Grothendieck now replaces a diagram in one category by a family of categories whose objects are transported along the arrows of the indexing category. The gain is substantial: a single theorem will approximate a diagram, a compatible family of objects, or a section of a fibred construction. His *cofibrant functor* means a Grothendieck opfibration, not a cofibration in a model category. For an arrow $a : i \to j$, choose transport $a_* : \mathcal{D}_i \to \mathcal{D}_j$. A section consists of objects X_i and maps $a_* X_i \to X_j$, satisfying unit and composition equations. These maps need not be invertible or cocartesian. The coherence isomorphisms of a cleavage are understood throughout.

To compare his cardinal conventions with current ones, put $\kappa = \pi^+$. A category "large before π" is κ-filtered: every diagram with at most π arrows admits a cocone. An object called π-accessible here is κ-presentable. Thus $\mathrm{Ind}_\pi \mathcal{D}_i$ in these pages is the usual $\mathrm{Ind}_\kappa \mathcal{D}_i$. This shift matters even when $\pi = \aleph_0$: Grothendieck is then asking for countable-diagram cocones, not merely finite-diagram cocones.

104. [Les numéros 4.10.1 à 4.10.4 sont utilisés deux fois.]

The completion of the opfibration is obtained by applying Ind_κ to every fibre and every transport functor. Its sections may be approximated pointwise without difficulty. The theorem asks whether those approximations can be made into sections before taking their colimit. The condition on one ordinal-shaped colimit supplies exactly the operation needed to turn successive approximations to the structure maps into an actual section.

There is a size qualification to the preliminary assertion on p. 244. Pointwise κ-presentable sections are κ-presentable when $|\operatorname{Mor} J| < \kappa$, as in the theorem immediately following. Without it the assertion is false. Take J discrete with $|J| \geq \kappa$ and fibres $\mathrm{Set} = \mathrm{Ind}_\kappa(\mathrm{Set}_{<\kappa})$. The constant singleton family is pointwise presentable. Index a diagram by the $< \kappa$-sized subsets $A \subset J$, putting a singleton in coordinates of A and the empty set elsewhere. This index is κ-filtered and the colimit is the constant singleton, but no stage receives a map from that family. The presentability claim therefore uses the forthcoming size bound; this does not affect Theorem 4.10.1 itself.

p. 92

Théorème 4.10.1. *Supposons qu'on ait*

$$\operatorname{card} \text{réd}\, J \leqslant \pi \,, \tag{4.10.1.1}$$

et qu'il existe un ordinal limite α, tel que :

1) $\operatorname{card} \alpha \leqslant \pi$.

2) Les limites inductives de type $I_\alpha = \{\beta \mid \beta < \alpha\}$ sont représentables dans les catégories $\mathcal{D}_j$ ($j \in \operatorname{Ob} J$), et pour toute flèche $\alpha : j \longrightarrow j'$ dans J [trop de α], $\alpha^* : \mathcal{D}_j \longrightarrow \mathcal{D}_{j'}$ commute auxdites limites. [page 245]

Alors le foncteur canonique pleinement fidèle (4.10.4) est une équivalence de catégories.

Ceci va être contenu dans le résultat un peu plus précis :

Corollaire 4.10.2. *Les hypothèses étant celles de 4.10.1, avec en plus*

$$\operatorname{card} J \leqslant \pi \tag{4.10.2.1}$$

(qui équivaut à (4.10.1.1) si J est réduite, cas auquel on peut évidemment se réduire dans 4.10.1). Soit $\varphi \in \operatorname{Ob} \underline{\Gamma}(\mathcal{N}/J)$, et considérons le foncteur canonique fidèle *:*

$$\underline{\Gamma}(\mathcal{D}/J)/\varphi \longrightarrow \prod_{j \in J} C_j/\varphi(j) \,. \tag{4.10.2.2}$$

Alors la catégorie source $\underline{\Gamma}(\mathcal{D}/J)/\varphi$ est grande devant π, et le foncteur envisagé est cofinal.

De ceci, le théorème résulte de façon essentiellement soritale, tout comme dans le cas particulier 4.9.3 précédent (cf. page 223) ([105]). [page 246]

Pour établir le corollaire, comme la catégorie but de (4.10.2.2) est grande devant π, comme produit de telles catégories (cf. 4.9.6.6, page 228), et le foncteur fidèle, il suffit en vertu de 4.9.6.3, 2°, de vérifier les conditions F 1 et F 3_π (page 225) pour le foncteur envisagé.

On va interpréter le foncteur (4.10.2.2) en termes d'une catégorie cofibrée

$$\mathcal{D}_\varphi \longrightarrow J \,, \tag{4.10.5}$$

dont les fibres sont les $\mathcal{D}_j/\varphi(j)$. On note que si $\alpha : j \longrightarrow j'$ est une flèche de J, alors on a

$$\begin{cases} \alpha_* : C_j \longrightarrow C_{j'} \\ \text{et} \quad \alpha_*(\varphi(j)) \longrightarrow \varphi(j') \,, \end{cases}$$

d'où un foncteur composé

p. 93
[page 247]

$$C_j/\varphi(j) \longrightarrow C_{j'}/\alpha^*(\varphi(j)) \longrightarrow C_{j'}/\varphi(j') \,,$$

105. Un peu court ! Il faudrait expliciter l'argument avec soin – il y en a pour deux ou trois pages !

avec transitivité à isomorphisme canonique près etc. On note que l'on a un isomorphisme canonique

$$\underline{\Gamma}(\mathcal{D}_\varphi/J) \;\simeq\; \underline{\Gamma}(\mathcal{D}/J)/\varphi\ ,$$

de sorte que le foncteur fidèle canonique (4.10.2.2) s'identifie au foncteur

$$\underline{\Gamma}(\mathcal{D}_\varphi/J) \longrightarrow \prod_{j\in \mathrm{Ob}\, J} (\mathcal{D}_\varphi)_j\ . \tag{4.10.6}$$

Commentary · Exposition [ms. 246]

From cofinal approximation to the equivalence

Grothendieck explicitly asks that the passage from the cofinality assertion to the representation theorem be written out. We do so because it also explains which information the section category retains. Let φ be a section of $\mathcal{N} = \mathrm{Ind}_\kappa(\mathcal{D}/J)$ and let

$$I = \Gamma(\mathcal{D}/J)/\varphi.$$

After replacing the fibres and J by small equivalent categories, this category is small: its objects comprise a small family of section data and maps in locally small fibres. Corollary 4.10.2 makes I κ-filtered and makes the evaluation functor $I \to \mathcal{D}_j/\varphi_j$ final for every j. Indeed it is the composite of the asserted final functor to the product and a final product projection. The canonical diagram $I \to \Gamma(\mathcal{D}/J)$ has a cocone to φ. At j its colimit is

$$\operatorname*{colim}_{(D\to\varphi)\in I} D_j \simeq \operatorname*{colim}_{(d\to\varphi_j)\in \mathcal{D}_j/\varphi_j} d \simeq \varphi_j.$$

These isomorphisms are compatible with every structure map, because the original cocone consists of section morphisms. Filtered colimits of sections are pointwise: the completed transport functors preserve them. Consequently the colimit section is φ, proving essential surjectivity.

For full faithfulness, let D be a section with values in $\mathcal{D}_j$. Maps from D to a section Y form the equalizer

$$\mathrm{Hom}_\Gamma(D,Y) \longrightarrow \prod_j \mathrm{Hom}(D_j, Y_j) \rightrightarrows \prod_{a:i\to j} \mathrm{Hom}(a_* D_i, Y_j).$$

Every object appearing on the left of these fibrewise hom functors belongs to the corresponding $\mathcal{D}_j$ and is therefore κ-presentable. Both products have fewer than κ factors. Since κ-filtered colimits of sets commute with κ-small limits, this equalizer preserves those colimits. The usual ind hom formula now gives full faithfulness of the comparison functor. This proves the omitted passage without assuming that arbitrary pointwise morphisms are natural.

The auxiliary opfibration $\mathcal{D}_\varphi$ records the same argument more economically. Its transport sends $(d \to \varphi_i)$ to $a_* d \to a_* \varphi_i \to \varphi_j$, and a section of it is precisely a section D together with $D \to \varphi$. A colimit of an ordinal chain in such a slice is computed in $\mathcal{D}_j$, with its induced map to φ_j. Lemma 4.10.4 ensures that this remains the same colimit after embedding in the ind-category. Thus no extra colimit hypothesis is lost in passing to the slices.

Le corollaire 4.10.2 sera alors une conséquence du résultat plus général suivant :

Corollaire 4.10.3. *Soit $\mathcal{E} \longrightarrow J$ un foncteur cofibrant, et π un cardinal infini. On suppose :*

a) $\mathrm{card}\, J \leqslant \pi$.

b) Les catégories fibres $\mathcal{E}_j$ sont équivalentes à des petites catégories, et grandes devant π.

c) Il existe un ordinal limite tel que :

1°) $\mathrm{card}\, \alpha \leqslant \pi$.

2°) Les limites inductives de type I_α sont représentables dans les catégories $\mathcal{E}_j$, et les foncteurs cochangement de base $\lambda_* : \mathcal{E}_j \longrightarrow \mathcal{E}_{j'}$ y commutent. (C'est le cas p.ex. si les limites inductives filtrantes dénombrables *sont représentables dans les $\mathcal{E}_j$, et si les α_* y commutent ; ou s'il existe un cardinal $\pi_0 \leqslant \pi$ et un cardinal $\pi_0' > \pi_0$, tel que les limites inductives grandes devant π_0 et de cardinal $\leqslant \pi_0'$ soient représentables dans les $\mathcal{E}_j$, et que les α_* y commutent.)* [page 248]

Sous ces conditions, le foncteur canonique fidèle

$$\underline{\Gamma}(\mathcal{E}/J) \longrightarrow \prod_{j\in \mathrm{Ob}\, J} \mathcal{E}_j \tag{4.10.3.1}$$

est cofinal, et la catégorie source est grande devant π.

Il est ± immédiat que les hypothèses de 4.10.1 impliquent que celles de 4.10.3 sont satisfaites pour $\mathcal{E} = \mathcal{D}_\varphi \longrightarrow J$ (4.10.5). Pour ceci, on utilise le [page 249]

Lemme 4.10.4. *Soient C une catégorie, et π un cardinal infini. Alors le foncteur d'inclusion $C \longrightarrow \mathcal{M} \overset{\text{déf}}{=} \mathrm{Ind}_\pi C$ commute aux $\varinjlim$ de cardinal $\leqslant \pi$.*

Soit en effet I telle que $\mathrm{card}\, I \leqslant \pi$, et $(x_i)_{i\in I}$ un système inductif sur C, y ayant une limite x. Soit d'autre part $X = \varinjlim_{I'} \xi_{i'}$ un objet de $\mathrm{Ind}_\pi C$, avec I' grande devant π. On a

donc p. 94

$$\begin{aligned}
\mathrm{Hom}_{\mathcal{M}}(x, X) &\simeq \varinjlim_{I'} \mathrm{Hom}(x, \xi_{i'}) \\
&\simeq \varinjlim_{I'} \varprojlim_{I} \mathrm{Hom}(x_i, \xi_{i'}) \\
&\overset{\text{SGA 4 I 9.8}}{\simeq} \varprojlim_{I} \underbrace{\varinjlim_{I'} \mathrm{Hom}(x_i, \xi_{i'})}_{\simeq \mathrm{Hom}(x_i, X)} \\
&\simeq \varprojlim_{I} \mathrm{Hom}(x_i, X)\ ,
\end{aligned}$$

ce qui prouve que x est aussi une $\varinjlim$ des x_i dans $\mathcal{M}$ tout entier.

Ainsi, on est réduit à prouver 4.10.3. Appliquant 4.9.6 2°, on est ramené à prouver, pour [250] le foncteur envisagé, les conditions F 1 et F 3_π (page 225).

Dans les arguments qui suivent, on va utiliser les conditions a), b) de 4.10.3, mais non, pour l'instant, la condition c.

Lemme 4.10.5. *Sous les conditions de 4.10.3 a) b) (à l'exclusion de c), pour que (4.10.3.1) soit cofinal et $\Gamma(\mathcal{E}/J)$ grande devant π, il faut et il suffit que le foncteur envisagé satisfasse à la condition F 1, rappelée ci-dessous, et à la variante F′ 1 de celle-ci énoncée ci-dessous :*

F 1) *Pour tout système $(X(j))_{j\in \mathrm{Ob}\, J}$ d'éléments $X(j) \in \mathrm{Ob}\, \mathcal{E}_j$, existe une* section *$G$ de $\mathcal{E}$ sur J, et un système de flèches*

$$X(j) \longrightarrow G(j) \qquad (j \in ObJ)\ . \tag{4.10.5.1}$$

F′1) *Sous les conditions de* F 1 *précédent, supposons de plus une section F de $\mathcal{E}$ sur J et un système de flèches* [251]

$$F(j) \longrightarrow X(j) \qquad (j \in \mathrm{Ob}\, J)\ ,$$

alors G et le système de flèches (4.10.5.1) peut être choisi de telle façon que les flèches composées

$$F(j) \longrightarrow X(j) \longrightarrow G(j)$$

définissent un homomorphisme de sections

$$F \longrightarrow G$$

(dans la catégorie $\underline{\Gamma}(\mathcal{E}/J)$).

([106]).

106. **N.B.** Les conditions F 1 et F′ 1 équivalent à dire que le foncteur envisagé satisfait aux conditions F1 et F3, vu encore (en vertu de 4.9.6.3, 1°) que $\underline{\Gamma}(\mathcal{E}/J)$ est *filtrante* sans plus, et le foncteur fidèle (4.10.3.1) cofinal. Cela implique déjà que $\underline{\Gamma}(\mathcal{E}/J)$ est grande devant π.

Commentary · Exposition [ms. 250]

Why one compatible family controls many families

The force of Lemma 4.10.5 is that a construction with one prescribed section suffices to establish the simultaneous lifting condition for at most π sections. The fibres supply enough room to remove all failures of naturality before the section construction starts.

Suppose sections F_a and F are given, together with arbitrary component maps $u_{a,j} : F_{a,j} \to F_j$, where $|A| \leq \pi$. For an arrow $b : i \to j$, compare the two maps from $b_*F_{a,i}$ to F_j: one uses the structure of F_a followed by $u_{a,j}$, the other uses $b_*u_{a,i}$ followed by the structure of F. There are at most π such pairs at each j. Since $\mathcal{E}_j$ is κ-filtered, they can all be equalized by one map $F_j \to X_j$. This use of transport is implicit in the source's arrows in the total category.

Apply F′1 to the section F and the maps $F_j \to X_j$. It produces a section G and maps $X_j \to G_j$ for which $F \to G$ is natural. Each $F_a \to G$ is now natural as well: its naturality square is the composite of the already equalized square ending at X_j and the naturality square for $F \to G$. Thus F3$_\pi$ holds. The faithful cofinality criterion gives both κ-filteredness of $\Gamma(\mathcal{E}/J)$ and finality of evaluation.

Conversely, suppose evaluation is final and its source is κ-filtered. F1 follows. Given $F_j \to X_j$, apply F1 to obtain $X_j \to H_j$ for some section H. The ordinary F3 condition for the resulting pointwise arrow $F \to H$ then gives a section map $H \to G$ whose composite from F is natural. This is F′1. The equivalence asserted in the lemma is therefore exact; it is special to this evaluation functor and the bound on the number of naturality equations.

Démonstration. Il faut voir que F 1 & F′ 1 équivaut à F 1 & F 3$_\pi$. L'implication $\Longleftarrow$ est ± immédiate, c'est de $\Longrightarrow$ que nous aurons besoin, donc

$$\text{F } 1 \;\&\; \text{F}' \; 1 \quad \Longrightarrow \quad \text{F } 3_\pi \; .$$

Pour ceci, réénonçons F 3$_\pi$ dans le contexte présent. On se donne une famille $(F_\alpha)_{\alpha \in A}$, p. 95
$\operatorname{card} A \leqslant \pi$, d'objets de $\underline{\Gamma}(\mathcal{E}/J)$, un objet F de la même catégorie, et pour tout j dans $\operatorname{Ob} J$, un système de flèches

$$F_\alpha(j) \xrightarrow{u_\alpha(j)} F(j) \; ,$$

mais sans supposer que pour α fixé, j variable

cela définisse un homomorphisme de sections $F_\alpha \longrightarrow F$. On cherche une section G de $\mathcal{E}$ sur [252]
J, et un homomorphisme de sections $F \longrightarrow G$, tels que pour tout α, le système des composés

$$F_\alpha(j) \longrightarrow F(j) \longrightarrow G(j)$$

définisse un homomorphisme de sections $F_\alpha \longrightarrow G_\alpha$. Soit donc $\lambda : j \longrightarrow j'$ une flèche de J, considérons les carrés (*non commutatifs*)

$$(*) \qquad \begin{array}{ccc} F_\alpha(j) & \xrightarrow{u_\alpha(j)} & F(j) \\ {\scriptstyle F_\alpha(\lambda)}\downarrow & ! & \downarrow{\scriptstyle F(\lambda)} \\ F_\alpha(j') & \xrightarrow{u_\alpha(j')} & F(j') \end{array}$$

(indexés par $\alpha \in A$). Pour j' fixé dans J, le nombre de ces carrés est majoré par $\operatorname{card} \operatorname{Ob}(J/j') \times \operatorname{card} A \leqslant \pi \cdot \pi = \pi$. Il s'ensuit, comme $\mathcal{E}_{j'}$ est grande devant π, qu'il existe un objet $X(j')$ de $\mathcal{E}_{j'}$, et une flèche $v(j') : F(j') \longrightarrow X(j')$ de telle façon que pour tout $\alpha \in A$, et tout $\lambda \in \operatorname{Ob} J/j'$, le carré déduit de $(*)$

en composant avec $F(j') \longrightarrow X(j')$ soit commutatif. On fait ce choix pour tout objet j' [253]
de J. Appliquons alors F′ 1 à F et au système de flèches $F(j) \xrightarrow{v(j)} X(j)$, on trouve G et un système de flèches

$$w(j) \; : \; X(j) \longrightarrow G(j)$$

de telle façon que les flèches composées

$$w(j)v(j): \qquad F(j) \longrightarrow G(j), \quad F(j) \to X(j) \to G(j)$$

définissent un homomorphisme de sections. Je dis que cet homomorphisme a la propriété voulue pour F 3_π, i.e. que pour tout $\alpha \in A$, le système des composés

$$F_\alpha(j) \longrightarrow F(j) \longrightarrow G(j)$$

(pour $j \in \mathrm{Ob}\, J$ variable) définit un homomorphisme de sections $F_\alpha \longrightarrow G$. En effet, il suffit pour cela de considérer, pour une flèche λ de J, le diagramme

p. 96 [254]

$$\begin{array}{ccccccc} F_\alpha(j) & \xrightarrow{u_\alpha(j)} & F(j) & \xrightarrow{v(j)} & X(j) & \xrightarrow{w(j)} & G(j) \\ \downarrow{\scriptstyle F_\alpha(\lambda)} & ! & \downarrow{\scriptstyle F(\lambda)} & & \text{comm.} & & \downarrow{\scriptstyle G(\lambda)} \\ F_\alpha(j') & \xrightarrow{u_\alpha(j')} & F(j') & \xrightarrow{v(j')} & X(j') & \xrightarrow{w(j')} & G(j') \,. \end{array}$$

Le lemme 4.10.5 étant prouvé, tout revient donc à prouver la condition F 1, et son complément F′ 1, qui se prouvera dans la même haleine.

4.10.6. On se donne donc le système d'objets

$$X(j) \in \mathrm{Ob}\, \mathcal{E}_j \qquad (j \in \mathrm{Ob}\, J)\,, \tag{4.10.6.1}$$

et le cas échéant, un objet F de $\underline{\Gamma}(\mathcal{E}/J)$, et un système de flèches

$$F(j) \xrightarrow{u_j} X(j)\,, \tag{4.6.10.2}$$

et on se propose de construire G de $\underline{\Gamma}(\mathcal{E}/J)$, et un système de flèches

$$X(j) \xrightarrow{w(j)} G(j)\,, \tag{4.10.6.3}$$

de telle façon que (si on a la donnée supplémentaire des $u(j)$) les flèches composées

$$F(j) \xrightarrow{u(j)} X(j) \xrightarrow{v(j)} G(j)$$

soient fonctorielles en j.

On va faire une construction transfinie sur les ordinaux α tels que [255]

$$\mathrm{card}\, \alpha \leqslant \pi$$

d'un système inductif

$$(X_\alpha(j))_\alpha$$

d'objets de $\prod_{j \in \mathrm{Ob}\, J} \mathcal{E}_j$. On va d'abord ignorer la donnée éventuelle (4.10.6.2). On posera

$$X_0(j) = X(j)\,, \tag{4.10.6.4}$$

et on va donner la prescription pour passer de X_α à $X_{\alpha+1}$. La construction inclura aussi, pour toute flèche $\lambda : j' \longrightarrow j$ de J et tout α, la donnée de

$$X_\alpha(j) \xrightarrow{t_\alpha(\lambda)} X_{\alpha+1}(j)\,, \qquad \text{flèche (dans } \mathcal{E}) \text{ au dessus de } \lambda, \tag{4.10.6.5}$$

et ces flèches, et les flèches de transition $X_\alpha(j) \xrightarrow{\tau_\alpha(j)} X_{\alpha+1}(j)$, doivent satisfaire aux trois conditions suivantes : p. 97 [256]

(i) Si λ est une flèche identique $\mathrm{id}_J : J \longrightarrow J$, alors $\boxed{t_\alpha(\lambda) = \tau_\alpha(j).}$

(ii) Dans le diagramme

$$\begin{array}{ccccccccc} X_\alpha(j') & \xrightarrow{\tau_\alpha(j')} & X_{\alpha+1}(j') & \longrightarrow & X_{\alpha+2}(j') & \longrightarrow & X_{\alpha+3}(j') & \longrightarrow \\ & \searrow^{t_\alpha(\lambda)} & \bullet & \searrow^{t_{\alpha+1}(\lambda)} & & \searrow & & \cdots\searrow \\ X_\alpha(j') & \longrightarrow & X_{\alpha+1}(j') & \xrightarrow{\tau_{\alpha+1}(j')} & X_{\alpha+2}(j') & \xrightarrow{\tau_{\alpha+2}(j')} & X_{\alpha+3}(j') & \longrightarrow \end{array},$$

$\tau_{\alpha+2}(j)$ "rend le carré $\bullet$ commutatif", i.e. égalise $t_{\alpha+1}(\lambda)\tau_\alpha(j')$ et $\tau_{\alpha+1}(j)t_\alpha(\lambda)$:

$$\boxed{\tau_{\alpha+2}(j)t_{\alpha+1}(\lambda)\tau_\alpha(j') \;=\; \tau_{\alpha+2}(j)\tau_{\alpha+1}(j)t_\alpha(\lambda)\;.}$$

(iii) Si on a deux flèches consécutives $\lambda : j'' \longrightarrow j'$, $\mu : j' \longrightarrow j$, on a donc le diagramme

$$\begin{array}{ccccccc} X_\alpha(j'') & \longrightarrow & X_{\alpha+1}(j'') & \longrightarrow & X_{\alpha+2}(j'') & \longrightarrow & X_{\alpha+3}(j'') \\ & \searrow^{t_\alpha(\lambda)} & & \searrow & & \searrow & \\ X_\alpha(j') & \xrightarrow{t_\alpha(\mu\lambda)} & X_{\alpha+1}(j') & \longrightarrow & X_{\alpha+2}(j') & \longrightarrow & X_{\alpha+3}(j') \\ & \searrow & & \searrow^{t_{\alpha+1}(\mu)} & & \searrow & \\ X_\alpha(j) & \longrightarrow & X_{\alpha+1}(j) & \xrightarrow{\tau_{\alpha+1}(j)} & X_{\alpha+2}(j) & \xrightarrow{\tau_{\alpha+2}(j)} & X_{\alpha+3}(j)\;, \end{array}$$

$$\boxed{\tau_{\alpha+2}(j)t_{\alpha+1}(\mu)t_\alpha(\lambda) \;=\; \tau_{\alpha+2}(j)\tau_{\alpha+1}(j)t_\alpha(\mu\lambda)}$$

Ainsi, pour définir les $X_{\alpha+1}(j)$, il faut tenir compte de toutes ces contraintes. Donc : [257]

1°) Pour j fixé, $X_{\alpha+1}(j)$ doit être but des flèches

$$X_\alpha(j') \xrightarrow{t_\alpha(\lambda)} X_{\alpha+1}(j)\;,$$

pour les $\lambda : j' \longrightarrow j$ dans $\mathrm{Ob}\, J/j$ – y compris pour $\lambda = \mathrm{id}_j$, auquel cas $t_\alpha(\lambda) : X_\alpha(j) \longrightarrow X_{\alpha+1}(j)$ jouera le rôle du morphisme de transition $\tau_\alpha(j)$.

2°) Si α est de la forme $\alpha' + 2$, i.e. s'il n'est ni ordinal limite, ni successeur d'un ordinal limite, alors on exige de plus, pour les morphismes de transition $\tau_\alpha(j)$, d'égaliser les couples de flèches prévus dans (ii) et (iii) ci-dessus.

Comme l'ensemble de ces données dans 1°), ou de ces contraintes dans 2°), est de cardinal $\leqslant \pi$, et que la catégorie $\mathcal{E}_j$ est grande devant π, on voit qu'on peut bel et bien trouver $X_{\alpha+1}(j)$ et des $t_\alpha(\lambda) : X_\alpha(j') \longrightarrow X_{\alpha+1}(j)$, de façon à satisfaire (si $\alpha = \alpha'+2$) aux contraintes rappelées dans 2°. Cela achève la description de la construction de $X_{\alpha+1}(\bullet)$, en termes de $X_\beta(\bullet)$ ($\beta \leqslant \alpha$) et des $t_\beta(\lambda)$ ($\beta < \alpha$). [258]

Il faut enfin définir $X_\alpha(\bullet)$ pour α un ordinal limite. Cette fois, on lui demande seulement de recevoir les $X_\beta(\bullet)$ pour $\beta < \alpha$. Comme l'ensemble I_α de ces β est de cardinal $\leqslant \pi$, et que les $\mathcal{E}_j$ sont grandes devant π, on peut bel et bien trouver les $X_\alpha(j)$.

On a donc prouvé l'essentiel du

p. 98 [259]

Commentary · Development (commentators') [ms. 258]

A transfinite compatibility condition must be added

Grothendieck's iterative construction has the right mechanism: add all needed arrows, and then move farther along the filtered system until the equations hold. At an internal limit ordinal, however, the displayed conditions on consecutive stages no longer compare the new arrows with all earlier ones. The inference on p. 262 from condition (ii) to a morphism of ind-objects needs a repair.

A counterexample to that inference. [ours] Let $J = B(\mathbb{Z}/2)$ and take the constant opfibration with fibre a small skeleton of countable sets. This fibre has a terminal object, hence is $\aleph_1$-filtered,

and it admits countable ordinal colimits. Set $X_\beta = \{0,1\}$ with identity transition maps for every countable ordinal β. For the nonidentity group element s, set $t_\beta(s) = \mathrm{id}$ when $\beta < \omega$, and set $t_\beta(s)$ equal to the transposition when $\beta \geq \omega$. Put $t_\beta(1) = \mathrm{id}$ throughout. Conditions (i)–(iii) hold: consecutive indices never cross from a finite ordinal to ω, and on either side the maps form a representation of $\mathbb{Z}/2$. But along the chain of length $\omega + \omega$ the induced maps from its constant stages would have to be both the identity and the transposition. They define no morphism of its ind-objects. This disproves the intermediate inference, not the section theorem.

Strengthened construction. [ours] Under 4.10.3(a),(b), fix a nonzero limit ordinal δ with $|\delta| \leq \pi$. Write $r_{\gamma\beta} : X_\beta \to X_\gamma$ for the transitions, and write a map above $a : i \to j$ as $t_\beta(a) : a_* X_\beta(i) \to X_{\beta+1}(j)$. The objects and maps can be chosen so that, for every $\beta < \gamma < \delta$,

$$t_\gamma(a)\, a_* r_{\gamma\beta} = r_{\gamma+1,\beta+1}\, t_\beta(a), \tag{B1}$$

and so that the unit condition holds and the composition equations become equal after a later transition. Prescribed maps $F_j \to X_0(j)$ from a section can also be made natural after the first step.

Here is the full induction. Having constructed the system through stage γ, first choose in each fibre a common target Y_j of $X_\gamma(j)$ and all $a_* X_\gamma(i)$ for arrows $a : i \to j$. This uses at most π objects. For every $\beta < \gamma$, the proposed two sides of (B1) are now parallel maps into Y_j. Postcompose the chosen maps by one equalizing arrow, simultaneously for every such pair. There are at most $\pi \cdot \pi = \pi$ pairs.

At this same step impose all previously formed composition equations. For $a : i \to j$, $b : j \to k$ and $\beta + 2 \leq \gamma$, compare, in $\mathcal{E}_k$, the maps obtained from

$$t_{\beta+1}(b)\, b_* t_\beta(a) \quad \text{and} \quad t_\beta(ba)$$

by transport to $X_\gamma(k)$ and then to Y_k. Again there are at most π pairs; add them to the equalizations already required. Coherence isomorphisms identify $(ba)_*$ with $b_* a_*$. Choose the identity transport map to be the transition itself. At the first step also equalize $t_0(a) a_* u_i$ and $r_{10}(j) u_j F(a)$ for every a, where $u_j : F_j \to X_0(j)$ is the optional datum. Filteredness supplies a single further target accommodating all these equations. Call it $X_{\gamma+1}(j)$.

At a limit stage $\gamma < \delta$, choose a cocone on the entire earlier chain in each fibre; this defines X_γ and the transitions $r_{\gamma\beta}$. Its diagram has at most π arrows. No colimit is required at this stage. At the next successor step (B1) is imposed for every earlier β, including those before this limit. This is precisely the additional condition absent from the printed construction. All equations already imposed survive postcomposition, so the induction is consistent.

Suppose now that the δ-chain colimits exist and transport preserves them. Put $G_j = \operatorname{colim}_{\beta<\delta} X_\beta(j)$. Equation (B1) makes the maps $a_* X_\beta(i) \to G_j$ a compatible cocone; preservation gives $a_* G_i \to G_j$. The composition equations hold on each stage after some further stage, and hence hold in G. The unit equations hold by construction. Thus G is a section receiving X_0, and the first-step equations make $F \to G$ natural when prescribed. This proves F1 and F′1, completing the general ordinal case of the theorem. For $\delta = \omega$, the source's consecutive-stage argument already suffices.

The parenthesis in 4.10.3(c) has a related endpoint issue. If $\pi_0 = \pi$, its hypotheses need not supply an ordinal of cardinal at most π. They still suffice for the conclusion: take $\delta = \pi_0^+$, which is at most π_0' and is large before π_0. Every proper initial segment has cardinal at most π, so the induction above runs through all stages below δ; only the final colimit has the larger size. Alternatively, imposing $\pi_0 < \pi$ makes the printed implication to condition (c) valid as stated.

We checked the related constructions in SGA 4, I, 9.21 and the modern limit results cited below. We did not find this particular defect in the displayed XVIII argument discussed there. The strengthened induction is supplied as a proof repair, with no claim of priority.

Lemme 4.10.7. *Sous les conditions de 4.10.3 a) b), et pour un système $X(\bullet) \in \mathrm{Ob} \prod_{j \in \mathrm{Ob}\, J} \mathcal{E}_j$ donné, on peut trouver un système inductif $(X_\alpha(i))_\alpha$ dans $\prod \mathcal{E}_j$, indexé par les ordinaux α telle que* $\operatorname{card} \alpha \leqslant \pi$, *et pour toute flèche $\lambda : j' \longrightarrow j$ dans J des flèches $t_\alpha(\lambda) : X_\alpha(j') \longrightarrow X_\alpha(j)$, de telle façon que l'on ait les conditions suivantes.*

1°) $X_0(\bullet) = X(\bullet)$.

2°) Les conditions (i) (ii) (iii) (p. 256).

3°) Si on se donne F dans $\operatorname{Ob}\underline{\Gamma}(\mathcal{E}/J)$, et des flèches $u(j) : F(j) \longrightarrow X(j) = X_0(j)$, on peut supposer que pour toute flèche $\lambda : j' \longrightarrow j$ dans J, on a

$$(*) \qquad \begin{array}{ccccc} F(j') & \xrightarrow{u(j')} & X_0(j') & & \\ {\scriptstyle F(\lambda)}\downarrow & & & \searrow^{t_0(\lambda)} & \\ F(j) & \xrightarrow{u(j)} & X_0(j) & \xrightarrow{\tau_0(j)} & X_1(j) \end{array} \qquad \text{commutatif.}$$

Il reste seulement à prouver ce dernier complément 3°. C'est une contrainte uniquement [260] sur la construction de $X_1(\bullet)$ et des $t_0(\lambda)$. Supposons donc que l'on ait construit $X_1(\bullet)$ sans tenir compte de cette contrainte. Pour j fixé, il faut donc "rendre commutatif" tous ces carrés, i.e. toutes ces double-flèches $(t_0(\lambda)u(j'), \tau_0(j)u(j)F(\lambda))$ de but $X_1(j)$. Comme l'ensemble de ces double-flèches est de cardinal $\operatorname{card}\operatorname{Ob}\mathcal{E}/j \leqslant \pi$, on peut trouver $X_1(j) \longrightarrow X_1'(j)$ dans $\mathcal{E}_j$, égalisant toutes ces double-flèches, donc quitte à remplacer $X_1(j)$ par $X_1'(j)$, on gagne.

Il est temps enfin de faire appel à l'hypothèse c) de 4.10.3. Soit donc α un ordinal limite, [261] comme stipulé dans cette condition. Comme $\operatorname{card}\alpha \leqslant \pi$, a fortiori $\operatorname{card}\beta \leqslant \pi$ si $\beta < \alpha$, on a donc un système inductif

$$(X_\beta(\bullet))_{\beta \in I_\alpha}$$

dans $\prod_{j\in J}\mathcal{E}_j$. Comme les $\varinjlim$ de type I_α existent dans $\mathcal{E}_j$, on peut donc considérer

$$G(j) \;=\; \varinjlim_{\beta\in I_\alpha} X_\beta(j) \;\in\; \operatorname{Ob}\mathcal{E}_j \;.$$

Si $\lambda : j' \longrightarrow j$ est une flèche de J, on en déduit une λ-flèche

$$G(\lambda) \;:\; G(j) = \varinjlim X_\beta(j) \longrightarrow G(j') \simeq \varinjlim X_{\beta+1}(j')$$

par $G(\lambda) = \varinjlim t_\beta(\lambda)$. Il faut faire attention que les $t_\beta(\lambda)$ ne définissent pas un homomorphisme [262] de systèmes projectifs, mais par la condition (ii), p. 256, ils définissent un homomorphisme d'ind-objets, donc

un homomorphisme sur les $\varinjlim$ ([106]). La condition (i) assure que $\lambda \longmapsto G(\lambda)$ transforme p. 99 identités en identités, et la condition (iii) que l'on a $G(\mu\lambda) = G(\mu)G(\lambda)$. Donc G apparaît comme une section de $\mathcal{E}$ sur J. On a d'autre part des flèches

$$X(j) \overset{\text{déf}}{=} X_0(j) \longrightarrow G(j) = \varinjlim_{\beta\in I_\alpha} X_\beta(j) \;,$$

ce qui prouve F (1). D'autre part, la condition 3°) du lemme prouve la validité de F′ (1). Cela achève la démonstration de 4.10.3, et par là de 4.10.1 et de 4.9.3.

[page 263]

Commentary · Examples [ms. 262]

A section is a module with an extra structure map

The section theorem becomes concrete over the walking arrow. Let $R \to S$ be a homomorphism of rings and let its transport functor be $S \otimes_R - : R\text{-Mod} \to S\text{-Mod}$. A section is a triple (M, N, θ) with $\theta : S \otimes_R M \to N$. A map of sections is a pair (f, g) satisfying $g\theta = \theta'(1 \otimes f)$. Thus the condition imposed by the section category is one familiar commutative square.

This category is equivalent to the category of left modules over

$$T = \begin{pmatrix} R & 0 \\ S & S \end{pmatrix},$$

106. Attention, c'est ici qu'on utilise tacitement le fait que le foncteur $\lambda_* : \mathcal{E}_{j'} \longrightarrow \mathcal{E}_j$ commute aux $\varinjlim$ de type I_α, ce qui implique que les inclusions $\mathcal{E}_j \hookrightarrow \mathcal{E}$ commutent à ce type de $\varinjlim$.

where the lower-left copy of S is the (S,R)-bimodule supplied by the ring map. The module corresponding to (M,N,θ) is $M \oplus N$, with action

$$\begin{pmatrix} r & 0 \\ s & t \end{pmatrix} (m,n) = (rm, \theta(s \otimes m) + tn).$$

Bilinearity of θ makes this action associative; conversely the two diagonal idempotents of T split any T-module into M and N, and its lower-left action recovers θ. These constructions are inverse on objects and maps.

Choose an infinite $\pi \geq |R|, |S|$ and put $\kappa = \pi^+$. A triple is the union of its subtriples whose two underlying sets have cardinal at most π. To construct one containing specified subsets of at most π elements, take the R-submodule M_0 they generate, then the S-submodule N_0 generated by the prescribed elements and the image of $S \otimes_R M_0$. Both sets still have cardinal at most π, since π is infinite. Any collection of at most π such subtriples is contained in another, by the same construction. Their poset is therefore κ-filtered and their colimit is the original triple.

Each small triple is κ-presentable. Indeed its T-module has at most π elements, hence a presentation with at most π generators and relations: use all its elements as generators and all relations in the free module, whose underlying set also has cardinal at most π. A map from this presentation into a κ-filtered colimit factors at one stage, and its relations and the equality of two such maps can be witnessed at one later stage. This directly verifies the presentability assertion. The construction illustrates both tasks in Grothendieck's proof: enlarge the objects enough to contain the structure map, then impose its equations.

Corollaire 4.10.8 (du théorème 4.10.1). *Soit $\mathcal{N} \longrightarrow J$ un foncteur cofibrant, tel que les fibres soient des catégories accessibles (4.8), les foncteurs cochangement de base $\lambda_* : \mathcal{N}_{j'} \longrightarrow \mathcal{N}_j$ accessibles, enfin J équivalente à une petite catégorie. Alors la catégorie $\mathcal{M} = \underline{\Gamma}(\mathcal{N}/J)$ est accessible. Plus précisément, soit π_0 un cardinal infini tel que les catégories $\mathcal{N}_j$ soient π_0-accessibles, et que les foncteurs λ_* soient π_0-accessibles, et soit π un cardinal infini tel que l'on ait :*

ⓐ $\pi^{\pi_0} = \pi$.

ⓑ $\forall\ \lambda : j' \longrightarrow j$ *dans* J, $\lambda_* : \mathcal{E}_j \longrightarrow \mathcal{E}_{j'}$ *applique* $(\mathcal{E}_j)_{\pi_0}$ *dans* $(\mathcal{E}_{j'})_{\pi_0}$.

ⓒ $\operatorname{card} \text{réd}\, J \leqslant \pi$.

Alors $\mathcal{M}$ est π-accessible, et plus précisément, on a

$$\mathcal{M} \ \simeq\ \operatorname{Ind}_\pi(\mathcal{M}_\pi)\ ,$$

où $\mathcal{M}_\pi$ est la sous-catégorie pleine de $\mathcal{M}$ formée des sections F de $\mathcal{M}$ sur J telles que l'on ait $F(j) \in (\mathcal{E}_j)_\pi\ \forall\ j \in \operatorname{Ob} J$, $(\mathcal{E}_j)_\pi$ désignant la sous-catégorie pleine de $\mathcal{E}_j$ formée des objets π-accessibles de $\mathcal{E}_j$. [page 264]

En effet, l'hypothèse au début implique l'existence de π_0 et de π satisfaisant les conditions dites. Les conditions a) et b) sur π, plus le fait que les foncteurs λ_* soient π_0-accessibles, impliquent que la sous-catégorie strictement pleine $\mathcal{N}_\pi$ de $\mathcal{N}$ dont les fibres sont les $(\mathcal{N}_j)_\pi$, est une sous-catégorie cofibrée, et que de plus $\mathcal{N}$ est J-équivalent à $\operatorname{Ind}_\pi(\mathcal{N}_\pi/J)$. D'autre part, les fibres de $\mathcal{N}_\pi/J$ sont stables par $\varinjlim$ grandes devant π_0 et de cardinal $\leqslant \pi$, donc la condition "ordinale" de 4.10.1 est vérifiée (en y faisant $\mathcal{D} = \mathcal{N}_\pi$). Comme de plus $\operatorname{card} J \leqslant \pi$, on en conclut par 4.10.1 que $\mathcal{M} \simeq \operatorname{Ind}_\pi(\mathcal{M}_\pi)$.

[page 265]

Commentary · Context [ms. 264]

The earlier language of cardinal filtrations

Grothendieck is returning to a problem already treated in the foundations of SGA 4: how a category assembled from small objects retains a small supply of generators after one imposes compatible structure. Exposé I first develops ind-objects and simultaneous presentations in Section 8, then cardinal filtrations and categories of sections in Section 9. The concrete antecedents are

8.8.2–8.8.5 and 9.22–9.24 (Artin et al., 1972). The references to SGA in this chapter therefore identify specific constructions, not merely a historical affinity.

The present argument makes a useful separation within that method. One cardinal bounds the arrows, equations, and chosen approximations. Another colimit shape supplies the completion of the successive corrections. Once those two roles have been isolated, a theorem about ordinary diagrams becomes a theorem about sections, because the correction is performed fibre by fibre and then checked against transport. The distinction between the category of all sections and its cartesian or cocartesian part must survive that passage: invertibility of a structure map is additional structure to enforce, not a consequence of ordinary naturality.

This is also a way to read the source's repeated requests to write out apparently formal steps. The same filtered system must witness an object, its maps, and their relations. A proof which merely chooses representatives of the separate objects has not yet constructed that system. The cofinality criterion and the iteration are precisely the two parts which supply it. They remain ordinary categorical arguments here; the homotopical relevance comes from applying them to categories of diagrams, sections, and specified classes of arrows.

Corollaire 4.10.9. *Soit $\mathcal{N} \longrightarrow J$ une "cofibration en paratopos" ([107]), i.e. un foncteur cofibrant tel que les fibres soient des catégories paratopos, et les foncteurs cochangement de base soient des foncteurs images inverses (resp. images directes) pour des morphismes de paratopos, i.e. qu'ils commutent aux (petites) $\varinjlim$ (resp. aux petites $\varprojlim$). Supposons de plus J équivalente à une petite catégorie. Alors $\underline{\Gamma}(\mathcal{N}/J)$ est une catégorie paratopos.*

En effet, c'est une catégorie accessible en vertu de 4.10.8, d'autre part l'hypothèse sur les λ_* implique aussitôt que $\underline{\Gamma}(\mathcal{N}/J)$ est stable par petites $\varprojlim$ (resp. par petites $\varprojlim$ `[plutôt` $\varinjlim$ `?]`). Donc $\underline{\Gamma}(\mathcal{N}/J)$ est une catégorie paratopos en vertu de 4.8.9 b $\Longrightarrow$ a (resp. c $\Longrightarrow$ a). p. 100

Corollaire 4.10.10. *Soit $\mathcal{N} \longrightarrow J$ une catégorie* fibrée *(non cofibrée!), telle que les fibres soient des catégories paratopos et les foncteurs changement de base soient des foncteurs images directes pour des morphismes de paratopos (i.e. des foncteurs accessibles, commutant aux petites $\varprojlim$). Supposons de plus J équivalente à une petite catégorie. Alors $\underline{\Gamma}(\mathcal{N}/J)$ est une catégorie paratopos.* [page 266]

Commentary · Exposition [ms. 265]

Paratoposes and the direction of pointwise limits

The application to paratoposes uses the characterization established earlier in this chapter: an accessible category with all small limits, equivalently an accessible category with all small colimits, is locally presentable. "Paratopos" here should not be read as asserting the additional exactness properties of a Grothendieck topos. Module categories, including the preceding example, already lie in the intended range.

In the direct-image alternative we retain the words "for a morphism of paratoposes": the transports are actual right adjoints and are therefore accessible. Their bare preservation of limits, mentioned in the parenthesis, is not by itself the hypothesis needed to invoke 4.10.8. Indeed, to see accessibility of a right adjoint R, take a small strongly generating family of presentable objects a in its target. Choose a regular cardinal bounding the presentability ranks of both the a and their images La under its left adjoint. Testing the filtered-colimit comparison for R by $\mathrm{Hom}(a, R-) \simeq \mathrm{Hom}(La, -)$ now makes it a bijection for every a, hence an isomorphism. This gives one common rank. The next corollary explicitly includes accessibility.

For ordinary sections of an opfibration, limits are pointwise whenever the fibres have them, with no preservation requirement on transport. Given a diagram of sections X^d, put $L_j = \lim_d X^d_j$. For $a : i \to j$, the maps $a_*L_i \to a_*X^d_i \to X^d_j$ form a cone and therefore induce $a_*L_i \to L_j$. Their unit and composition equations follow after composing with each limit projection. A cone of sections factors uniquely through the L_j, and the same projections verify naturality of the factorization. This proves the limit universal property in the section category.

By contrast, constructing colimits pointwise uses preservation of those colimits by a_*, since the needed arrow is $a_* \operatorname{colim} X^d_i \to \operatorname{colim} X^d_j$. Thus the colimit-preserving case of 4.10.9 gives pointwise colimits as well as limits. In the limit-preserving case, pointwise limits suffice, together

107. terme ambigu, cf. la double explication.

with accessibility, for local presentability. Both printed occurrences of $\varinjlim$ in the proof are defensible. The editors' tentative replacement of the second by $\varprojlim$ is not needed and does not follow from the stated preservation hypothesis. We retain the printed text and its question rather than turning a tentative editorial suggestion into a mathematical correction.

For a fibration the directions reverse. A section carries maps $X_i \to a^* X_j$. Colimits are pointwise without a preservation condition, while pointwise limits require that a^* preserve them. Consequently 4.10.10 follows either from accessibility of sections and pointwise limits, or by replacing each right-adjoint inverse-image functor by its left adjoint, obtaining the equivalent description by an opfibration. The arrow labels in its printed proof are inconsistent with the direction $a : i \to j$; the adjunction is correctly $a_! : \mathcal{N}_i \rightleftarrows \mathcal{N}_j : a^*$.

En effet, l'hypothèse implique que pour toute flèche $\lambda : j' \longrightarrow j$ dans J, le foncteur $\lambda_* : \mathcal{N}_{j'} \longrightarrow \mathcal{N}_j$ admet un adjoint à gauche $\lambda^* : \mathcal{N}_j \longrightarrow \mathcal{N}_{j'}$, d'où s'ensuit que $\mathcal{N} \longrightarrow J$ est aussi cofibrant, les foncteurs de cochangement de base commutant aux petites $\varinjlim$. Donc on peut appliquer 4.8.9, qui donne le résultat voulu.

Question 4.10.11. *Soit $\mathcal{N} \longrightarrow J$ un foncteur* fibrant, *tel que les $\mathcal{N}_j$ soient des catégories accessibles, et les foncteurs λ^* de changement de base accessibles, J étant équivalente à une petite catégorie. Je conjecture que sous ces conditions, $\underline{\Gamma}(\mathcal{N}/J) \overset{\text{déf}}{=} \mathrm{Hom}(J, \mathcal{N})$ est accessible,* [page 267] *et qu'il en est de même de* $\mathrm{Hom}\ \underline{\mathrm{cart}}_J(J, \mathcal{N})$.

C'est ce qui est prouvé dans SGA 4 I 9.24 sous des hypothèses légèrement plus fortes (à l'aide de l'ancienne notion trop forte à mon gré, de filtrations cardinales). Je serais étonné que la démonstration (un peu longuette) de loc. cit. ne s'applique pas au cas général. Mais on aimerait avoir un énoncé qui coiffe celui-ci et 4.10.8, dans le cas d'un foncteur $\mathcal{N} \longrightarrow J$ qui ne serait supposé ni fibrant ni cofibrant, mais …(???).

4.11 Application à l'existence des limites dans une catégorie de la forme $\mathrm{Ind}_{\pi_0}(C)$.

Proposition 4.11.1. *Soit C une catégorie équivalente à une petite catégorie, π un cardinal infini, posons $\mathcal{M} = \mathrm{Ind}_\pi(C)$. Soit Φ un ensemble de catégories de cardinal réduit $\leqslant \pi$. On se propose d'examiner si $\mathcal{M}$ est "stable par $\varinjlim$ de type Φ" (i.e. stable par $\varinjlim$ de type I, quel que soit $I \in \Phi$). Il suffit pour cela que C est stable par $\varinjlim$ de type Φ, et que de plus il existe un ordinal limite α avec* $\mathrm{card}\,\alpha \leqslant \pi$ *et tel que C soit stable par $\varinjlim$ de type I_α (p.ex. tel que $I_\alpha \in \Phi$). Inversement, si $\mathcal{M}$ est stable par $\varinjlim$ de type Φ, et si on suppose de plus C karoubienne (i.e. stable par facteurs directs), alors C est stable par $\varinjlim$ de type Φ. Dans l'un et l'autre cas, le foncteur d'inclusion $C \hookrightarrow \mathcal{M}$ commute aux $\varinjlim$ envisagées.* [page 268]

Commentary · Present state [ms. 267]

The question about fibred sections is settled

Grothendieck asks for accessibility of ordinary and cartesian sections under accessible inverse-image functors. Both assertions are now proved. The first construction is an oplax limit of the pseudofunctor $J^{\mathrm{op}} \to \mathrm{Acc}$ classifying the fibration; the second is its pseudolimit. After the ranks have been enlarged uniformly, Positselski (2024, Theorems 8.1, 8.3 and Corollary 9.2) give these conclusions and pointwise descriptions of the presentable objects. The hypotheses on a common smaller chain-colimit cardinal and on preservation of presentables in the pseudolimit theorem are essential to that precise fixed-rank statement. This supplies a direct modern answer to 4.10.11, rather than merely an analogy with it.

Here is how the categorical construction encodes the question. Form the product $P = \prod_j \mathcal{N}_j$. For each $a : i \to j$, a potential section needs an arrow $X_i \to a^* X_j$. An inserter adds all these arrows to an object of P. Equifiers then impose the equations for identity arrows and composable pairs. The result is the category of ordinary sections. To obtain cartesian sections, add inverses to the structure maps and equify the two composites with the identity maps. The coherence isomorphisms of the

pseudofunctor occur in the composition equations. This accounts for the actual section category, rather than a category in which compatibility is imposed only up to an unspecified isomorphism.

The required common cardinals can be chosen because there are only set-many fibres and transport functors. First choose a regular λ for which all fibres are accessible and all transports preserve λ-filtered colimits. A λ-chain is itself λ-filtered, so these chain colimits exist and are preserved. Then choose a sufficiently large regular $\kappa > \lambda$, with the appropriate sharp cardinal inequality, so that all fibres are κ-accessible, J is κ-small, and all transports preserve κ-presentables when needed. The uniformization mechanism developed in 4.12 explains the last step.

SGA 4, I, 9.22–9.24 already treats ordinary and cartesian sections, using cardinal filtrations and stronger closure assumptions (Artin et al., 1972). The new formulation separates the smaller colimits used to construct compatible structure from the larger filtered colimits used to present arbitrary objects. The wish for an encompassing theorem is realized by closure of accessible categories under the relevant products, inserters, equifiers and pseudolimits. It does not assert closure under arbitrary strict equalizers of accessible functors, nor does it give a hypothesis-free theorem for an arbitrary functor to J.

([108]).

Démonstration. a) Condition suffisante. On applique 4.9.3 pour conclure que pour toute catégorie J de cardinal réduit $\leqslant \pi$ (en particulier pour $J \in \Phi$), tout foncteur $\varphi : J \longrightarrow \mathcal{M} = \mathrm{Ind}_\pi(C)$ admet une représentation indicielle $\Phi : J \times I \longrightarrow C$, avec I grande devant π. Ceci, plus la stabilité de C par $\varinjlim$ de type J, suffit à établir l'existence de la $\varinjlim$ de φ. p. 101 [page 269]

La réciproque provient du fait que si C est karoubienne, alors son image essentielle dans $\mathcal{M}$ n'est autre que $\mathcal{M}_\pi$, laquelle est stable dans $\mathcal{M}$ par toute $\varinjlim$ de cardinal $\leqslant \pi$ représentable dans $\mathcal{M}$. Si $\mathcal{M}$ est stable par $\varinjlim$ de type Φ, il en est de même de $\mathcal{M}_\pi$, donc de C, q.e.d.

Corollaire 4.11.2. *Pour que $\mathcal{M}$ soit stable par $\varinjlim$ filtrantes (resp. par petites $\varinjlim$), il suffit que C le soit par $\varinjlim$ filtrantes (resp. par $\varinjlim$)* de cardinal $\leqslant \pi$, *et c'est aussi nécessaire si C est karoubienne.*

Supposons C stable par $\varinjlim$ filtrantes (resp. par $\varinjlim$ de cardinal $\leqslant \pi$, alors par 4.11.1 $\mathcal{M}$ est stable par ce même type de limites inductives. D'autre part, il est stable par $\varinjlim$ grandes devant π. Il suffit donc de prouver le [page 270]

Commentary · Exposition [ms. 269]

Transporting colimits through an ind-presentation

The representation theorem pays for itself immediately. Once a diagram $X : J \to \mathrm{Ind}_\kappa C$ is written as $X_j = \mathrm{colim}_{i\in I} D_{i,j}$ with I κ-filtered and every D_i a genuine C-valued diagram, the candidate colimit is forced:

$$Q = \mathrm{colim}_{i\in I} \left(\mathrm{colim}^C_{j\in J} D_{i,j} \right).$$

The inner colimits are preserved by $C \to \mathrm{Ind}_\kappa C$ because $|\operatorname{Mor} J| < \kappa$; the outer one exists by definition of the completion. For any Y in the completion,

$$\mathrm{Hom}(Q, Y) \simeq \lim_i \lim_j \mathrm{Hom}(D_{i,j}, Y) \simeq \lim_j \mathrm{Hom}(X_j, Y).$$

The middle interchange is an interchange of two limits, so it needs no extra exactness hypothesis. This proves 4.11.1's sufficiency, including the prescribed cocone.

The preservation assertion of Lemma 4.10.4 is equally specific. If $c = \mathrm{colim}^C_{j\in J} c_j$ and $Y = \mathrm{colim}_{a\in A} y_a$, then

$$\mathrm{Hom}(c, Y) \simeq \mathrm{colim}_a \lim_j \mathrm{Hom}(c_j, y_a) \simeq \lim_j \mathrm{colim}_a \mathrm{Hom}(c_j, y_a).$$

Here the second interchange does use κ-filteredness of A and κ-smallness of J. This distinction explains why no general colimit-preservation property of the ordinary Yoneda embedding is being claimed.

For necessity, an existing κ-small colimit of κ-presentables is again κ-presentable: its hom functor is a κ-small limit of hom functors, and the same interchange applies. The κ-presentable objects

108. Expliciter ici les corollaires des types 4.13.20, 4.13.21 (p. 341).

of $\mathrm{Ind}_\kappa C$ are retracts of objects of C. To see the retract statement, express a presentable X as a filtered colimit of representables and factor id_X through one stage. If C is idempotent complete, that retract lies in C, proving the converse. Idempotent completeness is used at precisely this point, not in the construction of Q.

Lemme 4.11.3. *Soit $\mathcal{M}$ une catégorie, π un cardinal. Supposons que $\mathcal{M}$ soit stable par $\varinjlim$ filtrantes de cardinal $\leqslant \pi$ (resp. par $\varinjlim$ de cardinal $\leqslant \pi$, resp. par sommes des cardinal $\leqslant \pi$),* et *par $\varinjlim$ grandes devant π. Alors $\mathcal{M}$ est stable par petites $\varinjlim$ filtrantes (resp. par petites $\varinjlim$, resp. par petites sommes).*

Ce lemme se prouve comme 4.4.14, page 97.

Reste la réciproque dans 4.11.2, qui résulte de l'assertion similaire dans 4.11.1.

Corollaire 4.11.4. *Quand $\mathcal{M} = \mathrm{Ind}_\pi C$ est-il stable par sommes ? S'il existe un ordinal α tel que* $\mathrm{card}\,\alpha \leqslant \pi$, *et que C soit stable par $\varinjlim$ de type I_α, alors il suffit que C soit stable par sommes de cardinal $\leqslant \pi$. Cette condition est aussi nécessaire si C est karoubienne.*

([109]).

Se prouve comme 4.11.2, en utilisant 4.11.3.

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Commentary · Exposition [ms. 270]

Building an arbitrary colimit from bounded pieces

The cardinal enlargement in Lemma 4.11.3 can be made quite explicit. Assume π is infinite, as it is in the application, and put $\kappa = \pi^+$. Every small category J is the union of its subcategories with at most π arrows. Their poset P is κ-filtered: the subcategory generated by the union of at most π such subcategories still has at most π arrows, since only finite composites are added. For a diagram $X : J \to \mathcal{M}$, suppose all the colimits $Q_K = \mathrm{colim}_K X$ exist for $K \in P$. A cocone from the diagram $K \mapsto Q_K$ to Y is exactly a cocone from X to Y. Every object and arrow of J occurs in some K, and two occurrences can be compared in a larger one. Thus $\mathrm{colim}_{K\in P} Q_K$ is the desired colimit over J.

When J is filtered and only filtered colimits are assumed for the bounded pieces, take instead its filtered subcategories of size at most π. Every set of at most π arrows is contained in one: repeatedly add cocones for pairs of objects and equalizers for parallel pairs, then close under composition. Countably many repetitions suffice; their union is filtered and still has at most π arrows. The same construction bounds a family of at most π such subcategories. Their indexing poset is therefore κ-filtered, and the preceding cocone argument applies unchanged. For coproducts use the poset of subsets of the indexing set of size at most π.

This also makes 4.11.4 stronger than its stated form. If C has coproducts indexed by sets of size at most π, then $\mathrm{Ind}_\kappa C$ has all small coproducts without an additional ordinal-colimit assumption. For a bounded family of ind-objects, take the product of their κ-filtered presentation categories, form the coproduct of the resulting objects of C at each index, and then take its filtered colimit. The hom universal property proves this is the coproduct. The preceding bounded-subset construction gives arbitrary coproducts. The converse under idempotent completeness follows from the presentability argument on p. 269. Thus the omission of the word "limit" before "ordinal" in 4.11.4 creates no counterexample to its conclusion.

Proposition 4.11.5 ([110]). *Soit Φ un ensemble de catégories I équivalentes à des catégories finies, et "rigides", i.e. pour tout $i \in \mathrm{Ob}\, I$, $\mathrm{Hom}_I(i,i) = \{\mathrm{id}_i\}$. Soient C une catégorie équivalente à une petite catégorie, π un cardinal infini, $\mathcal{M} = \mathrm{Ind}_\pi C$. Pour que $\mathcal{M}$ soit stable par $\varinjlim$ (resp. par $\varprojlim$) de type Φ, il* suffit *que C le soit. Dans le cas non respé, si C est karoubienne, cette condition est aussi nécessaire.*

Cette dernière assertion est contenue dans l'assertion similaire dans 4.11.1. Pour la première assertion, cela résulte de la façon habituelle du

Lemme 4.11.6 ([111]). *Soit J une catégorie équivalente à une catégorie finie, J rigide, C une catégorie équivalente à une petite catégorie, π un cardinal infini. Alors tout foncteur*

109. Pour autres variantes de 4.11.1, cf. 4.13.19 (p. 340) et corollaires.

110. ?, démonstration pas achevée.

111. ?, démonstration non explicitée est peut-être faux!

$J \longrightarrow \mathcal{M} \overset{\text{déf}}{=} \mathrm{Ind}_\pi C$ *admet une représentation indicielle* $J \times I \longrightarrow C$*, avec* I *grande devant* π*,* p. 102
en d'autres termes, le foncteur canonique pleinement fidèle [page 272]

$$\mathrm{Ind}_\pi(\mathrm{Hom}(J, C)) \longrightarrow \mathrm{Hom}(J, \mathcal{M})$$

déduit de l'inclusion $C \hookrightarrow \mathcal{M}$*, d'où* $\mathrm{Hom}(J, C) \hookrightarrow \mathrm{Hom}(J, \mathcal{M})$ *(cf. 4.9), est une équivalence de catégories.* (**N.B.** J'ignore s'il en est ainsi sans hypothèse de rigidité sur J.)

Démonstration. Cela se prouve comme le résultat similaire pour $\mathrm{Ind}(C)$ au lieu de $\mathrm{Ind}_\pi C$, cf. SGA 4 I 8.8.5.

[page 273]

Commentary · Exposition [ms. 272]

Finite rigid diagrams really do admit simultaneous presentations

The hesitation attached to 4.11.6 concerns a true statement. After taking a skeleton, a finite category with only identity endomorphisms admits an ordering of its objects in which every nonidentity arrow increases the order. Indeed a cycle would produce an endomorphism, and the identity-only condition would make every arrow in the cycle invertible, contradicting skeletal distinctness. The finite induction in SGA 4, I, 8.8.5 therefore extends to the present cardinal bound (Artin et al., 1972).

We give the induction with the required simultaneous constraints. Fix $X : J \to \mathrm{Ind}_\kappa C$ and let $S = C^J/X$. To show that S is κ-filtered, take a diagram $H : K \to S$ with $|\operatorname{Mor} K| < \kappa$. Construct a cocone $H \to D$ one vertex of J at a time. At a vertex j, include in C/X_j every $H_k(j)$, with the arrows prescribed by K, and every previously chosen $D(i)$ indexed by an incoming nonidentity arrow $a : i \to j$, mapped to X_j through $X(a)$. Include also the maps $H_k(i) \to D(i)$ and $H_k(i) \to H_k(j)$, and, for composable incoming arrows, the maps between previously chosen $D(i)$'s. This is a diagram with fewer than κ arrows. A cocone in the κ-filtered category C/X_j gives $D(j)$, the arrows $D(a)$, and the components $H_k(j) \to D(j)$. The imposed relations say exactly that composition is respected and that every $H_k \to D$ is natural. Since no nonidentity arrow leaves j for an earlier vertex, subsequent choices cannot alter this work. The induction supplies the desired cocone, including nonemptiness when K is empty.

The same induction can begin with arbitrary objects $c_j \to X_j$, adding them as isolated objects to the diagrams at the corresponding vertices. It proves the object-lifting condition for evaluation $S \to \prod_j C/X_j$. To see the required connectedness, take two lifts of a fixed family. Repeat the induction with the two section diagrams and with the additional pairs of arrows from each c_j; equalizing these pairs in C/X_j gives a common successor in the relevant comma category. Evaluation is therefore final. The density argument already given on p. 246 proves the equivalence $\mathrm{Ind}_\kappa(C^J) \simeq (\mathrm{Ind}_\kappa C)^J$.

The modern sharp result is Henry (2025b, Theorem 1.3): for all categories C, this equivalence holds exactly when J is essentially κ-small and well-founded. Finite rigid categories are such shapes. Thus the source's further question about dropping rigidity has a negative answer in general; a finite category with a nonidentity endomorphism is already outside the universal statement. The condition concerns a theorem for every C, not the impossibility of successful representation for one particular C.

Corollaire 4.11.7 ([112]). *Pour que* $\mathcal{M} = \mathrm{Ind}_\pi C$ *soit stable par* $\varinjlim$ *(resp. par* $\varprojlim$ *)* finies, *il* suffit *que* C *le soit. Et c'est aussi nécessaire dans le cas non respé, lorsqu'on suppose* C *karoubienne.*

4.12 Foncteurs accessibles entre catégories accessibles

Proposition 4.12.1 ([113]). *Soient* $\mathcal{M}, \mathcal{N}$ *deux catégories accessibles (4.8), et* $f : \mathcal{M} \longrightarrow \mathcal{N}$ *un foncteur accessible. Alors il existe des cardinaux infinis* π_0*,* π_1 *tels que* $\pi_1 \geqslant \pi_0$ *et que pour tout cardinal* $\pi \geqslant \pi_1$*, on ait*

$$f(\mathcal{M}_\pi) \subset \mathcal{N}_{\pi^{\pi_0}} , \tag{4.12.1}$$

112. sous réserve de validité de 4.11.5.
113. Remonterait à 4.8.

et en particulier

(4.12.1.2) $$f(\mathcal{M}_\pi) \subset \mathcal{N}_\pi \qquad \textit{si } \pi = \pi^{\pi_0}.$$

De façon plus précise, on peut prendre pour π_0 un cardinal utile pour $\mathcal{M}$ et tel que f soit [page 274]
π_0-accessible (p.ex. le plus petit cardinal de ce type). Soit alors π_1 un cardinal tel qu'on ait $f(\mathcal{M}_{\pi_0}) \subset \mathcal{N}_{\pi_1}$, $\pi_1 \geqslant \operatorname{card} \operatorname{réd} \mathcal{M}_{\pi_0}$, et enfin tel que $\mathcal{N}_{\pi_1}$ soit [une] *sous-catégorie strictement génératrice de $\mathcal{N}$. Le couple (π_0, π_1) satisfait aux conditions énoncées.*

Commentary · Exposition [ms. 273]

Why the finite limit and colimit corollaries follow

The preceding representation establishes both clauses of 4.11.5, but they use different universal-property arguments. For colimits, the iterated-colimit calculation on p. 269 applies directly. For limits, the inclusion $C \to \mathrm{Ind}_\kappa C$ preserves every existing limit in C: the ordinary Yoneda embedding preserves limits, and the completion is a full subcategory of the presheaf category.

Let $D : J \to \mathrm{Ind}_\kappa C$ have a simultaneous presentation $D_j = \mathrm{colim}_{i \in I} D_{i,j}$, with J finite rigid. If C has J-limits, put $L_i = \lim_j^C D_{i,j}$ and $L = \mathrm{colim}_i L_i$. For a representable $c \in C$, compactness and filtered-limit commutation in sets give

$$\mathrm{Hom}(c, L) \simeq \mathrm{colim}_i \lim_j \mathrm{Hom}(c, D_{i,j}) \simeq \lim_j \mathrm{Hom}(c, D_j).$$

For an arbitrary ind-object $Z = \mathrm{colim}_a c_a$, apply $\lim_a$ to this identity and commute the two limits. This proves $\mathrm{Hom}(Z, L) \simeq \lim_j \mathrm{Hom}(Z, D_j)$, hence $L = \lim_j D_j$.

Although an arbitrary finite category need not be rigid, finite limits are generated by finite products and equalizers, and finite colimits by finite coproducts and coequalizers. The corresponding shapes are finite rigid categories. The empty shape provides the terminal or initial object, and the parallel-pair shape provides equalizers or coequalizers. This proves Corollary 4.11.7 without silently applying 4.11.6 to a finite category with endomorphisms. The converse for finite colimits uses idempotent completeness as before; no analogous converse for finite limits is asserted.

Démonstration. On aura en effet

$$\mathcal{N}_1 \xleftarrow{\sim} \mathrm{Ind}_{\pi_0}(\mathcal{M}_{\pi_0})$$

et f commutant aux $\varinjlim$ grandes devant π_0, sera connu (à isomorphisme unique près) par sa restriction à $\mathcal{M}_{\pi_0}$, laquelle se factorise par $\mathcal{N}_{\pi_1}$,

$$f|\mathcal{M}_{\pi_0} : \mathcal{M}_{\pi_0} \longrightarrow \mathcal{N}_{\pi_1} .$$

Pour prouver (4.12.1.1), posant $\pi' = \pi^{\pi_0} \geqslant \pi$, et notant que $\pi'^{\pi_0} = \pi_0$ [plutôt $\pi'^{\pi_0} = \pi'$],

$$f(\mathcal{M}_\pi) \subset f(\mathcal{M}_{\pi'}) \overset{?}{\subset} \mathcal{N}_{\pi'} ,$$

on voit qu'il suffit de prouver (4.12.1.2) (pour π'). Donc on peut supposer $\pi = \pi^{\pi_0}$ et prouver p. 103
(4.12.1.2). Mais comme $\pi_1 \geqslant \operatorname{card} \operatorname{réd} \mathcal{M}_{\pi_0}$, on sait que si π est tel que $\pi \geqslant \pi_1$, $\pi^{\pi_0} = \pi$, alors $\mathcal{M}_\pi$ est formé des $\varinjlim_I x_i$, les $x_i \in \mathcal{N}_{\pi_0}$, et I grande devant π_0, et $\operatorname{card} I \leqslant \pi$.

On a alors (f étant π_0-accessible) [page 275]

$$f(x) = \varinjlim_I f(x_i) ,$$

les $f(x_i)$ sont dans $\mathcal{N}_{\pi_1}$, donc dans $\mathcal{N}_\pi$, or $\mathcal{N}_\pi$ stable par $\varinjlim$ de cardinal $\leqslant \pi$ (puisque $\mathcal{N}$ est engendrée par $\mathcal{N}_{\pi_1}$, formée d'objets π_1-accessibles et a fortiori π-accessibles, cf. SGA 4 I 9.9). D'où $f(x) \in \mathcal{N}_\pi$, q.e.d.

Corollaire 4.12.2. *Supposons que π_0 soit aussi utile pour $\mathcal{N}$, et que $\pi_1 \geqslant \operatorname{card} \operatorname{réd} \mathcal{N}_{\pi_0}$. Soit π un cardinal $\geqslant \pi_1$, et tel que $\pi^{\pi_0} = \pi$. Alors f s'identifie, à équivalence de catégories et à isomorphisme de foncteurs près, au foncteur*

$$\operatorname{Ind}(f_\pi) \ : \ \operatorname{Ind}_\pi(\mathcal{M}_\pi) \ \longrightarrow \ \operatorname{Ind}_\pi(\mathcal{N}_\pi) \ ,$$

où $f_\pi : \mathcal{M}_\pi \longrightarrow \mathcal{N}_\pi$ est le foncteur induit par f.

Proposition 4.12.3 ([114]). *Soient C une catégorie équivalente à une petite catégorie, $\mathcal{N}$ une $\mathfrak{U}$-catégorie, π_0 un cardinal infini. On suppose que $\mathcal{N}$ satisfait L_{π_0} (stabilité par $\varinjlim$ grandes devant π_0). Soit $\mathcal{M} = \operatorname{Ind}_{\pi_0} C$, $\operatorname{Hom}_{!\,\pi_0}(\mathcal{M},\mathcal{N})$ la sous-catégorie strictement pleine de* [page 276] *$\operatorname{Hom}(\mathcal{M},\mathcal{N})$ formée des $f : \mathcal{M} \longrightarrow \mathcal{N}$ π_0-accessibles. Alors le foncteur restriction*

$$\operatorname{Hom}_{!\,\pi_0}(\mathcal{M},\mathcal{N}) \ \longrightarrow \ \operatorname{Hom}(C,\mathcal{N}) \tag{4.12.3.1}$$

est une équivalence de catégories. On trouve un foncteur quasi-inverse comme composé

$$\operatorname{Hom}(C,\mathcal{N}) \ \longrightarrow \ \operatorname{Hom}(\underbrace{\operatorname{Ind}_{\pi_0}(C)}_{\mathcal{M}}, \operatorname{Ind}_{\pi_0}(\mathcal{N})) \ \longrightarrow \ \operatorname{Hom}(\mathcal{M},\mathcal{N}) \ , \tag{4.12.3.2}$$

où la deuxième flèche est le foncteur composition $F \longmapsto L \circ F$ avec le foncteur limite inductive $L : \operatorname{Ind}_{\pi_0}\mathcal{N} \longrightarrow \mathcal{N}$. Le foncteur composé (4.11.3.2) a une image contenue dans $\operatorname{Hom}_{!\,\pi_0}(\mathcal{M},\mathcal{N})$ et induit une équivalence entre cette dernière catégorie et $\operatorname{Hom}(C,\mathcal{N})$.

Commentary · Exposition [ms. 275]

An accessible functor eventually respects a common size bound

The point of 4.12.1 is to replace the qualitative word "accessible" by a controlled estimate on the images of small objects. Choose a useful π_0 for $\mathcal{M}$ such that f preserves colimits indexed by categories large before π_0. The essentially small category $\mathcal{M}_{\pi_0}$ has only set-many images under f. Their accessibility ranks can therefore be bounded by one π_1, enlarged also to dominate the size of a skeleton and a generating rank for $\mathcal{N}$.

For $\pi \geq \pi_1$ satisfying $\pi^{\pi_0} = \pi$, the earlier presentation theorem expresses every $x \in \mathcal{M}_\pi$ as a colimit of objects $x_i \in \mathcal{M}_{\pi_0}$ over a category I large before π_0 with $|\operatorname{Mor} I| \leq \pi$. Since f preserves this colimit,

$$f(x) \simeq \operatorname{colim}_{i \in I} f(x_i).$$

Every $f(x_i)$ is π-accessible, and a colimit of at most π such objects is π-accessible. The latter assertion follows by writing its hom functor as a π-bounded limit and commuting it with the filtered colimits used to define π-accessibility. Thus $f(\mathcal{M}_\pi) \subset \mathcal{N}_\pi$. For general π, put $\rho = \pi^{\pi_0}$. Cardinal arithmetic gives $\rho^{\pi_0} = \rho$, and monotonicity gives $\mathcal{M}_\pi \subset \mathcal{M}_\rho$, proving the advertised bound $f(\mathcal{M}_\pi) \subset \mathcal{N}_\rho$.

The printed $\mathcal{N}_1 \simeq \operatorname{Ind}_{\pi_0}(\mathcal{M}_{\pi_0})$ on p. 274 must mean $\mathcal{M} \simeq \operatorname{Ind}_{\pi_0}(\mathcal{M}_{\pi_0})$, and the x_i later belong to $\mathcal{M}_{\pi_0}$, not $\mathcal{N}_{\pi_0}$. These are present in the edition and are retained as source slips. The proof itself is the familiar uniformization argument. Applied to a small family of functors, one first bounds all their required ranks, then chooses a common fixed point; this is the step used for sections and cartesian sections.

Once the same π is useful for both categories, preservation of the appropriate filtered colimits says that f is determined by $f_\pi : \mathcal{M}_\pi \to \mathcal{N}_\pi$. Corollary 4.12.2 identifies it with the ind-extension of that small functor. The identification includes natural transformations, as the following universal property makes precise.

En d'autres termes, on peut dire que le foncteur d'inclusion canonique

$$C \ \longrightarrow \ \operatorname{Ind}_{\pi_0} C \ = \ \mathcal{M} \tag{4.12.3.3}$$

[page 277]

est 2-universel *pour les foncteurs de C dans des catégories $\mathcal{N}$ satisfaisant L_{π_0}, celles-ci étant considérées comme* 0*-objets d'une* 2*-catégorie où les* 1*-objets sont les foncteurs $\mathcal{N} \longrightarrow \mathcal{N}'$ π_0-accessibles, i.e. commutent aux $\varinjlim$ grandes devant π_0.*

114. remonte à 4.7.

Cet énoncé est l'analogue, pour C petite quelconque, de 4.4.16 (page 101), valable pour C stable par $\varinjlim$ de cardinal $\leqslant \pi_0$. Il y a bien sûr d'autres variantes, en relation avec la commutation à d'autres types de $\varinjlim$ ($\varinjlim$ filtrantes, sommes, limites inductives finies etc.) envisagées dans la section 4.11 précédente.

La démonstration de 4.12.3 est soritale.

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[page 278]

Corollaire 4.12.4. *Soient $\mathcal{M}$ et $\mathcal{N}$ deux catégories accessibles, et soit π_0 un cardinal infini tel que $\mathcal{M}$ soit π_0-accessible, i.e. que π_0 soit un cardinal utile pour $\mathcal{M}$. Soit $\pi_1 = \sup(\pi_0, \operatorname{card} \text{réd}\, \mathcal{M}_{\pi_0})$. Alors pour tout cardinal $\pi \geqslant \pi_1$ utile pour $\mathcal{M}$ (et en particulier, pour tout $\pi \geqslant \pi_1$ tel que $\pi^{\pi_0} = \pi$), la catégorie des foncteurs $\mathcal{M} \longrightarrow \mathcal{N}$ qui sont π-accessibles est elle-même accessible.*

Commentary · Exposition [ms. 277]

The free filtered-colimit completion includes natural transformations

Grothendieck's "2-universal" formulation records more than existence of an extension. Given $g : C \to \mathcal{N}$, where $\mathcal{N}$ admits κ-filtered colimits, define

$$\overline{g}(X) = \operatorname{colim}_{(c \to X) \in C/X} g(c), \qquad X \in \operatorname{Ind}_\kappa C.$$

The category C/X is small and κ-filtered. A map $X \to Y$ induces $C/X \to C/Y$, and hence a map between the displayed colimits. Identity maps and composition are respected by uniqueness in the colimit universal property. For $X = c$ representable, (c, id) is terminal in C/c, so $\overline{g}(c) \simeq g(c)$.

This extension preserves κ-filtered colimits. For a κ-filtered diagram X_i with colimit X, consider pairs $(i, c \to X_i)$. Their category is κ-filtered: fewer than κ objects and equations can first be moved to a common index i, using presentability of the representables, and then bounded inside C/X_i. The functor sending a pair to $c \to X$ is final. Every such map factors through some X_i, and two factorizations become equal at a later index; the same argument with fewer than κ constraints verifies the required compatibility. Therefore the displayed formula for $\overline{g}(X)$ is the iterated colimit of the formulas for $\overline{g}(X_i)$, proving preservation.

If F already preserves these colimits, the canonical expression $X = \operatorname{colim}_{C/X} c$ forces $F(X) \simeq \overline{F|_C}(X)$. Finally a transformation $g \to h$ induces $\overline{g} \to \overline{h}$ by applying its components to each presentation. Conversely a transformation between colimit-preserving extensions is forced by its restriction to C, since the maps from the presentation objects are a colimit cocone. Restriction is consequently fully faithful and essentially surjective, exactly the equivalence of categories stated in 4.12.3. This is the cardinal-bounded counterpart of the ordinary ind universal property in SGA 4, I, 8.7 (Artin et al., 1972).

En effet, on aura $\mathcal{M} \simeq \operatorname{Ind}_\pi \mathcal{M}_\pi$, d'où par 4.12.3

$$\operatorname{Hom}_{!\,\pi}(\mathcal{M}, \mathcal{N}) \xrightarrow{\sim} \operatorname{Hom}(\mathcal{M}_\pi, \mathcal{N}),$$

or $\operatorname{Hom}(\mathcal{M}_\pi, \mathcal{N})$ est accessible par 4.9.5 (p. 220). On peut préciser d'ailleurs ce point. Supposons π_0 choisi tel que π_0 soit utile non seulement pour $\mathcal{M}$, mais pour $\mathcal{N}$ également. Rappelons que l'on a $\operatorname{card} \operatorname{Fl} \mathcal{M}_\pi \leqslant 2^\pi$ (cf. 4.7.7, page 173).

Posons donc [page 279]

$$\pi' = 2^\pi,$$

on a donc $\pi'^{\pi_0} = 2^{\pi\pi_0} = 2^\pi = \pi'$, et $\operatorname{card} \text{réd}\, \mathcal{M}_\pi \leqslant \pi'$. Donc 4.9.5 nous donne que

$$\operatorname{Hom}(\mathcal{M}_\pi, \mathcal{N}) \simeq \operatorname{Ind}_{\pi'}(\operatorname{Hom}(\mathcal{M}_\pi, \mathcal{M}_{\pi'})),$$

d'où :

Commentary · Exposition [ms. 278]

Fixing an accessibility rank before forming a functor category

Grothendieck first fixes the amount of colimit preservation demanded of a functor. This makes its domain effectively small: a functor preserving the relevant filtered colimits is determined, including

its natural transformations, by its values on a small category of presentable objects. Write $\lambda = \pi^+$ and $A = \mathcal{M}_\pi$. Whenever $\mathcal{N}$ admits λ-filtered colimits, restriction gives

$$\mathrm{Fun}_{\lambda\text{-filt}}(\mathcal{M},\mathcal{N}) \simeq \mathrm{Fun}(A,\mathcal{N}).$$

The inverse is the extension $F(x) = \varinjlim_{(a\to x)\in A/x} F(a)$. The same formula extends a natural transformation, and its uniqueness follows because the cocone $a \to x$ presents x. Thus the equivalence concerns categories of functors, rather than merely their isomorphism classes. This is the ordinary categorical form of the universal property of Ind; compare Chorny and Rosický (2012, Remark 2.3(1)).

There is one hypothesis to restore in the proof as printed. Proposition 4.12.3 assumes $\mathcal{N}$ satisfies L_π, whereas the first statement of 4.12.4 chooses π_0 using $\mathcal{M}$ alone. The displayed restriction equivalence does not follow without the required colimits in $\mathcal{N}$. Choosing π_0 useful for both categories supplies this hypothesis, exactly as Grothendieck does in the paragraph leading to 4.12.5. This repairs the proof and gives the form used subsequently; we do not infer a counterexample to the broader assertion merely from this omitted hypothesis.

The cardinal 2^π in 4.12.5 has a different job from π. The latter controls how a functor is extended; the former is large enough to control all the objects and arrows in its small domain simultaneously. If a skeleton of A has at most 2^π arrows and $\pi \geq \pi_0$, then

$$(2^\pi)^{\pi_0} = 2^{\pi\pi_0} = 2^\pi.$$

Corollary 4.9.5 therefore applies to $\mathrm{Fun}(A,\mathcal{N})$. The $\mathcal{M}_{\pi'}$ printed in the preliminary display on p. 279 must be read mathematically as $\mathcal{N}_{\pi'}$, as the following corollary confirms. Also, "under the conditions of 4.12.4" retains the requirement that π be useful for $\mathcal{M}$; the shorter inequality $\pi \geq \pi_1$ in 4.12.5 does not by itself replace that requirement.

Corollaire 4.12.5. *Sous les conditions de 4.12.4, si π_0 est utile pour $\mathcal{M}$* et *pour $\mathcal{N}$, alors pour $\pi \geqslant \pi_1$, $\mathrm{Hom}_{!\,\pi}(\mathcal{M},\mathcal{N})$ est π'-accessible, où $\pi' = 2^\pi$, de façon plus précise, on a une équivalence de catégories*

$$\mathrm{Hom}_{!\,\pi}(\mathcal{M},\mathcal{N}) \simeq \mathrm{Ind}_{\pi'}(\mathrm{Hom}(\mathcal{M}_\pi,\mathcal{N}_{\pi'})).$$

Remarque 4.12.6. Si $\mathcal{M}$, $\mathcal{N}$ sont accessibles, la catégorie $\mathrm{Hom}_{\mathrm{acc}}(\mathcal{M},\mathcal{N})$ des foncteurs accessibles n'est par toujours accessible. Si $\mathcal{N}$ satisfait L_{π_0}, cette catégorie satisfait L_{π_0}, et les $\varinjlim$ grandes devant π_0 se calculent argument par argument.

Si on pouvait trouver une sous-catégorie (équivalente à une petite catégorie) $\mathcal{H}_\pi$ de [page 280]
$\mathcal{H} = \mathrm{Hom}_{\mathrm{acc}}(\mathcal{M},\mathcal{N})$, pour un cardinal $\pi \geqslant \pi_0$ telle que tout objet F de $\mathcal{H}$ s'écrive $\varinjlim_I F_i$, les F_i dans $\mathcal{H}_\pi$, I grande devant π, alors si les objets de $\mathcal{H}_\pi$ sont des foncteurs π'-accessibles de $\mathcal{M}$ dans $\mathcal{N}$, il en serait de même de tout F dans $\mathcal{H}$. Donc il existerait un cardinal infini π' tel que tout foncteur accessible de $\mathcal{M}$ dans $\mathcal{N}$ soit π'-accessible. Mais sauf erreur, si p.ex. $\mathcal{M} = \mathcal{N} = \mathrm{Ens}$, il n'existe pas de tel cardinal. Je ne veux pas m'arrêter ici sur cette question, et faire un contre-exemple.

[page 281]

Commentary · Examples [ms. 280]

Accessible endofunctors of sets have no common bound

Grothendieck's hesitation on p. 280 can be settled by the most elementary representable functors. For each set B, consider

$$H_B(X) = \mathrm{Hom}(B,X) = X^B.$$

If λ is regular and $|B| < \lambda$, this functor preserves λ-filtered colimits. A map $B \to \varinjlim_i X_i$ is represented by choosing a representative for each of its fewer than λ values and then a common stage. If two representatives define the same map, fewer than λ equalities have to be witnessed, and filteredness places all their witnesses at a further common stage. These are respectively surjectivity and injectivity of the comparison map.

The bound cannot be made independent of B. Fix an infinite cardinal π, let $|B| > \pi$, and order the subsets $T \subseteq B$ of cardinal at most π by inclusion. Their poset is π^+-directed: the union of at most π such subsets still has cardinal at most π. Their colimit is B. But the image of

$$\varinjlim_{T \in [B]^{\leq \pi}} T^B \longrightarrow B^B$$

consists precisely of the maps with image of cardinal at most π. It omits id_B. Thus H_B is accessible, for example at the regular cardinal $|B|^+$, but is not π^+-accessible.

This proves the stronger assertion about the category of all accessible endofunctors. Suppose that category were μ-accessible and choose a small set $\mathcal{G}$ of μ-presentable generators. There is a regular cardinal λ such that every $G \in \mathcal{G}$ preserves λ-filtered colimits. Pointwise small colimits of accessible set-valued functors remain accessible: choose one common preservation rank for the small diagram, and interchange its colimit with a filtered colimit in Set. Consequently these pointwise colimits are also colimits in the full category of accessible functors. Every object, being a μ-filtered colimit of objects of $\mathcal{G}$, would therefore preserve λ-filtered colimits. The preceding example contradicts this conclusion. Natural transformations cause no size difficulty: between two accessible functors they are determined on a common small subcategory of sufficiently presentable sets.

This completes the counterexample requested by the source. The broader replacement of small generating sets by generating classes is the subject of Chorny and Rosický (2012, Definition 2.1 and Examples 2.2); that theory gives a context for the size obstruction, rather than making the ordinary accessibility assertion true.

4.13 Ensembles accessibles d'objets d'une catégorie accessible

Définition 4.13.1. Soient $\mathcal{M}$ une catégorie accessible, S une partie de $\mathrm{Ob}\,\mathcal{M}$, $\underline{S}$ la sous-catégorie pleine de $\mathcal{M}$ définie par S. On dit que S est une *partie accessible* de $\mathrm{Ob}\,\mathcal{M}$, ou encore que $\underline{S}$ est une *sous-catégorie "accessible dans $\mathcal{M}$"*, si p. 105

a) $\underline{S}$ est strictement pleine, i.e. pour x, y deux objets isomorphes de $\mathcal{M}$, $x \in S$ implique $y \in S$,

b) $\underline{S}$ est accessible,

c) le foncteur d'inclusion $\underline{S} \longrightarrow \mathcal{M}$ est accessible.

Soient π_0 un cardinal utile pour $\mathcal{M}$, i.e. tel que $\mathcal{M} \simeq \mathrm{Ind}_{\pi_0}(\mathcal{M}_{\pi_0})$, soit $\pi_1 = \sup(\pi_0, \mathrm{card\,réd}\,\mathcal{M}_{\pi_0})$. Soit d'autre part π_0' un cardinal tel que

$\underline{S}$ (supposé accessible dans $\mathcal{M}$) soit π_0-*accessible*, i.e. π_0'-utile pour $\underline{S}$, et de plus tel que $f : \underline{S} \hookrightarrow \mathcal{M}$ soit π_0'-accessible. Soit $\pi_1' \geqslant \pi_0'$ un cardinal tel que $\underline{S}_{\pi_0'} \subset \mathcal{M}_{\pi_1'}$, $\pi_1' \geqslant \mathrm{card\,réd}\,\underline{S}_{\pi_0'}$. [page 282]
Il résulte de 4.12.1 que pour tout cardinal

$$(*) \qquad \pi \geqslant \pi_1', \qquad \text{tel que } \pi^{\pi_0'} = \pi,$$

on doit avoir

$$\underline{S}_\pi \subset \mathcal{M}_\pi, \qquad \text{donc } \underline{S}_\pi \subset \mathcal{M}_\pi \cap \underline{S},$$

d'ailleurs, comme $\underline{S} \hookrightarrow \mathcal{M}$ est π_0'-accessible et a fortiori π-accessible, et pleinement fidèle, il s'ensuit aussitôt que $\mathcal{M}_\pi \cap S \subset S_\pi$, d'où

$$S_\pi = \mathcal{M} \cap S.$$

D'ailleurs, la condition $(*)$ implique que π est utile pour $\underline{S}$, donc

$$\underline{S} \xleftarrow{\sim} \mathrm{Ind}_\pi(\underline{S}_\pi). \qquad \text{[page 283]}$$

(D'ailleurs, *si* $\pi_0' \geqslant \pi_0$, la relation $\pi^{\pi_0'} = \pi$ implique a fortiori $\pi^{\pi_0} = \pi$, et si $\pi_1' \geqslant \pi_1$, on voit que $(*)$ implique que π est utile aussi pour $\mathcal{M}$, i.e.

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_\pi(\mathcal{M}_\pi).$$

D'autre part, le foncteur d'inclusion $\underline{S} \hookrightarrow \mathcal{M}$ se déduit essentiellement de $\underline{S}_\pi \hookrightarrow \mathcal{M}_\pi$ en passant aux catégories Ind_π. ([116]).)

Commentary · Exposition [ms. 282]

Accessibility of a full subcategory is a relative assertion

Grothendieck now turns from functors to classes of objects specified by a property. His terminology "accessible in $\mathcal{M}$" includes three separate requirements: closure under isomorphism, accessibility of the full subcategory itself, and accessibility of its inclusion. In current language this is a replete, accessible, accessibly embedded full subcategory. Intrinsic accessibility alone does not control how its objects sit in the ambient category. This distinction will account for the extra bound needed in 4.13.2(c).

The equality of presentable objects on p. 282 is particularly useful. Let $i : \mathcal{S} \hookrightarrow \mathcal{M}$ be fully faithful and preserve λ-filtered colimits. If $s \in \mathcal{S}$ is λ-presentable in $\mathcal{M}$, then for every λ-filtered diagram D in $\mathcal{S}$,

$$\mathcal{S}(s, \varinjlim D) \cong \mathcal{M}(is, \varinjlim iD) \cong \varinjlim \mathcal{M}(is, iD) \cong \varinjlim \mathcal{S}(s, D).$$

Thus ambient presentability implies intrinsic presentability. The reverse implication requires the inclusion to preserve presentable objects as well as colimits. Uniformization supplies this after increasing the cardinal and bounding the ambient ranks of a small generating family. With those choices,

$$\mathcal{S}_\pi = \mathcal{S} \cap \mathcal{M}_\pi.$$

The missing subscript on the last $\mathcal{M}$ in the printed formula is therefore substantive notation, not a new description of S_π.

The subsequent correspondence between retract-closed subcategories of $\mathcal{M}_\pi$ and subcategories generated from them by π^+-filtered colimits follows from one familiar retract argument. If a π^+-presentable object s is a filtered colimit of objects a_i in the chosen small subcategory, its identity factors through some $a_i \to s$, so s is a retract of a_i. Conversely presentability is preserved by retracts. The natural closure in 4.13.3 is precisely the closure which makes this argument an equality of full subcategories. In 4.13.4 the printed words "non pleinement fidèle" contradict both 4.13.2 and this proof: the canonical functor under discussion is fully faithful.

Ceci prouve entre autres le "il faut" dans la proposition qui suit :

Proposition 4.13.2. *Soient $\mathcal{M}$ une catégorie accessible, π_0 un cardinal infini utile pour $\mathcal{M}$, S une partie de* $\mathrm{Ob}\,\mathcal{M}$, *$\underline{S}$ la sous-catégorie pleine de $\mathcal{M}$ correspondante.*

[page 284]

a) Pour que S soit accessible, il faut et il suffit qu'il existe un cardinal infini π et une partie $S(\pi)$ de $\mathcal{M}_\pi$, tels que S soit formé des objets de $\mathcal{M}$ qui sont isomorphes à un objet de la forme $\varinjlim_{i \in I} x_i$, avec les x_i dans $S(\pi)$ et I grande devant π. En d'autres termes, il faut et il suffit qu'il existe un cardinal infini π_1 et une sous-catégorie pleine $\underline{S}(\pi)$ de $\mathcal{M}_\pi$, tels que $\underline{S}$ soit l'image essentielle du foncteur canonique pleinement fidèle

$$\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M} \qquad (\xleftarrow{\sim} \mathrm{Ind}_\pi(\mathcal{M}) \ ([117])), \tag{4.13.2.1}$$

prolongeant l'inclusion canonique $\underline{S}(\pi) \hookrightarrow \mathcal{M}$ (cf. 4.12.3).

p. 106

b) Si S est accessible, on peut trouver des cardinaux π_0', π_1' tels que $\pi_1' \geqslant \pi_0' \geqslant \pi_0$ (où π_0 est un cardinal donné utile pour $\mathcal{M}$) et tels que tout cardinal $\pi \geqslant \pi_1'$ avec $\pi^{\pi_0'} = \pi$ satisfasse aux conditions de c), en prenant $S(\pi) = S \cap \mathrm{Ob}\,\mathcal{M}_\pi$, i.e. $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. Sous ces conditions, π est utile pour $\underline{S}$ et *pour $\mathcal{M}$, $S(\pi)$ est une sous-catégorie* strictement *pleine de $\mathcal{M}_\pi$, stable dans $\mathcal{M}_\pi$ par facteurs directs (ou ce qui revient au même, $\mathcal{M}_\pi$ étant karoubienne, une sous-catégorie karoubienne de $\mathcal{M}_\pi$).*

[page 285]

116. [Le contenu de cette parenthèse est] "facultatif".

117. si π utile pour $\mathcal{M}$.

c) On peut prendre π_0' le plus petit cardinal $\geqslant \pi_0$ utile pour $\underline{S}$ et tel que $\underline{S} \hookrightarrow \mathcal{M}$ soit π_0'-accessible, et $\pi_1' = \sup(\pi_0', \operatorname{card réd} \underline{S}_{\pi_0'})$.

La partie b) ainsi que c) a déjà été prouvée, ce qui implique le "il faut" dans la partie a). Il reste à prouver le "il suffit" dans a). Mais l'hypothèse implique que

$$\underline{S} \xleftarrow{\sim} \operatorname{Ind}_\pi(S_\pi),$$

ce qui implique que $\underline{S}$ est accessible et même π-accessible, et elle implique aussi que $\underline{S}$ est une sous-catégorie *strictement* pleine de $\mathcal{M}$. Il reste à voir

[que] le foncteur d'inclusion est accessible, en fait il est π-accessible, puisque (4.13.2.1) [page 286]
l'est (cf. 4.12.3).

Commentary · Development (commentators') [ms. 285]

The ambient rank cannot be omitted from the threshold

The purpose of 4.13.2(c) is to make the eventual comparison $\mathcal{S}_\pi = \mathcal{S} \cap \mathcal{M}_\pi$ effective. Its proposed threshold omits data which the preceding argument explicitly used. We give a small counterexample, so that the correction can be made without weakening the surrounding theory.

Counterexample to the threshold in 4.13.2(c). [ours] Let X be a set of cardinal greater than $2^{\aleph_0}$, let $\mathcal{M} = \mathcal{P}(X)$ ordered by inclusion, and let $\mathcal{S}$ be its full one-object subcategory $\{X\}$. Then $\mathcal{M}$ is $\aleph_0$-accessible in Grothendieck's convention, $\mathcal{S}$ is accessible at every infinite cardinal, and its inclusion preserves all nonempty filtered colimits. Nevertheless, for $\pi = 2^{\aleph_0}$, the sole object of $\mathcal{S}_\pi$ does not belong to $\mathcal{M}_\pi$.

Proof. Colimits in $\mathcal{P}(X)$ are unions. For every infinite ρ, a subset $Y \subseteq X$ is ρ^+-presentable precisely when $|Y| \leq \rho$. If this inequality holds and $Y \subseteq \bigcup_i Z_i$ for a ρ^+-directed family, select an index for each element of Y and then a common upper bound. This proves preservation of the relevant colimits by the representable, whose values are either empty or singleton. If $|Y| > \rho$, the diagram of its subsets of cardinal at most ρ is ρ^+-directed and has union Y, but none of its members contains Y. This proves the converse. Every subset is the union of its countable subsets, with an $\aleph_1$-directed indexing poset; hence $\mathcal{M}$ has the claimed accessibility. The corresponding assertions for the one-object category and its inclusion are immediate from constant diagrams. Finally $|X| > \pi$ gives the failure of ambient presentability. □

Here the cardinal prescribed in 4.13.2(c) is $\pi_0' = \aleph_0$ and $\sup(\pi_0', \operatorname{card}_{\mathrm{red}} \mathcal{S}_{\pi_0'}) = \aleph_0$, whereas $\pi^{\pi_0'} = \pi$. Thus its displayed numerical conditions hold and its proposed conclusion fails. The repair is to retain a cardinal θ such that

$$i(\mathcal{S}_{\pi_0'}) \subseteq \mathcal{M}_\theta$$

and to take π_1' at least θ, as well as the bounds already listed. To guarantee simultaneous usefulness for $\mathcal{M}$, also retain the ambient bound π_1 from p. 281. The existential assertion of 4.13.2(b) is unaffected.

This example is our explicit diagnostic of the printed estimate, not a claim of priority. We checked the uniformization discussion in Raptis (2020, §2.3), which explicitly separates preservation of filtered colimits from preservation of presentable objects, and the standard limits treatment in Positselski (2024); neither is being cited as a correction of this particular passage.

Définition 4.13.3. Soient $\mathcal{M}$, $S \subset \operatorname{Ob}\mathcal{M}$, $\underline{S} \subset \mathcal{M}$ comme dans 4.12.1, π un cardinal infini. On dit que S est une partie π-*accessible* de $\operatorname{Ob}\mathcal{M}$, ou que $\underline{S}$ est une sous-catégorie π-accessible dans $\mathcal{M}$, si S, π satisfont les conditions de 4.13.3 a), i.e. $\underline{S}$ est l'image essentielle de $\operatorname{Ind}_\pi(S(\pi))$, où $S(\pi)$ est une sous-catégorie pleine de $\mathcal{M}_\pi$.

Notons que, quitte à renforcer $S(\pi)$ par $S'(\pi) = \natural$-clôture de $S(\pi)$ dans $\mathcal{M}_\pi$, on peut supposer que $S(\pi)$ est une sous-catégorie *strictement pleine et karoubienne*

de $\mathcal{M}_\pi$. On sait alors que, $S(\pi)$ étant karoubienne, $S(\pi) \longrightarrow \underbrace{(\operatorname{Ind}_\pi(S(\pi)))}_{\pi}$ est une [page 287]

équivalence de catégories. Comme $\operatorname{Ind}_\pi(S(\pi)) \simeq \underline{S}$ et que $S(\pi)$ est strictement pleine dans $\mathcal{M}_\pi$, donc aussi dans $\mathcal{M}$ et a fortiori dans $\underline{S}$, que l'on a

(4.13.1) $\qquad \underline{S}(\pi) = \underline{S}_\pi, \qquad$ sous-catégorie de $\underline{S}$ des objets π-accessibles de S.

Je dis qu'on aura aussi

$$\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi. \tag{4.13.2}$$

Comme c est clair, il reste à voir que $\underline{S} \cap \mathcal{M}_\pi \subset \underline{S}(\pi)$, or il est immédiat que $\underline{S} \cap \mathcal{M}_\pi \subset S_\pi$, et on conclut par (4.13.1). On trouve ainsi :

Proposition 4.13.4. *Soit $\mathcal{M}$ une catégorie accessible, π un cardinal infini. Soit Σ_π l'ensemble des*

sous-catégories strictement pleines $\underline{S}(\pi)$ de $\mathcal{M}_\pi$ qui sont karoubiennes, ou ce qui revient au même, stables dans $\mathcal{M}_\pi$ par facteurs directs. Soit Σ'_π l'ensemble des sous-catégories de $\mathcal{M}$ qui sont π-accessibles dans $\mathcal{M}$. Considérons les applications [page 288] p. 107

$$\Sigma_\pi \underset{\beta_\pi}{\overset{\alpha_\pi}{\rightleftarrows}} \Sigma'_\pi \tag{4.13.4.1}$$

définies ainsi : Si $\underline{S}(\pi) \in \Sigma_\pi$, $\alpha_\pi(\underline{S}(\pi))$ est l'image essentielle du foncteur non pleinement fidèle $\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}$. Si $\underline{S} \in \Sigma'_\pi$, $\beta_\pi(\underline{S}) = \underline{S} \cap \mathcal{M}_\pi$. Les applications α_π, β_π sont des bijections, inverses l'une de l'autre.

On n'a pas eu à supposer que π soit utile pour $\mathcal{M}$, ni que π soit minoré par un cardinal utile pour $\mathcal{M}$, i.e. qu'on ait $\pi \geqslant \pi_0$, où π_0 est le plus petit cardinal utile pour $\mathcal{M}$.

4.13.5. Soit $\underline{S}$ une sous-catégorie *strictement pleine* de $\mathcal{M}$ (catégorie accessible), π'_0 un cardinal infini. Dire que $\underline{S}$ est π'_0-accessible dans $\mathcal{M}$ signifie que les conditions suivantes sont satisfaites. [page 289]

1) π'_0 est utile pour $\underline{S}$, i.e. $\underline{S} \xleftarrow{\sim} \mathrm{Ind}_{\pi'_0}(\underline{S}_{\pi'_0})$.
2) $\underline{S} \hookrightarrow \mathcal{M}$ est π'_0-accessible.
3) $\underline{S}_{\pi'_0} \hookrightarrow \mathcal{M}_{\pi'_0}$ (qui implique que $S_{\pi'_0}$ est équivalente à une petite catégorie).

Supposons ces conditions satisfaites, et soit π un cardinal $> \pi'_0$. Est-ce que $\underline{S}$ est π-accessible dans $\mathcal{M}$, i.e. les conditions similaires sont-elles satisfaites pour π ? C'est évident pour 2°), mais risque d'être faux pour 1°, 2° `[plutôt 3° ?]`. Mais faisons appel à 4.13.2 c). Posons

$$\pi'_1 = \sup(\pi'_0, \operatorname{card}\operatorname{réd} S_{\pi'_0}, \underbrace{\pi_0}_{\substack{\text{plus petit cardinal}\\ \text{infini utile}\\ \text{pour } \mathcal{M}}}).$$

Alors pour $\pi \geqslant \pi'_1$ tel que $\pi^{\pi'_0} = \pi$, $\underline{S}$ est π-accessible dans $\mathcal{M}$ (4.13.2 b et c). Supposons p.ex. [page 290]

$$\pi'_0 \geqslant \pi_0$$

(pratiquement ce sera toujours vérifié) et de plus

$$\pi'_0 \geqslant \operatorname{card}\operatorname{réd} \mathcal{M}_{\pi_0},$$

i.e. supposons

$$\pi'_0 \geqslant \sup(\pi_0, \operatorname{card}\operatorname{réd} \mathcal{M}_{\pi_0}) = \pi_1.$$

Alors

$$\pi'_1 = \sup(\pi'_0, \operatorname{card}\operatorname{réd} S_{\pi'_0}) \leqslant \sup(\pi'_0, \operatorname{card}\operatorname{réd} \mathcal{M}_{\pi'_0}) \leqslant 2^{(\pi'_0{}^{\pi_0})}$$

(4.7.7.1, p. 177). Donc on trouve ceci :

Corollaire 4.13.6. *Soit π'_0 un cardinal $\geqslant \pi_1 = \sup(\pi_0, \operatorname{card}\operatorname{réd} \mathcal{M}_{\pi_0})$ ([118]), et soit $\underline{S}$ une sous-catégorie π'_0-accessible dans $\mathcal{M}$. Alors pour tout cardinal π tel que $\pi \geqslant 2^{(\pi'_0{}^{\pi_0})}$, et tel que l'on ait $\pi^{\pi'_0} = \pi$ (i.e. π de la forme $c^{\pi'_0}$, avec c un cardinal $\geqslant 2^{(\pi'_0{}^{\pi_0})}$), $\underline{S}$ est π-accessible dans $\mathcal{M}$. (De plus, ces cardinaux sont réguliers pour $\underline{S}$ et pour $\mathcal{M}$, donc $\underline{S} \simeq \mathrm{Ind}_\pi(\underline{S}_\pi)$, $\mathcal{M} \simeq \mathrm{Ind}_\pi(\mathcal{M}_\pi)$).* p. 108 [page 291]

([119], [120]).

Proposition 4.13.7 ([121]). *Soient $\mathcal{M}$ une catégorie accessible, $(S_\alpha)_{\alpha\in A}$ une famille de parties de* $\mathrm{Ob}\,\mathcal{M}$, *indexée par un petit ensemble A, $(\underline{S}_\alpha)_{\alpha\in A}$ la famille des sous-catégories accessibles dans $\mathcal{M}$ correspondante. Soit $S=\bigcap_{\alpha\in A} S_\alpha$, $\underline{S}=\bigcap_{\alpha\in A}\underline{S}_\alpha$, de sorte que $\underline{S}$ est la sous-catégorie pleine de $\mathcal{M}$ définie par $S\subset \mathrm{Ob}\,\mathcal{M}$. Ceci posé, si les S_α sont accessibles, il en est de même de S – ou encore, si les $\underline{S}_\alpha$ sont accessibles dans $\mathcal{M}$, il en est de même de $\underline{S}$.*

Commentary · Later developments [ms. 290]

A homotopical use of accessible properties of arrows

The passage from accessible categories to accessible full subcategories is useful when the property under consideration belongs to a morphism. Regard a map as an object of the arrow category $\mathcal{M}^{[1]}$ and a commutative square as a morphism there. The phrase "the weak equivalences form an accessible subcategory" then has the same precise meaning as the relative accessibility discussed here: a suitable small family of weak equivalences presents every weak equivalence, and the inclusion preserves the specified filtered colimits.

A concrete later application is the treatment of weak equivalences in combinatorial model categories by Raptis and Rosický (2015, Section 2 and Proposition 2.2). After a functorial factorization, a map is a weak equivalence exactly when its fibration part is a trivial fibration. Their construction expresses the weak-equivalence category as a pseudopullback and obtains accessibility with an explicit enlarged rank. They then prove the sharper result that weak equivalences of simplicial sets form a finitely accessible category (Raptis and Rosický, 2015, Corollary 5.1). The finite conclusion is a further theorem, not a consequence of existence of some accessibility rank.

For the reader of these pages, the structural connection is the relevant one. A property is imposed by pulling back a suitable full subcategory along an accessible functor; several properties are imposed by intersection. The full subcategory retains its morphisms, so the resulting statement concerns coherent squares and their presentations, rather than only a set of admissible arrows. This is why bounds on the inclusion and on the images of presentable objects matter. It also explains why increasing the rank is a substantial part of the method: the homotopical construction may preserve filtered colimits long before it preserves the originally selected small objects. None of this identifies the ordinary cofinality arguments here with a theorem about homotopy colimits; their role is to provide the size control needed before those constructions are applied.

Cela résulte du résultat plus précis :

Corollaire 4.13.8 ([122]). *Avec les notations de 4.13.7 et 4.13.6, soit, pour tout $\alpha\in A$, π'_α un cardinal $\geqslant \pi_1$ tel que S_α soit π'_α-accessible. Soit*

$$\pi'_0=\sup_{\alpha\in A}\pi'_\alpha$$
$$\pi''_0=\sup(\operatorname{card} A,\sup_\alpha 2^{(\pi'^{\pi_0}_\alpha)})$$

et soit $\pi\geqslant\pi''_0$ un cardinal tel que $\pi^{\pi'_0}=\pi$, i.e. de la forme $c^{\pi'_0}$, avec $c\geqslant\pi'_0$. Alors S est π-accessible (i.e. $\underline{S}$ est π-accessible dans $\mathcal{M}$). En particulier, S est $2^{\pi''_0}$-accessible. ([123]).

Démonstration. Soit $\alpha\in A$. Comme $\pi'_\alpha\leqslant\pi'_0$ et $\pi^{\pi'_0}=\pi$, on a a fortiori [page 292]

$$\pi^{\pi'_\alpha}=\pi,\qquad \text{de plus } \pi\geqslant\pi''_0\geqslant 2^{(\pi^{\pi_0}_\alpha)},$$

donc en vertu de 4.13.6, $\underline{S}_\alpha$ est π-accessible dans $\mathcal{M}$. D'autre part, comme $\underline{S}_\alpha$ est π'_α-accessible, il est stable dans $\mathcal{M}$ par $\varinjlim$ grandes devant π'_α, a fortiori par $\varinjlim$ grandes devant π'_0 ($\geqslant\pi'_\alpha$). D'ailleurs, $\pi^{\pi'_0}=\pi$ implique $\pi\geqslant 2^{\pi_0}>\pi_0$. Donc 4.13.8 résulte du

118. **N.B.** π_0 est le plus petit cardinal utile pour $\mathcal{M}$.
119. Enchaîner ici avec 4.13.23 etc.
120. Mettre ici le lemme 4.16.7.11 signalé en marge p. 292.
121. Le rejeter après 4.13.23.
122. Cf. reformulation plus raisonnable et plus correcte page 302, 302 bis.
123. Ça a l'air un peu abracadabrant ! Il faut de plus supposer que π n'est pas cardinal limite !

Commentary · Later developments [ms. 291]

Intersections and the later theory of accessible subcategories

Grothendieck wants a class of admissible properties of objects which is closed under imposing several conditions at once. Proposition 4.13.7 supplies exactly that operation. Its force is that a small intersection still has a small generating family, although the generating families for its separate conditions need not intersect in a useful way at the initial cardinal. The proof therefore raises the cardinal first and then constructs approximations which satisfy all the conditions.

This is now a standard stability theorem. In the higher categorical setting, Lurie (2009, Definition 5.4.7.8 and Proposition 5.4.7.10) uses the same relative notion and proves closure under small intersections by a homotopy pullback involving the diagonal and a product of inclusions. The ordinary categorical counterpart belongs to the limit theory of accessible categories. A class-sized generalization, with the stronger embedding hypothesis needed there, is Chorny and Rosický (2012, Corollary 3.3). For explicit preservation ranks, Positselski (2024, Corollary 5.1 and Theorem 9.1) retains a smaller ordinal-colimit hypothesis while controlling the larger presentability rank.

These formulations illuminate the calculation which follows. The objects of $\mathcal{M}_\pi/x$ are small approximations equipped with a map to x. If x satisfies one of the properties, the approximations satisfying that property form a cofinal full subcategory. The problem is therefore to show that a small family of such cofinal subcategories has cofinal intersection. Cofinality alone does not suffice: in $\mathbb{N}$ the even and odd subsets are both cofinal and disjoint. Closure under a suitable family of colimits is the additional input which turns repeated approximation into simultaneous approximation.

Corollaire 4.13.9. *Soit $\pi > \pi_0$ un cardinal tel que $\underline{S}_\alpha$ soit π-accessible dans $\mathcal{M}$ pour tout $\alpha \in A$, et tel que* $\operatorname{card} A \leqslant \pi$. *Supposons de plus qu'il existe un cardinal π_0' tel que $\pi_0 \leqslant \pi_0' < \pi$, tel que les $\underline{S}_\alpha$ soient stables dans $\mathcal{M}$ par $\varinjlim$ grandes devant π_0'. Alors $\underline{S} = \bigcap_{\alpha\in A} \underline{S}_\alpha$ est π-accessible dans $\mathcal{M}$.*

([124]).

Posons, pour tout $\alpha \in A$,

$$\underline{S}_\alpha(\pi) \overset{\text{déf}}{=} \underline{S}_\alpha \cap \mathcal{M}_\pi$$
$$\underline{S}(\pi) \overset{\text{déf}}{=} \underline{S} \cap \mathcal{M}_\pi = \bigcap \underline{S}_\alpha(\pi).$$

Chaque $\underline{S}_\alpha(\pi)$ est stable dans $\mathcal{M}$ par $\varinjlim$ grandes devant π_0' et de cardinal $\leqslant \pi$ (puisqu'il en est ainsi de $\underline{S}_\alpha$ et de $\mathcal{M}_\pi$) (donc il en est de même de leur intersection

$\underline{S}(\pi)$. A fortiori, $\underline{S}(\pi)$ est une sous-catégorie strictement pleine et karoubienne de $\mathcal{M}_\pi$.) [page 293]

D'autre part, chaque $\underline{S}_\alpha$ étant stable dans $\mathcal{M}$ par $\varinjlim$ grandes devant π_0', il en est de même p. 109
de leur intersection $\underline{S}$. A fortiori, celle-ci est stable par $\varinjlim$ grandes devant π, donc le foncteur pleinement fidèle canonique $\operatorname{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}$ se factorise par S en

$$\operatorname{Ind}_\pi \underline{S}(\pi) \hookrightarrow \underline{S}\ ,$$

il faut prouver que ce foncteur est essentiellement surjectif. Soit donc $x \in S = \operatorname{Ob} \underline{S}$, tout revient à prouver que

$$(*) \qquad \varinjlim_{i\in \underline{S}(\pi)/x} x_i \xrightarrow{\sim} x\ ,$$

et que $\underline{S}(\pi)/x$ est grande devant π.

Notons que, comme $\pi_0' \geqslant \pi_0$, et $\pi^{\pi_0'} = \pi$, on aura $\pi^{\pi_0} = \pi$, donc π est cardinal utile pour $\mathcal{M}$, donc $I = \mathcal{M}_\pi/x$ est grande devant x, et on a

$$(**) \qquad \varinjlim_{i\in I=\mathcal{M}_\pi/x} x_i \xrightarrow{\sim} x\ .$$

Donc, si on admet que $\underline{S}(\pi)/x$ est grande devant π, le fait que $(*)$ soit un isomorphisme

124. Ce serait le moment de formuler ici le Lemme 4.16.7.11 (p. 488), qui devrait même remonter avec les premières généralités sur les parties π-accessibles dès avant 4.13.7 (immédiat après 4.13.6). [page 294]

équivaut à dire que le foncteur canonique

$$J \overset{\text{déf}}{=} \underline{S}(\pi)/x \longrightarrow \mathcal{M}_\pi \;=\; I$$

est *cofinal* (cf. 4.4.2, p. 79). Mais on voit tout de suite que ce foncteur est un foncteur d'inclusion d'une sous-catégorie strictement pleine (stable par $\varinjlim$ grandes devant π_0', de cardinal $\leqslant \pi$).

Notons maintenant que la flèche $(**)$ similaire à $(*)$, avec $\underline{S}(\pi)$ remplacée par $\underline{S}_\alpha(\pi)$, est bel et bien un isomorphisme, puisque $x \in \operatorname{Ob} \underline{S} \subset \operatorname{Ob} \underline{S}_\alpha$, et que $\underline{S}_\alpha$ est π-accessible. Cela s'interprète par le fait que le foncteur d'inclusion

$$J_\alpha \overset{\text{déf}}{=} \underline{S}_\alpha(\pi)/x \longrightarrow \mathcal{M}_\pi/x \;=\; I$$

est cofinal. C'est encore une inclusion d'une sous-catégorie J_α strictement pleine de I, stable par $\varinjlim$ grandes devant π_0' et de cardinal $\leqslant \pi$, et on a

$$J \;=\; \bigcap J_\alpha \,.$$

Donc la condition voulue résultera du

Lemme 4.13.10 ([125], [126]). *Soient π_0' et $\pi > \pi_0'$ des cardinaux infinis, I une catégorie équivalente à une petite catégorie, $(J_\alpha)_{\alpha\in A}$ une famille de sous-catégories J_α de I. On suppose :* [page 295]

ⓐ *I est grande devant π ([127]).*

ⓑ *I est stable par $\varinjlim$ de cardinal $\leqslant \pi$, et grandes devant π_0' ([128]).*

ⓒ *$\forall\, r \in A$, J_r est stable dans I par les limites inductives envisagés dans b) (en particulier, J_r est une sous-catégorie* strictement pleine *de I).* p. 110

ⓓ *$\forall\, r \in A$, J_r est une sous-catégorie* cofinale *de I (*grande devant π*).*

ⓔ $\operatorname{card} A \leqslant \pi$.

Soit $J = \bigcap_{r\in A} J_r$. Alors J est encore cofinale dans I, et grande devant π (et de plus stable dans I par $\varinjlim$ de cardinal $\leqslant \pi$ et grandes devant π_0').

En d'autres termes, si I est grande devant π, l'intersection J d'une famille $(J_\alpha)_{\alpha\in A}$ de sous-catégories strictement pleines cofinales et grandes devant π, est encore cofinale et grande devant π, *pourvu que* $\operatorname{card} A \leqslant \pi$, et que l'on postule une propriété de stabilité de I et des J_α dans I, pour les limites inductives de cardinal $\leqslant \pi$ et grandes devant un cardinal convenable $\pi_0' < \pi$ donné. [page 296]

4.13.10.1. Notons d'ailleurs que si $f : J \longrightarrow I$ est un foncteur cofinal *pleinement fidèle*, avec I grande devant π, alors J est automatiquement grande devant π. Cela résulte p.ex. de 4.9.6.3 2° (p. 225) par le critère c_π (vu que pour f pleinement fidèle, la condition F 3_π est tautologiquement satisfaite). On peut aussi le voir ainsi : Soit B un ensemble avec $\operatorname{card} B \leqslant \pi$, il faut prouver que $J \longrightarrow J^B$ est cofinal, or on a

$$\begin{array}{ccc} J & \xrightarrow{\text{cof. ?}} & J^B \\ {\scriptstyle\text{cof.}}\downarrow & & \downarrow{\scriptstyle\text{cof.}} \\ I & \xrightarrow{\text{cof.}} & I^B \,, \end{array}$$

où trois des quatre flèches du carré sont cofinales, $J^B \longrightarrow I^B$ l'étant comme produit de foncteurs cofinaux entre catégories *filtrantes* (cf. 4.9.6.5, page 227) – on sait en effet déjà que J est filtrante.

125. Il faut supposer que π ne soit pas un cardinal limite, pour pouvoir conclure que I est stable par $\varinjlim$ de type I_{α_π}, et les J_r stables dans I par ces limites.

126. Cf. forme plus générale du lemme dans 4.13.10.2, p. 302. [page 297]

127. Il suffit que I soit grande devant tout $\pi' < \pi$.

128. Il suffit ici la stabilité par $\varinjlim$ de type I_{α_π}, cf. page 302 [?].

Donc le composé

$$J \longrightarrow J^B \xrightarrow{\text{cof.}} I^B$$

est cofinal. Mais comme le deuxième foncteur $J^B \longrightarrow I^B$ est pleinement fidèle, il s'ensuit aussitôt que $J \longrightarrow J^B$ est cofinal (car $\xi\backslash J$ pour $\xi \in \mathrm{Ob}\, J^B$ est le même, à isomorphisme près, qu'on considère ξ comme objet de J^B, ou de I^B).

Ainsi dans l'énoncé de la condition d) du lemme, il est inutile de spécifier que J_α est grande devant π, et de même dans la conclusion sur J.

Comme $J \hookrightarrow I$ est pleinement fidèle, pour voir que ce foncteur est cofinal, il suffit de voir qu'il satisfait F 1, i.e. que pour tout $i \in \mathrm{Ob}\, I$, existe $j \in J$ avec $i \longrightarrow j$. On va construire j comme limite d'un système inductif ordinal d'objets i_α de I, indexé par les ordinaux α tels que $\mathrm{card}\,\alpha < \pi$. Désignons par α_π le plus petite ordinal tel que $\mathrm{card}\,\alpha_\pi = \pi$. L'ensemble p. 111 ordonné I_{α_π} des ordinaux $\alpha < \alpha_\pi$ étant grande devant π'_0 ([129]) et de cardinal π, on pourra [page 298] donc prendre la limite $\varinjlim_{\alpha \in I_{\alpha_0}} i_\alpha$ dans I.

Construction de $(i_\alpha)_{\alpha \in I_{\alpha_0}}$. Soit

$$\alpha_\pi \;=\; \text{plus petit ordinal } \alpha \text{ tel que } \mathrm{card}\,\alpha = \pi, \tag{1}$$

donc c'est l'ordinal caractérisé par

$$\begin{cases} \mathrm{card}\,\alpha_\pi \;=\; \pi \\ \alpha < \alpha_\pi \;\Longrightarrow\; \mathrm{card}\,\alpha \;<\; \pi\ , \end{cases} \tag{2}$$

ou encore par le fait que l'ensemble ordonné I_{α_π} des ordinaux $\alpha < \alpha_\pi$ est *parfaitement bien ordonné*. Ainsi, I_{α_π} *est grande devant* π' *pour tout cardinal* $\pi' < \pi$ ([130]). Dans les conditions b) c) de 4.3.10, il suffira de supposer I stable par $\varinjlim$ de type I_{α_π}, et les J_r ($r \in A$) stables dans I par ce type de limites.

Lemme 4.13.10.1. *Soit π un cardinal infini, A un ensemble non vide tel que* $\mathrm{card}\,A \leqslant \pi$. *Alors il existe une application* [page 299]

$$\varphi \;:\; I_{\alpha_\pi} \longrightarrow A$$

telle que pour tout $r \in A$, $\varphi^{-1}(\{r\})$ soit une partie cofinale de I_{α_π}, ou ce que est plus précis, $\mathrm{card}\,\varphi^{-1}(\{r\}) = \pi$.

Commentary · Exposition [ms. 298]

The ordinal calculation needs cofinality, not cardinality alone

Grothendieck chooses an ordinal long enough to revisit every condition, and then takes its colimit. Two distinct size arguments enter this construction. An ordinal β below the initial ordinal of cardinal π has cardinal less than π, so filteredness supplies a cocone at that stage. By contrast, the whole initial ordinal π is large before ρ precisely when

$$\mathrm{cf}(\pi) > \rho.$$

Indeed a subset of size at most ρ has an upper bound exactly when it is not cofinal. The statement on p. 298 that the ordinal is large before every cardinal below π is therefore equivalent to regularity of π. For $\pi = \aleph_\omega$, the countable sequence of initial ordinals $\aleph_n$ is a counterexample. The marginal warning is justified, but "limit cardinal" is too broad: a regular limit cardinal would cause no such failure.

There is a useful qualification for 4.13.8. Its equation $\pi^\rho = \pi$ already forces $\mathrm{cf}(\pi) > \rho$. To see this directly, suppose $\delta = \mathrm{cf}(\pi) \leq \rho$ and enumerate all maps $\delta \to \pi$ by indices below π. Partition those indices into δ sets E_ξ, each of cardinal less than π. At coordinate ξ, choose a value different from the values of all maps indexed by E_ξ. The resulting map differs from every map in the enumeration,

129. faux si π est un cardinal limite.

130. Faux si π est cardinal limite.

a contradiction. Thus the particular filtration needed in 4.13.8 follows from its fixed-point equation; the extra successor-cardinal requirement in footnote 123 is unnecessary there.

The ordinal lemma restated in 4.13.10.2 has an independent valid proof under its explicitly stated chain-colimit hypothesis. Partition the initial ordinal π into at most π subsets, each of cardinal π, and assign one part to each J_r. Every such part is cofinal and, with its induced order, has order type π. Recursively choose a compatible chain which, at the assigned stages, belongs to the corresponding J_r; at a limit stage take a cocone, then extend into the required cofinal subcategory. Start by extending the prescribed object into the first assigned subcategory, rather than requiring an arbitrary $i_0 = i$ already to belong to it. The final colimit is the colimit of its restriction to each assigned part, hence belongs to every J_r. This proves cofinality. If I is only large before every cardinal below π, the full subcategory inherits precisely those filtration bounds; inheritance of the stronger bound π requires that bound for I itself.

En effet, il suffit de considérer un ensemble A' de cardinal π (p.ex. $A' = I_{\alpha_\pi}$), l'application

$$\psi \;=\; \mathrm{pr}_1 \;:\; A \times A' \longrightarrow A\;,$$

et de se donner une bijection

$$A \times A' \xrightarrow{\sim} I_{\alpha_\pi}$$

laquelle existe puisque

$$\mathrm{card}\,(A \times A') \;=\; \mathrm{card}\,A \cdot \mathrm{card}\,A' \;=\; \mathrm{card}\,A' \;=\; \pi \;=\; \mathrm{card}\,I_{\alpha_\pi}\;,$$

puisque $0 \neq \mathrm{card}\,A \leqslant \mathrm{card}\,A' = \pi$ et π infini. On suppose choisie φ comme dans 4.13.10.1, et on construira le système inductif $(i_\alpha)_{\alpha \in I_{\alpha_\pi}}$ de telle façon qu'on ait

$$i_\alpha \;\in\; J_{\varphi(\alpha)}\;.$$

Posant

$$j \;=\; \varinjlim_{\alpha \in I_{\alpha_\pi}} i_\alpha\;,$$

pour tout r, comme p. 112 [page 300]

$$I_r \overset{\text{déf}}{=} \varphi^{-1}(r) \;\subset\; I_{\alpha_\pi}$$

est cofinal dans I_{α_π}, on aura donc

$$j \;=\; \varinjlim_{\alpha \in I_r} i_\alpha\;,$$

et en vertu de $(*)$, les i_α pour $\alpha \in I_r$ sont dans J_r, donc par passage à la limite, utilisant l'hypothèse c), on trouve que $j \in J_r$, donc

$$j \;\in\; \bigcap_{r \in A} J_r \;=\; J\;,$$

et comme on choisira

$$(**) \qquad i_0 \;=\; i\;,$$

on aura bien trouvé $i \longrightarrow j$ avec $i \in J$. Donc il suffit de construire le système inductif (i_α) satisfaisant $(*)$ et $(**)$.

Si le système inductif est construit jusqu'au cran $i_\alpha \in \mathrm{Ob}\,J_{\varphi(\alpha)}$, on continue par

$$i_\alpha \longrightarrow i_{\alpha+1} \;\in\; \mathrm{Ob}\,J_{\varphi(\alpha+1)}\;,$$

ce qui est possible puisque $J_{\varphi(\alpha+1)}$ est cofinale. Donc il reste à construire i_α pour α ordinal limite. Or l'ensemble des i_β pour $\beta < \alpha$ est de cardinal $< \pi$, et comme par a) I est grande [page 301]

devant π (il suffit qu'il le soit devant tout cardinal $< \pi$), il existe un $i \in \mathrm{Ob}\, I$ qui majore tous les i_β. Choisissons des

$$i_\beta \xrightarrow{v_\beta} i\ .$$

On n'aura pas a priori commutativité dans

$$i_{\beta'} \xrightarrow{\tau_{\beta,\beta'}} i_\beta \underset{v_{\beta'}}{\overset{v_\beta}{\rightrightarrows}} i\ ,$$

mais comme l'ensemble des double-flèches $i_{\beta'} \underset{v_{\beta'}}{\overset{v_\beta\tau_{\beta,\beta'}}{\rightrightarrows}} i$ est de cardinal $< \pi$, I étant grande devant π, il existe $i \longrightarrow i'$ qui les égalise tous. Comme $J_{\varphi(\alpha)}$ est cofinal dans I, il existe $i'' \in J_{\varphi(\alpha)}$, et $i \longrightarrow i''$. On prend $i_\alpha = i''$, avec les homomorphismes composés

$$i_\beta \xrightarrow{v_\beta} i \longrightarrow i' \longrightarrow i''$$
$$\tau_{\alpha,\beta}$$

comme morphismes de transition. Cela achève la démonstration de 4.13.10 et par là de 4.13.9, 4.13.8. p. 113 [page 302]

Remarque 4.13.10.2. Voici ce qu'on a eu à utiliser vraiment, pour établir la conclusion de 4.13.10, en termes de I, π, $(J_r)_{r\in A}$ (et sans introduire de π'_0).

(a′) I est grande devant tout cardinal $\pi' < \pi$.

(b′) I est stable par $\varinjlim$ de type I_{α_π}.

(c′) $\forall\, r \in A$, J_r est stable dans I par les $\varinjlim$ de type I_{α_π}.

(d′) $\forall\, r \in A$, J_r est cofinale dans I.

(e′) $\mathrm{card}\, A \leqslant \pi$.

(**N.B.** d′) et e′) ne sont autres que d) et e) de 4.3.10, tandis que a′), b′), c′) sont des versions affaiblies de a), b), c) respectivement.)

Commentary · Development (commentators') [ms. 302]

Simultaneous refinement removes the regularity restriction

The intersection argument can be arranged so that the number of conditions and the length of the construction play separate roles. This gives the conclusion of 4.13.10 for the hypotheses in its main text, without imposing the extra cofinality restriction introduced in its footnotes.

Simultaneous cofinal intersection lemma. [ours] Let $\rho < \pi$ be infinite cardinals. Suppose I is π^+-filtered and admits all ρ^+-filtered colimits indexed by categories with at most π arrows. Let $(J_r)_{r\in A}$, where $|A| \le \pi$, be replete full cofinal subcategories closed under these colimits in I. Then $\bigcap_r J_r$ is cofinal in I, is π^+-filtered, and is closed under the same colimits.

Proof. The empty family is immediate. Put $\kappa = \rho^+$, a regular cardinal with $\kappa \le \pi$, and fix $x_0 \in I$. Construct a chain $(x_\beta)_{\beta<\kappa}$. After choosing x_β, choose, for every r, an arrow

$$a_{\beta r} : x_\beta \longrightarrow y_{\beta r}, \qquad y_{\beta r} \in J_r.$$

The star consisting of these arrows has at most π arrows. Filteredness therefore supplies a cocone with vertex $x_{\beta+1}$ and arrows $b_{\beta r} : y_{\beta r} \to x_{\beta+1}$ satisfying

$$b_{\beta r} a_{\beta r} = t_{\beta+1,\beta} \quad \text{for every } r.$$

At a limit $\gamma < \kappa$, choose a cocone on the entire already constructed x-chain; its size is less than $\kappa \le \pi$. The resulting transition arrows $t_{\gamma\beta}$ are compatible by construction. No intermediate colimit is required.

For a fixed r define, when $\beta < \gamma$,

$$y_{\beta r} \xrightarrow{b_{\beta r}} x_{\beta+1} \xrightarrow{t_{\gamma,\beta+1}} x_\gamma \xrightarrow{a_{\gamma r}} y_{\gamma r}.$$

These maps compose correctly: the equality $b_{\gamma r}a_{\gamma r} = t_{\gamma+1,\gamma}$ reduces composition to the transition identity for the x-chain. The maps a and b consequently give mutually inverse maps between the colimits of the two chains; their composites are the maps induced by a shift of the indexing ordinal, hence the identities on those colimits. The ordinal κ is ρ^+-filtered and has at most π arrows. Thus $x = \varinjlim_{\beta<\kappa} x_\beta$ exists, and closure of J_r shows that the isomorphic colimit of the $y_{\beta r}$ lies in J_r. Repleteness gives $x \in J_r$ for every r. The map $x_0 \to x$ proves the first cofinality condition.

For completeness, a full subcategory of a filtered category which receives a map from every ambient object is cofinal: any two objects in a comma category have a common successor by taking a cocone in the ambient category and then extending into the subcategory; parallel arrows are equalized in the same way. The same argument for diagrams with at most π arrows shows that the intersection is π^+-filtered. Closure under the specified colimits is inherited from each J_r. □

Applied to $I = \mathcal{M}_\pi/x$, this also repairs 4.13.9 without its unstated fixed-point equation, for a nonempty family. Choose one r: since $x \in \mathcal{S}_r$ and $\mathcal{S}_r$ is π-accessible in $\mathcal{M}$, a π^+-filtered presentation of x inside $\mathcal{S}_r$ uses objects presentable in $\mathcal{M}$. Factoring any at-most-π diagram of maps from presentable objects through a common stage shows directly that $\mathcal{M}_\pi/x$ is π^+-filtered and presents x. Global usefulness of π for $\mathcal{M}$ is therefore unnecessary for these particular slices. For an empty family the fixed-rank assertion requires π useful for $\mathcal{M}$, since the intersection is then $\mathcal{M}$ itself.

We checked the intersection and pseudopullback formulations in Lurie (2009); Chorny and Rosický (2012); Positselski (2024); we did not find this exact simultaneous-star repair of the printed ordinal argument there. The lemma is recorded as an explicit completion of Grothendieck's method, without a priority claim. It uses the full π^+-filteredness of I for the stars. The weaker hypothesis suggested in footnote 127 does not support this proof when $|A| = \pi$.

4.13.8 (reformulation). *Soit pour $r \in A$, $\pi''_r \geqslant \pi'_r \geqslant \pi_0$ (= cardinal utile pour $\mathcal{M}$ donné) comme spécifié dans 4.13.2 b) (et c), soit $\pi' = \sup_{r\in A} \pi'_r$, $\pi'' = \sup_{r\in A} \pi''_r$, de sorte que pour tout cardinal π avec $\pi^{\pi'_0} = \pi \geqslant \pi''$, $\underline{S}_r$ soit π-accessible dans $\mathcal{M}$ (et π utile pour $\underline{S}_r$* et *pour $\mathcal{M}$). Soit c un cardinal* non limite *tel que $c \geqslant \operatorname{card} A$ et $c > \pi'$ (p.ex. le successeur de* $\sup(\operatorname{card} A, \pi')$*). Prenons π tel que*

$$\pi^{\pi'} \;=\; \pi \;\geqslant\; \sup(\pi'', c)\ .$$

Alors $\underline{S} = \bigcap_{r\in A} \underline{S}_r$ est π-accessible.

En effet, si α_c est le premier ordinal de cardinal c, comme c n'est pas limite, I_{α_c} est grande devant tout cardinal $c' < c$, en particulier devant π'. Or pour tout $x \in \operatorname{Ob}\mathcal{M}$, $I = \mathcal{M}_\pi/x$ et les $J_r = \underline{S}_r/x$ sont stables par $\varinjlim$ grandes devant π' et de cardinal $\leqslant \pi$, et les J_r stables dans I pour ces limites, donc il en est en particulier ainsi pour les $\varinjlim$ de type I_{α_c}. On peut donc appliquer le résultat 4.13.10.2 pour conclure que si les J_r sont cofinaux dans I (qui est grande devant π, π étant utile pour $\mathcal{M}$), alors $J = \underline{S}/x$ l'est également. Cela implique alors, comme on a vu, que $\underline{S}$ est π-accessible dans $\mathcal{M}$. [page 302 bis]

4.13.10.3. On peut reformuler la reformulation précédente, en utilisant la terminologie introduite plus bas (4.13.23) :

Soit c un cardinal, α_c le plus petit ordinal α tel que $\operatorname{card}\alpha = c$, *$\pi$ un cardinal $\geqslant c$. On suppose*

(∗) *π est utile pour $\mathcal{M}$, et $\mathcal{M}$ est stable par $\varinjlim$ de type I_{α_c}. (Cette deuxième condition est le cas p.ex. si c est* non limite *et $c > \pi_0$, π_0 étant un cardinal tel que $\mathcal{M}$ satisfait 2_{π_0}, p.ex. un cardinal* utile *pour $\mathcal{M}$ ([131])).*

Alors l'ensemble des sous-catégories $\underline{S}$ de $\mathcal{M}$ qui sont α_c-π-accessible dans $\mathcal{M}$ (i.e. qui sont π-accessibles et stables dans $\mathcal{M}$ par $\varinjlim$ de type I_{α_c}) est stable par intersection de cardinal $\leqslant c$. Plus généralement, si Φ est un ensemble de catégories telles que

131. Car alors I_{α_c} est grande devant π_0.

1°) $I \in \Phi \Longrightarrow \operatorname{card} I \leqslant \pi$ *et*

2°) $I_{\alpha_c} \in \Phi$,

alors l'ensemble des sous-catégories de $\mathcal{M}$ *qui sont* Φ-π*-accessibles dans* $\mathcal{M}$*, est stable par intersection de cardinal* $\leqslant c$.

Définition 4.13.11 ([132]). Soient $\mathcal{M}$ une catégorie accessible, π_0 un cardinal utile pour $\mathcal{M}$, $\pi > \pi_0$ un cardinal. Une catégorie I est dite π_0-π-*adaptée* si p. 114 [page 303]

a) I est équivalente à une petite catégorie,

b) I est grande devant π, et

c) I est stable par $\varinjlim$ de cardinal $\leqslant \pi$, et grandes devant π_0 ([133]).

Un système inductif $(x_i)_{i\in I}$ de type I dans $\mathcal{M}$ est dit π_0-π-*adapté*, si I est π_0-π-adaptée, i.e. satisfait les conditions a) b) c), et si de plus on a

d) le foncteur $i \longmapsto x_i : I \longrightarrow \mathcal{M}$ commute aux $\varinjlim$ (grandes devant π_0 et de cardinal $\leqslant \pi$) envisagées dans c), et

e) les x_i sont dans $\mathcal{M}_\pi$.

Proposition 4.13.12. *Soient* $\mathcal{M}$ *une catégorie* π_0*-accessible, et* $\pi > \pi_0$ *un cardinal utile pour* $\mathcal{M}$*. Alors tout* $x \in \operatorname{Ob}\mathcal{M}$ *admet une représentation comme* $x \xleftarrow{\sim} \varinjlim_{j\in J} x_j$*, où le système inductif* $(x_j)_{j\in J}$ *dans* $\mathcal{M}$ *est* π_0*-*π*-adapté. De façon précise, on peut prendre* $J = \mathcal{M}_\pi/x$.

En effet, la vérification des propriétés a) à d) pour le système inductif canonique, de $\varinjlim$ isomorphe à x, est immédiate. [page 304]

La proposition précédente ne dit pas si on peut prendre pour J un ensemble ordonné. Dans ce cas, la condition c) signifie que pour toute partie J' de J grande devant π_0 et de cardinal $\leqslant \pi$, J admet une borne supérieure i_J dans J, et la condition d) que l'on a alors $x_{i_J} \xleftarrow{\sim} \varinjlim_{i\in J'} x_i$.

L'idée naturelle, pour construire un système inductif π_0-π-adapté $(x_j)_{j\in J}$, avec J ordonnée, en partant de celui qu'on a canoniquement, avec $J = \mathcal{M}_\pi/x$, est de prendre pour I l'ensemble, ordonné par inclusion, des sous-catégories J' de J_0 (sous-catégorie pleine petite de J équivalente à J) qui sont de cardinal $\leqslant \pi$ et grandes devant π_0. (Comparer (4.7.3.3), page 165). Cet ensemble ordonné est bel et bien (petit et) *grand devant* π, si on suppose que $\pi^{\pi_0} = \pi$ (cf. démonstration 4.7.5, p. 168-171).

Lemme 4.13.13. *Soient* π_0*,* $\pi > \pi_0$ *des cardinaux infinis tels que* $\pi^{\pi_0} = \pi$*, et* J *une petite catégorie, grande devant* π*. Soit* I *l'ensemble des sous-catégories* J' *de* J *qui sont grandes devant* π_0*, et de cardinal* $\leqslant \pi$*. Alors* I *est* π_0*-*π*-adaptée.* [page 305]

Il reste à voir que pour tout ensemble ordonné A grand devant π_0 et de cardinal $\leqslant \pi$, si $r \longmapsto J_r$ est une famille croissante d'éléments de I indexée par A, alors

$$J \overset{\text{déf}}{=} \bigcup_{r\in A} J_r \;=\; \varinjlim_{r\in A} J_r$$

est dans I, i.e. est une sous-catégorie de I de cardinal $\leqslant \pi$ (ce qui est clair) et *grande devant* π_0. Cela va résulter du

Lemme 4.13.13.1. *Soit* $(J_r)_{r\in A}$ *un système inductif de catégories grandes devant* π_0*, indexé par un ensemble ordonné* A *grand devant* π_0*. Alors* $J = \varinjlim_{r\in A} J_r$ *est grande devant* π_0. p. 115

Il faut voir que si B est un ensemble de cardinal $\leqslant \pi_0$, alors le foncteur diagonal [page 306]

$$(*) \qquad \delta \;:\; J \longrightarrow J^B$$

132. Refondre avec 4.13.23 etc.

133. **N.B.** Si $\pi = \pi_0$, la condition c) équivaut à dire que I est stable par facteurs directs.

est cofinal. Or comme A est grande devant π_0, $\operatorname{card} B \leqslant \pi_0$, les produits indexés par B commutent aux $\varinjlim$ de type A (SGA 4 I 9.8 appliqué à Cat et à π_0). Donc la flèche $(*)$ s'interprète comme limite inductive des flèches similaires

$$(**) \qquad \delta_r \ : \ J_r \longrightarrow J_r^B \ ,$$

lesquelles sont cofinales, donc δ est cofinal comme limite inductive de foncteurs cofinaux.

Commentary · Exposition [ms. 305]

Replacing a presentation by a continuous poset of presentations

The adapted presentations are designed to retain more than a filtered colimit expression. They remember colimits of smaller filtered diagrams inside the indexing category, and require the diagram itself to preserve those colimits. This continuity is what will allow a property of x to be detected in any sufficiently well-behaved presentation of x.

For the canonical presentation $\mathcal{M}_\pi/x \to \mathcal{M}$, the verification is concrete. A diagram with at most π arrows has a colimit in $\mathcal{M}$ when it is large before π_0. Its colimit carries the induced map to x. Moreover it remains π^+-presentable: its representable functor is an at-most-π limit of representables, and those limits of sets commute with π^+-filtered colimits. Thus the colimit is still an object of $\mathcal{M}_\pi/x$ and is computed there as claimed.

To replace this indexing category by a poset, one must close small sets of arrows under enough cocone choices. Here is the size argument behind 4.13.13. Start with at most π arrows of a small π^+-filtered category J, adjoin identities and composites, and, for every diagram of size at most π_0 in the resulting subcategory, adjoin a chosen cocone in J. There are at most $\pi^{\pi_0} = \pi$ such diagrams. Repeat the operation along π_0^+ stages, which is permissible because $\pi_0^+ \leq \pi$. The union still has at most π arrows. Every diagram of size at most π_0 in the union occurs at one stage, since π_0^+ is regular, so a subsequent stage contains its cocone. This gives a subcategory large before π_0.

Applying the construction to the union of at most π prescribed subcategories proves that the poset I in 4.13.13 is large before π. Its smaller directed suprema are literal unions. If A is π_0^+-directed, $|A| \leq \pi$, and $J_r \subseteq J_s$ for $r \leq s$, then $\bigcup_r J_r$ has at most π arrows. A diagram of size at most π_0 in this union is contained in one J_r, where it has a cocone. The union is therefore again an object of I.

This also proves 4.13.13.1 without relying on an unstated stability theorem for final functors. For general transition functors, rather than inclusions, lift the objects and arrows of a small diagram in $\varinjlim_r J_r$ to a common stage. There are at most π_0 composition and identity equations to impose, so a further common stage realizes them all. Take a cocone there. Hence the filtered colimit category is again π_0^+-filtered.

Finally the transition maps between $x_{J'} = \varinjlim_{j \in J'} x_j$ are induced by inclusions of subcategories. For a directed union $J' = \bigcup_r J_r$, a cocone on $x|_{J'}$ is exactly a compatible family of cocones on the restrictions $x|_{J_r}$. This proves

$$\varinjlim_r x_{J_r} \cong x_{\bigcup_r J_r}$$

by the universal property, and gives the continuity asserted in 4.13.13.2. Every object and arrow of J is contained in one of these subcategories; the same cocone argument gives $\varinjlim_{J' \in I} x_{J'} \cong \varinjlim_{j \in J} x_j$. This is the precise reason the replacement preserves the object being presented.

Lemme 4.13.13.2. *Sous les hypothèses de 4.13.13, soient $\mathcal{M}$ une catégorie satisfaisant L_{π_0}, et $(x_j)_{j \in J}$ (où $J \longrightarrow \mathcal{M}$) un système inductif dans $\mathcal{M}$, indexé par J. Pour toute $J' \in I$, posons*

$$x_{J'} \ = \ \varinjlim_{j \in J'} x_j$$

(qui existe dans $\mathcal{M}$, puisque $\mathcal{M}$ satisfait à L_{π_0}). Alors les $(x_{J'})_{J' \in I}$ forment un système inductif dans $\mathcal{M}$, et $I \longrightarrow \mathcal{M}$ commute aux limites inductives grandes devant π_0 et de cardinal $\leqslant \pi$. Si $\mathcal{M}$ est π_0-accessible, et les x_j ($j \in \operatorname{Ob} J$) sont dans $\mathcal{M}_\pi$, alors $(x_{J'})_{J' \in I}$ est un système inductif π_0-π-adapté dans $\mathcal{M}$. [page 307]

Tout ceci est essentiellement trivial, et donne :

Corollaire 4.13.14 (de 4.13.12). *Sous les conditions de 4.13.12, si $\pi^{\pi_0} = \pi$, on peut pour tout $x \in \mathrm{Ob}\,\mathcal{M}$ trouver une représentation $x \simeq \varinjlim_{i\in I} x_i$, où $(x_i)_{i\in I}$ est un système inductif π_0-π-adapté avec I* ensemble ordonné. *De plus, si $\pi' > \pi$ et si $x \in \mathcal{M}_{\pi'}$, on peut prendre I tel qu'on ait*

$$\mathrm{card}\, I \ \leqslant\ \pi'^{\pi}\ .$$

Ce dernier complément est immédiat, via les majorations habituelles. [page 308]

Commentary · Examples [ms. 307]

Adapted systems in a powerset category

The ordered version becomes especially transparent in a powerset category. Let $\mathcal{M} = \mathcal{P}(X)$ and let $Y \subseteq X$. For any infinite π, take

$$I = [Y]^{\leq \pi}, \qquad D(T) = T.$$

The poset I is π^+-directed and its colimit is Y. A directed family of at most π members has supremum their union, again of cardinal at most π, and D preserves that supremum. Every value T is π^+-presentable. Thus this is an adapted presentation, with no cardinal exponent calculation needed in this example.

Fix $Z \subseteq X$ with $|Z| \leq \pi$ and let $\mathcal{S}$ consist of the subsets containing Z. Then

$$I_{\mathcal{S}} = \{T \subseteq Y : |T| \leq \pi,\ Z \subseteq T\}$$

is cofinal in I exactly when $Z \subseteq Y$. If $Z \subseteq Y$, the extension $T \subseteq T \cup Z$ proves cofinality. Conversely a single object of $I_{\mathcal{S}}$ implies $Z \subseteq Y$. This calculates the recognition condition of 4.13.15 in full: an eventual supply of small approximations satisfying the property is equivalent to the colimit satisfying it. The word "eventual" means cofinality, not that every sufficiently large index must work; in more general diagrams the good indices need not form a final segment.

Théorème 4.13.15 ([134]). *Soient $\mathcal{M}$ une catégorie π_0-accessible, et $\pi > \pi'_0 \geqslant \pi_0$ tels que $\pi^{\pi_0} = \pi$ ([135]). Soit $\underline{S}$ une sous-catégorie strictement pleine de $\mathcal{M}$, et soit $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. Les conditions suivantes sont équivalentes.*

ⓐ *$\underline{S}$ est π-accessible dans $\mathcal{M}$ (4.13.3) et $\underline{S}$ est stable dans $\mathcal{M}$ par limites inductives grandes devant π'_0.*

p. 116

a′) *$\underline{S}$ est π-accessible dans $\mathcal{M}$, et $\underline{S}(\pi)$ est stable dans $\mathcal{M}_\pi$ par limites inductives grandes devant π'_0 et de cardinal $\leqslant \pi$ ([136]).*

b) *Pour tout objet x de $\mathcal{M}$, on a ceci : la sous-catégorie strictement pleine $\underline{S}(\pi)/x$ de $J = \mathcal{M}_\pi/x$ est cofinale dans $\mathcal{M}_\pi/x$ si et seulement si $x \in \mathrm{Ob}\,\underline{S}$. De plus, $\underline{S}(\pi)$ est stable dans $\mathcal{M}_\pi$ pour $\varinjlim$ grandes devant π'_0 et de cardinal $\leqslant \pi$.*

ⓒ *Pour tout objet x de $\mathcal{M}$ et pour toute représentation $x \simeq \varinjlim_{i\in I} x_i$, où $(x_i)_{i\in I}$ est un système inductif π'_0-π-adapté, (4.13.11), on a ceci : la sous-catégorie strictement pleine $I_{\underline{S}}$ de I formée des $i \in \mathrm{Ob}\, I$ tels que $x_i \in \mathrm{Ob}\,\underline{S}$, est cofinale dans I si et seulement si $x \in \mathrm{Ob}\,\underline{S}$. De plus, $S(\pi)$ est stable dans $\mathcal{M}_\pi$ par $\varinjlim$ grandes devant π'_0 et de cardinal $\leqslant \pi$.*

[page 309]

c′) *Comme d)* `[plutôt c)]`*, avec I ensemble ordonné.*

Corollaire 4.13.16. *Les hypothèses sur $\mathcal{M}$, π_0, $\underline{S}$ étant celles de 4.13.15, pour qu'il existe des cardinaux $\pi > \pi'_0 \geqslant \pi_0$ satisfaisant les conditions $\pi^{\pi'_0} = \pi$ (a fortiori $\pi^{\pi_0} = \pi$) et les conditions*

134. Cf. version plus générale et plus symétrique du théorème 4.13.15 dans 4.13.22.

135. On peut se borner à supposer π utile pour $\mathcal{M}$ (au lieu de $\pi^{\pi_0} = \pi$), alors on trouve que a, a′, b, c sont équivalentes et impliquent c′, mais ne sont peut-être pas équivalentes à c′.

136. donner un nom à cette dernière condition, soit $\mathrm{L}(\pi'_0, \pi)$ (pour $\underline{S}(\pi)$ dans $\mathcal{M}_\pi$).

équivalentes a) à d′) [plutôt c′)] *de 4.13.15, il (faut et il) suffit que* $\underline{S}$ *soit accessible dans* $\mathcal{M}$*. De façon plus précise, si* $\underline{S}$ *est accessible dans* $\mathcal{M}$*, soit* $\pi_0' \geqslant \pi_0$ *un cardinal tel que* $\underline{S}$ *soit* π_0'*-accessible dans* $\mathcal{M}$*, et soit* π *un cardinal* $> \pi_0'$ *tel que* $\pi^{\pi_0'} = \pi$*. Alors* $(\underline{S}, \pi_0', \pi)$ *satisfait aux conditions équivalentes de 4.13.15.*

En effet, le "il faut" est évident sur la condition a) de 4.13.15. Pour le "il suffit" on note que $\underline{S}$ π_0'-accessible dans $\mathcal{M}$ implique que $\underline{S}$ est stable dans $\mathcal{M}$ par $\varinjlim$ grandes devant π_0'. Donc pour vérifier a), il suffit de vérifier que $\underline{S}$ est π-accessible dans $\overrightarrow{\mathcal{M}}$. Or comme $\underline{S}$ est [page 310]
π_0'-accessible et que $\pi > \pi_0'$ satisfait $\pi^{\pi_0'} = \pi$, on en conclut que $\underline{S}$ est $\underline{S}_\pi$-accessible, i.e. $\underline{S} \xleftarrow{\sim} \mathrm{Ind}_\pi(\underline{S}_\pi)$. Donc en vertu de 4.13.3, il reste à voir que $\underline{S}_\pi \subset \mathcal{M}_\pi$, ou ce qui revient au même, que $\underline{S}_\pi \subset \underline{S}(\pi) \overset{\text{déf}}{=} \underline{S} \cap \mathcal{M}_\pi$. (En fait, on aura même égalité, car il est clair que $\underline{S}(\pi) \subset \underline{S}_\pi \ldots$). Or $\underline{S}(\pi)$ est une sous-catégorie de $\underline{S}$ stable par $\varinjlim$ grandes devant π_0' et de cardinal $\leqslant \pi$ (puisque $\underline{S}$ et $\mathcal{M}_\pi$ sont stables par ces limites *dans* $\overrightarrow{\mathcal{M}}$, et que ces limites dans $\underline{S}$ et dans $\mathcal{M}$ sont itou), et $\underline{S}(\pi)$ contient $S_{\pi_0'}$. Donc $\underline{S}(\pi)$ contient $\underline{S}_\pi$, en vertu de 4.7.1 (page 161), implication c) $\Longrightarrow$ b′), appliquée à $\underline{S}$, π_0', π au lieu de $\mathcal{M}$, π_0, π. Cela prouve donc le corollaire 4.13.16.

Commentary · Development (commentators') [ms. 309]

The small-colimit condition does not imply the unbounded one

Theorem 4.13.15 attempts to recognize stability under all ${\pi_0'}^+$-filtered colimits from stability of the presentable objects under those of size at most π. This passage requires a relation between π and π_0', whereas the printed hypothesis controls only π_0. The distinction has mathematical consequences: the implication a′ $\Rightarrow$ a is false under the displayed hypotheses.

Counterexample to a′ $\Rightarrow$ a. [ours] Put $\rho = \aleph_1$, $\pi = \beth_{\omega_1}$, $\kappa = \rho^+$, and $\lambda = \pi^+$. The cardinal π is a strong limit of cofinality ρ. In particular $\pi^{\aleph_0} = \pi$: a countable sequence of ordinals below π is bounded, and for every $\theta < \pi$ one has $\theta^{\aleph_0} \leq 2^{\theta\aleph_0} < \pi$. Let

$$P = [\pi]^{\leq \rho}, \qquad \mathcal{M} = \mathcal{P}(P),$$

with both categories ordered by inclusion. Let A be the full subcategory of $\mathcal{M}$ consisting of subsets $Y \subseteq P$ with $|Y| \leq \pi$ which are nonempty and κ-directed in their induced order. Let $\mathcal{S}$ be the essential image of $\mathrm{Ind}_\pi(A)$ in $\mathcal{M}$.

The powerset calculation above shows that $\mathcal{M}$ is $\aleph_0$-accessible in the source convention and that its π-accessible objects are exactly its subsets of cardinal at most π. The inclusion of A consequently extends fully faithfully to $\mathrm{Ind}_\pi(A)$: for $Y \in A$, a map $Y \subseteq \bigcup_i Z_i$ into a λ-directed union factors through one Z_i. Thus $\mathcal{S}$ is π-accessible in $\mathcal{M}$. Its π-accessible objects are exactly A, since a λ-presentable object of an Ind-category is a retract of a generator, and a retract in a poset is the same object.

A κ-directed union of at most π members of A is again in A. Its cardinal is at most π. Any at-most-ρ family of its elements lies in one member of the diagram, where it has an upper bound. Hence A is closed under every κ-filtered colimit of cardinal at most π in $\mathcal{M}$. This proves condition a′ with $\pi_0 = \aleph_0$ and $\pi_0' = \rho$.

For each $z \in P$, the principal ideal $\downarrow z \subseteq P$ already belongs to A: it has largest element z and cardinal $2^{|z|} \leq 2^\rho < \pi$, since π is a strong limit. But the diagram

$$P \longrightarrow \mathcal{S}, \qquad z \longmapsto \downarrow z$$

is κ-directed and has union P, which does not lie in $\mathcal{S}$. To prove the last assertion, suppose P were a λ-directed union of members of A. The π-element subset $\{\{\alpha\} : \alpha < \pi\} \subseteq P$ would then lie in a single $Y \in A$. Its κ-directedness would make Y cofinal in P: bound the singletons belonging to any given at-most-ρ subset of π.

No cofinal subset of P has cardinal at most π. If $(y_\alpha)_{\alpha<\pi}$ were one, choose increasing cardinals $\pi_\xi < \pi$, $\xi < \rho$, cofinal in π. For each ξ choose

$$a_\xi \in \pi \setminus \bigcup_{\alpha < \pi_\xi} y_\alpha,$$

which is possible since the union has cardinal less than π. Then $\{a_\xi : \xi < \rho\} \in P$ is contained in none of the y_α, a contradiction. This proves failure of a. □

A sufficient repair is to strengthen the cardinal equation to $\pi^{\pi_0'} = \pi$, exactly the equation imposed in Corollary 4.13.16. Here is the missing unbounded-colimit argument under that repair. Write $A = \mathcal{S} \cap \mathcal{M}_\pi$. The bounded colimits assumed in a′ include chains of length $(\pi_0')^+ \leq \pi$. The simultaneous presentation theorem 4.9.3 therefore represents every diagram of cardinal at most π in $\mathrm{Ind}_\pi(A)$ by a π^+-filtered system of diagrams in A. For a ${\pi_0'}^+$-filtered such shape, take its colimit in A and then the filtered colimit; interchange of colimits proves that the inclusion $\mathcal{S} \hookrightarrow \mathcal{M}$ preserves the result.

For an arbitrarily large ${\pi_0'}^+$-filtered indexing category K, the closure construction of 4.13.13, now with π_0' in place of π_0, expresses K as the union of a π^+-directed system of ${\pi_0'}^+$-filtered subcategories of cardinal at most π. The equation $\pi^{\pi_0'} = \pi$ is precisely what bounds that construction. Compute the diagram's colimit first on these subcategories and then along their directed system. Both operations preserve membership in $\mathcal{S}$, proving a.

We checked the source's continuation: p. 337 calls a′ ⇒ a "sorital", and 4.13.19–21, pp. 340–341, states the bounded-to-unbounded extension without resolving this cardinal issue. Our counterexample concerns the printed theorem and those uses of that extension, not the valid existential recognition statement at sufficiently enlarged cardinals. We did not find this particular counterexample in the primary references checked; it is presented as a repair of the argument, without a priority claim.

Démonstration de 4.13.15. Voici les implications ± tautologiques,

(∗)
$$\mathrm{a} \Longrightarrow \mathrm{a}' \overset{\bullet}{\Longrightarrow} \mathrm{c} \rightrightarrows \begin{matrix} \mathrm{c}' \\ \mathrm{b}\ , \end{matrix}$$

que je vais expliciter un peu. On a a $\Longrightarrow$ a′ puisque si a est satisfait, $\underline{S}$ *et* $\mathcal{M}_\pi$ sont stables *dans* $\mathcal{M}$ par $\varinjlim$ grandes devant π_0' et de cardinal $\leqslant \pi$, donc aussi leur intersection $\underline{S}(\pi)$, qui est donc aussi stable dans $\mathcal{M}_\pi$ par lesdites limites. L'implication c $\Longrightarrow$ c′ est une tautologie, et c $\Longrightarrow$ b aussi, compte tenu de 4.13.12, qui nous dit que le système inductif canonique $(x_i)_{i\in I}$, avec $I = \mathcal{M}_\pi/x$, de limite x est π_0-π-adapté. (**N.B.** Comme $\pi^{\pi_0'} = \pi$ et $\pi_0' \geqslant \pi_0$, on a aussi $\pi^{\pi_0} = \pi$, donc π est utile pour $\mathcal{M}$, de sorte que 4.13.12 s'applique.) [page 311] p. 117

Reste à prouver la "± tautologie" a′ $\Longrightarrow$ c. Soit donc une représentation π_0-π-adaptée

$$x \ \simeq\ \varinjlim_{i\in I} x_i\ ,$$

prouvons que

$$x \ \in\ \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad I_{\underline{S}} \text{ cofinal dans } I.$$

L'implication $\Longleftarrow$ provient du fait que, $\underline{S}$ étant π-accessible dans $\mathcal{M}$, est stable dans $\mathcal{M}$ par $\varinjlim$ grandes devant π ([137]). Reste à prouver que si $x \in \mathrm{Ob}\,\underline{S}$, alors $I_{\underline{S}}$ est cofinal dans I. [page 312] Considérons donc le diagramme de catégories

$$\begin{array}{ccc} I & \xrightarrow{\ \varphi\ } & J = \mathcal{M}_\pi/x \\ \uparrow & & \uparrow{\scriptstyle \alpha} \\ I' = I_{\underline{S}} & \xrightarrow{\ \varphi'\ } & J' = \underline{S}(\pi)/x\ , \end{array}$$

c'est un carré cartésien ($I' = I_{\underline{S}}$ est l'image inverse par $\varphi : I \longrightarrow J$ de la sous-catégorie strictement pleine J' de J). On a les propriétés suivantes

1) I est grande devant π, et φ cofinal ($\Longrightarrow$ J grande devant π).
2) I stable par $\varinjlim$ grandes devant π_0' et de cardinal $\leqslant \pi$, et φ commute à ces $\varinjlim$.

137. On n'utilise ici, pour $\Longleftarrow$, de l'hypothèse $(x_i)_{i\in I}$ π_0-π-adapté, que le seul fait que I soit grande devant π.

3) J' est stable par $\varinjlim$ du type précédent et α y commute ([138]).

4) J' cofinal dans J.

On vient de conclure que $I' = \varphi^{-1}(J')$ est cofinale dans I. Comme c'est une sous-catégorie pleine (et même strictement pleine), il suffit de prouver la propriété F 1, i.e. que pour tout $i \in \mathrm{Ob}\, I$, existe $i' \in \mathrm{Ob}\, I'$ et une flèche $i \longrightarrow i'$. [page 313]

Commentary · Exposition [ms. 312]

What must be pulled back is a closed cofinal subcategory

The difficult implication in the recognition theorem is a′ ⇒ c. The canonical map from a presentation of x to $\mathcal{M}_\pi/x$ is cofinal because its values are presentable: maps from at most π presentable objects and their compatibility equations factor through a common stage. For $x \in \mathcal{S}$, the subcategory $\mathcal{S}(\pi)/x$ is cofinal for the same reason. One must then show that its inverse image in the chosen presentation is cofinal.

A strict inverse image of a cofinal subcategory need not be cofinal without further hypotheses. The inclusions of the even and odd natural numbers into $\mathbb{N}$ again give a counterexample. Grothendieck's extra continuity conditions are therefore doing real work: after alternating between the presentation and the good approximations, a colimit makes their two chains agree.

The elementary lifting fact used on pp. 321–325 is this. If $F : I \to J$ is final and I is filtered, an arrow $F(i) \to F(k)$ becomes the image of an arrow of I after postcomposition by some $F(k \to k')$. To prove it, put i and k over a common object of I. The two resulting arrows out of $F(i)$ become equal after a further transition by the filtered finality criterion. That equality is the required lift. The same criterion equalizes any prescribed small family of parallel arrows after first equalizing them in J and then lifting the resulting transition. This is how the construction corrects commutativity while retaining actual arrows in I.

The four compatibility conditions on pp. 316–318 should be read as two factorization identities and naturality for two transformations, one of them shifted by one index. Once the two factorization identities hold at a successor stage, naturality of u with the immediately preceding stage is automatic:

$$(u_{\alpha+1}v_\alpha)u_\alpha = u_{\alpha+1}(v_\alpha u_\alpha).$$

Thus the additional equalization proposed on p. 323 is harmless but redundant. Naturality of v with all earlier stages still has to be imposed, especially after a limit stage; this is the substantive correction performed on pp. 324–326. The final block gives the complete construction, including the step taken up on manuscript p. 329.

On va construire encore i' par construction transfinie, sur l'intervalle ordinal I_{α_π}, en construisant un système inductif

$$(i_\alpha)_{\alpha \in I_{\alpha_\pi}}\ , \qquad i_\alpha \in \mathrm{Ob}\, I\ ,$$

d'objets de I, et un système inductif $(j_\alpha)_{\alpha \in I_{\alpha_\pi}}$, $j_\alpha \in \mathrm{Ob}\, J'$, flèches de transition notées $\tau_{\beta,\alpha} : i_\alpha \longrightarrow i_\beta$, $\tau'_{\beta,\alpha} : j_\alpha \longrightarrow j_\beta$, enfin pour tout α dans I_{α_π} des flèches

(∗)
$$\varphi(i_\alpha) \xrightarrow{u_\alpha} j_\alpha \underset{\varphi(\tau_{\alpha+1,\alpha})}{\overset{v_\alpha}{\rightrightarrows}} \varphi(i_{\alpha+1}) \ (\ \overset{u_{\alpha+1}}{\dashrightarrow}\ j_{\alpha+1}\ )$$
(with $\tau'_{\alpha+1,\alpha} : j_\alpha \dashrightarrow j_{\alpha+1}$)

dans J, telles que

(∗∗)
$$\begin{cases} v_\alpha u_\alpha &= \varphi(\tau_{\alpha+1,\alpha}) \\ u_{\alpha+1}v_\alpha &= \tau'_{\alpha+1,\alpha} \end{cases} \qquad ([139])\ .$$ p. 118

138. C'est ici qu'intervient l'hypothèse de stabilité faite dans b), car d'après b), $\underline{S}$ est π-accessible dans $\mathcal{M}$.

139. cf. aussi (∗ ∗ ∗∗), page suivante. [page 314]

Montrons comment la construction d'un tel système d'objets et de flèches avec

$$(***) \qquad i_0 \;=\; i$$

va impliquer l'existence de $i' \in \mathrm{Ob}\, I'$ et de $i \longrightarrow i'$. En effet, comme I_{α_π} est grande devant π'_0 (il est grand devant tout cardinal $< \pi$), il existe

$$i' \overset{\text{déf}}{=} \varinjlim_{i \in I_{\alpha_\pi}} i_\alpha$$

dans I, et on a une flèche de transition

$$i \;=\; i_0 \longrightarrow i' \;,$$

donc il reste à prouver que $i' \in \mathrm{Ob}\, I'$, i.e. $\varphi(i') \in \mathrm{Ob}\, J'$. Or φ commutant aux $\varinjlim$ de type I_{α_π}, on aura

$$\varphi(i) \;=\; \varinjlim \varphi(i_\alpha) \;.$$

Il me faut encore des compatibilités

$$(****) \qquad \begin{array}{ccc} \varphi(i_\alpha) & \xrightarrow{u_\alpha} & j_\alpha \\ {\scriptstyle \varphi(\tau_{\beta,\alpha})}\downarrow & & \downarrow{\scriptstyle \tau'_{\beta,\alpha}} \\ \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\alpha \end{array} \quad \text{et} \quad \begin{array}{ccc} j_\alpha & \xrightarrow{v_\alpha} & \varphi(i_{\alpha+1}) \\ {\scriptstyle \tau'_{\beta,\alpha}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\beta+1,\alpha+1})} \\ j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \end{array}$$

oubliées tantôt.
Les compatibilités $(****)$ impliquent qu'on a des homomorphismes [page 315]

$$\varphi(i) \underset{v}{\overset{u}{\rightleftarrows}} j \qquad \text{où } j \overset{\text{déf}}{=} \varinjlim_{\alpha \in I_{\alpha_\pi}} j_\alpha \quad \text{(limite } \textit{dans } J', \text{ donc dans } J) \;,$$

et les compatibilités $(**)$ impliquent que

$$vu \;=\; \mathrm{id}_{\varphi(i)} \;, \qquad uv = \mathrm{id}_J \;,$$

donc que u, v sont des isomorphismes inverses l'un de l'autre. Comme J' est une sous-catégorie *strictement* pleine de J, il s'ensuit bien que $\varphi(i) \in J'$, i.e. $i \in I'$, q.e.d.

Donc tout revient à construire les

$$\underbrace{i_\alpha}_{\in \mathrm{Ob}\, I} \;,\; \tau_{\beta,\alpha} \;, \qquad \underbrace{j_\alpha}_{\in \mathrm{Ob}\, J'} \;,\; \tau'_{\beta,\alpha} \;, \qquad u_\alpha \;,\; v_\alpha \;,$$

satisfaisant aux contraintes $(**)$, $(***)$, $(****)$. On va faire la construction en trois étapes. p. 119

① Départ :

$$i_0 \;=\; i \;.$$

Comme J' est cofinal dans J, existe $j_0 \in \mathrm{Ob}\, J'$ et

$$\varphi(i_0) \xrightarrow{u_0} j_0 \;.$$

② Supposons que les i_β, j_β soient construits pour $\beta < \alpha$, ainsi que les $\tau_{\beta',\beta} : i_\beta \longrightarrow i_{\beta'}$, $\tau'_{\beta',\beta} : j_\beta \longrightarrow j_{\beta'}$ pour $\beta, \beta' < \alpha$, enfin les $u_\beta : \varphi(i_\beta) \longrightarrow j_\beta$, $v_\beta : j_\beta \longrightarrow \varphi(i_{\beta+1})$ pour $\beta < \alpha$ [page 316]

resp. pour $\beta+1<\alpha$, de façon à satisfaire aux contraintes $(**)$, $(***)$, i.e. commutativités dans

$$\left\{\begin{array}{ll}
\textcircled{1}\quad \varphi(i_\beta)\xrightarrow{u_\beta} j_\beta \xrightarrow{v_\beta}\varphi(i_{\beta+1}),\ \varphi(i_\beta)\xrightarrow{\varphi(\tau_{\beta+1,\beta})}\varphi(i_{\beta+1}) & \\
\textcircled{2}\quad j_\beta\xrightarrow{v_\beta}\varphi(i_{\beta+1})\xrightarrow{u_{\beta+1}} j_{\beta+1},\ j_\beta\xrightarrow{\tau'_{\beta+1,\beta}} j_{\beta+1} & \left.\begin{array}{l}\\\\\end{array}\right\}\ si\ \beta+1<\alpha \\
\textcircled{3}\quad \begin{array}{ccc}\varphi(i_\beta)&\xrightarrow{u_\beta}& j_\beta\\ {\scriptstyle\varphi(\tau_{\beta',\beta})}\downarrow & & \downarrow{\scriptstyle\tau'_{\beta',\beta}}\\ \varphi(i_{\beta'})&\xrightarrow{u_{\beta'}}& j_{\beta'}\end{array} & \text{si } \beta<\beta'<\alpha \\
\textcircled{4}\quad \begin{array}{ccc} j_\beta&\xrightarrow{v_\beta}&\varphi(i_{\beta+1})\\ {\scriptstyle\tau'_{\beta',\beta}}\downarrow & & \downarrow{\scriptstyle\varphi(\tau_{\beta'+1,\beta+1})}\\ j_{\beta'}&\xrightarrow{v_{\beta'}}&\varphi(i_{\beta'+1})\end{array} & \text{si } \beta<\beta' \text{ et } \beta'+1<\alpha\ ,
\end{array}\right.$$

il faut construire [page 317]

1) $i_\alpha\in\operatorname{Ob} I$, et un homomorphisme $(i_\beta)_{\beta\in I_\alpha}\longrightarrow i_\alpha$, i.e. des $\tau_{\alpha,\beta}: i_\beta\longrightarrow i_\alpha$ $(\beta\in I_\alpha)$, compatibles aux translations $\tau_{\beta',\beta}$ pour $\beta<\beta'<\alpha$. p. 120
2) $j_\alpha\in\operatorname{Ob} J'$, et un homomorphisme $(j_\beta)_{\beta\in I_\alpha}\longrightarrow j_\alpha$, i.e. des $\tau'_{\alpha,\beta}: j_\beta\longrightarrow j_\alpha$ $(\beta\in I_\alpha)$, compatibles aux translations $\tau'_{\beta',\beta}$ pour $\beta<\beta'<\alpha$.
3) Une flèche
$$u_\alpha:\varphi(i_\alpha)\longrightarrow j_\alpha\ .$$
4) Si α n'est pas ordinal limite, i.e. $\alpha=\alpha_0+1$, une flèche
$$v_{\alpha_0}: j_{\alpha_0}\longrightarrow\varphi(i_{\alpha_0+1})=\varphi(i_\alpha)\ ;$$

et ceci de façon à satisfaire aux contraintes ①②③④ quand on y suppose $\beta+1\leqslant\alpha$ (au lieu de $\beta'<\alpha$) dans ① et ②, et $\beta'\leqslant\alpha$ (au lieu de $\beta'<\alpha$) dans ③, et $\beta'+1\leqslant\alpha$ (au lieu de $\beta'+1<\alpha$) dans ④. Donc pour les contraintes 1°, 2°, 4°, il n'y a en tenir compte *que* si α n'est pas ordinal limite, donc $\alpha=\alpha_0+1$, et en y faisant

$$\beta'\ =\ \alpha_0\ .$$

Quant à la contrainte 3°, il faut en tenir compte pour $\beta'=\alpha$. Donc au total, il faut exiger [page 318]

$$(3_\alpha)(\beta)\qquad \begin{array}{ccc}\varphi(i_\beta)&\xrightarrow{u_\beta}& j_\beta\\ {\scriptstyle\varphi(\tau_{\alpha,\beta})}\downarrow & & \downarrow{\scriptstyle\tau'_{\alpha,\beta}}\\ \varphi(i_\alpha)&\xrightarrow{u_\alpha}& j_\beta\end{array}\qquad \text{commutatif pour } \beta<\alpha\ (^{140}),$$

(par restriction sur α), et si $\alpha=\alpha_0+1$ est ordinal non limite, il faut exiger de plus

$$(4_\alpha)(\beta)\qquad \begin{array}{ccc} j_\beta&\xrightarrow{v_\beta}&\varphi(i_{\beta+1})\\ {\scriptstyle\tau'_{\alpha_0,\beta}}\downarrow & & \downarrow{\scriptstyle\varphi(\tau_{\alpha,\beta+1})}\\ j_{\alpha_0}&\xrightarrow{v_{\alpha_0}}&\varphi(i_{\alpha_0+1})=\varphi(i_\alpha)\end{array}\qquad \beta<\alpha_0$$

140. **N.B.** Il suffit de l'exiger pour $\beta=\alpha_0$, dans le cas $\alpha=\alpha_0+1$, cf. ci-dessous, p. 323.

$$(1_\alpha) \qquad \varphi(i_{\alpha_0}) \xrightarrow{u_{\alpha_0}} j_{\alpha_0} \xrightarrow{\textcircled{v_{\alpha_0}}} \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \quad \text{(}\varphi(\textcircled{\tau_{\alpha,\alpha_0}})\text{)}$$

$$(2_\alpha) \qquad j_{\alpha_0} \xrightarrow{\textcircled{v_{\alpha_0}}} \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \xrightarrow{\textcircled{u_\alpha}} j_\alpha \ . \quad (\textcircled{\tau'_{\alpha,\alpha_0}})$$

p. 121

Dans ces compatibilités, j'ai cerclé les flèches $\tau_{\alpha,\beta}$, $\tau'_{\alpha,\beta}$ $(\beta < \alpha)$, $\Big(\ \tau_{\alpha,\beta+1}\ (\beta+1<\alpha)\ \Big)$, v_{α_0}, $\Big(\ \tau_{\alpha,\alpha_0}\ \Big)$, u_α, $\Big(\ \tau'_{\alpha,\alpha_0}\ \Big)$, qui ont été introduites dans les donnés 1) 2) 3) 4), page 317.

On va distinguer donc les deux cas. [page 319]

2 A) $\alpha = \alpha_0 + 1$.

Pour construire i_α et les $\tau_{\alpha,\beta} : i_\beta \longrightarrow i_\alpha$, il suffit de construire i_α et

$$\textcircled{\tau_{\alpha,\alpha_0}} \ : \ i_{\alpha_0} \ \longrightarrow \ i_\alpha = i_{\alpha_0+1} \ .$$

De même pour j_α et les $\tau'_{\alpha,\beta}$, il suffit de construire

$$\textcircled{\tau'_{\alpha,\alpha_0}} \ : \ j_{\alpha_0} \ \longrightarrow \ j_\alpha \ .$$

Il faut de plus se donner

$$j_{\alpha_0} \xrightarrow{\textcircled{v_{\alpha_0}}} \varphi(\underbrace{i_\alpha}_{=i_{\alpha_0+1}}) \xrightarrow{\textcircled{u_\alpha}} j_\alpha \ ,$$

donc en tout deux objets i_α, j_α (dans I et dans J' respectivement) et quatre flèches τ_{α,α_0}, τ'_{α,α_0}, v_{α_0}, u_α. Il faut quatre compatibilités (page 318), où les deux dernières $3_\alpha(\beta)$, $4_\alpha(\beta)$ dépendent d'un paramètre $\beta < \alpha$, i.e. $\beta \leqslant \alpha_0$, ou de $\beta < \alpha_0$.

On va d'abord s'occuper de deux premières compatibilités 1_α, 2_α, en termes de la donnée de [page 320 biffé] [page 321]

$$i_{\alpha_0} \ \in \ \mathrm{Ob}\, I \ , \qquad j_{\alpha_0} \ \in \ \mathrm{Ob}\, J' \ , \qquad u_{\alpha_0} \ : \ \varphi(i_{\alpha_0}) \ \longrightarrow \ j_{\alpha_0} \ .$$

Les deux compatibilités voulues pour les deux objets i_α, j_α et les quatre flèches τ_{α,α_0}, τ'_{α,α_0}, v_{α_0}, u_α sont indiquées dans le diagramme

$$(*) \qquad \varphi(i_{\alpha_0}) \xrightarrow{u_{\alpha_0}} j_{\alpha_0} \xrightarrow{v_{\alpha_0}\ ?} \varphi(\underbrace{i_\alpha}_{?}) \xrightarrow{u_\alpha\ ?} j_\alpha \ , \quad \tau'_{\alpha,\alpha_0} \text{ (}?\text{)}, \quad \varphi(\tau_{\alpha,\alpha_0})$$

On note que comme φ est cofinal, $\exists\ i_\alpha = i \in \mathrm{Ob}\, I$ et $j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i)$. Mais on veut que le composé

$$w \ : \ \varphi(i_{\alpha_0}) \xrightarrow{u_{\alpha_0}} j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i)$$

soit de la forme $\varphi(\tau)$, où $\tau = \tau_{\alpha,\alpha_0} : i_{\alpha_0} \longrightarrow i_\alpha$ est une flèche dans I. Comme φ est un foncteur cofinal de catégories filtrantes, on sait qu'on peut trouver $i \xrightarrow{\rho} i'$ dans I, tel que la flèche composée $\varphi(i_{\alpha_0}) \xrightarrow{w} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$ soit de la forme $\varphi(\tau)$. Donc quitte à remplacer i par i', et v_{α_0} par le composé $v'_{\alpha_0} : j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$, on peut supposer que $v_{\alpha_0} u_{\alpha_0}$ est de la p. 122 [page 322]

forme $\varphi(\tau)$. Cela nous donne donc en principe l'objet $i = i_\alpha$, et les flèches $v_{\alpha_0} : j_{\alpha_0} \longrightarrow \varphi(i)$ et $\tau_{\alpha,\alpha_0} : i_{\alpha_0} \longrightarrow i_\alpha$, de façon à satisfaire à la première compatibilité dans $(*)$. Il reste à construire j_α, τ'_{α,α_0}, u_α de façon à satisfaire à la deuxième compatibilité $(*)$. Or J' étant cofinal dans J, il existe j_α dans J' et une flèche $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$. D'autre part, comme $J' \longrightarrow J$ est pleinement fidèle, la flèche composée $j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i_\alpha) \xrightarrow{u_\alpha} j_\alpha$ dans J, entre objets de J', provient d'une unique flèche $\tau'_{\alpha,\alpha_0} : j_{\alpha_0} \longrightarrow j_\alpha$ dans J'.

Ainsi, on a construit les deux objets et les quatre flèches voulues, de façon à satisfaire [page 323] aux contraintes 1_α, 2_α de la page 318. Les contraintes $3_\alpha(\beta)$, $4_\alpha(\beta)$ sont-elles alors satisfaites automatiquement ?

$$\begin{array}{ccc} \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \\ \downarrow & & \downarrow \\ \varphi(i_{\alpha_0}) & \xrightarrow{u_{\alpha_0}} & j_{\alpha_0} \\ {\scriptstyle \varphi(\tau_{\alpha,\alpha_0})}\downarrow & & \downarrow{\scriptstyle \tau'_{\alpha,\alpha_0}} \\ \varphi(i_\alpha) & \xrightarrow{u_\alpha} & j_\alpha \end{array} \qquad \beta < \alpha, \text{ i.e. } \beta \leqslant \alpha_0 \ .$$

Le carré supérieur est commutatif par hypothèse de récurrence, pour que le carré total (composé) soit commutatif, il suffit que le carré inférieur le soit. On a donc une double-flèche $(\tau'_{\alpha,\alpha_0} u_{\alpha_0}, u_\alpha \varphi(\tau_{\alpha,\alpha_0})) : \varphi(i_\alpha) \rightrightarrows j_\alpha$, elle n'est peut-être pas triviale, mais on sait, J' étant filtrante, qu'on peut l'égaliser par une flèche $j_\alpha \longrightarrow \bar{j}_\alpha$ dans J'. Donc quitte à remplacer j_α par $\bar{j}_\alpha$, u_α par le composé $\bar{u}_\alpha : \varphi(i_0) \longrightarrow j_\alpha \longrightarrow \bar{j}_\alpha$ et τ'_{α,α_0} par le composé $\bar{\tau}'_{\alpha,\alpha_0} : j_{\alpha_0} \longrightarrow j_\alpha \longrightarrow \bar{j}_\alpha$ [page 324] (sans toucher à i_α, v_{α_0}, $\tau_{\alpha_0,\alpha}$), on peut obtenir la commutativité, d'où $3_\alpha(\beta)$.

Reste la contrainte $4_\alpha(\beta)$. Pour ceci, considérons le diagramme composite

$$\begin{array}{ccc} j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \\ {\scriptstyle \tau'_{\alpha_0,\beta}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\alpha,\beta+1})} \\ j_{\alpha_0} & \xrightarrow{v_{\alpha_0}} & \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \ . \end{array}$$

Si α_0 n'est pas un ordinal limite, mais de la forme $\alpha_1 + 1$, on peut encore se ramener au cas où $\beta = \alpha_1$, en procédant comme ci-dessus. Mais on peut procéder dans tous les cas, sans un tel artifice (!), en considérant les double-flèches

$$(*) \qquad j_\beta \underset{v_{\alpha_0}\tau'_{\alpha_0,\beta}}{\overset{\varphi(\tau_{\alpha,\beta+1})v_\beta}{\rightrightarrows}} \varphi(i_\alpha)$$

déduites du carré précédent. Comme l'ensemble de ces double-flèches est de cardinal $\operatorname{card} \alpha < \pi$, p. 123 et que J est grande devant π (il suffirait qu'il le soit pour les cardinaux $\pi' < \pi$), on peut [page 325] trouver une flèche $\varphi(i_\alpha) \xrightarrow{u} j$ dans J qui égalise toutes ces double-flèches. Comme φ est cofinal, on peut supposer que j est de la forme $\varphi(i)$. Comme φ est un foncteur cofinal de catégories filtrantes, il existe alors $i \xrightarrow{\rho} i'$ dans I, tel que le composé $\varphi(i_\alpha) \xrightarrow{u} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$ soit de la forme $\varphi(\sigma)$, pour $\sigma : i_\alpha \longrightarrow i'$. Au total, on a trouvé $\sigma : i_\alpha \longrightarrow i'$, tel que $\varphi(\sigma)$ $(= \varphi(\rho)u) : \varphi(i_\alpha) \longrightarrow \varphi(i')$ égalise toutes les double-flèches envisagées dans $(*)$, p. 324. Donc il s'impose de remplacer i_α par $\bar{i}_\alpha = i'$, v_{α_0} par le composé $j_{\alpha_0} \longrightarrow \varphi(i_\alpha) \xrightarrow{\varphi(\sigma)} \varphi(\bar{i}_\alpha)$, τ_{α,α_0} par le composé $\bar{\tau}_{\alpha,\alpha_0} : i_{\alpha_0} \xrightarrow{\tau_{\alpha,\alpha_0}} i_\alpha \xrightarrow{\sigma} \bar{i}_\alpha$. [page 326]

Pour résumer, voici la recette pour construire, d'abord $\boxed{i_\alpha,\ v_{\alpha_0},\ \tau_{\alpha,\alpha_0},}$ et *ensuite* $\boxed{j_\alpha,\ u_\alpha,\ \tau'_{\alpha,\alpha_0}}$.

On procède d'abord comme page 321, 322 pour construire le premier triple, de façon à satisfaire à la première compatibilité $(*)$ (p. 321), i.e. à 1_α. On "corrige" ensuite ce choix

de i_α, v_{α_0}, τ_{α,α_0}, en remplaçant i_α par $\bar{i}_\alpha$ pour $\sigma : i_\alpha \longrightarrow \bar{i}_\alpha$, v_{α_0} par $\bar{v}_{\alpha_0} = \varphi(\sigma)v_{\alpha_0}$, et τ_{α,α_0} par $\bar{\tau}_{\alpha,\alpha_0} = \sigma\tau_{\alpha,\alpha_0}$, de sorte à obtenir les compatibilités $4_\alpha(\beta)$ pour $\beta < \alpha_0$, comme on vient de l'indiquer. On ne touchera plus à ces choix de i_α, v_{α_0}, τ_{α,α_0}, dans la suite de la construction. On construit $(j_\alpha, u_\alpha, \tau'_{\alpha,\alpha_0})$ de façon à satisfaire à la deuxième compatibilité $(*)$ (p. 321), comme indiqué page 322. Ensuite, on corrige ce choix, en remplaçant j_α par $\bar{j}_\alpha$, [page 327]
avec $\rho : j_\alpha \longrightarrow \bar{j}_\alpha$, et u_α par $\bar{u}_\alpha = \rho u_\alpha$, τ'_{α,α_0} par $\bar{\tau}'_{\alpha,\alpha_0} = \rho\tau'_{\alpha,\alpha_0}$, de façon à satisfaire aux contraintes $3_\alpha(\beta)$ de la page 318.

Cela achève la construction dans le cas où $\alpha = \alpha_0 + 1$.

Cas 2 B : α est ordinal limite.

Il faut donc construire les données 1) 2) 3) de la page 317 (à l'exclusion de la donnée 4), de façon à satisfaire à la seule contrainte $3_\alpha(\beta)$.

On commence à construire i_α et les $\tau_{\beta,\alpha}$ dans I. C'est possible, car $I_\alpha = \{\beta \mid \beta < \alpha\}$ est de cardinal $\pi_\alpha < \pi$ – il suffit donc de supposer I grande devant les cardinaux $\pi' < \pi$.

On construit ensuite de même J_α dans J' et les $\tau'_{\beta,\alpha}$ $(\beta < \alpha)$, en utilisant le fait que J' est [page 328]
grande devant $\operatorname{card}\alpha = \operatorname{card} I_\alpha$. On doit de plus construire une flèche

$$u_\alpha \ : \ \varphi(i_\alpha) \ \longrightarrow \ j_\alpha \ .$$

C'est possible, car J' étant cofinale dans J, il existe j' dans J' et $\varphi(i_\alpha) \longrightarrow J'$, et J' étant filtrante, il existe $\bar{j}_\alpha$ et des flèches $j_\alpha \longrightarrow \bar{j}_\alpha$, $j' \nearrow \bar{j}_\alpha$, donc quitte à remplacer j_α par $\bar{j}_\alpha$, les $\tau'_{\alpha,\beta}$ par des $\bar{\tau}'_{\alpha,\beta}$, on trouve bien une flèche $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$. Il faut ensuite s'occuper des

Commentary · Exposition [ms. 328]

Completing the alternating construction at the page boundary

Grothendieck is finishing a construction which makes a closed cofinal class visible in every continuous presentation. These pages end on manuscript p. 328, before the last naturality equations have been equalized. The complete chapter continues on manuscript p. 329 and states the result as Lemma 4.13.17 on pp. 329–330. We give a proof here with a suitable regular ordinal in place of the possibly singular ordinal π. This is a completion of his proof, not an additional assertion of originality.

Let $F : I \to J$ be final, with I π^+-filtered, and let $J' \subseteq J$ be replete, full, and final. Let $\rho < \pi$ be infinite. Assume that I and J' admit κ-indexed chain colimits for $\kappa = \rho^+$, and that both F and the inclusion $J' \to J$ preserve them. Then $I' = F^{-1}(J')$ is final in I. These assumptions hold in the application on p. 312, taking $\rho = \pi'_0$; no equation $\pi^{\pi'_0} = \pi$ is needed for this implication.

Fix $i_0 \in I$. We construct chains $i_\alpha \in I$, $j_\alpha \in J'$ for $\alpha < \kappa$, with arrows

$$F(i_\alpha) \xrightarrow{u_\alpha} j_\alpha \xrightarrow{v_\alpha} F(i_{\alpha+1})$$

and transition maps $t_{\beta\alpha}$, $s_{\beta\alpha}$. Require

$$v_\alpha u_\alpha = F(t_{\alpha+1,\alpha}), \qquad u_{\alpha+1}v_\alpha = s_{\alpha+1,\alpha},$$

and naturality of $u_\alpha : F(i_\alpha) \to j_\alpha$ and $v_\alpha : j_\alpha \to F(i_{\alpha+1})$. Choose j_0, u_0 using finality of $J' \to J$.

At a successor stage, finality of F gives $j_\alpha \to F(k)$. The lifting fact explained above allows us to enlarge k and arrange that its composite with u_α is $F(t)$ for an actual $t : i_\alpha \to k$. For each $\beta < \alpha$, compare the two arrows

$$j_\beta \rightrightarrows F(k) : \quad F(t\, t_{\alpha,\beta+1})v_\beta, \qquad v_\alpha s_{\alpha\beta}.$$

There are at most π of them. The category J is π^+-filtered, as follows by applying finality to diagrams in I or the criterion 4.9.6.3. Equalize this family in J, then use finality and the lifting fact to replace the equalizing transition by $F(k \to \bar{k})$. Set $i_{\alpha+1} = \bar{k}$ and postcompose the already chosen arrows. This imposes naturality of v without disturbing its factorization identity. Choose $u_{\alpha+1} : F(i_{\alpha+1}) \to j_{\alpha+1}$ with $j_{\alpha+1} \in J'$, and define $s_{\alpha+1,\alpha} = u_{\alpha+1}v_\alpha$. Naturality of u follows from the first identity and earlier naturality; the remaining transition maps are composites.

At a nonzero limit $\gamma < \kappa$, choose compatible cocones $i_\beta \to i_\gamma$ and $j_\beta \to j_\gamma$ on the two preceding chains. This is possible by filteredness, since $|\gamma| \leq \rho < \pi$. Extend j_γ inside J' if necessary to obtain $u_\gamma : F(i_\gamma) \to j_\gamma$. For each $\beta < \gamma$, the two composites

$$F(i_\beta) \rightrightarrows j_\gamma : \quad s_{\gamma\beta} u_\beta, \qquad u_\gamma F(t_{\gamma\beta})$$

need not yet coincide. Equalize them simultaneously in J and then extend the target into J'. Postcomposition corrects u_γ and the whole j-cocone, preserving all its transition identities. This is exactly the operation begun on p. 328 and completed on p. 329. There is no v-naturality condition with target index γ at this step: its target is the shifted index $\alpha + 1$, so the new conditions are imposed at the next successor stage.

Take $i_* = \varinjlim_{\alpha<\kappa} i_\alpha$ and $j_* = \varinjlim_{\alpha<\kappa} j_\alpha$. Preservation and naturality give arrows $F(i_*) \rightleftarrows j_*$. Their composites are identities because they are induced by the transition maps of the two chains. Hence $F(i_*) \cong j_* \in J'$, and repleteness gives $i_* \in I'$. The map $i_0 \to i_*$ proves the first finality condition; fullness and filteredness imply full finality as above.

The source's i in the colimit identities on pp. 314–315 must mean this final i_*, not the initial object i_0. That distinction is essential: otherwise the proof would assert that the starting object already lies in I'. With the corrected notation, the construction proves a$'$ ⇒ c. The continuation of the complete chapter also derives b and c$'$ ⇒ a$'$ using the same lemma. The separate implication a$'$ ⇒ a has the cardinal defect described in the preceding development block; completing the alternating construction does not repair that different step.

Overview · XVIII-c

Presentations which retain structure. These pages turn the existence of small approximations into a method for handling objects with structure. An accessible category supplies filtered presentations of its objects; a diagram requires compatible presentations of its arrows and their equations. A section of a fibred category adds transport between different fibres. A full subcategory defined by a property requires approximations which already possess that property. The successive arguments address these problems with the same operations: choose a common stage, impose a bounded family of equations, and take a suitably long colimit. The cardinal estimates specify when this process can still be carried out among the chosen small objects.

These pages begin with Theorem 4.9.3 and end during the proof of Theorem 4.13.15, on manuscript page 328 (Grothendieck, 1990–1991n). They are printed pages 79–123 of the edition; the manuscript labels run from an opening continuation before 220 through 328, including 302 bis and the cancelled page 320. These systems should not be identified with one another. The complete chapter was consulted where it supplies the opening hypotheses or the continuation of an unfinished proof. The end of these pages is not itself a mathematical gap.

The cardinal convention. Grothendieck's phrase "grande devant π" means modern π^+-filteredness: every diagram with at most π arrows has a cocone. His π-accessible objects are therefore π^+-presentable objects, and his $\mathrm{Ind}_\pi C$ is the completion under π^+-filtered colimits. This translation matters even for $\pi = \aleph_0$, where countable constraints, rather than only finite constraints, are admitted. It also prevents an unnoticed assumption that π is regular. The adjective "regular" applied to a category's filtration elsewhere in the chapter denotes a separate categorical condition.

A second distinction concerns a full subcategory $\mathcal{S} \subseteq \mathcal{M}$. Intrinsic accessibility, accessibility of the inclusion, and preservation of presentable objects by that inclusion are different requirements. The source's fixed-rank notion of accessibility *in* $\mathcal{M}$ requires the last comparison as well. We keep the intrinsic notation $\mathcal{S}_\pi$ separate from $\mathcal{S} \cap \mathcal{M}_\pi$ until the hypotheses identify them.

Cofinality and simultaneous presentation. The first task is to show that a small diagram in $\mathrm{Ind}_\pi C$ comes from a filtered system of diagrams in C. Full faithfulness follows from expressing natural transformations as an equalizer of products of hom sets, provided the indexing category has at most π arrows. Essential surjectivity is the substantial part. The source supplies it when C has colimits of one suitable limit-ordinal shape. In the ensuing section theorem, that colimit turns approximately compatible structure maps into actual structure maps.

The cofinality discussion 4.9.6 isolates the mechanism. We use "final" for right-cofinal functors. A faithful functor can reflect an equality once the requisite arrows have been lifted after enlargement. Without faithfulness, eventual equalization of arrows having the same image suffices; the commentary proves this relative variant explicitly. It also completes the cardinal descent and the discrete-opfibration lemma. Common upper bounds, connectedness, and equalization are kept distinct, with worked examples showing why none can silently replace another.

One clause is missing from the printed Lemma 4.9.6.3(b′): filteredness of the comma categories. The faithful functor from a two-object discrete category to the terminal category disproves the formulation as printed. The proof on page 236 specifies the missing condition. The possible empty-category issue in b″ is instead a convention to clarify: its cardinal version already includes the empty family among the families to be bounded.

The finite-diagram question now has a definitive answer. The comparison works for every C exactly for essentially small enough well-founded shapes, in the sense of Henry (2025b, Theorem 1.3). The finite rigid case in 4.11.6 is included. A nontrivial finite group supplies an

obstruction, and the commentary calculates the signed-ordinal involution example at every infinite completion bound. This is a known construction, with its antecedent in SGA 4, I, 8.8.8, not a result claimed for the commentators (Artin et al., 1972).

Sections and the transfer of colimits. In 4.10 "cofibrant" describes a Grothendieck opfibration. Its sections are families X_i with maps $a_*X_i \to X_j$ for $a : i \to j$; those maps need not be cocartesian. The approximation category of sections over a given section maps finally to the product of the fibrewise approximation categories. Evaluating its canonical colimit therefore recovers the original section. We write out this passage, which the source asks to have made explicit, and prove full faithfulness separately.

The transfinite construction has a coherence defect as displayed. Compatibility of consecutive stages does not compare choices across an internal limit ordinal. A constant two-point diagram whose involution changes at ω satisfies the displayed conditions but gives no induced map on its constant colimit. The repair imposes compatibility with every earlier stage when constructing a new stage; there are still at most π equations. The completed induction proves the intended section theorem. A separate endpoint adjustment handles the parenthesis allowing an auxiliary cardinal equal to π.

The application to paratoposes concerns locally presentable categories, without an additional assertion of topos exactness. Ordinary sections of an opfibration have pointwise limits whenever the fibres have them: their structure maps are induced by the limit projections. Preservation by transport is needed for pointwise colimits. This distinction vindicates the printed limit argument in 4.10.9 and prevents the editors' tentative alternative from becoming an unwarranted correction.

Question 4.10.11 about ordinary and cartesian sections is settled by the later theory of accessible limits. Products, inserters, and equifiers encode arbitrary sections; requiring invertible structure maps gives the pseudolimit describing cartesian sections. After choosing common ranks, Positselski (2024, Theorems 8.1, 8.3 and Corollary 9.2) apply. Their fixed-rank conclusions retain hypotheses about a smaller chain-colimit cardinal, close to the distinction Grothendieck is already exploiting.

Section 4.11 transfers bounded colimits from C to $\mathrm{Ind}_\pi C$ by first presenting the entire diagram. The universal-property argument interchanges two limits of hom sets. The separate assertion that $C \to \mathrm{Ind}_\pi C$ preserves existing bounded colimits uses filtered-colimit/limit commutation in sets. Idempotent completeness enters in the converse, through the description of presentable ind-objects as retracts of objects of C. We also give the finite rigid induction and the reduction of finite limits and colimits to products and equalizers, or coproducts and coequalizers.

Accessible functors and accessible properties. Uniformization in 4.12 bounds the ranks of the images of a small generating family. At a sufficiently large fixed point $\pi^{\pi_0} = \pi$, an accessible functor carries π-accessible objects to π-accessible objects. Fixing a preservation rank reduces the functor category to one with small domain. The proof of 4.12.4 invokes the ind universal property without explicitly securing the required colimits in the target. Choosing the rank useful for both categories repairs the proof, as in 4.12.5. We distinguish this verified form from the broader first assertion rather than manufacture a counterexample from an omitted hypothesis.

There is no common rank for all accessible endofunctors of sets: the representables $X \mapsto X^B$ prove this as $|B|$ grows. Small colimits of accessible set-valued functors are pointwise, so the same example rules out a small accessible generating family for their entire category. This supplies the counterexample sought in Remark 4.12.6.

The existential theorems about accessible full subcategories and their small intersections remain valid. A later higher-categorical counterpart is Lurie (2009, Proposition 5.4.7.10). The effective threshold in 4.13.2(c), however, omits the ambient ranks of the generators. The

singleton subcategory $\{X\} \subseteq \mathcal{P}(X)$ gives a complete counterexample to that estimate. The repair retains the missing bound already used on page 282.

For intersections, an ordinal of cardinality π need not have cofinality greater than the smaller bound. We prove a replacement construction which attends to all the subcategories at each stage of a chain of regular length $\rho^{+} \leq \pi$. Its colimit belongs to every subcategory. This proves the main cofinal-intersection lemma without an unnecessary regularity restriction on π. A related argument repairs the cofinal-pullback construction in the final proof.

Theorem 4.13.15 needs a more substantial correction. Under its printed hypothesis $\pi^{\pi_0} = \pi$, condition a′ need not imply a): closure of the presentable part under $\pi_0'^{+}$-filtered colimits of size at most π need not give unrestricted closure. We construct a counterexample with $\pi_0 = \aleph_0$, $\pi_0' = \aleph_1$, and $\pi = \beth_{\omega_1}$, proving both the bounded closure and the failure of unrestricted closure. Strengthening the hypothesis to $\pi^{\pi_0'} = \pi$ gives a sufficient repair: the indexing categories can then be exhausted by suitably filtered subcategories of size at most π. The bounded membership criteria and the accessible-intersection theorem survive this failed implication.

satisfaisant aux contraintes (∗∗), (∗ ∗ ∗), (∗ ∗ ∗∗). On va faire la construction en trois étapes. p. 119

① Départ :

$$i_0 \;=\; i\ .$$

Comme J' est cofinal dans J, existe $j_0 \in \mathrm{Ob}\, J'$ et

$$\varphi(i_0) \xrightarrow{u_0} j_0\ .$$

② Supposons que les i_β, j_β soient construits pour $\beta < \alpha$, ainsi que les $\tau_{\beta',\beta} : i_\beta \longrightarrow i_{\beta'}$, $\tau'_{\beta',\beta} : j_\beta \longrightarrow j_{\beta'}$ pour $\beta, \beta' < \alpha$, enfin les $u_\beta : \varphi(i_\beta) \longrightarrow j_\beta$, $v_\beta : j_\beta \longrightarrow \varphi(i_{\beta+1})$ pour $\beta < \alpha$ resp. pour $\beta + 1 < \alpha$, de façon à satisfaire aux contraintes (∗∗), (∗ ∗ ∗∗), i.e. commutativités dans [page 316]

$$\begin{cases}
\left.\begin{array}{ll}
① & \varphi(i_\beta) \xrightarrow{u_\beta} j_\beta \xrightarrow{v_\beta} \varphi(i_{\beta+1}) \quad \big(= \varphi(\tau_{\beta+1,\beta})\big) \\
② & j_\beta \xrightarrow{v_\beta} \varphi(i_{\beta+1}) \xrightarrow{u_{\beta+1}} j_{\beta+1} \quad \big(= \tau'_{\beta+1,\beta}\big)
\end{array}\right\} \quad si\ \beta + 1 < \alpha \\
③ \quad \begin{array}{ccc} \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \\ {\scriptstyle \varphi(\tau_{\beta',\beta})}\downarrow & & \downarrow{\scriptstyle \tau'_{\beta',\beta}} \\ \varphi(i_{\beta'}) & \xrightarrow{u_{\beta'}} & j_{\beta'} \end{array} \quad \text{si } \beta < \beta' < \alpha \\
④ \quad \begin{array}{ccc} j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \\ {\scriptstyle \tau'_{\beta',\beta}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\beta'+1,\beta+1})} \\ j_{\beta'} & \xrightarrow{v_{\beta'}} & \varphi(i_{\beta'+1}) \end{array} \quad \text{si } \beta < \beta' \text{ et } \beta' + 1 < \alpha\ ,
\end{cases}$$

[page 317]

Commentary · Exposition [ms. 316]

The size convention and the purpose of the zigzag

Grothendieck is trying to recognize a large object by the small pieces of a chosen presentation. The difficulty is that two cofinal presentations need not have any pieces in common. The construction begun on the preceding manuscript pages supplies a common piece by alternately passing from one presentation to the other, and then taking a limit of the resulting process. This is the technical engine of the membership criterion in 4.13.22.

We first fix the translation of the cardinal notation. In this chapter a category is *grande devant* the infinite cardinal π when every diagram having at most π arrows admits a cocone. In present terminology it is λ-filtered, where $\lambda = \pi^+$. Accordingly, a π-accessible object is λ-presentable: its represented functor preserves λ-filtered colimits. A useful cardinal for $\mathcal{M}$ says that $\mathcal{M}$ is generated under these colimits by its essentially small full subcategory $\mathcal{M}_\pi$. We shall retain π in quotations and use λ when the regular-cardinal convention makes a statement easier to compare with modern literature. Replacing π by a regular cardinal with the same name would lose the distinction between diagrams of size $\leq \pi$ and diagrams of size $< \pi$.

Here $\varphi : I \longrightarrow J$ is final, I is filtered, and $J' \subset J$ is a full, isomorphism-closed, final subcategory. The desired conclusion is that the strict inverse image $I' = \varphi^{-1}(J')$ is final in I. It will be enough to reach an object of I' from each prescribed $i \in I$: a full subcategory of a filtered category with this property is itself filtered and final. This familiar criterion is the one recorded in The Stacks Project Authors (2026d, Lemma 4.19.3); throughout these pages cofinality has this ordinary categorical meaning.

Finality alone is insufficient. Let $J = \mathbb{N}$ with its usual order, take I to be the even natural numbers, and take J' to be the odd natural numbers. Both inclusions are final, but I' is empty. The additional hypothesis in 4.13.17 is therefore doing real work: there is a nonzero limit ordinal θ such that the relevant θ-chains have colimits, those colimits are preserved by φ and $J' \hookrightarrow J$, and the shorter initial segments admit enough cocones. The even–odd example has no colimit for its unbounded countable chain.

Starting at i, the displayed maps have the form

$$\varphi(i_\beta) \xrightarrow{u_\beta} j_\beta \xrightarrow{v_\beta} \varphi(i_{\beta+1}).$$

They do not assert that any intermediate $\varphi(i_\beta)$ belongs to J'. Instead, the two composites are required to be the transition maps of the two chains. At the final limit stage the chains consequently have isomorphic colimits in J. One colimit comes from I, the other belongs to J'; isomorphism-closure then puts the former in J'. This explains both the alternation and the insistence on a strictly full subcategory.

il faut construire

p. 120

1) $i_\alpha \in \mathrm{Ob}\, I$, et un homomorphisme $(i_\beta)_{\beta\in I_\alpha} \longrightarrow i_\alpha$, i.e. des $\tau_{\alpha,\beta} : i_\beta \longrightarrow i_\alpha$ $(\beta \in I_\alpha)$, compatibles aux translations $\tau_{\beta',\beta}$ pour $\beta < \beta' < \alpha$.
2) $j_\alpha \in \mathrm{Ob}\, J'$, et un homomorphisme $(j_\beta)_{\beta\in I_\alpha} \longrightarrow j_\alpha$, i.e. des $\tau'_{\alpha,\beta} : j_\beta \longrightarrow j_\alpha$ $(\beta \in I_\alpha)$, compatibles aux translations $\tau'_{\beta',\beta}$ pour $\beta < \beta' < \alpha$.
3) Une flèche
$$u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha \ .$$
4) Si α n'est pas ordinal limite, i.e. $\alpha = \alpha_0 + 1$, une flèche
$$v_{\alpha_0} : j_{\alpha_0} \longrightarrow \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \ ;$$

et ceci de façon à satisfaire aux contraintes ①②③④ quand on y suppose $\beta + 1 \leqslant \alpha$ (au lieu de $\beta' < \alpha$) dans ① et ②, et $\beta' \leqslant \alpha$ (au lieu de $\beta' < \alpha$) dans ③, et $\beta' + 1 \leqslant \alpha$ (au lieu de $\beta' + 1 < \alpha$) dans ④. Donc pour les contraintes 1°, 2°, 4°, il n'y a en tenir compte *que* si α n'est pas ordinal limite, donc $\alpha = \alpha_0 + 1$, et en y faisant

$$\beta' \ = \ \alpha_0 \ .$$

Quant à la contrainte 3°, il faut en tenir compte pour $\beta' = \alpha$. Donc au total, il faut exiger [page 318]

$(3_\alpha)(\beta)$
$$\begin{array}{ccc} \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \\ {\scriptstyle \varphi(\tau_{\alpha,\beta})}\downarrow & & \downarrow{\scriptstyle \tau'_{\alpha,\beta}} \\ \varphi(i_\alpha) & \xrightarrow{u_\alpha} & j_\beta \end{array}$$
commutatif pour $\beta < \alpha$ ([140]),

(par restriction sur α), et si $\alpha = \alpha_0 + 1$ est ordinal non limite, il faut exiger de plus

$(4_\alpha)(\beta)$
$$\begin{array}{ccc} j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \\ {\scriptstyle \tau'_{\alpha_0,\beta}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\alpha,\beta+1})} \\ j_{\alpha_0} & \xrightarrow{v_{\alpha_0}} & \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \end{array}$$
$\beta < \alpha_0$

(1_α)
$$\varphi(i_{\alpha_0}) \xrightarrow{u_{\alpha_0}} j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha)$$
$$\varphi(\tau_{\alpha,\alpha_0})$$

p. 121

$$(2_\alpha) \qquad j_{\alpha_0} \xrightarrow{\textcircled{v_{\alpha_0}}} \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \xrightarrow{\textcircled{u_\alpha}} j_\alpha \ .$$
$$\qquad\qquad\qquad\qquad\qquad\qquad \underset{\textcircled{\tau'_{\alpha,\alpha_0}}}{\curvearrowright}$$

Dans ces compatibilités, j'ai cerclé les flèches $\tau_{\alpha,\beta}$, $\tau'_{\alpha,\beta}$ $(\beta < \alpha)$, $\Big($ $\tau_{\alpha,\beta+1}$ $(\beta+1 < \alpha)$ $\Big)$, v_{α_0}, $\Big($ τ_{α,α_0} $\Big)$, u_α, $\Big($ τ'_{α,α_0} $\Big)$, qui ont été introduites dans les donnés 1) 2) 3) 4), page 317.

On va distinguer donc les deux cas. [page 319]

2 A) $\alpha = \alpha_0 + 1$.

Pour construire i_α et les $\tau_{\alpha,\beta} : i_\beta \longrightarrow i_\alpha$, il suffit de construire i_α et

$$\textcircled{\tau_{\alpha,\alpha_0}} \ : \ i_{\alpha_0} \ \longrightarrow \ i_\alpha = i_{\alpha_0+1} \ .$$

De même pour j_α et les $\tau'_{\alpha,\beta}$, il suffit de construire

$$\textcircled{\tau'_{\alpha,\alpha_0}} \ : \ j_{\alpha_0} \ \longrightarrow \ j_\alpha \ .$$

Il faut de plus se donner

$$j_{\alpha_0} \xrightarrow{\textcircled{v_{\alpha_0}}} \varphi(\underbrace{i_\alpha}_{=i_{\alpha_0+1}}) \xrightarrow{\textcircled{u_\alpha}} j_\alpha \ ,$$

donc en tout deux objets i_α, j_α (dans I et dans J' respectivement) et quatre flèches τ_{α,α_0}, τ'_{α,α_0}, v_{α_0}, u_α. Il faut quatre compatibilités (page 318), où les deux dernières $3_\alpha(\beta)$, $4_\alpha(\beta)$ dépendent d'un paramètre $\beta < \alpha$, i.e. $\beta \leqslant \alpha_0$, ou de $\beta < \alpha_0$.

On va d'abord s'occuper de deux premières compatibilités 1_α, 2_α, en termes de la donnée de [page 320 biffé] [page 321]

$$i_{\alpha_0} \ \in \ \mathrm{Ob}\, I \ , \qquad j_{\alpha_0} \ \in \ \mathrm{Ob}\, J' \ , \qquad u_{\alpha_0} \ : \ \varphi(i_{\alpha_0}) \ \longrightarrow \ j_{\alpha_0} \ .$$

Les deux compatibilités voulues pour les deux objets i_α, j_α et les quatre flèches τ_{α,α_0}, τ'_{α,α_0}, v_{α_0}, u_α sont indiquées dans le diagramme

$$(*) \qquad \varphi(i_{\alpha_0}) \xrightarrow{u_{\alpha_0}} j_{\alpha_0} \underset{v_{\alpha_0}\ ?}{\overset{\tau'_{\alpha,\alpha_0}}{\rightrightarrows}} \varphi(\underbrace{i_\alpha}_{?}) \xrightarrow{u_\alpha\ ?} j_\alpha \ .$$
$$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \underset{\varphi(\tau_{\alpha,\alpha_0})}{\curvearrowright}$$

On note que comme φ est cofinal, $\exists\ i_\alpha = i \in \mathrm{Ob}\, I$ et $j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i)$. Mais on veut que le composé

$$w \ : \ \varphi(i_{\alpha_0}) \ \xrightarrow{u_{\alpha_0}} \ j_{\alpha_0} \ \xrightarrow{v_{\alpha_0}} \ \varphi(i)$$

soit de la forme $\varphi(\tau)$, où $\tau = \tau_{\alpha,\alpha_0} : i_{\alpha_0} \longrightarrow i_\alpha$ est une flèche dans I. Comme φ est un foncteur cofinal de catégories filtrantes, on sait qu'on peut trouver $i \xrightarrow{\rho} i'$ dans I, tel que la flèche composée $\varphi(i_{\alpha_0}) \xrightarrow{w} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$ soit de la forme $\varphi(\tau)$. Donc quitte à remplacer i par i', et v_{α_0} par le composé $v'_{\alpha_0} : j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$, on peut supposer que $v_{\alpha_0} u_{\alpha_0}$ est de la forme $\varphi(\tau)$. Cela nous donne donc en principe l'objet $i = i_\alpha$, et les flèches $v_{\alpha_0} : j_{\alpha_0} \longrightarrow \varphi(i)$ et $\tau_{\alpha,\alpha_0} : i_{\alpha_0} \longrightarrow i_\alpha$, de façon à satisfaire à la première compatibilité dans $(*)$. Il reste à construire j_α, τ'_{α,α_0}, u_α de façon à satisfaire à la deuxième compatibilité $(*)$. Or J' étant cofinal dans J, il existe j_α dans J' et une flèche $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$. D'autre part, comme [page 323]

p. 122 [page 322]

$J' \longrightarrow J$ est pleinement fidèle, la flèche composée $j_{\alpha_0} \xrightarrow{v_{\alpha_0}} \varphi(i_\alpha) \xrightarrow{u_\alpha} j_\alpha$ dans J, entre objets de J', provient d'une unique flèche $\tau'_{\alpha,\alpha_0} : j_{\alpha_0} \longrightarrow j_\alpha$ dans J'.

Commentary · Exposition [ms. 322]

Finality supplies eventual lifting, not fullness

The successor construction needs to turn a map between images into the image of a map, after moving farther along the indexing category. Grothendieck invokes this several times. Writing out the small lemma makes clear why fullness of φ is unnecessary.

Suppose I is filtered and $\varphi : I \to J$ is final. Given $w : \varphi(a) \to \varphi(b)$, there exist arrows $r : b \to c$ and $t : a \to c$ in I such that

$$\varphi(r)w = \varphi(t).$$

Indeed, the objects $(a, 1_{\varphi(a)})$ and (b, w) of the comma category $(\varphi(a) \downarrow \varphi)$ are connected by a finite zigzag. Forgetting to I, choose a cocone on that finite zigzag. After applying φ, the equations defining its comma morphisms say that every composite from $\varphi(a)$ to the common target is the same. The two endpoints give t and r. This is a direct use of the connected-comma definition of finality, rather than a lifting property for individual arrows; compare The Stacks Project Authors (2026a, Definition 4.17.1).

Apply the lemma after first choosing $j_{\alpha_0} \to \varphi(b)$ by finality. Its composite with u_{α_0} then becomes $\varphi(t)$ after a further transition. This constructs i_α, v_{α_0} and τ_{α,α_0} with

$$v_{\alpha_0} u_{\alpha_0} = \varphi(\tau_{\alpha,\alpha_0}).$$

Finality of J' now supplies $u_\alpha : \varphi(i_\alpha) \to j_\alpha$ with $j_\alpha \in J'$. Fullness of $J' \hookrightarrow J$ permits us to define its transition arrow by

$$\tau'_{\alpha,\alpha_0} = u_\alpha v_{\alpha_0}.$$

Only this last passage uses fullness. It is useful to separate it from the earlier use of finality, since the proof would otherwise appear to assume that φ were fully faithful.

Ainsi, on a construit les deux objets et les quatre flèches voulues, de façon à satisfaire aux contraintes 1_α, 2_α de la page 318. Les contraintes $3_\alpha(\beta)$, $4_\alpha(\beta)$ sont-elles alors satisfaites automatiquement ?

$$\begin{array}{ccc} \varphi(i_\beta) & \xrightarrow{u_\beta} & j_\beta \\ \downarrow & & \downarrow \\ \varphi(i_{\alpha_0}) & \xrightarrow{u_{\alpha_0}} & j_{\alpha_0} \\ {\scriptstyle \varphi(\tau_{\alpha,\alpha_0})}\downarrow & & \downarrow{\scriptstyle \tau'_{\alpha,\alpha_0}} \\ \varphi(i_\alpha) & \xrightarrow{u_\alpha} & j_\alpha \end{array} \qquad \beta < \alpha, \text{ i.e. } \beta \leqslant \alpha_0 \ .$$

Le carré supérieur est commutatif par hypothèse de récurrence, pour que le carré total (composé) soit commutatif, il suffit que le carré inférieur le soit. On a donc une double-flèche $(\tau'_{\alpha,\alpha_0} u_{\alpha_0}, u_\alpha \varphi(\tau_{\alpha,\alpha_0})) : \varphi(i_\alpha) \rightrightarrows j_\alpha$, elle n'est peut-être pas triviale, mais on sait, J' étant filtrante, qu'on peut l'égaliser par une flèche $j_\alpha \longrightarrow \bar{j}_\alpha$ dans J'. Donc quitte à remplacer j_α par $\bar{j}_\alpha$, u_α par le composé $\bar{u}_\alpha : \varphi(i_0) \longrightarrow j_\alpha \longrightarrow \bar{j}_\alpha$ et τ'_{α,α_0} par le composé $\bar{\tau}'_{\alpha,\alpha_0} : j_{\alpha_0} \longrightarrow j_\alpha \longrightarrow \bar{j}_\alpha$ (sans toucher à i_α, v_{α_0}, $\tau_{\alpha_0,\alpha}$), on peut obtenir la commutativité, d'où $3_\alpha(\beta)$. [page 324]

Reste la contrainte $4_\alpha(\beta)$. Pour ceci, considérons le diagramme composite

$$\begin{array}{ccc} j_\beta & \xrightarrow{v_\beta} & \varphi(i_{\beta+1}) \\ {\scriptstyle \tau'_{\alpha_0,\beta}}\downarrow & & \downarrow{\scriptstyle \varphi(\tau_{\alpha,\beta+1})} \\ j_{\alpha_0} & \xrightarrow{v_{\alpha_0}} & \varphi(i_{\alpha_0+1}) = \varphi(i_\alpha) \ . \end{array}$$

140. **N.B.** Il suffit de l'exiger pour $\beta = \alpha_0$, dans le cas $\alpha = \alpha_0 + 1$, cf. ci-dessous, p. 323.

Si α_0 n'est pas un ordinal limite, mais de la forme $\alpha_1 + 1$, on peut encore se ramener au cas où $\beta = \alpha_1$, en procédant comme ci-dessus. Mais on peut procéder dans tous les cas, sans un tel artifice (!), en considérant les double-flèches

$$(*) \qquad j_\beta \underset{v_{\alpha_0}\tau'_{\alpha_0,\beta}}{\overset{\varphi(\tau_{\alpha,\beta+1})v_\beta}{\rightrightarrows}} \varphi(i_\alpha)$$

déduites du carré précédent. Comme l'ensemble de ces double-flèches est de cardinal $\operatorname{card}\alpha < \pi$, et que J est grande devant π (il suffirait qu'il le soit pour les cardinaux $\pi' < \pi$), on peut trouver une flèche $\varphi(i_\alpha) \xrightarrow{u} j$ dans J qui égalise toutes ces double-flèches. Comme φ est cofinal, on peut supposer que j est de la forme $\varphi(i)$. Comme φ est un foncteur cofinal de catégories filtrantes, il existe alors $i \xrightarrow{\rho} i'$ dans I, tel que le composé $\varphi(i_\alpha) \xrightarrow{u} \varphi(i) \xrightarrow{\varphi(\rho)} \varphi(i')$ soit de la forme $\varphi(\sigma)$, pour $\sigma : i_\alpha \longrightarrow i'$. Au total, on a trouvé $\sigma : i_\alpha \longrightarrow i'$, tel que $\varphi(\sigma)$ $(= \varphi(\rho)u) : \varphi(i_\alpha) \longrightarrow \varphi(i')$ égalise toutes les double-flèches envisagées dans $(*)$, p. 324. Donc il s'impose de remplacer i_α par $\bar{i}_\alpha = i'$, v_{α_0} par le composé $j_{\alpha_0} \longrightarrow \varphi(i_\alpha) \xrightarrow{\varphi(\sigma)} \varphi(\bar{i}_\alpha)$, τ_{α,α_0} par le composé $\bar{\tau}_{\alpha,\alpha_0} : i_{\alpha_0} \xrightarrow{\tau_{\alpha,\alpha_0}} i_\alpha \xrightarrow{\sigma} \bar{i}_\alpha$.

p. 123 [page 325] [page 326]

Pour résumer, voici la recette pour construire, d'abord $\boxed{i_\alpha,\ v_{\alpha_0},\ \tau_{\alpha,\alpha_0},}$ et *ensuite* $\boxed{j_\alpha,\ u_\alpha,\ \tau'_{\alpha,\alpha_0}}$.

On procède d'abord comme page 321, 322 pour construire le premier triple, de façon à satisfaire à la première compatibilité $(*)$ (p. 321), i.e. à 1_α. On "corrige" ensuite ce choix de i_α, v_{α_0}, τ_{α,α_0}, en remplaçant i_α par $\bar{i}_\alpha$ pour $\sigma : i_\alpha \longrightarrow \bar{i}_\alpha$, v_{α_0} par $\bar{v}_{\alpha_0} = \varphi(\sigma)v_{\alpha_0}$, et τ_{α,α_0} par $\bar{\tau}_{\alpha,\alpha_0} = \sigma\tau_{\alpha,\alpha_0}$, de sorte à obtenir les compatibilités $4_\alpha(\beta)$ pour $\beta < \alpha_0$, comme on vient de l'indiquer. On ne touchera plus à ces choix de i_α, v_{α_0}, τ_{α,α_0}, dans la suite de la construction. On construit $(j_\alpha, u_\alpha, \tau'_{\alpha,\alpha_0})$ de façon à satisfaire à la deuxième compatibilité $(*)$ (p. 321), comme indiqué page 322. Ensuite, on corrige ce choix, en remplaçant j_α par $\bar{j}_\alpha$, avec $\rho : j_\alpha \longrightarrow \bar{j}_\alpha$, et u_α par $\bar{u}_\alpha = \rho u_\alpha$, τ'_{α,α_0} par $\bar{\tau}'_{\alpha,\alpha_0} = \rho\tau'_{\alpha,\alpha_0}$, de façon à satisfaire aux contraintes $3_\alpha(\beta)$ de la page 318.

[page 327]

Cela achève la construction dans le cas où $\alpha = \alpha_0 + 1$.

Cas 2 B : α est ordinal limite.

Il faut donc construire les données 1) 2) 3) de la page 317 (à l'exclusion de la donnée 4), de façon à satisfaire à la seule contrainte $3_\alpha(\beta)$.

On commence à construire i_α et les $\tau_{\beta,\alpha}$ dans I. C'est possible, car $I_\alpha = \{\beta \mid \beta < \alpha\}$ est de cardinal $\pi_\alpha < \pi$ – il suffit donc de supposer I grande devant les cardinaux $\pi' < \pi$.

On construit ensuite de même J_α dans J' et les $\tau'_{\beta,\alpha}$ $(\beta < \alpha)$, en utilisant le fait que J' est grande devant $\operatorname{card}\alpha = \operatorname{card} I_\alpha$. On doit de plus construire une flèche

[page 328]

$$u_\alpha \ : \ \varphi(i_\alpha) \ \longrightarrow \ j_\alpha \ .$$

C'est possible, car J' étant cofinale dans J, il existe j' dans J' et $\varphi(i_\alpha) \longrightarrow J'$, et J' étant filtrante, il existe $\bar{j}_\alpha$ et des flèches $j_\alpha \longrightarrow \bar{j}_\alpha$, $j' \nearrow \bar{j}_\alpha$, donc quitte à remplacer j_α par $\bar{j}_\alpha$, les $\tau'_{\alpha,\beta}$ par des $\bar{\tau}'_{\alpha,\beta}$, on trouve bien une flèche $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$. Il faut ensuite s'occuper des

Commentary · Development (commentators') [ms. 327]

The successor compatibilities are already forced

Grothendieck checks the four displayed compatibilities by successive corrections. His objective is sound, but once the two adjacent composites have been fixed, the remaining squares at a successor stage already commute. The following calculation shortens the construction without changing its hypotheses.

Successor simplification. [ours] Let $\alpha = \delta + 1$, assume all preceding transition maps and u's satisfy the stated compatibilities, and choose

$$v_\delta u_\delta = \varphi(\tau_{\alpha,\delta}), \qquad u_\alpha v_\delta = \tau'_{\alpha,\delta}.$$

Extend the transition maps to earlier indices by composition. Then both $(3_\alpha)(\beta)$ and $(4_\alpha)(\beta)$ hold automatically.

For the first square, the old naturality relation for $\beta < \delta$, and the identity relation when $\beta = \delta$, give

$$\begin{aligned}\tau'_{\alpha,\beta} u_\beta &= u_\alpha v_\delta \tau'_{\delta,\beta} u_\beta \\ &= u_\alpha v_\delta u_\delta \varphi(\tau_{\delta,\beta}) \\ &= u_\alpha \varphi(\tau_{\alpha,\beta}).\end{aligned}$$

For the second, let $\beta < \delta$. Since $\beta + 1 \le \delta$, the earlier second compatibility and naturality give

$$\begin{aligned}v_\delta \tau'_{\delta,\beta} &= v_\delta \tau'_{\delta,\beta+1} u_{\beta+1} v_\beta \\ &= v_\delta u_\delta \varphi(\tau_{\delta,\beta+1}) v_\beta \\ &= \varphi(\tau_{\alpha,\beta+1}) v_\beta.\end{aligned}$$

The same equations cover $\beta + 1 = \delta$, using the identity transition. This proves the assertion.

Thus the extra equalizations at manuscript pages 323–327 can be omitted at successor stages. They are not a counterexample to the construction: applying them to an already equal pair causes no harm, provided the later choices are made in the order specified on page 326. In particular, the phrase suggesting that the pair on page 323 may differ should not be read as evidence of an unresolved obstruction.

At a genuine limit stage there is no preceding last object, and a simultaneous equalization still has a purpose. One chooses cocones on the two earlier chains, adds a map $\varphi(i_\alpha) \to j_\alpha$ after enlarging j_α, and equalizes the resulting family of parallel pairs. The arrows have sources in J, not necessarily in J', so this equalization is first performed in J; finality of J' then moves its target back into J'. The required filteredness of J follows from that of I and finality of φ.

We checked the cofinality and eventual-equality lemmas in The Stacks Project Authors (2026a,d), and Grothendieck's complete construction. We did not locate this particular simplification as a separately stated result. The marking records our proof in this commentary, without a claim of priority.

compatibilités $3_\alpha(\beta)$, donc des double-flèches p. 124

$$\varphi(i_\beta) \underset{u_\alpha \varphi(\tau_{\alpha,\beta})}{\overset{\tau'_{\alpha,\beta} u_\beta}{\rightrightarrows}} j_\alpha \ .$$

Comme l'ensemble de ces double-flèches est de cardinal $\operatorname{card} \alpha < \pi$, et que J' est grande devant ledit cardinal, on trouve donc une flèche [page 329]

$$\rho : j_\alpha \longrightarrow j$$

dans J qui les égalise toutes. Comme J' est cofinal, on peut supposer j dans J', soit $\bar{j}_\alpha$. Donc quitte à remplacer j_α par $\bar{j}_\alpha$, et les $\tau'_{\alpha,\beta}$ par $\bar{\tau}'_{\alpha,\beta} = \rho \tau'_{\alpha,\beta}$, enfin $u_\alpha : \varphi(i_\alpha) \longrightarrow j_\alpha$ par $\bar{u}_\alpha = \rho u_\alpha : \varphi(i_\alpha) \longrightarrow \bar{j}_\alpha$, on satisfait à toutes les contraintes voulues.

Je résume le résultat ("± tautologique"!) prouvé dans les pages 313-329.

Lemme-clef 4.13.17. *Considérons un carré cartésien dans* Cat

$$\begin{array}{ccc} I & \xrightarrow{\varphi} & J \\ \uparrow & & \uparrow \psi \\ I' & \longrightarrow & J' \ . \end{array}$$

On fait les hypothèses suivantes : [page 330]

1) φ et ψ sont cofinaux, ψ est l'inclusion d'une sous-catégorie strictement pleine J' de J, I est filtrant (donc J et J' aussi).

On cherche des conditions pour que l'inclusion de la sous-catégorie strictement pleine $I' \longrightarrow I$ soit cofinale (auquel cas on voit tout de suite que I' est filtrante, et $I' \longrightarrow J'$ est aussi cofinal). Pour ceci, on suppose qu'il existe un ordinal limite α_0 *ayant les propriétés supplémentaires suivantes :*

2) I et J' sont stables par $\varinjlim$ de type I_{α_0} $(= \{\beta \mid \beta$ ordinal $< \alpha_0\})$, et les foncteurs φ, ψ commutent à ces limites.

3) Pour tout $\alpha \in I_{\alpha_0}$, I est grande devant $\operatorname{card}\alpha$*. (Donc, φ étant cofinal, J est également grande devant* $\operatorname{card}\alpha$*, et de même, ψ étant pleinement fidèle, et cofinal, J' est grande devant* $\operatorname{card}\alpha$*.)*

Sous ces conditions, I' est cofinal dans I (donc aussi grande devant $\operatorname{card}\alpha$*, pour tout $\alpha \in I_{\alpha_0}$).*

La démonstration est celle donnée ci-dessus. On n'a pas eu à utiliser que l'ordinal $\alpha_0 = \alpha_\pi$ était de cette espèce particulière, seulement que c'était un cardinal limite.

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Commentary · Exposition [ms. 330] [page 331]

Taking the final colimit completes the cofinality proof

The statement of 4.13.17 collects exactly the ingredients needed to finish the alternating construction. It is useful to make the last passage explicit, because it is the point where approximate membership becomes actual membership.

Write θ for the limit ordinal in that lemma. For $\beta < \theta$, the preceding construction gives compatible chains (i_β) in I and (j_β) in J', with i_0 prescribed. At each limit stage $\alpha < \theta$ there are at most $|\alpha|$ objects and compatibility equations to accommodate. The filteredness assumption for this cardinal supplies their cocones and simultaneous equalizations. At successor stages the preceding block gives all the compatibilities. Transfinite recursion therefore constructs the entire two chains.

Take their colimits,

$$\bar{i} = \varinjlim_{\beta<\theta} i_\beta \quad \text{in } I, \qquad \bar{j} = \varinjlim_{\beta<\theta} j_\beta \quad \text{in } J'.$$

Both exist by hypothesis. The functors to J preserve them. The maps u_β induce $U : \varphi(\bar{i}) \to \bar{j}$. The maps v_β induce $V : \bar{j} \to \varphi(\bar{i})$: compatibility (4) is precisely what is needed to define this map. The composite VU agrees on $\varphi(i_\beta)$ with the transition to $\varphi(i_{\beta+1})$ and hence with the colimit structure map. Therefore $VU = 1$. Likewise $UV = 1$ by compatibility (2). There is always a next index because θ is a limit ordinal; the successor indices are cofinal in θ.

We have proved $\varphi(\bar{i}) \simeq \bar{j}$. Since J' is closed under isomorphisms, $\bar{i} \in I'$, and the first colimit map gives $i_0 \to \bar{i}$. As i_0 was arbitrary and I' is full in the filtered category I, the required finality follows. A full final subcategory of a category admitting cocones on diagrams of a specified size again admits such cocones: form a cocone in the ambient category, then move its vertex into the subcategory. This also proves the final parenthesis of 4.13.17.

There is a similarly direct verification that $I' \to J'$ is final. The composite $I' \to I \to J$ is final. For each $j' \in J'$, its comma category with that composite is identical to $(j' \downarrow (I' \to J'))$, because $J' \subset J$ is full. Its nonemptiness and connectedness give the assertion.

A cardinal remark in the subsequent application requires correction. The initial ordinal of cardinality π is not large before every smaller cardinal when π is singular: a cofinal subset of size $\operatorname{cf}(\pi) < \pi$ has no upper bound. There is a repair which does not enlarge π: use instead the initial ordinal $\theta = (\pi_0')^+$. Since $\pi > \pi_0'$, its cardinality is at most π, it is large before π_0', and the hypotheses of adaptedness and $\mathrm{L}(\pi_0', \pi)$ provide exactly the θ-colimits needed. Every shorter segment has size at most π_0', so the remaining hypotheses of the lemma hold. If one insists on using the initial ordinal of π, it suffices to assume $\operatorname{cf}(\pi) > \pi_0'$, for example by imposing $\pi^{\pi_0'} = \pi$. The stronger sentence on manuscript page 336 is false at singular cardinals, but the argument has the smaller-ordinal repair just given.

Ce lemme implique, comme on l'a vu à la page 312, l'implication b $\Longrightarrow$ d dans le diagramme d'implications $(*)$ page 310, lequel est donc à présent prouvé.

Pour achever de prouver le théorème 4.13.15, il reste à prouver les implications

$$\begin{array}{ccc} \mathrm{c}' & & \\ & \searrow & \\ & & \mathrm{a}' \Longrightarrow \mathrm{a}\ . \\ & \nearrow & \\ \mathrm{b} & & \end{array}$$

Démontrons donc (b ou c′) $\Longrightarrow$ a′. Il reste à prouver que $\underline{S}$ est π-accessible dans $\mathcal{M}$, i.e. que le foncteur pleinement fidèle

$$(*) \qquad \mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}$$

a pour image essentielle $\underline{S}$. Donc il faut prouver que si $x \in \mathrm{Ob}\,\mathcal{M}$, alors

$$(*) \qquad x \in \mathrm{Ob}\,\underline{S} \quad \overset{(?)}{\Longleftrightarrow} \quad \begin{array}{l} x \text{ est dans l'image essentielle de } \mathrm{Ind}_\pi(\underline{S}(\pi)), \text{ i.e. } x \\ \text{est isomorphe à } \varinjlim_{i\in I} x_i, \text{ les } x_i \text{ dans } \mathrm{Ob}\,\underline{S}(\pi),\ I \text{ grande} \\ \text{devant } \pi. \end{array}$$

(Supposons d'abord que $\underline{S}$ satisfasse b) ([141]). Donc on sait que [page 332]

$$(**) \qquad x \in \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad \underline{S}/x \text{ est cofinale dans } \mathcal{M}_\pi/x\ .$$

Nous voulons déduire l'équivalence $(*)$ de l'équivalence $(**)$.

Prouvons $\Longrightarrow$ dans $(*)$, i.e. supposons $x \in \mathrm{Ob}\,\underline{S}$, et prouvons que x est dans l'image essentielle de $\mathrm{Ind}_\pi\,\underline{S}(\pi)$. En vertu de $(**)$, $\underline{S}(\pi)/x$ est une sous-catégorie (strictement pleine) cofinale dans $\mathcal{M}_\pi/x$, donc comme $\mathcal{M}_\pi/x$ est grande devant π, $\underline{S}(\pi)/x$ l'est aussi. Comme on sait que

$$x \ = \varinjlim_{i\in\mathcal{M}_\pi/x} x_i$$

(π étant cardinal utile pour $\mathcal{M}$), il s'ensuit par cofinalité qu'on a aussi

$$x \ = \varinjlim_{i\in\underline{S}(\pi)/x} x_i\ ,$$

et comme $\underline{S}(\pi)/x$ est grande devant π, cela implique que x appartient à l'image essentielle de $\mathrm{Ind}_\pi(\underline{S}(\pi))$.

Prouvons $\Longleftarrow$ dans $(*)$, i.e. supposons que x soit dans l'image essentielle de $\mathrm{Ind}_\pi(\underline{S}(\pi))$,

$$x \ = \varinjlim_{i\in I} x_i\ , \qquad \text{les } x_i \text{ dans } \underline{S}(\pi),\ I \text{ grande devant } \pi. \qquad \begin{array}{r}\text{p. 126}\\ \text{[page 333]}\end{array}$$

Alors le foncteur composé $i \longmapsto x_i$,

$$I \ \longrightarrow\ \underline{S}(\pi)/x \ \hookrightarrow\ \mathcal{M}_\pi/x$$

est cofinal (cf. 4.4.2, page 79), donc $\underline{S}(\pi)/x \longrightarrow \mathcal{M}_\pi/x$ satisfait la condition F 1 (puisque le composé le satisfait), et comme $\underline{S}(\pi)/x \longrightarrow \mathcal{M}_\pi/x$ est pleinement fidèle, cela implique déjà que ce foncteur est cofinal. Donc par $(**)$ on en conclut $x \in \mathrm{Ob}\,\underline{S}$, q.e.d.

Supposons maintenant que S satisfasse c′ (au lieu de b), et prouvons encore l'équivalence $(*)$. Pour ceci, écrivons

$$\square \qquad x \ = \varinjlim_{i\in I} x_i \qquad \begin{cases} \text{les } x_i \in \mathrm{Ob}\,\mathcal{M}_\pi,\ I \textit{ ensemble ordonné},\ (x_i)_{i\in I} \\ \text{est } \pi_0\text{-}\pi\text{-adaptée } ([142]). \end{cases}$$

141. Maladroit, on peut traiter les deux cas b) et c) en même temps, cf. bas de page 333.

ce qui est possible puisque π est utile pour $\mathcal{M}$ $\Big)$. Définissons $I_{\underline{S}}$ comme dans l'énoncé de c), donc comme image inverse de $\underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x$ par $I \longrightarrow \mathcal{M}_\pi$. Par hypothèse d′) [plutôt c′)], on a

$$(**') \qquad x \in \operatorname{Ob} \underline{S} \qquad \Longleftrightarrow \qquad I_{\underline{S}} \text{ est cofinal dans } I \ .$$

En utilisant $(**')$, prouvons encore $(*)$ (page 331). En fait, on utilisera seulement qu'on a l'isomorphisme $\square$, avec I grande devant π, et qu'on a l'équivalence $(**')$, sans avoir à utiliser que I est ordonnée. Donc la démonstration vaut aussi dans le cas c). [page 334]

Prouvons $\Longrightarrow$ dans $(*)$, i.e. supposons $x \in \operatorname{Ob} \underline{S}$, prouvons que x est dans l'image essentielle de $\operatorname{Ind}_\pi(\underline{S}(\pi))$. En vertu de $(**')$, $I_{\underline{S}}$ est cofinale dans I, et comme $I_{\underline{S}} \longrightarrow I$ est une inclusion pleinement fidèle, il s'ensuit que $I_{\underline{S}}$ est également grande devant π, d'autre part on aura par cofinalité

$$x \ = \ \varinjlim_{i \in I_{\underline{S}}} x_i \ ,$$

où cette fois les x_i sont dans $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$, d'où résulte que x est dans l'image essentielle de $\operatorname{Ind}_\pi(\underline{S}(\pi))$.

Prouvons $\Longleftarrow$ dans $(*)$, i.e. supposons que l'on ait

$$x \ = \ \varinjlim_{j \in J} x_j \ , \qquad \text{les } x_j \text{ dans } \underline{S}(\pi), \ J \text{ grande devant } \pi \ .$$

p. 127

Commentary · Examples [ms. 334] [page 335]

A divisible group recognized through countable subgroups

The criterion is easiest to see when the small pieces are actual subobjects. Consider $\mathcal{M} = \mathrm{Ab}$ and $\pi = \aleph_0$, so that the modern regular cardinal is $\lambda = \aleph_1$. The objects of $\mathcal{M}_\pi$ are the countable abelian groups. A countable group has a presentation with countably many generators and relations, and a map from it into an $\aleph_1$-filtered colimit factors at one stage, with all relations and all equalities satisfied at one further stage. Conversely every group is the $\aleph_1$-filtered union of its countable subgroups; presentability then makes its identity factor through one countable subgroup, forcing the group itself to be countable.

Let $\mathcal{D} \subset \mathrm{Ab}$ be the full subcategory of divisible groups. For an abelian group X, put

$$I_X = \{A \leq X : |A| \leq \aleph_0\},$$

ordered by inclusion. This poset is $\aleph_1$-filtered: a countable family has a countable subgroup containing its union. It has colimits of countable chains, given by unions, and the inclusion into Ab preserves them. Its colimit is X.

If X is divisible, every countable subgroup A_0 is contained in a countable divisible subgroup. Having chosen A_n, choose for each $a \in A_n$ and each positive integer m one mth root of a in X, and let A_{n+1} be the subgroup generated by A_n and these roots. There are countably many choices, so A_{n+1} remains countable. The union $D = \bigcup_{n<\omega} A_n$ is divisible: an element belongs to some A_n and each required root belongs to A_{n+1}. Thus the countable divisible subgroups form a cofinal subposet of I_X.

Conversely, if that subposet is cofinal, any $x \in X$ lies in a countable divisible subgroup, by applying cofinality to $\langle x \rangle$. Every equation $my = x$ with $m \geq 1$ then has a solution in X. Hence

$$X \text{ divisible} \quad \Longleftrightarrow \quad \{A \in I_X : A \text{ divisible}\} \text{ cofinal in } I_X.$$

Divisibility is preserved by filtered colimits: represent an element at a stage, choose a root there, and take its image. The countable divisible groups form a set of isomorphism classes and are $\aleph_1$-presentable in $\mathcal{D}$, since the inclusion preserves these colimits. The preceding cofinal presentation

142. **N.B.** Ça existe, grâce à 4.13.14.

therefore proves that $\mathcal{D}$ is $\aleph_1$-accessible and accessibly embedded, with the indicated presentable objects. This establishes all the ingredients of the example, without invoking the classification of divisible groups.

The adaptedness condition cannot be discarded even here. The sequence

$$\mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{3} \mathbb{Z} \xrightarrow{4} \cdots$$

has colimit $\mathbb{Q}$, using the maps $m \mapsto m/(n+1)!$ at stage n. None of its stages is divisible. Its indexing poset ω is filtered but is not $\aleph_1$-filtered, and does not contain the colimit of its own unbounded countable chain. The countable-subgroup presentation has exactly the additional closure which permits the membership criterion to work.

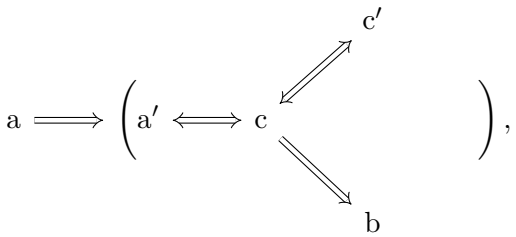
En invoquant à nouveau 4.4.2, on trouve que $J \longrightarrow \mathcal{M}_\pi/x$ est cofinal, et comme ce foncteur se factorise par le foncteur d'inclusion pleinement fidèle $\underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x$, il s'ensuit comme tantôt que ce dernier est cofinal. Nous allons admettre pour un moment que ceci implique que l'image inverse $I_{\underline{S}}$ de cette sous-catégorie strictement pleine, par le foncteur également cofinal $I \longrightarrow \mathcal{M}_\pi/x$, est cofinal dans I. Ceci étant, l'équivalence $(**')$ (p. 333) implique $x \in \mathrm{Ob}\,\underline{S}$, ce qu'on voulait.

Donc il reste à prouver que si

$$J' \;=\; \underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x \;=\; J$$

est cofinal, alors $I' = I_{\underline{S}} \longrightarrow I$ l'est aussi. Il semble qu'il faille bel et bien à nouveau invoquer les conditions de stabilité $\mathrm{L}(\pi_0, \pi)$ faites dans d$'$) `[plutôt c')]` ce qui n'avait pas été nécessaire pour le cas c) et le lemme-clef (un peu) vache 4.13.17, où on prendra (comme dans la démonstration écrite avant l'énoncé de cet lemme) $\alpha = \alpha_\pi \overset{\text{déf}}{=}$ plus petit ordinal α tel que $\mathrm{card}\,\alpha = \pi$. Donc I_α est de cardinal π, et grande devant tout cardinal $\pi' < \pi$, et en particulier devant π'_0. L'hypothèse 2°) de 4.13.17 est vérifiée pour I, par définition d'un système inductif π_0-π-adapté, et pour J', grâce à l'hypothèse $\mathrm{L}(\pi_0, \pi)$ sur $\underline{S}(\pi)$. De même, l'hypothèse 3°) (I grande devant tout cardinal $< \pi$) est satisfaite par définition des systèmes π_0-π-adaptés – I est même grande devant π. L'hypothèse 1°) a déjà été vue. [page 336]

On a donc prouvé (par voie circulaire)



`[et aussi b` $\Longrightarrow$ `c,]` en ayant à invoquer le lemme-clef vache pour les *deux* implications

$$\mathrm{a}' \implies \mathrm{c} \qquad \text{et} \qquad \mathrm{c}' \implies \mathrm{a}'$$

(donc pour deux implications *en sens inverses*, chose étrange …).

Il reste seulement encore à prouver l'implication [page 337]

$$\mathrm{a}' \implies \mathrm{a}\,,$$

ce qui est essentiellement sorital.

Cela achève la démonstration du critère d'accessibilité 4.13.15 d'une sous-catégorie pleine $\underline{S}$ de $\mathcal{M}$.

On va donner une variante un peu décantée de 4.13.15, qui (avec des hypothèses un peu plus faibles) capte l'essentiel de sa substance. L'énoncé 4.13.15 en est une conséquence quasi immédiate : p. 128 [page 338]

Corollaire 4.13.18 (de 4.13.15). *Soient $\mathcal{M}$ une catégorie accessible, π un cardinal utile pour $\mathcal{M}$, α_0 un* ordinal limite *tel que l'on ait* $\operatorname{card}\alpha_0 \leqslant \pi$. *Soit $I_{\alpha_0} = \{\alpha \mid \alpha < \alpha_0\}$, et supposons que $\mathcal{M}$ soit stable par $\varinjlim$ de type I_{α_0}.*

Soit alors $\underline{S}$ une sous-catégorie strictement pleine de $\mathcal{M}$, $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. Désignons par $L(\alpha_0)$ la condition : $\underline{S}(\pi)$ est stable dans $\mathcal{M}_\pi$ (ou ce qui revient au même, dans $\mathcal{M}$) par $\varinjlim$ de type I_{α_0}. Alors les conditions suivantes sont équivalentes.

a) $\underline{S}$ est π-accessible dans $\mathcal{M}$, et est stable dans $\mathcal{M}$ par $\varinjlim$ de type I_{α_0}.

a′) $\underline{S}$ est π-accessible et $\underline{S}(\pi)$ satisfait $L(\alpha_0)$.

b) $\underline{S}(\pi)$ satisfait $L(\alpha_0)$, et pour tout $x \in \operatorname{Ob}\mathcal{M}$, on a ceci : $x \in \operatorname{Ob}\underline{S}$ si et seulement si la sous-catégorie strictement pleine $\underline{S}(\pi)/x$ de $\mathcal{M}(\pi)/x$ est cofinale dans $\mathcal{M}(\pi)/x$ `[lire` *$\mathcal{M}_\pi/x$*`]`.

c) $\underline{S}(\pi)$ satisfait $L(\alpha_0)$, de plus pour tout $x \in \operatorname{Ob}\mathcal{M}$, et toute représentation

$$x \;\simeq\; \varinjlim_{i\in I} x_i$$

qui est "α_0-π-adaptée", i.e. telle que [page 339]

1°) I équivalente à une petite catégorie, et grande devant π,

2°) les $\varinjlim$ de type I_{α_0} existent dans I,

3°) le foncteur $I \longrightarrow \mathcal{M}$, $i \longmapsto x_i$, commute aux $\varinjlim$ de type I_{α_0}, et enfin

4°) les $x_i \in \operatorname{Ob}\mathcal{M}_\pi$

(comparer 4.13.11), on a ceci (où on désigne par $I_{\underline{S}}$ la sous-catégorie strictement pleine de I formée des i tels que $x_i \in \operatorname{Ob}\underline{S}$) : on a $x \in \operatorname{Ob}\underline{S}$ si et seulement si $I_{\underline{S}}$ est cofinale dans I.

*c′) (*Si *I_{α_0} est grande devant π_0, et $\pi^{\pi_0} = \pi$, pour pouvoir invoquer 4.13.14, cf. p. 335.) Comme c), mais en supposant I ordonnée.*

([143]).

Commentary · Exposition [ms. 338]

The canonical presentation and an arbitrary adapted one

Grothendieck's criterion compares two ways of saying that the small objects of $\underline{S}$ suffice to build x. In the canonical presentation, this means that $\underline{S}(\pi)/x$ is final in $\mathcal{M}_\pi/x$. In a chosen presentation it means that the indices already lying in $\underline{S}$ are final. The first description is intrinsic; the second can be used in calculations.

For a useful π, the canonical diagram $\mathcal{M}_\pi/x \to \mathcal{M}$ is λ-filtered and has colimit x, where $\lambda = \pi^+$. If $\underline{S}(\pi)/x$ is final, it supplies a λ-filtered presentation of x using objects of $\underline{S}(\pi)$, so x lies in the essential image of $\mathrm{Ind}_\lambda(\underline{S}(\pi))$. Conversely, a presentation $x = \varinjlim_i x_i$ by such objects gives a final functor $I \to \mathcal{M}_\pi/x$. For example, an arrow $a \to x$ with a presentable factors at a stage, and equality of two such factorizations holds at a later stage; these are the two required comma-category conditions. Its factorization through the full subcategory $\underline{S}(\pi)/x$ makes that subcategory final.

This equivalence alone does not permit pulling the subcategory back along every final presentation. That was precisely the failure in the even–odd example. Conditions on Φ, and on the preservation of its colimits by the indexing diagram, supply the hypotheses of 4.13.17. They transfer finality to

$$I_{\underline{S}} = \{i \in I : x_i \in \underline{S}\}.$$

The converse direction is then immediate from the closure of $\underline{S}$ under the filtered colimits in question.

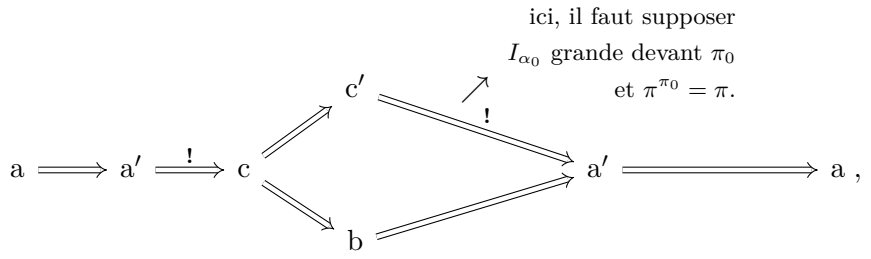
Here being π-accessible *in* $\mathcal{M}$ carries more information than abstract accessibility of $\underline{S}$. It identifies $\underline{S}$ with the full image of the Ind-completion of $\underline{S} \cap \mathcal{M}_\pi$ and ensures that the inclusion preserves the chosen filtered colimits. In particular, its small objects are measured in the ambient

143. **N.B.** Si on ne suppose par I_{α_0} grande devant π_0 et $\pi^{\pi_0} = \pi$, alors toutes les conditions de a) à c) (a, a′, b, c) sont équivalentes et *impliquent* c, mais ne lui sont peut-être pas équivalentes.

category. This distinction is made explicit in 4.13.26. It prevents one from treating any abstractly accessible full subcategory as if it automatically had the canonical presentations needed here.

The optional restriction to ordered indexing categories introduces a separate existence problem. The exponentiation hypotheses in c′ are used to construct sufficiently closed ordered presentations of controlled cardinality. They are not part of the elementary argument with the canonical slice. Thus the text correctly distinguishes the unconditional equivalence of a), a′), b), c) from the stronger assertion involving c′; the missing prime in footnote 143 is a typographical slip in the edition.

Démonstration. Elle se fait suivant le même diagramme logique que celle de 4.13.15,



où on a marqué du signe ! les deux implications qui sont "vaches", en ce sens qu'elles font appel au lemme-clef 4.13.17 (pages 329-330), qui se prouve par une construction transfinie un peu vissée. La démonstration des sept implications du diagramme est essentiellement celle donné dans les pages précédentes 311-337,

à l'exception de $a' \Longrightarrow a$, qui est un peu plus simple dans le cas actuel. On va la dégager sous forme de lemme un peu plus général. (**N.B.** La démonstration de $a' \Longrightarrow a$ dans 4.13.15 avait été omise comme "soritale" !, cf. page 337 …) p. 129 [page 340]

Lemme 4.13.19. *Soient $\mathcal{M}$ une catégorie accessible dans $\mathcal{M}$, π un cardinal utile pour $\mathcal{M}$ et tel que $\underline{S}$ soit π-accessible, i.e. $\underline{S}$ est l'image essentielle de $\mathrm{Ind}_\pi(\underline{S}(\pi))$ par le foncteur pleinement fidèle canonique $\mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}$, où $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. On suppose qu'il existe un ordinal limite α, tel que $\operatorname{card} \alpha \leqslant \pi$ et tel que $\mathcal{M}_\pi$ et $\underline{S}(\pi)$ $(= \underline{S}_\pi)$ soient stables par $\varinjlim$ de type I_α soient stables par $\varinjlim$ de type I_α, et que le foncteur d'inclusion $\underline{S}(\pi) \longrightarrow \mathcal{M}_\pi$ y commute. Soit Φ un ensemble de catégories telles que $\forall\ I \in \Phi$, on ait $\operatorname{card} \text{réd}\, I \leqslant \pi$, et supposons que $\mathcal{M}_\pi$ soit stable par $\varinjlim$ de type Φ, et que $\underline{S}(\pi)$ soit stable dans $\mathcal{M}_\pi$ par lesdites limites (ou ce qui revient au même, qu'il soit stable en lui-même par ces $\varinjlim$, et que le foncteur d'inclusion y commute). Alors $\mathcal{M}$ et $\underline{S}$ sont stables par $\varinjlim$ de type Φ, et le foncteur d'inclusion $\underline{S} \longrightarrow \mathcal{M}$ y commute, i.e. $\mathcal{M}$ est stable par ces limites, et $\underline{S}$ stable dans $\mathcal{M}$ par ces mêmes limites.*

Démonstration. On applique 4.11.1 (p. 267-268) séparément à $\mathcal{M} \simeq \mathrm{Ind}_\pi \mathcal{M}_\pi$ et à $\underline{S} \simeq \mathrm{Ind}_\pi \underline{S}(\pi)$, pour conclure que $\mathcal{M}$ et $\underline{S}$ sont stables séparément par $\varinjlim$ de type Φ. Il reste à voir que le foncteur $\underline{S} \longrightarrow \mathcal{M}$ y commute. Cela se voit comme dans la démonstration de 4.11.1, on obtient plus généralement ceci : [page 341]

Lemme 4.13.20 (cor. à 4.11.1). *Soient $f_0 : C \longrightarrow C'$ un foncteur entre catégories équivalentes à des petites catégories, π un cardinal infini, $f : \mathcal{M} \overset{\text{déf}}{=} \mathrm{Ind}_\pi C \longrightarrow \mathcal{M}' \overset{\text{déf}}{=} \mathrm{Ind}_\pi C'$ l'extension canonique $\mathrm{Ind}_\pi(f_0)$ de f_0. Soient Φ et α comme dans 4.11.1, et supposons que $I_\alpha \in \Phi$ et que f_0 commute aux $\varinjlim$ de type Φ. Alors f commute à ces mêmes limites inductives.* ([144]).

Corollaire 4.13.21. *Sous les conditions de 4.13.20, supposons que Φ contienne les catégories de cardinal $\leqslant \pi$ (resp. les catégories filtrantes de cardinal $\leqslant \pi$, resp. les catégories filtrantes grandes devant π_0 de cardinal $\leqslant \pi$, resp. les catégories discrètes de cardinal $\leqslant \pi$). Alors f commute aux petites $\varinjlim$ (resp. aux petites $\varinjlim$ filtrantes, resp. aux petites $\varinjlim$ filtrantes grandes devant π_0, resp. aux petites sommes). (Réciproque claire et "laissée au lecteur".)*

[page 342]

Se voit comme 4.4.14 et 4.4.14.1 (p. 97, 98).
Voici enfin, il me semble, la forme définitive (?) de 4.13.15, 4.13.18.

Commentary · Exposition [ms. 341]

Why a colimit calculation on small objects extends

The passage from a′ to a) requires more than saying that large objects are filtered colimits of small ones. A diagram of large objects also has arrows and relations; the small presentations must realize them simultaneously. The earlier representation theorem, used in 4.11.1, supplies this compatibility under the limit-ordinal hypothesis.

Let C and C' be the essentially small categories of 4.13.20 and put $\lambda = \pi^+$. For a diagram $D : J \to \mathrm{Ind}_\lambda(C)$ of one of the allowed shapes, its simultaneous presentation is a functor

$$X : J \times A \longrightarrow C$$

with A λ-filtered and $D(j) \simeq \varinjlim_{a\in A} X(j,a)$ naturally in j. The assumptions ensure that the J-colimits of $X(-,a)$ exist in C and represent the corresponding colimits in its Ind-completion. Consequently

$$\varinjlim_{j\in J} D(j) \simeq \varinjlim_{a\in A} \left(\varinjlim_{j\in J} X(j,a) \right).$$

This interchange uses only the universal property of colimits: both sides represent the set of cocones from the two-variable diagram to a prescribed target. It is not an interchange of a limit and a colimit.

If $f = \mathrm{Ind}_\lambda(f_0)$ and f_0 preserves these J-colimits, then

$$\begin{aligned} f\left(\varinjlim_{j} D(j) \right) &\simeq \varinjlim_{a} f_0 \left(\varinjlim_{j} X(j,a) \right) \\ &\simeq \varinjlim_{a} \varinjlim_{j} f_0 X(j,a) \\ &\simeq \varinjlim_{j} f(D(j)). \end{aligned}$$

All comparisons are natural, since they are induced by the same cocones. Apply this to the inclusion $\underline{S}(\pi) \hookrightarrow \mathcal{M}_\pi$ to obtain 4.13.19, including preservation by $\underline{S} \hookrightarrow \mathcal{M}$. This fills in the proof that page 337 calls sorital, while retaining the nontrivial simultaneous-presentation input rather than silently assuming it.

For the arbitrary-small-colimit assertion of 4.13.21, one can also see the size reduction directly. A small category is the directed union of its subcategories generated by at most π arrows. The poset of these subcategories is λ-filtered; each has at most π arrows because π is infinite. The colimit of a diagram over the whole category is the colimit, over this poset, of its restricted colimits. Preservation of the bounded colimits and of λ-filtered colimits therefore gives preservation of all small colimits. The versions with filtered shapes use filtered envelopes of the small subcategories; the later construction in 4.13.37–39 makes that step explicit.

For the variant restricted to categories large before π_0, that bounded-envelope argument is verified below with the sufficient additional hypothesis $\pi^{\pi_0} = \pi$. The bare formulation of 4.13.21 does not display this hypothesis. We have not established that its one ordinal-colimit assumption suffices to remove it, so we do not certify that particular fixed-rank inference in its full printed generality. This qualification concerns the third variant, not the explicit interchange calculation above.

The underlying extension principle already occurs in Artin et al. (1972, Exposé I, 8.7.2–8.7.3): a functor on C extends uniquely, up to canonical isomorphism, to a functor on $\mathrm{Ind}(C)$ preserving filtered colimits. Here the additional work controls the cardinal and the extra colimit shapes. It belongs to the same mathematical setting as the accessible-category theory of Makkai and Paré (1989), which predates these manuscript pages; it should not be described as a later discovery of that theory.

144. Il vaut mieux prendre des foncteurs $f_0 : C \longrightarrow \mathcal{N}$, et $\mathcal{N}$ [?] que f_0 commute aux $\varinjlim$ de type Φ, donc il en est de même de son extension $\mathrm{Ind}_\pi(C,\mathcal{N}) \longrightarrow \mathcal{N}$. (**N.B.** On suppose que $\mathcal{N}$ satisfait $\overrightarrow{L}_\pi$.)

Théorème 4.13.22 ([145]). *Soient $\mathcal{M}$ une catégorie accessible, π un cardinal infini, utile pour $\mathcal{M}$, et Φ un ensemble de catégories satisfaisant les deux conditions :*

1°) $J \in \Phi$ implique card réd $J \leqslant \pi$.

2°) Il existe un ordinal limite α_0 *avec* $I_{\alpha_0} \in \Phi$.

On suppose $\mathcal{M}$ stable par $\varinjlim$ de type Φ. Soient $\underline{S}$ une sous-catégorie strictement pleine *de $\mathcal{M}$, et $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$. Désignons par $L(\varphi)$ la condition sur $\underline{S}(\pi)$: $\underline{S}(\pi)$ est stable dans $\mathcal{M}$ (où* p. 130
ce qui revient au même, dans $\mathcal{M}_\pi$) par $\varinjlim$ de type Φ. Ceci posé, les conditions suivantes sur $\underline{S}$ sont équivalentes :

ⓐ *$\underline{S}$ est π-accessible dans $\mathcal{M}$, et stable dans $\mathcal{M}$ par $\varinjlim$ de type Φ.*

a′) *$\underline{S}$ est π-accessible dans $\mathcal{M}$, et $\underline{S}(\pi)$ satisfait à $L(\Phi)$.*

b) *$\underline{S}(\pi)$ satisfait à $L(\Phi)$, et pour tout $x \in \mathrm{Ob}\,\mathcal{M}$, on a ceci : $x \in \mathrm{Ob}\,\underline{S}$ si et seulement si la sous-catégorie strictement pleine $\underline{S}(\pi)/x$ de $\mathcal{M}(\pi)/x$ est cofinale dans $\mathcal{M}(\pi)/x$* `[lire` *$\mathcal{M}_\pi/x$*`]`. [page 343]

ⓒ *$\underline{S}(\pi)$ satisfait à $L(\Phi)$, et pour tout $x \in \mathrm{Ob}\,\mathcal{M}$ et toute représentation $x \simeq \varinjlim_{i \in I} x_i$ qui est Φ-π-adaptée (cf. définition plus bas), on a ceci : $x \in \mathrm{Ob}\,\underline{S}$ si et seulement si $I_{\underline{S}}$ est cofinale dans I, où $I_{\underline{S}}$ est la sous-catégorie strictement pleine de I formée des $i \in \mathrm{Ob}\,I$ tels que $x_i \in \mathrm{Ob}\,\underline{S}$ (ou encore, $x_i \in \mathrm{Ob}\,\underline{S}(\pi)$).* **N.B.** *On dit qu'un système inductif $(x_i)_{i \in I}$ dans $\mathcal{M}$ est Φ-π-adapté si*

1) I équivalente à une petite catégorie, et I grande devant π,

2) les $\varinjlim$ de type Φ existent dans I,

3) le foncteur $i \longmapsto x_i$ de I dans $\mathcal{M}$ commute aux $\varinjlim$ de type Φ, et enfin

4) les $x_i \in \mathrm{Ob}\,\mathcal{M}_\pi$.

c′) *(Si $\pi^{\pi_0} = \pi$, où π_0 est un cardinal $\leqslant \pi$ utile pour $\mathcal{M}$, et si $J \in \Phi$ implique que J est grande devant π_0.) Comme c), avec I* ordonnée.

([146], [147]).

Définition 4.13.23 ([148]). Soient $\mathcal{M}$ une catégorie accessible, π un cardinal, Φ un ensemble de catégories, satisfaisant les deux conditions :

a) $J \in \Phi \Longrightarrow$ card réd $J \leqslant \pi$.

b) Il existe un ordinal limite α_0, tel que $I_{\alpha_0} \in \Phi$ ([149]).

Soit S une partie de $\mathrm{Ob}\,\mathcal{M}$, $\underline{S}$ la sous-catégorie pleine qu'elle définit. On dit que S est une [page 344]
partie Φ-π-accessible de $\mathrm{Ob}\,\mathcal{M}$, ou que $\underline{S}$ est une *sous-catégorie Φ-π-accessible dans $\mathcal{M}$*, si les conditions suivantes sont satisfaites.

1°) π est utile pour $\mathcal{M}$, $\mathcal{M}$ est stable par $\varinjlim$ de type Φ ([150]).

2°) $\underline{S}$ est une sous-catégorie π-accessible dans $\mathcal{M}$.

3°) $\underline{S}$ est stable dans $\mathcal{M}$ par $\varinjlim$ de type Φ.

([151]).

N.B. Quand la condition préliminaire 1°) *sur* $\mathcal{M}$ est satisfaite, la validité des conditions 2°) 3°) équivaut donc à la condition que $\underline{S}$ soit *strictement* pleine dans $\mathcal{M}$, et que de plus $\underline{S}$ p. 131

145. cf. variante 4.13.36, page 358.

146. **N.B.** Quand les conditions restrictives ci-contre `[parenthèse dans c′)]` ne sont pas satisfaites, alors a, a′, b, c sont encore équivalentes, et impliquent c′, mais ne sont peut-être pas équivalentes à c′).

147. **N.B.** Il faudrait réénoncer les conditions annexes [?] (Si …), cf. p. 359-360.

148. **N.B.** Il y aurait intérêt à remonter cette définition (*sans* b), dès avant 4.13.15.

149. Si c'est pour poser une définition, on peut se passer de la condition b).

150. Condition préliminaire sur $\mathcal{M}$.

151. **N.B.** Quand $\Phi = \{I_{\alpha_0}\}$, on dit aussi que $\underline{S}$ est α_0-π-accessible dans $\mathcal{M}$. Si Φ est formé des catégories de cardinal $\leqslant \pi$ et grandes devant π_0, on dit que $\underline{S}$ est π_0-π-accessible dans $\mathcal{M}$.

satisfasse la condition a) du théorème 4.13.22. D'autre part, cette condition est équivalente à chacune des conditions a′, b, c de ce même théorème, et aussi à la dernière condition c′, sous réserve qu'on ait $\pi^{\pi_0} = \pi$ (pour un cardinal $\pi_0 < \pi$ utile pour $\mathcal{M}$) et que $J \in \Phi$ implique que J est grande devant π_0.

Conceptuellement, c'est la définition donnée dans 4.13.23, i.e. la condition ⓐ de 4.13.22, qui paraît la plus claire, ou sa variante immédiate a′. Mais du point de vue pratique, du travail avec les parties accessibles, c'est par propriété ⓒ qui paraît la plus commode. Le poids principal du théorème 4.13.22 est bien l'implication a $\Longrightarrow$ c (via a $\Longrightarrow$ a′ $\overset{!}{\Longrightarrow}$ c). Donc il faut [page 345]
surtout voir ce théorème comme donnant un critère remarquable d'appartenance à $\mathrm{Ob}\,\underline{S}$ d'un objet x de $\mathcal{M}$ qui est présenté par un système inductif Φ-π-adapté :

$$(*) \qquad x \in \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad I_{\underline{S}} \text{ cofinal},$$

et dans cette équivalence, c'est l'implication $\Longrightarrow$ qui est la partie non soritale, et remarquable.

Proposition 4.13.24. *Soient $\mathcal{M}$, π, Φ comme dans 4.13.23 ([152]), on suppose que la condition 1° de 4.13.23 est satisfaite (π utile pour $\mathcal{M}$, $\mathcal{M}$ stable par $\varinjlim$ de type Φ). Soit Σ'^{Φ}_{π} l'ensemble des sous-catégories $\underline{S}$ de $\mathcal{M}$ qui sont Φ-π-accessibles dans $\mathcal{M}$, et Σ^{Φ}_{π} l'ensemble des sous-catégories $\underline{S}(\pi)$ de $\mathcal{M}_\pi$ qui sont strictement pleines, et stables dans $\mathcal{M}_\pi$ par $\varinjlim$ de type Φ. (Ces sous-catégories sont forcément karoubiennes, car stables par $\varinjlim$ de type I_{α_0}, donc aussi par facteurs directs.) Alors les bijections (4.13.4.1) (page 288) induisent des bijections inverses l'une à l'autre*

$$\Sigma^{\Phi}_{\pi} \;\underset{\longleftarrow}{\longrightarrow}\; \Sigma'^{\Phi}_{\pi} \quad .$$

Commentary · Development (commentators') [ms. 345]

Retracts are essential in the small-subcategory correspondence

Grothendieck wants to replace a large accessible subcategory by its small part, so that operations on the former can be performed inside an essentially small category. Proposition 4.13.24 gives this correspondence when the small part is closed under the stated colimits and under retracts. The parenthetical observation supplies retract closure from an ordinal colimit. Its footnote, however, says that the ordinal hypothesis is unnecessary, without replacing it by a retract hypothesis.

Counterexample to the footnote as stated. [ours] Take $\mathcal{M} = \mathrm{Set}$, $\pi = \aleph_0$ and $\Phi = \varnothing$. Let C be the strictly full subcategory of $\mathcal{M}_\pi$ consisting of the two-element sets. It satisfies all the stated Φ-closure requirements, vacuously. Nevertheless the filtered-colimit closure of C contains a singleton, so taking its small part does not recover C.

For completeness, fix a two-element set B and an idempotent constant map $e : B \to B$. Index a diagram by the ordinal ω_1, put B at each index, and put e on every strict transition and the identity on identity transitions. This is a functor because $e^2 = e$. The ordinal ω_1 is $\aleph_1$-filtered: every countable set of its indices is bounded. Its colimit in sets is the image of e, a singleton. The singleton is also $\aleph_1$-presentable, hence belongs to the small part of the Ind-image, but not to C. This proves the failure of the asserted inverse correspondence under the footnote's weakened hypotheses.

The exact repair is to require C to be closed under retracts, or to retain a nonzero limit-ordinal shape among the closure assumptions. To verify the latter option, take any idempotent $e : x \to x$ in C and form the constant-object diagram along that ordinal, with e on strict transitions. If its colimit is $q : x \to y$, the compatible maps $e : x \to x$ induce $s : y \to x$ with $sq = e$. Since $qe = q$ and the colimit maps jointly determine a morphism out of y, one has $qs = 1_y$. Thus y splits e, and closure under this colimit puts its splitting object in C.

Conversely, when $C \subset \mathcal{M}_\pi$ is full and retract closed, every λ-presentable object of its Ind-image is a retract of an object of C: factor its identity through one stage of a λ-filtered presentation. Hence the small part of that image is exactly C. Combined with 4.13.19, this gives the stated correspondence with the intended additional-colimit closure.

The general role of idempotent completion in Ind-constructions is standard. Our contribution here is the explicit diagnosis and counterexample for the particular footnote, checked against the edition and the surrounding propositions. We have not found a published discussion of this manuscript slip.

152. **N.B.** La condition $I_{\alpha_0} \in \Phi$ n'est pas utile.

Proposition 4.13.25. *Soient $\mathcal{M}$, π, Φ comme dans 4.13.23, avec $I_{\alpha_\pi} \in \Phi$ (α_π étant le plus petit ordinal de cardinal Φ* [plutôt π]*). Alors l'ensemble Σ'^{Φ}_{π} des sous-catégories de $\mathcal{M}$ Φ-π-accessibles dans $\mathcal{M}$ est stable par intersection de familles de cardinal $\leqslant \pi$* ([153]). [page 346]

(**N.B.** La bijection $\Sigma^{\Phi}_{\pi} \xrightarrow{\sim} \Sigma'^{\Phi}_{\pi}$ de 4.13.24 respecte les relations d'ordre naturelles d'inclusion, donc transforme intersections en intersections.)

Se voit comme 4.13.7 (qui est contenu dans l'énoncé actuel), en utilisant le lemme 4.13.10.

4.13.26. Soient $\mathcal{M}$, π, Φ comme dans 4.13.23, et soit $\underline{S}$ une sous-catégorie *strictement pleine* de $\mathcal{M}$, $f : \underline{S} \longrightarrow \mathcal{M}$ le foncteur d'inclusion. La condition que $\underline{S}$ soit Φ-π-accessible dans $\mathcal{M}$ se décompose en :

1) π est utile pour $\mathcal{M}$ *et* pour $\underline{S}$.
2) $\mathcal{M}$ et $\underline{S}$ sont stables par $\varinjlim$ de type Φ.
3) f commute auxdites limites.
4) f est π-accessible.
5) $f(\underline{S}_\pi) \subset \mathcal{M}_\pi$.

Soit alors $\pi' > \pi$ un autre cardinal, quand peut-on assurer que si $\underline{S}$ est Φ-π-accessible dans $\mathcal{M}$, il soit aussi Φ-π'-accessible ? Les conditions 2) 3) 4) sont visiblement stables par passage de π à π'. Restent les conditions 1) et 5). On sait que si $\pi'^{\pi} = \pi'$, alors la condition 1) reste valable pour π'. Plus généralement, si π_0 est un cardinal $\leqslant \pi$ tel que π_0 soit utile pour $\mathcal{M}$ et pour $\underline{S}$, alors si $\pi' > \pi_0$ est tel que $\pi'^{\pi_0} = \pi'$, alors π' est aussi utile pour $\mathcal{M}$ et pour $\underline{S}$. Si on admet l'HCG (hypothèse du continu généralisée), il en est ainsi pour tout $\pi' > \pi_0$, donc pour tout $\pi' > \pi$. p. 132 [page 347]

Commentary · Exposition [ms. 346]

Changing the rank without changing the condition

Grothendieck wants a calculus of conditions on objects which remains usable when several constructions have to be carried out together. An accessible subcategory comes with a size bound, whereas an intersection, an inverse image, or a category of diagrams will usually require a larger common bound. The point of 4.13.26–27 is to arrange that enlargement once, including its effect on the compact generators, before using the calculus.

There are two sizes in the notation. Here "grande devant π" means that diagrams with at most π arrows have cocones; in the usual convention this is π^+-filteredness. Thus $\mathcal{M}_\pi$ is the subcategory of π^+-presentable objects, and a useful π supplies an essentially small family of these objects generating $\mathcal{M}$ under π^+-filtered colimits. The five conditions on the inclusion $f : \underline{S} \hookrightarrow \mathcal{M}$ separate preservation of colimits from preservation of presentability. The former becomes easier as the filteredness bound increases. The latter is a genuine uniformity requirement: sufficiently small generators of the source must have sufficiently small images in the target. It is this requirement which accounts for the cardinal of $\underline{S}_{\pi_0}$ in 4.13.27.

The explicit enlargement $\pi' = 2^c$ is particularly convenient. If c bounds π, the generating data, and π_0, then

$$(\pi')^{\pi_0} = (2^c)^{\pi_0} = 2^{c\cdot\pi_0} = 2^c = \pi'.$$

The argument therefore provides arbitrarily large usable bounds without a continuum hypothesis. In modern accounts this belongs to uniformization: accessible functors preserve presentable objects at suitable sufficiently large ranks, not at an arbitrarily prescribed larger rank. The distinction is visible in the rank comparison recalled in Brooke-Taylor and Rosický (2017, Section 2).

The proposed GCH shortcut requires a correction. Even under GCH, $\lambda > \mu$ need not imply $\lambda^\mu = \lambda$. Take $\mu = \aleph_0$ and $\lambda = \aleph_\omega$. To see the obstruction directly, partition a set of cardinal λ into countably many pieces A_n of cardinal less than λ. If all functions $\mathbf{N} \to \lambda$ could be enumerated as $(f_\alpha)_{\alpha<\lambda}$, choose $g(n)$ outside $\{f_\alpha(n) : \alpha \in A_n\}$. This is possible because $|A_n| < \lambda$. For every α, the coordinate corresponding to the piece containing α gives $g(n) \neq f_\alpha(n)$, contradicting the enumeration. Hence $\lambda^{\aleph_0} > \lambda$.

153. Si π n'est pas cardinal limite. Sinon, il faut dire : familles de cardinal $< \pi$. Cf. 4.13.10.3, où je donne un énoncé rectifié légèrement plus général.

A correct GCH version adds $\mathrm{cf}(\lambda) > \mu$. Every map $\mu \to \lambda$ then has bounded image. For an infinite $\nu < \lambda$, GCH gives $\nu^\mu \leq 2^{\max(\nu,\mu)} \leq \lambda$, so the union over all bounds still has cardinal at most λ; constant functions give the reverse inequality. This repairs the arithmetic statement. It does not require us to reject the category-theoretic conclusions obtained by choosing the larger bound freely: the power-of-two construction already supplies what their proofs need.

Reste la condition 5), pour laquelle une réponse est donnée par 4.12.1 (p. 273). Si $\pi_0 \leqslant \pi$ est choisi de façon que f soit π_0-accessible (p.ex. $\pi_0 = \pi$), et si

$$\pi'_1 = \mathrm{card\,réd}\,\underline{S}_{\pi_0} \qquad (\leqslant \mathrm{card\,réd}\,\underline{S}(\pi))$$
$$\pi''_1 = \sup(\pi'_1, \pi)\ ,$$

on voit que si $\pi' \geqslant \pi''_1$ (i.e. $\pi' \geqslant \pi'_1, \pi$) est tel que $\pi'^{\pi_0} = \pi'$, alors on a bien

$$f(\underline{S}_{\pi'}) \ \subset\ \mathcal{M}_{\pi'}\ .$$

Si donc de plus $\pi' > \pi \geqslant \pi_0$ (comme supposé au début), et si on suppose l'HCG, alors la [page 348]
condition $\pi'^{\pi_0} = \pi'$ est automatique, et il reste la seule condition $\pi' \geqslant \pi''_1$, i.e. en définitive $\pi' \geqslant \mathrm{card\,réd}\,\underline{S}_{\pi_0}$. Pour résumer :

Proposition 4.13.27. *Sous les conditions de 4.13.23 (sans condition b) sur Φ), soit $\underline{S}$ une sous-catégorie de $\mathcal{M}$ qui est Φ-π-adaptée. Soit $\pi_0 \leqslant \pi$ un cardinal qui est utile pour $\underline{S}$ et pour $\mathcal{M}$, et tel que $f : \underline{S} \longrightarrow \mathcal{M}$ soit π_0-accessible, i.e. tel que $\underline{S}$ soit stable dans $\mathcal{M}$ par $\varinjlim$ grandes devant π_0 (on peut p.ex. prendre $\pi_0 = \pi$). Alors pour tout cardinal $\pi' > \pi$ tel que*

a) $\pi'^{\pi_0} = \pi'$ et

b) $\pi' \geqslant \mathrm{card\,réd}\,\underline{S}_{\pi_0}$,

$\underline{S}$ est Φ-π'-accessible dans $\mathcal{M}$. Donc si $\pi' > \pi$ est tel que $\pi'^{\pi} = \pi'$, $\pi' > \mathrm{card\,réd}\,\underline{S}_{\pi}$, alors $\underline{S}$ est Φ-π'-accessible. (Il suffit p.ex. de prendre $\pi' = 2^c$, avec $c \geqslant \sup(\pi, \mathrm{card\,réd}\,\underline{S}(\pi))$.) Si on admet l'HCG, il suffit donc qu'on ait $\pi' > \sup(\pi, \mathrm{card\,réd}\,\underline{S}(\pi))$.

Proposition 4.13.28 ([154]). *Soient $\mathcal{M}$, π, Φ comme dans 4.13.23. Supposons de plus que* [page 349]
$J \in \Phi$ implique que J est filtrante. *Alors l'ensemble $\tilde{\Sigma}^\Phi_\pi$ des sous-ensembles Φ-π-accessibles de $\mathcal{M}$ est stable par réunions finies. Plus généralement, soit λ un cardinal $< \pi$ tel que $\forall\ J \in \Phi$, J soit grande devant λ. Alors l'ensemble $\tilde{\Sigma}^\Phi_\pi$ est stable par réunions $(S_r)_{r\in A}$, indexées par des ensembles A de cardinal $\leqslant \lambda$.*

Corollaire 4.13.29. *L'ensemble des parties accessibles de $\mathrm{Ob}\,\mathcal{M}$ est stable par "petites" réunions – i.e. si $(S_r)_{r\in A}$ est une petite famille de parties accessibles de $\mathrm{Ob}\,\mathcal{M}$, alors $\bigcup_{r\in A} S_r$ est aussi accessible.*

La démonstration de 4.13.28 repose sur le lemme suivant (je n'explicite par la démonstration) :

Lemme 4.13.30. *Soit I une catégorie grande devant π, et $(J_r)_{r\in A}$ une famille des sous-* p. 133
catégories pleines telles que $\mathrm{card}\,A \leqslant \pi$, telle que $\bigcup_{r\in A} \mathrm{Ob}\,J_r$ soit cofinal dans $\mathrm{Ord}(I)$, i.e. [page 350]
pour tout $i \in \mathrm{Ob}\,I\ \exists\ j \in \bigcup_{r\in A} \mathrm{Ob}\,J_r$, et une flèche $i \longrightarrow j$. Alors il existe $r \in A$ tel que J_r soit cofinal dans I.

Théorème 4.13.31. *Soient $\mathcal{M}$, $\mathcal{N}$ des catégories accessibles, $f : \mathcal{M} \longrightarrow \mathcal{N}$ un foncteur accessible, $T \subset \mathrm{Ob}\,\mathcal{N}$ une partie de $\mathcal{N}$, $\underline{T} \subset \mathcal{N}$ la sous-catégorie strictement pleine correspondante de $\mathcal{N}$, $S = f^{-1}(T)$, $\underline{S} = f^{-1}(\underline{T})$ la sous-catégorie pleine de $\mathcal{M}$ associée. Si T est accessible, alors S est accessible ; en d'autres termes, si $\underline{T}$ est accessible dans $\mathcal{N}$, $\underline{S}$ est accessible dans $\mathcal{M}$.*

Cela résulte de l'énoncé plus précis :

Corollaire 4.13.32. *Soient π_0 et $\pi > \pi_0$ des cardinaux ayant les propriétés suivantes :* [page 351]

154. À rejeter plus bas, après 4.13.31, qui est la première application importante de 4.13.22, alors que 4.13.28 se prouve sans avoir à utiliser 4.13.22 ou ses variantes.

a) $\underline{T}$ *est* π_0*-*π*-accessible dans* $\mathcal{N}$ *(4.13.23).*

b) π_0 *est utile pour* $\mathcal{M}$ *(il l'est pour* $\mathcal{N}$*, à cause de a)).*

c) $f(\mathcal{M}_\pi) \subset \mathcal{N}_\pi$*.*

d) $\mathcal{M}$ *satisfait à* L_{π_0}*, et* f *est* π_0*-accessible.*

(L'existence d'un tel couple (π_0, π) *résulte aussitôt de 4.13.16 et de 4.12.1.) Alors* $\underline{S}$ *est* π_0*-*π*-accessible dans* $\mathcal{N}$*.*

Commentary · Exposition [ms. 350]

A calculus of object conditions

Grothendieck's accessible parts are intended to behave like manageable predicates on the objects of a large category. The two operations now needed are disjunction and substitution: a small union combines alternative conditions, while an inverse image expresses a condition after applying a functor. These are not statements merely about the number of objects. They say that the resulting full subcategory again has compact generators and that its sufficiently filtered colimits are computed in the ambient category.

The omitted proof of Lemma 4.13.30 explains why disjunction works. Let I be π^+-filtered and suppose that no J_r is objectwise cofinal. Choose $i_r \in I$ for which no arrow $i_r \to j$ exists with $j \in J_r$. Since there are at most π indices r, choose a common upper bound i of the i_r. Cofinality of the union supplies an arrow $i \to j$ with $j \in J_s$ for some s. Its composite with $i_s \to i$ contradicts the choice of i_s. Consequently some J_s is objectwise cofinal.

For a full subcategory of a filtered category, objectwise cofinality supplies the usual categorical finality. Any finite diagram in J_s first has a cocone in I; an arrow from its vertex to an object of J_s gives a cocone in J_s, because the subcategory is full. The same argument in each comma category $(i \downarrow J_s)$ supplies a nonempty filtered category. This proves finality and justifies using J_s to compute colimits. Applied to the indices at which an adapted presentation belongs to a given member of a family, the lemma says that a sufficiently filtered colimit lying in their union can be computed wholly inside one member. Enlarging the common cardinal to bound a small family gives Corollary 4.13.29.

The inverse-image argument uses the stronger cofinal-presentation criterion established earlier. For $x \in f^{-1}(\underline{T})$, present x by the compact objects over it. The image presentation consists of compact objects of $\mathcal{N}$, once the common bound has been chosen to make f preserve presentability. Since its colimit lies in $\underline{T}$, the criterion supplies cofinally many stages already lying in $\underline{T}$. Their preimages are the desired compact generators of $f^{-1}(\underline{T})$. The argument proves essential surjectivity of

$$\mathrm{Ind}_\pi\big(f^{-1}(\underline{T}) \cap \mathcal{M}_\pi\big) \longrightarrow f^{-1}(\underline{T}).$$

Full faithfulness follows from compactness in $\mathcal{M}$.

There is a small rank omission in the quantitative formulation 4.13.32. Its proof uses that π is useful for $\mathcal{M}$, whereas condition b) literally says that π_0 is useful. The safe formulation includes the former condition, obtained, for example, by imposing $\pi^{\pi_0} = \pi$. This is harmless for the qualitative theorem, where the common rank is free to increase. The conclusion of the corollary also prints $\mathcal{N}$ where it must say $\mathcal{M}$; that is a source slip, not an error of transcription.

The present categorical formulation packages the same mechanism as a pseudopullback. Its objects are triples (x, t, θ) with $\theta : f(x) \xrightarrow{\sim} t$ and $t \in \underline{T}$. Because $\underline{T}$ is replete and full, forgetting t, θ is an equivalence with the strict full inverse image. The accessible pseudopullback theorem therefore applies. This is one instance of the limit theorem of Makkai and Paré (1989, Theorem 5.1.6); its precise relation to products, inserters and equifiers is discussed in Positselski (2024, Introduction and Section 5). The repleteness assumption is what permits passage from an isomorphism to literal membership, and should remain visible in either proof.

Comme f commute aux $\varinjlim$ grandes devant π_0 en vertu de d), et que $\underline{T}$ est stable par ces limites par a) (cf. 4.13.23, 3°), il en est donc de même de $\underline{S}$, i.e. le 3° de la définition 4.13.23 est satisfait. D'autre part, la condition 1° (π utile pour $\mathcal{M}$, $\mathcal{M}$ stable par $\varinjlim$ grandes devant π_0 et de cardinal $\leqslant \pi$) est vérifiée par b) et d). Donc il reste à vérifier la condition 2°, i.e. que

$\underline{S}$ est π-accessible. Posons

$$\underline{T}(\pi) = \underline{T} \cap \mathcal{N}_\pi$$
$$\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi \quad \underbrace{\subset}_{\text{grâce à c)}} \quad f^{-1}(\underline{T}(\pi)) \ .$$

Comme $\underline{S}$ est stable dans $\mathcal{M}$ par $\varinjlim$ grandes devant π_0 (lesquelles existent dans $\mathcal{M}$ par d), et a fortiori par $\varinjlim$ grandes devant π, on a un foncteur canonique pleinement fidèle

$$\mathrm{Ind}_\pi \underline{S}(\pi) \longrightarrow \underline{S} \ .$$

D'ailleurs, $\underline{T}$ étant strictement pleine dans $\mathcal{N}$, $\underline{S}$ est strictement pleine dans $\mathcal{M}$, et il reste [page 352] à vérifier que le foncteur précédent est essentiellement surjectif. Soit donc $x \in \mathrm{Ob}\,\underline{S}$. On aura, grâce à b),

$$x = \varinjlim_I x_i \ , \qquad x_i \in \mathcal{M}_\pi, \ I = \mathcal{M}_\pi/x \ .$$

D'ailleurs, le système inductif précédent est π_0-π-adapté, i.e. (p. 339) :

1) I équivalente à une petite catégorie, I grande devant π (résulte du fait que π est utile pour $\mathcal{M}$, cf. ⓑ).
2) Les limites inductives grandes devant π_0 est de cardinal $\leqslant \pi$ existent dans I (résulte de L_{π_0} sur $\mathcal{M}$ (cf. ⓓ)). p. 134
3) Le foncteur $\xi : I \longrightarrow \mathcal{M}$ y commute.
4) Les x_i sont dans $\mathcal{M}_\pi$.

Comme f commute aux $\varinjlim$ grandes devant π_0 (cf. ①), a fortiori à celles grandes devant π, on a donc

$$f(x) \simeq \varinjlim_I f(x_i) \ ,$$

je dis que le système inductif $(f(x_i))_{i\in I}$ dans $\mathcal{N}$ est π_0-π-adapté. Les conditions 1), 2) sont *ne varietur* (ne concernent que I), le foncteur $I \longrightarrow \mathcal{N}$ est le composé $I \xrightarrow{\xi} \mathcal{M} \xrightarrow{f} \mathcal{N}$, donc commute aux $\varinjlim$ grandes devant π_0 et de cardinal $\leqslant \pi$ ([155]), puisqu'il en est ainsi de ξ et de f, enfin les $f(x_i)$ sont dans $\mathcal{N}_\pi$ puisque par ⓒ $f(\mathcal{M}_\pi) \subset \mathcal{N}_\pi$.

Par ailleurs $f(x) \in \mathrm{Ob}\,\underline{T}$ (puisque $x \in \mathrm{Ob}\,\underline{S}$, où $\underline{S} \overset{\text{déf}}{=} f^{-1}(\underline{T})$). Donc on peut appliquer [page 353] 4.13.22 a $\Longrightarrow$ c à la situation $(\mathcal{N}, \pi_0, \pi, \underline{S})$ et au système inductif $(f(x_i))_{i\in I}$, pour conclure que la sous-catégorie I' de I formée des i tels que $f(x_i) \in \mathrm{Ob}\,\underline{S}$ `[plutôt Ob` $\underline{T}$`]`, i.e. tels que $x_i \in \underline{S}$, donc $x_i \in \underline{S}(\pi)$, est *cofinale* dans I. Donc on a

$$x \simeq \varinjlim_{i\in I'} x_i \ ,$$

avec cette fois les x_i dans $\underline{S}(\pi)$, et I' grande devant π (car cofinale et pleine dans I, qui est grande devant π), ce qui prouve que x est dans l'image essentielle de $\mathrm{Ind}_\pi(\underline{S}(\pi))$, q.e.d.

Question 4.13.33. On vient de voir que si $f : \mathcal{M} \longrightarrow \mathcal{N}$ est un foncteur accessible entre catégories accessibles, alors l'image inverse d'une partie accessible de $\mathrm{Ob}\,\mathcal{N}$ est une partie accessible de $\mathrm{Ob}\,\mathcal{M}$. Je conjecture que l'énoncé similaire, non pas pour les images directes, mais pour les *images essentielles*, est vraie, i.e. que pour toute sous-catégorie $\underline{S}$ [page 354] de $\mathcal{M}$ accessible dans $\mathcal{M}$, son image essentielle dans $\mathcal{N}$ est accessible dans $\mathcal{N}$ ([156]). Dans cette question, on peut supposer que $\underline{S} = \mathcal{M}$, et la question revient à celle-ci : tout foncteur accessible $f : \mathcal{M} \longrightarrow \mathcal{N}$ entre catégories accessibles se factorise-t-il en un foncteur (accessible)

155. cette commutation devrait avoir un nom, condition $\mathrm{L}(\pi_0, \pi)$ sur un foncteur.
156. Faux, même si on prend la $\natural$-clôture de l'image essentielle ("image $\natural$-essentielle").

essentiellement surjectif, et un foncteur accessible pleinement fidèle $\mathcal{M} \longrightarrow \underline{T} \longrightarrow \mathcal{N}$ (où $\underline{T}$ est elle-même accessible) ([157]) ? Si oui, cela me paraîtrait peut-être le résultat le plus profond (peut-être le seul véritablement profond) en théorie des catégories.

Commentary · Examples [ms. 353]

Why forgetting a retraction destroys the image theorem

The hope behind 4.13.33 is that an accessible construction might be followed by forgetting all its choices, while its essential image still retained a compact description. The footnotes reject that hope, including the attempted repair by closing under retracts. The following example makes both failures explicit. It is the split-monomorphism example of Rosický (2009, Example 3.5(1), author manuscript); the retract-closed formulation appears in Brooke-Taylor and Rosický (2017, Remark 3.3(2), accepted version).

Let R be the small category with two objects a, b, arrows $i : a \to b$ and $p : b \to a$, and relation $pi = 1_a$. The category Pos^R consists of split inclusions of posets with chosen retractions and maps preserving both pieces of data. Restriction along the walking arrow gives

$$F : \mathrm{Pos}^R \longrightarrow \mathrm{Pos}^{[1]}, \qquad (i, p) \longmapsto i.$$

These are accessible categories and F is accessible. One can check this without a general model theorem. A poset is the filtered union of its finite subposets. A finite collection of elements in an R-diagram generates a finite subdiagram, since R has only finitely many morphisms. The inherited orders are finite, and all operations and their equations restrict. Such finite diagrams are finitely presentable: maps and the finitely many order relations factor, and are eventually identified, in a filtered colimit. The same argument applies to the arrow category. Filtered colimits are computed pointwise, so restriction preserves them.

The replete full image $\mathcal{L}$ consists of split monomorphisms of posets, with every commutative square allowed as a morphism. Fix any infinite regular cardinal λ. For $\alpha < \lambda$, put

$$A_\alpha = \{\beta : \beta \leq \alpha\}, \qquad B_\alpha = A_\alpha \amalg \{\top\},$$

where $\top$ lies above every element, and let $j_\alpha : A_\alpha \hookrightarrow B_\alpha$ be the inclusion. It splits by sending $\top$ to α. The inclusions of initial segments, together with $\top \mapsto \top$, define a λ-chain in $\mathcal{L}$. Its index ordinal is λ-filtered. In the arrow category its colimit is

$$j : \lambda \hookrightarrow \lambda \amalg \{\top\}.$$

A monotone retraction would have to send $\top$ to an ordinal dominating every $\beta < \lambda$. No such ordinal belongs to λ. Thus j does not split. Since this construction works for every regular λ, the full image is not accessibly embedded.

Closing the image under retracts changes nothing. Suppose an arrow $j : A \to B$ is a retract of a split arrow $i : X \to Y$, with arrow-category maps $(u, v) : j \to i$ and $(r, s) : i \to j$ satisfying $ru = 1_A$, $sv = 1_B$. If $p : Y \to X$ retracts i, then

$$(rpv)j = rp(iu) = ru = 1_A.$$

Hence j already splits. This proves directly the stronger rejection recorded in footnote 156.

The reason the diagram above does not lift to Pos^R is concrete: the chosen retraction of the new top changes from α to β as the index increases. The squares preserve the inclusions but not their chosen retractions. Fullness of the essential image permits exactly these additional morphisms. In contrast, an Ind-completion of compact objects embeds fully faithfully before any choices are forgotten; this is why the image used in 4.13.34 behaves well.

Voici une question qui me paraît très voisine. Soient $\mathcal{M}$ une catégorie accessible, A une sous-catégories pleine petite. Supposons que $\mathcal{M}$ satisfasse à L_{π_0} (p.ex. π_0 utile pour $\mathcal{M}$), et soit $\underline{S}$ la plus petite sous-catégorie strictement pleine de $\mathcal{M}$ contenant A, et stable par limites grandes devant π_0. Je conjecture que $\underline{S}$ est accessible dans $\mathcal{M}$.

[page 355]

157. Il faut se borner tout au moins à essentiellement ♮-surjectif – mais même cela est trop optimiste.

De même, si Φ est un petit ensemble de petites catégories, tel que $\mathcal{M}$ soit stable par $\varinjlim$ de type Φ, on peut prendre l'enveloppe $\underline{S}$ de A pour les $\varinjlim$ grandes devant π_0 *et* les $\varinjlim$ de type Φ, et je conjecture encore que $\underline{S}$ est accessible. Dans cette direction, on a du moins le résultat très partiel suivant. p. 135

Proposition 4.13.34. *Soit $\mathcal{M}$ une catégorie accessible stable par petites $\varinjlim$ filtrantes, et soit $\underline{S}$ une sous-catégorie strictement pleine de $\mathcal{M}$. Conditions équivalentes :*

a) $\underline{S}$ est accessible dans $\mathcal{M}$, et stable dans $\mathcal{M}$ par petites $\varinjlim$ filtrantes.

b) Il existe une petite sous-catégorie pleine A de $\mathcal{M}$, telle que $\underline{S}$ soit la plus petite sous-catégorie strictement pleine de $\mathcal{M}$ contenant A et stable par petites $\varinjlim$ filtrantes.

[page 356]

Commentary · Development (commentators') [ms. 355]

The small-envelope conjectures at a fixed threshold

Grothendieck separates the false assertion about arbitrary essential images from a more controlled construction: start with a small collection of objects and close it under specified colimits already available in the ambient category. His proof of 4.13.34 treats all filtered colimits. The same method, combined with bounded completion of index categories, answers the two preceding envelope questions at a fixed filteredness threshold. We give the argument because the role of the size bound is otherwise easy to miss.

Proposition (small colimit envelopes). [ours] Let κ be an infinite regular cardinal. Let $\mathcal{M}$ be accessible, admitting all small κ-filtered colimits and the colimits of a small family Φ of small categories. If $A \subset \mathcal{M}$ is small, then its smallest replete full envelope closed under these colimits is accessible and accessibly embedded in $\mathcal{M}$.

We use the Ind extension result proved in 4.13.19: a small replete subcategory of $\mathcal{M}_\pi$, closed under a family of colimits of size at most π including a nonzero limit-ordinal shape, has a fully faithfully embedded Ind_π-completion closed under the same colimits. The closure under a limit-ordinal shape supplies the retract closure needed in applying that result.

Proof. Choose an infinite cardinal π which is useful for $\mathcal{M}$, with

$$\kappa \le \pi, \qquad \pi^{<\kappa} = \pi, \qquad A \subset \mathcal{M}_\pi, \qquad |\operatorname{Fl} I| \le \pi \quad (I \in \Phi).$$

Such a bound exists. Start above a usefulness bound for $\mathcal{M}$, above the presentability ranks of the small family A, and above the sizes of the shapes; then take a sufficiently large power 2^c. The exponentiation identity gives the displayed closure condition and the usual enlargement of usefulness.

Let Φ_π consist of the shapes in Φ and one representative of each κ-filtered category having at most π arrows. There is a set of such representatives. The subcategory $\mathcal{M}_\pi$ is closed under these colimits: a colimit with at most π arrows of π^+-presentable objects remains π^+-presentable, because π^+-filtered colimits of sets commute with limits indexed by at most π arrows. Let B be the smallest replete full subcategory of $\mathcal{M}_\pi$ containing A and closed under Φ_π-colimits. It is essentially small, being contained in $\mathcal{M}_\pi$.

We make its retract closure explicit. The ordinal κ is κ-filtered and has at most π arrows. For an idempotent $e : b \to b$, the diagram constant at b with transition e for every strict inequality has a colimit which splits e. Indeed, if $q : b \to c$ is its colimit map, then $qe = q$, while the compatible maps $e : b \to b$ induce $u : c \to b$ with $uq = e$; the colimit universal property gives $qu = 1_c$. Thus $c \in B$, and repleteness includes every ambient retract of b.

Consequently the canonical functor $\mathrm{Ind}_\pi B \to \mathcal{M}$ is fully faithful. Write $\mathcal{S}'$ for its replete image. It is accessible, its π^+-filtered colimits are computed in $\mathcal{M}$, and 4.13.19 shows that it is closed there under Φ_π-colimits. In particular it is already closed under the additional shapes Φ. It remains to pass from bounded κ-filtered diagrams to arbitrary small ones.

Here is the index-category argument with its cardinal estimate. If I is small and κ-filtered, every subcategory $K \subset I$ with at most π arrows is contained in a κ-filtered subcategory $\overline{K} \subset I$ with at most π arrows. For each diagram with fewer than κ arrows in the current subcategory, choose a cocone in I, adjoin its arrows and vertex, and close under finite composition. There are at most $\pi^{<\kappa} = \pi$ diagrams to treat, including their possible small shapes, so each stage still has at most π arrows. Iterate through κ stages, taking unions at limit stages. Since κ is regular, every diagram with fewer than κ arrows in the final union is contained in one earlier stage and acquires its cocone at the next. The final union is the required $\overline{K}$. The empty diagram is included, so nonemptiness is also assured.

Let P be the poset of all κ-filtered subcategories $J \subset I$ with at most π arrows. It is a set and is π^+-filtered. Indeed, the union of at most π such subcategories generates a subcategory with at most π arrows, and the preceding construction supplies an upper bound in P. Every object and arrow of I belongs to some member of P. These are actual subcategories, not full subcategories; taking full closure could add too many arrows.

Now let $D : I \to \mathcal{S}'$ be any small κ-filtered diagram. Each inner colimit below exists in $\mathcal{S}'$ by bounded closure, and the outer one exists by π^+-accessibility:

$$y = \varinjlim_{J \in P} \left(\varinjlim_{j \in J} D(j) \right).$$

Both are computed in $\mathcal{M}$. A compatible family of cocones on the restrictions $D|_J$ gives a unique cocone on D: to define its component at i, choose a J containing i; the result is independent of this choice because two such J have a common upper bound. Naturality follows by choosing a J containing the arrow in question. Conversely any cocone on D restricts compatibly. Hence y is the colimit of D, and $\mathcal{S}'$ is closed under all small κ-filtered colimits in $\mathcal{M}$.

Finally let $\mathcal{S}$ denote the envelope in the proposition. Every object of B belongs to $\mathcal{S}$ by its construction. Every π^+-filtered category is κ-filtered, so every object of $\mathcal{S}'$, being a π^+-filtered colimit of objects of B, belongs to $\mathcal{S}$. The reverse inclusion follows from the closure properties just proved for $\mathcal{S}'$. Therefore $\mathcal{S} = \mathcal{S}'$, proving the assertion. □

For the manuscript's two questions take $\kappa = \pi_0^+$, first with Φ empty and then with its specified additional shapes. The proof follows Grothendieck's own Ind and bounded-colimit method. We checked the cited primary papers on accessible constructions and images, together with the publisher descriptions of Makkai and Paré (1989) and Adámek and Rosický (1994); we did not locate this exact envelope formulation in the material inspected. The marker records the proof developed for this commentary, without asserting priority for the theorem.

On a a) $\Longrightarrow$ b), car si $\underline{S}$ est π-accessible dans $\mathcal{M}$, il est immédiat (sans autre condition de stabilité sur $\underline{S}$) que $\underline{S}$ est la plus petite sous-catégorie strictement pleine de $\mathcal{M}$ contenant $\underline{S}(\pi) \overset{\text{déf}}{=} \underline{S} \cap \mathcal{M}_\pi$ et stable par $\varinjlim$ grandes devant π. Si donc on sait déjà que $\underline{S}$ est même stable par *toutes* $\varinjlim$ filtrantes petites, il s'ensuit que $\underline{S}$ est la plus petite sous-catégorie de $\mathcal{M}$ ayant cette propriété de stabilité plus forte, et contenant A.

Prouvons b $\Longrightarrow$ a. Soit π un cardinal utile pour $\mathcal{M}$, et tel que $A \subset \mathcal{M}_\pi$. Soit B la plus petite sous-catégorie strictement pleine de $\mathcal{M}$ contenant A et stable par $\varinjlim$ filtrantes de cardinal $\leqslant \pi$. On a donc $B \hookrightarrow \mathcal{M}_\pi$ et $B \subset \underline{S}$, et $\underline{S}$ est aussi la L-enveloppe de B. D'ailleurs B est stable par facteurs directs. Considérons alors l'image essentielle $\underline{S}'$ de $\mathrm{Ind}_\pi(B)$ dans $\mathcal{M}$. On [page 357]
a $\underline{S}' \subset \underline{S}$, et tout revient à prouver que $\underline{S}' = \underline{S}$, d'où résultera en effet que $\underline{S}$ est π-accessible dans $\mathcal{M}$, a fortiori accessible dans $\mathcal{M}$, d'où a). Pour prouver que $\underline{S}' = \underline{S}$, par définition de $\underline{S}$ comme la L-enveloppe de A, il suffit de voir que $\underline{S}'$ est stable par petites $\varinjlim$ filtrantes. Or cela résulte du fait que $B \subset \mathcal{M}_\pi$ est stable dans $\mathcal{M}_\pi$ par $\varinjlim$ filtrantes de cardinal $\leqslant \pi$, et de 4.13.19, qui implique tout au moins, comme $B = \underline{S}' \cap \mathcal{M}_\pi$ (en vertu de 4.13.4) que $\underline{S}'$ est stable par $\varinjlim$ filtrantes *de cardinal* $\leqslant \pi$. Qu'il soit même stable par petites $\varinjlim$ filtrante quelconques, résulte de ceci, et du fait qu'il l'est par $\varinjlim$ grandes devant π. (Cf. 4.13.21 …)

Définition 4.13.35. Soient $\mathcal{M}$ une catégorie accessible, stable par petites $\varinjlim$ filtrantes, [page 358]
S une partie de $\mathrm{Ob}\,\mathcal{M}$, $\underline{S}$ la sous-catégorie pleine correspondante dans $\mathcal{M}$. On dit que S est *L-accessible*, ou que $\underline{S}$ est L-accessible dans $\mathcal{M}$, si $\underline{S}$ est accessible dans $\mathcal{M}$, et de plus stable par petites $\varinjlim$ filtrantes ([158]).

Proposition 4.13.36. *Soient $\mathcal{M}$, S, $\underline{S}$ comme dans 4.13.35, et soit π un cardinal utile pour $\mathcal{M}$. Supposons $\underline{S}$ stable dans $\mathcal{M}$ par petites $\varinjlim$ filtrantes, ou seulement $\underline{S}(\pi) \overset{déf}{=} \underline{S} \cap \mathcal{M}_\pi$ stable par petites $\varinjlim$ filtrantes de cardinal $\leqslant \pi$. Conditions équivalentes :*

158. **N.B.** L'ensemble des parties L-accessibles de $\mathrm{Ob}\,\mathcal{M}$ est stable par petites intersections, et par réunions *finies*. L'image inverse d'une partie L-accessible par un foncteur ([159]) $f : \mathcal{M} \longrightarrow \mathcal{N}$ de catégories accessibles est L-accessible.

159. un L_0-*foncteur* : qui commute aux petites $\varinjlim$ filtrantes.

a) $\underline{S}$ *est* π*-accessible. (Il sera automatiquement stable par petites* $\varinjlim$ *filtrantes, donc L-accessible.)*

b) Pour tout $x \in \mathrm{Ob}\,\mathcal{M}$*, on a* $x \in \mathrm{Ob}\,\underline{S}$ *si et seulement si* $\underline{S}(\pi)/x$ *est cofinale dans* $\mathcal{M}_\pi/x$. p. 136

c) Pour tout $x \in \mathrm{Ob}\,\mathcal{M}$*, et toute représentation* $x \simeq \varinjlim_{i\in I} x_i$ *qui est L-*π*-adaptée (cf. page* [page 359] *343, c), on a ceci :* $x \in \mathrm{Ob}\,S$ *si et seulement si* $I_{\underline{S}}$ *est cofinal dans* I*.*

c′) Comme c), avec I *ordonnée.*

c″) (Si on a choisi, pour tout $x \in \mathrm{Ob}\,\mathcal{M}$*, une représentation* π*-adaptée* $x \simeq \varinjlim_{I} x_i$*) :* $x \in \mathrm{Ob}\,\underline{S} \iff$ *l'ensemble* $I_{\underline{S}}$ *est cofinal dans* I*.*

Démonstration. C'est essentiellement l'énoncé de 4.13.22 dans le cas particulier où $\Phi =$ ensemble des catégories *filtrantes* de cardinal $\leqslant \pi$, mais où, pour simplifier, on a supposé d'avance que $\underline{S}(\pi)$ satisfait à $\mathrm{L}(\Phi)$. L'équivalence de a) b) c) est contenue dans 4.13.22, ainsi que c $\Longrightarrow$ c′. Mais pour c′ $\Longrightarrow$ a, l'hypothèse accessoire $\pi^{\pi_0} = \pi$ et que les $J \in \Phi$ seront grandes devant π_0, n'est pas satisfaite ici, et il faut un démonstration indépendante, calquée sur celle des pages 333-336. Il suffit pour cela, à l'endroit `[de]` □, page 333, d'invoquer le

Lemme 4.13.37. *Soient* $\mathcal{M}$ *une catégorie accessible stable par* $\varinjlim$ *filtrantes,* π *un cardinal utile pour* $\mathcal{M}$*. Alors pour tout* $x \in \mathrm{Ob}\,\mathcal{M}$*, on a une représentation* [page 360]

$$x \;=\; \varinjlim_{i\in I} x_i$$

L-π*-adaptée (i.e. avec*

1) I *équivalente à une petite catégorie, et* I *grande devant* π*,*

2) I *stable par* $\varinjlim$ *filtrantes de cardinal* $\leqslant \pi$*,*

3) $I \longrightarrow \mathcal{M}$ *commute à ces* $\varinjlim$ *,*

d) $x_i \in \mathrm{Ob}\,\mathcal{M}_\pi \;\forall\, i \in I$*),*

avec de plus I ordonné.

En effet, on part de

$$x \;=\; \varinjlim_{j\in J} x_j \;, \qquad J \;\simeq\; \underbrace{(\mathcal{M}_\pi/x)_0}_{\substack{\text{sous-catégorie réduite}\\ \text{équivalente à } \mathcal{M}_\pi/x}} \;.$$

On définit I comme l'ensemble des sous-catégories *filtrantes* J' de J, de cardinal `["cardinal" biffé]` $\leqslant \pi$, I ordonné par inclusion, et on pose, pour $J' \in I$,

$$x_{J'} \;=\; \varinjlim_{j\in J'} x_j \;.$$

Il est immédiat qu'on obtient ainsi un système inductif dans $\mathcal{M}$, indexé par I.

D'autre part, il est clair que I est petite, je dis que I est stable par $\varinjlim$ filtrantes de [page 361] cardinal $\leqslant \pi$, et que le foncteur $J' \longmapsto x_{J'} : I \longrightarrow \mathcal{M}$ y commute, enfin que les $x_{J'}$ sont dans $\mathcal{M}_\pi$. Il reste à voir que I est grande devant π, i.e. que si on a $(J'_r)_{r\in A}$, avec $\mathrm{card}\,A \leqslant \pi$, les J'_r dans I, alors $\exists\, J' \in I$ qui majore les J'_r. Soit K la sous-catégorie de J engendrée par les J'_r, on voit facilement que $\mathrm{card}\,K \leqslant \pi$. Tout `["tout" biffé]` revient à prouver :

Lemme 4.13.38 ([160]). *Soient* J *une catégorie grande devant* π*,* K *une sous-catégorie* p. 137 *de* J *de cardinal* $\leqslant \pi$*. Alors il existe une sous-catégorie* K' filtrante *de* J*, contenant* K*, de cardinal* $\leqslant \pi$*. Si on suppose que l'on n'est pas, dans le cas où* J *contient un objet maximal, et que* K *contienne tous les objets maximaux de* J*, on peut même prendre* K' *telle que* [page 362]

160. **N.B.** Pour la démonstration de 4.13.37, il suffit de prouver 4.13.38 dans le cas où J est ordonnée, auquel cas les contorsions de la fin sont inutiles.

$\operatorname{Ob} K' = \operatorname{Ob} K \cup \{e\}$, *où* e *est un* objet final *de* K', *et* K *est une sous-catégorie pleine de* K'. *(Donc si* $K' \neq K$, K' *est isomorphe au cône à droite construit sur* K, *"déduit de* K *en lui ajoutant un objet final".)*

Commentary · Exposition [ms. 361]

Making the presentation continuous

The ordered presentation in Lemma 4.13.37 has a purpose beyond replacing a category by a poset. Grothendieck needs the index poset to contain the bounded filtered colimits used to improve an approximation, and he needs the presentation itself to preserve them. The ordinary statement that every object is a filtered colimit of compact objects supplies neither requirement by itself.

There is a short proof of the bounded enlargement used here. In any filtered category J, start with a subcategory K having at most the infinite cardinal π arrows. For every pair of its objects choose a common target and arrows to it. For every parallel pair of its arrows choose an equalizing arrow. Adjoin these choices and close under finite composition; the resulting subcategory still has at most π arrows. Repeat this construction countably many times. The union K_∞ has at most π arrows and is filtered: every finite configuration occurs in one stage, and its required witnesses occur at the next. Include one object at the start if K is empty. This proves the first assertion of 4.13.38 under the weaker assumption that J is ordinarily filtered. The stronger ambient filteredness is needed elsewhere, including the one-vertex variant, but not for this bounded enlargement.

Apply this to the small skeleton J of $\mathcal{M}_\pi/x$. Let P be the poset of its filtered subcategories of size at most π. Given at most π members, generate their union under composition and apply the enlargement above. This makes P π^+-filtered. A filtered union of at most π members is itself filtered and still of size at most π; it is their least upper bound in P. Thus the required bounded filtered colimits in the index poset are simply unions.

For $J' \in P$, put $x_{J'} = \varinjlim_{j \in J'} x_j$. Each x_j is π^+-presentable, and the diagram has at most π arrows, so $x_{J'}$ is again π^+-presentable. The colimit comparison for a filtered union of subcategories is an isomorphism: a cocone on the union is exactly a compatible collection of cocones on its pieces. Therefore $J' \mapsto x_{J'}$ preserves the required colimits. The same description of cocones, now over all of P, gives $x \simeq \varinjlim_P x_{J'}$. This supplies all four conditions of an L-π-adapted presentation, including the continuity which drives the cofinal-membership criterion.

That continuity must also be read into condition c″ of 4.13.36, where the printed text abbreviates the phrase to "π-adaptée". To see what fails if it is omitted, let $\mathcal{M}$ be sets and $\underline{S}$ the full subcategory of sets of cardinal at most one. It is accessible and closed under filtered colimits. Over the ordinal π^+ take the constant object $\{0,1\}$, with every strict transition the map sending both points to 0. This is a π^+-filtered presentation by compact objects and its colimit is a singleton, but no stage lies in $\underline{S}$. At a limit stage the displayed object is still a two-element set, whereas the colimit of the earlier constant-transition diagram is a singleton. Precisely the missing continuity detects the defect.

Démonstration. Si K admet un objet final, on prend $K' = K$. Sinon, comme J est grande devant $\operatorname{card} \operatorname{Ob} K$, il existe un $j \in \operatorname{Ob} J$ qui majore tous les objets de K. Choisissons, pour tout $k \in \operatorname{Ob} K$, une flèche

$$\alpha_k \ : \ k \longrightarrow j \ .$$

Pour toute flèche

$$u \ : \ k' \longrightarrow k$$

dans K, considérons la double-flèche

$$(\alpha_k \circ u, \alpha_{k'}) \ : \ k' \rightrightarrows j \ .$$

L'ensemble de ces double-flèches, de but j, est de cardinal $\leqslant \operatorname{card} \operatorname{Fl} K \leqslant \pi$, donc J étant grande devant π, il existe une flèche $\rho : j \longrightarrow e$ dans J qui égalise toutes ces double-flèches. Soit $K' \subset J$, ayant comme objets $\operatorname{Ob} K \cup \{e\}$, et ayant comme flèches, en plus de celles de K, les flèches $\rho\alpha_k : k \longrightarrow e$ (pour $k \in \operatorname{Ob} K$), et id_e. *Si* $e \notin \operatorname{Ob} K$, on voit que c'est bien une [page 363]
sous-catégorie, admettant e comme objet final et K comme sous-catégorie pleine. Si on avait $e \in \operatorname{Ob} K$, et s'il existe un objet e' de J qui majore e et qui n'est pas dans K, on se ramène aussitôt au cas précédent, en remplaçant e par e'.

([161]). Donc il reste le cas où il existe un objet e dans K qui est un plus grand élément dans K, et tel que tout objet de J majorant e soit dans K. Cela implique, J étant filtrante, que e est un plus grand objet de J. (Car si $x \in J$, $\exists\, x'$ majorant commun de x et e, donc x' est dans K, donc majoré par e, qui majore donc x.) On voit comme tantôt qu'il existe une flèche $p : e \longrightarrow j$ dans J, qui égalise toutes les double-flèches $k \rightrightarrows e$ de but e *dans* K. Quitte à composer avec $j \longrightarrow e$, on peut supposer $j = e$. On a donc un endomorphisme p de e dans J, tel que $pu = pv$ pour toute double-flèche $k \overset{u}{\underset{v}{\rightrightarrows}} e$ dans K, de but e.

Pour $k \in \mathrm{Ob}\, K$, soit [page 364]

$$f_k \ : \ k \longrightarrow e$$

la valeur commune des pu, pour $u \in \mathrm{Hom}_K(k, e)$. On voit tout de suite que si on a $k' \overset{v}{\to} k$ dans K, alors

$$f_{k'} \ = \ f_k \circ v \ .$$

En particulier, prenant $k' = e$, on trouve p. 138

$$f_k \circ v \ = \ f_e \,(= \ p\ ) \qquad \text{si } v : e \longrightarrow k \ .$$

Soit donc K' la sous-catégorie de J ayant mêmes objets que K, et comme flèches, en plus de celles de K, les flèches composées $k \overset{u}{\to} e \overset{p^r}{\longrightarrow} e \overset{v}{\to} k'$, avec u, v dans K, et $r \geqslant 1$, i.e. les flèches composées $k \overset{f_k}{\longrightarrow} e \overset{p^{r'}}{\longrightarrow} e \overset{v}{\to} k'$, avec $k, k' \in \mathrm{Ob}\, K$, $v \in \mathrm{Fl}\, K$, $r' \geqslant 0$. Il faut voir que c'est stable par composition (pour avoir bien une sous-catégorie). C'est clair pour vu si u ou v sont dans $\mathrm{Fl}\, K$. Si `[ni]` u ni v ne le sont, on doit prendre un composé

$$k \xrightarrow{u_1} e \underset{r\geqslant 1}{\xrightarrow{p^r}} e \xrightarrow{u_2} k' \xrightarrow{v_1} e \underset{s\geqslant 1}{\xrightarrow{p^s}} e \xrightarrow{v_2} k'' \ ,$$

avec u_1, u_2, v_1, v_2 dans $\mathrm{Fl}\, K$, $r, s \geqslant 1$. Or on a

$$p(v_1 u_2) \ = \ p \ , \qquad \text{d'où} \qquad p^s(v_1 u_2) \ = \ p^{s-1}(p v_1 u_2) \ = \ p^{s-1} p \ = \ p^s \ ,$$

donc le composé est

$$k \xrightarrow{u_1} e \xrightarrow{p^{r+s}} e \xrightarrow{v_2} k'' \ ,$$

qui est donc bien dans $\mathrm{Fl}\, K'$.

Ainsi, K' est une sous-catégorie de J, mais est-elle filtrante ? La condition PS 1′ est évidente, puisque e est un plus grand élément de J, reste la condition PS 2 de la double-flèche. On voit aisément que cette condition est satisfaite dans K', si et seulement si elle est satisfaite pour toute double-flèche [page 365]

$$e \overset{p^r}{\underset{p^s}{\rightrightarrows}} e \qquad \text{avec } r, s \geqslant 0 \ .$$

(En effet, on se ramène aussitôt aux double-flèches de but e, et on note que toute flèche de K' de but e s'écrit comme un composé $k \overset{u}{\to} e \overset{p^r}{\longrightarrow} e \overset{v}{\to} e$, avec $r \geqslant 0$ et u, v des flèches dans K'. En composant avec $e \overset{p}{\to} e$, on trouve alors, comme $pv = pu$, $k \overset{u}{\to} e \overset{p^{r+1}}{\longrightarrow} e$, soit $k \overset{f_k}{\longrightarrow} e \overset{p^r}{\longrightarrow} e$.) Mais on voit aussitôt que la possibilité d'égaliser (p^r, p^s) dans K' équivaut à l'existence de $t \geqslant 0$ tel que $p^{r+t} = p^{s+t}$. Donc K' est filtrante si et seulement si $r \longmapsto p^r$ est une suite stationnaire (faire $r = 0$, $s = 1$, d'où $p^r = p^{r+1}$ pour r grand). Ce n'est pas forcément le cas. [page 366]

Posons donc $K = K_0$, $K' = K_1$, et construisons par induction une suite croissante de sous-catégories K_i ($i \in \mathbf{N}$), avec $\mathrm{Ob}\, K_i = \mathrm{Ob}\, J$, et (pour $i \geqslant 1$) K_i étant déduites de K_{i-1} en lui rajoutant une flèche $p_i : e \longrightarrow e$, comme ci-dessus. Ainsi toute double-flèche dans un K_i

161. La construction qui suit est contenue dans l'énoncé plus général et plus précis 4.13.39.

peut s'égaliser dans K_{i+1}. Prenons $K_\infty = \varinjlim K_i$, K_∞ est une sous-catégorie *filtrante* de J, contenant K et de cardinal $\leqslant \pi$, q.e.d.)

On peut préciser cette construction ainsi :

Corollaire 4.13.39. *Sous les conditions de 4.13.38, supposons que K contienne un objet e maximal dans J. Soit Σ l'ensemble des endomorphismes p de e dans J, tels que pour toute double-flèche $k \overset{u}{\underset{v}{\rightrightarrows}} e$ dans K, on ait $pu = pv$. Alors Σ est stable par composition, et il existe une partie non vide Σ_∞ [$= \Sigma_0$] de Σ, telle que pour $p, p' \in \Sigma_0$, existe $q \in \Sigma_0$ tel que $qp = qp'$ (i.e. telle que la sous-catégorie J_0 de J dont l'ensemble des objets est $\{e\}$, et l'ensemble des flèches $\{\mathrm{id}_e\} \cup \Sigma_\infty$, soit filtrante). Si Σ_0 est choisie ainsi, soit $\mathcal{F}$ l'ensemble réunion de* $\mathrm{Fl}\, K$ *et de l'ensemble des flèches composées** p. 139 [page 367]

$$(*) \qquad k \xrightarrow{u} e \xrightarrow{p} e \xrightarrow{v} k' \ ,$$

avec $u, v \in \mathrm{Fl}\, J$, $p \in \Sigma_\infty$. Alors $\mathcal{F}$ est l'ensemble des flèches d'une sous-catégorie K' de J, ayant mêmes objets que J, et admettant e pour plus grand élément. De plus, K' contient K et est filtrante.

Démonstration. Que Σ soit stable par composition est clair. Pour construire Σ_∞ ayant les propriétés énoncées, on note que, J étant grande devant π, Σ est non vide ([162]). On construit par récurrence une suite d'éléments

$$p_0 \ , \ p_1 \ , \ \cdots \ , \ p_n \ , \ \cdots$$

de Σ, en prenant pour p_0 un élément quelconque de Σ, et pour $i \geqslant 1$, prenant $p_i \in \Sigma$ tel que pour toute double-flèche (p, p') du sous-monoïde M_{i-1} de $\mathrm{End}_J(e)$ engendré par $(p_j)_{0 \leqslant j \leqslant i-1}$, on ait $p_i p = p_i p'$. Un tel p_i existe, car prenant l'ensemble des double-flèches de la forme précédente (p, p'), avec $p, p' \in M_{i-1}$, et de celles de la forme $(u, v) : k \rightrightarrows e$, avec $u, v \in \mathrm{Fl}\, J$, l'ensemble de toutes ces double-flèches est de cardinal $\leqslant \pi$ (M_{i-1} étant dénombrable, donc de cardinal $\leqslant \pi$), donc il existe une flèche $e \longrightarrow j$ dans J qui les égalise toutes (J étant grande devant π). Quitte à composer avec $j \longrightarrow e$ (e étant objet maximal de J), on trouve le $p_i : e \longrightarrow e$ cherché. [page 368]

Soit alors Σ_0 le sous-ensemble multiplicatif de Σ engendré par les p_i ($i \in \mathbf{N}$), limite inductive des ensembles similaires Σ_i engendrés par les p_j pour $j \in [0, i]$. Si p, p' sont dans Σ_∞, il sont dans un même Σ_i, donc on aura $p_{i+1} p = p_{i+1} p'$, ce qui prouve que Σ_∞ a la propriété filtrante exigée.

Prouvons maintenant que $\mathcal{F}$ est l'ensemble de flèches d'une sous-catégorie K' de J, ayant $\mathrm{Ob}\, K$ comme ensemble d'objets. Il reste à établir la stabilité par composition vu. Mais si u et v sont dans $\mathrm{Fl}\, K$, ou si l'un des deux est dans $\mathrm{Fl}\, K$, l'autre non (donc de la forme $(*)$), cette stabilité est évidente. Reste à voir le cas où ni [u] ni v sont dans $\mathrm{Fl}\, K$, donc sont tous deux de la forme $(*)$:

$$k \xrightarrow{u} e \xrightarrow{p} e \xrightarrow{v} k' \xrightarrow{u'} e \xrightarrow{p'} e \xrightarrow{v'} k'' \ ,$$

avec $u, v, u', v' \in \mathrm{Fl}\, K$, $p, p' \in \Sigma_\infty$. Comme $(u'v, \mathrm{id}_e)$ est une double-flèche de K de but e, on a p. 140 [page 369]

$$p' u' v \ = \ p' \mathrm{id}_e \ = \ p' \ ,$$

donc le composé en question s'écrit aussi

$$k \xrightarrow{u} e \xrightarrow{p'p} e \xrightarrow{v'} k'' \ ,$$

où $p'p \in \Sigma_\infty$ puisque Σ_∞ est stable par composition, donc il est dans $\mathcal{F}$, q.e.d.

162. cf. argument plus bas pour construction de p_i.

Il est clair que K' contient K, et ayant mêmes objets que K qui admet e comme plus grand élément, que e est un plus grand élément de K'. Reste à prouver que K' est filtrante. Comme K' a un plus grand élément, il reste à voir que toute double-flèche $k \rightrightarrows l$ dans K' peut être égalisée dans K'. Quitte à composer avec $l \longrightarrow e$, on peut supposer $l = e$. Or on voit que toute flèche $k \longrightarrow e$ dans K' est de la forme

$$k \xrightarrow{u} e \xrightarrow{p} e \xrightarrow{v} e \ ,$$

avec $u, v \in \mathrm{Fl}\, K$ et $p \in \Sigma_\infty \cup \{\mathrm{id}_e\}$. Quitte à composer avec $p_0 \in \Sigma_\infty$ fixé, cette flèche devient (puisque $p_0 v = p_0 \mathrm{id}_e = p_0$)

$$k \xrightarrow{u} e \xrightarrow{p_0 p} e \ , \qquad \text{i.e. } p_0 p u \ .$$

Pour une flèche $v'p'u' : k \longrightarrow e$, on trouve $p_0 p' u'$. Soit alors $q \in \Sigma_0$ tel que $q(p_0 p) = q(p_0 p')$, soit r (q existe puisque $p_0 p, p_0 p' \in \Sigma_\infty$). Donc $(q(p_0 p u), q(p_0 p' u')) = (ru, ru')$, et on a $ru = ru'$ puisque $r \in \Sigma_\infty$. Au total, qp_0 égalise les deux flèches vpu et $v'p'u'$, q.e.d.

[page 370]

Commentary · Exposition [ms. 369]

The exceptional vertex and its endomorphisms

Grothendieck pursues the one-vertex construction because it describes how much enlargement an approximation really needs. If a cocone vertex lies outside K, adding it with its compatible incoming arrows produces a terminal object immediately. If every possible vertex is already present, the obstruction lies in endomorphisms, and a weakly terminal object need not become terminal after a single equalization.

The exceptional-case proof needs one preliminary adjustment. A vertex e which is maximal in the reachability preorder of J receives arrows in J from every object of K, but those arrows need not belong to K. Adjoin one such arrow from each object and close under composition. This preserves the object set and the bound π on arrows. After this adjustment e is weakly terminal in K, and the formula $f_k = pu$ is defined for every k. The later normal-form argument then applies exactly as written, with its local J/K slips read correctly.

Here is the point of the powers of p. If p equalizes all K-arrows with target e, then a new arrow can be written $vp^r u$ with u, v in K. Products reduce because $p(v_1 u_2) = p$ for every K-endomorphism $v_1 u_2$ of e. But the powers $1, p, p^2, \ldots$ need not equalize one another. Filteredness of the resulting one-object part forces, by equalizing 1 and p, an equality $p^t = p^{t+1}$ at some stage. Conversely that equality equalizes every pair of powers after further multiplication. Thus the additional iteration is mathematically necessary.

In 4.13.39 the countable sequence p_i arranges this for all words already constructed. At stage $i+1$ one equalizes pairs in the monoid generated by $p_0, \ldots, p_i$, including the identity, together with the original parallel arrows of K ending at e. There are at most π pairs. Their simultaneous equalizer in J can be returned to e by maximality. Any two words occur at a finite stage, so a later p_i equalizes them. The resulting semigroup gives the filtered enlargement claimed. The proof uses arrows of K in its normal forms and retains $\mathrm{Ob}\, K$; the printed substitutions of J at these places are recorded in the correction register.

4.14 Critères d'accessibilité pour grosses catégories

Théorème 4.14.1. *Soient $\mathcal{C}$ une catégorie équivalente à une petite catégorie, Φ, Ψ deux petits ensembles de catégories, équivalentes à des petites catégories. Pour toute catégorie I, on désigne par $I_\bullet$ (resp. ${}_\bullet I$) le cône droit (resp. gauche) sur I, déduit de I en ajoutant un objet final (resp. initial) $e = e_I$. Soit Λ_1 (resp. Λ_2) un ensemble de couples (I, φ) (resp. (I, ψ)), où $I \in \Phi$ (resp. $I \in \Psi$), et où*

(4.14.1.1) $\qquad \varphi \ : \ I \longrightarrow \mathcal{C} \qquad$ *(resp.* $\psi \ : \ I \longrightarrow \mathcal{C}$ *)*

est un foncteur. Pour toute catégorie $\mathcal{N}$ stable ([163]), par $\varinjlim$ de type Φ et par $\varprojlim$ de type Ψ on désigne par

$$\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \subset \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N}) \tag{4.14.1.2}$$

la sous-catégorie strictement pleine de $\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ formée des foncteurs $f : \mathcal{C} \longrightarrow \mathcal{N}$ ayant la propriété suivante :

a) $\forall\ (I,\varphi) \in \Lambda_1$, le foncteur composé

$$I_\bullet \xrightarrow{\varphi} \mathcal{C} \xrightarrow{f} \mathcal{N}$$

est I-exact à gauche, i.e. fait de $(f\varphi)(e_I)$ une $\varinjlim_I$ de $f\varphi|I$.

p. 141
[page 371]

b) Pour tout $(I,\psi) \in \Lambda_2$, le foncteur composé

$$_\bullet I \xrightarrow{\psi} \mathcal{C} \xrightarrow{f} \mathcal{N}$$

est I-exact à droite, i.e. fait de $(f\psi)(e_I)$ une limite projective de $(f\psi)|I$.

Avec ces notations : si $\mathcal{N}$ est une catégorie accessible, il en est de même de $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$.

Commentary · Context [ms. 370]

Marked cones and categories of models

Grothendieck now turns the calculus of accessible parts into a way of constructing categories. A functor $\mathcal{C} \to \mathcal{N}$ carries raw diagram data; marking a cone or a cocone in $\mathcal{C}$ asks that some of those data satisfy a universal property. Section 4.14 treats the resulting categories of models uniformly. This is why the preceding theorem on inverse images matters: the condition that a canonical comparison be an isomorphism is an accessible condition on the functor.

The data in display (4.14.1.1) must include the whole marked cone. For a colimit condition one has $\varphi : I_\bullet \to \mathcal{C}$, not just $\varphi : I \to \mathcal{C}$; for a limit condition one has $\psi : {}_\bullet I \to \mathcal{C}$. The cone vertex is the object to which the universal property is attached. With the first of these data and an arbitrary $F : \mathcal{C} \to \mathcal{N}$, the comparison is

$$\varinjlim_{i \in I} F\varphi(i) \longrightarrow F\varphi(e_I).$$

The full subcategory of functors making this map invertible expresses exactly the desired colimit condition. The dual comparison expresses a limit condition. Taking the small intersection of these conditions gives the theorem, once accessibility of the comparison functors has been established. The hypothesis that $\mathcal{N}$ admits the indicated colimits and limits makes those comparison functors defined on the entire ambient functor category.

This is the language of sketches: a small category or graph together with specified diagrams, cones and cocones, whose models realize the specified universal constructions. The background is the study of categories presented by small data, including Gabriel and Ulmer (1971), and the theory of accessible categories developed by Makkai and Paré (1989). The latter appeared in 1989 and is contemporaneous with this manuscript; it should not be described simply as a later verification of Grothendieck's work. The comparison here concerns the mathematical formulations, without asserting a documented line of influence.

For a concrete example, take the one-sorted finite-product theory of groups. Its objects are the finite powers $1, x, x^2, \ldots$, and its maps are tuples of group terms modulo the group identities. Mark the finite-product cones. A product-preserving functor to sets chooses a set G, sends multiplication, inverse and unit terms to operations on G, and makes the term equalities precisely the group axioms. Conversely those operations determine the functor. A natural transformation is determined by its component at x, and naturality for the operation terms says exactly that this component is a group homomorphism. Thus the model category is the usual category of groups. Finite presentations supply

163. Cette propriété de stabilité n'est pas nécessaire pour la définition de $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$, et il est presque sûr que le théorème est valable sans faire cette hypothèse sur $\mathcal{N}$. Cf. 4.14.16, où on voit que c'est en tous cas inutile pour Ψ.

its compact generators; an arbitrary group is the filtered colimit of the finite generators-and-relations approximations mapping to it. This familiar example exhibits the full programme: the source encodes operations and equations, and the marked cones enforce the intended relation between the different arities.

The same method is useful for large source categories after restriction to compact objects, but then the chosen bound must control both the diagram shapes and the operations being imposed. This is a mathematical requirement on the restriction argument, not merely a matter of replacing a large set of formulas by a smaller notation.

Commentaire. On utilisera ce théorème surtout quand on a $\Lambda_2 = \emptyset$ ou $\Lambda_1 = \emptyset$, auquel cas on note $\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N})$, resp. $\underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N})$, la catégorie $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$. On appliquera le théorème surtout quand $\mathcal{C}$ elle-même est stable par $\varinjlim$ de type Φ et par $\varprojlim$ de type Ψ, et quand Λ_1 (resp. Λ_2) est déjà formé de foncteurs φ (resp. ψ) qui sont I-exacts à gauche (resp. à droite). Donc $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N})$ est alors formée des $f : \mathcal{C} \longrightarrow \mathcal{N}$ qui commutent à certaines $\varinjlim$ de type Φ, et à certaines $\varprojlim$ de type Ψ. En particulier, on trouve :

Corollaire 4.14.2. *Soient $\mathcal{C}$ une catégorie équivalente à une petite catégorie, $\mathcal{N}$ une $\mathfrak{U}$-catégorie, Φ et Ψ deux petits ensembles de catégories I, équivalentes à des petites catégories. On suppose $\mathcal{C}$ et $\mathcal{N}$ stables par $\varinjlim$ de type Φ, par $\varprojlim$ de type Ψ. Soit $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{C},\mathcal{N})$ la sous-catégorie strictement pleine de $\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ formée des $f : \mathcal{C} \longrightarrow \mathcal{N}$ qui commutent aux $\varinjlim$ de type Φ et aux $\varprojlim$ de type Ψ. Alors, si $\mathcal{N}$ est accessible, il en est de même de $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{C},\mathcal{N})$.* [page 372]

Voici des cas particuliers remarquables de ce corollaire :

Corollaire 4.14.3. *Soient $\mathcal{M}$, $\mathcal{N}$ deux catégories accessibles, et π_0 un cardinal infini, tel que $\mathcal{M}$ satisfasse à L_{π_0}. Soit $\underline{\mathrm{Hom}}_{\pi_0}(\mathcal{M},\mathcal{N})$ la sous-catégorie strictement pleine de $\underline{\mathrm{Hom}}(\mathcal{M},\mathcal{N})$ formée des foncteurs π_0-accessibles. Alors $\underline{\mathrm{Hom}}_{\pi_0}(\mathcal{M},\mathcal{N})$ est accessible.*

(Le cas où π_0 est utile pour $\mathcal{M}$ a déjà été traité.) [page 373]

Démonstration. Dans le cas général, soit $\pi > \pi_0$ un cardinal utile pour $\mathcal{M}$, de sorte qu'on a

$$\mathcal{M} \xleftarrow{\sim} \mathrm{Ind}_{\pi}(\mathcal{C}) \qquad \text{avec } \mathcal{C} = \mathcal{M}_{\pi} \ .$$

On peut supposer $\mathcal{M} = \mathrm{Ind}_{\pi}(\mathcal{C})$. Le fait que $\mathcal{M}$ satisfasse à L_{π_0} implique que $\mathcal{C}$ est stable par $\varinjlim$ grandes devant π_0, et de cardinal $\leqslant \pi$. Soit Φ l'ensemble des catégories de cette nature. Φ n'est pas petit, mais il existe $\Phi_0 \subset \Phi$ petit, tel que toute $I \in \Phi$ soit isomorphe à une $I_0 \in \Phi_0$, ce qui suffira pour la validité de 4.14.1 et 4.14.2 (où on fera ici $\Psi = \emptyset$). On peut considérer $\underline{\mathrm{Hom}}_{!\,\pi_0}(\mathcal{M},\mathcal{N})$ comme une sous-catégorie strictement pleine de $\underline{\mathrm{Hom}}_{!\,\pi}(\mathcal{M},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ (cf. 4.12.3, p. 275). Pour un foncteur $f : \mathcal{M} \longrightarrow \mathcal{N}$ qui est π-accessible, dire qu'il est même π_0-accessible revient à dire que sa restriction à $\mathcal{C}$, i.e. son image dans $\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$, commute aux $\varinjlim$ de type Φ. (Cf. 4.13.20, page 341 ([164]), où on prend $\alpha = \alpha_{\pi}$, notant que $J_{\alpha_{\pi}}$ grande devant π_0 puisque $\pi > \pi_0$).

Donc on trouve [page 374]

$$\underline{\mathrm{Hom}}_{!\,\pi_0}(\mathcal{M},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{\Phi}(\mathcal{C},\mathcal{N}) \ ,$$

et le corollaire 4.14.2 donne la conclusion voulue. p. 142

Commentary · Present state [ms. 374]

The target colimits in the functor-category argument

Grothendieck wants fixed-threshold accessible functors to form an accessible category even when their source is large. Restricting to a sufficiently large small subcategory of presentable objects is the right approach. It reduces preservation of all the relevant filtered colimits to preservation of a bounded family, provided the target has the colimits needed in the reduction.

Corollary 4.14.3 is stated with L_{π_0} only on $\mathcal{M}$. Its proof then invokes Corollary 4.14.2 for shapes which are π_0^+-filtered and have size at most π. That corollary assumes these colimits exist in $\mathcal{N}$.

164. La référence n'est pas tout à fait adéquate. Il faudrait réunir dans un paragraphe à part tous les énoncés de ce type, dont certains utilisent le théorème de représentation indicielle.

Accessibility of $\mathcal{N}$ by itself gives colimits above some sufficiently large threshold, which need not be the specified π_0. Choosing π useful for both categories repairs the equivalence with functors on $\mathcal{M}_\pi$, but does not supply the missing smaller-threshold colimits. Thus the printed deduction proves the assertion with the additional hypothesis $\mathcal{N} \models L_{\pi_0}$. The unrestricted case is not established by that deduction; it is part of the issue about colimit conditions without ambient colimit completeness explicitly raised in 4.14.16. We do not infer a counterexample to the unrestricted statement from this proof gap.

There is a separate, elementary correction to the sentence about the ordinal α_π. Its index category is π_0^+-filtered when $\mathrm{cf}(\pi) > \pi_0$, not simply when $\pi > \pi_0$. A power-of-two rank as above ensures the needed cofinality. In Corollary 4.14.4 the target filtered-colimit hypothesis is included, but the restriction to $\mathcal{M}_\pi$ should also be made after simultaneous uniformization of the selected limit and colimit functors. Merely bounding the sizes of their shapes does not guarantee that their values on presentable diagrams remain presentable. These rank adjustments leave the intended method intact and state precisely which version its argument proves.

Corollaire 4.14.4. *Soient $\mathcal{M}$, $\mathcal{N}$ deux catégories accessibles, Φ et Ψ deux ensembles des catégories comme dans 4.14.2, on suppose que $\mathcal{M}$ et $\mathcal{N}$ satisfassent L_{π_0}, et* [soient] *stables par $\varinjlim$ de type Φ et par $\varprojlim$ de type Ψ. Soit $\underline{\mathrm{Hom}}^{\Psi}_{\Phi,\pi_0}(\mathcal{M},\mathcal{N})$ la sous-catégorie strictement pleine de $\underline{\mathrm{Hom}}(\mathcal{M},\mathcal{N})$ formée des foncteurs π_0-accessibles qui commutent aux $\varinjlim$ de type Φ, et aux $\varprojlim$ de type Ψ. Alors $\underline{\mathrm{Hom}}^{\Psi}_{\Phi,\pi_0}(\mathcal{M},\mathcal{N})$ est accessible.*

Démonstration. Cette fois, on choisit $\pi > \pi_0$ tel qu'on ait $\operatorname{card}\text{réd}\, I \leqslant \pi$ pour tout I dans Φ ou dans Ψ. La catégorie envisagée s'interprète alors (posant encore $\mathcal{C} = \mathcal{M}_\pi$) comme [page 375]

$$\underline{\mathrm{Hom}}^{\Psi}_{\Phi,\pi_0}(\mathcal{M},\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}^{\Psi}_{\Phi'}(\mathcal{C},\mathcal{N}) \ ,$$

où Φ' est formée des catégories qui sont soit dans Φ, ou qui sont grandes devant π_0 et de cardinal $\leqslant \pi$. Il suffit donc d'appliquer encore 4.14.2.

Corollaire 4.14.5. *Considérons les trois ensembles suivants :*

a) Φ_1 est formée de toutes les catégories équivalentes à des petites catégories.

b) Φ_2 comme a), mais en restreignant aux catégories filtrantes.

c) Φ_3 formé des catégories discrètes, et de celles qui sont grandes devant π_0 (π_0 un cardinal infini donné).

Soit Φ l'un des ensembles de catégories Φ_1, Φ_2, Φ_3, $\Phi_2 \cup \Phi_3$ ou la réunion d'un de ceux-ci et d'un petit ensemble Φ_0 de catégories essentiellement petites ([165]). Soit Ψ un petit ensemble de catégories. Soient $\mathcal{M}$, $\mathcal{N}$ des catégories accessibles, stables par $\varinjlim$ de type Φ, et par $\varprojlim$ de type Ψ. Alors $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N})$ est accessible.

Démonstration. Soit $\pi_0' \geqslant \pi_0$ un cardinal utile pour $\mathcal{M}$, π un cardinal ayant les propriétés [page 376] suivantes.

a) $\pi \geqslant \sup(\pi_0', \operatorname{card}\text{réd}\, \mathcal{M}_{\pi_0'})$, $\pi > \pi_0$.

b) $\pi^{\pi_0'} = \pi$.

c) $I \in \Phi_0 \cup \Psi \Longrightarrow \operatorname{card}\text{réd}\, I \leqslant \pi$.

Soit $\mathcal{C} = \mathcal{M}_\pi$, d'où $\mathcal{M} \simeq \mathrm{Ind}_\pi(\mathcal{C})$. Dans tous les cas envisagés, Φ contient l'ensemble des catégories essentiellement petites qui sont grandes devant π_0, a fortiori celles qui sont grandes devant π, donc la catégorie $\underline{\mathrm{Hom}}^{\Psi}_{\Phi}$ s'interprète comme une sous-catégorie strictement pleine de

$$\underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N}) \ .$$

Si d'autre part Φ_π désigne l'ensemble des $I \in \Phi$ telles que $\operatorname{card} I \leqslant \pi$, on trouve

$$\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N}) \ \simeq \ \underline{\mathrm{Hom}}^{\Psi}_{\Phi_\pi}(\mathcal{C},\mathcal{N}) \ ,$$

p. 143

et on applique 4.14.2. [page 377]

Commentary · Exposition [ms. 376]

A bounded sketch for a large functor category

Grothendieck is trying to make preservation of universal constructions into an accessible condition. The domain may have a proper class of objects and the demand may concern all small colimits. The preceding work with cardinals serves to replace that demand by a set of tests on one essentially small category. This is why these corollaries belong to the same argument as the long cofinality constructions: a good presentation rank makes the large category recoverable from its bounded part.

Choose π useful for $\mathcal{M}$ and write $\lambda = \pi^+$, so that the objects of $\mathcal{M}_\pi$ are λ-presentable in modern terminology. Choose a small skeleton $\mathcal{A}$ of that subcategory. A functor preserving λ-filtered colimits is determined, including its natural transformations, by its restriction to $\mathcal{A}$. Its extension is expressed by

$$F(X) \simeq \varinjlim_{(A \to X) \in \mathcal{A}/X} F(A).$$

The choice of a common rank must ensure that the target has these filtered colimits. Preservation of additional shapes is then tested by comparison arrows associated to a set of bounded cones. One asks that these arrows become invertible. The category of candidates is therefore obtained from an accessible diagram category by inverse images of an accessible isomorphism subcategory and a small intersection. These are precisely the operations established in the preceding section.

The colimit assertion in 4.14.6 illustrates both the strength and the exact range of this method. If $\mathcal{M}$ and $\mathcal{N}$ are locally presentable, the category of cocontinuous functors between them is accessible by the bounded-sketch argument. It is also cocomplete. Given a small diagram (F_j) of cocontinuous functors, define $F(X) = \varinjlim_j F_j(X)$. For a small diagram (X_i) one has

$$\begin{aligned} F(\varinjlim_i X_i) &\simeq \varinjlim_j \varinjlim_i F_j(X_i) \\ &\simeq \varinjlim_i \varinjlim_j F_j(X_i) \simeq \varinjlim_i F(X_i). \end{aligned}$$

The interchange is just the universal property of a colimit over the product indexing category; no exactness condition on $\mathcal{N}$ enters. Thus this functor category is locally presentable. The first assertion of 4.14.6, with only accessibility of $\mathcal{N}$, is not justified by the cited corollary: the latter assumes the target colimits that the former omits. We retain that distinction in the audit instead of silently adding a hypothesis.

The tensor product in 4.14.8 uses the other side of the construction. Its objects are functors $\mathcal{M}^{op} \to \mathcal{N}$ carrying colimits in $\mathcal{M}$ to limits in $\mathcal{N}$. The relevant constraints are limit-cone constraints, and the argument on manuscript page 389 shows that these can be recognized even when other limits in the target do not exist. Here $\mathcal{N}$ is complete, so limits of such functors are computed pointwise: two successive limits commute. Consequently the tensor category is accessible and complete, hence locally presentable. This argument does not require the erroneous covariant representability assertion that follows.

A small calculation makes the construction concrete. Every set X is the coproduct of X copies of the singleton. Therefore a functor $F : \mathbf{Set}^{op} \to \mathcal{N}$ taking colimits to limits must satisfy $F(X) \simeq F(1)^X$. For a map $a : X \to Y$, its image sends a family indexed by Y to the family indexed by X obtained by composition with a. Conversely, for any $N \in \mathcal{N}$, the functor $X \mapsto N^X$ takes colimits of sets to limits: applying $\mathrm{Hom}(T, -)$ identifies the claim with the familiar contravariant representable functor $\mathrm{Hom}_{\mathbf{Set}}(-, \mathrm{Hom}(T, N))$. A natural transformation between these functors is determined by its component at 1, since projections to the singleton coordinates detect it. Hence $\mathbf{Set} \boxtimes \mathcal{N} \simeq \mathcal{N}$. The unit of the tensor construction is already visible in this elementary case.

En particulier, on trouve :

Corollaire 4.14.6. *Soient $\mathcal{M}$, $\mathcal{N}$ deux catégories accessibles, supposons $\mathcal{M}$ stable par petites $\varinjlim$ (i.e. $\mathcal{M}$ est une catégorie paratopos). Alors la catégorie $\underline{\mathrm{Hom}}_!(\mathcal{M}, \mathcal{N})$ des foncteurs $f : \mathcal{M} \longrightarrow \mathcal{N}$ qui commutent aux petites $\varinjlim$ est accessible. Si $\mathcal{N}$ est elle aussi un paratopos,*

165. On pourrait faire ici un énoncé qui englobe 4.14.4 en rajoutant un

d) Φ_4 formé des catégories grandes devant π_0,

et prendre Φ de la forme de Φ_1, Φ_2, Φ_4, Φ_3, $\Phi_3 \cup \Phi_2$, $\Phi_3 \cup \Phi_4$, ou la réunion de ceux-ci et d'un Φ_0.

$\underline{\mathrm{Hom}}_!(\mathcal{M},\mathcal{N})$ *est un paratopos.*

Démonstration. La première assertion est le cas particulier de 4.14.5 relatif au cas où $\Phi = \Phi_0$, $\Psi = \emptyset$. La deuxième assertion en résulte, puisque $\underline{\mathrm{Hom}}_!(\mathcal{M},\mathcal{N})$ est visiblement stable par petites $\varinjlim$, donc c'est un paratopos étant accessible (4.8.9, page 207).

Corollaire 4.14.7. *Soient Φ, Ψ, $\mathcal{M}$ comme dans 4.14.5. Alors pour toute catégorie accessible $\mathcal{N}$, la catégorie* [page 378]

$$\underline{\mathrm{Hom}}^{\Phi}_{\Psi}(\mathcal{M}^\circ,\mathcal{N}) \;\simeq\; (\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N}^\circ))^\circ$$

est accessible.

Démonstration. On procède comme pour 4.14.5, pour expliciter

$$\underline{\mathrm{Hom}}^{\Psi}_{\Phi}(\mathcal{M},\mathcal{N}^\circ) \xrightarrow{\sim} \underline{\mathrm{Hom}}^{\Psi}_{\Phi_\pi}(\mathcal{C},\mathcal{N}^\circ)\ ,$$

d'où

$$\underline{\mathrm{Hom}}^{\Phi}_{\Psi}(\mathcal{M}^\circ,\mathcal{N}) \xrightarrow{\sim} \underline{\mathrm{Hom}}^{\Phi_\pi}_{\Psi}(\mathcal{C}^\circ,\mathcal{N})\ ,$$

et on applique 4.14.2 à $\mathcal{C}^\circ$, Ψ, Φ_π.

Corollaire 4.14.8. *Soient $\mathcal{M}$ et $\mathcal{N}$ deux catégories paratopos. Alors*

$$\mathcal{M}\boxtimes\mathcal{N} \;\simeq\; \underline{\mathrm{Hom}}^!(\mathcal{M}^\circ,\mathcal{N}) \qquad \textit{(cf. 4.2 et 4.3)}$$

est un paratopos.

En effet, en vertu de 4.14.7 (avec $\Phi = \Phi_\pi$ [?], $\Psi = \emptyset$), c'est une catégorie accessible.

D'autre part, il est immédiat qu'elle est stable par petites $\varprojlim$, $\mathcal{N}$ l'étant. Donc on conclut [page 379] encore par 4.8.9 (page 207) que c'est une catégorie paratopos.

Corollaire 4.14.9. *Soient $\mathcal{M}$, $\mathcal{N}$ deux catégories paratopos, et $\underline{\mathrm{Hom}}^!(\mathcal{M},\mathcal{N})$ la catégorie des foncteurs $\mathcal{M}\longrightarrow\mathcal{N}$ qui commutent aux petites $\varprojlim$. On a alors une équivalence de catégories*

$$\begin{aligned}\underline{\mathrm{Hom}}^!(\mathcal{M},\mathcal{N}) &\xrightarrow{\sim} \underline{\mathrm{Hom}}_!(\mathcal{N},\mathcal{M})^\circ\\ f &\longmapsto \textit{adjoint à gauche de } f\ ,\end{aligned}$$

!! *et par suite la catégorie* **opposée** *à $\underline{\mathrm{Hom}}^!(\mathcal{M},\mathcal{N})$ est accessible.*

Ceci résulte du

Lemme 4.14.10 (pour mémoire). *Soient $\mathcal{M}$* `[une]` *catégorie pseudo-topos, $f:\mathcal{M}\longrightarrow\mathcal{N}$ un foncteur. Pour que f admette un adjoint à gauche (resp. à droite), il faut et il suffit que f commute aux petites $\varprojlim$ (resp. aux petites $\varinjlim$).*

Ceci équivaut au cas particulier : p. 144

Corollaire 4.14.11. *Soit $\mathcal{M}$ un paratopos, et f un foncteur covariant (resp. contravariant)* [page 380] *de $\mathcal{M}$ dans* Ens. *Pour que f soit représentable, il faut et il suffit que f commute aux $\varprojlim$.*

Ça a été vu – le cas des contrafoncteurs étant le plus évident (cf. 4.1 – il suffit que $\mathcal{M}$ soit un paratopos, et même que $\mathcal{M}$ admette une petite sous-catégorie pleine génératrice, cf. SGA 4 I 8.12.7 (pour l'énoncé dual). Dans les cas d'un foncteur covariant, la difficulté provient du fait que $\mathcal{M}$ n'admet pas forcément de petite sous-catégorie cogénératrice. Il faut utiliser l'existence d'une filtration cardinale (généralisée), cf. loc. cit. 8.3.12.)

Après tous ces beaux corollaires, on est mûr pour s'attaquer à la démonstration de 4.14.1 !

Commentary · Examples [ms. 380]

Continuity does not supply a solution set

Grothendieck wants the familiar passage between left and right adjoints to turn the preceding accessibility theorem into its dual. The formal passage is sound, but its input in 4.14.9–4.14.11 is too broad. A functor preserving all small limits need not have a left adjoint, even when its source and target are locally presentable. This is a genuine mathematical error in the printed statement, not a defect introduced by the transcription. The source's marginal mark beside 4.14.9 is retained without assigning it a meaning beyond what can be read on the page.

The obstruction was already available in the literature being cited: Artin et al. (1972, Exposé I, Example 8.12.9) attributes a continuous nonrepresentable functor on groups to H. Bass. Grothendieck himself had recalled this example and the need for an extra cogeneration hypothesis at the beginning of the chapter (Grothendieck, 1990–1991n, ms. 50–52). We give an explicit version so that the missing hypothesis has a visible effect. For every infinite cardinal κ choose a simple group S_κ of cardinality κ; one may take the group of even permutations of a set of that size having finite support. Simplicity follows from the finite alternating groups: a nonidentity element of a normal subgroup lies in a finite alternating subgroup on at least five letters, and its normal closure there contains a 3-cycle; conjugation gives all 3-cycles, which generate the finitary alternating group.

Define a covariant functor on groups by

$$B(G) = \prod_{\kappa \text{ infinite}} \operatorname{Hom}(S_\kappa, G).$$

This notation denotes a set, despite its apparent indexing by all infinite cardinals. A nontrivial homomorphism from a simple group is injective, so the factor indexed by $\kappa > |G|$ is a singleton. One can therefore define $B(G)$ by the product over $\kappa \le |G|$, with trivial coordinates understood elsewhere. Postcomposition defines B on homomorphisms and respects these omitted coordinates.

To verify continuity, take any small diagram (G_i) with limit G. There is a cardinal bounding the sizes of G and of every G_i. Above that cardinal all coordinates are trivial. Below it, each functor $\operatorname{Hom}(S_\kappa, -)$ preserves limits, and a set-indexed product of such functors does so as well. Thus the canonical map

$$B(\varprojlim_i G_i) \longrightarrow \varprojlim_i B(G_i)$$

is bijective. This checks all small shapes, including products with arbitrarily large small indexing sets.

To see why representability fails, suppose that an isomorphism $\operatorname{Hom}(H, -) \simeq B$ existed. By Yoneda it would be determined by an element $b \in B(H)$. Choose $\kappa > |H|$. The κ-coordinate of b is trivial, and therefore every element obtained from b by postcomposition has trivial κ-coordinate. But $B(S_\kappa)$ contains the family whose κ-coordinate is id_{S_κ} and whose other coordinates are trivial. That family cannot lie in the image of $\operatorname{Hom}(H, S_\kappa)$, a contradiction. Nor can B have a left adjoint, since an adjunction would give $B(G) \simeq \operatorname{Hom}(L(1), G)$.

The correct version of 4.14.9 uses accessible continuous functors. Between locally presentable categories these are exactly the right adjoints, and taking left adjoints gives

$$\operatorname{Fun}^R(\mathcal{M}, \mathcal{N}) \simeq \operatorname{Fun}^L(\mathcal{N}, \mathcal{M})^{op}.$$

The reversal on natural transformations is the mate correspondence. Accessibility supplies the solution-set requirement in the adjoint functor theorem. For a precise modern statement, including a bound on the sizes of limits that need to be tested once accessibility has been fixed, see Tendas (2022, Theorems 2.9 and 2.11).

The other half of 4.14.10 remains valid, even with its stated pseudo-topos hypothesis: a cocontinuous functor from a cocomplete locally small category with a small family generating by strict epimorphisms to a locally small category has a right adjoint. This is the dual of SGA 4 I 8.12.7–8.12.8. Locally presentable categories are a special case. Correspondingly, a contravariant set-valued functor taking colimits to limits is representable. There is no license to reverse this theorem without also reversing its size hypotheses. The reference printed as SGA 4 I 8.3.12 points toward 8.13.2, whose prorepresentability criterion explicitly assumes accessibility of the functor.

Commentary · Exposition [ms. 380]

Testing universal arrows on small generators

The proof that now begins returns to the part of the argument that works without the representability detour. Grothendieck encodes a universal property by a comparison arrow and then tests whether that arrow is invertible. This makes the preservation condition a property of an object in an accessible arrow category, where the inverse-image theorem can be used.

For a prescribed cocone, its comparison arrow is

$$\varinjlim_{i \in I} F\varphi(i) \longrightarrow F\varphi(e_I).$$

Its accessibility follows directly from pointwise colimits and interchange of colimits. For a prescribed cone, the comparison has the opposite orientation,

$$F\varphi(e'_I) \longrightarrow \varprojlim_{i\in I} F\varphi(i),$$

and a cardinal bound enters. Fix a regular λ larger than the number of arrows of I, and a small strong generating family $\mathcal{A}$ of λ-presentable objects in $\mathcal{N}$. If (D_j) is a λ-filtered diagram of I-diagrams, then for $A \in \mathcal{A}$ the comparison between $\operatorname{colim}_j \lim_i D_j(i)$ and $\lim_i \operatorname{colim}_j D_j(i)$ becomes

$$\varinjlim_j \varprojlim_i \operatorname{Hom}(A, D_j(i)) \longrightarrow \varprojlim_i \varinjlim_j \operatorname{Hom}(A, D_j(i)).$$

It is a bijection because λ-filtered colimits of sets commute with limits having fewer than λ arrows. The strong generators detect the original isomorphism. This explains both the size hypothesis and the correction of the printed $\varinjlim_I$ to $\varprojlim_I$ on manuscript page 383.

The isomorphism subcategory is especially transparent. The functor $X \mapsto \mathrm{id}_X$ is fully faithful from $\mathcal{N}$ to $\mathcal{N}^{[1]}$, preserves the available filtered colimits, and has as its replete essential image the invertible arrows. Thus this subcategory is accessible. For monomorphisms one uses the same small family differently: u is monic precisely when every $\operatorname{Hom}(A, u)$ is injective. A generating family suffices here, whereas detection of isomorphisms requires strong generation.

The finite-set calculation left to the reader in 4.14.13 can be made without any representation theorem. Replace an injection by an actual inclusion $A \subset B$. Consider the poset of pairs of finite subsets $A_0 \subset A$ and $B_0 \subset B$ with $A_0 \subset B_0$. It is directed under union, and the colimit of its arrows $A_0 \hookrightarrow B_0$ is $A \hookrightarrow B$. Each finite arrow is finitely presentable in the arrow category: its morphisms to a filtered colimit factor at one stage, and the one commutative-square equation holds at a later stage. Filtered colimits of injections in sets remain injective. Hence these finite injections form a small family of finitely presentable generators of the injection category. For a bijection, take a finite subset of its source and its image in the target; the same argument gives the isomorphism assertion.

Finally, full faithfulness matters in 4.14.14. The formula

$$\operatorname{Hom}_{\mathrm{Ind}(\mathcal{C})}(\varinjlim_i C_i, \varinjlim_j D_j) \simeq \varprojlim_i \varinjlim_j \operatorname{Hom}_{\mathcal{C}}(C_i, D_j)$$

shows that a full inclusion $\mathcal{D} \subset \mathcal{C}$ induces a fully faithful functor on Ind-completions. Together with preservation of filtered colimits, this identifies its essential image with an accessible embedded full subcategory. Mere accessibility of an arbitrary functor would not imply the corresponding assertion about its full image.

[page 381]

Démonstration de 4.14.1. On sait que $\mathcal{M} = \underline{\mathrm{Hom}}(\mathcal{C}, \mathcal{N})$ est accessible (4.9.5, p. 220), et $\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C}, \mathcal{N})$ est une sous-catégorie strictement pleine de cette catégorie. Il suffit de prouver que c'est une sous-catégorie accessible dans $\mathcal{M}$ (4.13.1, p. 281). Pour tout $\xi = (I, \varphi) \in \Lambda_1$, on a un foncteur

$$L_\xi \ : \ \mathcal{M} \ = \ \underline{\mathrm{Hom}}(\mathcal{C}, \mathcal{N}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{N}) \ ,$$

donné par

$$L_\xi(f) \ = \ \text{flèche canonique } \varinjlim_{i\in I} f\varphi(i) \longrightarrow f\varphi(e_I) \ .$$

Je dis que le foncteur L_ξ est *accessible.* En effet, soit π_0 un cardinal tel que $\mathcal{N}$ satisfasse à L_{π_0}, il en est donc de même de $\mathcal{M} = \underline{\mathrm{Hom}}(\mathcal{C}, \mathcal{N})$ et de $\underline{\mathrm{Fl}}(\mathcal{N}) \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(\mathbf{\Delta}_1, \mathcal{N})$, les limites en question se calculent argument par argument. Je dis que L_ξ est π_0-accessible, i.e. commute aux $\varinjlim$ grandes devant π_0. Il suffit de voir qu'il en est ainsi des foncteurs composés avec les foncteurs

$$\underline{\mathrm{Fl}}(\mathcal{N}) \ \underset{\rho}{\overset{\sigma}{\rightrightarrows}} \ \mathcal{N}$$

"source" et "but", i.e. des foncteurs

$$f \longmapsto \varinjlim_{i\in I} f\varphi(i) \qquad \text{et} \qquad f \longmapsto f\varphi(e_I) \ .$$

[page 382]

Or le premier foncteur, $f \longmapsto \varinjlim_{i\in I} f\varphi(i)$, est le composé de

$$f \longmapsto f\circ\varphi_I \ : \ \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}(I,\mathcal{N})$$

(lequel commute aux $\varinjlim$ qui sont représentables dans $\mathcal{N}$, et en particulier aux $\varinjlim$ grandes devant π_0) et du foncteur

$$\varinjlim_{I} \ : \ \underline{\mathrm{Hom}}(I,N) \longrightarrow \mathcal{N} \ ,$$

qui lui aussi commute aux limites inductives représentables dans $\mathcal{N}$, et en particulier à celles grandes devant π_0. Quant au foncteur $f \longmapsto (f\circ\varphi)(e_I) = f(\varphi(e_I))$, il est clair qu'il commute également aux $\varinjlim$ représentables dans $\mathcal{N}$, donc il est π_0-accessible.

Ceci posé, notons que p. 145

$$\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \ = \ \bigcap_{\xi\in\Lambda_1} \underline{\mathrm{Hom}}_{\{\xi\}}(\mathcal{C},\mathcal{N}) \ ,$$

et que $\underline{\mathrm{Hom}}_{\xi}(\mathcal{C},\mathcal{N})$ est l'image inverse, par le foncteur accessible L_ξ précédent, de la sous-catégorie strictement pleine $\underline{\mathrm{Is}}(\mathcal{N})$ de $\underline{\mathrm{Fl}}(\mathcal{N})$ formée des *isomorphismes* dans $\mathcal{N}$. Si on prouve que cette dernière est accessible dans $\underline{\mathrm{Fl}}(\mathcal{N})$, il s'ensuit que $\underline{\mathrm{Hom}}_{\xi}(\mathcal{C},\mathcal{N})$ est accessible dans $\mathcal{M}$ (4.3.31, page 350). Donc, Φ étant petite, il s'ensuit que $\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N})$ est accessible dans $\mathcal{M}$ (4.13.7, page 291). [page 383]

On prouve de la même façon que $\underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N})$ est accessible dans $\mathcal{M} = \mathrm{Hom}(\mathcal{C},\mathcal{N})$. Ici, on doit considérer les foncteurs

$$L'_\xi \ : \ \mathcal{M} = \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{N})$$

définis par

$$L'_\xi \ = \ \text{flèche canonique } f\varphi(e'_I) \longrightarrow \varprojlim_{i\in I} f\varphi(i) \ .$$

Pour prouver que ce foncteur est accessible, on procède comme tantôt, en prenant cependant π_0 assez grand pour qu'on ait

$$\begin{cases} I\in\Psi \implies \mathrm{card}\,\mathrm{réd}\, I \leqslant \pi_0. \\ \mathcal{N}_{\pi_0} \text{ est génératrice dans } \mathcal{N}. \end{cases}$$

Je dis qu'alors L'_ξ est π_0-accessible : en effet, en procédant comme tantôt pour L_ξ, on est ramené à voir que

$$\varinjlim_{I} \underline{\mathrm{Hom}}(I,\mathcal{N}) \longrightarrow \mathcal{N}$$

est π-accessible. Or cela est contenu dans SGA 4 I 9.8.

Ainsi, on a prouvé que les deux sous-catégories strictement pleines [page 384]

$$\underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \ , \qquad \underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N})$$

de $\mathcal{M} \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(\mathcal{C},\mathcal{N})$ sont accessibles dans $\mathcal{M}$. Il en est donc de même de leur intersection (4.13.7)

$$\underline{\mathrm{Hom}}^{\Lambda_2}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \ = \ \underline{\mathrm{Hom}}_{\Lambda_1}(\mathcal{C},\mathcal{N}) \cap \underline{\mathrm{Hom}}^{\Lambda_2}(\mathcal{C},\mathcal{N}) \ ,$$

q.e.d.

Il reste à prouver le

Lemme 4.14.12. *Soit $\mathcal{N}$ une catégorie accessible. Alors la catégorie $\underline{\mathrm{Fl}}(\mathcal{N}) \overset{déf}{=} \underline{\mathrm{Hom}}(\mathbf{\Delta}_1,\mathcal{N})$ est accessible (4.9.5), et la sous-catégorie strictement pleine $\underline{\mathrm{Is}}(\mathcal{N})$ formée des flèches inversibles dans $\mathcal{N}$ est accessible dans $\underline{\mathrm{Fl}}(\mathcal{N})$. (Il en est de même de la sous-catégorie strictement pleine $\underline{\mathrm{Mon}}(\mathcal{N})$ formée des monomorphismes dans $\mathcal{N}$.)*

Démonstration. Soit $\mathcal{C}$ une petite sous-catégorie strictement génératrice de $\mathcal{N}$, et considérons la petite famille des foncteurs

$$h_x \;=\; (y \longmapsto \mathrm{Hom}(x,y)) \;:\; \mathcal{N} \longrightarrow \mathrm{Ens}\ ,$$

d'où des foncteurs

$$\underline{\mathrm{Fl}}(h_\xi) \;:\; \underline{\mathrm{Fl}}(\mathcal{N}) \longrightarrow \underline{\mathrm{Fl}}(\mathrm{Ens})\ .$$

p. 146
[page 385]

Pour qu'une flèche $u : y \longrightarrow z$ dans $\mathcal{N}$ soit isomorphe (resp. monomorphe), il faut et il suffit que pour tout $x \in \mathrm{Ob}\,\mathcal{C}$, $h_x(u)$ le soit. Ainsi, $\underline{\mathrm{Is}}(\mathcal{N})$ (resp. $\underline{\mathrm{Mon}}(\mathcal{N})$) apparaît comme l'intersection des images inverses, par les foncteurs $\underline{\mathrm{Fl}}(h_\xi)$ ci-dessus, de la sous-catégorie $\underline{\mathrm{Is}}(\mathrm{Ens})$ (resp. $\underline{\mathrm{Mon}}(\mathrm{Ens})$) de $\underline{\mathrm{Fl}}(\mathrm{Ens})$. Cela nous ramène au cas où $\mathcal{N} = (\mathrm{Ens})$. Mais si $\mathcal{C}$ désigne la sous-catégorie strictement pleine de Ens formée des ensembles *finis*, ceux-ci forment une famille strictement génératrice formée d'objets de *présentation finie*, et comme Ens est stable par petites $\varinjlim$, donc par $\varinjlim$ filtrantes, il s'ensuit qu'on a

$$(*) \qquad \mathrm{Ind}(\mathcal{C}) \longrightarrow (\mathrm{Ens})\ ,$$

d'où aussitôt (SGA 4 I 8.8.5, p. 104)

[page 386]

$$(*) \qquad \mathrm{Ind}(\underline{\mathrm{Fl}}(\mathcal{C})) \xrightarrow{\sim} \underline{\mathrm{Fl}}(\mathrm{Ens})\ .$$

Donc le résultat d'accessibilité voulu résulte des deux lemmes suivants :

Lemme 4.14.13. *L'équivalence* $(*)$ *précédente induit des équivalences*

$$\begin{array}{rcl} \mathrm{Ind}(\underline{\mathrm{Is}}(\mathcal{C})) & \xrightarrow{\sim} & \underline{\mathrm{Is}}(\mathrm{Ens}) \\ \mathrm{Ind}(\underline{\mathrm{Mon}}(\mathcal{C})) & \xrightarrow{\sim} & \underline{\mathrm{Mon}}(\mathrm{Ens}) \end{array}$$

(où $\mathcal{C} = (\mathrm{Ensf})$ *est la sous-catégorie pleine de* (Ens) *formée des ensembles* finis*.)*

Lemme 4.14.14 ([166]). *Soit* $\mathcal{C}$ *une catégorie équivalente à une petite catégorie. Alors :*

a) $\mathcal{M} \overset{déf}{=} \mathrm{Ind}(\mathcal{C})$ *est une catégorie accessible.*

b) Pour toute sous-catégorie pleine $\mathcal{D}$ *dans* $\mathcal{C}$*, l'image essentielle de* $\mathrm{Ind}\,\mathcal{D}$ *dans* $\mathcal{M} = \mathrm{Ind}(\mathcal{C})$ *est une sous-catégorie L-accessible dans* $\mathcal{M}$*.*

Démonstration des deux lemmes immédiate et "laissée au lecteur". Le b) dans 4.14.14 provient du fait que le foncteur $\mathrm{Ind}(\mathcal{D}) \longrightarrow \mathrm{Ind}(\mathcal{C})$ provenant d'un foncteur $\mathcal{D} \longrightarrow \mathcal{C}$ (ici un foncteur d'inclusion) est un L-foncteur (i.e. commute aux petites $\varinjlim$ filtrantes), et a fortiori est accessible (et même $\aleph_0$-accessible). [page 387]

4.14.15. On ne peut s'empêcher de se demander si pour $\mathcal{N}$ accessible, la sous-catégorie strictement pleine $\underline{\mathrm{Ép}}(\mathcal{N})$ dans $\underline{\mathrm{Fl}}(\mathcal{N})$, formée des épimorphismes dans $\mathcal{N}$, est accessible dans $\mathcal{N}$. Il est immédiat que si $\mathcal{N}$ satisfait à L_{π_0}, alors $\underline{\mathrm{Ép}}(\mathcal{N})$ est stable dans $\underline{\mathrm{Fl}}(\mathcal{N})$ par les $\varinjlim$ de type L_{π_0}, a fortiori par facteurs directs. Elle aurait donc dû mal à ne pas être accessible dans $\underline{\mathrm{Fl}}(\mathcal{N})$. Notons quand même :

Corollaire 4.14.15.1 (de 4.14.13). *Soit* $\mathcal{N}$ *une catégorie accessible stable par sommes amalgamées* $y \amalg_x z$ ([167])*. Alors* $\underline{\mathrm{Ép}}(\mathcal{N})$ *est une sous-catégorie accessible dans* $\underline{\mathrm{Fl}}(\mathcal{N})$*.*

Considérons en effet le foncteur

p. 147
[page 388]

$$\begin{array}{rcl} \gamma \;:\; \underline{\mathrm{Fl}}(\mathcal{N}) & \longrightarrow & \underline{\mathrm{Fl}}(\mathcal{N}) \\ (x \longrightarrow y) & \longmapsto & (y \amalg_x y \xrightarrow{\text{codiag}} y)\ . \end{array}$$

166. **N.B.** À équivalence près, on trouve les catégories accessibles $\mathcal{M}$ stables par petites $\varinjlim$ filtrantes, et telles que $\mathcal{M}_{\mathrm{pf}}$ (sous-catégorie des objets de présentation *finie*) soit strictement génératrice.

167. Je suis presque sûr que cette hypothèse est superflue. Comparer page 390.

Ce foncteur commute aux types de $\varinjlim$ qui sont représentables dans $\mathcal{N}$, en particulier, il est accessible. D'autre part, $\underline{\text{Ép}}(\mathcal{N})$ est l'image inverse par γ de la sous-catégorie strictement pleine $\underline{\text{Is}}(\mathcal{N})$ de $\underline{\text{Fl}}(\mathcal{N})$. Donc le corollaire résulte du fait que par 4.14.13 cette dernière est accessible dans $\underline{\text{Fl}}(\mathcal{N})$.

Commentary · Examples [ms. 388]

Epimorphisms as degenerate pushout squares

Grothendieck turns to epimorphisms because they expose the asymmetry between testing arrows into an object and testing arrows out of it. A small generating family detects monomorphisms. There need not be a corresponding small cogenerating family to detect epimorphisms. The self-pushout replaces all those outgoing tests by one universal comparison.

Let $u : X \to Y$, and suppose its self-pushout $P = Y \amalg_X Y$ exists, with maps $i_1, i_2 : Y \to P$ and codiagonal $d : P \to Y$. If u is epic, the equality $i_1 u = i_2 u$ implies $i_1 = i_2$. Since $di_1 = di_2 = \mathrm{id}_Y$, one has $i_1 d = \mathrm{id}_P$ by the pushout universal property; hence d is invertible. Conversely, if d is invertible then $i_1 = i_2 = d^{-1}$. Given $a, b : Y \to Z$ with $au = bu$, the induced map $P \to Z$ shows that $a = b$. Thus u is epic exactly when d is invertible. This proves the inverse-image description in 4.14.15.1, and functoriality of pushouts makes its comparison functor accessible.

For sets, the self-pushout of u consists of two copies of Y, with corresponding points identified exactly where they belong to the image of u. The codiagonal is bijective precisely when no point remains doubled, that is, when u is surjective. A surjection $A \twoheadrightarrow B$ is a filtered colimit of finite surjections: index by finite subsets $A_0 \subset A$ and take $A_0 \twoheadrightarrow u(A_0)$. The finite subsets form a directed poset, their source union is A, and their image union is B. A square from a finite arrow factors through one stage, and its finitely many equations become valid at a later stage. Conversely, a finitely presentable surjection is a retract of one stage of this presentation, and therefore has finite source and target. Consequently the category of surjections of sets is finitely accessible with exactly the finite surjections as its finitely presentable objects.

The categorical formulation remains useful when epimorphism does not mean surjection of underlying sets. In commutative unital rings, $\mathbf{Z} \to \mathbf{Q}$ is epic: two ring maps out of $\mathbf{Q}$ agreeing on $\mathbf{Z}$ agree on every fraction, since the images of denominators have forced inverses. The corresponding codiagonal is the multiplication map

$$\mathbf{Q} \otimes_{\mathbf{Z}} \mathbf{Q} \longrightarrow \mathbf{Q}, \qquad r \otimes s \longmapsto rs.$$

Its inverse sends q to $q \otimes 1$. In the tensor product, $1 \otimes n = n \otimes 1$ for every nonzero integer n; inverting this equality gives $1 \otimes n^{-1} = n^{-1} \otimes 1$, and therefore $r \otimes s = rs \otimes 1$. This proves the claimed inverse directly, although $\mathbf{Z} \to \mathbf{Q}$ is far from surjective as a map of sets.

There are useful sharper results in algebraic settings. For a locally λ-presentable abelian category, Positselski (2024, Lemma 10.7) proves that its epimorphism category is locally λ-presentable, with precisely the epimorphisms between λ-presentable objects as its λ-presentables. This fixed-rank statement uses the abelian hypotheses and does not settle the unrestricted question on the preceding page.

4.14.16. Je présume que 4.14.1 reste valable sans supposer $\mathcal{N}$ stable par les $\varinjlim$ de type Φ et par les $\varprojlim$ de type Ψ. Si on a une essentiellement petite catégorie I, la question est de savoir si la sous-catégorie strictement pleine $\underline{\mathrm{Hom}}_{\square}(I_\bullet, \mathcal{N})$ de $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{N})$, formée des foncteurs satisfaisant la condition a) de 4.14.1, est accessible dans $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{N})$, et de même [page 389]
pour $\underline{\mathrm{Hom}}^{\square}({}_\bullet I, \mathcal{N})$. Pour ce deuxième cas, ça a l'air OK, car si $\mathcal{C}$ est une petite sous-catégorie strictement génératrice de $\mathcal{N}$, et si pour tout $x \in \mathrm{Ob}\ [\mathcal{C}]$ on considère encore le foncteur

$$h_x \ : \ \mathcal{N} \longrightarrow \mathrm{Ens}\ ,$$

d'où

$$\bar{h}_x \ : \ \underline{\mathrm{Hom}}({}_\bullet I, \mathcal{N}) \longrightarrow \underline{\mathrm{Hom}}({}_\bullet I, \mathrm{Ens})\ ,$$

on voit que $\underline{\mathrm{Hom}}^{\square}({}_\bullet I, \mathcal{N})$ est l'intersection des images inverses des sous-catégories $\underline{\mathrm{Hom}}^{\square}({}_\bullet I, \mathrm{Ens})$ de $\underline{\mathrm{Hom}}({}_\bullet I, \mathrm{Ens})$. Donc ces dernières étant accessibles dans $\underline{\mathrm{Hom}}({}_\bullet I, \mathrm{Ens})$,

par 4.14.1 ((Ens) étant stable par petites $\varprojlim$), il s'ensuit bien que $\underline{\mathrm{Hom}}^{\square}({}_{\bullet}I,\mathcal{N})$ l'est dans $\underline{\mathrm{Hom}}({}_{\bullet}I,\mathcal{N})$, q.e.d.

Le cas de $\underline{\mathrm{Hom}}_{\square}(I_{\bullet},\mathcal{N})$ dans $\underline{\mathrm{Hom}}(I_{\bullet},\mathcal{N})$ semble plus délicat, dans le cas où [page 390]

$$I = \quad \bullet \quad , \qquad \text{donc} \qquad I = \quad \bullet \quad ,$$

si l'énoncé envisagé est vrai, on en conclura que $\underline{\mathrm{Ép}}(\mathcal{N})$ est accessible dans $\underline{\mathrm{Fl}}(\mathcal{N})$, pour toute catégorie accessible, car c'est l'image inverse de $\underline{\mathrm{Hom}}_{\square}(I_{\bullet},\mathcal{N})$ par le foncteur canonique (accessible)

$$\begin{array}{ccc} \underline{\mathrm{Fl}}(\mathcal{N}) & \longrightarrow & \underline{\mathrm{Hom}}(I_{\bullet},\mathcal{N}) \\ (x \xrightarrow{u} y) & \longmapsto & \left(\begin{array}{ccccc} & & y & & \\ & \overset{u}{\nearrow} & & \overset{\mathrm{id}_y}{\searrow} & \\ x & & & & y \\ & \underset{u}{\searrow} & & \underset{\mathrm{id}_y}{\nearrow} & \\ & & y & & \end{array} \right) . \end{array}$$

C'est dire que la généralisation envisagée de 4.14.1 n'est pas oiseuse, mais est sans doute p. 148
utile.

[page 391]

Théorème 4.14.17 ([168]). *Soit $p : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur cofibrant ou fibrant, avec $\mathcal{C}$ essentiellement petite (i.e. équivalente à une petite catégorie), et tel que les catégories fibres soient accessibles, et les foncteurs cochangement de base (cas cofibrant) ou de changement de base (cas fibrant) associés aux flèches u de $\mathcal{C}$ soient accessibles. Alors la catégorie $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$ des sections de $\mathcal{E}$ sur $\mathcal{C}$ est accessible, et pour tout $s \in \mathrm{Ob}\,\mathcal{C}$, le foncteur "évaluation en s" $\varepsilon_s : \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \longrightarrow \mathcal{E}_s$ est accessible.*

Commentary · Development (commentators') [ms. 390]

Why the epimorphism question reaches set theory

Grothendieck's proposed generalization in 4.14.16 asks whether actual universal cones can be recognized accessibly without assuming that all cones of the same shape have universal objects. For limits, his representable test on page 389 proves the assertion. For colimits, the degenerate square on page 390 puts the epimorphism question inside the proposed generalization. The following consequence explains why closure under filtered colimits cannot by itself finish the proof.

Proposition (epimorphism accessibility forces co-wellpoweredness). [ours] Let $\mathcal{K}$ be accessible. If the full subcategory $\mathrm{Epi}(\mathcal{K}) \subset \mathcal{K}^{[1]}$ is accessible and accessibly embedded, then $\mathcal{K}$ is co-wellpowered: for each object X, its epimorphic quotients form a set up to isomorphism over X.

Proof. The domain functor $d : \mathrm{Epi}(\mathcal{K}) \to \mathcal{K}$ is accessible. Form its iso-comma category with the functor $\mathbf{1} \to \mathcal{K}$ selecting X. The latter functor is accessible, because a constant diagram indexed by a nonempty filtered category has its chosen value as colimit. By the iso-comma theorem 4.14.21, this category is accessible. Its objects are epimorphisms $e : A \to B$ equipped with an isomorphism $X \simeq A$, and it is equivalent to the full subcategory $\mathcal{Q}_X$ of $X/\mathcal{K}$ on the epimorphisms out of X.

Between any two objects of $\mathcal{Q}_X$ there is at most one morphism: if $he = ke$ and e is epic, then $h = k$. Thus $\mathcal{Q}_X$ is a preorder. We check directly that every accessible preorder is essentially small. Choose a set $\mathcal{G}$ of presentable generators. An object q determines the subset

$$D_q = \{g \in \mathcal{G} : g \longrightarrow q\}.$$

If $D_q \subset D_r$, express q as a filtered colimit of objects of $\mathcal{G}$. Every object in that presentation has a map to r, and these maps are automatically compatible because there is at most one. The colimit supplies $q \to r$. Equality of the two subsets therefore implies arrows both ways, hence an isomorphism. Isomorphism classes inject into the power set of $\mathcal{G}$, which is a set. Applying this to $\mathcal{Q}_X$ proves the proposition. □

168. Cf. généralisation dans [le] théorème page 422, th. 4.14.26.

The proof is supplied here as a direct consequence of the chapter's own machinery; the marker records our presentation and does not assert priority. We checked the literature on accessible epimorphism categories and on co-wellpoweredness. The argument is closely related to the use of the quotient preorder in Bagaria et al. (2015, Corollary 7.6); we did not locate this exact iso-comma formulation as a separate named result.

There is a definite foundational consequence. The discussion following that corollary, on published pages 574–575, records that co-wellpoweredness of every accessible category implies a proper class of measurable cardinals, and that a proper class of strongly compact cardinals suffices. Our proposition therefore gives the same necessary lower bound for the universal assertion that all accessible categories have accessible epimorphism categories. In particular, that universal assertion cannot be treated as an unconditional theorem of ZFC. This is an obstruction, not a determination of its precise consistency strength. The cited discussion also connects the strong-compact hypothesis to detectability of colimits through Makkai–Paré's Theorem 6.3.8(i).

The limit-cone argument has no such reversal of tests. For a small diagram $D : I \to \mathcal{K}$ with a proposed cone $x \to D$, choose a small dense subcategory $\mathcal{A}$ of presentables. The cone is limiting exactly when

$$\operatorname{Hom}(A, x) \longrightarrow \varprojlim_{i \in I} \operatorname{Hom}(A, D(i))$$

is bijective for every $A \in \mathcal{A}$. Density makes this criterion sufficient, and each test is accessible at a sufficiently large common rank. Hence the category of existing limit cones is an accessible intersection of inverse images of the corresponding set-valued cone category. The source thus resolves its limit question while leaving a substantially different colimit question.

Corollaire 4.14.18. *Soit π_0 un cardinal infini tel que les catégories fibres satisfassent la condition L_{π_0}, et que les foncteurs cochangement de base (resp. changement de base) entre icelles satisfassent également L_{π_0}. (Un tel π_0 existe.)*

ⓐ *Alors $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ satisfait la condition $\underline{\mathrm{L}}_{\pi_0}$, et les foncteurs ε_s aussi, i.e. les $\varinjlim$ grandes devant π_0 dans $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$ se calculent argument par argument.*

ⓑ *Soit un cardinal $\pi \geqslant \sup(\pi_0, \operatorname{card} \operatorname{réd} \mathcal{C})$. Pour qu'un objet f de $\mathcal{M}$ appartienne à $\mathcal{M}_\pi$, il* suffit *que $\forall\ s \in \operatorname{Ob} C$, $f(s) \overset{déf}{=} \varepsilon_s(x)$ [plutôt $\varepsilon_s(f)$] soit dans $\mathcal{E}_{s\,\pi}$. (Comparer 4.5.8, page 119.)*

ⓒ *Supposons pi_0 choisi de façon* [page 392]
à satisfaire les conditions supplémentaires suivantes : π_0 est utile pour $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$, et pour chacune des catégories fibres $\mathcal{E}_s$ ($s \in \operatorname{Ob} \mathcal{C}$) (un tel π_0 existe). Soit

$$\pi_1 \ = \ \sup(\pi_0, \operatorname{card} \operatorname{réd} \mathcal{C}, \operatorname{card} \operatorname{réd} \mathcal{M}_{\pi_0}) \ .$$

Alors pour tout cardinal $\pi \geqslant \pi_1$ et tel que $\pi^{\pi_0} = \pi$, on a

$$\mathcal{M}_\pi \ = \ \left\{ \begin{array}{l} \text{sous-catégorie strictement pleine de } \mathcal{M} \text{ formée des } f \in \\ \operatorname{Ob} \mathcal{M} \text{ telles que } f(s) \in \operatorname{Ob} \mathcal{E}_{s\,\pi} \ \forall\ s \in S \text{ [plutôt } s \in \\ \operatorname{Ob} C\text{]} . \end{array} \right.$$

Si de plus π est choisi tel qu'on ait

$$\pi \ \geqslant \ \sup_{s \in \operatorname{Ob} \mathcal{C}} \operatorname{card} \operatorname{réd} \mathcal{E}_{s\,\pi_0} \ ,$$

alors la sous-catégorie strictement pleine $\mathcal{E}(\pi)$ de $\mathcal{E}$ telle que $\operatorname{Ob} \mathcal{E}(\pi) = \bigcup_{s \in \operatorname{Ob} \mathcal{C}} \operatorname{Ob} \mathcal{E}_{s\,\pi}$ est une sous-catégorie $\mathcal{C}$-cofibrée (resp. $\mathcal{C}$-fibrée) de $\mathcal{E}$, et on a

$$\underline{\Gamma}(\mathcal{E}/\mathcal{C})_\pi \xleftarrow{\sim} \underline{\Gamma}(\mathcal{E}(\pi)/C),$$

donc (π étant utile pour $\mathcal{E}$) on a une équivalence de catégories

p. 149
$$\underline{\Gamma}(\mathcal{E}/\mathcal{C}) \xleftarrow{\sim} \operatorname{Ind}_\pi(\underline{\Gamma}(\mathcal{E}(\pi)/\mathcal{C})) \ .$$ [page 393]

Toutes ces assertions se prouvent facilement dans l'ordre où elles sont données, en utilisant 4.14.17 et les résultats généraux déjà prouvés. Bien sûr, on peut commencer par remplacer $\mathcal{C}$ par une sous-catégorie réduite $\mathcal{C}_0$ équivalente et $\mathcal{E}$ par $\mathcal{E}_0 = \mathcal{E}|\mathcal{C}$ `[plutôt` $\mathcal{E}|\mathcal{C}_0$`]`, de façon à se réduire (dans b) et c)) au cas où $\operatorname{card}\mathcal{C} \leqslant \pi$.

Commentary · Development (commentators') [ms. 392]

Sections and the two directions of transport

Grothendieck's section theorem assembles accessible categories over a small base. The objects are families in the fibres together with compatible transition arrows, so accessibility should follow by adjoining structure and imposing equations. The main theorem does hold. The presentability estimate in 4.14.18(b), however, needs a distinction between the two directions of transport that is absent from its printed statement.

First consider a fibration, and fix a cleavage. A section x consists of objects x_s and arrows $\alpha_a : x_s \to a^*x_t$ for $a : s \to t$, satisfying the unit and composition equations with the coherence isomorphisms of the cleavage. For sections x, y, their morphisms are the equalizer

$$\operatorname{Hom}_\Gamma(x,y) \longrightarrow \prod_{s\in \operatorname{Ob}\mathcal{C}} \operatorname{Hom}_{\mathcal{E}_s}(x_s, y_s) \rightrightarrows \prod_{a:s\to t} \operatorname{Hom}_{\mathcal{E}_s}(x_s, a^*y_t).$$

The two images of a family (h_s) at a are $a^*(h_t)\alpha_a$ and $\alpha_a^y h_s$. Take a regular λ exceeding the number of arrows of the base, suppose the transports preserve λ-filtered colimits, and suppose every x_s is λ-presentable. All terms on the right preserve those filtered colimits in the variable y, and the products and equalizer have fewer than λ entries. Interchange in sets proves that x is λ-presentable. This establishes the fibration half of 4.14.18(b).

For an opfibration, the second product instead has factors

$$\operatorname{Hom}_{\mathcal{E}_t}(a_*x_s, y_t).$$

Presentability of x_s does not guarantee presentability of a_*x_s merely because a_* is accessible. That missing implication cannot be repaired by the cardinality of the base.

Counterexample (pointwise presentability need not suffice). [ours] For every infinite regular λ, take the base [1], both fibres equal to **Set**, and the transition functor constant at a set B with $|B| = \lambda$. This functor preserves every filtered colimit: filtered categories are nonempty and connected. Its opfibration has section category

$$\mathbf{Set} \times (B/\mathbf{Set}),$$

where a section is $(X, B \to Y)$. The section $x = (\varnothing, B \to 1)$ has finite, hence λ-presentable, values in both fibres.

To test x itself, index by the poset of subsets $S \subset B$ of cardinality less than λ. This poset is λ-filtered, since the union of fewer than λ such subsets still has cardinality less than λ. Let B/S denote the quotient identifying all elements of S and leaving the other elements separate. The sections

$$x_S = (\varnothing, B \longrightarrow B/S)$$

form a λ-filtered diagram. Its colimit is x, because every pair of elements of B is identified at some stage. No map $x \to x_S$ exists over B: such a map would make $B \to B/S$ constant, although S has size less than $|B|$. Thus

$$\varinjlim_S \operatorname{Hom}(x, x_S) = \varnothing, \qquad \operatorname{Hom}(x, \varinjlim_S x_S) = \operatorname{Hom}(x,x) \neq \varnothing.$$

This disproves the opfibration half of 4.14.18(b) as written. For Grothendieck's indexing choose $\lambda = \pi^+$; the base is finite and the constant transition satisfies every condition L_{π_0} in the statement.

The corrected sufficient condition is explicit: require each transported object a_*x_s, as well as each x_s, to be λ-presentable. Choosing a common strong accessibility rank for the small family of transports ensures this condition for all pointwise presentable sections. After also enlarging the rank so that every evaluation preserves presentables, one obtains an eventual pointwise characterization. The printed numerical threshold in 4.14.18(c), however, does not itself ensure this enlargement.

Counterexample (the section skeleton does not bound evaluation). [ours] Let $\mathcal{P}(S)$ be the poset of subsets of a set S, ordered by inclusion. It is locally finitely presentable: unions compute colimits, finite subsets are finitely presentable, and every subset is the directed union of its finite subsets. Consider the [1]-opfibration given by the constant functor $\mathbf{1} \to \mathcal{P}(S)$ with value S. Its sections are pairs consisting of the unique object of $\mathbf{1}$ and an inclusion $S \subset Y \subset S$. There is therefore exactly one section, and its structure arrow is an isomorphism. Both the section category and its cocartesian subcategory are $\mathbf{1}$.

Choose $\pi_0 = \aleph_0$, $\pi = 2^{\aleph_0}$, and $|S| > \pi$. The fibres and section category admit presentations at π_0, and transport preserves every filtered colimit. The printed bound is $\pi_1 = \aleph_0$, since the section skeleton has one object and one arrow, and $\pi^{\pi_0} = \pi$. The unique section is presentable at every rank. Its value S is not π^+-presentable in $\mathcal{P}(S)$: the subsets of S of cardinality at most π form a π^+-filtered poset, their union is S, and no stage admits an inclusion from S. This refutes the first pointwise equality in 4.14.18(c), and also the pointwise equality in 4.14.20(c). The later additional bound involving the fibre skeletons excludes this particular example: the skeleton of $(\mathcal{P}(S))_{\pi_0}$ already contains all singletons.

These two examples were obtained by following the two missing size implications separately. We checked the fixed-rank results and hypotheses in Positselski (2024); we did not locate these exact counterexamples there or in the other sources consulted. The markers record the present constructions, without a claim of priority.

Cocartesian sections behave better even at a fixed rank. Their structure arrows identify $a_* x_s$ with x_t, so pointwise presentability already gives all the source objects needed by the equalizer. Thus the pointwise sufficient condition in 4.14.20(b) remains true for cocartesian sections, independently of the false unrestricted opfibration assertion. Cartesian sections likewise satisfy the fibration argument. Their filtered colimits are pointwise whenever transport preserves those colimits, because the induced structure comparisons remain isomorphisms. This accounts for the accessibility and reconstruction claims without conflating the two variance conventions.

Corollaire 4.14.19. *Sous les conditions de 4.14.17, la sous-catégorie* $\underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C})$ *de* $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ *(cas p cofibrant), resp. la sous-catégorie* $\underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}/\mathcal{C})$ *(cas p fibrant) est accessible dans* $\mathcal{M}$*, donc accessible et le foncteur d'inclusion* `[est]` *accessible.*

(**N.B.** Ces deux catégories sont strictement pleines dans $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$, donc l'assertion qu'elles soient accessibles dans celle-ci *équivaut* à la dernière assertion.)

Démonstration de 4.14.19. On peut supposer $\mathcal{C}$ réduite, donc petite. Considérons le cas où p est cofibrant. Pour toute flèche $\lambda : s \longrightarrow t$ dans $\mathcal{C}$, considérons le foncteur

$$\varphi_\lambda \ : \ \mathcal{M} \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_t)$$

défini par [page 394]

$$\varphi_\lambda(f) \ = \ (\lambda_*(f(s)) \xrightarrow{f(\lambda)^\natural} f(t)) \ .$$

C'est un foncteur accessible, car son composé avec les foncteurs source et but sont les foncteurs $\lambda_* \varepsilon_s$ et ε_t, donc accessibles, ε_s, ε_t et λ_* étant accessibles. Considérons la sous-catégorie strictement pleine

$$\underline{\mathrm{Is}}(\mathcal{E}_t) \ \subset \ \underline{\mathrm{Fl}}(\mathcal{E}_t) \ ,$$

on sait qu'elle est accessible dans $\underline{\mathrm{Fl}}(\mathcal{E}_t)$ (4.14.12). Donc son image inverse par φ_λ est accessible dans $\mathcal{M}$, et par suite aussi

$$\underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}/\mathcal{C}) \ \overset{\text{déf}}{=} \ \bigcap_{\lambda \in \mathrm{Ob}\, \mathcal{C}} \varphi_\lambda^{-1}(\underline{\mathrm{Is}}(\mathcal{E}_t))$$

comme intersection d'une *petite* famille de sous-catégories accessibles dans $\mathcal{M}$, q.e.d. Le cas où p est fibrant se traite de façon similaire, en introduisant les foncteurs

$$\psi_\lambda \ : \ \mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_s)$$

[page 395]

donnés par

$$\psi_\lambda(f) \;=\; (f(s) \xrightarrow{f(\lambda)^\flat} \lambda^* f(t)) \;.$$

Corollaire 4.14.20. *Soit π_0 un cardinal infini choisi comme dans 4.14.18 a).*

ⓐ *Alors $\mathcal{M}_{\mathrm{c}} = \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C})$ (resp. $\mathcal{M}_{\mathrm{c}} = \underline{\Gamma}_{\mathrm{cart}}$) est L_{π_0}-stable dans $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$, ou ce qui revient au même (comme elle est strictement pleine), elle satisfait L_{π_0}, et le foncteur d'inclusion satisfait à L_{π_0}. Les $\varinjlim$ grandes devant π_0 dans cette catégorie se calculent argument par argument.*

ⓑ *Pour tout cardinal $\pi \geqslant \pi_0$, tout objet de $\mathcal{M}_{\mathrm{c}}$ qui est π-accessible dans $\mathcal{M}$ l'est (tautologiquement) dans $\mathcal{M}_{\mathrm{c}}$. Si $\pi \geqslant \sup(\pi_0, \operatorname{card}\operatorname{réd}\mathcal{C})$, alors (par 4.14.18 b) tout objet f de $\mathcal{M}_{\mathrm{c}}$ tel que $f(s) \in \operatorname{Ob}\mathcal{E}_{s\,\pi}\ \forall\ s \in S$ [plutôt $s \in \operatorname{Ob} C$] est dans $\mathcal{M}_\pi$ et dans $\mathcal{M}_{\mathrm{c}\,\pi}$.*

ⓒ *Supposons π_0 choisi comme dans 4.14.18 c, et que de plus $\mathcal{M}_{\mathrm{c}}$ soit π_0-accessible dans $\mathcal{M}$. (Un tel π_0 existe.) On a donc, pour $\pi \geqslant \pi_1 = \sup(\pi_0, \operatorname{card}\operatorname{réd}\mathcal{C}, \operatorname{card}\operatorname{réd}\mathcal{M}_\pi)$ tel que $\pi^{\pi_0} = \pi$,* p. 150

$$(\mathcal{M}_{\mathrm{c}})_\pi \;=\; \mathcal{M}_\pi \cap \mathcal{M}_{\mathrm{c}} \;=\; \begin{array}{l}\text{Sous-catégorie strictement pleine de } \mathcal{M}_{\mathrm{c}} \text{ formée des } f \\ \text{telles que } f(s) \in \operatorname{Ob}\mathcal{E}_{s\,\pi}\ \forall\ s \in \operatorname{Ob}\mathcal{C}.\end{array}$$

Si de plus π est choisi tel qu'on ait $\pi \geqslant \sup_{s\in\operatorname{Ob}\mathcal{C}} \operatorname{card}\operatorname{réd}\mathcal{E}_{s\,\pi_0}$, alors [page 396]

$$\left\{\begin{array}{rcl} & \mathcal{M}_{\mathrm{c}\,\pi} & \xleftarrow{\sim} \quad \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}(\pi)/\mathcal{C}) \\ \textit{resp.} & \mathcal{M}_{\mathrm{c}\,\pi} & \xleftarrow{\sim} \quad \underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}(\pi)/\mathcal{C}) \;, \end{array}\right.$$

donc (π étant utile pour $\mathcal{M}_{\mathrm{c}}$) une équivalence de catégories

$$\begin{array}{rcl} \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C}) & \xleftarrow{\sim} & \operatorname{Ind}_\pi(\underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}(\pi)/\mathcal{C})) \\ \textit{resp.} \quad \underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}/\mathcal{C}) & \xleftarrow{\sim} & \operatorname{Ind}_\pi(\underline{\Gamma}_{\mathrm{cart}}(\mathcal{E}(\pi)/\mathcal{C})). \end{array}$$

Démonstration. Tout est clair dans l'ordre où c'est donné en utilisant les deux corollaires précédents.

Corollaire 4.14.21. *Considérons un diagramme de foncteurs*

$$(*) \qquad \begin{array}{ccccc} \mathcal{E}_1 & & & & \mathcal{E}_2 \\ & \searrow^{p_1} & & \swarrow_{p_2} & \\ & & \mathcal{E}_0 \;, & & \end{array}$$

les catégories et les foncteurs étant accessibles. [page 397]

a) Le produit 2-fibré

$$\mathcal{M}_0 \;=\; \mathcal{E}_1 \overset{2}{\times}_{\mathcal{E}_0} \mathcal{E}_2$$

est accessible, et les foncteurs canoniques $\mathcal{M} \longrightarrow \mathcal{E}_1$, $\mathcal{M} \longrightarrow \mathcal{E}_2$ sont accessibles.

b) ([169]). La catégorie $\mathcal{M}$ des triples (x_1, x_2, u) avec $x_1 \in \operatorname{Ob}\mathcal{E}_1$, $x_2 \in \operatorname{Ob}\mathcal{E}_2$, et

$$u \;:\; p_1(x_1) \;\longrightarrow\; p_2(x_2) \;,$$

est accessible, et les foncteurs canoniques $\mathcal{M} \longrightarrow \mathcal{E}_1$, $\mathcal{M} \longrightarrow \mathcal{E}_2$ sont accessibles.

c) La sous-catégorie strictement pleine $\mathcal{M}_0$ de $\mathcal{M}$ est accessible dans $\mathcal{M}$.

Démonstration. Soit $\mathcal{C}$ la catégorie

$$\begin{array}{ccccc} 1 & & & & 2 \\ & \searrow & & \swarrow & \\ & & 0 \;, & & \end{array}$$

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[page 398]

et $\mathcal{E}$ la catégorie cofibrée sur $\mathcal{C}$ définie par le diagramme $(*)$. On est sous les conditions de 4.14.17. La catégorie $\Gamma = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ s'interprète comme la catégorie des quintuplets $(x_0, x_1, x_2, u_1, u_2)$, avec $x_i \in \mathrm{Ob}\,\mathcal{E}$ pour $i \in \{0,1,2\}$, et $u_1, u_2 \in \mathrm{Fl}\,\mathcal{E}_0$,

$(**)$
$$\begin{array}{ccc} p_1(x_1) & & p_2(x_2) \\ & {\scriptstyle u_1}\searrow \quad \swarrow{\scriptstyle u_2} & \\ & x_0\ . & \end{array}$$

En vertu de 4.14.18, cette catégorie $\underline{\Gamma}$ est accessible. La catégorie $\mathcal{M}$ est équivalente à la sous-catégorie strictement pleine Γ_0 de Γ, formée des quintuplets pour lesquels on a $u_2 \in \underline{\mathrm{Is}}\mathcal{E}_0$. On a en effet des foncteurs quasi-inverses l'un de l'autre

$$\mathcal{M} \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \Gamma_1$$

donnés par

$$\begin{aligned} \alpha(x_1, x_2, u) &= (p_2(x_2), x_1, x_2, u, \mathrm{id}_{p_2(x_2)}) \\ \beta(x_0, x_1, x_2, u_1, u_2) &= (x_1, x_2, u_2^{-1}u_1)\ , \end{aligned}$$

on a alors

$$\begin{aligned} \beta\alpha &= \mathrm{id}_{\mathcal{M}} \\ \alpha\beta &\xrightarrow{\sim} \mathrm{id}_{\Gamma_1}\ . \end{aligned}$$

D'autre part, Γ_1 est évidemment accessible dans $\mathcal{M}$, comme image inverse de $\underline{\mathrm{Is}}(\mathcal{E}_0)$ (qui est accessible dans $\underline{\mathrm{Fl}}(\mathcal{E}_0)$ par le foncteur accessible [page 399]

$$\underline{\Gamma} \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_0)$$

donné par $(x_1, x_1, x_2, u_1, u_2) \longmapsto u_2$. Cela prouve que $\mathcal{M}$, qui est équivalente à Γ_1, est également accessible, et l'accessibilité du foncteur canonique $\mathcal{M} \longrightarrow \mathcal{E}_i$ $(i \in \{1,2\})$ résulte de celle de $\Gamma \longrightarrow \mathcal{E}_i$, impliquant celle du composé $\Gamma_1 \hookrightarrow \Gamma \longrightarrow \mathcal{E}_i$. Cela prouve donc b). Pour prouver a), il suffit de voir que $\mathcal{M}_0$ est accessible dans $\mathcal{M}$, or c'est encore l'image inverse de $\underline{\mathrm{Is}}(\mathcal{E}_0) \subset \underline{\mathrm{Fl}}(\mathcal{E}_0)$ par le foncteur accessible canonique

$$\mathcal{M} \longrightarrow \underline{\mathrm{Fl}}(\mathcal{E}_0)\ ,$$

donc $\mathcal{M}$ est bien accessible dans $\mathcal{M}$, comme image inverse d'une sous-catégorie accessible dans $\underline{\mathrm{Fl}}(\mathcal{E})$. On peut d'ailleurs aussi interpréter $\mathcal{M}$, à équivalence de catégories près, comme [page 400]

$$\mathcal{M}_0 \simeq \underline{\Gamma}_{\mathrm{cocart}}(\mathcal{E}/\mathcal{C})\ ,$$

les foncteurs canoniques $\mathcal{M}_0 \longrightarrow \mathcal{E}_i$ $(i \in \{0,1\})$ s'interprétant comme des foncteurs évaluation, ce qui donne a) directement, comme cas particulier de 4.14.19.

Commentary · Exposition [ms. 400]

A two-fibre product remembers the comparison isomorphism

Grothendieck closes this sequence of corollaries by giving a useful model for gluing accessible categories. Given $p_1 : \mathcal{E}_1 \to \mathcal{E}_0$ and $p_2 : \mathcal{E}_2 \to \mathcal{E}_0$, the ordinary comma category remembers an arrow $p_1x_1 \to p_2x_2$; its iso-comma subcategory requires that arrow to be invertible. The superscript 2 is substantive: equality of two objects in the base is replaced by a specified isomorphism.

The cospan section category makes the proof concrete. Its objects consist of $x_i \in \mathcal{E}_i$ and arrows $u_i : p_ix_i \to x_0$. Require u_2 to be invertible. One can then discard x_0 and replace u_1 by $u_2^{-1}u_1$. The inverse operation sends (x_1, x_2, u) to the quintuple with $x_0 = p_2x_2$ and $u_2 = \mathrm{id}$. The natural

169. Ceci est contenu dans 4.14.22.1 ci-dessus (p. 406), appliqué à $\Pi = \mathcal{E}_1 \times \mathcal{E}_2$, $\Pi' = \mathcal{E}_0$, et $F, G : \Pi \rightrightarrows \Pi'$ donnés par $F = p_1 \circ \mathrm{pr}_1$ et $G = p_2 \circ \mathrm{pr}_2$. D'autre part, a) et c) résultent facilement de b).

isomorphism between the composite and the original quintuple has component u_2 at the vertex 0 and identity components at $1, 2$. This checks the equivalence, including its action on morphisms, rather than merely comparing its object descriptions.

The condition on u_2 is an accessible inverse image of the isomorphism category of $\mathcal{E}_0$. The comma category is consequently accessible. Requiring the remaining comparison u to be an isomorphism is another such inverse image, yielding the iso-comma theorem. This proof uses only the opfibration section theorem already established earlier, so its use in the subsequent proof for fibrations creates no circularity.

As an example, take two copies of **1** mapping into **Set** by selecting the same set A. Their strict pullback is **1**. Their iso-comma category is the discrete category of permutations of A: an object is a bijection $u : A \to A$, and a morphism between two such objects must have identity components in both copies of **1**, so it exists exactly when the two bijections are equal. The set of comparison isomorphisms is therefore part of the categorical answer. Discarding it would change the result even in this elementary diagram.

For the cospan opfibration, a cocartesian section requires both u_1 and u_2 to be invertible. Its section category is thus equivalent to the same iso-comma category, exactly as in the final display of the source. The missing subscripts and exchanged names in the surrounding paragraphs are edition slips; none affects this argument once the three categories, the comma category, its iso-comma subcategory, and the cospan section category, are kept distinct.

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4.14.22. Démonstration de 4.14.17. Le cas où p est cofibrant a déjà été démontré (th. 4.10.8, p. 263). Nous allons voir que le cas où p est fibrant se ramène au théorème précédent (dans un cas très particulier), en utilisant à fond les résultats généraux sur les parties accessibles d'une catégorie accessible. Notons que les assertions a) et b) du corollaire 4.14.18 se prouvent de façon AQT `[âne qui trotte]` sans avoir à faire appel à 4.14.17. Le a) implique déjà que les foncteurs évaluation $\varepsilon_s : \mathcal{M} \longrightarrow \mathcal{E}_s$ sont accessibles. Il reste donc à prouver que $\mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$ est accessible, dans le cas où p est *fibrant*. On supposera de plus [page 401] que $\mathcal{C}$ est petite (quitte à remplacer $\mathcal{C}$ par une petite sous-catégorie pleine équivalente).

Soient

$$\begin{aligned} \Pi &= \prod_{s\in \mathrm{Ob}\,\mathcal{C}} \mathcal{E}_s \\ \Pi' &= \prod_{\lambda\in \mathrm{Fl}\,\mathcal{C}} \mathcal{E}_{s(\lambda)} \ , \end{aligned}$$

où $s(\lambda)$ désigne la *source* s de la flèche $\lambda : s \longrightarrow t$ dans $\mathcal{C}$. Π, Π' sont des catégories accessibles, comme (petits) produits de catégories accessibles. (Cas particulier trivial de 4.10.8, pour une catégorie d'indices discrète.) On va définir deux foncteurs

$$\Pi \underset{G}{\overset{F}{\rightrightarrows}} \Pi'$$

en définissant pour tout $\lambda : s_0 \longrightarrow t_0$ dans $\mathrm{Fl}\,\mathcal{C}$ deux foncteurs [page 402]

$$\left\{ \begin{array}{lcll} \Pi & \underset{G_\lambda}{\overset{F_\lambda}{\rightrightarrows}} & \mathcal{E}_{s(\lambda)=s_0} & \\ F_\lambda & = & \mathrm{pr}_{s_0} & \text{projection de } \Pi = \prod_s \mathcal{E}_s \text{ sur le facteur } \mathcal{E}_{s_0} \\ G_\lambda & = & \lambda^* \mathrm{pr}_{s_1} & \text{composé } \Pi \xrightarrow{\mathrm{pr}_{s_1}} \mathcal{E}_{s_1} \xrightarrow{\lambda^*} \mathcal{E}_{s_0} \\ & & & \texttt{[plutôt } \lambda^*\mathrm{pr}_{t_0} \texttt{ resp. } \Pi \xrightarrow{\mathrm{pr}_{t_0}} \mathcal{E}_{t_0} \xrightarrow{\lambda^*} \mathcal{E}_{s_0}\texttt{]} \end{array} \right. \qquad \left([170], [171]\right) .$$

170. Il est immédiat que les F_λ, G_λ, donc F et G, sont des foncteurs *accessibles*.

171. **N.B.** On suppose choisi pour tout $\lambda \in \mathrm{Fl}\,\mathcal{C}$ un foncteur image inverse $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$. On suppose que si λ est une flèche identique, id_s, `[on]` choisit $\lambda^* = \mathrm{id}_{\mathcal{E}_s}$.

Commentary · Exposition [ms. 401]

First choose the arrows, then impose the laws

Grothendieck is now extracting a reusable construction from the section theorem. A section contains objects in the fibres, arrows above the arrows of the base, and equations expressing functoriality. The first two sorts of data can be assembled before the equations are imposed. This separation explains why accessibility survives: the preliminary construction is an inserter, and the final constraints are a small family of equifiers.

Choose a normalized cleavage of the fibration. A prospective section is a family $X = (X_s)$ together with arrows

$$a_\lambda : X_s \longrightarrow \lambda^* X_t \qquad (\lambda : s \longrightarrow t).$$

For the moment these arrows need satisfy no laws. A morphism from (X, a) to (Y, b) is a family $h_s : X_s \to Y_s$ satisfying

$$\lambda^*(h_t) a_\lambda = b_\lambda h_s.$$

This is precisely the inserter of the two functors $F, G : \Pi \rightrightarrows \Pi'$ on the page. The word "diagram" here means the underlying directed graph: it does not already incorporate identities or composition. Consequently the preliminary category is larger than the section category even when the underlying objects have been fixed.

The cleavage need not be split. If $\lambda : s \to t$ and $\mu : t \to u$, cartesian factorisation supplies a canonical isomorphism

$$c_{\lambda,\mu} : \lambda^*\mu^* \xrightarrow{\sim} (\mu\lambda)^*.$$

The composition equation in fibre coordinates is

$$a_{\mu\lambda} = c_{\lambda,\mu,X_u}\, \lambda^*(a_\mu)\, a_\lambda.$$

The unit equation is $a_{\mathrm{id}_s} = \mathrm{id}_{X_s}$ with the chosen normalization. Associativity of these comparisons is inherited from composition in $\mathcal{E}$. There is therefore no extra assumption that all transition functors compose as literal equalities. Grothendieck's use of the actual arrow $f_1(\mu)f_1(\lambda)$ keeps this coherence implicit and correct.

To see what the cardinal estimates must control, choose a regular cardinal κ at which the fibres and the small family of transition functors admit and preserve filtered colimits. For a filtered diagram of prospective sections, put $X_s = \varinjlim_i X_{i,s}$. The structure arrow a_λ is induced by the compatible arrows $X_{i,s} \to \lambda^* X_{i,t}$ and the comparison $\varinjlim_i \lambda^* X_{i,t} \simeq \lambda^* X_t$. The equations for a genuine section pass to these colimits. This proves colimit preservation by the relevant comparison functors; accessibility of the equation loci additionally uses the preceding small-generation results. Stability under filtered colimits alone would not supply that conclusion.

There is also a useful formula for the morphisms between two sections. They form the equalizer

$$\mathrm{Hom}_\Gamma(X, Y) \longrightarrow \prod_s \mathrm{Hom}_{\mathcal{E}_s}(X_s, Y_s) \rightrightarrows \prod_{\lambda : s \to t} \mathrm{Hom}_{\mathcal{E}_s}(X_s, \lambda^* Y_t).$$

The two arrows send (h_s) respectively to $(\lambda^* h_t)a_\lambda$ and $b_\lambda h_s$. If κ exceeds the number of arrows of the base and every X_s is κ-presentable, the right-hand expression preserves κ-filtered colimits in Y, provided the transition functors do. This gives the pointwise sufficient condition for presentability directly. The converse requires a larger common bound; forgetting structure does not preserve presentability at every initial rank.

Soit $f \in \mathcal{M} = \underline{\Gamma}(\mathcal{E}/\mathcal{C})$, alors pour tout $\lambda \in \mathrm{Fl}\mathcal{C}$, f définit une flèche

$$\alpha_\lambda(f) \ : \quad \underbrace{f(s)}_{=F_\lambda(\varepsilon(f))} \quad \xrightarrow{f(\lambda)^\flat} \quad \underbrace{\lambda^*(f(t))}_{=G_\lambda(\varepsilon(f))} \ ,$$

fonctorielle en f. On a un foncteur

$$\varepsilon \ = \ (\varepsilon_s)_{s \in \mathrm{Ob}\,\mathcal{C}} \ : \ \mathcal{M} \ \longrightarrow \ \Pi \ ,$$

et pour tout $\lambda \in \mathrm{Fl}\mathcal{C}$ un homomorphisme fonctoriel

$$\alpha_\lambda \ : \ F_\lambda \circ \varepsilon \ \longrightarrow \ G_\lambda \circ \varepsilon \ ,$$

d'où un homomorphisme fonctoriel

$$\alpha \;=\; (\alpha_\lambda)_{\lambda\in \mathrm{Fl}\,\mathcal{C}} \;:\; F\circ\varepsilon \;\longrightarrow\; G\circ\varepsilon\;.$$

Soit alors $\mathcal{M}'$ la catégorie suivante : p. 153 [page 403]

$$\begin{array}{rcl}\mathcal{M}' & = & \text{catégorie des couples } (X,u), \text{ avec } X \text{ dans } \mathrm{Ob}\,\Pi,\\ & & \text{et } u : F(X)\longrightarrow G(X) \text{ une flèche dans } \Pi'.\end{array}$$

L'homomorphisme fonctoriel $\alpha : F\circ\varepsilon \longrightarrow G\circ\varepsilon$ définit un foncteur

$$\mathcal{M} \;\longrightarrow\; \mathcal{M}'\;,$$

et il est immédiat que ce foncteur est pleinement fidèle. En fait, la catégorie $\mathcal{M}'$ est isomorphe à la catégorie formée des couples (f_0, f_1) de deux applications

$$\begin{array}{rccccc} f_0 : & \mathrm{Ob}\,\mathcal{C} & \longrightarrow & \mathrm{Ob}\,\mathcal{E} & s & \longmapsto & f_0(s)\\ f_1 : & \mathrm{Fl}\,\mathcal{C} & \longrightarrow & \mathrm{Fl}\,\mathcal{E} & \lambda & \longmapsto & f_1(\lambda)\end{array}$$

de telle façon que f_1 et f_0 soient compatibles avec les applications source et but, i.e. qu'ils définissent un homomorphisme de *diagrammes* $f : \mathrm{diag}\,\mathcal{C} \longrightarrow \mathrm{diag}\,\mathcal{E}$, et que cet homomorphisme soit inverse à droite de $\mathrm{diag}(p) : \mathrm{diag}\,\mathcal{E} \longrightarrow \mathrm{diag}\,\mathcal{C}$.

Le foncteur précédent $\mathcal{M} \hookrightarrow \mathcal{M}'$ s'interprète comme le foncteur qui associe à toute sections f de $\mathcal{E}$ sur $\mathcal{C}$, le morphisme de diagrammes correspondant. C'est donc l'inclusion d'une sous-catégorie *strictement pleine* de $\mathcal{M}'$. Donc pour prouver que $\mathcal{M}$ est accessible, il suffit de prouver : [page 404]

a) $\mathcal{M}'$ est accessible.

b) $\mathcal{M}$ est accessible dans $\mathcal{M}'$.

Prouvons d'abord b) (en admettant a). La sous-catégorie $\mathcal{M}$ de $\mathcal{M}'$ est définie par deux types des conditions, sur un objet $f = (f_0, f_1)$ de $\mathcal{M}'$:

1°) $\forall\; s \in \mathrm{Ob}\,\mathcal{C},\; f_1(\mathrm{id}_s) = \mathrm{id}_{f_0(s)}$.

2°) Si $s \xrightarrow{\lambda} t \xrightarrow{\mu} u$ sont deux flèches composables dans $\mathcal{C}$, alors

$$f_1(\mu\lambda) \;=\; f_1(\mu)f_1(\lambda)\;.$$

Ces conditions s'explicitent l'une et `[l'autre par]` l'égalité d'une double-flèche dans $\mathcal{E}_s$, savoir [page 405]

$$f_0(s) \underset{\mathrm{id}_{f_0(s)}}{\overset{f_1(\mathrm{id}_s)}{\rightrightarrows}} f_0(s)$$

dans le cas 1°), et

$$f_0(s) \underset{(f_1(\mu)f_1(\lambda))^\flat}{\overset{f_1(\mu\lambda)^\flat}{\rightrightarrows}} (\mu\lambda)^*(f_0(u))$$

dans le cas 2°). Dans l'un et l'autre cas, on a un foncteur

$$\mathcal{M}' \xrightarrow{\gamma_s \text{ resp. } \gamma_{\lambda,\mu}} \underline{\mathrm{Hom}}(I, \mathcal{E}_s)$$

dans la catégorie des double-flèches de $\mathcal{E}_s$ (I désignant la catégorie double-flèche type $0 \rightrightarrows 1$), foncteur qui est (trivialement) accessible. Soit alors $I \xrightarrow{i} \mathbf{\Delta}_1$ le foncteur canonique, et p. 154

$$\underbrace{\underline{\mathrm{Hom}}(\mathbf{\Delta}_1, \mathcal{E}_s)}_{\simeq \underline{\mathrm{Fl}}(\mathcal{E}_s)} \;\hookrightarrow\; \underline{\mathrm{Hom}}(I, \mathcal{E}_s)$$

[page 406]

le foncteur $u \longmapsto u \circ i$, qui est un foncteur strictement pleine fidèle, ayant comme image la sous-catégorie des double-flèches (u, v) telles que $u = v$. Comme le foncteur d'inclusion est accessible et $\underline{\mathrm{Hom}}(\mathbf{\Delta}_1, \mathcal{E}_s)$ et $\underline{\mathrm{Hom}}(I, \mathcal{E}_s)$ accessibles (cas particulier de 4.14.1, ou aussi de 4.10.8!), on voit que la sous-catégorie strictement pleine de $\underline{\mathrm{Hom}}(I, \mathcal{E}_s)$ des double-flèches égalisées dans $\mathcal{E}_s$, est accessible dans $\underline{\mathrm{Hom}}(I, \mathcal{E}_s)$. Donc, son image inverse dans $\mathcal{M}'$ est accessible. Donc l'intersection de ces images inverses pour $s \in \mathrm{Ob}\,\mathcal{C}$ et $(\lambda, \mu) \in \mathrm{Fl}_2(\mathcal{C})$ variables, l'est également ($\mathcal{C}$ étant petite).

Donc tout revient à prouver que $\mathcal{M}'$ est accessible, ce qui est contenu dans le

Lemme 4.14.22.1. *Soient* Π, Π' *deux catégories accessibles,* $F, G : \Pi \rightrightarrows \Pi'$ *deux foncteurs accessibles, et* $\mathcal{M}'$ *la catégorie des couples* (X, u)*, avec* $X \in \mathrm{Ob}\,\Pi$ *et* $u : F(x) \longrightarrow G(x)$ *une flèche de* Π'*. Alors* $\mathcal{M}'$ *est accessible, et le foncteur canonique* $\mathcal{M}' \longrightarrow \Pi$ *est accessible.*

Démonstration du Lemme. La donnée de Π, Π', F, G définit une catégorie cofibrée $\mathcal{F}$ sur la catégorie $I = (0 \rightrightarrows 1)$ précédente. La catégorie $\underline{\Gamma}(\mathcal{F}/I)$ s'interprète, à isomorphisme près, comme la catégorie des triples (x, u, v) avec $x \in \mathrm{Ob}\,\Pi$, $x' \in \mathrm{Ob}\,\Pi'$, et u et v des flèches [page 407]

$$\begin{array}{ccc} F(x) & & G(x) \\ & {\scriptstyle u}\searrow \quad \swarrow{\scriptstyle v} & \\ & x' & \end{array}$$

dans Π'. La catégorie $\mathcal{M}'$ est équivalente à la sous-catégorie strictement pleine de $\underline{\Gamma}(\mathcal{F}/I)$, formée des triples (x, u, v) tels que v soit un isomorphisme, i.e. à la sous-catégorie de $\underline{\Gamma}(\mathcal{F}/I)$, image inverse de $\underline{\mathrm{Is}}(\Pi')$ par le foncteur $(x, u, v) \longmapsto v$,

$$\underline{\Gamma}(\mathcal{F}/I) \longrightarrow \underline{\mathrm{Fl}}(\Pi') .$$

Or par 4.10.8, $\underline{\Gamma}(\mathcal{F}/I)$ est accessible, et le foncteur précédent est bien sûr accessible. Enfin, par 4.14.12, $\underline{\mathrm{Is}}(\Pi')$ est accessible dans la catégorie accessible $\underline{\mathrm{Fl}}(\Pi')$. Donc son image inverse $\mathcal{M}'$ dans $\underline{\Gamma}(\mathcal{F}/I)$ est accessible dans $\underline{\Gamma}(\mathcal{F}/I)$, donc accessible. Donc il en est de même de la catégorie équivalente, q.e.d. [page 408]

Commentary · Later developments [ms. 407]

The inserter and the later limit theorems

The auxiliary lemma is the familiar inserter theorem. Its proof in the manuscript is especially useful because it avoids treating an inserter as an unexplained closure operation. A pair of arrows $FX \xrightarrow{u} Y \xleftarrow{v} GX$ becomes an inserter object when v is invertible, by replacing the pair with $v^{-1}u : FX \to GX$. Conversely, an inserter object gives such a pair with $Y = GX$ and $v = \mathrm{id}$. On morphisms the two constructions are compatible, and the resulting comparison is an equivalence. The conclusion therefore follows from the already established accessibility of opfibration sections and of the isomorphism locus.

The surrounding theory had already acquired a systematic form in Makkai and Paré (1989, §5.1). Positselski (2024, Theorems 3.1, 4.1, and 8.1–8.3) gives explicit rank control for equifiers, inserters, and lax limits under a smaller-chain-colimit hypothesis. These hypotheses explain why existence of some accessibility rank is easier than retaining a prescribed rank. This literature confirms the construction here; it does not turn an arbitrary strict equalizer of accessible functors into an accessible category.

A small example makes the distinction visible. Let both F and G be the identity of Set. Their inserter is the category of sets equipped with one endomorphism, equivalently sets with an action of the free monoid $\mathbb{N}$. The object $(\mathbb{N}, n \mapsto n + 1)$ is free on one generator. A morphism from it to (X, u) is uniquely specified by the image x of 0, because the image of n must be $u^n(x)$. Hence

$$\mathrm{Hom}\big((\mathbb{N}, +1), (X, u)\big) \simeq X.$$

Filtered colimits of these actions are computed on the underlying sets, so the free action is finitely presentable. Its underlying set is infinite and is not finitely presentable in Set. To verify the latter

assertion, express $\mathbb{N}$ as the directed union of its finite subsets: its identity map does not factor through a finite subset. Thus the inserter projection is accessible while failing to preserve finitely presentable objects. There is no conflict with the manuscript's eventual comparisons after enlarging the cardinal.

For higher categories the corresponding section statement is explicit in Lurie (2009, Proposition 5.4.7.11 and Remarks 5.4.7.13–14): accessible fibres and accessible transport over a small simplicial base give accessible categories of sections, with specified cartesian or cocartesian conditions allowed. The higher theorem includes coherence as part of the section space. The ordinary equations written here give its one-categorical counterpart.

Ainsi, grâce aux résultats généraux sur les parties accessibles des catégories accessibles, le théorème 4.14.17, dans le cas où p est fibrant, est ramené aux cas cofibrant, dans le cas très particulier où la catégorie d'indices est $I = (0 \rightrightarrows 1)$. On pourrait d'ailleurs faire une réduction similaire dans le cas où p est cofibrant, par une démonstration pratiquement identique. En d'autres termes, le théorème 4.10.8 se ramène au cas où la catégorie d'indices p. 155
est $I = (0 \rightrightarrows 1)$. Mais je suspecte que la démonstration du théorème dans ce cas particulier, n'est pas essentiellement plus simple que dans le cas général.

4.14.23. On peut se demander de trouver un énoncé assurant l'accessibilité d'une catégorie [page 409]
$\underline{\Gamma}(\mathcal{E}/\mathcal{C})$, qui coiffe les deux cas envisagés dans 4.14.17. Voici ce que je peux proposer. Je vais faire l'hypothèse suivante sur les foncteur $p : \mathcal{E} \longrightarrow \mathcal{C}$ (en plus de celle que $\mathcal{C}$ soit essentiellement petite) :

H) Pour tout $\lambda \in \mathrm{Fl}\,\mathcal{C}$, la catégorie $\mathcal{E}_\lambda = \mathcal{E} \times_{\mathcal{C}} (\mathbf{\Delta}_1, i_\lambda)$, où $i_\lambda : \mathbf{\Delta}_1 \longrightarrow \mathcal{C}$ est le foncteur défini par λ, est accessible, et le foncteur canonique $\mathcal{E}_\lambda \xrightarrow{p_\lambda} \mathbf{\Delta}_1$ est accessible.

Notons que si $\lambda : s \longrightarrow t$ dans $\mathcal{C}$ est tel que $\mathcal{E}_\lambda$ et p_λ soient accessibles, alors $\mathcal{E}_s$ et $\mathcal{E}_t$ sont accessible, car isomorphes aux catégories image inverse par $p_\lambda : \mathcal{E}_\lambda \longrightarrow \mathbf{\Delta}_1$ des sous-catégories $\{0\}$ resp. $\{1\}$ de Δ_1, lesquelles sont accessibles dans la catégorie accessible $\mathbf{\Delta}_1$, donc $\mathcal{E}_{\lambda_0}$ et [page 410]
$\mathcal{E}_{\lambda_1}$ sont accessibles dans $\mathcal{E}_\lambda$, donc sont accessibles, donc aussi $\mathcal{E}_s$ et $\mathcal{E}_t$. Ainsi, l'hypothèse H) implique que les catégories fibres sont accessibles. Notons de plus :

Lemme 4.14.23.1. *Supposons que $p : \mathcal{E} \longrightarrow \mathcal{C}$ soit cofibrant ou fibrant, et soit $(\lambda : s \longrightarrow t) \in \mathrm{Fl}(\mathcal{C})$. Pour que $\mathcal{E}_\lambda$ soit accessible et $\mathcal{E}_\lambda \xrightarrow{p_\lambda} \mathbf{\Delta}_1$ accessible, il faut et il suffit que $\mathcal{E}_s$, $\mathcal{E}_t$ soient accessibles et $\lambda_* : \mathcal{E}_s \longrightarrow \mathcal{E}_t$ resp. $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ soit un foncteur accessible.*

Démonstration. On peut supposer que $\mathcal{C} = \mathbf{\Delta}_1$, et que $\mathcal{E}$ est la catégorie cofibrée (resp. fibrée) sur $\mathbf{\Delta}_1$ définie par un foncteur $\varphi : \mathcal{E}_0 \longrightarrow \mathcal{E}_1$ (resp. $\psi : \mathcal{E}_1 \longrightarrow \mathcal{E}_0$), et à montrer que $\mathcal{E}$ et $p : \mathcal{E} \longrightarrow \mathbf{\Delta}_1$ sont accessibles si et seulement si $\mathcal{E}_0$ et $\mathcal{E}_1$ et φ (resp. ψ) le sont.

Prouvons le "il faut". On a déjà vu que si $\mathcal{E}$, p sont accessibles, $\mathcal{E}_0$ et $\mathcal{E}_1$ sont accessibles [page 411]
dans $\mathcal{E}_s$, donc accessibles, et les foncteurs d'inclusion

$$\mathcal{E}_0 \hookrightarrow \mathcal{E} \hookleftarrow \mathcal{E}_1$$

accessibles. Dire que p est cofibrant (resp. fibrant) signifie que le foncteur d'inclusion $\mathcal{E}_1 \hookrightarrow \mathcal{E}$ admet un adjoint à droite $\bar{\varphi}$ (resp. que le foncteur $\mathcal{E}_0 \hookrightarrow \mathcal{E}$ admet un adjoint à gauche $\tilde{\psi}$). Alors φ (resp. ψ) est le composé $\mathcal{E}_0 \longrightarrow \mathcal{E} \xrightarrow{\bar{\varphi}} \mathcal{E}_1$ (resp. le composé $\mathcal{E}_1 \hookrightarrow \mathcal{E} \xrightarrow{\tilde{\psi}} \mathcal{E}_1$). Donc pour prouver que φ (resp. ψ) est accessible, il suffit de prouver que $\bar{\varphi}$ (resp. $\tilde{\psi}$) l'est. Cela est un cas particulier du

Lemme 4.14.23.2. *Soient $\mathcal{E}$, $\mathcal{E}'$ deux catégories accessibles ([172]), et $\mathcal{E} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{E}'$ un couple de foncteurs adjoints entre $\mathcal{E}$ et $\mathcal{E}'$. Alors F et G sont accessibles.*

En effet, F est accessible, car il commute aux petites $\varinjlim$ quelconques, a fortiori il satisfait [page 412]
L_{π_0} (où π_0 est tel que $\mathcal{E}$, $\mathcal{E}'$ satisfassent L_{π_0}). Prouvons que G est accessible. Si S est un petit p. 156

172. **N.B.** Il suffit que $\mathcal{E}$ admette une petite sous-catégorie génératrice, et que les objets de $\mathcal{E}'$ soient accessibles.

ensemble d'éléments générateur dans $\mathcal{E}$, de sorte que la petite famille des foncteurs

$$h_\xi \;=\; (x \longmapsto \mathrm{Hom}_{\mathcal{E}}(\xi, x)) \;:\; \mathcal{E} \;\longrightarrow\; \mathrm{Ens}$$

est conservative, pour prouver que G est accessible, il suffit de prouver que les $h_\xi \circ G$ ($\xi \in S \subset \mathrm{Ob}\,\mathcal{E}$) le sont. Or

$$L_\xi \circ G(x') \;=\; \mathrm{Hom}(\xi, G(x')) \;\simeq\; \mathrm{Hom}_{\mathcal{E}'}(F(\xi), x')\ ,$$

et comme $F(\xi)$ est un objet accessible de $\mathcal{E}'$, le foncteur

$$x' \;\longmapsto\; (L_\xi \circ G)(x') \;\simeq\; \mathrm{Hom}_{\mathcal{E}'}(F(\xi), x')$$

est bien accessible, q.e.d.

Cela prouve le "il faut" dans 4.14.23.1. Prouvons le "il suffit", i.e. supposons $\mathcal{E}_s$, $\mathcal{E}_t$ accessibles, et φ (resp. ψ) accessible.

[page 413]

Commentary · Exposition [ms. 412]

Why both adjoints are accessible

Grothendieck needs the adjunction lemma to recover a transition functor from the total category over an interval. The lemma concerns the existence of a sufficiently large filtered-colimit rank. It does not assert that a right adjoint preserves all filtered colimits, or that the same rank works for every adjunction.

Write $F : \mathcal{A} \rightleftarrows \mathcal{B} : G$ with $F \dashv G$. Choose a small strong generating family S of presentable objects of $\mathcal{A}$. Since the family is small and all the objects $F(s)$ are presentable at some ranks, choose one regular κ such that both categories admit κ-filtered colimits and every s and $F(s)$ is κ-presentable. For a κ-filtered diagram (b_i), consider

$$q : \varinjlim_i G(b_i) \longrightarrow G(\varinjlim_i b_i).$$

For each $s \in S$, adjunction and the two presentability assertions give

$$\begin{aligned}
\mathrm{Hom}_{\mathcal{A}}(s, \varinjlim_i Gb_i) &\simeq \varinjlim_i \mathrm{Hom}_{\mathcal{A}}(s, Gb_i) \\
&\simeq \varinjlim_i \mathrm{Hom}_{\mathcal{B}}(Fs, b_i) \\
&\simeq \mathrm{Hom}_{\mathcal{B}}(Fs, \varinjlim_i b_i) \\
&\simeq \mathrm{Hom}_{\mathcal{A}}(s, G \varinjlim_i b_i).
\end{aligned}$$

This composite is $\mathrm{Hom}(s, q)$, so the strong generator implies that q is an isomorphism. The left adjoint preserves all colimits that exist, which proves its accessibility at once. The modern reference for the lemma is Adámek and Rosický (1994, Proposition 2.23); the argument just given makes the otherwise implicit common-cardinal choice explicit.

The word "generating" on page 412 must include the stated conservativity, and the calculation also uses presentability of the chosen generators. Both are available for an accessible category. Footnote 172 suggests a weaker hypothesis on $\mathcal{A}$; the printed argument does not explain how to dispense with preservation by these representables or with the relevant colimits. We use the lemma under its stated accessible-category hypotheses and do not certify that additional weakening.

Two orientation slips on page 411 should be read through this argument. For an opfibration classified by $\varphi : \mathcal{E}_0 \to \mathcal{E}_1$, extend φ to $\bar\varphi : \mathcal{E} \to \mathcal{E}_1$ by the identity on $\mathcal{E}_1$. Then

$$\mathrm{Hom}_{\mathcal{E}}(x, i_1 y) \simeq \mathrm{Hom}_{\mathcal{E}_1}(\bar\varphi x, y), \qquad \bar\varphi \dashv i_1.$$

For a fibration classified by $\psi : \mathcal{E}_1 \to \mathcal{E}_0$, the extension satisfies $i_0 \dashv \widetilde{\psi}$. The edition prints the opposite adjunction directions, as well as the wrong final fibre in the composite defining ψ. Correcting the types leaves the proof of Lemma 4.14.23.1 intact because the next lemma applies to either adjoint.

Dans les deux cas, $\mathcal{E}_0$ est un ouvert de $\mathcal{E}$, $\mathcal{E}_1$ est le fermé complémentaire, et $\mathcal{E}$ est la catégorie "recollée" à l'aide du bifoncteur

$$H \ : \ \mathcal{E}_0^\circ \times \mathcal{E}_1 \ \longrightarrow \ \mathrm{Ens}$$

donné par

$$\begin{cases} H(x_0, x_1) \ = \ \mathrm{Hom}(\varphi(x_0), x_1) \\ \qquad \text{resp.} \\ H(x_0, x_1) \ = \ \mathrm{Hom}(x_0, \psi(x_1)) \ , \end{cases}$$

suivant qu'on est dans le cas p cofibrant ou p fibrant. On va utiliser la

Proposition 4.14.24 ([173]). *Soit $\mathcal{E}$ une catégorie, $\mathcal{E}_0$ une sous-catégorie ouverte, $\mathcal{E}_1$ la sous-catégorie fermée complémentaire. Conditions équivalentes :*

a) $\mathcal{E}$ est accessible, et $\mathcal{E} \longrightarrow \mathbf{\Delta}_1$ (de fibres $\mathcal{E}_0$, $\mathcal{E}_1$) est accessible.

b) $\mathcal{E}$ est accessible, et $\mathcal{E}_0$, $\mathcal{E}_1$ sont accessibles dans $\mathcal{E}$.

c) $\mathcal{E}_0$, $\mathcal{E}_1$ sont accessibles, et pour tout $x_0 \in \mathrm{Ob}\,\mathcal{E}_0$, le foncteur

$$x_1 \ \longmapsto \ \underline{\mathrm{Hom}}_{\mathcal{E}}(x_0, x_1) \ : \ \mathcal{E}_1 \ \longrightarrow \ \mathrm{Ens}$$

est accessible, et il existe π tel que pour tout $x_1 \in \mathrm{Ob}\,\mathcal{E}_1$, le foncteur [page 414]

$$x_0 \ \longmapsto \ \underline{\mathrm{Hom}}_{\mathcal{E}}(x_0, x_1)^\circ \ : \ \mathcal{E}_0 \ \longrightarrow \ (\mathrm{Ens})^\circ$$

soit π-accessible.

Admettons ce lemme `[plutôt cette proposition]`, achevons la démonstration de 4.14.23.1. On suppose $\mathcal{E}_0$, $\mathcal{E}_1$ et φ (resp. ψ) accessibles, on veut prouver la condition a) de 4.14.23.3. Pour ceci, il suffit de prouver la condition c). Or on a, dans le cas p cofibrant, p. 157

$$\mathrm{Hom}_{\mathcal{E}}(x_0, x_1) \ \simeq \ \mathrm{Hom}_{\mathcal{E}_1}(\varphi(x_0), x_1) \ ,$$

c'est bien accessible en x_1, puisque $\mathcal{E}_1$ est accessible, et π-accessible en x_0 si φ l'est, comme composé de φ avec $\mathcal{E}_1 \longrightarrow \mathrm{Ens}$, $\mathrm{Hom}(-, x_1)$, qui l'est. Dans le cas p fibrant, on a

$$\mathrm{Hom}_{\mathcal{E}}(x_0, x_1) \ \simeq \ \mathrm{Hom}_{\mathcal{E}_0}(x_0, \psi(x_1)) \ ,$$

c'est accessible en x_1, comme le composé du foncteur accessible $\psi : \mathcal{E}_1 \longrightarrow \mathcal{E}_0$, et du foncteur également accessible $\mathrm{Hom}(x_0, -) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}$, qui est π-accessible en x_0 tautologiquement.

Pour achever la démonstration de 4.14.23.1, il reste donc à prouver la proposition 4.14.24. On a déjà noté qu'on a a $\Longrightarrow$ b, et b $\Longrightarrow$ c clair, puisque le foncteur envisagé $\mathcal{E} \longrightarrow \mathrm{Ens}$ est le composé $\mathcal{E}_1 \overset{\text{incl.}}{\hookrightarrow} \mathcal{E} \xrightarrow{\mathrm{Hom}_{\mathcal{E}}(x_0, -)} \mathrm{Ens}$, est accessible comme composé de deux foncteurs accessibles. Reste donc à prouver c $\Longrightarrow$ a. Soit donc π_0 un cardinal utile pour $\mathcal{E}_0$, et tel que [page 415] les foncteurs

$$x_0 \ \longmapsto \ H(x_0, x_1) \ : \ \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$$

soient π_0-accessibles, soit π un cardinal $\geqslant \sup(\pi_0, \mathrm{card\,réd}\,\mathcal{E}_{0\,\pi_0})$ tel que $\pi^{\pi_0} = \pi$ (donc utile pour $\mathcal{E}_0$), utile pour $\mathcal{E}_1$, et tel que les foncteurs

$$x_1 \ \longmapsto \ H(x_0, x_1) \ : \ \mathcal{E}_1 \longrightarrow \mathrm{Ens}$$

soient π-accessibles, pour $x_0 \in \mathrm{Ob}\,\mathcal{E}_{0\,\pi_0}$. Soit $\mathcal{E}(\pi)$ la sous-catégorie strictement pleine de $\mathcal{E}$, telle que

$$\mathrm{Ob}\,\mathcal{E}(\pi) \ = \ \mathrm{Ob}\,\mathcal{E}_{0\,\pi} \cup \mathrm{Ob}\,\mathcal{E}_{1\,\pi} \ .$$

Je dis que

173. cf. énoncés plus généraux et démonstration sympathique, 4.15.1 et 4.15.4 (p. 425, 432).

1°) $\mathcal{E}$ satisfait à L_π, et $\mathcal{E}_0$ et $\mathcal{E}_1$ sont stables dans $\mathcal{E}$ sous L_π, les foncteurs $\mathcal{E}_0 \longrightarrow \mathcal{E}$, $\mathcal{E}_1 \longrightarrow \mathcal{E}$, $\mathcal{E} \longrightarrow \mathbf{\Delta}_1$ satisfont L_π ([174]).

2°) $\mathcal{E}(\pi) = \mathcal{E}_\pi$ (d'où un foncteur pleinement fidèle

$$\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \longrightarrow \mathcal{E} \ ).$$

3°) Ledit foncteur est essentiellement surjectif, donc une équivalence de catégories. [page 416]

Notons que 2°) et 3°) impliquent que $\mathcal{E}$ est accessible, et 1°) que $\mathcal{E} \longrightarrow \mathbf{\Delta}_1$ est accessible, d'où la condition a) du lemme qu'il s'agit d'établir. Donc tout revient à prouver les 1°, 2°, 3° ci-dessus.

Commentary · Exposition [ms. 414]

A collage is controlled by its cross-Homs

The passage to a gluing bifunctor frees the argument from the existence of chosen lifts. An open subcategory here is a sieve, and its closed complement is a cosieve. Thus there are no arrows from $\mathcal{E}_1$ to $\mathcal{E}_0$. The remaining arrows between the two pieces are recorded by

$$H : \mathcal{E}_0^{\mathrm{op}} \times \mathcal{E}_1 \longrightarrow \mathrm{Set}.$$

The category reconstructed from this information is now called the collage of H. Its composition laws involving a cross-arrow are exactly the two functorial actions of H.

The two preservation conditions in Proposition 4.14.24 have different jobs. The contravariant condition says that a colimit formed in $\mathcal{E}_0$ remains universal when tested against objects of $\mathcal{E}_1$. The covariant condition says that an object of $\mathcal{E}_0$ eventually remains small when maps from it are tested on filtered diagrams in $\mathcal{E}_1$. The first condition needs a common rank for all target objects; the second can initially have a different rank for each source object. Accessibility permits us to bound that second family on a small generating subcategory.

Here is the colimit argument without the incidental notation slips. Fix a regular κ such that both fibres have κ-filtered colimits and each $H(-, y)$ converts those in $\mathcal{E}_0$ to limits. A diagram contained in $\mathcal{E}_0$ has its fibrewise colimit in the total category because it has the required universal maps to targets in either fibre. A diagram that has reached $\mathcal{E}_1$ stays there after passing to a cofinal tail. Its colimit in $\mathcal{E}_1$ is also a colimit in the total category: there are no maps from this nonempty diagram to an object of $\mathcal{E}_0$. These two cases exhaust a filtered diagram.

The smallness argument is equally concrete. An object $x \in \mathcal{E}_1$ is κ-presentable in the collage precisely when it is κ-presentable in its fibre. For $x \in \mathcal{E}_0$ one must additionally require that $H(x, -)$ preserve κ-filtered colimits. The calculation on page 419 obtains this latter requirement from a small presentation of x in $\mathcal{E}_0$. Contravariance gives

$$H(\varinjlim_j a_j, -) \simeq \varprojlim_j H(a_j, -),$$

and a sufficiently filtered colimit of sets commutes with this small limit. This is why the cardinal controlling the presentation, the variance condition, and the size of its indexing category must be chosen together.

Once these properties hold at a useful common cardinal, the union of the two small generating subcategories generates the collage. This proves accessibility itself, not merely closure under filtered colimits. The proof also explains the two familiar cases. For an opfibration, $H(x, y) = \mathrm{Hom}(\varphi x, y)$, and the relevant size of a lower-fibre object is the size of φx. For a fibration, $H(x, y) = \mathrm{Hom}(x, \psi y)$, and preservation by ψ supplies the corresponding comparison.

1°) Comme $\mathcal{E}_1$ est fermée dans $\mathcal{E}$, il est clair que $\mathcal{E}$ est stable par toutes $\varinjlim$ représentables dans $\mathcal{E}$, et plus précisément, que pour un foncteur $I \longrightarrow \mathcal{E}_1$, sa limite inductive dans $\mathcal{E}_1$ existe p. 158

174. il suffit que $\mathcal{E}_0$ et $\mathcal{E}_1$ satisfassent L_π, et que

$$x_0 \longmapsto H(x_0, x_1) \ : \ \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$$

satisfasse L_π $\forall\, x \in \mathrm{Ob}\,\mathcal{E}$.

si et seulement si elle existe dans $\mathcal{E}$, et les deux sont isomorphes. Supposons I grande devant π, et soit $\varphi : I \longrightarrow \mathcal{E}$, $i \longmapsto x_i$. Distinguons deux cas.

a) φ prend ses valeurs dans $\mathcal{E}_0$. Soit x_0 sa limite inductive *dans* $\mathcal{E}_0$. Je dis que c'est aussi une limite inductive dans $\mathcal{E}$. Il faut prouver que pour $x_1 \in \mathrm{Ob}\,\mathcal{E}_1$, la flèche

$$\mathrm{Hom}_{\mathcal{E}}(x_0, x_1) \longrightarrow \varprojlim_{i\in I} \mathrm{Hom}_{\mathcal{E}}(x_i, x_1)$$

est isomorphe. Or c'est le cas par hypothèse sur π, [page 417]

$$\xi \longmapsto H(\xi, x_1) \ : \ \mathcal{E}_0 \longrightarrow (\mathrm{Ens})^\circ$$

satisfait à L_π pour tout x_1 dans $\mathrm{Ob}\,\mathcal{E}_1$.

b) φ ne se factorise pas par $\mathcal{E}_0$, i.e. $\exists\, i_0 \in \mathrm{Ob}\, I$ avec $\varphi(i_0) \in \mathrm{Ob}\,\mathcal{E}_1$. Alors on a $\varphi(i) \in \mathrm{Ob}\,\mathcal{E}_1$ pour tout $i \in \mathrm{Ob}\, I[i_0]$, où $I[i_0]$ désigne la sous-catégorie strictement pleine de I formée des i tels que $i \geqslant i_0$. Cette catégorie est cofinale dans I, donc grande devant π comme I, et pour que $\varinjlim_{I} x_i$ existe, il faut et il suffit que $\varinjlim_{I[i_0]} x_i$ existe, ce qui est le cas d'après les cas a) déjà traité.

Cela prouve donc en même temps que $\mathcal{E}$ satisfait à L_π, et que $\mathcal{E}_0$ et $\mathcal{E}_1$ sont L_π-stables. La démonstration montre en même temps que le foncteur $p : \mathcal{E} \longrightarrow \mathbf{\Delta}_1$ satisfait à L_π.

2°) De ceci résulte aussitôt que $\mathcal{E}_\pi \subset \mathcal{E}(\pi)$. Prouvons qu'on a aussi

$$\mathcal{E}(\pi) \ \subset \ \mathcal{E}_\pi \ ,$$

et distinguons encore deux cas, pour $x \in \mathrm{Ob}\, E(\pi)$, suivant que x est dans $\mathcal{E}_{0\,\pi}$ ou $\mathcal{E}_{1\,\pi}$. S'il est [page 418] dans $\mathcal{E}_{1\,\pi}$, pour prouver que

$$(*) \qquad \mathrm{Hom}_{\mathcal{E}}(x, \underbrace{\varinjlim y_i}_{y}) \xleftarrow{\sim} \varinjlim_{i} \mathrm{Hom}_{\mathcal{E}}(x, y_i)$$

(pour un système inductif $(y_i)_{i\in I}$ dans $\mathcal{E}$ grand devant π), si les y_i sont dans $\mathcal{E}_0$, donc $y \in \mathcal{E}_0$, alors les deux membres sont vides, et c'est OK. Sinon, on peut supposer (quitte à remplacer I par un $I[i_0]$) que les x_i sont dans $\mathcal{E}_1$, et alors c'est OK aussi, puisque y est aussi la $\varinjlim$ dans $\mathcal{E}_1$, d'après 1°).

Si x est dans $\mathcal{E}_{0\,\pi}$, alors $(*)$ est clair si les y_i sont dans $\mathcal{E}_0$. Sinon, on peut supposer encore que les y_i sont dans $\mathcal{E}_0$, et $(*)$ s'écrit

$$H(x, \varinjlim y_i) \xleftarrow{\sim} \varinjlim H(x, y_i) \ ,$$

pour $x \in \mathrm{Ob}\,\mathcal{E}_{0\,\pi}$, les y_i dans $\mathcal{E}_1$, I grande devant π. En d'autres termes, il faut prouver que le foncteur

$$x_1 \longmapsto H(x_0, x_1) \ : \ \mathcal{E}_1 \longrightarrow \mathrm{Ens}$$

est π-accessible, quand x_0 est dans $\mathcal{E}_{0\,\pi}$. Or par hypothèse sur π, il est π-accessible si x_0 est dans $\mathcal{E}_{0\,\pi_0}$. Mais si x est dans $\mathcal{E}_{0\,\pi}$, on aura p. 159 [page 419]

$$x_0 \ = \ \varinjlim_{i\in I} x_{0\,i} \qquad \begin{cases} I \ \text{grande devant} \ \pi_0, \\ \text{les} \ x_{0\,i} \ \text{dans} \ \mathcal{E}_{0\,\pi_0}, \\ \mathrm{card}\, I \leqslant \pi \end{cases}$$

(car π_0 est utile pour $\mathcal{E}_0$ par hypothèse, et π satisfait $\pi^{\pi_0} = \pi \geqslant \mathrm{card\,réd}\,\mathcal{E}_{\pi_0}$). Comme $x_0 \longrightarrow H(x_0, x_1) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}$ satisfait à L_{π_0}, pour tout x_1 dans $\mathcal{E}_1$, on aura donc

$$H(x_0) \ \simeq \ \varprojlim_{i\in I} H(x_{0\,i})$$

(où pour x_0 dans $\mathcal{E}_0$, $H(x)$ désigne le foncteur $x_1 \longmapsto H(x_0, x_1) : \mathcal{E}_1 \longrightarrow (\mathrm{Ens})$). Comme les foncteurs $H(x_{0\,i})$ sont π-accessibles, et $\operatorname{card} I \leqslant \pi$, il s'ensuit que leur $\varprojlim_I H(x_0)$ est π-accessible (SGA 4 I 9.6 (ii)). Cela achève de prouver donc 2°).

3°). Pour prouver que $\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \longrightarrow \mathcal{E}$ est essentiellement surjectif, il suffit de voir que tout $x \in \mathrm{Ob}\,\mathcal{E}_0$ et tout $x \in \mathrm{Ob}\,\mathcal{E}_1$ est dans l'image essentielle. Or cela résulte de l'hypothèse [page 420] que π est utile pour $\mathcal{E}_0$ et $\mathcal{E}_1$, et de 1°) qui implique que $\mathcal{E}_0 \longrightarrow \mathcal{E}$ et $\mathcal{E}_1 \longrightarrow \mathcal{E}$ satisfont L_π.

Cela achève de prouver la proposition 4.14.24, et de même le résultat suivant, plus précis que l'implication c) $\Longrightarrow$ a).

Corollaire 4.14.25. *Soient $\mathcal{E}$ une catégorie, $\mathcal{E}_0$ une sous-catégorie ouverte, $\mathcal{E}_1$ la sous-catégorie fermée complémentaire,*

$$H \;=\; ((x_0, x_1) \longmapsto \mathrm{Hom}_{\mathcal{E}}(x_0, x_1)) \;:\; \mathcal{E}_0^\circ \times \mathcal{E}_1 \;\longrightarrow\; \mathrm{Ens}$$

le foncteur recollement.

1°) Soit π un cardinal infini. Pour ce que $\mathcal{E}$ et $\mathcal{E}_0$ satisfassent à L_π, et le foncteur d'inclusion $\mathcal{E}_0 \hookrightarrow \mathcal{E}$ itou, il faut et il suffit que $\mathcal{E}_0$ satisfasse à L_π, et que pour tout x_1 dans $\mathcal{E}_1$, le foncteur $x_0 \longmapsto H(x_0, x_1) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$ satisfasse à L_π. Alors $\mathcal{E}_1$ et $\mathcal{E}_1 \hookrightarrow \mathcal{E}$ satisfont aussi à L_π, et on a [page 421]

$$\mathcal{E}_\pi \;\subset\; \mathcal{E}(\pi) \overset{\text{déf}}{=} \begin{array}{l} \textit{sous-catégorie strictement pleine de } \mathcal{E} \textit{ telle que} \\ \mathrm{Ob}\,\mathcal{E}(\pi) = \mathrm{Ob}\,\mathcal{E}_{0\,\pi} \cup \mathrm{Ob}\,\mathcal{E}_{1\,\pi}. \end{array}$$

2°) Supposons que la condition c) de 4.14.24 soit satisfaite, et soit π_0 un cardinal infini tel que π_0 soit utile pour $\mathcal{E}_0$, et que les foncteurs $x_0 \longmapsto H(x_0, x_1) : \mathcal{E}_0 \longrightarrow \mathrm{Ens}^\circ$ définis par les $x_1 \in \mathrm{Ob}\,\mathcal{E}_1$ soient π_0-accessibles. Soit π un cardinal tel que $\pi^{\pi_0} = \pi \geqslant \sup(\pi_0, \operatorname{card} \text{réd}\,\mathcal{E}_{0\,\pi_0})$, et tel que pour tout $x_0 \in \mathrm{Ob}\,\mathcal{E}_{\pi_0}$, le foncteur $x_1 \longmapsto H(x_0, x_1) : \mathcal{E}_1 \longrightarrow \mathrm{Ens}$ soit π-accessible. Alors on a

$$\mathcal{E}_\pi \;=\; \mathcal{E}(\pi)\ .$$

Commentary · Present state [ms. 420]

The missing closed-fibre hypothesis

The first part of Corollary 4.14.25 is stronger than the preceding proof and is false as printed. The proof assumed that both fibres had the relevant filtered colimits. The right-hand side of the corollary mentions only $\mathcal{E}_0$ and the functors $H(-, y)$, then attempts to deduce colimit existence in $\mathcal{E}_1$. Nothing in those conditions controls a diagram entirely within the closed fibre.

For a nonempty counterexample, fix the manuscript cardinal π and put $\kappa = \pi^+$. Take $\mathcal{E}_0$ to be the terminal category, $\mathcal{E}_1$ the ordinal κ regarded as an ordered category, and H the constant singleton functor. The collage is the ordinal κ with an additional least object. The open fibre has all filtered colimits, and every $H(-, y)$ takes its filtered colimits to limits. Nevertheless the identity diagram $\kappa \to \mathcal{E}_1$ is κ-filtered and has no upper bound in the collage, hence no colimit. This also shows why nonemptiness of the open fibre does not rescue the statement.

The correction is to add that $\mathcal{E}_1$ satisfies L_π on the right-hand side. Sufficiency is precisely the two-case argument above. For necessity, assume the left-hand side. A nonempty filtered diagram in $\mathcal{E}_1$ cannot have a colimit in $\mathcal{E}_0$, because there is no cocone from it to such an object. Its colimit in $\mathcal{E}$ therefore lies in $\mathcal{E}_1$ and is universal there. The preservation condition on H follows by testing a colimit from $\mathcal{E}_0$ against an object of $\mathcal{E}_1$. The assertions about fibrewise small objects then follow by restricting the relevant representables.

This omission does not invalidate Proposition 4.14.24: there the cardinal was explicitly chosen useful for both fibres. It matters when the corollary is reused as a separate fixed-cardinal criterion, which is exactly what its formulation invites the reader to do.

3°) Si π_0 et π sont comme dans 2°), et si de plus π est utile pour $\mathcal{E}_1$, alors π est utile pour $\mathcal{E}$, donc (en vertu de 2°)

$$\mathcal{E} \xleftarrow{\sim} \mathrm{Ind}_\pi(\mathcal{E}(\pi)) \ .$$

Nous pouvons maintenant énoncer la généralisation cherchée de 4.14.17 :

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Théorème 4.14.26 (généralisation de 4.14.13). *Soit $p : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur, avec $\mathcal{C}$ petite. On suppose que p satisfait à l'hypothèse H) de la page 409, qui équivaut (moyennant 4.14.24) à la suivante :*

H′) $\forall\ s \in \mathrm{Ob}\,\mathcal{C}$, la catégorie fibre $\mathcal{E}_s$ est accessible, et pour tout $\lambda : s \longrightarrow t$ dans $\mathrm{Fl}\,\mathcal{C}$, le foncteur

$$\begin{cases} H_\lambda : \mathcal{E}_s^\circ \times \mathcal{E}_t & \longrightarrow & \mathrm{Ens} \\ \qquad (x_0, x_1) & \longmapsto & \mathrm{Hom}_\lambda(x_0, x_1) \end{cases}$$

satisfait les deux conditions suivantes :

a) $\forall\ x_0 \in \mathrm{Ob}\,\mathcal{E}_s^\circ$, $x_1 \longmapsto H_\lambda(x_0, x_1) : \mathcal{E}_t \longrightarrow \mathrm{Ens}$ est accessible.

b) Il existe un cardinal infini π_λ, tel que pour tout $x_1 \in \mathrm{Ob}\,\mathcal{E}_t$, le foncteur $x_0 \longmapsto H_\lambda(x_0, x_1) : \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$ soit π_λ-accessible.

Ceci posé, la catégorie $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$ des sections de $\mathcal{E}$ sur $\mathcal{C}$ est accessible, et les foncteurs $\varepsilon_s : \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \longrightarrow \mathcal{E}_s$ sont accessibles. On peut préciser de façon similaire à 4.14.18 (énoncé en forme laissé au lecteur).

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Démonstration de 4.14.26. Elle se fait exactement comme celle de 4.14.17 (p. 400 ff.), mais est conceptuellement plus simple. On suppose $\mathcal{C}$ petite, et on introduit les catégories

$$\Pi \ = \ \prod_{s \in \mathrm{Ob}\,\mathcal{C}} \mathcal{E}_s \ , \qquad \Pi' \ = \ \prod_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \mathcal{E}_\lambda \ ,$$

où

$$\mathcal{E}_\lambda \ = \ \mathcal{E} \times_{\mathcal{C}} (\mathbf{\Delta}_1, i_\lambda) \ ,$$

et les deux foncteurs canoniques

$$\Pi \underset{G}{\overset{F}{\rightrightarrows}} \Pi'$$

de composantes

$$\begin{aligned} F_\lambda \ &= \ i_\lambda \circ \mathrm{pr}_{s(\lambda)} \\ G_\lambda \ &= \ j_\lambda \circ \mathrm{pr}_{b(\lambda)} \ , \end{aligned}$$

où pour $\lambda : s \longrightarrow t$, on pose

$$s(\lambda) \ = \ s \ , \qquad b(\lambda) \ = \ t \ ,$$

et i_λ, j_λ sont les inclusions canoniques

$$\begin{array}{ccccc} \mathcal{E}_s & \xrightarrow{i_\lambda} & \mathcal{E}_\lambda & \xleftarrow{j_\lambda} & \mathcal{E}_t \ . \\ \big\downarrow \wr & & \big\downarrow & & \big\downarrow \wr \\ (\mathcal{E}_\lambda)_0 & & {\scriptstyle p_\lambda}\big\downarrow & & (\mathcal{E}_\lambda)_1 \\ & & \mathbf{\Delta}_1 & & \end{array}$$

On désigne par $\mathcal{M}'$ la catégorie

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[page 424]

$$\mathcal{M}' \ = \ \begin{array}{l} \text{catégorie des couples } (X, u), \text{ avec} \\ X \in \mathrm{Ob}\,\Pi,\ u : F(x) \longrightarrow G(x), \end{array}$$

qui est constructible par 4.14.22.1 est accessible (car Π et π' et les foncteurs F et G le sont, par l'hypothèse H) faite sur p). On peut l'interpréter comme la catégorie des sections de $\mathrm{Diag}(\mathcal{E})$ sur $\mathrm{Diag}(\mathcal{C})$. On a un foncteur strictement pleinement fidèle

$$\mathcal{M} \overset{\text{déf}}{=} \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \hookrightarrow \mathcal{M}' ,$$

qui permet d'interpréter $\mathcal{M}$ comme une sous-catégorie strictement pleine de $\mathcal{M}'$. On voit alors comme dans le cas de 4.14.17 que cette sous-catégorie est accessible dans $\mathcal{M}'$ – c'est essentiellement sorital. Il s'ensuit que $\mathcal{M}$ est accessible et que les foncteurs $\mathcal{M} \longrightarrow \mathcal{E}_s$ le sont aussi, comme composés de

$$\mathcal{M} \overset{\text{acc.}}{\hookrightarrow} \mathcal{M}' \overset{\text{acc.}}{\hookrightarrow} \Pi \xrightarrow{\mathrm{pr}_s \quad \text{acc.}} \mathcal{E}_s ,$$

Commentary · Exposition [ms. 423]

Sections without a fibration structure

Theorem 4.14.26 identifies how much of the section argument actually used cartesian or cocartesian transport. A single base arrow $\lambda : s \to t$ determines a collage $\mathcal{E}_\lambda$, whether or not a preferred lift exists. The hypotheses say that this collage and its endpoint inclusions are accessible. That is enough to choose the arrow data of a section by an inserter.

For a family $X = (X_s)$, an arrow

$$i_\lambda X_s \longrightarrow j_\lambda X_t \quad \text{in } \mathcal{E}_\lambda$$

is exactly an arrow $X_s \to X_t$ of $\mathcal{E}$ lying over λ. Thus the new inserter is again the category of graph sections. The remaining equations are literal identities and compositions in $\mathcal{E}$. No comparison isomorphisms from a cleavage occur.

One accessibility check is compressed into the word "sorital" on page 424. Choose a common regular cardinal at which the fibres, the interval categories, their endpoint inclusions, and the graph-section construction admit and preserve filtered colimits. Given a filtered diagram of graph sections, form all its object values in the fibres. The structural arrow over λ is the unique map induced by the corresponding colimit cocones in $\mathcal{E}_\lambda$. For composable λ and μ, composition of the induced arrows agrees with the arrow induced by the composites: precomposing with every object of the source colimit gives the same arrow, and the universal property gives uniqueness. Consequently the two functors recording $a_{\mu\lambda}$ and $a_\mu a_\lambda$ preserve these filtered colimits. The same argument handles identity arrows.

To impose their equality, regard the pair as a diagram of parallel arrows in $\mathcal{E}_{\mu\lambda}$, with fixed endpoint degrees. The category of equal pairs is accessible and accessibly embedded, just as on pages 405–406. There are only a set of objects and composable pairs in the small base. Pulling back these equation loci and intersecting them gives an accessible full subcategory of graph sections. Its evaluation functors are accessible by composition with the inserter projection and the product projections. The hooked arrow $\mathcal{M}' \hookrightarrow \Pi$ in the last display should therefore be an ordinary forgetful arrow; the argument only uses its accessibility.

This proof is local along arrows in a precise sense. It tests the categories of arrows over each edge, while functoriality supplies the relations among edges. It does not assert that the total category $\mathcal{E}$ is accessible for an arbitrary small base. The next section imposes an additional condition on idempotents in the base to address that separate conclusion.

q.e.d.

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4.15 Catégorie accessible sur une autre

Proposition 4.15.1. *Soit $f : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur avec $\mathcal{C}$ petite. On suppose de plus satisfaite la condition H (p. 409), qu'on explicite ici sous la forme équivalente H′ :*

H 1) Pour tout $s \in \mathrm{Ob}\,\mathcal{C}$, $\mathcal{E}_s$ est accessible.

H 2) Pour toute $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, existe un cardinal infini π_λ tel que pour tout $x_1 \in \mathrm{Ob}\,\mathcal{E}_t$, le foncteur

$$x_0 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) \ : \ \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$$

est π_λ-accessible ($\mathcal{E}_s$ satisfaisant de plus à L_{π_λ}).

H 3) Pour toute $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, et pour tout $x_0 \in \mathrm{Ob}\,\mathcal{E}_s$, le foncteur

$$x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) \ : \ \mathcal{E}_t \longrightarrow (\mathrm{Ens})$$

est accessible.

Alors les foncteurs d'inclusion $i_s : \mathcal{E}_s \longrightarrow \mathcal{E}$ sont accessibles. De plus, si $\mathcal{C}$ "est sans projecteurs" (i.e. si tout endomorphisme p d'un objet s de $\mathcal{C}$ tel que $p^2 = p$, est l'identité), alors $\mathcal{E}$ est accessible, et $f : \mathcal{E} \longrightarrow \mathcal{C}$ est accessible.

Cela résulte du résultat plus précis :

Commentary · Development (commentators') [ms. 425]

A presentability test and a necessary rank repair

Grothendieck's next aim is to reconstruct accessibility of the total category from its fibres. The explicit bounds in Corollary 4.15.2 are meant to make this reconstruction reusable. They omit one condition needed by the subsequent calculation: the initial presentation rank π_0 must already control the contravariant functors in H2. Bounding those ranks only by π_1 does not justify applying them to a π_0-filtered presentation. We make the issue concrete before giving a repair.

Fibrewise presentability test over a finite poset. [ours]

Let $p : \mathcal{A} \to P$ have finite poset base. Fix a regular infinite κ. Suppose every fibre $\mathcal{A}_s$ has κ-filtered colimits and, for $s \le t$ and $y \in \mathcal{A}_t$, the functor

$$\mathrm{Hom}_{s\le t}(-, y) : \mathcal{A}_s^{\mathrm{op}} \to \mathrm{Set}$$

sends these colimits to limits. Then the total category has κ-filtered colimits, computed in a cofinal fibre, and each fibre inclusion preserves them. An object $x \in \mathcal{A}_s$ is κ-presentable in $\mathcal{A}$ if and only if every functor $\mathrm{Hom}_{s\le t}(x, -)$, for $t \ge s$, preserves κ-filtered colimits. The case $t = s$ includes presentability in the fibre.

Proof. The image of a filtered diagram in the finite poset is a finite directed subset and therefore has a greatest element t. Choose an index mapping to t. The overcategory of that index is cofinal, and all its images equal t; the diagram has consequently reduced to one fibre. To show that a fibrewise colimit is universal in the total category, test maps to an object in every fibre. If there is no base arrow the relevant sets are empty; otherwise the assumed contravariant comparison is the required bijection. This proves colimit existence and preservation. If x is presentable in the total category, restrict its representable to a diagram in $\mathcal{A}_t$, using the just-proved preservation. Conversely reduce any target diagram to its cofinal fibre $\mathcal{A}_t$; the asserted preservation of the corresponding cross-Hom gives precisely the presentability comparison for x. This proves the test.

If the fibres are κ-accessible and every fibrewise κ-presentable object passes this test, the union of their small generating subcategories consists of total κ-presentables and generates $\mathcal{A}$. Hence $\mathcal{A}$ is κ-accessible. This last statement follows directly from the definition and does not require additional colimits.

Counterexample to the printed bound in 4.15.2(c). [ours]

Use Grothendieck's convention: π-presentable sets have cardinal at most π, corresponding to modern π^+-presentability. Put

$$|K| = \aleph_1, \qquad \pi_0 = \aleph_0, \qquad \pi_1 = 2^{\aleph_1}, \qquad \pi = \beth_{\omega_1}.$$

Take the opfibration over [1] classified by

$$\varphi : \mathrm{Set} \to \mathrm{Set}, \qquad X \longmapsto X^K.$$

Its cross-Hom is $H(X, Y) = \mathrm{Set}(X^K, Y)$. The functor φ preserves $\aleph_2$-filtered colimits: a K-indexed family and all equalities among its representatives are witnessed at one stage because $|K| < \aleph_2$. Thus H2 holds with $\pi_\lambda = \aleph_1$. Both fibres are useful at π_0, and their reduced π_0-presentable subcategories

have size at most $2^{\aleph_0}$. For countable X, the set X^K has size at most $2^{\aleph_1} = \pi_1$, so $H(X,-)$ is π_1-accessible. All three printed bounds of part (c) are satisfied.

The cardinal π is strong limit of cofinality $\aleph_1$. Every countable sequence in π is bounded, whence $\pi^{\aleph_0} = \pi$: for each bound $\nu < \pi$, one has $\nu^{\aleph_0} \leq 2^{\nu \cdot \aleph_0} < \pi$, and there are π possible bounds. In contrast, König's inequality gives $\pi^{\aleph_1} > \pi$. Choose a lower-fibre set X of size π. It is π-presentable in its fibre, but X^K has size greater than π. The π^+-filtered diagram of all subsets of X^K of size at most π, placed in the upper fibre, has colimit X^K. The identity of X^K does not factor through any of those subsets. Thus $H(X,-)$ does not preserve this colimit, and the fibrewise test proves that X is not π-presentable in the total category. The asserted inclusion $\mathcal{E}(\pi) \subseteq \mathcal{E}_\pi$ fails at the printed bounds.

The qualitative proposition is repaired by choosing π_0 useful for all fibres *and* at least every π_λ, then making the subsequent choices of π_1 and π as before. Now a π_0-filtered presentation may legitimately be passed through each contravariant $H_\lambda(-,y)$; the small-limit calculation on page 419 applies. This is also the missing justification in the continuation of the proof on manuscript page 430. We found no discussion of this precise printed rank omission in the accessible-category references checked here. The test is a direct reorganization of the manuscript's proof, and the counterexample concerns its formulation; neither is asserted as a priority claim.

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Corollaire 4.15.2 ([175]). *Les hypothèses sur $f : \mathcal{E} \longrightarrow \mathcal{C}$ sont celles de 4.15.1 ($\mathcal{C}$ petite, et f satisfait $H' = H$ 1) 2) 3)).*

a) Soit $s \in \mathrm{Ob}\,\mathcal{C}$ et soit π_1 un cardinal infini tel que pour toute $\lambda : s \longrightarrow t$ (flèche de $\mathcal{C}$ de source t), on ait $\pi_1 \geqslant \pi_\lambda$ (où π_λ est défini dans l'énoncé de H 2). Alors le foncteur $i_s : \mathcal{E}_s \longrightarrow \mathcal{E}$ est π_1-accessible.

b) On suppose $\mathcal{C}$ sans projecteurs. Soit π_1 un cardinal $\geqslant \sup(\pi_\lambda, \mathrm{card}\,\mathcal{C})$. Alors $\mathcal{E}$ satisfait L_{π_1}, et $f : \mathcal{E} \longrightarrow \mathcal{C}$ est π-accessible.

c) Supposons encore que $\mathcal{C}$ soit sans projecteurs (p.ex. une catégorie "rigide", p.ex. une catégorie préordonnée). Soient π_0, π_1 des cardinaux tels que :

① $\forall\, s \in \mathrm{Ob}\,\mathcal{C}$, π_0 est utile pour $\mathcal{E}_s$ (un tel π_0 existe grâce à H 1 et au fait que $\mathcal{C}$, donc $\mathrm{Ob}\,\mathcal{C}$ petits).

② $\pi_1 \geqslant \sup(\pi_0, \mathrm{card}\,\mathcal{C}, \sup_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda, \sup_{s \in \mathrm{Ob}\,\mathcal{C}} \mathrm{card\,réd}\,\mathcal{E}_{s\,\pi_0})$.

③ $\forall \lambda \in \mathrm{Fl}\,\mathcal{C}$, et $x_0 \in \mathrm{Ob}\,\mathcal{E}_{s,\pi_0}$, le foncteur $H_\lambda(x_0)$ de H 3, $x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) : \mathcal{E}_t \longrightarrow \mathrm{Ens}$ est π_1-accessible.

Soit π un cardinal tel que $\pi^{\pi_0} = \pi \geqslant \pi_1$. Alors par b), $\mathcal{E}$ satisfait L_{π_1}, donc L_π, et $f : \mathcal{E} \longrightarrow \mathcal{C}$ est π_1-accessible. De plus,

$$i_s(\mathcal{E}_{s\,\pi}) \;\subset\; \mathcal{E}_\pi \;,$$

ou encore

$$(*) \qquad \mathcal{E}(\pi) \;\subset\; \mathcal{E}_\pi \;,$$

où $\mathcal{E}(\pi)$ est la sous-catégorie pleine de $\mathcal{E}$ telle que

$$\mathrm{Ob}\,\mathcal{E}(\pi) \;=\; \bigcup_{s \in \mathrm{Ob}\,\mathcal{C}} \mathrm{Ob}\,(\mathcal{E}_{s\,\pi}) \;.$$

Comme $\mathcal{E}_\pi$ satisfait L_{π_1} par b), donc a fortiori L_π, on a donc par $(*)$ un foncteur pleinement fidèle

$$\mathrm{Ind}_\pi(\mathcal{E}_{s\,\pi}) \;\xrightarrow{\sim}\; \mathcal{E} \;,$$

et ce foncteur est une équivalence de catégories (donc $\mathcal{E}$ est accessible).

d) En fait, sous les hypothèses de c) sur $\mathcal{C}$, π_0, π_1, π, on a $\mathcal{E}(\pi) = \mathcal{E}_\pi$ (non seulement $\mathcal{E}(\pi) \subset \mathcal{E}_\pi$).

[page 427]

Démonstration de 4.15.1 et 4.15.2. Il est clair que 4.15.1 est contenu dans 4.15.2 a) et c). Nous allons donc prouver 4.15.2.

a) Soit $(x_i)_{i\in I}$ un système inductif dans $\mathcal{E}_s$, avec I grand devant π_0, et x sa limite inductive dans $\mathcal{E}_s$ (qui existe, car $\mathcal{E}_s$ satisfait à L_{π_λ}, donc à L_{π_0}). Il faut prouver que c'est une limite dans $\mathcal{E}$, donc que pour tout $y \in \mathrm{Ob}\,\mathcal{E}$, la flèche canonique

$$(*) \qquad \mathrm{Hom}_{\mathcal{E}}(x,y) \longrightarrow \varprojlim_I \mathrm{Hom}_{\mathcal{E}}(x_i,y)$$

est un isomorphisme. Or $\mathrm{Hom}_{\mathcal{E}}(x,y)$ et les $\mathrm{Hom}_{\mathcal{E}}(x_i,y)$ sont des objets de $\mathrm{Ens}/\mathrm{Hom}_{\mathcal{C}}(s,t)$, où $s=f(x)$, $t=f(y)$, et les flèches de transition sont des flèches dans $\mathrm{Ens}/\mathrm{Hom}_{\mathcal{C}}(s,t)$. Cela montre que la bijectivité de $(*)$ revient à celle de p. 163

$$(*) \qquad \mathrm{Hom}_{\lambda}(x,y) \longrightarrow \varprojlim_I \mathrm{Hom}_{\lambda}(x_i,y)$$

pour tout $\lambda \in \mathrm{Hom}_{\mathcal{C}}(x,y)$ [plutôt $\mathrm{Hom}_{\mathcal{C}}(f(x),f(y))$]. Pour λ fixé, et y fixé dans $\mathrm{Ob}\,\mathcal{E}_t$, la bijectivité dans $(*)$, pour tout système inductif grand devant π_1 dans $\mathcal{E}_s$, de limite x, signifie que le foncteur

$$x \longmapsto \mathrm{Hom}_{\lambda}(x,y) \,:\, \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$$

est π_0-accessible. Or c'est bien le cas, par hypothèse sur π_1, quel que soit la flèche $\lambda : s \longrightarrow t$ de source s, et quel que soit $y \in \mathrm{Ob}\,\mathcal{E}_t$.

Soit $(x_i)_{i\in I}$ un système inductif grand devant π_1, et $(s_i)_{i\in I}$ l'image de ce système dans $\mathcal{C}$. [page 428] Nous prouverons plus bas le

Lemme 4.15.3 ([176], [177]). *Soit $\mathcal{C}$ une catégorie, π un cardinal, tels que* $\mathrm{card}\,\mathcal{C} \leqslant \pi$, *et que $\mathcal{C}$ soit sans projecteurs. Alors pour tout système inductif $(s_i)_{i\in I}$ grand devant π dans $\mathcal{C}$, existe une partie cofinale J dans I telle que le système induit $(s_i)_{i\in J}$ soit* constant, *i.e. tel que les s_i soient égaux à un $s \in \mathcal{C}$, et que les morphismes de transition soient tous égaux à* id_s.

Mais attention, on s'appuie pour prouver 4.15.3 sur le théorème 4.8.13 (page 212-216), que je n'ai toujours pas prouvé. Dans le cas où $\mathcal{C}$ est rigide (i.e. $\mathrm{End}_{\mathcal{C}}(s) = \{\mathrm{id}_s\}\ \forall\ s \in \mathrm{Ob}\,\mathcal{C}$), la démonstration est cependant immédiate, cf. plus bas.

Pour prouver que $x = \varinjlim_I x_i$ existe dans $\mathcal{E}$, et commute à f, on peut remplacer I par une [page 429] sous-catégorie cofinale. Cela nous ramène, grâce au lemme, au cas où $(s_i)_{i\in I}$ est constant, donc au cas où le système inductif provient d'un système inductif dans une catégorie fibre $\mathcal{E}_s$. Mais par a), on sait que sa limite x dans $\mathcal{E}_s$ est une limite dans $\mathcal{E}$, d'où l'existence de cette dernière, et la commutation à f, puisque $\varinjlim s_i = s$ ($\varinjlim$ d'un système inductif filtrant constant).

Commentary · Examples [ms. 428]

What an idempotent in the base can obstruct

The restriction on the base is meant to force a sufficiently filtered diagram to become constant after a cofinal change of indices. If the base permits a nonidentity idempotent, such a diagram can instead settle on repeated applications of that idempotent. The simplest example shows why the local hypotheses alone do not guarantee accessibility of the total category.

Let B have one object $*$ and endomorphism monoid $\{1,e\}$ with $e^2 = e \neq 1$. Take $p = \mathrm{id}_B$. Its strict fibre is terminal. For either base arrow 1 or e, the pullback $B \times_B [1]$ is $[1]$, with its identity projection, so hypothesis H holds. Nevertheless B is not accessible. For any infinite regular κ, take the diagram indexed by the ordinal κ, constant on $*$, with every strictly increasing transition equal to e. This is a well-defined κ-filtered diagram because $e^2 = e$.

175. Pour a) et b) on n'utilise que H 1 et H 2, à l'exclusion de H 3.

176. Cf. démonstration page 452.

177. sous réserve de la validité de 4.8.13 (Mais c'est à présent établi, sous une forme suffisante, dans 4.15.10).

A cocone to $*$ is a family (a_i) with $a_j e = a_i$ whenever $i < j$. Since both $1e$ and ee equal e, and every index has a larger index, every a_i must be e. There is exactly one cocone. Were it colimiting, the map from $\mathrm{Hom}_B(*,*)$ to this singleton cocone set would be bijective. It sends both 1 and e to the same cocone and is not injective. There is no other possible colimit object. Hence B has no such colimit at any regular rank.

This is the elementary splitting obstruction: a colimit of the repeated-idempotent system would supply the missing image of e. The example explains the hypothesis without claiming it is necessary in every situation. A base with split nonidentity idempotents may be accessible, and additional total-category hypotheses can also supply the missing splittings.

For the general constant-subdiagram lemma the manuscript itself records a deferred dependency. We checked the continuation in the editors' complete Chapter XVIII: page 452 still uses 4.8.13, and Corollaries 4.15.10–11 on pages 462–463 finally justify the argument with the strict bound $\pi > \operatorname{card}\mathcal{C}$. Thus the qualitative conclusion of 4.15.1 is secured by increasing the chosen cardinal. The boundary equality $\pi = \operatorname{card}\mathcal{C}$ is not established by that continuation, and we do not report it as proved here. Separately, the two displayed slips on page 426 must be read as colimit existence in $\mathcal{E}$, and the equivalence $\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \simeq \mathcal{E}$ using all fibres. These pages end during the next presentability calculation.

c) Il faut prouver que si $x \in \mathrm{Ob}\,\mathcal{E}_s$ est π-accessible dans $\mathcal{E}_s$, il l'est dans $\mathcal{E}$, i.e. que pour tout système inductif $(y_i)_{i\in I}$ grande devant π, la flèche canonique

$$(**) \qquad \varinjlim_i \mathrm{Hom}(x, y_i) \longrightarrow \mathrm{Hom}(x, y)$$

est bijective. Or comme par 2°) $\pi \geqslant \pi_1 \geqslant \sup(\operatorname{card} S, \sup_\lambda \pi_\lambda)$, il résulte encore du lemme que, quitte à remplacer I par une sous-catégorie cofinale, on peut supposer que le système inductif $(y_i)_{i\in I}$ provient d'un système dans une catégorie fibre $\mathcal{E}_t$, et que y est sa $\varinjlim$ dans $\mathcal{E}_t$. Mais alors les ensembles $\mathrm{Hom}(x,y)$ et $\mathrm{Hom}(x,y_i)$, et les morphismes de transition entre celles-ci, sont encore dans $\mathrm{Ens}/\mathrm{Hom}(s,t)$, et on est ramené

Overview · XVIII-d

Accessible conditions, categories of structures, and gluing.

These pages turn accessibility into a working calculus. Their central question is how a condition on the objects of a large category can be recognized, transported, and imposed without losing a small description of the resulting category. The first part completes a criterion for accessible full subcategories. The second applies it to functors preserving specified cones and cocones, to categories of sections, and finally to categories lying over a small base. The progression is deliberate: once membership in a subcategory can be read on a suitable presentation, many apparently new constructions become instances of the same argument.

One convention must be fixed before comparing the manuscript with current accounts. Grothendieck calls an indexing category *grande devant* π when it admits cocones on diagrams having at most π arrows. Thus the associated regular cardinal in present terminology is $\lambda = \pi^+$, and his π-accessible objects are λ-presentable objects. A useful π provides an essentially small full subcategory $\mathcal{M}_\pi$ generating $\mathcal{M}$ under these filtered colimits. The stronger phrase accessible *in* an ambient category keeps track of the inclusion and of ambient presentability, not just an abstract equivalence class of accessible categories.

The opening argument is a transfinite alternation between two presentations. A final functor $\varphi : I \to J$ and a full replete final subcategory $J' \subset J$ do not in general have a final strict inverse image: the even and odd natural numbers in their common ordered set already show the problem. Grothendieck adds an ordinal-colimit hypothesis. Starting from a chosen object of I, he constructs alternating maps

$$\varphi(i_\beta) \longrightarrow j_\beta \longrightarrow \varphi(i_{\beta+1}), \qquad j_\beta \in J',$$

whose composites are transition maps. At the final limit, the two chains have isomorphic colimits. One colimit comes from I, and the other lies in J'; repleteness then gives an object of the inverse image beyond the prescribed starting object. Lemma 4.13.17 is therefore a genuine cofinality statement with an additional closure hypothesis. The commentary supplies its final comparison explicitly and shows that the successor-stage equalizations in the manuscript are redundant.

This lemma drives Theorem 4.13.22. Suppose $\underline{S}$ is an accessible full subcategory of $\mathcal{M}$ closed under the indicated colimit shapes. In an adapted presentation $x = \varinjlim_i x_i$, membership $x \in \underline{S}$ is equivalent to finality of the full subcategory of indices for which $x_i \in \underline{S}$. Adaptedness says that the indexing category contains the specified colimits and that the presenting diagram preserves them. It is what prevents a presentation from hiding the condition at every finite stage. The worked example of divisible abelian groups exhibits the criterion through countable divisible subgroups; the familiar presentation of $\mathbb{Q}$ by cyclic groups explains why an arbitrary filtered presentation is insufficient.

The subsequent calculus treats enlargement of the accessibility rank, small unions and intersections, and inverse images under accessible functors. It also exposes a boundary of the method. The essential full image of an accessible functor need not be accessibly embedded, even after closing under retracts. We work out the split-monomorphism example recorded in Rosický (2009); Brooke-Taylor and Rosický (2017): forgetting chosen retractions allows diagrams whose colimit no longer splits. This failure is distinct from the controlled envelope construction. Starting with a small family of objects and closing it under a fixed filteredness threshold and a small set of additional colimit shapes does yield an accessible, accessibly embedded subcategory. The commentary proves this by choosing a sufficiently closed cardinal, making bounded filtered envelopes of index categories, and identifying the result with a fully faithful Ind-image.

The historical framework is already visible in Artin et al. (1972, Exposé I, 8.7), where Ind-objects and their universal extension property are developed. The systematic accessible-category theory of Makkai and Paré (1989) was published in 1989, before these pages were

written. The relation is therefore a comparison of formulations, not a claim that this manuscript first announced a theory later discovered by those authors. The modern presentation separates products, inserters, equifiers, and pseudopullbacks; Grothendieck often constructs the same operations through accessible loci in categories of diagrams. Positselski (2024) makes especially clear how additional chain-colimit hypotheses permit control of a specified rank.

Section 4.14 begins with marked cones and cocones in a small category. A functor is required to take the marked data to genuine limits or colimits. When the target has the relevant universal constructions, each condition is the inverse image of an isomorphism locus under an accessible comparison functor. Their small intersection is again accessible. This is the language of sketches and categories of models. Restriction to presentable objects then allows certain large-source functor categories to be treated by the same method.

Here the hypotheses require particular care. The proof of the fixed-threshold functor-category assertion invokes target colimits not supplied by accessibility alone. We distinguish the theorem established with the additional target-colimit hypothesis from the more general assertion the manuscript leaves to its later question about colimit cones. We also distinguish the two adjoint theorems: cocontinuous functors from a locally presentable category have right adjoints, whereas a limit-preserving covariant functor need not have a left adjoint unless a solution-set or accessibility condition is added. The stronger claim at pages 379–380 fails already for the category of groups, a warning that Grothendieck himself recalled near the beginning of this chapter. The counterexample and the appropriate repair are explained with the primary references.

The question about epimorphisms has a different status. Grothendieck observes that epimorphisms are stable under the relevant filtered colimits and asks whether their full subcategory in the arrow category is always accessible. Closure alone does not answer that question. The commentary proves that accessibility and accessible embedding of this epimorphism locus would force the underlying accessible category to be co-wellpowered. Published results show that the universal co-wellpoweredness assertion already has a large-cardinal lower bound, while stronger large-cardinal assumptions give positive results (Bagaria et al., 2015). This identifies a real set-theoretic obstruction. It does not settle the exact strength of the epimorphism-locus assertion. The recent study of pure epimorphisms by Positselski (2026) concerns an additional purity condition; its publisher abstract was checked as a status lead, and no result about the unrestricted epimorphism class is inferred from it.

Categories of sections provide the main constructive application. For a fibration, a prospective section consists of a family of fibre objects and arrows $X_s \to \lambda^* X_t$. Those data form an inserter. Identity and composition impose a small family of equations, and cartesian or cocartesian requirements impose isomorphism conditions. The proof keeps track of a nonsplit cleavage through its canonical coherence isomorphisms. It is thereby a genuinely categorical construction, not a pointwise prescription with the functoriality omitted. The higher-category analogue is the accessible section theorem of Lurie (2009, Proposition 5.4.7.11).

The qualitative section theorem is robust; several printed rank assertions are not. For unrestricted sections of an opfibration, pointwise presentability need not imply presentability of the section unless transport has an appropriate presentability bound. Conversely, the category of sections can be very small while a forced component is large. The commentary gives explicit examples for both directions. Cocartesian sections admit a separate argument, because their structure isomorphisms identify the transported source with the target component. Keeping that distinction avoids discarding a correct corollary along with an incorrect proof reference.

The final pages replace fibrations by intervalwise accessible data. An arbitrary functor $p : \mathcal{E} \to \mathcal{C}$ determines, over each base arrow, a category obtained by gluing its two endpoint fibres. Its cross-Hom bifunctor

$$H : \mathcal{E}_0^{\mathrm{op}} \times \mathcal{E}_1 \to \mathrm{Set}$$

controls the gluing. Contravariant preservation makes fibrewise colimits universal in the total

category; covariant preservation makes source objects presentable at a suitable rank. This proves the accessible-section theorem without requiring chosen lifts. An additional hypothesis on idempotents in the base addresses accessibility of the total category itself.

Two corrections matter for reusing this last argument. Corollary 4.14.25 omits the colimit hypothesis on the closed fibre from one side of its equivalence; an ordinal example disproves the printed version. Corollary 4.15.2 also fails to put the contravariant preservation ranks below the initial presentation rank. The power-functor collage $X \mapsto X^{\aleph_1}$ gives a counterexample satisfying the printed cardinal bounds. Both qualitative constructions survive a common-rank choice with the missing requirements included. These pages end during the proof; the complete chapter later justifies the constant-subdiagram argument with the strict inequality $\pi > |\mathcal{C}|$. That later repair is reported at its exact manuscript locations.

The audit records these distinctions assertion by assertion: what the source proves, sketches, states, or asks; what is merely a slip in the edition; and what needs a mathematical repair. A few exact formulations remain uncertified, including the unrestricted fixed-rank third variant of 4.13.21 and the broad weakening suggested in footnote 172. These are identified as limits of the present verification rather than declared counterexamples or new open problems. Proofs marked *[ours]* are arguments supplied by the commentators, with the scope of the literature check stated and no claim of priority.

est π-accessible, quand x_0 est dans $\mathcal{E}_{0\,\pi}$. Or par hypothèse sur π, il est π-accessible si x_0 est dans $\mathcal{E}_{0\,\pi_0}$. Mais si x est dans $\mathcal{E}_{0\,\pi}$, on aura p. 159 [page 419]

$$x_0 \;=\; \varinjlim_{i\in I} x_{0\,i} \qquad \begin{cases} I \text{ grande devant } \pi_0, \\ \text{les } x_{0\,i} \text{ dans } \mathcal{E}_{0\,\pi_0}, \\ \operatorname{card} I \leqslant \pi \end{cases}$$

(car π_0 est utile pour $\mathcal{E}_0$ par hypothèse, et π satisfait $\pi^{\pi_0} = \pi \geqslant \operatorname{card} \text{réd}\mathcal{E}_{\pi_0}$). Comme $x_0 \longrightarrow H(x_0, x_1) : \mathcal{E}_0 \longrightarrow \text{Ens}$ satisfait à L_{π_0}, pour tout x_1 dans $\mathcal{E}_1$, on aura donc

$$H(x_0) \;\simeq\; \varprojlim_{i\in I} H(x_{0\,i})$$

(où pour x_0 dans $\mathcal{E}_0$, $H(x)$ désigne le foncteur $x_1 \longmapsto H(x_0, x_1) : \mathcal{E}_1 \longrightarrow (\text{Ens})$). Comme les foncteurs $H(x_{0\,i})$ sont π-accessibles, et $\operatorname{card} I \leqslant \pi$, il s'ensuit que leur $\varprojlim_I H(x_0)$ est π-accessible (SGA 4 I 9.6 (ii)). Cela achève de prouver donc 2°).

3°). Pour prouver que $\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \longrightarrow \mathcal{E}$ est essentiellement surjectif, il suffit de voir que tout $x \in \operatorname{Ob}\mathcal{E}_0$ et tout $x \in \operatorname{Ob}\mathcal{E}_1$ est dans l'image essentielle. Or cela résulte de l'hypothèse que π est utile pour $\mathcal{E}_0$ et $\mathcal{E}_1$, et de 1°) qui implique que $\mathcal{E}_0 \longrightarrow \mathcal{E}$ et $\mathcal{E}_1 \longrightarrow \mathcal{E}$ satisfont L_π. [page 420]

Commentary · Exposition [ms. 419]

The two cardinal bounds in the gluing argument

Grothendieck is finishing a construction of an accessible category from an open piece, a closed piece, and the arrows crossing from the first to the second. The point of the cardinal calculation is to make presentability in a piece agree with presentability in the whole category. The latter has an additional test: an object of the open piece must commute with sufficiently filtered colimits whose values lie in the closed piece. That test is measured by the gluing functor H.

We use the modern convention that a regular cardinal κ controls diagrams with fewer than κ arrows. Thus, for an infinite π, a diagram "grand devant π" is π^+-filtered, and an object called π-accessible here is π^+-presentable. A useful π gives a small family of such objects generating by those colimits. The shift by one successor is essential when interpreting the inequalities $\operatorname{card} I \leq \pi$ in the argument.

There are two distinct bounds. First, π_0 controls the presentation of an open object and the continuity of H in its contravariant variable. Second, π controls the number of terms in a presentation and the continuity of the mixed hom functors in their covariant variable. For a presentation $x_0 = \operatorname{colim}_{j\in J} a_j$ with $|\operatorname{Mor} J| \leq \pi$, the calculation is

$$\begin{aligned} H(x_0, \operatorname*{colim}_{i\in I} y_i) &\cong \lim_{j\in J} H(a_j, \operatorname*{colim}_{i\in I} y_i) \\ &\cong \lim_{j\in J} \operatorname*{colim}_{i\in I} H(a_j, y_i) \\ &\cong \operatorname*{colim}_{i\in I} H(x_0, y_i). \end{aligned}$$

The first and last identifications use continuity in the first variable at the rank of the J-presentation. The middle interchange uses the π^+-filteredness of I. To see the latter directly in sets, choose representatives for the at most π components of a compatible family, then choose one upper bound for their indices and for the at most π equations expressing compatibility. The same common-bound argument proves injectivity. This is the mechanism of SGA 4, I.9.6(ii) and I.9.8 (Artin et al., 1972).

This distinction will matter again on manuscript page 430. Increasing the rank at which a functor is known to preserve filtered colimits does not give continuity at the smaller rank used to present an object. Here the required smaller-rank continuity is explicitly assumed in Corollary 4.14.25(2); later it is inadvertently lost from the numerical hypotheses.

Cela achève de prouver la proposition 4.14.24, et de même le résultat suivant, plus précis que l'implication c) $\Longrightarrow$ a).

Corollaire 4.14.25. *Soient $\mathcal{E}$ une catégorie, $\mathcal{E}_0$ une sous-catégorie ouverte, $\mathcal{E}_1$ la sous-catégorie fermée complémentaire,*

$$H \;=\; ((x_0,x_1)\longmapsto \mathrm{Hom}_{\mathcal{E}}(x_0,x_1)) \;:\; \mathcal{E}_0^{\circ}\times\mathcal{E}_1 \;\longrightarrow\; \mathrm{Ens}$$

le foncteur recollement.

1°) Soit π un cardinal infini. Pour ce que $\mathcal{E}$ et $\mathcal{E}_0$ satisfassent à L_π, et le foncteur d'inclusion $\mathcal{E}_0\hookrightarrow\mathcal{E}$ itou, il faut et il suffit que $\mathcal{E}_0$ satisfasse à L_π, et que pour tout x_1 dans $\mathcal{E}_1$, le foncteur $x_0\longmapsto H(x_0,x_1):\mathcal{E}_0\longrightarrow\mathrm{Ens}^{\circ}$ satisfasse à L_π. Alors $\mathcal{E}_1$ et $\mathcal{E}_1\hookrightarrow\mathcal{E}$ satisfont aussi à L_π, et on a [page 421]

$$\mathcal{E}_\pi \;\subset\; \mathcal{E}(\pi) \overset{\text{déf}}{=} \begin{array}{l}\text{sous-catégorie strictement pleine de } \mathcal{E} \text{ telle que}\\ \mathrm{Ob}\,\mathcal{E}(\pi)=\mathrm{Ob}\,\mathcal{E}_{0\,\pi}\cup\mathrm{Ob}\,\mathcal{E}_{1\,\pi}.\end{array}$$

2°) Supposons que la condition c) de 4.14.24 soit satisfaite, et soit π_0 un cardinal infini tel que π_0 soit utile pour $\mathcal{E}_0$, et que les foncteurs $x_0\longmapsto H(x_0,x_1):\mathcal{E}_0\longrightarrow\mathrm{Ens}^{\circ}$ définis par les $x_1\in\mathrm{Ob}\,\mathcal{E}_1$ soient π_0-accessibles. Soit π un cardinal tel que $\pi^{\pi_0}=\pi\geqslant\sup(\pi_0,\mathrm{card\ réd}\mathcal{E}_{0\,\pi_0})$, et tel que pour tout $x_0\in\mathrm{Ob}\,\mathcal{E}_{\pi_0}$, le foncteur $x_1\longmapsto H(x_0,x_1):\mathcal{E}_1\longrightarrow\mathrm{Ens}$ soit π-accessible. Alors on a

$$\mathcal{E}_\pi \;=\; \mathcal{E}(\pi)\;.$$

Commentary · Examples [ms. 420]

The closed piece must have its colimits

The purpose of the first part of Corollary 4.14.25 is to separate existence of colimits from the presentability calculation. As printed, however, its right-hand condition says nothing about the existence of colimits in $\mathcal{E}_1$. This is a genuine omission in the edition. It cannot follow from continuity of the mixed hom functor.

Here is a finite example detecting the omission at every infinite rank. Let B have one object b and endomorphism monoid $\{1,p\}$, where $p^2=p\neq 1$. Put

$$\mathcal{E}=\{a\}\amalg B,\qquad \mathcal{E}_0=\{a\},\quad \mathcal{E}_1=B.$$

The gluing functor is constantly empty. For any nonempty filtered diagram, the limit of the constant empty-set diagram is empty, so the printed condition on H is satisfied. The terminal category $\mathcal{E}_0$ has all the required colimits.

To exhibit the failure in the closed piece, index a diagram by the ordinal π^+, send every index to b, and send every strict transition to p. This is a functor because $p^2=p$. The index is π^+-filtered. A cocone to b must have every leg equal to p: for each index i, choose $j>i$ and use $c_i=c_jp=p$. Thus there is exactly one cocone to b, whereas $B(b,b)$ has two elements. No object represents the cocone functor, and the diagram has no colimit in B or in $\mathcal{E}$.

The corrected criterion requires both pieces to admit L_π-colimits, and requires $H(-,x_1)$ to carry those in the open piece to limits. Its proof also explains why no further compatibility is needed. For a filtered diagram $D:I\to\mathcal{E}$, let I_1 consist of the indices whose values lie in the closed piece. If I_1 is nonempty, choose one of its objects. Every finite set of indices has an upper bound receiving an arrow from that chosen object, so I_1 is cofinal; the same argument with at most π indices gives the required higher filteredness. Its colimit is computed in $\mathcal{E}_1$. There are no arrows from the closed piece to the open one, so this is also a colimit in $\mathcal{E}$. If I_1 is empty, compute the colimit in $\mathcal{E}_0$; its universal property against targets in $\mathcal{E}_1$ is precisely the asserted continuity of H. Conversely, an existing total colimit of a nonempty diagram in $\mathcal{E}_1$ must lie in $\mathcal{E}_1$, since its cocone cannot point into the open piece.

3°) Si π_0 et π sont comme dans 2°), et si de plus π est utile pour $\mathcal{E}_1$, alors π est utile pour $\mathcal{E}$, donc (en vertu de 2°) p. 160

$$\mathcal{E}\;\xleftarrow{\sim}\;\mathrm{Ind}_\pi(\mathcal{E}(\pi))\;.$$ [page 422]

Nous pouvons maintenant énoncer la généralisation cherchée de 4.14.17 :

Théorème 4.14.26 (généralisation de 4.14.13). *Soit $p : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur, avec $\mathcal{C}$ petite. On suppose que p satisfait à l'hypothèse H) de la page 409, qui équivaut (moyennant 4.14.24) à la suivante :*

H′) $\forall\ s \in \mathrm{Ob}\,\mathcal{C}$, *la catégorie fibre $\mathcal{E}_s$ est accessible, et pour tout $\lambda : s \longrightarrow t$ dans $\mathrm{Fl}\,\mathcal{C}$, le foncteur*

$$\begin{cases} H_\lambda : \mathcal{E}_s^\circ \times \mathcal{E}_t & \longrightarrow & \mathrm{Ens} \\ \qquad (x_0, x_1) & \longmapsto & \mathrm{Hom}_\lambda(x_0, x_1) \end{cases}$$

satisfait les deux conditions suivantes :

a) $\forall\ x_0 \in \mathrm{Ob}\,\mathcal{E}_s^\circ$, $x_1 \longmapsto H_\lambda(x_0, x_1) : \mathcal{E}_t \longrightarrow \mathrm{Ens}$ *est accessible.*

b) Il existe un cardinal infini π_λ, tel que pour tout $x_1 \in \mathrm{Ob}\,\mathcal{E}_t$, le foncteur $x_0 \longmapsto H_\lambda(x_0, x_1) : \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$ soit π_λ-accessible.

Ceci posé, la catégorie $\underline{\Gamma}(\mathcal{E}/\mathcal{C})$ des sections de $\mathcal{E}$ sur $\mathcal{C}$ est accessible, et les foncteurs $\varepsilon_s : \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \longrightarrow \mathcal{E}_s$ sont accessibles. On peut préciser de façon similaire à 4.14.18 (énoncé en forme laissé au lecteur).

Démonstration de 4.14.26. Elle se fait exactement comme celle de 4.14.17 (p. 400 ff.), [page 423] mais est conceptuellement plus simple. On suppose $\mathcal{C}$ petite, et on introduit les catégories

$$\Pi \;=\; \prod_{s \in \mathrm{Ob}\,\mathcal{C}} \mathcal{E}_s\ , \qquad \Pi' \;=\; \prod_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \mathcal{E}_\lambda\ ,$$

où

$$\mathcal{E}_\lambda \;=\; \mathcal{E} \times_{\mathcal{C}} (\mathbf{\Delta}_1, i_\lambda)\ ,$$

et les deux foncteurs canoniques

$$\Pi \underset{G}{\overset{F}{\rightrightarrows}} \Pi'$$

de composantes

$$\begin{aligned} F_\lambda \;&=\; i_\lambda \circ \mathrm{pr}_{s(\lambda)} \\ G_\lambda \;&=\; j_\lambda \circ \mathrm{pr}_{b(\lambda)}\ , \end{aligned}$$

où pour $\lambda : s \longrightarrow t$, on pose

$$s(\lambda) \;=\; s\ , \qquad b(\lambda) \;=\; t\ ,$$

et i_λ, j_λ sont les inclusions canoniques

$$\begin{array}{ccccc} \mathcal{E}_s & \xrightarrow{\ i_\lambda\ } & \mathcal{E}_\lambda & \xleftarrow{\ j_\lambda\ } & \mathcal{E}_t\ . \\ \big\downarrow\wr & & \big\downarrow & & \big\downarrow\wr \\ (\mathcal{E}_\lambda)_0 & & {\scriptstyle p_\lambda} & & (\mathcal{E}_\lambda)_1 \\ & & \big\downarrow & & \\ & & \mathbf{\Delta}_1 & & \end{array}$$

Commentary · Exposition [ms. 422]

Sections retain the equations between arrows

Theorem 4.14.26 changes the task from assembling objects to assembling a diagram. A section chooses $X_s \in \mathcal{E}_s$ for every base object and a lift $u_\lambda : X_s \to X_t$ for every $\lambda : s \to t$. These choices must satisfy the unit and composition equations. Grothendieck first constructs all the choices and only then imposes the equations; this order makes the accessibility argument possible.

For each λ, the pullback $\mathcal{E}_\lambda$ is the collage of its two fibres with crossing hom functor H_λ. An arrow from the zero copy of X_s to the one copy of X_t is therefore exactly a lift of λ. The product Π

records all objects X_s, while the inserter of $F, G : \Pi \rightrightarrows \Pi'$ records these objects together with all arrows u_λ. A morphism in the inserter is a family $a_s : X_s \to Y_s$ satisfying

$$a_t u_\lambda = v_\lambda a_s \quad \text{for every } \lambda : s \to t.$$

These are already the naturality equations for a map of sections. What is still missing is functoriality of the chosen arrows themselves.

The asymmetry in H′ has a structural reason. A single cardinal must control the contravariant variable for all possible targets, because it ensures that colimits in a fibre remain colimits when one tests them against every object elsewhere. In the covariant variable one first tests only a small family of generators; their individual accessibility bounds can then be dominated by one larger cardinal. Confusing these two quantifiers would replace a genuine uniformity assumption by a pointwise statement.

This construction continues the older SGA 4 treatment of accessible section categories, particularly I.9.22–9.26, while allowing crossing hom functors which need not come from chosen cartesian transport (Artin et al., 1972). In current terminology, the argument belongs to the closure of accessible categories under products, inserters, and equifiers. Makkai–Paré's monograph appeared in 1989: its formalism was contemporary with this manuscript, rather than a subsequent solution. The relevant result is Theorem 5.1.6; an explicit modern use of precisely this sequence of constructions appears in the appendix of Ching and Riehl (2014). The use of a two-categorical limit theorem is substantive: arbitrary strict equalizers of functors between accessible categories do not enjoy the same general closure property.

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On désigne par $\mathcal{M}'$ la catégorie

$$\mathcal{M}' \;=\; \begin{array}{l}\text{catégorie des couples } (X, u), \text{ avec} \\ X \in \operatorname{Ob}\Pi,\ u : F(x) \longrightarrow G(x),\end{array}$$

qui est constructible par 4.14.22.1 est accessible (car Π et π' et les foncteurs F et G le sont, par l'hypothèse H) faite sur p). On peut l'interpréter comme la catégorie des sections de $\mathrm{Diag}(\mathcal{E})$ sur $\mathrm{Diag}(\mathcal{C})$. On a un foncteur strictement pleinement fidèle

$$\mathcal{M} \overset{\text{déf}}{=} \underline{\Gamma}(\mathcal{E}/\mathcal{C}) \;\hookrightarrow\; \mathcal{M}' \ ,$$

qui permet d'interpréter $\mathcal{M}$ comme une sous-catégorie strictement pleine de $\mathcal{M}'$. On voit alors comme dans le cas de 4.14.17 que cette sous-catégorie est accessible dans $\mathcal{M}'$ – c'est essentiellement sorital. Il s'ensuit que $\mathcal{M}$ est accessible et que les foncteurs $\mathcal{M} \longrightarrow \mathcal{E}_s$ le sont aussi, comme composés de

$$\mathcal{M} \overset{\text{acc.}}{\hookrightarrow} \mathcal{M}' \overset{\text{acc.}}{\hookrightarrow} \Pi \xrightarrow{\ \mathrm{pr}_s \ \text{ acc.}\ } \mathcal{E}_s \ ,$$

q.e.d.

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Commentary · Later developments [ms. 424]

Writing out the omitted equifiers

Grothendieck's description of the last step as "essentiellement sorital" can be made precise without adding an assumption to the theorem. For each object s, impose

$$u_{1_s} = 1_{X_s},$$

and, for each composable pair $s \xrightarrow{\lambda} t \xrightarrow{\mu} v$, impose

$$u_{\mu\lambda} = u_\mu u_\lambda.$$

Both sides are natural transformations between the same accessible functors on the inserter. For example, the two transformations in the second equation take values in the mixed-arrow collage $\mathcal{E}_{\mu\lambda}$, from its zero copy of X_s to its one copy of X_v. Naturality follows from the commutative squares

already built into morphisms of the inserter. The unit equations have the same description using $\mathcal{E}_{1_s}$.

An equifier is the full subcategory on the objects where two specified natural transformations agree. The smallness of $\mathcal{C}$ makes the collection of unit and composition equations small; a product gathers their target categories, and equifiers impose all the equations. Accessibility of these constructions and of their structural functors gives both assertions of the theorem. The notation $\mathcal{M}' \hookrightarrow \Pi$ in the last display should be read as a forgetful arrow: different arrow assignments may have the same underlying family of objects, and a family of object maps need not satisfy the naturality squares. The projection is accessible, which is all the proof uses.

For a concrete illustration, take $\mathcal{C} = B\mathbf{N}$, the one-object category freely generated by one endomorphism, and take $\mathcal{E} = \mathcal{C} \times \mathbf{Set}$. A section is a set X with an endomorphism T. Before imposing the equations, one has an unrelated family of self-maps u_n for $n \in \mathbf{N}$; afterward $u_0 = 1$ and $u_n = T^n$. The free section on one element is $(\mathbf{N}, n \mapsto n+1)$: an equivariant map from it is uniquely determined by the image of zero. Hence this section is finitely presentable, although its underlying set is infinite. Evaluation preserves filtered colimits, but it need not preserve finitely presentable objects at that same rank. The distinction between accessibility and strong accessibility therefore appears even in this elementary example.

The product–inserter–equifier method is part of the broader theory of flexible limits developed by Bird et al. (1989). Its relevance here is exact: the equations constrain specified arrows between already chosen objects, rather than requiring unrelated object-valued functors to become literally equal.

4.15 Catégorie accessible sur une autre

Proposition 4.15.1. *Soit $f : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur avec $\mathcal{C}$ petite. On suppose de plus satisfaite la condition H (p. 409), qu'on explicite ici sous la forme équivalente H′ :*

H 1) Pour tout $s \in \mathrm{Ob}\,\mathcal{C}$, $\mathcal{E}_s$ est accessible.

H 2) Pour toute $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, existe un cardinal infini π_λ tel que pour tout $x_1 \in \mathrm{Ob}\,\mathcal{E}_t$, le foncteur

$$x_0 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) \ : \ \mathcal{E}_s \longrightarrow (\mathrm{Ens})^\circ$$

est π_λ-accessible ($\mathcal{E}_s$ satisfaisant de plus à L_{π_λ}).

H 3) Pour toute $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, et pour tout $x_0 \in \mathrm{Ob}\,\mathcal{E}_s$, le foncteur

$$x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) \ : \ \mathcal{E}_t \longrightarrow (\mathrm{Ens})$$

est accessible.

Alors les foncteurs d'inclusion $i_s : \mathcal{E}_s \longrightarrow \mathcal{E}$ sont accessibles. De plus, si $\mathcal{C}$ "est sans projecteurs" (i.e. si tout endomorphisme p d'un objet s de $\mathcal{C}$ tel que $p^2 = p$, est l'identité), alors $\mathcal{E}$ est accessible, et $f : \mathcal{E} \longrightarrow \mathcal{C}$ est accessible.

Cela résulte du résultat plus précis :

Corollaire 4.15.2 ([175]). *Les hypothèses sur $f : \mathcal{E} \longrightarrow \mathcal{C}$ sont celles de 4.15.1 ($\mathcal{C}$ petite, et f satisfait $H' = H$ 1) 2) 3)).* p. 162 [page 426]

a) Soit $s \in \mathrm{Ob}\,\mathcal{C}$ et soit π_1 un cardinal infini tel que pour toute $\lambda : s \longrightarrow t$ (flèche de $\mathcal{C}$ de source t), on ait $\pi_1 \geqslant \pi_\lambda$ (où π_λ est défini dans l'énoncé de H 2). Alors le foncteur $i_s : \mathcal{E}_s \longrightarrow \mathcal{E}$ est π_1-accessible.

b) On suppose $\mathcal{C}$ sans projecteurs. Soit π_1 un cardinal $\geqslant \sup(\pi_\lambda, \mathrm{card}\,\mathcal{C})$. Alors $\mathcal{E}$ satisfait L_{π_1}, et $f : \mathcal{E} \longrightarrow \mathcal{C}$ est π-accessible.

c) Supposons encore que $\mathcal{C}$ soit sans projecteurs (p.ex. une catégorie "rigide", p.ex. une catégorie préordonnée). Soient π_0, π_1 des cardinaux tels que :

175. Pour a) et b) on n'utilise que H 1 et H 2, à l'exclusion de H 3.

① $\forall\ s \in \mathrm{Ob}\,\mathcal{C}$, π_0 est utile pour $\mathcal{E}_s$ *(un tel π_0 existe grâce à H 1 et au fait que $\mathcal{C}$, donc $\mathrm{Ob}\,\mathcal{C}$ petits).*

② $\pi_1 \geqslant \sup(\pi_0, \mathrm{card}\,\mathcal{C}, \sup_{\lambda\in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda, \sup_{s\in \mathrm{Ob}\,\mathcal{C}} \mathrm{card\ réd}\mathcal{E}_{s\,\pi_0})$.

③ $\forall \lambda \in \mathrm{Fl}\,\mathcal{C}$, *et* $x_0 \in \mathrm{Ob}\,\mathcal{E}_{s,\pi_0}$, *le foncteur* $H_\lambda(x_0)$ *de H 3,* $x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1) : \mathcal{E}_t \longrightarrow \mathrm{Ens}$ *est π_1-accessible.*

Soit π un cardinal tel que $\pi^{\pi_0} = \pi \geqslant \pi_1$. Alors par b), $\mathcal{E}$ satisfait L_{π_1}, donc L_π, et $f : \mathcal{E} \longrightarrow \mathcal{C}$ est π_1-accessible. De plus,

$$i_s(\mathcal{E}_{s\,\pi}) \ \subset\ \mathcal{E}_\pi\ ,$$

ou encore

$$(*) \qquad \mathcal{E}(\pi) \ \subset\ \mathcal{E}_\pi\ ,$$

où $\mathcal{E}(\pi)$ est la sous-catégorie pleine de $\mathcal{E}$ telle que

$$\mathrm{Ob}\,\mathcal{E}(\pi) \ =\ \bigcup_{s\in \mathrm{Ob}\,\mathcal{C}} \mathrm{Ob}\,(\mathcal{E}_{s\,\pi})\ .$$

Comme $\mathcal{E}_\pi$ satisfait L_{π_1} par b), donc a fortiori L_π, on a donc par $(*)$ un foncteur pleinement fidèle

$$\mathrm{Ind}_\pi(\mathcal{E}_{s\,\pi}) \ \xrightarrow{\sim}\ \mathcal{E}\ ,$$

et ce foncteur est une équivalence de catégories (donc $\mathcal{E}$ est accessible).

d) En fait, sous les hypothèses de c) sur $\mathcal{C}$, π_0, π_1, π, on a $\mathcal{E}(\pi) = \mathcal{E}_\pi$ (non seulement $\mathcal{E}(\pi) \subset \mathcal{E}_\pi$).

Démonstration de 4.15.1 et 4.15.2. Il est clair que 4.15.1 est contenu dans 4.15.2 a) et c). Nous allons donc prouver 4.15.2. [page 427]

a) Soit $(x_i)_{i\in I}$ un système inductif dans $\mathcal{E}_s$, avec I grand devant π_0, et x sa limite inductive dans $\mathcal{E}_s$ (qui existe, car $\mathcal{E}_s$ satisfait à L_{π_λ}, donc à L_{π_0}). Il faut prouver que c'est une limite dans $\mathcal{E}$, donc que pour tout $y \in \mathrm{Ob}\,\mathcal{E}$, la flèche canonique

$$(*) \qquad \mathrm{Hom}_{\mathcal{E}}(x, y) \ \longrightarrow\ \varprojlim_I \mathrm{Hom}_{\mathcal{E}}(x_i, y)$$

est un isomorphisme. Or $\mathrm{Hom}_{\mathcal{E}}(x, y)$ et les $\mathrm{Hom}_{\mathcal{E}}(x_i, y)$ sont des objets de $\mathrm{Ens}/\mathrm{Hom}_{\mathcal{C}}(s, t)$, où $s = f(x)$, $t = f(y)$, et les flèches de transition sont des flèches dans $\mathrm{Ens}/\mathrm{Hom}_{\mathcal{C}}(s, t)$. Cela montre que la bijectivité de $(*)$ revient à celle de p. 163

$$(*) \qquad \mathrm{Hom}_\lambda(x, y) \ \longrightarrow\ \varprojlim_I \mathrm{Hom}_\lambda(x_i, y)$$

pour tout $\lambda \in \mathrm{Hom}_{\mathcal{C}}(x, y)$ [plutôt $\mathrm{Hom}_{\mathcal{C}}(f(x), f(y))$]. Pour λ fixé, et y fixé dans $\mathrm{Ob}\,\mathcal{E}_t$, la bijectivité dans $(*)$, pour tout système inductif grand devant π_1 dans $\mathcal{E}_s$, de limite x, signifie que le foncteur

$$x \ \longmapsto\ \mathrm{Hom}_\lambda(x, y)\ :\ \mathcal{E}_s \ \longrightarrow\ (\mathrm{Ens})^\circ$$

est π_0-accessible. Or c'est bien le cas, par hypothèse sur π_1, quel que soit la flèche $\lambda : s \longrightarrow t$ de source s, et quel que soit $y \in \mathrm{Ob}\,\mathcal{E}_t$.

Soit $(x_i)_{i\in I}$ un système inductif grand devant π_1, et $(s_i)_{i\in I}$ l'image de ce système dans $\mathcal{C}$. Nous prouverons plus bas le [page 428]

Commentary · Exposition [ms. 427]

Colimits in a fibre and colimits in the total category

Grothendieck now asks whether the same local data control the total category itself. The first step, preservation of colimits by a fibre inclusion, is more elementary than the final theorem and does not need H3. Let $x = \operatorname{colim}_i x_i$ be a sufficiently filtered colimit in $\mathcal{E}_s$, and let $y \in \mathcal{E}_t$. The decomposition

$$\mathcal{E}(x,y) = \coprod_{\lambda \in \mathcal{C}(s,t)} \operatorname{Hom}_\lambda(x,y)$$

is respected by every transition map in the comparison with $\mathcal{E}(x_i, y)$. Since the index category is nonempty and connected, a compatible family in the limit has one fixed base arrow λ. Thus its universal property reduces to the separate universal properties of $H_\lambda(-,y)$, which H2 supplies. The occurrences of π_0 in this proof are to be read as π_1, the bound specified in part (a).

The second step explains the hypothesis on the base. If the image of a sufficiently filtered diagram can be reindexed to become constant with identity transitions, the diagram lies in one fibre, so the preceding calculation supplies its total colimit. A small base with no nonidentity idempotents has this stabilization property at a sufficiently large rank. The proof of the required small-category statement is deferred until manuscript pages 440–463. For the existential conclusion of Proposition 4.15.1 it is enough to use the strict cardinal bound proved there; no claim about an optimal boundary cardinal is needed.

The phrase "sans projecteurs" must be kept distinct from "karoubienne". A groupoid has no nonidentity idempotents, although it may have many automorphisms. A Karoubian category may have nonidentity idempotents, provided their images exist. A constant-object diagram whose strict transitions are one such idempotent need not become a diagram of identity maps inside its original fibre. The later difficulty with a Karoubian base is therefore a difficulty of transport to the image of an idempotent, rather than a failure of the fibrewise universal property just proved.

Lemme 4.15.3 ([176], [177]). *Soit $\mathcal{C}$ une catégorie, π un cardinal, tels que* $\operatorname{card}\mathcal{C} \leqslant \pi$, *et que $\mathcal{C}$ soit sans projecteurs. Alors pour tout système inductif $(s_i)_{i\in I}$ grand devant π dans $\mathcal{C}$, existe une partie cofinale J dans I telle que le système induit $(s_i)_{i\in J}$ soit* constant, *i.e. tel que les s_i soient égaux à un $s \in \mathcal{C}$, et que les morphismes de transition soient tous égaux à* id_s.

Mais attention, on s'appuie pour prouver 4.15.3 sur le théorème 4.8.13 (page 212-216), que je n'ai toujours pas prouvé. Dans le cas où $\mathcal{C}$ est rigide (i.e. $\operatorname{End}_{\mathcal{C}}(s) = \{\mathrm{id}_s\}\ \forall\ s \in \operatorname{Ob}\mathcal{C}$), la démonstration est cependant immédiate, cf. plus bas.

Pour prouver que $x = \varinjlim_I x_i$ existe dans $\mathcal{E}$, et commute à f, on peut remplacer I par une [page 429]
sous-catégorie cofinale. Cela nous ramène, grâce au lemme, au cas où $(s_i)_{i\in I}$ est constant, donc au cas où le système inductif provient d'un système inductif dans une catégorie fibre $\mathcal{E}_s$. Mais par a), on sait que sa limite x dans $\mathcal{E}_s$ est une limite dans $\mathcal{E}$, d'où l'existence de cette dernière, et la commutation à f, puisque $\varinjlim s_i = s$ ($\varinjlim$ d'un système inductif filtrant constant).

c) Il faut prouver que si $x \in \operatorname{Ob}\mathcal{E}_s$ est π-accessible dans $\mathcal{E}_s$, il l'est dans $\mathcal{E}$, i.e. que pour tout système inductif $(y_i)_{i\in I}$ grande devant π, la flèche canonique

$$(**) \qquad \varinjlim_i \operatorname{Hom}(x, y_i) \longrightarrow \operatorname{Hom}(x,y)$$

est bijective. Or comme par 2°) $\pi \geqslant \pi_1 \geqslant \sup(\operatorname{card} S, \sup_\lambda \pi_\lambda)$, il résulte encore du lemme que, quitte à remplacer I par une sous-catégorie cofinale, on peut supposer que le système inductif $(y_i)_{i\in I}$ provient d'un système dans une catégorie fibre $\mathcal{E}_t$, et que y est sa $\varinjlim$ dans $\mathcal{E}_t$. Mais alors les ensembles $\operatorname{Hom}(x,y)$ et $\operatorname{Hom}(x,y_i)$, et les morphismes de transition entre celles-ci, sont encore dans $\operatorname{Ens}/\operatorname{Hom}(s,t)$, et on est ramené

à prouver que pour tout $\lambda : s \longrightarrow t$, la flèche

$$\varinjlim_i \operatorname{Hom}_\lambda(x, y_i) \longrightarrow \operatorname{Hom}_\lambda(x,y) \qquad \text{p. 164 [page 430]}$$

176. Cf. démonstration page 452.

177. sous réserve de la validité de 4.8.13 (Mais c'est à présent établi, sous une forme suffisante, dans 4.15.10).

est bijective. Pour x et λ fixés, et des systèmes inductifs variables dans $\mathcal{E}_t$, grands devant π, cela signifie que le foncteur

$$H_\lambda(x) \;=\; (y \longmapsto \mathrm{Hom}_\lambda(x,y)) \;:\; \mathcal{E}_t \;\longrightarrow\; \mathrm{Ens}$$

est π-accessible, et ceci quel que soit

$$x \;\in\; \mathrm{Ob}\,\mathcal{E}_{s\,\pi}\ .$$

Or c'est le cas si $s \in \mathrm{Ob}\,\mathcal{E}_{s\,\pi_0}$, en vertu de l'hypothèse 3°) (alors $H_\lambda(x_0)$ est π_1-accessible, et a fortiori π-accessible). Il faut passer de là au cas où $x \in \mathrm{Ob}\,\mathcal{E}_{s\,\pi}$. Or comme π_0 est utile pour $\mathcal{E}_s$ par 1°, on a

$$x \;=\; \varinjlim_{j\in J} x_j\ , \qquad x_j \in \mathrm{Ob}\,\mathcal{E}_{s\,\pi_0},\ J \text{ grande devant } \pi_0\ ,$$

et comme par 2°, $\pi^{\pi_0} = \pi \geqslant \pi_1 \geqslant \sup(\pi_0, \mathrm{card\,réd}\,\mathcal{E}_{s\,\pi_0})$, on en conclut qu'on peut prendre J de sorte que

$$\mathrm{card}\,J \;\leqslant\; \pi\ .$$

D'autre part, en vertu de $\pi \geqslant \pi_1 \;\geqslant\; \pi_\lambda$, le foncteur

$$\begin{cases} x_0 & \longmapsto & H_\lambda(x_0) \\ \mathcal{E}_s & \longrightarrow & (\underline{\mathrm{Hom}}(\mathcal{E}_s, \mathrm{Ens}))^\circ \end{cases}$$

est π_0-accessible, donc on a

$$H_\lambda(x_0) \;\simeq\; \varprojlim_{j\in J} H_\lambda(x_j)\ .$$

Comme $x_j \;\in\; \mathrm{Ob}\,\mathcal{E}_{s\,\pi_0}$, $H(x_j)$ est un foncteur π_1-accessible, et comme $\mathrm{card}\,J \leqslant \pi$, la $\varprojlim_{j\in J} H_\lambda(x_j)$ est aussi π-accessible (cf. SGA 4 I 9.8). [page 431]

Commentary · Development (commentators') [ms. 430]

The numerical rank assertion needs another hypothesis

Grothendieck wants to pass from continuity on a small family of generators to continuity on every object of the chosen larger rank. This is a sound strategy, but the displayed implication on this page uses first-variable continuity at π_0 although H2 only provides it at π_λ. The inequality $\pi \geq \pi_1 \geq \pi_\lambda$ does not repair that mismatch. The following example shows that the numerical assertion of Corollary 4.15.2(c), as stated, is false.

A bounded-powerset counterexample. [ours] Put $\pi_0 = \aleph_0$ and $\pi_1 = \pi = 2^{\aleph_0}$. Let both fibres over Δ_1 be copies of **Set**. For a set A, write $P_{\leq\pi}(A)$ for the set of its subsets of cardinal at most π, with direct image as its action on maps. Define the total category by the usual hom sets within each fibre, no arrows from the second fibre to the first, and

$$\mathrm{Hom}_{0\to1}(A,B) = \mathbf{Set}(P_{\leq\pi}(A), B).$$

Functoriality of bounded powerset defines both compositions involving a crossing arrow, so this is a category over Δ_1.

We check every numerical assumption. Both fibres have useful cardinal $\aleph_0$ in Grothendieck's convention: every set is the $\aleph_1$-filtered union of its countable subsets. A skeleton of the category of countable sets has $2^{\aleph_0}$ morphisms, so the reduced-size bound holds. Cardinal arithmetic gives $\pi^{\pi_0} = \pi$.

The functor $P_{\leq\pi}$ preserves π^+-filtered colimits. Indeed, a subset of the colimit of size at most π can be lifted by choosing at most π representatives and one common stage. If two lifted subsets have the same image, choose matching elements and witnesses for their equalities; at most π witnesses suffice, and a common upper stage makes the two subsets equal. These two observations prove surjectivity and injectivity of the canonical comparison. Consequently H2 holds with $\pi_\lambda = \pi$ for

the crossing arrow. H3 holds for every A, since the mixed hom functor is representable by the set $P_{\leq\pi}(A)$. If A is countable, this representing set has cardinal at most $2^{\aleph_0} = \pi$, so the stronger H3 bound demanded in part (c) also holds. The identity-arrow conditions are automatic.

Now choose A of cardinal π in the first fibre. It is π-accessible there. But $P_{\leq\pi}(A) = P(A)$ has cardinal $2^\pi > \pi$. Index a diagram in the second fibre by all subsets $B \subseteq P(A)$ with $|B| \leq \pi$, ordered by inclusion. This poset is π^+-filtered, its union is $P(A)$, and this is also its total colimit. The identity map of $P(A)$ belongs to $\mathrm{Hom}_{0\to1}(A, P(A))$ and factors through no such B. Therefore A is not π-accessible in the total category. This disproves both the inclusion in part (c) and the equality in part (d).

The repair is simple and preserves Proposition 4.15.1. Choose π_0 to dominate all π_λ before making the subsequent choices of π_1 and π. Then the π_0-filtered generator presentation can be carried through $H_\lambda(-, y)$, and the limit–colimit calculation on manuscript page 419 applies. Such a common initial choice is available because the base is small. We checked the standard uniformization and two-categorical closure results in the sources cited here; we did not find this particular bounded-powerset diagnosis of the printed rank statement. The example is supplied to isolate the missing hypothesis, without a claim of priority.

Ainsi on a prouvé $\mathcal{E}_{s\,\pi} \subset \mathcal{E}(\pi)$ pour tout $s \in \mathrm{Ob}\,\mathcal{C}$, i.e. l'inclusion

$$\mathcal{E}(\pi) \ \subset \ \mathcal{E}_\pi \ ,$$

d'où le foncteur pleinement fidèle

$$(*) \qquad \mathrm{Ind}_\pi(\mathcal{E}(\pi)) \longrightarrow \mathcal{E} \ ,$$

comme $\mathcal{E}$ satisfait à L_π. Il reste à prouver que c'est essentiellement surjectif. Mais si $x \in \mathrm{Ob}\,\mathcal{E}$, $s = f(x)$, on a $x \in \mathrm{Ob}\,\mathcal{E}_s$, et comme π_0 est utile pour $\mathcal{E}_s$ par 1°), et $\pi^{\pi_0} = \pi > \pi_0$, π est également utile pour π, donc on a *dans* $\mathcal{E}_s$

$$x \ = \ \varinjlim_I x_i \ , \qquad \text{les } x_i \in \mathcal{E}_{s\,\pi} \subset \mathcal{E}(\pi),\ I \text{ grande devant } \pi \ ,$$

et d'après a) (comme $\pi \geqslant \pi_1 \geqslant \sup_{\lambda\in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda$) x est aussi limite des x_i dans $\mathcal{E}$, donc il est dans l'image essentielle de $(*)$. p. 165

d) Reste à prouver que l'on a $\mathcal{E}_\pi \ \subset \ \mathcal{E}(\pi)$. Mais on sait (par 4.8. …), comme $\mathcal{E} \ \simeq \ \mathrm{Ind}_\pi(\mathcal{E}(\pi))$, que $\mathcal{E}_\pi \ = \ \mathcal{E}(\pi)^\natural$, i.e. que pour $x \ \in \ \mathrm{Ob}\,\mathcal{E}_\pi$, $\exists\ y \ \in \ \mathcal{E}(\pi)$ et un projecteur p dans x, tel que x soit isomorphe à l'image de p dans $\mathcal{E}$. Or $f(p)$ est un projecteur dans $f(y) = s$, donc par hypothèse sur $\mathcal{C}$, $f(p) = \mathrm{id}_s$, i.e. p est un projecteur dans $\mathcal{E}_s$. Comme donc x est dans $\mathcal{E}_s$, et

même dans $\mathcal{E}_{s\,\pi}$, puisque $\mathcal{E}_{s\,\pi}$ est stable dans $\mathcal{E}_s$ par facteurs directs. Cela prouve que $x \in \mathrm{Ob}\,\mathcal{E}(\pi)$, q.e.d. [page 432]

On peut présenter 4.15.1 sous une forme plus sympathique.

Corollaire 4.15.4. *Soit $f : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur de $\mathfrak{U}$-catégories. Les conditions suivantes sont équivalentes :*

a) Pour tout foncteur $\mathcal{C}' \longrightarrow \mathcal{C}$, avec $\mathcal{C}'$ petite et sans projecteurs, $\mathcal{E}' \overset{déf}{=} \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ est accessible, et $f' : \mathcal{E}' \longrightarrow \mathcal{C}'$ est accessible.

b) Comme a), mais avec $\mathcal{C}' = \mathbf{\Delta}_1$, en d'autres termes, c'est la condition H p. 409 : pour toute $\lambda \in \mathrm{Fl}\,\mathcal{C}$, $\mathcal{E}_\lambda = \mathcal{E} \times_{\mathcal{C}} (\mathbf{\Delta}_1, i_\lambda)$ est accessible, et $\mathcal{E}_\lambda \longrightarrow \mathbf{\Delta}_1$ est accessible.

c) La condition $H' = H\ 1 + H\ 2 + H\ 3$, explicitée dans l'énoncé de 4.15.1.

Démonstration. Elle se fait circulaire,

$$\mathrm{a} \implies \mathrm{b} \implies \mathrm{c} \implies \mathrm{a} \ ,$$

l'implication a $\implies$ b étant tautologique, b $\implies$ c l'étant presque (cf. 4.14.24, a $\implies$ b $\implies$ c). En fait, les conditions b et c sont des conditions sur les catégories $\mathcal{E}_s \xrightarrow{f_\lambda} \mathbf{\Delta}_1$ pour $\lambda \in \mathrm{Fl}\,\mathcal{C}$, donc stables par changement de base. Pour prouver [page 433]

c $\Longrightarrow$ a, on peut donc supposer que $\mathcal{C}' = \mathcal{C}$, $\mathcal{C}' \longrightarrow \mathcal{C}$ le foncteur identique, et prouver que si c) est satisfaite et si $\mathcal{C}$ est petite et sans projecteurs, alors $\mathcal{E}$ est accessible et f accessible. Or c'est ce qu'affirme 4.15.1.

Commentary · Exposition [ms. 432]

Recovering the generators without choosing a splitting

The conclusion Grothendieck needs is that the small generators of the total category can be recognized in the fibres. After correcting the uniformity assumption, the difficult inclusion is $\mathcal{E}(\pi) \subseteq \mathcal{E}_\pi$. The reverse inclusion has a short proof which also avoids a strict-fibre ambiguity in the printed retract argument.

Suppose $x \in \mathcal{E}_s$ is π-accessible in the total category. Let $y = \operatorname{colim}_i y_i$ be an L_π-colimit inside $\mathcal{E}_s$. The inclusion preserves it by part (a), so

$$\operatorname*{colim}_i \mathcal{E}(x, y_i) \xrightarrow{\sim} \mathcal{E}(x, y).$$

Both sides map to the fixed set $\mathcal{C}(s, s)$, and the transition maps preserve that map. Taking the fibre over 1_s commutes with the colimit: a representative has base arrow 1_s precisely when its image does, because every transition lies over the identity. We obtain the required bijection for $\mathcal{E}_s(x, -)$. Thus x is already π-accessible in its own fibre.

In the printed proof the projector must be on the chosen object y, not on x. Moreover, a splitting of that vertical projector in the total category may have an image over an object merely isomorphic to s. One can replace it by the splitting in the accessible fibre, but the identity-component argument proves the assertion directly and works without selecting an isomorphic representative.

Corollary 4.15.4 then expresses the result in a useful local form. H′ is unchanged by base change, because each new crossing hom functor is one of the original ones. Testing the walking arrow detects exactly this data, and the corrected existential theorem assembles it over any small base without nonidentity idempotents. The restriction on the testing bases is part of the assertion; it does not license replacement of an arbitrary strict pullback by a pseudopullback.

Remarque 4.15.5. Si $f : \mathcal{E} \longrightarrow \mathcal{C}$ est un foncteur pleinement fidèle, il est immédiat que la condition H est satisfaite (car $\mathcal{E}_\lambda \longrightarrow \mathbf{\Delta}_1$ sera pleinement fidèle, donc $\mathcal{E}_\lambda$ est équivalente à e où à $\mathbf{\Delta}_1$, donc accessible, et tout foncteur de $\mathcal{E}_\lambda$ dans une catégorie quelconque est accessible …). Mais si on suppose $\mathcal{C}$ petite *sans* supposer $\mathcal{C}$ sans projecteurs, il ne s'ensuit pas que $\mathcal{E}$ soit accessible, ou ce qui revient au même (étant essentiellement petite), que $\mathcal{E}$ soit karoubienne. (L'exemple le plus trivial est celui où $\mathcal{E} = \mathcal{C}$, $f = \mathrm{id}_{\mathcal{C}}$, $\mathcal{E}$ n'est karoubienne que si $\mathcal{C}$ l'est !) Mais même en supposant $\mathcal{C}$ une petite catégorie karoubienne, il ne s'ensuit pas forcément que $\mathcal{E}$ l'est. En effet, *toute* petite catégorie $\mathcal{E}$ (karoubienne ou non) se plonge comme sous-catégorie pleine dans son enveloppe karoubienne $\mathcal{C}$ – il suffit donc de prendre $\mathcal{E}$ non karoubienne. L'exemple le plus économique est [page 434]

$$\mathrm{Ob}\,\mathcal{E} \;=\; \{0\}\;, \qquad \mathrm{Fl}(\mathcal{E}) \;=\; \{\mathrm{id}_0, p\}\;, \qquad \text{avec } p^2 = p \neq \mathrm{id}_0\;,$$

l'enveloppe de Karoubi de $\mathcal{E}$ est donnée par p. 166

$$P \;=\; \left\{ 1 \underset{\sigma}{\overset{\rho}{\leftrightarrows}} 0 \circlearrowright p \qquad \begin{array}{l} \rho\sigma \;=\; \mathrm{id}_1 \\ \sigma\rho \;=\; p \;\neq\; \mathrm{id}_0 \end{array} \qquad \Longrightarrow\; p^2 = p \right\}.$$

Cela soulève la question de trouver un énoncé similaire à 4.15.1, donnant des conditions serviables simples, plus fortes que H′ = H (1+2+3), sur $f : \mathcal{E} \longrightarrow \mathcal{C}$, pour que $\mathcal{E}$ soit accessible et f accessible, en supposant $\mathcal{C}$ *karoubienne*. Ou encore, des conditions nécessaires et suffisants serviables (plus fortes que 4.15.4) sur $f : \mathcal{E} \longrightarrow \mathcal{C}$, pour que toute *petite* catégorie *karoubienne* $\mathcal{C}'$, et tout foncteur $\mathcal{C}' \longrightarrow \mathcal{C}$, la catégorie $\mathcal{E}' = \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ soit accessible, et le foncteur $\mathcal{E}' \longrightarrow \mathcal{C}'$ accessible. Une idée naturelle, c'est qu'il suffirait que cette conclusion soit vraie quand $\mathcal{C}'$

est une des deux catégories $\mathbf{\Delta}_1$, P (où P est la catégorie à deux objets, à trois flèches non identiques, définie plus haut). Mais je doute qu'il en soit ainsi ([178]).

On va quand même essayer un énoncé un peu très particulier dans ce sens : [page 435]

Proposition 4.15.6 ([180]). *Soit $f : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur de $\mathfrak{U}$-catégories. On suppose :*

a) f est fibrant ou cofibrant ([181]).

b) Les catégories fibres $\mathcal{E}_s$ ($s \in \mathrm{Ob}\,\mathcal{C}$) sont accessibles, et pour toute flèche $\lambda : s \longrightarrow t$ dans $\mathcal{C}$, le foncteur changement de base $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ (resp. cochangement de base $\lambda_* : \mathcal{E}_s \longrightarrow \mathcal{E}_t$) est accessible.

c) $\mathcal{C}$ est essentiellement petite et karoubienne.

Alors $\mathcal{E}$ est accessible, et f est accessible.

Commentary · Examples [ms. 434]

Why a retract has to be lifted

Grothendieck's smallest example makes visible what the walking arrow does not test. Let B be the one-object category with $p^2 = p \neq 1$. Its Karoubi envelope P has objects $0, 1$, arrows $\sigma : 1 \to 0$ and $\rho : 0 \to 1$, and relations

$$\rho\sigma = 1_1, \qquad \sigma\rho = p.$$

The inclusion $B \hookrightarrow P$ is fully faithful. Every pullback along a walking arrow has total category equivalent to an empty category, a point, or Δ_1, and therefore satisfies the local accessibility condition H. Nevertheless B is not accessible: the long constant-object diagram with transition p has no colimit, as computed above. The base P is Karoubian, but its retract object has not been lifted into B.

This is why the next proposed test categories attach an incoming or outgoing arrow to a retract. The retract alone tests existence of an image; an incoming arrow tests maps into that image, and an outgoing arrow tests maps from it. Concretely, once a splitting $x \xrightarrow{r} z \xrightarrow{i} x$ with $ri = 1$ and $ir = p$ is fixed, an arrow $y \to z$ corresponds to an arrow $u : y \to x$ satisfying $pu = u$, by $v \mapsto iv$ and $u \mapsto ru$. The two composites are identities because $ri = 1$ and $pu = u$. Dually, arrows $z \to y$ correspond to arrows $u : x \to y$ satisfying $up = u$. These are the two three-object diagrams in footnote 178. Their later reformulation in §4.16.6 separates the global accessibility problem from the stronger requirement concerning all strict base changes.

The counterexample also explains why a (co)fibration is a promising extra assumption. Cartesian or cocartesian transport along the two arrows of a retract gives a way to move the problem into the fibre over its image. The proof below makes that move explicit; it resolves the doubts attached to Proposition 4.15.6 without pretending that its written proof is complete.

Démonstration. On a déjà vu que les conditions a) et b) impliquent H′, i.e. H (1+2+3). D'ailleurs, si $\mathcal{C}'$ est une catégorie et $\mathcal{C}' \xrightarrow{\sim} \mathcal{C}$ une équivalence, donc alors $\mathcal{E}' \overset{\text{déf}}{=} \mathcal{E} \times_{\mathcal{C}} \mathcal{C}' \longrightarrow \mathcal{E}$ est une équivalence, donc la conclusion voulue pour $\mathcal{E}$, f équivaut à la même conclusion pour $\mathcal{E}'$, $f' : \mathcal{E}' \longrightarrow \mathcal{C}'$. Donc on peut supposer que $\mathcal{C}$ est petite (et même réduite, si on y tient). En vertu de 4.15.2 a), on voit donc que les foncteurs d'inclusion $\mathcal{E}_s \longrightarrow \mathcal{E}$ sont accessibles (et même π_1-accessibles) dès que $\pi_1 \geqslant \sup_{\lambda \in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda$. On voit d'ailleurs

178. Ça devrait marcher par contre si on prend les catégories $0 \longrightarrow 1$, [page 436]

$$2 \longrightarrow \underbrace{1 \underset{\sigma}{\overset{\rho}{\leftrightarrows}} 0 \circlearrowright p}_{P} \quad \text{et} \quad 2 \longleftarrow \underbrace{1 \underset{\sigma}{\overset{\rho}{\leftrightarrows}} 0 \circlearrowright p}_{P} \qquad ([179]),$$

3 objets et 5 flèches non identiques

grâce à 4.15.7, page 440.

179. ces catégories représentent respectivement les foncteurs Cat $\longrightarrow$ Ens associant à une catégorie $\mathcal{C}$ les diagrammes dans $\mathcal{C}$ formés de (x, y, p, u), avec p projecteur dans p `[plutôt dans C]`, et $u : y \longrightarrow x^\natural$ (i.e. $u : y \longrightarrow x$ avec $u = pu$), resp. $u : x^\natural \longrightarrow y$ (i.e. $u : x \longrightarrow y$ avec $u = up$).

180. *Faux ?*

181. Le cas fibrant n'est pas prouvé, peut-être *faux* – et de même pour le cas cofibrant !

par AQT [âne qui trotte] qu'on peut trouver des cardinaux π_0, π_1 ayant les propriétés suivantes :

p. 167

(1°) π_0 est utile pour $\mathcal{E}_s$, $\forall\ s \in \mathrm{Ob}\,\mathcal{C}$.

2°) $\pi_1 \geqslant \sup(\pi_0, \mathrm{card}\,\mathcal{C}, \sup_{\lambda\in \mathrm{Fl}\,\mathcal{C}} \pi_\lambda, \sup_{s\in \mathrm{Ob}\,\mathcal{C}} \mathrm{card\,réd}\,\mathcal{E}_s)$.

(3°) $\forall\ (\lambda : s \longrightarrow t) \in \mathrm{Fl}\,\mathcal{C}$ et $x_0 \in \mathrm{Ob}\,\mathcal{E}_{s\,\pi_0}$, le foncteur

$$H_\lambda(x_0)\ :\ (x_1 \longmapsto \mathrm{Hom}_\lambda(x_0, x_1))\ :\ \mathcal{E}_t\ \longrightarrow\ (\mathrm{Ens})$$

est π_1-accessible. (**N.B.** Si f est cofibrant, cela signifie que $\forall\ \lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda : s \longrightarrow t$, $\boxed{\lambda_*(\mathcal{E}_{s\,\pi_0})\ \subset\ \mathcal{E}_{s\,\pi_1}}$. Si f est fibrant, il suffit d'exiger que $\forall\ \lambda \in \mathrm{Fl}\,\mathcal{C}$, $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ soit π_1-accessible.) Mais dans ce cas, on veut aussi $\boxed{\lambda^*(\mathcal{E}_{s\,\pi_0})\ \subset\ \mathcal{E}_{s\,\pi_1}}$.

(4°) Les foncteurs λ^* (cas f fibrant) resp. λ_* (cas f cofibrant) sont π_1-accessibles.

Soit alors π un cardinal tel que

$$(*) \qquad \pi\ =\ \pi^{\pi_0}\ \geqslant\ \pi_1\ ,$$

et soit $\mathcal{E}(\pi)$ la sous-catégorie (strictement) pleine de $\mathcal{E}$ définie par

$$\mathrm{Ob}\,\mathcal{E}(\pi)\ =\ \bigcup_{s\,\in\,\mathrm{Ob}\,\mathcal{C}} \mathrm{Ob}\,\mathcal{E}_{s\,\pi}\ .$$

Grâce à $(*)$, et à 1°, 4° et à

$$\begin{cases} \pi_1\ \geqslant\ \sup_s(\mathrm{card\,réd}\,\mathcal{E}_{s\,\pi_0}) & \text{(contenu dans 2°)} \\ \lambda^*(\mathcal{E}_{t\,\pi_0})\ \subset\ \mathcal{E}_{s\,\pi_1} \qquad \text{resp.} \qquad \lambda_*(\mathcal{E}_{s\,\pi_0})\ \subset\ \mathcal{E}_{t\,\pi_1} & \text{(contenu dans 3°)}\ , \end{cases}$$

on voit que $\mathcal{E}(\pi)$ est une sous-catégorie

fibrée, resp. une sous-catégorie cofibrée, sur $\mathcal{C}$ (le foncteur $\mathcal{E}(\pi) \longrightarrow \mathcal{E}$ étant donc cartésien resp. cocartésien). À $\mathcal{C}$-équivalence près, $\mathcal{E}$ se reconstitue alors à partir de $\mathcal{E}(\pi)$, en passant au $\mathrm{Ind}_\pi(\mathcal{E}_{s\,\pi})$, qui forment un système fibré resp. cofibré sur $\mathcal{C}$ par les foncteurs $\mathrm{Ind}_\pi(\lambda_\pi^*)$ resp. $\mathrm{Ind}_\pi(\lambda_*^\pi)$ … [page 437]

Prouvons que $\mathcal{E}$ satisfait à L_π et que $f : \mathcal{E} \longrightarrow \mathcal{C}$ est π-accessible ([182]). Comme $\mathcal{C}$ est karoubienne, et $\pi \geqslant \pi_1 \geqslant \mathrm{card}\,\mathcal{C}$, $\mathcal{C}$ satisfait à L_π et les ind-objets grands devant π de $\mathcal{C}$ sont essentiellement constants (4.8.13). Soit alors I grande devant π, et $I \longrightarrow \mathcal{E}$, $i \longmapsto x_i$, un foncteur, il faut construire une $\varinjlim$ dans $\mathcal{E}$, en utilisant la $\varinjlim$ déjà connue s des $s_i = f(x_i)$. On va prendre un x dans $\mathcal{E}_s$. Prenons d'abord le cas où f est cofibrant. On voit que les $\beta_i(x_i)$ forment un système inductif dans $\mathcal{E}_s$, puisque pour $i = j$, on a

$$\beta_{i\,*}(x_j)\ =\ \beta_{j\,*}\underbrace{\beta_{ji\,*}(x_i)}_{\longrightarrow\,x_j}\ \longrightarrow\ \beta_{j\,*}(x_j)\ .$$

$$\begin{array}{ccc} x_i & \xrightarrow{\quad\alpha_i\quad} & x\,? \\ s_i & \xrightarrow{\quad\beta_i\quad} & s \\ & {\scriptstyle\beta_{ji}}\searrow \quad \nearrow{\scriptstyle\beta_j} & \\ & s_j & \end{array}$$

On pose

$$x\ =\ \varinjlim_I \beta_{i\,*}(x_i)\ , \qquad \varinjlim\ \textit{dans}\ \mathcal{E}_s\ .$$

p. 168
[page 438]

182. Ici on utilise seulement $\pi \geqslant \mathrm{card\,réd}\,\mathcal{C}$.

On sait que c'est une $\varinjlim$ dans $\mathcal{E}$, donc que pour tout y dans $\mathcal{E}$, on a

$$\mathrm{Hom}(x,y) \xrightarrow{\sim} \varprojlim_I \mathrm{Hom}(\beta_{i*}(x_i),y) \ ,$$

ou ce qui revient au même, pour toute flèche $\lambda : s \longrightarrow t$, et y dans $\mathrm{Ob}\,\mathcal{E}_t$, on a

$$\underbrace{\mathrm{Hom}_\lambda(x,y)}_{=\mathrm{Hom}_{\mathcal{E}_t}(\lambda_* x,y)} \simeq \varprojlim \mathrm{Hom}_{\mathcal{E}_t}(\underbrace{\lambda_*\beta_{i*}(x_i)}_{(\lambda_i\beta_i)_*(x_i)},y) \ .$$

Cela résulte aussi du fait que λ_* est π-accessible (étant π_1-accessible) … On vérifie facilement que c'est une $\varinjlim$ des x_i dans $\mathcal{E}$. Et bien sûr, cette limite commute avec f par construction.

Mais dans le cas où f est fibrant, je ne vois comment faire pour construire une limite dans $\mathcal{E}_s$. Il est vrai qu'on peut supposer que tous les s_i sont égaux à s, mais les transitions β_{ji} ne sont pas l'identité, ni même des isomorphismes. Il faudrait prendre $\mathcal{C} = P$ pour essayer de comprendre – je ne vais pas insister ([183]).

Commentary · Exposition [ms. 438]

Constructing the missing colimit in a fibration

Grothendieck has already constructed total colimits in the opfibration case. For a fibration the direction of transport initially seems wrong, but the splitting of the stabilized base idempotent supplies exactly the missing direction. We give a direct proof, using the constant-idempotent reindexing established later in §4.15.7 and the sufficient strict bound in Corollary 4.15.11. This makes the proof independent of an unresolved optimal cardinal estimate.

Replace the fibration by an equivalent split fibration and choose a sufficiently large regular successor κ so that every fibre is κ-accessible, every reindexing functor preserves κ-filtered colimits, and the small-base stabilization theorem applies. This simultaneous choice uses only a small family of accessible functors. Reindex a given κ-filtered diagram cofinally by a κ-directed poset so that every base object is s and every strict transition is one idempotent p. If this poset has a greatest element, its value is the total colimit and the base colimit, and every hom comparison is immediate. Otherwise every index has a strictly larger index, since the poset is directed. Work in this latter case and split

$$s \xrightarrow{r} t \xrightarrow{i} s, \qquad ri = 1_t, \quad ir = p.$$

Write $U = i^* : \mathcal{E}_s \to \mathcal{E}_t$ and $R = r^* : \mathcal{E}_t \to \mathcal{E}_s$, so $UR = 1$ and $RU = p^*$. A strict transition $X_j \to X_k$ is represented by

$$a_{kj} : X_j \longrightarrow RUX_k.$$

Composition says $a_{\ell j} = RU(a_{\ell k})a_{kj}$ for $j < k < \ell$. Therefore $Z_j = UX_j$, with transition $z_{kj} = U(a_{kj})$, is a genuine diagram in $\mathcal{E}_t$. Form $Z = \mathrm{colim}_j Z_j$ there and denote its structural maps by c_j.

For $k > j$, the composite $R(c_k)a_{kj} : X_j \to RZ$ is independent of k: two choices admit a common later index, where the composition identity and $c_\ell z_{\ell k} = c_k$ give equality. These composites define a cocone from the original diagram to the total object Z, lying over r.

To prove universality, take a cocone to $Y \in \mathcal{E}_v$. Its base maps $\lambda_j : s \to v$ satisfy $\lambda_j = \lambda_k p$ for $j < k$. Choosing one further index shows that they are all equal to a map λ with $\lambda p = \lambda$. Hence $\lambda = \mu r$ for the unique $\mu = \lambda i : t \to v$. Its legs are maps $b_j : X_j \to R\mu^* Y$. Applying U gives a compatible family $Ub_j : Z_j \to \mu^* Y$. Conversely that compatible family recovers b_j from any later index by

$$b_j = R(Ub_k)a_{kj} \qquad (k > j).$$

These operations are inverse, by the cocone equation and $UR = 1$. Fibre cocones on Z_j correspond to maps $Z \to \mu^* Y$, so the original cocones correspond precisely to total maps $Z \to Y$. This proves both existence of the total colimit and its preservation by the projection.

The same transport also proves the Karoubian assertion in footnote 183. An idempotent on X over p is represented by $a : X \to RUX$ with $RU(a)a = a$. Then $b = Ua$ is a vertical idempotent on UX. Split $b = jq$, $qj = 1_Y$ in the accessible fibre $\mathcal{E}_t$. The maps represented by $Rq\,a : X \to RY$ and $j : Y \to UX$ give a splitting in the total category: their composites are 1_Y and the original idempotent. Thus the source's observation is correct, and transport explains it. [page 439]

Dans le cas où f est fibrant, admettons (comme hypothèse supplémentaire au besoin) que $\mathcal{E}$ satisfasse L et que f soit accessible, et prenons π assez grand pour que $\mathcal{E}$, f satisfassent L_π. Nous pouvons alors terminer la démonstration de 4.15.6. Il faut prouver

$$\mathcal{E}(\pi) \ \subset \ \mathcal{E}_\pi \ ,$$

donc que si $x \in \mathrm{Ob}\,\mathcal{E}_{s\,\pi}$, et $(y_i)_{i\in I}$ est un système inductif dans $\mathcal{E}$ grand devant π, de limite y, on a

$$\varinjlim \mathrm{Hom}_{\mathcal{E}}(x, y_i) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(x, y) \ .$$

Donc il faut prouver que toute flèche $x \xrightarrow{u} y$ se factorise, de façon essentiellement unique u_i, par un y_i. Mais la flèche $s \xrightarrow{\lambda = f(u)} t$ dans $\mathcal{C}$ se factorise, de façon essentiellement unique, par des $\lambda_i : s \longrightarrow t_i$. Donc on est ramené à trouver un λ_i-*morphisme* $x \longrightarrow y_i$ pour i grand, qui factorise. Dans le cas fibrant, cela signifie qu'on cherche $x \longrightarrow \lambda_i^*(y_i)$ qui factorise $x \longrightarrow \lambda^*(y)$, or les $\lambda_i^*(y_i)$ forment un système inductif dans $\mathcal{E}_s$, et sa limite *dans* $\mathcal{E}_s$ est $\lambda^*(y)$ (*à vérifier*). Comme x est dans $\mathcal{E}_{s\,\pi}$, l'hom `[omorphisme]` donné $x \longrightarrow \lambda^*(y)$ se factorise de façon essentiellement unique par un $\lambda_i^*(y_i)$. [page 440]

Commentary · Exposition [ms. 439]

Presentability after splitting the base idempotent

The preceding construction identifies the colimit whose existence was assumed on this page. It also verifies the pullback comparison marked "à vérifier", and proves the presentability statement needed for accessibility.

Keep the notation $Z_j = UX_j$, $Z = \operatorname{colim} Z_j$ and the splitting $s \xrightarrow{r} t \xrightarrow{i} s$. Let $x \in \mathcal{E}_u$ be κ-presentable in its fibre. A total arrow $x \to Z$ over $\nu : u \to t$ is a fibre arrow

$$x \longrightarrow \nu^* Z \cong \operatorname*{colim}_j \nu^* Z_j.$$

The displayed isomorphism follows from the chosen accessibility rank of ν^*. Fibre presentability factors this map through $\nu^* Z_j = (i\nu)^* X_j$, giving a total arrow $x \to X_j$ over $i\nu$. The cocone leg $X_j \to Z$ over r has fibre representative $d_j = R(c_k)a_{kj}$ for any $k > j$, and

$$U(d_j) = c_k U(a_{kj}) = c_k z_{kj} = c_j.$$

Consequently composition with that leg applies $\nu^* c_j$ to the chosen factorization, giving exactly the prescribed arrow to Z; its base is $ri\nu = \nu$. This proves surjectivity of the total hom comparison.

For injectivity, take two representatives whose composites to Z agree. Passing to a common index makes their base arrows have the same composite with r. Passing once more through a transition over $p = ir$ makes both base arrows equal to $i\nu$ for that common ν. They can now be interpreted as maps into the one fibre diagram $\nu^* Z_j$. Their equality at the colimit is witnessed at a later stage by the injectivity of the fibre hom comparison. This is exactly equality of the original total representatives after that transition.

Every fibrewise κ-presentable object is therefore κ-presentable in $\mathcal{E}$. Fibre inclusions preserve the relevant colimits: for a diagram in $\mathcal{E}_u$, test its fibre colimit against $Y \in \mathcal{E}_v$ separately over every $\lambda : u \to v$ using $\mathcal{E}_u(-, \lambda^* Y)$. Since the base is small, the union of small generating families in the fibres is small, and it generates all total objects. The fibration case of Proposition 4.15.6 follows, including accessibility of its projection.

Makkai–Paré's general theory treats accessible indexed categories and their total categories in §5.3 (Makkai and Paré, 1989). The proof here records directly the special small Karoubian-base situation Grothendieck needs; we do not use an unverified identification of a strict pullback with a two-categorical limit. For a (co)fibration, however, pullback along an equivalence of bases is safe because isomorphisms lift. This explains the reduction to a small skeleton at the start of his argument.

183. J'ai vérifié que si f est fibrant, alors $\mathcal{E}$ est *karoubienne*. Il y a donc de l'espoir …

Cette fois par contre, c'est le cas f cofibrant qui pose problème. En fait, je n'arrive pas à m'en tirer.

Ainsi, on trouve bien un foncteur

$$\mathrm{Ind}_\pi(\mathcal{E}(\pi)) \longrightarrow \mathcal{E}\ ,$$

qui est essentiellement surjectif, mais faute de savoir que $\mathcal{E}(\pi) \subset \mathcal{E}_\pi$ dans le cas cofibrant, il n'est pas clair que ce foncteur soit une équivalence de catégories. p. 169

Commentary · Exposition [ms. 440]

Completing the opfibration argument

The opfibration proof needs an additional transition after a fibre factorization. Choose a common strong rank κ for the direct-image functors: they then preserve κ-filtered colimits and κ-presentable objects. This is the uniformization recalled by Paré and Rosický (2013, Introduction and proof of Theorem 2.1). Reindex and split the base idempotent as above, and put

$$R = r_* : \mathcal{E}_s \to \mathcal{E}_t, \qquad U = i_* : \mathcal{E}_t \to \mathcal{E}_s,$$
$$RU = 1, \qquad UR = p_*.$$

A transition is represented by $a_{kj} : URX_j \to X_k$, with $a_{\ell j} = a_{\ell k} UR(a_{kj})$ for $j < k < \ell$. The diagram $Z_j = RX_j$ has transitions $z_{kj} = R(a_{kj})$, and its fibre colimit Z is the total colimit. To check this last assertion, base cocones again have one form μr. Maps out of Z over μ are maps $\mu_* Z \to Y$; preservation of the fibre colimit by μ_* identifies them with exactly the cocones $\mu_* RX_j \to Y$ arising from the original diagram. This completes the universal-property verification sketched on page 438.

Let $x \in \mathcal{E}_u$ be fibrewise κ-presentable. A total map $x \to Z$ over $\nu : u \to t$ is a map $h : \nu_* x \to Z$. Since $\nu_* x$ is κ-presentable, it factors through some $Z_j = RX_j$, say by $h_j : \nu_* x \to RX_j$. Applying U gives a map into URX_j, rather than into X_j. One later transition supplies the missing arrow:

$$(i\nu)_* x = U\nu_* x \xrightarrow{Uh_j} URX_j \xrightarrow{a_{kj}} X_k \qquad (k > j).$$

Its composite to Z is the desired total map. This proves surjectivity.

For injectivity, normalize two representatives to the same index and the same base arrow $i\nu$ by passing to a further transition. Write their fibre representatives as $b, b' : U\nu_* x \to X_j$. Equality at Z means equality of the two maps $Rb, Rb' : \nu_* x \to RX_j$ in the colimit. Fibre presentability gives a stage k where their composites with z_{kj} agree. After one additional transition to $\ell > k$, the original representatives become

$$a_{\ell k}\, U(z_{kj} Rb) \quad \text{and} \quad a_{\ell k}\, U(z_{kj} Rb'),$$

which are equal. The composition law above shows these are exactly their images in X_ℓ. Thus the total hom comparison is bijective.

Fibre generation and preservation of its colimits by the inclusions now give accessibility of the total category, and the displayed Ind functor is an equivalence at the chosen rank. The doubts in footnotes 180–181 concern an unfinished proof, not a false conclusion. The abstract result is consistent with Makkai–Paré's Theorem 5.4.7, which computes lax colimits of strong diagrams of accessible categories by taking the idempotent completion of the categorical lax colimit; this formulation is explicitly recalled by Paré and Rosický (2013, Introduction). We checked that attribution there rather than claiming to have inspected the unavailable book proof. The calculations above establish both orientations needed in this manuscript directly.

4.15.7. Il me faut reporter la démonstration du lemme 4.15.3 (p. 428), en admettant 4.8.13. (Non encore démontré – je vais revenir plus bas sur la démonstration de 4.8.13.) On va utiliser le

Lemme 4.15.7.1. *Soient $\mathcal{C}$ une catégorie, $\underline{x} = (x_i)_{i\in I}$ un ind-objet de $\mathcal{C}$. Pour que $\underline{x}$ soit essentiellement constant, i.e. isomorphe, dans* $\mathrm{Ind}(\mathcal{C})$*, au ind-objet constant défini par un objet*

de $\mathcal{C}$, il faut et il suffit qu'il existe un foncteur cofinal $\varphi : J \longrightarrow I$, avec J filtrant, de telle façon que le système réindexé $(x_j)_{j\in J}$, avec $x_j = x_{\varphi(j)}$, ait la propriété suivante : pour tout [page 441]
$j \in J$, on peut trouver un projecteur p_j dans x_j, de telle façon que :

1°) Pour toute flèche $\lambda : j \longrightarrow j'$ dans J, on ait commutativité dans

$$\begin{array}{ccc} x_j & \xrightarrow{x_\lambda} & x_{j'} \\ \downarrow{\scriptstyle p_j} & & \downarrow{\scriptstyle p_{j'}} \\ x_j & \xrightarrow{x_\lambda} & x_{j'} \end{array}$$

(où les x_λ sont les flèches de transition), i.e. $p_{j'} x_\lambda = x_\lambda p_j$.

2°) Pour tout $j \in \mathrm{Ob}\, J$, il existe une flèche $\lambda : j \longrightarrow j'$ dans J telle que $p_{j'} x_\lambda = x_\lambda p_j = x_\lambda$.

3°) Pour tout $j \in \mathrm{Ob}\, J$, le projecteur p_j dans x_j admet un image $x_j^\natural$.

4°) Pour toute flèche $\lambda : j \longrightarrow j'$ dans J, la flèche $x_\lambda^\natural : x_j^\natural \longrightarrow x_{j'}^\natural$ induite par x_λ (grâce à 1°) est un isomorphisme.

De plus, si on peut supposer J de la forme $i_0\backslash I$, le foncteur cofinal $J \longrightarrow I$ étant le foncteur canonique $i_0\backslash I \longrightarrow I$ (i_0 étant un objet fixé convenable de I) `[phrase incomplète]`

Notons tout de suite deux corollaires. [page 442]

Commentary · Exposition [ms. 441]

The retract inside an essentially constant system

Grothendieck wants to recover an actual stage of a large diagram from the ind-object it represents. The obstruction is already visible in a finite calculation: the limiting object may be a retract of a stage without being the whole stage. The projectors in 4.15.7.1 identify precisely the part which survives. We write $h_x = \mathrm{Hom}_{\mathcal{C}}(-, x)$ and

$$X = \varinjlim_{i \in I} h_{x_i} \quad \text{in} \quad [\mathcal{C}^{\mathrm{op}}, \mathrm{Ens}].$$

Essentially constant means that X is representable. It does not mean that the displayed diagram was already constant, or that its colimit inside $\mathcal{C}$, if one exists, is represented by the same presheaf.

Here is the proof with the two inverse maps kept in their correct order. Suppose $X \simeq h_z$, and write

$$\alpha : h_z \longrightarrow X, \qquad \beta : X \longrightarrow h_z, \qquad \beta\alpha = 1_{h_z}, \quad \alpha\beta = 1_X.$$

Yoneda gives a representative $\alpha_{i_0} : z \to x_{i_0}$ of α, while β gives compatible maps $\beta_i : x_i \to z$. The compatibility equation is $\beta_j \tau_{ji} = \beta_i$, for $\tau_{ji} : x_i \to x_j$. Several equations on manuscript pp. 449–450 reverse these composites; the reversals are printed so in the edition.

The upper slice $J = i_0\backslash I$ makes the chosen representative coherent. An object of J is an arrow $u : i_0 \to i$, and we put $\alpha_u = x(u)\alpha_{i_0}$. Then

$$\beta_i \alpha_u = 1_z, \qquad p_u = \alpha_u \beta_i, \qquad p_u^2 = p_u.$$

Thus p_u splits with image z, and a morphism of the slice acts as the identity on that image. Its naturality follows from the compatibility of both families. Finally $\alpha\beta = 1_X$ says that the class of p_u equals the class of 1_{x_i} in the appropriate filtered colimit of Hom sets. Equality there has a witness: some arrow $u \to v$ satisfies $x(u \to v)p_u = x(u \to v)$. Naturality gives the other absorbed composite. These are exactly the four conditions of the lemma.

For completeness, the slice has the required size of filteredness. A small diagram in $i_0\backslash I$ becomes a diagram in I after adjoining i_0 and its specified arrows. A cocone in I therefore gives a cocone in the slice. The same construction shows that the projection is final. In particular, if I admits cocones for diagrams of cardinal at most π_0, so does J.

Conversely, choose splittings $x_j \xrightarrow{r_j} z_j \xrightarrow{s_j} x_j$ of the stated projectors. Condition 1 makes the z_j a diagram, and condition 4 makes its arrows isomorphisms. A filtered category has trivial groupoid completion: common upper objects compare two paths, and equalization of parallel arrows removes their ambiguity. Hence the z_j are coherently isomorphic to one object z. Transporting r_j and s_j through this isomorphism produces maps $\beta : X \to h_z$ and $\alpha : h_z \to X$. The splitting identities give $\beta\alpha = 1$, while condition 2 gives $\alpha\beta = 1$ in the colimit. This also repairs the incomplete final sentence of the statement: one *can* choose the upper slice described above.

Corollaire 4.15.7.2. *Supposons que $\mathcal{C}$ soit sans projecteurs. Alors $\underline{x} = (x_i)_{i\in I}$ est un ind-objet essentiellement constant si et seulement s'il existe un foncteur cofinal, $J \longrightarrow I$, avec J filtrant ([184]), de telle façon que le système inductif réindexé $(x_j)_{j\in J}$ soit isomorphe* (dans la catégorie $\underline{\mathrm{Hom}}(J, \mathcal{C})$) *à un système inductif* constant *(i.e. défini par un foncteur constant $J \longrightarrow \mathcal{C}$: tous les x_j égaux à un même x, et les morphismes de transition des identités).*

La suffisance est claire, pour la nécessité on applique 4.15.7.1, qui nous dit qu'on peut réindexer de façon à satisfaire aux conditions 1°) à 4°) (pour des projecteurs p_j convenables dans les x_j). Mais l'hypothèse sur $\mathcal{C}$ implique que $p_j = \mathrm{id}_{x_j}\ \forall\, j \in \mathrm{Ob}\, J$, et les conditions 1°, 2°, 3° sont satisfaites tautologiquement pour un tel choix, la condition 4° signifie que les morphismes de transition sont des isomorphismes. Mais il est immédiat que pour un système inductif $(x_j)_{j\in J}$ donné, avec J 1-*connexe* (p.ex. J filtrant), la condition que les x_j sont des isomorphismes est nécessaire et suffisante pour que ce système soit isomorphe à un système inductif constant, d'où la conclusion.

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[page 443]

Corollaire 4.15.7.3. *Supposons que I soit grande devant π_0, où π_0 est un cardinal infini tel que $\pi_0 \geqslant \mathrm{card}\, \mathcal{C}$. Alors $(x_i)_{i\in I}$ est essentiellement constant si et seulement si on peut trouver un foncteur cofinal $J \longrightarrow I$ avec J grande devant π_0 (et ordonnée si on veut) ([185]), de telle façon que le système réindexé $(x_j)_{j\in J}$ satisfasse les conditions suivantes :*

1°) L'application $j \longmapsto x_j$ de $\mathrm{Ob}\, J$ dans $\mathrm{Ob}\, \mathcal{C}$ est constante, i.e. il existe un $x_0 \in \mathrm{Ob}\, \mathcal{C}$ tel que $x_j = x_0\ \forall\, j \in \mathrm{Ob}\, J$.

2°) Il existe un projecteur p dans x_0 tel que pour $j < j'$ dans J, on ait $\tau_{j',j} = p$.

3°) Le projecteur p admet une image dans $\mathcal{C}$.

Démonstration. Ces conditions impliquent évidemment celles de 4.15.7.1, en prenant $p_j = p\ \forall\, j \in \mathrm{Ob}\, J$, du moins si J n'admet de plus grand élément (ce qui fait une difficulté pour 2°). Mais si J admet un objet final, il en est de même de I, et il est clair alors que tout système inductif indexé par I est essentiellement constant. Donc, dans ces cas, les conditions 1°) 2°) 3°) ci-dessus impliquent que $\underline{x}$ est essentiellement constant. Inversement, supposons que [page 444]
$\underline{x}$ soit essentiellement constant, et prouvons que l'on peut réindexer comme indiqué. Le cas où I admet un objet final est trivial, on prendra $J = e$, $p = \mathrm{id}$. Supposons donc que I n'ait pas d'objet final. Comme I grande devant π_0 et $\pi_0 > \mathrm{card}\, \mathrm{Ob}\, \mathcal{C}$, on voit qu'il existe un $x_0 \in \mathrm{Ob}\, \mathcal{C}$ tel que la sous-catégorie pleine $I' = I_{x_0}$ de I, formée des i tels que $x_i = x_0$, soit cofinale dans I. Comme $(x_i)_{i\in I}$ est essentiellement constant, il en est de même de $(x_{i'})_{i'\in I'}$, et on peut lui appliquer le critère 4.15.7.1, avec $J \longrightarrow I'$ cofinal. La démonstration montrera de plus que si I' ||
est grand devant π_0 (cardinal infini sans autres conditions), on peut supposer J grande devant π_0 ([186]). De plus, il est clair qu'on peut supposer J ordonnée. Comme $\pi_0 \geqslant \mathrm{card}\, \mathrm{Hom}(x_0, x_0)$, on voit encore qu'il existe un projecteur p dans x_0 tel que la partie $J_1 = J'$ de J, formée des j tels que $p_j = p$, soit cofinale. Étant cofinale, elle est également grande devant π_0. Donc on peut supposer que tous les p_j sont égaux à un même p. Les conditions 1°-4° deviennent :

1°) $p\tau_{j'j} = \tau_{j'j}p$ pour $j \leqslant j'$.

2°) Pour tout j, existe $j' \geqslant j$ avec

$$p\tau_{j'j} \;=\; \tau_{j'j}p \;=\; \tau_{jj'}\ .$$

3°) L'image $x_0^\natural$ du projecteur p dans x_0 existe.

4°) L'endomorphisme $\tau^\natural_{j'j}$ de $x_0^\natural$ est un isomorphisme, pour $j \leqslant j'$.

Or on a le [page 445]

184. et on peut supposer $J = i_0\backslash I \xrightarrow{\text{can.}} I$

185. La démonstration (légèrement réarrangée) montre qu'on peut prendre pour J une sous-catégorie (pas pleine) d'une catégorie $i_1\backslash I$, pour un objet convenable i_1 de I. On a donc $\mathrm{card}\, J \leqslant \pi$ si $\mathrm{card}\, I \leqslant \pi$, π étant un cardinal infini donné.

186. en tenir compte dans la démonstration de 4.15.7.1.

Commentary · Context [ms. 442]

Constancy in the ind-category and constancy of a diagram

The distinction Grothendieck is using belongs to the original ind-object formalism. In SGA 4, Exposé I, 8.5.3, the quotation marks around a formal inductive limit indicate its formation in the ind-category; they prevent its identification with a colimit calculated in the original category. The same exposé, 8.7.8, locates the Karoubi envelope among retracts of representables (Artin et al., 1972). These pages return to that distinction because a small base category is supposed to stop a sufficiently filtered diagram from wandering. A surviving projector is the precise residue of that wandering.

Corollary 4.15.7.2 has the stronger hypothesis that there are *no nonidentity idempotents*. This is not another expression for idempotent completeness. A category of vector spaces is idempotent complete but has many projections. A groupoid has only identity idempotents and is therefore automatically idempotent complete. With the stronger hypothesis, every p_j in the preceding lemma is 1_{x_j}, and the image diagram is the original diagram.

There remains a small distinction even then. An isomorphism-valued diagram is naturally isomorphic to a constant diagram; its arrows need not literally be identities. To see the trivialization directly, fix j_0. For j, choose a common upper object k and put

$$\theta_j = x(j \to k)^{-1} x(j_0 \to k) : x_{j_0} \longrightarrow x_j.$$

Filteredness makes this independent of the common upper object and of the chosen arrows, and gives $x(\lambda)\theta_j = \theta_{j'}$. The family θ is the required natural isomorphism. The later thinning argument obtains literal identity arrows by passing to a further subsystem. Keeping these two operations apart will matter in Theorem 4.15.9, where the printed condition b) incorrectly demands identity arrows on an entire tail.

Commentary · Examples [ms. 443]

A two-element monoid which creates one missing retract

The projectors in the criterion are necessary even when the category is as small as possible. Let $\mathcal{C}$ have one object c and endomorphism monoid

$$M = \{1, p\}, \qquad p^2 = p, \qquad p \neq 1.$$

Choose any infinite regular ordinal κ, regard it as a directed poset, and form the diagram which has value c at every stage and transition p at every strict inequality. Composition holds because $p^2 = p$; identity arrows still have value 1.

The representable h_c has value M at the unique object. Every strict transition in its presheaf diagram is left multiplication by p, sending both elements to p. Consequently

$$X(c) = \varinjlim_{\alpha<\kappa} M = \{*\}.$$

The two elements of each stage become equal after one transition, and the resulting class is nonempty. The presheaf X is therefore the one-element right M-set, with both elements of M acting identically. It is a retract of h_c: the unique map $M \to \{*\}$ has the equivariant section $* \mapsto p$. Their composite on M is precisely left multiplication by p.

Yet X is not representable in $\mathcal{C}$, since its value has one element whereas the only representable has two. The ind-object is not essentially constant. The projectors commute, eventually absorb every transition, and induce identity maps on their images in the presheaf category; condition 3 of 4.15.7.1 fails exactly because that image does not belong to $\mathcal{C}$.

The Karoubi envelope repairs the defect by adjoining an object $d = (c, p)$. Explicitly,

$$\mathrm{Hom}(c, d) = \{p\}, \quad \mathrm{Hom}(d, c) = \{p\}, \quad \mathrm{End}(d) = \{p\},$$

where the last p is the identity of d. The composite $d \to c \to d$ is 1_d, and the reverse composite is the original endomorphism p of c. In this enlarged category the same diagram represents d.

Taking $\kappa = \pi^+$ makes the indexing poset grand devant π for any chosen infinite π. Thus increasing the cardinal cannot remove the need to split idempotents. It is the conclusion *Karoubi envelope*, rather than the original category, which survives without a hypothesis on projectors.

Lemme 4.15.7.4. *Soient $\mathcal{C}$ une catégorie, x_0 un objet de $\mathcal{C}$, p un projecteur dans x_0, τ un endomorphisme de x_0 satisfaisant les relations*

$$(*) \qquad \tau p \;=\; p\tau \;=\; \tau \;,$$

et tel que l'automorphisme de $x_0^\natural \overset{\text{déf}}{=} \operatorname{Im} p$ (défini p.ex. comme objet de $\mathcal{C}^\wedge$, $x_0^\natural = \operatorname{Ker}(\mathrm{id}_{x_0}, p)$ p. 171 *soit l'identité. Alors on a $\tau = p$.*

DÉMONSTRATION DE 4.15.7.4. On est ramené aussitôt, par le procédé standard, au cas où $\mathcal{C} = \mathrm{Ens}$, où la vérification est immédiate : les automorphismes τ de l'ensemble E $(= x)$ qui satisfait $(*)$ correspondent biunivoquement aux endomorphismes u de $E^\natural = pE$, via

$$\begin{cases} u \;\longmapsto\; \sigma u \rho \\ \tau \;\longmapsto\; \tau | E^\natural \end{cases} \qquad (\text{où} \quad E^\natural \underset{\rho}{\overset{\sigma}{\rightleftarrows}} E \quad \text{sont les flèches canoniques}) \;,$$

donc $\tau | E^\natural = \mathrm{id} \Longrightarrow u = \sigma\rho = p$.

Choisissons un $j_0 \in J$, et considérons les homomorphismes

$$\tau^\natural_{j,j_0} \;:\; x_0^\natural \longrightarrow x_0^\natural$$

induits par les τ_{j,j_0} avec $j \geqslant j_0$. Ici encore, comme l'ensemble des isomorphismes possibles est de

cardinal $\leqslant \pi_1$, et J/j_0 est grande devant π_0, on voit qu'il existe $J' \subset j_0\backslash J$, cofinale dans [page 446] $j_0\backslash J$, donc dans J, telle que pour $j \in J'$, les $\tau^\natural_{j,j_0}$ soient égaux à un même $\tau : x_0^\natural \xrightarrow{\sim} x_0^\natural$. Donc si $j', j'' \in J'$, avec $j' \leqslant j''$, on a

$$\underbrace{\tau_{j''j_0}}_{\tau} \;=\; \tau_{j''j'} \; \underbrace{\tau_{j'j_0}}_{\tau} \;,$$

d'où

$$\tau_{j''j'} \;=\; \mathrm{id}_{x_0^\natural} \;,$$

puisque la flèche τ est un isomorphisme. Donc le système inductif réindexé par J', la condition 4° est remplacée par la condition plus stricte $\boxed{4°\text{ bis) } \tau^\natural_{j',j} = \mathrm{id}_{x_0^\natural}.}$ Mais la condition 2° devient alors, grâce à 4.15.7.4,

$$\boxed{2°\text{ bis) Pour tout } j, \text{ existe } j' \geqslant j \text{ avec } \tau_{j',j} = p} \;.$$

Donc, on peut supposer 1°, 2° bis, 3°, 4° bis (on désigne à nouveau par J, au lieu de J', l'ensemble ordonné grand devant π_0 indexant).

Soit K l'ensemble ordonné défini par

[page 447]

$$\begin{cases} \operatorname{Ob} K \;=\; \operatorname{Ob} J \\ \text{si } j, j' \in \operatorname{Ob} J, \text{ on a } j \overset{K}{\leqslant} j' \text{ si et seulement si } j \overset{\mathrm{J}}{=} j', \text{ ou } j \overset{J}{<} j' \text{ et } \tau_{j',j} = p \,. \end{cases}$$

Il est immédiat que c'est une *relation d'ordre* sur J (car $p^2 = p$), plus fin que la relation d'ordre initiale. Le système inductif induit sur K satisfait aux conditions 1°, 2°, 3° de 4.15.7.3. D'autre part, $K \longrightarrow J$ est cofinal, donc aussi le composé $K \longrightarrow J \longrightarrow I$. Il reste à prouver que K est grande devant π_0, donc que toute partie $K' \subset K$ telle que $\operatorname{card} K' \leqslant \pi_0$ admet un majorant dans K. Elle admet un majorant *dans* J, soit j. Par hypothèse 2 bis, il existe $j' \geqslant j$ tel que $\tau_{j'j} = p$, et on peut supposer $j' > j$. (En effet, si $\tau_{j'j} = p$, on voit que la forme p. 172 initiale 2°) de la condition 2° bis que l'on a $\tau_{j''j} = p$ pour tout $j'' \geqslant j'$, donc si j n'est par [page 448] objet final de J, on peut supposer $j' > j$. Mais J n'a pas d'objet final, car sinon I en aurait

un, cas que l'on avait éliminé. Donc il existe j' tel que $j' \overset{K}{>} j$.) Or on a ceci (qu'on applique au cas $i \in K'$) :

$$i \leqslant j \ \text{ et } \ j \overset{K}{<} j' \qquad \Longrightarrow \qquad i \overset{K}{<} j' \ .$$

$$\underbrace{x_i}_{=x_0} \xrightarrow{\tau_{j,i}} x_j \xrightarrow{\tau_{j',j}=p} x_{j'} \ ,$$

$$\tau_{j'i} \ = \ \tau_{j'j}\tau_{ji} = \underbrace{p\tau_{ji}}_{\overset{1^\circ}{=}\tau_{ji}p} \overset{?}{=} \ p \ ,$$

on a en effet

$$\left.\begin{aligned} \tau_{j',i} \ &= \ p\tau_{j',i} \\ \tau_{j',i} \ &- \ \tau_{j',i}p \end{aligned}\right\} \quad \overset{\text{par } 4.15.7.4}{\Longrightarrow} \quad \tau_{j',i} \ = \ p \ .$$

Cela achève la démonstration du corollaire 4.15.7.3.

Corollaire 4.15.7.5 (de 4.15.7.3). *Soit $(x_i)_{i\in I}$ un système inductif essentiellement constant grand devant $\pi_0 \geqslant \operatorname{card}\mathcal{C}$, dans $\mathcal{C}$* sans projecteurs. *Alors il existe $J \longrightarrow I$ cofinal, avec J ordonné grand devant π_0, et tel que le système inductif induit sur J soit* constant.

Démontrons le lemme 4.15.7.1. Il faut expliciter l'existence de deux flèches dans $\mathrm{Ind}(\mathcal{C})$,

$$z \underset{\beta}{\overset{\alpha}{\rightleftarrows}} \text{"}\varinjlim_{i\in I}\text{"} x_i \ = \ \underline{x} \ ,$$

avec $\beta\alpha = \mathrm{id}_z$, $\alpha\beta = \mathrm{id}_{\underline{x}}$. La donnée

de x équivaut à celle d'un élément de $\varinjlim_{i\in I} \mathrm{Hom}(z, x_i)$, et est donc donné par un [page 449]

$$\alpha_{i_0} \ : \ z \longrightarrow x_{i_0} \ ,$$

la donnée de β équivaut à celle d'un élément de $\varprojlim_{i\in I} \mathrm{Hom}(x_i, z)$, i.e. de flèches

$$\beta_i \ : \ x_i \longrightarrow z \ ,$$

avec la condition de compatibilité habituelle. Quitte à remplacer I par I', avec $I' \longrightarrow I$ cofinal, on peut supposer I ordonnée (et grande devant π_0, si I initiale l'était), et la condition de compatibilité sur les β_i s'écrit

$$\beta_i\tau_{ji} \ = \ \beta_j \qquad \text{si } i \leqslant j \text{ dans } I,$$

où $\tau_{ji} : x_i \longrightarrow x_j$ désigne la flèche de transition. Il faut expliciter, en termes de ces données, les conditions

$$\alpha\beta \ = \ \mathrm{id}_z \ , \qquad \beta\alpha \ = \ \mathrm{id}_{\underline{x}} \ .$$

Commentary · Exposition [ms. 448]

A cofinal subsystem with one projector

The purpose of 4.15.7.3 is stronger than representability. Grothendieck needs a presentation whose arrows in the small base are literally controlled, so that a diagram upstairs can be studied in one fibre. The argument combines a cardinal pigeonhole principle with a careful deletion of arrows. Both operations preserve the filteredness needed later.

Throughout this argument, grand devant π_0 means that diagrams with at most π_0 morphisms admit a cocone. For a poset this says that every subset of cardinal at most π_0 has an upper bound; in modern terminology it is π_0^+-directed. Suppose such a poset I is coloured by a set A of at most π_0 colours. If no colour fibre were cofinal, we could choose, for every $a \in A$, an element i_a above which colour a never occurs. A common upper bound of all i_a would have no colour. Hence one fibre

is cofinal. A cofinal full subposet remains π_0^+-directed, since an upper bound in I can be moved into that subposet.

Replace the original index by a final π_0^+-directed poset, using the usual cofinal-poset theorem (also Lurie, 2009, Proposition 5.3.1.18 in its higher form). First apply this principle to the objects x_i, and then to the projectors supplied by 4.15.7.1. We obtain a fixed object x_0 and a fixed split idempotent p. Fixing a base index and applying the principle once more makes the isomorphisms from its image to the later images equal. Cancellation of these isomorphisms makes all induced transitions on $\operatorname{Im} p$ identities. There are at most π_0 possible image automorphisms: if $p = sr$ with $rs = 1$, the map $u \mapsto sur$ embeds their set into $\operatorname{End}_{\mathcal{C}}(x_0)$.

The elementary identity behind the remaining step is

$$\tau p = p\tau = \tau, \quad r\tau s = 1 \quad \Longrightarrow \quad \tau = sr\tau sr = s(r\tau s)r = p.$$

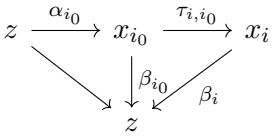
Thus the eventual absorption condition says that every index eventually maps by p. Define K on the same objects by retaining identities and only those strict arrows whose value is p. Transitivity follows from $p^2 = p$. To see that K remains π_0^+-directed, take at most π_0 objects and choose a common upper bound j in the old order. Choose $j' > j$ with $\tau_{j'j} = p$; the terminal-object case is treated separately by a singleton index. For each chosen i,

$$\tau_{j'i} = p\tau_{ji}.$$

This arrow is absorbed by p on both sides, and its action on the image is the identity. The displayed identity therefore gives $\tau_{j'i} = p$. Hence j' is an upper bound in K.

The inclusion $K \to J$ is final as a functor, not merely surjective on objects. For a fixed $j \in J$, the comma category j/K is nonempty; any two of its objects acquire a common upper object in K by the same construction. It is a directed poset and therefore connected. This is the missing explicit finality verification. The conclusion has exactly the form required: one object, one idempotent on every strict transition, and a sufficiently directed indexing category. The printed π_1 and lower slice J/j_0 on manuscript p. 446 should be read as π_0 and $j_0\backslash J$; they are retained in the source text and registered separately.

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La relation $\alpha\beta = \mathrm{id}_z$ équivaut à

$$(*_{i_0}) \qquad \beta_{i_0}\alpha_{i_0} \;=\; \mathrm{id}_z \;,$$

elle implique d'ailleurs, en posant pour $i \geqslant i_0$,

$$\alpha_i \;\overset{\text{déf}}{=}\; \tau_{i,i_0}\alpha_{i_0} \;:\; z \longrightarrow x_i \;,$$

que l'on a pour $i \geqslant i_0$

$$(*_i) \qquad \beta_i\alpha_i \;=\; \mathrm{id}_z \qquad \forall\, i \geqslant i_0 \;.$$

Posons alors

$$p_i \;=\; \alpha_i\beta_i \;:\; x_i \longrightarrow x_i \;, \qquad \text{[page 450]}$$

il est clair qu'on a

$$\begin{cases} p_i^2 \;=\; p_i \qquad (p_i \text{ un projecteur}) \\ \alpha_i \;:\; z \longrightarrow x_i \qquad \text{induit un isomorphisme} \\ \qquad z \xrightarrow{\sim} x_i^\natural \;\overset{\text{déf}}{=}\; \operatorname{Im} p_i \;=\; \operatorname{Ker}(\mathrm{id}_{x_i}, p_i) \;. \end{cases}$$

L'ensemble ordonné cofinal cherché J est $i_0\backslash I$, il est filtrant, et est grand devant π_0 si I l'est. À vrai dire, la réduction au cas I ordonné est inutile, on peut prendre directement $J = i_0\backslash I$,

il est immédiat que J est filtrante (I l'étant), et $i_0\backslash I \longrightarrow I$ cofinal, enfin que si I est grande devant un cardinal infini π_0, il en est de même de $i_0\backslash I$.

Exprimons que $\beta\alpha = \mathrm{id}_{\underline{x}}$; or on voit que $\beta\alpha$ est défini par l'endomorphisme $(p_j)_{j\in J}$ du système inductif $(x_j)_{j\in J}$ – le fait que ce soit bien un endomorphisme de ce système, i.e. qu'on ait la relation 1°) de 4.15.7.1,

$$p_{j'}x_\lambda \;=\; x_\lambda p_\lambda \qquad \text{si } \lambda : j \longrightarrow j' \text{ dans } J,$$

est immédiat, ça s'écrit

$$\alpha_{j'}\underbrace{\beta_{j'}x_\lambda}_{\beta_j} \;=\; \underbrace{x_\lambda\alpha_j}_{\alpha_{j'}}\beta_j\ , \qquad \text{or les deux membres sont égaux à } \alpha_{j'}\beta_j.$$

Dire que c'est l'identité, est exprimé par la condition 2°) sur $(x_j, p_j)_{j\in J}$. La condition 3° est clair par construction, p_j admettant une image isomorphe à z par α_j. D'autre part, on voit [tout] de suite que pour une flèche $\lambda : j \longrightarrow j'$ de J, on a commutativité dans p. 174 [page 451]

$$\begin{array}{ccc} x_j & \xrightarrow{x_\lambda} & x_{j'} \\ \uparrow & & \uparrow \\ x_j^\natural & \xrightarrow{x_\lambda} & x_{j'}^\natural \\ \wr\uparrow{\scriptstyle \overline{\alpha_j}} & & \wr\uparrow{\scriptstyle \overline{\alpha_{j'}}} \\ z & \xrightarrow[\sim]{\mathrm{id}_z} & z\ , \end{array}$$

où $\overline{\alpha_j}$, $\overline{\alpha_{j'}}$ sont les isomorphismes définis par α_j, $\alpha_{j'}$ respectivement. Cela prouve que $x_\lambda^\natural$ est un isomorphisme.

Cela prouve que si $(x_i)_{i\in I}$ est essentiellement constant, il existe $J \longrightarrow I$ comme indiqué. L'inverse est également vrai, car on peut supposer $J = I$, puisque $(x_i)_{i\in I}$ et $(x_j)_{j\in J}$ définissent des ind-objets isomorphes. Alors le système inductif des $x_j^\natural$ a des morphismes de transition des isomorphismes, et comme J est filtrante donc 1-connexe, on en déduit facilement qu'il est isomorphe à un système inductif constant, de valeur z. En choisissant un tel isomorphisme [page 452]

$$z \xrightarrow[\sim]{\overline{\alpha_i}} x_i^\natural\ ,$$

on trouve $z \xrightarrow{\alpha} \underline{x}$, et le système des p_i définit des

$$x_i \xrightarrow{\overline{p_i}} x_i^\natural \xrightarrow{\overline{\alpha_j}^{-1}} z\ , \qquad \beta_i \overset{\text{déf}}{=} \overline{\alpha_j}^{-1}\overline{p_i}$$

et on voit facilement que l'on aura

$$\beta\alpha \;=\; id_z\ , \qquad \alpha\beta \;=\; \mathrm{id}_{\underline{x}}\ ,$$

q.e.d.

Je peux maintenant prouver 4.15.3. C'est en effet essentiellement la conjonction des 4.15.7.5 (p. 448), et de 4.8.13, qui nous dit essentiellement que si $\mathcal{C}$ est *karoubienne* (condition vérifiée si $\mathcal{C}$ est sans projecteurs), alors tout système inductif dans $\mathcal{C}$, grand devant π_0, où $\pi_0 \geqslant \mathrm{card}\ \mathrm{réd}\,\mathcal{C}$, est essentiellement constant,

q.e.d.

4.15.8. Je veux revenir ici sur la démonstration inachevée de 4.8.13 (p. 212-216). Je reprends la démonstration où je l'avais laissée (p. 215). Il faut prouver que p. 175 [page 453]

$$F(\xi) \;=\; \varinjlim_{i\in I} M_i$$

est de cardinal $\leqslant \pi_0$. J'ai annoncé qu'on peut se ramener au cas où M est un monoïde de monomorphismes, i.e. que $uv = uv' \Longrightarrow v = v'$ (pour $u, v, v' \in M$). Je vais expliciter cet argument.

Pour tout $i \in I$, soit

$$N_i \;=\; \left\{ u \in M \;\middle|\; \begin{array}{l} \text{l'ensemble } I_{i,u} \text{ des } j \in I \text{ tels} \\ j \geqslant i,\ \alpha_{ji} = u \text{ est cofinal} \\ \text{dans } I, \text{ i.e. pour tout } j_0 \in I, \\ \text{existe } j \in I_{i,u}, \text{ avec } j \geqslant j_0 \end{array} \right\}. \tag{4.15.8.1}$$

Comme I, donc $i\backslash I$ est grande devant π_0, et $\operatorname{card} M \leqslant \pi_0$, on voit aussitôt que

$$N_i \;\neq\; \emptyset\ . \tag{4.15.8.2}$$

(Si d'autre part $i' \geqslant i$, on trouve

$$N_i \;=\; N_{i'}\alpha_{i'i}\ . \tag{4.15.8.3}$$

L'inclusion $N_{i'}\alpha_{i'i} \subset N_i$ est ± tautologique, prouvons $N_i \subset N_{i'}\alpha_{i'i}$. Soit $u \in N_i$, l'ensemble $J = I_{i,u} \cap (i'\backslash I)$ des $j \geqslant i'$ tels que $\alpha_{ji} = u$ est cofinal dans I, donc grand devant π_0.

$$i \xrightarrow[u=\alpha_{ji}]{} i' \xrightarrow{\alpha_{ji'}} j$$

Donc il existe un $v \in M$ tel que l'ensemble J' de $j \in J$ tels que $\alpha_{ji'} = v$ soit cofinal dans J, donc dans I. Comme cet ensemble J' est contenu dans $I_{i',v}$, cela implique que $I_{i',v}$ est cofinal dans I, i.e. $v \in N_{i'}$. On a d'autre part, pour $j \in J'$, [page 454]

$$\underbrace{\alpha_{ji'}}_{v}\alpha_{i'i} \;=\; \underbrace{\alpha_{ji}}_{u}\ ,$$

d'où $u = v\alpha_{i'i}$ avec $v \in N_{i'}$, d'où $u \in N_i\alpha_{i'i}$, q.e.d.)

Pour $i \in I$, soit

$$\alpha_i \;:\; \underbrace{\operatorname{Hom}(\xi, \underbrace{x_i}_{=\xi})}_{=M} \longrightarrow \underbrace{\operatorname{Hom}(\xi, F) \;=\; F(\xi)}_{\simeq \varinjlim_i \operatorname{Hom}(\xi, x_i)}$$

le morphisme canonique. On trouve ceci : p. 176

(4.15.8.4) Si $u \in N_i$, alors la relation d'équivalence R_i dans M définie par α_i ($v \overset{R_i}{\sim} v' \Longleftrightarrow \alpha_i(v) = \alpha_i(v')$) est égale à celle définie par l'application $v \longmapsto uv$ de M dans M, i.e.

$$\alpha_i(v) \;=\; \alpha_i(v') \qquad \Longleftrightarrow \qquad uv \;=\; uv'\ .$$

C'est immédiate, puisque $I_{i,u}$ étant cofinal dans I, on a $\alpha_i(v) = \alpha_i(v')$ si et seulement si $\exists\ j \in I_{i,u}$ avec $\alpha_{ji}v = \alpha_{ji}v'$, or $\alpha_{ji} = u$.

Pour $u \in M$, notons R_u la relation d'équivalence dans M définie par $v \longmapsto uv$. L'ensemble des relations d'équivalence dans $[M]$ obtenu ainsi est de cardinal $\leqslant \operatorname{card} M \leqslant \pi_0$, donc l'ensemble des relations d'équivalences R_i ($i \in I$) est lui aussi de cardinal $\leqslant \pi_0$. Cela montre, π_0 étant grand devant π_0, qu'il existe une relation d'équivalence de la forme R_u ($u \in M$), et un ensemble cofinal I' dans I, tel que $\forall\ i \in I'$ on ait $R_i = R_u$. Cela montre que, quitte à remplacer I par l'ensemble ordonné cofinal I', qu'on peut supposer que l'on a [page 455]

$$R_i \;=\; R_u \qquad \forall\, i \in I\ . \tag{4.15.8.5}$$

Soit alors

$$E \;=\; M/R_u \;=\; M/R_i \qquad (\forall\, i \in I)\ ,$$

on voit donc que les $\alpha_{ji} \in M$ (pour $i \leqslant j$) sont tels que l'application

$$\tau_{\alpha_{ji}} \;:\; v \;\longmapsto\; \alpha_{ji} v \;:\; M \;\longrightarrow\; M$$

est compatible avec la relation d'équivalence R_u, donc passe au quotient en une application

$$\overline{\alpha_{ji}} \;:\; E \;\longmapsto\; E\ ,$$

laquelle est d'ailleurs *injective.*

Soit M' le sous-monoïde de M formé des $u \in M$ qui ont la propriété de stabilité précédente, ainsi que la propriété d'injectivité qu'on vient de dire, et $\overline{M'}$ le monoïde image de M' dans le monoïde

$$\mathrm{Mon}(E) \;\subset\; \mathrm{End}_{\mathrm{Ens}}(E)$$

des applications injectives de E dans lui-même. On voit donc que le système inductif donné, défini par les α_{ji}, prend ses valeurs dans le sous-monoïde M' de M (i.e. de la sous-catégorie correspondante de la catégorie B_M), et que la limite inductive $F(\xi)$ qui nous intéresse (pour prouver $\mathrm{card}\, F(\xi) \leqslant \pi_0$) est isomorphe à la limite inductive du système inductif d'un système d'ensembles tous égaux à E, les morphismes de transition étant les $\overline{\alpha_{ji}} \in \overline{M'} \subset \mathrm{Mon}(E)$. De plus, on aura $\mathrm{card}\, \overline{M'} \leqslant \mathrm{card}\, M \leqslant \pi_0$. Le résultat $\mathrm{card}\, F(\xi) \leqslant \pi_0$ voulu résultera alors du [page 456]

Commentary · Exposition [ms. 455]

Removing identifications before counting new elements

The obstruction to a cardinal estimate is that a transition may both identify old elements and introduce new ones. Grothendieck separates these effects. The sets N_i record the transition maps which recur cofinally; each such map already detects every identification made after stage i.

For $u \in N_i$ and $v, v' \in M$, equality of the two classes in $\varinjlim M_j$ is witnessed at some later stage. Moving further into the cofinal fibre where the transition from i equals u gives

$$[v]_i = [v']_i \quad\Longleftrightarrow\quad uv = uv'.$$

The converse follows at any stage in that fibre. Thus the kernel relation R_i is one of the at most $|M|$ relations R_u. The cofinal-colour argument gives a subsystem with $R_i = R$ independent of i.

Passing from M to $E = M/R$ now removes exactly the identifications which the final colimit would make. The induced transition maps of E are injective: equality of the images of $[v]$ and $[v']$ implies equality of their final classes, hence vRv'. The total colimit is unchanged, because no additional equivalence has been imposed. What remains to be counted is actual growth by inclusions.

There is a useful bookkeeping point in this construction. The elements of M which preserve R and act injectively on E form a *submonoid* M_1. Its image

$$\overline{M_1} \subseteq \mathrm{Inj}(E, E)$$

is a quotient of M_1, has cardinal at most $|M|$, and is left cancellative. It need not be a quotient of the whole monoid M. Consequently the reduction on these pages proves a sufficient criterion involving such subquotients. The restricted quotient wording in 4.15.9 b) is not justified by this reduction alone.

The distinction does not affect the strict-cardinal conclusion. Its counting argument applies to *every* left-cancellative monoid below the chosen bound, including these subquotients. It does matter if one tries to read 4.15.9 as a sharp equivalence for one fixed category and one fixed cardinal. We retain the useful reduction while recording exactly what its proof supplies.

Lemme-clef 4.15.8.1. *Soit M un monoïde de monomorphismes (i.e. tel que $uv = uv' \Longrightarrow v = v'$, pour $u, v, v' \in M$), et soit $(\alpha_{ji})_{i \leqslant j}$ un système inductif dans B_M, indexé par un ensemble ordonné I grand devant π_0, où π_0 est un cardinal infini tel que* $\mathrm{card}\, M \leqslant \pi_0$. *Alors il existe un $i_0 \in I$ tel que pour tout $i \geqslant i_0$, $\alpha_{i_0,i}$ soit un isomorphisme.* p. 177 [page 457]

Commentary · Examples [ms. 456]

An ordinary filtered sequence which really grows

The increased filteredness assumption is doing real work. To measure it against a familiar sequence, take the one-object category with endomorphism monoid $(\mathbf{N}, +)$, whose identity is 0. At every integer stage put the unique object, and give the transition $i \to j$ the value $j - i$. Every strict transition is a nonunit.

Evaluating its ind-object at the unique object gives the sequence of regular right $\mathbf{N}$-sets

$$\mathbf{N} \xrightarrow{n \mapsto n+1} \mathbf{N} \xrightarrow{n \mapsto n+1} \mathbf{N} \longrightarrow \cdots .$$

Its colimit is $\mathbf{Z}$, with the usual action by addition of nonnegative integers. An explicit isomorphism sends the class of a at stage i to $a - i$. The transition relation preserves this difference. Every integer occurs by choosing i large enough, and two representatives with the same difference become equal at any stage above their two indices. This verifies both surjectivity and injectivity of the description.

The resulting right $\mathbf{N}$-set is not representable. The sole representable is $\mathbf{N}$, and an equivariant map $\mathbf{Z} \to \mathbf{N}$ cannot exist: if its value at 0 were b, equivariance would force the value at $-n$ to be $b - n$, which is negative for $n > b$. In particular $\mathbf{Z}$ is not a retract of the representable. Equivalently, the only idempotent of the additive monoid $\mathbf{N}$ is 0, so its Karoubi envelope adds nothing.

This sequence is filtered in the ordinary sense, but its index $\mathbf{N}$ is not grand devant $\aleph_0$: the whole countable set of indices has no upper bound. The next proposition shows that a diagram in the same monoid indexed by an $\aleph_1$-directed poset must have a constant tail. There is no contradiction with the explicit colimit above. The regular cardinal attached to the word filtered, and the source's successor convention, determine which of the two phenomena occurs.

Si on applique ce lemme au cas de $\overline{M'}$ et du $\overline{\alpha_{ji}}$, on trouve donc qu'il existe un $i_0 \in I$ tel que les $\overline{\alpha_{ji_0}}$ soient des isomorphismes. Cela implique que $F(\xi) \simeq \varinjlim_{i} \underbrace{E_i}_{E}$ est isomorphe à E_{i_0}, donc est de cardinal égal à $\operatorname{card} E_{i_0} = \operatorname{card} E \leqslant \pi_0$, ce qui achèvera la démonstration.

D'ailleurs, l'énoncé 4.8.13 implique 4.15.8.1, car dans le système inductif des $(M_i = M, \alpha_{ji})$, de $\varinjlim$ égale à $F(\xi)$, s'il n'existait *pas* de i_0 comme stipulé dans 4.15.8.1, alors on en conclurait aisément que l'on aurait $\operatorname{card} F(\xi) > \pi_0$, contrairement à 4.8.13. Nous allons dégager ceci dans un lemme :

Lemme 4.15.8.2. *Soit $(E_i)_{i \in I}$ un système inductif d'ensembles, indexé par un ensemble ordonné I grand devant π_0. On suppose qu'il existe $i_0 \in I$ tel que* [page 458]

a) $i \geqslant i_0$ *implique* $\operatorname{card} E_i = \pi_0$, *et*

b) $j \geqslant i \geqslant i_0$ *implique que* $u_{ji} : E_i \longrightarrow E_j$ *est un monomorphisme. Supposons de plus*

c) $\forall\, i \geqslant i_0\ \exists\, j > i$ *tel que* u_{ji} *ne soit* pas *un isomorphisme (i.e. ne soit pas surjectif). Soit* $E_\infty = \varinjlim_{i \in I} E_i$. *Alors* $\operatorname{card} E_\infty > \pi_0$.

Ce lemme implique donc l'*équivalence* de 4.8.13 et du "lemme-clef" 4.15.8.1. Pour prouver 4.15.8.2, nous allons prouver quelque chose d'un peu plus précis :

Commentary · Development (commentators') [ms. 457]

A boundary case controlled by right cancellation

Grothendieck has reduced his remaining difficulty to a monoid of injections. The source obtains a strict cardinal bound below. At the boundary, a recurring transition supplies an additional way to conclude if it can also be cancelled on the right. We record that limited extension, since it identifies the algebraic obstacle more precisely than the cardinal count alone.

Proposition: a cancellable recurring transition. [ours] Let π be infinite, let I be a π^+-directed poset, and let (a_{ji}) be a diagram in a left-cancellative monoid M of cardinal at most π. Suppose that, for some i_*, a cofinally recurring value u of a_{ji_*} is right cancellable: $vu = wu$ implies $v = w$. Then there is j_0 such that every a_{ij_0}, $i \geq j_0$, is a unit. On a cofinal full subposet all transitions are identities.

Proof. Let $J = \{j \geq i_* : a_{ji_*} = u\}$, which is cofinal by assumption. If $j \leq k$ belong to J, the cocycle identity gives

$$u = a_{ki_*} = a_{kj} a_{ji_*} = a_{kj} u.$$

Right cancellation gives $a_{kj} = 1$. Choose $j_0 \in J$. Given $i \geq j_0$, choose $k \in J$ with $k \geq i$. Then $a_{ki}a_{ij_0} = a_{kj_0} = 1$. In a left-cancellative monoid, $ab = 1$ also implies $ba = 1$, since $a(ba) = a = a1$. Thus a_{ij_0} is a unit. Moreover, for $j \geq i \geq j_0$,

$$a_{ji} = a_{jj_0}a_{ij_0}^{-1}$$

is a unit. The full subsystem on J already has identity transitions. This proves all assertions.

Corollary: two-sided cancellation. [ours] The key lemma 4.15.8.1 holds with $|M| \leq \pi$, including equality, if M is cancellative on both sides. More generally, if $|\operatorname{Mor}\mathcal{C}| \leq \pi$ and every endomorphism monoid of $\mathcal{C}$ is cancellative on both sides, every π^+-filtered diagram in $\mathcal{C}$ is essentially constant.

Proof. In the monoid case the cofinal-colour principle supplies a recurring u, and every u is right cancellable. For a general category, first replace the index by a final π^+-directed poset. Apply that principle to its at most π object values, obtaining a cofinal full subposet with one value x. The diagram now takes values in $\operatorname{End}(x)$, and the proposition supplies a further cofinal subsystem with identity transitions. Its presheaf colimit is h_x.

We checked the small-category and Cauchy-completion treatments cited below, together with searches for the left-cancellative stabilization formulation; we did not find this exact recurring-value statement in the literature consulted. The proof is an elementary continuation of Grothendieck's cofinal-fibre argument, with no claim of priority. It does not settle the general left-cancellative case: from $vu = u$ one cannot cancel u on the right without the extra hypothesis.

Corollaire 4.15.8.3. *Sous les conditions de 4.15.8.2, soit α_0 le premier ordinal tel que* $\operatorname{card}\alpha_0 > \pi_0$, *et soit I_{α_0} l'ensemble des ordinaux $\beta < \alpha$ (ensemble ordonné qui est grand devant π_0). Il existe alors une application strictement croissante*

$$I_{\alpha_0} \longrightarrow I\ , \qquad \alpha \longmapsto i_\alpha\ , \qquad \textit{appliquant } 0 \textit{ dans } i_0\ ,$$

tel que pour tout $\alpha \in I_{\alpha_0}$, on ait [page 459]

$$\underbrace{u_{i_{\alpha+1},i_\alpha}}_{\substack{\textit{c'est un } \text{monomorphisme,}\\ \textit{d'après b, vu que } i_{\alpha+1} > i_\alpha \geqslant i_0}} \qquad \textit{n'est pas un isomorphisme } ([187]).$$

On a alors

$$E'_\infty \overset{\text{déf}}{=} \underbrace{\varinjlim_{\alpha\in I_{\alpha_0}} E_{i_\alpha}}\ \hookrightarrow\ \varinjlim_{i\in I} E_i \overset{\text{déf}}{=} E_\infty$$

et

$$\operatorname{card} E'_\alpha \ \geqslant\ \underbrace{\operatorname{card} I_{\alpha_0}}_{=\ \textit{successeur de } \pi_0}\ >\ \pi_0\ ,$$

d'où $\operatorname{card} E_\infty \geqslant \operatorname{card} E'_\infty > \pi_0$.

Démonstration. La construction des i_α par récurrence transfinie est immédiate, en utilisant c) et le fait que I est grande devant π_0, et que pour tout $\alpha \in I_{\alpha_0}$, l'ensemble I_α des $\beta < \alpha$ est de cardinal $\leqslant \pi_0$. Le fait que $E'_\infty \longrightarrow E_\infty$ soit injectif résulte de b). Le fait que [p. 178]

$$\operatorname{card} E'_\infty \ \geqslant\ \operatorname{card} I_{\alpha_0}$$

résulte du fait que E'_∞ est réunion croissante de sous-ensembles identifiés à E_{i_α}, et pour tout α on peut trouver un [page 460]

$$\xi_\alpha \ \in\ E_{i_\alpha} \setminus \bigcup_{\beta<\alpha} E_{i_\beta}\ .$$

187. *mieux* : $\forall\ \alpha \in I_{\alpha_0}\ \varinjlim_{\beta\in I_\alpha} E_{i_\beta} \longrightarrow E_{i_\alpha}$ n'est pas surjectif.

L'application

$$\begin{aligned} \alpha &\longmapsto \xi_\alpha \\ I_{\alpha_0} &\longrightarrow E'_\infty \end{aligned}$$

est injective, donc $\operatorname{card} E'_\infty \geqslant \operatorname{card} I_{\alpha_0}$.

Corollaire 4.15.8.4. *Pour un M et un π_0 donnés, pour que le lemme-clef 4.15.8.1 soit valable, il faut et il suffit qu'il le soit dans le cas où $I = I_{\alpha_0}$, α_0 étant le premier ordinal tel que $\operatorname{card} \alpha_0 > \alpha_0$. Il revient au même qu'on ne puisse trouver de système inductif $(\alpha_{ji})_{j \geqslant i}$ indexé par I_{α_0}, à valeurs dans M, tel que pour $i < j$, on ait α_{ji} non inversible dans M.*

Je vais résumer ce qui a été obtenu dans un énoncé récapitulatif : [page 461]

Commentary · Exposition [ms. 460]

The transfinite growth argument, including limit stages

The point of 4.15.8.2 is to turn the failure of stabilization into too many distinct elements. Once transitions are injective, an element genuinely added at one stage can never disappear. The only care needed is at limit ordinals, where successive strict growth by itself does not guarantee that the next stage adds something beyond the whole preceding union.

Put $\kappa = \pi_0^+$. Starting at the prescribed i_0, choose a chain i_α for $\alpha < \kappa$. At a successor stage use hypothesis c) to choose a nonsurjective transition. At a nonzero limit α, the earlier indices form a set of cardinal at most π_0, so they have an upper bound b. Pass from b to a stage i_α by a nonsurjective transition. Its image contains the image of every earlier stage, and is a proper subset of E_{i_α}. This produces precisely the strengthened assertion in footnote 187:

$$\varinjlim_{\beta<\alpha} E_{i_\beta} \longrightarrow E_{i_\alpha} \quad \text{is not surjective.}$$

At $\alpha = 0$ the domain is empty and the source hypothesis $|E_{i_0}| = \pi_0$ makes the assertion valid.

Choose ξ_α outside the preceding union. Injectivity ensures that all these elements remain distinct in the chain colimit. That colimit injects into the original colimit: two representatives which become equal in the latter are equal at a common later stage, and injectivity then identifies them already at a stage of the chain above their two indices. Consequently the original colimit contains at least κ elements.

The cardinal equality $|E_i| = \pi_0$ is not used to prove this lower bound, apart from the harmless nonemptiness at the first stage. This makes the lemma applicable to a monoid of cardinal *at most* π_0, as required just above it. One can alternatively start counting at successor stages and omit the initial nonemptiness assumption altogether.

For a left-cancellative monoid, left multiplication by a is surjective exactly when a is a unit: a preimage of 1 is a right inverse, which is also a left inverse by cancellation. Thus nonunits give precisely the nonsurjective injections needed here. A product containing a nonunit cannot be a unit in such a monoid. The selected chain therefore has every strict transition noninvertible, proving the reduction in 4.15.8.4 to the ordinal π_0^+. Its printed condition $|\alpha_0| > \alpha_0$ is impossible; the intended condition is $|\alpha_0| > \pi_0$.

Théorème 4.15.9. *Soient $\mathcal{C}$ une petite catégorie, π un cardinal tel que $\pi \geqslant \operatorname{card} \mathcal{C}$. Les conditions suivantes sont équivalentes (et je conjecture qu'elles sont toujours satisfaites) :*

a) $\mathcal{M} \overset{déf}{=} \operatorname{Ind}_\pi \mathcal{C}$ est réduit à $\mathcal{M}_\pi$.

a′) (Si $\mathcal{C}$ est karoubienne.) Tout système inductif $\underline{x} = (x_i)_{i \in I}$ dans $\mathcal{C}$, grand devant π, est essentiellement constant.

a″) Pour tout système inductif $(x_i)_{i \in I}$ dans $\mathcal{C}$, grand devant π, il existe un foncteur cofinal $J \longrightarrow I$, avec J ensemble ordonné grand devant π, et tel que le système réindexé $(x_j)_{j \in J}$ satisfasse les conditions suivantes :

1) Tous les x_j sont égaux à un même élément x de $\mathcal{C}$.

2) Il existe un projecteur p dans x, tel que l'on ait $\alpha_{j'j} = p$ pour $j < j'$ $(j, j' \in J)$, où $\alpha_{j'j} : x_j \longrightarrow x_{j'}$ est la flèche de transition.

b) Pour tout $x \in \mathrm{Ob}\,\mathcal{C}$, et pour tout monoïde quotient M' de $M = \mathrm{End}_{\mathcal{C}}(x)$, tel que M' soit un monoïde de monomorphismes (i.e. $uv = uv' \Longrightarrow v = v'$ si $u, v, v' \in M'$), et pour tout système inductif grand devant π à valeurs dans $B_{M'}$ (définie par un système d'éléments $(\alpha_{ji})_{i \leqslant j,\ i,j \in I}$ d'un ensemble ordonné I grand devant π, avec $\alpha_{kj}\alpha_{ji} = \alpha_{ki}$ pour $i \leqslant j \leqslant k$), il existe $i_0 \in I$ tel que $\alpha_{ii_0} = 1$ pour $i \geqslant i_0$, i.e. $\alpha_{ji} = 1$ pour $j \geqslant i \geqslant i_0$. [page 462]

b′) M' étant un monoïde de monomorphismes comme dans b) (donc $\mathrm{card}\, M' \leqslant \mathrm{card}\, \mathcal{C} \leqslant \pi$*), soit de plus α_0 le plus petit ordinal tel que* $\mathrm{card}\, \alpha_0 > \pi$*, et soit $I = I_{\alpha_0}$ l'ensemble ordonné des ordinaux $\alpha < \alpha_0$ (lequel ensemble est grand devant π). Il n'existe* pas *de système inductif dans $B_{M'}$, tel que pour $i < j$ dans I, α_{ji} soit un élément non inversible dans M' (donc $u \longmapsto \alpha_{ji}u : M' \longrightarrow M'$ est injectif mais non surjectif).*

Signalons tout de suite le p. 179

Corollaire 4.15.10. *Les conditions équivalentes précédentes sont satisfaites, si* $\pi > \mathrm{card}\, \mathcal{C}$.

En effet, soit α_1 le plus petit ordinal tel que $\mathrm{card}\, \alpha_1 > \mathrm{card}\, \mathcal{C}$. Donc $\pi_1 \overset{\text{déf}}{=} \mathrm{card}\, \alpha_1$ est le successeur de $\pi_0 \overset{\text{déf}}{=} \mathrm{card}\, \mathcal{C}$, et on a $\pi \geqslant \pi_1$, donc (si α_0 est comme explicité dans b′ ci-dessus) $\mathrm{card}\, \alpha_0 > \pi_1$, donc $\alpha_1 < \alpha_0$. Soit (avec les notations de b′), pour $\alpha \in I$ [plutôt $\alpha \in I_{\alpha_0}$], i.e. $\alpha < \alpha_0$,

$$M_\alpha \;=\; \varinjlim_{i<\alpha} \underbrace{M_i}_{=M} \;,$$

on voit donc, comme les α_{ji} sont injectifs et non surjectifs, que

$$\mathrm{card}\, M_\alpha \;\geqslant\; \mathrm{card}\, \alpha \;,$$

et que pour $\alpha < \beta$, on a [463]

$$M_\alpha \;\hookrightarrow\; M_\beta \qquad \text{injectif} \;.$$

Appliquons ceci au cas où $\alpha = \alpha_1$, donc on trouve

$$\mathrm{card}\, M_{\alpha_1} \;\geqslant\; \mathrm{card}\, \alpha_1 \;=\; \pi_1 \;,$$

d'autre part, en prenant $\beta = \alpha_1 + 1$, d'où $M_\beta \simeq M$, on trouve une injection

$$M_{\alpha_1} \;\hookrightarrow\; M \;,$$

d'où

$$\pi_1 \;=\; \mathrm{card}\, M_{\alpha_1} \;\leqslant\; \mathrm{card}\, M \;=\; \pi_0 \;,$$

une contradiction, ce qui établit b′.

Commentary · Examples [ms. 462]

Invertibility is the invariant condition

Grothendieck wants a diagram to stop producing new ind-objects. This is invariant under an isomorphism of diagrams. Literal equality of its transition arrows with identities is not invariant in that way, and the distinction produces a small counterexample to condition b) as printed.

Let $G = \{1, g\}$ with $g^2 = 1$, let $\mathcal{C} = BG$, and let $I = \pi^+$ for any infinite cardinal π. Choose $h_\alpha \in G$ by taking $h_\alpha = g$ at successor ordinals and $h_\alpha = 1$ at zero and nonzero limit ordinals. Both values occur arbitrarily far out. Define

$$a_{\beta\alpha} = h_\beta h_\alpha^{-1} \qquad (\alpha \leq \beta).$$

Then $a_{\gamma\beta}a_{\beta\alpha} = a_{\gamma\alpha}$ and $a_{\alpha\alpha} = 1$, so this is a diagram in BG. The maps h_α give a natural isomorphism from the constant diagram to it. Its ind-object is therefore representable.

There is no α_0 such that every $a_{\beta\alpha_0}$ is 1 for $\beta \geq \alpha_0$: that would make $h_\beta = h_{\alpha_0}$ throughout a tail. Thus condition b) fails. Condition b′ holds automatically, because a group has no noninvertible elements. Conditions a), a′), and a″) hold as well: every filtered group-valued diagram trivializes, and the cofinal-colour argument makes it literally constant on a suitable subsystem. The equivalence asserted in 4.15.9 is therefore false in its printed form even for this two-arrow category.

The appropriate replacement for the tail condition is

$$\text{there exists } i_0 \text{ such that } a_{ii_0} \in (M')^\times \text{ for every } i \geq i_0.$$

The cocycle identity then makes all later transitions units. If literal identities are desired, one either changes the trivialization or restricts to a cofinal fibre of the finitely or cardinally many trivializing maps. This is exactly the distinction already handled correctly in 4.15.7.2–5.

The edition has $= 1$ in both places, so the transcription is retained. Corollary 4.15.10 inherits that literal defect when it says that *all* the preceding conditions hold. Its proof establishes the intended exclusion of nonunit chains, and its substantive conclusion, Corollary 4.15.11, remains valid. The separate subquotient issue noted above must also be respected when using the monoid reformulation.

Corollaire 4.15.11. *Soit $\mathcal{C}$ une catégorie essentiellement petite, π un cardinal tel que $\pi > \operatorname{card}\operatorname{réd}\mathcal{C}$. Soit $\mathcal{M} \overset{\text{déf}}{=} \operatorname{Ind}_\pi \mathcal{C}$, alors $\mathcal{M} = \mathcal{M}_\pi$, et c'est une enveloppe de Karoubi de $\mathcal{C}$.*

On peut supposer $\mathcal{C}$ réduite, donc $\operatorname{card}\mathcal{C} < \pi$, et on applique le corollaire précédent.

Ainsi, le résultat litigieux 4.8.13 (p. 212) est finalement prouvé sous une forme légèrement plus faible, avec $\pi > \operatorname{card}\operatorname{réd}\mathcal{C}$ au lieu de $\pi \geqslant \operatorname{card}\operatorname{réd}\mathcal{C}$. Du même coup, les applications de 4.8.13 qu'on a fait jusqu'à présent (notamment le lemme 4.15.3).

[464]

Commentary · Exposition [ms. 463]

Why a larger filteredness bound leaves only retracts

The counting proof reaches the result needed for accessibility: a sufficiently small category acquires no objects under sufficiently filtered colimits except its missing retracts. We first spell out the source's strict inequality. Write $\mu = |\operatorname{Mor}\mathcal{C}| < \pi$ and first suppose μ infinite. A chain of nonunits indexed by π^+, in any left-cancellative subquotient monoid M of cardinal at most μ, would give a partial colimit at μ^+ with at least μ^+ elements. That partial colimit injects into the next stage, a copy of M. The contradiction is

$$\mu^+ \leq \left| \varinjlim_{\alpha<\mu^+} M_\alpha \right| \leq |M| \leq \mu.$$

The equalities in the source's last display are unnecessary; these inequalities are the valid argument. If μ is finite, the assertion is immediate because an injective self-map of a finite set is surjective.

There is also a direct presheaf verification which explains the resulting retract. Let X be a π^+-filtered colimit of representables and let $\mathcal{E}_X$ be its category of elements, with objects (c, x) for $x \in X(c)$. A family of at most π such elements, including at most π relations between them, factors through one stage of the presentation: representables preserve the colimit on Hom, and π^+-filteredness makes the choices and relations simultaneous. Hence $\mathcal{E}_X$ is π^+-filtered.

If $\mathcal{E}_X$ had more than μ objects, select μ^+ of them. Their cocone would have distinct underlying arrows of $\mathcal{C}$: an arrow $f : c \to d$ and an element $y \in X(d)$ determine its source element uniquely as $X(f)(y)$. This contradicts $|\operatorname{Mor}\mathcal{C}| = \mu$. It follows that $\mathcal{E}_X$ has at most μ objects and at most μ^2 arrows. This bound is at most π, including the finite case, because π is infinite.

Apply its filteredness to its entire identity diagram, obtaining a cocone to (d, y). Write its component at (c, x) as $u_{c,x} : c \to d$. These components define a natural transformation

$$s : X \longrightarrow h_d, \qquad s_c(x) = u_{c,x}.$$

Naturality is the cocone equation. The element y defines $t : h_d \to X$, and the fact that every component lies in $\mathcal{E}_X$ says $t_c s_c(x) = X(u_{c,x})(y) = x$. Thus $ts = 1_X$: X is a retract of a representable. Conversely every split idempotent image occurs as a sufficiently filtered colimit by the constant-object construction in the worked example, using the desired regular ordinal as index. This proves

$$\operatorname{Ind}_\pi(\mathcal{C}) \simeq \operatorname{Kar}(\mathcal{C}) \qquad \text{when } \pi > |\operatorname{Mor}\mathcal{C}|.$$

The same proof sharpens the bound when each Hom set has cardinal less than π and $|\operatorname{Ob}\mathcal{C}| \leq \pi$. A family of π distinct elements of one $X(c)$ would force π distinct arrows in one $\operatorname{Hom}(c, d)$, which is impossible. Hence every $|X(c)| < \pi$, the category of elements has at most π arrows, and its identity diagram again has a cocone. This covers some equality cases, without assuming that it covers all of them.

Commentary · Later developments [ms. 463]

Small accessible categories and Cauchy completion

The conclusion belongs to a broader description of accessibility by completion. Makkai and Paré's 1989 monograph develops accessible categories through categorical model theory and mixed sketches (Makkai and Paré, 1989). The result here is especially close to the Cauchy-completion side of that theory: retracts of representables are the objects left after the filteredness bound is raised sufficiently.

A modern ordinary formulation is Tendas (2022, Lemmas 2.5–2.6 and Corollary 2.8): if the Hom sets have cardinal less than β, a γ-flat presheaf for $\gamma > \beta$ is small enough to be Cauchy, provided the object set also lies below γ. This makes the two cardinal bounds explicit. The paper's Theorem 3.7 extends the equivalence between small accessibility and Cauchy completeness to enriched categories.

For ∞-categories, Lurie (2009, Lemma 5.4.2.4 and Corollaries 5.4.3.5–5.4.3.6) identifies compact objects in an ind-completion with the idempotent completion, proves the sufficiently-large-bound collapse, and characterizes small accessible ∞-categories by idempotent completeness. Coherent idempotents replace the ordinary projectors. These statements preserve Grothendieck's conclusion while expressing the required redundancy through the sizes of mapping spaces.

Commentary · Present state [ms. 463]

The strict bound and the boundary question

Grothendieck finishes with the assertion he has actually proved: the bound may be chosen strictly larger than the reduced cardinal of the category. That is sufficient for the qualitative assertion that a small idempotent-complete category is accessible. It is also sufficient for earlier applications in which the chosen cardinal can be enlarged.

The stronger assertion at $\pi = |\operatorname{Mor}\mathcal{C}|$ must be kept separate. The checked modern references give a strict separation between the local Hom bound and the filteredness bound; their general statements do not automatically remove this boundary. The preceding proofs settle it when all Hom sets have cardinal less than π, and for categories with two-sided cancellative endomorphism monoids. We have not settled the unrestricted equality case, and did not find an exact resolution of that sharper formulation in the sources consulted. This records the limit of this commentary, rather than declaring a new open problem.

There is consequently no need to weaken the valid conclusion of 4.15.11, and no justification for silently strengthening its cardinal inequality. The audit also records the two logically separate defects in 4.15.9: the false literal identity condition, and the use of quotients where the preceding construction supplies subquotients.

4.16 Exemples et questions

4.16.0. J'avais cru que tout foncteur "raisonnable" (ou plus précisément, tout foncteur "logiquement accessible") entre catégories accessibles devrait être accessible. Ce n'est malheureusement pas vrai, puisqu'un composé de deux foncteurs Hom

$$\mathrm{Ens} \longrightarrow \mathrm{Ens}\ , \qquad E \longmapsto \underbrace{\mathrm{Hom}(\mathrm{Hom}(E,I),J)}_{J^{(I^E)}}$$

est non accessible, si I, J sont tous deux de cardinal $\geqslant 2$. Prenant p.ex. $I = J = \{0,1\}$, on trouve le foncteur p. 180

$$E \longmapsto \mathfrak{P}(\mathfrak{P}(E))\ ,$$

qui n'est pas accessible.

4.16.1. Deuxième désillusion : j'avais cru que pour toute catégories accessible $\mathcal{M}$, toute sous-catégorie $\mathcal{N}$ strictement pleine et stable par facteurs directs "raisonnable" (p.ex. "logiquement accessible") devrait être accessible dans $\mathcal{M}$. Il n'en est rien, même dans le cas particulier de l'image inverse de la sous-catégorie accessible dans Ens [465]

$$\underbrace{\mathrm{Ens}^*}_{=\{E\in\mathrm{Ens}\,|\,E\neq\emptyset\}} \subset \mathrm{Ens}$$

par un contrafoncteur accessible *représentable*

$$\begin{array}{ccc} \mathcal{M} & \longrightarrow & (\mathrm{Ens})^\circ \\ x & \longmapsto & \mathrm{Hom}_{\mathcal{M}}(x,z)\ , \end{array}$$

i.e. la sous-catégorie ouverte (donc strictement pleine) $\mathcal{M}_z$ de $\mathcal{M}$ formée des objets x tels que $\mathrm{Hom}_{\mathcal{M}}(x,z)\neq\emptyset$, n'est par toujours accessible dans $\mathcal{M}$. Ici, je ne suis pas parvenu à un exemple aussi élémentaire (cf. appendice) – j'ai pris $\mathcal{M}=(k\text{-alg. comm.})$, k un corps infini et z l'objet initial k de $\mathcal{M}$ – j'ai trouvé que $\mathcal{M}_z$ n'est pas L_π-stable dans $\mathcal{M}$, quelque soit le cardinal π – mais cet exemple utilisait que $\mathfrak{U}=\mathfrak{U}_1$, plus petit univers contenant $\mathbf{N}$. J'ignore si pour tout autre univers, on peut trouver un corps k de cardinal $\pi\in\mathfrak{U}$ assez grand, de façon que tout π-ultrafiltre sur un ensemble $E\in\mathfrak{U}$ soit trivial. Peut-être un tel exemple, où non la [466]
construction du couple $(\mathcal{M},\mathcal{N})$, mais la démonstration du fait que $\mathcal{N}$ n'est pas accessible dans $\mathcal{M}$, fait appel à des hypothèses particulières sur l'univers ambiant, ne doit pas être considéré comme (tout à fait !) "logiquement accessible". Pour être vraiment probant, il faudrait un exemple de nature plus élémentaire.

Commentary · Examples [ms. 464]

A double dual which defeats every rank

Grothendieck begins his examples by testing the hope that a familiar formula should automatically define an accessible functor. The double dual is a particularly useful test: each of its two steps is representable with the appropriate variance, but their composite can remember genuinely global information. The issue is visible without cardinal arithmetic beyond the definition of filteredness.

We use a regular cardinal κ in the modern convention: a category is κ-filtered when every diagram with fewer than κ arrows has a cocone. Thus a system grand devant π in the manuscript is π^+-filtered. Fix distinct $a,b\in I$ and distinct $c,d\in J$. The functor under discussion is

$$D(X)=\mathrm{Set}(\mathrm{Set}(X,I),J),\qquad D(u)(\varphi)(g)=\varphi(g\circ u).$$

This action on maps matters. When $I=J=2$, the notation $\mathfrak{PP}(X)$ describes its objects, but does not specify its action by itself.

For any infinite regular κ, choose a set X of cardinal κ. Its subsets of cardinal less than κ, ordered by inclusion, form a κ-directed system with union X. Define $\Phi: I^X\to J$ by setting $\Phi(g)=d$ when g is constantly a, and $\Phi(g)=c$ otherwise. If $S\subset X$ has cardinal less than κ, every member of the image of $D(S)\to D(X)$ depends only on $g|_S$. But the constant function a and the function which is a everywhere except at one point of $X\setminus S$, where it is b, have the same restriction to S and different values under Φ. Therefore Φ belongs to none of these images, and

$$\mathrm{colim}_{S\in[X]^{<\kappa}} D(S)\longrightarrow D(X)$$

is not surjective. This proves the assertion for every possible accessibility rank.

The calculation identifies the obstruction more precisely than growth of cardinalities would. An element of the double dual need not have a support of bounded size. Accessibility would supply one uniform bound, because every element would have to come from a stage in the displayed presentation. The failure concerns that uniformity, even though each individual value of the functor is an ordinary set.

Il reste la possibilité qu'il y ait un théorème métamathématique qui assure que tout couple $(\mathcal{M},\mathcal{N})$ logiquement accessible, avec $\mathcal{M}$ accessible, et $\mathcal{N}$ sous-catégorie strictement pleine L_π-stable dans $\mathcal{M}$ pour π assez grand, soit accessible dans $\mathcal{M}$ ([188]). Ceci s'appliquerait en particulier au cas d'une sous-catégorie *fermée* : toute sous-catégorie fermée (qui soit

188. C'est faux, même pour une sous-catégorie fermée, cf. 4.16.7.10 ci-dessous.

logiquement accessible) serait accessible dans $\mathcal{M}$. L'exemple-test le plus simple serait celui du complémentaire fermé $\mathcal{M}'_z$ de la sous-catégorie ouverte $\mathcal{M}_z$ ci-dessus ([189]), i.e. la catégorie des $x \in \mathrm{Ob}\,\mathcal{M}$ tels que $\mathrm{Hom}_{\mathcal{M}}(x,z) = \emptyset$, ou encore l'image inverse de la sous-catégorie fermée [467]
accessible

$$e \;=\; \{\emptyset\} \;\subset\; (\mathrm{Ens})$$

réduite à $\{\emptyset\}$, par le foncteur représentable $x \longmapsto \mathrm{Hom}(x,z) : \mathcal{M} \longrightarrow \mathrm{Ens}$. Il se pourrait donc qu'une telle catégorie soit toujours accessible dans $\mathcal{M}$. Mais j'ai été incapable de le prouver, p. 181
même dans le cas particulier ci-dessus, où $\mathcal{M} = (k\text{-alg. comm.})$, k étant un corps commutatif infini. La difficulté est (dans tous les cas) la suivante : trouver, pour z fixé, un cardinal $\pi > \pi_0$ tel que pour tout $x \in \mathrm{Ob}\,\mathcal{M}$, et tout système (ou *un* système) π_0-π-adapté $(x_i)_{i\in I}$ de limite inductive x_0 (π_0 un cardinal donné, utile pour $\mathcal{M}$), *si* $\mathrm{Hom}_{\mathcal{M}}(x,z)$ ($\simeq \varprojlim \mathrm{Hom}_{\mathcal{M}}(x_i,z)$) est vide, alors $\exists\, i_0 \in I$ tel que $\mathrm{Hom}_{\mathcal{M}}(x_{i_0},z)$ soit déjà vide. Dans l'exemple $\mathcal{M} = (k\text{-alg. comm.})$, il faut donc prouver que l'on peut trouver un cardinal π tel que pour tout k-algèbre A avec $\mathrm{Hom}_{\mathcal{M}}(A,k) = \emptyset$, il existe une sous-$k$-algèbre A_i de cardinal $\leqslant \pi$, telle que $\mathrm{Hom}_k(A_i,k) = \emptyset$ ([190]). J'ai tendance à croire que ça doit être vrai (et même sans supposer que k soit un corps, [468]
et aussi en prenant la condition $\mathrm{Hom}_{\mathcal{M}}(A,Z) = \emptyset$, où Z est un objet quelconque de $\mathcal{M}$). Ce genre de questions semble étonnement délicat …

Commentary · Exposition [ms. 466]

A point requires compatible choices

Grothendieck next moves the same difficulty into a category of algebraic objects. For a fixed $z \in \mathcal{M}$, the subcategory

$$\mathcal{T}_z = \{x : \mathrm{Hom}_{\mathcal{M}}(x,z) \neq \varnothing\}$$

is a sieve: an arrow $y \to x$ followed by a point $x \to z$ gives a point of y. Its complement $\mathcal{S}_z$ is a cosieve. Both are replete and closed under retracts. These are the meanings of open and closed throughout the closing pages; no topology on a space of objects is being introduced.

The representable contrafunctor sends every existing colimit to a limit of sets,

$$\mathrm{Hom}(\mathrm{colim}_i\, x_i, z) \cong \lim_i \mathrm{Hom}(x_i, z).$$

The existence of an element in every set on the right need not provide a compatible family. Consequently, preservation of these limits by the representable functor does not show that $\mathcal{T}_z$ is closed under the corresponding colimits. Describing $\mathrm{Hom}(-,z)$ as an accessible functor to Set^{op} means preservation of sufficiently filtered colimits into that target; it does not assert that Set^{op} is an accessible category.

The appendix supplies the concrete example with $\mathcal{M} = \mathsf{CAlg}_k$ and $z = k$. Its hypothesis on complete ultrafilters is part of the example. Under that hypothesis, for every infinite $\pi \geq |k|$ there is an algebra A with no map to k, although every subalgebra of cardinal at most π has such a map. The explicit reduced-product construction is proved in the appendix commentary. Since the poset of these subalgebras is π^+-directed, this proves that $\mathcal{T}_k$ is closed under no sufficiently large filtered-colimit class. Proposition 4.16.7.8 then transfers the failure of accessibility to its complementary cosieve $\mathcal{S}_k$.

The distinction explains the later marginal corrections. The no-point class is automatically closed under every nonempty colimit: one structure map from an object with no point would contradict a point on the colimit. Nevertheless, that closure does not furnish a small family of generators. The missing condition is eventually isolated in Corollary 4.16.7.3 as a small weakly initial family in the cosieve.

4.16.2. Il est possible, plus généralement, que la question suivante ait une réponse affirmative : Soit

$$f \;:\; \mathcal{M} \longrightarrow \mathcal{N}^\circ$$

un foncteur accessible, avec $\mathcal{M}$, $\mathcal{N}$ catégories accessibles, et soit $\underline{T}$ une sous-catégorie de $\mathcal{N}$ accessible dans $\mathcal{N}$, et telle qu'il existe un cardinal π ayant les propriétés suivantes : $\mathcal{N}^\circ$ satisfait

189. **N.B.** elles est accessible dans $\mathcal{M}$ si et seulement si $\mathcal{M}_z$ l'est !
190. C'est faux, cf. annotation marginale p. 466.

L_π (i.e. $\mathcal{N}$ stable par $\varprojlim$ grandes devant π_1). Alors $\underline{S} \overset{\text{déf}}{=} f^{-1}(\underline{T})$ est-elle accessible dans $\mathcal{M}$ ([191])? Ceci généralise le cas $\mathcal{N} = (\mathrm{Ens})$, $\underline{T} = \{\emptyset\}$, f foncteur représenté par $z \in \mathrm{Ob}\,\mathcal{M}$, envisagé tout à l'heure. Avec le même type d'exemple, mais en prenant $\mathcal{N} = (\mathrm{Ens})^*$ au lieu [469]
de $\mathcal{N} = \{\emptyset\}$, on voit que la condition supplémentaire de stabilité sur $\underline{T}$ n'est pas superflue. (**N.B.** Elle sont à assurer que $\underline{S}$ est L_π-stable dans $\mathcal{M}$ pour π grand.)

4.16.3. Questions apparentée : j'avais espéré que pour un foncteur accessible entre catégories accessibles

$$f : \mathcal{N} \longrightarrow \mathcal{M} ,$$

la $\natural$-image essentielle, formée des $y \in \mathrm{Ob}\,\mathcal{M}$ qui sont isomorphes à un facteur direct d'un objet de la forme $f(x)$, avec $x \in \mathrm{Ob}\,\mathcal{N}$, serait accessible dans $\mathcal{N}$. L'exemple ci-dessus (nullement élémentaire!) de la sous-catégorie $\mathcal{M}_z$ de $\mathcal{M}$ (formée des $x \in \mathrm{Ob}\,\mathcal{M}$ tels que $\mathrm{Hom}_{\mathcal{M}}(x, z) \neq \emptyset$) montre qu'il n'en est rien. En effet, $\mathcal{M}_z$ est l'image essentielle, et aussi l'image $\natural$-essentielle, du foncteur

$$\mathcal{M}/z \longrightarrow \mathcal{M} ,$$

or $\mathcal{M}/z$ est accessible, et le foncteur précédent accessible, si $\mathcal{M}$ est accessible (vérification par AQT `[âne qui trotte]`, cf. plus bas, énoncé plus général).

On a l'impression que la raison pour laquelle on a de tels contre-exemples, c'est que l'on a [470]
des $\natural$-images essentielles qui ne sont L_π-stables dans $\mathcal{M}$ pour aucun cardinal $\pi \in \mathfrak{U}$. Cela amène à modifier la conjecture ainsi : soit π un cardinal tel que $\mathcal{M}$ satisfasse à L_π, et soit $\underline{S}(\pi)$ la plus p. 182
petite sous-catégorie strictement pleine de $\mathcal{M}$, L_π-stable dans $\mathcal{M}$, et qui contienne l'image essentielle de f. (On pourrait appeler $\underline{S}(\pi)$ la L_π-image essentielle.) Alors $\underline{S}(\pi)$ est accessible dans $\mathcal{M}$. De plus, la famille (décroissante) des $\underline{S}(\pi)$, pour π variable, est stationnaire, i.e. il existe un cardinal π tel que $\pi' \geqslant \pi$ implique $\underline{S}(\pi') = \underline{S}(\pi)$. Il est clair alors que $\underline{S}(\pi)$ est la plus petite sous-catégorie de $\mathcal{M}$ qui soit accessible dans $\mathcal{M}$ et qui contienne l'image essentielle de f. La même question se pose pour un foncteur $g^\circ : \mathcal{N}^\circ \longrightarrow \mathcal{M}$, où $g : \mathcal{N} \longrightarrow \mathcal{M}^\circ$ est un foncteur accessible, $\mathcal{N}$ et $\mathcal{M}$ accessibles. Je conjecture que la réponse est encore affirmative.
[471]

Commentary · Development (commentators') [ms. 470]

Filtered closures need not settle at a final rank

Grothendieck tries to repair the failure of essential images by closing them under sufficiently filtered colimits. This does recover an accessible category at suitable ranks. The additional prediction that these closures eventually stop changing is stronger, and the appendix gives a way to disprove it.

The closure at a common strong rank. [ours] Let $F : \mathcal{A} \to \mathcal{B}$ be an accessible functor. Choose a regular κ such that both categories are κ-accessible and F preserves both κ-filtered colimits and κ-presentable objects. Let $\mathcal{C}$ be the small full subcategory of $\mathcal{B}_\kappa$ on one chosen representative of each isomorphism class in $F(\mathcal{A}_\kappa)$, with no additional objects. Then the smallest replete full κ-filtered-colimit-closed subcategory $\mathcal{S}_\kappa$ containing $F(\mathcal{A})$ is

$$\mathcal{S}_\kappa \simeq \mathrm{Ind}_\kappa(\mathcal{C}),$$

where this Ind notation uses the modern regular-cardinal convention. In particular, it is accessible and accessibly embedded in $\mathcal{B}$.

Indeed, extend $\mathcal{C} \hookrightarrow \mathcal{B}$ by κ-filtered colimits. For presentations $X = \mathrm{colim}_i\, c_i$ and $Y = \mathrm{colim}_j\, d_j$, compactness gives

$$\mathrm{Hom}_{\mathcal{B}}(X, Y) \cong \lim_i \mathrm{colim}_j\, \mathrm{Hom}_{\mathcal{C}}(c_i, d_j).$$

This is precisely the Ind hom formula, so the extension is fully faithful and its image is closed under κ-filtered colimits. Every $F(a)$ lies in that image, because a presentation of a by $\mathcal{A}_\kappa$ is preserved by F. Conversely, any closed full subcategory containing $F(\mathcal{A})$ contains $\mathcal{C}$ and all its κ-filtered colimits. This proves both the formula and minimality. The existence of arbitrarily large common strong ranks is the usual uniformization theorem; its formulation is recalled in the introduction of Paré and

191. Faux, avec $\mathcal{N} = \mathrm{Ens}$ et un foncteur représentable $\mathcal{M} \longrightarrow \mathrm{Ens}^\circ$, prenant $I = \{\emptyset\}$, cf. annotations marginales p. 466, 467.

Rosický (2013). The proof here applies that theorem rather than asserting the formula at every initially chosen rank.

Failure of eventual stabilization. [ours] Work under the universe hypothesis of the appendix: some infinite π_0 excludes all nonprincipal π_0^+-complete ultrafilters on small sets. Choose an infinite field k with $|k| \geq \pi_0$, put $\mathcal{B} = \mathsf{CAlg}_k$, and take $F : \mathcal{B}/k \to \mathcal{B}$. Its essential image is the sieve $\mathcal{T}$ of algebras admitting a k-point. For every regular rank κ, the appendix construction, with a cardinal bound at least κ, supplies a κ-filtered diagram in $\mathcal{T}$ whose colimit A has no k-point. Thus $A \in \mathcal{S}_\kappa \setminus \mathcal{T}$.

Choose a larger regular μ for which A is μ-presentable. The full subcategory

$$\mathcal{Z}_A = \{B : \mathrm{Hom}(A, B) = \varnothing\}$$

contains $\mathcal{T}$: an arrow $A \to B \to k$ would be a forbidden point. It is closed under μ-filtered colimits, because $\mathrm{Hom}(A, -)$ preserves them and a filtered colimit of empty sets is empty. By minimality, $\mathcal{S}_\mu \subseteq \mathcal{Z}_A$. But $A \notin \mathcal{Z}_A$, as its identity is a morphism. Consequently

$$A \in \mathcal{S}_\kappa \setminus \mathcal{S}_\mu.$$

There can be no final stationary rank. This also works when both ranks are chosen among common strong ranks, where the individual closures are accessible by the preceding argument.

The two assertions in the manuscript thus have different answers. We checked the accessible-colimit literature cited here and the large-cardinal image theorem discussed in the appendix; we did not find this particular application to the decreasing closures in those sources. The argument is presented as a development of these pages, without a claim of priority. The next development treats the contravariant stabilization assertion separately, using the same reduced products.

Commentary · Development (commentators') [ms. 470]

The contravariant closures also need not stabilize

The contravariant version of the conjecture can be tested with the very products used in the appendix. The following argument concerns stationarity; it does not assume, or establish, accessibility of each individual closure.

Conditional failure of contravariant stationarity. [ours] Under the universe hypothesis of Proposition 4 of the appendix, there are accessible categories $\mathcal{N}, \mathcal{M}$ and a functor $g : \mathcal{N} \to \mathcal{M}^{op}$ preserving all colimits for which the L_π-closures of the image of g^{op} never become stationary.

Choose an infinite field $k \in \mathfrak{U}$ with $|k| \geq \pi_0$, where π_0 is the cardinal in that universe hypothesis. Take $\mathcal{N} = \mathrm{Set}$, $\mathcal{M} = \mathsf{CAlg}_k$, and $g(E) = k^E$, with pullback of functions on arrows. The product universal property gives

$$\mathrm{Hom}_k(A, k^E) \cong \mathrm{Set}(E, \mathrm{Hom}_k(A, k)).$$

Consequently g takes every colimit of sets to the corresponding limit of algebras, and therefore preserves colimits as a functor to $\mathcal{M}^{op}$. This is the accessibility convention required in the manuscript; no accessibility of $\mathcal{M}^{op}$ is being asserted. Write $\mathcal{S}(\pi)$ for the smallest full replete subcategory containing all k^E and closed under π^+-filtered colimits in $\mathcal{M}$.

Fix an infinite π, choose $|E| > \pi$, and put

$$B_\pi = k^E / I_\pi, \qquad I_\pi = \{f : |\operatorname{supp} f| \leq \pi\}.$$

The ideal is proper, and restriction gives

$$B_\pi \cong \varinjlim_{F \in [E]^{\leq \pi}} k^{E \setminus F}.$$

The indexing poset is π^+-directed, so $B_\pi \in \mathcal{S}(\pi)$. It has no k-point. Indeed, the appendix's ultrafilter calculation says that every k-algebra map $k^E \to k$ is evaluation at a coordinate: the corresponding ultrafilter is $|k|^+$-complete, hence principal under the hypothesis. A point of B_π would therefore give evaluation at some $e \in E$. But the characteristic function of $\{e\}$ maps to zero in B_π and to one under that evaluation, a contradiction.

Choose $\pi' > \max\{\pi, |k|, |B_\pi|\}$. Then B_π is π'^{+}-presentable in CAlg_k: use one generator for each element and at most π' relations recording the algebra operations and constants. Let 0 denote the zero algebra, which is terminal, and consider the full replete class

$$\mathcal{Z} = \{0\} \ \cup \ \{A : \mathrm{Hom}_k(B_\pi, A) = \varnothing\}.$$

Every product k^J belongs to $\mathcal{Z}$. If $J \neq \varnothing$, a map $B_\pi \to k^J$ followed by a coordinate projection would be a forbidden k-point; if $J = \varnothing$, the product is 0.

The class $\mathcal{Z}$ is closed under π'^{+}-filtered colimits. For a diagram with no zero value, presentability of B_π makes the hom set into its colimit empty. If a zero value occurs, its zero-valued indices form a nonempty cofinal cosieve, since a unital map from 0 to an algebra forces that algebra to be zero. The colimit is then zero. By minimality, $\mathcal{S}(\pi') \subseteq \mathcal{Z}$. However, B_π is nonzero and has an identity map, so $B_\pi \notin \mathcal{Z}$. Thus

$$B_\pi \in \mathcal{S}(\pi) \setminus \mathcal{S}(\pi').$$

As π was arbitrary, there is no final stationary rank. The zero-algebra case is necessary because the contravariant image includes the empty product. This supplies a separate negative answer to the stabilization assertion for contravariant images, under the stated universe hypothesis. We did not find this application of the appendix's calculation in the literature checked; no claim of priority is intended.

Soit $\mathcal{M}$ une catégorie satisfaisant la condition L, et admettant une petite sous-catégorie pleine strictement génératrice $\mathcal{C}$, formée d'objets accessibles. On a donc un foncteur pleinement fidèle et accessible

$$\mathcal{M} \ \hookrightarrow \ \mathcal{C}^\wedge \ ,$$

$\mathcal{M}$ L_π-stable dans $\mathcal{C}^\wedge$ pour π convenable. Si $\overline{\mathcal{M}}$ est l'image essentielle, de sorte que $\mathcal{M} \simeq \overline{\mathcal{M}}$, c'est donc une sous-catégorie strictement pleine L_π-stable dans $\mathcal{C}^\wedge$. Donc on s'attend à ce qu'elle soit accessible dans $\mathcal{C}^\wedge$ (si la situation décrite est "logiquement accessible"), ou ce qui revient au même, que $\mathcal{M}$ soit accessible. (Comparer 4.8.1, p. 187.) Mais il ne doit pas être difficile ([192]), en procédant à un nombre $\pi_{\mathfrak{U}}$ de choix (où $\pi_{\mathfrak{U}} = \mathrm{card}\,\mathfrak{U}$), de faire quand même des exemples ("logiquement inaccessibles") d'une sous-catégorie strictement pleine $\mathcal{M}$ de $\mathcal{C}^\wedge$ qui soit L_π-stable, mais non accessible dans $\mathcal{C}^\wedge$. Cette $\mathcal{M}$ admet donc une petite sous-catégorie strictement génératrice formée d'objets accessibles ([193]) et satisfait à L_π, sans pourtant être accessible.

[472]

Commentary · Development (commentators') [ms. 471]

A dense family of presentables need not suffice

The restricted Yoneda argument at the start of this paragraph is correct. The difficulty in footnotes 192–193 is that an arbitrary full subcategory of a presheaf category need not contain its representables. We can repair the proposed counterexample by adjoining their retracts to a suitable nonaccessible cosieve. This gives an example satisfying the stronger hypothesis actually stated here, under the same universe assumption as the appendix.

Conditional counterexample. [ours] Assume that some infinite $\pi_0 \in \mathfrak{U}$ has the property that every π_0^{+}-complete ultrafilter on a $\mathfrak{U}$-small set is principal. There is a locally small category $\mathcal{M}$ with κ-filtered colimits and a small dense full subcategory consisting of κ-presentable objects, for one regular κ, although $\mathcal{M}$ is not accessible.

Choose an infinite field $k \in \mathfrak{U}$ with $|k| \geq \pi_0$. Put $\mathcal{A} = \mathsf{CAlg}_k$, and let $\mathcal{C}$ be the small full category of polynomial algebras $k[x_1, \ldots, x_n]$, for finite n. The restricted nerve

$$N : \mathcal{A} \longrightarrow \mathcal{P} = \mathrm{Set}^{\mathcal{C}^{op}}$$

is fully faithful and preserves filtered colimits. Indeed, its value on the algebra with n generators is the nth power of the underlying set. Naturality with respect to polynomial substitutions says exactly

192. OK en prenant un *fermé* $\underline{S}$ non accessible dans $\mathcal{M}$ d'une catégorie accessible $\mathcal{M}$, qu'on plonge dans un $\mathcal{C}^\wedge$. Mais ces $\underline{S}$ n'a *pas* une petite famille strictement génératrice !

193. Ce n'est pas clair, il faudrait pour cela que le foncteur d'inclusion soit un adjoint à gauche !

that a transformation between two such nerves preserves the algebra operations and constants. Finite powers commute with filtered colimits.

Set $Z = N(k)$ and

$$\mathcal{S} = \{X \in \mathcal{P} : \mathrm{Hom}_{\mathcal{P}}(X, Z) = \varnothing\}.$$

This is a full replete cosieve, closed under every nonempty colimit. It is not accessible in $\mathcal{P}$. Otherwise its inverse image under the accessible functor N would be accessible in $\mathcal{A}$, contrary to Proposition 4 of the appendix together with Corollary 4.16.7.9. Here the inverse-image theorem applies without a strict-pullback defect: the inclusion of a full replete subcategory is an isofibration, so its strict pullback is equivalent to the accessible pseudopullback. The latter is constructed by products, inserters and equifiers, using the closure theorem discussed in Ching and Riehl (2014, Appendix A). Full faithfulness identifies this inverse image with the algebras having no k-point.

Let $\mathcal{T}$ consist of the retracts of representable presheaves, including all isomorphic copies. It is essentially small, since retracts are classified by idempotents on representables, and is closed under retracts. Every member of $\mathcal{T}$ maps to Z: each representable has such a map by evaluating its polynomial generators at zero, and a retract inherits one. Consequently $\mathcal{S} \cap \mathcal{T} = \varnothing$.

We verify the required filtered closure of $\mathcal{T}$ explicitly. Choose an infinite $\mu \geq |\operatorname{Mor} \mathcal{C}|$ and put $\kappa = \mu^{++}$. Every object of $\mathcal{T}$ has at most μ elements in total across its components. If X is the presheaf colimit of a κ-filtered diagram in $\mathcal{T}$, then $|X(c)| \leq \mu$ for each c: otherwise μ^+ distinct elements would have representatives at one common stage, since $\mu^+ < \kappa$, contradicting that stage's bound. Thus X also has at most μ elements in total. Its category of elements has at most μ arrows, so its canonical expression as a colimit of representables makes X κ-presentable. Applying presentability to its original filtered presentation factors 1_X through a stage. Hence X is a retract of that stage and belongs to $\mathcal{T}$. Notice the strict inequality $\mu^+ < \kappa$ in this argument.

Now take the full replete subcategory

$$\mathcal{M} = \mathcal{S} \cup \mathcal{T} \subseteq \mathcal{P}.$$

It is closed under κ-filtered colimits. A diagram taking values only in $\mathcal{T}$ was just handled. If it has a value in $\mathcal{S}$, its $\mathcal{S}$-valued indices form a nonempty cosieve in the filtered indexing category, hence a cofinal filtered subcategory; its colimit belongs to $\mathcal{S}$. Thus these colimits in $\mathcal{M}$ are computed in $\mathcal{P}$.

The representables form a small dense full subcategory of $\mathcal{M}$. In fact, the restricted nerve is precisely the full inclusion $\mathcal{M} \hookrightarrow \mathcal{P}$, by Yoneda. This proves genuine density, rather than just the weaker generator property. Each representable is κ-presentable in $\mathcal{M}$, because its hom functor is evaluation and the relevant colimits are computed pointwise.

Finally, suppose $\mathcal{M}$ were accessible. Choose a regular $\lambda \geq \kappa$ at which it is accessible, with a small family $\mathcal{G}$ of λ-presentable generators. In a λ-filtered presentation of $X \in \mathcal{S}$ by $\mathcal{G}$, not all stages can belong to $\mathcal{T}$, since $\mathcal{T}$ is closed under such colimits. The $\mathcal{S}$-stages are therefore nonempty and cofinal. They present X using the small family $\mathcal{G} \cap \mathcal{S}$, whose members remain λ-presentable in $\mathcal{S}$. This makes $\mathcal{S}$ accessible; its inclusion into $\mathcal{P}$ preserves filtered colimits, contradicting the earlier conclusion. This is the mechanism of Corollary 4.16.7.9 in the present example.

The construction uses the appendix's explicit universe hypothesis, and does not establish failure in every universe. It repairs the dense-family issue without a universe-sized sequence of arbitrary choices. We did not find this particular adjoining-of-retracts construction in the literature checked for this commentary; it is supplied as a development of the manuscript, without a claim of priority.

4.16.4 ([194]). Soit $\mathcal{M}$ une catégorie telle que $\mathcal{M}$ et $\mathcal{M}^\circ$ soient accessibles. Je conjecture qu'alors $\mathcal{M}$ est essentiellement petite. Plus généralement, supposons $\mathcal{M}$ accessible, et que $\mathcal{M}$ admette une petite sous-catégorie qui soit *strictement co*génératrice. Alors $\mathcal{M}$ est-elle essentiellement petite ? Plus généralement, si $\mathcal{M}$ admet une petite sous-catégorie strictement génératrice, et aussi une petite sous-catégorie strictement cogénératrice, i.e. si $\mathcal{M}$ et $\mathcal{M}^\circ$ sont toutes deux équivalentes à des sous-catégories pleines d'une $\mathcal{C}^\wedge$, avec $\mathcal{C}$ petite, alors $\mathcal{M}$ est-elle essentiellement petite ?

4.16.5. Soit I une petite catégorie, $I_\bullet$ le cône droit sur I (obtenu en rajoutant un objet final e), $\mathcal{M}$ une catégorie accessible,

$$\underline{\mathrm{Hom}}_{\square}(I_\bullet, \mathcal{M}) \hookrightarrow \underline{\mathrm{Hom}}(I_\bullet, \mathcal{M})$$

194. descendre après 4.16.5.

la sous-catégorie strictement pleine de la catégorie (accessible) $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{M})$, formée des foncteurs f qui font de $f(e)$ la limite inductive dans $\mathcal{M}$ des $f(i)$ ($i \in \mathrm{Ob}\, I$). Cette sous-catégorie est L_π-stable pour π grand, donc il y a lieu de conjecturer qu'elle est accessible [473] dans $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{M})$. (Je rappelle qu'on a vu dans 4.14 que c'est OK si les $\varinjlim$ de type I sont représentables dans $\mathcal{M}$. Et qu'une réponse affirmative, dans le cas où p. 183



implique que la sous-catégorie $\underline{\mathrm{Ép}}(\mathcal{M})$ dans $\underline{\mathrm{Fl}}(\mathcal{M})$ est accessible dans $\underline{\mathrm{Fl}}(\mathcal{M})$.)

Commentary · Examples [ms. 472]

Isbell's large category seen from both sides

Grothendieck's three smallness questions ask whether having enough descriptions by small objects from both directions leaves room for a large category. Isbell had already constructed a decisive example. Let PInj have sets as objects and partial bijections as morphisms: a map $X \dashrightarrow Y$ is a bijection from a subset of X onto a subset of Y. Reversing that bijection defines an isomorphism $\mathsf{PInj}^{op} \cong \mathsf{PInj}$. Isbell's example is recorded by Avery and Leinster (2021, Example 8.7), who attribute the one-object dense and codense subcategory to Isbell (1968, Example 1). We verify the properties relevant here directly.

First, filtered colimits exist and finite sets are finitely presentable. To construct the colimit of a filtered diagram, adjoin a basepoint to every set and send an undefined value of a partial map to the basepoint. Take the colimit of this diagram of pointed sets. Each structure map is still injective away from the basepoint: if two elements from one stage become equal to the same nonbasepoint element, they become equal at a common later stage, where the transition is injective away from the basepoint. Removing the colimit basepoint therefore gives a partial bijection from each original stage.

For the universal property, a cocone of partial bijections gives a pointed-set map out of this colimit. If two distinct nonbasepoint classes had the same nonbasepoint image, represent them at a common stage; injectivity of that stage's cocone map would make them equal. Hence the induced map is a partial bijection, and its uniqueness follows from the pointed-set universal property. This proves the colimit assertion without assuming that all transitions are everywhere defined.

A partial bijection from a finite set K to this colimit has finitely many defined values. Represent those values at one common stage; their distinct nonbasepoint images ensure that the representatives are distinct and nonbasepoint. Leave the other elements of K undefined. This lifts the map to that stage. Two such lifts which agree in the colimit agree at a later stage, by applying the filtered equality criterion to their finitely many values, including the basepoint. Thus K is finitely presentable. Finally, every set is the filtered union of its finite subsets along total inclusions. There is a small skeleton of finite sets, so PInj is finitely accessible. Its self-duality proves the same for its opposite.

To see the stronger density assertion, let $D = \{1, 2\}$. Its partial-bijection endomorphism monoid has seven elements: the empty map, four singleton-domain maps, and two permutations. A natural transformation

$$\mathsf{PInj}(D, X) \longrightarrow \mathsf{PInj}(D, Y)$$

means a function equivariant for precomposition by these seven endomorphisms. Commutation with the empty map forces the empty map to be preserved. Commutation with the two partial identity maps shows that the image of a map defined only at 1 is again defined only at 1, or empty; this defines a partial function $h : X \dashrightarrow Y$. Commutation with the transposition gives the same h in the second coordinate. For distinct $x, x' \in X$, apply the transformation to the map $D \to X$ taking the two points to x, x'. Its image must still be a partial bijection. Hence, whenever both $h(x)$ and $h(x')$ are defined, they are distinct. Thus h is a partial bijection. The partial identities now show that the transformation is exactly postcomposition with h. Its uniqueness is detected by singleton-domain maps. The restricted nerve at D is therefore fully faithful, proving density of this one-object full subcategory. Self-duality proves codensity.

Sets of distinct cardinalities are not isomorphic in PInj. There are therefore universe-many isomorphism classes, while the dense and codense test category is finite. All three smallness assertions fail. No large-cardinal hypothesis is used in this example, and the conclusion is about accessibility, not about the much stronger assumption of having all small limits or colimits.

N.B. Si $\mathcal{M}$ admet une petite sous-catégorie *co*génératrice, alors la question est justiciable de la conjecture 4.16.2 relative à un foncteur accessible $\mathcal{M} \longrightarrow \mathcal{N}^\circ$, ici $\underline{\mathrm{Hom}}(I_\bullet, \mathcal{M}) \longrightarrow \underline{\mathrm{Hom}}(I^\circ \times \mathcal{C}, \mathrm{Ens})^\circ \ldots$

4.16.6. Il reste la question de l'accessibilité relative d'une catégorie $\mathcal{E}$ sur une petite catégorie $\mathcal{C}$. Je pense à présent, grâce à 4.15.9, et son corollaire 4.15.10, que l'énoncé suivant doit "tomber" par AQT `[âne qui trotte]` :

Proposition 4.16.6.1. *Soit $f : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur, avec $\mathcal{C}$ petite catégorie karoubienne. On suppose que f satisfait la condition $H' = H$ (1+2+3) de 4.15.1 (page 415), où il suffit H 2, H 3, pour toute flèche λ* non identique *de $\mathcal{C}$. On suppose de plus que pour une catégorie $\mathcal{C}'$ d'un des deux types qu'on va décrire, et un foncteur $\mathcal{C}' \longrightarrow \mathcal{C}$, $\mathcal{E}' = \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ soit accessible, et $\mathcal{E}' \longrightarrow \mathcal{C}'$ accessible.*

[474]

Commentary · Development (commentators') [ms. 473]

Existing pushouts can form a nonaccessible class

Grothendieck wants the collection of universal cocones itself to have a manageable presentation. When all diagrams of a fixed shape have colimits, this is true: restriction identifies the category of colimit cocones with the accessible diagram category. The higher-categorical version has exactly this existence hypothesis in Lurie (2009, Example 5.4.7.9). The hypothesis becomes consequential when only some diagrams have colimits.

A counterexample from the appendix. [ours] Assume the appendix's universe hypothesis with cutoff π_0, and choose an infinite field k with $|k| \geq \pi_0$. There is a finitely accessible category $\mathcal{E}$ for which the full replete subcategory of pushout squares is not accessible and accessibly embedded in $\mathcal{E}^{\square}$.

To construct it, start with $\mathcal{A} = \mathsf{CAlg}_k$ and adjoin a new object $*$, with

$$\mathrm{Hom}_{\mathcal{E}}(A, *) = \mathrm{Hom}_k(A, k), \qquad \mathrm{Hom}_{\mathcal{E}}(*, A) = \varnothing, \qquad \mathrm{End}_{\mathcal{E}}(*) = \{1\}.$$

Composition of crossing maps is composition in $\mathcal{A}$. Thus $\mathcal{A}$ is the open piece of a collage and $\{*\}$ its closed piece. The new object and the old algebra k are distinct: in particular, there is no arrow from $*$ to k.

We first show that $\mathcal{E}$ is finitely accessible. A filtered diagram with a $*$-valued stage has a cofinal full subcategory of $*$-valued stages. Indeed, a common successor of such a stage and any other stage must again have value $*$, since no arrow leaves $*$ for the open piece. That subcategory is filtered, and its image diagram is constant with identity transitions. Its colimit is $*$. A filtered diagram entirely in $\mathcal{A}$ has its usual colimit there, which is also a colimit in $\mathcal{E}$: cocones to $*$ are precisely compatible maps to k, and representability identifies them with maps from the algebraic colimit to k. These two cases also show that each finitely presentable algebra remains finitely presentable in $\mathcal{E}$, and that $*$ is finitely presentable. A small skeleton of finitely presentable algebras together with $*$ generates $\mathcal{E}$ under filtered colimits. Hence $\mathcal{E}$ is finitely accessible.

For an algebra A, form the commutative square

$$\begin{array}{ccc} k & \longrightarrow & A \times k \\ \downarrow & & \downarrow{\scriptstyle \mathrm{pr}_2} \\ * & \xrightarrow[1]{} & *. \end{array}$$

Here the right vertical arrow means the crossing map specified by the algebra homomorphism $A \times k \to k$. Every cocone on the upper-left span has vertex $*$, because its leg from $*$ forces that vertex. Its remaining choice is a map $A \times k \to k$; compatibility with the leg from k is automatic for k-algebra maps. Since k is a field, the two complementary central idempotents of $A \times k$ show that

$$\mathrm{Hom}_k(A \times k, k) \cong \mathrm{Hom}_k(A, k) \amalg \{\mathrm{pr}_2\}.$$

The category of these cocones is discrete, because $*$ has only its identity endomorphism. The displayed canonical cocone is initial precisely when this set is a singleton, or equivalently when $\mathrm{Hom}_k(A,k)=\varnothing$.

The assignment $A\mapsto$ this square is an accessible functor $\mathcal{A}\to\mathcal{E}^{\square}$. Finite products in k-algebras commute with filtered colimits, the inclusion of the open piece preserves them by the preceding calculation, and the two constant $*$ components also preserve filtered colimits. If the full replete subcategory of pushout squares were accessible and accessibly embedded, its inverse image under this functor would be so as well. This last assertion is the accessible pseudopullback theorem, applied to a replete full inclusion, which is an isofibration; see Bourke and Lack (2024, Remark 4.9). The inverse image, however, is exactly the nonaccessible cosieve

$$\{A:\mathrm{Hom}_k(A,k)=\varnothing\}$$

proved nonaccessible in the appendix and Proposition 4.16.7.8. This contradiction establishes the claim.

The example also respects the source's observation that these cocones are closed under sufficiently filtered colimits: universal colimits commute with such pointwise colimits. It is the small generating family which fails. We did not find this collage realization in the checked accounts of accessible categories or in the all-colimits theorem cited above; the proof is given in full as a conditional development, without a priority claim.

The separate implication concerning epimorphisms is correct. For $u:A\to B$, the square with top and left edge u and bottom and right edge 1_B is a pushout exactly when u is epic: a pair of maps out of B agreeing after u must then agree. This gives the accessible inverse-image argument mentioned in the manuscript. It does not reverse the implication or remove the existence hypothesis from the preceding positive theorem.

(*On prend pour $\mathcal{C}'$ la catégorie*

$$\mathcal{Q}=\left\{\begin{array}{l}
\xi_0=\xi^{\natural}\ \underset{\rho}{\overset{\sigma}{\rightleftarrows}}\ \xi\circlearrowright p\qquad \rho\sigma = \mathrm{id}_{\xi_0}\\
\qquad\qquad\qquad\quad \downarrow{\scriptstyle u=u_0\rho}\qquad\quad \sigma\rho = p\ ,\quad d'où\ \left\{\begin{array}{rcl} p^2 &=& p\\ p\sigma &=& \sigma\\ \rho p &=& \rho\end{array}\right.\\
\qquad\quad {\scriptstyle u_0}\searrow\qquad\qquad\\
\qquad\qquad\ \eta\qquad\qquad\quad u = u_0\rho\ ,\quad d'où\ \left\{\begin{array}{rcl} up &=& u\\ u\sigma &=& u_0\end{array}\right.
\end{array}\right.$$

(trois objets ξ, ξ_0, η et cinq flèches non identiques σ, ρ, p, u_0, u, liés par les relations ci-dessus), ou la catégorie (isomorphe à Q°)

$$\mathcal{Q}'=\left\{\begin{array}{l}
\xi_0\ \underset{\rho}{\overset{\sigma}{\rightleftarrows}}\ \xi\circlearrowright p\qquad \left.\begin{array}{rcl}\rho\sigma &=& \mathrm{id}_{\xi_0}\\ \sigma\rho &=& p\end{array}\right\}\implies \left\{\begin{array}{rcl} p^2 &=& p\\ p\sigma &=& \sigma\\ \rho p &=& \rho\end{array}\right.\\
\quad {\scriptstyle u_0}\nwarrow\ \uparrow{\scriptstyle u}\\
\qquad\ \eta\qquad\qquad u = \sigma u_0\qquad\implies \left\{\begin{array}{rcl} pu &=& u\\ \rho u &=& u_0\end{array}\right.
\end{array}\right.$$

On se limite aux foncteurs $\mathcal{Q}\longrightarrow\mathcal{C}$, $\mathcal{Q}'\longrightarrow\mathcal{C}$ tels que p soit transformé en une flèche non identique *de $\mathcal{C}$ ([195]) (donc la condition relative à $\mathcal{Q}$, $\mathcal{Q}'$ devient vide, si $\mathcal{C}$ est sans projecteurs).*)

Sous ces conditions, $\mathcal{E}$ est accessible, et $f:\mathcal{E}\longrightarrow\mathcal{C}$ est accessible.

On trouve alors le

p. 184
[page 475]

195. Je ne suis pas sûr si c'est raisonnable – à vérifier avec soin !

Commentary · Development (commentators') [ms. 474]

What the two split-idempotent tests supply

Grothendieck introduces $\mathcal{Q}$ and $\mathcal{Q}'$ to isolate the obstruction left after large filtered diagrams have been made constant in the base. One test attaches a target to a split idempotent; the other attaches a source. This is exactly the distinction between constructing a colimit and proving that a chosen object is presentable.

The finite-test assertion of Proposition 4.16.6.1. [ours] Let $f : \mathcal{E} \to \mathcal{C}$ satisfy the hypotheses stated there, with $\mathcal{C}$ small and Karoubian. Then $\mathcal{E}$ and f are accessible. In this assertion about the total category itself, it is enough to use the $\mathcal{Q}$ and $\mathcal{Q}'$ tests whose distinguished idempotent has nonidentity image in $\mathcal{C}$.

Here are the rank choices needed before the construction. The hypothesis H supplies accessible fibres, accessible pullbacks over arrows, and preservation of sufficiently filtered fibre colimits in those pullbacks. For a test category B equal to Δ_1, $\mathcal{Q}$, or $\mathcal{Q}'$, its fibre inclusions into $\mathcal{E}_B$ are accessible: maps from a fibre colimit to any other fibre are computed by the contravariant mixed hom functors of H2. This includes arrows of B whose images are identities, where the assertion is ordinary representability inside one fibre. There are only a small number of the indicated tests. Choose a common sufficiently large strong accessibility rank κ for their categories $\mathcal{E}_B$, their projections $\mathcal{E}_B \to B$, all fibres, and all fibre inclusions. Enlarge it to a suitable successor rank beyond the strict cardinal bound of Corollary 4.15.11. The uniformization used here is the same as in the preceding accessibility arguments; no claim at a prescribed smaller rank is needed.

Let $D : J \to \mathcal{E}$ be κ-filtered. The normal form proved near pp. 443–463 permits a cofinal, still κ-directed reindexing for which all base objects are one object s and every strict base transition is one idempotent $p : s \to s$. If the new index has a last object, its colimit is already a stage and there is nothing to construct. Otherwise every index has a strict successor. If $p = 1_s$, H2 shows directly that the colimit in $\mathcal{E}_s$ is a colimit in $\mathcal{E}$. Suppose then that $p \neq 1_s$, and split it in $\mathcal{C}$:

$$s \xrightarrow{r} t \xrightarrow{i} s, \qquad ri = 1_t, \quad ir = p.$$

Write K for the full split-idempotent subcategory of $\mathcal{Q}$ on ξ, ξ_0. The diagram lifts to $\mathcal{E}_K$ with all index objects labelled ξ. Extend $K \to \mathcal{C}$ to $\mathcal{Q} \to \mathcal{C}$ by sending η to t and u_0 to 1_t. In $\mathcal{Q}$ the colimit of this base diagram is ξ_0, with structure arrows ρ. Indeed, the strict successors force any cocone arrow a to satisfy $ap = a$, and these arrows factor uniquely through ρ. Accessibility of $\mathcal{E}_{\mathcal{Q}}$ supplies its colimit, and preservation by the projection puts its vertex over ξ_0: this finite category has no nontrivial isomorphisms. We obtain $z \in \mathcal{E}_t$ with cocone maps over r. Fullness of K restricts the universal property to $\mathcal{E}_K$, so this is also a colimit there.

To check all targets in $\mathcal{E}$, take $y \in \mathcal{E}_c$. The base arrows of any cocone $D \to y$ are constant, say $a : s \to c$, and satisfy $ap = a$. Hence $a = \mu r$ for the unique $\mu = ai : t \to c$. Use now the $\mathcal{Q}$ test with $\eta \mapsto c$ and $u_0 \mapsto \mu$. Its independently existing colimit again projects to ξ_0 and restricts to the same colimit in $\mathcal{E}_K$. The canonical isomorphism to z is vertical because $\mathrm{End}_K(\xi_0) = \{1\}$. Its universal property gives exactly one arrow $z \to y$ over μ producing the original cocone. This proves the total universal property and, at the same time, preservation of this colimit by f.

We next prove presentability, since existence of colimits alone does not prove accessibility. Let x be κ-presentable in a fibre $\mathcal{E}_c$, and let $x \to z$ lie over $\nu : c \to t$. In the $\mathcal{Q}'$ test put $\eta \mapsto c$, $\xi_0 \mapsto t$, $\xi \mapsto s$, $u_0 \mapsto \nu$, and $u \mapsto i\nu$. The object labelled (x, η) is κ-presentable in $\mathcal{E}_{\mathcal{Q}'}$ by the strong rank choice. Accessibility gives a colimit of the lifted diagram there, and its projection is again ξ_0. Restricting to the same full K identifies this colimit vertically with (z, ξ_0). Consequently $x \to z$ factors through some D_j, giving surjectivity of the hom-colimit comparison in $\mathcal{E}$.

For injectivity, consider two maps from x to stages with the same composite to z. Their base maps $a, b : c \to s$ have $ra = rb = \nu$. After advancing each stage once, their bases become $pa = pb = i\nu$. They are now maps in the same $\mathcal{Q}'$ test. Presentability there makes them equal at a common later stage. When $p = 1_s$, precisely this argument uses the corresponding Δ_1 test instead. The case with a final index is immediate. Thus every fibrewise κ-presentable object is κ-presentable in $\mathcal{E}$.

The union over the small base of small skeletons of these fibrewise objects is small. Every object of $\mathcal{E}$ is a κ-filtered colimit of them inside its own fibre, hence also in $\mathcal{E}$ by H2. This proves accessibility and completes the finite-test assertion. We did not locate this exact $\mathcal{Q}, \mathcal{Q}'$ formulation in the checked literature; the proof supplies the missing argument rather than attributing it to a later author. The restriction on p just justified is specific to this theorem. It cannot be carried unchanged through arbitrary strict base changes, as the next example shows.

Corollaire 4.16.6.2. *Soit $f : \mathcal{E} \longrightarrow \mathcal{C}$ un foncteur. Conditions équivalentes :*

a) Pour toute catégorie $\mathcal{C}'$ petite *et* karoubienne*, et tout foncteur $\mathcal{C}' \longrightarrow \mathcal{C}$, $\mathcal{E}' = \mathcal{E} \times_{\mathcal{C}} \mathcal{C}'$ est accessible, et le foncteur $\mathcal{E}' \longrightarrow \mathcal{C}'$ est accessible.*

b) Comme a), mais $\mathcal{C}'$ étant une des trois catégories Δ_1, $\mathcal{Q}$, $\mathcal{Q}'$ (où $\mathcal{Q}$, $\mathcal{Q}'$ sont définies dans 4.15.6.1 ci-dessus).

b′) ([196]). (Si $\mathcal{C}$ est sans projecteurs.) Comme b), mais en se limitant au cas $\mathcal{C}' = \mathbf{\Delta}_1$ (à l'exclusion de $\mathcal{Q}$, $\mathcal{Q}'$).

Corollaire 4.16.6.3. *Supposons $f : \mathcal{E} \longrightarrow \mathcal{C}$ fibrant ou cofibrant. Alors les conditions équivalentes précédentes impliquent que :*

1°) Les $\mathcal{E}_s$ sont accessibles ($s \in \mathrm{Ob}\,\mathcal{C}$).

2°) Pour toute flèche $\lambda : s \longrightarrow t$ dans $\mathcal{C}$, le foncteur $\lambda^* : \mathcal{E}_t \longrightarrow \mathcal{E}_s$ (resp. $\mathcal{E}_s \longrightarrow \mathcal{E}_t$) est accessible.

D'autre part, l'ensemble de ces conditions 1°, 2°) équivaut à la validité de b) ci-dessus, dans le cas $\mathcal{C}' = \mathbf{\Delta}_1$, ou encore à la validité de a) ci-dessus, quand $\mathcal{C}'$ est sans projecteurs. [page 476]

Commentary · Development (commentators') [ms. 475]

A strict pullback can expose a hidden projector

The uncertain clause b′ in Corollary 4.16.6.2 asks to discard both split-idempotent tests when the base has no nonidentity projectors. A strict pullback remembers more than the images of its arrows in that base. A projector upstairs may map to an identity downstairs, and this is exactly the case which the shortened criterion misses.

Counterexample to b′ ⇒a. [ours] Let $\mathcal{C}$ be the walking isomorphism, with objects $0, 1$ and exactly one arrow between each ordered pair. It has no nonidentity idempotents. Let $f : e \to \mathcal{C}$ select 0. For any functor $\Delta_1 \to \mathcal{C}$, the strict pullback is empty, a point, or Δ_1, according to which endpoints map to 0. Each is accessible and its projection to Δ_1 preserves sufficiently filtered colimits. Thus the printed condition b′ holds.

Let R be the walking split idempotent, with two objects a, b and arrows

$$a \xrightarrow{r} b \xrightarrow{i} a, \qquad ri = 1_b, \quad ir = p \neq 1_a.$$

Its only nonidentity endomorphism is p, and it is split, so R is a small Karoubian category. Send a to 0, b to 1, and r, i to the inverse isomorphisms of $\mathcal{C}$. Then p is sent to 1_0. The strict pullback $e \times_{\mathcal{C}} R$ has one object with endomorphism monoid $\{1, p\}$, where $p^2 = p \neq 1$.

This one-object category is not accessible. For any infinite regular κ, index a diagram by the ordinal κ, put the sole object at every stage, and put p on every strict transition. A cocone to the sole object must have every component equal to p. If it were a colimit cocone, the two endomorphisms 1 and p of its vertex would induce distinct cocones. They induce the same one, since $1p = pp = p$. Thus the required κ-filtered colimit does not exist, for any κ.

Condition a) therefore fails. The unrestricted equivalence a)⇔b) is repaired by retaining all $\mathcal{Q}$ and $\mathcal{Q}'$ tests, including those which send p to an identity. Indeed, applying Proposition 4.16.6.1 to a pullback over a small Karoubian $\mathcal{C}'$ may involve a nonidentity projector of $\mathcal{C}'$ whose image in $\mathcal{C}$ is an identity. The example was checked directly against the generators and compositions of these finite categories. We found no statement of this particular counterexample in the consulted sources; it is a correction of the queried implication, without a claim of priority.

Question 4.16.6.4. *Les conditions 1° et 2° précédentes impliquent-elles celles de 4.16.6.2 ? En d'autres termes, si la catégorie base $\mathcal{C}$ est égale à $\mathcal{Q}$ ou à $\mathcal{Q}'$, alors ces conditions impliquent-elles que $\mathcal{E}$ est accessible, et $f : \mathcal{E} \longrightarrow \mathcal{C}$ accessible ?*

Ça semble idiot de ne pas connaître la réponse !

4.16.6.5. J'ai envie d'appeler catégorie *accessible sur $\mathcal{C}$*, une catégorie $\mathcal{E}$ sur $\mathcal{C}$ satisfaisant les conditions équivalentes de 4.16.6.1 – du moins dans le cas où on suppose $\mathcal{C}$ petite. Pour

196. ?, à vérifier avec soin !

que cette terminologie soit vraiment satisfaisant, il faudrait que l'on sache prouver que ces conditions impliquent ceci : pour toute catégorie accessible $\mathcal{M}$ et tout foncteur accessible $\mathcal{M} \longrightarrow \mathcal{C}$, la catégorie $\mathcal{E} \times_{\mathcal{C}} \mathcal{M}$ est accessible, et le foncteur $\mathcal{E} \times_{\mathcal{C}} \mathcal{M} \longrightarrow \mathcal{M}$ est accessible.

L'idée, c'est qu'on appellerait (sous hypothèse de petitesse) *catégorie accessible* $\mathcal{E}$ sur une autre $\mathcal{C}$, une catégorie ayant la propriété qu'on vient de dire – mais en supposant maintenant de plus que $\mathcal{C}$ satisfasse à L_π, j'imagine, pour tout foncteur accessible $\mathcal{M} \longrightarrow \mathcal{C}$, avec $\mathcal{M}$ accessible, $\mathcal{E} \times_{\mathcal{C}} \mathcal{M}$ est accessible et le foncteur canonique $\mathcal{E} \times_{\mathcal{C}} \mathcal{M} \longrightarrow \mathcal{E}$ est accessible. [page 476bis]

Commentary · Development (commentators') [ms. 476bis]

Relative accessibility and lifting isomorphisms

Grothendieck wants relative accessibility to survive replacement of the base by an accessible category. The corrected finite tests do give this property. Their strict-pullback formulation also forces an additional feature which explains why a bare accessible functor is insufficient.

A criterion over a small Karoubian base. [ours] For $f : \mathcal{E} \to \mathcal{C}$ with $\mathcal{C}$ small and Karoubian, the following are equivalent: every strict pullback along a small Karoubian $\mathcal{C}' \to \mathcal{C}$ is accessible and has accessible projection to $\mathcal{C}'$; and $\mathcal{E}$ and f are accessible and f is an isofibration. Here isofibration means that for every $x \in \mathcal{E}$ and every isomorphism $u : f(x) \to t$ in $\mathcal{C}$, there is an isomorphism $\widetilde{u} : x \to y$ with $f(\widetilde{u}) = u$ and $f(y) = t$.

For necessity, use $\mathcal{C}' = \mathcal{C}$ to obtain accessibility of $\mathcal{E}$ and f. For the lifting assertion, use the walking split idempotent R from the preceding block, mapping its two objects to $f(x)$ and t, its arrows r, i to u, u^{-1}, and its projector p to $1_{f(x)}$. In the accessible strict pullback there is an idempotent $(1_x, p)$ on (x, a). An accessible category splits every idempotent: a sufficiently long regular-ordinal diagram with that idempotent on all strict transitions has a colimit, and its universal property gives the splitting. The only object of R at which p can split is b, since the monoid $\{1, p\}$ at a does not split it internally. The splitting object is therefore (y, b) for some $f(y) = t$. Its two splitting arrows project in $\mathcal{E}$ to inverse isomorphisms, because the first component of the original idempotent was 1_x. The arrow over r is the required lift of u.

For sufficiency, a small Karoubian category is accessible at a sufficiently large rank, and any functor between two such categories preserves sufficiently filtered colimits: those colimits reduce to splittings of idempotents, which every functor preserves. The iso-comma category of the accessible functors $\mathcal{E} \to \mathcal{C} \leftarrow \mathcal{C}'$ is accessible and its projections are accessible, by the pseudopullback theorem. Because f is an isofibration, its strict pullback is equivalent to this iso-comma category. Explicitly, an object $(x, c', \alpha : f(x) \cong g(c'))$ can be replaced, using a lift of α, by one with equal base objects and identity comparison. The inclusion of the strict pullback is fully faithful and this replacement proves essential surjectivity. This proves sufficiency.

The same argument now allows any accessible $\mathcal{M} \to \mathcal{C}$ in place of $\mathcal{C}'$. Its strict pullback is accessible, and both projections are accessible. This answers the intended base-change question of p. 476, and also reconciles the two different projections mentioned on p. 476 bis. The assertion here is over the stated small Karoubian base; the proposal for a general nonsmall base is a further definition.

The standard closure theorem used in this proof is discussed precisely in Bourke and Lack (2024, Examples 4.8 and Remark 4.9). The identification of Grothendieck's strict test property with isomorphism lifting is our deduction; we did not find this formulation in those references. For actual Grothendieck fibrations and opfibrations, isomorphisms already lift, and the total-category proof supplied near pp. 438–440 gives the remaining accessibility. Thus Question 4.16.6.4 has an affirmative answer in both orientations. The distinction between strict pullbacks and pseudopullbacks accounts for the failure of b′ without making that structured result problematic.

[page 477]

4.16.7. Retour sur les sous-catégories fermées d'une catégorie accessible (4.16.1).

Proposition 4.16.7.1. *Soient $\mathcal{M}$ une catégorie accessible, $(\underline{S}_i)_{i\in I}$ une famille de sous-catégories fermées accessibles dans $\mathcal{M}$. On ne suppose que* $\operatorname{card} I$ *soit petit. Soit $\underline{S}$ la sous-catégorie strictement pleine de $\mathcal{M}$ telle que* $\operatorname{Ob} \underline{S} = \bigcup_{i\in I} \operatorname{Ob} \underline{S}_i$. *Alors pour que $\underline{S}$ soit accessible dans $\mathcal{M}$, il faut et il suffit qu'il existe une* petite *partie $J \subset I$, telle que* $\operatorname{Ob} \underline{S} = \bigcup_{i\in J} \underline{S}_i$. *(En* p. 185

particulier, si I est ordonnée, et $i \longmapsto \underline{S}_i$ croissante et grande devant tout petit cardinal, il faut et il suffit qu'il existe un $i \in I$ tel que $\underline{S} = \underline{S}_i$, i.e. tel que $j \geqslant i$ implique $\underline{S}_j = \underline{S}_i$.)

C'est évidemment suffisant, en vertu de 4.13… Prouvons inversement que si $\underline{S}$ est accessible dans $\mathcal{M}$, alors la condition précédente est satisfaite. Soit π un cardinal $\geqslant \pi_0$ tel que $\underline{S}$ soit π-accessible dans $\mathcal{M}$, donc

$$\underline{S} \xleftarrow{\sim} \operatorname{Ind}_\pi \underline{S}(\pi) , \qquad \text{où } \underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi.$$

Soit $\underline{S}(\pi)_0$ une sous-catégorie réduite de $\underline{S}(\pi)$, telle que $\underline{S}(\pi)_0 \hookrightarrow \underline{S}(\pi)$ soit une équivalence, [page 478] de sorte que $\underline{S}(\pi)_0$ est petite. Pour tout $x \in \operatorname{Ob} \underline{S}(\pi)_0$, soit $i_x \in I$ tel que $x \in \operatorname{Ob} \underline{S}_{i_x}$. Soit J l'ensemble de tous les i_x, que est une petite partie de I, et soit $\underline{S}_J$ la sous-catégorie strictement pleine de $\mathcal{M}$ ayant pour ensemble d'objets $\bigcup_{j\in J} \operatorname{Ob} \underline{S}_j$. Donc $\underline{S}(\pi)_0$ est contenue dans $\underline{S}_J$, donc aussi $\underline{S}(\pi)$, puisque $\underline{S}_J$ est strictement pleine :

$$\underline{S}(\pi) \hookrightarrow \underline{S}_J .$$

D'ailleurs $\underline{S}_J$ est accessible dans $\mathcal{M}$ (loc. cit.), et est fermée dans $\mathcal{M}$ comme réunion de sous-catégories fermées, *a fortiori elle satisfait à L_π*. Donc, comme $\underline{S}(\pi) \subset \underline{S}_J$, le foncteur essentiellement surjectif $\operatorname{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \underline{S}$ se factorise par $\underline{S}_J$. Donc $\underline{S}_J \hookrightarrow \underline{S}$ est essentiellement surjectif, et comme $\underline{S}_J$ est strictement pleine, on en conclut $\underline{S}_J = \underline{S}$, q.e.d.

Proposition 4.16.7.2. [page 479] *Soient $\mathcal{M}$ une catégorie accessible, x un objet de $\mathcal{M}$, et ${}_x\mathcal{M}$ la sous-catégorie fermée de $\mathcal{M}$ engendrée par x, formée des $y \in \operatorname{Ob} \mathcal{M}$ tels que $\operatorname{Hom}(x, y) \neq \emptyset$, i.e. l'image essentielle de $x\backslash\mathcal{M} \longrightarrow \mathcal{M}$. Alors ${}_x\mathcal{M}$ est une sous-catégorie de $\mathcal{M}$ accessible dans $\mathcal{M}$.*

En effet, c'est l'image inverse de la sous-catégorie Ens^* accessible dans Ens par le foncteur représentable (donc accessible)

$$h_x \ : \ y \longmapsto \operatorname{Hom}(x, y) \ : \ \mathcal{M} \longrightarrow \mathrm{Ens} ,$$

donc est accessible dans $\mathcal{M}$ en vertu de 4.13…

Corollaire 4.16.7.3. *Soit $\mathcal{M}$ une catégorie accessible, $\underline{S}$ une sous-catégorie fermée. Pour que $\underline{S}$ soit accessible, il faut et il suffit qu'il existe une* petite *partie J de $\underline{S}$, telle que $\operatorname{Ob} \underline{S} = \bigcup_{x\in J} \operatorname{Ob}({}_x\mathcal{M})$.*

En effet, on a

$$\operatorname{Ob} \underline{S} = \bigcup_{x \in \operatorname{Ob} \underline{S}} \operatorname{Ob}({}_x\mathcal{M}) ,$$

i.e. $\underline{S}$ apparaît comme réunion de la famille (pas nécessairement petite) de sous-catégories [page 480] fermées ${}_x\mathcal{M}$ de $\mathcal{M}$ (indexées par $x \in \operatorname{Ob} \underline{S}$), accessibles dans $\mathcal{M}$ en vertu de 4.16.7.2. Donc notre assertion est un cas particulier de 4.16.7.1.

Commentary · Exposition [ms. 479]

Closed subcategories and small weakly initial families

Grothendieck's return to closed subcategories isolates the generation condition which the earlier examples lacked. A cosieve $\mathcal{S}$ is closed under any existing nonempty colimit of its objects: choosing one stage already gives an arrow from $\mathcal{S}$ into the colimit. Its accessibility therefore depends on how many different starting objects are required to reach every object of $\mathcal{S}$.

A family $A \subseteq \operatorname{Ob} \mathcal{S}$ generates the cosieve precisely when every $s \in \mathcal{S}$ receives a map from some $a \in A$. This is a weakly initial family; no uniqueness of the map is required. If it is small, choose a common useful rank at which all its objects are presentable. Let $\mathcal{T}$ be the complementary sieve. A sufficiently filtered diagram in $\mathcal{T}$ cannot have its colimit in $\mathcal{S}$: a map from one of the presentable generators to that colimit would factor through a stage, placing that stage in the cosieve. Thus $\mathcal{T}$ is closed under those colimits, and the canonical presentations used below prove accessibility of both parts.

Conversely, if $\mathcal{S}$ is accessible in $\mathcal{M}$, take a small skeleton of its presentable generators at a common useful rank. Every object of $\mathcal{S}$ is a nonempty filtered colimit of these generators, so receives a map from one of them. This proves Corollary 4.16.7.3 directly and explains why its small family is the right condition.

The union criterion follows with the same economy. If a union of accessible cosieves is accessible, choose one member of the family containing each object of a small generating skeleton of the union. These choices use only a small subfamily. Its union is itself a cosieve, hence contains every colimit generated by that skeleton and is the whole union. Conversely, the union of the small generating families of a small subfamily is again small. In particular, a union indexed by an order directed beyond every small cardinal is accessible exactly when a single stage already contains it. No finiteness or countability assertion is hidden in the word small: it is always relative to the fixed universe.

p. 186

On peut aussi, pour toute partie petite J de $\mathrm{Ob}\,\underline{S}$, considérer la sous-catégorie fermée $\underline{S}_J \subset \underline{S}$ de $\mathcal{M}$ engendrée par J. L'ensemble I des parties J en question, ordonné par inclusion, est grand devant tout cardinal $\pi \in \mathfrak{U}$, et $\underline{S}$ est réunion croissante des $\underline{S}_J$. D'autre part, la proposition nous dit que $\underline{S}$ est accessible dans $\mathcal{M}$ si et seulement s'il existe un $J \in I$, tel que $\underline{S}_J = \underline{S}$, i.e. tel que $\underline{S}_J = \underline{S}_{J'}$ pour $J \subset J'$.

Proposition 4.16.7.4. *Soit $\mathcal{M}$ une catégorie accessible. Les conditions suivantes sont équivalentes :*

a) Il existe une sous-catégorie $\underline{S}$ de $\mathcal{M}$ fermée dans $\mathcal{M}$, et non accessible dans $\mathcal{M}$.

b) Soit I l'ensemble des ordinaux α tels que $\mathrm{card}\,\alpha \in \mathfrak{U}$*, i.e. des ordinaux α qui sont dans $\mathfrak{U}$. Donc $I = I_{\alpha_0}$, où α_0 est le plus petit ordinal qui n'appartienne pas à $\mathfrak{U}$, et I est grand devant tout petit cardinal π, i.e. devant tout cardinal dans $\mathfrak{U}$ ([197]). La condition envisagée est la suivante :* [page 481]

Il existe une suite transfini

$$\underline{x} \;=\; (x_\alpha)_{\alpha\in I}$$

d'éléments de $\mathrm{Ob}\,\mathcal{M}$*, telle que l'on ait*

$$\mathrm{Hom}(x_\beta, x_\alpha) \;=\; \emptyset \qquad\qquad \textit{si } \beta < \alpha.$$

De façon précise, pour une telle suite transfinie, soit $\underline{S}$ ou ${}_{\underline{x}}\mathcal{M}$ la sous-catégorie fermée de $\mathcal{M}$ engendrée par la famille des x_α, i.e. telle que $\mathrm{Ob}\,\underline{S} = \bigcup_{\alpha\in I} \mathrm{Ob}\ {}_{x_\alpha}\mathcal{M} = \bigcup_{\alpha\in I} x_\alpha \setminus \mathrm{Ob}\,\mathcal{M}$. Alors ${}_{\underline{x}}\mathcal{M}$ est une sous-catégorie fermée non accessible dans $\mathcal{M}$, et toute sous-catégorie fermée non accessible dans $\mathcal{M}$ s'obtient de cette façon.

Démonstration. Considérons une suite $\underline{x} = (x_\alpha)_{\alpha\in I}$ comme dessus. Si J est une petite partie de I, il existe un $\alpha \in I$ qui majore les $\beta \in J$, donc $\mathrm{Hom}(x_\beta, x_\alpha) = \emptyset$, pour $\beta \in J$, ce qui signifie que x_α n'appartient à aucune des $\underline{S}_\beta = {}_{x_\beta}\mathcal{M}$, pour $\beta \in J$. Donc $\underline{S}$ n'est par réunion d'une petite sous-famille de la famille des ${}_{x_\alpha}\mathcal{M}$, donc en vertu de 4.16.7.1, $\underline{S}$ n'est par accessible dans $\mathcal{M}$. [page 482]

Inversement, soit $\underline{S}$ une telle sous-catégorie dans $\mathcal{M}$. Considérons un bon ordre sur $\mathrm{Ob}\,\underline{S}$ (objets s_α), et construisons la suite des x_α ($\geqslant s_\alpha$) par récurrence transfinie ([198]).

1) On choisit $x_0 =$ premier élément de $\mathrm{Ob}\,\underline{S}$ (existe, car $\mathrm{Ob}\,\underline{S} \neq \emptyset$, i.e. $\underline{S} \neq \emptyset$, car sinon, $\underline{S}$ serait accessible dans $\mathcal{M}$).

2) Si les x_β ($\beta < \alpha$) sont construits, on prend pour x_α le premier élément de $\underline{S}$ majorant les x_β et qui n'appartient pas à la sous-catégorie fermée de $\mathcal{M}$ engendrée par les x_β ($\beta < \alpha$). (Cette sous-catégorie fermée est $\subset \underline{S}$, et $\neq \underline{S}$, car $\underline{S}$ n'est pas accessible dans $\mathcal{M}$.) On a donc bien

$$\mathrm{Hom}(x_\beta, x_\alpha) \;=\; \emptyset \qquad\qquad \text{si } \beta < \alpha.$$

p. 187

197. introduire $I = I_{\mathfrak{U}}$ avant l'énoncé de la proposition. [page 483]

198. **N.B.** On a $\mathrm{card}\,\underline{S} \notin \mathfrak{U}$, sinon, $\underline{S}$ serait petite, donc accessible dans $\mathcal{M}$, contrairement à l'hypothèse.

Ainsi, la récurrence transfinie se poursuit sur I tout entier. On voit par récurrence transfinie qu'on a, pour tout $\alpha \in I$,

$$s_\alpha \leqslant x_\alpha$$

(card $x_\alpha > x_\beta \geqslant s_\beta$ pour $\beta < \alpha$, donc $x_\alpha > s_\beta$ pour tout $\beta < \alpha$, ce qui équivaut à $x_\alpha \geqslant s_\alpha$). De plus, on trouve aussi, par récurrence transfinie, que

$$(*) \qquad \beta < \alpha \quad \Longrightarrow \quad \begin{array}{l} s_\beta \text{ appartient à la sous-catégorie} \\ \text{fermée } \underline{S}_\alpha \text{ de } \mathcal{M} \text{ engendrée par} \\ \text{les } x_{\beta'} \text{ avec } \beta' < \alpha. \end{array} ,$$

puisque par construction s_α est le *premier* des éléments de $\underline{S}$ qui n'est *pas* dans ladite sous-catégorie $\underline{S}_\alpha$ de $\mathcal{M}$. Cela montre que tout s_β appartient à la sous-catégorie fermée ${}_{\underline{x}}\mathcal{M}$ de $\mathcal{M}$ engendrée par la famille de *tous* les x_α, donc on trouve que $\underline{S} \subset {}_{\underline{x}}\mathcal{M} \subset \underline{S}$, donc $\underline{S} = {}_{\underline{x}}\mathcal{M}$, q.e.d.

Proposition 4.16.7.5. *Pour un univers $\mathfrak{U}$ donné (contenant $\mathcal{N}$ comme élément), les conditions suivantes sont équivalentes :*

1) Il existe une catégorie $\mathfrak{U}$-accessible $\mathcal{M}$ ayant une sous-catégorie fermée non accessible dans $\mathcal{M}$. [page 484]

b) Il existe une catégorie $\mathcal{C} \in \mathfrak{U}$, et une suite transfinie d'éléments de

$$\mathcal{C}^\wedge \overset{\text{déf}}{=} \underline{\mathrm{Hom}}(\mathcal{C}, \mathrm{Ens}_\mathfrak{U}) ,$$

$\underline{x} = (x_\alpha)_{\alpha \in I}$, indexée par l'ensemble de tous les ordinaux $\alpha \in \mathfrak{U}$, et telle que

$$\mathrm{Hom}(x_\alpha, x_\beta) = \emptyset \qquad \textit{si } \alpha < \beta.$$

Commentary · Exposition [ms. 483]

The transfinite construction and its size bound

Grothendieck turns failure of a small weakly initial family into a sequence with no arrows pointing forward. The order used in the recursive construction is an external well order of representatives. It is not the preorder defined by existence of a morphism; confusing them would make the displayed condition contradict the instruction to choose a larger object.

Write $\mathfrak{U} = V_\theta$ for the fixed infinite Grothendieck universe. An accessible $\mathfrak{U}$-category has at most θ isomorphism classes. Indeed, a small dense subcategory embeds it fully faithfully in a presheaf category on a $\mathfrak{U}$-small category. For every cardinal bound below θ, only a $\mathfrak{U}$-small set of isomorphism classes of such presheaves has values within that bound; taking all bounds gives at most θ classes. We may therefore work with a skeleton of a cosieve $\mathcal{S}$, rather than a possibly much larger set of duplicate objects. If $\mathcal{S}$ is nonaccessible, this skeleton cannot have cardinal less than θ, by the small-family criterion. Enumerate it as $(s_\alpha)_{\alpha<\theta}$.

At stage α, let $\mathcal{S}_\alpha$ be the cosieve generated by the already chosen x_β, $\beta < \alpha$. This is generated by a small set, hence is not all of $\mathcal{S}$. Choose x_α to be the first element of the well order outside $\mathcal{S}_\alpha$. Then

$$\mathrm{Hom}(x_\beta, x_\alpha) = \varnothing \qquad (\beta < \alpha).$$

The indices of the chosen objects are strictly increasing, since every object preceding the least previously uncovered object was already covered. By induction, before stage α every s_β with $\beta < \alpha$ is covered: if one were still uncovered, the next least-uncovered index would be less than α, contrary to the strictly increasing sequence of prior choices. Thus the resulting family generates all of $\mathcal{S}$. In the printed explanation of this step, the least uncovered object is x_α, not s_α.

Conversely, let $(x_\alpha)_{\alpha<\theta}$ have no forward morphisms and let $\mathcal{S}$ be its generated cosieve. If it had a small weakly initial family (a_j), each a_j would receive a map from some x_{β_j}. The small set of ordinals β_j has a bound below θ. A later x_α would have to receive a map from one a_j, giving the forbidden composite $x_{\beta_j} \to a_j \to x_\alpha$. This proves nonaccessibility and both directions of Proposition 4.16.7.4.

The same full presheaf embedding proves Proposition 4.16.7.5, since it preserves the emptiness of hom sets. It does not prove the additional assertion about a sequence consisting entirely of fields: the algebras constructed in the appendix need not be fields. The following development at p. 485 answers that stronger question by a separate graph-to-field construction and identifies its exact dependence on the universe boundary.

Démonstration. On a b) $\Longrightarrow$ a) en prenant $\mathcal{M} = \mathcal{C}^\wedge$, en vertu de la proposition précédente. On voit de même a) $\Longrightarrow$ b), en plongeant $\mathcal{M}$ dans une catégorie $\mathcal{C}^\wedge$ ($\mathcal{C}$ étant une petite sous-catégorie strictement génératrice de $\mathcal{M}$).

Je n'ai pas été capable ([199]) de construire une suite $(x_\alpha)_{\alpha\in I}$ comme exigée dans cette proposition, ni de prouver qu'une telle suite n'existe pas, même dans un cas particulier tel que $\mathcal{M} = (k\text{-alg. comm.})$, k un anneau commutatif. Je n'ai pas même prouvé qu'on ne puisse [page 485] trouver une suite transfinie de *corps* k_α, ayant les propriétés dites.

A fortiori, je n'ai pu tirer au clair la question si une sous-catégorie $\underline{\mathrm{L}}$-stable d'une catégorie $\mathcal{M}$ accessible est toujours accessible dans $\mathcal{M}$ ([200]). Il se pourrait qu'il n'y ait pas de tel cas. Il se pourrait aussi qu'il y en ait, même avec une sous-catégorie fermée (laquelle est stable par tout type de $\varinjlim$ représentable dans $\mathcal{M}$, sauf la $\varinjlim$ sur une catégorie d'indices vide …).

Pour les sous-catégories *ouvertes*, je rappelle que j'ai construit (dans l'appendice) un p. 188 exemple d'une telle sous-catégorie, non accessible dans $\mathcal{M}$. Au sujet des sous-catégories ouvertes, je signale ici l'énoncé (quasi tautologique) :

Proposition 4.16.7.6 ([201]). *Soient $\mathcal{M}$ une catégorie accessible, $\underline{S}$ une sous-catégorie* [page 486] ouverte *de $\mathcal{M}$. Pour que $\underline{S}$ soit accessible dans $\mathcal{M}$, il faut et il suffit qu'elle soit $\underline{L}$-stable dans $\mathcal{M}$, i.e. qu'il existe un cardinal π_0 tel que $\mathcal{M}$ satisfasse à L_{π_0}, et que $\underline{S}$ soit stable dans $\mathcal{M}$ par $\varinjlim$ de type L_{π_0}. De façon plus précise, si cette condition est satisfaite, alors pour tout cardinal π tel que $\pi \geqslant \pi_0$ et que π soit utile pour $\mathcal{M}$, $\underline{S}$ est π-accessible dans $\mathcal{M}$. Ou encore : si π est un cardinal utile pour $\mathcal{M}$, alors $\underline{S}$ est π-accessible dans $\mathcal{M}$ si et seulement s'il est L_π-stable dans $\mathcal{M}$.*

Commentary · Development (commentators') [ms. 485]

The field question detects a Vopenka universe

The question about fields has an answer depending on the chosen universe. The later example with arbitrary k-algebras does not by itself answer this narrower question. A field functor from graph theory provides the missing passage.

Universe-relative field criterion. [ours] Let $\mathfrak{U} = V_\kappa$, with κ strongly inaccessible and $\omega \in \mathfrak{U}$. There exists a sequence $(k_\alpha)_{\alpha<\kappa}$ of fields in $\mathfrak{U}$ such that

$$\mathrm{Hom}(k_\alpha, k_\beta) = \varnothing \quad (\alpha < \beta)$$

if and only if κ is not a Vopěnka cardinal. If such a sequence exists, one can choose characteristic zero throughout and make the Hom sets empty for *all* distinct indices.

Here a Vopěnka cardinal means that every κ-element family of structures in V_κ in a fixed small language contains distinct members with an elementary embedding between them. All external subsets of V_κ are allowed as families. This is stronger than merely asking V_κ to satisfy an internally definable axiom scheme. A fixed language with one binary and one unary relation suffices (Brooke-Taylor, 2011, Definitions 1–3).

First suppose κ is Vopěnka. The proposed fields are pairwise nonisomorphic. For every infinite $\mu < \kappa$, at most $2^\mu < \kappa$ isomorphism types of fields have size at most μ: put their operations on subsets of μ and count the finite-arity tables. Regularity of κ therefore lets us thin the sequence to length κ with strictly increasing field cardinalities and indices. At each stage, the previous suprema are still below κ, whereas fewer than κ indices have bounded field size. Backwards homomorphisms

199. Mais si, finalement, cf. 4.16.7.10 plus bas.

200. C'est faux, cf. plus bas.

201. Cette proposition et son corollaire sont contenus dans 4.16.7.8 ci-dessous, il n'y a pas lieu d'en faire un énoncé séparé.

in this subsequence are impossible because field homomorphisms are injective. Forward ones are excluded by hypothesis. The Vopěnka property nevertheless supplies an elementary embedding between two distinct fields, hence a field homomorphism, a contradiction.

Conversely, suppose κ is not Vopěnka. Choose κ structures $M_\alpha \in V_\kappa$ in the fixed finite relational language with no elementary embedding between distinct members. For every formula $\varphi(\bar{x})$ add the relation R_φ of its true tuples, including negated formulas and sentences. This countable relational expansion has the property that any injection preserving all its relations is elementary: R_φ preserves truth and $R_{\neg\varphi}$ preserves falsity.

We give the graph encoding explicitly, to match the precise graph morphisms needed below. Make a core graph with an element vertex v_a for every $a \in M$, a tuple vertex $t_{\varphi,\bar{a}}$ for each true $R_\varphi(\bar{a})$, and a position vertex $s_{\varphi,\bar{a},j}$ for each coordinate. The core edges are exactly

$$t_{\varphi,\bar{a}} \;-\; s_{\varphi,\bar{a},j} \;-\; v_{a_j}.$$

This core is bipartite: elements and tuple vertices form one part, positions the other. Assign one type to elements, a distinct type to each formula's tuple vertices, and a distinct type to each formula-coordinate pair. There are countably many types; enumerate them by $n \in \omega$.

At every core vertex of type n, attach two cycles of lengths $4n+5$ and $4n+7$, identifying one vertex of each cycle with that core vertex and otherwise using fresh vertices. Every odd simple cycle in the resulting graph is one of these attached cycles. A simple cycle cannot enter and leave an attached block through its sole attachment vertex without repeating that vertex; a cycle confined to the core is even.

An induced-subgraph embedding must send each tag cycle to a tag cycle of the same length. The two cycles at a source core vertex meet exactly in that vertex; their images meet in its image. As the target tag cycles of these two lengths intersect only when they are attached at the same core vertex, that image is a core vertex of the same type. Thus every core type is preserved. The image of a tuple vertex has the same formula type, and its unique position neighbor of type (φ, j) forces the corresponding coordinate element to be the image of a_j. The induced injection on elements preserves every R_φ, and is elementary. The graphs $G(M_\alpha)$ consequently have no induced-subgraph embeddings between distinct members.

Their sizes are at most $\max(|M_\alpha|, \aleph_0)$: finite tuples, countably many relation symbols, and the finite tags do not exceed that bound. Now apply Miller et al. (2018, Theorem 1.1, Section 5.A and Remark 5.1). They construct a fully faithful functor from symmetric irreflexive graphs with induced-subgraph embeddings to fields with field homomorphisms. Its values have characteristic zero and size $\max(|G|, \aleph_0)$. One detail of their proof must be read with finite support in view: the real-specialization argument in Lemma 5.3(ii) is applied to the finite-vertex subfield containing the parameters of an alleged point and the indicated vertices i, j. An arbitrarily large whole field need not embed in $\mathbb{R}$, but that finite-parameter argument proves the stated exclusion at every cardinality. Therefore

$$k_\alpha = F(G(M_\alpha))$$

belongs to $\mathfrak{U}$ and has no homomorphism to k_β for $\alpha \neq \beta$. Standard presentations of cardinality below κ may be chosen inside V_κ. This completes the proof.

In particular, such a field sequence exists for the smallest universe containing ω. Its first inaccessible height cannot be Vopěnka. Indeed the characterization by extendibility below κ gives an elementary embedding $j : V_\eta \to V_\zeta$ of critical point $\delta < \kappa$, with $\eta > \delta + \omega$ (Brooke-Taylor, 2011, Proposition 7). The derived ultrafilter

$$\{X \subseteq \delta : \delta \in j(X)\}$$

is nonprincipal and δ-complete, so δ is measurable and hence inaccessible, contradicting first inaccessibility. At a Vopěnka universe the first half of the argument excludes the sequence.

This is a deduction from established theorems, with the reduction supplied in detail and no claim of priority. The field functor is a substantial algebraic construction: its cardinal preservation and full faithfulness, rather than merely the existence of fields with few automorphisms, are what the argument needs. The manuscript's question should accordingly be recorded as dependent on the universe, rather than left as an unqualified currently open field problem.

Cette dernière assertion résulte aussitôt du corollaire suivant, lui-même tautologique.

Corollaire 4.16.7.7. *Soit π un cardinal comme indiqué ci-dessus, soit $\underline{S}(\pi) = \underline{S} \cap \mathcal{M}_\pi$, et soit $x \in \mathrm{Ob}\,\mathcal{M}$ et*

$$x \simeq \varinjlim_{i \in I} x_i \qquad \begin{cases} x_i \in \mathrm{Ob}\,\mathcal{M}_\pi \\ I \text{ grande devant } \pi \end{cases}$$

une représentation de x comme image essentielle d'un objet de $\mathrm{Ind}_\pi \mathcal{M}_\pi$. Alors $x \in \mathrm{Ob}\,\underline{S} \iff x_i \in \mathrm{Ob}\,\underline{S}(\pi)\ \forall\, i \in I$.

J'ai finalement tiré au clair la question de l'accessibilité des sous-catégories ouvertes et des sous-catégories fermées d'une catégorie accessible : [page 487]

Proposition 4.16.7.8. *Soient $\mathcal{M}$ une catégorie accessible, $\underline{S}$ une sous-catégorie fermée de $\mathcal{M}$, $\underline{T}$ la sous-catégorie ouverte complémentaire de $\underline{S}$, π un cardinal infini utile pour $\mathcal{M}$. Les conditions suivantes sont équivalentes :*

a) $\underline{S}$ est π-accessible dans $\mathcal{M}$.

a′) $\underline{S}$ est le fermé engendré par $\underline{S}(\pi) \overset{\text{déf}}{=} \underline{S} \cap \mathcal{M}_\pi$, i.e. pour $x \in \mathrm{Ob}\,\mathcal{M}$, on a $x \in \mathrm{Ob}\,\underline{S}$ si et seulement si $\exists\, \xi \in \mathrm{Ob}\,\underline{S}(\pi)$ tel que $\mathrm{Hom}_{\mathcal{M}}(\xi, x) \neq \emptyset$.

b) $\underline{T}$ est π-accessible dans $\mathcal{M}$.

b′) $\underline{T}$ est $\underline{L}_\pi$-stable dans $\mathcal{M}$.

Notons tout de suite le

Corollaire 4.16.7.9. *Pour $\mathcal{M}$, $\underline{S}$, $\underline{T}$ comme dessus, conditions équivalentes :*

a) $\underline{S}$ est accessible dans $\mathcal{M}$.

a′) Il existe une petite *partie A de $\mathrm{Ob}\,\mathcal{M}$, telle que $\underline{S}$ soit la sous-catégorie fermée ${}_A\mathcal{M}$ engendrée par A, i.e. $\underline{S} = \bigcup_{\xi \in A} {}_\xi\mathcal{M}$, où ${}_\xi\mathcal{M}$ est la sous-catégorie fermée de $\mathcal{M}$ formée des $x \in \mathrm{Ob}\,\mathcal{M}$ tel que $\mathrm{Hom}(\xi, x) \neq \emptyset$.*

b) $\underline{T}$ est accessible dans $\mathcal{M}$. [page 488]

Commentary · Exposition [ms. 487]

One canonical slice proves the open–closed criterion

Grothendieck's final criterion makes the algebraic obstruction usable. It says that the inaccessible no-point cosieve and the failure of filtered-colimit closure of the point-admitting sieve are two manifestations of one phenomenon. The proof takes place in a single category of presentable objects over a fixed target; several omitted slice denominators in the printed proof obscure this otherwise exact argument.

Fix a useful π, write $\kappa = \pi^+$, and let $\mathcal{A}$ be a small skeleton of $\mathcal{M}_\pi$. For $x \in \mathcal{M}$, the canonical category $\mathcal{A}/x$ is κ-filtered, and its tautological diagram has colimit x. If $\mathcal{S}$ is a replete cosieve and $\mathcal{T}$ its complementary sieve, put

$$J_x = (\mathcal{S} \cap \mathcal{A})/x, \qquad K_x = (\mathcal{T} \cap \mathcal{A})/x.$$

These are a cosieve and its complementary sieve in $\mathcal{A}/x$, not in $\mathcal{S}(\pi)$. A nonempty full cosieve in a filtered category is cofinal. Given an object j in the cosieve and any object a of the ambient category, a common successor belongs to the cosieve; common successors and equalizations similarly verify connectedness of the relevant comma categories. For the stronger κ-filtered assertion, first cocone any fewer-than-κ diagram in the ambient category and then move its vertex into the cosieve. In contrast, a proper complementary sieve cannot be cofinal, since an object in the cosieve has no arrow into it. Hence

$$J_x \neq \varnothing \iff J_x \text{ cofinal}, \qquad K_x \text{ cofinal} \iff J_x = \varnothing.$$

For completeness, the Ind-image criterion used here has a direct proof. If x is a κ-filtered colimit of objects of a full $\mathcal{B} \subseteq \mathcal{A}$, every map $a \to x$ with $a \in \mathcal{A}$ factors through a stage by presentability. Equalities between two factorizations hold at a later stage. Therefore the presentation functor is

cofinal in $\mathcal{A}/x$, and so is the full subcategory $\mathcal{B}/x$ through which it factors. Conversely, if $\mathcal{B}/x$ is cofinal, restriction of the canonical presentation expresses x as its colimit. A full cofinal subcategory of this κ-filtered category is again κ-filtered: cocone a small diagram in the ambient category and use fullness after moving the vertex into the subcategory. This proves Lemma 4.16.7.11 with its slice variables restored.

Apply the criterion to $\mathcal{B} = \mathcal{S} \cap \mathcal{A}$. The condition that the Ind-image be exactly $\mathcal{S}$ becomes

$$x \in \mathcal{S} \quad \Longleftrightarrow \quad J_x \neq \varnothing,$$

which says precisely that $\mathcal{S}$ is generated by its π-presentable objects. Taking complements turns this into

$$x \in \mathcal{T} \quad \Longleftrightarrow \quad K_x \text{ is cofinal in } \mathcal{A}/x,$$

the identical Ind-image criterion for $\mathcal{T}$. Finally, if $\mathcal{T}$ is closed under the relevant filtered colimits, every presentation of an object in $\mathcal{T}$ has all its stages in $\mathcal{T}$, by the sieve condition, and conversely a presentation with all stages there has colimit there by closure. This proves its accessibility and gives the equivalence with b′.

Passing to a common larger useful rank yields Corollary 4.16.7.9. This proof explains both why the theorem is valid and why it settles the appendix's abandoned question about the complementary cosieve. The only changes required in the printed argument are the missing slice or presentability subscripts, the names of the Ind functor, and the final reference to Proposition 4.16.7.6.

p. 189

b′) $\underline{T}$ est $\underline{L}$-stable dans $\mathcal{M}$.

Corollaire 4.16.7.10. *Soit $\mathcal{M}$ une catégorie accessible. Les conditions suivantes sont équivalentes ([202]).*

a) Il existe une sous-catégorie fermée de $\mathcal{M}$, non accessible dans $\mathcal{M}$.

b) Il existe une sous-catégorie ouverte dans $\mathcal{M}$, non accessible dans $\mathcal{M}$.

c) Il existe une suite transfinie

$$\underline{x} \;=\; (x_\alpha)_{\alpha \in I}$$

d'objets de $\mathcal{M}$, indexée par l'ensemble I des ordinaux de l'univers de référence $\mathfrak{U}$, telle que l'on ait

$$\mathrm{Hom}(x_\alpha, x_\beta) \;=\; \emptyset \qquad \textit{pour } \alpha < \beta.$$

En effet, l'équivalence de a) et b) est claire par le corollaire précédent, et celle de a) et c) n'est autre que 4.16.7.4.

Démonstration de 4.16.7.8. Nous aurons à utiliser le

Lemme 4.16.7.11. *Soient $\mathcal{M}$ une catégorie accessible, π un cardinal utile pour $\mathcal{M}$, $\underline{S}$ une sous-catégorie pleine de $\mathcal{M}$, $\underline{S}(\pi) \overset{\text{déf}}{=} \underline{S} \cap \mathcal{M}_\pi$, d'où un foncteur pleinement fidèle*

$$\varphi \;:\; \mathrm{Ind}_\pi(\underline{S}(\pi)) \longrightarrow \mathcal{M}\ .$$

a) Soit $x \in \mathrm{Ob}\,\mathcal{M}$. Pour que x appartienne à l'image essentielle du foncteur ψ, il faut et il suffit que la sous-catégorie pleine $\underline{S}(\pi)/x$ de la catégorie $\mathcal{M}_\pi/x$ (qui est filtrante et même grande devant π) soit cofinale (donc aussi grande devant π). [page 489]

b) Pour que $\underline{S}$ soit π-accessible dans $\mathcal{M}$, il faut et il suffit que pour tout $x \in \mathrm{Ob}\,\mathcal{M}$, on ait

$$x \;\in\; \mathrm{Ob}\,\underline{S} \qquad \Longleftrightarrow \qquad \underline{S}(\pi)/x \textit{ est cofinale dans } \mathcal{M}_\pi/x$$

(ou encore, que $\underline{S}$ satisfasse à L_π, et que pour tout x dans $\mathcal{M}$, on ait

$$x \;\in \mathrm{Ob}\,\underline{S} \qquad \Longrightarrow \qquad \underline{S}(\pi)/x \textit{ cofinale dans } \mathcal{M}_\pi/x.)$$

202. Exemple où ces conditions sont satisfaites : $\mathcal{M} = (k\text{-alg. comm.})$, k un corps commutatif infini, $\mathfrak{U}$ étant le plus petit univers tel que $\mathbf{N} \in \mathfrak{U}$.

Commentary · Examples [ms. 488]

The criterion in sets and in a pointed category

The distinction between an arbitrary full subcategory and a sieve or cosieve is restrictive enough to make the final criterion concrete. In Set, a nonempty cosieve containing $\varnothing$ is all of Set, because $\varnothing$ maps to every set. If a cosieve contains a nonempty set X, the map $X \to 1$ puts 1 in the cosieve, and the maps $1 \to Y$ put every nonempty Y there. Thus the only replete cosieves are

$$\varnothing, \qquad \mathsf{Set}_{\neq\varnothing}, \qquad \mathsf{Set}.$$

The middle one is generated by 1 and its complementary sieve consists only of $\varnothing$. The former has the usual filtered presentations by nonempty finite subsets; the latter is the terminal category and is closed under filtered colimits. This computes both sides of the criterion at the smallest relevant ranks.

In a pointed category with zero morphisms, any two objects admit a map. Hence every nonempty cosieve, and every nonempty sieve, is the whole category. For example, the no-map construction cannot give a nontrivial open–closed decomposition of abelian groups: every group has its zero map to every fixed group. This is why commutative unital k-algebras are a useful test here. Their morphisms must preserve the unit and the specified copy of k, so the absence of a map to k is a genuine algebraic condition.

Démonstration. La partie b) est une conséquence tautologique de a), que je vais prouver. Dire que x est dans l'image essentielle de $\underline{S}$, signifie que

$$(*) \qquad x \simeq \varinjlim_{i\in I} x_i \ ,$$

avec I grande devant π et les x_i dans $\underline{S}(\pi)$. On sait que $(*)$ implique (I étant filtrante, et $\mathcal{M}$ étant équivalente à $\mathrm{Ind}_\pi(\mathcal{M})$ puisque π utile pour $\mathcal{M}$) que $I \longrightarrow \mathcal{M}_\pi/x$ est cofinale (4.4.2, page 79). Or ce foncteur se factorise par le foncteur d'inclusion pleinement fidèle $\underline{S}(\pi)/x \hookrightarrow \mathcal{M}_\pi/x$, lequel est cofinal si et seulement s'il satisfait la condition F 1. Il est immédiat que cette condition est satisfaite quand elle est satisfaite pour le foncteur composé en question, ce qui est bien le cas par hypothèse. p. 190

Inversement, supposons que $\underline{S}(\pi)/x$ soit cofinale dans $\mathcal{M}_\pi/x$. Comme on a [page 490]

$$x \xleftarrow{\sim} \varinjlim_{\mathcal{M}_\pi/x} x_i \ ,$$

π étant utile pour $\mathcal{M}$, on aura aussi par cofinalité

$$x \xleftarrow{\sim} \varinjlim_{\underline{S}(\pi)/\pi} x_i \ ,$$

or $\mathcal{M}_\pi$ étant grande devant π, il en est de même de la sous-catégorie pleine cofinale $\underline{S}(\pi)/x$. Donc x est dans l'image essentielle de $\mathrm{Ind}_\pi \underline{S}(\pi)$, q.e.d.

Lemme 4.16.7.12. *Soient I une catégorie filtrante, J une sous-catégorie fermée, K la sous-catégorie ouverte complémentaire. On a alors les équivalences*

$$J \neq \emptyset \iff J \text{ cofinale} \iff K \text{ non cofinale} \iff K \neq I \ .$$

Démonstration immédiate.

Je reviens à la démonstration de 4.16.7.8. On applique le lemme 4.16.7.11 à $\mathcal{M}$, π, et aux deux sous-catégories $\underline{S}$, $\underline{T}$ de $\mathcal{M}$, en notant que pour $x \in \mathrm{Ob}\,\mathcal{M}$, $\underline{S}(\pi)/x$ est une sous-catégorie [page 491] fermée de $\underline{S}(\pi)$, dont le complémentaire ouvert est $\underline{T}(\pi)/x$, de sorte qu'on peut appliquer le

lemme 4.16.7.12. On trouve :

$$\underline{S}\ \pi\text{-accessible} \overset{4.16.7.11}{\iff} \forall\, x \in \mathrm{Ob}\,\mathcal{M},\ \text{on a } (x \in \mathrm{Ob}\,\underline{S} \iff \underline{S}(\pi)/x \text{ cofinale dans } \mathcal{M}_\pi/x)$$
$$\iff \forall\, x \in \mathrm{Ob}\,\mathcal{M},\ \text{on a } \Big(\underbrace{x \notin \mathrm{Ob}\,\underline{S}}_{\text{(i.e. } x \in \mathrm{Ob}\,\underline{T}\text{, par 4.16.7.12)}} \iff \underbrace{\underline{S}(\pi)/x \text{ non cofinale dans } \mathcal{M}_\pi/x}_{\text{(i.e. } \underline{T}(\pi)/x \text{ cofinale dans } \mathcal{M}_\pi\text{, par 4.16.7.12)}} \Big)$$
$$\overset{4.16.7.11}{\iff} \underline{T} \text{ est } \pi\text{-accessible.}$$

On a donc l'équivalence de a) et b). D'autre part, $\underline{S}(\pi)/x$ cofinale dans $\mathcal{M}_\pi/x$ signifie simplement $\underline{S}(\pi)/x \neq \emptyset$ (4.16.7.12), donc $\underline{S}$ π-accessible signifie que $\underline{S}$ est formée des $x \in \mathrm{Ob}\,\mathcal{M}$ tels que $\underline{S}(\pi) \neq \emptyset$, ce qui signifie aussi que $\underline{S}$ est le fermé dans $\mathcal{M}$ engendré par $\underline{S}(\pi)$, d'où a $\iff$ a′. D'autre part, l'équivalence b $\iff$ b′ n'est autre que 4.6.7.6 (dernière formulation).

[page 492]

4.17 Retour sur les paratopos, et sur les opérations $\mathcal{M} \boxtimes \mathcal{N}$, $\underline{\mathrm{Hom}}_!(\mathcal{M}, \mathcal{N})$

[Ce paragraphe n'existe pas.]

p. 191

[page 1]

Appendice

Ultraproduits de corps, et catégories accessibles

Soit $(k_\xi)_{\xi \in E}$ une famille de corps, indexé par un ensemble E. (Soit Φ un filtre non trivial dans E, i.e. une partie $\Phi \subset \mathfrak{P}(E)$ telle que :

a) $\emptyset \notin \Phi$, i.e. $A \in \Phi \Longrightarrow A \neq \emptyset$.

b) $A, B \in \Phi \Longrightarrow A \cap B \in \Phi$.

c) Si $A \in \Phi$, $A \subset B$, alors $B \in \Phi$.

d) $\Phi \neq \emptyset$ ($\iff E \in \Phi$).

(Cela revient à dire que Φ est une partie fermée de l'ensemble ordonné $\mathfrak{P}(E)$, *filtrante à gauche*, et ne contient pas le plus petit élément de $\mathfrak{P}(E)$, i.e. non égale à $\mathfrak{P}(E)$ tout entier.))

Pour toute partie J de E, posons

$$k_J = \prod_{\xi \in J} k_\xi ,$$

on a donc un système projectif

$$(k_J)_{J \in \mathfrak{P}(E)} , \qquad \text{i.e. } J \subset J' \text{ donne } k_{J'} \longrightarrow k_J ,$$

et le système

$$(k_J)_{J \in \Phi}$$

est un système inductif d'annaux, indexé par l'ensemble ordonné filtrant Φ^o. Posons

$$k_\Phi = \varinjlim_{J \in \Phi} k_J .$$

Si Ψ est un filtre plus fin que Φ, i.e. tel que $\Psi \supset \Phi$, on trouve un homomorphisme canonique

[page 2]

$$k_\Phi \longrightarrow k_\Psi ,$$

donc si $\mathcal{F}(E) = \mathcal{F}$ est l'ensemble des filtres sur E, ordonnés par inclusion, on trouve un système inductif

$$\Phi \longmapsto k_\Phi \ : \ \mathcal{F} \longrightarrow \mathrm{Ann} \ .$$

On n'a pas utilisé encore que les k_ξ sont des corps, en fait il suffirait que ce soient des objets d'une catégorie $\mathcal{A}$ stable par petits produis et par petits $\varinjlim$ filtrantes. Prenons maintenant $\mathcal{A} = \mathrm{Ann}$ (catégorie des anneaux unitaires, non nécessairement commutatifs), et supposons que les k_ξ soient des corps.

Proposition 1. *Supposons que les k_ξ ($\xi \in E$) soient des* corps ([203]). *Pour que k_Φ soit un corps, il faut et il suffit que Φ soit un ultrafiltre, i.e. que pour toute décomposition $E = A \amalg B$, on ait $A \in \Phi$ ou $B \in \Phi$.* (NB Si $A \in \Phi$, on a $B \notin \Phi$, puisque sinon, on aurait $A \cap B \in \Phi$, donc $\emptyset \in \Phi$, absurde.) p. 192

Nécessité : Soit, pour toute partie A de E,

$$f_A \in k_E \qquad \text{défini par} \qquad f_A(\xi) \ = \ \begin{cases} 1_{k_\xi} & \text{si} \quad \xi \in A \\ 0_{k_\xi} & \text{si} \quad \xi \in E \backslash A \ . \end{cases}$$

Soit [page 3]

$$f_{A,\Phi} \ \in k_\Phi$$

l'image de f_A par l'homomorphisme canonique $k_E \longrightarrow k_\Phi$. Si $E = A \amalg B$, on aura

$$f_A \ + \ f_B \ = \ 1 \ , \qquad f_A \, f_B \ = \ 0 \ ,$$

d'où

$$f_{A,\Phi} \ + \ f_{B,\Phi} \ = \ 1 \ , \qquad f_{A,\Phi} \, f_{B,\Phi} \ = \ 0 \ .$$

Si k_Φ est un corps, cela implique qu'on a $f_{A,\Phi} = 0$ ou $f_{B,\Phi} = 0$. Mais si p.ex. $f_{A,\Phi} = 0$, on voit que cela signifie qu'il existe un $J \in \Phi$ tel que $f_A | J = 0$, ce qui signifie aussi que $J \subset B$. De même, si $f_{B,\Phi} = 0$, on aura $J \subset A$. Donc Φ est un ultrafiltre.

Commentary · Exposition [ms. 2]

Reduced products as germs on a filter

Grothendieck turns to fields because they offer a concrete way of separating the existence of many local solutions from the existence of one compatible solution. The categorical construction comes first: a filter specifies which parts of a product may be discarded, and a filtered colimit makes those restrictions into identifications. This is the reduced product; it is called an ultraproduct only when the filter is an ultrafilter.

Write $R = \prod_{\xi \in E} k_\xi$. Every element of k_Φ has a representative in R, since a section on $J \in \Phi$ extends by zero outside J. Two representatives f, g have the same image precisely when their equalizer belongs to Φ. Consequently

$$k_\Phi \cong R/I_\Phi, \qquad I_\Phi = \{f \in R : Z(f) \in \Phi\}, \qquad Z(f) = \{\xi : f(\xi) = 0\}.$$

This formula includes the order reversal in the indexing category: shrinking J moves forward in Φ^{op}. It also proves directly that I_Φ is a two-sided ideal. Intersect two sets on which two functions vanish to handle addition; multiplying a vanishing function by an arbitrary function preserves its zero set. Properness of the filter gives $1 \notin I_\Phi$.

The proof of Proposition 1 uses exactly the dichotomy supplied by an ultrafilter. If $[g] \neq 0$, its zero set is not in Φ. For an ultrafilter the complementary support A is therefore in Φ. Define $h(\xi) = g(\xi)^{-1}$ on A, and put $h(\xi) = 1$ elsewhere. Both gh and hg agree with 1 on A, so $[h]$ is a two-sided inverse. This proof applies to division rings. In the reverse direction, the images of the complementary characteristic functions $e_A, e_{E \backslash A}$ are orthogonal idempotents summing to 1. A division ring has no idempotents other than $0, 1$, so one of the two sets belongs to the filter.

203. NB Cette condition est nécessaire si on veut que k_Φ soit un corps pour tout ultrafiltre.

Several formulas in this short proof look like failures of transcription but are already printed in the edition. The conditions involving $f \neq \emptyset$ and $g(\xi) \neq \emptyset$ concern nonzero ring elements. The representative to invert is $g(\xi)$, and the concluding identity must mean $ff' = f'f = 1$. The later sentence saying that every k_Φ is a field needs $\Phi \in \mathcal{F}_0$. For example, $E = \{0, 1\}$ and $\Phi = \{E\}$ give $k_\Phi = k \times k$. These slips are retained in the source and interpreted by the argument above.

The field assertion is also an instance of Łoś's theorem: the axioms for division rings are first-order sentences, and ultraproducts preserve them. The modern construction and the theorem, including their proofs, can be read in Keisler (2010, Definitions 2.4–2.5 and Theorem 3.1). Grothendieck's elementary proof is especially useful here because it keeps visible the characteristic functions which will classify all maps to the ground field.

Inversement, supposons que Φ soit un ultrafiltre. Prouvons que k_Φ est un corps, i.e. que pour $f \in k_\Phi$, $f \neq \emptyset \Longrightarrow f$ inversible. Notons que f provient d'un élément d'un k_J ($J \in \Phi$), donc est l'image canonique d'un $g \in k_E$. Dire que $f = 0$ signifie que $\exists\ J \in \Phi$ tel que $g|J = 0$. L'inverse $f \neq 0$ signifie donc que pour tout $J \in \Phi$, on a $g|J \neq \emptyset$.
Soit alors A l'ensemble des $\xi \in E$ tel que $g(\xi) \neq \emptyset$, $B = E\backslash A$ l'ensemble des $\xi \in E$ tels que [page 4]
$g(\xi) \neq \emptyset$. On a donc $B \notin \Phi$ puisque $f \neq 0$, et comme Φ est un ultrafiltre, on a donc $A \in \Phi$. Mais soit $g' \in k_E$ défini par $g'(\xi) = f(\xi)^{-1}$ si $\xi \in A$, $g'(\xi) = 1$ si $\xi \in B$, et soit f' l'image de g' dans k_Φ. On voit alors qu'on a

$$f\ f' \ =\ f' \ =\ f \ =\ 1\ ,$$

ce qui montre bien que k_Φ est un corps.
Supposons maintenant que tous les k_ξ ($\xi \in E$) soient égaux à un même corps k. Alors les k_J forment un système projectif de k-algèbres, les k_Φ un système inductif de k-algèbres. Si k est commutatif, il en est de même des k_J, k_Φ, et en particulier pour Φ un ultrafiltre, k_Φ est une *extension* du corps k.
Revenons au cas général, mais les k_ξ commutatifs. Soit $\mathcal{F}_0 \subset \mathcal{F}$ l'ensemble des ultrafiltres sur E. Notons que pour tout $\Phi \in \mathcal{F}$, $k_E \longrightarrow k_\Phi$ est surjectif. Donc si $\Phi \in \mathcal{F}$, k_Φ s'identifie à un corps quotient de l'anneau k_E.
Pour tout anneau $\mathcal{A}$, soit $\mathrm{Max}(\mathcal{A})$ le spectre maximal, i.e. l'ensemble des idéaux maximaux p. 193
de $\mathcal{A}$. On trouve donc une application canonique [page 5]

$$(*) \qquad \begin{cases} \mathcal{F}_0 & \longrightarrow & \mathrm{Max}(k_E) \\ \Phi & \longmapsto & \mathrm{Ker}(k_E \longrightarrow k_\Phi) \ \overset{\text{déf}}{=}\ \mathfrak{m}_\Phi\ . \end{cases}$$

Proposition 2. *L'application précédente est* bijective.
Prouvons d'abord qu'elle est injective, cela résulte aussitôt du
Corollaire 1 ([204]). *Soit, pour $\Phi \in \mathcal{F}_0$,*

$$\chi_\Phi\ :\ k_E \longrightarrow k_\Phi$$

l'application canonique. Alors pour $A \in \mathfrak{P}(E)$, on a

$$A\ \notin\ \Phi \qquad \Longleftrightarrow \qquad \chi_\Phi(f_A)\ =\ 0\ , \quad \textit{i.e. } f_A \in \mathfrak{m}_\Phi$$

(i.e. $A \in \Phi \iff \chi_\Phi(f_A) \neq 0$, i.e. $f_A \notin \mathfrak{m}_\Phi$) ([205]).
En effet, $\chi_\Phi(f_A) = 0$ signifie qu'il existe $B \in \Phi$ tel que $f_A|B = 0$, i.e. tel que $B \subset E\backslash A$; mais cela équivaut à $E\backslash A \in \Phi$, donc à $A \notin \Phi$ puisque Φ est un ultrafiltre, q.e.d.

204. **NB** Pour $\Phi \in \mathcal{F}$ quelconque, $\mathfrak{m}_\Phi \overset{\text{déf}}{=} \mathrm{Ker}(\chi_\Phi : k_E \longrightarrow k_\Phi)$ détermine Φ, puisque

$$A\ \in\ \Phi \qquad \Longleftrightarrow \qquad \chi_\Phi(f_A)\ =\ 1 \qquad \text{[page 6]}$$

(mais il faut supposer les $K_\xi \neq 0$, i.e. $0_\xi \neq 1_\xi$ ...).

205. **NB** On a donc

$$(\chi_\Phi(f_A)\ \neq\ 0) \qquad \Longleftrightarrow \qquad (\chi_\Phi(f_A)\ =\ 1)\ .$$

Reste à prouver que l'application $(*)$ est surjective. Soit donc $\mathfrak{m}$ un idéal maximal de k_E, et soit

$$\Phi \overset{\text{déf}}{=} \{A \in \mathfrak{P}(E) \mid \chi(f_A) = 1\} ,$$

où $\chi : k_E \longrightarrow k_E/\mathfrak{m} = \kappa(\mathfrak{m})$ est l'application canonique. Je dis que Φ est un ultrafiltre. En effet :

a) $\emptyset \notin \Phi$ car $f_\emptyset = 0$, d'où $\chi(f_\emptyset) = 0 \neq 1$ dans k_Φ.

b) Si $A, B \in \Phi$, on a $A \cap B \in \Phi$, car $f_{A\cap B} = f_A \cdot f_B$, d'où $\chi(f_{A\cap B}) = \chi(f_A)\chi(f_B) = 1$.

c) Si $A \subset B$ et $A \in \Phi$, alors $B \in \Phi$. En effet, on a $f_A = f_A f_B$, d'où $\chi(f_A) = \chi(f_A)\chi(f_B)$, et comme $\chi(f_A) = 1$, on trouve $\chi(f_B) = 1$.

d) On a $E \in \Phi$, i.e. $\chi(f_E) = 1$, or $f_E = 1$, donc $\chi(f_E) = 1$.

Ainsi, Φ est un filtre – on n'a utilisé que le fait que $A/\mathfrak{m}$ est un anneau $\neq 0$ (pour conclure que $\emptyset \notin \Phi$). Prouvons que si $A/\mathfrak{m}$ est intègre, Φ est un *ultrafiltre*, i.e. :

e) Si $A \notin \Phi$, alors $B \overset{\text{déf}}{=} E\backslash A \in \Phi$.

En effet, on a $f_A + f_B = 1$, $f_A f_B = 0$, d'où

ⓐ $\chi(f_A) + \chi(f_B) = 1$, et

ⓑ $\chi(f_A)\chi(f_B) = 0$. p. 194

$\chi(f_A) \neq 1$, on conclut de ⓐ $\chi(f_B) \neq 0$, d'où par ⓑ $\chi(f_A) = 0$, d'où de ⓐ $\chi(f_B) = 1$, d'où $B \in \Phi$.

Prouvons enfin que $\mathfrak{m} = \mathfrak{m}_\Phi$. Comme $\mathfrak{m}_\Phi$ est un idéal maximal, il suffit de prouver [page 7]

$$\mathfrak{m}_\Phi \subset \mathfrak{m} .$$

Or si $A \in \mathfrak{P}(E)$ est telle que $A \notin \Phi$, ou $f_A \in \mathfrak{m}_\Phi$ (cor. 1), on a aussi $f_A \in \mathfrak{m}$. Donc il suffit de prouver le

Lemme. *Soit $\Phi \in \mathcal{F}_0$. Alors $\mathfrak{m}_\Phi$ est l'idéal engendré par les f_A pour $A \in \mathfrak{P}(E)\backslash\Phi$. Plus précisément, les f_A sont dans $\mathfrak{m}_\Phi$, et tout $f \in \mathfrak{m}_\Phi$ est de la forme ff_A, avec A comme dessus.*

En effet, $f \in \mathfrak{m}_\Phi$ signifie qu'il existe $B \in \Phi$ tel que $f|B = 0$. Mais cela signifie aussi que $f = ff_A$, où $A = E\backslash A$ [plutôt $A = E\backslash B$], d'où $A \notin \Phi$.

Cela prouve la proposition 2, et en même temps :

Corollaire 2. *Tout anneau quotient intègre de $k_E = \prod_{\xi\in E} k_\xi$ est un corps.*

Notons maintenant que si $\Phi \in \mathcal{F}$ est un filtre quelconque sur E, alors k_Φ est un anneau quotient de k_E, donc

$$(*) \qquad \mathrm{Max}(k_\Phi) \hookrightarrow \underbrace{\mathrm{Max}(k_E)}_{\simeq \mathcal{F}_0} .$$

Ceci posé, on trouve :

Corollaire 3 ([206]). *L'inclusion précédente identifie* $\mathrm{Max}(k_\Phi)$ *à l'ensemble des ultrafiltres Ψ sur E qui sont* plus fins *que Φ.*

En effet, si Ψ est un tel ultrafiltre, on a le diagramme commutatif

$$(**) \qquad k_E \xrightarrow{\chi_\Phi} k_\Phi \twoheadrightarrow k_\Psi \quad , \qquad \text{(via } \chi_\Psi\text{)} \qquad [\text{page 8}]$$

ce qui montre que Ψ [plutôt $\mathfrak{m}_\Psi$] est dans l'image de $(*)$. Inversement, supposons que $\Psi \in \mathcal{F}_0$ provienne de $\mathrm{Max}(k_\Phi)$, i.e. que l'on ait $\mathfrak{m}_\Phi \subset \mathfrak{m}_\Psi$ (où $\mathfrak{m}_\Phi$ est le noyau de χ_Φ), ou encore qu'on ait une factorisation $(**)$ de χ_Ψ. Prouvons que Ψ est plus fin que Φ, i.e. que $\Psi \supset \Phi$. Soit donc $A \in \Phi$, prouvons que l'on a $A \in \Psi$. Mais $A \in \Phi$ implique [le] fait que $\chi_\Phi(f_A) = 1$, donc son image dans k_Ψ est égale à 1, donc on a $A \in \Psi$ par le corollaire 1.

206. Marche pour tout système d'anneaux $(k_\xi)_{\xi\in E}$ *non nuls*, et $\Phi, \Psi \in \mathcal{F}$, Ψ plus fin que Φ, i.e. $\Phi \subset \Psi$ si et seulement si $\mathfrak{m}_\Phi \subset \mathfrak{m}_\Psi$ (pas la peine que Ψ soit un ultrafiltre).

Commentary · Context [ms. 7]

Idempotents recover the spectrum of a product

Grothendieck needs more than the existence of some residue fields: he needs to know exactly which quotients can return the original field. The characteristic idempotents make this classification possible. They form a Boolean algebra inside the product ring, and each prime quotient must decide every Boolean alternative.

Let $R = \prod_E k_\xi$ with commutative field factors. For $f \in R$, put $f^*(\xi) = f(\xi)^{-1}$ when $f(\xi) \neq 0$, and $f^*(\xi) = 0$ otherwise. Then

$$ff^* = e_{\mathrm{supp}(f)}, \qquad f = f^2 f^*.$$

In particular R is a commutative von Neumann regular, or absolutely flat, ring. If $R \to D$ is a nonzero domain quotient, a nonzero image $\bar{f}$ satisfies $\bar{f} = \bar{f}^2 \bar{f}^*$; cancellation gives $\bar{f}\bar{f}^* = 1$. Thus every domain quotient is a field. This is a second proof of Corollary 2 which isolates the ring-theoretic reason that all prime ideals are maximal.

For an ultrafilter U, the ideal I_U is maximal by Proposition 1. Conversely, if $\mathfrak{p}$ is prime, put

$$U_\mathfrak{p} = \{A \subseteq E : e_A \notin \mathfrak{p}\}.$$

In the domain $R/\mathfrak{p}$, each idempotent is 0 or 1. The identities $e_{A\cap B} = e_A e_B$ and $e_{E\setminus A} = 1 - e_A$ show that $U_\mathfrak{p}$ is an ultrafilter. If $Z(f) \in U_\mathfrak{p}$, then $e_{\mathrm{supp}(f)} \in \mathfrak{p}$ and $f = f e_{\mathrm{supp}(f)} \in \mathfrak{p}$. Hence $I_{U_\mathfrak{p}} \subseteq \mathfrak{p}$; maximality of the former gives equality. This proves the asserted bijection without assuming what it is meant to establish about the residue field.

There is a useful topological interpretation. Under this bijection the principal open subset $D(f) \subseteq \operatorname{Spec} R$ is

$$\{U : \mathrm{supp}(f) \in U\}.$$

It is both open and closed, with complement determined by the zero set of f. Thus $\operatorname{Spec} R$ is the Stone space of the Boolean algebra $\mathcal{P}(E)$; its points are ultrafilters and its principal points are the coordinates of E. This is the Boolean representation viewpoint originating with Stone (1936). The proof just given supplies the particular identification used here; the reference is historical, not a substitute for the proof.

For a proper filter Φ, passing from R to R/I_Φ restricts this space to

$$\{U : U \supseteq \Phi\} = \bigcap_{A\in\Phi} \{U : A \in U\}.$$

Indeed $I_\Phi \subseteq I_U$ if and only if $\Phi \subseteq U$, as one sees by applying ideal inclusion to $1 - e_A$. This also explains the scope of footnote 206. The order-recovery argument works for arbitrary nonzero coefficient rings. The assertion that all prime quotients are fields uses the field hypothesis and is not part of that generalization.

Supposons à présent que tous les k_ξ sont égaux [à] un même anneau $\boxed{k \neq 0}$. Donc k_J, k_Φ sont des 'anneaux sous k' $k \longrightarrow k_J \longrightarrow k_\Phi$, d'ailleurs $k \longrightarrow k_J$ (si $J \neq \emptyset$) et $k \longrightarrow k_\Phi$ sont injectifs. On se pose la question quand $k \longrightarrow k_\Phi$ est *bijectif*, i.e. surjectif. Comme $k_\Phi \simeq k_E/\mathfrak{m}_\Phi$, cela signifie donc que $k_E = k + \mathfrak{m}_\Phi$, i.e. que tout $f \in k_E$ s'écrit p. 195 [page 9]

$$f \;=\; c \;+\; g \qquad \begin{cases} c \;\in\; k\ , & \text{i.e. } c(\xi) = \underbrace{c_0}_{\in k}\ \forall\, \xi \in E\ , \\ g \;\in\; \mathfrak{m}_\Phi & \text{i.e. } \exists\, A \in \Phi \text{ avec } g|A = 0\ . \end{cases}$$

Mais pour $f \in k_E$, $c \in k$ et $A \in \Phi$ fixés, l'existence de $g \in k_E$ telle que $g|A = 0$ et $f = c + g$ équivaut à $f|A = c_A$ (fonction constante dans A, de valeur c), i.e. $A \subset f^{-1}(\{c\})$. Pour f, c fixés, l'existence de $A \in \Phi$ satisfaisant la relation équivaut à $f^{-1}(\{c\}) \in \Phi$. Et pour f fixé, l'existence de c équivaut à celle d'une fibre E_c de $f : E \longrightarrow k$, qui soit dans Φ. Ainsi :

Proposition 3. *Soit $\Phi \in \mathcal{F}$. Pour que $k \longrightarrow k_\Phi$ soit surjectif, il faut et il suffit que pour toute famille de parties* essentiellement disjointes *$(E_\xi)_{\xi\in k}$ de E indexée par k, de réunion E, il*

existe un $\xi \in k$ tel que $E_\xi \in \Phi$. A fortiori, il faut *que Φ soit un ultrafiltre (puisque* $\operatorname{card} k \geqslant 2$ *par hypothèse).*

En effet, la donnée d'une telle famille $(E_\xi)_{\xi\in k}$ équivaut à la donnée d'une application [page 10]
$E \longrightarrow k$, i.e. d'un élément de f_E.

Scholie. La propriété qu'on vient d'expliciter pour un filtre Φ sur E, ne dépend de k que par l'intermédiaire de $\pi = \operatorname{card} k$. On peut aussi l'énoncer en disant que pour toute *partition* $(E_i)_{i\in I}$ de E en un ensemble I des parties tel que $\operatorname{card} I \leqslant \pi$, il existe un (et un seul) des E_i qui soit dans Φ. On dit alors que Φ est un *π-ultrafiltre.*

Si $\pi \geqslant 2$ est fini, cette propriété équivaut à dire que Φ est un ultrafiltre. Par ailleurs, quel que soit π, les ultrafiltres Φ *triviaux* (i.e. tels qu'il existe $\xi \in E$ avec $\{\xi\} \in \Phi$, ou encore tel que $\Phi = \{A \in \mathfrak{P}(E) \mid \xi \in A\}$) sont des π-ultrafiltres. Si π est infini, il n'est pas clair qu'il existe des π-ultrafiltres non triviaux. On sait qu'il n'en existe sur E *que* si $\operatorname{card} E \notin \mathfrak{U}_1$, où $\mathfrak{U}_1$ est le plus petit univers tel que $\mathbf{N} \in \mathfrak{U}_1$, i.e. tel que $\mathfrak{U}$ contienne comme élément un ensemble [page 11]
infini.

Commentary · Exposition [ms. 10]

The successor hidden in a pi-ultrafilter

The ultrapower becomes trivial precisely when every function to the coefficient ring is almost constant. Grothendieck packages this condition as a property of the ultrafilter alone, so that the size of the field can later be chosen to exclude it. A small shift in cardinal convention is essential in comparing his terminology with modern set theory.

For an infinite cardinal π, his π-ultrafilter is an ultrafilter closed under intersections of *at most* π sets. With the usual convention that κ-complete means closed under intersections of fewer than κ sets, it is therefore a π^+-complete ultrafilter. To verify this equivalence, suppose first that the intersection of a family $(A_i)_{i<\delta}$ in U, with $\delta \leq \pi$, is not in U. Partition E into the first failures

$$(E \setminus A_i) \cap \bigcap_{j<i} A_j \quad (i < \delta), \qquad \text{and} \qquad \bigcap_{i<\delta} A_i.$$

Every cell lies outside U, contradicting the partition condition. Conversely, if a partition has at most π cells and none belongs to U, all their complements belong to U; completeness would put the empty intersection of the complements in U. This proves both implications, including the case of singular π.

Let $\theta = |k|$. Proposition 3 says exactly that

$$k \longrightarrow k^E/U \text{ is onto} \quad \Longleftrightarrow \quad U \text{ selects a fibre of every map } E \longrightarrow k.$$

For infinite k this is θ^+-completeness. For finite nonzero k every ultrafilter has the property, because a finite partition always has one selected cell. Consequently every ultrapower of a finite field is canonically the field itself, while a nonprincipal ultrapower of an infinite field can be a proper extension. The elementary embedding supplied by Łoś's theorem does not decide surjectivity; compare Keisler (2010, Corollary 3.3).

A concrete infinite example keeps the two questions apart. Take $k = \mathbb{Q}$, $E = \mathbb{N}$, and a nonprincipal ultrafilter U. The class $t = [n \mapsto n]$ differs from every constant, because each fibre of the function $n \mapsto n$ is finite. More strongly, for every nonzero polynomial $P \in \mathbb{Q}[X]$, the zero set of $P(n)$ is finite, so $P(t) \neq 0$. Thus $\mathbb{Q}^{\mathbb{N}}/U$ contains a transcendental element. It is an elementary extension of $\mathbb{Q}$, but it admits no $\mathbb{Q}$-algebra map back to $\mathbb{Q}$: any field homomorphism would be injective and would have to send the transcendental t to an element of $\mathbb{Q}$.

The later filter-extension terminology used by Brooke-Taylor and Rosický (2017, Definition 2.3) follows the fewer-than convention. We will keep θ^+ visible when moving from Grothendieck's partition condition to those compactness principles.

Si $\mathfrak{U} = \mathfrak{U}_1$, et si k est un corps infini, p.ex. $k = \mathbf{Q}$, alors pour tout ensemble $E \in \mathfrak{U}$ et pour tout ultrafiltre *non trivial* Φ sur E, k_E est une extension *non triviale* de k. Mais si $\mathfrak{U} = \mathfrak{U}_\alpha$ est un univers quelconque, et π le premier cardinal dans $\mathfrak{U}$ qui n'appartienne pas à un univers

$\mathfrak{U}' \subset \mathfrak{U}$ strictement plus petit, je présume que pour tout $E \in \mathfrak{U}$, tout π-ultrafiltre sur E est trivial.

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Application à un exemple d'une sous-catégorie non accessible.

Soit $\mathcal{M}$ une catégorie $\mathfrak{U}$-accessible, et $a \in \mathrm{Ob}\,\mathcal{M}$, d'où un foncteur

$$\begin{aligned} f^o \ : \ \mathcal{M}^o \ &\longrightarrow \ \mathrm{Ens} \\ x \ &\longmapsto \ \mathrm{Hom}_{\mathcal{M}}(x,a) \overset{\text{déf}}{=} a(x) \ , \end{aligned}$$

foncteur commutant aux $\varprojlim$ quelconques, donc tel que $f : \mathcal{M} \longrightarrow (\mathrm{Ens})^o$ commute aux $\varinjlim$ quelconques, et en particulier f est accessible. Considérons la sous-catégorie strictement pleine

[page 12]

$$\underbrace{\mathrm{Ens}^*}_{=\mathrm{Ens}\setminus\{\emptyset\}} \ \subset \ \mathrm{Ens}$$

des ensembles non vides, on sait qu'elle est accessible, mais elle n'est par stable par $\varprojlim$ de type L_π, quelque grand qu'on prenne π.

Commentary · Present state [ms. 11]

The universe hypothesis and its precise boundary

Grothendieck's first universe is chosen to exclude complete nonprincipal ultrafilters. This is a mathematical restriction on the available cardinals, not a harmless change in the notation for smallness. The link with measurable cardinals can be established directly, and it also clarifies the conjecture at the end of the page.

Suppose that U is a nonprincipal π^+-complete ultrafilter on E, with π infinite. Let μ be the least cardinal for which E has a partition into μ sets, none belonging to U. Such a partition exists, by singletons, and $\mu > \pi$. Pushing U forward along this partition gives an ultrafilter V on μ. It contains no singleton. Minimality of μ shows that U, and hence V, is closed under intersections of fewer than μ members: a failed intersection would yield a partition into fewer than μ null pieces by the first-failure construction. Thus V is a nonprincipal μ-complete ultrafilter, and μ is measurable.

For completeness, the inaccessibility needed here follows from the same argument. Every subset of μ of size less than μ is V-null. If μ were singular, a partition into $\mathrm{cf}(\mu)$ bounded pieces would contradict completeness. If $\mu \le 2^\lambda$ for some $\lambda < \mu$, assign distinct subsets of λ to the elements of μ. For each coordinate of λ, V chooses one of the two membership alternatives. The intersection of these fewer than μ chosen sets belongs to V, but contains at most one element, another contradiction. Hence μ is regular and a strong limit. This is the classical measure obstruction associated with Ulam (1930); compare the complete-ultrafilter discussion in Keisler (2010, Section 5). The argument is included because the universe boundary is decisive for the example.

If $\mathfrak{U} = V_\Omega$ is the least Grothendieck universe containing an infinite set as an element, Ω is the first inaccessible cardinal. A set $E \in \mathfrak{U}$ has cardinality less than Ω, whereas the construction above would produce an inaccessible $\mu \le |E|$. Therefore every such complete ultrafilter is principal. More generally, the hypothesis of Proposition 4 holds in V_Ω exactly when the measurable cardinals below Ω are bounded. In one direction an unbounded measurable cardinal supplies a sufficiently complete nonprincipal measure. In the other, a nonprincipal π_0^+-complete ultrafilter would produce a measurable cardinal above the chosen bound π_0.

The final conjecture on this page has a positive answer *when its proposed cardinal exists.* Suppose π belongs to no strictly smaller infinite universe inside $\mathfrak{U}$. A nonprincipal π-ultrafilter would produce a measurable $\mu > \pi$ in $\mathfrak{U}$. Then V_μ is a smaller universe containing π, which is impossible. There need not, however, be a first cardinal with the stated property: when the smaller inaccessible cardinals are cofinal in Ω, every cardinal below Ω belongs to a smaller universe. This separates the valid conditional assertion from an unstated existence assumption.

No conclusion about all universes follows from the first-universe example. A universe containing a sufficiently large strongly compact cardinal permits the opposite compactness behavior, as the final commentary block will show. The elementary algebraic construction remains the same; the supply of ultrafilters changes.

Considérons

$$\underbrace{\underline{S} \;=\; \underline{S}_a}_{\overset{\text{déf}}{=} f^{-1}(\mathrm{Ens}^*)} \;\subset\; \mathcal{M}\,,$$

sous-catégorie strictement pleine de $\mathcal{M}$ formée des $x \in \mathrm{Ob}\,\mathcal{M}$ tels que

$$a(x) \;\neq\; \emptyset\,, \qquad \text{i.e. } x \leqslant a \text{ pour le préordre canonique de } \mathcal{M}\,.$$

C'est donc une sous-catégorie *ouverte* de $\mathcal{M}$. Son complémentaire fermé $\underline{S}' = \underline{S}'_a$ est la sous-catégorie fermée image inverse, par f, de la sous-catégorie strictement pleine de Ens réduite au seul objet $\{\emptyset\}$. Cette sous-catégorie de Ens est la catégorie finale type. Elle est stable dans Ens par $\varinjlim$ quelconques, et par $\varprojlim$ filtrantes quelconques. Ainsi, $\underline{S}'$ est stable dans $\mathcal{M}$ par $\varinjlim$ filtrantes quelconques. Ainsi, $\underline{S}'$ est stable dans $\mathcal{M}$ par $\varinjlim$ filtrantes quelconques, en particulier, elle est L_π-stable pour π assez grand. Explicitons : [page 13]

$$\begin{array}{rl} \underbrace{\underline{S}' \;=\; \underline{S}'_a}\;\subset\;\mathcal{M} & \\ \Big\| \text{ déf} & \text{sous-catégorie fermée de } \mathcal{M} \\ f^{-1}(\{\emptyset\}) \;= & \text{formée des } x \in \mathrm{Ob}\,\mathcal{M} \text{ tels} \\ & \text{que } a(x) = \emptyset. \end{array}$$

Commentary · Exposition [ms. 12]

Existence of points is the missing compatibility condition

Grothendieck is testing a tempting inference: a representable functor behaves perfectly with respect to limits, so perhaps the property that its values are nonempty should also behave well. The obstruction is that an inverse limit asks for compatible elements. Separate nonemptiness supplies no such compatibility.

For any diagram with colimit A, the universal property gives

$$\mathrm{Hom}_{\mathcal{M}}(A, a) \cong \varprojlim_i \mathrm{Hom}_{\mathcal{M}}(A_i, a).$$

The full subcategory of objects admitting a map to a is therefore a sieve: if $B \to A$ and $A \to a$ exist, so does $B \to a$. Its complement is a cosieve. A nonempty colimit of objects of the complement cannot admit a map to a, since composing with even one structure map would contradict membership of that stage. Dually, a nonempty limit of objects in the sieve lies in the sieve, by composing any projection with a map to a. These observations prove the closure and retract assertions without choosing a compatible family of points.

Even highly directed inverse systems of nonempty sets can have empty limit. Given an infinite π, choose $|E| > \pi$ and let $I = [E]^{\leq \pi}$, ordered by inclusion. It is π^+-directed. The inverse system $X_F = E \setminus F$, with inclusion maps for $F \subseteq G$, has nonempty terms, while

$$\varprojlim_{F \in I} X_F = \bigcap_{F \in I} (E \setminus F) = \varnothing.$$

For the categorical counterexample, Grothendieck realizes the same disappearance of coordinates inside products of fields. Complete nonprincipal ultrafilters would create additional points not given by coordinates; the universe hypothesis removes precisely those additional points.

The statement that the contravariant representable is accessible needs its original convention. It sends existing colimits in $\mathcal{M}$ to limits of sets, or equivalently preserves them as a functor to $\mathrm{Set}^{\mathrm{op}}$. This does not assert that $\mathrm{Set}^{\mathrm{op}}$ is an accessible category in the modern sense. The example concerns the sieve of existence of a point, not a failure of representability to preserve its appropriate limits.

Comme $\underline{S}'$, étant fermée dans $\mathcal{M}$, est stable par $\varinjlim$ non vides et en particulier L_π-stable pour π grand, il y a plus de raison de s'attendre à ce que $\underline{S}'$ soit accessible dans $\mathcal{M}$ que

pour son complémentaire $\underline{S}$, dont il n'est pas clair qu'il ait cette propriété de stabilité, mais seulement stabilité par $\varprojlim$ non vides, et en particulier par facteurs directs.

En fait, nous allons voir qu'il peut arriver que $\underline{S}$ et $\underline{S}'$ soient tous deux non accessibles dans $\mathcal{M}$:

Proposition 4. *Soient $\mathfrak{U}$ un univers, contenant un cardinal infini π_0 tel que pour tout $E \in \mathfrak{U}$, tout π_0-ultrafiltre sur E soit trivial. Soit alors $k \in \mathfrak{U}$ un corps de cardinal $\pi_0' \geqslant \pi_0$, et soit $\mathcal{M}$ la catégorie des k-algèbres commutatives. Donc $\mathcal{M}$ est un $\mathfrak{U}$-paratopos, admettant k* p. 197
comme objet initial. Soit $\underline{S}'$ la sous-catégorie strictement pleine, stable par petites $\varinjlim$ non [page 14]
vides, formée des objets A de $\mathcal{M}$ tels que $\mathrm{Hom}_{\mathcal{M}}(A,k) = \emptyset$, et $\underline{S}$ la sous-catégorie strictement pleine complémentaire, formée des objets A tels que $\mathrm{Hom}_{\mathcal{M}}(A,k) = \emptyset$. Alors $\underline{S}'$ est une sous-catégorie fermée *dans $\mathcal{M}$, et $\underline{S}$ la sous-catégorie ouverte complémentaire, donc $\underline{S}$ (tout comme $\underline{S}'$) est stable par facteurs directs. Ceci posé :*

a) Il n'existe pas de cardinal π tel que $\underline{S}$ soit L_π-stable dans $\mathcal{M}$, a fortiori $\underline{S}$ n'est pas accessible dans $\mathcal{M}$.

b) ([207]). Bien que $\underline{S}'$ soit L_0-stable dans $\mathcal{M}$ (i.e. stable par $\varinjlim$ filtrantes, et même, on l'a dit, par toute $\varinjlim$ non vide), $\underline{S}'$ n'est pourtant pas accessible dans $\mathcal{M}$.

Démonstration. L'assertion a) va résulter du

Corollaire 1. *Sous les conditions de la proposition 4, soient $E \in \mathfrak{U}$ un ensemble infini, Φ_0 le filtre des complémentaires des parties finies de E. Alors $\mathrm{Hom}_{\mathcal{M}}(k_{\Phi_0}, k) = \emptyset$, a fortiori, pour tout filtre Φ sur E plus fin que Φ_0, on a $\mathrm{Hom}_{\mathcal{M}}(k_\Phi, k) = \emptyset$.*

On a posé ici, bien sûr, pour tout filtre Φ sur E, [page 15]

$$k_\Phi \;=\; \varinjlim_{J \in \Phi} k_J \ .$$

On sait que pour toute k-algèbre commutative $\mathcal{A}$, $\mathrm{Hom}(\mathcal{A}, k)$ est en correspondance 1-1 avec la partie de $\mathrm{Max}(\mathcal{A})$ formée des idéaux maximaux $\mathfrak{m}$ de $\mathcal{A}$ tels que $k \longrightarrow \mathcal{A}/\mathfrak{m}$ soit un isomorphisme. Si $\mathcal{A}$ est de la forme k_Φ, Φ un filtre sur E, cet ensemble est en correspondance biunivoque avec l'ensemble des π_0'-ultrafiltres Ψ sur E plus fins que Φ, où $\pi_0' = \mathrm{card}(k)$ (proposition 3, et cor. 3 de prop. 2, page 7). Comme $\pi_0' \geqslant \pi_0$, un tel ultrafiltre est trivial par hypothèse. Donc on trouve

$$(*) \qquad \mathrm{Hom}_{\mathcal{M}}(k_\Phi, k) \;\simeq\; \text{ensemble des } \xi \in E \text{ tels que } \{\xi\} = \textstyle\bigcap_{A \in \Phi} A \ .$$

Mais dire que Φ est plus fin que Φ_0 signifie aussi que pour tout $\xi \in E$, on a $E \backslash \{\xi\} \in \Phi$, i.e. $\exists\, A \in \Phi$ avec $A \subset E \backslash \{\xi\}$, i.e. $\{\xi\} \notin \bigcap_{A \in \Phi} A$. Donc on a alors $\mathrm{Hom}_{\mathcal{M}}(k_\Phi, k) = \emptyset$.

La conclusion du corollaire 1 s'exprime par la relation

$$k_\Phi \;\notin\; \mathrm{Ob}\,\underline{S} \ , \qquad \text{i.e. } k_\Phi \in \mathrm{Ob}\,\underline{S}' \ ,$$

d'autre part, on a par définition

$$k_\Phi \;=\; \varinjlim_{J \in \Phi} k_J \ ,$$

et il est clair (comme ces $J \in \Phi$ sont $\neq \emptyset$), que les k_J en question sont dans $\underline{S}$. Ainsi, pour un filtre Φ comme dans le corollaire 1, on trouve un système inductif d'objets de $\underline{S}$, indexé [page 16]
par Φ^o, dont la limite inductive n'est pas dans $\underline{S}$. Donc le a) de la proposition 4 se déduit en notant que pour tout cardinal $\pi \in \mathfrak{U}$, on peut trouver un ensemble $E \in \mathfrak{U}$, et un filtre Φ sur E plus fin que Φ_0, tel que l'ensemble ordonné Φ^o soit grand devant π. Pour trouver un tel Φ, il suffit de prendre E de cardinal $\pi' > \pi$, et pour Φ l'ensemble des complémentaires des parties de E de cardinal $\leqslant \pi$, de sorte que Φ^o est isomorphe à l'ensemble ordonné des parties de E de cardinal $\leqslant \pi$, lequel est bien grand devant π. Cela prouve le a) de la proposition 4. p. 198

207. ?, pas prouvé.

Prouvons maintenant le b). Notons le

Lemme. *Soit $\mathcal{M}$ une catégorie accessible stable par $\varinjlim$ de type I_α, α un ordinal limite donné, $\underline{S}'$ une sous-catégorie fermée, π un cardinal utile pour $\mathcal{M}$, tel que* $\operatorname{card}\alpha \leqslant \pi$. *Pour* [page 17] *tout $x \in \operatorname{Ob}\underline{S}'$, choisissons une représentation ω-π-adaptée (ω étant le premier ordinal infini)*

$$x \simeq \varinjlim_{i\in I_x} x_i$$

(donc I_x grande devant π, stable par $\varinjlim$ de type I_α, $i \longmapsto x_i : I \longrightarrow \mathcal{M}$ commute auxdites limites, enfin les $x_i \in \operatorname{Ob}\mathcal{M}_\pi$). Pour que $\underline{S}'$ soit π-accessible dans $\mathcal{M}$, il faut et il suffit que pour tout $x \in \operatorname{Ob}\underline{S}'$, il existe un $i \in I_x$ tel que $x_i \in \operatorname{Ob}\underline{S}'$.

Commentary · Examples [ms. 16]

A directed family of products loses every rational point

The proof of Proposition 4(a) becomes quite explicit once the filter is written in terms of the coordinates it allows us to discard. Fix an infinite π, choose E of cardinality greater than π, and set

$$\Phi_\pi = \{E \setminus F : F \subseteq E,\ |F| \leq \pi\}, \qquad A_\pi = k^E/I_\pi, \quad I_\pi = \{f : |\operatorname{supp}(f)| \leq \pi\}.$$

The set Φ_π is a proper filter, and its opposite order is π^+-directed: the union of at most π exceptional subsets still has size at most π. Its reduced product is exactly A_π. Restriction of functions gives

$$A_\pi \cong \varinjlim_{F\in[E]^{\leq\pi}} k^{E\setminus F}.$$

Every stage has a k-point, evaluation at any surviving coordinate.

Now let $\theta = |k|$, and assume the hypothesis of Proposition 4. A k-algebra map $A_\pi \to k$ is necessarily surjective, since it fixes all constants. Its kernel therefore gives a maximal ideal, hence an ultrafilter U extending Φ_π. The residue field is the ultrapower k^E/U, and the structural map from k to this field must be onto. Proposition 3 says that U is θ^+-complete. Since $\theta \geq \pi_0$, the universe hypothesis makes U principal. But Φ_π contains $E \setminus \{\xi\}$ for every ξ, so no principal ultrafilter extends it. This contradiction proves that A_π has no k-point.

The argument proves failure of closure for every allowed π; it does not merely show a failure of ordinary filtered-colimit closure. This is why it rules out accessibility of the inclusion at every eventual rank. It also identifies the exact role of the field hypothesis: a map to k has a field quotient, whose maximal ideal can be classified by an ultrafilter.

The formula $(*)$ on manuscript page 15 should be interpreted as

$$\operatorname{Hom}_{k\text{-alg}}(k_\Phi, k) \cong \bigcap_{A\in\Phi} A$$

under the principal-ultrafilter hypothesis. Each member of the intersection gives an evaluation map. The printed condition $\{\xi\} = \bigcap\Phi$ would fail already for $\Phi = \{E\}$ with two coordinates. The corrected membership condition, rather than singleton equality, is exactly what the proof uses.

En effet, notons que $\underline{S}'$ étant fermée, est stable dans $\mathcal{M}$ par toute $\varinjlim$ non vide représentable dans $\mathcal{M}$, et en particulier par $\varinjlim$ de type I_α. Donc on peut appliquer 4.13.22 (p. 342) à $\mathcal{M}$, $\underline{S}'$, π et à $\Phi = \{I_\omega\}$. Alors a) $\Longrightarrow$ c) implique que si $\underline{S}'$ est π-accessible dans $\mathcal{M}$, alors pour tout $x \in \operatorname{Ob}\underline{S}'$, l'ensemble $I_{x,S'}$ des $i \in I_x$ tels que $x_i \in \operatorname{Ob}\underline{S}$ est cofinal dans I_x. Cela implique qu'il est non vide. (Et réciproquement d'ailleurs, $\underline{S}'$ étant un fermé dans $\mathcal{M}$, puisque si $x_i \in \underline{S}'$, alors pour $j \in \operatorname{Ob} I_x$ avec $i \leqslant j$, on a $x_j \in \underline{S}'$, i.e. $I_{x,\underline{S}'}$ contient l'image essentielle de $i\backslash I_x$, laquelle est cofinale dans I …) Cela prouve le 'il faut'. Prouvons le 'il suffit'. Donc prouvons que $\underline{S}'$ est égale à l'image essentielle de $\operatorname{Ind}_\pi(\underline{S}'(\pi))$. Elle la contient, puisque $\underline{S}'$ est stable par $\varinjlim$ de type L_π.

Donc il reste à prouver que tout $x \in \operatorname{Ob}\underline{S}'$ est dans l'image essentielle. Or l'hypothèse [page 18] nous assure que $x \simeq \varinjlim_{i\in I_x} x_i$ est isomorphe à $\varinjlim_{i\in I_{x,\underline{S}'}} x_i$ (car $I_{x,\underline{S}'}$ est cofinale dans I_x), où cette

fois les x_i sont dans $\underline{S}'$. Comme I_x grande devant π, de même la sous-catégorie pleine cofinale $I_{x,\underline{S}'}$, donc on gagne.

Corollaire. *Sous les conditions précédentes, si $\underline{S}'$ est π-accessible, il en est de même de l'ouvert complémentaire $\underline{S}$,* pourvu *que $\underline{S}$ soit stable dans $\mathcal{M}$ par $\varinjlim$ de type L_π.*

Il faut prouver que $\underline{S}$ est égale à l'image essentielle de $\mathrm{Ind}_\pi \underline{S}(\pi)$. L'hypothèse 'pourvu' assure que cette image essentielle est $\subset \underline{S}$. Inversement, tout [?] $x \in \mathrm{Ob}\,\underline{S}$ est dans cette image essentielle. On aura

$$x \simeq \varinjlim_{i\in I} x_i\ , \qquad x_i \in \mathrm{Ob}\,\mathcal{M}_\pi,\ I = \mathcal{M}_\pi/x,\ \text{grande devant } \pi\ ,$$

et comme $x \in \mathrm{Ob}\,\underline{S}$ et $\underline{S}$ ouverte, les x_i sont dans $\underline{S}$, donc dans $\underline{S}(\pi)$, OK.

Dans le cas qui nous occupe, où $\mathcal{M} = (k\text{-alg.})$, comme $\underline{S}$ n'est justement *pas* stable dans $\mathcal{M}$ par L_π (quel que soit π) (c'est cela qui nous faisait conclure que $\underline{S}$ n'est pas accessible dans $\mathcal{M}$), on ne peut donc conclure de cette inaccessibilité de $\underline{S}$, celle de $\underline{S}'$. Notons le [page 19]

Commentary · Present state [ms. 18]

The main text completes the abandoned argument

Grothendieck distinguishes the two possible failures with care. The open class of algebras with points fails a closure condition. The closed class of algebras without points already has every nonempty colimit, so its possible failure of accessibility must concern the absence of a small family of generators. The appendix reaches this distinction but does not finish the second argument; footnote 207 records that fact.

The main text has already supplied the missing implication. Proposition 4.16.7.8 and Corollary 4.16.7.9 identify accessibility of a cosieve with accessibility of its complementary sieve. Hence Proposition 4(a), under its stated universe hypothesis, proves Proposition 4(b) as well. The final question in the appendix must be read with this later result in view; it is not an unresolved assertion under those hypotheses.

Here is the mechanism, in the notation needed for the last page. Let $\pi \geq |k|$ be infinite. A k-algebra of cardinality at most π has a presentation with at most π generators and relations: take one generator for each element and record its addition, multiplication and scalar relations. It is consequently π^+-presentable. Every algebra is the π^+-directed union of its subalgebras of cardinality at most π.

If the closed class S' were generated by its ambient π^+-presentable objects, an object $A \in S'$ would receive a map from one such object $B \in S'$. That map factors through a sufficiently large stage of any π^+-filtered presentation of A. Since S' is a cosieve, that stage lies in S'. Conversely, one stage in S' makes the full subcategory of S'-stages a nonempty cosieve in the filtered indexing category, hence cofinal; these stages then present A inside S'. This proves the appendix's criterion without its additional ordinal-continuity apparatus.

The corollary on manuscript page 18 does not perform this argument: its explicit proviso assumes closure of S, and that closure is known to fail. Grothendieck correctly refuses the inference based on that corollary alone. The stronger equivalence in the main text is what repairs the proof. The next block will exhibit the missing small-generator obstruction directly in the very reduced products he proposes.

Corollaire. *Soit π un cardinal infini. Pour que la sous-catégorie fermée $\underline{S}'$ de $\mathcal{M}$ (formée de k-algèbres commutatives A telles que $\mathrm{Hom}_{\mathcal{M}}(A,k) = \emptyset$) soit π-accessible, il faut et il suffit que pour tout k-algèbre A (de cardinal $> \pi$), telle que $\mathrm{Hom}_{\mathcal{M}}(A,k) = \emptyset$, existe une sous-algèbre A_i* de cardinal $\leqslant \pi$, *telle que $\mathrm{Hom}_{\mathcal{M}}(A_i,k) = \emptyset$.* p. 199

(On applique le lemme, en prenant pour α le premier ordinal infini.)

Question. *Cela est-il vérifié, en prenant simplement $\pi \geqslant \mathrm{card}\,k$?* La question de l'existence d'un cardinal π, ayant la propriété indiquée plus haut, me paraît délicate, et j'ignore la réponse. Si on veut construire un contre-exemple pour tout π, les développements précédents suggèrent de prendre pour A un produit infini d'extensions non triviales de k, ou un anneau k_Φ comme dans la démonstration de la proposition 4 a).

Commentary · Development (commentators') [ms. 19]

The proposed reduced products answer every bound

Grothendieck's final suggestion is especially effective because the reduced product can have many local evaluations even when it has no global point. The following calculation realizes that suggestion under the exact hypothesis of Proposition 4. We did not find this explicit small-subalgebra calculation in the sources checked here; its conclusion about nonaccessibility already follows from his own Corollary 4.16.7.9.

Small subalgebras of a co-small reduced power. [ours] Let k be an infinite field with $|k| \geq \pi_0$ in the universe of Proposition 4, and let $\pi \geq |k|$ be an infinite cardinal in that universe. Take $E = \pi^+$ and

$$A = k^E/I_\pi, \qquad I_\pi = \{f \in k^E : |\operatorname{supp}(f)| \leq \pi\}.$$

Then A has no k-point, but every k-subalgebra $B \subseteq A$ of cardinality at most π has a k-point.

Proof. The preceding product construction proves that A has no point. The ideal is proper because $|E| > \pi$, and it is an ideal because finite unions of supports of size at most π again have that size.

To construct a point on B, choose a representative $g_b \in k^E$ of each $b \in B$. For the embedded constants $c \in k$, choose the constant representative $g_c(\xi) = c$; constants are distinct in A, so these choices are consistent. Since addition and multiplication hold in the quotient, each of

$$g_{b+b'} - g_b - g_{b'}, \qquad g_{bb'} - g_b g_{b'} \quad (b, b' \in B)$$

has support of cardinality at most π. There are at most $|B|^2 \leq \pi$ such exceptional supports. Their union F has cardinality at most π. Choose $\xi \in E \setminus F$. Then

$$h : B \longrightarrow k, \qquad h(b) = g_b(\xi)$$

preserves addition and multiplication, and it fixes every constant. In particular $h(1) = 1$, and h is a k-algebra homomorphism. No compatibility between the initially chosen representatives was required; discarding the common exceptional set makes them compatible at one coordinate. This proves the assertion. □

The poset of these subalgebras is π^+-directed: the algebra generated by the union of at most π of them still has cardinality at most π. Their union is A. Thus the same algebra is both a directed colimit of products with points and a directed union of small subalgebras with points. The first presentation defeats closure of S; the second defeats the small-obstruction criterion for S'.

In particular, under Proposition 4's hypothesis the answer to the final question is negative for *every* cardinal bound, not only for $\pi = |k|$. This establishes the statement marked "pas prouvé" in footnote 207 by a direct argument. The conditional nature of the conclusion is retained: the proof that A has no point uses the absence of sufficiently complete nonprincipal ultrafilters. The local evaluation argument itself needs no such hypothesis.

Commentary · Development (commentators') [ms. 19]

The proposed first threshold fails without a universe hypothesis

The previous example depends on the universe only when excluding a global point. There is a different obstruction at the proposed smallest threshold which works for every infinite field and in any ambient universe containing the construction. It simply asks a field to accommodate more distinct elements than it has.

Pairwise distinct variables. [ours] Let k be an infinite field, $\theta = |k|$, and let J have cardinality θ^+. Put

$$B_J = k[x_j \mid j \in J][(x_i - x_j)^{-1} \mid i \neq j].$$

Then B_J has no k-point, while every k-subalgebra of B_J of cardinality at most θ has one.

Proof. A homomorphism $B_J \to k$ would assign values $a_j \in k$ with $a_i - a_j$ invertible for every distinct pair. Thus $j \mapsto a_j$ would inject J into k, which is impossible. The localization is nevertheless nonzero: it localizes the polynomial domain at nonzero elements.

Let $C \subseteq B_J$ have cardinality at most θ. Each element of C is represented by a polynomial fraction involving finitely many variables and finitely many inverted differences. Taking the union of these finite supports gives $J_0 \subseteq J$ with $|J_0| \leq \theta$ and $C \subseteq B_{J_0}$. Choose an injection $J_0 \hookrightarrow k$. Evaluation

at these distinct values makes every inverted difference nonzero, hence extends to $B_{J_0} \to k$. Its restriction is the required point of C. □

We checked this formulation against the reduced-product and accessible-image sources cited in the appendix; we did not locate this particular presentation there. The construction is an elementary cardinality obstruction, with no claim of priority. It disproves the suggestion that the bound $\pi = |k|$ should suffice. It does not exclude a larger bound: when $\pi \geq \theta^+$, the displayed algebra itself is a small obstruction of cardinality θ^+.

Commentary · Later developments [ms. 19]

A compact cardinal restores a uniform obstruction bound

Grothendieck's example anticipates a central feature of later accessibility theory: the existence of small witnesses can depend on compactness principles stronger than the existence of Grothendieck universes. Here the positive direction can be proved directly, so that the point of contact with the general theory remains explicit.

Let $\kappa > |k|$ be a regular uncountable cardinal such that every κ-complete filter extends to a $|k|^+$-complete ultrafilter. A strongly compact $\kappa > |k|$ satisfies this condition. In the terminology of Brooke-Taylor and Rosický (2017, Definition 2.3), the weaker assumption is $L_{|k|^+,\omega}$-compactness. Assume the cardinal and the sets used below belong to the chosen universe.

We claim that if every k-subalgebra of an algebra A of cardinality less than κ has a k-point, then A has a point. Let I be the set of those subalgebras. Regularity of κ and $|k| < \kappa$ imply that I is κ-directed: the subalgebra generated by fewer than κ of its members again has cardinality less than κ. The filter generated by the upper cones in I is κ-complete and proper. Extend it to a $|k|^+$-complete ultrafilter U. For each $B \in I$, choose a homomorphism $h_B : B \to k$.

For $a \in A$, define a function on I by $B \mapsto h_B(a)$ when $a \in B$, and give it the arbitrary value 0 otherwise. The upper cone of subalgebras containing a belongs to U. Since the function has at most $|k|$ values, completeness gives a unique value $h(a) \in k$ attained on a member of U. For $a, b \in A$, intersect the upper cone containing both with the three sets selecting $h(a), h(b), h(a+b)$. This intersection belongs to U and is nonempty. At any B in it,

$$h(a+b) = h_B(a+b) = h_B(a) + h_B(b) = h(a) + h(b).$$

The same argument proves multiplicativity. For a constant $c \in k$, all h_B take the value c, so $h(c) = c$. Hence $h : A \to k$ is a k-point. Taking the contrapositive yields a subalgebra of size less than κ witnessing the absence of a point.

For a finite field the same proof needs only the ordinary ultrafilter lemma. Use the directed poset of finitely generated subalgebras and an ultrafilter extending its upper-cone filter. Every function to the finite set k is constant on a member of this ultrafilter, without any additional completeness assumption. Therefore every algebra without a k-point has a finitely generated subalgebra without a k-point. Such a subalgebra has cardinality at most $\aleph_0$, so the final question has a positive answer with $\pi = \aleph_0$ for finite k, in every universe under consideration.

This is the usual complete-ultrafilter compactness argument specialized to Grothendieck's question, not a new compactness principle. It makes the relation to accessible images precise. The class S of algebras with a k-point is the full image of the domain functor

$$(\mathrm{CAlg}_k)/k \longrightarrow \mathrm{CAlg}_k.$$

It is closed under subobjects, so it is also that functor's powerful image. Theorem 3.4 of Brooke-Taylor and Rosický (2017), extending the work of Makkai and Paré, gives accessibility and accessible embedding of powerful images under appropriately large compactness assumptions. The elementary argument above proves the needed witness property in this particular case; Corollary 4.16.7.9 then also gives accessibility of the complementary cosieve.

The alternatives established here have definite hypotheses. In a universe with the cutoff assumed by Proposition 4, no uniform obstruction bound exists. In a universe containing a sufficiently large compact cardinal, such a bound exists. The exact weakest set-theoretic assumption for this one fixed field is not identified by these arguments, and the general powerful-image theorem should not be cited as an exact converse for this particular example.

Overview · XVIII-e

These final pages of Chapter XVIII examine how far accessibility survives constructions which are elementary at the level of categories but delicate at the level of cardinal bounds (Grothendieck, 1990–1991n). The passage begins inside the proof of a gluing theorem, moves through categories of sections and categories over a small base, and ends by testing the limits of the theory against subcategories defined by the existence of a morphism. The appended calculation with products of fields supplies the decisive examples. The announced section 4.17 on paratoposes contains no mathematical text in the edition; the reader should not look to that heading for an additional theorem.

One convention governs almost every argument. An indexing category grand devant an infinite cardinal π admits cocones for diagrams with at most π arrows. Thus the source's L_π means existence or preservation of π^+-filtered colimits in the modern regular-cardinal convention. The same successor occurs in a π-ultrafilter: its intersections may have at most π factors. A useful cardinal also controls a small family of presentable objects. These are separate requirements. An accessible category has arbitrarily large suitable ranks, but a proof using a rank chosen in advance must verify the additional uniformity it needs.

The gluing construction explains why these distinctions matter. A category over the walking arrow has an open fibre $\mathcal{E}_0$, a closed fibre $\mathcal{E}_1$, and a bifunctor $H(x_0, x_1)$ describing arrows between them. Presentability in a fibre need not be presentability in the glued category: the crossing Hom sets have to preserve the same filtered colimits. The calculations in SGA 4, Exposé I, 9.6 and 9.8, provide the small-limit interchange used in the proof (Artin et al., 1972). The audit identifies two necessary repairs. Corollary 4.14.25(1) needs existence of the relevant colimits in the closed fibre. Corollary 4.15.2 needs a common initial rank controlling the stated functors before its subsequent cardinal enlargement. Explicit counterexamples show that these are hypotheses with mathematical content. The existential accessibility theorem survives the repaired choice of ranks.

The categories of sections express the same problem in a more organized form. Products collect the objects, inserters collect the transition maps, and equifiers impose their equations. This relates the source directly to the accessible-category closure results of Makkai and Paré, whose book appeared in 1989, and to the contemporary theory of flexible limits. The detailed modern construction in Ching and Riehl (2014, Appendix A) also explains why this machinery became useful in homotopical algebra. For the total category of a fibration or an opfibration, the commentary gives separate proofs of colimit existence and presentability. The two orientations require different uses of cartesian or cocartesian factorization; merely reversing a statement about filtered colimits would not supply either proof.

The small-base argument is the technical centre of the passage. A sufficiently filtered diagram in a small category can be reduced to one object and one recurrent idempotent. Its formal ind-object is the image of that idempotent, so it becomes representable exactly when the surviving projector splits. The walking unsplit projector, with monoid $\{1, p\}$ and $p^2 = p$, is the simplest example. The source eventually proves the strict-bound conclusion

$$\mathrm{Ind}_\pi(\mathcal{C}) \simeq \mathrm{Kar}(\mathcal{C}) \qquad (\pi > |\operatorname{Mor}\mathcal{C}|).$$

Here the notation on the left retains the source's successor convention. The commentary supplies both its growth argument and a presheaf proof. The literal identity condition in 4.15.9 is false, even for a two-element group; eventual invertibility and an appropriate trivialization are the invariant conclusions. A separate monoid reduction produces a quotient of a submonoid, not necessarily a quotient of the whole monoid. Neither defect invalidates the strict-bound theorem. The general equality case remains unsettled here, although two-sided cancellation and a stricter local Hom bound give affirmative special cases. Modern comparisons with Cauchy completion and coherent idempotents are provided by Tendas (2022) and Lurie (2009, Sections 5.4.2–5.4.3).

The later questions expose several different ways in which accessibility can fail. The double-dual functor on sets fails to preserve filtered colimits at every rank. Isbell's category of partial bijections is self-dual and finitely accessible, yet has a one-object finite dense and codense subcategory and is not essentially small; this answers all three questions in 4.16.4 negatively (Avery and Leinster, 2021, Example 8.7). Other failures depend on the universe. Under the appendix's hypothesis, the commentary proves that the decreasing filtered-colimit closures of an accessible functor's image need not stabilize. A separate argument treats contravariant images. Adjoining retracts of representables to a nonaccessible cosieve also produces a category with a small dense family of presentable objects and sufficiently filtered colimits which is nevertheless not accessible. This repairs the precise defect which Grothendieck notices in his own proposed construction.

The question about existing colimits needs equal care. If all diagrams of a fixed shape have colimits, their colimit cocones form an accessible category, as in Lurie (2009, Example 5.4.7.9). The commentary constructs, under the appendix's hypothesis, a finitely accessible category whose existing pushout squares form a nonaccessible full subcategory. It is a collage obtained by adjoining one object to the category of k-algebras. Its pushout condition detects exactly the absence of a k-point. Thus closure under filtered colimits alone does not produce a small generating family.

The finite tests in 4.16.6 provide a positive result of a different kind. We prove the two split-idempotent tests by first constructing a colimit over the split image and then checking maps out of fibrewise presentable objects. A strict pullback can nevertheless retain an unsplit projector whose image in the base is an identity; the walking isomorphism gives a counterexample to the proposed shortcut b′. With all the tests retained, the intended relative property has a concise description. Over a small Karoubian base $\mathcal{C}$, it is equivalent to accessibility of $\mathcal{E}$ and $f : \mathcal{E} \to \mathcal{C}$, together with the lifting of every prescribed base isomorphism. Such an f is an isofibration. Its strict pullbacks are equivalent to accessible pseudopullbacks, so the proposed change from small test bases to accessible test bases is valid. The standard closure theorem and its applications are discussed in Bourke and Lack (2024, Examples 4.8 and Remark 4.9, published version).

The closing open–closed calculus supplies the conceptual explanation of the counterexamples. Open means a full sieve, closed means a full cosieve. A cosieve is accessible precisely when it has a small weakly initial family, and precisely when its complementary sieve is closed under sufficiently filtered colimits. The canonical slice of presentable objects over x proves these equivalences: a nonempty cosieve in that filtered slice is cofinal, whereas a proper complementary sieve cannot be. This argument repairs the omitted slice denominators in the edition and completes the implication left unfinished in the appendix. The transfinite no-forward-morphism construction is also justified with an explicit universe bound. For the additional question restricted to fields, the answer is exact: in V_κ such a κ-long sequence exists if and only if κ is not a Vopěnka cardinal, with external families allowed. The commentary proves the reduction using a typed graph construction and the fully faithful functor of Miller et al. (2018).

The appendix identifies every maximal ideal of a product of fields with an ultrafilter. A map from the product back to an infinite ground field k requires $|k|^+$-completeness. Its universe hypothesis says that sufficiently complete nonprincipal ultrafilters do not exist on small sets, equivalently that measurable cardinals below the universe boundary are bounded. Under that hypothesis, for every infinite $\pi \geq |k|$ the algebra

$$k^{\pi^+}/\{f : |\operatorname{supp}(f)| \leq \pi\}$$

has no k-point, while every subalgebra of size at most π has one. A complete proof evaluates representatives outside their common exceptional set. A second construction disproves the proposed bound $\pi = |k|$ for every infinite field without the universe hypothesis. Conversely,

a sufficiently large compact cardinal restores a uniform obstruction bound; for finite fields the ordinary ultrafilter lemma already gives a finitely generated obstruction. The comparison with powerful images uses Brooke-Taylor and Rosický (2017), without asserting an exact large-cardinal converse for this one fixed-field problem.

Chapter XIX

Modèles (4)

XIX · Chapter XIX · manuscript pages 1–50

p. (non numérotée)

LES DÉRIVATEURS

ALEXANDRE GROTHENDIECK

Chapitre XIX

Modèles (4)

p. (non numérotée)

Ce texte a été déchiffré et transcrit en LaTeX 2ε par M. Künzer. Il a été édité par M. Künzer, J. Malgoire, et G. Maltsiniotis. La transcription est aussi fidèle que possible au manuscrit. Seules quelques corrections mineures évidentes ont été effectuées. Pour les rares ajouts ou commentaires des éditeurs, ainsi que pour la numérotation originale des pages du manuscrit, les caractères de machine à écrire `[typewriter]` entre crochets sont utilisés.

Cette édition est provisoire. Les remarques, commentaires et corrections sont bienvenus. Envoyer un message à :

[**e-mail address removed**]

G. Maltsiniotis

1 Rappels sur les couples de Quillen (couples de Quillen spéciaux)

p. 1 [page 1]

Je vais reprendre les notations et la terminologie pour les relations de LLP `[left lifting property]` et RLP `[right lifting property]` de Quillen. J'écris (pour une catégorie $\mathcal{M}$, et $\Phi, \Psi \subset \mathrm{Fl}(\mathcal{M})$)

$$\Phi \;\text{⌐⟶}\; \Psi \qquad \Longleftrightarrow \qquad (\Phi \subset \Psi^*) \qquad \Longleftrightarrow \qquad (\Psi \subset \Phi_*)\ , \tag{1.1}$$

i.e. Φ a la LLP par rapport à Ψ, ou encore Ψ la RLP par rapport à Φ. Je dis aussi que (Φ, Ψ) est un *couple* Q-*orthogonal.* Cette notion n'est pas symétrique, mais je me permets parfois (exceptionnellement) d'écrire

$$\Psi \;\text{⟵⌙}\; \Phi \qquad \text{comme synonyme de} \qquad \Phi \;\text{⌐⟶}\; \Psi\ . \tag{1.2}$$

Φ est Q-*orthogonal à gauche* de Ψ, Ψ est Q-*orthogonal à droite* à Φ, et la Q-orthogonalité à gauche (resp. à droite) est notée par le demi-sommet de flèche pointant vers le bas (resp. vers le haut). J'écris

$$\begin{cases} \Phi \;\text{⟵⌙}\; \Psi & \Longleftrightarrow \quad (\Phi = \Psi^*) \\ \Phi \;\text{⌐⟶}\; \Psi & \Longleftrightarrow \quad (\Psi = \Phi_*) \\ \Phi \;\text{⟵⌙}\; \Psi & \Longleftrightarrow \quad (\Phi = \Psi^* \ \text{ et } \ \Psi = \Phi_*)\ . \end{cases} \tag{1.3}$$

Ces trois cas de Q-orthogonalité renforcée sont donc ceux où on a de plus $\Phi = \tilde{\Phi}$ resp. $\Psi = \bar{\Psi}$ resp. $\Phi = \tilde{\Phi}$ et $\Psi = \bar{\Psi}$. On dit que Φ est Q-*clos à gauche* si $\Phi = \tilde{\Phi}$, que Ψ est Q-*clos à droite* si $\Psi = \bar{\Psi}$, que le couple Q-orthogonal (Φ, Ψ) est Q-*clos à gauche* (resp. *à droite*) si Φ (resp. Ψ) est Q-clos à gauche (resp. à droite). On dit qu'il est Q-*clos* si les deux conditions

sont satisfaites simultanément. Quand aucune confusion n'est à craindre, il m'arrivera de [page 2] dire simplement que Φ est Q-clos, ou que Ψ est Q-clos, sans spécifier si c'est à gauche où à droite. Mais pour un couple Q-orthogonal (Φ, Ψ) (ou je préfère l'écrire $\Phi \;\text{⌐⟶}\; \Psi$), Q-clos signifie toujours Q-clos à gauche et à droite.

Commentary · Exposition [ms. 2]

Orthogonality, closure, and factorization

Grothendieck begins by separating the ingredients that will later have to survive passage to a category of diagrams. A lifting property is easy to transport along an adjunction. A factorization requires a construction. The purpose of the elaborate arrows is to prevent the existence of that construction from being smuggled into the lifting notation.

For classes of arrows L, R in a locally small category, write $L \perp R$ when every commutative square with left side in L and right side in R has a diagonal filler. The filler need not be unique. In modern notation,

$$^{\square}R = \{i : i \perp R\}, \qquad L^{\square} = \{p : L \perp p\}.$$

These are the source's R^* and L_*. They form an antitone Galois connection: $L \subseteq {}^{\square}R$ if and only if $R \subseteq L^{\square}$. Consequently

$$\overline{L} = {}^{\square}(L^{\square}), \qquad \overline{R} = ({}^{\square}R)^{\square}$$

are extensive, monotone, idempotent operations. Grothendieck writes a small tilde for the first closure and an overbar for the second. An arrow with two complete heads means the stronger pair of identities $L = {}^{\square}R$ and $R = L^{\square}$. A transverse stroke adds the assertion that every f factors as pi with $i \in L$ and $p \in R$. Together these are precisely the data of a weak factorization system, without a chosen functorial factorization.

There is a genuine logical slip in the sentence following (1.3). For an arbitrary orthogonal pair, separate closedness of L and R does not imply that they are one another's orthogonals. Indeed, in

Set take both classes to be the isomorphisms. Each is closed under its appropriate double-orthogonal operation, because

$$\,^{\square}\mathrm{Fl} = \mathrm{Is}, \quad \mathrm{Is}^{\square} = \mathrm{Fl}, \quad \,^{\square}\mathrm{Is} = \mathrm{Fl}, \quad \mathrm{Fl}^{\square} = \mathrm{Is}.$$

For example, an arrow having the left lifting property against itself has a two-sided inverse, by using the square with identity horizontal sides; this proves the first identity. The other three follow in the same way or directly from invertibility. Thus $(\mathrm{Is}, \mathrm{Is})$ is separately closed and orthogonal, but is not a mutually maximal orthogonal pair unless the category is a groupoid.

The repair is to retain the explicit equalities of (1.3) as the meaning of a closed *pair*. There is no difficulty with the later terminology for a Quillen pair: once factorization is available, Proposition 1.3 shows that separate retract-closedness forces these equalities. The distinction matters precisely before factorization, where this chapter's questions lie. We therefore retain the printed sentence, record the defect in the audit, and use the mutually orthogonal meaning whenever a two-headed arrow occurs below.

Quand un couple Q-orthogonal $\Phi \longmapsto \Psi$ a la propriété de factorisation de Quillen, je l'indique par la barre verticale croisant l'horizontale, ainsi

$$\Phi \;\rightarrowtail\!\!\!\!\!\mid\; \Psi\ , \tag{1.4}$$

et on définit de même le sens des graphismes

$$\begin{cases} \Phi & \rightarrowtail\!\!\!\!\!\mid & \Psi \\ \Phi & \mid\!\!\!\!\!\leftarrowtail & \Psi \\ \Phi & \rightarrowtail\!\!\!\!\!\mid\!\!\!\!\!\rightarrow & \Psi \\ \Phi & \leftarrow\!\!\!\!\!\mid\!\!\!\!\!\rightarrow & \Psi\ . \end{cases} \tag{1.3}$$

Un couple Q-*orthogonal* Q-*factorisant* $\Phi \rightarrowtail\!\!\!\!\!\!\rightarrow \Psi$ est appelé un *précouple de Quillen.* S'il est de plus Q-clos, i.e. si on a $\Phi \longleftrightarrow \Psi$, on dit que c'est un *couple de Quillen.* p. 2

Φ est dit Q-*saturé à gauche* s'il contient les isomorphismes et est stable par cochangement de base, par (petite) composition transfinie, et par facteurs directs. Dualement Ψ est dit Q-*saturé à droite* s'il

contient les isomorphismes, est stable par changement de base, par composition transfinie, et par facteurs directs. Si Φ est Q-saturé à gauche, il est stable par (petites) sommes directes, i.e. si les $f_i : A_i \longrightarrow B_i$ ($i \in I$, I petit) sont dans Φ, $f = \coprod f_i : \coprod A_i \longrightarrow \coprod B_i$ aussi, et de même pour (petites) sommes amalgamées finies ou infinies (quand on se place dans un $A\backslash\mathcal{M}$ au lieu de $\mathcal{M}$). On a l'énoncé dual pour Ψ, quand Ψ est Q-saturé à droite. [page 3]

En fait, il y a des notions plus faibles de Q-saturation que celle que je viens de dire : Q-*saturation simple* (à gauche : Φ contient les isomorphismes, stable par composition binaire, par cochangement de base), Q-*saturation transfinie* (on exige de plus stabilité par composition transfinie), enfin Q-*saturation totale,* ou Q-saturation tout court (quand on exige de plus la stabilité par facteurs directs). On a les implications

$$\text{(1.6)} \qquad \begin{cases} \text{Q-clos à gauche} \\ \Downarrow \\ \text{Q-saturé (total)} \\ \Downarrow \\ \text{Q-saturé transfini} \\ \Downarrow \\ \text{Q-saturé simple.} \end{cases}$$

Pour $\Phi \subset \mathrm{Fl}(\mathcal{M})$, on a sa Q-clôture (à gauche) $\tilde{\Phi}$ (qui est la plus petite partie Q-close à gauche de $\mathrm{Fl}(\mathcal{M})$ contenant Φ) et de même son Q-saturé (total) à gauche, son Q-saturé transfini, son Q-saturé simple

avec les inclusions tautologiques [page 4]

$$\text{(1.7)} \qquad \Phi \subset \underbrace{\Phi'}_{\substack{\text{Q-saturé} \\ \text{à gauche simple}}} \subset \underbrace{\Phi''}_{\substack{\text{Q-saturé} \\ \text{à gauche transfini}}} \subset \underbrace{\Phi'''}_{\substack{\text{Q-saturé} \\ \text{à gauche total}}} \subset \underbrace{\tilde{\Phi}}_{\substack{\text{Q-clôture} \\ \text{à gauche}}},$$

et dualement pour Ψ. Entre Φ et Φ' il y a encore un Φ'_0,

$$\text{(1.8)} \qquad \Phi \subset \Phi'_0 \subset \Phi',$$

qui est le Q-*saturé pour cochangement de base* (contenant aussi, par définition, tous les isomorphismes). Pour construire tous ces saturés, on procède ainsi :

Commentary · Exposition [ms. 4]

What cellular closure does and does not supply

The four closures organize a constructive question: how much of a left lifting class can be built by attaching specified arrows? The elementary operations are chosen because every left lifting class survives them. For a pushout of i, a lifting square is equivalent, by the pushout property, to a lifting square against i. For a composite $i_2 i_1$, solve the lifting problem first against i_1 and then against i_2. For a continuous ordinal sequence $(X_\alpha)_{\alpha \leq \lambda}$, compatible fillers are constructed at successor stages and are assembled at a limit stage by the colimit property. The resulting map $X_\lambda \to E$ is the required filler for the transfinite composite. Finally, a lifting problem against a retract of an arrow is a retract of a lifting problem against that arrow, so a filler restricts along the retraction.

These arguments prove the implications in (1.6), including the dual assertions, whenever the indicated colimits or limits exist. They also explain why small coproducts need not be added as an independent closure operation. Well-order the indexing set S of a family $(A_s \to B_s)$. Starting with

$\coprod_s A_s$, replace the summand A_s by B_s at stage s, using a pushout. At a limit stage take the colimit. The final map is $\coprod_s A_s \to \coprod_s B_s$. The same construction in an undercategory treats amalgamated coproducts. This use of a well-order belongs to the usual set-theoretic choice convention.

The formula for Φ'' on the next page rests on a useful flattening argument. A pushout of a transfinite composite is the transfinite composite of the successive pushouts. A transfinite composite of transfinite composites can be indexed by the ordinal sum of the component ordinals. Thus the transfinite composites of pushouts of the original generators already form a class closed under pushout and transfinite composition. Identity maps allow stages at which nothing is attached. This proves the asserted description of Φ'' without first adding an unspecified hierarchy of repeated cellular operations.

What remains at this point is the relation with retracts. Closure under retracts alone gives no factorization, and the notation Φ''' initially means the smallest class stable under *all* the listed operations. Grothendieck will identify it with the retracts of Φ'' under small-generation hypotheses, and simultaneously identify both with the entire lifting class. The first of these equalities in fact has a general proof, given in the next block; the second is what the small object argument adds. This distinction prevents a positive answer to the elementary closure question from being mistaken for a factorization theorem. A modern reference for the latter is Hovey (1999a, Theorem 2.1.14 and Corollary 2.1.15).

1) Φ'_0 = ensemble des flèches $i : A \longrightarrow B$ s'insérant dans un carré cocartésien p. 3

$$\begin{array}{ccc} A_0 & \xrightarrow{i_0} & B_0 \\ \downarrow & & \downarrow \\ A & \xrightarrow{i} & B\ , \end{array}$$

avec $i_0 \in \Phi$ ou i_0 une identité (auquel cas on trouve pour i les isomorphismes).

2) Φ' = ensemble des flèches composées multiples

$$i_n\, i_{n-1}\ \cdots\ i_1\ ,$$

avec $i_\alpha \in \Phi'_0$.

3) Φ'' = ensemble des composés transfinis de flèches $\in \Phi'_0$.

On a envie de poser

$$\Phi''' \ = \ \Phi''^{\natural} \ \overset{\text{déf}}{=}\ \text{ensemble des facteurs directs d'éléments de } \Phi'',$$

et c'est un fait que $\Phi''^{\natural} \subset \Phi'''$ trivialement, mais il n'est par clair en général qu'on ait égalité, faute de pouvoir montrer que $\Phi''^{\natural}$ est stable, non seulement par facteurs directs, mais aussi par cochangement de base et par composition transfinie. Cependant, rappelons : [page 5]

Commentary · Later developments [ms. 5]

Retracts of cellular arrows are closed under transfinite composition

The uncertainty expressed here has a positive answer without a small generating set. If $\mathcal{M}$ is cocomplete and $\mathcal{C}$ contains the isomorphisms and is stable under pushout and transfinite composition, then $\mathrm{Retr}(\mathcal{C})$ has the same properties. This is Makkai and Rosický (2014b, Lemma 2.2). We give the explicit compatible-retraction construction, since it isolates exactly the issue Grothendieck raises.

First every retract of a map in $\mathcal{C}$ can be presented as a retract with its domain fixed. Suppose $f : A \to B$ is a retract of $g : X \to Y$, with maps $a : A \to X$, $b : X \to A$, $c : B \to Y$, $d : Y \to B$ satisfying

$$ba = 1_A, \quad dc = 1_B, \quad ga = cf, \quad fb = dg.$$

Push g out along b, obtaining $k : A \to P = A \amalg_X Y$ in $\mathcal{C}$. If $q : Y \to P$ is the other pushout map, put $s = qc : B \to P$. The maps $f : A \to B$ and $d : Y \to B$ induce $r : P \to B$. Then $sf = k$, $rk = f$, and $rs = 1_B$. Thus f is a retract of k in $A\backslash\mathcal{M}$.

Now let $(X_\alpha)_{\alpha \le \lambda}$ be a continuous chain with each $f_\alpha : X_\alpha \to X_{\alpha+1}$ in $\mathrm{Retr}(\mathcal{C})$. Choose fixed-domain presentations

$$X_\alpha \xrightarrow{h_\alpha} Z_\alpha, \quad s'_\alpha : X_{\alpha+1} \to Z_\alpha, \quad r'_\alpha : Z_\alpha \to X_{\alpha+1},$$

with $h_\alpha \in \mathcal{C}$, $s'_\alpha f_\alpha = h_\alpha$, $r'_\alpha h_\alpha = f_\alpha$, and $r'_\alpha s'_\alpha = 1$. Construct a second continuous chain Y_α and compatible retractions $X_\alpha \xrightarrow{s_\alpha} Y_\alpha \xrightarrow{r_\alpha} X_\alpha$, starting with $Y_0 = X_0$. At a successor stage set

$$Y_{\alpha+1} = Y_\alpha \amalg_{X_\alpha} Z_\alpha,$$

using s_α and h_α for the pushout. Its structure map $g_\alpha : Y_\alpha \to Y_{\alpha+1}$ lies in $\mathcal{C}$. Define $s_{\alpha+1}$ by the inclusion of Z_α composed with s'_α. Define $r_{\alpha+1}$ by $f_\alpha r_\alpha$ on Y_α and r'_α on Z_α; these agree on X_α. The equations

$$r_{\alpha+1}s_{\alpha+1} = 1, \quad g_\alpha s_\alpha = s_{\alpha+1}f_\alpha, \quad r_{\alpha+1}g_\alpha = f_\alpha r_\alpha$$

follow directly. At a limit stage take colimits of both the chains and these equations. The composite $X_0 \to X_\lambda$ is consequently a retract, with domain fixed, of $Y_0 \to Y_\lambda \in \mathcal{C}$.

Pushout stability follows by pushing out a fixed-domain retraction. Stability under retracts follows by composing retraction diagrams. Taking $\mathcal{C} = \Phi''$ therefore proves $\Phi''' = \mathrm{Retr}(\Phi'')$ in every cocomplete category, even for a proper class of generators. It does not prove $\Phi''' = {}^{\square}(\Phi^{\square})$: that last equality still needs a mechanism, such as Theorem 1.1, for constructing the appropriate factorizations.

Théorème 1.1. *Supposons $\mathcal{M}$ stable par petites $\varinjlim$, et que ses éléments soient accessibles. Supposons de plus que Φ soit petit, plus généralement, qu'il existe une partie $\Phi_0 \subset \Phi$ petite telle que $\Phi \subset \tilde{\Phi_0}$, donc*

$$\Psi \overset{\text{déf}}{=} \Phi_* = (\Phi_0)_* \qquad [1]. \tag{1.1.1}$$

Sous ces conditions, si Φ_0'' désigne le Q*-saturé à gauche transfini de Φ_0, alors on a $\Phi_0'' \rightarrowtail \Psi$, i.e. toute flèche f de $\mathcal{M}$ se factorise en $f = pi$, avec $i \in \Phi_0''$, $p \in \Psi$ [2]. A fortiori, si Φ'' désigne le* Q*-saturé à gauche transfini de Φ, on a*

$$\Phi'' \rightarrowtail \Psi .$$

Commentary · Exposition [ms. 5]

The small object argument and the retract that finishes it

Theorem 1.1 is the engine of the chapter. It recalls the transfinite construction developed in Grothendieck (1990–1991k, Chapter XV, §1, Theorem 1). Its force is not merely that sufficiently many cells exist, but that a *set* of generators admits one common bound on the stages needed to see all its lifting problems. Read "accessible object" here as an object whose maps into sufficiently filtered colimits are detected at a stage; for this argument the weaker condition of smallness relative to the relevant cellular sequences suffices. Local smallness is understood in the chosen universe. These hypotheses do not assert that the entire category is locally presentable.

Put $S = \Phi_0$ and let $f : X \to Y$. Choose an infinite regular cardinal κ such that every domain of an arrow of S is κ-small for the sequences being used. Such a common bound exists because S is a set. Starting from $X_0 = X$, form at stage $\alpha + 1$ the pushout

$$\begin{array}{ccc} \coprod_{\sigma} A_\sigma & \longrightarrow & X_\alpha \\ \downarrow & & \downarrow \\ \coprod_{\sigma} B_\sigma & \longrightarrow & X_{\alpha+1}, \end{array}$$

where σ runs over all commutative squares $s : A_\sigma \to B_\sigma$ against $X_\alpha \to Y$, with $s \in S$. There is a set of such squares, by local smallness and the smallness of S. Their lower maps induce $X_{\alpha+1} \to Y$. At a limit stage take the colimit. Choose the length λ so that the sequence is sufficiently filtered for every domain in S, for example a regular cardinal larger than a common smallness bound.

The composite $i : X \to X_\lambda$ lies in S'': attaching a coproduct of cells can be serialized as on the preceding page. The induced $p : X_\lambda \to Y$ has the right lifting property against S. Indeed, in a square from $s : A \to B$ to p, the map $A \to X_\lambda$ factors through some X_α. Together with the fixed map $B \to Y$, it is one of the squares attached at the next stage. The corresponding copy of B in $X_{\alpha+1}$ gives the filler after composition with $X_{\alpha+1} \to X_\lambda$. The equation with Y already holds at stage α, since all maps are over Y. This proves $f = pi$. Attaching *all* squares makes the construction functorial once colimit choices are fixed; a morphism of arrows sends each indexed square to an indexed square at the next stage.

The retract argument is independent of this construction and deserves to be explicit. Suppose $f : A \to B$ has the left lifting property against $p : E \to B$ and $f = pi$ for $i : A \to E$. A filler in

$$\begin{array}{ccc} A & \xrightarrow{i} & E \\ {\scriptstyle f}\downarrow & \overset{s}{\nearrow} & \downarrow{\scriptstyle p} \\ B & \xrightarrow[1_B]{} & B \end{array}$$

satisfies $sf = i$ and $ps = 1_B$. Hence $(1_A, s) : f \to i$ and $(1_A, p) : i \to f$ exhibit f as a retract of i in the arrow category. In particular, for any factorizing orthogonal pair (L, R),

$$^{\square}R = \mathrm{Retr}(L), \qquad L^{\square} = \mathrm{Retr}(R).$$

1. **N.B.** On a donc $\Phi \rightarrowtail \Psi$, a fortiori $\Phi_0 \rightarrowtail \Psi$.
2. Il y a même factorisation *fonctorielle* en f, propriété qui mérite une notation, p.ex. $\Phi_0'' \rightarrowtail\!\!\!\!\!\rightarrow \Psi$.

The reverse inclusions follow from stability of lifting under retracts. The double-orthogonal closures equal these classes because orthogonality is unchanged by taking retracts. This proves Proposition 1.3 and, applied to $(S'', S^{\square})$, Corollary 1.2. It also proves Corollary 1.4: factor an arrow of L as a map in W followed by a map in R, then use the same square. If W is retract-closed and (L, R) itself factorizes, the converse follows from $L \subseteq W$.

Corollaire 1.2. *Sous les conditions du théorème, on a*

$$\Phi''' \text{ (Q-saturé total à gauche de } \Phi) \;=\; \underbrace{\Phi''^{\natural}}_{\substack{\text{ensemble des}\\ \text{facteurs directs d'éléments}\\ \text{de } \Phi''}} \;=\; \tilde{\Phi}\,. \tag{1.2.1}$$

Plus généralement (en apparence seulement!), si Φ_1 *est une partie de* $\mathrm{Fl}(\mathcal{M})$ *qui est* Q*-saturée à gauche transfinie, et telle que*

$$\Phi_0 \;\subset\; \Phi_1 \;\subset\; \tilde{\Phi} \;=\; \tilde{\Phi_0}\,,$$

p. 4

alors [page 6]

a) *on a* $\Phi_1 \rightarrowtail \Psi$ *et*

b) $\Phi_1^{\sim} = \Phi^{\sim} = \Phi_1^{\natural}$.

Ce corollaire résulte du théorème en vertu de XV prop. 4.2, dont je rappelle ici l'essentiel :

Proposition 1.3. *Soit* $\Phi \rightarrowtail \Psi$. *Alors* $\Phi^{\sim} = \Phi^{\natural} = \Psi^{*}$, *et dualement on a* $\Psi^{-} = \Psi^{\natural} = \Phi_{*}$.

Notons aussi, dans cet ordre d'idées :

Corollaire 1.4. *Soit* $\Phi \longrightarrow \Psi$, *et soit* W *une partie quelconque de* $\mathrm{Fl}(\mathcal{M})$. *Pour qu'on ait* $\Phi \subset W^{\natural}$, *il suffit que toute flèche de* $\mathcal{M}$ *se factorise en* $f = pi$, *avec* $i \in W$, $p \in \Psi$. *(Et cette condition est évidemment aussi nécessaire si* $\Phi \rightarrowtail \Psi$, *et* W *stable par facteurs directs.)*

On dit qu'une partie Φ_0 de Φ est Q-*génératrice à gauche* de Φ si elle satisfait la condition

$$\Phi \;\subset \Phi_0^{\sim}\,, \tag{1.9}$$

et on dit que Φ est une partie *spéciale* si Φ admet une *petite* partie Q-génératrice à gauche. Si $\Phi_0 \subset \Phi$ est petit, et si $\mathcal{M}$ satisfait les conditions du théorème 1.1, alors Φ_0 est une partie Q-génératrice (à gauche) si et seulement si son Q-saturé total (à gauche) Φ_0''' contient Φ (condition en apparence plus forte que $\Phi \subset \Phi_0^{\sim}$, puisque en général on a $\Phi_0''' \subset \Phi_0^{\sim}$). Cela signifie donc aussi, via 1.2, que tout élément de Φ est facteur direct d'un composé transfini de [page 7]
flèches déduites de

$$\Phi_0 \cup (\text{Identités de } \mathcal{M})$$

par cochangement de base.

Je rappelle encore le sorite ± tautologique des foncteurs adjoints :

Proposition 1.5. *Soit*

$$\mathcal{M}' \underset{v^{*}}{\overset{v_{!}}{\leftarrow\!\!\!\rightarrow}} \mathcal{M} \tag{1.5.1}$$

un couple de foncteurs adjoints, soient

$$\Phi \;\subset\; \mathrm{Fl}(\mathcal{M})\,, \qquad \Psi' \;\subset\; \mathrm{Fl}(\mathcal{M}')\,,$$

alors on a

$$(v_{!}(\Phi) \longrightarrow \Psi') \qquad \Longleftrightarrow \qquad (\Phi \longrightarrow v^{*}(\Psi'))\,. \tag{1.5.2}$$

p. 5

Corollaire 1.6. *Pour toute partie* Φ *de* $\mathrm{Fl}(\mathcal{M})$,

$$(v_!(\Phi))_* \;=\; (v^*)^{-1}(\Phi_*)\ . \tag{1.6.1}$$

En effet, appliquant 1.5 à Φ et à $\{f'\}$, où $f' \in \mathrm{Fl}(\mathcal{M}')$, on trouve

$$f' \in (v_!(\Phi))_* \qquad \Longleftrightarrow \qquad v^*(f') \in \Phi_*\ ,$$

ce qui équivaut à (1.6.1). Dualement :

Corollaire 1.7. *Pour toute partie* Ψ' *de* $\mathcal{M}'$*, on a*

$$(v^*(\Psi'))^* \;=\; (v_!)^{-1}(\Psi'^*). \tag{1.7.1}$$

En effet, si $f \in \mathrm{Fl}(\mathcal{M})$, on a par (1.5.2)

$$f \in (v^*(\Psi'))^* \qquad \Longleftrightarrow \qquad v_!(f) \in \Psi^*\ .$$

(3).

Un couple Q-orthogonal $\Phi \rightarrowtail \Psi$ est dit *spécial* s'il satisfait les conditions suivantes. [page 8]

(1.10)
- a) $\mathcal{M}$ est stable par petites $\varinjlim$, les objets de $\mathcal{M}$ sont accessibles. (On dit alors que $\mathcal{M}$ est "spéciale".)
- b) Le couple est clos : $\Phi \longleftrightarrow \Psi$.
- c) Φ est *spéciale.*

Cela implique donc, par 1.1, que $\Phi \longleftrightarrow \Psi$ est un couple de Quillen. Si $\mathcal{M}$ satisfait a), les couples Q-orthogonaux spéciaux dans $\mathcal{M}$, ou encore les couples de Quillen spéciaux, correspondent donc biunivoquement aux parties Φ de $\mathrm{Fl}(\mathcal{M})$ qui sont *spéciales* et closes à gauche (ou ce qui revient au même, Q-saturées à gauche), en associant à une telle Φ le couple $\Phi \longleftrightarrow \Phi_*$.

Commentary · Exposition [ms. 7]

An adjunction transports the whole lifting problem

The next passage prepares the construction of diagrams by proving a statement more economical than a transfer theorem. If $L = v_!$ is left adjoint to $R = v^*$, a commutative square from Li to p' corresponds to a commutative square from i to Rp'. This correspondence includes the diagonal fillers: a map from the codomain of Li to the domain of p' corresponds to a map from the codomain of i to the domain of Rp', and the two triangle equations correspond by naturality of the adjunction. Thus

$$Li \perp p' \quad \Longleftrightarrow \quad i \perp Rp'.$$

There is no need for unique fillers, and no factorization hypothesis enters this equivalence.

Applying it one arrow at a time gives

$$(L\Phi)^{\square} = R^{-1}(\Phi^{\square}), \qquad {}^{\square}(R\Psi') = L^{-1}({}^{\square}\Psi').$$

Here the inverse image means inverse image of a class of arrows, rather than an inverse functor. The unprimed Ψ^* in the proof of Corollary 1.7 is a printed slip: its ambient category forces Ψ'^*.

The same calculation explains why a left adjoint preserves generation by a set even when it does not literally preserve a chosen list of cellular presentations. If $\Phi \subseteq {}^{\square}(\Phi_0^{\square})$, then $\Phi_0^{\square} \subseteq \Phi^{\square}$; inverse image under R gives

$$(L\Phi_0)^{\square} \subseteq (L\Phi)^{\square},$$

and taking left orthogonals gives $L\Phi \subseteq {}^{\square}((L\Phi_0)^{\square})$. This is Corollary 1.9. It needs neither accessibility nor cocompleteness. The extra hypotheses of Corollary 1.8 are used only after this calculation: the

3. donner ici 1.9 page 9.

small object argument in the target produces the factorizations and identifies the left class with the retracts of the relative $L\Phi_0$-cell complexes.

This order of reasoning is helpful throughout the chapter. First identify the desired right class as an inverse image. Then identify a small family that defines it. Only then apply a factorization theorem in the target. In particular, the source category's "specialness" in Corollary 1.8 is stronger than is required for the transport of generators itself; Grothendieck explicitly observes this when extracting Corollary 1.9. The later transfer of a *model structure* will require an additional condition on weak equivalences. That condition is absent here because a single weak factorization system has no class of weak equivalences to preserve.

Ceci posé :

Corollaire 1.8. *Supposons que $\mathcal{M}$, $\mathcal{M}'$ satisfassent la condition* a) *de* (1.10), *et soit* (Φ, Ψ) || *un couple de Quillen spécial dans $\mathcal{M}$. Alors* $\Big(\Phi' = v_!(\Phi)^\sim, \Psi' = (v^*)^{-1}(\Psi)\Big)$ *est un couple de Quillen spécial dans $\mathcal{M}'$.*

En effet, par le corollaire 1.6 on a

$$\Phi'_* \;=\; (v_!(\Phi))_* \;=\; (v^*)^{-1}(\Psi) \;=\; \Psi' \;,$$

donc $\Phi' \longleftrightarrow \Psi'$. Il reste à prouver que Φ' est spécial. Soit Φ_0 une partie petite de Φ telle que $\Phi \subset \Phi_0^\sim$, posons $\Phi'_0 \;=\; v_!(\Phi_0) \subset \Phi'$, il suffit de voir qu'on a

[page 9]

$$(*) \qquad v_!(\Phi) \;\subset\; v_!(\Phi_0)^\sim \;,$$

d'où résultera qu'on a $\Phi' \overset{\text{déf}}{=} (v_!(\Phi))^\sim \subset v_!(\Phi_0)^\sim$. Mais on a

$$v_!(\Phi_0)^\sim \;=\; (v_!(\Phi_0)_*)^* \;,$$

et

$$v_!(\Phi_0)_* \;=\; (v^*)^{-1}(\underbrace{\Phi_{0*}}_{\Psi}) \;=\; (v^*)^{-1}(\Psi) \;\overset{\text{déf}}{=}\; \Psi' \;,$$

p. 6

donc

$$v_!(\Phi_0)^\sim \;=\; \Psi'^* \;=\; \Phi' \;=\; v_!(\Phi)^\sim \;,$$

d'où $(*)$. Cet argument ne fait pas appel à l'hypothèse "spéciale" sur $\mathcal{M}$, et peut s'expliciter en le

Corollaire 1.9. [4] *Sous les conditions générales de* 1.5 *($v_! : \mathcal{M} \longrightarrow \mathcal{M}'$ étant un foncteur admettant un adjoint à droite), soient Φ_0, Φ deux parties de* $\mathrm{Fl}(\mathcal{M})$ *telles que $\Phi \subset \Phi_0^\sim$ (ou ce qui revient au même, $\Phi^\sim \subset \Phi_0^\sim$). Alors $v_!(\Phi) \subset v_!(\Phi_0)^\sim$, i.e. $v_!(\Phi)^\sim \subset v_!(\Phi_0)^\sim$.*

En effet, cette relation équivaut aussi à $v_!(\Phi_0)_* \;\subset\; v_!(\Phi)_*$, c'est-à-dire (par 1.6) $(v^*)^{-1}(\Phi_{0*}) \;\subset\; (v^*)^{-1}(\Phi_*)$, et est impliqué par $\Phi_{0*} \;\subset\; \Phi_*$. Cette dernière d'autre part équivaut à l'hypothèse $\Phi^\sim \subset \Phi_0^\sim$, q.e.d.

Si Ψ est une partie de $\mathrm{Fl}(\mathcal{M})$, on en déduit pour toute I dans Cat une partie

[page 10]

$$(1.11) \qquad \begin{cases} \Psi(I) \subset \mathrm{Fl}(\mathcal{M}(I)), \quad \text{avec } \; \mathcal{M}(I) = \underline{\mathrm{Hom}}(I^\circ, \mathcal{M}) \;, \\ f \in \Psi(I) \iff f(i) \in \Psi \;\; \forall\, i \in \mathrm{Ob}\, I \;. \end{cases}$$

Si on introduit la catégorie discrète I_0 sous-jacente à $\mathrm{Ob}\, I$, et le foncteur canonique

$$(1.12) \qquad v_I = v \;:\; I_0 \longrightarrow I \;,$$

on trouve un couple de foncteurs adjoints

$$(1.13) \qquad \mathcal{M}(I) \underset{v^*}{\overset{v_!}{\leftrightarrows}} \mathcal{M}(I_0) \;\simeq\; \mathcal{M}^{I_0} \;,$$

$$\Big(\underset{v_*}{\overset{\cdots\cdots}{\leftarrow}} \Big)$$

4. devrait passer avant 1.8.

et on a

$$\Psi(I) \;=\; (v^*)^{-1}(\Psi(I_0))\ .$$

Supposons maintenant qu'on ait

$$\left\{\begin{array}{rcl} \Psi & = & \Phi_* \\ \text{d'où}\ \ \Psi(I_0) & = & \Phi(I_0)_* \end{array}\right. \tag{1.14}$$

(vérification immédiate), on conclut alors par 1.6

$$\Psi(I) \;=\; (v_!(\Phi(I_0)))_* \;=\; (\Phi\langle I\rangle)_*\ , \tag{1.15}$$

avec

$$\Phi\langle I\rangle \;\overset{\text{déf}}{=}\; v_!(\Phi(I_0))^\sim \;=\; \Psi(I)^*\ . \tag{1.16}$$

Si on a un foncteur

$$u\ :\ I \longrightarrow I'\ , \tag{1.17}$$

d'où un triple de foncteurs adjoints

$$\left\{\ \mathcal{M}(I) \underset{\longrightarrow}{\overset{u_!}{\underset{u_*}{\overset{\longrightarrow}{\longleftarrow}}}} \mathcal{M}(I')\ \right. , \tag{1.18}$$

p. 7
[page 11]

on aura de façon ± tautologique

$$\left\{\begin{array}{rcl} u_!(\Phi\langle I\rangle) & \subset & \Phi\langle I'\rangle \\ u^*(\Psi(I')) & \subset & \Psi(I)\ . \end{array}\right. \tag{1.19}$$

Le corollaire 1.8 donne :

Commentary · Exposition [ms. 11]

Free diagrams and the projective construction

The reason for passing through the discrete category I_0 is that independent lifting problems can then be solved component by component. The left Kan extension freely inserts the naturality required by the arrows of I. Throughout this chapter $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$. For an object i of I, the left adjoint to evaluation at i is

$$i_!A(j) = \coprod_{I(j,i)} A.$$

Thus the free diagram on a family (A_i) has value $\coprod_i \coprod_{I(j,i)} A_i$ at j. This formula fixes the variance: it is the representable presheaf $I(-,i)$, not $I(i,-)$, that occurs.

Suppose (Φ, Ψ) is a mutually orthogonal pair and $S \subseteq \Phi$ generates it. The set

$$S_I = \{i_!s \mid i \in \mathrm{Ob}\, I,\ s \in S\}$$

generates the projective pair. Indeed, p has the right lifting property against $i_!s$ precisely when $p(i)$ has the right lifting property against s. Hence

$$S_I^{\square} = \Psi(I), \qquad {}^{\square}(S_I^{\square}) = \Phi\langle I\rangle.$$

This makes the smallness in Proposition 1.10 concrete. One need not use the much larger class of all objectwise Φ-arrows on the discrete category. If the domain A of s is small, then $i_!A$ is small for pointwise colimits because

$$\mathrm{Hom}(i_!A, X) = \mathrm{Hom}(A, X(i)).$$

The same common cardinal as in the original small object argument therefore suffices for these generators.

There is a small scope error in the preliminary equation (1.14). Its equality $\Psi(I_0) = \Phi(I_0)^{\square}$ is valid for the closed left classes used in Proposition 1.10: identities in Φ allow one to isolate a chosen coordinate, using identity arrows on the domains of the right-hand maps at all other coordinates. It is not valid for an arbitrary Φ merely satisfying $\Psi = \Phi^{\square}$. For instance, let $\Phi = \{\varnothing \to 1\}$ in Set and let I_0 have two objects. The arrow $(\varnothing \to \varnothing, \varnothing \to 1)$ has the right lifting property against the sole member $(\varnothing \to 1, \varnothing \to 1)$ of $\Phi(I_0)$, because no square can be formed in the first coordinate. Its second component is not surjective. Adding identities to Φ repairs this preliminary assertion; it is automatic in all the subsequent applications to a closed pair.

The source also asserts that the diagram category itself is special. Cocompleteness is pointwise. To check its object-smallness condition, fix $X : I^{\mathrm{op}} \to \mathcal{M}$ and choose one regular cardinal dominating the ranks of its set of values and the cardinality of the set of arrows of I. A natural transformation out of X is a family of component maps subject to one equation for each arrow of I. In a sufficiently filtered colimit, all these component maps occur at one stage, and all their naturality equations hold together at a later stage. Equality of two transformations is detected similarly. This proves smallness of X. If one uses the stronger filtered-presentability convention for accessible objects, the same argument is expressed by the equalizer of the two products computing the set of natural transformations.

Restriction plainly preserves the objectwise class $\Psi(I')$. Its left adjoint consequently preserves $\Phi\langle I\rangle$, by the preceding lifting correspondence. The dual construction uses objectwise left arrows $\Phi(I)$ and defines $\Psi[I] = \Phi(I)^{\square}$. This is the injective pair; its factorization is a different question. Any displayed right Kan extension is understood where its required limits exist. Cocompleteness and smallness of individual objects alone do not imply completeness: the ordered class of ordinals, with arrows $\alpha \to \beta$ for $\alpha \leq \beta$, has all small suprema and small individual objects, but no terminal object. This observation keeps the existence of the right adjoints separate from the lifting calculations that use them.

Proposition 1.10. *Supposons que $\Phi \longleftarrow\!\!\!\!\!\!+\!\!\!\!\!\!\longrightarrow \Psi$ soit un couple de Quillen* spécial *dans $\mathcal{M}$, i.e. de la forme (Φ, Φ_*), avec Φ partie Q-close à gauche spéciale dans $\mathcal{M}$ ($\mathcal{M}$ satisfaisant la condition* (1.10. a)*). Alors pour toute I dans* Cat, *$(\Phi\langle I\rangle, \Psi(I))$ est un couple spécial dans $\mathcal{M}(I)$, en particulier il satisfait au théorème de factorisation.*

Considérons la construction duale, qui nous fait introduire, à partir d'un couple $\Phi \longleftrightarrow \Psi$ dans $\mathcal{M}$, le couple

$$\Phi(I) \longleftrightarrow \Psi[I] \ , \tag{1.20}$$

où $\Phi(I)$ est défini par la formule générale (1.11) (où Ψ est remplacé par Φ), et où $\Psi[I]$ est défini soit comme $\Phi(I)_*$, soit comme

$$\Psi[I] \ = \ v_*(\Psi(I_0))^- \ = \ \Phi(I)_* \ . \tag{1.21}$$

On a alors, pour $u : I \longrightarrow I'$ [page 12]

$$\begin{cases} u^*(\Phi(I')) & \subset \quad \Phi(I) \\ u_*(\Psi[I]) & \subset \quad \Psi[I'] \ . \end{cases} \tag{1.22}$$

Il se pose encore la question si le couple $(\Phi(I), \Psi[I])$ satisfait au théorème de factorisation, sous des hypothèses convenables sur (Φ, Ψ). Je peux seulement hasarder la

Question-conjecture 1.11. *Supposons que (Φ, Ψ) soit un couple de Quillen* spécial, *en est-il de même de $(\Phi(I), \Psi[I])$ pour toute I dans* (Cat) *? En d'autres termes, si $\mathcal{M}$ est spéciale, condition* (1.10. a)*, et si Φ est une partie Q-close à gauche* spéciale *de $\mathcal{M}$, i.e. admettant une petite partie Q-génératrice, est-il vrai que pour toute I dans* Cat, *$\Phi(I)$ est également spéciale ? Je conjecture que oui, en faisant au besoin des hypothèses supplémentaires sur $\mathcal{M}$ (existence d'une filtration cardinale ?). P.ex. si Φ_0 est Q-génératrice dans Φ, et petite, alors $\Phi_0(I)$ est-elle Q-génératrice dans $\Phi(I)$?*

Commentary · Later developments [ms. 12]

The injective question under local presentability

Grothendieck's first question in 1.11 has a precise affirmative answer after the additional hypothesis of local presentability. In a locally presentable category, a weakly saturated class generated by a set remains generated by a set when imposed objectwise on a small diagram category. This is exactly Lurie (2009, Lemma A.2.8.3); its proof is supplied in the more general Lemma A.3.3.3. The corresponding statement for combinatorial categories, without introducing weak equivalences, is Makkai and Rosický (2014b, Corollary 3.4), using the numbering of the 2013 arXiv version. These statements apply directly to a cofibrantly generated weak factorization system and prove that $(\Phi(I), \Psi[I])$ is again cofibrantly generated.

The additional hypothesis is substantive. A category being cocomplete with individually accessible objects does not, by itself, specify a set of presentable objects generating the whole category under filtered colimits. The cited theorems therefore settle the locally presentable form of the question, not its entire literal generality. Our literature check did not establish a theorem with exactly that weaker generality. The final sentence of 1.11 asks for more than existence of *some* generating set: it proposes the particular set $\Phi_0(I)$. That proposal fails, as the following calculation shows.

Commentary · Examples [ms. 12]

Fixed points detect the missing generators

The difficulty with the proposed generators can be seen without homotopy theory. Work in Set with its usual choice convention. The single map $\varnothing \to 1$ generates the weak factorization system of injections and surjections: its right lifting property is surjectivity, injections lift against surjections by choosing preimages on their complements, and the standard factorization is

$$X \longrightarrow X \amalg Y \xrightarrow{[f,1_Y]} Y.$$

Take $I = BG$ for a nontrivial group G, say $G = C_2$. The diagram category consists of right G-sets. The class $\Phi_0(I)$ contains only the map from the empty G-set to the one-point trivial G-set. A G-map $p : X \to Y$ has its right lifting property if and only if $X^G \to Y^G$ is surjective: a map from the trivial point chooses a fixed point.

Now consider $p : \varnothing \to G$, where G acts on itself freely. Both fixed-point sets are empty, so p passes the proposed test. But the objectwise monomorphism $\varnothing \to G$ does not lift against p: in the square whose lower horizontal map is 1_G, a filler would be a map $G \to \varnothing$. Thus $\Phi_0(I)$ does not Q-generate $\Phi(I)$.

The correct generating family in this example is

$$\{\varnothing \to G/H \mid H \leq G\},$$

with right-coset or left-coset notation adjusted to the chosen action convention. An equivariant injection is obtained by attaching the orbits of its invariant complement; conversely, such attachments are injections, and injections are preserved under retracts. The right class consists of maps surjective on H-fixed points for every subgroup H. To see this directly, a map from a transitive orbit to Y is a point fixed by the stabilizer of that orbit, and a lift is a fixed preimage. Choosing such lifts orbit by orbit solves an arbitrary equivariant lifting problem against an injection. The single proposed test sees only $H = G$ and misses the free orbit $H = \{1\}$. This fully accounts for the failure; it does not contradict existence of the injective pair.

Commentary · Development (commentators') [ms. 12]

Exactly when the original singleton still generates

The preceding example suggests a sharper answer to Grothendieck's last sentence: one can characterize all diagram shapes for which his proposed singleton works. The following elementary result is our development of that sentence. We checked the generator constructions of Lurie (2009, Lemma A.2.8.3) and Makkai and Rosický (2014b, Corollary 3.4); we did not locate this precise characterization there. It is a calculation with constant presheaves, not a claim of priority.

Proposition (the constant-generator test). [ours] Let I be a small category, let Δ denote the constant-presheaf functor, and let $\Phi_0 = \{\varnothing \to 1\}$ in Set. Then $\Phi_0(I)$ Q-generates all monomorphisms in $\mathrm{Set}^{I^{\mathrm{op}}}$ if and only if I is empty or equivalent to the terminal category.

Proof. The class $\Phi_0(I)$ consists of the single map $\varnothing \to \Delta 1$: all its component domains and codomains are prescribed, and their transition maps are forced. A pushout of a coproduct of copies of this map adds a constant summand. Consequently every relative cell complex starting from the initial presheaf is $\varnothing \to \Delta S$ for some set S. This includes transfinite cell complexes, since colimits of the successive constant summands are computed pointwise. The small object argument applies, since the initial presheaf is small. Therefore every object P for which $\varnothing \to P$ belongs to the generated left class is a retract of some ΔS.

Suppose now that this left class contains every monomorphism and that I is nonempty. For every $i \in I$, the representable $h_i = I(-, i)$ is then a retract of a constant presheaf. Choose natural transformations

$$h_i \xrightarrow{a} \Delta S_i \xrightarrow{b} h_i, \qquad ba = 1_{h_i}.$$

For any arrow $\alpha : j \to k$ of I, naturality gives

$$a_j h_i(\alpha) = a_k, \qquad h_i(\alpha) b_k = b_j.$$

It follows that $h_i(\alpha)$ has inverse $b_k a_j$: composing in either order and using the two displayed equations yields the appropriate identity. Since this holds for every i, every map

$$I(k, i) \longrightarrow I(j, i), \qquad t \longmapsto t\alpha$$

is bijective. This forces α to be invertible. Explicitly, surjectivity with $i = j$ gives $\beta : k \to j$ with $\beta\alpha = 1_j$; injectivity with $i = k$ then gives $\alpha\beta = 1_k$, since $(\alpha\beta)\alpha = \alpha = 1_k\alpha$. Thus I is a groupoid.

Moreover, S_i is nonempty because $h_i(i)$ contains 1_i and maps into S_i. For every j, the map $b_j : S_i \to I(j, i)$ therefore shows that $I(j, i)$ is nonempty. The groupoid is connected. Finally, an endomorphism $\gamma : i \to i$ acts trivially on $h_i(i)$: naturality gives $a_i h_i(\gamma) = a_i$, and a_i is injective because it has a left inverse. Applying this identity to 1_i gives $\gamma = 1_i$. Every isotropy group is therefore trivial. A connected groupoid with trivial isotropy has exactly one arrow between any two objects, so it is equivalent to the terminal category.

Conversely, if I is equivalent to the terminal category, evaluation at any object is an equivalence $\mathrm{Set}^{I^{\mathrm{op}}} \simeq \mathrm{Set}$ and takes $\varnothing \to \Delta 1$ to $\varnothing \to 1$. Equivalences preserve and reflect lifting properties and monomorphisms, so the stated generation follows from the set calculation. If I is empty, the presheaf category has one object and one arrow; both the generated left class and its class of monomorphisms consist of that arrow. This proves all cases. □

The statement concerns the proposed *unchanged* generator. Even a discrete category with two objects fails the test: it asks one attachment to add a point simultaneously in both coordinates. The affirmative injective-generation theorem supplies a different set, large enough to express the independent and compatible attachments required by the chosen shape.

Proposition 1.12 [5]. *Soit* $\Phi \longleftrightarrow \Psi$ *dans* $\mathcal{M}$, *soit* $u : I \longrightarrow I'$ *une flèche dans* Cat, *d'où*

$$\left\{ \mathcal{M}(I) \underset{u_*}{\overset{u_!}{\underset{\longleftarrow}{\overset{\longrightarrow}{\longrightarrow}}}} \overset{u^*}{\longleftarrow} \mathcal{M}(I') \right. . \tag{1.12.1}$$

On a alors (1.19), *i.e.*

p. 8

$$\left\{ \begin{array}{rcl} u_!(\Phi\langle I\rangle) & \subset & \Phi\langle I'\rangle \\ u^*(\Psi(I')) & \subset & \Psi(I) \end{array} \right. , \tag{1.13.1}$$

et de plus l'équivalence

$$u^*(\Phi(I')) \subset \Phi\langle I'\rangle \qquad \Longleftrightarrow \qquad u_*(\Psi(I)) \subset \Psi(I') \ . \tag{1.13.2}$$

Ces relations équivalentes sont satisfaites dans chacun des trois cas [6] *:*

5. cf. XVII 6.4 (p. 117), mais où les notations pour $\Phi\langle I\rangle$, $\Psi(I)$ sont inverses.
6. **N.B.** On a $1^\circ \Longrightarrow 3^\circ$, $2^\circ \Longrightarrow 3^\circ$.

1°) *u est coétale.*

2°) *I est discrète.*

3°) *$\forall\, i' \in \operatorname{Ob} I'$, toute composante connexe de I/i' admet un objet final.*

[page 13]

Commentary · Exposition [ms. 12]

The comma-category test for preserving projective arrows

Proposition 1.12 identifies shapes for which restriction preserves the projective left class. This is a useful exception to the usual asymmetry: restriction always preserves objectwise right arrows, but it need not preserve the left arrows defined by a lifting property against them.

The printed formula (1.13.2) contains two slips that have to be corrected in interpreting the statement. Its left side should read

$$(*) \qquad u^*\big(\Phi\langle I'\rangle\big) \subseteq \Phi\langle I\rangle,$$

not $u^*(\Phi(I')) \subseteq \Phi\langle I'\rangle$. The latter has the wrong ambient category on the right and the wrong, objectwise, class on the left. The intended equivalence follows by applying the lifting correspondence to the adjunction $u^* \dashv u_*$:

$$\begin{aligned} u^*\Phi\langle I'\rangle \subseteq {}^{\square}\Psi(I) &\iff \Phi\langle I'\rangle \perp u_*\Psi(I) \\ &\iff u_*\Psi(I) \subseteq \Psi(I'). \end{aligned}$$

The source's established mutual-orthogonality identities are exactly what the last step needs.

To prove the sufficient condition, let $K_{i'} = (u \downarrow i')$: its objects are pairs $(i, u(i) \to i')$. For presheaves, the pointwise right Kan extension is

$$(u_*X)(i') = \lim_{K_{i'}^{\mathrm{op}}} X(i).$$

Suppose each connected component c of $K_{i'}$ has a terminal object $t_c = (i_c, u(i_c) \to i')$. Then t_c is initial in the opposite category, and a limit over that component is evaluation at i_c. Therefore

$$(u_*X)(i') \cong \prod_c X(i_c).$$

For an objectwise arrow in $\Psi(I)$, the induced arrow is a product of arrows in Ψ. Any right lifting class is closed under products: project a lifting square to each factor, choose its fillers, and combine them by the product property. The empty product gives an identity of a terminal object. Thus u_* preserves $\Psi(I)$ and (*) follows. This supplies the proof left implicit in the statement.

If I is discrete, every $K_{i'}$ is discrete and each nonempty component is a singleton. For a coétale functor in the source's convention, lifting a map into i' identifies a terminal lift in each component of $K_{i'}$. These are the two advertised special cases of the third condition. All formulas use the relevant products or right Kan extensions where they exist. No factorization theorem and no accessibility hypothesis are needed for this argument; those enter only when one wants the orthogonal pairs themselves to give replacement factorizations.

2 Rappels sur les triples de Quillen clos et les triples de Quillen spéciaux

Soit $(\mathcal{M}, W)$ une *catégorie de modèles,* i.e. $\mathcal{M}$ munie d'un localiseur $W \subset \mathrm{Fl}(\mathcal{M})$, i.e. une partie satisfaisant

(2.1) W contient les isomorphismes, et si f, g sont des flèches composables, si deux parmi f, g, gf sont dans W, la troisième aussi *(saturation faible).*

Je rappelle que la donnée d'un *triple de Quillen clos* (W, C, F), ou d'une structure de Quillen close, sur $\mathcal{M}$, ayant W comme ensemble de quasi-isomorphismes, revient à la donnée de deux couples de Quillen

$$C \longleftarrow\!\!\!\shortmid\!\!\!\longrightarrow TF\ , \qquad TC \longleftarrow\!\!\!\shortmid\!\!\!\longrightarrow F\ , \tag{2.2}$$

satisfaisant

$$TC \subset C \cap W\ , \qquad TF \subset F \cap W\ , \tag{2.3}$$

plus une des trois conditions équivalentes :

$$\begin{cases} \text{a)} & TC = C \cap W & \text{(i.e. } C \cap W \subset TC\text{)} \\ \text{b)} & TF = F \cap W & \text{(i.e. } F \cap W \subset TF\text{)} \\ \text{c)} & W = TF \circ TC\ . & \end{cases} \tag{2.4}$$

Rappelons aussi que ceci implique que W *est fortement saturée,* et a fortiori que $W = W^{\natural}$, i.e. W stable par facteurs directs.

Rappelons aussi qu'en vertu de (1.4), on a :

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[page 14]

Commentary · Exposition [ms. 13]

How the two factorizations recognize weak equivalences

The purpose of the second section is to add compatibility with a localizer to the machinery already established for one weak factorization system. Grothendieck's "category of models" at (2.1) means a category with a specified two-out-of-three class. It should not be silently identified with a modern model category. The model structure enters through the two factorizing pairs

$$(C, TF), \qquad (TC, F),$$

with $TC \subseteq C$ and $TF \subseteq F$. The letters T stand for the arrows intended to be trivial, and that intention has to be checked against W. For the equivalence in (2.4), the source explicitly assumes $TC \subseteq C \cap W$ and $TF \subseteq F \cap W$ before asking whether these inclusions are equalities.

The equivalence of the three conditions (2.4) is a short but instructive use of both factorizations. Suppose first that $TC = C \cap W$. If $f \in W$, factor $f = pi$ using (C, TF). Since $p \in W$, two-out-of-three gives $i \in C \cap W = TC$, hence $W \subseteq TF \circ TC$. The reverse inclusion follows because both factors belong to W. Conversely, if $W = TF \circ TC$ and $f \in C \cap W$, choose such a factorization. Since $f \perp p$, the retract argument makes f a retract of $i \in TC$, so $f \in TC$. This proves equivalence of (a) and (c). Using the other factorization and the dual retract argument proves equivalence with (b).

Formula (2.5) is another application of the same idea. To test $TC \subseteq W$ when W is retract-closed, it suffices to factor each arrow of TC as a W-arrow followed by an F-arrow; the left lifting property makes the original arrow a retract of the first factor. If $TC \subseteq W$ already holds, the (TC, F)-factorization supplies this for all arrows. The dual test is a factorization $f = pi$ with $i \in C$ and $p \in W$. The printed last clause "$i \in W$, $f \in C$" is incorrect and is retained as an edition slip, with its intended dual stated here.

The "special" condition adds small generators for both left classes, together with the ambient smallness hypotheses that make their small object arguments work. It is closely related to cofibrant generation in contemporary terminology, but stronger ambient hypotheses are explicitly included here. It is not synonymous with "combinatorial": local presentability is not part of (2.6). The recalled strong saturation of W uses the ambient model-category convention: Grothendieck (1990–1991e, Chapter XIII, §3, manuscript pp. 27–29) explicitly assumes M0, existence of finite limits and finite colimits, and invokes Quillen's saturation result. Strong saturation means that precisely the arrows of W become invertible in $\mathcal{M}[W^{-1}]$; retract-closedness follows since a retract of an isomorphism is an isomorphism. A modern reference is Hovey (1999a, Theorem 1.2.10(iv)). The audit records this as a recalled theorem with its inherited ambient hypotheses, not as a gap in the present proof.

(2.5) Étant donnés les deux couples (2.2), avec $TC \subset C$ (équivalent à $TF \subset F$), pour qu'on ait la première relation (2.3), i.e. $TC \subset W$, il faut, et il suffit si $W = W^\natural$, que toute flèche f de $\mathcal{M}$ (ou seulement toute flèche de TC) se factorise en $f = pi$, avec $i \in W$, $p \in F$. Dualement, pour qu'on ait $TF \subset W$, il faut et il suffit si $W = W^\natural$, que toute flèche f de $\mathcal{M}$ (ou seulement toute flèche de TF) se factorise en $f = pi$, avec $i \in W$, $f \in C$.

On dit que la structure de Quillen (W, C, F) est *spéciale* si les deux couples de Quillen (2.3) sont spéciaux, i.e. si les conditions suivantes sont satisfaites.

(2.6) a) $\mathcal{M}$ est stable par petites $\varinjlim$, et ses éléments sont accessibles ($\mathcal{M}$ "spéciale").

b) Il existe des *petites* parties $C_0 \subset C$, $TC_0 \subset TC$ qui soient Q-génératrices à gauche, i.e. telles que l'on ait

$$\left\{\begin{array}{lllllllll} C & \subset & C_0^\sim\,, & \text{i.e.} & C & = & C_0^\sim\,, & \text{i.e.} & TF = C_{0*}\,; \\ TC & \subset & (TC_0)^\sim\,, & \text{i.e.} & TC & = & (TC_0)^\sim\,, & \text{i.e.} & F = (TC_0)_*\,. \end{array}\right.$$

Ainsi, la donnée d'un tel triple de Quillen *spécial* dans $\mathcal{M}$, quand $\mathcal{M}$ satisfait (2.6. a), revient à la donnée des deux parties [page 15]

$$TC \subset C \tag{2.7}$$

de $\mathrm{Fl}(\mathcal{M})$, satisfaisant les conditions :

(2.8)

- C, TC "spéciales"
  - a) C, TC sont Q-closes à gauche.
  - b) C, TC sont Q-engendrées à gauche par des *petites* parties C_0, TC_0 de C, TC respectivement.
- c) $TC \subset W$, $TF \overset{\text{déf}}{=} C_* \subset W$.
- d) On a l'une des trois conditions équivalentes supplémentaires de (2.4), i.e. :

$$\left\{\begin{array}{lllllllll} \alpha) & TC & = & C \cap W\,, & \text{i.e.} & C \cap W & \subset & TC\,. \\ \beta) & TF & = & F \cap W\,, & \text{i.e.} & F \cap W & \subset & TF\,. \\ \gamma) & W & = & TF \circ TC\,. & & & & \end{array}\right.$$

Commentary · Exposition [ms. 15]

The extra condition in transferring a model structure

Before Proposition 2.1, it helps to identify what can go wrong. The adjunction transfers the two lifting relations formally, and small generation can provide both factorizations. Nevertheless, a map built from the transferred trivial generators might cease to be a weak equivalence after applying the right adjoint. The condition $TC' \subseteq W'$ rules out exactly that possibility.

Write $L = v_!$ and $R = v^*$. With

$$W' = R^{-1}W, \quad F' = R^{-1}F, \quad TF' = R^{-1}TF,$$

the adjunction identifies

$$C' = {}^{\square}TF', \qquad TC' = {}^{\square}F'.$$

Thus both pairs are already mutually orthogonal, although not yet known to factorize. Since $TC \subseteq C$, monotonicity gives $TC' \subseteq C'$, and inverse image gives $TF' \subseteq F'$. The class W' satisfies two-out-of-three because R is a functor. Once the two factorizations exist, the equation $TF' = F' \cap W'$ is built into the definitions; the previous block therefore reduces the entire model-structure question to $TC' \subseteq W'$.

The equivalent condition (2.1.5) has a useful practical interpretation: factor every target arrow into a weak equivalence followed by a member of F', without insisting in advance that the first arrow is in TC'. If the arrow being factored already belongs to TC', its left lifting property makes it a retract of that weak equivalence. Since W' is retract-closed, it belongs to W'. Conversely, a (TC', F')-factorization works once $TC' \subseteq W'$ is known.

The convenient stronger condition is $RL(TC) \subseteq TC$. It tests the images of the original trivial left arrows before taking their left closure. Cases A and B show two mechanisms by which that condition passes to the closure. Case C, completed on manuscript page 19, instead transports a factorization of Rf' through a fully faithful adjunction. This is why the three cases have different hypotheses. A right adjoint generally does not preserve pushouts, so the hypothesis of Case B must not be suppressed in a purported general transfer theorem.

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Proposition 2.1. *Soit*

$$\mathcal{M}' \underset{v^*}{\overset{v_!}{\leftrightarrows}} \mathcal{M} \tag{2.11}$$

un couple de foncteurs adjoints, $\mathcal{M}$ étant muni d'une structure de Quillen close (W, C, F). Posons

$$\left\{\begin{array}{rcl} W' & = & (v^*)^{-1}(W) \\ F' & = & (v^*)^{-1}(F) \\ TF' & = & F' \cap W' \;=\; (v^*)^{-1}(TF) \\ C' & = & v_!(C)^\sim \\ TC' & = & v_!(TC)^\sim \ . \end{array}\right. \tag{2.1.2}$$

1°) *On a alors*

$$\left\{\begin{array}{ccccccc} C' & \longleftrightarrow & TF' \ , & & TC' & \longleftrightarrow & F' \ , \\ TC' & \subset & C' \ , & & TF' & \subset & F' \ . \end{array}\right. \tag{2.1.3}$$

Pour que les données (2.1.2) *dans $\mathcal{M}'$ y correspondent à un triple de Quillen (W', C', F'), il faut et il suffit que les conditions suivantes soient satisfaites :* [page 16]

$$\left\{\begin{array}{ll} \text{a)} & \textit{Factorisation pour } C' \longleftrightarrow TF' \textit{ et } TC' \longleftrightarrow F', \textit{ i.e. les relations} \\ & C' \longleftarrow\!\!\!\shortmid\!\!\!\longrightarrow TF' \textit{ et } TC' \longleftarrow\!\!\!\shortmid\!\!\!\longrightarrow F', \\ \text{b)} & TC' \subset W', \end{array}\right. \tag{2.1.4}$$

ou encore, au lieu de b), *la condition équivalente* (en présence de a)) :

$$\left\{\begin{array}{ll} \text{b}') & \textit{Toute flèche } f' \textit{ dans } \mathcal{M}' \textit{ (non seulement } f' \in TC') \textit{ se factorise en} \\ & f' = p'i', \textit{ avec } i' \in W', \ p' \in F'. \end{array}\right. \tag{2.1.5}$$

2°) *Supposons* (2.1.4. a) *satisfaite (factorisation). Alors, pour que (W', C', F') soit un triple de Quillen clos, il* suffit *que l'on ait*

$$\text{b}'') \qquad \boxed{v^* v_!(TC) \ \subset \ TC} \ , \tag{2.1.6}$$

et que de plus on soit dans l'un des trois cas suivants :

A) *v^* admet un adjoint à droite v_*.*

B) *v^* commute aux petites $\varinjlim$, (W, C, F) est spécial, et $\mathcal{M}'$ est spéciale, i.e. satisfait à la condition* (2.6. a).

C) *v^* est pleinement fidèle, et $v^* v_!(F) \subset F$.* p. 11 [page 17]

Notons tout de suite le

Corollaire 2.2. *Supposons que (W, C, F) soit un triple de Quillen spécial, que $\mathcal{M}'$ soit spéciale, que v^* commute aux petites $\varinjlim$, enfin que l'on ait* (2.1.6)*, i.e. $v^* v_!(TC) \subset TC$. Alors (W', C', F') est un triple de Quillen spécial.*

En effet, en vertu du 2.1. 2°, cas B, pour voir que (W', C', F') soit un triple de Quillen clos, il est à vérifier (2.1.6. b″). Mais cela résulte de 1.8, appliqué aux deux couples de Quillen spéciaux (C, TF), (TC, F) dans $\mathcal{M}$: les couples (C', TC') et (TC', F') sont des couples de Quillen spéciaux, ce qui prouve en même temps que le triple (W', C', F') est spécial, q.e.d.

Démontrons 2.1. Les premières deux relations (2.1.3) sont contenues dans 1.6, les relations $TC' \subset C'$, $TF' \subset F'$ sont tautologiques. Le critère (2.1.4) pour que (W', C', F') soit un triple de Quillen clos, est contenu dans le rappel de la page 13, compte tenu que l'on sait déjà que $TF' = F' \cap W'$ (relation (2.4. b)). Que la condition (2.1.4. b), en présence de (2.1.4. a), soit équivalente à la condition (2.1.5. b′), est contenu dans la première assertion (2.5), en notant que l'on a $W' = W^{\pitchfork}$ (ce qui résulte de la définition de W' et de $W = W^{\pitchfork}$).

Il reste à prouver 2°), i.e. que la condition b) est satisfaite moyennant a) et b″) (2.1.6) dans chacun des trois cas A), B), C).

Cas A. Par définition de $W' = (v^*)^{-1}(W)$, pour prouver que $TC' \subset W'$, il suffit de prouver

$$v^*(TC') \ \subset \ W' \ ,$$

et a fortiori, il suffit de prouver

$$v^*(TC') \ \subset \ TC \ , \qquad \text{i.e. } v^*(TC') \rightarrowtail F \ . \tag{2.1.7}$$

Or par 1.5, appliqué au couple de foncteurs adjoints (v^*, v_*), cette relation équivaut à [page 18]

$$TC' \rightarrowtail v_*(F) \ .$$

Comme $TC' \overset{\text{déf}}{=} v_!(TC)^{\sim}$, ceci équivaut à

$$v_!(TC) \rightarrowtail v_*(F) \ ,$$

ou encore (en réappliquant 1.5 à $(v_!(TC), F)$ au lieu de (TC', F)), à

$$v^* v_!(TC) \rightarrowtail F \ , \qquad \text{i.e. } v^* v_!(TC) \subset \underbrace{TC}_{= F^*} \ ,$$

or c'est là l'hypothèse b″).

Cas B. Il suffit encore de prouver 2.1.7. Or, du fait que $v_!(TC)$ admet un petit ensemble Q-générateur à gauche, $v_!(TC)^{\sim}$ est la clôture de $v_!(TC)$ pour les opérations de cochangement de base, de composition transfinie, de facteurs directs. Comme v^* commute à toutes ces opérations (comme il commute aux petites $\varinjlim$), et comme TC dans $\mathrm{Fl}(\mathcal{M})$ est stable par ces opérations, on voit donc que pour montrer que v^* applique $v_!(TC)^{\sim}$ dans TC, il suffit de voir qu'il applique $v_!(TC)$ dans TC, ce qui est encore l'hypothèse b″).

Commentary · Exposition [ms. 18]

Two ways of passing from generators to their closure

The proofs of Cases A and B establish the stronger conclusion $R(TC') \subseteq TC$. They are worth comparing because one uses only lifting properties, whereas the other uses the cellular description supplied by small generation.

In Case A, R has a right adjoint $V = v_*$. For each $p \in F$, the condition $RL(TC) \perp p$ is equivalent to $L(TC) \perp Vp$. Taking the left double-orthogonal closure of $L(TC)$ does not change which arrows have the right lifting property against it; hence $TC' \perp Vp$. Applying the adjunction once more gives $R(TC') \perp p$. Since $TC = {}^{\square}F$, this proves the desired containment. This argument does not require R to preserve any colimit and does not require the ambient categories to be locally presentable.

In Case B, the small object argument identifies TC' with the retracts of cellular composites of a small family inside $L(TC)$. The functor R preserves pushouts and transfinite composites by the stated colimit hypothesis; every functor preserves retract diagrams. Since TC is closed under these operations and $RL(TC) \subseteq TC$, it follows that $R(TC') \subseteq TC$. One may equally use the full weakly saturated closure of $L(TC)$: Corollaries 1.8–1.9 show that its small generating family has the same left closure.

Corollary 2.2 combines this argument with Corollary 1.8, which gives both target factorizations and their small generators. Its printed proof contains two local slips. The first transferred pair is (C', TF'), not (C', TC'). Moreover, the remaining condition supplied by Corollary 1.8 is factorization (2.1.4.a), not (2.1.6.b″), which is already an explicit hypothesis of Corollary 2.2. In the subsequent Case A proof, $R(TC')$ is a class in $\mathcal{M}$, so the intermediate target must be W, not W'. With these corrections of interpretation, the proof establishes the corollary exactly as stated.

Cas C. On va vérifier b) sous la forme équivalente b′). Soit donc $f' \in \mathrm{Fl}(\mathcal{M}')$, $f' : X' \longrightarrow Y'$, posons p. 12

$$f \;=\; v^*(f'), \qquad \text{[page 19]}$$

et factorisons f en pi, avec $i \in TC$, $p \in F$, d'où

$$v_!v^*(f') = v_!(f) = v_!(p)v_!(i)\ .$$

Mais v^* étant pleinement fidèle, $v_!v^* \xrightarrow{\sim}$ id, donc on trouve une factorisation de f' en $v_!(p)v_!(i)$. Il suffit donc de voir que

$$v_!(p) \;\in\; F'\ , \qquad\qquad v_!(i) \;\in\; W'\ ,$$

ce qui équivaut à

$$v^*v_!(p) \in F\ , \qquad\qquad v^*v_!(i) \in W\ ,$$

et résulte des hypothèses que $v^*v_!(F) \subset F$ et $v^*v_!(TC) \subset TC \subset W$, q.e.d.

N.B. J'ai noté les cas A) et C) par acquis de conscience, sans application précise en vue. Le cas le plus utile sera justiciable du cas B), et mérite d'être réexplicité en un

Commentary · Exposition [ms. 19]

Free diagrams and the projective model structure

The concluding Case C can first be made completely explicit. Write $L = v_!$, $R = v^*$ and $\varepsilon : LR \to 1$ for the counit. Since R is fully faithful, ε is an isomorphism. For $f' : X' \to Y'$, choose in $\mathcal{M}$ a factorization

$$RX' \xrightarrow{i} Z \xrightarrow{p} RY', \qquad i \in TC, \quad p \in F.$$

Define

$$i' = L(i)\varepsilon_{X'}^{-1} : X' \longrightarrow LZ, \qquad p' = \varepsilon_{Y'}L(p) : LZ \longrightarrow Y'.$$

Naturality of the counit gives $p'i' = f'$. Moreover $R(i') = RL(i)R(\varepsilon_{X'}^{-1})$ lies in TC, since $RL(TC) \subseteq TC$ and TC is stable under composition with isomorphisms. Likewise $R(p') = R(\varepsilon_{Y'})RL(p)$ lies in F, using the other hypothesis $RL(F) \subseteq F$. Thus $i' \in W' = R^{-1}W$ and $p' \in F' = R^{-1}F$, proving the factorization condition (2.1.5). The two factorizing pairs required separately in Proposition 2.1 remain part of its hypotheses; this argument supplies their compatibility with weak equivalences.

Grothendieck now puts the transfer argument to its principal use: a homotopy theory of coefficients should supply a homotopy theory of diagrams of every small shape. The weak equivalences must be tested at the objects of the indexing category, since this is what allows a diagram to retain its homotopy coherence while its individual values are replaced. The freedom lies in choosing the auxiliary cofibrations and fibrations. For homotopy left Kan extensions, the useful choice fixes the fibrations componentwise and lets lifting determine the cofibrations.

Write $\mathcal{M}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})$, as in the source, and use $L^{\square}$ and ${}^{\square}R$ for right and left lifting classes. The intended formulas are

$$C_{\mathrm{proj}}(I) = {}^{\square}(F(I) \cap W(I)), \qquad TC_{\mathrm{proj}}(I) = {}^{\square}F(I).$$

The first equality in Theorem 2.3 as printed has $F(I)$ where $TF(I)$ is required. This is visible in the edition and is therefore retained in the transcription. Its correction follows already from (1.16), applied to (C, TF). With the printed reading, the alleged cofibrations would be precisely the trivial cofibrations, and even the case $I = e$ would give the wrong class.

The free-diagram adjunction makes the construction concrete. For $i \in I$, evaluation at i has a left adjoint

$$(i_!A)(k) = \coprod_{I(k,i)} A.$$

A lifting problem against $i_!a$ is exactly a lifting problem against a at i. If S and T generate the two lifting pairs in $\mathcal{M}$, the small families

$$\{i_!s \mid i \in \operatorname{Ob} I,\ s \in S\}, \qquad \{i_!t \mid i \in \operatorname{Ob} I,\ t \in T\}$$

therefore detect the componentwise trivial fibrations and fibrations. The required smallness is inherited because evaluation preserves the colimits used in the small object argument. Thus a domain small for a sufficiently long cellular construction in $\mathcal{M}$ remains small after applying $i_!$. The common bound may depend on I and on the generating families; no bound independent of every small category is needed.

To see why the second family produces weak equivalences, evaluate a cell attachment at k. It becomes a coproduct of pushouts of maps in T, indexed by the sets $I(k, i)$. Trivial cofibrations in $\mathcal{M}$ are closed under coproducts, pushouts, transfinite composites and retracts. The evaluated relative cell map is consequently a trivial cofibration. The transfer criterion applies. This is the projective model structure of present terminology; the modern combinatorial statement appears in Groth (2013, Theorem 1.28). The word "special" in this chapter includes small generating families and the stated accessibility of objects, but it does not by itself assert that the underlying category is locally presentable.

The rest of Theorem 2.3 is formal once these factorizations exist, but one step deserves to be made explicit. Suppose L preserves cofibrations and trivial cofibrations and that $f : X \to Y$ is a weak equivalence between cofibrant objects. Factor

$$(f, 1_Y) : X \amalg Y \xrightarrow{k} Z \xrightarrow{p} Y$$

with k a cofibration and p a trivial fibration. Let $i = k|_X$ and $j = k|_Y$. Both are cofibrations because X and Y are cofibrant. Since $pi = f$ and $pj = 1_Y$, two-out-of-three makes i and j trivial cofibrations. Now $L(p)L(j) = 1$, while $L(j)$ is weak; hence $L(p)$ is weak, and so is $L(f) = L(p)L(i)$. This proves exactly the homotopy invariance needed to define $u_!^W$ on cofibrant diagrams. It also explains why taking the ordinary Kan extension of an arbitrary representative is insufficient.

Théorème 2.3[7]. *Soit (W, C, F) un triple de Quillen spécial dans $\mathcal{M}$. Pour toute I dans* Cat*, considérons le triple $(W(I), C\langle I\rangle, F(I))$ dans $\mathcal{M}(I) = \underline{\mathrm{Hom}}(I^\circ, \mathcal{M})$, où les notations sont celles de page 10, donc $W(I)$, $F(I)$ et $TF(I)$ se définissent fibre par fibre, et $C\langle I\rangle = (F(I))^* = (v_{I\,!}(C(I_0)))^\sim$*[8].

1°) *Pour toute I dans* Cat*, $(W(I), C\langle I\rangle, F(I))$ est un triple de Quillen spécial dans $\mathcal{M}(I)$.* [page 20]

2°) *Si $u : I \longrightarrow I'$ est une flèche dans* Cat*, d'où*

$$\mathcal{M}(I) \underset{u^*}{\overset{u_!}{\rightleftarrows}} \mathcal{M}(I'), \tag{2.3.1}$$

alors

$$\left\{\begin{array}{rcl} u^*(W(I')) & \subset & W(I) \\ u^*(F(I')) & \subset & F(I) \\ u^*(TF(I')) & \subset & TF(I) \end{array}\right\} \textit{tautologie}$$
$$\left\{\begin{array}{rcl} u_!(C\langle I\rangle) & \subset & C\langle I'\rangle \\ u_!(TC\langle I\rangle) & \subset & TC\langle I'\rangle\ , \end{array}\right. \tag{2.3.2}$$

7. cf. scholie 2.10, qui devrait précéder le théorème 2.3.

8. où $v_I : I_0 \longrightarrow I$ (I_0 catégorie discrète ayant mêmes objets que I).

enfin, $u_!$ induit un foncteur

$$\mathcal{M}(I)_c \longrightarrow \mathcal{M}(I')_c \qquad (2.3.3)$$

[9]

compatible aux localiseurs $W(I)_c = W(I)|\mathcal{M}(I)_c$ *et* $W(I')_c = W(I')|\mathcal{M}(I')_c$, *donc induit un foncteur*

$$u_!^W \ : \ \mathrm{Ho}_W(I) \longrightarrow \mathrm{Ho}_W(I') \ . \qquad (2.3.4)$$

Ce foncteur est un adjoint à gauche du foncteur u_W^*, p. 13

$$u_W^* \ : \ \mathrm{Ho}_W(I') \longrightarrow \mathrm{Ho}_W(I) \ , \qquad (2.3.5)$$

induit par u^* (3.3.1).

Démonstration de 2.3.

Prouvons 1°). On considère comme dans 1.10 le foncteur

$$v = v_I \ : \ I_0 \longrightarrow I$$

et le couple de foncteurs adjoints

$$\mathcal{M}(I) \underset{v^*}{\overset{v_!}{\leftrightarrows}} \mathcal{M}(I_0) \ , \qquad (2.3.6)$$

auquel on peut appliquer 2.2. Il reste à vérifier seulement [page 21]

$$v^* v_!(TC) \ \subset \ TC \ , \qquad (2.3.7)$$

or on a, pour $X = (X_i)_{i\in I_0} \in \mathcal{M}(I_0)$,

$$\begin{aligned} v_!(X)(i) \ &\simeq \ \coprod_{j\in I_0} v_{j\,!}(X_j)(i) \\ &\simeq \ \coprod_{j\in I_0} \ \underbrace{X_j^{\mathrm{Hom}(i,j)}}_{\substack{\text{somme directe} \\ \text{de } \mathrm{Hom}(i,j) \text{ copies}}} \ , \end{aligned}$$

et la relation (2.3.7) résulte du fait que TC, étant Q-clos à gauche, est stable par petites sommes directes.

Commentary · Examples [ms. 21]

A constant diagram whose homotopy colimit is a classifying space

The free objects in the preceding proof have a geometric meaning: they introduce the room needed to carry the relations in a diagram through a replacement. A diagram may have cofibrant values and still fail to be cofibrant as a diagram. Group actions give a complete calculation of the difference.

Let G be a nontrivial group and let $I = BG$ be the category with one object and automorphism group G. A contravariant diagram of simplicial sets on BG is a simplicial set with a right G-action. Give simplicial sets their usual Kan model structure and give these action diagrams the projective structure. The ordinary colimit is the orbit simplicial set. In particular the colimit of the point with trivial action is a point. We calculate its derived colimit directly.

Let $\mathcal{E}_G$ be the category whose objects are the elements of G and which has exactly one morphism between any ordered pair of objects. Right translation gives a free action on this category. Its nerve EG has

$$(EG)_n = G^{n+1},$$

with faces deleting an entry and degeneracies repeating an entry; G acts diagonally on the right. The category $\mathcal{E}_G$ has an initial object, indeed every object is initial, so its nerve is contractible. An

9. Sous-catégories des objets cofibrants.

explicit contraction is obtained by choosing the identity element and inserting it as an extra first vertex. The augmentation $EG \to *$ is therefore a weak equivalence of the underlying simplicial sets. Since $\mathcal{E}_G$ is a groupoid, its nerve is Kan: a missing face in a horn is supplied by composition and inversion. Thus the augmentation is also a fibration.

It is also a projective cofibrant replacement. To verify cofibrancy without assuming an equivariant model theorem, filter EG by its skeleta. The action on its nondegenerate simplices is free: an element fixing the tuple $(g_0, \dots, g_n)$ must fix g_0, hence is the identity. After choosing one representative in every orbit of nondegenerate n-simplices, the n-skeleton is obtained from the preceding skeleton by attaching copies of

$$G \times \partial\Delta^n \longrightarrow G \times \Delta^n.$$

These are the free diagrams on the generating cofibrations of simplicial sets. Starting from the empty diagram proves that EG is projectively cofibrant.

Its orbit simplicial set is the ordinary bar construction BG. There is one vertex, an edge for every group element, and a two-simplex for every composable pair. Its fundamental group has generators $[g]$, $g \in G$, with relations

$$[1] = 1, \qquad [g][h] = [gh],$$

so the evident map identifies it with G. Thus the derived colimit of the constant point is

$$\operatorname{hocolim}_{BG^{\mathrm{op}}} * \simeq EG/G \simeq BG,$$

which cannot be contractible for nontrivial G. For $G = C_2$ the first homology group is $\mathbb{Z}/2$, whereas the first homology of the ordinary orbit point vanishes. The calculation also proves that the trivial-action point was not projectively cofibrant: otherwise the weak equivalence $EG \to *$ between cofibrant diagrams would remain a weak equivalence after taking orbits, contrary to the calculation.

Every individual simplicial set in this example is cofibrant. What the replacement corrects is therefore the action, including its coherent relations, rather than the values of the diagram. This is the concrete content of the angle brackets in $C\langle I\rangle$. The same distinction will reappear with arrows reversed when injectively fibrant diagrams are used to compute right Kan extensions. The general diagram conventions and their Kan adjunctions are recorded in Lurie (2009, Definitions A.2.8.1 and Proposition A.2.8.7); the calculation above gives the elementary free resolution behind this instance.

Prouvons 2°). Les formules (2.3.2) sont essentiellement tautologiques, les deux dernières résultent de la troisième et de la deuxième, en vertu de 1.5 appliqué au couple de foncteurs adjoints $(u_!, u^*)$. La relation $u_!(C\langle I\rangle) \subset C\langle I'\rangle$, et le fait que $u_!$ transforme objet initial en objet initial, impliquent (2.3.3). Que $u_!$ restreint à $\mathcal{M}(I)_c$ soit compatible aux localiseurs provient de $u_!(TC\langle I\rangle) \subset TC\langle I'\rangle$, et du fait que toute flèche f dans $\mathcal{M}(I)_c$ qui est dans $W(I)$, se décompose en $f = pi$, avec $i \in TC$ et p inverse à gauche d'une $j \in TC\langle I\rangle$. Ceci vu, joint aux deux premières formules (2.3.2) montre qu'on est sous les conditions du théorème d'adjonction de Quillen, d'où l'adjonction de $u_!^W$ et de u_W^*, q.e.d. [page 22]

Corollaire 2.4. *Soit W un localiseur dans $\mathcal{M}$, tel qu'il existe un triple de Quillen spécial (W, C, F) incluant W. Alors le prédérivateur* Ho_W *sur* Cat *satisfait aux axiomes Dér 1 (décomposition), Dér 2 (localisation), Dér 3 (objet final), Der 4 bis (existence des $u_!^W$), Dér 5 bis (calcul des fibres des $u_!^W$), Dér 6 bis (exactitude à gauche, i.e. exactitude à gauche des carrés W-cocartésiens dans les* $\mathrm{Ho}_W(I)$*), Dér 6 (exactitude à droite des carrés W-cartésiens).*

On vient de prouver Dér 4 bis, et Dér 5 bis est pratiquement contenu dans le mode d'obtenir de $u_!$, compte tenu de proposition 1.12, qui implique que pour X dans $\mathcal{M}(I)_c$, et i' dans I', l'image inverse de X dans $\mathcal{M}(i'\backslash I)$ est dans $\mathcal{M}(i'\backslash I)_c$ …

L'axiome de décomposition est satisfait pour tout prédérivateur de la forme $\mathbf{D}_{\mathcal{M},W}$. p. 14
L'axiome d'exactitude à gauche Dér 5 bis est valable, pourvu que les $(\mathcal{M}(I), W(I))$ s'insèrent [page 23]
dans une structure à cofibrations, ce qui est le cas. De même pour l'axiome Dér 2 de localisation, pourvu que de plus W soit fortement saturé, ce qui est également le cas. Enfin, l'axiome de l'objet final dit que si I admet un objet final, alors

$$\mathrm{Ho}_W(e) \longrightarrow \mathrm{Ho}_W(I)$$

est pleinement fidèle, ou encore que le morphisme d'adjonction

$$u_! u^*(\xi) \longrightarrow \xi$$

est un isomorphisme. Mais en vertu de 1.12, cas 3°, dans ce cas u^* applique $\mathcal{M}(e)_c = \mathcal{M}_c$ dans $\mathcal{M}(I)_c$, donc le calcul du couple de foncteurs adjoints $u_!^W$, u_W^* peut se faire en utilisant les restrictions de $u_!$, u^* à $\mathcal{M}(I)_c$ et $\mathcal{M}_c$ respectivement. Comme le morphisme d'adjonction $u_!^c u_c^* \longrightarrow \mathrm{id}$ est un isomorphisme, il en est donc de même de $u_!^W u_W^* \longrightarrow \mathrm{id}$, qui s'en déduit en localisant, q.e.d.

Commentary · Exposition [ms. 23]

The comma formula and the first derivator axioms

Grothendieck wants more than an adjoint on each localized diagram category. He wants its values to be calculable from the part of the diagram lying over a chosen object. This local calculation is what makes the construction a derivator and permits the subsequent base-change arguments. The opposites in his convention must be retained throughout the calculation.

Let $u : I \to J$, $j \in J$, and let $q : j \downarrow u \to I$ forget the arrow $j \to u(i)$. For a contravariant diagram X, the ordinary formula is

$$(u_! X)(j) = \mathrm{colim}_{(j\downarrow u)^{\mathrm{op}}} q^* X.$$

To derive this equality, one must know that the same representatives compute both sides. Put $E = j \downarrow u$. The functor $q : E \to I$ is a discrete opfibration: an arrow $i \to i'$ carries $(i, a : j \to u(i))$ to $(i', u(i \to i')a)$, uniquely. For every $i \in I$ and object $A \in \mathcal{M}$, this unique-lifting property gives

$$q^*(i_! A) \cong \coprod_{e \in q^{-1}(i)} e_! A.$$

Indeed, at $e' \in E$, the indexing sets on the two sides are identified by the bijection

$$\coprod_{e \in q^{-1}(i)} E(e', e) \xrightarrow{\sim} I(q(e'), i).$$

Each arrow on the right has exactly one lift with source e', whose target is in the indicated fibre. Uniqueness also makes the bijection compatible with precomposition, so it defines an isomorphism of diagrams, natural in A. The same formula holds for arrows a in place of objects A.

Apply it to the two generating families in the preceding block. Restriction sends each generating projective cofibration to a coproduct of generating projective cofibrations in $\mathcal{M}(E)$, and likewise for trivial cofibrations. It preserves colimits, computed pointwise, and every functor preserves retract diagrams. It consequently preserves the cellular and retract closures of both families, hence all projective cofibrations and trivial cofibrations. It also preserves the initial object, so q^*QX is projectively cofibrant whenever QX is. This proof uses precisely the cocompleteness and generation hypotheses already available here.

Apply the ordinary formula to a projectively cofibrant replacement $QX \to X$. On the left it computes $u_!^W X$; on the right it computes the homotopy colimit of q^*X. Hence

$$(u_!^W X)_j \cong \mathrm{hocolim}_{(j\downarrow u)^{\mathrm{op}}} q^* X.$$

The isomorphism is the canonical comparison obtained from the comma square, since the ordinary comparison was used before localization. This last point matters: an abstract isomorphism between the two objects would not establish the base-change axiom.

The argument for a terminal object in I is similarly precise. If t is terminal, then $I(i, t)$ is a singleton for each i, so $t_! A$ is the constant diagram on A. Thus the constant-diagram functor takes cofibrant objects to projectively cofibrant diagrams. Since I is nonempty and connected, the colimit of a constant diagram is A, and the counit is an isomorphism already before localization. Its derived counit is consequently an isomorphism. This establishes full faithfulness of constant coefficients in the form claimed by Dér 3.

There is one overstatement in the paragraph being commented on. Decomposition for arbitrary small disjoint unions does not follow for an entirely arbitrary relative category merely from the

formula $\mathbf{D}_{\mathcal{M},W}(I) = \mathcal{M}(I)[W(I)^{-1}]$. Localization need not commute with infinite products: a single morphism in a product localization has a finite zigzag presentation whose length is uniform over all components. The commentary on Chapter VI, manuscript p. 131, addresses that obstruction, and should be consulted here. In the present model-category situation the conclusion is valid. Componentwise replacements and the description of maps between cofibrant-fibrant objects by homotopy classes supply uniform representatives and uniform homotopies. The same description proves that evaluation detects isomorphisms: a representative with weak-equivalence components is a weak equivalence in the diagram model structure.

Finally, the two exactness axioms concern finite homotopy pushouts and pullbacks. They do not require that arbitrary right Kan extensions have already been constructed. The cofibration and fibration structures provide the required finite calculus. The reference to "Dér 5 bis" for left exactness on manuscript page 23 should read "Dér 6 bis"; the statement on page 22 has the right numbering. The author supplies the omitted right-exactness argument himself on page 24. These are local repairs in the exposition, not failures of Corollary 2.4.

2.5. Pour achever de construire Ho_W comme un dérivateur sur Cat tout entier, il faut donc établir Dér 4, Dér 5 : existence des u_*^W, et calcul des fibres de u_*^W. Quant à Dér 6, il est acquis automatiquement, du fait que chaque $(\mathcal{M}(I), W(I))$ s'insère dans une structure de fibration $(W(I), F(I))$. (Oubli dans la démonstration de 2.4.) L'idée naturelle est d'utiliser à présent la structure de triple [page 24]

$$(2.5.1) \qquad (W(I), C(I), \quad \underbrace{F[I]}_{\overset{\text{déf}}{=} TC(I)_*} \quad) \ ,$$

mais la difficulté est de prouver la factorisation pour un couple

$$(2.5.2) \qquad C(I) \longleftrightarrow TF[I] \ , \qquad \underbrace{TC(I)}_{\overset{\text{déf}}{=} C(I)\cap W(I)} \qquad \longleftrightarrow F[I] \ ,$$

faute de pouvoir montrer, dans les conditions générales de 2.3 seulement (triple de Quillen spécial) que $C(I)$ et $TC(I)$ sont également spéciaux. (Cf. question 1.11.) Mais moyennant des conditions convenables sur $\mathcal{M}$, on arrivera à s'en tirer si $C = mono$. L'énoncé pertinent, un peu technique, se trouve dans XV cor. 7.4 (p. 98-99), que je vais rappeler ici.

Commentary · Context [ms. 24]

Why monomorphisms enter the construction

The obstruction to right Kan extensions is now a problem about generating cofibrations. The source already knows how to generate the projective classes by free diagrams. The injective classes are prescribed componentwise, and a small generating family for them is not produced by simply taking the old generators componentwise. Grothendieck returns to monomorphisms because subobjects provide a way to control the size of successive attachments. This is the same geometry of unions and injective objects that underlies the older theory of sheaves and abelian categories.

The six conditions in Theorem 2.6 separate this size control from the exactness needed to glue subobjects. Cocompleteness and accessibility of individual objects supply the small object argument. The cardinal filtration adds a supply of objects bounded in size; it must not be silently absorbed into the word "special." Left exactness of filtered colimits allows intersections to pass through increasing unions. Condition (e) says that the union of two subobjects can be formed by their pushout over the intersection. Condition (f) ensures that attaching such a piece preserves monicity even when the attaching map is not itself monic. These hypotheses explain why the subsequent proof can remain inside the category of subobjects of a fixed target.

The direction of the limit in (2.6.1d) requires an editorial correction. The edition prints a projective-limit arrow. The cited antecedent, XV, Theorem 6.1(d), manuscript page 76, has inductive limits, and XVIII, Proposition 4.6.7, manuscript page 157, likewise gives closure of the cardinal stages under inductive limits of bounded size (Grothendieck, 1990–1991k,-). This is also what the intended examples require. For the ordinary filtration of sets by cardinality at most π,

a coproduct of at most π sets of that size still has size at most π. By contrast, the product of π copies of a two-element set has size $2^\pi > \pi$. We preserve the printed sign and use the intended inductive-limit condition when explaining the theorem.

The examples in the footnotes give a useful test of the exactness assumptions. In an abelian category the pushout of subobjects $A_1, A_2 \subset B$ over their intersection is their sum $A_1 + A_2$, which embeds in B. If $i : A_0 \hookrightarrow A_1$ is monic and $f : A_0 \to A_2$ is arbitrary, the pushout is

$$(A_1 \oplus A_2)/\operatorname{im}(i, -f).$$

The map from A_2 into this quotient is monic: an element of the intersection of $0 \oplus A_2$ with $\operatorname{im}(i, -f)$ must come from an element killed by i, hence from zero. The same proof can be phrased with kernels and images in any abelian category. Thus these conditions extend beyond topoi.

For groups the effective-union condition fails visibly. Inside $C_2 \times C_2$, take the two coordinate subgroups. Their intersection is trivial, while their pushout in groups is $C_2 * C_2$. The canonical map to $C_2 \times C_2$ kills the commutator of the two nonidentity generators. That commutator is a nonempty reduced word in the free product, so the map is not injective. The appeal to the absence of enough injectives in groups is therefore not an arbitrary warning: the precise gluing mechanism used by the theorem has already broken. The cited SGA passage establishes that there are groups which embed in no injective group (Artin et al., 1972, Exposé I, Example 8.12.9).

The tentative footnote about removing condition (d) should be read alongside Grothendieck's own refinement in XVIII, 4.6.7. There he replaces the excessive demand on all quotients by factorization into an effective epimorphism followed by a monomorphism, recovering the weaker quotient condition actually used in XV. This is a specified replacement of hypotheses; it does not justify the projective-limit sign printed here, nor does it establish a small generating set for every accessible class of monomorphisms.

p. 15
[page 25]

Théorème 2.6. *Soit $\mathcal{M}$ une catégorie satisfaisant les conditions suivantes.*

(2.6.1)

a) *$\mathcal{M}$ est "spéciale", i.e. stable par petites $\varinjlim$, et ses éléments sont accessibles.*

b) *Les limites inductives filtrantes dans $\mathcal{M}$ sont exactes à gauche (et dans $\mathcal{M}$ les $\varprojlim$ finies sont représentables – résulte de* a *et* c*).*

c) *$\mathcal{M}$ admet une filtration cardinale $(\mathrm{Filt}^\pi)_{\pi \geq \pi_0}$.*

d) *Les Filt^π sont stables par $\varprojlim$ indexées par des catégories I telles que $\operatorname{card} \mathrm{Fl}(I) \leq \pi$, et par quotients ($Z \in \mathrm{Filt}^\pi$ et $Z \longrightarrow Z'$ épimorphisme implique $Z' \in \mathrm{Filt}^\pi$).*

e) *Si A_1, A_2 sont deux sous-objets d'un objet B de $\mathcal{M}$, et $A_0 = A_1 \cap A_2$, alors $A_1 \amalg_{A_0} A_2 \longrightarrow B$ est un monomorphisme.*

f) *Si on a un carré cocartésien*

$$\begin{array}{ccc} A_0 & \xrightarrow{i} & A_1 \\ \downarrow & & \downarrow \\ A_2 & \xrightarrow{i'} & A_3 \end{array}$$

avec i monomorphisme, alors i' est un monomorphisme.

[10,11,12,13]. *(On dira alors que $\mathcal{M}$ est* "très spéciale"*). Soit de plus*

$$W \subset \mathrm{Fl}(\mathcal{M}) \tag{2.6.2}$$

un localiseur dans $\mathcal{M}$, qui satisfait les conditions

(2.6.3)
- a) *W est stable par $\varinjlim$ filtrantes* [14].
- b) *W est de plus L_1-accessible, i.e. il existe une* petite *partie $W_0 \subset W$ telle que W soit la L_1-enveloppe de W, i.e. que W soit la plus petite partie de $\mathrm{Fl}(\mathcal{M})$ contenant W_0, et stable par $\varinjlim$ filtrantes.*

Posons alors

p. 16

$$C = \mathrm{Mono}(\mathcal{M}), \qquad TC = C \cap W\ , \qquad TF = C_*\ , \qquad F = (TC)_*\ . \tag{2.6.4}$$

[page 26]

Sous ces conditions :

1°) *$C \leftarrow\!\!\!\!\!\!\rightarrow TF$, plus précisément $C \rightarrowtail TF$ est un couple de Quillen spécial.*

2°) *Pour que $TC \rightarrowtail F$ soit un couple de Quillen, il faut et il suffit que TC soit stable par cochangement de base, et alors c'est un couple de Quillen spécial.*

3°) *Pour que (W, C, F) soit un triple de Quillen (lequel sera nécessairement spécial par* 1°, 2°*, et en particulier clos), il faut et il suffit que les conditions suivantes soient satisfaites.*

(2.6.5) [15]
- a) TC ($\overset{\text{déf}}{=} W \cap \mathrm{Mono}$) *est stable par cochangement de base.*
- b) TF ($\overset{\text{déf}}{=} (\mathrm{Mono})_*$) = *"applications de type injectif"* $\subset W$.

Et la condition b) *équivaut encore à :*

(2.6.6)
- b′) *Toute flèche f de $\mathcal{M}$ (ou seulement de TF) se factorise en $f = pi$, avec i monomorphisme, $p \in W$.*

Corollaire 2.7. *$\mathcal{M}$ et W étant comme dans* (2.6.1), (2.6.3)*, pour que (W, C, F) soit un triple de Quillen* propre à gauche *(i.e. $C \subset \mathrm{Cof}_W$), il faut et il suffit que W satisfasse la condition* (2.6.5. b) *(i.e. W contient les flèches de type injectif), et de plus*

$C \subset \mathrm{Cof}_W$, *i.e. la condition :*

a′) *Pour tout diagramme cocartésien*

[page 27]

$$\begin{array}{ccc} A_0 & \xrightarrow{i} & A_1 \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ A_2 & \xrightarrow{i'} & A_3 \end{array} \tag{2.7.7}$$

avec i monomorphisme, si $f \in W$, alors $f' \in W$. (Stabilité de W par cochangement de base par monomorphismes.)

10. **N.B.** a + c signifie aussi que $\mathcal{M}$ est une "catégorie $\mathfrak{U}$-paratopos", et implique l'existence d'une filtration cardinale satisfaisant d) (sauf l'assertion concernant les quotients Z' de Z). Donc il s'agit d'un $\mathfrak{U}$-paratopos satisfaisant de plus b) e) f). (Est-ce alors un topos ? *Non,* car e), f) sont O.K. dans les *catégories abéliennes !*)

11. **N.B.** La condition d) est sans doute inutile, du moins si les épimorphismes sont effectifs, car d) sera satisfaite en tous cas pour $\pi = 2^c$, avec $c \geq \pi_0$.

12. **N.B.** La condition e) n'est *pas* satisfaite pour la catégorie (Gr), et le théorème est en défaut pour cette catégorie, laquelle n'admet *pas* "assez d'injectifs", cf. SGA 4 I 8.12.9. Par contre, e) et f) sont satisfaites par les catégories *abéliennes.*

13. Cf. XVIII 4.6.7 (p. 157) pour une formulation plus sympathique de l'ensemble des conditions a) à f) pour $\mathcal{M}$.

14. **N.B.** a) implique déjà que $W = W^{\natural}$.

15. **N.B.** Ces conditions (2.6.5) équivalent à dire que (W, C) est une structure de catégorie à cofibrations sur $\mathcal{M}$.

Sous ces conditions, ce triple de Quillen est spécial, et a fortiori clos.

Ce corollaire résulte de 2.6. 3°), puisque $C \subset \mathrm{Cof}_W$ implique $TC \overset{\text{déf}}{=} C \cap W = C \cap W^{\mathrm{univ}}$ (car $\mathrm{Cof}_W \cap W = W^{\mathrm{univ}}$), donc que TC est stable par cochangement de base, d'où la condition (2.6.5. a).

Commentary · Exposition [ms. 27]

The recognition argument and its essential cancellation

Grothendieck's criterion is designed so that the difficult set-theoretic work produces two factorizations, after which the model axioms follow from a short argument. Keeping these stages distinct also prevents the recalled theorem from being confused with a broader assertion about arbitrary accessible classes of monomorphisms. The latter is not what Theorem 2.6 needs.

Set $C = \mathrm{Mono}$, $TC = C \cap W$, $TF = C^{\square}$ and $F = (TC)^{\square}$. Suppose that the two asserted weak factorization systems have been obtained and that $TF \subset W$. Then $TF \subset F$ because $TC \subset C$. We verify the remaining equality $F \cap W = TF$. Given $f \in F \cap W$, use the first factorization system to write

$$f = pi, \qquad i \in C, \quad p \in TF.$$

Since $p, f \in W$, two-out-of-three puts i in TC. Since f has the right lifting property against i, the usual diagonal makes f a retract of p in the arrow category. Right lifting classes are retract-closed, so $f \in TF$. The reverse inclusion has already been observed. Hence the two weak factorization systems have exactly the intersections required for a closed Quillen triple.

The alternative condition (2.6.6) is the same retract argument in a different position. If $TF \subset W$, factorization with respect to (C, TF) supplies a monomorphism followed by a weak equivalence. Conversely suppose each $f \in TF$ factors as $f = pi$ with i monic and $p \in W$. The right lifting property of f against i makes f a retract of p, so $f \in W$, provided W is retract-closed. That last property is already forced by (2.6.3a): if f is a retract of $g \in W$, split the associated idempotent $e : g \to g$ by the chain

$$g \xrightarrow{e} g \xrightarrow{e} g \xrightarrow{e} \cdots .$$

Its colimit in the arrow category is f. Filtered-colimit closure therefore gives $f \in W$.

The historical proof of the generating-set assertion should be located accurately. XV, Theorem 6.1 carries the author's warning that its proof is invalid. The surviving argument is the separate Lemma 7.3 and its application in 7.1–7.4. Starting with $A \hookrightarrow B$ in TC, one chooses a bounded subobject $B_0 \subset B$ for which $A \cap B_0 \hookrightarrow B_0$ lies in TC and adds it to A. The density argument permits B_0 to contain a prescribed bounded piece of B; when $A \neq B$, choose that piece not contained in A. Thus the union strictly enlarges A. Effective unions keep $A \cup B_0 \to B$ monic; pushout stability makes $A \to A \cup B_0$ weak. Since $A \to B$ is weak, cancellation makes $A \cup B_0 \to B$ weak. Repeat this construction, taking unions at limit stages. Filtered-colimit closure keeps the resulting map to B in TC at such stages. Since B has only a set of subobjects, strict enlargement cannot continue indefinitely: the process reaches B. The cancellation step is the extra ingredient missing for an arbitrary accessible mono class (Grothendieck, 1990–1991k, §7, especially Lemma 7.3 and Corollary 7.4).

A later independent formulation is Smith's recognition theorem, proved in Beke (2000, Theorem 1.7): with a locally presentable underlying category and a chosen set generating the cofibrations, accessibility supplies the solution-set condition for weak equivalences, and pushout/transfinite closure of the trivial cofibrations completes the construction. Beke's Proposition 1.12 identifies hypotheses ensuring that all monomorphisms admit such generators. This provides a modern route to the intended very-special case without appealing to the withdrawn general claim.

Corollary 2.7 records an additional condition, left properness. Its pushout square tests whether a weak equivalence remains weak after attachment along a monomorphism. When the attached map itself is both a monomorphism and a weak equivalence, the two-out-of-three argument in the source identifies it with a universally weak map, whose every cobase change is weak. Hence left properness supplies the pushout stability of TC required by Theorem 2.6. The recognition theorem needs that stability; the stronger properness property is singled out for applications rather than inserted into the definition of a model structure.

2.8. On dit qu'une *structure de Quillen* (W, C, F) sur $\mathcal{M}$ est *très spéciale,* si elle est p. 17

obtenue comme au théorème 2.1., i.e. si on a les conditions :

(2.8.1)
$$\begin{cases} \text{a) } \mathcal{M} \text{ est très spéciale, i.e. satisfait aux conditions (2.6.1).} \\ \text{b) } W \text{ est stable par petites } \varinjlim \text{ filtrantes, et est accessible (conditions (2.6.3)).} \\ \text{c) } C = \text{Mono.} \end{cases}$$

Donc pour qu'un localiseur W dans $\mathcal{M}$ corresponde à une structure de Quillen très spéciale dans $\mathcal{M}$ (laquelle sera forcément unique), il faut et il suffit que $\mathcal{M}$ et W satisfassent les conditions (2.8.1. a) b)), et que de plus on ait : [page 28]

(2.8.2)
$$\begin{cases} \text{a) } TC \overset{\text{déf}}{=} \text{Mono} \cap W \text{ est stable par cochangement de base.} \\ \text{b) } TF \overset{\text{déf}}{=} (\text{Mono})_* \subset W, \text{ i.e. } W \text{ contient les flèches de type injectif dans } \mathcal{M}. \end{cases}$$

[16].

Commentary · Later developments [ms. 28]

Injective structures and the role of accessibility

Grothendieck has reached a class of coefficient categories for which the second half of the diagram theory can be constructed. Theorem 2.9 uses componentwise cofibrations, so restriction along every functor preserves both cofibrations and trivial cofibrations. The right Kan adjoint is then a right Quillen functor. This is the injective model structure; the square brackets in $F[I]$ indicate that its fibrations have been determined by a lifting property on whole diagrams.

There are two sorts of permanence hidden in the phrase "par AQT." The exactness conditions are evaluated at objects of I: limits, colimits, intersections, pushouts and the maps from unions to ambient objects are computed there. Monicity is also componentwise. For the converse direction of that assertion, apply the defining monomorphism test to free diagrams $i_!A$: monicity of $X \to Y$ in the diagram category says that $\mathcal{M}(A, X_i) \to \mathcal{M}(A, Y_i)$ is injective for every A, hence $X_i \to Y_i$ is monic. The existence of the free adjoint is what makes this argument valid in the stated generality.

The size statements require a common bound. One chooses a cardinal larger than the arrow set of I and enlarges the bounds used for the coefficients and for W. A small diagram then involves only a bounded family of objects, structure maps and equations. A sufficiently large accessible presentation can carry all these data at one stage. Equalities of composites are imposed at a later common stage, and the filtered system of such choices recovers the original diagram. Thus the bounds are allowed to grow with the shape. The source's cardinal-filtration hypothesis is doing real work here; accessibility of each individual object alone would not justify this conclusion. A precise antecedent for the diagram-filtration step is Artin et al. (1972, Exposé I, Corollary 9.24), with the indexing category held fixed and the cardinal enlarged as prescribed there.

In the later combinatorial formulation, Lurie (2009, Lemma A.2.8.3) proves that the componentwise class associated to a set-generated weakly saturated class is again set-generated in a small diagram category. Proposition A.2.8.2 then gives both projective and injective model structures. This settles the obstacle of page 24 when local presentability is imposed; it should not be quoted as a theorem about every historical "special" category with that hypothesis omitted. For presheaves with an accessible localizer and monomorphism cofibrations, the corresponding statement is Cisinski (2006a, Corollary 1.4.22). These references supply exact later versions of the passage from coefficients to diagrams.

The derived right adjunction itself can be checked without new size arguments. Since u^* preserves the componentwise trivial cofibrations, adjunction shows that u_* preserves injective fibrations. Applying the same reasoning to all componentwise cofibrations gives preservation of trivial fibrations. It preserves the terminal object, hence fibrant objects. The dual of the Brown argument above shows that it preserves weak equivalences between those objects. Localizing its restriction to fibrant objects defines u_*^W, and the derived unit and counit give

$$u_W^* \dashv u_*^W.$$

16. On dit aussi que le localiseur W est très spécial, s'il définit une structure de Quillen très spéciale.

The replacement needed for the pointwise formula can also be specified. Let $q : E = (u \downarrow j) \to I$ forget the arrow $u(i) \to j$. This projection is a discrete fibration: an arrow $i' \to i$ uniquely lifts into an object $(i, u(i) \to j)$ by precomposition. Consequently its left Kan extension on contravariant diagrams is

$$(q_! Y)(i) \cong \coprod_{e \in q^{-1}(i)} Y(e).$$

Coproducts preserve cofibrations and trivial cofibrations, so $q_!$ preserves both componentwise left classes. The adjunction $q_! \dashv q^*$ therefore makes q^* a right Quillen functor for the injective structures. In particular, if $X \to RX$ is an injective fibrant replacement, then q^*RX is injectively fibrant and $q^*X \to q^*RX$ is a weak equivalence. Applying the ordinary right Kan formula to RX now gives

$$(u_*^W X)_j \cong \mathrm{holim}_{(u\downarrow j)^{\mathrm{op}}} q^* X.$$

Here the comma category has objects $u(i) \to j$, opposite to the comma category used for the left extension. This accounts for the two different fibre formulas in the derivator axioms.

The two model structures have the same weak equivalences and therefore produce the same localized diagram category. Their purpose is computational: one makes the left adjoints homotopical on cofibrant objects, the other makes the right adjoints homotopical on fibrant objects. The special monomorphism hypotheses guarantee these choices in the source's proof. They are not an invariant property that every presentation of the resulting derivator must share.

On trouve maintenant le

Théorème 2.9. *Soit* $(W, C = \mathrm{Mono}, F = (TC)_* = (W \cap \mathrm{Mono})_*)$ *un triple de Quillen très spécial dans une catégorie* $\mathcal{M}$ (2.8). *Alors :*

1°) *Pour toute* I *dans* Cat, $(W(I), C(I), F[I])$ *est un triple de Quillen très spécial dans* $\mathcal{M}(I)$, *et on a en particulier*

$$C(I) \;=\; \mathrm{Mono}(\mathcal{M}(I))\ , \tag{2.9.1}$$

et de plus

$$F[I] \cap W(I) \;=\; (TF)[I]\ . \tag{2.9.2}$$

2°) *Pour toute flèche* $u : I \longrightarrow I'$ *dans* Cat, *d'où le couple de foncteurs adjoints*

$$\mathcal{M}(I) \underset{u_*}{\overset{u^*}{\leftrightarrows}} \mathcal{M}(I')\quad , \tag{2.9.3}$$

on a

$$\begin{cases} u^*(W(I')) & \subset & W(I) \\ u^*(C(I')) & \subset & C(I) \\ u^*(TC(I')) & \subset & TC(I) \\ u_*(F[I]) & \subset & F[I'] \\ u_*(TF[I]) & \subset & TF[I']\ , \end{cases} \tag{2.9.4}$$

enfin u_* *induit un foncteur*

$$u_*^{\mathrm{f}} \;:\; \mathcal{M}(I)_{\mathrm{f}} \longrightarrow \mathcal{M}(I')_{\mathrm{f}} \tag{2.9.5}$$

(17)

compatible aux localiseurs $W(I)_{\mathrm{f}}$ *et* $W(I')_{\mathrm{f}}$ *induits par* $W(I)$, $W(I')$ *respectivement, donc induit un foncteur* p. 18 [page 29]

$$u_*^W \;:\; \mathrm{Ho}_W(I) \longrightarrow Ho_W(I')\ . \tag{2.9.6}$$

Ce foncteur est un adjoint à droite *du foncteur*

$$u_W^* \;:\; Ho_W(I') \longrightarrow \mathrm{Ho}_W(I) \tag{2.9.7}$$

induit par u^* (2.9.3).

17. sous-catégories pleines des objets fibrants.

Démonstration de 2.9. Pour prouver 1°), il faut vérifier d'abord que les conditions préliminaires (2.8.1. a) b)) sur $\mathcal{M}$, W sont stables par passage de $\mathcal{M}$, W à $\mathcal{M}(I)$, $W(I)$, ce qui se fait par AQT `[âne qui trotte]`. D'ailleurs on vérifie tout de suite que

$$C(I) \;=\; \mathrm{Mono}(\mathcal{M}(I))\ ,$$

et le théorème 2.6 (1°, 2°) implique alors que

$$\begin{array}{ccccc} C(I) & \longmapsto & C(I)_* & \overset{\text{déf}}{=} & TF[I] \\ \underbrace{TC(I)}_{\overset{\text{déf}}{=}C(I)\cap W(I)} & \longmapsto & (TC(I))_* & \overset{\text{déf}}{=} & F[I] \end{array}$$

sont des couples de Quillen spéciaux. (En notant pour le deuxième, que $TC(I)$ est bien stable par cochangement de base, TC l'étant.) Pour vérifier que l'on a un triple de Quillen très spécial, il reste en vertu de 2.8 à vérifier que [page 30]

$$TF[I] \;\subset\; W(I)\ ,$$

ce qui s'écrit

$$v^*(TF[I]) \;\subset\; W(I_0)\ ,$$

où

$$v \;:\; I_0 \longrightarrow I$$

est défini comme dans (1.12) (p. 10). En fait, on aura même

$$v^*(TF[I]) \;=\; v^*v_*(TF(I_0)) \;\subset\; \underbrace{TF(I_0)}_{=F(I_0)\cap W(I_0)}\ ,$$

et la démonstration est duale de celle de

$$v^*v_!(TC(I_0)) \;\subset\; TC(I_0)$$

(cf. (2.3.7), p. 21), elle ne dépend pas des questions de factorisation dans $\mathcal{M}(I)$, et provient ici de la stabilité de TF par petits produits. Cela prouve 1°.

Commentary · Development (commentators') [ms. 30]

The cofree image is smaller than the lifting class

Grothendieck needs the inclusion $TF[I] \subset W(I)$ to finish the injective model structure. The intended conclusion is correct, but the equality involving cofree diagrams on this page is stronger than the adjunction permits. We spell out both the direct repair and a finite example distinguishing the two classes.

First there is a short proof of the necessary inclusion. For $v : I_0 \to I$, the free-diagram functor $v_!$ takes componentwise monomorphisms to monomorphisms. Indeed, at i it is a coproduct of the given component maps, and these coproducts are monic under the hypotheses in use. One can also see the coproduct assertion by expressing a coproduct of monomorphisms as a transfinite succession of pushouts, using closure of monomorphisms under filtered colimits at limit stages. If $p \in TF[I] = \mathrm{Mono}(\mathcal{M}(I))^{\square}$, adjunction now gives

$$v^*p \in \mathrm{Mono}(\mathcal{M}(I_0))^{\square} = TF(I_0).$$

Since $TF \subset W$, this proves the required inclusion. No factorization of p and no assertion that it is cofree have been used.

Finite separation of the two image classes. [ours] In Set, with weak equivalences all maps and cofibrations the injections, take $I = (0 \to 1)$. Then

$$v^*(TF[I]) \neq v^*v_*(TF(I_0)).$$

Here the model structure is generated by the usual injection–surjection weak factorization system; it satisfies the intended, inductive-limit version of the very-special hypotheses.

Proof. A contravariant I-diagram is a map $X_1 \to X_0$. Let X_0 have two elements and X_1 three, and choose a surjection $r : X_1 \twoheadrightarrow X_0$. Let $p : X \to 1$ be the map to the terminal diagram. We show directly that it has the right lifting property against every componentwise injection $A \hookrightarrow B$. Regard A_i as a subset of B_i. Given a map $A \to X$, first extend $A_0 \to X_0$ to $h_0 : B_0 \to X_0$ by sending all new elements to one fixed element of X_0. Choose a section $s : X_0 \to X_1$ of r. On $B_1 \setminus A_1$, define

$$h_1(b) = s\big(h_0(r_B(b))\big),$$

and on A_1 retain the specified map. The two definitions have disjoint domains and agree with the required compatibility on A_1. Thus $rh_1 = h_0 r_B$, giving a lift of diagrams. Hence $p \in TF[I]$.

For a family (E_0, E_1), the cofree diagram is

$$(v_*E)_0 = E_0, \qquad (v_*E)_1 = E_0 \times E_1,$$

with the projection as its structure map. Consequently the restriction of a cofree arrow has component source sizes $|E_0|$ and $|E_0||E_1|$. If it were isomorphic, after restriction, to p, these sizes would be 2 and 3. The equation $2|E_1| = 3$ has no solution. Therefore v^*p is not in the cofree image class, even up to isomorphism. This proves the assertion. □

The product argument in the source does establish

$$v^*v_*(TF(I_0)) \subset TF(I_0),$$

because right lifting classes are closed under products. It does not replace the direct lifting proof for the larger class $TF[I]$. In a closure formulation one must retain the appropriate right-orthogonal saturation; deleting it creates exactly the mistaken equality.

We checked the diagram-model discussions of Lurie (2009, §A.2.8) and Cisinski (2006a, Corollary 1.4.22); both use lifting to establish the relevant preservation statements. We did not find this particular finite illustration there. The example is recorded as a local development of the printed argument, with no claim of priority for the general distinction between cofree objects and a right lifting class.

La démonstration de 2° est maintenant la duale de celle de (2.3. 2°). Comme cela nous fait répéter deux fois la même démonstration, cela montre qu'on n'a pas "monté le sorite" tout à fait avec assez de soin. Donc à titre de rattrapage, j'ajoute un

p. 19 [page 31]

Scholie 2.10. *Soit (W, C, F) un triple de Quillen clos dans $\mathcal{M}$. Pour I dans* Cat*, pour que $(W(I), C\langle I\rangle, F(I))$ soit un triple de Quillen clos pour $\mathcal{M}(I)$, il faut et il suffit que les couples $C\langle I\rangle \longleftrightarrow TF(I)$ et $TC\langle I\rangle \longleftrightarrow F(I)$ soient factorisants. Supposons que cette condition soit satisfaite pour I et pour I', et soit $u : I \longrightarrow I'$ une flèche de I dans I'. Alors les assertions de* 2.3. 2° *sont valables, en particulier on trouve un foncteur $u_!^W : \mathrm{Ho}_W(I) \longrightarrow \mathrm{Ho}_W(I')$, adjoint à gauche de u_W^** [18]*. Ces résultats se dualisent aussitôt (par $\mathcal{M} \longmapsto \mathcal{M}^\circ$, $I \longmapsto I^\circ$) pour les triples $(W(I), C(I), F[I])$, et pour la construction de u_*^W, adjoint à droite de u_W^*. En particulier, pour que le triple précédent soit un triple de Quillen (forcément clos), il faut et il suffit que les deux couples $C(I) \longleftrightarrow TF[I]$ et $TC(I) \longleftrightarrow F[I]$ soient factorisants ; et quand cette condition est satisfaite pour I et pour I' (pour une flèche $u : I \longrightarrow I'$ dans* Cat *donnée), alors u_*^W existe et se construit dualement de la construction de $u_!^W$ explicitée dans* 2.3. 2°.

[page 32]

Corollaire 2.11. *Le prédérivateur $\mathbf{D}_{\mathcal{M},W}$ défini pour un localiseur* très spécial *W dans $\mathcal{M}$ est un dérivateur en plein sens du terme, i.e. il satisfait à tous les axiomes Dér 1 à Dér 6 (y inclus Dér 4 bis et Dér 5 bis et Dér 6 bis) des dérivateurs.*

N.B. À ma propre surprise, je n'ai pas été capable de trouver un cas d'un localiseur W qui définisse un dérivateur sur Cat tout entier (avec existence de $u_!^W$, u_*^W), *que* pour W un localiseur très spécial, donc s'insérant dans un triple de Quillen très spécial – ou quand on peut s'y ramener de la façon standard. (Par exemple dans le cas où $\mathcal{M} = \mathrm{Cat}$, W un

18. Dire aussi que le calcul des fibres de $u_!^W(X)$, prévu dans Dér 5 bis, est OK.

localiseur fondamental satisfaisant Loc 6 a) et Loc 8, 8 bis) [19] – les monomorphismes dans W se refusent formellement à être stable par cochangement de base !) J'ignore totalement si le résultat mirobolant "prouvé" par Anderson est vrai ou faux. Ça a l'air presque trop beau pour être vrai …D'ailleurs, je ne connais pas de structure de Quillen spéciale qui ne soit très spéciale – désolé !

Commentary · Present state [ms. 32]

The existence question after Chapter XIX

The final note measures the distance between the construction just completed and the general statement Grothendieck hoped for. Theorem 2.11 provides both Kan extensions by imposing enough smallness and exactness to construct two model structures on every diagram category. The subsequent uncertainty concerns how much of this apparatus is necessary. That existence question now has a definitive answer for ordinary complete and cocomplete Quillen model categories.

Cisinski (2003, Theorem 6.11) proves the existence and pointwise calculation of homotopy right Kan extensions for a Quillen model category admitting all small limits; applying the dual result supplies the left extensions when all small colimits exist. This gives the associated derivator on all small categories, without requiring the original category to be combinatorial or the diagram categories to possess the two projective and injective model structures used here. The same general statement is recorded in Groth (2013, Theorem 1.32). Groth's Proposition 1.30 gives a shorter proof for combinatorial models and establishes the usual strength axiom in that setting. These later results settle the mathematical question; they do not certify the details of the historical proof attributed to Anderson.

There is already a very small distinction between "special" and "very special." On Set take

$$W = \mathrm{Iso}, \qquad C = F = \mathrm{Fl}(\mathrm{Set}).$$

The two weak factorization systems are $(\mathrm{Fl}, \mathrm{Iso})$ and $(\mathrm{Iso}, \mathrm{Fl})$. The first is generated by $\varnothing \to 1$ and $2 \to 1$: right lifting against the first means surjectivity, and right lifting against the second means injectivity. The empty generating family gives the second system. All sets are accessible at a suitable cardinal and all small colimits exist, so this is special in the chapter's sense. It is not very special because $2 \to 1$ is a cofibration but not a monomorphism. Its associated derivator is the represented diagram theory of sets. This elementary example separates a chosen model presentation from the special construction with all monomorphisms as cofibrations.

The warning about Cat points to a different, genuine obstruction. Here is a complete finite construction for the ordinary weak-homotopy localizer. Let P be the poset of nonempty proper faces of a tetrahedron. Its nerve is the barycentric subdivision of the boundary sphere, so $|NP|$ is homeomorphic to S^2. For each nonidentity arrow a of P, take a copy of the walking arrow $[1]$ and its inclusion in the groupoid J with two objects and one isomorphism between them. Form the disjoint unions

$$A = \coprod_a [1], \qquad B = \coprod_a J,$$

and map $A \to P$ by the chosen arrows. The map $A \to B$ is a monomorphism of categories: it is injective on objects and faithful. It is also a weak homotopy equivalence, since it is a disjoint union of maps between contractible nerves.

The pushout $P \amalg_A B$ is the ordinary groupoid completion of P, by the universal property of freely adjoining inverses to all arrows. The fundamental groupoid of the nerve of a category is its groupoid completion: its generators are the one-simplices, and its two-simplex relations are exactly composition. Since S^2 is connected and simply connected, this groupoid is equivalent to the terminal groupoid. Thus the pushout map

$$P \longrightarrow P \amalg_A B$$

induces $S^2 \to *$ on homotopy types and is not a weak equivalence, as its effect on $H_2(-;\mathbb{Z})$ already shows. All categories used to specify the pushout are finite. The claim on page 32 that weak monomorphisms in Cat need not survive pushout therefore has a concrete content; one cannot obtain the usual homotopy theory of categories by simply declaring every monomorphism to be a cofibration.

19. axiome des carrés cocartésiens (version directe), axiome des limites inductives, accessibilité.

Scholie 2.10 isolates what the direct proof actually requires: the two lifting pairs must factor at the diagram shapes being used. For its pointwise assertion one must also justify the corresponding comma-shape calculation, as done above when all the projective structures are available. Cisinski's theorem supplies the general alternative when that direct model-structure construction is unavailable. The categorical limits and colimits remain essential hypotheses; this is the distinction the examples in the following section must continue to respect.

[page 33]

3 Cas des catégories à cofibrations, des catégories à fibrations

Je rappelle les axiomes d'une *structure à cofibrations* (W, C) sur une catégorie $\mathcal{M}$.

Cof_0 : W est un localiseur.

Cof_1 : C contient les isomorphismes, est stable par cochangement de base, par composition.

Cof_2 : $C \cap W$ est stable par cochangement de base (et aussi par composition, et contient les isomorphismes, en vertu de Cof_0, Cof_1).

Cof_3 : Toute flèche f dans $\mathcal{M}$ se factorise en $f = pi$, avec $i \in C$, $p \in W$.

(Axiomes de K. S. Brown, légèrement généralisés par Anderson). Il y a un "axiome de continuité" ou *"axiome transfini"* facultatif, qui est le suivant : p. 20

Cof_4 : $\mathcal{M}$ stable par petites $\varinjlim$ filtrantes, et C et $TC = C \cap W$ sont stables par composition transfinie.

Ainsi, Cof_1, Cof_2, Cof_4 peuvent se résumer en disant que C et TC sont Q-saturés transfiniment à gauche.

La structure de cofibration est dite *spéciale* si elle satisfait Cof_4, si $\mathcal{M}$ est spéciale, et si C admet une *petite* partie C_0 qui soit Q-génératrice à gauche, i.e. telle que

$$C \ \subset \ C_0{}^{\sim}, \qquad \text{i.e.} \quad \tilde{C} = C_0{}^{\sim}. \tag{3.1}$$

Alors, par le théorème de factorisation 1.1, on a [page 34]

$$C \ \leftarrow\!\!\!\!\!\rightarrow \ TF \ \overset{\text{déf}}{=} \ C_* \ ,$$

et l'axiome de factorisation Cof_3 implique (cf. 1.4) $TF \subset W^{\natural}$. Plus généralement, on a :

Proposition 3.1. *Supposons que $\mathcal{M}$ soit spéciale (stable par petites $\varinjlim$, et éléments accessibles), que (W, C) satisfasse à Cof_0, Cof_1, et que C soit stable par composition transfinie (ce qui, joint à Cof_1, s'énonce en disant que C est Q-saturé transfini à gauche), enfin que C admette une petite partie C_0 Q-génératrice à gauche, de sorte que $C \leftarrow\!\!\!\!\!\rightarrow TF \overset{déf}{=} C_*$. Alors pour que l'axiome de factorisation Cof_3 soit satisfait, il suffit qu'on ait*

$$TF \ \subset \ W \ , \tag{3.1.1}$$

et il faut qu'on ait $TF \subset W^{\natural}$, donc si $W = W^{\natural}$, l'axiome Cof_3 équivaut à (3.1.1).

Commentary · Exposition [ms. 34]

What one factorization can accomplish

Grothendieck now removes the fibrations from the starting data. His purpose is to isolate what is actually needed to construct homotopy colimits: the ability to replace a map by a cofibration, together with the invariance of gluing along an acyclic cofibration. The resulting structure is deliberately asymmetric. It controls homotopy cofibres without committing to a second factorization, and therefore without assuming that every lifting class already belongs to a model structure.

Write $TC = C \cap W$ and, in modern notation, write $S^{\perp}$ for the maps having the right lifting property against S. The symbol $TF = C_* = C^{\perp}$ initially names a lifting class; its membership in W

is a separate assertion. Proposition 3.1 keeps this distinction correctly. If the small object argument gives a factorization $f = pi$ with $i \in C$ and $p \in C^{\perp}$, then $C^{\perp} \subset W$ immediately supplies Cof_3. Conversely, apply Cof_3 to a map $f \in C^{\perp}$, obtaining $f = pi$, $i \in C$, $p \in W$. The square with left side i, right side f, and identity on the common source has a diagonal s satisfying $si = 1$ and $fs = p$. In the arrow category, $(i, 1)$ and $(s, 1)$ exhibit f as a retract of p. This proves exactly $C^{\perp} \subset W^{\pitchfork}$; replacing the latter class by W requires retract closure. No additional lifting axiom has entered the argument.

The two closure requirements on C and TC have different tasks. Closure under cobase change gives the elementary gluing operation. Closure under transfinite composition permits a diagram to be assembled through its successive stages. Closure under retracts then turns a factorization into a characterization by lifting. In the terminology of these pages, Q-closed means equality with ${}^{\perp}(S^{\perp})$, whereas Q-saturated describes specified operations on arrows. Theorem 1.1 identifies them under a small generation hypothesis; section 3 ends by asking how far that hypothesis can be removed.

The historical starting point is Brown's theory of categories of fibrant objects, used here in its dual form (Brown, 1973). Theorem-Scholie 2 of Chapter VI, manuscript p. 108, already places this asymmetry in the development of the manuscript. What is new here is its systematic application to categories of diagrams. Grothendieck's assertion that a special cofibration structure has functorial factorizations uses the functorial version of the small object construction from section 1. It is a conclusion of that construction, not part of the definition of an arbitrary cofibration category.

La structure de cofibration est dite *semi-close* si C est Q-clos à gauche, i.e. si on a $C = C^{\sim}$ (ce qui implique que C est stable par composition transfinie), *close* si de plus $TC = (TC)^{\sim}$ (ce qui implique l'axiome transfini Cof_4).

Je ne reviens pas ici sur la théorie de K. S. Brown des catégories à cofibrations. C'est [page 35]
à ma connaissance l'ensemble d'axiomes le plus économique connu, pour assurer d'une part une bonne prise sur Ho_W (par une notion correspondante d'homotopie d'applications, et un calcul de fractions dans $\Pi\mathcal{M}_c$), et d'autre part l'existence de la suite *exacte* de cofibrations, donc une théorie de cofibres W-homotopiques. (**N.B.** Quand la structure est *propre,* i.e. $C \subset \mathrm{Cof}_W$, alors l'existence de cette structure revient à dire simplement que "W admet assez de W-cofibrations", i.e. que toute $f \in \mathrm{Fl}(\mathcal{M})$ se factorise en pi, avec $i \in \mathrm{Cof}_W$, $p \in W$; et on peut donc prendre $C = \mathrm{Cof}_W$. Dans le cas où W est stable par petites $\varinjlim$ filtrantes, $C = \mathrm{Cof}_W$ et $TC = W^{\mathrm{univ}}$ satisfont de plus l'axiome transfini Cof_4.) Enfin, K. S. Brown montre que toute flèche $i : X \longrightarrow Y$ dans C, *avec X, Y cofibrants* (i.e. $\emptyset_{\mathcal{M}} \longrightarrow X$, $\emptyset_{\mathcal{M}} \longrightarrow Y$ dans C) est dans Cof_W ("Lemme de Brown").

Commentary · Examples [ms. 35]

Brown excision and the scope of properness

Grothendieck wants gluing to depend only on the weak equivalences, which explains the intrinsic class Cof_W. One qualification is necessary in the final sentence on this page. Brown excision gives homotopy invariance of a pushout along a cofibration when the weak-equivalence comparison is between cofibrant objects. It does not say that a cofibration with cofibrant endpoints belongs to the ambient, universally defined class Cof_W.

This is a distinction in the manuscript's own terminology. Chapter XIII, ms. p. 8, defines Cof_W by two consecutive pushout squares: after any initial cobase change, pushing out a weak equivalence must again give a weak equivalence (Grothendieck, 1990–1991e, Ch. XIII, ms. p. 8). Thus it requires invariance on the entire undercategory. The safe form of Brown's assertion is obtained by restricting the coefficient category and the comparison maps to cofibrant objects. Alternatively, one may assume $C \subset \mathrm{Cof}_W$, the properness hypothesis already distinguished in the preceding parenthesis.

A concrete obstruction makes the difference visible. In simplicial associative unital $\mathbb{Z}$-algebras, let $A = \mathbb{Z}/p$ be constant and choose a simplicial free algebra resolution $B \longrightarrow A$. Adjoining a free noncommuting generator has underlying simplicial abelian group

$$B\langle x\rangle \cong \bigoplus_{n\geq 1} B^{\otimes_{\mathbb{Z}} n}.$$

The summand for $n = 2$ has first homotopy group $\mathrm{Tor}_1^{\mathbb{Z}}(A, A) \cong \mathbb{Z}/p$, whereas $A\langle x\rangle$ is constant. Hence $B\langle x\rangle \longrightarrow A\langle x\rangle$ is not a weak equivalence. But $\mathbb{Z} \longrightarrow \mathbb{Z}\langle x\rangle$ is a free cofibration with cofibrant

source and target. Its pushout functor is precisely adjoining x, so it is not a W-cofibration. This is the mechanism of Rezk (2002, Example 2.11, arXiv version). The error concerns the unrestricted formulation of the sentence, not Brown's excision theorem and not the finite-diagram constructions that follow.

Je m'intéresse ici à la question d'étendre la structure à cofibrations à une catégorie de la forme $\mathcal{M}(I)$, et d'utiliser cette extension pour construire les foncteurs $u_!^W$. Notons déjà que la théorie de K. S. Brown, avec les compléments d'Anderson [20], donne essentiellement ceci : [page 36] p. 21

Proposition 3.2. *Soit $W \subset \mathrm{Fl}(\mathcal{M})$ un localiseur associé à une structure à cofibrations (C, W) dans $\mathcal{M}$.*

1°) *Sur la catégorie* Diag_0 *des ensembles ordonnés* finis*, ce prédérivateur satisfait aux axiomes Dér 1 (décomposition), Dér 2 (localisation), Dér 3 (objet final), Dér 4 bis (existence de $u_!^W$), Dér 5 bis (calcul des fibres de $u_!^W$), Dér 6 bis (exactitude à gauche).*

2°) *C'est encore vrai sur la sous-catégorie* Diag_1 *dans* Cat *plus grande, formée des ensembles ordonnés I* noethériens *tels que l'espace topologique associé à I° soit noethérien (toute suite croissante d'objets de I, ou d'ouverts de I°, est stationnaire).*

3°) *Enfin, si l'axiome transfini Cof_4 est satisfait, i.e. si C et TC sont stables par composition transfinie, alors la conclusion de* 1° *reste valable sur la sous-catégorie pleine* Diag_2 *de* Cat *formée des ensembles ordonnés qui sont noethériens sans plus.*

Corollaire 3.3. *Supposons que l'on ait Cof_3 sous la forme plus forte d'une factorisation $f = pi$* fonctorielle *en f, ce qui est le cas en particulier si la structure à cofibrations est spéciale.* [page 37] *Alors pour toute I dans* Cat*, $\mathcal{M}(I)$ muni de $(W(I), C(I))$ est une catégorie à cofibrations, laquelle satisfait l'axiome transfini Cof_4 si (W, C) y satisfait. Donc on peut lui appliquer les résultats de* 3.2*, qui impliquent notamment que les axiomes Dér 1, 2, 3 (?) sont valables sur* Cat *tout entier, et que $u_!^W$ existe quand u est de la forme $\mathrm{id}_I \times v$, avec $v : J \longrightarrow K$ une flèche dans* Diag_1 *(resp. dans* Diag_2*, si l'axiome transfini est satisfait).*

3.4. La clef de la théorie est dans l'extension de la structure à cofibrations (W, C) de $\mathcal{M}$ à une catégorie $\mathcal{M}(I)$, où I est un ensemble ordonné, en $(W(I), C_I)$. Les faits principaux sont énoncés dans XVII, dans les théorèmes p. 26-27, p. 50, p. 60. C'est quand I est *noethérien,* et qu'on suppose soit que $\mathrm{Esp}(I^\circ)$ est également noethérien (p.ex. I fini), soit que (W, C) satisfait l'axiome transfini Cof_4, qu'on prouve que $(W(I), C_I)$ est également une structure à cofibrations sur $\mathcal{M}(I)$.

Commentary · Exposition [ms. 37]

Attaching a diagram in the order of its dependencies

Grothendieck's noetherian hypothesis makes an inductive construction possible: the value at an object can be attached after the values on which it depends have been constructed. With his convention $\mathcal{M}(I) = \mathrm{Fun}(I^\circ, \mathcal{M})$, it is I° that serves as a direct indexing category. In a finite example this is the familiar Reedy construction, although the manuscript allows transfinite well-founded orders as well.

For a direct category J and a diagram $X : J \longrightarrow \mathcal{M}$, let

$$L_j X = \operatorname*{colim}_{(k \to j),\ k \neq j} X_k$$

be the latching object, when the indicated colimit exists. A map $f : X \longrightarrow Y$ is a Reedy cofibration when

$$X_j \amalg_{L_j X} L_j Y \longrightarrow Y_j$$

belongs to C for every j. The relative latching object incorporates all the compatibility already imposed at preceding objects. Merely requiring $X_j \longrightarrow Y_j$ to be a cofibration would omit those compatibility conditions.

20. revus et arrangés par mes soins.

For the arrow $0 \to 1$, the two tests are $X_0 \to Y_0 \in C$ and $X_1 \amalg_{X_0} Y_0 \to Y_1 \in C$. Given a map of arrows $X \to Y$, first factor its component at 0 as $X_0 \to Z_0 \to Y_0$. Next factor the induced map $X_1 \amalg_{X_0} Z_0 \to Y_1$ as a cofibration followed by a weak equivalence. Its intermediate object defines Z_1, and the pushout supplies $Z_0 \to Z_1$. The resulting $X \to Z$ is a Reedy cofibration and $Z \to Y$ is objectwise a weak equivalence. The same construction applies successively at every vertex of a finite direct category. Transfinite stages require colimits and the continuity condition Cof_4.

The identity $C_I \cap W(I) = (TC)_I$ records a second induction. In the arrow example, if the component at 0 is acyclic, its pushout $X_1 \to X_1 \amalg_{X_0} Y_0$ is acyclic. Consequently the relative latching map is a weak equivalence exactly when $X_1 \to Y_1$ is one, by two-out-of-three. This proves both directions of the identity at the second vertex. The general induction repeats this argument with the latching diagram; at a limit stage Cof_4 supplies the required continuity. This is why the equality, rather than the bare definition of C_I, is central to homotopy left Kan extension.

Corollary 3.3 uses a different construction. A functorial factorization in $\mathcal{M}$ can be applied to all components and all transition maps of an arbitrary diagram at once. Cof_0, Cof_1, and Cof_2 then hold objectwise, and functoriality supplies Cof_3 in $\mathcal{M}(I)$. This gives finite homotopy-colimit operations inside every shifted diagram category. It does not itself make the ordinary colimit functor preserve objectwise acyclic cofibrations. That additional issue is exactly what the projective classes $C\langle I\rangle$ are meant to address.

Un fait important pour la construction des foncteurs $u_!^W$, pour $u : I \longrightarrow I'$ une flèche [page 38]
entre catégories ordonnées noethériennes, est la relation

$$C_I \cap W(I) \;=\; (TC)_I \;, \tag{3.4.1}$$

d'où résulte que l'on a, pour u comme dessus,

$$u_!(C_I \cap W(I)) \;\subset\; C_{I'} \cap W(I') \;, \tag{3.4.2}$$

ingrédient essentiel de la construction du foncteur $u_!^W$. Pour étendre cette construction au cas d'une flèche $\pm$ quelconque dans Cat, il se pose donc le problème d'étendre la définition de C_I, et les formules (3.4.1) et (3.4.2), au cas de catégories $\pm$ quelconques dans Cat. Or on a prouvé :

Proposition 3.5. *Si $\Phi = \Psi^*$ est une partie* Q-close *à gauche de* $\mathrm{Fl}(\mathcal{M})$*, et I un ensemble ordonné noethérien, Φ_I l'extension de Φ en une partie de* $\mathrm{Fl}(\mathcal{M}(I))$ *définie par* Anderson*, alors on a*

$$\Phi_I \;=\; \Psi(I)^* \;=\; v_!(\Phi(I_0))^\sim \;, \tag{3.5.1}$$

où $v : I_0 \longrightarrow I$ est le foncteur canonique de la catégorie discrète définie par $\mathrm{Ob}\, I$*.*

C'est (avec des notations différentes) le lemme 6.2.5 (p. 107) de XVII, dont je ne reprends pas ici la démonstration. p. 22

Ceci donne la clef d'une définition de Φ_I comme $\Phi\langle I\rangle$ (avec les notations des paragraphes 1, 2), pour I quelconque dans Cat,

$$\Phi_I \;\overset{\text{déf}}{=}\; \Phi\langle I\rangle \;\overset{\text{déf}}{=}\; \Psi(I)^* \;=\; v_!(\Phi(I_0))^\sim \;, \tag{3.5.2}$$ [page 39]

où $\Psi = \Phi_*$, et où on suppose Φ Q-clos à gauche, i.e. $\Phi = \Psi^*$. Dans le cas où on suppose seulement Φ Q-saturé transfini sans plus, on aurait envie de poser

$$\Phi_I \;\overset{\text{déf}}{=}\; \text{Q-saturé transfini de } v_!(\Phi(I_0)) \;, \tag{3.5.3}$$

et je présume que pour I ordonné noethérien, cela coïncide avec la définition d'Anderson. Mais je vais me borner pour la suite au cas où $C = \Phi$ est Q-clos à gauche.

Revenons alors au cas d'une structure à cofibrations (W, C) dans $\mathcal{M}$, en supposant C et $TC = C \cap W$ Q-clos à gauche. Définissons pour toute I dans Cat $C_I = C\langle I\rangle$ comme dans (3.5.2). On a alors, pour $u : I \longrightarrow I'$ une flèche dans Cat,

$$u_!(C\langle I\rangle) \;\subset\; C\langle I'\rangle \;, \tag{3.5.4}$$

mais on n'y a guère de raison d'espérer avoir aussi (3.4.2), sans disposer de la formule

$$C\langle I\rangle \cap W(I) \overset{?}{=} (TC)\langle I\rangle \qquad [21]. \tag{3.5.5}$$

On a prouvé que cette formule est valable si I est ordonné noethérien (XVII th. 2.1, p. 26-27). Cela nous amène à la [page 40]

Question 3.6. *Soit $\mathcal{M}$ une catégorie stable par petites limites, et soit (W, C) une structure à cofibrations dans $\mathcal{M}$ telle que C et $TC = C \cap W$ soient Q-clos à gauche. Soit I un objet de* Cat*, a-t-on alors*

$$C\langle I\rangle \cap W(I) \;=\; (TC)\langle I\rangle\,? \tag{3.6.1}$$

Commentary · Exposition [ms. 40]

Lifting properties retain the coherence

Grothendieck replaces a geometric attachment rule by a characterization that makes sense for every small category. Put $TF = C^{\perp}$, $F = (TC)^{\perp}$, and

$$C\langle I\rangle = {}^{\perp}TF(I), \qquad (TC)\langle I\rangle = {}^{\perp}F(I).$$

The lifting problem here is a problem in the category of diagrams. A diagonal must commute with every arrow of I; a collection of independently chosen componentwise diagonals is insufficient. This is the coherence retained by the projective lifting class.

Let $v : I_0 \to I$ include the discrete category of objects. The adjunction $v_! \dashv v^*$ gives

$$(v_!\Phi(I_0))^{\perp} = \{p : v^*p \in \Phi(I_0)^{\perp}\} = \Psi(I),$$

where $\Psi = \Phi^{\perp}$ and Φ is Q-closed. The last equality follows because a lifting problem in a discrete family is solved component by component; equivalently, one tests a single component and puts identity maps in the others. Taking left orthogonals proves (3.5.2). The same adjunction calculation for $u_! \dashv u^*$ gives (3.5.4), since restriction preserves the objectwise right class. These arguments require existence of the ordinary left Kan extension, not a factorization system on diagrams.

The specification of Ψ matters, just as in the arbitrary-test-class defect in (1.14) discussed earlier. If Proposition 3.5 is read as allowing an arbitrary presentation $\Phi = {}^{\perp}\Psi$, its first equality needs qualification. In sets, take $\Psi = \{\emptyset \to 1\}$. Then Φ consists of the maps with nonempty source together with $1_{\emptyset}$. On the direct arrow category $J = [1]$, the diagram map

$$f : (\emptyset \longrightarrow 1) \longrightarrow (1 \xrightarrow{1} 1)$$

fails the relative latching condition at 0. Yet it lifts vacuously against $\Psi(J)$: that class contains only the map from the constant empty diagram to the constant singleton diagram, and the source of f has no map to the former. Thus ${}^{\perp}\Psi(J)$ is too large for this nonclosed Ψ. The canonical choice $\Psi = \Phi^{\perp}$, explicitly imposed in (3.5.2), removes this issue. It leaves the intended construction and the subsequent propositions intact.

For a well-founded direct indexing order, the latching description and this lifting description agree by induction on objects: at stage j, the already constructed diagonals determine a map out of the relative latching object, and its lifting property supplies the next component. The converse can be checked explicitly for the posets in Proposition 3.5. Given a lifting problem from the relative latching map at j to $p : A \to B$, use the diagram map which is 1_A at strict predecessors of j, is p at j, and is the identity of the terminal object elsewhere. The map $L_jY \to A$ supplies its predecessors. A coherent diagonal then gives exactly the requested relative lift. The canonical right class $\Phi^{\perp}$ contains all identities, as this test requires. This proves the comparison under the stated existence assumptions.

There is also a direct cellular verification of the conjecture (3.5.3) whenever the small colimits needed for its transfinite construction exist. Assume that Φ contains isomorphisms and is closed under pushouts and small transfinite composition. The following is the direct-poset version of the standard Reedy cellular construction; compare Riehl and Verity (2014, Proposition 6.3 and Corollary 6.7). Let $J = I^{\circ}$ be the well-founded poset and well order its objects so that every strict

predecessor is earlier. Write $j_!$ for the left adjoint of evaluation, so $(j_!A)_k = A$ if $j \leq k$ and is initial otherwise. A map $j_!a$, $a \in \Phi$, has relative latching map a at j and an identity elsewhere.

The class of maps whose relative latching maps lie in Φ is closed under pushouts, since forming a relative latching map takes a pushout of diagram maps to a pushout of that relative map. For transfinite closure, let $(X^\alpha)_{\alpha\leq\lambda}$ be a continuous chain of such maps. For fixed j put

$$Q_\alpha = X^\alpha_j \amalg_{L_jX^\alpha} L_jX^\lambda.$$

Each $Q_\alpha \to Q_{\alpha+1}$ is a pushout of the relative latching map of $X^\alpha \to X^{\alpha+1}$. The Q-chain is continuous because latching objects are colimits. Its starting object is the relative-latching domain of $X^0 \to X^\lambda$, and $Q_\lambda = X^\lambda_j$. Transfinite closure of Φ therefore proves the assertion. Consequently this relative-latching class contains the transfinite saturation of all the maps $j_!a$.

For the reverse inclusion, take $f : X \to Y$ with all relative latching maps in Φ. Starting from X, successively attach the value Y_j at each object j. When j is reached, all strict predecessors already equal their values in Y, and the current value at j is

$$X_j \amalg_{L_jX} L_jY.$$

Push out the free diagram map $j_!(X_j \amalg_{L_jX} L_jY \to Y_j)$ along its evaluation-adjoint map to the current diagram. This replaces the value at j by Y_j and leaves every earlier value fixed. At limit ordinals take colimits. The resulting composite ends at Y, since every component is eventually fixed to Y_j. Each step is a pushout of $j_!a$ for some $a \in \Phi$. Finally these one-object free maps belong to $v_!\Phi(I_0)$ by putting identity maps of the initial object in the other components; conversely a whole family is a coproduct of such one-object maps, formed by successive pushouts. This proves precisely the proposed transfinite-saturation description, without adding retract closure. It is a presentation by a class of cells, not an assertion that a small set of them suffices. If the required colimits have not been assumed, that existence qualification must remain explicit.

The easy inclusion in (3.5.5) is structural. Under the small-completeness hypothesis of 3.6, evaluation at an object j has a right adjoint whose values are products indexed by the arrows into j in I°. Right lifting classes are closed under products, so this right adjoint sends a map of F to an objectwise F-map. If $f \in {}^\perp F(I)$, adjunction therefore implies $f_j \in {}^\perp F = TC$. Hence projective TC-cofibrations are objectwise in TC, and therefore in $W(I)$; also $TC \subset C$ gives $(TC)\langle I\rangle \subset C\langle I\rangle$. The difficult inclusion says that an objectwise weak equivalence which already satisfies one coherent lifting condition satisfies the stronger one. Existence of abstract homotopy Kan extensions, established later, does not by itself identify these two specifically prescribed classes of strict diagrams.

On a vu au paragraphe précédent qu'il en est ainsi si (W, C) s'insère dans un triple de Quillen clos $(W, C, F = (TC)_*)$, et quand on suppose de plus que les couples

$$C\langle I\rangle \longleftrightarrow TF(I)\ , \qquad TC\langle I\rangle \longleftrightarrow F(I)$$

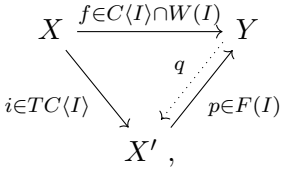
sont factorisant, auquel cas $(W(I), C\langle I\rangle, F(I))$ est une structure de Quillen close dans $\mathcal{M}(I)$. D'ailleurs, (W, C, F) est une structure de Quillen (nécessairement close) si et seulement si les couples $C \longleftrightarrow \underbrace{TF}_{\overset{\text{déf}}{=}C_*}$, $TC \longleftrightarrow \underbrace{F}_{\overset{\text{déf}}{=}(TC)_*}$ eux-mêmes sont factorisants, et si de plus $TF \subset W$ (ce qui est automatique si $W = W^\natural$, car on aura en tous cas, par Cof_4, $TF \subset W^\natural$). p. 23

Mais peut-on s'en tirer avec moins, pour pouvoir conclure à (3.6.1) ? Ne suffirait-il pas de savoir que $TC\langle I\rangle \longleftrightarrow F(I)$ est factorisant (ce qui n'a guère de chances d'être vrai pour une I ± générale, si on n'a déjà $TC \leftrightarrows F$, factorisation dans $\mathcal{M}$ lui-même). C'est bien le cas :

Proposition 3.7. *Sous les conditions générales de* (3.6) *sur $\mathcal{M}$, W, C, I, supposons que le couple $TC\langle I\rangle \longleftrightarrow F(I)$ (où $F \overset{déf}{=} (TC)_*$) soit factorisant, i.e. $TC\langle I\rangle \leftrightarrow\!\!\!\!\rightarrow F(I)$. Alors la formule* (3.6.1) *est vraie.* [page 41]

En effet, si $f \in C\langle I\rangle \cap W(I)$, on factorise f en pi, avec $i \in TC\langle I\rangle$, $p \in F(I)$. Comme $f, i \in W(I)$, on aura $p \in W(I)$, i.e. $p \in TF(I)$, et à cause de $\underbrace{C\langle I\rangle}_{\ni f} \longleftrightarrow \underbrace{TF(I)}_{\ni p}$, il existe

$p : Y \longrightarrow X'$ telle que $pq = \mathrm{id}_Y$, $qf = i$,



donc f est facteur direct de $i \in TC\langle I\rangle$. Comme $TC\langle I\rangle$ est stable par facteurs directs, on gagne.

Corollaire 3.8. *Soit (W, C, F) une structure de Quillen close dans $\mathcal{M}$, et supposons que $TC = C \cap W$ soit une partie Q-spéciale de $\mathcal{M}$, i.e. le couple $TC \leftarrow\!\!\!\!\!\rightarrow F$ un couple spécial (i.e. $\mathcal{M}$ est* spéciale*, et il existe une petite partie TC_0 Q-génératrice à gauche pour TC). Alors pour toute I dans* Cat*, on a*

$$C\langle I\rangle \cap W(I) \;=\; TC\langle I\rangle \;. \tag{3.8.1}$$

Commentary · Exposition [ms. 41]

The lifting lemma inside the retract argument

Grothendieck has found that only one diagram factorization is needed for (3.6.1). The proof is sound, but the line identifying a weak equivalence in F with a map lifting against C uses a short lemma that is useful beyond the present proposition.

Suppose (W, C) satisfies Cof_0–Cof_3, and set $F = (C \cap W)^{\perp}$. Then

$$F \cap W \subset C^{\perp}.$$

To prove this, consider a square from $a : A \to B$ in C to $p : X \to Y$ in $F \cap W$, with top map $h : A \to X$ and bottom map $k : B \to Y$. Form $Z = X \amalg_A B$, write $j : X \to Z$, and let $g : Z \to Y$ be induced by p and k. Factor $g = q\ell$ by Cof_3, where $\ell : Z \to T$ is in C and $q : T \to Y$ is in W. The composite ℓj belongs to C, and $q\ell j = p$ shows by two-out-of-three that $\ell j \in TC$. Since $p \in F$, the square with left side ℓj, right side p, top 1_X, and bottom q admits $s : T \to X$ satisfying $s\ell j = 1_X$ and $ps = q$. If $b : B \to Z$ is the other pushout map, then $s\ell b$ is the desired lift. This proves the inclusion without a second factorization in $\mathcal{M}$, and without retract closure of W.

Now take $f : X \to Y$ in $C\langle I\rangle \cap W(I)$ and factor it as $f = pi$ with $i \in (TC)\langle I\rangle$ and $p \in F(I)$. The already established easy inclusion makes i an objectwise weak equivalence, hence $p \in F(I) \cap W(I)$. Applying the preceding lemma at every object gives $p \in TF(I) = C^{\perp}(I)$. The defining coherent lifting property of f produces a natural map $q : Y \to X'$ with $qf = i$ and $pq = 1_Y$. The arrow-category maps $(1_X, q)$ and $(1_X, p)$ exhibit f as a retract of i. Left lifting classes are retract closed, so $f \in (TC)\langle I\rangle$.

The printed prose writes $p : Y \to X'$ for the new diagonal; the diagram and both identities show that q is intended. The earlier reference on ms. p. 40 to Cof_4 in the assertion $TF \subset W^{\natural}$ is likewise a reference slip: the relevant axiom is Cof_3, as in Proposition 3.1. Neither slip changes the argument. Corollary 3.8 is then exactly the small object theorem applied to TC in diagram categories. It proves the equality without assuming that C has a small generating set; the separate difficulty is obtaining the other factorization needed for a complete diagram model structure.

[page 42]

En apparence, on n'a pas eu à supposer que C soit elle aussi une partie Q-spéciale de $\mathrm{Fl}(\mathcal{M})$. Mais comme on a besoin de la factorisation pour $C \longleftrightarrow TF$ (pour avoir au moins une structure de cofibrations sur $\mathcal{M}$), on n'a guère le moyen de l'établir sans supposer C spéciale. Sans compter que pour pouvoir expliciter (3.8.1) pour la construction de $u_!^W$, on a besoin de savoir que $(W(I), C\langle I\rangle)$ définit bien une structure à cofibrations sur $\mathcal{M}(I)$, ce qui ne peut guère se faire (il me semble) sans avoir la factorisation pour $C\langle I\rangle \longleftrightarrow TF(I)$ également ; chose que je n'arrive à établir pour un I général dans Cat, que moyennant l'hypothèse de spécialité sur C également.

3.9. Pour en avoir le cœur net, il faudrait quelques exemples concrets et illuminants, du type suivant :

a) `[Il n'y a pas de b).]` Structure de Quillen close (W, C, F) sur $\mathcal{M}$, telle que pour le prédérivateur associé, $u_!^W$ n'existe pas toujours, ou u_*^W n'existe par toujours. Je pense notamment au cas d'une flèche $u : I \longrightarrow I'$ avec $I' = e$ (présumant que si dans ce cas, $u_!^W$ resp. u_*^W existe, alors il existe dans tous les cas). Un tel exemple serait encore plus

instructif, bien sûr, si $\mathcal{M}$ est stable par petites $\varinjlim$, que ses objets soient accessibles etc. et que W soit L-accessible (i.e. stable par petites $\varinjlim$ filtrantes, et de plus accessible …). Le premier candidat pour I auquel je pense serait un $\overrightarrow{B}_G$ (G un groupe, p.ex. $G = \pm 1$), c'est en effet l'exemple aux antipodes des ensembles ordonnés, lesquels, eux, ne semblent pas faire trop de problème (moyennant l'hypothèse noethérienne tout au moins). p. 24 [page 43]

Les exemples suivants ne sont pas aussi convaincants que ça, hélas!

Exemple 3.10. Soit $\mathcal{M}$ une catégorie quelconque, alors les couples $(\mathrm{Fl}(\mathcal{M}), \mathrm{Is}(\mathcal{M}))$ et $(\mathrm{Is}(\mathcal{M}), \mathrm{Fl}(\mathcal{M}))$ sont tous deux des couples de Quillen,

$$\mathrm{Fl}(\mathcal{M}) \longleftarrow\!\!\!\!\!\!\longrightarrow \mathrm{Is}(\mathcal{M}) \ , \qquad \mathrm{Is}(\mathcal{M}) \longleftarrow\!\!\!\!\!\!\longrightarrow \mathrm{Fl}(\mathcal{M}) \ . \tag{3.10.1}$$

Il s'ensuit que si on pose

$$W \ = \ \mathrm{Is}(\mathcal{M}) \ , \qquad C \ = \ F \ = \ \mathrm{Fl}(\mathcal{M}) \ , \tag{3.10.2}$$

(W, C, F) est un triple de Quillen clos [22]. D'ailleurs pour toute I dans Cat, on a $W(I) = \mathrm{Is}(\mathcal{M}(I))$, donc

$$\mathrm{Ho}_W(I) \ = \ \mathcal{M}(I) \ . \tag{3.10.3}$$

Donc la question de l'existence des foncteurs $u_!^W$, u_*^W se réduit à celle des foncteurs $u_!$, u_* habituels, adjoints à gauche et à droite de $u^* : \mathcal{M}(I') \longrightarrow \mathcal{M}(I)$.

Commentary · Present state [ms. 43]

Finite completeness and the modern model category theorem

Example 3.10 isolates the completeness issue. Every map factors as itself followed by an identity, or as an identity followed by itself. A map with the lifting property against every map lifts against itself, and that diagonal is its inverse. Consequently the two displayed pairs really are closed factorization pairs in any category. With finite limits and colimits they give the historical, finitely complete version of Quillen's axioms. Their localization does nothing, so existence of homotopy Kan extensions is precisely existence of ordinary Kan extensions. For a functor between finite categories, the relevant comma categories are finite and the assumed finite limits and colimits suffice. For arbitrary indexing categories those assumptions say nothing about the required infinite constructions. The small-generation calculation for the corresponding example in sets is already given in the commentary to ms. p. 32.

The terminology now usually includes *all* small limits and colimits in the definition of a model category. Under that convention the requested example in 3.9 does not exist. Cisinski (2003, Theorem 6.11) proves the derivator assertion for an arbitrary Quillen model category with the required small limits; duality gives the assertion about colimits. Neither cofibrant generation nor local presentability is required. The distinction matters: this theorem supplies homotopy Kan extensions without asserting that the specific projective Quillen structure written in Question 3.6 exists on every diagram category. The bounded-complex example below therefore addresses the older finite-completeness convention, while Question 3.6 retains its more specific lifting-class content.

Grothendieck's suggestion that terminal projections should suffice anticipates the pointwise Kan-extension formulas. In the established derivator framework the value of a Kan extension is a homotopy limit or colimit over a comma category. For an arbitrary prederivator, however, adjoints to terminal projections alone do not supply the coherent pointwise construction or its base-change isomorphisms. Those are additional assertions proved by the model-category theorem, rather than a formal consequence of representability at the terminal category.

[page 44]

On sait bien que l'existence de ces foncteurs pour u donné, fait appel à l'existence de certaines $\varinjlim$ resp. $\varprojlim$ dans $\mathcal{M}$; p.ex. si $I' = e$, il est nécessaire et suffisant que les $\varinjlim$ (resp. $\varprojlim$) de type I° soient représentables dans $\mathcal{M}$. Mais avec les hypothèses faites sur $\mathcal{M}$, il n'y a pas de problème si $u : I \longrightarrow I'$ est un foncteur entre catégories *finies.* On va donner un exemple où il n'en est plus ainsi.

Exemple 3.11. Soit k un anneau commutatif, et $\mathcal{M} = K^{\mathrm{b}}(k\text{-Ab})$ la catégorie des complexes de cochaînes bornés de k-modules, W l'ensemble des quasi-isomorphismes, donc

$$\mathrm{Ho}_W \;=\; D^{\mathrm{b}}(k\text{-Ab})\ ,$$

catégorie dérivée bornée de la catégorie abélienne des k-modules. On sait, en utilisant l'existence d'assez de projectifs dans k-Ab, que cette structure est associée à une structure de Quillen close (W, C, F), décrite par Quillen dans Homotopical Algebra – à cela près qu'il prend [page 45]
$D^-(k)$ au lieu de $D^{\mathrm{b}}(k)$. Peut-être, pour vérifier les axiomes, doit-on supposer que k est de dimension cohomologique finie.

Prenons $I = B_G$, où G est un groupe (p.ex. $G = \{\pm 1\}$), et $u : I \longrightarrow e$. Alors $\mathcal{M}(I)$ est la catégorie des complexes bornés de $k[G]$-modules. On voit que $u_!^W$ resp. u_*^W existe si et seulement si pour tout tel complexe borné, ou simplement pour un $k[G]$-module sans plus M, le complexe $L\Gamma^G(M)$ resp. $R\Gamma_G(M)$ appartient à $D^{\mathrm{b}}(k)$, ou ce qui revient au même, si et p. 25
seulement si on a

$$H_i(G, M) \;=\; 0 \quad \text{pour } i \gg 0\ ,$$
$$\text{resp.} \quad H^i(G, M) \;=\; 0 \quad \text{pour } i \gg 0\ ,$$

pour tout $k[G]$-module M. Ce n'est pratiquement jamais le cas; quand on prend p.ex. $k = \mathbf{Z}$ et G groupe fini, cela n'est le cas que si $G = e$. L'exemple le plus économique me paraît $G = \pm 1$, $k = \mathbf{F}_2$.

Commentary · Examples [ms. 45]

A finite group can force an unbounded resolution

The notation $K^{\mathrm{b}}(k\text{-Ab})$ is explicitly defined here as the category of bounded complexes and chain maps. It is what is now usually denoted $C^{\mathrm{b}}(k\text{-Mod})$, before quotienting by chain homotopy. The sentence about enough projectives needs the qualification which Grothendieck immediately adds: an unbounded projective resolution does not provide a bounded cofibrant replacement. It is enough for his example to take the proposed field $k = \mathbf{F}_2$, where every bounded acyclic complex is contractible.

Here is a complete verification of the finite Quillen structure in that case. Put W = quasi-isomorphisms, C = degreewise injections, and F = degreewise surjections. A bounded contractible complex of vector spaces is a finite direct sum of disk complexes and is both projective and injective in the category of complexes. Thus a surjective quasi-isomorphism, whose kernel is contractible, has the right lifting property against every injection; dually an injective quasi-isomorphism has the left lifting property against every surjection. For a bounded complex X, the cone $I(X) = \mathrm{Cone}(1_X)$ is contractible and has a natural degreewise injection $X \to I(X)$. Dually $P(Y) = \mathrm{Cone}(1_Y)[-1]$ has a natural degreewise surjection $P(Y) \to Y$. For $f : X \to Y$ the factorizations

$$X \xrightarrow{(f, i_X)} Y \oplus I(X) \longrightarrow Y, \qquad X \longrightarrow X \oplus P(Y) \xrightarrow{(f, p_Y)} Y$$

have types $(C, F \cap W)$ and $(C \cap W, F)$ respectively. Retract closure and two-out-of-three are immediate from degreewise linear algebra and cohomology. All objects and constructions remain bounded. Finite limits and colimits exist, whereas, for example, a coproduct of $k[n]$ over all integers cannot be bounded: its universal injections and retractions would force nonzero cohomology in every degree.

For $G = C_2$ write $A = k[G] = k[\epsilon]/(\epsilon^2)$, with $\epsilon = g - 1$. Since BG has one object, a diagram of bounded complexes is exactly a bounded complex of A-modules. There is no uniform-boundedness

22. au moins si on admet que $\mathcal{M}$ est stable par $\varinjlim$ et $\varprojlim$ *finies.*

issue here; for a general category with infinitely many objects, an objectwise bounded diagram need not be bounded uniformly. Thus the relevant restriction functor is

$$p^* : D^{\mathrm{b}}(k) \longrightarrow D^{\mathrm{b}}(A), \qquad V \longmapsto V \text{ with trivial } G\text{-action.}$$

The exact projective resolution

$$\cdots \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \xrightarrow{\epsilon} A \longrightarrow k \longrightarrow 0$$

is exact because $\ker(\epsilon) = \epsilon A$. Tensoring it with k, or applying $\mathrm{Hom}_A(-, k)$, kills every differential. Consequently

$$\mathrm{Tor}^A_n(k,k) = k \quad \text{and} \quad \mathrm{Ext}^n_A(k,k) = k \qquad (n \geq 0).$$

This proves nonexistence of the adjoints themselves, not merely failure of one proposed construction. If R were right adjoint to p^*, then

$$H^n(Rk) = \mathrm{Hom}_{D(k)}(k[-n], Rk) \cong \mathrm{Hom}_{D(A)}(k[-n], k) = k$$

for every $n \geq 0$, contradicting boundedness. If L were left adjoint, then

$$\mathrm{Hom}_{D(k)}(Lk, k[n]) \cong \mathrm{Hom}_{D(A)}(k, k[n]) = k$$

for every $n \geq 0$. Over a field, the left side vanishes outside a finite range for every bounded Lk, another contradiction.

The source's general boundedness criterion can also be justified. The unbounded derived adjunctions are tensoring with the trivial module and derived invariants. If they take every module to a bounded complex, finite truncation triangles show that they do so for every bounded complex. Their restrictions are then the required adjoints. Conversely, the shifts of k detect all cohomology of the proposed right adjoint. For the left adjoint, test maps into shifts of an injective cogenerator of k-modules: these detect the homology of the derived tensor product and force its boundedness. For $k = \mathbf{Z}$ and a nontrivial finite group, choose a subgroup C_p of prime order and induce its trivial $\mathbf{F}_p$-module to G. The cyclic periodic resolution and the induction/coinduction identities give nonzero homology and cohomology in arbitrarily large degrees; see Brown (n.d., Lecture 3). Thus the stated finite-group obstruction is valid. Its mechanism is the arbitrarily long composable strings in BG, despite the finiteness of its object and arrow sets.

Donc ici $u^W_!$ et u^W_* n'existent pas, pour $I \longrightarrow e$ avec I catégorie finie. Mais aussi $\mathcal{M}$ n'est pas stable par petites $\varinjlim$. [page 46]

Exemple 3.12. Soit $\mathcal{M}$ une catégorie muni d'un localiseur W fortement saturé, tel que $\mathcal{M} \longrightarrow \underbrace{\mathcal{M}W^{-1}}_{=\mathcal{H}}$ admette un adjoint – à gauche disons. On a donc une situation de deux foncteurs adjoints

$$\mathcal{H} \underset{v^* \text{ localisation}}{\overset{v_! \text{ pleinement fidèle}}{\rightleftarrows}} \mathcal{M} \quad , \tag{3.12.1}$$

et $W = (v^*)^{-1}(\mathrm{Iso}(\mathcal{H}))$. On trouve que pour toute I dans Cat, on a

$$\mathrm{HO}_W(I) \;\simeq\; \mathcal{H}(I) \overset{\text{déf}}{=} \mathrm{Hom}(I^\circ, \mathcal{H}). \tag{3.12.2}$$

On peut donc songer à faire un exemple où dans $\mathcal{M}$ les petites $\varinjlim$ et $\varprojlim$ existent, mais non dans $\mathcal{H}$, de sorte que les $u^W_!$ ou u^W_* n'existent pas toujours. Mais ce semble difficile à construire, vu que dans la situation envisagé, on voit facilement que les petites $\varprojlim$ existent dans $\mathcal{H}$. Donc si dans $\mathcal{H}$ il existe une petite famille cogénératrice (et pour ceci, il suffit qu'une telle famille existe dans $\mathcal{M}$), il y aura également des petites $\varinjlim$, en vertu de SGA 4 I 8.12.7. De même, elles existent si $\mathcal{H}$ est une sous-catégorie pleine de $\mathcal{M}$ *stable par petites* $\varinjlim$, et il paraît pratiquement impossible de faire un exemple (3.12.1) sans que $\mathcal{H}$ ait cette propriété de stabilité. Sans compter qu'il y a même la perplexité suivante : [page 47]

! |

Question 3.13. *Existe-t-il une $\mathfrak{U}$-catégorie ($\mathfrak{U}$ un univers donné) où les petites $\varinjlim$ soient représentables, sans qu'il en soit de même des petites $\varprojlim$?* **N.B.** Une telle catégorie n'admet pas de petite famille génératrice, a fortiori elle n'est pas équivalente à une petite catégorie.

Commentary · Exposition [ms. 47]

A coreflective localization inherits both kinds of limits

The proposed difficulty in Example 3.12 is removed by a formal adjunction argument. Write $L : \mathcal{H} \to \mathcal{M}$ for the fully faithful left adjoint and $R : \mathcal{M} \to \mathcal{H}$ for the localization. The unit is an isomorphism. The counit $\varepsilon_X : LRX \to X$ belongs to W, since $R\varepsilon_X$ is an isomorphism by the triangle identity. These facts hold objectwise in every diagram category. Therefore L^I and R^I descend to inverse equivalences

$$\mathcal{M}(I)[W(I)^{-1}] \simeq \mathcal{H}(I),$$

which proves (3.12.2) directly and explains why the example becomes a question about ordinary completeness.

If $\mathcal{M}$ is complete and $D : J \to \mathcal{H}$ is small, then

$$R\big(\varprojlim_J LD\big)$$

is its limit in $\mathcal{H}$: applying $\mathrm{Hom}_{\mathcal{H}}(H, -)$ and the adjunction identifies the required universal property with the limit universal property in $\mathcal{M}$. If $\mathcal{M}$ is cocomplete, let $X = \mathrm{colim}_J LD$. Every cocone map $LD(j) \to X$ has a unique factorization through $\varepsilon_X : LRX \to X$, because its source lies in the image of L. These factorizations form a cocone, hence induce $g : X \to LRX$ with $\varepsilon_X g = 1_X$. Postcomposition with ε_X is a bijection on maps from any object in the image of L. Applying this with source LRX gives $g\varepsilon_X = 1_{LRX}$. Thus ε_X is an isomorphism and X belongs to the essential image of L. In particular that image is automatically closed under all small colimits, and $R\,\mathrm{colim}\, LD$ is the colimit in $\mathcal{H}$.

Consequently a full coreflective subcategory of a bicomplete category is bicomplete. No cogenerating family is needed, and the envisaged counterexample cannot have the adjunction (3.12.1). The source's auxiliary observation about cogenerators is nevertheless correct: if $\{C_s\}$ conservatively detects arrows in $\mathcal{M}$ by $\mathrm{Hom}_{\mathcal{M}}(-, C_s)$, then $\{RC_s\}$ does so in $\mathcal{H}$, by adjunction and full faithfulness of L. The cited representability criterion, Artin et al. (1972, Exposé I, 8.12.7), supplies another sufficient argument when that hypothesis is available; the colimit argument above shows why it is unnecessary here.

Commentary · Examples [ms. 47]

All small suprema without a largest ordinal

Question 3.13 has a direct affirmative answer, with its universe distinction essential. Let $\mathrm{Ord}_{\mathfrak{U}}$ have the ordinals belonging to $\mathfrak{U}$ as objects, with one arrow $\alpha \to \beta$ when $\alpha \le \beta$. It is locally $\mathfrak{U}$-small: every Hom-set is empty or a singleton. For a $\mathfrak{U}$-small diagram its colimit is the supremum of the ordinals occurring in the diagram. This supremum belongs to $\mathfrak{U}$ by the universe axioms; for the empty diagram it is 0. There is no terminal object, since $\alpha + 1$ is again an object and is larger than α. Thus all small colimits exist but even the empty limit does not. The category is necessarily large relative to $\mathfrak{U}$.

The note about generators is correct with the convention used in SGA 4. In Artin et al. (1972, Exposé I, 7.1), a generating family means that the associated represented functors jointly *reflect isomorphisms*. Given any small family of ordinals $\{\alpha_s\}$, put $\beta = \sup_s \alpha_s$. Every map

$$\mathrm{Hom}(\alpha_s, \beta) \longrightarrow \mathrm{Hom}(\alpha_s, \beta + 1)$$

is a bijection between singletons, but $\beta \to \beta + 1$ is not an isomorphism. Hence there is no small generating family in this conservative sense. A merely faithful family of Hom-functors would be a different convention: every poset has at most one arrow with given endpoints, so faithfulness alone says almost nothing here. It would be incorrect either to reject Grothendieck's note using that weaker convention or to conclude from this example that a small cocomplete category need not be complete.

[page 48]

Question 3.14. Anderson prétend que pour le prédérivateur défini par une catégorie à cofibrations satisfaisant l'axiome Cof_4 transfini, les foncteurs $u_!^W$ existent. Sa démonstration est totalement canulée, néanmoins je n'arrive pas à faire de contre-exemple. Le résultat qu'il annonce serait-il vrai? Je ne connais pas même d'exemple d'un localiseur W, dans une catégorie $\mathcal{M}$ *avec petites* $\varinjlim$ *et petites* $\varprojlim$ *,* tel que $u_!^W$ ou u_*^W n'existent pas toujours – mais je suis persuadé qu'il y a des exemples (même avec $\mathcal{M}$ une petite catégorie?). J'en suis moins convaincu si on suppose W stable par $\varinjlim$ filtrantes, et au besoin, de plus, que les éléments de $\mathcal{M}$ sont accessibles (plus éventuellement d'autres propriétés catégoriques anodines sur $\mathcal{M}$, telles l'existence d'une filtration cardinale etc.) p. 26 [page 49]

Commentary · Later developments [ms. 48]

The cofibration assertion is true under the stated continuity hypothesis

The first question in 3.14 now has a positive answer. Modern cofibration-category theory separates this assertion from the existence of projective Quillen structures. In the presence of the initial object and pushouts in the standing framework, Cof_4 implies the coproduct and sequential-colimit axioms of an Anderson–Brown–Cisinski cofibration category. To see the coproduct assertion, well order a small family of cofibrant objects, attach its members one at a time by pushout from the initial object, and take colimits at limit stages. The resulting transfinite composite is the coproduct. Applied to a family of cofibrations, the same construction proves closure of cofibrations under coproducts; for acyclic cofibrations use the asserted transfinite closure of TC. Countable sequential composites are already a special case of Cof_4.

Rădulescu-Banu (2009, Lemma 1.6.2 and Theorem 10.3.2) gives the corresponding axioms and proves that every such cofibration category defines a strong derivator with all homotopy left Kan extensions. His Theorem 10.3.3 includes arbitrary Quillen model categories. A complementary formulation uses derivable categories: the dual of Cisinski (2010, Corollary 6.21) supplies the colimit theorem, while that corollary itself treats the limit direction in Cisinski's conventions. These results require neither a small set of generating cofibrations nor functorial factorization. They settle the existence question raised here, without retroactively validating the proof which Grothendieck criticizes.

The broader request in the same paragraph drops the cofibration structure entirely. It has a different answer, even when the underlying category and localizer have every finiteness and accessibility property explicitly mentioned here. The following finite example makes that distinction precise.

Commentary · Development (commentators') [ms. 48]

A finite complete lattice whose localization loses binary limits

An eight-object counterexample to the broader request in 3.14. [ours] There is a finite complete and cocomplete category P, with a strongly saturated localizer W stable under filtered colimits and accessible as a full subcategory of the arrow category, for which neither homotopy Kan extension along the discrete two-object category $D \to e$ exists. This elementary construction was prepared for this commentary. A targeted search for localizations of finite and Boolean lattices and failures of products or coproducts did not establish an attribution for this exact presentation. No priority claim is made. This is not a counterexample to the cofibration-category theorem just cited.

Let P be the Boolean lattice of subsets of $\{1,2,3\}$, ordered by inclusion, and abbreviate subsets by their strings of elements. Put $a = 1$, $b = 12$, and let W consist of the identities and $s : a \to b$. The two-out-of-three property holds: $a < b$ is a cover, so a factorization of its only nonidentity weak equivalence has an identity factor. The underlying category has all small limits and colimits, computed by intersections and unions.

We first compute the localization as a category, not just its object classes. Adjoin $s^{-1} : b \to a$. Any composable word containing two such inverses has the intervening forward path $a \to b$, which is s; it therefore reduces to a word with at most one inverse. Thus a morphism $x \to y$ is represented either by an inclusion or by

$$x \longrightarrow b \xrightarrow{s^{-1}} a \longrightarrow y.$$

If both forms occur, then $x \subseteq b$, $a \subseteq y$, and $x \subseteq y$. Either $x \subseteq a$, or x contains 2 and therefore $b \subseteq y$. In the first case the inverse cancels on the source side; in the second it cancels on the target side. The two forms consequently represent the same arrow. The localization is thin, with relation

$$x \preceq y \quad \Longleftrightarrow \quad x \subseteq y \ \text{ or } \ (x \subseteq 12 \text{ and } 1 \subseteq y).$$

This also verifies its universal property: for any functor F inverting s, the new arrow must be sent to $F(a \to y)F(s)^{-1}F(x \to b)$; the reductions just proved verify compositions and uniqueness. Only a and b become isomorphic. A skeleton Q therefore has seven objects, and W is exactly the inverse image of its isomorphisms, hence is strongly saturated.

In Q the common upper bounds of 2 and 3 are 13, 23, and 123. The first two are incomparable, so no least upper bound exists. Dually the common lower bounds of 13 and 23 are $\emptyset$, 2, and 3, and there is no greatest one. Thus Q lacks both a binary coproduct and a binary product. Localization of the product relative category $(P \times P, W \times W)$ gives $Q \times Q$: one may first invert $(w, 1)$ and then $(1, w)$, and these arrows generate the same inversions as $W \times W$. The restriction along $D \to e$ consequently identifies with the diagonal $Q \to Q \times Q$. Its left and right adjoints would give exactly the missing binary coproducts and products. Both therefore fail to exist.

Finally, all the accessibility qualifications can be checked directly. A filtered diagram in a finite poset has a finite directed image, hence a largest value which occurs at some stage and is its colimit. Every object of P is therefore finitely presentable. The same observation applies to the finite arrow category $P^{[1]}$. A filtered diagram all of whose objects lie in W has its colimit equal to one of those objects, hence in W. The full subcategory on W is itself finitely accessible: it has filtered colimits, every object is finitely presentable, and its finite set of objects supplies a generating set. The inclusion preserves filtered colimits. A cardinal filtration is trivial for this finite category. Thus the counterexample persists under the explicit additional hypotheses in the final sentences of 3.14. What fails is the extra homotopical structure supplied by the cofibration axioms, not any smallness or ordinary completeness condition.

Question 3.15. Cela me ramène à la question particulièrement élusive et irritante, de l'existence des petits ensembles Q-générateurs, dans une partie Φ de $\mathrm{Fl}(\mathcal{M})$, quand on suppose (disons) que $\mathcal{M}$ est stable par petites $\varprojlim$, que ses éléments sont accessibles et qu'il admet une filtration cardinale. Ainsi on a la question : une partie L-accessible de $\mathrm{Fl}(\mathcal{M})$, qu'on suppose au besoin Q-saturé à gauche (ou ce qui revient au même, contenant les isomorphismes, et stable par cochangement de base, par composition et par facteurs directs) admet-elle une *petite partie* Φ_0 Q-génératrice à gauche?

Commentary · Development (commentators') [ms. 49]

An accessible class with no small Q-generating subset

A negative answer to the Q-saturated form of Question 3.15. [ours] Let $\mathcal{K}$ be the category of directed multigraphs, with sets of edges and vertices and two functions $s, t : E \to V$, and let Φ be its class of pure monomorphisms. Then $\mathcal{K}$ is bicomplete, every object is accessible, and cardinality supplies a cardinal filtration. The full subcategory of $\mathcal{K}^{[1]}$ on Φ is accessible and is closed under all filtered colimits. The class Φ contains the isomorphisms and is closed under pushouts, transfinite compositions, and retracts, yet it has no small Q-generating subset. The non-generation theorem is due to Cox et al. (2026a, Theorem 5.1); the detailed translation to the hypotheses of this manuscript is the development supplied here. We make no priority claim for that translation.

Write $\mathcal{K} = \mathbf{Set}^{\mathcal{E}}$, where $\mathcal{E}$ has objects e, v, the distinct arrows $s, t : e \to v$, and identities. Limits and colimits are computed on the two sets, and finite graphs are finitely presentable. More generally a graph of cardinality less than a regular infinite cardinal λ is λ-presentable: maps out of it and equalities between such maps involve fewer than λ elements and equations, all of which occur at a common stage of a λ-filtered diagram. Thus every graph is presentable at some rank, and the subcategories with bounded cardinality are essentially small.

For an inclusion $A \subseteq B$, purity means that every finite system of equations in vertices and edges, with parameters from A, which has a solution in B, has a solution in A. Equivalently, in a commutative square from an arrow $P \to Q$ of finitely presentable graphs to $A \to B$, the top map $P \to A$ extends to $Q \to A$; equality with the given bottom map into B is not required. These are the equivalent descriptions in Cox et al. (2026a, Definition 2.7 and Remark 2.8).

We verify accessibility of the *full arrow subcategory*, whose maps are arbitrary commuting squares. This is distinct from the category with graphs as objects and only pure maps as morphisms. Given a pure inclusion $A \subseteq B$ and countable subsets in its domain and codomain, first close them under s, t and ensure that the chosen part of A is included in the chosen part of B. Suppose countable subgraphs $A_n \subseteq A$ and $B_n \subseteq B$, with $A_n \subseteq B_n$, have been chosen. There are only countably many finite equation systems with parameters in A_n. For each such system that has a solution in B_n, choose a solution in A, using purity of $A \subseteq B$. Adjoin all these witnesses to A_n, close under s, t, and adjoin their images to B_n. Repeat for $n < \omega$. The unions $A' \subseteq B'$ are countable. Any finite system over A' with a solution in B' has all its parameters and witnesses in a common stage, and was therefore given a solution in A' at the next stage. Hence $A' \to B'$ is pure.

The poset of countable pure subarrows of $A \to B$ is $\aleph_1$-directed: for a countable collection take the unions of their domains and codomains as seeds and apply the preceding closure construction. Its union is the whole arrow, since any chosen vertex or edge can be included in the seeds. There is a set of isomorphism types of countable graph arrows. Each is $\aleph_1$-presentable in $\mathcal{K}^{[1]}$, by the same element-and-equation argument already used for graphs, with the two components of the arrow included as additional functions. In particular every pure arrow is a filtered colimit of a fixed small family of bounded-size pure arrows, exactly the filtered-colimit-envelope condition used by the manuscript's earlier accessibility language.

We must also check preservation of the relevant colimits. Let $f_i : A_i \to B_i$ be a filtered diagram of pure arrows with arbitrary commuting-square transition maps. A square from finite $P \to Q$ into its colimit factors through a common stage, and its required equality holds at some later stage, because P, Q are finitely presentable. Purity at that stage gives $Q \to A_i$ extending the top map; composing into the colimit proves purity of the colimit arrow. Filtered colimits of sets preserve injections, so it is again a monomorphism. Thus the full subcategory on Φ has filtered colimits created in $\mathcal{K}^{[1]}$. Its countable arrows remain $\aleph_1$-presentable in that full subcategory, and the preceding directed presentation proves that it is $\aleph_1$-accessible. This verifies both parts of the manuscript's L-accessibility requirement, rather than inferring them from accessibility of a different category of pure maps.

For completeness, the closure conditions on Φ are also those required in the question. Composition follows by solving the same finite system first in the intermediate graph and then in the source. A retract of a pure arrow is pure: carry a parameter system and its solution along the section of the retract diagram, solve it in the larger source, and carry the solution back along the retraction. Arbitrary pushouts of pure monomorphisms are pure, as recorded in Cox et al. (2026a, Fact 2.9(iii)). Finally transfinite closure follows from composition at successor stages and the filtered-colimit argument at limit stages, applied to the arrows from the fixed starting object. Thus Φ is Q-saturated in the sense of the question, and in fact has the additional transfinite stability used elsewhere in the chapter.

The cited classification says that pure monomorphisms in $\mathbf{Set}^{\mathcal{E}}$ are cofibrantly generated precisely when, for every span $y \leftarrow x \to z$ in $\mathcal{E}$, one leg factors through the other. The span $v \xleftarrow{s} e \xrightarrow{t} v$ fails this test: the only endomorphism of v is its identity, and $s \neq t$. Hence Φ is not cofibrantly generated.

To connect that conclusion exactly to Grothendieck's terminology, suppose a small subset $S \subseteq \Phi$ were Q-generating. Its defining property is

$$\Phi \subseteq {}^{\perp}(S^{\perp}).$$

The small object argument applies in the locally presentable category $\mathcal{K}$. It identifies ${}^{\perp}(S^{\perp})$ with $\mathrm{cof}(S)$, the class of retracts of transfinite composites of pushouts of maps in S. The closure properties just proved give $\mathrm{cof}(S) \subseteq \Phi$. Both inclusions force equality, contradicting the classification theorem. This proves the asserted negative answer with no additional set-theoretic principle. The example concerns Q-*saturation*; it does not silently strengthen the question to the different hypothesis that Φ is already Q-*closed*. Nor does the argument identify accessibility of an arrow class with combinatoriality of a model structure, since no model structure has been supplied.

D'autre part, je n'ai pas à présent construit un seul exemple d'un couple $\Phi \longleftrightarrow \Psi$, dans une catégorie $\mathcal{M}$, qui ne soit factorisant! Le cas où $\mathcal{M}$ est spéciale, était bien sûr le plus intéressant. Visiblement, dans tout ça ce sont les exemples qui manquent. Le clef d'une compréhension semble bien être dans les questions de factorisation entre ensembles $\Phi \longleftrightarrow \Psi$. Il faudrait en faire une étude systématique, mais je recule devant la perspective d'une digression qui me ! | [page 50]

prendrait peut-être encore des semaines!

Commentary · Examples [ms. 50]

Mutual lifting classes need not factor every arrow

The failure of a mutually orthogonal pair to admit factorizations is already illustrated by Taylor (1999, Example 5.7.10). The following rational-cut presentation gives a small elementary instance, with no priority claim; it does not meet the stronger condition that the ambient category be special. Regard $\mathbf{Q}$ as a category under its usual order, and fix an irrational real number r, for example $\sqrt{2}$. Define $\mathcal{L}$ to contain the identities and the nonidentity arrows $a \to b$ with $b < r$. Define $\mathcal{R}$ to contain the identities and the nonidentity arrows $x \to y$ with $r < x$.

In a poset a lifting problem from $a \to b$ to $x \to y$ has a solution exactly when $b \leq x$, provided the square exists. Nonidentity arrows in the displayed classes have $b < r < x$, so $\mathcal{L} \perp \mathcal{R}$. Conversely, if a nonidentity $a \to b$ is not in $\mathcal{L}$, then $b > r$. Choose a rational x with $\max(a, r) < x < b$. The square from $a \to b$ to $x \to b \in \mathcal{R}$, with bottom identity, has no lift. Hence ${}^{\perp}\mathcal{R} = \mathcal{L}$. If a nonidentity $x \to y$ is not in $\mathcal{R}$, then $x < r$. Choose rational b with $x < b < \min(y, r)$. The square from $x \to b \in \mathcal{L}$ to $x \to y$, with top identity, has no lift. Thus $\mathcal{L}^{\perp} = \mathcal{R}$.

Nevertheless, an arrow $a \to b$ with $a < r < b$ has no factorization of the indicated type. An intermediate object z with both factors nonidentity would have to satisfy simultaneously $z < r$ and $r < z$; choosing either factor to be an identity also fails. This is a pair of mutually closed lifting classes, but not a weak factorization system. The rational poset is not cocomplete, so the example answers the unrestricted request only. It leaves the final, more demanding special-ambient question logically separate from the counterexample.

Overview · XIX

Chapter XIX: factorization, diagram categories, and the existence of derived Kan extensions.

Grothendieck has reached the point at which his formalism must acquire its examples. Given a category $\mathcal{M}$ and a class W of weak equivalences, the intended coefficient system is

$$\mathbb{D}_{\mathcal{M},W}(I) = \mathrm{Fun}(I^{\mathrm{op}}, \mathcal{M})[W(I)^{-1}], \qquad u^* : \mathbb{D}(J) \longrightarrow \mathbb{D}(I).$$

The problem is to construct both adjoints to restriction, prove their pointwise formulas, and explain which hypotheses on the models are doing the work. The chapter's fifty manuscript pages therefore move from lifting and factorization to model structures on diagrams, and then return to the places where that method appears to stop. In the edition the chapter occupies printed pages 1–26, after two preliminary leaves.

The historical motivation precedes the terminology of this chapter. In *Pursuing Stacks*, §69, Grothendieck relates the passage from a single derived category to a system of derived diagram categories to the inadequacy of the former for the variance and trace constructions arising in SGA 5. His letter to Thomason of 2 April 1991 connects the long investigation of accessible categories in Chapter XVIII with the foundations needed for this coefficient formalism (Grothendieck, 1983, 1991a). Thus accessibility enters here for a concrete reason: the homotopical operations must be constructed using sets of generators and controlled colimits before they can be recognized intrinsically after localization.

The two levels of closure. The opening pages isolate the relation $i \perp p$ of admitting a diagonal in every lifting square. It gives an antitone correspondence between classes of arrows, hence the closures ${}^{\perp}(S^{\perp})$ and $({}^{\perp}T)^{\perp}$. Pushouts, transfinite compositions, and retracts describe a second closure process. The small object argument identifies the two when a set of arrows generates the lifting condition and their domains have the required smallness. The retract argument is the decisive step: factor an arrow already having the left lifting property, then use its lift against the right factor to express it as a retract of the constructed cellular arrow. This is why the distinction between an orthogonal pair and a factorizing pair must remain visible. The source's initial identification of separate closedness of two orthogonal classes with mutual determination of the classes is too strong; the factorization hypothesis supplies the missing implication later.

Grothendieck's word "special" must also retain its stated meaning. His ambient category has small colimits and each object is presentable at some cardinal; his special left classes have a small set of lifting generators. This does not itself supply the small generating subcategory required for local presentability. Modern statements about combinatorial model categories may consequently be used only after the additional ambient hypothesis has been made explicit.

The asymmetry of projective and injective diagrams. For contravariant diagrams, evaluation at i has a left adjoint $i_!A(j) = \coprod_{\mathrm{Hom}_I(j,i)} A$. Applying these adjoints to generating cofibrations and generating trivial cofibrations constructs the projective structure. Its fibrations and weak equivalences are pointwise; its cofibrations encode compatibility with the arrows of the indexing category. Theorem 2.3 then derives $u_!$, and Proposition 1.12 supplies the restriction property needed to compute its values on comma categories. In the variance used here the formulas are

$$(\mathbf{L}u_!X)_j \simeq \operatorname*{hocolim}_{(j\downarrow u)^{\mathrm{op}}} X, \qquad (\mathbf{R}u_*X)_j \simeq \operatorname*{holim}_{(u\downarrow j)^{\mathrm{op}}} X.$$

Writing the opposite categories is necessary: the manuscript's diagrams are presheaves, whereas much modern derivator notation uses covariant diagrams.

For the injective structure the cofibrations are pointwise, and it is the right class whose compatibility is no longer pointwise. One cannot obtain small generators merely by placing the

old generators at every component. Question 1.11 has a positive answer for locally presentable ambient categories, by the generation theorem of Lurie (2009, Lemma A.2.8.3), but its final suggested choice of generators already fails for sets with a nontrivial group action. The commentary gives the sharper elementary criterion for when the sole generator $\varnothing \to 1$ can be used unchanged on a diagram category, with a complete proof and an explicit statement of the literature checked.

Theorems 2.6 and 2.9 secure the injective construction in Grothendieck's "very special" situation, in which the cofibrations are all monomorphisms. The geometric conditions on unions and pushouts, together with accessibility of W, are the mechanism behind the recognition theorem. Here two-out-of-three does essential work: after building a bounded trivial subextension, it keeps the remaining extension in W. This is a substantially stronger situation than an arbitrary accessible class of monomorphisms. The proof recalled from Chapter XV must be read through its corrected argument in §7, not through the explicitly abandoned general claim in §6. The cardinal-stage condition printed with a projective-limit arrow in XIX is identified in the apparatus as a source slip: the cited formulations require closure under the corresponding small inductive limits.

The latching interpretation in section 3 returns to the same care about testing classes. Proposition 3.5 needs the canonical right class, which the source explicitly chooses in (3.5.2); an arbitrary presentation by right tests can give a false equality. The commentary supplies the counterexample and the valid comparison. It also proves the classwise cellular description conjectured in (3.5.3) for noetherian posets, with the required colimits stated, and supplies the short lifting lemma implicit in Proposition 3.7.

What later existence theorems change. The existence of a convenient diagram model structure and the existence of derived Kan extensions are different assertions. Cisinski proves the latter for an arbitrary Quillen model category with small limits; duality supplies the colimit statement when small colimits exist (Cisinski, 2003, Theorem 6.11). Consequently the bicomplete Quillen case of Question 3.9 admits no counterexample of the requested kind. For combinatorial models, both diagram model structures give a direct and symmetric proof; Groth (2013, Theorem 1.28 and Proposition 1.30) presents this route. The later existence theorem does not retroactively prove every factorization assertion contemplated in the manuscript, and it does not make the adjective "very special" necessary.

The final section shows why the size of the permitted diagrams matters. Its bounded-complex example is effective precisely because bounded derived categories need not contain the result of a homotopy limit indexed by a finite category with arbitrarily long composable strings. For $k = \mathbf{F}_2$ and $G = C_2$, the resolution of the trivial $k[G]$-module is periodic and gives nonzero group homology and cohomology in every nonnegative degree. Testing a hypothetical adjoint to the trivial-action functor against shifts proves its nonexistence within the bounded category. This supplies the required obstruction without identifying a bounded category with a bicomplete modern model category. The distinction between cofibration-category excision on cofibrant objects and the manuscript's absolute W-cofibrations is equally necessary in reading the Brown lemma invoked on manuscript page 35.

Some of the closing searches have elementary answers. A fully faithful coreflective image in a bicomplete category is itself bicomplete: the coreflector constructs limits, and the image is closed under colimits. Thus the proposed adjoint-localization construction in Example 3.12 cannot produce its hoped-for failure. The category of ordinals small relative to the fixed universe has all small suprema and no terminal object, answering Question 3.13 with the universe convention made explicit. Its lack of a small conservative generating family is consistent with the convention of SGA 4. A rationally ordered category cut at an irrational number supplies a mutually lifting-closed pair which does not factor arrows across the cut; it does not have the cocompleteness required of a special ambient category.

The broader localizer question in 3.14 admits a small counterexample. Localize the

Boolean lattice of subsets of $\{1,2,3\}$ at the single cover $\{1\} \subset \{1,2\}$. The resulting seven-object skeleton has neither a least upper bound for $\{2\},\{3\}$ nor a greatest lower bound for $\{1,3\},\{2,3\}$. The localizer is strongly saturated, accessible, and stable under filtered colimits, and the original lattice is complete and cocomplete. The commentary computes the localization and proves each assertion. This answers the broad question even with the additional smallness hypotheses. The cofibration-category question in the same paragraph has the opposite answer: the transfinite continuity axiom supplies the coproduct and sequential-colimit hypotheses of modern cofibration-category theory, which yields all homotopy left Kan extensions (Rădulescu-Banu, 2009, Theorem 10.3.2).

Question 3.15 also benefits from separating the closures introduced at the start. The class of pure monomorphisms in directed multigraphs is accessible, closed under filtered colimits, pushouts, transfinite composition, and retracts. It nevertheless has no small set of Quillen generators. The non-generation theorem is due to Cox, Feigert, Kamsma, Mazari-Armida, and Rosický, in their 2026 classification for presheaf categories (Cox et al., 2026a, Theorem 5.1); the commentary verifies the translation into the hypotheses stated here and supplies the small-subdiagram argument for accessibility. This answers the question for a Q-saturated class. It does not identify that class with its double lifting orthogonal, and the stronger question obtained by imposing that additional condition is not silently substituted for the question Grothendieck actually wrote.

The commentary retains the edition's slips as source evidence, distinguishes them from errors of transcription, and states repairs at the point where they are needed. The claim register records the source's proof status separately from later theorems or counterexamples. Mathematical dependencies on Chapters XIII, XV, XVII, and XVIII are checked against the relevant original passages or explicitly identified as recalled results.

Register of claims

The audit of each part records the definitions, axioms, propositions, claims, conjectures and questions of the text, each with the status the text gives it: proved, sketched, stated, assumed, conjectured, false (the assertion, or the inference as written, is false) or open. The table counts the entries of each part by status; the list after it gives every entry whose status is false, open or conjectured, with its manuscript page and its identifier in the audit.

Part	proved	sketched	stated	assumed	conjectured	false	open	total
I-a	53	24	136	19	7	14	0	253
I-b	72	37	101	22	9	12	0	253
II-a	50	21	145	1	5	20	3	245
II-b	67	45	108	4	0	21	9	254
III-V	54	27	127	8	11	12	10	249
VI-a	88	29	137	19	16	23	0	312
VI-b	115	18	59	17	9	15	17	250
VI-c	88	27	72	9	8	5	5	214
VII-a	92	19	74	2	0	12	0	199
VII-b	54	16	82	3	5	17	7	184
VII-c	70	26	62	4	30	35	11	238
VIII-X	11	19	104	5	24	12	10	185
XI-XIIa	124	35	90	43	12	9	11	324
XIIb-a	91	23	70	7	0	18	12	221
XIIb-b	74	11	36	4	3	4	1	133
XIIc-a	51	21	89	14	11	9	19	214
XIIc-b	63	20	31	4	3	5	1	127
XIIapp	11	31	73	4	4	17	5	145
XIII-a	56	19	172	27	44	12	17	347
XIII-b	33	13	63	17	5	11	9	151
XIV	18	6	27	8	2	4	0	65
XV-a	88	23	66	17	1	28	2	225
XV-b	86	31	125	16	7	15	9	289
XVI	79	35	137	31	17	10	28	337
XVII-a	100	14	53	9	0	9	4	189
XVII-b	110	9	42	7	2	6	4	180
XVIII-a	141	33	95	0	6	41	2	318
XVIII-b	144	28	97	0	8	19	13	309
XVIII-c	112	37	51	3	4	15	2	224
XVIII-d	100	46	50	5	5	16	1	223
XVIII-e	98	26	52	7	6	20	13	222
XIX	62	17	148	7	4	11	20	269
total	2455	786	2774	343	268	477	245	7348

I-a · Chapter I, part 1 of 2, manuscript pages 1–66

ms. 3 CONJECTURED I-a-E020 – For a monotone map of finite ordinals, restriction on A-valued presheaves has a second adjoint on each side when A is pointed and has finite

limits and colimits.

ms. 3 FALSE I-a-E021 – All restriction functors associated with all maps between two fixed finite ordinals occur in the same successive adjunction chain.

ms. 3 CONJECTURED I-a-E022 – For [n] and [n+1], the face and degeneracy restrictions form a chain of 2n+3 terms, extendible by two terms at each end to 2n+7.

ms. 4 FALSE I-a-E024 – A category in which every arrow is monic and every isomorphism is an identity is a poset.

ms. 15 FALSE I-a-E066 – The abelianization morphism preserves all right Kan extensions.

ms. 15 FALSE I-a-E067 – A homotopy pullback of spaces along a fibration is sent by abelianization to a homotopy pullback.

ms. 25 FALSE I-a-M007 – The mixed interchange map might be invertible in additive derivators.

ms. 29 CONJECTURED I-a-M021 – There should be a stable or triangulated derivator theory in which left and right exact morphisms coincide.

ms. 32 FALSE I-a-M030 – The withdrawn Der5 asserts that a strict hot-equivalence has fully faithful restriction, equivalently $1 \to f_* f^*$ is invertible.

ms. 32 FALSE I-a-M032 – The dual withdrawn Der5 asserts $f_! f^* \to 1$ invertible for every costrict hot-equivalence.

ms. 32 FALSE I-a-M033 – The withdrawn argument would make f_*, respectively $f_!$, localization functors.

ms. 47 FALSE I-a-M060 – The same cohomology invariance would follow if instead all (α/x) were contractible.

ms. 47 FALSE I-a-M062 – The terminal-object condition above would substantially strengthen cohomological cofinality.

ms. 47 FALSE I-a-M064 – The condition $\alpha_! \varepsilon_{X'} \simeq \varepsilon_X$ follows from either orientation of cofinality.

ms. 47–48 FALSE I-a-M067 – Corollary2 repeats the claim that strict hot-equivalence implies $F \to f_* f^* F$ invertible for arbitrary F.

ms. 48 FALSE I-a-M071 – The preceding slice calculation proves the withdrawn full-faithfulness corollary by opposite cofinality.

ms. 49 FALSE I-a-L003 – The same sufficient base-change criterion holds with $h_{y'}$ homotopy aspherical (all $h_{y'}/(x,u)$ contractible).

ms. 58 CONJECTURED I-a-L035 – If right base change for a fixed f holds for every base change and every derivator, then f is proper.

ms. 58 CONJECTURED I-a-L036 – If base change along a fixed p commutes with right direct images for every f and every coefficient theory, then p is smooth.

ms. 58 CONJECTURED I-a-L037 – Testing HOTAB and HOT should suffice to detect the universal proper/smooth base-change properties.

ms. 59 CONJECTURED I-a-L039 – For f, the four conditions universal right base change, universal left-mate base change, coasphericity of X_y->X/y, and contractibility of every fibre of x\X->f(x)\Y are equivalent.

I-b · Chapter I, part 2 of 2, manuscript pages 67–126

ms. 78 CONJECTURED I-b-A069 – The doubled-arrow construction probably does not give the desired counterexample to base-change stability.

ms. 79 FALSE I-b-B079-03 – Every universally cohomological D-fibration preserves all constant-coefficient D-equivalences under pullback.

ms. 79 FALSE I-b-B079-04 – Every universally homological D-fibration preserves all constant-coefficient D-equivalences under pullback.

ms. 79 FALSE I-b-B079-07 – The implication (b)⇒(a) nevertheless holds for all derivators D_i of i-truncated homotopy types.

ms. 81 FALSE I-b-B081-04 – Universal cohomological/homological D-fibrations as subsequently redefined are closed under composition.

ms. 84 CONJECTURED I-b-B084-01 – For a universal D-equivalence f:X→Y, restriction f^*:D(Y)→D(X) is fully faithful.

ms. 84 FALSE I-b-B084-06 – For D-fibrations over S, a D-equivalence of total categories must be a D-equivalence on each fibre.

ms. 84 FALSE I-b-B084-07 – The converse of Proposition6(a) holds whenever S is D-contractible.

ms. 87 FALSE I-b-B087-05 – A bundle over the circle is determined up to unique isomorphism by a specified monodromy automorphism of its fibre.

ms. 93 FALSE I-b-B093-04 – Conditions(a)–(c) alone imply that the inclusion of a fibre of(E×Z)/G→E/G is not a D-equivalence.

ms. 93 FALSE I-b-B093-05 – If additionally H*(Z;k)^G=k, then X=(E×Z)/G is D(k)-aspherical.

ms. 95 FALSE I-b-B095-01 – The orbit of the six-object poset has the displayed globular relations p1^-p0^-=p1^-p0^+ and p1^+p0^-=p1^+p0^+.

ms. 96 CONJECTURED I-b-B096-04 – Under Der3a′, the revised weakly-contractible-base test condition is equivalent to each of the universal one-sided fibration conditions.

ms. 102 FALSE Ib-C003 – If f is Hot-proper then both X_y→X/y and X/y→X/y′ are Hot-equivalences, universally.

ms. 107 CONJECTURED Ib-C021 – Footnote55: for HOT, cohomological equivalence, homological equivalence, and constant-coefficient equivalence coincide.

ms. 109–110 CONJECTURED Ib-C026 – Every Hot-equivalence induces an equivalence on locally constant HOT coefficients and hence preserves their cohomology and homology.

ms. 115 CONJECTURED Ib-C048 – For HOT all five equivalence notions coincide with nerve weak equivalence.

ms. 119 CONJECTURED Ib-C069 – For HOT, D-fibrations do preserve D-equivalences by pullback.

ms. 122–123 FALSE Ib-C082 – With compatible internal Homs, every unqualified D-equivalence is 1-complete.

ms. 125–126 CONJECTURED Ib-C092 – HOT satisfies c″ and its equivalences are precisely nerve weak equivalences.

ms. 126 CONJECTURED Ib-C096 – There should be a single intrinsic HOT-fibration notion on all small categories; the homotopy derivator must be constructed to justify the coefficient route.

II-a · Chapter II, part 1 of 2, manuscript pages 1–53

ms. 3–4 CONJECTURED II-A-013 – For homotopical cofinality, replace the connectedness criterion on comma categories by weak contractibility of their nerves.

ms. 5 FALSE II-A-017 – The manuscript invokes Der5 to infer $F \to u_* u^* F$ invertible for all F from homotopy strictness, i.e. contractibility of each u/c.

ms. 5 FALSE II-A-018 – The dual asserted consequence is $u_! u^* F \to F$ invertible for all F when all c/u are contractible.

ms. 5 OPEN II-A-019 – If each c/u is weakly contractible, is $H^i(C, F) \to H^i(C\prime, u^* F)$ an isomorphism for every abelian presheaf F and every i?

ms. 7 CONJECTURED II-A-025 – For $K_u(c) = \pi_0(c/u)$ there is a natural isomorphism $\lim u^* F \cong \mathrm{Nat}(K_u, F)$.

ms. 7–8 CONJECTURED II-A-026 – $\mathbf{R}\Gamma(C\prime, u^* F) \simeq \mathbf{R}\,\mathrm{Hom}_{C;k}(\mathbf{L}u_! k_{C\prime}, F)$ in D(k).

ms. 8 CONJECTURED II-A-028 – For space-valued presheaves, $\mathbf{R}\Gamma(C\prime, u^* F) \simeq \mathbf{R}\Gamma(C, \underline{\mathrm{Hom}}(u_! 1, F))$.

ms. 9 FALSE II-A-032 – The proposed unsectioned identity is $u_* u^* F \simeq \underline{\mathrm{Hom}}(u_! 1, F)$.

ms. 10 FALSE II-A-034 – The proposed general formula is $u_* \underline{\mathrm{Hom}}(G\prime, u^* F) \simeq \underline{\mathrm{Hom}}(u_! G\prime, F)$.

ms. 10 OPEN II-A-036 – There is a canonical projection comparison from adjunction, whose invertibility is in question.

ms. 12–13 FALSE II-A-049 – The first list claims equivalence between local irreducibility of Top(X), local irreducibility of Esp(X), irreducibility of components, and irreducibility of all slices X/x.

ms. 12–13 FALSE II-A-051 – The second list a–e again includes weak cofiltering of connected components among conditions equivalent to local irreducibility of Top(X).

ms. 13 FALSE II-A-054 – The prose identifies local irreducibility of the presheaf topos with PS1 plus PS2, or componentwise cofiltering.

ms. 15 OPEN II-A-060 – Find a topos-theoretic condition equivalent to PS1+PS2 and ensuring every connected étale category has weakly contractible nerve.

ms. 16 FALSE II-A-066 – The NB claims PS1 is equivalent to componentwise weak cofiltering.

ms. 18 FALSE II-B-006 – Total 0-connectedness alone implies W_infty-asphericity.

ms. 19 FALSE II-B-019 – The representable-base conditions (c,c-prime) imply the strong equivalent conditions (a,a-prime,b,d).

ms. 21 FALSE II-B-024 – For every pair of connected U,V over an arbitrary object W, U times_W V is nonempty.

ms. 24 FALSE II-B-033 – Condition 2.6(c-prime) implies PS2.

ms. 28 FALSE II-B-045 – If all slices X/x are cofiltered, then X must be locally cofiltered.

ms. 36;36ter CONJECTURED II-C-003 – Does T(A) imply that A is filtered?

ms. 36 FALSE II-C-006 – Proposition2.19: a nonempty category with cofinal diagonal satisfies T(A).

ms. 36 FALSE II-C-008 – A totally0-connected category has weakly contractible nerve.

ms. 36 FALSE II-C-009 – Corollary to2.19: Delta^op satisfies T despite not being filtered.

ms. 36bis–36ter FALSE II-C-015 – In the proposed proof of2.20, a weak equivalence i:X→ E is used to infer that X/H is weakly contractible by a reference to ChapterXII.

ms. 51 FALSE II-C-074 – The source asserts that M-connectedness is equivalent to the slice-limit comparison for coefficients on I/i restricted from I.

ms. 51 FALSE II-C-076 – The converse to4.8 holds for ordered I without restrictions on M.

ms. 52 FALSE II-C-078 – Proposition4.1(d prime) as printed: all coslice functors i\I prime→i\I are cofinal is equivalent to full faithful restriction.

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ms. 45 OPEN IIb-A004 – Find a common cofinality statement encompassing 3.4 and XVIII Proposition 4.9.6.3.

ms. 51 OPEN IIb-A026 – Does M-connectedness imply M-cofinality of all slice maps, when slice diagrams need not extend globally?

ms. 52 FALSE IIb-A031 – The printed condition that every i\ I′→ i\ I is cofinal is equivalent to full faithfulness of restriction.

ms. 54-56 OPEN IIb-A038 – Does invertibility of y_i→ u_*u^*y_i for all representables imply W_0-connectedness?

ms. 59 FALSE IIb-A052 – Are W_0-connected functors precisely ordinary localization functors, equivalently are conservative W_0-connected functors equivalences?

ms. 63bis-64bis FALSE IIb-A069 – If binary coproduct on D(e) is faithful, left Kan extension along every discrete fibration is faithful.

ms. 64bis OPEN IIb-A073 – Can a D-connected functor fail to be D-aspherical (the literal printed question)?

ms. 64bis OPEN IIb-A074 – Can D-connectedness fail to imply D-biasphericity when both coproduct and product conservativity are absent?

ms. 65 FALSE IIb-A080 – D-connectedness is an exception to closure of the seven classes under composition.

ms. 46 FALSE IIb-A085 – Nonemptiness of every comma category i\ I′ is equivalent to F0 together with F1.

ms. 67 FALSE IIb-B013 – For every base-change- and composition-stable L, N(S(N(L)))=N(L).

ms. 67 FALSE IIb-B015 – For every A contained in W, S(N(S(A)))=S(A).

ms. 77 FALSE IIb-B050 – The printed second line (5.20) states A^natural=sup_{p in A} p^natural.

ms. 79 FALSE IIb-B057 – For a Galois extension L/K, the fixed-point polarity has intermediate fields as its closed subsets of L and all subgroups as its closed subsets of G.

ms. 81 FALSE IIb-B062 – The lifting relation on classes is not induced by a relation on individual arrows.

ms. 81-82 FALSE IIb-B064 – If phi records (left edge,right edge), the lifting condition for classes (Phi right,Psi left) is phi inverse(Phi times Psi) subset D0 as printed.

ms. 83 FALSE IIb-B071 – The condition in (6.3) is equivalent to saying A is stable under pullback along L.

ms. 84 FALSE IIb-B072 – With the initial definitions, P and Q are closed under arbitrary unions.

ms. 85 FALSE IIb-B078 – The pullback-stability of T_W(A) proves T_W(A) belongs to the Q defined in (6.2).

ms. 86 FALSE IIb-B088 – The prose before (6.8) identifies all closed pairs with pairs whose two members are composition-closed.

ms. 89 OPEN IIb-B100 – Find an effective intrinsic characterization of T_W(W_asph) and T_W(W_coas).

ms. 89 OPEN IIb-B101 – Define Net_W=Stab(W_bias) and seek a characterization using bilocal factorization categories.

ms. 89 OPEN IIb-B102 – Determine Stab(Equ), the stabilizer of categorical equivalences.

ms. 89 FALSE IIb-B103 – The printed final inclusion says Equ subset P(W) for a fundamental localizer W.

ms. 90 FALSE IIb-C009 – Remark 7.2, literally: every discrete fibration is locally filtered.

ms. 91 OPEN IIb-C019 – Remark 7.4(1): are locally cofiltered or precofiltered functors stable under arbitrary base change?

ms. 92 FALSE IIb-C022 – Remark 7.4(3), literal intermediate assertion: a cofiltered functor is W_0-aspherical.

ms. 92-95 FALSE IIb-C027 – Proposition 7.4(c), literally: for any open U⊂ X/x let V be the lower open generated by the image of all X/x; then U→ V is cofiltered (respectively weakly cofiltered).

ms. 98-99 FALSE IIb-C063 – The abbreviated total-connectedness criterion C^{op}→ C^{op}× C^{op} final, without nonemptiness.

ms. 99 FALSE IIb-C065 – The parenthetical locally cofiltered = components filtered, and the label PS*1 = the x\ C satisfy PS*1′, in the first closing diagram.

III-V · Chapter III, manuscript pages 1–12; Chapter IV, manuscript pages 73–83; Chapter V, manuscript pages 1–52

ms. III:1 OPEN III-002 – Seek a covariant tensor Hot × $\mathcal{A} \to \mathcal{A}$ and a contravariant-in-space cotensor $\mathrm{Hot}^{op} \times \mathcal{A} \to \mathcal{A}$.

ms. III:1 CONJECTURED III-004 – For each space the two operations should extend from $\mathbf{D}(e)$ to morphisms $\mathbf{D} \to \mathbf{D}$.

ms. III:3 OPEN III-010 – The comparison $\mathrm{Hot}_0 \to \mathrm{Hot}$ may be fully faithful for a suitable domain of sufficiently small homotopy types.

ms. III:4 CONJECTURED III-013 – A nerve weak equivalence $f : X' \to X$ is a **D**-equivalence.

ms. III:4 OPEN III-015 – An equivalence of categories in Diag is a **D**-equivalence.

ms. III:5 FALSE III-018 – Tensor and cotensor by the unpointed circle are suspension and based loops.

ms. III:7 CONJECTURED III-026 – The functor $\mathrm{Hom}_{(-)}(M, N)$ appears representable by an object of Hot_0, or at least by a space.

ms. III:8 OPEN III-029 – For a Quillen model presentation, the external mapping space satisfies the representation formula (10).

ms. III:10 FALSE III-034 – Equation (14) specializes (13) to a circle and identifies it with suspension–loop adjunction.

ms. III:11 OPEN III-043 – Footnote 2 requests an explicit verification of the tensor-cotensor adjunction for complexes.

ms. III:11 CONJECTURED III-044 – A candidate external Hom is the space underlying the nonpositive truncation of $\mathbf{R}\,\mathrm{Hom}_k(M, N)$.

ms. III:12 OPEN III-045 – Footnote 3 asks whether $[\xi, H] \cong \mathrm{Hom}_{\mathrm{Hotab}}(\xi_{\mathrm{ab}}, H)$.

ms. III:12 CONJECTURED III-051 – The better external Hom might take values in $D^-(\mathbb{Z})$, retaining more than its nonpositive truncation.

ms. IV:73 OPEN IV-004 – Are coherent diagrams with isomorphic skeletons necessarily isomorphic?

ms. IV:78 OPEN IV-018 – Seek canonical coherent lifts of the unit $G \to f_* f^* G$ and counit $f^* f_* F \to F$, and dually for $f_! \dashv f^*$.

ms. IV:82 CONJECTURED IV-030 – The collage with its structural maps is characterized by the universal triple description, up to compatible canonical isomorphism.

ms. IV:82 OPEN IV-034 – Seek $\mathbf{D}(K) \to \mathbf{D}(I \times [1])$ whose underlying arrow is u.

ms. IV:82 OPEN IV-035 – Seek $\mathbf{D}(K) \to \mathbf{D}(J \times [1])$ whose underlying arrow is the adjoint v.

ms. V:8–9 FALSE V-P035 – Raw concatenation has shape Φ_{m+n} whenever m, n have the same parity.

ms. V:13 FALSE V-P047 – Checking only simultaneous increments $(m, n) \mapsto (m+1, n+1)$ suffices for the proposed compatibility.

ms. V:15–16,24 FALSE V-P052 – Fixed-endpoint path categories are full subcategories of the unqualified functor categories.

ms. V:25–26 CONJECTURED V-P081 – A nonempty category of endpoint-preserving functors between generalized intervals is connected.

ms. V:26 CONJECTURED V-P082 – The same nonempty endpoint-preserving functor category is contractible.

ms. V:37 FALSE V-I026 – The localization $\varphi : \mathrm{Cat}\,\mathrm{hot} \to \mathrm{Hot}$ has a right adjoint ψ, called a Kan envelope.

ms. V:46 CONJECTURED V-I057 – Essential constancy should almost never occur in practice.

ms. V:47 FALSE V-I059 – If such an ind-map induces an isomorphism in Ind(Cathot), its union map is necessarily a weak equivalence.

ms. V:47 FALSE V-I060 – A weak equivalence between the envelope unions is an isomorphism in Cathot.

ms. V:49 CONJECTURED V-I064 – The required minimal lengths of stagewise homotopies may tend to infinity, obstructing a global categorical homotopy.

ms. V:49 FALSE V-I065 – For a Thomason fibrant K′, are categorically homotopic functors X→K′ always connected by one natural transformation?

ms. V:49-50 CONJECTURED V-I066 – Different reasonable theories of paths might yield canonically and transitively isomorphic unions in Cathot, as well as isomorphic ind-objects.

ms. V:52 FALSE V-I071 – This restricted Yoneda functor is conservative.

ms. V:52 FALSE V-I072 – Bijectivity on unbased homotopy classes from all spheres already detects weak equivalence.

ms. V:52 FALSE V-I073 – Is the restricted Yoneda functor Hot→Fun(Hot_0^op,Ens) fully faithful?

VI-a · Chapter VI, part 1 of 3, manuscript pages 1–70

ms. 4, 8 FALSE VIa-039 – The initial comparison theorem asserts that all functors of the thirteen-category diagram induce equivalences after the localizations specified on ms7, identifying strict-fibrewise diagram theory with coarse Cat/S theory.

ms. 5 CONJECTURED VIa-040 – The higher coherence discarded by ordinary localization should be recovered in relative internal Hom objects interpreted as ∞-groupoid stacks over S.

ms. 6, 33 FALSE VIa-048 – Read literally for arbitrary W⊇W∞, the source identifies transitions being Hot-equivalences with transitions belonging to W (ms6), and identifies Hot_W-valued local coefficients exactly with the index-0 split categories defined by W∞-transitions (ms33).

ms. 14 CONJECTURED VIa-067 – Replacing Φ and Ψ by $\Phi'{=}\alpha'\Phi_0$ and $\Psi'{=}\beta'\Psi_0$ is proposed as a way to make the initial localized equivalence argument work.

ms. 14 FALSE VIa-068 – For an arbitrary W satisfying the initially stated localizer assumptions, the source identifies total and strict-fibrewise W-equivalence on Parf(S), Fibhot(S), $\mathrm{Fib}_0(S)$ and $\mathrm{Cofib}_0(S)$.

ms. 21 FALSE VIa-097 – Footnote 13 literally states that if X is a homotopy fibration over S then $\Psi(X)$ belongs to $\mathrm{Fibsc}_0(S)$.

ms. 22 CONJECTURED VIa-103 – Desired numbered assertion (2): for f:X→Y over S with X,Y perfect, or merely homotopy fibrations, total f∈W should imply f_s∈W for every s.

ms. 23, 45 CONJECTURED VIa-108 – The source conjectures on ms23 that perfect-base-change stability should hold for the finite homotopy-truncation localizers W_i, and on ms45 asks whether they satisfy new L5, L6 and L6 bis.

ms. 23 FALSE VIa-111 – From a commutative square with vertical weak homotopy equivalences and lower horizontal map f′ merely in W, the source concludes that the upper fibre map f′_{s′} is also a weak homotopy equivalence.

ms. 26 FALSE VIa-123 – Every Grothendieck fibration over S is isomorphic over S to a split fibration, so Fibsc(S)→Fib(S) is essentially surjective as an ordinary functor; dually for opfibrations.

ms. 26 FALSE VIa-124 – Restricting Fib(S) to Cartesian functors makes Fibsc(S)→Fib(S) fully faithful and an equivalence of ordinary categories; the cocartesian dual assertion also holds.

ms. 26-27 CONJECTURED VIa-125 – After restricting morphisms from arbitrary functors over S to coherent splitting-preserving ones and simultaneously restricting weak equivalences from coarse to fibrewise ones, the resulting localized categories might still be equivalent.

ms. 30-31 FALSE VIa-135 – In the surrounding contravariant Φ construction, its transport square is the printed covariant square with fibres X/s and X/t and transport u_*:X_s→X_t.

ms. 32-35 CONJECTURED VIa-141 – The nine nonsplit categories of the p8 diagram, with coarse W-equivalences, should model the full subcategory of locally constant objects of HOT_W(S); passing from an arbitrary diagram to its total category and then C should give its locally constant replacement.

ms. 33 FALSE VIa-143 – The category Cofibsc(S) is the outer opposite category (Fibsc(S^op))^op.

ms. 33, 35, 37 CONJECTURED VIa-145 – The functors Fibsc_0(S)→Fib_0(S) and Cofibsc_0(S)→Cofib_0(S) should induce equivalences after localization, with quasi-inverses represented by Φ_0 and Ψ_0, under the standing fibre/coarse identification hypothesis on internal objects.

ms. 33-35, 37 CONJECTURED VIa-146 – The inclusions Fibsc_0(S)→Fibsc(S) and Cofibsc_0(S)→Cofibsc(S), localized fibrewise, should be fully faithful, with essential images precisely the locally constant objects of the respective contravariant and covariant diagram homotopy categories.

ms. 33-34 FALSE VIa-147 – The fine covariant theory and its locally constant inclusion are to be written using (HOT_W(S^op))^op, as printed in the p33–34 discussion.

ms. 34-35 CONJECTURED VIa-148 – After identifying the coarse localization with local systems, forgetting the strict diagram structure should induce retractions from HOT_W(S) and HOT_W(S^op) onto their locally constant full subcategories.

ms. 38 FALSE VIa-156 – The equation $k\psi \cong \mathrm{id}_B$ implies that k is full on Hom_A(a,a′) for arbitrary objects a,a′ of A.

ms. 38 FALSE VIa-157 – Under $k\psi \cong \mathrm{id}_B$, failure of full faithfulness of ψ is equivalent to failure of faithfulness of k.

ms. 38 conjectured VIa-160 – The essential image of $\bar{\psi}'$ is expected to consist precisely of the locally constant objects of HOT_W(S).

ms. 41 conjectured VIa-173 – Left-fine W-equivalences should be preserved by smooth base change T→S, extending their stability under discrete-fibration base change.

ms. 41 conjectured VIa-174 – Right-fine W-equivalences should be preserved by proper base change T→S.

ms. 42, 44 false VIa-180 – Proposition 1(1), which compares X/s→Y/s over an independent base S, is simply the original absolute axiom L4.

ms. 43 conjectured VIa-186 – The integral-homology localizer is suspected also to fail the total-to-fibre implication for maps between homotopy fibrations.

ms. 43, 45 false VIa-188 – L1, L3 quater, relative L4 bis, and the new L5 alone force W=W∞, with no component-preservation hypothesis.

ms. 44 conjectured VIa-191 – For the fundamental localizers under consideration, L6—or even the new L5 alone—should imply L6 bis.

ms. 45 false VIa-200 – Saturation, the terminal-object axiom, relative L4 bis, and the new L5 alone imply W⊆W∞, without any component condition.

ms. 47 false VIa-208 – On Liss(S), Fib(S), and Fibsc(S), the coarse W-class equals the right-fine W-class; dually on Prop(S), Cofib(S), and Cofibsc(S), it equals the left-fine W-class.

ms. 49, 53-57 conjectured VIa-213 – Does the coarse weak-equivalence class also give the asserted comparison for the split categories Fibsc(S) and Cofibsc(S), whose morphisms must preserve the chosen splitting?

ms. 50 false VIa-218 – The common left-fine localization of Cat/S, Prop(S), Cofib(S), and Cofibsc(S) is isomorphic to HOT_W(S^op)^op.

ms. 51 false VIa-224 – The editorial replacement HOT_W^o(S):=HOT_W(S^op)^op is the correct target convention for the covariant comparison functor j.

ms. 51-52, 64-69 conjectured VIa-226 – The canonical functors i:HOT_W^lc(S)→HOT_W(S) and j:HOT_W^lc(S)→HOT_W(S^op) should be fully faithful under the p45 package L1, L3 quater, L4 bis, L7 and the new L5.

ms. 55 false VIa-238 – A Cartesian functor between split fibrations is therefore compatible with their specified splittings.

ms. 55-57 false VIa-239 – The pullback retractions γ_X:Φ(X)→X form a strictly natural transformation on the full category Fib(S), allowing every functor over S and imposing no Cartesian condition on the morphisms.

ms. 56 false VIa-242 – For every X→S and split fibration Y→S, restriction along η_X gives a bijection Hom_cart/S(ΦX,Y)≅Hom_S(X,UY), where the left-hand side includes all Cartesian functors.

ms. 65-67 false VIa-267 – To make Cbar a right adjoint to ρ, the required adjunction arrows should have the initially drawn directions id_coarse→ρCbar and Cbarρ→id_fine.

ms. 65 false VIb-A006 – The attempted unit/counit directions before p.67 do not yet give $\rho \dashv \bar{C}$.

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ms. 71 false VIb-A027 – The editorial outer-opposite correction asserts $\mathrm{HOT}^{o}_{W}(S) \simeq (\mathrm{HOT}_{W}(S^{op}))^{op}$ while retaining the ordinary localization definitions.

ms. 71 FALSE VIb-A031 – Footnote46 conjectures that coarse and fibrewise localized theories differ whenever S is not equivalent to a discrete category.

ms. 74 FALSE VIb-A041 – The covariant underlying diagram for arbitrary X/S is printed as $s \mapsto \mathrm{hot}_W(s\backslash X)$.

ms. 82 OPEN VIb-A070 – Can the coarse comparison be obtained from a Quillen-type replacement without choosing a functorial factorization?

ms. 82 OPEN VIb-A071 – Can fine right and left relative equivalences in M/S be defined intrinsically from a general category of models (M,W)?

ms. 82 FALSE VIb-R001 – The prose identifies local-isomorphism tests with W_S^d and co-local-isomorphism tests with W_S^g.

ms. 83 CONJECTURED VIb-A073 – The suggested smooth and Zariski-like localizations should have the same relevant invariants, and dually proper and co-Zariski-like localizations should have the same cohomology.

ms. 84 FALSE VIb-A075 – The compatibility $i_{CX} = C(i_X)$ cannot be formal under the hypotheses already available.

ms. 84 OPEN VIb-A077 – Does $i_{CX} = C(i_X)$ hold after inverting universal weak equivalences over S?

ms. 84 OPEN VIb-A078 – Does duality reverse the order of the two adjunctions in this comparison?

ms. 85 OPEN VIb-A079 – A finer localizer on M/S contained in W_S gives a localization from the fine quotient to the coarse quotient; constructing a section and adjunction requires suitable fibrant representatives.

ms. 87 OPEN VIb-A086 – To construct a quasi-inverse one seeks a functor gamma:M/S→(Fib S)[W_S^{-1}] sending X to C(X) and W_S to isomorphisms.

ms. 94 CONJECTURED VIb-B015 – Replacing local-isomorphism tests by all smooth morphisms gives the same left relative localizer.

ms. 94 OPEN VIb-B017 – Is closure of the test class $\mathcal{L}$ under composition needed?

ms. 97–98 CONJECTURED VIb-B028 – The section i is expected to be fully faithful.

ms. 98 CONJECTURED VIb-B029 – The localization rho is expected to be left adjoint to i.

ms. 105 FALSE VIb-B049 – The source attributes $(x,c) \mapsto (x, 1_{f(x)}, c)$ to i_{CX} and $(x,c) \mapsto (x, c, 1_{\beta(c)})$ to $C(i_X)$, respectively.

ms. 108 FALSE VIb-B057 – Footnote59 asserts that the stability hypotheses on Fib are used only in II and only in case A/A-prime.

ms. 115 FALSE VIb-B085 – The editorial correction inserts an outer opposite in $\mathrm{HOT}^{o}_{W}(S) = \mathrm{HOT}_W(S^{op})^{op}$.

ms. 120 FALSE VIb-C009 – The path replacement $C_S(X)$ represents the same object of $\mathrm{HOT}_W(S)$ as arbitrary $X \to S$.

ms. 120 FALSE VIb-C011 – Corollary 2, second isomorphism: $f^* q_S(X) \cong q_{S'}(C_S(X) \times_S S')$ for arbitrary $X \to S$.

ms. 120 OPEN VIb-C012 – Which changes of base $f : S' \to S$ allow strict pullback to calculate derived restriction on every $X \to S$?

ms. 122 CONJECTURED VIb-C016 – Proper $f : S' \to S$ induces a strict pullback functor preserving W^{d}_S.

ms. 122 CONJECTURED VIb-C017 – Smooth $f : S' \to S$ induces a strict pullback functor preserving W^{g}_S.

ms. 123 OPEN VIb-C019 – Is preservation of right-fine weak equivalences by pullback equivalent to W-properness of the base functor?

ms. 124 OPEN VIb-C024 – If an arrow of a chosen category over S' becomes invertible in its coarse localization, must it belong to $W_{S'}$?

ms. 124 CONJECTURED VIb-C025 – For proper change of base, the same comparison criterion can be formulated on all of Cat/S.

ms. 126 FALSE VIb-C028 – An arrow invertible after an arbitrary localization necessarily belonged to the original class of arrows being inverted.

ms. 126 OPEN VIb-C031 – Does a Hot-fibration which is not proper furnish a pullback that fails to preserve the right-fine class on all categories over the base?

ms. 127 OPEN VIb-C032 – Are there functors other than Hot-fibrations or W-fibrations for which restriction commutes with local-constant reflection?

ms. 128 CONJECTURED VIb-C035 – The varying-base assignment $S \mapsto \mathrm{HOT}^{\mathrm{lc}}_W(S)$ is not expected to be a derivator.

ms. 128 OPEN VIb-C036 – Do restriction functors between locally constant categories have left and right adjoints?

ms. 130 OPEN VIb-C042 – For an arbitrary relative category (M,W), are isomorphisms in localized diagram categories detected by evaluation?

ms. 130–132 OPEN VIb-C043 – Does $\mathrm{HOT}_W(\coprod_i S_i) \to \prod_i \mathrm{HOT}_W(S_i)$ remain an equivalence for arbitrary small index sets?

ms. 131 OPEN VIb-C045 – For arbitrary relative categories (F_i,W_i), is $(\prod_i F_i)[(\prod_i W_i)^{-1}] \to \prod_i (F_i[W_i^{-1}])$ an equivalence?

ms. 132 CONJECTURED VIb-C048 – Uniform short presentations of morphisms could prove the infinite coproduct axiom for HOT.

ms. 137 OPEN VIb-C061 – Does a Hot-fibration f ensure that direct image preserves locally constant types?

ms. 142 FALSE VIb-C070 – Pullback of the right-fine replacement map i_X along arbitrary S-prime→S is again right-fine.

ms. 142–143 FALSE VIb-C071 – The counit may always be defined by projection composed with the inverse of the base-changed replacement map.

ms. 142 FALSE VIc-A007 – Question: does the pullback i_X times_S id_{S′}:X times_S S′ -> Phi_S(X) times_S S′ necessarily belong to W_S^d?

ms. 142 FALSE VIc-A008 – Proposed formula: beta_X=pr_1^X (i_X times_S id_{S′})^{-1} defines the counit at every representative X over S.

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ms. 151 FALSE VIc-A021 – Proposed product formula: f_!f^*(xi)=[S′/S] times_S xi for every base functor f and every coefficient xi.

ms. 152 FALSE VIc-A025 – False Corollary1: if only one of X,Y is smooth, then [X times_S Y] is necessarily the product [X] times_S [Y].

ms. 152 FALSE VIc-A026 – Every smooth X -> S induces by pullback a functor Cat/S -> Cat/X preserving all right-fine equivalences.

ms. 159 CONJECTURED VIc-A052 – The strengthened Der3b′_2 says that every f in W induces an equivalence f_0^*:HOT_W^lc(S)->HOT_W^lc(S′).

ms. 159 CONJECTURED VIc-A054 – Full faithfulness of f_0^* is expected to follow from the enhanced cohomological equivalence condition by applying internal Hom and cohomological direct images.

ms. 159 conjectured VIc-A055 – The construction of the right adjoints f_* and of internal Hom is expected to proceed together.

ms. 163 conjectured VIc-B011 – Properness should be equivalent to preservation of the right-fine localizers by pullback.

ms. 172–173 conjectured VIc-B034 – There may exist substantial homotopy theories HOT_W for classes W strictly smaller than W∞ while W(1,2,3) hold.

ms. 180–181 false VIc-B065 – Intermediate full subcategories between Parf_omega S and Fib_W S might yield distinct fine localizations.

ms. 184 open VIc-C010 – Does the old restricted W(4b′) alone force $\lambda : \mathrm{HOT}^{lc}_{W}(S) \to \mathrm{HOT}^{lc!}_{W}(S)$ to be an equivalence?

ms. 185 conjectured VIc-C015 – The new Cartesian fibre-detection axiom should imply the strongest W(4a) preservation condition, at least for the strong saturation of W.

ms. 186 open VIc-C019 – Do $\mathrm{Parf}_W S \hookrightarrow \mathrm{Liss}_{0,W} S$ and $\mathrm{Parf}_W S \hookrightarrow \mathrm{Prop}_{0,W} S$ induce equivalences after inverting fibrewise W-equivalences, without W(4b)?

ms. 186 open VIc-C020 – The same fine-localization comparison is asked with $\mathrm{Parf}_\omega S$ in place of $\mathrm{Parf}_W S$.

ms. 186 open VIc-C022 – Does $(\mathrm{Bifib} S)[(W_S^f)^{-1}] \to (\mathrm{Fib}_W S)[(W_S^f)^{-1}]$ become essentially surjective, or an equivalence, even assuming W(4a)?

ms. 187 open VIc-C024 – Does Phi carry W-proper objects, or proper weakly W-fibred objects, to W-proper objects?

ms. 189 conjectured VIc-C028 – If both arrows in the span are open embeddings, or both closed embeddings, is the total category W-equivalent to the ordinary amalgamated sum $X_a \amalg_{X_0} X_b$?

ms. 201 conjectured VIc-C077 – If X is weakly W-fibred, then $[X \times_S Y] \to [X] \times_S [Y]$ should be invertible for every Y, without smoothness of Y.

ms. 202 false VIc-C083 – The general product formula should fail if neither X is W-proper nor Y W-smooth, even when X is weakly W-fibred.

VII-a · Chapter VII, part 1 of 3, manuscript pages 1–57

ms. 4-5 false VIIa-A013 – As printed: composable chains correspond to full simplex subcategories, and maximal chains of length i correspond exactly to object-injective functors $\Delta_i \to I$.

ms. 5 false VIIa-A018 – Proposition 1(b), literally: every endofunctor of an interval fixing both endpoints is the identity.

ms. 10 false VIIa-A031 – Proposition 3 as printed: with $A = (I/s) \setminus \{s\}$ and $B = (s\backslash I) \setminus \{s\}$, s is critical iff $I \setminus \{s\} = A \amalg B$.

ms. 11 false VIIa-A037 – Footnote 4, literally: every subcategory of I which is an interval is necessarily full.

ms. 13 false VIIa-A046 – As printed, the length n of the path is $\mathrm{card}\, I$.

ms. 33 false VIIa.B030 – A constant functor at every non-endpoint vertex of a target interval factors through exactly two critical intervals.

ms. 34 false VIIa.B033 – For increasing f, the full subcategory on the image critical vertices equals the ambient integer interval [f(s_0),f(s_r)].

ms. 36 false VIIa.B046 – The global source critical intervals contained in f^{-1}(J') are exactly all critical intervals of the preimage.

ms. 42 FALSE VIIa-C009 – Each target transition segment [i-1,i] equals the unique image block [t_{j-1},t_j] which crosses it.

ms. 45 FALSE VIIa-C017 – The object map is constant on every integer-connected component of the set of vertices incident to a collapsed source transition.

ms. 47 FALSE VIIa-C030 – The printed object-map table sends the third fibre block to l_2 and its general block to l_i.

ms. 50 FALSE VIIa-C036 – The same object concatenation extends to a bifunctor $\underline{\mathrm{Typ}} \times \underline{\mathrm{Typ}} \to \underline{\mathrm{Typ}}$, where Typ has all functors of interval categories.

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ms. 60 FALSE VIIb-A032 – The parenthetical restriction of the possibility of assigning an X-morphism to a path to type Δ_+ alone is too narrow.

ms. 63 FALSE VIIb-A040 – The displayed uncorrected map from a constant a-path to $(b \leftarrow a \rightarrow b)$ is a map in $\underline{Ch}_{-+}(b,b)$.

ms. 69 FALSE VIIb-A055 – An equality established only in the groupoid completion Pi(X) implies the needed equality in the relative localization Pi_Sigma(X).

ms. 76 FALSE VIIb-B012 – For a word with r alternating runs beginning with sign epsilon, its last run has sign $(-1)^r\varepsilon$.

ms. 77 FALSE VIIb-B015 – The concatenated interval $I_{\tau\tau'}$ is the pushout of I_τ and $I_{\tau'}$ over their two-element endpoint category $\{0,1\}$.

ms. 79–80 OPEN VIIb-B019 – Does the fibrewise/cartesian construction extend to a functor on the total ordinary path categories?

ms. 85 FALSE VIIb-B034 – The collapsing remainder in the alpha-th source run has type $\Delta_{\varepsilon_\alpha}^{l_{\alpha-1}}$.

ms. 86 FALSE VIIb-B036 – The quickest map is a terminal object because it is numerically ahead of every other map.

ms. 86 CONJECTURED VIIb-B037 – The quickest map is the greatest admissible map for pointwise numerical order.

ms. 86 OPEN VIIb-B038 – Is every nonempty increasing endpoint-preserving interval mapping category contractible?

ms. 93 FALSE VIIb-B052 – Reversal gives an isomorphism $P(\tau,\tau') \to P((\tau')^o,\tau^o)$.

ms. 90 FALSE VIIb-B057 – Applying the endpoint-fixed induction to the subinterval $[1,n-1] \subseteq [0,n]$ reduces a map whose active edges have one sign and include the first and last edges to a map with only those two active edges.

ms. 90 FALSE VIIb-B058 – A first source run that is not a single edge has length2.

ms. 102 FALSE VIIb-C024 – The displayed transition $C_{t_n} \to C_{t_{n-1}}$ has the variance required by the sequential direct system.

ms. 102–103 OPEN VIIb-C026 – Compare the total category C, the various sequential colimits C_∞, and the ind-object H_C, in Cat hot and in Hot; determine dependence on the chosen cofinal chain.

ms. 103 FALSE VIIb-C028 – The thin reflection of the slice satisfies $(T/t_0)_o = P/t_0 = \{t : t \le t_0\}$.

ms. 103 FALSE VIIb-C029 – Under only the stated preorder filtration, passage from T to T/t_0 automatically gives another filtered indexing preorder and leaves the same ind-object construction available.

ms. 103–104 conjectured VIIb-C030 – If the total category were canonically recovered from the ind-object, restriction to a preorder-cofinal chain would preserve its homotopy type.

ms. 104–105 open VIIb-C031 – For a left-cofinal base change $\alpha : T' \to T$ preserving the indicated ind-diagram, is $C \times_T T' \to C$ a Cat-hot isomorphism or at least a weak equivalence of nerves?

ms. 107 open VIIb-C040 – Are the sets $\mathrm{Cat\,hot}(S, C)$ or $\mathrm{Hot}(S, C)$ naturally identified with $H(S)$ beyond S=e?

ms. 107 conjectured VIIb-C041 – An isomorphism on the preceding component sets need not extend to higher homotopy groups or to a Hot equivalence of the compared objects.

ms. 112–113 conjectured VIIb-C053 – For the motivating path fibration, Grothendieck suspects that $C_\infty \to C$ is generally not a Hot isomorphism.

ms. 113 false VIIb-C054 – A loop-space multiplication cannot in general be represented by a strictly associative simplicial monoid, so such strict multiplication would obstruct the proposed comparison.

ms. 115 false VIIb-C060 – For an arbitrary chosen cofinal sequence of types, concatenation has target stage n+m: $P_{t_n}(a, b) \times P_{t_m}(b, c) \to P_{t_{n+m}}(a, c)$.

ms. 108–109 open VIIc-A003 – For T=N^op, does the canonical localization C→colim_n C_n induce a weak equivalence?

ms. 109 open VIIc-A006 – Is the canonical morphism C_∞→C always an isomorphism in Hot?

ms. 112–113 conjectured VIIc-A015 – The comparison C_∞→Ch_Σ(X;a,b) is conjectured not to be a weak equivalence in general.

ms. 113 false VIIc-A016 – The based loop object Ω(BX,a) cannot in general be modeled by a strictly associative simplicial monoid.

ms. 114 false VIIc-A020 – The object of the finite-test presheaf category associated with C_∞ equals the filtered colimit of the stage-representable presheaves.

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ms. 117–118 false VIIc-A029 – Every natural transformation between cofinal sequences induces a weak equivalence C_∞(t′)→C_∞(t) under the general section10 hypotheses.

ms. 119 open VIIc-A034 – Does the induced Hot morphism depend only on the endpoints t,t′ and not on the chosen natural transformation?

ms. 119 false VIIc-A035 – On finite-test presheaves every comparison is the canonical identification with the presheaf associated to H_T.

ms. 120 false VIIc-A038 – The map colim_n Hom_Hot(ξ,C_n)→Hom_Hot(ξ,C_∞) is bijective for every ξ∈Hot.

ms. 121 false VIIc-A042 – Bijectivity on all unpointed finite homotopy test sets implies that a map is an isomorphism in Hot.

ms. 121 false VIIc-A043 – Hom_Hot(Y,−) commutes with filtered colimits whenever Y is a finite category.

ms. 121,124 open VIIc-A046 – Are all comparison functors invertible already in Cat hot?

ms. 123 false VIIc-A050 – The ind-identification implies that all induced comparison maps on C_∞ are isomorphisms in Hot.

ms. 123 false VIIc-A053 – The restriction H_0 of H to the source's finite-category test subcategory is represented by the presheaf h(C_∞).

ms. 124–125 OPEN VIIc-A056 – Does a binary path composition C_∞(a,b)×C_∞(b,c)→C_∞(a,c) exist canonically in Hot?

ms. 126 FALSE VIIc-A063 – Two unrelated interval types can be compared by their categorical product τ_0×τ'_0 as another interval type.

ms. 127 OPEN VIIc-A065 – Is the constructed binary operation associative in Hot?

ms. 127 OPEN VIIc-A066 – Is Hot→widehat(Hot_0) faithful, or even fully faithful?

ms. 127 OPEN VIIc-A069 – Are the canonical maps H→C_∞ epimorphisms in Ind Hot, and do they remain so after products?

ms. 128 CONJECTURED VIIc-A072 – For a non-Kan simplicial set K, the strict endpoint fibres of K^{Δ^1}→K×K need not be the derived path spaces.

ms. 128 CONJECTURED VIIc-A073 – A correct simple path construction for finite simplicial sets would imply a finite computation of all their homotopy groups.

ms. 128 CONJECTURED VIIc-A074 – For Σ=Fl(X), a weak equivalence X→Xbar should induce Ch_∞(X;a,b)≃Ch_∞(Xbar;a,b).

ms. 129 CONJECTURED VIIc-B002 – The external-Hom parameter functor F_{a,b}(Y)=Hom_{D(Y)}(a_Y,b_Y) should send nerve weak equivalences of Y to bijections.

ms. 129 FALSE VIIc-B005 – The source infers under its locality hypothesis that the finite-parameter evaluation of H(a,b) is [Y,Ch_{infty,Sigma}(a,b)] in Hot, using finite-category compactness.

ms. 130 CONJECTURED VIIc-B006 – A nerve weak equivalence f:X→Xprime should induce weak equivalences Ch_infty(X;a,b)→Ch_infty(Xprime;fa,fb), with all arrows allowed.

ms. 130 FALSE VIIc-B009 – For Thomason-fibrant X, fixed-endpoint paths of type Delta_+ might suffice to model Ch_infty(X;a,b).

ms. 130 OPEN VIIc-B011 – Is Hot→Hot_0^wedge faithful, so that finite tests would prove associativity?

ms. 132 CONJECTURED VIIc-B018 – Noncontractible components of Omega(X,a) would make the invariance condition almost never hold.

ms. 133 FALSE VIIc-B022 – For a fixed finite circle category Y, degrees realized by functors Y→Ch_infty(X;a,a) are bounded by the number of its arrows, hence by four in the displayed example.

ms. 133 FALSE VIIc-B023 – Footnote28 proposes the sharper bound -2≤degree≤2 for the displayed four-arrow circle.

ms. 133 CONJECTURED VIIc-B024 – By enlarging alternating circle models Y_n, one can realize degrees of arbitrarily large magnitude.

ms. 134 FALSE VIIc-B026 – The proposed bound on degrees proves that pi_0 Fun(Y_2,Omega)→pi_0 Fun(Y_n,Omega) is not bijective.

ms. 134 CONJECTURED VIIc-B027 – The same finite-size obstruction should apply whenever higher homotopy groups of X provide infinitely many classes.

ms. 134 CONJECTURED VIIc-B028 – Products of alternating intervals with their boundary identified should provide categorical models of S^i.

ms. 134 FALSE VIIc-B029 – A fixed finite categorical sphere Y_n should realize only a bounded subset of pi_i Omega(X,a).

ms. 134–135 CONJECTURED VIIc-B030 – Every homotopy class should be represented after choosing a sufficiently large categorical sphere and a degree-one map to a smaller model.

ms. 135 FALSE VIIc-B034 – The absolute comparison already fails for Y=[1].

ms. 136 FALSE VIIc-B035 – For any category Y and Thomason-fibrant C, BFun(Y,C) represents RMap(BY,BC).

ms. 136 FALSE VIIc-B036 – If C is Thomason-fibrant then Fun(Y,C) is Thomason-fibrant for arbitrary Y.

ms. 136 FALSE VIIc-B037 – Footnote29 requires Thomason cofibrations to remain cofibrations after product with arbitrary Y.

ms. 136 FALSE VIIc-B038 – For fibrant C and arbitrary Z,Y, pi_0 Fun(Z×Y,C) computes [B(Z×Y),BC].

ms. 136 CONJECTURED VIIc-B040 – Categorical path objects should have the correct homotopy type without first making X Thomason-fibrant.

ms. 139 FALSE VIIc-B050 – For every f, bar f in this factorization should be both fibrant and a weak equivalence.

ms. 139 FALSE VIIc-B051 – The map alpha should be a Thomason cofibration.

ms. 139 FALSE VIIc-B052 – The map bar f is expected to be fibrant in the model-category sense.

ms. 139 CONJECTURED VIIc-B053 – Item1: p:Ch_infty(X)→X×X should be smooth, proper and a HOT-fibration.

ms. 139 CONJECTURED VIIc-B055 – Item2 additionally asserts contractible fibres for sigma and beta, hence trivial HOT-fibrations.

ms. 139 CONJECTURED VIIc-B056 – Item3: bar f:C(f)→X should be smooth, proper and a HOT-fibration for every f.

ms. 140–141 CONJECTURED VIIc-B062 – For a path c:a→b, transport should give an isomorphism bar c:hot(C(f)_a)→hot(C(f)_b).

ms. 140–141 CONJECTURED VIIc-B063 – Transport should be compatible with concatenation of base paths.

ms. 141 CONJECTURED VIIc-B066 – Transport should depend only on the homotopy class of c in the path category.

ms. 141 CONJECTURED VIIc-B067 – The fibre homotopy types and transport should define a functor Pi(X)→Hot.

ms. 141 CONJECTURED VIIc-B068 – The monodromy construction should apply to every HOT-fibration, not just this mapping-path replacement.

ms. 141 CONJECTURED VIIc-B069 – For a D-fibration and coefficient A∈D(e), fibrewise D-homology and D-cohomology should give two Pi(X)-functors to D(e).

ms. 141–142 OPEN VIIc-B070 – Footnotes33–34 ask for the effect of transport on components of the fibres.

ms. 142 CONJECTURED VIIc-B072 – The nerve of Ch_infty(X) over NX×NX should be weakly equivalent in the slice to a simplicial path object.

ms. 142 CONJECTURED VIIc-B073 – The fixed-endpoint categorical path categories should have canonically the same homotopy types as simplicial path spaces.

ms. 142 OPEN VIIc-B074 – Does a functorial classical simplicial path object exist for every simplicial set?

ms. 145 FALSE VIIc-B084 – The significant edge indices of the t_n stage are [-2i+1,2i], with 4i edges, as printed.

ms. 146 FALSE VIIc-B088 – For even n, tau_{n,m}=(+-)^{(m-n)/2} if m is even, and the printed (+-)^{(m-n)/2-1}+ if m is odd.

ms. 147 FALSE VIIc-B089 – For odd n, tau_{n,m}=(-+)^{(m-n)/2} if m is odd, and the printed (-+)^{(m-n)/2-1}- if m is even.

ms. 147 FALSE VIIc-B090 – For n<m the initial sign is (-1)^n and the terminal sign is printed as (-1)^m.

ms. 150 OPEN VIIc-C004 – Is the restricted Yoneda functor Hot → $\mathrm{Hot}_0^{\wedge}$ faithful, so that equality of the two finite-test images implies $T^2 = 1$?

ms. 151 CONJECTURED VIIc-C016 – The finite-stage composition operations should make the comparison square commute on associated ind-objects.

ms. 152 FALSE VIIc-C017 – If the comma transport becomes an isomorphism in the unbased finite-test presheaf category, it must be an isomorphism in Hot.

ms. 153 CONJECTURED VIIc-C023 – The mapping-path projection $\bar{f} : C(f) \to X$ should be smooth, proper, and a homotopy fibration.

ms. 155 CONJECTURED VIIc-C031 – The finite-stage v_* are compatible up to categorical homotopy and induce a morphism of ind-objects in Cat hot, then a map of finite-test presheaves.

ms. 155 CONJECTURED VIIc-C033 – The stagewise v^* likewise induce an ind-morphism and hence a map on finite-test presheaves.

ms. 156 CONJECTURED VIIc-C034 – The two ind-morphisms v_*, v^* are inverse, because the stagewise backtracking operations are homotopic to the identity.

ms. 156 CONJECTURED VIIc-C035 – The ind-object comparison square with the comma transport $\bar{v}$ commutes.

ms. 156 FALSE VIIc-C037 – Invertibility of $\bar{v}$ after restriction to finite unbased tests implies its invertibility in Hot.

ms. 156 OPEN VIIc-C038 – $\bar{f}$ is a trivial homotopy fibration if and only if f is a weak equivalence; a categorical long exact sequence seems necessary to prove it.

ms. 157 CONJECTURED VIIc-C039 – The canonical maps $\underline{\mathrm{Ch}}_\infty(X) \to \underline{\mathrm{Ch}}(X)$ and $\underline{\mathrm{Ch}}_\infty(X; a, b) \to \underline{\mathrm{Ch}}(X; a, b)$ in Hot are isomorphisms.

ms. 157 CONJECTURED VIIc-C040 – The total comparison is an isomorphism in the relative homotopy category over $X \times X$, giving the fixed-endpoint comparisons.

ms. 157 FALSE VIIc-C042 – $\underline{\mathrm{Ch}}(X) \to X \times X$ is both smooth and proper.

ms. 158 FALSE VIIc-C045 – $\mathcal{C}_{\mathbf{N}^o} \to X \times X$ is smooth and proper.

ms. 161 FALSE VIIc-C058 – Smoothness of $\underline{\mathrm{Ch}}(X^o) \to X^o \times X^o$ implies properness of $\underline{\mathrm{Ch}}(X) \to X \times X$.

ms. 166 FALSE VIIc-C073 – The composite $j'i' : X \to \mathcal{C}_{\mathbf{N}^o}$ is aspherical.

ms. 168 FALSE VIIc-C078 – The padded formula $\varphi(c', w') = (\widetilde{c}', \widetilde{w}')$ defines a functor from the sequential lift category to the sequential fixed-endpoint slice.

ms. 169 FALSE VIIc-C081 – The proposed sequential endofunctor $\psi\varphi$ admits a natural transformation to the identity.

ms. 169 FALSE VIIc-C083 – Because the system of interval types is self-opposite, smoothness for X^o proves properness of $\mathcal{C}_{\mathbf{N}^o}(X) \to X \times X$.

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ms. VIII:4 CONJECTURED VIII-016 – For X→S→T and every groupoid fibration Γ on X, (fp)_!Γ≃f_!p_!Γ, naturally and coherently.

ms. VIII:7 CONJECTURED VIII-024 – For a cartesian square with invertible γ, the comparison f′_!q*G→p*f_!G is an equivalence if f is 1-smooth or p is 1-proper.

ms. VIII:7 FALSE VIII-029 – For i∈{0,1,2}, i-faithfulness of f* is equivalent to i-faithfulness of Γ(T,H)→Γ(S,f*H) for every H, taking only G=e_T.

ms. VIII:8 CONJECTURED VIII-036 – If f is (-1)-proper and (-1)-aspherical, or (-1)-smooth and (-1)-coaspherical, then f* is 0-faithful.

ms. VIII:8 CONJECTURED VIII-037 – If f is 0-proper and 0-aspherical, or 0-smooth and 0-coaspherical, then f* is 1-faithful.

ms. VIII:8 CONJECTURED VIII-038 – If f is 1-proper and 1-aspherical, or 1-smooth and 1-coaspherical, then f* is 2-faithful.

ms. VIII:8–9 CONJECTURED VIII-040 – Under either hypothesis of Theorem 3 with i=2, G on S is in the essential image of f* exactly when G restricted to each strict fibre S_t is equivalent to a constant groupoid fibration.

ms. VIII:9 FALSE VIII-041 – For 1-connected Z, a groupoid fibration on Z is constant up to fibred equivalence if and only if every transport functor is an equivalence.

ms. VIII:9 FALSE VIII-043 – Under Theorem 3 with i=2, G descends along f if and only if every f-vertical arrow induces an equivalence on its fibres.

ms. IX:4 FALSE IX-011 – Arbitrary arrows u:a-prime->a and v:b->b-prime induce an isomorphism of Hom objects.

ms. IX:4 FALSE IX-013 – The displayed Ind Cat hot bifunctor has domain M_Pi x M_Pi.

ms. IX:9-10 OPEN IX-R01 – For the full subcategory Hot_0 of types of finite categories, is the restricted Yoneda functor psi:Hot -> PSh(Hot_0) fully faithful (and likewise Ind psi)?

ms. IX:13 OPEN IX-037 – Ch composition is associative in Hot.

ms. IX:14 CONJECTURED IX-040 – Quillen pi_1(f,g) agrees with arrows between f and g in the fundamental groupoid of Ch_infty(a,b).

ms. IX:15 CONJECTURED IX-041 – The homotopy groupoid construction extends naturally from actual arrows to all stabilized zigzags.

ms. IX:15 OPEN IX-044 – The path compositions should lift to pointed Hom objects with associative composition and units.

ms. IX:16 FALSE IX-046 – There are canonical arrows in both directions between these zero representatives, identity at every vertex except degree zero.

ms. IX:17 CONJECTURED IX-049 – The resulting exact path pairings are associative in the pointed homotopy category and perhaps in pointed Cat hot.

ms. IX:17 FALSE IX-050 – Cartesian pointed spaces instantiate the earlier monoidal axiom with pi_0=Hom(unit,-).

ms. IX:17 FALSE IX-051 – Forgetting basepoints preserves the corrected monoidal product.

ms. IX:17 CONJECTURED IX-052 – Pointed CH should exist in Ind Hot_bullet and Ind Cat hot_bullet.

ms. IX:18 FALSE IX-054 – The displayed truncated path a->0<-b represents the desired distinguished element.

ms. IX:18 OPEN IX-055 – The zero object should remain a zero object in M[W^{-1}].

ms. IX:18 OPEN IX-056 – Find conditions under which localization preserves finite coproducts and finite products.

ms. IX:18 CONJECTURED IX-059 – An additional five-lemma-type condition may be needed for the additive localization.

ms. IX:19 CONJECTURED IX-061 – The external Hom of an additive relative category should lift to D^{<=0}(k-Mod).

ms. IX:19 conjectured IX-065 – Derived Hom objects should admit associative composition over derived tensor and units.

ms. IX:21 conjectured IX-070 – A full theory of strong derived k-bilinear Hom composition is expected.

ms. IX:21 false IX-071 – pi_1(a,b) is written pi_1(a,b;0_a,0_b).

ms. IX:21 conjectured IX-072 – Quillen pi_1(a,b) is canonically pi_1 of the pointed Ch_infty(a,b).

ms. IX:22 conjectured IX-080 – The two formulations of test (b) are equivalent up to minor qualifications.

ms. IX:22 conjectured IX-081 – Tests (a) and (b) are a decisive criterion for a homotopy theory in M.

ms. IX:23 false IX-082 – The stated tests cannot hold for a finite M except in the trivial cases.

ms. IX:23 false IX-083 – If M is empty, its localization is equivalent to the point.

ms. IX:23 conjectured IX-084 – Test (c) asks for Homext(Sigma a,b)=Homext(a,Omega b)=Omega Homext(a,b) as pointed homotopy types.

ms. IX:23 conjectured IX-086 – External Hom preserves finite products in the second variable and sends finite coproducts in the first variable to products.

ms. IX:23 conjectured IX-087 – Omega should extend to a left exact endomorphism of the derivator, and Sigma dually.

ms. IX:23 conjectured IX-088 – Test (b) is necessary for the loop-space equivalences in test (c).

ms. IX:24 conjectured IX-090 – The essential condition on the path factorization is that p be a fibration in a suitable sense.

ms. IX:25 open IX-091 – A cofibration condition on i may be useful; fibrancy of S is queried.

ms. IX:25 open IX-093 – The constructed object should represent pi_1 Ch_infty(T,S) naturally in T after localization.

ms. X:3 conjectured X022 – The categorical assertions of Chapter I that use only weak equivalences and their formal properties should extend from W_D to abstract localizers W.

ms. X:3 open X023 – Under suitable conditions on W, the pointwise localization A |-> Cat^{A^op}[(W^{A^op})^{-1}] should be a derivator.

ms. X:3 open X024 – One should be able to realize a given localizer W as W_D for a derivator constructed from (Cat,W).

ms. X:5 conjectured X038 – The derivators useful in practice seem to satisfy La.

ms. X:5 open X042 – Do Lb and Lc hold for every W_D under consideration, and what are their concrete conditions on W?

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ms. XI:129 false XI-017 – For an arbitrary locally constant nonabelian F, lifting $\eta \in$H0(Y,R^1f_*F) is governed by a fixed-band obstruction in H2(Y,f_*F), as written.

ms. XII:1 open XII-I-001 – Find intrinsic conditions on W implying W$\subseteq$ W_∞, without defining those conditions through W_∞.

ms. XII:5 OPEN XII-I-028 – Are Ch(X)→ X× X and Ch_∞(X)→ X× X W-fibrations for a general W?

ms. XII:5 OPEN XII-I-032 – Does the left test alone imply preservation over W-aspherical bases without additional axioms?

ms. XII:13 OPEN XII-I-057 – Do W(1)-W(5) already imply W=W_∞, or is an additional descent/limit axiom necessary?

ms. XII:14 CONJECTURED XII-I-071 – Even the strongest W(4), with W(1)-W(3) and W(5), probably does not force W=W_infty.

ms. XII:15 CONJECTURED XII-I-076 – W(1)-W(6), using the weakest fibre axiom, imply W=W_infty.

ms. XII:16 OPEN XII-I-077 – Perhaps closure under retracts must be added to W(1) for the equality conjecture.

ms. XII:17 FALSE XII-I-084 – The formula pi_n(F)=coker(pi_{n+1}(E)→ pi_{n+1}(B)) is asserted without excluding n=0.

ms. XII:18,20 FALSE XII-II-002 – W_D⊇W_infty and W(4b") imply W_D=W_infty without another stated qualification; the analogous inference is repeated for W in the HOT_W discussion.

ms. XII:19 FALSE XII-II-005 – The left/right conditions are the same as the earlier D-cohomological/D-homological fibrations.

ms. XII:19 CONJECTURED XII-II-007 – Every genuinely useful derivator/localizer should satisfy the indicated constancy condition.

ms. XII:20 OPEN XII-II-023 – Did the preceding construction and properties of f_! for HOT_W actually use the working assumption W_infty⊆W?

ms. XII:26 CONJECTURED XII-III-A-024 – Aspherical functors should be preserved by smooth base change, and coaspherical functors by proper base change.

ms. XII:36 FALSE XII-III-A-057 – It suffices, to define functorial covariant transport universally, to require only that the terminal-fibre inclusion in every pullback over [1] belongs to W.

ms. XII:36 FALSE XII-III-A-058 – The analogous initial-fibre edge-only condition suffices to define universal contravariant transport.

ms. XII:40 CONJECTURED XII-III-B-003 – Among the provisional fibration notions, only ultrafibrations appear likely to be stable under composition.

ms. XII:40 OPEN XII-III-B-005 – Find minimal hypotheses on W making the stronger fibration variants equivalent.

ms. XII:40 OPEN XII-III-B-006 – Does the interval endpoint condition a0 alone imply the remaining conditions for W∞?

ms. XII:40 CONJECTURED XII-III-B-008 – W-perfect⇒hot-strong W-fibration (circled implication).

ms. XII:40 CONJECTURED XII-III-B-009 – Footnote 48: the hot-strong condition should be equivalent to both the left and right fibration conditions.

ms. XII:42 OPEN XII-III-B-017 – If f is W-perfect and S categorically contractible, is X_s→X in W?

ms. XII:44 FALSE XII-III-B-025 – The first displayed transport is F∘f→F∘g for α:f→g.

ms. XII:45 FALSE XII-III-B-026 – The classifying functor F itself is 'in W fibre by fibre'.

ms. XII:46 CONJECTURED XII-III-B-032 – These results should use at most full W(2) rather than W(2 bis).

ms. XII:48 CONJECTURED XII-III-B-040 – The stronger complement: pullback of an arbitrary categorical homotopy equivalence along the family is a W-map.

ms. XII:54 OPEN XII-IV-011 – Does W(4c′) imply W(4c)?

ms. XII:54 OPEN XII-IV-012 – Does W(4c) imply W(4b)?

ms. XII:54 CONJECTURED XII-IV-014 – The weaker split test W(4b′) should suffice for the same conclusion.

ms. XII:54 CONJECTURED XII-IV-015 – W(4b″), perhaps with W(6 bis), should likewise suffice.

ms. XII:54 CONJECTURED XII-IV-017 – The equivalence should also include W(4b′) and W(4b″).

ms. XII:58 FALSE XII-IV-030 – Proposition 3: under W(4a), a strict Cartesian map of split fibrations with W-fibrant fibre maps is W-fibrant.

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ms. 59 FALSE XIIba-A-003 – With x<=y iff Hom_X(x,y) is nonempty and opens the lower sets, the source asserts x<=y iff x belongs to the closure of {y}.

ms. 61 FALSE XIIba-A-012 – The induced triple represents on all triples Y=(Y,V,S) the functor Hom(V,U) x Hom(S,T), as asserted in displayed (*).

ms. 61 FALSE XIIba-A-013 – The displayed unqualified functor-category version (**) of the same representability is an isomorphism.

ms. 76 OPEN XIIba-A-053 – Does W(7bis) alone imply that W-asphericity of the intersection and union forces W-asphericity of both pieces?

ms. 78 FALSE XIIba-A-058 – The ordinary weak homotopy equivalences W_infty satisfy the full axiom W(7).

ms. 79-80 FALSE XIIba-A-062 – Reverse excision is asserted to be a special property of HOT, while the dual derivator fails it.

ms. 94–97 OPEN XIIb-a-B035 – Does total W-asphericity of H imply total W-asphericity of U/H?

ms. 97–100 FALSE XIIb-a-B045 – Assuming W(7 bis), a union V=V′∪V″ of two sieves is claimed to be totally W-aspherical whenever V′,V″,V′∩V″ are totally W-aspherical, by preservation of the subobject square under f!.

ms. 98–100 FALSE XIIb-a-B047 – Under the standing W(7 bis) hypothesis, the assertion of Proposition 9(b) is repaired by categorical retractions V′→V′∩V″ and V″→V′∩V″.

ms. 98 FALSE XIIb-a-B048 – Under the same gluing hypothesis, the analogous union assertion is true for a pushout square of subpresheaves in an arbitrary presheaf category merely because the three smaller objects are totally W-aspherical.

ms. 98 FALSE XIIb-a-B049 – Under the same hypotheses, one of the two categorical retractions already suffices to imply the asserted total-asphericity conclusion.

ms. 99 FALSE XIIb-a-B051 – The existence of those categorical retractions implies that f! sends the square of the associated subterminal presheaves to a square of monomorphisms.

ms. 99 FALSE XIIb-a-B052 – The same retractions imply f!(V′∩V″)=f!V′∩f!V″.

ms. 99–100 FALSE XIIb-a-B053 – Consequently the four categories of elements of these Kan extensions form a Cartesian and cocartesian square of sieve inclusions.

ms. 100 FALSE XIIb-a-B054 – Assuming W(7 bis), the outgoing fork Ψ=(0→a,0→b) is totally W-aspherical.

ms. 100 FALSE XIIb-a-B055 – Assuming W(7 bis), every finite ordered category with an initial object is totally W-aspherical.

ms. 100 FALSE XIIb-a-B056 – A category is totally W-aspherical only if it has a final object.

ms. 103 OPEN XIIba-C005 – If Y is W_∞-aspherical, must $\underline{\mathrm{Ch}}(Y; y_0, -)$ be totally W_∞-aspherical?

ms. 104 OPEN XIIba-C009 – Can one similarly find a sieve pushout with $p \in W$ but $q \notin W$?

ms. 111 OPEN XIIba-C035 – Must a sieve that is a W-cofibration, or even a couniversal W-equivalence, be W-dwyerienne?

ms. 111 OPEN XIIba-C036 – Must such a sieve even satisfy the original Dwyer right-adjoint condition?

ms. 111 FALSE XIIba-C037 – The source asserts the converse to the W-dwyerienne criterion when the complementary category T is terminal.

ms. 111 FALSE XIIba-C038 – Under the cited gluing framework, i and all its changes of complementary category $T_1 \to T$ are W-cofibrations iff i is W-dwyerienne.

ms. 111–112 OPEN XIIba-C039 – Is there a useful Dwyer notion for arbitrary full inclusions which includes both the sieve and cosieve cases?

ms. 112 FALSE XIIba-C040 – The proof says that W-dwyerienne stability makes i' W-aspherical and in W.

ms. 113 OPEN XIIba-C045 – Are W-dwyerienne and totally W-dwyerienne inclusions stable under composition?

ms. 114 OPEN XIIba-C047 – Are strictly W-aspherical functors and totally W-aspherical functors stable under composition?

ms. 114 OPEN XIIba-C048 – Seek a class of sieve W-cofibrations stable under the endpoint pushout product with $\{0, 2n\} \to \Phi_n$, n>=2.

ms. 119 OPEN XIIba-C064 – Request: make the map $(\lambda_\alpha)_!$ in the preceding transition formula explicit.

ms. 114 (continued before 115) OPEN XIIbb-A001 – Find a class of sieve inclusions that are W-cofibrations and is closed under the endpoint pushout-product U×Phi_n union X×{0,2n} -> X×Phi_n for every n>=2.

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ms. 136 OPEN XIIb-b-B009 – Is phi_Q in W equivalent to tilde(phi)_Q in W, so that smooth and proper integrals define the same W-homotopy-cocartesian squares?

ms. 139 FALSE XIIb-b-B014 – The indexing category Gamma is the ordered square [1]x[1] with an initial object adjoined.

ms. 143 CONJECTURED XIIb-b-B025 – The canonical isomorphisms among the four expanded diagrams should form a transitive system.

ms. 144 CONJECTURED XIIb-b-B029 – For arbitrary diagram shapes the four expanded comparisons have the same canonical-isomorphism relations as in Proposition 14.

ms. 145–146 CONJECTURED XIIb-b-B033 – Can phi_{Q_0}:int_lis F->P fail to belong to W while there exists a strict cocone P->Y with composite phi_Q in W?

ms. 155 FALSE XIIbb-C009 – The arrows x_0→α_i(x_0) displayed in the source are the cartesian arrows in the smooth integral.

ms. 156 FALSE XIIbb-C013 – The displayed diamond A_0(u)→A_1(u),A_2(u)→(E/x_0)_u is a commutative W-cocartesian square to which W(7 bis) can be applied directly.

ms. 157 FALSE XIIbb-C021 – For x in any of the three strata, local W-smoothness of φ_Q is equivalent to local W-smoothness of f_i without any additional hypotheses on the other strata.

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ms. 172 OPEN A011 – Compare the homotopy-group theories defined using $\underline{\mathrm{Ch}}$ and $\underline{\mathrm{Ch}}_\infty$.

ms. 172 FALSE A012 – Theorem1: isomorphisms of homotopy groups through n, either at all basepoints or at one per component, are equivalent to isomorphisms of local-coefficient cohomology through n, with nonabelian H1.

ms. 172 FALSE A013 – Footnote112 suggests that the preceding converse already fails for n=2 unless source and target are simply connected.

ms. 174 OPEN A019 – Does a W-fibration E->X with a section canonically determine a functor X->Hot_{bullet,W} sending all arrows to isomorphisms?

ms. 176-177 FALSE A022 – Under the lower-degree homotopy and cohomology comparison hypotheses, pi_n(f) is an isomorphism iff H^n(f;F) is an isomorphism for all abelian local systems F.

ms. 177-178 OPEN A024 – For a homomorphism of the categorical loop monoids, a homotopy-group comparison at the units extends to every component.

ms. 179 FALSE A029 – Each arrow in the corrected implication chain is conjectured strict.

ms. 183 OPEN A042 – Does the comparison of the universal-cover E2 rows for q<=n suffice for the desired total-cohomology comparison through n?

ms. 187 FALSE A050 – The good fundamental localizer W_n for n-types is defined by a_n.

ms. 179 OPEN A062 – Footnote117 asks whether c_n implies b_n even for simply connected X,Y and conjectures counterexamples because only the Hurewicz composite from pi_nZ is seen.

ms. 194 OPEN B006 – In what sense are fixed-base or varying-base functorial factorizations unique?

ms. 196-197 OPEN B011 – Is there a construction-independent canonical comparison φ_C≅φ_C′, independent of using CC′ or C′C?

ms. 201 FALSE B022 – The existence of these common targets, or of the natural binary comparison maps, proves C_S is filtered.

ms. 202 CONJECTURED B024 – The three binary-to-ternary and three unary-to-ternary comparisons obtained by inserting coaugmentations agree.

ms. 202-203 CONJECTURED B025 – For any ordered partial product of a finite list of replacements there is a unique comparison to the full ordered product obtained by the permitted unit insertions and associativity maps.

ms. 203 OPEN B026 – Do these coherences imply that C_S is 1-connected?

ms. 204 FALSE B028 – The one-object category B_G of an arbitrary group G is a strict monoidal category of the indicated kind.

ms. 205 FALSE B032 – The functor labelled (*) is an equivalence.

ms. 208 FALSE B039 – Under only the assumptions of footnote 119, Fib_W is stable under retracts in the arrow category.

ms. 209 OPEN B042 – Construct inverse images on Ho_W^(S) for arbitrary base maps, coherently in S.

ms. 216 CONJECTURED B056 – Axioms a and b will usually be satisfied and proved together.

ms. 216 CONJECTURED B058 – For Cat, W(4a) supplies the fibre-localization comparison; footnote 122 suggests W(4b) should suffice.

ms. 217-218 OPEN B062 – Do axioms a and b alone imply the intermediate-full-subcategory assertion c?

ms. 218 CONJECTURED B065 – In practical cases verification of a should also establish c.

ms. 219 CONJECTURED C004 – For the main models of ordinary homotopy types one expects a natural notion of local isomorphism.

ms. 221 CONJECTURED C012 – A satisfactory theory for varying S should have $f^* : \mathrm{Ho}^{\mathcal{L}}_W(S) \to \mathrm{Ho}^{\mathcal{L}}_W(S')$ for every $f : S' \to S$.

ms. 221 CONJECTURED C013 – The localization of $\mathrm{Fib}_W(S)$ should identify the locally constant full subcategory of the finer relative theory, and usually not all of it.

ms. 222 OPEN C017 – The provisional definition does not visibly imply stability under all pullbacks.

ms. 222 OPEN C023 – Is $\widehat{\mathcal{L}} = \mathrm{Liss}_W$?

ms. 224-225 OPEN C030 – Does a smooth/proper complementarity occur in general models of homotopy types?

ms. 225 CONJECTURED C032 – The smooth/proper duality may extend to all presheaf models and to topological spaces.

ms. 225-226 CONJECTURED C033 – Proposed data: classes Liss and Prop containing isomorphisms and stable under composition and pullback, perhaps with $\mathrm{Liss} \cap \mathrm{Prop} \subseteq \mathrm{Fib}_W$.

ms. 226 CONJECTURED C038 – Proposed axiom (c): the two dual adjunction properties from ChapterVI theorem-scholie2 should be added.

ms. 226 OPEN C039 – Can the hypotheses for those adjunction formulas be simplified?

ms. 226-227 OPEN C040 – Construct $f_!$ and f_* for the locally constant, proper and smooth relative theories; $f_! \dashv f^*$ is expected to be comparatively direct.

ms. 227 OPEN C041 – Seek a cofiber theory dual to the homotopy-fibre theory for (Cat,W).

ms. 232-233 OPEN C057 – Does every ordinary pushout along a W-cofibration give a W-cocartesian square, in the earlier integration-based definition?

ms. 228 OPEN XIIcb-A001 – Every functor $f : X \to Y$ admits a factorization $X \xrightarrow{i_f} \widetilde{Y} \xrightarrow{p_f} Y$ with $i_f \in \mathrm{Cof}_W$ and $p_f \in W$.

ms. 232–233 OPEN XIIcb-A019 – Is every strict pushout along a W-cofibration W-cocartesian in the categorical-integral sense of XII, definition 3, ms.131?

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ms. 236 footnote 127 CONJECTURED XIIcb-A029 – The composition-and-cancellation statement should hold in every model setting with sufficient cofibrations.

ms. 238 FALSE XIIcb-A038 – For $\{1, 2, 1'\} \hookrightarrow V$, the slices at $0, 0'$ are annotated as Δ^1 and Φ.

ms. 240 conjectured XIIcb-B004 – For the proposed fibre theory, it should suffice that every W-fibration admit arbitrary pullbacks, rather than that every pair of maps admit a pullback.

ms. 240 conjectured XIIcb-B006 – The author suspects that the left factorization axiom gives the entire good relative fibre theory only if every X→ e is a W-fibration.

ms. 269–270 false XIIcb-C020 – The reduction of all three cube implications to the case where Y is the terminal square uses only the direct, pushout-stability half of W(7 bis).

ms. 272 open XIIcb-C028 – Does the factorization hypothesis alone suffice to construct the required theory when not every object is fibrant over the terminal object?

ms. 273 false XIIcb-C035 – The conservative-pushout axiom is equivalent to the conjunction of Cons1 and literal cartesian Cons2.

ms. 274 false XIIcb-C037 – Under only the theorem's general cofiber/factorization hypotheses, a terminal object and closure of aspherical objects under retracts, f_0,f_3∈ W imply f_1,f_2∈ W for every map of W-cocartesian squares.

ms. 274 false XIIcb-C038 – Corollary1: the universal map-of-squares implication f_0,f_3∈ W⇒f_1,f_2∈ W is equivalent to the object condition Z_0,Z_3 aspherical⇒Z_1,Z_2 aspherical, and to its special case Z_0=e.

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ms. 1 false XIIapp-O002 – If b and c are weakly terminal and their categorical product b×c exists, BX is weakly contractible.

ms. 3 false XIIapp-O015 – H^p(T,F)=0 for every abelian presheaf F and every p≥2.

ms. 3 false XIIapp-O017 – H^2(T,F)=coker(F(i_1)-F(i_2)), with H^1 evidently intended in the displayed two-term formula.

ms. 4 false XIIapp-O021 – H^n(X_G;M)≃H^1(T,R^{n-1}f_*M)≃coker(H^{n-1}(G;M)→H^{n-1}(G²;M)).

ms. 4 false XIIapp-O025 – H²(BX_G;M)≃Hom_mon(G,M).

ms. 4 false XIIapp-O026 – X_G is 2-connected if and only if the universal abelian group of G is zero.

ms. 4 false XIIapp-O027 – If G is a nonzero abelian group, then X_G is not 2-connected.

ms. 4 open XIIapp-O029 – If y_b⊔_{y_a}y_c=1_X, must X at least be simply connected?

ms. 5 open XIIapp-O030 – Are totally W∞-aspherical categories precisely categories with a terminal object?

ms. 6 false XIIapp-O035 – X_G is simply connected, but not 2-connected whenever its universal abelian group is nontrivial.

ms. 5 false XIIapp-R001 – Every totally W∞-aspherical category, in the universal left-Kan-extension sense used here, has a terminal object.

ms. 5 conjectured XIIapp-R002 – The suspected terminal-object characterization would mean that the existing notion of open Dwyer immersion cannot be substantially improved.

ms. 9 false XIIapp-C007 – There is a retraction r:V→ D fixing D and sending b,c to a.

ms. 9 false XIIapp-C008 – The alleged retraction gives a natural transformation ir⇒ id_V and a categorical homotopy equivalence D→ V.

ms. 9 false XIIapp-C010 – BV has the homotopy type of S^1 and is not simply connected.

ms. 10 false XIIapp-C018 – The singleton-pushout condition can hold while Z is not simply connected.

ms. 11 FALSE XIIapp-C029 – The inclusion Z_0=(0_0⇉ a_0)→ Z is a homotopy equivalence by an evident retraction.

ms. 15 FALSE XIIapp-C062 – pi_1(V_0 prime) is isomorphic to Z.

ms. 15 FALSE XIIapp-C063 – V_0 prime is non-W-aspherical whenever W is contained in W_1, by its alleged nontrivial fundamental group.

ms. 17 OPEN XIIapp-G05 – The crossing arrows define a natural transformation $ip \Rightarrow jq$ and hence a homotopy-commutative square with vertices E, A, B, X; is it W-cocartesian?

ms. 17 OPEN XIIapp-G11 – Under $i \in W$ or $j \in W$ and W-asphericity of X, is the same square still W-cocartesian, in particular for $W = \mathbf{W}_\infty$?

ms. 17 FALSE XIIapp-G12 – For $W = \mathbf{W}_\infty$, under $i \in W$ (respectively $j \in W$) and W-asphericity of X, W-cocartesianness is equivalent to $q \in W$ (respectively $p \in W$).

ms. 17 OPEN XIIapp-G14 – If $A \simeq e$ and X, B are W-aspherical, must the attaching comma category $a\backslash B$ be W-aspherical, equivalently must j be W-coaspherical?

ms. 18 CONJECTURED XIIapp-G23 – The covering-descent argument should extend to higher homotopy degrees through n-gerbes.

ms. 18 CONJECTURED XIIapp-G26 – For every open–closed decomposition, $\varphi \in \mathbf{W}_\infty$.

ms. 18 CONJECTURED XIIapp-G27 – If $W \supseteq \mathbf{W}_\infty$ satisfies Loc (6a′), then the open–closed square is W-cocartesian.

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ms. 6 CONJECTURED XIIIa-A025 – The small object construction should require only pushouts and filtered, or ordinal, colimits instead of full cocompleteness.

ms. 6–7 CONJECTURED XIIIa-A026 – There should be a notion of accessible arrow class Φ implying existence of a set I⊆Φ with Φ⊆{}^□(I^□).

ms. 7 CONJECTURED XIIIa-A027 – The proposed accessible classes should be preserved by Φ↦Φ_* and Ψ↦Ψ^*.

ms. 7 CONJECTURED XIIIa-A028 – If a left/right lifting closed pair is accessible in a sense supplying a set of left generators, it has Quillen factorization.

ms. 12 CONJECTURED XIIIa-A052 – The weak axioms M0–M4,M′,M″ may be insufficient to obtain exact suspension and cosuspension sequences.

ms. 13 CONJECTURED XIIIa-A058 – The equality TC=C∩ W should imply the required exactness of W-cartesian squares and cosuspension sequences.

ms. 13 CONJECTURED XIIIa-A059 – The equality TF=F∩ W should imply the required exactness of W-cocartesian squares and suspension sequences.

ms. 14 CONJECTURED XIIIa-A061 – A relative category may admit distinct left and right pre-Q structures even when no full Quillen structure with those weak equivalences is available.

ms. 15 CONJECTURED XIIIa-A069 – Under accessibility of M, conditions (a),(a′),(a″),(b),(b′) are proposed to be equivalent.

ms. 15 CONJECTURED XIIIa-A071 – Theorem-scholie(3): a closed right pre-Q structure exists iff (b′) holds and A_W=(W^univ)^□∩ W_univ is right lifting closed.

ms. 16 CONJECTURED XIIIa-A072 – The preliminary hypotheses, accessibility of M, and the proposed equivalent recognition conditions are stable under (M,W)↦(M^I,W^I) for every small diagram category I.

ms. 16 OPEN XIIIa-A073 – Is the extra right-closedness condition in item 3 stable under passage to diagram categories?

ms. 16 OPEN XIIIa-A074 – Do the conditions in item 2 imply the extra condition in item 3, making the two criteria equivalent?

ms. 16 CONJECTURED XIIIa-A078 – (a′) should imply (a), after requiring lifting closedness of the quadruple.

ms. 17 FALSE XIIIa-A080 – The closure properties proved in item 1 imply that Cof_W and W^univ are left lifting closed.

ms. 19 CONJECTURED XIIIa-A085 – Filtered-colimit stability of W should make W accessible in a suitable sense, and then imply small generation of C∩ W when C is small-generated.

ms. 20 CONJECTURED XIIIa-A089 – Conjectural lemma(1), fibration version: if Φ is left lifting closed and Φ^□⊆ Fib_W, there exists a set Φ0⊆Φ such that Φ0^□⊆ Fib_W.

ms. 20 CONJECTURED XIIIa-A090 – Conjectural lemma(1), weak-equivalence version intended: if Φ is left lifting closed and Φ^□⊆ W, there exists a set Φ0⊆Φ such that Φ0^□⊆ W.

ms. 20 CONJECTURED XIIIa-A091 – Conjectural lemma(2): the intersection of two set-generated left lifting closed classes in a locally presentable category is again set-generated.

ms. 20 CONJECTURED XIIIa-A092 – An intrinsic class such as Fib_W, defined from an accessible W, should itself be accessible and yield the compactness lemma.

ms. 22 FALSE XIIIa-B003 – Part 1 of the conjectural theorem implies that Cof_W and W^univ are left lifting classes.

ms. 25 OPEN XIIIa-B016 – Does existence of some closed Quillen triple imply either the stronger closure conditions or the maximal compatibility A_W=(Cof_W)_*?

ms. 26 CONJECTURED XIIIa-B019 – Group I identifies weak Quillen existence, left exact weak Quillen existence, and the two orthogonality inclusions (*).

ms. 26 CONJECTURED XIIIa-B020 – Group II identifies right exact weak Quillen existence, condition (*) plus right closure of A_W, closed Quillen existence, and the displayed maximal-TC construction.

ms. 26 FALSE XIIIa-B021 – With TC=W^univ, F=TC_*, TF=F∩ W_univ, define C=(TF)_*.

ms. 26 CONJECTURED XIIIa-B022 – Group III identifies the maximal closed model candidate with (*) plus A_W=(Cof_W)_*.

ms. 26 OPEN XIIIa-B027 – Could the identity $\overline{W_{\mathrm{univ}}}$∩Fib_W=W_univ imply III⇔IV?

ms. 26 FALSE XIIIa-B029 – The displayed intersection (W^univ)_*∩$\overline{W_{\mathrm{univ}}}$ contains D→ e for discrete D without qualification.

ms. 27 FALSE XIIIa-B041 – If C,F,W are retract closed, left exactness is equivalent to right exactness for the original designated classes and forces all four classes closed.

ms. 28 FALSE XIIIa-B044 – The preceding saturation produces a structure de Quillen in the biexact sense introduced immediately afterwards.

ms. 30 CONJECTURED XIIIa-B060 – For arbitrary weak structures, right Q-homotopy is an equivalence relation whenever the target is fibrant, without exactness.

ms. 30 CONJECTURED XIIIa-B061 – On fibrant-cofibrant objects, the right and left homotopies coincide and are independent of the chosen cylinder, as suggested by Baues.

ms. 30–31 CONJECTURED XIIIa-B062 – Under left exactness, the fibrant-cofibrant quotient equivalent to Ho_W embeds as a fully faithful right adjoint to localization of the right-homotopy quotient of cofibrant objects.

ms. 32 OPEN XIIIa-B067 – Does left exactness also suffice for exactness of W-homotopy-cocartesian squares?

ms. 32 FALSE XIIIa-B068 – Exactness alone may fail to imply either cartesian or cocartesian homotopy exactness.

ms. 33–34 CONJECTURED XIIIa-B069 – If exact weak structures exist on (M,W) and on (M(Ψ),W(Ψ)) for the cospan indexing category, W-homotopy-cartesian squares are W-exact and give loop exact sequences.

ms. 33 CONJECTURED XIIIa-B071 – The dual exact-structure hypothesis for M(Φ) gives cocartesian exactness and pointed suspension exact sequences.

ms. 34 CONJECTURED XIIIa-B072 – For simultaneous homotopical algebra in all M(I), it suffices to have an exact weak Quillen structure on each relative diagram category.

ms. 37 OPEN XIIIa-B086 – For any arbitrary arrow class Φ, evaluation might send Φ(I)_* into Φ_*.

ms. 40 CONJECTURED XIIIa-B097 – The author expects filtered-colimit stability of W to be practically necessary for this strategy.

ms. 41 CONJECTURED XIIIa-C003 – Closure of W under all filtered colimits is proposed as the practical way to establish the required diagram factorizations.

ms. 42–43 CONJECTURED XIIIa-C011 – Condition a implies existence, for every small I, of a closed left-exact weak Quillen structure on Fun(I^op,M) with objectwise cofibrations.

ms. 42–43 CONJECTURED XIIIa-C012 – Condition a-prime implies existence of the prescribed injective structures for all small I.

ms. 42–43 CONJECTURED XIIIa-C013 – Condition a-double-prime implies existence of the prescribed closed left-exact weak structures for all small I.

ms. 42–43 CONJECTURED XIIIa-C014 – Condition b implies existence of the prescribed closed left-exact weak structures for all small I.

ms. 44 CONJECTURED XIIIa-C020 – The author expects a-double-prime to imply all other conditions of Theorem 2.

ms. 44 CONJECTURED XIIIa-C021 – The author expects the small-set condition a-double-prime to be the practical verification method, notably for M=Cat.

ms. 48 CONJECTURED XIIIa-C041 – The converse componentwise condition for injective fibrations need not hold.

ms. 48–49 CONJECTURED XIIIa-C044 – Restriction also preserves homotopy cartesian squares and cosuspension exact sequences.

ms. 49 FALSE XIIIa-C045 – The ordinary functor u_* preserves arbitrary homotopy cartesian squares merely because it preserves F and TF.

ms. 49 CONJECTURED XIIIa-C046 – A right homotopy Kan extension generally does not preserve homotopy cocartesian squares.

ms. 49 FALSE XIIIa-C047 – The ordinary u_! should preserve arbitrary homotopy cocartesian squares.

ms. 50 OPEN XIIIa-C049 – Do the preceding assumptions ensure the existence of homotopy adjoints u_!^W and u_*^W, even for finite posets?

ms. 52 OPEN XIIIa-C068 – What quillenization or derived-functor theory is available for unbounded complexes in a general abelian category?

ms. 53 FALSE XIIIa-C071 – The same filtered-colimit closure holds for Ch^+(A), Ch^-(A), and Ch^b(A).

ms. 53 OPEN XIIIa-C075 – Does the right derived Kan extension exist on the whole unbounded derived category?

ms. 53 CONJECTURED XIIIa-C076 – A left homotopy Kan extension need not preserve D^+ without a homological-dimension restriction.

ms. 53–54 conjectured XIIIa-C077 – For Grothendieck A, a left derived Kan extension is expected on D^-(A(I-prime)) with values in D^-(A(I)).

ms. 54 open XIIIa-C078 – Does left derived Kan extension exist on all of D(A(I-prime)) without finite cohomological dimension?

ms. 55 open XIIIa-C083 – Which derivator axioms follow from quillenizability alone?

ms. 55–56 conjectured XIIIa-C087 – The strong local-system axiom asks that u in the chosen fundamental localizer induce an equivalence on homotopy categories of locally constant diagrams.

ms. 56 conjectured XIIIa-C092 – Quillenizability alone seems too weak to ensure interesting homotopy Kan extensions.

ms. 59 false XIIIa-C101 – The parenthesis invokes finite colimits to justify the finite limits required by finite-index u_*.

ms. 60 conjectured XIIIa-C109 – A poset whose associated opposite Alexandrov space is noetherian and sober is admissible.

ms. 52 open XIIIb-Q013 – What hypotheses suffice for quillenisability of unbounded complexes, and what can be done without enough injectives or projectives?

ms. 53 false XIIIb-Q015 – If A is accessible and has filtered colimits, all the listed complex categories have the same closure properties.

ms. 53–54 open XIIIb-Q019 – Does u_* extend to a right adjoint on unbounded derived categories of diagrams?

ms. 53–54 conjectured XIIIb-Q020 – For globally bounded-above diagrams of Grothendieck coefficients, one expects a left adjoint Lu_! to restriction preserving upper bounds.

ms. 54 open XIIIb-Q021 – Does a left adjoint to restriction exist on the entire unbounded derived diagram category without a finite cohomological-dimension hypothesis?

ms. 54 footnote32 open XIIIb-Q024 – Can the square-based structure of an unpointed homotopy theory be axiomatized?

ms. 55 open XIIIb-Q026 – Which derivator axioms follow from quillenisability alone?

ms. 55–56 conjectured XIIIb-Q030 – A strongest proposed local-constancy axiom says that u in a chosen fundamental localizer induces HO_W^lc(I)≃HO_W^lc(I′).

ms. 56 open XIIIb-Q032 – How can the local-constancy equivalence be verified directly from (M,W), before a theory of u_! or u_* has been built?

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ms. 69 footnote37 open XIIIb-Q066 – Can the representing object of the preceding pointwise quotient be chosen functorially in localized G′ without admissibility?

ms. 71 false XIIIb-Q070 – Representability of F↦Hom(HO(I′))(u*F,G′) gives a left adjoint of u*.

ms. 71 false XIIIb-Q071 – The left adjoint just obtained is computed by u_* on fibrant G′.

ms. 72 open XIIIb-Q073 – Do all functors u admit homotopy right images under the earlier large-category hypotheses; can a counterexample be found?

ms. 72–73 open XIIIb-Q074 – Is a small product of W-maps between fibrant objects again a W-map?

ms. 73 open XIIIb-Q076 – Does inverse limit preserve levelwise W-maps between such fibrant towers?

ms. 73–74 OPEN XIIIb-Q078 – Can the product assertion fail even for a closed model category with the additional accessibility and filtered-colimit exactness hypotheses listed?

ms. 74 OPEN XIIIb-Q079 – Is the one-object category BG admissible for a group G?

ms. 74 OPEN XIIIb-Q080 – Is the full subcategory of Δ on [0] and [1] admissible?

ms. 74–75 FALSE XIIIb-Q081 – For the dual construction assume a closed right-exact structure with TF=F∩C.

ms. 75 CONJECTURED XIIIb-Q084 – These classes should define a uniform right quillenisability on all small categories.

ms. 76 CONJECTURED XIIIb-Q087 – If the projective construction succeeds, left homotopy Kan extensions exist for maps of finite posets and the indicated noetherian sober extensions.

ms. 76 CONJECTURED XIIIb-Q088 – Closure of W under filtered colimits should make discrete and left-filtered categories such as N^op admissible for left images.

ms. 76 CONJECTURED XIIIb-Q089 – A poset which is a filtered union of closed noetherian sober subposets should be right-admissible under the projected dual hypotheses.

ms. 76 OPEN XIIIb-Q090 – What happens to left admissibility for BG?

ms. 77 FALSE XIIIb-C07-007 – Under GCH, (pi')^pi=pi' for every pair of infinite cardinals pi'>pi.

ms. 85 FALSE XIIIb-C07-027 – The printed formula asserts Z_sup A ≅ lim_A Z_T for the covariant partial-colimit diagram.

ms. 89 FALSE XIIIb-C07-037 – The three conditions a_pi, b_pi, c_pi are equivalent.

ms. 90 FALSE XIIIb-C07-040 – To deduce membership of an arbitrary pi^+-directed colimit of M^pi-objects, it is enough to apply b_pi directly because I_M=I.

ms. 94 FALSE XIIIb-C07-054 – For every C_i' admitted by the printed image-size definition, Z_{C_i'}=colim_{C_i'} Z belongs to Filt^pi(E).

ms. 94 FALSE XIIIb-C07-055 – With the printed tilde C_i, the diagram (Z_{C_i'}) is perfectly adapted at pi.

ms. 97 OPEN XIIIb-C07-068 – If X has both a perfectly adapted Filt^pi presentation and a pi^+-directed presentation by M^pi-objects, can one show its adapted M-stages cofinal?

ms. 97 CONJECTURED XIIIb-C07-069 – After replacing pi by a larger cardinal pi' (suggested pi'=pi^+), a_pi should imply b_{pi'}.

ms. 86 (glossary discrepancy) FALSE XIIIb-G036 – The glossary states that a strict cardinal filtration has every stage Filt^π(E) closed under all filtered colimits.

ms. 86 (glossary discrepancy) FALSE XIIIb-G037 – The final glossary paragraph calls a perfectly adapted representation the colimit of a 'well adapted' system.

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ms. 4 CONJECTURED XIV.016 – The preceding stability axioms appear to exhaust the useful stability properties, so further such properties should be consequences.

ms. 7 FALSE XIV.030 – If the underlying category is additive, the h-cartesian and h-cocartesian classes should coincide.

ms. 8 CONJECTURED XIV.035 – In some sense the general case ought to reduce to the elementary ordinary-pullback case.

ms. 8 FALSE XIV.036 – The proposed second family of left quadrangulated examples is the localized category with the homotopy-image class of squares, for localizers admitting homotopy fibres.

ms. 8 FALSE XIV.037 – Proposition 1 is expected to hold under further hypotheses to be identified, including every presentation quillenizable on finite poset diagram shapes.

ms. 8 FALSE XIV.039 – Proposition 1, necessary direction: if $\mathrm{ho}_W(X)$ belongs to the designated h-cartesian class, the chosen strict square X is W-cartesian.

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ms. 7–8 FALSE XV-A-012 – If every domain of a generator is π-accessible and $I_\alpha = \{\beta < \alpha\}$ is grand devant π, the source asserts $p_\alpha(f) \in I^{\perp}$.

ms. 9–10 FALSE XV-A-024 – The proposed first stage should satisfy $i_1(f) \in C_0''$ for every arrow f, and the construction is asserted to do so.

ms. 11 FALSE XV-A-033 – If the domains of I are accessible, choose a common π; the source claims $p_\alpha(f) \in I^{\perp}$ for every ordinal grand devant π.

ms. 17 FALSE XV3-006 – GCH implies pi^pi_0=pi for every pi>pi_0.

ms. 17 CONJECTURED XV3-008 – The endpoint filtration Fl^pi(M) should itself be a cardinal filtration of the arrow category.

ms. 19 FALSE XV3-015 – Proposition 3.1(2), (3.8): for r,s>=max(pi_1,C_1), A in Filt^r and X in Filt^s imply |Hom(A,X)|<=s^(r^pi_0).

ms. 19 FALSE XV3-018 – GCH implies pi^pi_prime=pi for infinite pi>pi_prime>=1 and hence gives (3.7 bis)-(3.9 bis) for every r>pi_0.

ms. 20 FALSE XV3-020 – The category of pairs (A,B subset A) gives equality in the arrow-isomorphism bound (3.7 bis) for the indicated filtration.

ms. 20 FALSE XV3-022 – |Hom(A_i,X_j)|<=C_1 because C_1 counts arrow-isomorphism classes at level pi_1.

ms. 21 FALSE XV3-024 – For a fixed poset I of size at most mu=r^pi_0, the choices of all objects A_i up to isomorphism can be bounded by |I| times C_0<=mu.

ms. 21 FALSE XV3-025 – For fixed objects A_i, choices of all transition maps can be bounded by |I times I| times C_0<=mu.

ms. 21 FALSE XV3-026 – A cocone from a fixed I-diagram to X has at most |I| times s^mu choices.

ms. 23 FALSE XV3-031 – Lemma 3.2 literally states r^rprime=2^rprime whenever rprime>=r, without an infinitude or lower-bound qualification.

ms. 23 FALSE XV3-032 – Corollary 1 claims |Hom(A,X)|<=2^(r^pi_0) for A,X in Filt^pi and r>=max(pi_1,C_1).

ms. 27 FALSE XV3-048 – Theorem 2(1): p_alpha(f) lies in Psi when alpha is large relative to r^pi_0.

ms. 27 FALSE XV3-049 – The initial ordinal of any cardinal rprime>r^pi_0 may be chosen to satisfy the large-cofinality condition.

ms. 35 FALSE XV4-025 – In the proof of Corollary2, $C \cap W \longleftrightarrow TC$.

ms. 36 FALSE XV4-030 – For two precouples with $TC \subset C \cap W$ and $TF \subset F \cap W$, the equalities $TC = C \cap W$, $TF = F \cap W$, $W = TF \circ TC$ and the assertion that (W, C, F) is closed are all equivalent.

ms. 40 FALSE XV4-039 – Under only the hypotheses of Proposition4.4.2, Cof_W cap W equals W^univ, the maps all of whose cobase changes belong to W.

ms. 40 FALSE XV4-041 – Under only the Proposition4.4.2 hypotheses, Cof_W cap W is Q-stable on the left.

ms. 40–41 FALSE XV4-042 – Under the hypotheses of Corollary4.4.4, C=tilde(C0) equals Cof_W.

ms. 40–41 FALSE XV4-043 – Under only the hypotheses of Corollary4.4.4, TC=C cap W equals C cap W^univ and is contained in W^univ.

ms. 41 FALSE XV4-045 – TC is Q-stable on the left under only Corollary4.4.4 hypotheses.

ms. 43 OPEN XV4-054 – Given suitable small C0 and accessible W, does there exist a small TC0⊂C cap W generating C cap W under left lifting closure?

ms. 46 FALSE XV5-015 – Footnote 17: (d_2) should imply (d_1).

ms. 52 FALSE XV5-033 – Proposition 5.6 as first stated: at a rank where bounded objects are π-accessible, condition (5.3) alone is equivalent to b_π.

ms. 53 FALSE XV5-036 – The proof applies (5.3) directly to the image J of the monomorphic stage indices in $\mathrm{Sub}_\pi(X)$.

ms. 58 FALSE XV5-042 – Printed condition (cprime): M is the smallest class closed under filtered colimits, with no stated requirement to contain M_0.

ms. 58–59 OPEN XV5-045 – Footnote 22 asks whether (cprime) implies (c) without the special object-accessibility assumption on pi.

ms. 62–63 FALSE XV5-053 – The printed lemma and display assert that the unrestricted $\mathrm{Sub}_\pi(Y)$ maps canonically and cofinally into M^π/Y.

ms. 64 FALSE XV5-057 – The projection in Corollary 5.11 is described parenthetically as fibrant.

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ms. 58–59 OPEN XVb-A009 – Without requiring every Filt^pi object to be pi-accessible, footnote22 asserts (a)⇔(b), (a)⇒(c′), (b)⇒(c), and tentatively (c′)⇒(c).

ms. 62–63 FALSE XVb-A021 – As printed, all of Ss_pi(Y) is a cofinal subcategory of M^pi/Y.

ms. 72 FALSE XVb-A041 – The individual good-index sets I_{M_alpha} are parenthetically asserted to be filtered without a cofinality hypothesis.

ms. 75 OPEN XVb-A049 – The exactness assumption(e) might be eliminated from the general machinery.

ms. 76 FALSE XVb-A056 – If C is an L-accessible class of monomorphisms containing all isomorphisms under(a)–(e), there is a small C_0⊂C with C⊂tilde(C_0).

ms. 76–77 FALSE XVb-A057 – For any pi witnessing C accessibility,6.1 further asserts C⊂tilde(C^pi), with C^pi=C∩Fl^pi(M).

ms. 77 FALSE XVb-A059 – Under the proposed unconditional hypotheses of6.1,6.2 asserts C_*=C_{0*} and tilde(C)=tilde(C_0) for a small generating C_0.

ms. 79 FALSE XVb-A066 – For every nonisomorphism f:A→B in C, there exists A⊊A_1⊂B with A→A_1∈tilde(C^pi) and A_1→B∈C.

ms. 83 OPEN XVb-A078 – The proof needs the comparison j:C′=A amalg_{A′}B′→B to lie in C, and does not even know that it is monic.

ms. 83–84 FALSE XVb-A080 – The diagram of C_i and gamma_i is called pi-adapted after checking continuity, without establishing that its objects lie in the bounded filtration stage.

ms. 84 FALSE XVb-A081 – Since colim gamma_i is an isomorphism in C and C is pi-accessible, the indices with gamma_i∈C are claimed cofinal.

ms. 84 FALSE XVb-A082 – The attempted proof concludes Lemma6.4 and Theorem6.1 by choosing an index with both alpha_i∈C^pi and gamma_i∈C.

ms. 84–85 OPEN XVb-B001 – The generation assertion of §6 may be false: an accessible or L-accessible class need not be the left Q-closure of a set.

ms. 85 CONJECTURED XVb-B002 – For a fundamental localizer W, the classes Cof_W and W^{univ} in Cat should be accessible, in particular for W=W_infty.

ms. 85 OPEN XVb-B003 – If C=tilde(C) is closed under filtered colimits and generated on the left by a set, and W is L-accessible, is C intersect W generated on the left by a set, even when C consists of monomorphisms?

ms. 90 FALSE XVb-B026 – For every monomorphism i:A->B, there is a generator i_prime mapping to it whose pushout strictly enlarges A inside B.

ms. 99 FALSE XVb-B046 – In the cartesian square (7.4.3), if the left vertical f belongs to F and the bottom p belongs to W, then the top q belongs to W.

ms. 102 OPEN XVb-B056 – Effective unions of subobjects may fail in Cat.

ms. 104 OPEN XVb-B066 – Is the presheaf model structure necessarily also right proper?

ms. 106 FALSE XVb-B073 – Footnote38(d_prime): j_A sends every W-equivalence of categories to a universally W_A-equivalent map of presheaves, and this is equivalent to the local test condition.

ms. 112–114 FALSE XVb-B089 – Lemma8.7: finite cocompleteness, two-out-of-three for W, composition/pushout stability of C, a fold factorization with left factor in C and right factor in W, and the stated coproduct inclusions suffice for mapping-cylinder factorizations C then W.

ms. 115 CONJECTURED XVb-B096 – If the connected-component axiom fails in the presence of the other listed axioms, perhaps W is the class of all functors.

ms. 116 CONJECTURED XVb-C010 – Loc(3bis) should suffice when tested only on split fibred categories and split cartesian morphisms, assuming Loc(1,2).

ms. 121 OPEN XVb-C033 – Can coproduct reflection f1⊔f2∈W⇒f1,f2∈W be proved with only the preceding basic assumptions?

ms. 122 OPEN XVb-C037 – Footnote 47: the converse, arbitrary coproduct preservation under only Loc(1,2,3,6c′), is not known to the author.

ms. 125 CONJECTURED XVb-C051 – Loc(8) and Loc(5bis) are conjectured independent under the basic localizer axioms.

ms. 125 CONJECTURED XVb-C052 – The author expects every localizer useful in practice to satisfy connectivity and decomposition, while these are not needed in preceding general developments.

ms. 125 CONJECTURED XVb-C053 – Connectivity may matter when recovering contractibility and asphericity structures from one another.

ms. 130 CONJECTURED XVb-C068 – The author expects connectivity to hold automatically for localizers that will be introduced.

ms. 133 bis FALSE XVb-C084 – The editorial bracket says (c′)⇒(c) is the trivial implication and the quotient argument below proves (c)⇒(c′).

ms. 136 FALSE XVb-C104 – 10.1(3b) with current Loc(1ter,2,3,5b) read literally: every f∈W induces an equivalence on locally constant objects.

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ms. 9 FALSE XVI-A038 – With filtered-colimit stability, finite Loc (4a) is equivalent to full Loc (4).

ms. 10-11 FALSE XVI-A045 – Lemma2.5: product closure for an arbitrary subset W implies preservation of binary, finite, or arbitrary products by localization.

ms. 17' CONJECTURED XVI-A089 – The same forward/reverse distinction holds for all finite n-equivalence classes W_n.

ms. 17',22-23,25 OPEN XVI-A092 – Does (6-prime a) imply (6a), hence provide the desired intrinsic cofibration theory?

ms. 23 OPEN XVI-A112 – Does (6-prime a) alone make trivial Dwyer inclusions stable under every pushout?

ms. 23-24 OPEN XVI-A114 – Is a square W-homotopy cocartesian exactly when the integral of its span maps to its corner by a W-map?

ms. 24 OPEN XVI-A116 – Every presheaf monomorphism is in Cof_WA under the stated excision hypotheses.

ms. 27 CONJECTURED XVI-A130 – Loc (6c) is probably satisfied only when W is retract closed.

ms. 29 CONJECTURED XVI-B013 – W-smooth maps and W-proper maps, hence W-perfect maps, should be closed under retracts.

ms. 29 OPEN XVI-B014 – Fibf_W and F_W need not be closed under composition; examples should be supplied.

ms. 37 CONJECTURED XVI-B026 – Loc(7), or even Loc(7b), probably does not imply Loc(6a) or Loc(6′a); possible left Kan extensions without left exactness are suggested.

ms. 40 FALSE XVI-B039 – B(F)⇒C(F) for arbitrary F⊂Fl(Cat).

ms. 50 FALSE XVI-B059 – A(Parf_{W0}) is the assertion Parf_{W0}⊂W.

ms. 52–53 OPEN XVI-B063 – Objectwise universally equivalent replacements alone may not suffice to transfer B from G to F.

ms. 57 FALSE XVI-B072 – For arbitrary weakly W-fibrant input, the infinite-path unit is universally a W0-equivalence whenever W0 satisfies Loc8bis.

ms. 64–65 OPEN XVI-B084 – Does D imply A without sequential-colimit stability, perhaps with the extra condition printed Loc3ter?

ms. 65 OPEN XVI-B085 – Does D_W0 imply A under the printed assumption that W and W0 satisfy Loc8bis?

ms. 65 CONJECTURED XVI-B086 – The minimum weak fundamental localizer Wω almost certainly fails Loc8 and even Loc8bis.

ms. 65 OPEN XVI-B087 – It is not clear whether Wω satisfies F; if it fails F then it also fails D and D⇒A is vacuous there.

ms. 66 CONJECTURED XVI-B088 – Does F imply A under Loc1ter,3bis,8bis?

ms. 66–67 OPEN XVI-B089 – Does enough intrinsic W-cofibrations imply Loc6a?

ms. 70 OPEN XVI-B098 – Without Loc8bis it remains unclear whether the arrows in the eight-condition diagram are strict.

ms. 70 OPEN XVI-B099 – The truncation localizers bold W_n should be tested for Loc7F.

ms. 28–29 CONJECTURED XVI-B100 – The Wω-perfect maps and the locally constant Wω-Grothendieck fibrations should play a role dual to Dwyer maps.

ms. 73 FALSE XVI-C005 – Neither W_ω nor Parf_{W_ω} is closed under filtered colimits.

ms. 79 conjectured XVI-C019 – All fundamental localizers used in contemporary examples appear to satisfy Loc(8) and Loc(9).

ms. 79 false XVI-C021 – W_∞ satisfies Loc(1 ter,2,3 bis,4,5,6,7,8,9).

ms. 85–86 conjectured XVI-C034 – A future descent axiom Loc10 should characterize W∞ uniquely together with Loc5 and D_{W_ω}.

ms. 86–87 conjectured XVI-C036 – There exists a fundamental localizer strictly contained in W∞; in particular Wω≠W∞.

ms. 87 open XVI-C037 – The smallest fundamental localizer satisfying D_{W_ω} might be strictly smaller than W∞.

ms. 87–88 false XVI-C039 – The listed axioms define a localizer tilde W∞⊂W∞, which consequently satisfies Loc5.

ms. 87–88 conjectured XVI-C040 – The proposed tilde W∞ is strictly smaller than W∞.

ms. 88 conjectured XVI-C042 – Hot_{tilde W∞} might model Whitehead simple homotopy types.

ms. 88 open XVI-C043 – A Whitehead invariant might be defined on pointed Hot_{tilde W∞}, provisionally as an element of K(Z[π1]).

ms. 88–89 conjectured XVI-C044 – Elementary Whitehead expansion/collapse of a finite simplicial complex should induce an isomorphism of its face-poset model in Hot_{tilde W∞}.

ms. 88; fn43 false XVI-C045 – Barycentric subdivision leaves the strict categorical homotopy type of a finite face poset unchanged.

ms. 91;124 open XVI-C048 – Whether n-equivalence localizers W_n have enough intrinsic W_n-fibrations is asked; knowing Loc7F alone supplies no canonical perfect-path calculation.

ms. 92–93 conjectured XVI-C050 – Simplicial approximation suggests that homotopy-equivalent polyhedra have suitably subdivided models that are categorically homotopy equivalent.

ms. 93–95 open XVI-C052 – Is the maximum map sd(I)→I always a categorical homotopy equivalence?

ms. 96 conjectured XVI-C055 – A fundamental localizer strictly below W∞ might retain simple homotopy types and support Whitehead and higher Whitehead invariants.

ms. 96 open XVI-C056 – Before a finer localizer is constructed, determine whether the intended Whitehead invariant is invariant under categorical homotopy equivalence of finite poset models.

ms. 97–98 open XVI-C057 – For which W are (Cat,W,Cof_W) and (Cat,W,Fib_W) Brown cofibration/fibration categories?

ms. 97–98 open XVI-C058 – For which W and u do derived direct images u_!,u_* exist, with exact homotopy pushout/pullback squares?

ms. 98 open XVI-C059 – Which derivator axioms does HOT_W satisfy, on which diagram domain, and when is it a full derivator on all small categories?

ms. 98 open XVI-C060 – When can W be the weak-equivalence class of a closed Quillen model structure on Cat, preferably a proper one?

ms. 99–100 open XVI-C063 – The explicit integrated-span construction models W-homotopy pushouts; the author asks whether its identification needs Loc3bis.

ms. 100 conjectured XVI-C065 – Loc3bis alone probably does not imply the source's homotopy-pushout exactness axiom Der6*.

ms. 102 false XVI-C068 – D_{W_ω} implies W⊂W∞.

ms. 105–106 OPEN XVI-C073 – Do Loc6a,Loc8,Loc9 imply enough intrinsic fibrations, Loc7F?

ms. 106–107 OPEN XVI-C074 – Can infinite paths in simplicial presheaves make localized right homotopy operations concrete without Loc7?

ms. 121 OPEN XVI-C097 – Does Loc6a,Loc8,Loc9 imply the coproduct decomposition axiom Loc4?

ms. 122–123 OPEN XVI-C100 – Let V be the least fundamental localizer satisfying6′a,8, with or without3bis. Is V accessible, and is V strictly contained in W∞?

ms. 123–124 OPEN XVI-C102 – Does Thomason's model structure extend to special localizers or to other broader fundamental localizers?

ms. 124 CONJECTURED XVI-C104 – For special localizers, Loc6a may admit a direct elementary proof without the full Thomason construction.

ms. 124 OPEN XVI-C105 – Do the suggested homological or cohomological equivalence localizers satisfy Loc7F?

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ms. 3 FALSE XVIIa-A011 – The printed reduction says alpha_i∈TC is equivalent to alpha_i∈C in the proof of the reverse implication.

ms. 7 FALSE XVIIa-A022 – Footnote 5 claims that cofibrancy of X and Y is unnecessary in Proposition 2 and both its corollaries, including all conclusions of Corollary 2.

ms. 19 FALSE XVIIa-A047 – The proof says that the case I'_0≠I is covered by induction on the size of I, with the alternative giving a terminal object i0.

ms. 23 OPEN XVIIa-A053 – Does u∈C_I together with local finiteness imply u_{K,I/i}∈C for every i and every lower ideal K⊂I/i?

ms. 24 FALSE XVIIa-A057 – The printed condition (I/i⊂P⇒i∈P) forces P=I for artinian I.

ms. 27,38 OPEN XVIIa-B008 – Question: can the artinian hypothesis be removed from Theorem 2.1(3), with C_I understood in the strong sense?

ms. 32 FALSE XVIIa-B021 – The intermediate cube in the proof of Lemma 2.6 is totally cocartesian, if this is read as saying all six faces are pushouts.

ms. 33–34 FALSE XVIIa-B022 – Lemma 2.7 as printed: every increasing ordinal sequence of opens yields the relative map as a transfinite composite of successor relative maps, including an extra initial factor.

ms. 55 FALSE XVIIa-C021 – Footnote 17 asserts that the candidate in (3.3.1) exists even when I prime is an arbitrary small category, without distinguishing the finite and cocomplete branches.

ms. 56–57 OPEN XVIIa-C025 – Is the constructed candidate u_!^W left adjoint to the localized restriction u_W^*?

ms. 58–59 FALSE XVIIa-C028 – The preceding density and homotopicality hypotheses alone imply that the transported candidate L is left adjoint to the localized G.

ms. 66 FALSE XVIIa-C052 – The cofibrant-replacement Kan-extension candidate extends to arbitrary small targets under only the finite-colimit assumptions of the finite-source branch.

ms. 67 OPEN XVIIa-C054 – Does this candidate for arbitrary target admit the desired adjunction, given that C(u) may no longer be a poset and the established diagram cofibration structure is unavailable there?

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ms. 64,69 FALSE XVIIb-A065 – Theorem 3.7, formulated for covariant diagrams on artinian posets, immediately proves the corresponding restriction of the presheaf prederivator on the same artinian posets.

ms. 66 FALSE XVIIb-A017 – For arbitrary target category I' and artinian-poset source I, the displayed formula $(u_!X)(j) = \operatorname{colim}_{(u\downarrow j)} X$ still constructs a weak-equivalence-preserving functor on cofibrant diagrams without further hypotheses.

ms. 67 OPEN XVIIb-A019 – Without a suitable cofibration structure on diagrams over non-posets, the mapping-cylinder proof of Theorem 3.7 has not yet established the adjunction for arbitrary target shapes.

ms. 68 FALSE XVIIb-A021 – The inference attributed here to Anderson, from ordinary full faithfulness of restriction to full faithfulness of localized restriction without further control of derived unit/counit, is not a valid formal implication.

ms. 70–71 OPEN XVIIb-A026 – Find hypotheses on a strict cofibration category guaranteeing all small left homotopy Kan extensions and their pointwise formula.

ms. 79 OPEN XVIIb-A056 – Under suitable additional assumptions, seek $L(TC) \subset TC' \subset \widetilde{L(TC)}$, or the stronger $\widetilde{C'} \cap W' \subset \widetilde{L(TC)}$.

ms. 81 OPEN XVIIb-B002 – Under what additional assumptions does $F' = (v^*)^{-1}(F)$ hold for the transferred cofibration structure?

ms. 82 FALSE XVIIb-B013 – Footnote 23 proposes the literal equality $v_!(TC(I_0)) = TC(I)$ under stronger special-model hypotheses.

ms. 82 CONJECTURED XVIIb-B014 – Footnote 23 proposes $F(I) = (v^*)^{-1}F(I_0)$ under stronger special-model hypotheses.

ms. 97 CONJECTURED XVIIb-B055 – The informal expectation that special Quillen triples probably occur only when W is stable under all small filtered colimits.

ms. 101 FALSE XVIIb-C013 – As printed, $C(I) = C_I$ and $TC(I) = TC_I$ whenever I is artinian, with $\mathcal{M}(I) = \mathrm{Fun}(I^{op}, \mathcal{M})$.

ms. 116 FALSE XVIIb-C044 – The proof of6.3 completes the equality assertion of6.1(d).

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ms. 53 FALSE XVIII-A50-014 – The operation subsequently denoted $X \boxtimes Y$ was initially proposed as a bicategorical product of pseudo-topoi.

ms. 60–61 FALSE XVIII-A50-033 – Equation4.2.8 claims restriction $\mathrm{Cont}(S\text{-hat}^{op}, N) \to [S^{op}, N]$ is an equivalence for arbitrary N.

ms. 61 FALSE XVIII-A50-034 – The unrestricted functor category $[S\text{-hat}^{op}, N]$ appearing twice on p61 is identified with the continuous-functor category.

ms. 63 OPEN XVIII-A50-037 – The remaining tensor problem is to construct a left adjoint to its fully faithful restricted-nerve embedding.

ms. 63 CONJECTURED XVIII-A50-038 – Lemma4.2.3: for arbitrary pseudo-topos M,N and s∈M,t∈N, evaluation F↦F(s,t) on M⊠N is representable by s⊠t.

ms. 64 FALSE XVIII-A50-040 – Elementary tensors should decompose as products of two objects depending separately and cocontinuously on the factors.

ms. 66 OPEN XVIII-A66-003 – Does the tensor of arbitrary pseudo-topoi remain a pseudo-topos?

ms. 66 FALSE XVIII-A66-004 – Lemma 4.3.2: for complete $\mathcal{E}$, pointwise corepresentability of $\alpha : \mathcal{E} \to X^\wedge$ is sufficient for a left adjoint.

ms. 66 FALSE XVIII-A66-006 – The displayed evaluation $\xi \mapsto \mathrm{Hom}_{X^\wedge}(x, \alpha\xi)$ has domain $\mathcal{E}^\circ$.

ms. 67 FALSE XVIII-A67-004 – Lemma 4.3.3(b): the images under the reflector of a small dense family in $\mathcal{F}$ form a small dense family in $\mathcal{F}$.

ms. 76 FALSE XVIII-A76-003 – This proposed I is grand devant π.

ms. 76 FALSE XVIII-A76-006 – "On a donc prouvé PTA⇒c."

ms. 78 FALSE XVIII-A78-004 – It remains to prove that $\mathrm{Ind}_\pi C$ is closed under colimits in ordinary $\mathrm{Ind}\, C$.

ms. 82 FALSE XVIII-A82-001 – Proposition4.4.4: condition(b) implies condition(c).

ms. 82 FALSE XVIII-A82-008 – Alternative(e′): finality of one diagonal $J \to J^{A_0}$ with $|A_0| = \pi$ is sufficient without further hypotheses.

ms. 82 FALSE XVIII-A82-009 – The attempted F2 argument using three terminal subcategories proves finality without the added hypothesis.

ms. 82 FALSE XVIII-A82-010 – Corollary4.4.5: $F \in \mathrm{Ind}'_\pi C$ if and only if C/F satisfies all four conditions of4.4.4.

ms. 85 FALSE XVIII-A85-002 – The index $I = \mathrm{Fun}(J, C)/\varphi$ is an ordered filtered set.

ms. 88 FALSE XVIII-A88-004 – The preceding lemma is equivalent to closure of the restricted ind-category under all pi^+-filtered colimits of its objects.

ms. 94 CONJECTURED XVIII-A91-011 – Question/negative expectation: even if C has colimits of cardinal at most π, evaluation $C^J/\varphi \to C/\varphi(j)$ need not be final for $\varphi : J \to \mathrm{Ind}(C)$.

ms. 102 FALSE XVIII-A91-029 – Claim as written: unrestricted restriction $\mathrm{Fun}(M, E) \to \mathrm{Fun}(C, E)$ is fully faithful because C strictly generates M.

ms. 106 FALSE XVIII-A91-037 – Claim encoded in the diagram: restriction sends every filtered-colimit-preserving functor Ind(C)→E to a functor Ind_pi(C)→E preserving all size≤pi colimits.

ms. 107 FALSE XVIII-A91-042 – Proposition4.5.2 as first stated: conditions a,a′,a″,b,c,c′ are equivalent for every pi≥pi1.

ms. 108 FALSE XVIII-A91-046 – Claim implicit in(4.5.2.2): presheaves F∈M that are pi-accessible in the ambient presheaf category are closed under all ambient colimits of size≤pi.

ms. 109 FALSE XVIII-A91-049 – The proposed proof of c⇒a invokes SGA4 I9.16 after constructing the claimed cardinal filtration.

ms. 109 FALSE XVIII-A91-051 – Corollary4.5.3 as first stated: (Filt^pi)_{pi≥pi1} is a cardinal filtration of M.

ms. 110 FALSE XVIII-A91-054 – Corollary4.5.4 as first stated: Filt^pi is closed under every colimit in M indexed by a category with at most pi arrows.

ms. 111 FALSE XVIII-A91-057 – The earlier proof of4.3.6(b⇒c) is asserted to have already established this same pointwise filtration with the desired cardinal bounds.

ms. 113 FALSE XVIII-A91-062 – Lemma4.5.5 as first stated: whenever pi>pi0 and J is pi0^+-filtered, every set of at most pi arrows of J lies in a pi0^+-filtered subcategory with at most pi arrows.

ms. 114 FALSE XVIII-A91-064 – Claim as printed: closing a set Φ of at most pi arrows under identities and finite compositions produces a subcategory with at most pi0 arrows, although pi>pi0.

ms. 114 FALSE XVIII-A91-066 – Claim as written: [S]^{≤pi0} is the smallest pi0^+-directed subposet of P(S) containing all singleton subsets.

ms. 115 FALSE XVIII-A91-069 – The cardinality calculation for [S]^{≤pi0} is taken to prove that a directed hull in P(S) must have size pi^{pi0}.

ms. 115 FALSE XVIII-A91-070 – The source concludes that its original equivalences can hold only for cardinals with pi^{pi0}=pi.

ms. 117–118 FALSE XVIII-A116-009 – The proof claims that a fixed Filt^pi is closed under all colimits large relative to pi0.

ms. 120 FALSE XVIII-A116-014 – The right adjoint to evaluation at c on covariant functors C→E is claimed to have value E^{Hom(c,d)} at d.

ms. 126–129 FALSE XVIII-A116-031 – Proposition4.5.13(1): the six conditions a,a-prime,a-double-prime,b,b-prime,b-double-prime are equivalent.

ms. 126 FALSE XVIII-A116-036 – The definition of Filt-prime^pi via the supposed common condition P(pi) in4.5.13 is unambiguous.

ms. 127 CONJECTURED XVIII-A116-047 – Footnote50 suggests that condition C on M is equivalent to the analogous condition on C for filtered diagrams of at most pi arrows.

ms. 127–128 CONJECTURED XVIII-A116-049 – Outside A,B,C, the one-step colimit classes D_pi may fail to form a cardinal filtration.

ms. 133 FALSE XVIII-A116-042 – The proof infers D_pi=M_pi from generation under pi^+-filtered colimits, solely by inclusion D_pi⊂M_pi and Lemma4.5.14.

ms. 134 FALSE XVIII-A116-058 – Lemma4.5.14,b⇒c: pi^+-filtered generation by a strictly full M0 implies M_pi⊂M0.

ms. 134 FALSE XVIII-A116-059 – Lemma4.5.14 final assertion: under M0⊂M_pi and strict generation by M_pi, conditions a,b,b-prime,c are equivalent to M0=M_pi.

ms. 134 CONJECTURED XVIII-A116-060 – Condition B implies condition C for the paratoposes under discussion.

ms. 134–135 CONJECTURED XVIII-A116-061 – Condition C does not imply condition B.

ms. 135 FALSE XVIII-A116-064 – If G0*G1→G is surjective then (G0→G,G1→G) is a strict epimorphic family in groups and Cat.

ms. 135 FALSE XVIII-A116-065 – Footnote53 retains strict epimorphy for the special axes H→H×H with H nontrivial.

ms. 137 FALSE XVIII-A116-069 – Corollary4.5.16: under C, an object covered by a universally epimorphic family of<=pi objects in D_pi belongs to D_pi.

ms. 137 FALSE XVIII-A116-070 – A universally epimorphic outer family composed with universal strict epimorphic inner families is universally strict epimorphic.

ms. 138 FALSE XVIII-A116-074 – The one-step class of arbitrary colimits of<=pi-arrow diagrams in C need not equal M_pi.

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ms. 127 CONJECTURED XVIIIb-A010 – The pullback-commutation hypothesis on M should be equivalent to its analogue on C for filtered colimits of cardinal at most pi.

ms. 127–128 CONJECTURED XVIIIb-A011 – Outside the three sufficient hypotheses the Filt-prime filtration may fail to be a cardinal filtration.

ms. 133–134 FALSE XVIIIb-A025 – Lemma4.5.14(b-prime) implies M_pi is contained in M0.

ms. 133–134 FALSE XVIIIb-A026 – If M0 is contained in M_pi and M_pi is dense, all four conditions of4.5.14 are equivalent to M0=M_pi.

ms. 134–135 conjectured XVIIIb-A028 – Universal strict-epimorphism hypotheses B should imply filtered pullback exactness C, whereas C need not imply B.

ms. 135 false XVIIIb-A031 – A surjection G0*G1→G makes the two corresponding group maps a strict epimorphic family in Gr and Cat.

ms. 135 false XVIIIb-A032 – The two coordinate inclusions H→H×H form a strict epimorphic family, as claimed by the special-case exemption in footnote53.

ms. 142 false XVIIIb-A051 – If card-red(C0)<=pi1, then that closure has skeletal morphism cardinal <=pi1^(pi0).

ms. 143 false XVIIIb-A053 – That iterated closure has skeletal morphism cardinal <=pi1^(pi0).

ms. 145–146 false XVIIIb-A058 – With pi1=card-red(S), one has card-red(M_pi)<=pi1^pi for pi>=pi0.

ms. 145 false XVIIIb-A059 – The equality pi1^pi=2^pi holds whenever pi>=pi0, as printed in the parenthesis of4.6.2.1.

ms. 147 false XVIIIb-A063 – Full faithfulness of Ind_pi0(C)→M alone implies that every object of C is pi0-accessible in M.

ms. 149–150 open XVIIIb-A076 – Do e_pi0, or merely alpha(pi0) and beta(pi0), plus cocompleteness and a small strict generator imply a pi0-paratopos at the same cardinal?

ms. 150 false XVIIIb-A079 – Being a paratopos is equivalent to being Ind-prime_pi0(S) for some arbitrary small S, as written without a completeness condition.

ms. 154 conjectured XVIIIb-A088 – Conjecture4.6.6(a): condition L plus a small strictly generating subcategory of accessible objects should imply a suitable Ind presentation/cardinal filtration.

ms. 155–156 conjectured XVIIIb-A093 – If these closures form a cardinal filtration, they should equal M_pi; their existence does not by itself make M_pi/X filtered.

ms. 159–160 open XVIIIb-A106 – Are there useful examples of(a)–(f) beyond topoi and abelian categories and their elementary combinations?

ms. 160 open XVIIIb-A108 – Internal groups in groups are abelian groups; could analogous constructions M boxtimes M provide further paratopos examples?

ms. 162–164 false XVIIIb-B014 – (a″) implies (c′), without a size hypothesis on C.

ms. 167 false XVIIIb-B029 – If $\pi = \pi_1 = \pi_0$, then $\mathrm{Filt}^{\pi} = \mathrm{Filt}'^{\pi}$ is the subcategory of representables and agrees with $C^{\natural}$ only when C is Karoubian.

ms. 175 open XVIIIb-B048 – Is Filt^{π} closed under retracts at $\pi^{\pi_0} = \pi$, equivalently can the missing (c)=> (b) be proved?

ms. 177 false XVIIIb-B055 – The initial level need not be retract-closed; specifically $\mathrm{Filt}'^{\pi_0} = \mathrm{Filt}^{\pi_0} \simeq e$ for non-Karoubian C because every small highly filtered category is cofinal to e.

ms. 178–179 false XVIIIb-B059 – If every slice $\mathcal{F}^{\pi}/F$ is π^{+}-filtered, the realization $\mathrm{Ind}_{\pi}(\mathcal{F}^{\pi}) \to \mathcal{M}$ is an equivalence.

ms. 181–182 conjectured XVIIIb-B070 – Canonical generalized filtrations of Ind categories need not have every larger cardinal useful; some may be superfluous.

ms. 184 open XVIIIb-B079 – Can both exponents π^{π_0} in Lemma 4.7.19 be replaced by π_0?

ms. 185 open XVIIIb-B082 – Are the sharper bounds 2^{π} and $(\pi')^{\pi}$ valid for the general intrinsic filtration of Lemma 4.7.19?

ms. 180 false XVIIIb-B098 – The initial level $\mathcal{F}^{\pi_0}$ of any accessible generalized filtration is a generating subcategory of M.

ms. 189 OPEN XVIIIb-C008 – The condition that some family of arrows from C to every X is strict epimorphic may differ from the all-arrows density criterion.

ms. 190-191 FALSE XVIIIb-C010 – The feared failure of density under passage from C to a containing full subcategory C-prime would invalidate earlier arguments.

ms. 199-200 OPEN XVIIIb-C018 – Essential surjectivity of Ind_{pi0}(S)->M does not follow merely by the asserted construction; saturating S under existing bounded colimits is proposed but not shown sufficient.

ms. 202 OPEN XVIIIb-C024 – 4.8.6(1): find a category satisfying4.8.1(a) at pi0 but not the Ind-presentation condition at pi0.

ms. 202 OPEN XVIIIb-C026 – 4.8.6(2): find serviceable additional assumptions making the dense-presentable-generator condition imply an Ind-presentation at the same rank.

ms. 203 OPEN XVIIIb-C027 – 4.8.6(3): is Filt^pi in Ind_{pi0}(C) closed under retracts for pi>pi0, at least when pi^pi0=pi?

ms. 205 OPEN XVIIIb-C032 – 4.8.6(4): find nonuseful cardinals, preferably arbitrarily large ones, for one accessible category.

ms. 205 OPEN XVIIIb-C033 – 4.8.6(4): find useful cardinals that fail quantitative regularity, preferably arbitrarily large ones.

ms. 209 FALSE XVIIIb-C046 – The displayed formula X=colim_J x_j=product_{Ob J}x_j computes the projective limit.

ms. 210-211 FALSE XVIIIb-C049 – 4.8.11 first sentence: on an accessible finitely complete M, every accessible finite-limit-preserving functor M->Set is representable.

ms. 213 CONJECTURED XVIIIb-C056 – Footnote91 conjecture: if M and M-op are both accessible, then M is essentially small.

ms. 220 CONJECTURED XVIIIb-C071 – 4.9.4(b): without the closure hypotheses the comparison can fail essential surjectivity even for J=B(Z/2).

ms. 224 FALSE XVIIIb-C082 – 4.9.6.3(b-prime) as printed: connectedness and equalization of parallel pairs in I-prime alone imply condition(a).

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ms. 220 CONJECTURED XVIIIc-A006 – Without an additional closure hypothesis on C, the comparison Ind_pi([J,C])->[J,Ind_pi C] need not be essentially surjective even for J=B(Z/2).

ms. 224 FALSE XVIIIc-A024 – The printed condition b-prime (I-prime connected and satisfying PF2, with no comma condition) is equivalent to a and b.

ms. 244 FALSE XVIIIc-B005 – Every section with values in D_j is κ-presentable in Γ(N/J).

ms. 248 FALSE XVIIIc-B014 – Existence and preservation of colimits large before π_0 of size$\leq\pi_0'$, with $\pi_0\leq\pi<\kappa$ and $\pi_0'>\pi_0$, imply the stated ordinal hypothesis $|\delta|\leq\pi$.

ms. 262 FALSE XVIIIc-B030 – The local consecutive-stage condition (ii) implies the t_β(a) define a morphism of ind-objects for arbitrary limit δ.

ms. 266-267 CONJECTURED XVIIIc-B041 – Question 4.10.11(a): for a small-base fibration with accessible fibres and accessible inverse images, Γ(N/J) is accessible.

ms. 267 CONJECTURED XVIIIc-B042 – Question 4.10.11(b): the category of cartesian sections under the same hypotheses is accessible.

ms. 267 OPEN XVIIIc-B043 – Find a common formulation covering fibred and opfibred sections, and possibly more general functors to the base.

ms. 272 OPEN XVIIIc-B060 – Does 4.11.6 remain true without rigidity?

ms. 280 conjectured XVIIIc-C008 – For M=N=Set, accessible functors have no common preservation cardinal.

ms. 285 false XVIIIc-C015 – 4.13.2(c): one may take pi0' minimal useful for S and for preservation by its inclusion, and pi1'=max(pi0',card_red S_{pi0'}), without bounding ambient ranks.

ms. 288 false XVIIIc-C019 – 4.13.4: the canonical functor Ind_pi(A)→M used in the classification is non-fully-faithful.

ms. 291 false XVIIIc-C028 – Footnote123: the fixed-point formulation4.13.8 additionally requires pi to be a non-limit cardinal.

ms. 292 false XVIIIc-C029 – The fixed-point equation pi^{pi0'}=pi entails pi≥2^{pi0}>pi0'.

ms. 293 false XVIIIc-C033 – The hypotheses of4.13.9 imply pi^{pi0'}=pi and hence global usefulness of pi for M.

ms. 298 false XVIIIc-C039 – The initial ordinal of cardinal pi is rho^+-filtered for every rho<pi.

ms. 300 false XVIIIc-C043 – The construction may satisfy both i0=i for arbitrary i and i_alpha∈J_{phi(alpha)} for every alpha without adjustment.

ms. 308-309 false XVIIIc-C062 – 4.13.15: under pi>pi0'≥pi0 and pi^{pi0}=pi, conditions a,a',b,c,c' are equivalent.

ms. 308 false XVIIIc-C068 – Footnote135: merely requiring usefulness of pi keeps a,a',b,c equivalent, without necessarily c'.

ms. 314 false XVIIIc-C076 – The initial ordinal pi is sufficiently filtered for taking the final alternating-chain colimits under pi>pi0' alone.

ms. 322-323 false XVIIIc-C079 – After the two successor factorization identities are imposed, naturality of u with the immediately preceding stage may still fail.

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ms. 336 false XVIII-d-A021 – The initial ordinal of cardinality π is large before every cardinal $\pi' < \pi$.

ms. 345 false XVIII-d-A041 – In Proposition 4.13.24, the condition that Φ contain a nonzero limit ordinal can simply be omitted with no replacement.

ms. 347–348 false XVIII-d-B006 – GCH makes $(\pi')^{\pi_0} = \pi'$ automatic for every infinite $\pi' > \pi_0$.

ms. 353–354 conjectured XVIII-d-B020 – Question 4.13.33 conjectures that the replete full essential image of every accessible functor between accessible categories is accessible and accessibly embedded; equivalently the functor factors through such a full image.

ms. 354 conjectured XVIII-d-B022 – If $\mathcal{M}$ is accessible with π_0^+-filtered colimits and A is a small full subcategory, the smallest replete full subcategory containing A and closed under those colimits is conjectured accessible and accessibly embedded.

ms. 355 conjectured XVIII-d-B023 – The same small-envelope conjecture is made after additionally closing under a small family Φ of small colimit shapes which exist in $\mathcal{M}$.

ms. 370 conjectured XVIII-d-B052 – Footnote 163 conjectures that the target completeness assumptions in Theorem 4.14.1 can be dropped; it points forward to 4.14.16 for removal of the limit-shape assumption.

ms. 373 FALSE XVIII-d-B057 – The ordinal of cardinal pi is pi0-plus-filtered merely because pi>pi0.

ms. 379 FALSE XVIII-d-C009 – Corollary 4.14.9 as printed: every small-limit-preserving functor between locally presentable categories has a left adjoint, and taking left adjoints gives an equivalence to the opposite category of cocontinuous functors.

ms. 379 FALSE XVIII-d-C011 – Lemma 4.14.10, left-adjoint half: a functor from a pseudo-topos (a cocomplete locally small category with a small family generating by strict epimorphisms) to a locally small category has a left adjoint if and only if it preserves all small limits.

ms. 380 FALSE XVIII-d-C013 – Corollary 4.14.11, covariant half: every small-limit-preserving $F : \mathcal{M} \to \mathbf{Set}$ on a locally presentable category is representable.

ms. 387 OPEN XVIII-d-C031 – Question 4.14.15: for every accessible $\mathcal{N}$, is the full category of epimorphisms accessible in $\mathcal{N}^{[1]}$?

ms. 388–390 CONJECTURED XVIII-d-C037 – Conjecture 4.14.16: one may remove the target existence assumptions in 4.14.1 and still obtain accessibility of the subcategory of diagrams that satisfy the prescribed universal cone conditions.

ms. 391 FALSE XVIII-d-C044 – Corollary 4.14.18(b), opfibration case: if pi≥max(pi_0,cardred C), a section whose values are pi^+-presentable is itself pi^+-presentable.

ms. 392 FALSE XVIII-d-C046 – Corollary 4.14.18(c): after choosing pi_0 useful for the fibres and the section category, pi_1=max(pi_0,cardred C,cardred M_pi_0), and pi≥pi_1 with pi^{pi_0}=pi, the pi-presentable sections are exactly the pointwise pi-presentable sections.

ms. 395–396 FALSE XVIII-d-C057 – Corollary 4.14.20(c): under the printed threshold and pi^{pi_0}=pi, intrinsic presentables of the cartesian/cocartesian section category are exactly the ambient presentables in it, and exactly its pointwise presentable sections.

ms. 411 FALSE XVIII-d-D017 – As printed, the inclusion E_1→E in the opfibration has a right adjoint, and the inclusion E_0→E in the fibration has a left adjoint.

ms. 420 FALSE XVIII-d-D031 – Corollary 4.14.25(1), literally: E and E_0 have π-large filtered colimits and E_0→E preserves them iff E_0 has them and all H(−,x_1) preserve them contravariantly; it then asserts existence of the same colimits in E_1.

ms. 426–429 FALSE XVIII-d-D049 – Under these bounds, if $\pi^{\pi_0}=\pi\geq\pi_1$, each π-presentable object of a fibre is π-presentable in E.

ms. 426 FALSE XVIII-d-D050 – As printed in 4.15.2(c), the subcategory E_π itself satisfies L_{π_1}, hence L_π.

ms. 426 FALSE XVIII-d-D051 – As printed in 4.15.2(c), Ind_π(E_{s,π})→E is an equivalence for a single fixed fibre s.

ms. 426 FALSE XVIII-d-D052 – Corollary 4.15.2(d): under the same hypotheses E(π)=E_π, not just E(π)⊂E_π.

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ms. 420–421 FALSE XVIIIe-A005 – 4.14.25(1), literally: $\mathcal{E}$ and $\mathcal{E}_0$ admit L_π-colimits and $\mathcal{E}_0 \to \mathcal{E}$ preserves them iff $\mathcal{E}_0$ admits them and each $H(-, x_1)$ takes them to limits.

ms. 426,429–431 FALSE XVIIIe-A023 – 4.15.2(c), rank conclusion: under its printed hypotheses every fibrewise pi-presentable object is pi-presentable in the total category.

ms. 426 FALSE XVIIIe-A024 – The displayed functor Ind_pi(E_{s,pi})->E is an equivalence.

ms. 426,431–432 FALSE XVIIIe-A025 – 4.15.2(d): E(pi)=E_pi at the stated numerical rank.

ms. 430 FALSE XVIIIe-A032 – The inequalities pi>=pi1>=pi_lambda imply that H_lambda(-,y) is pi0-accessible.

ms. 434 OPEN XVIIIe-A042 – Find usable necessary and sufficient local conditions on f ensuring accessibility of every pullback over a small Karoubian category.

ms. 434 OPEN XVIIIe-A043 – It might suffice to test only Delta1 and the walking retract P; the author expresses doubt.

ms. 434 CONJECTURED XVIIIe-A044 – Testing Delta1 and the two three-object categories Q=(2->1->0 with the retract0<->1) and Q-prime=(2<-1<->0) should suffice.

ms. 438 OPEN XVIIIe-A054 – The analogous construction for a fibration is not supplied; the author suggests testing the walking retract P.

ms. 456–457 CONJECTURED XVIIIe-B029 – Key lemma4.15.8.1: for a left-cancellative monoid M with $|M| \le \pi_0$, every ${\pi_0}^+$-directed diagram in BM has a tail whose transition elements are units.

ms. 461–462 FALSE XVIIIe-B040 – Condition(b) asserts literal identity of every sufficiently late transition in every diagram valued in a left-cancellative quotient of End_C(x).

ms. 462 CONJECTURED XVIIIe-B042 – Condition(b′) excludes π^+-long diagrams in the relevant left-cancellative monoids with all strict transitions noninvertible.

ms. 461–462 FALSE XVIIIe-B043 – Theorem4.15.9 declares all of(a),(a′),(a″),(b),(b′) equivalent as printed.

ms. 461 CONJECTURED XVIIIe-B044 – Grothendieck conjectures that Ind_π C is the idempotent completion whenever $|C| \le \pi$.

ms. 462–463 FALSE XVIIIe-B045 – Corollary4.15.10 says all printed equivalent conditions hold when $\pi > |C|$.

ms. 465-466 OPEN XVIIIe-R005 – For every universe U there might be an infinite cardinal pi in U such that every ultrafilter on a U-small set selecting a cell of every partition into at most pi pieces is principal.

ms. 466 FALSE XVIIIe-R006 – Every suitably definable replete full subcategory of an accessible category closed under sufficiently filtered colimits is accessible and accessibly embedded.

ms. 467-468 FALSE XVIIIe-R008 – For fixed z, failure of Hom(x,z) to be nonempty should be witnessed by a uniformly bounded member of a suitable presentable-object presentation of x.

ms. 467-468 OPEN XVIIIe-R009 – For every infinite field k, one may take pi=|k| in the small-subalgebra obstruction criterion for absence of k-points.

ms. 468-469 OPEN XVIIIe-R010 – If f:M->N^op preserves sufficiently filtered colimits and T is accessible in N and stable under the corresponding cofiltered limits in N, is f^{-1}(T) accessible in M?

ms. 470 CONJECTURED XVIIIe-R014 – The full L_pi-colimit closure S(pi) of the image of an accessible functor is accessible in M.

ms. 470 FALSE XVIIIe-R015 – The decreasing family S(pi) stabilizes for all sufficiently large pi.

ms. 470 FALSE XVIIIe-R017 – The same accessibility and eventual-stationarity claims should hold for the image of g^op:N^op->M when g:N->M^op preserves sufficiently filtered colimits.

ms. 471 FALSE XVIIIe-R019 – A full subcategory of a presheaf category closed under sufficiently filtered colimits automatically inherits a small dense subcategory of accessible objects.

ms. 471 OPEN XVIIIe-R020 – A category with a small dense family of presentable objects and sufficiently filtered colimits is accessible.

ms. 472 FALSE XVIIIe-R021 – If M and M^op are accessible, then M is essentially small.

ms. 472 OPEN XVIIIe-R022 – An accessible category with a small codense subcategory is essentially small.

ms. 472 OPEN XVIIIe-R023 – A locally small category with small dense and codense subcategories is essentially small.

ms. 473 FALSE XVIIIe-R025 – The subcategory of existing I-colimit cocones in an accessible M is accessible and accessibly embedded even if some I-diagrams have no colimit.

ms. 475 FALSE XVIIIe-R032 – Corollary4.16.6.2(b'): if C has no nonidentity idempotents, the Delta1-tests alone imply (a).

ms. 476 OPEN XVIIIe-R035 – Question4.16.6.4: do accessible fibres and transition functors of a fibration or opfibration imply accessibility after Q and Q' base change?

ms. 476-476bis OPEN XVIIIe-R037 – If E is accessible over small Karoubian C, then for every accessible M and accessible M->C, the strict pullback E x_C M and both projections are accessible.

ms. 484-485 OPEN XVIIIe-R047 – It is asked whether there is a U-ordinal-length family of fields with no morphisms from an earlier field to a later one.

ms. 4 FALSE XVIIIe-C011 – The printed identity in the inverse proof is ff′=f′=f=1.

ms. 4 FALSE XVIIIe-C013 – The last sentence asserts that k_Φ is a field quotient for every $\Phi\in\mathcal{F}$.

ms. 11 CONJECTURED XVIIIe-C030 – If π is the least cardinal in U belonging to no strictly smaller infinite universe, every π-ultrafilter on a U-small set is principal.

ms. 14 FALSE XVIIIe-C038 – The source defines both complementary classes S and S′ by Hom(A,k)=∅.

ms. 15 FALSE XVIIIe-C044 – Under the principal-ultrafilter hypothesis the source identifies k-points with coordinates satisfying {xi}=intersection(Φ).

ms. 19 OPEN XVIIIe-C053 – Does the preceding small-subalgebra witness property hold simply at $\pi\geq|\mathrm{k}|$? Does any such U-small bound exist?

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ms. 1 FALSE XIX-A007 – The strengthened equalities of (1.3) are claimed to be equivalent to adding separate double-orthogonal closedness of the corresponding classes to $\Phi \perp \Psi$.

ms. 4–5 OPEN XIX-A026 – Is Retr(Φ'') stable under pushout and transfinite composition, and therefore equal to Φ''', without small generation?

ms. 10 FALSE XIX-A052 – If $\Psi = \Phi^{\square}$, then $\Psi(I_0) = \Phi(I_0)^{\square}$.

ms. 12 CONJECTURED XIX-A060 – Question-conjecture 1.11: if (Φ, Ψ) is special, is the injective pair $(\Phi(I), \Psi[I])$ special for every small I, perhaps after an additional cardinal-filtration hypothesis?

ms. 12 OPEN XIX-A061 – In 1.11, if Φ_0 is a small generating family for Φ, does the particular class $\Phi_0(I)$ generate $\Phi(I)$?

ms. 12 FALSE XIX-A062 – Proposition 1.12, formula (1.13.2), prints $u^*(\Phi(I')) \subseteq \Phi\langle I'\rangle \iff u_*\Psi(I) \subseteq \Psi(I')$.

ms. 14 FALSE XIX-A074 – The printed dual criterion in (2.5) uses a factorization $f = pi$ with $i \in W$ and $f \in C$.

ms. 17 FALSE XIX-A087 – The Case A proof says that $TC' \subseteq W'$ is equivalent to $R(TC') \subseteq W'$.

ms. 10 FALSE XIX-A090 – The equalities (1.15)–(1.16) identify $(v_!\Phi(I_0))^{\square}$ with $\Psi(I)$ and its left orthogonal with $\Phi\langle I\rangle$ under the sole displayed assumption $\Psi = \Phi^{\square}$.

ms. 19 FALSE XIX-B003 – The displayed equality in Theorem 2.3 identifies $C\langle I\rangle$ with ${}^{\square}F(I)$.

ms. 23 FALSE XIX-B021 – Decomposition holds for every prederivator of the form $\mathbf{D}_{\mathcal{M},W}$, without hypotheses on the relative category.

ms. 24 OPEN XIX-B024 – For a merely special triple, the source has not established small generating sets or factorizations for the componentwise classes $C(I)$ and $TC(I)$.

ms. 25 CONJECTURED XIX-B035 – Condition (d) should be redundant, at least when epimorphisms are effective, after choosing suitable cardinals of the form 2^c.

ms. 30 FALSE XIX-B062 – $v^*(TF[I]) = v^*v_*(TF(I_0))$ as image classes of arrows.

ms. 32 OPEN XIX-B070 – Can special model structures outside the very special class define derivators on all small shapes?

ms. 32 OPEN XIX-B072 – The author expresses uncertainty about the all-shapes homotopy-Kan-extension theorem attributed to Anderson.

ms. 35 FALSE XIX-C023 – Every cofibration between cofibrant objects belongs to ambient Cof_W by Brown's lemma.

ms. 38 FALSE XIX-C038 – Proposition3.5: if Φ=^perpΨ is Q-closed, then for noetherian I the Anderson class Φ_I equals ^perp(Ψ(I)).

ms. 39 CONJECTURED XIX-C043 – For only transfinitely Q-saturated Φ, conjecturally the transfinite Q-saturation of v!Φ(I_0) agrees with Anderson's noetherian extension.

ms. 39 OPEN XIX-C045 – The desired identity is C<I>∩W(I)=(TC)<I>.

ms. 40 OPEN XIX-C048 – Question3.6 asks the identity for every small I, under small completeness of M and Q-closedness of C and TC.

ms. 40 OPEN XIX-C053 – It may suffice for Question3.6 that only (TC)<I>–F(I) factor.

ms. 42 OPEN XIX-C062 – Question3.9 requests a closed Quillen structure whose prederivator lacks some u!^W.

ms. 42 OPEN XIX-C063 – Question3.9 also requests such a structure lacking some u*^W.

ms. 42 CONJECTURED XIX-C064 – The author presumes testing u:I→e should suffice for general Kan extensions.

ms. 43 OPEN XIX-C065 – The requested counterexample would be stronger with cocomplete accessible ambient M and L-accessible W.

ms. 46–47 OPEN XIX-C092 – A counterexample might have bicomplete M but H lack some small colimits.

ms. 47 OPEN XIX-C096 – Question3.13: can a locally U-small category have all U-small colimits but lack some U-small limits?

ms. 48 OPEN XIX-C099 – Question3.14: a cofibration category satisfying Cof_4 should have u!^W for all small u.

ms. 48 OPEN XIX-C100 – The author requests a localizer on bicomplete M whose prederivator lacks some left or right Kan extension, perhaps with M small.

ms. 48–49 OPEN XIX-C101 – The author is less convinced such a counterexample exists when W is filtered-colimit stable and M has accessible objects and a cardinal filtration.

ms. 49 OPEN XIX-C102 – Question3.15 asks whether an L-accessible arrow class in the stated accessible complete ambient category has a small left Q-generating subset.

ms. 49 OPEN XIX-C103 – Question3.15 allows the extra assumption that Φ is left Q-saturated: isomorphisms, cobase change, composition, and retracts.

ms. 49–50 open XIX-C105 – The author requests any mutually Q-orthogonal closed pair that does not factor all arrows.

ms. 49–50 open XIX-C106 – The same nonfactorization request with special ambient M is of particular interest.

Transcription corrections

The corrections to the transcription, counted by kind for each part: readings of the transcription corrected against the edition (transcription), slips of the edition kept in the text with their intended reading recorded (edition slip kept), and doubtful readings settled against the edition (doubt resolved).

Part	transcription	edition slip kept	doubt resolved	total
I-a	8	29	5	42
I-b	3	69	3	75
II-a	11	65	1	77
II-b	3	73	4	80
III-V	14	114	6	134
VI-a	11	43	5	59
VI-b	2	67	3	72
VI-c	1	43	3	47
VII-a	19	81	2	102
VII-b	6	53	0	59
VII-c	3	45	7	55
VIII-X	0	29	3	32
XI-XIIa	1	72	6	79
XIIb-a	0	48	2	50
XIIb-b	1	27	2	30
XIIc-a	2	41	4	47
XIIc-b	3	38	0	41
XIIapp	2	21	3	26
XIII-a	0	33	2	35
XIII-b	5	28	1	34
XIV	0	2	1	3
XV-a	2	37	5	44
XV-b	4	42	2	48
XVI	10	80	6	96
XVII-a	1	71	4	76
XVII-b	5	52	2	59
XVIII-a	0	65	8	73
XVIII-b	3	104	8	115
XVIII-c	7	121	7	135
XVIII-d	0	121	2	123
XVIII-e	1	129	5	135
XIX	6	36	10	52
total	134	1879	122	2135

Results proved along the way that we did not find in the literature

Some statements had to be proved along the way, while checking Grothendieck's text and the returns of the research prompts, because we could not find them in the sources we searched. They are collected here so that a reader can check them, find the literature we missed, and see where they bear on the manuscript. Our search was limited, so some of them may well be known; whoever knows a reference is asked to tell us, and the entry will cite it. No priority is claimed. Each entry gives the statement, where the proof is, who proved it (model and effort setting), whether anyone has checked it yet, and the pages of the manuscript it bears on. Statements that turned out to be known are listed at the end with the reference.
Editorial scope. The inventory was started on 12 November 2026 by Claude Opus 5.5 (effort max). This appendix was assembled by GPT-6 Astra (OpenAI, ChatGPT Pro mode). Each numbered section preserves the inventory statement and the available cited proof, with mathematical glyphs and scripts typeset for reading. Proofs and source comments remain attributed to their source authors. The paragraph headed "GPT-6 Astra's reading" is an editorial reading of the supplied argument; it is not the separate peer review requested for later work. Missing cited evidence is stated explicitly. Original source identifiers are preserved. The source extracts and the exact glyph mapping are included among the build records. No material from STRATA, PATHS, BISITE or RESIDUE is added beyond the inventory's own references.

Notation from the cited sources. A universe U is fixed; categories are small in that universe. $BX = |NX|$ denotes the classifying space, W_∞ the functors inducing weak homotopy equivalences, and **Hot** the homotopy category. In $\mathrm{Cat}_{\mathrm{hot}}$, morphisms are $\pi_0 \operatorname{Fun}(X, K)$. The functor Π_1 inverts every arrow, with unit $\eta_X : X \to \Pi_1 X$. For a monoid M, $U(M)$ is its universal group. Brackets $[A, Y]$ denote free homotopy classes. Original labels such as Theorem 2.1 and G.4 refer to the cited source, not to the numbering of this appendix.

1. Finite monoids, finite dimension, and bounded homology

Statement recorded in the inventory.

Finite monoids: finite type, finite dimension and bounded homology coincide. For a finite monoid M: BM has the homotopy type of a finite complex ⇔ of a finite-dimensional complex ⇔ $\mathrm{H_k}$(M; Z) = 0 for all large k; then the universal group U(M) is trivial. Corollaries: a "Swan theorem" for finite monoids (U(M) ≠ 1 forces homology in infinitely many degrees; p-locally too); finite cohomological dimension forces U(M) = 1; with Theorem C of Ben-Bassat (arXiv:2610.38760v1) the homotopy types of finite monoids with bounded homology are exactly the finite simply connected complexes.
Bearing on the manuscript. V ms. 51–52 (Hot_0, "types d'homotopie de type fini"); sharpens G.4 of the CATHOT return, the counterexample to Ben-Bassat's Conjecture 7.1
Proved by. Claude Fable 5.2, effort max, 12 Nov 2026

Checked by (inventory record). not yet (Pro check prompt drafted)

Cited source. `pro_reviews_max_20261112/1_CATHOT/OWN_RESULTS.md`, section 2

Statement and proof in the supplied source

Let M be a finite monoid. Facts used, all proved in `G6_VERDICT.md`: $\pi_1(\mathrm{BM}) = \mathrm{U}(\mathrm{M})$; η: $\mathrm{M} \to \mathrm{U}(\mathrm{M})$ is onto; for an idempotent e of the minimal ideal, $\mathrm{H} = \mathrm{eMe}$ is a finite group and there is a map $\mathrm{j_e}$: $\mathrm{BH} \to \mathrm{BM}$ (through the Karoubi envelope) inducing the surjection $\eta|\mathrm{H}$ on π_1. Also $\mathrm{H_k}(\mathrm{BM}; \mathrm{Z}) = \mathrm{Tor_k}^{\mathrm{ZM}}(\mathrm{Z}, \mathrm{Z})$ is finitely generated for every k (the bar complex is finitely generated in each degree).

Theorem 2.1. If $\mathrm{H_k}(\mathrm{M}; \mathrm{Z}) = 0$ for all large k, then $\mathrm{U}(\mathrm{M}) = 1$. More precisely, if a prime p divides $|\mathrm{U}(\mathrm{M})|$, then $\mathrm{H_k}(\mathrm{M}; \mathrm{F_p}) \neq 0$ for infinitely many k.

Proof. If $\mathrm{H_k}(\mathrm{M}; \mathrm{Z}) = 0$ for $\mathrm{k} \geq \mathrm{n}$, then $\mathrm{H^q}(\mathrm{BM}; \mathrm{Z}) = \mathrm{Hom}(\mathrm{H_q}, \mathrm{Z}) \oplus \mathrm{Ext}(\mathrm{H_{q-1}}, \mathrm{Z}) = 0$ for $\mathrm{q} > \mathrm{n}$. $\pi_1(\mathrm{BM}) = \mathrm{U}(\mathrm{M})$ is finite. Theorem 1.1 applied to $\mathrm{j_e}$: $\mathrm{BH} \to \mathrm{BM}$ says $\eta|\mathrm{H}$ is trivial; it is onto; so $\mathrm{U}(\mathrm{M}) = 1$. For the second statement: if $\mathrm{H_k}(\mathrm{M}; \mathrm{F_p}) = 0$ for all $\mathrm{k} \geq \mathrm{k_0}$, then $\mathrm{H_k}(\mathrm{M}; \mathrm{Z}) \otimes \mathrm{F_p} = 0$ and $\mathrm{Tor}(\mathrm{H_{k-1}}(\mathrm{M}; \mathrm{Z}), \mathrm{F_p}) = 0$ for $\mathrm{k} \geq \mathrm{k_0}$, so the finitely generated groups $\mathrm{H_k}(\mathrm{M}; \mathrm{Z})$ have zero localization at p for $\mathrm{k} \geq \mathrm{k_0}$, and $\mathrm{H^q}(\mathrm{BM}; \mathrm{Z_{(p)}}) = 0$ for $\mathrm{q} > \mathrm{k_0}$. Theorem 1.2 applied to $\mathrm{j_e}$ gives $\mathrm{p} \nmid |\eta(\mathrm{H})| = |\mathrm{U}(\mathrm{M})|$. ■

Theorem 2.2. For a finite monoid M the following are equivalent: (i) BM has the homotopy type of a finite CW complex; (ii) BM has the homotopy type of a finite-dimensional CW complex; (iii) $\mathrm{H_k}(\mathrm{M}; \mathrm{Z}) = 0$ for all large k. If they hold, $\mathrm{U}(\mathrm{M}) = 1$ and BM is simply connected.

Proof. (i) ⇒ (ii) ⇒ (iii) are clear. (iii) ⇒ (i): by 2.1, BM is simply connected; its homology groups are finitely generated and vanish above some degree n; by Hatcher, *Algebraic Topology*, Proposition 4C.1, a simply connected CW complex with finitely generated homology is homotopy equivalent to a CW complex with one k-cell for each Z summand of $\mathrm{H_k}$ and one k-cell and one (k+1)-cell for each finite cyclic summand of $\mathrm{H_k}$: a finite complex of dimension $\leq \mathrm{n} + 1$. ■

Corollary 2.3 (Swan's theorem for finite monoids). A finite monoid with $\mathrm{U}(\mathrm{M}) \neq 1$ has $\mathrm{H_k}(\mathrm{M}; \mathrm{Z}) \neq 0$ for infinitely many k; if p divides $|\mathrm{U}(\mathrm{M})|$, then $\mathrm{H_k}(\mathrm{M}; \mathrm{F_p}) \neq 0$ for infinitely many k. (For a finite group this is Swan's theorem.)

Corollary 2.4. If the trivial left (or right) ZM-module Z has a projective resolution of finite length (M has finite left or right cohomological dimension), then $\mathrm{U}(\mathrm{M}) = 1$ and BM has the homotopy type of a finite simply connected CW complex.

Corollary 2.5 (with Theorem C of arXiv:2610.38760v1; recent, unrefereed). The homotopy types of the classifying spaces of finite monoids with bounded integral homology are exactly the homotopy types of finite simply connected CW complexes.

GPT-6 Astra's reading

The argument explicitly connects the surjective group map $eMe \to U(M)$ to a map of classifying spaces through the Karoubi envelope; section 12 below reproduces that construction. The bounded-homology implication then uses the cohomological result reproduced in section 2, and the finite-generation input is supplied by the degreewise finite bar complex. The finite-complex conclusion invokes Hatcher's Proposition 4C.1; the positive realization assertion in Corollary 2.5 invokes Theorem C of the stated, unrefereed preprint. The proof records these dependencies rather than proving them here. I found no additional missing inference in the displayed chain. In the universal-coefficient line the direct-sum decomposition is noncanonical, which does not affect the vanishing implication used. The computational observation following the source corollaries is not needed for the proof and is not reproduced here.

2. Maps from finite-group classifying spaces under a cohomological bound

Statement recorded in the inventory.

Miller's Theorem B under a cohomological bound. A map $BG \to X$, G finite, X connected with finite (or residually finite, or linear) π_1 and $H^q(X; Z) = 0$ for $q > n$, is trivial on π_1; p-local form with $Z_{(p)}$ coefficients.
Bearing on the manuscript. the tool for result 1

Proved by. Claude Fable 5.2, effort max
Checked by (inventory record). not yet

Cited source. `pro_reviews_max_20261112/1_CATHOT/OWN_RESULTS.md`, section 1

Statement and proof in the supplied source

Theorem 1.1. Let X be a connected CW complex with finite fundamental group π, and suppose $H^q(X; Z) = 0$ for all $q > n$. Let G be a finite group and $f: BG \to X$ a map. Then $f_*: G \to \pi$ is trivial.

Theorem 1.2 (p-local form). Let X be a connected CW complex with finite fundamental group π, p a prime, and suppose $H^q(X; Z_{(p)}) = 0$ for all $q > n$. Then for every finite group G and every map $f: BG \to X$, the order of the image of $f_*: G \to \pi$ is prime to p.

Miller's Theorem B (Ann. of Math. 120 (1984), p. 40) has "X finite-dimensional" as hypothesis and no condition on $\pi_1(X)$. Here the dimension hypothesis is weakened to a bound on ordinary cohomology with constant coefficients, at the price of assuming $\pi_1(X)$ finite (residually finite, or linear over C, is enough for 1.1; see the remark at the end). The proof is Miller's and Atiyah's, with the covering-space step replaced by the regular representation.

K-theory used. $K^0(Y) = [Y, Z \times BU]$ for CW complexes Y; it is a ring, contravariant in Y. For a CW complex Y put $F^q(Y) = \ker(K^0(Y) \to K^0(Y^{(q-1)}))$. (K1) $F^a \cdot F^b \subset F^{a+b}$. Proof: if $x \in F^a$, $y \in F^b$, lift them to $\widetilde{x} \in K^0(Y, Y^{(a-1)})$, $\widetilde{y} \in K^0(Y, Y^{(b-1)})$; the external product $\widetilde{x} \boxtimes \widetilde{y}$ lies in $K^0(Y \times Y, Y^{(a-1)} \times Y \cup Y \times Y^{(b-1)})$, and the (a+b−1)-skeleton of $Y \times Y$ is contained in $Y^{(a-1)} \times Y \cup Y \times Y^{(b-1)}$; a cellular approximation $\Delta\prime$ of the diagonal maps $Y^{(a+b-1)}$ into it, so $(xy)|Y^{(a+b-1)} = \Delta\prime^*(x \boxtimes y)|\ldots = 0$. (K2) If $H^q(Y; Z) = 0$ for all $q > n$, then $F^q(Y) = F^{q+1}(Y)$ for all $q > n$; hence $F^{n+1}(Y) = \bigcap_q F^q(Y)$. Proof: let $x \in F^q$, $q > n$. Then $x|Y^{(q)}$ comes from a class $c \in K^0(Y^{(q)}, Y^{(q-1)}) = C^q(Y; \widetilde{K}^0(S^q))$, the cellular cochains with coefficients Z (q even) or 0 (q odd). Since $x|Y^{(q)}$ extends over $Y^{(q+1)}$, the image of c in $K^1(Y^{(q+1)}, Y^{(q)})$ vanishes; that image is the cellular coboundary δc. So c is a cocycle, hence ($H^q = 0$) a coboundary $c = \delta b$ with $b \in K^{-1}(Y^{(q-1)}, Y^{(q-2)})$, and the image of δb in $K^0(Y^{(q)})$ is zero because $K^{-1}(Y^{(q-1)}) \to K^0(Y^{(q)}, Y^{(q-1)}) \to K^0(Y^{(q)})$ is exact. So $x|Y^{(q)} = 0$, that is $x \in F^{q+1}$. (K3) A cellular map $g: Y\prime \to Y$ satisfies $g^*F^q(Y) \subset F^q(Y\prime)$. (K1)–(K3) hold verbatim for the p-localized theory $K_{(p)}$ (represented by the p-localization of the spectrum KU, coefficients $Z_{(p)}$ in even degrees), with $H^q(-; Z_{(p)})$ in (K2); and $K^0{}_{(p)}(Y) = K^0(Y) \otimes Z_{(p)}$ for finite Y.

Atiyah's theorem used (Publ. Math. IHES 9 (1961)). For a finite group G let $\mathcal{K}^0(B_G) = \lim_m K^0(B_G{}^{(m)})$ over the finite skeleta of a model of BG with finite skeleta, and α: $R(G) \to \mathcal{K}^0(B_G)$ the map sending a representation to its associated bundle. (7.2): α induces an isomorphism of the I(G)-adic completion $R(G)\hat{}$ onto $\mathcal{K}^*(B_G)$. (6.10): the kernel of $R(G) \to R(G)\hat{}$ is the set of virtual characters vanishing on all elements of prime-power order; (6.11): it is zero for a p-group. Two models of BG with finite skeleta are related by cellular homotopy equivalences, so "$\alpha(\theta)$ restricts to zero on every finite skeleton" does not depend on the model.

Proof of 1.1. Make f cellular; $BG = |NG|$ has finite skeleta. Let ρ: $\pi \to GL_d(C)$ be the regular representation ($d = |\pi|$), $V = \widetilde{X} \times_\pi C^d$ the flat bundle on X, $v = [V] - d \in K^0(X)$. Since X is connected, $v \in F^1(X)$; by (K1), $v^{n+1} \in F^{n+1}(X) = \bigcap_q F^q(X)$ by (K2). By

(K3), $f^*(v^{n+1})$ restricts to zero on every skeleton of BG. Now f^*V is the flat bundle of the representation $\rho\circ\varphi$ of G, $\varphi = f_*$; so $\alpha((\rho\varphi - d)^{n+1}) = 0$ in $\mathcal{K}^0(B_G)$. By (7.2) and (6.10) the virtual character $(\chi_{\rho\varphi} - d)^{n+1}$ vanishes at every $g \in G$ of prime-power order: $\chi_\rho(\varphi(g)) = d$. The regular character takes the value d only at 1, so $\varphi(g) = 1$ for every g of prime-power order. Every element of a finite group is a product of elements of prime-power order (its primary components). Hence $\varphi = 1$. ■

Proof of 1.2. Suppose p divides $|\varphi(G)|$. Take $u \in \varphi(G)$ of order p and g with $\varphi(g) = u$; if g has order p^a m with $p \nmid m$ ($a \geq 1$) and $mm' \equiv 1 \bmod p^a$, then $c = g^{mm'}$ has p-power order and $\varphi(c) = u$. Put $C = \langle c\rangle$ and restrict f to BC (cellular; take for BC the infinite lens space, whose odd skeleta are lens spaces L^{2j+1}). As above, with $K_{(p)}$: the image of v^{n+1} in $K^0{}_{(p)}(X)$ lies in $\bigcap_q F^q$, so $\alpha((\rho\varphi|C - d)^{n+1})$ goes to zero in $K^0(L^{2j+1}) \otimes Z_{(p)}$ for all j. The group $\widetilde{K}^0(L^{2j+1})$ is a finite p-group (Atiyah–Hirzebruch: $\widetilde{H}^{\mathrm{even}}(L^{2j+1}; Z)$ consists of copies of $Z/|C|$), so $K^0(L^{2j+1}) \to K^0(L^{2j+1}) \otimes Z_{(p)}$ is injective. Hence $\alpha((\rho\varphi|C - d)^{n+1}) = 0$ in $\mathcal{K}^0(B_C)$, and by (7.2), (6.11) (C is a p-group; for cyclic groups Atiyah's (8.1) suffices) $(\rho\varphi|C - d)^{n+1} = 0$ in R(C). Its value at c is $(\chi_\rho(u) - d)^{n+1} = (-d)^{n+1} \neq 0$. Contradiction. ■

Remark. In 1.1 the regular representation can be replaced by any finite-dimensional complex representation ρ of $\pi_1(X)$, finite or not: the same proof gives $\rho(f_*(g)) = 1$ for all $g \in G$ (a matrix of finite order with trace equal to its size is the identity). So for arbitrary $\pi_1(X)$ the image $f_*(G)$ lies in the intersection of the kernels of all finite-dimensional complex representations, and f_* is trivial if $\pi_1(X)$ is residually finite or linear.

GPT-6 Astra's reading

The proof states the skeletal-filtration facts it needs and derives the character obstruction from the regular representation. The essential point in (K2) is the exactness of the skeletal relative-K-theory sequence: the cellular coboundary representing the relative class maps to zero in absolute K-theory. This establishes the filtration inclusion actually used without asserting convergence of an unrestricted Atiyah–Hirzebruch spectral sequence. The p-local proof reduces to finite lens-space skeleta, where the reduced K-groups are finite p-groups and localization is injective. The residual-finiteness and linearity sentence uses the stated finite-dimensional representation argument. I found no gap in these deductions from the explicitly quoted Atiyah completion and character-kernel results; those external results remain cited inputs.

3. Kan–Thurston without small quotients and the ind-category statement

Statement recorded in the inventory.

A Kan–Thurston theorem without small quotients, and E.17. Every connected CW complex W has an acyclic map $BG \to W$ such that every homomorphism from G to a group of cardinality $< \lambda$ factors through $\pi_1(W)$. Hence a U-small ind-object of $\mathrm{Cat}_{\mathrm{hot}}$ is $W\infty$-invariant iff it is ind-isomorphic to an ind-object of groupoids.

Bearing on the manuscript. V ms. 46 is a theorem for "local", false for "fibrant"; VII ms. 129–132 and IX ms. 22–23: invariant path ind-categories see only Π_1

Proved by. Claude Fable 5.2, effort max (the reduction also in the earlier review, effort medium)

Checked by (inventory record). not yet; the Kan–Thurston cardinality bound was quoted from memory

Cited source. `pro_reviews_max_20261112/1_CATHOT/OWN_RESULTS.md`, section 3

Statement and proof in the supplied source

The return leaves open (E.17): if a sequence $K = (K_n)$ in $\mathrm{Cat}_{\mathrm{hot}}$ has a W∞-invariant presheaf $H_K = \mathrm{colim}_n\, \pi_0\mathrm{Fun}(-, K_n)$, is K isomorphic in $\mathrm{Ind}(\mathrm{Cat}_{\mathrm{hot}})$ to a sequence of groupoids? It reduces this to (E.17c): H_K inverts every $\eta_X\colon X \to \Pi_1 X$. The answer is yes.

Input from the return (verified): B.7 (for every group H and every infinite cardinal $\lambda \geq |H|$ there is a non-abelian simple, integrally acyclic group $A \supset H$ with $|A| = \lambda$); A.4 and A.7; C.1, C.4; (E.17a). Input from the literature: the Kan–Thurston theorem (D. M. Kan, W. P. Thurston, *Every connected space has the homology of a K(π,1)*, Topology 15 (1976), 253–258): for every connected CW complex E there are a group T and a map $BT \to E$ inducing isomorphisms on homology with every local coefficient system on E (an *acyclic* map); T can be taken of cardinality at most $\max(\aleph_0$, number of cells of E). I could not open the paper (the publisher's open archive refused automated access); the statement is quoted from memory and agrees with the summary in Palmer–Wu, arXiv:2510.16879v1, Introduction. It is used only in the successor step for cells of dimension ≥ 3 below. Standard facts on acyclic maps (Hausmann–Husemoller): the homotopy fibre of an acyclic map is acyclic; acyclic maps are stable under homotopy pull-back and composition; an acyclic map $f\colon A \to B$ of connected CW complexes is onto on π_1 with perfect kernel N, and for every space Y, $f^*\colon [B, Y] \to [A, Y]$ is injective with image the classes of maps g with $\pi_1(g)(N) = 1$.

Theorem 3.1. Let W be a connected CW complex with one 0-cell, $\pi = \pi_1(W)$, and λ an infinite cardinal not less than the number of cells of W. There are a group G with $|G| \leq \lambda$ and a map $t\colon BG \to W$ such that (i) t is acyclic (an isomorphism on homology with every local coefficient system on W); in particular $\tau = \pi_1(t)\colon G \to \pi$ is onto; (ii) every homomorphism from G to a group of cardinality $< \lambda$ factors through τ. Consequently, for every space Y all of whose fundamental groups have cardinality $< \lambda$, $t^*\colon [W, Y] \to [BG, Y]$ is a bijection.

Proof. Enumerate the cells of dimension ≥ 2 by an ordinal, in non-decreasing order of dimension: $W_1 = W^{(1)}$, $W_{\alpha+1} = W_\alpha \cup_{a_\alpha} e^{m_\alpha+1}$ with $a_\alpha\colon S^{m_\alpha} \to W_\alpha$, $m_\alpha \geq 1$, unions at limit ordinals. We build groups G_α, injective homomorphisms $G_\alpha \to G_\beta$ ($\alpha < \beta$) and maps $t_\alpha\colon BG_\alpha \to W_\alpha$, with $t_\beta|BG_\alpha = t_\alpha$, satisfying (i) and (ii) for (G_α, W_α), with $\tau_\alpha = \pi_1(t_\alpha)\colon G_\alpha \to \pi_\alpha = \pi_1(W_\alpha)$.

Start. G_1 is the free group on the 1-cells and $t_1\colon BG_1 \to W_1$ a homotopy equivalence; (ii) is empty.

Successor step. Let $a = a_\alpha\colon S^m \to W_\alpha$. Claim: there are a group $H\prime$, an acyclic map $q\colon BH\prime \to S^m$ and a homomorphism $u\colon H\prime \to G_\alpha$ with $t_\alpha \circ Bu \simeq a \circ q$. For $m = 1$: $H\prime = Z$, $q = \mathrm{id}$, $u(1) = g$ with $\tau_\alpha(g)$ the class of the loop a (τ_α is onto). For $m \geq 2$: let E be the homotopy pull-back of $S^m \to W_\alpha \leftarrow BG_\alpha$; $E \to S^m$ is acyclic and E is connected; by Kan–Thurston choose $H\prime$ and an acyclic map $BH\prime \to E$; q is the composite to S^m, and the composite $BH\prime \to E \to BG_\alpha$ is homotopic to Bu for a homomorphism u. In both cases $|H\prime| \leq \lambda$. By B.7 choose simple acyclic groups $A\prime$, $A\prime\prime$ of cardinality λ and embeddings $i\prime\colon H\prime \to A\prime$, $i\prime\prime\colon H\prime \to A\prime\prime$. Put $P = G_\alpha \times A\prime$ and $v = (u, i\prime)\colon H\prime \to P$, which is injective, and

$G_{\alpha+1} = P *_{H\prime} A\prime\prime$ (amalgamated along v and $i\prime\prime$).

P and $A\prime\prime$ embed in the amalgam, so $G_\alpha \to G_{\alpha+1}$, $g \mapsto (g, 1)$, is injective, and $BG_{\alpha+1}$ is the double mapping cylinder of $BP \leftarrow BH\prime \to BA\prime\prime$. Define $t_{\alpha+1}$ on it by $t_\alpha \circ \mathrm{pr}_1$ on BP, by the cone of q on the mapping cylinder of $BH\prime \to BA\prime\prime$ ($BA\prime\prime$ going to the centre of the new cell), and by the chosen homotopy $t_\alpha \circ Bu \simeq a \circ q$ on the other cylinder; then $t_{\alpha+1}|BG_\alpha = t_\alpha$. *Acyclicity:* for a local system L on $W_{\alpha+1}$ compare the Mayer–Vietoris sequences of the two double mapping cylinders. On BP: $H_*(G_\alpha \times A\prime; L) = H_*(G_\alpha; L)$ because $A\prime$ is acyclic and acts trivially on L (Lyndon–Hochschild–Serre), and t_α is acyclic. On $BH\prime$: q is acyclic. On $BA\prime\prime$: L is constant on the cell and $A\prime\prime$ is acyclic. The five lemma gives (i). *No small quotients:* let $\rho\colon G_{\alpha+1} \to \Lambda$ with $|\Lambda| < \lambda$. ρ is trivial on $A\prime\prime$ and on $1 \times A\prime$ (simple groups of cardinality λ). For $h \in H\prime$, $v(h) = (u(h), 1)(1, i\prime(h))$ is identified with $i\prime\prime(h) \in A\prime\prime$, so $\rho(u(h), 1) = 1$. By (ii) for G_α, $\rho|G_\alpha = \overline{\rho} \circ \tau_\alpha$, and $\overline{\rho}$ kills $\tau_\alpha(u(H\prime))$, which is trivial for $m \geq 2$ and is

the cyclic subgroup generated by the class of the attaching loop for m = 1. By van Kampen $\overline{\rho}$ factors through $\pi_{\alpha+1} = \pi_1(W_{\alpha+1})$. Since $G_{\alpha+1}$ is generated by G_α, A′, A″, and $\tau_{\alpha+1}$ is trivial on A′ and A″ and equals $G_\alpha \to \pi_\alpha \to \pi_{\alpha+1}$ on G_α, ρ factors through $\tau_{\alpha+1}$.

Limit step. $G_\gamma = \bigcup G_\alpha$, $t_\gamma = \bigcup t_\alpha$. Homology with local coefficients commutes with these unions, and a homomorphism from G_γ to Λ factors through colim $\pi_\alpha = \pi_\gamma$.

The consequence. Let $N = \ker \tau$. A map g: $BG \to Y$ induces $G \to \pi_1(Y, g(*))$, which factors through τ by (ii); so every class in [BG, Y] kills N, and the universal property of the acyclic map t gives the bijection. ■

Theorem 3.2 (E.17 is true). Let I be a U-small filtered category and K: $I \to \mathrm{Cat}_{\mathrm{hot}}$ a diagram of U-small categories. Suppose $H_K(X) = \mathrm{colim}_i\, \pi_0\mathrm{Fun}(X, K_i)$ sends every functor in W∞ between U-small categories to a bijection. Then H_K sends every unit η_X: $X \to \Pi_1 X$ to a bijection; equivalently (E.17a) the canonical morphism $K \to (\Pi_1 K_i)_{i\in I}$ is an isomorphism in $\mathrm{Ind}(\mathrm{Cat}_{\mathrm{hot}})$. So a U-small ind-object of $\mathrm{Cat}_{\mathrm{hot}}$ is W∞-invariant if and only if it is isomorphic to an ind-object of groupoids, and then $H_K(X) = \mathrm{colim}_i\, [BX, B\Pi_1 K_i]$ depends only on $\Pi_1 X$.

Proof. Let μ be an infinite cardinal with $|\mathrm{Mor}\, K_i| \le \mu$ for all i. Let $H'(Y) = \mathrm{colim}_i\, [Y, BK_i]$ on Hot. By A.7 the canonical map c_X: $H_K(X) \to H'(BX)$ is a bijection for every U-small X, naturally in X. Fix X. Choose a base object in each component c of X, let π_c be the fundamental group there, and let $\lambda > \mu$ be an infinite cardinal not less than the number of cells of BX. By 3.1 (applied to a model of the component BX_c with one 0-cell) choose groups G_c, surjections p_c: $G_c \to \pi_c$ and acyclic maps t_c: $BG_c \to BX_c$ inducing p_c, such that every homomorphism from G_c to a group of cardinality $< \lambda$ factors through p_c. Let $\mathcal{G} = \coprod_c BG_c$ and $\mathcal{P} = \coprod_c B\pi_c$, with p: $\mathcal{G} \to \mathcal{P}$ and the equivalence of groupoids ε: $\mathcal{P} \to \Pi_1 X$. (a) p^*: $\mathrm{Fun}(\mathcal{P}, K_i) \to \mathrm{Fun}(\mathcal{G}, K_i)$ is an isomorphism of categories for every i: a functor $\mathcal{G} \to K_i$ is a family of objects k_c with homomorphisms $G_c \to \mathrm{Aut}(k_c)$, and $|\mathrm{Aut}(k_c)| \le \mu < \lambda$, so these factor uniquely through p_c; a natural transformation is a family of arrows satisfying conditions that are the same for G_c and for its quotient π_c. Hence $H_K(p)$ is a bijection, and so is H′(Bp) (c is natural). (b) $t = \coprod t_c$: $B\mathcal{G} \to BX$ induces bijections $[BX, BK_i] \to [B\mathcal{G}, BK_i]$ for every i, by the last statement of 3.1 (the fundamental groups of BK_i have cardinality $\le \mu$). Hence H′(t) is a bijection. (c) $B\eta_X \circ t = B\varepsilon \circ Bp$ in Hot: both are maps from $B\mathcal{G}$ into the classifying space of a groupoid inducing the same functor on fundamental groupoids up to isomorphism, and C.1 applies. So $H'(t)\circ H'(B\eta_X) = H'(Bp)\circ H'(B\varepsilon)$ is a bijection and H′(t) is a bijection; therefore $H'(B\eta_X)$ is a bijection, and by naturality of c so is $H_K(\eta_X)$. The rest is (E.17a), that is, A.4 for the reflection $\Pi_1 \dashv j$ of C.4. ■

Corollary 3.3 (what this says about V ms. 44–46, VII ms. 129–132, IX ms. 22–23). (a) Grothendieck's sentence at V ms. 46, "cela équivaut à la condition qu'il puisse se représenter par un système inductif sur N formé de catégories de Thomason–Kan", is a theorem when "catégorie de Thomason–Kan" is read as "local category" (C.3): a sequence (K_n) has a W∞-invariant functor exactly when it is ind-isomorphic to the sequence of groupoids $(\Pi_1 K_n)$. With "Thomason-fibrant" it is false (C.6). (b) Consequently an invariant functor only sees fundamental groupoids: $H(X) = H(\Pi_1 X)$. For the path ind-categories $H(a, b) = (\mathrm{Ch}_{n,\Sigma}(a, b))_n$ of a U-small (M, Σ), the "condition préalable" of VII ms. 129–132 and test (b) of IX ms. 22, imposed on all of Cat_U, forces $\mathrm{Hom}_{M(Y)\Sigma_Y^{-1}}(a_Y, b_Y)$ to depend only on $\Pi_1 Y$; in particular it equals $\mathrm{Hom}_{M\Sigma^{-1}}(a, b)$ for every simply connected Y. (This uses Grothendieck's description of these Hom-sets as $\mathrm{colim}_n\, \pi_0\mathrm{Fun}(Y, \mathrm{Ch}_{n,\Sigma}(a, b))$, V ms. 45, which I did not re-derive.) So for small (M, Σ) the condition can hold only when the derived mapping spaces are, as ind-objects, 1-types: a precise form of his remark that the condition "n'est pratiquement jamais satisfaite" (VII ms. 132) and of his doubt at IX ms. 23. (c) The filtered case of D.8 of the return (the functor [B−, Y] is a U-small filtered colimit of representables only if Y is a 1-type) follows: if [B−, Y] = H_K, then by 3.2 the map $[P_1 Y', Y] \to [Y', Y]$ is bijective for all Y′; for $Y' = Y$ the identity of Y factors through $Y \to P_1 Y$, so Y is a retract in Hot of a 1-type, hence a 1-type. (d) The size restriction on I is necessary (D.7 of the return).

GPT-6 Astra's reading

The source supplies the amalgamated-product construction and the factorization of every homomorphism to a smaller group. It explicitly identifies two places needing a careful independent check: the successor-step comparison with local coefficients, and the cardinal bound in the Kan–Thurston input. The cited cardinal estimate was recalled from memory, and its proof is not present in the supplied files. The construction also treats compatible classifying-space models and limit unions without spelling out their replacement maps. Thus the written argument has a specific citation and model-compatibility gap at the claimed cardinal bound and its repeated use. The deduction of Theorem 3.2 from Theorem 3.1 is written out, including the functor-category isomorphism induced by $G_c \twoheadrightarrow \pi_c$, but it also cites the return's localization results A.4, A.7 and E.17a. Corollary 3.3(b) expressly relies on the manuscript's path-category formula without re-deriving it. These are the source's dependencies; this reading does not supply missing proofs or promote its conclusion to an independently checked theorem.

4. Uncountably many homotopy classes of maps to the two-sphere

Statement recorded in the inventory.

Uncountably many maps to S^2. If X is connected with $H^1(X; Z) = 0$ and S^n is a homotopy retract of X for infinitely many n with $\pi_n(S^3) \neq 0$, then $[X, S^2]$ is uncountable.
Bearing on the manuscript. D.6 of CATHOT; V ms. 43–46 (with hunch H3)

Proved by. Claude Fable 5.2, effort max
Checked by (inventory record). not yet

Cited source. `pro_reviews_max_20261112/1_CATHOT/OWN_RESULTS.md`, section 4

Statement and proof in the supplied source

Proposition 4.1. Let X be a connected CW complex with $H^1(X; Z) = 0$. Suppose that for infinitely many n with $\pi_n(S^3) \neq 0$ the sphere S^n is a homotopy retract of X: there are maps $s_n\colon S^n \to X$ and $f_n\colon X \to S^n$ with $f_n \circ s_n$ of degree ± 1. Then $[X, S^3]$ and $[X, S^2]$ are uncountable.

($\pi_n(S^3) \neq 0$ for $n = 2p$, p prime, by Serre: the p-primary component of $\pi_{2p}(S^3)$ is Z/p.)

Proof. Let N be the set of such n. Take f_n and s_n cellular for the cell structure of S^n with one 0-cell and one n-cell; then $f_n(X^{(n-1)})$ is the base point. Choose pointed maps $a_n\colon S^n \to S^3$ representing non-zero classes, with base point $1 \in S^3 = SU(2)$, and put $g_n = a_n \circ f_n$. For $A \subset N$ define $g_A(x) = \prod_{n \in A} g_n(x)$, the product in increasing order in the group S^3; on $X^{(k)}$ only the factors with $n \leq k$ differ from 1, so g_A is continuous. Let $A \neq A'$ and $n = \min(A \triangle A')$, say $n \in A$. On $X^{(n)}$: $g_A = h \cdot g_n$ and $g_{A'} = h$, with h the product of the common factors of index $< n$. If $g_A \simeq g_{A'}$, then $h \cdot g_n \simeq h$ on $X^{(n)}$, hence $g_n|X^{(n)}$ is null-homotopic (multiply pointwise by h^{-1}), hence $g_n \circ s_n = a_n \circ f_n \circ s_n$ is null-homotopic; but it represents $\pm$ the class of a_n (free and pointed classes agree in S^3). So $A \mapsto [g_A]$ is injective on the subsets of N. For S^2: the Hopf map $S^3 \to S^2$ is a principal S^1-bundle, so two maps $X \to S^3$ with homotopic images in S^2 differ, after a homotopy, by a map $X \to S^1$, which is null-homotopic because $H^1(X; Z) = 0$; thus $[X, S^3] \to [X, S^2]$ is injective. ■

GPT-6 Astra's reading

The infinite product is locally finite on every CW skeleton, so the weak topology of the CW complex supplies its continuity. The least differing index compares the two maps on the

corresponding skeleton; the cellular representative of s_n lands there, and multiplication by the inverse of the common initial product gives the stated contradiction. The Hopf-bundle lifting argument then identifies two lifts differing by a map to S^1, whose class vanishes by $H^1(X;\mathbb{Z}) = 0$. I found the proof complete from the stated hypotheses and standard cellular approximation and bundle homotopy lifting. It does not construct a finite category satisfying the retract hypothesis; the unlisted speculative finite-category consequence is not added to this appendix.

5. The thirty-four-arrow lower bound for the projective plane

Statement recorded in the inventory.

Posets of type RP² need at least 34 arrows (the return proved 28).
Bearing on the manuscript. CATHOT G.8

Proved by. Claude Fable 5.2, effort max
Checked by (inventory record). not yet

Cited source. `pro_reviews_max_20261112/1_CATHOT/OWN_RESULTS.md`, section 5

Statement and proof in the supplied source

The return (G.8) proves $28 \le a_{pos}(RP^2) \le 49$ for the least number of arrows (pairs $x \le y$) of a poset whose order complex has the homotopy type of RP^2.

Proposition 5.1. $a_{pos}(RP^2) \ge 34$.

Proof. Let P be such a poset with the least number of arrows; it is finite. As in G.8, its comparability graph has no vertex of degree ≤ 1 (a vertex with one neighbour is a beat point and can be removed), and P has at least 13 points (Cianci–Ottina, arXiv:1512.06088v1, Corollary 5.8). If P has exactly 13 points it has 49 arrows (their Theorem 5.10). Otherwise $f_0 \ge 14$. The order complex Δ is a flag complex with $\widetilde{H}_2(\Delta; F_2) \ne 0$. By the argument of G.7 over F_2 (take a full subcomplex L minimal with $\widetilde{H}_2(L; F_2) \ne 0$; Mayer–Vietoris gives $\widetilde{H}_1(\mathrm{lk}_L\, v; F_2) \ne 0$ for each vertex v of L, and links in flag complexes are flag), every vertex of L has at least 4 neighbours in L and L has at least 6 vertices. Summing degrees: $2f_1 \ge 4|L| + 2(f_0 - |L|)$, so $f_1 \ge f_0 + |L| \ge f_0 + 6$, and the number of arrows is $f_0 + f_1 \ge 2f_0 + 6 \ge 34$. ■

GPT-6 Astra's reading

The bound combines the cited thirteen-point classification with a degree count that is explicitly proved. Minimality in arrow count removes degree-one beat points. A vertex-minimal full subcomplex with nonzero second mod-two homology has at least six vertices and degree at least four at each of them; the remaining comparability vertices have degree at least two. The sum gives $f_0 + f_1 \ge 2f_0 + 6 \ge 34$ outside the thirteen-point case. The argument is complete once the cited Cianci–Ottina object bound and classification are supplied. It proves the lower bound recorded here and does not identify the global arrow minimum.

6. The reported census of small categories

Statement recorded in the inventory.

Census of small categories. The fewest arrows for finite categories of types S^1, S^2, S^3 are 4, 5, 7; S^1: only the parallel pair; S^2: only the monoid P_2; every smallest S^3 is a monoid. Grothendieck's own S^2 model (XII app. ms. 10–11) has 10 arrows.
Bearing on the manuscript. XII app. ms. 8–11

Proved by. Claude Fable 5.2, effort max (exhaustive enumeration)
Checked by (inventory record). not yet
Cited source. `pro_reviews_max_20261112/1_CATHOT/OWN_RESULTS.md`, section 6

Statement and proof in the supplied source

Statement 6.1. (a) The numbers of monoid structures on {0, …, n−1} with identity 0 are 1, 2, 11, 156, 4,122, 208,672 for n = 1, …, 6, in 1, 2, 6, 27, 156, 1,373 classes up to isomorphism and anti-isomorphism. (b) The triples (H_1, H_2, H_3) of BM that occur for monoids of order 6 are (0,0,0) [1,159 classes], (0,Z,0) [2], (0,Z,Z) [1], (0,Z,Z/2) [2], (0,Z/2,0) [1], (Z/2,0,Z/2) [172], (Z/3,0,Z/3) [22], (Z/4,0,Z/4) [6], (Z/5,0,Z/5) [1], (Z/6,0,Z/6) [1], (Z/2,0,Z/6) [1], ($(Z/2)^2$,Z/2,$(Z/2)^3$) [5]; for order 5: (0,0,0) [126], (0,Z,0) [1, the monoid P_2], (Z/2,0,Z/2) [22], (Z/3,0,Z/3) [4], (Z/4,0,Z/4) [1], (Z/5,0,Z/5) [1], ($(Z/2)^2$,Z/2,$(Z/2)^3$) [1]. (c) Connected categories with k ≥ 2 objects and N arrows, as labelled tables with arrows sorted by (source, target): N = 3: 2; N = 4: 11; N = 5: 76 + 6 (k = 2, 3); N = 6: 977 + 54; N = 7: 22,798 + 498 + 32 (k = 2, 3, 4). None has (H_1, H_2, H_3) = (0, 0, Z); none with N ≤ 5 has H_2 = Z; exactly the two labellings of the parallel pair have (Z, 0, 0) with N = 4; none with N ≤ 7 has (Z/2, 0, 0). (d) Hence: the categories of type S^1, S^2, S^3 with the fewest arrows have 4, 5, 7 arrows; for S^1 it is the parallel pair, for S^2 the monoid P_2 (unique up to isomorphism), and for S^3 every such category is a monoid of order 7 (P_3 is one). (e) The six classes of monoids of order 6 with $H_1 = 0$ and $H_2 \neq 0$ have mod-p Betti numbers in degrees 1–5 as listed in `work/code/higher_homology_order6.out`; none has the Betti numbers of CP^∞ (0,1,0,1,0).

Method: backtracking over composition tables with associativity checks (`catlib.py`), canonical forms under relabelling and transposition, exact Smith normal forms of the normalized nerve boundaries (python-flint). Checks of the method: the counts in (a) are the known ones; group homology of C_k, ${C_2}^2$, S_3; the nerves of the parallel pair, of Grothendieck's category Z of XII app. ms. 10–11 (homology of S^2), of R_2 (RP^2), of P_2 and P_3.

GPT-6 Astra's reading

The supplied file contains the census counts and the algorithm description, but the referenced programs, enumeration certificates and output files are absent from the archive. Consequently the exhaustive lower bounds and uniqueness claims cannot be reproduced or independently certified from these attachments. The homology tests described are sufficient to exclude the listed sphere types when the enumeration is exhaustive; homology alone would not certify the positive homotopy types. The source points to separately proved models P_2, P_3 and the parallel pair for existence, but does not include an enumeration certificate in this section. The asserted ten-arrow size of Grothendieck's model is recorded by NEW_RESULTS; this section's source statement does not give its complete composition table. This entry is therefore an exact account of a reported computational result with a missing computational proof record, not a fresh exhaustive verification.

7. The reversible localizer defined by two-local spheres

Statement recorded in the inventory.

A reversible fundamental localizer not detected by spectra. W_S (homotopy cofibre null for all 2-local spheres S^n, n ≥ 2) satisfies W(7), contains $RP^\infty \to e$ and not $RP^2 \to e$, while every reversible localizer containing $RP^\infty \to e$ has RP^2 in its stable closure.
Bearing on the manuscript. POLES F.14 (1); XII §5 and XVI §3: the reversibility axiom does not single out homology theories

Proved by. Claude Fable 5.2, effort max
Checked by (inventory record). not yet (prompt REVERSIBLE$_{\text{LOCALIZERS}}$ drafted)
Cited source. `pro_reviews_max_20261112/DERIVATEURS_EXPERT/RESULT_REVERSIBLE_NOT_STABLE.md`, sections 1–4

Statement and proof in the supplied source

1. Statement

Notation as in POLES: a fundamental localizer is a class W of functors satisfying identities, two out of three, the retraction rule, the terminal-object axiom and the relative Theorem A. W is *reversible* when it satisfies Loc (6′ b) (XVI ms. 17′; W(7 bis) "⇐" of XII ms. 76), equivalently (POLES B.5) when $g \in W \Leftrightarrow C(g)$ is W-aspherical, C(g) the unreduced homotopy cofibre of Bg, pointed at the cone point. For a localizing subcategory L of Ho(Sp), $R(L) = \{g : \Sigma^\infty C(g) \in L\}$ (POLES F.2), and $T(W) = \mathrm{loc}\{\Sigma^\infty C(g) : g \in W\}$.

Let S_n denote the 2-localization of the sphere S^n, and let

$$\mathfrak{N} = \{ \text{ pointed spaces } X : \mathrm{Map}_*(X, S_n) \text{ is weakly contractible for every } n \geq 2 \},$$
$$W_S = \{ \text{ functors } g : C(g) \in \mathfrak{N} \}.$$

Theorem. (a) W_S is a fundamental localizer, it is reversible (so it satisfies Loc (6 a, b, c), W(7), W(7 bis)), and it is not Fl(Cat). (b) $RP^\infty \to e$ is in W_S and $RP^2 \to e$ is not. (c) Every reversible fundamental localizer W that contains $RP^\infty \to e$ has $\Sigma^\infty RP^2 \in T(W)$, so $RP^2 \to e \in RT(W)$. Hence $W_S \neq RT(W_S)$: W_S is not of the form R(L), and in particular is not a homology localizer W_E. The same holds for every reversible localizer between the one generated by $RP^\infty \to e$ and W_S.

So the reversible cofibration axiom of Grothendieck does not force a localizer to be "stable" or homological.

2. The class $\mathfrak{N}$ is closed under desuspension

Lemma 1. If $\Sigma X \in \mathfrak{N}$ then $X \in \mathfrak{N}$.

Proof. James: for every $n \geq 1$ there is, after localization at 2, a fibre sequence $S^n \to \Omega S^{n+1} \to \Omega S^{2n+1}$ (E followed by the Hopf–James invariant H); so for $n \geq 2$ a fibre sequence

$$\Omega^2 S_{2n+1} \text{ –P–> } S_n \text{ –E–> } \Omega S_{n+1}.$$

Let $\varphi\colon X \to S_n$ be pointed. $E\varphi$ is adjoint to a map $\Sigma X \to S_{n+1}$, which is null because $\mathrm{Map}_*(\Sigma X, S_{n+1})$ is weakly contractible. So $\varphi \simeq P\psi$ with $\psi\colon X \to \Omega^2 S_{2n+1}$, and ψ is adjoint to an element of $\pi_1(\mathrm{Map}_*(\Sigma X, S_{2n+1}), \mathrm{const}) = 0$. So φ is null: $\pi_0 \mathrm{Map}_*(X, S_n) = *$. For $j \geq 1$, $\pi_j(\mathrm{Map}_*(X, S_n), \mathrm{const}) = \pi_{j-1}(\mathrm{Map}_*(\Sigma X, S_n), \mathrm{const}) = 0$. ■

Lemma 2. (i) $\mathfrak{N}$ is closed under pointed homotopy colimits and retracts and contains the contractible spaces. (ii) For a cofibre sequence $X \to Y \to C$ of pointed spaces, if two of X, Y, C are in $\mathfrak{N}$, so is the third. (iii) $X \in \mathfrak{N} \Leftrightarrow \Sigma X \in \mathfrak{N}$. (iv) If $X \in \mathfrak{N}$ then $X \wedge Y_+ \in \mathfrak{N}$ and, if also $Y \in \mathfrak{N}$, $X \times Y \in \mathfrak{N}$.

Proof. (i) $\mathrm{Map}_*(\mathrm{hocolim}\, X_i, Z) = \mathrm{holim}\, \mathrm{Map}_*(X_i, Z)$. (ii) Use the fibre sequences $\mathrm{Map}_*(C, Z) \to \mathrm{Map}_*(Y, Z) \to \mathrm{Map}_*(X, Z)$ and $\mathrm{Map}_*(\Sigma X, Z) \to \mathrm{Map}_*(C, Z) \to \mathrm{Map}_*(Y, Z)$, with $\mathrm{Map}_*(\Sigma X, Z) = \Omega\, \mathrm{Map}_*(X, Z)$: $X, C \in \mathfrak{N} \Rightarrow Y \in \mathfrak{N}$; $X, Y \in \mathfrak{N} \Rightarrow C \in \mathfrak{N}$; $Y, C \in \mathfrak{N} \Rightarrow \Sigma X \in \mathfrak{N}$ (from $C \to \Sigma X \to \Sigma Y$), and then $X \in \mathfrak{N}$ by Lemma 1. (iii) ΣX is the cofibre of $X \to *$; Lemma 1. (iv) $\mathrm{Map}_*(X \wedge Y_+, Z) = \mathrm{Map}(Y, \mathrm{Map}_*(X, Z))$; then $Y \to X \times Y \to X \wedge Y_+$. ■

3. W_S is a reversible fundamental localizer

For f: $X \to Y$, g: $Y \to Z$ there is a cofibre sequence $C(f) \to C(gf) \to C(g)$ whose connecting map is $C(g) \to \Sigma Y \to \Sigma C(f)$.

— Identities: C(id) is a cone. Two out of three: Lemma 2 (ii).

— Retraction rule. Let $ri = id_X$, $ir \in W_S$, $D = C(i)$. From $C(i) \to C(ri) \simeq * \to C(r)$: $C(r) \simeq \Sigma D$. In $C(r) \to C(ir) \to C(i)$ –δ–> $\Sigma C(r)$ the map δ factors through the connecting map ∂: $C(i) \to \Sigma X$ of i, and $\partial \simeq (\Sigma r)(\Sigma i)\partial \simeq *$ because $(\Sigma i)\partial \simeq *$. So $\Sigma C(ir) \simeq \Sigma^2 D \vee \Sigma D$. As $C(ir) \in \mathfrak{N}$, the retract ΣD is in $\mathfrak{N}$, so $D \in \mathfrak{N}$ (Lemma 1): $i \in W_S$, and $r \in W_S$ by two out of three.

— A category with a terminal object has contractible nerve.

— Relative Theorem A. For u: $A \to B$ over C with all $u/c \in W_S$: $BA \simeq \mathrm{hocolim}_c B(A/c)$, $BB \simeq \mathrm{hocolim}_c B(B/c)$ naturally, and C(Bu) is the pointed homotopy colimit over C of the $C(B(u/c))$ (the unpointed one, with the section BC of cone points collapsed). Lemma 2 (i).

— Reversible: $X \to e \in W_S \Leftrightarrow \Sigma X \in \mathfrak{N} \Leftrightarrow X \in \mathfrak{N}$ (Lemma 2 (iii)); so the W_S-aspherical categories are those with nerve in $\mathfrak{N}$ (for any base point), and $g \in W_S \Leftrightarrow C(g) \in \mathfrak{N} \Leftrightarrow C(g)$ is W_S-aspherical. By POLES B.5 this is Loc (6 b), and Loc (6 c), W(7) follow.

— $S^2 \to e \notin W_S$ (the identity of S_2), so $W_S \neq \mathrm{Fl}(\mathrm{Cat})$. Acyclic spaces are in $\mathfrak{N}$ (the S_n are simply connected, hence HZ-local), in accordance with $W_Z \subset W_S$ (POLES B.6).

4. RP^∞, RP^2, and the stable side

(b) Miller's theorem (the Sullivan conjecture): $\mathrm{Map}_*(BG, K)$ is weakly contractible for G finite and K a finite complex; for $K = S^n$ and the 2-localization S_n use the arithmetic square (RP^∞ is rationally acyclic and maps from $BZ/2$ into a simply connected space see only its 2-completion). So $RP^\infty \in \mathfrak{N}$. On the other hand $[RP^2, S_2] = H^2(RP^2; Z_{(2)}) = Z/2 \neq 0$, so $RP^2 \notin \mathfrak{N}$.

(c) Let W be reversible with $RP^\infty \to e \in W$, A its class of aspherical pointed spaces, $L = T(W)$. Then $\Sigma^\infty A \subset L$, L is a tensor ideal (POLES F.13), and: 1. $\Sigma^\infty K(Z/2, n) \in L$ for all n: $K(Z/2, n+1)$ is the bar construction of $K(Z/2, n)$, a pointed homotopy colimit of finite products, and $\Sigma^\infty(X \times Y) = \Sigma^\infty X \vee \Sigma^\infty Y \vee \Sigma^\infty(X \wedge Y)$. Hence $HZ/2 = \mathrm{colim}_n \Sigma^{-n} \Sigma^\infty K(Z/2, n) \in L$. 2. Snaith: $\Sigma^\infty Q(RP^\infty) \simeq \bigvee_j \Sigma^\infty D_j(RP^\infty)$, and $\Sigma^\infty D_j X = (\Sigma^\infty X)^{\wedge j}{}_{h\Sigma_j} \in L$. Kahn–Priddy: the 2-localization of $Q_0 S^0$ is a retract, as a space, of $Q(RP^\infty)$. So $\Sigma^\infty (Q_0 S^0)_{(2)} \in L$. Its iterated bar constructions are the spaces of the spectrum $\tau_{\geq 1} S_{(2)}$; as in 1, $\tau_{\geq 1} S_{(2)} \in L$. 3. Smashing $\tau_{\geq 1} S_{(2)} \to S_{(2)} \to HZ_{(2)}$ with the Moore spectrum: $(\tau_{\geq 1} S_{(2)})/2 \to S/2 \to HZ/2$. So $S/2 \in L$, i.e. $\Sigma^\infty RP^2 \in L$, and $RP^2 \to e \in R(L) = RT(W)$. Applied to $W = W_S$: $RP^2 \to e \in RT(W_S) \setminus W_S$. ■

Remark. Step (c) shows $\mathrm{loc}(\Sigma^\infty RP^\infty) = \mathrm{loc}(S/2)$, the 2-torsion spectra; so RT of the reversible localizer generated by $RP^\infty \to e$ is the localizer of $Z[1/2]$-homology equivalences.

GPT-6 Astra's reading

The note supplies the desuspension argument, the localizer-axiom verification and the stable-closure argument. Its decisive outside inputs are the two-local James EHP fibre sequence, Miller's theorem together with the arithmetic-square passage to two-local spheres, Kahn–Priddy in the stated space-retraction form, and Snaith's splitting. Those inputs are named but not proved here; in particular the arithmetic-square passage and the use of iterated bar constructions are compressed. I found no algebraic discrepancy in the displayed cofibre and stable-closure steps, but the precise localization and retraction formulations need independent source checking before this is described as a completed peer review. The source's accessibility

discussion is expressly unproved and lies outside the recorded theorem; it is omitted rather than turned into an additional assertion. The relation between W_S and $RT(W_S)$ is exhibited by the specific map $\mathbb{RP}^2 \to e$ and the source's explicit construction, not inferred from a categorical label.

8. The two-cone counterexample to the reverse axiom

Statement recorded in the inventory.

XII Théorème 2 (ms. 78) is false for W∞: X = A ⋆ {t, s}, A connected, acyclic, not aspherical. Everything in XII that uses the strong form of W(7 bis) holds for homology localizers only.

Bearing on the manuscript. XII Th. 4 Cor. 4 and 5, Cor. of ms. 85, Prop. 7, ms. 105, XIIapp ms. 7, 15

Proved by. Claude Fable 5.2, effort max (also TOTASPH A.13 of the return)

Checked by (inventory record). Pro return + expert agree

Cited source. `pro_reviews_max_20261112/DERIVATEURS_EXPERT/FINDINGS.md`, F-G1; `pro_reviews_max_20261112/2_TOTAL_ASPHERICITY/REVIEW.md`, A.12–A.13

Statement and proof in the supplied source

F-G1 (mine). **XII Théorème 2 (ms. 78), "W∞ satisfait à W(7)", is false**, and so is the reverse half of W(7 bis) for W∞: X = A ⋆ {t, s} with A connected, acyclic, not aspherical. Grothendieck did not write the verification ("je devrais sans doute réparer cette omission"). Everything in XII that uses "W(7 bis) forme forte" (his own footnotes 104, 105 mark these places: Th. 4 Cor. 4 and 5; also the Corollary of ms. 85, Prop. 7, and the uses at ms. 105, XIIapp ms. 7 and 15) holds for homology localizers and not for W∞. (Also in TOTASPH A.13; it is the starting point of POLES.)

Proof details recorded in the supplied TOTASPH review

Own check of the data: 3 + 17 + 15 simplices; arrows 35 + 34 + 45 + 45 = 159; nerve ranks (35, 124, 90). $d_2 = [[5, -2], [-3, 1]]$, determinant −1 (exponent sums of a^5b^{-3} and $b^3(ab)^{-2}$). In A_5: a = (1 4 3 2 5), b = (1 2 3), ab = (1 5)(3 4) (I composed them), so both relators die and $a \neq 1$. Hatcher's Example 2.38 is the binary icosahedral presentation.

X = A ⋆ {u, t}: U = A ∪ {u}, V = A ∪ {t} are sieves with contractible nerves, BX = ΣBA is contractible (simply connected and acyclic), U ∩ V = A is not. I had found the same counterexample independently while reading XII ms. 78. **This refutes XII Théorème 2 (ms. 78)** as stated; only the direct halves hold. Consequence I add: in II ms. 36ter (Contre-exemple 2.20) the step "i ∈ W∞ ⇒ X/H aspherical" is not available; one gets only that X/H is connected and acyclic. The counterexample 2.20 survives (take H connected with ΣB(Δ/H) not contractible).

GPT-6 Astra's reading

The source's two-cone construction makes the failure concrete: both cones and their union have contractible classifying spaces, while their intersection has the specified nontrivial fundamental group. The supplied TOTASPH review gives the relation matrix and an explicit nontrivial permutation representation for an acyclic example. The suspension is simply connected and acyclic, so its contractibility follows from Hurewicz and Whitehead. This proves the counterexample to the reverse axiom once the category model of the presentation

complex is supplied; the review reports that model's counts but the full attaching data are outside the cited FINDINGS paragraph. The list of later manuscript dependencies is a source attribution, not a separate proof of each later statement. The inventory adds the word "only" to its homology-localizer sentence; the cited FINDINGS paragraph says "for homology localizers and not for W_∞." The two-cone proof supplies the stated failure for W_∞ but gives no classification excluding every other localizer. The inventory's stronger word is preserved in its exact statement and flagged here as unsupported by that proof.

9. The reported trefoil argument for failure of Loc (7 F)

Statement recorded in the inventory.

A direct proof that $\mathbf{W_Z}$ and $\mathbf{W_k}$ fail Loc (7 F) (trefoil complement over a base of type S^1).

Bearing on the manuscript. XVI ms. 105 (Grothendieck: "Je ne hasarderais aucune conjecture")

Proved by. Claude Fable 5.2, effort max

Checked by (inventory record). not yet

Cited source. `pro_reviews_max_20261112/DERIVATEURS_EXPERT/FINDINGS.md`, F3-5; `pro_reviews_max_20261112/3_POLES/REVIEW.md`, C.5

Statement and proof in the supplied source

F3-5 (mine). Direct proof that $\mathbf{W_Z}$ and $\mathbf{W_k}$ fail (7 F) (trefoil complement over a base of type S^1; the Alexander module appears in H_1 of the base change). Independent of Cisinski's properness theory.

The available proof sketch

Independent confirmation for $\mathbf{W_Z}$ and $\mathbf{W_k}$, k a field (my argument before reading, `work/checks_3/own_thoughts_before_reading.md` item 7, not using E.2): if $e \to E \to S$ with $BS \simeq S^1$, E acyclic and $p \in \mathrm{Fib}_W$, the local system $s \mapsto H_0(E/s)$ must be $Z[t, t^{-1}]$, and base change along a functor $Y' \to S$ realizing the map from the trefoil complement to S^1 gives $H_1(E \times_S Y') \twoheadrightarrow Z[t, t^{-1}]/(t^2 - t + 1) \neq 0$, although $E \times_S Y' \to E$ should be a homology isomorphism. This answers XVI 7.5 (ms. 105): **Loc (6 a, 8, 9) does not imply Loc (7 F)**; and it answers the wish of XVI ms. 124 ("regarder quelques exemples (équivalences homologiques ou cohomologiques, W_n) … si Loc (7 F) est satisfait").

GPT-6 Astra's reading

The cited FINDINGS entry gives a claim and one-line description, not a full proof. The supplied POLES review adds the trefoil outline reproduced below, including the Alexander module $\mathbb{Z}[t, t^{-1}]/(t^2 - t + 1)$. Its derivation of the fibre local system, the comparison from strict categorical base change to the stated homology calculation, and the claimed surjection are not written out. The review refers for those steps to a working note absent from the archive. Thus the direct proof as supplied has a definite missing middle step. The field-coefficient version is also asserted without its corresponding derivation. This appendix preserves that limitation and does not replace the missing proof with an invented argument.

10. Totality and the fixed-origin path category

Statement recorded in the inventory.

For Y simply connected, Ch(Y; y_0, −) total ⇒ Y total.
Bearing on the manuscript. XII ms. 102–103, 112

Proved by. Claude Fable 5.2, effort max
Checked by (inventory record). not yet

Cited source. `pro_reviews_max_20261112/DERIVATEURS_EXPERT/FINDINGS.md`, F2-11

Statement and proof in the supplied source

F2-11 (mine). C005 (XII ms. 103) follows from Grothendieck's own remarks: the endpoint functor of Ch(Y; y_0, −) is final when Y is 1-connected (ms. 102–103), totality passes to final images (Lemma, ms. 112), and Ψ is aspherical and not total (his footnote at ms. 103). General form: for Y simply connected, Ch(Y; y_0, −) total ⇒ Y total. Open: the converse, starting with Y = e (the category of interval types).

GPT-6 Astra's reading

The proof is the composition of two identified manuscript facts: the endpoint functor is final for the specified simply connected target, and total asphericity passes to final images. The source identifies both manuscript locations, and the supplied review repeats the deduction. These two cited facts establish exactly the displayed implication. Their internal proofs are not reproduced in FINDINGS; this is a short proof by reference rather than a standalone construction of the endpoint category. The converse is not asserted here.

11. Exact colimits, total asphericity, and injective ordinals

Statement recorded in the inventory.

Grothendieck's conjecture of II ms. 36ter ("totally W∞-aspherical ⇒ filtered") is false (Δ_{inj}); with the bridge from exactness of abelian colimits to total asphericity (TOTASPH C.5–C.6). The category and the key fact are Isbell–Mitchell (Bull. AMS 79, 1973, Theorem 3); the bridge is not found there.
Bearing on the manuscript. II ms. 36–36ter

Proved by. GPT-6 Astra (effort Pro, ChatGPT Pro mode); proof of C.9 taken from our Open Maths lesson "Derived colimits over a small category" (20 Oct 2026)
Checked by (inventory record). Claude Fable 5.2, effort max

Cited source. `pro_returns_cited/TOTASPH_sections_C.tex`, C.1–C.10; `pro_reviews_max_20261112/2_TOTAL_ASPHERICITY/REVIEW.md`, C.5–C.10

Statement and proof in the supplied source

An intrinsic criterion and the failure of Lead 4

Source definition 1 (Proved: the action criterion)**.** *Let R be a nonzero commutative ring with identity. We write $\mathcal{P}_R(A)$ for the following property of a small category A: for every covariant functor $H : A \to \mathbf{Set}$, each connected component E of the category of elements $\mathrm{El}_A(H)$ is R-acyclic, meaning that $H_0(BE; R) = R$ and $H_j(BE; R) = 0$ for $j > 0$. No condition is imposed on the empty category of elements. We write $\mathcal{P}_\infty(A)$ when every such component is weakly contractible. These definitions use covariant, rather than presheaf, categories of elements: an arrow $(a, h) \to (a', h')$ is an arrow $v : a \to a'$ with $H(v)(h) = h'$.*

Source lemma 1 (Proved: homological Theorem A in the final variance)**.** *Let $f : A \to B$ be a functor. If every $b\backslash A$ is nonempty and R-acyclic, then $Bf : BA \to BB$ induces an isomorphism on homology with coefficients in R. More generally, if every $b\backslash A$ is nonempty and its reduced R-homology vanishes in degrees $0, \dots, d-1$, for an integer $d \geq 1$, then Bf induces isomorphisms in degrees less than d and an epimorphism in degree d.*

Proof. We use the bisimplicial set whose (p, q)-simplices are strings

$$b_0 \longrightarrow \cdots \longrightarrow b_p \longrightarrow f(a_0), \qquad a_0 \longrightarrow \cdots \longrightarrow a_q.$$

The forgetful map to NA is a weak equivalence after diagonal realization: for a fixed string in A, the remaining simplicial direction is the nerve of $B/f(a_0)$, which has terminal object $\mathrm{id}_{f(a_0)}$. The other projection is to NB. Taking normalized R-chains in both directions and filtering by p gives the spectral sequence of this projection. Its row in degree $q = 0$ is the chain complex of NB, because each comma category is connected; the rows in degrees $1, \dots, d-1$ vanish after taking vertical homology. The first possible additional row has degree d. Thus the projection induces isomorphisms below degree d and an epimorphism in degree d. The first-quadrant filtration is finite in each total degree, so it converges without a boundedness or finiteness assumption on the categories. If all positive rows vanish, it induces isomorphisms in every degree. The two projections commute with the canonical map Bf up to the homotopy supplied by the displayed arrow $b_p \to f(a_0)$; consequently the comparison just computed is the map induced by f. □

Source lemma 2 (Proved: inheritance by categories of elements)**.** *If A satisfies $\mathcal{P}_R$, then every connected component E of $\mathrm{El}_A(H)$, for every $H : A \to \mathbf{Set}$, also satisfies $\mathcal{P}_R$. The corresponding assertion holds for $\mathcal{P}_\infty$.*

Proof. The elements lying in E define a subfunctor $H_E \subseteq H$, because arrows preserve connected components. Let $K : E \to \mathbf{Set}$ be any covariant functor. Define

$$L(a) = \coprod_{h \in H_E(a)} K(a, h).$$

For an arrow $v : a \to a'$ in A, its restriction to the summand indexed by h is the map $K(v) : K(a, h) \to K(a', H(v)h)$, landing in the indicated summand of $L(a')$. This defines a covariant functor $L : A \to \mathbf{Set}$. The map

$$((a, h), k) \longmapsto (a, (h, k))$$

is an isomorphism of categories $\mathrm{El}_E(K) \cong \mathrm{El}_A(L)$, including their specified arrows. Its connected components therefore have the required property. □

Source lemma 3 (Proved: integral acyclicity detects every cyclic subgroup)**.** *Suppose that a connected, nonempty small category E satisfies $\mathcal{P}_\mathbb{Z}$. Then BE is weakly contractible. Consequently $\mathcal{P}_\mathbb{Z}(A)$ and $\mathcal{P}_\infty(A)$ are equivalent for every small category A.*

Proof. The constant singleton functor on E shows that BE is integrally acyclic. Fix an object e_0 and let ΠE be the groupoid obtained by inverting all arrows of E. The usual edge-path presentation of the fundamental group identifies $\mathrm{Aut}_{\Pi E}(e_0)$ with $G = \pi_1(BE, e_0)$. For any subgroup $K \leq G$, define

$$H_K(e) = \mathrm{Hom}_{\Pi E}(e_0, e)/K,$$

where K acts by precomposition, and an arrow $v : e \to e'$ acts by postcomposition with its image in ΠE. All these maps are bijections. The projection $\mathrm{El}_E(H_K) \to E$ is therefore a categorical covering: each arrow has a unique lift with either a prescribed source or a prescribed target. On nerves this gives the usual covering of BE whose monodromy is the transitive

coset action of G. It is connected, and its fundamental group, at the lift represented by id_{e_0}, is precisely K. The covering classification is also recorded in Hatcher (2002, Theorem 1.38). These assertions follow directly here by lifting the edge-path presentation; a loop lifts to a loop exactly when its class belongs to K.

Take K to be the cyclic subgroup generated by an arbitrary element of G. The hypothesis gives

$$0 = H_1(B\,\mathrm{El}_E(H_K);\mathbb{Z}) = K_{\mathrm{ab}} = K.$$

The identification of first homology with abelianization is Hatcher (2002, Theorem 2A.1). Thus G is trivial. A simply connected acyclic CW complex is weakly contractible: if a first nonzero homotopy group existed, the Hurewicz theorem would identify it with a nonzero homology group (Hatcher, 2002, Theorem 4.32). A nerve realization is a CW complex, so this applies to BE.

For general A satisfying $\mathcal{P}_{\mathbb{Z}}$, apply Lemma 2 and the preceding argument to each component E of each category of elements. The converse follows from homotopy invariance of integral homology. □

Source theorem 1 (Proved: characterization of total asphericity)**.** *For every small category A, the following conditions are equivalent:*

1. *A is totally W_∞-aspherical in the sense of XII, Definition 1.*
2. *A is totally $W_{\mathbb{Z}}$-aspherical.*
3. *A is nonempty and connected, and satisfies $\mathcal{P}_{\mathbb{Z}}$.*
4. *A is nonempty and connected, and satisfies $\mathcal{P}_\infty$.*

In particular, this is a criterion in terms of all covariant set-valued diagrams on A, involving no quantification over target categories or final functors. Moreover every connected component of every $\mathrm{El}_A(H)$ is itself totally W_∞-aspherical whenever A is totally W_∞-aspherical.

Proof. The implication (1)⇒(2) follows because a weak homotopy equivalence induces an integral homology isomorphism. Assume (2). The identity functor is final, so A is integrally acyclic and hence nonempty and connected. For any connected nonempty category of elements, the collage of Part B is a final image of A. Its homotopy-pushout description gives the homology exact sequence

$$\cdots \to \widetilde{H}_{j+1}(BA_H;\mathbb{Z}) \to \widetilde{H}_j(B\,\mathrm{El}_A\,H;\mathbb{Z}) \to \widetilde{H}_j(BA;\mathbb{Z}) \to \cdots .$$

Both outside terms vanish. For a component of an arbitrary H, first replace H by its component subfunctor. This proves (3). Lemma 3 proves (3)⇒(4).

Assume (4), and let $f : A \to B$ be any ordinary final functor. For $b \in B$, the category $b\backslash A$ is the category of elements of the covariant functor $a \mapsto \mathrm{Hom}_B(b, f(a))$. It is connected and nonempty by finality, and therefore weakly contractible by the hypothesis. Quillen's Theorem A (Quillen, 1973, Section 1, Theorem A) implies that the specified map Bf is a weak homotopy equivalence. The constant singleton functor in (4) makes BA weakly contractible. Thus BB is weakly contractible, proving (1) by the comprehensive-factorization criterion. The final assertion follows from Lemma 2 and (4). □

Source proposition 1 (Proved: exact integral colimits are sufficient)**.** *Let A be nonempty and connected. If $\mathrm{colim}_A : \mathbf{Ab}^A \to \mathbf{Ab}$ is exact, then A is totally W_∞-aspherical.*

Proof. For $H : A \to \mathbf{Set}$, define $\Sigma_H : \mathbf{Ab}^{\mathrm{El}_A(H)} \to \mathbf{Ab}^A$ by

$$(\Sigma_H F)(a) = \bigoplus_{h \in H(a)} F(a,h),$$

with the transition maps specified in the proof of Lemma 2. It is exact because direct sums of abelian groups are exact. The universal property of a cocone gives the canonical natural isomorphism

$$\operatorname*{colim}_{\mathrm{El}_A(H)} F \cong \operatorname*{colim}_{A} \Sigma_H F.$$

Thus the colimit functor on each category of elements is exact, and the same is true on any one of its components, by extending diagrams by zero on the other components. The bar resolution computing derived colimits identifies $L_j \operatorname{colim}_E \mathbb{Z}$ with $H_j(BE;\mathbb{Z})$: its term in degree j is the free abelian group on strings of j composable arrows of E, and its differential is the alternating sum of the nerve faces. Exactness therefore makes every component E integrally acyclic. Now apply Theorem 1. □

An explicit nonfiltered totally aspherical category

Source definition 2 (Proved: the category of injective ordinals)**.** *Let Δ_{inj} have objects $[n] = \{0, \dots, n\}$ for $n \geq 0$, and all strictly increasing functions as arrows, with ordinary composition of functions. This is a countable small category. In the next construction $\mathbb{Z}[S]$ denotes the free abelian group on a set S, and the augmentation of a finite linear combination of arrows is the sum of its coefficients.*

Historical source wording. The review corrects the attribution below to Isbell–Mitchell (1973), Theorem 3; the source wording is retained.

The exact-colimit example is attributed to Isbell (1974) in Husainov (2002, Section 2.3, after Theorem 2.5); its bibliographic record was verified, but the original article's body was not available in this run. We do not use that unavailable body as a proof. The following explicit verification uses polynomial interpolation. A CC0 AI-authored exposition containing the construction was found during the literature search (Claude Opus 5.5, 2026, Section 4); we independently verify its algebraic identities here and by the accompanying program `cd_affine_polynomials.py`. Its provenance is recorded in the report, and it is not used as an authority.

Source lemma 4 (Proved: interpolation on coordinate hyperplanes)**.** *Let $F_0, \dots, F_n$ be polynomials in $\mathbb{Z}[x_0, \dots, x_n, u]$ such that $F_i - F_j \in (x_i, x_j)$ for every pair i, j. There exists C in this polynomial ring with $C - F_i \in (x_i)$ for every i. If all F_i are homogeneous of the same degree, C can be chosen homogeneous of that degree.*

Proof. For a subset $S \subseteq \{0, \dots, n\}$ let ρ_S set $x_i = 0$ for $i \in S$. For $S \neq \varnothing$, the value $\Phi_S = \rho_S(F_i)$ is independent of the choice of $i \in S$: the difference of two choices belongs to (x_i, x_j) and is killed by ρ_S. Set

$$C = \sum_{\varnothing \neq S \subseteq \{0,\dots,n\}} (-1)^{|S|+1} \Phi_S.$$

Modulo (x_j) the terms indexed by S and $S \cup \{j\}$ cancel whenever $j \notin S$ and $S \neq \varnothing$. The remaining term is $\Phi_{\{j\}}$, congruent to F_j. All substitutions preserve homogeneity. □

Source lemma 5 (Proved: normalized affine cones)**.** *For every $n \geq 0$ there are an integer $M \geq n$ and*

$$t \in \mathbb{Z}[\mathrm{Hom}_{\Delta_{\mathrm{inj}}}([n], [M])]$$

of augmentation 1 such that $th = th'$ whenever $h, h' : [m] \to [n]$ are parallel arrows. Here composition is extended bilinearly in the free preadditive category. For $n \geq 1$ one can take $M = 2^{n+1} - 2$.

Proof. We prove a polynomial assertion simultaneously for all integers $k \geq 1$. For variables $z_0, \dots, z_{n+1}$, let D_i identify z_i and z_{i+1} and relabel the remaining variables in order as $y_0, \dots, y_n$. We claim that there is a homogeneous integral polynomial $C_{n,k}$ with value 1 when all variables are 1, such that $y_i^k D_i(C_{n,k})$ is independent of i.

For $n = 0$, use $C_{0,k} = 1$. Assume the assertion for $n-1$ and all k. Let $Q = C_{n-1,2k}$, expressed in the variables $y_0, \dots, y_n$, and prescribe

$$D_i(C_{n,k}) = F_i := Q \prod_{l \neq i} y_l^k.$$

These prescriptions agree on every pairwise intersection of the hyperplanes $z_i = z_{i+1}$. Indeed, for $i < j$, one must check $D_{j-1}(F_i) = D_i(F_j)$. If $j = i+1$, both omitted-variable products become $\prod_l w_l^k$ and the images of Q agree. If $j \geq i+2$, put $p = i$, $q = j-1$. The two expressions are respectively

$$w_q^{2k} D_q(Q) \prod_{l \neq p,q} w_l^k, \qquad w_p^{2k} D_p(Q) \prod_{l \neq p,q} w_l^k,$$

and are equal by induction. Choose homogeneous polynomial lifts of the F_i. The change of coordinates $x_i = z_i - z_{i+1}$, $u = z_{n+1}$ is invertible over $\mathbb{Z}$; the kernels of the D_i are the coordinate ideals (x_i). Lemma 4 therefore supplies $C_{n,k}$. It has value 1 at the all-ones point, and the required common expression is $Q \prod_l y_l^k$. Its degree satisfies

$$d(n,k) = d(n-1,2k) + nk = k(2^{n+1} - n - 2).$$

For an injection $g : [n] \to [M]$, let its gaps be

$$e_0 = g(0), \quad e_i = g(i) - g(i-1) - 1 \ (1 \leq i \leq n), \quad e_{n+1} = M - g(n).$$

This gives a bijection between injections and monomials of degree $M - n$ in $z_0, \dots, z_{n+1}$. Extend it linearly, writing P_t for the polynomial of an arrow combination t. If $\delta^i : [n-1] \to [n]$ omits i, deleting $g(i)$ merges two gaps and the deleted point; hence

$$P_{t\delta^i} = y_i D_i(P_t).$$

Take $P_t = C_{n,1}$ and $M = n + d(n,1)$. Its augmentation is 1, and all $t\delta^i$ agree.

We finally check that this equalizes every parallel pair, rather than just the cofaces. If their common value is s, the relation $\delta^{b+1}\delta^a = \delta^a\delta^b$ for $a \leq b$ gives $s\delta^a = s\delta^b$. Induction on n therefore shows that s equalizes all parallel injections into $[n-1]$. Every proper injection into $[n]$ factors through a coface; factoring two such injections proves $th = th'$. The identity is the only injection $[n] \to [n]$. The case $n = 0$ is immediate. □

Source theorem 2 (Refuted: total asphericity need not imply filteredness)**.** *The category Δ_{inj} is totally W_∞-aspherical and has exact colimits of abelian-group-valued diagrams, but is not filtered. Consequently Lead 4 for W_∞ and manuscript row II-C-003 have a negative answer for arbitrary small categories.*

Proof. The affine category $\mathrm{Aff}_{\mathbb{Z}}(\Delta_{\mathrm{inj}})$, defined and analyzed in Theorem D.4 of the source return, has the same objects and all integral arrow combinations of augmentation 1. Any two objects map to a common larger ordinal. Given two parallel affine arrows $\alpha, \beta : [m] \to [n]$, Lemma 5 provides t for which th is independent of the ordinary injection $h : [m] \to [n]$. Since the coefficients of both α and β sum to 1, we get $t\alpha = t\beta$. Thus the affine category is filtered. Theorem D.4 of the source return proves exactness of integral colimits. The original category is nonempty and connected, so Proposition 1 proves total W_∞-asphericity.

There are two distinct injections $a, b : [0] \to [1]$, taking 0 to 0 and 1 respectively. Every arrow $c : [1] \to [n]$ is injective, so $ca \neq cb$. Hence this parallel pair has no equalizer by postcomposition, and the category is not filtered. This verifies every hypothesis and the failure in a completely specified countable category; no finite-to-infinite inference is involved. □

The supplied reviewer on attribution and the exact-colimit step

I verified the return's proof line by line (gap monomials: $P_{t\delta^i} = y_i\, D_i(P_t)$; the compatibility $D_{j-1}(F_i) = D_i(F_j)$ in both cases; value 1 at the all-ones point; degree $k(2^{n+1} - n - 2)$), and the case $n = 1$ by hand: $t = (1,2) - (0,2) + (0,1)$, $t\delta^0 = t\delta^1 = [1]$. I also opened the 1973 note (free at ams.org), which the return did not: it states the theorem and sketches a different, combinatorial proof. The label "Proved" should read "Known (Isbell–Mitchell 1973, Th. 3); independent proof".

$\mathrm{Aff}_{\mathbb{Z}}(\Delta_{\mathrm{inj}})$ filtered $\Rightarrow$ colim exact: I checked this direction directly ($\mathrm{colim}_A\, F = \mathrm{colim}_{\mathrm{Aff}\, A}\, \widetilde{F}$, since $x - (\Sigma\, n_c\, c)x = \Sigma\, n_c\, (x - cx)$); it is the "if" part of Isbell–Mitchell's Theorem 1. Then C.6. Not filtered: d^0, d^1: $[0] \to [1]$ are never equalized. **So the conjecture Grothendieck states at II ms. 36ter is false, and the counterexample is the Isbell–Mitchell category of 1973**; what is new is the identification of total asphericity (C.5) that makes the old example apply. Independent evidence of mine: for several explicit H on Δ_{inj} I computed El(H) by hand before reading the proof (subfunctors and quotients of h^0, h^1; all components came out contractible).

GPT-6 Astra's reading

The supplied text writes out the covering-space argument that converts the category-of-elements acyclicity condition into weak contractibility, and it gives the exact direct-sum functor that transfers exact colimits to categories of elements. It then supplies the polynomial affine-cone construction and the two injections which no postcomposition can equalize. The source excerpt retains two dependencies outside its boundaries: the collage model from Part B and the affine-category exactness theorem from Part D. The supplied review checks both and gives the short colimit relation used in the latter; its text is reproduced as an accompanying source note. I found the bridge C.5–C.6 and the displayed polynomial identities internally consistent. The theorem that the affine category of injective ordinals is filtered is credited to Isbell–Mitchell, 1973, Theorem 3, as the review requires. This is an independent written proof with recorded dependencies, not a claim of priority for that theorem.

12. The projective-plane counterexample for finite monoids

Statement recorded in the inventory.

Counterexample to Conjecture 7.1 of arXiv:2610.38760v1: RP^2 is not BM for a finite monoid M (a finite monoid whose classifying space is finite-dimensional has trivial universal group).

Bearing on the manuscript. sent to the author on 12 Nov 2026

Proved by. GPT-6 Astra (effort Pro)

Checked by (inventory record). Claude Fable 5.2 (effort medium, 11 Nov; again at effort max, 12 Nov)

Cited source. `pro_returns_cited/CATHOT_sections_G.tex`, G.4 and G.6; `pro_reviews_max_20261112/1_CATHOT/G6_VERDICT.md`, theorem and verification

Statement and proof in the supplied source

Source G.4. A finite monoid cannot have a nonsimply connected finite-dimensional type *Proved*

Let M be a finite monoid. Its universal group $U(M) = \pi_1(BM)$ is finite. If BM is homotopy equivalent to a finite-dimensional CW complex, then $U(M) = 1$. In particular neither S^1 nor $\mathbb{RP}^2$ is the classifying-space type of a finite monoid.

Proof. Let $\eta : M \to U(M)$ be the universal homomorphism. For $a \in M$, finiteness gives $a^r = a^s$ for some $r < s$, so $\eta(a)^{s-r} = 1$. Every inverse of a generator is therefore a nonnegative power of that generator. It follows that every word in the generators and their inverses equals $\eta(m)$ for some $m \in M$. Thus η is surjective, and $U(M)$ is finite. The identification with $\pi_1(BM)$ follows directly from the edge-and-triangle presentation: the generators are the nonidentity elements and the relations are $[a][b] = [ab]$.

Choose an idempotent e in the minimal nonempty two-sided ideal I of M. Such an ideal exists by finiteness, and a sufficiently large power of any of its elements is idempotent. We verify that $H = eMe$, with identity e, is a group. For $x \in eMe$, minimality gives $MxM = I$, so $e = axb$ for some $a, b \in M$. Multiplying by e at both ends yields

$$e = (eae)x(ebe)$$

in the finite monoid eMe. In any finite monoid a product is invertible only if each factor is invertible: a right inverse makes right multiplication injective, hence surjective, and gives a two-sided inverse; the dual argument treats a left inverse. Thus x is invertible. The group H is finite, and the homomorphism $H \to U(M)$ is surjective since $\eta(eme) = \eta(m)$.

There is a canonical morphism $BH \to BM$ in **Hot** even though the inclusion $H \subset M$ sends 1_H to e, not necessarily to 1_M. To type this map, let $\mathrm{Kar}(M)$ be the category of idempotents of M with

$$\mathrm{Hom}_{\mathrm{Kar}(M)}(f, g) = gMf.$$

The full one-object subcategory at 1_M is M and the full one-object subcategory at e is H. Every object is a retract of 1_M. The inclusion $j : M \to \mathrm{Kar}(M)$ is a weak equivalence: for each object f, the comma category j/f is a retract of $j/1_M$, which has a terminal object. A retract of a contractible space is contractible, and Quillen's Theorem A applies. Hence the typed morphism is

$$BH \longrightarrow B\,\mathrm{Kar}(M) \xleftarrow[\simeq]{Bj} BM.$$

It induces the surjection $H \to U(M)$ on fundamental groups, after the basepoint path given by the retract arrows is chosen.

Suppose $h : BM \to Z$ is a homotopy equivalence with Z a connected finite-dimensional CW complex. Miller's theorem states that every map from the classifying space of a torsion group to Z induces the zero map on fundamental groups (Miller, 1984, Theorem B, p. 40). Apply it to the composite $BH \to BM \xrightarrow{h} Z$. Its map on fundamental groups is both zero and surjective. Consequently $\pi_1(Z) = U(M) = 1$. Notice that this step uses the precise fundamental-group theorem, with no assumption that Z was simply connected. □

Source G.6. Finite fundamental group is not sufficient for finite monoids *Refuted*

The assertion that every connected finite CW complex with finite fundamental group is the classifying space of a finite monoid is false.

Proof. Take $\mathbb{RP}^2$, whose fundamental group is $\mathbb{Z}/2$. G.4 excludes any finite monoid realization. In particular this refutes Conjecture 7.1 in the verified version of Ben-Bassat (2026); it does not refute that paper's simply connected Theorem C. The explicit finite categorical and poset models below verify that the obstruction concerns the one-object restriction. □

The supplied verdict extquotesingle s verification of the finite-group step

The proof of step 4, so that nothing is taken on authority. Let f: BG → X with G finite, X a connected CW complex of dimension d, and suppose $f_* = \varphi$ is non-trivial on π_1. Pick c ∈ G of prime-power order p^k with $\varphi(c) \neq 1$ (a primary component of an element not in the kernel), C = ⟨c⟩, N = φ(C) ≠ 1, X′ → X the covering with $\pi_1(X') = N$, f′: BC → X′ the lift, ψ a character of N of order p, L the flat line bundle of ψ on X′. Then f'^*L is the line bundle of $\chi = \psi \circ \varphi|C$ on BC, and χ has order p. In $K^0(X')$, $([L] - 1)^{d+1} = 0$, because the skeletal filtration is multiplicative and X′ has dimension d (Miller argues with Lusternik–Schnirelmann category). So $(\chi - 1)^{d+1}$ restricts to zero in K^0 of every finite skeleton of BC. By Atiyah, *Characters and cohomology of finite groups*, Publ. Math. IHES 9 (1961), Proposition (8.1) for cyclic groups (or Theorem (7.2), p. 46: $\widehat{\alpha}$: $R(G)\hat{} \to \mathcal{K}^*(B_G)$ is a topological isomorphism) and Proposition (6.11), p. 45 (for a p-group $R(G) \to R(G)\hat{}$ is a monomorphism), $(\chi - 1)^{d+1} = 0$ in R(C). But its value at c is $(\zeta - 1)^{d+1} \neq 0$ with ζ a primitive p-th root of unity. Contradiction. (Miller's own computation: $K^0(BZ_{mn}) = Z[[x]]/((1+x)^{mn} - 1)$ and the image $(1+x)^n - 1$ of x_m is not nilpotent; I verified this by mapping x to ζ − 1 in $Z_p[\zeta]$ for ζ of order p^c, p^c the exact power of a prime p | m in mn.)

GPT-6 Astra's reading

The proof specifies the exact morphism from the maximal subgroup's classifying space to that of the monoid through its Karoubi envelope, and computes the induced surjection on fundamental groups. Miller's stated theorem makes the same induced map trivial for a finite-dimensional target; the two conclusions force $U(M) = 1$. The fundamental group of $\mathbb{RP}^2$ then supplies the counterexample. I found the source proof complete from Quillen's Theorem A and Miller's stated theorem. The supplied verdict adds a K-theoretic verification of the latter finite-group case. The separately cited EMAIL_CLAIMS_CHECK file is absent, so its purported additional checks cannot be reproduced here. The refutation is tied to the conjecture as quoted in the supplied version-1 record; no assertion about a later version or publication status is added.

Statements found to be known

The inventory closes by recording the following prior references, so that these statements are not presented as priority claims.

1. The filteredness of the affine category of injective ordinals: J. R. Isbell and B. Mitchell, "Exact colimits," *Bulletin of the American Mathematical Society* **79** (1973), 994–996, Theorem 3. This is the prior theorem used in section 11; the bridge to total asphericity is the additional connection recorded there. The source review also identifies the intermediate CC0 Open Maths lesson, "Derived colimits over a small category" (20 October 2026), as the provenance of the return's polynomial proof.
2. Uniqueness in Nielsen realization: W. J. Harvey, "On branch loci in Teichmüller space," *Transactions of the American Mathematical Society* **153** (1971), 387–399. NEW_RESULTS records that the La Longue Marche expert located this reference; its proof and precise formulation are outside the supplied new-results proof files.
3. The identification recorded as "localiseur spécial" and accessible fundamental localizer: D.-C. Cisinski, *Les préfaisceaux comme modèles des types d'homotopie*, Astérisque 308 (2006), Corollary 4.2.22, as cited in NEW_RESULTS and the supplied expert FINDINGS F3-4. The exact axiom translation is not added as a new appendix theorem.

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Henning Krause. Morphisms determined by objects and flat covers. *Forum Mathematicum*, 28(3):425–435, 2016. doi: 10.1515/forum-2014-0115. URL https://www.math.uni-bielefeld.de/~hkrause/flatcover.pdf. verified: author manuscript dated 23 March 2014, p. 2 (right minimality), p. 6 (minimal weak kernels, paragraph before Proposition 1.12); publication metadata verified on the author's publications page.

Henning Krause. Homological theory of representations. Author manuscript dated 15 September 2021, 2021. URL https://www.math.uni-bielefeld.de/~hkrause/HomTheRep.pdf. verified: Lemma 1.1.6 on descent of adjunctions through localization and Lemma 2.2.1 with its proof on additive localization under finite direct-sum closure; this entry cites the dated author manuscript.

Alexander Kurz and Jiří Rosický. Weak factorizations, fractions and homotopies. *Applied Categorical Structures*, 13(2):141–160, 2005. doi: 10.1007/s10485-004-6730-z. URL https://alexhkurz.github.io/papers/WEAK/weak.pdf. verified: publication metadata in the Chapman University author repository. Definition 3.1, Observation 3.3 and Example 3.6 checked in the author preprint dated 21 October 2004, pp. 5–9; these are preprint pages, not journal pagination. The commentary uses the literal right lifting property for multigraphs, including each specified target edge, rather than the abbreviated adjacency wording in Example 3.6.

Ioannis Lagkas Nikolos. *Levelwise Modules and Localization in Derivators*. PhD thesis, University of California, Los Angeles, 2018. URL https://escholarship.org/uc/item/8

6h4g32m. verified: thesis metadata, Corollary 4.2.5, Proposition 4.2.8 and its Der2 proof on printed pages 84–86, and Remark 4.2.19 on printed page 95 in the institutional PDF.

François Lamarche. Path functors in Cat. verified, 2013. URL https://www.researchgate.net/publication/281566170_Path_Functors_in_Cat.

Tom Leinster. *Basic Category Theory*, volume 143 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2014. URL https://arxiv.org/pdf/1612.09375. verified: Exercise 6.3.23(d), printed page 168 of the author PDF, asserting that a finite category with finite products is equivalent to a complete ordered set; the version read is arXiv:1612.09375v2, 26 August 2025; the commentary supplies the finite-arrow counting proof.

Tobias Lenz. Homotopy (pre)derivators of cofibration categories and quasicategories. *Algebraic & Geometric Topology*, 18(6):3601–3646, 2018. doi: 10.2140/agt.2018.18.3601. URL https://msp.org/agt/2018/18-6/agt-v18-n6-p12-s.pdf. verified: published version, Definition 1.5 and Warning 1.7 on ABC axioms; Definition 1.33; Example 1.49; Proposition 3.32 and concluding comparison argument of Theorem 3.33; Corollary 4.1 and proof. Earlier auxiliary constructions used in the comparison were checked selectively, not re-proved.

Michael Lieberman, Jiří Rosický, and Sebastien Vasey. Sizes and filtrations in accessible categories, 2019. URL https://arxiv.org/abs/1902.06777. verified: arXiv:1902.06777v3, 5 June 2019, Definitions 2.5 and 2.10, Fact 2.11; preprint numbering.

Michael Lieberman, Jiří Rosický, and Sebastien Vasey. Sizes and filtrations in accessible categories. *Israel Journal of Mathematics*, 238(1):243–278, 2020. doi: 10.1007/s11856-020-2018-8. URL https://svasey.com/academic-homepage-may-2020/papers/lrv-st-accessible/lrv-st-accessible_v3.pdf. verified: arXiv:1902.06777v3, dated 5 June 2019, Definitions 2.1 and 2.5, Section 8 through Theorem 8.8, in particular Definition 8.1, Lemma 8.4 and its following paragraph on club agreement. Numbering in the commentary refers to this author version; journal metadata checked separately.

A. J. Lindenhovius. Grothendieck topologies on posets, 2014. URL https://arxiv.org/pdf/1405.4408v2. verified: version 2, dated 3 December 2014, Definition B.1 and Proposition B.5 with its proof, pages 49–50, for the lower Alexandrov topology. The commentary reverses the order to match the manuscript.

Zhen Lin Low. The heart of a combinatorial model category. *Theory and Applications of Categories*, 31(2):31–62, 2016. URL https://arxiv.org/abs/1402.6659. verified: Theorem 4.4, Remark 4.5, Theorem 5.9 and Proposition 5.12 in arXiv:1402.6659v4, posted 6 January 2016 (PDF title page carries 6 January 2015); published metadata checked at the journal. The cited numbering is that of this preprint.

Fernando Lucatelli Nunes and Lurdes Sousa. On lax epimorphisms and the associated factorization. *Journal of Pure and Applied Algebra*, 226(12):107126, 2022. doi: 10.1016/j.jpaa.2022.107126. URL https://arxiv.org/abs/2109.09836. verified.

Jacob Lurie. *Higher Topos Theory*, volume 170 of *Annals of Mathematics Studies*. Princeton University Press, 2009. URL https://www.math.ias.edu/~lurie/papers/HTT.pdf. verified.

Jacob Lurie. Higher algebra. Author's online manuscript, 18 September 2017, 2017. URL https://www.math.ias.edu/~lurie/papers/HA.pdf. verified: Proposition 1.1.3.4 and proof, pp. 31–32; Propositions 1.3.4.24 and 1.3.4.25, p. 119.

Michael Makkai and Robert Paré. *Accessible Categories: The Foundations of Categorical Model Theory*, volume 104 of *Contemporary Mathematics*. American Mathematical Society, 1989. doi: 10.1090/conm/104. URL https://doi.org/10.1090/conm/104. verified: metadata and publisher's description only; used for historical orientation, not as the source of an unchecked numbered theorem.

Michael Makkai and Jiří Rosický. Cellular categories. *Journal of Pure and Applied Algebra*, 218(9):1652–1664, 2014a. doi: 10.1016/j.jpaa.2014.01.005. URL https://archive.math.muni.cz/project/amega/files/data/publications/Rosicky/cc4.pdf. verified: author preprint dated 21 November 2013, Definition 2.3, Theorem 3.2 and its proof, Corollary 3.4 and its proof, and Remark 3.8; numbering refers to this preprint; journal metadata checked against publisher.

Mihály Makkai and Jiří Rosický. Cellular categories. *Journal of Pure and Applied Algebra*, 218:1652–1664, 2014b. URL https://arxiv.org/abs/1304.7572. verified: journal metadata on the arXiv record; Definition 2.1, Lemma 2.2, and Corollary 3.4 and its proof in arXiv:1304.7572v1 (29 April 2013). All result numbers cited refer to that preprint; the retract-closure argument is reconstructed explicitly in the commentary.

Georges Maltsiniotis. Structures d'asphéricité, foncteurs lisses, et fibrations. *Annales mathématiques Blaise Pascal*, 12(1):1–39, 2005. doi: 10.5802/ambp.194. URL https://www.numdam.org/article/AMBP_2005__12_1_1_0.pdf. verified.

Georges Maltsiniotis. Le théorème de Quillen, d'adjonction des foncteurs dérivés, revisité. *Comptes Rendus Mathématique*, 344(9):549–552, 2007. doi: 10.1016/j.crma.2007.03.011. URL https://webusers.imj-prg.fr/~georges.maltsiniotis/ps/adjquill.pdf. verified: author's seven-page preprint, definitions of absolute derived functors on pp. 2–3 and the main unnumbered theorem on pp. 3–4 with the construction of derived unit and counit; journal metadata checked on the author's publication list. The linked version is the longer 2006 preprint, not the four-page journal layout.

Georges Maltsiniotis. Carrés exacts homotopiques, et dérivateurs. *Cahiers de Topologie et Géométrie Différentielle Catégoriques*, 53(1):3–63, 2012. URL https://arxiv.org/abs/1101.4144. verified: arXiv:1101.4144v1, 21 January 2011, Sections 1.1 and 1.5.2–1.5.5; Propositions 2.12–2.13 and their proofs; Theorem 4.32 and its proof. The commentary uses the numbering of this 49-page preprint, not unverified pagination in the 2012 publication; publication metadata checked against the journal record.

Georges Maltsiniotis. La théorie de l'homotopie de Grothendieck. Author's expanded provisional version, March 2020, 2020. URL https://webusers.imj-prg.fr/~georges.maltsiniotis/ps/prstnewprov2020.pdf. verified.

Marco Manetti and Francesco Meazzini. Formal deformation theory in left-proper model categories. *New York Journal of Mathematics*, 25:1259–1311, 2019. URL https://nyjm.albany.edu/j/2019/25-52v.pdf. verified: published version, Definition 2.1 and Lemma 2.4 on pp. 1263–1264, and the historical attribution to Grothendieck on p. 1264; Definition 2.9 uses a different, stronger meaning of flatness.

Andrew Marks. Set theory. Author's lecture notes, updated 31 January 2026, 2026. URL https://math.berkeley.edu/~marks/notes/set_theory_notes_4.pdf. verified: Theorems 10.11–10.12, Corollary 10.13, Theorem 11.7 and Exercise 11.8, printed pages 45–49; proof of the cofinality obstruction checked.

Takao Matumoto, Norihiko Minami, and Masahiro Sugawara. On the set of free homotopy classes and Brown's construction. *Hiroshima Mathematical Journal*, 14(2):359–369, 1984. doi:

10.32917/hmj/1206133043. URL https://projecteuclid.org/euclid.hmj/1206133043. unverified.

J. Peter May, Inna Zakharevich, and Marc Stephan. The homotopy theory of equivariant posets. *Cahiers de topologie et géométrie différentielle catégoriques*, 58(2):82–114, 2017. URL https://cahierstgdc.com/wp-content/uploads/2018/01/May-al._58-2.pdf. verified: published author order and metadata; Lemma 3.6 on subdivided horn Dwyer maps, Lemma 4.7 and its full pushout proof, and Lemma 4.8 and its proof on preservation of fixed points. The author preprint has a different author order and numbers the last lemma 4.9; all citations here use the published version.

Aaron Mazel-Gee. Hammocks and fractions in relative ∞-categories. *Journal of Homotopy and Related Structures*, 13(2):321–383, 2018. doi: 10.1007/s40062-017-0184-0. URL https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/mazel-gee18.pdf. verified.

Aaron Mazel-Gee. The universality of the Rezk nerve. *Algebraic & Geometric Topology*, 19 (7):3217–3260, 2019. doi: 10.2140/agt.2019.19.3217. URL https://msp.org/agt/2019/19-7/agt-v19-n7-p01-s.pdf. verified.

Haynes Miller. The Sullivan conjecture on maps from classifying spaces. *Annals of Mathematics*, 120(1):39–87, 1984. Theorem B, p. 40; Theorem 10.1 and proof, pp. 84–85. Reference as specified in the supplied CATHOT return and verdict.

Russell Miller, Bjorn Poonen, Hans Schoutens, and Alexandra Shlapentokh. A computable functor from graphs to fields. *The Journal of Symbolic Logic*, 83(1):326–348, 2018. doi: 10.1017/jsl.2017.50. URL https://math.mit.edu/~poonen/papers/graphs-to-fields.pdf. verified: publisher metadata; Theorem 1.1, the graph-morphism convention in Section 1.A, Section 5.A, Remark 5.1, and Section 6 in the author-hosted version received 11 June 2017. The construction uses characteristic zero and preserves infinite cardinalities; Lemma 5.3's real-field argument for arbitrary cardinalities is applied to finite subgraphs containing the finitely many parameters.

James S. Milne. Fields and galois theory. Author's course notes, version 5.10, September 2022, 2022. URL https://www.jmilne.org/math/CourseNotes/FT.pdf. verified: Proposition 7.12 and Theorem 7.13 with proofs, pp. 97–98 of version 5.10; finite and infinite Galois correspondence distinguished.

John Milnor. Whitehead torsion. *Bulletin of the American Mathematical Society*, 72(3): 358–426, 1966. doi: 10.1090/S0002-9904-1966-11484-2. URL https://webhomes.maths.ed.ac.uk/~v1ranick/papers/milnorwh.pdf. verified: scan of the published article, Section 6 definition of the group Wh, Section 7 definitions of relative torsion and torsion of a homotopy equivalence, including the mapping-cylinder formulation and zero-torsion criterion. No use is made of the algebraic claims flagged by Milnor's added-in-proof warning on p. 360.

Sidney A. Morris. *Pontryagin Duality and the Structure of Locally Compact Abelian Groups*. Cambridge University Press, 1977. doi: 10.1017/CBO9780511600722. verified: Chapter 7, Proposition 38 and its proof from character separation, in the publisher's primary-text extract; chapter pp. 90–103 and publication metadata checked.

Robert Paré and Jiří Rosický. Colimits of accessible categories. *Mathematical Proceedings of the Cambridge Philosophical Society*, 155(1):47–50, 2013. URL https://arxiv.org/abs/1110.0767. verified: arXiv:1110.0767v1, Introduction, uniform strong accessibility and statement of Makkai–Paré Theorem 5.4.7; proof of Theorem 2.1 and Corollary 3.2

for simultaneous choice of ranks. Published metadata checked against the arXiv journal reference.

Dmitri Pavlov. The enriched Thomason model structure on 2-categories. *Journal of Pure and Applied Algebra*, 228(5):107496, 2024. doi: 10.1016/j.jpaa.2023.107496. URL https://arxiv.org/abs/2208.02954. verified: Definition 2.6, Proposition 2.9, and Propositions 4.1–4.3 with proofs in arXiv:2208.02954v4, 16 November 2023; author identifies v4 as the journal version except formatting and style; journal year follows the arXiv journal-reference field.

Jorge Picado and Aleš Pultr. *Frames and Locales: Topology without Points*. Frontiers in Mathematics. Birkhäuser, 2012. doi: 10.1007/978-3-0348-0154-6. URL https://www.mat.uc.pt/~picado/publicat/bookpage.html. verified: publisher and author metadata; Chapter II, Proposition 6.4.4, on sobriety of the Alexandrov space of a poset, checked in an indexed excerpt of the digitized first-edition text; finiteness argument is supplied in the commentary.

Andrew M. Pitts. On product and change of base for toposes. *Cahiers de topologie et géométrie différentielle catégoriques*, 26(1):43–61, 1985. URL https://www.numdam.org/item/CTGDC_1985__26_1_43_0.pdf. verified: Proposition 1.5, Definition 2.2, Theorem 2.3 and its proof, printed pages 47–52; the theorem identifies the tensor of Grothendieck topoi with their product in the indexed setting. Here the base is specialized to sets.

Pietro Polesello and Ingo Waschkies. Higher monodromy. *Homology, Homotopy and Applications*, 7(1):109–150, 2005. doi: 10.4310/HHA.2005.v7.n1.a7. URL https://arxiv.org/abs/math/0407507. verified: journal metadata against the journal's EMIS archive; mathematical passages checked in arXiv:math/0407507v1, Corollary 2.2.4, Theorem 2.2.5 and section 2.3, including the description of abelian gerbes by degree-two cohomology; theorem references in the commentary use this arXiv version.

Kate Ponto and Michael Shulman. The linearity of traces in monoidal categories and bicategories. *Theory and Applications of Categories*, 31(23):594–689, 2016. URL https://tac.mta.ca/tac/volumes/31/23/31-23abs.html. verified.

Leonid Positselski. Notes on limits of accessible categories. *Cahiers de Topologie et Géométrie Différentielle Catégoriques*, 65(4):390–437, 2024. URL https://cahierstgdc.com/wp-content/uploads/2024/10/Positetski-LXV-4.pdf. verified: published PDF, Corollary 5.1 with proof, Theorem 6.1, the definitions and proofs in Section 8, Theorems 8.1–8.3 and 9.1, and Corollary 9.2. The different preservation hypotheses for chain colimits, filtered colimits and presentable objects were checked. Metadata checked against the journal volume page. Numbers refer to the published version.

Leonid Positselski. On pure monomorphisms and pure epimorphisms in accessible categories. *Theory and Applications of Categories*, 45(7):246–288, 2026. URL https://tac.mta.ca/tac/volumes/45/7/45-07abs.html. verified: publisher metadata and abstract only, including publication on 30 January 2026. Screened in the current-status search because it treats pure epimorphisms; no result about all epimorphisms in arbitrary accessible categories is attributed to it.

Daniel Quillen. Higher algebraic K-theory: I. In Hyman Bass, editor, *Algebraic K-Theory I: Higher K-Theories*, volume 341 of *Lecture Notes in Mathematics*, pages 85–147. Springer, Berlin, 1973. doi: 10.1007/BFb0067053. URL https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/bass-seattle.pdf. verified.

Daniel G. Quillen. *Homotopical Algebra*, volume 43 of *Lecture Notes in Mathematics*. Springer, 1967. doi: 10.1007/BFb0097438. URL https://aareyanmanzoor.github.io/assets/books/homotopical-algebra.pdf. verified: original publication metadata and the 2022 TEXromancers retypesetting, Chapter I, Section 2, Proposition 1.2.1 and Theorem 1.2, especially equation (5) in its proof; Chapter II, Section 2, final paragraph identifying the fundamental groupoid of a simplicial mapping complex. Numbering cited is that retypesetting, not an asserted original pagination.

Andrei Rădulescu-Banu. Cofibrations in homotopy theory. verified: version 4, sections 1.3–1.4, including the Brown factorization and functor criteria in Lemmas 1.3.1 and 1.3.4 and the gluing/properness results in Lemmas 1.4.1–1.4.3; used as a modern one-sided formulation, not to add retract closure or functorial factorization to the manuscript hypotheses, 2009. URL https://arxiv.org/abs/math/0610009v4.

George Raptis. Homotopy theory of posets. *Homology, Homotopy and Applications*, 12(2): 211–230, 2010. doi: 10.4310/HHA.2010.v12.n2.a7. URL https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/raptis-poset.pdf. verified: Proposition 2.4 and the following pseudo-Dwyer discussion, Theorem 2.6 and its proof, and the noncartesian example on pp. 220–221; published version.

George Raptis. Notes on combinatorial model categories, 2020. URL https://www.graptismath.net/files/notes-comb-model-cat.pdf. verified: author-hosted notes, originally lectures in July 2014 and revised in 2020; Section 2.3, Proposition 2.7 and Theorem 2.8 read. These cite the uniformization and inverse-image results rather than giving their complete proofs.

George Raptis and Jiří Rosický. The accessibility rank of weak equivalences. *Theory and Applications of Categories*, 30(19):687–703, 2015. URL https://www.tac.mta.ca/tac/volumes/30/19/30-19abs.html. verified: journal metadata, Proposition 2.2 and the preceding pseudopullback argument, the statement of Theorem 4.2, and Corollary 5.1 with proof in arXiv:1403.3042v2, dated 12 May 2015. Proposition 2.2 of that author version is Proposition 2.3 on page 690 of the journal article. The full proof of Theorem 4.2 was not rechecked.

Olivier Renaudin. Plongement de certaines théories homotopiques de Quillen dans les dérivateurs. *Journal of Pure and Applied Algebra*, 213(10):1916–1935, 2009. doi: 10.1016/j.jpaa.2009.02.014. URL https://arxiv.org/abs/math/0603339. verified.

Charles Rezk. A model category for categories. verified: primary six-page note, Section 2 and Theorem 3.1; date and September 2000 addendum checked against the author's publication list, 1996. URL https://ncatlab.org/nlab/files/Rezk_ModelCategoryForCategories.pdf.

Charles Rezk. Fibrations and homotopy colimits of simplicial sheaves. verified: Section 2 definition of sharp maps and Propositions 2.2 and 2.7, pp. 3–4 of the author's manuscript; arXiv metadata checked, 1998. URL https://rezk.web.illinois.edu/rezk-sharp-maps.pdf.

Charles Rezk. Every homotopy theory of simplicial algebras admits a proper model. *Topology and its Applications*, 119(1):65–94, 2002. doi: 10.1016/S0166-8641(01)00057-8. URL https://arxiv.org/abs/math/0003065. verified: arXiv math/0003065v1, Example 2.11 and the adjoining-a-generator calculation; example numbering cited in the commentary is that of the preprint, not the journal version.

Emily Riehl. *Categorical Homotopy Theory*, volume 24 of *New Mathematical Monographs*. Cambridge University Press, 2014. doi: 10.1017/CBO9781107261457. URL https://emilyriehl.github.io/files/cathtpy.pdf. verified.

Emily Riehl. Category theory in context. Author's online second-edition revision; first edition published by Dover in 2016, n.d. URL https://emilyriehl.github.io/files/context.pdf. verified.

Emily Riehl and Dominic Verity. The theory and practice of Reedy categories. *Theory and Applications of Categories*, 29(9):256–301, 2014. URL https://math.jhu.edu/~eriehl/reedy.pdf. verified: author-hosted 43-page version, Proposition 6.3 and its proof, Corollary 6.7, Lemma 7.1 and Observation 7.2; journal metadata checked against the journal abstract page. The cited proposition and lemma numbers are those of the author-hosted version; additionally checked: published version dated 13 June 2014, read from https://emilyriehl.github.io/files/reedy.pdf (47 PDF pages, distinct from the 43-page April preprint at math.jhu.edu); Definition 4.14, Observation 5.1, Example 5.2, Definition 5.3, and Proposition 6.3, including the relative latching formula, composition identity, continuity convention, and cellular presentation. Journal pagination, rather than the earlier April preprint pagination, is used.

Emily Riehl and Dominic Verity. Kan extensions and the calculus of modules for ∞-categories. *Algebraic & Geometric Topology*, 17(1):189–271, 2017. doi: 10.2140/agt.2017.17.189. URL https://emilyriehl.github.io/files/equipment.pdf. verified.

Emily Riehl and Dominic Verity. *Elements of ∞-Category Theory*, volume 194 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2022. doi: 10.1017/9781108936880. URL https://emilyriehl.github.io/files/elements.pdf. verified: author's prepublication PDF, Lemma 1.2.19 and its proof, Appendix C.1, especially Lemmas C.1.10–C.1.11; numbering is that of this PDF; published metadata and DOI checked against the Cambridge book page and front matter.

Jiří Rosický. On combinatorial model categories. *Applied Categorical Structures*, 17:303–316, 2009. doi: 10.1007/s10485-008-9171-2. URL https://www.math.muni.cz/~rosicky/papers/comb2.pdf. verified: author manuscript, Example 3.5(1), printed manuscript page 8, on split monomorphisms of posets as a full accessible image; publisher metadata checked separately. Example numbering cited is from the author manuscript.

Jiří Rosický. Accessible model categories. *Applied Categorical Structures*, 25(2):187–196, 2017. doi: 10.1007/s10485-015-9419-6. URL https://arxiv.org/pdf/1503.05010v2. verified: Example 2.8 and its proof, Definition 4.1 and Remark 4.2 in arXiv v2 (22 July 2015), with published volume, pages and DOI checked on the publisher's page. The non-generation proof cites Christensen–Hovey; that cited theorem was not independently re-proved here.

Jiří Rosický. Remarks on combinatorial and accessible model categories. *Theory and Applications of Categories*, 37(9):266–275, 2021. URL https://tac.mta.ca/tac/volumes/37/9/37-09.pdf. verified: published revision of 18 May 2021, Introduction and Section 3, especially Remarks 3.2 and 3.5, Examples 3.7 and Theorem 3.9; the distinction between full images and accessible arrow subcategories was checked against the corrected version.

Jonathan Rubin. Elmendorf constructions for G-categories and G-posets. verified: manuscript dated December 30, 2020, Section 2.3, Theorem 2.6 and the following explicit natural comparison. The theorem is attributed there to Thomason (1979), Theorem 1.2. This entry identifies the version actually consulted, rather than assuming the numbering of the subsequent publication, 2020. URL https://par.nsf.gov/servlets/purl/10393703.

Helmut H. Schaefer and Manfred P. Wolff. *Topological Vector Spaces*, volume 3 of *Graduate Texts in Mathematics*. Springer, New York, 2 edition, 1999. doi: 10.1007/978-1-4612-1468-7. URL https://link.springer.com/book/10.1007/978-1-4612-1468-7. verified: Chapter IV, section 1, Theorem 1.5, p. 126, and its separation proof in the indexed primary text; publisher metadata checked; the full text was not consulted.

Stefan Schwede and Brooke Shipley. Equivalences of monoidal model categories. *Algebraic & Geometric Topology*, 3:287–334, 2003. doi: 10.2140/agt.2003.3.287. URL https://homepages.math.uic.edu/~bshipley/monoidalequi.final.pdf. verified: Section 2.3 of the author version dated 11 February 2003, the shuffle and Alexander–Whitney identities and homotopies, and equation (2.8) giving the monoidal comparison for denormalization; publication metadata also checked.

Christian Serpé. Resolution of unbounded complexes in Grothendieck categories. *Journal of Pure and Applied Algebra*, 177(1):103–112, 2003. doi: 10.1016/S0022-4049(02)00075-0. URL https://www.sciencedirect.com/science/article/pii/S0022404902000750. verified: publisher metadata and abstract only; the general existence theorem was independently checked in the Stacks Project, Tag 079P, and Beke's Proposition 3.13.

Jean-Pierre Serre. Homologie singulière des espaces fibrés: Applications. *Annals of Mathematics*, 54(3):425–505, 1951. doi: 10.2307/1969485. URL https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/Serre-HSEF.pdf. verified: original published scan; introduction pp. 425–426, Chapter II section 8 Proposition 6 on the fibre-cohomology local system, Chapter IV sections 1 and 5, Proposition 7 on contractibility of based paths; Chapter IV section 3 Proposition 3 and its proof on H-space simplicity.

Jean-Pierre Serre. Faisceaux algébriques cohérents. *Annals of Mathematics*, 61(2):197–278, 1955. doi: 10.2307/1969915. URL https://www.college-de-france.fr/media/jean-pierre-serre/UPL5435398796951750634_Serre_FAC.pdf. verified.

Boris Shoikhet. On Evrard's homotopy fibrant replacement of a functor. *Theory and Applications of Categories*, 31:989–1015, 2016. URL https://arxiv.org/abs/1412.0317. verified: arXiv:1412.0317v4, 7 November 2016, Theorems 1.8, 1.9 and 2.8, Proposition 1.14, Sections 1.5 and 2.1; numbering cited is that of this version.

Michael Shulman. Flat functors and morphisms of sites. Author's post, The n-Category Café, June 2011, 2011. URL https://golem.ph.utexas.edu/category/2011/06/flat_functors_and_morphisms_of.html. verified: indexed text of the author's opening discussion distinguishing Set-valued flatness and representable flatness; full page retrieval timed out, so no further claims rely on its contents.

Michael Shulman. Reedy categories and their generalizations, 2015. URL https://home.sandiego.edu/~shulman/papers/reedy.pdf. verified: author-hosted version dated 29 June 2015, Definition 4.6, Theorem 4.7 and its proof, and Remark 4.8, pp. 15–16; the ordinal-valued degree convention in the introduction was also checked. No universal monomorphism-recognition theorem for posets was found in these inspected passages.

Michael Shulman. The derivator of setoids. *Cahiers de Topologie et Géométrie Différentielle Catégoriques*, 64(1):29–96, 2023. URL https://cahierstgdc.com/wp-content/uploads/2023/01/SHULMAN-LXIV-1.pdf. verified: publisher issue metadata and the discussion on printed p. 33 identifying the classical truth-value derivator with the two-element lattice and with homotopy minus-one-types; this citation is not an attribution of our free-cycle counterexample.

Nicolas Spaltenstein. Resolutions of unbounded complexes. *Compositio Mathematica*, 65(2): 121–154, 1988. URL https://www.numdam.org/article/CM_1988__65_2_121_0.pdf. verified: introduction, definitions of K-projective and K-injective complexes, Theorems C and D; these concern modules and sheaves, not arbitrary abelian categories.

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